GenESyS User Documentation

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1 Editing an EFSM Network in the graphical interface 1

1.1 Create or attach a Network . . . 1
1.2 Add states and transitions . . . 1
1.3 Edit adapter bindings . . . 2
1.4 Save and reopen . . . 2
1.5 Current scope . . . 2

2 Cellular Automata Viewer 3

3 GenESyS Launcher and Per-User Runtime Dispatcher 5

3.1 Purpose . . . 5
3.2 Architecture . . . 5
3.3 CMake targets and preset . . . 6
3.4 Install contract for future packaging . . . 6
3.5 XDG directories . . . 7
3.6 Configuration and precedence . . . 7
3.7 Application selection . . . 8
3.8 Active runtime validation . . . 8
3.9 <tt>genesys-worker</tt> and legacy <tt>genesys-web</tt> . . . 8
3.10 MCP . . . 9
3.11 Version rules . . . 9
3.12 Platform detection . . . 9
3.13 Remote manifest schema . . . 9
3.14 Network behavior . . . 10
3.15 Integrity and signature verification . . . 10
3.16 Secure archive extraction . . . 11
3.17 Runtime validation and atomic activation . . . 11
3.18 Concurrency and retention . . . 12
3.19 Logging . . . 12
3.20 Launcher CLI . . . 12
3.21 Failure and fallback behavior . . . 12
3.22 Tests . . . 13
3.23 Known limitations and future Debian integration . . . 13

4 Genesys MCP Connector 15

4.1 Prerequisites . . . 15
4.2 Install . . . 15
4.3 Run . . . 15
4.4 Claude Desktop integration . . . 15
4.5 Claude Code integration . . . 16
4.6 Workflow . . . 16
4.7 Available tools (18) . . . 16

5 Genesys Worker Application 17

5.1 Current scope . . . 17
5.2 Worker cycle Stage 1 (new cycle) . . . 17
5.3 Worker cycle Stage 2 . . . 18
5.4 Worker cycle Stage 3 . . . 18
5.5 Worker cycle Stage 4 . . . 18
5.6 Worker cycle Stage 5 . . . 19
5.7 Worker cycle Stage 6 . . . 19
5.8 How to run . . . 19
5.9 Security notice . . . 20

6 Customization Points **21**
6.1 Header <tt>gtest.h</tt> . . . 21
6.1.1 The following macros can be defined: . . . 21
6.2 Header <tt>gtest-port.h</tt> . . . 21
6.2.1 Logging: . . . 21
6.2.2 Threading: . . . 21
6.2.3 Underlying library support features . . . 22
6.2.4 Exporting API symbols: . . . 22
6.3 Header <tt>gtest-printers.h</tt> . . . 22

7 Análise técnica do parser (Bison/Flex) **23**
7.1 Audit Status (WiP20261) . . . 23
7.2 Escopo . . . 23
7.3 Registro histórico (pré-reauditoria) . . . 23
7.4 Reclassificação dos achados históricos . . . 23
7.4.1 Conversão hexadecimal . . . 23
7.4.2 Audit status . . . 23
7.4.3 Evidence . . . 24
7.4.4 Remaining gaps . . . 24
7.4.5 Tokenização de <tt>min</tt> / <tt>max</tt> . . . 24
7.4.6 Audit status . . . 24
7.4.7 Evidence . . . 24
7.4.8 Remaining gaps . . . 24
7.4.9 Regex de <tt>EntitiesWIP</tt> . . . 24
7.4.10 Audit status . . . 24
7.4.11 Evidence . . . 24
7.4.12 Remaining gaps . . . 24
7.4.13 Regra isolada <tt>T</tt> no lexer . . . 24
7.4.14 Audit status . . . 24
7.4.15 Evidence . . . 25
7.4.16 Remaining gaps . . . 25
7.4.17 Token <tt>ILLEGAL</tt> antes inútil . . . 25
7.4.18 Audit status . . . 25

7.4.19 Evidence . . . 25
7.4.20 Remaining gaps . . . 25
7.4.21 Estrutura de precedência/associatividade da gramática . . . 25
7.4.22 Audit status . . . 25
7.4.23 Evidence . . . 25
7.4.24 Remaining gaps . . . 25
7.5 Pontos ainda pendentes (ou parcialmente mitigados) . . . 26
7.5.1 Funções semânticas não implementadas totalmente (<tt>fDISC</tt>, <tt>fLASTINQ</tt>, <tt>fRESSEIZES</tt>) . . . 26
7.5.2 Audit status . . . 26
7.5.3 Evidence . . . 26
7.5.4 Remaining gaps . . . 26
7.5.5 Avaliação de <tt>FORM</tt> . . . 26
7.5.6 Audit status . . . 26
7.5.7 Evidence . . . 26
7.5.8 Remaining gaps . . . 26
7.5.9 Risco de null-deref em atribuições/leituras ligadas a evento atual . . . 26
7.5.10 Audit status . . . 26
7.5.11 Evidence . . . 27
7.5.12 Remaining gaps . . . 27
7.5.13 Conflitos LR atuais . . . 27
7.5.14 Audit status . . . 27
7.5.15 Evidence . . . 27
7.5.16 Remaining gaps . . . 27
7.6 Recomendações arquiteturais que seguem válidas . . . 27
7.7 Remaining Work . . . 27

8 Kernel Simulator Unit Tests - Build & Run Guide **29**
8.1 What to use (recommended) . . . 29
8.2 CLion setup (single-click workflow) . . . 29
8.3 Command line equivalent . . . 30
8.4 Executables included in the run target . . . 30
8.5 Notes . . . 30

9 Architecture notes for <tt>source/tools</tt> **31**
9.1 1. Architectural overview . . . 31
9.2 2. Current package responsibilities . . . 31
9.3 3. Approved target decomposition . . . 31
9.3.1 Data analysis . . . 31
9.3.2 Fitting . . . 31
9.3.3 Hypothesis testing . . . 31
9.3.4 Distribution abstractions . . . 32
9.3.5 Numerical tools . . . 32

9.3.6 Traits/binding . . . 32
9.4 4. Legacy compatibility strategy . . . 32
9.5 5. Planned migration order . . . 32
9.6 6. AI Assistant component . . . 32
9.6.1 Role . . . 32
9.6.2 Stage 1 (complete) — Interface and implementation scaffold . . . 33
9.6.3 Stage 2 (complete) — Plugin introspection . . . 33
9.6.4 Stage 3 (complete) — AI provider HTTP clients . . . 33
9.6.5 Stage 4 (complete) — Prompt analysis and model plan generation . . . 33
9.6.6 Stage 5 (complete) — Model construction . . . 34
9.6.7 Stage 6 (complete) — Simulation execution and result collection . . . 34
9.6.8 Planned stages . . . 34
9.6.9 Dependency constraints . . . 34
9.7 7. Stability notes and non-goals for this phase . . . 34

10 tools package **35**
10.1 1. Purpose of the tools package . . . 35
10.2 2. Current major abstractions . . . 35
10.3 3. Current limitations . . . 35
10.4 4. Planned evolution . . . 35
10.5 5. Relationship with kernel/statistics . . . 36
10.6 6. AI Assistant abstraction . . . 36
10.7 7. Current status by topic . . . 36

11 Todo List **37**

12 Namespace Index **39**
12.1 Namespace List . . . 39

13 Hierarchical Index **41**
13.1 Class Hierarchy . . . 41

14 Class Index **65**
14.1 Class List . . . 65

15 File Index **93**
15.1 File List . . . 93

16 Namespace Documentation **105**
16.1 EFSMNetworkDiagramCodec Namespace Reference . . . 105
16.1.1 Detailed Description . . . 105
16.2 generate_kernel_simulator_test_stubs Namespace Reference . . . 105
16.2.1 Detailed Description . . . 106
16.3 genesys_mcp Namespace Reference . . . 106
16.3.1 Detailed Description . . . 106

16.4 genesys_mcp.__main__ Namespace Reference . . . 106
16.4.1 Detailed Description . . . 106
16.5 genesys_mcp.client Namespace Reference . . . 106
16.5.1 Detailed Description . . . 106
16.6 genesys_mcp.server Namespace Reference . . . 106
16.6.1 Detailed Description . . . 107
16.6.2 Function Documentation . . . 107
16.6.2.1 genesys_auth() . . . 107
16.6.2.2 genesys_configure_simulation() . . . 107
16.6.2.3 genesys_create_job() . . . 108
16.6.2.4 genesys_get_model() . . . 108
16.6.2.5 genesys_health() . . . 108
16.6.2.6 genesys_import_model_language() . . . 108
16.6.2.7 genesys_info() . . . 109
16.6.2.8 genesys_job_result() . . . 109
16.6.2.9 genesys_job_status() . . . 109
16.6.2.10 genesys_load_model() . . . 109
16.6.2.11 genesys_new_model() . . . 109
16.6.2.12 genesys_run_job() . . . 110
16.6.2.13 genesys_run_simulation() . . . 110
16.6.2.14 genesys_save_model() . . . 110
16.6.2.15 genesys_simulation_status() . . . 110
16.6.2.16 genesys_step_simulation() . . . 110
16.6.2.17 genesys_worker_capabilities() . . . 111
16.6.2.18 genesys_worker_info() . . . 111
16.6.3 Variable Documentation . . . 111
16.6.3.1 mcp . . . 111
16.7 GuiCrashDiagnostics Namespace Reference . . . 111
16.7.1 Detailed Description . . . 111
16.8 QCP Namespace Reference . . . 111
16.8.1 Detailed Description . . . 112
16.8.2 Enumeration Type Documentation . . . 112
16.8.2.1 AntialiasedElement . . . 112
16.8.2.2 ExportPen . . . 113
16.8.2.3 Interaction . . . 113
16.8.2.4 MarginSide . . . 114
16.8.2.5 PlottingHint . . . 114
16.8.2.6 ResolutionUnit . . . 114
16.8.2.7 SelectionRectMode . . . 115
16.8.2.8 SelectionType . . . 115
16.8.2.9 SignDomain . . . 115

17 Class Documentation **117**

17.1 Access Class Reference . . . 117
17.1.1 Detailed Description . . . 122
17.1.2 of Cells Number of contiguous conveyor cells the entity requires for . . . 122
17.1.3 Member Function Documentation . . . 122
17.1.3.1 _check() . . . 122
17.1.3.2 _loadInstance() . . . 122
17.1.3.3 _onDispatchEvent() . . . 123
17.1.3.4 _saveInstance() . . . 123
17.1.3.5 show() . . . 123
17.2 AddUndoCommand Class Reference . . . 124
17.3 AIAssistant Class Reference . . . 125
17.3.1 Member Function Documentation . . . 130
17.3.1.1 _check() . . . 130
17.3.1.2 _loadInstance() . . . 130
17.3.1.3 _onDispatchEvent() . . . 130
17.3.1.4 _saveInstance() . . . 130
17.3.1.5 show() . . . 131
17.4 AIAssistant_if Class Reference . . . 131
17.4.1 Detailed Description . . . 133
17.4.1.1 Security model . . . 134
17.4.1.2 Planned evolution . . . 134
17.4.1.3 Thread safety . . . 134
17.4.2 Member Function Documentation . . . 134
17.4.2.1 addKnowledgeItem() . . . 134
17.4.2.2 analyzePrompt() . . . 134
17.4.2.3 buildModel() . . . 134
17.4.2.4 checkReady() . . . 134
17.4.2.5 clearApiKey() . . . 135
17.4.2.6 clearDiagnostics() . . . 135
17.4.2.7 clearKnowledgeBase() . . . 135
17.4.2.8 collectResults() . . . 135
17.4.2.9 configureSimulation() . . . 135
17.4.2.10 createModelPlan() . . . 136
17.4.2.11 execute() . . . 136
17.4.2.12 exportAuditLog() . . . 136
17.4.2.13 exportKnowledgeBaseAsMarkdown() . . . 136
17.4.2.14 getApiKeyEnvironmentVariable() . . . 136
17.4.2.15 getAuditEntries() . . . 136
17.4.2.16 getAuditLogPath() . . . 137
17.4.2.17 getBaseUrl() . . . 137
17.4.2.18 getConfiguration() . . . 137

17.4.2.19 getKnowledgeBaseSummary() . . . 137
17.4.2.20 getKnowledgeItems() . . . 137
17.4.2.21 getLastDiagnostics() . . . 137
17.4.2.22 getLastError() . . . 137
17.4.2.23 getMaskedApiKey() . . . 137
17.4.2.24 getMaxOutputTokens() . . . 138
17.4.2.25 getModelName() . . . 138
17.4.2.26 getProvider() . . . 138
17.4.2.27 getProviderClient() . . . 138
17.4.2.28 getProviderName() . . . 138
17.4.2.29 getReasoningEffort() . . . 138
17.4.2.30 getSimulatorFacade() . . . 138
17.4.2.31 getState() . . . 138
17.4.2.32 getTemperature() . . . 138
17.4.2.33 getTimeoutSeconds() . . . 138
17.4.2.34 hasApiKey() . . . 139
17.4.2.35 hasProviderClient() . . . 139
17.4.2.36 hasSimulatorFacade() . . . 139
17.4.2.37 loadApiKeyFromEnvironment() . . . 139
17.4.2.38 refreshKnowledgeBaseFromSimulator() . . . 139
17.4.2.39 runSimulation() . . . 139
17.4.2.40 setApiKey() . . . 140
17.4.2.41 setApiKeyEnvironmentVariable() . . . 140
17.4.2.42 setBaseUrl() . . . 140
17.4.2.43 setConfiguration() . . . 140
17.4.2.44 setMaxOutputTokens() . . . 140
17.4.2.45 setModelName() . . . 140
17.4.2.46 setProvider() . . . 141
17.4.2.47 setProviderClient() . . . 141
17.4.2.48 setProviderName() . . . 141
17.4.2.49 setReasoningEffort() . . . 141
17.4.2.50 setSimulatorFacade() . . . 141
17.4.2.51 setTemperature() . . . 141
17.4.2.52 setTimeoutSeconds() . . . 141
17.5 AIAssistantConfiguration Struct Reference . . . 142
17.5.1 Detailed Description . . . 142
17.5.2 Member Data Documentation . . . 143
17.5.2.1 dryRun . . . 143
17.5.2.2 sandboxEnabled . . . 143
17.6 AIAssistantDefaultImpl Class Reference . . . 143
17.6.1 Detailed Description . . . 146
17.6.1.1 Security model . . . 146

17.6.1.2 Ownership . . . 146
17.6.2 Member Function Documentation . . . 147
17.6.2.1 addKnowledgeItem() . . . 147
17.6.2.2 analyzePrompt() . . . 147
17.6.2.3 buildModel() . . . 147
17.6.2.4 checkReady() . . . 147
17.6.2.5 clearApiKey() . . . 148
17.6.2.6 clearDiagnostics() . . . 148
17.6.2.7 clearKnowledgeBase() . . . 148
17.6.2.8 collectResults() . . . 148
17.6.2.9 configureSimulation() . . . 148
17.6.2.10 createAndAttachProviderClient() . . . 148
17.6.2.11 createModelPlan() . . . 149
17.6.2.12 execute() . . . 149
17.6.2.13 exportAuditLog() . . . 149
17.6.2.14 exportKnowledgeBaseAsMarkdown() . . . 149
17.6.2.15 getApiKeyEnvironmentVariable() . . . 149
17.6.2.16 getAuditEntries() . . . 150
17.6.2.17 getAuditLogPath() . . . 150
17.6.2.18 getBaseUrl() . . . 150
17.6.2.19 getConfiguration() . . . 150
17.6.2.20 getKnowledgeBaseSummary() . . . 150
17.6.2.21 getKnowledgeItems() . . . 150
17.6.2.22 getLastDiagnostics() . . . 150
17.6.2.23 getLastError() . . . 150
17.6.2.24 getMaskedApiKey() . . . 151
17.6.2.25 getMaxOutputTokens() . . . 151
17.6.2.26 getModelName() . . . 151
17.6.2.27 getProvider() . . . 151
17.6.2.28 getProviderClient() . . . 151
17.6.2.29 getProviderName() . . . 151
17.6.2.30 getReasoningEffort() . . . 151
17.6.2.31 getSimulatorFacade() . . . 151
17.6.2.32 getState() . . . 151
17.6.2.33 getTemperature() . . . 151
17.6.2.34 getTimeoutSeconds() . . . 152
17.6.2.35 hasApiKey() . . . 152
17.6.2.36 hasProviderClient() . . . 152
17.6.2.37 hasSimulatorFacade() . . . 152
17.6.2.38 loadApiKeyFromEnvironment() . . . 152
17.6.2.39 refreshKnowledgeBaseFromSimulator() . . . 152
17.6.2.40 runSimulation() . . . 153

17.6.2.41 setApiKey() . . . 153
17.6.2.42 setApiKeyEnvironmentVariable() . . . 153
17.6.2.43 setBaseUrl() . . . 153
17.6.2.44 setConfiguration() . . . 153
17.6.2.45 setMaxOutputTokens() . . . 153
17.6.2.46 setModelName() . . . 154
17.6.2.47 setProvider() . . . 154
17.6.2.48 setProviderClient() . . . 154
17.6.2.49 setProviderName() . . . 154
17.6.2.50 setReasoningEffort() . . . 154
17.6.2.51 setSimulatorFacade() . . . 154
17.6.2.52 setTemperature() . . . 154
17.6.2.53 setTimeoutSeconds() . . . 155
17.7 AIAssistantRequest Struct Reference . . . 155
17.7.1 Detailed Description . . . 155
17.7.2 Member Data Documentation . . . 155
17.7.2.1 modelPlan . . . 155
17.8 AIAssistantResponse Struct Reference . . . 156
17.8.1 Detailed Description . . . 156
17.9 AIAssistantWindow Class Reference . . . 156
17.10 AIAuditLog Class Reference . . . 157
17.10.1 Detailed Description . . . 157
17.10.2 Member Function Documentation . . . 158
17.10.2.1 exportCsv() . . . 158
17.10.2.2 loadRecentEntries() . . . 158
17.11 AIConversationConfiguration Struct Reference . . . 158
17.12 AIConversationResponse Struct Reference . . . 158
17.13 AIConversationService Class Reference . . . 159
17.14 AIProviderClient_if Class Reference . . . 159
17.14.1 Detailed Description . . . 160
17.14.1.1 Security contract . . . 160
17.14.1.2 Ownership . . . 160
17.14.2 Member Function Documentation . . . 160
17.14.2.1 clearApiKey() . . . 160
17.14.2.2 getBaseUrl() . . . 160
17.14.2.3 getProviderName() . . . 160
17.14.2.4 getTimeoutSeconds() . . . 160
17.14.2.5 hasApiKey() . . . 160
17.14.2.6 isAvailable() . . . 160
17.14.2.7 send() . . . 161
17.14.2.8 setApiKey() . . . 161
17.14.2.9 setBaseUrl() . . . 161

17.14.2.10 setTimeoutSeconds() . . . 161
17.15 Airport_Extended1 Class Reference . . . 161
17.15.1 Member Function Documentation . . . 163
17.15.1.1 main() . . . 163
17.16 AirportSecurityExample Class Reference . . . 163
17.16.1 Member Function Documentation . . . 164
17.16.1.1 main() . . . 164
17.17 AirportSecurityExampleExtended Class Reference . . . 165
17.17.1 Member Function Documentation . . . 166
17.17.1.1 main() . . . 166
17.18 AISecretStore Class Reference . . . 166
17.18.1 Detailed Description . . . 166
17.18.1.1 Usage . . . 167
17.18.2 Member Function Documentation . . . 167
17.18.2.1 isAvailable() . . . 167
17.18.2.2 load() . . . 167
17.18.2.3 remove() . . . 167
17.18.2.4 save() . . . 167
17.19 AISupport Class Reference . . . 168
17.19.1 Member Function Documentation . . . 172
17.19.1.1 _check() . . . 172
17.19.1.2 _initBetweenReplications() . . . 172
17.19.1.3 _loadInstance() . . . 172
17.19.1.4 _saveInstance() . . . 172
17.19.1.5 show() . . . 172
17.20 AND Class Reference . . . 173
17.20.1 Member Function Documentation . . . 179
17.20.1.1 _loadInstance() . . . 179
17.20.1.2 _saveInstance() . . . 179
17.20.1.3 Build() . . . 179
17.21 AnElectronicAssemblyAndTestSystem Class Reference . . . 179
17.21.1 Member Function Documentation . . . 181
17.21.1.1 main() . . . 181
17.22 AnimationAttribute Class Reference . . . 181
17.23 AnimationCounter Class Reference . . . 182
17.24 AnimationEntity Class Reference . . . 183
17.25 AnimationEvent Class Reference . . . 185
17.26 AnimationExpression Class Reference . . . 186
17.27 AnimationPlaceholder Class Reference . . . 187
17.28 AnimationPlot Class Reference . . . 188
17.29 AnimationQueue Class Reference . . . 190
17.30 AnimationQueueDisplay Class Reference . . . 190

17.31 AnimationResource Class Reference . . . 192
17.32 AnimationStation Class Reference . . . 193
17.33 AnimationStatistics Class Reference . . . 194
17.34 AnimationTimer Class Reference . . . 195
17.35 AnimationTransition Class Reference . . . 196
17.36 AnimationVariable Class Reference . . . 198
17.37 AnthropicProviderClientImpl Class Reference . . . 199
17.37.1 Detailed Description . . . 201
17.37.1.1 API key . . . 201
17.37.2 Member Function Documentation . . . 201
17.37.2.1 _authorizationHeader() . . . 201
17.37.2.2 _buildRequestBody() . . . 202
17.37.2.3 _extraRequestHeaders() . . . 202
17.37.2.4 _parseResponseBody() . . . 202
17.37.2.5 getProviderName() . . . 202
17.37.2.6 isAvailable() . . . 202
17.37.2.7 setApiVersion() . . . 202
17.37.2.8 setThinkingBudgetTokens() . . . 202
17.38 ApiRouter Class Reference . . . 203
17.38.1 Detailed Description . . . 203
17.38.2 Constructor & Destructor Documentation . . . 203
17.38.2.1 ApiRouter() . . . 203
17.38.3 Member Function Documentation . . . 203
17.38.3.1 handle() . . . 203
17.39 Assembly_Line Class Reference . . . 204
17.39.1 Member Function Documentation . . . 205
17.39.1.1 main() . . . 205
17.40 testing::internal::AssertHelper Class Reference . . . 205
17.41 Assign Class Reference . . . 205
17.41.1 Detailed Description . . . 210
17.41.2 Member Function Documentation . . . 211
17.41.2.1 _check() . . . 211
17.41.2.2 _createEditableDataDefinitions() . . . 211
17.41.2.3 _loadInstance() . . . 211
17.41.2.4 _onDispatchEvent() . . . 211
17.41.2.5 _saveInstance() . . . 211
17.41.2.6 show() . . . 211
17.42 Assignment Class Reference . . . 212
17.42.1 Detailed Description . . . 212
17.43 Attribute Class Reference . . . 212
17.43.1 Detailed Description . . . 217
17.43.2 Member Function Documentation . . . 217

17.43.2.1 _check() . . . 217
17.43.2.2 _loadInstance() . . . 217
17.43.2.3 _saveInstance() . . . 217
17.43.2.4 getInitialValuesText() . . . 218
17.43.2.5 setInitialValuesText() . . . 218
17.43.2.6 show() . . . 218
17.44 AuditEntry Struct Reference . . . 218
17.44.1 Detailed Description . . . 219
17.45 BacteriaColony Class Reference . . . 219
17.45.1 Detailed Description . . . 226
17.45.2 Member Function Documentation . . . 226
17.45.2.1 _check() . . . 226
17.45.2.2 _createAttachedAttributes() . . . 226
17.45.2.3 _initBetweenReplications() . . . 226
17.45.2.4 _loadInstance() . . . 226
17.45.2.5 _onDispatchEvent() . . . 226
17.45.2.6 _saveInstance() . . . 227
17.45.2.7 show() . . . 227
17.46 BacteriaColonyViewerGuiExtensionPlugin Class Reference . . . 227
17.46.1 Member Function Documentation . . . 228
17.46.1.1 displayName() . . . 228
17.46.1.2 extensionId() . . . 228
17.46.1.3 registerContributions() . . . 228
17.46.1.4 requiredModelPlugins() . . . 228
17.46.1.5 version() . . . 228
17.47 BacteriaSignalGrid Class Reference . . . 229
17.47.1 Detailed Description . . . 233
17.47.2 Member Function Documentation . . . 233
17.47.2.1 _check() . . . 233
17.47.2.2 _loadInstance() . . . 233
17.47.2.3 _saveInstance() . . . 233
17.47.2.4 show() . . . 233
17.48 BacteriaColony::BacteriumState Struct Reference . . . 233
17.49 Banking_Transactions Class Reference . . . 234
17.49.1 Member Function Documentation . . . 236
17.49.1.1 main() . . . 236
17.50 BacteriaColony::BarrierSegment Struct Reference . . . 236
17.51 BaseGenesysTerminalApplication Class Reference . . . 236
17.51.1 Member Function Documentation . . . 237
17.51.1.1 main() . . . 237
17.51.1.2 setDefaultEventHandlers() . . . 238
17.52 BaseGenesysWebApplication Class Reference . . . 238

17.52.1 Detailed Description . . . 239
17.52.2 Member Function Documentation . . . 239
17.52.2.1 main() . . . 239
17.53 BaseMVController Class Reference . . . 239
17.53.1 Detailed Description . . . 240
17.54 yy::genesyspp_parser::basic_symbol< Base > Struct Template Reference . . . 240
17.54.1 Detailed Description . . . 242
17.55 Batch Class Reference . . . 242
17.55.1 Detailed Description . . . 247
17.55.2 Member Function Documentation . . . 247
17.55.2.1 _check() . . . 247
17.55.2.2 _createAttachedAttributes() . . . 247
17.55.2.3 _createEditableDataDefinitions() . . . 248
17.55.2.4 _createNonEditableDataDefinitions() . . . 248
17.55.2.5 _loadInstance() . . . 248
17.55.2.6 _onDispatchEvent() . . . 248
17.55.2.7 _saveInstance() . . . 248
17.55.2.8 show() . . . 248
17.56 testing::internal::TemplateSel< Tmpl >::Bind< T > Struct Template Reference . . . 249
17.57 ModalModelBindingsUndoCommand::Bindings Struct Reference . . . 249
17.58 BioCompartment Class Reference . . . 249
17.58.1 Detailed Description . . . 253
17.58.2 Member Function Documentation . . . 253
17.58.2.1 _check() . . . 253
17.58.2.2 _createEditableDataDefinitions() . . . 254
17.58.2.3 _loadInstance() . . . 254
17.58.2.4 _saveInstance() . . . 254
17.58.2.5 show() . . . 254
17.59 BioKineticLawExpression Class Reference . . . 254
17.59.1 Detailed Description . . . 254
17.60 BioNetwork Class Reference . . . 255
17.60.1 Detailed Description . . . 259
17.60.2 Member Function Documentation . . . 259
17.60.2.1 _check() . . . 259
17.60.2.2 _createEditableDataDefinitions() . . . 259
17.60.2.3 _initBetweenReplications() . . . 259
17.60.2.4 _loadInstance() . . . 260
17.60.2.5 _saveInstance() . . . 260
17.60.2.6 show() . . . 260
17.61 BioNetworkAwareGuiExtensionPlugin Class Reference . . . 260
17.61.1 Member Function Documentation . . . 261
17.61.1.1 displayName() . . . 261

17.61.1.2 extensionId() . . . 261
17.61.1.3 registerContributions() . . . 261
17.61.1.4 requiredModelPlugins() . . . 261
17.61.1.5 version() . . . 261
17.62 BioNetworkEditorGuiExtensionPlugin Class Reference . . . 262
17.62.1 Member Function Documentation . . . 262
17.62.1.1 displayName() . . . 262
17.62.1.2 extensionId() . . . 262
17.62.1.3 registerContributions() . . . 263
17.62.1.4 requiredModelPlugins() . . . 263
17.62.1.5 version() . . . 263
17.63 BioNetworkExecutionGuiExtensionPlugin Class Reference . . . 263
17.63.1 Member Function Documentation . . . 264
17.63.1.1 displayName() . . . 264
17.63.1.2 extensionId() . . . 264
17.63.1.3 registerContributions() . . . 264
17.63.1.4 requiredModelPlugins() . . . 264
17.63.1.5 version() . . . 264
17.64 BioParameter Class Reference . . . 264
17.64.1 Detailed Description . . . 268
17.64.2 Member Function Documentation . . . 268
17.64.2.1 _check() . . . 268
17.64.2.2 _loadInstance() . . . 269
17.64.2.3 _saveInstance() . . . 269
17.64.2.4 show() . . . 269
17.65 BioParameterManagerGuiExtensionPlugin Class Reference . . . 269
17.65.1 Member Function Documentation . . . 270
17.65.1.1 displayName() . . . 270
17.65.1.2 extensionId() . . . 270
17.65.1.3 registerContributions() . . . 270
17.65.1.4 requiredModelPlugins() . . . 270
17.65.1.5 version() . . . 270
17.66 BioParameterSensitivityEntry Struct Reference . . . 271
17.67 BioReaction Class Reference . . . 271
17.67.1 Detailed Description . . . 276
17.67.2 Constructor & Destructor Documentation . . . 276
17.67.2.1 BioReaction() . . . 276
17.67.3 Member Function Documentation . . . 276
17.67.3.1 _check() . . . 276
17.67.3.2 _createAttachedAttributes() . . . 276
17.67.3.3 _loadInstance() . . . 276
17.67.3.4 _saveInstance() . . . 277

17.67.3.5 addModifier() . . . 277
17.67.3.6 addProduct() . . . 277
17.67.3.7 addReactant() . . . 277
17.67.3.8 getKineticLawExpression() . . . 277
17.67.3.9 getModifiers() . . . 278
17.67.3.10 GetPluginInformation() . . . 278
17.67.3.11 getProducts() . . . 278
17.67.3.12 getRateConstant() . . . 278
17.67.3.13 getRateConstantParameterName() . . . 278
17.67.3.14 getReactants() . . . 278
17.67.3.15 getReverseKineticLawExpression() . . . 279
17.67.3.16 getReverseRateConstant() . . . 279
17.67.3.17 getReverseRateConstantParameterName() . . . 279
17.67.3.18 isReversible() . . . 279
17.67.3.19 LoadInstance() . . . 279
17.67.3.20 NewInstance() . . . 279
17.67.3.21 resolveRateConstant() . . . 280
17.67.3.22 resolveReverseRateConstant() . . . 280
17.67.3.23 setKineticLawExpression() . . . 280
17.67.3.24 setRateConstant() . . . 280
17.67.3.25 setRateConstantParameterName() . . . 280
17.67.3.26 setReverseKineticLawExpression() . . . 281
17.67.3.27 setReverseRateConstant() . . . 281
17.67.3.28 setReverseRateConstantParameterName() . . . 281
17.67.3.29 setReversible() . . . 281
17.67.3.30 show() . . . 281
17.68 BioReactionKineticsGuiExtensionPlugin Class Reference . . . 282
17.68.1 Member Function Documentation . . . 282
17.68.1.1 displayName() . . . 282
17.68.1.2 extensionId() . . . 282
17.68.1.3 registerContributions() . . . 283
17.68.1.4 requiredModelPlugins() . . . 283
17.68.1.5 version() . . . 283
17.69 BioReactionRateSample Struct Reference . . . 283
17.70 BioReactionRateTimeCourse Struct Reference . . . 283
17.71 BioReactionStoichiometryGuiExtensionPlugin Class Reference . . . 284
17.71.1 Member Function Documentation . . . 284
17.71.1.1 displayName() . . . 284
17.71.1.2 extensionId() . . . 284
17.71.1.3 registerContributions() . . . 285
17.71.1.4 requiredModelPlugins() . . . 285
17.71.1.5 version() . . . 285

17.72 BioRunnerCommand Class Reference . . . 285
17.72.1 Detailed Description . . . 290
17.72.2 Member Function Documentation . . . 290
17.72.2.1 _check() . . . 290
17.72.2.2 _createEditableDataDefinitions() . . . 290
17.72.2.3 _loadInstance() . . . 290
17.72.2.4 _onDispatchEvent() . . . 290
17.72.2.5 _saveInstance() . . . 291
17.72.2.6 show() . . . 291
17.73 BioSBMLBridge Class Reference . . . 291
17.74 BioSBMLExportResult Struct Reference . . . 291
17.75 BioSBMLImportResult Struct Reference . . . 292
17.76 BioSBMLInteroperabilityGuiExtensionPlugin Class Reference . . . 292
17.76.1 Member Function Documentation . . . 293
17.76.1.1 displayName() . . . 293
17.76.1.2 extensionId() . . . 293
17.76.1.3 registerContributions() . . . 293
17.76.1.4 requiredModelPlugins() . . . 293
17.76.1.5 version() . . . 293
17.77 BioSensitivityScan Struct Reference . . . 293
17.78 BioSimulate Class Reference . . . 293
17.78.1 Detailed Description . . . 298
17.78.2 Member Function Documentation . . . 299
17.78.2.1 _check() . . . 299
17.78.2.2 _createEditableDataDefinitions() . . . 299
17.78.2.3 _loadInstance() . . . 299
17.78.2.4 _onDispatchEvent() . . . 299
17.78.2.5 _saveInstance() . . . 299
17.78.2.6 show() . . . 299
17.79 BioSimulationAnalysis Class Reference . . . 300
17.80 BioSimulationResult Class Reference . . . 300
17.81 BioSimulationResultsGuiExtensionPlugin Class Reference . . . 301
17.81.1 Member Function Documentation . . . 301
17.81.1.1 displayName() . . . 301
17.81.1.2 extensionId() . . . 301
17.81.1.3 registerContributions() . . . 302
17.81.1.4 requiredModelPlugins() . . . 302
17.81.1.5 version() . . . 302
17.82 BioSimulationSample Struct Reference . . . 302
17.83 BioSimulationSpeciesAmount Struct Reference . . . 302
17.84 BioSimulatorRunner Class Reference . . . 303
17.84.1 Member Function Documentation . . . 307

17.84.1.1 _check() . . . 307
17.84.1.2 _loadInstance() . . . 307
17.84.1.3 _saveInstance() . . . 307
17.84.1.4 show() . . . 307
17.85 BioSpecies Class Reference . . . 308
17.85.1 Detailed Description . . . 311
17.85.2 Member Function Documentation . . . 312
17.85.2.1 _check() . . . 312
17.85.2.2 _initBetweenReplications() . . . 312
17.85.2.3 _loadInstance() . . . 312
17.85.2.4 _saveInstance() . . . 312
17.85.2.5 show() . . . 312
17.86 BioSpeciesDerivative Struct Reference . . . 312
17.87 BioSpeciesManagerGuiExtensionPlugin Class Reference . . . 313
17.87.1 Member Function Documentation . . . 313
17.87.1.1 displayName() . . . 313
17.87.1.2 extensionId() . . . 313
17.87.1.3 registerContributions() . . . 314
17.87.1.4 requiredModelPlugins() . . . 314
17.87.1.5 version() . . . 314
17.88 BioStateProjectionComponent Class Reference . . . 314
17.88.1 Detailed Description . . . 319
17.88.2 Member Function Documentation . . . 319
17.88.2.1 _check() . . . 319
17.88.2.2 _createEditableDataDefinitions() . . . 319
17.88.2.3 _loadInstance() . . . 320
17.88.2.4 _onDispatchEvent() . . . 320
17.88.2.5 _saveInstance() . . . 320
17.88.2.6 show() . . . 320
17.89 BioSteadyState Class Reference . . . 320
17.89.1 Detailed Description . . . 325
17.89.2 Member Function Documentation . . . 326
17.89.2.1 _check() . . . 326
17.89.2.2 _createAttachedAttributes() . . . 326
17.89.2.3 _loadInstance() . . . 326
17.89.2.4 _onDispatchEvent() . . . 326
17.89.2.5 _saveInstance() . . . 326
17.89.2.6 show() . . . 326
17.90 BioSteadyStateCheck Struct Reference . . . 327
17.91 BioStoichiometryMatrix Struct Reference . . . 327
17.92 Book_Cap02_Example01 Class Reference . . . 327
17.92.1 Member Function Documentation . . . 329

17.92.1.1 main() . . . 329
17.93 Boundary_Closed Class Reference . . . 329
17.93.1 Member Function Documentation . . . 330
17.93.1.1 getNeighborCell() . . . 330
17.94 Boundary_Fixed Class Reference . . . 330
17.94.1 Member Function Documentation . . . 331
17.94.1.1 getNeighborCell() . . . 331
17.95 BoundaryCondition Class Reference . . . 332
17.95.1 Member Function Documentation . . . 333
17.95.1.1 getNeighborCell() . . . 333
17.96 Buffer Class Reference . . . 333
17.96.1 Detailed Description . . . 338
17.96.2 Member Function Documentation . . . 338
17.96.2.1 _addSimulationControl() . . . 338
17.96.2.2 _check() . . . 338
17.96.2.3 _createEditableDataDefinitions() . . . 339
17.96.2.4 _getParserChangesInformation() . . . 339
17.96.2.5 _initBetweenReplications() . . . 339
17.96.2.6 _loadInstance() . . . 339
17.96.2.7 _onDispatchEvent() . . . 339
17.96.2.8 _saveInstance() . . . 339
17.96.2.9 show() . . . 339
17.97 BufferFIFO Class Reference . . . 340
17.97.1 Member Function Documentation . . . 341
17.97.1.1 main() . . . 341
17.98 yy::genesyspp_parser::by_kind Struct Reference . . . 341
17.98.1 Detailed Description . . . 342
17.98.2 Member Function Documentation . . . 343
17.98.2.1 kind() . . . 343
17.98.3 Member Data Documentation . . . 343
17.98.3.1 kind_ . . . 343
17.99 QCPAxisPainterPrivate::CachedLabel Struct Reference . . . 343
17.100 QCPLabelPainterPrivate::CachedLabel Struct Reference . . . 343
17.101 ModelLifecycleController::Callbacks Struct Reference . . . 343
17.102 SimulationEventController::Callbacks Struct Reference . . . 344
17.103 Capacitor Class Reference . . . 344
17.103.1 Member Function Documentation . . . 349
17.103.1.1 _loadInstance() . . . 349
17.103.1.2 _saveInstance() . . . 350
17.103.1.3 Build() . . . 350
17.104 testing::internal::CartesianProductGenerator< T > Class Template Reference . . . 350
17.104.1 Member Function Documentation . . . 351

17.104.1.1 Begin() . . . 351
17.104.1.2 End() . . . 351
17.105 testing::internal::CartesianProductHolder< Gen > Class Template Reference . . . 351
17.106 Cell Class Reference . . . 352
17.107 CellCycleCheckpointComponent Class Reference . . . 353
17.107.1 Detailed Description . . . 358
17.107.2 Member Function Documentation . . . 359
17.107.2.1 _check() . . . 359
17.107.2.2 _createEditableDataDefinitions() . . . 359
17.107.2.3 _initBetweenReplications() . . . 359
17.107.2.4 _loadInstance() . . . 359
17.107.2.5 _onDispatchEvent() . . . 359
17.107.2.6 _saveInstance() . . . 359
17.107.2.7 show() . . . 360
17.108 CellDivisionEvent Class Reference . . . 360
17.108.1 Detailed Description . . . 365
17.108.2 Member Function Documentation . . . 365
17.108.2.1 _check() . . . 365
17.108.2.2 _createEditableDataDefinitions() . . . 366
17.108.2.3 _initBetweenReplications() . . . 366
17.108.2.4 _loadInstance() . . . 366
17.108.2.5 _onDispatchEvent() . . . 366
17.108.2.6 _saveInstance() . . . 366
17.108.2.7 show() . . . 366
17.109 CellFateDecisionComponent Class Reference . . . 367
17.109.1 Detailed Description . . . 372
17.109.2 Member Function Documentation . . . 372
17.109.2.1 _check() . . . 372
17.109.2.2 _createEditableDataDefinitions() . . . 372
17.109.2.3 _initBetweenReplications() . . . 372
17.109.2.4 _loadInstance() . . . 372
17.109.2.5 _onDispatchEvent() . . . 372
17.109.2.6 _saveInstance() . . . 373
17.109.2.7 show() . . . 373
17.110 CellGrowthComponent Class Reference . . . 373
17.110.1 Detailed Description . . . 378
17.110.2 Member Function Documentation . . . 378
17.110.2.1 _check() . . . 378
17.110.2.2 _createEditableDataDefinitions() . . . 378
17.110.2.3 _initBetweenReplications() . . . 378
17.110.2.4 _loadInstance() . . . 378
17.110.2.5 _onDispatchEvent() . . . 379

17.110.2.6 _saveInstance() . . . 379
17.110.2.7 show() . . . 379
17.111 CellularAutomata Class Reference . . . 380
17.112 CellularAutomata_1DTimed Class Reference . . . 381
17.112.1 Detailed Description . . . 382
17.112.2 Member Function Documentation . . . 382
17.112.2.1 applyLocalRule() . . . 382
17.113 CellularAutomata_Classic Class Reference . . . 382
17.113.1 Detailed Description . . . 384
17.113.2 Member Function Documentation . . . 384
17.113.2.1 applyLocalRule() . . . 384
17.114 CellularAutomataBase Class Reference . . . 384
17.114.1 Member Function Documentation . . . 385
17.114.1.1 applyLocalRule() . . . 385
17.115 CellularAutomataComp Class Reference . . . 386
17.115.1 Detailed Description . . . 391
17.115.2 Member Function Documentation . . . 391
17.115.2.1 _check() . . . 391
17.115.2.2 _initBetweenReplications() . . . 391
17.115.2.3 _loadInstance() . . . 392
17.115.2.4 _onDispatchEvent() . . . 392
17.115.2.5 _saveInstance() . . . 392
17.115.2.6 show() . . . 392
17.116 CellularAutomataDemoBuilder Class Reference . . . 392
17.117 CellularAutomataDemoModel Struct Reference . . . 393
17.118 CellularAutomataDemoSettings Struct Reference . . . 393
17.119 CellularAutomataEvent Class Reference . . . 394
17.120 CellularAutomataGuiExtensionPlugin Class Reference . . . 395
17.120.1 Member Function Documentation . . . 395
17.120.1.1 displayName() . . . 395
17.120.1.2 extensionId() . . . 395
17.120.1.3 registerContributions() . . . 396
17.120.1.4 requiredModelPlugins() . . . 396
17.120.1.5 version() . . . 396
17.121 CellularAutomataLatticeWidget Class Reference . . . 396
17.122 CellularAutomataStateColorMap Class Reference . . . 397
17.123 CellularAutomataViewerController Class Reference . . . 397
17.124 CellularAutomataViewerWindow Class Reference . . . 399
17.125 ChatMessage Struct Reference . . . 399
17.125.1 Detailed Description . . . 400
17.126 Clone Class Reference . . . 400
17.126.1 Detailed Description . . . 405

17.126.2 Member Function Documentation . . . 405
17.126.2.1 _check() . . . 405
17.126.2.2 _createInternalStatisticReporters() . . . 405
17.126.2.3 _initBetweenReplications() . . . 405
17.126.2.4 _loadInstance() . . . 405
17.126.2.5 _onDispatchEvent() . . . 405
17.126.2.6 _saveInstance() . . . 406
17.126.2.7 show() . . . 406
17.127 CodeEditor Class Reference . . . 406
17.128 testing::internal::CodeLocation Struct Reference . . . 407
17.129 Collector_if Class Reference . . . 407
17.129.1 Detailed Description . . . 408
17.129.2 Member Function Documentation . . . 408
17.129.2.1 addValue() . . . 408
17.129.2.2 clear() . . . 409
17.129.2.3 getLastValue() . . . 409
17.129.2.4 numElements() . . . 409
17.129.2.5 setAddValueHandler() . . . 409
17.129.2.6 setClearHandler() . . . 409
17.130 CollectorDatafile_if Class Reference . . . 410
17.130.1 Detailed Description . . . 411
17.130.2 Member Function Documentation . . . 411
17.130.2.1 getDataFilename() . . . 411
17.130.2.2 getNextValue() . . . 411
17.130.2.3 getValue() . . . 411
17.130.2.4 seekFirstValue() . . . 412
17.130.2.5 setDataFilename() . . . 412
17.131 CollectorDatafileDefaultImpl1 Class Reference . . . 412
17.131.1 Detailed Description . . . 413
17.131.2 Member Function Documentation . . . 414
17.131.2.1 addValue() . . . 414
17.131.2.2 clear() . . . 414
17.131.2.3 getDataFilename() . . . 414
17.131.2.4 getLastValue() . . . 414
17.131.2.5 getNextValue() . . . 414
17.131.2.6 getValue() . . . 414
17.131.2.7 numElements() . . . 414
17.131.2.8 seekFirstValue() . . . 414
17.131.2.9 setAddValueHandler() . . . 414
17.131.2.10 setClearHandler() . . . 415
17.131.2.11 setDataFilename() . . . 415
17.132 CollectorDefaultImpl1 Class Reference . . . 415

17.132.1 Detailed Description . . . 416
17.132.2 Member Function Documentation . . . 416
17.132.2.1 addValue() . . . 416
17.132.2.2 clear() . . . 416
17.132.2.3 getLastValue() . . . 416
17.132.2.4 numElements() . . . 416
17.132.2.5 setAddValueHandler() . . . 416
17.132.2.6 setClearHandler() . . . 417
17.133 GroProgramRuntime::ColonyMutation Struct Reference . . . 417
17.134 ColoredPetriNetNetwork Class Reference . . . 417
17.134.1 Detailed Description . . . 422
17.134.2 Member Function Documentation . . . 423
17.134.2.1 _activate() . . . 423
17.134.2.2 _check() . . . 423
17.134.2.3 _initBetweenReplications() . . . 423
17.134.2.4 _loadInstance() . . . 423
17.134.2.5 _saveInstance() . . . 423
17.134.2.6 show() . . . 423
17.135 ComboBoxEnum Class Reference . . . 424
17.136 GroProgramIr::Command Struct Reference . . . 424
17.137 Util::CommandResult Class Reference . . . 425
17.138 CompartmentExchangeComponent Class Reference . . . 425
17.138.1 Detailed Description . . . 431
17.138.2 Member Function Documentation . . . 431
17.138.2.1 _check() . . . 431
17.138.2.2 _createEditableDataDefinitions() . . . 431
17.138.2.3 _initBetweenReplications() . . . 431
17.138.2.4 _loadInstance() . . . 431
17.138.2.5 _onDispatchEvent() . . . 432
17.138.2.6 _saveInstance() . . . 432
17.138.2.7 show() . . . 432
17.139 CppCompiler::CompilationResult Class Reference . . . 432
17.140 CompilerThread Class Reference . . . 433
17.141 ComponentItem Struct Reference . . . 433
17.142 CompositeState Class Reference . . . 434
17.142.1 Member Function Documentation . . . 435
17.142.1.1 clone() . . . 435
17.142.1.2 equals() . . . 435
17.142.1.3 toString() . . . 435
17.142.1.4 typeName() . . . 435
17.143 CompositeStateSet Class Reference . . . 435
17.143.1 Member Function Documentation . . . 436

17.143.1.1 contains() . . . 436
17.143.1.2 createDefaultState() . . . 436
17.143.1.3 isComposite() . . . 436
17.143.1.4 name() . . . 437
17.144 HypothesisTester_if::ConfidenceInterval Class Reference . . . 437
17.144.1 Detailed Description . . . 437
17.145 Connection Struct Reference . . . 437
17.145.1 Detailed Description . . . 438
17.146 ConnectionChannel Struct Reference . . . 438
17.146.1 Detailed Description . . . 438
17.147 ConnectionManager Class Reference . . . 438
17.147.1 Detailed Description . . . 439
17.147.2 Member Function Documentation . . . 439
17.147.2.1 connections() . . . 439
17.147.2.2 getConnectionAtPort() . . . 439
17.147.2.3 getCurrentInputConnectionsSize() . . . 439
17.147.2.4 getCurrentOutputConnectionsSize() . . . 440
17.147.2.5 getFrontConnection() . . . 440
17.147.2.6 getMaxInputConnections() . . . 440
17.147.2.7 getMaxOutputConnections() . . . 440
17.147.2.8 getMinInputConnections() . . . 440
17.147.2.9 getMinOutputConnections() . . . 440
17.147.2.10 insert() [1/2] . . . 440
17.147.2.11 insert() [2/2] . . . 441
17.147.2.12 insertAtPort() . . . 441
17.147.2.13 remove() . . . 441
17.147.2.14 removeAtPort() . . . 441
17.147.2.15 setMaxInputConnections() . . . 442
17.147.2.16 setMaxOutputConnections() . . . 442
17.147.2.17 setMinInputConnections() . . . 442
17.147.2.18 setMinOutputConnections() . . . 442
17.147.2.19 size() . . . 442
17.148 testing::internal::ConstCharPtr Struct Reference . . . 443
17.149 Optimizer_if::ConstraintDefinition Struct Reference . . . 443
17.149.1 Detailed Description . . . 443
17.150 testing::internal::ConstRef< T > Struct Template Reference . . . 443
17.151 testing::internal::ConstRef< T & > Struct Template Reference . . . 443
17.152 testing::internal::ContainerPrinter Struct Reference . . . 444
17.153 yy::genesyspp_parser::context Class Reference . . . 444
17.153.1 Member Function Documentation . . . 444
17.153.1.1 expected_tokens() . . . 444
17.154 ContinuousDistribution_if Class Reference . . . 444

17.154.1 Detailed Description . . . 445
17.155 ContinuousModel Class Reference . . . 446
17.155.1 Member Function Documentation . . . 447
17.155.1.1 Simulate() . . . 447
17.156 ContinuousSystemComponent Class Reference . . . 447
17.156.1 Detailed Description . . . 452
17.156.2 Member Function Documentation . . . 453
17.156.2.1 _check() . . . 453
17.156.2.2 _createEditableDataDefinitions() . . . 453
17.156.2.3 _loadInstance() . . . 453
17.156.2.4 _onDispatchEvent() . . . 453
17.156.2.5 _saveInstance() . . . 453
17.156.2.6 show() . . . 453
17.157 testing::internal::ConvertibleToIntegerPrinter Struct Reference . . . 454
17.158 testing::internal::ConvertibleToStringViewPrinter Struct Reference . . . 454
17.159 Conveyor Class Reference . . . 454
17.159.1 Detailed Description . . . 458
17.159.2 Member Function Documentation . . . 458
17.159.2.1 _check() . . . 458
17.159.2.2 _createEditableDataDefinitions() . . . 458
17.159.2.3 _initBetweenReplications() . . . 459
17.159.2.4 _loadInstance() . . . 459
17.159.2.5 _saveInstance() . . . 459
17.159.2.6 show() . . . 459
17.160 CoreGuiExtensionPlugin Class Reference . . . 459
17.160.1 Member Function Documentation . . . 460
17.160.1.1 displayName() . . . 460
17.160.1.2 extensionId() . . . 460
17.160.1.3 registerContributions() . . . 460
17.160.1.4 requiredModelPlugins() . . . 460
17.160.1.5 version() . . . 460
17.161 Counter Class Reference . . . 461
17.161.1 Detailed Description . . . 464
17.161.2 Member Function Documentation . . . 465
17.161.2.1 _check() . . . 465
17.161.2.2 _initBetweenReplications() . . . 465
17.161.2.3 _loadInstance() . . . 465
17.161.2.4 _saveInstance() . . . 465
17.161.2.5 clear() . . . 465
17.161.2.6 getCountValue() . . . 465
17.161.2.7 getParent() . . . 465
17.161.2.8 incCountValue() . . . 466

17.161.2.9 show() . . . 466
17.162 CountingSampler Class Reference . . . 466
17.162.1 Member Function Documentation . . . 468
17.162.1.1 getRNGparameters() . . . 468
17.162.1.2 random() . . . 468
17.162.1.3 sampleBernoulli() . . . 468
17.162.1.4 sampleBeta() [1/2] . . . 468
17.162.1.5 sampleBeta() [2/2] . . . 468
17.162.1.6 sampleBinomial() . . . 469
17.162.1.7 sampleDiscrete() [1/2] . . . 469
17.162.1.8 sampleDiscrete() [2/2] . . . 469
17.162.1.9 sampleErlang() . . . 469
17.162.1.10 sampleExponential() . . . 469
17.162.1.11 sampleGamma() . . . 469
17.162.1.12 sampleGeometric() . . . 470
17.162.1.13 sampleGumbell() . . . 470
17.162.1.14 sampleLogNormal() . . . 470
17.162.1.15 sampleNormal() . . . 470
17.162.1.16 sampleTriangular() . . . 470
17.162.1.17 sampleUniform() . . . 470
17.162.1.18 sampleWeibull() . . . 470
17.162.1.19 setRNGparameters() . . . 471
17.163 CPNArc Class Reference . . . 471
17.163.1 Detailed Description . . . 475
17.163.2 Member Function Documentation . . . 475
17.163.2.1 _check() . . . 475
17.163.2.2 _loadInstance() . . . 475
17.163.2.3 _saveInstance() . . . 475
17.163.2.4 show() . . . 475
17.164 CPNTransition Class Reference . . . 476
17.164.1 Detailed Description . . . 480
17.164.2 Member Function Documentation . . . 481
17.164.2.1 _check() . . . 481
17.164.2.2 _loadInstance() . . . 481
17.164.2.3 _saveInstance() . . . 481
17.164.2.4 show() . . . 481
17.165 CppCompiler Class Reference . . . 482
17.165.1 Member Function Documentation . . . 486
17.165.1.1 _check() . . . 486
17.165.1.2 _initBetweenReplications() . . . 486
17.165.1.3 _loadInstance() . . . 486
17.165.1.4 _saveInstance() . . . 486

17.165.1.5 show() . . . 487
17.166 CppForG Class Reference . . . 487
17.166.1 Detailed Description . . . 492
17.166.2 Member Function Documentation . . . 492
17.166.2.1 _check() . . . 492
17.166.2.2 _createNonEditableDataDefinitions() . . . 492
17.166.2.3 _initBetweenReplications() . . . 492
17.166.2.4 _loadInstance() . . . 492
17.166.2.5 _onDispatchEvent() . . . 493
17.166.2.6 _saveInstance() . . . 493
17.166.2.7 show() . . . 493
17.167 CppModelExporter Class Reference . . . 493
17.167.1 Detailed Description . . . 494
17.167.2 Member Function Documentation . . . 494
17.167.2.1 addCppCodeLine() . . . 494
17.168 CppSerializer Class Reference . . . 494
17.168.1 Member Function Documentation . . . 495
17.168.1.1 dump() . . . 495
17.168.1.2 for_each() . . . 495
17.168.1.3 get() . . . 496
17.168.1.4 load() . . . 496
17.168.1.5 newPersistenceRecord() . . . 496
17.168.1.6 put() . . . 496
17.169 Create Class Reference . . . 497
17.169.1 Detailed Description . . . 503
17.169.2 Member Function Documentation . . . 503
17.169.2.1 _check() . . . 503
17.169.2.2 _createEditableDataDefinitions() . . . 503
17.169.2.3 _createInternalStatisticReporters() . . . 503
17.169.2.4 _initBetweenReplications() . . . 503
17.169.2.5 _loadInstance() . . . 504
17.169.2.6 _onDispatchEvent() . . . 504
17.169.2.7 _saveInstance() . . . 504
17.169.2.8 show() . . . 504
17.170 SimulatorSessionService::CreateSessionResult Struct Reference . . . 504
17.170.1 Detailed Description . . . 505
17.171 QtBoolPropertyManagerPrivate::Data Struct Reference . . . 505
17.172 QtDatePropertyManagerPrivate::Data Struct Reference . . . 505
17.173 QtDateTimePropertyManagerPrivate::Data Struct Reference . . . 505
17.174 QtDoublePropertyManagerPrivate::Data Struct Reference . . . 505
17.175 QtEnumPropertyManagerPrivate::Data Struct Reference . . . 506
17.176 QtFlagPropertyManagerPrivate::Data Struct Reference . . . 506

17.177 QtIntPropertyManagerPrivate::Data Struct Reference . . . 506
17.178 QtPointFPropertyManagerPrivate::Data Struct Reference . . . 507
17.179 QtRectFPropertyManagerPrivate::Data Struct Reference . . . 507
17.180 QtRectPropertyManagerPrivate::Data Struct Reference . . . 507
17.181 QtSizeFPropertyManagerPrivate::Data Struct Reference . . . 507
17.182 QtSizePropertyManagerPrivate::Data Struct Reference . . . 507
17.183 QtStringPropertyManagerPrivate::Data Struct Reference . . . 508
17.184 QtTimePropertyManagerPrivate::Data Struct Reference . . . 508
17.185 DataAnalyser_if Class Reference . . . 508
17.185.1 Detailed Description . . . 508
17.186 DataAnalyzerWindow Class Reference . . . 509
17.187 DataComponentEditor Class Reference . . . 510
17.188 DataComponentProperty Class Reference . . . 511
17.189 DataDefinitionItem Struct Reference . . . 512
17.190 DataDefinitionRelationItem Struct Reference . . . 512
17.191 ModelInspectorController::DataDefinitionSelection Struct Reference . . . 513
17.192 AnimationPlot::DataSeries Struct Reference . . . 513
17.193 DataSet_if Class Reference . . . 514
17.193.1 Detailed Description . . . 514
17.194 Decide Class Reference . . . 514
17.194.1 Detailed Description . . . 519
17.194.2 Member Function Documentation . . . 519
17.194.2.1 _check() . . . 519
17.194.2.2 _createInternalStatisticReporters() . . . 519
17.194.2.3 _loadInstance() . . . 519
17.194.2.4 _onDispatchEvent() . . . 519
17.194.2.5 _saveInstance() . . . 520
17.194.2.6 show() . . . 520
17.195 Delay::DEFAULT_VALUES Struct Reference . . . 520
17.196 ModalModelDefault::DEFAULT_VALUES Struct Reference . . . 520
17.197 Seize::DEFAULT_VALUES Struct Reference . . . 521
17.198 SourceModelComponent::DEFAULT_VALUES Struct Reference . . . 521
17.199 testing::internal::DefaultGlobalTestPartResultReporter Class Reference . . . 521
17.200 SamplerDefaultImpl1::DefaultImpl1RNG_Parameters Struct Reference . . . 522
17.200.1 Detailed Description . . . 523
17.201 testing::internal::DefaultNameGenerator Struct Reference . . . 523
17.202 DefaultNetwork Class Reference . . . 523
17.202.1 Detailed Description . . . 528
17.202.2 Member Function Documentation . . . 528
17.202.2.1 _activate() . . . 528
17.202.2.2 _check() . . . 528
17.202.2.3 _createInternalStatisticReporters() . . . 528

17.202.2.4 _initBetweenReplications() . . . 529
17.202.2.5 _loadInstance() . . . 529
17.202.2.6 _saveInstance() . . . 529
17.202.2.7 activate() . . . 529
17.202.2.8 show() . . . 529
17.203 DefaultNode Class Reference . . . 529
17.203.1 Detailed Description . . . 533
17.203.2 Constructor & Destructor Documentation . . . 534
17.203.2.1 DefaultNode() . . . 534
17.203.3 Member Function Documentation . . . 534
17.203.3.1 _loadInstance() . . . 534
17.203.3.2 _saveInstance() . . . 534
17.203.3.3 show() . . . 534
17.204 DefaultNodeTransition Class Reference . . . 535
17.205 testing::internal::DefaultPerThreadTestPartResultReporter Class Reference . . . 536
17.206 Delay Class Reference . . . 536
17.206.1 Detailed Description . . . 542
17.206.2 Member Function Documentation . . . 542
17.206.2.1 _check() . . . 542
17.206.2.2 _createAttachedAttributes() . . . 542
17.206.2.3 _createInternalStatisticReporters() . . . 542
17.206.2.4 _loadInstance() . . . 542
17.206.2.5 _onDispatchEvent() . . . 542
17.206.2.6 _saveInstance() . . . 542
17.206.2.7 show() . . . 543
17.207 DelayProbe Class Reference . . . 543
17.208 DeleteUndoCommand Class Reference . . . 548
17.209 dialogBreakpoint Class Reference . . . 549
17.210 DialogFind Class Reference . . . 550
17.211 DialogModelInformation Class Reference . . . 551
17.212 DialogPluginManager Class Reference . . . 552
17.212.1 Constructor & Destructor Documentation . . . 553
17.212.1.1 DialogPluginManager() . . . 553
17.213 DialogReportIssue Class Reference . . . 553
17.213.1 Detailed Description . . . 554
17.214 DialogSelectCounter Class Reference . . . 555
17.215 DialogSelectVariable Class Reference . . . 556
17.216 DialogSimulationConfigure Class Reference . . . 557
17.216.1 Constructor & Destructor Documentation . . . 557
17.216.1.1 DialogSimulationConfigure() . . . 557
17.216.2 Member Function Documentation . . . 558
17.216.2.1 accept . . . 558

17.216.2.2 reject . . . 558
17.217 DialogSystemPreferences Class Reference . . . 558
17.218 DialogTimerConfigure Class Reference . . . 559
17.219 DialogUtilityController Class Reference . . . 559
17.219.1 Detailed Description . . . 560
17.219.2 Member Function Documentation . . . 561
17.219.2.1 onActionAboutReportIssueTriggered() . . . 561
17.219.2.2 onActionEditReplaceTriggered() . . . 561
17.219.2.3 onActionToolsParserGrammarCheckerTriggered() . . . 561
17.220 DiffEquations Class Reference . . . 561
17.220.1 Detailed Description . . . 566
17.220.2 Member Function Documentation . . . 567
17.220.2.1 _check() . . . 567
17.220.2.2 _createEditableDataDefinitions() . . . 567
17.220.2.3 _loadInstance() . . . 567
17.220.2.4 _onDispatchEvent() . . . 567
17.220.2.5 _saveInstance() . . . 567
17.220.2.6 getEquations() . . . 567
17.220.2.7 show() . . . 568
17.221 DiffusionField Class Reference . . . 568
17.221.1 Member Function Documentation . . . 572
17.221.1.1 _check() . . . 572
17.221.1.2 _initBetweenReplications() . . . 572
17.221.1.3 _loadInstance() . . . 573
17.221.1.4 _saveInstance() . . . 573
17.221.1.5 show() . . . 573
17.222 DiffusionMethodOfLinesSystem Class Reference . . . 573
17.222.1 Member Function Documentation . . . 574
17.222.1.1 dimension() . . . 574
17.222.1.2 evaluate() . . . 575
17.223 DiffusionSimulate Class Reference . . . 575
17.223.1 Detailed Description . . . 580
17.223.2 Member Function Documentation . . . 580
17.223.2.1 _check() . . . 580
17.223.2.2 _createEditableDataDefinitions() . . . 580
17.223.2.3 _loadInstance() . . . 580
17.223.2.4 _onDispatchEvent() . . . 580
17.223.2.5 _saveInstance() . . . 581
17.223.2.6 show() . . . 581
17.224 Diode Class Reference . . . 582
17.224.1 Member Function Documentation . . . 587
17.224.1.1 _loadInstance() . . . 587

17.224.1.2 _saveInstance() . . . 587
17.224.1.3 Build() . . . 587
17.225 DirectedAcyclicGraphNetwork Class Reference . . . 588
17.225.1 Detailed Description . . . 594
17.225.2 Member Function Documentation . . . 594
17.225.2.1 _check() . . . 594
17.225.2.2 addEdge() . . . 594
17.226 DirectedGraphNetwork Class Reference . . . 594
17.226.1 Detailed Description . . . 601
17.226.2 Member Function Documentation . . . 601
17.226.2.1 _edgeConnects() . . . 601
17.226.2.2 _oppositeNode() . . . 601
17.226.2.3 getIncomingEdges() . . . 601
17.226.2.4 getOutgoingEdges() . . . 601
17.226.2.5 getPredecessors() . . . 601
17.226.2.6 getSuccessors() . . . 601
17.226.2.7 hasCycle() . . . 601
17.226.2.8 inDegree() . . . 601
17.226.2.9 isDirected() . . . 602
17.226.2.10 outDegree() . . . 602
17.226.2.11 topologicalOrder() . . . 602
17.227 DiscreteDistribution_if Class Reference . . . 602
17.227.1 Detailed Description . . . 603
17.228 genesys::launcher::DispatchController Class Reference . . . 603
17.229 Dispose Class Reference . . . 603
17.229.1 Detailed Description . . . 609
17.229.2 Member Function Documentation . . . 609
17.229.2.1 _check() . . . 609
17.229.2.2 _createAttachedAttributes() . . . 609
17.229.2.3 _createInternalStatisticReporters() . . . 609
17.229.2.4 _initBetweenReplications() . . . 609
17.229.2.5 _loadInstance() . . . 609
17.229.2.6 _onDispatchEvent() . . . 609
17.229.2.7 _saveInstance() . . . 610
17.229.2.8 show() . . . 610
17.230 Distance Class Reference . . . 610
17.230.1 Detailed Description . . . 614
17.230.2 Member Function Documentation . . . 614
17.230.2.1 _check() . . . 614
17.230.2.2 _loadInstance() . . . 614
17.230.2.3 _saveInstance() . . . 614
17.230.2.4 show() . . . 615

17.231 DistanceEntry Class Reference . . . 615
17.231.1 Detailed Description . . . 615
17.232 Distribution_if Class Reference . . . 615
17.232.1 Detailed Description . . . 616
17.233 DataAnalyzerWindow::DoePreviewPlot Class Reference . . . 616
17.234 DormandPrince54OdeSolver Class Reference . . . 617
17.234.1 Member Function Documentation . . . 618
17.234.1.1 advance() . . . 618
17.235 testing::internal::DoubleSequence< plus_one, T, sizeofT > Struct Template Reference . . . 618
17.236 testing::internal::DoubleSequence< false, IndexSequence< I... >, sizeofT > Struct Template Reference . . . 618
17.237 testing::internal::DoubleSequence< true, IndexSequence< I... >, sizeofT > Struct Template Reference . . . 619
17.238 genesys::launcher::DownloadResult Struct Reference . . . 619
17.239 DrawingItem Struct Reference . . . 619
17.240 DropOff Class Reference . . . 619
17.240.1 Detailed Description . . . 624
17.240.2 Member Function Documentation . . . 624
17.240.2.1 _check() . . . 624
17.240.2.2 _loadInstance() . . . 625
17.240.2.3 _onDispatchEvent() . . . 625
17.240.2.4 _saveInstance() . . . 625
17.240.2.5 show() . . . 625
17.241 DummyComponent Class Reference . . . 625
17.241.1 Detailed Description . . . 630
17.241.2 Constructor & Destructor Documentation . . . 630
17.241.2.1 DummyComponent() . . . 630
17.241.3 Member Function Documentation . . . 630
17.241.3.1 _loadInstance() . . . 630
17.241.3.2 _onDispatchEvent() . . . 631
17.241.3.3 _saveInstance() . . . 631
17.241.3.4 show() . . . 631
17.242 DummyElement Class Reference . . . 631
17.242.1 Member Function Documentation . . . 635
17.242.1.1 _check() . . . 635
17.242.1.2 _loadInstance() . . . 635
17.242.1.3 _saveInstance() . . . 635
17.242.1.4 show() . . . 635
17.243 EditCommandController Class Reference . . . 636
17.243.1 Detailed Description . . . 636
17.244 EditorFactoryPrivate< Editor > Class Template Reference . . . 637
17.245 EFSMInitialStateUndoCommand Class Reference . . . 637
17.246 EFSMNetwork Class Reference . . . 638

17.246.1 Detailed Description . . . 643
17.246.2 Member Function Documentation . . . 643
17.246.2.1 _activate() . . . 643
17.246.2.2 _check() . . . 643
17.246.2.3 _initBetweenReplications() . . . 643
17.246.2.4 _loadInstance() . . . 644
17.246.2.5 _saveInstance() . . . 644
17.246.2.6 detachTransition() . . . 644
17.246.2.7 restoreTransition() . . . 644
17.246.2.8 show() . . . 644
17.247 EFSMNetworkDiagramLayout Struct Reference . . . 644
17.247.1 Detailed Description . . . 645
17.248 EFSMStateMembershipUndoCommand Class Reference . . . 645
17.249 EFSMStatePropertiesUndoCommand Class Reference . . . 646
17.250 EFSMTransition Class Reference . . . 647
17.250.1 Member Function Documentation . . . 649
17.250.1.1 canFire() . . . 649
17.250.1.2 execute() . . . 649
17.251 EFSMTransitionGraphicsItem Class Reference . . . 649
17.251.1 Detailed Description . . . 650
17.252 EFSMTransitionPropertiesUndoCommand Class Reference . . . 650
17.253 EFSMTransitionUndoCommand Class Reference . . . 651
17.253.1 Detailed Description . . . 652
17.254 testing::internal::ElemFromList< N, T > Struct Template Reference . . . 652
17.255 testing::internal::ElemFromListImpl< typename > Struct Template Reference . . . 652
17.256 testing::internal::ElemFromListImpl< IndexSequence< I... > > Struct Template Reference . . . 652
17.257 testing::EmptyTestEventListener Class Reference . . . 653
17.257.1 Member Function Documentation . . . 654
17.257.1.1 OnEnvironmentsSetUpEnd() . . . 654
17.257.1.2 OnEnvironmentsSetUpStart() . . . 654
17.257.1.3 OnEnvironmentsTearDownEnd() . . . 654
17.257.1.4 OnEnvironmentsTearDownStart() . . . 654
17.257.1.5 OnTestCaseEnd() . . . 654
17.257.1.6 OnTestCaseStart() . . . 654
17.257.1.7 OnTestDisabled() . . . 654
17.257.1.8 OnTestEnd() . . . 654
17.257.1.9 OnTestIterationEnd() . . . 654
17.257.1.10 OnTestIterationStart() . . . 655
17.257.1.11 OnTestPartResult() . . . 655
17.257.1.12 OnTestProgramEnd() . . . 655
17.257.1.13 OnTestProgramStart() . . . 655
17.257.1.14 OnTestStart() . . . 655

17.257.1.15 OnTestSuiteEnd() . . . 655
17.257.1.16 OnTestSuiteStart() . . . 655
17.258 Enter Class Reference . . . 655
17.258.1 Detailed Description . . . 660
17.258.2 Member Function Documentation . . . 661
17.258.2.1 _check() . . . 661
17.258.2.2 _createEditableDataDefinitions() . . . 661
17.258.2.3 _createInternalStatisticReporters() . . . 661
17.258.2.4 _loadInstance() . . . 661
17.258.2.5 _onDispatchEvent() . . . 661
17.258.2.6 _saveInstance() . . . 662
17.258.2.7 show() . . . 662
17.259 Entity Class Reference . . . 662
17.259.1 Detailed Description . . . 666
17.259.2 Member Function Documentation . . . 666
17.259.2.1 _check() . . . 666
17.259.2.2 _loadInstance() . . . 667
17.259.2.3 _saveInstance() . . . 667
17.259.2.4 entityNumber() . . . 667
17.259.2.5 getAttributeValue() [1/2] . . . 667
17.259.2.6 getAttributeValue() [2/2] . . . 667
17.259.2.7 getEntityType() . . . 668
17.259.2.8 getEntityTypeName() . . . 668
17.259.2.9 setAttributeValue() [1/2] . . . 668
17.259.2.10 setAttributeValue() [2/2] . . . 668
17.259.2.11 setEntityType() . . . 668
17.259.2.12 setEntityTypeName() . . . 669
17.259.2.13 show() . . . 669
17.260 EntityGroup Class Reference . . . 669
17.260.1 Member Function Documentation . . . 673
17.260.1.1 _check() . . . 673
17.260.1.2 _createAttachedAttributes() . . . 673
17.260.1.3 _createInternalStatisticReporters() . . . 673
17.260.1.4 _initBetweenReplications() . . . 673
17.260.1.5 _loadInstance() . . . 673
17.260.1.6 _saveInstance() . . . 674
17.260.1.7 show() . . . 674
17.261 EntityType Class Reference . . . 674
17.261.1 Detailed Description . . . 678
17.261.2 Member Function Documentation . . . 678
17.261.2.1 _createInternalStatisticReporters() . . . 678
17.261.2.2 _initBetweenReplications() . . . 678

17.261.2.3 _loadInstance() . . . 679
17.261.2.4 _saveInstance() . . . 679
17.261.2.5 addGetStatisticsCollector() . . . 679
17.261.2.6 show() . . . 679
17.262 PersistenceRecord::Entry Struct Reference . . . 679
17.262.1 Detailed Description . . . 680
17.263 testing::Environment Class Reference . . . 680
17.264 EukaryoticCellCycleComponent Class Reference . . . 680
17.264.1 Detailed Description . . . 686
17.264.2 Member Function Documentation . . . 686
17.264.2.1 _check() . . . 686
17.264.2.2 _createEditableDataDefinitions() . . . 687
17.264.2.3 _initBetweenReplications() . . . 687
17.264.2.4 _loadInstance() . . . 687
17.264.2.5 _onDispatchEvent() . . . 687
17.264.2.6 _saveInstance() . . . 687
17.264.2.7 show() . . . 687
17.265 Event Class Reference . . . 688
17.265.1 Detailed Description . . . 688
17.265.2 Member Function Documentation . . . 688
17.265.2.1 getComponent() . . . 688
17.265.2.2 getComponentinputPortNumber() . . . 689
17.265.2.3 getEntity() . . . 689
17.265.2.4 getTime() . . . 689
17.265.2.5 setEntity() . . . 689
17.266 Exact Class Reference . . . 689
17.267 Example_BasicOrderShipping Class Reference . . . 690
17.267.1 Member Function Documentation . . . 692
17.267.1.1 main() . . . 692
17.268 Example_PortModel Class Reference . . . 692
17.268.1 Member Function Documentation . . . 693
17.268.1.1 main() . . . 693
17.269 Example_PublicTransport Class Reference . . . 694
17.269.1 Member Function Documentation . . . 695
17.269.1.1 main() . . . 695
17.270 CompartmentExchangeComponent::ExchangeRule Struct Reference . . . 695
17.271 GroProgramRuntime::ExecutionResult Struct Reference . . . 696
17.272 Exit Class Reference . . . 696
17.272.1 Detailed Description . . . 701
17.272.2 of Cells Number of contiguous conveyor cells the entity will relinquish. . . . 701
17.272.3 Member Function Documentation . . . 701
17.272.3.1 _check() . . . 701

17.272.3.2 _loadInstance() . . . 701
17.272.3.3 _onDispatchEvent() . . . 701
17.272.3.4 _saveInstance() . . . 702
17.272.3.5 show() . . . 702
17.273 ExperimentManager Class Reference . . . 702
17.273.1 Detailed Description . . . 702
17.274 ExperimentManager_if Class Reference . . . 703
17.274.1 Detailed Description . . . 703
17.274.2 Member Function Documentation . . . 703
17.274.2.1 addTraceSimulationHandler() . . . 703
17.274.2.2 getScenarios() . . . 704
17.274.2.3 startExperiment() . . . 704
17.274.2.4 startSimulationOfScenario() . . . 704
17.274.2.5 stopExperiment() . . . 704
17.275 ExperimentManagerDefaultImpl1 Class Reference . . . 704
17.275.1 Member Function Documentation . . . 705
17.275.1.1 addTraceSimulationHandler() . . . 705
17.275.1.2 getScenarios() . . . 705
17.275.1.3 startExperiment() . . . 705
17.275.1.4 startSimulationOfScenario() . . . 706
17.275.1.5 stopExperiment() . . . 706
17.276 ExpressionBuilder Class Reference . . . 706
17.277 ExpressionCategory Struct Reference . . . 707
17.278 ExpressionParameter Struct Reference . . . 707
17.279 ExpressionTemplate Struct Reference . . . 707
17.280 genesys::launcher::ExtractionResult Struct Reference . . . 707
17.281 FactorialDesign Class Reference . . . 707
17.281.1 Member Function Documentation . . . 708
17.281.1.1 calculateResultsMean() . . . 708
17.281.1.2 calculateStatistics() . . . 709
17.281.1.3 checkFeasibility() . . . 709
17.281.1.4 createColumnLabels() . . . 709
17.281.1.5 createSignTable() . . . 709
17.281.1.6 generateIndexCombinations() . . . 709
17.281.1.7 generateInputs() . . . 709
17.281.1.8 linearRegression() . . . 709
17.281.1.9 showImpacts() . . . 710
17.281.1.10 showLinearRegressionModel() . . . 710
17.281.1.11 showResiduals() . . . 710
17.281.1.12 showSignTable() . . . 710
17.282 Failure Class Reference . . . 710
17.282.1 Detailed Description . . . 715

17.282.2 Member Function Documentation . . . 715
17.282.2.1 _check() . . . 715
17.282.2.2 _initBetweenReplications() . . . 715
17.282.2.3 _loadInstance() . . . 715
17.282.2.4 _saveInstance() . . . 715
17.282.2.5 show() . . . 716
17.283 FakeModelPersistence Class Reference . . . 716
17.283.1 Member Function Documentation . . . 717
17.283.1.1 getFormatedField() . . . 717
17.283.1.2 getOption() . . . 717
17.283.1.3 hasChanged() . . . 717
17.283.1.4 load() . . . 717
17.283.1.5 save() . . . 718
17.283.1.6 setHasChanged() . . . 718
17.283.1.7 setOption() . . . 718
17.284 FakeModelPersistenceB Class Reference . . . 718
17.284.1 Member Function Documentation . . . 719
17.284.1.1 getFormatedField() . . . 719
17.284.1.2 getOption() . . . 719
17.284.1.3 hasChanged() . . . 719
17.284.1.4 load() . . . 719
17.284.1.5 save() . . . 720
17.284.1.6 setHasChanged() . . . 720
17.284.1.7 setOption() . . . 720
17.285 FakeSystemCommandExecutor Class Reference . . . 720
17.285.1 Member Function Documentation . . . 721
17.285.1.1 run() . . . 721
17.286 testing::internal::FallbackPrinter Struct Reference . . . 721
17.287 File Class Reference . . . 721
17.287.1 Detailed Description . . . 726
17.287.2 Member Function Documentation . . . 726
17.287.2.1 _check() . . . 726
17.287.2.2 _getParserChangesInformation() . . . 726
17.287.2.3 _loadInstance() . . . 726
17.287.2.4 _saveInstance() . . . 726
17.287.2.5 show() . . . 726
17.288 testing::internal::FindFirstPrinter< T, E, Printer, Printers > Struct Template Reference . . . 727
17.289 testing::internal::FindFirstPrinter< T, decltype(Printer::PrintValue(std::declval< const T & >(), nullptr)), Printer, Printers... > Struct Template Reference . . . 727
17.290 FindLineEdit Class Reference . . . 727
17.291 Fitter_if Class Reference . . . 728
17.291.1 Detailed Description . . . 729

17.291.2 Member Function Documentation . . . 729
17.291.2.1 getDataFilename() . . . 729
17.291.2.2 setDataFilename() . . . 729
17.292 FitterDefaultImpl Class Reference . . . 730
17.292.1 Detailed Description . . . 731
17.292.2 Member Function Documentation . . . 731
17.292.2.1 fitAll() . . . 731
17.292.2.2 fitBeta() . . . 731
17.292.2.3 fitErlang() . . . 731
17.292.2.4 fitExpo() . . . 731
17.292.2.5 fitNormal() . . . 732
17.292.2.6 fitTriangular() . . . 732
17.292.2.7 fitUniform() . . . 732
17.292.2.8 fitWeibull() . . . 732
17.292.2.9 getDataFilename() . . . 732
17.292.2.10 isNormalDistributed() . . . 732
17.292.2.11 setDataFilename() . . . 732
17.293 FitterDummyImpl Class Reference . . . 732
17.293.1 Detailed Description . . . 733
17.293.2 Member Function Documentation . . . 734
17.293.2.1 fitAll() . . . 734
17.293.2.2 fitBeta() . . . 734
17.293.2.3 fitErlang() . . . 734
17.293.2.4 fitExpo() . . . 734
17.293.2.5 fitNormal() . . . 734
17.293.2.6 fitTriangular() . . . 734
17.293.2.7 fitUniform() . . . 735
17.293.2.8 fitWeibull() . . . 735
17.293.2.9 getDataFilename() . . . 735
17.293.2.10 isNormalDistributed() . . . 735
17.293.2.11 setDataFilename() . . . 735
17.294 testing::internal::FlatTuple< T > Class Template Reference . . . 735
17.295 testing::internal::FlatTupleBase< Derived, Idx > Struct Template Reference . . . 736
17.296 testing::internal::FlatTupleBase< FlatTuple< T... >, IndexSequence< Idx... > > Struct Template Reference . . . 736
17.297 testing::internal::FlatTupleConstructTag Struct Reference . . . 737
17.298 testing::internal::FlatTupleElemBase< Derived, I > Struct Template Reference . . . 737
17.299 testing::internal::FlatTupleElemBase< FlatTuple< T... >, I > Struct Template Reference . . . 738
17.300 testing::internal::FloatingPoint< RawType > Class Template Reference . . . 738
17.300.1 Member Data Documentation . . . 739
17.300.1.1 kFractionBitCount . . . 739
17.300.1.2 kFractionBitMask . . . 739

17.301 testing::internal::FormatForComparison< ToPrint, OtherOperand > Class Template Reference . 739
17.302 testing::internal::FormatForComparison< ToPrint[N], OtherOperand > Class Template Reference 739
17.303 Formula Class Reference 740
17.303.1 Detailed Description 744
17.303.2 Constructor & Destructor Documentation 744
17.303.2.1 Formula() 744
17.303.3 Member Function Documentation 744
17.303.3.1 _check() 744
17.303.3.2 _createAttachedAttributes() 744
17.303.3.3 _loadInstance() 745
17.303.3.4 _saveInstance() 745
17.303.3.5 getExpression() 745
17.303.3.6 GetPluginInformation() 745
17.303.3.7 getValue() 745
17.303.3.8 LoadInstance() 746
17.303.3.9 NewInstance() 746
17.303.3.10 setExpression() 746
17.303.3.11 show() 746
17.303.3.12 size() 747
17.304 FSMState Class Reference 747
17.304.1 Member Function Documentation 752
17.304.1.1 _loadInstance() 752
17.304.1.2 _saveInstance() 752
17.305 FtsZPolymerizationComponent Class Reference 752
17.305.1 Detailed Description 757
17.305.2 Member Function Documentation 757
17.305.2.1 _check() 757
17.305.2.2 _createEditableDataDefinitions() 757
17.305.2.3 _initBetweenReplications() 758
17.305.2.4 _loadInstance() 758
17.305.2.5 _onDispatchEvent() 758
17.305.2.6 _saveInstance() 758
17.305.2.7 show() 758
17.306 FullSimulationOfComplexModel Class Reference 759
17.306.1 Member Function Documentation 760
17.306.1.1 main() 760
17.307 testing::internal::FunctionPointerPrinter Struct Reference 760
17.308 ParserManager::GenerateNewParserResult Struct Reference 760
17.308.1 Detailed Description 761
17.309 testing::internal::GenerateTypeList< T > Struct Template Reference 761
17.310 GenesysApplication_if Class Reference 762
17.310.1 Member Function Documentation 763

17.310.1.1 main() . . . 763
17.311 genesys_mcp.client.GenesysClient Class Reference . . . 763
17.311.1 Detailed Description . . . 764
17.312 genesys_mcp.client.GenesysClientError Class Reference . . . 764
17.312.1 Detailed Description . . . 765
17.313 GenesysKnowledgeBaseSummary Struct Reference . . . 765
17.313.1 Detailed Description . . . 765
17.314 GenesysKnowledgeItem Struct Reference . . . 765
17.314.1 Detailed Description . . . 766
17.315 genesyspp_driver Class Reference . . . 766
17.316 yy::genesyspp_parser Class Reference . . . 767
17.316.1 Detailed Description . . . 769
17.316.2 Member Function Documentation . . . 770
17.316.2.1 error() . . . 770
17.316.2.2 operator()() . . . 770
17.316.2.3 parse() . . . 770
17.316.2.4 symbol_name() . . . 770
17.317 GenesysPropertyDescriptor Struct Reference . . . 771
17.318 GenesysPropertyIntrospection Class Reference . . . 772
17.319 GenesysShell Class Reference . . . 772
17.319.1 Member Function Documentation . . . 774
17.319.1.1 main() . . . 774
17.320 GenesysShell_if Class Reference . . . 774
17.321 GeneticCircuit Class Reference . . . 775
17.321.1 Detailed Description . . . 779
17.321.2 Member Function Documentation . . . 779
17.321.2.1 _check() . . . 779
17.321.2.2 _loadInstance() . . . 779
17.321.2.3 _saveInstance() . . . 780
17.321.2.4 show() . . . 780
17.322 GeneticCircuitGuiExtensionPlugin Class Reference . . . 780
17.322.1 Member Function Documentation . . . 781
17.322.1.1 displayName() . . . 781
17.322.1.2 extensionId() . . . 781
17.322.1.3 registerContributions() . . . 781
17.322.1.4 requiredModelPlugins() . . . 781
17.322.1.5 version() . . . 781
17.323 GeneticCircuitPart Class Reference . . . 782
17.323.1 Detailed Description . . . 786
17.323.2 Member Function Documentation . . . 786
17.323.2.1 _check() . . . 786
17.323.2.2 _loadInstance() . . . 786

17.323.2.3 _saveInstance() . . . 786
17.323.2.4 show() . . . 786
17.324 GeneticCircuitSimulate Class Reference . . . 786
17.324.1 Detailed Description . . . 792
17.324.2 Member Function Documentation . . . 792
17.324.2.1 _check() . . . 792
17.324.2.2 _createEditableDataDefinitions() . . . 792
17.324.2.3 _loadInstance() . . . 792
17.324.2.4 _onDispatchEvent() . . . 792
17.324.2.5 _saveInstance() . . . 792
17.324.2.6 show() . . . 792
17.325 GeneticCircuitSimulationSummary Struct Reference . . . 793
17.326 GeneticExpressionStep Class Reference . . . 793
17.326.1 Detailed Description . . . 798
17.326.2 Member Function Documentation . . . 798
17.326.2.1 _check() . . . 798
17.326.2.2 _createEditableDataDefinitions() . . . 798
17.326.2.3 _loadInstance() . . . 798
17.326.2.4 _onDispatchEvent() . . . 798
17.326.2.5 _saveInstance() . . . 799
17.326.2.6 show() . . . 799
17.327 GeneticRegulation Class Reference . . . 799
17.327.1 Detailed Description . . . 803
17.327.2 Member Function Documentation . . . 803
17.327.2.1 _check() . . . 803
17.327.2.2 _loadInstance() . . . 803
17.327.2.3 _saveInstance() . . . 804
17.327.2.4 show() . . . 804
17.328 GenSerializer Class Reference . . . 804
17.328.1 Member Function Documentation . . . 805
17.328.1.1 dump() . . . 805
17.328.1.2 for_each() . . . 806
17.328.1.3 get() . . . 806
17.328.1.4 load() . . . 806
17.328.1.5 newPersistenceRecord() . . . 806
17.328.1.6 put() . . . 807
17.329 genesys::launcher::GpgvSignatureVerifier Class Reference . . . 807
17.329.1 Member Function Documentation . . . 808
17.329.1.1 verify() . . . 808
17.330 GraphEdge Class Reference . . . 808
17.330.1 Detailed Description . . . 812
17.330.2 Member Function Documentation . . . 813

17.330.2.1 _check() . . . 813
17.330.2.2 _loadInstance() . . . 813
17.330.2.3 _saveInstance() . . . 813
17.330.2.4 show() . . . 813
17.331 GraphicalAnimateExpression Class Reference . . . 814
17.332 GraphicalAssociation Class Reference . . . 814
17.333 GraphicalAutomaticPositioningStrategy Class Reference . . . 814
17.333.1 Constructor & Destructor Documentation . . . 815
17.333.1.1 ~GraphicalAutomaticPositioningStrategy() . . . 815
17.333.2 Member Function Documentation . . . 815
17.333.2.1 dataDefinitionPosition() . . . 815
17.334 GraphicalAutomaticPositioningStrategyFactory Class Reference . . . 815
17.334.1 Detailed Description . . . 815
17.335 GraphicalComponentPort Class Reference . . . 816
17.335.1 Constructor & Destructor Documentation . . . 817
17.335.1.1 GraphicalComponentPort() . . . 817
17.336 GraphicalConnection Class Reference . . . 817
17.336.1 Detailed Description . . . 819
17.336.2 Constructor & Destructor Documentation . . . 819
17.336.2.1 GraphicalConnection() . . . 819
17.336.3 Member Function Documentation . . . 819
17.336.3.1 boundingRect() . . . 819
17.336.3.2 paint() . . . 820
17.336.3.3 updateDimensionsAndPosition() . . . 820
17.337 GraphicalConnectionStyle Class Reference . . . 820
17.338 GraphicalContextMenuController Class Reference . . . 820
17.338.1 Detailed Description . . . 821
17.339 GraphicalDataDefinitionLayout Class Reference . . . 821
17.339.1 Detailed Description . . . 821
17.339.2 Member Function Documentation . . . 821
17.339.2.1 arcPosition() . . . 821
17.340 GraphicalDiagramConnection Class Reference . . . 822
17.341 GraphicalImageAnimation Class Reference . . . 823
17.342 GraphicalModelBuilder Class Reference . . . 823
17.342.1 Detailed Description . . . 824
17.342.2 Constructor & Destructor Documentation . . . 824
17.342.2.1 GraphicalModelBuilder() . . . 824
17.342.3 Member Function Documentation . . . 824
17.342.3.1 generateGraphicalModelFromModel() . . . 824
17.342.3.2 recursivalyGenerateGraphicalModelFromModel() . . . 825
17.342.3.3 setModelLevelFilter() . . . 825
17.342.3.4 synchronizeGraphicalDataDefinitionsLayer() . . . 825

17.343 GraphicalModelComponent Class Reference . . . 825
17.343.1 Detailed Description . . . 830
17.343.2 Member Function Documentation . . . 830
17.343.2.1 renderContext() . . . 830
17.343.2.2 sceneEvent() . . . 830
17.344 GraphicalModelDataDefinition Class Reference . . . 830
17.344.1 Detailed Description . . . 832
17.345 GraphicalModelEvent Class Reference . . . 832
17.345.1 Detailed Description . . . 833
17.346 GraphicalModelItemRenderContext Struct Reference . . . 833
17.347 GraphicalModelItemRenderer Class Reference . . . 834
17.348 GraphicalModelItemRenderStrategy Class Reference . . . 834
17.349 GraphicalModelSerializer Class Reference . . . 834
17.349.1 Detailed Description . . . 835
17.349.2 Constructor & Destructor Documentation . . . 835
17.349.2.1 GraphicalModelSerializer() . . . 835
17.349.3 Member Function Documentation . . . 835
17.349.3.1 efsmNetworkDiagramLayout() . . . 835
17.349.3.2 loadGraphicalModel() . . . 836
17.349.3.3 saveGraphicalModel() . . . 836
17.349.3.4 saveTextModel() . . . 836
17.349.3.5 setEFSMNetworkDiagramLayout() . . . 836
17.350 GraphNetwork Class Reference . . . 836
17.350.1 Detailed Description . . . 842
17.350.2 Member Function Documentation . . . 842
17.350.2.1 _check() . . . 842
17.350.2.2 _initBetweenReplications() . . . 842
17.350.2.3 _loadInstance() . . . 842
17.350.2.4 _saveInstance() . . . 842
17.350.2.5 show() . . . 843
17.351 GraphNode Class Reference . . . 843
17.351.1 Detailed Description . . . 847
17.352 GraphPathResult Struct Reference . . . 848
17.352.1 Detailed Description . . . 848
17.353 GraphSearchTreeEntry Struct Reference . . . 848
17.354 GraphTopologicalOrderResult Struct Reference . . . 849
17.355 GraphTraversalResult Struct Reference . . . 849
17.355.1 Detailed Description . . . 849
17.356 GraphvizModelExporter Class Reference . . . 849
17.356.1 Detailed Description . . . 850
17.356.2 Member Function Documentation . . . 850
17.356.2.1 adjustDotName() . . . 850

17.356.2.2 createModelImage() . . . 850
17.357 ModelGraphicsScene::GRID Struct Reference . . . 851
17.358 GroProgram Class Reference . . . 851
17.358.1 Detailed Description . . . 855
17.358.2 Member Function Documentation . . . 855
17.358.2.1 _check() . . . 855
17.358.2.2 _loadInstance() . . . 855
17.358.2.3 _saveInstance() . . . 856
17.358.2.4 show() . . . 856
17.359 GroProgramAst Struct Reference . . . 856
17.359.1 Detailed Description . . . 857
17.360 GroProgramCompiler Class Reference . . . 857
17.360.1 Detailed Description . . . 857
17.361 GroProgramIr Struct Reference . . . 857
17.361.1 Detailed Description . . . 857
17.362 GroProgramParser Class Reference . . . 858
17.362.1 Detailed Description . . . 858
17.363 GroProgramRuntime Class Reference . . . 858
17.363.1 Detailed Description . . . 859
17.363.2 Member Function Documentation . . . 859
17.363.2.1 validateSignalDeclarationPlacement() . . . 859
17.364 GroProgramRuntimeState Struct Reference . . . 859
17.364.1 Detailed Description . . . 859
17.365 GroupItem Struct Reference . . . 859
17.366 GroupUndoCommand Class Reference . . . 860
17.367 testing::internal::GTestFlagSaver Class Reference . . . 860
17.368 testing::internal::GTestLog Class Reference . . . 860
17.369 testing::internal::GTestMutexLock Class Reference . . . 861
17.370 GuiActionContribution Struct Reference . . . 861
17.371 GuiDockContribution Struct Reference . . . 861
17.372 GuiExtensionConfigurationDialog Class Reference . . . 862
17.373 GuiExtensionConfigurationGuiExtensionPlugin Class Reference . . . 863
17.373.1 Member Function Documentation . . . 863
17.373.1.1 displayName() . . . 863
17.373.1.2 extensionId() . . . 863
17.373.1.3 registerContributions() . . . 864
17.373.1.4 requiredModelPlugins() . . . 864
17.373.1.5 version() . . . 864
17.374 GuiExtensionManager Class Reference . . . 864
17.375 GuiExtensionPlugin Class Reference . . . 865
17.376 GuiExtensionPluginAutoRegistrar Class Reference . . . 866
17.377 GuiExtensionPluginCatalog Class Reference . . . 866

17.378 GuiExtensionPresentationMetadata Struct Reference . . . 866
17.379 GuiExtensionRegistry Class Reference . . . 866
17.380 GuiExtensionRuntimeContext Struct Reference . . . 867
17.381 GuiScopeTrace Class Reference . . . 867
17.381.1 Detailed Description . . . 867
17.381.2 Constructor & Destructor Documentation . . . 867
17.381.2.1 GuiScopeTrace() . . . 867
17.382 GuiThemeManager Class Reference . . . 869
17.382.1 Detailed Description . . . 869
17.382.2 Member Function Documentation . . . 869
17.382.2.1 applyApplicationTheme() . . . 869
17.382.2.2 applyModelGraphicsTheme() . . . 869
17.383 GuiWindowContribution Struct Reference . . . 870
17.384 Half_Adder Class Reference . . . 870
17.384.1 Member Function Documentation . . . 871
17.384.1.1 main() . . . 871
17.385 HarmonicOscillator Class Reference . . . 872
17.385.1 Member Function Documentation . . . 872
17.385.1.1 dimension() . . . 872
17.385.1.2 evaluate() . . . 872
17.386 testing::internal::HasDebugStringAndShortDebugString< T > Class Template Reference . . . 873
17.386.1 Member Data Documentation . . . 873
17.386.1.1 value . . . 873
17.387 DataAnalyzerWindow::HistogramPreview Class Reference . . . 873
17.388 HppLatticeGasState Class Reference . . . 874
17.388.1 Member Function Documentation . . . 875
17.388.1.1 clone() . . . 875
17.388.1.2 equals() . . . 875
17.388.1.3 getValue() . . . 875
17.388.1.4 setValue() . . . 875
17.388.1.5 toString() . . . 875
17.388.1.6 typeName() . . . 876
17.389 HppLatticeGasStateSet Class Reference . . . 876
17.389.1 Member Function Documentation . . . 877
17.389.1.1 contains() . . . 877
17.389.1.2 createDefaultState() . . . 877
17.389.1.3 enumerateStates() . . . 877
17.389.1.4 isComposite() . . . 877
17.389.1.5 isDiscrete() . . . 877
17.389.1.6 isFinite() . . . 877
17.389.1.7 name() . . . 877
17.389.1.8 parseState() . . . 877

17.390 HttpProviderClientBase Class Reference . . . 878
17.390.1 Detailed Description . . . 879
17.390.1.1 Transport design . . . 880
17.390.1.2 Subclass responsibilities . . . 880
17.390.1.3 curl dependency . . . 880
17.390.2 Member Function Documentation . . . 880
17.390.2.1 _authorizationHeader() . . . 880
17.390.2.2 _buildRequestBody() . . . 880
17.390.2.3 _extraRequestHeaders() . . . 880
17.390.2.4 _parseResponseBody() . . . 881
17.390.2.5 clearApiKey() . . . 881
17.390.2.6 getBaseUrl() . . . 881
17.390.2.7 getTimeoutSeconds() . . . 881
17.390.2.8 hasApiKey() . . . 881
17.390.2.9 isAvailable() . . . 881
17.390.2.10 send() . . . 881
17.390.2.11 setApiKey() . . . 882
17.390.2.12 setBaseUrl() . . . 882
17.390.2.13 setTimeoutSeconds() . . . 882
17.390.3 Member Data Documentation . . . 882
17.390.3.1 _apiKey . . . 882
17.391 HttpRequest Struct Reference . . . 882
17.391.1 Detailed Description . . . 883
17.391.2 Member Function Documentation . . . 883
17.391.2.1 header() . . . 883
17.392 HttpResponse Struct Reference . . . 883
17.392.1 Detailed Description . . . 883
17.393 HypothesisTester_if Class Reference . . . 884
17.393.1 Detailed Description . . . 885
17.394 HypothesisTesterDefaultImpl1 Class Reference . . . 885
17.394.1 Detailed Description . . . 887
17.394.2 Member Function Documentation . . . 887
17.394.2.1 averageConfidenceInterval() [1/2] . . . 887
17.394.2.2 averageConfidenceInterval() [2/2] . . . 888
17.394.2.3 averageDifferenceConfidenceInterval() . . . 888
17.394.2.4 estimateSampleSize() . . . 888
17.394.2.5 proportionConfidenceInterval() [1/4] . . . 888
17.394.2.6 proportionConfidenceInterval() [2/4] . . . 888
17.394.2.7 proportionConfidenceInterval() [3/4] . . . 888
17.394.2.8 proportionConfidenceInterval() [4/4] . . . 889
17.394.2.9 proportionDifferenceConfidenceInterval() . . . 889
17.394.2.10 testAverage() [1/4] . . . 889

17.394.2.11 testAverage() [2/4] . . . 889
17.394.2.12 testAverage() [3/4] . . . 889
17.394.2.13 testAverage() [4/4] . . . 890
17.394.2.14 testProportion() [1/4] . . . 890
17.394.2.15 testProportion() [2/4] . . . 890
17.394.2.16 testProportion() [3/4] . . . 890
17.394.2.17 testProportion() [4/4] . . . 890
17.394.2.18 testVariance() [1/4] . . . 890
17.394.2.19 testVariance() [2/4] . . . 891
17.394.2.20 testVariance() [3/4] . . . 891
17.394.2.21 testVariance() [4/4] . . . 891
17.394.2.22 varianceConfidenceInterval() [1/2] . . . 891
17.394.2.23 varianceConfidenceInterval() [2/2] . . . 891
17.394.2.24 varianceRatioConfidenceInterval() . . . 891
17.395 genesys::launcher::IArchiveExtractor Class Reference . . . 892
17.396 testing::internal::Ignore< size_t > Struct Template Reference . . . 892
17.397 testing::internal::IgnoredValue Class Reference . . . 892
17.398 testing::internal::IndexSequence< Is > Struct Template Reference . . . 892
17.399 genesys::launcher::INetworkTransport Class Reference . . . 893
17.400 IntegerStateSet Class Reference . . . 893
17.400.1 Member Function Documentation . . . 894
17.400.1.1 contains() . . . 894
17.400.1.2 createDefaultState() . . . 894
17.400.1.3 enumerateStates() . . . 895
17.400.1.4 isDiscrete() . . . 895
17.400.1.5 isFinite() . . . 895
17.400.1.6 name() . . . 895
17.400.1.7 parseState() . . . 895
17.401 InternalEvent Class Reference . . . 895
17.401.1 Member Function Documentation . . . 896
17.401.1.1 show() . . . 896
17.402 genesys::launcher::IProcessLauncher Class Reference . . . 897
17.403 testing::internal::is_proxy_type_list< typename > Struct Template Reference . . . 897
17.404 testing::internal::is_proxy_type_list< ProxyTypeList< Ts... > > Struct Template Reference . . . 898
17.405 testing::internal::IsHashTable< T > Struct Template Reference . . . 899
17.406 genesys::launcher::ISignatureVerifier Class Reference . . . 899
17.407 testing::internal::IsRecursiveContainer< C > Struct Template Reference . . . 900
17.408 testing::internal::IsRecursiveContainerImpl< C, bool > Struct Template Reference . . . 900
17.409 testing::internal::IsRecursiveContainerImpl< C, false > Struct Template Reference . . . 901
17.410 testing::internal::IsRecursiveContainerImpl< C, true > Struct Template Reference . . . 901
17.411 IssueReportDraft Struct Reference . . . 902
17.411.1 Detailed Description . . . 902

17.412 IssueReportRelayService Class Reference . . . 902
17.412.1 Detailed Description . . . 902
17.413 IssueReportRuntimeContext Struct Reference . . . 903
17.413.1 Detailed Description . . . 903
17.414 IssueReportSubmissionResult Struct Reference . . . 903
17.414.1 Detailed Description . . . 903
17.415 genesys::launcher::IUpdateUi Class Reference . . . 904
17.416 Json Struct Reference . . . 904
17.416.1 Member Data Documentation . . . 904
17.416.1.1 fields . . . 904
17.416.1.2 sequence . . . 905
17.417 JsonSerializer Class Reference . . . 905
17.417.1 Member Function Documentation . . . 906
17.417.1.1 dump() . . . 906
17.417.1.2 for_each() . . . 906
17.417.1.3 get() . . . 906
17.417.1.4 load() . . . 907
17.417.1.5 newPersistenceRecord() . . . 907
17.417.1.6 put() . . . 907
17.418 KernelAccessorProbe Struct Reference . . . 907
17.419 Label Class Reference . . . 908
17.419.1 Member Function Documentation . . . 911
17.419.1.1 _check() . . . 911
17.419.1.2 _loadInstance() . . . 911
17.419.1.3 _saveInstance() . . . 912
17.419.1.4 show() . . . 912
17.420 QCPLabelPainterPrivate::LabelData Struct Reference . . . 912
17.421 Lattice Class Reference . . . 913
17.421.1 Detailed Description . . . 914
17.422 genesys::launcher::LauncherConfig Struct Reference . . . 914
17.423 genesys::launcher::LauncherController Class Reference . . . 914
17.424 genesys::launcher::LauncherLogger Class Reference . . . 915
17.425 genesys::launcher::LauncherUpdateResult Struct Reference . . . 915
17.426 Leave Class Reference . . . 915
17.426.1 Detailed Description . . . 920
17.426.2 of Cells Number of contiguous cells the entity requires. . . . 921
17.426.3 Member Function Documentation . . . 921
17.426.3.1 _check() . . . 921
17.426.3.2 _createEditableDataDefinitions() . . . 921
17.426.3.3 _createInternalStatisticReporters() . . . 921
17.426.3.4 _loadInstance() . . . 921
17.426.3.5 _onDispatchEvent() . . . 921

17.426.3.6 _saveInstance() . . . 922
17.426.3.7 show() . . . 922
17.427 LicenceManager Class Reference . . . 922
17.427.1 Detailed Description . . . 922
17.428 LineNumberArea Class Reference . . . 923
17.429 List< T > Class Template Reference . . . 923
17.429.1 Detailed Description . . . 925
17.429.2 Member Data Documentation . . . 925
17.429.2.1 _sortFunc . . . 925
17.430 ListObservable< T > Class Template Reference . . . 925
17.430.1 Detailed Description . . . 928
17.431 LocalProviderClientImpl Class Reference . . . 928
17.431.1 Detailed Description . . . 930
17.431.1.1 Tested server compatibility . . . 931
17.431.2 Member Function Documentation . . . 931
17.431.2.1 _authorizationHeader() . . . 931
17.431.2.2 _buildRequestBody() . . . 931
17.431.2.3 _parseResponseBody() . . . 931
17.431.2.4 getProviderName() . . . 931
17.431.2.5 isAvailable() . . . 931
17.432 LocalRule Class Reference . . . 932
17.432.1 Member Function Documentation . . . 933
17.432.1.1 applyRule() . . . 933
17.433 LocalRule_CopyCurrentState Class Reference . . . 934
17.433.1 Member Function Documentation . . . 935
17.433.1.1 applyRule() . . . 935
17.434 LocalRule_Elementary Class Reference . . . 935
17.434.1 Member Function Documentation . . . 936
17.434.1.1 applyRule() . . . 936
17.435 LocalRule_FlorestalFire Class Reference . . . 936
17.435.1 Member Function Documentation . . . 937
17.435.1.1 applyRule() . . . 937
17.436 LocalRule_GameOfLife Class Reference . . . 938
17.436.1 Member Function Documentation . . . 939
17.436.1.1 applyRule() . . . 939
17.437 LocalRule_Growty Class Reference . . . 939
17.437.1 Member Function Documentation . . . 940
17.437.1.1 applyRule() . . . 940
17.438 LocalRule_HppLatticeGas Class Reference . . . 940
17.438.1 Member Function Documentation . . . 941
17.438.1.1 applyRule() . . . 941
17.439 LocalRule_SandRock_Student Class Reference . . . 942

17.439.1 Member Function Documentation . . . 943
17.439.1.1 applyRule() [1/2] . . . 943
17.439.1.2 applyRule() [2/2] . . . 943
17.440 LocalRule_ShowNeighbordhood Class Reference . . . 943
17.440.1 Member Function Documentation . . . 944
17.440.1.1 applyRule() . . . 944
17.441 yy::location Class Reference . . . 944
17.441.1 Detailed Description . . . 945
17.442 Loja01 Class Reference . . . 946
17.442.1 Member Function Documentation . . . 947
17.442.1.1 main() . . . 947
17.443 testing::internal::internal_stream_operator_without_lexical_name_lookup::LookupBlocker Struct Reference . . . 947
17.444 LSODE Class Reference . . . 947
17.444.1 Detailed Description . . . 952
17.444.2 Member Function Documentation . . . 953
17.444.2.1 _check() . . . 953
17.444.2.2 _loadInstance() . . . 953
17.444.2.3 _onDispatchEvent() . . . 953
17.444.2.4 _saveInstance() . . . 953
17.444.2.5 show() . . . 953
17.445 LSODEInitialState Struct Reference . . . 954
17.446 LSODESample Struct Reference . . . 954
17.447 MainWindow Class Reference . . . 954
17.447.1 Detailed Description . . . 956
17.448 testing::internal::MakeIndexSequenceImpl< N > Struct Template Reference . . . 956
17.449 testing::internal::MakeIndexSequenceImpl< 0 > Struct Reference . . . 957
17.450 genesys::launcher::ManifestParseResult Struct Reference . . . 958
17.451 testing::internal::MarkAsIgnored Struct Reference . . . 959
17.452 MarkovChain Class Reference . . . 959
17.452.1 Member Function Documentation . . . 964
17.452.1.1 _check() . . . 964
17.452.1.2 _createEditableDataDefinitions() . . . 964
17.452.1.3 _initBetweenReplications() . . . 964
17.452.1.4 _loadInstance() . . . 964
17.452.1.5 _onDispatchEvent() . . . 964
17.452.1.6 _saveInstance() . . . 965
17.452.1.7 show() . . . 965
17.453 MarkovChainNetwork Class Reference . . . 965
17.453.1 Detailed Description . . . 970
17.453.2 Member Function Documentation . . . 970
17.453.2.1 _activate() . . . 970

17.453.2.2 _check() . . . 970
17.453.2.3 _initBetweenReplications() . . . 970
17.453.2.4 _loadInstance() . . . 970
17.453.2.5 _saveInstance() . . . 970
17.453.2.6 show() . . . 972
17.454 MarkovState Class Reference . . . 972
17.454.1 Detailed Description . . . 976
17.455 MarkovChainNetwork::MarkovTransition Class Reference . . . 977
17.456 MassActionOdeSystem Class Reference . . . 977
17.456.1 Detailed Description . . . 978
17.456.2 Member Function Documentation . . . 978
17.456.2.1 dimension() . . . 978
17.456.2.2 evaluate() . . . 978
17.457 Match Class Reference . . . 978
17.457.1 Detailed Description . . . 983
17.457.2 Member Function Documentation . . . 984
17.457.2.1 _check() . . . 984
17.457.2.2 _createEditableDataDefinitions() . . . 984
17.457.2.3 _loadInstance() . . . 984
17.457.2.4 _onDispatchEvent() . . . 984
17.457.2.5 _saveInstance() . . . 984
17.457.2.6 show() . . . 984
17.458 testing::Message Class Reference . . . 985
17.459 MetabolicFluxBalance Class Reference . . . 985
17.459.1 Detailed Description . . . 990
17.459.1.1 LP solver dependency (GLPK) . . . 990
17.459.2 Member Function Documentation . . . 990
17.459.2.1 _check() . . . 990
17.459.2.2 _createEditableDataDefinitions() . . . 990
17.459.2.3 _loadInstance() . . . 991
17.459.2.4 _onDispatchEvent() . . . 991
17.459.2.5 _saveInstance() . . . 991
17.459.2.6 show() . . . 991
17.460 MetabolicFluxBalanceSolver Class Reference . . . 991
17.460.1 Detailed Description . . . 992
17.461 MetabolicNetwork Class Reference . . . 992
17.461.1 Detailed Description . . . 996
17.461.2 Member Function Documentation . . . 996
17.461.2.1 _check() . . . 996
17.461.2.2 _createEditableDataDefinitions() . . . 996
17.461.2.3 _loadInstance() . . . 996
17.461.2.4 _saveInstance() . . . 996

17.461.2.5 show() . . . 996
17.462 MetabolicReaction Class Reference . . . 997
17.462.1 Detailed Description . . . 1001
17.462.2 Member Function Documentation . . . 1001
17.462.2.1 _check() . . . 1001
17.462.2.2 _createEditableDataDefinitions() . . . 1001
17.462.2.3 _loadInstance() . . . 1001
17.462.2.4 _saveInstance() . . . 1001
17.462.2.5 show() . . . 1001
17.463 MetabolicStateProjectionComponent Class Reference . . . 1002
17.463.1 Detailed Description . . . 1008
17.463.2 Member Function Documentation . . . 1008
17.463.2.1 _check() . . . 1008
17.463.2.2 _createEditableDataDefinitions() . . . 1008
17.463.2.3 _loadInstance() . . . 1008
17.463.2.4 _onDispatchEvent() . . . 1008
17.463.2.5 _saveInstance() . . . 1008
17.463.2.6 show() . . . 1009
17.464 MetabolicSubmodelComponent Class Reference . . . 1009
17.464.1 Detailed Description . . . 1014
17.464.2 Member Function Documentation . . . 1014
17.464.2.1 _check() . . . 1014
17.464.2.2 _createEditableDataDefinitions() . . . 1014
17.464.2.3 _initBetweenReplications() . . . 1014
17.464.2.4 _loadInstance() . . . 1014
17.464.2.5 _onDispatchEvent() . . . 1015
17.464.2.6 _saveInstance() . . . 1015
17.464.2.7 show() . . . 1015
17.465 generate_kernel_simulator_test_stubs.MethodEntry Class Reference . . . 1015
17.466 ModalModelBindingsUndoCommand Class Reference . . . 1016
17.467 ModalModelDefault Class Reference . . . 1016
17.467.1 Detailed Description . . . 1022
17.467.2 Member Function Documentation . . . 1022
17.467.2.1 _check() . . . 1022
17.467.2.2 _createAttachedAttributes() . . . 1022
17.467.2.3 _initBetweenReplications() . . . 1022
17.467.2.4 _loadInstance() . . . 1022
17.467.2.5 _onDispatchEvent() . . . 1022
17.467.2.6 _saveInstance() . . . 1022
17.467.2.7 show() . . . 1023
17.468 Model Class Reference . . . 1023
17.468.1 Detailed Description . . . 1024

17.468.2 Constructor & Destructor Documentation . . . 1025
17.468.2.1 Model() . . . 1025
17.468.3 Member Function Documentation . . . 1025
17.468.3.1 check() . . . 1025
17.468.3.2 checkExpression() . . . 1025
17.468.3.3 checkReferencesToDataDefinitions() . . . 1025
17.468.3.4 collectDataDefinitionsRemovedWith() . . . 1026
17.468.3.5 createEntity() . . . 1026
17.468.3.6 getComponentManager() . . . 1026
17.468.3.7 getControls() . . . 1026
17.468.3.8 getDataManager() . . . 1027
17.468.3.9 getFutureEvents() . . . 1027
17.468.3.10 getId() . . . 1027
17.468.3.11 getInfos() . . . 1027
17.468.3.12 getLevel() . . . 1027
17.468.3.13 getOnEventManager() . . . 1027
17.468.3.14 getParentSimulator() . . . 1028
17.468.3.15 getPersistence() . . . 1028
17.468.3.16 getResponses() . . . 1028
17.468.3.17 getSimulation() . . . 1028
17.468.3.18 getTracer() . . . 1028
17.468.3.19 hasChanged() . . . 1028
17.468.3.20 insert() . . . 1028
17.468.3.21 load() . . . 1029
17.468.3.22 parseExpression() [1/2] . . . 1029
17.468.3.23 parseExpression() [2/2] . . . 1029
17.468.3.24 remove() . . . 1030
17.468.3.25 removeEntity() . . . 1030
17.468.3.26 save() . . . 1030
17.468.3.27 sendEntityToComponent() [1/2] . . . 1030
17.468.3.28 sendEntityToComponent() [2/2] . . . 1031
17.468.3.29 setHasChanged() . . . 1031
17.468.3.30 setTracer() . . . 1031
17.468.3.31 showLanguage() . . . 1031
17.469 ModelChecker_if Class Reference . . . 1031
17.469.1 Detailed Description . . . 1032
17.469.2 Member Function Documentation . . . 1032
17.469.2.1 checkActivationCode() . . . 1032
17.469.2.2 checkAll() . . . 1032
17.469.2.3 checkConnected() . . . 1033
17.469.2.4 checkLimits() . . . 1033
17.469.2.5 checkSymbols() . . . 1033

17.470 ModelCheckerDefaultImpl1 Class Reference . . . 1033
17.470.1 Member Function Documentation . . . 1034
17.470.1.1 checkActivationCode() . . . 1034
17.470.1.2 checkAll() . . . 1034
17.470.1.3 checkConnected() . . . 1034
17.470.1.4 checkLimits() . . . 1035
17.470.1.5 checkSymbols() . . . 1035
17.471 ModelComponent Class Reference . . . 1035
17.471.1 Detailed Description . . . 1039
17.471.2 Constructor & Destructor Documentation . . . 1040
17.471.2.1 ModelComponent() . . . 1040
17.471.3 Member Function Documentation . . . 1040
17.471.3.1 _loadInstance() . . . 1040
17.471.3.2 _onDispatchEvent() . . . 1040
17.471.3.3 _saveInstance() . . . 1041
17.471.3.4 Check() . . . 1041
17.471.3.5 connectTo() . . . 1041
17.471.3.6 CreateInternalData() . . . 1042
17.471.3.7 DispatchEvent() . . . 1042
17.471.3.8 getConnectionManager() . . . 1042
17.471.3.9 getDescription() . . . 1042
17.471.3.10 hasBreakpointAt() . . . 1042
17.471.3.11 LoadInstance() . . . 1043
17.471.3.12 remove() . . . 1043
17.471.3.13 SaveInstance() . . . 1043
17.471.3.14 setDescription() . . . 1043
17.471.3.15 show() . . . 1044
17.471.4 Member Data Documentation . . . 1044
17.471.4.1 _connections . . . 1044
17.472 ModelComponentManager Class Reference . . . 1044
17.472.1 Detailed Description . . . 1045
17.472.2 Member Function Documentation . . . 1045
17.472.2.1 begin() . . . 1045
17.472.2.2 clear() . . . 1045
17.472.2.3 end() . . . 1045
17.472.2.4 find() [1/2] . . . 1046
17.472.2.5 find() [2/2] . . . 1046
17.472.2.6 front() . . . 1046
17.472.2.7 getAllComponents() . . . 1046
17.472.2.8 getNumberOfComponents() . . . 1046
17.472.2.9 getSourceComponents() . . . 1047
17.472.2.10 getTransferInComponents() . . . 1047

17.472.2.11 hasChanged() . . . 1047
17.472.2.12 insert() . . . 1047
17.472.2.13 next() . . . 1047
17.472.2.14 remove() . . . 1047
17.472.2.15 setHasChanged() . . . 1048
17.473 ModelDataDefinition Class Reference . . . 1048
17.473.1 Detailed Description . . . 1051
17.473.2 Constructor & Destructor Documentation . . . 1051
17.473.2.1 ModelDataDefinition() . . . 1051
17.473.3 Member Function Documentation . . . 1052
17.473.3.1 _addSimulationControl() . . . 1052
17.473.3.2 _check() . . . 1052
17.473.3.3 _createAttachedAttributes() . . . 1052
17.473.3.4 _createEditableDataDefinitions() . . . 1052
17.473.3.5 _createInternalAndAttachedData() . . . 1052
17.473.3.6 _createNonEditableDataDefinitions() . . . 1052
17.473.3.7 _getParserChangesInformation() . . . 1052
17.473.3.8 _getSaveDefaultsOption() . . . 1053
17.473.3.9 _initBetweenReplications() . . . 1053
17.473.3.10 _loadInstance() . . . 1053
17.473.3.11 _saveInstance() . . . 1053
17.473.3.12 Check() . . . 1054
17.473.3.13 CreateInternalData() . . . 1054
17.473.3.14 CreateRelatedDataElements() . . . 1054
17.473.3.15 defineTraceLevelSpecific() . . . 1055
17.473.3.16 getAttachedData() . . . 1055
17.473.3.17 getClassname() . . . 1055
17.473.3.18 getId() . . . 1055
17.473.3.19 getInternalData() [1/2] . . . 1055
17.473.3.20 getInternalData() [2/2] . . . 1055
17.473.3.21 getLabel() . . . 1056
17.473.3.22 getLevel() . . . 1056
17.473.3.23 getName() . . . 1056
17.473.3.24 getParentModel() . . . 1056
17.473.3.25 getSimulationControls() . . . 1056
17.473.3.26 getTraceLevelSpecific() . . . 1057
17.473.3.27 hasChanged() . . . 1057
17.473.3.28 InitBetweenReplications() . . . 1057
17.473.3.29 isReportStatistics() . . . 1057
17.473.3.30 isTraceLevelSpecificEnabled() . . . 1057
17.473.3.31 LoadInstance() . . . 1057
17.473.3.32 NewInstance() . . . 1058

17.473.3.33 SaveInstance() . . . 1058
17.473.3.34 setHasChanged() . . . 1058
17.473.3.35 setLabel() . . . 1058
17.473.3.36 setModelLevel() . . . 1059
17.473.3.37 setName() . . . 1059
17.473.3.38 setReportStatistics() . . . 1059
17.473.3.39 setTraceLevelSpecific() . . . 1059
17.473.3.40 setTraceLevelSpecificEnabled() . . . 1059
17.473.3.41 show() . . . 1060
17.473.3.42 traceSimulation() . . . 1060
17.474 ModelDataDefinitionAssociations Class Reference . . . 1060
17.474.1 Detailed Description . . . 1061
17.475 ModelDataManager Class Reference . . . 1061
17.475.1 Detailed Description . . . 1062
17.475.2 Member Function Documentation . . . 1063
17.475.2.1 check() [1/2] . . . 1063
17.475.2.2 check() [2/2] . . . 1063
17.475.2.3 getDataDefinition() [1/2] . . . 1063
17.475.2.4 getDataDefinition() [2/2] . . . 1064
17.475.2.5 getDataDefinitionClassnames() . . . 1064
17.475.2.6 getNumberOfDataDefinitions() [1/2] . . . 1064
17.475.2.7 getNumberOfDataDefinitions() [2/2] . . . 1064
17.475.2.8 getRankOf() . . . 1064
17.475.2.9 insert() [1/2] . . . 1066
17.475.2.10 insert() [2/2] . . . 1066
17.475.2.11 remove() [1/2] . . . 1066
17.475.2.12 remove() [2/2] . . . 1066
17.476 ModelEvent Class Reference . . . 1067
17.476.1 Detailed Description . . . 1067
17.477 ModelGraphicsScene Class Reference . . . 1067
17.477.1 Detailed Description . . . 1075
17.477.2 Member Function Documentation . . . 1076
17.477.2.1 notifyGraphicalModelChange() . . . 1076
17.477.2.2 setModelLevelFilter() . . . 1076
17.478 ModelGraphicsView Class Reference . . . 1076
17.478.1 Constructor & Destructor Documentation . . . 1078
17.478.1.1 ModelGraphicsView() . . . 1078
17.478.2 Member Function Documentation . . . 1079
17.478.2.1 notifySceneGraphicalModelEventHandler() . . . 1079
17.479 SimulatorSessionService::ModelImportResult Struct Reference . . . 1079
17.479.1 Detailed Description . . . 1079
17.480 ModelInfo Class Reference . . . 1080

17.480.1 Detailed Description . . . 1080
17.480.2 Member Function Documentation . . . 1080
17.480.2.1 getAnalystName() . . . 1080
17.480.2.2 getDescription() . . . 1081
17.480.2.3 getName() . . . 1081
17.480.2.4 getProjectTitle() . . . 1081
17.480.2.5 getVersion() . . . 1081
17.480.2.6 hasChanged() . . . 1081
17.480.2.7 loadInstance() . . . 1081
17.480.2.8 saveInstance() . . . 1081
17.480.2.9 setAnalystName() . . . 1082
17.480.2.10 setDescription() . . . 1082
17.480.2.11 setName() . . . 1082
17.480.2.12 setProjectTitle() . . . 1082
17.480.2.13 setVersion() . . . 1082
17.481 SimulatorSessionService::ModelInfoResult Struct Reference . . . 1083
17.481.1 Detailed Description . . . 1083
17.482 ModelInspectorController Class Reference . . . 1083
17.482.1 Detailed Description . . . 1084
17.483 ModelLanguageSynchronizer Class Reference . . . 1084
17.483.1 Detailed Description . . . 1084
17.483.2 Constructor & Destructor Documentation . . . 1085
17.483.2.1 ModelLanguageSynchronizer() . . . 1085
17.483.3 Member Function Documentation . . . 1085
17.483.3.1 actualizeModelSimLanguage() . . . 1085
17.483.3.2 setSimulationModelBasedOnText() . . . 1085
17.484 ModelLifecycleController Class Reference . . . 1085
17.484.1 Detailed Description . . . 1086
17.485 ModelManager Class Reference . . . 1086
17.485.1 Detailed Description . . . 1087
17.485.2 Constructor & Destructor Documentation . . . 1088
17.485.2.1 ModelManager() . . . 1088
17.485.3 Member Function Documentation . . . 1088
17.485.3.1 canGoNext() . . . 1088
17.485.3.2 canGoPrevious() . . . 1088
17.485.3.3 createFromLanguage() . . . 1088
17.485.3.4 current() . . . 1088
17.485.3.5 currentIndex() . . . 1088
17.485.3.6 front() . . . 1089
17.485.3.7 hasModel() . . . 1089
17.485.3.8 indexOf() . . . 1089
17.485.3.9 insert() . . . 1089

17.485.3.10 last() . . . 1089
17.485.3.11 loadModel() . . . 1090
17.485.3.12 modelAt() . . . 1090
17.485.3.13 models() . . . 1090
17.485.3.14 newModel() . . . 1090
17.485.3.15 next() . . . 1090
17.485.3.16 previous() . . . 1091
17.485.3.17 remove() . . . 1091
17.485.3.18 saveModel() . . . 1091
17.485.3.19 setCurrent() . . . 1091
17.485.3.20 size() . . . 1091
17.486 SimulatorSessionService::ModelPersistenceResult Struct Reference . . . 1092
17.486.1 Detailed Description . . . 1092
17.487 ModelPluginInspectorGuiExtensionPlugin Class Reference . . . 1092
17.487.1 Member Function Documentation . . . 1093
17.487.1.1 displayName() . . . 1093
17.487.1.2 extensionId() . . . 1093
17.487.1.3 registerContributions() . . . 1093
17.487.1.4 requiredModelPlugins() . . . 1093
17.487.1.5 version() . . . 1093
17.488 ModelSimulation Class Reference . . . 1094
17.488.1 Detailed Description . . . 1095
17.488.2 Member Function Documentation . . . 1095
17.488.2.1 isRunning() . . . 1095
17.488.2.2 start() . . . 1095
17.489 MolecularSpecies Class Reference . . . 1095
17.489.1 Detailed Description . . . 1099
17.489.2 Member Function Documentation . . . 1100
17.489.2.1 _check() . . . 1100
17.489.2.2 _initBetweenReplications() . . . 1100
17.489.2.3 _loadInstance() . . . 1100
17.489.2.4 _saveInstance() . . . 1100
17.489.2.5 show() . . . 1100
17.490 GroProgramRuntime::MotionMutation Struct Reference . . . 1100
17.491 Move Class Reference . . . 1101
17.491.1 Detailed Description . . . 1105
17.491.2 Member Function Documentation . . . 1106
17.491.2.1 _check() . . . 1106
17.491.2.2 _createAttachedAttributes() . . . 1106
17.491.2.3 _createEditableDataDefinitions() . . . 1106
17.491.2.4 _createInternalStatisticReporters() . . . 1106
17.491.2.5 _loadInstance() . . . 1106

17.491.2.6 _onDispatchEvent() . . . 1106
17.491.2.7 _saveInstance() . . . 1106
17.491.2.8 show() . . . 1107
17.492 MoveUndoCommand Class Reference . . . 1107
17.493 testing::internal::Mutex Class Reference . . . 1108
17.494 dialogBreakpoint::MVCResult Class Reference . . . 1108
17.495 GroProgramAst::NamedProgram Struct Reference . . . 1108
17.496 GroProgramIr::NamedProgramDefinition Struct Reference . . . 1108
17.497 testing::internal::NameGeneratorSelector< Provided > Struct Template Reference . . . 1108
17.498 NAND Class Reference . . . 1109
17.498.1 Member Function Documentation . . . 1114
17.498.1.1 _loadInstance() . . . 1114
17.498.1.2 _saveInstance() . . . 1114
17.498.1.3 Build() . . . 1114
17.499 testing::internal::NativeArray< Element > Class Template Reference . . . 1115
17.500 NaturalStateSet Class Reference . . . 1115
17.500.1 Member Function Documentation . . . 1116
17.500.1.1 name() . . . 1116
17.501 Neighborhood Class Reference . . . 1117
17.501.1 Detailed Description . . . 1118
17.501.2 Member Function Documentation . . . 1118
17.501.2.1 getNeighbors() . . . 1118
17.502 Neighborhood_Center Class Reference . . . 1118
17.502.1 Member Function Documentation . . . 1120
17.502.1.1 getNeighbors() . . . 1120
17.503 Neighborhood_Moore Class Reference . . . 1120
17.503.1 Member Function Documentation . . . 1121
17.503.1.1 getNeighbors() . . . 1121
17.503.1.2 show() . . . 1121
17.504 Neighborhood_VonNeumann Class Reference . . . 1122
17.504.1 Member Function Documentation . . . 1123
17.504.1.1 getNeighbors() . . . 1123
17.504.1.2 show() . . . 1123
17.505 NetworkActivationFrame Class Reference . . . 1123
17.505.1 Detailed Description . . . 1124
17.506 NetworkActivationResult Class Reference . . . 1124
17.506.1 Detailed Description . . . 1124
17.507 NetworkPortValue Struct Reference . . . 1124
17.507.1 Detailed Description . . . 1125
17.508 genesys::launcher::NetworkRequestOptions Struct Reference . . . 1125
17.509 genesys::launcher::NetworkResponse Struct Reference . . . 1125
17.510 ParserManager::NewParser Struct Reference . . . 1125

17.510.1 Detailed Description . . . 1125
17.511 NMOS Class Reference . . . 1126
17.511.1 Member Function Documentation . . . 1131
17.511.1.1 _loadInstance() . . . 1131
17.511.1.2 _saveInstance() . . . 1131
17.511.1.3 Build() . . . 1131
17.512 testing::internal::None Struct Reference . . . 1132
17.513 NOR Class Reference . . . 1132
17.513.1 Member Function Documentation . . . 1138
17.513.1.1 _loadInstance() . . . 1138
17.513.1.2 _saveInstance() . . . 1138
17.513.1.3 Build() . . . 1138
17.514 NOT Class Reference . . . 1139
17.514.1 Member Function Documentation . . . 1144
17.514.1.1 _loadInstance() . . . 1144
17.514.1.2 _saveInstance() . . . 1144
17.514.1.3 Build() . . . 1144
17.515 obj_t Class Reference . . . 1145
17.516 Optimizer_if::ObjectiveDefinition Struct Reference . . . 1145
17.516.1 Detailed Description . . . 1145
17.517 ObjectPropertyBrowser Class Reference . . . 1146
17.518 OctaveRunner Class Reference . . . 1149
17.518.1 Detailed Description . . . 1153
17.518.2 Member Function Documentation . . . 1154
17.518.2.1 _check() . . . 1154
17.518.2.2 _loadInstance() . . . 1154
17.518.2.3 _saveInstance() . . . 1154
17.518.2.4 show() . . . 1154
17.519 ODEfunction Class Reference . . . 1154
17.520 ODESolver Class Reference . . . 1155
17.520.1 Member Function Documentation . . . 1159
17.520.1.1 _check() . . . 1159
17.520.1.2 _createEditableDataDefinitions() . . . 1159
17.520.1.3 _initBetweenReplications() . . . 1159
17.520.1.4 _loadInstance() . . . 1159
17.520.1.5 _saveInstance() . . . 1159
17.520.1.6 show() . . . 1160
17.521 OdeSolver_if Class Reference . . . 1160
17.521.1 Detailed Description . . . 1160
17.522 OdeSolverFactory Class Reference . . . 1161
17.523 OdeSystem_if Class Reference . . . 1161
17.523.1 Detailed Description . . . 1161

17.524 OLD_ODEelement Class Reference . . . 1162
17.524.1 Member Function Documentation . . . 1165
17.524.1.1 _check() . . . 1165
17.524.1.2 _loadInstance() . . . 1166
17.524.1.3 _saveInstance() . . . 1166
17.524.1.4 show() . . . 1166
17.525 OnEventManager Class Reference . . . 1166
17.525.1 Detailed Description . . . 1167
17.526 OpenAIProviderClientImpl Class Reference . . . 1168
17.526.1 Detailed Description . . . 1170
17.526.1.1 API key . . . 1170
17.526.2 Constructor & Destructor Documentation . . . 1170
17.526.2.1 OpenAIProviderClientImpl() . . . 1170
17.526.3 Member Function Documentation . . . 1170
17.526.3.1 _authorizationHeader() . . . 1170
17.526.3.2 _buildRequestBody() . . . 1170
17.526.3.3 _extraRequestHeaders() . . . 1171
17.526.3.4 _parseResponseBody() . . . 1171
17.526.3.5 getProviderName() . . . 1171
17.526.3.6 isAvailable() . . . 1171
17.527 OperatingSystem02 Class Reference . . . 1172
17.527.1 Member Function Documentation . . . 1173
17.527.1.1 main() . . . 1173
17.528 OperatingSystem03 Class Reference . . . 1173
17.528.1 Member Function Documentation . . . 1175
17.528.1.1 main() . . . 1175
17.529 Optimizer_if::OptimizationSettings Struct Reference . . . 1175
17.529.1 Detailed Description . . . 1175
17.530 Optimizer_if Class Reference . . . 1175
17.530.1 Detailed Description . . . 1177
17.530.2 Member Function Documentation . . . 1177
17.530.2.1 addConstraint() . . . 1177
17.530.2.2 addObjective() . . . 1178
17.530.2.3 checkReady() . . . 1178
17.530.2.4 clearConstraints() . . . 1178
17.530.2.5 clearObjectives() . . . 1178
17.530.2.6 getAvailableControls() . . . 1178
17.530.2.7 getAvailableResponses() . . . 1178
17.530.2.8 getBestSolutions() . . . 1178
17.530.2.9 getConstraints() . . . 1178
17.530.2.10 getCurrentBestSolution() . . . 1179
17.530.2.11 getCurrentIteration() . . . 1179

17.530.2.12 getExecutionState() . . . 1179
17.530.2.13 getModel() . . . 1179
17.530.2.14 getModelFilename() . . . 1179
17.530.2.15 getObjectives() . . . 1179
17.530.2.16 getSelectedControls() . . . 1179
17.530.2.17 getSelectedResponses() . . . 1179
17.530.2.18 getSettings() . . . 1179
17.530.2.19 getTotalSimulations() . . . 1179
17.530.2.20 pause() . . . 1180
17.530.2.21 resetResults() . . . 1180
17.530.2.22 resume() . . . 1180
17.530.2.23 setModel() . . . 1180
17.530.2.24 setModelFilename() . . . 1180
17.530.2.25 setSelectedControls() . . . 1180
17.530.2.26 setSelectedResponses() . . . 1180
17.530.2.27 setSettings() . . . 1180
17.530.2.28 start() . . . 1181
17.530.2.29 step() . . . 1181
17.530.2.30 stop() . . . 1181
17.531 OptimizerDefaultImpl1 Class Reference . . . 1181
17.531.1 Detailed Description . . . 1183
17.531.2 Member Function Documentation . . . 1184
17.531.2.1 addConstraint() . . . 1184
17.531.2.2 addObjective() . . . 1184
17.531.2.3 checkReady() . . . 1184
17.531.2.4 clearConstraints() . . . 1184
17.531.2.5 clearObjectives() . . . 1184
17.531.2.6 getAvailableControls() . . . 1184
17.531.2.7 getAvailableResponses() . . . 1184
17.531.2.8 getBestSolutions() . . . 1185
17.531.2.9 getConstraints() . . . 1185
17.531.2.10 getCurrentBestSolution() . . . 1185
17.531.2.11 getCurrentIteration() . . . 1185
17.531.2.12 getExecutionState() . . . 1185
17.531.2.13 getModel() . . . 1185
17.531.2.14 getModelFilename() . . . 1185
17.531.2.15 getObjectives() . . . 1185
17.531.2.16 getSelectedControls() . . . 1185
17.531.2.17 getSelectedResponses() . . . 1186
17.531.2.18 getSettings() . . . 1186
17.531.2.19 getTotalSimulations() . . . 1186
17.531.2.20 pause() . . . 1186

17.531.2.21 resetResults() . . . 1186
17.531.2.22 resume() . . . 1186
17.531.2.23 setModel() . . . 1186
17.531.2.24 setModelFilename() . . . 1186
17.531.2.25 setSelectedControls() . . . 1187
17.531.2.26 setSelectedResponses() . . . 1187
17.531.2.27 setSettings() . . . 1187
17.531.2.28 start() . . . 1187
17.531.2.29 step() . . . 1187
17.531.2.30 stop() . . . 1187
17.532 OptimizerWindow Class Reference . . . 1188
17.533 OR Class Reference . . . 1189
17.533.1 Member Function Documentation . . . 1194
17.533.1.1 _loadInstance() . . . 1194
17.533.1.2 _saveInstance() . . . 1194
17.533.1.3 Build() . . . 1194
17.534 testing::internal::OsStackTraceGetter Class Reference . . . 1195
17.534.1 Member Function Documentation . . . 1195
17.534.1.1 CurrentStackTrace() . . . 1195
17.534.1.2 UponLeavingGTest() . . . 1196
17.535 testing::internal::OsStackTraceGetterInterface Class Reference . . . 1196
17.535.1 Member Data Documentation . . . 1196
17.535.1.1 kElidedFramesMarker . . . 1196
17.536 testing::internal::ParameterizedTestFactory< TestClass > Class Template Reference . . . 1197
17.536.1 Member Function Documentation . . . 1197
17.536.1.1 CreateTest() . . . 1197
17.537 testing::internal::ParameterizedTestSuiteInfo< TestSuite > Class Template Reference . . . 1198
17.537.1 Member Function Documentation . . . 1199
17.537.1.1 GetTestSuiteName() . . . 1199
17.537.1.2 GetTestSuiteTypeId() . . . 1199
17.537.1.3 RegisterTests() . . . 1199
17.538 testing::internal::ParameterizedTestSuiteInfoBase Class Reference . . . 1199
17.539 testing::internal::ParameterizedTestSuiteRegistry Class Reference . . . 1200
17.540 testing::internal::ParamGenerator< T > Class Template Reference . . . 1200
17.541 testing::internal::ParamGeneratorInterface< T > Class Template Reference . . . 1201
17.542 testing::internal::ParamIterator< T > Class Template Reference . . . 1201
17.543 testing::internal::ParamIteratorInterface< T > Class Template Reference . . . 1202
17.544 Parser_if Class Reference . . . 1202
17.544.1 Detailed Description . . . 1202
17.544.2 Member Function Documentation . . . 1203
17.544.2.1 getErrorMessage() . . . 1203
17.544.2.2 getParser() . . . 1203

17.544.2.3 getSampler() . . . 1203
17.544.2.4 parse() [1/2] . . . 1203
17.544.2.5 parse() [2/2] . . . 1203
17.544.2.6 setSampler() . . . 1204
17.545 ParserAttributeEventInjector Struct Reference . . . 1204
17.546 ParserAttributeInternalEventOwner Class Reference . . . 1205
17.547 ParserAttributeProcessObserver Struct Reference . . . 1210
17.548 ParserChangesInformation Class Reference . . . 1210
17.549 ParserDefaultImpl1 Class Reference . . . 1211
17.549.1 Member Function Documentation . . . 1212
17.549.1.1 getErrorMessage() . . . 1212
17.549.1.2 parse() [1/2] . . . 1212
17.549.1.3 parse() [2/2] . . . 1212
17.550 ParserDefaultImpl2 Class Reference . . . 1213
17.550.1 Constructor & Destructor Documentation . . . 1214
17.550.1.1 ParserDefaultImpl2() . . . 1214
17.550.1.2 ~ParserDefaultImpl2() . . . 1214
17.550.2 Member Function Documentation . . . 1214
17.550.2.1 getErrorMessage() . . . 1214
17.550.2.2 getParser() . . . 1214
17.550.2.3 getSampler() . . . 1214
17.550.2.4 parse() [1/2] . . . 1215
17.550.2.5 parse() [2/2] . . . 1215
17.550.2.6 setSampler() . . . 1215
17.551 ParserExpressionsTest Class Reference . . . 1216
17.551.1 Member Function Documentation . . . 1217
17.551.1.1 SetUp() . . . 1217
17.552 ParserManager Class Reference . . . 1217
17.552.1 Member Function Documentation . . . 1217
17.552.1.1 connectNewParser() . . . 1217
17.552.1.2 generateNewParser() . . . 1217
17.553 PasteUndoCommand Class Reference . . . 1218
17.554 PathwayStressResponseComponent Class Reference . . . 1219
17.554.1 Detailed Description . . . 1224
17.554.2 Member Function Documentation . . . 1224
17.554.2.1 _check() . . . 1224
17.554.2.2 _createEditableDataDefinitions() . . . 1224
17.554.2.3 _initBetweenReplications() . . . 1224
17.554.2.4 _loadInstance() . . . 1224
17.554.2.5 _onDispatchEvent() . . . 1225
17.554.2.6 _saveInstance() . . . 1225
17.554.2.7 show() . . . 1225

17.555 Persistence_if Class Reference . . . 1225
17.555.1 Detailed Description . . . 1226
17.555.2 Member Function Documentation . . . 1226
17.555.2.1 getFormatedField() . . . 1226
17.555.2.2 getOption() . . . 1227
17.555.2.3 hasChanged() . . . 1227
17.555.2.4 load() . . . 1227
17.555.2.5 save() . . . 1227
17.555.2.6 setHasChanged() . . . 1227
17.555.2.7 setOption() . . . 1227
17.556 PersistenceDefaultImpl2 Class Reference . . . 1228
17.556.1 Member Function Documentation . . . 1229
17.556.1.1 getFormatedField() . . . 1229
17.556.1.2 getOption() . . . 1229
17.556.1.3 hasChanged() . . . 1229
17.556.1.4 load() . . . 1229
17.556.1.5 save() . . . 1229
17.556.1.6 setHasChanged() . . . 1229
17.556.1.7 setOption() . . . 1229
17.557 PersistencePartialLoadImpl1 Class Reference . . . 1230
17.557.1 Member Function Documentation . . . 1231
17.557.1.1 load() . . . 1231
17.558 PersistenceRecord Class Reference . . . 1231
17.558.1 Detailed Description . . . 1232
17.558.2 Member Function Documentation . . . 1232
17.558.2.1 insert() . . . 1232
17.559 PersistentObject_base Class Reference . . . 1233
17.559.1 Detailed Description . . . 1234
17.559.2 Member Function Documentation . . . 1234
17.559.2.1 _loadInstance() . . . 1234
17.559.2.2 _saveInstance() . . . 1234
17.560 PetriPlace Class Reference . . . 1235
17.560.1 Member Function Documentation . . . 1240
17.560.1.1 _loadInstance() . . . 1240
17.560.1.2 _saveInstance() . . . 1240
17.561 PickableStationItem Class Reference . . . 1240
17.562 PickStation Class Reference . . . 1241
17.562.1 Detailed Description . . . 1246
17.562.2 Member Function Documentation . . . 1246
17.562.2.1 _addSimulationControl() . . . 1246
17.562.2.2 _check() . . . 1246
17.562.2.3 _createEditableDataDefinitions() . . . 1246

17.562.2.4 _getParserChangesInformation() . . . 1246
17.562.2.5 _initBetweenReplications() . . . 1246
17.562.2.6 _loadInstance() . . . 1247
17.562.2.7 _onDispatchEvent() . . . 1247
17.562.2.8 _saveInstance() . . . 1247
17.562.2.9 show() . . . 1247
17.563 PickUp Class Reference . . . 1247
17.563.1 Detailed Description . . . 1253
17.564 genesys::launcher::PlatformInfo Struct Reference . . . 1253
17.565 Plugin Class Reference . . . 1253
17.565.1 Detailed Description . . . 1254
17.565.2 Constructor & Destructor Documentation . . . 1254
17.565.2.1 Plugin() [1/2] . . . 1254
17.565.2.2 Plugin() [2/2] . . . 1254
17.565.3 Member Function Documentation . . . 1255
17.565.3.1 getPluginInfo() . . . 1255
17.565.3.2 isIsValidPlugin() . . . 1255
17.565.3.3 loadAndInsertNew() . . . 1255
17.565.3.4 loadNew() . . . 1255
17.565.3.5 newInstance() . . . 1255
17.565.3.6 show() . . . 1257
17.566 PluginCatalogController Class Reference . . . 1257
17.566.1 Detailed Description . . . 1258
17.567 PluginConnector_if Class Reference . . . 1258
17.567.1 Detailed Description . . . 1259
17.567.2 Member Function Documentation . . . 1259
17.567.2.1 check() . . . 1259
17.567.2.2 connect() . . . 1259
17.567.2.3 disconnect() [1/2] . . . 1259
17.567.2.4 disconnect() [2/2] . . . 1260
17.567.2.5 find() . . . 1260
17.568 PluginConnectorDummyImpl1 Class Reference . . . 1260
17.568.1 Detailed Description . . . 1261
17.568.2 Member Function Documentation . . . 1261
17.568.2.1 check() . . . 1261
17.568.2.2 connect() . . . 1261
17.568.2.3 disconnect() [1/2] . . . 1262
17.568.2.4 disconnect() [2/2] . . . 1262
17.568.2.5 find() . . . 1262
17.569 PluginConnectorStaticImpl1 Class Reference . . . 1263
17.569.1 Detailed Description . . . 1263
17.569.2 Member Function Documentation . . . 1264

17.569.2.1 check() . . . 1264
17.569.2.2 connect() . . . 1264
17.569.2.3 disconnect() [1/2] . . . 1264
17.569.2.4 disconnect() [2/2] . . . 1264
17.569.2.5 find() . . . 1265
17.570 PluginInformation Class Reference . . . 1265
17.571 PluginInsertionOptions Struct Reference . . . 1266
17.571.1 Detailed Description . . . 1266
17.572 PluginLoadIssue Class Reference . . . 1266
17.572.1 Detailed Description . . . 1267
17.573 PluginManager Class Reference . . . 1267
17.573.1 Detailed Description . . . 1269
17.573.2 Constructor & Destructor Documentation . . . 1269
17.573.2.1 PluginManager() . . . 1269
17.573.3 Member Function Documentation . . . 1269
17.573.3.1 getDataDefinitionPluginTypenames() . . . 1269
17.574 PMOS Class Reference . . . 1270
17.574.1 Member Function Documentation . . . 1275
17.574.1.1 _loadInstance() . . . 1275
17.574.1.2 _saveInstance() . . . 1275
17.574.1.3 Build() . . . 1275
17.575 testing::internal::PointerPrinter Struct Reference . . . 1276
17.576 GroProgramRuntime::PopulationMutation Struct Reference . . . 1276
17.577 yy::position Class Reference . . . 1276
17.577.1 Detailed Description . . . 1277
17.578 testing::PrintToStringParamName Struct Reference . . . 1277
17.579 ProbabilityDistribution Class Reference . . . 1277
17.579.1 Detailed Description . . . 1278
17.579.2 Member Function Documentation . . . 1278
17.579.2.1 inverseChi2() . . . 1278
17.579.2.2 inverseFFisherSnedecor() . . . 1278
17.579.2.3 inverseNormal() . . . 1279
17.579.2.4 inverseTStudent() . . . 1279
17.580 ProbabilityDistributionBase Class Reference . . . 1279
17.580.1 Detailed Description . . . 1280
17.580.2 Member Function Documentation . . . 1280
17.580.2.1 _betaFunction() . . . 1280
17.580.2.2 _factorial() . . . 1280
17.580.2.3 _gammaFunction() . . . 1280
17.580.2.4 beta() . . . 1280
17.580.2.5 chi2() . . . 1280
17.580.2.6 erlang() . . . 1280

17.580.2.7 exponential() . . . 1281
17.580.2.8 fisherSnedecor() . . . 1281
17.580.2.9 gamma() . . . 1281
17.580.2.10 logNormal() . . . 1281
17.580.2.11 normal() . . . 1281
17.580.2.12 poisson() . . . 1281
17.580.2.13 triangular() . . . 1281
17.580.2.14 tStudent() . . . 1281
17.580.2.15 uniform() . . . 1282
17.580.2.16 weibull() . . . 1282
17.581 MetabolicFluxBalanceSolver::Problem Struct Reference . . . 1282
17.582 Process Class Reference . . . 1282
17.582.1 Detailed Description . . . 1288
17.582.2 Member Function Documentation . . . 1288
17.582.2.1 _check() . . . 1288
17.582.2.2 _createEditableDataDefinitions() . . . 1288
17.582.2.3 _createNonEditableDataDefinitions() . . . 1288
17.582.2.4 _loadInstance() . . . 1288
17.582.2.5 _onDispatchEvent() . . . 1288
17.582.2.6 _saveInstance() . . . 1289
17.582.2.7 show() . . . 1289
17.583 genesys::launcher::ProcessLauncher Class Reference . . . 1289
17.583.1 Member Function Documentation . . . 1290
17.583.1.1 launchReplacingProcess() . . . 1290
17.584 OptimizerWindow::ProgressPlot Class Reference . . . 1290
17.585 BioStateProjectionComponent::Projection Struct Reference . . . 1291
17.586 MetabolicStateProjectionComponent::Projection Struct Reference . . . 1291
17.587 EFSMStatePropertiesUndoCommand::Properties Struct Reference . . . 1292
17.588 EFSMTransitionPropertiesUndoCommand::Properties Struct Reference . . . 1292
17.589 PropertyEditor Class Reference . . . 1292
17.589.1 Member Function Documentation . . . 1293
17.589.1.1 setModelBlock() . . . 1293
17.590 PropertyEditorController Class Reference . . . 1293
17.590.1 Detailed Description . . . 1294
17.591 PropertyEditorGenesys Class Reference . . . 1294
17.592 PropertyGenesys Class Reference . . . 1294
17.593 PropertyGenesysBase Class Reference . . . 1296
17.593.1 Detailed Description . . . 1298
17.594 testing::internal::ProtobufPrinter Struct Reference . . . 1298
17.595 ProviderRequest Struct Reference . . . 1298
17.595.1 Detailed Description . . . 1298
17.596 ProviderResponse Struct Reference . . . 1298

17.596.1 Detailed Description . . . 1299
17.597 testing::internal::ProxyTypeList< Ts > Struct Template Reference . . . 1299
17.598 PythonForG Class Reference . . . 1300
17.598.1 Member Function Documentation . . . 1304
17.598.1.1 _check() . . . 1304
17.598.1.2 _createNonEditableDataDefinitions() . . . 1305
17.598.1.3 _initBetweenReplications() . . . 1305
17.598.1.4 _loadInstance() . . . 1305
17.598.1.5 _onDispatchEvent() . . . 1305
17.598.1.6 _saveInstance() . . . 1305
17.598.1.7 show() . . . 1305
17.599 PythonRuntime Class Reference . . . 1306
17.599.1 Member Function Documentation . . . 1310
17.599.1.1 _check() . . . 1310
17.599.1.2 _loadInstance() . . . 1310
17.599.1.3 _saveInstance() . . . 1310
17.599.1.4 show() . . . 1310
17.600 QCPAbstractItem Class Reference . . . 1311
17.600.1 Detailed Description . . . 1313
17.600.2 Clipping . . . 1314
17.600.3 Using items . . . 1314
17.600.4 Creating own items . . . 1314
17.600.4.1 Allowing the item to be positioned . . . 1314
17.600.4.2 The draw function . . . 1314
17.600.4.3 The selectTest function . . . 1315
17.600.4.4 Providing anchors . . . 1315
17.600.5 Constructor & Destructor Documentation . . . 1315
17.600.5.1 QCPAbstractItem() . . . 1315
17.600.6 Member Function Documentation . . . 1315
17.600.6.1 anchor() . . . 1315
17.600.6.2 anchors() . . . 1315
17.600.6.3 applyDefaultAntialiasingHint() . . . 1315
17.600.6.4 clipRect() . . . 1316
17.600.6.5 deselectEvent() . . . 1316
17.600.6.6 draw() . . . 1316
17.600.6.7 hasAnchor() . . . 1316
17.600.6.8 position() . . . 1316
17.600.6.9 positions() . . . 1316
17.600.6.10 selectEvent() . . . 1316
17.600.6.11 selectionCategory() . . . 1317
17.600.6.12 selectionChanged . . . 1317
17.600.6.13 selectTest() . . . 1317

17.600.6.14 setClipAxisRect() . . . 1317
17.600.6.15 setClipToAxisRect() . . . 1318
17.600.6.16 setSelectable() . . . 1318
17.600.6.17 setSelected() . . . 1318
17.601 QCPAbstractLegendItem Class Reference . . . 1318
17.601.1 Detailed Description . . . 1322
17.601.2 Constructor & Destructor Documentation . . . 1322
17.601.2.1 QCPAbstractLegendItem() . . . 1322
17.601.3 Member Function Documentation . . . 1323
17.601.3.1 applyDefaultAntialiasingHint() . . . 1323
17.601.3.2 clipRect() . . . 1323
17.601.3.3 deselectEvent() . . . 1323
17.601.3.4 draw() . . . 1323
17.601.3.5 selectEvent() . . . 1323
17.601.3.6 selectionCategory() . . . 1323
17.601.3.7 selectionChanged . . . 1323
17.601.3.8 selectTest() . . . 1323
17.601.3.9 setFont() . . . 1324
17.601.3.10 setSelectable() . . . 1324
17.601.3.11 setSelected() . . . 1324
17.601.3.12 setSelectedFont() . . . 1324
17.601.3.13 setSelectedTextColor() . . . 1324
17.601.3.14 setTextColor() . . . 1324
17.602 QCPAbstractPaintBuffer Class Reference . . . 1325
17.602.1 Detailed Description . . . 1325
17.602.2 Constructor & Destructor Documentation . . . 1326
17.602.2.1 QCPAbstractPaintBuffer() . . . 1326
17.602.3 Member Function Documentation . . . 1326
17.602.3.1 clear() . . . 1326
17.602.3.2 donePainting() . . . 1326
17.602.3.3 draw() . . . 1326
17.602.3.4 reallocateBuffer() . . . 1326
17.602.3.5 setDevicePixelRatio() . . . 1326
17.602.3.6 setInvalidated() . . . 1327
17.602.3.7 setSize() . . . 1327
17.602.3.8 startPainting() . . . 1327
17.603 QCPAbstractPlottable Class Reference . . . 1327
17.603.1 Detailed Description . . . 1331
17.603.2 Creating own plottables . . . 1331
17.603.3 Constructor & Destructor Documentation . . . 1332
17.603.3.1 QCPAbstractPlottable() . . . 1332
17.603.4 Member Function Documentation . . . 1332

17.603.4.1 addToLegend() [1/2] . . . 1332
17.603.4.2 addToLegend() [2/2] . . . 1332
17.603.4.3 applyDefaultAntialiasingHint() . . . 1333
17.603.4.4 clipRect() . . . 1333
17.603.4.5 coordsToPixels() [1/2] . . . 1333
17.603.4.6 coordsToPixels() [2/2] . . . 1333
17.603.4.7 deselectEvent() . . . 1333
17.603.4.8 draw() . . . 1333
17.603.4.9 getKeyRange() . . . 1333
17.603.4.10 getValueRange() . . . 1334
17.603.4.11 interface1D() . . . 1334
17.603.4.12 pixelsToCoords() [1/2] . . . 1334
17.603.4.13 pixelsToCoords() [2/2] . . . 1334
17.603.4.14 removeFromLegend() [1/2] . . . 1335
17.603.4.15 removeFromLegend() [2/2] . . . 1335
17.603.4.16 rescaleAxes() . . . 1335
17.603.4.17 rescaleKeyAxis() . . . 1335
17.603.4.18 rescaleValueAxis() . . . 1336
17.603.4.19 selectableChanged . . . 1336
17.603.4.20 selected() . . . 1336
17.603.4.21 selectEvent() . . . 1336
17.603.4.22 selection() . . . 1336
17.603.4.23 selectionCategory() . . . 1336
17.603.4.24 selectionChanged [1/2] . . . 1336
17.603.4.25 selectionChanged [2/2] . . . 1337
17.603.4.26 selectionDecorator() . . . 1337
17.603.4.27 selectTest() . . . 1337
17.603.4.28 setAntialiasedFill() . . . 1337
17.603.4.29 setAntialiasedScatters() . . . 1338
17.603.4.30 setBrush() . . . 1338
17.603.4.31 setKeyAxis() . . . 1338
17.603.4.32 setName() . . . 1338
17.603.4.33 setPen() . . . 1338
17.603.4.34 setSelectable() . . . 1339
17.603.4.35 setSelection() . . . 1339
17.603.4.36 setSelectionDecorator() . . . 1339
17.603.4.37 setValueAxis() . . . 1339
17.604 QCPAbstractPlottable1D< DataType > Class Template Reference . . . 1340
17.604.1 Detailed Description . . . 1343
17.604.2 Constructor & Destructor Documentation . . . 1344
17.604.2.1 QCPAbstractPlottable1D() . . . 1344
17.604.3 Member Function Documentation . . . 1344

17.604.3.1 dataCount() . . . 1344
17.604.3.2 dataMainKey() . . . 1344
17.604.3.3 dataMainValue() . . . 1344
17.604.3.4 dataPixelPosition() . . . 1344
17.604.3.5 dataSortKey() . . . 1344
17.604.3.6 dataValueRange() . . . 1345
17.604.3.7 drawPolyline() . . . 1345
17.604.3.8 findBegin() . . . 1345
17.604.3.9 findEnd() . . . 1345
17.604.3.10 getDataSegments() . . . 1346
17.604.3.11 interface1D() . . . 1346
17.604.3.12 selectTest() . . . 1346
17.604.3.13 selectTestRect() . . . 1346
17.604.3.14 sortKeyIsMainKey() . . . 1346
17.605 QCPAxis Class Reference . . . 1347
17.605.1 Detailed Description . . . 1352
17.605.2 Member Enumeration Documentation . . . 1352
17.605.2.1 AxisType . . . 1352
17.605.2.2 LabelSide . . . 1352
17.605.2.3 ScaleType . . . 1352
17.605.2.4 SelectablePart . . . 1353
17.605.3 Constructor & Destructor Documentation . . . 1353
17.605.3.1 QCPAxis() . . . 1353
17.605.4 Member Function Documentation . . . 1353
17.605.4.1 applyDefaultAntialiasingHint() . . . 1353
17.605.4.2 coordToPixel() . . . 1353
17.605.4.3 deselectEvent() . . . 1353
17.605.4.4 draw() . . . 1354
17.605.4.5 getPartAt() . . . 1354
17.605.4.6 graphs() . . . 1354
17.605.4.7 grid() . . . 1354
17.605.4.8 items() . . . 1354
17.605.4.9 marginSideToAxisType() . . . 1354
17.605.4.10 mouseMoveEvent() . . . 1354
17.605.4.11 mousePressEvent() . . . 1355
17.605.4.12 mouseReleaseEvent() . . . 1355
17.605.4.13 moveRange() . . . 1355
17.605.4.14 opposite() . . . 1355
17.605.4.15 orientation() [1/2] . . . 1356
17.605.4.16 orientation() [2/2] . . . 1356
17.605.4.17 pixelOrientation() . . . 1356
17.605.4.18 pixelToCoord() . . . 1356

17.605.4.19 plottables() . . . 1356
17.605.4.20 rangeChanged [1/2] . . . 1356
17.605.4.21 rangeChanged [2/2] . . . 1357
17.605.4.22 rescale() . . . 1357
17.605.4.23 scaleRange() [1/2] . . . 1357
17.605.4.24 scaleRange() [2/2] . . . 1357
17.605.4.25 scaleTypeChanged . . . 1357
17.605.4.26 selectableChanged . . . 1357
17.605.4.27 selectEvent() . . . 1358
17.605.4.28 selectionCategory() . . . 1358
17.605.4.29 selectionChanged . . . 1358
17.605.4.30 selectTest() . . . 1358
17.605.4.31 setBasePen() . . . 1358
17.605.4.32 setLabel() . . . 1359
17.605.4.33 setLabelColor() . . . 1359
17.605.4.34 setLabelFont() . . . 1359
17.605.4.35 setLabelPadding() . . . 1359
17.605.4.36 setLowerEnding() . . . 1359
17.605.4.37 setNumberFormat() . . . 1359
17.605.4.38 setNumberPrecision() . . . 1360
17.605.4.39 setOffset() . . . 1360
17.605.4.40 setPadding() . . . 1360
17.605.4.41 setRange() [1/3] . . . 1361
17.605.4.42 setRange() [2/3] . . . 1361
17.605.4.43 setRange() [3/3] . . . 1361
17.605.4.44 setRangeLower() . . . 1361
17.605.4.45 setRangeReversed() . . . 1361
17.605.4.46 setRangeUpper() . . . 1361
17.605.4.47 setScaleRatio() . . . 1362
17.605.4.48 setScaleType() . . . 1362
17.605.4.49 setSelectableParts() . . . 1362
17.605.4.50 setSelectedBasePen() . . . 1362
17.605.4.51 setSelectedLabelColor() . . . 1362
17.605.4.52 setSelectedLabelFont() . . . 1363
17.605.4.53 setSelectedParts() . . . 1363
17.605.4.54 setSelectedSubTickPen() . . . 1363
17.605.4.55 setSelectedTickLabelColor() . . . 1363
17.605.4.56 setSelectedTickLabelFont() . . . 1363
17.605.4.57 setSelectedTickPen() . . . 1364
17.605.4.58 setSubTickLength() . . . 1364
17.605.4.59 setSubTickLengthIn() . . . 1364
17.605.4.60 setSubTickLengthOut() . . . 1364

17.605.4.61 setSubTickPen() . . . 1364
17.605.4.62 setSubTicks() . . . 1364
17.605.4.63 setTicker() . . . 1365
17.605.4.64 setTickLabelColor() . . . 1365
17.605.4.65 setTickLabelFont() . . . 1365
17.605.4.66 setTickLabelPadding() . . . 1365
17.605.4.67 setTickLabelRotation() . . . 1365
17.605.4.68 setTickLabels() . . . 1366
17.605.4.69 setTickLabelSide() . . . 1366
17.605.4.70 setTickLength() . . . 1366
17.605.4.71 setTickLengthIn() . . . 1366
17.605.4.72 setTickLengthOut() . . . 1366
17.605.4.73 setTickPen() . . . 1366
17.605.4.74 setTicks() . . . 1367
17.605.4.75 setUpperEnding() . . . 1367
17.605.4.76 ticker() . . . 1367
17.605.4.77 wheelEvent() . . . 1367
17.606 QCPAxisPainterPrivate Class Reference . . . 1368
17.606.1 Constructor & Destructor Documentation . . . 1369
17.606.1.1 QCPAxisPainterPrivate() . . . 1369
17.607 QCPAxisRect Class Reference . . . 1370
17.607.1 Detailed Description . . . 1374
17.607.2 Constructor & Destructor Documentation . . . 1374
17.607.2.1 QCPAxisRect() . . . 1374
17.607.3 Member Function Documentation . . . 1375
17.607.3.1 addAxes() . . . 1375
17.607.3.2 addAxis() . . . 1375
17.607.3.3 applyDefaultAntialiasingHint() . . . 1375
17.607.3.4 axes() [1/2] . . . 1375
17.607.3.5 axes() [2/2] . . . 1375
17.607.3.6 axis() . . . 1376
17.607.3.7 axisCount() . . . 1376
17.607.3.8 bottom() . . . 1376
17.607.3.9 bottomLeft() . . . 1376
17.607.3.10 bottomRight() . . . 1376
17.607.3.11 calculateAutoMargin() . . . 1376
17.607.3.12 center() . . . 1376
17.607.3.13 draw() . . . 1376
17.607.3.14 elements() . . . 1377
17.607.3.15 graphs() . . . 1377
17.607.3.16 height() . . . 1377
17.607.3.17 insetLayout() . . . 1377

17.607.3.18 items() . . . 1377
17.607.3.19 layoutChanged() . . . 1377
17.607.3.20 left() . . . 1377
17.607.3.21 mouseMoveEvent() . . . 1378
17.607.3.22 mousePressEvent() . . . 1378
17.607.3.23 mouseReleaseEvent() . . . 1378
17.607.3.24 plottables() . . . 1379
17.607.3.25 rangeDragAxes() . . . 1379
17.607.3.26 rangeDragAxis() . . . 1379
17.607.3.27 rangeZoomAxes() . . . 1379
17.607.3.28 rangeZoomAxis() . . . 1379
17.607.3.29 rangeZoomFactor() . . . 1379
17.607.3.30 removeAxis() . . . 1380
17.607.3.31 right() . . . 1380
17.607.3.32 setBackground() [1/3] . . . 1380
17.607.3.33 setBackground() [2/3] . . . 1380
17.607.3.34 setBackground() [3/3] . . . 1380
17.607.3.35 setBackgroundScaled() . . . 1381
17.607.3.36 setBackgroundScaledMode() . . . 1381
17.607.3.37 setRangeDrag() . . . 1381
17.607.3.38 setRangeDragAxes() [1/3] . . . 1381
17.607.3.39 setRangeDragAxes() [2/3] . . . 1382
17.607.3.40 setRangeDragAxes() [3/3] . . . 1382
17.607.3.41 setRangeZoom() . . . 1382
17.607.3.42 setRangeZoomAxes() [1/3] . . . 1382
17.607.3.43 setRangeZoomAxes() [2/3] . . . 1382
17.607.3.44 setRangeZoomAxes() [3/3] . . . 1383
17.607.3.45 setRangeZoomFactor() [1/2] . . . 1383
17.607.3.46 setRangeZoomFactor() [2/2] . . . 1383
17.607.3.47 setupFullAxesBox() . . . 1383
17.607.3.48 size() . . . 1384
17.607.3.49 top() . . . 1384
17.607.3.50 topLeft() . . . 1384
17.607.3.51 topRight() . . . 1384
17.607.3.52 update() . . . 1384
17.607.3.53 wheelEvent() . . . 1384
17.607.3.54 width() . . . 1384
17.607.3.55 zoom() [1/2] . . . 1385
17.607.3.56 zoom() [2/2] . . . 1385
17.608 QCPAxisTicker Class Reference . . . 1385
17.608.1 Detailed Description . . . 1386
17.608.2 Creating own axis tickers . . . 1387

17.608.3 Member Enumeration Documentation . . . 1387
17.608.3.1 TickStepStrategy . . . 1387
17.608.4 Constructor & Destructor Documentation . . . 1387
17.608.4.1 QCPAxisTicker() . . . 1387
17.608.5 Member Function Documentation . . . 1387
17.608.5.1 createTickVector() . . . 1387
17.608.5.2 generate() . . . 1387
17.608.5.3 getSubTickCount() . . . 1388
17.608.5.4 getTickLabel() . . . 1388
17.608.5.5 getTickStep() . . . 1388
17.608.5.6 setTickCount() . . . 1388
17.608.5.7 setTickOrigin() . . . 1388
17.608.5.8 setTickStepStrategy() . . . 1388
17.609 QCPAxisTickerDateTime Class Reference . . . 1389
17.609.1 Detailed Description . . . 1391
17.609.2 Constructor & Destructor Documentation . . . 1391
17.609.2.1 QCPAxisTickerDateTime() . . . 1391
17.609.3 Member Function Documentation . . . 1391
17.609.3.1 createTickVector() . . . 1391
17.609.3.2 dateTimeToKey() [1/2] . . . 1391
17.609.3.3 dateTimeToKey() [2/2] . . . 1392
17.609.3.4 getSubTickCount() . . . 1392
17.609.3.5 getTickLabel() . . . 1392
17.609.3.6 getTickStep() . . . 1392
17.609.3.7 keyToDateTime() . . . 1392
17.609.3.8 setDateTimeFormat() . . . 1392
17.609.3.9 setDateTimeSpec() . . . 1393
17.609.3.10 setTickOrigin() [1/2] . . . 1394
17.609.3.11 setTickOrigin() [2/2] . . . 1394
17.609.3.12 setTimeZone() . . . 1394
17.610 QCPAxisTickerFixed Class Reference . . . 1394
17.610.1 Detailed Description . . . 1396
17.610.2 Member Enumeration Documentation . . . 1396
17.610.2.1 ScaleStrategy . . . 1396
17.610.3 Constructor & Destructor Documentation . . . 1396
17.610.3.1 QCPAxisTickerFixed() . . . 1396
17.610.4 Member Function Documentation . . . 1397
17.610.4.1 getTickStep() . . . 1397
17.610.4.2 setScaleStrategy() . . . 1397
17.610.4.3 setTickStep() . . . 1397
17.611 QCPAxisTickerLog Class Reference . . . 1397
17.611.1 Detailed Description . . . 1399

17.611.2 Constructor & Destructor Documentation . . . 1399
17.611.2.1 QCPAxisTickerLog() . . . 1399
17.611.3 Member Function Documentation . . . 1399
17.611.3.1 createTickVector() . . . 1399
17.611.3.2 getSubTickCount() . . . 1399
17.611.3.3 setLogBase() . . . 1399
17.611.3.4 setSubTickCount() . . . 1400
17.612 QCPAxisTickerPi Class Reference . . . 1400
17.612.1 Detailed Description . . . 1402
17.612.2 Member Enumeration Documentation . . . 1402
17.612.2.1 FractionStyle . . . 1402
17.612.3 Constructor & Destructor Documentation . . . 1402
17.612.3.1 QCPAxisTickerPi() . . . 1402
17.612.4 Member Function Documentation . . . 1402
17.612.4.1 getSubTickCount() . . . 1402
17.612.4.2 getTickLabel() . . . 1402
17.612.4.3 getTickStep() . . . 1403
17.612.4.4 setFractionStyle() . . . 1403
17.612.4.5 setPeriodicity() . . . 1403
17.612.4.6 setPiSymbol() . . . 1403
17.612.4.7 setPiValue() . . . 1403
17.613 QCPAxisTickerText Class Reference . . . 1403
17.613.1 Detailed Description . . . 1405
17.613.2 Constructor & Destructor Documentation . . . 1405
17.613.2.1 QCPAxisTickerText() . . . 1405
17.613.3 Member Function Documentation . . . 1405
17.613.3.1 addTick() . . . 1405
17.613.3.2 addTicks() [1/2] . . . 1405
17.613.3.3 addTicks() [2/2] . . . 1406
17.613.3.4 clear() . . . 1406
17.613.3.5 createTickVector() . . . 1406
17.613.3.6 getSubTickCount() . . . 1406
17.613.3.7 getTickLabel() . . . 1406
17.613.3.8 getTickStep() . . . 1407
17.613.3.9 setSubTickCount() . . . 1407
17.613.3.10 setTicks() [1/2] . . . 1407
17.613.3.11 setTicks() [2/2] . . . 1407
17.613.3.12 ticks() . . . 1407
17.614 QCPAxisTickerTime Class Reference . . . 1407
17.614.1 Detailed Description . . . 1409
17.614.2 Member Enumeration Documentation . . . 1410
17.614.2.1 TimeUnit . . . 1410

17.614.3 Constructor & Destructor Documentation . . . 1410
17.614.3.1 QCPAxisTickerTime() . . . 1410
17.614.4 Member Function Documentation . . . 1410
17.614.4.1 getSubTickCount() . . . 1410
17.614.4.2 getTickLabel() . . . 1410
17.614.4.3 getTickStep() . . . 1410
17.614.4.4 setFieldWidth() . . . 1410
17.614.4.5 setTimeFormat() . . . 1411
17.615 QCPBars Class Reference . . . 1411
17.615.1 Detailed Description . . . 1416
17.615.2 Changing the appearance . . . 1416
17.615.3 Usage . . . 1416
17.615.4 Member Enumeration Documentation . . . 1416
17.615.4.1 WidthType . . . 1416
17.615.5 Constructor & Destructor Documentation . . . 1417
17.615.5.1 QCPBars() . . . 1417
17.615.6 Member Function Documentation . . . 1417
17.615.6.1 addData() [1/2] . . . 1417
17.615.6.2 addData() [2/2] . . . 1417
17.615.6.3 barAbove() . . . 1417
17.615.6.4 barBelow() . . . 1418
17.615.6.5 data() . . . 1418
17.615.6.6 dataPixelPosition() . . . 1418
17.615.6.7 draw() . . . 1418
17.615.6.8 drawLegendIcon() . . . 1418
17.615.6.9 getKeyRange() . . . 1418
17.615.6.10 getValueRange() . . . 1419
17.615.6.11 moveAbove() . . . 1419
17.615.6.12 moveBelow() . . . 1419
17.615.6.13 selectTest() . . . 1419
17.615.6.14 selectTestRect() . . . 1420
17.615.6.15 setBarsGroup() . . . 1420
17.615.6.16 setBaseValue() . . . 1420
17.615.6.17 setData() [1/2] . . . 1420
17.615.6.18 setData() [2/2] . . . 1421
17.615.6.19 setStackingGap() . . . 1421
17.615.6.20 setWidth() . . . 1421
17.615.6.21 setWidthType() . . . 1421
17.616 QCPBarsData Class Reference . . . 1421
17.616.1 Detailed Description . . . 1422
17.616.2 Constructor & Destructor Documentation . . . 1422
17.616.2.1 QCPBarsData() [1/2] . . . 1422

17.616.2.2 QCPBarsData() [2/2] . . . 1422
17.616.3 Member Function Documentation . . . 1422
17.616.3.1 fromSortKey() . . . 1422
17.616.3.2 mainKey() . . . 1422
17.616.3.3 mainValue() . . . 1422
17.616.3.4 sortKey() . . . 1423
17.616.3.5 sortKeyIsMainKey() . . . 1423
17.616.3.6 valueRange() . . . 1423
17.617 QCPBarsGroup Class Reference . . . 1423
17.617.1 Detailed Description . . . 1425
17.617.2 Usage . . . 1425
17.617.3 Example . . . 1425
17.617.4 Member Enumeration Documentation . . . 1425
17.617.4.1 SpacingType . . . 1425
17.617.5 Constructor & Destructor Documentation . . . 1425
17.617.5.1 QCPBarsGroup() . . . 1425
17.617.6 Member Function Documentation . . . 1426
17.617.6.1 append() . . . 1426
17.617.6.2 bars() [1/2] . . . 1426
17.617.6.3 bars() [2/2] . . . 1426
17.617.6.4 clear() . . . 1426
17.617.6.5 contains() . . . 1426
17.617.6.6 insert() . . . 1426
17.617.6.7 isEmpty() . . . 1427
17.617.6.8 remove() . . . 1427
17.617.6.9 setSpacing() . . . 1427
17.617.6.10 setSpacingType() . . . 1427
17.617.6.11 size() . . . 1427
17.618 QCPColorGradient Class Reference . . . 1427
17.618.1 Detailed Description . . . 1429
17.618.2 Member Enumeration Documentation . . . 1429
17.618.2.1 ColorInterpolation . . . 1429
17.618.2.2 GradientPreset . . . 1429
17.618.2.3 NanHandling . . . 1430
17.618.3 Constructor & Destructor Documentation . . . 1430
17.618.3.1 QCPColorGradient() [1/2] . . . 1430
17.618.3.2 QCPColorGradient() [2/2] . . . 1430
17.618.4 Member Function Documentation . . . 1430
17.618.4.1 clearColorStops() . . . 1430
17.618.4.2 colorize() [1/2] . . . 1431
17.618.4.3 colorize() [2/2] . . . 1431
17.618.4.4 inverted() . . . 1431

17.618.4.5 loadPreset() . . . 1431
17.618.4.6 setColorInterpolation() . . . 1432
17.618.4.7 setColorStopAt() . . . 1432
17.618.4.8 setColorStops() . . . 1432
17.618.4.9 setLevelCount() . . . 1432
17.618.4.10 setNanColor() . . . 1432
17.618.4.11 setNanHandling() . . . 1432
17.618.4.12 setPeriodic() . . . 1433
17.619 QCPColorMap Class Reference . . . 1433
17.619.1 Detailed Description . . . 1437
17.619.2 Changing the appearance . . . 1437
17.619.3 Transparency . . . 1437
17.619.4 Usage . . . 1437
17.619.5 Constructor & Destructor Documentation . . . 1438
17.619.5.1 QCPColorMap() . . . 1438
17.619.6 Member Function Documentation . . . 1438
17.619.6.1 data() . . . 1438
17.619.6.2 dataRangeChanged . . . 1438
17.619.6.3 dataScaleTypeChanged . . . 1438
17.619.6.4 draw() . . . 1438
17.619.6.5 drawLegendIcon() . . . 1438
17.619.6.6 getKeyRange() . . . 1439
17.619.6.7 getValueRange() . . . 1439
17.619.6.8 gradientChanged . . . 1439
17.619.6.9 rescaleDataRange() . . . 1439
17.619.6.10 selectTest() . . . 1440
17.619.6.11 setColorScale() . . . 1440
17.619.6.12 setData() . . . 1441
17.619.6.13 setDataRange() . . . 1441
17.619.6.14 setDataScaleType() . . . 1441
17.619.6.15 setGradient() . . . 1441
17.619.6.16 setInterpolate() . . . 1441
17.619.6.17 setTightBoundary() . . . 1441
17.619.6.18 updateLegendIcon() . . . 1442
17.620 QCPColorMapData Class Reference . . . 1442
17.620.1 Detailed Description . . . 1443
17.620.2 Constructor & Destructor Documentation . . . 1444
17.620.2.1 QCPColorMapData() [1/2] . . . 1444
17.620.2.2 QCPColorMapData() [2/2] . . . 1444
17.620.3 Member Function Documentation . . . 1444
17.620.3.1 alpha() . . . 1444
17.620.3.2 cellToCoord() . . . 1444

17.620.3.3 clear() . . . 1445
17.620.3.4 clearAlpha() . . . 1445
17.620.3.5 coordToCell() . . . 1445
17.620.3.6 fill() . . . 1445
17.620.3.7 fillAlpha() . . . 1445
17.620.3.8 isEmpty() . . . 1445
17.620.3.9 operator=() . . . 1445
17.620.3.10 recalculateDataBounds() . . . 1446
17.620.3.11 setAlpha() . . . 1446
17.620.3.12 setCell() . . . 1446
17.620.3.13 setData() . . . 1446
17.620.3.14 setKeyRange() . . . 1447
17.620.3.15 setKeySize() . . . 1447
17.620.3.16 setRange() . . . 1447
17.620.3.17 setSize() . . . 1447
17.620.3.18 setValueRange() . . . 1448
17.620.3.19 setValueSize() . . . 1448
17.621 QCPColorScale Class Reference . . . 1448
17.621.1 Detailed Description . . . 1452
17.621.2 Constructor & Destructor Documentation . . . 1452
17.621.2.1 QCPColorScale() . . . 1452
17.621.3 Member Function Documentation . . . 1452
17.621.3.1 applyDefaultAntialiasingHint() . . . 1452
17.621.3.2 axis() . . . 1452
17.621.3.3 colorMaps() . . . 1453
17.621.3.4 dataRangeChanged . . . 1453
17.621.3.5 dataScaleTypeChanged . . . 1453
17.621.3.6 gradientChanged . . . 1453
17.621.3.7 mouseMoveEvent() . . . 1453
17.621.3.8 mousePressEvent() . . . 1453
17.621.3.9 mouseReleaseEvent() . . . 1454
17.621.3.10 rescaleDataRange() . . . 1454
17.621.3.11 setBarWidth() . . . 1454
17.621.3.12 setDataRange() . . . 1454
17.621.3.13 setDataScaleType() . . . 1455
17.621.3.14 setGradient() . . . 1455
17.621.3.15 setLabel() . . . 1455
17.621.3.16 setRangeDrag() . . . 1455
17.621.3.17 setRangeZoom() . . . 1455
17.621.3.18 setType() . . . 1455
17.621.3.19 update() . . . 1456
17.621.3.20 wheelEvent() . . . 1456

17.622 QCPColorScaleAxisRectPrivate Class Reference . . . 1457
17.622.1 Constructor & Destructor Documentation . . . 1461
17.622.1.1 QCPColorScaleAxisRectPrivate() . . . 1461
17.622.2 Member Function Documentation . . . 1461
17.622.2.1 calculateAutoMargin() . . . 1461
17.622.2.2 draw() . . . 1461
17.622.2.3 mouseMoveEvent() . . . 1462
17.622.2.4 mousePressEvent() . . . 1462
17.622.2.5 mouseReleaseEvent() . . . 1462
17.622.2.6 update() . . . 1463
17.622.2.7 wheelEvent() . . . 1463
17.623 QCPCurve Class Reference . . . 1463
17.623.1 Detailed Description . . . 1468
17.623.2 Changing the appearance . . . 1468
17.623.3 Usage . . . 1468
17.623.4 Member Enumeration Documentation . . . 1469
17.623.4.1 LineStyle . . . 1469
17.623.5 Constructor & Destructor Documentation . . . 1469
17.623.5.1 QCPCurve() . . . 1469
17.623.6 Member Function Documentation . . . 1469
17.623.6.1 addData() [1/4] . . . 1469
17.623.6.2 addData() [2/4] . . . 1469
17.623.6.3 addData() [3/4] . . . 1470
17.623.6.4 addData() [4/4] . . . 1470
17.623.6.5 data() . . . 1470
17.623.6.6 draw() . . . 1470
17.623.6.7 drawLegendIcon() . . . 1470
17.623.6.8 getKeyRange() . . . 1470
17.623.6.9 getValueRange() . . . 1471
17.623.6.10 selectTest() . . . 1471
17.623.6.11 setData() [1/3] . . . 1471
17.623.6.12 setData() [2/3] . . . 1472
17.623.6.13 setData() [3/3] . . . 1472
17.623.6.14 setLineStyle() . . . 1472
17.623.6.15 setScatterSkip() . . . 1472
17.623.6.16 setScatterStyle() . . . 1473
17.624 QCPCurveData Class Reference . . . 1473
17.624.1 Detailed Description . . . 1473
17.624.2 Constructor & Destructor Documentation . . . 1474
17.624.2.1 QCPCurveData() [1/2] . . . 1474
17.624.2.2 QCPCurveData() [2/2] . . . 1474
17.624.3 Member Function Documentation . . . 1474

17.624.3.1 fromSortKey() . . . 1474
17.624.3.2 mainKey() . . . 1474
17.624.3.3 mainValue() . . . 1474
17.624.3.4 sortKey() . . . 1474
17.624.3.5 sortKeyIsMainKey() . . . 1474
17.624.3.6 valueRange() . . . 1474
17.625 QCPDataContainer< DataType > Class Template Reference . . . 1475
17.625.1 Detailed Description . . . 1476
17.625.2 Requirements for the DataType template parameter . . . 1476
17.625.3 Constructor & Destructor Documentation . . . 1477
17.625.3.1 QCPDataContainer() . . . 1477
17.625.4 Member Function Documentation . . . 1477
17.625.4.1 add() [1/3] . . . 1477
17.625.4.2 add() [2/3] . . . 1477
17.625.4.3 add() [3/3] . . . 1477
17.625.4.4 at() . . . 1478
17.625.4.5 begin() . . . 1478
17.625.4.6 clear() . . . 1478
17.625.4.7 constBegin() . . . 1478
17.625.4.8 constEnd() . . . 1478
17.625.4.9 dataRange() . . . 1478
17.625.4.10 end() . . . 1478
17.625.4.11 findBegin() . . . 1479
17.625.4.12 findEnd() . . . 1479
17.625.4.13 isEmpty() . . . 1479
17.625.4.14 keyRange() . . . 1479
17.625.4.15 limitIteratorsToDataRange() . . . 1480
17.625.4.16 remove() [1/2] . . . 1480
17.625.4.17 remove() [2/2] . . . 1480
17.625.4.18 removeAfter() . . . 1480
17.625.4.19 removeBefore() . . . 1480
17.625.4.20 set() [1/2] . . . 1481
17.625.4.21 set() [2/2] . . . 1481
17.625.4.22 setAutoSqueeze() . . . 1481
17.625.4.23 size() . . . 1481
17.625.4.24 sort() . . . 1481
17.625.4.25 squeeze() . . . 1481
17.625.4.26 valueRange() . . . 1482
17.625.5 Friends And Related Symbol Documentation . . . 1482
17.625.5.1 qcpLessThanSortKey() . . . 1482
17.626 QCPDataRange Class Reference . . . 1482
17.626.1 Detailed Description . . . 1483

17.626.2 Constructor & Destructor Documentation . . . 1483
17.626.2.1 QCPDataRange() [1/2] . . . 1483
17.626.2.2 QCPDataRange() [2/2] . . . 1483
17.626.3 Member Function Documentation . . . 1483
17.626.3.1 adjusted() . . . 1483
17.626.3.2 bounded() . . . 1484
17.626.3.3 contains() . . . 1484
17.626.3.4 expanded() . . . 1484
17.626.3.5 intersection() . . . 1484
17.626.3.6 intersects() . . . 1484
17.626.3.7 isEmpty() . . . 1484
17.626.3.8 isValid() . . . 1485
17.626.3.9 length() . . . 1485
17.626.3.10 setBegin() . . . 1485
17.626.3.11 setEnd() . . . 1485
17.626.3.12 size() . . . 1485
17.626.4 Friends And Related Symbol Documentation . . . 1485
17.626.4.1 operator<<() . . . 1485
17.627 QCPDataSelection Class Reference . . . 1486
17.627.1 Detailed Description . . . 1486
17.627.2 Iterating over a data selection . . . 1487
17.627.3 Constructor & Destructor Documentation . . . 1487
17.627.3.1 QCPDataSelection() [1/2] . . . 1487
17.627.3.2 QCPDataSelection() [2/2] . . . 1487
17.627.4 Member Function Documentation . . . 1487
17.627.4.1 addDataRange() . . . 1487
17.627.4.2 clear() . . . 1487
17.627.4.3 contains() . . . 1487
17.627.4.4 dataPointCount() . . . 1487
17.627.4.5 dataRange() . . . 1488
17.627.4.6 dataRangeCount() . . . 1488
17.627.4.7 dataRanges() . . . 1488
17.627.4.8 enforceType() . . . 1488
17.627.4.9 intersection() [1/2] . . . 1488
17.627.4.10 intersection() [2/2] . . . 1488
17.627.4.11 inverse() . . . 1489
17.627.4.12 isEmpty() . . . 1489
17.627.4.13 operator+=() [1/2] . . . 1489
17.627.4.14 operator+=() [2/2] . . . 1489
17.627.4.15 operator-=() [1/2] . . . 1489
17.627.4.16 operator-=() [2/2] . . . 1489
17.627.4.17 operator==() . . . 1489

17.627.4.18 simplify() . . . 1489
17.627.4.19 span() . . . 1490
17.627.5 Friends And Related Symbol Documentation . . . 1490
17.627.5.1 operator+ [1/4] . . . 1490
17.627.5.2 operator+ [2/4] . . . 1490
17.627.5.3 operator+ [3/4] . . . 1490
17.627.5.4 operator+ [4/4] . . . 1490
17.627.5.5 operator- [1/4] . . . 1490
17.627.5.6 operator- [2/4] . . . 1490
17.627.5.7 operator- [3/4] . . . 1490
17.627.5.8 operator- [4/4] . . . 1491
17.627.5.9 operator<<() . . . 1491
17.628 QCPErrorBars Class Reference . . . 1491
17.628.1 Detailed Description . . . 1495
17.628.2 Changing the appearance . . . 1495
17.628.3 Member Enumeration Documentation . . . 1496
17.628.3.1 ErrorType . . . 1496
17.628.4 Constructor & Destructor Documentation . . . 1496
17.628.4.1 QCPErrorBars() . . . 1496
17.628.5 Member Function Documentation . . . 1496
17.628.5.1 addData() [1/4] . . . 1496
17.628.5.2 addData() [2/4] . . . 1496
17.628.5.3 addData() [3/4] . . . 1497
17.628.5.4 addData() [4/4] . . . 1497
17.628.5.5 data() . . . 1497
17.628.5.6 dataCount() . . . 1497
17.628.5.7 dataMainKey() . . . 1497
17.628.5.8 dataMainValue() . . . 1497
17.628.5.9 dataPixelPosition() . . . 1498
17.628.5.10 dataSortKey() . . . 1498
17.628.5.11 dataValueRange() . . . 1498
17.628.5.12 draw() . . . 1498
17.628.5.13 drawLegendIcon() . . . 1498
17.628.5.14 findBegin() . . . 1498
17.628.5.15 findEnd() . . . 1499
17.628.5.16 getKeyRange() . . . 1499
17.628.5.17 getValueRange() . . . 1499
17.628.5.18 interface1D() . . . 1500
17.628.5.19 selectTest() . . . 1500
17.628.5.20 selectTestRect() . . . 1500
17.628.5.21 setData() [1/3] . . . 1500
17.628.5.22 setData() [2/3] . . . 1500

17.628.5.23 setData() [3/3] . . . 1501
17.628.5.24 setDataPlottable() . . . 1501
17.628.5.25 setErrorType() . . . 1501
17.628.5.26 setSymbolGap() . . . 1501
17.628.5.27 setWhiskerWidth() . . . 1501
17.628.5.28 sortKeyIsMainKey() . . . 1502
17.629 QCPErrorBarsData Class Reference . . . 1502
17.629.1 Detailed Description . . . 1502
17.629.2 Constructor & Destructor Documentation . . . 1502
17.629.2.1 QCPErrorBarsData() [1/3] . . . 1502
17.629.2.2 QCPErrorBarsData() [2/3] . . . 1502
17.629.2.3 QCPErrorBarsData() [3/3] . . . 1503
17.630 QCPFinancial Class Reference . . . 1503
17.630.1 Detailed Description . . . 1508
17.630.2 Changing the appearance . . . 1508
17.630.3 Usage . . . 1508
17.630.4 Member Enumeration Documentation . . . 1508
17.630.4.1 ChartStyle . . . 1508
17.630.4.2 WidthType . . . 1509
17.630.5 Constructor & Destructor Documentation . . . 1509
17.630.5.1 QCPFinancial() . . . 1509
17.630.6 Member Function Documentation . . . 1509
17.630.6.1 addData() [1/2] . . . 1509
17.630.6.2 addData() [2/2] . . . 1509
17.630.6.3 data() . . . 1510
17.630.6.4 draw() . . . 1510
17.630.6.5 drawLegendIcon() . . . 1510
17.630.6.6 getKeyRange() . . . 1510
17.630.6.7 getValueRange() . . . 1510
17.630.6.8 selectTest() . . . 1511
17.630.6.9 selectTestRect() . . . 1511
17.630.6.10 setBrushNegative() . . . 1511
17.630.6.11 setBrushPositive() . . . 1512
17.630.6.12 setChartStyle() . . . 1512
17.630.6.13 setData() [1/2] . . . 1512
17.630.6.14 setData() [2/2] . . . 1512
17.630.6.15 setPenNegative() . . . 1512
17.630.6.16 setPenPositive() . . . 1513
17.630.6.17 setTwoColored() . . . 1513
17.630.6.18 setWidth() . . . 1513
17.630.6.19 setWidthType() . . . 1513
17.630.6.20 timeSeriesToOhlc() . . . 1513

17.631 QCPFinancialData Class Reference . . . 1514
17.631.1 Detailed Description . . . 1514
17.631.2 Constructor & Destructor Documentation . . . 1514
17.631.2.1 QCPFinancialData() [1/2] . . . 1514
17.631.2.2 QCPFinancialData() [2/2] . . . 1515
17.631.3 Member Function Documentation . . . 1515
17.631.3.1 fromSortKey() . . . 1515
17.631.3.2 mainKey() . . . 1515
17.631.3.3 mainValue() . . . 1515
17.631.3.4 sortKey() . . . 1515
17.631.3.5 sortKeyIsMainKey() . . . 1515
17.631.3.6 valueRange() . . . 1515
17.632 QCPGraph Class Reference . . . 1516
17.632.1 Detailed Description . . . 1521
17.632.2 Changing the appearance . . . 1521
17.632.2.1 Filling under or between graphs . . . 1521
17.632.3 Member Enumeration Documentation . . . 1521
17.632.3.1 LineStyle . . . 1521
17.632.4 Constructor & Destructor Documentation . . . 1521
17.632.4.1 QCPGraph() . . . 1521
17.632.5 Member Function Documentation . . . 1522
17.632.5.1 addData() [1/2] . . . 1522
17.632.5.2 addData() [2/2] . . . 1522
17.632.5.3 data() . . . 1522
17.632.5.4 draw() . . . 1522
17.632.5.5 drawLegendIcon() . . . 1522
17.632.5.6 getKeyRange() . . . 1523
17.632.5.7 getValueRange() . . . 1523
17.632.5.8 getVisibleDataBounds() . . . 1523
17.632.5.9 selectTest() . . . 1523
17.632.5.10 setAdaptiveSampling() . . . 1524
17.632.5.11 setChannelFillGraph() . . . 1524
17.632.5.12 setData() [1/2] . . . 1524
17.632.5.13 setData() [2/2] . . . 1525
17.632.5.14 setLineStyle() . . . 1525
17.632.5.15 setScatterSkip() . . . 1525
17.632.5.16 setScatterStyle() . . . 1525
17.633 QCPGraphData Class Reference . . . 1525
17.633.1 Detailed Description . . . 1526
17.633.2 Constructor & Destructor Documentation . . . 1526
17.633.2.1 QCPGraphData() [1/2] . . . 1526
17.633.2.2 QCPGraphData() [2/2] . . . 1526

17.633.3 Member Function Documentation . . . 1526
17.633.3.1 fromSortKey() . . . 1526
17.633.3.2 mainKey() . . . 1527
17.633.3.3 mainValue() . . . 1527
17.633.3.4 sortKey() . . . 1527
17.633.3.5 sortKeyIsMainKey() . . . 1527
17.633.3.6 valueRange() . . . 1527
17.634 QCPGrid Class Reference . . . 1527
17.634.1 Detailed Description . . . 1530
17.634.2 Constructor & Destructor Documentation . . . 1530
17.634.2.1 QCPGrid() . . . 1530
17.634.3 Member Function Documentation . . . 1530
17.634.3.1 applyDefaultAntialiasingHint() . . . 1530
17.634.3.2 draw() . . . 1530
17.634.3.3 setAntialiasedSubGrid() . . . 1530
17.634.3.4 setAntialiasedZeroLine() . . . 1531
17.634.3.5 setPen() . . . 1531
17.634.3.6 setSubGridPen() . . . 1531
17.634.3.7 setSubGridVisible() . . . 1531
17.634.3.8 setZeroLinePen() . . . 1531
17.635 QCPItemAnchor Class Reference . . . 1531
17.635.1 Detailed Description . . . 1532
17.635.2 Constructor & Destructor Documentation . . . 1533
17.635.2.1 QCPItemAnchor() . . . 1533
17.635.3 Member Function Documentation . . . 1533
17.635.3.1 pixelPosition() . . . 1533
17.635.3.2 toQCPItemPosition() . . . 1533
17.636 QCPItemBracket Class Reference . . . 1533
17.636.1 Detailed Description . . . 1537
17.636.2 Member Enumeration Documentation . . . 1537
17.636.2.1 BracketStyle . . . 1537
17.636.3 Constructor & Destructor Documentation . . . 1537
17.636.3.1 QCPItemBracket() . . . 1537
17.636.4 Member Function Documentation . . . 1537
17.636.4.1 anchorPixelPosition() . . . 1537
17.636.4.2 draw() . . . 1538
17.636.4.3 selectTest() . . . 1538
17.636.4.4 setLength() . . . 1538
17.636.4.5 setPen() . . . 1538
17.636.4.6 setSelectedPen() . . . 1539
17.636.4.7 setStyle() . . . 1539
17.637 QCPItemCurve Class Reference . . . 1539

17.637.1 Detailed Description . . . 1542
17.637.2 Constructor & Destructor Documentation . . . 1542
17.637.2.1 QCPItemCurve() . . . 1542
17.637.3 Member Function Documentation . . . 1543
17.637.3.1 draw() . . . 1543
17.637.3.2 selectTest() . . . 1543
17.637.3.3 setHead() . . . 1543
17.637.3.4 setPen() . . . 1543
17.637.3.5 setSelectedPen() . . . 1544
17.637.3.6 setTail() . . . 1544
17.638 QCPItemEllipse Class Reference . . . 1544
17.638.1 Detailed Description . . . 1547
17.638.2 Constructor & Destructor Documentation . . . 1548
17.638.2.1 QCPItemEllipse() . . . 1548
17.638.3 Member Function Documentation . . . 1548
17.638.3.1 anchorPixelPosition() . . . 1548
17.638.3.2 draw() . . . 1548
17.638.3.3 selectTest() . . . 1548
17.638.3.4 setBrush() . . . 1549
17.638.3.5 setPen() . . . 1549
17.638.3.6 setSelectedBrush() . . . 1549
17.638.3.7 setSelectedPen() . . . 1549
17.639 QCPItemLine Class Reference . . . 1549
17.639.1 Detailed Description . . . 1553
17.639.2 Constructor & Destructor Documentation . . . 1553
17.639.2.1 QCPItemLine() . . . 1553
17.639.3 Member Function Documentation . . . 1553
17.639.3.1 draw() . . . 1553
17.639.3.2 selectTest() . . . 1553
17.639.3.3 setHead() . . . 1554
17.639.3.4 setPen() . . . 1554
17.639.3.5 setSelectedPen() . . . 1554
17.639.3.6 setTail() . . . 1554
17.640 QCPItemPixmap Class Reference . . . 1554
17.640.1 Detailed Description . . . 1558
17.640.2 Constructor & Destructor Documentation . . . 1558
17.640.2.1 QCPItemPixmap() . . . 1558
17.640.3 Member Function Documentation . . . 1558
17.640.3.1 anchorPixelPosition() . . . 1558
17.640.3.2 draw() . . . 1558
17.640.3.3 selectTest() . . . 1558
17.640.3.4 setPen() . . . 1559

17.640.3.5 setPixmap() . . . 1559
17.640.3.6 setScaled() . . . 1559
17.640.3.7 setSelectedPen() . . . 1559
17.641 QCPItemPosition Class Reference . . . 1560
17.641.1 Detailed Description . . . 1562
17.641.2 Member Enumeration Documentation . . . 1562
17.641.2.1 PositionType . . . 1562
17.641.3 Constructor & Destructor Documentation . . . 1563
17.641.3.1 QCPItemPosition() . . . 1563
17.641.4 Member Function Documentation . . . 1563
17.641.4.1 parentAnchor() . . . 1563
17.641.4.2 pixelPosition() . . . 1563
17.641.4.3 setAxes() . . . 1563
17.641.4.4 setAxisRect() . . . 1563
17.641.4.5 setCoords() [1/2] . . . 1563
17.641.4.6 setCoords() [2/2] . . . 1564
17.641.4.7 setParentAnchor() . . . 1564
17.641.4.8 setParentAnchorX() . . . 1564
17.641.4.9 setParentAnchorY() . . . 1564
17.641.4.10 setPixelPosition() . . . 1565
17.641.4.11 setType() . . . 1565
17.641.4.12 setTypeX() . . . 1565
17.641.4.13 setTypeY() . . . 1565
17.641.4.14 toQCPItemPosition() . . . 1566
17.641.4.15 type() . . . 1566
17.642 QCPItemRect Class Reference . . . 1566
17.642.1 Detailed Description . . . 1569
17.642.2 Constructor & Destructor Documentation . . . 1570
17.642.2.1 QCPItemRect() . . . 1570
17.642.3 Member Function Documentation . . . 1570
17.642.3.1 anchorPixelPosition() . . . 1570
17.642.3.2 draw() . . . 1570
17.642.3.3 selectTest() . . . 1570
17.642.3.4 setBrush() . . . 1571
17.642.3.5 setPen() . . . 1571
17.642.3.6 setSelectedBrush() . . . 1571
17.642.3.7 setSelectedPen() . . . 1571
17.643 QCPItemStraightLine Class Reference . . . 1571
17.643.1 Detailed Description . . . 1574
17.643.2 Constructor & Destructor Documentation . . . 1575
17.643.2.1 QCPItemStraightLine() . . . 1575
17.643.3 Member Function Documentation . . . 1575

17.643.3.1 draw() . . . 1575
17.643.3.2 selectTest() . . . 1575
17.643.3.3 setPen() . . . 1575
17.643.3.4 setSelectedPen() . . . 1576
17.644 QCPItemText Class Reference . . . 1576
17.644.1 Detailed Description . . . 1580
17.644.2 Constructor & Destructor Documentation . . . 1580
17.644.2.1 QCPItemText() . . . 1580
17.644.3 Member Function Documentation . . . 1580
17.644.3.1 anchorPixelPosition() . . . 1580
17.644.3.2 draw() . . . 1580
17.644.3.3 selectTest() . . . 1581
17.644.3.4 setBrush() . . . 1581
17.644.3.5 setColor() . . . 1581
17.644.3.6 setFont() . . . 1581
17.644.3.7 setPadding() . . . 1582
17.644.3.8 setPen() . . . 1582
17.644.3.9 setPositionAlignment() . . . 1582
17.644.3.10 setRotation() . . . 1582
17.644.3.11 setSelectedBrush() . . . 1582
17.644.3.12 setSelectedColor() . . . 1582
17.644.3.13 setSelectedFont() . . . 1582
17.644.3.14 setSelectedPen() . . . 1583
17.644.3.15 setText() . . . 1583
17.644.3.16 setTextAlignment() . . . 1583
17.645 QCPItemTracer Class Reference . . . 1583
17.645.1 Detailed Description . . . 1587
17.645.2 Member Enumeration Documentation . . . 1587
17.645.2.1 TracerStyle . . . 1587
17.645.3 Constructor & Destructor Documentation . . . 1587
17.645.3.1 QCPItemTracer() . . . 1587
17.645.4 Member Function Documentation . . . 1588
17.645.4.1 draw() . . . 1588
17.645.4.2 selectTest() . . . 1588
17.645.4.3 setBrush() . . . 1588
17.645.4.4 setGraph() . . . 1588
17.645.4.5 setGraphKey() . . . 1589
17.645.4.6 setInterpolating() . . . 1589
17.645.4.7 setPen() . . . 1589
17.645.4.8 setSelectedBrush() . . . 1589
17.645.4.9 setSelectedPen() . . . 1590
17.645.4.10 setSize() . . . 1590

17.645.4.11 setStyle() . . . 1590
17.645.4.12 updatePosition() . . . 1590
17.646 QCPLabelPainterPrivate Class Reference . . . 1591
17.646.1 Member Enumeration Documentation . . . 1593
17.646.1.1 AnchorMode . . . 1593
17.646.1.2 AnchorReferenceType . . . 1593
17.646.1.3 AnchorSide . . . 1593
17.646.2 Constructor & Destructor Documentation . . . 1593
17.646.2.1 QCPLabelPainterPrivate() . . . 1593
17.647 QCPLayer Class Reference . . . 1593
17.647.1 Detailed Description . . . 1595
17.647.2 Default layers . . . 1595
17.647.3 Controlling the rendering order via layers . . . 1595
17.647.4 Replotting only a specific layer . . . 1595
17.647.5 Member Enumeration Documentation . . . 1596
17.647.5.1 LayerMode . . . 1596
17.647.6 Constructor & Destructor Documentation . . . 1596
17.647.6.1 QCPLayer() . . . 1596
17.647.7 Member Function Documentation . . . 1596
17.647.7.1 children() . . . 1596
17.647.7.2 index() . . . 1596
17.647.7.3 replot() . . . 1596
17.647.7.4 setMode() . . . 1597
17.647.7.5 setVisible() . . . 1597
17.648 QCPLayerable Class Reference . . . 1597
17.648.1 Detailed Description . . . 1600
17.648.2 Constructor & Destructor Documentation . . . 1600
17.648.2.1 QCPLayerable() . . . 1600
17.648.3 Member Function Documentation . . . 1601
17.648.3.1 layerChanged . . . 1601
17.648.3.2 mouseDoubleClickEvent() . . . 1601
17.648.3.3 mouseMoveEvent() . . . 1601
17.648.3.4 mousePressEvent() . . . 1601
17.648.3.5 mouseReleaseEvent() . . . 1602
17.648.3.6 parentLayerable() . . . 1602
17.648.3.7 realVisibility() . . . 1602
17.648.3.8 selectTest() . . . 1602
17.648.3.9 setAntialiased() . . . 1603
17.648.3.10 setLayer() [1/2] . . . 1603
17.648.3.11 setLayer() [2/2] . . . 1603
17.648.3.12 setVisible() . . . 1604
17.648.3.13 wheelEvent() . . . 1604

17.649 QCPLayout Class Reference . . . 1604
17.649.1 Detailed Description . . . 1608
17.649.2 Constructor & Destructor Documentation . . . 1608
17.649.2.1 QCPLayout() . . . 1608
17.649.3 Member Function Documentation . . . 1608
17.649.3.1 clear() . . . 1608
17.649.3.2 elementAt() . . . 1609
17.649.3.3 elementCount() . . . 1609
17.649.3.4 elements() . . . 1609
17.649.3.5 remove() . . . 1609
17.649.3.6 removeAt() . . . 1609
17.649.3.7 simplify() . . . 1610
17.649.3.8 sizeConstraintsChanged() . . . 1610
17.649.3.9 take() . . . 1610
17.649.3.10 takeAt() . . . 1610
17.649.3.11 update() . . . 1610
17.650 QCPLayoutElement Class Reference . . . 1611
17.650.1 Detailed Description . . . 1613
17.650.2 Member Enumeration Documentation . . . 1614
17.650.2.1 SizeConstraintRect . . . 1614
17.650.2.2 UpdatePhase . . . 1614
17.650.3 Constructor & Destructor Documentation . . . 1614
17.650.3.1 QCPLayoutElement() . . . 1614
17.650.4 Member Function Documentation . . . 1614
17.650.4.1 applyDefaultAntialiasingHint() . . . 1614
17.650.4.2 draw() . . . 1614
17.650.4.3 elements() . . . 1614
17.650.4.4 layout() . . . 1615
17.650.4.5 maximumOuterSizeHint() . . . 1615
17.650.4.6 minimumOuterSizeHint() . . . 1615
17.650.4.7 outerRect() . . . 1615
17.650.4.8 parentPlotInitialized() . . . 1615
17.650.4.9 rect() . . . 1616
17.650.4.10 selectTest() . . . 1616
17.650.4.11 setAutoMargins() . . . 1616
17.650.4.12 setMarginGroup() . . . 1616
17.650.4.13 setMargins() . . . 1617
17.650.4.14 setMaximumSize() [1/2] . . . 1617
17.650.4.15 setMaximumSize() [2/2] . . . 1617
17.650.4.16 setMinimumMargins() . . . 1617
17.650.4.17 setMinimumSize() [1/2] . . . 1617
17.650.4.18 setMinimumSize() [2/2] . . . 1618

17.650.4.19 setOuterRect() . . . 1618
17.650.4.20 setSizeConstraintRect() . . . 1618
17.650.4.21 update() . . . 1618
17.651 QCPLayoutGrid Class Reference . . . 1618
17.651.1 Detailed Description . . . 1623
17.651.2 Member Enumeration Documentation . . . 1624
17.651.2.1 FillOrder . . . 1624
17.651.3 Constructor & Destructor Documentation . . . 1624
17.651.3.1 QCPLayoutGrid() . . . 1624
17.651.4 Member Function Documentation . . . 1624
17.651.4.1 addElement() [1/2] . . . 1624
17.651.4.2 addElement() [2/2] . . . 1624
17.651.4.3 columnCount() . . . 1625
17.651.4.4 element() . . . 1625
17.651.4.5 elementAt() . . . 1625
17.651.4.6 elementCount() . . . 1625
17.651.4.7 elements() . . . 1625
17.651.4.8 expandTo() . . . 1626
17.651.4.9 hasElement() . . . 1626
17.651.4.10 indexToRowCol() . . . 1626
17.651.4.11 insertColumn() . . . 1626
17.651.4.12 insertRow() . . . 1626
17.651.4.13 maximumOuterSizeHint() . . . 1627
17.651.4.14 minimumOuterSizeHint() . . . 1627
17.651.4.15 rowColToIndex() . . . 1627
17.651.4.16 rowCount() . . . 1627
17.651.4.17 setColumnSpacing() . . . 1627
17.651.4.18 setColumnStretchFactor() . . . 1628
17.651.4.19 setColumnStretchFactors() . . . 1628
17.651.4.20 setFillOrder() . . . 1628
17.651.4.21 setRowSpacing() . . . 1629
17.651.4.22 setRowStretchFactor() . . . 1629
17.651.4.23 setRowStretchFactors() . . . 1629
17.651.4.24 setWrap() . . . 1629
17.651.4.25 simplify() . . . 1629
17.651.4.26 take() . . . 1630
17.651.4.27 takeAt() . . . 1630
17.651.4.28 updateLayout() . . . 1630
17.652 QCPLayoutInset Class Reference . . . 1630
17.652.1 Detailed Description . . . 1634
17.652.2 Member Enumeration Documentation . . . 1634
17.652.2.1 InsetPlacement . . . 1634

17.652.3 Constructor & Destructor Documentation . . . 1635
17.652.3.1 QCPLayoutInset() . . . 1635
17.652.4 Member Function Documentation . . . 1635
17.652.4.1 addElement() [1/2] . . . 1635
17.652.4.2 addElement() [2/2] . . . 1635
17.652.4.3 elementAt() . . . 1635
17.652.4.4 elementCount() . . . 1636
17.652.4.5 insetAlignment() . . . 1636
17.652.4.6 insetPlacement() . . . 1636
17.652.4.7 insetRect() . . . 1636
17.652.4.8 selectTest() . . . 1636
17.652.4.9 setInsetAlignment() . . . 1636
17.652.4.10 setInsetPlacement() . . . 1636
17.652.4.11 setInsetRect() . . . 1637
17.652.4.12 simplify() . . . 1637
17.652.4.13 take() . . . 1637
17.652.4.14 takeAt() . . . 1637
17.652.4.15 updateLayout() . . . 1637
17.653 QCPLegend Class Reference . . . 1638
17.653.1 Detailed Description . . . 1644
17.653.2 Member Enumeration Documentation . . . 1644
17.653.2.1 SelectablePart . . . 1644
17.653.3 Constructor & Destructor Documentation . . . 1645
17.653.3.1 QCPLegend() . . . 1645
17.653.4 Member Function Documentation . . . 1645
17.653.4.1 addItem() . . . 1645
17.653.4.2 applyDefaultAntialiasingHint() . . . 1645
17.653.4.3 clearItems() . . . 1645
17.653.4.4 deselectEvent() . . . 1645
17.653.4.5 draw() . . . 1645
17.653.4.6 hasItem() . . . 1645
17.653.4.7 hasItemWithPlottable() . . . 1645
17.653.4.8 item() . . . 1646
17.653.4.9 itemCount() . . . 1646
17.653.4.10 itemWithPlottable() . . . 1646
17.653.4.11 parentPlotInitialized() . . . 1646
17.653.4.12 removeItem() [1/2] . . . 1646
17.653.4.13 removeItem() [2/2] . . . 1647
17.653.4.14 selectedItems() . . . 1647
17.653.4.15 selectEvent() . . . 1647
17.653.4.16 selectionCategory() . . . 1647
17.653.4.17 selectionChanged . . . 1647

17.653.4.18 selectTest() . . . 1647
17.653.4.19 setBorderPen() . . . 1648
17.653.4.20 setBrush() . . . 1648
17.653.4.21 setFont() . . . 1648
17.653.4.22 setIconBorderPen() . . . 1648
17.653.4.23 setIconSize() [1/2] . . . 1648
17.653.4.24 setIconSize() [2/2] . . . 1648
17.653.4.25 setIconTextPadding() . . . 1648
17.653.4.26 setSelectableParts() . . . 1648
17.653.4.27 setSelectedBorderPen() . . . 1649
17.653.4.28 setSelectedBrush() . . . 1649
17.653.4.29 setSelectedFont() . . . 1649
17.653.4.30 setSelectedIconBorderPen() . . . 1649
17.653.4.31 setSelectedParts() . . . 1649
17.653.4.32 setSelectedTextColor() . . . 1650
17.653.4.33 setTextColor() . . . 1650
17.654 QCPLineEnding Class Reference . . . 1650
17.654.1 Detailed Description . . . 1651
17.654.2 Member Enumeration Documentation . . . 1651
17.654.2.1 EndingStyle . . . 1651
17.654.3 Constructor & Destructor Documentation . . . 1651
17.654.3.1 QCPLineEnding() [1/2] . . . 1651
17.654.3.2 QCPLineEnding() [2/2] . . . 1652
17.654.4 Member Function Documentation . . . 1652
17.654.4.1 realLength() . . . 1652
17.654.4.2 setInverted() . . . 1652
17.654.4.3 setLength() . . . 1652
17.654.4.4 setStyle() . . . 1652
17.654.4.5 setWidth() . . . 1652
17.655 QCPMarginGroup Class Reference . . . 1653
17.655.1 Detailed Description . . . 1654
17.655.2 Example . . . 1654
17.655.3 Constructor & Destructor Documentation . . . 1654
17.655.3.1 QCPMarginGroup() . . . 1654
17.655.4 Member Function Documentation . . . 1654
17.655.4.1 clear() . . . 1654
17.655.4.2 elements() . . . 1654
17.655.4.3 isEmpty() . . . 1655
17.656 QCPPaintBufferPixmap Class Reference . . . 1655
17.656.1 Detailed Description . . . 1656
17.656.2 Constructor & Destructor Documentation . . . 1656
17.656.2.1 QCPPaintBufferPixmap() . . . 1656

17.656.3 Member Function Documentation . . . 1656
17.656.3.1 clear() . . . 1656
17.656.3.2 draw() . . . 1656
17.656.3.3 reallocateBuffer() . . . 1657
17.656.3.4 startPainting() . . . 1657
17.657 QCPPainter Class Reference . . . 1657
17.657.1 Detailed Description . . . 1658
17.657.2 Member Enumeration Documentation . . . 1659
17.657.2.1 PainterMode . . . 1659
17.657.3 Constructor & Destructor Documentation . . . 1659
17.657.3.1 QCPPainter() [1/2] . . . 1659
17.657.3.2 QCPPainter() [2/2] . . . 1659
17.657.4 Member Function Documentation . . . 1659
17.657.4.1 begin() . . . 1659
17.657.4.2 drawLine() . . . 1659
17.657.4.3 makeNonCosmetic() . . . 1660
17.657.4.4 restore() . . . 1660
17.657.4.5 save() . . . 1660
17.657.4.6 setAntialiasing() . . . 1660
17.657.4.7 setMode() . . . 1660
17.657.4.8 setModes() . . . 1660
17.657.4.9 setPen() [1/3] . . . 1661
17.657.4.10 setPen() [2/3] . . . 1661
17.657.4.11 setPen() [3/3] . . . 1661
17.658 QCPPlottableInterface1D Class Reference . . . 1661
17.658.1 Detailed Description . . . 1662
17.658.2 Member Function Documentation . . . 1663
17.658.2.1 dataCount() . . . 1663
17.658.2.2 dataMainKey() . . . 1663
17.658.2.3 dataMainValue() . . . 1663
17.658.2.4 dataPixelPosition() . . . 1663
17.658.2.5 dataSortKey() . . . 1663
17.658.2.6 dataValueRange() . . . 1663
17.658.2.7 findBegin() . . . 1664
17.658.2.8 findEnd() . . . 1664
17.658.2.9 selectTestRect() . . . 1664
17.658.2.10 sortKeyIsMainKey() . . . 1665
17.659 QCPPlottableLegendItem Class Reference . . . 1665
17.659.1 Detailed Description . . . 1669
17.659.2 Constructor & Destructor Documentation . . . 1669
17.659.2.1 QCPPlottableLegendItem() . . . 1669
17.659.3 Member Function Documentation . . . 1669

17.659.3.1 draw() . . . 1669
17.659.3.2 minimumOuterSizeHint() . . . 1669
17.660 QCPPolarAxisAngular Class Reference . . . 1670
17.660.1 Detailed Description . . . 1676
17.660.2 Member Enumeration Documentation . . . 1677
17.660.2.1 LabelMode . . . 1677
17.660.2.2 SelectablePart . . . 1677
17.660.3 Constructor & Destructor Documentation . . . 1677
17.660.3.1 QCPPolarAxisAngular() . . . 1677
17.660.4 Member Function Documentation . . . 1677
17.660.4.1 addRadialAxis() . . . 1677
17.660.4.2 applyDefaultAntialiasingHint() . . . 1677
17.660.4.3 bottom() . . . 1678
17.660.4.4 bottomLeft() . . . 1678
17.660.4.5 bottomRight() . . . 1678
17.660.4.6 center() . . . 1678
17.660.4.7 coordToPixel() . . . 1678
17.660.4.8 draw() . . . 1678
17.660.4.9 elements() . . . 1678
17.660.4.10 getPartAt() . . . 1678
17.660.4.11 height() . . . 1679
17.660.4.12 insetLayout() . . . 1679
17.660.4.13 left() . . . 1679
17.660.4.14 mouseMoveEvent() . . . 1679
17.660.4.15 mousePressEvent() . . . 1679
17.660.4.16 mouseReleaseEvent() . . . 1680
17.660.4.17 moveRange() . . . 1680
17.660.4.18 pixelToCoord() . . . 1680
17.660.4.19 radialAxes() . . . 1680
17.660.4.20 radialAxis() . . . 1680
17.660.4.21 radialAxisCount() . . . 1681
17.660.4.22 removeRadialAxis() . . . 1681
17.660.4.23 rescale() . . . 1681
17.660.4.24 right() . . . 1681
17.660.4.25 scaleRange() [1/2] . . . 1681
17.660.4.26 scaleRange() [2/2] . . . 1681
17.660.4.27 selectionCategory() . . . 1682
17.660.4.28 selectTest() . . . 1682
17.660.4.29 setBackground() [1/3] . . . 1682
17.660.4.30 setBackground() [2/3] . . . 1682
17.660.4.31 setBackground() [3/3] . . . 1682
17.660.4.32 setBackgroundScaled() . . . 1683

17.660.4.33 setBackgroundScaledMode() . . . 1683
17.660.4.34 setBasePen() . . . 1683
17.660.4.35 setLabel() . . . 1683
17.660.4.36 setLabelColor() . . . 1683
17.660.4.37 setLabelFont() . . . 1684
17.660.4.38 setLabelPadding() . . . 1684
17.660.4.39 setNumberFormat() . . . 1684
17.660.4.40 setNumberPrecision() . . . 1684
17.660.4.41 setRange() [1/3] . . . 1685
17.660.4.42 setRange() [2/3] . . . 1685
17.660.4.43 setRange() [3/3] . . . 1685
17.660.4.44 setRangeLower() . . . 1685
17.660.4.45 setRangeReversed() . . . 1685
17.660.4.46 setRangeUpper() . . . 1685
17.660.4.47 setSelectableParts() . . . 1686
17.660.4.48 setSelectedBasePen() . . . 1686
17.660.4.49 setSelectedLabelColor() . . . 1686
17.660.4.50 setSelectedLabelFont() . . . 1686
17.660.4.51 setSelectedParts() . . . 1686
17.660.4.52 setSelectedSubTickPen() . . . 1687
17.660.4.53 setSelectedTickLabelColor() . . . 1687
17.660.4.54 setSelectedTickLabelFont() . . . 1687
17.660.4.55 setSelectedTickPen() . . . 1687
17.660.4.56 setSubTickLength() . . . 1687
17.660.4.57 setSubTickLengthIn() . . . 1687
17.660.4.58 setSubTickLengthOut() . . . 1688
17.660.4.59 setSubTickPen() . . . 1688
17.660.4.60 setSubTicks() . . . 1688
17.660.4.61 setTicker() . . . 1688
17.660.4.62 setTickLabelColor() . . . 1688
17.660.4.63 setTickLabelFont() . . . 1689
17.660.4.64 setTickLabelPadding() . . . 1689
17.660.4.65 setTickLabelRotation() . . . 1689
17.660.4.66 setTickLabels() . . . 1689
17.660.4.67 setTickLength() . . . 1689
17.660.4.68 setTickLengthIn() . . . 1689
17.660.4.69 setTickLengthOut() . . . 1690
17.660.4.70 setTickPen() . . . 1690
17.660.4.71 setTicks() . . . 1690
17.660.4.72 size() . . . 1690
17.660.4.73 top() . . . 1690
17.660.4.74 topLeft() . . . 1690

17.660.4.75 topRight() . . . 1690
17.660.4.76 update() . . . 1691
17.660.4.77 wheelEvent() . . . 1691
17.660.4.78 width() . . . 1691
17.661 QCPPolarAxisRadial Class Reference . . . 1691
17.661.1 Detailed Description . . . 1696
17.661.2 Member Enumeration Documentation . . . 1697
17.661.2.1 AngleReference . . . 1697
17.661.2.2 ScaleType . . . 1698
17.661.2.3 SelectablePart . . . 1698
17.661.3 Constructor & Destructor Documentation . . . 1698
17.661.3.1 QCPPolarAxisRadial() . . . 1698
17.661.4 Member Function Documentation . . . 1698
17.661.4.1 applyDefaultAntialiasingHint() . . . 1698
17.661.4.2 coordToPixel() . . . 1699
17.661.4.3 deselectEvent() . . . 1699
17.661.4.4 draw() . . . 1699
17.661.4.5 getPartAt() . . . 1699
17.661.4.6 mouseMoveEvent() . . . 1699
17.661.4.7 mousePressEvent() . . . 1699
17.661.4.8 mouseReleaseEvent() . . . 1700
17.661.4.9 moveRange() . . . 1700
17.661.4.10 pixelToCoord() . . . 1700
17.661.4.11 rangeChanged [1/2] . . . 1700
17.661.4.12 rangeChanged [2/2] . . . 1701
17.661.4.13 rescale() . . . 1701
17.661.4.14 scaleRange() [1/2] . . . 1701
17.661.4.15 scaleRange() [2/2] . . . 1701
17.661.4.16 scaleTypeChanged . . . 1701
17.661.4.17 selectableChanged . . . 1701
17.661.4.18 selectEvent() . . . 1702
17.661.4.19 selectionCategory() . . . 1702
17.661.4.20 selectionChanged . . . 1702
17.661.4.21 selectTest() . . . 1702
17.661.4.22 setBasePen() . . . 1702
17.661.4.23 setLabel() . . . 1703
17.661.4.24 setLabelColor() . . . 1703
17.661.4.25 setLabelFont() . . . 1703
17.661.4.26 setLabelPadding() . . . 1703
17.661.4.27 setNumberFormat() . . . 1703
17.661.4.28 setNumberPrecision() . . . 1704
17.661.4.29 setRange() [1/3] . . . 1704

17.661.4.30 setRange() [2/3] . . . 1704
17.661.4.31 setRange() [3/3] . . . 1704
17.661.4.32 setRangeLower() . . . 1704
17.661.4.33 setRangeReversed() . . . 1705
17.661.4.34 setRangeUpper() . . . 1705
17.661.4.35 setScaleType() . . . 1705
17.661.4.36 setSelectableParts() . . . 1705
17.661.4.37 setSelectedBasePen() . . . 1705
17.661.4.38 setSelectedLabelColor() . . . 1706
17.661.4.39 setSelectedLabelFont() . . . 1706
17.661.4.40 setSelectedParts() . . . 1706
17.661.4.41 setSelectedSubTickPen() . . . 1706
17.661.4.42 setSelectedTickLabelColor() . . . 1706
17.661.4.43 setSelectedTickLabelFont() . . . 1707
17.661.4.44 setSelectedTickPen() . . . 1707
17.661.4.45 setSubTickLength() . . . 1707
17.661.4.46 setSubTickLengthIn() . . . 1707
17.661.4.47 setSubTickLengthOut() . . . 1707
17.661.4.48 setSubTickPen() . . . 1707
17.661.4.49 setSubTicks() . . . 1708
17.661.4.50 setTicker() . . . 1708
17.661.4.51 setTickLabelColor() . . . 1708
17.661.4.52 setTickLabelFont() . . . 1708
17.661.4.53 setTickLabelPadding() . . . 1708
17.661.4.54 setTickLabelRotation() . . . 1708
17.661.4.55 setTickLabels() . . . 1709
17.661.4.56 setTickLength() . . . 1709
17.661.4.57 setTickLengthIn() . . . 1709
17.661.4.58 setTickLengthOut() . . . 1709
17.661.4.59 setTickPen() . . . 1709
17.661.4.60 setTicks() . . . 1709
17.661.4.61 ticker() . . . 1710
17.661.4.62 wheelEvent() . . . 1710
17.662 QCPPolarGraph Class Reference . . . 1710
17.662.1 Detailed Description . . . 1714
17.662.2 Member Enumeration Documentation . . . 1714
17.662.2.1 LineStyle . . . 1714
17.662.3 Constructor & Destructor Documentation . . . 1714
17.662.3.1 QCPPolarGraph() . . . 1714
17.662.4 Member Function Documentation . . . 1715
17.662.4.1 applyDefaultAntialiasingHint() . . . 1715
17.662.4.2 clipRect() . . . 1715

17.662.4.3 coordsToPixels() . . . 1715
17.662.4.4 deselectEvent() . . . 1715
17.662.4.5 draw() . . . 1715
17.662.4.6 selectEvent() . . . 1715
17.662.4.7 selectionCategory() . . . 1715
17.662.4.8 selectTest() . . . 1715
17.662.4.9 setAntialiasedFill() . . . 1716
17.662.4.10 setAntialiasedScatters() . . . 1716
17.662.4.11 setBrush() . . . 1716
17.662.4.12 setData() [1/2] . . . 1716
17.662.4.13 setData() [2/2] . . . 1717
17.662.4.14 setKeyAxis() . . . 1717
17.662.4.15 setLineStyle() . . . 1717
17.662.4.16 setName() . . . 1717
17.662.4.17 setPen() . . . 1718
17.662.4.18 setScatterStyle() . . . 1718
17.662.4.19 setSelectable() . . . 1718
17.662.4.20 setSelection() . . . 1718
17.662.4.21 setValueAxis() . . . 1718
17.663 QCPPolarGrid Class Reference . . . 1719
17.663.1 Detailed Description . . . 1721
17.663.2 Member Enumeration Documentation . . . 1721
17.663.2.1 GridType . . . 1721
17.663.3 Constructor & Destructor Documentation . . . 1721
17.663.3.1 QCPPolarGrid() . . . 1721
17.663.4 Member Function Documentation . . . 1721
17.663.4.1 applyDefaultAntialiasingHint() . . . 1721
17.663.4.2 draw() . . . 1722
17.663.4.3 setAngularPen() . . . 1722
17.663.4.4 setAngularSubGridPen() . . . 1722
17.663.4.5 setAntialiasedSubGrid() . . . 1722
17.663.4.6 setAntialiasedZeroLine() . . . 1722
17.664 QCPPolarLegendItem Class Reference . . . 1722
17.664.1 Detailed Description . . . 1726
17.664.2 Member Function Documentation . . . 1727
17.664.2.1 draw() . . . 1727
17.664.2.2 minimumOuterSizeHint() . . . 1727
17.665 QCPRange Class Reference . . . 1727
17.665.1 Detailed Description . . . 1728
17.665.2 Constructor & Destructor Documentation . . . 1728
17.665.2.1 QCPRange() [1/2] . . . 1728
17.665.2.2 QCPRange() [2/2] . . . 1728

17.665.3 Member Function Documentation . . . 1728
17.665.3.1 bounded() . . . 1728
17.665.3.2 center() . . . 1729
17.665.3.3 contains() . . . 1729
17.665.3.4 expand() [1/2] . . . 1729
17.665.3.5 expand() [2/2] . . . 1729
17.665.3.6 expanded() [1/2] . . . 1729
17.665.3.7 expanded() [2/2] . . . 1730
17.665.3.8 normalize() . . . 1730
17.665.3.9 operator*=() . . . 1730
17.665.3.10 operator+=() . . . 1730
17.665.3.11 operator-=() . . . 1730
17.665.3.12 operator/=() . . . 1730
17.665.3.13 sanitizedForLinScale() . . . 1730
17.665.3.14 sanitizedForLogScale() . . . 1730
17.665.3.15 size() . . . 1731
17.665.3.16 validRange() [1/2] . . . 1731
17.665.3.17 validRange() [2/2] . . . 1731
17.665.4 Friends And Related Symbol Documentation . . . 1731
17.665.4.1 operator* [1/2] . . . 1731
17.665.4.2 operator* [2/2] . . . 1731
17.665.4.3 operator+ [1/2] . . . 1731
17.665.4.4 operator+ [2/2] . . . 1731
17.665.4.5 operator- . . . 1732
17.665.4.6 operator/ . . . 1732
17.665.4.7 operator<<() . . . 1732
17.665.5 Member Data Documentation . . . 1732
17.665.5.1 maxRange . . . 1732
17.665.5.2 minRange . . . 1732
17.666 QCPScatterStyle Class Reference . . . 1733
17.666.1 Detailed Description . . . 1734
17.666.2 Specifying a scatter style . . . 1734
17.666.3 Leaving the color/pen up to the plottable . . . 1734
17.666.4 Custom shapes and pixmaps . . . 1734
17.666.5 Member Enumeration Documentation . . . 1734
17.666.5.1 ScatterProperty . . . 1734
17.666.5.2 ScatterShape . . . 1734
17.666.6 Constructor & Destructor Documentation . . . 1735
17.666.6.1 QCPScatterStyle() [1/7] . . . 1735
17.666.6.2 QCPScatterStyle() [2/7] . . . 1735
17.666.6.3 QCPScatterStyle() [3/7] . . . 1735
17.666.6.4 QCPScatterStyle() [4/7] . . . 1736

17.666.6.5 QCPScatterStyle() [5/7] . . . 1736
17.666.6.6 QCPScatterStyle() [6/7] . . . 1736
17.666.6.7 QCPScatterStyle() [7/7] . . . 1736
17.666.7 Member Function Documentation . . . 1736
17.666.7.1 applyTo() . . . 1736
17.666.7.2 drawShape() [1/2] . . . 1737
17.666.7.3 drawShape() [2/2] . . . 1737
17.666.7.4 isNone() . . . 1737
17.666.7.5 isPenDefined() . . . 1737
17.666.7.6 setBrush() . . . 1737
17.666.7.7 setCustomPath() . . . 1738
17.666.7.8 setFromOther() . . . 1738
17.666.7.9 setPen() . . . 1738
17.666.7.10 setPixmap() . . . 1738
17.666.7.11 setShape() . . . 1738
17.666.7.12 setSize() . . . 1738
17.666.7.13 undefinePen() . . . 1739
17.667 QCPSelectionDecorator Class Reference . . . 1739
17.667.1 Detailed Description . . . 1740
17.667.2 Constructor & Destructor Documentation . . . 1740
17.667.2.1 QCPSelectionDecorator() . . . 1740
17.667.3 Member Function Documentation . . . 1740
17.667.3.1 applyBrush() . . . 1740
17.667.3.2 applyPen() . . . 1741
17.667.3.3 copyFrom() . . . 1741
17.667.3.4 drawDecoration() . . . 1741
17.667.3.5 getFinalScatterStyle() . . . 1741
17.667.3.6 setBrush() . . . 1741
17.667.3.7 setPen() . . . 1741
17.667.3.8 setScatterStyle() . . . 1742
17.667.3.9 setUsedScatterProperties() . . . 1742
17.668 QCPSelectionDecoratorBracket Class Reference . . . 1742
17.668.1 Detailed Description . . . 1744
17.668.2 Member Enumeration Documentation . . . 1744
17.668.2.1 BracketStyle . . . 1744
17.668.3 Constructor & Destructor Documentation . . . 1745
17.668.3.1 QCPSelectionDecoratorBracket() . . . 1745
17.668.4 Member Function Documentation . . . 1745
17.668.4.1 drawBracket() . . . 1745
17.668.4.2 drawDecoration() . . . 1745
17.668.4.3 setBracketBrush() . . . 1745
17.668.4.4 setBracketHeight() . . . 1745

17.668.4.5 setBracketPen() . . . 1745
17.668.4.6 setBracketStyle() . . . 1746
17.668.4.7 setBracketWidth() . . . 1746
17.668.4.8 setTangentAverage() . . . 1746
17.668.4.9 setTangentToData() . . . 1746
17.669 QCPSelectionRect Class Reference . . . 1746
17.669.1 Detailed Description . . . 1749
17.669.2 Constructor & Destructor Documentation . . . 1749
17.669.2.1 QCPSelectionRect() . . . 1749
17.669.3 Member Function Documentation . . . 1749
17.669.3.1 accepted . . . 1749
17.669.3.2 applyDefaultAntialiasingHint() . . . 1750
17.669.3.3 cancel() . . . 1750
17.669.3.4 canceled . . . 1750
17.669.3.5 changed . . . 1750
17.669.3.6 draw() . . . 1750
17.669.3.7 isActive() . . . 1750
17.669.3.8 range() . . . 1750
17.669.3.9 setBrush() . . . 1751
17.669.3.10 setPen() . . . 1751
17.669.3.11 started . . . 1751
17.670 QCPStatisticalBox Class Reference . . . 1751
17.670.1 Detailed Description . . . 1756
17.670.2 Changing the appearance . . . 1756
17.670.3 Usage . . . 1756
17.670.4 Constructor & Destructor Documentation . . . 1757
17.670.4.1 QCPStatisticalBox() . . . 1757
17.670.5 Member Function Documentation . . . 1757
17.670.5.1 addData() [1/2] . . . 1757
17.670.5.2 addData() [2/2] . . . 1757
17.670.5.3 data() . . . 1757
17.670.5.4 draw() . . . 1758
17.670.5.5 drawLegendIcon() . . . 1758
17.670.5.6 drawStatisticalBox() . . . 1758
17.670.5.7 getKeyRange() . . . 1758
17.670.5.8 getValueRange() . . . 1758
17.670.5.9 selectTest() . . . 1759
17.670.5.10 selectTestRect() . . . 1759
17.670.5.11 setData() [1/2] . . . 1759
17.670.5.12 setData() [2/2] . . . 1760
17.670.5.13 setMedianPen() . . . 1760
17.670.5.14 setOutlierStyle() . . . 1760

17.670.5.15 setWhiskerAntialiased() . . . 1760
17.670.5.16 setWhiskerBarPen() . . . 1760
17.670.5.17 setWhiskerPen() . . . 1760
17.670.5.18 setWhiskerWidth() . . . 1761
17.670.5.19 setWidth() . . . 1761
17.671 QCPStatisticalBoxData Class Reference . . . 1761
17.671.1 Detailed Description . . . 1762
17.671.2 Constructor & Destructor Documentation . . . 1762
17.671.2.1 QCPStatisticalBoxData() [1/2] . . . 1762
17.671.2.2 QCPStatisticalBoxData() [2/2] . . . 1762
17.671.3 Member Function Documentation . . . 1762
17.671.3.1 fromSortKey() . . . 1762
17.671.3.2 mainKey() . . . 1763
17.671.3.3 mainValue() . . . 1763
17.671.3.4 sortKey() . . . 1763
17.671.3.5 sortKeyIsMainKey() . . . 1763
17.671.3.6 valueRange() . . . 1763
17.672 QCPTextElement Class Reference . . . 1763
17.672.1 Detailed Description . . . 1767
17.672.2 Constructor & Destructor Documentation . . . 1767
17.672.2.1 QCPTextElement() [1/5] . . . 1767
17.672.2.2 QCPTextElement() [2/5] . . . 1768
17.672.2.3 QCPTextElement() [3/5] . . . 1768
17.672.2.4 QCPTextElement() [4/5] . . . 1768
17.672.2.5 QCPTextElement() [5/5] . . . 1768
17.672.3 Member Function Documentation . . . 1768
17.672.3.1 applyDefaultAntialiasingHint() . . . 1768
17.672.3.2 clicked . . . 1768
17.672.3.3 deselectEvent() . . . 1769
17.672.3.4 doubleClicked . . . 1769
17.672.3.5 draw() . . . 1769
17.672.3.6 maximumOuterSizeHint() . . . 1769
17.672.3.7 minimumOuterSizeHint() . . . 1769
17.672.3.8 mouseDoubleClickEvent() . . . 1769
17.672.3.9 mousePressEvent() . . . 1770
17.672.3.10 mouseReleaseEvent() . . . 1770
17.672.3.11 selectEvent() . . . 1770
17.672.3.12 selectionChanged . . . 1770
17.672.3.13 selectTest() . . . 1770
17.672.3.14 setFont() . . . 1770
17.672.3.15 setSelectable() . . . 1771
17.672.3.16 setSelected() . . . 1771

17.672.3.17 setSelectedFont() . . . 1771
17.672.3.18 setSelectedTextColor() . . . 1771
17.672.3.19 setText() . . . 1771
17.672.3.20 setTextColor() . . . 1771
17.672.3.21 setTextFlags() . . . 1772
17.673 QCPVector2D Class Reference . . . 1772
17.673.1 Detailed Description . . . 1773
17.673.2 Constructor & Destructor Documentation . . . 1773
17.673.2.1 QCPVector2D() [1/4] . . . 1773
17.673.2.2 QCPVector2D() [2/4] . . . 1773
17.673.2.3 QCPVector2D() [3/4] . . . 1773
17.673.2.4 QCPVector2D() [4/4] . . . 1773
17.673.3 Member Function Documentation . . . 1774
17.673.3.1 angle() . . . 1774
17.673.3.2 distanceSquaredToLine() [1/2] . . . 1774
17.673.3.3 distanceSquaredToLine() [2/2] . . . 1774
17.673.3.4 distanceToStraightLine() . . . 1774
17.673.3.5 dot() . . . 1774
17.673.3.6 isNull() . . . 1774
17.673.3.7 length() . . . 1775
17.673.3.8 lengthSquared() . . . 1775
17.673.3.9 normalize() . . . 1775
17.673.3.10 normalized() . . . 1775
17.673.3.11 operator*=() . . . 1775
17.673.3.12 operator+=() . . . 1775
17.673.3.13 operator-=() . . . 1775
17.673.3.14 operator/=() . . . 1775
17.673.3.15 perpendicular() . . . 1776
17.673.3.16 setX() . . . 1776
17.673.3.17 setY() . . . 1776
17.673.3.18 toPoint() . . . 1776
17.673.3.19 toPointF() . . . 1776
17.673.4 Friends And Related Symbol Documentation . . . 1776
17.673.4.1 operator<<() . . . 1776
17.674 QCustomPlot Class Reference . . . 1777
17.674.1 Detailed Description . . . 1781
17.674.2 Member Enumeration Documentation . . . 1782
17.674.2.1 LayerInsertMode . . . 1782
17.674.2.2 RefreshPriority . . . 1782
17.674.3 Constructor & Destructor Documentation . . . 1782
17.674.3.1 QCustomPlot() . . . 1782
17.674.4 Member Function Documentation . . . 1782

17.674.4.1 addGraph() . . . 1782
17.674.4.2 addLayer() . . . 1783
17.674.4.3 afterLayout . . . 1783
17.674.4.4 afterReplot . . . 1783
17.674.4.5 axisClick . . . 1784
17.674.4.6 axisDoubleClick . . . 1784
17.674.4.7 axisRect() . . . 1784
17.674.4.8 axisRectAt() . . . 1784
17.674.4.9 axisRectCount() . . . 1785
17.674.4.10 axisRects() . . . 1785
17.674.4.11 beforeReplot . . . 1785
17.674.4.12 clearGraphs() . . . 1785
17.674.4.13 clearItems() . . . 1785
17.674.4.14 clearPlottables() . . . 1785
17.674.4.15 currentLayer() . . . 1786
17.674.4.16 deselectAll() . . . 1786
17.674.4.17 graph() [1/2] . . . 1786
17.674.4.18 graph() [2/2] . . . 1786
17.674.4.19 graphCount() . . . 1786
17.674.4.20 hasInvalidatedPaintBuffers() . . . 1786
17.674.4.21 hasItem() . . . 1787
17.674.4.22 hasPlottable() . . . 1787
17.674.4.23 item() [1/2] . . . 1787
17.674.4.24 item() [2/2] . . . 1787
17.674.4.25 itemAt() [1/2] . . . 1787
17.674.4.26 itemAt() [2/2] . . . 1788
17.674.4.27 itemClick . . . 1788
17.674.4.28 itemCount() . . . 1788
17.674.4.29 itemDoubleClick . . . 1788
17.674.4.30 layer() [1/2] . . . 1788
17.674.4.31 layer() [2/2] . . . 1789
17.674.4.32 layerCount() . . . 1789
17.674.4.33 layoutElementAt() . . . 1789
17.674.4.34 legendClick . . . 1789
17.674.4.35 legendDoubleClick . . . 1789
17.674.4.36 mouseDoubleClick . . . 1790
17.674.4.37 mouseMove . . . 1790
17.674.4.38 mousePress . . . 1790
17.674.4.39 mouseRelease . . . 1790
17.674.4.40 mouseWheel . . . 1790
17.674.4.41 moveLayer() . . . 1790
17.674.4.42 plotLayout() . . . 1791

17.674.4.43 plottable() [1/2] . . . 1791
17.674.4.44 plottable() [2/2] . . . 1791
17.674.4.45 plottableAt() [1/2] . . . 1791
17.674.4.46 plottableAt() [2/2] . . . 1791
17.674.4.47 plottableClick . . . 1792
17.674.4.48 plottableCount() . . . 1792
17.674.4.49 plottableDoubleClick . . . 1792
17.674.4.50 removeGraph() [1/2] . . . 1792
17.674.4.51 removeGraph() [2/2] . . . 1792
17.674.4.52 removeItem() [1/2] . . . 1793
17.674.4.53 removeItem() [2/2] . . . 1793
17.674.4.54 removeLayer() . . . 1793
17.674.4.55 removePlottable() [1/2] . . . 1793
17.674.4.56 removePlottable() [2/2] . . . 1793
17.674.4.57 replot() . . . 1793
17.674.4.58 replotTime() . . . 1794
17.674.4.59 rescaleAxes() . . . 1794
17.674.4.60 saveBmp() . . . 1794
17.674.4.61 saveJpg() . . . 1795
17.674.4.62 savePdf() . . . 1796
17.674.4.63 savePng() . . . 1796
17.674.4.64 saveRastered() . . . 1797
17.674.4.65 selectedAxes() . . . 1798
17.674.4.66 selectedGraphs() . . . 1798
17.674.4.67 selectedItems() . . . 1798
17.674.4.68 selectedLegends() . . . 1798
17.674.4.69 selectedPlottables() . . . 1798
17.674.4.70 selectionChangedByUser . . . 1799
17.674.4.71 selectionRect() . . . 1799
17.674.4.72 setAntialiasedElement() . . . 1799
17.674.4.73 setAntialiasedElements() . . . 1799
17.674.4.74 setAutoAddPlottableToLegend() . . . 1799
17.674.4.75 setBackground() [1/3] . . . 1800
17.674.4.76 setBackground() [2/3] . . . 1800
17.674.4.77 setBackground() [3/3] . . . 1800
17.674.4.78 setBackgroundScaled() . . . 1800
17.674.4.79 setBackgroundScaledMode() . . . 1801
17.674.4.80 setBufferDevicePixelRatio() . . . 1801
17.674.4.81 setCurrentLayer() [1/2] . . . 1801
17.674.4.82 setCurrentLayer() [2/2] . . . 1801
17.674.4.83 setInteraction() . . . 1801
17.674.4.84 setInteractions() . . . 1802

17.674.4.85 setMultiSelectModifier() . . . 1802
17.674.4.86 setNoAntialiasingOnDrag() . . . 1803
17.674.4.87 setNotAntialiasedElement() . . . 1803
17.674.4.88 setNotAntialiasedElements() . . . 1803
17.674.4.89 setOpenGl() . . . 1803
17.674.4.90 setPlottingHint() . . . 1804
17.674.4.91 setPlottingHints() . . . 1804
17.674.4.92 setSelectionRect() . . . 1804
17.674.4.93 setSelectionRectMode() . . . 1804
17.674.4.94 setSelectionTolerance() . . . 1805
17.674.4.95 setViewport() . . . 1805
17.674.4.96 toPainter() . . . 1805
17.674.4.97 toPixmap() . . . 1806
17.674.5 Member Data Documentation . . . 1806
17.674.5.1 legend . . . 1806
17.674.5.2 xAxis . . . 1806
17.674.5.3 xAxis2 . . . 1806
17.674.5.4 yAxis . . . 1806
17.674.5.5 yAxis2 . . . 1807
17.675 QtAbstractEditorFactory< PropertyManager > Class Template Reference . . . 1807
17.675.1 Constructor & Destructor Documentation . . . 1808
17.675.1.1 QtAbstractEditorFactory() . . . 1808
17.675.2 Member Function Documentation . . . 1809
17.675.2.1 addPropertyManager() . . . 1809
17.675.2.2 connectPropertyManager() . . . 1809
17.675.2.3 createEditor() [1/2] . . . 1809
17.675.2.4 createEditor() [2/2] . . . 1809
17.675.2.5 disconnectPropertyManager() . . . 1810
17.675.2.6 managerDestroyed() . . . 1810
17.675.2.7 propertyManager() . . . 1810
17.675.2.8 propertyManagers() . . . 1810
17.675.2.9 removePropertyManager() . . . 1810
17.676 QtAbstractEditorFactoryBase Class Reference . . . 1811
17.676.1 Constructor & Destructor Documentation . . . 1812
17.676.1.1 QtAbstractEditorFactoryBase() . . . 1812
17.676.2 Member Function Documentation . . . 1812
17.676.2.1 createEditor() . . . 1812
17.677 QtAbstractPropertyBrowser Class Reference . . . 1813
17.677.1 Constructor & Destructor Documentation . . . 1814
17.677.1.1 QtAbstractPropertyBrowser() . . . 1814
17.677.1.2 ~QtAbstractPropertyBrowser() . . . 1814
17.677.2 Member Function Documentation . . . 1814

17.677.2.1 addProperty . . . 1814
17.677.2.2 clear() . . . 1814
17.677.2.3 createEditor() . . . 1815
17.677.2.4 currentItem() . . . 1815
17.677.2.5 currentItemChanged . . . 1815
17.677.2.6 insertProperty . . . 1815
17.677.2.7 itemChanged() . . . 1815
17.677.2.8 itemInserted() . . . 1816
17.677.2.9 itemRemoved() . . . 1816
17.677.2.10 items() . . . 1816
17.677.2.11 properties() . . . 1816
17.677.2.12 removeProperty . . . 1817
17.677.2.13 setCurrentItem() . . . 1817
17.677.2.14 setFactoryForManager() . . . 1817
17.677.2.15 topLevelItem() . . . 1817
17.677.2.16 topLevelItems() . . . 1817
17.677.2.17 unsetFactoryForManager() . . . 1818
17.678 QtAbstractPropertyBrowserPrivate Class Reference . . . 1818
17.679 QtAbstractPropertyManager Class Reference . . . 1819
17.679.1 Constructor & Destructor Documentation . . . 1820
17.679.1.1 QtAbstractPropertyManager() . . . 1820
17.679.1.2 ~QtAbstractPropertyManager() . . . 1820
17.679.2 Member Function Documentation . . . 1821
17.679.2.1 addProperty() . . . 1821
17.679.2.2 clear() . . . 1821
17.679.2.3 createProperty() . . . 1821
17.679.2.4 hasValue() . . . 1821
17.679.2.5 initializeProperty() . . . 1821
17.679.2.6 properties() . . . 1822
17.679.2.7 propertyChanged . . . 1822
17.679.2.8 propertyDestroyed . . . 1822
17.679.2.9 propertyInserted . . . 1822
17.679.2.10 propertyRemoved . . . 1823
17.679.2.11 uninitializeProperty() . . . 1823
17.679.2.12 valueIcon() . . . 1823
17.679.2.13 valueText() . . . 1823
17.680 QtAbstractPropertyManagerPrivate Class Reference . . . 1824
17.681 QtBoolEdit Class Reference . . . 1824
17.682 QtBoolPropertyManager Class Reference . . . 1825
17.682.1 Constructor & Destructor Documentation . . . 1827
17.682.1.1 QtBoolPropertyManager() . . . 1827
17.682.1.2 ~QtBoolPropertyManager() . . . 1827

17.682.2 Member Function Documentation . . . 1827
17.682.2.1 initializeProperty() . . . 1827
17.682.2.2 setValue . . . 1827
17.682.2.3 uninitializeProperty() . . . 1827
17.682.2.4 value() . . . 1827
17.682.2.5 valueChanged . . . 1828
17.682.2.6 valueIcon() . . . 1828
17.682.2.7 valueText() . . . 1828
17.683 QtBoolPropertyManagerPrivate Class Reference . . . 1828
17.684 QtBrowserItem Class Reference . . . 1828
17.684.1 Member Function Documentation . . . 1829
17.684.1.1 browser() . . . 1829
17.684.1.2 children() . . . 1829
17.684.1.3 parent() . . . 1829
17.684.1.4 property() . . . 1829
17.685 QtBrowserItemPrivate Class Reference . . . 1829
17.686 QtButtonPropertyBrowser Class Reference . . . 1830
17.686.1 Constructor & Destructor Documentation . . . 1832
17.686.1.1 QtButtonPropertyBrowser() . . . 1832
17.686.1.2 ~QtButtonPropertyBrowser() . . . 1832
17.686.2 Member Function Documentation . . . 1832
17.686.2.1 collapsed . . . 1832
17.686.2.2 expanded . . . 1832
17.686.2.3 isExpanded() . . . 1832
17.686.2.4 itemChanged() . . . 1833
17.686.2.5 itemInserted() . . . 1833
17.686.2.6 itemRemoved() . . . 1833
17.686.2.7 setExpanded() . . . 1833
17.687 QtButtonPropertyBrowserPrivate Class Reference . . . 1833
17.688 QtCharEdit Class Reference . . . 1834
17.689 QtCharEditorFactory Class Reference . . . 1835
17.689.1 Constructor & Destructor Documentation . . . 1836
17.689.1.1 QtCharEditorFactory() . . . 1836
17.689.1.2 ~QtCharEditorFactory() . . . 1836
17.690 QtCharEditorFactoryPrivate Class Reference . . . 1837
17.691 QtCharPropertyManager Class Reference . . . 1838
17.691.1 Constructor & Destructor Documentation . . . 1839
17.691.1.1 QtCharPropertyManager() . . . 1839
17.691.1.2 ~QtCharPropertyManager() . . . 1839
17.691.2 Member Function Documentation . . . 1839
17.691.2.1 initializeProperty() . . . 1839
17.691.2.2 setValue . . . 1840

17.691.2.3 uninitializeProperty() . . . 1840
17.691.2.4 value() . . . 1840
17.691.2.5 valueChanged . . . 1840
17.691.2.6 valueText() . . . 1840
17.692 QtCharPropertyManagerPrivate Class Reference . . . 1840
17.693 QtCheckBoxFactory Class Reference . . . 1841
17.693.1 Constructor & Destructor Documentation . . . 1842
17.693.1.1 QtCheckBoxFactory() . . . 1842
17.693.1.2 ~QtCheckBoxFactory() . . . 1842
17.694 QtCheckBoxFactoryPrivate Class Reference . . . 1843
17.695 QtColorEditorFactory Class Reference . . . 1844
17.695.1 Constructor & Destructor Documentation . . . 1846
17.695.1.1 QtColorEditorFactory() . . . 1846
17.695.1.2 ~QtColorEditorFactory() . . . 1846
17.696 QtColorEditorFactoryPrivate Class Reference . . . 1846
17.697 QtColorEditWidget Class Reference . . . 1848
17.698 QtColorPropertyManager Class Reference . . . 1849
17.698.1 Constructor & Destructor Documentation . . . 1850
17.698.1.1 QtColorPropertyManager() . . . 1850
17.698.1.2 ~QtColorPropertyManager() . . . 1850
17.698.2 Member Function Documentation . . . 1850
17.698.2.1 initializeProperty() . . . 1850
17.698.2.2 setValue . . . 1850
17.698.2.3 subIntPropertyManager() . . . 1851
17.698.2.4 uninitializeProperty() . . . 1851
17.698.2.5 value() . . . 1851
17.698.2.6 valueChanged . . . 1851
17.698.2.7 valueIcon() . . . 1851
17.698.2.8 valueText() . . . 1851
17.699 QtColorPropertyManagerPrivate Class Reference . . . 1852
17.700 QtCursorDatabase Class Reference . . . 1853
17.701 QtCursorEditorFactory Class Reference . . . 1853
17.701.1 Constructor & Destructor Documentation . . . 1855
17.701.1.1 QtCursorEditorFactory() . . . 1855
17.701.1.2 ~QtCursorEditorFactory() . . . 1855
17.702 QtCursorEditorFactoryPrivate Class Reference . . . 1855
17.703 QtCursorPropertyManager Class Reference . . . 1856
17.703.1 Constructor & Destructor Documentation . . . 1857
17.703.1.1 QtCursorPropertyManager() . . . 1857
17.703.1.2 ~QtCursorPropertyManager() . . . 1857
17.703.2 Member Function Documentation . . . 1858
17.703.2.1 initializeProperty() . . . 1858

17.703.2.2 setValue . . . 1858
17.703.2.3 uninitializeProperty() . . . 1858
17.703.2.4 value() . . . 1858
17.703.2.5 valueChanged . . . 1858
17.703.2.6 valueIcon() . . . 1858
17.703.2.7 valueText() . . . 1859
17.704 QtCursorPropertyManagerPrivate Class Reference . . . 1859
17.705 QtDateEdit Class Reference . . . 1859
17.706 QtDateEditFactory Class Reference . . . 1860
17.706.1 Constructor & Destructor Documentation . . . 1862
17.706.1.1 QtDateEditFactory() . . . 1862
17.706.1.2 ~QtDateEditFactory() . . . 1862
17.707 QtDateEditFactoryPrivate Class Reference . . . 1862
17.708 QtDatePropertyManager Class Reference . . . 1864
17.708.1 Constructor & Destructor Documentation . . . 1865
17.708.1.1 QtDatePropertyManager() . . . 1865
17.708.1.2 ~QtDatePropertyManager() . . . 1865
17.708.2 Member Function Documentation . . . 1865
17.708.2.1 initializeProperty() . . . 1865
17.708.2.2 maximum() . . . 1866
17.708.2.3 minimum() . . . 1866
17.708.2.4 rangeChanged . . . 1866
17.708.2.5 setMaximum . . . 1866
17.708.2.6 setMinimum . . . 1866
17.708.2.7 setRange . . . 1867
17.708.2.8 setValue . . . 1867
17.708.2.9 uninitializeProperty() . . . 1867
17.708.2.10 value() . . . 1867
17.708.2.11 valueChanged . . . 1867
17.708.2.12 valueText() . . . 1868
17.709 QtDatePropertyManagerPrivate Class Reference . . . 1868
17.710 QtDateTimeEdit Class Reference . . . 1868
17.711 QtDateTimeEditFactory Class Reference . . . 1870
17.711.1 Constructor & Destructor Documentation . . . 1871
17.711.1.1 QtDateTimeEditFactory() . . . 1871
17.711.1.2 ~QtDateTimeEditFactory() . . . 1871
17.712 QtDateTimeEditFactoryPrivate Class Reference . . . 1872
17.713 QtDateTimePropertyManager Class Reference . . . 1873
17.713.1 Constructor & Destructor Documentation . . . 1874
17.713.1.1 QtDateTimePropertyManager() . . . 1874
17.713.1.2 ~QtDateTimePropertyManager() . . . 1874
17.713.2 Member Function Documentation . . . 1874

17.713.2.1 initializeProperty() . . . 1874
17.713.2.2 setValue . . . 1875
17.713.2.3 uninitializeProperty() . . . 1875
17.713.2.4 value() . . . 1875
17.713.2.5 valueChanged . . . 1875
17.713.2.6 valueText() . . . 1875
17.714 QtDateTimePropertyManagerPrivate Class Reference . . . 1875
17.715 QtDoubleEdit Class Reference . . . 1876
17.716 QtDoublePropertyManager Class Reference . . . 1877
17.716.1 Constructor & Destructor Documentation . . . 1878
17.716.1.1 QtDoublePropertyManager() . . . 1878
17.716.1.2 ~QtDoublePropertyManager() . . . 1878
17.716.2 Member Function Documentation . . . 1879
17.716.2.1 decimals() . . . 1879
17.716.2.2 decimalsChanged . . . 1879
17.716.2.3 initializeProperty() . . . 1879
17.716.2.4 maximum() . . . 1879
17.716.2.5 minimum() . . . 1879
17.716.2.6 rangeChanged . . . 1879
17.716.2.7 setDecimals . . . 1880
17.716.2.8 setMaximum . . . 1880
17.716.2.9 setMinimum . . . 1880
17.716.2.10 setRange . . . 1880
17.716.2.11 setSingleStep . . . 1880
17.716.2.12 setValue . . . 1881
17.716.2.13 singleStep() . . . 1881
17.716.2.14 singleStepChanged . . . 1881
17.716.2.15 uninitializeProperty() . . . 1881
17.716.2.16 value() . . . 1881
17.716.2.17 valueChanged . . . 1882
17.716.2.18 valueText() . . . 1882
17.717 QtDoublePropertyManagerPrivate Class Reference . . . 1882
17.718 QtDoubleSpinBoxFactory Class Reference . . . 1883
17.718.1 Constructor & Destructor Documentation . . . 1884
17.718.1.1 QtDoubleSpinBoxFactory() . . . 1884
17.718.1.2 ~QtDoubleSpinBoxFactory() . . . 1884
17.719 QtDoubleSpinBoxFactoryPrivate Class Reference . . . 1885
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17.720.1 Constructor & Destructor Documentation . . . 1888
17.720.1.1 QtEnumEditorFactory() . . . 1888
17.720.1.2 ~QtEnumEditorFactory() . . . 1888
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17.722.1 Constructor & Destructor Documentation . . . 1891
17.722.1.1 QtEnumPropertyManager() . . . 1891
17.722.1.2 ~QtEnumPropertyManager() . . . 1891
17.722.2 Member Function Documentation . . . 1891
17.722.2.1 enumIcons() . . . 1891
17.722.2.2 enumIconsChanged . . . 1892
17.722.2.3 enumNames() . . . 1892
17.722.2.4 enumNamesChanged . . . 1892
17.722.2.5 initializeProperty() . . . 1892
17.722.2.6 setEnumIcons . . . 1892
17.722.2.7 setEnumNames . . . 1892
17.722.2.8 setValue . . . 1893
17.722.2.9 uninitializeProperty() . . . 1893
17.722.2.10 value() . . . 1893
17.722.2.11 valueChanged . . . 1893
17.722.2.12 valueIcon() . . . 1893
17.722.2.13 valueText() . . . 1893
17.723 QtEnumPropertyManagerPrivate Class Reference . . . 1894
17.724 QtEnumPropertyType Class Reference . . . 1894
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17.725.1 Constructor & Destructor Documentation . . . 1896
17.725.1.1 QtFlagPropertyManager() . . . 1896
17.725.1.2 ~QtFlagPropertyManager() . . . 1896
17.725.2 Member Function Documentation . . . 1896
17.725.2.1 flagNames() . . . 1896
17.725.2.2 flagNamesChanged . . . 1896
17.725.2.3 initializeProperty() . . . 1896
17.725.2.4 setFlagNames . . . 1896
17.725.2.5 setValue . . . 1897
17.725.2.6 subBoolPropertyManager() . . . 1897
17.725.2.7 uninitializeProperty() . . . 1897
17.725.2.8 value() . . . 1897
17.725.2.9 valueChanged . . . 1897
17.725.2.10 valueText() . . . 1897
17.726 QtFlagPropertyManagerPrivate Class Reference . . . 1898
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17.728.1 Constructor & Destructor Documentation . . . 1900
17.728.1.1 QtFontEditorFactory() . . . 1900
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17.729 QtFontEditorFactoryPrivate Class Reference . . . 1901

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17.731.1.2 ~QtFontPropertyManager() . . . 1904
17.731.2 Member Function Documentation . . . 1904
17.731.2.1 initializeProperty() . . . 1904
17.731.2.2 setValue . . . 1904
17.731.2.3 subBoolPropertyManager() . . . 1905
17.731.2.4 subEnumPropertyManager() . . . 1905
17.731.2.5 subIntPropertyManager() . . . 1905
17.731.2.6 uninitializeProperty() . . . 1905
17.731.2.7 value() . . . 1905
17.731.2.8 valueChanged . . . 1906
17.731.2.9 valueIcon() . . . 1906
17.731.2.10 valueText() . . . 1906
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17.733.2 Member Function Documentation . . . 1909
17.733.2.1 itemChanged() . . . 1909
17.733.2.2 itemInserted() . . . 1909
17.733.2.3 itemRemoved() . . . 1909
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17.735.1.2 ~QtGroupPropertyManager() . . . 1911
17.735.2 Member Function Documentation . . . 1911
17.735.2.1 hasValue() . . . 1911
17.735.2.2 initializeProperty() . . . 1911
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17.736 QtGroupPropertyType Class Reference . . . 1912
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17.738.1.2 ~QtIntPropertyManager() . . . 1915
17.738.2 Member Function Documentation . . . 1915
17.738.2.1 initializeProperty() . . . 1915

17.738.2.2 maximum() . . . 1915
17.738.2.3 minimum() . . . 1915
17.738.2.4 rangeChanged . . . 1915
17.738.2.5 setMaximum . . . 1915
17.738.2.6 setMinimum . . . 1916
17.738.2.7 setRange . . . 1916
17.738.2.8 setSingleStep . . . 1916
17.738.2.9 setValue . . . 1916
17.738.2.10 singleStep() . . . 1916
17.738.2.11 singleStepChanged . . . 1917
17.738.2.12 uninitializeProperty() . . . 1917
17.738.2.13 value() . . . 1917
17.738.2.14 valueChanged . . . 1917
17.738.2.15 valueText() . . . 1917
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17.742.1.2 ~QtKeySequencePropertyManager() . . . 1923
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17.742.2.1 initializeProperty() . . . 1923
17.742.2.2 setValue . . . 1923
17.742.2.3 uninitializeProperty() . . . 1924
17.742.2.4 value() . . . 1924
17.742.2.5 valueChanged . . . 1924
17.742.2.6 valueText() . . . 1924
17.743 QtKeySequencePropertyManagerPrivate Class Reference . . . 1924
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17.745 QtLineEditFactoryPrivate Class Reference . . . 1927
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17.746.1.2 ~QtLocalePropertyManager() . . . 1929
17.746.2 Member Function Documentation . . . 1929

17.746.2.1 initializeProperty() . . . 1929
17.746.2.2 setValue . . . 1930
17.746.2.3 subEnumPropertyManager() . . . 1930
17.746.2.4 uninitializeProperty() . . . 1930
17.746.2.5 value() . . . 1930
17.746.2.6 valueChanged . . . 1930
17.746.2.7 valueText() . . . 1930
17.747 QtLocalePropertyManagerPrivate Class Reference . . . 1931
17.748 QtMetaEnumProvider Class Reference . . . 1931
17.749 QtMetaEnumWrapper Class Reference . . . 1932
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17.750.1.1 download() . . . 1933
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17.751.1 Constructor & Destructor Documentation . . . 1935
17.751.1.1 QtPointFPropertyManager() . . . 1935
17.751.1.2 ~QtPointFPropertyManager() . . . 1935
17.751.2 Member Function Documentation . . . 1935
17.751.2.1 decimals() . . . 1935
17.751.2.2 decimalsChanged . . . 1936
17.751.2.3 initializeProperty() . . . 1936
17.751.2.4 setDecimals . . . 1936
17.751.2.5 setValue . . . 1936
17.751.2.6 subDoublePropertyManager() . . . 1936
17.751.2.7 uninitializeProperty() . . . 1937
17.751.2.8 value() . . . 1937
17.751.2.9 valueChanged . . . 1937
17.751.2.10 valueText() . . . 1937
17.752 QtPointFPropertyManagerPrivate Class Reference . . . 1938
17.753 QtPointPropertyManager Class Reference . . . 1939
17.753.1 Constructor & Destructor Documentation . . . 1940
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17.753.1.2 ~QtPointPropertyManager() . . . 1940
17.753.2 Member Function Documentation . . . 1940
17.753.2.1 initializeProperty() . . . 1940
17.753.2.2 setValue . . . 1940
17.753.2.3 subIntPropertyManager() . . . 1941
17.753.2.4 uninitializeProperty() . . . 1941
17.753.2.5 value() . . . 1941
17.753.2.6 valueChanged . . . 1941
17.753.2.7 valueText() . . . 1941
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17.755.1 Constructor & Destructor Documentation . . . 1944
17.755.1.1 ~QtProperty() . . . 1944
17.755.1.2 QtProperty() . . . 1944
17.755.2 Member Function Documentation . . . 1944
17.755.2.1 addSubProperty() . . . 1944
17.755.2.2 descriptionToolTip() . . . 1944
17.755.2.3 hasValue() . . . 1944
17.755.2.4 insertSubProperty() . . . 1945
17.755.2.5 isEnabled() . . . 1945
17.755.2.6 isModified() . . . 1945
17.755.2.7 propertyManager() . . . 1945
17.755.2.8 propertyName() . . . 1945
17.755.2.9 removeSubProperty() . . . 1945
17.755.2.10 setDescriptionToolTip() . . . 1946
17.755.2.11 setEnabled() . . . 1946
17.755.2.12 setModified() . . . 1946
17.755.2.13 setPropertyName() . . . 1946
17.755.2.14 setStatusTip() . . . 1946
17.755.2.15 setValueToolTip() . . . 1946
17.755.2.16 setWhatsThis() . . . 1947
17.755.2.17 statusTip() . . . 1947
17.755.2.18 subProperties() . . . 1947
17.755.2.19 valueIcon() . . . 1947
17.755.2.20 valueText() . . . 1947
17.755.2.21 valueToolTip() . . . 1948
17.755.2.22 whatsThis() . . . 1948
17.756 QtPropertyBrowserUtils Class Reference . . . 1948
17.757 QtPropertyEditorDelegate Class Reference . . . 1949
17.758 QtPropertyEditorView Class Reference . . . 1950
17.759 QtPropertyPrivate Class Reference . . . 1951
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17.760.1 Constructor & Destructor Documentation . . . 1953
17.760.1.1 QtRectFPropertyManager() . . . 1953
17.760.1.2 ~QtRectFPropertyManager() . . . 1953
17.760.2 Member Function Documentation . . . 1953
17.760.2.1 constraint() . . . 1953
17.760.2.2 constraintChanged . . . 1954
17.760.2.3 decimals() . . . 1954
17.760.2.4 decimalsChanged . . . 1954
17.760.2.5 initializeProperty() . . . 1954
17.760.2.6 setConstraint . . . 1954

17.760.2.7 setDecimals . . . 1955
17.760.2.8 setValue . . . 1955
17.760.2.9 subDoublePropertyManager() . . . 1955
17.760.2.10 uninitializeProperty() . . . 1955
17.760.2.11 value() . . . 1955
17.760.2.12 valueChanged . . . 1955
17.760.2.13 valueText() . . . 1956
17.761 QtRectFPropertyManagerPrivate Class Reference . . . 1956
17.762 QtRectPropertyManager Class Reference . . . 1957
17.762.1 Constructor & Destructor Documentation . . . 1959
17.762.1.1 QtRectPropertyManager() . . . 1959
17.762.1.2 ~QtRectPropertyManager() . . . 1959
17.762.2 Member Function Documentation . . . 1959
17.762.2.1 constraint() . . . 1959
17.762.2.2 constraintChanged . . . 1959
17.762.2.3 initializeProperty() . . . 1959
17.762.2.4 setConstraint . . . 1959
17.762.2.5 setValue . . . 1960
17.762.2.6 subIntPropertyManager() . . . 1960
17.762.2.7 uninitializeProperty() . . . 1960
17.762.2.8 value() . . . 1960
17.762.2.9 valueChanged . . . 1960
17.762.2.10 valueText() . . . 1961
17.763 QtRectPropertyManagerPrivate Class Reference . . . 1961
17.764 QtScrollBarFactory Class Reference . . . 1962
17.764.1 Constructor & Destructor Documentation . . . 1964
17.764.1.1 QtScrollBarFactory() . . . 1964
17.764.1.2 ~QtScrollBarFactory() . . . 1964
17.765 QtScrollBarFactoryPrivate Class Reference . . . 1964
17.766 QtSizeFPropertyManager Class Reference . . . 1966
17.766.1 Constructor & Destructor Documentation . . . 1967
17.766.1.1 QtSizeFPropertyManager() . . . 1967
17.766.1.2 ~QtSizeFPropertyManager() . . . 1967
17.766.2 Member Function Documentation . . . 1967
17.766.2.1 decimals() . . . 1967
17.766.2.2 decimalsChanged . . . 1968
17.766.2.3 initializeProperty() . . . 1968
17.766.2.4 maximum() . . . 1968
17.766.2.5 minimum() . . . 1968
17.766.2.6 rangeChanged . . . 1968
17.766.2.7 setDecimals . . . 1969
17.766.2.8 setMaximum . . . 1969

17.766.2.9 setMinimum . . . 1969
17.766.2.10 setRange . . . 1969
17.766.2.11 setValue . . . 1969
17.766.2.12 subDoublePropertyManager() . . . 1970
17.766.2.13 uninitializeProperty() . . . 1970
17.766.2.14 value() . . . 1970
17.766.2.15 valueChanged . . . 1970
17.766.2.16 valueText() . . . 1970
17.767 QtSizeFPropertyManagerPrivate Class Reference . . . 1971
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17.768.1 Constructor & Destructor Documentation . . . 1973
17.768.1.1 QtSizePolicyPropertyManager() . . . 1973
17.768.1.2 ~QtSizePolicyPropertyManager() . . . 1973
17.768.2 Member Function Documentation . . . 1973
17.768.2.1 initializeProperty() . . . 1973
17.768.2.2 setValue . . . 1973
17.768.2.3 subEnumPropertyManager() . . . 1974
17.768.2.4 subIntPropertyManager() . . . 1974
17.768.2.5 uninitializeProperty() . . . 1974
17.768.2.6 value() . . . 1974
17.768.2.7 valueChanged . . . 1974
17.768.2.8 valueText() . . . 1974
17.769 QtSizePolicyPropertyManagerPrivate Class Reference . . . 1975
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17.770.1 Constructor & Destructor Documentation . . . 1977
17.770.1.1 QtSizePropertyManager() . . . 1977
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17.770.2 Member Function Documentation . . . 1977
17.770.2.1 initializeProperty() . . . 1977
17.770.2.2 maximum() . . . 1978
17.770.2.3 minimum() . . . 1978
17.770.2.4 rangeChanged . . . 1978
17.770.2.5 setMaximum . . . 1978
17.770.2.6 setMinimum . . . 1978
17.770.2.7 setRange . . . 1979
17.770.2.8 setValue . . . 1979
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17.770.2.10 uninitializeProperty() . . . 1979
17.770.2.11 value() . . . 1979
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17.777.2 Member Function Documentation . . . 1990
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17.777.2.3 regExpChanged . . . 1991
17.777.2.4 setRegExp . . . 1991
17.777.2.5 setValue . . . 1991
17.777.2.6 uninitializeProperty() . . . 1991
17.777.2.7 value() . . . 1991
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17.777.2.9 valueText() . . . 1992
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17.782.2.1 initializeProperty() . . . 1998
17.782.2.2 setValue . . . 1999
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17.784.3.10 itemRemoved() . . . 2005
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17.789.2.5 attributeValue() . . . 2015
17.789.2.6 createProperty() . . . 2016
17.789.2.7 enumTypeId() . . . 2016
17.789.2.8 flagTypeId() . . . 2016
17.789.2.9 groupTypeId() . . . 2016
17.789.2.10 hasValue() . . . 2016
17.789.2.11 iconMapTypeId() . . . 2017
17.789.2.12 initializeProperty() . . . 2017
17.789.2.13 isPropertyTypeSupported() . . . 2017
17.789.2.14 propertyType() . . . 2017
17.789.2.15 setAttribute . . . 2017
17.789.2.16 setValue . . . 2018
17.789.2.17 uninitializeProperty() . . . 2018
17.789.2.18 value() . . . 2018
17.789.2.19 valueChanged . . . 2018
17.789.2.20 valueIcon() . . . 2018
17.789.2.21 valueText() . . . 2019
17.789.2.22 valueType() [1/2] . . . 2019
17.789.2.23 valueType() [2/2] . . . 2019
17.789.2.24 variantProperty() . . . 2019
17.790 QtVariantPropertyManagerPrivate Class Reference . . . 2019
17.791 QtVariantPropertyPrivate Class Reference . . . 2021
17.792 Quadrature_if Class Reference . . . 2021
17.792.1 Detailed Description . . . 2021
17.793 Queue Class Reference . . . 2022
17.793.1 Detailed Description . . . 2026
17.793.2 Member Function Documentation . . . 2026
17.793.2.1 _check() . . . 2026
17.793.2.2 _createInternalStatisticReporters() . . . 2026
17.793.2.3 _getParserChangesInformation() . . . 2026
17.793.2.4 _initBetweenReplications() . . . 2026
17.793.2.5 _loadInstance() . . . 2027
17.793.2.6 _saveInstance() . . . 2027
17.793.2.7 show() . . . 2027

17.794 QueueableItem Class Reference . . . 2027
17.795 testing::internal::Random Class Reference . . . 2028
17.796 testing::internal::RangeGenerator< T, IncrementT > Class Template Reference . . . 2028
17.796.1 Member Function Documentation . . . 2029
17.796.1.1 Begin() . . . 2029
17.796.1.2 End() . . . 2029
17.797 testing::internal::RawBytesPrinter Struct Reference . . . 2029
17.798 testing::internal::RE Class Reference . . . 2030
17.799 MassActionOdeSystem::Reaction Struct Reference . . . 2030
17.800 RealIntervalStateSet Class Reference . . . 2030
17.800.1 Member Function Documentation . . . 2031
17.800.1.1 contains() . . . 2031
17.800.1.2 createDefaultState() . . . 2031
17.800.1.3 isContinuous() . . . 2031
17.800.1.4 name() . . . 2032
17.800.1.5 parseState() . . . 2032
17.801 RealState Class Reference . . . 2032
17.801.1 Member Function Documentation . . . 2033
17.801.1.1 clone() . . . 2033
17.801.1.2 equals() . . . 2033
17.801.1.3 getValue() . . . 2033
17.801.1.4 setValue() . . . 2033
17.801.1.5 toString() . . . 2033
17.801.1.6 typeName() . . . 2033
17.802 Record Class Reference . . . 2034
17.802.1 Detailed Description . . . 2039
17.802.2 Member Function Documentation . . . 2039
17.802.2.1 _check() . . . 2039
17.802.2.2 _createInternalStatisticReporters() . . . 2039
17.802.2.3 _initBetweenReplications() . . . 2039
17.802.2.4 _loadInstance() . . . 2039
17.802.2.5 _onDispatchEvent() . . . 2039
17.802.2.6 _saveInstance() . . . 2039
17.802.2.7 setDatasetDescription() . . . 2040
17.802.2.8 setDatasetName() . . . 2040
17.802.2.9 setExpressionName() . . . 2040
17.802.2.10 setRandomVariableName() . . . 2040
17.802.2.11 setVariableType() . . . 2040
17.802.2.12 show() . . . 2040
17.803 RecordingContinuousSystemComponent Class Reference . . . 2041
17.803.1 Member Function Documentation . . . 2046
17.803.1.1 _onDispatchEvent() [1/2] . . . 2046

17.803.1.2 _onDispatchEvent() [2/2] . . . 2046
17.804 RecordingLSODE Class Reference . . . 2047
17.804.1 Member Function Documentation . . . 2052
17.804.1.1 _onDispatchEvent() . . . 2052
17.805 Rectifier Class Reference . . . 2053
17.805.1 Member Function Documentation . . . 2054
17.805.1.1 main() . . . 2054
17.806 testing::internal::RelationToSourceCopy Struct Reference . . . 2054
17.807 testing::internal::RelationToSourceReference Struct Reference . . . 2054
17.808 Release Class Reference . . . 2054
17.808.1 Detailed Description . . . 2059
17.808.2 Member Function Documentation . . . 2059
17.808.2.1 _check() . . . 2059
17.808.2.2 _createEditableDataDefinitions() . . . 2060
17.808.2.3 _initBetweenReplications() . . . 2060
17.808.2.4 _loadInstance() . . . 2060
17.808.2.5 _onDispatchEvent() . . . 2060
17.808.2.6 _saveInstance() . . . 2060
17.808.2.7 show() . . . 2060
17.809 Remove Class Reference . . . 2061
17.809.1 Detailed Description . . . 2066
17.809.2 Member Function Documentation . . . 2066
17.809.2.1 _check() . . . 2066
17.809.2.2 _createEditableDataDefinitions() . . . 2066
17.809.2.3 _loadInstance() . . . 2066
17.809.2.4 _onDispatchEvent() . . . 2066
17.809.2.5 _saveInstance() . . . 2066
17.809.2.6 show() . . . 2067
17.810 Resistor Class Reference . . . 2067
17.810.1 Member Function Documentation . . . 2072
17.810.1.1 _loadInstance() . . . 2072
17.810.1.2 _saveInstance() . . . 2073
17.810.1.3 Build() . . . 2073
17.811 Resource Class Reference . . . 2073
17.811.1 Detailed Description . . . 2078
17.811.2 Constructor & Destructor Documentation . . . 2078
17.811.2.1 Resource() . . . 2078
17.811.3 Member Function Documentation . . . 2078
17.811.3.1 _check() . . . 2078
17.811.3.2 _createAttachedAttributes() . . . 2078
17.811.3.3 _initBetweenReplications() . . . 2078
17.811.3.4 _loadInstance() . . . 2078

17.811.3.5 _saveInstance() . . . 2079
17.811.3.6 show() . . . 2079
17.812 ResourceAllocationComponent Class Reference . . . 2079
17.812.1 Detailed Description . . . 2084
17.812.2 Member Function Documentation . . . 2084
17.812.2.1 _check() . . . 2084
17.812.2.2 _createEditableDataDefinitions() . . . 2084
17.812.2.3 _initBetweenReplications() . . . 2084
17.812.2.4 _loadInstance() . . . 2085
17.812.2.5 _onDispatchEvent() . . . 2085
17.812.2.6 _saveInstance() . . . 2085
17.812.2.7 show() . . . 2085
17.813 ResourceTestProbe Class Reference . . . 2085
17.814 GroProgramParser::Result Struct Reference . . . 2086
17.815 Sampler_if::RNG_Parameters Struct Reference . . . 2086
17.815.1 Detailed Description . . . 2087
17.816 RootFinder_if Class Reference . . . 2087
17.816.1 Detailed Description . . . 2087
17.817 Route Class Reference . . . 2087
17.817.1 Detailed Description . . . 2092
17.817.2 Member Function Documentation . . . 2092
17.817.2.1 _check() . . . 2092
17.817.2.2 _createAttachedAttributes() . . . 2093
17.817.2.3 _createEditableDataDefinitions() . . . 2093
17.817.2.4 _createInternalStatisticReporters() . . . 2093
17.817.2.5 _loadInstance() . . . 2093
17.817.2.6 _onDispatchEvent() . . . 2093
17.817.2.7 _saveInstance() . . . 2093
17.817.2.8 show() . . . 2093
17.818 RSimulator Class Reference . . . 2094
17.818.1 Detailed Description . . . 2099
17.818.2 Member Function Documentation . . . 2099
17.818.2.1 _check() . . . 2099
17.818.2.2 _createEditableDataDefinitions() . . . 2100
17.818.2.3 _loadInstance() . . . 2100
17.818.2.4 _onDispatchEvent() . . . 2100
17.818.2.5 _saveInstance() . . . 2100
17.818.2.6 show() . . . 2100
17.819 RSimulatorRunner Class Reference . . . 2100
17.819.1 Detailed Description . . . 2105
17.819.2 Member Function Documentation . . . 2105
17.819.2.1 _check() . . . 2105

17.819.2.2 _loadInstance() . . . 2106
17.819.2.3 _saveInstance() . . . 2106
17.819.2.4 show() . . . 2106
17.820 RungeKutta4OdeSolver Class Reference . . . 2106
17.820.1 Detailed Description . . . 2107
17.820.2 Member Function Documentation . . . 2107
17.820.2.1 advance() . . . 2107
17.821 genesys::launcher::RuntimeInstallResult Struct Reference . . . 2107
17.822 genesys::launcher::RuntimePaths Class Reference . . . 2107
17.823 genesys::launcher::RuntimeSelection Struct Reference . . . 2108
17.824 genesys::launcher::RuntimeSelector Class Reference . . . 2108
17.825 Sample Struct Reference . . . 2109
17.826 Sampler_if Class Reference . . . 2109
17.826.1 Detailed Description . . . 2110
17.826.2 Member Function Documentation . . . 2110
17.826.2.1 getRNGparameters() . . . 2110
17.826.2.2 random() . . . 2110
17.826.2.3 sampleBernoulli() . . . 2110
17.826.2.4 sampleBeta() [1/2] . . . 2111
17.826.2.5 sampleBeta() [2/2] . . . 2111
17.826.2.6 sampleBinomial() . . . 2111
17.826.2.7 sampleDiscrete() [1/2] . . . 2111
17.826.2.8 sampleDiscrete() [2/2] . . . 2111
17.826.2.9 sampleErlang() . . . 2111
17.826.2.10 sampleExponential() . . . 2112
17.826.2.11 sampleGamma() . . . 2112
17.826.2.12 sampleGeometric() . . . 2112
17.826.2.13 sampleGumbell() . . . 2112
17.826.2.14 sampleLogNormal() . . . 2112
17.826.2.15 sampleNormal() . . . 2112
17.826.2.16 sampleTriangular() . . . 2112
17.826.2.17 sampleUniform() . . . 2113
17.826.2.18 sampleWeibull() . . . 2113
17.826.2.19 setRNGparameters() . . . 2113
17.827 SamplerDefaultImpl1 Class Reference . . . 2113
17.827.1 Detailed Description . . . 2115
17.827.2 Member Function Documentation . . . 2115
17.827.2.1 getRNGparameters() . . . 2115
17.827.2.2 random() . . . 2115
17.827.2.3 sampleBernoulli() . . . 2115
17.827.2.4 sampleBeta() [1/2] . . . 2115
17.827.2.5 sampleBeta() [2/2] . . . 2115

17.827.2.6 sampleBinomial() . . . 2116
17.827.2.7 sampleDiscrete() [1/2] . . . 2116
17.827.2.8 sampleDiscrete() [2/2] . . . 2116
17.827.2.9 sampleErlang() . . . 2116
17.827.2.10 sampleExponential() . . . 2116
17.827.2.11 sampleGamma() . . . 2116
17.827.2.12 sampleGeometric() . . . 2116
17.827.2.13 sampleGumbell() . . . 2117
17.827.2.14 sampleLogNormal() . . . 2117
17.827.2.15 sampleNormal() . . . 2117
17.827.2.16 sampleTriangular() . . . 2117
17.827.2.17 sampleUniform() . . . 2117
17.827.2.18 sampleWeibull() . . . 2117
17.827.2.19 setRNGparameters() . . . 2117
17.828 ScenarioExperiment_if Class Reference . . . 2118
17.829 SceneToolController Class Reference . . . 2118
17.829.1 Detailed Description . . . 2120
17.829.2 Constructor & Destructor Documentation . . . 2120
17.829.2.1 SceneToolController() . . . 2120
17.829.3 Member Function Documentation . . . 2120
17.829.3.1 onActionDrawLineTriggered() . . . 2120
17.829.3.2 onActionShowGridTriggered() . . . 2121
17.829.3.3 onActionZoomActualSizeTriggered() . . . 2121
17.829.3.4 onActionZoomAllTriggered() . . . 2121
17.829.3.5 onHorizontalSliderZoomGraphicalValueChanged() . . . 2121
17.830 SchedulableItem Class Reference . . . 2121
17.830.1 Detailed Description . . . 2121
17.831 Schedule Class Reference . . . 2122
17.831.1 Member Function Documentation . . . 2125
17.831.1.1 _addSimulationControl() . . . 2125
17.831.1.2 _check() . . . 2126
17.831.1.3 _getParserChangesInformation() . . . 2126
17.831.1.4 _initBetweenReplications() . . . 2126
17.831.1.5 _loadInstance() . . . 2126
17.831.1.6 _saveInstance() . . . 2126
17.831.1.7 show() . . . 2126
17.832 testing::ScopedTrace Class Reference . . . 2127
17.833 Search Class Reference . . . 2127
17.833.1 Detailed Description . . . 2132
17.833.2 Member Function Documentation . . . 2132
17.833.2.1 _check() . . . 2132
17.833.2.2 _createEditableDataDefinitions() . . . 2133

17.833.2.3 _loadInstance() . . . 2133
17.833.2.4 _onDispatchEvent() . . . 2133
17.833.2.5 _saveInstance() . . . 2133
17.833.2.6 show() . . . 2133
17.834 Segment Class Reference . . . 2134
17.834.1 Detailed Description . . . 2138
17.834.2 Member Function Documentation . . . 2138
17.834.2.1 _check() . . . 2138
17.834.2.2 _loadInstance() . . . 2138
17.834.2.3 _saveInstance() . . . 2138
17.834.2.4 show() . . . 2138
17.835 SegmentStep Class Reference . . . 2138
17.835.1 Detailed Description . . . 2139
17.836 SeizableItem Class Reference . . . 2139
17.837 Seize Class Reference . . . 2140
17.837.1 Detailed Description . . . 2145
17.837.2 Member Function Documentation . . . 2145
17.837.2.1 _check() . . . 2145
17.837.2.2 _createEditableDataDefinitions() . . . 2146
17.837.2.3 _initBetweenReplications() . . . 2146
17.837.2.4 _loadInstance() . . . 2146
17.837.2.5 _onDispatchEvent() . . . 2146
17.837.2.6 _saveInstance() . . . 2146
17.837.2.7 show() . . . 2146
17.838 Separate Class Reference . . . 2147
17.838.1 Detailed Description . . . 2151
17.838.2 Member Function Documentation . . . 2152
17.838.2.1 _check() . . . 2152
17.838.2.2 _loadInstance() . . . 2152
17.838.2.3 _onDispatchEvent() . . . 2152
17.838.2.4 _saveInstance() . . . 2152
17.838.2.5 show() . . . 2152
17.839 Sequence Class Reference . . . 2153
17.839.1 Detailed Description . . . 2156
17.839.2 Member Function Documentation . . . 2157
17.839.2.1 _check() . . . 2157
17.839.2.2 _createAttachedAttributes() . . . 2157
17.839.2.3 _loadInstance() . . . 2157
17.839.2.4 _saveInstance() . . . 2157
17.839.2.5 show() . . . 2157
17.840 SequenceStep Class Reference . . . 2158
17.840.1 Member Function Documentation . . . 2159

17.840.1.1 _loadInstance() . . . 2159
17.840.1.2 _saveInstance() . . . 2159
17.841 Serializer Class Reference . . . 2159
17.841.1 Detailed Description . . . 2160
17.841.2 Member Function Documentation . . . 2160
17.841.2.1 dump() . . . 2160
17.841.2.2 for_each() . . . 2161
17.841.2.3 get() . . . 2161
17.841.2.4 load() . . . 2161
17.841.2.5 newPersistenceRecord() . . . 2161
17.841.2.6 put() . . . 2161
17.842 SessionContext Struct Reference . . . 2163
17.842.1 Detailed Description . . . 2163
17.843 SessionManager Class Reference . . . 2163
17.843.1 Detailed Description . . . 2164
17.843.2 Constructor & Destructor Documentation . . . 2164
17.843.2.1 SessionManager() . . . 2164
17.843.3 Member Function Documentation . . . 2164
17.843.3.1 createSession() . . . 2164
17.843.3.2 getSessionByToken() . . . 2164
17.844 Set Class Reference . . . 2165
17.844.1 Detailed Description . . . 2169
17.844.2 Member Function Documentation . . . 2169
17.844.2.1 _check() . . . 2169
17.844.2.2 _createAttachedAttributes() . . . 2169
17.844.2.3 _getParserChangesInformation() . . . 2169
17.844.2.4 _loadInstance() . . . 2169
17.844.2.5 _saveInstance() . . . 2170
17.844.2.6 createElementSetOfType() . . . 2170
17.844.2.7 setAllowedElementTypes() . . . 2170
17.844.2.8 setSetOfType() . . . 2170
17.844.2.9 show() . . . 2170
17.845 Signal Class Reference . . . 2171
17.845.1 Detailed Description . . . 2175
17.845.2 Member Function Documentation . . . 2175
17.845.2.1 _check() . . . 2175
17.845.2.2 _createEditableDataDefinitions() . . . 2176
17.845.2.3 _initBetweenReplications() . . . 2176
17.845.2.4 _loadInstance() . . . 2176
17.845.2.5 _onDispatchEvent() . . . 2176
17.845.2.6 _saveInstance() . . . 2176
17.845.2.7 show() . . . 2176

17.846 SignalData Class Reference . . . 2177
17.846.1 Member Function Documentation . . . 2181
17.846.1.1 _check() . . . 2181
17.846.1.2 _initBetweenReplications() . . . 2181
17.846.1.3 _loadInstance() . . . 2181
17.846.1.4 _saveInstance() . . . 2181
17.846.1.5 show() . . . 2181
17.847 GroProgramRuntime::SignalMutation Struct Reference . . . 2181
17.848 SimpleHttpServer Class Reference . . . 2182
17.848.1 Detailed Description . . . 2182
17.848.2 Constructor & Destructor Documentation . . . 2182
17.848.2.1 SimpleHttpServer() . . . 2182
17.848.3 Member Function Documentation . . . 2182
17.848.3.1 serve() . . . 2182
17.849 SimulatorSessionService::SimulationActionResult Struct Reference . . . 2183
17.849.1 Detailed Description . . . 2183
17.850 SimulationCommandController Class Reference . . . 2183
17.850.1 Detailed Description . . . 2184
17.851 SimulatorSessionService::SimulationConfigInput Struct Reference . . . 2184
17.851.1 Detailed Description . . . 2184
17.852 SimulatorSessionService::SimulationConfigResult Struct Reference . . . 2185
17.852.1 Detailed Description . . . 2185
17.853 SimulationControl Class Reference . . . 2185
17.853.1 Detailed Description . . . 2188
17.853.2 Member Function Documentation . . . 2188
17.853.2.1 getChildSimulationControls() . . . 2188
17.854 SimulationControlBool Class Reference . . . 2189
17.854.1 Member Function Documentation . . . 2191
17.854.1.1 getValue() . . . 2191
17.854.1.2 setValue() . . . 2191
17.855 SimulationControlDouble Class Reference . . . 2192
17.855.1 Member Function Documentation . . . 2194
17.855.1.1 getValue() . . . 2194
17.855.1.2 setValue() . . . 2194
17.856 SimulationControlGeneric< T > Class Template Reference . . . 2195
17.856.1 Member Function Documentation . . . 2197
17.856.1.1 getValue() . . . 2197
17.856.1.2 setValue() . . . 2197
17.857 SimulationControlGenericClass< T, M, C > Class Template Reference . . . 2198
17.857.1 Member Function Documentation . . . 2200
17.857.1.1 createObjectInstance() . . . 2200
17.857.1.2 ensureObjectInstance() . . . 2200

17.857.1.3 getChildSimulationControls() . . . 2201
17.857.1.4 getReferencedModelDataDefinition() . . . 2201
17.857.1.5 getStrValues() . . . 2201
17.857.1.6 getValue() . . . 2201
17.857.1.7 hasObjectInstance() . . . 2201
17.857.1.8 isInlineObjectProperty() . . . 2201
17.857.1.9 isModelDataDefinitionReference() . . . 2201
17.857.1.10 setValue() . . . 2201
17.857.1.11 supportsExistingObjectSelection() . . . 2202
17.857.1.12 supportsInlineExpansion() . . . 2202
17.857.1.13 supportsObjectCreation() . . . 2202
17.858 SimulationControlGenericClassNotDC< T, M, C > Class Template Reference . . . 2202
17.858.1 Member Function Documentation . . . 2205
17.858.1.1 createObjectInstance() . . . 2205
17.858.1.2 ensureObjectInstance() . . . 2205
17.858.1.3 getChildSimulationControls() . . . 2205
17.858.1.4 getValue() . . . 2205
17.858.1.5 hasObjectInstance() . . . 2205
17.858.1.6 isInlineObjectProperty() . . . 2205
17.858.1.7 setValue() . . . 2206
17.858.1.8 supportsInlineExpansion() . . . 2206
17.858.1.9 supportsObjectCreation() . . . 2206
17.859 SimulationControlGenericEnum< T, E > Class Template Reference . . . 2206
17.859.1 Member Function Documentation . . . 2209
17.859.1.1 getStrValues() . . . 2209
17.859.1.2 getValue() . . . 2209
17.859.1.3 setValue() . . . 2209
17.860 SimulationControlGenericList< T, M, C > Class Template Reference . . . 2209
17.860.1 Member Function Documentation . . . 2212
17.860.1.1 getStrValues() . . . 2212
17.860.1.2 getValue() . . . 2212
17.860.1.3 setValue() . . . 2212
17.860.1.4 supportsListEditor() . . . 2212
17.861 SimulationControlGenericListPointer< T, M, C > Class Template Reference . . . 2213
17.861.1 Member Function Documentation . . . 2215
17.861.1.1 createNewListElement() . . . 2215
17.861.1.2 createNewListElementOfType() . . . 2215
17.861.1.3 getChildSimulationControls() . . . 2216
17.861.1.4 getCreatableListElementTypes() . . . 2216
17.861.1.5 getCurrentListElementType() . . . 2216
17.861.1.6 getListElementModelDataDefinition() . . . 2216
17.861.1.7 getStrValues() . . . 2216

17.861.1.8 getValue() . . . 2216
17.861.1.9 setCurrentListElementType() . . . 2216
17.861.1.10 setValue() . . . 2216
17.861.1.11 supportsListEditor() . . . 2217
17.861.1.12 supportsNewListElementCreation() . . . 2217
17.862 SimulationControlInt Class Reference . . . 2217
17.862.1 Member Function Documentation . . . 2220
17.862.1.1 getValue() . . . 2220
17.862.1.2 setValue() . . . 2220
17.863 SimulationController Class Reference . . . 2220
17.863.1 Detailed Description . . . 2220
17.863.2 Constructor & Destructor Documentation . . . 2221
17.863.2.1 SimulationController() . . . 2221
17.863.3 Member Function Documentation . . . 2221
17.863.3.1 currentSimulation() . . . 2221
17.863.3.2 ensureReady() . . . 2221
17.863.3.3 hasCurrentModelSimulation() . . . 2221
17.864 SimulationControlSourceCodeString Class Reference . . . 2222
17.864.1 Member Function Documentation . . . 2225
17.864.1.1 getValue() . . . 2225
17.864.1.2 setValue() . . . 2225
17.864.1.3 validateProposedValue() . . . 2225
17.865 SimulationControlString Class Reference . . . 2225
17.865.1 Member Function Documentation . . . 2228
17.865.1.1 getValue() . . . 2228
17.865.1.2 setValue() . . . 2228
17.865.1.3 validateProposedValue() . . . 2228
17.866 SimulationControlTimeUnit Class Reference . . . 2228
17.866.1 Member Function Documentation . . . 2231
17.866.1.1 getValue() . . . 2231
17.866.1.2 setValue() . . . 2231
17.867 SimulationControlUInt Class Reference . . . 2231
17.867.1 Member Function Documentation . . . 2234
17.867.1.1 getValue() . . . 2234
17.867.1.2 setValue() . . . 2234
17.868 SimulationControlUShort Class Reference . . . 2234
17.868.1 Member Function Documentation . . . 2237
17.868.1.1 getValue() . . . 2237
17.868.1.2 setValue() . . . 2237
17.869 SimulationEvent Class Reference . . . 2237
17.869.1 Detailed Description . . . 2237
17.870 SimulationEventController Class Reference . . . 2238

17.870.1 Detailed Description . . . 2238
17.871 SimulationExperiment Class Reference . . . 2239
17.871.1 Detailed Description . . . 2239
17.872 SimulationReporter_if Class Reference . . . 2239
17.872.1 Detailed Description . . . 2240
17.872.2 Member Function Documentation . . . 2240
17.872.2.1 showReplicationStatistics() . . . 2240
17.872.2.2 showSimulationControls() . . . 2240
17.872.2.3 showSimulationResponses() . . . 2240
17.872.2.4 showSimulationStatistics() . . . 2240
17.873 SimulationReporterDefaultImpl1 Class Reference . . . 2240
17.873.1 Detailed Description . . . 2241
17.873.2 Member Function Documentation . . . 2241
17.873.2.1 showReplicationStatistics() . . . 2241
17.873.2.2 showSimulationControls() . . . 2242
17.873.2.3 showSimulationResponses() . . . 2242
17.873.2.4 showSimulationStatistics() . . . 2242
17.874 SimulationResponse Class Reference . . . 2242
17.874.1 Detailed Description . . . 2244
17.875 SimulationResponseDouble Class Reference . . . 2244
17.875.1 Member Function Documentation . . . 2246
17.875.1.1 getValue() . . . 2246
17.876 SimulationResultsDataset Struct Reference . . . 2246
17.876.1 Member Function Documentation . . . 2246
17.876.1.1 formatKindName() . . . 2246
17.876.1.2 isRecordFormat() . . . 2246
17.876.1.3 replications() . . . 2246
17.876.1.4 values() . . . 2247
17.876.1.5 valuesForReplication() . . . 2247
17.877 SimulationResultsDatasetParser Class Reference . . . 2247
17.877.1 Member Function Documentation . . . 2247
17.877.1.1 loadFromTextFile() . . . 2247
17.878 SimulationResultsObservation Struct Reference . . . 2247
17.879 SimulationScenario Class Reference . . . 2247
17.879.1 Detailed Description . . . 2248
17.879.2 Member Function Documentation . . . 2248
17.879.2.1 getResponseValues() . . . 2248
17.880 SimulatorSessionService::SimulationStatusResult Struct Reference . . . 2248
17.880.1 Detailed Description . . . 2249
17.881 Simulator Class Reference . . . 2249
17.881.1 Detailed Description . . . 2249
17.881.2 Constructor & Destructor Documentation . . . 2250

17.881.2.1 Simulator() . . . 2250
17.881.3 Member Function Documentation . . . 2250
17.881.3.1 getExperimentManager() . . . 2250
17.881.3.2 getLicenceManager() . . . 2250
17.881.3.3 getModelManager() . . . 2250
17.881.3.4 getName() . . . 2250
17.881.3.5 getParserManager() . . . 2250
17.881.3.6 getPluginManager() . . . 2250
17.881.3.7 getTraceManager() . . . 2251
17.881.3.8 getVersion() . . . 2251
17.881.3.9 getVersionNumber() . . . 2251
17.882 SimulatorFacade Class Reference . . . 2251
17.882.1 Detailed Description . . . 2257
17.883 SimulatorSessionService::SimulatorInfoResult Struct Reference . . . 2257
17.883.1 Detailed Description . . . 2257
17.884 SimulatorSessionService Class Reference . . . 2257
17.884.1 Detailed Description . . . 2259
17.884.2 Constructor & Destructor Documentation . . . 2259
17.884.2.1 SimulatorSessionService() . . . 2259
17.884.3 Member Function Documentation . . . 2260
17.884.3.1 configureSimulation() . . . 2260
17.884.3.2 createSession() . . . 2260
17.884.3.3 createWorkerJob() . . . 2260
17.884.3.4 getSimulationStatus() . . . 2260
17.884.3.5 getWorkerCapabilities() . . . 2261
17.884.3.6 getWorkerInfo() . . . 2261
17.884.3.7 getWorkerJob() . . . 2261
17.884.3.8 getWorkerJobResult() . . . 2261
17.884.3.9 importModelFromLanguage() . . . 2262
17.884.3.10 loadModel() . . . 2262
17.884.3.11 runSimulation() . . . 2262
17.884.3.12 runWorkerJob() . . . 2262
17.884.3.13 saveCurrentModel() . . . 2263
17.884.3.14 stepSimulation() . . . 2263
17.884.3.15 tryCreateModel() . . . 2263
17.884.3.16 tryGetCurrentModelInfo() . . . 2264
17.884.3.17 tryGetSimulatorInfo() . . . 2264
17.885 SinkModelComponent Class Reference . . . 2264
17.885.1 Detailed Description . . . 2269
17.885.2 Member Function Documentation . . . 2269
17.885.2.1 _check() . . . 2269
17.885.2.2 _loadInstance() . . . 2269

17.885.2.3 _saveInstance() . . . 2270
17.886 yy::genesyspp_parser::stack< T, S >::slice Class Reference . . . 2270
17.886.1 Detailed Description . . . 2270
17.887 Smart_AddingResource Class Reference . . . 2271
17.887.1 Member Function Documentation . . . 2272
17.887.1.1 main() . . . 2272
17.888 Smart_AlternatingEntityCreation Class Reference . . . 2272
17.888.1 Member Function Documentation . . . 2274
17.888.1.1 main() . . . 2274
17.889 Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime Class Reference . . . 2274
17.889.1 Member Function Documentation . . . 2276
17.889.1.1 main() . . . 2276
17.890 Smart_ArrivalsEntityTypeVsAttribute Class Reference . . . 2276
17.890.1 Member Function Documentation . . . 2278
17.890.1.1 main() . . . 2278
17.891 Smart_AssignExample Class Reference . . . 2278
17.891.1 Member Function Documentation . . . 2279
17.891.1.1 main() . . . 2279
17.892 Smart_AssignWriteSeizes Class Reference . . . 2280
17.892.1 Member Function Documentation . . . 2281
17.892.1.1 main() . . . 2281
17.893 Smart_Attribute_Variable Class Reference . . . 2281
17.893.1 Member Function Documentation . . . 2283
17.893.1.1 main() . . . 2283
17.894 Smart_AutomaticStatisticsCollection Class Reference . . . 2283
17.894.1 Member Function Documentation . . . 2285
17.894.1.1 main() . . . 2285
17.895 Smart_BacteriaColony_GRO Class Reference . . . 2285
17.895.1 Member Function Documentation . . . 2286
17.895.1.1 main() . . . 2286
17.896 Smart_BasicModeling Class Reference . . . 2287
17.896.1 Member Function Documentation . . . 2288
17.896.1.1 main() . . . 2288
17.897 Smart_BatchAndSeparate Class Reference . . . 2288
17.897.1 Member Function Documentation . . . 2290
17.897.1.1 main() . . . 2290
17.898 Smart_BatchSeparate Class Reference . . . 2290
17.898.1 Member Function Documentation . . . 2291
17.898.1.1 main() . . . 2291
17.899 Smart_BioKineticLawRegulation Class Reference . . . 2292
17.899.1 Member Function Documentation . . . 2293
17.899.1.1 main() . . . 2293

17.900 Smart_BioNetworkAnalysis Class Reference . . . 2293
17.900.1 Member Function Documentation . . . 2295
17.900.1.1 main() . . . 2295
17.901 Smart_BioReversibleKineticLaw Class Reference . . . 2295
17.901.1 Member Function Documentation . . . 2296
17.901.1.1 main() . . . 2296
17.902 Smart_BioReversibleMassAction Class Reference . . . 2297
17.902.1 Member Function Documentation . . . 2298
17.902.1.1 main() . . . 2298
17.903 Smart_Buffer Class Reference . . . 2298
17.903.1 Member Function Documentation . . . 2300
17.903.1.1 main() . . . 2300
17.904 Smart_CellularAutomata Class Reference . . . 2300
17.904.1 Member Function Documentation . . . 2301
17.904.1.1 main() . . . 2301
17.905 Smart_CellularAutomata1 Class Reference . . . 2302
17.905.1 Member Function Documentation . . . 2303
17.905.1.1 main() . . . 2303
17.906 Smart_Clone Class Reference . . . 2303
17.906.1 Member Function Documentation . . . 2305
17.906.1.1 main() . . . 2305
17.907 Smart_ContinuousFlowEntities Class Reference . . . 2305
17.907.1 Member Function Documentation . . . 2306
17.907.1.1 main() . . . 2306
17.908 Smart_CppForG Class Reference . . . 2307
17.908.1 Member Function Documentation . . . 2308
17.908.1.1 main() . . . 2308
17.909 Smart_Create Class Reference . . . 2308
17.909.1 Member Function Documentation . . . 2310
17.909.1.1 main() . . . 2310
17.910 Smart_DecideNWayByChance Class Reference . . . 2310
17.910.1 Member Function Documentation . . . 2311
17.910.1.1 main() . . . 2311
17.911 Smart_DefaultModalModel Class Reference . . . 2312
17.911.1 Member Function Documentation . . . 2313
17.911.1.1 main() . . . 2313
17.912 Smart_DefiningAttributesAsStrings Class Reference . . . 2313
17.912.1 Member Function Documentation . . . 2315
17.912.1.1 main() . . . 2315
17.913 Smart_DefiningControlLogic Class Reference . . . 2315
17.913.1 Member Function Documentation . . . 2316
17.913.1.1 main() . . . 2316

17.914 Smart_DefiningResourceCapacity Class Reference . . . 2317
17.914.1 Member Function Documentation . . . 2318
17.914.1.1 main() . . . 2318
17.915 Smart_Delay Class Reference . . . 2318
17.915.1 Member Function Documentation . . . 2320
17.915.1.1 main() . . . 2320
17.916 Smart_DelayBasedOnReplication Class Reference . . . 2320
17.916.1 Member Function Documentation . . . 2322
17.916.1.1 main() . . . 2322
17.916.1.2 TestEventHandler() . . . 2322
17.917 Smart_Dummy Class Reference . . . 2322
17.917.1 Member Function Documentation . . . 2324
17.917.1.1 main() . . . 2324
17.918 Smart_EntitiesProcessedByPriority Class Reference . . . 2324
17.918.1 Member Function Documentation . . . 2326
17.918.1.1 main() . . . 2326
17.919 Smart_EvaluatingConditionsBeforeEnteringQueue Class Reference . . . 2326
17.919.1 Member Function Documentation . . . 2328
17.919.1.1 main() . . . 2328
17.920 Smart_Expression Class Reference . . . 2328
17.920.1 Member Function Documentation . . . 2329
17.920.1.1 main() . . . 2329
17.921 Smart_Failures Class Reference . . . 2330
17.921.1 Member Function Documentation . . . 2331
17.921.1.1 main() . . . 2331
17.922 Smart_FSMState Class Reference . . . 2331
17.922.1 Member Function Documentation . . . 2333
17.922.1.1 main() . . . 2333
17.923 Smart_GroColonyGrowth Class Reference . . . 2333
17.923.1 Member Function Documentation . . . 2334
17.923.1.1 main() . . . 2334
17.924 Smart_GroColonyLifecycle Class Reference . . . 2335
17.924.1 Member Function Documentation . . . 2336
17.924.1.1 main() . . . 2336
17.925 Smart_HoldSearchRemove Class Reference . . . 2336
17.925.1 Member Function Documentation . . . 2338
17.925.1.1 main() . . . 2338
17.926 Smart_InventoryAndHoldingCosts Class Reference . . . 2338
17.926.1 Member Function Documentation . . . 2340
17.926.1.1 main() . . . 2340
17.927 Smart_LSODE Class Reference . . . 2340
17.927.1 Member Function Documentation . . . 2341

17.927.1.1 main() . . . 2341
17.928 Smart_MarkovChain Class Reference . . . 2342
17.928.1 Member Function Documentation . . . 2343
17.928.1.1 main() . . . 2343
17.929 Smart_MaxArrivalsField Class Reference . . . 2343
17.929.1 Member Function Documentation . . . 2345
17.929.1.1 main() . . . 2345
17.930 Smart_ModalModel Class Reference . . . 2345
17.930.1 Member Function Documentation . . . 2346
17.930.1.1 main() . . . 2346
17.931 Smart_ModalModelDefaultNetwork Class Reference . . . 2347
17.931.1 Member Function Documentation . . . 2348
17.931.1.1 main() . . . 2348
17.932 Smart_ModalModelFSM Class Reference . . . 2348
17.932.1 Member Function Documentation . . . 2350
17.932.1.1 main() . . . 2350
17.933 Smart_ModalModelPetriNet Class Reference . . . 2350
17.933.1 Member Function Documentation . . . 2351
17.933.1.1 main() . . . 2351
17.934 Smart_ModelInfoModelSimulation Class Reference . . . 2352
17.934.1 Member Function Documentation . . . 2353
17.934.1.1 main() . . . 2353
17.935 Smart_ModelRunUntil1000Parts Class Reference . . . 2353
17.935.1 Member Function Documentation . . . 2355
17.935.1.1 main() . . . 2355
17.936 Smart_ModuleDisplayVariables Class Reference . . . 2355
17.936.1 Member Function Documentation . . . 2356
17.936.1.1 main() . . . 2356
17.937 Smart_ODE Class Reference . . . 2357
17.937.1 Member Function Documentation . . . 2358
17.937.1.1 main() . . . 2358
17.938 Smart_OnEvent Class Reference . . . 2358
17.938.1 Member Function Documentation . . . 2360
17.938.1.1 main() . . . 2360
17.938.1.2 onBreakpointHandler() . . . 2360
17.938.1.3 onEntityCreateHandler() . . . 2360
17.938.1.4 onEntityMoveHandler() . . . 2360
17.938.1.5 onEntityRemoveHandler() . . . 2360
17.938.1.6 onProcessEventHandler() . . . 2360
17.938.1.7 onReplicationEndHandler() . . . 2360
17.938.1.8 onReplicationStartHandler() . . . 2360
17.938.1.9 onReplicationStepHandler() . . . 2360

17.938.1.10 onSimulationEndHandler() . . . 2361
17.938.1.11 onSimulationPausedHandler() . . . 2361
17.938.1.12 onSimulationResumeHandler() . . . 2361
17.938.1.13 onSimulationStartHandler() . . . 2361
17.939 Smart_OverlappingResources Class Reference . . . 2361
17.939.1 Member Function Documentation . . . 2363
17.939.1.1 main() . . . 2363
17.940 Smart_ParallelProcessingOfEntities Class Reference . . . 2363
17.940.1 Member Function Documentation . . . 2365
17.940.1.1 main() . . . 2365
17.941 Smart_Parser Class Reference . . . 2365
17.941.1 Member Function Documentation . . . 2366
17.941.1.1 main() . . . 2366
17.942 Smart_ParserModelFunctions Class Reference . . . 2367
17.942.1 Member Function Documentation . . . 2368
17.942.1.1 main() . . . 2368
17.943 Smart_PetriPlace Class Reference . . . 2368
17.943.1 Member Function Documentation . . . 2370
17.943.1.1 main() . . . 2370
17.944 Smart_PlacingEntitiesInQueueSets Class Reference . . . 2370
17.944.1 Member Function Documentation . . . 2371
17.944.1.1 main() . . . 2371
17.945 Smart_Plugin Class Reference . . . 2372
17.945.1 Member Function Documentation . . . 2373
17.945.1.1 main() . . . 2373
17.946 Smart_PriorityExample Class Reference . . . 2373
17.946.1 Member Function Documentation . . . 2375
17.946.1.1 main() . . . 2375
17.947 Smart_Process Class Reference . . . 2375
17.947.1 Member Function Documentation . . . 2376
17.947.1.1 main() . . . 2376
17.948 Smart_ProcessArena Class Reference . . . 2377
17.948.1 Member Function Documentation . . . 2378
17.948.1.1 main() . . . 2378
17.949 Smart_ProcessSet Class Reference . . . 2378
17.949.1 Member Function Documentation . . . 2380
17.949.1.1 main() . . . 2380
17.950 Smart_PythonForG Class Reference . . . 2380
17.950.1 Member Function Documentation . . . 2381
17.950.1.1 main() . . . 2381
17.951 Smart_R_Simulator Class Reference . . . 2381
17.951.1 Detailed Description . . . 2383

17.951.2 Member Function Documentation . . . 2383
17.951.2.1 main() . . . 2383
17.952 Smart_Record Class Reference . . . 2384
17.952.1 Member Function Documentation . . . 2385
17.952.1.1 main() . . . 2385
17.953 Smart_Record_Arena Class Reference . . . 2385
17.953.1 Member Function Documentation . . . 2387
17.953.1.1 main() . . . 2387
17.954 Smart_RemovingAndReorderingEntitiesInAQueue Class Reference . . . 2387
17.954.1 Member Function Documentation . . . 2389
17.954.1.1 main() . . . 2389
17.955 Smart_ResourceCosting Class Reference . . . 2389
17.955.1 Member Function Documentation . . . 2390
17.955.1.1 main() . . . 2390
17.956 Smart_ResourceScheduleCosting Class Reference . . . 2391
17.956.1 Member Function Documentation . . . 2392
17.956.1.1 main() . . . 2392
17.957 Smart_ResourceSets Class Reference . . . 2392
17.957.1 Member Function Documentation . . . 2394
17.957.1.1 main() . . . 2394
17.958 Smart_RouteStation Class Reference . . . 2394
17.958.1 Member Function Documentation . . . 2395
17.958.1.1 main() . . . 2395
17.959 Smart_SeizeDelayRelease Class Reference . . . 2396
17.959.1 Member Function Documentation . . . 2397
17.959.1.1 main() . . . 2397
17.960 Smart_SeizeDelayReleaseMany Class Reference . . . 2397
17.960.1 Member Function Documentation . . . 2399
17.960.1.1 main() . . . 2399
17.961 Smart_SeizeDelayReleaseNoDataDefs Class Reference . . . 2399
17.961.1 Member Function Documentation . . . 2401
17.961.1.1 main() . . . 2401
17.962 Smart_SeizingMultipleSimultaneosly Class Reference . . . 2401
17.962.1 Member Function Documentation . . . 2402
17.962.1.1 main() . . . 2402
17.963 Smart_SelectingRouteBasedOnProbability Class Reference . . . 2403
17.963.1 Member Function Documentation . . . 2404
17.963.1.1 main() . . . 2404
17.964 Smart_SelectingShorterQueue Class Reference . . . 2404
17.964.1 Member Function Documentation . . . 2406
17.964.1.1 main() . . . 2406
17.965 Smart_Sequence Class Reference . . . 2406

17.965.1 Member Function Documentation . . . 2407
17.965.1.1 main() . . . 2407
17.966 Smart_SimulationControlResponse Class Reference . . . 2408
17.966.1 Member Function Documentation . . . 2409
17.966.1.1 main() . . . 2409
17.967 Smart_SynchronizingParallelEntities Class Reference . . . 2409
17.967.1 Member Function Documentation . . . 2411
17.967.1.1 main() . . . 2411
17.968 Smart_ValueAdded Class Reference . . . 2411
17.968.1 Member Function Documentation . . . 2412
17.968.1.1 main() . . . 2412
17.969 Smart_WaitForSignal Class Reference . . . 2413
17.969.1 Member Function Documentation . . . 2414
17.969.1.1 main() . . . 2414
17.970 Smart_WaitScanCondition Class Reference . . . 2414
17.970.1 Member Function Documentation . . . 2416
17.970.1.1 main() . . . 2416
17.971 Smart_WaitSignal Class Reference . . . 2416
17.971.1 Member Function Documentation . . . 2417
17.971.1.1 main() . . . 2417
17.972 WebWorkerRuntime::Snapshot Struct Reference . . . 2417
17.972.1 Detailed Description . . . 2418
17.973 MetabolicFluxBalanceSolver::Solution Struct Reference . . . 2418
17.974 Optimizer_if::SolutionSummary Struct Reference . . . 2418
17.974.1 Detailed Description . . . 2418
17.975 Solver_if Class Reference . . . 2419
17.975.1 Detailed Description . . . 2419
17.976 SolverDefaultImpl1 Class Reference . . . 2420
17.976.1 Detailed Description . . . 2421
17.976.2 Member Function Documentation . . . 2421
17.976.2.1 derivate() [1/4] . . . 2421
17.976.2.2 derivate() [2/4] . . . 2421
17.976.2.3 derivate() [3/4] . . . 2421
17.976.2.4 derivate() [4/4] . . . 2421
17.976.2.5 getMaxSteps() . . . 2422
17.976.2.6 getPrecision() . . . 2422
17.976.2.7 integrate() [1/4] . . . 2422
17.976.2.8 integrate() [2/4] . . . 2422
17.976.2.9 integrate() [3/4] . . . 2422
17.976.2.10 integrate() [4/4] . . . 2422
17.976.2.11 setMaxSteps() . . . 2423
17.976.2.12 setPrecision() . . . 2423

17.977 SortFile Class Reference . . . 2423
17.977.1 Detailed Description . . . 2423
17.977.2 Member Function Documentation . . . 2423
17.977.2.1 setDataFilename() . . . 2423
17.977.2.2 sort() . . . 2423
17.978 SourceCodeString Class Reference . . . 2424
17.979 SourceModelComponent Class Reference . . . 2424
17.979.1 Detailed Description . . . 2430
17.979.2 Member Function Documentation . . . 2430
17.979.2.1 _check() . . . 2430
17.979.2.2 _createAttachedAttributes() . . . 2430
17.979.2.3 _createEditableDataDefinitions() . . . 2430
17.979.2.4 _initBetweenReplications() . . . 2430
17.979.2.5 _loadInstance() . . . 2430
17.979.2.6 _saveInstance() . . . 2430
17.979.2.7 getEntitiesCreated() . . . 2431
17.979.2.8 getEntitiesPerCreation() . . . 2431
17.979.2.9 getEntityType() . . . 2431
17.979.2.10 getFirstCreation() . . . 2431
17.979.2.11 getMaxCreations() . . . 2431
17.979.2.12 getTimeBetweenCreationsExpression() . . . 2431
17.979.2.13 getTimeUnit() . . . 2432
17.979.2.14 setEntitiesCreated() . . . 2432
17.979.2.15 setEntitiesPerCreation() . . . 2432
17.979.2.16 setEntityType() . . . 2432
17.979.2.17 setEntityTypeName() . . . 2432
17.979.2.18 setFirstCreation() . . . 2432
17.979.2.19 setMaxCreations() [1/2] . . . 2433
17.979.2.20 setMaxCreations() [2/2] . . . 2433
17.979.2.21 setTimeBetweenCreationsExpression() . . . 2433
17.979.2.22 setTimeUnit() . . . 2433
17.979.2.23 show() . . . 2433
17.980 SparseValueStore Class Reference . . . 2434
17.980.1 Detailed Description . . . 2435
17.981 MassActionOdeSystem::Species Struct Reference . . . 2435
17.982 SPICECircuit Class Reference . . . 2435
17.982.1 Detailed Description . . . 2440
17.982.2 Member Function Documentation . . . 2440
17.982.2.1 _createNonEditableDataDefinitions() . . . 2440
17.982.2.2 _loadInstance() . . . 2440
17.982.2.3 _onDispatchEvent() . . . 2440
17.982.2.4 _saveInstance() . . . 2441

17.982.2.5 show() . . . 2441
17.983 SPICENode Class Reference . . . 2441
17.983.1 Detailed Description . . . 2446
17.983.2 Member Function Documentation . . . 2446
17.983.2.1 _loadInstance() . . . 2446
17.983.2.2 _onDispatchEvent() . . . 2446
17.983.2.3 _saveInstance() . . . 2446
17.983.2.4 show() . . . 2446
17.984 SPICERunner Class Reference . . . 2447
17.984.1 Member Function Documentation . . . 2451
17.984.1.1 _check() . . . 2451
17.984.1.2 _loadInstance() . . . 2451
17.984.1.3 _saveInstance() . . . 2451
17.984.1.4 show() . . . 2452
17.985 Start Class Reference . . . 2452
17.985.1 Detailed Description . . . 2457
17.985.2 Member Function Documentation . . . 2457
17.985.2.1 _check() . . . 2457
17.985.2.2 _loadInstance() . . . 2457
17.985.2.3 _onDispatchEvent() . . . 2457
17.985.2.4 _saveInstance() . . . 2458
17.985.2.5 show() . . . 2458
17.986 StartAIAssistant Class Reference . . . 2458
17.986.1 Member Function Documentation . . . 2460
17.986.1.1 main() . . . 2460
17.987 State Class Reference . . . 2460
17.988 GroProgramAst::Statement Struct Reference . . . 2461
17.989 EFSMNetworkDiagramLayout::StatePosition Struct Reference . . . 2461
17.990 StateSet Class Reference . . . 2461
17.991 StateSet_Enumerable Class Reference . . . 2463
17.991.1 Member Function Documentation . . . 2464
17.991.1.1 contains() . . . 2464
17.991.1.2 createDefaultState() . . . 2464
17.991.1.3 enumerateStates() . . . 2464
17.991.1.4 isDiscrete() . . . 2464
17.991.1.5 isFinite() . . . 2464
17.991.1.6 name() . . . 2464
17.991.1.7 parseState() . . . 2464
17.992 Station Class Reference . . . 2465
17.992.1 Detailed Description . . . 2469
17.992.2 Member Function Documentation . . . 2469
17.992.2.1 _check() . . . 2469

17.992.2.2 _createAttachedAttributes() . . . 2469
17.992.2.3 _createInternalStatisticReporters() . . . 2469
17.992.2.4 _initBetweenReplications() . . . 2469
17.992.2.5 _loadInstance() . . . 2469
17.992.2.6 _saveInstance() . . . 2469
17.992.2.7 show() . . . 2470
17.993 StationTestProbe Class Reference . . . 2470
17.994 Statistics_if Class Reference . . . 2474
17.994.1 Detailed Description . . . 2475
17.994.2 Member Function Documentation . . . 2476
17.994.2.1 average() . . . 2476
17.994.2.2 confidenceLevel() . . . 2476
17.994.2.3 getCollector() . . . 2476
17.994.2.4 halfWidthConfidenceInterval() . . . 2476
17.994.2.5 max() . . . 2476
17.994.2.6 min() . . . 2476
17.994.2.7 newSampleSize() . . . 2477
17.994.2.8 numElements() . . . 2477
17.994.2.9 setCollector() . . . 2477
17.994.2.10 setConfidenceLevel() . . . 2477
17.994.2.11 stddeviation() . . . 2477
17.994.2.12 variance() . . . 2478
17.994.2.13 variationCoef() . . . 2478
17.995 StatisticsCollector Class Reference . . . 2478
17.995.1 Detailed Description . . . 2482
17.995.2 Member Function Documentation . . . 2482
17.995.2.1 _check() . . . 2482
17.995.2.2 _loadInstance() . . . 2482
17.995.2.3 _saveInstance() . . . 2482
17.995.2.4 getParent() . . . 2483
17.995.2.5 getStatistics() . . . 2483
17.995.2.6 show() . . . 2483
17.996 StatisticsDatafile_if Class Reference . . . 2483
17.996.1 Detailed Description . . . 2485
17.996.2 Member Function Documentation . . . 2485
17.996.2.1 centil() . . . 2485
17.996.2.2 decil() . . . 2485
17.996.2.3 histogramClassFrequency() . . . 2485
17.996.2.4 histogramClassLowerLimit() . . . 2486
17.996.2.5 histogramNumClasses() . . . 2486
17.996.2.6 mediane() . . . 2486
17.996.2.7 mode() . . . 2486

17.996.2.8 quartil() . . . 2487
17.996.2.9 setHistogramNumClasses() . . . 2487
17.997 StatisticsDatafileDefaultImpl1 Class Reference . . . 2487
17.997.1 Detailed Description . . . 2489
17.997.2 Member Function Documentation . . . 2489
17.997.2.1 average() . . . 2489
17.997.2.2 centil() . . . 2489
17.997.2.3 confidenceLevel() . . . 2489
17.997.2.4 decil() . . . 2489
17.997.2.5 getCollector() . . . 2490
17.997.2.6 halfWidthConfidenceInterval() . . . 2490
17.997.2.7 histogramClassFrequency() . . . 2490
17.997.2.8 histogramClassLowerLimit() . . . 2490
17.997.2.9 histogramNumClasses() . . . 2490
17.997.2.10 max() . . . 2490
17.997.2.11 mediane() . . . 2490
17.997.2.12 min() . . . 2490
17.997.2.13 mode() . . . 2490
17.997.2.14 newSampleSize() . . . 2491
17.997.2.15 numElements() . . . 2491
17.997.2.16 quartil() . . . 2491
17.997.2.17 setCollector() . . . 2491
17.997.2.18 setConfidenceLevel() . . . 2491
17.997.2.19 setHistogramNumClasses() . . . 2491
17.997.2.20 stddeviation() . . . 2491
17.997.2.21 variance() . . . 2491
17.997.2.22 variationCoef() . . . 2491
17.998 StatisticsDefaultImpl1 Class Reference . . . 2492
17.998.1 Detailed Description . . . 2493
17.998.2 Constructor & Destructor Documentation . . . 2493
17.998.2.1 StatisticsDefaultImpl1() . . . 2493
17.998.3 Member Function Documentation . . . 2493
17.998.3.1 average() . . . 2493
17.998.3.2 confidenceLevel() . . . 2493
17.998.3.3 getCollector() . . . 2493
17.998.3.4 halfWidthConfidenceInterval() . . . 2493
17.998.3.5 max() . . . 2494
17.998.3.6 min() . . . 2494
17.998.3.7 newSampleSize() . . . 2494
17.998.3.8 numElements() . . . 2494
17.998.3.9 setCollector() . . . 2494
17.998.3.10 setConfidenceLevel() . . . 2494

17.998.3.11 stddeviation() . . . 2494
17.998.3.12 variance() . . . 2494
17.998.3.13 variationCoef() . . . 2494
17.999 StochasticReactionComponent Class Reference . . . 2495
17.999.1 Detailed Description . . . 2500
17.999.2 Member Function Documentation . . . 2500
17.999.2.1 _check() . . . 2500
17.999.2.2 _createEditableDataDefinitions() . . . 2500
17.999.2.3 _initBetweenReplications() . . . 2500
17.999.2.4 _loadInstance() . . . 2500
17.999.2.5 _onDispatchEvent() . . . 2500
17.999.2.6 _saveInstance() . . . 2500
17.999.2.7 show() . . . 2501
17.1000 StochasticReactionRule Class Reference . . . 2501
17.1000.1 Detailed Description . . . 2505
17.1000.2 Member Function Documentation . . . 2505
17.1000.2.1 _check() . . . 2505
17.1000.2.2 _loadInstance() . . . 2505
17.1000.2.3 _saveInstance() . . . 2506
17.1000.2.4 show() . . . 2506
17.1001 StochasticTranscription Class Reference . . . 2506
17.1001.1 Detailed Description . . . 2511
17.1001.2 Member Function Documentation . . . 2511
17.1001.2.1 _check() . . . 2511
17.1001.2.2 _createEditableDataDefinitions() . . . 2512
17.1001.2.3 _initBetweenReplications() . . . 2512
17.1001.2.4 _loadInstance() . . . 2512
17.1001.2.5 _onDispatchEvent() . . . 2512
17.1001.2.6 _saveInstance() . . . 2512
17.1001.2.7 show() . . . 2512
17.1002 StochasticTranslation Class Reference . . . 2513
17.1002.1 Detailed Description . . . 2518
17.1002.2 Member Function Documentation . . . 2518
17.1002.2.1 _check() . . . 2518
17.1002.2.2 _createEditableDataDefinitions() . . . 2518
17.1002.2.3 _initBetweenReplications() . . . 2518
17.1002.2.4 _loadInstance() . . . 2518
17.1002.2.5 _onDispatchEvent() . . . 2518
17.1002.2.6 _saveInstance() . . . 2518
17.1002.2.7 show() . . . 2520
17.1003 BioReaction::StoichiometricTerm Struct Reference . . . 2520
17.1004 MassActionOdeSystem::StoichiometricTerm Struct Reference . . . 2520

17.1005 MetabolicReaction::StoichiometricTerm Struct Reference . . . 2520
17.1006 StochasticReactionRule::StoichiometricTerm Struct Reference . . . 2521
17.1007 Stop Class Reference . . . 2521
17.1007.1 Detailed Description . . . 2526
17.1007.2 Member Function Documentation . . . 2526
17.1007.2.1 _check() . . . 2526
17.1007.2.2 _loadInstance() . . . 2526
17.1007.2.3 _onDispatchEvent() . . . 2526
17.1007.2.4 _saveInstance() . . . 2526
17.1007.2.5 show() . . . 2527
17.1008 Storage Class Reference . . . 2527
17.1008.1 Detailed Description . . . 2531
17.1008.2 Member Function Documentation . . . 2531
17.1008.2.1 _check() . . . 2531
17.1008.2.2 _getParserChangesInformation() . . . 2531
17.1008.2.3 _initBetweenReplications() . . . 2531
17.1008.2.4 _loadInstance() . . . 2532
17.1008.2.5 _saveInstance() . . . 2532
17.1008.2.6 show() . . . 2532
17.1009 Store Class Reference . . . 2532
17.1009.1 Detailed Description . . . 2537
17.1009.2 Member Function Documentation . . . 2537
17.1009.2.1 _check() . . . 2537
17.1009.2.2 _createEditableDataDefinitions() . . . 2538
17.1009.2.3 _loadInstance() . . . 2538
17.1009.2.4 _onDispatchEvent() . . . 2538
17.1009.2.5 _saveInstance() . . . 2538
17.1009.2.6 show() . . . 2538
17.1010 testing::internal::internal_stream_operator_without_lexical_name_lookup::StreamPrinter Struct Reference . . . 2539
17.1011 testing::internal::String Class Reference . . . 2539
17.1012 Submodel Class Reference . . . 2539
17.1012.1 Detailed Description . . . 2544
17.1012.2 Member Function Documentation . . . 2544
17.1012.2.1 _check() . . . 2544
17.1012.2.2 _loadInstance() . . . 2544
17.1012.2.3 _onDispatchEvent() . . . 2545
17.1012.2.4 _saveInstance() . . . 2545
17.1012.2.5 show() . . . 2545
17.1013 testing::internal::SuiteApiResolver< T > Struct Template Reference . . . 2546
17.1014 yy::genesyspp_parser::symbol_kind Struct Reference . . . 2546
17.1014.1 Detailed Description . . . 2547

17.1014.2 Member Enumeration Documentation . . . 2547
17.1014.2.1 symbol_kind_type . . . 2547
17.1015 yy::genesyspp_parser::symbol_type Struct Reference . . . 2547
17.1015.1 Detailed Description . . . 2550
17.1016 yy::genesyspp_parser::syntax_error Struct Reference . . . 2550
17.1016.1 Detailed Description . . . 2551
17.1017 SystemCommandExecutor_if Class Reference . . . 2551
17.1017.1 Detailed Description . . . 2552
17.1017.2 Member Function Documentation . . . 2552
17.1017.2.1 run() . . . 2552
17.1018 SystemCommandResult Struct Reference . . . 2552
17.1018.1 Detailed Description . . . 2553
17.1019 SystemDependency Class Reference . . . 2553
17.1020 SystemDependencyCheckEntry Class Reference . . . 2553
17.1020.1 Detailed Description . . . 2554
17.1021 SystemDependencyCheckResult Class Reference . . . 2554
17.1021.1 Detailed Description . . . 2554
17.1022 SystemDependencyInstallEntry Class Reference . . . 2554
17.1022.1 Detailed Description . . . 2555
17.1023 SystemDependencyInstallResult Class Reference . . . 2555
17.1023.1 Detailed Description . . . 2555
17.1024 SystemDependencyResolver Class Reference . . . 2555
17.1024.1 Detailed Description . . . 2556
17.1025 SystemPreferences Class Reference . . . 2556
17.1025.1 Detailed Description . . . 2558
17.1025.2 Member Function Documentation . . . 2558
17.1025.2.1 load() . . . 2558
17.1025.2.2 pushRecentModelFile() . . . 2558
17.1025.2.3 save() . . . 2558
17.1026 SystemShellCommandExecutor Class Reference . . . 2558
17.1026.1 Detailed Description . . . 2559
17.1026.2 Member Function Documentation . . . 2559
17.1026.2.1 run() . . . 2559
17.1027 genesys::launcher::TarZstdArchiveExtractor Class Reference . . . 2560
17.1027.1 Member Function Documentation . . . 2560
17.1027.1.1 extract() . . . 2560
17.1028 testing::internal::Templates< Head_, Tail_ > Struct Template Reference . . . 2561
17.1029 testing::internal::Templates< Head_ > Struct Template Reference . . . 2561
17.1030 testing::internal::TemplateSel< Tmpl > Struct Template Reference . . . 2561
17.1031 testing::Test Class Reference . . . 2561
17.1032 testing::TestEventListener Class Reference . . . 2562
17.1033 testing::TestEventListeners Class Reference . . . 2563

17.1034 testing::internal::TestFactoryBase Class Reference . . . 2563
17.1035 testing::internal::TestFactoryImpl< TestClass > Class Template Reference . . . 2564
17.1035.1 Member Function Documentation . . . 2564
17.1035.1.1 CreateTest() . . . 2564
17.1036 testing::TestInfo Class Reference . . . 2565
17.1037 testing::internal::TestMetaFactory< TestSuite > Class Template Reference . . . 2565
17.1037.1 Member Function Documentation . . . 2566
17.1037.1.1 CreateTestFactory() . . . 2566
17.1038 testing::internal::TestMetaFactoryBase< ParamType > Class Template Reference . . . 2566
17.1039 testing::TestParamInfo< ParamType > Struct Template Reference . . . 2566
17.1040 testing::TestProperty Class Reference . . . 2567
17.1041 testing::internal::TestPropertyKeyIs Class Reference . . . 2567
17.1042 HypothesisTester_if::TestResult Class Reference . . . 2567
17.1042.1 Detailed Description . . . 2567
17.1043 testing::TestResult Class Reference . . . 2568
17.1044 testing::internal::TestResultAccessor Class Reference . . . 2568
17.1045 TestSimulationControl Class Reference . . . 2569
17.1045.1 Member Function Documentation . . . 2571
17.1045.1.1 getValue() . . . 2571
17.1045.1.2 setValue() . . . 2571
17.1046 TestSimulationResponse Class Reference . . . 2572
17.1046.1 Member Function Documentation . . . 2573
17.1046.1.1 getValue() . . . 2573
17.1047 testing::TestSuite Class Reference . . . 2573
17.1048 testing::TestWithParam< T > Class Template Reference . . . 2574
17.1049 testing::internal::ThreadLocal< T > Class Template Reference . . . 2575
17.1050 QCPAxisPainterPrivate::TickLabelData Struct Reference . . . 2575
17.1051 Token Struct Reference . . . 2576
17.1052 yy::genesyspp_parser::token Struct Reference . . . 2576
17.1052.1 Detailed Description . . . 2576
17.1053 TokenService Class Reference . . . 2577
17.1054 TraceConsoleController Class Reference . . . 2577
17.1054.1 Detailed Description . . . 2577
17.1055 TraceErrorEvent Class Reference . . . 2578
17.1056 TraceEvent Class Reference . . . 2579
17.1057 testing::internal::TraceInfo Struct Reference . . . 2579
17.1058 TraceManager Class Reference . . . 2579
17.1058.1 Detailed Description . . . 2581
17.1058.2 Member Function Documentation . . . 2581
17.1058.2.1 addTraceErrorHandler() [1/2] . . . 2581
17.1058.2.2 addTraceErrorHandler() [2/2] . . . 2581
17.1058.2.3 addTraceHandler() [1/2] . . . 2581

17.1058.2.4 addTraceHandler() [2/2] . . . 2581
17.1058.2.5 addTraceReportHandler() [1/2] . . . 2582
17.1058.2.6 addTraceReportHandler() [2/2] . . . 2582
17.1058.2.7 addTraceSimulationHandler() [1/2] . . . 2582
17.1058.2.8 addTraceSimulationHandler() [2/2] . . . 2582
17.1058.2.9 errorMessages() . . . 2582
17.1058.2.10 getParentSimulator() . . . 2582
17.1058.2.11 getTraceLevel() . . . 2583
17.1058.2.12 isTraceSimulationRuleAllAllowed() . . . 2583
17.1058.2.13 setTraceLevel() . . . 2583
17.1058.2.14 setTraceSimulationRuleAllAllowed() . . . 2583
17.1059 TraceSimulationEvent Class Reference . . . 2584
17.1060 TraceSimulationProcess Class Reference . . . 2584
17.1060.1 Detailed Description . . . 2585
17.1061 TraitsGUI< T > Struct Template Reference . . . 2585
17.1061.1 Detailed Description . . . 2586
17.1062 TraitsGUI< GAssociation > Struct Reference . . . 2586
17.1063 TraitsGUI< GComponentPort > Struct Reference . . . 2586
17.1064 TraitsGUI< GConnection > Struct Reference . . . 2586
17.1065 TraitsGUI< GMainWindow > Struct Reference . . . 2587
17.1066 TraitsGUI< GModelComponent > Struct Reference . . . 2587
17.1067 TraitsGUI< GModelDataDefinition > Struct Reference . . . 2587
17.1068 TraitsGUI< GModelGraphicPic > Struct Reference . . . 2588
17.1069 TraitsGUI< GScene > Struct Reference . . . 2588
17.1070 TraitsGUI< GView > Struct Reference . . . 2588
17.1071 TraitsKernel< T > Struct Template Reference . . . 2588
17.1072 TraitsKernel< Collector_if > Struct Reference . . . 2589
17.1073 TraitsKernel< Model > Struct Reference . . . 2589
17.1074 TraitsKernel< ModelChecker_if > Struct Reference . . . 2589
17.1075 TraitsKernel< ModelComponent > Struct Reference . . . 2589
17.1076 TraitsKernel< ModelDataDefinition > Struct Reference . . . 2589
17.1077 TraitsKernel< Parser_if > Struct Reference . . . 2590
17.1078 TraitsKernel< Persistence_if > Struct Reference . . . 2590
17.1079 TraitsKernel< PluginConnector_if > Struct Reference . . . 2590
17.1080 TraitsKernel< Sampler_if > Struct Reference . . . 2590
17.1081 TraitsKernel< SimulationReporter_if > Struct Reference . . . 2590
17.1082 TraitsKernel< Simulator > Struct Reference . . . 2591
17.1083 TraitsKernel< Statistics_if > Struct Reference . . . 2591
17.1084 TraitsKernel< StatisticsDatafile_if > Struct Reference . . . 2591
17.1085 TraitsTools< T > Struct Template Reference . . . 2591
17.1085.1 Detailed Description . . . 2591
17.1086 TraitsTools< AIAssistant_if > Struct Reference . . . 2592

17.1086.1 Detailed Description . . . 2592
17.1087 TraitsTools< Fitter_if > Struct Reference . . . 2592
17.1087.1 Detailed Description . . . 2592
17.1088 TraitsTools< HypothesisTester_if > Struct Reference . . . 2592
17.1088.1 Detailed Description . . . 2592
17.1089 TraitsTools< Solver_if > Struct Reference . . . 2592
17.1089.1 Detailed Description . . . 2593
17.1090 EFSMNetworkDiagramLayout::TransitionLayout Struct Reference . . . 2593
17.1091 Transporter Class Reference . . . 2593
17.1091.1 Member Function Documentation . . . 2597
17.1091.1.1 _check() . . . 2597
17.1091.1.2 _createEditableDataDefinitions() . . . 2597
17.1091.1.3 _initBetweenReplications() . . . 2598
17.1091.1.4 _loadInstance() . . . 2598
17.1091.1.5 _saveInstance() . . . 2598
17.1091.1.6 show() . . . 2598
17.1092 Transporter::TravelCompletion Struct Reference . . . 2599
17.1093 testing::internal::TrueWithString Struct Reference . . . 2599
17.1094 std::tuple_size< testing::internal::FlatTuple< Ts... > > Struct Template Reference . . . 2600
17.1095 testing::internal::TypeIdHelper< T > Class Template Reference . . . 2600
17.1096 testing::internal::TypeParameterizedTest< Fixture, TestSel, Types > Class Template Reference 2601
17.1097 testing::internal::TypeParameterizedTest< Fixture, TestSel, internal::None > Class Template Reference . . . 2601
17.1098 testing::internal::TypeParameterizedTestSuite< Fixture, Tests, Types > Class Template Reference 2601
17.1099 testing::internal::TypeParameterizedTestSuite< Fixture, internal::None, Types > Class Template Reference . . . 2601
17.1100 testing::internal::TypeParameterizedTestSuiteRegistry Class Reference . . . 2601
17.1101 testing::internal::Types< Head_, Tail_ > Struct Template Reference . . . 2602
17.1102 testing::internal::Types< Head_ > Struct Template Reference . . . 2602
17.1103 testing::internal::TypeWithSize< size > Class Template Reference . . . 2602
17.1104 testing::internal::TypeWithSize< 4 > Class Reference . . . 2602
17.1105 testing::internal::TypeWithSize< 8 > Class Reference . . . 2602
17.1106 UngroupUndoCommand Class Reference . . . 2603
17.1107 testing::UnitTest Class Reference . . . 2603
17.1108 testing::internal::UnitTestImpl Class Reference . . . 2604
17.1109 testing::internal::UnitTestOptions Class Reference . . . 2606
17.1110 testing::internal::UniversalPrinter< T > Class Template Reference . . . 2606
17.1111 testing::internal::UniversalPrinter< const T > Class Template Reference . . . 2607
17.1112 testing::internal::UniversalPrinter< T & > Class Template Reference . . . 2608
17.1113 testing::internal::UniversalPrinter< T[N]> Class Template Reference . . . 2608
17.1114 testing::internal::UniversalTersePrinter< T > Class Template Reference . . . 2608
17.1115 testing::internal::UniversalTersePrinter< char * > Class Reference . . . 2608
17.1116 testing::internal::UniversalTersePrinter< char16_t * > Class Reference . . . 2609

17.1117 testing::internal::UniversalTersePrinter< char32_t * > Class Reference . . . 2610
17.1118 testing::internal::UniversalTersePrinter< const char * > Class Reference . . . 2611
17.1119 testing::internal::UniversalTersePrinter< const char16_t * > Class Reference . . . 2612
17.1120 testing::internal::UniversalTersePrinter< const char32_t * > Class Reference . . . 2612
17.1121 testing::internal::UniversalTersePrinter< T & > Class Template Reference . . . 2613
17.1122 testing::internal::UniversalTersePrinter< T[N]> Class Template Reference . . . 2613
17.1123 testing::internal::UniversalTersePrinter< wchar_t * > Class Reference . . . 2613
17.1124 Unstore Class Reference . . . 2613
17.1124.1 Detailed Description . . . 2618
17.1124.2 Member Function Documentation . . . 2618
17.1124.2.1 _check() . . . 2618
17.1124.2.2 _createEditableDataDefinitions() . . . 2619
17.1124.2.3 _loadInstance() . . . 2619
17.1124.2.4 _onDispatchEvent() . . . 2619
17.1124.2.5 _saveInstance() . . . 2619
17.1124.2.6 show() . . . 2619
17.1125 genesys::launcher::UpdateChecker Class Reference . . . 2620
17.1126 genesys::launcher::UpdateCheckResult Struct Reference . . . 2620
17.1127 genesys::launcher::UpdateDialog Class Reference . . . 2621
17.1127.1 Member Function Documentation . . . 2621
17.1127.1.1 beginDownload() . . . 2621
17.1127.1.2 confirmUpdate() . . . 2622
17.1127.1.3 finishDownload() . . . 2622
17.1127.1.4 showError() . . . 2622
17.1127.1.5 showInfo() . . . 2622
17.1127.1.6 updateDownloadProgress() . . . 2622
17.1128 genesys::launcher::UpdateDownloader Class Reference . . . 2622
17.1129 genesys::launcher::UpdateManifest Struct Reference . . . 2623
17.1130 genesys::launcher::UpdateManifestParser Class Reference . . . 2623
17.1131 genesys::launcher::UpdatePlatform Struct Reference . . . 2624
17.1132 genesys::launcher::UpdateVerifier Class Reference . . . 2624
17.1133 genesys::launcher::UserRuntimeManager Class Reference . . . 2624
17.1134 Util Class Reference . . . 2624
17.1134.1 Member Function Documentation . . . 2626
17.1134.1.1 convertEnumToStr() [1/2] . . . 2626
17.1134.1.2 convertEnumToStr() [2/2] . . . 2627
17.1134.1.3 DirSeparator() . . . 2627
17.1134.1.4 ExecuteCommand() . . . 2627
17.1134.1.5 FileDelete() . . . 2627
17.1134.1.6 FileExists() . . . 2627
17.1134.1.7 FilenameFromFullFilename() . . . 2628
17.1134.1.8 GenerateNewId() . . . 2628

17.1134.1.9 GenerateNewIdOfType() [1/2] . . . 2628
17.1134.1.10 GenerateNewIdOfType() [2/2] . . . 2628
17.1134.1.11 GetLastIdOfType() . . . 2629
17.1134.1.12 Indent() . . . 2629
17.1134.1.13 List2str() . . . 2629
17.1134.1.14 ListFiles() . . . 2629
17.1134.1.15 Map2str() [1/2] . . . 2630
17.1134.1.16 Map2str() [2/2] . . . 2630
17.1134.1.17 PathFromFullFilename() . . . 2630
17.1134.1.18 ResetIdOfType() . . . 2630
17.1134.1.19 RunningPath() . . . 2631
17.1134.1.20 SepKeyVal() . . . 2631
17.1134.1.21 SetIndent() . . . 2631
17.1134.1.22 SetW() . . . 2631
17.1134.1.23 StrAllocation() . . . 2631
17.1134.1.24 StrIndex() . . . 2632
17.1134.1.25 StrReplace() . . . 2632
17.1134.1.26 StrReplaceSpecialChars() . . . 2632
17.1134.1.27 StrTimeUnitLong() . . . 2633
17.1134.1.28 StrTimeUnitShort() . . . 2633
17.1134.1.29 StrTruncIfInt() [1/2] . . . 2633
17.1134.1.30 StrTruncIfInt() [2/2] . . . 2633
17.1134.1.31 TimeUnitConvert() . . . 2634
17.1134.1.32 Trim() . . . 2634
17.1134.1.33 Trimwithin() . . . 2634
17.1134.1.34 TypeOf() . . . 2634
17.1135 yy::genesyspp_parser::value_type Class Reference . . . 2635
17.1135.1 Detailed Description . . . 2635
17.1135.2 Member Function Documentation . . . 2636
17.1135.2.1 build() [1/2] . . . 2636
17.1135.2.2 build() [2/2] . . . 2636
17.1135.2.3 move() . . . 2636
17.1135.2.4 swap() . . . 2636
17.1136 testing::internal::ValueArray< Ts > Class Template Reference . . . 2636
17.1137 testing::internal::ValuesInIteratorRangeGenerator< T > Class Template Reference . . . 2637
17.1137.1 Member Function Documentation . . . 2637
17.1137.1.1 Begin() . . . 2637
17.1137.1.2 End() . . . 2638
17.1138 Variable Class Reference . . . 2638
17.1138.1 Detailed Description . . . 2643
17.1138.2 Member Function Documentation . . . 2643
17.1138.2.1 _check() . . . 2643

17.1138.2.2 _initBetweenReplications() . . . 2643
17.1138.2.3 _loadInstance() . . . 2643
17.1138.2.4 _saveInstance() . . . 2644
17.1138.2.5 getInitialValuesText() . . . 2644
17.1138.2.6 setInitialValuesText() . . . 2644
17.1138.2.7 show() . . . 2644
17.1139 genesys::launcher::VerificationResult Struct Reference . . . 2644
17.1140 genesys::launcher::Version Class Reference . . . 2644
17.1141 Vpulse Class Reference . . . 2645
17.1141.1 Member Function Documentation . . . 2650
17.1141.1.1 _loadInstance() . . . 2650
17.1141.1.2 _saveInstance() . . . 2651
17.1141.1.3 Build() . . . 2651
17.1142 Vsine Class Reference . . . 2651
17.1142.1 Member Function Documentation . . . 2656
17.1142.1.1 _loadInstance() . . . 2656
17.1142.1.2 _saveInstance() . . . 2657
17.1142.1.3 Build() . . . 2657
17.1143 Vsource Class Reference . . . 2657
17.1143.1 Member Function Documentation . . . 2662
17.1143.1.1 _loadInstance() . . . 2662
17.1143.1.2 _saveInstance() . . . 2663
17.1143.1.3 Build() . . . 2663
17.1144 Wait Class Reference . . . 2663
17.1144.1 Detailed Description . . . 2668
17.1144.2 Member Function Documentation . . . 2668
17.1144.2.1 _check() . . . 2668
17.1144.2.2 _createEditableDataDefinitions() . . . 2668
17.1144.2.3 _createNonEditableDataDefinitions() . . . 2668
17.1144.2.4 _initBetweenReplications() . . . 2669
17.1144.2.5 _loadInstance() . . . 2669
17.1144.2.6 _onDispatchEvent() . . . 2669
17.1144.2.7 _saveInstance() . . . 2669
17.1144.2.8 show() . . . 2669
17.1145 Waiting Class Reference . . . 2670
17.1146 WaitingResource Class Reference . . . 2670
17.1146.1 Member Function Documentation . . . 2671
17.1146.1.1 show() . . . 2671
17.1147 WcmBioNetworkBridge Class Reference . . . 2671
17.1147.1 Detailed Description . . . 2673
17.1147.2 Member Function Documentation . . . 2673
17.1147.2.1 main() . . . 2673

17.1148 WcmCompartmentTransportWholeCellMvp Class Reference . . . 2673
17.1148.1 Detailed Description . . . 2675
17.1148.2 Member Function Documentation . . . 2675
17.1148.2.1 main() . . . 2675
17.1149 WcmFluxBalanceWholeCellMvp Class Reference . . . 2675
17.1149.1 Detailed Description . . . 2677
17.1149.2 Member Function Documentation . . . 2677
17.1149.2.1 main() . . . 2677
17.1150 WcmGeneExpression Class Reference . . . 2677
17.1150.1 Detailed Description . . . 2679
17.1150.2 Member Function Documentation . . . 2679
17.1150.2.1 main() . . . 2679
17.1151 WcmLifecycleStress Class Reference . . . 2679
17.1151.1 Detailed Description . . . 2681
17.1151.2 Member Function Documentation . . . 2681
17.1151.2.1 main() . . . 2681
17.1152 WcmMgenitaliumKarr2012 Class Reference . . . 2681
17.1152.1 Detailed Description . . . 2683
17.1152.2 Member Function Documentation . . . 2683
17.1152.2.1 main() . . . 2683
17.1153 WcmPathwayStressWholeCellMvp Class Reference . . . 2684
17.1153.1 Detailed Description . . . 2685
17.1153.2 Member Function Documentation . . . 2685
17.1153.2.1 main() . . . 2685
17.1154 WcmSimpleSsa Class Reference . . . 2685
17.1154.1 Detailed Description . . . 2687
17.1154.2 Member Function Documentation . . . 2687
17.1154.2.1 main() . . . 2687
17.1155 WcmWholeCellMvp Class Reference . . . 2687
17.1155.1 Detailed Description . . . 2689
17.1155.2 Member Function Documentation . . . 2689
17.1155.2.1 main() . . . 2689
17.1156 WcmYeastDidacticWholeCellMvp Class Reference . . . 2689
17.1156.1 Detailed Description . . . 2691
17.1156.2 Member Function Documentation . . . 2691
17.1156.2.1 main() . . . 2691
17.1157 WcmYeastEukaryoticWholeCellMvp Class Reference . . . 2691
17.1157.1 Detailed Description . . . 2693
17.1157.2 Member Function Documentation . . . 2693
17.1157.2.1 main() . . . 2693
17.1158 WebWorkerDialog Class Reference . . . 2693
17.1158.1 Detailed Description . . . 2694

17.1159 WebWorkerRuntime Class Reference . . . 2694
17.1159.1 Detailed Description . . . 2695
17.1159.2 Member Function Documentation . . . 2695
17.1159.2.1 restart() . . . 2695
17.1159.2.2 start() . . . 2696
17.1160 WholeCellParameterReader Class Reference . . . 2696
17.1160.1 Detailed Description . . . 2696
17.1161 WholeCellState Class Reference . . . 2696
17.1161.1 Detailed Description . . . 2701
17.1161.2 Member Function Documentation . . . 2701
17.1161.2.1 _check() . . . 2701
17.1161.2.2 _initBetweenReplications() . . . 2701
17.1161.2.3 _loadInstance() . . . 2701
17.1161.2.4 _saveInstance() . . . 2701
17.1161.2.5 show() . . . 2702
17.1162 QtButtonPropertyBrowserPrivate::WidgetItem Struct Reference . . . 2702
17.1163 QtGroupBoxPropertyBrowserPrivate::WidgetItem Struct Reference . . . 2703
17.1164 testing::WithParamInterface< T > Class Template Reference . . . 2703
17.1165 SimulatorSessionService::WorkerCapabilitiesResult Struct Reference . . . 2704
17.1165.1 Detailed Description . . . 2704
17.1166 SimulatorSessionService::WorkerInfoResult Struct Reference . . . 2704
17.1166.1 Detailed Description . . . 2705
17.1167 WorkerJob Struct Reference . . . 2705
17.1167.1 Detailed Description . . . 2705
17.1168 SimulatorSessionService::WorkerJobCreationResult Struct Reference . . . 2706
17.1168.1 Detailed Description . . . 2706
17.1169 SimulatorSessionService::WorkerJobInfoResult Struct Reference . . . 2706
17.1169.1 Detailed Description . . . 2706
17.1170 WorkerJobManager Class Reference . . . 2707
17.1170.1 Detailed Description . . . 2707
17.1170.2 Member Function Documentation . . . 2707
17.1170.2.1 createQueuedJob() . . . 2707
17.1170.2.2 getJob() . . . 2707
17.1170.2.3 setMessage() . . . 2708
17.1170.2.4 setSnapshotFilename() . . . 2708
17.1170.2.5 setState() . . . 2708
17.1170.2.6 setTerminalResult() . . . 2709
17.1171 SimulatorSessionService::WorkerJobQueryResult Struct Reference . . . 2709
17.1171.1 Detailed Description . . . 2709
17.1172 SimulatorSessionService::WorkerJobResultInfo Struct Reference . . . 2710
17.1172.1 Detailed Description . . . 2710
17.1173 SimulatorSessionService::WorkerJobResultQueryResult Struct Reference . . . 2710

17.1173.1 Detailed Description . . . 2711
17.1174 SimulatorSessionService::WorkerJobRunResult Struct Reference . . . 2711
17.1174.1 Detailed Description . . . 2711
17.1175 WorkerJobTerminalResult Struct Reference . . . 2711
17.1175.1 Detailed Description . . . 2712
17.1176 Write Class Reference . . . 2712
17.1176.1 Detailed Description . . . 2717
17.1176.2 Member Function Documentation . . . 2717
17.1176.2.1 _check() . . . 2717
17.1176.2.2 _initBetweenReplications() . . . 2717
17.1176.2.3 _loadInstance() . . . 2717
17.1176.2.4 _onDispatchEvent() . . . 2717
17.1176.2.5 _saveInstance() . . . 2718
17.1176.2.6 show() . . . 2718
17.1177 XmlSerializer Class Reference . . . 2718
17.1177.1 Member Function Documentation . . . 2719
17.1177.1.1 dump() . . . 2719
17.1177.1.2 for_each() . . . 2719
17.1177.1.3 get() . . . 2720
17.1177.1.4 load() . . . 2720
17.1177.1.5 newPersistenceRecord() . . . 2720
17.1177.1.6 put() . . . 2720
17.1178 XNOR Class Reference . . . 2721
17.1178.1 Member Function Documentation . . . 2727
17.1178.1.1 _loadInstance() . . . 2727
17.1178.1.2 _saveInstance() . . . 2727
17.1178.1.3 Build() . . . 2727
17.1179 XOR Class Reference . . . 2728
17.1179.1 Member Function Documentation . . . 2733
17.1179.1.1 _loadInstance() . . . 2733
17.1179.1.2 _saveInstance() . . . 2733
17.1179.1.3 Build() . . . 2733
17.1180 yy_buffer_state Struct Reference . . . 2734
17.1180.1 Member Data Documentation . . . 2734
17.1180.1.1 yy_bs_column . . . 2734
17.1180.1.2 yy_bs_lineno . . . 2734
17.1181 yy_trans_info Struct Reference . . . 2734

18 File Documentation **2735**

18.1 BaseGenesysTerminalApplication.h . . . 2735
18.2 GenesysApplication_if.h . . . 2736
18.3 AIAssistantWindow.h . . . 2736

18.4 AddUndoCommand.h . . . 2737
18.5 DeleteUndoCommand.h . . . 2738
18.6 EFSMInitialStateUndoCommand.h . . . 2738
18.7 EFSMStateMembershipUndoCommand.h . . . 2739
18.8 EFSMStatePropertiesUndoCommand.h . . . 2739
18.9 EFSMTransitionPropertiesUndoCommand.h . . . 2740
18.10 EFSMTransitionUndoCommand.h . . . 2740
18.11 GroupUndoCommand.h . . . 2741
18.12 ModalModelBindingsUndoCommand.h . . . 2741
18.13 MoveUndoCommand.h . . . 2742
18.14 PasteUndoCommand.h . . . 2742
18.15 UngroupUndoCommand.h . . . 2742
18.16 AnimationCounter.h . . . 2743
18.17 AnimationPlaceholder.h . . . 2743
18.18 AnimationQueue.h . . . 2745
18.19 AnimationTimer.h . . . 2746
18.20 AnimationTransition.h . . . 2746
18.21 AnimationVariable.h . . . 2747
18.22 BaseMVController.h . . . 2748
18.23 CellularAutomataDemoBuilder.h . . . 2749
18.24 CellularAutomataLatticeWidget.h . . . 2751
18.25 CellularAutomataStateColorMap.h . . . 2751
18.26 CellularAutomataViewerController.h . . . 2752
18.27 CellularAutomataViewerWindow.h . . . 2753
18.28 CodeEditor.h . . . 2753
18.29 LineNumberArea.h . . . 2754
18.30 DialogUtilityController.h . . . 2754
18.31 EditCommandController.h . . . 2755
18.32 GraphicalContextMenuController.h . . . 2756
18.33 ModelInspectorController.h . . . 2757
18.34 ModelLifecycleController.h . . . 2757
18.35 PluginCatalogController.h . . . 2758
18.36 PropertyEditorController.h . . . 2759
18.37 SceneToolController.h . . . 2760
18.38 SimulationCommandController.h . . . 2761
18.39 SimulationController.h . . . 2762
18.40 SimulationEventController.h . . . 2762
18.41 TraceConsoleController.h . . . 2763
18.42 dialogBreakpoint.h . . . 2764
18.43 DialogFind.h . . . 2764
18.44 Dialogmodelinformation.h . . . 2765
18.45 dialogpluginmanager.h . . . 2766

18.46 DialogReportIssue.h . . . 2767
18.47 DialogSelectCounter.h . . . 2767
18.48 DialogSelectVariable.h . . . 2768
18.49 dialogsimulationconfigure.h . . . 2768
18.50 dialogsystempreferences.h . . . 2769
18.51 DialogTimerConfigure.h . . . 2769
18.52 BacteriaColonyViewerCoordinates.h . . . 2770
18.53 GuiExtensionConfigurationDialog.h . . . 2770
18.54 GuiExtensionContracts.h . . . 2771
18.55 GuiExtensionManager.h . . . 2772
18.56 GuiExtensionPluginCatalog.h . . . 2773
18.57 EFSMTransitionGraphicsItem.h . . . 2773
18.58 GraphicalAnimateExpression.h . . . 2773
18.59 GraphicalAssociation.h . . . 2774
18.60 GraphicalComponentPort.h . . . 2774
18.61 GraphicalConnection.h . . . 2774
18.62 GraphicalConnectionStyle.h . . . 2775
18.63 GraphicalDiagramConnection.h . . . 2776
18.64 GraphicalImageAnimation.h . . . 2776
18.65 GraphicalModelComponent.h . . . 2777
18.66 GraphicalModelDataDefinition.h . . . 2779
18.67 GraphicalModelItemRenderStrategy.h . . . 2780
18.68 ModelGraphicsScene.h . . . 2781
18.69 ModelGraphicsView.h . . . 2786
18.70 GuiCrashDiagnostics.h . . . 2788
18.71 GuiScopeTrace.h . . . 2788
18.72 guithememanager.h . . . 2788
18.73 applications/gui/genesys/mainwindow.cpp File Reference . . . 2789
18.73.1 Detailed Description . . . 2790
18.74 mainwindow.h . . . 2790
18.75 applications/gui/genesys/mainwindow_controller.cpp File Reference . . . 2797
18.75.1 Detailed Description . . . 2798
18.76 applications/gui/genesys/mainwindow_modelrepresentations.cpp File Reference . . . 2798
18.76.1 Detailed Description . . . 2799
18.77 applications/gui/genesys/mainwindow_scene.cpp File Reference . . . 2799
18.77.1 Detailed Description . . . 2799
18.78 applications/gui/genesys/mainwindow_simulator.cpp File Reference . . . 2800
18.78.1 Detailed Description . . . 2800
18.79 MovedGuiCompatibilityIncludes.h . . . 2800
18.80 ComboBoxEnum.h . . . 2801
18.81 DataComponentEditor.h . . . 2801
18.82 DataComponentProperty.h . . . 2802

18.83 ObjectPropertyBrowser.h . . . 2803
18.84 PropertyEditor.h . . . 2806
18.85 qtbuttonpropertybrowser.h . . . 2806
18.86 qteditorfactory.h . . . 2807
18.87 qtgroupboxpropertybrowser.h . . . 2812
18.88 qtpropertybrowser.h . . . 2813
18.89 qtpropertybrowserutils_p.h . . . 2816
18.90 qtpropertymanager.h . . . 2820
18.91 qttreepropertybrowser.h . . . 2830
18.92 qtvariantproperty.h . . . 2831
18.93 qcustomplot.h . . . 2833
18.94 CppModelExporter.h . . . 2932
18.95 EFSMNetworkDiagramCodec.h . . . 2932
18.96 EFSMNetworkDiagramLayout.h . . . 2933
18.97 EFSMNetworkDiagramProjection.h . . . 2933
18.98 GraphicalAutomaticPositioningStrategy.h . . . 2933
18.99 GraphicalDataDefinitionLayout.h . . . 2934
18.100 GraphicalModelBuilder.h . . . 2934
18.101 GraphicalModelSerializer.h . . . 2935
18.102 GraphvizModelExporter.h . . . 2936
18.103 IssueReportRelayService.h . . . 2937
18.104 ModelLanguageSynchronizer.h . . . 2937
18.105 ToolLauncher.h . . . 2938
18.106 systempreferences.h . . . 2938
18.107 DataAnalyzerWindow.h . . . 2939
18.108 ExpressionBuilder.h . . . 2941
18.109 OptimizerWindow.h . . . 2942
18.110 TraitsGUI.h . . . 2944
18.111 UtilGUI.h . . . 2945
18.112 WebWorkerDialog.h . . . 2946
18.113 ArchiveExtractor.h . . . 2946
18.114 DispatchController.h . . . 2947
18.115 LauncherConfig.h . . . 2947
18.116 LauncherController.h . . . 2947
18.117 LauncherLogger.h . . . 2948
18.118 NetworkTransport.h . . . 2948
18.119 PlatformInfo.h . . . 2949
18.120 ProcessLauncher.h . . . 2949
18.121 RuntimePaths.h . . . 2950
18.122 RuntimeSelector.h . . . 2950
18.123 UpdateChecker.h . . . 2951
18.124 UpdateDialog.h . . . 2952

18.125 UpdateDownloader.h . . . 2953
18.126 UpdateManifest.h . . . 2953
18.127 UpdateVerifier.h . . . 2954
18.128 UserRuntimeManager.h . . . 2955
18.129 Version.h . . . 2955
18.130 Airport_Extended1.h . . . 2956
18.131 AirportSecurityExample.h . . . 2956
18.132 AirportSecurityExampleExtended.h . . . 2956
18.133 Assembly_Line.h . . . 2956
18.134 Banking_Transactions.h . . . 2957
18.135 Example_BasicOrderShipping.h . . . 2957
18.136 Example_PortModel.h . . . 2957
18.137 Example_PublicTransport.h . . . 2958
18.138 Smart_AddingResource.h . . . 2958
18.139 Smart_AlternatingEntityCreation.h . . . 2958
18.140 Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime.h . . . 2959
18.141 Smart_ArrivalsEntityTypeVsAttribute.h . . . 2959
18.142 Smart_AssignExample.h . . . 2960
18.143 Smart_AutomaticStatisticsCollection.h . . . 2960
18.144 Smart_BasicModeling.h . . . 2960
18.145 Smart_BatchAndSeparate.h . . . 2961
18.146 Smart_ContinuousFlowEntities.h . . . 2961
18.147 Smart_Create.h . . . 2961
18.148 Smart_DecideNWayByChance.h . . . 2962
18.149 Smart_DefiningAttributesAsStrings.h . . . 2962
18.150 Smart_DefiningControlLogic.h . . . 2962
18.151 Smart_DefiningResourceCapacity.h . . . 2963
18.152 Smart_DelayBasedOnReplication.h . . . 2963
18.153 Smart_EntitiesProcessedByPriority.h . . . 2964
18.154 Smart_EvaluatingConditionsBeforeEnteringQueue.h . . . 2964
18.155 Smart_Expression.h . . . 2964
18.156 Smart_InventoryAndHoldingCosts.h . . . 2965
18.157 Smart_MaxArrivalsField.h . . . 2965
18.158 Smart_ModelRunUntil1000Parts.h . . . 2965
18.159 Smart_ModuleDisplayVariables.h . . . 2966
18.160 Smart_OverlappingResources.h . . . 2966
18.161 Smart_ParallelProcessingOfEntities.h . . . 2966
18.162 Smart_PlacingEntitiesInQueueSets.h . . . 2967
18.163 Smart_PriorityExample.h . . . 2967
18.164 Smart_ProcessArena.h . . . 2968
18.165 Smart_Record_Arena.h . . . 2968
18.166 Smart_RemovingAndReorderingEntitiesInAQueue.h . . . 2968

18.167 Smart_ResourceCosting.h . . . 2969
18.168 Smart_ResourceScheduleCosting.h . . . 2969
18.169 Smart_ResourceSets.h . . . 2969
18.170 Smart_SeizingMultipleSimultaneosly.h . . . 2970
18.171 Smart_SelectingRouteBasedOnProbability.h . . . 2970
18.172 Smart_SelectingShorterQueue.h . . . 2970
18.173 Smart_SynchronizingParallelEntities.h . . . 2971
18.174 Smart_ValueAdded.h . . . 2971
18.175 Smart_WaitForSignal.h . . . 2971
18.176 Book_Cap02_Example01.h . . . 2972
18.177 StartAIAssistant.h . . . 2972
18.178 Smart_MarkovChain.h . . . 2972
18.179 Smart_BacteriaColony_GRO.h . . . 2973
18.180 Smart_BioKineticLawRegulation.h . . . 2973
18.181 Smart_BioNetworkAnalysis.h . . . 2973
18.182 Smart_BioReversibleKineticLaw.h . . . 2974
18.183 Smart_BioReversibleMassAction.h . . . 2974
18.184 Smart_GroColonyGrowth.h . . . 2974
18.185 Smart_GroColonyLifecycle.h . . . 2974
18.186 Smart_LSODE.h . . . 2975
18.187 Smart_ODE.h . . . 2975
18.188 Smart_Buffer.h . . . 2975
18.189 Smart_Clone.h . . . 2976
18.190 Smart_Delay.h . . . 2976
18.191 Smart_Failures.h . . . 2977
18.192 Smart_HoldSearchRemove.h . . . 2977
18.193 Smart_Process.h . . . 2977
18.194 Smart_ProcessSet.h . . . 2977
18.195 Smart_SeizeDelayRelease.h . . . 2978
18.196 Smart_SeizeDelayReleaseMany.h . . . 2978
18.197 Smart_SeizeDelayReleaseNoDataDefs.h . . . 2979
18.198 Smart_CppForG.h . . . 2979
18.199 Smart_PythonForG.h . . . 2979
18.200 Smart_R_Simulator.h . . . 2979
18.201 Smart_BatchSeparate.h . . . 2980
18.202 Smart_Record.h . . . 2980
18.203 Smart_AssignWriteSeizes.h . . . 2981
18.204 Smart_Attribute_Variable.h . . . 2981
18.205 Smart_ModelInfoModelSimulation.h . . . 2981
18.206 Smart_Parser.h . . . 2982
18.207 Smart_ParserModelFunctions.h . . . 2982
18.208 Smart_Plugin.h . . . 2982

18.209 Smart_RouteStation.h . . . 2983
18.210 Smart_Sequence.h . . . 2983
18.211 Smart_CellularAutomata.h . . . 2984
18.212 Smart_CellularAutomata1.h . . . 2984
18.213 Smart_DefaultModalModel.h . . . 2984
18.214 Smart_Dummy.h . . . 2985
18.215 Smart_FSMState.h . . . 2985
18.216 Smart_Modalmodel.h . . . 2985
18.217 Smart_ModalModelDefaultNetwork.h . . . 2985
18.218 Smart_ModalModelFSM.h . . . 2986
18.219 Smart_ModalModelPetriNet.h . . . 2986
18.220 Smart_PetriPlace.h . . . 2986
18.221 Smart_OnEvent.h . . . 2986
18.222 Smart_SimulationControlResponse.h . . . 2987
18.223 Smart_WaitScanCondition.h . . . 2987
18.224 Smart_WaitSignal.h . . . 2987
18.225 AnElectronicAssemblyAndTestSystem.h . . . 2988
18.226 BufferFIFO.h . . . 2988
18.227 ContinuousModel.h . . . 2989
18.228 FullSimulationOfComplexModel.h . . . 2989
18.229 Half_Adder.h . . . 2990
18.230 Loja01.h . . . 2990
18.231 OperatingSystem02.h . . . 2990
18.232 OperatingSystem03.h . . . 2991
18.233 Rectifier.h . . . 2991
18.234 WcmBioNetworkBridge.h . . . 2991
18.235 WcmCompartmentTransportWholeCellMvp.h . . . 2992
18.236 WcmFluxBalanceWholeCellMvp.h . . . 2992
18.237 WcmGeneExpression.h . . . 2992
18.238 WcmLifecycleStress.h . . . 2992
18.239 WcmMgenitaliumKarr2012.h . . . 2992
18.240 WcmPathwayStressWholeCellMvp.h . . . 2993
18.241 WcmSimpleSsa.h . . . 2993
18.242 WcmWholeCellMvp.h . . . 2993
18.243 WcmYeastDidacticWholeCellMvp.h . . . 2993
18.244 WcmYeastEukaryoticWholeCellMvp.h . . . 2993
18.245 GenesysShell.h . . . 2994
18.246 GenesysShell_if.h . . . 2995
18.247 ApiRouter.h . . . 2996
18.248 TokenService.h . . . 2996
18.249 BaseGenesysWebApplication.h . . . 2997
18.250 HttpRequest.h . . . 2997

18.251 HttpResponse.h . . . 2997
18.252 SimpleHttpServer.h . . . 2997
18.253 SimulatorSessionService.h . . . 2998
18.254 WebWorkerRuntime.h . . . 3000
18.255 SessionContext.h . . . 3001
18.256 SessionManager.h . . . 3002
18.257 WorkerJob.h . . . 3002
18.258 WorkerJobManager.h . . . 3002
18.259 gtest-assertion-result.h . . . 3003
18.260 gtest-death-test.h . . . 3006
18.261 gtest-matchers.h . . . 3010
18.262 gtest-message.h . . . 3023
18.263 gtest-param-test.h . . . 3026
18.264 gtest-printers.h . . . 3032
18.265 gtest-printers.h . . . 3045
18.266 gtest-spi.h . . . 3045
18.267 gtest-test-part.h . . . 3048
18.268 gtest-typed-test.h . . . 3051
18.269 gtest.h . . . 3055
18.270 gtest.h . . . 3084
18.271 gtest_pred_impl.h . . . 3085
18.272 gtest_prod.h . . . 3088
18.273 gtest-port.h . . . 3089
18.274 gtest-port.h . . . 3089
18.275 gtest-death-test-internal.h . . . 3121
18.276 gtest-filepath.h . . . 3125
18.277 gtest-internal.h . . . 3127
18.278 gtest-param-util.h . . . 3145
18.279 gtest-port-arch.h . . . 3158
18.280 gtest-string.h . . . 3159
18.281 gtest-type-util.h . . . 3161
18.282 gtest-assertion-result.cc . . . 3164
18.283 gtest-death-test.cc . . . 3165
18.284 gtest-filepath.cc . . . 3185
18.285 gtest-internal-inl.h . . . 3189
18.286 gtest-matchers.cc . . . 3205
18.287 gtest-port.cc . . . 3206
18.288 gtest-printers.cc . . . 3223
18.289 gtest-test-part.cc . . . 3230
18.290 gtest-typed-test.cc . . . 3231
18.291 gtest.cc . . . 3233
18.292 ConnectionManager.h . . . 3316

18.293 DefineGetterSetter.h . . . 3317
18.294 Attribute.h . . . 3317
18.295 Counter.h . . . 3318
18.296 Entity.h . . . 3318
18.297 EntityType.h . . . 3319
18.298 SparseValueStore.h . . . 3320
18.299 StatisticsCollector.h . . . 3326
18.300 Event.h . . . 3326
18.301 ExperimentManager.h . . . 3328
18.302 ExperimentManagerDefaultImpl1.h . . . 3329
18.303 ExperimetManager_if.h . . . 3329
18.304 GenesysPropertyIntrospection.h . . . 3330
18.305 LicenceManager.h . . . 3331
18.306 Model.h . . . 3331
18.307 ModelChecker_if.h . . . 3333
18.308 ModelCheckerDefaultImpl1.h . . . 3333
18.309 ModelComponent.h . . . 3334
18.310 ModelComponentManager.h . . . 3335
18.311 ModelDataDefinition.h . . . 3335
18.312 ModelDataDefinitionAssociations.h . . . 3338
18.313 ModelDataManager.h . . . 3339
18.314 ModelInfo.h . . . 3340
18.315 ModelManager.h . . . 3340
18.316 ModelSimulation.h . . . 3341
18.317 OnEventManager.h . . . 3343
18.318 Parser_if.h . . . 3349
18.319 ParserChangesInformation.h . . . 3349
18.320 ParserDefaultImpl1.h . . . 3350
18.321 ParserDefaultImpl2.h . . . 3351
18.322 ParserManager.h . . . 3351
18.323 Persistence.h . . . 3352
18.324 CppSerializer.h . . . 3353
18.325 GenSerializer.h . . . 3353
18.326 JsonSerializer.h . . . 3354
18.327 Persistence_if.h . . . 3354
18.328 PersistenceDefaultImpl2.h . . . 3355
18.329 PersistencePartialLoadImpl1.h . . . 3355
18.330 Serializer.h . . . 3355
18.331 XmlSerializer.h . . . 3356
18.332 Plugin.h . . . 3356
18.333 PluginConnector_if.h . . . 3357
18.334 PluginInformation.h . . . 3357

18.335 PluginManager.h . . . 3359
18.336 PropertyGenesys.h . . . 3361
18.337 ScenarioExperiment_if.h . . . 3362
18.338 SimulationControlAndResponse.h . . . 3362
18.339 SimulationExperiment.h . . . 3377
18.340 SimulationReporter_if.h . . . 3377
18.341 SimulationReporterDefaultImpl1.h . . . 3378
18.342 SimulationScenario.h . . . 3378
18.343 Simulator.h . . . 3379
18.344 SimulatorFacade.h . . . 3380
18.345 SinkModelComponent.h . . . 3398
18.346 SourceModelComponent.h . . . 3398
18.347 SystemDependencyResolver.h . . . 3399
18.348 TraceManager.h . . . 3401
18.349 Collector_if.h . . . 3403
18.350 CollectorDatafile_if.h . . . 3404
18.351 CollectorDatafileDefaultImpl1.h . . . 3404
18.352 CollectorDefaultImpl1.h . . . 3405
18.353 Sampler_if.h . . . 3406
18.354 SamplerDefaultImpl1.h . . . 3406
18.355 SorttFile.h . . . 3412
18.356 Statistics_if.h . . . 3412
18.357 StatisticsDataFile_if.h . . . 3413
18.358 StatisticsDataFileDefaultImpl.h . . . 3413
18.359 StatisticsDefaultImpl1.h . . . 3415
18.360 TraitsKernel.h . . . 3416
18.361 Exact.h . . . 3418
18.362 List.h . . . 3419
18.363 ListObservable.h . . . 3423
18.364 Util.h . . . 3426
18.365 Genesys++-driver.h . . . 3428
18.366 parser/GenesysParser.h File Reference . . . 3429
18.366.1 Detailed Description . . . 3431
18.367 GenesysParser.h . . . 3431
18.368 parser/location.hh File Reference . . . 3469
18.368.1 Detailed Description . . . 3471
18.368.2 Function Documentation . . . 3471
18.368.2.1 operator<<() [1/2] . . . 3471
18.368.2.2 operator<<() [2/2] . . . 3471
18.369 location.hh . . . 3471
18.370 obj_t.h . . . 3474
18.371 position.hh . . . 3475

18.372 stack.hh . . . 3475
18.373 AIAssistant.h . . . 3475
18.374 MarkovChain.h . . . 3476
18.375 BacteriaColony.h . . . 3477
18.376 BioRunnerCommand.h . . . 3481
18.377 BioSimulate.h . . . 3482
18.378 BioSteadyState.h . . . 3483
18.379 GeneticCircuitSimulate.h . . . 3484
18.380 GeneticExpressionStep.h . . . 3485
18.381 MetabolicFluxBalance.h . . . 3486
18.382 ContinuousSystemComponent.h . . . 3487
18.383 DiffEquations.h . . . 3487
18.384 DiffusionSimulate.h . . . 3489
18.385 LSODE.h . . . 3490
18.386 OLD_ODEelement.h . . . 3490
18.387 Decide.h . . . 3491
18.388 DropOff.h . . . 3492
18.389 PickUp.h . . . 3493
18.390 Remove.h . . . 3493
18.391 Search.h . . . 3494
18.392 QueueableItem.h . . . 3495
18.393 SeizableItem.h . . . 3496
18.394 Buffer.h . . . 3498
18.395 Clone.h . . . 3499
18.396 Delay.h . . . 3499
18.397 Process.h . . . 3500
18.398 Release.h . . . 3501
18.399 Seize.h . . . 3502
18.400 SPICECircuit.h . . . 3504
18.401 SPICENode.h . . . 3511
18.402 CppForG.h . . . 3512
18.403 PythonForG.h . . . 3513
18.404 RSimulator.h . . . 3513
18.405 Batch.h . . . 3514
18.406 Separate.h . . . 3515
18.407 Record.h . . . 3516
18.408 Write.h . . . 3517
18.409 Assign.h . . . 3518
18.410 Create.h . . . 3519
18.411 Dispose.h . . . 3520
18.412 Access.h . . . 3520
18.413 Enter.h . . . 3521

18.414 Exit.h . . . 3522
18.415 Leave.h . . . 3523
18.416 Move.h . . . 3523
18.417 PickableStationItem.h . . . 3524
18.418 PickStation.h . . . 3525
18.419 Route.h . . . 3526
18.420 Start.h . . . 3527
18.421 Stop.h . . . 3528
18.422 Store.h . . . 3528
18.423 Unstore.h . . . 3529
18.424 Boundary_Closed.h . . . 3530
18.425 Boundary_Fixed.h . . . 3531
18.426 BoundaryCondition.h . . . 3531
18.427 Cell.h . . . 3532
18.428 CellularAutomata.h . . . 3533
18.429 CellularAutomata_1DTimed.h . . . 3533
18.430 CellularAutomata_Classic.h . . . 3534
18.431 CellularAutomataAllHeaders.h . . . 3534
18.432 CellularAutomataBase.h . . . 3535
18.433 Lattice.h . . . 3535
18.434 LocalRule.h . . . 3536
18.435 LocalRule_Elementary.h . . . 3536
18.436 LocalRule_FlorestalFire.h . . . 3537
18.437 LocalRule_GameOfLife.h . . . 3538
18.438 LocalRule_Growty.h . . . 3538
18.439 LocalRule_HppLatticeGas.h . . . 3539
18.440 LocalRule_SandRock.h . . . 3540
18.441 LocalRule_SandRock_Student.h . . . 3542
18.442 LocalRule_ShowNeighbordhood.h . . . 3544
18.443 Neighborhood.h . . . 3545
18.444 Neighborhood_Center.h . . . 3546
18.445 Neighborhood_Moore.h . . . 3546
18.446 Neighborhood_VonNeumann.h . . . 3547
18.447 State.h . . . 3548
18.448 State_Numerical.h . . . 3550
18.449 StateSet.h . . . 3551
18.450 StateSet_Enumerable.h . . . 3553
18.451 CellularAutomataComp.h . . . 3553
18.452 DefaultNode.h . . . 3555
18.453 DefaultTransitionExtensions.h . . . 3557
18.454 FSMState.h . . . 3557
18.455 ModalModelDefault.h . . . 3557

18.456 PetriPlace.h . . . 3558
18.457 Submodel.h . . . 3559
18.458 Match.h . . . 3559
18.459 Signal.h . . . 3560
18.460 Wait.h . . . 3561
18.461 DummyComponent.h . . . 3562
18.462 BioStateProjectionComponent.h . . . 3563
18.463 CellCycleCheckpointComponent.h . . . 3564
18.464 CellDivisionEvent.h . . . 3565
18.465 CellFateDecisionComponent.h . . . 3566
18.466 CellGrowthComponent.h . . . 3567
18.467 CompartmentExchangeComponent.h . . . 3568
18.468 EukaryoticCellCycleComponent.h . . . 3569
18.469 FtsZPolymerizationComponent.h . . . 3571
18.470 MetabolicStateProjectionComponent.h . . . 3572
18.471 MetabolicSubmodelComponent.h . . . 3574
18.472 PathwayStressResponseComponent.h . . . 3575
18.473 ResourceAllocationComponent.h . . . 3576
18.474 StochasticReactionComponent.h . . . 3577
18.475 StochasticTranscription.h . . . 3578
18.476 StochasticTranslation.h . . . 3579
18.477 AISupport.h . . . 3580
18.478 BacteriaSignalGrid.h . . . 3581
18.479 BioNetwork.h . . . 3582
18.480 BioParameter.h . . . 3583
18.481 BioReaction.h . . . 3584
18.482 BioSBMLBridge.h . . . 3585
18.483 BioSimulatorRunner.h . . . 3585
18.484 BioSpecies.h . . . 3587
18.485 GeneticCircuit.h . . . 3588
18.486 GeneticCircuitPart.h . . . 3588
18.487 GeneticRegulation.h . . . 3589
18.488 GroProgram.h . . . 3590
18.489 GroProgramAst.h . . . 3595
18.490 GroProgramCompiler.h . . . 3595
18.491 GroProgramIr.h . . . 3596
18.492 GroProgramParser.h . . . 3597
18.493 GroProgramRuntime.h . . . 3597
18.494 MetabolicNetwork.h . . . 3599
18.495 MetabolicReaction.h . . . 3600
18.496 DiffusionField.h . . . 3601
18.497 ODESolver.h . . . 3602

18.498 Failure.h . . . 3604
18.499 Queue.h . . . 3605
18.500 Resource.h . . . 3607
18.501 Schedule.h . . . 3609
18.502 CppCompiler.h . . . 3610
18.503 OctaveRunner.h . . . 3612
18.504 PythonRuntime.h . . . 3613
18.505 RSimulatorRunner.h . . . 3614
18.506 SPICERunner.h . . . 3615
18.507 EntityGroup.h . . . 3617
18.508 File.h . . . 3618
18.509 AssignmentItem.h . . . 3619
18.510 Formula.h . . . 3620
18.511 Label.h . . . 3620
18.512 Set.h . . . 3621
18.513 Variable.h . . . 3622
18.514 Conveyor.h . . . 3623
18.515 Distance.h . . . 3623
18.516 Segment.h . . . 3624
18.517 Sequence.h . . . 3625
18.518 Station.h . . . 3626
18.519 Storage.h . . . 3627
18.520 Transporter.h . . . 3628
18.521 ColoredPetriNetNetwork.h . . . 3628
18.522 CPNArc.h . . . 3629
18.523 CPNTransition.h . . . 3630
18.524 DefaultNetwork.h . . . 3631
18.525 DirectedAcyclicGraphNetwork.h . . . 3631
18.526 DirectedGraphNetwork.h . . . 3632
18.527 EFSMNetwork.h . . . 3632
18.528 GraphEdge.h . . . 3633
18.529 GraphNetwork.h . . . 3634
18.530 GraphNode.h . . . 3636
18.531 MarkovChainNetwork.h . . . 3636
18.532 MarkovState.h . . . 3637
18.533 NetworkActivation.h . . . 3637
18.534 SignalData.h . . . 3638
18.535 DummyElement.h . . . 3639
18.536 BioCompartment.h . . . 3640
18.537 MolecularSpecies.h . . . 3641
18.538 StochasticReactionRule.h . . . 3641
18.539 WholeCellState.h . . . 3642

18.540 PluginConnectorDummyImpl1.h . . . 3644
18.541 PluginConnectorStaticImpl1.h . . . 3644
18.542 AIAssistant_if.h . . . 3644
18.543 AIAssistantDefaultImpl.h . . . 3647
18.544 AIAuditLog.h . . . 3649
18.545 AIConversationService.h . . . 3650
18.546 AIProviderClient_if.h . . . 3651
18.547 AISecretStore.h . . . 3652
18.548 AnthropicProviderClientImpl.h . . . 3652
18.549 HttpProviderClientBase.h . . . 3653
18.550 LocalProviderClientImpl.h . . . 3654
18.551 OpenAIProviderClientImpl.h . . . 3654
18.552 BioKineticLawExpression.h . . . 3655
18.553 BioSimulationAnalysis.h . . . 3658
18.554 BioSimulationResult.h . . . 3663
18.555 GlpkFluxBalanceSolver.h . . . 3665
18.556 MassActionOdeSystem.h . . . 3667
18.557 MetabolicFluxBalanceSolver.h . . . 3669
18.558 WholeCellParameterReader.h . . . 3673
18.559 DiffusionMethodOfLinesSystem.h . . . 3675
18.560 DormandPrince54OdeSolver.h . . . 3677
18.561 OdeSolver_if.h . . . 3679
18.562 OdeSolverFactory.h . . . 3679
18.563 OdeSystem_if.h . . . 3681
18.564 Quadrature_if.h . . . 3681
18.565 RootFinder_if.h . . . 3681
18.566 RungeKutta4OdeSolver.h . . . 3681
18.567 Solver_if.h . . . 3682
18.568 SolverDefaultImpl1.h . . . 3682
18.569 FactorialDesign.h . . . 3683
18.570 Optimizer_if.h . . . 3684
18.571 OptimizerDefaultImpl1.h . . . 3686
18.572 ContinuousDistribution_if.h . . . 3687
18.573 DataAnalyser_if.h . . . 3687
18.574 DataSet_if.h . . . 3688
18.575 DiscreteDistribution_if.h . . . 3688
18.576 Distribution_if.h . . . 3688
18.577 Fitter_if.h . . . 3688
18.578 FitterDefaultImpl.h . . . 3689
18.579 FitterDummyImpl.h . . . 3696
18.580 HypothesisTester_if.h . . . 3697
18.581 HypothesisTesterDefaultImpl1.h . . . 3699

18.582 ProbabilityDistribution.h 3700
18.583 ProbabilityDistributionBase.h 3700
18.584 SimulationResultsDataset.h 3701
18.585 TraitsTools.h 3702

Index **3703**

Chapter 1

Editing an EFSM Network in the graphical interface

This guide describes the EFSM Network editor integrated into the GenESyS Qt6 GUI. Menu labels below are the English labels in the current GUI. The editor uses the model's existing `ModalModelDefault`, `EFSMNetwork`, states, and transitions; EFSM relations are not process-flow connections.

1.1 Create or attach a Network

1. Open a process model and select a `ModalModelDefault` component in its graphics view.
2. From the **Model** menu, choose **Create EFSM Network** and enter a unique Network name. Canceling the dialog leaves the model unchanged.
3. To attach an existing Network instead, choose **Select EFSM Network** and select the entry identified by its name and model ID. Selecting a new Network replaces the adapter's association; the previously registered Network remains in the model.
4. With the adapter selected, choose **Open EFSM Network**. The Network opens in a tab titled `Model > EFSM Network: NetworkName`. Adapters that refer to the same Network open the same context.

1.2 Add states and transitions

1. In the Network tab, choose **Model** > **Create EFSM State** and enter a unique state name. Repeat for each state.
2. Select one state to create a self-loop, or select two states to create a directed relation. Choose **Model** > **Create EFSM Transition**, then select the source and destination in the dialog. The same source/destination pair can have parallel transitions; a transition name may be left empty.
3. Move a state by dragging it. This changes its diagram position and updates the drawn edges.
4. Select a state and use **Model** > **Set Selected EFSM State as Initial** to change the initial state.
5. Select one state and choose **Model** > **Edit EFSM State...** to change its name, entry action, exit action, or final-state flag. Select one transition and choose **Model** > **Edit EFSM Transition...** to edit its name, guard, output, input event, priority, trigger event, or probability expression.
6. In a Network tab, the normal **Edit** > **Delete** action removes selected EFSM items. Removing a state also removes its incident transitions. Undo/redo applies to supported EFSM creation, deletion, movement, and property edits.

The expression fields are stored by the domain APIs. This editor does not add a parse-only syntax check; the current expression parser can evaluate an expression, so the GUI does not execute it merely to validate input.

1.3 Edit adapter bindings

Return to the process-model tab, select the `ModalModelDefault`, and choose **Model** > **Edit EFSM Bindings...** to edit the Network's supported input and output bindings. The binding dialog is available only when an EFSM Network is attached.

1.4 Save and reopen

Save the model using the graphical `.gui` format to keep both the mathematical model and its diagram layout. The `.gen` data remains authoritative for the Network, states, transitions, and their properties; `.gui` stores graphical positions and edge presentation. Reopen the `.gui` file and use **Open EFSM Network** again to restore the diagram. Older `.gui` files without EFSM layout data remain loadable; the editor assigns deterministic positions when saved positions are unavailable.

1.5 Current scope

The integrated editor provides a diagram editor for EFSM Networks. It does not provide a separate EFSM execution animation or a new runtime. Simulation semantics remain in the existing GenESyS kernel. The guide records the tested menu and dialog workflow; it does not claim that every possible expression is syntactically or semantically validated by the GUI.

Chapter 2

Cellular Automata Viewer

This folder contains the first standalone cellular-automata tool for the Genesys Qt GUI.

Current scope:

- 2D demo built from the existing cellular-automata plugin classes
- UI selectors for rule, neighborhood, and state preset
- Forest Fire preset with four enumerated states
- mouse editing with a selectable paint state
- discrete stepping plus timer-driven auto-run
- JSON save/load for lattice size, time, and cell states

Known limitations in this first pass:

- the viewer is intentionally demo-focused, not a universal CA builder yet
- more advanced neighborhood/radius combinations remain plugin-side follow-up work

Chapter 3

GenESyS Launcher and Per-User Runtime Dispatcher

3.1 Purpose

`source/applications/launcher` provides the stable invocation layer required for a future Debian installation to coexist with independently updated GenESyS runtimes in each user's home directory.

This directory builds two executables:

- `genesys-launcher`: interactive GUI entry point and the **only** process in this domain permitted to check for remote runtime updates;
- `genesys-dispatch`: non-interactive application dispatcher. It never checks for updates, never uses the network, never displays an update dialog, and never modifies the active runtime.

The launcher is intentionally independent of the simulator kernel, parser and plugin libraries. The focused `launcher-app` preset disables those subsystems.

3.2 Architecture

```
future Debian installation
  |
  +-- stable genesys-launcher / genesys-dispatch
  +-- stable system fallback runtime
  |     `-- <system-runtime-root>/bin/<application>
  `-- per-user runtime support
        |
        +-- $XDG_DATA_HOME/genesys/apps/<version>/
        `-- $XDG_DATA_HOME/genesys/current -> apps/<version>
```

A future package may expose public `/usr/bin/genesys-*` wrappers, but this implementation does **not** create or replace those wrappers. Debian package restructuring is a separate integration phase.

3.3 CMake targets and preset

The root option is:
`GENESYS_BUILD_LAUNCHER_APPLICATION`

Focused tests use:
`GENESYS_BUILD_LAUNCHER_TESTS`

The application CMake targets are:
```
genesys-launcher
genesys-dispatch
```

Shared internal libraries are:
```
genesys_launcher_runtime   # Qt6::Core only
genesys_launcher_update    # runtime + Qt6::Network
```

The dedicated preset is:
```
cmake --preset launcher-app
cmake --build --preset launcher-app
ctest --preset launcher-app --output-on-failure
```

The preset builds the launcher, dispatcher and focused tests without building the GenESyS kernel, parser, plugins, worker or main GUI.

3.4 Install contract for future packaging

CMake installs only the stable internal executables:
```
${CMAKE_INSTALL_LIBEXECDIR}/genesys/genesys-launcher
${CMAKE_INSTALL_LIBEXECDIR}/genesys/genesys-dispatch
```

It deliberately does **not** install public `/usr/bin` wrappers in this change.

The configurable system fallback root defaults conceptually to:
```
/usr/libexec/genesys/system/
  `-- bin/
      +-- genesys-gui
      +-- genesys-shell
      +-- genesys-worker
      `-- genesys-mcp
```

The path is centralized as the CMake cache variable `GENESYS_SYSTEM_RUNTIME_ROOT`; tests override it with temporary directories.

Other build-time integration variables are:

- `GENESYS_LAUNCHER_VERSION` — stable launcher contract version;
- `GENESYS_SYSTEM_PACKAGE_VERSION` — base installation version used for manifest compatibility checks;
- `GENESYS_SYSTEM_CONFIG_PATH` — defaults to `/etc/genesys/update.conf`;
- `GENESYS_UPDATE_KEYRING_PATH` — trusted OpenPGP keyring used by `gpgv`.

The repository currently has no canonical project-wide CMake version. Therefore these launcher/package version inputs default to `0.0.0` rather than creating a competing global version source. The future Debian/release integration must provide authoritative values at build time.

3.5 XDG directories

User paths are resolved centrally by `RuntimePaths`.

When the corresponding XDG variable is an absolute path:

```
configuration: $XDG_CONFIG_HOME/genesys/update.conf
cache:         $XDG_CACHE_HOME/genesys/
data:          $XDG_DATA_HOME/genesys/
```

Fallbacks when the XDG variable is absent are:

```
~/.config/genesys/update.conf
~/.cache/genesys/
~/.local/share/genesys/
```

Runtime layout:

```
~/.local/share/genesys/apps/<version>/
~/.local/share/genesys/current
```

Download cache:

```
~/.cache/genesys/downloads/
```

Launcher log:

```
~/.local/share/genesys/logs/launcher.log
```

If `HOME` is unavailable or unusable, per-user runtime selection/update is disabled and the dispatcher can use only the configured system fallback.

3.6 Configuration and precedence

The launcher reads:

```
/etc/genesys/update.conf
$XDG_CONFIG_HOME/genesys/update.conf
```

The user configuration is loaded first and explicitly supplied system values are applied last. Consequently the system policy wins for every key that it defines.

Supported `[updates]` keys are:

```
[updates]
enabled=true
channel=stable
allow_user_runtime=true
require_signature=true
check_on_gui_startup=true
fallback_to_system=true
minimum_check_interval_hours=12
max_user_versions=2
manifest_url=https://example.invalid/update-manifest.json
```

Remote updating is fail-safe by default: **no remote check occurs unless explicit effective policy exists and provides a valid HTTPS `manifest_url`.** A source-tree/development build therefore does not unexpectedly contact GitHub.

Invalid system security values are handled conservatively. In particular, invalid `enabled`, `allow_user_runtime`, or `require_signature` values cannot silently enable an unsafe path.

3.7 Application selection

`genesys-dispatch` requires an explicit application:

```
genesys-dispatch --app genesys-shell -- <arguments>
genesys-dispatch --app genesys-worker -- <arguments>
genesys-dispatch --app genesys-mcp -- <arguments>
```

The dispatcher preserves the subsequent argument vector; it does not construct a shell command. On Linux, `ProcessLauncher` uses `execv()` so the stable dispatcher process is replaced by the selected application and the environment/current working directory are inherited.

Selection is deterministic:

1. read effective policy;
2. if user runtimes are permitted, inspect `current`;
3. use the requested user-runtime application only when the runtime and executable satisfy validation;
4. otherwise use the system application when fallback is permitted;
5. return a controlled error if neither exists.

The dispatcher never downloads, activates, cleans, or changes `current`.

3.8 Active runtime validation

A user runtime is eligible only when:

- `current` is a symbolic link;
- it resolves to a direct child of the configured `apps` root;
- the target does not escape the authorized data tree;
- `VERSION` exists and parses as a numeric dotted version;
- `MANIFEST.json` exists and has schema version 1/product `genesys-simulator`;
- the internal manifest version matches `VERSION`;
- the requested application is declared by the internal manifest;
- the corresponding `bin/<application>` is a regular executable file;
- the executable is not a symlink and resolves inside the runtime `bin` directory.

Any failed condition causes a logged fallback rather than execution of an untrusted path.

3.9 <tt>genesys-worker</tt> and legacy <tt>genesys-web</tt>

The current CMake worker target is `genesys_worker_app` and its output is `genesys-worker`. The existing Debian tree still contains `genesys-web` naming.

The canonical application name for the launcher contract is therefore `genesys-worker`.

For a request for `genesys-web`, selection is explicitly ordered as:

```
1. genesys-web, when that executable is explicitly present and declared;
2. genesys-worker as the compatibility alias;
3. system fallback under the same ordered rule.
```

No other application receives an implicit alias.

3.10 MCP

`source/applications/mcp` is an existing Python MCP server and publishes the entry point:

```
genesys-mcp = genesys_mcp.server:main
```

The launcher does not rewrite it in C++ and does not install Python packages.

If `bin/genesys-mcp` is declared and present in an active runtime, the dispatcher executes it like any other application. Otherwise it tries the system fallback. If neither exists, `genesys-dispatch --app genesys-mcp` returns a controlled message directing the user to `genesys-gui` or to a distribution that includes MCP.

3.11 Version rules

`Version` accepts dotted non-negative numeric components, including a leading `v` when parsing a release tag:

```
2026.1.0
2026.1.2
v2026.2.0
```

Comparisons pad missing trailing components with zero (`2026.1 == 2026.1.0`). Non-numeric prerelease syntax is intentionally rejected because it is outside the current update contract.

A remote candidate older than the active/base version is rejected as a downgrade.

3.12 Platform detection

On Linux the updater reads `/etc/os-release` and combines normalized `ID`, `VERSION_ID`, and the actual CPU architecture.

Example:

```
ubuntu-24.04-x86_64
```

Linux is not assumed to mean Ubuntu. If platform identification fails or the exact key is absent from the manifest, no update is installed and normal runtime/system selection remains available.

3.13 Remote manifest schema

The parser supports schema version 1 and requires:

```
{
  "schema_version": 1,
  "product": "genesys-simulator",
  "channel": "stable",
  "version": "2026.2.1",
  "release_tag": "v2026.2.1",
  "published_at": "2026-08-20T00:00:00Z",
  "minimum_launcher_version": "2026.1.0",
  "minimum_system_package_version": "2026.1.0",
  "platforms": {
    "ubuntu-24.04-x86_64": {
      "kind": "user-runtime",
      "url": "https://example.invalid/genesys-runtime.tar.zst",
      "sha256": "<64 lowercase/uppercase hexadecimal characters>",
      "signature_url": "https://example.invalid/genesys-runtime.tar.zst.sig",
      "size": 123,
      "applications": [
        "genesys-gui",
        "genesys-shell",
        "genesys-worker",
        "genesys-mcp"
      ]
    }
  }
}
```

The parser rejects unsupported schema/product/channel/platform, invalid versions, downgrade, invalid/non-↩ HTTPS URLs, invalid SHA-256, invalid sizes, duplicate/invalid/empty application lists, and unsatisfied minimum launcher/system-package versions. GitHub HTML is never scraped.

3.14 Network behavior

Only the launcher update library links `Qt6::Network`.

`QtNetworkTransport` enforces:

- HTTPS-only requests for production update operations;
- bounded response sizes;
- finite transfer timeouts;
- a bounded redirect count;
- Qt's no-less-safe redirect policy;
- normal TLS validation (TLS errors are never ignored);
- cancellation;
- streaming downloads to a destination instead of concatenating shell commands or trusting a remote filename.

The manifest is limited to 1 MiB. Runtime bundle size is bounded by the signed manifest and currently capped at 2 GiB by the manifest parser. Signature downloads are limited to 1 MiB.

Tests use an injected `INetworkTransport`; they do not access GitHub or any public network.

3.15 Integrity and signature verification

SHA-256 verification is mandatory for every runtime bundle. The expected digest is supplied by the validated remote manifest. Digest comparison is performed over decoded fixed-size digest bytes without early exit.

When `require_signature=true`, `GpgvSignatureVerifier` invokes `gpgv` directly with an argument vector:

```
gpgv --keyring <trusted-keyring> <signature> <bundle>
```

No shell is involved. Signature-required mode fails closed when the trusted keyring, bundle/signature, `gpgv`, or a valid signature is unavailable.

This repository change does not choose or install the institutional/release signing key. Future Debian integration must install the public keyring and declare `gpgv`/archive runtime dependencies as appropriate. The code does not weaken signature-required policy while that deployment step is pending.

3.16 Secure archive extraction

Runtime archives are expected to be `tar.zst`.

`TarZstdArchiveExtractor` invokes the system `tar` directly, never through a shell. Before extraction it inspects both path and verbose listings.

The current implementation intentionally accepts only:

- regular files;
- directories.

It rejects:

- absolute paths;
- `./..` path components and traversal;
- escaped/backslash path forms in the archive listing;
- symlinks;
- hardlinks;
- device nodes;
- FIFOs and other special entries;
- setuid/setgid entries.

Extraction uses `--no-same-owner` and `--no-same-permissions`, targets a fresh `.partial` directory, and the resulting tree is scanned again for links/special entries/root escape before runtime validation.

Future Debian packaging must ensure compatible `tar` + zstd support is installed. Missing tooling causes a controlled update failure and leaves the existing runtime/system installation usable.

3.17 Runtime validation and atomic activation

The installation flow is:

```
download .part
  -> verify exact size
  -> verify SHA-256
  -> verify OpenPGP signature when required
  -> promote verified cache file
  -> extract to apps/<version>.partial
  -> validate extracted runtime
  -> rename to apps/<version>
  -> create temporary current symlink
  -> atomic rename over current
```

The extracted runtime must contain:

```
VERSION
MANIFEST.json
bin/
```

The internal manifest must match the remote version and application set. The complete runtime contract currently requires at least:

```
genesys-gui
genesys-shell
genesys-worker
genesys-mcp
```

Every declared application must exist as a non-symlink regular executable in `bin/`.

`current` is never changed before extraction and validation finish. On Unix the temporary symlink is atomically renamed over the previous `current` symlink. A non-symlink `current` entry is never overwritten.

3.18 Concurrency and retention

`UserRuntimeManager` uses `QLockFile` under the GenESyS user data root. The launcher acquires the lock before download finalization and holds it through verification, installation, activation and cache cleanup. It is released before the selected GenESyS application is executed.

Runtime cleanup is best-effort and respects `max_user_versions`. It always preserves:

- the newly active runtime;
- the previously active runtime as immediate rollback.

Therefore rollback safety can retain two versions even when an administrative `max_user_versions` value of 1 is supplied.

3.19 Logging

The default launcher log is:
`$XDG_DATA_HOME/genesys/logs/launcher.log`

Each line contains a UTC timestamp, launcher version, event and bounded diagnostic detail. Relevant events include selection/fallback, update checks, downloads, verification, extraction/activation errors, activation and retention.

The implementation does not log configuration contents, tokens, credentials or secrets. The log rotates to `.1` when it reaches approximately 1 MiB.

3.20 Launcher CLI

Supported operations are intentionally small:

```
genesys-launcher --app genesys-gui --interactive-update
genesys-launcher --check-only
genesys-launcher --no-update
genesys-launcher --version
```

`--check-only` reports through console output and never installs a runtime. `--no-update` bypasses the update path and launches according to the selector. Normal `genesys-gui` startup checks only when `check_on_gui_startup=true` and all other effective update policy requirements permit it.

The launcher refuses remote update modes for non-GUI applications. Non-interactive applications must use `genesys-dispatch`.

3.21 Failure and fallback behavior

Update failure is not application failure. Network errors, invalid manifests, checksum/signature errors, archive errors, disk/write errors, invalid runtime trees, lock contention and activation errors all leave the previously active `current` untouched.

After update handling, the launcher selects and starts the active user GUI or the system fallback. The dispatcher similarly selects user/system applications without attempting an update.

On Linux `ProcessLauncher` attempts the system fallback if execution of a selected user executable returns an execution error such as `EACCES` (including a home mounted `noexec`).

3.22 Tests

Focused Google Test coverage uses only temporary roots and local test doubles. It covers:

- version parsing/comparison/downgrade;
- user/system configuration and fail-safe defaults;
- XDG defaults/overrides/missing HOME;
- remote manifest contract and minimum versions;
- runtime selection/fallback;
- `genesys-web` compatibility and MCP;
- SHA-256 and signature-required fail-closed paths;
- unsafe archive listings;
- `.partial` installation and atomic activation;
- preservation of the previous `current` on failures;
- retention/rollback;
- dispatcher argument forwarding through an injected process launcher;
- update-check interval behavior;
- a complete local fake manifest -> download -> checksum -> signature -> extract -> validate -> activate -> select cycle.

The focused GitHub Actions workflow additionally records dynamic-link evidence showing that `genesys-dispatch` does not link `Qt6::Network` or `Qt6::Widgets`.

3.23 Known limitations and future Debian integration

This change intentionally does not:

- rewrite `debian/control` or `.install` files;
- replace `/usr/bin/genesys-gui` or create final wrappers;
- move current application binaries into the system fallback tree;
- install the OpenPGP public keyring;
- publish/sign release runtime bundles;
- convert the Python MCP server to C++;
- perform a real `dpkg` installation or laboratory test.

The next Debian phase must map the existing application outputs into the system fallback directory, install public wrappers that call this stable launcher/dispatcher, reconcile legacy `genesys-web` packaging with canonical `genesys-worker`, package the existing Python `genesys-mcp` when desired, install update policy/key material/tool dependencies, and validate the complete package lifecycle as a non-root laboratory user.

Real Debian install, user-without-sudo, `HOME` mounted `noexec`, and UFSC laboratory behavior remain `needs-local-validation` until that packaging phase exists.

Chapter 4

Genesys MCP Connector

Python MCP server that exposes GenESyS simulation capabilities to AI assistants (Claude Desktop, Claude Code, and any MCP-compatible client).

4.1 Prerequisites

- Python 3.10+
- `genesys-worker` running (see `source/applications/worker/README.md`)

4.2 Install

```
cd source/applications/mcp
pip install -e .
```

4.3 Run

```
# via entry point
genesys-mcp

# or via module
python -m genesys_mcp
```

The server communicates over stdio (default FastMCP transport). Set `GENESYS_URL` to point to a non-default worker app address:

```
GENESYS_URL=http://192.168.1.10:8080 genesys-mcp
```

4.4 Claude Desktop integration

Add to `~/.config/claude/claude_desktop_config.json` (Linux) or `~/Library/Application Support/Claude/claude_desktop_config.json` (macOS):

```
{
  "mcpServers": {
    "genesys": {
      "command": "genesys-mcp",
      "env": {
        "GENESYS_URL": "http://localhost:8080"
      }
    }
```

```
  }
}
```

If `genesys-mcp` is not on PATH (e.g. installed in a venv), use the full path:

```
{
  "mcpServers": {
    "genesys": {
      "command": "/path/to/venv/bin/genesys-mcp",
      "env": {
        "GENESYS_URL": "http://localhost:8080"
      }
    }
  }
}
```

4.5 Claude Code integration

```
claude mcp add genesys -- genesys-mcp
```

Or with a custom URL:

```
claude mcp add genesys -e GENESYS_URL=http://localhost:8080 -- genesys-mcp
```

4.6 Workflow

All tools that interact with the simulator require a session token. Call `genesys_auth` once per conversation and pass the returned token to every subsequent tool.

```
1. genesys_auth          → { token: "..." }
2. genesys_new_model     (token)
3. genesys_import_model_language  (token, model_text)
4. genesys_configure_simulation   (token, replications=10, length=480.0)
5. genesys_run_simulation         (token)
6. genesys_create_job    (token) → { jobId: "..." }
7. genesys_run_job       (token, jobId)
8. genesys_job_result    (token, jobId)
```

4.7 Available tools (18)

Tool	Auth required	Description
`genesys_health`	No	Check worker app reachability
`genesys_auth`	No	Create session, returns token
`genesys_info`	Yes	Simulator version and licence
`genesys_new_model`	Yes	Create empty model
`genesys_get_model`	Yes	Describe current model
`genesys_save_model`	Yes	Save model to workspace file
`genesys_load_model`	Yes	Load model from workspace file
`genesys_import_model_language`	Yes	Import model from text
`genesys_simulation_status`	Yes	Current simulation state
`genesys_configure_simulation`	Yes	Set replication/length/warmup params
`genesys_run_simulation`	Yes	Run simulation synchronously
`genesys_step_simulation`	Yes	Advance one event step
`genesys_worker_info`	No	Worker identity metadata
`genesys_worker_capabilities`	No	Worker capability flags
`genesys_create_job`	Yes	Create job from current model
`genesys_job_status`	Yes	Query job state
`genesys_run_job`	Yes	Execute job synchronously
`genesys_job_result`	Yes	Retrieve job result summary

Chapter 5

Genesys Worker Application

First functional implementation of the **Worker** application type for GenESyS.

5.1 Current scope

- CMake target: `genesys_worker_app` (enabled with `-DGENESYS_BUILD_WORKER_APPLICATION=ON`).
- Executable name: `genesys-worker`.
- Layered architecture for HTTP transport, API routing, session/token, and simulator service.
- Stateful session management with one `Simulator` per session and per-session workspace directories.
- Endpoints:
  - `GET /health`
  - `POST /api/v1/auth/session`
  - `GET /api/v1/simulator/info` (requires `Authorization: Bearer <token>`)
  - `POST /api/v1/models` (requires `Authorization: Bearer <token>`)
  - `GET /api/v1/models/current` (requires `Authorization: Bearer <token>`)
  - `POST /api/v1/models/save` (requires `Authorization: Bearer <token>`)
  - `POST /api/v1/models/load` (requires `Authorization: Bearer <token>`)
  - `GET /api/v1/simulation/status` (requires `Authorization: Bearer <token>`)
  - `POST /api/v1/simulation/config` (requires `Authorization: Bearer <token>`)
  - `POST /api/v1/simulation/run` (requires `Authorization: Bearer <token>`)
  - `POST /api/v1/simulation/step` (requires `Authorization: Bearer <token>`)

5.2 Worker cycle Stage 1 (new cycle)

The Worker application is now also being evolved as a **worker API** that can be discovered by a future distributed orchestrator application.

This stage adds two public worker discovery endpoints (no Bearer token required):

- `GET /api/v1/worker/info`
- `GET /api/v1/worker/capabilities`

These routes expose worker identity metadata and current implementation capability flags. They are intentionally limited to discovery only and do not include job submission, polling, or background execution in this stage.

5.3 Worker cycle Stage 2

This stage starts worker model-ingress support with one authenticated and session-scoped endpoint:

- `POST /api/v1/worker/models/import-language` (requires `Authorization: Bearer <token>`)

The request body must be plain text GenESyS model language/specification. The worker imports this text into the session simulator as the current model.

This stage still does **not** include multipart/binary upload, distributed job submission, polling, or background orchestration.

5.4 Worker cycle Stage 3

This stage introduces a minimal worker job abstraction with authenticated, session-scoped endpoints:

- `POST /api/v1/worker/jobs` (requires `Authorization: Bearer <token>`)
- `GET /api/v1/worker/jobs/{jobId}` (requires `Authorization: Bearer <token>`)

Behavior in this stage:

- job creation requires a valid current model in the session;
- each job is created in in-memory storage with state `queued`;
- job creation persists a session-workspace snapshot file (for example `job_<id>.gen`) to decouple future execution from live model mutations;
- job metadata can be queried by id from the same authenticated session.

This stage still does **not** execute jobs, provide result retrieval, add polling loops, or introduce background worker threads.

5.5 Worker cycle Stage 4

This stage adds explicit, synchronous execution of existing worker jobs:

- `POST /api/v1/worker/jobs/{jobId}/run` (requires `Authorization: Bearer <token>`)

Behavior in this stage:

- execution is triggered only for an existing job id;
- job state is updated in memory through `queued -> running -> finished|failed`;
- execution is synchronous and performed by the session simulator using the job snapshot file;
- `GET /api/v1/worker/jobs/{jobId}` remains the basic state inspection/polling mechanism after run requests.

This stage still does **not** include background execution, queue worker loops, cancellation, streaming updates, or a dedicated job result endpoint.

5.6 Worker cycle Stage 5

This stage adds terminal result retrieval for previously executed worker jobs:

- `GET /api/v1/worker/jobs/{jobId}/result` (requires `Authorization: Bearer <token>`)

Behavior in this stage:

- result retrieval is available only after the job reaches a terminal state (`finished` or `failed`);
- queued or running jobs return a conflict response because no terminal result exists yet;
- the endpoint returns a minimal structured simulation summary persisted at the end of synchronous execution;
- execution remains synchronous and non-background (`POST /run` still performs the run inline);
- cancellation is still not available;
- distributed orchestration remains out of scope.

5.7 Worker cycle Stage 6

This stage hardens HTTP transport behavior for existing worker and session APIs without changing route contracts:

- request reading is now incremental (`recv` loop) instead of single-shot reads;
- request assembly is bounded by a transport limit of **1 MiB** (larger payloads return `413 Payload Too Large`);
- socket receive/send timeouts are configured to **5 seconds** per client connection;
- when timeout happens before a complete request is assembled, the server returns `408 Request Timeout`;
- transport-level error payloads remain JSON-shaped and consistent with current server error responses.

This stage is explicitly operational hardening only. Application-layer endpoints and worker cycle stages 1-5 contracts remain unchanged.

5.8 How to run

```
cmake -S . -B build/worker-debug -G Ninja -DGENESYS_BUILD_WORKER_APPLICATION=ON
cmake --build build/worker-debug --target genesys_worker_app
./build/worker-debug/source/applications/worker/genesys-worker --port 8080
```

Optional parameters:

- `--port <N>`: HTTP port (default 8080).
- `--max-requests <N>`: exit after N requests (useful for scripted smoke tests).

5.9 Security notice

Session creation in Stage 1 is provisional and stateful in memory. Generated tokens are opaque random strings and **not** production-grade authentication.

For Stage 3 model persistence:

- Save/load operations are restricted to each session workspace directory.
- API requests accept only a `filename` basename (no paths).
- Filenames must match `[A-Za-z0-9._-]` and cannot include `..`, `/`, or `\`.

For Stage 4 simulation configuration:

- `POST /api/v1/simulation/config` currently uses a minimal contract parser without external JSON libraries.
- All configuration fields are required in the request body: `numberOfReplications`, `replicationLength`, `warmUpPeriod`, `pauseOnEvent`, `pauseOnReplication`, `initializeStatistics`, `initializeSystem`.

For Stage 5 simulation execution:

- `POST /api/v1/simulation/run` and `POST /api/v1/simulation/step` are synchronous wrappers over `ModelSimulation::start()` and `ModelSimulation::step()`.
- This stage intentionally does not introduce asynchronous/background simulation control endpoints.

Chapter 6

Customization Points

The custom directory is an injection point for custom user configurations.

6.1 Header <tt>gtest.h</tt>

6.1.1 The following macros can be defined:

- `GTEST_OS_STACK_TRACE_GETTER_` - The name of an implementation of `OsStackTraceGetter↩Interface`.
- `GTEST_CUSTOM_TEMPDIR_FUNCTION_` - An override for `testing::TempDir()`. See `testing↩::TempDir` for semantics and signature.

6.2 Header <tt>gtest-port.h</tt>

The following macros can be defined:

6.2.1 Logging:

- `GTEST_LOG_(severity)`
- `GTEST_CHECK_(condition)`
- Functions `LogToStderr()` and `FlushInfoLog()` have to be provided too.

6.2.2 Threading:

- `GTEST_HAS_NOTIFICATION_` - Enabled if Notification is already provided.
- `GTEST_HAS_MUTEX_AND_THREAD_LOCAL_` - Enabled if `Mutex` and `ThreadLocal` are already provided. Must also provide `GTEST_DECLARE_STATIC_MUTEX_(mutex)` and `GTEST_DEFINE_↩STATIC_MUTEX_(mutex)`
- `GTEST_EXCLUSIVE_LOCK_REQUIRED_(locks)`
- `GTEST_LOCK_EXCLUDED_(locks)`

6.2.3 Underlying library support features

- GTEST_HAS_CXXABI_H_

6.2.4 Exporting API symbols:

- GTEST_API_ - Specifier for exported symbols.

6.3 Header <tt>gtest-printers.h</tt>

- See documentation at gtest/gtest-printers.h for details on how to define a custom printer.

Chapter 7

Análise técnica do parser (Bison/Flex)

7.1 Audit Status (WiP20261)

- Branch audited: `WiP20261`
- Audit scope: reauditoria do lexer/gramática e ações semânticas em `source/parser/parserBison↩Flex`
- Status legend: `DONE`, `PARTIAL`, `OPEN`, `UNCERTAIN`, `SUPERSEDED`
- Data desta reauditoria documental: `2026-04-10`

7.2 Escopo

Arquivos revalidados:

- `source/parser/parserBisonFlex/bisonparser.yy`
- `source/parser/parserBisonFlex/lexerparser.ll`
- `source/parser/parserBisonFlex/bisonparser.report` (não encontrado no checkout atual)

7.3 Registro histórico (pré-reauditoria)

O texto original deste documento apontava conflitos LR elevados, erros de tokenização e lacunas semânticas. Parte desse diagnóstico foi superada por mudanças posteriores; parte ainda permanece relevante.

7.4 Reclassificação dos achados históricos

7.4.1 Conversão hexadecimal

7.4.2 Audit status

`DONE`

7.4.3 Evidence

- O lexer converte hex com `std::strtoll(yytext, &endPtr, 16)` e emite `NUMH`.

7.4.4 Remaining gaps

- Sem gap direto neste item.

7.4.5 Tokenização de <tt>min</tt> / <tt>max</tt>

7.4.6 Audit status

`DONE`

7.4.7 Evidence

- `min` retorna `mathMIN` e `max` retorna `mathMAX`.

7.4.8 Remaining gaps

- Sem gap direto neste item.

7.4.9 Regex de <tt>EntitiesWIP</tt>

7.4.10 Audit status

`DONE`

7.4.11 Evidence

- Regra atual usa padrão de palavra completa `[eE][nN][tT][iI][tT][iI][eE][sS][wW][iI][pP]`.

7.4.12 Remaining gaps

- Sem gap direto neste item.

7.4.13 Regra isolada <tt>T</tt> no lexer

7.4.14 Audit status

`SUPERSEDED`

7.4.15 Evidence

- A regra isolada não aparece mais no lexer atual.

7.4.16 Remaining gaps

- Sem gap direto neste item.

7.4.17 Token <tt>ILLEGAL</tt> antes inútil

7.4.18 Audit status

DONE

7.4.19 Evidence

- A gramática atual inclui `expression: ... | illegal { $$.valor = -1; }`, tornando o caminho alcançável.

7.4.20 Remaining gaps

- Avaliar qualidade de recuperação/diagnóstico de erro além da alcançabilidade.

7.4.21 Estrutura de precedência/associatividade da gramática

7.4.22 Audit status

DONE

7.4.23 Evidence

- Estrutura por níveis explícitos: `logicalOr -> logicalXor -> logicalAnd -> logicalNot -> relational -> additive -> multiplicative -> power -> unary -> primary`.
- Potência em `power: unary POWER power`, indicando associatividade à direita.

7.4.24 Remaining gaps

- Não foi possível revalidar numericamente conflitos LR no estado atual sem `bisonparser.report`.

7.5 Pontos ainda pendentes (ou parcialmente mitigados)

7.5.1 Funções semânticas não implementadas totalmente (<tt>fDISC</tt>, <tt>fLASTINQ</tt>, <tt>fRESSEIZES</tt>)

7.5.2 Audit status

OPEN

7.5.3 Evidence

- `fDISC` em `probFunction` contém TODO e retorno placeholder (`sampleDiscrete(0,0)`).
- `fLASTINQ` possui ação vazia com comentário de não implementado.
- `fRESSEIZES` também está com comentário de não implementado.

7.5.4 Remaining gaps

- Implementar semântica real dessas funções e cobrir com testes de expressão.

7.5.5 Avaliação de <tt>FORM</tt>

7.5.6 Audit status

OPEN

7.5.7 Evidence

- Regras de `FORM` continuam comentando problema sério de reentrada e retornam `0.0`.

7.5.8 Remaining gaps

- Resolver avaliação de fórmula (reentrância/arquitetura de avaliação) para evitar retorno silencioso inválido.

7.5.9 Risco de null-deref em atribuições/leituras ligadas a evento atual

7.5.10 Audit status

PARTIAL

7.5.11 Evidence

- Há guardas em alguns caminhos (`if (getCurrentEvent() != nullptr && getCurrentEvent()->getEntity() != nullptr)`), mas ainda existem acessos diretos como `fIDENT` e outras ações usando `getCurrentEvent()->getEntity()` sem guarda uniforme.

7.5.12 Remaining gaps

- Padronizar guarda robusta em todos os pontos de acesso a evento/entidade corrente.

7.5.13 Conflitos LR atuais

7.5.14 Audit status

`UNCERTAIN`

7.5.15 Evidence

- `source/parser/parserBisonFlex/bisonparser.report` não estava disponível no checkout atual, então não houve recontagem atual de conflitos shift/reduce e reduce/reduce.

7.5.16 Remaining gaps

- Gerar novo report do Bison e atualizar números concretos de conflitos.

7.6 Recomendações arquiteturais que seguem válidas

1. Formalizar tratamento de funções não implementadas com erro explícito (em vez de `0`/placeholder silencioso).
2. Consolidar estratégia de null-safety para contexto de evento corrente.
3. Reintroduzir rotina de auditoria de conflitos LR no pipeline (geração e versionamento de report ou artefato equivalente).
4. Fortalecer testes de regressão de parser para operadores, funções de plugin e cenários de erro.

7.7 Remaining Work

- `OPEN` — Implementar semântica real de `fDISC`.
- `OPEN` — Implementar semântica real de `fLASTINQ`.
- `OPEN` — Implementar semântica real de `fRESSEIZES`.
- `OPEN` — Resolver avaliação de `FORM` sem retorno neutro silencioso.
- `PARTIAL` — Eliminar caminhos restantes de null-deref em `getCurrentEvent()->getEntity()`.
- `UNCERTAIN` — Recontar conflitos LR com novo `bisonparser.report`.

Chapter 8

Kernel Simulator Unit Tests - Build & Run Guide

This document describes a **ready-to-use CMake/CLion setup** for running all unit tests related to the Kernel Simulator package.

8.1 What to use (recommended)

Use these presets from `CMakePresets.json`:

- **Configure preset:** `tests-kernel-unit`
- **Build preset:** `tests-kernel-unit`
- **Build target (executed by the preset):** `genesys_kernel_unit_tests_run`

The `genesys_kernel_unit_tests_run` target:

1. Builds all kernel unit-test executables;
2. Runs all of them automatically (including the generated method-inventory test executable).

8.2 CLion setup (single-click workflow)

1. Open **Settings/Preferences** > **Build, Execution, Deployment** > **CMake**.
2. Enable **Load CMake presets** (if not already enabled).
3. Select **Configure preset = `tests-kernel-unit`**.
4. In the run/build target selector, choose **Build preset = `tests-kernel-unit`**.
5. Build that preset.

Expected behavior: CLion builds everything required and then runs all kernel unit-test binaries automatically in sequence.

8.3 Command line equivalent

```
cmake --preset tests-kernel-unit
cmake --build --preset tests-kernel-unit
```

This is equivalent to running the single target `genesys_kernel_unit_tests_run`.

8.4 Executables included in the run target

The run target executes the following unit-test executables:

- `genesys_test_util`
- `genesys_test_simulator_support`
- `genesys_test_simulator_runtime`
- `genesys_test_support_modelmanager`
- `genesys_test_support_persistence`
- `genesys_test_runtime_pluginmanager`
- `genesys_test_support_oneventmanager`
- `genesys_test_support_parsermanager`
- `genesys_test_support_tracemanager`
- `genesys_test_support_connectionmanager`
- `genesys_test_support_simulationscenario`
- `genesys_test_support_experimentmanager`
- `genesys_test_kernel_simulator_method_inventory` (auto-generated inventory scaffold)

8.5 Notes

- The generated inventory test executable is built from:
  - generator script: `source/tests/unit/tools/generate_kernel_simulator_test_stubs.py`
  - generated source: `source/tests/unit/generated/test_kernel_simulator_method_inventory.generated.cpp`
- If headers in `source/kernel/simulator` change, the generated inventory test file is automatically regenerated during build.

Chapter 9

Architecture notes for <tt>source/tools</tt>

9.1 1. Architectural overview

The `tools` package provides domain-support services for statistical analysis and numerical procedures. It exposes interfaces used by higher-level workflows and retains legacy implementations for compatibility.

9.2 2. Current package responsibilities

- Dataset-oriented analysis orchestration (`DataAnalyser_if`).
- Simulation result dataset loading, including Genesys `Record` text output with replication metadata.
- Distribution fitting contracts (`Fitter_if`).
- Parametric hypothesis testing (`HypothesisTester_if`, default impl 1).
- Distribution mathematical utilities and quantile inversion.
- Numerical integration and derivation utilities via legacy solver abstractions.
- Binding of abstractions to implementations through traits.

9.3 3. Approved target decomposition

9.3.1 Data analysis

- Keep façade role in `DataAnalyser_if`.
- Clarify ownership and lifecycle of returned collaborating services.

9.3.2 Fitting

- Preserve legacy fitting API.
- Transition from dummy implementation to a concrete default implementation in phased manner (FITTER-1 delivered baseline behavior, FITTER-2 added functional Beta and Weibull fitting in `FitterDefaultImpl`, and FITTER-3 promotes `TraitsTools<Fitter_if>` to `FitterDefaultImpl`).

9.3.3 Hypothesis testing

- Maintain current contract and refine implementation completeness, especially two-population methods.
- HYPTEST-1 alignment keeps the current pooled-vs-Welch heuristic, but documents it as a TODO while consolidating one-population p-value coherence and the two-proportion-difference confidence interval formula.

- HYPTEST-2 final alignment updates one-population proportion confidence intervals to the normal-approximation formulation (including finite-population correction when `N` is provided).

9.3.4 Distribution abstractions

- Introduce `Distribution_if` and specialized continuous/discrete interfaces.
- Keep legacy static façade while migration is in progress.

9.3.5 Numerical tools

- Split legacy solver responsibilities into dedicated interfaces:
  - `Quadrature_if`
  - `RootFinder_if`
  - `OdeSystem_if`
  - `OdeSolver_if`

9.3.6 Traits/binding

- Keep current traits stable.
- Expand only when safe concrete implementations exist.

9.4 4. Legacy compatibility strategy

- Do not break existing public signatures in this phase.
- Preserve compatibility by keeping legacy classes available while promoting only validated trait selections.
- Add new abstractions as header-only contracts until implementations are mature.

9.5 5. Planned migration order

1. Stabilize documentation and explicit contracts for existing headers.
2. Introduce new interface headers without forcing behavior changes.
3. Add concrete implementations incrementally behind existing traits.
4. Migrate trait bindings once implementations are production-safe.
5. Deprecate legacy conflated abstractions only after full replacement is validated.

9.6 6. AI Assistant component

9.6.1 Role

`AIAssistant_if` is a new high-level tool abstraction that will allow users to describe a simulation system in natural language and have the assistant construct, configure and run a GenESyS model automatically.

9.6.2 Stage 1 (complete) — Interface and implementation scaffold

- `AIAssistant_if`: abstract interface with enumerations (`AIProvider`, `ExecutionState`, `ReasoningEffort`), data structures (`AIAssistantConfiguration`, `AIAssistantRequest`, `AIAssistantResponse`, `GenesysKnowledgeItem`, `GenesysKnowledgeBaseSummary`) and method groups for configuration, security, SimulatorFacade integration, knowledge base, provider client, and workflow.
- `AIAssistantDefaultImpl`: concrete class implementing all interface methods.
- Security model: API key stored in memory only; no public getter; clearApiKey() overwrites the buffer before clearing; getMaskedApiKey() exposes only the last four characters. Permission flags (allowNetworkAccess, allowModelMutation, allowSimulationExecution, allowFileSystemWrites) default to false (deny-by-default).
- `TraitsTools<AIAssistant_if>` bound to `AIAssistantDefaultImpl`.

9.6.3 Stage 2 (complete) — Plugin introspection

- `refreshKnowledgeBaseFromSimulator()` populates the knowledge base with:
  - A static catalog of 15 well-known GenESyS building blocks (Create, Dispose, Queue, Seize, Release, Process, Delay, Decide, Assign, Record, Resource, Entity, Statistics, Variable).
  - Dynamic introspection via `SimulatorFacade::firstPlugin()` / `nextPlugin()`: for each loaded `Plugin`, reads `PluginInformation` fields (typename, isComponent, isSource/isSink, input/output topology, descriptionHelp, languageTemplate).
  - Duplicate entries between the static catalog and dynamic plugins are silently skipped.
- Returns `true` even when no facade is attached (static catalog is still useful).

9.6.4 Stage 3 (complete) — AI provider HTTP clients

- `AIProviderClient_if`: abstract interface for a single chat-completion HTTP call. Structures: `ChatMessage`, `ProviderRequest`, `ProviderResponse`.
- `HttpProviderClientBase`: abstract base implementing popen+curl transport. Security: API key written to a 0600 temp file (`mkstemp`), never on the command line. Shared JSON utilities: `_escapeJsonString`, `_extractJsonString`, `_extractJsonInt`, `_buildMessagesArray`, `_extractFirstArrayElement`.
- `OpenAIProviderClientImpl`: OpenAI chat completions; supports o-series reasoning models via `highReasoningMode`; optional Organization and Project headers.
- `AnthropicProviderClientImpl`: Anthropic Messages API; separates system message from conversation turns; extended-thinking budget for `highReasoningMode`.
- `LocalProviderClientImpl`: Ollama/LM Studio/llama.cpp; OpenAI-compatible format; no API key required.
- `AIAssistant_if` extended with `setProviderClient`, `getProviderClient`, `hasProviderClient`. `AIAssistantDefaultImpl` also exposes `createAndAttachProviderClient()` which creates the right client from configuration and takes ownership.

9.6.5 Stage 4 (complete) — Prompt analysis and model plan generation

- `analyzePrompt()`: builds a system prompt injecting the GenESyS knowledge base, calls the provider client, and returns the LLM's structured analysis.
- `createModelPlan()`: uses a structured system prompt requesting a Markdown model plan (entities, resources, queues, arrival, processing steps, statistics, simulation parameters) and returns it in `AIAssistantResponse::generatedPlan`.
- Both methods enforce `allowNetworkAccess == true` and check `checkReady()`.
- Diagnostics include provider name, model, input/output token counts, and HTTP status.

9.6.6 Stage 5 (complete) — Model construction

- `buildModel()`: makes an LLM call requesting a GenESyS `.gen` model language file. The system prompt includes the `.gen` format specification, a complete example, and the knowledge base. If `AIAssistantRequest::modelPlan` is non-empty it is included as the model specification. The generated `.gen` content is written to a `0600` temp file (`mkstemp`), loaded via `SimulatorFacade::loadModel()`, and then deleted. `AIAssistantResponse::generatedModelLanguage` holds the generated file text.
- `configureSimulation()`: after a model is loaded, makes a small deterministic (temperature=0) LLM call to extract simulation parameters as JSON from `AIAssistantRequest::experimentDescription`. Applies non-zero values to the current model via `simSetNumberOfReplications`, `simSetReplicationLength`, `simSetWarmUpPeriod`. Falls back gracefully when no provider is available (`.gen` parameters remain active).
- `HttpProviderClientBase` JSON utilities promoted to `public static` so that `AIAssistantDefaultImpl` can call `_extractJsonInt`, `_extractJsonDouble`, and `_extractJsonString` directly; `_extractJsonDouble` added for floating-point fields.
- `AIAssistantRequest::modelPlan` field added; `execute()` populates it from `createModelPlan()` output and passes the enriched request to `buildModel()`.

9.6.7 Stage 6 (complete) — Simulation execution and result collection

- `runSimulation()`: verifies `allowSimulationExecution` and non-null SimulatorFacade, checks not already running, then calls `SimulatorFacade::simStart()` (synchronous). Reports number of completed replications in the response.
- `collectResults()`: iterates `SimulatorFacade::modelGetResponses()` using `List<Simulation Response*>::front()/next()` and renders a Markdown table of metric name → value.
- `execute()` updated to call steps in the correct order with proper plan propagation: analyzePrompt → createModelPlan → buildModel (with enriched plan) → configureSimulation → runSimulation → collectResults.

9.6.8 Planned stages

- **Stage 7** — GUI integration: prompt panel, plan review, graphical model creation.
- **Stage 8** — Security and governance: OS secret store, sandbox, dry-run mode, audit log.

9.6.9 Dependency constraints

- `AIAssistant_if.h` uses only a forward declaration of `SimulatorFacade` to avoid pulling the full kernel chain into every including translation unit.
- `AIAssistantDefaultImpl.cpp` includes `kernel/simulator/SimulatorFacade.h` directly.
- No external libraries are introduced in Stage 1.

9.7 7. Stability notes and non-goals for this phase

- FITTER-2 consolidated additional fitting algorithms in `FitterDefaultImpl` (beta and weibull) on top of FITTER-1 families; FITTER-3 now switches the default trait binding so `TraitsTools<Fitter_if>` points to `FitterDefaultImpl`.
- `FitterDummyImpl` is intentionally preserved as a legacy/documental placeholder and is no longer the default fitter binding.
- No full completion of hypothesis-testing theory coverage.
- No numerical refactor of legacy solver internals.
- No cross-package behavior changes outside `source/tools`.

Chapter 10

tools package

10.1 1. Purpose of the tools package

The `source/tools` package hosts statistical and numerical support abstractions used by simulation and analysis flows. It provides interfaces and legacy implementations for fitting, hypothesis testing, probability utilities, and numerical solvers.

10.2 2. Current major abstractions

- `DataAnalyser_if`: high-level façade to orchestrate dataset-oriented analysis services.
- `SimulationResultsDatasetParser`: loader for numeric result files, including Genesys `Record` outputs split by replication and optional time/value columns.
- `Fitter_if`: fitting contract for distribution parameter inference from sample data.
- `HypothesisTester_if`: parametric inference API (confidence intervals and tests).
- `ProbabilityDistributionBase` / `ProbabilityDistribution`: static math façade for PDF/↩ PMF and inverse/quantile routines.
- `Solver_if`: legacy numerical contract mixing integration and derivation/advancement.
- `TraitsTools`: traits registry binding abstractions to concrete implementations.

10.3 3. Current limitations

- Fitting baseline was expanded in FITTER-2: `FitterDefaultImpl` now also provides functional Beta (scaled) and Weibull fitting, still preserving controlled-failure behavior when constraints are not met.
- `FitterDummyImpl` is preserved as a legacy placeholder/documental implementation, but it is no longer the default trait binding after FITTER-3.
- Some hypothesis-testing paths, especially two-population paths, remain partially consolidated.
- Distribution APIs are static utilities, not yet an OO hierarchy with reusable distribution objects.
- Solver abstraction conflates quadrature and ODE-like concerns.

10.4 4. Planned evolution

- Introduce cohesive interfaces for dataset, distributions, quadrature, root finding, and ODE solving.
- Continue hardening `FitterDefaultImpl` as the promoted default fitting implementation.
- Evolve traits coverage to include newly stabilized abstractions.
- Keep legacy interfaces during migration to avoid behavior breaks.

10.5 5. Relationship with kernel/statistics

source/tools consumes kernel/statistics contracts (collectors, samplers and data files) but this phase does not modify kernel or statistics code. The package remains a consumer and adapter layer over those existing contracts.

10.6 6. AI Assistant abstraction

source/tools includes an AI-assisted simulation model-building tool across four completed stages:

- AIAssistant_if: interface defining configuration, security, knowledge base, provider client injection, workflow and diagnostics contracts.
- AIAssistantDefaultImpl: concrete implementation covering all six complete stages:
  - Full configuration management and secure API key handling.
  - Knowledge base: static catalog of 15 well-known components plus dynamic plugin introspection.
  - createAndAttachProviderClient() factory for owned provider clients.
  - analyzePrompt() / createModelPlan() — LLM-backed prompt analysis and plan generation.
  - buildModel() — LLM generates a .gen file, loaded via SimulatorFacade::loadModel().
  - configureSimulation() — LLM extracts simulation parameters; applies via simSet*().
  - runSimulation() — calls SimulatorFacade::simStart() (synchronous).
  - collectResults() — formats modelGetResponses() as a Markdown results table.
  - execute() — orchestrates the full pipeline with plan propagation between steps.
- AIProviderClient_if: interface for a single HTTP chat-completion call.
- HttpProviderClientBase: abstract base with popen+curl transport; API key stored in 0600 temp file, never on command line.
- OpenAIProviderClientImpl: OpenAI chat completions with o-series reasoning support.
- AnthropicProviderClientImpl: Anthropic Messages API with extended-thinking support.
- LocalProviderClientImpl: Ollama/LM Studio/llama.cpp — no API key required.
- TraitsTools<AIAssistant_if>: binds the abstraction to AIAssistantDefaultImpl.

See source/tools/ARCHITECTURE_tools.md for the full 8-stage roadmap.

10.7 7. Current status by topic

- **Fitting**: interface defined; FitterDefaultImpl is functional for uniform/triangular/normal/exponential/erlang/beta/weibull with binary dataset loading and SSE-CDF comparison and is now the default TraitsTools<Fitter_if> binding (FITTER-3). FitterDummyImpl remains available as legacy placeholder.
- **Hypothesis testing**: functional baseline exists in HypothesisTesterDefaultImpl1, with known partial areas.
  - HYPTEST-1 alignment update: proportion-difference CI now follows the classical two-proportion formula, and one-population average/variance tests now compute p-values with Student-t/chi-square-coherent CDF paths.
  - HYPTEST-2 final alignment update: one-population proportion confidence intervals (with and without finite-population correction) now use the large-sample normal quantile formulation.
- **Simulation result datasets**: SimulationResultsDatasetParser reads text datasets and Genesys Record files, preserving replication ids and optional time columns so GUI analysis can expose pooled and per-replication scopes.
- **Probability distributions**: mathematical static base and inverse façade available, with internal numeric dependencies.
- **Numerical solvers**: legacy Solver_if + SolverDefaultImpl1 remain the compatible baseline.

Chapter 11

Todo List

Member AIAssistant_if::refreshKnowledgeBaseFromSimulator (std::string *message=nullptr)=0

(Stage 2): Implement full plugin introspection via SimulatorFacade.

Class AIAssistantDefaultImpl

annotations.

Class GraphicalConnection

Evaluate decoupling model metadata from this item to keep this class purely visual.

Member GraphicalConnection::GraphicalConnection (GraphicalComponentPort *sourceGraphicalPort, GraphicalComponentPort *destinationGraphicalPort, unsigned int portSourceConnection=0, unsigned int portDestinationConnection=0, QColor color=Qt::black, QGraphicsItem *parent=nullptr)

Extrair para método dedicado `_bindPorts()` para legibilidade.

Member ModelGraphicsScene::notifyGraphicalModelChange (GraphicalModelEvent::EventType eventType, GraphicalModelEvent::EventObjectType eventObjectType, QGraphicsItem *item)

Consider broadcasting to all attached views when multi-view editing is supported.

Member ModelGraphicsView::notifySceneGraphicalModelEventHandler (const GraphicalModelEvent &modelGraphicsEvent)

Replace callback with signal/slot to improve composability.

actualize property editor?

Member SimulationController::ensureReady (bool checkModel, bool modelChecked, const std::function< bool()> &runModelCheck, const std::function< bool()> &syncModelFromText) const

Replace callback-based contract by dedicated interfaces to improve testability.

Chapter 12

Namespace Index

12.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

EFSMNetworkDiagramCodec
: Encodes/decodes the "Graphical EFSM Network Diagrams" `.gui` overlay section 105

generate_kernel_simulator_test_stubs 105

genesys_mcp 106

genesys_mcp.__main__ 106

genesys_mcp.client 106

genesys_mcp.server 106

GuiCrashDiagnostics
: Crash diagnostics helpers for the Qt GUI process 111

QCP 111

Chapter 13

Hierarchical Index

13.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

AIAssistant_if . . . 131
  AIAssistantDefaultImpl . . . 143
AIAssistantConfiguration . . . 142
AIAssistantRequest . . . 155
AIAssistantResponse . . . 156
AIAuditLog . . . 157
AIConversationConfiguration . . . 158
AIConversationResponse . . . 158
AIConversationService . . . 159
AIProviderClient_if . . . 159
  HttpProviderClientBase . . . 878
    AnthropicProviderClientImpl . . . 199
    LocalProviderClientImpl . . . 928
    OpenAIProviderClientImpl . . . 1168
AISecretStore . . . 166
AnimationQueue . . . 190
ApiRouter . . . 203
testing::internal::AssertHelper . . . 205
Assignment . . . 212
AuditEntry . . . 218
BacteriaColony::BacteriumState . . . 233
BacteriaColony::BarrierSegment . . . 236
Base
  yy::genesyspp_parser::basic_symbol< Base > . . . 240
BaseMVController . . . 239
yy::genesyspp_parser::basic_symbol< by_state > . . . 240
testing::internal::TemplateSel< Tmpl >::Bind< T > . . . 249
ModalModelBindingsUndoCommand::Bindings . . . 249
BioKineticLawExpression . . . 254
BioParameterSensitivityEntry . . . 271
BioReactionRateSample . . . 283
BioReactionRateTimeCourse . . . 283
BioSBMLBridge . . . 291
BioSBMLExportResult . . . 291
BioSBMLImportResult . . . 292
BioSensitivityScan . . . 293
BioSimulationAnalysis . . . 300
BioSimulationResult . . . 300
BioSimulationSample . . . 302
BioSimulationSpeciesAmount . . . 302

BioSpeciesDerivative . . . 312
BioSteadyStateCheck . . . 327
BioStoichiometryMatrix . . . 327
BoundaryCondition . . . 332
Boundary_Closed . . . 329
Boundary_Fixed . . . 330
yy::genesyspp_parser::by_kind . . . 341
yy::genesyspp_parser::basic_symbol< by_kind > . . . 240
yy::genesyspp_parser::symbol_type . . . 2547
QCPAxisPainterPrivate::CachedLabel . . . 343
QCPLabelPainterPrivate::CachedLabel . . . 343
ModelLifecycleController::Callbacks . . . 343
SimulationEventController::Callbacks . . . 344
testing::internal::CartesianProductHolder< Gen > . . . 351
Cell . . . 352
CellularAutomata . . . 380
CellularAutomataBase . . . 384
CellularAutomata_1DTimed . . . 381
CellularAutomata_Classic . . . 382
CellularAutomataDemoBuilder . . . 392
CellularAutomataDemoModel . . . 393
CellularAutomataDemoSettings . . . 393
CellularAutomataEvent . . . 394
CellularAutomataStateColorMap . . . 397
ChatMessage . . . 399
testing::internal::CodeLocation . . . 407
Collector_if . . . 407
CollectorDatafile_if . . . 410
CollectorDatafileDefaultImpl1 . . . 412
CollectorDefaultImpl1 . . . 415
GroProgramRuntime::ColonyMutation . . . 417
GroProgramIr::Command . . . 424
Util::CommandResult . . . 425
CppCompiler::CompilationResult . . . 432
CompilerThread . . . 433
ComponentItem . . . 433
HypothesisTester_if::ConfidenceInterval . . . 437
Connection . . . 437
ConnectionChannel . . . 438
ConnectionManager . . . 438
testing::internal::ConstCharPtr . . . 443
Optimizer_if::ConstraintDefinition . . . 443
testing::internal::ConstRef< T > . . . 443
testing::internal::ConstRef< T & > . . . 443
testing::internal::ContainerPrinter . . . 444
yy::genesyspp_parser::context . . . 444
testing::internal::ConvertibleToIntegerPrinter . . . 454
testing::internal::ConvertibleToStringViewPrinter . . . 454
CppModelExporter . . . 493
SimulatorSessionService::CreateSessionResult . . . 504
QtBoolPropertyManagerPrivate::Data . . . 505
QtDatePropertyManagerPrivate::Data . . . 505
QtDateTimePropertyManagerPrivate::Data . . . 505
QtDoublePropertyManagerPrivate::Data . . . 505
QtEnumPropertyManagerPrivate::Data . . . 506
QtFlagPropertyManagerPrivate::Data . . . 506
QtIntPropertyManagerPrivate::Data . . . 506

QtPointFPropertyManagerPrivate::Data . . . 507
QtRectFPropertyManagerPrivate::Data . . . 507
QtRectPropertyManagerPrivate::Data . . . 507
QtSizeFPropertyManagerPrivate::Data . . . 507
QtSizePropertyManagerPrivate::Data . . . 507
QtStringPropertyManagerPrivate::Data . . . 508
QtTimePropertyManagerPrivate::Data . . . 508
DataAnalyser_if . . . 508
DataDefinitionItem . . . 512
DataDefinitionRelationItem . . . 512
ModelInspectorController::DataDefinitionSelection . . . 513
AnimationPlot::DataSeries . . . 513
DataSet_if . . . 514
Delay::DEFAULT_VALUES . . . 520
ModalModelDefault::DEFAULT_VALUES . . . 520
Seize::DEFAULT_VALUES . . . 521
SourceModelComponent::DEFAULT_VALUES . . . 521
testing::internal::DefaultNameGenerator . . . 523
DefaultNodeTransition . . . 535
EFSMTransition . . . 647
DialogUtilityController . . . 559
genesys::launcher::DispatchController . . . 603
DistanceEntry . . . 615
Distribution_if . . . 615
ContinuousDistribution_if . . . 444
DiscreteDistribution_if . . . 602
testing::internal::DoubleSequence< plus_one, T, sizeofT > . . . 618
testing::internal::DoubleSequence< false, IndexSequence< I... >, sizeofT > . . . 618
testing::internal::DoubleSequence< true, IndexSequence< I... >, sizeofT > . . . 619
genesys::launcher::DownloadResult . . . 619
DrawingItem . . . 619
EditCommandController . . . 636
EditorFactoryPrivate< Editor > . . . 637
EditorFactoryPrivate< QComboBox > . . . 637
QtEnumEditorFactoryPrivate . . . 1888
EditorFactoryPrivate< QKeySequenceEdit > . . . 637
QtKeySequenceEditorFactoryPrivate . . . 1920
EditorFactoryPrivate< QScrollBar > . . . 637
QtScrollBarFactoryPrivate . . . 1964
EditorFactoryPrivate< QSlider > . . . 637
QtSliderFactoryPrivate . . . 1983
EditorFactoryPrivate< QtBoolEdit > . . . 637
QtCheckBoxFactoryPrivate . . . 1843
EditorFactoryPrivate< QtCharEdit > . . . 637
QtCharEditorFactoryPrivate . . . 1837
EditorFactoryPrivate< QtColorEditWidget > . . . 637
QtColorEditorFactoryPrivate . . . 1846
EditorFactoryPrivate< QtDateEdit > . . . 637
QtDateEditFactoryPrivate . . . 1862
EditorFactoryPrivate< QtDateTimeEdit > . . . 637
QtDateTimeEditFactoryPrivate . . . 1872
EditorFactoryPrivate< QtDoubleEdit > . . . 637
QtDoubleSpinBoxFactoryPrivate . . . 1885
EditorFactoryPrivate< QtFontEditWidget > . . . 637
QtFontEditorFactoryPrivate . . . 1901

EditorFactoryPrivate< QtIntEdit > . . . 637
QtSpinBoxFactoryPrivate . . . 1987
EditorFactoryPrivate< QtStringEdit > . . . 637
QtLineEditFactoryPrivate . . . 1927
EditorFactoryPrivate< QtTimeEdit > . . . 637
QtTimeEditFactoryPrivate . . . 1996
EFSMNetworkDiagramLayout . . . 644
testing::internal::ElemFromList< N, T > . . . 652
testing::internal::ElemFromList< I, T... > . . . 652
testing::internal::ElemFromListImpl< typename > . . . 652
testing::internal::ElemFromListImpl< IndexSequence< I... > > . . . 652
PersistenceRecord::Entry . . . 679
testing::Environment . . . 680
Event . . . 688
InternalEvent . . . 895
Exact . . . 689
Exception
genesys_mcp.client.GenesysClientError . . . 764
CompartmentExchangeComponent::ExchangeRule . . . 695
GroProgramRuntime::ExecutionResult . . . 696
ExperimentManager . . . 702
ExperimentManager_if . . . 703
ExperimentManagerDefaultImpl1 . . . 704
ExpressionCategory . . . 707
ExpressionParameter . . . 707
ExpressionTemplate . . . 707
genesys::launcher::ExtractionResult . . . 707
FactorialDesign . . . 707
testing::internal::FallbackPrinter . . . 721
std::false_type
testing::internal::IsRecursiveContainerImpl< C, false > . . . 901
testing::internal::is_proxy_type_list< typename > . . . 897
testing::internal::FindFirstPrinter< T, E, Printer, Printers > . . . 727
testing::internal::FindFirstPrinter< T, decltype(Printer::PrintValue(std::declval< const T & >(), nullptr)), Printer, Printers... > . . . 727
Fitter_if . . . 728
FitterDefaultImpl . . . 730
FitterDummyImpl . . . 732
testing::internal::FlatTupleBase< Derived, Idx > . . . 736
testing::internal::FlatTupleBase< FlatTuple< T... >, MakeIndexSequence< sizeof...(T)>::type > . . . 736
testing::internal::FlatTuple< Ts... > . . . 735
testing::internal::FlatTuple< T > . . . 735
testing::internal::FlatTupleConstructTag . . . 737
testing::internal::FlatTupleElemBase< Derived, I > . . . 737
testing::internal::FlatTupleElemBase< FlatTuple< T... >, I > . . . 738
testing::internal::FlatTupleElemBase< FlatTuple< T... >, Idx > . . . 737
testing::internal::FlatTupleBase< FlatTuple< T... >, IndexSequence< Idx... > > . . . 736
testing::internal::FloatingPoint< RawType > . . . 738
testing::internal::FormatForComparison< ToPrint, OtherOperand > . . . 739
testing::internal::FormatForComparison< ToPrint[N], OtherOperand > . . . 739
testing::internal::FunctionPointerPrinter . . . 760
ParserManager::GenerateNewParserResult . . . 760
testing::internal::GenerateTypeList< T > . . . 761
GenesysApplication_if . . . 762
BaseGenesysTerminalApplication . . . 236
AirportSecurityExample . . . 163

AirportSecurityExampleExtended . . . 165
Airport_Extended1 . . . 161
AnElectronicAssemblyAndTestSystem . . . 179
Assembly_Line . . . 204
Banking_Transactions . . . 234
Book_Cap02_Example01 . . . 327
BufferFIFO . . . 340
ContinuousModel . . . 446
Example_BasicOrderShipping . . . 690
Example_PortModel . . . 692
Example_PublicTransport . . . 694
FullSimulationOfComplexModel . . . 759
GenesysShell . . . 772
Half_Adder . . . 870
Loja01 . . . 946
OperatingSystem02 . . . 1172
OperatingSystem03 . . . 1173
Rectifier . . . 2053
Smart_AddingResource . . . 2271
Smart_AlternatingEntityCreation . . . 2272
Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime . . . 2274
Smart_ArrivalsEntityTypeVsAttribute . . . 2276
Smart_AssignExample . . . 2278
Smart_AssignWriteSeizes . . . 2280
Smart_Attribute_Variable . . . 2281
Smart_AutomaticStatisticsCollection . . . 2283
Smart_BacteriaColony_GRO . . . 2285
Smart_BasicModeling . . . 2287
Smart_BatchAndSeparate . . . 2288
Smart_BatchSeparate . . . 2290
Smart_BioKineticLawRegulation . . . 2292
Smart_BioNetworkAnalysis . . . 2293
Smart_BioReversibleKineticLaw . . . 2295
Smart_BioReversibleMassAction . . . 2297
Smart_Buffer . . . 2298
Smart_CellularAutomata . . . 2300
Smart_CellularAutomata1 . . . 2302
Smart_Clone . . . 2303
Smart_ContinuousFlowEntities . . . 2305
Smart_CppForG . . . 2307
Smart_Create . . . 2308
Smart_DecideNWayByChance . . . 2310
Smart_DefaultModalModel . . . 2312
Smart_DefiningAttributesAsStrings . . . 2313
Smart_DefiningControlLogic . . . 2315
Smart_DefiningResourceCapacity . . . 2317
Smart_Delay . . . 2318
Smart_DelayBasedOnReplication . . . 2320
Smart_Dummy . . . 2322
Smart_EntitiesProcessedByPriority . . . 2324
Smart_EvaluatingConditionsBeforeEnteringQueue . . . 2326
Smart_Expression . . . 2328
Smart_FSMState . . . 2331
Smart_Failures . . . 2330
Smart_GroColonyGrowth . . . 2333
Smart_GroColonyLifecycle . . . 2335
Smart_HoldSearchRemove . . . 2336
Smart_InventoryAndHoldingCosts . . . 2338

Smart_LSODE . . . 2340
Smart_MarkovChain . . . 2342
Smart_MaxArrivalsField . . . 2343
Smart_ModalModel . . . 2345
Smart_ModalModelDefaultNetwork . . . 2347
Smart_ModalModelFSM . . . 2348
Smart_ModalModelPetriNet . . . 2350
Smart_ModelInfoModelSimulation . . . 2352
Smart_ModelRunUntil1000Parts . . . 2353
Smart_ModuleDisplayVariables . . . 2355
Smart_ODE . . . 2357
Smart_OnEvent . . . 2358
Smart_OverlappingResources . . . 2361
Smart_ParallelProcessingOfEntities . . . 2363
Smart_Parser . . . 2365
Smart_ParserModelFunctions . . . 2367
Smart_PetriPlace . . . 2368
Smart_PlacingEntitiesInQueueSets . . . 2370
Smart_Plugin . . . 2372
Smart_PriorityExample . . . 2373
Smart_Process . . . 2375
Smart_ProcessArena . . . 2377
Smart_ProcessSet . . . 2378
Smart_PythonForG . . . 2380
Smart_R_Simulator . . . 2381
Smart_Record . . . 2384
Smart_Record_Arena . . . 2385
Smart_RemovingAndReorderingEntitiesInAQueue . . . 2387
Smart_ResourceCosting . . . 2389
Smart_ResourceScheduleCosting . . . 2391
Smart_ResourceSets . . . 2392
Smart_RouteStation . . . 2394
Smart_SeizeDelayRelease . . . 2396
Smart_SeizeDelayReleaseMany . . . 2397
Smart_SeizeDelayReleaseNoDataDefs . . . 2399
Smart_SeizingMultipleSimultaneosly . . . 2401
Smart_SelectingRouteBasedOnProbability . . . 2403
Smart_SelectingShorterQueue . . . 2404
Smart_Sequence . . . 2406
Smart_SimulationControlResponse . . . 2408
Smart_SynchronizingParallelEntities . . . 2409
Smart_ValueAdded . . . 2411
Smart_WaitForSignal . . . 2413
Smart_WaitScanCondition . . . 2414
Smart_WaitSignal . . . 2416
StartAIAssistant . . . 2458
WcmBioNetworkBridge . . . 2671
WcmCompartmentTransportWholeCellMvp . . . 2673
WcmFluxBalanceWholeCellMvp . . . 2675
WcmGeneExpression . . . 2677
WcmLifecycleStress . . . 2679
WcmMgenitaliumKarr2012 . . . 2681
WcmPathwayStressWholeCellMvp . . . 2684
WcmSimpleSsa . . . 2685
WcmWholeCellMvp . . . 2687
WcmYeastDidacticWholeCellMvp . . . 2689
WcmYeastEukaryoticWholeCellMvp . . . 2691
BaseGenesysWebApplication . . . 238

GenesysShell_if . . . 774
genesys_mcp.client.GenesysClient . . . 763
GenesysKnowledgeBaseSummary . . . 765
GenesysKnowledgeItem . . . 765
genesyspp_driver . . . 766
yy::genesyspp_parser . . . 767
GenesysPropertyDescriptor . . . 771
GenesysPropertyIntrospection . . . 772
GeneticCircuitSimulationSummary . . . 793
GraphicalAssociation . . . 814
GraphicalAutomaticPositioningStrategy . . . 814
GraphicalAutomaticPositioningStrategyFactory . . . 815
GraphicalConnectionStyle . . . 820
GraphicalContextMenuController . . . 820
GraphicalDataDefinitionLayout . . . 821
GraphicalModelBuilder . . . 823
GraphicalModelEvent . . . 832
GraphicalModelItemRenderContext . . . 833
GraphicalModelItemRenderer . . . 834
GraphicalModelItemRenderStrategy . . . 834
GraphicalModelSerializer . . . 834
GraphPathResult . . . 848
GraphSearchTreeEntry . . . 848
GraphTopologicalOrderResult . . . 849
GraphTraversalResult . . . 849
GraphvizModelExporter . . . 849
ModelGraphicsScene::GRID . . . 851
GroProgramAst . . . 856
GroProgramCompiler . . . 857
GroProgramIr . . . 857
GroProgramParser . . . 858
GroProgramRuntime . . . 858
GroProgramRuntimeState . . . 859
GroupItem . . . 859
testing::internal::GTestFlagSaver . . . 860
testing::internal::GTestLog . . . 860
testing::internal::GTestMutexLock . . . 861
GuiActionContribution . . . 861
GuiDockContribution . . . 861
GuiExtensionManager . . . 864
GuiExtensionPlugin . . . 865
BacteriaColonyViewerGuiExtensionPlugin . . . 227
BioNetworkAwareGuiExtensionPlugin . . . 260
BioNetworkEditorGuiExtensionPlugin . . . 262
BioNetworkExecutionGuiExtensionPlugin . . . 263
BioParameterManagerGuiExtensionPlugin . . . 269
BioReactionKineticsGuiExtensionPlugin . . . 282
BioReactionStoichiometryGuiExtensionPlugin . . . 284
BioSBMLInteroperabilityGuiExtensionPlugin . . . 292
BioSimulationResultsGuiExtensionPlugin . . . 301
BioSpeciesManagerGuiExtensionPlugin . . . 313
CellularAutomataGuiExtensionPlugin . . . 395
CoreGuiExtensionPlugin . . . 459
GeneticCircuitGuiExtensionPlugin . . . 780
GuiExtensionConfigurationGuiExtensionPlugin . . . 863
ModelPluginInspectorGuiExtensionPlugin . . . 1092
GuiExtensionPluginAutoRegistrar . . . 866
GuiExtensionPluginCatalog . . . 866

GuiExtensionPresentationMetadata . . . 866
GuiExtensionRegistry . . . 866
GuiExtensionRuntimeContext . . . 867
GuiScopeTrace . . . 867
GuiThemeManager . . . 869
GuiWindowContribution . . . 870
testing::internal::HasDebugStringAndShortDebugString< T > . . . 873
HttpRequest . . . 882
HttpResponse . . . 883
HypothesisTester_if . . . 884
    HypothesisTesterDefaultImpl1 . . . 885
genesys::launcher::IArchiveExtractor . . . 892
    genesys::launcher::TarZstdArchiveExtractor . . . 2560
testing::internal::Ignore< size_t > . . . 892
testing::internal::IgnoredValue . . . 892
testing::internal::IndexSequence< Is > . . . 892
testing::internal::IndexSequence<> . . . 892
    testing::internal::MakeIndexSequenceImpl< 0 > . . . 957
genesys::launcher::INetworkTransport . . . 893
    genesys::launcher::QtNetworkTransport . . . 1933
std::integral_constant
    std::tuple_size< testing::internal::FlatTuple< Ts... > > . . . 2600
genesys::launcher::IProcessLauncher . . . 897
    genesys::launcher::ProcessLauncher . . . 1289
testing::internal::IsHashTable< T > . . . 899
genesys::launcher::ISignatureVerifier . . . 899
    genesys::launcher::GpgvSignatureVerifier . . . 807
testing::internal::IsRecursiveContainerImpl< C, bool > . . . 900
testing::internal::IsRecursiveContainerImpl< C, true > . . . 901
IssueReportDraft . . . 902
IssueReportRelayService . . . 902
IssueReportRuntimeContext . . . 903
IssueReportSubmissionResult . . . 903
genesys::launcher::IUpdateUi . . . 904
    genesys::launcher::UpdateDialog . . . 2621
Json . . . 904
KernelAccessorProbe . . . 907
QCPLabelPainterPrivate::LabelData . . . 912
Lattice . . . 913
genesys::launcher::LauncherConfig . . . 914
genesys::launcher::LauncherController . . . 914
genesys::launcher::LauncherLogger . . . 915
genesys::launcher::LauncherUpdateResult . . . 915
LicenceManager . . . 922
List< T > . . . 923
    ListObservable< T > . . . 925
List< Assignment * > . . . 923
List< ConstraintDefinition > . . . 923
List< Counter * > . . . 923
List< CPNArc * > . . . 923
List< CPNTransition * > . . . 923
List< DefaultNodeTransition * > . . . 923
List< DistanceEntry * > . . . 923
List< double > . . . 923
List< EFSMTransition * > . . . 923
List< Entity * > . . . 923

List< Event * > . . . 923
List< Failure * > . . . 923
List< FSMState * > . . . 923
List< MarkovChainNetwork::MarkovTransition * > . . . 923
List< MarkovState * > . . . 923
List< Model * > . . . 923
List< ModelComponent * > . . . 923
List< ModelDataDefinition * > . . . 923
List< modelEventHandler > . . . 923
List< modelEventHandlerMethod > . . . 923
List< ObjectiveDefinition > . . . 923
List< ODEfunction * > . . . 923
List< PairSignalDataEventHandler * > . . . 923
List< PetriPlace * > . . . 923
List< PickableStationItem * > . . . 923
List< Plugin * > . . . 923
List< PluginLoadIssue > . . . 923
List< Queue * > . . . 923
List< Resource * > . . . 923
List< SchedulableItem * > . . . 923
List< SegmentStep * > . . . 923
List< SeizableItem * > . . . 923
List< SequenceStep * > . . . 923
List< ShellCommand * > . . . 923
List< SimulationControl * > . . . 923
List< simulationEventHandler > . . . 923
List< simulationEventHandlerMethod > . . . 923
List< SimulationExperiment * > . . . 923
List< SimulationResponse * > . . . 923
List< SolutionSummary > . . . 923
List< SortedResourceEventHandler * > . . . 923
List< SparseValueStore * > . . . 923
List< StatisticsCollector * > . . . 923
List< std::string > . . . 923
List< traceErrorListener > . . . 923
List< traceErrorListenerMethod > . . . 923
List< traceListener > . . . 923
List< traceListenerMethod > . . . 923
List< traceSimulationListener > . . . 923
List< traceSimulationListenerMethod > . . . 923
List< void * > . . . 923
List< Waiting * > . . . 923
LocalRule . . . 932
LocalRule_CopyCurrentState . . . 934
LocalRule_Elementary . . . 935
LocalRule_FlorestalFire . . . 936
LocalRule_GameOfLife . . . 938
LocalRule_Growty . . . 939
LocalRule_HppLatticeGas . . . 940
LocalRule_SandRock_Student . . . 942
LocalRule_SandRock_Student . . . 942
LocalRule_ShowNeighbordhood . . . 943
yy::location . . . 944
testing::internal::internal_stream_operator_without_lexical_name_lookup::LookupBlocker . . . 947
LSODEInitialState . . . 954
LSODESample . . . 954
genesys::launcher::ManifestParseResult . . . 958
testing::internal::MarkAsIgnored . . . 959

MarkovChainNetwork::MarkovTransition . . . 977
testing::Message . . . 985
MetabolicFluxBalanceSolver . . . 991
generate_kernel_simulator_test_stubs.MethodEntry . . . 1015
Model . . . 1023
ModelChecker_if . . . 1031
ModelCheckerDefaultImpl1 . . . 1033
ModelComponentManager . . . 1044
ModelDataDefinitionAssociations . . . 1060
ModelDataManager . . . 1061
ModelEvent . . . 1067
SimulatorSessionService::ModelImportResult . . . 1079
ModelInfo . . . 1080
SimulatorSessionService::ModelInfoResult . . . 1083
ModelInspectorController . . . 1083
ModelLanguageSynchronizer . . . 1084
ModelLifecycleController . . . 1085
ModelManager . . . 1086
SimulatorSessionService::ModelPersistenceResult . . . 1092
ModelSimulation . . . 1094
GroProgramRuntime::MotionMutation . . . 1100
testing::internal::Mutex . . . 1108
dialogBreakpoint::MVCResult . . . 1108
GroProgramAst::NamedProgram . . . 1108
GroProgramIr::NamedProgramDefinition . . . 1108
testing::internal::NameGeneratorSelector< Provided > . . . 1108
testing::internal::NativeArray< Element > . . . 1115
Neighborhood . . . 1117
Neighborhood_Center . . . 1118
Neighborhood_Moore . . . 1120
Neighborhood_VonNeumann . . . 1122
NetworkActivationFrame . . . 1123
NetworkActivationResult . . . 1124
NetworkPortValue . . . 1124
genesys::launcher::NetworkRequestOptions . . . 1125
genesys::launcher::NetworkResponse . . . 1125
ParserManager::NewParser . . . 1125
testing::internal::None . . . 1132
obj_t . . . 1145
Optimizer_if::ObjectiveDefinition . . . 1145
ODEfunction . . . 1154
OdeSolver_if . . . 1160
DormandPrince54OdeSolver . . . 617
RungeKutta4OdeSolver . . . 2106
OdeSolverFactory . . . 1161
OdeSystem_if . . . 1161
HarmonicOscillator . . . 872
DiffusionMethodOfLinesSystem . . . 573
HarmonicOscillator . . . 872
MassActionOdeSystem . . . 977
OnEventManager . . . 1166
Optimizer_if::OptimizationSettings . . . 1175
Optimizer_if . . . 1175
OptimizerDefaultImpl1 . . . 1181
testing::internal::OsStackTraceGetterInterface . . . 1196
testing::internal::OsStackTraceGetter . . . 1195
testing::internal::ParameterizedTestSuiteInfoBase . . . 1199

testing::internal::ParameterizedTestSuiteInfo< TestSuite > . . . 1198
testing::internal::ParameterizedTestSuiteRegistry . . . 1200
testing::internal::ParamGenerator< T > . . . 1200
testing::internal::ParamGeneratorInterface< T > . . . 1201
testing::internal::RangeGenerator< T, IncrementT > . . . 2028
testing::internal::ValuesInIteratorRangeGenerator< T > . . . 2637
testing::internal::ParamGeneratorInterface< ParamType > . . . 1201
testing::internal::ParamGeneratorInterface<::std::tuple< T... > > . . . 1201
testing::internal::CartesianProductGenerator< T > . . . 350
testing::internal::ParamIterator< T > . . . 1201
testing::internal::ParamIteratorInterface< T > . . . 1202
testing::internal::ParamIteratorInterface< ParamType > . . . 1202
Parser_if . . . 1202
ParserDefaultImpl1 . . . 1211
ParserDefaultImpl2 . . . 1213
ParserAttributeEventInjector . . . 1204
ParserAttributeProcessObserver . . . 1210
ParserChangesInformation . . . 1210
ParserManager . . . 1217
Persistence_if . . . 1225
FakeModelPersistence . . . 716
FakeModelPersistenceB . . . 718
PersistenceDefaultImpl2 . . . 1228
PersistencePartialLoadImpl1 . . . 1230
PersistenceRecord . . . 1231
PersistentObject_base . . . 1233
ModelDataDefinition . . . 1048
AISupport . . . 168
Attribute . . . 212
ParserAttributeInternalEventOwner . . . 1205
Variable . . . 2638
BacteriaSignalGrid . . . 229
BioCompartment . . . 249
BioNetwork . . . 255
BioParameter . . . 264
BioReaction . . . 271
BioSimulatorRunner . . . 303
BioSpecies . . . 308
CPNArc . . . 471
Conveyor . . . 454
Counter . . . 461
CppCompiler . . . 482
DefaultNetwork . . . 523
ColoredPetriNetNetwork . . . 417
EFSMNetwork . . . 638
GraphNetwork . . . 836
DirectedGraphNetwork . . . 594
DirectedAcyclicGraphNetwork . . . 588
MarkovChainNetwork . . . 965
DefaultNode . . . 529
CPNTransition . . . 476
FSMState . . . 747
GraphNode . . . 843
MarkovState . . . 972
PetriPlace . . . 1235
DiffusionField . . . 568

Distance . . . 610
DummyElement . . . 631
Entity . . . 662
EntityGroup . . . 669
EntityType . . . 674
Failure . . . 710
File . . . 721
Formula . . . 740
GeneticCircuit . . . 775
GeneticCircuitPart . . . 782
GeneticRegulation . . . 799
GraphEdge . . . 808
GroProgram . . . 851
Label . . . 908
MetabolicNetwork . . . 992
MetabolicReaction . . . 997
ModelComponent . . . 1035
AIAssistant . . . 125
Access . . . 117
Assign . . . 205
BacteriaColony . . . 219
Batch . . . 242
BioRunnerCommand . . . 285
BioSimulate . . . 293
BioStateProjectionComponent . . . 314
BioSteadyState . . . 320
Buffer . . . 333
CellCycleCheckpointComponent . . . 353
CellDivisionEvent . . . 360
CellFateDecisionComponent . . . 367
CellGrowthComponent . . . 373
CellularAutomataComp . . . 386
Clone . . . 400
CompartmentExchangeComponent . . . 425
ContinuousSystemComponent . . . 447
RecordingContinuousSystemComponent . . . 2041
RecordingContinuousSystemComponent . . . 2041
CppForG . . . 487
Decide . . . 514
Delay . . . 536
DelayProbe . . . 543
DelayProbe . . . 543
DiffEquations . . . 561
DiffusionSimulate . . . 575
DropOff . . . 619
DummyComponent . . . 625
Enter . . . 655
EukaryoticCellCycleComponent . . . 680
Exit . . . 696
FtsZPolymerizationComponent . . . 752
GeneticCircuitSimulate . . . 786
GeneticExpressionStep . . . 793
LSODE . . . 947
RecordingLSODE . . . 2047
Leave . . . 915
MarkovChain . . . 959
Match . . . 978
MetabolicFluxBalance . . . 985

MetabolicStateProjectionComponent . . . 1002
MetabolicSubmodelComponent . . . 1009
ModalModelDefault . . . 1016
Move . . . 1101
PathwayStressResponseComponent . . . 1219
PickStation . . . 1241
Process . . . 1282
PythonForG . . . 1300
RSimulator . . . 2094
Record . . . 2034
Release . . . 2054
Remove . . . 2061
PickUp . . . 1247
ResourceAllocationComponent . . . 2079
Route . . . 2087
SPICECircuit . . . 2435
AND . . . 173
Capacitor . . . 344
Diode . . . 582
NAND . . . 1109
NMOS . . . 1126
NOR . . . 1132
NOT . . . 1139
OR . . . 1189
PMOS . . . 1270
Resistor . . . 2067
Vpulse . . . 2645
Vsine . . . 2651
Vsource . . . 2657
XNOR . . . 2721
XOR . . . 2728
SPICENode . . . 2441
Search . . . 2127
Seize . . . 2140
Separate . . . 2147
Signal . . . 2171
SinkModelComponent . . . 2264
Dispose . . . 603
SourceModelComponent . . . 2424
Create . . . 497
Start . . . 2452
StochasticReactionComponent . . . 2495
StochasticTranscription . . . 2506
StochasticTranslation . . . 2513
Stop . . . 2521
Store . . . 2532
Submodel . . . 2539
Unstore . . . 2613
Wait . . . 2663
Write . . . 2712
MolecularSpecies . . . 1095
ODESolver . . . 1155
OLD_ODEelement . . . 1162
OctaveRunner . . . 1149
PythonRuntime . . . 1306
Queue . . . 2022
RSimulatorRunner . . . 2100
Resource . . . 2073

SPICERunner . . . 2447
Schedule . . . 2122
Segment . . . 2134
Sequence . . . 2153
Set . . . 2165
SignalData . . . 2177
Station . . . 2465
StationTestProbe . . . 2470
StatisticsCollector . . . 2478
StochasticReactionRule . . . 2501
Storage . . . 2527
Transporter . . . 2593
WholeCellState . . . 2696
SequenceStep . . . 2158
PickableStationItem . . . 1240
genesys::launcher::PlatformInfo . . . 1253
Plugin . . . 1253
PluginCatalogController . . . 1257
PluginConnector_if . . . 1258
PluginConnectorDummyImpl1 . . . 1260
PluginConnectorStaticImpl1 . . . 1263
PluginInformation . . . 1265
PluginInsertionOptions . . . 1266
PluginLoadIssue . . . 1266
PluginManager . . . 1267
testing::internal::PointerPrinter . . . 1276
GroProgramRuntime::PopulationMutation . . . 1276
yy::position . . . 1276
testing::PrintToStringParamName . . . 1277
ProbabilityDistributionBase . . . 1279
ProbabilityDistribution . . . 1277
MetabolicFluxBalanceSolver::Problem . . . 1282
BioStateProjectionComponent::Projection . . . 1291
MetabolicStateProjectionComponent::Projection . . . 1291
EFSMStatePropertiesUndoCommand::Properties . . . 1292
EFSMTransitionPropertiesUndoCommand::Properties . . . 1292
PropertyEditorController . . . 1293
PropertyEditorGenesys . . . 1294
PropertyGenesysBase . . . 1296
PropertyGenesys . . . 1294
SimulationResponse . . . 2242
SimulationControl . . . 2185
SimulationControlBool . . . 2189
SimulationControlDouble . . . 2192
SimulationControlGeneric< T > . . . 2195
SimulationControlGenericClass< T, M, C > . . . 2198
SimulationControlGenericClassNotDC< T, M, C > . . . 2202
SimulationControlGenericEnum< T, E > . . . 2206
SimulationControlGenericList< T, M, C > . . . 2209
SimulationControlGenericListPointer< T, M, C > . . . 2213
SimulationControlInt . . . 2217
SimulationControlSourceCodeString . . . 2222
SimulationControlString . . . 2225
SimulationControlTimeUnit . . . 2228
SimulationControlUInt . . . 2231
SimulationControlUShort . . . 2234
TestSimulationControl . . . 2569

SimulationResponseDouble . . . 2244
TestSimulationResponse . . . 2572
testing::internal::ProtobufPrinter . . . 1298
ProviderRequest . . . 1298
ProviderResponse . . . 1298
testing::internal::ProxyTypeList< Ts > . . . 1299
QCPAbstractPaintBuffer . . . 1325
QCPPaintBufferPixmap . . . 1655
QCPAxisPainterPrivate . . . 1368
QCPAxisTicker . . . 1385
QCPAxisTickerDateTime . . . 1389
QCPAxisTickerFixed . . . 1394
QCPAxisTickerLog . . . 1397
QCPAxisTickerPi . . . 1400
QCPAxisTickerText . . . 1403
QCPAxisTickerTime . . . 1407
QCPBarsData . . . 1421
QCPColorGradient . . . 1427
QCPColorMapData . . . 1442
QCPCurveData . . . 1473
QCPDataContainer< DataType > . . . 1475
QCPDataRange . . . 1482
QCPDataSelection . . . 1486
QCPErrorBarsData . . . 1502
QCPFinancialData . . . 1514
QCPGraphData . . . 1525
QCPItemAnchor . . . 1531
QCPItemPosition . . . 1560
QCPLabelPainterPrivate . . . 1591
QCPLineEnding . . . 1650
QCPPlottableInterface1D . . . 1661
QCPAbstractPlottable1D< QCPBarsData > . . . 1340
QCPBars . . . 1411
QCPAbstractPlottable1D< QCPCurveData > . . . 1340
QCPCurve . . . 1463
QCPAbstractPlottable1D< QCPFinancialData > . . . 1340
QCPFinancial . . . 1503
QCPAbstractPlottable1D< QCPGraphData > . . . 1340
QCPGraph . . . 1516
QCPAbstractPlottable1D< QCPStatisticalBoxData > . . . 1340
QCPStatisticalBox . . . 1751
QCPAbstractPlottable1D< DataType > . . . 1340
QCPErrorBars . . . 1491
QCPRange . . . 1727
QCPScatterStyle . . . 1733
QCPSelectionDecorator . . . 1739
QCPSelectionDecoratorBracket . . . 1742
QCPStatisticalBoxData . . . 1761
QCPVector2D . . . 1772
QDialog
DialogFind . . . 550
DialogModelInformation . . . 551
DialogPluginManager . . . 552
DialogReportIssue . . . 553
DialogSelectCounter . . . 555
DialogSelectVariable . . . 556

DialogSimulationConfigure . . . 557
DialogSystemPreferences . . . 558
DialogTimerConfigure . . . 559
ExpressionBuilder . . . 706
GuiExtensionConfigurationDialog . . . 862
WebWorkerDialog . . . 2693
dialogBreakpoint . . . 549
QGraphicsLineItem
GraphicalDiagramConnection . . . 822
QGraphicsObject
GraphicalAnimateExpression . . . 814
GraphicalComponentPort . . . 816
GraphicalConnection . . . 817
GraphicalModelDataDefinition . . . 830
GraphicalModelComponent . . . 825
QGraphicsPathItem
EFSMTransitionGraphicsItem . . . 649
QGraphicsPixmapItem
GraphicalImageAnimation . . . 823
QGraphicsRectItem
AnimationCounter . . . 182
AnimationPlaceholder . . . 187
AnimationAttribute . . . 181
AnimationEntity . . . 183
AnimationEvent . . . 185
AnimationExpression . . . 186
AnimationPlot . . . 188
AnimationQueueDisplay . . . 190
AnimationResource . . . 192
AnimationStation . . . 193
AnimationStatistics . . . 194
AnimationTimer . . . 195
AnimationVariable . . . 198
QGraphicsScene
ModelGraphicsScene . . . 1067
QGraphicsView
ModelGraphicsView . . . 1076
QItemDelegate
QtPropertyEditorDelegate . . . 1949
QLineEdit
FindLineEdit . . . 727
QMainWindow
AIAssistantWindow . . . 156
CellularAutomataViewerWindow . . . 399
DataAnalyzerWindow . . . 509
MainWindow . . . 954
OptimizerWindow . . . 1188
QObject
CellularAutomataViewerController . . . 397
ComboBoxEnum . . . 424
DataComponentEditor . . . 510
DataComponentProperty . . . 511
QCPBarsGroup . . . 1423
QCPLayer . . . 1593
QCPLayerable . . . 1597
QCPAbstractItem . . . 1311
QCPItemBracket . . . 1533
QCPItemCurve . . . 1539

QCPItemEllipse . . . 1544
QCPItemLine . . . 1549
QCPItemPixmap . . . 1554
QCPItemRect . . . 1566
QCPItemStraightLine . . . 1571
QCPItemText . . . 1576
QCPItemTracer . . . 1583
QCPAbstractPlottable . . . 1327
QCPAbstractPlottable1D< QCPBarsData > . . . 1340
QCPAbstractPlottable1D< QCPCurveData > . . . 1340
QCPAbstractPlottable1D< QCPFinancialData > . . . 1340
QCPAbstractPlottable1D< QCPGraphData > . . . 1340
QCPAbstractPlottable1D< QCPStatisticalBoxData > . . . 1340
QCPAbstractPlottable1D< DataType > . . . 1340
QCPColorMap . . . 1433
QCPErrorBars . . . 1491
QCPAxis . . . 1347
QCPGrid . . . 1527
QCPLayoutElement . . . 1611
QCPAbstractLegendItem . . . 1318
QCPPlottableLegendItem . . . 1665
QCPPolarLegendItem . . . 1722
QCPAxisRect . . . 1370
QCPColorScaleAxisRectPrivate . . . 1457
QCPColorScale . . . 1448
QCPLayout . . . 1604
QCPLayoutGrid . . . 1618
QCPLegend . . . 1638
QCPLayoutInset . . . 1630
QCPPolarAxisAngular . . . 1670
QCPTextElement . . . 1763
QCPPolarAxisRadial . . . 1691
QCPPolarGraph . . . 1710
QCPPolarGrid . . . 1719
QCPSelectionRect . . . 1746
QCPMarginGroup . . . 1653
QtAbstractEditorFactoryBase . . . 1811
QtAbstractEditorFactory< QtCharPropertyManager > . . . 1807
QtCharEditorFactory . . . 1835
QtAbstractEditorFactory< QtBoolPropertyManager > . . . 1807
QtCheckBoxFactory . . . 1841
QtAbstractEditorFactory< QtColorPropertyManager > . . . 1807
QtColorEditorFactory . . . 1844
QtAbstractEditorFactory< QtCursorPropertyManager > . . . 1807
QtCursorEditorFactory . . . 1853
QtAbstractEditorFactory< QtDatePropertyManager > . . . 1807
QtDateEditFactory . . . 1860
QtAbstractEditorFactory< QtDateTimePropertyManager > . . . 1807
QtDateTimeEditFactory . . . 1870
QtAbstractEditorFactory< QtDoublePropertyManager > . . . 1807
QtDoubleSpinBoxFactory . . . 1883
QtAbstractEditorFactory< QtEnumPropertyManager > . . . 1807
QtEnumEditorFactory . . . 1886
QtAbstractEditorFactory< QtFontPropertyManager > . . . 1807
QtFontEditorFactory . . . 1899
QtAbstractEditorFactory< QtKeySequencePropertyManager > . . . 1807

QtKeySequenceEditorFactory . . . 1918
QtAbstractEditorFactory< QtStringPropertyManager > . . . 1807
QtLineEditFactory . . . 1925
QtAbstractEditorFactory< QtIntPropertyManager > . . . 1807
QtScrollBarFactory . . . 1962
QtSliderFactory . . . 1981
QtSpinBoxFactory . . . 1985
QtAbstractEditorFactory< QtTimePropertyManager > . . . 1807
QtTimeEditFactory . . . 1994
QtAbstractEditorFactory< QtVariantPropertyManager > . . . 1807
QtVariantEditorFactory . . . 2007
QtAbstractEditorFactory< PropertyManager > . . . 1807
QtAbstractPropertyManager . . . 1819
QtBoolPropertyManager . . . 1825
QtCharPropertyManager . . . 1838
QtColorPropertyManager . . . 1849
QtCursorPropertyManager . . . 1856
QtDatePropertyManager . . . 1864
QtDateTimePropertyManager . . . 1873
QtDoublePropertyManager . . . 1877
QtEnumPropertyManager . . . 1890
QtFlagPropertyManager . . . 1894
QtFontPropertyManager . . . 1903
QtGroupPropertyManager . . . 1910
QtIntPropertyManager . . . 1913
QtKeySequencePropertyManager . . . 1922
QtLocalePropertyManager . . . 1928
QtPointFPropertyManager . . . 1934
QtPointPropertyManager . . . 1939
QtRectFPropertyManager . . . 1952
QtRectPropertyManager . . . 1957
QtSizeFPropertyManager . . . 1966
QtSizePolicyPropertyManager . . . 1972
QtSizePropertyManager . . . 1976
QtStringPropertyManager . . . 1989
QtTimePropertyManager . . . 1997
QtVariantPropertyManager . . . 2013
QtMetaEnumWrapper . . . 1932
QPainter
QCPPainter . . . 1657
QPlainTextEdit
CodeEditor . . . 406
QtAbstractPropertyBrowserPrivate . . . 1818
QtAbstractPropertyManagerPrivate . . . 1824
QtBoolPropertyManagerPrivate . . . 1828
QtBrowserItem . . . 1828
QtBrowserItemPrivate . . . 1829
QtButtonPropertyBrowserPrivate . . . 1833
QtCharPropertyManagerPrivate . . . 1840
QtColorPropertyManagerPrivate . . . 1852
QtCursorDatabase . . . 1853
QtCursorEditorFactoryPrivate . . . 1855
QtCursorPropertyManagerPrivate . . . 1859
QtDatePropertyManagerPrivate . . . 1868
QtDateTimePropertyManagerPrivate . . . 1875
QtDoublePropertyManagerPrivate . . . 1882
QtEnumPropertyManagerPrivate . . . 1894
QtEnumPropertyType . . . 1894

QtFlagPropertyManagerPrivate . . . 1898
QtFlagPropertyType . . . 1898
QtFontPropertyManagerPrivate . . . 1906
QtGroupBoxPropertyBrowserPrivate . . . 1909
QtGroupPropertyType . . . 1912
QtIntPropertyManagerPrivate . . . 1917
QtKeySequencePropertyManagerPrivate . . . 1924
QtLocalePropertyManagerPrivate . . . 1931
QtMetaEnumProvider . . . 1931
QtPointFPropertyManagerPrivate . . . 1938
QtPointPropertyManagerPrivate . . . 1942
QtProperty . . . 1943
    QtVariantProperty . . . 2010
QtPropertyBrowserUtils . . . 1948
QtPropertyPrivate . . . 1951
QtRectFPropertyManagerPrivate . . . 1956
QtRectPropertyManagerPrivate . . . 1961
QTreeWidget
    PropertyEditor . . . 1292
    QtPropertyEditorView . . . 1950
QtSizeFPropertyManagerPrivate . . . 1971
QtSizePolicyPropertyManagerPrivate . . . 1975
QtSizePropertyManagerPrivate . . . 1980
QtStringPropertyManagerPrivate . . . 1992
QtTimePropertyManagerPrivate . . . 1999
QtTreePropertyBrowserPrivate . . . 2006
QtVariantEditorFactoryPrivate . . . 2009
QtVariantPropertyManagerPrivate . . . 2019
QtVariantPropertyPrivate . . . 2021
Quadrature_if . . . 2021
QueueableItem . . . 2027
QUndoCommand
    AddUndoCommand . . . 124
    DeleteUndoCommand . . . 548
    EFSMInitialStateUndoCommand . . . 637
    EFSMStateMembershipUndoCommand . . . 645
    EFSMStatePropertiesUndoCommand . . . 646
    EFSMTransitionPropertiesUndoCommand . . . 650
    EFSMTransitionUndoCommand . . . 651
    GroupUndoCommand . . . 860
    ModalModelBindingsUndoCommand . . . 1016
    MoveUndoCommand . . . 1107
    PasteUndoCommand . . . 1218
    UngroupUndoCommand . . . 2603
QVariantAnimation
    AnimationTransition . . . 196
QWidget
    CellularAutomataLatticeWidget . . . 396
    DataAnalyzerWindow::DoePreviewPlot . . . 616
    DataAnalyzerWindow::HistogramPreview . . . 873
    LineNumberArea . . . 923
    OptimizerWindow::ProgressPlot . . . 1290
    QCustomPlot . . . 1777
    QtAbstractPropertyBrowser . . . 1813
        QtButtonPropertyBrowser . . . 1830
        QtGroupBoxPropertyBrowser . . . 1907
        QtTreePropertyBrowser . . . 2000
            ObjectPropertyBrowser . . . 1146

QtBoolEdit . . . 1824
QtCharEdit . . . 1834
QtColorEditWidget . . . 1848
QtDateEdit . . . 1859
QtDateTimeEdit . . . 1868
QtDoubleEdit . . . 1876
QtFontEditWidget . . . 1902
QtIntEdit . . . 1912
QtStringEdit . . . 1988
QtTimeEdit . . . 1992
testing::internal::Random . . . 2028
testing::internal::RawBytesPrinter . . . 2029
testing::internal::RE . . . 2030
MassActionOdeSystem::Reaction . . . 2030
testing::internal::RelationToSourceCopy . . . 2054
testing::internal::RelationToSourceReference . . . 2054
ResourceTestProbe . . . 2085
GroProgramParser::Result . . . 2086
Sampler_if::RNG_Parameters . . . 2086
SamplerDefaultImpl1::DefaultImpl1RNG_Parameters . . . 522
RootFinder_if . . . 2087
std::runtime_error
yy::genesyspp_parser::syntax_error . . . 2550
genesys::launcher::RuntimeInstallResult . . . 2107
genesys::launcher::RuntimePaths . . . 2107
genesys::launcher::RuntimeSelection . . . 2108
genesys::launcher::RuntimeSelector . . . 2108
Sample . . . 2109
Sampler_if . . . 2109
CountingSampler . . . 466
SamplerDefaultImpl1 . . . 2113
ScenarioExperiment_if . . . 2118
SceneToolController . . . 2118
SchedulableItem . . . 2121
testing::ScopedTrace . . . 2127
SegmentStep . . . 2138
SeizableItem . . . 2139
Serializer . . . 2159
CppSerializer . . . 494
GenSerializer . . . 804
JsonSerializer . . . 905
XmlSerializer . . . 2718
SessionContext . . . 2163
SessionManager . . . 2163
GroProgramRuntime::SignalMutation . . . 2181
SimpleHttpServer . . . 2182
SimulatorSessionService::SimulationActionResult . . . 2183
SimulationCommandController . . . 2183
SimulatorSessionService::SimulationConfigInput . . . 2184
SimulatorSessionService::SimulationConfigResult . . . 2185
SimulationController . . . 2220
SimulationEvent . . . 2237
SimulationEventController . . . 2238
SimulationExperiment . . . 2239
SimulationReporter_if . . . 2239
SimulationReporterDefaultImpl1 . . . 2240
SimulationResultsDataset . . . 2246

SimulationResultsDatasetParser . . . 2247
SimulationResultsObservation . . . 2247
SimulationScenario . . . 2247
SimulatorSessionService::SimulationStatusResult . . . 2248
Simulator . . . 2249
SimulatorFacade . . . 2251
SimulatorSessionService::SimulatorInfoResult . . . 2257
SimulatorSessionService . . . 2257
yy::genesyspp_parser::stack< T, S >::slice . . . 2270
WebWorkerRuntime::Snapshot . . . 2417
MetabolicFluxBalanceSolver::Solution . . . 2418
Optimizer_if::SolutionSummary . . . 2418
Solver_if . . . 2419
SolverDefaultImpl1 . . . 2420
SortFile . . . 2423
SourceCodeString . . . 2424
SparseValueStore . . . 2434
MassActionOdeSystem::Species . . . 2435
State . . . 2460
CompositeState . . . 434
HppLatticeGasState . . . 874
RealState . . . 2032
GroProgramAst::Statement . . . 2461
EFSMNetworkDiagramLayout::StatePosition . . . 2461
StateSet . . . 2461
CompositeStateSet . . . 435
HppLatticeGasStateSet . . . 876
IntegerStateSet . . . 893
NaturalStateSet . . . 1115
RealIntervalStateSet . . . 2030
StateSet_Enumerable . . . 2463
Statistics_if . . . 2474
StatisticsDatafile_if . . . 2483
StatisticsDatafileDefaultImpl1 . . . 2487
StatisticsDefaultImpl1 . . . 2492
BioReaction::StoichiometricTerm . . . 2520
MassActionOdeSystem::StoichiometricTerm . . . 2520
MetabolicReaction::StoichiometricTerm . . . 2520
StochasticReactionRule::StoichiometricTerm . . . 2521
testing::internal::internal_stream_operator_without_lexical_name_lookup::StreamPrinter . . . 2539
testing::internal::String . . . 2539
yy::genesyspp_parser::symbol_kind . . . 2546
SystemCommandExecutor_if . . . 2551
FakeSystemCommandExecutor . . . 720
SystemShellCommandExecutor . . . 2558
SystemCommandResult . . . 2552
SystemDependency . . . 2553
SystemDependencyCheckEntry . . . 2553
SystemDependencyCheckResult . . . 2554
SystemDependencyInstallEntry . . . 2554
SystemDependencyInstallResult . . . 2555
SystemDependencyResolver . . . 2555
SystemPreferences . . . 2556
T
testing::internal::SuiteApiResolver< T > . . . 2546
testing::internal::Templates< Head_, Tail_ > . . . 2561
testing::internal::Templates< Head_ > . . . 2561

testing::internal::TemplateSel< Tmpl > . . . 2561
testing::Test . . . 2561
ParserExpressionsTest . . . 1216
testing::TestWithParam< T > . . . 2574
testing::TestEventListener . . . 2562
testing::EmptyTestEventListener . . . 653
testing::TestEventListeners . . . 2563
testing::internal::TestFactoryBase . . . 2563
testing::internal::ParameterizedTestFactory< TestClass > . . . 1197
testing::internal::TestFactoryImpl< TestClass > . . . 2564
testing::TestInfo . . . 2565
testing::internal::TestMetaFactoryBase< ParamType > . . . 2566
testing::internal::TestMetaFactoryBase< TestSuite::ParamType > . . . 2566
testing::internal::TestMetaFactory< TestSuite > . . . 2565
testing::TestParamInfo< ParamType > . . . 2566
TestPartResultReporterInterface
testing::internal::DefaultGlobalTestPartResultReporter . . . 521
testing::internal::DefaultPerThreadTestPartResultReporter . . . 536
testing::TestProperty . . . 2567
testing::internal::TestPropertyKeyIs . . . 2567
HypothesisTester_if::TestResult . . . 2567
testing::TestResult . . . 2568
testing::internal::TestResultAccessor . . . 2568
testing::TestSuite . . . 2573
testing::internal::ThreadLocal< T > . . . 2575
testing::internal::ThreadLocal< std::vector< testing::internal::TraceInfo > > . . . 2575
testing::internal::ThreadLocal< TestPartResultReporterInterface * > . . . 2575
QCPAxisPainterPrivate::TickLabelData . . . 2575
Token . . . 2576
yy::genesyspp_parser::token . . . 2576
TokenService . . . 2577
TraceConsoleController . . . 2577
TraceEvent . . . 2579
TraceErrorEvent . . . 2578
TraceSimulationEvent . . . 2584
TraceSimulationProcess . . . 2584
testing::internal::TraceInfo . . . 2579
TraceManager . . . 2579
TraitsGUI< T > . . . 2585
TraitsGUI< GAssociation > . . . 2586
TraitsGUI< GComponentPort > . . . 2586
TraitsGUI< GConnection > . . . 2586
TraitsGUI< GMainWindow > . . . 2587
TraitsGUI< GModelComponent > . . . 2587
TraitsGUI< GModelDataDefinition > . . . 2587
TraitsGUI< GModelGraphicPic > . . . 2588
TraitsGUI< GScene > . . . 2588
TraitsGUI< GView > . . . 2588
TraitsKernel< T > . . . 2588
TraitsKernel< Collector_if > . . . 2589
TraitsKernel< Model > . . . 2589
TraitsKernel< ModelChecker_if > . . . 2589
TraitsKernel< ModelComponent > . . . 2589
TraitsKernel< ModelDataDefinition > . . . 2589
TraitsKernel< Parser_if > . . . 2590
TraitsKernel< Persistence_if > . . . 2590
TraitsKernel< PluginConnector_if > . . . 2590

TraitsKernel< Sampler_if > . . . 2590
TraitsKernel< SimulationReporter_if > . . . 2590
TraitsKernel< Simulator > . . . 2591
TraitsKernel< Statistics_if > . . . 2591
TraitsKernel< StatisticsDatafile_if > . . . 2591
TraitsTools< T > . . . 2591
TraitsTools< AIAssistant_if > . . . 2592
TraitsTools< Fitter_if > . . . 2592
TraitsTools< HypothesisTester_if > . . . 2592
TraitsTools< Solver_if > . . . 2592
EFSMNetworkDiagramLayout::TransitionLayout . . . 2593
Transporter::TravelCompletion . . . 2599
std::true_type
testing::internal::is_proxy_type_list< ProxyTypeList< Ts... > > . . . 898
testing::internal::TrueWithString . . . 2599
DoubleSequence::type
testing::internal::MakeIndexSequenceImpl< N > . . . 956
IsRecursiveContainerImpl::type
testing::internal::IsRecursiveContainer< C > . . . 900
testing::internal::TypeIdHelper< T > . . . 2600
testing::internal::TypeParameterizedTest< Fixture, TestSel, Types > . . . 2601
testing::internal::TypeParameterizedTest< Fixture, TestSel, internal::None > . . . 2601
testing::internal::TypeParameterizedTestSuite< Fixture, Tests, Types > . . . 2601
testing::internal::TypeParameterizedTestSuite< Fixture, internal::None, Types > . . . 2601
testing::internal::TypeParameterizedTestSuiteRegistry . . . 2601
testing::internal::Types< Head_, Tail_ > . . . 2602
testing::internal::Types< Head_ > . . . 2602
testing::internal::TypeWithSize< size > . . . 2602
testing::internal::TypeWithSize< 4 > . . . 2602
testing::internal::TypeWithSize< 8 > . . . 2602
testing::internal::TypeWithSize< sizeof(RawType)> . . . 2602
testing::UnitTest . . . 2603
testing::internal::UnitTestImpl . . . 2604
testing::internal::UnitTestOptions . . . 2606
testing::internal::UniversalPrinter< T > . . . 2606
testing::internal::UniversalPrinter< const T > . . . 2607
testing::internal::UniversalPrinter< T & > . . . 2608
testing::internal::UniversalPrinter< T[N]> . . . 2608
testing::internal::UniversalTersePrinter< T > . . . 2608
testing::internal::UniversalTersePrinter< const char * > . . . 2611
testing::internal::UniversalTersePrinter< char * > . . . 2608
testing::internal::UniversalTersePrinter< const char16_t * > . . . 2612
testing::internal::UniversalTersePrinter< char16_t * > . . . 2609
testing::internal::UniversalTersePrinter< const char32_t * > . . . 2612
testing::internal::UniversalTersePrinter< char32_t * > . . . 2610
testing::internal::UniversalTersePrinter< T & > . . . 2613
testing::internal::UniversalTersePrinter< T[N]> . . . 2613
testing::internal::UniversalTersePrinter< wchar_t * > . . . 2613
genesys::launcher::UpdateChecker . . . 2620
genesys::launcher::UpdateCheckResult . . . 2620
genesys::launcher::UpdateDownloader . . . 2622
genesys::launcher::UpdateManifest . . . 2623
genesys::launcher::UpdateManifestParser . . . 2623
genesys::launcher::UpdatePlatform . . . 2624
genesys::launcher::UpdateVerifier . . . 2624
genesys::launcher::UserRuntimeManager . . . 2624
Util . . . 2624

yy::genesyspp_parser::value_type 2635
testing::internal::ValueArray< Ts > 2636
genesys::launcher::VerificationResult 2644
genesys::launcher::Version 2644
Waiting 2670
    WaitingResource 2670
WebWorkerRuntime 2694
WholeCellParameterReader 2696
QtButtonPropertyBrowserPrivate::WidgetItem 2702
QtGroupBoxPropertyBrowserPrivate::WidgetItem 2703
testing::WithParamInterface< T > 2703
    testing::TestWithParam< T > 2574
SimulatorSessionService::WorkerCapabilitiesResult 2704
SimulatorSessionService::WorkerInfoResult 2704
WorkerJob 2705
SimulatorSessionService::WorkerJobCreationResult 2706
SimulatorSessionService::WorkerJobInfoResult 2706
WorkerJobManager 2707
SimulatorSessionService::WorkerJobQueryResult 2709
SimulatorSessionService::WorkerJobResultInfo 2710
SimulatorSessionService::WorkerJobResultQueryResult 2710
SimulatorSessionService::WorkerJobRunResult 2711
WorkerJobTerminalResult 2711
yy_buffer_state 2734
yy_trans_info 2734

Chapter 14

Class Index

14.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

Access . . . 117
AddUndoCommand . . . 124
AIAssistant . . . 125
AIAssistant_if
    Abstract interface for the GenESyS AI Assistant tool . . . 131
AIAssistantConfiguration
    Full configuration record for the AI assistant . . . 142
AIAssistantDefaultImpl
    Initial concrete scaffold for the GenESyS AI Assistant . . . 143
AIAssistantRequest
    Input to a single AI assistant request . . . 155
AIAssistantResponse
    Output produced by a single AI assistant operation . . . 156
AIAssistantWindow . . . 156
AIAuditLog
    Append-only audit log for AI assistant operations . . . 157
AIConversationConfiguration . . . 158
AIConversationResponse . . . 158
AIConversationService . . . 159
AIProviderClient_if
    Abstract interface for AI provider HTTP clients . . . 159
Airport_Extended1 . . . 161
AirportSecurityExample . . . 163
AirportSecurityExampleExtended . . . 165
AISecretStore
    Static helper for storing API keys in the OS keyring . . . 166
AISupport . . . 168
AND . . . 173
AnElectronicAssemblyAndTestSystem . . . 179
AnimationAttribute . . . 181
AnimationCounter . . . 182
AnimationEntity . . . 183
AnimationEvent . . . 185
AnimationExpression . . . 186
AnimationPlaceholder . . . 187
AnimationPlot . . . 188
AnimationQueue . . . 190
AnimationQueueDisplay . . . 190
AnimationResource . . . 192
AnimationStation . . . 193

AnimationStatistics . . . 194
AnimationTimer . . . 195
AnimationTransition . . . 196
AnimationVariable . . . 198
AnthropicProviderClientImpl
AI provider client for the Anthropic Messages API . . . 199
ApiRouter
Routes HTTP requests to Web API operations . . . 203
Assembly_Line . . . 204
testing::internal::AssertHelper . . . 205
Assign
Component that evaluates one or more (destination, expression) pairs and writes each result into an Attribute or a Variable on the entity/model . . . 205
Assignment
One (destination, expression) pair used by Assign and by Sequence's per-step assignments . 212
Attribute
Data definition declaring one named, per-entity attribute (its dimensionality and sparse initial values), shared across every entity created in the model . . . 212
AuditEntry
One entry in the AI assistant audit log . . . 218
BacteriaColony
First GenESyS component for Gro-inspired bacteria colony simulation . . . 219
BacteriaColonyViewerGuiExtensionPlugin . . . 227
BacteriaSignalGrid
Reusable discrete signal-field definition for bacteria colonies . . . 229
BacteriaColony::BacteriumState . . . 233
Banking_Transactions . . . 234
BacteriaColony::BarrierSegment . . . 236
BaseGenesysTerminalApplication . . . 236
BaseGenesysWebApplication
Entry point for the standalone Worker application flavor of Genesys . . . 238
BaseMVController
Compatibility base class that binds the Qt UI and the simulator model . . . 239
yy::genesyspp_parser::basic_symbol< Base > . . . 240
Batch
Accumulates entities in an internal queue until a batch is complete, then dispatches one representative entity for the group . . . 242
testing::internal::TemplateSel< Tmpl >::Bind< T > . . . 249
ModalModelBindingsUndoCommand::Bindings . . . 249
BioCompartment
Defines a biological compartment for biochemical and whole-cell models . . . 249
BioKineticLawExpression . . . 254
BioNetwork . . . 255
BioNetworkAwareGuiExtensionPlugin . . . 260
BioNetworkEditorGuiExtensionPlugin . . . 262
BioNetworkExecutionGuiExtensionPlugin . . . 263
BioParameter . . . 264
BioParameterManagerGuiExtensionPlugin . . . 269
BioParameterSensitivityEntry . . . 271
BioReaction . . . 271
BioReactionKineticsGuiExtensionPlugin . . . 282
BioReactionRateSample . . . 283
BioReactionRateTimeCourse . . . 283
BioReactionStoichiometryGuiExtensionPlugin . . . 284
BioRunnerCommand
Component wrapper over BioSimulatorRunner command execution . . . 285
BioSBMLBridge . . . 291
BioSBMLExportResult . . . 291

BioSBMLImportResult . . . 292
BioSBMLInteroperabilityGuiExtensionPlugin . . . 292
BioSensitivityScan . . . 293
BioSimulate
Component that triggers deterministic BioNetwork simulation . . . 293
BioSimulationAnalysis . . . 300
BioSimulationResult . . . 300
BioSimulationResultsGuiExtensionPlugin . . . 301
BioSimulationSample . . . 302
BioSimulationSpeciesAmount . . . 302
BioSimulatorRunner . . . 303
BioSpecies . . . 308
BioSpeciesDerivative . . . 312
BioSpeciesManagerGuiExtensionPlugin . . . 313
BioStateProjectionComponent
Projects selected BioSpecies values into a WholeCellState . . . 314
BioSteadyState
Component that checks steady-state diagnostics over BioNetwork results . . . 320
BioSteadyStateCheck . . . 327
BioStoichiometryMatrix . . . 327
Book_Cap02_Example01 . . . 327
Boundary_Closed . . . 329
Boundary_Fixed . . . 330
BoundaryCondition . . . 332
Buffer . . . 333
BufferFIFO . . . 340
yy::genesyspp_parser::by_kind
Type access provider for token (enum) based symbols . . . 341
QCPAxisPainterPrivate::CachedLabel . . . 343
QCPLabelPainterPrivate::CachedLabel . . . 343
ModelLifecycleController::Callbacks . . . 343
SimulationEventController::Callbacks . . . 344
Capacitor . . . 344
testing::internal::CartesianProductGenerator< T > . . . 350
testing::internal::CartesianProductHolder< Gen > . . . 351
Cell . . . 352
CellCycleCheckpointComponent
Evaluates lifecycle checkpoints and advances whole-cell step time . . . 353
CellDivisionEvent . . . 360
CellFateDecisionComponent
Routes entities according to the current whole-cell lifecycle state . . . 367
CellGrowthComponent . . . 373
CellularAutomata . . . 380
CellularAutomata_1DTimed
Complete uniform CA specific for unidimensional lattices where 2nd dimension is time (usefull for elementary CA) . . . 381
CellularAutomata_Classic
Complete uniform CA . . . 382
CellularAutomataBase . . . 384
CellularAutomataComp . . . 386
CellularAutomataDemoBuilder . . . 392
CellularAutomataDemoModel . . . 393
CellularAutomataDemoSettings . . . 393
CellularAutomataEvent . . . 394
CellularAutomataGuiExtensionPlugin . . . 395
CellularAutomataLatticeWidget . . . 396
CellularAutomataStateColorMap . . . 397
CellularAutomataViewerController . . . 397

CellularAutomataViewerWindow . . . 399
ChatMessage
One message in a multi-turn conversation with an LLM . . . 399
Clone
Creates _numClonesExpression copies of the incoming entity (same EntityType and Attribute values), sending clones out port 1 and the original out port 0 . . . 400
CodeEditor . . . 406
testing::internal::CodeLocation . . . 407
Collector_if
Interface responsible for collecting observations of a stochastic variable . . . 407
CollectorDatafile_if
Collector interface specialized in file-backed value storage . . . 410
CollectorDatafileDefaultImpl1
Default file-backed implementation of CollectorDatafile_if . . . 412
CollectorDefaultImpl1
Default in-memory implementation of Collector_if . . . 415
GroProgramRuntime::ColonyMutation . . . 417
ColoredPetriNetNetwork
Pragmatic fixed-inscription Colored Petri Net subset . . . 417
ComboBoxEnum . . . 424
GroProgramIr::Command . . . 424
Util::CommandResult . . . 425
CompartmentExchangeComponent
Moves metabolite proxies between whole-cell regions . . . 425
CppCompiler::CompilationResult . . . 432
CompilerThread . . . 433
ComponentItem . . . 433
CompositeState . . . 434
CompositeStateSet . . . 435
HypothesisTester_if::ConfidenceInterval
Confidence interval [inferior, superior] and its half-width e0 . . . 437
Connection . . . 437
ConnectionChannel
Defines an input or output port that connects a ModelComponent to another . . . 438
ConnectionManager
Maintains outgoing connections from a component to downstream components . . . 438
testing::internal::ConstCharPtr . . . 443
Optimizer_if::ConstraintDefinition
Feasibility restriction represented as an expression . . . 443
testing::internal::ConstRef< T > . . . 443
testing::internal::ConstRef< T & > . . . 443
testing::internal::ContainerPrinter . . . 444
yy::genesyspp_parser::context . . . 444
ContinuousDistribution_if
Interface for continuous distributions . . . 444
ContinuousModel . . . 446
ContinuousSystemComponent
Component that advances a continuous ODE system when entities arrive . . . 447
testing::internal::ConvertibleToIntegerPrinter . . . 454
testing::internal::ConvertibleToStringViewPrinter . . . 454
Conveyor
Minimal conveyor data definition built on top of one Segment path . . . 454
CoreGuiExtensionPlugin . . . 459
Counter
Model datum used to accumulate event counts (or weighted counts) . . . 461
CountingSampler . . . 466
CPNArc
Directed bipartite arc used by ColoredPetriNetNetwork . . . 471

CPNTransition
Transition node used by ColoredPetriNetNetwork . . . 476
CppCompiler . . . 482
CppForG . . . 487
CppModelExporter
Service responsible for C++ code representation of the current simulation model . . . 493
CppSerializer . . . 494
Create
Source component that generates entities and starts them into the model, driven by a parsed inter-arrival expression, a Schedule, or a Formula . . . 497
SimulatorSessionService::CreateSessionResult
Contains session creation identifiers returned to API clients . . . 504
QtBoolPropertyManagerPrivate::Data . . . 505
QtDatePropertyManagerPrivate::Data . . . 505
QtDateTimePropertyManagerPrivate::Data . . . 505
QtDoublePropertyManagerPrivate::Data . . . 505
QtEnumPropertyManagerPrivate::Data . . . 506
QtFlagPropertyManagerPrivate::Data . . . 506
QtIntPropertyManagerPrivate::Data . . . 506
QtPointFPropertyManagerPrivate::Data . . . 507
QtRectFPropertyManagerPrivate::Data . . . 507
QtRectPropertyManagerPrivate::Data . . . 507
QtSizeFPropertyManagerPrivate::Data . . . 507
QtSizePropertyManagerPrivate::Data . . . 507
QtStringPropertyManagerPrivate::Data . . . 508
QtTimePropertyManagerPrivate::Data . . . 508
DataAnalyser_if
High-level façade for applied statistical analysis over datasets . . . 508
DataAnalyzerWindow . . . 509
DataComponentEditor . . . 510
DataComponentProperty . . . 511
DataDefinitionItem . . . 512
DataDefinitionRelationItem . . . 512
ModelInspectorController::DataDefinitionSelection . . . 513
AnimationPlot::DataSeries . . . 513
DataSet_if
Abstract view of a numeric dataset . . . 514
Decide
Routes an entity to the first output port whose parser expression evaluates truthy, or to a trailing "else" port when none do . . . 514
Delay::DEFAULT_VALUES . . . 520
ModalModelDefault::DEFAULT_VALUES . . . 520
Seize::DEFAULT_VALUES . . . 521
SourceModelComponent::DEFAULT_VALUES . . . 521
testing::internal::DefaultGlobalTestPartResultReporter . . . 521
SamplerDefaultImpl1::DefaultImpl1RNG_Parameters
Parameter bundle for the default RNG algorithm . . . 522
testing::internal::DefaultNameGenerator . . . 523
DefaultNetwork
Base class for a formalism-agnostic network model of computation (EFSM, DTMC, Coloured Petri Net, ...) that may be shared by several `ModalModel` process adapters . . . 523
DefaultNode . . . 529
DefaultNodeTransition . . . 535
testing::internal::DefaultPerThreadTestPartResultReporter . . . 536
Delay
Holds an entity in place for an evaluated expression's worth of time, crediting that time to one of five allocation categories . . . 536
DelayProbe . . . 543

DeleteUndoCommand . . . 548
dialogBreakpoint . . . 549
DialogFind . . . 550
DialogModelInformation . . . 551
DialogPluginManager . . . 552
DialogReportIssue
Modal form used to collect a user-facing issue report before relay submission . . . 553
DialogSelectCounter . . . 555
DialogSelectVariable . . . 556
DialogSimulationConfigure . . . 557
DialogSystemPreferences . . . 558
DialogTimerConfigure . . . 559
DialogUtilityController
Controller that orchestrates auxiliary dialogs and utility actions . . . 559
DiffEquations . . . 561
DiffusionField . . . 568
DiffusionMethodOfLinesSystem . . . 573
DiffusionSimulate
Component that triggers a DiffusionField solve when entities arrive . . . 575
Diode . . . 582
DirectedAcyclicGraphNetwork
Directed graph specialization that enforces the DAG invariant . . . 588
DirectedGraphNetwork
Directed mathematical graph specialization . . . 594
DiscreteDistribution_if
Interface for discrete distributions . . . 602
genesys::launcher::DispatchController . . . 603
Dispose
Sink component that ends an entity's life in the model, optionally recording its total time in system before removing it . . . 603
Distance
Data definition holding a set of direct station-to-station distances, used by free-path (non-guided) `Transporter` units to compute travel time . . . 610
DistanceEntry
One directed (from-station, to-station, length) entry of a Distance set, used to look up a free-path Transporter's travel length between two stations . . . 615
Distribution_if
Base abstraction for a theoretical probability distribution . . . 615
DataAnalyzerWindow::DoePreviewPlot . . . 616
DormandPrince54OdeSolver . . . 617
testing::internal::DoubleSequence< plus_one, T, sizeofT > . . . 618
testing::internal::DoubleSequence< false, IndexSequence< I... >, sizeofT > . . . 618
testing::internal::DoubleSequence< true, IndexSequence< I... >, sizeofT > . . . 619
genesys::launcher::DownloadResult . . . 619
DrawingItem . . . 619
DropOff
Placeholder for releasing a subset of a batched group's members . . . 619
DummyComponent . . . 625
DummyElement . . . 631
EditCommandController
Controller for edit/clipboard/grouping command orchestration in the scene . . . 636
EditorFactoryPrivate< Editor > . . . 637
EFSMInitialStateUndoCommand . . . 637
EFSMNetwork
Extended finite-state-machine network specialization . . . 638
EFSMNetworkDiagramLayout
GUI-only presentation layout for one EFSMNetwork diagram . . . 644
EFSMStateMembershipUndoCommand . . . 645

EFSMStatePropertiesUndoCommand . . . 646
EFSMTransition . . . 647
EFSMTransitionGraphicsItem
Read-only graphical projection of one mathematical EFSM transition . . . 649
EFSMTransitionPropertiesUndoCommand . . . 650
EFSMTransitionUndoCommand . . . 651
testing::internal::ElemFromList< N, T > . . . 652
testing::internal::ElemFromListImpl< typename > . . . 652
testing::internal::ElemFromListImpl< IndexSequence< I... > > . . . 652
testing::EmptyTestEventListener . . . 653
Enter . . . 655
Entity
Runtime entity (token) flowing through the model during simulation . . . 662
EntityGroup . . . 669
EntityType
Data definition describing one class of entity that Create-family components may instantiate at simulation time . . . 674
PersistenceRecord::Entry
"Raw" entry, users should treat this as an opaque type since payloads could be in any format . 679
testing::Environment . . . 680
EukaryoticCellCycleComponent
Coordinates a didactic eukaryotic cell cycle over a shared WholeCellState . . . 680
Event . . . 688
Exact . . . 689
Example_BasicOrderShipping . . . 690
Example_PortModel . . . 692
Example_PublicTransport . . . 694
CompartmentExchangeComponent::ExchangeRule . . . 695
GroProgramRuntime::ExecutionResult . . . 696
Exit . . . 696
ExperimentManager
Manages higher-level simulation experiments . . . 702
ExperimentManager_if
Interface that coordinates execution of scenario-based simulation experiments . . . 703
ExperimentManagerDefaultImpl1 . . . 704
ExpressionBuilder . . . 706
ExpressionCategory . . . 707
ExpressionParameter . . . 707
ExpressionTemplate . . . 707
genesys::launcher::ExtractionResult . . . 707
FactorialDesign . . . 707
Failure
Data definition describing a resource breakdown pattern (whole resource failure regardless of capacity) attached to one or more Resource instances . . . 710
FakeModelPersistence . . . 716
FakeModelPersistenceB . . . 718
FakeSystemCommandExecutor . . . 720
testing::internal::FallbackPrinter . . . 721
File
Data definition identifying an external file and its access mode (read/write/append/read-write), used by file-driven components and data definitions . . . 721
testing::internal::FindFirstPrinter< T, E, Printer, Printers > . . . 727
testing::internal::FindFirstPrinter< T, decltype(Printer::PrintValue(std::declval< const T & >(), nullptr)), Printer, Printers... > 727
FindLineEdit . . . 727
Fitter_if
Interface for fitting theoretical distributions to real datasets . . . 728

FitterDefaultImpl
Functional implementation of Fitter_if consolidated through FITTER-2 . . . 730
FitterDummyImpl
Placeholder implementation of Fitter_if . . . 732
testing::internal::FlatTuple< T > . . . 735
testing::internal::FlatTupleBase< Derived, Idx > . . . 736
testing::internal::FlatTupleBase< FlatTuple< T... >, IndexSequence< Idx... > > . . . 736
testing::internal::FlatTupleConstructTag . . . 737
testing::internal::FlatTupleElemBase< Derived, I > . . . 737
testing::internal::FlatTupleElemBase< FlatTuple< T... >, I > . . . 738
testing::internal::FloatingPoint< RawType > . . . 738
testing::internal::FormatForComparison< ToPrint, OtherOperand > . . . 739
testing::internal::FormatForComparison< ToPrint[N], OtherOperand > . . . 739
Formula
Data definition holding one or more named parser expressions, optionally indexed to represent a 1D/2D array of expressions . . . 740
FSMState . . . 747
FtsZPolymerizationComponent . . . 752
FullSimulationOfComplexModel . . . 759
testing::internal::FunctionPointerPrinter . . . 760
ParserManager::GenerateNewParserResult
Result bundle for parser generation requests . . . 760
testing::internal::GenerateTypeList< T > . . . 761
GenesysApplication_if . . . 762
genesys_mcp.client.GenesysClient . . . 763
genesys_mcp.client.GenesysClientError . . . 764
GenesysKnowledgeBaseSummary
Aggregate summary of the current knowledge base state . . . 765
GenesysKnowledgeItem
One entry in the GenESyS knowledge base used to inform the AI assistant . . . 765
genesyspp_driver . . . 766
yy::genesyspp_parser
A Bison parser . . . 767
GenesysPropertyDescriptor . . . 771
GenesysPropertyIntrospection . . . 772
GenesysShell . . . 772
GenesysShell_if . . . 774
GeneticCircuit . . . 775
GeneticCircuitGuiExtensionPlugin . . . 780
GeneticCircuitPart . . . 782
GeneticCircuitSimulate . . . 786
GeneticCircuitSimulationSummary . . . 793
GeneticExpressionStep . . . 793
GeneticRegulation . . . 799
GenSerializer . . . 804
genesys::launcher::GpgvSignatureVerifier . . . 807
GraphEdge
Mathematical graph edge between two GraphNode endpoints . . . 808
GraphicalAnimateExpression . . . 814
GraphicalAssociation . . . 814
GraphicalAutomaticPositioningStrategy . . . 814
GraphicalAutomaticPositioningStrategyFactory
Factory for the automatic positioning strategy selected from GUI preferences . . . 815
GraphicalComponentPort . . . 816
GraphicalConnection
Graphical edge that represents model-level component connections . . . 817
GraphicalConnectionStyle . . . 820

GraphicalContextMenuController
Builds context menus for the graphical model view using existing main-window actions 820
GraphicalDataDefinitionLayout
Pure helper that computes arc placement for graphical data definitions 821
GraphicalDiagramConnection 822
GraphicalImageAnimation 823
GraphicalModelBuilder
Service that reconstructs graphical components/connections from kernel model data 823
GraphicalModelComponent
Visual component item that extends a data definition with ports and internal data 825
GraphicalModelDataDefinition
Visual representation of a model data definition in the scene 830
GraphicalModelEvent
Lightweight event envelope used to notify GUI changes in the graphics scene 832
GraphicalModelItemRenderContext 833
GraphicalModelItemRenderer 834
GraphicalModelItemRenderStrategy 834
GraphicalModelSerializer
Persistence service for saving/loading graphical and textual model representations 834
GraphNetwork
Mathematical graph specialization of DefaultNetwork 836
GraphNode
Mathematical graph vertex used by GraphNetwork 843
GraphPathResult
Explicit graph path result 848
GraphSearchTreeEntry 848
GraphTopologicalOrderResult 849
GraphTraversalResult
Result of a deterministic graph traversal 849
GraphvizModelExporter
Service that builds DOT content and renders model images for the GUI 849
ModelGraphicsScene::GRID 851
GroProgram
Reusable Gro source program attached to a GenESyS model 851
GroProgramAst
Minimal parsed representation of a Gro source program 856
GroProgramCompiler
Converts parsed Gro AST into the first non-executable semantic IR 857
GroProgramIr
Initial semantic representation compiled from GroProgramAst 857
GroProgramParser
Plugin-side parser boundary for reusable Gro source programs 858
GroProgramRuntime
Executes the supported Gro IR commands without simulator binding 858
GroProgramRuntimeState
Mutable state used by the plugin-side Gro runtime helper 859
GroupItem 859
GroupUndoCommand 860
testing::internal::GTestFlagSaver 860
testing::internal::GTestLog 860
testing::internal::GTestMutexLock 861
GuiActionContribution 861
GuiDockContribution 861
GuiExtensionConfigurationDialog 862
GuiExtensionConfigurationGuiExtensionPlugin 863
GuiExtensionManager 864
GuiExtensionPlugin 865
GuiExtensionPluginAutoRegistrar 866

GuiExtensionPluginCatalog . . . 866
GuiExtensionPresentationMetadata . . . 866
GuiExtensionRegistry . . . 866
GuiExtensionRuntimeContext . . . 867
GuiScopeTrace
Emits a scoped trace entry for a GUI method and measures its runtime . . . 867
GuiThemeManager
Applies the visual theme used by the Qt application and model views . . . 869
GuiWindowContribution . . . 870
Half_Adder . . . 870
HarmonicOscillator . . . 872
testing::internal::HasDebugStringAndShortDebugString< T > . . . 873
DataAnalyzerWindow::HistogramPreview . . . 873
HppLatticeGasState . . . 874
HppLatticeGasStateSet . . . 876
HttpProviderClientBase
Abstract base class for HTTP-based AI provider clients . . . 878
HttpRequest
Lightweight representation of one parsed HTTP request . . . 882
HttpResponse
Minimal HTTP response payload produced by the web API router . . . 883
HypothesisTester_if
Interface for classical parametric inference . . . 884
HypothesisTesterDefaultImpl1
Current default implementation of HypothesisTester_if . . . 885
genesys::launcher::IArchiveExtractor . . . 892
testing::internal::Ignore< size_t > . . . 892
testing::internal::IgnoredValue . . . 892
testing::internal::IndexSequence< Is > . . . 892
genesys::launcher::INetworkTransport . . . 893
IntegerStateSet . . . 893
InternalEvent . . . 895
genesys::launcher::IProcessLauncher . . . 897
testing::internal::is_proxy_type_list< typename > . . . 897
testing::internal::is_proxy_type_list< ProxyTypeList< Ts... > > . . . 898
testing::internal::IsHashTable< T > . . . 899
genesys::launcher::ISignatureVerifier . . . 899
testing::internal::IsRecursiveContainer< C > . . . 900
testing::internal::IsRecursiveContainerImpl< C, bool > . . . 900
testing::internal::IsRecursiveContainerImpl< C, false > . . . 901
testing::internal::IsRecursiveContainerImpl< C, true > . . . 901
IssueReportDraft
User-authored issue report collected by the GUI form before transport . . . 902
IssueReportRelayService
HTTPS client service that sends issue reports to a secure relay endpoint . . . 902
IssueReportRuntimeContext
Runtime diagnostics captured by the GUI after the user approves submission . . . 903
IssueReportSubmissionResult
Relay submission outcome returned to the controller layer . . . 903
genesys::launcher::IUpdateUi . . . 904
Json . . . 904
JsonSerializer . . . 905
KernelAccessorProbe . . . 907
Label . . . 908
QCPLabelPainterPrivate::LabelData . . . 912
Lattice
N-dimensional lattice . . . 913
genesys::launcher::LauncherConfig . . . 914

genesys::launcher::LauncherController . . . 914
genesys::launcher::LauncherLogger . . . 915
genesys::launcher::LauncherUpdateResult . . . 915
Leave . . . 915
LicenceManager . . . 922
LineNumberArea . . . 923
List< T >
Lightweight wrapper around `std::list` with convenience helpers used by the kernel . . . 923
ListObservable< T >
Observable-oriented list variant used by kernel structures . . . 925
LocalProviderClientImpl
AI provider client for local model servers (Ollama, LM Studio, etc.) . . . 928
LocalRule . . . 932
LocalRule_CopyCurrentState . . . 934
LocalRule_Elementary . . . 935
LocalRule_FlorestalFire . . . 936
LocalRule_GameOfLife . . . 938
LocalRule_Growty . . . 939
LocalRule_HppLatticeGas . . . 940
LocalRule_SandRock_Student . . . 942
LocalRule_ShowNeighbordhood . . . 943
yy::location
Two points in a source file . . . 944
Loja01 . . . 946
testing::internal::internal_stream_operator_without_lexical_name_lookup::LookupBlocker . . . 947
LSODE . . . 947
LSODEInitialState . . . 954
LSODESample . . . 954
MainWindow
Main Qt window that composes controllers/services and preserves legacy GUI entry points . . . 954
testing::internal::MakeIndexSequenceImpl< N > . . . 956
testing::internal::MakeIndexSequenceImpl< 0 > . . . 957
genesys::launcher::ManifestParseResult . . . 958
testing::internal::MarkAsIgnored . . . 959
MarkovChain . . . 959
MarkovChainNetwork
Finite time-homogeneous Discrete-Time Markov Chain network . . . 965
MarkovState
State node used by MarkovChainNetwork DTMC models . . . 972
MarkovChainNetwork::MarkovTransition . . . 977
MassActionOdeSystem
ODE system generated from biochemical reactions using mass-action kinetics . . . 977
Match
Brings together one entity from each of several input queues, optionally requiring a matching attribute value, and releases them in sync . . . 978
testing::Message . . . 985
MetabolicFluxBalance . . . 985
MetabolicFluxBalanceSolver . . . 991
MetabolicNetwork . . . 992
MetabolicReaction . . . 997
MetabolicStateProjectionComponent
Projects flux-balance outputs into a WholeCellState . . . 1002
MetabolicSubmodelComponent . . . 1009
generate_kernel_simulator_test_stubs.MethodEntry . . . 1015
ModalModelBindingsUndoCommand . . . 1016
ModalModelDefault . . . 1016
Model . . . 1023
ModelChecker_if . . . 1031

ModelCheckerDefaultImpl1 . . . 1033
ModelComponent . . . 1035
ModelComponentManager
Owns and organizes the set of `ModelComponent` objects in a model . . . 1044
ModelDataDefinition . . . 1048
ModelDataDefinitionAssociations . . . 1060
ModelDataManager
Central registry for `ModelDataDefinition` instances owned by a model . . . 1061
ModelEvent . . . 1067
ModelGraphicsScene
Scene that owns and manipulates the graphical representation of a model . . . 1067
ModelGraphicsView . . . 1076
SimulatorSessionService::ModelImportResult
Contains model import output and error state . . . 1079
ModelInfo . . . 1080
SimulatorSessionService::ModelInfoResult
Describes the currently active model information for a session . . . 1083
ModelInspectorController
Controller for model inspection views and scene-selection synchronization . . . 1083
ModelLanguageSynchronizer
Service layer that synchronizes textual model language with kernel model state . . . 1084
ModelLifecycleController
Controller for model/application lifecycle flows delegated from MainWindow . . . 1085
ModelManager
The ModelManager class . . . 1086
SimulatorSessionService::ModelPersistenceResult
Contains save/load model operation output and error state . . . 1092
ModelPluginInspectorGuiExtensionPlugin . . . 1092
ModelSimulation
Controls the mature, currently functional simulation-execution layer of a model . . . 1094
MolecularSpecies . . . 1095
GroProgramRuntime::MotionMutation . . . 1100
Move
Minimal transporter movement component . . . 1101
MoveUndoCommand . . . 1107
testing::internal::Mutex . . . 1108
dialogBreakpoint::MVCResult . . . 1108
GroProgramAst::NamedProgram . . . 1108
GroProgramIr::NamedProgramDefinition . . . 1108
testing::internal::NameGeneratorSelector< Provided > . . . 1108
NAND . . . 1109
testing::internal::NativeArray< Element > . . . 1115
NaturalStateSet . . . 1115
Neighborhood
Defines the complete neighborhood of a cell . . . 1117
Neighborhood_Center . . . 1118
Neighborhood_Moore . . . 1120
Neighborhood_VonNeumann . . . 1122
NetworkActivationFrame
Input side of one `DefaultNetwork::activate()` call: the presence and value of every declared input port for this activation only . . . 1123
NetworkActivationResult
Output side of one `DefaultNetwork::activate()` call: the presence and value of every declared output port produced by that activation . . . 1124
NetworkPortValue
Presence/value pair carried by one network port during one activation . . . 1124
genesys::launcher::NetworkRequestOptions . . . 1125
genesys::launcher::NetworkResponse . . . 1125

ParserManager::NewParser
Paths associated with a generated parser artifact set . . . 1125
NMOS . . . 1126
testing::internal::None . . . 1132
NOR . . . 1132
NOT . . . 1139
obj_t . . . 1145
Optimizer_if::ObjectiveDefinition
Configurable objective definition . . . 1145
ObjectPropertyBrowser . . . 1146
OctaveRunner
Model data definition that executes Octave code through Rscript . . . 1149
ODEfunction . . . 1154
ODESolver . . . 1155
OdeSolver_if
ODE solver abstraction . . . 1160
OdeSolverFactory . . . 1161
OdeSystem_if
Interface for first-order ODE systems dy/dt = f(t, y) . . . 1161
OLD_ODEelement . . . 1162
OnEventManager . . . 1166
OpenAIProviderClientImpl
AI provider client for the OpenAI Chat Completions API . . . 1168
OperatingSystem02 . . . 1172
OperatingSystem03 . . . 1173
Optimizer_if::OptimizationSettings
Generic optimization settings independent of a specific algorithm . . . 1175
Optimizer_if
Interface for simulation-based optimization engines (OptQuest-like) . . . 1175
OptimizerDefaultImpl1
Initial concrete scaffold for an OptQuest-like optimizer . . . 1181
OptimizerWindow . . . 1188
OR . . . 1189
testing::internal::OsStackTraceGetter . . . 1195
testing::internal::OsStackTraceGetterInterface . . . 1196
testing::internal::ParameterizedTestFactory< TestClass > . . . 1197
testing::internal::ParameterizedTestSuiteInfo< TestSuite > . . . 1198
testing::internal::ParameterizedTestSuiteInfoBase . . . 1199
testing::internal::ParameterizedTestSuiteRegistry . . . 1200
testing::internal::ParamGenerator< T > . . . 1200
testing::internal::ParamGeneratorInterface< T > . . . 1201
testing::internal::ParamIterator< T > . . . 1201
testing::internal::ParamIteratorInterface< T > . . . 1202
Parser_if
Interface for evaluating model expressions and stochastic functions . . . 1202
ParserAttributeEventInjector . . . 1204
ParserAttributeInternalEventOwner . . . 1205
ParserAttributeProcessObserver . . . 1210
ParserChangesInformation . . . 1210
ParserDefaultImpl1 . . . 1211
ParserDefaultImpl2 . . . 1213
ParserExpressionsTest . . . 1216
ParserManager . . . 1217
PasteUndoCommand . . . 1218
PathwayStressResponseComponent
Applies sustained pathway-collapse responses to a whole-cell state . . . 1219
Persistence_if
Interface for model serialization/deserialization services . . . 1225

PersistenceDefaultImpl2 . . . 1228
PersistencePartialLoadImpl1 . . . 1230
PersistenceRecord
Serialization/Deserialization structure which behaves like an associative array . . . 1231
PersistentObject_base
Dynamic serializable interface . . . 1233
PetriPlace . . . 1235
PickableStationItem . . . 1240
PickStation . . . 1241
PickUp
Removes a consecutive run of entities from a queue, appending them to the incoming entity's group . . . 1247
genesys::launcher::PlatformInfo . . . 1253
Plugin . . . 1253
PluginCatalogController
Controller responsible for plugin catalog presentation and plugin palette behavior . . . 1257
PluginConnector_if
Interface for dynamic discovery/loading/unloading of plugin libraries . . . 1258
PluginConnectorDummyImpl1
In-memory plugin connector that resolves built-in plugin metadata . . . 1260
PluginConnectorStaticImpl1
Static plugin connector stub used by the kernel interface . . . 1263
PluginInformation . . . 1265
PluginInsertionOptions
Optional policy hooks used while inserting a plugin . . . 1266
PluginLoadIssue
Diagnostic record for a plugin candidate that could not be loaded . . . 1266
PluginManager
Manages discovery, loading, lookup and instantiation of simulation plugins . . . 1267
PMOS . . . 1270
testing::internal::PointerPrinter . . . 1276
GroProgramRuntime::PopulationMutation . . . 1276
yy::position
A point in a source file . . . 1276
testing::PrintToStringParamName . . . 1277
ProbabilityDistribution
Static façade for inverse/CDF-related distribution utilities . . . 1277
ProbabilityDistributionBase
Mathematical base library for distribution-related functions . . . 1279
MetabolicFluxBalanceSolver::Problem . . . 1282
Process
Composite component that always chains an internal Seize, Delay and Release to model "seize a resource, hold it for a while, release it" . . . 1282
genesys::launcher::ProcessLauncher . . . 1289
OptimizerWindow::ProgressPlot . . . 1290
BioStateProjectionComponent::Projection . . . 1291
MetabolicStateProjectionComponent::Projection . . . 1291
EFSMStatePropertiesUndoCommand::Properties . . . 1292
EFSMTransitionPropertiesUndoCommand::Properties . . . 1292
PropertyEditor . . . 1292
PropertyEditorController
Controller that orchestrates property editor state and scene-selection coupling . . . 1293
PropertyEditorGenesys . . . 1294
PropertyGenesys . . . 1294
PropertyGenesysBase
Base metadata for kernel-side control/response abstractions . . . 1296
testing::internal::ProtobufPrinter . . . 1298

ProviderRequest
Input to a single LLM chat-completion call . . . 1298
ProviderResponse
Output from a single LLM chat-completion call . . . 1298
testing::internal::ProxyTypeList< Ts > . . . 1299
PythonForG . . . 1300
PythonRuntime . . . 1306
QCPAbstractItem
The abstract base class for all items in a plot . . . 1311
QCPAbstractLegendItem
The abstract base class for all entries in a QCPLegend . . . 1318
QCPAbstractPaintBuffer
The abstract base class for paint buffers, which define the rendering backend . . . 1325
QCPAbstractPlottable
The abstract base class for all data representing objects in a plot . . . 1327
QCPAbstractPlottable1D< DataType >
A template base class for plottables with one-dimensional data . . . 1340
QCPAxis
Manages a single axis inside a QCustomPlot . . . 1347
QCPAxisPainterPrivate . . . 1368
QCPAxisRect
Holds multiple axes and arranges them in a rectangular shape . . . 1370
QCPAxisTicker
The base class tick generator used by QCPAxis to create tick positions and tick labels . . . 1385
QCPAxisTickerDateTime
Specialized axis ticker for calendar dates and times as axis ticks . . . 1389
QCPAxisTickerFixed
Specialized axis ticker with a fixed tick step . . . 1394
QCPAxisTickerLog
Specialized axis ticker suited for logarithmic axes . . . 1397
QCPAxisTickerPi
Specialized axis ticker to display ticks in units of an arbitrary constant, for example pi . . . 1400
QCPAxisTickerText
Specialized axis ticker which allows arbitrary labels at specified coordinates . . . 1403
QCPAxisTickerTime
Specialized axis ticker for time spans in units of milliseconds to days . . . 1407
QCPBars
A plottable representing a bar chart in a plot . . . 1411
QCPBarsData
Holds the data of one single data point (one bar) for QCPBars . . . 1421
QCPBarsGroup
Groups multiple QCPBars together so they appear side by side . . . 1423
QCPColorGradient
Defines a color gradient for use with e.g. QCPColorMap . . . 1427
QCPColorMap
A plottable representing a two-dimensional color map in a plot . . . 1433
QCPColorMapData
Holds the two-dimensional data of a QCPColorMap plottable . . . 1442
QCPColorScale
A color scale for use with color coding data such as QCPColorMap . . . 1448
QCPColorScaleAxisRectPrivate . . . 1457
QCPCurve
A plottable representing a parametric curve in a plot . . . 1463
QCPCurveData
Holds the data of one single data point for QCPCurve . . . 1473
QCPDataContainer< DataType >
The generic data container for one-dimensional plottables . . . 1475

QCPDataRange
Describes a data range given by begin and end index . . . 1482
QCPDataSelection
Describes a data set by holding multiple QCPDataRange instances . . . 1486
QCPErrorBars
A plottable that adds a set of error bars to other plottables . . . 1491
QCPErrorBarsData
Holds the data of one single error bar for QCPErrorBars . . . 1502
QCPFinancial
A plottable representing a financial stock chart . . . 1503
QCPFinancialData
Holds the data of one single data point for QCPFinancial . . . 1514
QCPGraph
A plottable representing a graph in a plot . . . 1516
QCPGraphData
Holds the data of one single data point for QCPGraph . . . 1525
QCPGrid
Responsible for drawing the grid of a QCPAxis . . . 1527
QCPItemAnchor
An anchor of an item to which positions can be attached to . . . 1531
QCPItemBracket
A bracket for referencing/highlighting certain parts in the plot . . . 1533
QCPItemCurve
A curved line from one point to another . . . 1539
QCPItemEllipse
An ellipse . . . 1544
QCPItemLine
A line from one point to another . . . 1549
QCPItemPixmap
An arbitrary pixmap . . . 1554
QCPItemPosition
Manages the position of an item . . . 1560
QCPItemRect
A rectangle . . . 1566
QCPItemStraightLine
A straight line that spans infinitely in both directions . . . 1571
QCPItemText
A text label . . . 1576
QCPItemTracer
Item that sticks to QCPGraph data points . . . 1583
QCPLabelPainterPrivate . . . 1591
QCPLayer
A layer that may contain objects, to control the rendering order . . . 1593
QCPLayerable
Base class for all drawable objects . . . 1597
QCPLayout
The abstract base class for layouts . . . 1604
QCPLayoutElement
The abstract base class for all objects that form the layout system . . . 1611
QCPLayoutGrid
A layout that arranges child elements in a grid . . . 1618
QCPLayoutInset
A layout that places child elements aligned to the border or arbitrarily positioned . . . 1630
QCPLegend
Manages a legend inside a QCustomPlot . . . 1638
QCPLineEnding
Handles the different ending decorations for line-like items . . . 1650

QCPMarginGroup
A margin group allows synchronization of margin sides if working with multiple layout elements 1653
QCPPaintBufferPixmap
A paint buffer based on QPixmap, using software raster rendering . . . 1655
QCPPainter
QPainter subclass used internally . . . 1657
QCPPlottableInterface1D
Defines an abstract interface for one-dimensional plottables . . . 1661
QCPPlottableLegendItem
A legend item representing a plottable with an icon and the plottable name . . . 1665
QCPPolarAxisAngular
The main container for polar plots, representing the angular axis as a circle . . . 1670
QCPPolarAxisRadial
The radial axis inside a radial plot . . . 1691
QCPPolarGraph
A radial graph used to display data in polar plots . . . 1710
QCPPolarGrid
The grid in both angular and radial dimensions for polar plots . . . 1719
QCPPolarLegendItem
A legend item for polar plots . . . 1722
QCPRange
Represents the range an axis is encompassing . . . 1727
QCPScatterStyle
Represents the visual appearance of scatter points . . . 1733
QCPSelectionDecorator
Controls how a plottable's data selection is drawn . . . 1739
QCPSelectionDecoratorBracket
A selection decorator which draws brackets around each selected data segment . . . 1742
QCPSelectionRect
Provides rect/rubber-band data selection and range zoom interaction . . . 1746
QCPStatisticalBox
A plottable representing a single statistical box in a plot . . . 1751
QCPStatisticalBoxData
Holds the data of one single data point for QCPStatisticalBox . . . 1761
QCPTextElement
A layout element displaying a text . . . 1763
QCPVector2D
Represents two doubles as a mathematical 2D vector . . . 1772
QCustomPlot
The central class of the library. This is the QWidget which displays the plot and interacts with the user . . . 1777
QtAbstractEditorFactory< PropertyManager > . . . 1807
QtAbstractEditorFactoryBase . . . 1811
QtAbstractPropertyBrowser . . . 1813
QtAbstractPropertyBrowserPrivate . . . 1818
QtAbstractPropertyManager . . . 1819
QtAbstractPropertyManagerPrivate . . . 1824
QtBoolEdit . . . 1824
QtBoolPropertyManager . . . 1825
QtBoolPropertyManagerPrivate . . . 1828
QtBrowserItem . . . 1828
QtBrowserItemPrivate . . . 1829
QtButtonPropertyBrowser . . . 1830
QtButtonPropertyBrowserPrivate . . . 1833
QtCharEdit . . . 1834
QtCharEditorFactory . . . 1835
QtCharEditorFactoryPrivate . . . 1837
QtCharPropertyManager . . . 1838

QtCharPropertyManagerPrivate . . . 1840
QtCheckBoxFactory . . . 1841
QtCheckBoxFactoryPrivate . . . 1843
QtColorEditorFactory . . . 1844
QtColorEditorFactoryPrivate . . . 1846
QtColorEditWidget . . . 1848
QtColorPropertyManager . . . 1849
QtColorPropertyManagerPrivate . . . 1852
QtCursorDatabase . . . 1853
QtCursorEditorFactory . . . 1853
QtCursorEditorFactoryPrivate . . . 1855
QtCursorPropertyManager . . . 1856
QtCursorPropertyManagerPrivate . . . 1859
QtDateEdit . . . 1859
QtDateEditFactory . . . 1860
QtDateEditFactoryPrivate . . . 1862
QtDatePropertyManager . . . 1864
QtDatePropertyManagerPrivate . . . 1868
QtDateTimeEdit . . . 1868
QtDateTimeEditFactory . . . 1870
QtDateTimeEditFactoryPrivate . . . 1872
QtDateTimePropertyManager . . . 1873
QtDateTimePropertyManagerPrivate . . . 1875
QtDoubleEdit . . . 1876
QtDoublePropertyManager . . . 1877
QtDoublePropertyManagerPrivate . . . 1882
QtDoubleSpinBoxFactory . . . 1883
QtDoubleSpinBoxFactoryPrivate . . . 1885
QtEnumEditorFactory . . . 1886
QtEnumEditorFactoryPrivate . . . 1888
QtEnumPropertyManager . . . 1890
QtEnumPropertyManagerPrivate . . . 1894
QtEnumPropertyType . . . 1894
QtFlagPropertyManager . . . 1894
QtFlagPropertyManagerPrivate . . . 1898
QtFlagPropertyType . . . 1898
QtFontEditorFactory . . . 1899
QtFontEditorFactoryPrivate . . . 1901
QtFontEditWidget . . . 1902
QtFontPropertyManager . . . 1903
QtFontPropertyManagerPrivate . . . 1906
QtGroupBoxPropertyBrowser . . . 1907
QtGroupBoxPropertyBrowserPrivate . . . 1909
QtGroupPropertyManager . . . 1910
QtGroupPropertyType . . . 1912
QtIntEdit . . . 1912
QtIntPropertyManager . . . 1913
QtIntPropertyManagerPrivate . . . 1917
QtKeySequenceEditorFactory . . . 1918
QtKeySequenceEditorFactoryPrivate . . . 1920
QtKeySequencePropertyManager . . . 1922
QtKeySequencePropertyManagerPrivate . . . 1924
QtLineEditFactory . . . 1925
QtLineEditFactoryPrivate . . . 1927
QtLocalePropertyManager . . . 1928
QtLocalePropertyManagerPrivate . . . 1931
QtMetaEnumProvider . . . 1931
QtMetaEnumWrapper . . . 1932

genesys::launcher::QtNetworkTransport . . . 1933
QtPointFPropertyManager . . . 1934
QtPointFPropertyManagerPrivate . . . 1938
QtPointPropertyManager . . . 1939
QtPointPropertyManagerPrivate . . . 1942
QtProperty . . . 1943
QtPropertyBrowserUtils . . . 1948
QtPropertyEditorDelegate . . . 1949
QtPropertyEditorView . . . 1950
QtPropertyPrivate . . . 1951
QtRectFPropertyManager . . . 1952
QtRectFPropertyManagerPrivate . . . 1956
QtRectPropertyManager . . . 1957
QtRectPropertyManagerPrivate . . . 1961
QtScrollBarFactory . . . 1962
QtScrollBarFactoryPrivate . . . 1964
QtSizeFPropertyManager . . . 1966
QtSizeFPropertyManagerPrivate . . . 1971
QtSizePolicyPropertyManager . . . 1972
QtSizePolicyPropertyManagerPrivate . . . 1975
QtSizePropertyManager . . . 1976
QtSizePropertyManagerPrivate . . . 1980
QtSliderFactory . . . 1981
QtSliderFactoryPrivate . . . 1983
QtSpinBoxFactory . . . 1985
QtSpinBoxFactoryPrivate . . . 1987
QtStringEdit . . . 1988
QtStringPropertyManager . . . 1989
QtStringPropertyManagerPrivate . . . 1992
QtTimeEdit . . . 1992
QtTimeEditFactory . . . 1994
QtTimeEditFactoryPrivate . . . 1996
QtTimePropertyManager . . . 1997
QtTimePropertyManagerPrivate . . . 1999
QtTreePropertyBrowser . . . 2000
QtTreePropertyBrowserPrivate . . . 2006
QtVariantEditorFactory . . . 2007
QtVariantEditorFactoryPrivate . . . 2009
QtVariantProperty . . . 2010
QtVariantPropertyManager . . . 2013
QtVariantPropertyManagerPrivate . . . 2019
QtVariantPropertyPrivate . . . 2021
Quadrature_if
One-dimensional numerical quadrature abstraction . . . 2021
Queue
Data definition holding waiting entities for a component (a Seize-family component, Process, Batch, etc.) and defining their ranking rule . . . 2022
QueueableItem . . . 2027
testing::internal::Random . . . 2028
testing::internal::RangeGenerator< T, IncrementT > . . . 2028
testing::internal::RawBytesPrinter . . . 2029
testing::internal::RE . . . 2030
MassActionOdeSystem::Reaction . . . 2030
RealIntervalStateSet . . . 2030
RealState . . . 2032
Record
Evaluates a parser expression each time an entity passes through and records the value into a StatisticsCollector and/or an external file . . . 2034

RecordingContinuousSystemComponent . . . 2041
RecordingLSODE . . . 2047
Rectifier . . . 2053
testing::internal::RelationToSourceCopy . . . 2054
testing::internal::RelationToSourceReference . . . 2054
Release
Gives up control of one or more previously seized resources, immediately unblocking any entity waiting for them . . . 2054
Remove
Removes one or more entities at a given rank (or rank range) from a Queue or an EntityGroup and forwards them . . . 2061
Resistor . . . 2067
Resource
Data definition representing a seizable capacity pool (equipment, staff, etc.), its costing, its optional capacity schedule, and its associated Failure definitions . . . 2073
ResourceAllocationComponent . . . 2079
ResourceTestProbe . . . 2085
GroProgramParser::Result . . . 2086
Sampler_if::RNG_Parameters
Encapsulates generator-specific configuration/state parameters . . . 2086
RootFinder_if
Root-finding abstraction for scalar equations . . . 2087
Route . . . 2087
RSimulator
Component that sends configured R commands to an internal RSimulatorRunner . . . 2094
RSimulatorRunner
Model data definition that executes R code through Rscript . . . 2100
RungeKutta4OdeSolver
Fixed-step fourth-order Runge-Kutta solver for first-order ODE systems . . . 2106
genesys::launcher::RuntimeInstallResult . . . 2107
genesys::launcher::RuntimePaths . . . 2107
genesys::launcher::RuntimeSelection . . . 2108
genesys::launcher::RuntimeSelector . . . 2108
Sample . . . 2109
Sampler_if
Abstraction for random-number generation and probability-distribution sampling . . . 2109
SamplerDefaultImpl1
Default pseudo-random sampler implementation used by the kernel . . . 2113
ScenarioExperiment_if . . . 2118
SceneToolController
Controller for scene visualization, drawing tools, alignment, and zoom orchestration . . . 2118
SchedulableItem
One (expression, duration, rule) triple inside a Schedule . . . 2121
Schedule . . . 2122
testing::ScopedTrace . . . 2127
Search
Scans a Queue or an EntityGroup rank range for the first entity satisfying a condition, saving the found rank (or 0) to an attribute . . . 2127
Segment
Data definition holding the ordered sequence of stations (and the length between consecutive stations) that make up one Conveyor's physical path . . . 2134
SegmentStep
One (station, length-to-next-station) step of a Segment's ordered conveyor path . . . 2138
SeizableItem . . . 2139
Seize
Allocates one or more resources (or resource-set members) to an entity, queueing it if they are not simultaneously available . . . 2140

Separate
Releases the original member entities of a temporary Batch group back into the model 2147
Sequence
Data definition holding an ordered list of visitation steps (SequenceStep), each naming a Station or Label an entity should reach next, with optional attribute/variable assignments per step 2153
SequenceStep 2158
Serializer
Serialization/Deserialization interface for simulation models 2159
SessionContext
Owns the state that belongs to one authenticated web session 2163
SessionManager
Creates, stores and resolves authenticated web sessions 2163
Set
Data definition grouping an ordered list of ModelDataDefinition members of one concrete type, used by selection-rule-based components (e.g. Resource sets consumed by Seize-family components) 2165
Signal
Broadcasts a signal value, releasing up to _limitExpression entities waiting at any Wait (Arena Hold) component in WaitForSignal mode with a matching signal 2171
SignalData 2177
GroProgramRuntime::SignalMutation 2181
SimpleHttpServer
Minimal blocking HTTP/1.1 server for single-request-per-connection handling 2182
SimulatorSessionService::SimulationActionResult
Contains the result of a simulation action such as run or step 2183
SimulationCommandController
Controller that orchestrates UI simulation commands over SimulationController guards 2183
SimulatorSessionService::SimulationConfigInput
Defines user-provided simulation parameters to apply to the current model 2184
SimulatorSessionService::SimulationConfigResult
Contains the result of applying simulation configuration 2185
SimulationControl
Read/write kernel-side simulation control abstraction 2185
SimulationControlBool 2189
SimulationControlDouble 2192
SimulationControlGeneric< T > 2195
SimulationControlGenericClass< T, M, C > 2198
SimulationControlGenericClassNotDC< T, M, C > 2202
SimulationControlGenericEnum< T, E > 2206
SimulationControlGenericList< T, M, C > 2209
SimulationControlGenericListPointer< T, M, C > 2213
SimulationControlInt 2217
SimulationController
Controller that validates simulation command readiness for MainWindow wrappers 2220
SimulationControlSourceCodeString 2222
SimulationControlString 2225
SimulationControlTimeUnit 2228
SimulationControlUInt 2231
SimulationControlUShort 2234
SimulationEvent 2237
SimulationEventController
Controller for simulation event callbacks, UI updates, and handler wiring 2238
SimulationExperiment
Represents a higher-level simulation experiment definition 2239
SimulationReporter_if
Interface for reporting simulation results and experiment-facing outputs 2239
SimulationReporterDefaultImpl1 2240

SimulationResponse
Read-only kernel-side simulation response abstraction . . . 2242
SimulationResponseDouble . . . 2244
SimulationResultsDataset . . . 2246
SimulationResultsDatasetParser . . . 2247
SimulationResultsObservation . . . 2247
SimulationScenario
Represents a candidate simulation scenario within the unfinished experiment layer . . . 2247
SimulatorSessionService::SimulationStatusResult
Captures the current simulation execution state . . . 2248
Simulator . . . 2249
SimulatorFacade
High-level non-owning facade over Simulator and its main composed services . . . 2251
SimulatorSessionService::SimulatorInfoResult
Describes simulator identity metadata for a token-scoped session . . . 2257
SimulatorSessionService
Encapsulates session-scoped simulator operations exposed by the Web API . . . 2257
SinkModelComponent . . . 2264
yy::genesyspp_parser::stack< T, S >::slice
Present a slice of the top of a stack . . . 2270
Smart_AddingResource . . . 2271
Smart_AlternatingEntityCreation . . . 2272
Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime . . . 2274
Smart_ArrivalsEntityTypeVsAttribute . . . 2276
Smart_AssignExample . . . 2278
Smart_AssignWriteSeizes . . . 2280
Smart_Attribute_Variable . . . 2281
Smart_AutomaticStatisticsCollection . . . 2283
Smart_BacteriaColony_GRO . . . 2285
Smart_BasicModeling . . . 2287
Smart_BatchAndSeparate . . . 2288
Smart_BatchSeparate . . . 2290
Smart_BioKineticLawRegulation . . . 2292
Smart_BioNetworkAnalysis . . . 2293
Smart_BioReversibleKineticLaw . . . 2295
Smart_BioReversibleMassAction . . . 2297
Smart_Buffer . . . 2298
Smart_CellularAutomata . . . 2300
Smart_CellularAutomata1 . . . 2302
Smart_Clone . . . 2303
Smart_ContinuousFlowEntities . . . 2305
Smart_CppForG . . . 2307
Smart_Create . . . 2308
Smart_DecideNWayByChance . . . 2310
Smart_DefaultModalModel . . . 2312
Smart_DefiningAttributesAsStrings . . . 2313
Smart_DefiningControlLogic . . . 2315
Smart_DefiningResourceCapacity . . . 2317
Smart_Delay . . . 2318
Smart_DelayBasedOnReplication . . . 2320
Smart_Dummy . . . 2322
Smart_EntitiesProcessedByPriority . . . 2324
Smart_EvaluatingConditionsBeforeEnteringQueue . . . 2326
Smart_Expression . . . 2328
Smart_Failures . . . 2330
Smart_FSMState . . . 2331
Smart_GroColonyGrowth . . . 2333
Smart_GroColonyLifecycle . . . 2335

Smart_HoldSearchRemove . . . 2336
Smart_InventoryAndHoldingCosts . . . 2338
Smart_LSODE . . . 2340
Smart_MarkovChain . . . 2342
Smart_MaxArrivalsField . . . 2343
Smart_ModalModel . . . 2345
Smart_ModalModelDefaultNetwork . . . 2347
Smart_ModalModelFSM . . . 2348
Smart_ModalModelPetriNet . . . 2350
Smart_ModelInfoModelSimulation . . . 2352
Smart_ModelRunUntil1000Parts . . . 2353
Smart_ModuleDisplayVariables . . . 2355
Smart_ODE . . . 2357
Smart_OnEvent . . . 2358
Smart_OverlappingResources . . . 2361
Smart_ParallelProcessingOfEntities . . . 2363
Smart_Parser . . . 2365
Smart_ParserModelFunctions . . . 2367
Smart_PetriPlace . . . 2368
Smart_PlacingEntitiesInQueueSets . . . 2370
Smart_Plugin . . . 2372
Smart_PriorityExample . . . 2373
Smart_Process . . . 2375
Smart_ProcessArena . . . 2377
Smart_ProcessSet . . . 2378
Smart_PythonForG . . . 2380
Smart_R_Simulator
Builds a minimal Create -> RSimulator -> Dispose model . . . 2381
Smart_Record . . . 2384
Smart_Record_Arena . . . 2385
Smart_RemovingAndReorderingEntitiesInAQueue . . . 2387
Smart_ResourceCosting . . . 2389
Smart_ResourceScheduleCosting . . . 2391
Smart_ResourceSets . . . 2392
Smart_RouteStation . . . 2394
Smart_SeizeDelayRelease . . . 2396
Smart_SeizeDelayReleaseMany . . . 2397
Smart_SeizeDelayReleaseNoDataDefs . . . 2399
Smart_SeizingMultipleSimultaneosly . . . 2401
Smart_SelectingRouteBasedOnProbability . . . 2403
Smart_SelectingShorterQueue . . . 2404
Smart_Sequence . . . 2406
Smart_SimulationControlResponse . . . 2408
Smart_SynchronizingParallelEntities . . . 2409
Smart_ValueAdded . . . 2411
Smart_WaitForSignal . . . 2413
Smart_WaitScanCondition . . . 2414
Smart_WaitSignal . . . 2416
WebWorkerRuntime::Snapshot
Captures a thread-safe snapshot of the worker state . . . 2417
MetabolicFluxBalanceSolver::Solution . . . 2418
Optimizer_if::SolutionSummary
Captures one candidate or best-known optimization solution . . . 2418
Solver_if
Legacy compatibility interface for numerical routines . . . 2419
SolverDefaultImpl1
Legacy default implementation of Solver_if . . . 2420

SortFile
Utility class that sorts numeric values persisted in a data file . . . 2423
SourceCodeString . . . 2424
SourceModelComponent
Kernel base class for components that inject entities into the model (only Create derives from it so far), generalizing the fields Arena exposes on its "Create module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 31-32): Entity Type (_entityType), First Creation (_firstCreation), Entities per Arrival (_entitiesPerCreation), Max Arrivals (_maxCreationsExpression, a parsed expression rather than a static integer) and the time-between-creations expression/time unit . . . 2424
SparseValueStore
Sparse N-dimensional numeric store using GenESyS textual index keys . . . 2434
MassActionOdeSystem::Species . . . 2435
SPICECircuit . . . 2435
SPICENode . . . 2441
SPICERunner . . . 2447
Start . . . 2452
StartAIAssistant . . . 2458
State . . . 2460
GroProgramAst::Statement . . . 2461
EFSMNetworkDiagramLayout::StatePosition . . . 2461
StateSet . . . 2461
StateSet_Enumerable . . . 2463
Station
Data definition representing one physical/logical processing location, tracking the entities currently in it and their sojourn time . . . 2465
StationTestProbe . . . 2470
Statistics_if
Interface for statistical synthesis over values provided by a Collector_if . . . 2474
StatisticsCollector . . . 2478
StatisticsDatafile_if
Extension of Statistics_if for metrics that require full data access . . . 2483
StatisticsDatafileDefaultImpl1
Default implementation of StatisticsDatafile_if for file-backed samples . . . 2487
StatisticsDefaultImpl1
Default online statistics implementation over a Collector_if source . . . 2492
StochasticReactionComponent . . . 2495
StochasticReactionRule . . . 2501
StochasticTranscription . . . 2506
StochasticTranslation . . . 2513
BioReaction::StoichiometricTerm . . . 2520
MassActionOdeSystem::StoichiometricTerm . . . 2520
MetabolicReaction::StoichiometricTerm . . . 2520
StochasticReactionRule::StoichiometricTerm . . . 2521
Stop . . . 2521
Storage
Data definition representing a named storage location with explicit area/capacity/density accounting . . . 2527
Store
Placeholder for adding an entity to a Storage (§5.11) . . . 2532
testing::internal::internal_stream_operator_without_lexical_name_lookup::StreamPrinter . . . 2539
testing::internal::String . . . 2539
Submodel . . . 2539
testing::internal::SuiteApiResolver< T > . . . 2546
yy::genesyspp_parser::symbol_kind
Symbol kinds . . . 2546
yy::genesyspp_parser::symbol_type
"External" symbols: returned by the scanner . . . 2547

yy::genesyspp_parser::syntax_error
Syntax errors thrown from user actions . . . 2550
SystemCommandExecutor_if
Minimal command execution interface used to make system dependency checks testable . . . 2551
SystemCommandResult
Captures the result of executing a system command used by dependency checks . . . 2552
SystemDependency . . . 2553
SystemDependencyCheckEntry
One dependency entry produced by a system dependency preflight . . . 2553
SystemDependencyCheckResult
Full preflight result for the system dependencies declared by one plugin . . . 2554
SystemDependencyInstallEntry
One command result from an attempted system dependency installation . . . 2554
SystemDependencyInstallResult
Aggregates commands executed while trying to satisfy missing system dependencies . . . 2555
SystemDependencyResolver
Evaluates and optionally installs plugin system dependencies . . . 2555
SystemPreferences
Persistent application preferences used by the Qt GUI . . . 2556
SystemShellCommandExecutor
Shell-based executor used by production plugin dependency preflight . . . 2558
genesys::launcher::TarZstdArchiveExtractor . . . 2560
testing::internal::Templates< Head_, Tail_ > . . . 2561
testing::internal::Templates< Head_ > . . . 2561
testing::internal::TemplateSel< Tmpl > . . . 2561
testing::Test . . . 2561
testing::TestEventListener . . . 2562
testing::TestEventListeners . . . 2563
testing::internal::TestFactoryBase . . . 2563
testing::internal::TestFactoryImpl< TestClass > . . . 2564
testing::TestInfo . . . 2565
testing::internal::TestMetaFactory< TestSuite > . . . 2565
testing::internal::TestMetaFactoryBase< ParamType > . . . 2566
testing::TestParamInfo< ParamType > . . . 2566
testing::TestProperty . . . 2567
testing::internal::TestPropertyKeyIs . . . 2567
HypothesisTester_if::TestResult
Result of a parametric hypothesis test . . . 2567
testing::TestResult . . . 2568
testing::internal::TestResultAccessor . . . 2568
TestSimulationControl . . . 2569
TestSimulationResponse . . . 2572
testing::TestSuite . . . 2573
testing::TestWithParam< T > . . . 2574
testing::internal::ThreadLocal< T > . . . 2575
QCPAxisPainterPrivate::TickLabelData . . . 2575
Token . . . 2576
yy::genesyspp_parser::token
Token kinds . . . 2576
TokenService . . . 2577
TraceConsoleController
Controller that routes simulator trace events to GUI text widgets . . . 2577
TraceErrorEvent . . . 2578
TraceEvent . . . 2579
testing::internal::TraceInfo . . . 2579
TraceManager . . . 2579
TraceSimulationEvent . . . 2584
TraceSimulationProcess . . . 2584

TraitsGUI< T >
GUI-specific trait constants used to tune default rendering behavior . . . 2585
TraitsGUI< GAssociation > . . . 2586
TraitsGUI< GComponentPort > . . . 2586
TraitsGUI< GConnection > . . . 2586
TraitsGUI< GMainWindow > . . . 2587
TraitsGUI< GModelComponent > . . . 2587
TraitsGUI< GModelDataDefinition > . . . 2587
TraitsGUI< GModelGraphicPic > . . . 2588
TraitsGUI< GScene > . . . 2588
TraitsGUI< GView > . . . 2588
TraitsKernel< T > . . . 2588
TraitsKernel< Collector_if > . . . 2589
TraitsKernel< Model > . . . 2589
TraitsKernel< ModelChecker_if > . . . 2589
TraitsKernel< ModelComponent > . . . 2589
TraitsKernel< ModelDataDefinition > . . . 2589
TraitsKernel< Parser_if > . . . 2590
TraitsKernel< Persistence_if > . . . 2590
TraitsKernel< PluginConnector_if > . . . 2590
TraitsKernel< Sampler_if > . . . 2590
TraitsKernel< SimulationReporter_if > . . . 2590
TraitsKernel< Simulator > . . . 2591
TraitsKernel< Statistics_if > . . . 2591
TraitsKernel< StatisticsDatafile_if > . . . 2591
TraitsTools< T >
Registry-like traits to bind abstractions to concrete tools classes . . . 2591
TraitsTools< AIAssistant_if > . . . 2592
TraitsTools< Fitter_if > . . . 2592
TraitsTools< HypothesisTester_if > . . . 2592
TraitsTools< Solver_if > . . . 2592
EFSMNetworkDiagramLayout::TransitionLayout . . . 2593
Transporter . . . 2593
Transporter::TravelCompletion . . . 2599
testing::internal::TrueWithString . . . 2599
std::tuple_size< testing::internal::FlatTuple< Ts... > > . . . 2600
testing::internal::TypeIdHelper< T > . . . 2600
testing::internal::TypeParameterizedTest< Fixture, TestSel, Types > . . . 2601
testing::internal::TypeParameterizedTest< Fixture, TestSel, internal::None > . . . 2601
testing::internal::TypeParameterizedTestSuite< Fixture, Tests, Types > . . . 2601
testing::internal::TypeParameterizedTestSuite< Fixture, internal::None, Types > . . . 2601
testing::internal::TypeParameterizedTestSuiteRegistry . . . 2601
testing::internal::Types< Head_, Tail_ > . . . 2602
testing::internal::Types< Head_ > . . . 2602
testing::internal::TypeWithSize< size > . . . 2602
testing::internal::TypeWithSize< 4 > . . . 2602
testing::internal::TypeWithSize< 8 > . . . 2602
UngroupUndoCommand . . . 2603
testing::UnitTest . . . 2603
testing::internal::UnitTestImpl . . . 2604
testing::internal::UnitTestOptions . . . 2606
testing::internal::UniversalPrinter< T > . . . 2606
testing::internal::UniversalPrinter< const T > . . . 2607
testing::internal::UniversalPrinter< T & > . . . 2608
testing::internal::UniversalPrinter< T[N]> . . . 2608
testing::internal::UniversalTersePrinter< T > . . . 2608
testing::internal::UniversalTersePrinter< char * > . . . 2608
testing::internal::UniversalTersePrinter< char16_t * > . . . 2609

testing::internal::UniversalTersePrinter< char32_t * > . . . 2610
testing::internal::UniversalTersePrinter< const char * > . . . 2611
testing::internal::UniversalTersePrinter< const char16_t * > . . . 2612
testing::internal::UniversalTersePrinter< const char32_t * > . . . 2612
testing::internal::UniversalTersePrinter< T & > . . . 2613
testing::internal::UniversalTersePrinter< T[N]> . . . 2613
testing::internal::UniversalTersePrinter< wchar_t * > . . . 2613
Unstore
Placeholder for removing an entity from a Storage (§5.11) . . . 2613
genesys::launcher::UpdateChecker . . . 2620
genesys::launcher::UpdateCheckResult . . . 2620
genesys::launcher::UpdateDialog . . . 2621
genesys::launcher::UpdateDownloader . . . 2622
genesys::launcher::UpdateManifest . . . 2623
genesys::launcher::UpdateManifestParser . . . 2623
genesys::launcher::UpdatePlatform . . . 2624
genesys::launcher::UpdateVerifier . . . 2624
genesys::launcher::UserRuntimeManager . . . 2624
Util . . . 2624
yy::genesyspp_parser::value_type . . . 2635
testing::internal::ValueArray< Ts > . . . 2636
testing::internal::ValuesInIteratorRangeGenerator< T > . . . 2637
Variable
Data definition for a named value (scalar or indexed array) that is global to the whole model, as opposed to Attribute's per-entity values . . . 2638
genesys::launcher::VerificationResult . . . 2644
genesys::launcher::Version . . . 2644
Vpulse . . . 2645
Vsine . . . 2651
Vsource . . . 2657
Wait
Holds an entity in a queue until a signal arrives, a condition becomes true, or it is explicitly removed elsewhere . . . 2663
Waiting . . . 2670
WaitingResource . . . 2670
WcmBioNetworkBridge
Demonstrates a minimal deterministic/stochastic whole-cell bridge . . . 2671
WcmCompartmentTransportWholeCellMvp
Whole-cell MVP with compartment-aware metabolic transport . . . 2673
WcmFluxBalanceWholeCellMvp
Whole-cell MVP driven by flux-balance metabolism . . . 2675
WcmGeneExpression . . . 2677
WcmLifecycleStress
Demonstrates starvation and death routing in a whole-cell stress scenario . . . 2679
WcmMgenitaliumKarr2012 . . . 2681
WcmPathwayStressWholeCellMvp
Whole-cell stress example driven by sustained low pathway activity . . . 2684
WcmSimpleSsa . . . 2685
WcmWholeCellMvp
Minimal whole-cell MVP combining metabolism, expression, growth, and division . . . 2687
WcmYeastDidacticWholeCellMvp
Didactic yeast-oriented whole-cell MVP built on GEM-inspired metabolism . . . 2689
WcmYeastEukaryoticWholeCellMvp
Didactic eukaryotic whole-cell MVP inspired by budding yeast . . . 2691
WebWorkerDialog
Lightweight control window for the embedded GenESyS web worker . . . 2693
WebWorkerRuntime
Owns the embedded worker runtime used by the GUI and the standalone worker app . . . 2694

WholeCellParameterReader 2696
WholeCellState 2696
QtButtonPropertyBrowserPrivate::WidgetItem 2702
QtGroupBoxPropertyBrowserPrivate::WidgetItem 2703
testing::WithParamInterface< T > 2703
SimulatorSessionService::WorkerCapabilitiesResult
Describes currently supported worker features 2704
SimulatorSessionService::WorkerInfoResult
Describes worker identity metadata for distributed discovery 2704
WorkerJob
Stores metadata for a worker job snapshot owned by the web worker 2705
SimulatorSessionService::WorkerJobCreationResult
Contains worker job creation output and error state 2706
SimulatorSessionService::WorkerJobInfoResult
Describes worker job metadata exposed by worker stage 3 endpoints 2706
WorkerJobManager
Owns worker job records in memory and assigns unique identifiers 2707
SimulatorSessionService::WorkerJobQueryResult
Contains worker job lookup output and error state 2709
SimulatorSessionService::WorkerJobResultInfo
Describes terminal worker job result data exposed by worker stage 5 endpoint 2710
SimulatorSessionService::WorkerJobResultQueryResult
Contains worker job terminal result lookup output and error state 2710
SimulatorSessionService::WorkerJobRunResult
Contains worker job run output and error state 2711
WorkerJobTerminalResult
Stores the terminal simulation summary persisted for a finished or failed worker job 2711
Write
Writes one or more literal/expression text lines to the screen (trace output) or to a file 2712
XmlSerializer 2718
XNOR 2721
XOR 2728
yy_buffer_state 2734
yy_trans_info 2734

Chapter 15

File Index

15.1 File List

Here is a list of all documented files with brief descriptions:

applications/BaseGenesysTerminalApplication.h . . . 2735
applications/GenesysApplication_if.h . . . 2736
applications/gui/ai_assistant/AIAssistantWindow.h . . . 2736
applications/gui/genesys/BaseMVController.h . . . 2748
applications/gui/genesys/GuiCrashDiagnostics.h . . . 2788
applications/gui/genesys/GuiScopeTrace.h . . . 2788
applications/gui/genesys/guithememanager.h . . . 2788
applications/gui/genesys/mainwindow.cpp
Composition-root partition of MainWindow implementation . . . 2789
applications/gui/genesys/mainwindow.h . . . 2790
applications/gui/genesys/mainwindow_controller.cpp
Controller-delegation partition of MainWindow implementation . . . 2797
applications/gui/genesys/mainwindow_modelrepresentations.cpp
Model representation/persistence wrapper partition of MainWindow implementation . . . 2798
applications/gui/genesys/mainwindow_scene.cpp
Scene callback and scene synchronization partition of MainWindow implementation . . . 2799
applications/gui/genesys/mainwindow_simulator.cpp
Simulator trace/event wrapper partition of MainWindow implementation . . . 2800
applications/gui/genesys/MovedGuiCompatibilityIncludes.h . . . 2800
applications/gui/genesys/qcustomplot.h . . . 2833
applications/gui/genesys/systempreferences.h . . . 2938
applications/gui/genesys/TraitsGUI.h . . . 2944
applications/gui/genesys/UtilGUI.h . . . 2945
applications/gui/genesys/actions/AddUndoCommand.h . . . 2737
applications/gui/genesys/actions/DeleteUndoCommand.h . . . 2738
applications/gui/genesys/actions/EFSMInitialStateUndoCommand.h . . . 2738
applications/gui/genesys/actions/EFSMStateMembershipUndoCommand.h . . . 2739
applications/gui/genesys/actions/EFSMStatePropertiesUndoCommand.h . . . 2739
applications/gui/genesys/actions/EFSMTransitionPropertiesUndoCommand.h . . . 2740
applications/gui/genesys/actions/EFSMTransitionUndoCommand.h . . . 2740
applications/gui/genesys/actions/GroupUndoCommand.h . . . 2741
applications/gui/genesys/actions/ModalModelBindingsUndoCommand.h . . . 2741
applications/gui/genesys/actions/MoveUndoCommand.h . . . 2742
applications/gui/genesys/actions/PasteUndoCommand.h . . . 2742
applications/gui/genesys/actions/UngroupUndoCommand.h . . . 2742
applications/gui/genesys/animations/AnimationCounter.h . . . 2743
applications/gui/genesys/animations/AnimationPlaceholder.h . . . 2743
applications/gui/genesys/animations/AnimationQueue.h . . . 2745
applications/gui/genesys/animations/AnimationTimer.h . . . 2746
applications/gui/genesys/animations/AnimationTransition.h . . . 2746

applications/gui/genesys/animations/AnimationVariable.h . . . 2747
applications/gui/genesys/cellular_automata/CellularAutomataDemoBuilder.h . . . 2749
applications/gui/genesys/cellular_automata/CellularAutomataLatticeWidget.h . . . 2751
applications/gui/genesys/cellular_automata/CellularAutomataStateColorMap.h . . . 2751
applications/gui/genesys/cellular_automata/CellularAutomataViewerController.h . . . 2752
applications/gui/genesys/cellular_automata/CellularAutomataViewerWindow.h . . . 2753
applications/gui/genesys/codeeditor/CodeEditor.h . . . 2753
applications/gui/genesys/codeeditor/LineNumberArea.h . . . 2754
applications/gui/genesys/controllers/DialogUtilityController.h . . . 2754
applications/gui/genesys/controllers/EditCommandController.h . . . 2755
applications/gui/genesys/controllers/GraphicalContextMenuController.h . . . 2756
applications/gui/genesys/controllers/ModelInspectorController.h . . . 2757
applications/gui/genesys/controllers/ModelLifecycleController.h . . . 2757
applications/gui/genesys/controllers/PluginCatalogController.h . . . 2758
applications/gui/genesys/controllers/PropertyEditorController.h . . . 2759
applications/gui/genesys/controllers/SceneToolController.h . . . 2760
applications/gui/genesys/controllers/SimulationCommandController.h . . . 2761
applications/gui/genesys/controllers/SimulationController.h . . . 2762
applications/gui/genesys/controllers/SimulationEventController.h . . . 2762
applications/gui/genesys/controllers/TraceConsoleController.h . . . 2763
applications/gui/genesys/dialogs/dialogBreakpoint.h . . . 2764
applications/gui/genesys/dialogs/DialogFind.h . . . 2764
applications/gui/genesys/dialogs/Dialogmodelinformation.h . . . 2765
applications/gui/genesys/dialogs/dialogpluginmanager.h . . . 2766
applications/gui/genesys/dialogs/DialogReportIssue.h . . . 2767
applications/gui/genesys/dialogs/DialogSelectCounter.h . . . 2767
applications/gui/genesys/dialogs/DialogSelectVariable.h . . . 2768
applications/gui/genesys/dialogs/dialogsimulationconfigure.h . . . 2768
applications/gui/genesys/dialogs/dialogsystempreferences.h . . . 2769
applications/gui/genesys/dialogs/DialogTimerConfigure.h . . . 2769
applications/gui/genesys/extensions/BacteriaColonyViewerCoordinates.h . . . 2770
applications/gui/genesys/extensions/GuiExtensionConfigurationDialog.h . . . 2770
applications/gui/genesys/extensions/GuiExtensionContracts.h . . . 2771
applications/gui/genesys/extensions/GuiExtensionManager.h . . . 2772
applications/gui/genesys/extensions/GuiExtensionPluginCatalog.h . . . 2773
applications/gui/genesys/graphicals/EFSMTransitionGraphicsItem.h . . . 2773
applications/gui/genesys/graphicals/GraphicalAnimateExpression.h . . . 2773
applications/gui/genesys/graphicals/GraphicalAssociation.h . . . 2774
applications/gui/genesys/graphicals/GraphicalComponentPort.h . . . 2774
applications/gui/genesys/graphicals/GraphicalConnection.h . . . 2774
applications/gui/genesys/graphicals/GraphicalConnectionStyle.h . . . 2775
applications/gui/genesys/graphicals/GraphicalDiagramConnection.h . . . 2776
applications/gui/genesys/graphicals/GraphicalImageAnimation.h . . . 2776
applications/gui/genesys/graphicals/GraphicalModelComponent.h . . . 2777
applications/gui/genesys/graphicals/GraphicalModelDataDefinition.h . . . 2779
applications/gui/genesys/graphicals/GraphicalModelItemRenderStrategy.h . . . 2780
applications/gui/genesys/graphicals/ModelGraphicsScene.h . . . 2781
applications/gui/genesys/graphicals/ModelGraphicsView.h . . . 2786
applications/gui/genesys/propertyeditor/ComboBoxEnum.h . . . 2801
applications/gui/genesys/propertyeditor/DataComponentEditor.h . . . 2801
applications/gui/genesys/propertyeditor/DataComponentProperty.h . . . 2802
applications/gui/genesys/propertyeditor/ObjectPropertyBrowser.h . . . 2803
applications/gui/genesys/propertyeditor/PropertyEditor.h . . . 2806
applications/gui/genesys/propertyeditor/qtpropertybrowser/qtbuttonpropertybrowser.h . . . 2806
applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h . . . 2807
applications/gui/genesys/propertyeditor/qtpropertybrowser/qtgroupboxpropertybrowser.h . . . 2812
applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.h . . . 2813
applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils_p.h . . . 2816

applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h . . . 2820
applications/gui/genesys/propertyeditor/qtpropertybrowser/qttreepropertybrowser.h . . . 2830
applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.h . . . 2831
applications/gui/genesys/services/CppModelExporter.h . . . 2932
applications/gui/genesys/services/EFSMNetworkDiagramCodec.h . . . 2932
applications/gui/genesys/services/EFSMNetworkDiagramLayout.h . . . 2933
applications/gui/genesys/services/EFSMNetworkDiagramProjection.h . . . 2933
applications/gui/genesys/services/GraphicalAutomaticPositioningStrategy.h . . . 2933
applications/gui/genesys/services/GraphicalDataDefinitionLayout.h . . . 2934
applications/gui/genesys/services/GraphicalModelBuilder.h . . . 2934
applications/gui/genesys/services/GraphicalModelSerializer.h . . . 2935
applications/gui/genesys/services/GraphvizModelExporter.h . . . 2936
applications/gui/genesys/services/IssueReportRelayService.h . . . 2937
applications/gui/genesys/services/ModelLanguageSynchronizer.h . . . 2937
applications/gui/genesys/services/ToolLauncher.h . . . 2938
applications/gui/genesys/tools/dataanalyzer/DataAnalyzerWindow.h . . . 2939
applications/gui/genesys/tools/expressionbuilder/ExpressionBuilder.h . . . 2941
applications/gui/genesys/tools/optimizer/OptimizerWindow.h . . . 2942
applications/gui/httpworker/WebWorkerDialog.h . . . 2946
applications/launcher/ArchiveExtractor.h . . . 2946
applications/launcher/DispatchController.h . . . 2947
applications/launcher/LauncherConfig.h . . . 2947
applications/launcher/LauncherController.h . . . 2947
applications/launcher/LauncherLogger.h . . . 2948
applications/launcher/NetworkTransport.h . . . 2948
applications/launcher/PlatformInfo.h . . . 2949
applications/launcher/ProcessLauncher.h . . . 2949
applications/launcher/RuntimePaths.h . . . 2950
applications/launcher/RuntimeSelector.h . . . 2950
applications/launcher/UpdateChecker.h . . . 2951
applications/launcher/UpdateDialog.h . . . 2952
applications/launcher/UpdateDownloader.h . . . 2953
applications/launcher/UpdateManifest.h . . . 2953
applications/launcher/UpdateVerifier.h . . . 2954
applications/launcher/UserRuntimeManager.h . . . 2955
applications/launcher/Version.h . . . 2955
applications/modelSpecific/arenaExamples/Airport_Extended1.h . . . 2956
applications/modelSpecific/arenaExamples/AirportSecurityExample.h . . . 2956
applications/modelSpecific/arenaExamples/AirportSecurityExampleExtended.h . . . 2956
applications/modelSpecific/arenaExamples/Assembly_Line.h . . . 2956
applications/modelSpecific/arenaExamples/Banking_Transactions.h . . . 2957
applications/modelSpecific/arenaExamples/Example_BasicOrderShipping.h . . . 2957
applications/modelSpecific/arenaExamples/Example_PortModel.h . . . 2957
applications/modelSpecific/arenaExamples/Example_PublicTransport.h . . . 2958
applications/modelSpecific/arenaSmarts/Smart_AddingResource.h . . . 2958
applications/modelSpecific/arenaSmarts/Smart_AlternatingEntityCreation.h . . . 2958
applications/modelSpecific/arenaSmarts/Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime.h . 2959
applications/modelSpecific/arenaSmarts/Smart_ArrivalsEntityTypeVsAttribute.h . . . 2959
applications/modelSpecific/arenaSmarts/Smart_AssignExample.h . . . 2960
applications/modelSpecific/arenaSmarts/Smart_AutomaticStatisticsCollection.h . . . 2960
applications/modelSpecific/arenaSmarts/Smart_BasicModeling.h . . . 2960
applications/modelSpecific/arenaSmarts/Smart_BatchAndSeparate.h . . . 2961
applications/modelSpecific/arenaSmarts/Smart_ContinuousFlowEntities.h . . . 2961
applications/modelSpecific/arenaSmarts/Smart_Create.h . . . 2961
applications/modelSpecific/arenaSmarts/Smart_DecideNWayByChance.h . . . 2962
applications/modelSpecific/arenaSmarts/Smart_DefiningAttributesAsStrings.h . . . 2962
applications/modelSpecific/arenaSmarts/Smart_DefiningControlLogic.h . . . 2962
applications/modelSpecific/arenaSmarts/Smart_DefiningResourceCapacity.h . . . 2963

applications/modelSpecific/arenaSmarts/Smart_DelayBasedOnReplication.h . . . 2963
applications/modelSpecific/arenaSmarts/Smart_EntitiesProcessedByPriority.h . . . 2964
applications/modelSpecific/arenaSmarts/Smart_EvaluatingConditionsBeforeEnteringQueue.h . . . 2964
applications/modelSpecific/arenaSmarts/Smart_Expression.h . . . 2964
applications/modelSpecific/arenaSmarts/Smart_InventoryAndHoldingCosts.h . . . 2965
applications/modelSpecific/arenaSmarts/Smart_MaxArrivalsField.h . . . 2965
applications/modelSpecific/arenaSmarts/Smart_ModelRunUntil1000Parts.h . . . 2965
applications/modelSpecific/arenaSmarts/Smart_ModuleDisplayVariables.h . . . 2966
applications/modelSpecific/arenaSmarts/Smart_OverlappingResources.h . . . 2966
applications/modelSpecific/arenaSmarts/Smart_ParallelProcessingOfEntities.h . . . 2966
applications/modelSpecific/arenaSmarts/Smart_PlacingEntitiesInQueueSets.h . . . 2967
applications/modelSpecific/arenaSmarts/Smart_PriorityExample.h . . . 2967
applications/modelSpecific/arenaSmarts/Smart_ProcessArena.h . . . 2968
applications/modelSpecific/arenaSmarts/Smart_Record_Arena.h . . . 2968
applications/modelSpecific/arenaSmarts/Smart_RemovingAndReorderingEntitiesInAQueue.h . . . 2968
applications/modelSpecific/arenaSmarts/Smart_ResourceCosting.h . . . 2969
applications/modelSpecific/arenaSmarts/Smart_ResourceScheduleCosting.h . . . 2969
applications/modelSpecific/arenaSmarts/Smart_ResourceSets.h . . . 2969
applications/modelSpecific/arenaSmarts/Smart_SeizingMultipleSimultaneosly.h . . . 2970
applications/modelSpecific/arenaSmarts/Smart_SelectingRouteBasedOnProbability.h . . . 2970
applications/modelSpecific/arenaSmarts/Smart_SelectingShorterQueue.h . . . 2970
applications/modelSpecific/arenaSmarts/Smart_SynchronizingParallelEntities.h . . . 2971
applications/modelSpecific/arenaSmarts/Smart_ValueAdded.h . . . 2971
applications/modelSpecific/arenaSmarts/Smart_WaitForSignal.h . . . 2971
applications/modelSpecific/book/Book_Cap02_Example01.h . . . 2972
applications/modelSpecific/smarts/AI/StartAIAssistant.h . . . 2972
applications/modelSpecific/smarts/AnalyticalModeling/Smart_MarkovChain.h . . . 2972
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BacteriaColony_GRO.h . . . 2973
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioKineticLawRegulation.h . . . 2973
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioNetworkAnalysis.h . . . 2973
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioReversibleKineticLaw.h . . . 2974
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioReversibleMassAction.h . . . 2974
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_GroColonyGrowth.h . . . 2974
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_GroColonyLifecycle.h . . . 2974
applications/modelSpecific/smarts/Continuous/Smart_LSODE.h . . . 2975
applications/modelSpecific/smarts/Continuous/Smart_ODE.h . . . 2975
applications/modelSpecific/smarts/DiscreteProcessing/Smart_Buffer.h . . . 2975
applications/modelSpecific/smarts/DiscreteProcessing/Smart_Clone.h . . . 2976
applications/modelSpecific/smarts/DiscreteProcessing/Smart_Delay.h . . . 2976
applications/modelSpecific/smarts/DiscreteProcessing/Smart_Failures.h . . . 2977
applications/modelSpecific/smarts/DiscreteProcessing/Smart_HoldSearchRemove.h . . . 2977
applications/modelSpecific/smarts/DiscreteProcessing/Smart_Process.h . . . 2977
applications/modelSpecific/smarts/DiscreteProcessing/Smart_ProcessSet.h . . . 2977
applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayRelease.h . . . 2978
applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayReleaseMany.h . . . 2978
applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayReleaseNoDataDefs.h . . . 2979
applications/modelSpecific/smarts/ExternalIntegration/Smart_CppForG.h . . . 2979
applications/modelSpecific/smarts/ExternalIntegration/Smart_PythonForG.h . . . 2979
applications/modelSpecific/smarts/ExternalIntegration/Smart_R_Simulator.h . . . 2979
applications/modelSpecific/smarts/Grouping/Smart_BatchSeparate.h . . . 2980
applications/modelSpecific/smarts/InputOutput/Smart_Record.h . . . 2980
applications/modelSpecific/smarts/Logic/Smart_AssignWriteSeizes.h . . . 2981
applications/modelSpecific/smarts/Logic/Smart_Attribute_Variable.h . . . 2981
applications/modelSpecific/smarts/Logic/Smart_ModelInfoModelSimulation.h . . . 2981
applications/modelSpecific/smarts/Logic/Smart_Parser.h . . . 2982
applications/modelSpecific/smarts/Logic/Smart_ParserModelFunctions.h . . . 2982
applications/modelSpecific/smarts/Logic/Smart_Plugin.h . . . 2982
applications/modelSpecific/smarts/MaterialHandling/Smart_RouteStation.h . . . 2983

applications/modelSpecific/smarts/MaterialHandling/Smart_Sequence.h . . . 2983
applications/modelSpecific/smarts/ModalModel/Smart_CellularAutomata.h . . . 2984
applications/modelSpecific/smarts/ModalModel/Smart_CellularAutomata1.h . . . 2984
applications/modelSpecific/smarts/ModalModel/Smart_DefaultModalModel.h . . . 2984
applications/modelSpecific/smarts/ModalModel/Smart_Dummy.h . . . 2985
applications/modelSpecific/smarts/ModalModel/Smart_FSMState.h . . . 2985
applications/modelSpecific/smarts/ModalModel/Smart_Modalmodel.h . . . 2985
applications/modelSpecific/smarts/ModalModel/Smart_ModalModelDefaultNetwork.h . . . 2985
applications/modelSpecific/smarts/ModalModel/Smart_ModalModelFSM.h . . . 2986
applications/modelSpecific/smarts/ModalModel/Smart_ModalModelPetriNet.h . . . 2986
applications/modelSpecific/smarts/ModalModel/Smart_PetriPlace.h . . . 2986
applications/modelSpecific/smarts/Synchronization/Smart_OnEvent.h . . . 2986
applications/modelSpecific/smarts/Synchronization/Smart_SimulationControlResponse.h . . . 2987
applications/modelSpecific/smarts/Synchronization/Smart_WaitScanCondition.h . . . 2987
applications/modelSpecific/smarts/Synchronization/Smart_WaitSignal.h . . . 2987
applications/modelSpecific/teaching/AnElectronicAssemblyAndTestSystem.h . . . 2988
applications/modelSpecific/teaching/BufferFIFO.h . . . 2988
applications/modelSpecific/teaching/ContinuousModel.h . . . 2989
applications/modelSpecific/teaching/FullSimulationOfComplexModel.h . . . 2989
applications/modelSpecific/teaching/Half_Adder.h . . . 2990
applications/modelSpecific/teaching/OperatingSystem02.h . . . 2990
applications/modelSpecific/teaching/OperatingSystem03.h . . . 2991
applications/modelSpecific/teaching/Rectifier.h . . . 2991
applications/modelSpecific/teaching/Loja/Loja01.h . . . 2990
applications/modelSpecific/wcm/WcmBioNetworkBridge.h . . . 2991
applications/modelSpecific/wcm/WcmCompartmentTransportWholeCellMvp.h . . . 2992
applications/modelSpecific/wcm/WcmFluxBalanceWholeCellMvp.h . . . 2992
applications/modelSpecific/wcm/WcmGeneExpression.h . . . 2992
applications/modelSpecific/wcm/WcmLifecycleStress.h . . . 2992
applications/modelSpecific/wcm/WcmMgenitaliumKarr2012.h . . . 2992
applications/modelSpecific/wcm/WcmPathwayStressWholeCellMvp.h . . . 2993
applications/modelSpecific/wcm/WcmSimpleSsa.h . . . 2993
applications/modelSpecific/wcm/WcmWholeCellMvp.h . . . 2993
applications/modelSpecific/wcm/WcmYeastDidacticWholeCellMvp.h . . . 2993
applications/modelSpecific/wcm/WcmYeastEukaryoticWholeCellMvp.h . . . 2993
applications/shell/GenesysShell.h . . . 2994
applications/shell/GenesysShell_if.h . . . 2995
applications/worker/BaseGenesysWebApplication.h . . . 2997
applications/worker/api/ApiRouter.h . . . 2996
applications/worker/auth/TokenService.h . . . 2996
applications/worker/http/HttpRequest.h . . . 2997
applications/worker/http/HttpResponse.h . . . 2997
applications/worker/http/SimpleHttpServer.h . . . 2997
applications/worker/service/SimulatorSessionService.h . . . 2998
applications/worker/service/WebWorkerRuntime.h . . . 3000
applications/worker/session/SessionContext.h . . . 3001
applications/worker/session/SessionManager.h . . . 3002
applications/worker/worker/WorkerJob.h . . . 3002
applications/worker/worker/WorkerJobManager.h . . . 3002
gtest/gtest/gtest-assertion-result.h . . . 3003
gtest/gtest/gtest-death-test.h . . . 3006
gtest/gtest/gtest-matchers.h . . . 3010
gtest/gtest/gtest-message.h . . . 3023
gtest/gtest/gtest-param-test.h . . . 3026
gtest/gtest/gtest-printers.h . . . 3032
gtest/gtest/gtest-spi.h . . . 3045
gtest/gtest/gtest-test-part.h . . . 3048
gtest/gtest/gtest-typed-test.h . . . 3051

gtest/gtest/gtest.h . . . 3055
gtest/gtest/gtest_pred_impl.h . . . 3085
gtest/gtest/gtest_prod.h . . . 3088
gtest/gtest/internal/gtest-death-test-internal.h . . . 3121
gtest/gtest/internal/gtest-filepath.h . . . 3125
gtest/gtest/internal/gtest-internal.h . . . 3127
gtest/gtest/internal/gtest-param-util.h . . . 3145
gtest/gtest/internal/gtest-port-arch.h . . . 3158
gtest/gtest/internal/gtest-port.h . . . 3089
gtest/gtest/internal/gtest-string.h . . . 3159
gtest/gtest/internal/gtest-type-util.h . . . 3161
gtest/gtest/internal/custom/gtest-port.h . . . 3089
gtest/gtest/internal/custom/gtest-printers.h . . . 3045
gtest/gtest/internal/custom/gtest.h . . . 3084
gtest/src/gtest-assertion-result.cc . . . 3164
gtest/src/gtest-death-test.cc . . . 3165
gtest/src/gtest-filepath.cc . . . 3185
gtest/src/gtest-internal-inl.h . . . 3189
gtest/src/gtest-matchers.cc . . . 3205
gtest/src/gtest-port.cc . . . 3206
gtest/src/gtest-printers.cc . . . 3223
gtest/src/gtest-test-part.cc . . . 3230
gtest/src/gtest-typed-test.cc . . . 3231
gtest/src/gtest.cc . . . 3233
kernel/TraitsKernel.h . . . 3416
kernel/simulator/ConnectionManager.h . . . 3316
kernel/simulator/DefineGetterSetter.h . . . 3317
kernel/simulator/Event.h . . . 3326
kernel/simulator/ExperimentManager.h . . . 3328
kernel/simulator/ExperimentManagerDefaultImpl1.h . . . 3329
kernel/simulator/ExperimetManager_if.h . . . 3329
kernel/simulator/GenesysPropertyIntrospection.h . . . 3330
kernel/simulator/LicenceManager.h . . . 3331
kernel/simulator/OnEventManager.h . . . 3343
kernel/simulator/Parser_if.h . . . 3349
kernel/simulator/ParserChangesInformation.h . . . 3349
kernel/simulator/ParserDefaultImpl1.h . . . 3350
kernel/simulator/ParserDefaultImpl2.h . . . 3351
kernel/simulator/ParserManager.h . . . 3351
kernel/simulator/Persistence.h . . . 3352
kernel/simulator/Plugin.h . . . 3356
kernel/simulator/PluginConnector_if.h . . . 3357
kernel/simulator/PluginInformation.h . . . 3357
kernel/simulator/PluginManager.h . . . 3359
kernel/simulator/PropertyGenesys.h . . . 3361
kernel/simulator/ScenarioExperiment_if.h . . . 3362
kernel/simulator/SimulationControlAndResponse.h . . . 3362
kernel/simulator/SimulationExperiment.h . . . 3377
kernel/simulator/SimulationReporter_if.h . . . 3377
kernel/simulator/SimulationReporterDefaultImpl1.h . . . 3378
kernel/simulator/SimulationScenario.h . . . 3378
kernel/simulator/Simulator.h . . . 3379
kernel/simulator/SimulatorFacade.h . . . 3380
kernel/simulator/SinkModelComponent.h . . . 3398
kernel/simulator/SourceModelComponent.h . . . 3398
kernel/simulator/SystemDependencyResolver.h . . . 3399
kernel/simulator/TraceManager.h . . . 3401
kernel/simulator/essentialPlugins/Attribute.h . . . 3317

kernel/simulator/essentialPlugins/Counter.h . . . 3318
kernel/simulator/essentialPlugins/Entity.h . . . 3318
kernel/simulator/essentialPlugins/EntityType.h . . . 3319
kernel/simulator/essentialPlugins/SparseValueStore.h . . . 3320
kernel/simulator/essentialPlugins/StatisticsCollector.h . . . 3326
kernel/simulator/model/Model.h . . . 3331
kernel/simulator/model/ModelChecker_if.h . . . 3333
kernel/simulator/model/ModelCheckerDefaultImpl1.h . . . 3333
kernel/simulator/model/ModelComponent.h . . . 3334
kernel/simulator/model/ModelComponentManager.h . . . 3335
kernel/simulator/model/ModelDataDefinition.h . . . 3335
kernel/simulator/model/ModelDataDefinitionAssociations.h . . . 3338
kernel/simulator/model/ModelDataManager.h . . . 3339
kernel/simulator/model/ModelInfo.h . . . 3340
kernel/simulator/model/ModelManager.h . . . 3340
kernel/simulator/model/ModelSimulation.h . . . 3341
kernel/simulator/persistence/CppSerializer.h . . . 3353
kernel/simulator/persistence/GenSerializer.h . . . 3353
kernel/simulator/persistence/JsonSerializer.h . . . 3354
kernel/simulator/persistence/Persistence_if.h . . . 3354
kernel/simulator/persistence/PersistenceDefaultImpl2.h . . . 3355
kernel/simulator/persistence/PersistencePartialLoadImpl1.h . . . 3355
kernel/simulator/persistence/Serializer.h . . . 3355
kernel/simulator/persistence/XmlSerializer.h . . . 3356
kernel/statistics/Collector_if.h . . . 3403
kernel/statistics/CollectorDatafile_if.h . . . 3404
kernel/statistics/CollectorDatafileDefaultImpl1.h . . . 3404
kernel/statistics/CollectorDefaultImpl1.h . . . 3405
kernel/statistics/Sampler_if.h . . . 3406
kernel/statistics/SamplerDefaultImpl1.h . . . 3406
kernel/statistics/SorttFile.h . . . 3412
kernel/statistics/Statistics_if.h . . . 3412
kernel/statistics/StatisticsDataFile_if.h . . . 3413
kernel/statistics/StatisticsDataFileDefaultImpl.h . . . 3413
kernel/statistics/StatisticsDefaultImpl1.h . . . 3415
kernel/util/Exact.h . . . 3418
kernel/util/List.h . . . 3419
kernel/util/ListObservable.h . . . 3423
kernel/util/Util.h . . . 3426
parser/Genesys++-driver.h . . . 3428
parser/GenesysParser.h . . . 3429
parser/location.hh . . . 3469
parser/obj_t.h . . . 3474
parser/position.hh . . . 3475
parser/stack.hh . . . 3475
plugins/PluginConnectorDummyImpl1.h . . . 3644
plugins/PluginConnectorStaticImpl1.h . . . 3644
plugins/components/AI/AIAssistant.h . . . 3475
plugins/components/AnalyticalModeling/MarkovChain.h . . . 3476
plugins/components/BiochemicalSimulation/BacteriaColony.h . . . 3477
plugins/components/BiochemicalSimulation/BioRunnerCommand.h . . . 3481
plugins/components/BiochemicalSimulation/BioSimulate.h . . . 3482
plugins/components/BiochemicalSimulation/BioSteadyState.h . . . 3483
plugins/components/BiochemicalSimulation/GeneticCircuitSimulate.h . . . 3484
plugins/components/BiochemicalSimulation/GeneticExpressionStep.h . . . 3485
plugins/components/BiochemicalSimulation/MetabolicFluxBalance.h . . . 3486
plugins/components/Continuous/ContinuousSystemComponent.h . . . 3487
plugins/components/Continuous/DiffEquations.h . . . 3487

plugins/components/Continuous/DiffusionSimulate.h . . . 3489
plugins/components/Continuous/LSODE.h . . . 3490
plugins/components/Continuous/OLD_ODEelement.h . . . 3490
plugins/components/Decisions/Decide.h . . . 3491
plugins/components/Decisions/DropOff.h . . . 3492
plugins/components/Decisions/PickUp.h . . . 3493
plugins/components/Decisions/Remove.h . . . 3493
plugins/components/Decisions/Search.h . . . 3494
plugins/components/DiscreteProcessing/Buffer.h . . . 3498
plugins/components/DiscreteProcessing/Clone.h . . . 3499
plugins/components/DiscreteProcessing/Delay.h . . . 3499
plugins/components/DiscreteProcessing/Process.h . . . 3500
plugins/components/DiscreteProcessing/Release.h . . . 3501
plugins/components/DiscreteProcessing/Seize.h . . . 3502
plugins/components/DiscreteProcessing/auxiliar/QueueableItem.h . . . 3495
plugins/components/DiscreteProcessing/auxiliar/SeizableItem.h . . . 3496
plugins/components/ElectronicsSimulation/SPICECircuit.h . . . 3504
plugins/components/ElectronicsSimulation/SPICENode.h . . . 3511
plugins/components/ExternalIntegration/CppForG.h . . . 3512
plugins/components/ExternalIntegration/PythonForG.h . . . 3513
plugins/components/ExternalIntegration/RSimulator.h . . . 3513
plugins/components/Grouping/Batch.h . . . 3514
plugins/components/Grouping/Separate.h . . . 3515
plugins/components/InputOutput/Record.h . . . 3516
plugins/components/InputOutput/Write.h . . . 3517
plugins/components/Logic/Assign.h . . . 3518
plugins/components/Logic/Create.h . . . 3519
plugins/components/Logic/Dispose.h . . . 3520
plugins/components/MaterialHandling/Access.h . . . 3520
plugins/components/MaterialHandling/Enter.h . . . 3521
plugins/components/MaterialHandling/Exit.h . . . 3522
plugins/components/MaterialHandling/Leave.h . . . 3523
plugins/components/MaterialHandling/Move.h . . . 3523
plugins/components/MaterialHandling/PickableStationItem.h . . . 3524
plugins/components/MaterialHandling/PickStation.h . . . 3525
plugins/components/MaterialHandling/Route.h . . . 3526
plugins/components/MaterialHandling/Start.h . . . 3527
plugins/components/MaterialHandling/Stop.h . . . 3528
plugins/components/MaterialHandling/Store.h . . . 3528
plugins/components/MaterialHandling/Unstore.h . . . 3529
plugins/components/ModalModel/CellularAutomataComp.h . . . 3553
plugins/components/ModalModel/DefaultNode.h . . . 3555
plugins/components/ModalModel/DefaultTransitionExtensions.h . . . 3557
plugins/components/ModalModel/FSMState.h . . . 3557
plugins/components/ModalModel/ModalModelDefault.h . . . 3557
plugins/components/ModalModel/PetriPlace.h . . . 3558
plugins/components/ModalModel/Submodel.h . . . 3559
plugins/components/ModalModel/CellularAutomata/Boundary_Closed.h . . . 3530
plugins/components/ModalModel/CellularAutomata/Boundary_Fixed.h . . . 3531
plugins/components/ModalModel/CellularAutomata/BoundaryCondition.h . . . 3531
plugins/components/ModalModel/CellularAutomata/Cell.h . . . 3532
plugins/components/ModalModel/CellularAutomata/CellularAutomata.h . . . 3533
plugins/components/ModalModel/CellularAutomata/CellularAutomata_1DTimed.h . . . 3533
plugins/components/ModalModel/CellularAutomata/CellularAutomata_Classic.h . . . 3534
plugins/components/ModalModel/CellularAutomata/CellularAutomataAllHeaders.h . . . 3534
plugins/components/ModalModel/CellularAutomata/CellularAutomataBase.h . . . 3535
plugins/components/ModalModel/CellularAutomata/Lattice.h . . . 3535
plugins/components/ModalModel/CellularAutomata/LocalRule.h . . . 3536

plugins/components/ModalModel/CellularAutomata/LocalRule_Elementary.h . . . 3536
plugins/components/ModalModel/CellularAutomata/LocalRule_FlorestalFire.h . . . 3537
plugins/components/ModalModel/CellularAutomata/LocalRule_GameOfLife.h . . . 3538
plugins/components/ModalModel/CellularAutomata/LocalRule_Growty.h . . . 3538
plugins/components/ModalModel/CellularAutomata/LocalRule_HppLatticeGas.h . . . 3539
plugins/components/ModalModel/CellularAutomata/LocalRule_SandRock.h . . . 3540
plugins/components/ModalModel/CellularAutomata/LocalRule_SandRock_Student.h . . . 3542
plugins/components/ModalModel/CellularAutomata/LocalRule_ShowNeighbordhood.h . . . 3544
plugins/components/ModalModel/CellularAutomata/Neighborhood.h . . . 3545
plugins/components/ModalModel/CellularAutomata/Neighborhood_Center.h . . . 3546
plugins/components/ModalModel/CellularAutomata/Neighborhood_Moore.h . . . 3546
plugins/components/ModalModel/CellularAutomata/Neighborhood_VonNeumann.h . . . 3547
plugins/components/ModalModel/CellularAutomata/State.h . . . 3548
plugins/components/ModalModel/CellularAutomata/State_Numerical.h . . . 3550
plugins/components/ModalModel/CellularAutomata/StateSet.h . . . 3551
plugins/components/ModalModel/CellularAutomata/StateSet_Enumerable.h . . . 3553
plugins/components/Synchronization/Match.h . . . 3559
plugins/components/Synchronization/Signal.h . . . 3560
plugins/components/Synchronization/Wait.h . . . 3561
plugins/components/Template/DummyComponent.h . . . 3562
plugins/components/WholeCellModeling/BioStateProjectionComponent.h . . . 3563
plugins/components/WholeCellModeling/CellCycleCheckpointComponent.h . . . 3564
plugins/components/WholeCellModeling/CellDivisionEvent.h . . . 3565
plugins/components/WholeCellModeling/CellFateDecisionComponent.h . . . 3566
plugins/components/WholeCellModeling/CellGrowthComponent.h . . . 3567
plugins/components/WholeCellModeling/CompartmentExchangeComponent.h . . . 3568
plugins/components/WholeCellModeling/EukaryoticCellCycleComponent.h . . . 3569
plugins/components/WholeCellModeling/FtsZPolymerizationComponent.h . . . 3571
plugins/components/WholeCellModeling/MetabolicStateProjectionComponent.h . . . 3572
plugins/components/WholeCellModeling/MetabolicSubmodelComponent.h . . . 3574
plugins/components/WholeCellModeling/PathwayStressResponseComponent.h . . . 3575
plugins/components/WholeCellModeling/ResourceAllocationComponent.h . . . 3576
plugins/components/WholeCellModeling/StochasticReactionComponent.h . . . 3577
plugins/components/WholeCellModeling/StochasticTranscription.h . . . 3578
plugins/components/WholeCellModeling/StochasticTranslation.h . . . 3579
plugins/data/AI/AISupport.h . . . 3580
plugins/data/BiochemicalSimulation/BacteriaSignalGrid.h . . . 3581
plugins/data/BiochemicalSimulation/BioNetwork.h . . . 3582
plugins/data/BiochemicalSimulation/BioParameter.h . . . 3583
plugins/data/BiochemicalSimulation/BioReaction.h . . . 3584
plugins/data/BiochemicalSimulation/BioSBMLBridge.h . . . 3585
plugins/data/BiochemicalSimulation/BioSimulatorRunner.h . . . 3585
plugins/data/BiochemicalSimulation/BioSpecies.h . . . 3587
plugins/data/BiochemicalSimulation/GeneticCircuit.h . . . 3588
plugins/data/BiochemicalSimulation/GeneticCircuitPart.h . . . 3588
plugins/data/BiochemicalSimulation/GeneticRegulation.h . . . 3589
plugins/data/BiochemicalSimulation/GroProgram.h . . . 3590
plugins/data/BiochemicalSimulation/GroProgramAst.h . . . 3595
plugins/data/BiochemicalSimulation/GroProgramCompiler.h . . . 3595
plugins/data/BiochemicalSimulation/GroProgramIr.h . . . 3596
plugins/data/BiochemicalSimulation/GroProgramParser.h . . . 3597
plugins/data/BiochemicalSimulation/GroProgramRuntime.h . . . 3597
plugins/data/BiochemicalSimulation/MetabolicNetwork.h . . . 3599
plugins/data/BiochemicalSimulation/MetabolicReaction.h . . . 3600
plugins/data/Continuous/DiffusionField.h . . . 3601
plugins/data/Continuous/ODESolver.h . . . 3602
plugins/data/DiscreteProcessing/Failure.h . . . 3604
plugins/data/DiscreteProcessing/Queue.h . . . 3605

plugins/data/DiscreteProcessing/Resource.h . . . 3607
plugins/data/DiscreteProcessing/Schedule.h . . . 3609
plugins/data/ExternalIntegration/CppCompiler.h . . . 3610
plugins/data/ExternalIntegration/OctaveRunner.h . . . 3612
plugins/data/ExternalIntegration/PythonRuntime.h . . . 3613
plugins/data/ExternalIntegration/RSimulatorRunner.h . . . 3614
plugins/data/ExternalIntegration/SPICERunner.h . . . 3615
plugins/data/Grouping/EntityGroup.h . . . 3617
plugins/data/InputOutput/File.h . . . 3618
plugins/data/Logic/AssignmentItem.h . . . 3619
plugins/data/Logic/Formula.h . . . 3620
plugins/data/Logic/Label.h . . . 3620
plugins/data/Logic/Set.h . . . 3621
plugins/data/Logic/Variable.h . . . 3622
plugins/data/MaterialHandling/Conveyor.h . . . 3623
plugins/data/MaterialHandling/Distance.h . . . 3623
plugins/data/MaterialHandling/Segment.h . . . 3624
plugins/data/MaterialHandling/Sequence.h . . . 3625
plugins/data/MaterialHandling/Station.h . . . 3626
plugins/data/MaterialHandling/Storage.h . . . 3627
plugins/data/MaterialHandling/Transporter.h . . . 3628
plugins/data/ModalModel/ColoredPetriNetNetwork.h . . . 3628
plugins/data/ModalModel/CPNArc.h . . . 3629
plugins/data/ModalModel/CPNTransition.h . . . 3630
plugins/data/ModalModel/DefaultNetwork.h . . . 3631
plugins/data/ModalModel/DirectedAcyclicGraphNetwork.h . . . 3631
plugins/data/ModalModel/DirectedGraphNetwork.h . . . 3632
plugins/data/ModalModel/EFSMNetwork.h . . . 3632
plugins/data/ModalModel/GraphEdge.h . . . 3633
plugins/data/ModalModel/GraphNetwork.h . . . 3634
plugins/data/ModalModel/GraphNode.h . . . 3636
plugins/data/ModalModel/MarkovChainNetwork.h . . . 3636
plugins/data/ModalModel/MarkovState.h . . . 3637
plugins/data/ModalModel/NetworkActivation.h . . . 3637
plugins/data/Synchronization/SignalData.h . . . 3638
plugins/data/Template/DummyElement.h . . . 3639
plugins/data/WholeCellModeling/BioCompartment.h . . . 3640
plugins/data/WholeCellModeling/MolecularSpecies.h . . . 3641
plugins/data/WholeCellModeling/StochasticReactionRule.h . . . 3641
plugins/data/WholeCellModeling/WholeCellState.h . . . 3642
tools/TraitsTools.h . . . 3702
tools/AIAssistant/AIAssistant_if.h . . . 3644
tools/AIAssistant/AIAssistantDefaultImpl.h . . . 3647
tools/AIAssistant/AIAuditLog.h . . . 3649
tools/AIAssistant/AIConversationService.h . . . 3650
tools/AIAssistant/AIProviderClient_if.h . . . 3651
tools/AIAssistant/AISecretStore.h . . . 3652
tools/AIAssistant/AnthropicProviderClientImpl.h . . . 3652
tools/AIAssistant/HttpProviderClientBase.h . . . 3653
tools/AIAssistant/LocalProviderClientImpl.h . . . 3654
tools/AIAssistant/OpenAIProviderClientImpl.h . . . 3654
tools/Biochemical/BioKineticLawExpression.h . . . 3655
tools/Biochemical/BioSimulationAnalysis.h . . . 3658
tools/Biochemical/BioSimulationResult.h . . . 3663
tools/Biochemical/GlpkFluxBalanceSolver.h . . . 3665
tools/Biochemical/MassActionOdeSystem.h . . . 3667
tools/Biochemical/MetabolicFluxBalanceSolver.h . . . 3669
tools/Biochemical/WholeCellParameterReader.h . . . 3673

tools/Continuous/DiffusionMethodOfLinesSystem.h . . . 3675
tools/Continuous/DormandPrince54OdeSolver.h . . . 3677
tools/Continuous/OdeSolver_if.h . . . 3679
tools/Continuous/OdeSolverFactory.h . . . 3679
tools/Continuous/OdeSystem_if.h . . . 3681
tools/Continuous/Quadrature_if.h . . . 3681
tools/Continuous/RootFinder_if.h . . . 3681
tools/Continuous/RungeKutta4OdeSolver.h . . . 3681
tools/Continuous/Solver_if.h . . . 3682
tools/Continuous/SolverDefaultImpl1.h . . . 3682
tools/FactorialDesign/FactorialDesign.h . . . 3683
tools/Optimization/Optimizer_if.h . . . 3684
tools/Optimization/OptimizerDefaultImpl1.h . . . 3686
tools/Statistics/ContinuousDistribution_if.h . . . 3687
tools/Statistics/DataAnalyser_if.h . . . 3687
tools/Statistics/DataSet_if.h . . . 3688
tools/Statistics/DiscreteDistribution_if.h . . . 3688
tools/Statistics/Distribution_if.h . . . 3688
tools/Statistics/Fitter_if.h . . . 3688
tools/Statistics/FitterDefaultImpl.h . . . 3689
tools/Statistics/FitterDummyImpl.h . . . 3696
tools/Statistics/HypothesisTester_if.h . . . 3697
tools/Statistics/HypothesisTesterDefaultImpl1.h . . . 3699
tools/Statistics/ProbabilityDistribution.h . . . 3700
tools/Statistics/ProbabilityDistributionBase.h . . . 3700
tools/Statistics/SimulationResultsDataset.h . . . 3701

Chapter 16

Namespace Documentation

16.1 EFSMNetworkDiagramCodec Namespace Reference

Encodes/decodes the "Graphical EFSM Network Diagrams" `.gui` overlay section.

Functions

- void **writeSection** (QTextStream &out, Simulator *simulator, const QHash< unsigned long, EFSMNetworkDiagramLayout > &layouts)

  Writes one record block per network present in `layouts` whose EFSMNetwork (and, per entry, states/transitions) still resolve against `simulator`'s current model. Entries that no longer resolve are silently skipped (not written), never causing a save failure.
- QHash< unsigned long, EFSMNetworkDiagramLayout > **parseSection** (const QStringList &lines, Model *model)

  Parses previously collected section lines (already stripped of the leading "# " comment prefix) and resolves them against the just-loaded `model`. Unknown schema versions, malformed lines and unresolved state/transition references are skipped with a trace diagnostic; no kernel object is ever created or mutated by this function.

Variables

- constexpr int **kSupportedSchemaVersion** = 1

  Current supported schema version for the diagram record.
- constexpr const char * **kSectionName** = "Graphical EFSM Network Diagrams"

  `.gui` section name, matching the existing `# <Section Name>` convention.

16.1.1 Detailed Description

Encodes/decodes the "Graphical EFSM Network Diagrams" `.gui` overlay section.
This is a bounded, formalism-specific codec deliberately kept separate from the generic `GraphicalModelSerializer` so the shared serializer stays agnostic of concrete network formalisms (EFSM, Graph, DTMC, CPN). It writes the smallest versioned record that can recover, deterministically: the identity of the associated EFSMNetwork, the identity and position of its states, and the validated sequence identity for its transitions (including an exact empty-name comparison; see `EFSMNetworkDiagramLayout`). It never reads or writes mathematical state, and it never creates kernel objects – malformed or unresolved entries are skipped, not substituted.

16.2 generate_kernel_simulator_test_stubs Namespace Reference

Classes

- class MethodEntry

Functions

- str **_sanitize_identifier** (str text)
- bool **_is_method_candidate** (str line)
- list[MethodEntry] **parse_header** (pathlib.Path path)
- str **generate_test_source** (list[MethodEntry] methods)
- None **main** ()

Variables

- **CLASS_RE** = re.compile(r"^\s*class\s+([A-Za-z_]\w*)\b")
- **METHOD_RE** = re.compile(r"\(")

16.2.1 Detailed Description

```
Generate gtest inventory tests for every method declaration found in
source/kernel/simulator headers.

The goal is to provide a scalable scaffold so each class/method has a
corresponding unit-test entry that can be progressively implemented.
```

16.3 genesys_mcp Namespace Reference

Namespaces

- namespace __main__
- namespace client
- namespace server

16.3.1 Detailed Description

```
Genesys Simulator MCP Connector.
```

16.4 genesys_mcp.__main__ Namespace Reference

16.4.1 Detailed Description

```
Allow running the MCP server with: python -m genesys_mcp
```

16.5 genesys_mcp.client Namespace Reference

Classes

- class GenesysClient
- class GenesysClientError

16.5.1 Detailed Description

```
HTTP client wrapping the Genesys Worker Application REST API.
```

16.6 genesys_mcp.server Namespace Reference

Functions

- GenesysClient **_client** (str base_url, str|None token=None)
- dict[str, str] **_err** (GenesysClientError e)
- dict[str, str] **_conn_err** (httpx.RequestError e)
- dict[str, Any] genesys_health (str base_url=_DEFAULT_URL)

- dict[str, Any] genesys_auth (str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_info (str token, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_new_model (str token, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_get_model (str token, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_save_model (str token, str filename, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_load_model (str token, str filename, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_import_model_language (str token, str model_text, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_simulation_status (str token, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_configure_simulation (str token, int number_of_replications, float replication_length, float warm_up_period=0.0, bool pause_on_event=False, bool pause_on_replication=False, bool initialize_↩ statistics=True, bool initialize_system=True, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_run_simulation (str token, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_step_simulation (str token, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_worker_info (str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_worker_capabilities (str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_create_job (str token, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_job_status (str token, str job_id, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_run_job (str token, str job_id, str base_url=_DEFAULT_URL)
- dict[str, Any] genesys_job_result (str token, str job_id, str base_url=_DEFAULT_URL)
- None **main** ()

Variables

- mcp
- **_DEFAULT_URL** = os.environ.get("GENESYS_URL", "http://localhost:8080")

16.6.1 Detailed Description

```
Genesys MCP Server -- exposes GenESyS simulation capabilities via MCP.
```

16.6.2 Function Documentation

16.6.2.1 genesys_auth()

```
dict[str, Any] genesys_mcp.server.genesys_auth (
            str  base_url = _DEFAULT_URL )

Create a new Genesys session and return the bearer token.

Returns {"token": "<bearer-token>", ...}.
Pass the token to all subsequent tools as the `token` parameter.
```

16.6.2.2 genesys_configure_simulation()

```
dict[str, Any] genesys_mcp.server.genesys_configure_simulation (
            str token,
            int number_of_replications,
            float replication_length,
            float  warm_up_period = 0.0,
            bool  pause_on_event = False,
            bool  pause_on_replication = False,
            bool  initialize_statistics = True,
            bool  initialize_system = True,
            str  base_url = _DEFAULT_URL )
```

```
Configure simulation parameters before running.

Args:
    token: Bearer token from genesys_auth.
    number_of_replications: How many independent replications to run.
    replication_length: Simulated time length of each replication.
    warm_up_period: Simulated time before statistics start being collected.
    pause_on_event: Pause simulation after each event (step-by-step mode).
    pause_on_replication: Pause between replications.
    initialize_statistics: Reset statistics counters at start of each replication.
    initialize_system: Reset system state at start of each replication.
    base_url: Genesys worker app base URL.
```

16.6.2.3 genesys_create_job()

```
dict[str, Any] genesys_mcp.server.genesys_create_job (
            str token,
            str  base_url = _DEFAULT_URL )

Create a worker job from the session's current model.

The job is created in 'queued' state. A snapshot of the model is
persisted so later execution is decoupled from live model mutations.
Returns {"jobId": "<id>", "state": "queued", ...}.
Requires a valid session token from genesys_auth.
```

16.6.2.4 genesys_get_model()

```
dict[str, Any] genesys_mcp.server.genesys_get_model (
            str token,
            str  base_url = _DEFAULT_URL )

Return a description of the session's current simulation model.

Requires a valid session token from genesys_auth.
```

16.6.2.5 genesys_health()

```
dict[str, Any] genesys_mcp.server.genesys_health (
            str  base_url = _DEFAULT_URL )

Check that the Genesys worker application is reachable.

Returns the health status object from GET /health.
No authentication required.
```

16.6.2.6 genesys_import_model_language()

```
dict[str, Any] genesys_mcp.server.genesys_import_model_language (
            str token,
            str model_text,
            str  base_url = _DEFAULT_URL )

Import a simulation model from GenESyS model language text.

`model_text` is the full plain-text GenESyS model specification.
The imported model becomes the session's current model.
Requires a valid session token from genesys_auth.
```

16.6.2.7 genesys_info()

```
dict[str, Any] genesys_mcp.server.genesys_info (
            str token,
            str  base_url = _DEFAULT_URL )
```

Return simulator version, name, and licence information.

Requires a valid session token from genesys_auth.

16.6.2.8 genesys_job_result()

```
dict[str, Any] genesys_mcp.server.genesys_job_result (
            str token,
            str job_id,
            str  base_url = _DEFAULT_URL )
```

Retrieve the simulation result summary for a finished or failed job.

Only available after the job reaches a terminal state.
Requires a valid session token from genesys_auth.

16.6.2.9 genesys_job_status()

```
dict[str, Any] genesys_mcp.server.genesys_job_status (
            str token,
            str job_id,
            str  base_url = _DEFAULT_URL )
```

Return the current state of a worker job (queued/running/finished/failed).

Requires a valid session token from genesys_auth.

16.6.2.10 genesys_load_model()

```
dict[str, Any] genesys_mcp.server.genesys_load_model (
            str token,
            str filename,
            str  base_url = _DEFAULT_URL )
```

Load a model from a file inside the session workspace.

`filename` must be a basename (no path separators).
Requires a valid session token from genesys_auth.

16.6.2.11 genesys_new_model()

```
dict[str, Any] genesys_mcp.server.genesys_new_model (
            str token,
            str  base_url = _DEFAULT_URL )
```

Create a new empty simulation model in the current session.

The new model becomes the session's current model.
Requires a valid session token from genesys_auth.

16.6.2.12 genesys_run_job()

```
dict[str, Any] genesys_mcp.server.genesys_run_job (
            str token,
            str job_id,
            str  base_url = _DEFAULT_URL )
```

Execute a worker job synchronously.

The job must be in 'queued' state. This call blocks until the simulation
finishes. Poll genesys_job_status or call genesys_job_result afterwards.
Requires a valid session token from genesys_auth.

16.6.2.13 genesys_run_simulation()

```
dict[str, Any] genesys_mcp.server.genesys_run_simulation (
            str token,
            str  base_url = _DEFAULT_URL )
```

Run the current simulation synchronously and return a summary.

Configure simulation parameters first with genesys_configure_simulation.
Requires a valid session token from genesys_auth.

16.6.2.14 genesys_save_model()

```
dict[str, Any] genesys_mcp.server.genesys_save_model (
            str token,
            str filename,
            str  base_url = _DEFAULT_URL )
```

Save the current model to a file inside the session workspace.

`filename` must be a basename (no path separators). Only alphanumeric
characters, dots, hyphens, and underscores are allowed.
Requires a valid session token from genesys_auth.

16.6.2.15 genesys_simulation_status()

```
dict[str, Any] genesys_mcp.server.genesys_simulation_status (
            str token,
            str  base_url = _DEFAULT_URL )
```

Return the current simulation status (state, replication, clock, etc.).

Requires a valid session token from genesys_auth.

16.6.2.16 genesys_step_simulation()

```
dict[str, Any] genesys_mcp.server.genesys_step_simulation (
            str token,
            str  base_url = _DEFAULT_URL )
```

Advance the simulation by one event step.

Useful for step-by-step inspection. Configure with pause_on_event=True first.
Requires a valid session token from genesys_auth.

16.6.2.17 genesys_worker_capabilities()

```
dict[str, Any] genesys_mcp.server.genesys_worker_capabilities (
            str  base_url = _DEFAULT_URL )

Return the current worker's capability flags.

Describes what simulation features this worker instance supports.
No authentication required.
```

16.6.2.18 genesys_worker_info()

```
dict[str, Any] genesys_mcp.server.genesys_worker_info (
            str  base_url = _DEFAULT_URL )

Return worker identity metadata (id, version, role).

No authentication required.
```

16.6.3 Variable Documentation

16.6.3.1 mcp

```
genesys_mcp.server.mcp
```

Initial value:

```
00001 =  FastMCP(
00002     "Genesys Simulator",
00003     instructions=(
00004         "Use these tools to create, configure, and run GenESyS discrete-event "
00005         "simulation models. Always call genesys_auth first to obtain a session "
00006         "token before using any other tool."
00007     ),
00008 )
```

16.7 GuiCrashDiagnostics Namespace Reference

Crash diagnostics helpers for the Qt GUI process.

Functions

- void **installFatalSignalHandlers** ()

  Installs temporary fatal-signal handlers that print a crash backtrace.

16.7.1 Detailed Description

Crash diagnostics helpers for the Qt GUI process.
This namespace keeps temporary signal-handling utilities isolated from the rest of the GUI runtime so they can be installed only when needed during crash investigation.

16.8 QCP Namespace Reference

Enumerations

- enum ResolutionUnit { ruDotsPerMeter , ruDotsPerCentimeter , ruDotsPerInch }
- enum ExportPen { epNoCosmetic , epAllowCosmetic }
- enum SignDomain { sdNegative , sdBoth , sdPositive }
- enum MarginSide {
  msLeft = 0x01 , msRight = 0x02 , msTop = 0x04 , msBottom = 0x08 ,
  msAll = 0xFF , msNone = 0x00 }

- enum AntialiasedElement {
  aeAxes = 0x0001 , aeGrid = 0x0002 , aeSubGrid = 0x0004 , aeLegend = 0x0008 ,
  aeLegendItems = 0x0010 , aePlottables = 0x0020 , aeItems = 0x0040 , aeScatters = 0x0080 ,
  aeFills = 0x0100 , aeZeroLine = 0x0200 , aeOther = 0x8000 , aeAll = 0xFFFF ,
  aeNone = 0x0000 }
- enum PlottingHint { phNone = 0x000 , phFastPolylines = 0x001 , phImmediateRefresh = 0x002 , phCacheLabels = 0x004 }
- enum Interaction {
  iNone = 0x000 , iRangeDrag = 0x001 , iRangeZoom = 0x002 , iMultiSelect = 0x004 ,
  iSelectPlottables = 0x008 , iSelectAxes = 0x010 , iSelectLegend = 0x020 , iSelectItems = 0x040 ,
  iSelectOther = 0x080 , iSelectPlottablesBeyondAxisRect = 0x100 }
- enum SelectionRectMode { srmNone , srmZoom , srmSelect , srmCustom }
- enum SelectionType {
  stNone , stWhole , stSingleData , stDataRange ,
  stMultipleDataRanges }

Functions

- bool **isInvalidData** (double value)
- bool **isInvalidData** (double value1, double value2)
- void **setMarginValue** (QMargins &margins, QCP::MarginSide side, int value)
- int **getMarginValue** (const QMargins &margins, QCP::MarginSide side)

Variables

- const QMetaObject **staticMetaObject**

16.8.1 Detailed Description

The QCP Namespace contains general enums, QFlags and functions used throughout the QCustomPlot library.
It provides QMetaObject-based reflection of its enums and flags via *QCP::staticMetaObject*.

16.8.2 Enumeration Type Documentation

16.8.2.1 AntialiasedElement

`enum QCP::AntialiasedElement`

Defines what objects of a plot can be forcibly drawn antialiased/not antialiased. If an object is neither forcibly drawn antialiased nor forcibly drawn not antialiased, it is up to the respective element how it is drawn. Typically it provides a *setAntialiased* function for this.
`AntialiasedElements` is a flag of or-combined elements of this enum type.

See also

QCustomPlot::setAntialiasedElements, QCustomPlot::setNotAntialiasedElements

Enumerator

aeAxes	`0x0001` Axis base line and tick marks
aeGrid	`0x0002` Grid lines
aeSubGrid	`0x0004` Sub grid lines
aeLegend	`0x0008` Legend box
aeLegendItems	`0x0010` Legend items
aePlottables	`0x0020` Main lines of plottables
aeItems	`0x0040` Main lines of items
aeScatters	`0x0080` Scatter symbols of plottables (excluding scatter symbols of type ssPixmap)
aeFills	`0x0100` Borders of fills (e.g. under or between graphs)

Enumerator

aeZeroLine	0x0200 Zero-lines, see QCPGrid::setZeroLinePen
aeOther	0x8000 Other elements that don't fit into any of the existing categories
aeAll	0xFFFF All elements
aeNone	0x0000 No elements

16.8.2.2 ExportPen

enum QCP::ExportPen

Defines how cosmetic pens (pens with numerical width 0) are handled during export.

See also

QCustomPlot::savePdf

Enumerator

epNoCosmetic	Cosmetic pens are converted to pens with pixel width 1 when exporting.
epAllowCosmetic	Cosmetic pens are exported normally (e.g. in PDF exports, cosmetic pens always appear as 1 pixel on screen, independent of viewer zoom level)

16.8.2.3 Interaction

enum QCP::Interaction

Defines the mouse interactions possible with QCustomPlot.
Interactions is a flag of or-combined elements of this enum type.

See also

QCustomPlot::setInteractions

Enumerator

iNone	0x000 None of the interactions are possible
iRangeDrag	0x001 Axis ranges are draggable (see QCPAxisRect::setRangeDrag, QCPAxisRect::setRangeDragAxes)
iRangeZoom	0x002 Axis ranges are zoomable with the mouse wheel (see QCPAxisRect::setRangeZoom, QCPAxisRect::setRangeZoomAxes)
iMultiSelect	0x004 The user can select multiple objects by holding the modifier set by QCustomPlot::setMultiSelectModifier while clicking
iSelectPlottables	0x008 Plottables are selectable (e.g. graphs, curves, bars,... see QCPAbstractPlottable)
iSelectAxes	0x010 Axes are selectable (or parts of them, see QCPAxis::setSelectableParts)
iSelectLegend	0x020 Legends are selectable (or their child items, see QCPLegend::setSelectableParts)
iSelectItems	0x040 Items are selectable (Rectangles, Arrows, Textitems, etc. see QCPAbstractItem)
iSelectOther	0x080 All other objects are selectable (e.g. your own derived layerables, other layout elements,...)
iSelectPlottablesBeyondAxisRect	0x100 When performing plottable selection/hit tests, this flag extends the sensitive area beyond the axis rect

16.8.2.4 MarginSide

enum QCP::MarginSide

Defines the sides of a rectangular entity to which margins can be applied.

See also

QCPLayoutElement::setAutoMargins, QCPAxisRect::setAutoMargins

Enumerator

msLeft	0x01 left margin
msRight	0x02 right margin
msTop	0x04 top margin
msBottom	0x08 bottom margin
msAll	0xFF all margins
msNone	0x00 no margin

16.8.2.5 PlottingHint

enum QCP::PlottingHint

Defines plotting hints that control various aspects of the quality and speed of plotting.

See also

QCustomPlot::setPlottingHints

Enumerator

phNone	0x000 No hints are set
phFastPolylines	joins, thus is most effective for pen sizes larger than 1. It is only used for solid line pens. 0x001 Graph/Curve lines are drawn with a faster method. This reduces the quality especially of the line segment
phImmediateRefresh	This is set by default to prevent the plot from freezing on fast consecutive replots (e.g. user drags ranges with mouse). 0x002 causes an immediate repaint() instead of a soft update() when QCustomPlot::replot() is called with parameter QCustomPlot::rpRefreshHint.
phCacheLabels	0x004 axis (tick) labels will be cached as pixmaps, increasing replot performance.

16.8.2.6 ResolutionUnit

enum QCP::ResolutionUnit

Defines the different units in which the image resolution can be specified in the export functions.

See also

QCustomPlot::savePng, QCustomPlot::saveJpg, QCustomPlot::saveBmp, QCustomPlot::saveRastered

Enumerator

ruDotsPerMeter	Resolution is given in dots per meter (dpm)
ruDotsPerCentimeter	Resolution is given in dots per centimeter (dpcm)
ruDotsPerInch	Resolution is given in dots per inch (DPI/PPI)

16.8.2.7 SelectionRectMode

`enum QCP::SelectionRectMode`

Defines the behaviour of the selection rect.

See also

QCustomPlot::setSelectionRectMode, QCustomPlot::selectionRect, QCPSelectionRect

Enumerator

srmNone	The selection rect is disabled, and all mouse events are forwarded to the underlying objects, e.g. for axis range dragging.
srmZoom	When dragging the mouse, a selection rect becomes active. Upon releasing, the axes that are currently set as range zoom axes (QCPAxisRect::setRangeZoomAxes) will have their ranges zoomed accordingly.
srmSelect	When dragging the mouse, a selection rect becomes active. Upon releasing, plottable data points that were within the selection rect are selected, if the plottable's selectability setting permits. (See data selection mechanism for details.)
srmCustom	When dragging the mouse, a selection rect becomes active. It is the programmer's responsibility to connect according slots to the selection rect's signals (e.g. QCPSelectionRect::accepted) in order to process the user interaction.

16.8.2.8 SelectionType

`enum QCP::SelectionType`

Defines the different ways a plottable can be selected. These images show the effect of the different selection types, when the indicated selection rect was dragged:

td	td	td	td	td

See also

QCPAbstractPlottable::setSelectable, QCPDataSelection::enforceType

Enumerator

stNone	The plottable is not selectable.
stWhole	Selection behaves like stMultipleDataRanges, but if there are any data points selected, the entire plottable is drawn as selected.
stSingleData	One individual data point can be selected at a time.
stDataRange	Multiple contiguous data points (a data range) can be selected.
stMultipleDataRanges	Any combination of data points/ranges can be selected.

16.8.2.9 SignDomain

`enum QCP::SignDomain`

Represents negative and positive sign domain, e.g. for passing to QCPAbstractPlottable::getKeyRange and QCPAbstractPlottable::getValueRange.

This is primarily needed when working with logarithmic axis scales, since only one of the sign domains can be visible at a time.

Enumerator

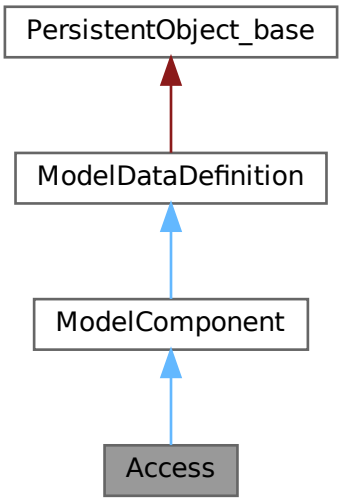
sdNegative	The negative sign domain, i.e. numbers smaller than zero.
sdBoth	Both sign domains, including zero, i.e. all numbers.
sdPositive	The positive sign domain, i.e. numbers greater than zero.

Chapter 17

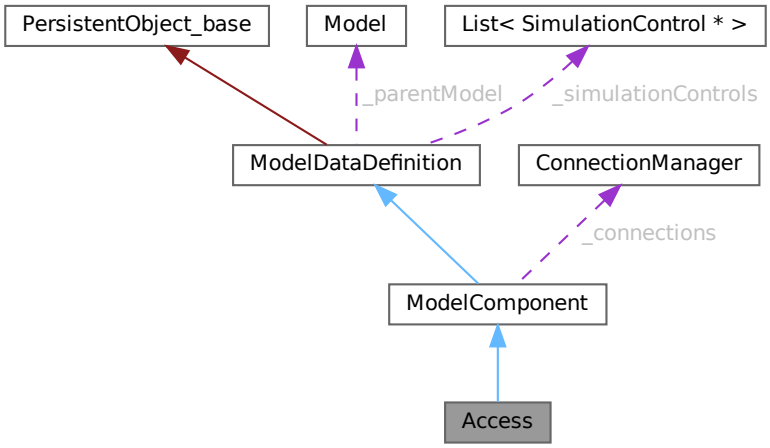
Class Documentation

17.1 Access Class Reference

`#include <plugins/components/MaterialHandling/Access.h>`
Inheritance diagram for Access:



Collaboration diagram for Access:



Public Member Functions

- **Access** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setConveyor** (Conveyor ∗conveyor)
- Conveyor ∗ **getConveyor** () const
- void **setQuantityExpression** (std::string quantityExpression)
- std::string **getQuantityExpression** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent ∗comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1.1 Detailed Description

Access module DESCRIPTION The Access module allocates one or more cells of a conveyor to an entity for movement from one station to another. Once the entity has control of the cells on the conveyor, it may then be conveyed to the next station. When an entity arrives at an Access module, it will wait until the appropriate number of contiguous cells on the conveyor are empty and aligned with the entity's station location. TYPICAL USES Parts accessing a conveyor to be sent to a paint booth Glass accessing a conveyor to be transferred to a cutting station PROMPTS Prompt Description Name Unique name of the module that will be displayed in the flowchart. Conveyor Name Name of the conveyor that the entity desires.

17.1.2 of Cells Number of contiguous conveyor cells the entity requires for

movement on the conveyor. Queue Type Determines the type of queue used to hold the entities, either an individual Queue, a queue Set, and Internal queue or an Attribute or Expression that evaluate to the queue name. Queue Name Name of the queue that will hold the entity until it accesses the conveyor. Set Name Name of the set of queues. Set Index Defines the index into the queue set. Note that this is the index into the set and not the name of the queue in the set. For example, the only valid entries for a queue set containing three members is an expression that evaluates to 1, 2, or 3. Attribute Name Defines the name of the attribute that stores the queue name to which entities will reside. Expression Defines the name of the expression that stores the queue name to which entities will reside.

17.1.3 Member Function Documentation

17.1.3.1 _check()

```
bool Access::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1.3.2 _loadInstance()

```
bool Access::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.1.3.3 _onDispatchEvent()

```
void Access::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.1.3.4 _saveInstance()

```
void Access::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.1.3.5 show()

```
std::string Access::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

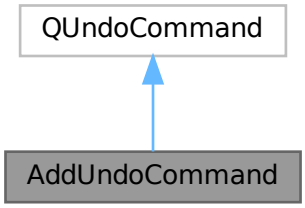
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

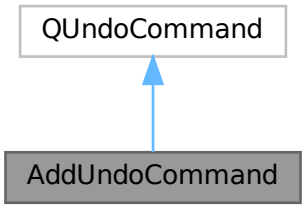
- plugins/components/MaterialHandling/Access.h
- plugins/components/MaterialHandling/Access.cpp

17.2 AddUndoCommand Class Reference

Inheritance diagram for AddUndoCommand:



Collaboration diagram for AddUndoCommand:



Public Member Functions

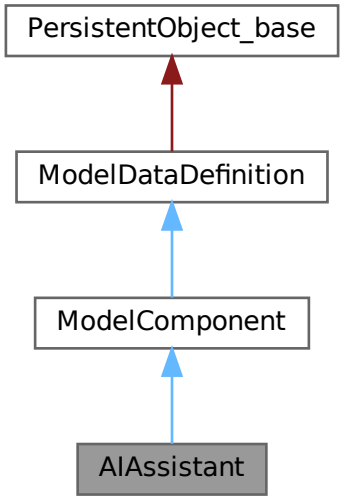
- **AddUndoCommand** (QGraphicsItem ∗item, ModelGraphicsScene ∗scene, QUndoCommand ∗parent=nullptr)
- void **undo** () override
- void **redo** () override

The documentation for this class was generated from the following files:

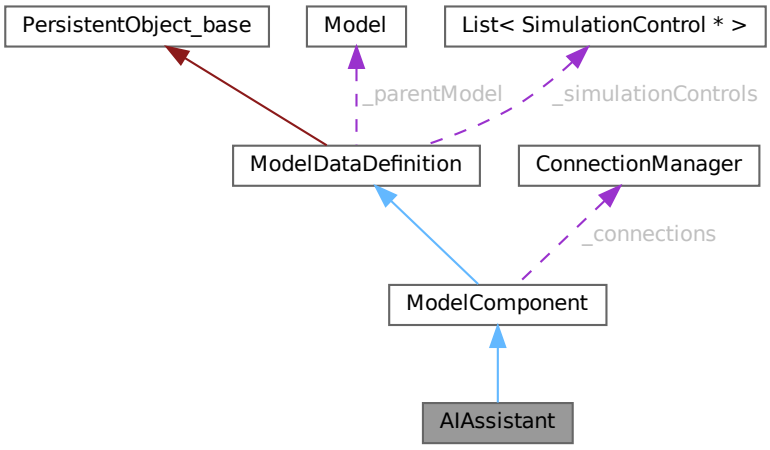
- applications/gui/genesys/actions/AddUndoCommand.h
- applications/gui/genesys/actions/AddUndoCommand.cpp

17.3 AIAssistant Class Reference

Inheritance diagram for AIAssistant:



Collaboration diagram for AIAssistant:



Public Member Functions

- **AIAssistant** (Model *model, std::string name="")
- void **setAISupport** (AISupport *support)
- AISupport * **getAISupport** () const
- void **setAISupportName** (std::string value)
- std::string **getAISupportName** () const

- void **setPromptTemplate** (std::string value)
- std::string **getPromptTemplate** () const
- void **setConversationScope** (std::string value)
- std::string **getConversationScope** () const
- void **setConversationKeyExpression** (std::string value)
- std::string **getConversationKeyExpression** () const
- void **setActionMappings** (std::string value)
- std::string **getActionMappings** () const
- void **setDelayExpression** (std::string value)
- std::string **getDelayExpression** () const
- void **setDefaultOutputPort** (unsigned int value)
- unsigned int **getDefaultOutputPort** () const
- void **setErrorOutputPort** (unsigned int value)
- unsigned int **getErrorOutputPort** () const
- void **setAllowDirectOutputPort** (bool value)
- bool **isAllowDirectOutputPort** () const
- void **setMinimumConfidence** (double value)
- double **getMinimumConfidence** () const
- void **setSaveConfidenceAttribute** (std::string value)
- std::string **getSaveConfidenceAttribute** () const
- void **setSaveSuccessAttribute** (std::string value)
- std::string **getSaveSuccessAttribute** () const
- std::string **getLastPrompt** () const
- std::string **getLastResponse** () const
- std::string **getLastError** () const
- bool **renderPrompt** (std::string &result, std::string &error)
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.3.1 Member Function Documentation

17.3.1.1 _check()

```
bool AIAssistant::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.3.1.2 _loadInstance()

```
bool AIAssistant::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.3.1.3 _onDispatchEvent()

```
void AIAssistant::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.3.1.4 _saveInstance()

```
void AIAssistant::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.3.1.5 show()

```
std::string AIAssistant::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

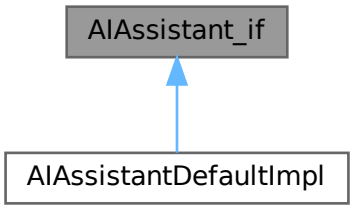
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/AI/AIAssistant.h
- plugins/components/AI/AIAssistant.cpp

17.4 AIAssistant_if Class Reference

Abstract interface for the GenESyS AI Assistant tool.
`#include <tools/AIAssistant/AIAssistant_if.h>`
Inheritance diagram for AIAssistant_if:



Public Member Functions

- virtual void setConfiguration (const AIAssistantConfiguration &configuration)=0

  Replaces the entire configuration record.
- virtual AIAssistantConfiguration getConfiguration () const =0

  Returns a copy of the current configuration record.
- virtual void setProvider (AIProvider provider)=0

  Sets the AI backend provider.
- virtual AIProvider getProvider () const =0

  Returns the currently configured AI provider.
- virtual void setProviderName (const std::string &providerName)=0

  Sets a human-readable provider label.
- virtual std::string getProviderName () const =0

Returns the human-readable provider label.

- virtual void setModelName (const std::string &modelName)=0

  Sets the provider model identifier (e.g. "gpt-4o", "claude-opus-4-7").
- virtual std::string getModelName () const =0

  Returns the provider model identifier.
- virtual void setBaseUrl (const std::string &baseUrl)=0

  Sets the base URL; required when provider == CustomHttp.
- virtual std::string getBaseUrl () const =0

  Returns the currently configured base URL.
- virtual void setReasoningEffort (ReasoningEffort effort)=0

  Sets the desired reasoning effort hint.
- virtual ReasoningEffort getReasoningEffort () const =0

  Returns the currently configured reasoning effort.
- virtual void setTemperature (double temperature)=0

  Sets the sampling temperature (0.0 = deterministic, 1.0 = creative).
- virtual double getTemperature () const =0

  Returns the configured sampling temperature.
- virtual void setMaxOutputTokens (unsigned int maxOutputTokens)=0

  Sets the maximum number of output tokens to request.
- virtual unsigned int getMaxOutputTokens () const =0

  Returns the maximum output tokens limit.
- virtual void setTimeoutSeconds (unsigned int timeoutSeconds)=0

  Sets the HTTP request timeout in seconds.
- virtual unsigned int getTimeoutSeconds () const =0

  Returns the HTTP request timeout in seconds.
- virtual void setApiKey (const std::string &apiKey)=0

  Stores the API key in memory for the duration of the session.
- virtual void clearApiKey ()=0

  Erases the in-memory API key.
- virtual bool hasApiKey () const =0

  Returns `true` when a non-empty API key is held in memory.
- virtual std::string getMaskedApiKey () const =0

  Returns a redacted representation of the current API key.
- virtual void setApiKeyEnvironmentVariable (const std::string &variableName)=0

  Stores the name of the environment variable that holds the API key.
- virtual std::string getApiKeyEnvironmentVariable () const =0

  Returns the configured environment-variable name.
- virtual bool loadApiKeyFromEnvironment ()=0

  Reads the API key from the configured environment variable.
- virtual void setSimulatorFacade (SimulatorFacade *simulatorFacade)=0

  Attaches a non-owning SimulatorFacade pointer.
- virtual SimulatorFacade * getSimulatorFacade () const =0

  Returns the currently attached SimulatorFacade pointer, or `nullptr`.
- virtual bool hasSimulatorFacade () const =0

  Returns `true` when a non-null SimulatorFacade is attached.
- virtual void setProviderClient (AIProviderClient_if *client)=0

  Injects an external provider client (non-owning).
- virtual AIProviderClient_if * getProviderClient () const =0

  Returns the currently attached provider client, or `nullptr`.
- virtual bool hasProviderClient () const =0

  Returns `true` when a non-null provider client is attached.

- virtual void clearKnowledgeBase ()=0

  Removes all items from the knowledge base.
- virtual void addKnowledgeItem (const GenesysKnowledgeItem &item)=0

  Appends one item to the knowledge base.
- virtual std::vector< GenesysKnowledgeItem > getKnowledgeItems () const =0

  Returns a copy of all registered knowledge items.
- virtual GenesysKnowledgeBaseSummary getKnowledgeBaseSummary () const =0

  Returns an aggregate summary of the knowledge base.
- virtual std::string exportKnowledgeBaseAsMarkdown () const =0

  Renders the knowledge base as a Markdown document.
- virtual bool refreshKnowledgeBaseFromSimulator (std::string *message=nullptr)=0

  Populates the knowledge base by introspecting the attached SimulatorFacade.
- virtual bool checkReady (std::string *message=nullptr) const =0

  Checks whether the assistant has enough configuration to operate.
- virtual AIAssistantResponse analyzePrompt (const AIAssistantRequest &request)=0

  Parses and summarises a user prompt without calling any external service.
- virtual AIAssistantResponse createModelPlan (const AIAssistantRequest &request)=0

  Produces a structured plan mapping the prompt to GenESyS components.
- virtual AIAssistantResponse buildModel (const AIAssistantRequest &request)=0

  Constructs a GenESyS model via SimulatorFacade using an LLM-generated .gen file.
- virtual AIAssistantResponse configureSimulation (const AIAssistantRequest &request)=0

  Overrides ModelSimulation parameters on the current model.
- virtual AIAssistantResponse runSimulation (const AIAssistantRequest &request)=0

  Executes the configured simulation synchronously via SimulatorFacade.
- virtual AIAssistantResponse collectResults (const AIAssistantRequest &request)=0

  Collects and formats simulation results from the current model.
- virtual AIAssistantResponse execute (const AIAssistantRequest &request)=0

  High-level dispatcher that orchestrates the full pipeline based on request flags.
- virtual ExecutionState getState () const =0

  Returns the current lifecycle state.
- virtual std::string getLastError () const =0

  Returns the error message from the most recent failed operation.
- virtual std::string getLastDiagnostics () const =0

  Returns technical diagnostic output from the most recent operation.
- virtual void clearDiagnostics ()=0

  Clears both the last error and diagnostics strings.
- virtual std::vector< AuditEntry > getAuditEntries (unsigned int maxEntries=200) const =0

  Returns up to `maxEntries` most-recent audit entries (oldest first).
- virtual std::string getAuditLogPath () const =0

  Returns the absolute path of the audit log file.
- virtual unsigned int exportAuditLog (const std::string &csvPath) const =0

  Exports all audit entries to a CSV file.

17.4.1 Detailed Description

Abstract interface for the GenESyS AI Assistant tool.
The AI Assistant translates natural-language prompts into GenESyS simulation models, configures experiments and optionally executes them. This interface defines the full contract; concrete implementations supply the provider integration logic.

17.4.1.1 Security model

- API keys are accepted only through setApiKey() or loadApiKeyFromEnvironment().
- No method returns the raw key value (getApiKey() does not exist by design).
- getMaskedApiKey() exposes only the last four characters.
- All capability-affecting operations are guarded by explicit permission flags in AIAssistantConfiguration. The deny-by-default posture means nothing is sent to external services, no model is mutated, and no simulation is run unless the caller has explicitly enabled each permission.

17.4.1.2 Planned evolution

See source/tools/ARCHITECTURE_tools.md for the multi-stage roadmap.

17.4.1.3 Thread safety

Implementations are not required to be thread-safe in this version.

17.4.2 Member Function Documentation

17.4.2.1 addKnowledgeItem()

```
virtual void AIAssistant_if::addKnowledgeItem (
            const GenesysKnowledgeItem & item )  [pure virtual]
```

Appends one item to the knowledge base.
Implemented in AIAssistantDefaultImpl.

17.4.2.2 analyzePrompt()

```
virtual AIAssistantResponse AIAssistant_if::analyzePrompt (
            const AIAssistantRequest & request )  [pure virtual]
```

Parses and summarises a user prompt without calling any external service.
Stage 3 will implement actual LLM calls. Current stub returns NotImplemented.
Implemented in AIAssistantDefaultImpl.

17.4.2.3 buildModel()

```
virtual AIAssistantResponse AIAssistant_if::buildModel (
            const AIAssistantRequest & request )  [pure virtual]
```

Constructs a GenESyS model via SimulatorFacade using an LLM-generated .gen file.
Requires `allowModelMutation` == `true`, `allowNetworkAccess` == `true`, a valid SimulatorFacade, and an attached provider client.
Behaviour (Stage 5):

1. If `request.modelPlan` is non-empty, it is used as the model specification. Otherwise, the `request` description fields are used directly.
2. An LLM call is made requesting a GenESyS model language (.gen) file.
3. The generated .gen content is written to a temp file and loaded via SimulatorFacade::loadModel(). The temp file is deleted immediately after loading.
4. `AIAssistantResponse::generatedModelLanguage` holds the generated .gen text.

Implemented in AIAssistantDefaultImpl.

17.4.2.4 checkReady()

```
virtual bool AIAssistant_if::checkReady (
            std::string * message = nullptr ) const  [pure virtual]
```

Checks whether the assistant has enough configuration to operate.

Parameters

message	Optional; receives a description of the first unsatisfied condition, or a confirmation message when ready.

Returns

`true` when all minimum requirements are met for prompt analysis.

Checks performed (in order):

1. Provider is not `AIProvider::Undefined`.
2. Model name is not empty.
3. For OpenAI/Anthropic/CustomHttp: an API key is held in memory or an environment-variable name is configured.
4. For CustomHttp: baseUrl is not empty.

Note: SimulatorFacade and permission flags are checked separately when model-mutation or simulation-execution operations are requested.
Implemented in AIAssistantDefaultImpl.

17.4.2.5 clearApiKey()

```
virtual void AIAssistant_if::clearApiKey ( )  [pure virtual]
```

Erases the in-memory API key.
Implementations should overwrite the key buffer before clearing to reduce the window of exposure. Note that standard C++ provides no guarantee that the compiler will not optimise away the overwrite; platform-specific secure zeroing (explicit_bzero, SecureZeroMemory) should be used in security-critical contexts.
Implemented in AIAssistantDefaultImpl.

17.4.2.6 clearDiagnostics()

```
virtual void AIAssistant_if::clearDiagnostics ( )  [pure virtual]
```

Clears both the last error and diagnostics strings.
Implemented in AIAssistantDefaultImpl.

17.4.2.7 clearKnowledgeBase()

```
virtual void AIAssistant_if::clearKnowledgeBase ( )  [pure virtual]
```

Removes all items from the knowledge base.
Implemented in AIAssistantDefaultImpl.

17.4.2.8 collectResults()

```
virtual AIAssistantResponse AIAssistant_if::collectResults (
            const AIAssistantRequest & request )  [pure virtual]
```

Collects and formats simulation results from the current model.
Iterates SimulatorFacade::modelGetResponses() and renders them as a Markdown table in `AIAssistantResponse::message`
Implemented in AIAssistantDefaultImpl.

17.4.2.9 configureSimulation()

```
virtual AIAssistantResponse AIAssistant_if::configureSimulation (
            const AIAssistantRequest & request )  [pure virtual]
```

Overrides ModelSimulation parameters on the current model.
Requires `allowModelMutation` == `true` and a valid SimulatorFacade.
Behaviour (Stage 5):

1. Makes a small LLM call to extract structured simulation parameters from `request.experiment↩ Description` (replications, replication length, warm-up period, time units).
2. Applies the extracted values via SimulatorFacade::simSet∗() methods.
3. Parameters already set by a loaded .gen file are overridden only when the LLM extraction yields non-zero values.

Implemented in AIAssistantDefaultImpl.

17.4.2.10 createModelPlan()

```
virtual AIAssistantResponse AIAssistant_if::createModelPlan (
            const AIAssistantRequest & request ) [pure virtual]
```

Produces a structured plan mapping the prompt to GenESyS components.
Stage 4 will implement prompt-to-plan translation. Current stub returns NotImplemented.
Implemented in AIAssistantDefaultImpl.

17.4.2.11 execute()

```
virtual AIAssistantResponse AIAssistant_if::execute (
            const AIAssistantRequest & request ) [pure virtual]
```

High-level dispatcher that orchestrates the full pipeline based on request flags.
Calls analyzePrompt, createModelPlan, buildModel, configureSimulation, runSimulation and collectResults in sequence according to the enabled flags in `request`. Stops at the first failure.
Implemented in AIAssistantDefaultImpl.

17.4.2.12 exportAuditLog()

```
virtual unsigned int AIAssistant_if::exportAuditLog (
            const std::string & csvPath ) const [pure virtual]
```

Exports all audit entries to a CSV file.

Parameters

csvPath	Destination file path.

Returns

Number of data rows written (excluding the header).

Implemented in AIAssistantDefaultImpl.

17.4.2.13 exportKnowledgeBaseAsMarkdown()

```
virtual std::string AIAssistant_if::exportKnowledgeBaseAsMarkdown ( ) const [pure virtual]
```

Renders the knowledge base as a Markdown document.
Intended to be injected into the system prompt so the AI model knows which GenESyS building blocks are available.
Implemented in AIAssistantDefaultImpl.

17.4.2.14 getApiKeyEnvironmentVariable()

```
virtual std::string AIAssistant_if::getApiKeyEnvironmentVariable ( ) const [pure virtual]
```

Returns the configured environment-variable name.
Implemented in AIAssistantDefaultImpl.

17.4.2.15 getAuditEntries()

```
virtual std::vector< AuditEntry > AIAssistant_if::getAuditEntries (
            unsigned int maxEntries = 200 ) const [pure virtual]
```

Returns up to `maxEntries` most-recent audit entries (oldest first).
Reads from the persistent audit log file. Returns an empty vector when the file does not yet exist or cannot be read.
Implemented in AIAssistantDefaultImpl.

17.4.2.16 getAuditLogPath()

```
virtual std::string AIAssistant_if::getAuditLogPath ( ) const  [pure virtual]
```

Returns the absolute path of the audit log file.
Implemented in AIAssistantDefaultImpl.

17.4.2.17 getBaseUrl()

```
virtual std::string AIAssistant_if::getBaseUrl ( ) const  [pure virtual]
```

Returns the currently configured base URL.
Implemented in AIAssistantDefaultImpl.

17.4.2.18 getConfiguration()

```
virtual AIAssistantConfiguration AIAssistant_if::getConfiguration ( ) const  [pure virtual]
```

Returns a copy of the current configuration record.
The returned struct never contains the API key value.
Implemented in AIAssistantDefaultImpl.

17.4.2.19 getKnowledgeBaseSummary()

```
virtual GenesysKnowledgeBaseSummary AIAssistant_if::getKnowledgeBaseSummary ( ) const  [pure
virtual]
```

Returns an aggregate summary of the knowledge base.
Implemented in AIAssistantDefaultImpl.

17.4.2.20 getKnowledgeItems()

```
virtual std::vector< GenesysKnowledgeItem > AIAssistant_if::getKnowledgeItems ( ) const  [pure
virtual]
```

Returns a copy of all registered knowledge items.
Implemented in AIAssistantDefaultImpl.

17.4.2.21 getLastDiagnostics()

```
virtual std::string AIAssistant_if::getLastDiagnostics ( ) const  [pure virtual]
```

Returns technical diagnostic output from the most recent operation.
Never contains API key values. Safe to log for debugging.
Implemented in AIAssistantDefaultImpl.

17.4.2.22 getLastError()

```
virtual std::string AIAssistant_if::getLastError ( ) const  [pure virtual]
```

Returns the error message from the most recent failed operation.
Implemented in AIAssistantDefaultImpl.

17.4.2.23 getMaskedApiKey()

```
virtual std::string AIAssistant_if::getMaskedApiKey ( ) const  [pure virtual]
```

Returns a redacted representation of the current API key.
Only the last four characters are exposed, preceded by "****". Returns an empty string when no key is set. Example: "****abcd".
Implemented in AIAssistantDefaultImpl.

17.4.2.24 getMaxOutputTokens()

```
virtual unsigned int AIAssistant_if::getMaxOutputTokens ( ) const  [pure virtual]
```

Returns the maximum output tokens limit.
Implemented in AIAssistantDefaultImpl.

17.4.2.25 getModelName()

```
virtual std::string AIAssistant_if::getModelName ( ) const  [pure virtual]
```

Returns the provider model identifier.
Implemented in AIAssistantDefaultImpl.

17.4.2.26 getProvider()

```
virtual AIProvider AIAssistant_if::getProvider ( ) const  [pure virtual]
```

Returns the currently configured AI provider.
Implemented in AIAssistantDefaultImpl.

17.4.2.27 getProviderClient()

```
virtual AIProviderClient_if * AIAssistant_if::getProviderClient ( ) const  [pure virtual]
```

Returns the currently attached provider client, or `nullptr`.
Implemented in AIAssistantDefaultImpl.

17.4.2.28 getProviderName()

```
virtual std::string AIAssistant_if::getProviderName ( ) const  [pure virtual]
```

Returns the human-readable provider label.
Implemented in AIAssistantDefaultImpl.

17.4.2.29 getReasoningEffort()

```
virtual ReasoningEffort AIAssistant_if::getReasoningEffort ( ) const  [pure virtual]
```

Returns the currently configured reasoning effort.
Implemented in AIAssistantDefaultImpl.

17.4.2.30 getSimulatorFacade()

```
virtual SimulatorFacade * AIAssistant_if::getSimulatorFacade ( ) const  [pure virtual]
```

Returns the currently attached SimulatorFacade pointer, or `nullptr`.
Implemented in AIAssistantDefaultImpl.

17.4.2.31 getState()

```
virtual ExecutionState AIAssistant_if::getState ( ) const  [pure virtual]
```

Returns the current lifecycle state.
Implemented in AIAssistantDefaultImpl.

17.4.2.32 getTemperature()

```
virtual double AIAssistant_if::getTemperature ( ) const  [pure virtual]
```

Returns the configured sampling temperature.
Implemented in AIAssistantDefaultImpl.

17.4.2.33 getTimeoutSeconds()

```
virtual unsigned int AIAssistant_if::getTimeoutSeconds ( ) const  [pure virtual]
```

Returns the HTTP request timeout in seconds.
Implemented in AIAssistantDefaultImpl.

17.4.2.34 hasApiKey()

```
virtual bool AIAssistant_if::hasApiKey ( ) const  [pure virtual]
```

Returns `true` when a non-empty API key is held in memory.
Implemented in AIAssistantDefaultImpl.

17.4.2.35 hasProviderClient()

```
virtual bool AIAssistant_if::hasProviderClient ( ) const  [pure virtual]
```

Returns `true` when a non-null provider client is attached.
Implemented in AIAssistantDefaultImpl.

17.4.2.36 hasSimulatorFacade()

```
virtual bool AIAssistant_if::hasSimulatorFacade ( ) const  [pure virtual]
```

Returns `true` when a non-null SimulatorFacade is attached.
Implemented in AIAssistantDefaultImpl.

17.4.2.37 loadApiKeyFromEnvironment()

```
virtual bool AIAssistant_if::loadApiKeyFromEnvironment ( )  [pure virtual]
```

Reads the API key from the configured environment variable.

Returns

`true` when the variable exists and holds a non-empty value.

Does nothing and returns `false` when apiKeyEnvironmentVariable is empty or the variable is not defined in the process environment. The loaded value is kept only in memory; it is never written to disk.
Implemented in AIAssistantDefaultImpl.

17.4.2.38 refreshKnowledgeBaseFromSimulator()

```
virtual bool AIAssistant_if::refreshKnowledgeBaseFromSimulator (
            std::string * message = nullptr )  [pure virtual]
```

Populates the knowledge base by introspecting the attached SimulatorFacade.

Parameters

message	Optional; receives a human-readable status or error message.

Returns

`true` when introspection succeeded and at least one item was added.

This method is a planned feature (Stage 2). In the current implementation it is a safe stub that returns `false` with a descriptive message.
Planned behaviour (Stage 2):

- Query SimulatorFacade::getDataDefinitionPluginTypenames() for registered types.
- Enumerate ModelComponent and ModelDataDefinition subtypes.
- Build GenesysKnowledgeItem entries with typenames, descriptions, and usage examples.

Todo (Stage 2): Implement full plugin introspection via SimulatorFacade.

Implemented in AIAssistantDefaultImpl.

17.4.2.39 runSimulation()

```
virtual AIAssistantResponse AIAssistant_if::runSimulation (
            const AIAssistantRequest & request )  [pure virtual]
```

Executes the configured simulation synchronously via SimulatorFacade.
Requires `allowSimulationExecution == true` and a valid SimulatorFacade. Calls SimulatorFacade↩ ::simStart() (blocking) and returns after all replications complete. `AIAssistantResponse::message` reports the number of replications run.
Implemented in AIAssistantDefaultImpl.

17.4.2.40 setApiKey()

```
virtual void AIAssistant_if::setApiKey (
            const std::string & apiKey )  [pure virtual]
```

Stores the API key in memory for the duration of the session.
The key is never written to disk or included in any log output. It is intentionally not retrievable via a getter.
Security guidance for production deployments:

- Prefer loadApiKeyFromEnvironment() over embedding keys in source code.
- For long-running services, consider OS keyring / secret-store integration (planned for Stage 8; not implemented in this version).

Implemented in AIAssistantDefaultImpl.

17.4.2.41 setApiKeyEnvironmentVariable()

```
virtual void AIAssistant_if::setApiKeyEnvironmentVariable (
            const std::string & variableName )  [pure virtual]
```

Stores the name of the environment variable that holds the API key.
This is the variable *name*, not the value. Call loadApiKeyFromEnvironment() to actually read the variable and load it into memory.
Implemented in AIAssistantDefaultImpl.

17.4.2.42 setBaseUrl()

```
virtual void AIAssistant_if::setBaseUrl (
            const std::string & baseUrl )  [pure virtual]
```

Sets the base URL; required when provider == CustomHttp.
The URL should point to a chat-completions-compatible endpoint. Example: "http://localhost:11434/v1" for a local Ollama server.
Implemented in AIAssistantDefaultImpl.

17.4.2.43 setConfiguration()

```
virtual void AIAssistant_if::setConfiguration (
            const AIAssistantConfiguration & configuration )  [pure virtual]
```

Replaces the entire configuration record.
Does not transfer the API key, which must be set separately via setApiKey() or loadApiKeyFromEnvironment().
Implemented in AIAssistantDefaultImpl.

17.4.2.44 setMaxOutputTokens()

```
virtual void AIAssistant_if::setMaxOutputTokens (
            unsigned int maxOutputTokens )  [pure virtual]
```

Sets the maximum number of output tokens to request.
Implemented in AIAssistantDefaultImpl.

17.4.2.45 setModelName()

```
virtual void AIAssistant_if::setModelName (
            const std::string & modelName )  [pure virtual]
```

Sets the provider model identifier (e.g. "gpt-4o", "claude-opus-4-7").
Implemented in AIAssistantDefaultImpl.

17.4.2.46 setProvider()

```
virtual void AIAssistant_if::setProvider (
            AIProvider provider )  [pure virtual]
```

Sets the AI backend provider.
Implemented in AIAssistantDefaultImpl.

17.4.2.47 setProviderClient()

```
virtual void AIAssistant_if::setProviderClient (
            AIProviderClient_if * client )  [pure virtual]
```

Injects an external provider client (non-owning).
The assistant does not take ownership; the caller is responsible for the lifetime of the injected client. The previously held client pointer (if any) is released without deletion.
To let the assistant create and own its own client based on the current AIAssistantConfiguration, use createAnd↩ AttachProviderClient() in AIAssistantDefaultImpl instead.
Implemented in AIAssistantDefaultImpl.

17.4.2.48 setProviderName()

```
virtual void AIAssistant_if::setProviderName (
            const std::string & providerName )  [pure virtual]
```

Sets a human-readable provider label.
Implemented in AIAssistantDefaultImpl.

17.4.2.49 setReasoningEffort()

```
virtual void AIAssistant_if::setReasoningEffort (
            ReasoningEffort effort )  [pure virtual]
```

Sets the desired reasoning effort hint.
Implemented in AIAssistantDefaultImpl.

17.4.2.50 setSimulatorFacade()

```
virtual void AIAssistant_if::setSimulatorFacade (
            SimulatorFacade * simulatorFacade )  [pure virtual]
```

Attaches a non-owning SimulatorFacade pointer.
The assistant does not take ownership of the facade. The caller is responsible for keeping the facade alive for the duration of any operation that uses it.
Implemented in AIAssistantDefaultImpl.

17.4.2.51 setTemperature()

```
virtual void AIAssistant_if::setTemperature (
            double temperature )  [pure virtual]
```

Sets the sampling temperature (0.0 = deterministic, 1.0 = creative).
Not all providers support this parameter; implementations may ignore it when the provider uses a fixed sampling scheme.
Implemented in AIAssistantDefaultImpl.

17.4.2.52 setTimeoutSeconds()

```
virtual void AIAssistant_if::setTimeoutSeconds (
            unsigned int timeoutSeconds )  [pure virtual]
```

Sets the HTTP request timeout in seconds.
Implemented in AIAssistantDefaultImpl.
The documentation for this class was generated from the following file:

- tools/AIAssistant/AIAssistant_if.h

17.5 AIAssistantConfiguration Struct Reference

Full configuration record for the AI assistant.
`#include <tools/AIAssistant/AIAssistant_if.h>`

Public Attributes

- AIProvider **provider** = AIProvider::Undefined
- std::string **providerName**

  Human-readable label (e.g. "OpenAI GPT-4o").
- std::string **modelName**

  Provider model identifier (e.g. "gpt-4o").
- std::string **baseUrl**

  Required when provider == CustomHttp.
- std::string **apiKeyEnvironmentVariable**

  Name of env-var holding the API key, not the key itself.
- std::string **organizationId**

  Optional; passed in request headers when non-empty.
- std::string **projectId**

  Optional; passed in request headers when non-empty.
- ReasoningEffort **reasoningEffort** = ReasoningEffort::Medium
- double **temperature** = 0.2
- unsigned int **maxOutputTokens** = 4096
- unsigned int **timeoutSeconds** = 120
- bool **allowNetworkAccess** = false

  Permit outbound HTTP calls to the AI provider.
- bool **allowModelMutation** = false

  Permit creation or modification of GenESyS models.
- bool **allowSimulationExecution** = false

  Permit starting simulation replications.
- bool **allowFileSystemWrites** = false

  Permit writing model/result files to disk.
- bool sandboxEnabled = false

  When `true`, clears the current model before every buildModel() call.
- bool dryRun = false

  When `true`, workflow methods simulate their actions without mutating the simulator state or consuming network quota beyond what is needed for planning.

17.5.1 Detailed Description

Full configuration record for the AI assistant.
Security contract:

- `apiKeyEnvironmentVariable` holds only the *name* of an environment variable, never the secret value itself.
- The actual API key must be supplied via AIAssistant_if::setApiKey() or loaded at runtime via AIAssistant_if::loadApiKeyFromEnvironment().
- This struct must not be serialised to disk without first redacting the providerName field if it could be confused with a secret, and ensuring no key value has been stored here.

Permission flags default to `false` (deny-by-default). Each capability that touches external systems or mutates simulator state must be explicitly enabled before use.

17.5.2 Member Data Documentation

17.5.2.1 dryRun

`bool AIAssistantConfiguration::dryRun = false`
When `true`, workflow methods simulate their actions without mutating the simulator state or consuming network quota beyond what is needed for planning.

- buildModel() — generates the .gen text via LLM but does NOT load it.
- configureSimulation() — extracts parameters via LLM but does NOT apply them.
- runSimulation() — reports what would run without calling simStart().
- collectResults() — reports the count of available responses without rendering.
- analyzePrompt() / createModelPlan() — always execute regardless of this flag.

Use dry-run to verify the LLM output before committing changes.

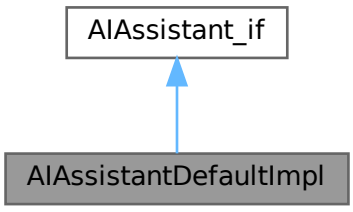
17.5.2.2 sandboxEnabled

`bool AIAssistantConfiguration::sandboxEnabled = false`
When `true`, clears the current model before every buildModel() call.
Sandbox mode provides a clean-room environment: any existing model elements are discarded so the LLM always starts from an empty canvas. This prevents the AI from accidentally inheriting components from a model the user was editing. The cleared state is permanent; no undo is provided.
The documentation for this struct was generated from the following file:

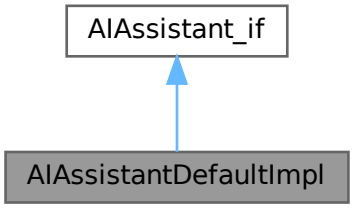
- tools/AIAssistant/AIAssistant_if.h

17.6 AIAssistantDefaultImpl Class Reference

Initial concrete scaffold for the GenESyS AI Assistant.
`#include <tools/AIAssistant/AIAssistantDefaultImpl.h>`
Inheritance diagram for AIAssistantDefaultImpl:



Collaboration diagram for AIAssistantDefaultImpl:



Public Member Functions

- **AIAssistantDefaultImpl** (const AIAssistantDefaultImpl &)=delete
- AIAssistantDefaultImpl & **operator=** (const AIAssistantDefaultImpl &)=delete
- virtual void setConfiguration (const AIAssistantConfiguration &configuration) override

  Replaces the entire configuration record.
- virtual AIAssistantConfiguration getConfiguration () const override

  Returns a copy of the current configuration record.
- virtual void setProvider (AIProvider provider) override

  Sets the AI backend provider.
- virtual AIProvider getProvider () const override

  Returns the currently configured AI provider.
- virtual void setProviderName (const std::string &providerName) override

  Sets a human-readable provider label.
- virtual std::string getProviderName () const override

  Returns the human-readable provider label.
- virtual void setModelName (const std::string &modelName) override

  Sets the provider model identifier (e.g. "gpt-4o", "claude-opus-4-7").
- virtual std::string getModelName () const override

  Returns the provider model identifier.
- virtual void setBaseUrl (const std::string &baseUrl) override

  Sets the base URL; required when provider == CustomHttp.
- virtual std::string getBaseUrl () const override

  Returns the currently configured base URL.
- virtual void setReasoningEffort (ReasoningEffort effort) override

  Sets the desired reasoning effort hint.
- virtual ReasoningEffort getReasoningEffort () const override

  Returns the currently configured reasoning effort.
- virtual void setTemperature (double temperature) override

  Sets the sampling temperature (0.0 = deterministic, 1.0 = creative).
- virtual double getTemperature () const override

  Returns the configured sampling temperature.
- virtual void setMaxOutputTokens (unsigned int maxOutputTokens) override

  Sets the maximum number of output tokens to request.
- virtual unsigned int getMaxOutputTokens () const override

  Returns the maximum output tokens limit.

- virtual void setTimeoutSeconds (unsigned int timeoutSeconds) override

  Sets the HTTP request timeout in seconds.
- virtual unsigned int getTimeoutSeconds () const override

  Returns the HTTP request timeout in seconds.
- virtual void setApiKey (const std::string &apiKey) override

  Stores the API key in memory for the duration of the session.
- virtual void clearApiKey () override

  Erases the in-memory API key.
- virtual bool hasApiKey () const override

  Returns `true` when a non-empty API key is held in memory.
- virtual std::string getMaskedApiKey () const override

  Returns a redacted representation of the current API key.
- virtual void setApiKeyEnvironmentVariable (const std::string &variableName) override

  Stores the name of the environment variable that holds the API key.
- virtual std::string getApiKeyEnvironmentVariable () const override

  Returns the configured environment-variable name.
- virtual bool loadApiKeyFromEnvironment () override

  Reads the API key from the configured environment variable.
- virtual void setSimulatorFacade (SimulatorFacade ∗simulatorFacade) override

  Attaches a non-owning SimulatorFacade pointer.
- virtual SimulatorFacade ∗ getSimulatorFacade () const override

  Returns the currently attached SimulatorFacade pointer, or `nullptr`.
- virtual bool hasSimulatorFacade () const override

  Returns `true` when a non-null SimulatorFacade is attached.
- virtual void setProviderClient (AIProviderClient_if ∗client) override

  Injects an external provider client (non-owning).
- virtual AIProviderClient_if ∗ getProviderClient () const override

  Returns the currently attached provider client, or `nullptr`.
- virtual bool hasProviderClient () const override

  Returns `true` when a non-null provider client is attached.
- bool createAndAttachProviderClient ()

  Creates an owned provider client from the current configuration and attaches it, replacing any previously held client pointer.
- virtual void clearKnowledgeBase () override

  Removes all items from the knowledge base.
- virtual void addKnowledgeItem (const GenesysKnowledgeItem &item) override

  Appends one item to the knowledge base.
- virtual std::vector< GenesysKnowledgeItem > getKnowledgeItems () const override

  Returns a copy of all registered knowledge items.
- virtual GenesysKnowledgeBaseSummary getKnowledgeBaseSummary () const override

  Returns an aggregate summary of the knowledge base.
- virtual std::string exportKnowledgeBaseAsMarkdown () const override

  Renders the knowledge base as a Markdown document.
- virtual bool refreshKnowledgeBaseFromSimulator (std::string ∗message=nullptr) override

  Populates the knowledge base by introspecting the attached SimulatorFacade.
- virtual bool checkReady (std::string ∗message=nullptr) const override

  Checks whether the assistant has enough configuration to operate.
- virtual AIAssistantResponse analyzePrompt (const AIAssistantRequest &request) override

  Parses and summarises a user prompt without calling any external service.
- virtual AIAssistantResponse createModelPlan (const AIAssistantRequest &request) override

  Produces a structured plan mapping the prompt to GenESyS components.

- virtual AIAssistantResponse buildModel (const AIAssistantRequest &request) override

  Constructs a GenESyS model via SimulatorFacade using an LLM-generated .gen file.
- virtual AIAssistantResponse configureSimulation (const AIAssistantRequest &request) override

  Overrides ModelSimulation parameters on the current model.
- virtual AIAssistantResponse runSimulation (const AIAssistantRequest &request) override

  Executes the configured simulation synchronously via SimulatorFacade.
- virtual AIAssistantResponse collectResults (const AIAssistantRequest &request) override

  Collects and formats simulation results from the current model.
- virtual AIAssistantResponse execute (const AIAssistantRequest &request) override

  High-level dispatcher that orchestrates the full pipeline based on request flags.
- virtual ExecutionState getState () const override

  Returns the current lifecycle state.
- virtual std::string getLastError () const override

  Returns the error message from the most recent failed operation.
- virtual std::string getLastDiagnostics () const override

  Returns technical diagnostic output from the most recent operation.
- virtual void clearDiagnostics () override

  Clears both the last error and diagnostics strings.
- virtual std::vector< AuditEntry > getAuditEntries (unsigned int maxEntries) const override

  Returns up to `maxEntries` most-recent audit entries (oldest first).
- virtual std::string getAuditLogPath () const override

  Returns the absolute path of the audit log file.
- virtual unsigned int exportAuditLog (const std::string &csvPath) const override

  Exports all audit entries to a CSV file.

17.6.1 Detailed Description

Initial concrete scaffold for the GenESyS AI Assistant.
This class implements AIAssistant_if and provides:

- Full configuration management including secure API key handling.
- A knowledge base backed by a simple in-memory vector.
- A checkReady() implementation that validates minimum configuration.
- Safe stubs for all workflow methods (analyzePrompt, buildModel, etc.) that return ExecutionState::Not↩ Implemented until the corresponding stage is implemented.

Current status:

- Stage 1: configuration, security, knowledge base, diagnostics — complete.
- Stage 2 and above: intentionally deferred; see

  Todo annotations.

17.6.1.1 Security model

- The API key is held in a private `std::string` member, never written to disk or included in trace output.
- No getter exposes the key value; only getMaskedApiKey() is available.
- Copy construction and copy assignment are deleted to prevent accidental key propagation through object copying.

17.6.1.2 Ownership

- The SimulatorFacade pointer is non-owning; this class does not delete it.

17.6.2 Member Function Documentation

17.6.2.1 addKnowledgeItem()

```
void AIAssistantDefaultImpl::addKnowledgeItem (
            const GenesysKnowledgeItem & item )  [override], [virtual]
```

Appends one item to the knowledge base.
Implements AIAssistant_if.

17.6.2.2 analyzePrompt()

```
AIAssistantResponse AIAssistantDefaultImpl::analyzePrompt (
            const AIAssistantRequest & request )  [override], [virtual]
```

Parses and summarises a user prompt without calling any external service.
Stage 3 will implement actual LLM calls. Current stub returns NotImplemented.
Implements AIAssistant_if.

17.6.2.3 buildModel()

```
AIAssistantResponse AIAssistantDefaultImpl::buildModel (
            const AIAssistantRequest & request )  [override], [virtual]
```

Constructs a GenESyS model via SimulatorFacade using an LLM-generated .gen file.
Requires `allowModelMutation` == `true`, `allowNetworkAccess` == `true`, a valid SimulatorFacade, and an attached provider client.
Behaviour (Stage 5):

1. If `request.modelPlan` is non-empty, it is used as the model specification. Otherwise, the `request` description fields are used directly.
2. An LLM call is made requesting a GenESyS model language (.gen) file.
3. The generated .gen content is written to a temp file and loaded via SimulatorFacade::loadModel(). The temp file is deleted immediately after loading.
4. `AIAssistantResponse::generatedModelLanguage` holds the generated .gen text.

Implements AIAssistant_if.

17.6.2.4 checkReady()

```
bool AIAssistantDefaultImpl::checkReady (
            std::string * message = nullptr ) const  [override], [virtual]
```

Checks whether the assistant has enough configuration to operate.

Parameters

message	Optional; receives a description of the first unsatisfied condition, or a confirmation message when ready.

Returns

`true` when all minimum requirements are met for prompt analysis.

Checks performed (in order):

1. Provider is not `AIProvider::Undefined`.
2. Model name is not empty.
3. For OpenAI/Anthropic/CustomHttp: an API key is held in memory or an environment-variable name is configured.
4. For CustomHttp: baseUrl is not empty.

Note: SimulatorFacade and permission flags are checked separately when model-mutation or simulation-execution operations are requested.
Implements AIAssistant_if.

17.6.2.5 clearApiKey()

```
void AIAssistantDefaultImpl::clearApiKey ( )  [override], [virtual]
```

Erases the in-memory API key.
Implementations should overwrite the key buffer before clearing to reduce the window of exposure. Note that standard C++ provides no guarantee that the compiler will not optimise away the overwrite; platform-specific secure zeroing (explicit_bzero, SecureZeroMemory) should be used in security-critical contexts.
Implements AIAssistant_if.

17.6.2.6 clearDiagnostics()

```
void AIAssistantDefaultImpl::clearDiagnostics ( )  [override], [virtual]
```

Clears both the last error and diagnostics strings.
Implements AIAssistant_if.

17.6.2.7 clearKnowledgeBase()

```
void AIAssistantDefaultImpl::clearKnowledgeBase ( )  [override], [virtual]
```

Removes all items from the knowledge base.
Implements AIAssistant_if.

17.6.2.8 collectResults()

```
AIAssistantResponse AIAssistantDefaultImpl::collectResults (
            const AIAssistantRequest & request )  [override], [virtual]
```

Collects and formats simulation results from the current model.
Iterates SimulatorFacade::modelGetResponses() and renders them as a Markdown table in `AIAssistantResponse::message`
Implements AIAssistant_if.

17.6.2.9 configureSimulation()

```
AIAssistantResponse AIAssistantDefaultImpl::configureSimulation (
            const AIAssistantRequest & request )  [override], [virtual]
```

Overrides ModelSimulation parameters on the current model.
Requires `allowModelMutation` == `true` and a valid SimulatorFacade.
Behaviour (Stage 5):

1. Makes a small LLM call to extract structured simulation parameters from `request.experiment↩ Description` (replications, replication length, warm-up period, time units).
2. Applies the extracted values via SimulatorFacade::simSet∗() methods.
3. Parameters already set by a loaded .gen file are overridden only when the LLM extraction yields non-zero values.

Implements AIAssistant_if.

17.6.2.10 createAndAttachProviderClient()

```
bool AIAssistantDefaultImpl::createAndAttachProviderClient ( )
```

Creates an owned provider client from the current configuration and attaches it, replacing any previously held client pointer.

Returns

`true` when a suitable client was created and attached.

The assistant takes ownership of the created client and deletes it in the destructor or on the next call to createAndAttachProviderClient(). Clients injected via setProviderClient() are never deleted by the assistant.
Provider selection is based on AIAssistantConfiguration::provider:

- AIProvider::OpenAI → OpenAIProviderClientImpl
- AIProvider::Anthropic → AnthropicProviderClientImpl
- AIProvider::Local → LocalProviderClientImpl
- AIProvider::CustomHttp → LocalProviderClientImpl with custom base URL
- AIProvider::Undefined → returns false, no client created

17.6.2.11 createModelPlan()

```
AIAssistantResponse AIAssistantDefaultImpl::createModelPlan (
            const AIAssistantRequest & request )  [override], [virtual]
```

Produces a structured plan mapping the prompt to GenESyS components.
Stage 4 will implement prompt-to-plan translation. Current stub returns NotImplemented.
Implements AIAssistant_if.

17.6.2.12 execute()

```
AIAssistantResponse AIAssistantDefaultImpl::execute (
            const AIAssistantRequest & request )  [override], [virtual]
```

High-level dispatcher that orchestrates the full pipeline based on request flags.
Calls analyzePrompt, createModelPlan, buildModel, configureSimulation, runSimulation and collectResults in sequence according to the enabled flags in `request`. Stops at the first failure.
Implements AIAssistant_if.

17.6.2.13 exportAuditLog()

```
unsigned int AIAssistantDefaultImpl::exportAuditLog (
            const std::string & csvPath ) const  [override], [virtual]
```

Exports all audit entries to a CSV file.

Parameters

csvPath	Destination file path.

Returns

Number of data rows written (excluding the header).

Implements AIAssistant_if.

17.6.2.14 exportKnowledgeBaseAsMarkdown()

```
std::string AIAssistantDefaultImpl::exportKnowledgeBaseAsMarkdown ( ) const  [override], [virtual]
```

Renders the knowledge base as a Markdown document.
Intended to be injected into the system prompt so the AI model knows which GenESyS building blocks are available.
Implements AIAssistant_if.

17.6.2.15 getApiKeyEnvironmentVariable()

```
std::string AIAssistantDefaultImpl::getApiKeyEnvironmentVariable ( ) const  [override], [virtual]
```

Returns the configured environment-variable name.
Implements AIAssistant_if.

17.6.2.16 getAuditEntries()

```
std::vector< AuditEntry > AIAssistantDefaultImpl::getAuditEntries (
            unsigned int maxEntries ) const  [override], [virtual]
```

Returns up to `maxEntries` most-recent audit entries (oldest first).
Reads from the persistent audit log file. Returns an empty vector when the file does not yet exist or cannot be read.
Implements AIAssistant_if.

17.6.2.17 getAuditLogPath()

```
std::string AIAssistantDefaultImpl::getAuditLogPath ( ) const  [override], [virtual]
```

Returns the absolute path of the audit log file.
Implements AIAssistant_if.

17.6.2.18 getBaseUrl()

```
std::string AIAssistantDefaultImpl::getBaseUrl ( ) const  [override], [virtual]
```

Returns the currently configured base URL.
Implements AIAssistant_if.

17.6.2.19 getConfiguration()

```
AIAssistantConfiguration AIAssistantDefaultImpl::getConfiguration ( ) const  [override], [virtual]
```

Returns a copy of the current configuration record.
The returned struct never contains the API key value.
Implements AIAssistant_if.

17.6.2.20 getKnowledgeBaseSummary()

```
GenesysKnowledgeBaseSummary AIAssistantDefaultImpl::getKnowledgeBaseSummary ( ) const  [override],
[virtual]
```

Returns an aggregate summary of the knowledge base.
Implements AIAssistant_if.

17.6.2.21 getKnowledgeItems()

```
std::vector< GenesysKnowledgeItem > AIAssistantDefaultImpl::getKnowledgeItems ( ) const  [override],
[virtual]
```

Returns a copy of all registered knowledge items.
Implements AIAssistant_if.

17.6.2.22 getLastDiagnostics()

```
std::string AIAssistantDefaultImpl::getLastDiagnostics ( ) const  [override], [virtual]
```

Returns technical diagnostic output from the most recent operation.
Never contains API key values. Safe to log for debugging.
Implements AIAssistant_if.

17.6.2.23 getLastError()

```
std::string AIAssistantDefaultImpl::getLastError ( ) const  [override], [virtual]
```

Returns the error message from the most recent failed operation.
Implements AIAssistant_if.

17.6.2.24 getMaskedApiKey()

`std::string AIAssistantDefaultImpl::getMaskedApiKey ( ) const [override], [virtual]`
Returns a redacted representation of the current API key.
Only the last four characters are exposed, preceded by "****". Returns an empty string when no key is set. Example: "****abcd".
Implements AIAssistant_if.

17.6.2.25 getMaxOutputTokens()

`unsigned int AIAssistantDefaultImpl::getMaxOutputTokens ( ) const [override], [virtual]`
Returns the maximum output tokens limit.
Implements AIAssistant_if.

17.6.2.26 getModelName()

`std::string AIAssistantDefaultImpl::getModelName ( ) const [override], [virtual]`
Returns the provider model identifier.
Implements AIAssistant_if.

17.6.2.27 getProvider()

`AIProvider AIAssistantDefaultImpl::getProvider ( ) const [override], [virtual]`
Returns the currently configured AI provider.
Implements AIAssistant_if.

17.6.2.28 getProviderClient()

`AIProviderClient_if * AIAssistantDefaultImpl::getProviderClient ( ) const [override], [virtual]`
Returns the currently attached provider client, or `nullptr`.
Implements AIAssistant_if.

17.6.2.29 getProviderName()

`std::string AIAssistantDefaultImpl::getProviderName ( ) const [override], [virtual]`
Returns the human-readable provider label.
Implements AIAssistant_if.

17.6.2.30 getReasoningEffort()

`ReasoningEffort AIAssistantDefaultImpl::getReasoningEffort ( ) const [override], [virtual]`
Returns the currently configured reasoning effort.
Implements AIAssistant_if.

17.6.2.31 getSimulatorFacade()

`SimulatorFacade * AIAssistantDefaultImpl::getSimulatorFacade ( ) const [override], [virtual]`
Returns the currently attached SimulatorFacade pointer, or `nullptr`.
Implements AIAssistant_if.

17.6.2.32 getState()

`ExecutionState AIAssistantDefaultImpl::getState ( ) const [override], [virtual]`
Returns the current lifecycle state.
Implements AIAssistant_if.

17.6.2.33 getTemperature()

`double AIAssistantDefaultImpl::getTemperature ( ) const [override], [virtual]`
Returns the configured sampling temperature.
Implements AIAssistant_if.

17.6.2.34 getTimeoutSeconds()

```
unsigned int AIAssistantDefaultImpl::getTimeoutSeconds ( ) const  [override], [virtual]
```

Returns the HTTP request timeout in seconds.
Implements AIAssistant_if.

17.6.2.35 hasApiKey()

```
bool AIAssistantDefaultImpl::hasApiKey ( ) const  [override], [virtual]
```

Returns `true` when a non-empty API key is held in memory.
Implements AIAssistant_if.

17.6.2.36 hasProviderClient()

```
bool AIAssistantDefaultImpl::hasProviderClient ( ) const  [override], [virtual]
```

Returns `true` when a non-null provider client is attached.
Implements AIAssistant_if.

17.6.2.37 hasSimulatorFacade()

```
bool AIAssistantDefaultImpl::hasSimulatorFacade ( ) const  [override], [virtual]
```

Returns `true` when a non-null SimulatorFacade is attached.
Implements AIAssistant_if.

17.6.2.38 loadApiKeyFromEnvironment()

```
bool AIAssistantDefaultImpl::loadApiKeyFromEnvironment ( )  [override], [virtual]
```

Reads the API key from the configured environment variable.

Returns

`true` when the variable exists and holds a non-empty value.

Does nothing and returns `false` when apiKeyEnvironmentVariable is empty or the variable is not defined in the process environment. The loaded value is kept only in memory; it is never written to disk.
Implements AIAssistant_if.

17.6.2.39 refreshKnowledgeBaseFromSimulator()

```
bool AIAssistantDefaultImpl::refreshKnowledgeBaseFromSimulator (
            std::string * message = nullptr )  [override], [virtual]
```

Populates the knowledge base by introspecting the attached SimulatorFacade.

Parameters

message	Optional; receives a human-readable status or error message.

Returns

`true` when introspection succeeded and at least one item was added.

This method is a planned feature (Stage 2). In the current implementation it is a safe stub that returns `false` with a descriptive message.
Planned behaviour (Stage 2):

- Query SimulatorFacade::getDataDefinitionPluginTypenames() for registered types.
- Enumerate ModelComponent and ModelDataDefinition subtypes.
- Build GenesysKnowledgeItem entries with typenames, descriptions, and usage examples.

Todo (Stage 2): Implement full plugin introspection via SimulatorFacade.

Implements AIAssistant_if.

17.6.2.40 runSimulation()

```
AIAssistantResponse AIAssistantDefaultImpl::runSimulation (
            const AIAssistantRequest & request )  [override], [virtual]
```

Executes the configured simulation synchronously via SimulatorFacade.
Requires `allowSimulationExecution == true` and a valid SimulatorFacade. Calls SimulatorFacade↩ ::simStart() (blocking) and returns after all replications complete. `AIAssistantResponse::message` reports the number of replications run.
Implements AIAssistant_if.

17.6.2.41 setApiKey()

```
void AIAssistantDefaultImpl::setApiKey (
            const std::string & apiKey )  [override], [virtual]
```

Stores the API key in memory for the duration of the session.
The key is never written to disk or included in any log output. It is intentionally not retrievable via a getter.
Security guidance for production deployments:

- Prefer loadApiKeyFromEnvironment() over embedding keys in source code.
- For long-running services, consider OS keyring / secret-store integration (planned for Stage 8; not implemented in this version).

Implements AIAssistant_if.

17.6.2.42 setApiKeyEnvironmentVariable()

```
void AIAssistantDefaultImpl::setApiKeyEnvironmentVariable (
            const std::string & variableName )  [override], [virtual]
```

Stores the name of the environment variable that holds the API key.
This is the variable *name*, not the value. Call loadApiKeyFromEnvironment() to actually read the variable and load it into memory.
Implements AIAssistant_if.

17.6.2.43 setBaseUrl()

```
void AIAssistantDefaultImpl::setBaseUrl (
            const std::string & baseUrl )  [override], [virtual]
```

Sets the base URL; required when provider == CustomHttp.
The URL should point to a chat-completions-compatible endpoint. Example: "http://localhost:11434/v1" for a local Ollama server.
Implements AIAssistant_if.

17.6.2.44 setConfiguration()

```
void AIAssistantDefaultImpl::setConfiguration (
            const AIAssistantConfiguration & configuration )  [override], [virtual]
```

Replaces the entire configuration record.
Does not transfer the API key, which must be set separately via setApiKey() or loadApiKeyFromEnvironment().
Implements AIAssistant_if.

17.6.2.45 setMaxOutputTokens()

```
void AIAssistantDefaultImpl::setMaxOutputTokens (
            unsigned int maxOutputTokens )  [override], [virtual]
```

Sets the maximum number of output tokens to request.
Implements AIAssistant_if.

17.6.2.46 setModelName()

```
void AIAssistantDefaultImpl::setModelName (
            const std::string & modelName )  [override], [virtual]
```

Sets the provider model identifier (e.g. "gpt-4o", "claude-opus-4-7").
Implements AIAssistant_if.

17.6.2.47 setProvider()

```
void AIAssistantDefaultImpl::setProvider (
            AIProvider provider )  [override], [virtual]
```

Sets the AI backend provider.
Implements AIAssistant_if.

17.6.2.48 setProviderClient()

```
void AIAssistantDefaultImpl::setProviderClient (
            AIProviderClient_if * client )  [override], [virtual]
```

Injects an external provider client (non-owning).
The assistant does not take ownership; the caller is responsible for the lifetime of the injected client. The previously held client pointer (if any) is released without deletion.
To let the assistant create and own its own client based on the current AIAssistantConfiguration, use createAndAttachProviderClient() in AIAssistantDefaultImpl instead.
Implements AIAssistant_if.

17.6.2.49 setProviderName()

```
void AIAssistantDefaultImpl::setProviderName (
            const std::string & providerName )  [override], [virtual]
```

Sets a human-readable provider label.
Implements AIAssistant_if.

17.6.2.50 setReasoningEffort()

```
void AIAssistantDefaultImpl::setReasoningEffort (
            ReasoningEffort effort )  [override], [virtual]
```

Sets the desired reasoning effort hint.
Implements AIAssistant_if.

17.6.2.51 setSimulatorFacade()

```
void AIAssistantDefaultImpl::setSimulatorFacade (
            SimulatorFacade * simulatorFacade )  [override], [virtual]
```

Attaches a non-owning SimulatorFacade pointer.
The assistant does not take ownership of the facade. The caller is responsible for keeping the facade alive for the duration of any operation that uses it.
Implements AIAssistant_if.

17.6.2.52 setTemperature()

```
void AIAssistantDefaultImpl::setTemperature (
            double temperature )  [override], [virtual]
```

Sets the sampling temperature (0.0 = deterministic, 1.0 = creative).
Not all providers support this parameter; implementations may ignore it when the provider uses a fixed sampling scheme.
Implements AIAssistant_if.

17.6.2.53 setTimeoutSeconds()

```
void AIAssistantDefaultImpl::setTimeoutSeconds (
            unsigned int timeoutSeconds )  [override], [virtual]
```

Sets the HTTP request timeout in seconds.
Implements AIAssistant_if.
The documentation for this class was generated from the following files:

- tools/AIAssistant/AIAssistantDefaultImpl.h
- tools/AIAssistant/AIAssistantDefaultImpl.cpp

17.7 AIAssistantRequest Struct Reference

Input to a single AI assistant request.
`#include <tools/AIAssistant/AIAssistant_if.h>`

Public Attributes

- std::string **userPrompt**

  Free-text prompt from the user.
- std::string **systemContext**

  Optional extra context to prepend to the system message.
- std::string **modelDescription**

  Description of the system to model (may be empty).
- std::string **experimentDescription**

  Description of the simulation experiment.
- std::string modelPlan

  Structured model plan (Markdown) to guide model construction.
- bool **buildModel** = false

  Request model construction via SimulatorFacade.
- bool **configureSimulation** = false

  Request ModelSimulation configuration.
- bool **runSimulation** = false

  Request simulation execution.
- bool **collectResults** = false

  Request result collection and reporting.

17.7.1 Detailed Description

Input to a single AI assistant request.
Not all fields need to be populated for every request. The assistant interprets the populated subset and ignores empty strings. Boolean flags signal intent; the assistant may reject them if the corresponding AIAssistant↩ Configuration::allow∗ flag is not set.

17.7.2 Member Data Documentation

17.7.2.1 modelPlan

```
std::string AIAssistantRequest::modelPlan
```

Structured model plan (Markdown) to guide model construction.
When non-empty, buildModel() uses this plan directly without calling the LLM a second time to produce a new plan. Populated automatically by execute() from the output of createModelPlan(). Callers may also supply a previously generated or manually written plan.
The documentation for this struct was generated from the following file:

- tools/AIAssistant/AIAssistant_if.h

17.8 AIAssistantResponse Struct Reference

Output produced by a single AI assistant operation.
`#include <tools/AIAssistant/AIAssistant_if.h>`

Public Attributes

- bool **success** = false
- ExecutionState **state** = ExecutionState::NotImplemented
- std::string **message**

  Human-readable outcome summary or error description.
- std::string **generatedPlan**

  Intermediate structured plan (Markdown or JSON fragment).
- std::string **generatedModelLanguage**

  GenESyS model language text, when generated.
- std::string **diagnostics**

  Technical detail for debugging; not shown to end users.

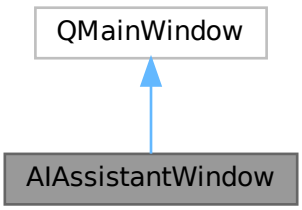
17.8.1 Detailed Description

Output produced by a single AI assistant operation.
The documentation for this struct was generated from the following file:

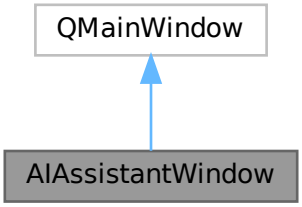
- tools/AIAssistant/AIAssistant_if.h

17.9 AIAssistantWindow Class Reference

Inheritance diagram for AIAssistantWindow:



Collaboration diagram for AIAssistantWindow:



Public Member Functions

- **AIAssistantWindow** (Simulator *simulator, QWidget *parent=nullptr)

The documentation for this class was generated from the following files:

- applications/gui/ai_assistant/AIAssistantWindow.h
- applications/gui/ai_assistant/AIAssistantWindow.cpp

17.10 AIAuditLog Class Reference

Append-only audit log for AI assistant operations.
`#include <tools/AIAssistant/AIAuditLog.h>`

Public Member Functions

- **AIAuditLog** (const std::string &logFilePath="")
- **AIAuditLog** (const AIAuditLog &)=delete
- AIAuditLog & **operator=** (const AIAuditLog &)=delete
- void **append** (const AuditEntry &entry)

  Appends one entry to the log file (creates directory if needed).
- std::vector< AuditEntry > loadRecentEntries (unsigned int maxEntries=200) const

  Reads up to `maxEntries` most-recent entries.
- std::string **logFilePath** () const

  Returns the absolute path of the log file.
- unsigned int exportCsv (const std::string &csvPath) const

  Exports all log entries to a CSV file.

Static Public Member Functions

- static std::string **defaultLogFilePath** ()

  Returns the platform-default log file path.

17.10.1 Detailed Description

Append-only audit log for AI assistant operations.
Entries are written in JSON Lines format to a file under $XDG_DATA_HOME/genesys/ (typically ~/.local/share/genesys/ai_audit.log). The log never contains API key values. The directory is created on first write.

17.10.2 Member Function Documentation

17.10.2.1 exportCsv()

```
unsigned int AIAuditLog::exportCsv (
            const std::string & csvPath ) const
```

Exports all log entries to a CSV file.

Returns

Number of rows written (excluding the header).

17.10.2.2 loadRecentEntries()

```
std::vector< AuditEntry > AIAuditLog::loadRecentEntries (
            unsigned int maxEntries = 200 ) const
```

Reads up to `maxEntries` most-recent entries.

Returns

Entries in chronological order (oldest first).

The documentation for this class was generated from the following files:

- tools/AIAssistant/AIAuditLog.h
- tools/AIAssistant/AIAuditLog.cpp

17.11 AIConversationConfiguration Struct Reference

Public Attributes

- AIConversationProvider **provider** = AIConversationProvider::Local
- std::string **model** = "llama3.2"
- std::string **baseUrl**
- std::string **apiKeyEnvironmentVariable**
- std::string **systemInstructions**
- double **temperature** = 0.2
- unsigned int **maxOutputTokens** = 512
- unsigned int **timeoutSeconds** = 120
- bool **highReasoningMode** = false
- bool **conversationEnabled** = true
- unsigned int **maxHistoryMessages** = 20

The documentation for this struct was generated from the following file:

- tools/AIAssistant/AIConversationService.h

17.12 AIConversationResponse Struct Reference

Public Attributes

- bool **success** = false
- std::string **content**
- std::string **errorMessage**
- int **httpStatusCode** = 0
- unsigned int **inputTokens** = 0
- unsigned int **outputTokens** = 0

The documentation for this struct was generated from the following file:

- tools/AIAssistant/AIConversationService.h

17.13 AIConversationService Class Reference

Public Member Functions

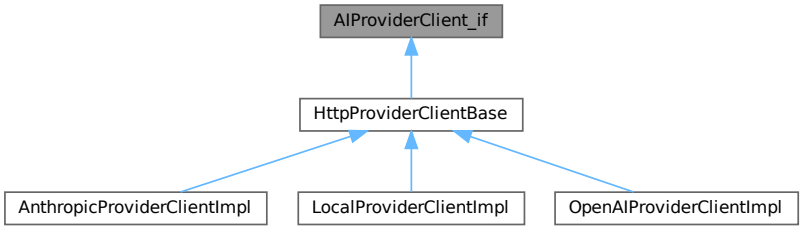
- void **setConfiguration** (const AIConversationConfiguration &configuration)
- const AIConversationConfiguration & **getConfiguration** () const
- bool **initialize** (std::string &errorMessage)
- void **setProviderClient** (std::unique_ptr< AIProviderClient_if > client)
- AIConversationResponse **send** (const std::string &conversationKey, const std::string &prompt)
- void **clearConversation** (const std::string &conversationKey)
- void **clearAllConversations** ()
- bool **isInitialized** () const

The documentation for this class was generated from the following files:

- tools/AIAssistant/AIConversationService.h
- tools/AIAssistant/AIConversationService.cpp

17.14 AIProviderClient_if Class Reference

Abstract interface for AI provider HTTP clients.
`#include <tools/AIAssistant/AIProviderClient_if.h>`
Inheritance diagram for AIProviderClient_if:



Public Member Functions

- virtual std::string getProviderName () const =0

  Returns a human-readable provider name (e.g. "OpenAI", "Anthropic").
- virtual bool isAvailable () const =0

  Returns `true` when this client can attempt to send requests.
- virtual void setBaseUrl (const std::string &url)=0

  Sets the base URL for the provider API endpoint.
- virtual std::string getBaseUrl () const =0

  Returns the currently configured base URL.
- virtual void setTimeoutSeconds (unsigned int seconds)=0

  Sets the HTTP request timeout in seconds.
- virtual unsigned int getTimeoutSeconds () const =0

  Returns the configured HTTP request timeout.
- virtual void setApiKey (const std::string &apiKey)=0

  Stores the API key in memory.
- virtual void clearApiKey ()=0

  Overwrites (best-effort) and clears the in-memory API key.
- virtual bool hasApiKey () const =0

Returns `true` when a non-empty API key is held.

- virtual ProviderResponse send (const ProviderRequest &request)=0

  Sends a chat-completion request and returns the response.

17.14.1 Detailed Description

Abstract interface for AI provider HTTP clients.
A provider client wraps the HTTP call to one AI backend (OpenAI, Anthropic, a local Ollama server, etc.) and converts GenESyS-internal request/response objects to and from the provider's wire format.

17.14.1.1 Security contract

- The API key is stored in-memory only; no getter exposes it.
- clearApiKey() overwrites the buffer before releasing it.
- Implementations must not write the key to logs or trace output.

17.14.1.2 Ownership

Provider client objects do not own the assistants that use them. Callers that inject clients into AIAssistant_if are responsible for keeping the client alive for the duration of any ongoing request.

17.14.2 Member Function Documentation

17.14.2.1 clearApiKey()

```
virtual void AIProviderClient_if::clearApiKey ( )  [pure virtual]
```

Overwrites (best-effort) and clears the in-memory API key.
Implemented in HttpProviderClientBase.

17.14.2.2 getBaseUrl()

```
virtual std::string AIProviderClient_if::getBaseUrl ( ) const  [pure virtual]
```

Returns the currently configured base URL.
Implemented in HttpProviderClientBase.

17.14.2.3 getProviderName()

```
virtual std::string AIProviderClient_if::getProviderName ( ) const  [pure virtual]
```

Returns a human-readable provider name (e.g. "OpenAI", "Anthropic").
Implemented in AnthropicProviderClientImpl, LocalProviderClientImpl, and OpenAIProviderClientImpl.

17.14.2.4 getTimeoutSeconds()

```
virtual unsigned int AIProviderClient_if::getTimeoutSeconds ( ) const  [pure virtual]
```

Returns the configured HTTP request timeout.
Implemented in HttpProviderClientBase.

17.14.2.5 hasApiKey()

```
virtual bool AIProviderClient_if::hasApiKey ( ) const  [pure virtual]
```

Returns `true` when a non-empty API key is held.
Implemented in HttpProviderClientBase.

17.14.2.6 isAvailable()

```
virtual bool AIProviderClient_if::isAvailable ( ) const  [pure virtual]
```

Returns `true` when this client can attempt to send requests.
A client is considered available when it holds an API key (if required by the provider) and a reachable base URL.
Implemented in AnthropicProviderClientImpl, HttpProviderClientBase, LocalProviderClientImpl, and OpenAIProviderClientImpl.

17.14.2.7 send()

```
virtual ProviderResponse AIProviderClient_if::send (
            const ProviderRequest & request )  [pure virtual]
```

Sends a chat-completion request and returns the response.
This call is synchronous and may block for up to timeoutSeconds. On transport failure `ProviderResponse::success` is `false` and `ProviderResponse::errorMessage` is populated.
Implemented in HttpProviderClientBase.

17.14.2.8 setApiKey()

```
virtual void AIProviderClient_if::setApiKey (
            const std::string & apiKey )  [pure virtual]
```

Stores the API key in memory.
Never write the key to disk or logs. Prefer loading from an environment variable via the owning AIAssistant_if rather than embedding keys in source.
Implemented in HttpProviderClientBase.

17.14.2.9 setBaseUrl()

```
virtual void AIProviderClient_if::setBaseUrl (
            const std::string & url )  [pure virtual]
```

Sets the base URL for the provider API endpoint.
Implemented in HttpProviderClientBase.

17.14.2.10 setTimeoutSeconds()

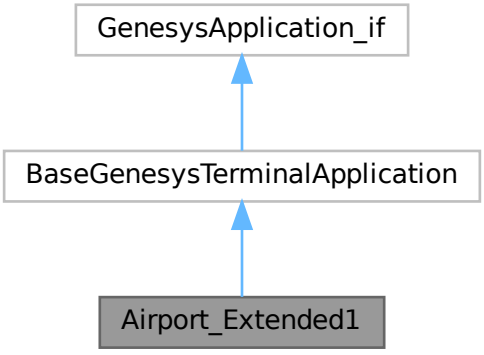
```
virtual void AIProviderClient_if::setTimeoutSeconds (
            unsigned int seconds )  [pure virtual]
```

Sets the HTTP request timeout in seconds.
Implemented in HttpProviderClientBase.
The documentation for this class was generated from the following file:

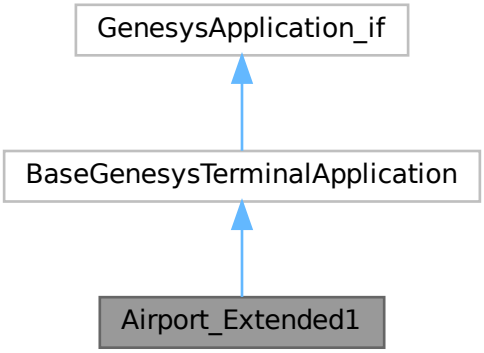
- tools/AIAssistant/AIProviderClient_if.h

17.15 Airport_Extended1 Class Reference

Inheritance diagram for Airport_Extended1:



Collaboration diagram for Airport_Extended1:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.15.1 Member Function Documentation

17.15.1.1 main()

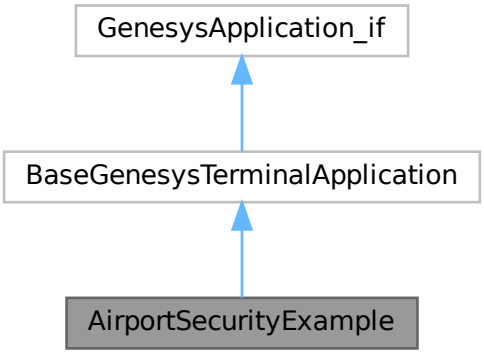
```
int Airport_Extended1::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

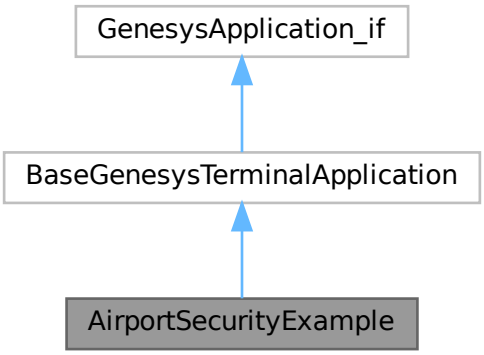
- applications/modelSpecific/arenaExamples/Airport_Extended1.h
- applications/modelSpecific/arenaExamples/Airport_Extended1.cpp

17.16 AirportSecurityExample Class Reference

Inheritance diagram for AirportSecurityExample:



Collaboration diagram for AirportSecurityExample:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.16.1 Member Function Documentation

17.16.1.1 main()

```
int AirportSecurityExample::main (
            int argc,
            char ** argv )  [override], [virtual]
```

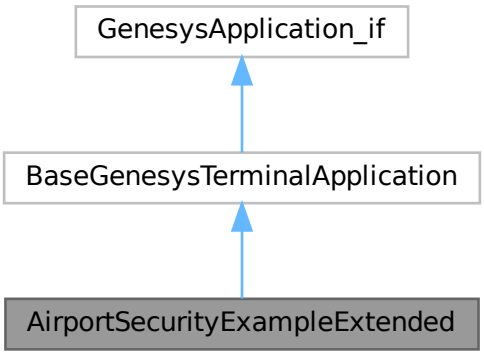
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

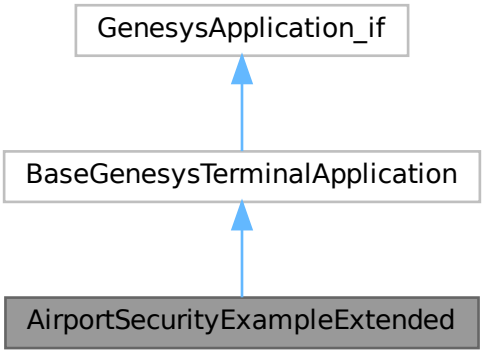
- applications/modelSpecific/arenaExamples/AirportSecurityExample.h
- applications/modelSpecific/arenaExamples/AirportSecurityExample.cpp

17.17 AirportSecurityExampleExtended Class Reference

Inheritance diagram for AirportSecurityExampleExtended:



Collaboration diagram for AirportSecurityExampleExtended:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.17.1 Member Function Documentation

17.17.1.1 main()

```
int AirportSecurityExampleExtended::main (
            int argc,
            char ** argv )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

- applications/modelSpecific/arenaExamples/AirportSecurityExampleExtended.h
- applications/modelSpecific/arenaExamples/AirportSecurityExampleExtended.cpp

17.18 AISecretStore Class Reference

Static helper for storing API keys in the OS keyring.

`#include <tools/AIAssistant/AISecretStore.h>`

Static Public Member Functions

- static bool isAvailable ()

  Returns `true` if the keyring backend is reachable on this system.
- static bool save (const std::string &service, const std::string &account, const std::string &secret)

  Saves `secret` under the given `service` and `account` labels.
- static std::optional< std::string > load (const std::string &service, const std::string &account)

  Loads the secret stored under `service` and `account`.
- static bool remove (const std::string &service, const std::string &account)

  Removes the stored secret.

17.18.1 Detailed Description

Static helper for storing API keys in the OS keyring.

On Linux, uses the `secret-tool` command-line utility (part of the `libsecret-tools` package) to access the GNOME Keyring / KWallet service via the freedesktop Secret Service protocol.

All methods return a failure indicator (`false` or `std::nullopt`) when `secret-tool` is not installed; callers should fall back to env-var loading.

API keys are never written to disk in plaintext; they live only in the session keyring managed by the OS.

17.18.1.1 Usage

```
const std::string svc = "genesys-ai-assistant";
const std::string acct = "openai";
if (AISecretStore::isAvailable()) {
    AISecretStore::save(svc, acct, rawKey);
    auto loaded = AISecretStore::load(svc, acct);
}
```

17.18.2 Member Function Documentation

17.18.2.1 isAvailable()

```
bool AISecretStore::isAvailable ( )  [static]
```

Returns `true` if the keyring backend is reachable on this system.
Checks that `secret-tool` is on PATH and that the D-Bus session bus is available (required by the Secret Service protocol).

17.18.2.2 load()

```
std::optional< std::string > AISecretStore::load (
            const std::string & service,
            const std::string & account )  [static]
```

Loads the secret stored under `service` and `account`.

Returns

The secret, or `std::nullopt` when not found or unavailable.

17.18.2.3 remove()

```
bool AISecretStore::remove (
            const std::string & service,
            const std::string & account )  [static]
```

Removes the stored secret.

Returns

`true` on success.

17.18.2.4 save()

```
bool AISecretStore::save (
            const std::string & service,
            const std::string & account,
            const std::string & secret )  [static]
```

Saves `secret` under the given `service` and `account` labels.

Returns

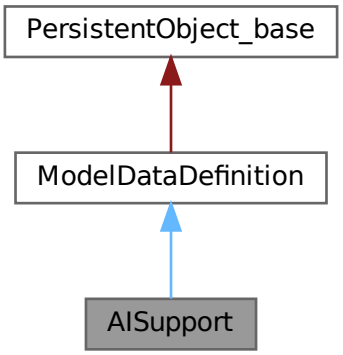
`true` on success, `false` if the keyring is unavailable or the operation failed.

The documentation for this class was generated from the following files:

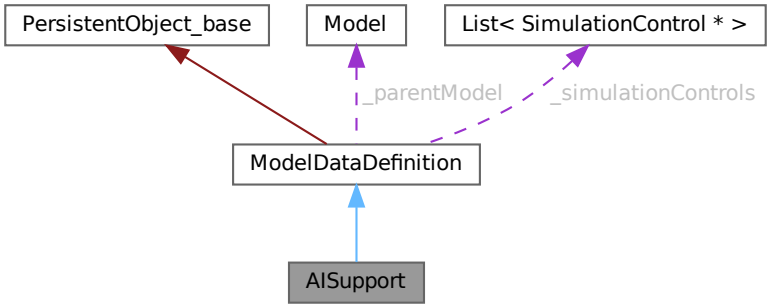
- tools/AIAssistant/AISecretStore.h
- tools/AIAssistant/AISecretStore.cpp

17.19 AISupport Class Reference

Inheritance diagram for AISupport:



Collaboration diagram for AISupport:



Public Member Functions

- **AISupport** (Model *model, std::string name="")
- AIConversationResponse **converse** (const std::string &conversationKey, const std::string &prompt)
- void **clearConversation** (const std::string &conversationKey)
- void **clearAllConversations** ()
- void **setProvider** (std::string value)
- std::string **getProvider** () const
- void **setModel** (std::string value)
- std::string **getModel** () const
- void **setBaseUrl** (std::string value)
- std::string **getBaseUrl** () const
- void **setApiKeyEnvironmentVariable** (std::string value)
- std::string **getApiKeyEnvironmentVariable** () const
- void **setSystemInstructions** (std::string value)
- std::string **getSystemInstructions** () const

- void **setTemperature** (double value)
- double **getTemperature** () const
- void **setMaxOutputTokens** (unsigned int value)
- unsigned int **getMaxOutputTokens** () const
- void **setTimeoutSeconds** (unsigned int value)
- unsigned int **getTimeoutSeconds** () const
- void **setHighReasoningMode** (bool value)
- bool **isHighReasoningMode** () const
- void **setConversationEnabled** (bool value)
- bool **isConversationEnabled** () const
- void **setMaxHistoryMessages** (unsigned int value)
- unsigned int **getMaxHistoryMessages** () const
- void **setClearHistoryBetweenReplications** (bool value)
- bool **isClearHistoryBetweenReplications** () const
- unsigned long **getCallCount** () const
- unsigned long **getFailureCount** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.19.1 Member Function Documentation

17.19.1.1 _check()

```
bool AISupport::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.19.1.2 _initBetweenReplications()

```
void AISupport::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.19.1.3 _loadInstance()

```
bool AISupport::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.19.1.4 _saveInstance()

```
void AISupport::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.19.1.5 show()

```
std::string AISupport::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

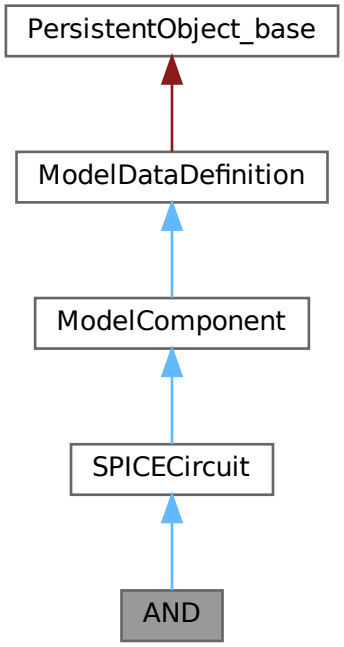
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

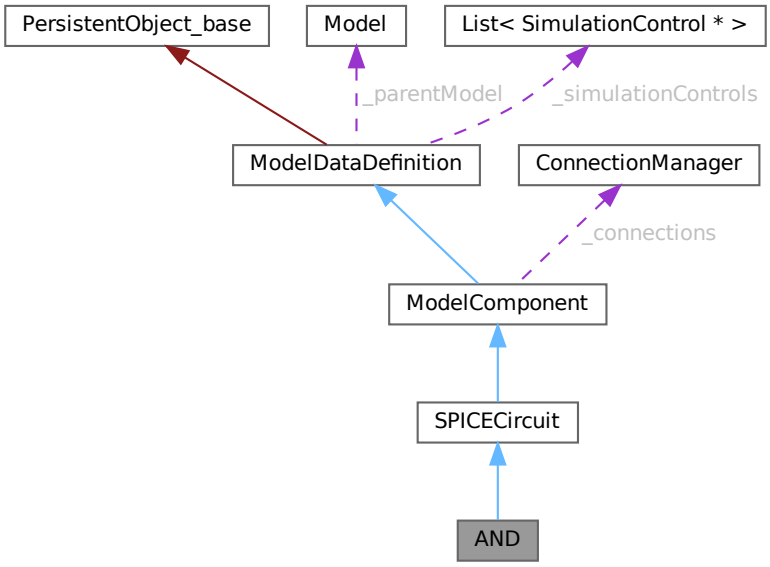
- plugins/data/AI/AISupport.h
- plugins/data/AI/AISupport.cpp

17.20 AND Class Reference

Inheritance diagram for AND:



Collaboration diagram for AND:



Public Member Functions

- **AND** (Model *model, std::string name="")
- void Build () override
- void **setPMOSWidth** (float PMOSWidth)
- void **setPMOSLength** (float length)
- void **setNMOSWidth** (float NMOSWidth)
- void **setNMOSLength** (float NMOSLength)
- void **setElectricalModel** (std::string electricalModel)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

Connects this component to another component.

- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.

- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden

- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden

- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.

- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes

- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)

- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.20.1 Member Function Documentation

17.20.1.1 _loadInstance()

```
bool AND::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.20.1.2 _saveInstance()

```
void AND::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.20.1.3 Build()

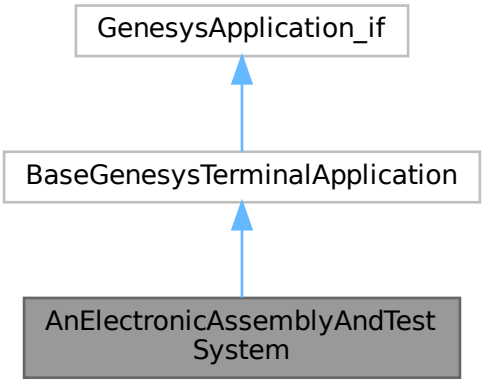
```
void AND::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.
The documentation for this class was generated from the following files:

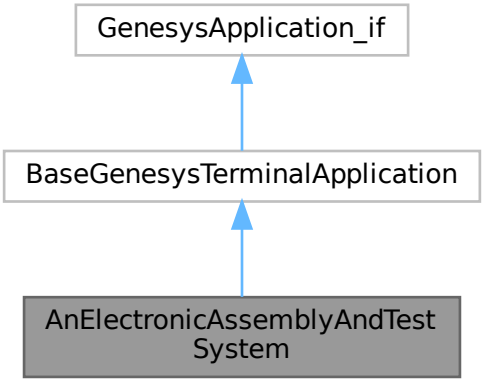
- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.21 AnElectronicAssemblyAndTestSystem Class Reference

Inheritance diagram for AnElectronicAssemblyAndTestSystem:



Collaboration diagram for AnElectronicAssemblyAndTestSystem:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.21.1 Member Function Documentation

17.21.1.1 main()

```
int AnElectronicAssemblyAndTestSystem::main (
            int argc,
            char ** argv )  [override], [virtual]
```

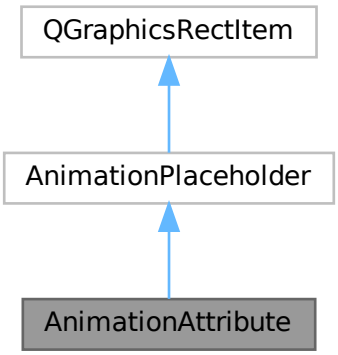
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

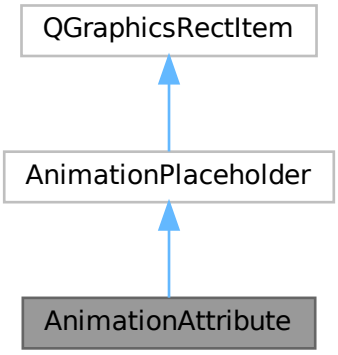
- applications/modelSpecific/teaching/AnElectronicAssemblyAndTestSystem.h
- applications/modelSpecific/teaching/AnElectronicAssemblyAndTestSystem.cpp

17.22 AnimationAttribute Class Reference

Inheritance diagram for AnimationAttribute:



Collaboration diagram for AnimationAttribute:



Additional Inherited Members

Public Member Functions inherited from AnimationPlaceholder

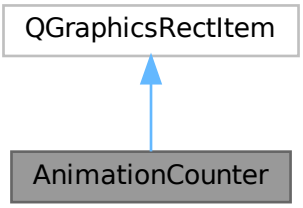
- **AnimationPlaceholder** (const QString &animationType)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override
- QString **getAnimationType** () const
- QString **getTargetName** () const
- void **setTargetName** (const QString &targetName)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)
- bool **isDrawingInicialized** () const
- bool **isDrawingFinalized** () const

The documentation for this class was generated from the following files:

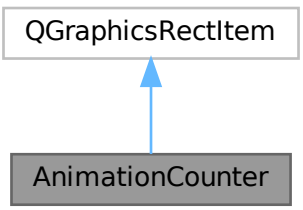
- applications/gui/genesys/animations/AnimationPlaceholder.h
- applications/gui/genesys/animations/AnimationPlaceholder.cpp

17.23 AnimationCounter Class Reference

Inheritance diagram for AnimationCounter:



Collaboration diagram for AnimationCounter:



Public Member Functions

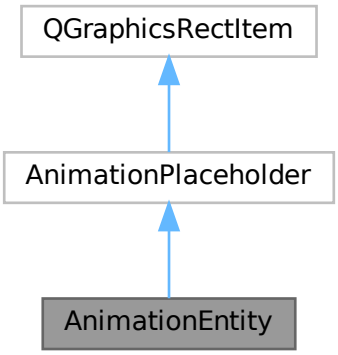
- void **paint** (QPainter *painter, const QStyleOptionGraphicsItem *option, QWidget *widget=nullptr) override
- double **getValue** ()
- Counter * **getCounter** ()
- void **setValue** (double value)
- void **setCounter** (Counter *myCounter)
- void **setIdCounter** (int id)
- void **setNameCounter** (std::string name)
- void **startDrawing** (QGraphicsSceneMouseEvent *event)
- void **continueDrawing** (QGraphicsSceneMouseEvent *event)
- void **stopDrawing** (QGraphicsSceneMouseEvent *event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent *event)
- bool **isDrawingInicialized** ()
- bool **isDrawingFinalized** ()
- void **setWhenLoaded** (QList< Counter * > *counters)

The documentation for this class was generated from the following files:

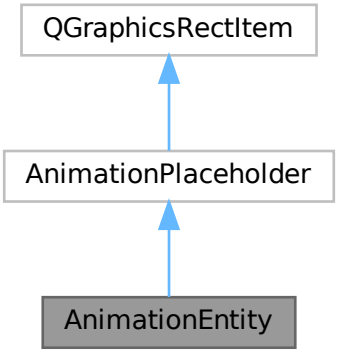
- applications/gui/genesys/animations/AnimationCounter.h
- applications/gui/genesys/animations/AnimationCounter.cpp

17.24 AnimationEntity Class Reference

Inheritance diagram for AnimationEntity:



Collaboration diagram for AnimationEntity:



Additional Inherited Members

Public Member Functions inherited from AnimationPlaceholder

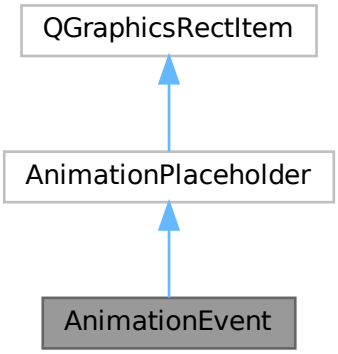
- **AnimationPlaceholder** (const QString &animationType)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override
- QString **getAnimationType** () const
- QString **getTargetName** () const
- void **setTargetName** (const QString &targetName)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)
- bool **isDrawingInicialized** () const
- bool **isDrawingFinalized** () const

The documentation for this class was generated from the following files:

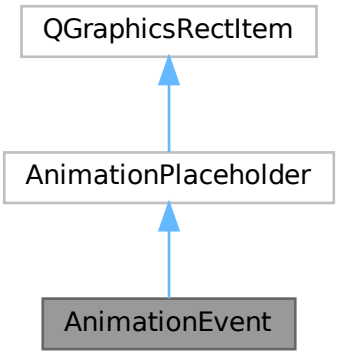
- applications/gui/genesys/animations/AnimationPlaceholder.h
- applications/gui/genesys/animations/AnimationPlaceholder.cpp

17.25 AnimationEvent Class Reference

Inheritance diagram for AnimationEvent:



Collaboration diagram for AnimationEvent:



Additional Inherited Members

Public Member Functions inherited from AnimationPlaceholder

- **AnimationPlaceholder** (const QString &animationType)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override
- QString **getAnimationType** () const
- QString **getTargetName** () const
- void **setTargetName** (const QString &targetName)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)

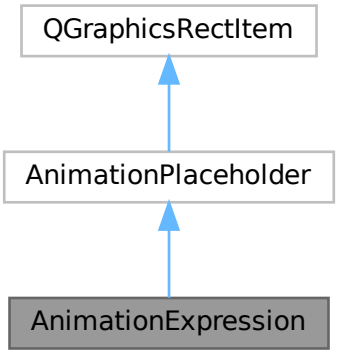
- bool **isDrawingInicialized** () const
- bool **isDrawingFinalized** () const

The documentation for this class was generated from the following files:

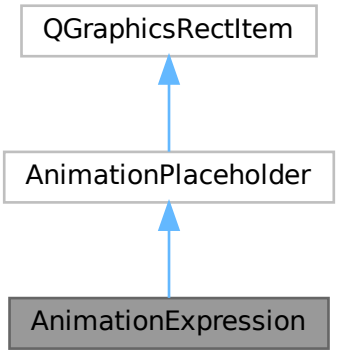
- applications/gui/genesys/animations/AnimationPlaceholder.h
- applications/gui/genesys/animations/AnimationPlaceholder.cpp

17.26 AnimationExpression Class Reference

Inheritance diagram for AnimationExpression:



Collaboration diagram for AnimationExpression:



Additional Inherited Members

Public Member Functions inherited from AnimationPlaceholder

- **AnimationPlaceholder** (const QString &animationType)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override

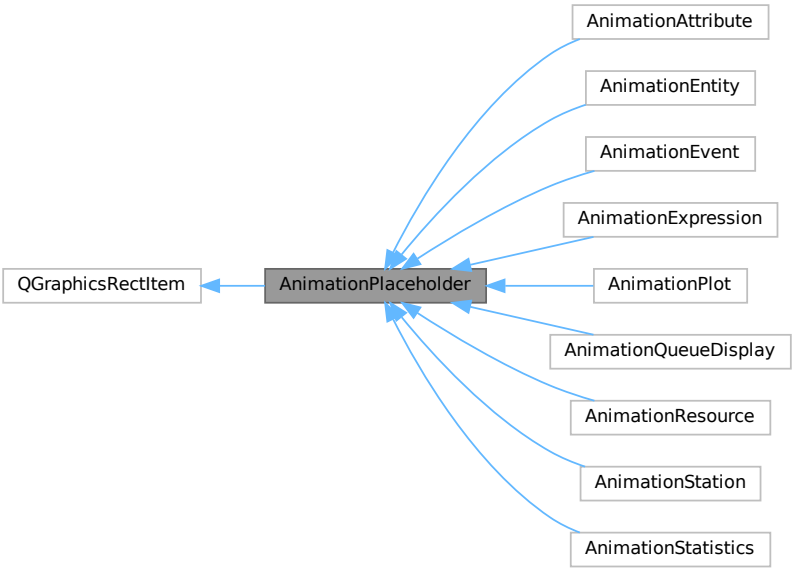
- QString **getAnimationType** () const
- QString **getTargetName** () const
- void **setTargetName** (const QString &targetName)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)
- bool **isDrawingInicialized** () const
- bool **isDrawingFinalized** () const

The documentation for this class was generated from the following files:

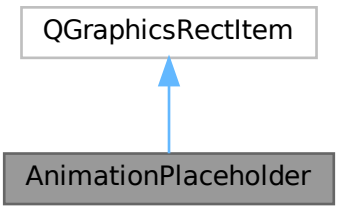
- applications/gui/genesys/animations/AnimationPlaceholder.h
- applications/gui/genesys/animations/AnimationPlaceholder.cpp

17.27 AnimationPlaceholder Class Reference

Inheritance diagram for AnimationPlaceholder:



Collaboration diagram for AnimationPlaceholder:



Public Member Functions

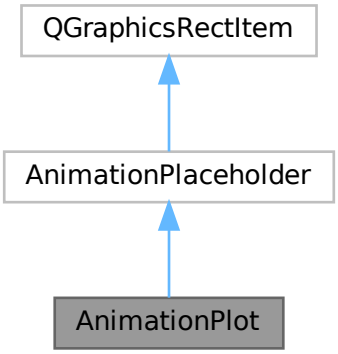
- **AnimationPlaceholder** (const QString &animationType)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override
- QString **getAnimationType** () const
- QString **getTargetName** () const
- void **setTargetName** (const QString &targetName)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)
- bool **isDrawingInicialized** () const
- bool **isDrawingFinalized** () const

The documentation for this class was generated from the following files:

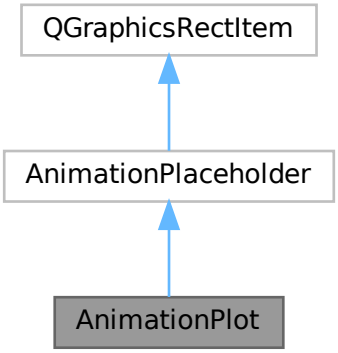
- applications/gui/genesys/animations/AnimationPlaceholder.h
- applications/gui/genesys/animations/AnimationPlaceholder.cpp

17.28 AnimationPlot Class Reference

Inheritance diagram for AnimationPlot:



Collaboration diagram for AnimationPlot:



Classes

- struct DataSeries

Public Types

- enum class **PlotType** { **Line** = 0 , **Bar** = 1 }

Public Member Functions

- QString **getTitle** () const
- void **setTitle** (const QString &title)
- QString **getYAxisTitle** () const
- void **setYAxisTitle** (const QString &title)
- QString **getXAxisTitle** () const
- void **setXAxisTitle** (const QString &title)
- PlotType **getPlotType** () const
- void **setPlotType** (PlotType type)
- QString **getDatasetsText** () const
- void **setDatasetsText** (const QString &datasetsText)
- bool **getShowGridLines** () const
- void **setShowGridLines** (bool show)
- bool **getShowTicks** () const
- void **setShowTicks** (bool show)
- double **getXAxisMin** () const
- void **setXAxisMin** (double value)
- double **getXAxisMax** () const
- void **setXAxisMax** (double value)
- double **getYAxisMin** () const
- void **setYAxisMin** (double value)
- double **getYAxisMax** () const
- void **setYAxisMax** (double value)
- void **clearValues** ()
- void **appendSample** (Model ∗model)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override

Public Member Functions inherited from AnimationPlaceholder

- **AnimationPlaceholder** (const QString &animationType)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override
- QString **getAnimationType** () const
- QString **getTargetName** () const
- void **setTargetName** (const QString &targetName)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)
- bool **isDrawingInicialized** () const
- bool **isDrawingFinalized** () const

The documentation for this class was generated from the following files:

- applications/gui/genesys/animations/AnimationPlaceholder.h
- applications/gui/genesys/animations/AnimationPlaceholder.cpp

17.29 AnimationQueue Class Reference

Public Member Functions

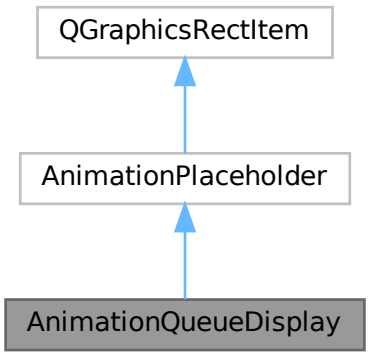
- **AnimationQueue** (ModelGraphicsScene ∗myScene, ModelComponent ∗component)
- GraphicalModelComponent ∗ **getGraphicalComponent** () const
- void **addAnimationQueue** (bool visivible)
- void **removeAnimationQueue** ()
- void **verifyAddAnimationQueue** (bool visivible)
- void **verifyRemoveAnimationQueue** ()
- QPointF **calculatePositionImageQueue** (unsigned int indexQueue, unsigned int sizeQueue, unsigned int width, unsigned int height)

The documentation for this class was generated from the following files:

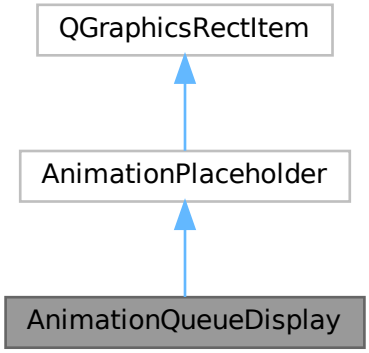
- applications/gui/genesys/animations/AnimationQueue.h
- applications/gui/genesys/animations/AnimationQueue.cpp

17.30 AnimationQueueDisplay Class Reference

Inheritance diagram for AnimationQueueDisplay:



Collaboration diagram for AnimationQueueDisplay:



Additional Inherited Members

Public Member Functions inherited from AnimationPlaceholder

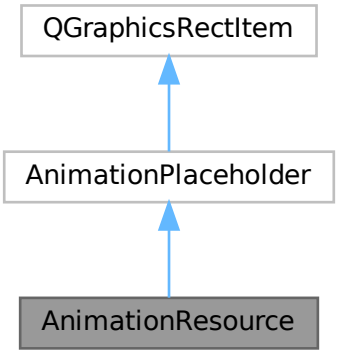
- **AnimationPlaceholder** (const QString &animationType)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override
- QString **getAnimationType** () const
- QString **getTargetName** () const
- void **setTargetName** (const QString &targetName)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)
- bool **isDrawingInicialized** () const
- bool **isDrawingFinalized** () const

The documentation for this class was generated from the following files:

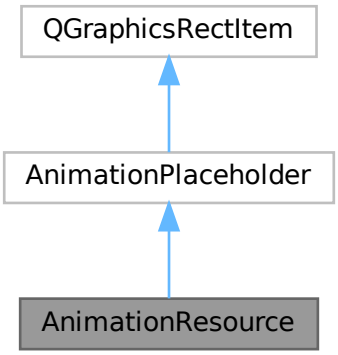
- applications/gui/genesys/animations/AnimationPlaceholder.h
- applications/gui/genesys/animations/AnimationPlaceholder.cpp

17.31 AnimationResource Class Reference

Inheritance diagram for AnimationResource:



Collaboration diagram for AnimationResource:



Additional Inherited Members

Public Member Functions inherited from AnimationPlaceholder

- **AnimationPlaceholder** (const QString &animationType)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override
- QString **getAnimationType** () const
- QString **getTargetName** () const
- void **setTargetName** (const QString &targetName)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)

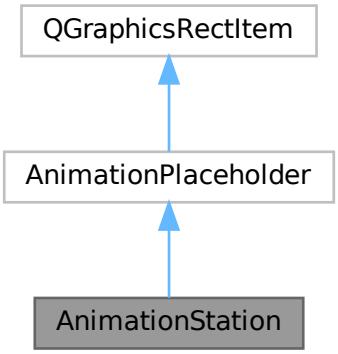
- bool **isDrawingInicialized** () const
- bool **isDrawingFinalized** () const

The documentation for this class was generated from the following files:

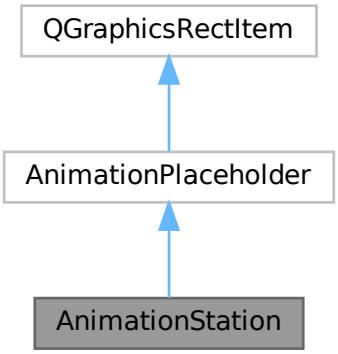
- applications/gui/genesys/animations/AnimationPlaceholder.h
- applications/gui/genesys/animations/AnimationPlaceholder.cpp

17.32 AnimationStation Class Reference

Inheritance diagram for AnimationStation:



Collaboration diagram for AnimationStation:



Additional Inherited Members

Public Member Functions inherited from AnimationPlaceholder

- **AnimationPlaceholder** (const QString &animationType)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override

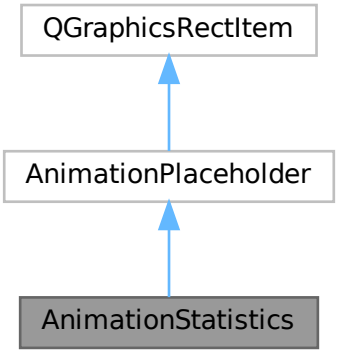
- QString **getAnimationType** () const
- QString **getTargetName** () const
- void **setTargetName** (const QString &targetName)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)
- bool **isDrawingInicialized** () const
- bool **isDrawingFinalized** () const

The documentation for this class was generated from the following files:

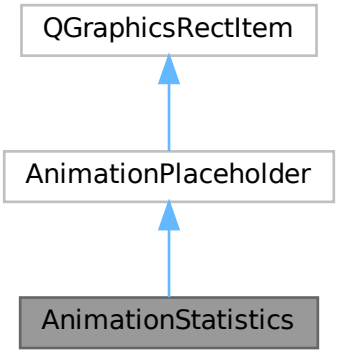
- applications/gui/genesys/animations/AnimationPlaceholder.h
- applications/gui/genesys/animations/AnimationPlaceholder.cpp

17.33 AnimationStatistics Class Reference

Inheritance diagram for AnimationStatistics:



Collaboration diagram for AnimationStatistics:



Additional Inherited Members

Public Member Functions inherited from AnimationPlaceholder

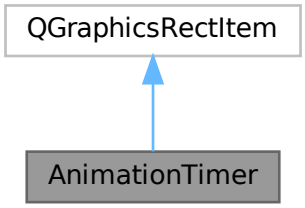
- **AnimationPlaceholder** (const QString &animationType)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override
- QString **getAnimationType** () const
- QString **getTargetName** () const
- void **setTargetName** (const QString &targetName)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)
- bool **isDrawingInicialized** () const
- bool **isDrawingFinalized** () const

The documentation for this class was generated from the following files:

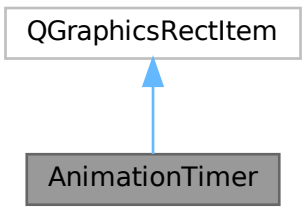
- applications/gui/genesys/animations/AnimationPlaceholder.h
- applications/gui/genesys/animations/AnimationPlaceholder.cpp

17.34 AnimationTimer Class Reference

Inheritance diagram for AnimationTimer:



Collaboration diagram for AnimationTimer:



Public Member Functions

- **AnimationTimer** (ModelGraphicsScene ∗myScene)
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override
- double **getTime** ()
- unsigned int **getInitialHours** ()
- unsigned int **getInitialMinutes** ()
- unsigned int **getInitialSeconds** ()
- unsigned int **getHours** ()
- unsigned int **getMinutes** ()
- unsigned int **getSeconds** ()
- Util::TimeFormat **getTimeFormat** ()
- void **setTime** (double value)
- void **setInitialHours** (unsigned int initialHours)
- void **setInitialMinutes** (unsigned int initialMinutes)
- void **setInitialSeconds** (unsigned int initialSeconds)
- void **setHours** (unsigned int hours)
- void **setMinutes** (unsigned int minutes)
- void **setSeconds** (unsigned int seconds)
- void **setTimeFormat** (Util::TimeFormat timeFormat)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)
- void **convertSeconds** ()
- bool **isDrawingInicialized** ()
- bool **isDrawingFinalized** ()

Static Public Member Functions

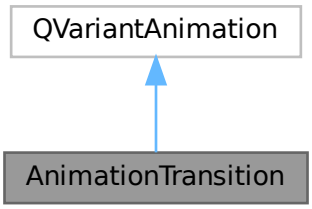
- static void **setConversionFactorToSeconds** (double factor)

The documentation for this class was generated from the following files:

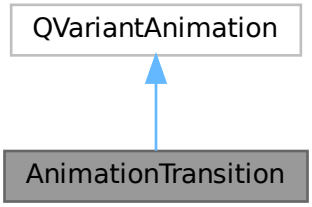
- applications/gui/genesys/animations/AnimationTimer.h
- applications/gui/genesys/animations/AnimationTimer.cpp

17.35 AnimationTransition Class Reference

Inheritance diagram for AnimationTransition:



Collaboration diagram for AnimationTransition:



Public Slots

- void **onAnimationValueChanged** (const QVariant &value)
- void **onAnimationFinished** ()

Public Member Functions

- **AnimationTransition** (ModelGraphicsScene ∗myScene, ModelComponent ∗graphicalStartComponent, ModelComponent ∗graphicalEndComponent, bool viewSimulation)
- GraphicalModelComponent ∗ **getGraphicalStartComponent** () const
- GraphicalModelComponent ∗ **getGraphicalEndComponent** () const
- GraphicalConnection ∗ **getGraphicalConnection** () const
- QList< QPointF > **getPointsForAnimation** () const
- GraphicalImageAnimation ∗ **getImageAnimation** () const
- unsigned int **getPortNumber** () const
- bool **isReadyToRun** () const
- void **setImageAnimation** (GraphicalImageAnimation ∗imageAnimation)
- void **startAnimation** ()
- void **stopAnimation** ()
- void **restartAnimation** ()
- void **configureAnimation** ()
- void **updateDurationIfNeeded** ()
- void **connectValueChangedSignal** ()
- void **connectFinishedSignal** ()

Static Public Member Functions

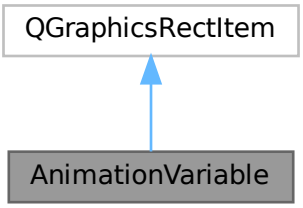
- static double **getTimeExecution** ()
- static void **setTimeExecution** (double timeExecution)
- static void **setPause** (bool pause)
- static void **setRunning** (bool running)

The documentation for this class was generated from the following files:

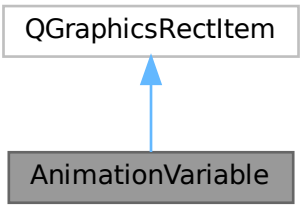
- applications/gui/genesys/animations/AnimationTransition.h
- applications/gui/genesys/animations/AnimationTransition.cpp

17.36 AnimationVariable Class Reference

Inheritance diagram for AnimationVariable:



Collaboration diagram for AnimationVariable:



Public Member Functions

- void **paint** (QPainter *painter, const QStyleOptionGraphicsItem *option, QWidget *widget=nullptr) override
- double **getValue** ()
- ModelDataDefinition * **getDataDefinition** () const
- Variable * **getVariable** ()
- Attribute * **getAttribute** ()
- QString **getTargetName** () const
- int **getTargetId** () const
- unsigned int **getDimensionCount** () const
- unsigned int **getDimensionSize** (unsigned int dimensionIndex) const
- unsigned int **getSliceIndex** (unsigned int dimensionIndex) const
- void **setValue** (double value)
- void **setDataDefinition** (ModelDataDefinition *definition)
- void **setVariable** (Variable *variable)
- void **setAttribute** (Attribute *attribute)
- void **setModel** (Model *model)
- void **setTargetName** (std::string name)
- void **setIdVariable** (int id)
- void **setSliceIndex** (unsigned int dimensionIndex, unsigned int value)
- void **setWhenLoaded** (QList< ModelDataDefinition * > *definitions)

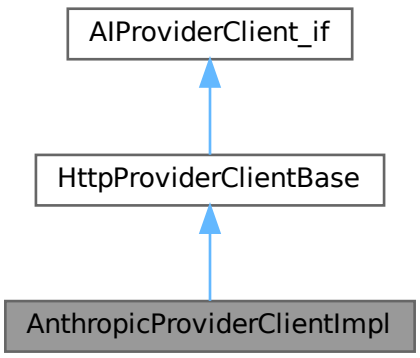
- void **refreshValue** (Entity ∗entity=nullptr)
- void **startDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **continueDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **stopDrawing** (QGraphicsSceneMouseEvent ∗event)
- void **adjustSizeAndPosition** (QGraphicsSceneMouseEvent ∗event)
- bool **isDrawingInicialized** ()
- bool **isDrawingFinalized** ()

The documentation for this class was generated from the following files:

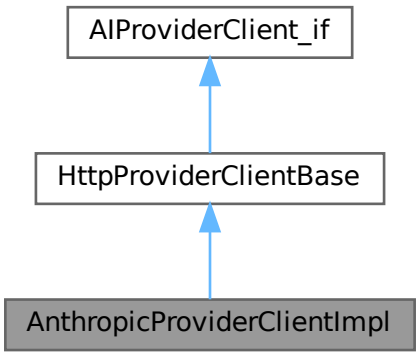
- applications/gui/genesys/animations/AnimationVariable.h
- applications/gui/genesys/animations/AnimationVariable.cpp

17.37 AnthropicProviderClientImpl Class Reference

AI provider client for the Anthropic Messages API.
`#include <tools/AIAssistant/AnthropicProviderClientImpl.h>`
Inheritance diagram for AnthropicProviderClientImpl:



Collaboration diagram for AnthropicProviderClientImpl:



Public Member Functions

- **AnthropicProviderClientImpl** ()

  Constructs the client with the standard Anthropic base URL.
- std::string getProviderName () const override

  Returns a human-readable provider name (e.g. "OpenAI", "Anthropic").
- bool isAvailable () const override

  Returns `true` when an API key is set and a base URL is configured.
- void setApiVersion (const std::string &version)

  Sets the Anthropic API version header value.
- std::string **getApiVersion** () const

  Returns the configured Anthropic API version string.
- void setThinkingBudgetTokens (unsigned int budget)

  Sets the extended-thinking budget in tokens (default: 10000).
- unsigned int **getThinkingBudgetTokens** () const

  Returns the thinking budget in tokens.

Public Member Functions inherited from HttpProviderClientBase

- **HttpProviderClientBase** (const HttpProviderClientBase &)=delete
- HttpProviderClientBase & **operator=** (const HttpProviderClientBase &)=delete
- bool isAvailable () const override

  Returns `true` when this client can attempt to send requests.
- void setBaseUrl (const std::string &url) override

  Sets the base URL for the provider API endpoint.
- std::string getBaseUrl () const override

  Returns the currently configured base URL.
- void setTimeoutSeconds (unsigned int seconds) override

  Sets the HTTP request timeout in seconds.
- unsigned int getTimeoutSeconds () const override

  Returns the configured HTTP request timeout.
- void setApiKey (const std::string &apiKey) override

  Stores the API key in memory.
- void clearApiKey () override

  Overwrites (best-effort) and clears the in-memory API key.
- bool hasApiKey () const override

  Returns `true` when a non-empty API key is held.
- ProviderResponse send (const ProviderRequest &request) override

  Executes the HTTP request via curl and returns the provider response.

Protected Member Functions

- std::string _buildRequestBody (const ProviderRequest &request) const override

  Serialises `request` to the provider's JSON wire format.
- ProviderResponse _parseResponseBody (const std::string &rawBody, int httpStatus) const override

  Deserialises `rawBody` (the full curl stdout) into a ProviderResponse.
- std::string _authorizationHeader () const override

  Returns the Authorization header line without the "header = " prefix.
- std::vector< std::string > _extraRequestHeaders () const override

  Returns additional HTTP headers to send with every request.

Additional Inherited Members

Static Public Member Functions inherited from HttpProviderClientBase

- static std::string **_escapeJsonString** (const std::string &s)

  Escapes a plain string for safe embedding inside a JSON string value.
- static std::string **_extractJsonString** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON string field by key. Returns an empty string when not found or value is not a quoted string.
- static int **_extractJsonInt** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON integer field by key. Returns 0 when not found.
- static double **_extractJsonDouble** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON number field (int or float) by key. Returns 0.0 when not found.
- static std::string **_buildMessagesArray** (const std::vector< ChatMessage > &messages)

  Serialises a vector of ChatMessage objects to a JSON array string.
- static std::string **_extractFirstArrayElement** (const std::string &json, const std::string &arrayKey)

  Extracts the raw substring of the first element of a named JSON array. Returns an empty string when not found.

Protected Attributes inherited from HttpProviderClientBase

- std::string **_baseUrl**
- unsigned int **_timeoutSeconds** = 30
- std::string _apiKey

  In-memory API key.

17.37.1 Detailed Description

AI provider client for the Anthropic Messages API.
Wraps the Anthropic `/v1/messages` endpoint (Claude 3/4 model family). Default base URL: `https://api.anthropic.com/v1/messages`
Wire format differences from the OpenAI schema:

- Auth header: `x-api-key: <key>` (not `Authorization: Bearer`).
- Required extra header: `anthropic-version: 2023-06-01`.
- Request body: `max_tokens` (required), `system` is a top-level field (not a message in the array), `messages` contains only user/assistant turns.
- Response: content is in `content[0].text`, tokens in `usage.input_tokens` / `usage.output_tokens`.
- Extended thinking: when `highReasoningMode` is true, a `thinking.type=enabled` budget block is added to the request and the text is extracted from the first content block whose `type` is `"text"`.

17.37.1.1 API key

Sent as `x-api-key: <key>`. Written to the curl config file at mode 0600; never placed on the command line.

17.37.2 Member Function Documentation

17.37.2.1 _authorizationHeader()

`std::string AnthropicProviderClientImpl::_authorizationHeader ( ) const [override], [protected], [virtual]`
Returns the Authorization header line without the "header = " prefix.
Example return values:

- "Authorization: Bearer sk-..." (OpenAI, LocalProvider)

- "x-api-key: sk-ant-..." (Anthropic)

Called only from within send(); the value is written to a 0600 temp file and never logged.
Implements HttpProviderClientBase.

17.37.2.2 _buildRequestBody()

```
std::string AnthropicProviderClientImpl::_buildRequestBody (
            const ProviderRequest & request ) const  [override], [protected], [virtual]
```

Serialises `request` to the provider's JSON wire format.
Implements HttpProviderClientBase.

17.37.2.3 _extraRequestHeaders()

```
std::vector< std::string > AnthropicProviderClientImpl::_extraRequestHeaders ( ) const  [override],
[protected], [virtual]
```

Returns additional HTTP headers to send with every request.
Default implementation returns an empty vector. Anthropic overrides to add "anthropic-version: 2023-06-01".
Each element must be a complete header line, e.g. "anthropic-version: 2023-06-01".
Reimplemented from HttpProviderClientBase.

17.37.2.4 _parseResponseBody()

```
ProviderResponse AnthropicProviderClientImpl::_parseResponseBody (
            const std::string & rawBody,
            int httpStatus ) const  [override], [protected], [virtual]
```

Deserialises `rawBody` (the full curl stdout) into a ProviderResponse.

Parameters

rawBody	Complete HTTP response body as received from curl.
httpStatus	HTTP status code captured from curl exit / response line, or 0 when unavailable.

Implements HttpProviderClientBase.

17.37.2.5 getProviderName()

```
std::string AnthropicProviderClientImpl::getProviderName ( ) const  [override], [virtual]
```

Returns a human-readable provider name (e.g. "OpenAI", "Anthropic").
Implements AIProviderClient_if.

17.37.2.6 isAvailable()

```
bool AnthropicProviderClientImpl::isAvailable ( ) const  [override], [virtual]
```

Returns `true` when an API key is set and a base URL is configured.
Implements AIProviderClient_if.

17.37.2.7 setApiVersion()

```
void AnthropicProviderClientImpl::setApiVersion (
            const std::string & version )
```

Sets the Anthropic API version header value.
Defaults to "2023-06-01" which is the stable version as of mid-2026. Override only when Anthropic releases a newer stable version that changes the wire format.

17.37.2.8 setThinkingBudgetTokens()

```
void AnthropicProviderClientImpl::setThinkingBudgetTokens (
            unsigned int budget )
```

Sets the extended-thinking budget in tokens (default: 10000).
Used only when ProviderRequest::highReasoningMode is true. Must be less than ProviderRequest::maxTokens.
The documentation for this class was generated from the following files:

- tools/AIAssistant/AnthropicProviderClientImpl.h
- tools/AIAssistant/AnthropicProviderClientImpl.cpp

17.38 ApiRouter Class Reference

Routes HTTP requests to Web API operations.
`#include <applications/worker/api/ApiRouter.h>`

Public Member Functions

- ApiRouter (SimulatorSessionService &simulatorService)

  Creates an API router bound to the simulator session service.
- HttpResponse handle (const HttpRequest &request) const

  Handles one HTTP request and returns the corresponding response.

17.38.1 Detailed Description

Routes HTTP requests to Web API operations.

17.38.2 Constructor & Destructor Documentation

17.38.2.1 ApiRouter()

```
ApiRouter::ApiRouter (
            SimulatorSessionService & simulatorService )  [explicit]
```

Creates an API router bound to the simulator session service.
Dispatches HTTP routes for the Genesys web API.

Parameters

simulatorService	Service used to execute API operations.

The router keeps request parsing, token extraction, payload decoding and JSON formatting in one place so the HTTP server can stay transport-only.

17.38.3 Member Function Documentation

17.38.3.1 handle()

```
HttpResponse ApiRouter::handle (
            const HttpRequest & request ) const
```

Handles one HTTP request and returns the corresponding response.
Resolves a request into one of the public API operations.

Parameters

request	HTTP request containing method, path, headers, and body.

Returns

HTTP response with status, content type, and JSON payload.

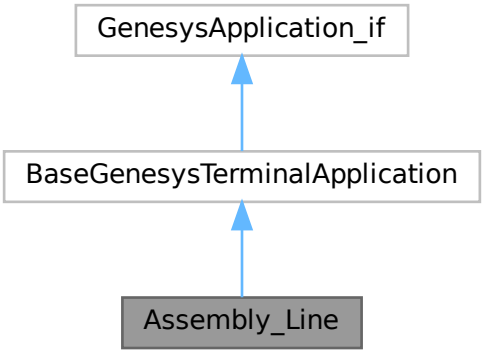
The route chain stays intentionally explicit so the generated PDF makes the supported endpoints and their validation rules easy to audit.

The documentation for this class was generated from the following files:

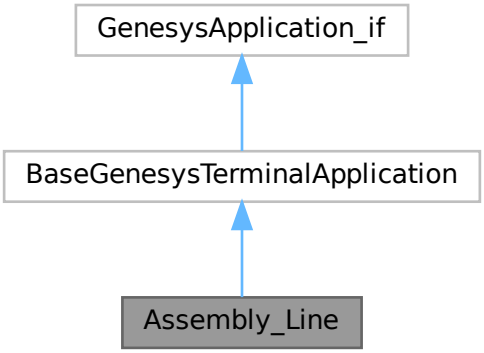
- applications/worker/api/ApiRouter.h
- applications/worker/api/ApiRouter.cpp

17.39 Assembly_Line Class Reference

Inheritance diagram for Assembly_Line:



Collaboration diagram for Assembly_Line:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.39.1 Member Function Documentation

17.39.1.1 main()

```
int Assembly_Line::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/arenaExamples/Assembly_Line.h
- applications/modelSpecific/arenaExamples/Assembly_Line.cpp

17.40 testing::internal::AssertHelper Class Reference

Public Member Functions

- **AssertHelper** (TestPartResult::Type type, const char ∗file, int line, const char ∗message)
- void **operator=** (const Message &message) const

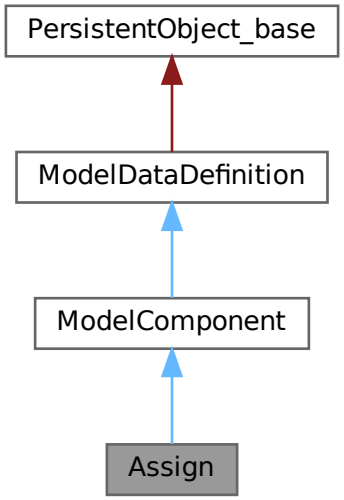
The documentation for this class was generated from the following files:

- gtest/gtest/gtest.h
- gtest/src/gtest.cc

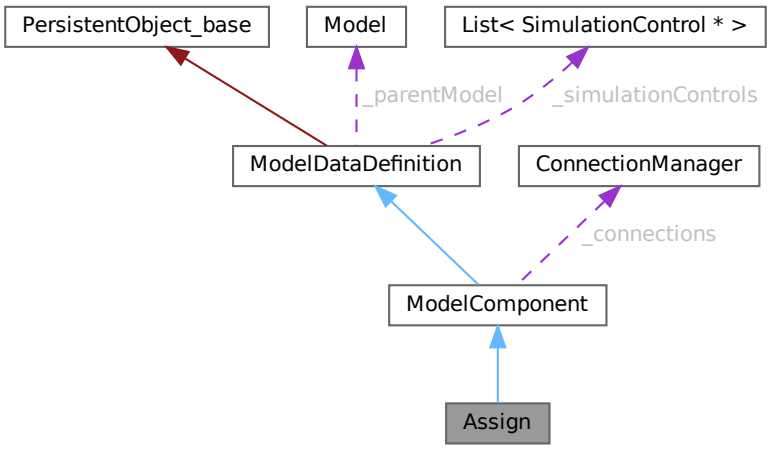
17.41 Assign Class Reference

Component that evaluates one or more (destination, expression) pairs and writes each result into an Attribute or a Variable on the entity/model.
`#include <plugins/components/Logic/Assign.h>`

Inheritance diagram for Assign:



Collaboration diagram for Assign:



Public Member Functions

- **Assign** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- List< Assignment ∗ > ∗ **getAssignments** () const
- void **addAssignment** (Assignment ∗newAssignment)
- void **removeAssignment** (Assignment ∗assignment)

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.41.1 Detailed Description

Component that evaluates one or more (destination, expression) pairs and writes each result into an Attribute or a Variable on the entity/model.
Arena correspondence: the "Assign module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 40-41), which supports five assignment kinds: Attribute, Variable (optionally a 1D/2D array via Row/↩ Column), Entity Type, Entity Picture, and Other (system variables such as resource capacity or simulation end time).
Each entry in `_assignments` (see Assignment, in `plugins/data/Logic/AssignmentItem.h`) only distinguishes Attribute vs. Variable destinations; an optional `name[index]` syntax on the destination string is the closest correspondence to Arena's separate Row/Column fields.
Known difference from Arena: there is no Entity Type, Entity Picture, or Other (system-variable) assignment kind — `_onDispatchEvent()` only has Attribute and Variable branches. See `docs/ai_↩ assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.4.

17.41.2 Member Function Documentation

17.41.2.1 _check()

```
bool Assign::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.41.2.2 _createEditableDataDefinitions()

```
void Assign::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
Reimplemented from ModelDataDefinition.

17.41.2.3 _loadInstance()

```
bool Assign::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.41.2.4 _onDispatchEvent()

```
void Assign::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.41.2.5 _saveInstance()

```
void Assign::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.41.2.6 show()

```
std::string Assign::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/Logic/Assign.h
- plugins/components/Logic/Assign.cpp

17.42 Assignment Class Reference

One (destination, expression) pair used by Assign and by Sequence's per-step assignments.
`#include <plugins/data/Logic/AssignmentItem.h>`

Public Member Functions

- **Assignment** (Model *model, std::string destination, std::string expression="1", bool isAttributeNot↩ Variable=true)
- **Assignment** (std::string destination, std::string expression, bool isAttributeNotVariable=true)
- void **setDestination** (std::string _destination)
- std::string **getDestination** () const
- void **setExpression** (std::string _expression)
- std::string **getExpression** () const
- void **setAttributeNotVariable** (bool isAttributeNotVariable)
- bool **isAttributeNotVariable** () const
- void **setChangeCallback** (std::function< void(Assignment &)> callback)
- std::string **getName** () const
- List< SimulationControl * > * **getSimulationControls** () const
- void **ensureSimulationControls** (Model *model)
- void **_addSimulationControl** (SimulationControl *control)
- std::string **getTypeDC** ()
- bool **loadInstance** (PersistenceRecord *fields, unsigned int parentIndex)
- void **saveInstance** (PersistenceRecord *fields, unsigned int parentIndex, bool saveDefault)

17.42.1 Detailed Description

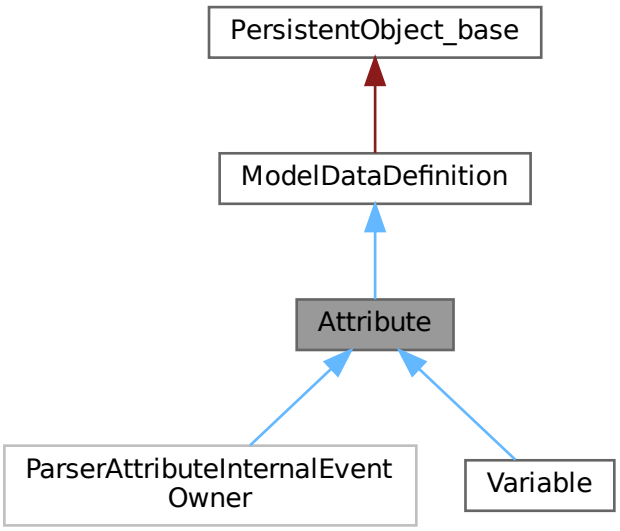
One (destination, expression) pair used by Assign and by Sequence's per-step assignments.
`destination` is a free-form name, optionally with a `[index]` suffix (parsed by the owning component, e.g. `Assign::_destinationBaseName()` / `_destinationIndex()`); `_isAttributeNotVariable` selects whether it names an Attribute or a Variable. `expression` is evaluated through the parser at dispatch time. There is no dedicated destination kind for an entity's EntityType, its picture, or an arbitrary "system variable" — see Assign for the corresponding Arena divergence.
The documentation for this class was generated from the following files:

- plugins/data/Logic/AssignmentItem.h
- plugins/data/Logic/AssignmentItem.cpp

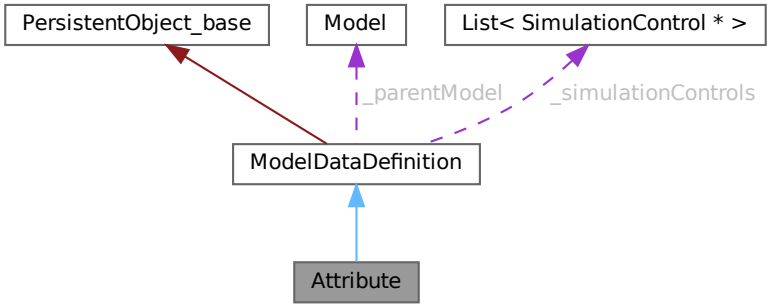
17.43 Attribute Class Reference

Data definition declaring one named, per-entity attribute (its dimensionality and sparse initial values), shared across every entity created in the model.
`#include <kernel/simulator/essentialPlugins/Attribute.h>`

Inheritance diagram for Attribute:



Collaboration diagram for Attribute:



Public Member Functions

- **Attribute** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< Attribute >())

  Creates an entity attribute definition in the model.
- virtual ~**Attribute** () override

  Releases sparse initial values owned by this definition.
- virtual std::string show () override

  Returns a textual representation of this data definition.
- virtual std::string getInitialValuesText () const

  Returns the initial values serialized in bracket notation.
- virtual void setInitialValuesText (std::string valuesText)

Parses bracket notation and replaces the initial value store.

- double **getInitialValue** (std::string index="")

  Reads the initial sparse value, returning 0.0 when the index is absent.
- void **setInitialValue** (double value, std::string index="")

  Writes the initial sparse value at the scalar or indexed position.
- void **setInitialValues** (const std::vector< std::pair< std::string, double > > values)

  Replaces/extends initial sparse values from textual index/value pairs.
- void **copyInitialValuesFrom** (const Attribute &source)

  Copies the full initial sparse store from another attribute.
- void **insertDimentionSize** (unsigned int size)

  Appends one dimension size to the attribute definition.
- std::list< unsigned int > * **getDimensionSizes** () const

  Returns dimension sizes for compatibility with existing callers.
- std::map< std::string, double > * **getInitialValues** () const

  Returns mutable initial sparse values for compatibility with existing callers.
- SparseValueStore * **getInitialValueStore** ()

  Returns the sparse initial value store used by this attribute.
- bool **isInitialValuesTextValid** () const

  Returns whether the last textual initial value is syntactically valid.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.43.1 Detailed Description

Data definition declaring one named, per-entity attribute (its dimensionality and sparse initial values), shared across every entity created in the model.
Arena correspondence: the "Attribute module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", p. 43), which defines an attribute's rows/columns, data type and initial value(s). Attribute values are per-entity (each Entity instance holds its own store), which is the same distinction Arena makes between per-entity Attributes and global Variables — see Variable, which extends this class to add global (model-wide) storage.
Initial values are kept as a sparse, indexed store (SparseValueStore) addressed by a scalar or textual index, and can also be edited through the bracket-notation text form (`getInitialValuesText()`/`setInitialValuesText()`).
Known difference from Arena: only a numeric (`double`) value type is modeled; Arena's "String" data type has no equivalent here.

17.43.2 Member Function Documentation

17.43.2.1 _check()

```
bool Attribute::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.
Reimplemented in Variable.

17.43.2.2 _loadInstance()

```
bool Attribute::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.
Reimplemented in Variable.

17.43.2.3 _saveInstance()

```
void Attribute::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

Reimplemented in Variable.

17.43.2.4 getInitialValuesText()

```
std::string Attribute::getInitialValuesText ( ) const  [virtual]
```

Returns the initial values serialized in bracket notation.
Reimplemented in Variable.

17.43.2.5 setInitialValuesText()

```
void Attribute::setInitialValuesText (
            std::string valuesText )  [virtual]
```

Parses bracket notation and replaces the initial value store.
Reimplemented in Variable.

17.43.2.6 show()

```
std::string Attribute::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
Reimplemented in Variable.
The documentation for this class was generated from the following files:

- kernel/simulator/essentialPlugins/Attribute.h
- kernel/simulator/essentialPlugins/Attribute.cpp

17.44 AuditEntry Struct Reference

One entry in the AI assistant audit log.
`#include <tools/AIAssistant/AIAuditLog.h>`

Public Attributes

- std::chrono::system_clock::time_point **timestamp**
- std::string **operation**

  e.g. "analyzePrompt", "buildModel"
- std::string **provider**

  e.g. "OpenAI", "Anthropic", "Local"
- std::string **modelName**

  LLM model identifier.
- std::string **promptPreview**

  First 120 chars of user-visible input — no keys.
- bool **success** = false
- bool **dryRun** = false
- long long **durationMs** = 0
- std::string **errorSummary**

  Empty on success; capped at 200 chars.

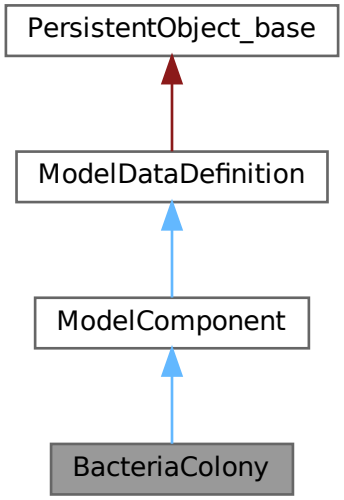
17.44.1 Detailed Description

One entry in the AI assistant audit log.
Never contains API key values. The promptPreview field is limited to the first 120 characters of the user-visible prompt text.
The documentation for this struct was generated from the following file:

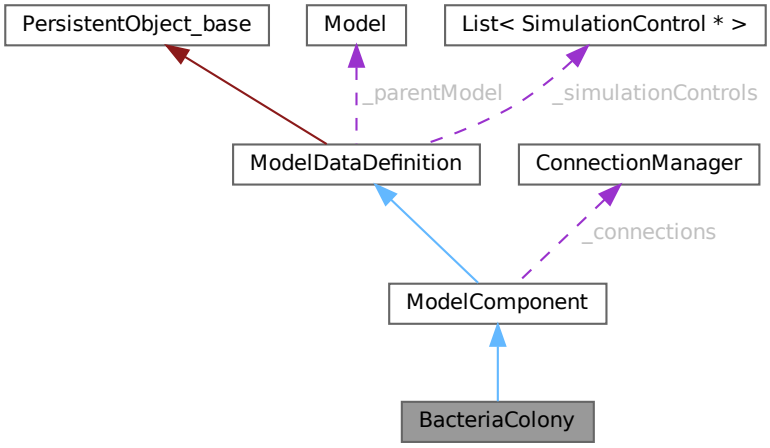
- tools/AIAssistant/AIAuditLog.h

17.45 BacteriaColony Class Reference

First GenESyS component for Gro-inspired bacteria colony simulation.
`#include <plugins/components/BiochemicalSimulation/BacteriaColony.h>`
Inheritance diagram for BacteriaColony:



Collaboration diagram for BacteriaColony:



Classes

- struct BacteriumState
- struct BarrierSegment

Public Member Functions

- **BacteriaColony** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setGroProgram** (GroProgram *groProgram)

  Sets the reusable Gro program used by this colony.
- GroProgram * **getGroProgram** () const

  Returns the reusable Gro program used by this colony.
- void **setSimulationStep** (double simulationStep)

  Sets the internal colony time increment used by one colony step.
- double **getSimulationStep** () const

  Returns the internal colony time increment.
- void **setNumSteps** (unsigned int numSteps)

  Sets the number of colony steps processed before forwarding one entity.
- unsigned int **getNumSteps** () const

  Returns the number of colony steps processed before forwarding one entity.
- void **setColonyTimeUnit** (Util::TimeUnit colonyTimeUnit)

  Sets the time unit used by the colony time-related attributes.
- Util::TimeUnit **getColonyTimeUnit** () const

  Returns the time unit used by the colony time-related attributes.
- double **getColonyTime** () const

  Returns the current colony time from the enclosing ModelSimulation.
- void **setInitialPopulation** (unsigned int initialPopulation)

  Sets the initial bacteria population summary for each replication.
- unsigned int **getInitialPopulation** () const

  Returns the initial bacteria population summary.

- unsigned int **getPopulationSize** () const

  Returns the current bacteria population summary.
- void **setAutomaticDivisionEnabled** (bool enabled)

  Enables optional automatic volume-threshold division.
- bool **getAutomaticDivisionEnabled** () const

  Returns whether automatic division is enabled.
- void **setDivisionThresholdVolume** (double threshold)

  Sets the volume threshold that triggers automatic division.
- double **getDivisionThresholdVolume** () const

  Returns the automatic-division volume threshold.
- void **setBioNetwork** (BioNetwork *bioNetwork)

  Sets the optional BioNetwork reused as the colony biochemical state.
- BioNetwork * **getBioNetwork** () const

  Returns the optional BioNetwork reused as the colony biochemical state.
- void **setSignalGrid** (BacteriaSignalGrid *signalGrid)

  Sets the reusable signal-grid definition used by this colony.
- BacteriaSignalGrid * **getSignalGrid** () const

  Returns the reusable signal-grid definition used by this colony.
- std::size_t **getInternalBacteriaCount** () const

  Returns the number of internal bacteria currently owned by the colony.
- const BacteriumState & **getBacteriumState** (std::size_t index) const

  Returns one internal bacterium state by zero-based index.
- double **getBacteriumAge** (std::size_t index) const

  Returns the age of one internal bacterium at the current colony time.
- double **getBacteriumPositionX** (std::size_t index) const

  Returns the continuous X coordinate used by the viewer.
- double **getBacteriumPositionY** (std::size_t index) const

  Returns the continuous Y coordinate used by the viewer.
- double **getBacteriumDirectionRadians** (std::size_t index) const

  Returns the bacterium direction in radians.
- double **getBacteriumVolume** (std::size_t index) const

  Returns the current bacterium volume.
- double **getBacteriumSize** (std::size_t index) const

  Returns the current bacterium size used for rendering.
- double **getBacteriumGfp** (std::size_t index) const

  Returns the current GFP amount used for rendering.
- double **getBacteriumRfp** (std::size_t index) const

  Returns the current RFP amount used for rendering.
- double **getBacteriumRuntimeVariableValue** (std::size_t index, const std::string &variableName) const

  Returns the last stored value for one bacterium-scoped runtime variable.
- bool **hasBacteriumRuntimeVariable** (std::size_t index, const std::string &variableName) const

  Tells whether one bacterium currently stores one runtime scalar variable.
- double **getRuntimeVariableValue** (const std::string &variableName) const

  Returns the last stored value for one runtime scalar variable.
- bool **hasRuntimeVariable** (const std::string &variableName) const

  Tells whether the colony currently stores one runtime scalar variable.
- double **getSignalValueAt** (unsigned int x, unsigned int y) const

  Returns the runtime signal value stored at one grid coordinate.
- unsigned int **getSignalChannelCount** () const

  Returns the number of currently available signal channels, including the legacy first channel.
- double **getAdditionalSignalValueAt** (unsigned int channel, unsigned int x, unsigned int y) const

Returns the runtime value for one additional (handle >= 2) signal channel at one grid coordinate.

- std::vector< std::vector< double > > **getSignalMatrix** () const

  Returns the current runtime signal field as a row-major matrix.
- std::string **getSignalMatrixDump** (std::size_t maxRows=0, std::size_t maxColumns=0) const

  Returns a textual dump of the current runtime signal field.
- double **getBacteriumLocalSignal** (std::size_t index) const

  Returns the runtime signal value seen by one bacterium at its current cell.
- void **setChemostatMode** (bool enabled)

  Enables or disables the colony chemostat mode used by Gro microfluidics.
- bool **getChemostatMode** () const

  Returns whether the colony chemostat mode is enabled.
- void **addBarrier** (double x1, double y1, double x2, double y2)

  Adds one barrier segment to the colony environment.
- const std::vector< BarrierSegment > & **getBarriers** () const

  Returns the current barrier segments used by the viewer.
- const std::vector< double > & **getMappedCellValues** () const

  Returns the latest per-cell values produced by map_to_cells, if any.
- const std::string & **getMappedCellExpression** () const

  Returns the expression used by the latest map_to_cells call, if any.
- void **setGridWidth** (unsigned int gridWidth)

  Sets the discrete spatial grid width reserved for the first colony model.
- unsigned int **getGridWidth** () const

  Returns the discrete spatial grid width.
- void **setGridHeight** (unsigned int gridHeight)

  Sets the discrete spatial grid height reserved for the first colony model.
- unsigned int **getGridHeight** () const

  Returns the discrete spatial grid height.
- GroProgramRuntime::ExecutionResult **executeGroProgram** ()

  Executes the configured Gro program once against the current colony state.

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.45.1 Detailed Description

First GenESyS component for Gro-inspired bacteria colony simulation.
BacteriaColony owns bacteria and signal-field runtime state while using the enclosing ModelSimulation time as the single colony clock. The component references a reusable GroProgram and executes the first supported runtime commands through plugin-side parser/runtime helpers.

17.45.2 Member Function Documentation

17.45.2.1 _check()

```
bool BacteriaColony::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.45.2.2 _createAttachedAttributes()

```
void BacteriaColony::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.45.2.3 _initBetweenReplications()

```
void BacteriaColony::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.45.2.4 _loadInstance()

```
bool BacteriaColony::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.45.2.5 _onDispatchEvent()

```
void BacteriaColony::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.45.2.6 _saveInstance()

```
void BacteriaColony::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.45.2.7 show()

```
std::string BacteriaColony::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

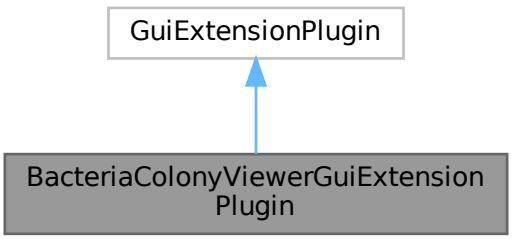
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

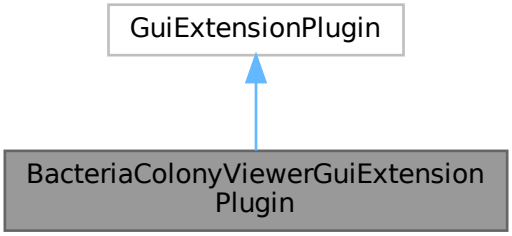
- plugins/components/BiochemicalSimulation/BacteriaColony.h
- plugins/components/BiochemicalSimulation/BacteriaColony.cpp

17.46 BacteriaColonyViewerGuiExtensionPlugin Class Reference

Inheritance diagram for BacteriaColonyViewerGuiExtensionPlugin:



Collaboration diagram for BacteriaColonyViewerGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.46.1 Member Function Documentation

17.46.1.1 displayName()

```
std::string BacteriaColonyViewerGuiExtensionPlugin::displayName ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.46.1.2 extensionId()

```
std::string BacteriaColonyViewerGuiExtensionPlugin::extensionId ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.46.1.3 registerContributions()

```
void BacteriaColonyViewerGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.46.1.4 requiredModelPlugins()

```
std::vector< std::string > BacteriaColonyViewerGuiExtensionPlugin::requiredModelPlugins ( )
const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.46.1.5 version()

```
std::string BacteriaColonyViewerGuiExtensionPlugin::version ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

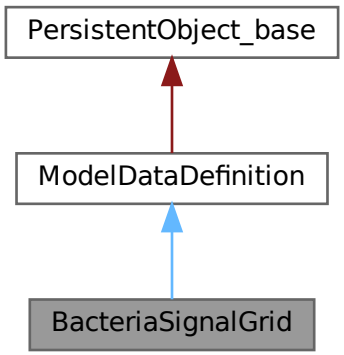
- applications/gui/genesys/extensions/BacteriaColonyViewerGuiExtensionPlugin.cpp

17.47 BacteriaSignalGrid Class Reference

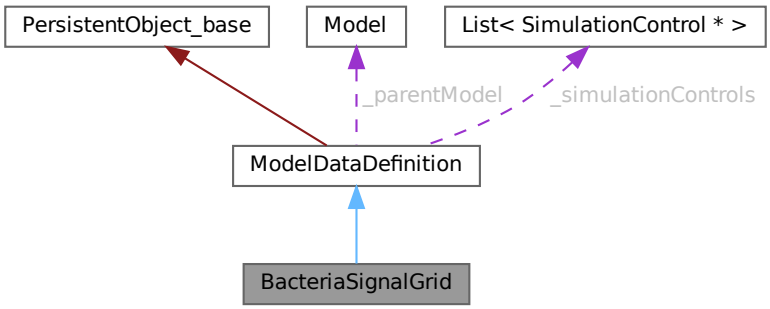
Reusable discrete signal-field definition for bacteria colonies.

`#include <plugins/data/BiochemicalSimulation/BacteriaSignalGrid.h>`

Inheritance diagram for BacteriaSignalGrid:



Collaboration diagram for BacteriaSignalGrid:



Public Member Functions

- **BacteriaSignalGrid** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **setWidth** (unsigned int width)
- unsigned int **getWidth** () const
- void **setHeight** (unsigned int height)
- unsigned int **getHeight** () const
- void **setInitialSignal** (double initialSignal)
- double **getInitialSignal** () const
- void **setDiffusionRate** (double diffusionRate)
- double **getDiffusionRate** () const
- void **setDecayRate** (double decayRate)

- double **getDecayRate** () const
- void **setInitialValues** (std::string initialValues)
- std::string **getInitialValues** () const
- unsigned int **getCellCount** () const
- bool **buildInitialField** (std::vector< double > &values, std::string &errorMessage) const

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()

- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model ∗ **_parentModel**
- List< SimulationControl ∗ > ∗ **_simulationControls** = new List<SimulationControl∗>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.47.1 Detailed Description

Reusable discrete signal-field definition for bacteria colonies.
The data definition persists the spatial configuration and initial scalar values of one signal field. Colonies copy this configuration into their runtime state so model configuration and simulation mutation stay separated.

17.47.2 Member Function Documentation

17.47.2.1 _check()

```
bool BacteriaSignalGrid::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.47.2.2 _loadInstance()

```
bool BacteriaSignalGrid::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.47.2.3 _saveInstance()

```
void BacteriaSignalGrid::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.47.2.4 show()

```
std::string BacteriaSignalGrid::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/BiochemicalSimulation/BacteriaSignalGrid.h
- plugins/data/BiochemicalSimulation/BacteriaSignalGrid.cpp

17.48 BacteriaColony::BacteriumState Struct Reference

Public Attributes

- unsigned int **id** = 0
- unsigned int **parentId** = 0
- unsigned int **generation** = 0

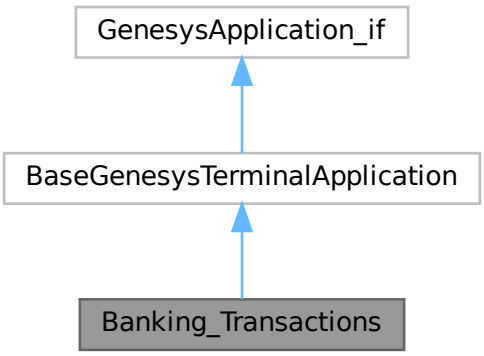
- std::string **programName** = ""
- std::vector< double > **programArguments**
- unsigned int **divisionCount** = 0
- double **birthTime** = 0.0
- double **lastUpdateTime** = 0.0
- double **lastDivisionTime** = 0.0
- double **positionX** = 0.0
- double **positionY** = 0.0
- double **directionRadians** = 0.0
- double **volume** = 1.0
- double **size** = 1.0
- double **gfp** = 0.0
- double **rfp** = 0.0
- double **yfp** = 0.0
- double **cfp** = 0.0
- double **speed** = 0.1
- unsigned int **tickCount** = 0
- unsigned int **gridX** = 0
- unsigned int **gridY** = 0
- bool **hasExplicitGridPosition** = false
- bool **positionInitialized** = false
- bool **justDivided** = false
- bool **daughter** = false
- bool **alive** = true
- std::map< std::string, double > **runtimeVariables**

The documentation for this struct was generated from the following file:

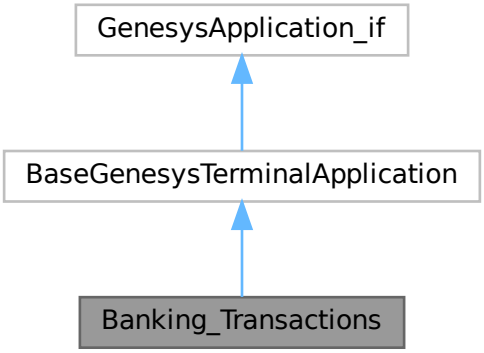
- plugins/components/BiochemicalSimulation/BacteriaColony.h

17.49 Banking_Transactions Class Reference

Inheritance diagram for Banking_Transactions:



Collaboration diagram for Banking_Transactions:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.49.1 Member Function Documentation

17.49.1.1 main()

```
int Banking_Transactions::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/arenaExamples/Banking_Transactions.h
- applications/modelSpecific/arenaExamples/Banking_Transactions.cpp

17.50 BacteriaColony::BarrierSegment Struct Reference

Public Attributes

- double **x1** = 0.0
- double **y1** = 0.0
- double **x2** = 0.0
- double **y2** = 0.0

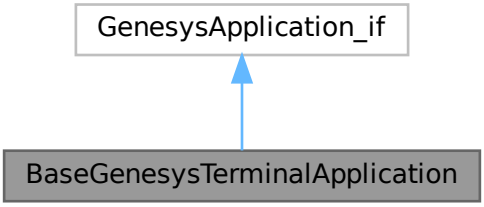
The documentation for this struct was generated from the following file:

- plugins/components/BiochemicalSimulation/BacteriaColony.h

17.51 BaseGenesysTerminalApplication Class Reference

Inherits GenesysApplication_if.
Inherited by AirportSecurityExample, AirportSecurityExampleExtended, Airport_Extended1, AnElectronicAssemblyAndTestSystem, Assembly_Line, Banking_Transactions, Book_Cap02_Example01, BufferFIFO, ContinuousModel, Example_BasicOrderShipping, Example_PortModel, Example_PublicTransport, FullSimulationOfComplexModel, GenesysShell, Half_Adder, Loja01, OperatingSystem02, OperatingSystem03, Rectifier, Smart_AddingResource, Smart_AlternatingEntityCreation, Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime, Smart_ArrivalsEntityTypeVsAttribute, Smart_AssignExample, Smart_AssignWriteSeizes, Smart_Attribute_Variable, Smart_AutomaticStatisticsCollection, Smart_BacteriaColony_GRO, Smart_BasicModeling, Smart_BatchAndSeparate, Smart_BatchSeparate, Smart_BioKineticLawRegulation, Smart_BioNetworkAnalysis, Smart_BioReversibleKineticLaw, Smart_BioReversibleMassAction, Smart_Buffer, Smart_CellularAutomata, Smart_CellularAutomata1, Smart_Clone, Smart_ContinuousFlowEntities, Smart_CppForG, Smart_Create, Smart_DecideNWayByChance, Smart_DefaultModalModel, Smart_DefiningAttributesAsStrings, Smart_DefiningControlLogic, Smart_DefiningResourceCapacity, Smart_Delay, Smart_DelayBasedOnReplication, Smart_Dummy, Smart_EntitiesProcessedByPriority, Smart_EvaluatingConditionsBeforeEnteringQueue, Smart_Expression, Smart_FSMState, Smart_Failures, Smart_GroColonyGrowth, Smart_GroColonyLifecycle, Smart_HoldSearchRemove, Smart_InventoryAndHoldingCosts, Smart_LSODE, Smart_MarkovChain, Smart_MaxArrivalsField, Smart_ModalModel, Smart_ModalModelDefaultNetwork, Smart_ModalModelFSM, Smart_ModalModelPetriNet, Smart_ModelInfoModelSimulation, Smart_ModelRunUntil1000Parts, Smart_ModuleDisplayVariables, Smart_ODE, Smart_OnEvent, Smart_OverlappingResources, Smart_ParallelProcessingOfEntities, Smart_Parser, Smart_ParserModelFunctions, Smart_PetriPlace, Smart_PlacingEntitiesInQueue, Smart_Plugin, Smart_PriorityExample, Smart_Process, Smart_ProcessArena, Smart_ProcessSet, Smart_PythonForG, Smart_R_Simulator, Smart_Record, Smart_Record_Arena, Smart_RemovingAndReorderingEntitiesInAQueue, Smart_ResourceCosting, Smart_ResourceScheduleCosting, Smart_ResourceSets, Smart_RouteStation, Smart_SeizeDelayRelease, Smart_SeizeDelayReleaseMany, Smart_SeizeDelayReleaseNoDataDefs, Smart_SeizingMultipleSimulta, Smart_SelectingRouteBasedOnProbability, Smart_SelectingShorterQueue, Smart_Sequence, Smart_SimulationControlResponse, Smart_SynchronizingParallelEntities, Smart_ValueAdded, Smart_WaitForSignal, Smart_WaitScanCondition, Smart_WaitSignal, StartAIAssistant, WcmBioNetworkBridge, WcmCompartmentTransportWholeCellMvp, WcmFluxBalanceWholeCellMvp, WcmGeneExpression, WcmLifecycleStress, WcmMgenitaliumKarr2012, WcmPathwayStressWholeCellMvp, WcmSimpleSsa, WcmWholeCellMvp, WcmYeastDidacticWholeCellMvp, and WcmYeastEukaryoticWholeCellMvp.

Collaboration diagram for BaseGenesysTerminalApplication:



Public Member Functions

- virtual int main (int argc, char **argv) override
- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Protected Member Functions

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.51.1 Member Function Documentation

17.51.1.1 main()

```
int BaseGenesysTerminalApplication::main (
            int argc,
            char ** argv )  [override], [virtual]
```

Implements GenesysApplication_if.

Reimplemented in Airport_Extended1, Assembly_Line, Banking_Transactions, Example_BasicOrderShipping, Example_PortModel, Example_PublicTransport, Smart_AddingResource, Smart_AlternatingEntityCreation, Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime, Smart_AutomaticStatisticsCollection, Smart_ContinuousFlowEntities, Smart_Create, Smart_DecideNWayByChance, Smart_DefiningAttributesAsStrings, Smart_DefiningControlLogic, Smart_DefiningResourceCapacity, Smart_EntitiesProcessedByPriority, Smart_EvaluatingConditionsBeforeEnteringQueue, Smart_Expression, Smart_MaxArrivalsField, Smart_ModelRunUntil1000Parts, Smart_ModuleDisplayVariables,

Smart_OverlappingResources, Smart_ParallelProcessingOfEntities, Smart_PlacingEntitiesInQueueSets, Smart_PriorityExample, Smart_ProcessArena, Smart_Record_Arena, Smart_RemovingAndReorderingEntitiesInAQueue, Smart_ResourceSets, Smart_SeizingMultipleSimultaneosly, Smart_SelectingRouteBasedOnProbability, Smart_SelectingShorterQueue, Smart_SynchronizingParallelEntities, Smart_WaitForSignal, Smart_MarkovChain, Smart_Buffer, Smart_Clone, Smart_Delay, Smart_Failures, Smart_HoldSearchRemove, Smart_Process, Smart_ProcessSet, Smart_SeizeDelayRelease, Smart_SeizeDelayReleaseMany, Smart_SeizeDelayReleaseNoDataDefs, Smart_R_Simulator, Smart_Record, Smart_Attribute_Variable, Smart_ModelInfoModelSimulation, Smart_Parser, Smart_ParserModelFunctions, Smart_RouteStation, Smart_CellularAutomata, Smart_CellularAutomata1, Smart_DefaultModalModel, Smart_Dummy, Smart_OnEvent, Smart_WaitScanCondition, Smart_WaitSignal, FullSimulationOfComplexModel, Half_Adder, and Rectifier.

17.51.1.2 setDefaultEventHandlers()

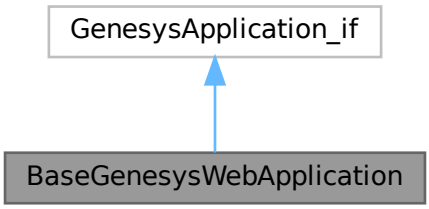
```
void BaseGenesysTerminalApplication::setDefaultEventHandlers (
            OnEventManager * oem )
```

Add default method to handle simulator events so information is plotted to output screen
The documentation for this class was generated from the following files:

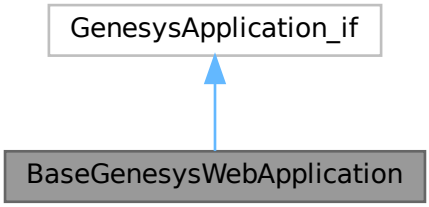
- applications/BaseGenesysTerminalApplication.h
- applications/BaseGenesysTerminalApplication.cpp

17.52 BaseGenesysWebApplication Class Reference

Entry point for the standalone Worker application flavor of Genesys.
`#include <applications/worker/BaseGenesysWebApplication.h>`
Inheritance diagram for BaseGenesysWebApplication:



Collaboration diagram for BaseGenesysWebApplication:



Public Member Functions

- int main (int argc, char **argv) override

  Parses command-line arguments, starts the HTTP server and runs the worker app.

17.52.1 Detailed Description

Entry point for the standalone Worker application flavor of Genesys.
The worker variant bootstraps a session manager, HTTP server and router, then keeps processing requests until the configured request limit is reached or a transport error occurs.

17.52.2 Member Function Documentation

17.52.2.1 main()

```
int BaseGenesysWebApplication::main (
            int argc,
            char ** argv )  [override], [virtual]
```

Parses command-line arguments, starts the HTTP server and runs the worker app.
Boots the web stack and keeps the request loop alive until shutdown.

Parameters

argc	Number of command-line arguments.
argv	Command-line argument vector.

Returns

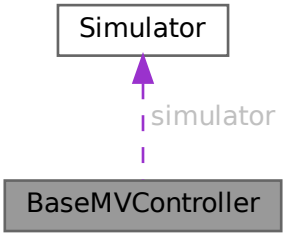
Zero on normal termination; non-zero on startup or server failure.

This entry point wires token management, session creation, worker-job storage, API routing and the embedded HTTP server into one runtime pipeline.
Implements GenesysApplication_if.
The documentation for this class was generated from the following files:

- applications/worker/BaseGenesysWebApplication.h
- applications/worker/BaseGenesysWebApplication.cpp

17.53 BaseMVController Class Reference

Compatibility base class that binds the Qt UI and the simulator model.
`#include <applications/gui/genesys/BaseMVController.h>`
Collaboration diagram for BaseMVController:



Protected Attributes

- Ui::MainWindow * **ui**

  MainWindow UI built from the Qt Designer form.
- Simulator * **simulator**

  Active simulator instance owned elsewhere by the application.

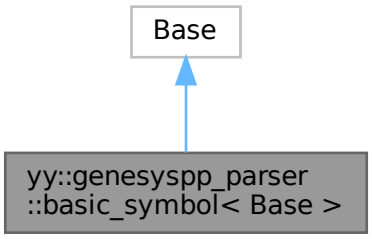
17.53.1 Detailed Description

Compatibility base class that binds the Qt UI and the simulator model.
Legacy GUI controllers in Genesys share the same two core pointers: the main window UI object and the active simulator instance. This base class keeps that pairing in one place so higher-level controllers can reuse it.
The documentation for this class was generated from the following files:

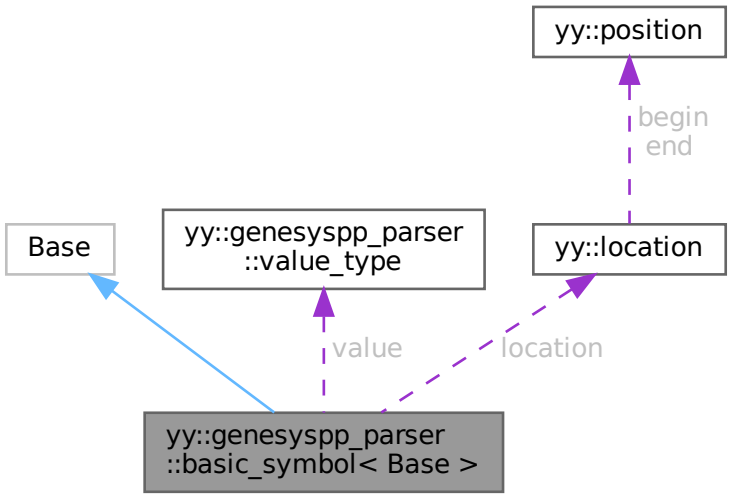
- applications/gui/genesys/BaseMVController.h
- applications/gui/genesys/BaseMVController.cpp

17.54 yy::genesyspp_parser::basic_symbol< Base > Struct Template Reference

`#include <parser/GenesysParser.h>`
Inheritance diagram for yy::genesyspp_parser::basic_symbol< Base >:



Collaboration diagram for yy::genesyspp_parser::basic_symbol< Base >:



Public Types

- typedef Base **super_type**

  Alias to Base.

Public Member Functions

- **basic_symbol** () YY_NOEXCEPT

  Default constructor.
- **basic_symbol** (const basic_symbol &that)

  Copy constructor.
- **basic_symbol** (typename Base::kind_type t, const location_type &l)

  Constructors for typed symbols.
- **basic_symbol** (typename Base::kind_type t, const obj_t &v, const location_type &l)
- **basic_symbol** (typename Base::kind_type t, const std::string &v, const location_type &l)
- ~**basic_symbol** ()

  Destroy the symbol.
- void **clear** () YY_NOEXCEPT

  Destroy contents, and record that is empty.
- std::string **name** () const YY_NOEXCEPT

  The user-facing name of this symbol.
- symbol_kind_type **type_get** () const YY_NOEXCEPT

  Backward compatibility (Bison 3.6).
- bool **empty** () const YY_NOEXCEPT

  Whether empty.
- void **move** (basic_symbol &s)

  Destructive move, s is emptied into this.

Public Attributes

- value_type **value**

  The semantic value.
- location_type **location**

  The location.

17.54.1 Detailed Description

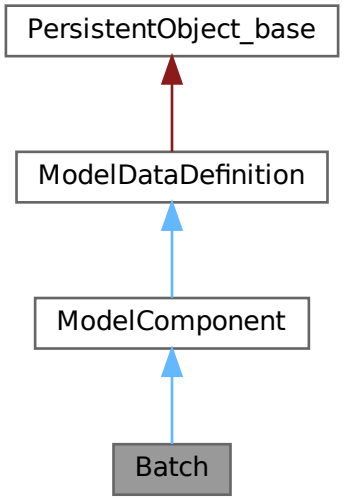
template<**typename Base**>
struct yy::genesyspp_parser::basic_symbol< **Base** >

A complete symbol.
Expects its Base type to provide access to the symbol kind via kind ().
Provide access to semantic value and location.
The documentation for this struct was generated from the following file:

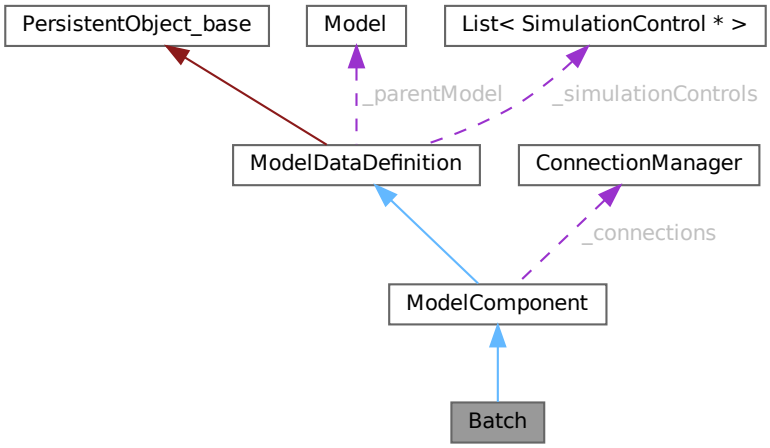
- parser/GenesysParser.h

17.55 Batch Class Reference

Accumulates entities in an internal queue until a batch is complete, then dispatches one representative entity for the group.
`#include <plugins/components/Grouping/Batch.h>`
Inheritance diagram for Batch:



Collaboration diagram for Batch:



Public Types

- enum class **BatchType** : int { **Temporary** = 0 , **Permanent** = 1 , **num_elements** = 2 }
- enum class **Rule** : int { **Any** = 0 , **ByAttribute** = 1 , **num_elements** = 2 }
- enum class **GroupedAttribs** : int { **FirstEntity** = 0 , **LastEntity** = 1 , **SumAttributes** = 2 , **num_elements** = 3 }

Public Member Functions

- **Batch** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setGroupedEntityType** (EntityType ∗groupedEntityType)
- void **setGroupedEntityTypeName** (std::string groupedEntityTypeName)
- EntityType ∗ **getGroupedEntityType** () const
- void **setBatchType** (Batch::BatchType batchType)
- Batch::BatchType **getBatchType** () const
- void **setAttributeName** (std::string attributeName)
- std::string **getAttributeName** () const
- void **setBatchSize** (std::string batchSize)
- std::string **getBatchSize** () const
- void **setRule** (Batch::Rule _rule)
- Batch::Rule **getRule** () const
- void **setGroupedAttributes** (Batch::GroupedAttribs _groupedAttributes)
- Batch::GroupedAttribs **getGroupedAttributes** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

new public user methods for this component

- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static std::string **convertEnumToStr** (BatchType type)
- static std::string **convertEnumToStr** (Rule rule)
- static std::string **convertEnumToStr** (GroupedAttribs attribs)
- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _createNonEditableDataDefinitions () override
- virtual void _createEditableDataDefinitions () override
- virtual void _createAttachedAttributes () override
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions

- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.55.1 Detailed Description

Accumulates entities in an internal queue until a batch is complete, then dispatches one representative entity for the group.
Arena correspondence: the "Batch module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", p. 38). `BatchType` (Temporary/Permanent), `Rule` (Any/ByAttribute) + `_attributeName`, and `GroupedAttribs` (FirstEntity/LastEntity/ SumAttributes) map directly onto Arena's Type, Rule/Attribute Name, and Save Criterion (First/Last/Sum) fields; `_batchSize` is a parsed expression generalizing Arena's static number; `_groupedEntityType` is the Representative Entity type. A temporary batch is later split back into its original members by Separate.

17.55.2 Member Function Documentation

17.55.2.1 _check()

```
bool Batch::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.55.2.2 _createAttachedAttributes()

```
void Batch::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.55.2.3 _createEditableDataDefinitions()

```
void Batch::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.55.2.4 _createNonEditableDataDefinitions()

```
void Batch::_createNonEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.55.2.5 _loadInstance()

```
bool Batch::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.55.2.6 _onDispatchEvent()

```
void Batch::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.55.2.7 _saveInstance()

```
void Batch::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.55.2.8 show()

```
std::string Batch::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/Grouping/Batch.h
- plugins/components/Grouping/Batch.cpp

17.56 testing::internal::TemplateSel< Tmpl >::Bind< T > Struct Template Reference

Public Types

- typedef Tmpl< T > **type**

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-type-util.h

17.57 ModalModelBindingsUndoCommand::Bindings Struct Reference

Public Attributes

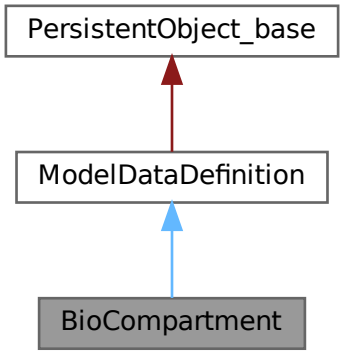
- std::vector< std::string > **inputs**
- std::vector< std::string > **outputs**

The documentation for this struct was generated from the following file:

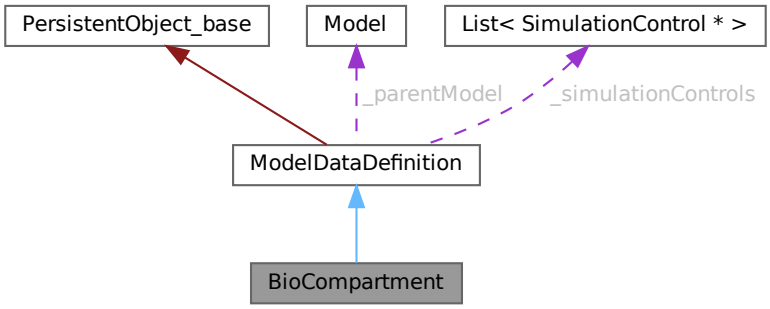
- applications/gui/genesys/actions/ModalModelBindingsUndoCommand.h

17.58 BioCompartment Class Reference

Defines a biological compartment for biochemical and whole-cell models.
`#include <plugins/data/WholeCellModeling/BioCompartment.h>`
Inheritance diagram for BioCompartment:



Collaboration diagram for BioCompartment:



Public Member Functions

- **BioCompartment** (Model *model, std::string name="")
- void **setCompartmentType** (std::string type)
- std::string **getCompartmentType** () const
- void **setParentCompartmentName** (std::string parentName)
- std::string **getParentCompartmentName** () const
- void **setMembraneBounded** (bool membraneBounded)
- bool **isMembraneBounded** () const
- void **setVolumeFraction** (double volumeFraction)
- double **getVolumeFraction** () const
- void **setCopyNumber** (unsigned int copyNumber)
- unsigned int **getCopyNumber** () const
- void **setRole** (std::string role)
- std::string **getRole** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes

- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.58.1 Detailed Description

Defines a biological compartment for biochemical and whole-cell models.
BioCompartment represents a named physical or logical cellular region such as cytosol, nucleus, mitochondrion, bud, membrane, or extracellular medium. It is a ModelDataDefinition and therefore stores structural metadata used by other plugins, but it does not execute simulation behavior itself.
The class is intentionally generic so it can support both prokaryotic and eukaryotic cells. Parent-child relationships allow hierarchical structures such as:

- extracellular -> cell surface
- cytosol -> nucleus
- cytosol -> mitochondrion
- cytosol -> bud

17.58.2 Member Function Documentation

17.58.2.1 _check()

```
bool BioCompartment::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.58.2.2 _createEditableDataDefinitions()

```
void BioCompartment::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.58.2.3 _loadInstance()

```
bool BioCompartment::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.58.2.4 _saveInstance()

```
void BioCompartment::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.58.2.5 show()

```
std::string BioCompartment::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/WholeCellModeling/BioCompartment.h
- plugins/data/WholeCellModeling/BioCompartment.cpp

17.59 BioKineticLawExpression Class Reference

```
#include <tools/Biochemical/BioKineticLawExpression.h>
```

Public Types

- using **SymbolResolver** = std::function< bool(const std::string &name, double &value)>

Public Member Functions

- bool **evaluate** (const std::string &expression, SymbolResolver resolver, double &value, std::string &error↩ Message) const

17.59.1 Detailed Description

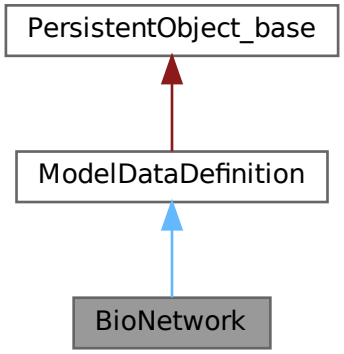
Small expression evaluator for biochemical kinetic laws.

The evaluator intentionally avoids the global GenESyS parser because the generated parser does not currently expose BioSpecies and BioParameter as kinetic-law symbols. Supported syntax is numeric literals, identifiers, parentheses, unary +/- operators, +, -, ∗, /, ^, and a small set of math functions.
The documentation for this class was generated from the following file:

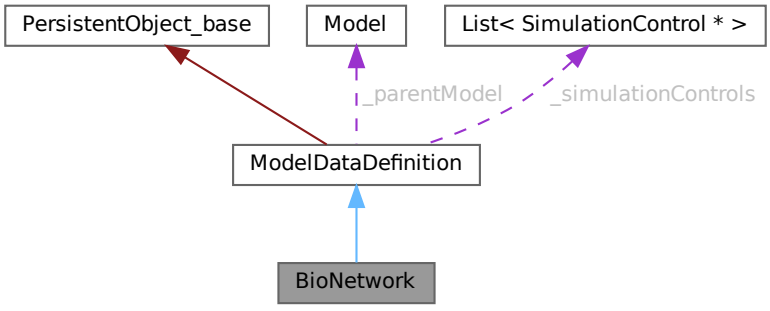
- tools/Biochemical/BioKineticLawExpression.h

17.60 BioNetwork Class Reference

`#include <plugins/data/BiochemicalSimulation/BioNetwork.h>`
Inheritance diagram for BioNetwork:



Collaboration diagram for BioNetwork:



Public Member Functions

- **BioNetwork** (Model ∗model, std::string name="")
- void **setStartTime** (double startTime)
- double **getStartTime** () const
- void **setStopTime** (double stopTime)
- double **getStopTime** () const
- void **setStepSize** (double stepSize)

- double **getStepSize** () const
- void **setCurrentTime** (double currentTime)
- double **getCurrentTime** () const
- void **setAutoSchedule** (bool autoSchedule)
- bool **getAutoSchedule** () const
- void **setLastStatus** (std::string lastStatus)
- std::string **getLastStatus** () const
- void **setLastErrorMessage** (std::string lastErrorMessage)
- std::string **getLastErrorMessage** () const
- void **setLastResponsePayload** (std::string lastResponsePayload)
- std::string **getLastResponsePayload** () const
- void **addSpecies** (std::string speciesName)
- void **addReaction** (std::string reactionName)
- void **clearSpecies** ()
- void **clearReactions** ()
- const std::vector< std::string > & **getSpeciesNames** () const
- const std::vector< std::string > & **getReactionNames** () const
- const BioSimulationResult & **getLastSimulationResult** () const
- bool **getSpeciesTimeCourseDataset** (const std::string &speciesName, SimulationResultsDataset *dataset, std::string *errorMessage=nullptr) const
- bool **buildOdeSystemForAnalysis** (MassActionOdeSystem *system, std::string &errorMessage) const
- bool **getStoichiometryMatrix** (BioStoichiometryMatrix *matrix, std::string *errorMessage=nullptr) const
- bool **getReactionRateTimeCourse** (BioReactionRateTimeCourse *timeCourse, std::string *errorMessage=nullptr) const
- bool **checkLastSampleSteadyState** (double tolerance, BioSteadyStateCheck *check, std::string *errorMessage=nullptr) const
- bool **scanLocalParameterSensitivity** (double relativeStep, double absoluteStep, BioSensitivityScan *scan, std::string *errorMessage=nullptr) const
- bool **simulate** (std::string &errorMessage)
- bool **simulate** (double startTime, double stopTime, double stepSize, std::string &errorMessage)
- bool **advanceOneStep** (std::string &errorMessage)
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createEditableDataDefinitions () override

  Refreshes editable associations to referenced biochemical definitions.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions

- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.60.1 Detailed Description

Native deterministic biochemical network runner.
BioNetwork advances BioSpecies with fixed-step RK4 over a mass-action ODE system. Networks may define explicit BioSpecies and BioReaction membership by name; when a membership list is empty, the runner falls back to discovering every registered definition of that type in the owning model for compatibility with existing models.

17.60.2 Member Function Documentation

17.60.2.1 _check()

```
bool BioNetwork::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.60.2.2 _createEditableDataDefinitions()

```
void BioNetwork::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Refreshes editable associations to referenced biochemical definitions.
BioNetwork stores model membership by name for persistence. During model checking these associations keep referenced species, reactions, and parameters from being pruned as orphan ModelDataDefinitions.
Reimplemented from ModelDataDefinition.

17.60.2.3 _initBetweenReplications()

```
void BioNetwork::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.60.2.4 _loadInstance()

```
bool BioNetwork::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.60.2.5 _saveInstance()

```
void BioNetwork::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.60.2.6 show()

```
std::string BioNetwork::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

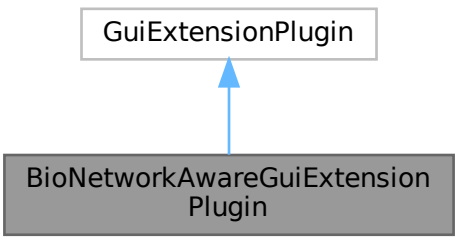
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

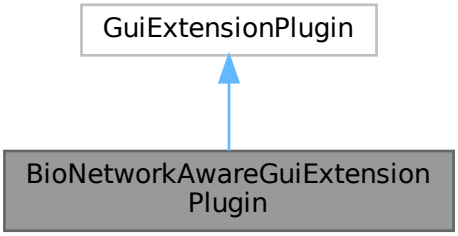
- plugins/data/BiochemicalSimulation/BioNetwork.h
- plugins/data/BiochemicalSimulation/BioNetwork.cpp

17.61 BioNetworkAwareGuiExtensionPlugin Class Reference

Inheritance diagram for BioNetworkAwareGuiExtensionPlugin:



Collaboration diagram for BioNetworkAwareGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.61.1 Member Function Documentation

17.61.1.1 displayName()

```
std::string BioNetworkAwareGuiExtensionPlugin::displayName ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.61.1.2 extensionId()

```
std::string BioNetworkAwareGuiExtensionPlugin::extensionId ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.61.1.3 registerContributions()

```
void BioNetworkAwareGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.61.1.4 requiredModelPlugins()

```
std::vector< std::string > BioNetworkAwareGuiExtensionPlugin::requiredModelPlugins ( ) const
[inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.61.1.5 version()

```
std::string BioNetworkAwareGuiExtensionPlugin::version ( ) const  [inline], [override], [virtual]
```

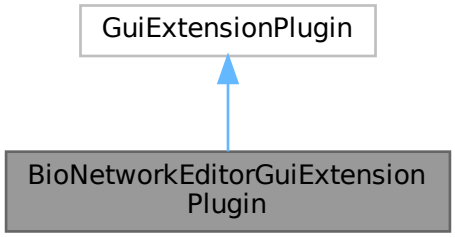
Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

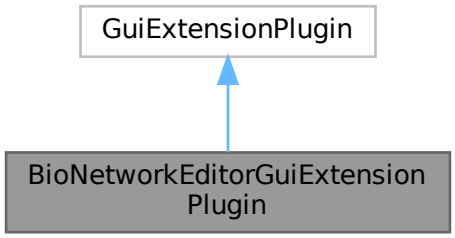
- applications/gui/genesys/extensions/BioNetworkAwareGuiExtensionPlugin.cpp

17.62 BioNetworkEditorGuiExtensionPlugin Class Reference

Inheritance diagram for BioNetworkEditorGuiExtensionPlugin:



Collaboration diagram for BioNetworkEditorGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.62.1 Member Function Documentation

17.62.1.1 displayName()

```
std::string BioNetworkEditorGuiExtensionPlugin::displayName ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.62.1.2 extensionId()

```
std::string BioNetworkEditorGuiExtensionPlugin::extensionId ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.62.1.3 registerContributions()

```
void BioNetworkEditorGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.62.1.4 requiredModelPlugins()

```
std::vector< std::string > BioNetworkEditorGuiExtensionPlugin::requiredModelPlugins ( ) const
[inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.62.1.5 version()

```
std::string BioNetworkEditorGuiExtensionPlugin::version ( ) const  [inline], [override], [virtual]
```

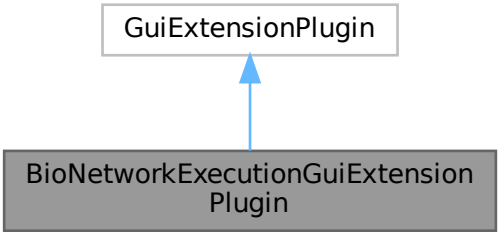
Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

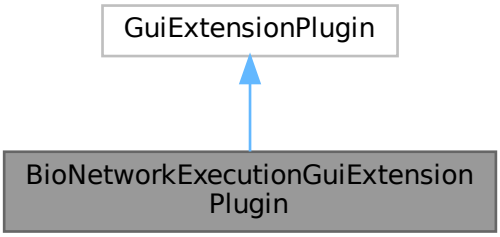
- applications/gui/genesys/extensions/BioNetworkEditorGuiExtensionPlugin.cpp

17.63 BioNetworkExecutionGuiExtensionPlugin Class Reference

Inheritance diagram for BioNetworkExecutionGuiExtensionPlugin:



Collaboration diagram for BioNetworkExecutionGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.63.1 Member Function Documentation

17.63.1.1 displayName()

```
std::string BioNetworkExecutionGuiExtensionPlugin::displayName ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.63.1.2 extensionId()

```
std::string BioNetworkExecutionGuiExtensionPlugin::extensionId ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.63.1.3 registerContributions()

```
void BioNetworkExecutionGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.63.1.4 requiredModelPlugins()

```
std::vector< std::string > BioNetworkExecutionGuiExtensionPlugin::requiredModelPlugins ( )
const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.63.1.5 version()

```
std::string BioNetworkExecutionGuiExtensionPlugin::version ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

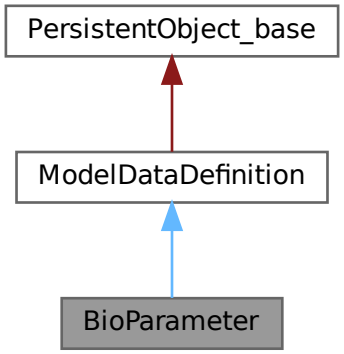
The documentation for this class was generated from the following file:

- applications/gui/genesys/extensions/BioNetworkExecutionGuiExtensionPlugin.cpp

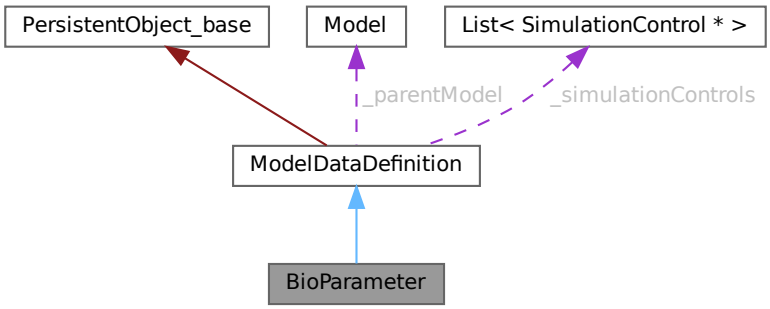
17.64 BioParameter Class Reference

```
#include <plugins/data/BiochemicalSimulation/BioParameter.h>
```

Inheritance diagram for BioParameter:



Collaboration diagram for BioParameter:



Public Member Functions

- **BioParameter** (Model *model, std::string name="")
- void **setValue** (double value)
- double **getValue** () const
- void **setUnit** (std::string unit)
- std::string **getUnit** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)

- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.64.1 Detailed Description

Scalar biochemical parameter, typically used as a kinetic constant.
BioReaction may reference a BioParameter by name. Parameter lookup remains a model-data responsibility, so missing references are reported by BioReaction validation rather than by this value object.

17.64.2 Member Function Documentation

17.64.2.1 _check()

```
bool BioParameter::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.64.2.2 _loadInstance()

```
bool BioParameter::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.64.2.3 _saveInstance()

```
void BioParameter::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.64.2.4 show()

```
std::string BioParameter::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

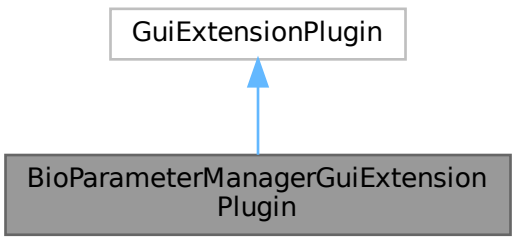
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

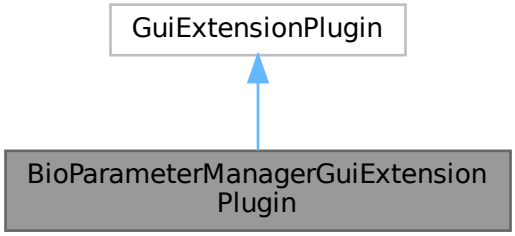
- plugins/data/BiochemicalSimulation/BioParameter.h
- plugins/data/BiochemicalSimulation/BioParameter.cpp

17.65 BioParameterManagerGuiExtensionPlugin Class Reference

Inheritance diagram for BioParameterManagerGuiExtensionPlugin:



Collaboration diagram for BioParameterManagerGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.65.1 Member Function Documentation

17.65.1.1 displayName()

```
std::string BioParameterManagerGuiExtensionPlugin::displayName ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.65.1.2 extensionId()

```
std::string BioParameterManagerGuiExtensionPlugin::extensionId ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.65.1.3 registerContributions()

```
void BioParameterManagerGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.65.1.4 requiredModelPlugins()

```
std::vector< std::string > BioParameterManagerGuiExtensionPlugin::requiredModelPlugins ( )
const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.65.1.5 version()

```
std::string BioParameterManagerGuiExtensionPlugin::version ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

- applications/gui/genesys/extensions/BioParameterManagerGuiExtensionPlugin.cpp

17.66 BioParameterSensitivityEntry Struct Reference

Public Attributes

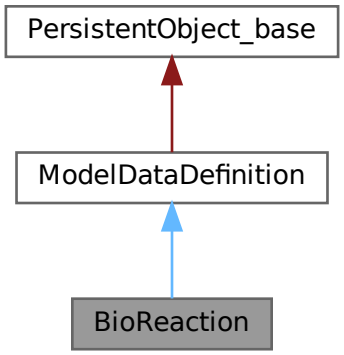
- std::string **parameterName**
- double **baseValue** = 0.0
- double **delta** = 0.0
- double **maxAbsoluteSensitivity** = 0.0
- std::vector< BioSpeciesDerivative > **derivativeSensitivities**

The documentation for this struct was generated from the following file:

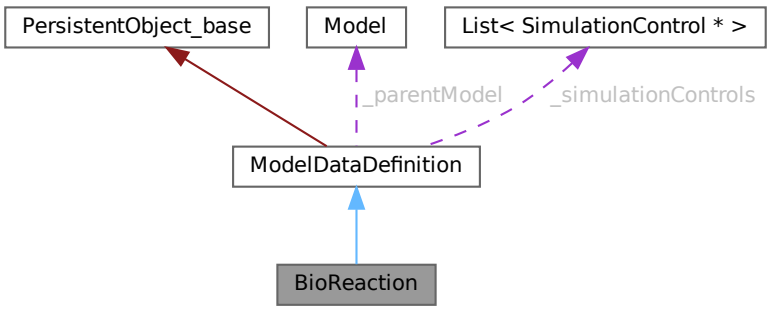
- tools/Biochemical/BioSimulationAnalysis.h

17.67 BioReaction Class Reference

`#include <plugins/data/BiochemicalSimulation/BioReaction.h>`
Inheritance diagram for BioReaction:



Collaboration diagram for BioReaction:



Classes

- struct StoichiometricTerm

Public Member Functions

- BioReaction (Model *model, std::string name="")

  Creates an empty biochemical reaction definition.
- virtual ~**BioReaction** () override=default

  Releases reaction-owned resources.
- void addReactant (std::string speciesName, double stoichiometry=1.0)

  Adds a reactant species to the reaction.
- void addProduct (std::string speciesName, double stoichiometry=1.0)

  Adds a product species to the reaction.
- void addModifier (std::string speciesName)

  Adds a modifier species that influences the reaction rate.
- void **clearReactants** ()

  Removes all reactants from the reaction.
- void **clearProducts** ()

  Removes all products from the reaction.
- void **clearModifiers** ()

  Removes all modifiers from the reaction.
- const std::vector< StoichiometricTerm > & getReactants () const

  Returns the registered reactants.
- const std::vector< StoichiometricTerm > & getProducts () const

  Returns the registered products.
- const std::vector< std::string > & getModifiers () const

  Returns the registered modifiers.
- void setRateConstant (double rateConstant)

  Sets the forward rate constant.
- double getRateConstant () const

  Returns the forward rate constant.
- void setRateConstantParameterName (std::string rateConstantParameterName)

  Stores the name of the parameter used for the forward rate constant.
- std::string getRateConstantParameterName () const

  Returns the forward-rate parameter name.
- double resolveRateConstant () const

  Resolves the effective forward rate constant.
- void setReverseRateConstant (double reverseRateConstant)

  Sets the reverse rate constant.
- double getReverseRateConstant () const

  Returns the reverse rate constant.
- void setReverseRateConstantParameterName (std::string reverseRateConstantParameterName)

  Stores the name of the parameter used for the reverse rate constant.
- std::string getReverseRateConstantParameterName () const

  Returns the reverse-rate parameter name.
- double resolveReverseRateConstant () const

  Resolves the effective reverse rate constant.
- void setKineticLawExpression (std::string kineticLawExpression)

  Sets the forward kinetic-law expression.
- std::string getKineticLawExpression () const

  Returns the forward kinetic-law expression.
- void setReverseKineticLawExpression (std::string reverseKineticLawExpression)

  Sets the reverse kinetic-law expression.
- std::string getReverseKineticLawExpression () const

Returns the reverse kinetic-law expression.

- void setReversible (bool reversible)

  Marks the reaction as reversible or irreversible.
- bool isReversible () const

  Indicates whether the reaction is reversible.
- virtual std::string show () override

  Returns a textual representation of the reaction.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields)

  Loads a biochemical reaction from serialized fields.
- static PluginInformation * GetPluginInformation ()

  Returns plugin metadata for this reaction type.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new empty reaction instance.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.67.1 Detailed Description

Irreversible biochemical reaction definition.
Reactants, products, and modifiers are stored by species name so the reaction can be persisted independently from runtime pointers. Modifiers participate in kinetic laws without being consumed or produced. Reactions may omit reactants for synthesis or omit products for degradation, but must define at least one reactant or product. The current execution contract supports direct rate constants, a BioParameter reference, or an optional kinetic-law expression that resolves BioSpecies/BioParameter names at BioNetwork execution time. Reversible reactions use either a separate reverse kinetic-law expression or a reverse mass-action rate constant over the product side of the reaction.

17.67.2 Constructor & Destructor Documentation

17.67.2.1 BioReaction()

```
BioReaction::BioReaction (
            Model * model,
            std::string name = "" )
```

Creates an empty biochemical reaction definition.

Parameters

model	Parent model.
name	Initial reaction name.

17.67.3 Member Function Documentation

17.67.3.1 _check()

```
bool BioReaction::_check (
            std::string & errorMessage ) [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.67.3.2 _createAttachedAttributes()

```
void BioReaction::_createAttachedAttributes ( ) [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.67.3.3 _loadInstance()

```
bool BioReaction::_loadInstance (
            PersistenceRecord * fields ) [override], [protected], [virtual]
```

must be overriden by derived classes

Reimplemented from ModelDataDefinition.

17.67.3.4 _saveInstance()

```
void BioReaction::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.67.3.5 addModifier()

```
void BioReaction::addModifier (
            std::string speciesName )
```

Adds a modifier species that influences the reaction rate.

Parameters

speciesName	Modifier species name.

17.67.3.6 addProduct()

```
void BioReaction::addProduct (
            std::string speciesName,
            double stoichiometry = 1.0 )
```

Adds a product species to the reaction.

Parameters

speciesName	Product species name.
stoichiometry	Product stoichiometry.

17.67.3.7 addReactant()

```
void BioReaction::addReactant (
            std::string speciesName,
            double stoichiometry = 1.0 )
```

Adds a reactant species to the reaction.

Parameters

speciesName	Reactant species name.
stoichiometry	Reactant stoichiometry.

17.67.3.8 getKineticLawExpression()

```
std::string BioReaction::getKineticLawExpression ( ) const
```

Returns the forward kinetic-law expression.

Returns

Forward kinetic-law expression.

17.67.3.9 getModifiers()

```
const std::vector< std::string > & BioReaction::getModifiers ( ) const
```

Returns the registered modifiers.

Returns

Constant reference to the modifier names.

17.67.3.10 GetPluginInformation()

```
PluginInformation * BioReaction::GetPluginInformation ( )  [static]
```

Returns plugin metadata for this reaction type.

Returns

Plugin information block.

17.67.3.11 getProducts()

```
const std::vector< BioReaction::StoichiometricTerm > & BioReaction::getProducts ( ) const
```

Returns the registered products.

Returns

Constant reference to the product terms.

17.67.3.12 getRateConstant()

```
double BioReaction::getRateConstant ( ) const
```

Returns the forward rate constant.

Returns

Forward rate constant.

17.67.3.13 getRateConstantParameterName()

```
std::string BioReaction::getRateConstantParameterName ( ) const
```

Returns the forward-rate parameter name.

Returns

Parameter name used to resolve the forward rate constant.

17.67.3.14 getReactants()

```
const std::vector< BioReaction::StoichiometricTerm > & BioReaction::getReactants ( ) const
```

Returns the registered reactants.

Returns

Constant reference to the reactant terms.

17.67.3.15 getReverseKineticLawExpression()

`std::string BioReaction::getReverseKineticLawExpression ( ) const`
Returns the reverse kinetic-law expression.

Returns

Reverse kinetic-law expression.

17.67.3.16 getReverseRateConstant()

`double BioReaction::getReverseRateConstant ( ) const`
Returns the reverse rate constant.

Returns

Reverse rate constant.

17.67.3.17 getReverseRateConstantParameterName()

`std::string BioReaction::getReverseRateConstantParameterName ( ) const`
Returns the reverse-rate parameter name.

Returns

Parameter name used to resolve the reverse rate constant.

17.67.3.18 isReversible()

`bool BioReaction::isReversible ( ) const`
Indicates whether the reaction is reversible.

Returns

`true` when the reaction has a reverse branch.

17.67.3.19 LoadInstance()

```
ModelDataDefinition * BioReaction::LoadInstance (
            Model * model,
            PersistenceRecord * fields )  [static]
```
Loads a biochemical reaction from serialized fields.

Parameters

model	Parent model.
fields	Serialized fields.

Returns

Newly created reaction instance.

17.67.3.20 NewInstance()

```
ModelDataDefinition * BioReaction::NewInstance (
            Model * model,
            std::string name = "" )  [static]
```
Creates a new empty reaction instance.

Parameters

model	Parent model.
name	Initial reaction name.

Returns

Newly created reaction instance.

17.67.3.21 resolveRateConstant()

```
double BioReaction::resolveRateConstant ( ) const
```

Resolves the effective forward rate constant.

Returns

Forward rate constant after parameter resolution.

17.67.3.22 resolveReverseRateConstant()

```
double BioReaction::resolveReverseRateConstant ( ) const
```

Resolves the effective reverse rate constant.

Returns

Reverse rate constant after parameter resolution.

17.67.3.23 setKineticLawExpression()

```
void BioReaction::setKineticLawExpression (
            std::string kineticLawExpression )
```

Sets the forward kinetic-law expression.

Parameters

kineticLawExpression	Expression text.

17.67.3.24 setRateConstant()

```
void BioReaction::setRateConstant (
            double rateConstant )
```

Sets the forward rate constant.

Parameters

rateConstant	Forward rate constant value.

17.67.3.25 setRateConstantParameterName()

```
void BioReaction::setRateConstantParameterName (
            std::string rateConstantParameterName )
```

Stores the name of the parameter used for the forward rate constant.

Parameters

rateConstantParameterName	Parameter name.

17.67.3.26 setReverseKineticLawExpression()

```
void BioReaction::setReverseKineticLawExpression (
            std::string reverseKineticLawExpression )
```

Sets the reverse kinetic-law expression.

Parameters

reverseKineticLawExpression	Expression text.

17.67.3.27 setReverseRateConstant()

```
void BioReaction::setReverseRateConstant (
            double reverseRateConstant )
```

Sets the reverse rate constant.

Parameters

reverseRateConstant	Reverse rate constant value.

17.67.3.28 setReverseRateConstantParameterName()

```
void BioReaction::setReverseRateConstantParameterName (
            std::string reverseRateConstantParameterName )
```

Stores the name of the parameter used for the reverse rate constant.

Parameters

reverseRateConstantParameterName	Parameter name.

17.67.3.29 setReversible()

```
void BioReaction::setReversible (
            bool reversible )
```

Marks the reaction as reversible or irreversible.

Parameters

reversible	New reversible flag.

17.67.3.30 show()

```
std::string BioReaction::show ( )  [override], [virtual]
```

Returns a textual representation of the reaction.

Returns

Human-readable reaction summary.

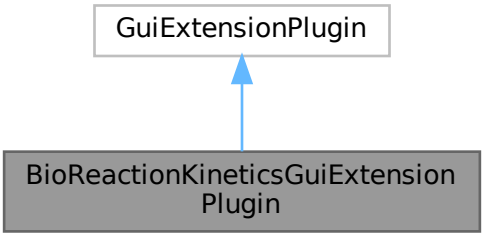
Reimplemented from ModelDataDefinition.

The documentation for this class was generated from the following files:

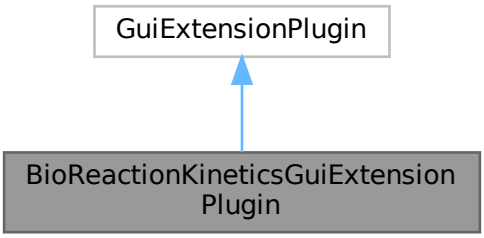
- plugins/data/BiochemicalSimulation/BioReaction.h
- plugins/data/BiochemicalSimulation/BioReaction.cpp

17.68 BioReactionKineticsGuiExtensionPlugin Class Reference

Inheritance diagram for BioReactionKineticsGuiExtensionPlugin:



Collaboration diagram for BioReactionKineticsGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.68.1 Member Function Documentation

17.68.1.1 displayName()

```
std::string BioReactionKineticsGuiExtensionPlugin::displayName ( ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.68.1.2 extensionId()

```
std::string BioReactionKineticsGuiExtensionPlugin::extensionId ( ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.68.1.3 registerContributions()

```
void BioReactionKineticsGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.68.1.4 requiredModelPlugins()

```
std::vector< std::string > BioReactionKineticsGuiExtensionPlugin::requiredModelPlugins ( )
const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.68.1.5 version()

```
std::string BioReactionKineticsGuiExtensionPlugin::version ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

- applications/gui/genesys/extensions/BioReactionKineticsGuiExtensionPlugin.cpp

17.69 BioReactionRateSample Struct Reference

Public Attributes

- double **time** = 0.0
- std::vector< double > **forwardRates**
- std::vector< double > **reverseRates**
- std::vector< double > **netRates**

The documentation for this struct was generated from the following file:

- tools/Biochemical/BioSimulationAnalysis.h

17.70 BioReactionRateTimeCourse Struct Reference

Public Member Functions

- bool **hasReaction** (const std::string &reactionName) const
- bool **toDataset** (const std::string &reactionName, BioReactionRateKind kind, SimulationResultsDataset *dataset, std::string *errorMessage=nullptr) const

Public Attributes

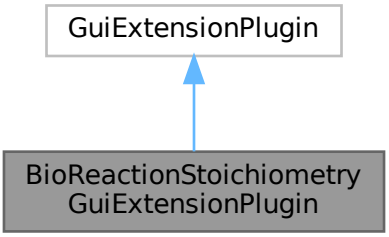
- std::vector< std::string > **reactionNames**
- std::vector< BioReactionRateSample > **samples**

The documentation for this struct was generated from the following file:

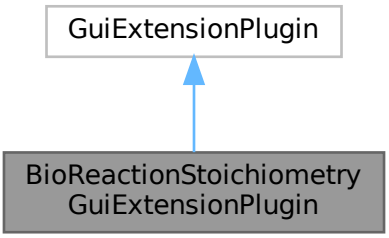
- tools/Biochemical/BioSimulationAnalysis.h

17.71 BioReactionStoichiometryGuiExtensionPlugin Class Reference

Inheritance diagram for BioReactionStoichiometryGuiExtensionPlugin:



Collaboration diagram for BioReactionStoichiometryGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.71.1 Member Function Documentation

17.71.1.1 displayName()

`std::string BioReactionStoichiometryGuiExtensionPlugin::displayName ( ) const [inline], [override], [virtual]`

Implements GuiExtensionPlugin.

17.71.1.2 extensionId()

`std::string BioReactionStoichiometryGuiExtensionPlugin::extensionId ( ) const [inline], [override], [virtual]`

Implements GuiExtensionPlugin.

17.71.1.3 registerContributions()

```
void BioReactionStoichiometryGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.71.1.4 requiredModelPlugins()

```
std::vector< std::string > BioReactionStoichiometryGuiExtensionPlugin::requiredModelPlugins (
) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.71.1.5 version()

```
std::string BioReactionStoichiometryGuiExtensionPlugin::version ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

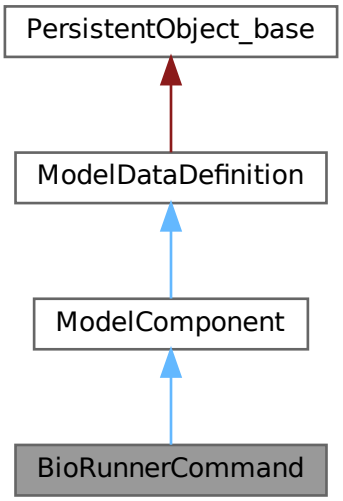
- applications/gui/genesys/extensions/BioReactionStoichiometryGuiExtensionPlugin.cpp

17.72 BioRunnerCommand Class Reference

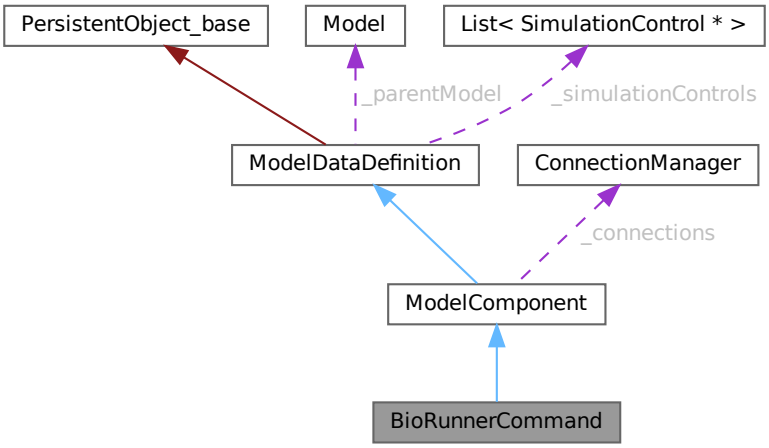
Component wrapper over BioSimulatorRunner command execution.

`#include <plugins/components/BiochemicalSimulation/BioRunnerCommand.h>`

Inheritance diagram for BioRunnerCommand:



Collaboration diagram for BioRunnerCommand:



Public Member Functions

- **BioRunnerCommand** (Model ∗model, std::string name="")
- void **setRunner** (BioSimulatorRunner ∗runner)
- BioSimulatorRunner ∗ **getRunner** () const
- void **setCommand** (std::string command)
- std::string **getCommand** () const
- void **setLastSucceeded** (bool lastSucceeded)
- bool **getLastSucceeded** () const
- void **setLastStatus** (std::string lastStatus)
- std::string **getLastStatus** () const
- void **setLastMessage** (std::string lastMessage)
- std::string **getLastMessage** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent ∗comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.72.1 Detailed Description

Component wrapper over BioSimulatorRunner command execution.
BioRunnerCommand dispatches one configured command into a referenced BioSimulatorRunner whenever an entity arrives and stores the last runner status/payload metadata for downstream model logic.

17.72.2 Member Function Documentation

17.72.2.1 _check()

```
bool BioRunnerCommand::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.72.2.2 _createEditableDataDefinitions()

```
void BioRunnerCommand::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.72.2.3 _loadInstance()

```
bool BioRunnerCommand::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.72.2.4 _onDispatchEvent()

```
void BioRunnerCommand::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.72.2.5 _saveInstance()

```
void BioRunnerCommand::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.72.2.6 show()

```
std::string BioRunnerCommand::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/BiochemicalSimulation/BioRunnerCommand.h
- plugins/components/BiochemicalSimulation/BioRunnerCommand.cpp

17.73 BioSBMLBridge Class Reference

Static Public Member Functions

- static bool **importFromString** (Model *model, const std::string &sbmlText, const std::string &preferred↩ NetworkName, BioSBMLImportResult *result, std::string &errorMessage)
- static bool **exportToString** (Model *model, const std::string &requestedNetworkName, std::string *sbmlText, BioSBMLExportResult *result, std::string &errorMessage)

The documentation for this class was generated from the following files:

- plugins/data/BiochemicalSimulation/BioSBMLBridge.h
- plugins/data/BiochemicalSimulation/BioSBMLBridge.cpp

17.74 BioSBMLExportResult Struct Reference

Public Attributes

- std::string **networkName**
- unsigned int **speciesExported** = 0
- unsigned int **parametersExported** = 0
- unsigned int **reactionsExported** = 0
- std::vector< std::string > **warnings**

The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/BioSBMLBridge.h

17.75 BioSBMLImportResult Struct Reference

Public Attributes

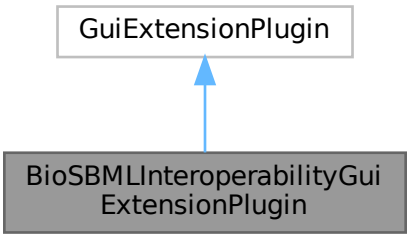
- std::string **networkName**
- unsigned int **speciesImported** = 0
- unsigned int **parametersImported** = 0
- unsigned int **reactionsImported** = 0
- std::vector< std::string > **warnings**

The documentation for this struct was generated from the following file:

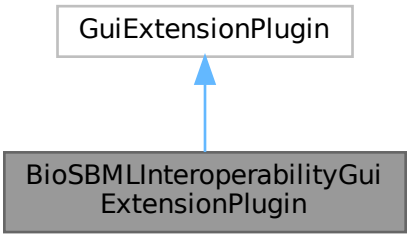
- plugins/data/BiochemicalSimulation/BioSBMLBridge.h

17.76 BioSBMLInteroperabilityGuiExtensionPlugin Class Reference

Inheritance diagram for BioSBMLInteroperabilityGuiExtensionPlugin:



Collaboration diagram for BioSBMLInteroperabilityGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.76.1 Member Function Documentation

17.76.1.1 displayName()

```
std::string BioSBMLInteroperabilityGuiExtensionPlugin::displayName ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.76.1.2 extensionId()

```
std::string BioSBMLInteroperabilityGuiExtensionPlugin::extensionId ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.76.1.3 registerContributions()

```
void BioSBMLInteroperabilityGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.76.1.4 requiredModelPlugins()

```
std::vector< std::string > BioSBMLInteroperabilityGuiExtensionPlugin::requiredModelPlugins ( )
const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.76.1.5 version()

```
std::string BioSBMLInteroperabilityGuiExtensionPlugin::version ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

- applications/gui/genesys/extensions/BioSBMLInteroperabilityGuiExtensionPlugin.cpp

17.77 BioSensitivityScan Struct Reference

Public Attributes

- double **time** = 0.0
- std::vector< std::string > **speciesNames**
- std::vector< BioParameterSensitivityEntry > **entries**

The documentation for this struct was generated from the following file:

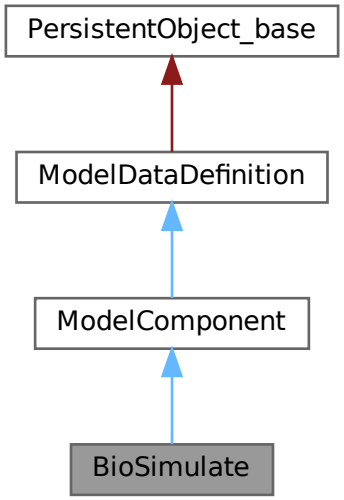
- tools/Biochemical/BioSimulationAnalysis.h

17.78 BioSimulate Class Reference

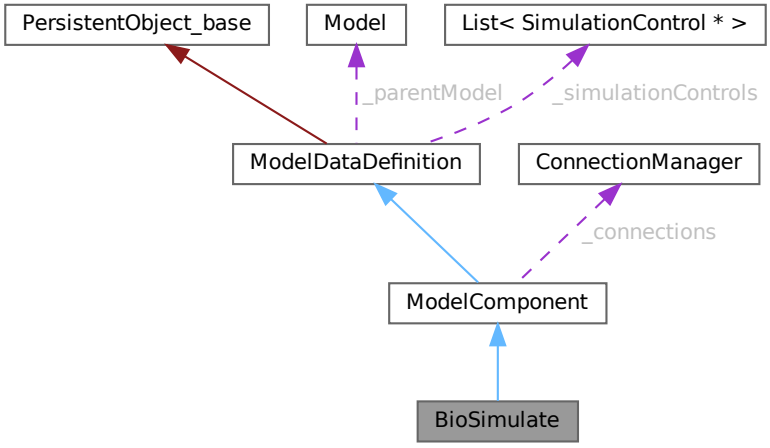
Component that triggers deterministic BioNetwork simulation.

```
#include <plugins/components/BiochemicalSimulation/BioSimulate.h>
```

Inheritance diagram for BioSimulate:



Collaboration diagram for BioSimulate:



Public Member Functions

- **BioSimulate** (Model *model, std::string name="")
- void **setBioNetwork** (BioNetwork *bioNetwork)
- BioNetwork * **getBioNetwork** () const
- void **setUseNetworkTimeWindow** (bool useNetworkTimeWindow)
- bool **getUseNetworkTimeWindow** () const
- void **setStartTime** (double startTime)
- double **getStartTime** () const

- void **setStopTime** (double stopTime)
- double **getStopTime** () const
- void **setStepSize** (double stepSize)
- double **getStepSize** () const
- void **setLastSucceeded** (bool lastSucceeded)
- bool **getLastSucceeded** () const
- void **setLastMessage** (std::string lastMessage)
- std::string **getLastMessage** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.78.1 Detailed Description

Component that triggers deterministic BioNetwork simulation.
BioSimulate delegates simulation execution to a referenced BioNetwork. Each arriving entity can either run the network with its own configured time window or with component-provided start/stop/step values.

17.78.2 Member Function Documentation

17.78.2.1 _check()

```
bool BioSimulate::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.78.2.2 _createEditableDataDefinitions()

```
void BioSimulate::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.78.2.3 _loadInstance()

```
bool BioSimulate::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.78.2.4 _onDispatchEvent()

```
void BioSimulate::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.78.2.5 _saveInstance()

```
void BioSimulate::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.78.2.6 show()

```
std::string BioSimulate::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/BiochemicalSimulation/BioSimulate.h
- plugins/components/BiochemicalSimulation/BioSimulate.cpp

17.79 BioSimulationAnalysis Class Reference

Static Public Member Functions

- static BioStoichiometryMatrix **buildStoichiometryMatrix** (const MassActionOdeSystem &system)
- static bool **buildReactionRateTimeCourse** (const MassActionOdeSystem &system, const BioSimulationResult &result, BioReactionRateTimeCourse *timeCourse, std::string *errorMessage=nullptr)
- static bool **checkSteadyState** (const MassActionOdeSystem &system, const BioSimulationSample &sample, double tolerance, BioSteadyStateCheck *check, std::string *errorMessage=nullptr)
- static bool **scanLocalParameterSensitivity** (const MassActionOdeSystem &system, const BioSimulationSample &sample, double relativeStep, double absoluteStep, BioSensitivityScan *scan, std::string *error↩ Message=nullptr)

The documentation for this class was generated from the following file:

- tools/Biochemical/BioSimulationAnalysis.h

17.80 BioSimulationResult Class Reference

Public Member Functions

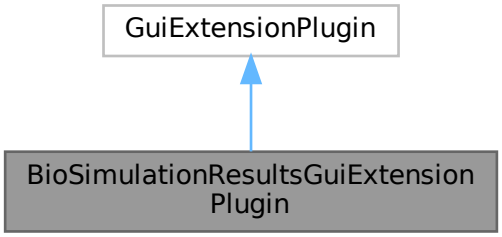
- void **clear** ()
- void **configure** (std::string networkName, double startTime, double stopTime, double stepSize, const std↩ ::vector< std::string > &speciesNames)
- void **appendSample** (double time, const std::vector< BioSimulationSpeciesAmount > &speciesAmounts)
- bool **empty** () const
- unsigned int **sampleCount** () const
- const std::string & **getNetworkName** () const
- double **getStartTime** () const
- double **getStopTime** () const
- double **getStepSize** () const
- const std::vector< std::string > & **getSpeciesNames** () const
- const std::vector< BioSimulationSample > & **getSamples** () const
- bool **hasSpecies** (const std::string &speciesName) const
- bool **toDataset** (const std::string &speciesName, SimulationResultsDataset *dataset, std::string *error↩ Message=nullptr) const
- std::vector< SimulationResultsDataset > **toDatasets** () const

The documentation for this class was generated from the following file:

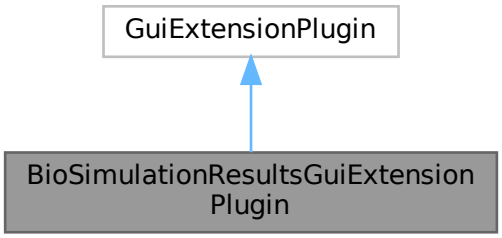
- tools/Biochemical/BioSimulationResult.h

17.81 BioSimulationResultsGuiExtensionPlugin Class Reference

Inheritance diagram for BioSimulationResultsGuiExtensionPlugin:



Collaboration diagram for BioSimulationResultsGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.81.1 Member Function Documentation

17.81.1.1 displayName()

```
std::string BioSimulationResultsGuiExtensionPlugin::displayName ( ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.81.1.2 extensionId()

```
std::string BioSimulationResultsGuiExtensionPlugin::extensionId ( ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.81.1.3 registerContributions()

```
void BioSimulationResultsGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.81.1.4 requiredModelPlugins()

```
std::vector< std::string > BioSimulationResultsGuiExtensionPlugin::requiredModelPlugins ( )
const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.81.1.5 version()

```
std::string BioSimulationResultsGuiExtensionPlugin::version ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

- applications/gui/genesys/extensions/BioSimulationResultsGuiExtensionPlugin.cpp

17.82 BioSimulationSample Struct Reference

Public Attributes

- double **time** = 0.0
- std::vector< BioSimulationSpeciesAmount > **species**

The documentation for this struct was generated from the following file:

- tools/Biochemical/BioSimulationResult.h

17.83 BioSimulationSpeciesAmount Struct Reference

Public Attributes

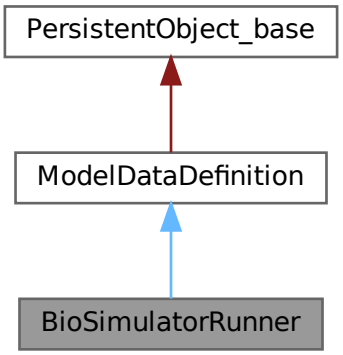
- std::string **speciesName**
- double **amount** = 0.0

The documentation for this struct was generated from the following file:

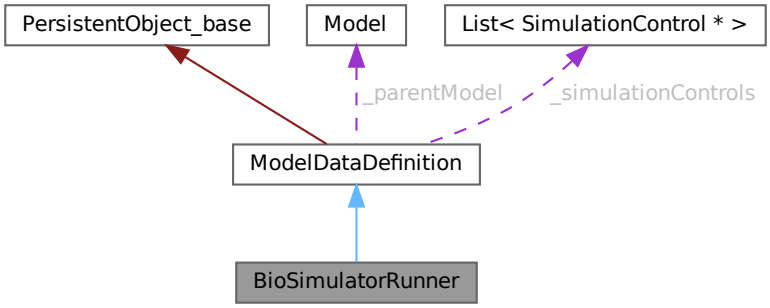
- tools/Biochemical/BioSimulationResult.h

17.84 BioSimulatorRunner Class Reference

Inheritance diagram for BioSimulatorRunner:



Collaboration diagram for BioSimulatorRunner:



Public Member Functions

- **BioSimulatorRunner** (Model *model, std::string name="")
- void **setBackend** (std::string backend)
- std::string **getBackend** () const
- void **setModelSourceType** (std::string modelSourceType)
- std::string **getModelSourceType** () const
- void **setModelSource** (std::string modelSource)
- std::string **getModelSource** () const
- void **setCommand** (std::string command)
- std::string **getCommand** () const
- void **setLastStatus** (std::string lastStatus)
- std::string **getLastStatus** () const
- void **setLastErrorMessage** (std::string lastErrorMessage)
- std::string **getLastErrorMessage** () const
- void **setLastResponsePayload** (std::string lastResponsePayload)

- std::string **getLastResponsePayload** () const
- void **setLastResponseFilename** (std::string lastResponseFilename)
- std::string **getLastResponseFilename** () const
- void **setTargetBioNetworkName** (std::string targetBioNetworkName)
- std::string **getTargetBioNetworkName** () const
- void **setWorkingDirectory** (std::string workingDirectory)
- std::string **getWorkingDirectory** () const
- void **setWorkingInputFilename** (std::string workingInputFilename)
- std::string **getWorkingInputFilename** () const
- void **setWorkingOutputFilename** (std::string workingOutputFilename)
- std::string **getWorkingOutputFilename** () const
- void **setEndpointOrLibrary** (std::string endpointOrLibrary)
- std::string **getEndpointOrLibrary** () const
- void **setTimeoutSeconds** (unsigned int timeoutSeconds)
- unsigned int **getTimeoutSeconds** () const
- void **setAutoValidateModel** (bool autoValidateModel)
- bool **getAutoValidateModel** () const
- bool **executeCommand** (std::string &errorMessage)
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.84.1 Member Function Documentation

17.84.1.1 _check()

```
bool BioSimulatorRunner::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.84.1.2 _loadInstance()

```
bool BioSimulatorRunner::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.84.1.3 _saveInstance()

```
void BioSimulatorRunner::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.84.1.4 show()

```
std::string BioSimulatorRunner::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

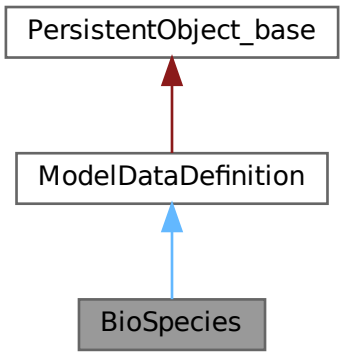
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

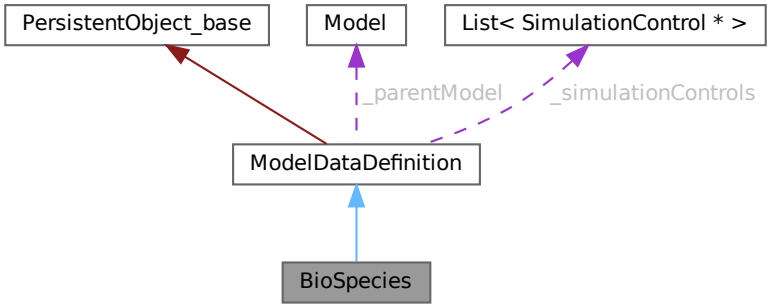
- plugins/data/BiochemicalSimulation/BioSimulatorRunner.h
- plugins/data/BiochemicalSimulation/BioSimulatorRunner.cpp

17.85 BioSpecies Class Reference

`#include <plugins/data/BiochemicalSimulation/BioSpecies.h>`
Inheritance diagram for BioSpecies:



Collaboration diagram for BioSpecies:



Public Member Functions

- **BioSpecies** (Model *model, std::string name="")
- void **setInitialAmount** (double initialAmount)
- double **getInitialAmount** () const
- void **setAmount** (double amount)
- double **getAmount** () const
- void **setConstant** (bool constant)
- bool **isConstant** () const
- void **setBoundaryCondition** (bool boundaryCondition)
- bool **isBoundaryCondition** () const
- void **setUnit** (std::string unit)
- std::string **getUnit** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)

- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.85.1 Detailed Description

Biochemical state variable used by BioReaction and BioNetwork.
A BioSpecies stores an initial amount for replication reset and a current amount for simulation. Constant and boundary-condition flags follow SBML-like semantics: BioNetwork still uses the current amount as kinetic input, but it must not overwrite the species amount while advancing the ODE state.

17.85.2 Member Function Documentation

17.85.2.1 _check()

```
bool BioSpecies::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.85.2.2 _initBetweenReplications()

```
void BioSpecies::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.85.2.3 _loadInstance()

```
bool BioSpecies::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.85.2.4 _saveInstance()

```
void BioSpecies::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.85.2.5 show()

```
std::string BioSpecies::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/BiochemicalSimulation/BioSpecies.h
- plugins/data/BiochemicalSimulation/BioSpecies.cpp

17.86 BioSpeciesDerivative Struct Reference

Public Attributes

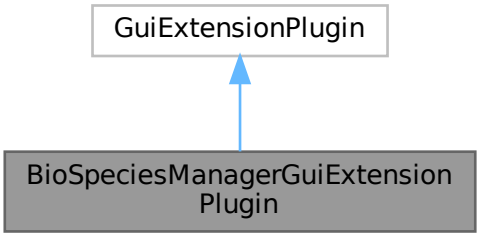
- std::string **speciesName**
- double **derivative** = 0.0

The documentation for this struct was generated from the following file:

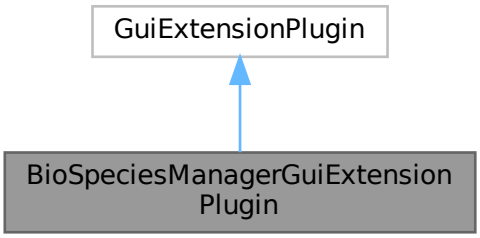
- tools/Biochemical/BioSimulationAnalysis.h

17.87 BioSpeciesManagerGuiExtensionPlugin Class Reference

Inheritance diagram for BioSpeciesManagerGuiExtensionPlugin:



Collaboration diagram for BioSpeciesManagerGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.87.1 Member Function Documentation

17.87.1.1 displayName()

```
std::string BioSpeciesManagerGuiExtensionPlugin::displayName ( ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.87.1.2 extensionId()

```
std::string BioSpeciesManagerGuiExtensionPlugin::extensionId ( ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.87.1.3 registerContributions()

```
void BioSpeciesManagerGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.87.1.4 requiredModelPlugins()

```
std::vector< std::string > BioSpeciesManagerGuiExtensionPlugin::requiredModelPlugins ( ) const
[inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.87.1.5 version()

```
std::string BioSpeciesManagerGuiExtensionPlugin::version ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

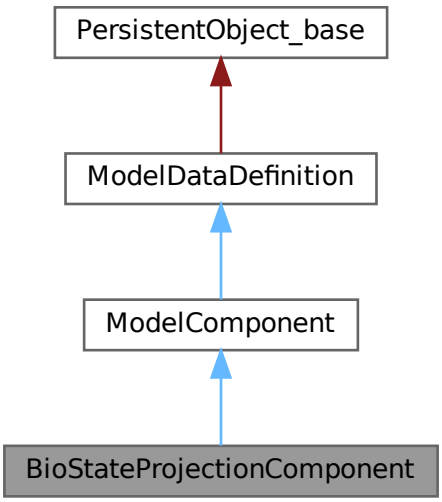
- applications/gui/genesys/extensions/BioSpeciesManagerGuiExtensionPlugin.cpp

17.88 BioStateProjectionComponent Class Reference

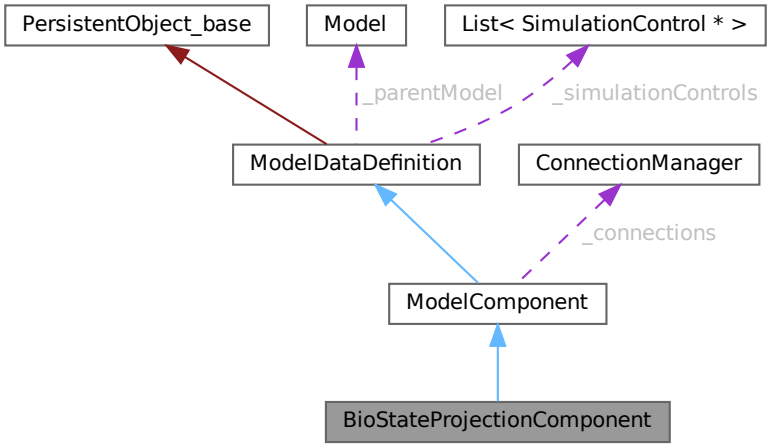
Projects selected BioSpecies values into a WholeCellState.

```
#include <plugins/components/WholeCellModeling/BioStateProjectionComponent.↩
h>
```

Inheritance diagram for BioStateProjectionComponent:



Collaboration diagram for BioStateProjectionComponent:



Classes

- struct Projection

Public Types

- enum class **ProjectionKind** { **MoleculeCount** , **MetaboliteAmount** }

Public Member Functions

- **BioStateProjectionComponent** (Model ∗model, std::string name="")
- void **setWholeCellState** (WholeCellState ∗state)
- WholeCellState ∗ **getWholeCellState** () const
- void **clearProjections** ()
- void **addMoleculeProjection** (const std::string &sourceSpeciesName, const std::string &targetKey="", double scale=1.0, double offset=0.0)
- void **addMetaboliteProjection** (const std::string &sourceSpeciesName, const std::string &targetKey="", double scale=1.0, double offset=0.0)
- unsigned int **getProjectionCount** () const
- void **setLastSucceeded** (bool lastSucceeded)
- bool **getLastSucceeded** () const
- void **setLastMessage** (std::string lastMessage)
- std::string **getLastMessage** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component

- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component

- static bool Check (ModelComponent *component)

  Validates a component instance.

- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.

- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.

- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.

- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.

- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.

- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.

- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions

- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.88.1 Detailed Description

Projects selected BioSpecies values into a WholeCellState.
BioStateProjectionComponent bridges the biochemical and whole-cell layers by copying chosen BioSpecies amounts into WholeCellState molecule counts or metabolite pools at simulation event boundaries. It is intentionally small and explicit: the caller decides which source species feed which whole-cell fields and which scale/offset transform applies to each projection.

17.88.2 Member Function Documentation

17.88.2.1 _check()

```
bool BioStateProjectionComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.88.2.2 _createEditableDataDefinitions()

```
void BioStateProjectionComponent::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.88.2.3 _loadInstance()

```
bool BioStateProjectionComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.88.2.4 _onDispatchEvent()

```
void BioStateProjectionComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.88.2.5 _saveInstance()

```
void BioStateProjectionComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.88.2.6 show()

```
std::string BioStateProjectionComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

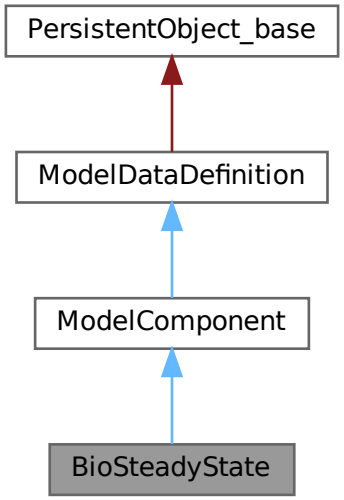
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/WholeCellModeling/BioStateProjectionComponent.h
- plugins/components/WholeCellModeling/BioStateProjectionComponent.cpp

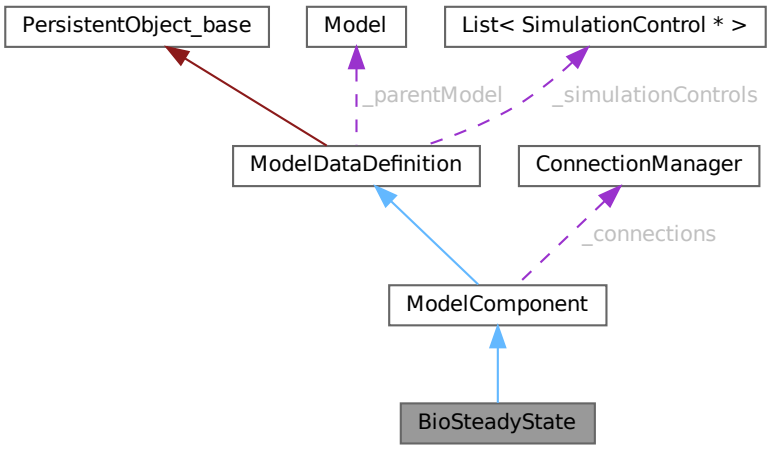
17.89 BioSteadyState Class Reference

Component that checks steady-state diagnostics over BioNetwork results.
`#include <plugins/components/BiochemicalSimulation/BioSteadyState.h>`

Inheritance diagram for BioSteadyState:



Collaboration diagram for BioSteadyState:



Public Member Functions

- **BioSteadyState** (Model *model, std::string name="")
- void **setBioNetwork** (BioNetwork *bioNetwork)
- BioNetwork * **getBioNetwork** () const
- void **setTolerance** (double tolerance)
- double **getTolerance** () const
- void **setRunSimulationBeforeCheck** (bool runSimulationBeforeCheck)
- bool **getRunSimulationBeforeCheck** () const

- void **setLastSteady** (bool lastSteady)
- bool **getLastSteady** () const
- void **setLastMaxAbsoluteDerivative** (double lastMaxAbsoluteDerivative)
- double **getLastMaxAbsoluteDerivative** () const
- void **setLastMessage** (std::string lastMessage)
- std::string **getLastMessage** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.89.1 Detailed Description

Component that checks steady-state diagnostics over BioNetwork results.
BioSteadyState can optionally trigger one simulation run before evaluating steady-state through BioNetwork analysis helpers.

17.89.2 Member Function Documentation

17.89.2.1 _check()

```
bool BioSteadyState::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.89.2.2 _createAttachedAttributes()

```
void BioSteadyState::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.89.2.3 _loadInstance()

```
bool BioSteadyState::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.89.2.4 _onDispatchEvent()

```
void BioSteadyState::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.89.2.5 _saveInstance()

```
void BioSteadyState::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.89.2.6 show()

```
std::string BioSteadyState::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/BiochemicalSimulation/BioSteadyState.h
- plugins/components/BiochemicalSimulation/BioSteadyState.cpp

17.90 BioSteadyStateCheck Struct Reference

Public Attributes

- bool **steady** = false
- double **maxAbsoluteDerivative** = 0.0
- double **tolerance** = 0.0
- std::vector< BioSpeciesDerivative > **derivatives**

The documentation for this struct was generated from the following file:

- tools/Biochemical/BioSimulationAnalysis.h

17.91 BioStoichiometryMatrix Struct Reference

Public Member Functions

- double **coefficient** (unsigned int speciesIndex, unsigned int reactionIndex) const

Public Attributes

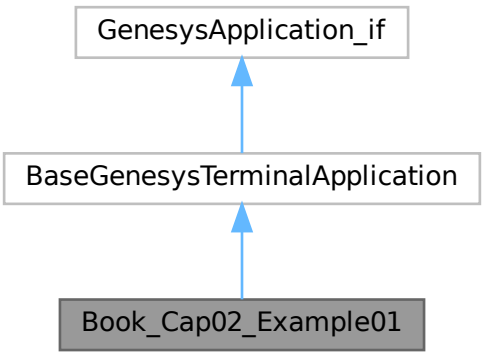
- std::vector< std::string > **speciesNames**
- std::vector< std::string > **reactionNames**
- std::vector< std::vector< double > > **coefficients**

The documentation for this struct was generated from the following file:

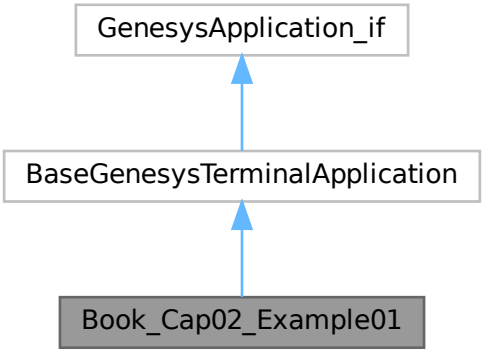
- tools/Biochemical/BioSimulationAnalysis.h

17.92 Book_Cap02_Example01 Class Reference

Inheritance diagram for Book_Cap02_Example01:



Collaboration diagram for Book_Cap02_Example01:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.92.1 Member Function Documentation

17.92.1.1 main()

```
int Book_Cap02_Example01::main (
            int argc,
            char ** argv )  [override], [virtual]
```

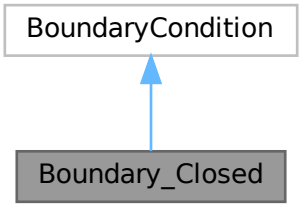
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

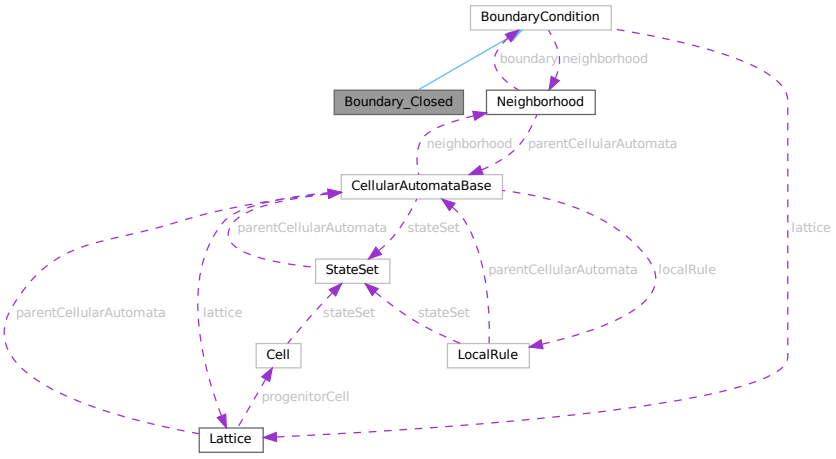
- applications/modelSpecific/book/Book_Cap02_Example01.h
- applications/modelSpecific/book/Book_Cap02_Example01.cpp

17.93 Boundary_Closed Class Reference

Inheritance diagram for Boundary_Closed:



Collaboration diagram for Boundary_Closed:



Public Member Functions

- **Boundary_Closed** (Lattice *lattice=nullptr, Neighborhood *neighborhood=nullptr)
- **Boundary_Closed** (const Boundary_Closed &orig)
- virtual Cell * getNeighborCell (std::vector< int > cellNDimPosition, std::vector< int > neighborNDimPosition) override

Pure virtual method that has to be overiden by derived class, responsable for defining the missing neighbor of a cell.

- void **setFixedState** (const State &fixedState)
- const State & **getFixedState** () const

Public Member Functions inherited from BoundaryCondition

- **BoundaryCondition** (Lattice *lattice=nullptr, Neighborhood *neighborhood=nullptr)
- **BoundaryCondition** (const BoundaryCondition &orig)
- virtual std::string **show** ()
- void **setNeighborhood** (Neighborhood *neighborhood)
- Neighborhood * **getNeighborhood** () const
- void **setLattice** (Lattice *lattice)
- Lattice * **getLattice** () const

Additional Inherited Members

Protected Attributes inherited from BoundaryCondition

- Lattice * **lattice**
- Neighborhood * **neighborhood**

17.93.1 Member Function Documentation

17.93.1.1 getNeighborCell()

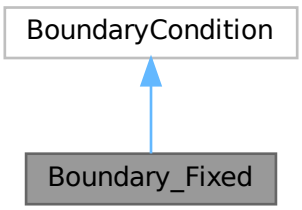
```
virtual Cell * Boundary_Closed::getNeighborCell (
            std::vector< int > cellPosition,
            std::vector< int > neighborCellPosition )  [inline], [override], [virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for defining the missing neighbor of a cell.
Implements BoundaryCondition.
The documentation for this class was generated from the following file:

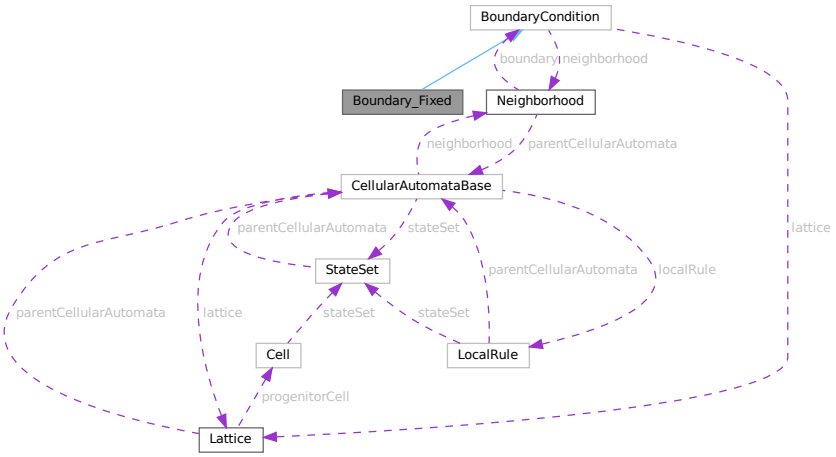
- plugins/components/ModalModel/CellularAutomata/Boundary_Closed.h

17.94 Boundary_Fixed Class Reference

Inheritance diagram for Boundary_Fixed:



Collaboration diagram for Boundary_Fixed:



Public Member Functions

- **Boundary_Fixed** (Lattice *lattice=nullptr, Neighborhood *neighborhood=nullptr)
- **Boundary_Fixed** (const Boundary_Fixed &orig)
- virtual Cell * getNeighborCell (std::vector< int > cellNDimPosition, std::vector< int > neighborNDimPosition) override

  Pure virtual method that has to be overiden by derived class, responsable for defining the missing neighbor of a cell.
- void **setFixedState** (const State &fixedState)
- const State & **getFixedState** () const

Public Member Functions inherited from BoundaryCondition

- **BoundaryCondition** (Lattice *lattice=nullptr, Neighborhood *neighborhood=nullptr)
- **BoundaryCondition** (const BoundaryCondition &orig)
- virtual std::string **show** ()
- void **setNeighborhood** (Neighborhood *neighborhood)
- Neighborhood * **getNeighborhood** () const
- void **setLattice** (Lattice *lattice)
- Lattice * **getLattice** () const

Additional Inherited Members

Protected Attributes inherited from BoundaryCondition

- Lattice * **lattice**
- Neighborhood * **neighborhood**

17.94.1 Member Function Documentation

17.94.1.1 getNeighborCell()

```
virtual Cell * Boundary_Fixed::getNeighborCell (
            std::vector< int > cellPosition,
            std::vector< int > neighborCellPosition )  [inline], [override], [virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for defining the missing neighbor of a cell.

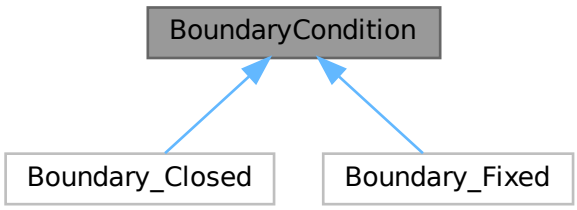
Implements BoundaryCondition.

The documentation for this class was generated from the following file:

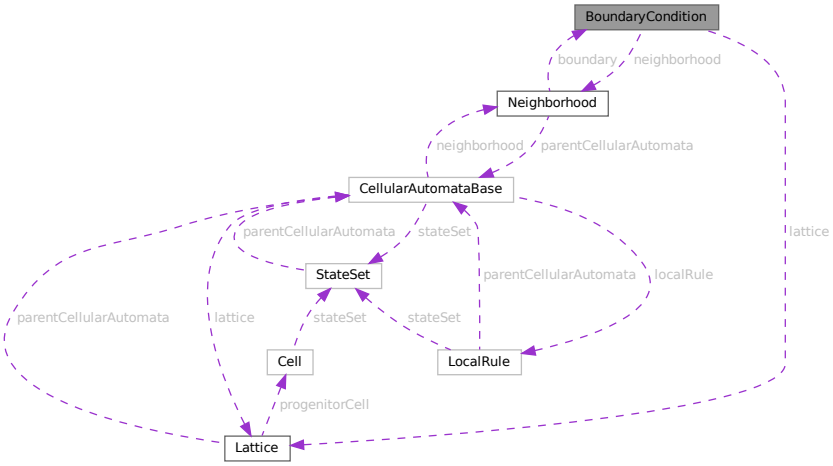
- plugins/components/ModalModel/CellularAutomata/Boundary_Fixed.h

17.95 BoundaryCondition Class Reference

Inheritance diagram for BoundaryCondition:



Collaboration diagram for BoundaryCondition:



Public Member Functions

- **BoundaryCondition** (Lattice *lattice=nullptr, Neighborhood *neighborhood=nullptr)
- **BoundaryCondition** (const BoundaryCondition &orig)
- virtual std::string **show** ()
- virtual Cell * getNeighborCell (std::vector< int > cellPosition, std::vector< int > neighborCellPosition)=0

  Pure virtual method that has to be overiden by derived class, responsable for defining the missing neighbor of a cell.
- void **setNeighborhood** (Neighborhood *neighborhood)
- Neighborhood * **getNeighborhood** () const
- void **setLattice** (Lattice *lattice)
- Lattice * **getLattice** () const

Protected Attributes

- Lattice * **lattice**
- Neighborhood * **neighborhood**

17.95.1 Member Function Documentation

17.95.1.1 getNeighborCell()

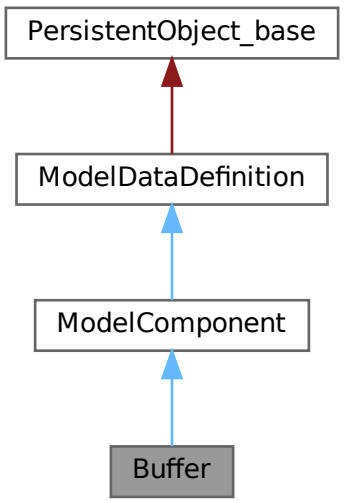
```
virtual Cell * BoundaryCondition::getNeighborCell (
            std::vector< int > cellPosition,
            std::vector< int > neighborCellPosition )  [pure virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for defining the missing neighbor of a cell.
Implemented in Boundary_Closed, and Boundary_Fixed.
The documentation for this class was generated from the following file:

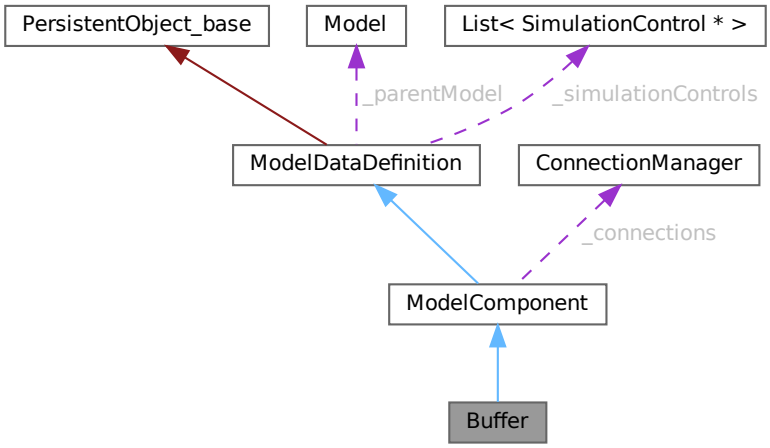
- plugins/components/ModalModel/CellularAutomata/BoundaryCondition.h

17.96 Buffer Class Reference

`#include <plugins/components/DiscreteProcessing/Buffer.h>`
Inheritance diagram for Buffer:



Collaboration diagram for Buffer:



Public Types

- enum class **AdvanceOn** : int { **NewArrivals** = 0 , **Signal** = 1 , **num_elements** = 2 }
- enum class **ArrivalOnFullBufferRule** : int { **Dispose** = 0 , **SendToBulkPort** = 1 , **ReplaceLastPosition** = 2 , **num_elements** = 3 }

Public Member Functions

- **Buffer** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- Buffer::ArrivalOnFullBufferRule **getarrivalOnFullBufferRule** () const
- void **setArrivalOnFullBufferRule** (Buffer::ArrivalOnFullBufferRule newArrivalOnFullBufferRule)
- Buffer::AdvanceOn **getadvanceOn** () const
- void **setAdvanceOn** (Buffer::AdvanceOn newAdvanceOn)
- unsigned int **getcapacity** () const
- void **setCapacity** (unsigned int newCapacity)
- SignalData * **getsignal** () const
- void **setSignal** (SignalData *newSignal)

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static std::string **convertEnumToStr** (AdvanceOn advance)
- static std::string **convertEnumToStr** (ArrivalOnFullBufferRule arrival)
- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation () override
- virtual void _initBetweenReplications () override
- virtual void _addSimulationControl (SimulationControl *property) override
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions

- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.96.1 Detailed Description

This component ...

17.96.2 Member Function Documentation

17.96.2.1 _addSimulationControl()

```
void Buffer::_addSimulationControl (
            SimulationControl * property )  [override], [protected], [virtual]
```

This method is necessary only for those components that instantiate internal elements that must exist before simulation starts and even before model checking. That's the case of components that have internal StatisticsCollectors, since others components may refer to them as expressions (as in "TVAG(ThisCSTAT)") and therefore the modeldatum must exist before checking such expression
Reimplemented from ModelDataDefinition.

17.96.2.2 _check()

```
bool Buffer::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.96.2.3 _createEditableDataDefinitions()

```
void Buffer::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```
Reimplemented from ModelDataDefinition.

17.96.2.4 _getParserChangesInformation()

```
ParserChangesInformation * Buffer::_getParserChangesInformation ( )  [override], [protected],
[virtual]
```
This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
Reimplemented from ModelDataDefinition.

17.96.2.5 _initBetweenReplications()

```
void Buffer::_initBetweenReplications ( )  [override], [protected], [virtual]
```
Reimplemented from ModelDataDefinition.

17.96.2.6 _loadInstance()

```
bool Buffer::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```
virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.96.2.7 _onDispatchEvent()

```
void Buffer::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```
Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.96.2.8 _saveInstance()

```
void Buffer::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```
Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.96.2.9 show()

```
std::string Buffer::show ( )  [override], [virtual]
```
virtual public methods

Returns a textual representation of this component.

Returns

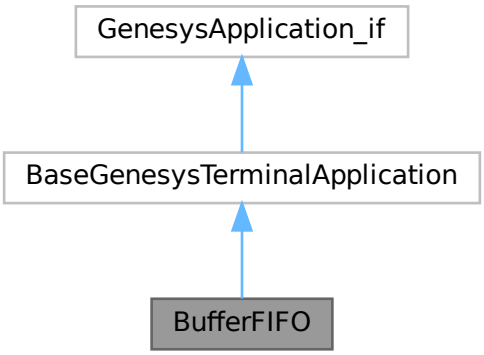
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

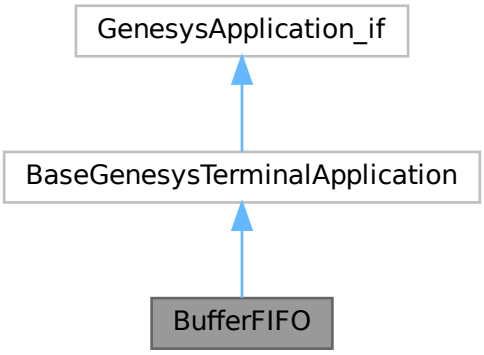
- plugins/components/DiscreteProcessing/Buffer.h
- plugins/components/DiscreteProcessing/Buffer.cpp

17.97 BufferFIFO Class Reference

Inheritance diagram for BufferFIFO:



Collaboration diagram for BufferFIFO:



Public Member Functions

- virtual int main (int argc, char **argv)

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.97.1 Member Function Documentation

17.97.1.1 main()

```
virtual int BufferFIFO::main (
            int argc,
            char ** argv )  [inline], [virtual]
```

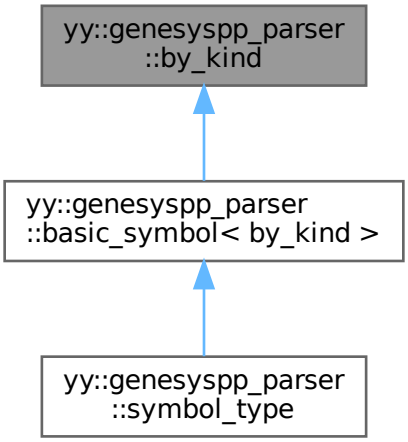
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following file:

- applications/modelSpecific/teaching/BufferFIFO.h

17.98 yy::genesyspp_parser::by_kind Struct Reference

Type access provider for token (enum) based symbols.
`#include <parser/GenesysParser.h>`

Inheritance diagram for yy::genesyspp_parser::by_kind:



Public Types

- typedef token_kind_type **kind_type**

  The symbol kind as needed by the constructor.

Public Member Functions

- **by_kind** () YY_NOEXCEPT

  Default constructor.
- **by_kind** (const by_kind &that) YY_NOEXCEPT

  Copy constructor.
- **by_kind** (kind_type t) YY_NOEXCEPT

  Constructor from (external) token numbers.
- void **clear** () YY_NOEXCEPT

  Record that this symbol is empty.
- void **move** (by_kind &that)

  Steal the symbol kind from that.
- symbol_kind_type kind () const YY_NOEXCEPT
- symbol_kind_type **type_get** () const YY_NOEXCEPT

  Backward compatibility (Bison 3.6).

Public Attributes

- symbol_kind_type kind_

17.98.1 Detailed Description

Type access provider for token (enum) based symbols.

17.98.2 Member Function Documentation

17.98.2.1 kind()

`genesyspp_parser::symbol_kind_type yy::genesyspp_parser::by_kind::kind ( ) const [inline]`
The (internal) type number (corresponding to *type*). *empty* when empty.

17.98.3 Member Data Documentation

17.98.3.1 kind_

`symbol_kind_type yy::genesyspp_parser::by_kind::kind_`
The symbol kind. *S_YYEMPTY* when empty.
The documentation for this struct was generated from the following file:

- parser/GenesysParser.h

17.99 QCPAxisPainterPrivate::CachedLabel Struct Reference

Public Attributes

- QPointF **offset**
- QPixmap **pixmap**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/qcustomplot.h

17.100 QCPLabelPainterPrivate::CachedLabel Struct Reference

Public Attributes

- QPoint **offset**
- QPixmap **pixmap**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/qcustomplot.h

17.101 ModelLifecycleController::Callbacks Struct Reference

Public Attributes

- std::function< void(const std::string &)> **insertCommandInConsole**
- std::function< void(Model *)> **initUiForNewModel**
- std::function< void()> **actualizeActions**
- std::function< void()> **actualizeTabPanes**
- std::function< void(bool)> **actualizeModelTextHasChanged**
- std::function< bool()> **checkModel**
- std::function< bool()> **setSimulationModelBasedOnText**
- std::function< void()> **clearModelEditors**
- std::function< bool(QString)> **saveGraphicalModel**
- std::function< bool(QFile *, QString)> **saveTextModel**
- std::function< Model *(std::string)> **loadGraphicalModel**
- std::function< void()> **connectSceneSignals**
- std::function< void(const char *)> **disconnectSceneSignals**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/controllers/ModelLifecycleController.h

17.102 SimulationEventController::Callbacks Struct Reference

Public Attributes

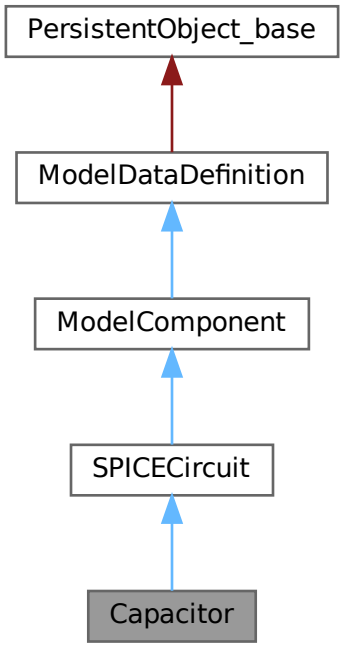
- std::function< void() > **actualizeActions**
- std::function< void(SimulationEvent *) > **actualizeSimulationEvents**
- std::function< void(bool) > **actualizeDebugEntities**
- std::function< void(bool) > **actualizeDebugVariables**
- std::function< void(SimulationEvent *) > **actualizeGraphicalModel**

The documentation for this struct was generated from the following file:

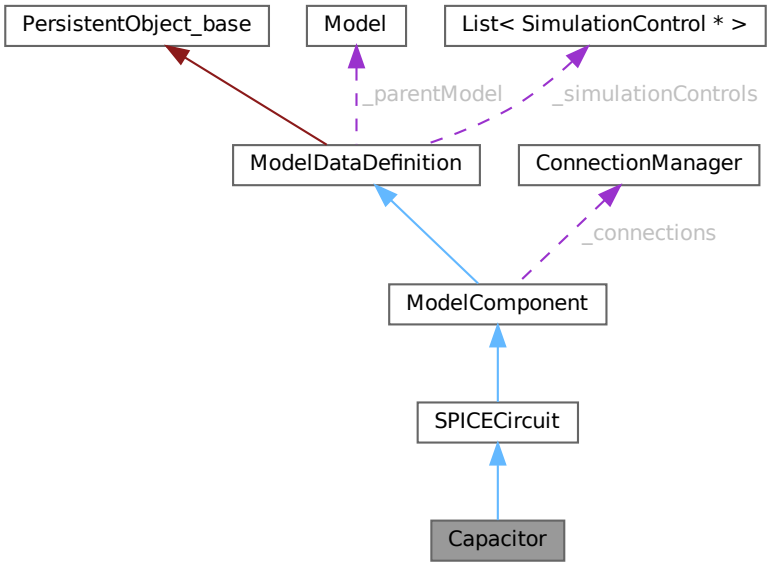
- applications/gui/genesys/controllers/SimulationEventController.h

17.103 Capacitor Class Reference

Inheritance diagram for Capacitor:



Collaboration diagram for Capacitor:



Public Member Functions

- **Capacitor** (Model ∗model, std::string name="")
- void **setCapacitance** (float capacitance)
- void Build () override

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model ∗model, std::string name="")

  constructors
- void **setRunner** (SPICERunner ∗runner)

  new public user methods for this components
- SPICERunner ∗ **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)

- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager ∗ _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model ∗ **_parentModel**
- List< SimulationControl ∗ > ∗ **_simulationControls** = new List<SimulationControl∗>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.103.1 Member Function Documentation

17.103.1.1 _loadInstance()

```
bool Capacitor::_loadInstance (
            PersistenceRecord * fields ) [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.103.1.2 _saveInstance()

```
void Capacitor::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.103.1.3 Build()

```
void Capacitor::Build ( )  [override], [virtual]
```

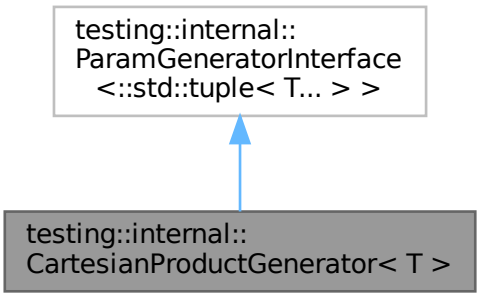
Reimplemented from SPICECircuit.

The documentation for this class was generated from the following files:

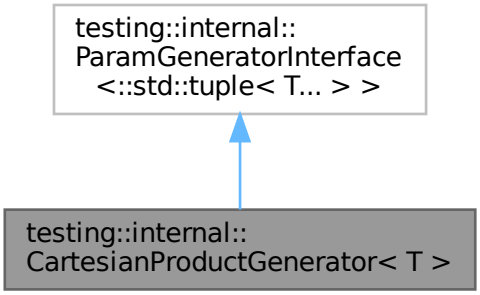
- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.104 testing::internal::CartesianProductGenerator< T > Class Template Reference

Inheritance diagram for testing::internal::CartesianProductGenerator< T >:



Collaboration diagram for testing::internal::CartesianProductGenerator< T >:



Public Types

- typedef ::std::tuple< T... > **ParamType**

Public Types inherited from testing::internal::ParamGeneratorInterface<::std::tuple< T... > >

- typedef ::std::tuple< T... > **ParamType**

Public Member Functions

- **CartesianProductGenerator** (const std::tuple< ParamGenerator< T >... > &g)
- ParamIteratorInterface< ParamType > * Begin () const override
- ParamIteratorInterface< ParamType > * End () const override

17.104.1 Member Function Documentation

17.104.1.1 Begin()

```
template<typename... T>
ParamIteratorInterface< ParamType > * testing::internal::CartesianProductGenerator< T >::↩
Begin ( ) const [inline], [override], [virtual]
```

Implements testing::internal::ParamGeneratorInterface<::std::tuple< T... > >.

17.104.1.2 End()

```
template<typename... T>
ParamIteratorInterface< ParamType > * testing::internal::CartesianProductGenerator< T >::End
( ) const [inline], [override], [virtual]
```

Implements testing::internal::ParamGeneratorInterface<::std::tuple< T... > >.

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.105 testing::internal::CartesianProductHolder< Gen > Class Template Reference

Public Member Functions

- **CartesianProductHolder** (const Gen &... g)

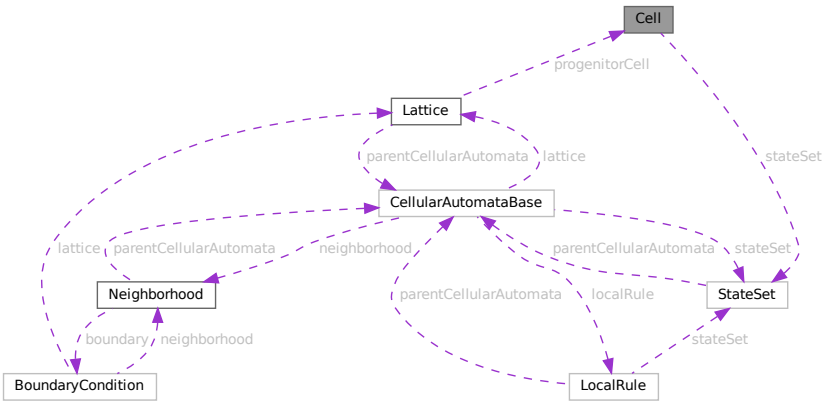
- template<typename... T>
  operator ParamGenerator<::std::tuple< T... > > () const

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.106 Cell Class Reference

Collaboration diagram for Cell:



Public Member Functions

- **Cell** (StateSet *stateSet=nullptr, long cellNum=0, std::vector< int > position={})
- **Cell** (long cellNum, std::vector< int > position)
- **Cell** (const Cell &orig)
- Cell & **operator=** (const Cell &orig)
- virtual std::string **show** ()
- bool **updateState** ()

  update cell state (currentState = nextState)
- bool **isUpdatePending** ()

  true if nextState differs from currentState
- virtual void **draw** ()

  virtual method that can be override by derived classes
- void **setCellNumber** (long cellNum)

  cell rank position in the lattice vector
- long **getCellNumber** () const
- void **setPosition** (std::vector< int > position)
- std::vector< int > **getPosition** () const

  n-dimensional position
- void **setStateSet** (StateSet *stateSet)
- StateSet * **getStateSet** () const
- const State & **getPreviousState** () const
- bool **setCurrentState** (const State &currentState)
- const State & **getCurrentState** () const
- bool **setNextState** (const State &nextState)
- const State & **getNextState** () const
- void **setNeighbors** (std::vector< Cell * > neighbors)
- std::vector< Cell * > **getNeighbors** () const

Protected Member Functions

- bool **acceptsState** (const State &state) const

Protected Attributes

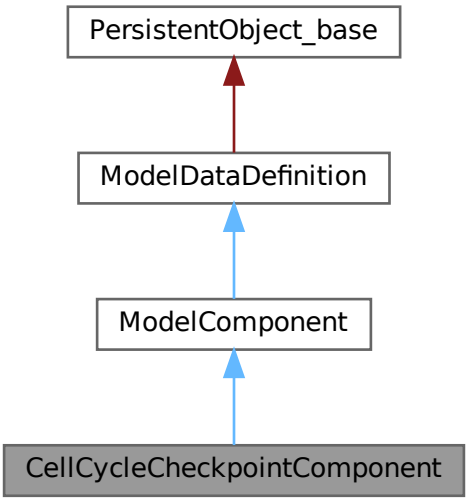
- std::unique_ptr< State > **previousState**
- std::unique_ptr< State > **currentState**
- std::unique_ptr< State > **nextState**
- StateSet * **stateSet** = nullptr
- bool **updatePending**
- std::vector< int > **position**
- long **cellNum**
- std::vector< Cell * > **neighbors**

The documentation for this class was generated from the following files:

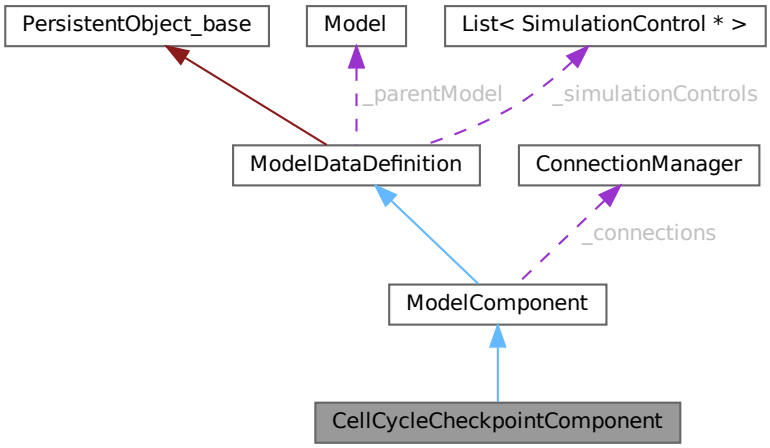
- plugins/components/ModalModel/CellularAutomata/Cell.h
- plugins/components/ModalModel/CellularAutomata/Cell.cpp

17.107 CellCycleCheckpointComponent Class Reference

Evaluates lifecycle checkpoints and advances whole-cell step time.
`#include <plugins/components/WholeCellModeling/CellCycleCheckpointComponent.h>`
Inheritance diagram for CellCycleCheckpointComponent:



Collaboration diagram for CellCycleCheckpointComponent:



Public Member Functions

- **CellCycleCheckpointComponent** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)
- WholeCellState * **getWholeCellState** () const
- void **setDeltaT** (double deltaT)
- double **getDeltaT** () const
- void **setAdvanceWholeCellClock** (bool advance)
- bool **getAdvanceWholeCellClock** () const
- void **setEnergyMetaboliteKey** (std::string key)
- std::string **getEnergyMetaboliteKey** () const
- void **setCompartmentEnergyRegion** (std::string region)
- std::string **getCompartmentEnergyRegion** () const
- void **setCompartmentEnergyMetaboliteKey** (std::string key)
- std::string **getCompartmentEnergyMetaboliteKey** () const
- void **setStarvationAtpThreshold** (double threshold)
- double **getStarvationAtpThreshold** () const
- void **setCompartmentStarvationThreshold** (double threshold)
- double **getCompartmentStarvationThreshold** () const
- void **setLethalStarvationSteps** (unsigned int steps)
- unsigned int **getLethalStarvationSteps** () const
- void **setDivisionMassThreshold** (double threshold)
- double **getDivisionMassThreshold** () const
- void **setFtsZRingKey** (std::string key)
- std::string **getFtsZRingKey** () const
- void **setFtsZThreshold** (double threshold)
- double **getFtsZThreshold** () const
- void **setCriticalPathwayActivityKey** (std::string key)
- std::string **getCriticalPathwayActivityKey** () const
- void **setCriticalPathwayActivityThreshold** (double threshold)
- double **getCriticalPathwayActivityThreshold** () const
- unsigned int **getStarvationStreak** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.107.1 Detailed Description

Evaluates lifecycle checkpoints and advances whole-cell step time.
CellCycleCheckpointComponent centralizes two concerns that otherwise become scattered across individual submodels:

- explicit advancement of WholeCellState current time and step count; and
- annotation of a coarse lifecycle phase based on energy availability, division readiness, and viability.

The component does not execute metabolism or division itself. It only reads WholeCellState and writes back lifecycle metadata such as:

- `LifecyclePhase` = `newborn`, `growth`, `division_ready`, `starved`, `dead`
- `Viable`

This keeps WholeCellState as a passive data container and concentrates the event-boundary checkpoint logic in a dedicated ModelComponent.

17.107.2 Member Function Documentation

17.107.2.1 _check()

```
bool CellCycleCheckpointComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.107.2.2 _createEditableDataDefinitions()

```
void CellCycleCheckpointComponent::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.107.2.3 _initBetweenReplications()

```
void CellCycleCheckpointComponent::_initBetweenReplications ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.107.2.4 _loadInstance()

```
bool CellCycleCheckpointComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.107.2.5 _onDispatchEvent()

```
void CellCycleCheckpointComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.107.2.6 _saveInstance()

```
void CellCycleCheckpointComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.107.2.7 show()

`std::string CellCycleCheckpointComponent::show ( ) [override], [virtual]`
virtual public methods
Returns a textual representation of this component.

Returns

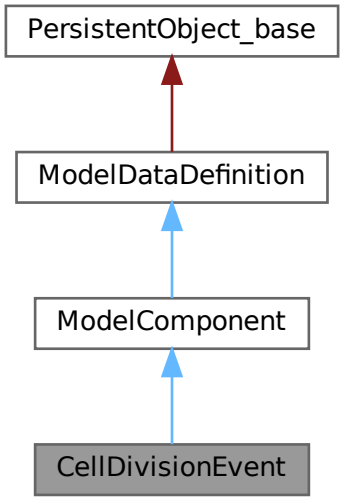
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

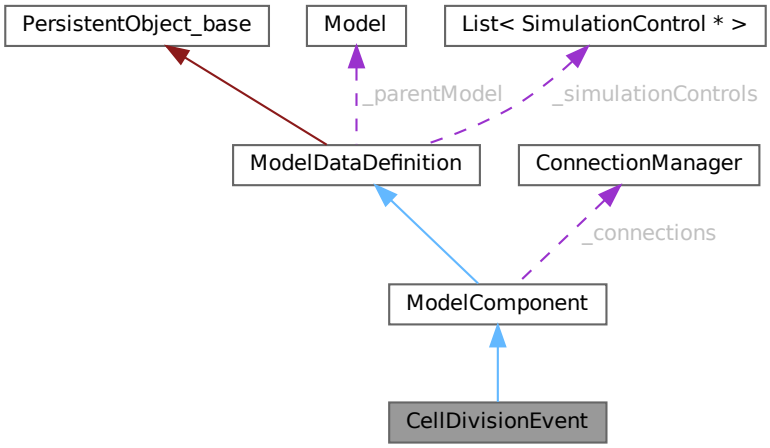
- plugins/components/WholeCellModeling/CellCycleCheckpointComponent.h
- plugins/components/WholeCellModeling/CellCycleCheckpointComponent.cpp

17.108 CellDivisionEvent Class Reference

`#include <plugins/components/WholeCellModeling/CellDivisionEvent.h>`
Inheritance diagram for CellDivisionEvent:



Collaboration diagram for CellDivisionEvent:



Public Member Functions

- **CellDivisionEvent** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)
- WholeCellState * **getWholeCellState** () const
- void **setDivisionMassThreshold** (double threshold)
- double **getDivisionMassThreshold** () const
- void **setFtsZThreshold** (double threshold)
- double **getFtsZThreshold** () const
- void **setFtsZRingKey** (std::string key)
- std::string **getFtsZRingKey** () const
- void **setRandomSeed** (unsigned int seed)
- unsigned int **getRandomSeed** () const
- bool **getLastDivided** () const
- int **getDivisionCount** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)

- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.108.1 Detailed Description

Discrete cell division event for whole-cell model simulations.
Implements the M. genitalium CellDivision process (Karr et al. 2012): Trigger: cell mass >= divisionMass↩ Threshold (default 2 × cellInitialDryWeight) AND (ftsZRingCompletion key in WholeCellState >= ftsZThreshold OR ftsZThreshold == 0, i.e., mass-only trigger)
On division:

- Each integer molecule count is binomially partitioned between daughters (each copy independently assigned to daughter 1 with p=0.5)
- This simulated cell becomes daughter 1
- Cell mass and volume are set to half the pre-division values
- Simulation step count is reset to 0 (new generation)
- WholeCellState lifecycle metadata is updated to reflect the new generation

Output ports: Port 0: entity forwarded when no division occurred this step (normal flow) Port 1: entity forwarded immediately after a division event (optional handler)
The division mass threshold is read from WholeCellState metabolite pool entry "param.cellDivisionMassThreshold" when set by loadParameters(). Override with the divisionMassThreshold property for manual configuration.
FtsZ ring completion is read from WholeCellState molecule count key ftsZRingKey (default "FtsZ_ring_completion", expected range 0–1000 as integer per-mille value; divide by 1000 for fraction).

17.108.2 Member Function Documentation

17.108.2.1 _check()

```
bool CellDivisionEvent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.108.2.2 _createEditableDataDefinitions()

```
void CellDivisionEvent::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.108.2.3 _initBetweenReplications()

```
void CellDivisionEvent::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.108.2.4 _loadInstance()

```
bool CellDivisionEvent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.108.2.5 _onDispatchEvent()

```
void CellDivisionEvent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.108.2.6 _saveInstance()

```
void CellDivisionEvent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.108.2.7 show()

```
std::string CellDivisionEvent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

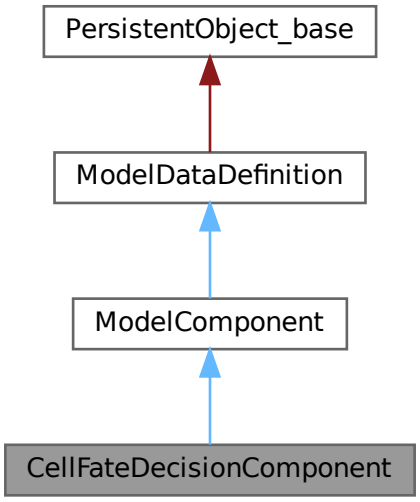
- plugins/components/WholeCellModeling/CellDivisionEvent.h
- plugins/components/WholeCellModeling/CellDivisionEvent.cpp

17.109 CellFateDecisionComponent Class Reference

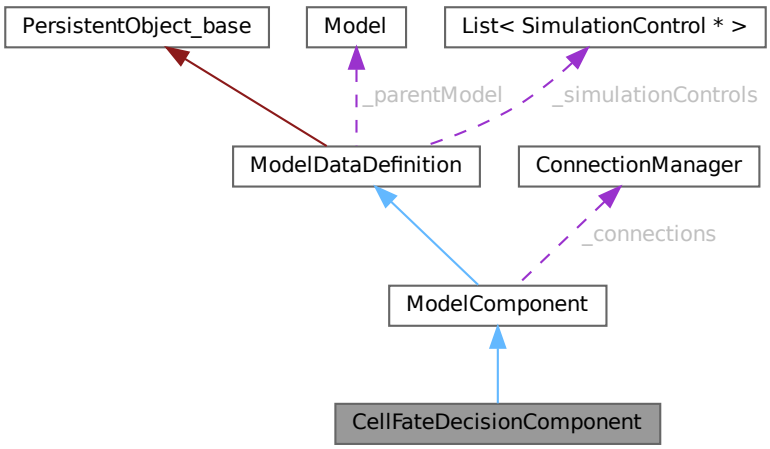
Routes entities according to the current whole-cell lifecycle state.

`#include <plugins/components/WholeCellModeling/CellFateDecisionComponent.h>`

Inheritance diagram for CellFateDecisionComponent:



Collaboration diagram for CellFateDecisionComponent:



Public Member Functions

- **CellFateDecisionComponent** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)
- WholeCellState * **getWholeCellState** () const

- void **setDivisionReadyPhase** (std::string phase)
- std::string **getDivisionReadyPhase** () const
- void **setStarvedPhase** (std::string phase)
- std::string **getStarvedPhase** () const
- void **setArrestedPhase** (std::string phase)
- std::string **getArrestedPhase** () const
- void **setDeadPhase** (std::string phase)
- std::string **getDeadPhase** () const
- unsigned int **getLastRoutedPort** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.109.1 Detailed Description

Routes entities according to the current whole-cell lifecycle state.
CellFateDecisionComponent converts the coarse lifecycle annotation stored in WholeCellState into explicit simulation branching:

- port 0: viable default flow
- port 1: division-ready flow
- port 2: starved-but-still-viable flow
- port 3: dead or non-viable flow

The component does not mutate biological state. It only reads viability and lifecycle phase from WholeCellState and routes the entity to the appropriate downstream component.

17.109.2 Member Function Documentation

17.109.2.1 _check()

```
bool CellFateDecisionComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.109.2.2 _createEditableDataDefinitions()

```
void CellFateDecisionComponent::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.109.2.3 _initBetweenReplications()

```
void CellFateDecisionComponent::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.109.2.4 _loadInstance()

```
bool CellFateDecisionComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.109.2.5 _onDispatchEvent()

```
void CellFateDecisionComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.109.2.6 _saveInstance()

```
void CellFateDecisionComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.109.2.7 show()

```
std::string CellFateDecisionComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

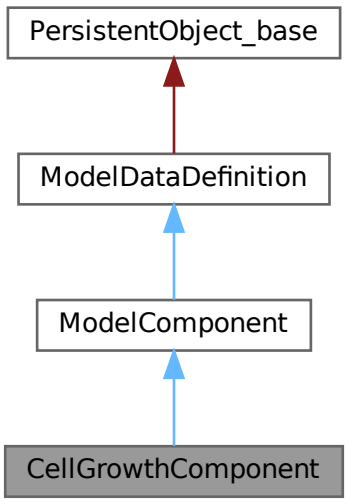
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/WholeCellModeling/CellFateDecisionComponent.h
- plugins/components/WholeCellModeling/CellFateDecisionComponent.cpp

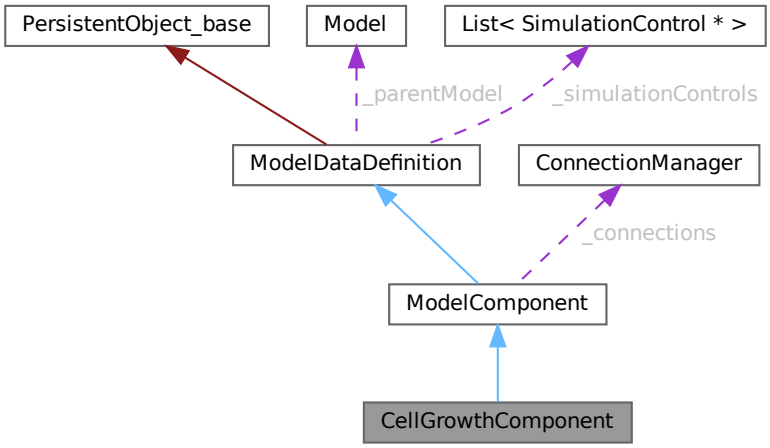
17.110 CellGrowthComponent Class Reference

```
#include <plugins/components/WholeCellModeling/CellGrowthComponent.h>
```

Inheritance diagram for CellGrowthComponent:



Collaboration diagram for CellGrowthComponent:



Public Member Functions

- **CellGrowthComponent** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)
- WholeCellState * **getWholeCellState** () const
- void **setGrowthRate** (double rate)
- double **getGrowthRate** () const
- void **setEnergyMetaboliteKey** (std::string key)
- std::string **getEnergyMetaboliteKey** () const
- void **setEnergyHalfSaturation** (double value)
- double **getEnergyHalfSaturation** () const
- void **setDensity** (double density)
- double **getDensity** () const
- void **setDeltaT** (double dt)
- double **getDeltaT** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)

- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.110.1 Detailed Description

Continuous cell growth applied at each simulation step.
Implements exponential mass growth: mass(t+Δt) = mass(t) $\times$ (1 + growthRate $\times$ Δt) volume(fL) = (mass_g $\times$ 1e-3 / density_kg_m3) $\times$ 1e15
Parameter defaults from CovertLab/WholeCell parameters.json (MIT): growthRate = 2.1393e-05 /s (states.↩ MetabolicReaction.meanInitialGrowthRate) density = 1100.0 kg/m^3 (states.Geometry.density) deltaT = 60.0 s (must match simulation clock step)

17.110.2 Member Function Documentation

17.110.2.1 _check()

```
bool CellGrowthComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.110.2.2 _createEditableDataDefinitions()

```
void CellGrowthComponent::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.110.2.3 _initBetweenReplications()

```
void CellGrowthComponent::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.110.2.4 _loadInstance()

```
bool CellGrowthComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.110.2.5 _onDispatchEvent()

```
void CellGrowthComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.110.2.6 _saveInstance()

```
void CellGrowthComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.110.2.7 show()

```
std::string CellGrowthComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

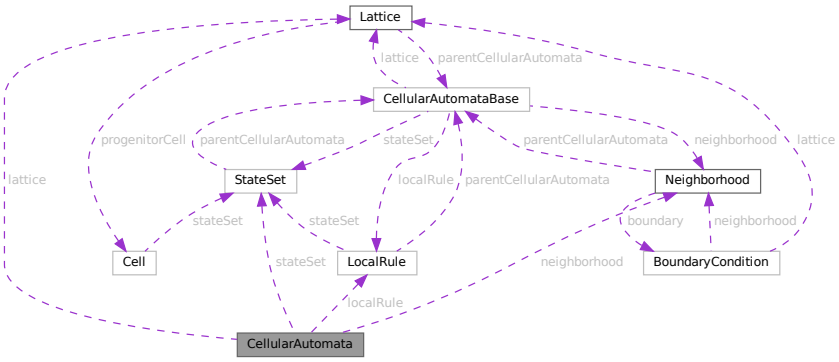
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/WholeCellModeling/CellGrowthComponent.h
- plugins/components/WholeCellModeling/CellGrowthComponent.cpp

17.111 CellularAutomata Class Reference

Collaboration diagram for CellularAutomata:



Public Member Functions

- **CellularAutomata** (std::string name="", Lattice ∗lattice=nullptr, StateSet ∗stateSet=nullptr, Neighborhood ∗neighborhood=nullptr, LocalRule ∗localRule=nullptr)
- **CellularAutomata** (const CellularAutomata &orig)
- virtual bool **init** ()

  Init lattice and cells.
- virtual void **step** ()

  Create next configuration c(t+1) = G(c(t))
- virtual void **start** ()

  Run (step) until end of simulation.
- virtual void **pause** ()

  Pase the "run".
- template<typename Class >
  void **addObserver** (Class ∗object, void(Class::∗function)(CellularAutomataEvent))
- std::string **getName** () const
- void **setName** (const std::string name)
- void **setLocalRule** (LocalRule ∗localRule)
- LocalRule ∗ **getLocalRule** () const
- void **setNeighborhood** (Neighborhood ∗neighborhood)
- Neighborhood ∗ **getNeighborhood** () const
- void **setStateSet** (StateSet ∗stateSet)
- StateSet ∗ **getStateSet** () const
- void **setLattice** (Lattice ∗lattice)
- Lattice ∗ **getLattice** () const
- void **setSimulationLength** (unsigned int simulationLength)
- unsigned int **getSimulationLength** () const
- unsigned int **getSimulatedTime** () const
- bool **isRunning** () const

Protected Member Functions

- void **notifyObservers** (CellularAutomataEvent event)

  Invoke observers then something (an event) happes.
- virtual void **applyLocalRule** ()=0

  Pure virtual method that has to be overriden by derived classes.

Protected Attributes

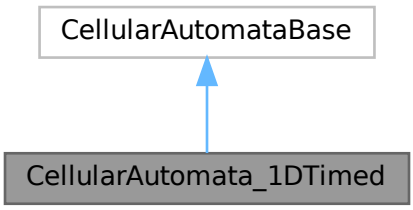
- Lattice * **lattice**
- StateSet * **stateSet**
- Neighborhood * **neighborhood**
- LocalRule * **localRule**
- std::string **name**
- unsigned int **simulatedTime**
- unsigned int **simulationLength**
- bool **running**

The documentation for this class was generated from the following files:

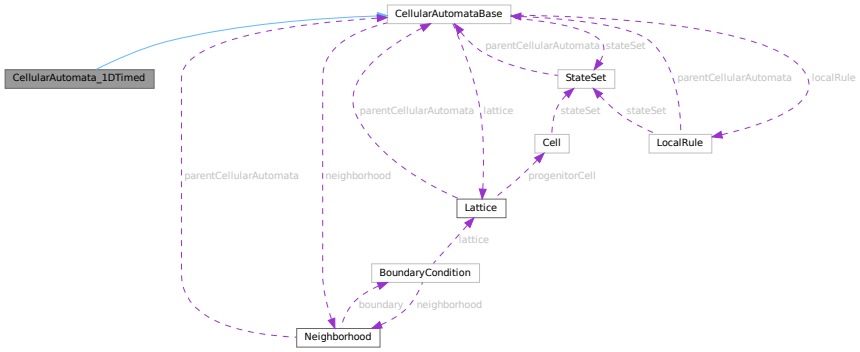
- plugins/components/ModalModel/CellularAutomata/CellularAutomata.h
- plugins/components/ModalModel/CellularAutomata/CellularAutomata.cpp

17.112 CellularAutomata_1DTimed Class Reference

Complete uniform CA specific for unidimensional lattices where 2nd dimension is time (usefull for elementary CA)
`#include <plugins/components/ModalModel/CellularAutomata/CellularAutomata_↩ 1DTimed.h>`
Inheritance diagram for CellularAutomata_1DTimed:



Collaboration diagram for CellularAutomata_1DTimed:



Public Member Functions

- **CellularAutomata_1DTimed** (Lattice *lattice=nullptr, StateSet *stateSet=nullptr, Neighborhood *neighborhood=nullptr, LocalRule *localRule=nullptr)
- **CellularAutomata_1DTimed** (const CellularAutomata_1DTimed &orig)

Public Member Functions inherited from CellularAutomataBase

- **CellularAutomataBase** (Lattice *lattice=nullptr, StateSet *stateSet=nullptr, Neighborhood *neighborhood=nullptr, LocalRule *localRule=nullptr)
- **CellularAutomataBase** (const CellularAutomataBase &orig)
- virtual bool **init** ()

  Init lattice and cells.
- virtual void **step** ()

  Create next configuration c(t+1) = G(c(t))
- void **setLocalRule** (LocalRule *localRule)
- LocalRule * **getLocalRule** () const
- void **setNeighborhood** (Neighborhood *neighborhood)
- Neighborhood * **getNeighborhood** () const
- void **setStateSet** (StateSet *stateSet)
- StateSet * **getStateSet** () const
- void **setLattice** (Lattice *lattice)
- Lattice * **getLattice** () const

Protected Member Functions

- virtual void applyLocalRule () override

  Pure virtual method that has to be overriden by derived classes.

Additional Inherited Members

Protected Attributes inherited from CellularAutomataBase

- Lattice * **lattice**
- StateSet * **stateSet**
- Neighborhood * **neighborhood**
- LocalRule * **localRule**

17.112.1 Detailed Description

Complete uniform CA specific for unidimensional lattices where 2nd dimension is time (usefull for elementary CA)

17.112.2 Member Function Documentation

17.112.2.1 applyLocalRule()

```
virtual void CellularAutomata_1DTimed::applyLocalRule ( )  [inline], [override], [protected],
[virtual]
```

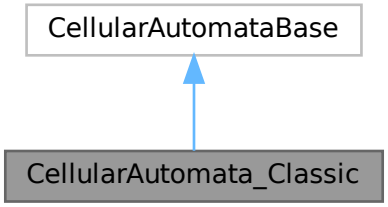
Pure virtual method that has to be overriden by derived classes.
Implements CellularAutomataBase.
The documentation for this class was generated from the following file:

- plugins/components/ModalModel/CellularAutomata/CellularAutomata_1DTimed.h

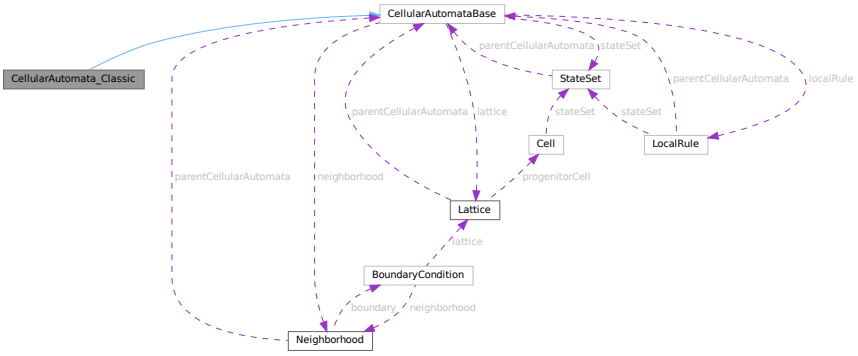
17.113 CellularAutomata_Classic Class Reference

Complete uniform CA.
`#include <plugins/components/ModalModel/CellularAutomata/CellularAutomata_↩ Classic.h>`

Inheritance diagram for CellularAutomata_Classic:



Collaboration diagram for CellularAutomata_Classic:



Public Member Functions

- **CellularAutomata_Classic** (Lattice ∗lattice=nullptr, StateSet ∗stateSet=nullptr, Neighborhood ∗neighborhood=nullptr, LocalRule ∗localRule=nullptr)
- **CellularAutomata_Classic** (const CellularAutomata_Classic &orig)

Public Member Functions inherited from CellularAutomataBase

- **CellularAutomataBase** (Lattice ∗lattice=nullptr, StateSet ∗stateSet=nullptr, Neighborhood ∗neighborhood=nullptr, LocalRule ∗localRule=nullptr)
- **CellularAutomataBase** (const CellularAutomataBase &orig)
- virtual bool **init** ()

  Init lattice and cells.
- virtual void **step** ()

  Create next configuration c(t+1) = G(c(t))
- void **setLocalRule** (LocalRule ∗localRule)
- LocalRule ∗ **getLocalRule** () const
- void **setNeighborhood** (Neighborhood ∗neighborhood)
- Neighborhood ∗ **getNeighborhood** () const
- void **setStateSet** (StateSet ∗stateSet)
- StateSet ∗ **getStateSet** () const
- void **setLattice** (Lattice ∗lattice)
- Lattice ∗ **getLattice** () const

Protected Member Functions

- virtual void applyLocalRule () override

  Pure virtual method that has to be overriden by derived classes.

Additional Inherited Members

Protected Attributes inherited from CellularAutomataBase

- Lattice * **lattice**
- StateSet * **stateSet**
- Neighborhood * **neighborhood**
- LocalRule * **localRule**

17.113.1 Detailed Description

Complete uniform CA.

17.113.2 Member Function Documentation

17.113.2.1 applyLocalRule()

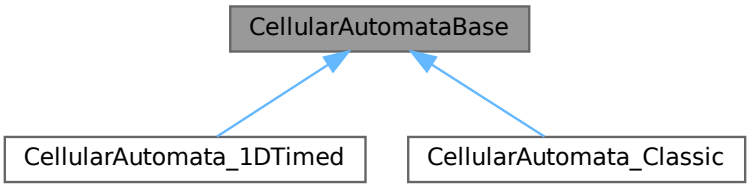
```
virtual void CellularAutomata_Classic::applyLocalRule ( )  [inline], [override], [protected],
[virtual]
```

Pure virtual method that has to be overriden by derived classes.
< overrides pure virtual method
Implements CellularAutomataBase.
The documentation for this class was generated from the following file:

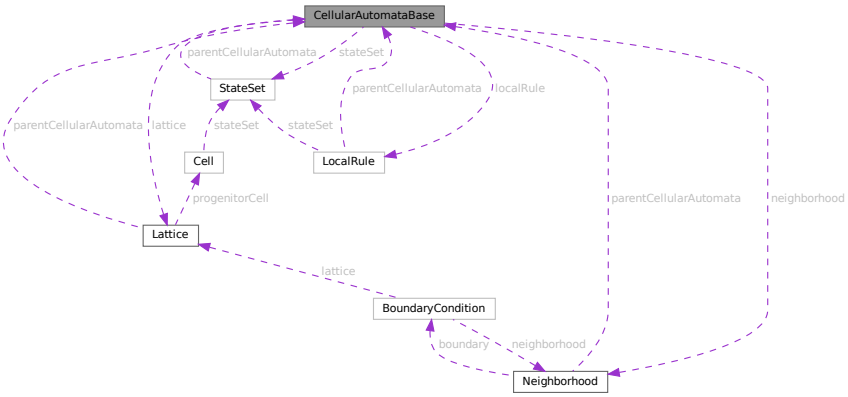
- plugins/components/ModalModel/CellularAutomata/CellularAutomata_Classic.h

17.114 CellularAutomataBase Class Reference

Inheritance diagram for CellularAutomataBase:



Collaboration diagram for CellularAutomataBase:



Public Member Functions

- **CellularAutomataBase** (Lattice *lattice=nullptr, StateSet *stateSet=nullptr, Neighborhood *neighborhood=nullptr, LocalRule *localRule=nullptr)
- **CellularAutomataBase** (const CellularAutomataBase &orig)
- virtual bool **init** ()

  Init lattice and cells.
- virtual void **step** ()

  Create next configuration c(t+1) = G(c(t))
- void **setLocalRule** (LocalRule *localRule)
- LocalRule * **getLocalRule** () const
- void **setNeighborhood** (Neighborhood *neighborhood)
- Neighborhood * **getNeighborhood** () const
- void **setStateSet** (StateSet *stateSet)
- StateSet * **getStateSet** () const
- void **setLattice** (Lattice *lattice)
- Lattice * **getLattice** () const

Protected Member Functions

- virtual void applyLocalRule ()=0

  Pure virtual method that has to be overriden by derived classes.

Protected Attributes

- Lattice * **lattice**
- StateSet * **stateSet**
- Neighborhood * **neighborhood**
- LocalRule * **localRule**

17.114.1 Member Function Documentation

17.114.1.1 applyLocalRule()

`void CellularAutomataBase::applyLocalRule ( ) [protected], [pure virtual]`

Pure virtual method that has to be overriden by derived classes.

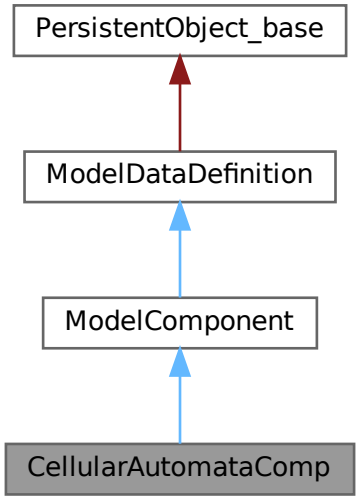
Implemented in CellularAutomata_1DTimed, and CellularAutomata_Classic.

The documentation for this class was generated from the following files:

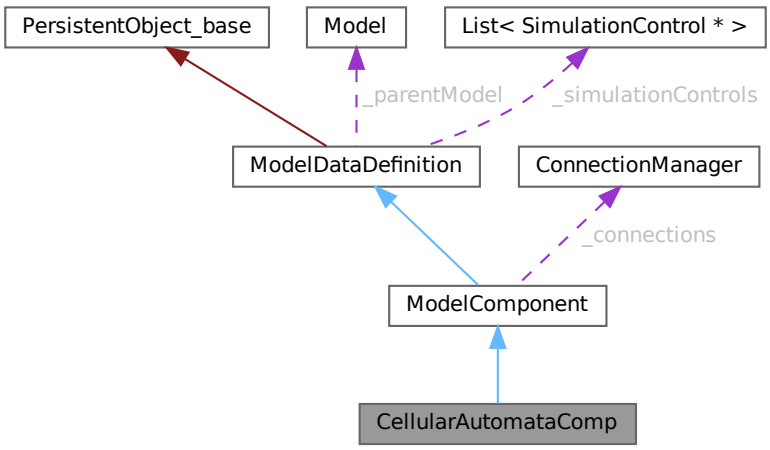
- plugins/components/ModalModel/CellularAutomata/CellularAutomataBase.h
- plugins/components/ModalModel/CellularAutomata/CellularAutomataBase.cpp

17.115 CellularAutomataComp Class Reference

`#include <plugins/components/ModalModel/CellularAutomataComp.h>`
Inheritance diagram for CellularAutomataComp:



Collaboration diagram for CellularAutomataComp:



Public Types

- enum class CellularAutomataType : int {
  CLASSIC = 1 , **TIMED_1D** = 2 , **ASYNCHRONOUS** = 3 , **NONUNIFORMRULE** = 4 ,
  NONUNIFORMNEIGHBOOR = 5 , **USERDEFINED** = 6 }

  enums

- enum class **LatticeType** : int {
  RETICULAR = 1 , **TRIANGULAR** = 2 , **HEXAGONAL** = 3 , **NETWORK** = 4 ,
  USERDEFINED = 5 }
- enum class **NeighboorhoodType** : int {
  CENTERED = 1 , **BACKWARD** = 2 , **FORWARD** = 3 , **VONNEUMANN** = 4 ,
  MOORE = 5 , **USERDEFINED** = 6 }
- enum class **BoundaryType** : int {
  FIXED = 1 , **CLOSED** = 2 , **REFLEXIVE** = 3 , **ADIABATIC** = 4 ,
  USERDEFINED = 5 }
- enum class **StateSetType** : int {
  ENUMERATED = 1 , **INTEGERBASED** = 2 , **BITBASED** = 3 , **DOUBLEBASED** = 4 ,
  USERDEFINED = 5 }
- enum class **LocalRuleType** : int {
  ELEMENTAR_CA = 1 , **GAME_OF_LIFE** = 2 , **BIASED_COMPETITION** = 3 , **HPP** = 4 ,
  SAND_ROCK = 5 , **USERDEFINED** = 99 }
- enum class **AutomataType** : int { **Temporary** = 0 , **Permanent** = 1 }

Public Member Functions

- **CellularAutomataComp** (Model *model, std::string name="")

  constructors
- CellularAutomataComp::CellularAutomataType **getCellularAutomataType** () const

  new public user methods for this component
- void **setCellularAutomataType** (CellularAutomataComp::CellularAutomataType newCellularAutomataType)
- CellularAutomataComp::LatticeType **getLatticeType** () const
- void **setLatticeType** (CellularAutomataComp::LatticeType newLatticeStructure)
- CellularAutomataComp::NeighboorhoodType **getNeighboorhoodType** () const
- void **setNeighboorhoodType** (CellularAutomataComp::NeighboorhoodType newNeighboorhood)
- CellularAutomataComp::BoundaryType **geBoundaryType** () const
- void **setBoundaryType** (CellularAutomataComp::BoundaryType newBoundary)
- CellularAutomataComp::StateSetType **getStateSetType** () const
- void **setStateSetType** (CellularAutomataComp::StateSetType newStateSetType)
- Lattice * **getlattice** () const
- Neighborhood * **getNeighboorhood** () const
- StateSet * **getStateSet** () const
- virtual std::string show () override

  virtual public methods
- CellularAutomataComp::LocalRuleType **getlocalRuleType** () const
- void **setLocalRuleType** (CellularAutomataComp::LocalRuleType newLocalRuleType)
- LocalRule * **getlocalRule** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual bool _check (std::string *errorMessage)

  virtual protected methods that could be overriden by derived classes, if needed
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)

- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.115.1 Detailed Description

This component ...

17.115.2 Member Function Documentation

17.115.2.1 _check()

```
bool CellularAutomataComp::_check (
            std::string * errorMessage )  [protected], [virtual]
```

virtual protected methods that could be overriden by derived classes, if needed
This method is called by ModelChecker during model check. The component should check itself to verify if user parameters are ok (ex: correct syntax for the parser) and everithing in its parameters allow the model too run without errors in this component

17.115.2.2 _initBetweenReplications()

```
void CellularAutomataComp::_initBetweenReplications ( )  [override], [protected], [virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
This method is called by ModelSimulation when initianting the replication. The model should set all value for a new replication (Ex: setting back to 0 any internal counter, clearing lists, etc.
Reimplemented from ModelDataDefinition.

17.115.2.3 _loadInstance()

```
bool CellularAutomataComp::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.115.2.4 _onDispatchEvent()

```
void CellularAutomataComp::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

This method is only for ModelComponents, not ModelDataElements.
Implements ModelComponent.

17.115.2.5 _saveInstance()

```
void CellularAutomataComp::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.115.2.6 show()

```
std::string CellularAutomataComp::show ( )  [override], [virtual]
```

virtual public methods
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/ModalModel/CellularAutomataComp.h
- plugins/components/ModalModel/CellularAutomataComp.cpp

17.116 CellularAutomataDemoBuilder Class Reference

Static Public Member Functions

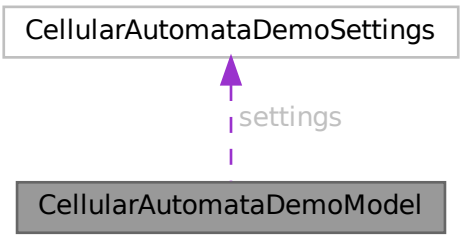
- static std::unique_ptr< CellularAutomataDemoModel > **buildDemo** (CellularAutomataDemoSettings settings={})

The documentation for this class was generated from the following files:

- applications/gui/genesys/cellular_automata/CellularAutomataDemoBuilder.h
- applications/gui/genesys/cellular_automata/CellularAutomataDemoBuilder.cpp

17.117 CellularAutomataDemoModel Struct Reference

Collaboration diagram for CellularAutomataDemoModel:



Public Member Functions

- **CellularAutomataDemoModel** (CellularAutomataDemoSettings settings={})
- int **columns** () const
- int **rows** () const
- bool **isValidPosition** (int x, int y) const
- Cell * **cellAt** (int x, int y) const
- long **stateAt** (int x, int y) const
- QString **stateTextAt** (int x, int y) const
- bool **setStateAt** (int x, int y, long value)
- void **fill** (long value)
- void **seedDefaultPattern** ()
- std::vector< long > **availablePaintStates** () const

Public Attributes

- CellularAutomataDemoSettings **settings**
- std::vector< State > **ownedStates**
- std::vector< State * > **statePointers**
- std::unique_ptr< CellularAutomata_Classic > **automaton**
- std::unique_ptr< Lattice > **lattice**
- std::unique_ptr< StateSet > **stateSet**
- std::unique_ptr< BoundaryCondition > **boundary**
- std::unique_ptr< Neighborhood > **neighborhood**
- std::unique_ptr< LocalRule > **localRule**
- std::unique_ptr< LocalRule_CopyCurrentState > **identityLocalRule**

The documentation for this struct was generated from the following files:

- applications/gui/genesys/cellular_automata/CellularAutomataDemoBuilder.h
- applications/gui/genesys/cellular_automata/CellularAutomataDemoBuilder.cpp

17.118 CellularAutomataDemoSettings Struct Reference

Public Attributes

- QSize **latticeSize** = QSize{100, 100}
- CellularAutomataRulePreset **rulePreset** = CellularAutomataRulePreset::SandRock

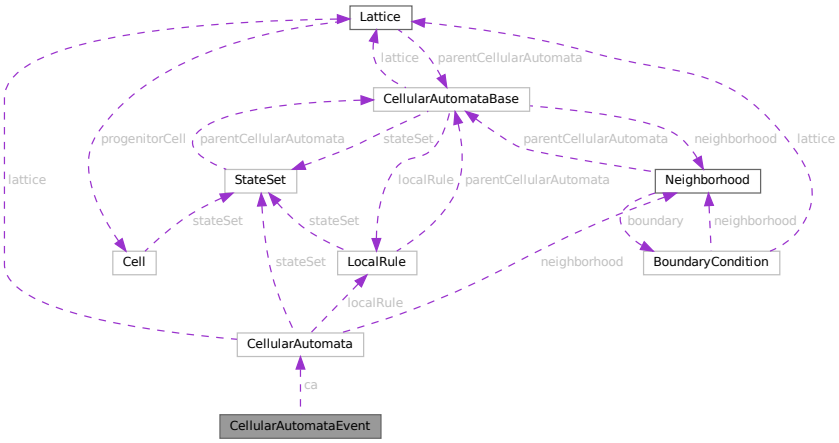
- CellularAutomataNeighborhoodPreset **neighborhoodPreset** = CellularAutomataNeighborhoodPreset::↩ Moore
- CellularAutomataBoundaryPreset **boundaryPreset** = CellularAutomataBoundaryPreset::Fixed
- CellularAutomataStatePreset **statePreset** = CellularAutomataStatePreset::Enumerated

The documentation for this struct was generated from the following file:

- applications/gui/genesys/cellular_automata/CellularAutomataDemoBuilder.h

17.119 CellularAutomataEvent Class Reference

Collaboration diagram for CellularAutomataEvent:



Public Member Functions

- **CellularAutomataEvent** (CellularAutomata ∗ca)

Public Attributes

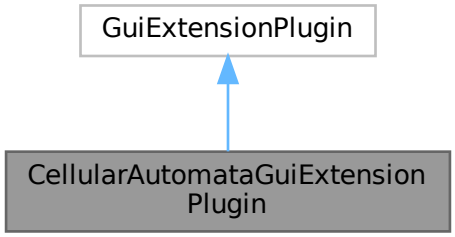
- CellularAutomata ∗ **ca**

The documentation for this class was generated from the following file:

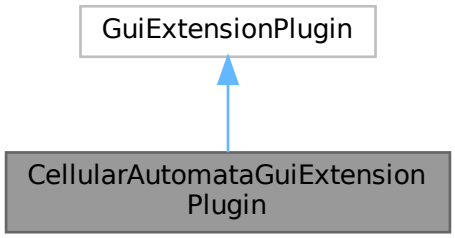
- plugins/components/ModalModel/CellularAutomata/CellularAutomata.h

17.120 CellularAutomataGuiExtensionPlugin Class Reference

Inheritance diagram for CellularAutomataGuiExtensionPlugin:



Collaboration diagram for CellularAutomataGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.120.1 Member Function Documentation

17.120.1.1 displayName()

```
std::string CellularAutomataGuiExtensionPlugin::displayName ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.120.1.2 extensionId()

```
std::string CellularAutomataGuiExtensionPlugin::extensionId ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.120.1.3 registerContributions()

```
void CellularAutomataGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.120.1.4 requiredModelPlugins()

```
std::vector< std::string > CellularAutomataGuiExtensionPlugin::requiredModelPlugins ( ) const
[inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.120.1.5 version()

```
std::string CellularAutomataGuiExtensionPlugin::version ( ) const  [inline], [override], [virtual]
```

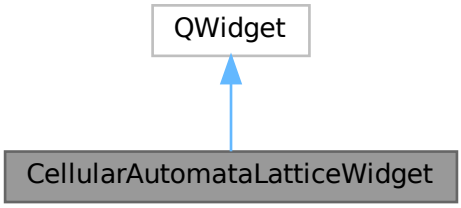
Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

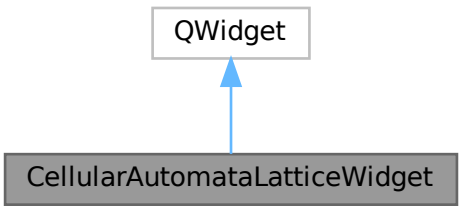
- applications/gui/genesys/extensions/CellularAutomataGuiExtensionPlugin.cpp

17.121 CellularAutomataLatticeWidget Class Reference

Inheritance diagram for CellularAutomataLatticeWidget:



Collaboration diagram for CellularAutomataLatticeWidget:



Public Member Functions

- **CellularAutomataLatticeWidget** (QWidget *parent=nullptr)
- void **setController** (CellularAutomataViewerController *controller)

Protected Member Functions

- void **paintEvent** (QPaintEvent ∗event) override
- void **mousePressEvent** (QMouseEvent ∗event) override
- void **mouseMoveEvent** (QMouseEvent ∗event) override
- void **mouseReleaseEvent** (QMouseEvent ∗event) override
- void **contextMenuEvent** (QContextMenuEvent ∗event) override
- QSize **minimumSizeHint** () const override

The documentation for this class was generated from the following files:

- applications/gui/genesys/cellular_automata/CellularAutomataLatticeWidget.h
- applications/gui/genesys/cellular_automata/CellularAutomataLatticeWidget.cpp

17.122 CellularAutomataStateColorMap Class Reference

Static Public Member Functions

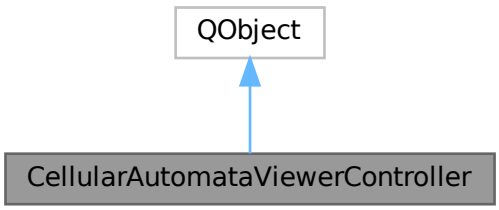
- static QColor **colorForState** (long state)
- static QColor **colorForStateText** (const QString &stateText, long fallbackState)
- static QColor **backgroundColor** ()
- static QColor **gridLineColor** ()
- static QColor **emptyTextColor** ()

The documentation for this class was generated from the following file:

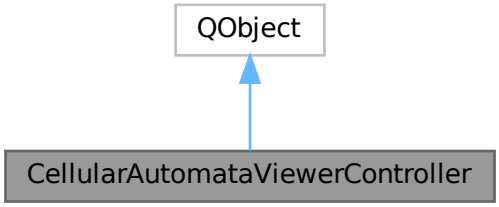
- applications/gui/genesys/cellular_automata/CellularAutomataStateColorMap.h

17.123 CellularAutomataViewerController Class Reference

Inheritance diagram for CellularAutomataViewerController:



Collaboration diagram for CellularAutomataViewerController:



Signals

- void **modelChanged** ()
- void **settingsChanged** ()
- void **timeChanged** (unsigned int time)
- void **runningChanged** (bool running)
- void **paintStateChanged** ()
- void **statusMessage** (const QString &message)

Public Member Functions

- **CellularAutomataViewerController** (QObject *parent=nullptr)
- CellularAutomataDemoModel * **model** () const
- CellularAutomataDemoSettings **settings** () const
- QSize **latticeSize** () const
- bool **hasDemo** () const
- unsigned int **simulatedTime** () const
- bool **isRunning** () const
- QString **configurationSummary** () const
- std::vector< long > **availablePaintStates** () const
- long **cellState** (int x, int y) const
- bool **setCellState** (int x, int y, long value)
- void **fill** (long value)
- bool **hasSelectedPaintState** () const
- std::optional< long > **selectedPaintState** () const
- QString **selectedPaintStateText** () const
- void **setSelectedPaintState** (long state)
- void **clearSelectedPaintState** ()
- int **autoStepIntervalMs** () const
- void **setAutoStepIntervalMs** (int intervalMs)
- void **rebuildDemo** (CellularAutomataDemoSettings settings={})
- void **setSettings** (CellularAutomataDemoSettings settings)
- void **setRulePreset** (CellularAutomataRulePreset preset)
- void **setNeighborhoodPreset** (CellularAutomataNeighborhoodPreset preset)
- void **setBoundaryPreset** (CellularAutomataBoundaryPreset preset)
- void **setStatePreset** (CellularAutomataStatePreset preset)
- void **resetToInitialState** ()
- void **step** ()
- void **start** ()

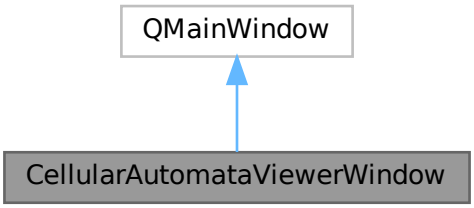
- void **pause** ()
- void **stop** ()
- bool **saveConfiguration** (const QString &filePath, QString ∗errorMessage=nullptr) const
- bool **loadConfiguration** (const QString &filePath, QString ∗errorMessage=nullptr)

The documentation for this class was generated from the following files:

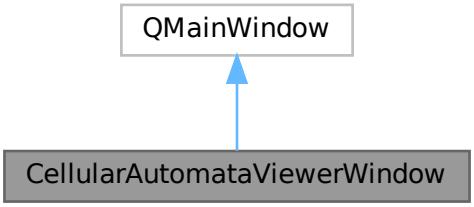
- applications/gui/genesys/cellular_automata/CellularAutomataViewerController.h
- applications/gui/genesys/cellular_automata/CellularAutomataViewerController.cpp

17.124 CellularAutomataViewerWindow Class Reference

Inheritance diagram for CellularAutomataViewerWindow:



Collaboration diagram for CellularAutomataViewerWindow:



Public Member Functions

- **CellularAutomataViewerWindow** (QWidget ∗parent=nullptr)

The documentation for this class was generated from the following files:

- applications/gui/genesys/cellular_automata/CellularAutomataViewerWindow.h
- applications/gui/genesys/cellular_automata/CellularAutomataViewerWindow.cpp

17.125 ChatMessage Struct Reference

One message in a multi-turn conversation with an LLM.
`#include <tools/AIAssistant/AIProviderClient_if.h>`

Public Attributes

- std::string **role**

  "system", "user", or "assistant".
- std::string **content**

  Text content of the message.

17.125.1 Detailed Description

One message in a multi-turn conversation with an LLM.
Role conventions follow the OpenAI and Anthropic chat-completion APIs:

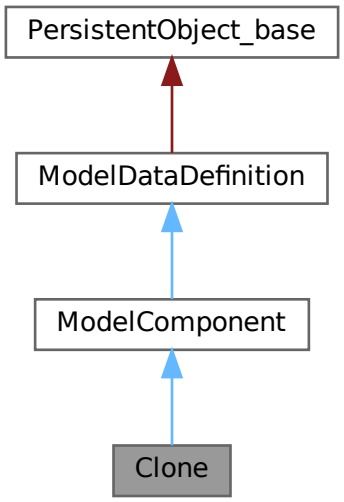
- "system" — instructions/persona injected before the conversation.
- "user" — message from the human turn.
- "assistant" — previous model replies (for multi-turn context).

The documentation for this struct was generated from the following file:

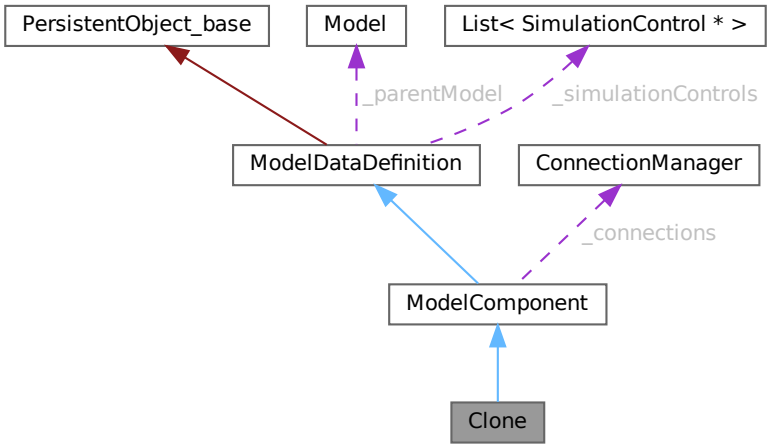
- tools/AIAssistant/AIProviderClient_if.h

17.126 Clone Class Reference

Creates `_numClonesExpression` copies of the incoming entity (same EntityType and Attribute values), sending clones out port 1 and the original out port 0.
`#include <plugins/components/DiscreteProcessing/Clone.h>`
Inheritance diagram for Clone:



Collaboration diagram for Clone:



Public Member Functions

- **Clone** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setNumClonesExpression** (std::string numClones)
- std::string **getNumClonesExpression** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createInternalStatisticReporters () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()

- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager ∗ _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.126.1 Detailed Description

Creates `_numClonesExpression` copies of the incoming entity (same EntityType and Attribute values), sending clones out port 1 and the original out port 0.
Arena correspondence: confirmed by the maintainer (2026-08-30) as the intended GenESyS counterpart to the "Duplicate Original" Type of Arena's "Separate module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", p. 39) — see Separate, whose own `_onDispatchEvent()` only implements Arena's other Type ("Split Existing Batch"). `_numClonesExpression` matches Arena's "# of Duplicates".
Known difference from Arena: no "Percent Cost to Duplicates" — splitting accumulated category cost/time between the original and its clones is not implemented (see `docs/ai_assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.3 and §6.19).

17.126.2 Member Function Documentation

17.126.2.1 _check()

```
bool Clone::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.126.2.2 _createInternalStatisticReporters()

```
void Clone::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.126.2.3 _initBetweenReplications()

```
void Clone::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.126.2.4 _loadInstance()

```
bool Clone::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.126.2.5 _onDispatchEvent()

```
void Clone::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.126.2.6 _saveInstance()

```
void Clone::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.126.2.7 show()

```
std::string Clone::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

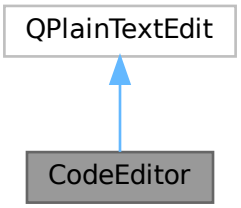
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

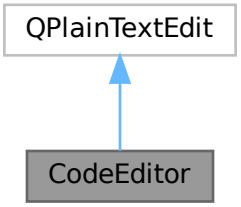
- plugins/components/DiscreteProcessing/Clone.h
- plugins/components/DiscreteProcessing/Clone.cpp

17.127 CodeEditor Class Reference

Inheritance diagram for CodeEditor:



Collaboration diagram for CodeEditor:



Public Member Functions

- **CodeEditor** (QWidget ∗parent=nullptr)
- void **lineNumberAreaPaintEvent** (QPaintEvent ∗event)
- int **lineNumberAreaWidth** ()

Protected Member Functions

- void **resizeEvent** (QResizeEvent ∗event) override

The documentation for this class was generated from the following files:

- applications/gui/genesys/codeeditor/CodeEditor.h
- applications/gui/genesys/codeeditor/CodeEditor.cpp

17.128 testing::internal::CodeLocation Struct Reference

Public Member Functions

- **CodeLocation** (const std::string &a_file, int a_line)

Public Attributes

- std::string **file**
- int **line**

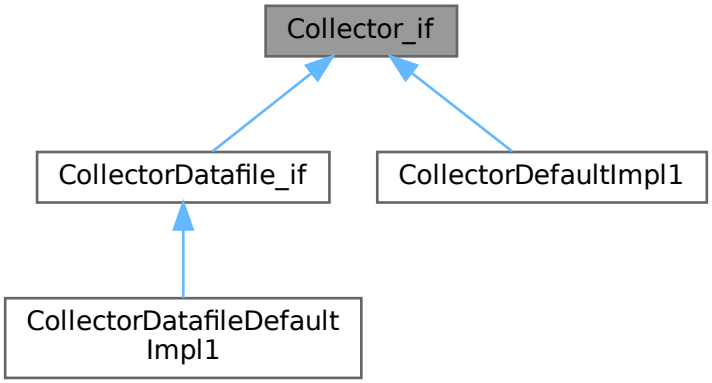
The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.129 Collector_if Class Reference

Interface responsible for collecting observations of a stochastic variable.
`#include <kernel/statistics/Collector_if.h>`

Inheritance diagram for Collector_if:



Public Member Functions

- virtual void clear ()=0

  clear
- virtual void addValue (double value, double weight=1)=0

  addValue
- virtual double getLastValue ()=0

  getLastValue
- virtual unsigned long numElements ()=0

  numElements
- virtual void setAddValueHandler (CollectorAddValueHandler addValueHandler)=0

  setAddValueHandler
- virtual void setClearHandler (CollectorClearHandler clearHandler)=0

  setClearHandler

17.129.1 Detailed Description

Interface responsible for collecting observations of a stochastic variable.
A collector stores values generated during simulation and optionally supports callbacks for integration with online statistics, tracing or UI updates.

17.129.2 Member Function Documentation

17.129.2.1 addValue()

```
virtual void Collector_if::addValue (
            double value,
            double weight = 1 )  [pure virtual]
```

addValue

Parameters

value	
weight	

Adds a new observed value (with optional weight).
Implemented in CollectorDatafileDefaultImpl1, and CollectorDefaultImpl1.

17.129.2.2 clear()

```
virtual void Collector_if::clear ( )  [pure virtual]
```

clear
Clears all collected values and resets internal state.
Implemented in CollectorDatafileDefaultImpl1, and CollectorDefaultImpl1.

17.129.2.3 getLastValue()

```
virtual double Collector_if::getLastValue ( )  [pure virtual]
```

getLastValue

Returns

Returns the last collected value.
Implemented in CollectorDatafileDefaultImpl1, and CollectorDefaultImpl1.

17.129.2.4 numElements()

```
virtual unsigned long Collector_if::numElements ( )  [pure virtual]
```

numElements

Returns

Returns the number of stored observations.
Implemented in CollectorDatafileDefaultImpl1, and CollectorDefaultImpl1.

17.129.2.5 setAddValueHandler()

```
virtual void Collector_if::setAddValueHandler (
            CollectorAddValueHandler addValueHandler )  [pure virtual]
```

setAddValueHandler

Parameters

addValueHandler	

Registers a callback triggered after each added value.
Implemented in CollectorDatafileDefaultImpl1, and CollectorDefaultImpl1.

17.129.2.6 setClearHandler()

```
virtual void Collector_if::setClearHandler (
            CollectorClearHandler clearHandler )  [pure virtual]
```

setClearHandler

Parameters

clearHandler	

Registers a callback triggered when the collector is cleared.
Implemented in CollectorDatafileDefaultImpl1, and CollectorDefaultImpl1.
The documentation for this class was generated from the following file:

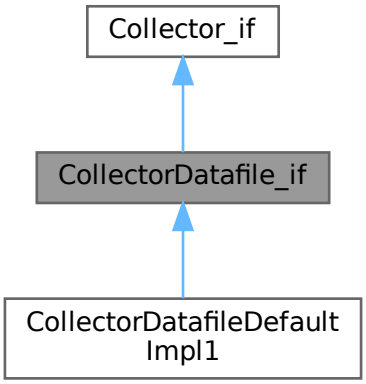
- kernel/statistics/Collector_if.h

17.130 CollectorDatafile_if Class Reference

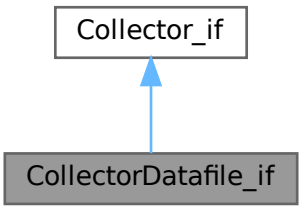
Collector interface specialized in file-backed value storage.

`#include <kernel/statistics/CollectorDatafile_if.h>`

Inheritance diagram for CollectorDatafile_if:



Collaboration diagram for CollectorDatafile_if:



Public Member Functions

- virtual double getValue (unsigned int rank)=0

  getValue
- virtual void seekFirstValue ()=0

  seekFirstValue
- virtual double getNextValue ()=0

  getNextValue
- virtual std::string getDataFilename ()=0

  getDataFilename
- virtual void setDataFilename (std::string filename)=0

  setDataFilename

Public Member Functions inherited from Collector_if

- virtual void clear ()=0

  clear
- virtual void addValue (double value, double weight=1)=0

  addValue
- virtual double getLastValue ()=0

  getLastValue
- virtual unsigned long numElements ()=0

  numElements
- virtual void setAddValueHandler (CollectorAddValueHandler addValueHandler)=0

  setAddValueHandler
- virtual void setClearHandler (CollectorClearHandler clearHandler)=0

  setClearHandler

17.130.1 Detailed Description

Collector interface specialized in file-backed value storage.
Besides standard collection operations, implementations provide sequential/random access to persisted samples and filename management for the underlying dataset.

17.130.2 Member Function Documentation

17.130.2.1 getDataFilename()

```
virtual std::string CollectorDatafile_if::getDataFilename ( )  [pure virtual]
```

getDataFilename

Returns

Returns the filename used to persist collected values.
Implemented in CollectorDatafileDefaultImpl1.

17.130.2.2 getNextValue()

```
virtual double CollectorDatafile_if::getNextValue ( )  [pure virtual]
```

getNextValue

Returns

Get the next value in the file and advances the pointer

Implemented in CollectorDatafileDefaultImpl1.

17.130.2.3 getValue()

```
virtual double CollectorDatafile_if::getValue (
            unsigned int rank )  [pure virtual]
```

getValue

Parameters

rank	

Returns

Get a value from a specific position

Implemented in CollectorDatafileDefaultImpl1.

17.130.2.4 seekFirstValue()

```
virtual void CollectorDatafile_if::seekFirstValue ( )  [pure virtual]
```

seekFirstValue
Set the pointer to the first value in the file
Implemented in CollectorDatafileDefaultImpl1.

17.130.2.5 setDataFilename()

```
virtual void CollectorDatafile_if::setDataFilename (
            std::string filename )  [pure virtual]
```

setDataFilename

Parameters

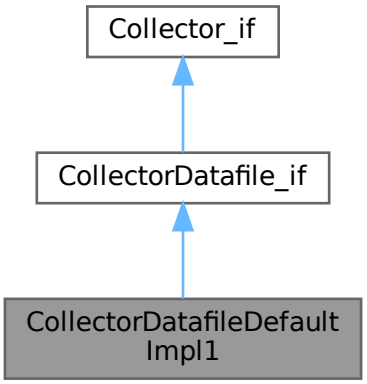
filename	

Sets/updates the filename used to store collected data.
Implemented in CollectorDatafileDefaultImpl1.
The documentation for this class was generated from the following file:

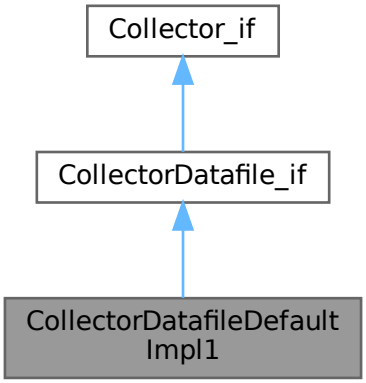
- kernel/statistics/CollectorDatafile_if.h

17.131 CollectorDatafileDefaultImpl1 Class Reference

Default file-backed implementation of `CollectorDatafile_if`.
`#include <kernel/statistics/CollectorDatafileDefaultImpl1.h>`
Inheritance diagram for CollectorDatafileDefaultImpl1:



Collaboration diagram for CollectorDatafileDefaultImpl1:



Public Member Functions

- **CollectorDatafileDefaultImpl1** ()

  Creates an empty file-backed collector instance.
- virtual void clear () override

  Clears collector state and resets file cursor/content as implemented.
- virtual void addValue (double value, double weight=1) override

  Appends a value (and optional weight) to the underlying storage.
- virtual double getLastValue () override

  Returns the most recently collected value.
- virtual unsigned long numElements () override

  Returns the number of collected elements.
- virtual double getValue (unsigned int num) override

  Returns the value stored at ordinal position `num`.
- virtual double getNextValue () override

  Returns the next value from current file cursor and advances cursor.
- virtual void seekFirstValue () override

  Repositions internal cursor to the first stored value.
- virtual std::string getDataFilename () override

  Returns the filename currently used as data backing store.
- virtual void setDataFilename (std::string filename) override

  Sets the filename used as data backing store.
- virtual void setAddValueHandler (CollectorAddValueHandler addValueHandler) override

  Registers callback invoked whenever a value is added.
- virtual void setClearHandler (CollectorClearHandler clearHandler) override

  Registers callback invoked whenever the collector is cleared.

17.131.1 Detailed Description

Default file-backed implementation of `CollectorDatafile_if`.
Stores collected values in an associated data file and keeps lightweight counters/cache for last value and number of elements.

17.131.2 Member Function Documentation

17.131.2.1 addValue()

```
void CollectorDatafileDefaultImpl1::addValue (
            double value,
            double weight = 1 )  [override], [virtual]
```

Appends a value (and optional weight) to the underlying storage.
Implements Collector_if.

17.131.2.2 clear()

```
void CollectorDatafileDefaultImpl1::clear ( )  [override], [virtual]
```

Clears collector state and resets file cursor/content as implemented.
Implements Collector_if.

17.131.2.3 getDataFilename()

```
std::string CollectorDatafileDefaultImpl1::getDataFilename ( )  [override], [virtual]
```

Returns the filename currently used as data backing store.
Implements CollectorDatafile_if.

17.131.2.4 getLastValue()

```
double CollectorDatafileDefaultImpl1::getLastValue ( )  [override], [virtual]
```

Returns the most recently collected value.
Implements Collector_if.

17.131.2.5 getNextValue()

```
double CollectorDatafileDefaultImpl1::getNextValue ( )  [override], [virtual]
```

Returns the next value from current file cursor and advances cursor.
Implements CollectorDatafile_if.

17.131.2.6 getValue()

```
double CollectorDatafileDefaultImpl1::getValue (
            unsigned int num )  [override], [virtual]
```

Returns the value stored at ordinal position `num`.
Implements CollectorDatafile_if.

17.131.2.7 numElements()

```
unsigned long CollectorDatafileDefaultImpl1::numElements ( )  [override], [virtual]
```

Returns the number of collected elements.
Implements Collector_if.

17.131.2.8 seekFirstValue()

```
void CollectorDatafileDefaultImpl1::seekFirstValue ( )  [override], [virtual]
```

Repositions internal cursor to the first stored value.
Implements CollectorDatafile_if.

17.131.2.9 setAddValueHandler()

```
void CollectorDatafileDefaultImpl1::setAddValueHandler (
            CollectorAddValueHandler addValueHandler )  [override], [virtual]
```

Registers callback invoked whenever a value is added.
Implements Collector_if.

17.131.2.10 setClearHandler()

```
void CollectorDatafileDefaultImpl1::setClearHandler (
            CollectorClearHandler clearHandler )  [override], [virtual]
```

Registers callback invoked whenever the collector is cleared.
Implements Collector_if.

17.131.2.11 setDataFilename()

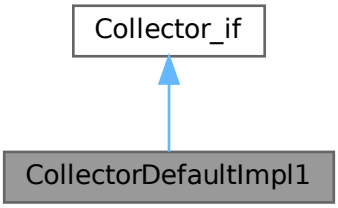
```
void CollectorDatafileDefaultImpl1::setDataFilename (
            std::string filename )  [override], [virtual]
```

Sets the filename used as data backing store.
Implements CollectorDatafile_if.
The documentation for this class was generated from the following files:

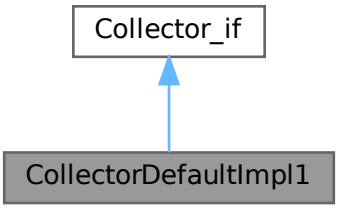
- kernel/statistics/CollectorDatafileDefaultImpl1.h
- kernel/statistics/CollectorDatafileDefaultImpl1.cpp

17.132 CollectorDefaultImpl1 Class Reference

Default in-memory implementation of `Collector_if`.
`#include <kernel/statistics/CollectorDefaultImpl1.h>`
Inheritance diagram for CollectorDefaultImpl1:



Collaboration diagram for CollectorDefaultImpl1:



Public Member Functions

- **CollectorDefaultImpl1** ()

Creates an empty collector.

- virtual void clear () override

  Clears all collected data and resets counters.

- virtual void addValue (double value, double weight=1) override

  Adds a new observation to the collector.

- virtual double getLastValue () override

  Returns the latest observed value.

- virtual unsigned long numElements () override

  Returns total number of observed values.

- virtual void setAddValueHandler (CollectorAddValueHandler addValueHandler) override

  Registers callback notified after each `addValue` call.

- virtual void setClearHandler (CollectorClearHandler clearHandler) override

  Registers callback notified when `clear` is invoked.

17.132.1 Detailed Description

Default in-memory implementation of `Collector_if`.
Maintains minimal aggregate state (last value and count) and supports optional callbacks for add/clear events.

17.132.2 Member Function Documentation

17.132.2.1 addValue()

```
void CollectorDefaultImpl1::addValue (
            double value,
            double weight = 1 )  [override], [virtual]
```

Adds a new observation to the collector.
Implements Collector_if.

17.132.2.2 clear()

```
void CollectorDefaultImpl1::clear ( )  [override], [virtual]
```

Clears all collected data and resets counters.
Implements Collector_if.

17.132.2.3 getLastValue()

```
double CollectorDefaultImpl1::getLastValue ( )  [override], [virtual]
```

Returns the latest observed value.
Implements Collector_if.

17.132.2.4 numElements()

```
unsigned long CollectorDefaultImpl1::numElements ( )  [override], [virtual]
```

Returns total number of observed values.
Implements Collector_if.

17.132.2.5 setAddValueHandler()

```
void CollectorDefaultImpl1::setAddValueHandler (
            CollectorAddValueHandler addValueHandler )  [override], [virtual]
```

Registers callback notified after each `addValue` call.
Implements Collector_if.

17.132.2.6 setClearHandler()

```
void CollectorDefaultImpl1::setClearHandler (
            CollectorClearHandler clearHandler )  [override], [virtual]
```

Registers callback notified when `clear` is invoked.
Implements Collector_if.
The documentation for this class was generated from the following files:

- kernel/statistics/CollectorDefaultImpl1.h
- kernel/statistics/CollectorDefaultImpl1.cpp

17.133 GroProgramRuntime::ColonyMutation Struct Reference

Public Types

- enum class **Type** {
  Reset , **SpawnSeed** , **SetChemostatMode** , **AddBarrier** ,
  MapToCells , **SetSignalAt** , **SetSignalRect** , **SetSignalGridWidth** ,
  SetSignalGridHeight , **GetSignalMatrix** , **DumpSignalField** , **EnsureSignalChannel** }

Public Attributes

- Type **type** = Type::Reset
- std::vector< std::string > **arguments**
- std::vector< double > **numericArguments**
- std::string **expressionText**
- std::string **signalDeclarationScope**
- std::size_t **previewRows** = 0
- std::size_t **previewColumns** = 0

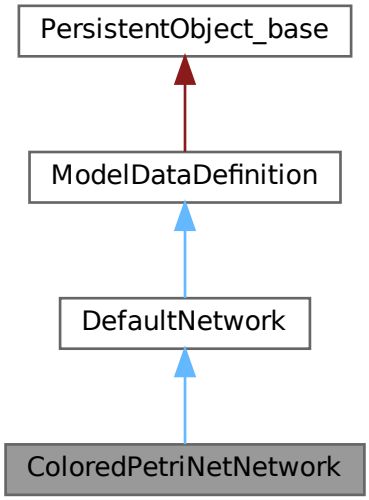
The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/GroProgramRuntime.h

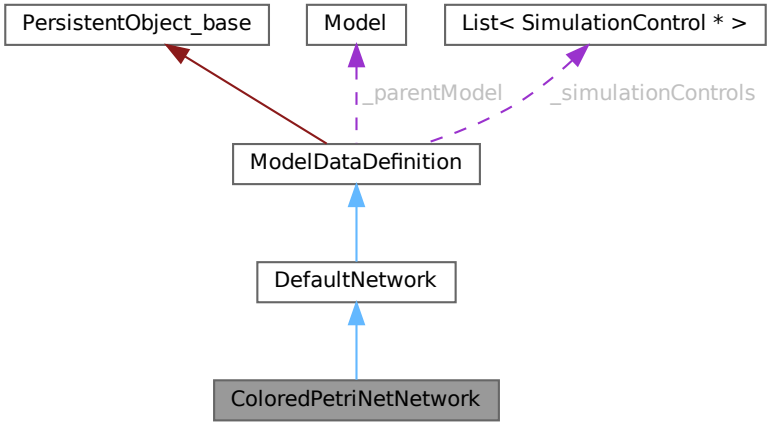
17.134 ColoredPetriNetNetwork Class Reference

Pragmatic fixed-inscription Colored Petri Net subset.
`#include <plugins/data/ModalModel/ColoredPetriNetNetwork.h>`

Inheritance diagram for ColoredPetriNetNetwork:



Collaboration diagram for ColoredPetriNetNetwork:



Public Types

- enum class **FiringMode** : unsigned int { **SingleDeterministic** = 0 }

Public Member Functions

- **ColoredPetriNetNetwork** (Model ∗model, std::string name="")
- void **addPlace** (PetriPlace ∗place)
- void **removePlace** (PetriPlace ∗place)
- List< PetriPlace ∗ > ∗ **getPlaces** () const

- void **addTransition** (CPNTransition *transition)
- void **removeTransition** (CPNTransition *transition)
- List< CPNTransition * > * **getTransitions** () const
- bool **addArc** (CPNArc *arc)
- void **removeArc** (CPNArc *arc)
- List< CPNArc * > * **getArcs** () const
- std::vector< CPNArc * > **getInputArcs** (CPNTransition *transition) const
- std::vector< CPNArc * > **getOutputArcs** (CPNTransition *transition) const
- bool **isEnabled** (CPNTransition *transition) const
- CPNTransition * **firstEnabledTransition** () const
- bool **fire** (CPNTransition *transition)
- void **setInitialTokens** (PetriPlace *place, std::string color, unsigned int quantity)
- unsigned int **getInitialTokens** (PetriPlace *place, std::string color="default") const
- void **captureCurrentMarkingAsInitial** ()
- void **restoreInitialMarking** ()
- FiringMode **getFiringMode** () const
- void **setFiringMode** (FiringMode firingMode)
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from DefaultNetwork

- **DefaultNetwork** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< DefaultNetwork >())
- unsigned int **addInputPort** (const std::string &portName)

  Declares a new input port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **addOutputPort** (const std::string &portName)

  Declares a new output port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **getNumInputPorts** () const
- unsigned int **getNumOutputPorts** () const
- std::string **getInputPortName** (unsigned int index) const
- std::string **getOutputPortName** (unsigned int index) const
- int **getInputPortIndex** (const std::string &portName) const

  Returns the input port index for `portName`, or -1 when not declared.
- int **getOutputPortIndex** (const std::string &portName) const

  Returns the output port index for `portName`, or -1 when not declared.
- double **getActivationCount** () const

  Number of times activate() has been called since the last replication reset.
- NetworkActivationResult activate (const NetworkActivationFrame &frame)

  Runs exactly one network activation for `frame`.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from DefaultNetwork

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual NetworkActivationResult _activate (const NetworkActivationFrame &frame) override

  Formalism-specific activation behavior.

Protected Member Functions inherited from DefaultNetwork

- virtual void _createInternalStatisticReporters () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)

- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.134.1 Detailed Description

Pragmatic fixed-inscription Colored Petri Net subset.
The topology is formally bipartite: places and transitions are nodes, and CPNArc objects connect exactly one place with one transition. The initial subset supports symbolic colors with unsigned multiplicities, fixed arc inscriptions, optional transition guards through the GenESyS parser, and a deterministic single firing per activation.

17.134.2 Member Function Documentation

17.134.2.1 _activate()

```
NetworkActivationResult ColoredPetriNetNetwork::_activate (
            const NetworkActivationFrame & frame )  [override], [protected], [virtual]
```

Formalism-specific activation behavior.
The base implementation performs no state change and returns a result sized to `getNumOutputPorts()` with every output absent. Specializations override this to implement EFSM transition firing, one DTMC step, one CPN firing step, etc. Must not receive or depend on `Entity*` (architecture reference §8.4, §19).
Reimplemented from DefaultNetwork.

17.134.2.2 _check()

```
bool ColoredPetriNetNetwork::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from DefaultNetwork.

17.134.2.3 _initBetweenReplications()

```
void ColoredPetriNetNetwork::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from DefaultNetwork.

17.134.2.4 _loadInstance()

```
bool ColoredPetriNetNetwork::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from DefaultNetwork.

17.134.2.5 _saveInstance()

```
void ColoredPetriNetNetwork::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from DefaultNetwork.

17.134.2.6 show()

```
std::string ColoredPetriNetNetwork::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

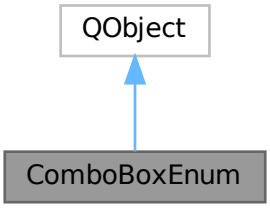
Human-readable summary used by traces and debug output.

Reimplemented from DefaultNetwork.
The documentation for this class was generated from the following files:

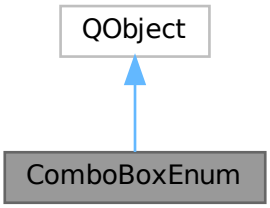
- plugins/data/ModalModel/ColoredPetriNetNetwork.h
- plugins/components/ModalModel/ColoredPetriNetNetwork.cpp

17.135 ComboBoxEnum Class Reference

Inheritance diagram for ComboBoxEnum:



Collaboration diagram for ComboBoxEnum:



Public Types

- using **AfterChange** = std::function< void()>

Public Member Functions

- **ComboBoxEnum** (PropertyEditorGenesys *editor, SimulationControl *property, AfterChange afterChange={})
- void **open_box** ()
- void **configure_options** (SimulationControl *property)

The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/ComboBoxEnum.h
- applications/gui/genesys/propertyeditor/ComboBoxEnum.cpp

17.136 GroProgramIr::Command Struct Reference

Public Types

- enum class **Kind** { **RawStatement** , **FunctionCall** , **Assignment** , **IfStatement** }

Public Member Functions

- bool **isFunctionCall** () const
- bool **isAssignment** () const
- bool **isIfStatement** () const

Public Attributes

- Kind **kind** = Kind::RawStatement
- std::string **sourceText** = ""
- std::string **functionName** = ""
- std::vector< std::string > **arguments**
- std::string **assignmentTarget** = ""
- std::string **expressionText** = ""
- bool **assignmentOnlyIfUnset** = false
- std::vector< Command > **thenCommands**
- std::vector< Command > **elseCommands**

The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/GroProgramIr.h

17.137 Util::CommandResult Class Reference

Public Attributes

- bool **success** = false
- std::string **commandStdOutput** = ""
- std::string **commandErrOutput** = ""
- std::string **workingDirectory** = "./"
- std::string **destinationPath** = "./"

The documentation for this class was generated from the following file:

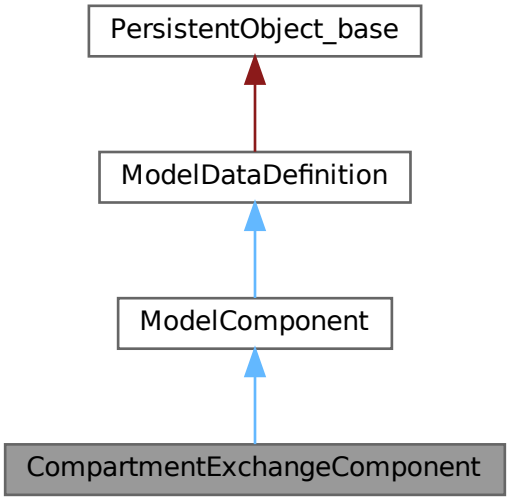
- kernel/util/Util.h

17.138 CompartmentExchangeComponent Class Reference

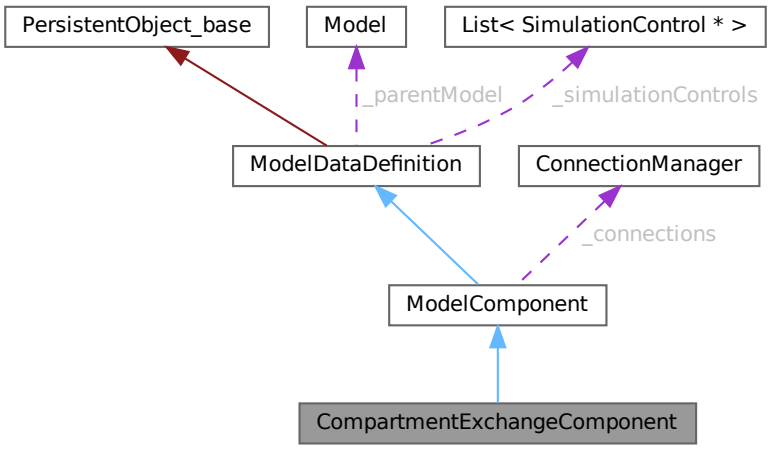
Moves metabolite proxies between whole-cell regions.

`#include <plugins/components/WholeCellModeling/CompartmentExchangeComponent.h>`

Inheritance diagram for CompartmentExchangeComponent:



Collaboration diagram for CompartmentExchangeComponent:



Classes

- struct ExchangeRule

Public Member Functions

- **CompartmentExchangeComponent** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)
- WholeCellState * **getWholeCellState** () const

- void **setSourceRegion** (std::string region)
- std::string **getSourceRegion** () const
- void **setSourceMetaboliteKey** (std::string key)
- std::string **getSourceMetaboliteKey** () const
- void **setTargetRegion** (std::string region)
- std::string **getTargetRegion** () const
- void **setTargetMetaboliteKey** (std::string key)
- std::string **getTargetMetaboliteKey** () const
- void **setExchangeFraction** (double fraction)
- double **getExchangeFraction** () const
- void **setBaselineAmount** (double amount)
- double **getBaselineAmount** () const
- void **setDriverPathwayKey** (std::string key)
- std::string **getDriverPathwayKey** () const
- void **setDriverScale** (double scale)
- double **getDriverScale** () const
- void **setMaxTransferAmount** (double amount)
- double **getMaxTransferAmount** () const
- void **setConserveMass** (bool conserve)
- bool **getConserveMass** () const
- void **clearExchangeRules** ()
- void **addExchangeRule** (const std::string &label, const std::string &sourceRegion, const std::string &sourceMetaboliteKey, const std::string &targetRegion, const std::string &targetMetaboliteKey, double exchangeFraction, double baselineAmount=0.0, const std::string &driverPathwayKey="", double driver↩ Scale=1.0, double maxTransferAmount=0.0, bool conserveMass=true)
- unsigned int **getExchangeRuleCount** () const
- const std::vector< ExchangeRule > & **getExchangeRules** () const
- double **getLastTransferAmount** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.138.1 Detailed Description

Moves metabolite proxies between whole-cell regions.
CompartmentExchangeComponent transfers a fraction of a source metabolite pool in WholeCellState into a target pool on each dispatched event. The component can represent didactic transport, exchange, or redistribution policies between global and compartment-scoped pools without embedding that logic directly into examples.
Source and target regions follow this convention:

- empty region name: global WholeCellState metabolite pool;
- non-empty region name: compartment-scoped pool.

17.138.2 Member Function Documentation

17.138.2.1 _check()

```
bool CompartmentExchangeComponent::_check (
            std::string & errorMessage ) [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.138.2.2 _createEditableDataDefinitions()

```
void CompartmentExchangeComponent::_createEditableDataDefinitions ( ) [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.138.2.3 _initBetweenReplications()

```
void CompartmentExchangeComponent::_initBetweenReplications ( ) [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.138.2.4 _loadInstance()

```
bool CompartmentExchangeComponent::_loadInstance (
            PersistenceRecord * fields ) [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.138.2.5 _onDispatchEvent()

```
void CompartmentExchangeComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.138.2.6 _saveInstance()

```
void CompartmentExchangeComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.138.2.7 show()

```
std::string CompartmentExchangeComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/WholeCellModeling/CompartmentExchangeComponent.h
- plugins/components/WholeCellModeling/CompartmentExchangeComponent.cpp

17.139 CppCompiler::CompilationResult Class Reference

Public Attributes

- bool **success** = false
- std::string **compilationStdOutput** = ""
- std::string **compilationErrOutput** = ""
- std::string **destinationPath** = ""
- unsigned int **lineNumber** = 0

The documentation for this class was generated from the following file:

- plugins/data/ExternalIntegration/CppCompiler.h

17.140 CompilerThread Class Reference

Public Member Functions

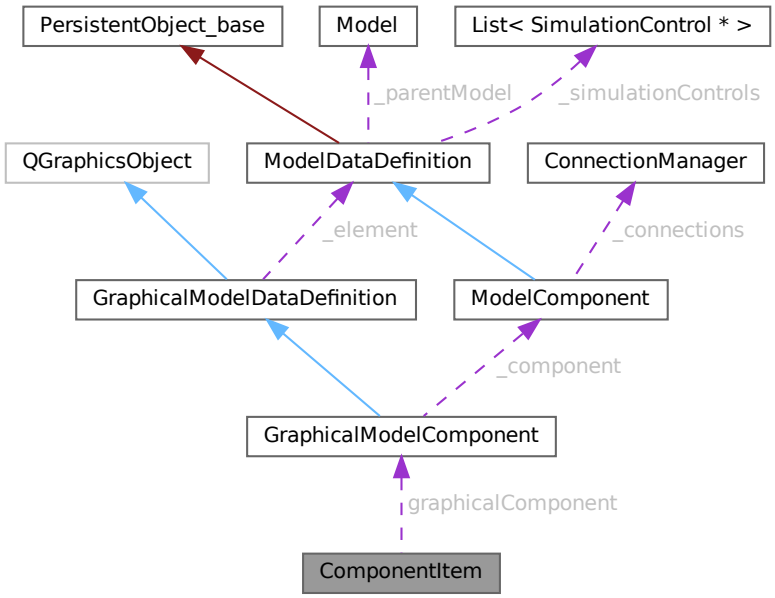
- void **operator()** ()

The documentation for this class was generated from the following file:

- plugins/data/ExternalIntegration/CppCompiler.h

17.141 ComponentItem Struct Reference

Collaboration diagram for ComponentItem:



Public Attributes

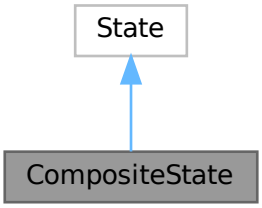
- GraphicalModelComponent * **graphicalComponent**
- QPointF **initialPosition**
- QList< GraphicalConnection * > **inputConnections**
- QList< GraphicalConnection * > **outputConnections**

The documentation for this struct was generated from the following file:

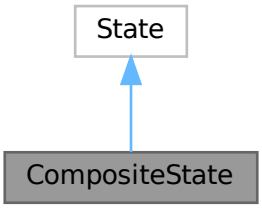
- applications/gui/genesys/actions/AddUndoCommand.h

17.142 CompositeState Class Reference

Inheritance diagram for CompositeState:



Collaboration diagram for CompositeState:



Public Types

- using **Field** = std::pair< std::string, std::unique_ptr< State > >

Public Member Functions

- **CompositeState** (const CompositeState &orig)
- CompositeState & **operator=** (const CompositeState &orig)
- std::unique_ptr< State > clone () const override
- bool equals (const State &other) const override
- std::string toString () const override
- std::string typeName () const override
- bool **addField** (std::string name, const State &state)
- const std::vector< Field > & **fields** () const

Public Member Functions inherited from State

- **State** (long value)
- **State** (const State &orig)
- virtual std::string **show** () const
- virtual long **getValue** () const
- virtual void **setValue** (long value)

- template<typename T >
  const T * **as** () const noexcept
- template<typename T >
  T * **as** () noexcept

Additional Inherited Members

Protected Attributes inherited from State

- long **_value**

17.142.1 Member Function Documentation

17.142.1.1 clone()

```
std::unique_ptr< State > CompositeState::clone ( ) const  [inline], [override], [virtual]
```

Reimplemented from State.

17.142.1.2 equals()

```
bool CompositeState::equals (
            const State & other ) const  [inline], [override], [virtual]
```

Reimplemented from State.

17.142.1.3 toString()

```
std::string CompositeState::toString ( ) const  [inline], [override], [virtual]
```

Reimplemented from State.

17.142.1.4 typeName()

```
std::string CompositeState::typeName ( ) const  [inline], [override], [virtual]
```

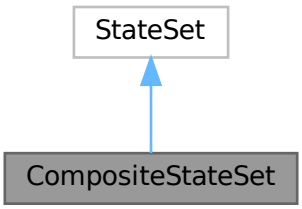
Reimplemented from State.

The documentation for this class was generated from the following file:

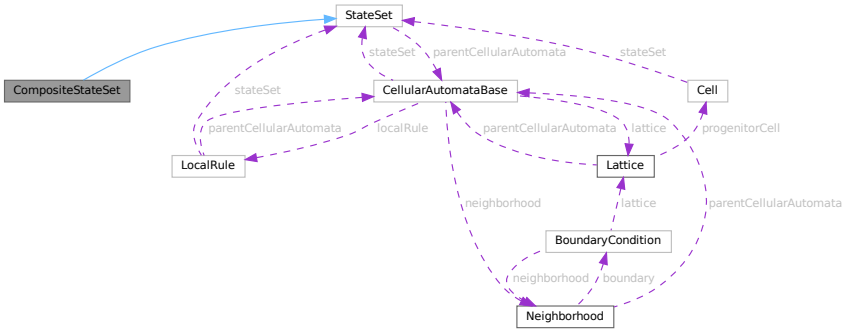
- plugins/components/ModalModel/CellularAutomata/State.h

17.143 CompositeStateSet Class Reference

Inheritance diagram for CompositeStateSet:



Collaboration diagram for CompositeStateSet:



Public Member Functions

- **CompositeStateSet** (CellularAutomataBase ∗parentCellularAutomata)
- bool contains (const State &state) const override
- std::unique_ptr< State > createDefaultState () const override
- bool isComposite () const override
- std::string name () const override

Public Member Functions inherited from StateSet

- **StateSet** (CellularAutomataBase ∗parentCellularAutomata)
- **StateSet** (const StateSet &orig)
- virtual std::unique_ptr< State > **parseState** (const std::string &text) const
- virtual bool **isFinite** () const
- virtual bool **isDiscrete** () const
- virtual bool **isContinuous** () const
- virtual std::vector< std::unique_ptr< State > > **enumerateStates** () const

Additional Inherited Members

Protected Attributes inherited from StateSet

- CellularAutomataBase ∗ **parentCellularAutomata**

17.143.1 Member Function Documentation

17.143.1.1 contains()

```
bool CompositeStateSet::contains (
            const State & state ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.143.1.2 createDefaultState()

```
std::unique_ptr< State > CompositeStateSet::createDefaultState ( ) const  [inline], [override],
[virtual]
```

Reimplemented from StateSet.

17.143.1.3 isComposite()

```
bool CompositeStateSet::isComposite ( ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.143.1.4 name()

```
std::string CompositeStateSet::name ( ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

The documentation for this class was generated from the following file:

- plugins/components/ModalModel/CellularAutomata/StateSet.h

17.144 HypothesisTester_if::ConfidenceInterval Class Reference

Confidence interval [inferior, superior] and its half-width e0.

```
#include <tools/Statistics/HypothesisTester_if.h>
```

Public Member Functions

- **ConfidenceInterval** (double inferiorLimit, double superiorLimit, double e0)
- double **inferiorLimit** ()
- double **superiorLimit** ()
- double **halfWidth** ()

17.144.1 Detailed Description

Confidence interval [inferior, superior] and its half-width e0.

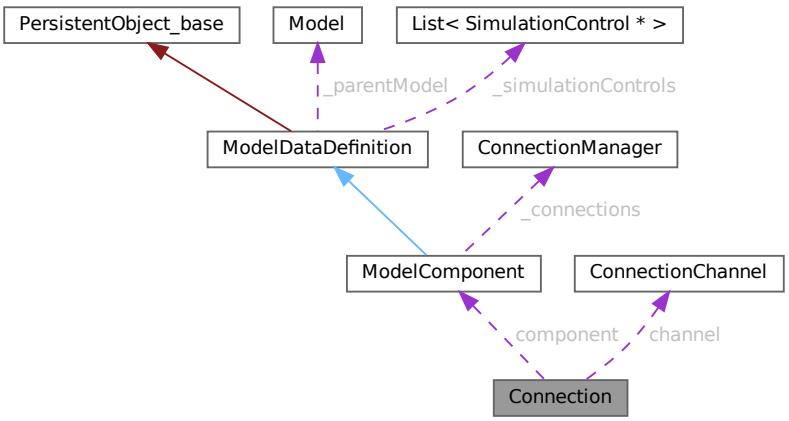
The documentation for this class was generated from the following file:

- tools/Statistics/HypothesisTester_if.h

17.145 Connection Struct Reference

```
#include <kernel/simulator/ConnectionManager.h>
```

Collaboration diagram for Connection:



Public Attributes

- ModelComponent * **component**
- ConnectionChannel **channel**

17.145.1 Detailed Description

Defines a connection between two or more ModelComponents. A (receiver) component's inputPort may receive connection from one or more (sender) components' outputPorts. A (sender) component's outputPort is connected to one and only one (receiver) component's inputPort.
The documentation for this struct was generated from the following file:

- kernel/simulator/ConnectionManager.h

17.146 ConnectionChannel Struct Reference

The ConnectionChannel class defines an input or output port that connects a ModelComponent to another.
`#include <kernel/simulator/ConnectionManager.h>`

Public Attributes

- unsigned int **portNumber** = 0
- std::string **portDescription** = ""

17.146.1 Detailed Description

The ConnectionChannel class defines an input or output port that connects a ModelComponent to another.
The documentation for this struct was generated from the following file:

- kernel/simulator/ConnectionManager.h

17.147 ConnectionManager Class Reference

Maintains outgoing connections from a component to downstream components.
`#include <kernel/simulator/ConnectionManager.h>`

Public Member Functions

- **ConnectionManager** ()

  Creates an empty connection manager with zero constraints.
- unsigned int size ()

  size
- Connection * getFrontConnection ()

  getFrontConnection
- Connection * getConnectionAtPort (unsigned int rank)

  getConnectionAtPort
- void insert (ModelComponent *component, unsigned int inputPortNumber=0)

  Inserts a new connection to a target component.
- void insert (Connection *connection)

  insert
- void insertAtPort (unsigned int port, Connection *connection)

  insertAtPort
- void remove (Connection *connection)

  remove
- void removeAtPort (unsigned int port)

  removeAtPort
- std::map< unsigned int, Connection * > * connections () const

  Returns the internal output-port to connection map.
- unsigned int getCurrentOutputConnectionsSize () const

  Returns the number of currently configured outgoing connections.

- void setMaxOutputConnections (unsigned int _maxOutputConnections)

  Sets the maximum number of outgoing connections allowed.
- unsigned int getMaxOutputConnections () const

  Returns the maximum number of outgoing connections allowed.
- void setMinOutputConnections (unsigned int _minOutputConnections)

  Sets the minimum number of outgoing connections allowed.
- unsigned int getMinOutputConnections () const

  Returns the minimum number of outgoing connections allowed.
- unsigned int getCurrentInputConnectionsSize () const

  Returns the number of currently configured incoming connections.
- void setMaxInputConnections (unsigned int _maxInputConnections)

  Sets the maximum number of incoming connections allowed.
- unsigned int getMaxInputConnections () const

  Returns the maximum number of incoming connections allowed.
- void setMinInputConnections (unsigned int _minInputConnections)

  Sets the minimum number of incoming connections allowed.
- unsigned int getMinInputConnections () const

  Returns the minimum number of incoming connections allowed.

17.147.1 Detailed Description

Maintains outgoing connections from a component to downstream components.
A connection maps an output rank (port) to a destination component/input port. The manager also stores min/max input/output constraints used during validation.

17.147.2 Member Function Documentation

17.147.2.1 connections()

```
std::map< unsigned int, Connection * > * ConnectionManager::connections ( ) const
```

Returns the internal output-port to connection map.

Returns

Map of output ports to outgoing connections.

17.147.2.2 getConnectionAtPort()

```
Connection * ConnectionManager::getConnectionAtPort (
            unsigned int rank )
```

getConnectionAtPort

Parameters

rank	

Returns

Connection bound to output rank `rank`.

17.147.2.3 getCurrentInputConnectionsSize()

```
unsigned int ConnectionManager::getCurrentInputConnectionsSize ( ) const
```

Returns the number of currently configured incoming connections.

Returns

Current input connection count.

17.147.2.4 getCurrentOutputConnectionsSize()

```
unsigned int ConnectionManager::getCurrentOutputConnectionsSize ( ) const
```

Returns the number of currently configured outgoing connections.

Returns

Current output connection count.

17.147.2.5 getFrontConnection()

```
Connection * ConnectionManager::getFrontConnection ( )
```

getFrontConnection

Returns

Connection at the first available output rank.

17.147.2.6 getMaxInputConnections()

```
unsigned int ConnectionManager::getMaxInputConnections ( ) const
```

Returns the maximum number of incoming connections allowed.

Returns

Maximum incoming connection count.

17.147.2.7 getMaxOutputConnections()

```
unsigned int ConnectionManager::getMaxOutputConnections ( ) const
```

Returns the maximum number of outgoing connections allowed.

Returns

Maximum outgoing connection count.

17.147.2.8 getMinInputConnections()

```
unsigned int ConnectionManager::getMinInputConnections ( ) const
```

Returns the minimum number of incoming connections allowed.

Returns

Minimum incoming connection count.

17.147.2.9 getMinOutputConnections()

```
unsigned int ConnectionManager::getMinOutputConnections ( ) const
```

Returns the minimum number of outgoing connections allowed.

Returns

Minimum outgoing connection count.

17.147.2.10 insert() [1/2]

```
void ConnectionManager::insert (
            Connection * connection )
```

insert

Parameters

connection	

Inserts a prebuilt connection object.

17.147.2.11 insert() [2/2]

```
void ConnectionManager::insert (
            ModelComponent * component,
            unsigned int inputPortNumber = 0 )
```

Inserts a new connection to a target component.
Creates and inserts a connection to `component` at input port `inputPortNumber`. When the port is not specified, the connection uses the default input port 0.

Parameters

component	Target component.
inputPortNumber	Target input port number.

17.147.2.12 insertAtPort()

```
void ConnectionManager::insertAtPort (
            unsigned int port,
            Connection * connection )
```

insertAtPort

Parameters

port	
connection	

Forces insertion/replacement at a specific output port.

17.147.2.13 remove()

```
void ConnectionManager::remove (
            Connection * connection )
```

remove

Parameters

connection	

Removes a connection by pointer match.

17.147.2.14 removeAtPort()

```
void ConnectionManager::removeAtPort (
            unsigned int port )
```

removeAtPort

Parameters

port	

Removes the connection associated with output port `port`.

17.147.2.15 setMaxInputConnections()

```
void ConnectionManager::setMaxInputConnections (
            unsigned int _maxInputConnections )
```

Sets the maximum number of incoming connections allowed.

Parameters

_maxInputConnections	Maximum incoming connection count.

17.147.2.16 setMaxOutputConnections()

```
void ConnectionManager::setMaxOutputConnections (
            unsigned int _maxOutputConnections )
```

Sets the maximum number of outgoing connections allowed.

Parameters

_maxOutputConnections	Maximum outgoing connection count.

17.147.2.17 setMinInputConnections()

```
void ConnectionManager::setMinInputConnections (
            unsigned int _minInputConnections )
```

Sets the minimum number of incoming connections allowed.

Parameters

_minInputConnections	Minimum incoming connection count.

17.147.2.18 setMinOutputConnections()

```
void ConnectionManager::setMinOutputConnections (
            unsigned int _minOutputConnections )
```

Sets the minimum number of outgoing connections allowed.

Parameters

_minOutputConnections	Minimum outgoing connection count.

17.147.2.19 size()

```
unsigned int ConnectionManager::size ( )
```

size

Returns

Number of configured output connections.

The documentation for this class was generated from the following files:

- kernel/simulator/ConnectionManager.h
- kernel/simulator/ConnectionManager.cpp

17.148 testing::internal::ConstCharPtr Struct Reference

Public Member Functions

- **ConstCharPtr** (const char ∗str)
- **operator bool** () const

Public Attributes

- const char ∗ **value**

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.149 Optimizer_if::ConstraintDefinition Struct Reference

Feasibility restriction represented as an expression.
`#include <tools/Optimization/Optimizer_if.h>`

Public Attributes

- std::string **name**
- std::string **expression**
- bool **enabled** = true

17.149.1 Detailed Description

Feasibility restriction represented as an expression.
A constraint is considered satisfied when expression != 0 and violated when expression == 0.
The documentation for this struct was generated from the following file:

- tools/Optimization/Optimizer_if.h

17.150 testing::internal::ConstRef< T > Struct Template Reference

Public Types

- typedef const T & **type**

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-port.h

17.151 testing::internal::ConstRef< T & > Struct Template Reference

Public Types

- typedef T & **type**

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-port.h

17.152 testing::internal::ContainerPrinter Struct Reference

Static Public Member Functions

- template<typename T , typename = typename std::enable_if< (sizeof (IsContainerTest<T>(0)) == sizeof (IsContainer)) && !Is↩ RecursiveContainer<T>::value>::type>
  static void **PrintValue** (const T &container, std::ostream *os)

The documentation for this struct was generated from the following file:

- gtest/gtest/gtest-printers.h

17.153 yy::genesyspp_parser::context Class Reference

Public Member Functions

- **context** (const genesyspp_parser &yyparser, const symbol_type &yyla)
- const symbol_type & **lookahead** () const YY_NOEXCEPT
- symbol_kind_type **token** () const YY_NOEXCEPT
- const location_type & **location** () const YY_NOEXCEPT
- int expected_tokens (symbol_kind_type yyarg[ ], int yyargn) const

17.153.1 Member Function Documentation

17.153.1.1 expected_tokens()

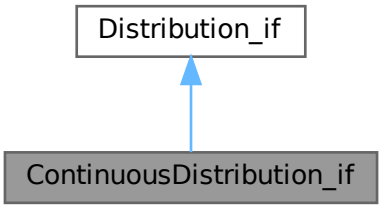
```
int yy::genesyspp_parser::context::expected_tokens (
            symbol_kind_type yyarg[],
            int yyargn ) const
```

Put in YYARG at most YYARGN of the expected tokens, and return the number of tokens stored in YYARG. If YYARG is null, return the number of expected tokens (guaranteed to be less than YYNTOKENS).
The documentation for this class was generated from the following files:

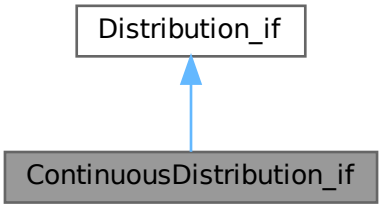
- parser/GenesysParser.h
- parser/GenesysParser.cpp

17.154 ContinuousDistribution_if Class Reference

Interface for continuous distributions.
`#include <tools/Statistics/ContinuousDistribution_if.h>`
Inheritance diagram for ContinuousDistribution_if:



Collaboration diagram for ContinuousDistribution_if:



Public Member Functions

- virtual double **pdf** (double x) const =0
- virtual double **cdf** (double x) const =0
- virtual double **quantile** (double probability) const =0

Public Member Functions inherited from Distribution_if

- virtual std::string **name** () const =0

17.154.1 Detailed Description

Interface for continuous distributions.
Mathematical meaning:

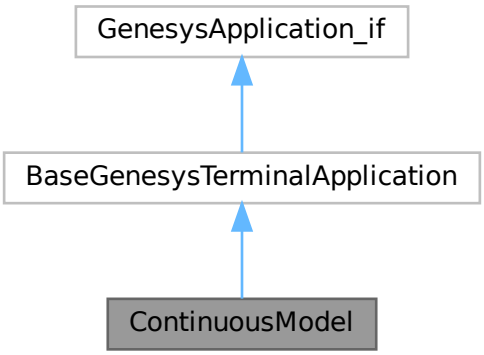
- pdf(x): probability density function value at x.
- cdf(x): cumulative probability P(X <= x).
- quantile(p): inverse CDF for p in [0, 1].

The documentation for this class was generated from the following file:

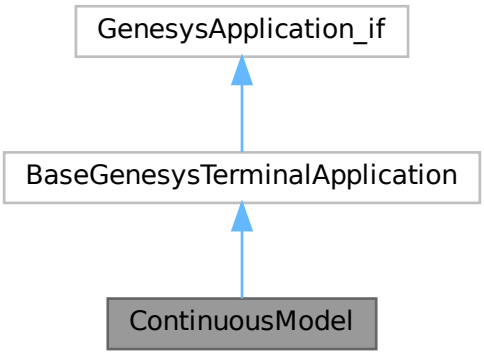
- tools/Statistics/ContinuousDistribution_if.h

17.155 ContinuousModel Class Reference

Inheritance diagram for ContinuousModel:



Collaboration diagram for ContinuousModel:



Public Member Functions

- virtual void Simulate (Simulator ∗genesys, Model ∗model, PluginManager ∗plugins) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual int main (int argc, char ∗∗argv) override
- void **setDefaultTraceHandlers** (TraceManager ∗tm)
- void setDefaultEventHandlers (OnEventManager ∗oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.155.1 Member Function Documentation

17.155.1.1 Simulate()

```
void ContinuousModel::Simulate (
            Simulator * genesys,
            Model * model,
            PluginManager * plugins )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

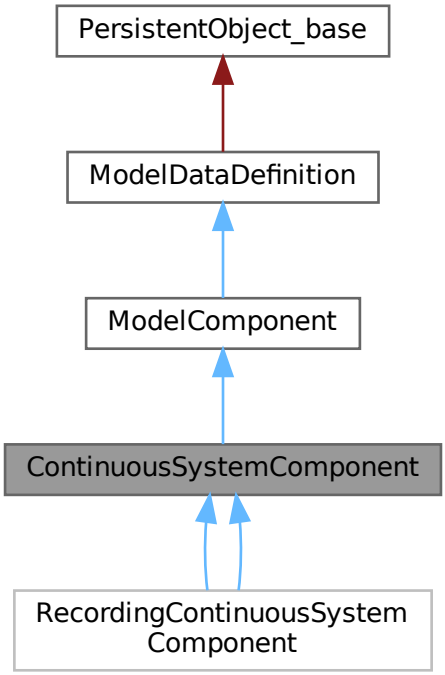
- applications/modelSpecific/teaching/ContinuousModel.h
- applications/modelSpecific/teaching/ContinuousModel.cpp

17.156 ContinuousSystemComponent Class Reference

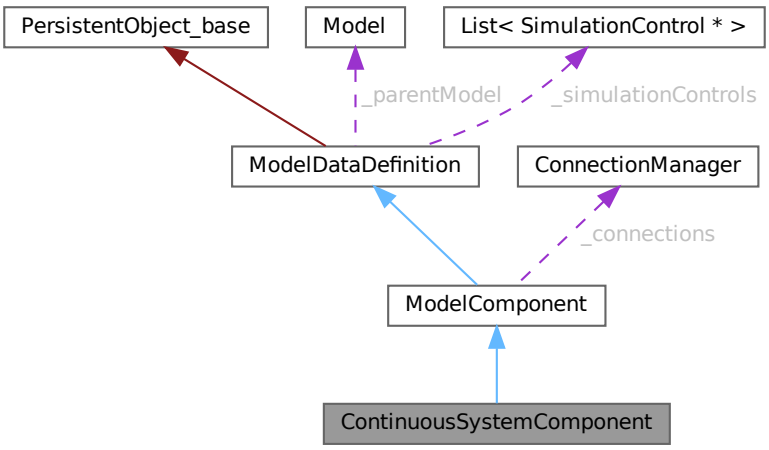
Component that advances a continuous ODE system when entities arrive.

```
#include <plugins/components/Continuous/ContinuousSystemComponent.h>
```

Inheritance diagram for ContinuousSystemComponent:



Collaboration diagram for ContinuousSystemComponent:



Public Member Functions

- **ContinuousSystemComponent** (Model *model, std::string name="")
- virtual std::string show () override

virtual public methods

- void **setOdeSolver** (ODESolver *solver)
- ODESolver * **getOdeSolver** () const
- void **setOdeSolverName** (std::string name)
- std::string **getOdeSolverName** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static void **SaveInstance** (PersistenceRecord *fields, ModelComponent *component)
- static bool **Check** (ModelComponent *component)
- static void **CreateInternalData** (ModelComponent *component)
- static void **DispatchEvent** (Event *event)

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.156.1 Detailed Description

Component that advances a continuous ODE system when entities arrive.
ContinuousSystemComponent holds a reference to an ODESolver data definition and triggers integration each time an entity enters through port 0. The solved state is written back to the ODESolver so downstream components can read the updated values. The entity is forwarded through output port 0 after integration completes.

17.156.2 Member Function Documentation

17.156.2.1 _check()

```
bool ContinuousSystemComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.156.2.2 _createEditableDataDefinitions()

```
void ContinuousSystemComponent::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.156.2.3 _loadInstance()

```
bool ContinuousSystemComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.156.2.4 _onDispatchEvent()

```
void ContinuousSystemComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.
Reimplemented in RecordingContinuousSystemComponent, and RecordingContinuousSystemComponent.

17.156.2.5 _saveInstance()

```
void ContinuousSystemComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.156.2.6 show()

```
std::string ContinuousSystemComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/Continuous/ContinuousSystemComponent.h
- plugins/components/Continuous/ContinuousSystemComponent.cpp

17.157 testing::internal::ConvertibleToIntegerPrinter Struct Reference

Static Public Member Functions

- static void **PrintValue** (internal::BiggestInt value, ::std::ostream *os)

The documentation for this struct was generated from the following file:

- gtest/gtest/gtest-printers.h

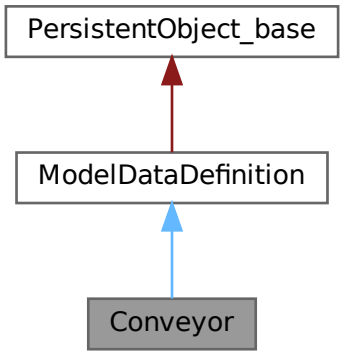
17.158 testing::internal::ConvertibleToStringViewPrinter Struct Reference

The documentation for this struct was generated from the following file:

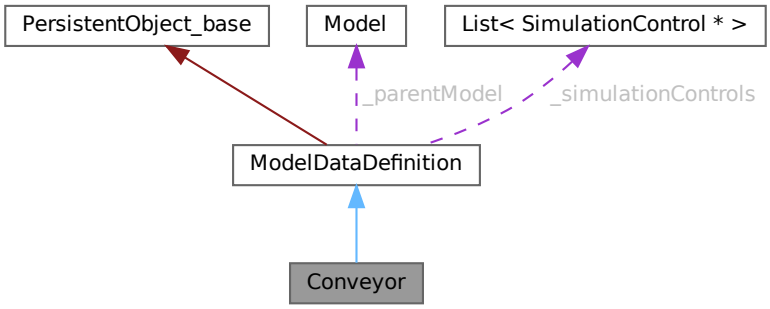
- gtest/gtest/gtest-printers.h

17.159 Conveyor Class Reference

Minimal conveyor data definition built on top of one Segment path.
`#include <plugins/data/MaterialHandling/Conveyor.h>`
Inheritance diagram for Conveyor:



Collaboration diagram for Conveyor:



Public Member Functions

- **Conveyor** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **setSegment** (Segment *segment)
- Segment * **getSegment** () const
- void **setVelocity** (double velocity)
- double **getVelocity** () const
- void **setCapacity** (unsigned int capacity)
- unsigned int **getCapacity** () const
- void **setInitiallyActive** (bool initiallyActive)
- bool **isInitiallyActive** () const
- void **setActive** (bool active)
- bool **isActive** () const
- unsigned int **getCurrentAllocation** () const
- bool **access** (unsigned int quantity=1)
- bool **exit** (unsigned int quantity=1)
- double **getDistanceBetween** (const std::string &fromStationName, const std::string &toStationName) const

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

Validates a data definition instance.

- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)

- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.159.1 Detailed Description

Minimal conveyor data definition built on top of one Segment path.
This implementation intentionally keeps only the semantics required by the current Advanced Transfer closeout: active/inactive state, a single ordered path (Segment), velocity and a simplified concurrent-allocation capacity. It does not model Arena's cell-by-cell occupancy contract.

17.159.2 Member Function Documentation

17.159.2.1 _check()

```
bool Conveyor::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.159.2.2 _createEditableDataDefinitions()

```
void Conveyor::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.159.2.3 _initBetweenReplications()

```
void Conveyor::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.159.2.4 _loadInstance()

```
bool Conveyor::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.159.2.5 _saveInstance()

```
void Conveyor::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.159.2.6 show()

```
std::string Conveyor::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

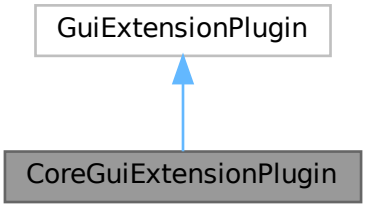
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

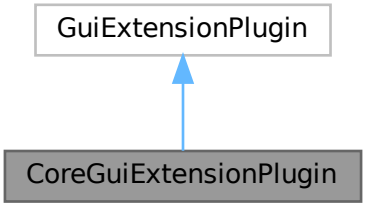
- plugins/data/MaterialHandling/Conveyor.h
- plugins/data/MaterialHandling/Conveyor.cpp

17.160 CoreGuiExtensionPlugin Class Reference

Inheritance diagram for CoreGuiExtensionPlugin:



Collaboration diagram for CoreGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.160.1 Member Function Documentation

17.160.1.1 displayName()

```
std::string CoreGuiExtensionPlugin::displayName ( ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.160.1.2 extensionId()

```
std::string CoreGuiExtensionPlugin::extensionId ( ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.160.1.3 registerContributions()

```
void CoreGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.160.1.4 requiredModelPlugins()

```
std::vector< std::string > CoreGuiExtensionPlugin::requiredModelPlugins ( ) const  [inline],
[override], [virtual]
```

Implements GuiExtensionPlugin.

17.160.1.5 version()

```
std::string CoreGuiExtensionPlugin::version ( ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

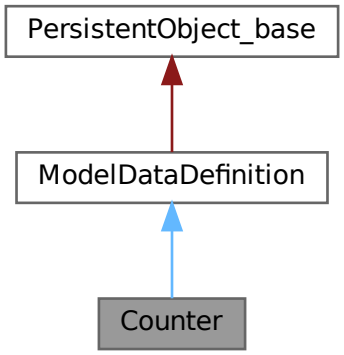
- applications/gui/genesys/extensions/CoreGuiExtensionPlugin.cpp

17.161 Counter Class Reference

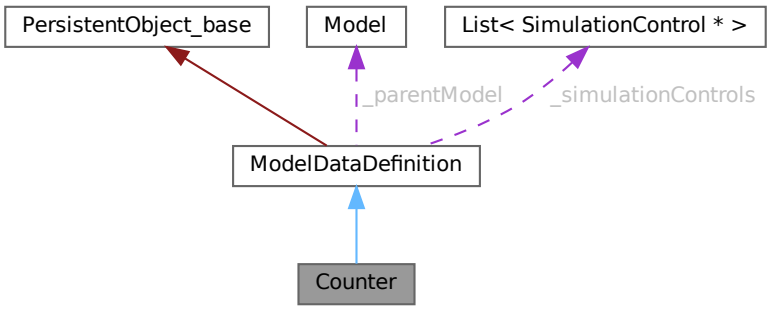
Model datum used to accumulate event counts (or weighted counts).

`#include <kernel/simulator/essentialPlugins/Counter.h>`

Inheritance diagram for Counter:



Collaboration diagram for Counter:



Public Member Functions

- **Counter** (Model *model, std::string name="", ModelDataDefinition *parent=nullptr)
  Creates a counter model datum with optional parent relationship.
- virtual std::string show () override
  Returns a textual representation of this data definition.
- void clear ()
  clear
- void incCountValue (double value=1.0)
  incCountValue
- double getCountValue () const
  getCountValue
- ModelDataDefinition * getParent () const
  getParent

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)

- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.161.1 Detailed Description

Model datum used to accumulate event counts (or weighted counts).
The counter value starts cleared for each replication (depending on model lifecycle hooks) and can be incremented by arbitrary amounts.

17.161.2 Member Function Documentation

17.161.2.1 _check()

```
bool Counter::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.161.2.2 _initBetweenReplications()

```
void Counter::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.161.2.3 _loadInstance()

```
bool Counter::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.161.2.4 _saveInstance()

```
void Counter::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.161.2.5 clear()

```
void Counter::clear ( )
```

clear
Resets the internal count value to zero.

17.161.2.6 getCountValue()

```
double Counter::getCountValue ( ) const
```

getCountValue

Returns

Current accumulated count value.

17.161.2.7 getParent()

```
ModelDataDefinition * Counter::getParent ( ) const
```

getParent

Returns

Parent model datum associated with this counter (if any).

17.161.2.8 incCountValue()

```
void Counter::incCountValue (
            double value = 1.0 )
```

incCountValue

Parameters

value	

Adds `value` to the internal count (default increment is 1).

17.161.2.9 show()

```
std::string Counter::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

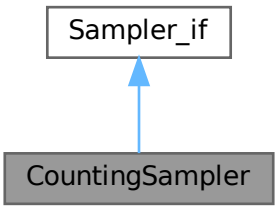
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

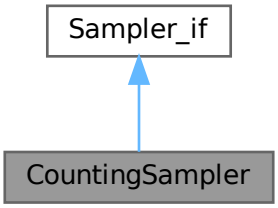
- kernel/simulator/essentialPlugins/Counter.h
- kernel/simulator/essentialPlugins/Counter.cpp

17.162 CountingSampler Class Reference

Inheritance diagram for CountingSampler:



Collaboration diagram for CountingSampler:



Public Member Functions

- **CountingSampler** (int *destroyedCounter, bool *destroyedFlag=nullptr)
- double random () override

  Generates a pseudo-random value uniformly distributed in the range [0,1).
- double sampleBeta (double, double, double infLimit, double) override

  Samples a Beta-distributed value scaled to an arbitrary output interval.
- double sampleBeta (double, double) override

  Samples a standard Beta-distributed value in [0,1].
- double sampleErlang (double, int, double offset=0.0) override

  Samples from an Erlang distribution using the provided mean and number of phases.
- double sampleExponential (double, double offset=0.0) override

  Samples from an Exponential distribution with the provided mean.
- double sampleGamma (double, double, double offset=0.0) override

  Samples from a Gamma distribution using the provided shape/scale parameters.
- double sampleGumbell (double mode, double) override

  Samples from a Gumbel distribution using mode and scale.
- double sampleLogNormal (double, double, double offset=0.0) override

  Samples from a Lognormal distribution parameterized by mean and standard deviation.
- double sampleNormal (double mean, double) override

  Samples from a Normal distribution.
- double sampleTriangular (double min, double, double) override

  Samples from a Triangular distribution.
- double sampleUniform (double min, double) override

  Samples from a continuous Uniform distribution in [min, max].
- double sampleWeibull (double, double) override

  Samples from a Weibull distribution.
- double sampleBinomial (int, double) override

  Samples from a Binomial distribution with `trials` and success probability `p`.
- double sampleBernoulli (double) override

  Samples from a Bernoulli distribution (0/1).
- double sampleDiscrete (double, double,...) override

  Samples from a discrete distribution using a variadic list of probability/value pairs.
- double sampleDiscrete (double *, double *, int) override

  Samples from a discrete distribution using probability/value arrays.
- double sampleGeometric (double) override

Samples from a Geometric distribution with success probability p.

- void setRNGparameters (RNG_Parameters *) override

  Updates the pseudo-random generator internal parameters.

- RNG_Parameters * getRNGparameters () const override

  Returns the current pseudo-random generator parameters.

17.162.1 Member Function Documentation

17.162.1.1 getRNGparameters()

```
RNG_Parameters * CountingSampler::getRNGparameters ( ) const  [inline], [override], [virtual]
```

Returns the current pseudo-random generator parameters.
Implements Sampler_if.

17.162.1.2 random()

```
double CountingSampler::random ( )  [inline], [override], [virtual]
```

Generates a pseudo-random value uniformly distributed in the range [0,1).

Returns

Base pseudo-random value used by the other distribution samplers.

Implements Sampler_if.

17.162.1.3 sampleBernoulli()

```
double CountingSampler::sampleBernoulli (
            double p )  [inline], [override], [virtual]
```

Samples from a Bernoulli distribution (0/1).
Implements Sampler_if.

17.162.1.4 sampleBeta() [1/2]

```
double CountingSampler::sampleBeta (
            double alpha,
            double beta )  [inline], [override], [virtual]
```

Samples a standard Beta-distributed value in [0,1].
Implements Sampler_if.

17.162.1.5 sampleBeta() [2/2]

```
double CountingSampler::sampleBeta (
            double alpha,
            double beta,
            double infLimit,
            double supLimit )  [inline], [override], [virtual]
```

Samples a Beta-distributed value scaled to an arbitrary output interval.

Parameters

alpha	Alpha shape parameter (>0).
beta	Beta shape parameter (>0).
infLimit	Lower bound of the output interval.
supLimit	Upper bound of the output interval.

Returns

Sampled value within [infLimit, supLimit].

Implements Sampler_if.

17.162.1.6 sampleBinomial()

```
double CountingSampler::sampleBinomial (
            int trials,
            double p )  [inline], [override], [virtual]
```

Samples from a Binomial distribution with `trials` and success probability `p`.
Implements Sampler_if.

17.162.1.7 sampleDiscrete() [1/2]

```
double CountingSampler::sampleDiscrete (
            double * prob,
            double * value,
            int size )  [inline], [override], [virtual]
```

Samples from a discrete distribution using probability/value arrays.
Implements Sampler_if.

17.162.1.8 sampleDiscrete() [2/2]

```
double CountingSampler::sampleDiscrete (
            double prob,
            double value,
             ...  )  [inline], [override], [virtual]
```

Samples from a discrete distribution using a variadic list of probability/value pairs.
Implements Sampler_if.

17.162.1.9 sampleErlang()

```
double CountingSampler::sampleErlang (
            double mean,
            int M,
            double offset = 0.0 )  [inline], [override], [virtual]
```

Samples from an Erlang distribution using the provided mean and number of phases.
Implements Sampler_if.

17.162.1.10 sampleExponential()

```
double CountingSampler::sampleExponential (
            double mean,
            double offset = 0.0 )  [inline], [override], [virtual]
```

Samples from an Exponential distribution with the provided mean.
Implements Sampler_if.

17.162.1.11 sampleGamma()

```
double CountingSampler::sampleGamma (
            double alpha,
            double beta,
            double offset = 0.0 )  [inline], [override], [virtual]
```

Samples from a Gamma distribution using the provided shape/scale parameters.
Implements Sampler_if.

17.162.1.12 sampleGeometric()

```
double CountingSampler::sampleGeometric (
            double p )  [inline], [override], [virtual]
```

Samples from a Geometric distribution with success probability p.
Implements Sampler_if.

17.162.1.13 sampleGumbell()

```
double CountingSampler::sampleGumbell (
            double mode,
            double scale )  [inline], [override], [virtual]
```

Samples from a Gumbel distribution using mode and scale.
Implements Sampler_if.

17.162.1.14 sampleLogNormal()

```
double CountingSampler::sampleLogNormal (
            double mean,
            double stddev,
            double offset = 0.0 )  [inline], [override], [virtual]
```

Samples from a Lognormal distribution parameterized by mean and standard deviation.
Implements Sampler_if.

17.162.1.15 sampleNormal()

```
double CountingSampler::sampleNormal (
            double mean,
            double stddev )  [inline], [override], [virtual]
```

Samples from a Normal distribution.
Implements Sampler_if.

17.162.1.16 sampleTriangular()

```
double CountingSampler::sampleTriangular (
            double min,
            double mode,
            double max )  [inline], [override], [virtual]
```

Samples from a Triangular distribution.
Implements Sampler_if.

17.162.1.17 sampleUniform()

```
double CountingSampler::sampleUniform (
            double min,
            double max )  [inline], [override], [virtual]
```

Samples from a continuous Uniform distribution in [min, max].
Implements Sampler_if.

17.162.1.18 sampleWeibull()

```
double CountingSampler::sampleWeibull (
            double alpha,
            double scale )  [inline], [override], [virtual]
```

Samples from a Weibull distribution.
Implements Sampler_if.

17.162.1.19 setRNGparameters()

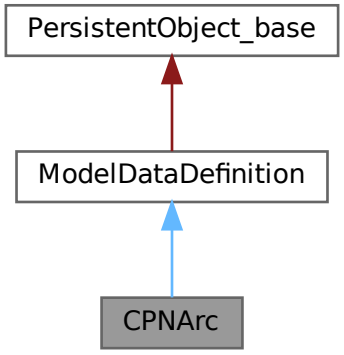
```
void CountingSampler::setRNGparameters (
            RNG_Parameters * param )  [inline], [override], [virtual]
```

Updates the pseudo-random generator internal parameters.
Implements Sampler_if.
The documentation for this class was generated from the following file:

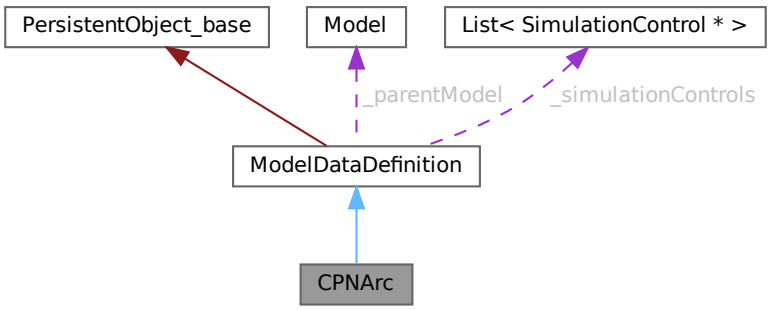
- tests/unit/test_parser_expressions.cpp

17.163 CPNArc Class Reference

Directed bipartite arc used by ColoredPetriNetNetwork.
`#include <plugins/data/ModalModel/CPNArc.h>`
Inheritance diagram for CPNArc:



Collaboration diagram for CPNArc:



Public Types

- enum class **Direction** : unsigned int { **PlaceToTransition** = 0 , **TransitionToPlace** = 1 }

Public Member Functions

- **CPNArc** (Model *model, std::string name="")
- **CPNArc** (Model *model, PetriPlace *place, CPNTransition *transition, Direction direction, std::string name="")
- void **setPlace** (PetriPlace *place)
- PetriPlace * **getPlace** () const
- void **setTransition** (CPNTransition *transition)
- CPNTransition * **getTransition** () const
- void **setDirection** (Direction direction)
- Direction **getDirection** () const
- bool **isInputArc** () const
- bool **isOutputArc** () const
- void **setInscription** (std::string color, unsigned int quantity)
- unsigned int **getInscription** (std::string color="default") const
- const std::map< std::string, unsigned int > & **getInscriptions** () const
- void **clearInscriptions** ()
- std::string **getPlaceName** () const
- std::string **getTransitionName** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.163.1 Detailed Description

Directed bipartite arc used by ColoredPetriNetNetwork.
The initial CPN subset uses fixed inscriptions represented as a multiset from symbolic color name to quantity. Variable bindings and typed token values are deliberately deferred.

17.163.2 Member Function Documentation

17.163.2.1 _check()

```
bool CPNArc::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.163.2.2 _loadInstance()

```
bool CPNArc::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.163.2.3 _saveInstance()

```
void CPNArc::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.163.2.4 show()

```
std::string CPNArc::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

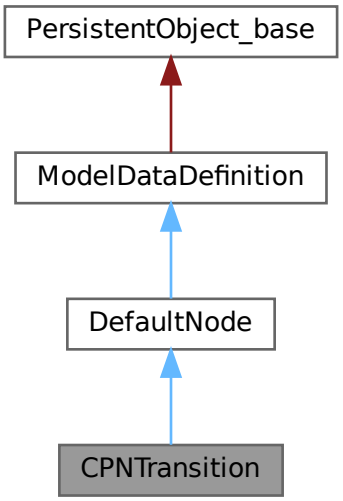
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

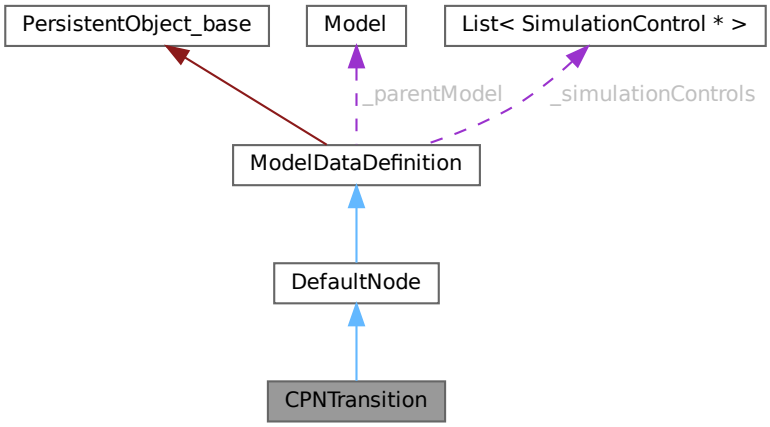
- plugins/data/ModalModel/CPNArc.h
- plugins/components/ModalModel/CPNArc.cpp

17.164 CPNTransition Class Reference

Transition node used by ColoredPetriNetNetwork.
`#include <plugins/data/ModalModel/CPNTransition.h>`
Inheritance diagram for CPNTransition:



Collaboration diagram for CPNTransition:



Public Member Functions

- **CPNTransition** (Model ∗model, std::string name="")
- void **setGuardExpression** (std::string guardExpression)
- std::string **getGuardExpression** () const
- void **setPriority** (unsigned int priority)
- unsigned int **getPriority** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from DefaultNode

- DefaultNode (Model ∗model, std::string componentTypename, std::string name)

  constructors
- **DefaultNode** (Model ∗model, std::string name="")
- void **addTransition** (DefaultNodeTransition ∗transition)

  new public user methods for this component
- void **removeTransition** (DefaultNodeTransition ∗transition)
- List< DefaultNodeTransition ∗ > ∗ **getTransitions** () const
- void **setInitialNode** (bool initialNode)
- bool **isInitialNode** () const
- void **setFinalNode** (bool finalNode)
- bool **isFinalNode** () const

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model ∗model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ∼**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from DefaultNode

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)

- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.164.1 Detailed Description

Transition node used by ColoredPetriNetNetwork.
This is a Petri-net transition node, not a source-to-destination edge. Incidence is represented by explicit CPNArc objects in the owning network.

17.164.2 Member Function Documentation

17.164.2.1 _check()

```
bool CPNTransition::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.164.2.2 _loadInstance()

```
bool CPNTransition::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from DefaultNode.

17.164.2.3 _saveInstance()

```
void CPNTransition::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from DefaultNode.

17.164.2.4 show()

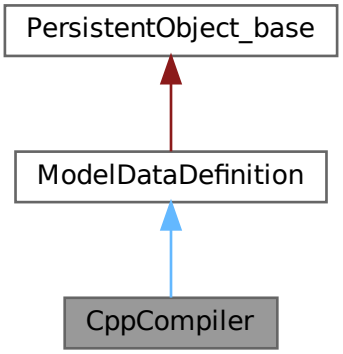
```
std::string CPNTransition::show ( )  [override], [virtual]
```

virtual public methods
Reimplemented from DefaultNode.
The documentation for this class was generated from the following files:

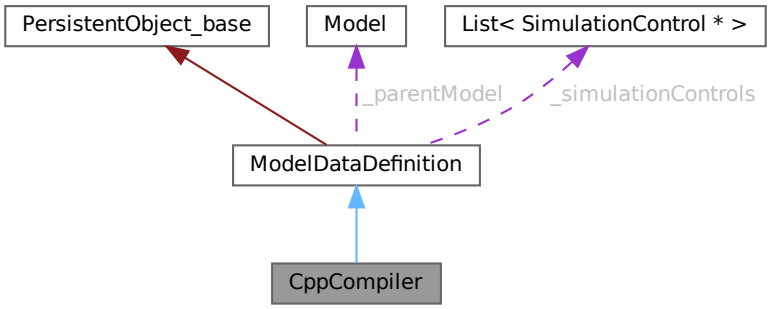
- plugins/data/ModalModel/CPNTransition.h
- plugins/components/ModalModel/CPNTransition.cpp

17.165 CppCompiler Class Reference

Inheritance diagram for CppCompiler:



Collaboration diagram for CppCompiler:



Classes

- class CompilationResult

Public Member Functions

- **CppCompiler** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **setSourceFilename** (std::string sourceFilename)
- std::string **getSourceFilename** () const
- bool **IsLibraryLoaded** () const
- void **setFlagsExecutable** (std::string _flagsExecutable)
- std::string **getFlagsExecutable** () const
- void **setLibraryLoaded** (bool libraryLoaded)
- void **setFlagsStaticLibrary** (std::string _flagsStaticLibrary)

- std::string **getFlagsStaticLibrary** () const
- void **setFlagsDynamicLibrary** (std::string _flagsDynamicLibrary)
- std::string **getFlagsDynamicLibrary** () const
- void **setFlagsGeneral** (std::string _flagsGeneral)
- std::string **getFlagsGeneral** () const
- void **setCompilerCommand** (std::string _compilerCommand)
- std::string **getCompilerCommand** () const
- void **setOutputDir** (std::string _outputDir)
- std::string **getOutputDir** () const
- void **setTempDir** (std::string _tempDir)
- std::string **getTempDir** () const
- void **setOutputFilename** (std::string _outputFilename)
- std::string **getOutputFilename** () const
- void **setObjectFiles** (std::string _objectFiles)
- std::string **getObjectFiles** () const
- CompilationResult **compileToExecutable** ()
- CompilationResult **compileToDynamicLibrary** ()
- CompilationResult **compileToStaticLibrary** ()
- bool **loadLibrary** (std::string &errorMessage)
- bool **unloadLibrary** ()
- void * **getDynamicLibraryHandler** () const

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

could be overriden by derived classes

- virtual void _initBetweenReplications () override
- CompilationResult **_invokeCompiler** (std::string command)
- std::string **_read** (std::string filename)

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)

- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.165.1 Member Function Documentation

17.165.1.1 _check()

```
bool CppCompiler::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Validate minimal compiler configuration before use.
Reimplemented from ModelDataDefinition.

17.165.1.2 _initBetweenReplications()

```
void CppCompiler::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.165.1.3 _loadInstance()

```
bool CppCompiler::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Load persisted compiler configuration.
The structure intentionally follows the same pattern used by other ModelDataDefinition implementations to simplify future maintenance.
Reimplemented from ModelDataDefinition.

17.165.1.4 _saveInstance()

```
void CppCompiler::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Persist compiler configuration.
Keep field names symmetric to _loadInstance for predictable serialization behavior and easier backward-compatible migrations.
Reimplemented from ModelDataDefinition.

17.165.1.5 show()

```
std::string CppCompiler::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

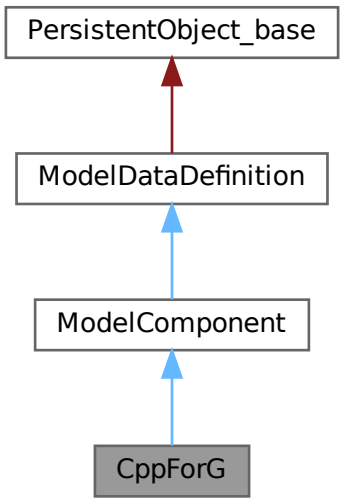
Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/ExternalIntegration/CppCompiler.h
- plugins/data/ExternalIntegration/CppCompiler.cpp

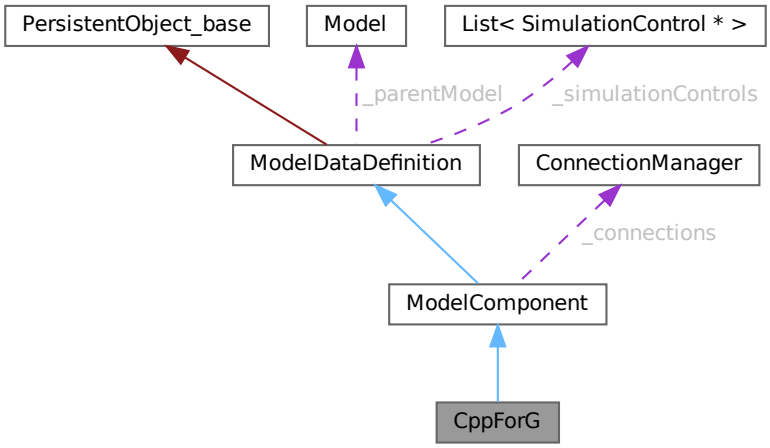
17.166 CppForG Class Reference

```
#include <plugins/components/ExternalIntegration/CppForG.h>
```

Inheritance diagram for CppForG:



Collaboration diagram for CppForG:



Public Member Functions

- **CppForG** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setCppCompiler** (CppCompiler *_cppCompiler)
- CppCompiler * **getCppCompiler** () const
- void **setInitBetweenReplicationCode** (std::string _initBetweenReplicationCode)
- std::string **getInitBetweenReplicationCode** () const
- void **setOnDispatchEventCode** (std::string _onDispatchEventCode)
- std::string **getOnDispatchEventCode** () const
- void **setIncludesCode** (std::string _includesCode)
- std::string **getIncludesCode** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createNonEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.166.1 Detailed Description

This component ...

17.166.2 Member Function Documentation

17.166.2.1 _check()

```
bool CppForG::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.166.2.2 _createNonEditableDataDefinitions()

```
void CppForG::_createNonEditableDataDefinitions ( )  [override], [protected], [virtual]
```

This method is necessary only for those components that instantiate internal elements that must exist before simulation starts and even before model checking. That's the case of components that have internal StatisticsCollectors, since others components may refer to them as expressions (as in "TVAG(ThisCSTAT)") and therefore the modeldatum must exist before checking such expression
Reimplemented from ModelDataDefinition.

17.166.2.3 _initBetweenReplications()

```
void CppForG::_initBetweenReplications ( )  [override], [protected], [virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
Reimplemented from ModelDataDefinition.

17.166.2.4 _loadInstance()

```
bool CppForG::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.166.2.5 _onDispatchEvent()

```
void CppForG::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.166.2.6 _saveInstance()

```
void CppForG::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.166.2.7 show()

```
std::string CppForG::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/ExternalIntegration/CppForG.h
- plugins/components/ExternalIntegration/CppForG.cpp

17.167 CppModelExporter Class Reference

Service responsible for C++ code representation of the current simulation model.
`#include <applications/gui/genesys/services/CppModelExporter.h>`

Public Member Functions

- **CppModelExporter** (Simulator *simulator, QPlainTextEdit *cppCodeEditor)

  Creates the exporter service for the C++ model-representation pane.
- std::string addCppCodeLine (const std::string &line, unsigned int indent=0) const

  Formats one C++ output line with the requested indentation level.
- void **actualizeModelCppCode** () const

  Rebuilds the full C++ model representation shown in the GUI editor pane.

17.167.1 Detailed Description

Service responsible for C++ code representation of the current simulation model.
The refactoring keeps C++ representation generation outside MainWindow by delegating this concern to a dedicated service. MainWindow wrappers call into this class when model or UI state changes require regeneration.
Responsibilities:

- format indented C++ output lines used by exporter routines;
- rebuild the full C++ representation shown in the GUI code pane.

Boundaries:

- it does not compile or execute generated code;
- it does not persist files directly;
- it does not manage lifecycle/simulation/scene controller flows.

17.167.2 Member Function Documentation

17.167.2.1 addCppCodeLine()

```
std::string CppModelExporter::addCppCodeLine (
            const std::string & line,
            unsigned int indent = 0 ) const
```

Formats one C++ output line with the requested indentation level.

Returns

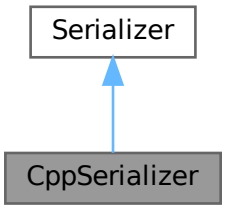
Formatted line that can be appended to the representation buffer.

The documentation for this class was generated from the following files:

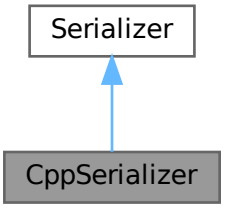
- applications/gui/genesys/services/CppModelExporter.h
- applications/gui/genesys/services/CppModelExporter.cpp

17.168 CppSerializer Class Reference

Inheritance diagram for CppSerializer:



Collaboration diagram for CppSerializer:



Public Member Functions

- **CppSerializer** (Model *model)
- PersistenceRecord * newPersistenceRecord () override

  Creates an empty persistence record.
- bool dump (std::ostream &output) override

  Writes formatted contents to a given output stream.
- bool load (std::istream &input) override

  Attempts to load contents from a given input stream.
- bool get (const std::string &name, PersistenceRecord *entry) override

  Tries to fetch an entry with the given label as a key.
- bool put (const std::string name, const std::string type, const Util::identification id, PersistenceRecord *fields) override

  Upserts an entry into this structure.
- int for_each (std::function< int(const std::string &) > delegate) override

  Iterates over this structure's entries, ordered by ID.

17.168.1 Member Function Documentation

17.168.1.1 dump()

```
bool CppSerializer::dump (
            std::ostream & output )  [override], [virtual]
```

Writes formatted contents to a given output stream.

Parameters

output	output stream to write to.

Returns

whether the current state of this structure's contents was deemed valid for serialization.

Implements Serializer.

17.168.1.2 for_each()

```
int CppSerializer::for_each (
            std::function< int(const std::string &) > delegate )  [override], [virtual]
```

Iterates over this structure's entries, ordered by ID.

Parameters

delegate	delegate which is applied to every entry-identifying key, until it returns a non-zero value.

Returns

either zero, or the first non-zero value returned by the delegate.

Implements Serializer.

17.168.1.3 get()

```
bool CppSerializer::get (
            const std::string & name,
            PersistenceRecord * entry )  [override], [virtual]
```

Tries to fetch an entry with the given label as a key.

Parameters

name	a label which uniquely identifies an entry.
entry	(out-param) a pointer to a caller-owned entry, which is filled on success.

Returns

whether the entry was found and the out-param was actually written to.

Implements Serializer.

17.168.1.4 load()

```
bool CppSerializer::load (
            std::istream & input )  [override], [virtual]
```

Attempts to load contents from a given input stream.

Returns

whether the input was well-formatted and loading was successful.

Implements Serializer.

17.168.1.5 newPersistenceRecord()

```
PersistenceRecord * CppSerializer::newPersistenceRecord ( )  [override], [virtual]
```

Creates an empty persistence record.
Although the returned record can be freed normally with `delete`, care must be taken to only use records which come from a implementation, when configuring that implementation's contents.
Implements Serializer.

17.168.1.6 put()

```
bool CppSerializer::put (
            const std::string name,
            const std::string type,
            const Util::identification id,
            PersistenceRecord * fields )  [override], [virtual]
```

Upserts an entry into this structure.

Parameters

name	a label which uniquely identifies the entry.

Parameters

type	a type name.
id	a component/datadef ID number.
fields	a persistence record for the entry, whose contents will be copied if valid.

Returns

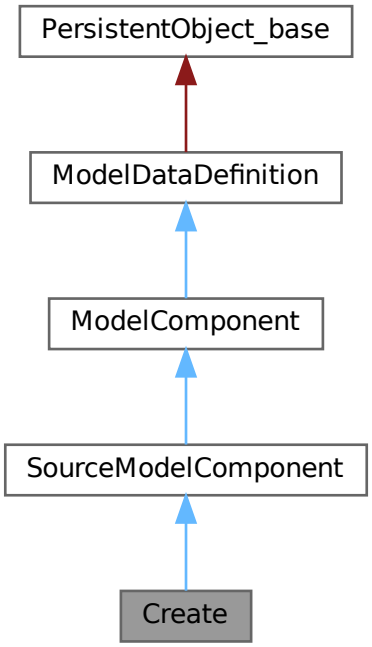
whether the entry was valid and was actually copied into this structure.

Implements Serializer.
The documentation for this class was generated from the following files:

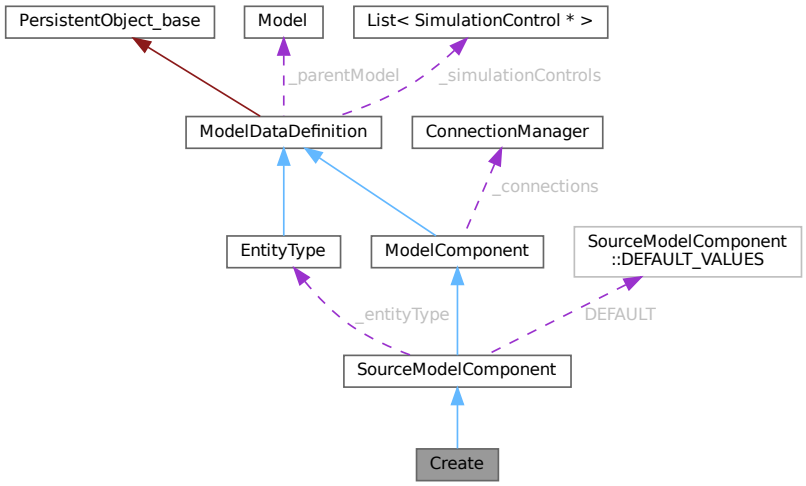
- kernel/simulator/persistence/CppSerializer.h
- kernel/simulator/persistence/CppSerializer.cpp

17.169 Create Class Reference

Source component that generates entities and starts them into the model, driven by a parsed inter-arrival expression, a Schedule, or a Formula.
`#include <plugins/components/Logic/Create.h>`
Inheritance diagram for Create:



Collaboration diagram for Create:



Public Member Functions

- **Create** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setTimeBetweenCreationsFormula** (Formula ∗_timeBetweenCreationsFormula)
- Formula ∗ **getTimeBetweenCreationsFormula** () const
- void **setTimeBetweenCreationsSchedule** (Schedule ∗_timeBetweenCreationsSchedule)
- Schedule ∗ **getTimeBetweenCreationsSchedule** () const
- double **testePropertyCreateDouble** () const
- void **setTestePropertyCreateDouble** (double newTestePropertyCreateDouble)

Public Member Functions inherited from SourceModelComponent

- **SourceModelComponent** (Model ∗model, std::string componentTypename, std::string name="")
- void setFirstCreation (double _firstCreation)

  setFirstCreation
- double getFirstCreation () const

  getFirstCreation
- void setEntityType (EntityType ∗_entityType)

  setEntityType
- void setEntityTypeName (std::string entityTypeName)

  setEntityTypeName
- EntityType ∗ getEntityType () const

  getEntityType
- void setTimeUnit (Util::TimeUnit _timeUnit)

  setTimeUnit
- Util::TimeUnit getTimeUnit () const

  getTimeUnit
- void setTimeBetweenCreationsExpression (std::string _timeBetweenCreations, Util::TimeUnit _time↩ Unit=Util::TimeUnit::unknown)

  setTimeBetweenCreationsExpression

- std::string getTimeBetweenCreationsExpression () const

  getTimeBetweenCreationsExpression
- void setMaxCreations (unsigned long _maxCreations)

  setMaxCreations
- void setMaxCreations (std::string _maxCreationsExpression)

  setMaxCreations
- std::string getMaxCreations () const

  getMaxCreations
- unsigned int getEntitiesCreated () const

  getEntitiesCreated
- void setEntitiesCreated (unsigned int _entitiesCreated)

  setEntitiesCreated
- void setEntitiesPerCreation (unsigned int _entitiesPerCreation)

  setEntitiesPerCreation
- unsigned int getEntitiesPerCreation () const

  getEntitiesPerCreation

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component

- static bool Check (ModelComponent *component)

  Validates a component instance.

- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.

- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _initBetweenReplications () override
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createInternalStatisticReporters () override
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from SourceModelComponent

- virtual void _createAttachedAttributes ()

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SourceModelComponent

- EntityType * **_entityType** = nullptr
- const struct SourceModelComponent::DEFAULT_VALUES **DEFAULT**
- double **_firstCreation** = DEFAULT.firstCreation
- unsigned int **_entitiesPerCreation** = DEFAULT.entitiesPerCreation
- std::string **_maxCreationsExpression** = DEFAULT.maxCreationsExpression
- std::string **_timeBetweenCreationsExpression** = DEFAULT.timeBetweenCreationsExpression
- Util::TimeUnit **_timeBetweenCreationsTimeUnit** = DEFAULT.timeBetweenCreationsTimeUnit
- unsigned int **_entitiesCreatedSoFar** = 0

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.169.1 Detailed Description

Source component that generates entities and starts them into the model, driven by a parsed inter-arrival expression, a Schedule, or a Formula.
Arena correspondence: the "Create module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 31-32). Entity Type, First Creation, Entities per Arrival and Max Arrivals are generalized on the base class SourceModelComponent; see that header for their accessors. Arena's Type field (Random/Schedule/↩ Constant/Expression) is not a stored enum here — Random/Constant/Expression collapse into one parsed time-between-creations expression (SourceModelComponent's default "`EXPO(1.0)`" already matches Arena's default Random behavior), while `_timeBetweenCreationsSchedule` keeps Schedule as a separate, mutually exclusive alternative (`_check()` enforces exactly one active source). `_timeBetweenCreationsFormula` adds a Formula-backed alternative beyond what Arena exposes.
See `docs/ai_assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.1 for the full audited comparison, including an open question about whether Arena's VA/NVA/Wait/Transfer/Other per-entity cost/time accounting has any GenESyS runtime equivalent.

17.169.2 Member Function Documentation

17.169.2.1 _check()

```
bool Create::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from SourceModelComponent.

17.169.2.2 _createEditableDataDefinitions()

```
void Create::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from SourceModelComponent.

17.169.2.3 _createInternalStatisticReporters()

```
void Create::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.169.2.4 _initBetweenReplications()

```
void Create::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from SourceModelComponent.

17.169.2.5 _loadInstance()

```
bool Create::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SourceModelComponent.

17.169.2.6 _onDispatchEvent()

```
void Create::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.169.2.7 _saveInstance()

```
void Create::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SourceModelComponent.

17.169.2.8 show()

```
std::string Create::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from SourceModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/Logic/Create.h
- plugins/components/Logic/Create.cpp

17.170 SimulatorSessionService::CreateSessionResult Struct Reference

Contains session creation identifiers returned to API clients.
`#include <applications/worker/service/SimulatorSessionService.h>`

Public Attributes

- std::string **sessionId**
- std::string **accessToken**

17.170.1 Detailed Description

Contains session creation identifiers returned to API clients.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.171 QtBoolPropertyManagerPrivate::Data Struct Reference

Public Attributes

- bool **val**
- bool **initVal**
- bool **isInitialed**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.172 QtDatePropertyManagerPrivate::Data Struct Reference

Public Member Functions

- QDate **minimumValue** () const
- QDate **maximumValue** () const
- void **setMinimumValue** (QDate newMinVal)
- void **setMaximumValue** (QDate newMaxVal)

Public Attributes

- QDate **val** { QDate::currentDate() }
- QDate **minVal** { QDate(1752, 9, 14) }
- QDate **maxVal** { QDate(9999, 12, 31) }
- QDate **initVal** { QDate::currentDate() }
- bool **isInitialed**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.173 QtDateTimePropertyManagerPrivate::Data Struct Reference

Public Attributes

- QDateTime **val** { QDateTime() }
- QDateTime **initVal** { QDateTime() }
- bool **isInitialed** { false }

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.174 QtDoublePropertyManagerPrivate::Data Struct Reference

Public Member Functions

- double **minimumValue** () const
- double **maximumValue** () const
- void **setMinimumValue** (double newMinVal)
- void **setMaximumValue** (double newMaxVal)

Public Attributes

- double **val** { 0 }
- double **initVal** { 0 }
- double **minVal** { -DBL_MAX }
- double **maxVal** { DBL_MAX }
- double **singleStep** { 1 }
- int **decimals** { 2 }
- bool **isInitialed** { false }

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.175 QtEnumPropertyManagerPrivate::Data Struct Reference

Public Attributes

- int **val** { -1 }
- QStringList **enumNames**
- QMap< int, QIcon > **enumIcons**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.176 QtFlagPropertyManagerPrivate::Data Struct Reference

Public Attributes

- int **val** { -1 }
- QStringList **flagNames**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.177 QtIntPropertyManagerPrivate::Data Struct Reference

Public Member Functions

- int **minimumValue** () const
- int **maximumValue** () const
- void **setMinimumValue** (int newMinVal)
- void **setMaximumValue** (int newMaxVal)

Public Attributes

- int **val** { 0 }
- int **initVal** { 0 }
- int **minVal** { -INT_MAX }
- int **maxVal** { INT_MAX }
- int **singleStep** { 1 }
- bool **isInitialed** { false }

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.178 QtPointFPropertyManagerPrivate::Data Struct Reference

Public Attributes

- QPointF **val**
- int **decimals** { 2 }

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.179 QtRectFPropertyManagerPrivate::Data Struct Reference

Public Attributes

- QRectF **val** { 0, 0, 0, 0 }
- QRectF **constraint**
- int **decimals** { 2 }

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.180 QtRectPropertyManagerPrivate::Data Struct Reference

Public Attributes

- QRect **val** { 0, 0, 0, 0 }
- QRect **constraint**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.181 QtSizeFPropertyManagerPrivate::Data Struct Reference

Public Member Functions

- QSizeF **minimumValue** () const
- QSizeF **maximumValue** () const
- void **setMinimumValue** (const QSizeF &newMinVal)
- void **setMaximumValue** (const QSizeF &newMaxVal)

Public Attributes

- QSizeF **val** { 0, 0 }
- QSizeF **minVal** { 0, 0 }
- QSizeF **maxVal** { std::numeric_limits<qreal>::max(), std::numeric_limits<qreal>::max() }
- int **decimals** { 2 }

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.182 QtSizePropertyManagerPrivate::Data Struct Reference

Public Member Functions

- QSize **minimumValue** () const
- QSize **maximumValue** () const
- void **setMinimumValue** (const QSize &newMinVal)
- void **setMaximumValue** (const QSize &newMaxVal)

Public Attributes

- QSize **val** { 0, 0 }
- QSize **minVal** { 0, 0 }
- QSize **maxVal** { INT_MAX, INT_MAX }

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.183 QtStringPropertyManagerPrivate::Data Struct Reference

Public Attributes

- QString **val**
- QString **initVal**
- bool **isInitialed**
- QRegularExpression **regExp**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.184 QtTimePropertyManagerPrivate::Data Struct Reference

Public Attributes

- QTime **val** { QTime() }
- QTime **initVal** { QTime() }
- bool **isInitialed** { false }

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.185 DataAnalyser_if Class Reference

High-level façade for applied statistical analysis over datasets.
`#include <tools/Statistics/DataAnalyser_if.h>`

Public Member Functions

- virtual bool **loadDataSet** (std::string datafilename)=0
- virtual bool **saveDataSet** (std::string datasetname)=0
- virtual void **newDataSet** (std::string datasetname, std::string datafilename)=0
- virtual Fitter_if * **fitter** ()=0
- virtual Sampler_if * **sampler** ()=0
- virtual ExperimentManager_if * **experimenter** ()=0
- virtual HypothesisTester_if * **tester** ()=0

17.185.1 Detailed Description

High-level façade for applied statistical analysis over datasets.
Purpose:

- Coordinate dataset lifecycle and analysis services in a single entry point.

Architectural role:

- Facade that orchestrates fitting, sampling, experiment execution and hypothesis testing.

Ownership/lifetime:

- The returned pointers from fitter(), sampler(), experimenter() and tester() are non-owning views. Concrete implementations must document whether those services are internally owned singletons, members, or externally injected objects.

Implementation status:

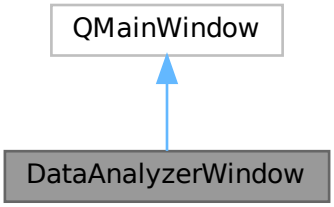
- Interface contract is stable and legacy-compatible.
- A consolidated default concrete implementation is still pending.

The documentation for this class was generated from the following file:

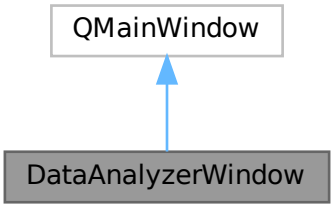
- tools/Statistics/DataAnalyser_if.h

17.186 DataAnalyzerWindow Class Reference

Inheritance diagram for DataAnalyzerWindow:



Collaboration diagram for DataAnalyzerWindow:



Classes

- class DoePreviewPlot
- class HistogramPreview

Public Member Functions

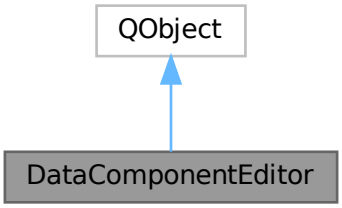
- **DataAnalyzerWindow** (Simulator *simulator, QString &lastDataAnalyzerPath, QWidget *parent=nullptr)

The documentation for this class was generated from the following files:

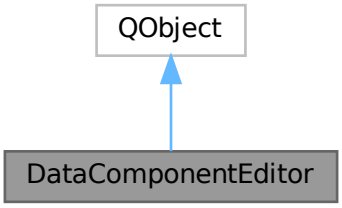
- applications/gui/genesys/tools/dataanalyzer/DataAnalyzerWindow.h
- applications/gui/genesys/tools/dataanalyzer/DataAnalyzerWindow.cpp

17.187 DataComponentEditor Class Reference

Inheritance diagram for DataComponentEditor:



Collaboration diagram for DataComponentEditor:



Public Types

- using **AfterChange** = std::function< void()>

Public Member Functions

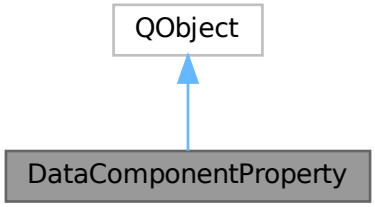
- **DataComponentEditor** (PropertyEditorGenesys *editor, SimulationControl *property, AfterChange after↩ Change={})
- **DataComponentEditor** (PropertyEditorGenesys *editor, List< SimulationControl * > *properties, After↩ Change afterChange={})
- void **open_window** (SimulationControl *property)
- void **open_window** (List< SimulationControl * > *properties)
- void **configure_properties** (SimulationControl *property)
- void **configure_properties** (List< SimulationControl * > *properties)

The documentation for this class was generated from the following files:

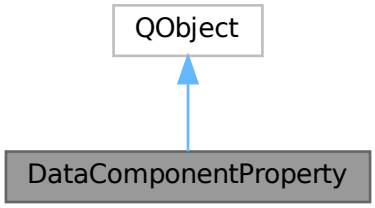
- applications/gui/genesys/propertyeditor/DataComponentEditor.h
- applications/gui/genesys/propertyeditor/DataComponentEditor.cpp

17.188 DataComponentProperty Class Reference

Inheritance diagram for DataComponentProperty:



Collaboration diagram for DataComponentProperty:



Public Types

- using **AfterChange** = std::function< void()>

Public Member Functions

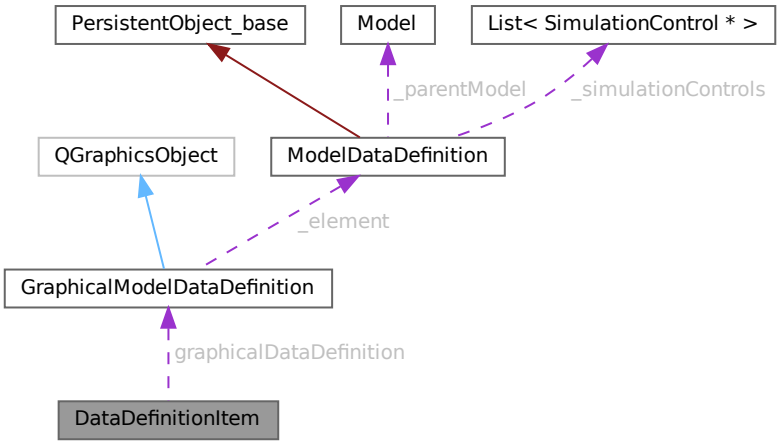
- **DataComponentProperty** (PropertyEditorGenesys *editor, SimulationControl *property, bool necessary↩ Config, AfterChange afterChange={})
- void **open_window** ()
- void **config_values** ()
- bool **isInList** (const std::string &value) const

The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/DataComponentProperty.h
- applications/gui/genesys/propertyeditor/DataComponentProperty.cpp

17.189 DataDefinitionItem Struct Reference

Collaboration diagram for DataDefinitionItem:



Public Attributes

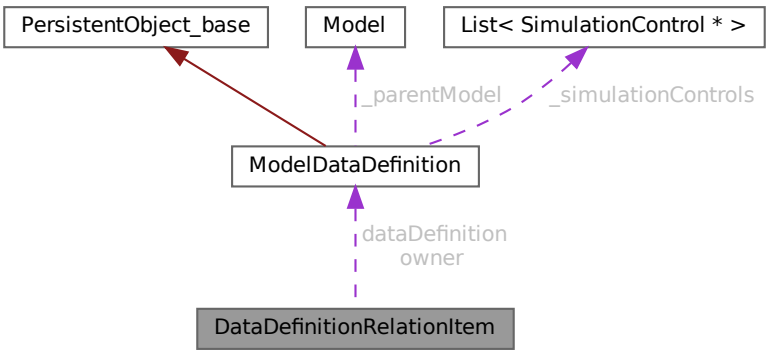
- GraphicalModelDataDefinition * **graphicalDataDefinition**
- QPointF **initialPosition**
- bool **initiallySelected**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/actions/DeleteUndoCommand.h

17.190 DataDefinitionRelationItem Struct Reference

Collaboration diagram for DataDefinitionRelationItem:



Public Attributes

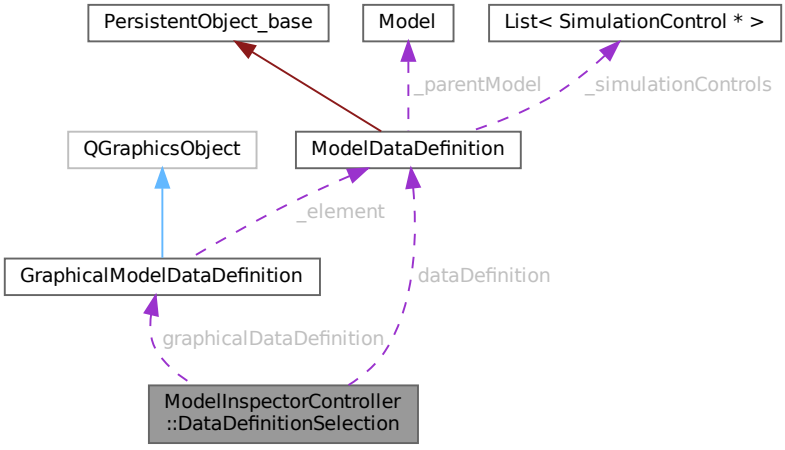
- ModelDataDefinition * **owner**
- ModelDataDefinition * **dataDefinition**
- std::string **key**
- bool **internal**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/actions/DeleteUndoCommand.h

17.191 ModelInspectorController::DataDefinitionSelection Struct Reference

Collaboration diagram for ModelInspectorController::DataDefinitionSelection:



Public Attributes

- bool **handled** = false
- ModelDataDefinition * **dataDefinition** = nullptr
- GraphicalModelDataDefinition * **graphicalDataDefinition** = nullptr

The documentation for this struct was generated from the following file:

- applications/gui/genesys/controllers/ModelInspectorController.h

17.192 AnimationPlot::DataSeries Struct Reference

Public Attributes

- QString **label**
- QString **expression**
- QColor **color** = QColor(31, 119, 180)
- qreal **lineWidth** = 2.0
- QVector< double > **values**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/animations/AnimationPlaceholder.h

17.193 DataSet_if Class Reference

Abstract view of a numeric dataset.
`#include <tools/Statistics/DataSet_if.h>`

Public Member Functions

- virtual unsigned int **size** () const =0
- virtual double **value** (unsigned int index) const =0
- virtual bool **isLoaded** () const =0
- virtual std::string **sourceDescription** () const =0

17.193.1 Detailed Description

Abstract view of a numeric dataset.
Purpose:

- Represent sample data loaded in memory or abstracted from persistent source.

Ownership/lifetime:

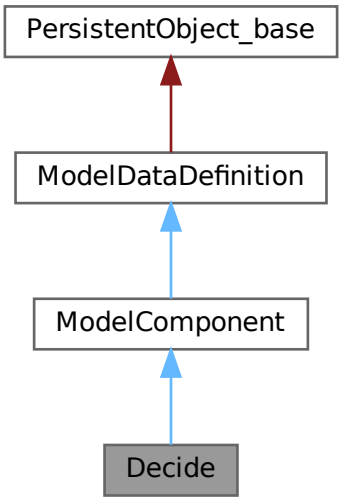
- Implementations own storage/source handles; clients use this interface as a non-owning access contract.

The documentation for this class was generated from the following file:

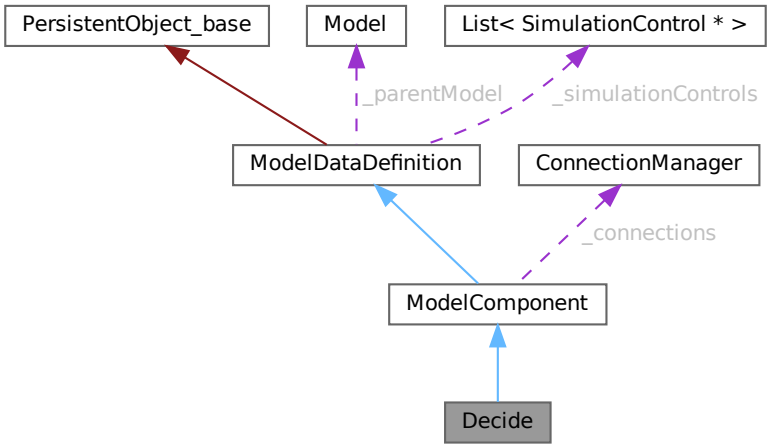
- tools/Statistics/DataSet_if.h

17.194 Decide Class Reference

Routes an entity to the first output port whose parser expression evaluates truthy, or to a trailing "else" port when none do.
`#include <plugins/components/Decisions/Decide.h>`
Inheritance diagram for Decide:



Collaboration diagram for Decide:



Public Member Functions

- **Decide** (Model *model, std::string name="")
- List< std::string > * **getConditions** () const
- void **addConditions** (std::string newCondition)
- void **removeConditions** (std::string condition)
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createInternalStatisticReporters () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.194.1 Detailed Description

Routes an entity to the first output port whose parser expression evaluates truthy, or to a trailing "else" port when none do.
Arena correspondence: the "Decide module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 36-37), which offers a structured 2-way/N-way, by-Condition/by-Chance dialog (Named/Is/Value with Row/Column for array conditions, or a Percentages field for Chance branching).
GenESyS collapses all of that into one ordered list of boolean parser expressions (`_conditions`), one per output port — a deliberate generalization, not a missing capability. There is no dedicated percentage-based Chance input; expressing chance-based branching (e.g. "60% true") is left to the modeler as a boolean expression over a random draw. Each port has its own entity-count Counter when reporting is enabled.

17.194.2 Member Function Documentation

17.194.2.1 _check()

```
bool Decide::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.194.2.2 _createInternalStatisticReporters()

```
void Decide::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.194.2.3 _loadInstance()

```
bool Decide::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.194.2.4 _onDispatchEvent()

```
void Decide::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.194.2.5 _saveInstance()

```
void Decide::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.194.2.6 show()

```
std::string Decide::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/Decisions/Decide.h
- plugins/components/Decisions/Decide.cpp

17.195 Delay::DEFAULT_VALUES Struct Reference

Public Attributes

- const std::string **delayExpression** = "1.0"
- const Util::TimeUnit **delayTimeUnit** = Util::TimeUnit::second
- const Util::AllocationType **allocation** = Util::AllocationType::Wait

The documentation for this struct was generated from the following file:

- plugins/components/DiscreteProcessing/Delay.h

17.196 ModalModelDefault::DEFAULT_VALUES Struct Reference

Public Attributes

- const std::string **timeDelayExpressionPerDispatch** = "0"
- const Util::TimeUnit **timeDelayPerDispatchTimeUnit** = Util::TimeUnit::second
- const unsigned int **maxTransitionsPerDispatch** = 1

The documentation for this struct was generated from the following file:

- plugins/components/ModalModel/ModalModelDefault.h

17.197 Seize::DEFAULT_VALUES Struct Reference

Public Attributes

- const Util::AllocationType **allocationType** = Util::AllocationType::Others
- const unsigned short **priority** = 0
- const std::string **priorityExpression** = ""
- const unsigned int **seizeRequestSize** = 1

The documentation for this struct was generated from the following file:

- plugins/components/DiscreteProcessing/Seize.h

17.198 SourceModelComponent::DEFAULT_VALUES Struct Reference

Public Attributes

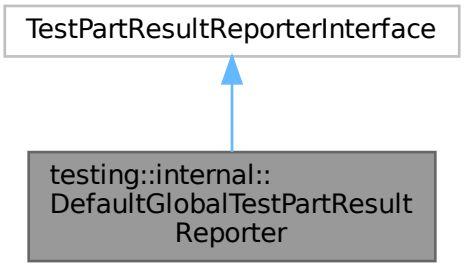
- const double **firstCreation** = 0.0
- const unsigned int **entitiesPerCreation** = 1
- const std::string **maxCreationsExpression** = std::to_string(std::numeric_limits<unsigned int>::max())
- const std::string **timeBetweenCreationsExpression** = "EXPO(1.0)"
- const Util::TimeUnit **timeBetweenCreationsTimeUnit** = Util::TimeUnit::second
- const std::string **entityTypename** = "entitytype"

The documentation for this struct was generated from the following file:

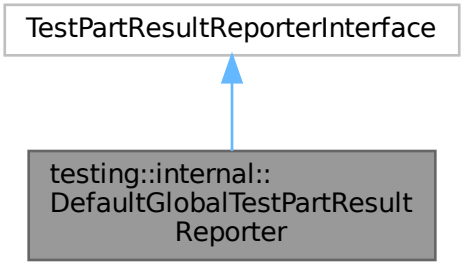
- kernel/simulator/SourceModelComponent.h

17.199 testing::internal::DefaultGlobalTestPartResultReporter Class Reference

Inheritance diagram for testing::internal::DefaultGlobalTestPartResultReporter:



Collaboration diagram for testing::internal::DefaultGlobalTestPartResultReporter:



Public Member Functions

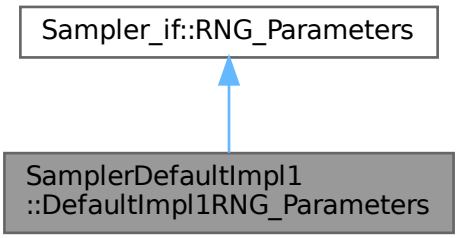
- **DefaultGlobalTestPartResultReporter** (UnitTestImpl ∗unit_test)
- void **ReportTestPartResult** (const TestPartResult &result) override

The documentation for this class was generated from the following files:

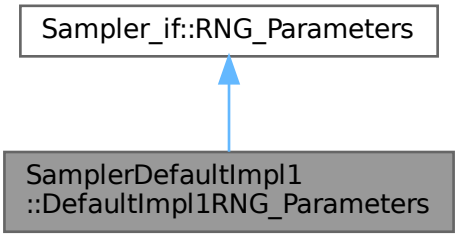
- gtest/src/gtest-internal-inl.h
- gtest/src/gtest.cc

17.200 SamplerDefaultImpl1::DefaultImpl1RNG_Parameters Struct Reference

Parameter bundle for the default RNG algorithm.
`#include <kernel/statistics/SamplerDefaultImpl1.h>`
Inheritance diagram for SamplerDefaultImpl1::DefaultImpl1RNG_Parameters:



Collaboration diagram for SamplerDefaultImpl1::DefaultImpl1RNG_Parameters:



Public Attributes

- uint32_t **seed** = 16021974

  Initial seed/state.
- uint32_t **a** = 279470273u

  Multiplicative factor in the linear congruential generator.
- uint32_t **m** = 0xfffffffb

  Modulus used by the generator.
- uint32_t **c** = 0

  Increment term used by the generator.

17.200.1 Detailed Description

Parameter bundle for the default RNG algorithm.
The documentation for this struct was generated from the following file:

- kernel/statistics/SamplerDefaultImpl1.h

17.201 testing::internal::DefaultNameGenerator Struct Reference

Static Public Member Functions

- template<typename T >
  static std::string **GetName** (int i)

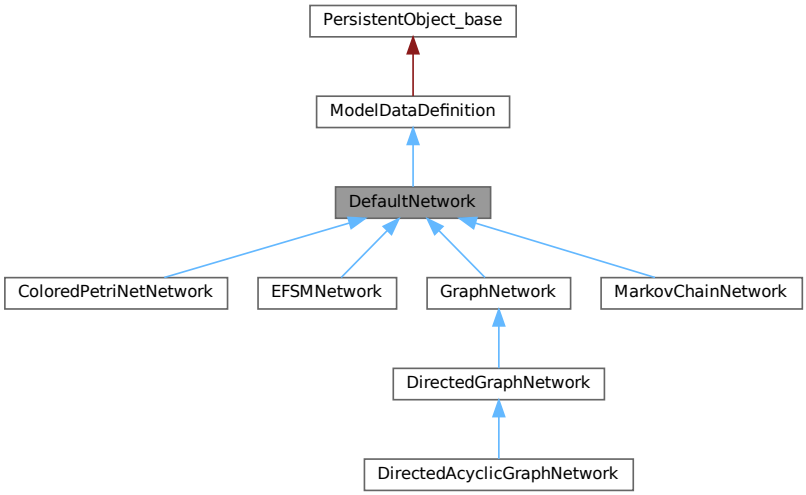
The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

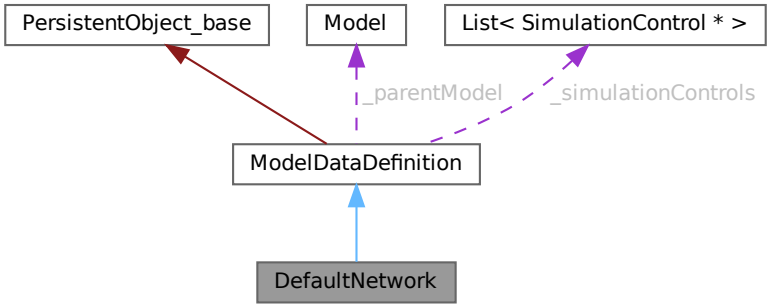
17.202 DefaultNetwork Class Reference

Base class for a formalism-agnostic network model of computation (EFSM, DTMC, Coloured Petri Net, ...) that may be shared by several `ModalModel` process adapters.
`#include <plugins/data/ModalModel/DefaultNetwork.h>`

Inheritance diagram for DefaultNetwork:



Collaboration diagram for DefaultNetwork:



Public Member Functions

- **DefaultNetwork** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< DefaultNetwork >())
- virtual std::string show () override

  Returns a textual representation of this data definition.
- unsigned int **addInputPort** (const std::string &portName)

  Declares a new input port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **addOutputPort** (const std::string &portName)

  Declares a new output port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **getNumInputPorts** () const
- unsigned int **getNumOutputPorts** () const
- std::string **getInputPortName** (unsigned int index) const

- std::string **getOutputPortName** (unsigned int index) const
- int **getInputPortIndex** (const std::string &portName) const

  Returns the input port index for `portName`, or -1 when not declared.
- int **getOutputPortIndex** (const std::string &portName) const

  Returns the output port index for `portName`, or -1 when not declared.
- double **getActivationCount** () const

  Number of times activate() has been called since the last replication reset.
- NetworkActivationResult activate (const NetworkActivationFrame &frame)

  Runs exactly one network activation for `frame`.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createInternalStatisticReporters () override
- virtual NetworkActivationResult _activate (const NetworkActivationFrame &frame)

  Formalism-specific activation behavior.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.202.1 Detailed Description

Base class for a formalism-agnostic network model of computation (EFSM, DTMC, Coloured Petri Net, ...) that may be shared by several `ModalModel` process adapters.
This is the `DefaultNetwork` described in `docs/ai_assistants/reference/GENESYS_MODAL_MODEL_NETWORK_ARCHITECTURE.md` §4.2, §6-§9: a `ModelDataDefinition` (not a `ModelComponent`) that owns its declared input/output port schema, its own persistent runtime state, and an activation counter, and that never receives a GenESyS `Entity*`. Several attached `ModalModel` components may reference and activate the same `DefaultNetwork` instance; the network is the sole source of truth for its state (architecture reference §5, §17-§18).
This base class only implements the infrastructure genuinely common to every formalism: port schema declaration/lookup, the activation counter, and the activation entry point. It intentionally has no notion of nodes, transitions, markings or probabilities — those belong to a specialization (`EFSMNetwork`, `MarkovChainNetwork`, `ColoredPetriNetNetwork`, ...) that overrides `_activate()`.
`activate()` always increments the activation counter, even when the specialization's `_activate()` produces no state change and no present outputs (architecture reference §21) — the counter counts activations, not transitions/firings.

17.202.2 Member Function Documentation

17.202.2.1 _activate()

```
NetworkActivationResult DefaultNetwork::_activate (
            const NetworkActivationFrame & frame )  [protected], [virtual]
```

Formalism-specific activation behavior.
The base implementation performs no state change and returns a result sized to `getNumOutputPorts()` with every output absent. Specializations override this to implement EFSM transition firing, one DTMC step, one CPN firing step, etc. Must not receive or depend on `Entity*` (architecture reference §8.4, §19).
Reimplemented in ColoredPetriNetNetwork, EFSMNetwork, and MarkovChainNetwork.

17.202.2.2 _check()

```
bool DefaultNetwork::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.
Reimplemented in ColoredPetriNetNetwork, DirectedAcyclicGraphNetwork, EFSMNetwork, GraphNetwork, and MarkovChainNetwork.

17.202.2.3 _createInternalStatisticReporters()

```
void DefaultNetwork::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.202.2.4 _initBetweenReplications()

```
void DefaultNetwork::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.202.2.5 _loadInstance()

```
bool DefaultNetwork::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.
Reimplemented in ColoredPetriNetNetwork, EFSMNetwork, GraphNetwork, and MarkovChainNetwork.

17.202.2.6 _saveInstance()

```
void DefaultNetwork::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.
Reimplemented in ColoredPetriNetNetwork, EFSMNetwork, GraphNetwork, and MarkovChainNetwork.

17.202.2.7 activate()

```
NetworkActivationResult DefaultNetwork::activate (
            const NetworkActivationFrame & frame )
```

Runs exactly one network activation for `frame`.
Increments the activation counter unconditionally, then delegates formalism-specific behavior to the protected virtual `_activate()`. The base implementation of `_activate()` produces an all-absent result, so a `DefaultNetwork` used directly (without a specialization) is a valid, inert network.

17.202.2.8 show()

```
std::string DefaultNetwork::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

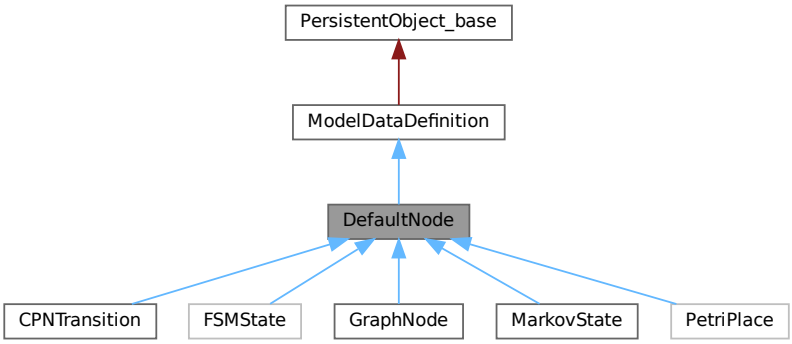
Reimplemented from ModelDataDefinition.
Reimplemented in ColoredPetriNetNetwork, EFSMNetwork, GraphNetwork, and MarkovChainNetwork.
The documentation for this class was generated from the following files:

- plugins/data/ModalModel/DefaultNetwork.h
- plugins/data/ModalModel/DefaultNetwork.cpp

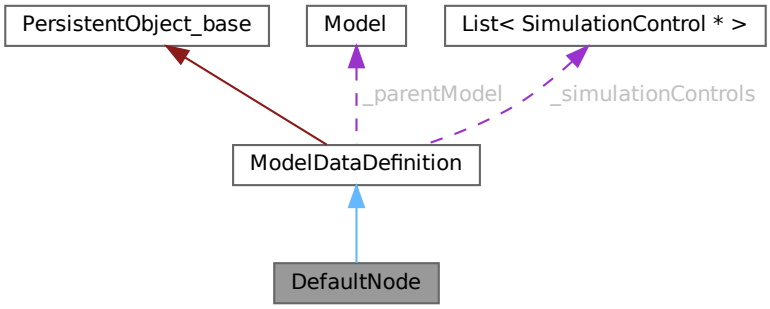
17.203 DefaultNode Class Reference

```
#include <plugins/components/ModalModel/DefaultNode.h>
```

Inheritance diagram for DefaultNode:



Collaboration diagram for DefaultNode:



Public Member Functions

- DefaultNode (Model *model, std::string componentTypename, std::string name)

  constructors
- **DefaultNode** (Model *model, std::string name="")
- void **addTransition** (DefaultNodeTransition *transition)

  new public user methods for this component
- void **removeTransition** (DefaultNodeTransition *transition)
- List< DefaultNodeTransition * > * **getTransitions** () const
- void **setInitialNode** (bool initialNode)
- bool **isInitialNode** () const
- void **setFinalNode** (bool finalNode)
- bool **isFinalNode** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)

- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.203.1 Detailed Description

This component ...

17.203.2 Constructor & Destructor Documentation

17.203.2.1 DefaultNode()

```
DefaultNode::DefaultNode (
            Model * model,
            std::string componentTypename,
            std::string name )
```

constructors
Externalize function GetPluginInformation to be accessible throught dynamic linked library.

17.203.3 Member Function Documentation

17.203.3.1 _loadInstance()

```
bool DefaultNode::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelDataDefinition.
Reimplemented in FSMState, PetriPlace, and CPNTransition.

17.203.3.2 _saveInstance()

```
void DefaultNode::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.
Reimplemented in FSMState, PetriPlace, and CPNTransition.

17.203.3.3 show()

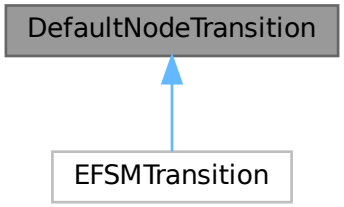
```
std::string DefaultNode::show ( )  [override], [virtual]
```

virtual public methods
Reimplemented from ModelDataDefinition.
Reimplemented in CPNTransition.
The documentation for this class was generated from the following files:

- plugins/components/ModalModel/DefaultNode.h
- plugins/components/ModalModel/DefaultNode.cpp

17.204 DefaultNodeTransition Class Reference

Inheritance diagram for DefaultNodeTransition:



Public Types

- enum class **TransitionKind** : int { **DETERMINISTIC** = 0 , **PROBABILISTIC** = 1 }

Public Member Functions

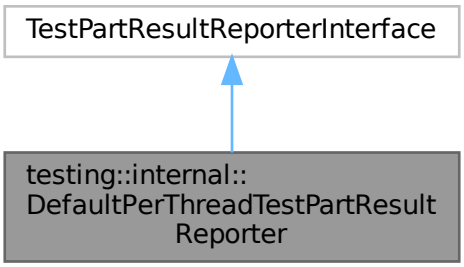
- **DefaultNodeTransition** (DefaultNode ∗source, DefaultNode ∗destination, std::string name="")
- void **setSource** (DefaultNode ∗source)
- DefaultNode ∗ **getSource** () const
- void **setDestination** (DefaultNode ∗destination)
- DefaultNode ∗ **getDestination** () const
- void **setName** (std::string name)
- std::string **getName** () const
- void **setGuardExpression** (std::string guardExpression)
- std::string **getGuardExpression** () const
- void **setOutputExpression** (std::string outputExpression)
- std::string **getOutputExpression** () const
- void **setInputEvent** (std::string inputEvent)
- std::string **getInputEvent** () const
- void **setPriority** (unsigned int priority)
- unsigned int **getPriority** () const
- void **setProbability** (double probability)
- double **getProbability** () const
- void **setTransitionKind** (TransitionKind transitionKind)
- TransitionKind **getTransitionKind** () const
- virtual bool **canFire** (Model ∗model, Entity ∗entity) const
- virtual void **execute** (Model ∗model, Entity ∗entity) const

The documentation for this class was generated from the following files:

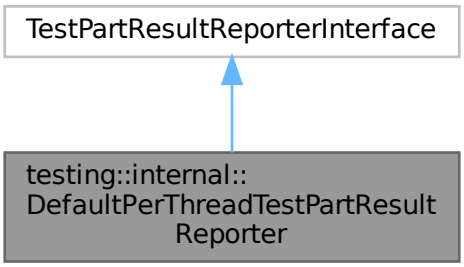
- plugins/components/ModalModel/DefaultNode.h
- plugins/components/ModalModel/DefaultNode.cpp

17.205 testing::internal::DefaultPerThreadTestPartResultReporter Class Reference

Inheritance diagram for testing::internal::DefaultPerThreadTestPartResultReporter:



Collaboration diagram for testing::internal::DefaultPerThreadTestPartResultReporter:



Public Member Functions

- **DefaultPerThreadTestPartResultReporter** (UnitTestImpl *unit_test)
- void **ReportTestPartResult** (const TestPartResult &result) override

The documentation for this class was generated from the following files:

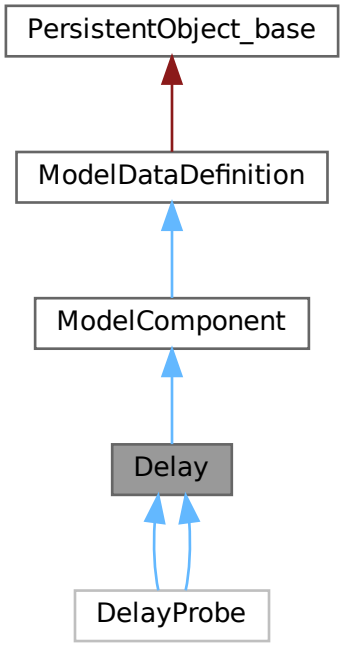
- gtest/src/gtest-internal-inl.h
- gtest/src/gtest.cc

17.206 Delay Class Reference

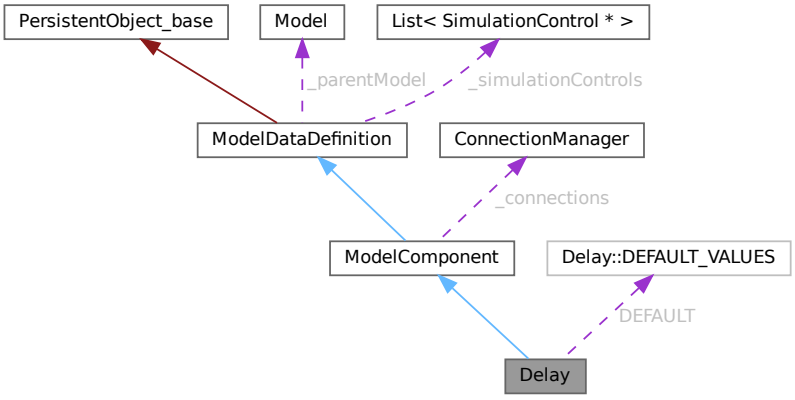
Holds an entity in place for an evaluated expression's worth of time, crediting that time to one of five allocation categories.

`#include <plugins/components/DiscreteProcessing/Delay.h>`

Inheritance diagram for Delay:



Collaboration diagram for Delay:



Classes

- struct DEFAULT_VALUES

Public Member Functions

- **Delay** (Model *model, std::string name="")

- void **setDelayExpression** (std::string _delayExpression, Util::TimeUnit _delayTimeUnit=Util::TimeUnit↩ ::unknown)
- std::string **delayExpression** () const
- void **setDelay** (double delay)
- double **delay** () const
- void **setDelayTimeUnit** (Util::TimeUnit _delayTimeUnit)
- Util::TimeUnit **delayTimeUnit** () const
- void **setAllocation** (Util::AllocationType allocation)
- Util::AllocationType **getAllocation** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Public Attributes

- const struct Delay::DEFAULT_VALUES **DEFAULT**

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createInternalStatisticReporters () override
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)

- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Friends

- class **DelayProbe**

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.206.1 Detailed Description

Holds an entity in place for an evaluated expression's worth of time, crediting that time to one of five allocation categories.
Arena correspondence: the "Delay module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", pp. 53-54). `_allocation` (a `Util::AllocationType` — ValueAdded/NonValueAdded/Transfer/Wait/Others, matching Arena's five categories exactly) is the closest correspondence to Arena's "Allocation" field.
Confirmed behavior: `_onDispatchEvent()` credits the evaluated delay **time** to the configured category twice — into a `<EntityType>.<Category>Time` StatisticsCollector (when the EntityType reports statistics) and into a running `Entity.Total<Category>Time` attribute on the entity itself. This is the mechanism referenced by `docs/ai_assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.3.
Known difference from Arena: only **time** is allocated; no code path converts accumulated category time into category **cost** (Arena's "costs are calculated and allocated as well").

17.206.2 Member Function Documentation

17.206.2.1 _check()

```
bool Delay::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.206.2.2 _createAttachedAttributes()

```
void Delay::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.206.2.3 _createInternalStatisticReporters()

```
void Delay::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.206.2.4 _loadInstance()

```
bool Delay::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.206.2.5 _onDispatchEvent()

```
void Delay::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.206.2.6 _saveInstance()

```
void Delay::_saveInstance (
```

```
PersistenceRecord * fields,
bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.206.2.7 show()

```
std::string Delay::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

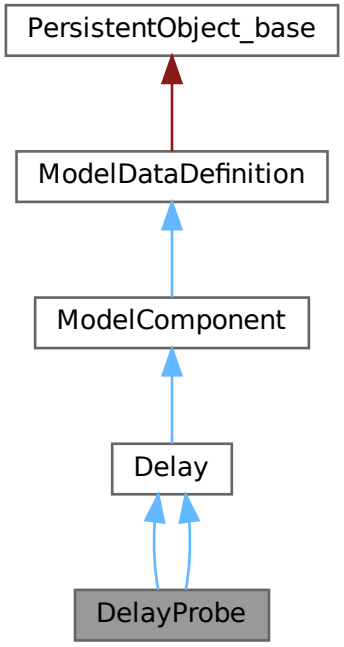
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

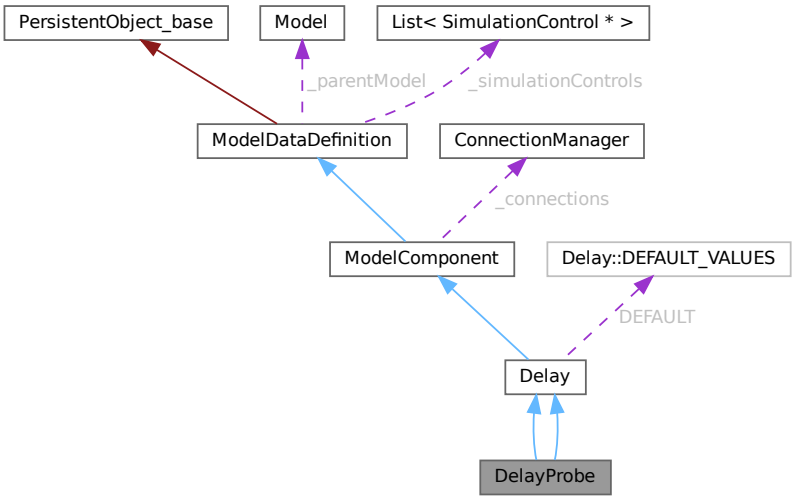
- plugins/components/DiscreteProcessing/Delay.h
- plugins/components/DiscreteProcessing/Delay.cpp

17.207 DelayProbe Class Reference

Inheritance diagram for DelayProbe:



Collaboration diagram for DelayProbe:



Public Member Functions

- **DelayProbe** (Model *model, const std::string &name="")
- void **CreateInternalAndAttachedDataProbe** ()
- StatisticsCollector * **WaitTimeStatisticsCollectorProbe** () const
- void **AddWaitTimeValueProbe** (double waitTime)
- bool **HasAttachedDataProbe** (const std::string &key) const

Public Member Functions inherited from Delay

- **Delay** (Model *model, std::string name="")
- void **setDelayExpression** (std::string _delayExpression, Util::TimeUnit _delayTimeUnit=Util::TimeUnit↩ ::unknown)
- std::string **delayExpression** () const
- void **setDelay** (double delay)
- double **delay** () const
- void **setDelayTimeUnit** (Util::TimeUnit _delayTimeUnit)
- Util::TimeUnit **delayTimeUnit** () const
- void **setAllocation** (Util::AllocationType allocation)
- Util::AllocationType **getAllocation** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

Connects this component to another component.

- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.

- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Additional Inherited Members

Static Public Member Functions inherited from Delay

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Public Attributes inherited from Delay

- const struct Delay::DEFAULT_VALUES **DEFAULT**

Protected Member Functions inherited from Delay

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createInternalStatisticReporters () override
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions

- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

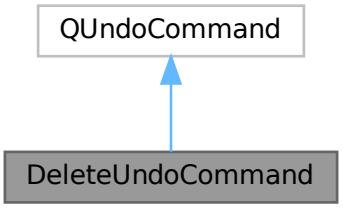
- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

The documentation for this class was generated from the following file:

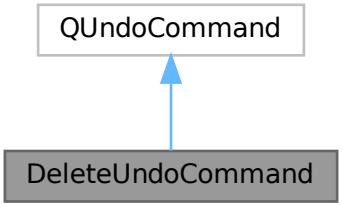
- tests/unit/test_simulator_runtime.cpp

17.208 DeleteUndoCommand Class Reference

Inheritance diagram for DeleteUndoCommand:



Collaboration diagram for DeleteUndoCommand:



Public Member Functions

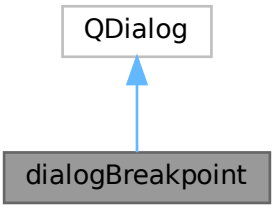
- **DeleteUndoCommand** (QList< QGraphicsItem * > items, ModelGraphicsScene *scene, QUndoCommand *parent=nullptr)
- void **undo** () override
- void **redo** () override

The documentation for this class was generated from the following files:

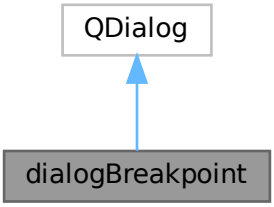
- applications/gui/genesys/actions/DeleteUndoCommand.h
- applications/gui/genesys/actions/DeleteUndoCommand.cpp

17.209 dialogBreakpoint Class Reference

Inheritance diagram for dialogBreakpoint:



Collaboration diagram for dialogBreakpoint:



Classes

- class MVCResult

Public Member Functions

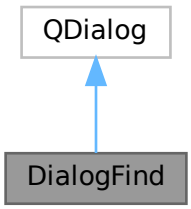
- void **setMVCModel** (Simulator ∗simulator)
- dialogBreakpoint::MVCResult ∗ **getMVCResult** ()

The documentation for this class was generated from the following files:

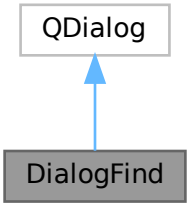
- applications/gui/genesys/dialogs/dialogBreakpoint.h
- applications/gui/genesys/dialogs/dialogBreakpoint.cpp

17.210 DialogFind Class Reference

Inheritance diagram for DialogFind:



Collaboration diagram for DialogFind:



Public Slots

- void **findClicked** ()
- void **nextClicked** ()
- void **previousClicked** ()

Public Member Functions

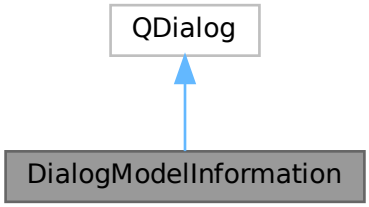
- **DialogFind** (QWidget ∗parent=nullptr, ModelGraphicsScene ∗scene=nullptr)
- QString **getFindText** ()

The documentation for this class was generated from the following files:

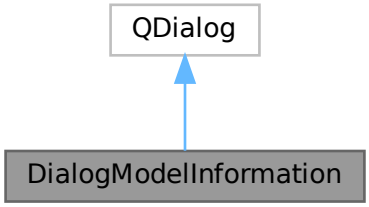
- applications/gui/genesys/dialogs/DialogFind.h
- applications/gui/genesys/dialogs/DialogFind.cpp

17.211 DialogModelInformation Class Reference

Inheritance diagram for DialogModelInformation:



Collaboration diagram for DialogModelInformation:



Public Slots

- void **accept** () override
- void **reject** () override

Public Member Functions

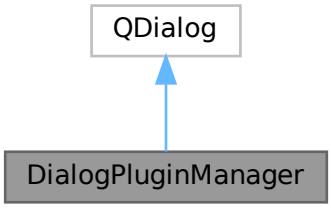
- **DialogModelInformation** (QWidget ∗parent=nullptr)
- void **setModelInfo** (ModelInfo ∗modelInfo)

The documentation for this class was generated from the following files:

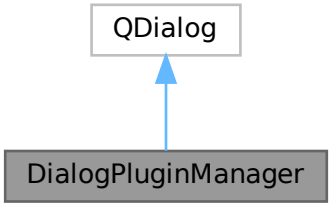
- applications/gui/genesys/dialogs/Dialogmodelinformation.h
- applications/gui/genesys/dialogs/Dialogmodelinformation.cpp

17.212 DialogPluginManager Class Reference

Inheritance diagram for DialogPluginManager:



Collaboration diagram for DialogPluginManager:



Public Member Functions

- DialogPluginManager (QWidget ∗parent=nullptr)

  Builds the plugin manager dialog and prepares the loaded/problem tables.
- void **setSimulator** (Simulator ∗simulator)
- void **setPluginCatalogRefreshCallback** (std::function< void()> refreshCallback)
- void **showProblemPluginsTab** ()

  Selects the problems tab when the dialog is opened for startup diagnostics.

17.212.1 Constructor & Destructor Documentation

17.212.1.1 DialogPluginManager()

```
DialogPluginManager::DialogPluginManager (
            QWidget * parent = nullptr )  [explicit]
```

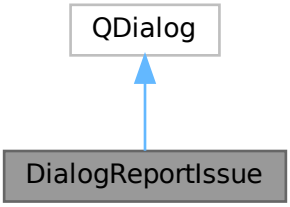
Builds the plugin manager dialog and prepares the loaded/problem tables.
The dialog is responsible for plugin insertion/removal, dependency diagnostics, and recovery of blocked plugins that require system-package installation before they can load.
The documentation for this class was generated from the following files:

- applications/gui/genesys/dialogs/dialogpluginmanager.h
- applications/gui/genesys/dialogs/dialogpluginmanager.cpp

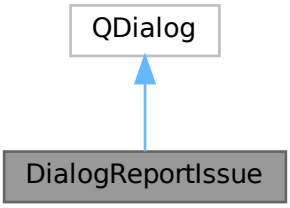
17.213 DialogReportIssue Class Reference

Modal form used to collect a user-facing issue report before relay submission.
`#include <applications/gui/genesys/dialogs/DialogReportIssue.h>`

Inheritance diagram for DialogReportIssue:



Collaboration diagram for DialogReportIssue:



Public Member Functions

- **DialogReportIssue** (QWidget *parent=nullptr)
- IssueReportDraft **draft** () const

  Returns a normalized draft from the current form fields.
- void **setRelayConfigurationStatus** (const QString &endpoint, bool configured, const QString &detail)

  Updates relay availability text and enables/disables the send action accordingly.

17.213.1 Detailed Description

Modal form used to collect a user-facing issue report before relay submission.
Responsibilities:

- collect category, title, description, and optional diagnostic consent flags;
- expose a validated IssueReportDraft for the controller layer;
- reflect whether the report relay is configured and available.

Boundaries:

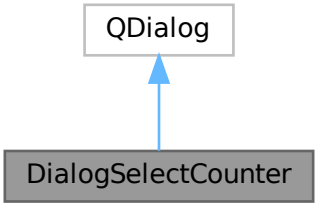
- it does not perform HTTP transport;
- it does not talk to GitHub directly;
- it does not inspect application runtime state on its own.

The documentation for this class was generated from the following files:

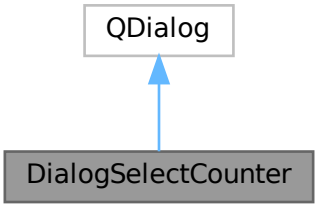
- applications/gui/genesys/dialogs/DialogReportIssue.h
- applications/gui/genesys/dialogs/DialogReportIssue.cpp

17.214 DialogSelectCounter Class Reference

Inheritance diagram for DialogSelectCounter:



Collaboration diagram for DialogSelectCounter:



Public Member Functions

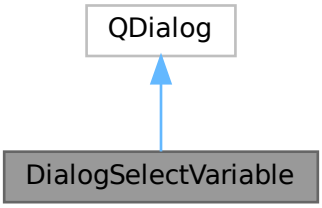
- **DialogSelectCounter** (QWidget *parent=nullptr)
- Counter * **selectedIndex** () const
- void **setCounterItems** (QList< Counter * > *counters, Counter *counter)

The documentation for this class was generated from the following files:

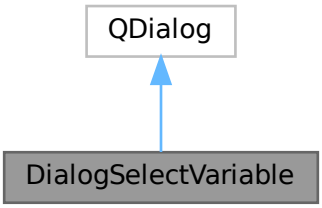
- applications/gui/genesys/dialogs/DialogSelectCounter.h
- applications/gui/genesys/dialogs/DialogSelectCounter.cpp

17.215 DialogSelectVariable Class Reference

Inheritance diagram for DialogSelectVariable:



Collaboration diagram for DialogSelectVariable:



Public Member Functions

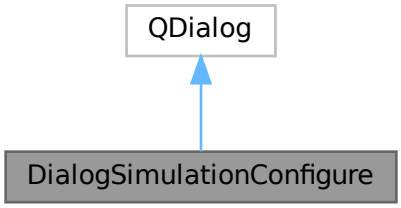
- **DialogSelectVariable** (QWidget *parent=nullptr)
- ModelDataDefinition * **selectedIndex** () const
- void **setVariableItems** (QList< ModelDataDefinition * > *definitions, ModelDataDefinition *definition)

The documentation for this class was generated from the following files:

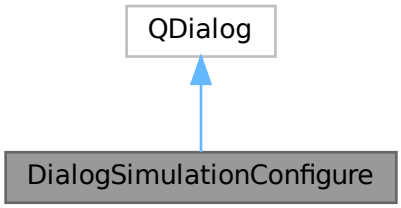
- applications/gui/genesys/dialogs/DialogSelectVariable.h
- applications/gui/genesys/dialogs/DialogSelectVariable.cpp

17.216 DialogSimulationConfigure Class Reference

Inheritance diagram for DialogSimulationConfigure:



Collaboration diagram for DialogSimulationConfigure:



Public Slots

- void accept () override

  Applies the edited configuration back to the model only when the dialog is accepted.
- void reject () override

  Asks for confirmation before discarding unsaved changes.

Public Member Functions

- DialogSimulationConfigure (QWidget ∗parent=nullptr)

  Builds the simulation-configuration dialog and seeds the static combobox content.
- void **setModelSimulation** (ModelSimulation ∗modelSimulation)
- void **setExperimentManager** (ExperimentManager ∗experimentManager)
- void **setParallelizationSettings** (bool ∗enabled, int ∗localThreads, int ∗batchSize)

17.216.1 Constructor & Destructor Documentation

17.216.1.1 DialogSimulationConfigure()

```
DialogSimulationConfigure::DialogSimulationConfigure (
            QWidget * parent = nullptr )  [explicit]
```

Builds the simulation-configuration dialog and seeds the static combobox content.

The dialog currently owns a placeholder entry for distributed parallelization while the local simulation and parallelization settings are still exposed as editable GUI state.

17.216.2 Member Function Documentation

17.216.2.1 accept

`void DialogSimulationConfigure::accept ( ) [override], [slot]`

Applies the edited configuration back to the model only when the dialog is accepted.
This keeps the original values intact until the user confirms the dialog, while still supporting partial ownership of the parallelization fields when no model simulation is currently attached.

17.216.2.2 reject

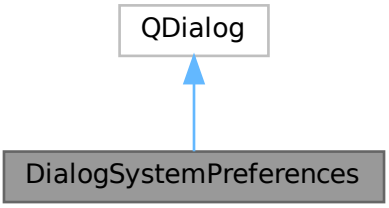
`void DialogSimulationConfigure::reject ( ) [override], [slot]`

Asks for confirmation before discarding unsaved changes.
The dialog checks the current widget state against the snapshot captured at load time so the user does not lose pending configuration edits accidentally.
The documentation for this class was generated from the following files:

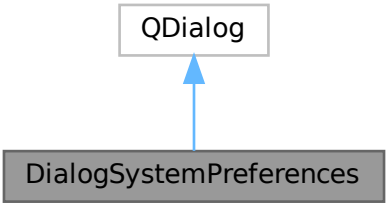
- applications/gui/genesys/dialogs/dialogsimulationconfigure.h
- applications/gui/genesys/dialogs/dialogsimulationconfigure.cpp

17.217 DialogSystemPreferences Class Reference

Inheritance diagram for DialogSystemPreferences:



Collaboration diagram for DialogSystemPreferences:



Public Member Functions

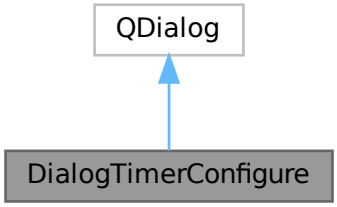
- **DialogSystemPreferences** (Simulator *simulator=nullptr, QWidget *parent=nullptr)

The documentation for this class was generated from the following files:

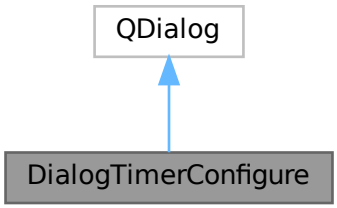
- applications/gui/genesys/dialogs/dialogsystempreferences.h
- applications/gui/genesys/dialogs/dialogsystempreferences.cpp

17.218 DialogTimerConfigure Class Reference

Inheritance diagram for DialogTimerConfigure:



Collaboration diagram for DialogTimerConfigure:



Public Member Functions

- **DialogTimerConfigure** (QWidget *parent=nullptr)
- void **setAnimationTimer** (AnimationTimer *animationTimer)
- void **updateAnimationTimer** ()

The documentation for this class was generated from the following files:

- applications/gui/genesys/dialogs/DialogTimerConfigure.h
- applications/gui/genesys/dialogs/DialogTimerConfigure.cpp

17.219 DialogUtilityController Class Reference

Controller that orchestrates auxiliary dialogs and utility actions.
`#include <applications/gui/genesys/controllers/DialogUtilityController.h>`

Public Member Functions

- **DialogUtilityController** (MainWindow *ownerWidget, Simulator *simulator, Ui::MainWindow *ui, ModelGraphicsView *graphicsView, std::function< void()> showMessageNotImplemented, std::function< void(bool)> actualizeDebugBreakpoints, std::function< bool()> createModelImage, std::function< void()> actualizeActions, std::function< void()> actualizeTabPanes, std::function< void()> reloadPluginCatalog, std::function< ModelGraphicsScene *()> currentScene, double &optimizerPrecision, unsigned int &optimizerMaxSteps, bool &parallelizationEnabled, int &parallelizationThreads, int &parallelizationBatchSize, QString &lastDataAnalyzerPath)

  Builds the utility controller that backs legacy MainWindow dialog wrappers.
- void **onActionAboutAboutTriggered** ()

  Opens the About dialog while preserving legacy action routing.
- void **onActionAboutLicenceTriggered** ()

  Opens the license dialog from the compatibility façade action slot.
- void **onActionAboutGetInvolvedTriggered** ()

  Opens the get-involved/help contribution dialog flow.
- void onActionAboutReportIssueTriggered ()

  Opens the report-issue workflow and submits it through the configured relay.
- void **onActionEditFindTriggered** ()

  Runs find workflow for the currently focused text editor pane.
- void onActionEditReplaceTriggered ()

  Runs replace workflow for the currently focused text editor pane.
- void onActionToolsParserGrammarCheckerTriggered ()

  Opens parser grammar checker utility without altering model state.
- void **onActionToolsOptimizatorTriggered** ()

  Opens optimization settings dialog and stores lightweight preferences.
- void **onActionToolsAIAssistantTriggered** ()

  Opens the AI Assistant panel for LLM-driven simulation model generation.
- void **onActionToolsDataAnalyzerTriggered** ()

  Launches the Data Analyzer workstation using persisted last-path compatibility state.
- void **onActionViewConfigureTriggered** ()

  Opens view configuration dialog and applies delegated scene/UI refreshes.
- void **onActionSimulatorPreferencesTriggered** ()

  Opens simulator preferences utility dialog.
- void **onActionSimulatorsPluginManagerTriggered** (bool showProblemPlugins=false)

  Opens plugin manager dialog from the compatibility action surface.
- void **onPushButtonBreakpointInsertClicked** ()

  Inserts a breakpoint entry based on current scene selection.
- void **onPushButtonBreakpointRemoveClicked** ()

  Removes selected breakpoint entries and refreshes debug breakpoint pane.
- void **onPushButtonExportClicked** ()

  Exports current debug/trace context using legacy push-button route.
- void **onActionParallelizationTriggered** ()

  Opens parallelization configuration and persists run-preference flags.

17.219.1 Detailed Description

Controller that orchestrates auxiliary dialogs and utility actions.
This Phase-11 controller gathers non-core menu/toolbar actions that were still in MainWindow, such as About dialogs, breakpoint utilities, optimization/preferences dialogs, data analyzer launch flow, and parallelization toggles. Responsibilities:

- execute dialog-driven utility actions delegated from MainWindow wrappers;

- access shared UI/simulator state through narrow injected callbacks/references;
- keep compatibility behavior for debug breakpoints and auxiliary tooling actions.

Boundaries:

- it does not own simulator/model lifecycle;
- it does not manage scene command pipeline or property editor internals;
- it acts as a controller-level compatibility wrapper, not as a domain service.

17.219.2 Member Function Documentation

17.219.2.1 onActionAboutReportIssueTriggered()

`void DialogUtilityController::onActionAboutReportIssueTriggered ( )`
Opens the report-issue workflow and submits it through the configured relay.
Opens the GUI issue-report flow, captures runtime context, and submits the payload to the configured relay.
The report draft can include the current model text, console tail, simulation and report logs, the latest process log, and an optional screenshot. The submission result is then surfaced back to the user, including a browser link when the relay returns a created GitHub issue URL.

17.219.2.2 onActionEditReplaceTriggered()

`void DialogUtilityController::onActionEditReplaceTriggered ( )`
Runs replace workflow for the currently focused text editor pane.
Opens the editor replacement dialog and preserves the current find/replace workflow.
The dialog keeps the last search terms, supports case-sensitive matching, wraps the search to the start of the document, and performs single or bulk replacement directly on the code editor without detaching the current selection state.

17.219.2.3 onActionToolsParserGrammarCheckerTriggered()

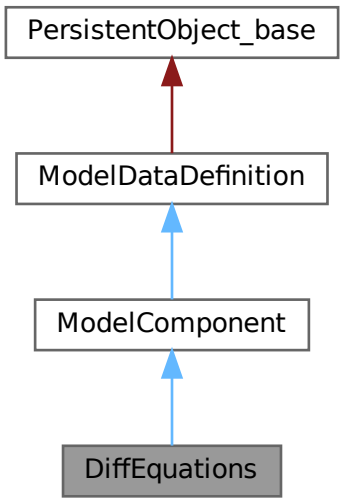
`void DialogUtilityController::onActionToolsParserGrammarCheckerTriggered ( )`
Opens parser grammar checker utility without altering model state.
Builds the parser diagnostics window with grammar overview, DOT export, catalog, and console tabs.
The dialog combines a collapsible grammar map, a DOT text editor, a searchable function catalog, a parser console, and a lightweight model symbol browser so the user can inspect and exercise the expression language from one place.
The documentation for this class was generated from the following files:

- applications/gui/genesys/controllers/DialogUtilityController.h
- applications/gui/genesys/controllers/DialogUtilityController.cpp

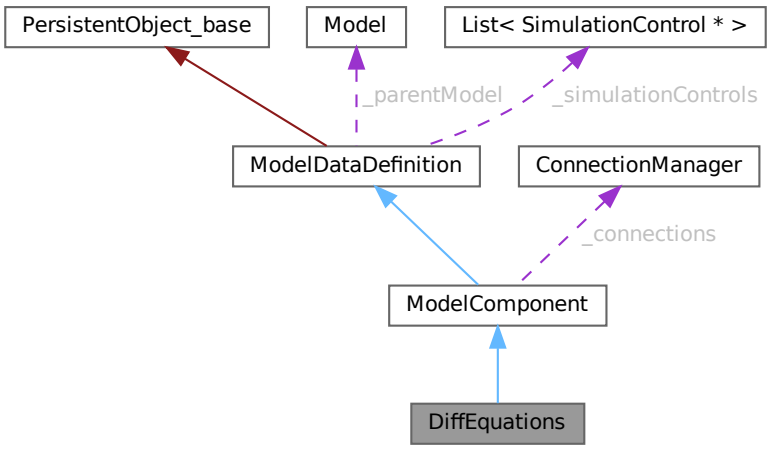
17.220 DiffEquations Class Reference

`#include <plugins/components/Continuous/DiffEquations.h>`

Inheritance diagram for DiffEquations:



Collaboration diagram for DiffEquations:



Public Member Functions

- **DiffEquations** (Model ∗model, std::string name="")
  constructors
- List< std::string > ∗ getEquations () const
  new public user methods for this component
- List< std::string > ∗ **getStateVariableNames** () const
- double **getprecision** () const
- void **setPrecision** (double newPrecision)

- double **getmaxSteps** () const
- void **setMaxSteps** (double newMaxSteps)
- double **getfinalTime** () const
- void **setFinalTime** (double newFinalTime)
- List< double > * **getvaluesOnFinalTime** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual bool _check (std::string &errorMessage) override

  virtual protected methods that could be overriden by derived classes, if needed
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.220.1 Detailed Description

This component ...

17.220.2 Member Function Documentation

17.220.2.1 _check()

```
bool DiffEquations::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

virtual protected methods that could be overriden by derived classes, if needed
This method is called by ModelChecker during model check. The component should check itself to verify if user parameters are ok (ex: correct syntax for the parser) and everithing in its parameters allow the model too run without errors in this component
Reimplemented from ModelDataDefinition.

17.220.2.2 _createEditableDataDefinitions()

```
void DiffEquations::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
This method is called by ModelSimulation when initianting the replication. The model should set all value for a new replication (Ex: setting back to 0 any internal counter, clearing lists, etc.
This method is not used yet. It should be usefull for new UIs
Reimplemented from ModelDataDefinition.

17.220.2.3 _loadInstance()

```
bool DiffEquations::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.220.2.4 _onDispatchEvent()

```
void DiffEquations::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

This method is only for ModelComponents, not ModelDataElements.
Implements ModelComponent.

17.220.2.5 _saveInstance()

```
void DiffEquations::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.220.2.6 getEquations()

```
List< std::string > * DiffEquations::getEquations ( ) const
```

new public user methods for this component
equations should be in terms of state variable names and t

17.220.2.7 show()

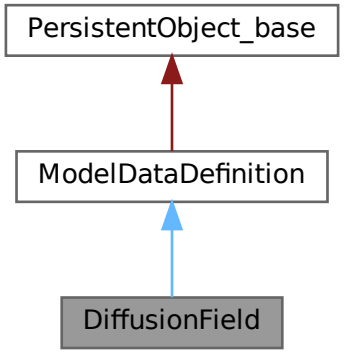
```
std::string DiffEquations::show ( )  [override], [virtual]
```

virtual public methods
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

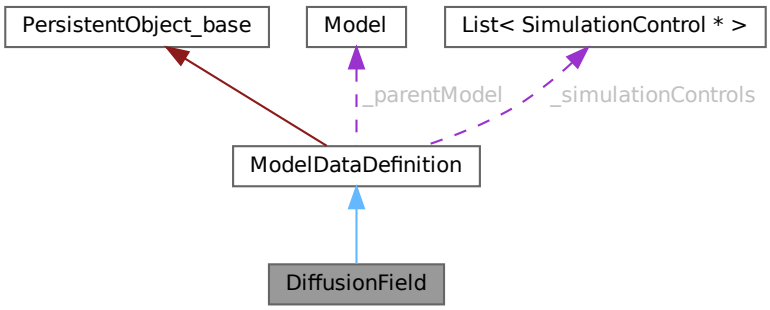
- plugins/components/Continuous/DiffEquations.h
- plugins/components/Continuous/DiffEquations.cpp

17.221 DiffusionField Class Reference

Inheritance diagram for DiffusionField:



Collaboration diagram for DiffusionField:



Public Member Functions

- **DiffusionField** (Model *model, std::string name="")
- void **setDimensions** (unsigned int dimensions)
- unsigned int **getDimensions** () const
- void **setPointsPerDimension** (unsigned int pointsPerDimension)

- unsigned int **getPointsPerDimension** () const
- void **setDomainLength** (double domainLength)
- double **getDomainLength** () const
- void **setDiffusionCoefficient** (double diffusionCoefficient)
- double **getDiffusionCoefficient** () const
- void **setBoundaryCondition** (std::string boundaryCondition)
- std::string **getBoundaryCondition** () const
- void **setInitialCondition** (std::string initialCondition)
- std::string **getInitialCondition** () const
- void **setInitialParameter** (double initialParameter)
- double **getInitialParameter** () const
- void **setStartTime** (double startTime)
- double **getStartTime** () const
- void **setStopTime** (double stopTime)
- double **getStopTime** () const
- void **setStepSize** (double stepSize)
- double **getStepSize** () const
- void **setCurrentTime** (double currentTime)
- double **getCurrentTime** () const
- void **setAutoSchedule** (bool autoSchedule)
- bool **getAutoSchedule** () const
- void **setOdeSolver** (std::string odeSolver)
- std::string **getOdeSolver** () const
- void **setLastStatus** (std::string lastStatus)
- std::string **getLastStatus** () const
- void **setLastErrorMessage** (std::string lastErrorMessage)
- std::string **getLastErrorMessage** () const
- double **getTotalMass** () const
- double **getMaxValue** () const
- double **getL2Norm** () const
- const std::vector< double > & **getField** () const
- double **getFieldValue** (const std::vector< unsigned int > &multiIndex) const

  Field value at a multi-index (size must equal Dimensions); 0 if invalid.
- bool **simulate** (std::string &errorMessage)
- bool **simulate** (double startTime, double stopTime, double stepSize, std::string &errorMessage)
- bool **advanceOneStep** (std::string &errorMessage)
- std::string **showFieldAsAscii** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)

- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.221.1 Member Function Documentation

17.221.1.1 _check()

```
bool DiffusionField::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.221.1.2 _initBetweenReplications()

```
void DiffusionField::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.221.1.3 _loadInstance()

```
bool DiffusionField::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.221.1.4 _saveInstance()

```
void DiffusionField::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.221.1.5 show()

```
std::string DiffusionField::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

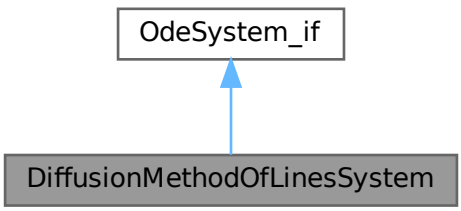
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

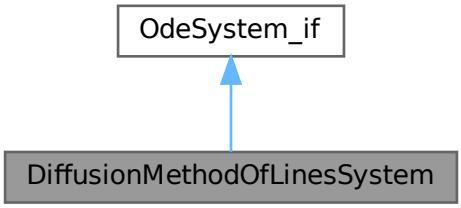
- plugins/data/Continuous/DiffusionField.h
- plugins/data/Continuous/DiffusionField.cpp

17.222 DiffusionMethodOfLinesSystem Class Reference

Inheritance diagram for DiffusionMethodOfLinesSystem:



Collaboration diagram for DiffusionMethodOfLinesSystem:



Public Types

- enum class **Boundary** { **Dirichlet** , **Neumann** }

Public Member Functions

- **DiffusionMethodOfLinesSystem** (std::vector< unsigned int > shape, std::vector< double > spacing, double diffusionCoefficient, Boundary boundary)
- unsigned int dimension () const override
- void evaluate (double, const double *y, double *dydt) const override
- bool **isValid** () const
- unsigned int **totalNodes** () const
- const std::vector< unsigned int > & **shape** () const
- const std::vector< double > & **spacing** () const
- double **diffusionCoefficient** () const
- Boundary **boundary** () const
- unsigned int **coordinate** (unsigned int node, unsigned int d) const
- unsigned int **nodeIndex** (const std::vector< unsigned int > &multiIndex) const
- bool **isBoundaryNode** (unsigned int node) const
- void **fillSineModes** (const std::vector< unsigned int > &modes, double amplitude, double *y) const
- void **fillGaussian** (double amplitude, double sigma, double *y) const
- double **sineModeDecayRate** (const std::vector< unsigned int > &modes) const
- double **totalMass** (const double *y) const
- double **l2Norm** (const double *y) const
- double **maxValue** (const double *y) const

Static Public Attributes

- static constexpr double **kPi** = 3.14159265358979323846

17.222.1 Member Function Documentation

17.222.1.1 dimension()

```
unsigned int DiffusionMethodOfLinesSystem::dimension ( ) const  [inline], [override], [virtual]
```

Implements OdeSystem_if.

17.222.1.2 evaluate()

```
void DiffusionMethodOfLinesSystem::evaluate (
            double ,
            const double * y,
            double * dydt ) const  [inline], [override], [virtual]
```

Implements OdeSystem_if.

The documentation for this class was generated from the following file:

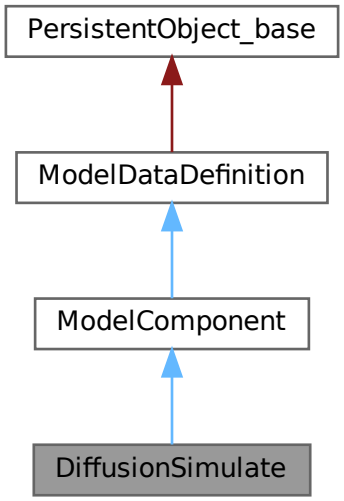
- tools/Continuous/DiffusionMethodOfLinesSystem.h

17.223 DiffusionSimulate Class Reference

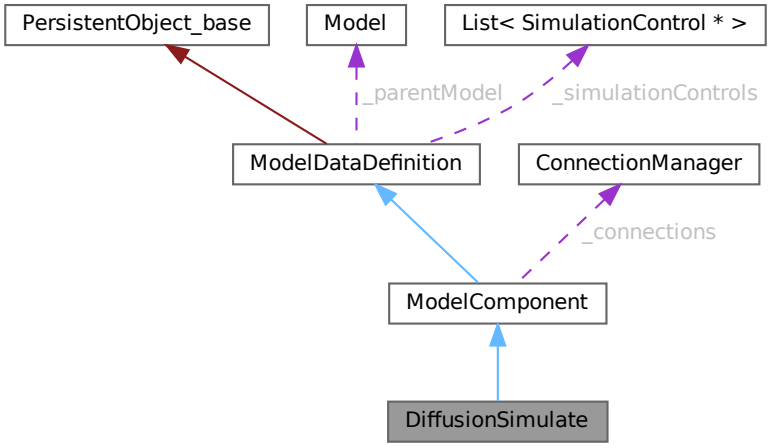
Component that triggers a DiffusionField solve when entities arrive.

`#include <plugins/components/Continuous/DiffusionSimulate.h>`

Inheritance diagram for DiffusionSimulate:



Collaboration diagram for DiffusionSimulate:



Public Member Functions

- **DiffusionSimulate** (Model *model, std::string name="")
- void **setDiffusionField** (DiffusionField *diffusionField)
- DiffusionField * **getDiffusionField** () const
- void **setUseFieldTimeWindow** (bool useFieldTimeWindow)
- bool **getUseFieldTimeWindow** () const
- void **setStartTime** (double startTime)
- double **getStartTime** () const
- void **setStopTime** (double stopTime)
- double **getStopTime** () const
- void **setStepSize** (double stepSize)
- double **getStepSize** () const
- void **setLastSucceeded** (bool lastSucceeded)
- bool **getLastSucceeded** () const
- void **setLastMessage** (std::string lastMessage)
- std::string **getLastMessage** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.223.1 Detailed Description

Component that triggers a DiffusionField solve when entities arrive.
DiffusionSimulate delegates the continuous diffusion solve to a referenced DiffusionField (a ModelDataDefinition in plugins/data/Continuous). Each arriving entity runs the field either with its own configured time window or with component-provided start/stop/step values, then forwards the entity. This is the discrete->continuous bridge: a discrete-event component drives the continuous PDE subsystem. For event-clock-synchronized advance, enable AutoSchedule on the DiffusionField instead, so it steps once per kernel InternalEvent.

17.223.2 Member Function Documentation

17.223.2.1 _check()

```
bool DiffusionSimulate::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.223.2.2 _createEditableDataDefinitions()

```
void DiffusionSimulate::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.223.2.3 _loadInstance()

```
bool DiffusionSimulate::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.223.2.4 _onDispatchEvent()

```
void DiffusionSimulate::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.

Parameters

inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.223.2.5 _saveInstance()

```
void DiffusionSimulate::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.223.2.6 show()

```
std::string DiffusionSimulate::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

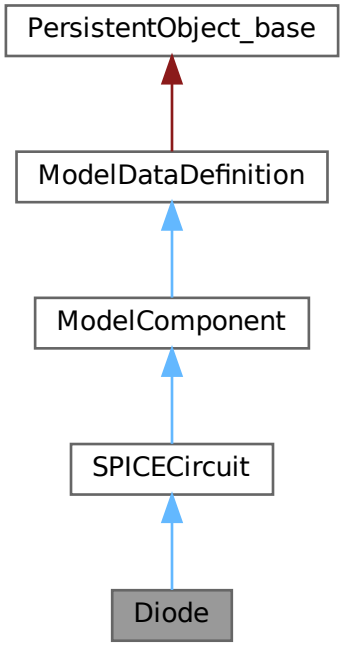
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

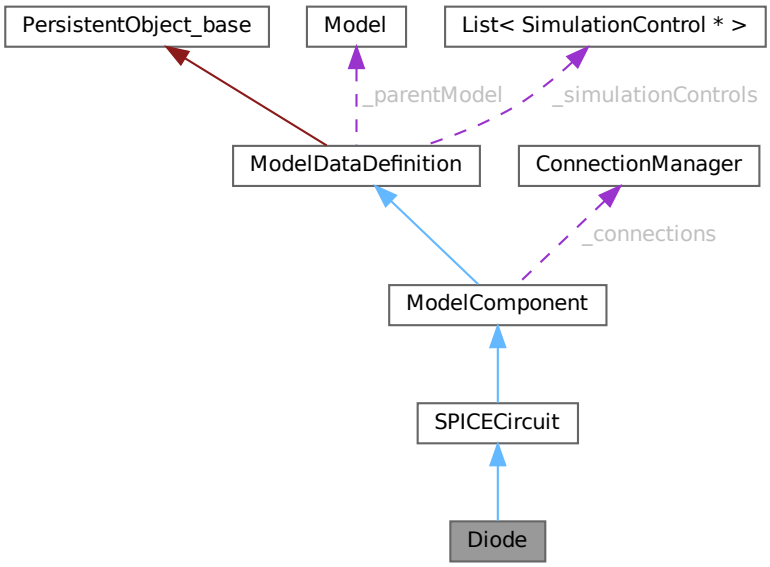
- plugins/components/Continuous/DiffusionSimulate.h
- plugins/components/Continuous/DiffusionSimulate.cpp

17.224 Diode Class Reference

Inheritance diagram for Diode:



Collaboration diagram for Diode:



Public Member Functions

- **Diode** (Model *model, std::string name="")
- void Build () override
- void **setElectricalModel** (std::string electricalModel)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component

- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.224.1 Member Function Documentation

17.224.1.1 _loadInstance()

```
bool Diode::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.224.1.2 _saveInstance()

```
void Diode::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.224.1.3 Build()

```
void Diode::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.
The documentation for this class was generated from the following files:

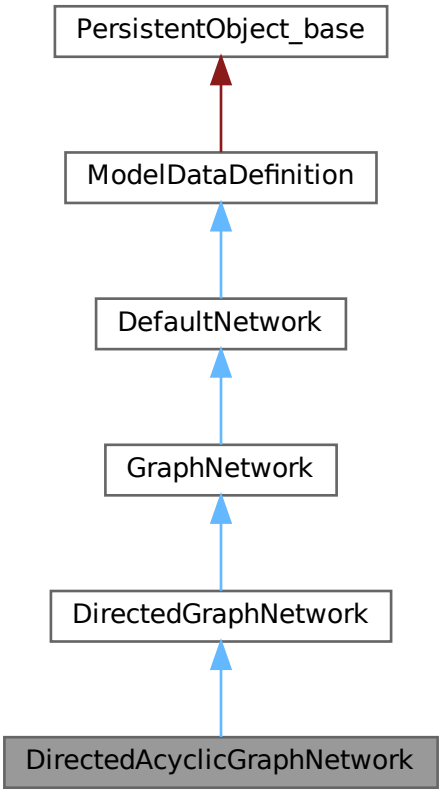
- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.225 DirectedAcyclicGraphNetwork Class Reference

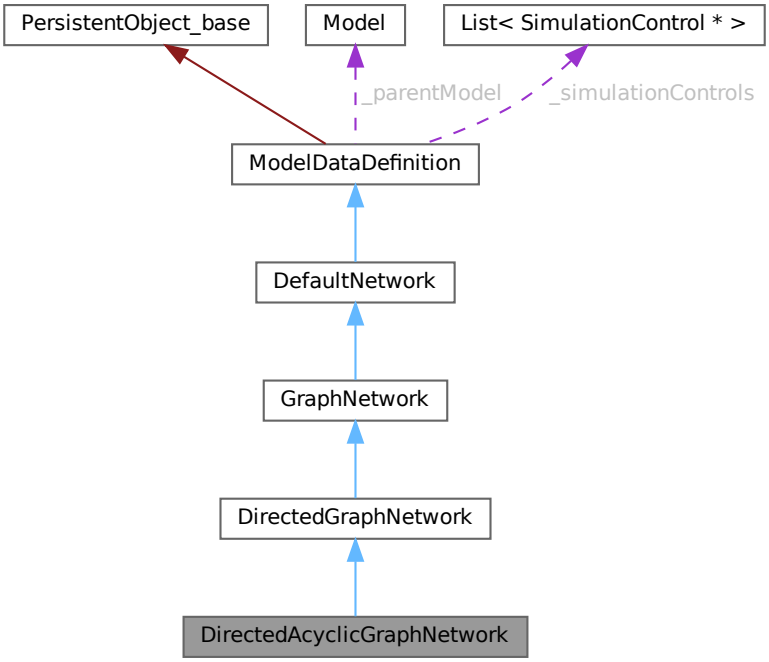
Directed graph specialization that enforces the DAG invariant.

`#include <plugins/data/ModalModel/DirectedAcyclicGraphNetwork.h>`

Inheritance diagram for DirectedAcyclicGraphNetwork:



Collaboration diagram for DirectedAcyclicGraphNetwork:



Public Member Functions

- **DirectedAcyclicGraphNetwork** (Model *model, std::string name="")
- virtual bool addEdge (GraphEdge *edge) override

Public Member Functions inherited from DirectedGraphNetwork

- **DirectedGraphNetwork** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< DirectedGraphNetwork >())
- virtual bool isDirected () const override
- virtual std::vector< GraphNode * > getPredecessors (GraphNode *node) const override
- virtual std::vector< GraphNode * > getSuccessors (GraphNode *node) const override
- virtual std::vector< GraphEdge * > getIncomingEdges (GraphNode *node) const override
- virtual std::vector< GraphEdge * > getOutgoingEdges (GraphNode *node) const override
- virtual unsigned int inDegree (GraphNode *node) const override
- virtual unsigned int outDegree (GraphNode *node) const override
- std::vector< std::vector< GraphNode * > > **stronglyConnectedComponents** () const
- virtual bool hasCycle () const override
- virtual GraphTopologicalOrderResult topologicalOrder () const override

Public Member Functions inherited from GraphNetwork

- **GraphNetwork** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< GraphNetwork >())
- virtual bool **addNode** (GraphNode *node)
- virtual bool **removeNode** (GraphNode *node)
- bool **containsNode** (GraphNode *node) const
- GraphNode * **findNodeByName** (const std::string &name) const

- std::vector< GraphNode * > **getNodes** () const
- virtual bool **removeEdge** (GraphEdge *edge)
- bool **containsEdge** (GraphEdge *edge) const
- GraphEdge * **findEdgeByName** (const std::string &name) const
- std::vector< GraphEdge * > **getEdges** () const
- bool **hasEdge** (GraphNode *source, GraphNode *destination) const
- std::vector< GraphEdge * > **getEdgesBetween** (GraphNode *source, GraphNode *destination) const
- std::vector< GraphNode * > **getNeighbors** (GraphNode *node) const
- std::vector< GraphEdge * > **getIncidentEdges** (GraphNode *node) const
- unsigned int **degree** (GraphNode *node) const
- GraphTraversalResult **breadthFirstSearch** (GraphNode *source) const
- GraphTraversalResult **depthFirstSearch** (GraphNode *source) const
- bool **hasPath** (GraphNode *source, GraphNode *destination) const
- std::vector< GraphNode * > **reachableNodes** (GraphNode *source) const
- GraphPathResult **shortestPathUnweighted** (GraphNode *source, GraphNode *destination) const
- GraphPathResult **shortestPathDijkstra** (GraphNode *source, GraphNode *destination) const
- std::vector< std::vector< GraphNode * > > **connectedComponents** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from DefaultNetwork

- **DefaultNetwork** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< DefaultNetwork >())
- unsigned int **addInputPort** (const std::string &portName)

  Declares a new input port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **addOutputPort** (const std::string &portName)

  Declares a new output port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **getNumInputPorts** () const
- unsigned int **getNumOutputPorts** () const
- std::string **getInputPortName** (unsigned int index) const
- std::string **getOutputPortName** (unsigned int index) const
- int **getInputPortIndex** (const std::string &portName) const

  Returns the input port index for `portName`, or -1 when not declared.
- int **getOutputPortIndex** (const std::string &portName) const

  Returns the output port index for `portName`, or -1 when not declared.
- double **getActivationCount** () const

  Number of times activate() has been called since the last replication reset.
- NetworkActivationResult activate (const NetworkActivationFrame &frame)

  Runs exactly one network activation for `frame`.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from DirectedGraphNetwork

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from GraphNetwork

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from DefaultNetwork

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from DirectedGraphNetwork

- virtual bool _edgeConnects (GraphEdge *edge, GraphNode *source, GraphNode *destination) const override
- virtual GraphNode * _oppositeNode (GraphEdge *edge, GraphNode *node) const override

Protected Member Functions inherited from GraphNetwork

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from DefaultNetwork

- virtual void _createInternalStatisticReporters () override
- virtual NetworkActivationResult _activate (const NetworkActivationFrame &frame)

  Formalism-specific activation behavior.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.225.1 Detailed Description

Directed graph specialization that enforces the DAG invariant.
Edges that would create a directed cycle are rejected on insertion, and the model checker validates the invariant again for corrupted persistence.

17.225.2 Member Function Documentation

17.225.2.1 _check()

```
bool DirectedAcyclicGraphNetwork::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from GraphNetwork.

17.225.2.2 addEdge()

```
bool DirectedAcyclicGraphNetwork::addEdge (
            GraphEdge * edge )  [override], [virtual]
```

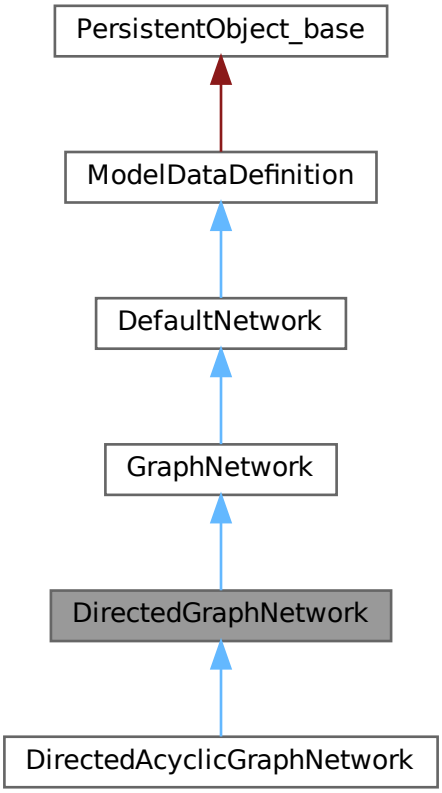
Reimplemented from GraphNetwork.
The documentation for this class was generated from the following files:

- plugins/data/ModalModel/DirectedAcyclicGraphNetwork.h
- plugins/components/ModalModel/DirectedAcyclicGraphNetwork.cpp

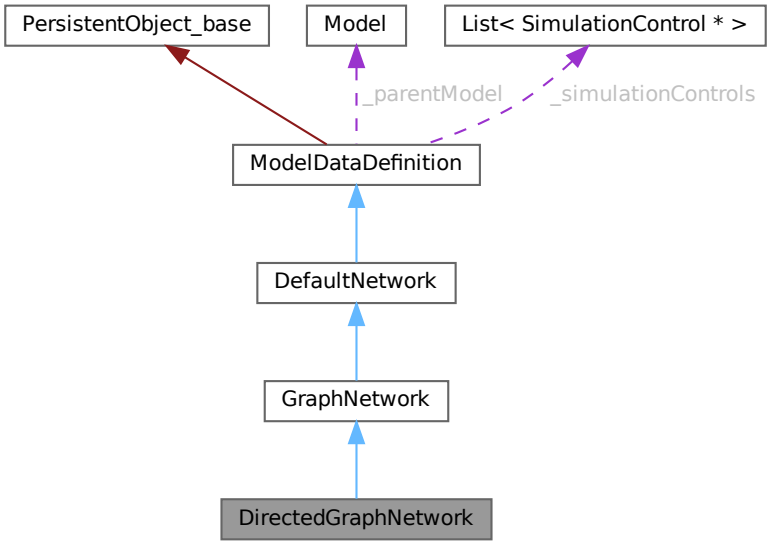
17.226 DirectedGraphNetwork Class Reference

Directed mathematical graph specialization.
`#include <plugins/data/ModalModel/DirectedGraphNetwork.h>`

Inheritance diagram for DirectedGraphNetwork:



Collaboration diagram for DirectedGraphNetwork:



Public Member Functions

- **DirectedGraphNetwork** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< DirectedGraphNetwork >())
- virtual bool isDirected () const override
- virtual std::vector< GraphNode * > getPredecessors (GraphNode *node) const override
- virtual std::vector< GraphNode * > getSuccessors (GraphNode *node) const override
- virtual std::vector< GraphEdge * > getIncomingEdges (GraphNode *node) const override
- virtual std::vector< GraphEdge * > getOutgoingEdges (GraphNode *node) const override
- virtual unsigned int inDegree (GraphNode *node) const override
- virtual unsigned int outDegree (GraphNode *node) const override
- std::vector< std::vector< GraphNode * > > **stronglyConnectedComponents** () const
- virtual bool hasCycle () const override
- virtual GraphTopologicalOrderResult topologicalOrder () const override

Public Member Functions inherited from GraphNetwork

- **GraphNetwork** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< GraphNetwork >())
- virtual bool **addNode** (GraphNode *node)
- virtual bool **removeNode** (GraphNode *node)
- bool **containsNode** (GraphNode *node) const
- GraphNode * **findNodeByName** (const std::string &name) const
- std::vector< GraphNode * > **getNodes** () const
- virtual bool **addEdge** (GraphEdge *edge)
- virtual bool **removeEdge** (GraphEdge *edge)
- bool **containsEdge** (GraphEdge *edge) const
- GraphEdge * **findEdgeByName** (const std::string &name) const
- std::vector< GraphEdge * > **getEdges** () const
- bool **hasEdge** (GraphNode *source, GraphNode *destination) const
- std::vector< GraphEdge * > **getEdgesBetween** (GraphNode *source, GraphNode *destination) const

- std::vector< GraphNode * > **getNeighbors** (GraphNode *node) const
- std::vector< GraphEdge * > **getIncidentEdges** (GraphNode *node) const
- unsigned int **degree** (GraphNode *node) const
- GraphTraversalResult **breadthFirstSearch** (GraphNode *source) const
- GraphTraversalResult **depthFirstSearch** (GraphNode *source) const
- bool **hasPath** (GraphNode *source, GraphNode *destination) const
- std::vector< GraphNode * > **reachableNodes** (GraphNode *source) const
- GraphPathResult **shortestPathUnweighted** (GraphNode *source, GraphNode *destination) const
- GraphPathResult **shortestPathDijkstra** (GraphNode *source, GraphNode *destination) const
- std::vector< std::vector< GraphNode * > > **connectedComponents** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from DefaultNetwork

- **DefaultNetwork** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< DefaultNetwork >())
- unsigned int **addInputPort** (const std::string &portName)

  Declares a new input port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **addOutputPort** (const std::string &portName)

  Declares a new output port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **getNumInputPorts** () const
- unsigned int **getNumOutputPorts** () const
- std::string **getInputPortName** (unsigned int index) const
- std::string **getOutputPortName** (unsigned int index) const
- int **getInputPortIndex** (const std::string &portName) const

  Returns the input port index for `portName`, or -1 when not declared.
- int **getOutputPortIndex** (const std::string &portName) const

  Returns the output port index for `portName`, or -1 when not declared.
- double **getActivationCount** () const

  Number of times activate() has been called since the last replication reset.
- NetworkActivationResult activate (const NetworkActivationFrame &frame)

  Runs exactly one network activation for `frame`.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from GraphNetwork

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from DefaultNetwork

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _edgeConnects (GraphEdge *edge, GraphNode *source, GraphNode *destination) const override
- virtual GraphNode * _oppositeNode (GraphEdge *edge, GraphNode *node) const override

Protected Member Functions inherited from GraphNetwork

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from DefaultNetwork

- virtual void _createInternalStatisticReporters () override
- virtual NetworkActivationResult _activate (const NetworkActivationFrame &frame)

  Formalism-specific activation behavior.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.226.1 Detailed Description

Directed mathematical graph specialization.
In DirectedGraphNetwork an edge u->v makes v a successor of u and u a predecessor of v. It does not imply the reverse relation. Self-loops and parallel edges are supported.

17.226.2 Member Function Documentation

17.226.2.1 _edgeConnects()

```
bool DirectedGraphNetwork::_edgeConnects (
            GraphEdge * edge,
            GraphNode * source,
            GraphNode * destination ) const  [override], [protected], [virtual]
```

Reimplemented from GraphNetwork.

17.226.2.2 _oppositeNode()

```
GraphNode * DirectedGraphNetwork::_oppositeNode (
            GraphEdge * edge,
            GraphNode * node ) const  [override], [protected], [virtual]
```

Reimplemented from GraphNetwork.

17.226.2.3 getIncomingEdges()

```
std::vector< GraphEdge * > DirectedGraphNetwork::getIncomingEdges (
            GraphNode * node ) const  [override], [virtual]
```

Reimplemented from GraphNetwork.

17.226.2.4 getOutgoingEdges()

```
std::vector< GraphEdge * > DirectedGraphNetwork::getOutgoingEdges (
            GraphNode * node ) const  [override], [virtual]
```

Reimplemented from GraphNetwork.

17.226.2.5 getPredecessors()

```
std::vector< GraphNode * > DirectedGraphNetwork::getPredecessors (
            GraphNode * node ) const  [override], [virtual]
```

Reimplemented from GraphNetwork.

17.226.2.6 getSuccessors()

```
std::vector< GraphNode * > DirectedGraphNetwork::getSuccessors (
            GraphNode * node ) const  [override], [virtual]
```

Reimplemented from GraphNetwork.

17.226.2.7 hasCycle()

```
bool DirectedGraphNetwork::hasCycle ( ) const  [override], [virtual]
```

Reimplemented from GraphNetwork.

17.226.2.8 inDegree()

```
unsigned int DirectedGraphNetwork::inDegree (
            GraphNode * node ) const  [override], [virtual]
```

Reimplemented from GraphNetwork.

17.226.2.9 isDirected()

```
bool DirectedGraphNetwork::isDirected ( ) const  [override], [virtual]
```

Reimplemented from GraphNetwork.

17.226.2.10 outDegree()

```
unsigned int DirectedGraphNetwork::outDegree (
            GraphNode * node ) const  [override], [virtual]
```

Reimplemented from GraphNetwork.

17.226.2.11 topologicalOrder()

```
GraphTopologicalOrderResult DirectedGraphNetwork::topologicalOrder ( ) const  [override],
[virtual]
```

Reimplemented from GraphNetwork.

The documentation for this class was generated from the following files:

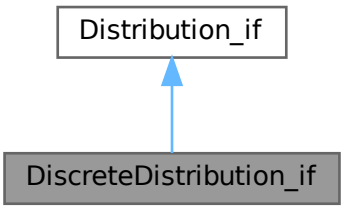
- plugins/data/ModalModel/DirectedGraphNetwork.h
- plugins/components/ModalModel/DirectedGraphNetwork.cpp

17.227 DiscreteDistribution_if Class Reference

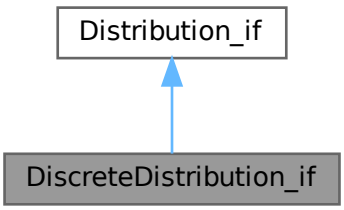
Interface for discrete distributions.

`#include <tools/Statistics/DiscreteDistribution_if.h>`

Inheritance diagram for DiscreteDistribution_if:



Collaboration diagram for DiscreteDistribution_if:



Public Member Functions

- virtual double **pmf** (unsigned int k) const =0
- virtual double **cdf** (unsigned int k) const =0
- virtual unsigned int **quantile** (double probability) const =0

Public Member Functions inherited from Distribution_if

- virtual std::string **name** () const =0

17.227.1 Detailed Description

Interface for discrete distributions.
Mathematical meaning:

- pmf(k): probability mass function value at k.
- cdf(k): cumulative probability P(X <= k).
- quantile(p): smallest k such that CDF(k) >= p.

The documentation for this class was generated from the following file:

- tools/Statistics/DiscreteDistribution_if.h

17.228 genesys::launcher::DispatchController Class Reference

Public Member Functions

- **DispatchController** (const RuntimeSelector &selector, IProcessLauncher &processLauncher)
- int **dispatch** (const QString &application, const QStringList &arguments, const LauncherConfig &config, QString *error=nullptr) const

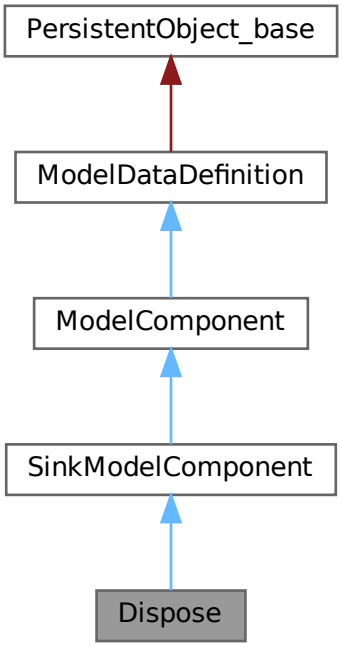
The documentation for this class was generated from the following files:

- applications/launcher/DispatchController.h
- applications/launcher/DispatchController.cpp

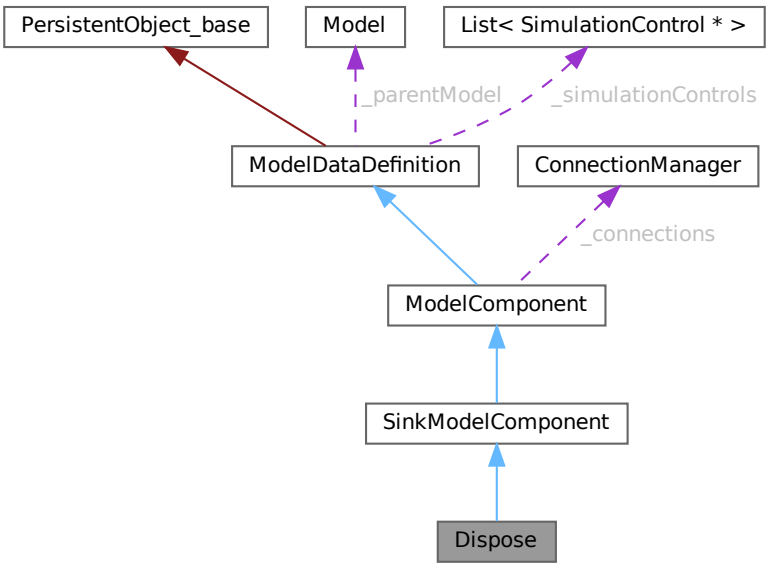
17.229 Dispose Class Reference

Sink component that ends an entity's life in the model, optionally recording its total time in system before removing it.
`#include <plugins/components/Logic/Dispose.h>`

Inheritance diagram for Dispose:



Collaboration diagram for Dispose:



Public Member Functions

- **Dispose** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from SinkModelComponent

- **SinkModelComponent** (Model *model, std::string componentTypename, std::string name="")

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _initBetweenReplications () override
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createInternalStatisticReporters () override
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.229.1 Detailed Description

Sink component that ends an entity's life in the model, optionally recording its total time in system before removing it.
Arena correspondence: the "Dispose module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", p. 32). `isReportStatistics()` (inherited) is the closest correspondence to Arena's "Record Entity Statistics" checkbox.
Known difference from Arena: only a single collapsed `TotalTimeInSystem` statistic is produced per EntityType (gated by that EntityType's own `isReportStatistics()`); Arena's separate value-added/non-value-added/wait/transfer/other time-and-cost breakdown has no confirmed GenESyS runtime equivalent — see `docs/ai_assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.2-§6.3.

17.229.2 Member Function Documentation

17.229.2.1 _check()

```
bool Dispose::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from SinkModelComponent.

17.229.2.2 _createAttachedAttributes()

```
void Dispose::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.229.2.3 _createInternalStatisticReporters()

```
void Dispose::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.229.2.4 _initBetweenReplications()

```
void Dispose::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.229.2.5 _loadInstance()

```
bool Dispose::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SinkModelComponent.

17.229.2.6 _onDispatchEvent()

```
void Dispose::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.229.2.7 _saveInstance()

```
void Dispose::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SinkModelComponent.

17.229.2.8 show()

```
std::string Dispose::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

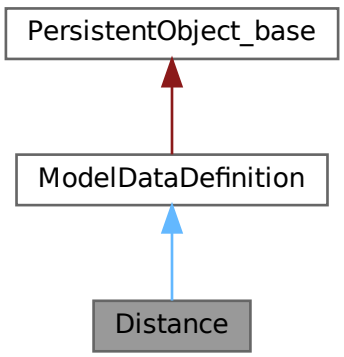
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

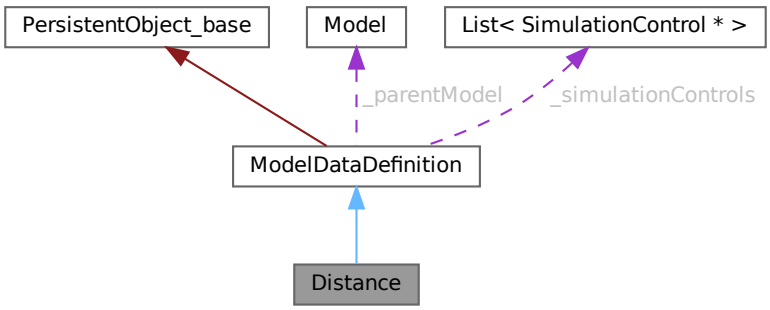
- plugins/components/Logic/Dispose.h
- plugins/components/Logic/Dispose.cpp

17.230 Distance Class Reference

Data definition holding a set of direct station-to-station distances, used by free-path (non-guided) `Transporter` units to compute travel time.
`#include <plugins/data/MaterialHandling/Distance.h>`
Inheritance diagram for Distance:



Collaboration diagram for Distance:



Public Member Functions

- **Distance** (Model ∗model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- List< DistanceEntry ∗ > ∗ **getEntries** () const
- void **insertEntry** (DistanceEntry ∗entry)
- double **getDistanceBetween** (const std::string &fromStationName, const std::string &toStationName) const

  Returns the direct distance between two stations, or -1 when no matching entry exists.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model ∗model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ∼**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model ∗ getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition ∗ getInternalData (const std::string &name) const

Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

not just an easy access to trace manager, but wrappers to check if specific trace level applies

- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.230.1 Detailed Description

Data definition holding a set of direct station-to-station distances, used by free-path (non-guided) `Transporter` units to compute travel time.
Arena correspondence: the "Distance module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Transfer Panel", p. 106), which defines a Distance Set of (Beginning Station, Ending Station, Distance) triples for free-path transporters.
Known limitation of this initial implementation: only a direct from/to (or to/from, when `bidirectional`) lookup is supported — there is no multi-hop shortest-path resolution when two stations are not directly listed.

17.230.2 Member Function Documentation

17.230.2.1 _check()

```
bool Distance::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.230.2.2 _loadInstance()

```
bool Distance::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.230.2.3 _saveInstance()

```
void Distance::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.230.2.4 show()

`std::string Distance::show ( ) [override], [virtual]`
Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/MaterialHandling/Distance.h
- plugins/data/MaterialHandling/Distance.cpp

17.231 DistanceEntry Class Reference

One directed (from-station, to-station, length) entry of a Distance set, used to look up a free-path Transporter's travel length between two stations.
`#include <plugins/data/MaterialHandling/Distance.h>`

Public Member Functions

- **DistanceEntry** (std::string fromStationName, std::string toStationName, double length, bool bidirectional=true)

Public Attributes

- std::string **fromStationName**
- std::string **toStationName**
- double **length**
- bool **bidirectional**

17.231.1 Detailed Description

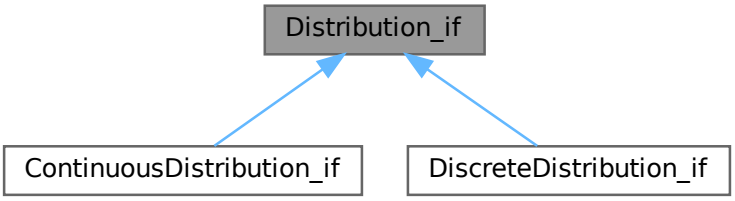
One directed (from-station, to-station, length) entry of a Distance set, used to look up a free-path Transporter's travel length between two stations.
The documentation for this class was generated from the following file:

- plugins/data/MaterialHandling/Distance.h

17.232 Distribution_if Class Reference

Base abstraction for a theoretical probability distribution.
`#include <tools/Statistics/Distribution_if.h>`

Inheritance diagram for Distribution_if:



Public Member Functions

- virtual std::string **name** () const =0

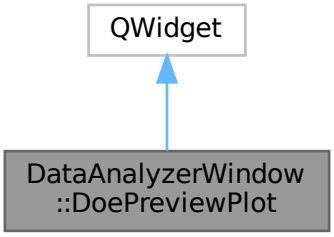
17.232.1 Detailed Description

Base abstraction for a theoretical probability distribution.
The documentation for this class was generated from the following file:

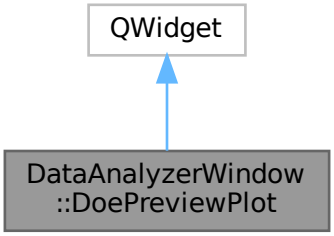
- tools/Statistics/Distribution_if.h

17.233 DataAnalyzerWindow::DoePreviewPlot Class Reference

Inheritance diagram for DataAnalyzerWindow::DoePreviewPlot:



Collaboration diagram for DataAnalyzerWindow::DoePreviewPlot:



Public Member Functions

- **DoePreviewPlot** (QWidget *parent=nullptr)
- void **setMode** (const QString &mode)

Protected Member Functions

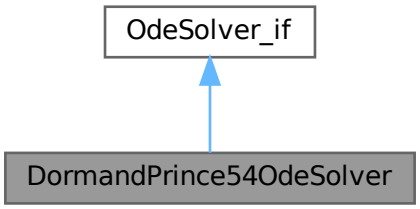
- void **paintEvent** (QPaintEvent *) override

The documentation for this class was generated from the following file:

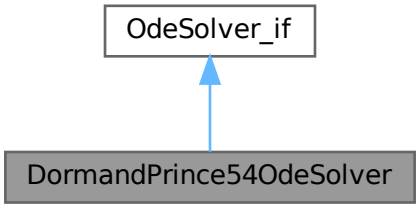
- applications/gui/genesys/tools/dataanalyzer/DataAnalyzerWindow.cpp

17.234 DormandPrince54OdeSolver Class Reference

Inheritance diagram for DormandPrince54OdeSolver:



Collaboration diagram for DormandPrince54OdeSolver:



Public Member Functions

- **DormandPrince54OdeSolver** (double relativeTolerance=1e-6, double absoluteTolerance=1e-9, double safetyFactor=0.9, double minScale=0.2, double maxScale=10.0, unsigned int maxSubsteps=100000u)
- double **getRelativeTolerance** () const
- double **getAbsoluteTolerance** () const
- bool advance (const OdeSystem_if &system, double t0, double dt, const double *y0, double *y1) const override

17.234.1 Member Function Documentation

17.234.1.1 advance()

```
bool DormandPrince54OdeSolver::advance (
            const OdeSystem_if & system,
            double t0,
            double dt,
            const double * y0,
            double * y1 ) const  [inline], [override], [virtual]
```

Implements OdeSolver_if.

The documentation for this class was generated from the following file:

- tools/Continuous/DormandPrince54OdeSolver.h

17.235 testing::internal::DoubleSequence< plus_one, T, sizeofT > Struct Template Reference

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.236 testing::internal::DoubleSequence< false, IndexSequence< I... >, sizeofT > Struct Template Reference

Public Types

- using **type** = IndexSequence< I...,(sizeofT+I)... >

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.237 testing::internal::DoubleSequence< true, IndexSequence< I... >, sizeofT > Struct Template Reference

Public Types

- using **type** = IndexSequence< I...,(sizeofT+I)..., 2 ∗sizeofT >

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.238 genesys::launcher::DownloadResult Struct Reference

Public Attributes

- bool **ok** = false
- bool **cancelled** = false
- QString **bundlePartPath**
- QString **signaturePartPath**
- QString **error**

The documentation for this struct was generated from the following file:

- applications/launcher/UpdateDownloader.h

17.239 DrawingItem Struct Reference

Public Attributes

- QGraphicsItem ∗ **myDrawingItem**
- QPointF **initialPosition**

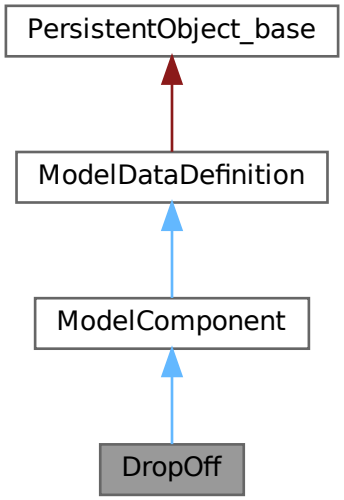
The documentation for this struct was generated from the following file:

- applications/gui/genesys/actions/AddUndoCommand.h

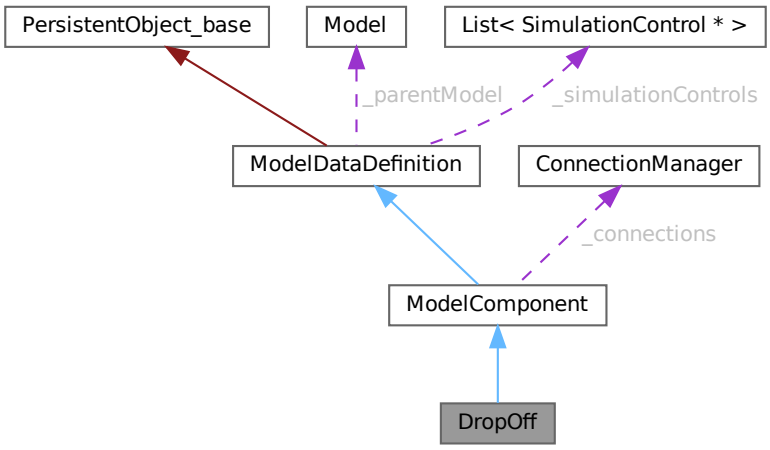
17.240 DropOff Class Reference

Placeholder for releasing a subset of a batched group's members.
`#include <plugins/components/Decisions/DropOff.h>`

Inheritance diagram for DropOff:



Collaboration diagram for DropOff:



Public Member Functions

- **DropOff** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setQuantityExpression** (std::string quantityExpression)
- std::string **getQuantityExpression** () const
- void **setStartingRankExpression** (std::string startingRankExpression)
- std::string **getStartingRankExpression** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.240.1 Detailed Description

Placeholder for releasing a subset of a batched group's members.
Arena correspondence: the "Dropoff module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 54).

Warning

Incomplete stub template: this class declares no fields at all, and `_onDispatchEvent()` unconditionally forwards the entity without interacting with any EntityGroup. See `docs/ai_↩ assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.17.

17.240.2 Member Function Documentation

17.240.2.1 _check()

```
bool DropOff::_check (
            std::string & errorMessage ) [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.240.2.2 _loadInstance()

```
bool DropOff::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.240.2.3 _onDispatchEvent()

```
void DropOff::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.240.2.4 _saveInstance()

```
void DropOff::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.240.2.5 show()

```
std::string DropOff::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

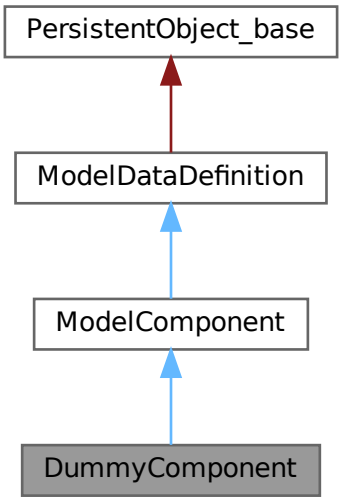
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/Decisions/DropOff.h
- plugins/components/Decisions/DropOff.cpp

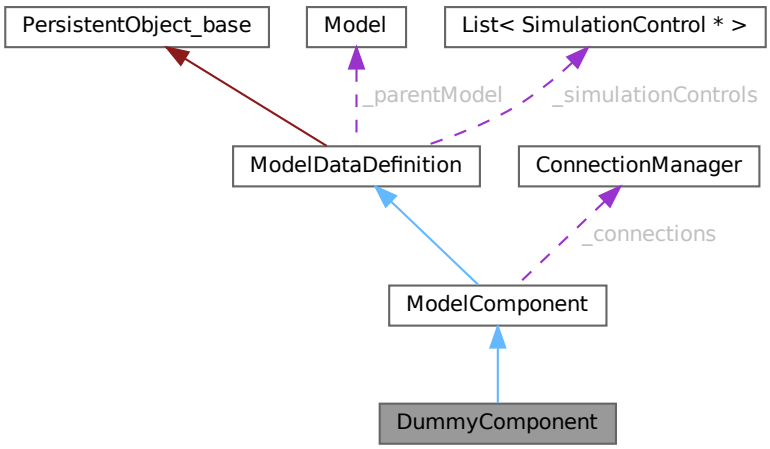
17.241 DummyComponent Class Reference

```
#include <plugins/components/Template/DummyComponent.h>
```

Inheritance diagram for DummyComponent:



Collaboration diagram for DummyComponent:



Public Member Functions

- DummyComponent (Model *model, std::string name="")
  constructors
- virtual std::string show () override
  new public user methods for this component

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

constructors

- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.241.1 Detailed Description

This component ...

17.241.2 Constructor & Destructor Documentation

17.241.2.1 DummyComponent()

```
DummyComponent::DummyComponent (
            Model * model,
            std::string name = "" )
```

constructors
Externalize function GetPluginInformation to be accessible throught dynamic linked library.

17.241.3 Member Function Documentation

17.241.3.1 _loadInstance()

```
bool DummyComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.241.3.2 _onDispatchEvent()

```
void DummyComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

This method is only for ModelComponents, not ModelDataElements.
Implements ModelComponent.

17.241.3.3 _saveInstance()

```
void DummyComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.241.3.4 show()

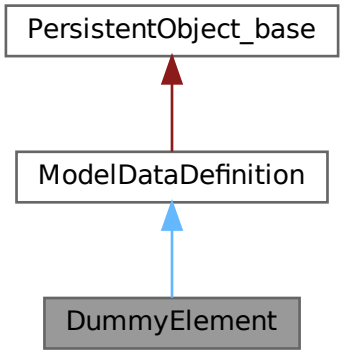
```
std::string DummyComponent::show ( )  [override], [virtual]
```

new public user methods for this component
virtual public methods
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

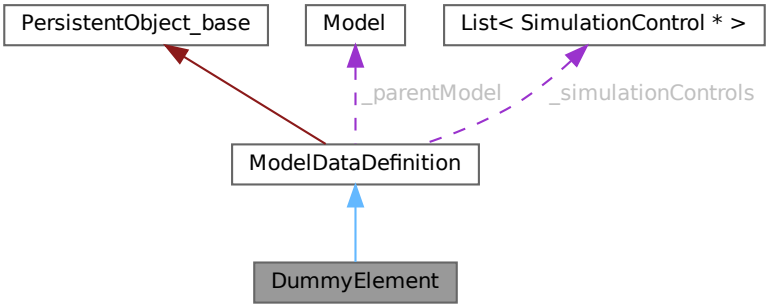
- plugins/components/Template/DummyComponent.h
- plugins/components/Template/DummyComponent.cpp

17.242 DummyElement Class Reference

Inheritance diagram for DummyElement:



Collaboration diagram for DummyElement:



Public Member Functions

- **DummyElement** (Model *model, std::string name="")

  constructors
- void **setSomeString** (const std::string &someString)

  new public user methods for this component
- std::string **getSomeString** () const
- void **setSomeUint** (unsigned int someUint)
- unsigned int **getSomeUint** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

not just an easy access to trace manager, but wrappers to check if specific trace level applies

- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.242.1 Member Function Documentation

17.242.1.1 _check()

```
bool DummyElement::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.242.1.2 _loadInstance()

```
bool DummyElement::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelDataDefinition.

17.242.1.3 _saveInstance()

```
void DummyElement::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.242.1.4 show()

```
std::string DummyElement::show ( )  [override], [virtual]
```

virtual public methods
Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/Template/DummyElement.h
- plugins/data/Template/DummyElement.cpp

17.243 EditCommandController Class Reference

Controller for edit/clipboard/grouping command orchestration in the scene.
`#include <applications/gui/genesys/controllers/EditCommandController.h>`

Public Member Functions

- **EditCommandController** (Simulator *simulator, ModelGraphicsView *graphicsView, std::function< void()> actualizeActions, bool *cut, QList< GraphicalModelComponent * > **gmcCopies, QList< GraphicalConnection * > **portsCopies, QList< QGraphicsItem * > **drawCopy, QList< QGraphics↩ ItemGroup * > **groupCopy)

  Creates the edit-command controller used by MainWindow compatibility slots.
- void **onActionEditUndoTriggered** () const

  Delegates undo command handling without changing scene undo semantics.
- void **onActionEditRedoTriggered** () const

  Delegates redo command handling without changing scene undo semantics.
- void **onActionEditCutTriggered** () const

  Delegates cut command handling while preserving compatibility copy buffers.
- void **onActionEditCopyTriggered** () const

  Delegates copy command handling while preserving compatibility copy buffers.
- void **onActionEditPasteTriggered** () const

  Delegates paste command handling while preserving compatibility copy buffers.
- void **onActionEditDeleteTriggered** () const

  Delegates delete command handling while preserving undo integration.
- void **onActionEditGroupTriggered** () const

  Delegates group command handling through scene grouping APIs.
- void **onActionEditUngroupTriggered** () const

  Delegates ungroup command handling through scene grouping APIs.
- void **onActionViewGroupTriggered** () const

  Delegates view-group command handling through scene grouping APIs.
- void **onActionViewUngroupTriggered** () const

  Delegates view-ungroup command handling through scene grouping APIs.
- void **saveItemForCopy** (QList< GraphicalModelComponent * > *gmcList, QList< GraphicalConnection * > *connList) const

  Stores eligible selected items in compatibility clipboard structures.
- void **helpCopy** () const

  Performs deep-copy preparation for subsequent paste command execution.

17.243.1 Detailed Description

Controller for edit/clipboard/grouping command orchestration in the scene.
This Phase-9 controller keeps MainWindow slots as compatibility wrappers while centralizing clipboard buffers, grouping operations, and undo-aware edit command dispatch against the current graphical scene.
Responsibilities:

- execute undo/redo/cut/copy/paste/delete/group/ungroup command flows;
- preserve legacy clipboard structures used by graphical components and drawings;

- trigger action refresh callbacks after command execution.

Boundaries:

- it does not own scene lifetime (resolved from injected graphics view);
- it does not persist clipboard state across sessions;
- it does not manage simulation events, model lifecycle, or property-editor synchronization.

The documentation for this class was generated from the following files:

- applications/gui/genesys/controllers/EditCommandController.h
- applications/gui/genesys/controllers/EditCommandController.cpp

17.244 EditorFactoryPrivate< Editor > Class Template Reference

Public Types

- typedef QList< Editor * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< Editor *, QtProperty * > **EditorToPropertyMap**

Public Member Functions

- Editor * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, Editor *e)
- void **slotEditorDestroyed** (QObject *object)

Public Attributes

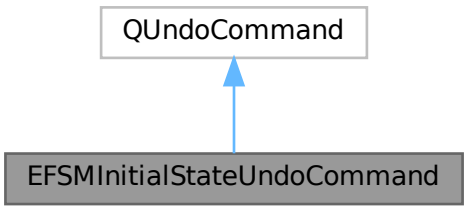
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

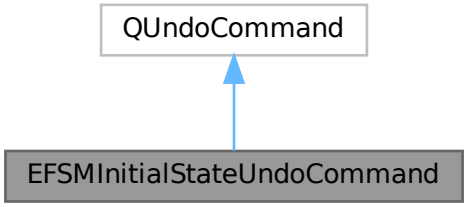
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.245 EFSMInitialStateUndoCommand Class Reference

Inheritance diagram for EFSMInitialStateUndoCommand:



Collaboration diagram for EFSMInitialStateUndoCommand:



Public Member Functions

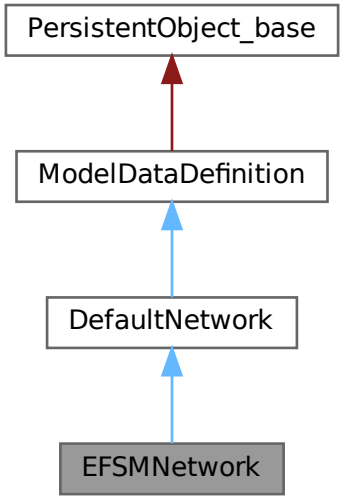
- **EFSMInitialStateUndoCommand** (EFSMNetwork *network, FSMState *state, QUndoCommand *parent=nullptr)
- void **undo** () override
- void **redo** () override

The documentation for this class was generated from the following files:

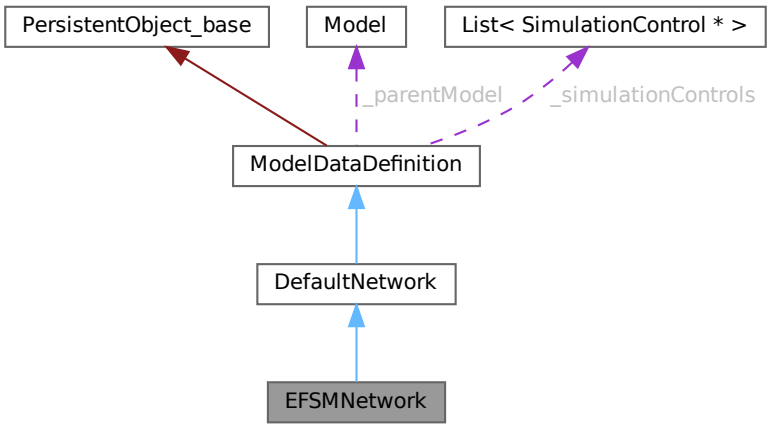
- applications/gui/genesys/actions/EFSMInitialStateUndoCommand.h
- applications/gui/genesys/actions/EFSMInitialStateUndoCommand.cpp

17.246 EFSMNetwork Class Reference

Extended finite-state-machine network specialization.
`#include <plugins/data/ModalModel/EFSMNetwork.h>`
Inheritance diagram for EFSMNetwork:



Collaboration diagram for EFSMNetwork:



Public Types

- enum class **ConflictPolicy** : unsigned int { **MODEL_ERROR** = 0 , **NONDETERMINISTIC_CHOICE** = 1 , **DETERMINISTIC_PRIORITY** = 2 }

Public Member Functions

- **EFSMNetwork** (Model ∗model, std::string name="")
- void **addState** (FSMState ∗state)
- void **removeState** (FSMState ∗state)
- List< FSMState ∗ > ∗ **getStates** () const
- void **addTransition** (EFSMTransition ∗transition)
- void **removeTransition** (EFSMTransition ∗transition)
- std::unique_ptr< EFSMTransition > detachTransition (EFSMTransition ∗transition, unsigned int ∗ordinal=nullptr)
- bool restoreTransition (EFSMTransition ∗transition, unsigned int ordinal)
- List< EFSMTransition ∗ > ∗ **getTransitions** () const
- void **setInitialState** (FSMState ∗state)
- FSMState ∗ **getInitialState** () const
- FSMState ∗ **getCurrentState** () const
- void **setCurrentState** (FSMState ∗state)
- void **setConflictPolicy** (ConflictPolicy policy)
- ConflictPolicy **getConflictPolicy** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from DefaultNetwork

- **DefaultNetwork** (Model ∗model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< DefaultNetwork >())
- unsigned int **addInputPort** (const std::string &portName)

  Declares a new input port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **addOutputPort** (const std::string &portName)

  Declares a new output port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).

- unsigned int **getNumInputPorts** () const
- unsigned int **getNumOutputPorts** () const
- std::string **getInputPortName** (unsigned int index) const
- std::string **getOutputPortName** (unsigned int index) const
- int **getInputPortIndex** (const std::string &portName) const

  Returns the input port index for `portName`, or -1 when not declared.
- int **getOutputPortIndex** (const std::string &portName) const

  Returns the output port index for `portName`, or -1 when not declared.
- double **getActivationCount** () const

  Number of times activate() has been called since the last replication reset.
- NetworkActivationResult activate (const NetworkActivationFrame &frame)

  Runs exactly one network activation for `frame`.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::string **convertConflictPolicyToString** (ConflictPolicy policy)
- static ConflictPolicy **convertStringToConflictPolicy** (const std::string &policy)

Static Public Member Functions inherited from DefaultNetwork

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

could be overriden by derived classes

- virtual void _initBetweenReplications () override
- virtual NetworkActivationResult _activate (const NetworkActivationFrame &frame) override

  Formalism-specific activation behavior.

Protected Member Functions inherited from DefaultNetwork

- virtual void _createInternalStatisticReporters () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)

- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.246.1 Detailed Description

Extended finite-state-machine network specialization.
`EFSMNetwork` owns the modal state machine semantics that used to live in `ModalModelDefault`: the current state, the initial state, state nodes and EFSM transitions. A `ModalModelDefault` only adapts process entities to this network through explicit input/output bindings.
Transition ownership: `addTransition()` and persistence reload take exclusive ownership of heap-allocated `EFSMTransition` objects and delete them from the network destructor / reload path. Callers must not stack-allocate transitions that are registered with the network and must not delete them separately.
This first implementation supports one deterministic activation step: evaluate outgoing transitions of the current state, select the enabled transition with the lowest priority value, update the current state, and expose the transition output expression on output port 0 when present.

17.246.2 Member Function Documentation

17.246.2.1 _activate()

```
NetworkActivationResult EFSMNetwork::_activate (
            const NetworkActivationFrame & frame )  [override], [protected], [virtual]
```

Formalism-specific activation behavior.
The base implementation performs no state change and returns a result sized to `getNumOutputPorts()` with every output absent. Specializations override this to implement EFSM transition firing, one DTMC step, one CPN firing step, etc. Must not receive or depend on `Entity*` (architecture reference §8.4, §19).
Reimplemented from DefaultNetwork.

17.246.2.2 _check()

```
bool EFSMNetwork::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from DefaultNetwork.

17.246.2.3 _initBetweenReplications()

```
void EFSMNetwork::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from DefaultNetwork.

17.246.2.4 _loadInstance()

```
bool EFSMNetwork::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from DefaultNetwork.

17.246.2.5 _saveInstance()

```
void EFSMNetwork::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from DefaultNetwork.

17.246.2.6 detachTransition()

```
std::unique_ptr< EFSMTransition > EFSMNetwork::detachTransition (
            EFSMTransition * transition,
            unsigned int * ordinal = nullptr )
```

Detaches an owned transition without destroying it. The returned owner can safely retain the transition for an undo command. `ordinal` receives its former persisted sequence position. Returns empty if it is not registered.

17.246.2.7 restoreTransition()

```
bool EFSMNetwork::restoreTransition (
            EFSMTransition * transition,
            unsigned int ordinal )
```

Reattaches a detached transition at its prior sequence position. Ownership transfers to the network only when this returns true.

17.246.2.8 show()

```
std::string EFSMNetwork::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from DefaultNetwork.
The documentation for this class was generated from the following files:

- plugins/data/ModalModel/EFSMNetwork.h
- plugins/components/ModalModel/EFSMNetwork.cpp

17.247 EFSMNetworkDiagramLayout Struct Reference

GUI-only presentation layout for one EFSMNetwork diagram.
`#include <applications/gui/genesys/services/EFSMNetworkDiagramLayout.h>`

Classes

- struct StatePosition
- struct TransitionLayout

Public Member Functions

- bool **isEmpty** () const

Public Attributes

- unsigned long **networkId** = 0
- QString **networkName**
- QVector< StatePosition > **states**
- QVector< TransitionLayout > **transitions**

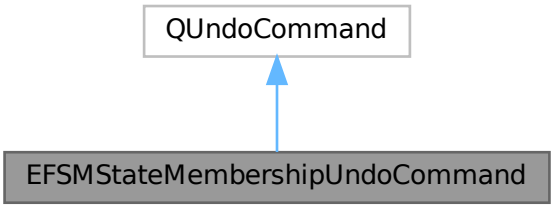
17.247.1 Detailed Description

GUI-only presentation layout for one EFSMNetwork diagram.
This is purely visual/presentation data: node positions and optional transition control/label points. It never carries mathematical state (current/initial state, guards, actions, priorities, probabilities) – that remains the exclusive responsibility of `EFSMNetwork`/`FSMState`/ `EFSMTransition`. A missing or empty layout is valid; the future viewer falls back to deterministic automatic positioning.
`EFSMTransition` has no persistent identity in the kernel (unlike `FSMState`, which is a `ModelDataDefinition`). The graphical entry uses `ordinal` as the zero-based index in the complete `getTransitions()` list. That sequence is written and reconstructed by the EFSM `.gen` persistence contract and distinguishes parallel and unnamed transitions without using display names. Source/destination names remain integrity checks. The identity is stable only while transition order is unchanged.
The documentation for this struct was generated from the following file:

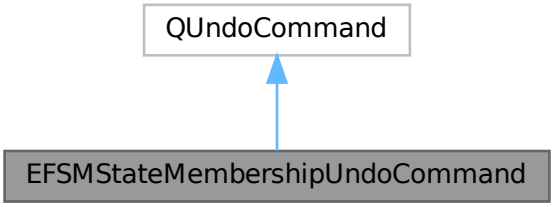
- applications/gui/genesys/services/EFSMNetworkDiagramLayout.h

17.248 EFSMStateMembershipUndoCommand Class Reference

Inheritance diagram for EFSMStateMembershipUndoCommand:



Collaboration diagram for EFSMStateMembershipUndoCommand:



Public Member Functions

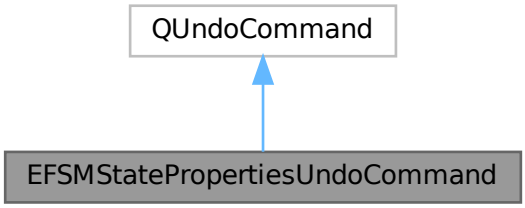
- **EFSMStateMembershipUndoCommand** (Model *model, EFSMNetwork *network, std::unique_ptr< FSMState > state, QUndoCommand *parent=nullptr)
- **EFSMStateMembershipUndoCommand** (Model *model, EFSMNetwork *network, FSMState *state, QUndoCommand *parent=nullptr)
- void **undo** () override
- void **redo** () override

The documentation for this class was generated from the following files:

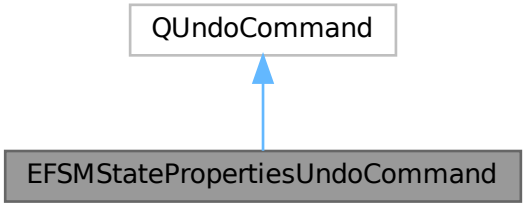
- applications/gui/genesys/actions/EFSMStateMembershipUndoCommand.h
- applications/gui/genesys/actions/EFSMStateMembershipUndoCommand.cpp

17.249 EFSMStatePropertiesUndoCommand Class Reference

Inheritance diagram for EFSMStatePropertiesUndoCommand:



Collaboration diagram for EFSMStatePropertiesUndoCommand:



Classes

- struct Properties

Public Member Functions

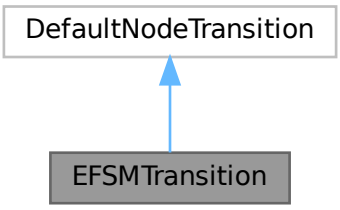
- **EFSMStatePropertiesUndoCommand** (EFSMNetwork *network, FSMState *state, Properties updated, QUndoCommand *parent=nullptr)
- void **undo** () override
- void **redo** () override

The documentation for this class was generated from the following files:

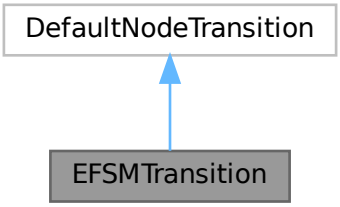
- applications/gui/genesys/actions/EFSMStatePropertiesUndoCommand.h
- applications/gui/genesys/actions/EFSMStatePropertiesUndoCommand.cpp

17.250 EFSMTransition Class Reference

Inheritance diagram for EFSMTransition:



Collaboration diagram for EFSMTransition:



Public Member Functions

- **EFSMTransition** (DefaultNode ∗source, DefaultNode ∗destination, std::string name="")
- void **setTriggerEvent** (std::string triggerEvent)
- std::string **getTriggerEvent** () const
- void **setProbabilityExpression** (std::string probabilityExpression)
- std::string **getProbabilityExpression** () const
- virtual bool canFire (Model ∗model, Entity ∗entity) const override
- virtual void execute (Model ∗model, Entity ∗entity) const override

Public Member Functions inherited from DefaultNodeTransition

- **DefaultNodeTransition** (DefaultNode ∗source, DefaultNode ∗destination, std::string name="")
- void **setSource** (DefaultNode ∗source)
- DefaultNode ∗ **getSource** () const
- void **setDestination** (DefaultNode ∗destination)
- DefaultNode ∗ **getDestination** () const
- void **setName** (std::string name)
- std::string **getName** () const
- void **setGuardExpression** (std::string guardExpression)
- std::string **getGuardExpression** () const
- void **setOutputExpression** (std::string outputExpression)
- std::string **getOutputExpression** () const
- void **setInputEvent** (std::string inputEvent)
- std::string **getInputEvent** () const
- void **setPriority** (unsigned int priority)
- unsigned int **getPriority** () const
- void **setProbability** (double probability)
- double **getProbability** () const
- void **setTransitionKind** (TransitionKind transitionKind)
- TransitionKind **getTransitionKind** () const

Additional Inherited Members

Public Types inherited from DefaultNodeTransition

- enum class **TransitionKind** : int { **DETERMINISTIC** = 0 , **PROBABILISTIC** = 1 }

17.250.1 Member Function Documentation

17.250.1.1 canFire()

```
bool EFSMTransition::canFire (
            Model * model,
            Entity * entity ) const  [override], [virtual]
```

Reimplemented from DefaultNodeTransition.

17.250.1.2 execute()

```
void EFSMTransition::execute (
            Model * model,
            Entity * entity ) const  [override], [virtual]
```

Reimplemented from DefaultNodeTransition.

The documentation for this class was generated from the following files:

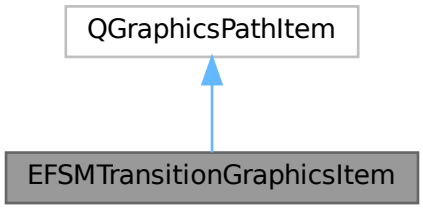
- plugins/components/ModalModel/DefaultTransitionExtensions.h
- plugins/components/ModalModel/DefaultTransitionExtensions.cpp

17.251 EFSMTransitionGraphicsItem Class Reference

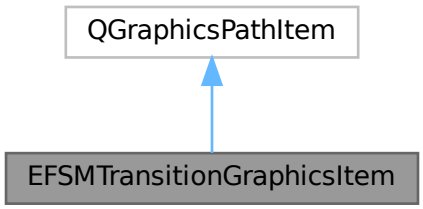
Read-only graphical projection of one mathematical EFSM transition.

`#include <applications/gui/genesys/graphicals/EFSMTransitionGraphicsItem.h>`

Inheritance diagram for EFSMTransitionGraphicsItem:



Collaboration diagram for EFSMTransitionGraphicsItem:



Public Member Functions

- **EFSMTransitionGraphicsItem** (EFSMTransition *transition, int ordinal, QGraphicsItem *parent=nullptr)
- EFSMTransition * **transition** () const
- int **ordinal** () const
- QString **transitionLabel** () const
- void **setGeometry** (const QPointF &sourceCenter, const QPointF &destinationCenter, int parallelRank, int parallelCount)

Protected Member Functions

- void **paint** (QPainter *painter, const QStyleOptionGraphicsItem *option, QWidget *widget=nullptr) override

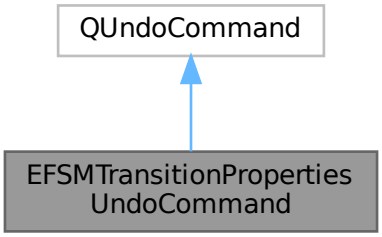
17.251.1 Detailed Description

Read-only graphical projection of one mathematical EFSM transition.
The item observes the transition and never owns or mutates it. It is kept separate from GraphicalConnection because an EFSM relation is not an Entity flow connection.
The documentation for this class was generated from the following files:

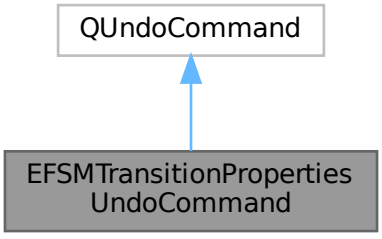
- applications/gui/genesys/graphicals/EFSMTransitionGraphicsItem.h
- applications/gui/genesys/graphicals/EFSMTransitionGraphicsItem.cpp

17.252 EFSMTransitionPropertiesUndoCommand Class Reference

Inheritance diagram for EFSMTransitionPropertiesUndoCommand:



Collaboration diagram for EFSMTransitionPropertiesUndoCommand:



Classes

- struct Properties

Public Member Functions

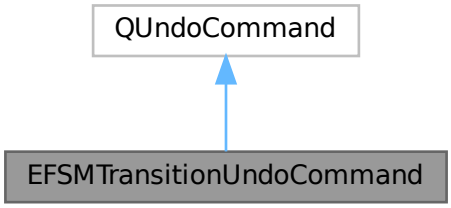
- **EFSMTransitionPropertiesUndoCommand** (EFSMNetwork *network, EFSMTransition *transition, Properties updated, QUndoCommand *parent=nullptr)
- void **undo** () override
- void **redo** () override

The documentation for this class was generated from the following files:

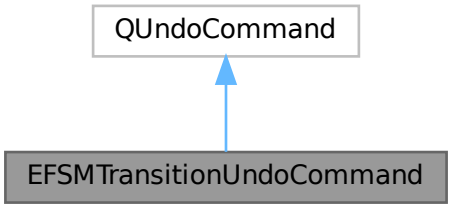
- applications/gui/genesys/actions/EFSMTransitionPropertiesUndoCommand.h
- applications/gui/genesys/actions/EFSMTransitionPropertiesUndoCommand.cpp

17.253 EFSMTransitionUndoCommand Class Reference

`#include <applications/gui/genesys/actions/EFSMTransitionUndoCommand.h>`
Inheritance diagram for EFSMTransitionUndoCommand:



Collaboration diagram for EFSMTransitionUndoCommand:



Public Member Functions

- **EFSMTransitionUndoCommand** (EFSMNetwork *network, std::unique_ptr< EFSMTransition > transition, QUndoCommand *parent=nullptr)
- **EFSMTransitionUndoCommand** (EFSMNetwork *network, EFSMTransition *transition, QUndoCommand *parent=nullptr)

- void **undo** () override
- void **redo** () override

17.253.1 Detailed Description

Undoable insertion or removal of one EFSM transition.
The Network owns an attached transition. This command owns it with a unique_ptr only while detached, so undo never retains a pointer that the Network has destroyed.
The documentation for this class was generated from the following files:

- applications/gui/genesys/actions/EFSMTransitionUndoCommand.h
- applications/gui/genesys/actions/EFSMTransitionUndoCommand.cpp

17.254 testing::internal::ElemFromList< N, T > Struct Template Reference

Public Types

- using **type** = decltype(ElemFromListImpl< typename MakeIndexSequence< N >::type >::Apply(static_↩ cast< T(*)()>(nullptr)...))

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.255 testing::internal::ElemFromListImpl< typename > Struct Template Reference

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.256 testing::internal::ElemFromListImpl< IndexSequence< I... > > Struct Template Reference

Static Public Member Functions

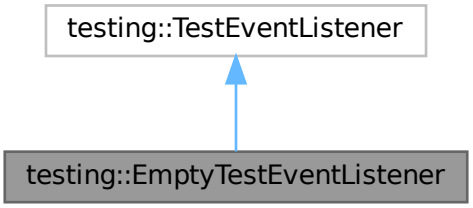
- template<typename R >
  static R **Apply** (Ignore< 0 *I >..., R(*)(),...)

The documentation for this struct was generated from the following file:

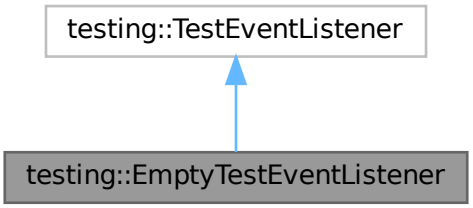
- gtest/gtest/internal/gtest-internal.h

17.257 testing::EmptyTestEventListener Class Reference

Inheritance diagram for testing::EmptyTestEventListener:



Collaboration diagram for testing::EmptyTestEventListener:



Public Member Functions

- void OnTestProgramStart (const UnitTest &) override
- void OnTestIterationStart (const UnitTest &, int) override
- void OnEnvironmentsSetUpStart (const UnitTest &) override
- void OnEnvironmentsSetUpEnd (const UnitTest &) override
- void OnTestSuiteStart (const TestSuite &) override
- void OnTestCaseStart (const TestCase &) override
- void OnTestStart (const TestInfo &) override
- void OnTestDisabled (const TestInfo &) override
- void OnTestPartResult (const TestPartResult &) override
- void OnTestEnd (const TestInfo &) override
- void OnTestSuiteEnd (const TestSuite &) override
- void OnTestCaseEnd (const TestCase &) override
- void OnEnvironmentsTearDownStart (const UnitTest &) override
- void OnEnvironmentsTearDownEnd (const UnitTest &) override
- void OnTestIterationEnd (const UnitTest &, int) override
- void OnTestProgramEnd (const UnitTest &) override

17.257.1 Member Function Documentation

17.257.1.1 OnEnvironmentsSetUpEnd()

```
void testing::EmptyTestEventListener::OnEnvironmentsSetUpEnd (
            const UnitTest &  )  [inline], [override], [virtual]
```

Implements testing::TestEventListener.

17.257.1.2 OnEnvironmentsSetUpStart()

```
void testing::EmptyTestEventListener::OnEnvironmentsSetUpStart (
            const UnitTest &  )  [inline], [override], [virtual]
```

Implements testing::TestEventListener.

17.257.1.3 OnEnvironmentsTearDownEnd()

```
void testing::EmptyTestEventListener::OnEnvironmentsTearDownEnd (
            const UnitTest &  )  [inline], [override], [virtual]
```

Implements testing::TestEventListener.

17.257.1.4 OnEnvironmentsTearDownStart()

```
void testing::EmptyTestEventListener::OnEnvironmentsTearDownStart (
            const UnitTest &  )  [inline], [override], [virtual]
```

Implements testing::TestEventListener.

17.257.1.5 OnTestCaseEnd()

```
void testing::EmptyTestEventListener::OnTestCaseEnd (
            const TestCase &  )  [inline], [override], [virtual]
```

Reimplemented from testing::TestEventListener.

17.257.1.6 OnTestCaseStart()

```
void testing::EmptyTestEventListener::OnTestCaseStart (
            const TestCase &  )  [inline], [override], [virtual]
```

Reimplemented from testing::TestEventListener.

17.257.1.7 OnTestDisabled()

```
void testing::EmptyTestEventListener::OnTestDisabled (
            const TestInfo &  )  [inline], [override], [virtual]
```

Reimplemented from testing::TestEventListener.

17.257.1.8 OnTestEnd()

```
void testing::EmptyTestEventListener::OnTestEnd (
            const TestInfo &  )  [inline], [override], [virtual]
```

Implements testing::TestEventListener.

17.257.1.9 OnTestIterationEnd()

```
void testing::EmptyTestEventListener::OnTestIterationEnd (
            const UnitTest & ,
            int  )  [inline], [override], [virtual]
```

Implements testing::TestEventListener.

17.257.1.10 OnTestIterationStart()

```
void testing::EmptyTestEventListener::OnTestIterationStart (
            const UnitTest & ,
            int  )  [inline], [override], [virtual]
```

Implements testing::TestEventListener.

17.257.1.11 OnTestPartResult()

```
void testing::EmptyTestEventListener::OnTestPartResult (
            const TestPartResult &  )  [inline], [override], [virtual]
```

Implements testing::TestEventListener.

17.257.1.12 OnTestProgramEnd()

```
void testing::EmptyTestEventListener::OnTestProgramEnd (
            const UnitTest &  )  [inline], [override], [virtual]
```

Implements testing::TestEventListener.

17.257.1.13 OnTestProgramStart()

```
void testing::EmptyTestEventListener::OnTestProgramStart (
            const UnitTest &  )  [inline], [override], [virtual]
```

Implements testing::TestEventListener.

17.257.1.14 OnTestStart()

```
void testing::EmptyTestEventListener::OnTestStart (
            const TestInfo &  )  [inline], [override], [virtual]
```

Implements testing::TestEventListener.

17.257.1.15 OnTestSuiteEnd()

```
void testing::EmptyTestEventListener::OnTestSuiteEnd (
            const TestSuite &  )  [inline], [override], [virtual]
```

Reimplemented from testing::TestEventListener.

17.257.1.16 OnTestSuiteStart()

```
void testing::EmptyTestEventListener::OnTestSuiteStart (
            const TestSuite &  )  [inline], [override], [virtual]
```

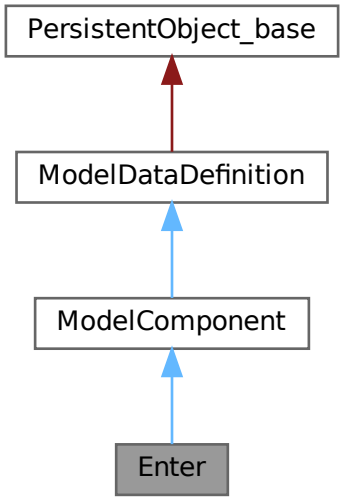
Reimplemented from testing::TestEventListener.
The documentation for this class was generated from the following file:

- gtest/gtest/gtest.h

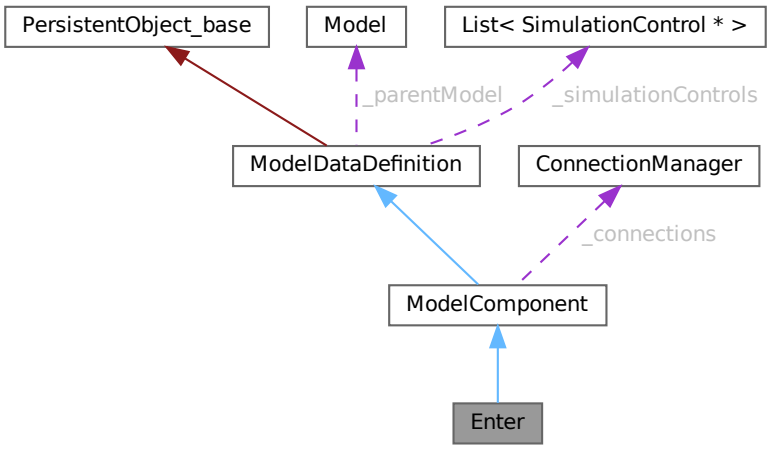
17.258 Enter Class Reference

```
#include <plugins/components/MaterialHandling/Enter.h>
```

Inheritance diagram for Enter:



Collaboration diagram for Enter:



Public Member Functions

- **Enter** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setStation** (Station ∗_station)
- void **setStationName** (std::string stationName)
- Station ∗ **getStation** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createInternalStatisticReporters () override
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.258.1 Detailed Description

Enter module DESCRIPTION The Enter module defines a station (or a set of stations) corresponding to a physical or logical location where processing occurs. When an entity arrives at an Enter module, an unloading delay may occur and any transfer device used to transfer the entity to the Enter module's station may be released. The station (or each station within the defined set) has a matching Activity Area that is used to report all times and costs accrued by the entities in this station. This Activity Area's name is the same as the station. If a parent Activity Area is defined, then it also accrues any times and costs by the entities in this station. TYPICAL USES The start of a part's production in a series of parallel processes where the part's forklift needs to be released The start of a document's processing after the document has been created where the mail clerk resource needs to be released PROMPTS Prompt Description Name Unique name of the module that will be displayed in the flowchart Station Type Type of station, either a single Station or station Set. Station Name Name of the individual station. A given station can only exist once within a model. Parent Activity Area Name of the Activity Area's parent. Associated Intersection Name of the intersection associated with this station in a guided transporter network. Report Statistics Specifies whether or not statistics will automatically be collected and stored in the report database for this station and its corresponding activity area. Set Name Name of the station set. A given station set can only exist once within a model. Save Attribute Specifies the attribute to be used to store the index into the station set for an entity entering this module. Set Members This repeat group permits you to define the individual stations that are to be members of the specified station set. A station set must have at least one member station. Active when Station

Type is Set. Station Name This field indicates the name of a station that is to be a member of this station set. A given station can only exist within a model once. Therefore, an individual station can only be the member of one station set, and that individual station may not be the name of a station in another module. Parent Activity Area Name of the Activity Area's parent for the station set member. Associated Intersection Name of the intersection associated with this station set in a guided transporter network. Report Statistics Specifies whether or not statistics will automatically be collected and stored in the report database for this station set member and its corresponding activity area. Allocation Type of category to which the entity's incurred delay time and cost will be added. Delay This field defines the delay that will be experienced by entities immediately upon arrival at the station. Units Time units used for the delay time. Transfer In If a resource, transporter, or conveyor was used to transfer the entity to this station, this can be used to release, free, or exit the device. If Release Resource is selected, the specified resource is released. If Free Transporter is selected, the specified transporter is freed. If Exit Conveyor is selected, the specified conveyor is exited. Transporter Name Name of the transporter to be freed upon arrival to the station. Active when Transfer Name is Free Transporter. Unit Number Unit number of the transporter if the transporter is multicapacity. Conveyor Name Name of the conveyor to exit upon arrival to the station. Resource Type Type of allocation, either single Resource or resource Set. Resource Name Name of the resource to release. Active when Transfer Name is Release Resource. Set Name Name of the resource set from which the resource is to be released. Release Rule Determines which member of the set is to be released, either the Last Member Seized, First Member Seized, or Specific Member. Set Index Index into the set that determines which member of the set is to be released. Attribute Name Name of the attribute that determines the instance number of the resource to release. Expression Expression value that determines the instance number of the resource to release.

17.258.2 Member Function Documentation

17.258.2.1 _check()

```
bool Enter::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.258.2.2 _createEditableDataDefinitions()

```
void Enter::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.258.2.3 _createInternalStatisticReporters()

```
void Enter::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.258.2.4 _loadInstance()

```
bool Enter::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.258.2.5 _onDispatchEvent()

```
void Enter::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.258.2.6 _saveInstance()

```
void Enter::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.258.2.7 show()

```
std::string Enter::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

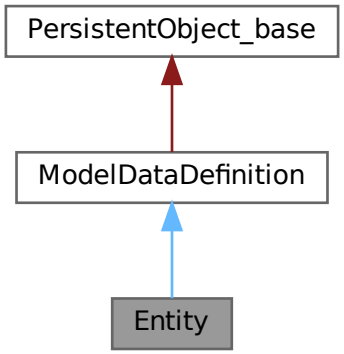
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

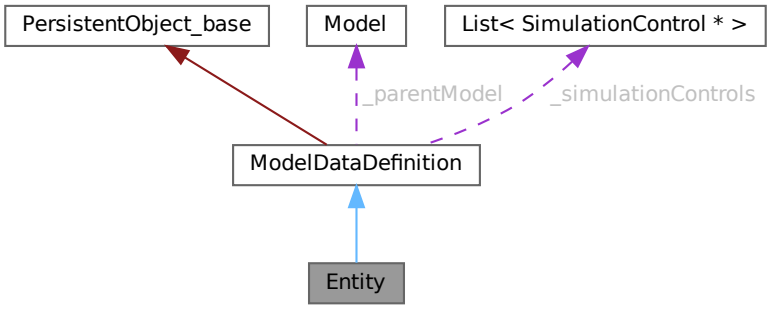
- plugins/components/MaterialHandling/Enter.h
- plugins/components/MaterialHandling/Enter.cpp

17.259 Entity Class Reference

Runtime entity (token) flowing through the model during simulation.
`#include <kernel/simulator/essentialPlugins/Entity.h>`
Inheritance diagram for Entity:



Collaboration diagram for Entity:



Public Member Functions

- virtual std::string show () override

  Returns a textual representation of this data definition.
- void setEntityTypeName (std::string entityTypeName)

  setEntityTypeName
- std::string getEntityTypeName () const

  getEntityTypeName
- void setEntityType (EntityType *entityType)

  setEntityType
- EntityType * getEntityType () const

  getEntityType
- double getAttributeValue (std::string attributeName, std::string index="")

  getAttributeValue
- double getAttributeValue (Util::identification attributeID, std::string index="")

  getAttributeValue
- void setAttributeValue (std::string attributeName, double value, std::string index="", bool createIfNot↩ Found=false)

  setAttributeValue
- void setAttributeValue (Util::identification attributeID, double value, std::string index="")

  setAttributeValue
- void **copyAttributeValues** (std::string destinationAttributeName, std::string sourceAttributeName)

  Copies all sparse values from one attribute to another for this entity.
- Util::identification entityNumber () const

  entityNumber

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Friends

- class **Model**

Additional Inherited Members

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.259.1 Detailed Description

Runtime entity (token) flowing through the model during simulation.
This is the live, per-replication object created and destroyed by `Model` for each simulated item (a "job", "customer", "part", etc.). It owns the sparse per-entity attribute values (see Attribute) and a pointer to its EntityType, but it is not itself the data definition an end user edits before running the model.
Arena correspondence: the configurable "Entity module" (name, initial picture, holding cost/hour and initial VA/↩ NVA/waiting/transfer/other costs; Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 44-45) is a data definition and corresponds to EntityType, not to this class. This class is the closest correspondence to what Arena calls an "entity" at simulation time: an instance created from an entity type, carrying attribute values and a runtime identification number.
Construction/destruction are restricted to `Model` so that entity lifetime always stays under simulator control.

17.259.2 Member Function Documentation

17.259.2.1 _check()

```
bool Entity::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.259.2.2 _loadInstance()

```
bool Entity::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.259.2.3 _saveInstance()

```
void Entity::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.259.2.4 entityNumber()

```
Util::identification Entity::entityNumber ( ) const
```

entityNumber

Returns

Unique runtime identifier for this entity instance.

17.259.2.5 getAttributeValue() [1/2]

```
double Entity::getAttributeValue (
            std::string attributeName,
            std::string index = "" )
```

getAttributeValue

Parameters

index	
attributeName	

Returns

Current attribute value for the provided name/index.

17.259.2.6 getAttributeValue() [2/2]

```
double Entity::getAttributeValue (
            Util::identification attributeID,
            std::string index = "" )
```

getAttributeValue

Parameters

index	
attributeID	

Returns

Current attribute value for the provided id/index.

17.259.2.7 getEntityType()

```
EntityType * Entity::getEntityType ( ) const
```

getEntityType

Returns

Pointer to the associated entity type.

17.259.2.8 getEntityTypeName()

```
std::string Entity::getEntityTypeName ( ) const
```

getEntityTypeName

Returns

Name of the associated entity type.

17.259.2.9 setAttributeValue() [1/2]

```
void Entity::setAttributeValue (
            std::string attributeName,
            double value,
            std::string index = "",
            bool createIfNotFound = false )
```

setAttributeValue

Parameters

index	
attributeName	
value	
createIfNotFound	

Assigns an attribute value by name and optionally creates missing attributes.

17.259.2.10 setAttributeValue() [2/2]

```
void Entity::setAttributeValue (
            Util::identification attributeID,
            double value,
            std::string index = "" )
```

setAttributeValue

Parameters

index	
attributeID	
value	

Assigns an attribute value by attribute identifier.

17.259.2.11 setEntityType()

```
void Entity::setEntityType (
```

```
            EntityType * entityType )
```
setEntityType

Parameters

entityType	

Directly assigns the entity type pointer.

17.259.2.12 setEntityTypeName()

```
void Entity::setEntityTypeName (
            std::string entityTypeName )
```
setEntityTypeName

Parameters

entityTypeName	

Sets entity type by name (indirect lookup through model data manager).

17.259.2.13 show()

```
std::string Entity::show ( )  [override], [virtual]
```
Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

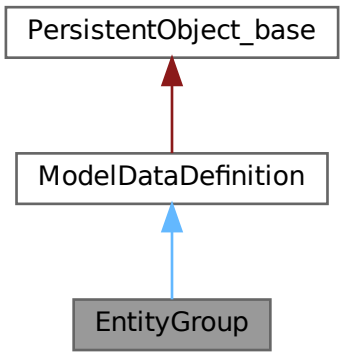
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

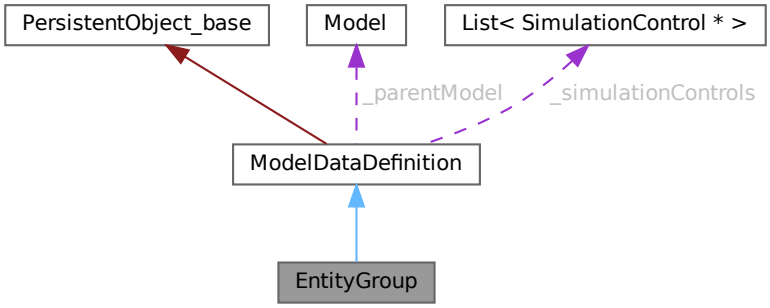
- kernel/simulator/essentialPlugins/Entity.h
- kernel/simulator/essentialPlugins/Entity.cpp

17.260 EntityGroup Class Reference

Inheritance diagram for EntityGroup:



Collaboration diagram for EntityGroup:



Public Member Functions

- **EntityGroup** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **insertElement** (unsigned int idKey, Entity *modeldatum)
- void **removeElement** (unsigned int idKey, Entity *modeldatum)
- List< Entity * > * **getGroup** (unsigned int idKey)

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createInternalStatisticReporters () override
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

not just an easy access to trace manager, but wrappers to check if specific trace level applies

- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.260.1 Member Function Documentation

17.260.1.1 _check()

```
bool EntityGroup::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.260.1.2 _createAttachedAttributes()

```
void EntityGroup::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.260.1.3 _createInternalStatisticReporters()

```
void EntityGroup::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.260.1.4 _initBetweenReplications()

```
void EntityGroup::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.260.1.5 _loadInstance()

```
bool EntityGroup::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.260.1.6 _saveInstance()

```
void EntityGroup::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.260.1.7 show()

```
std::string EntityGroup::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

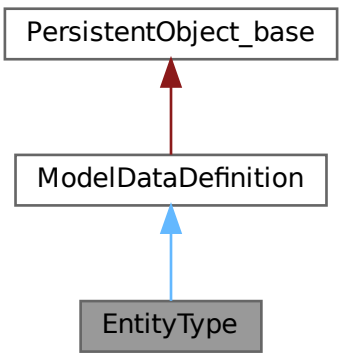
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

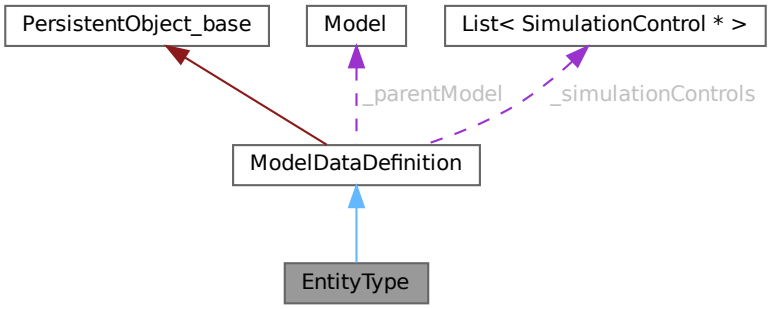
- plugins/data/Grouping/EntityGroup.h
- plugins/data/Grouping/EntityGroup.cpp

17.261 EntityType Class Reference

Data definition describing one class of entity that Create-family components may instantiate at simulation time.
`#include <kernel/simulator/essentialPlugins/EntityType.h>`
Inheritance diagram for EntityType:



Collaboration diagram for EntityType:



Public Member Functions

- **EntityType** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **setInitialWaitingCost** (double _initialWaitingCost)
- double **initialWaitingCost** () const
- void **setInitialOtherCost** (double _initialOtherCost)
- double **initialOtherCost** () const
- void **setInitialNVACost** (double _initialNVACost)
- double **initialNVACost** () const
- void **setInitialVACost** (double _initialVACost)
- double **initialVACost** () const
- void **setInitialPicture** (std::string _initialPicture)
- std::string **initialPicture** () const
- StatisticsCollector * addGetStatisticsCollector (std::string name)

  addGetStatisticsCollector

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.

- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.

- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.

- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.

- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes

- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

- virtual void _initBetweenReplications () override
- virtual void _createInternalStatisticReporters () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes

- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.261.1 Detailed Description

Data definition describing one class of entity that Create-family components may instantiate at simulation time.
Arena correspondence: the "Entity module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 44-45), which defines an entity type's initial picture, holding cost/hour and initial value-added/non-value-added/waiting/transfer/other cost attributes.
Confirmed in this class: initial picture (`initialPicture`) and initial value-added, non-value-added, waiting and other costs. `setInitialWaitingCost()/initialWaitingCost()` and the sibling VA/NVA/Other accessors are the closest correspondence to Arena's four cost categories.
Known differences from Arena: there is no explicit "Holding Cost/Hour" field, and Arena's fifth cost category ("↩ Initial Transfer Cost") has no dedicated accessor here; a maintainer should decide whether to add them or keep the current four-category cost model. Runtime entities created from an EntityType are instances of Entity, not of this class.

17.261.2 Member Function Documentation

17.261.2.1 _createInternalStatisticReporters()

```
void EntityType::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.261.2.2 _initBetweenReplications()

```
void EntityType::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.261.2.3 _loadInstance()

```
bool EntityType::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.261.2.4 _saveInstance()

```
void EntityType::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.261.2.5 addGetStatisticsCollector()

```
StatisticsCollector * EntityType::addGetStatisticsCollector (
            std::string name )
```

addGetStatisticsCollector

Parameters

name	

Returns

17.261.2.6 show()

```
std::string EntityType::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- kernel/simulator/essentialPlugins/EntityType.h
- kernel/simulator/essentialPlugins/EntityType.cpp

17.262 PersistenceRecord::Entry Struct Reference

"Raw" entry, users should treat this as an opaque type since payloads could be in any format.
`#include <kernel/simulator/Persistence.h>`

Public Types

- enum **Kind** { **text** , **numeric** }

Public Attributes

- std::string **first**
- Payload **second**
- Kind **kind** {Kind::text}

17.262.1 Detailed Description

"Raw" entry, users should treat this as an opaque type since payloads could be in any format.
The documentation for this struct was generated from the following file:

- kernel/simulator/Persistence.h

17.263 testing::Environment Class Reference

Public Member Functions

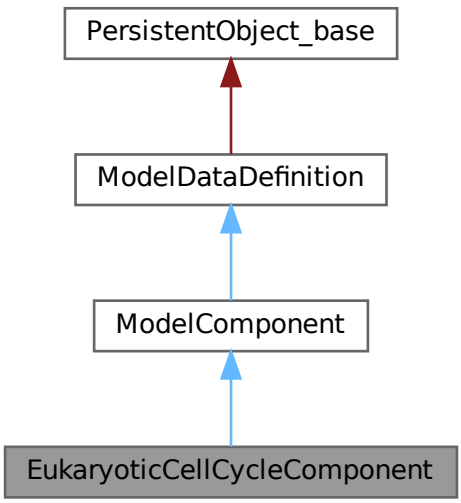
- virtual void **SetUp** ()
- virtual void **TearDown** ()

The documentation for this class was generated from the following file:

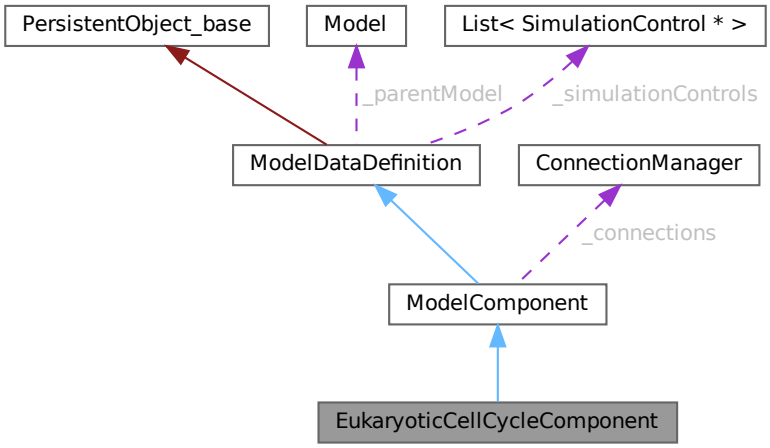
- gtest/gtest/gtest.h

17.264 EukaryoticCellCycleComponent Class Reference

Coordinates a didactic eukaryotic cell cycle over a shared WholeCellState.
`#include <plugins/components/WholeCellModeling/EukaryoticCellCycleComponent.h>`
Inheritance diagram for EukaryoticCellCycleComponent:



Collaboration diagram for EukaryoticCellCycleComponent:



Public Member Functions

- **EukaryoticCellCycleComponent** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)
- WholeCellState * **getWholeCellState** () const
- void **setDeltaT** (double deltaT)
- double **getDeltaT** () const
- void **setAdvanceWholeCellClock** (bool advance)
- bool **getAdvanceWholeCellClock** () const
- void **setCytosolCompartmentName** (std::string name)
- std::string **getCytosolCompartmentName** () const
- void **setNucleusCompartmentName** (std::string name)
- std::string **getNucleusCompartmentName** () const
- void **setMitochondrionCompartmentName** (std::string name)
- std::string **getMitochondrionCompartmentName** () const
- void **setBudCompartmentName** (std::string name)
- std::string **getBudCompartmentName** () const
- void **setEnergyMetaboliteKey** (std::string key)
- std::string **getEnergyMetaboliteKey** () const
- void **setCytosolicEnergyMetaboliteKey** (std::string key)
- std::string **getCytosolicEnergyMetaboliteKey** () const
- void **setMitochondrialEnergyMetaboliteKey** (std::string key)
- std::string **getMitochondrialEnergyMetaboliteKey** () const
- void **setBudEnergyMetaboliteKey** (std::string key)
- std::string **getBudEnergyMetaboliteKey** () const
- void **setGrowthFluxKey** (std::string key)
- std::string **getGrowthFluxKey** () const
- void **setRespirationFluxKey** (std::string key)
- std::string **getRespirationFluxKey** () const
- void **setBudProgressKey** (std::string key)
- std::string **getBudProgressKey** () const
- void **setDnaReplicationProgressKey** (std::string key)
- std::string **getDnaReplicationProgressKey** () const

- void **setSpindleAssemblyProgressKey** (std::string key)
- std::string **getSpindleAssemblyProgressKey** () const
- void **setMitoticExitProgressKey** (std::string key)
- std::string **getMitoticExitProgressKey** () const
- void **setGlobalAtpThreshold** (double threshold)
- double **getGlobalAtpThreshold** () const
- void **setCytosolicAtpThreshold** (double threshold)
- double **getCytosolicAtpThreshold** () const
- void **setMitochondrialAtpThreshold** (double threshold)
- double **getMitochondrialAtpThreshold** () const
- void **setBudAtpThreshold** (double threshold)
- double **getBudAtpThreshold** () const
- void **setGrowthFluxThreshold** (double threshold)
- double **getGrowthFluxThreshold** () const
- void **setRespirationFluxThreshold** (double threshold)
- double **getRespirationFluxThreshold** () const
- void **setBudProgressRate** (double rate)
- double **getBudProgressRate** () const
- void **setDnaReplicationRate** (double rate)
- double **getDnaReplicationRate** () const
- void **setSpindleAssemblyRate** (double rate)
- double **getSpindleAssemblyRate** () const
- void **setMitoticExitRate** (double rate)
- double **getMitoticExitRate** () const
- void **setBudProgressThreshold** (double threshold)
- double **getBudProgressThreshold** () const
- void **setDnaReplicationThreshold** (double threshold)
- double **getDnaReplicationThreshold** () const
- void **setSpindleAssemblyThreshold** (double threshold)
- double **getSpindleAssemblyThreshold** () const
- void **setMitoticExitThreshold** (double threshold)
- double **getMitoticExitThreshold** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.264.1 Detailed Description

Coordinates a didactic eukaryotic cell cycle over a shared WholeCellState.
EukaryoticCellCycleComponent provides an explicit, event-driven lifecycle layer for budding-yeast-like whole-cell models. It interprets shared state variables and compartment-specific energy proxies, then annotates WholeCellState with a coarse-grained eukaryotic cycle:

- `newborn`
- `g1_budding`
- `s_phase`
- `g2_phase`
- `m_phase`
- `division_ready`
- `arrested`
- `dead`

The component does not execute metabolism, transport, expression, or division itself. Instead, it consumes their outputs from WholeCellState and advances four progress variables stored as pathway activities:

- bud growth progress
- DNA replication progress
- spindle assembly progress
- mitotic exit progress

This design keeps structural biology in BioCompartment, mutable system state in WholeCellState, and dynamic checkpoint logic in a dedicated ModelComponent.

17.264.2 Member Function Documentation

17.264.2.1 _check()

```
bool EukaryoticCellCycleComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.264.2.2 _createEditableDataDefinitions()

```
void EukaryoticCellCycleComponent::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.264.2.3 _initBetweenReplications()

```
void EukaryoticCellCycleComponent::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.264.2.4 _loadInstance()

```
bool EukaryoticCellCycleComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.264.2.5 _onDispatchEvent()

```
void EukaryoticCellCycleComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.264.2.6 _saveInstance()

```
void EukaryoticCellCycleComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.264.2.7 show()

```
std::string EukaryoticCellCycleComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

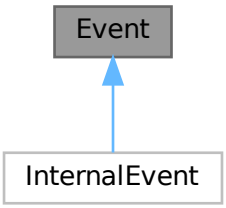
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/WholeCellModeling/EukaryoticCellCycleComponent.h
- plugins/components/WholeCellModeling/EukaryoticCellCycleComponent.cpp

17.265 Event Class Reference

`#include <kernel/simulator/Event.h>`
Inheritance diagram for Event:



Public Member Functions

- **Event** (double time, Entity *entity, ModelComponent *component, unsigned int componentinputPort↩ Number=0)
- **Event** (double time, Entity *entity, Connection *connection)
- double getTime () const

  getTime
- ModelComponent * getComponent () const

  getComponent
- Entity * getEntity () const

  getEntity
- unsigned int getComponentinputPortNumber () const

  getComponentinputPortNumber
- void setEntity (Entity *_entity)

  setEntity
- virtual std::string **show** ()

Protected Attributes

- double **_time**

17.265.1 Detailed Description

An an instantaneaous event, triggered at a certain moment by an entity upon reaching a component. The simulated time advances in discrete points in time and that are the instants that an event is triggered.

17.265.2 Member Function Documentation

17.265.2.1 getComponent()

`ModelComponent * Event::getComponent ( ) const`
getComponent

Returns

17.265.2.2 getComponentinputPortNumber()

```
unsigned int Event::getComponentinputPortNumber ( ) const
```

getComponentinputPortNumber

Returns

17.265.2.3 getEntity()

```
Entity * Event::getEntity ( ) const
```

getEntity

Returns

17.265.2.4 getTime()

```
double Event::getTime ( ) const
```

getTime

Returns

17.265.2.5 setEntity()

```
void Event::setEntity (
            Entity * _entity )
```

setEntity

Parameters

_entity	

The documentation for this class was generated from the following files:

- kernel/simulator/Event.h
- kernel/simulator/Event.cpp

17.266 Exact Class Reference

Public Member Functions

- **Exact** (int num, int den=1)
- const Exact & **operator+** (const Exact &right)
- const Exact & **operator+=** (const Exact &right)
- const Exact & **operator-=** (const Exact &right)
- const Exact & **operator*=** (const Exact &right)
- const Exact & **operator/=** (const Exact &right)
- const Exact & **operator=** (const Exact &right)
- const Exact & **operator=** (const int &right)

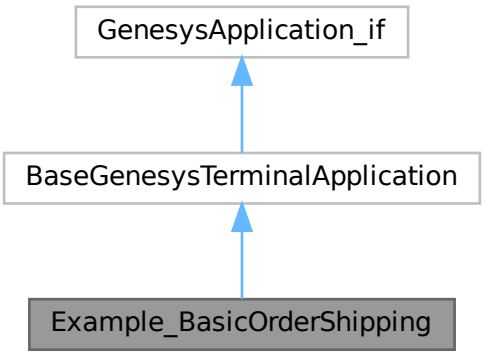
- bool **operator==** (const Exact &right)
- bool **operator==** (const double &right)
- bool **operator<** (const Exact &right)
- bool **operator<** (const double &right)
- bool **operator<=** (const Exact &right)
- bool **operator<=** (const double &right)
- bool **operator>** (const Exact &right)
- bool **operator>** (const double &right)
- bool **operator>=** (const Exact &right)
- bool **operator>=** (const double &right)
- void **Simplify** ()
- double **toFloat** ()

The documentation for this class was generated from the following file:

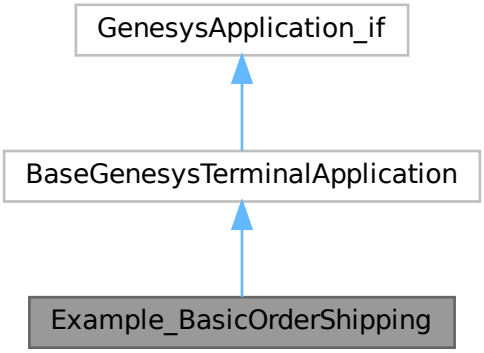
- kernel/util/Exact.h

17.267 Example_BasicOrderShipping Class Reference

Inheritance diagram for Example_BasicOrderShipping:



Collaboration diagram for Example_BasicOrderShipping:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.267.1 Member Function Documentation

17.267.1.1 main()

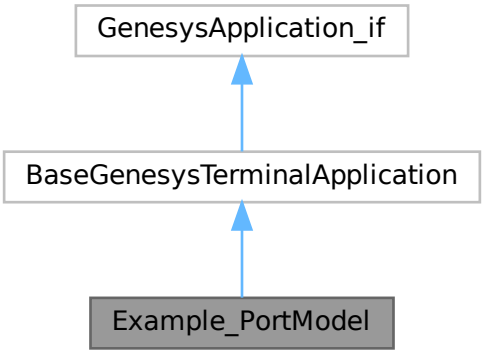
```
int Example_BasicOrderShipping::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instantiates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

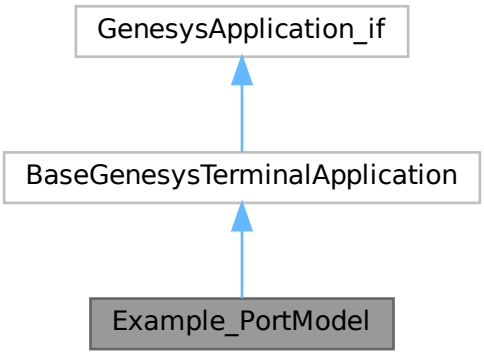
- applications/modelSpecific/arenaExamples/Example_BasicOrderShipping.h
- applications/modelSpecific/arenaExamples/Example_BasicOrderShipping.cpp

17.268 Example_PortModel Class Reference

Inheritance diagram for Example_PortModel:



Collaboration diagram for Example_PortModel:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.268.1 Member Function Documentation

17.268.1.1 main()

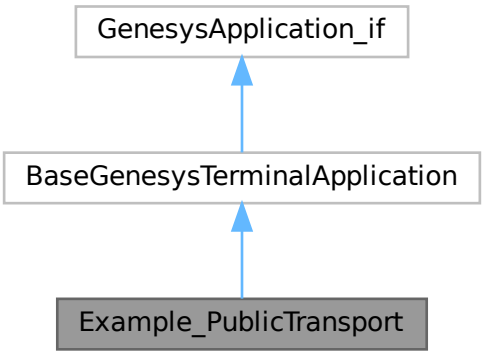
```
int Example_PortModel::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

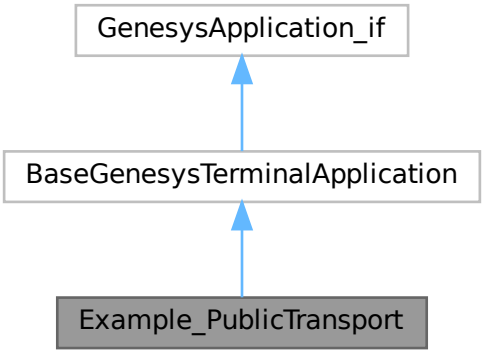
- applications/modelSpecific/arenaExamples/Example_PortModel.h
- applications/modelSpecific/arenaExamples/Example_PortModel.cpp

17.269 Example_PublicTransport Class Reference

Inheritance diagram for Example_PublicTransport:



Collaboration diagram for Example_PublicTransport:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.269.1 Member Function Documentation

17.269.1.1 main()

```
int Example_PublicTransport::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instantiates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/arenaExamples/Example_PublicTransport.h
- applications/modelSpecific/arenaExamples/Example_PublicTransport.cpp

17.270 CompartmentExchangeComponent::ExchangeRule Struct Reference

Public Attributes

- std::string **label**
- std::string **sourceRegion**
- std::string **sourceMetaboliteKey**
- std::string **targetRegion**
- std::string **targetMetaboliteKey**
- double **exchangeFraction** = 0.0
- double **baselineAmount** = 0.0
- std::string **driverPathwayKey**
- double **driverScale** = 1.0
- double **maxTransferAmount** = 0.0
- bool **conserveMass** = true

The documentation for this struct was generated from the following file:

- plugins/components/WholeCellModeling/CompartmentExchangeComponent.h

17.271 GroProgramRuntime::ExecutionResult Struct Reference

Public Attributes

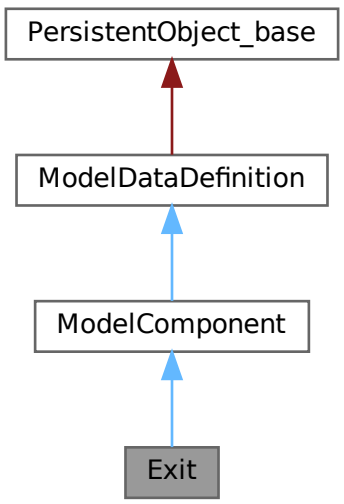
- bool **succeeded** = true
- std::string **errorMessage** = ""
- unsigned int **executedCommands** = 0
- std::vector< PopulationMutation > **populationMutations**
- std::vector< SignalMutation > **signalMutations**
- std::vector< MotionMutation > **motionMutations**
- std::vector< ColonyMutation > **colonyMutations**
- std::vector< double > **mappedCellValues**
- std::string **mappedCellExpression**
- std::vector< double > **signalMatrixValues**
- unsigned int **signalMatrixWidth** = 0
- unsigned int **signalMatrixHeight** = 0
- std::string **signalFieldDump**
- std::vector< std::string > **messages**
- std::map< std::string, double > **assignedVariables**
- std::vector< std::string > **unsupportedCommands**
- std::vector< std::string > **skippedRawStatements**

The documentation for this struct was generated from the following file:

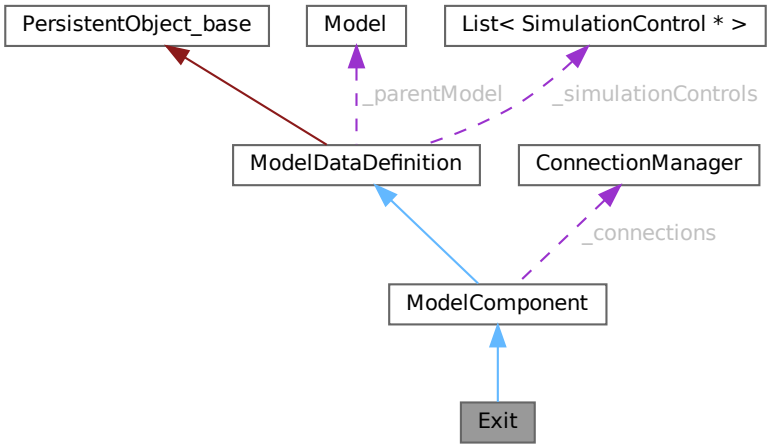
- plugins/data/BiochemicalSimulation/GroProgramRuntime.h

17.272 Exit Class Reference

`#include <plugins/components/MaterialHandling/Exit.h>`
Inheritance diagram for Exit:



Collaboration diagram for Exit:



Public Member Functions

- **Exit** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setConveyor** (Conveyor *conveyor)
- Conveyor * **getConveyor** () const
- void **setQuantityExpression** (std::string quantityExpression)
- std::string **getQuantityExpression** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.272.1 Detailed Description

Exit module DESCRIPTION The Exit module releases the entity's cells on the specified conveyor. If another entity is waiting in queue for the conveyor at the same station when the cells are released, it will then access the conveyor. TYPICAL USES Cases exit a conveyor for packing Bad parts are removed from the conveyor and disposed Passengers remove luggage from the baggage claim conveyor PROMPTS Prompt Description Name Unique name of the module that will be displayed in the flowchart. Conveyor Name Name of the conveyor on which the entity will exit. If left blank, the previously accessed conveyor is assumed.

17.272.2 of Cells Number of contiguous conveyor cells the entity will relinquish.

17.272.3 Member Function Documentation

17.272.3.1 _check()

```
bool Exit::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.272.3.2 _loadInstance()

```
bool Exit::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.272.3.3 _onDispatchEvent()

```
void Exit::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.272.3.4 _saveInstance()

```
void Exit::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.272.3.5 show()

```
std::string Exit::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/MaterialHandling/Exit.h
- plugins/components/MaterialHandling/Exit.cpp

17.273 ExperimentManager Class Reference

Manages higher-level simulation experiments.
`#include <kernel/simulator/ExperimentManager.h>`

Public Member Functions

- **ExperimentManager** (Simulator *simulator)
- SimulationExperiment * **newSimulationExperiment** ()
- void **insert** (SimulationExperiment *experiment)
- void **remove** (SimulationExperiment *experiment)
- void **setCurrent** (SimulationExperiment *experiment)
- bool **saveSimulationExperiment** (std::string filename)
- bool **loadSimulationExperiment** (std::string filename)
- unsigned int **size** ()
- SimulationExperiment * **front** ()
- SimulationExperiment * **current** ()
- SimulationExperiment * **next** ()
- List< SimulationExperiment * > * **getExperiments** () const

17.273.1 Detailed Description

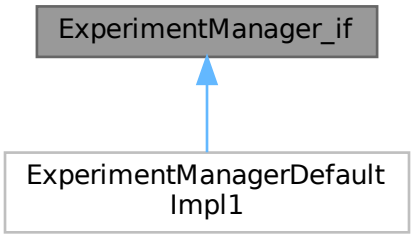
Manages higher-level simulation experiments.
Historically, GenESyS initially supported only the basic experiment controls available through ModelSimulation, especially the number of replications and the replication length.
This class represents a later attempt to move more advanced experiment design concerns into the simulation kernel, including higher-level experiment sets, scenario management and future Process Analyzer-like workflows.

This layer is still under development and should be treated as an incomplete experimental abstraction on top of the already functional ModelSimulation mechanism.
The documentation for this class was generated from the following files:

- kernel/simulator/ExperimentManager.h
- kernel/simulator/ExperimentManager.cpp

17.274 ExperimentManager_if Class Reference

Interface that coordinates execution of scenario-based simulation experiments.
`#include <kernel/simulator/ExperimetManager_if.h>`
Inheritance diagram for ExperimentManager_if:



Public Member Functions

- virtual List< SimulationScenario * > * getScenarios () const =0

  Returns the list of scenarios that compose the experiment.
- virtual void startSimulationOfScenario (SimulationScenario *scenario)=0

  Starts simulation for a specific scenario.
- virtual void startExperiment ()=0

  Starts full experiment execution (all enabled scenarios).
- virtual void stopExperiment ()=0

  Requests interruption of the currently running experiment.
- virtual void addTraceSimulationHandler (traceSimulationProcessListener traceSimulationProcessListener)=0

  Registers a callback to receive simulation progress traces for the experiment.

17.274.1 Detailed Description

Interface that coordinates execution of scenario-based simulation experiments.
An experiment manager owns a set of scenarios, starts/stops scenario runs and exposes progress tracing hooks so higher-level tools can monitor experiment status.

17.274.2 Member Function Documentation

17.274.2.1 addTraceSimulationHandler()

```
virtual void ExperimentManager_if::addTraceSimulationHandler (
            traceSimulationProcessListener traceSimulationProcessListener )  [pure virtual]
```

Registers a callback to receive simulation progress traces for the experiment.
Implemented in ExperimentManagerDefaultImpl1.

17.274.2.2 getScenarios()

```
virtual List< SimulationScenario * > * ExperimentManager_if::getScenarios ( ) const  [pure
virtual]
```

Returns the list of scenarios that compose the experiment.
Implemented in ExperimentManagerDefaultImpl1.

17.274.2.3 startExperiment()

```
virtual void ExperimentManager_if::startExperiment ( )  [pure virtual]
```

Starts full experiment execution (all enabled scenarios).
Implemented in ExperimentManagerDefaultImpl1.

17.274.2.4 startSimulationOfScenario()

```
virtual void ExperimentManager_if::startSimulationOfScenario (
            SimulationScenario * scenario )  [pure virtual]
```

Starts simulation for a specific scenario.
Implemented in ExperimentManagerDefaultImpl1.

17.274.2.5 stopExperiment()

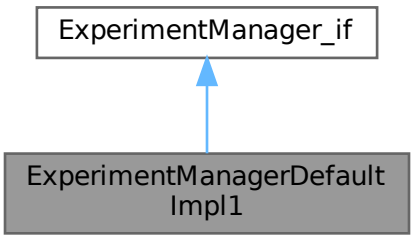
```
virtual void ExperimentManager_if::stopExperiment ( )  [pure virtual]
```

Requests interruption of the currently running experiment.
Implemented in ExperimentManagerDefaultImpl1.
The documentation for this class was generated from the following file:

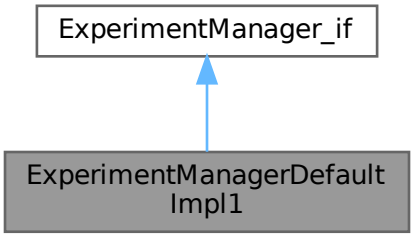
- kernel/simulator/ExperimetManager_if.h

17.275 ExperimentManagerDefaultImpl1 Class Reference

Inheritance diagram for ExperimentManagerDefaultImpl1:



Collaboration diagram for ExperimentManagerDefaultImpl1:



Public Member Functions

- virtual List< SimulationScenario * > * getScenarios () const override

  Returns the list of scenarios that compose the experiment.
- virtual void startSimulationOfScenario (SimulationScenario *scenario) override

  Starts simulation for a specific scenario.
- virtual void startExperiment () override

  Starts full experiment execution (all enabled scenarios).
- virtual void stopExperiment () override

  Requests interruption of the currently running experiment.
- virtual void addTraceSimulationHandler (traceSimulationProcessListener traceSimulationProcessListener) override

  Registers a callback to receive simulation progress traces for the experiment.

17.275.1 Member Function Documentation

17.275.1.1 addTraceSimulationHandler()

```
void ExperimentManagerDefaultImpl1::addTraceSimulationHandler (
            traceSimulationProcessListener traceSimulationProcessListener ) [override], [virtual]
```

Registers a callback to receive simulation progress traces for the experiment.
Implements ExperimentManager_if.

17.275.1.2 getScenarios()

```
List< SimulationScenario * > * ExperimentManagerDefaultImpl1::getScenarios ( ) const [override],
[virtual]
```

Returns the list of scenarios that compose the experiment.
Implements ExperimentManager_if.

17.275.1.3 startExperiment()

```
void ExperimentManagerDefaultImpl1::startExperiment ( ) [override], [virtual]
```

Starts full experiment execution (all enabled scenarios).
Implements ExperimentManager_if.

17.275.1.4 startSimulationOfScenario()

```
void ExperimentManagerDefaultImpl1::startSimulationOfScenario (
            SimulationScenario * scenario )  [override], [virtual]
```

Starts simulation for a specific scenario.
Implements ExperimentManager_if.

17.275.1.5 stopExperiment()

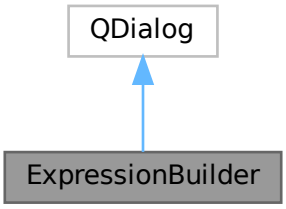
```
void ExperimentManagerDefaultImpl1::stopExperiment ( )  [override], [virtual]
```

Requests interruption of the currently running experiment.
Implements ExperimentManager_if.
The documentation for this class was generated from the following files:

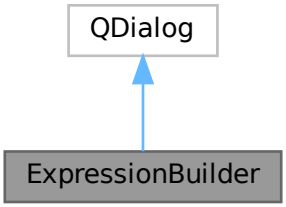
- kernel/simulator/ExperimentManagerDefaultImpl1.h
- kernel/simulator/ExperimentManagerDefaultImpl1.cpp

17.276 ExpressionBuilder Class Reference

Inheritance diagram for ExpressionBuilder:



Collaboration diagram for ExpressionBuilder:



Public Member Functions

- **ExpressionBuilder** (QWidget ∗parent=nullptr)

The documentation for this class was generated from the following files:

- applications/gui/genesys/tools/expressionbuilder/ExpressionBuilder.h
- applications/gui/genesys/tools/expressionbuilder/ExpressionBuilder.cpp

17.277 ExpressionCategory Struct Reference

Public Attributes

- QString **name**
- QList< ExpressionTemplate > **expressions**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/tools/expressionbuilder/ExpressionBuilder.h

17.278 ExpressionParameter Struct Reference

Public Attributes

- QString **name**
- QString **defaultValue**
- QString **description**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/tools/expressionbuilder/ExpressionBuilder.h

17.279 ExpressionTemplate Struct Reference

Public Attributes

- QString **name**
- QString **pattern**
- QString **description**
- QList< ExpressionParameter > **parameters**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/tools/expressionbuilder/ExpressionBuilder.h

17.280 genesys::launcher::ExtractionResult Struct Reference

Public Attributes

- bool **ok** = false
- QString **error**

The documentation for this struct was generated from the following file:

- applications/launcher/ArchiveExtractor.h

17.281 FactorialDesign Class Reference

Public Member Functions

- **FactorialDesign** (int k, int p, int r)
- bool checkFeasibility ()

  Checks the feasibility of the factorial design.
- void generateIndexCombinations ()

  Generates combinations of indexes for the Factorial Design of Experiments.
- void createColumnLabels ()

  Adds column labels for the sign table.

- void generateInputs ()

  Generates input vectors of experiments to populate the sign table.
- void calculateResultsMean ()

  Calculates the mean of results for each set of experiments from all replications.
- void createSignTable ()

  Creates the sign table for a Factorial Design of Experiments.
- void calculateStatistics ()

  Calculates statistical measures.
- double linearRegression (const std::vector< int > &xs)

  Performs linear regression based on a set of input factors.
- void showSignTable ()

  Displays the sign table.
- void showImpacts ()

  Displays the impacts of factors, interaction of factors and experimental error.
- void showLinearRegressionModel ()

  Displays the linear regression model.
- void showResiduals ()

  Displays the residuals for the experimental results.
- const std::vector< std::vector< int > > & **getIndexCombinations** () const
- const std::vector< std::string > & **getColumnLabels** () const
- const std::vector< std::vector< int > > & **getInputs** () const
- const std::vector< double > & **getResultsMean** () const
- const std::vector< std::tuple< int, std::vector< int >, std::vector< int >, double > > & **getSignTable** () const
- const std::vector< double > & **getEffectSums** () const
- const std::vector< double > & **getEffectAverages** () const
- const std::vector< int > & **getSquareSums** () const
- double **getSquareSumError** () const
- double **getSquareSumTotal** () const
- const std::vector< double > & **getFactorVariations** () const
- double **getErrorVariation** () const
- void **setResultsMatrix** (const std::vector< std::vector< int > > &matrix)

Public Attributes

- int **k**
- int **p**
- int **r**
- int **NUM_FACTORS**
- int **NUM_ROWS**
- std::vector< std::vector< int > > **resultsMatrix**

17.281.1 Member Function Documentation

17.281.1.1 calculateResultsMean()

```
void FactorialDesign::calculateResultsMean ( )
```

Calculates the mean of results for each set of experiments from all replications.
This method computes the mean of results for each set of experiments from all replications, which are stored in the 'resultsMatrix'.
The resulting mean values are stored in the 'resultsMean' member variable.

17.281.1.2 calculateStatistics()

```
void FactorialDesign::calculateStatistics ( )
```

Calculates statistical measures.

This method computes various statistical measures, including effect sums, effect averages, square sums, square sum of errors, square sum total, factor variations, and error variation. These measures are essential for assessing the significance of experimental factors and understanding the variability in the experimental results. The calculated statistics are stored in corresponding member variables for further analysis.

17.281.1.3 checkFeasibility()

```
bool FactorialDesign::checkFeasibility ( )
```

Checks the feasibility of the factorial design.

This method evaluates the feasibility of a factorial design based on the number of interactions.

Returns

true if the factorial design is feasible, false otherwise.

17.281.1.4 createColumnLabels()

```
void FactorialDesign::createColumnLabels ( )
```

Adds column labels for the sign table.

This method creates labels for experimental factors (F1, F2, ..., FN), interaction terms, and results (R1, R2, ..., Rr, R∼) based on the combinations of factor levels generated earlier. Additionally, it adds aliases for confounded factors and ensures appropriate labeling for each column in the table.

The resulting column labels are stored in the 'columnLabels' member variable.

17.281.1.5 createSignTable()

```
void FactorialDesign::createSignTable ( )
```

Creates the sign table for a Factorial Design of Experiments.

This method creates a sign table containing inputs and results of all experiments. It calculates the input for each interaction as the product of the inputs of its factors.

The resulting signTable is a vector of tuples, each representing a row in the sign table. Each tuple contains information about an experiment, facilitating further analysis and interpretation of the experimental results.

17.281.1.6 generateIndexCombinations()

```
void FactorialDesign::generateIndexCombinations ( )
```

Generates combinations of indexes for the Factorial Design of Experiments.

This method generates combinations of indexes with lengths ranging from 1 to NUM_FACTORS. The generated combinations are precomputed and utilized to access various data structures. These structures are essential for constructing the sign table and performing statistical calculations.

17.281.1.7 generateInputs()

```
void FactorialDesign::generateInputs ( )
```

Generates input vectors of experiments to populate the sign table.

This method creates input vectors representing the experimental conditions in a Factorial Design. Each input vector is a combination of factor levels, where the presence of a factor at a high level is represented by '1', whereas a low level is represented by '-1'. The generated input vectors cover all possible combinations of factor levels based on a binary counting pattern.

The resulting input vectors are stored in the 'inputs' member variable.

17.281.1.8 linearRegression()

```
double FactorialDesign::linearRegression (
            const std::vector< int > & xs )
```

Performs linear regression based on a set of input factors.

This method calculates the predicted result (y) using a linear regression model. The model is constructed using the effect averages for each factor in the Factorial Design, and the input factors provided in the 'xs' vector. The predicted result is the sum of the intercept (effect average for the first factor) and the product of each input factor with its corresponding effect average.

Parameters

xs	A vector representing the levels of input factors for which the result is predicted.

Returns

The predicted result (y) based on the linear regression model.

17.281.1.9 showImpacts()

```
void FactorialDesign::showImpacts ( )
```

Displays the impacts of factors, interaction of factors and experimental error.
This method prints the impacts of individual factors, interaction of factors and experimental error as calculated from the statistical measures. The impacts are represented as percentages of total variability, and provide insights into the significance of each factor and the contribution of experimental error to the overall variability.

17.281.1.10 showLinearRegressionModel()

```
void FactorialDesign::showLinearRegressionModel ( )
```

Displays the linear regression model.
This method prints the linear regression model for predicting the result variable based on the inputs of factors and interactions. The model is presented in the form of a mathematical expression, where coefficients are the intercept and the averages of effects for each factor, including interactions.

17.281.1.11 showResiduals()

```
void FactorialDesign::showResiduals ( )
```

Displays the residuals for the experimental results.
This method calculates and prints the residuals, which represent the differences between the observed experimental results and the values predicted by the linear regression model.

17.281.1.12 showSignTable()

```
void FactorialDesign::showSignTable ( )
```

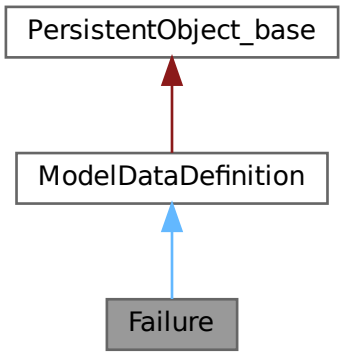
Displays the sign table.
This method prints the sign table, organized in a tabular format, to the console. The table includes column labels, experiment numbers, input combinations and results. The output is formatted to align columns, providing a clear and readable representation of the experimental conditions and their corresponding information.
The documentation for this class was generated from the following files:

- tools/FactorialDesign/FactorialDesign.h
- tools/FactorialDesign/FactorialDesign.cpp

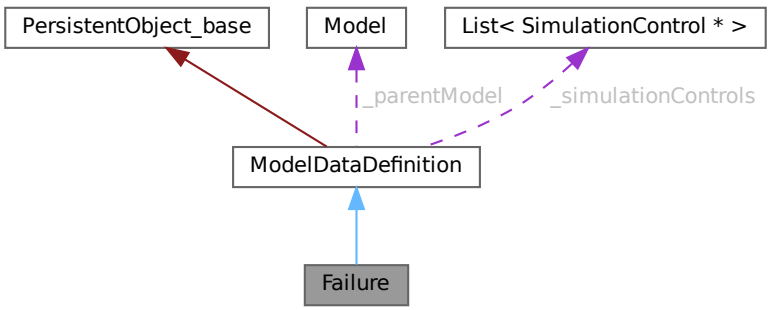
17.282 Failure Class Reference

Data definition describing a resource breakdown pattern (whole resource failure regardless of capacity) attached to one or more Resource instances.
`#include <plugins/data/DiscreteProcessing/Failure.h>`

Inheritance diagram for Failure:



Collaboration diagram for Failure:



Public Types

- enum class **FailureType** : int { **COUNT** = 0 , **TIME** = 1 , **num_elements** = 2 }
- enum class **FailureRule** : int { **IGNORE** = 0 , **PREEMPT** = 1 , **WAIT** = 2 , **num_elements** = 3 }

Public Member Functions

- **Failure** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **checkIsGoingToFailByCount** (Resource *resource)
- void **setFailureType** (FailureType _failureType)
- Failure::FailureType **getFailureType** () const
- void **setCountExpression** (std::string countExpression)
- std::string **getCountExpression** () const
- void **setDownTimeTimeUnit** (Util::TimeUnit downTimeTimeUnit)
- Util::TimeUnit **getDownTimeTimeUnit** () const
- void **setDownTimeExpression** (std::string downTimeExpression)

- std::string **getDownTimeExpression** () const
- void **setUpTimeTimeUnit** (Util::TimeUnit upTimeTimeUnit)
- Util::TimeUnit **getUpTimeTimeUnit** () const
- void **setUpTimeExpression** (std::string upTimeExpression)
- std::string **getUpTimeExpression** () const
- void **setFailureRule** (FailureRule _failureRule)
- FailureRule **getFailureRule** () const
- List< Resource * > * **falingResources** () const
- void **addResource** (Resource *newResource)
- void **removeResource** (Resource *resource)

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static std::string **convertEnumToStr** (FailureType type)
- static std::string **convertEnumToStr** (FailureRule rule)
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.282.1 Detailed Description

Data definition describing a resource breakdown pattern (whole resource failure regardless of capacity) attached to one or more Resource instances.
Arena correspondence: the "Failure module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", pp. 71-72). `FailureType` (COUNT/TIME) mirrors Arena's count-based/time-based failure types; `_countExpression`, `_upTimeExpression/_upTimeTimeUnit` and `_downTimeExpression/_downTimeTimeUnit` are the closest correspondence to Arena's Count, Up Time and Down Time fields (generalized to parsed expressions rather than static numbers). `FailureRule` (IGNORE/PREEMPT/WAIT) mirrors Arena's Failure Rule as configured per resource/failure pair.
Known difference from Arena: there is no "Uptime in this State only" field, so a time-based failure's up-time clock cannot be scoped to a single resource state (e.g. only while Busy); it always runs against total simulated time.

17.282.2 Member Function Documentation

17.282.2.1 _check()

```
bool Failure::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Validate failure expressions according to selected failure mode.
Reimplemented from ModelDataDefinition.

17.282.2.2 _initBetweenReplications()

```
void Failure::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.282.2.3 _loadInstance()

```
bool Failure::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Load Failure persistence fields.
This method mirrors _saveInstance field names to preserve a deterministic serialization contract.
Reimplemented from ModelDataDefinition.

17.282.2.4 _saveInstance()

```
void Failure::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Persist Failure configuration fields.
Reimplemented from ModelDataDefinition.

17.282.2.5 show()

```
std::string Failure::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

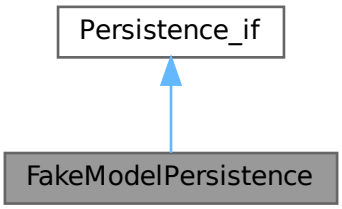
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

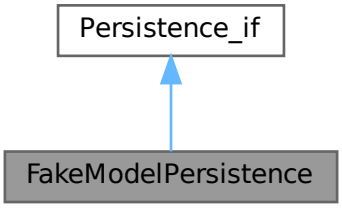
- plugins/data/DiscreteProcessing/Failure.h
- plugins/data/DiscreteProcessing/Failure.cpp

17.283 FakeModelPersistence Class Reference

Inheritance diagram for FakeModelPersistence:



Collaboration diagram for FakeModelPersistence:



Public Member Functions

- bool save (std::string) override

  Serializes and saves the model to the provided filename.
- bool load (std::string) override

  Loads and deserializes a model from file.
- bool hasChanged () override

  Indicates whether persistent state changed since the last save/load.
- void setHasChanged (bool) override

  Updates the persistent-state changed flag.
- bool getOption (Persistence_if::Options) override

  Returns whether a persistence option is currently enabled.
- void setOption (Persistence_if::Options, bool) override

  Enables or disables a persistence option.
- std::string getFormatedField (PersistenceRecord ∗) override

  Formats a set of fields as text according to the persistence format.

Additional Inherited Members

Public Types inherited from Persistence_if

- enum class **Options** : int {
  SAVEDEFAULTS = 1 , **HIDEIDKEY** = 2 , **HIDETYPEKEY** = 4 , **HIDENAMEKEY** = 8 ,
  SORTALPHLY = 16 }

17.283.1 Member Function Documentation

17.283.1.1 getFormatedField()

```
std::string FakeModelPersistence::getFormatedField (
            PersistenceRecord * fields ) [inline], [override], [virtual]
```

Formats a set of fields as text according to the persistence format.
Implements Persistence_if.

17.283.1.2 getOption()

```
bool FakeModelPersistence::getOption (
            Persistence_if::Options option ) [inline], [override], [virtual]
```

Returns whether a persistence option is currently enabled.
Implements Persistence_if.

17.283.1.3 hasChanged()

```
bool FakeModelPersistence::hasChanged ( ) [inline], [override], [virtual]
```

Indicates whether persistent state changed since the last save/load.
Implements Persistence_if.

17.283.1.4 load()

```
bool FakeModelPersistence::load (
            std::string filename ) [inline], [override], [virtual]
```

Loads and deserializes a model from file.
Implements Persistence_if.

17.283.1.5 save()

```
bool FakeModelPersistence::save (
            std::string filename )  [inline], [override], [virtual]
```

Serializes and saves the model to the provided filename.
Implements Persistence_if.

17.283.1.6 setHasChanged()

```
void FakeModelPersistence::setHasChanged (
            bool hasChanged )  [inline], [override], [virtual]
```

Updates the persistent-state changed flag.
Implements Persistence_if.

17.283.1.7 setOption()

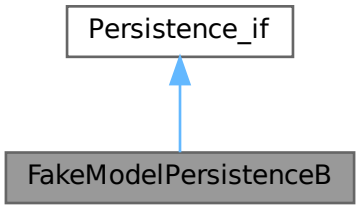
```
void FakeModelPersistence::setOption (
            Persistence_if::Options option,
            bool value )  [inline], [override], [virtual]
```

Enables or disables a persistence option.
Implements Persistence_if.
The documentation for this class was generated from the following file:

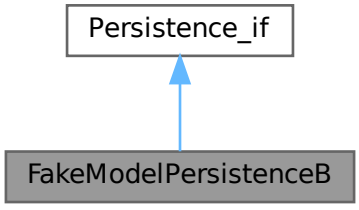
- tests/unit/test_simulator_support.cpp

17.284 FakeModelPersistenceB Class Reference

Inheritance diagram for FakeModelPersistenceB:



Collaboration diagram for FakeModelPersistenceB:



Public Member Functions

- bool save (std::string) override

  Serializes and saves the model to the provided filename.
- bool load (std::string) override

  Loads and deserializes a model from file.
- bool hasChanged () override

  Indicates whether persistent state changed since the last save/load.
- void setHasChanged (bool) override

  Updates the persistent-state changed flag.
- bool getOption (Persistence_if::Options) override

  Returns whether a persistence option is currently enabled.
- void setOption (Persistence_if::Options, bool) override

  Enables or disables a persistence option.
- std::string getFormatedField (PersistenceRecord ∗) override

  Formats a set of fields as text according to the persistence format.

Additional Inherited Members

Public Types inherited from Persistence_if

- enum class **Options** : int {
  SAVEDEFAULTS = 1 , **HIDEIDKEY** = 2 , **HIDETYPEKEY** = 4 , **HIDENAMEKEY** = 8 ,
  SORTALPHLY = 16 }

17.284.1 Member Function Documentation

17.284.1.1 getFormatedField()

```
std::string FakeModelPersistenceB::getFormatedField (
            PersistenceRecord * fields ) [inline], [override], [virtual]
```

Formats a set of fields as text according to the persistence format.
Implements Persistence_if.

17.284.1.2 getOption()

```
bool FakeModelPersistenceB::getOption (
            Persistence_if::Options option ) [inline], [override], [virtual]
```

Returns whether a persistence option is currently enabled.
Implements Persistence_if.

17.284.1.3 hasChanged()

```
bool FakeModelPersistenceB::hasChanged ( ) [inline], [override], [virtual]
```

Indicates whether persistent state changed since the last save/load.
Implements Persistence_if.

17.284.1.4 load()

```
bool FakeModelPersistenceB::load (
            std::string filename ) [inline], [override], [virtual]
```

Loads and deserializes a model from file.
Implements Persistence_if.

17.284.1.5 save()

```
bool FakeModelPersistenceB::save (
            std::string filename )  [inline], [override], [virtual]
```

Serializes and saves the model to the provided filename.
Implements Persistence_if.

17.284.1.6 setHasChanged()

```
void FakeModelPersistenceB::setHasChanged (
            bool hasChanged )  [inline], [override], [virtual]
```

Updates the persistent-state changed flag.
Implements Persistence_if.

17.284.1.7 setOption()

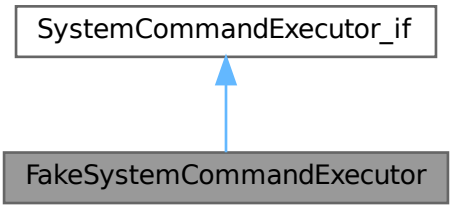
```
void FakeModelPersistenceB::setOption (
            Persistence_if::Options option,
            bool value )  [inline], [override], [virtual]
```

Enables or disables a persistence option.
Implements Persistence_if.
The documentation for this class was generated from the following file:

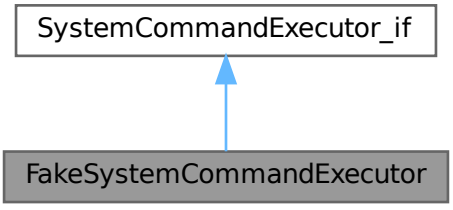
- tests/unit/test_support_persistence.cpp

17.285 FakeSystemCommandExecutor Class Reference

Inheritance diagram for FakeSystemCommandExecutor:



Collaboration diagram for FakeSystemCommandExecutor:



Public Member Functions

- SystemCommandResult run (const std::string &command) override

  Runs a shell command and returns its observable result.

Public Member Functions inherited from SystemCommandExecutor_if

- virtual ~**SystemCommandExecutor_if** ()=default

  Destroys the command executor abstraction.

Public Attributes

- std::map< std::string, SystemCommandResult > **results**
- std::vector< std::string > **commands**

17.285.1 Member Function Documentation

17.285.1.1 run()

```
SystemCommandResult FakeSystemCommandExecutor::run (
            const std::string & command )  [inline], [override], [virtual]
```

Runs a shell command and returns its observable result.
Implements SystemCommandExecutor_if.
The documentation for this class was generated from the following file:

- tests/unit/test_simulator_support.cpp

17.286 testing::internal::FallbackPrinter Struct Reference

Static Public Member Functions

- template<typename T >
  static void **PrintValue** (const T &, ::std::ostream *os)

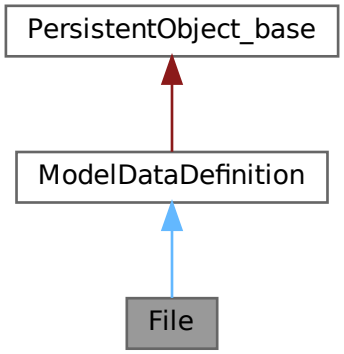
The documentation for this struct was generated from the following file:

- gtest/gtest/gtest-printers.h

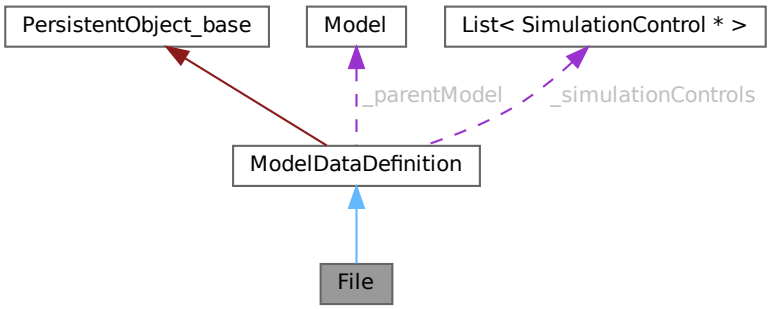
17.287 File Class Reference

Data definition identifying an external file and its access mode (read/write/append/read-write), used by file-driven components and data definitions.
`#include <plugins/data/InputOutput/File.h>`

Inheritance diagram for File:



Collaboration diagram for File:



Public Types

- enum class **AccessMode** : unsigned int { **Read** = 0 , **Write** = 1 , **Append** = 2 , **ReadWrite** = 3 }

Public Member Functions

- **File** (Model ∗model, std::string name="")
- void **setSystemFilename** (std::string systemFilename)
- std::string **getSystemFilename** () const
- std::string **getFilenameOnly** () const
- std::string **getPathOnly** () const
- void **setAccessMode** (AccessMode accessMode)
- AccessMode **getAccessMode** () const
- void **setAccessModeAsString** (std::string accessMode)
- std::string **getAccessModeAsString** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition ∗ **LoadInstance** (Model ∗model, PersistenceRecord ∗fields)
- static PluginInformation ∗ **GetPluginInformation** ()
- static ModelDataDefinition ∗ **NewInstance** (Model ∗model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition ∗ LoadInstance (Model ∗model, PersistenceRecord ∗fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition ∗ NewInstance (Model ∗model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord ∗fields, ModelDataDefinition ∗modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition ∗modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition ∗modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition ∗modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition ∗modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord ∗fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord ∗fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual ParserChangesInformation ∗ _getParserChangesInformation () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl ∗control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)

- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.287.1 Detailed Description

Data definition identifying an external file and its access mode (read/write/append/read-write), used by file-driven components and data definitions.
Arena correspondence: the "File module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", pp. 72-73), which backs the ReadWrite, Variable and Expression modules with a much richer set of structured data sources (typed file access, ADO connection strings, fixed/free/Fortran record structure, recordsets, SQL command text, named Excel ranges, Access table names, etc.).
Known difference from Arena: this class only models a plain operating-system file name and an `AccessMode` (Read/Write/Append/ ReadWrite); none of Arena's structured recordset/ADO/spreadsheet/database concepts have a GenESyS equivalent, so this correspondence is partial and covers only unformatted file access.

17.287.2 Member Function Documentation

17.287.2.1 _check()

```
bool File::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.287.2.2 _getParserChangesInformation()

```
ParserChangesInformation * File::_getParserChangesInformation ( )  [override], [protected],
[virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes Return parser deltas required by File.
File currently does not require parser extensions.
Reimplemented from ModelDataDefinition.

17.287.2.3 _loadInstance()

```
bool File::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.287.2.4 _saveInstance()

```
void File::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.287.2.5 show()

```
std::string File::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/InputOutput/File.h
- plugins/data/InputOutput/File.cpp

17.288 testing::internal::FindFirstPrinter< T, E, Printer, Printers > Struct Template Reference

The documentation for this struct was generated from the following file:

- gtest/gtest/gtest-printers.h

17.289 testing::internal::FindFirstPrinter< T, decltype(Printer::PrintValue(std::declval< const T & >(), nullptr)), Printer, Printers... > Struct Template Reference

Public Types

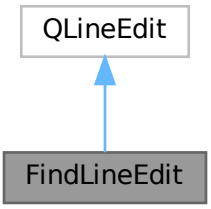
- using **type** = Printer

The documentation for this struct was generated from the following file:

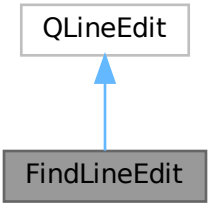
- gtest/gtest/gtest-printers.h

17.290 FindLineEdit Class Reference

Inheritance diagram for FindLineEdit:



Collaboration diagram for FindLineEdit:



Public Member Functions

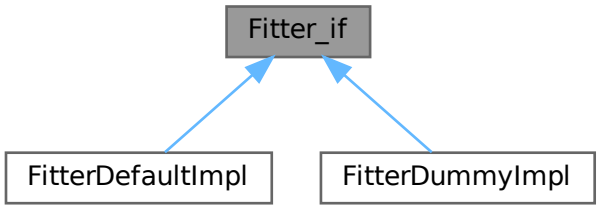
- **FindLineEdit** (QWidget ∗parent=nullptr)

The documentation for this class was generated from the following file:

- applications/gui/genesys/dialogs/DialogFind.h

17.291 Fitter_if Class Reference

Interface for fitting theoretical distributions to real datasets.
`#include <tools/Statistics/Fitter_if.h>`
Inheritance diagram for Fitter_if:



Public Member Functions

- virtual bool **isNormalDistributed** (double confidencelevel)=0
- virtual void **fitUniform** (double ∗sqrerror, double ∗min, double ∗max)=0
- virtual void **fitTriangular** (double ∗sqrerror, double ∗min, double ∗mo, double ∗max)=0
- virtual void **fitNormal** (double ∗sqrerror, double ∗avg, double ∗stddev)=0
- virtual void **fitExpo** (double ∗sqrerror, double ∗avg1)=0
- virtual void **fitErlang** (double ∗sqrerror, double ∗avg, double ∗m)=0
- virtual void **fitBeta** (double ∗sqrerror, double ∗alpha, double ∗beta, double ∗infLimit, double ∗supLimit)=0
- virtual void **fitWeibull** (double ∗sqrerror, double ∗alpha, double ∗scale)=0
- virtual void **fitAll** (double ∗sqrerror, std::string ∗name)=0
- virtual void setDataFilename (std::string dataFilename)=0

Defines the dataset source filename used by fitting operations.

- virtual std::string getDataFilename ()=0

  Retrieves the currently configured dataset source filename.

17.291.1 Detailed Description

Interface for fitting theoretical distributions to real datasets.
Purpose:

- Infer parameters of candidate probability distributions from sample data.

Architectural role:

- Core fitting abstraction consumed by analysis workflows.

Mathematical meaning:

- Each fit∗ method estimates family parameters and writes an adherence/error measure in `sqrerror` (typically squared-error-like criterion).
- fitAll() selects the best candidate according to implementation-defined criterion.
- isNormalDistributed() should be interpreted as a normality adherence check under the given confidence level.

Preconditions:

- Output pointers must be valid non-null addresses.
- Data source must be configured through setDataFilename() before fitting.

Postconditions:

- Output pointers are filled with estimated values when operation succeeds.

Failure modes:

- Concrete implementations may return conservative defaults and/or report failure when insufficient data or unsupported models are provided.

Legacy compatibility:

- The pointer-based output API is intentionally preserved as a legacy public contract; it is not necessarily the final long-term shape.

17.291.2 Member Function Documentation

17.291.2.1 getDataFilename()

```
virtual std::string Fitter_if::getDataFilename ( )  [pure virtual]
```

Retrieves the currently configured dataset source filename.
Implemented in FitterDummyImpl, and FitterDefaultImpl.

17.291.2.2 setDataFilename()

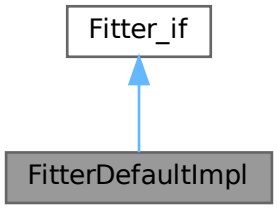
```
virtual void Fitter_if::setDataFilename (
            std::string dataFilename )  [pure virtual]
```

Defines the dataset source filename used by fitting operations.
Implemented in FitterDummyImpl, and FitterDefaultImpl.
The documentation for this class was generated from the following file:

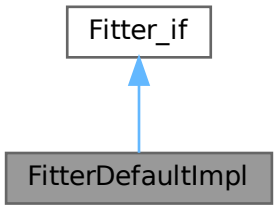
- tools/Statistics/Fitter_if.h

17.292 FitterDefaultImpl Class Reference

Functional implementation of Fitter_if consolidated through FITTER-2.
`#include <tools/Statistics/FitterDefaultImpl.h>`
Inheritance diagram for FitterDefaultImpl:



Collaboration diagram for FitterDefaultImpl:



Public Member Functions

- virtual bool isNormalDistributed (double confidencelevel) override
- virtual void fitUniform (double *sqrerror, double *min, double *max) override
- virtual void fitTriangular (double *sqrerror, double *min, double *mo, double *max) override
- virtual void fitNormal (double *sqrerror, double *avg, double *stddev) override
- virtual void fitExpo (double *sqrerror, double *avg1) override
- virtual void fitErlang (double *sqrerror, double *avg, double *m) override
- virtual void fitBeta (double *sqrerror, double *alpha, double *beta, double *infLimit, double *supLimit) override
- virtual void fitWeibull (double *sqrerror, double *alpha, double *scale) override
- virtual void fitAll (double *sqrerror, std::string *name) override
- virtual void setDataFilename (std::string dataFilename) override

  Defines the dataset source filename used by fitting operations.
- virtual std::string getDataFilename () override

  Retrieves the currently configured dataset source filename.

17.292.1 Detailed Description

Functional implementation of Fitter_if consolidated through FITTER-2.
Current scope:

- Supports real dataset loading from binary file with raw sequential doubles (compatible with CollectorDatafileDefaultImpl1 persistence format).
- Implements fitting for Uniform, Triangular, Normal, Exponential, Erlang, Beta (scaled) and Weibull.

Error metric:

- Uses SSE between empirical CDF points p_i = (i+0.5)/n and fitted model CDF.
- The same criterion is used consistently by fit∗ methods and fitAll().

Failure convention:

- sqrerror = +infinity.
- returned parameters = quiet_NaN().
- fitAll name = "no-valid-fit" or "invalid-dataset".
- isNormalDistributed() = false on invalid/insufficient data.

17.292.2 Member Function Documentation

17.292.2.1 fitAll()

```
virtual void FitterDefaultImpl::fitAll (
            double * sqrerror,
            std::string * name )  [inline], [override], [virtual]
```

Implements Fitter_if.

17.292.2.2 fitBeta()

```
virtual void FitterDefaultImpl::fitBeta (
            double * sqrerror,
            double * alpha,
            double * beta,
            double * infLimit,
            double * supLimit )  [inline], [override], [virtual]
```

Implements Fitter_if.

17.292.2.3 fitErlang()

```
virtual void FitterDefaultImpl::fitErlang (
            double * sqrerror,
            double * avg,
            double * m )  [inline], [override], [virtual]
```

Implements Fitter_if.

17.292.2.4 fitExpo()

```
virtual void FitterDefaultImpl::fitExpo (
            double * sqrerror,
            double * avg1 )  [inline], [override], [virtual]
```

Implements Fitter_if.

17.292.2.5 fitNormal()

```
virtual void FitterDefaultImpl::fitNormal (
            double * sqrerror,
            double * avg,
            double * stddev )  [inline], [override], [virtual]
```

Implements Fitter_if.

17.292.2.6 fitTriangular()

```
virtual void FitterDefaultImpl::fitTriangular (
            double * sqrerror,
            double * min,
            double * mo,
            double * max )  [inline], [override], [virtual]
```

Implements Fitter_if.

17.292.2.7 fitUniform()

```
virtual void FitterDefaultImpl::fitUniform (
            double * sqrerror,
            double * min,
            double * max )  [inline], [override], [virtual]
```

Implements Fitter_if.

17.292.2.8 fitWeibull()

```
virtual void FitterDefaultImpl::fitWeibull (
            double * sqrerror,
            double * alpha,
            double * scale )  [inline], [override], [virtual]
```

Implements Fitter_if.

17.292.2.9 getDataFilename()

```
virtual std::string FitterDefaultImpl::getDataFilename ( )  [inline], [override], [virtual]
```

Retrieves the currently configured dataset source filename.
Implements Fitter_if.

17.292.2.10 isNormalDistributed()

```
virtual bool FitterDefaultImpl::isNormalDistributed (
            double confidencelevel )  [inline], [override], [virtual]
```

Implements Fitter_if.

17.292.2.11 setDataFilename()

```
virtual void FitterDefaultImpl::setDataFilename (
            std::string dataFilename )  [inline], [override], [virtual]
```

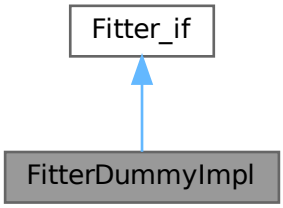
Defines the dataset source filename used by fitting operations.
Implements Fitter_if.
The documentation for this class was generated from the following file:

- tools/Statistics/FitterDefaultImpl.h

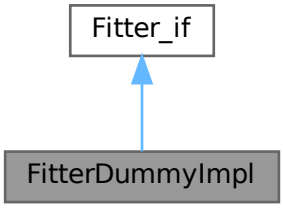
17.293 FitterDummyImpl Class Reference

Placeholder implementation of Fitter_if.
`#include <tools/Statistics/FitterDummyImpl.h>`

Inheritance diagram for FitterDummyImpl:



Collaboration diagram for FitterDummyImpl:



Public Member Functions

- **FitterDummyImpl** (const FitterDummyImpl &orig)
- bool isNormalDistributed (double confidencelevel)
- void fitUniform (double ∗sqrerror, double ∗min, double ∗max)
- void fitTriangular (double ∗sqrerror, double ∗min, double ∗mo, double ∗max)
- void fitNormal (double ∗sqrerror, double ∗avg, double ∗stddev)
- void fitExpo (double ∗sqrerror, double ∗avg1)
- void fitErlang (double ∗sqrerror, double ∗avg, double ∗m)
- void fitBeta (double ∗sqrerror, double ∗alpha, double ∗beta, double ∗infLimit, double ∗supLimit)
- void fitWeibull (double ∗sqrerror, double ∗alpha, double ∗scale)
- void fitAll (double ∗sqrerror, std::string ∗name)
- void setDataFilename (std::string dataFilename)

  Defines the dataset source filename used by fitting operations.
- std::string getDataFilename ()

  Retrieves the currently configured dataset source filename.

17.293.1 Detailed Description

Placeholder implementation of Fitter_if.
Purpose:

- Keep binary/source compatibility while a functional fitter implementation is not fully available.

Implementation status:

- Dummy/stub component; not a functional fitting engine.

Planned evolution:

- To be replaced as default by FitterDefaultImpl when fitting logic is consolidated.

17.293.2 Member Function Documentation

17.293.2.1 fitAll()

```
void FitterDummyImpl::fitAll (
            double * sqrerror,
            std::string * name )  [virtual]
```

Implements Fitter_if.

17.293.2.2 fitBeta()

```
void FitterDummyImpl::fitBeta (
            double * sqrerror,
            double * alpha,
            double * beta,
            double * infLimit,
            double * supLimit )  [virtual]
```

Implements Fitter_if.

17.293.2.3 fitErlang()

```
void FitterDummyImpl::fitErlang (
            double * sqrerror,
            double * avg,
            double * m )  [virtual]
```

Implements Fitter_if.

17.293.2.4 fitExpo()

```
void FitterDummyImpl::fitExpo (
            double * sqrerror,
            double * avg1 )  [virtual]
```

Implements Fitter_if.

17.293.2.5 fitNormal()

```
void FitterDummyImpl::fitNormal (
            double * sqrerror,
            double * avg,
            double * stddev )  [virtual]
```

Implements Fitter_if.

17.293.2.6 fitTriangular()

```
void FitterDummyImpl::fitTriangular (
            double * sqrerror,
            double * min,
            double * mo,
            double * max )  [virtual]
```

Implements Fitter_if.

17.293.2.7 fitUniform()

```
void FitterDummyImpl::fitUniform (
            double * sqrerror,
            double * min,
            double * max )  [virtual]
```

Implements Fitter_if.

17.293.2.8 fitWeibull()

```
void FitterDummyImpl::fitWeibull (
            double * sqrerror,
            double * alpha,
            double * scale )  [virtual]
```

Implements Fitter_if.

17.293.2.9 getDataFilename()

```
std::string FitterDummyImpl::getDataFilename ( )  [virtual]
```

Retrieves the currently configured dataset source filename.
Implements Fitter_if.

17.293.2.10 isNormalDistributed()

```
bool FitterDummyImpl::isNormalDistributed (
            double confidencelevel )  [virtual]
```

Implements Fitter_if.

17.293.2.11 setDataFilename()

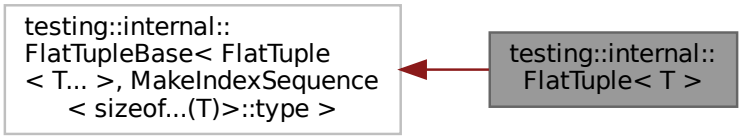
```
void FitterDummyImpl::setDataFilename (
            std::string dataFilename )  [virtual]
```

Defines the dataset source filename used by fitting operations.
Implements Fitter_if.
The documentation for this class was generated from the following files:

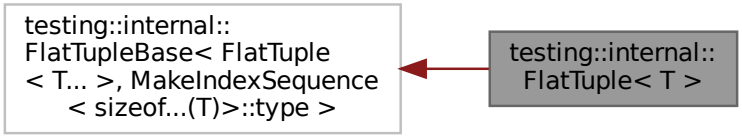
- tools/Statistics/FitterDummyImpl.h
- tools/Statistics/FitterDummyImpl.cpp

17.294 testing::internal::FlatTuple< T > Class Template Reference

Inheritance diagram for testing::internal::FlatTuple< T >:



Collaboration diagram for testing::internal::FlatTuple< T >:



Public Member Functions

- template<typename... Args>
  FlatTuple (FlatTupleConstructTag tag, Args &&... args)

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

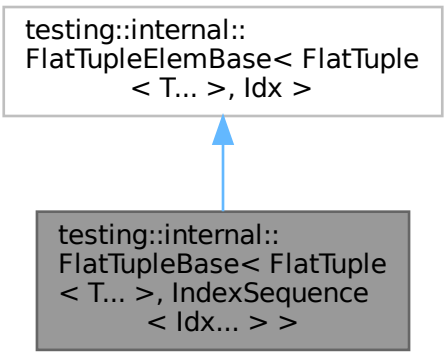
17.295 testing::internal::FlatTupleBase< Derived, Idx > Struct Template Reference

The documentation for this struct was generated from the following file:

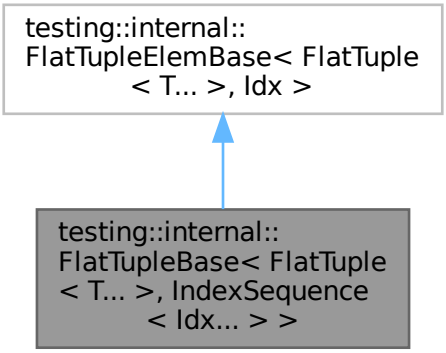
- gtest/gtest/internal/gtest-internal.h

17.296 testing::internal::FlatTupleBase< FlatTuple< T... >, IndexSequence< Idx... > > Struct Template Reference

Inheritance diagram for testing::internal::FlatTupleBase< FlatTuple< T... >, IndexSequence< Idx... > >:



Collaboration diagram for testing::internal::FlatTupleBase< FlatTuple< T... >, IndexSequence< Idx... > >:



Public Types

- using **Indices** = IndexSequence< Idx... >

Public Member Functions

- template<typename... Args>
  FlatTupleBase (FlatTupleConstructTag, Args &&... args)
- template<size_t I>
  const ElemFromList< I, T... >::type & **Get** () const
- template<size_t I>
  ElemFromList< I, T... >::type & **Get** ()
- template<typename F >
  auto **Apply** (F &&f) -> decltype(std::forward< F >(f)(this->Get< Idx >()...))
- template<typename F >
  auto **Apply** (F &&f) const -> decltype(std::forward< F >(f)(this->Get< Idx >()...))

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.297 testing::internal::FlatTupleConstructTag Struct Reference

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

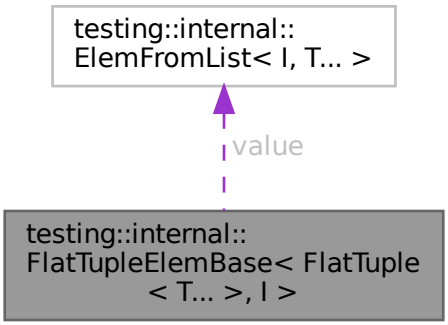
17.298 testing::internal::FlatTupleElemBase< Derived, I > Struct Template Reference

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.299 testing::internal::FlatTupleElemBase< FlatTuple< T... >, I > Struct Template Reference

Collaboration diagram for testing::internal::FlatTupleElemBase< FlatTuple< T... >, I >:



Public Types

- using **value_type** = typename ElemFromList< I, T... >::type

Public Member Functions

- template<typename Arg >
  FlatTupleElemBase (FlatTupleConstructTag, Arg &&t)

Public Attributes

- value_type **value**

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.300 testing::internal::FloatingPoint< RawType > Class Template Reference

Public Types

- typedef TypeWithSize< sizeof(RawType)>::UInt **Bits**

Public Member Functions

- **FloatingPoint** (const RawType &x)
- const Bits & **bits** () const
- Bits **exponent_bits** () const
- Bits **fraction_bits** () const
- Bits **sign_bit** () const
- bool **is_nan** () const
- bool **AlmostEquals** (const FloatingPoint &rhs) const
- float **Max** ()
- double **Max** ()

Static Public Member Functions

- static RawType **ReinterpretBits** (const Bits bits)
- static RawType **Infinity** ()
- static RawType **Max** ()

Static Public Attributes

- static const size_t **kBitCount** = 8 * sizeof(RawType)
- static const size_t kFractionBitCount
- static const size_t **kExponentBitCount** = kBitCount - 1 - kFractionBitCount
- static const Bits **kSignBitMask** = static_cast<Bits>(1) << (kBitCount - 1)
- static const Bits kFractionBitMask
- static const Bits **kExponentBitMask** = ~(kSignBitMask | kFractionBitMask)
- static const uint32_t **kMaxUlps** = 4

17.300.1 Member Data Documentation

17.300.1.1 kFractionBitCount

```
template<typename RawType >
const size_t testing::internal::FloatingPoint< RawType >::kFractionBitCount  [static]
```

Initial value:

```
=
      std::numeric_limits<RawType>::digits - 1
```

17.300.1.2 kFractionBitMask

```
template<typename RawType >
const Bits testing::internal::FloatingPoint< RawType >::kFractionBitMask  [static]
```

Initial value:

```
= ~static_cast<Bits>(0) <<
                                          (kExponentBitCount + 1)
```

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.301 testing::internal::FormatForComparison< ToPrint, OtherOperand > Class Template Reference

Static Public Member Functions

- ::std::string **Format** (const ToPrint &value)

The documentation for this class was generated from the following file:

- gtest/gtest/gtest-printers.h

17.302 testing::internal::FormatForComparison< ToPrint[N], OtherOperand > Class Template Reference

Static Public Member Functions

- ::std::string **Format** (const ToPrint *value)

The documentation for this class was generated from the following file:

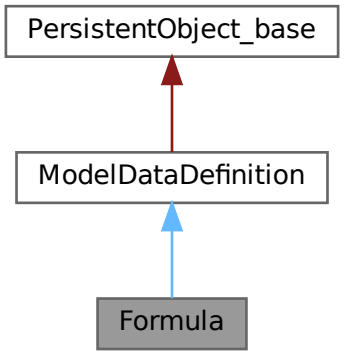
- gtest/gtest/gtest-printers.h

17.303 Formula Class Reference

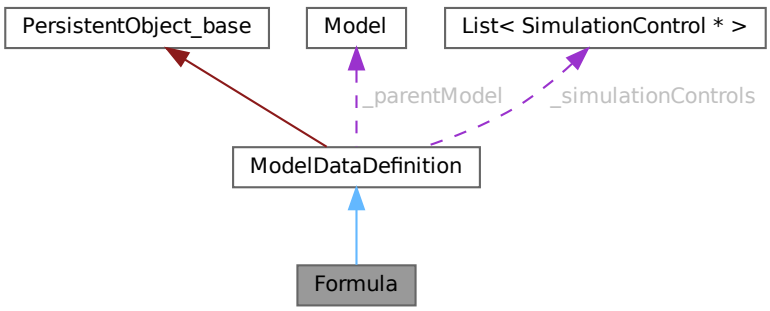
Data definition holding one or more named parser expressions, optionally indexed to represent a 1D/2D array of expressions.

`#include <plugins/data/Logic/Formula.h>`

Inheritance diagram for Formula:



Collaboration diagram for Formula:



Public Member Functions

- Formula (Model *model, std::string name="")

  Creates an empty formula definition.
- virtual ~**Formula** () override

  Releases formula-owned resources.
- virtual std::string show () override

  Returns a textual representation of the formula.
- unsigned int size ()

  Returns the number of indexed expressions stored in this formula.
- void setExpression (std::string formulaExpression, std::string index="")

  Stores or updates the expression associated with an index.

- std::string getExpression (std::string index="")

  Returns the expression associated with an index.
- double getValue (std::string index)

  Evaluates the expression associated with an index.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * GetPluginInformation ()

  Returns plugin metadata for this formula type.
- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields)

  Loads a formula from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new empty formula instance.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createNonEditableDataDefinitions ()

- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.303.1 Detailed Description

Data definition holding one or more named parser expressions, optionally indexed to represent a 1D/2D array of expressions.
Arena correspondence: the "Expression module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 70), which defines named expressions (optionally arrays) that may combine numbers, distributions, operators, attributes and variables, and are referenced by name elsewhere in the model. `setExpression()`/`getExpression()` with a textual index are the closest correspondence to Arena's Rows/Columns indexing; `getValue()` evaluates the stored expression through the parser.
Known differences from Arena: there is no declared Data Type (Real / String / Native) and no external-file linkage (File Name, Recordset) to source expression values from outside the model.

17.303.2 Constructor & Destructor Documentation

17.303.2.1 Formula()

```
Formula::Formula (
            Model * model,
            std::string name = "" )
```

Creates an empty formula definition.

Parameters

model	Parent model.
name	Initial formula name.

17.303.3 Member Function Documentation

17.303.3.1 _check()

```
bool Formula::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.303.3.2 _createAttachedAttributes()

```
void Formula::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes @ToDo: NOT AN ATTRIBUTE
Reimplemented from ModelDataDefinition.

17.303.3.3 _loadInstance()

```
bool Formula::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Load formula expressions map.
Reimplemented from ModelDataDefinition.

17.303.3.4 _saveInstance()

```
void Formula::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Persist formula expressions map.
Reimplemented from ModelDataDefinition.

17.303.3.5 getExpression()

```
std::string Formula::getExpression (
            std::string index = "" )
```

Returns the expression associated with an index.

Parameters

index	Optional index key used for vectorized formulas.

Returns

Stored formula text, or an empty string when absent.

17.303.3.6 GetPluginInformation()

```
PluginInformation * Formula::GetPluginInformation ( )  [static]
```

Returns plugin metadata for this formula type.

Returns

Plugin information block.

17.303.3.7 getValue()

```
double Formula::getValue (
            std::string index )
```

Evaluates the expression associated with an index.

Parameters

index	Optional index key used for vectorized formulas.

Returns

Numeric value of the indexed expression.

17.303.3.8 LoadInstance()

```
ModelDataDefinition * Formula::LoadInstance (
            Model * model,
            PersistenceRecord * fields )  [static]
```

Loads a formula from serialized fields.

Parameters

model	Parent model.
fields	Serialized fields.

Returns

Newly created formula instance.

17.303.3.9 NewInstance()

```
ModelDataDefinition * Formula::NewInstance (
            Model * model,
            std::string name = "" )  [static]
```

Creates a new empty formula instance.

Parameters

model	Parent model.
name	Initial formula name.

Returns

Newly created formula instance.

17.303.3.10 setExpression()

```
void Formula::setExpression (
            std::string formulaExpression,
            std::string index = "" )
```

Stores or updates the expression associated with an index.

Parameters

formulaExpression	Formula text.
index	Optional index key used for vectorized formulas.

17.303.3.11 show()

```
std::string Formula::show ( )  [override], [virtual]
```

Returns a textual representation of the formula.

Returns

Human-readable formula summary.

Reimplemented from ModelDataDefinition.

17.303.3.12 size()

```
unsigned int Formula::size ( )
```

Returns the number of indexed expressions stored in this formula.

Returns

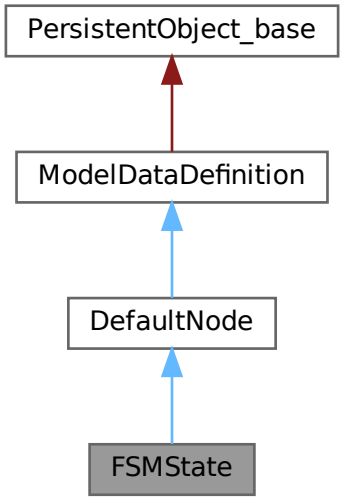
Number of expression slots.

The documentation for this class was generated from the following files:

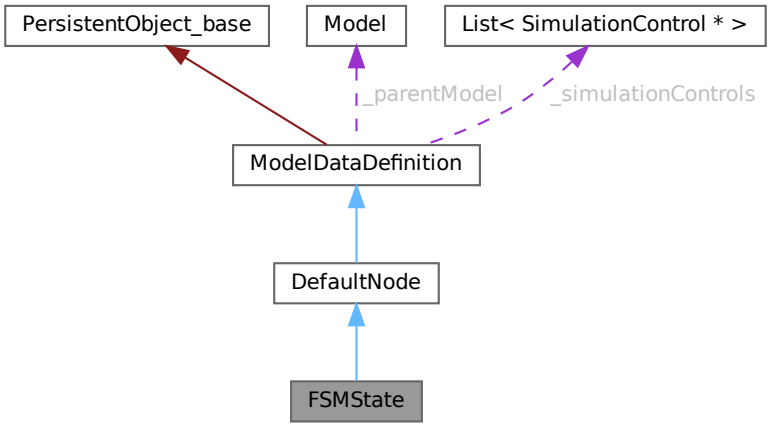
- plugins/data/Logic/Formula.h
- plugins/data/Logic/Formula.cpp

17.304 FSMState Class Reference

Inheritance diagram for FSMState:



Collaboration diagram for FSMState:



Public Member Functions

- **FSMState** (Model ∗model, std::string name="")
- void **setEntryActionExpression** (std::string expression)
- std::string **getEntryActionExpression** () const
- void **setExitActionExpression** (std::string expression)
- std::string **getExitActionExpression** () const

Public Member Functions inherited from DefaultNode

- DefaultNode (Model ∗model, std::string componentTypename, std::string name)

  constructors
- **DefaultNode** (Model ∗model, std::string name="")
- void **addTransition** (DefaultNodeTransition ∗transition)

  new public user methods for this component
- void **removeTransition** (DefaultNodeTransition ∗transition)
- List< DefaultNodeTransition ∗ > ∗ **getTransitions** () const
- void **setInitialNode** (bool initialNode)
- bool **isInitialNode** () const
- void **setFinalNode** (bool finalNode)
- bool **isFinalNode** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model ∗model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ∼**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from DefaultNode

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)

- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.304.1 Member Function Documentation

17.304.1.1 _loadInstance()

```
bool FSMState::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from DefaultNode.

17.304.1.2 _saveInstance()

```
void FSMState::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

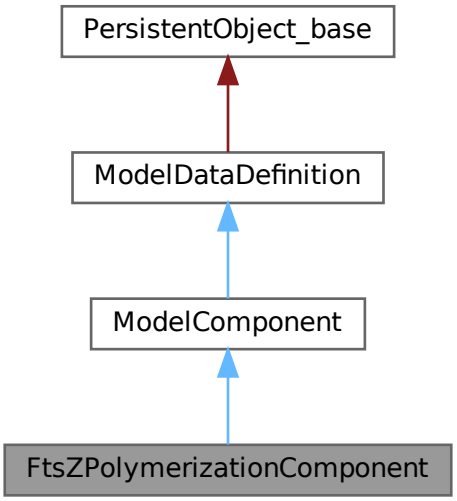
fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from DefaultNode.
The documentation for this class was generated from the following files:

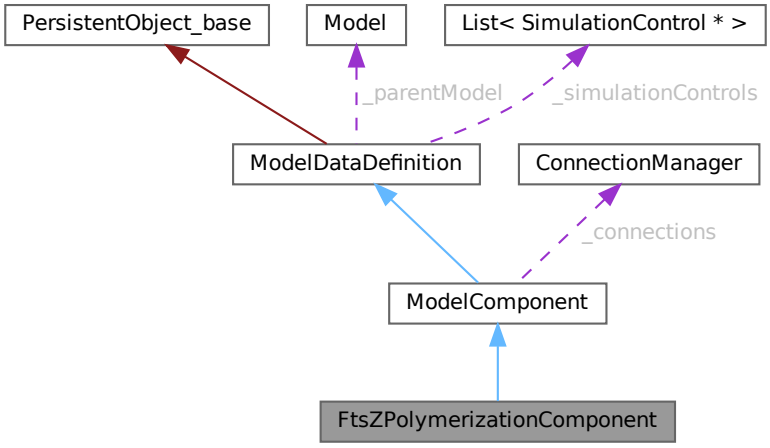
- plugins/components/ModalModel/FSMState.h
- plugins/components/ModalModel/FSMState.cpp

17.305 FtsZPolymerizationComponent Class Reference

`#include <plugins/components/WholeCellModeling/FtsZPolymerizationComponent.h>`
Inheritance diagram for FtsZPolymerizationComponent:



Collaboration diagram for FtsZPolymerizationComponent:



Public Member Functions

- **FtsZPolymerizationComponent** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)
- WholeCellState * **getWholeCellState** () const
- void **setActivationFwd** (double rate)
- double **getActivationFwd** () const
- void **setActivationRev** (double rate)
- double **getActivationRev** () const
- void **setVolumeNorm** (double norm)
- double **getVolumeNorm** () const
- void **setDeltaT** (double dt)
- double **getDeltaT** () const
- void **setFtsZRingKey** (std::string key)
- std::string **getFtsZRingKey** () const
- void **setFtsZMonomerKey** (std::string key)
- std::string **getFtsZMonomerKey** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)

- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.305.1 Detailed Description

FtsZ ring polymerization dynamics for whole-cell division timing.
Implements a simplified two-state Euler ODE for the FtsZ ring completion fraction f ∈ [0,1], stored as a per-mille integer (0–1000) in WholeCellState under the key ftsZRingKey (default "FtsZ_ring_completion").
Monomer availability is read from WholeCellState under ftsZMonomerKey (default "FtsZ_monomer"). The reference count N_ref is captured on the first _onDispatchEvent call (N_m at t=0) and held constant; the fraction N_m/N_ref scales the polymerization rate (lower monomer count = slower growth).
ODE (per step): f_new = f + (k_poly * phi * (1-f) - k_depoly * f) * deltaT where k_poly = activationFwd * volumeNorm [/s] k_depoly = activationRev * volumeNorm [/s] phi = N_m / N_ref [dimensionless monomer availability]
Default volumeNorm = 3.94e-5 is calibrated so that k_poly ≈ 4.3e-5/s, which gives ∼35% ring completion at t ≈ 12 000 s — consistent with the M. genitalium cell cycle. activationFwd and activationRev are read from WholeCellState "param.ftsZActivationFwd"/"param.ftsZActivationRev" when loadParameters() has been called; otherwise the component uses its own stored defaults.

17.305.2 Member Function Documentation

17.305.2.1 _check()

```
bool FtsZPolymerizationComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.305.2.2 _createEditableDataDefinitions()

```
void FtsZPolymerizationComponent::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.305.2.3 _initBetweenReplications()

```
void FtsZPolymerizationComponent::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.305.2.4 _loadInstance()

```
bool FtsZPolymerizationComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.305.2.5 _onDispatchEvent()

```
void FtsZPolymerizationComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.305.2.6 _saveInstance()

```
void FtsZPolymerizationComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.305.2.7 show()

```
std::string FtsZPolymerizationComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

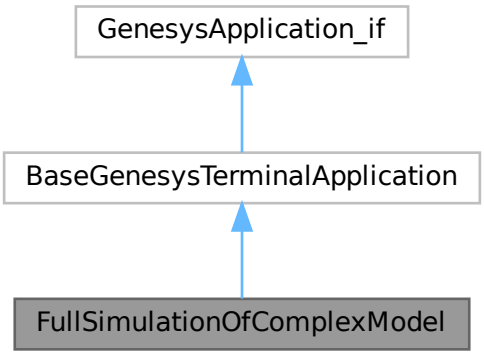
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

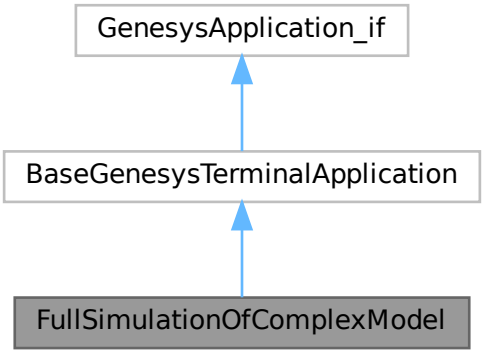
- plugins/components/WholeCellModeling/FtsZPolymerizationComponent.h
- plugins/components/WholeCellModeling/FtsZPolymerizationComponent.cpp

17.306 FullSimulationOfComplexModel Class Reference

Inheritance diagram for FullSimulationOfComplexModel:



Collaboration diagram for FullSimulationOfComplexModel:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.306.1 Member Function Documentation

17.306.1.1 main()

```
int FullSimulationOfComplexModel::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/teaching/FullSimulationOfComplexModel.h
- applications/modelSpecific/teaching/FullSimulationOfComplexModel.cpp

17.307 testing::internal::FunctionPointerPrinter Struct Reference

Static Public Member Functions

- template<typename T , typename = typename std::enable_if< std::is_function<T>::value>::type>
  static void **PrintValue** (T ∗p, ::std::ostream ∗os)

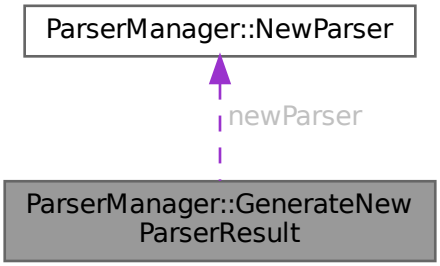
The documentation for this struct was generated from the following file:

- gtest/gtest/gtest-printers.h

17.308 ParserManager::GenerateNewParserResult Struct Reference

Result bundle for parser generation requests.
`#include <kernel/simulator/ParserManager.h>`

Collaboration diagram for ParserManager::GenerateNewParserResult:



Public Attributes

- bool **result** = false
- std::string **bisonMessages**
- std::string **lexMessages**
- std::string **compilationMessages**
- NewParser **newParser**

17.308.1 Detailed Description

Result bundle for parser generation requests.
The documentation for this struct was generated from the following file:

- kernel/simulator/ParserManager.h

17.309 testing::internal::GenerateTypeList< T > Struct Template Reference

Public Types

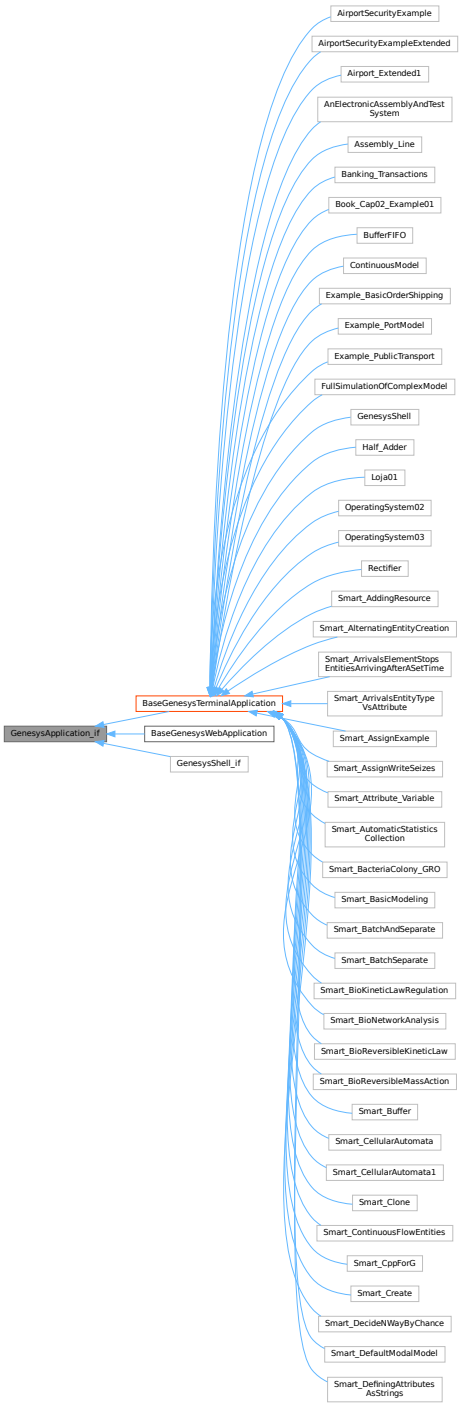
- using **type** = typename proxy::type

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-type-util.h

17.310 GenesysApplication_if Class Reference

Inheritance diagram for GenesysApplication_if:



Public Member Functions

- virtual int main (int argc, char **argv)=0

17.310.1 Member Function Documentation

17.310.1.1 main()

```
virtual int GenesysApplication_if::main (
            int argc,
            char ** argv )  [pure virtual]
```

Implemented in Airport_Extended1, Assembly_Line, Banking_Transactions, Example_BasicOrderShipping, Example_PortModel, Example_PublicTransport, Smart_AddingResource, Smart_AlternatingEntityCreation, Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime, Smart_AutomaticStatisticsCollection, Smart_ContinuousFlowEntities, Smart_Create, Smart_DecideNWayByChance, Smart_DefiningAttributesAsStrings, Smart_DefiningControlLogic, Smart_DefiningResourceCapacity, Smart_EntitiesProcessedByPriority, Smart_EvaluatingConditionsBeforeEnteringQueue, Smart_Expression, Smart_MaxArrivalsField, Smart_ModelRunUntil1000Parts, Smart_ModuleDisplayVariables, Smart_OverlappingResources, Smart_ParallelProcessingOfEntities, Smart_PlacingEntitiesInQueueSets, Smart_PriorityExample, Smart_ProcessArena, Smart_Record_Arena, Smart_RemovingAndReorderingEntitiesInAQueue, Smart_ResourceSets, Smart_SeizingMultipleSimultaneosly, Smart_SelectingRouteBasedOnProbability, Smart_SelectingShorterQueue, Smart_SynchronizingParallelEntities, Smart_WaitForSignal, Smart_MarkovChain, Smart_Buffer, Smart_Clone, Smart_Delay, Smart_Failures, Smart_HoldSearchRemove, Smart_Process, Smart_ProcessSet, Smart_SeizeDelayRelease, Smart_SeizeDelayReleaseMany, Smart_SeizeDelayReleaseNoDataDefs, Smart_R_Simulator, Smart_Record, Smart_Attribute_Variable, Smart_ModelInfoModelSimulation, Smart_Parser, Smart_ParserModelFunctions, Smart_RouteStation, Smart_CellularAutomata, Smart_CellularAutomata1, Smart_DefaultModalModel, Smart_Dummy, Smart_OnEvent, Smart_WaitScanCondition, Smart_WaitSignal, FullSimulationOfComplexModel, Half_Adder, Rectifier, and BaseGenesysWebApplication.

The documentation for this class was generated from the following file:

- applications/GenesysApplication_if.h

17.311 genesys_mcp.client.GenesysClient Class Reference

Public Member Functions

- None **__init__** (self, str base_url="http://localhost:8080", float timeout=30.0)
- dict[str, Any] **health** (self)
- dict[str, Any] **create_session** (self)
- dict[str, Any] **simulator_info** (self)
- dict[str, Any] **new_model** (self)
- dict[str, Any] **get_model** (self)
- dict[str, Any] **save_model** (self, str filename)
- dict[str, Any] **load_model** (self, str filename)
- dict[str, Any] **simulation_status** (self)
- dict[str, Any] **configure_simulation** (self, int number_of_replications, float replication_length, float warm↩ _up_period, bool pause_on_event=False, bool pause_on_replication=False, bool initialize_statistics=True, bool initialize_system=True)
- dict[str, Any] **run_simulation** (self)
- dict[str, Any] **step_simulation** (self)
- dict[str, Any] **worker_info** (self)
- dict[str, Any] **worker_capabilities** (self)
- dict[str, Any] **import_model_language** (self, str model_text)
- dict[str, Any] **create_job** (self)
- dict[str, Any] **get_job** (self, str job_id)
- dict[str, Any] **run_job** (self, str job_id)
- dict[str, Any] **get_job_result** (self, str job_id)
- None **close** (self)
- GenesysClient **__enter__** (self)
- None **__exit__** (self, ∗object _)

Protected Member Functions

- dict[str, str] **_headers** (self)
- Any **_get** (self, str path)
- Any **_post** (self, str path, Any body=None)
- Any **_post_text** (self, str path, str text)

Protected Attributes

- **_base_url**
- **_http**
- **_token**

17.311.1 Detailed Description

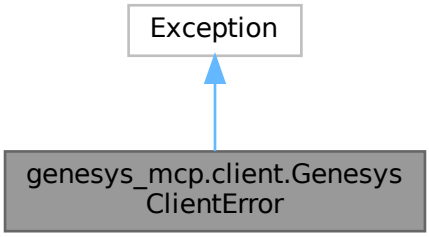
```
Stateful HTTP client for one Genesys worker-app session.
```

The documentation for this class was generated from the following file:

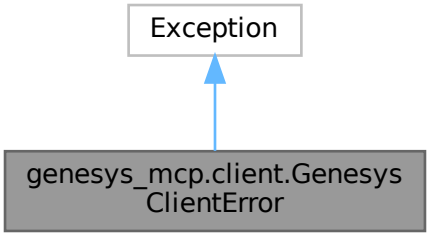
- applications/mcp/genesys_mcp/client.py

17.312 genesys_mcp.client.GenesysClientError Class Reference

Inheritance diagram for genesys_mcp.client.GenesysClientError:



Collaboration diagram for genesys_mcp.client.GenesysClientError:



Public Member Functions

- None **__init__** (self, int status_code, str message)

Public Attributes

- **status_code**

17.312.1 Detailed Description

```
Raised when the Genesys REST API returns an error.
```

The documentation for this class was generated from the following file:

- applications/mcp/genesys_mcp/client.py

17.313 GenesysKnowledgeBaseSummary Struct Reference

Aggregate summary of the current knowledge base state.
`#include <tools/AIAssistant/AIAssistant_if.h>`

Public Attributes

- unsigned int **numberOfPlugins** = 0
- unsigned int **numberOfComponents** = 0
- unsigned int **numberOfDataDefinitions** = 0
- std::string **summaryMarkdown**

  Full knowledge base rendered as a Markdown table.

17.313.1 Detailed Description

Aggregate summary of the current knowledge base state.
The documentation for this struct was generated from the following file:

- tools/AIAssistant/AIAssistant_if.h

17.314 GenesysKnowledgeItem Struct Reference

One entry in the GenESyS knowledge base used to inform the AI assistant.
`#include <tools/AIAssistant/AIAssistant_if.h>`

Public Attributes

- std::string **name**

  Short identifier (e.g. "Queue", "Create", "Entity").
- std::string **category**

  Broad category: "Component", "DataDefinition", "Plugin", etc.
- std::string **typeName**

  Exact C++ typename used by the kernel (e.g. "Queue").
- std::string **description**

  One-paragraph description of purpose and behaviour.
- std::string **usage**

  Typical usage pattern or construction example.
- std::string **constraints**

  Known constraints, limitations, or required connection rules.

17.314.1 Detailed Description

One entry in the GenESyS knowledge base used to inform the AI assistant.
Knowledge items describe available simulation building blocks so that the assistant can map natural-language descriptions to concrete GenESyS types. Items are populated from:

- Static documentation embedded at build time.
- Dynamic introspection of connected plugins via SimulatorFacade (Stage 2, see AIAssistant_if::refreshKnowledgeBaseFromSimu

The documentation for this struct was generated from the following file:

- tools/AIAssistant/AIAssistant_if.h

17.315 genesyspp_driver Class Reference

Public Member Functions

- **genesyspp_driver** (Model *model, Sampler_if *sampler, bool throws=false)
- **genesyspp_driver** (const genesyspp_driver &other)
- genesyspp_driver & **operator=** (const genesyspp_driver &other)
- **genesyspp_driver** (genesyspp_driver &&other) noexcept
- genesyspp_driver & **operator=** (genesyspp_driver &&other) noexcept
- void **scan_begin_file** ()
- void **scan_end_file** ()
- void **scan_begin_str** ()
- void **scan_end_str** ()
- int **parse_file** (const std::string &f)
- int **parse_str** (const std::string &str)
- void **error** (const yy::location &l, const std::string &m)
- void **error** (const std::string &m)
- double **getResult** ()
- void **setResult** (double value)
- bool **getThrowsException** ()
- void **setThrowsException** (bool throws)
- void **setErrorMessage** (std::string message)
- std::string **getErrorMessage** ()
- Model * **getModel** ()
- std::string **getFile** ()
- void **setFile** (std::string f)
- std::string **getStrToParse** ()
- void **setStrToParse** (std::string str)
- void **setSampler** (Sampler_if *_sampler)
- Sampler_if * **getSampler** () const
- void **setRegisterReferedDataElements** (bool value)
- bool **getRegisterReferedDataElements** ()
- std::map< std::string, std::list< std::string > * > * **getReferedDataElements** ()
- void **clearReferedDataElements** ()
- void **addRefered** (std::pair< std::string, std::string > referedElement)
- SimulationResponse * **findSimulationResponse** (const std::string &name) const
- SimulationControl * **findSimulationControl** (const std::string &name) const
- double **getSimulationResponseValueAsDouble** (const std::string &name) const
- double **getSimulationControlValueAsDouble** (const std::string &name) const
- double **stringToDoubleOrWarn** (const std::string &sourceType, const std::string &symbolName, const std↩ ::string &valueText) const
- void **traceWarning** (const std::string &message) const

The documentation for this class was generated from the following files:

- parser/Genesys++-driver.h
- parser/Genesys++-driver.cpp
- parser/Genesys++-scanner.cpp

17.316 yy::genesyspp_parser Class Reference

A Bison parser.

`#include <parser/GenesysParser.h>`

Classes

- struct basic_symbol
- struct by_kind

  Type access provider for token (enum) based symbols.
- class context
- struct symbol_kind

  Symbol kinds.
- struct symbol_type

  "External" symbols: returned by the scanner.
- struct syntax_error

  Syntax errors thrown from user actions.
- struct token

  Token kinds.
- class value_type

Public Types

- typedef value_type **semantic_type**

  Backward compatibility (Bison 3.8).
- typedef location **location_type**

  Symbol locations.
- typedef token::token_kind_type **token_kind_type**

  Token kind, as returned by yylex.
- typedef token_kind_type **token_type**

  Backward compatibility alias (Bison 3.6).
- typedef symbol_kind::symbol_kind_type **symbol_kind_type**

  (Internal) symbol kind.
- typedef by_kind **by_type**

  Backward compatibility for a private implementation detail (Bison 3.6).
- typedef int **debug_level_type**

  Type for debugging levels.

Public Member Functions

- **genesyspp_parser** (genesyspp_driver &driver_yyarg)

  Build a parser object.
- int operator() ()
- virtual int parse ()
- std::ostream & **debug_stream** () const YY_ATTRIBUTE_PURE

  The current debugging stream.
- void **set_debug_stream** (std::ostream &)

  Set the current debugging stream.
- debug_level_type **debug_level** () const YY_ATTRIBUTE_PURE

  The current debugging level.
- void **set_debug_level** (debug_level_type l)

  Set the current debugging level.
- virtual void error (const location_type &loc, const std::string &msg)
- void **error** (const syntax_error &err)

  Report a syntax error.

Static Public Member Functions

- static std::string symbol_name (symbol_kind_type yysymbol)
- static symbol_type **make_END** (const location_type &l)
- static symbol_type **make_YYerror** (const location_type &l)
- static symbol_type **make_YYUNDEF** (const location_type &l)
- static symbol_type **make_NUMD** (const obj_t &v, const location_type &l)
- static symbol_type **make_NUMH** (const obj_t &v, const location_type &l)
- static symbol_type **make_CTEZERO** (const obj_t &v, const location_type &l)
- static symbol_type **make_oLE** (const obj_t &v, const location_type &l)
- static symbol_type **make_oGE** (const obj_t &v, const location_type &l)
- static symbol_type **make_oEQ** (const obj_t &v, const location_type &l)
- static symbol_type **make_oNE** (const obj_t &v, const location_type &l)
- static symbol_type **make_oAND** (const obj_t &v, const location_type &l)
- static symbol_type **make_oOR** (const obj_t &v, const location_type &l)
- static symbol_type **make_oNAND** (const obj_t &v, const location_type &l)
- static symbol_type **make_oXOR** (const obj_t &v, const location_type &l)
- static symbol_type **make_oNOT** (const obj_t &v, const location_type &l)
- static symbol_type **make_fSIN** (const obj_t &v, const location_type &l)
- static symbol_type **make_fCOS** (const obj_t &v, const location_type &l)
- static symbol_type **make_fROUND** (const obj_t &v, const location_type &l)
- static symbol_type **make_fMOD** (const obj_t &v, const location_type &l)
- static symbol_type **make_fTRUNC** (const obj_t &v, const location_type &l)
- static symbol_type **make_fFRAC** (const obj_t &v, const location_type &l)
- static symbol_type **make_fEXP** (const obj_t &v, const location_type &l)
- static symbol_type **make_fSQRT** (const obj_t &v, const location_type &l)
- static symbol_type **make_fLOG** (const obj_t &v, const location_type &l)
- static symbol_type **make_fLN** (const obj_t &v, const location_type &l)
- static symbol_type **make_mathMIN** (const obj_t &v, const location_type &l)
- static symbol_type **make_mathMAX** (const obj_t &v, const location_type &l)
- static symbol_type **make_fVAL** (const obj_t &v, const location_type &l)
- static symbol_type **make_fEVAL** (const obj_t &v, const location_type &l)
- static symbol_type **make_fLENG** (const obj_t &v, const location_type &l)
- static symbol_type **make_fRND1** (const obj_t &v, const location_type &l)
- static symbol_type **make_fEXPO** (const obj_t &v, const location_type &l)
- static symbol_type **make_fNORM** (const obj_t &v, const location_type &l)
- static symbol_type **make_fUNIF** (const obj_t &v, const location_type &l)
- static symbol_type **make_fWEIB** (const obj_t &v, const location_type &l)
- static symbol_type **make_fLOGN** (const obj_t &v, const location_type &l)
- static symbol_type **make_fGAMM** (const obj_t &v, const location_type &l)
- static symbol_type **make_fERLA** (const obj_t &v, const location_type &l)
- static symbol_type **make_fTRIA** (const obj_t &v, const location_type &l)
- static symbol_type **make_fBETA** (const obj_t &v, const location_type &l)
- static symbol_type **make_fDISC** (const obj_t &v, const location_type &l)
- static symbol_type **make_fTNOW** (const obj_t &v, const location_type &l)
- static symbol_type **make_fTFIN** (const obj_t &v, const location_type &l)
- static symbol_type **make_fMAXREP** (const obj_t &v, const location_type &l)
- static symbol_type **make_fNUMREP** (const obj_t &v, const location_type &l)
- static symbol_type **make_fIDENT** (const obj_t &v, const location_type &l)
- static symbol_type **make_simulEntitiesWIP** (const obj_t &v, const location_type &l)
- static symbol_type **make_cIF** (const obj_t &v, const location_type &l)
- static symbol_type **make_cELSE** (const obj_t &v, const location_type &l)
- static symbol_type **make_cFOR** (const obj_t &v, const location_type &l)
- static symbol_type **make_cTO** (const obj_t &v, const location_type &l)
- static symbol_type **make_cDO** (const obj_t &v, const location_type &l)

- static symbol_type **make_ATRIB** (const obj_t &v, const location_type &l)
- static symbol_type **make_CSTAT** (const obj_t &v, const location_type &l)
- static symbol_type **make_COUNTER** (const obj_t &v, const location_type &l)
- static symbol_type **make_SIMRESP** (const obj_t &v, const location_type &l)
- static symbol_type **make_SIMCTRL** (const obj_t &v, const location_type &l)
- static symbol_type **make_fTAVG** (const obj_t &v, const location_type &l)
- static symbol_type **make_fCOUNT** (const obj_t &v, const location_type &l)
- static symbol_type **make_ILLEGAL** (const obj_t &v, const location_type &l)
- static symbol_type **make_RESOURCE** (const obj_t &v, const location_type &l)
- static symbol_type **make_fNR** (const obj_t &v, const location_type &l)
- static symbol_type **make_fMR** (const obj_t &v, const location_type &l)
- static symbol_type **make_fIRF** (const obj_t &v, const location_type &l)
- static symbol_type **make_fRESSEIZES** (const obj_t &v, const location_type &l)
- static symbol_type **make_fSTATE** (const obj_t &v, const location_type &l)
- static symbol_type **make_fSETSUM** (const obj_t &v, const location_type &l)
- static symbol_type **make_fRESUTIL** (const obj_t &v, const location_type &l)
- static symbol_type **make_QUEUE** (const obj_t &v, const location_type &l)
- static symbol_type **make_fNQ** (const obj_t &v, const location_type &l)
- static symbol_type **make_fFIRSTINQ** (const obj_t &v, const location_type &l)
- static symbol_type **make_fLASTINQ** (const obj_t &v, const location_type &l)
- static symbol_type **make_fSAQUE** (const obj_t &v, const location_type &l)
- static symbol_type **make_fAQUE** (const obj_t &v, const location_type &l)
- static symbol_type **make_fENTATRANK** (const obj_t &v, const location_type &l)
- static symbol_type **make_SET** (const obj_t &v, const location_type &l)
- static symbol_type **make_fNUMSET** (const obj_t &v, const location_type &l)
- static symbol_type **make_VARI** (const obj_t &v, const location_type &l)
- static symbol_type **make_FORM** (const obj_t &v, const location_type &l)
- static symbol_type **make_fNUMGR** (const obj_t &v, const location_type &l)
- static symbol_type **make_fATRGR** (const obj_t &v, const location_type &l)
- static symbol_type **make_LPAREN** (const location_type &l)
- static symbol_type **make_RPAREN** (const location_type &l)
- static symbol_type **make_LBRACKET** (const location_type &l)
- static symbol_type **make_RBRACKET** (const location_type &l)
- static symbol_type **make_PLUS** (const location_type &l)
- static symbol_type **make_MINUS** (const location_type &l)
- static symbol_type **make_STAR** (const location_type &l)
- static symbol_type **make_POWER** (const location_type &l)
- static symbol_type **make_SLASH** (const location_type &l)
- static symbol_type **make_LESS** (const location_type &l)
- static symbol_type **make_GREATER** (const location_type &l)
- static symbol_type **make_ASSIGN** (const location_type &l)
- static symbol_type **make_COMMA** (const location_type &l)

Static Public Attributes

- static const symbol_kind_type **YYNTOKENS** = symbol_kind::YYNTOKENS

  The number of tokens.

17.316.1 Detailed Description

A Bison parser.

17.316.2 Member Function Documentation

17.316.2.1 error()

```
void yy::genesyspp_parser::error (
            const location_type & loc,
            const std::string & msg )  [virtual]
```

Report a syntax error.

Parameters

loc	where the syntax error is found.
msg	a description of the syntax error.

17.316.2.2 operator()()

```
int yy::genesyspp_parser::operator() ( )
```

Parse. An alias for parse ().

Returns

0 iff parsing succeeded.

17.316.2.3 parse()

```
int yy::genesyspp_parser::parse ( )  [virtual]
```

Parse.

Returns

0 iff parsing succeeded.

Length of the RHS of the rule being reduced.
The lookahead symbol.
The locations where the error started and ended.
The return value of parse ().

17.316.2.4 symbol_name()

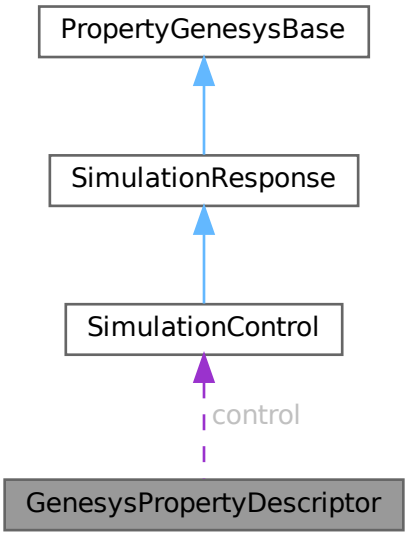
```
std::string yy::genesyspp_parser::symbol_name (
            symbol_kind_type yysymbol )  [static]
```

The user-facing name of the symbol whose (internal) number is YYSYMBOL. No bounds checking.
The documentation for this class was generated from the following files:

- parser/GenesysParser.h
- parser/GenesysParser.cpp

17.317 GenesysPropertyDescriptor Struct Reference

Collaboration diagram for GenesysPropertyDescriptor:



Public Attributes

- SimulationControl * **control** = nullptr
- std::string **ownerClassName**
- std::string **ownerElementName**
- std::string **displayName**
- std::string **technicalTypeName**
- GenesysPropertyKind **kind** = GenesysPropertyKind::Unknown
- bool **readOnly** = true
- bool **isList** = false
- bool **isClass** = false
- bool **isEnum** = false
- bool **isInlineObject** = false
- bool **isModelDataDefinitionReference** = false
- bool **supportsInlineExpansion** = false
- bool **supportsListEditor** = false
- bool **supportsExistingObjectSelection** = false
- bool **supportsObjectCreation** = false
- bool **supportsNewListElementCreation** = false
- SimulationControlEditorHint **editorHint** = SimulationControlEditorHint::Default
- std::string **currentValue**
- std::vector< std::string > **choices**
- std::string **currentReferenceType**
- std::vector< std::string > **creatableReferenceTypes**
- std::string **currentListElementType**
- std::vector< std::string > **creatableListElementTypes**

The documentation for this struct was generated from the following file:

- kernel/simulator/GenesysPropertyIntrospection.h

17.318 GenesysPropertyIntrospection Class Reference

Static Public Member Functions

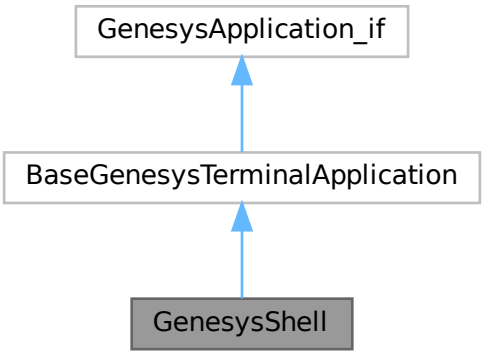
- static GenesysPropertyDescriptor **describe** (SimulationControl *control)
- static std::vector< GenesysPropertyDescriptor > **describe** (List< SimulationControl * > *controls)
- static GenesysPropertyKind **deduceKind** (const SimulationControl *control)
- static bool **setValue** (SimulationControl *control, const std::string &value, bool remove=false, std::string *errorMessage=nullptr)
- static std::vector< SimulationControl * > **children** (SimulationControl *control, int index=0)

The documentation for this class was generated from the following files:

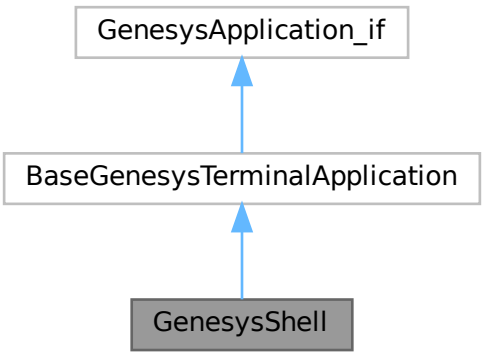
- kernel/simulator/GenesysPropertyIntrospection.h
- kernel/simulator/GenesysPropertyIntrospection.cpp

17.319 GenesysShell Class Reference

Inheritance diagram for GenesysShell:



Collaboration diagram for GenesysShell:



Public Member Functions

- **GenesysShell** (const GenesysShell &orig)
- virtual int main (int argc, char **argv) override
- void **cmdScript** ()
- void **cmdExecuteScript** ()
- void **cmdHelp** ()
- void **cmdQuit** ()
- void **cmdBash** ()
- void **cmdFacade** ()
- void **cmdLicence** ()
- void **cmdInfo** ()
- void **cmdSim** ()
- void **cmdData** ()
- void **cmdComponent** ()
- void **cmdExperiment** ()
- void **cmdParser** ()
- void **cmdTraceLevel** ()
- void **cmdPlugin** ()
- void **cmdSimulation** ()
- void **cmdReplication** ()
- void **cmdModel** ()
- void **cmdModelLoad** ()
- void **cmdModelSave** ()

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.319.1 Member Function Documentation

17.319.1.1 main()

```
int GenesysShell::main (
            int argc,
            char ** argv )  [override], [virtual]
```

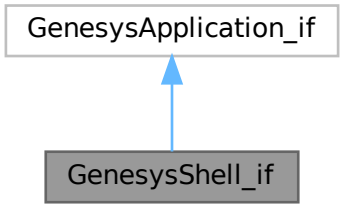
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

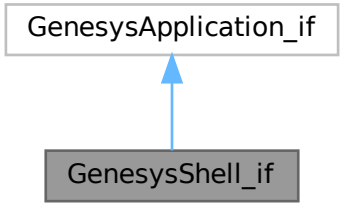
- applications/shell/GenesysShell.h
- applications/shell/GenesysShell.cpp

17.320 GenesysShell_if Class Reference

Inheritance diagram for GenesysShell_if:



Collaboration diagram for GenesysShell_if:



Public Member Functions

- virtual void **openModel** (std::string filename)=0
- virtual void **saveModelAs** (std::string filename)=0
- virtual void **saveModel** ()=0
- virtual void **listElements** ()=0
- virtual void **listComponents** ()=0
- virtual void **listHosts** ()=0
- virtual void **listPlugins** ()=0
- virtual void **deleteTraceFiles** ()=0

- virtual void **traceLevel** (TraceManager::TraceLevel tracelevel)=0
- virtual void **addPlugin** (std::string filename)=0
- virtual void **addFromFile** (std::string filename)=0
- virtual void **readCommandsFromFile** (std::string filename)=0
- virtual void **redirectTrace** (std::string trace, std::string dest, std::string filename)=0
- virtual void **closeModel** ()=0
- virtual void **createModel** ()=0
- virtual void **execLinuxCommand** (std::string command)=0
- virtual void **verboseMode** (bool on)=0
- virtual void **check** ()=0
- virtual void **getGenesysInfo** ()=0
- virtual void **getCommandLine** ()=0
- virtual void **sendFile** (std::string filename, std::string hostname, std::string portname)=0
- virtual void **setActivationCode** (std::string code)=0
- virtual void **receiveFile** (std::string filename)=0
- virtual void **startSimulation** ()=0
- virtual void **stepSimulation** ()=0
- virtual void **stopSimulation** ()=0
- virtual void **showInit** ()=0
- virtual void **showHelp** ()=0
- virtual void **showHostName** ()=0

Public Member Functions inherited from GenesysApplication_if

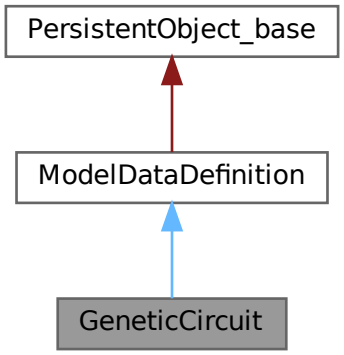
- virtual int main (int argc, char **argv)=0

The documentation for this class was generated from the following file:

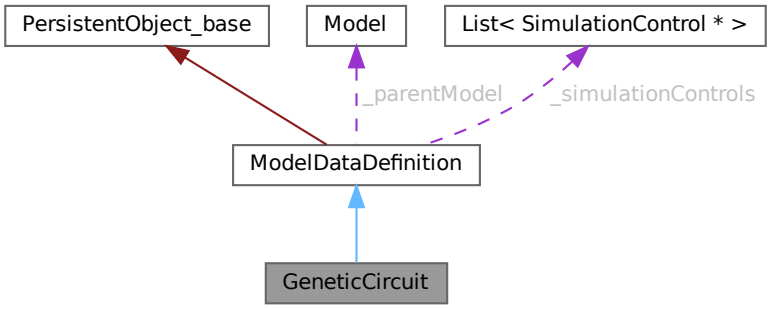
- applications/shell/GenesysShell_if.h

17.321 GeneticCircuit Class Reference

`#include <plugins/data/BiochemicalSimulation/GeneticCircuit.h>`
Inheritance diagram for GeneticCircuit:



Collaboration diagram for GeneticCircuit:



Public Member Functions

- **GeneticCircuit** (Model *model, std::string name="")
- void **addPart** (std::string partName)
- void **addRegulation** (std::string regulationName)
- void **clearParts** ()
- void **clearRegulations** ()
- const std::vector< std::string > & **getPartNames** () const
- const std::vector< std::string > & **getRegulationNames** () const
- void **setHostOrganism** (std::string hostOrganism)
- std::string **getHostOrganism** () const
- void **setCompartment** (std::string compartment)
- std::string **getCompartment** () const
- void **setEnabled** (bool enabled)
- bool **isEnabled** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition ∗modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition ∗modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord ∗fields) override

  must be overriden by derived classes

- virtual void _saveInstance (PersistenceRecord ∗fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation ∗ _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl ∗control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.321.1 Detailed Description

Aggregate genetic circuit definition.
A GeneticCircuit stores an ordered sequence of genetic parts and a set of regulatory relationships by name so it can be persisted independently from runtime object pointers.

17.321.2 Member Function Documentation

17.321.2.1 _check()

```
bool GeneticCircuit::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.321.2.2 _loadInstance()

```
bool GeneticCircuit::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.321.2.3 _saveInstance()

```
void GeneticCircuit::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.321.2.4 show()

```
std::string GeneticCircuit::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

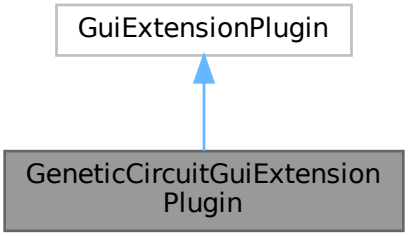
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

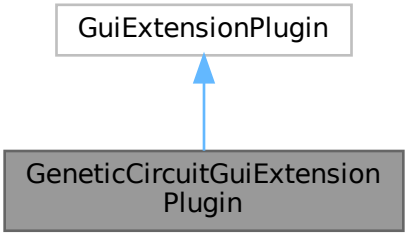
- plugins/data/BiochemicalSimulation/GeneticCircuit.h
- plugins/data/BiochemicalSimulation/GeneticCircuit.cpp

17.322 GeneticCircuitGuiExtensionPlugin Class Reference

Inheritance diagram for GeneticCircuitGuiExtensionPlugin:



Collaboration diagram for GeneticCircuitGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.322.1 Member Function Documentation

17.322.1.1 displayName()

```
std::string GeneticCircuitGuiExtensionPlugin::displayName ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.322.1.2 extensionId()

```
std::string GeneticCircuitGuiExtensionPlugin::extensionId ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.322.1.3 registerContributions()

```
void GeneticCircuitGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.322.1.4 requiredModelPlugins()

```
std::vector< std::string > GeneticCircuitGuiExtensionPlugin::requiredModelPlugins ( ) const
[inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.322.1.5 version()

```
std::string GeneticCircuitGuiExtensionPlugin::version ( ) const  [inline], [override], [virtual]
```

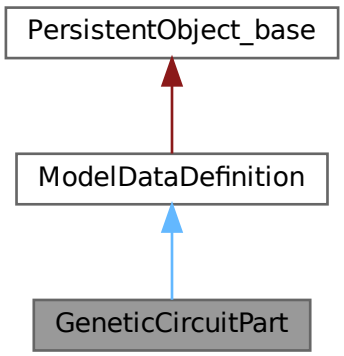
Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

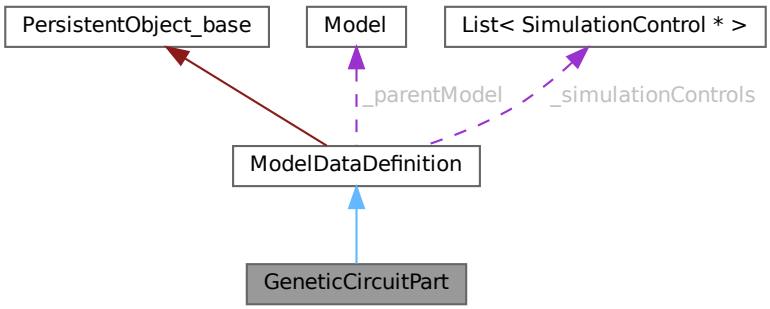
- applications/gui/genesys/extensions/GeneticCircuitGuiExtensionPlugin.cpp

17.323 GeneticCircuitPart Class Reference

`#include <plugins/data/BiochemicalSimulation/GeneticCircuitPart.h>`
Inheritance diagram for GeneticCircuitPart:



Collaboration diagram for GeneticCircuitPart:



Public Member Functions

- **GeneticCircuitPart** (Model *model, std::string name="")
- void **setPartType** (std::string partType)
- std::string **getPartType** () const
- void **setSequence** (std::string sequence)
- std::string **getSequence** () const
- void **setProductSpeciesName** (std::string productSpeciesName)
- std::string **getProductSpeciesName** () const
- void **setCopyNumber** (double copyNumber)
- double **getCopyNumber** () const
- void **setBasalExpressionRate** (double basalExpressionRate)
- double **getBasalExpressionRate** () const
- void **setDegradationRate** (double degradationRate)
- double **getDegradationRate** () const

- void **setEnabled** (bool enabled)
- bool **isEnabled** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()

- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model ∗ **_parentModel**
- List< SimulationControl ∗ > ∗ **_simulationControls** = new List<SimulationControl∗>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.323.1 Detailed Description

Genetic circuit structural part definition.
This data definition represents a reusable genetic part (promoter, RBS, CDS, terminator, etc.) and optionally links it to the BioSpecies produced by the part expression flow.

17.323.2 Member Function Documentation

17.323.2.1 _check()

```
bool GeneticCircuitPart::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.323.2.2 _loadInstance()

```
bool GeneticCircuitPart::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.323.2.3 _saveInstance()

```
void GeneticCircuitPart::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.323.2.4 show()

```
std::string GeneticCircuitPart::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

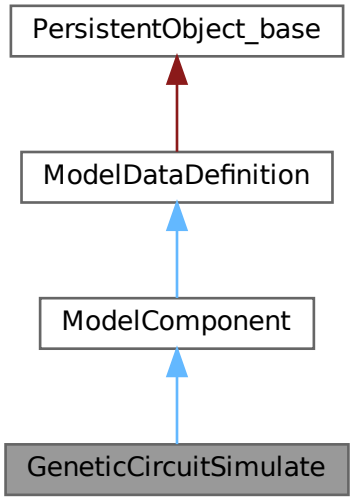
Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/BiochemicalSimulation/GeneticCircuitPart.h
- plugins/data/BiochemicalSimulation/GeneticCircuitPart.cpp

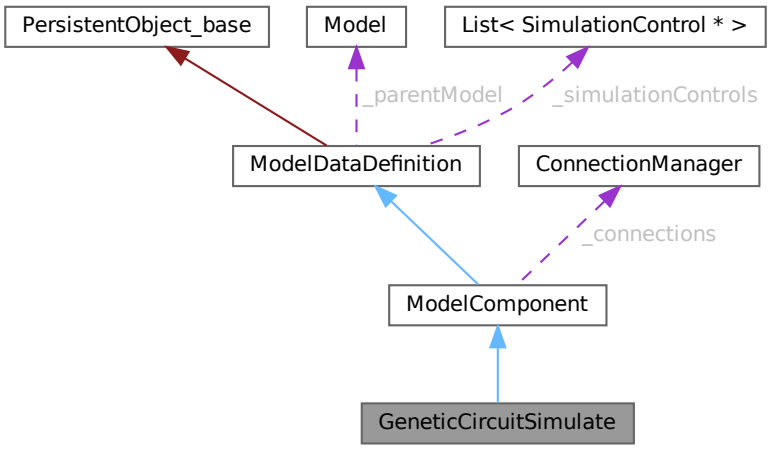
17.324 GeneticCircuitSimulate Class Reference

```
#include <plugins/components/BiochemicalSimulation/GeneticCircuitSimulate.↩
h>
```

Inheritance diagram for GeneticCircuitSimulate:



Collaboration diagram for GeneticCircuitSimulate:



Public Member Functions

- **GeneticCircuitSimulate** (Model *model, std::string name="")
- void **setGeneticCircuit** (GeneticCircuit *geneticCircuit)
- GeneticCircuit * **getGeneticCircuit** () const
- void **setStartTime** (double startTime)
- double **getStartTime** () const
- void **setStopTime** (double stopTime)
- double **getStopTime** () const

- void **setStepSize** (double stepSize)
- double **getStepSize** () const
- void **setApplyRegulation** (bool applyRegulation)
- bool **getApplyRegulation** () const
- void **setLastSucceeded** (bool lastSucceeded)
- bool **getLastSucceeded** () const
- void **setLastSampleCount** (unsigned int lastSampleCount)
- unsigned int **getLastSampleCount** () const
- void **setLastTotalExpression** (double lastTotalExpression)
- double **getLastTotalExpression** () const
- void **setLastMessage** (std::string lastMessage)
- std::string **getLastMessage** () const
- bool **simulate** (std::string &errorMessage)
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static bool **simulateCircuit** (Model *model, GeneticCircuit *geneticCircuit, double startTime, double stop↩ Time, double stepSize, bool applyRegulation, GeneticCircuitSimulationSummary *summary, std::string &errorMessage)

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.

- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)

- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.324.1 Detailed Description

Component that simulates a GeneticCircuit over a time window.

17.324.2 Member Function Documentation

17.324.2.1 _check()

```
bool GeneticCircuitSimulate::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.324.2.2 _createEditableDataDefinitions()

```
void GeneticCircuitSimulate::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.324.2.3 _loadInstance()

```
bool GeneticCircuitSimulate::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.324.2.4 _onDispatchEvent()

```
void GeneticCircuitSimulate::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.324.2.5 _saveInstance()

```
void GeneticCircuitSimulate::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.324.2.6 show()

```
std::string GeneticCircuitSimulate::show ( )  [override], [virtual]
```

virtual public methods

Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/BiochemicalSimulation/GeneticCircuitSimulate.h
- plugins/components/BiochemicalSimulation/GeneticCircuitSimulate.cpp

17.325 GeneticCircuitSimulationSummary Struct Reference

Public Attributes

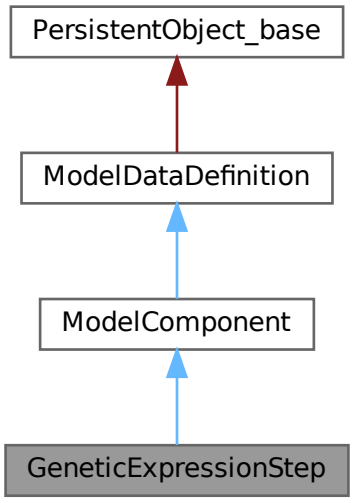
- bool **succeeded** = false
- unsigned int **sampleCount** = 0
- double **totalExpression** = 0.0
- std::string **message** = ""

The documentation for this struct was generated from the following file:

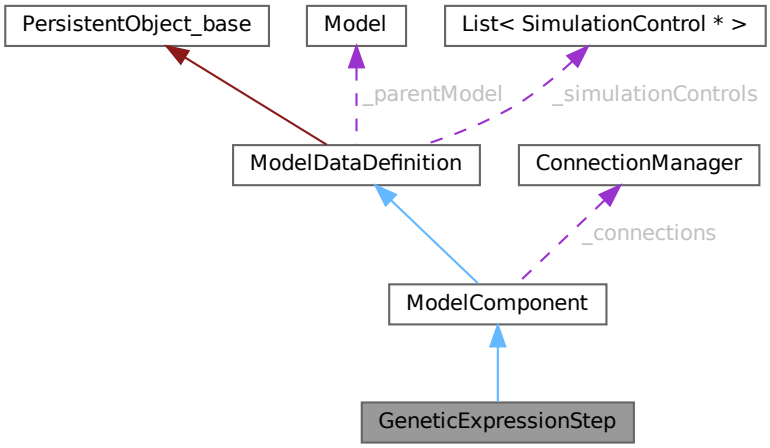
- plugins/components/BiochemicalSimulation/GeneticCircuitSimulate.h

17.326 GeneticExpressionStep Class Reference

`#include <plugins/components/BiochemicalSimulation/GeneticExpressionStep.h>`
Inheritance diagram for GeneticExpressionStep:



Collaboration diagram for GeneticExpressionStep:



Public Member Functions

- **GeneticExpressionStep** (Model ∗model, std::string name="")
- void **setGeneticCircuit** (GeneticCircuit ∗geneticCircuit)
- GeneticCircuit ∗ **getGeneticCircuit** () const
- void **setTimeStep** (double timeStep)
- double **getTimeStep** () const
- void **setApplyRegulation** (bool applyRegulation)
- bool **getApplyRegulation** () const
- void **setLastSucceeded** (bool lastSucceeded)
- bool **getLastSucceeded** () const
- void **setLastTotalExpression** (double lastTotalExpression)
- double **getLastTotalExpression** () const
- void **setLastMessage** (std::string lastMessage)
- std::string **getLastMessage** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.326.1 Detailed Description

Component that executes one expression update step over a GeneticCircuit.

17.326.2 Member Function Documentation

17.326.2.1 _check()

```
bool GeneticExpressionStep::_check (
            std::string & errorMessage ) [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.326.2.2 _createEditableDataDefinitions()

```
void GeneticExpressionStep::_createEditableDataDefinitions ( ) [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.326.2.3 _loadInstance()

```
bool GeneticExpressionStep::_loadInstance (
            PersistenceRecord * fields ) [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.326.2.4 _onDispatchEvent()

```
void GeneticExpressionStep::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber ) [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.326.2.5 _saveInstance()

```
void GeneticExpressionStep::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.326.2.6 show()

```
std::string GeneticExpressionStep::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

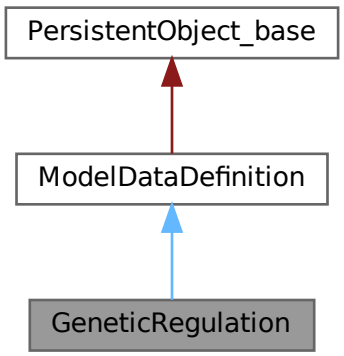
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/BiochemicalSimulation/GeneticExpressionStep.h
- plugins/components/BiochemicalSimulation/GeneticExpressionStep.cpp

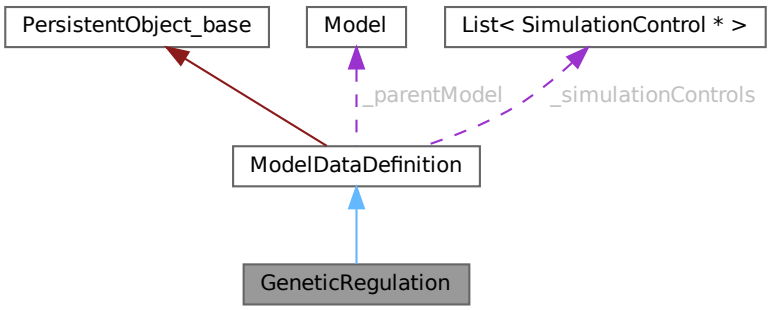
17.327 GeneticRegulation Class Reference

```
#include <plugins/data/BiochemicalSimulation/GeneticRegulation.h>
```

Inheritance diagram for GeneticRegulation:



Collaboration diagram for GeneticRegulation:



Public Member Functions

- **GeneticRegulation** (Model *model, std::string name="")
- void **setRegulatorSpeciesName** (std::string regulatorSpeciesName)
- std::string **getRegulatorSpeciesName** () const
- void **setTargetPartName** (std::string targetPartName)
- std::string **getTargetPartName** () const
- void **setRegulationType** (std::string regulationType)
- std::string **getRegulationType** () const
- void **setHillCoefficient** (double hillCoefficient)
- double **getHillCoefficient** () const
- void **setDissociationConstant** (double dissociationConstant)
- double **getDissociationConstant** () const
- void **setMaxFoldChange** (double maxFoldChange)
- double **getMaxFoldChange** () const
- void **setLeakiness** (double leakiness)
- double **getLeakiness** () const
- void **setEnabled** (bool enabled)
- bool **isEnabled** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.

- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()

- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.327.1 Detailed Description

Transcriptional regulation relationship for genetic circuits.

17.327.2 Member Function Documentation

17.327.2.1 _check()

```
bool GeneticRegulation::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.327.2.2 _loadInstance()

```
bool GeneticRegulation::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.327.2.3 _saveInstance()

```
void GeneticRegulation::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.327.2.4 show()

```
std::string GeneticRegulation::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

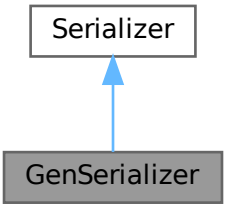
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

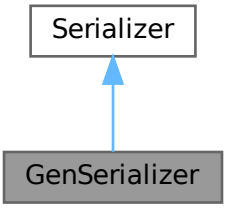
- plugins/data/BiochemicalSimulation/GeneticRegulation.h
- plugins/data/BiochemicalSimulation/GeneticRegulation.cpp

17.328 GenSerializer Class Reference

Inheritance diagram for GenSerializer:



Collaboration diagram for GenSerializer:



Public Member Functions

- **GenSerializer** (Model *model)
- PersistenceRecord * newPersistenceRecord () override

  Creates an empty persistence record.
- bool dump (std::ostream &output) override

  Writes formatted contents to a given output stream.
- bool load (std::istream &input) override

  Attempts to load contents from a given input stream.
- bool get (const std::string &name, PersistenceRecord *entry) override

  Tries to fetch an entry with the given label as a key.
- bool put (const std::string name, const std::string type, const Util::identification id, PersistenceRecord *fields) override

  Upserts an entry into this structure.
- int for_each (std::function< int(const std::string &) > delegate) override

  Iterates over this structure's entries, ordered by ID.

Friends

- class **PersistenceDefaultImpl2**

17.328.1 Member Function Documentation

17.328.1.1 dump()

```
bool GenSerializer::dump (
            std::ostream & output )  [override], [virtual]
```

Writes formatted contents to a given output stream.

Parameters

output	output stream to write to.

Returns

whether the current state of this structure's contents was deemed valid for serialization.

Implements Serializer.

17.328.1.2 for_each()

```
int GenSerializer::for_each (
            std::function< int(const std::string &) > delegate )  [override], [virtual]
```

Iterates over this structure's entries, ordered by ID.

Parameters

delegate	delegate which is applied to every entry-identifying key, until it returns a non-zero value.

Returns

either zero, or the first non-zero value returned by the delegate.

Implements Serializer.

17.328.1.3 get()

```
bool GenSerializer::get (
            const std::string & name,
            PersistenceRecord * entry )  [override], [virtual]
```

Tries to fetch an entry with the given label as a key.

Parameters

name	a label which uniquely identifies an entry.
entry	(out-param) a pointer to a caller-owned entry, which is filled on success.

Returns

whether the entry was found and the out-param was actually written to.

Implements Serializer.

17.328.1.4 load()

```
bool GenSerializer::load (
            std::istream & input )  [override], [virtual]
```

Attempts to load contents from a given input stream.

Returns

whether the input was well-formatted and loading was successful.

Implements Serializer.

17.328.1.5 newPersistenceRecord()

```
PersistenceRecord * GenSerializer::newPersistenceRecord ( )  [override], [virtual]
```

Creates an empty persistence record.
Although the returned record can be freed normally with `delete`, care must be taken to only use records which come from a implementation, when configuring that implementation's contents.
Implements Serializer.

17.328.1.6 put()

```
bool GenSerializer::put (
            const std::string name,
            const std::string type,
            const Util::identification id,
            PersistenceRecord * fields )  [override], [virtual]
```

Upserts an entry into this structure.

Parameters

name	a label which uniquely identifies the entry.
type	a type name.
id	a component/datadef ID number.
fields	a persistence record for the entry, whose contents will be copied if valid.

Returns

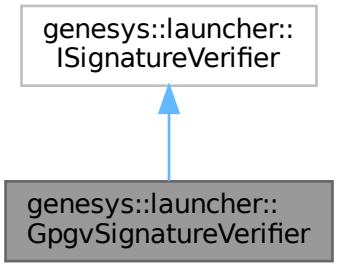
whether the entry was valid and was actually copied into this structure.

Implements Serializer.
The documentation for this class was generated from the following files:

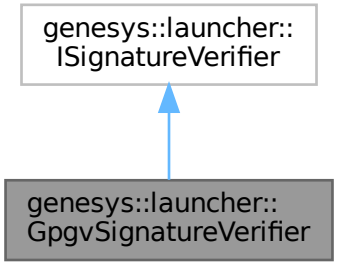
- kernel/simulator/persistence/GenSerializer.h
- kernel/simulator/persistence/GenSerializer.cpp

17.329 genesys::launcher::GpgvSignatureVerifier Class Reference

Inheritance diagram for genesys::launcher::GpgvSignatureVerifier:



Collaboration diagram for genesys::launcher::GpgvSignatureVerifier:



Public Member Functions

- **GpgvSignatureVerifier** (QString keyringPath, LauncherLogger *logger=nullptr, int timeoutMs=15000)
- VerificationResult verify (const QString &dataPath, const QString &signaturePath) override

17.329.1 Member Function Documentation

17.329.1.1 verify()

```
VerificationResult genesys::launcher::GpgvSignatureVerifier::verify (
            const QString & dataPath,
            const QString & signaturePath )  [override], [virtual]
```

Implements genesys::launcher::ISignatureVerifier.

The documentation for this class was generated from the following files:

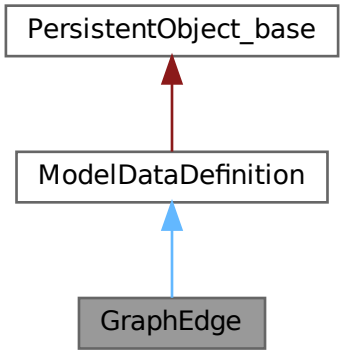
- applications/launcher/UpdateVerifier.h
- applications/launcher/UpdateVerifier.cpp

17.330 GraphEdge Class Reference

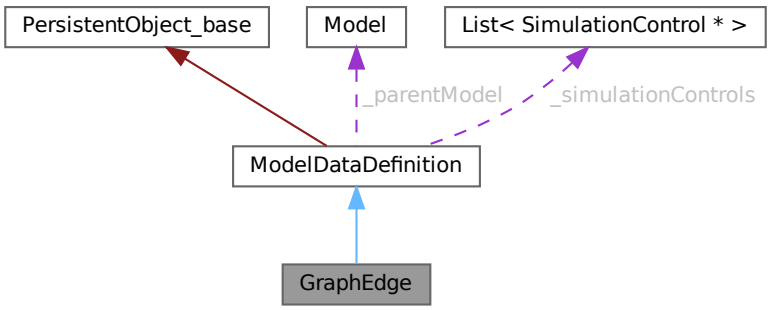
Mathematical graph edge between two GraphNode endpoints.

`#include <plugins/data/ModalModel/GraphEdge.h>`

Inheritance diagram for GraphEdge:



Collaboration diagram for GraphEdge:



Public Member Functions

- **GraphEdge** (Model ∗model, std::string name="")
- **GraphEdge** (Model ∗model, GraphNode ∗source, GraphNode ∗destination, std::string name="")
- void **setSource** (GraphNode ∗source)
- GraphNode ∗ **getSource** () const
- void **setDestination** (GraphNode ∗destination)
- GraphNode ∗ **getDestination** () const
- void **setDirected** (bool directed)
- bool **isDirected** () const
- void **setWeight** (double weight)
- void **clearWeight** ()
- bool **hasWeight** () const
- double **getWeight** (double defaultWeight=1.0) const
- std::string **getSourceNodeName** () const
- std::string **getDestinationNodeName** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)

- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.330.1 Detailed Description

Mathematical graph edge between two GraphNode endpoints.
GraphEdge is not a GenESyS process Connection and does not imply Entity flow. It supports directed or undirected interpretation, optional numeric weight, self-loops and parallel edges. A GraphNetwork decides whether the edge is interpreted as directed or undirected when the edge is inserted.

17.330.2 Member Function Documentation

17.330.2.1 _check()

```
bool GraphEdge::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.330.2.2 _loadInstance()

```
bool GraphEdge::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.330.2.3 _saveInstance()

```
void GraphEdge::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.330.2.4 show()

```
std::string GraphEdge::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

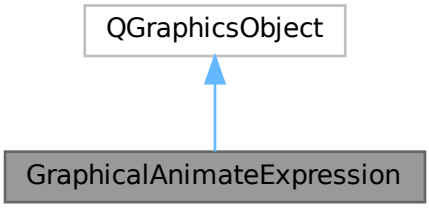
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

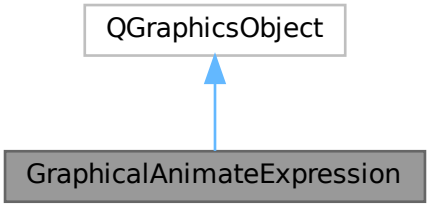
- plugins/data/ModalModel/GraphEdge.h
- plugins/components/ModalModel/GraphEdge.cpp

17.331 GraphicalAnimateExpression Class Reference

Inheritance diagram for GraphicalAnimateExpression:



Collaboration diagram for GraphicalAnimateExpression:



The documentation for this class was generated from the following files:

- applications/gui/genesys/graphicals/GraphicalAnimateExpression.h
- applications/gui/genesys/graphicals/GraphicalAnimateExpression.cpp

17.332 GraphicalAssociation Class Reference

The documentation for this class was generated from the following files:

- applications/gui/genesys/graphicals/GraphicalAssociation.h
- applications/gui/genesys/graphicals/GraphicalAssociation.cpp

17.333 GraphicalAutomaticPositioningStrategy Class Reference

Public Member Functions

- virtual ~GraphicalAutomaticPositioningStrategy ()=default

  Strategy interface for automatic placement of graphical data definitions.
- virtual QPointF **initialComponentLayoutOrigin** () const =0

  Returns the origin used when laying out the first graphical component.
- virtual QPointF dataDefinitionPosition (const QRectF &anchorBounds, const QSizeF &childSize, int index, int count, bool upperArc, int radialLayer) const =0

  Computes the position of one child item around an anchor rectangle.

17.333.1 Constructor & Destructor Documentation

17.333.1.1 ~GraphicalAutomaticPositioningStrategy()

```
virtual GraphicalAutomaticPositioningStrategy::~GraphicalAutomaticPositioningStrategy ( )
[virtual], [default]
```

Strategy interface for automatic placement of graphical data definitions.
Implementations decide how child graphical items are laid out around an anchor item. The GUI uses this abstraction so layout policy can be selected from preferences without changing the reconstruction code that consumes the strategy.

17.333.2 Member Function Documentation

17.333.2.1 dataDefinitionPosition()

```
virtual QPointF GraphicalAutomaticPositioningStrategy::dataDefinitionPosition (
            const QRectF & anchorBounds,
            const QSizeF & childSize,
            int index,
            int count,
            bool upperArc,
            int radialLayer ) const  [pure virtual]
```

Computes the position of one child item around an anchor rectangle.

Parameters

anchorBounds	Bounds of the item being decorated.
childSize	Size of the child item to place.
index	Zero-based child index in the current arc.
count	Number of children to place in the same arc.
upperArc	True when the child should be positioned above the anchor.
radialLayer	Additional radial layer for nested attachments.

Returns

Scene position where the child should be placed.

The documentation for this class was generated from the following file:

- applications/gui/genesys/services/GraphicalAutomaticPositioningStrategy.h

17.334 GraphicalAutomaticPositioningStrategyFactory Class Reference

Factory for the automatic positioning strategy selected from GUI preferences.
`#include <applications/gui/genesys/services/GraphicalAutomaticPositioning↩ Strategy.h>`

Static Public Member Functions

- static std::unique_ptr< GraphicalAutomaticPositioningStrategy > **createFromPreferences** ()

  Creates the preferred automatic positioning strategy instance.

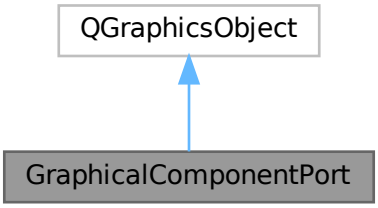
17.334.1 Detailed Description

Factory for the automatic positioning strategy selected from GUI preferences.
The documentation for this class was generated from the following files:

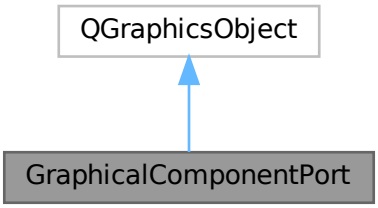
- applications/gui/genesys/services/GraphicalAutomaticPositioningStrategy.h
- applications/gui/genesys/services/GraphicalAutomaticPositioningStrategy.cpp

17.335 GraphicalComponentPort Class Reference

Inheritance diagram for GraphicalComponentPort:



Collaboration diagram for GraphicalComponentPort:



Public Member Functions

- GraphicalComponentPort (GraphicalModelComponent ∗componentGraph, bool isInputPort, unsigned int portNum=0, QGraphicsItem ∗parent=nullptr)

  Visual port attached to a graphical component for connection routing.
- QRectF **boundingRect** () const override

  Returns the local bounding rectangle used for hit testing and painting.
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget) override

  Paints the port handle using the component and selection styling.
- qreal **width** () const

  Returns the logical width used by the port geometry.
- qreal **height** () const

  Returns the logical height used by the port geometry.
- void **addGraphicalConnection** (GraphicalConnection ∗connection)

  Registers one connection attached to this port.
- void **removeGraphicalConnection** (GraphicalConnection ∗connection)

  Unregisters one connection attached to this port.
- unsigned int **portNum** () const

  Returns the port number within the owning component.
- bool **isInputPort** () const

Returns true when this is an input port.

- GraphicalModelComponent * **graphicalComponent** () const

  Returns the owning graphical component.

- QList< GraphicalConnection * > * **getConnections** () const

  Returns the list of connections currently attached to this port.

Protected Member Functions

- virtual QVariant **itemChange** (QGraphicsItem::GraphicsItemChange change, const QVariant &value) override

  Keeps the port aligned when its parent component moves or is reparented.

17.335.1 Constructor & Destructor Documentation

17.335.1.1 GraphicalComponentPort()

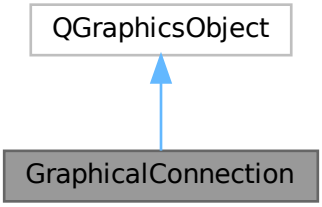
```
GraphicalComponentPort::GraphicalComponentPort (
            GraphicalModelComponent * componentGraph,
            bool isInputPort,
            unsigned int portNum = 0,
            QGraphicsItem * parent = nullptr )
```

Visual port attached to a graphical component for connection routing.
Ports are the connection endpoints used by the scene to attach `GraphicalConnection` items to a component. The port item keeps track of connected edges and exposes enough geometry for the scene to render and route links consistently.
The documentation for this class was generated from the following files:

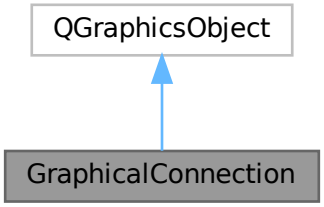
- applications/gui/genesys/graphicals/GraphicalComponentPort.h
- applications/gui/genesys/graphicals/GraphicalComponentPort.cpp

17.336 GraphicalConnection Class Reference

Graphical edge that represents model-level component connections.
`#include <applications/gui/genesys/graphicals/GraphicalConnection.h>`
Inheritance diagram for GraphicalConnection:



Collaboration diagram for GraphicalConnection:



Public Types

- enum class **ConnectionType** : int { **HORIZONTAL** = 1 , **VERTICAL** = 2 , **DIRECT** = 3 , **USERDEFINED** = 4 }

Public Member Functions

- GraphicalConnection (GraphicalComponentPort ∗sourceGraphicalPort, GraphicalComponentPort ∗destination↩ GraphicalPort, unsigned int portSourceConnection=0, unsigned int portDestinationConnection=0, QColor color=Qt::black, QGraphicsItem ∗parent=nullptr)

  Creates a graphical connection between source and destination ports.
- **GraphicalConnection** (const GraphicalConnection &orig)

  Copy constructor used by restoration and clipboard flows.
- virtual ∼**GraphicalConnection** ()

  Releases the connection item and its auxiliary routing state.
- QRectF boundingRect () const override

  Returns item bounding rectangle for Qt painting system.
- QPainterPath **shape** () const override

  Returns the precise clickable shape around the visible connection path.
- void paint (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget) override

  Paints connection path and selection handles.
- Connection ∗ **getSource** () const

  Returns auxiliary source Connection metadata.
- Connection ∗ **getDestination** () const

  Returns auxiliary destination Connection metadata.
- GraphicalComponentPort ∗ **getSourceGraphicalPort** ()

  Returns source graphical port.
- GraphicalComponentPort ∗ **getDestinationGraphicalPort** ()

  Returns destination graphical port.
- GraphicalModelComponent ∗ **getSourceGraphicalComponent** () const

  Returns source graphical component resolved from source port when available.
- GraphicalModelComponent ∗ **getDestinationGraphicalComponent** () const

  Returns destination graphical component resolved from destination port when available.
- void updateDimensionsAndPosition ()

  Recomputes local geometry based on current port positions.
- GraphicalConnection::ConnectionType **connectionType** () const

  Returns configured connection routing type.
- unsigned int **getPortSourceConnection** () const

Returns source output port index associated with model connection.

- unsigned int **getPortDestinationConnection** () const

  Returns destination input port index associated with model connection.

- void **setConnectionType** (GraphicalConnection::ConnectionType newConnectionType)

  Sets routing type used during painting.

- QList< QPointF > **getPoints** () const

  Returns sampled scene points used for serialization/inspection.

- bool **usesCurvedStyle** () const

  Returns true when the current visual style uses curved model connection paths.

- QPainterPath **animationPathForImage** (qreal imageWidth, qreal imageHeight) const

  Returns the scene path followed by entity animation in modern connection style.

Protected Member Functions

- virtual bool **sceneEvent** (QEvent *event) override

17.336.1 Detailed Description

Graphical edge that represents model-level component connections.
`GraphicalConnection` links two `GraphicalComponentPort` objects and keeps auxiliary `Connection` metadata synchronized with the simulator model.
The item is also used for the diagram-level routing that the GUI derives from the model structure when rendering connections inside the scene.

Todo Evaluate decoupling model metadata from this item to keep this class purely visual.

17.336.2 Constructor & Destructor Documentation

17.336.2.1 GraphicalConnection()

```
GraphicalConnection::GraphicalConnection (
            GraphicalComponentPort * sourceGraphicalPort,
            GraphicalComponentPort * destinationGraphicalPort,
            unsigned int portSourceConnection = 0,
            unsigned int portDestinationConnection = 0,
            QColor color = Qt::black,
            QGraphicsItem * parent = nullptr )
```

Creates a graphical connection between source and destination ports.
Bloco 1: inicialização de metadados de conexão. Mantemos wrappers de Connection para facilitar sincronização com kernel.
Bloco 2: configuração visual/interativa do item.
Bloco 3: cálculo inicial de geometria da conexão.
Bloco 4: registro bidirecional nas portas para atualização quando houver movimento.

Todo Extrair para método dedicado `_bindPorts()` para legibilidade.

17.336.3 Member Function Documentation

17.336.3.1 boundingRect()

```
QRectF GraphicalConnection::boundingRect ( ) const  [override]
```

Returns item bounding rectangle for Qt painting system.
Bloco 1: calcula margem extra baseada na porta para reduzir clipping visual.
Bloco 2: retorna retângulo local com margem.

17.336.3.2 paint()

```
void GraphicalConnection::paint (
            QPainter * painter,
            const QStyleOptionGraphicsItem * option,
            QWidget * widget )  [override]
```

Paints connection path and selection handles.
Bloco 2: normaliza coordenadas locais considerando orientação relativa.
Bloco 3: gera path de desenho com base no tipo de roteamento.
Bloco 4: renderiza o path final da conexão.
Bloco 5: desenha “handles” quando selecionado.

17.336.3.3 updateDimensionsAndPosition()

```
void GraphicalConnection::updateDimensionsAndPosition ( )
```

Recomputes local geometry based on current port positions.
Bloco 1: coleta de geometria absoluta de portas.
Bloco 2: projeta esta conexão para um sistema de coordenadas local mínimo.
Bloco 3: mantém apenas atualização geométrica local.
The documentation for this class was generated from the following files:

- applications/gui/genesys/graphicals/GraphicalConnection.h
- applications/gui/genesys/graphicals/GraphicalConnection.cpp

17.337 GraphicalConnectionStyle Class Reference

Public Types

- enum class **RouteType** { **Horizontal** , **Vertical** , **Direct** , **UserDefined** }

Static Public Member Functions

- static bool **usesModernCurves** ()
- static QPainterPath **modelConnectionPath** (const QPointF &source, const QPointF &destination, RouteType routeType, bool modernCurves)
- static QPainterPath **diagramConnectionPath** (const QPointF &source, const QPointF &destination, bool modernCurves)
- static QPointF **pointAtProgress** (const QPainterPath &path, qreal progress)

The documentation for this class was generated from the following files:

- applications/gui/genesys/graphicals/GraphicalConnectionStyle.h
- applications/gui/genesys/graphicals/GraphicalConnectionStyle.cpp

17.338 GraphicalContextMenuController Class Reference

Builds context menus for the graphical model view using existing main-window actions.
`#include <applications/gui/genesys/controllers/GraphicalContextMenuController.h>`

Public Member Functions

- **GraphicalContextMenuController** (ModelGraphicsView *graphicsView, Ui::MainWindow *ui, QAction *openSubmodelAction, std::function< ModelGraphicsScene *()> currentScene, std::function< void()> actualizeActions)

  Creates the controller without taking ownership of UI or scene objects.
- void **handleGraphicsViewContextMenu** (QContextMenuEvent *event)

  Shows the context menu matching the item or background under the mouse.

17.338.1 Detailed Description

Builds context menus for the graphical model view using existing main-window actions.
The controller intentionally reuses QAction instances from the main window so shortcuts, enabled state, icons, checked state, and command handlers remain centralized.
The documentation for this class was generated from the following files:

- applications/gui/genesys/controllers/GraphicalContextMenuController.h
- applications/gui/genesys/controllers/GraphicalContextMenuController.cpp

17.339 GraphicalDataDefinitionLayout Class Reference

Pure helper that computes arc placement for graphical data definitions.
`#include <applications/gui/genesys/services/GraphicalDataDefinitionLayout.h>`

Static Public Member Functions

- static QPointF arcPosition (const QRectF &anchorBounds, const QSizeF &childSize, int index, int count, bool upperArc, int radialLayer=0)

  Computes a child position along the upper or lower arc around an anchor item.

17.339.1 Detailed Description

Pure helper that computes arc placement for graphical data definitions.
This helper keeps the geometry math independent from the GUI controllers and scene code so layout policies can be reused in both live rebuilds and persisted-state restoration.

17.339.2 Member Function Documentation

17.339.2.1 arcPosition()

```
QPointF GraphicalDataDefinitionLayout::arcPosition (
            const QRectF & anchorBounds,
            const QSizeF & childSize,
            int index,
            int count,
            bool upperArc,
            int radialLayer = 0 )  [static]
```

Computes a child position along the upper or lower arc around an anchor item.

Parameters

anchorBounds	Bounds of the anchor graphical item.
childSize	Size of the child item to place.
index	Zero-based child index.
count	Total number of children in the arc.
upperArc	True for the upper arc, false for the lower arc.
radialLayer	Additional radial offset used for nested attachment rings.

Returns

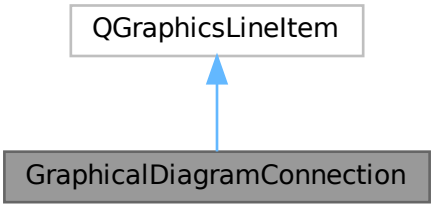
Scene position for the requested child.

The documentation for this class was generated from the following files:

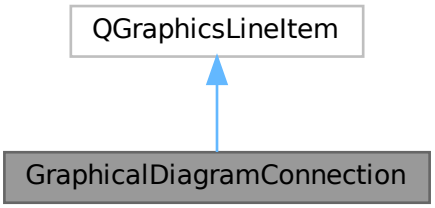
- applications/gui/genesys/services/GraphicalDataDefinitionLayout.h
- applications/gui/genesys/services/GraphicalDataDefinitionLayout.cpp

17.340 GraphicalDiagramConnection Class Reference

Inheritance diagram for GraphicalDiagramConnection:



Collaboration diagram for GraphicalDiagramConnection:



Public Types

- enum class **ConnectionType** { **ATTACHED** , **INTERNAL** }

Public Member Functions

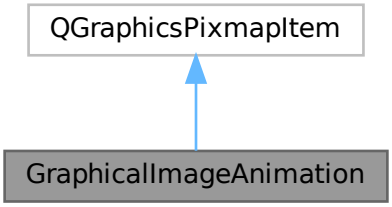
- **GraphicalDiagramConnection** (QGraphicsItem ∗dataDefinition, QGraphicsItem ∗linkedTo, ConnectionType type)
- void **refreshGeometry** ()
- QRectF **boundingRect** () const override
- QPainterPath **shape** () const override
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget=nullptr) override
- QGraphicsItem ∗ **getDataDefinition** ()
- QGraphicsItem ∗ **getLinkedDataDefinition** ()
- ConnectionType **getConnectionType** ()

The documentation for this class was generated from the following files:

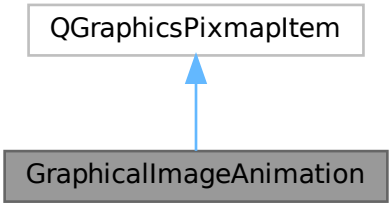
- applications/gui/genesys/graphicals/GraphicalDiagramConnection.h
- applications/gui/genesys/graphicals/GraphicalDiagramConnection.cpp

17.341 GraphicalImageAnimation Class Reference

Inheritance diagram for GraphicalImageAnimation:



Collaboration diagram for GraphicalImageAnimation:



Public Member Functions

- **GraphicalImageAnimation** (const QPointF startPoint, unsigned int width, unsigned int height, const QString imageName)
- QString **getImagePath** () const
- QString **getImageName** () const
- QPointF **getStartPoint** () const
- unsigned int **getWidth** () const
- unsigned int **getHeight** () const
- void **setImageName** (const QString &imageName)
- void **setStartPoint** (const QPointF &startPoint)
- void **setWidth** (unsigned int imageWidth)
- void **setHeight** (unsigned int imageHeight)
- void **updateImage** ()

The documentation for this class was generated from the following files:

- applications/gui/genesys/graphicals/GraphicalImageAnimation.h
- applications/gui/genesys/graphicals/GraphicalImageAnimation.cpp

17.342 GraphicalModelBuilder Class Reference

Service that reconstructs graphical components/connections from kernel model data.

`#include <applications/gui/genesys/services/GraphicalModelBuilder.h>`

Public Member Functions

- GraphicalModelBuilder (Simulator ∗simulator, ModelGraphicsView ∗graphicsView, ModelGraphicsScene ∗scene, std::map< std::string, QColor > ∗pluginCategoryColor, QTextEdit ∗console)

  Creates the reconstruction service used by load/rebuild compatibility flows.
- void setModelLevelFilter (unsigned int modelLevel)

  Restricts this builder and the target scene to one ModelDataDefinition level.
- void **clearModelLevelFilter** ()

  Clears the level restriction and restores full-model rebuild semantics.
- void recursivalyGenerateGraphicalModelFromModel (ModelComponent ∗component, List< ModelComponent ∗ > ∗visited, std::map< ModelComponent ∗, GraphicalModelComponent ∗ > ∗map, int ∗x, int ∗y, int ∗ymax, int sequenceInLine)

  Recursively rebuilds graphical items and links from one model-component branch.
- void generateGraphicalModelFromModel ()

  Rebuilds the full scene model representation from kernel source components.

Static Public Member Functions

- static void synchronizeGraphicalDataDefinitionsLayer (Simulator ∗simulator, ModelGraphicsScene ∗scene)

  Synchronizes graphical data definitions while preserving existing layout.

17.342.1 Detailed Description

Service that reconstructs graphical components/connections from kernel model data.
This builder is used after model loading to recreate the scene representation while keeping MainWindow as a compatibility façade. It acts as a model-representation bridge between the kernel component graph and GUI graphical items.
Responsibilities:

- recursively create graphical nodes/connections for component branches;
- generate a full scene representation from model source components;
- apply plugin-category visual metadata needed during reconstruction.

Boundaries:

- it does not persist files or parse textual model language;
- it does not manage selection/property editor/simulation command flows;
- it operates as a reconstruction service, not a controller.

17.342.2 Constructor & Destructor Documentation

17.342.2.1 GraphicalModelBuilder()

```
GraphicalModelBuilder::GraphicalModelBuilder (
            Simulator * simulator,
            ModelGraphicsView * graphicsView,
            ModelGraphicsScene * scene,
            std::map< std::string, QColor > * pluginCategoryColor,
            QTextEdit * console )
```

Creates the reconstruction service used by load/rebuild compatibility flows.
Builds the graphical reconstruction service with explicit dependencies.

17.342.3 Member Function Documentation

17.342.3.1 generateGraphicalModelFromModel()

```
void GraphicalModelBuilder::generateGraphicalModelFromModel ( )
```

Rebuilds the full scene model representation from kernel source components.
Rebuilds the full scene representation from the current kernel model.

17.342.3.2 recursivalyGenerateGraphicalModelFromModel()

```
void GraphicalModelBuilder::recursivalyGenerateGraphicalModelFromModel (
            ModelComponent * component,
            List< ModelComponent * > * visited,
            std::map< ModelComponent *, GraphicalModelComponent * > * map,
            int * x,
            int * y,
            int * ymax,
            int sequenceInLine )
```

Recursively rebuilds graphical items and links from one model-component branch.
Recursively rebuilds graphical items and connections from one branch.

17.342.3.3 setModelLevelFilter()

```
void GraphicalModelBuilder::setModelLevelFilter (
            unsigned int modelLevel )
```

Restricts this builder and the target scene to one ModelDataDefinition level.
Root model tabs use level 0 today. Submodel tabs will use the owner component/data definition level, so rebuilds can be reused without mixing visual scopes.

17.342.3.4 synchronizeGraphicalDataDefinitionsLayer()

```
void GraphicalModelBuilder::synchronizeGraphicalDataDefinitionsLayer (
            Simulator * simulator,
            ModelGraphicsScene * scene )  [static]
```

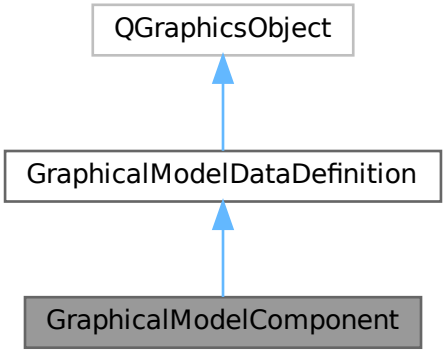
Synchronizes graphical data definitions while preserving existing layout.
Synchronizes the data-definition layer against the current kernel model.
A non-root level discovers its directly registered ModelDataDefinitions even without process components or attached/internal links. Root-level visibility retains the established process-association behavior. This does not draw mathematical Network transitions or change domain object ownership.
The documentation for this class was generated from the following files:

- applications/gui/genesys/services/GraphicalModelBuilder.h
- applications/gui/genesys/services/GraphicalModelBuilder.cpp

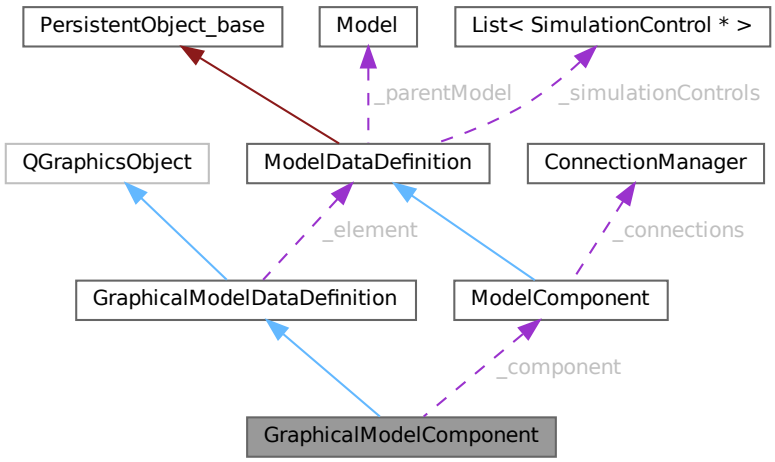
17.343 GraphicalModelComponent Class Reference

Visual component item that extends a data definition with ports and internal data.
`#include <applications/gui/genesys/graphicals/GraphicalModelComponent.h>`

Inheritance diagram for GraphicalModelComponent:



Collaboration diagram for GraphicalModelComponent:



Public Member Functions

- **GraphicalModelComponent** (Plugin *plugin, ModelComponent *component, QPointF position, QColor color=Qt::blue, QGraphicsItem *parent=nullptr)

  Creates a graphical component bound to a kernel component and plugin metadata.
- **GraphicalModelComponent** (const GraphicalModelComponent &orig)

  Copy constructor used by clipboard and restoration flows.
- virtual ~**GraphicalModelComponent** ()

  Releases the component item, including its generated port items and queues.
- QRectF **boundingRect** () const override

  Returns the rectangle used for painting and hit testing the component body.
- void **paint** (QPainter *painter, const QStyleOptionGraphicsItem *option, QWidget *widget) override

Paints the component body, ports, and selection indicators.

- ModelComponent * **getComponent** () const

  Returns the kernel component represented by this item.

- qreal **getHeight** () const

  Returns the current item height derived from the render strategy.

- QPointF **getOldPosition** () const

  Returns the previous scene position used by undo and restoration flows.

- void **setOldPosition** (QPointF oldPosition)

  Stores the previous scene position used by undo and restoration flows.

- QList< GraphicalComponentPort * > **getGraphicalInputPorts** () const

  Returns the input ports generated for this component.

- QList< GraphicalComponentPort * > **getGraphicalOutputPorts** () const

  Returns the output ports generated for this component.

- QColor **getColor** () const

  Returns the component color used by the renderer.

- void **setColor** (const QColor &color)

  Updates the component color used by the renderer.

- QList< ModelDataDefinition * > * **getInternalData** () const

  Returns the list of internal data definitions attached to the component.

- QList< ModelDataDefinition * > * **getAttachedData** () const

  Returns the list of externally attached data definitions.

- EntityType * **getEntityType** () const

  Returns the entity type synthesized or associated with the component.

- void **setEntityType** (EntityType *entityType)

  Sets the entity type synthesized or associated with the component.

- unsigned int **getOcupiedInputPorts** () const

  Returns how many input ports are currently occupied.

- unsigned int **getOcupiedOutputPorts** () const

  Returns how many output ports are currently occupied.

- void **setOcupiedInputPorts** (unsigned int value)

  Stores the number of occupied input ports tracked by the layout code.

- void **setOcupiedOutputPorts** (unsigned int value)

  Stores the number of occupied output ports tracked by the layout code.

- QString **getAnimationImageName** ()

  Returns the animation image identifier used for queue visualization.

- void **setAnimationImageName** (QString name)

  Updates the animation image identifier used by the renderer.

- QMap< Queue *, QPair< unsigned int, unsigned int > > * **getMapQueue** ()

  Returns the queue-to-metadata map used by the queue overlay renderer.

- QList< QList< GraphicalImageAnimation * > * > * **getImagesQueue** ()

  Returns the grouped queue image lists used by the renderer.

- void **verifyQueue** ()

  Verifies whether the component currently references a queue animation context.

- bool **hasQueue** ()

  Returns true when a queue animation context is available.

- unsigned int **getIndexQueue** (Queue *queue)

  Returns the queue index assigned during queue-overlay population.

- unsigned int **getSizeQueue** (Queue *queue)

  Returns the queue size recorded during queue-overlay population.

- void **populateMapQueue** (QList< Queue * > queues)

  Populates the queue overlay metadata from the supplied queue list.

- bool **insertImageQueue** (Queue *queue, GraphicalImageAnimation *image)

  Inserts one animation image entry for the given queue.
- QList< GraphicalImageAnimation * > * **removeImageQueue** (Queue *queue, unsigned int quantity↩ Removed)

  Removes a batch of animation images for the given queue.
- bool **insertImageQueue** (GraphicalImageAnimation *image)

  Inserts one animation image using the current queue context.
- GraphicalImageAnimation * **removeImageQueue** ()

  Removes the most recently inserted queue animation image.
- void **actualizeMapQueue** (Queue *queue)

  Refreshes the queue metadata after the queue model changes.
- void **visivibleImageQueue** (bool visivible)

  Shows or hides the queue image overlay.
- void **clearQueues** ()

  Clears the cached queue overlay state.

Public Member Functions inherited from GraphicalModelDataDefinition

- **GraphicalModelDataDefinition** (Plugin *plugin, ModelDataDefinition *element, QPointF position, QColor color=Qt::blue, QGraphicsItem *parent=nullptr)

  Creates one visual item bound to a kernel data definition.
- **GraphicalModelDataDefinition** (const GraphicalModelDataDefinition &orig)

  Copy constructor used by scene-copy and restore flows.
- virtual ~**GraphicalModelDataDefinition** ()

  Releases the graphical item and any owned helper state.
- QRectF **boundingRect** () const override

  Returns the item bounds used by Qt for painting and hit testing.
- QPainterPath **shape** () const override

  Returns the precise shape used for selection and collision checks.
- void **paint** (QPainter *painter, const QStyleOptionGraphicsItem *option, QWidget *widget) override

  Paints the data-definition body, label, and selection frame.
- ModelDataDefinition * **getDataDefinition** () const

  Returns the kernel data definition represented by this item.
- bool **isEditableInPropertyEditor** () const

  Returns whether the property editor may edit this item in place.
- void **setEditableInPropertyEditor** (bool editable)

  Marks the item as editable or read-only in the property editor.
- QPointF **getOldPosition** () const

  Returns the last persisted scene position before a move operation.
- void **setOldPosition** (qreal x, qreal y)

  Stores the old scene position used by undo and restoration flows.
- QColor **getColor** () const

  Returns the color used to paint the item.
- void **setColor** (QColor newColor)

  Sets the color used to paint the item.
- qreal **getHeight** () const

  Returns the height used by the current render strategy.

Static Public Member Functions

- static bool **compareQueuesById** (const Queue *a, const Queue *b)

  Compares queues by identifier so queue overlays can be ordered deterministically.

Protected Member Functions

- virtual bool sceneEvent (QEvent *event) override
- QVariant **itemChange** (GraphicsItemChange change, const QVariant &value) override
- GraphicalModelItemRenderContext renderContext () const override

Protected Member Functions inherited from GraphicalModelDataDefinition

- QColor **myrgba** (uint64_t color)

Protected Attributes

- qreal **_width** = TraitsGUI<GModelComponent>::width
- qreal **_height** = _width * TraitsGUI<GModelComponent>::heightProportion
- unsigned int **_margin** = TraitsGUI<GModelComponent>::margin
- unsigned int **_selWidth** = TraitsGUI<GModelComponent>::selectionWidth
- ModelComponent * **_component**
- QColor **_color**
- QPointF **_oldPosition**
- qreal **_stretchPosTop** = TraitsGUI<GModelComponent>::stretchPos
- qreal **_stretchPosBottom** = TraitsGUI<GModelComponent>::stretchPos
- qreal **_stretchPosLeft** = TraitsGUI<GModelComponent>::stretchPos
- qreal **_stretchPosRigth** = TraitsGUI<GModelComponent>::stretchPos
- qreal **_stretchRigth** = TraitsGUI<GModelComponent>::stretch
- qreal **_stretchLeft** = TraitsGUI<GModelComponent>::stretch
- qreal **_stretchRigthMidle** = TraitsGUI<GModelComponent>::stretch
- qreal **_stretchLeftMidle** = TraitsGUI<GModelComponent>::stretch
- qreal **_stretchTop** = TraitsGUI<GModelComponent>::stretch
- qreal **_stretchBottom** = TraitsGUI<GModelComponent>::stretch
- qreal **_stretchTopMidle** = TraitsGUI<GModelComponent>::stretch
- qreal **_stretchBottomMidle** = TraitsGUI<GModelComponent>::stretch

Protected Attributes inherited from GraphicalModelDataDefinition

- qreal **_width** = TraitsGUI<GModelDataDefinition>::width
- qreal **_height** = _width * TraitsGUI<GModelDataDefinition>::heightProportion
- unsigned int **_margin** = TraitsGUI<GModelDataDefinition>::margin
- unsigned int **_selWidth** = TraitsGUI<GModelDataDefinition>::selectionWidth
- ModelDataDefinition * **_element**
- bool **_editableInPropertyEditor** = false
- QColor **_color**
- QPointF **_oldPosition**
- qreal **_stretchPosTop** = TraitsGUI<GModelDataDefinition>::stretchPos
- qreal **_stretchPosBottom** = TraitsGUI<GModelDataDefinition>::stretchPos
- qreal **_stretchPosLeft** = TraitsGUI<GModelDataDefinition>::stretchPos
- qreal **_stretchPosRigth** = TraitsGUI<GModelDataDefinition>::stretchPos
- qreal **_stretchRigth** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchLeft** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchRigthMidle** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchLeftMidle** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchTop** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchBottom** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchTopMidle** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchBottomMidle** = TraitsGUI<GModelDataDefinition>::stretch

17.343.1 Detailed Description

Visual component item that extends a data definition with ports and internal data.
The scene uses this class for kernel model components that need both the visual body and the connection ports used to route edges between components.

17.343.2 Member Function Documentation

17.343.2.1 renderContext()

```
GraphicalModelItemRenderContext GraphicalModelComponent::renderContext ( ) const  [override],
[protected], [virtual]
```

Reimplemented from GraphicalModelDataDefinition.

17.343.2.2 sceneEvent()

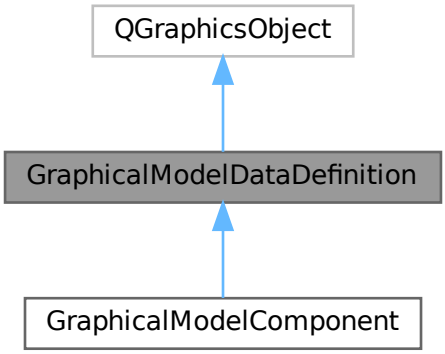
```
bool GraphicalModelComponent::sceneEvent (
            QEvent * event )  [override], [protected], [virtual]
```

Reimplemented from GraphicalModelDataDefinition.
The documentation for this class was generated from the following files:

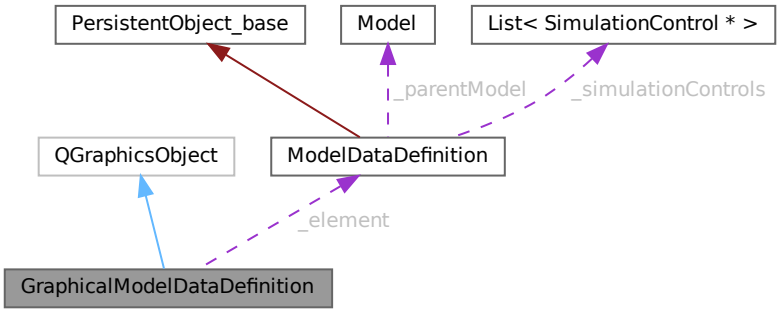
- applications/gui/genesys/graphicals/GraphicalModelComponent.h
- applications/gui/genesys/graphicals/GraphicalModelComponent.cpp

17.344 GraphicalModelDataDefinition Class Reference

Visual representation of a model data definition in the scene.
`#include <applications/gui/genesys/graphicals/GraphicalModelDataDefinition.h>`
Inheritance diagram for GraphicalModelDataDefinition:



Collaboration diagram for GraphicalModelDataDefinition:



Public Member Functions

- **GraphicalModelDataDefinition** (Plugin ∗plugin, ModelDataDefinition ∗element, QPointF position, QColor color=Qt::blue, QGraphicsItem ∗parent=nullptr)

  Creates one visual item bound to a kernel data definition.
- **GraphicalModelDataDefinition** (const GraphicalModelDataDefinition &orig)

  Copy constructor used by scene-copy and restore flows.
- virtual ∼**GraphicalModelDataDefinition** ()

  Releases the graphical item and any owned helper state.
- QRectF **boundingRect** () const override

  Returns the item bounds used by Qt for painting and hit testing.
- QPainterPath **shape** () const override

  Returns the precise shape used for selection and collision checks.
- void **paint** (QPainter ∗painter, const QStyleOptionGraphicsItem ∗option, QWidget ∗widget) override

  Paints the data-definition body, label, and selection frame.
- ModelDataDefinition ∗ **getDataDefinition** () const

  Returns the kernel data definition represented by this item.
- bool **isEditableInPropertyEditor** () const

  Returns whether the property editor may edit this item in place.
- void **setEditableInPropertyEditor** (bool editable)

  Marks the item as editable or read-only in the property editor.
- QPointF **getOldPosition** () const

  Returns the last persisted scene position before a move operation.
- void **setOldPosition** (qreal x, qreal y)

  Stores the old scene position used by undo and restoration flows.
- QColor **getColor** () const

  Returns the color used to paint the item.
- void **setColor** (QColor newColor)

  Sets the color used to paint the item.
- qreal **getHeight** () const

  Returns the height used by the current render strategy.

Protected Member Functions

- QColor **myrgba** (uint64_t color)
- virtual GraphicalModelItemRenderContext **renderContext** () const
- virtual bool **sceneEvent** (QEvent ∗event) override

Protected Attributes

- qreal **_width** = TraitsGUI<GModelDataDefinition>::width
- qreal **_height** = _width * TraitsGUI<GModelDataDefinition>::heightProportion
- unsigned int **_margin** = TraitsGUI<GModelDataDefinition>::margin
- unsigned int **_selWidth** = TraitsGUI<GModelDataDefinition>::selectionWidth
- ModelDataDefinition * **_element**
- bool **_editableInPropertyEditor** = false
- QColor **_color**
- QPointF **_oldPosition**
- qreal **_stretchPosTop** = TraitsGUI<GModelDataDefinition>::stretchPos
- qreal **_stretchPosBottom** = TraitsGUI<GModelDataDefinition>::stretchPos
- qreal **_stretchPosLeft** = TraitsGUI<GModelDataDefinition>::stretchPos
- qreal **_stretchPosRigth** = TraitsGUI<GModelDataDefinition>::stretchPos
- qreal **_stretchRigth** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchLeft** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchRigthMidle** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchLeftMidle** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchTop** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchBottom** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchTopMidle** = TraitsGUI<GModelDataDefinition>::stretch
- qreal **_stretchBottomMidle** = TraitsGUI<GModelDataDefinition>::stretch

17.344.1 Detailed Description

Visual representation of a model data definition in the scene.
This is the base graphical item used for non-component data definitions. Derived classes such as `GraphicalModelComponent` extend it with ports, internal data, and richer layout.
The documentation for this class was generated from the following files:

- applications/gui/genesys/graphicals/GraphicalModelDataDefinition.h
- applications/gui/genesys/graphicals/GraphicalModelDataDefinition.cpp

17.345 GraphicalModelEvent Class Reference

Lightweight event envelope used to notify GUI changes in the graphics scene.
`#include <applications/gui/genesys/graphicals/ModelGraphicsScene.h>`

Public Types

- enum class **EventType** : int {
  CREATE = 1 , **REMOVE** = 2 , **EDIT** = 3 , **CLONE** = 4 ,
  OTHER = 9 }
- enum class **EventObjectType** : int {
  COMPONENT = 1 , **DATADEFINITION** = 2 , **CONNECTION** = 3 , **DRAWING** = 4 ,
  ANIMATION = 5 , **PLOT** = 6 , **OTHER** = 9 }

Public Member Functions

- **GraphicalModelEvent** (GraphicalModelEvent::EventType eventType, GraphicalModelEvent::EventObject↩ Type eventObjectType, QGraphicsItem *item)

  Creates one event payload for the given item and event categories.

Public Attributes

- GraphicalModelEvent::EventType **eventType**
- GraphicalModelEvent::EventObjectType **eventObjectType**
- QGraphicsItem * **item**

17.345.1 Detailed Description

Lightweight event envelope used to notify GUI changes in the graphics scene.
The documentation for this class was generated from the following file:

- applications/gui/genesys/graphicals/ModelGraphicsScene.h

17.346 GraphicalModelItemRenderContext Struct Reference

Public Types

- enum class **ItemKind** { **DataDefinition** , **Component** }

Public Attributes

- ItemKind **kind** = ItemKind::DataDefinition
- QRectF **bounds**
- QColor **fillColor**
- QString **primaryText**
- QString **secondaryText**
- QString **tertiaryText**
- QString **semanticClassName**
- QString **semanticSubtype**
- bool **selected** = false
- bool **breakpoint** = false
- qreal **width** = 0.0
- qreal **height** = 0.0
- int **penWidth** = 1
- int **raise** = 0
- int **margin** = 0
- int **selectionWidth** = 0
- qreal **stretchPosTop** = 0.5
- qreal **stretchPosBottom** = 0.5
- qreal **stretchPosLeft** = 0.5
- qreal **stretchPosRight** = 0.5
- qreal **stretchRight** = 0.0
- qreal **stretchLeft** = 0.0
- qreal **stretchRightMiddle** = 0.0
- qreal **stretchLeftMiddle** = 0.0
- qreal **stretchTop** = 0.0
- qreal **stretchBottom** = 0.0
- qreal **stretchTopMiddle** = 0.0
- qreal **stretchBottomMiddle** = 0.0
- uint64_t **borderColor** = 0
- uint64_t **raisedColor** = 0
- uint64_t **sunkenColor** = 0
- uint64_t **textColor** = 0
- uint64_t **textShadowColor** = 0
- uint64_t **selectionColor** = 0
- uint64_t **breakpointColor** = 0

The documentation for this struct was generated from the following file:

- applications/gui/genesys/graphicals/GraphicalModelItemRenderStrategy.h

17.347 GraphicalModelItemRenderer Class Reference

Static Public Member Functions

- static void **paint** (QPainter *painter, const GraphicalModelItemRenderContext &context)
- static QPainterPath **shape** (const GraphicalModelItemRenderContext &context)
- static const GraphicalModelItemRenderStrategy & **currentStrategy** ()

The documentation for this class was generated from the following files:

- applications/gui/genesys/graphicals/GraphicalModelItemRenderStrategy.h
- applications/gui/genesys/graphicals/GraphicalModelItemRenderStrategy.cpp

17.348 GraphicalModelItemRenderStrategy Class Reference

Public Member Functions

- virtual const char * **name** () const =0
- virtual void **paint** (QPainter *painter, const GraphicalModelItemRenderContext &context) const =0
- virtual QPainterPath **shape** (const GraphicalModelItemRenderContext &context) const =0

The documentation for this class was generated from the following file:

- applications/gui/genesys/graphicals/GraphicalModelItemRenderStrategy.h

17.349 GraphicalModelSerializer Class Reference

Persistence service for saving/loading graphical and textual model representations.
`#include <applications/gui/genesys/services/GraphicalModelSerializer.h>`

Public Member Functions

- GraphicalModelSerializer (Simulator *simulator, QWidget *ownerWidget, QPlainTextEdit *modelTextEditor, ModelGraphicsView *graphicsView, QSlider *zoomSlider, QAction *actionShowGrid, QAction *action↩ ShowRule, QAction *actionShowSnap, QAction *actionShowGuides, QAction *actionShowInternalElements, QAction *actionShowEditableElements, QAction *actionShowAttachedElements, QAction *actionShow↩ RecursiveElements, QTextEdit *console, QString *modelFilename, std::function< void()> clearModel↩ Editors, std::function< void()> rebuildGraphicalModelFromModel, std::function< void()> applyShow↩ InternalElements, std::function< void()> applyShowEditableElements, std::function< void()> applyShow↩ AttachedElements, std::shared_ptr< EFSMNetworkDiagramLayoutsByModel > efsmNetworkDiagram↩ Layouts=nullptr)

  Creates the persistence service that backs MainWindow lifecycle wrappers.
- bool saveTextModel (QFile *saveFile, const QString &data) const

  Saves textual model language using the established compatibility format.
- bool saveGraphicalModel (const QString &filename) const

  Saves full graphical `.gui` persistence sections without format changes.
- Model * loadGraphicalModel (const std::string &filename) const

  Loads `.gui/.gen` model files and restores persisted graphical/UI state.
- void setEFSMNetworkDiagramLayout (Model *model, unsigned long networkId, const EFSMNetworkDiagramLayout &layout)

  Sets the GUI-only EFSM diagram layout for a root model and Network on the next `saveGraphicalModel()` call. An empty layout removes any previously stored entry for that model/network pair. This never touches kernel mathematical state; see `EFSMNetworkDiagramLayout`.
- EFSMNetworkDiagramLayout efsmNetworkDiagramLayout (Model *model, unsigned long networkId) const

  Returns the EFSM diagram layout held for this model and Network id. Returns a default-constructed (empty) layout when none is known.
- void **forgetEFSMNetworkLayouts** (Model *model)

  Drops cached GUI layouts when their owning root model is closed.

17.349.1 Detailed Description

Persistence service for saving/loading graphical and textual model representations.
This Phase-2 extraction removes serialization concerns from MainWindow while preserving file compatibility through wrapper delegation. It bridges GUI state (scene/view/actions/editors) and kernel model loading/saving workflows.
Responsibilities:

- save textual model content with the existing line-based format;
- save full graphical .gui state sections used by legacy compatibility;
- load persisted model files and restore graphical/editor UI state via callbacks.

Boundaries:

- it does not own widgets or simulator lifetime;
- it does not run simulation commands or trace rendering;
- it coordinates persistence/serialization only.

17.349.2 Constructor & Destructor Documentation

17.349.2.1 GraphicalModelSerializer()

```
GraphicalModelSerializer::GraphicalModelSerializer (
            Simulator * simulator,
            QWidget * ownerWidget,
            QPlainTextEdit * modelTextEditor,
            ModelGraphicsView * graphicsView,
            QSlider * zoomSlider,
            QAction * actionShowGrid,
            QAction * actionShowRule,
            QAction * actionShowSnap,
            QAction * actionShowGuides,
            QAction * actionShowInternalElements,
            QAction * actionShowEditableElements,
            QAction * actionShowAttachedElements,
            QAction * actionShowRecursiveElements,
            QTextEdit * console,
            QString * modelFilename,
            std::function< void()> clearModelEditors,
            std::function< void()> rebuildGraphicalModelFromModel,
            std::function< void()> applyShowInternalElements,
            std::function< void()> applyShowEditableElements,
            std::function< void()> applyShowAttachedElements,
            std::shared_ptr< EFSMNetworkDiagramLayoutsByModel > efsmNetworkDiagramLayouts =
nullptr )
```

Creates the persistence service that backs MainWindow lifecycle wrappers.
Builds the serializer with explicit, narrow dependencies from MainWindow.

17.349.3 Member Function Documentation

17.349.3.1 efsmNetworkDiagramLayout()

```
EFSMNetworkDiagramLayout GraphicalModelSerializer::efsmNetworkDiagramLayout (
            Model * model,
            unsigned long networkId ) const
```

Returns the EFSM diagram layout held for this model and Network id. Returns a default-constructed (empty) layout when none is known.
Returns the EFSM diagram layout currently held for the given network.

17.349.3.2 loadGraphicalModel()

```
Model * GraphicalModelSerializer::loadGraphicalModel (
            const std::string & filename ) const
```

Loads `.gui/.gen` model files and restores persisted graphical/UI state.
Loads `.gui` or `.gen` model files and restores persisted GUI state.

Returns

Loaded kernel model pointer or nullptr when loading fails.

17.349.3.3 saveGraphicalModel()

```
bool GraphicalModelSerializer::saveGraphicalModel (
            const QString & filename ) const
```

Saves full graphical `.gui` persistence sections without format changes.
Persists the complete graphical model, including view options and overlays.

Returns

true when serialization succeeds.

17.349.3.4 saveTextModel()

```
bool GraphicalModelSerializer::saveTextModel (
            QFile * saveFile,
            const QString & data ) const
```

Saves textual model language using the established compatibility format.
Writes model text line by line using the legacy persistence format.

Returns

true when write succeeds.

17.349.3.5 setEFSMNetworkDiagramLayout()

```
void GraphicalModelSerializer::setEFSMNetworkDiagramLayout (
            Model * model,
            unsigned long networkId,
            const EFSMNetworkDiagramLayout & layout )
```

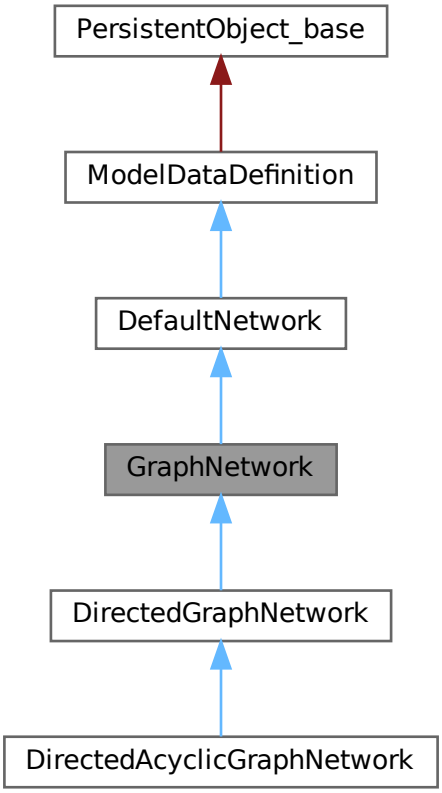
Sets the GUI-only EFSM diagram layout for a root model and Network on the next `saveGraphicalModel()` call. An empty layout removes any previously stored entry for that model/network pair. This never touches kernel mathematical state; see `EFSMNetworkDiagramLayout`.
Sets the GUI-only EFSM diagram layout to persist for the given network.
The documentation for this class was generated from the following files:

- applications/gui/genesys/services/GraphicalModelSerializer.h
- applications/gui/genesys/services/GraphicalModelSerializer.cpp

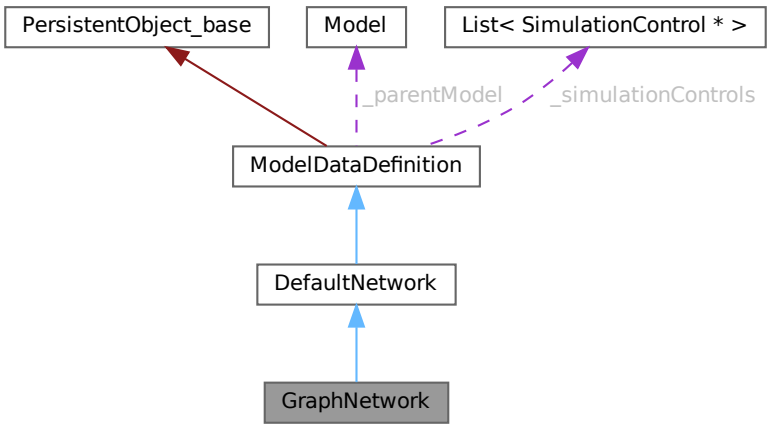
17.350 GraphNetwork Class Reference

Mathematical graph specialization of DefaultNetwork.
`#include <plugins/data/ModalModel/GraphNetwork.h>`

Inheritance diagram for GraphNetwork:



Collaboration diagram for GraphNetwork:



Public Member Functions

- **GraphNetwork** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< GraphNetwork >())
- virtual bool **isDirected** () const
- virtual bool **addNode** (GraphNode *node)
- virtual bool **removeNode** (GraphNode *node)
- bool **containsNode** (GraphNode *node) const
- GraphNode * **findNodeByName** (const std::string &name) const
- std::vector< GraphNode * > **getNodes** () const
- virtual bool **addEdge** (GraphEdge *edge)
- virtual bool **removeEdge** (GraphEdge *edge)
- bool **containsEdge** (GraphEdge *edge) const
- GraphEdge * **findEdgeByName** (const std::string &name) const
- std::vector< GraphEdge * > **getEdges** () const
- bool **hasEdge** (GraphNode *source, GraphNode *destination) const
- std::vector< GraphEdge * > **getEdgesBetween** (GraphNode *source, GraphNode *destination) const
- std::vector< GraphNode * > **getNeighbors** (GraphNode *node) const
- std::vector< GraphEdge * > **getIncidentEdges** (GraphNode *node) const
- unsigned int **degree** (GraphNode *node) const
- virtual std::vector< GraphNode * > **getPredecessors** (GraphNode *node) const
- virtual std::vector< GraphNode * > **getSuccessors** (GraphNode *node) const
- virtual std::vector< GraphEdge * > **getIncomingEdges** (GraphNode *node) const
- virtual std::vector< GraphEdge * > **getOutgoingEdges** (GraphNode *node) const
- virtual unsigned int **inDegree** (GraphNode *node) const
- virtual unsigned int **outDegree** (GraphNode *node) const
- GraphTraversalResult **breadthFirstSearch** (GraphNode *source) const
- GraphTraversalResult **depthFirstSearch** (GraphNode *source) const
- bool **hasPath** (GraphNode *source, GraphNode *destination) const
- std::vector< GraphNode * > **reachableNodes** (GraphNode *source) const
- GraphPathResult **shortestPathUnweighted** (GraphNode *source, GraphNode *destination) const
- GraphPathResult **shortestPathDijkstra** (GraphNode *source, GraphNode *destination) const
- std::vector< std::vector< GraphNode * > > **connectedComponents** () const
- virtual bool **hasCycle** () const
- virtual GraphTopologicalOrderResult **topologicalOrder** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from DefaultNetwork

- **DefaultNetwork** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< DefaultNetwork >())
- unsigned int **addInputPort** (const std::string &portName)

  Declares a new input port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **addOutputPort** (const std::string &portName)

  Declares a new output port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **getNumInputPorts** () const
- unsigned int **getNumOutputPorts** () const
- std::string **getInputPortName** (unsigned int index) const
- std::string **getOutputPortName** (unsigned int index) const
- int **getInputPortIndex** (const std::string &portName) const

  Returns the input port index for `portName`, or -1 when not declared.
- int **getOutputPortIndex** (const std::string &portName) const

Returns the output port index for `portName`, or -1 when not declared.

- double **getActivationCount** () const

  Number of times activate() has been called since the last replication reset.

- NetworkActivationResult activate (const NetworkActivationFrame &frame)

  Runs exactly one network activation for `frame`.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from DefaultNetwork

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual bool **_edgeConnects** (GraphEdge *edge, GraphNode *source, GraphNode *destination) const
- virtual GraphNode * **_oppositeNode** (GraphEdge *edge, GraphNode *node) const

Protected Member Functions inherited from DefaultNetwork

- virtual void _createInternalStatisticReporters () override
- virtual NetworkActivationResult _activate (const NetworkActivationFrame &frame)

  Formalism-specific activation behavior.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.350.1 Detailed Description

Mathematical graph specialization of DefaultNetwork.
GraphNetwork represents G=(V,E) and algorithms over that topology. It is not a process network of Model↩ Component+Connection, and its inherited activation remains inert: a pure graph has no current vertex or firing/movement semantics. Dynamic traversal, random walks and routing belong to future explicit specializations/adapters.
Algorithms use insertion-order adjacency for deterministic behavior: BFS/DFS/connected components/cycle detection are O(V+E); unweighted shortest path is BFS O(V+E); Dijkstra with the simple implementation here is O(V^2+E). Temporary visited, predecessor, queue, stack and distance data are local to each algorithm call and are never persisted.

17.350.2 Member Function Documentation

17.350.2.1 _check()

```
bool GraphNetwork::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from DefaultNetwork.
Reimplemented in DirectedAcyclicGraphNetwork.

17.350.2.2 _initBetweenReplications()

```
void GraphNetwork::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from DefaultNetwork.

17.350.2.3 _loadInstance()

```
bool GraphNetwork::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from DefaultNetwork.

17.350.2.4 _saveInstance()

```
void GraphNetwork::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from DefaultNetwork.

17.350.2.5 show()

`std::string GraphNetwork::show ( ) [override], [virtual]`
Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

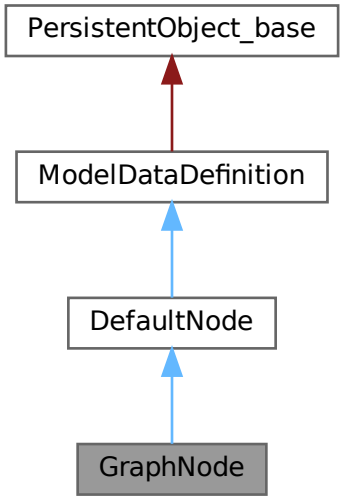
Human-readable summary used by traces and debug output.

Reimplemented from DefaultNetwork.
The documentation for this class was generated from the following files:

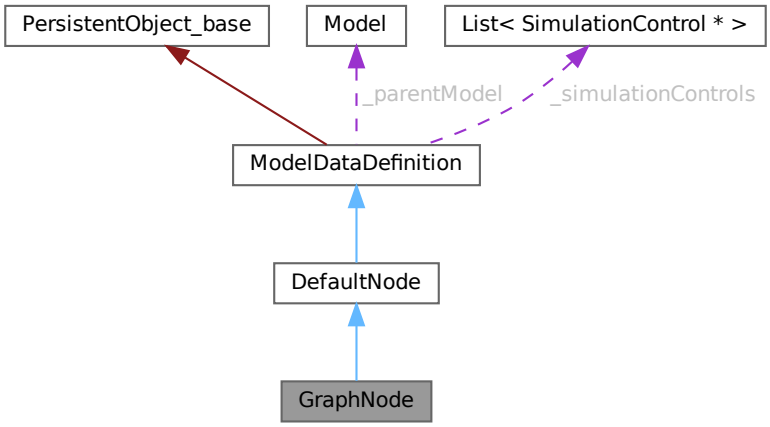
- plugins/data/ModalModel/GraphNetwork.h
- plugins/components/ModalModel/GraphNetwork.cpp

17.351 GraphNode Class Reference

Mathematical graph vertex used by GraphNetwork.
`#include <plugins/data/ModalModel/GraphNode.h>`
Inheritance diagram for GraphNode:



Collaboration diagram for GraphNode:



Public Member Functions

- **GraphNode** (Model ∗model, std::string name="")

Public Member Functions inherited from DefaultNode

- DefaultNode (Model ∗model, std::string componentTypename, std::string name)

  constructors
- **DefaultNode** (Model ∗model, std::string name="")
- void **addTransition** (DefaultNodeTransition ∗transition)

  new public user methods for this component
- void **removeTransition** (DefaultNodeTransition ∗transition)
- List< DefaultNodeTransition ∗ > ∗ **getTransitions** () const
- void **setInitialNode** (bool initialNode)
- bool **isInitialNode** () const
- void **setFinalNode** (bool finalNode)
- bool **isFinalNode** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model ∗model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ∼**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from DefaultNode

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Additional Inherited Members

Protected Member Functions inherited from DefaultNode

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.351.1 Detailed Description

Mathematical graph vertex used by GraphNetwork.
A GraphNode is a model data definition, not a process ModelComponent. It reuses DefaultNode's persistable identity/name/flag infrastructure without inheriting process-flow Connection semantics.
The documentation for this class was generated from the following files:

- plugins/data/ModalModel/GraphNode.h
- plugins/components/ModalModel/GraphNode.cpp

17.352 GraphPathResult Struct Reference

Explicit graph path result.
`#include <plugins/data/ModalModel/GraphNetwork.h>`

Public Attributes

- bool **reachable** = false
- double **distance** = 0.0
- std::vector< GraphNode * > **nodes**
- std::vector< GraphEdge * > **edges**
- std::string **errorMessage** = ""

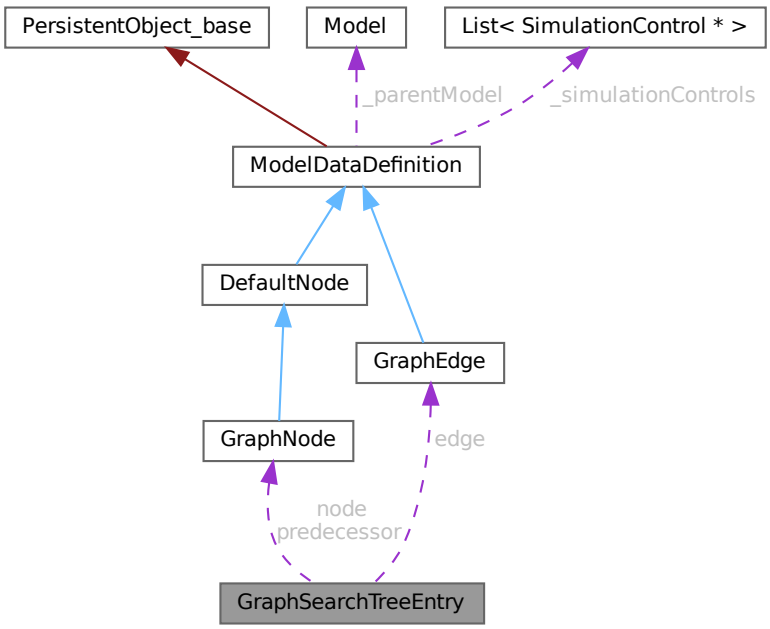
17.352.1 Detailed Description

Explicit graph path result.
`reachable=false` represents absence of a path. `errorMessage` reports rejected operations such as Dijkstra over negative-weight edges.
The documentation for this struct was generated from the following file:

- plugins/data/ModalModel/GraphNetwork.h

17.353 GraphSearchTreeEntry Struct Reference

Collaboration diagram for GraphSearchTreeEntry:



Public Attributes

- GraphNode * **node** = nullptr
- GraphNode * **predecessor** = nullptr
- GraphEdge * **edge** = nullptr

- unsigned int **distance** = 0

The documentation for this struct was generated from the following file:

- plugins/data/ModalModel/GraphNetwork.h

17.354 GraphTopologicalOrderResult Struct Reference

Public Attributes

- bool **acyclic** = false
- std::vector< GraphNode * > **order**
- std::string **errorMessage** = ""

The documentation for this struct was generated from the following file:

- plugins/data/ModalModel/GraphNetwork.h

17.355 GraphTraversalResult Struct Reference

Result of a deterministic graph traversal.
`#include <plugins/data/ModalModel/GraphNetwork.h>`

Public Member Functions

- bool **visited** (GraphNode *node) const
- GraphNode * **predecessorOf** (GraphNode *node) const
- GraphEdge * **edgeTo** (GraphNode *node) const
- unsigned int **distanceTo** (GraphNode *node) const

Public Attributes

- std::vector< GraphNode * > **visitOrder**
- std::vector< GraphSearchTreeEntry > **tree**

17.355.1 Detailed Description

Result of a deterministic graph traversal.
The traversal order follows node/edge insertion order. Temporary search structures are not persisted in GraphNetwork.
The documentation for this struct was generated from the following files:

- plugins/data/ModalModel/GraphNetwork.h
- plugins/components/ModalModel/GraphNetwork.cpp

17.356 GraphvizModelExporter Class Reference

Service that builds DOT content and renders model images for the GUI.
`#include <applications/gui/genesys/services/GraphvizModelExporter.h>`

Public Member Functions

- **GraphvizModelExporter** (Simulator *simulator, QLabel *modelGraphicLabel, QCheckBox *showStatistics, QCheckBox *showEditable, QCheckBox *showShared, QCheckBox *showRecursive, QCheckBox *showLevels, std::function< bool()> ensureModelSynchronized)

  Creates the Graphviz exporter service used by MainWindow compatibility wrappers.
- std::string adjustDotName (std::string name) const

Normalizes identifiers to Graphviz-safe names.

- void **insertTextInDot** (std::string text, unsigned int compLevel, unsigned int compRank, std::map< unsigned int, std::map< unsigned int, std::list< std::string > * > * > *dotmap, bool isNode=false) const

  Inserts a DOT line into the hierarchical rank/level map used for image generation.

- void **recursiveCreateModelGraphicPicture** (ModelDataDefinition *componentOrData, std::list< ModelDataDefinition * > *visited, std::map< unsigned int, std::map< unsigned int, std::list< std::string > * > * > *dotmap) const

  Recursively traverses model/data nodes to build DOT content fragments.

- bool createModelImage () const

  Generates and displays the model image after synchronization preconditions.

17.356.1 Detailed Description

Service that builds DOT content and renders model images for the GUI.
This service encapsulates Graphviz-oriented model representation logic extracted from MainWindow. It keeps image-generation behavior stable while MainWindow exposes compatibility wrappers used by actions and tabs.
Responsibilities:

- normalize identifiers for DOT compatibility;
- build hierarchical DOT fragments from model/data-definition traversal;
- generate and display model images after ensuring text/model synchronization.

Boundaries:

- it does not own model synchronization policy (uses injected callback);
- it does not manage simulation commands, traces, or lifecycle actions;
- it provides export/representation service behavior, not controller orchestration.

17.356.2 Member Function Documentation

17.356.2.1 adjustDotName()

```
std::string GraphvizModelExporter::adjustDotName (
            std::string name ) const
```

Normalizes identifiers to Graphviz-safe names.

Parameters

name	Raw label/name from model representation.

Returns

DOT-safe identifier preserving compatibility naming intent.

17.356.2.2 createModelImage()

```
bool GraphvizModelExporter::createModelImage ( ) const
```

Generates and displays the model image after synchronization preconditions.

Returns

true when DOT generation and image rendering complete successfully.

The documentation for this class was generated from the following files:

- applications/gui/genesys/services/GraphvizModelExporter.h
- applications/gui/genesys/services/GraphvizModelExporter.cpp

17.357 ModelGraphicsScene::GRID Struct Reference

Public Member Functions

- void **clear** ()

Public Attributes

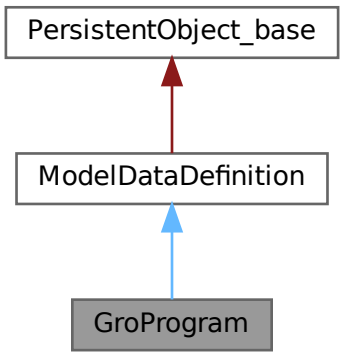
- unsigned int **interval**
- QPen **pen**
- std::list< QGraphicsLineItem * > * **lines**
- bool **visible**

The documentation for this struct was generated from the following files:

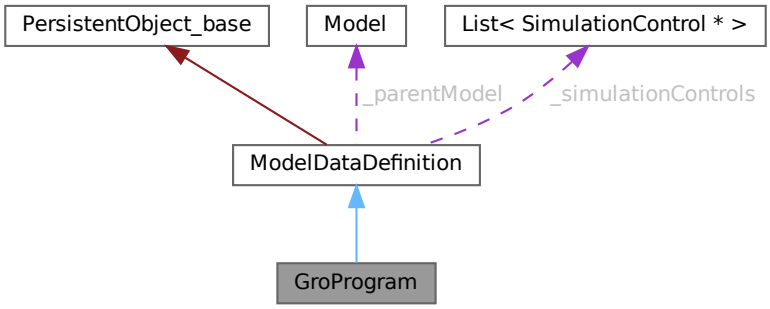
- applications/gui/genesys/graphicals/ModelGraphicsScene.h
- applications/gui/genesys/graphicals/ModelGraphicsScene.cpp

17.358 GroProgram Class Reference

Reusable Gro source program attached to a GenESyS model.
`#include <plugins/data/BiochemicalSimulation/GroProgram.h>`
Inheritance diagram for GroProgram:



Collaboration diagram for GroProgram:



Public Member Functions

- **GroProgram** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **setSourceCode** (std::string sourceCode)

  Replaces the stored Gro source text.
- void **setSourceCodeProperty** (SourceCodeString sourceCode)

  Replaces the stored Gro source text through the property-editor wrapper type.
- std::string **getSourceCode** () const

  Returns the stored Gro source text.
- SourceCodeString **getSourceCodeProperty** () const

  Returns the stored Gro source text wrapped for the property editor.
- bool **createDefaultGroProgram** (const std::string &filename="")

  Creates a small starter Gro program and optionally writes it to a file.
- bool **validateSyntax** (std::string &errorMessage) const

  Performs a permissive lexical sanity check for the stored source.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.

- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)

- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.358.1 Detailed Description

Reusable Gro source program attached to a GenESyS model.
GroProgram is intentionally a ModelDataDefinition so several biological components can share the same Gro source text. This first implementation only stores the source and performs a small lexical sanity check; complete Gro parsing and executable semantics belong to later plugin-side helper classes.

17.358.2 Member Function Documentation

17.358.2.1 _check()

```
bool GroProgram::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.358.2.2 _loadInstance()

```
bool GroProgram::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.358.2.3 _saveInstance()

```
void GroProgram::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.358.2.4 show()

```
std::string GroProgram::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/BiochemicalSimulation/GroProgram.h
- plugins/data/BiochemicalSimulation/GroProgram.cpp

17.359 GroProgramAst Struct Reference

Minimal parsed representation of a Gro source program.
`#include <plugins/data/BiochemicalSimulation/GroProgramAst.h>`

Classes

- struct NamedProgram
- struct Statement

Public Types

- enum class **SourceForm** { **RawStatements** , **ProgramBlock** }

Public Member Functions

- bool **isProgramBlock** () const
- bool **hasNamedPrograms** () const

Public Attributes

- SourceForm **sourceForm** = SourceForm::RawStatements
- std::string **programName** = ""
- std::vector< std::string > **programParameters**
- std::string **bodySource** = ""
- std::vector< Statement > **statements**
- std::vector< NamedProgram > **namedPrograms**

17.359.1 Detailed Description

Minimal parsed representation of a Gro source program.
This AST intentionally models only the first integration target: either a recognized `program name() { ... }` block or a permissive raw statement list. Execution semantics and typed Gro constructs belong to later phases.
The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/GroProgramAst.h

17.360 GroProgramCompiler Class Reference

Converts parsed Gro AST into the first non-executable semantic IR.
`#include <plugins/data/BiochemicalSimulation/GroProgramCompiler.h>`

Public Member Functions

- GroProgramIr **compile** (const GroProgramAst &ast) const

  Builds a lightweight semantic IR from a parsed Gro AST.

17.360.1 Detailed Description

Converts parsed Gro AST into the first non-executable semantic IR.
The documentation for this class was generated from the following files:

- plugins/data/BiochemicalSimulation/GroProgramCompiler.h
- plugins/data/BiochemicalSimulation/GroProgramCompiler.cpp

17.361 GroProgramIr Struct Reference

Initial semantic representation compiled from GroProgramAst.
`#include <plugins/data/BiochemicalSimulation/GroProgramIr.h>`

Classes

- struct Command
- struct NamedProgramDefinition

Public Member Functions

- bool **isProgramBlock** () const
- bool **hasNamedProgram** (const std::string &name) const

Public Attributes

- GroProgramAst::SourceForm **sourceForm** = GroProgramAst::SourceForm::RawStatements
- std::string **programName** = ""
- std::vector< Command > **commands**
- std::map< std::string, NamedProgramDefinition > **namedPrograms**

17.361.1 Detailed Description

Initial semantic representation compiled from GroProgramAst.
The IR keeps the parsed program in a runtime-oriented shape. It can represent function calls, assignments, and basic conditional blocks without pushing any knowledge into the simulator kernel itself.
The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/GroProgramIr.h

17.362 GroProgramParser Class Reference

Plugin-side parser boundary for reusable Gro source programs.
`#include <plugins/data/BiochemicalSimulation/GroProgramParser.h>`

Classes

- struct Result

Public Member Functions

- Result **parse** (const std::string &sourceCode) const

  Parses a Gro source string using the current lexical boundary.

17.362.1 Detailed Description

Plugin-side parser boundary for reusable Gro source programs.
This helper currently preserves the permissive lexical validation that used to live inside GroProgram. It is intentionally small so later phases can grow a real grammar, AST, and semantic checks without changing GroProgram storage.
The documentation for this class was generated from the following files:

- plugins/data/BiochemicalSimulation/GroProgramParser.h
- plugins/data/BiochemicalSimulation/GroProgramParser.cpp

17.363 GroProgramRuntime Class Reference

Executes the supported Gro IR commands without simulator binding.
`#include <plugins/data/BiochemicalSimulation/GroProgramRuntime.h>`

Classes

- struct ColonyMutation
- struct ExecutionResult
- struct MotionMutation
- struct PopulationMutation
- struct SignalMutation

Public Types

- enum class **PopulationMutationType** { **Grow** , **Divide** , **Die** , **SetPopulation** }
- enum class **SignalMutationType** { **Emit** , **Consume** , **Set** }
- enum class **MotionMutationType** { **Run** , **Tumble** }

Public Member Functions

- ExecutionResult **execute** (const GroProgramIr &ir, GroProgramRuntimeState &state) const

  Executes supported commands and reports unsupported commands.

Static Public Member Functions

- static bool validateSignalDeclarationPlacement (const GroProgramIr &ir, std::string &errorMessage)
- static bool **evaluateExpression** (const std::string &expressionText, const GroProgramRuntimeState &state, double &value, std::string &errorMessage)

  Evaluates one scalar Gro expression against one runtime state.

17.363.1 Detailed Description

Executes the supported Gro IR commands without simulator binding.
This helper is intentionally independent from BacteriaColony and the GenESyS event scheduler. It now supports persistent scalar variables, arithmetic expressions, and basic `if/else` control flow while still keeping the runtime local to the biological plugins.

17.363.2 Member Function Documentation

17.363.2.1 validateSignalDeclarationPlacement()

```
bool GroProgramRuntime::validateSignalDeclarationPlacement (
            const GroProgramIr & ir,
            std::string & errorMessage )  [static]
```

Reject structurally conditional signal declarations across all IR scopes.
The documentation for this class was generated from the following files:

- plugins/data/BiochemicalSimulation/GroProgramRuntime.h
- plugins/data/BiochemicalSimulation/GroProgramRuntime.cpp

17.364 GroProgramRuntimeState Struct Reference

Mutable state used by the plugin-side Gro runtime helper.
`#include <plugins/data/BiochemicalSimulation/GroProgramRuntime.h>`

Public Attributes

- double **colonyTime** = 0.0
- double **simulationStep** = 1.0
- unsigned int **populationSize** = 1
- unsigned int **tickCount** = 0
- std::map< std::string, double > **contextVariables**
- std::map< std::string, double > **variables**
- std::function< bool(double &, std::string &)> **sampleUnitInterval**
- unsigned int **signalDeclarationOrdinal** = 0
- unsigned int **knownSignalChannelCount** = 0
- std::string **signalDeclarationScope**

17.364.1 Detailed Description

Mutable state used by the plugin-side Gro runtime helper.
The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/GroProgramRuntime.h

17.365 GroupItem Struct Reference

Public Attributes

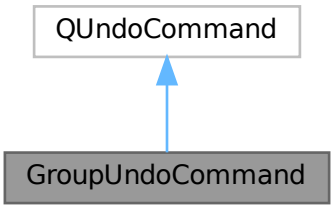
- QGraphicsItemGroup * **group**
- QList< ComponentItem > **myComponentItems**
- QPointF **initialPosition**

The documentation for this struct was generated from the following file:

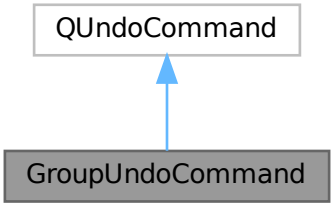
- applications/gui/genesys/actions/AddUndoCommand.h

17.366 GroupUndoCommand Class Reference

Inheritance diagram for GroupUndoCommand:



Collaboration diagram for GroupUndoCommand:



Public Member Functions

- **GroupUndoCommand** (QList< GraphicalModelComponent * > graphicalComponents, ModelGraphicsScene *scene, QUndoCommand *parent=nullptr)
- void **undo** () override
- void **redo** () override

The documentation for this class was generated from the following files:

- applications/gui/genesys/actions/GroupUndoCommand.h
- applications/gui/genesys/actions/GroupUndoCommand.cpp

17.367 testing::internal::GTestFlagSaver Class Reference

The documentation for this class was generated from the following file:

- gtest/src/gtest-internal-inl.h

17.368 testing::internal::GTestLog Class Reference

Public Member Functions

- **GTestLog** (GTestLogSeverity severity, const char *file, int line)

- ::std::ostream & **GetStream** ()

The documentation for this class was generated from the following files:

- gtest/gtest/internal/gtest-port.h
- gtest/src/gtest-port.cc

17.369 testing::internal::GTestMutexLock Class Reference

Public Member Functions

- **GTestMutexLock** (Mutex ∗)

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-port.h

17.370 GuiActionContribution Struct Reference

Public Attributes

- std::string **id**
- std::string **text**
- std::string **menuPath**
- std::string **toolBarId**
- std::string **statusTip**
- std::string **shortcut**
- std::function< bool(const GuiExtensionRuntimeContext &)> **isVisible**
- std::function< bool(const GuiExtensionRuntimeContext &)> **isEnabled**
- std::function< void(const GuiExtensionRuntimeContext &)> **trigger**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/extensions/GuiExtensionContracts.h

17.371 GuiDockContribution Struct Reference

Public Attributes

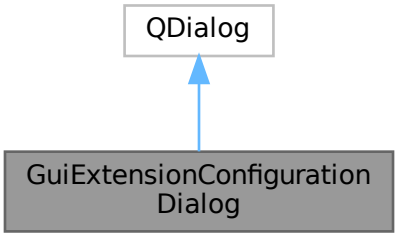
- std::string **id**
- std::string **title**
- std::string **areaHint**
- std::function< bool(const GuiExtensionRuntimeContext &)> **isVisible**
- std::function< QWidget ∗(const GuiExtensionRuntimeContext &)> **createWidget**

The documentation for this struct was generated from the following file:

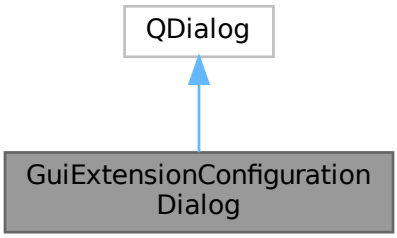
- applications/gui/genesys/extensions/GuiExtensionContracts.h

17.372 GuiExtensionConfigurationDialog Class Reference

Inheritance diagram for GuiExtensionConfigurationDialog:



Collaboration diagram for GuiExtensionConfigurationDialog:



Public Member Functions

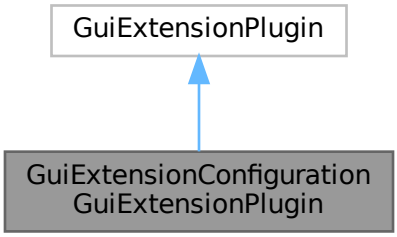
- **GuiExtensionConfigurationDialog** (QWidget ∗parent, std::vector< const GuiExtensionPlugin ∗ > plugins, std::function< void()> onSaved={})

The documentation for this class was generated from the following files:

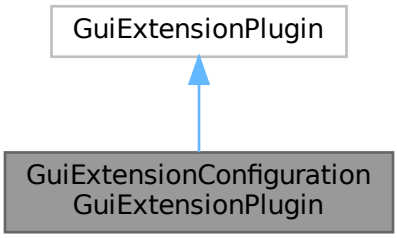
- applications/gui/genesys/extensions/GuiExtensionConfigurationDialog.h
- applications/gui/genesys/extensions/GuiExtensionConfigurationDialog.cpp

17.373 GuiExtensionConfigurationGuiExtensionPlugin Class Reference

Inheritance diagram for GuiExtensionConfigurationGuiExtensionPlugin:



Collaboration diagram for GuiExtensionConfigurationGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.373.1 Member Function Documentation

17.373.1.1 displayName()

```
std::string GuiExtensionConfigurationGuiExtensionPlugin::displayName ( ) const  [inline],
[override], [virtual]
```

Implements GuiExtensionPlugin.

17.373.1.2 extensionId()

```
std::string GuiExtensionConfigurationGuiExtensionPlugin::extensionId ( ) const  [inline],
[override], [virtual]
```

Implements GuiExtensionPlugin.

17.373.1.3 registerContributions()

```
void GuiExtensionConfigurationGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.373.1.4 requiredModelPlugins()

```
std::vector< std::string > GuiExtensionConfigurationGuiExtensionPlugin::requiredModelPlugins (
) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.373.1.5 version()

```
std::string GuiExtensionConfigurationGuiExtensionPlugin::version ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

- applications/gui/genesys/extensions/GuiExtensionConfigurationGuiExtensionPlugin.cpp

17.374 GuiExtensionManager Class Reference

Public Member Functions

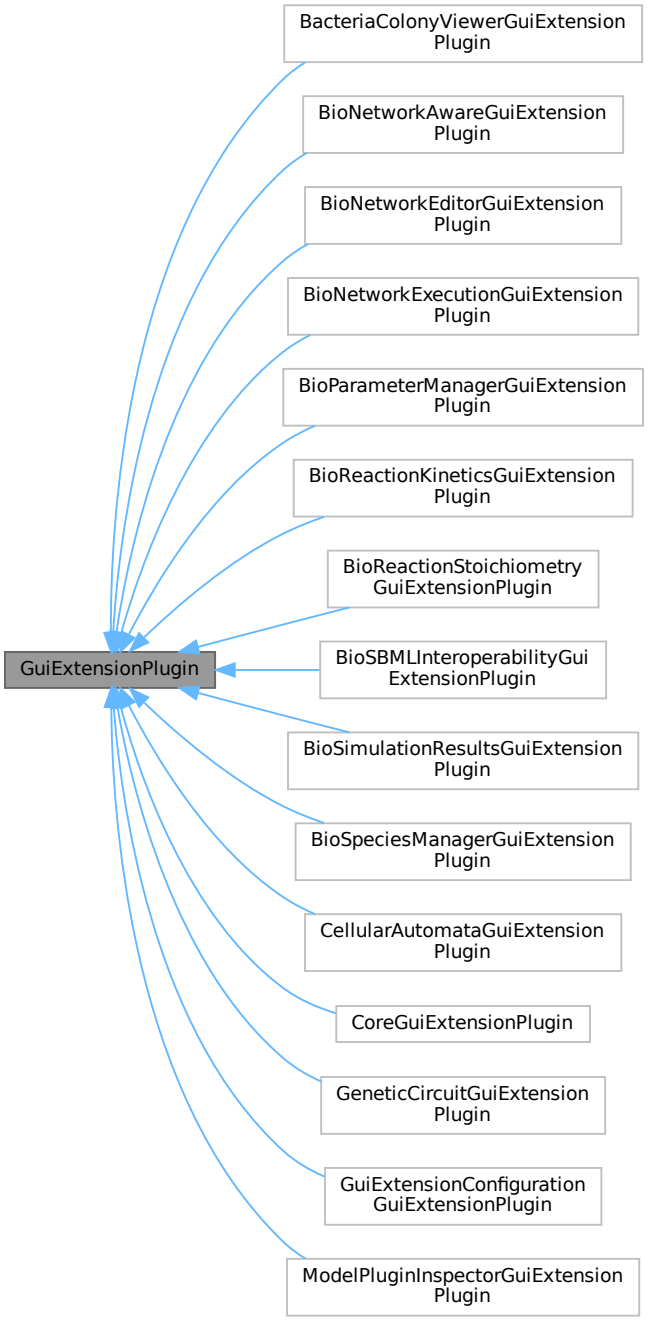
- **GuiExtensionManager** (QMainWindow ∗mainWindow)
- void **setPlugins** (std::vector< const GuiExtensionPlugin ∗ > plugins)
- void **setLoadedModelPluginIds** (std::vector< std::string > modelPluginIds)
- void **rebuild** (const GuiExtensionRuntimeContext &context)
- void **clear** ()

The documentation for this class was generated from the following files:

- applications/gui/genesys/extensions/GuiExtensionManager.h
- applications/gui/genesys/extensions/GuiExtensionManager.cpp

17.375 GuiExtensionPlugin Class Reference

Inheritance diagram for GuiExtensionPlugin:



Public Member Functions

- virtual std::string **extensionId** () const =0
- virtual std::string **displayName** () const =0
- virtual std::string **version** () const =0
- virtual std::vector< std::string > **requiredModelPlugins** () const =0
- virtual void **registerContributions** (GuiExtensionRegistry *registry) const =0

The documentation for this class was generated from the following file:

- applications/gui/genesys/extensions/GuiExtensionContracts.h

17.376 GuiExtensionPluginAutoRegistrar Class Reference

Public Member Functions

- **GuiExtensionPluginAutoRegistrar** (const GuiExtensionPlugin *plugin)

The documentation for this class was generated from the following files:

- applications/gui/genesys/extensions/GuiExtensionPluginCatalog.h
- applications/gui/genesys/extensions/GuiExtensionPluginCatalog.cpp

17.377 GuiExtensionPluginCatalog Class Reference

Static Public Member Functions

- static const std::vector< const GuiExtensionPlugin * > & **plugins** ()
- static std::vector< const GuiExtensionPlugin * > **resolvedPlugins** ()
- static GuiExtensionPresentationMetadata **presentationMetadata** (const std::string &extensionId)
- static void **registerPlugin** (const GuiExtensionPlugin *plugin)

The documentation for this class was generated from the following files:

- applications/gui/genesys/extensions/GuiExtensionPluginCatalog.h
- applications/gui/genesys/extensions/GuiExtensionPluginCatalog.cpp

17.378 GuiExtensionPresentationMetadata Struct Reference

Public Attributes

- std::string **category**
- std::string **group**
- int **priority** = 0

The documentation for this struct was generated from the following file:

- applications/gui/genesys/extensions/GuiExtensionPluginCatalog.h

17.379 GuiExtensionRegistry Class Reference

Public Member Functions

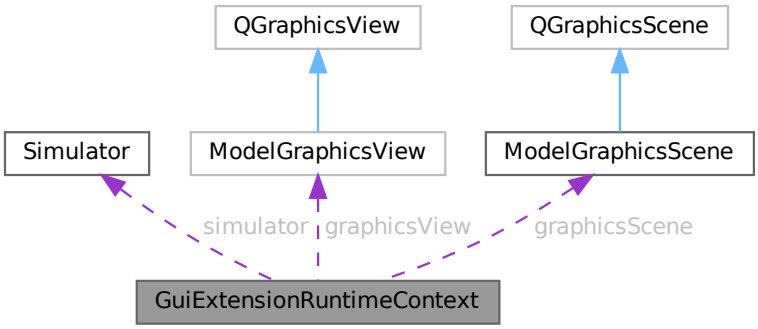
- void **addAction** (GuiActionContribution contribution)
- void **addDock** (GuiDockContribution contribution)
- void **addWindow** (GuiWindowContribution contribution)
- const std::vector< GuiActionContribution > & **actions** () const
- const std::vector< GuiDockContribution > & **docks** () const
- const std::vector< GuiWindowContribution > & **windows** () const
- void **clear** ()

The documentation for this class was generated from the following file:

- applications/gui/genesys/extensions/GuiExtensionContracts.h

17.380 GuiExtensionRuntimeContext Struct Reference

Collaboration diagram for GuiExtensionRuntimeContext:



Public Attributes

- Simulator * **simulator** = nullptr
- QMainWindow * **mainWindow** = nullptr
- Ui::MainWindow * **ui** = nullptr
- ModelGraphicsView * **graphicsView** = nullptr
- ModelGraphicsScene * **graphicsScene** = nullptr

The documentation for this struct was generated from the following file:

- applications/gui/genesys/extensions/GuiExtensionContracts.h

17.381 GuiScopeTrace Class Reference

Emits a scoped trace entry for a GUI method and measures its runtime.
`#include <applications/gui/genesys/GuiScopeTrace.h>`

Public Member Functions

- GuiScopeTrace (const char *methodName, const void *thisPtr)

  Starts a scoped GUI trace entry for a critical method invocation.
- ~**GuiScopeTrace** ()

  Ends the scoped trace and logs the elapsed execution time.

17.381.1 Detailed Description

Emits a scoped trace entry for a GUI method and measures its runtime.
Instances are usually created on the stack at the beginning of a method so the constructor logs entry and the destructor logs the elapsed time when the scope exits.

17.381.2 Constructor & Destructor Documentation

17.381.2.1 GuiScopeTrace()

```
GuiScopeTrace::GuiScopeTrace (
            const char * methodName,
            const void * thisPtr )
```

Starts a scoped GUI trace entry for a critical method invocation.

Parameters

methodName	Name of the traced method.
thisPtr	Pointer to the current object, when available.

The documentation for this class was generated from the following files:

- applications/gui/genesys/GuiScopeTrace.h
- applications/gui/genesys/GuiScopeTrace.cpp

17.382 GuiThemeManager Class Reference

Applies the visual theme used by the Qt application and model views.
`#include <applications/gui/genesys/guithememanager.h>`

Static Public Member Functions

- static void applyApplicationTheme (QApplication ∗application)

  Applies application-wide theme settings to the Qt application object.
- static void applyModelGraphicsTheme (ModelGraphicsView ∗graphicsView)

  Applies the model-graphics theme to a specific graphics view.

17.382.1 Detailed Description

Applies the visual theme used by the Qt application and model views.
The theme manager centralizes palette-level adjustments so the application and graphics scene remain visually consistent when preferences change.

17.382.2 Member Function Documentation

17.382.2.1 applyApplicationTheme()

```
void GuiThemeManager::applyApplicationTheme (
            QApplication * application )  [static]
```

Applies application-wide theme settings to the Qt application object.

Parameters

application	Target QApplication instance.

17.382.2.2 applyModelGraphicsTheme()

```
void GuiThemeManager::applyModelGraphicsTheme (
            ModelGraphicsView * graphicsView )  [static]
```

Applies the model-graphics theme to a specific graphics view.

Parameters

graphicsView	Target graphics view.

The documentation for this class was generated from the following files:

- applications/gui/genesys/guithememanager.h
- applications/gui/genesys/guithememanager.cpp

17.383 GuiWindowContribution Struct Reference

Public Attributes

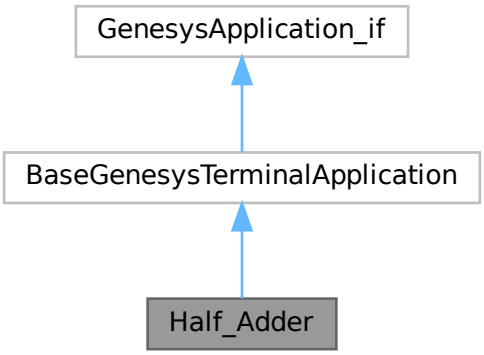
- std::string **id**
- std::string **menuPath**
- std::string **text**
- std::function< bool(const GuiExtensionRuntimeContext &)> **isVisible**
- std::function< void(const GuiExtensionRuntimeContext &)> **open**

The documentation for this struct was generated from the following file:

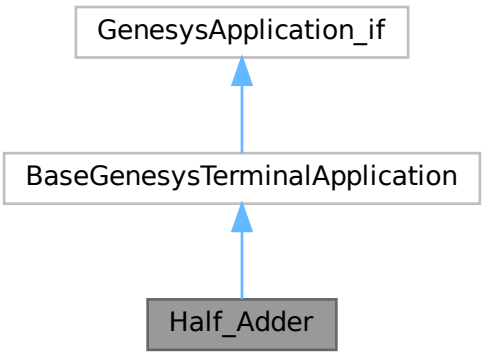
- applications/gui/genesys/extensions/GuiExtensionContracts.h

17.384 Half_Adder Class Reference

Inheritance diagram for Half_Adder:



Collaboration diagram for Half_Adder:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.384.1 Member Function Documentation

17.384.1.1 main()

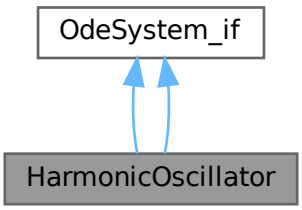
```
int Half_Adder::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

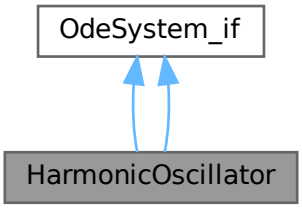
- applications/modelSpecific/teaching/Half_Adder.h
- applications/modelSpecific/teaching/Half_Adder.cpp

17.385 HarmonicOscillator Class Reference

Inheritance diagram for HarmonicOscillator:



Collaboration diagram for HarmonicOscillator:



Public Member Functions

- unsigned int dimension () const override
- void evaluate (double t, const double *y, double *dydt) const override

17.385.1 Member Function Documentation

17.385.1.1 dimension()

```
unsigned int HarmonicOscillator::dimension ( ) const  [inline], [override], [virtual]
```

Implements OdeSystem_if.

17.385.1.2 evaluate()

```
void HarmonicOscillator::evaluate (
            double t,
            const double * y,
            double * dydt ) const  [inline], [override], [virtual]
```

Implements OdeSystem_if.

The documentation for this class was generated from the following file:

- tests/test_ode_solver_numerical_validation.cpp

17.386 testing::internal::HasDebugStringAndShortDebugString< T > Class Template Reference

Static Public Attributes

- static constexpr bool value

17.386.1 Member Data Documentation

17.386.1.1 value

```
template<typename T >
constexpr bool testing::internal::HasDebugStringAndShortDebugString< T >::value  [static],
[constexpr]
```

Initial value:

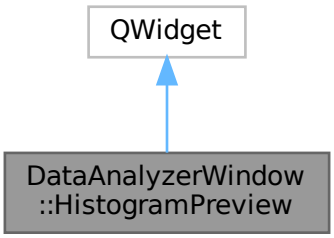
```
=
      HasDebugStringType::value && HasShortDebugStringType::value
```

The documentation for this class was generated from the following file:

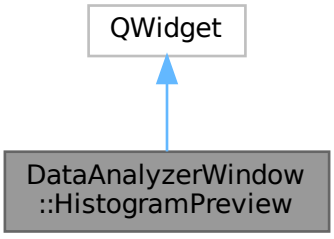
- gtest/gtest/internal/gtest-internal.h

17.387 DataAnalyzerWindow::HistogramPreview Class Reference

Inheritance diagram for DataAnalyzerWindow::HistogramPreview:



Collaboration diagram for DataAnalyzerWindow::HistogramPreview:



Public Member Functions

- **HistogramPreview** (QWidget *parent=nullptr)
- void **setValues** (const QList< double > &values)

Protected Member Functions

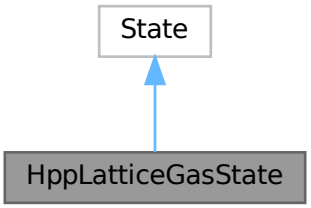
- void **paintEvent** (QPaintEvent *) override

The documentation for this class was generated from the following file:

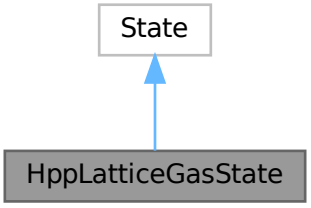
- applications/gui/genesys/tools/dataanalyzer/DataAnalyzerWindow.cpp

17.388 HppLatticeGasState Class Reference

Inheritance diagram for HppLatticeGasState:



Collaboration diagram for HppLatticeGasState:



Public Types

- enum **Direction** : size_t { **North** = 0 , **East** = 1 , **South** = 2 , **West** = 3 }

Public Member Functions

- **HppLatticeGasState** (std::array< bool, 4 > occupations)
- std::unique_ptr< State > clone () const override
- bool equals (const State &other) const override
- std::string toString () const override

- std::string typeName () const override
- long getValue () const override
- void setValue (long value) override
- bool **has** (Direction direction) const
- void **set** (Direction direction, bool occupied)
- const std::array< bool, 4 > & **occupations** () const

Public Member Functions inherited from State

- **State** (long value)
- **State** (const State &orig)
- virtual std::string **show** () const
- template<typename T >
  const T * **as** () const noexcept
- template<typename T >
  T * **as** () noexcept

Static Public Member Functions

- static long **maskFromOccupations** (const std::array< bool, 4 > &occupations)

Additional Inherited Members

Protected Attributes inherited from State

- long **_value**

17.388.1 Member Function Documentation

17.388.1.1 clone()

```
std::unique_ptr< State > HppLatticeGasState::clone ( ) const  [inline], [override], [virtual]
```

Reimplemented from State.

17.388.1.2 equals()

```
bool HppLatticeGasState::equals (
            const State & other ) const  [inline], [override], [virtual]
```

Reimplemented from State.

17.388.1.3 getValue()

```
long HppLatticeGasState::getValue ( ) const  [inline], [override], [virtual]
```

Reimplemented from State.

17.388.1.4 setValue()

```
void HppLatticeGasState::setValue (
            long value )  [inline], [override], [virtual]
```

Reimplemented from State.

17.388.1.5 toString()

```
std::string HppLatticeGasState::toString ( ) const  [inline], [override], [virtual]
```

Reimplemented from State.

17.388.1.6 typeName()

```
std::string HppLatticeGasState::typeName ( ) const  [inline], [override], [virtual]
```

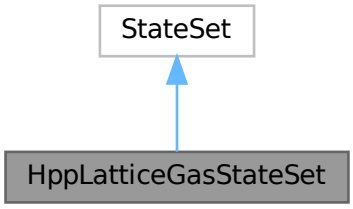
Reimplemented from State.

The documentation for this class was generated from the following file:

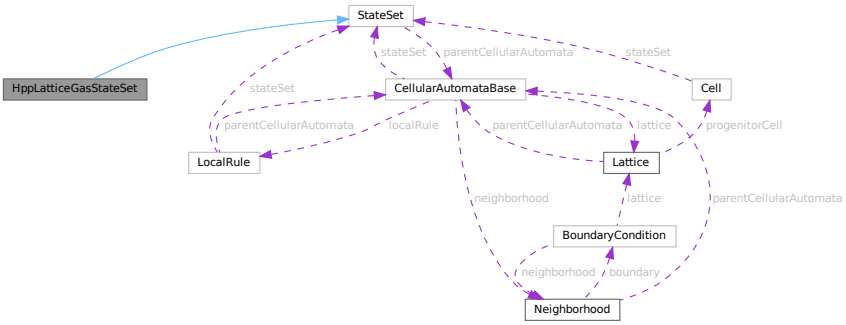
- plugins/components/ModalModel/CellularAutomata/State.h

17.389 HppLatticeGasStateSet Class Reference

Inheritance diagram for HppLatticeGasStateSet:



Collaboration diagram for HppLatticeGasStateSet:



Public Member Functions

- **HppLatticeGasStateSet** (CellularAutomataBase *parentCellularAutomata)
- bool contains (const State &state) const override
- std::unique_ptr< State > createDefaultState () const override
- std::unique_ptr< State > parseState (const std::string &text) const override
- bool isFinite () const override
- bool isDiscrete () const override
- bool isComposite () const override
- std::vector< std::unique_ptr< State > > enumerateStates () const override
- std::string name () const override

Public Member Functions inherited from StateSet

- **StateSet** (CellularAutomataBase *parentCellularAutomata)
- **StateSet** (const StateSet &orig)
- virtual bool **isContinuous** () const

Additional Inherited Members

Protected Attributes inherited from StateSet

- CellularAutomataBase * **parentCellularAutomata**

17.389.1 Member Function Documentation

17.389.1.1 contains()

```
bool HppLatticeGasStateSet::contains (
            const State & state ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.389.1.2 createDefaultState()

```
std::unique_ptr< State > HppLatticeGasStateSet::createDefaultState ( ) const  [inline], [override],
[virtual]
```

Reimplemented from StateSet.

17.389.1.3 enumerateStates()

```
std::vector< std::unique_ptr< State > > HppLatticeGasStateSet::enumerateStates ( ) const
[inline], [override], [virtual]
```

Reimplemented from StateSet.

17.389.1.4 isComposite()

```
bool HppLatticeGasStateSet::isComposite ( ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.389.1.5 isDiscrete()

```
bool HppLatticeGasStateSet::isDiscrete ( ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.389.1.6 isFinite()

```
bool HppLatticeGasStateSet::isFinite ( ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.389.1.7 name()

```
std::string HppLatticeGasStateSet::name ( ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.389.1.8 parseState()

```
std::unique_ptr< State > HppLatticeGasStateSet::parseState (
            const std::string & text ) const  [inline], [override], [virtual]
```

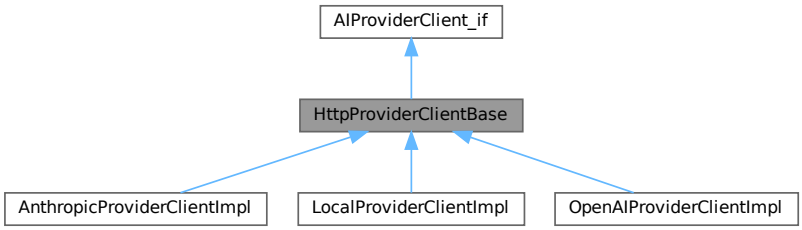
Reimplemented from StateSet.

The documentation for this class was generated from the following file:

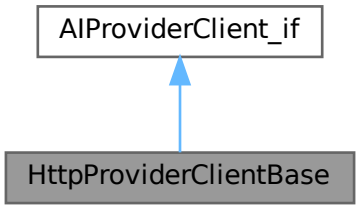
- plugins/components/ModalModel/CellularAutomata/StateSet.h

17.390 HttpProviderClientBase Class Reference

Abstract base class for HTTP-based AI provider clients.
`#include <tools/AIAssistant/HttpProviderClientBase.h>`
Inheritance diagram for HttpProviderClientBase:



Collaboration diagram for HttpProviderClientBase:



Public Member Functions

- **HttpProviderClientBase** (const HttpProviderClientBase &)=delete
- HttpProviderClientBase & **operator=** (const HttpProviderClientBase &)=delete
- bool isAvailable () const override

  Returns `true` when this client can attempt to send requests.
- void setBaseUrl (const std::string &url) override

  Sets the base URL for the provider API endpoint.
- std::string getBaseUrl () const override

  Returns the currently configured base URL.
- void setTimeoutSeconds (unsigned int seconds) override

  Sets the HTTP request timeout in seconds.
- unsigned int getTimeoutSeconds () const override

  Returns the configured HTTP request timeout.
- void setApiKey (const std::string &apiKey) override

  Stores the API key in memory.
- void clearApiKey () override

  Overwrites (best-effort) and clears the in-memory API key.
- bool hasApiKey () const override

  Returns `true` when a non-empty API key is held.

- ProviderResponse send (const ProviderRequest &request) override

  Executes the HTTP request via curl and returns the provider response.

Public Member Functions inherited from AIProviderClient_if

- virtual std::string getProviderName () const =0

  Returns a human-readable provider name (e.g. "OpenAI", "Anthropic").

Static Public Member Functions

- static std::string **_escapeJsonString** (const std::string &s)

  Escapes a plain string for safe embedding inside a JSON string value.
- static std::string **_extractJsonString** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON string field by key. Returns an empty string when not found or value is not a quoted string.
- static int **_extractJsonInt** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON integer field by key. Returns 0 when not found.
- static double **_extractJsonDouble** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON number field (int or float) by key. Returns 0.0 when not found.
- static std::string **_buildMessagesArray** (const std::vector< ChatMessage > &messages)

  Serialises a vector of ChatMessage objects to a JSON array string.
- static std::string **_extractFirstArrayElement** (const std::string &json, const std::string &arrayKey)

  Extracts the raw substring of the first element of a named JSON array. Returns an empty string when not found.

Protected Member Functions

- virtual std::string _buildRequestBody (const ProviderRequest &request) const =0

  Serialises `request` to the provider's JSON wire format.
- virtual ProviderResponse _parseResponseBody (const std::string &rawBody, int httpStatus) const =0

  Deserialises `rawBody` (the full curl stdout) into a ProviderResponse.
- virtual std::string _authorizationHeader () const =0

  Returns the Authorization header line without the "header = " prefix.
- virtual std::vector< std::string > _extraRequestHeaders () const

  Returns additional HTTP headers to send with every request.

Protected Attributes

- std::string **_baseUrl**
- unsigned int **_timeoutSeconds** = 30
- std::string _apiKey

  In-memory API key.

17.390.1 Detailed Description

Abstract base class for HTTP-based AI provider clients.
Implements the common parts of AIProviderClient_if that are shared across all cloud and local provider backends:

- In-memory API key management with the same security contract as AIAssistant_if.
- Base URL and timeout configuration.
- A synchronous send() that builds a curl command, writes the API key and request body to temporary files, invokes curl via popen(), and returns the raw response body to the subclass for parsing.

17.390.1.1 Transport design

The HTTP request is executed by shelling out to the system `curl` binary. This avoids any C++ HTTP library dependency while remaining portable. Security considerations:

- The API key is written to a temporary curl config file created with mkstemp() and restricted to mode 0600. The key is never passed on the command line, keeping it out of /proc/<pid>/cmdline and shell history.
- Both the body temp file and the config temp file are unlinked immediately after the curl invocation begins (unlink-before-pclose pattern).

17.390.1.2 Subclass responsibilities

Concrete implementations must override:

- _buildRequestBody() — serialise ProviderRequest to the provider's JSON format.
- _parseResponseBody() — deserialise the raw HTTP body to ProviderResponse.
- _authorizationHeader() — return the full auth header line (e.g. "Authorization: Bearer <key>").
- _extraRequestHeaders() — return additional provider-specific headers (default: none).

17.390.1.3 curl dependency

curl must be installed and on PATH at runtime. If curl is unavailable, send() returns a ProviderResponse with success=false and a descriptive error message.

17.390.2 Member Function Documentation

17.390.2.1 _authorizationHeader()

```
virtual std::string HttpProviderClientBase::_authorizationHeader ( ) const  [protected], [pure virtual]
```

Returns the Authorization header line without the "header = " prefix.
Example return values:

- "Authorization: Bearer sk-..." (OpenAI, LocalProvider)
- "x-api-key: sk-ant-..." (Anthropic)

Called only from within send(); the value is written to a 0600 temp file and never logged.
Implemented in AnthropicProviderClientImpl, LocalProviderClientImpl, and OpenAIProviderClientImpl.

17.390.2.2 _buildRequestBody()

```
virtual std::string HttpProviderClientBase::_buildRequestBody (
            const ProviderRequest & request ) const  [protected], [pure virtual]
```

Serialises `request` to the provider's JSON wire format.
Implemented in AnthropicProviderClientImpl, LocalProviderClientImpl, and OpenAIProviderClientImpl.

17.390.2.3 _extraRequestHeaders()

```
std::vector< std::string > HttpProviderClientBase::_extraRequestHeaders ( ) const  [protected], [virtual]
```

Returns additional HTTP headers to send with every request.
Default implementation returns an empty vector. Anthropic overrides to add "anthropic-version: 2023-06-01".
Each element must be a complete header line, e.g. "anthropic-version: 2023-06-01".
Reimplemented in AnthropicProviderClientImpl, and OpenAIProviderClientImpl.

17.390.2.4 _parseResponseBody()

```
virtual ProviderResponse HttpProviderClientBase::_parseResponseBody (
            const std::string & rawBody,
            int httpStatus ) const  [protected], [pure virtual]
```

Deserialises `rawBody` (the full curl stdout) into a ProviderResponse.

Parameters

rawBody	Complete HTTP response body as received from curl.
httpStatus	HTTP status code captured from curl exit / response line, or 0 when unavailable.

Implemented in AnthropicProviderClientImpl, LocalProviderClientImpl, and OpenAIProviderClientImpl.

17.390.2.5 clearApiKey()

```
void HttpProviderClientBase::clearApiKey ( )  [override], [virtual]
```

Overwrites (best-effort) and clears the in-memory API key.
Implements AIProviderClient_if.

17.390.2.6 getBaseUrl()

```
std::string HttpProviderClientBase::getBaseUrl ( ) const  [override], [virtual]
```

Returns the currently configured base URL.
Implements AIProviderClient_if.

17.390.2.7 getTimeoutSeconds()

```
unsigned int HttpProviderClientBase::getTimeoutSeconds ( ) const  [override], [virtual]
```

Returns the configured HTTP request timeout.
Implements AIProviderClient_if.

17.390.2.8 hasApiKey()

```
bool HttpProviderClientBase::hasApiKey ( ) const  [override], [virtual]
```

Returns `true` when a non-empty API key is held.
Implements AIProviderClient_if.

17.390.2.9 isAvailable()

```
bool HttpProviderClientBase::isAvailable ( ) const  [override], [virtual]
```

Returns `true` when this client can attempt to send requests.
A client is considered available when it holds an API key (if required by the provider) and a reachable base URL.
Implements AIProviderClient_if.
Reimplemented in LocalProviderClientImpl, and OpenAIProviderClientImpl.

17.390.2.10 send()

```
ProviderResponse HttpProviderClientBase::send (
            const ProviderRequest & request )  [override], [virtual]
```

Executes the HTTP request via curl and returns the provider response.
Steps:

1. Calls _buildRequestBody() to serialise the request.
2. Writes the body JSON to a temp file.
3. Writes a curl config file with URL, headers (including auth), data reference, and timeout.
4. Invokes curl via popen(), capturing stdout.

5. Unlinks both temp files.
6. Calls _parseResponseBody() on the captured output.

Implements AIProviderClient_if.

17.390.2.11 setApiKey()

```
void HttpProviderClientBase::setApiKey (
            const std::string & apiKey )  [override], [virtual]
```

Stores the API key in memory.
Never write the key to disk or logs. Prefer loading from an environment variable via the owning AIAssistant_if rather than embedding keys in source.
Implements AIProviderClient_if.

17.390.2.12 setBaseUrl()

```
void HttpProviderClientBase::setBaseUrl (
            const std::string & url )  [override], [virtual]
```

Sets the base URL for the provider API endpoint.
Implements AIProviderClient_if.

17.390.2.13 setTimeoutSeconds()

```
void HttpProviderClientBase::setTimeoutSeconds (
            unsigned int seconds )  [override], [virtual]
```

Sets the HTTP request timeout in seconds.
Implements AIProviderClient_if.

17.390.3 Member Data Documentation

17.390.3.1 _apiKey

```
std::string HttpProviderClientBase::_apiKey  [protected]
```

In-memory API key.
Security contract (matches AIAssistant_if):

- Never written to disk.
- Never included in log or trace output.
- Not accessible via a public getter.
- Overwritten (best-effort) before clearing in clearApiKey().

The documentation for this class was generated from the following files:

- tools/AIAssistant/HttpProviderClientBase.h
- tools/AIAssistant/HttpProviderClientBase.cpp

17.391 HttpRequest Struct Reference

Lightweight representation of one parsed HTTP request.
`#include <applications/worker/http/HttpRequest.h>`

Public Member Functions

- std::string header (const std::string &name) const

  Returns a header value using a case-insensitive lookup.

Public Attributes

- std::string **method**

  HTTP method token as received from the request line.
- std::string **path**

  Request target path used by the API router.
- std::string **body**

  Raw body payload passed to JSON parsers and handlers.
- std::unordered_map< std::string, std::string > **headers**

  Lowercase header map used for case-insensitive lookups.

17.391.1 Detailed Description

Lightweight representation of one parsed HTTP request.
The worker application uses a small transport surface, so the request keeps only the pieces needed by the router: method, path, raw body and normalized header lookup.

17.391.2 Member Function Documentation

17.391.2.1 header()

```
std::string HttpRequest::header (
            const std::string & name ) const  [inline]
```

Returns a header value using a case-insensitive lookup.

Parameters

name	Header name to search.

Returns

Header value when present, or an empty string otherwise.

The documentation for this struct was generated from the following file:

- applications/worker/http/HttpRequest.h

17.392 HttpResponse Struct Reference

Minimal HTTP response payload produced by the web API router.
`#include <applications/worker/http/HttpResponse.h>`

Public Attributes

- int **status** = 200

  HTTP status code returned to the client.
- std::string **contentType** = "application/json"

  MIME type advertised in the response.
- std::string **body**

  Raw response body, usually JSON.
- std::unordered_map< std::string, std::string > **headers**

  Arbitrary response headers written by the router or handlers.

17.392.1 Detailed Description

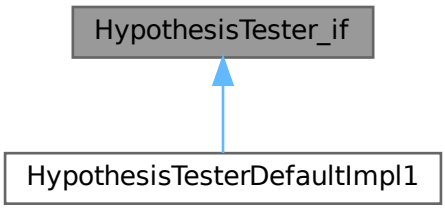
Minimal HTTP response payload produced by the web API router.

The response mirrors the small transport model used by the embedded server: status, content type, body, and a simple header map.
The documentation for this struct was generated from the following file:

- applications/worker/http/HttpResponse.h

17.393 HypothesisTester_if Class Reference

Interface for classical parametric inference.
`#include <tools/Statistics/HypothesisTester_if.h>`
Inheritance diagram for HypothesisTester_if:



Classes

- class ConfidenceInterval

  Confidence interval [inferior, superior] and its half-width e0.
- class TestResult

  Result of a parametric hypothesis test.

Public Types

- enum H1Comparition { **DIFFERENT** = 1 , **LESS_THAN** = 2 , **GREATER_THAN** = 3 }

  Alternative-hypothesis comparison mode. DIFFERENT: two-sided; LESS_THAN: left-tailed; GREATER_THAN: right-tailed.

Public Member Functions

- virtual HypothesisTester_if::ConfidenceInterval **averageConfidenceInterval** (double avg, double stddev, unsigned int n, double confidenceLevel)=0
- virtual HypothesisTester_if::ConfidenceInterval **proportionConfidenceInterval** (double prop, unsigned int n, double confidenceLevel)=0
- virtual HypothesisTester_if::ConfidenceInterval **proportionConfidenceInterval** (double prop, unsigned int n, int N, double confidenceLevel)=0
- virtual HypothesisTester_if::ConfidenceInterval **varianceConfidenceInterval** (double var, unsigned int n, double confidenceLevel)=0
- virtual HypothesisTester_if::ConfidenceInterval **averageDifferenceConfidenceInterval** (double avg1, double stddev1, unsigned int n1, double avg2, double stddev2, unsigned int n2, double confidenceLevel)=0
- virtual HypothesisTester_if::ConfidenceInterval **proportionDifferenceConfidenceInterval** (double avg1, double stddev1, unsigned int n1, double avg2, double stddev2, unsigned int n2, double confidenceLevel)=0
- virtual HypothesisTester_if::ConfidenceInterval **varianceRatioConfidenceInterval** (double var1, unsigned int n1, double var2, unsigned int n2, double confidenceLevel)=0
- virtual HypothesisTester_if::ConfidenceInterval **averageConfidenceInterval** (std::string sampleData↩ Filename, double confidenceLevel)=0

- virtual HypothesisTester_if::ConfidenceInterval **proportionConfidenceInterval** (std::string sampleDataFilename, checkProportionFunction function, double confidenceLevel)=0
- virtual HypothesisTester_if::ConfidenceInterval **proportionConfidenceInterval** (std::string sampleDataFilename, checkProportionFunction function, double N, double confidenceLevel)=0
- virtual HypothesisTester_if::ConfidenceInterval **varianceConfidenceInterval** (std::string sampleDataFilename, double confidenceLevel)=0
- virtual unsigned int **estimateSampleSize** (double avg, double stddev, double desiredE0, double confidenceLevel)=0
- virtual HypothesisTester_if::TestResult **testAverage** (double avg, double stddev, unsigned int n, double avgSample, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0
- virtual HypothesisTester_if::TestResult **testProportion** (double prop, unsigned int n, double proptest, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0
- virtual HypothesisTester_if::TestResult **testVariance** (double var, unsigned int n, double vartest, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0
- virtual HypothesisTester_if::TestResult **testAverage** (double avg1, double stddev1, unsigned int n1, double avg2, double stddev2, unsigned int n2, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0
- virtual HypothesisTester_if::TestResult **testProportion** (double prop1, unsigned int n1, double prop2, unsigned int n2, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0
- virtual HypothesisTester_if::TestResult **testVariance** (double var1, unsigned int n1, double var2, unsigned int n2, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0
- virtual HypothesisTester_if::TestResult **testAverage** (std::string sampleDataFilename, double avgSample, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0
- virtual HypothesisTester_if::TestResult **testProportion** (std::string sampleDataFilename, checkProportionFunction function, double proptest, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0
- virtual HypothesisTester_if::TestResult **testVariance** (std::string sampleDataFilename, double vartest, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0
- virtual HypothesisTester_if::TestResult **testAverage** (std::string firstSampleDataFilename, std::string secondSampleDataFilename, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0
- virtual HypothesisTester_if::TestResult **testProportion** (std::string firstSampleDataFilename, std::string secondSampleDataFilename, checkProportionFunction function, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0
- virtual HypothesisTester_if::TestResult **testVariance** (std::string firstSampleDataFilename, std::string secondSampleDataFilename, double confidenceLevel, HypothesisTester_if::H1Comparition comp)=0

17.393.1 Detailed Description

Interface for classical parametric inference.
Purpose:

- Provide confidence intervals and hypothesis tests from parameters or dataset files.

Mathematical meaning:

- Contract follows classical statistical inference for one-population and two-population analyses.

Confidence level contract:

- All confidenceLevel parameters are expected to be probabilities in [0, 1].

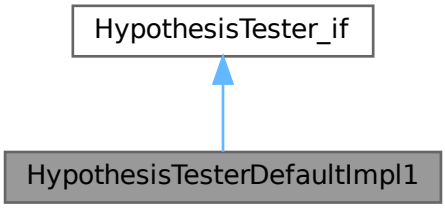
The documentation for this class was generated from the following file:

- tools/Statistics/HypothesisTester_if.h

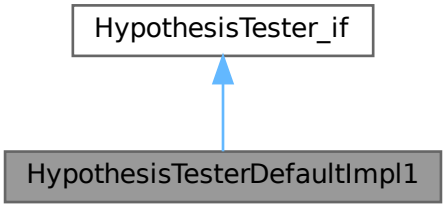
17.394 HypothesisTesterDefaultImpl1 Class Reference

Current default implementation of HypothesisTester_if.
`#include <tools/Statistics/HypothesisTesterDefaultImpl1.h>`

Inheritance diagram for HypothesisTesterDefaultImpl1:



Collaboration diagram for HypothesisTesterDefaultImpl1:



Public Member Functions

- HypothesisTester_if::ConfidenceInterval averageConfidenceInterval (double avg, double stddev, unsigned int n, double confidenceLevel) override
- HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval (double prop, unsigned int n, double confidenceLevel) override
- HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval (double prop, unsigned int n, int N, double confidenceLevel) override
- HypothesisTester_if::ConfidenceInterval varianceConfidenceInterval (double var, unsigned int n, double confidenceLevel) override
- HypothesisTester_if::ConfidenceInterval averageDifferenceConfidenceInterval (double avg1, double stddev1, unsigned int n1, double avg2, double stddev2, unsigned int n2, double confidenceLevel) override
- HypothesisTester_if::ConfidenceInterval proportionDifferenceConfidenceInterval (double avg1, double stddev1, unsigned int n1, double avg2, double stddev2, unsigned int n2, double confidenceLevel) override
- HypothesisTester_if::ConfidenceInterval varianceRatioConfidenceInterval (double var1, unsigned int n1, double var2, unsigned int n2, double confidenceLevel) override
- HypothesisTester_if::ConfidenceInterval averageConfidenceInterval (std::string sampleDataFilename, double confidenceLevel) override
- HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval (std::string sampleDataFilename, checkProportionFunction function, double confidenceLevel) override
- HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval (std::string sampleDataFilename, checkProportionFunction function, double N, double confidenceLevel) override
- HypothesisTester_if::ConfidenceInterval varianceConfidenceInterval (std::string sampleDataFilename, double confidenceLevel) override

- unsigned int estimateSampleSize (double avg, double stddev, double desiredE0, double confidenceLevel) override
- HypothesisTester_if::TestResult testAverage (double avg, double stddev, unsigned int n, double avgSample, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override
- HypothesisTester_if::TestResult testProportion (double prop, unsigned int n, double proptest, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override
- HypothesisTester_if::TestResult testVariance (double var, unsigned int n, double vartest, double confidence↩ Level, HypothesisTester_if::H1Comparition comp) override
- HypothesisTester_if::TestResult testAverage (double avg1, double stddev1, unsigned int n1, double avg2, double stddev2, unsigned int n2, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override
- HypothesisTester_if::TestResult testProportion (double prop1, unsigned int n1, double prop2, unsigned int n2, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override
- HypothesisTester_if::TestResult testVariance (double var1, unsigned int n1, double var2, unsigned int n2, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override
- HypothesisTester_if::TestResult testAverage (std::string sampleDataFilename, double avgSample, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override
- HypothesisTester_if::TestResult testProportion (std::string sampleDataFilename, checkProportionFunction function, double proptest, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override
- HypothesisTester_if::TestResult testVariance (std::string sampleDataFilename, double vartest, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override
- HypothesisTester_if::TestResult testAverage (std::string firstSampleDataFilename, std::string second↩ SampleDataFilename, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override
- HypothesisTester_if::TestResult testProportion (std::string firstSampleDataFilename, std::string second↩ SampleDataFilename, checkProportionFunction function, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override
- HypothesisTester_if::TestResult testVariance (std::string firstSampleDataFilename, std::string second↩ SampleDataFilename, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override

Additional Inherited Members

Public Types inherited from HypothesisTester_if

- enum H1Comparition { **DIFFERENT** = 1 , **LESS_THAN** = 2 , **GREATER_THAN** = 3 }

  Alternative-hypothesis comparison mode. DIFFERENT: two-sided; LESS_THAN: left-tailed; GREATER_THAN: right-tailed.

17.394.1 Detailed Description

Current default implementation of HypothesisTester_if.
Architectural role:

- Best available legacy implementation in the tools package.

Implementation status:

- Partially consolidated. One-population methods are mostly available; two-population routines still require theoretical and implementation alignment.

Planned evolution:

- Continue converging to a fully consistent implementation based on formal classical hypothesis-testing theory.

17.394.2 Member Function Documentation

17.394.2.1 averageConfidenceInterval() [1/2]

```
HypothesisTester_if::ConfidenceInterval HypothesisTesterDefaultImpl1::averageConfidence↩
Interval (
            double avg,
```

```
            double stddev,
            unsigned int n,
            double confidenceLevel )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.2 averageConfidenceInterval() [2/2]

```
HypothesisTester_if::ConfidenceInterval HypothesisTesterDefaultImpl1::averageConfidence↩
Interval (
            std::string sampleDataFilename,
            double confidenceLevel )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.3 averageDifferenceConfidenceInterval()

```
HypothesisTester_if::ConfidenceInterval HypothesisTesterDefaultImpl1::averageDifference↩
ConfidenceInterval (
            double avg1,
            double stddev1,
            unsigned int n1,
            double avg2,
            double stddev2,
            unsigned int n2,
            double confidenceLevel )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.4 estimateSampleSize()

```
unsigned int HypothesisTesterDefaultImpl1::estimateSampleSize (
            double avg,
            double stddev,
            double desiredE0,
            double confidenceLevel )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.5 proportionConfidenceInterval() [1/4]

```
HypothesisTester_if::ConfidenceInterval HypothesisTesterDefaultImpl1::proportionConfidence↩
Interval (
            double prop,
            unsigned int n,
            double confidenceLevel )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.6 proportionConfidenceInterval() [2/4]

```
HypothesisTester_if::ConfidenceInterval HypothesisTesterDefaultImpl1::proportionConfidence↩
Interval (
            double prop,
            unsigned int n,
            int N,
            double confidenceLevel )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.7 proportionConfidenceInterval() [3/4]

```
HypothesisTester_if::ConfidenceInterval HypothesisTesterDefaultImpl1::proportionConfidence↩
Interval (
            std::string sampleDataFilename,
```

```
            checkProportionFunction function,
            double confidenceLevel )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.8 proportionConfidenceInterval() [4/4]

```
HypothesisTester_if::ConfidenceInterval HypothesisTesterDefaultImpl1::proportionConfidence↩
Interval (
            std::string sampleDataFilename,
            checkProportionFunction function,
            double N,
            double confidenceLevel )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.9 proportionDifferenceConfidenceInterval()

```
HypothesisTester_if::ConfidenceInterval HypothesisTesterDefaultImpl1::proportionDifference↩
ConfidenceInterval (
            double avg1,
            double stddev1,
            unsigned int n1,
            double avg2,
            double stddev2,
            unsigned int n2,
            double confidenceLevel )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.10 testAverage() [1/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testAverage (
            double avg,
            double stddev,
            unsigned int n,
            double avgSample,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.11 testAverage() [2/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testAverage (
            double avg1,
            double stddev1,
            unsigned int n1,
            double avg2,
            double stddev2,
            unsigned int n2,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.12 testAverage() [3/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testAverage (
            std::string firstSampleDataFilename,
            std::string secondSampleDataFilename,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.13 testAverage() [4/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testAverage (
            std::string sampleDataFilename,
            double avgSample,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.14 testProportion() [1/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testProportion (
            double prop,
            unsigned int n,
            double proptest,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.15 testProportion() [2/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testProportion (
            double prop1,
            unsigned int n1,
            double prop2,
            unsigned int n2,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.16 testProportion() [3/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testProportion (
            std::string firstSampleDataFilename,
            std::string secondSampleDataFilename,
            checkProportionFunction function,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.17 testProportion() [4/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testProportion (
            std::string sampleDataFilename,
            checkProportionFunction function,
            double proptest,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.18 testVariance() [1/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testVariance (
            double var,
            unsigned int n,
            double vartest,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.19 testVariance() [2/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testVariance (
            double var1,
            unsigned int n1,
            double var2,
            unsigned int n2,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.20 testVariance() [3/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testVariance (
            std::string firstSampleDataFilename,
            std::string secondSampleDataFilename,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.21 testVariance() [4/4]

```
HypothesisTester_if::TestResult HypothesisTesterDefaultImpl1::testVariance (
            std::string sampleDataFilename,
            double vartest,
            double confidenceLevel,
            HypothesisTester_if::H1Comparition comp )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.22 varianceConfidenceInterval() [1/2]

```
HypothesisTester_if::ConfidenceInterval HypothesisTesterDefaultImpl1::varianceConfidence↩
Interval (
            double var,
            unsigned int n,
            double confidenceLevel )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.23 varianceConfidenceInterval() [2/2]

```
HypothesisTester_if::ConfidenceInterval HypothesisTesterDefaultImpl1::varianceConfidence↩
Interval (
            std::string sampleDataFilename,
            double confidenceLevel )  [override], [virtual]
```

Implements HypothesisTester_if.

17.394.2.24 varianceRatioConfidenceInterval()

```
HypothesisTester_if::ConfidenceInterval HypothesisTesterDefaultImpl1::varianceRatioConfidence↩
Interval (
            double var1,
            unsigned int n1,
            double var2,
            unsigned int n2,
            double confidenceLevel )  [override], [virtual]
```

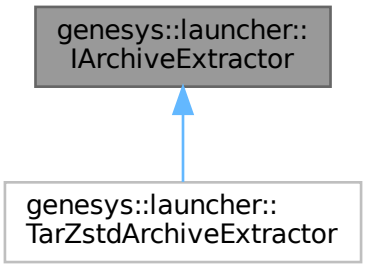
Implements HypothesisTester_if.

The documentation for this class was generated from the following files:

- tools/Statistics/HypothesisTesterDefaultImpl1.h
- tools/Statistics/HypothesisTesterDefaultImpl1.cpp

17.395 genesys::launcher::IArchiveExtractor Class Reference

Inheritance diagram for genesys::launcher::IArchiveExtractor:



Public Member Functions

- virtual ExtractionResult **extract** (const QString &archivePath, const QString &destinationRoot)=0

The documentation for this class was generated from the following file:

- applications/launcher/ArchiveExtractor.h

17.396 testing::internal::Ignore< size_t > Struct Template Reference

Public Member Functions

- **Ignore** (...)

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.397 testing::internal::IgnoredValue Class Reference

Public Member Functions

- template<typename T , typename std::enable_if<!std::is_convertible< T, Sink >::value, int >::type = 0>
  IgnoredValue (const T &)

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.398 testing::internal::IndexSequence< Is > Struct Template Reference

Public Types

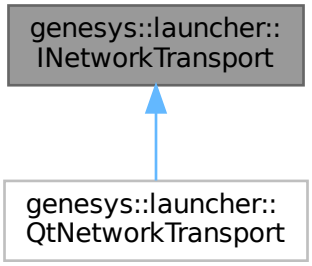
- using **type** = IndexSequence

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.399 genesys::launcher::INetworkTransport Class Reference

Inheritance diagram for genesys::launcher::INetworkTransport:



Public Member Functions

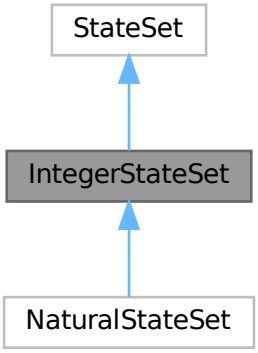
- virtual NetworkResponse **download** (const QUrl &url, QIODevice &destination, const NetworkRequestOptions &options, ProgressCallback progress={}, CancellationCallback cancelled={})=0
- virtual std::pair< NetworkResponse, QByteArray > **get** (const QUrl &url, const NetworkRequestOptions &options, ProgressCallback progress={}, CancellationCallback cancelled={})

The documentation for this class was generated from the following files:

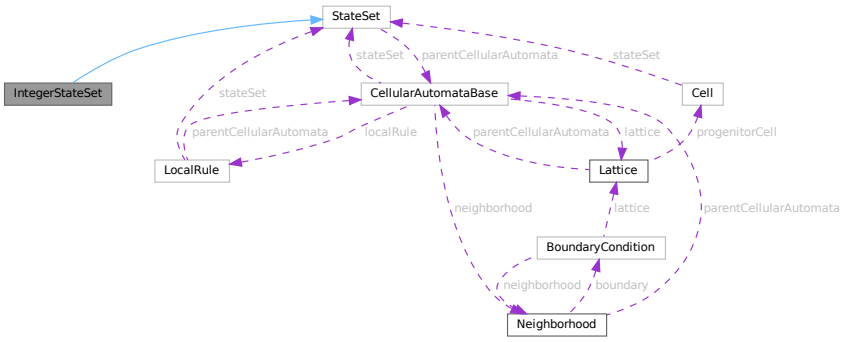
- applications/launcher/NetworkTransport.h
- applications/launcher/NetworkTransport.cpp

17.400 IntegerStateSet Class Reference

Inheritance diagram for IntegerStateSet:



Collaboration diagram for IntegerStateSet:



Public Member Functions

- **IntegerStateSet** (CellularAutomataBase *parentCellularAutomata, std::optional< long > minimum=std::nullopt, std::optional< long > maximum=std::nullopt)
- bool contains (const State &state) const override
- std::unique_ptr< State > createDefaultState () const override
- std::unique_ptr< State > parseState (const std::string &text) const override
- bool isFinite () const override
- bool isDiscrete () const override
- std::vector< std::unique_ptr< State > > enumerateStates () const override
- std::string name () const override

Public Member Functions inherited from StateSet

- **StateSet** (CellularAutomataBase *parentCellularAutomata)
- **StateSet** (const StateSet &orig)
- virtual bool **isContinuous** () const
- virtual bool **isComposite** () const

Additional Inherited Members

Protected Attributes inherited from StateSet

- CellularAutomataBase * **parentCellularAutomata**

17.400.1 Member Function Documentation

17.400.1.1 contains()

```
bool IntegerStateSet::contains (
            const State & state ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.400.1.2 createDefaultState()

```
std::unique_ptr< State > IntegerStateSet::createDefaultState ( ) const  [inline], [override],
[virtual]
```

Reimplemented from StateSet.

17.400.1.3 enumerateStates()

```
std::vector< std::unique_ptr< State > > IntegerStateSet::enumerateStates ( ) const  [inline],
[override], [virtual]
```

Reimplemented from StateSet.

17.400.1.4 isDiscrete()

```
bool IntegerStateSet::isDiscrete ( ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.400.1.5 isFinite()

```
bool IntegerStateSet::isFinite ( ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.400.1.6 name()

```
std::string IntegerStateSet::name ( ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.400.1.7 parseState()

```
std::unique_ptr< State > IntegerStateSet::parseState (
            const std::string & text ) const  [inline], [override], [virtual]
```

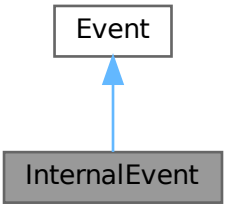
Reimplemented from StateSet.

The documentation for this class was generated from the following file:

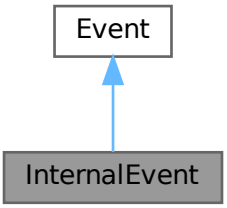
- plugins/components/ModalModel/CellularAutomata/StateSet.h

17.401 InternalEvent Class Reference

Inheritance diagram for InternalEvent:



Collaboration diagram for InternalEvent:



Public Member Functions

- **InternalEvent** (double time, std::string description)
- template<typename Class >
  void **setEventHandler** (Class *object, void(Class::*function)(void *), void *parameter)
- virtual std::string show () override
- const InternalEventHandler & **eventHandler** () const
- const std::string & **description** () const
- void **dispatchEvent** () const
- void * **object** () const
- void * **parameter** () const

Public Member Functions inherited from Event

- **Event** (double time, Entity *entity, ModelComponent *component, unsigned int componentinputPort↩ Number=0)
- **Event** (double time, Entity *entity, Connection *connection)
- double getTime () const

  getTime
- ModelComponent * getComponent () const

  getComponent
- Entity * getEntity () const

  getEntity
- unsigned int getComponentinputPortNumber () const

  getComponentinputPortNumber
- void setEntity (Entity *_entity)

  setEntity

Additional Inherited Members

Protected Attributes inherited from Event

- double **_time**

17.401.1 Member Function Documentation

17.401.1.1 show()

```
virtual std::string InternalEvent::show ( )  [inline], [override], [virtual]
```

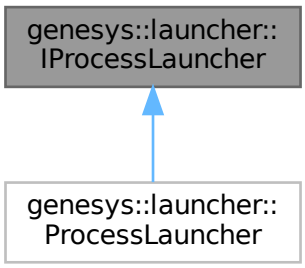
Reimplemented from Event.

The documentation for this class was generated from the following file:

- kernel/simulator/Event.h

17.402 genesys::launcher::IProcessLauncher Class Reference

Inheritance diagram for genesys::launcher::IProcessLauncher:



Public Member Functions

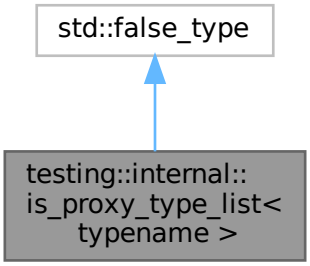
- virtual int **launchReplacingProcess** (const RuntimeSelection &selection, const QStringList &arguments, QString ∗error=nullptr)=0

The documentation for this class was generated from the following file:

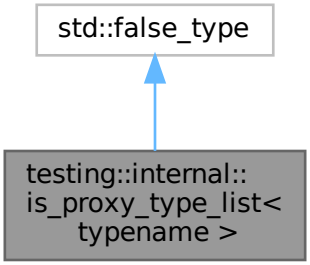
- applications/launcher/ProcessLauncher.h

17.403 testing::internal::is_proxy_type_list< typename > Struct Template Reference

Inheritance diagram for testing::internal::is_proxy_type_list< typename >:



Collaboration diagram for testing::internal::is_proxy_type_list< typename >:

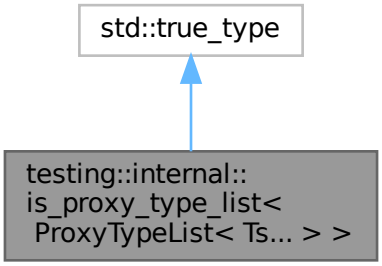


The documentation for this struct was generated from the following file:

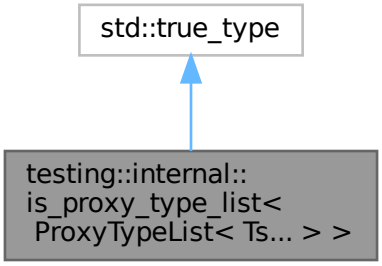
- gtest/gtest/internal/gtest-type-util.h

17.404 testing::internal::is_proxy_type_list< ProxyTypeList< Ts... > > Struct Template Reference

Inheritance diagram for testing::internal::is_proxy_type_list< ProxyTypeList< Ts... > >:



Collaboration diagram for testing::internal::is_proxy_type_list< ProxyTypeList< Ts... > >:



The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-type-util.h

17.405 testing::internal::IsHashTable< T > Struct Template Reference

Static Public Attributes

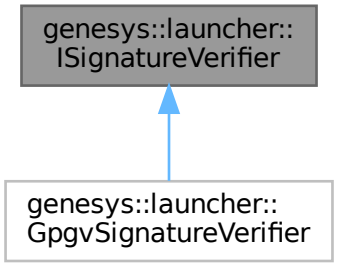
- static const bool **value** = sizeof(test<T>(nullptr, nullptr)) == sizeof(int)

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.406 genesys::launcher::ISignatureVerifier Class Reference

Inheritance diagram for genesys::launcher::ISignatureVerifier:



Public Member Functions

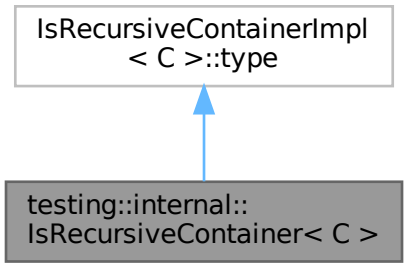
- virtual VerificationResult **verify** (const QString &dataPath, const QString &signaturePath)=0

The documentation for this class was generated from the following file:

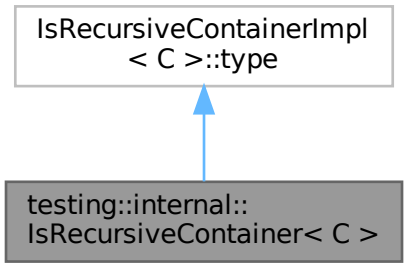
- applications/launcher/UpdateVerifier.h

17.407 testing::internal::IsRecursiveContainer< C > Struct Template Reference

Inheritance diagram for testing::internal::IsRecursiveContainer< C >:



Collaboration diagram for testing::internal::IsRecursiveContainer< C >:



The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

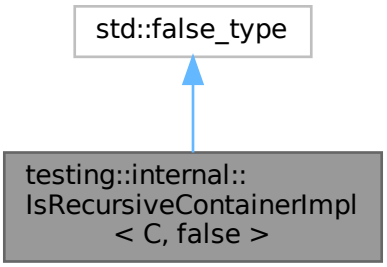
17.408 testing::internal::IsRecursiveContainerImpl< C, bool > Struct Template Reference

The documentation for this struct was generated from the following file:

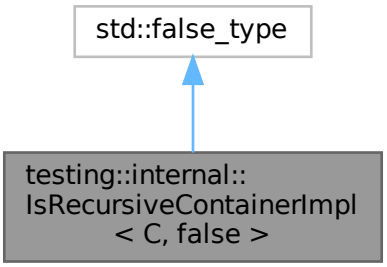
- gtest/gtest/internal/gtest-internal.h

17.409 testing::internal::IsRecursiveContainerImpl< C, false > Struct Template Reference

Inheritance diagram for testing::internal::IsRecursiveContainerImpl< C, false >:



Collaboration diagram for testing::internal::IsRecursiveContainerImpl< C, false >:



The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.410 testing::internal::IsRecursiveContainerImpl< C, true > Struct Template Reference

Public Types

- using **value_type** = decltype(*std::declval< typename C::const_iterator >())
- using **type** = std::is_same< typename std::remove_const< typename std::remove_reference< value_type >::type >::type, C >

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.411 IssueReportDraft Struct Reference

User-authored issue report collected by the GUI form before transport.
`#include <applications/gui/genesys/services/IssueReportRelayService.h>`

Public Attributes

- QString **categoryKey**
- QString **categoryLabel**
- QString **title**
- QString **summary**
- QString **reproductionSteps**
- QString **expectedBehavior**
- QString **observedBehavior**
- QString **additionalContext**
- QString **contactEmail**
- bool **includeScreenshot** = false
- bool **includeLogTail** = true
- bool **includeModelText** = false

17.411.1 Detailed Description

User-authored issue report collected by the GUI form before transport.
The documentation for this struct was generated from the following file:

- applications/gui/genesys/services/IssueReportRelayService.h

17.412 IssueReportRelayService Class Reference

HTTPS client service that sends issue reports to a secure relay endpoint.
`#include <applications/gui/genesys/services/IssueReportRelayService.h>`

Static Public Member Functions

- static QString **configuredEndpoint** ()
- static QString **optionalClientKey** ()
- static int **requestTimeoutMs** ()
- static bool **isEndpointConfigured** (QString *reason=nullptr)
- static IssueReportSubmissionResult **submit** (const IssueReportDraft &draft, const IssueReportRuntimeContext &runtimeContext)

17.412.1 Detailed Description

HTTPS client service that sends issue reports to a secure relay endpoint.
Responsibilities:

- validate relay configuration from environment;
- serialize the GUI report and runtime diagnostics into JSON;
- submit the payload via HTTPS and normalize the relay response.

Boundaries:

- it does not talk directly to GitHub with a secret embedded in the client;
- it does not own UI widgets or present dialogs;
- it assumes an external relay exists and enforces server-side policy.

The documentation for this class was generated from the following files:

- applications/gui/genesys/services/IssueReportRelayService.h
- applications/gui/genesys/services/IssueReportRelayService.cpp

17.413 IssueReportRuntimeContext Struct Reference

Runtime diagnostics captured by the GUI after the user approves submission.
`#include <applications/gui/genesys/services/IssueReportRelayService.h>`

Public Attributes

- QString **applicationName**
- QString **applicationVersion**
- QString **qtVersion**
- QString **operatingSystem**
- QString **currentModelName**
- QString **currentModelSimulanguage**
- QString **consoleTail**
- QString **simulationTail**
- QString **reportsTail**
- QString **processLogTail**
- QByteArray **screenshotPngBase64**

17.413.1 Detailed Description

Runtime diagnostics captured by the GUI after the user approves submission.
The documentation for this struct was generated from the following file:

- applications/gui/genesys/services/IssueReportRelayService.h

17.414 IssueReportSubmissionResult Struct Reference

Relay submission outcome returned to the controller layer.
`#include <applications/gui/genesys/services/IssueReportRelayService.h>`

Public Attributes

- bool **success** = false
- int **httpStatus** = 0
- int **issueNumber** = 0
- QString **issueUrl**
- QString **errorMessage**
- QString **responseBody**

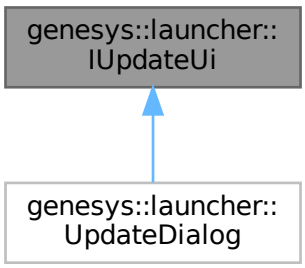
17.414.1 Detailed Description

Relay submission outcome returned to the controller layer.
The documentation for this struct was generated from the following file:

- applications/gui/genesys/services/IssueReportRelayService.h

17.415 genesys::launcher::IUpdateUi Class Reference

Inheritance diagram for genesys::launcher::IUpdateUi:



Public Member Functions

- virtual bool **confirmUpdate** (const Version &currentVersion, const Version &candidateVersion, const QString &releaseNotes)=0
- virtual void **beginDownload** (qint64 totalBytes)=0
- virtual bool **updateDownloadProgress** (qint64 receivedBytes, qint64 totalBytes)=0
- virtual void **finishDownload** ()=0
- virtual void **showError** (const QString &message)=0
- virtual void **showInfo** (const QString &message)=0

The documentation for this class was generated from the following file:

- applications/launcher/UpdateDialog.h

17.416 Json Struct Reference

Public Types

- enum **Kind** {
  null , **boolean** , **number** , **string** ,
  array , **object** }

Public Attributes

- enum Json::Kind **kind**
- double **value**
- std::string **text**
- std::vector< std::unique_ptr< Json > > sequence
- std::unordered_map< std::string, std::unique_ptr< Json > > fields

17.416.1 Member Data Documentation

17.416.1.1 fields

```
std::unordered_map<std::string, std::unique_ptr<Json> > Json::fields
```

Initial value:

```
{
    }
```

17.416.1.2 sequence

```
std::vector<std::unique_ptr<Json> > Json::sequence
```

Initial value:

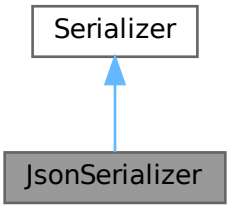
```
{
    }
```

The documentation for this struct was generated from the following file:

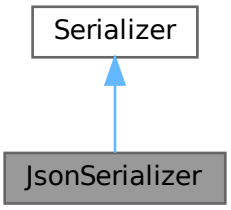
- kernel/simulator/persistence/JsonSerializer.cpp

17.417 JsonSerializer Class Reference

Inheritance diagram for JsonSerializer:



Collaboration diagram for JsonSerializer:



Public Member Functions

- **JsonSerializer** (Model ∗model)
- PersistenceRecord ∗ newPersistenceRecord () override

  Creates an empty persistence record.
- bool dump (std::ostream &output) override

  Writes formatted contents to a given output stream.
- bool load (std::istream &input) override

  Attempts to load contents from a given input stream.
- bool get (const std::string &name, PersistenceRecord ∗entry) override

  Tries to fetch an entry with the given label as a key.

- bool put (const std::string name, const std::string type, const Util::identification id, PersistenceRecord *fields) override

  Upserts an entry into this structure.
- int for_each (std::function< int(const std::string &) > delegate) override

  Iterates over this structure's entries, ordered by ID.

17.417.1 Member Function Documentation

17.417.1.1 dump()

```
bool JsonSerializer::dump (
            std::ostream & output )  [override], [virtual]
```

Writes formatted contents to a given output stream.

Parameters

output	output stream to write to.

Returns

whether the current state of this structure's contents was deemed valid for serialization.

Implements Serializer.

17.417.1.2 for_each()

```
int JsonSerializer::for_each (
            std::function< int(const std::string &) > delegate )  [override], [virtual]
```

Iterates over this structure's entries, ordered by ID.

Parameters

delegate	delegate which is applied to every entry-identifying key, until it returns a non-zero value.

Returns

either zero, or the first non-zero value returned by the delegate.

Implements Serializer.

17.417.1.3 get()

```
bool JsonSerializer::get (
            const std::string & name,
            PersistenceRecord * entry )  [override], [virtual]
```

Tries to fetch an entry with the given label as a key.

Parameters

name	a label which uniquely identifies an entry.
entry	(out-param) a pointer to a caller-owned entry, which is filled on success.

Returns

whether the entry was found and the out-param was actually written to.

Implements Serializer.

17.417.1.4 load()

```
bool JsonSerializer::load (
            std::istream & input )  [override], [virtual]
```

Attempts to load contents from a given input stream.

Returns

whether the input was well-formatted and loading was successful.

Implements Serializer.

17.417.1.5 newPersistenceRecord()

```
PersistenceRecord * JsonSerializer::newPersistenceRecord ( )  [override], [virtual]
```

Creates an empty persistence record.
Although the returned record can be freed normally with `delete`, care must be taken to only use records which come from a implementation, when configuring that implementation's contents.
Implements Serializer.

17.417.1.6 put()

```
bool JsonSerializer::put (
            const std::string name,
            const std::string type,
            const Util::identification id,
            PersistenceRecord * fields )  [override], [virtual]
```

Upserts an entry into this structure.

Parameters

name	a label which uniquely identifies the entry.
type	a type name.
id	a component/datadef ID number.
fields	a persistence record for the entry, whose contents will be copied if valid.

Returns

whether the entry was valid and was actually copied into this structure.

Implements Serializer.
The documentation for this class was generated from the following files:

- kernel/simulator/persistence/JsonSerializer.h
- kernel/simulator/persistence/JsonSerializer.cpp

17.418 KernelAccessorProbe Struct Reference

Public Member Functions

- std::string **getName** () const
- void **setName** (std::string newName)

Public Attributes

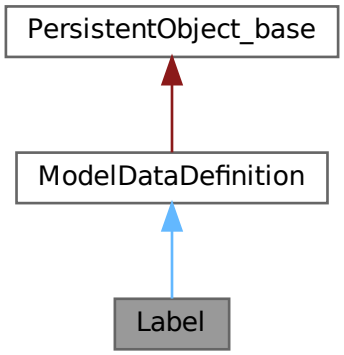
- std::string **name** = "alpha"

The documentation for this struct was generated from the following file:

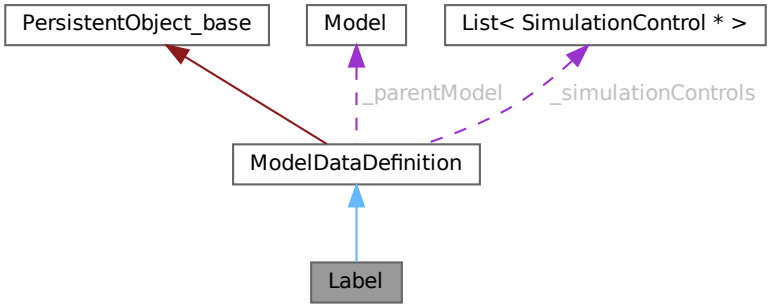
- tests/unit/test_simulator_support.cpp

17.419 Label Class Reference

Inheritance diagram for Label:



Collaboration diagram for Label:



Public Member Functions

- **Label** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **setLabel** (std::string _label)
- std::string **getLabel** () const
- void **setEnterIntoLabelComponent** (ModelComponent *enteringLabelComponent)
- ModelComponent * **getEnterIntoLabelComponent** () const
- void **sendEntityToLabelComponent** (Entity *entity, double timeDelay)

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)

- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.419.1 Member Function Documentation

17.419.1.1 _check()

```
bool Label::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.419.1.2 _loadInstance()

```
bool Label::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.419.1.3 _saveInstance()

```
void Label::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.419.1.4 show()

```
std::string Label::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/Logic/Label.h
- plugins/data/Logic/Label.cpp

17.420 QCPLabelPainterPrivate::LabelData Struct Reference

Public Attributes

- AnchorSide **side**
- double **rotation**
- QTransform **transform**
- QString **basePart**
- QString **expPart**
- QString **suffixPart**
- QRect **baseBounds**
- QRect **expBounds**
- QRect **suffixBounds**
- QRect **totalBounds**
- QRect **rotatedTotalBounds**
- QFont **baseFont**
- QFont **expFont**
- QColor **color**

The documentation for this struct was generated from the following file:

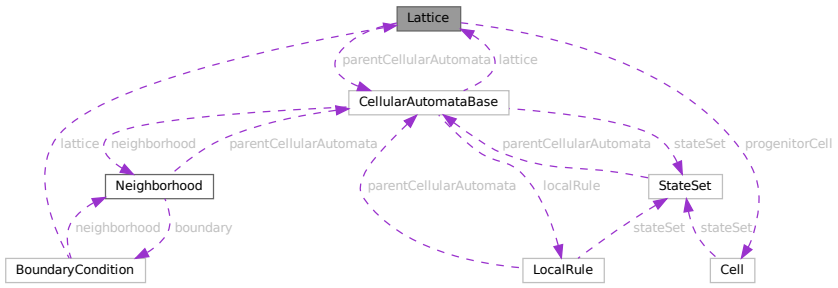
- applications/gui/genesys/qcustomplot.h

17.421 Lattice Class Reference

n-dimensional lattice

`#include <plugins/components/ModalModel/CellularAutomata/Lattice.h>`

Collaboration diagram for Lattice:



Public Member Functions

- **Lattice** (CellularAutomataBase *parentCellularAutomata, Cell *progenitorCell=nullptr, std::vector< unsigned short > dimensions={}, LatticeType latticeType=LatticeType::RETICULAR)
- **Lattice** (const Lattice &orig)
- virtual std::string **show** ()
- bool **init** ()

  init all cells
- virtual long **cellNDimPosition2Number** (const std::vector< int > position)

  Maps n-dimensional to cell number.
- virtual std::vector< int > **cellNumber2NDimPosition** (const long cellNumber)

  Maps cell number to n-dimensional.
- Cell * **getCell** (const long cellNumber)
- Cell * **getCell** (const std::vector< int > position)
- void **setCell** (const std::vector< int > position, Cell *cell)
- bool **setCellState** (std::vector< int > position, State *state=nullptr, Cell *cell=nullptr)
- bool **setCellState** (long cellNumber, State *state=nullptr, Cell *cell=nullptr)
- void **setProgenitorCell** (Cell *progenitorCell)

  set cell used to fill the lattice
- Cell * **getProgenitorCell** () const
- bool **setNumCells** (unsigned short dimension, unsigned int numCells, bool createDimensions=true)
- unsigned short **getNumDimensions** ()
- unsigned int **getNumCell** (unsigned short dimension)
- LatticeType **getLatticeType** () const
- unsigned long **getCellsSize** () const
- unsigned long **getTotalCells** () const
- unsigned long **calculateTotalCells** ()
- bool **isInit** () const
- void **setDimensions** (std::vector< unsigned short > dimensions)
- std::vector< unsigned short > **getDimensions** () const
- unsigned short **getDimension** (unsigned short dimensionNumber) const
- std::vector< Cell * > **getCells** () const

Protected Attributes

- CellularAutomataBase * **parentCellularAutomata**
- std::vector< Cell * > **cells**
- std::vector< unsigned short > **dimensions**
- Cell * **progenitorCell**
- LatticeType **latticeType**
- bool **hasBeenInit**
- unsigned long **totalCells**

17.421.1 Detailed Description

n-dimensional lattice
The documentation for this class was generated from the following files:

- plugins/components/ModalModel/CellularAutomata/Lattice.h
- plugins/components/ModalModel/CellularAutomata/Lattice.cpp

17.422 genesys::launcher::LauncherConfig Struct Reference

Static Public Member Functions

- static LauncherConfig **load** (const QString &systemConfigPath, const QString &userConfigPath)

Public Attributes

- bool **updatesEnabled** = false
- QString **channel** = QStringLiteral("stable")
- bool **allowUserRuntime** = true
- bool **requireSignature** = true
- bool **checkOnGuiStartup** = true
- bool **fallbackToSystem** = true
- int **minimumCheckIntervalHours** = 12
- int **maxUserVersions** = 2
- QUrl **manifestUrl**
- bool **hasExplicitPolicy** = false
- QStringList **warnings**

The documentation for this struct was generated from the following files:

- applications/launcher/LauncherConfig.h
- applications/launcher/LauncherConfig.cpp

17.423 genesys::launcher::LauncherController Class Reference

Public Member Functions

- **LauncherController** (LauncherConfig config, UpdateChecker &checker, UpdateDownloader &downloader, ISignatureVerifier *signatureVerifier, UserRuntimeManager &runtimeManager, IUpdateUi &ui, LauncherLogger *logger=nullptr)
- LauncherUpdateResult **checkAndMaybeUpdate** (const Version &currentVersion, bool interactive, bool forceCheck, bool checkOnly)

The documentation for this class was generated from the following files:

- applications/launcher/LauncherController.h
- applications/launcher/LauncherController.cpp

17.424 genesys::launcher::LauncherLogger Class Reference

Public Member Functions

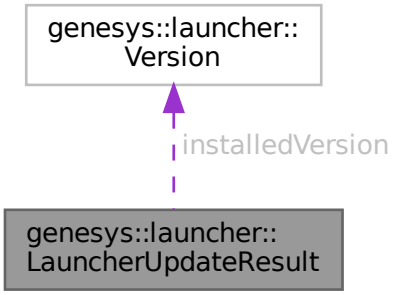
- **LauncherLogger** (QString path, qint64 maxBytes=1024 ∗1024)
- void **log** (const QString &event, const QString &detail={})

The documentation for this class was generated from the following files:

- applications/launcher/LauncherLogger.h
- applications/launcher/LauncherLogger.cpp

17.425 genesys::launcher::LauncherUpdateResult Struct Reference

Collaboration diagram for genesys::launcher::LauncherUpdateResult:



Public Attributes

- bool **checkCompleted** = false
- bool **updateAvailable** = false
- bool **installed** = false
- Version **installedVersion**
- QString **message**

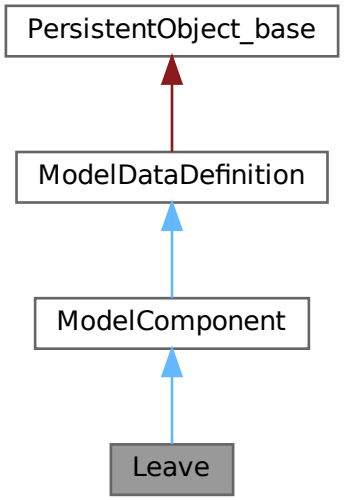
The documentation for this struct was generated from the following file:

- applications/launcher/LauncherController.h

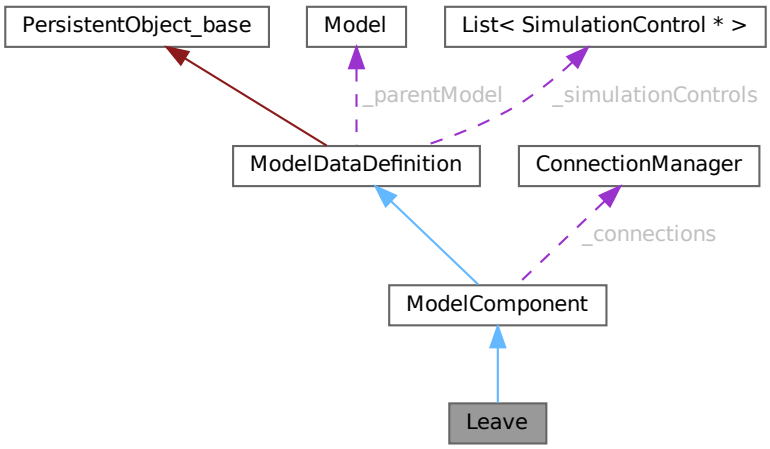
17.426 Leave Class Reference

```
#include <plugins/components/MaterialHandling/Leave.h>
```

Inheritance diagram for Leave:



Collaboration diagram for Leave:



Public Member Functions

- **Leave** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setStation** (Station ∗_station)
- void **setStationName** (std::string stationName)
- Station ∗ **getStation** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _createInternalStatisticReporters () override
- virtual void _createEditableDataDefinitions () override
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.426.1 Detailed Description

Leave module DESCRIPTION The Leave module is used to transfer an entity to a station or module. An entity may be transferred in two ways. It can be transferred to a module that defines a station by referencing the station and routing, conveying, or transporting to that station, or a graphical connection can be used to transfer an entity to another module. When an entity arrives at a Leave module, it may wait to obtain a transfer device (resource, transporter, or conveyor). When the transfer device has been obtained, the entity may experience a loading delay. Finally, the entity is transferred from this module to a destination module or station. TYPICAL USES The end of a part's production in a series of parallel processes where the part needs a forklift to be transferred to shipping PROMPTS Prompt Description Name Unique name of the module that will be displayed in the flowchart. Allocation Type of category to which the entity's incurred delay time and cost will be added. Delay Specifies a load time incurred after getting a transfer device. Units Time units used for the delay time. Transfer Out Determines whether a resource (Seize Resource), transporter (Request Transporter), or conveyor (Access Conveyor) is required prior to transferring the entity out of this module. Priority Indicates the priority of the module when either seizing a resource or requesting a transporter when there are entities waiting for that resource/transporter from other modules. This field is not visible when the Transfer Type is None or Access Conveyor. Transporter Name Name of the transporter to request. Queue Type Type of queue, either a single Queue, queue Set, Internal queue, Attribute, or Expression. Queue Name Name of the individual queue. Queue Set Name Name of the queue set. Set Index Defines the index into the queue set. Note that this is the index into the set and not the name of the queue in the set. Queue

Attribute Name The attribute name that will be evaluated to indicate which queue is to be used. Queue Expression The expression that will be evaluated to indicate which queue is to be used. Selection Rule Method of selecting among available transporters in a set. Cyclical will cycle through available members. Random will randomly select a member. Preferred Order will always select the first available member. Specific Member requires an input attribute value to specify which member of the set (previously saved in the Save Attribute field). Largest Distance selects the transporter farthest away, and Smallest Distance selects the closest transporter. Save Attribute Attribute name used to store the index number into the set of the member that is chosen. This attribute can later be referenced with the Specific Member selection rule. Active when Transfer Out is Request Transporter. Index Set Attribute name whose value identifies the index number into the set of the member requested. The entity must have a value for the attribute before utilizing this option. Resource Type Type of resource for seizing, either specifying a particular Resource, selecting from a pool of resources (that is, a resource Set), Attribute, or Expression. Resource Name Name of the resource to seize. Conveyor Name Name of the conveyor to access.

17.426.2 of Cells Number of contiguous cells the entity requires.

Connect Type Determines if the entity is to Route, Convey, or Transport to another station or Connect to another module. Move Time Time to route from this module to the destination station. Units Time units used for the move time. Station Type The entity's destination station type either an individual Station, a station based on an Attribute or Expression value, or By Sequence. Station Name Name of the individual destination station. Attribute Name The attribute name that will be evaluated to indicate the station. Expression The expression that will be evaluated to indicate the station.

17.426.3 Member Function Documentation

17.426.3.1 _check()

```
bool Leave::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.426.3.2 _createEditableDataDefinitions()

```
void Leave::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.426.3.3 _createInternalStatisticReporters()

```
void Leave::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.426.3.4 _loadInstance()

```
bool Leave::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.426.3.5 _onDispatchEvent()

```
void Leave::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.426.3.6 _saveInstance()

```
void Leave::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.426.3.7 show()

```
std::string Leave::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/MaterialHandling/Leave.h
- plugins/components/MaterialHandling/Leave.cpp

17.427 LicenceManager Class Reference

```
#include <kernel/simulator/LicenceManager.h>
```

Public Member Functions

- **LicenceManager** (Simulator *simulator)
- const std::string **showLicence** () const
- const std::string **showLimits** () const
- const std::string **showActivationCode** () const
- bool **lookforActivationCode** ()
- bool **insertActivationCode** ()
- void **removeActivationCode** ()
- unsigned int **getModelComponentsLimit** ()
- unsigned int **getModelDatasLimit** ()
- unsigned int **getEntityLimit** ()
- unsigned int **getHostsLimit** ()
- unsigned int **getThreadsLimit** ()

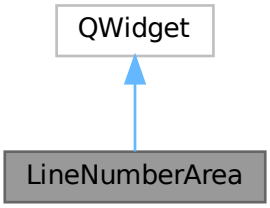
17.427.1 Detailed Description

LicenceManager just prints a licence agreement and checks for some resource limits.
The documentation for this class was generated from the following files:

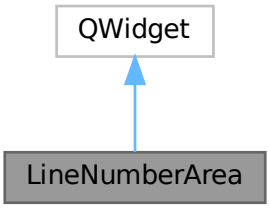
- kernel/simulator/LicenceManager.h
- kernel/simulator/LicenceManager.cpp

17.428 LineNumberArea Class Reference

Inheritance diagram for LineNumberArea:



Collaboration diagram for LineNumberArea:



Public Member Functions

- **LineNumberArea** (CodeEditor *editor)
- QSize **sizeHint** () const override

Protected Member Functions

- void **paintEvent** (QPaintEvent *event) override

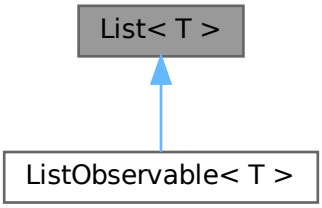
The documentation for this class was generated from the following file:

- applications/gui/genesys/codeeditor/LineNumberArea.h

17.429 List< T > Class Template Reference

Lightweight wrapper around `std::list` with convenience helpers used by the kernel.
`#include <kernel/util/List.h>`

Inheritance diagram for List< T >:



Public Types

- using **CompFunct** = std::function< bool(const T, const T) >

Public Member Functions

- **List** ()

  Creates an empty list and positions the internal iterator at the beginning.
- **List** (const List< T > &origin)

  Creates a copy of the source list.
- List< T > & **operator=** (const List< T > &origin)
- unsigned int **size** ()

  Returns the number of stored elements.
- bool **empty** ()

  Indicates whether the list is empty.
- void **clear** ()

  Removes all elements from the list.
- void **pop_front** ()

  Removes the first element from the list.
- template<class Compare >
  void **sort** (Compare comp)

  Sorts elements using the provided comparator.
- std::list< T > * **list** () const

  Returns direct access to the encapsulated `std::list` structure.
- T **create** ()

  Creates a new default element of type `T`.
- template<typename U >
  T **create** (U arg)

  Creates a new `T` element using a construction argument.
- std::string **show** ()

  Generates a textual representation of elements for debugging.
- std::list< T >::iterator **find** (T element)

  Searches for an element and returns an iterator to it (or `end()`).
- void **insert** (T element)

  Inserts an element preserving the current ordering policy.
- void **remove** (T element)

  Removes all occurrences of the provided element.

- void **setAtRank** (unsigned int rank, T element)

  Replaces (or appends) an element at a specific rank.
- T **getAtRank** (unsigned int rank)

  Returns the element at the provided rank.
- T **next** ()

  Advances the internal iterator and returns the next element.
- T **front** ()

  Moves to the beginning and returns the first element.
- T **last** ()

  Moves to the end and returns the last element.
- T **previous** ()

  Moves the internal iterator backward and returns the previous element.
- T **current** ()

  Returns the element at the current internal iterator position.
- void **setSortFunc** (CompFunct _sortFunc)

  Sets the comparison function used for ordered insertions.

Protected Attributes

- std::list< T > * **_list**
- CompFunct _sortFunc
- std::list< T >::iterator **_it**

  Default function: insert at the end of the list.

17.429.1 Detailed Description

template<typename T>
class List< T >

Lightweight wrapper around `std::list` with convenience helpers used by the kernel.
The class centralizes insertion/removal, rank-based access and iterator-like navigation patterns heavily used in simulator internals and plugin infrastructure.

17.429.2 Member Data Documentation

17.429.2.1 _sortFunc

```
template<typename T >
CompFunct List< T >::_sortFunc  [protected]
```

Initial value:

```
{[](const T, const T) {
            return false;
        }}
```

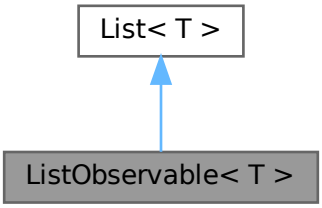
The documentation for this class was generated from the following files:

- applications/gui/genesys/services/GraphicalModelBuilder.h
- kernel/util/List.h

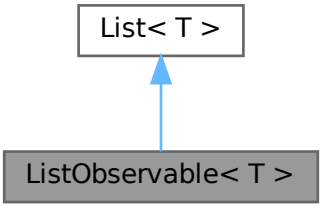
17.430 ListObservable< T > Class Template Reference

Observable-oriented list variant used by kernel structures.
`#include <kernel/util/ListObservable.h>`

Inheritance diagram for ListObservable< T >:



Collaboration diagram for ListObservable< T >:



Public Types

- using **CompFunct** = std::function< bool(const T, const T) >

Public Types inherited from List< T >

- using **CompFunct** = std::function< bool(const T, const T) >

Public Member Functions

- **ListObservable** ()

  Initializes an empty observable list.
- void **clear** ()

  Removes all elements from the observable list.
- void **pop_front** ()

  Removes the first element from the observable list.
- void **insert** (T element)

  Inserts an element respecting the configured ordering.
- void **remove** (T element)

  Removes an element from the observable list.
- void **setAtRank** (unsigned int rank, T element)

  Sets the element at a specific list index.

Public Member Functions inherited from List< T >

- **List** ()

  Creates an empty list and positions the internal iterator at the beginning.
- **List** (const List< T > &origin)

  Creates a copy of the source list.
- List< T > & **operator=** (const List< T > &origin)
- unsigned int **size** ()

  Returns the number of stored elements.
- bool **empty** ()

  Indicates whether the list is empty.
- void **clear** ()

  Removes all elements from the list.
- void **pop_front** ()

  Removes the first element from the list.
- template<class Compare >
  void **sort** (Compare comp)

  Sorts elements using the provided comparator.
- std::list< T > * **list** () const

  Returns direct access to the encapsulated `std::list` structure.
- T **create** ()

  Creates a new default element of type `T`.
- template<typename U >
  T **create** (U arg)

  Creates a new `T` element using a construction argument.
- std::string **show** ()

  Generates a textual representation of elements for debugging.
- std::list< T >::iterator **find** (T element)

  Searches for an element and returns an iterator to it (or `end()`).
- void **insert** (T element)

  Inserts an element preserving the current ordering policy.
- void **remove** (T element)

  Removes all occurrences of the provided element.
- void **setAtRank** (unsigned int rank, T element)

  Replaces (or appends) an element at a specific rank.
- T **getAtRank** (unsigned int rank)

  Returns the element at the provided rank.
- T **next** ()

  Advances the internal iterator and returns the next element.
- T **front** ()

  Moves to the beginning and returns the first element.
- T **last** ()

  Moves to the end and returns the last element.
- T **previous** ()

  Moves the internal iterator backward and returns the previous element.
- T **current** ()

  Returns the element at the current internal iterator position.
- void **setSortFunc** (CompFunct _sortFunc)

  Sets the comparison function used for ordered insertions.

Additional Inherited Members

Protected Attributes inherited from List< T >

- std::list< T > * **_list**
- CompFunct _sortFunc
- std::list< T >::iterator **_it**

  Default function: insert at the end of the list.

17.430.1 Detailed Description

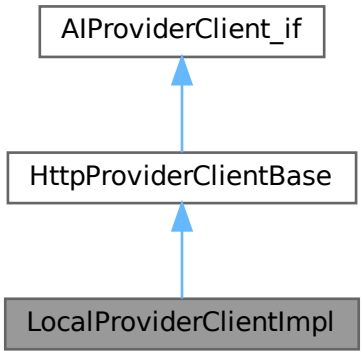
template<typename T>
class ListObservable< T >

Observable-oriented list variant used by kernel structures.
This template specializes `List` behavior for contexts where list mutations are relevant to external observers/managers, while preserving familiar list helpers.
The documentation for this class was generated from the following file:

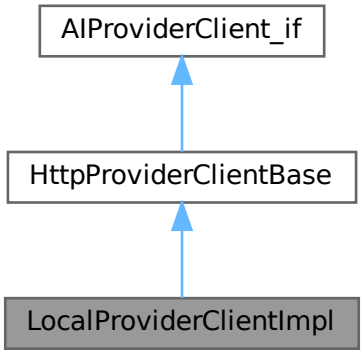
- kernel/util/ListObservable.h

17.431 LocalProviderClientImpl Class Reference

AI provider client for local model servers (Ollama, LM Studio, etc.).
`#include <tools/AIAssistant/LocalProviderClientImpl.h>`
Inheritance diagram for LocalProviderClientImpl:



Collaboration diagram for LocalProviderClientImpl:



Public Member Functions

- **LocalProviderClientImpl** ()

  Constructs the client targeting the default Ollama endpoint.
- std::string getProviderName () const override

  Returns a human-readable provider name (e.g. "OpenAI", "Anthropic").
- bool isAvailable () const override

  Returns `true` when a base URL is configured.

Public Member Functions inherited from HttpProviderClientBase

- **HttpProviderClientBase** (const HttpProviderClientBase &)=delete
- HttpProviderClientBase & **operator=** (const HttpProviderClientBase &)=delete
- void setBaseUrl (const std::string &url) override

  Sets the base URL for the provider API endpoint.
- std::string getBaseUrl () const override

  Returns the currently configured base URL.
- void setTimeoutSeconds (unsigned int seconds) override

  Sets the HTTP request timeout in seconds.
- unsigned int getTimeoutSeconds () const override

  Returns the configured HTTP request timeout.
- void setApiKey (const std::string &apiKey) override

  Stores the API key in memory.
- void clearApiKey () override

  Overwrites (best-effort) and clears the in-memory API key.
- bool hasApiKey () const override

  Returns `true` when a non-empty API key is held.
- ProviderResponse send (const ProviderRequest &request) override

  Executes the HTTP request via curl and returns the provider response.

Protected Member Functions

- std::string _buildRequestBody (const ProviderRequest &request) const override

  Serialises `request` to the provider's JSON wire format.
- ProviderResponse _parseResponseBody (const std::string &rawBody, int httpStatus) const override

  Deserialises `rawBody` (the full curl stdout) into a ProviderResponse.
- std::string _authorizationHeader () const override

  Returns the Authorization header when an API key is set, empty otherwise.

Protected Member Functions inherited from HttpProviderClientBase

- virtual std::vector< std::string > _extraRequestHeaders () const

  Returns additional HTTP headers to send with every request.

Additional Inherited Members

Static Public Member Functions inherited from HttpProviderClientBase

- static std::string **_escapeJsonString** (const std::string &s)

  Escapes a plain string for safe embedding inside a JSON string value.
- static std::string **_extractJsonString** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON string field by key. Returns an empty string when not found or value is not a quoted string.
- static int **_extractJsonInt** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON integer field by key. Returns 0 when not found.
- static double **_extractJsonDouble** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON number field (int or float) by key. Returns 0.0 when not found.
- static std::string **_buildMessagesArray** (const std::vector< ChatMessage > &messages)

  Serialises a vector of ChatMessage objects to a JSON array string.
- static std::string **_extractFirstArrayElement** (const std::string &json, const std::string &arrayKey)

  Extracts the raw substring of the first element of a named JSON array. Returns an empty string when not found.

Protected Attributes inherited from HttpProviderClientBase

- std::string **_baseUrl**
- unsigned int **_timeoutSeconds** = 30
- std::string _apiKey

  In-memory API key.

17.431.1 Detailed Description

AI provider client for local model servers (Ollama, LM Studio, etc.).
Targets any OpenAI-compatible chat completions endpoint reachable locally. Default base URL: `http://localhost:11434/v1/chat/completions` (Ollama default).
Differences from OpenAIProviderClientImpl:

- No API key is required; isAvailable() returns true when the base URL is set.
- When a non-empty API key IS set (some local servers require a token), it is sent as `Authorization: Bearer <key>`. When no key is set, the Authorization header is omitted.
- Extended reasoning (highReasoningMode) adds `"stream": false` only; no reasoning_effort parameter (not supported by most local servers).
- The JSON request/response format otherwise mirrors OpenAI chat completions.

17.431.1.1 Tested server compatibility

- Ollama (≥0.1.29): set the base URL to `http://localhost:11434/v1/chat/completions`.
- LM Studio local server: set base URL to `http://localhost:1234/v1/chat/completions`.
- llama.cpp server: set base URL to `http://localhost:8080/v1/chat/completions`.

17.431.2 Member Function Documentation

17.431.2.1 _authorizationHeader()

```
std::string LocalProviderClientImpl::_authorizationHeader ( ) const  [override], [protected],
[virtual]
```

Returns the Authorization header when an API key is set, empty otherwise.
Implements HttpProviderClientBase.

17.431.2.2 _buildRequestBody()

```
std::string LocalProviderClientImpl::_buildRequestBody (
            const ProviderRequest & request ) const  [override], [protected], [virtual]
```

Serialises `request` to the provider's JSON wire format.
Implements HttpProviderClientBase.

17.431.2.3 _parseResponseBody()

```
ProviderResponse LocalProviderClientImpl::_parseResponseBody (
            const std::string & rawBody,
            int httpStatus ) const  [override], [protected], [virtual]
```

Deserialises `rawBody` (the full curl stdout) into a ProviderResponse.

Parameters

rawBody	Complete HTTP response body as received from curl.
httpStatus	HTTP status code captured from curl exit / response line, or 0 when unavailable.

Implements HttpProviderClientBase.

17.431.2.4 getProviderName()

```
std::string LocalProviderClientImpl::getProviderName ( ) const  [override], [virtual]
```

Returns a human-readable provider name (e.g. "OpenAI", "Anthropic").
Implements AIProviderClient_if.

17.431.2.5 isAvailable()

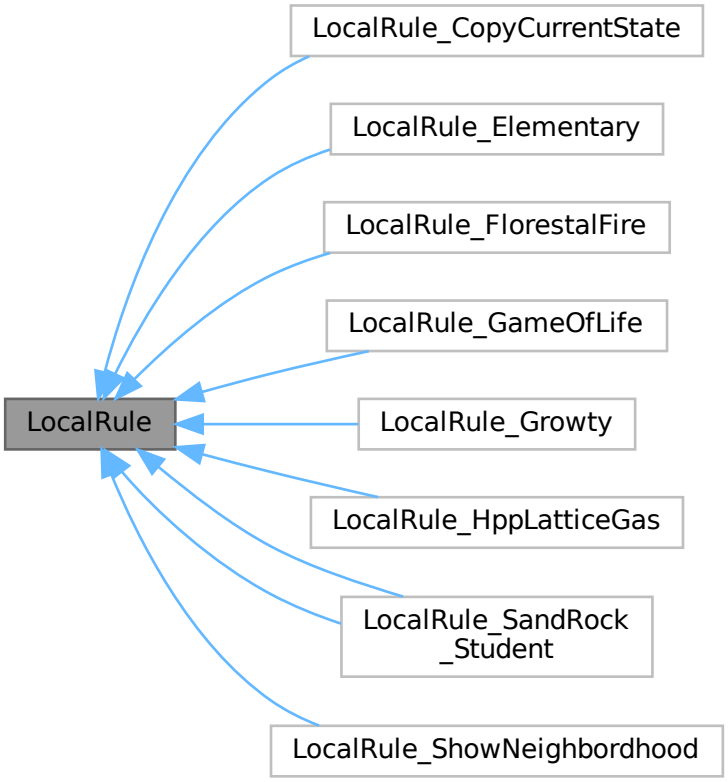
```
bool LocalProviderClientImpl::isAvailable ( ) const  [override], [virtual]
```

Returns `true` when a base URL is configured.
Unlike cloud providers, a local server does not require an API key.
Reimplemented from HttpProviderClientBase.
The documentation for this class was generated from the following files:

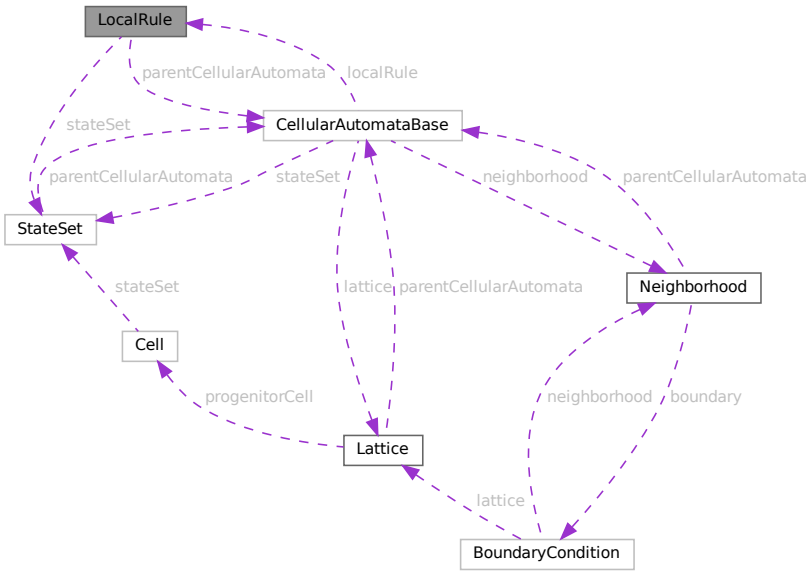
- tools/AIAssistant/LocalProviderClientImpl.h
- tools/AIAssistant/LocalProviderClientImpl.cpp

17.432 LocalRule Class Reference

Inheritance diagram for LocalRule:



Collaboration diagram for LocalRule:



Public Member Functions

- **LocalRule** (CellularAutomataBase ∗parentCellularAutomata, StateSet ∗stateSet=nullptr)
- **LocalRule** (const LocalRule &orig)
- virtual void applyRule (Cell ∗cell)=0

  Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
- void **setStateSet** (StateSet ∗stateSet)

Protected Attributes

- CellularAutomataBase ∗ **parentCellularAutomata**
- StateSet ∗ **stateSet**

17.432.1 Member Function Documentation

17.432.1.1 applyRule()

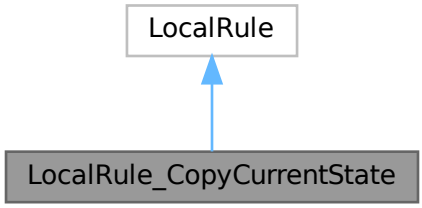
```
virtual void LocalRule::applyRule (
            Cell * cell )  [pure virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
Implemented in LocalRule_ShowNeighbordhood, LocalRule_CopyCurrentState, LocalRule_Elementary, LocalRule_FlorestalFire, LocalRule_GameOfLife, LocalRule_Growty, LocalRule_HppLatticeGas, LocalRule_SandRock_Student, and LocalRule_SandRock_Student.
The documentation for this class was generated from the following file:

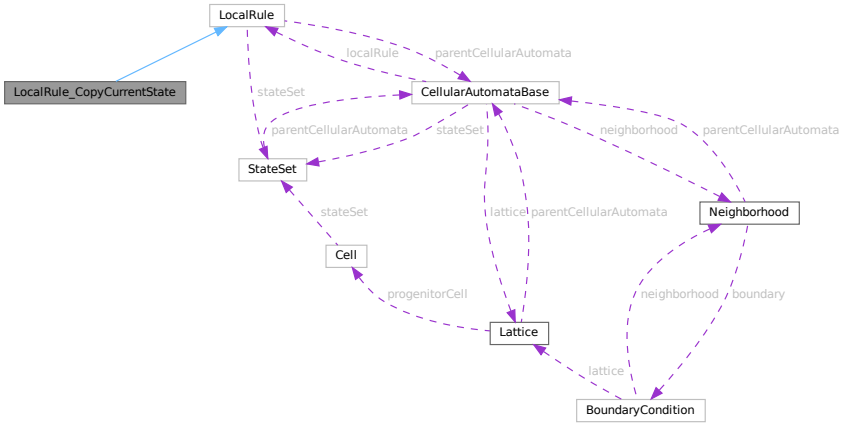
- plugins/components/ModalModel/CellularAutomata/LocalRule.h

17.433 LocalRule_CopyCurrentState Class Reference

Inheritance diagram for LocalRule_CopyCurrentState:



Collaboration diagram for LocalRule_CopyCurrentState:



Public Member Functions

- **LocalRule_CopyCurrentState** (CellularAutomataBase ∗parentCellularAutomata)
- void applyRule (Cell ∗cell) override

  Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.

Public Member Functions inherited from LocalRule

- **LocalRule** (CellularAutomataBase ∗parentCellularAutomata, StateSet ∗stateSet=nullptr)
- **LocalRule** (const LocalRule &orig)
- void **setStateSet** (StateSet ∗stateSet)

Additional Inherited Members

Protected Attributes inherited from LocalRule

- CellularAutomataBase ∗ **parentCellularAutomata**
- StateSet ∗ **stateSet**

17.433.1 Member Function Documentation

17.433.1.1 applyRule()

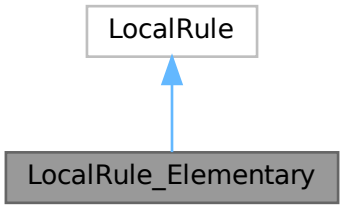
```
void LocalRule_CopyCurrentState::applyRule (
            Cell * cell )  [inline], [override], [virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
Implements LocalRule.
The documentation for this class was generated from the following file:

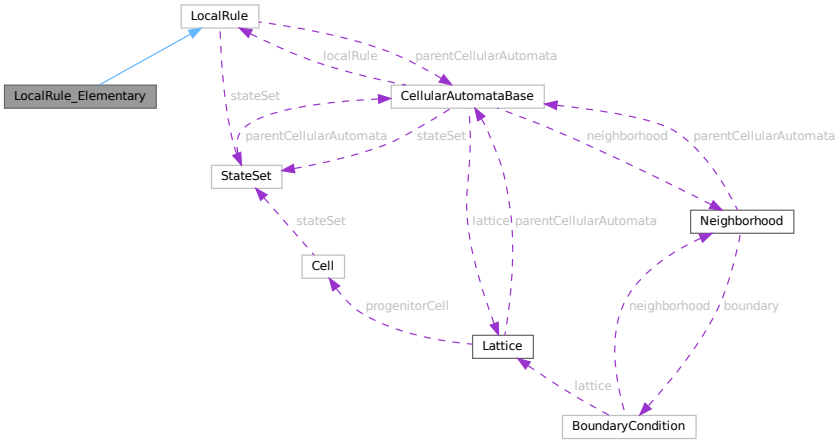
- applications/gui/genesys/cellular_automata/CellularAutomataDemoBuilder.h

17.434 LocalRule_Elementary Class Reference

Inheritance diagram for LocalRule_Elementary:



Collaboration diagram for LocalRule_Elementary:



Public Member Functions

- **LocalRule_Elementary** (CellularAutomataBase ∗parentCellularAutomata, uint8_t ruleNumber=30, StateSet ∗stateSet=nullptr)
- **LocalRule_Elementary** (const LocalRule_Elementary &orig)
- virtual void applyRule (Cell ∗cell) override

  Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.

- uint8_t **getRuleNumber** () const
- void **setRuleNumber** (const uint8_t ruleNumber)

Public Member Functions inherited from LocalRule

- **LocalRule** (CellularAutomataBase *parentCellularAutomata, StateSet *stateSet=nullptr)
- **LocalRule** (const LocalRule &orig)
- void **setStateSet** (StateSet *stateSet)

Additional Inherited Members

Protected Attributes inherited from LocalRule

- CellularAutomataBase * **parentCellularAutomata**
- StateSet * **stateSet**

17.434.1 Member Function Documentation

17.434.1.1 applyRule()

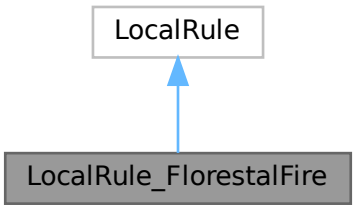
```
virtual void LocalRule_Elementary::applyRule (
            Cell * cell )  [inline], [override], [virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
Implements LocalRule.
The documentation for this class was generated from the following file:

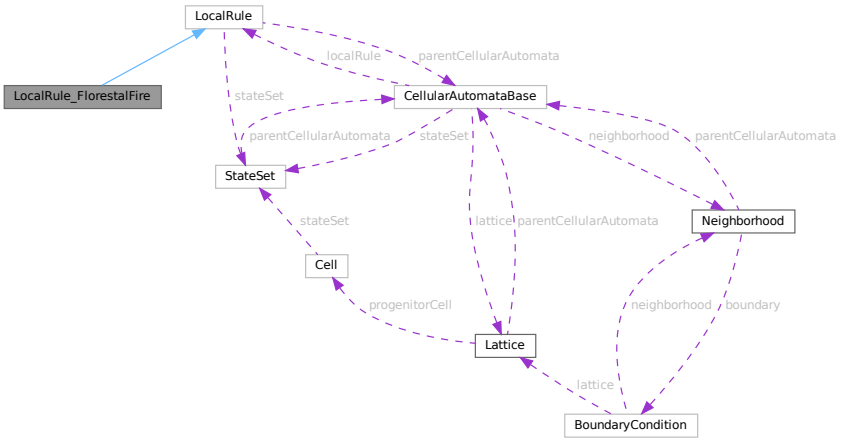
- plugins/components/ModalModel/CellularAutomata/LocalRule_Elementary.h

17.435 LocalRule_FlorestalFire Class Reference

Inheritance diagram for LocalRule_FlorestalFire:



Collaboration diagram for LocalRule_FlorestalFire:



Public Member Functions

- **LocalRule_FlorestalFire** (CellularAutomataBase ∗parentCellularAutomata)
- **LocalRule_FlorestalFire** (const LocalRule_FlorestalFire &orig)
- virtual void applyRule (Cell ∗cell) override

  Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.

Public Member Functions inherited from LocalRule

- **LocalRule** (CellularAutomataBase ∗parentCellularAutomata, StateSet ∗stateSet=nullptr)
- **LocalRule** (const LocalRule &orig)
- void **setStateSet** (StateSet ∗stateSet)

Additional Inherited Members

Protected Attributes inherited from LocalRule

- CellularAutomataBase ∗ **parentCellularAutomata**
- StateSet ∗ **stateSet**

17.435.1 Member Function Documentation

17.435.1.1 applyRule()

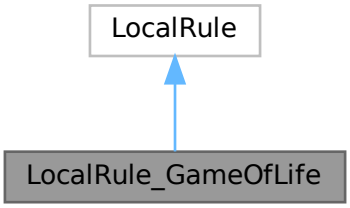
```
virtual void LocalRule_FlorestalFire::applyRule (
            Cell * cell )  [inline], [override], [virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
Implements LocalRule.
The documentation for this class was generated from the following file:

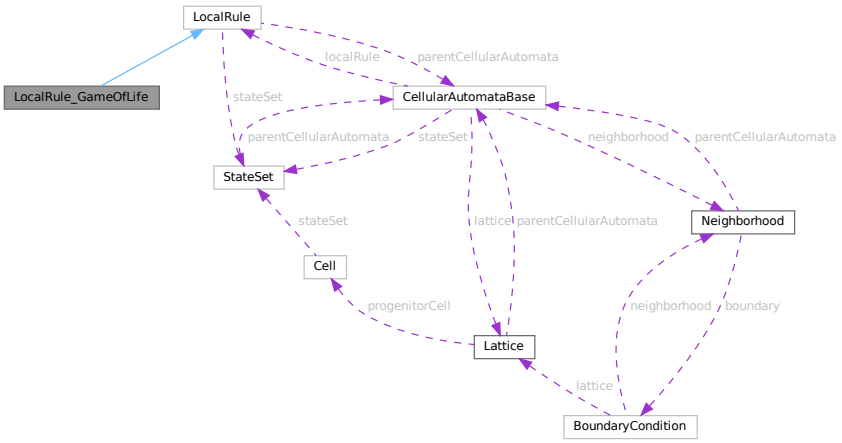
- plugins/components/ModalModel/CellularAutomata/LocalRule_FlorestalFire.h

17.436 LocalRule_GameOfLife Class Reference

Inheritance diagram for LocalRule_GameOfLife:



Collaboration diagram for LocalRule_GameOfLife:



Public Member Functions

- **LocalRule_GameOfLife** (CellularAutomataBase ∗parentCellularAutomata)
- **LocalRule_GameOfLife** (const LocalRule_GameOfLife &orig)
- virtual void applyRule (Cell ∗cell) override

  Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.

Public Member Functions inherited from LocalRule

- **LocalRule** (CellularAutomataBase ∗parentCellularAutomata, StateSet ∗stateSet=nullptr)
- **LocalRule** (const LocalRule &orig)
- void **setStateSet** (StateSet ∗stateSet)

Additional Inherited Members

Protected Attributes inherited from LocalRule

- CellularAutomataBase ∗ **parentCellularAutomata**
- StateSet ∗ **stateSet**

17.436.1 Member Function Documentation

17.436.1.1 applyRule()

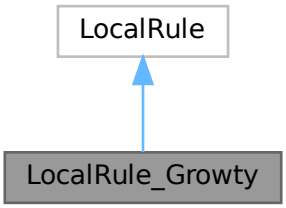
```
virtual void LocalRule_GameOfLife::applyRule (
            Cell * cell )  [inline], [override], [virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
Implements LocalRule.
The documentation for this class was generated from the following file:

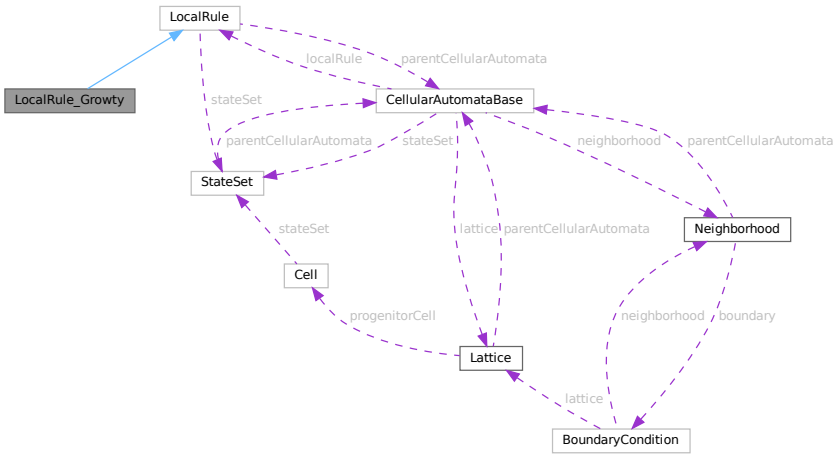
- plugins/components/ModalModel/CellularAutomata/LocalRule_GameOfLife.h

17.437 LocalRule_Growty Class Reference

Inheritance diagram for LocalRule_Growty:



Collaboration diagram for LocalRule_Growty:



Public Member Functions

- **LocalRule_Growty** (CellularAutomataBase ∗parentCellularAutomata)
- **LocalRule_Growty** (const LocalRule_Growty &orig)
- virtual void applyRule (Cell ∗cell) override

  Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.

Public Member Functions inherited from LocalRule

- **LocalRule** (CellularAutomataBase *parentCellularAutomata, StateSet *stateSet=nullptr)
- **LocalRule** (const LocalRule &orig)
- void **setStateSet** (StateSet *stateSet)

Additional Inherited Members

Protected Attributes inherited from LocalRule

- CellularAutomataBase * **parentCellularAutomata**
- StateSet * **stateSet**

17.437.1 Member Function Documentation

17.437.1.1 applyRule()

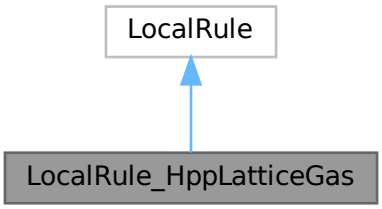
```
virtual void LocalRule_Growty::applyRule (
            Cell * cell )  [inline], [override], [virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
Implements LocalRule.
The documentation for this class was generated from the following file:

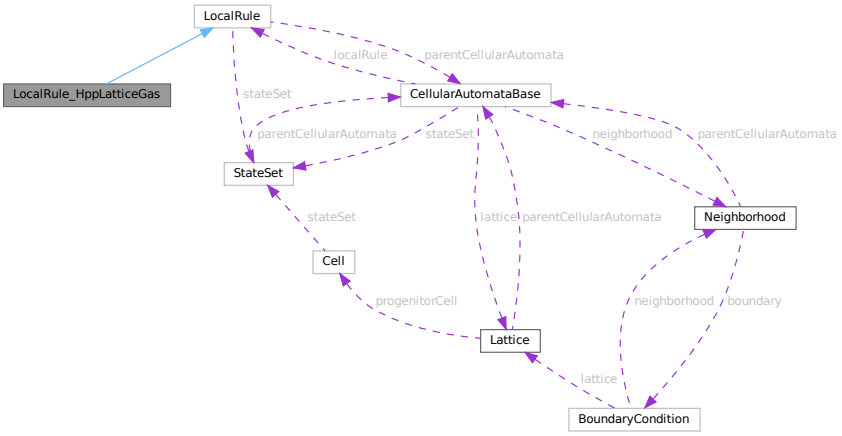
- plugins/components/ModalModel/CellularAutomata/LocalRule_Growty.h

17.438 LocalRule_HppLatticeGas Class Reference

Inheritance diagram for LocalRule_HppLatticeGas:



Collaboration diagram for LocalRule_HppLatticeGas:



Public Member Functions

- **LocalRule_HppLatticeGas** (CellularAutomataBase ∗parentCellularAutomata)
- void applyRule (Cell ∗cell) override

  Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.

Public Member Functions inherited from LocalRule

- **LocalRule** (CellularAutomataBase ∗parentCellularAutomata, StateSet ∗stateSet=nullptr)
- **LocalRule** (const LocalRule &orig)
- void **setStateSet** (StateSet ∗stateSet)

Additional Inherited Members

Protected Attributes inherited from LocalRule

- CellularAutomataBase ∗ **parentCellularAutomata**
- StateSet ∗ **stateSet**

17.438.1 Member Function Documentation

17.438.1.1 applyRule()

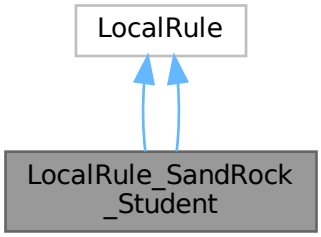
```
void LocalRule_HppLatticeGas::applyRule (
            Cell * cell )  [inline], [override], [virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
Implements LocalRule.
The documentation for this class was generated from the following file:

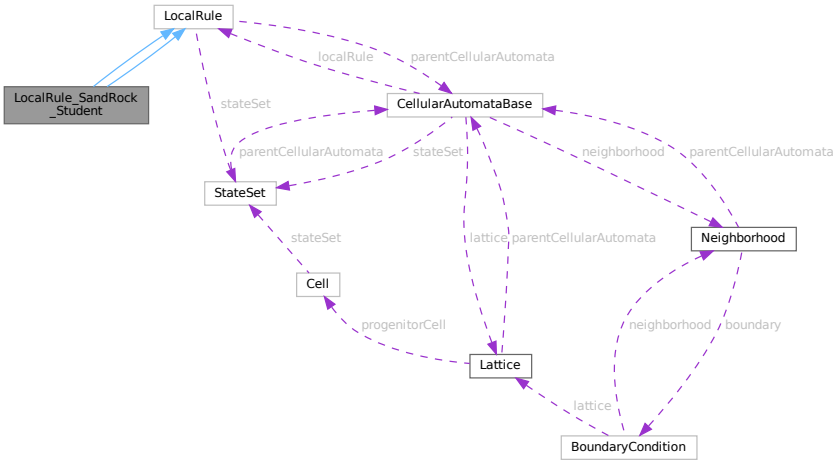
- plugins/components/ModalModel/CellularAutomata/LocalRule_HppLatticeGas.h

17.439 LocalRule_SandRock_Student Class Reference

Inheritance diagram for LocalRule_SandRock_Student:



Collaboration diagram for LocalRule_SandRock_Student:



Public Member Functions

- **LocalRule_SandRock_Student** (CellularAutomataBase ∗parentCellularAutomata)
- **LocalRule_SandRock_Student** (const LocalRule_SandRock_Student &orig)
- virtual void applyRule (Cell ∗cell) override

  Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
- **LocalRule_SandRock_Student** (CellularAutomataBase ∗parentCellularAutomata)
- **LocalRule_SandRock_Student** (const LocalRule_SandRock_Student &orig)
- virtual void applyRule (Cell ∗cell) override

  Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.

Public Member Functions inherited from LocalRule

- **LocalRule** (CellularAutomataBase ∗parentCellularAutomata, StateSet ∗stateSet=nullptr)
- **LocalRule** (const LocalRule &orig)
- void **setStateSet** (StateSet ∗stateSet)

Static Public Attributes

- static constexpr long **EMPTY** = 0
- static constexpr long **SAND** = 1
- static constexpr long **ROCK** = 2

Additional Inherited Members

Protected Attributes inherited from LocalRule

- CellularAutomataBase * **parentCellularAutomata**
- StateSet * **stateSet**

17.439.1 Member Function Documentation

17.439.1.1 applyRule() [1/2]

```
virtual void LocalRule_SandRock_Student::applyRule (
            Cell * cell )  [inline], [override], [virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
Implements LocalRule.

17.439.1.2 applyRule() [2/2]

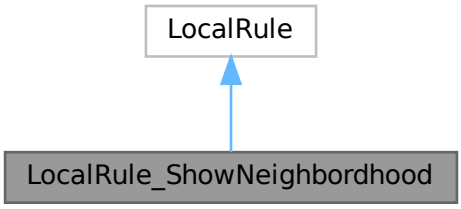
```
virtual void LocalRule_SandRock_Student::applyRule (
            Cell * cell )  [inline], [override], [virtual]
```

Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
Implements LocalRule.
The documentation for this class was generated from the following files:

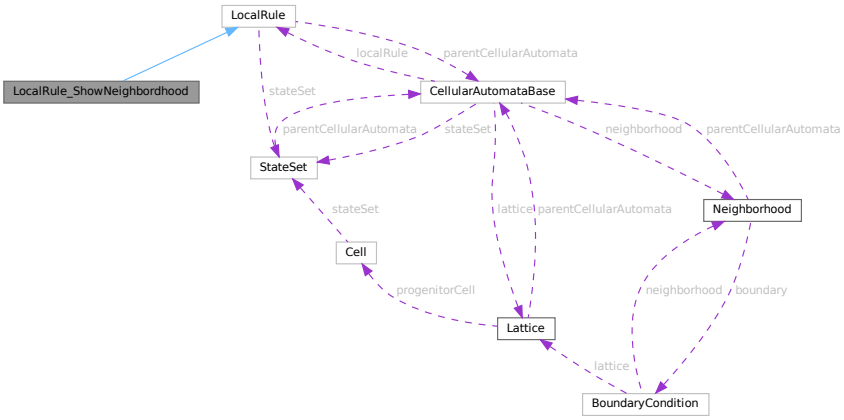
- plugins/components/ModalModel/CellularAutomata/LocalRule_SandRock.h
- plugins/components/ModalModel/CellularAutomata/LocalRule_SandRock_Student.h

17.440 LocalRule_ShowNeighbordhood Class Reference

Inheritance diagram for LocalRule_ShowNeighbordhood:



Collaboration diagram for LocalRule_ShowNeighbordhood:



Public Member Functions

- **LocalRule_ShowNeighbordhood** (CellularAutomataBase ∗parentCellularAutomata)
- **LocalRule_ShowNeighbordhood** (const LocalRule_ShowNeighbordhood &orig)
- virtual void applyRule (Cell ∗cell)

  Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.

Public Member Functions inherited from LocalRule

- **LocalRule** (CellularAutomataBase ∗parentCellularAutomata, StateSet ∗stateSet=nullptr)
- **LocalRule** (const LocalRule &orig)
- void **setStateSet** (StateSet ∗stateSet)

Additional Inherited Members

Protected Attributes inherited from LocalRule

- CellularAutomataBase ∗ **parentCellularAutomata**
- StateSet ∗ **stateSet**

17.440.1 Member Function Documentation

17.440.1.1 applyRule()

```
virtual void LocalRule_ShowNeighbordhood::applyRule (
            Cell * cell )  [inline], [virtual]
```

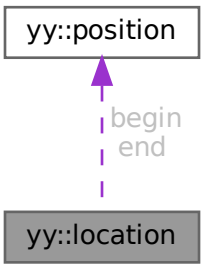
Pure virtual method that has to be overiden by derived class, responsable for applying the local rule to a cell.
Implements LocalRule.
The documentation for this class was generated from the following file:

- plugins/components/ModalModel/CellularAutomata/LocalRule_ShowNeighbordhood.h

17.441 yy::location Class Reference

Two points in a source file.
`#include <parser/location.hh>`

Collaboration diagram for yy::location:



Public Types

- typedef position::filename_type **filename_type**

  Type for file name.
- typedef position::counter_type **counter_type**

  Type for line and column numbers.

Public Member Functions

- **location** (const position &b, const position &e)

  Construct a location from b to e.
- **location** (const position &p=position())

  Construct a 0-width location in p.
- **location** (filename_type *f, counter_type l=1, counter_type c=1)

  Construct a 0-width location in f, l, c.
- void **initialize** (filename_type *f=YY_NULLPTR, counter_type l=1, counter_type c=1)

  Initialization.

Line and Column related manipulators

- void **step** ()

  Reset initial location to final location.
- void **columns** (counter_type count=1)

  Extend the current location to the COUNT next columns.
- void **lines** (counter_type count=1)

  Extend the current location to the COUNT next lines.

Public Attributes

- position **begin**

  Beginning of the located region.
- position **end**

  End of the located region.

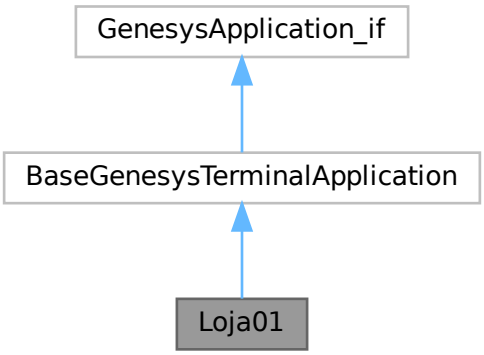
17.441.1 Detailed Description

Two points in a source file.
The documentation for this class was generated from the following file:

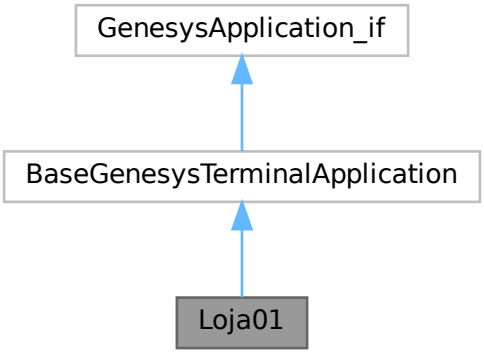
- parser/location.hh

17.442 Loja01 Class Reference

Inheritance diagram for Loja01:



Collaboration diagram for Loja01:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.442.1 Member Function Documentation

17.442.1.1 main()

```
int Loja01::main (
            int argc,
            char ** argv )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

- applications/modelSpecific/teaching/Loja/Loja01.h
- applications/modelSpecific/teaching/Loja/Loja01.cpp

17.443 testing::internal::internal_stream_operator_without_lexical_↩ name_lookup::LookupBlocker Struct Reference

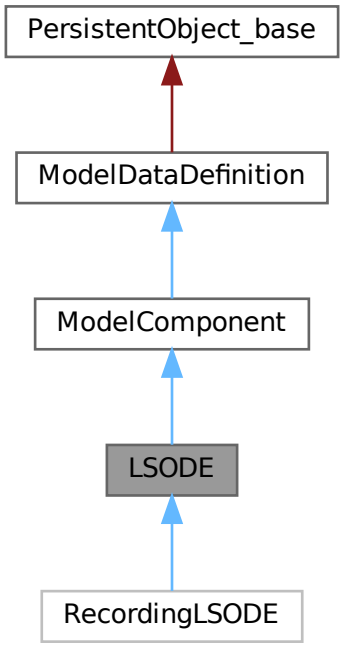
The documentation for this struct was generated from the following file:

- gtest/gtest/gtest-printers.h

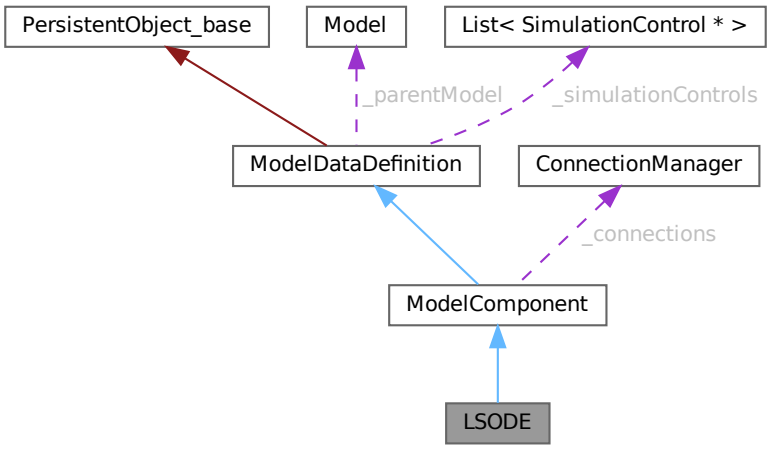
17.444 LSODE Class Reference

```
#include <plugins/components/Continuous/LSODE.h>
```

Inheritance diagram for LSODE:



Collaboration diagram for LSODE:



Public Member Functions

- **LSODE** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods

- void **setTimeVariable** (Variable *_timeVariable)
- Variable * **getTimeVariable** () const
- void **setStep** (double _step)
- double **getStep** () const
- void **setVariable** (Variable *_variables)
- Variable * **getVariable** () const
- List< std::string > * **getDiffEquations** () const
- void **addDiffEquation** (std::string newDiffEquation)
- void **removeDiffEquation** (std::string diffEquation)
- void **setFilename** (std::string filename)
- std::string **getFileName** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.444.1 Detailed Description

This component ...

17.444.2 Member Function Documentation

17.444.2.1 _check()

```
bool LSODE::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.444.2.2 _loadInstance()

```
bool LSODE::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.444.2.3 _onDispatchEvent()

```
void LSODE::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.
Reimplemented in RecordingLSODE.

17.444.2.4 _saveInstance()

```
void LSODE::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.444.2.5 show()

```
std::string LSODE::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

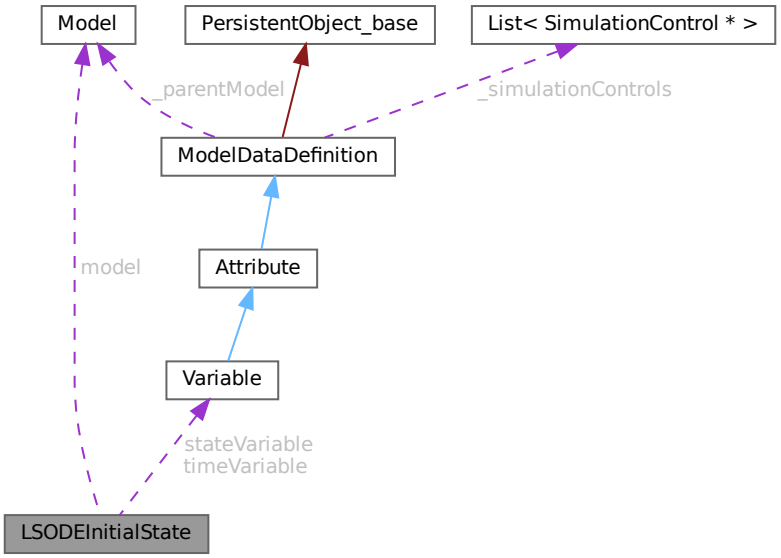
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/Continuous/LSODE.h
- plugins/components/Continuous/LSODE.cpp

17.445 LSODEInitialState Struct Reference

Collaboration diagram for LSODEInitialState:



Public Member Functions

- void **OnReplicationStart** (SimulationEvent ∗)

Public Attributes

- Model ∗ **model**
- Variable ∗ **timeVariable**
- Variable ∗ **stateVariable**

The documentation for this struct was generated from the following file:

- tests/test_lsode.cpp

17.446 LSODESample Struct Reference

Public Attributes

- double **t**
- double **x**
- double **v**
- double **energy**

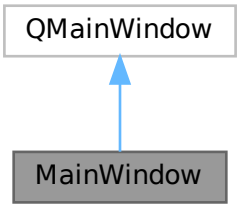
The documentation for this struct was generated from the following file:

- tests/test_lsode.cpp

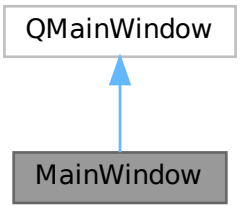
17.447 MainWindow Class Reference

Main Qt window that composes controllers/services and preserves legacy GUI entry points.

`#include <applications/gui/genesys/mainwindow.h>`
Inheritance diagram for MainWindow:



Collaboration diagram for MainWindow:



Public Member Functions

- **MainWindow** (QWidget *parent=nullptr)

  Builds the composition root, wiring controllers/services and legacy Qt surface.
- ~**MainWindow** ()

  Releases UI resources while preserving existing Qt ownership semantics.
- ModelGraphicsScene * **myScene** () const

  Returns the current model graphics scene used by delegated controllers/services.
- bool **graphicalModelHasChanged** () const

  Indicates whether graphical representation has unsaved modifications.
- void **setGraphicalModelHasChanged** (bool graphicalModelHasChanged)

  Updates graphical-change flag used by persistence/action synchronization.
- void **unselectDrawIcons** ()

  Compatibility wrapper that clears currently selected draw tool actions.
- bool **checkSelectedDrawIcons** ()

  Compatibility wrapper that reports whether any draw tool action is selected.
- void **refreshGuiExtensions** ()

  Public refresh hook for graphical extension contributions.
- void **refreshPluginCatalog** ()

  Public refresh hook for plugin catalog tree.

Protected Member Functions

- void **closeEvent** (QCloseEvent ∗event) override

Friends

- class **SimulationEventController**

17.447.1 Detailed Description

Main Qt window that composes controllers/services and preserves legacy GUI entry points.
MainWindow remains the composition root of GenesysQtGUI: it owns the generated UI, wires extracted controllers/services, and exposes the original Qt slot surface expected by actions, widgets, and kernel callback registration. In the refactored architecture it primarily acts as a compatibility façade that delegates operational responsibilities to focused collaborators.
Responsibilities:

- bootstrap UI construction and dependency wiring for controllers/services;
- host compatibility wrappers used by menus/toolbars/scene callbacks;
- coordinate cross-cutting state that still spans multiple controllers.

Boundaries:

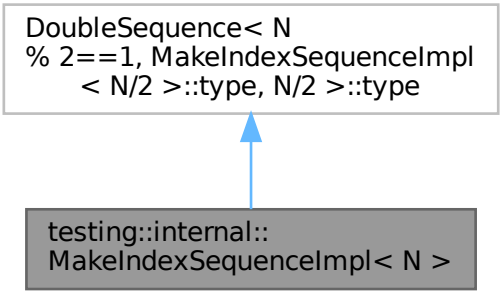
- detailed domain workflows (simulation commands, lifecycle, persistence, scene tools, property editing, trace/event rendering) are delegated to dedicated controller/service classes whenever extraction already exists;
- MainWindow does not replace kernel ownership semantics.

The documentation for this class was generated from the following files:

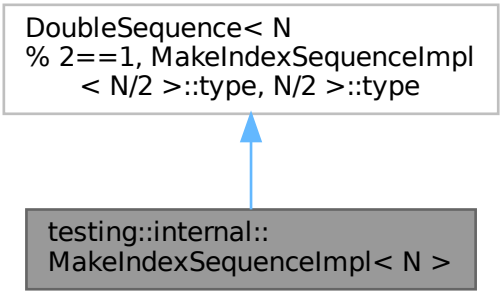
- applications/gui/genesys/mainwindow.h
- applications/gui/genesys/mainwindow.cpp
- applications/gui/genesys/mainwindow_controller.cpp
- applications/gui/genesys/mainwindow_modelrepresentations.cpp
- applications/gui/genesys/mainwindow_scene.cpp
- applications/gui/genesys/mainwindow_simulator.cpp

17.448 testing::internal::MakeIndexSequenceImpl< N > Struct Template Reference

Inheritance diagram for testing::internal::MakeIndexSequenceImpl< N >:



Collaboration diagram for testing::internal::MakeIndexSequenceImpl< N >:

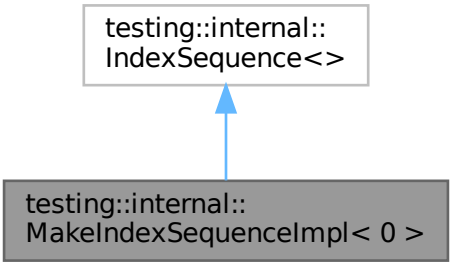


The documentation for this struct was generated from the following file:

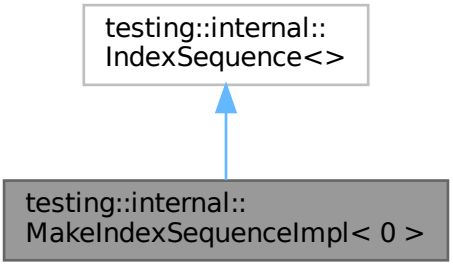
- gtest/gtest/internal/gtest-internal.h

17.449 testing::internal::MakeIndexSequenceImpl< 0 > Struct Reference

Inheritance diagram for testing::internal::MakeIndexSequenceImpl< 0 >:



Collaboration diagram for testing::internal::MakeIndexSequenceImpl< 0 >:



Additional Inherited Members

Public Types inherited from testing::internal::IndexSequence<>

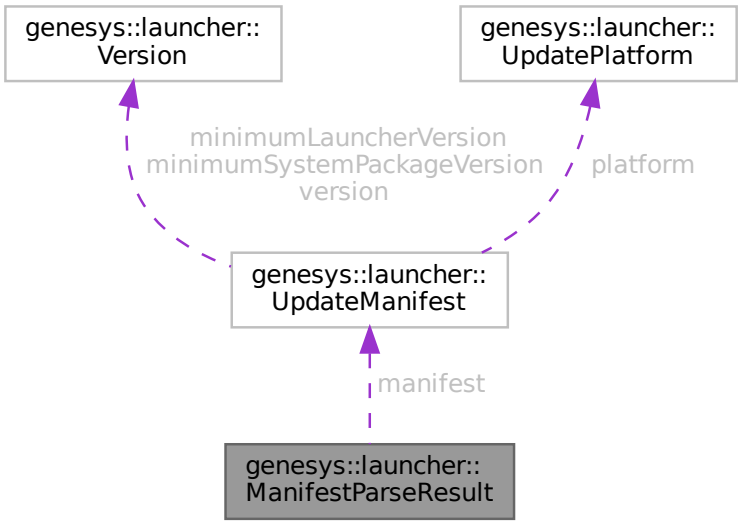
- using **type** = IndexSequence

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.450 genesys::launcher::ManifestParseResult Struct Reference

Collaboration diagram for genesys::launcher::ManifestParseResult:



Public Attributes

- bool **ok** = false

- UpdateManifest **manifest**
- QString **error**

The documentation for this struct was generated from the following file:

- applications/launcher/UpdateManifest.h

17.451 testing::internal::MarkAsIgnored Struct Reference

Public Member Functions

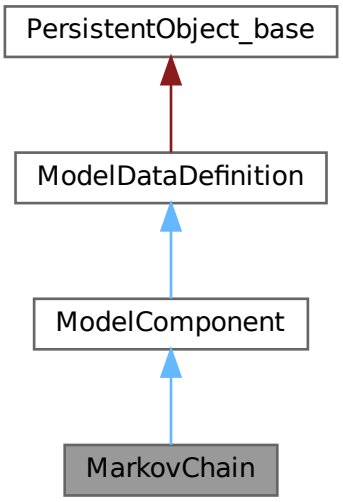
- **MarkAsIgnored** (const char *test_suite)

The documentation for this struct was generated from the following files:

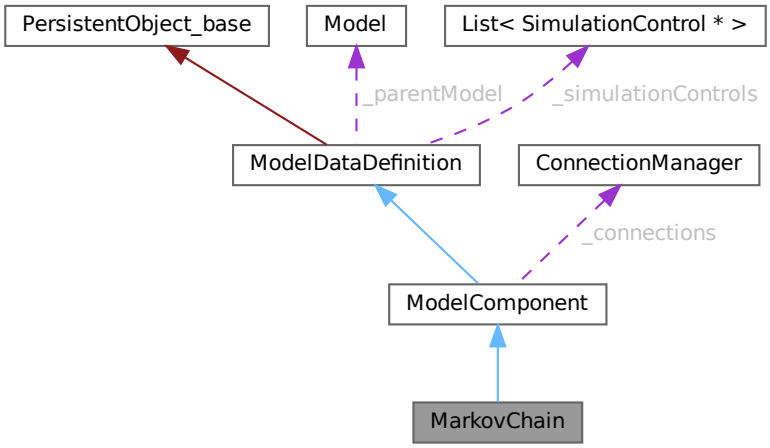
- gtest/gtest/internal/gtest-param-util.h
- gtest/src/gtest.cc

17.452 MarkovChain Class Reference

Inheritance diagram for MarkovChain:



Collaboration diagram for MarkovChain:



Public Member Functions

- **MarkovChain** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setTransitionProbabilityMatrix** (ModelDataDefinition *transitionMatrix)
- ModelDataDefinition * **getTransitionMatrix** () const
- void **setInitialDistribution** (ModelDataDefinition *initialDistribution)
- ModelDataDefinition * **getInitialDistribution** () const
- void **setCurrentState** (ModelDataDefinition *currentState)
- ModelDataDefinition * **getCurrentState** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _initBetweenReplications () override
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.452.1 Member Function Documentation

17.452.1.1 _check()

```
bool MarkovChain::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.452.1.2 _createEditableDataDefinitions()

```
void MarkovChain::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.452.1.3 _initBetweenReplications()

```
void MarkovChain::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.452.1.4 _loadInstance()

```
bool MarkovChain::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.452.1.5 _onDispatchEvent()

```
void MarkovChain::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.452.1.6 _saveInstance()

```
void MarkovChain::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.452.1.7 show()

```
std::string MarkovChain::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

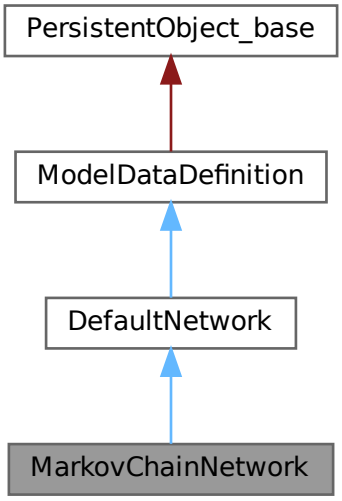
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

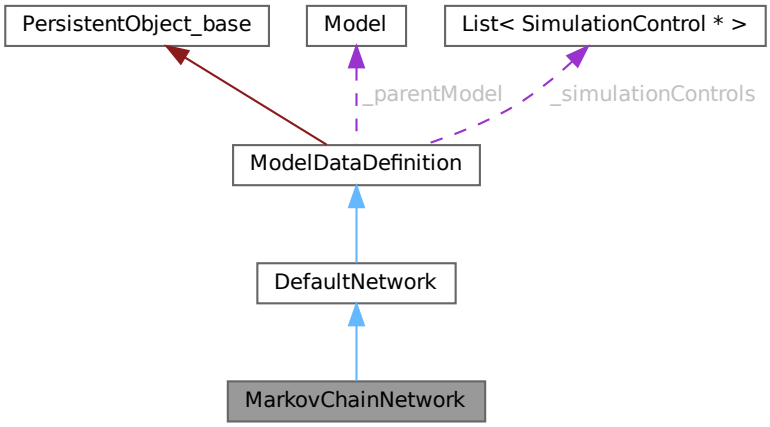
- plugins/components/AnalyticalModeling/MarkovChain.h
- plugins/components/AnalyticalModeling/MarkovChain.cpp

17.453 MarkovChainNetwork Class Reference

Finite time-homogeneous Discrete-Time Markov Chain network.
`#include <plugins/data/ModalModel/MarkovChainNetwork.h>`
Inheritance diagram for MarkovChainNetwork:



Collaboration diagram for MarkovChainNetwork:



Classes

- class MarkovTransition

Public Member Functions

- **MarkovChainNetwork** (Model *model, std::string name="")
- void **addState** (MarkovState *state)
- void **removeState** (MarkovState *state)
- List< MarkovState * > * **getStates** () const
- void **addTransition** (MarkovTransition *transition)
- void **removeTransition** (MarkovTransition *transition)
- List< MarkovTransition * > * **getTransitions** () const
- std::vector< MarkovTransition * > **getOutgoingTransitions** (MarkovState *state) const
- void **setInitialState** (MarkovState *state)
- MarkovState * **getInitialState** () const
- void **setCurrentState** (MarkovState *state)
- MarkovState * **getCurrentState** () const
- unsigned int **getStateIndex** (MarkovState *state) const
- MarkovState * **getStateAt** (unsigned int index) const
- double **getProbabilityTolerance** () const
- void **setProbabilityTolerance** (double tolerance)
- void **resetSampler** ()
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from DefaultNetwork

- **DefaultNetwork** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< DefaultNetwork >())
- unsigned int **addInputPort** (const std::string &portName)

  Declares a new input port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).
- unsigned int **addOutputPort** (const std::string &portName)

Declares a new output port; returns its 0-based index. Rejects a duplicate name (returns the existing index instead of creating a duplicate).

- unsigned int **getNumInputPorts** () const
- unsigned int **getNumOutputPorts** () const
- std::string **getInputPortName** (unsigned int index) const
- std::string **getOutputPortName** (unsigned int index) const
- int **getInputPortIndex** (const std::string &portName) const

  Returns the input port index for `portName`, or -1 when not declared.
- int **getOutputPortIndex** (const std::string &portName) const

  Returns the output port index for `portName`, or -1 when not declared.
- double **getActivationCount** () const

  Number of times activate() has been called since the last replication reset.
- NetworkActivationResult activate (const NetworkActivationFrame &frame)

  Runs exactly one network activation for `frame`.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from DefaultNetwork

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual NetworkActivationResult _activate (const NetworkActivationFrame &frame) override

  Formalism-specific activation behavior.

Protected Member Functions inherited from DefaultNetwork

- virtual void _createInternalStatisticReporters () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.453.1 Detailed Description

Finite time-homogeneous Discrete-Time Markov Chain network.
One activation performs exactly one Markov step from the current state by sampling the outgoing transition probabilities of that state using the GenESyS sampler infrastructure. The current state belongs to the network, not to a process Entity attribute.
Transition ownership: `addTransition()` and persistence reload take exclusive ownership of heap-allocated `MarkovTransition` objects and delete them from the network destructor / reload / remove paths.

17.453.2 Member Function Documentation

17.453.2.1 _activate()

```
NetworkActivationResult MarkovChainNetwork::_activate (
            const NetworkActivationFrame & frame )  [override], [protected], [virtual]
```

Formalism-specific activation behavior.
The base implementation performs no state change and returns a result sized to `getNumOutputPorts()` with every output absent. Specializations override this to implement EFSM transition firing, one DTMC step, one CPN firing step, etc. Must not receive or depend on `Entity*` (architecture reference §8.4, §19).
Reimplemented from DefaultNetwork.

17.453.2.2 _check()

```
bool MarkovChainNetwork::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from DefaultNetwork.

17.453.2.3 _initBetweenReplications()

```
void MarkovChainNetwork::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from DefaultNetwork.

17.453.2.4 _loadInstance()

```
bool MarkovChainNetwork::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from DefaultNetwork.

17.453.2.5 _saveInstance()

```
void MarkovChainNetwork::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from DefaultNetwork.

17.453.2.6 show()

`std::string MarkovChainNetwork::show ( ) [override], [virtual]`
Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

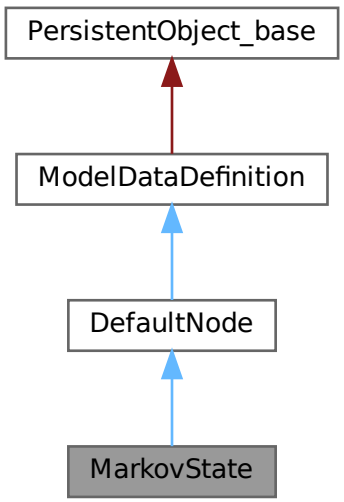
Human-readable summary used by traces and debug output.

Reimplemented from DefaultNetwork.
The documentation for this class was generated from the following files:

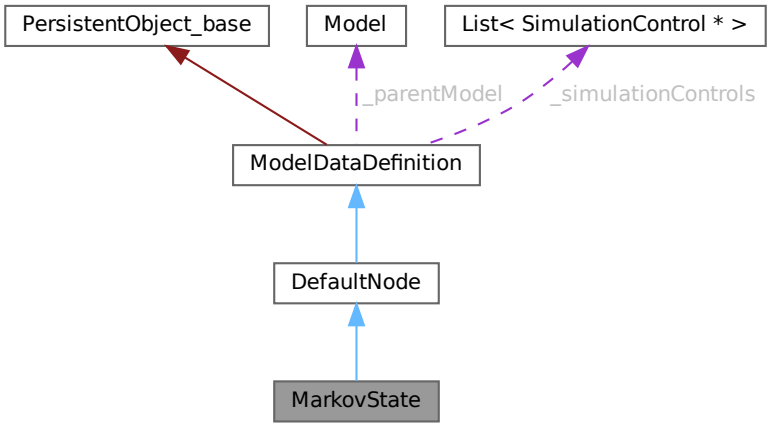
- plugins/data/ModalModel/MarkovChainNetwork.h
- plugins/components/ModalModel/MarkovChainNetwork.cpp

17.454 MarkovState Class Reference

State node used by MarkovChainNetwork DTMC models.
`#include <plugins/data/ModalModel/MarkovState.h>`
Inheritance diagram for MarkovState:



Collaboration diagram for MarkovState:



Public Member Functions

- **MarkovState** (Model ∗model, std::string name="")

Public Member Functions inherited from DefaultNode

- DefaultNode (Model ∗model, std::string componentTypename, std::string name)

  constructors
- **DefaultNode** (Model ∗model, std::string name="")
- void **addTransition** (DefaultNodeTransition ∗transition)

  new public user methods for this component
- void **removeTransition** (DefaultNodeTransition ∗transition)
- List< DefaultNodeTransition ∗ > ∗ **getTransitions** () const
- void **setInitialNode** (bool initialNode)
- bool **isInitialNode** () const
- void **setFinalNode** (bool finalNode)
- bool **isFinalNode** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model ∗model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ∼**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from DefaultNode

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Additional Inherited Members

Protected Member Functions inherited from DefaultNode

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model ∗ **_parentModel**
- List< SimulationControl ∗ > ∗ **_simulationControls** = new List<SimulationControl∗>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.454.1 Detailed Description

State node used by MarkovChainNetwork DTMC models.
MarkovState is a ModelDataDefinition, not a process ModelComponent. It represents one finite state in a time-homogeneous discrete-time Markov chain.
The documentation for this class was generated from the following files:

- plugins/data/ModalModel/MarkovState.h
- plugins/components/ModalModel/MarkovState.cpp

17.455 MarkovChainNetwork::MarkovTransition Class Reference

Public Member Functions

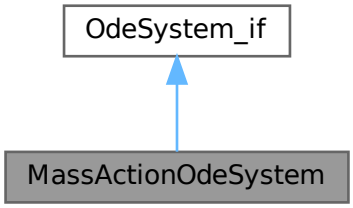
- **MarkovTransition** (MarkovState *source=nullptr, MarkovState *destination=nullptr, double probability=1.0, std::string name="")
- void **setSource** (MarkovState *source)
- MarkovState * **getSource** () const
- void **setDestination** (MarkovState *destination)
- MarkovState * **getDestination** () const
- void **setProbability** (double probability)
- double **getProbability** () const
- void **setName** (std::string name)
- std::string **getName** () const

The documentation for this class was generated from the following files:

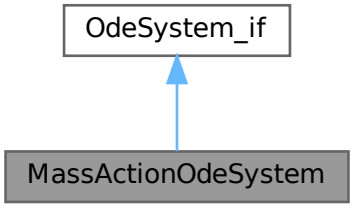
- plugins/data/ModalModel/MarkovChainNetwork.h
- plugins/components/ModalModel/MarkovChainNetwork.cpp

17.456 MassActionOdeSystem Class Reference

ODE system generated from biochemical reactions using mass-action kinetics.
`#include <tools/Biochemical/MassActionOdeSystem.h>`
Inheritance diagram for MassActionOdeSystem:



Collaboration diagram for MassActionOdeSystem:



Classes

- struct Reaction
- struct Species
- struct StoichiometricTerm

Public Member Functions

- **MassActionOdeSystem** (std::vector< Species > species, std::vector< Reaction > reactions)
- unsigned int dimension () const override
- void evaluate (double, const double *y, double *dydt) const override
- const std::vector< Species > & **species** () const
- const std::vector< Reaction > & **reactions** () const
- double **evaluateForwardRate** (unsigned int reactionIndex, const double *y) const
- double **evaluateReverseRate** (unsigned int reactionIndex, const double *y) const
- double **evaluateNetRate** (unsigned int reactionIndex, const double *y) const

17.456.1 Detailed Description

ODE system generated from biochemical reactions using mass-action kinetics.

17.456.2 Member Function Documentation

17.456.2.1 dimension()

```
unsigned int MassActionOdeSystem::dimension ( ) const  [inline], [override], [virtual]
```

Implements OdeSystem_if.

17.456.2.2 evaluate()

```
void MassActionOdeSystem::evaluate (
            double ,
            const double * y,
            double * dydt ) const  [inline], [override], [virtual]
```

Implements OdeSystem_if.

The documentation for this class was generated from the following file:

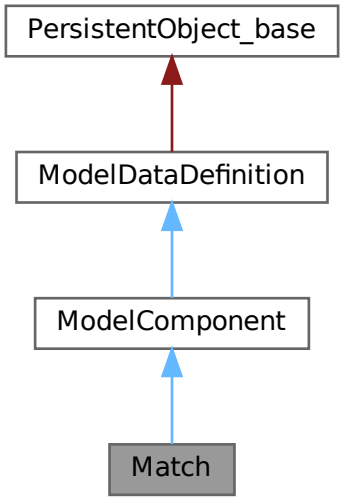
- tools/Biochemical/MassActionOdeSystem.h

17.457 Match Class Reference

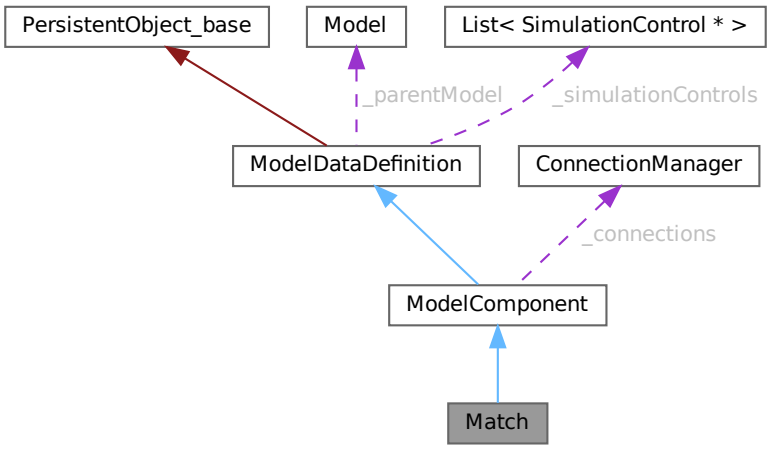
Brings together one entity from each of several input queues, optionally requiring a matching attribute value, and releases them in sync.

```
#include <plugins/components/Synchronization/Match.h>
```

Inheritance diagram for Match:



Collaboration diagram for Match:



Public Types

- enum class **Rule** : int { **Any** = 0 , **ByAttribute** = 1 , **num_elements** = 2 }

Public Member Functions

- **Match** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- void **setRule** (Match::Rule _rule)
- Match::Rule **getRule** () const
- void **setAttributeName** (std::string _attributeName)
- std::string **getAttributeName** () const
- void **setMatchSize** (std::string _matchSize)
- std::string **getMatchSize** () const
- void **setNumberOfQueues** (unsigned int _numberOfQueues)
- unsigned int **getNumberOfQueues** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static std::string **convertEnumToStr** (Rule rule)
- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _createEditableDataDefinitions () override
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)

- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.457.1 Detailed Description

Brings together one entity from each of several input queues, optionally requiring a matching attribute value, and releases them in sync.
Arena correspondence: the "Match module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 56). `Rule` (`Any/ByAttribute`) and `_attributeName` match Arena's Type/Attribute Name; `_matchSize` matches Number to Match; `_numberOfQueues` generalizes Arena's fixed five-queue limit to an arbitrary count.

17.457.2 Member Function Documentation

17.457.2.1 _check()

```
bool Match::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.457.2.2 _createEditableDataDefinitions()

```
void Match::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.457.2.3 _loadInstance()

```
bool Match::_loadInstance (
            PersistenceRecord * fields )  [override], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.457.2.4 _onDispatchEvent()

```
void Match::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.457.2.5 _saveInstance()

```
void Match::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.457.2.6 show()

```
std::string Match::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/Synchronization/Match.h
- plugins/components/Synchronization/Match.cpp

17.458 testing::Message Class Reference

Public Member Functions

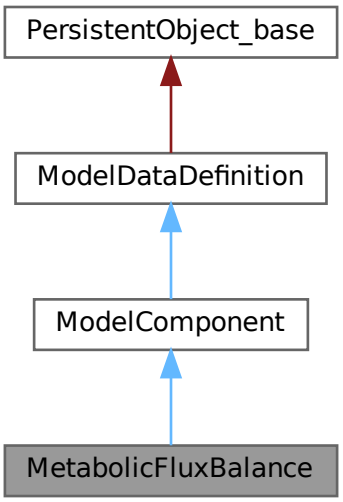
- **Message** (const Message &msg)
- **Message** (const char *str)
- template<typename T >
  Message & **operator**<< (const T &val)
- template<typename T >
  Message & **operator**<< (T *const &pointer)
- Message & **operator**<< (BasicNarrowIoManip val)
- Message & **operator**<< (bool b)
- Message & **operator**<< (const wchar_t *wide_c_str)
- Message & **operator**<< (wchar_t *wide_c_str)
- std::string **GetString** () const

The documentation for this class was generated from the following files:

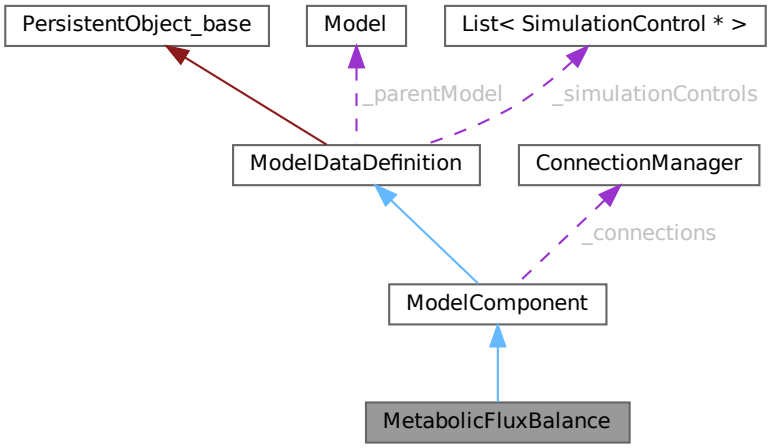
- gtest/gtest/gtest-message.h
- gtest/src/gtest.cc

17.459 MetabolicFluxBalance Class Reference

`#include <plugins/components/BiochemicalSimulation/MetabolicFluxBalance.h>`
Inheritance diagram for MetabolicFluxBalance:



Collaboration diagram for MetabolicFluxBalance:



Public Member Functions

- **MetabolicFluxBalance** (Model ∗model, std::string name="")
- void **setMetabolicNetwork** (MetabolicNetwork ∗metabolicNetwork)
- MetabolicNetwork ∗ **getMetabolicNetwork** () const
- void **setObjectiveReactionName** (std::string objectiveReactionName)
- std::string **getObjectiveReactionName** () const
- void **setLastSucceeded** (bool lastSucceeded)
- bool **getLastSucceeded** () const
- void **setLastObjectiveValue** (double lastObjectiveValue)
- double **getLastObjectiveValue** () const
- void **setLastMessage** (std::string lastMessage)
- std::string **getLastMessage** () const
- const std::map< std::string, double > & **getLastFluxes** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component

- static bool Check (ModelComponent *component)

  Validates a component instance.

- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.

- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.

- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.

- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.

- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.

- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.

- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden

- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.

- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.459.1 Detailed Description

Component that evaluates flux-balance analysis (FBA) over a MetabolicNetwork.

17.459.1.1 LP solver dependency (GLPK)

MetabolicFluxBalance requires GLPK for metabolic networks with more than a few dozen reactions (e.g., M. genitalium: 641 reactions; yeast-GEM: 4131 reactions). Without GLPK, a built-in basis-enumeration fallback is used, which is only practical for small toy networks.

17.459.1.1.1 Installation

Ubuntu/Debian: sudo apt-get install libglpk-dev Fedora/RHEL: sudo dnf install glpk-devel macOS (Brew): brew install glpk

17.459.1.1.2 Re-configuration after installation

cmake –preset terminal-app # or any other preset cmake –build build/terminal-app –target genesys_plugins_↩components
CMake will print at configure time: GenESyS: GLPK found — MetabolicFluxBalance ... will use GLPK LP solver or: GenESyS: GLPK not found — ... (suitable for small models only; install libglpk-dev ...)
The preprocessor guard is GENESYS_HAVE_GLPK. The GLPK solver wrapper is in source/tools/Biochemical/GlpkFluxBalanceSolver

17.459.2 Member Function Documentation

17.459.2.1 _check()

```
bool MetabolicFluxBalance::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.459.2.2 _createEditableDataDefinitions()

```
void MetabolicFluxBalance::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.459.2.3 _loadInstance()

```
bool MetabolicFluxBalance::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.459.2.4 _onDispatchEvent()

```
void MetabolicFluxBalance::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.459.2.5 _saveInstance()

```
void MetabolicFluxBalance::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.459.2.6 show()

```
std::string MetabolicFluxBalance::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/BiochemicalSimulation/MetabolicFluxBalance.h
- plugins/components/BiochemicalSimulation/MetabolicFluxBalance.cpp

17.460 MetabolicFluxBalanceSolver Class Reference

```
#include <tools/Biochemical/MetabolicFluxBalanceSolver.h>
```

Classes

- struct Problem
- struct Solution

Static Public Member Functions

- static Solution **solve** (const Problem &problem)

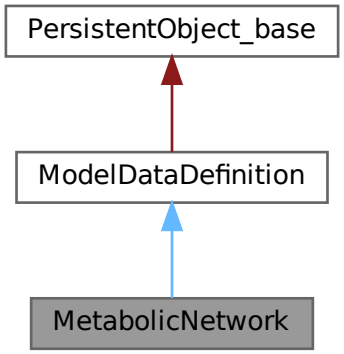
17.460.1 Detailed Description

Small LP solver for flux-balance problems with equality constraints and variable bounds.
It solves: maximize/minimize c^T v subject to A v = 0 lb <= v <= ub
using basis enumeration over bounded vertices. This is practical for the small metabolic models currently exercised by the runtime tests and provides a concrete solver-backed replacement for the previous stub objective logic.
The documentation for this class was generated from the following file:

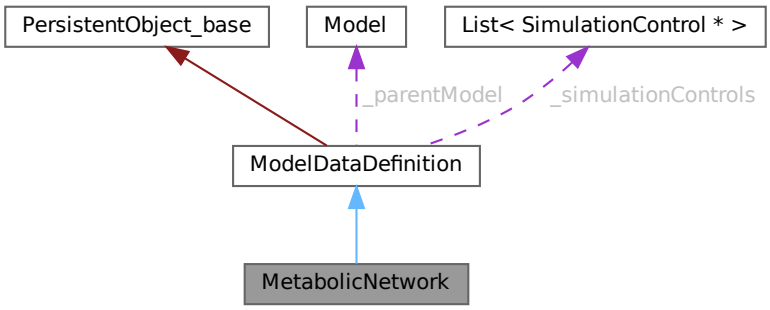
- tools/Biochemical/MetabolicFluxBalanceSolver.h

17.461 MetabolicNetwork Class Reference

`#include <plugins/data/BiochemicalSimulation/MetabolicNetwork.h>`
Inheritance diagram for MetabolicNetwork:



Collaboration diagram for MetabolicNetwork:



Public Member Functions

- **MetabolicNetwork** (Model *model, std::string name="")
- void **addReaction** (std::string reactionName)
- void **addExchangeSpecies** (std::string speciesName)
- void **clearReactions** ()
- void **clearExchangeSpecies** ()
- const std::vector< std::string > & **getReactionNames** () const
- const std::vector< std::string > & **getExchangeSpeciesNames** () const
- void **setObjectiveReactionName** (std::string objectiveReactionName)
- std::string **getObjectiveReactionName** () const
- void **setObjectiveSense** (std::string objectiveSense)
- std::string **getObjectiveSense** () const
- void **setCompartment** (std::string compartment)
- std::string **getCompartment** () const
- void **setEnabled** (bool enabled)
- bool **isEnabled** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.461.1 Detailed Description

Aggregate metabolic network definition.
A MetabolicNetwork stores the reaction membership and objective metadata for future flux-balance style components.

17.461.2 Member Function Documentation

17.461.2.1 _check()

```
bool MetabolicNetwork::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.461.2.2 _createEditableDataDefinitions()

```
void MetabolicNetwork::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.461.2.3 _loadInstance()

```
bool MetabolicNetwork::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.461.2.4 _saveInstance()

```
void MetabolicNetwork::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.461.2.5 show()

```
std::string MetabolicNetwork::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.

Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

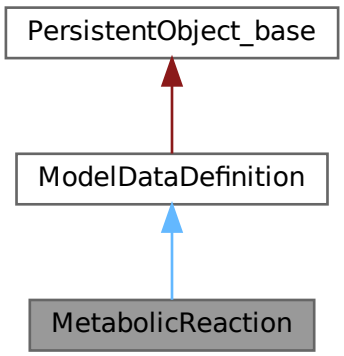
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

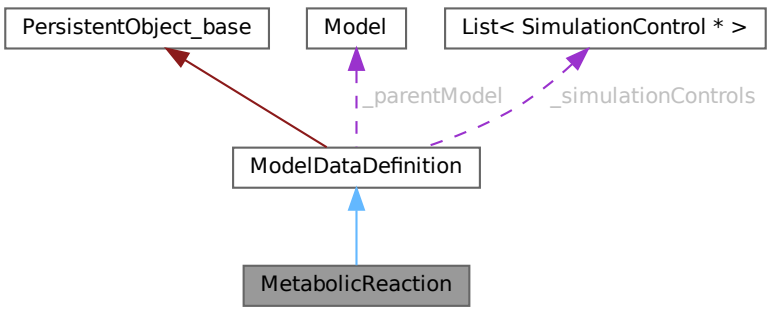
- plugins/data/BiochemicalSimulation/MetabolicNetwork.h
- plugins/data/BiochemicalSimulation/MetabolicNetwork.cpp

17.462 MetabolicReaction Class Reference

`#include <plugins/data/BiochemicalSimulation/MetabolicReaction.h>`
Inheritance diagram for MetabolicReaction:



Collaboration diagram for MetabolicReaction:



Classes

- struct StoichiometricTerm

Public Member Functions

- **MetabolicReaction** (Model *model, std::string name="")
- void **addReactant** (std::string speciesName, double stoichiometry=1.0)
- void **addProduct** (std::string speciesName, double stoichiometry=1.0)
- void **clearReactants** ()
- void **clearProducts** ()
- const std::vector< StoichiometricTerm > & **getReactants** () const
- const std::vector< StoichiometricTerm > & **getProducts** () const
- void **setLowerBound** (double lowerBound)
- double **getLowerBound** () const
- void **setUpperBound** (double upperBound)
- double **getUpperBound** () const
- void **setObjectiveCoefficient** (double objectiveCoefficient)
- double **getObjectiveCoefficient** () const
- void **setReversible** (bool reversible)
- bool **isReversible** () const
- void **setGeneRule** (std::string geneRule)
- std::string **getGeneRule** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.462.1 Detailed Description

Constraint-based metabolic reaction definition.

17.462.2 Member Function Documentation

17.462.2.1 _check()

```
bool MetabolicReaction::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.462.2.2 _createEditableDataDefinitions()

```
void MetabolicReaction::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.462.2.3 _loadInstance()

```
bool MetabolicReaction::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.462.2.4 _saveInstance()

```
void MetabolicReaction::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.462.2.5 show()

```
std::string MetabolicReaction::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

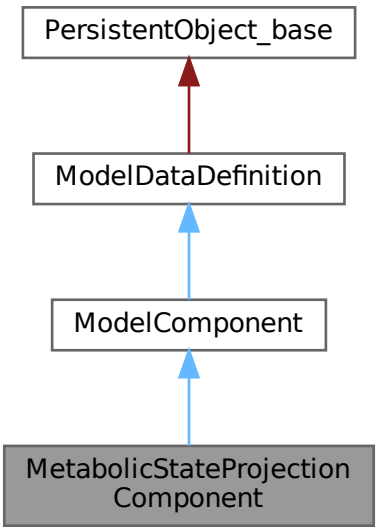
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

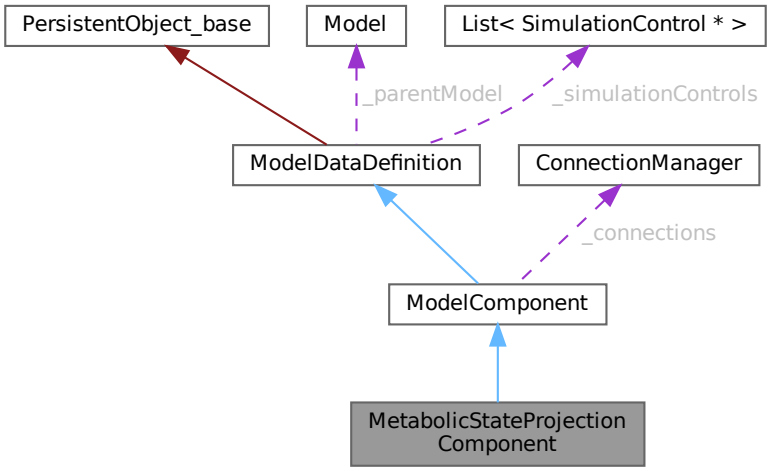
- plugins/data/BiochemicalSimulation/MetabolicReaction.h
- plugins/data/BiochemicalSimulation/MetabolicReaction.cpp

17.463 MetabolicStateProjectionComponent Class Reference

Projects flux-balance outputs into a WholeCellState.
`#include <plugins/components/WholeCellModeling/MetabolicStateProjection↩Component.h>`
Inheritance diagram for MetabolicStateProjectionComponent:



Collaboration diagram for MetabolicStateProjectionComponent:



Classes

- struct Projection

Public Types

- enum class **ProjectionKind** { **MetaboliteAmount** , **CompartmentMetaboliteAmount** , **PathwayActivity** }
- enum class **ProjectionUpdateMode** { **Overwrite** , **Accumulate** , **Turnover** }

Public Member Functions

- **MetabolicStateProjectionComponent** (Model ∗model, std::string name="")
- void **setWholeCellState** (WholeCellState ∗state)
- WholeCellState ∗ **getWholeCellState** () const
- void **setFluxBalanceComponent** (MetabolicFluxBalance ∗component)
- MetabolicFluxBalance ∗ **getFluxBalanceComponent** () const
- void **setFluxBalanceComponentName** (std::string componentName)
- std::string **getFluxBalanceComponentName** () const
- void **clearObjectiveProjection** ()
- void **setObjectiveAsMetabolite** (const std::string &targetKey, double scale=1.0, double offset=0.↩ 0, ProjectionUpdateMode updateMode=ProjectionUpdateMode::Overwrite, double turnoverFraction=0.0)
- void **setObjectiveAsCompartmentMetabolite** (const std::string &compartmentName, const std::string &targetKey, double scale=1.0, double offset=0.0, ProjectionUpdateMode updateMode=ProjectionUpdate↩ Mode::Overwrite, double turnoverFraction=0.0)
- void **setObjectiveAsPathwayActivity** (const std::string &targetKey, double scale=1.0, double offset=0.↩ 0, ProjectionUpdateMode updateMode=ProjectionUpdateMode::Overwrite, double turnoverFraction=0.0)
- bool **hasObjectiveProjection** () const
- void **clearFluxProjections** ()
- void **addMetaboliteProjection** (const std::string &reactionName, const std::string &targetKey, double scale=1.0, double offset=0.0, ProjectionUpdateMode updateMode=ProjectionUpdateMode::Overwrite, double turnoverFraction=0.0)
- void **addCompartmentMetaboliteProjection** (const std::string &reactionName, const std::string &compartmentName, const std::string &targetKey, double scale=1.0, double offset=0.0, ProjectionUpdate↩ Mode updateMode=ProjectionUpdateMode::Overwrite, double turnoverFraction=0.0)

- void **addPathwayProjection** (const std::string &reactionName, const std::string &targetKey, double scale=1.0, double offset=0.0, ProjectionUpdateMode updateMode=ProjectionUpdateMode::Overwrite, double turnoverFraction=0.0)
- unsigned int **getProjectionCount** () const
- void **setLastSucceeded** (bool lastSucceeded)
- bool **getLastSucceeded** () const
- void **setLastMessage** (std::string lastMessage)
- std::string **getLastMessage** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.463.1 Detailed Description

Projects flux-balance outputs into a WholeCellState.
MetabolicStateProjectionComponent bridges FBA-style submodels with the whole-cell shared state. It reads the latest objective value and reaction fluxes computed by a MetabolicFluxBalance component and writes selected projections into WholeCellState as:

- global metabolite proxies;
- compartment-specific metabolite proxies;
- pathway activity summaries.

This component intentionally does not solve the metabolic optimization itself. It only transfers selected results to the whole-cell layer so that growth, checkpoints, and gene-expression submodels can consume them.

17.463.2 Member Function Documentation

17.463.2.1 _check()

```
bool MetabolicStateProjectionComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.463.2.2 _createEditableDataDefinitions()

```
void MetabolicStateProjectionComponent::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.463.2.3 _loadInstance()

```
bool MetabolicStateProjectionComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.463.2.4 _onDispatchEvent()

```
void MetabolicStateProjectionComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.463.2.5 _saveInstance()

```
void MetabolicStateProjectionComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.463.2.6 show()

`std::string MetabolicStateProjectionComponent::show ( ) [override], [virtual]`
virtual public methods
Returns a textual representation of this component.

Returns

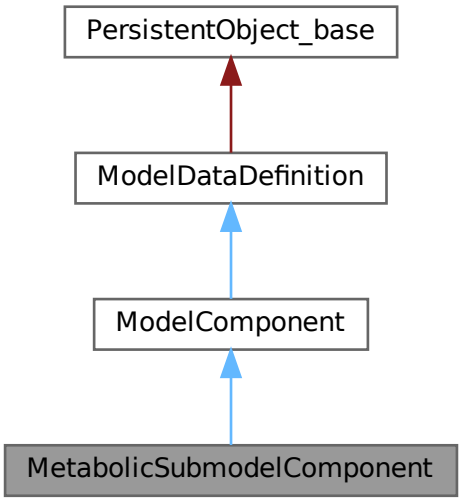
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

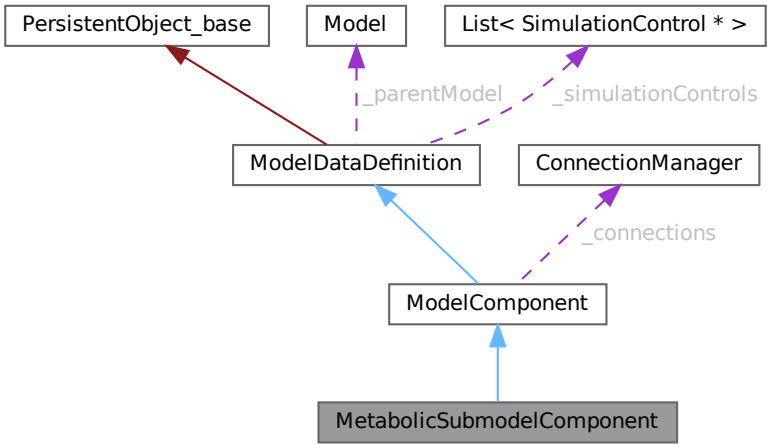
- plugins/components/WholeCellModeling/MetabolicStateProjectionComponent.h
- plugins/components/WholeCellModeling/MetabolicStateProjectionComponent.cpp

17.464 MetabolicSubmodelComponent Class Reference

`#include <plugins/components/WholeCellModeling/MetabolicSubmodelComponent.h>`
Inheritance diagram for MetabolicSubmodelComponent:



Collaboration diagram for MetabolicSubmodelComponent:



Public Member Functions

- **MetabolicSubmodelComponent** (Model ∗model, std::string name="")
- void **setWholeCellState** (WholeCellState ∗state)
- WholeCellState ∗ **getWholeCellState** () const
- void **setAtpYieldRate** (double rate)
- double **getAtpYieldRate** () const
- void **setConsumptionFraction** (double frac)
- double **getConsumptionFraction** () const
- void **setAtpPoolMax** (double maxPool)
- double **getAtpPoolMax** () const
- void **setDeltaT** (double dt)
- double **getDeltaT** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent ∗comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.464.1 Detailed Description

Minimal metabolic submodel that produces ATP and NADPH each step.
Replaces a full FBA step with a volume-proportional energy production model calibrated for M. genitalium (Karr et al. 2012):
ATP_produced = atpYieldRate × cellVolume(fL) × deltaT [mmol] ATP_pool(new) = min(ATP_pool + ATP_produced − ATP_consumed, atpPoolMax)
ATP consumption is estimated as a fixed fraction of the current pool (consumption rate = consumptionFraction × ATP_pool).
Results are written to the WholeCellState metabolite pool under keys "ATP" and "ADP". Downstream components can read "ATP" to gate reactions.
For a full FBA integration, replace this component with a proper MetabolicFluxBalance + network wiring. This component is intentionally minimal so that Phases 2–5 can be validated without GLPK-specific setup.

17.464.2 Member Function Documentation

17.464.2.1 _check()

```
bool MetabolicSubmodelComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.464.2.2 _createEditableDataDefinitions()

```
void MetabolicSubmodelComponent::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.464.2.3 _initBetweenReplications()

```
void MetabolicSubmodelComponent::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.464.2.4 _loadInstance()

```
bool MetabolicSubmodelComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.464.2.5 _onDispatchEvent()

```
void MetabolicSubmodelComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.464.2.6 _saveInstance()

```
void MetabolicSubmodelComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.464.2.7 show()

```
std::string MetabolicSubmodelComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/WholeCellModeling/MetabolicSubmodelComponent.h
- plugins/components/WholeCellModeling/MetabolicSubmodelComponent.cpp

17.465 generate_kernel_simulator_test_stubs.MethodEntry Class Reference

Static Public Attributes

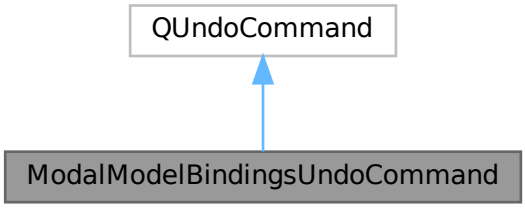
- str **header**
- str **class_name**
- str **signature**

The documentation for this class was generated from the following file:

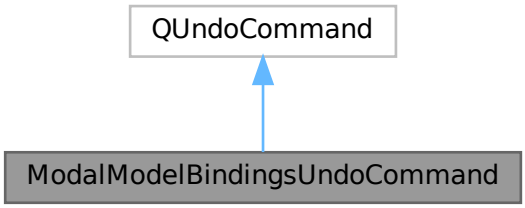
- tests/unit/tools/generate_kernel_simulator_test_stubs.py

17.466 ModalModelBindingsUndoCommand Class Reference

Inheritance diagram for ModalModelBindingsUndoCommand:



Collaboration diagram for ModalModelBindingsUndoCommand:



Classes

- struct Bindings

Public Member Functions

- **ModalModelBindingsUndoCommand** (ModalModelDefault ∗adapter, DefaultNetwork ∗network, Bindings updated, QUndoCommand ∗parent=nullptr)
- void **undo** () override
- void **redo** () override

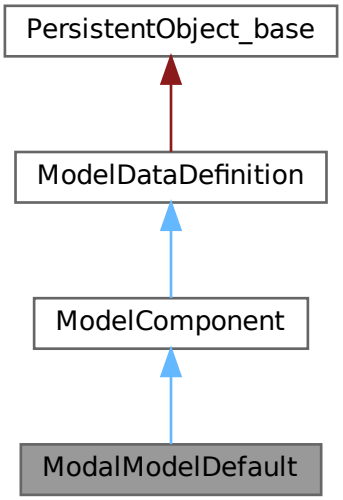
The documentation for this class was generated from the following files:

- applications/gui/genesys/actions/ModalModelBindingsUndoCommand.h
- applications/gui/genesys/actions/ModalModelBindingsUndoCommand.cpp

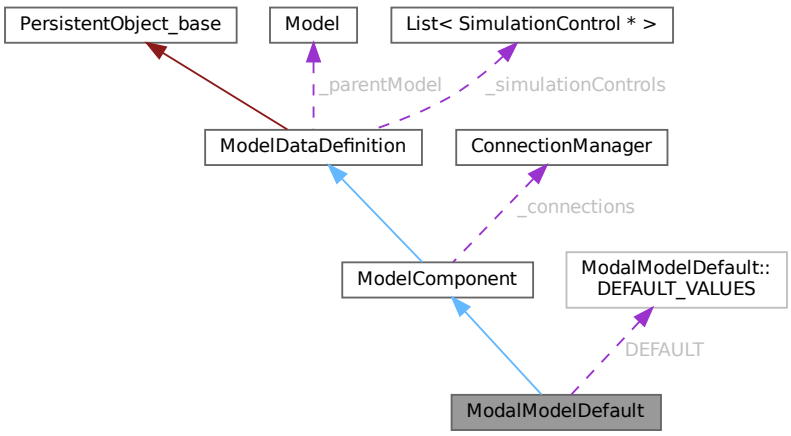
17.467 ModalModelDefault Class Reference

`#include <plugins/components/ModalModel/ModalModelDefault.h>`

Inheritance diagram for ModalModelDefault:



Collaboration diagram for ModalModelDefault:



Classes

- struct DEFAULT_VALUES

Public Member Functions

- **ModalModelDefault** (Model ∗model, std::string name="")
- virtual void **setMaxTransitionsPerDispatch** (unsigned int maxTransitionsPerDispatch)
- virtual unsigned int **getMaxTransitionsPerDispatch** () const
- std::string **getTimeDelayExpressionPerDispatch** ()

- void **setTimeDelayExpressionPerDispatch** (const std::string time_delay_expression_per_dispatch)
- Util::TimeUnit **getTimeDelayPerDispatchTimeUnit** ()
- void **setTimeDelayPerDispatchTimeUnit** (const Util::TimeUnit time_delay_per_dispatch_time_unit)
- DefaultNetwork * **getNetwork** () const
- void **setNetwork** (DefaultNetwork *network)
- std::string **getNetworkName** () const
- void **setNetworkName** (const std::string &networkName)
- void **setInputBinding** (unsigned int port, const std::string &expression)
- std::string **getInputBinding** (unsigned int port) const
- void **setOutputBinding** (unsigned int port, const std::string &attributeName)
- std::string **getOutputBinding** (unsigned int port) const
- virtual void **addOutputExpressionReference** (ModelDataDefinition *expressionReference)
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Protected Attributes

- const struct ModalModelDefault::DEFAULT_VALUES **DEFAULT**
- unsigned int **_maxTransitionsPerDispatch** = DEFAULT.maxTransitionsPerDispatch
- std::string **_timeDelayExpressionPerDispatch** = DEFAULT.timeDelayExpressionPerDispatch
- Util::TimeUnit **_timeDelayPerDispatchTimeUnit** = DEFAULT.timeDelayPerDispatchTimeUnit

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.467.1 Detailed Description

Process-world adapter that activates an attached DefaultNetwork.
Network topology and formalism state live in the attached network. This component only maps entity arrivals to network activations and routes network outputs back into the process model.

17.467.2 Member Function Documentation

17.467.2.1 _check()

```
bool ModalModelDefault::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.467.2.2 _createAttachedAttributes()

```
void ModalModelDefault::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.467.2.3 _initBetweenReplications()

```
void ModalModelDefault::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.467.2.4 _loadInstance()

```
bool ModalModelDefault::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.467.2.5 _onDispatchEvent()

```
void ModalModelDefault::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.467.2.6 _saveInstance()

```
void ModalModelDefault::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.467.2.7 show()

```
std::string ModalModelDefault::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/ModalModel/ModalModelDefault.h
- plugins/components/ModalModel/ModalModelDefault.cpp

17.468 Model Class Reference

```
#include <kernel/simulator/model/Model.h>
```

Public Member Functions

- Model (Simulator *simulator, unsigned int level=0)

  Creates a model bound to a simulator.
- virtual ~**Model** ()

  Releases model-owned runtime state.
- bool save (std::string filename)

  Saves the model to a file.
- bool load (std::string filename)

  Loads the model from a file.
- bool check ()

  Validates the model structure and configuration.
- void **clear** ()

  Clears the current model content.
- std::string showLanguage ()

  Returns the model specification in the source language.
- void **show** ()

  Displays the model contents to the configured trace output.
- bool insert (ModelDataDefinition *elemOrComp)

  Inserts a model data definition or component into the model.
- void remove (ModelDataDefinition *elemOrComp)

  Removes a model data definition or component from the model.
- std::list< ModelDataDefinition * > collectDataDefinitionsRemovedWith (const std::list< ModelDataDefinition * > &roots) const

  Computes the data definitions that should be removed together with a set of roots.
- Entity * createEntity (std::string name, bool insertIntoModel=true)

  Creates an entity owned by this model.
- void removeEntity (Entity *entity)

  Removes an entity owned by this model.
- void sendEntityToComponent (Entity *entity, Connection *connection, double timeDelay=0.0)

  Schedules an entity transfer through a specific connection.
- void sendEntityToComponent (Entity *entity, ModelComponent *component, double timeDelay=0.0, unsigned int componentinputPortNumber=0)

Schedules an entity transfer to a specific component and input port.

- double parseExpression (const std::string expression)

  Parses and evaluates an expression.
- double parseExpression (const std::string expression, bool &success, std::string &errorMessage)

  Parses and evaluates an expression while reporting success or failure.
- bool checkExpression (const std::string expression, const std::string expressionName, std::string &error↩ Message)

  Checks whether an expression is syntactically valid.
- void checkReferencesToDataDefinitions (std::string expression, std::map< std::string, std::list< std::string > * > *referencedDataDefinitions)

  Collects data-definition references found in an expression.
- Util::identification getId () const

  Returns the unique model identifier.
- bool hasChanged () const

  Indicates whether the model has pending changes.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for the model and its owned persistent objects.
- OnEventManager * getOnEventManager () const

  Returns the event manager that coordinates model-level events.
- ModelDataManager * getDataManager () const

  Returns the data manager for model data definitions.
- ModelComponentManager * getComponentManager () const

  Returns the component manager for model components.
- ModelInfo * getInfos () const

  Returns the model metadata object.
- Simulator * getParentSimulator () const

  Returns the parent simulator that owns this model.
- ModelSimulation * getSimulation () const

  Returns the simulation controller associated with this model.
- List< Event * > * getFutureEvents () const

  Returns the future-event queue.
- List< SimulationControl * > * getControls () const

  Returns the simulation controls exposed by the model.
- List< SimulationResponse * > * getResponses () const

  Returns the simulation responses exposed by the model.
- void setTracer (TraceManager *_traceManager)

  Sets the trace manager used by this model.
- TraceManager * getTracer () const

  Returns the trace manager used by this model.
- Persistence_if * getPersistence () const

  Returns the persistence interface used by this model.
- unsigned int getLevel () const

  Returns the hierarchy level of the model.

17.468.1 Detailed Description

Model is probably the most important class of Genesys kernel. It represents a discrete event-driven simulation model. Each model is responsible for controlling its own simulation, ie, for sequentially processing events and collecting statistical results. A model is mainly represented by a collection of components (ModelComponent), adequately configurated and connected, and a collection of under layered modeldatum (ModelDataDefinition).

17.468.2 Constructor & Destructor Documentation

17.468.2.1 Model()

```
Model::Model (
            Simulator * simulator,
            unsigned int level = 0 )
```

Creates a model bound to a simulator.

Parameters

simulator	Parent simulator that owns the model.
level	Hierarchy level of the model within the simulator.

17.468.3 Member Function Documentation

17.468.3.1 check()

```
bool Model::check ( )
```

Validates the model structure and configuration.
Checks the integrity and consistency of the model, may correct some inconsistencies, and returns whether the model is ready to be simulated.

Returns

`true` when the model is ready to simulate.

17.468.3.2 checkExpression()

```
bool Model::checkExpression (
            const std::string expression,
            const std::string expressionName,
            std::string & errorMessage )
```

Checks whether an expression is syntactically valid.
This method is invoked by ModelComponents and ModelDatas in their private _check() methods to validate user-defined expressions.

Parameters

expression	Expression text.
expressionName	Logical name used in diagnostics.
errorMessage	Output error message when validation fails.

Returns

`true` when the expression is valid.

17.468.3.3 checkReferencesToDataDefinitions()

```
void Model::checkReferencesToDataDefinitions (
            std::string expression,
            std::map< std::string, std::list< std::string > * > * referencedDataDefinitions
)
```

Collects data-definition references found in an expression.

Parameters

expression	Expression text.

Parameters

referencedDataDefinitions	Output map of referenced data definitions by name.

17.468.3.4 collectDataDefinitionsRemovedWith()

```
std::list< ModelDataDefinition * > Model::collectDataDefinitionsRemovedWith (
            const std::list< ModelDataDefinition * > & roots ) const
```

Computes the data definitions that should be removed together with a set of roots.

Parameters

roots	Root objects selected for removal.

Returns

Flattened list of dependent data definitions to remove as well.

17.468.3.5 createEntity()

```
Entity * Model::createEntity (
            std::string name,
            bool insertIntoModel = true )
```

Creates an entity owned by this model.

Parameters

name	Entity name.
insertIntoModel	If `true`, inserts the entity into the model immediately.

Returns

Newly created entity.

17.468.3.6 getComponentManager()

```
ModelComponentManager * Model::getComponentManager ( ) const
```

Returns the component manager for model components.

Returns

Component manager instance.

17.468.3.7 getControls()

```
List< SimulationControl * > * Model::getControls ( ) const
```

Returns the simulation controls exposed by the model.
These are the values that can be externally controlled and usually correspond to input parameters that must be changed for an experimental design.

Returns

List of writable simulation controls.

17.468.3.8 getDataManager()

```
ModelDataManager * Model::getDataManager ( ) const
```

Returns the data manager for model data definitions.
Provides access to the class that manages the basic elements of the simulation model, such as queues, resources, variables and similar model data definitions.

Returns

Data manager instance.

17.468.3.9 getFutureEvents()

```
List< Event * > * Model::getFutureEvents ( ) const
```

Returns the future-event queue.
The future events list is chronologically sorted. Events are scheduled by components while processing other events, and a replication evolves by sequentially processing the first event in this list. The list is initialized with the events first described by source components.

Returns

Chronologically ordered list of scheduled events.

17.468.3.10 getId()

```
Util::identification Model::getId ( ) const
```

Returns the unique model identifier.

Returns

Model identifier.

17.468.3.11 getInfos()

```
ModelInfo * Model::getInfos ( ) const
```

Returns the model metadata object.

Returns

Model info instance.

17.468.3.12 getLevel()

```
unsigned int Model::getLevel ( ) const
```

Returns the hierarchy level of the model.
Provides access to the class that performs the trace of simulation and replications.

Returns

Model level.

17.468.3.13 getOnEventManager()

```
OnEventManager * Model::getOnEventManager ( ) const
```

Returns the event manager that coordinates model-level events.
Provides access to the class that manages events generated by the model, such as the beginning of a new simulation or replication and the processing of an event.

Returns

On-event manager instance.

17.468.3.14 getParentSimulator()

```
Simulator * Model::getParentSimulator ( ) const
```

Returns the parent simulator that owns this model.

Returns

Parent simulator.

17.468.3.15 getPersistence()

```
Persistence_if * Model::getPersistence ( ) const
```

Returns the persistence interface used by this model.

Returns

Persistence interface instance.

17.468.3.16 getResponses()

```
List< SimulationResponse * > * Model::getResponses ( ) const
```

Returns the simulation responses exposed by the model.
These are the exits or simulation results that can be read externally and usually correspond to statistics gathered for an experimental design.

Returns

List of readable simulation responses.

17.468.3.17 getSimulation()

```
ModelSimulation * Model::getSimulation ( ) const
```

Returns the simulation controller associated with this model.
Provides access to the class that manages the model simulation.

Returns

Simulation controller instance.

17.468.3.18 getTracer()

```
TraceManager * Model::getTracer ( ) const
```

Returns the trace manager used by this model.

Returns

Trace manager instance.

17.468.3.19 hasChanged()

```
bool Model::hasChanged ( ) const
```

Indicates whether the model has pending changes.

Returns

`true` when the model or one of its owned objects changed.

17.468.3.20 insert()

```
bool Model::insert (
            ModelDataDefinition * elemOrComp )
```

Inserts a model data definition or component into the model.
This is a generic entry point for either ComponentManager->insert() or ModelDataManager->insert().

Parameters

elemOrComp	Object to insert.

Returns

`true` when the object was accepted.

17.468.3.21 load()

```
bool Model::load (
            std::string filename )
```

Loads the model from a file.

Parameters

filename	Input filename.

Returns

`true` when loading succeeds.

17.468.3.22 parseExpression() [1/2]

```
double Model::parseExpression (
            const std::string expression )
```

Parses and evaluates an expression.
The parser always returns a double, even when the expression has syntax errors, in which case the result is 0.

Parameters

expression	Expression text.

Returns

Numeric result of the expression, or 0 on syntax failure.

17.468.3.23 parseExpression() [2/2]

```
double Model::parseExpression (
            const std::string expression,
            bool & success,
            std::string & errorMessage )
```

Parses and evaluates an expression while reporting success or failure.
The parser always returns a double, even when the expression has syntax errors, in which case the result is 0. This overload also reports whether an error occurred.

Parameters

expression	Expression text.
success	Output flag that receives the parsing result.
errorMessage	Output error message when parsing fails.

Returns

Numeric result of the expression, or 0 on syntax failure.

17.468.3.24 remove()

```
void Model::remove (
            ModelDataDefinition * elemOrComp )
```

Removes a model data definition or component from the model.
This is a generic entry point for either ComponentManager->remove() or ModelDataManager->remove().

Parameters

elemOrComp	Object to remove.

17.468.3.25 removeEntity()

```
void Model::removeEntity (
            Entity * entity )
```

Removes an entity owned by this model.

Parameters

entity	Entity to remove.

17.468.3.26 save()

```
bool Model::save (
            std::string filename )
```

Saves the model to a file.

Parameters

filename	Output filename.

Returns

`true` when saving succeeds.

17.468.3.27 sendEntityToComponent() [1/2]

```
void Model::sendEntityToComponent (
            Entity * entity,
            Connection * connection,
            double timeDelay = 0.0 )
```

Schedules an entity transfer through a specific connection.
Used by components to send entities to the next connected component, or by the model itself while processing an event.

Parameters

entity	Entity to transfer.
connection	Destination connection.
timeDelay	Delay applied before the entity reaches the destination.

17.468.3.28 sendEntityToComponent() [2/2]

```
void Model::sendEntityToComponent (
            Entity * entity,
            ModelComponent * component,
            double timeDelay = 0.0,
            unsigned int componentinputPortNumber = 0 )
```

Schedules an entity transfer to a specific component and input port.
Used by components to send entities to the next connected component, or by the model itself while processing an event.

Parameters

entity	Entity to transfer.
component	Destination component.
timeDelay	Delay applied before the entity reaches the destination.
componentinputPortNumber	Destination input port number.

17.468.3.29 setHasChanged()

```
void Model::setHasChanged (
            bool hasChanged )
```

Updates the changed flag for the model and its owned persistent objects.

Parameters

hasChanged	New changed-state value.

17.468.3.30 setTracer()

```
void Model::setTracer (
            TraceManager * _traceManager )
```

Sets the trace manager used by this model.

Parameters

_traceManager	Trace manager instance.

17.468.3.31 showLanguage()

```
std::string Model::showLanguage ( )
```

Returns the model specification in the source language.

Returns

Textual model specification.

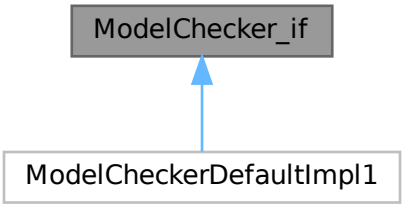
The documentation for this class was generated from the following files:

- kernel/simulator/model/Model.h
- kernel/simulator/model/Model.cpp
- tests/unit/test_simulator_support.cpp
- tests/unit/test_support_modelmanager.cpp

17.469 ModelChecker_if Class Reference

`#include <kernel/simulator/model/ModelChecker_if.h>`

Inheritance diagram for ModelChecker_if:



Public Member Functions

- virtual bool checkAll ()=0

  checkAll
- virtual bool checkConnected ()=0

  checkConnected
- virtual bool checkSymbols ()=0

  checkSymbols
- virtual bool checkActivationCode ()=0

  checkActivationCode
- virtual bool checkLimits ()=0

  checkLimits

17.469.1 Detailed Description

The ModelChecker is responsable for verifying the model consistency, fixing inconsistencies wheneaver possible

17.469.2 Member Function Documentation

17.469.2.1 checkActivationCode()

```
virtual bool ModelChecker_if::checkActivationCode ( )  [pure virtual]
```

checkActivationCode

Returns

Checks if the installed version has acquired a valid activation code for commercial use

Implemented in ModelCheckerDefaultImpl1.

17.469.2.2 checkAll()

```
virtual bool ModelChecker_if::checkAll ( )  [pure virtual]
```

checkAll

Returns

Invoques all other checks and returns true only if all of them returned true

Implemented in ModelCheckerDefaultImpl1.

17.469.2.3 checkConnected()

```
virtual bool ModelChecker_if::checkConnected ( )  [pure virtual]
```

checkConnected

Returns

Checks if components are consistently connected to other to form a valid process-oriented model, describing how entities proceed to the flow

Implemented in ModelCheckerDefaultImpl1.

17.469.2.4 checkLimits()

```
virtual bool ModelChecker_if::checkLimits ( )  [pure virtual]
```

checkLimits

Returns

Checks if the model (its components and elements) fits the limits based on the activation code

Implemented in ModelCheckerDefaultImpl1.

17.469.2.5 checkSymbols()

```
virtual bool ModelChecker_if::checkSymbols ( )  [pure virtual]
```

checkSymbols

Returns

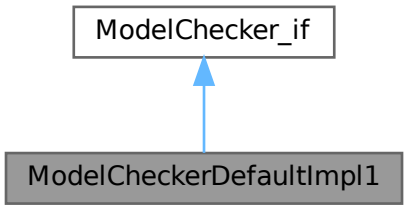
Checks if user-defined strings for symbols required by components, usually expressions or functions, are valid or references existing and valid elements.

Implemented in ModelCheckerDefaultImpl1.
The documentation for this class was generated from the following file:

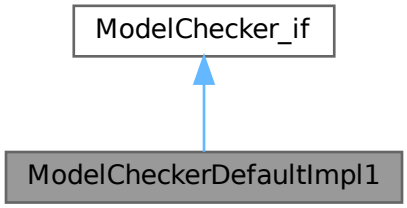
- kernel/simulator/model/ModelChecker_if.h

17.470 ModelCheckerDefaultImpl1 Class Reference

Inheritance diagram for ModelCheckerDefaultImpl1:



Collaboration diagram for ModelCheckerDefaultImpl1:



Public Member Functions

- **ModelCheckerDefaultImpl1** (Model *model)
- virtual bool checkAll () override

  checkAll
- virtual bool checkConnected () override

  checkConnected
- virtual bool checkSymbols () override

  checkSymbols
- virtual bool checkActivationCode () override

  checkActivationCode
- virtual bool checkLimits () override

  checkLimits

17.470.1 Member Function Documentation

17.470.1.1 checkActivationCode()

```
bool ModelCheckerDefaultImpl1::checkActivationCode ( )  [override], [virtual]
```

checkActivationCode

Returns

Checks if the installed version has acquired a valid activation code for commercial use

Implements ModelChecker_if.

17.470.1.2 checkAll()

```
bool ModelCheckerDefaultImpl1::checkAll ( )  [override], [virtual]
```

checkAll

Returns

Invoques all other checks and returns true only if all of them returned true

Implements ModelChecker_if.

17.470.1.3 checkConnected()

```
bool ModelCheckerDefaultImpl1::checkConnected ( )  [override], [virtual]
```

checkConnected

Returns

Checks if components are consistently connected to other to form a valid process-oriented model, describing how entities proceed to the flow

Implements ModelChecker_if.

17.470.1.4 checkLimits()

```
bool ModelCheckerDefaultImpl1::checkLimits ( )  [override], [virtual]
```

checkLimits

Returns

Checks if the model (its components and elements) fits the limits based on the activation code

Implements ModelChecker_if.

17.470.1.5 checkSymbols()

```
bool ModelCheckerDefaultImpl1::checkSymbols ( )  [override], [virtual]
```

checkSymbols

Returns

Checks if user-defined strings for symbols required by components, usually expressions or functions, are valid or references existing and valid elements.

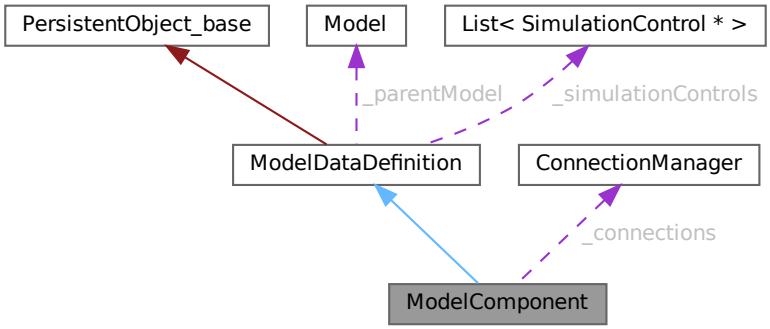
Implements ModelChecker_if.
The documentation for this class was generated from the following files:

- kernel/simulator/model/ModelCheckerDefaultImpl1.h
- kernel/simulator/model/ModelCheckerDefaultImpl1.cpp

17.471 ModelComponent Class Reference

`#include <kernel/simulator/model/ModelComponent.h>`
Inherits ModelDataDefinition.
Inherited by AIAssistant, Access, Assign, BacteriaColony, Batch, BioRunnerCommand, BioSimulate, BioStateProjectionComponent, BioSteadyState, Buffer, CellCycleCheckpointComponent, CellDivisionEvent, CellFateDecisionComponent, CellGrowthComponent, CellularAutomataComp, Clone, CompartmentExchangeComponent, ContinuousSystemComponent, CppForG, Decide, Delay, DiffEquations, DiffusionSimulate, DropOff, DummyComponent, Enter, EukaryoticCellCycleComponent, Exit, FtsZPolymerizationComponent, GeneticCircuitSimulate, GeneticExpressionStep, LSODE, Leave, MarkovChain, Match, MetabolicFluxBalance, MetabolicStateProjectionComponent, MetabolicSubmodelComponent, ModalModelDefault, Move, PathwayStressResponseComponent, PickStation, Process, PythonForG, RSimulator, Record, Release, Remove, ResourceAllocationComponent, Route, SPICECircuit, SPICENode, Search, Seize, Separate, Signal, SinkModelComponent, SourceModelComponent, Start, StochasticReactionComponent, StochasticTranscription, StochasticTranslation, Stop, Store, Submodel, Unstore, Wait, and Write.

Collaboration diagram for ModelComponent:



Public Member Functions

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component

- static bool Check (ModelComponent *component)

  Validates a component instance.

- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.

- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.

- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber)=0

  Handles a dispatched event for this component type.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Protected Attributes

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.471.1 Detailed Description

A component of the model is a block that represents a specific behavior to be simulated. The behavior is triggered when an entity arrives at the component, which corresponds to the occurrence of an event. A simulation model corresponds to a set of interconnected components to form the process by which the entity is submitted.

Parameters

model	The model this component belongs to

17.471.2 Constructor & Destructor Documentation

17.471.2.1 ModelComponent()

```
ModelComponent::ModelComponent (
            Model * model,
            std::string componentTypename,
            std::string name = "" )
```

constructors
Creates a model component and registers it in the component manager.

Parameters

model	Parent model.
componentTypename	Runtime type name for the concrete component.
name	Initial component name.

17.471.3 Member Function Documentation

17.471.3.1 _loadInstance()

```
bool ModelComponent::_loadInstance (
            PersistenceRecord * fields ) [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelDataDefinition.
Reimplemented in SinkModelComponent, SourceModelComponent, AIAssistant, MarkovChain, BacteriaColony, BioRunnerCommand, BioSimulate, BioSteadyState, GeneticCircuitSimulate, GeneticExpressionStep, MetabolicFluxBalance, ContinuousSystemComponent, DiffEquations, DiffusionSimulate, LSODE, Decide, DropOff, Remove, Search, Buffer, Clone, Delay, Process, Release, Seize, SPICECircuit, Resistor, Vsource, Vpulse, Vsine, Capacitor, Diode, PMOS, NMOS, NOT, NAND, AND, NOR, OR, XOR, XNOR, SPICENode, CppForG, PythonForG, RSimulator, Batch, Separate, Record, Write, Assign, Create, Dispose, Access, Enter, Exit, Leave, Move, PickStation, Route, Start, Stop, Store, Unstore, CellularAutomataComp, ModalModelDefault, Submodel, Match, Signal, Wait, DummyComponent, BioStateProjectionComponent, CellCycleCheckpointComponent, CellDivisionEvent, CellFateDecisionComponent, CellGrowthComponent, CompartmentExchangeComponent, EukaryoticCellCycleComponent, FtsZPolymerizationComponent, MetabolicStateProjectionComponent, MetabolicSubmodelCompone PathwayStressResponseComponent, ResourceAllocationComponent, StochasticReactionComponent, StochasticTranscription, and StochasticTranslation.

17.471.3.2 _onDispatchEvent()

```
virtual void ModelComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber ) [protected], [pure virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implemented in AIAssistant, MarkovChain, BacteriaColony, BioRunnerCommand, BioSimulate, BioSteadyState,

GeneticCircuitSimulate, GeneticExpressionStep, MetabolicFluxBalance, ContinuousSystemComponent, DiffEquations, DiffusionSimulate, LSODE, Decide, DropOff, Remove, Search, Buffer, Clone, Delay, Process, Release, Seize, SPICECircuit, SPICENode, CppForG, PythonForG, RSimulator, Batch, Separate, Record, Write, Assign, Create, Dispose, Access, Enter, Exit, Leave, Move, PickStation, Route, Start, Stop, Store, Unstore, CellularAutomataComp, ModalModelDefault, Submodel, Match, Signal, Wait, DummyComponent, BioStateProjectionComponent, CellCycleCheckpointComponent, CellDivisionEvent, CellFateDecisionComponent, CellGrowthComponent, CompartmentExchangeComponent, EukaryoticCellCycleComponent, FtsZPolymerizationComponent, MetabolicStateProjectionComp MetabolicSubmodelComponent, PathwayStressResponseComponent, ResourceAllocationComponent, StochasticReactionCompone StochasticTranscription, StochasticTranslation, RecordingContinuousSystemComponent, RecordingContinuousSystemComponent, and RecordingLSODE.

17.471.3.3 _saveInstance()

```
void ModelComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.
Reimplemented in SinkModelComponent, SourceModelComponent, AIAssistant, MarkovChain, BacteriaColony, BioRunnerCommand, BioSimulate, BioSteadyState, GeneticCircuitSimulate, GeneticExpressionStep, MetabolicFluxBalance, ContinuousSystemComponent, DiffEquations, DiffusionSimulate, LSODE, Decide, DropOff, Remove, Search, Buffer, Clone, Delay, Process, Release, Seize, SPICECircuit, Resistor, Vsource, Vpulse, Vsine, Capacitor, Diode, PMOS, NMOS, NOT, NAND, AND, NOR, OR, XOR, XNOR, SPICENode, CppForG, PythonForG, RSimulator, Batch, Separate, Record, Write, Assign, Create, Dispose, Access, Enter, Exit, Leave, Move, PickStation, Route, Start, Stop, Store, Unstore, CellularAutomataComp, ModalModelDefault, Submodel, Match, Signal, Wait, DummyComponent, BioStateProjectionComponent, CellCycleCheckpointComponent, CellDivisionEvent, CellFateDecisionComponent, CellGrowthComponent, CompartmentExchangeComponent, EukaryoticCellCycleComponent, FtsZPolymerizationComponent, MetabolicStateProjectionComponent, MetabolicSubmodelCompone PathwayStressResponseComponent, ResourceAllocationComponent, StochasticReactionComponent, StochasticTranscription, and StochasticTranslation.

17.471.3.4 Check()

```
bool ModelComponent::Check (
            ModelComponent * component )  [static]
```

Validates a component instance.

Parameters

component	Component to validate.

Returns

`true` when the component is valid.

17.471.3.5 connectTo()

```
void ModelComponent::connectTo (
            ModelComponent * component,
            unsigned int inputPortNumber = 0 )
```

Connects this component to another component.

Parameters

component	Target component.
inputPortNumber	Input port number of the target component.

17.471.3.6 CreateInternalData()

```
void ModelComponent::CreateInternalData (
            ModelComponent * component ) [static]
```

Creates internal data required by a component instance.

Parameters

component	Component whose related data should be created.

17.471.3.7 DispatchEvent()

```
void ModelComponent::DispatchEvent (
            Event * event ) [static]
```

Dispatches a simulation event to the target component.
This method triggers the simulation behavior of the component. It is invoked when an event corresponding to this component is taken from the future events list or when an entity arrives through a connection.

Parameters

event	Event to process.

17.471.3.8 getConnectionManager()

```
ConnectionManager * ModelComponent::getConnectionManager ( ) const
```

new public user methods for this component
Returns the connection manager owned by this component.
This manager stores the components directly connected to the output. Usually a component has a single output, but it may have none (such as Dispose) or more than one (such as Decide). Along with each connected component, the manager stores the input port number where the entity will be sent. Usually components have a single input, but they may have none (such as Create) or more than one (such as Match).

Returns

Connection manager that stores outgoing connections.

17.471.3.9 getDescription()

```
std::string ModelComponent::getDescription ( ) const
```

Returns the free-form description associated with this component.

Returns

Description text.

17.471.3.10 hasBreakpointAt()

```
bool ModelComponent::hasBreakpointAt ( )
```

Indicates whether a simulation breakpoint is set on this component.

Returns

`true` when the simulation breakpoint list contains this component.

17.471.3.11 LoadInstance()

```
static ModelComponent * ModelComponent::LoadInstance (
            Model * model,
            PersistenceRecord * fields )  [static]
```

static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
Loads a component instance from serialized fields.

Parameters

model	Parent model that will own the component.
fields	Serialized fields for the component.

Returns

Newly created component instance.

17.471.3.12 remove()

```
void ModelComponent::remove (
            ModelComponent * comp )
```

Removes a connected component from this component.

Parameters

comp	Component to remove from the outgoing connection set.

17.471.3.13 SaveInstance()

```
void ModelComponent::SaveInstance (
            PersistenceRecord * fields,
            ModelComponent * component )  [static]
```

Serializes a component instance into a persistence record.

Parameters

fields	Output persistence record.
component	Component to serialize.

17.471.3.14 setDescription()

```
void ModelComponent::setDescription (
            std::string _description )
```

Updates the free-form description shown by the GUI.

Parameters

_description	New description text.

17.471.3.15 show()

```
std::string ModelComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelDataDefinition.
Reimplemented in SourceModelComponent, AIAssistant, MarkovChain, BacteriaColony, BioRunnerCommand, BioSimulate, BioSteadyState, GeneticCircuitSimulate, GeneticExpressionStep, MetabolicFluxBalance, ContinuousSystemComponent, DiffEquations, DiffusionSimulate, LSODE, Decide, DropOff, Remove, Search, Buffer, Clone, Delay, Process, Release, Seize, SPICECircuit, SPICENode, CppForG, PythonForG, RSimulator, Batch, Separate, Record, Write, Assign, Create, Dispose, Access, Enter, Exit, Leave, Move, PickStation, Route, Start, Stop, Store, Unstore, CellularAutomataComp, ModalModelDefault, Submodel, Match, Signal, Wait, DummyComponent, BioStateProjectionComponent, CellCycleCheckpointComponent, CellDivisionEvent, CellFateDecisionComponent, CellGrowthComponent, CompartmentExchangeComponent, EukaryoticCellCycleComponent, FtsZPolymerizationComponent, MetabolicStateProjectionComponent, MetabolicSubmodelComponent, PathwayStressResponseComponent, ResourceAllocationComponent, StochasticReactionComponent, StochasticTranscription, and StochasticTranslation.

17.471.4 Member Data Documentation

17.471.4.1 _connections

```
ConnectionManager* ModelComponent::_connections = new ConnectionManager()  [protected]
```

virtual protected methods that could be overriden by derived classes, if needed
This method is called by ModelChecker during model check. The component should check itself to verify if user parameters are ok (ex: correct syntax for the parser) and everithing in its parameters allow the model too run without errors in this component
This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
This method is called by ModelSimulation when initianting the replication. The model should set all value for a new replication (Ex: setting back to 0 any internal counter, clearing lists, etc.
Legacy compatibility hook. Prefer the granular create* methods and the composed association manager.
This method is not used yet. It should be usefull for new UIs
The documentation for this class was generated from the following files:

- kernel/simulator/model/ModelComponent.h
- kernel/simulator/model/ModelComponent.cpp

17.472 ModelComponentManager Class Reference

Owns and organizes the set of `ModelComponent` objects in a model.
`#include <kernel/simulator/model/ModelComponentManager.h>`

Public Member Functions

- **ModelComponentManager** (Model *model)
  Creates a component manager attached to a model.
- bool insert (ModelComponent *comp)
  insert
- void remove (ModelComponent *comp)
  remove
- ModelComponent * find (std::string name)
  find
- ModelComponent * find (Util::identification id)

find

- void clear ()

  clear
- unsigned int getNumberOfComponents ()

  getNumberOfComponents
- std::list< ModelComponent * >::iterator begin ()

  begin
- std::list< ModelComponent * >::iterator end ()

  end
- ModelComponent * front ()

  front
- ModelComponent * next ()

  next
- bool hasChanged () const

  hasChanged
- void setHasChanged (bool _hasChanged)

  setHasChanged
- std::list< SourceModelComponent * > * getSourceComponents ()

  getSourceComponents
- std::list< ModelComponent * > * getTransferInComponents ()

  getTransferInComponents
- std::list< ModelComponent * > * getAllComponents () const

  getAllComponents

17.472.1 Detailed Description

Owns and organizes the set of `ModelComponent` objects in a model.
Provides insertion/removal, lookup and iteration helpers while tracking structural changes relevant to model validation/persistence workflows.

17.472.2 Member Function Documentation

17.472.2.1 begin()

`std::list< ModelComponent * >::iterator ModelComponentManager::begin ( )`
begin

Returns

Iterator to the first component.

17.472.2.2 clear()

`void ModelComponentManager::clear ( )`
clear
Removes all managed components from the model.

17.472.2.3 end()

`std::list< ModelComponent * >::iterator ModelComponentManager::end ( )`
end

Returns

Iterator past the last component.

17.472.2.4 find() [1/2]

```
ModelComponent * ModelComponentManager::find (
            std::string name )
```

find

Parameters

name	

Returns

Pointer to the component, or `nullptr` when not found.

17.472.2.5 find() [2/2]

```
ModelComponent * ModelComponentManager::find (
            Util::identification id )
```

find

Parameters

id	

Returns

Pointer to the component, or `nullptr` when not found.

17.472.2.6 front()

```
ModelComponent * ModelComponentManager::front ( )
```

front

Returns

First component in iteration order.

17.472.2.7 getAllComponents()

```
std::list< ModelComponent * > * ModelComponentManager::getAllComponents ( ) const
```

getAllComponents

Returns

17.472.2.8 getNumberOfComponents()

```
unsigned int ModelComponentManager::getNumberOfComponents ( )
```

getNumberOfComponents

Returns

Number of managed components.

17.472.2.9 getSourceComponents()

```
std::list< SourceModelComponent * > * ModelComponentManager::getSourceComponents ( )
```

getSourceComponents

Returns

17.472.2.10 getTransferInComponents()

```
std::list< ModelComponent * > * ModelComponentManager::getTransferInComponents ( )
```

getTransferInComponents

Returns

17.472.2.11 hasChanged()

```
bool ModelComponentManager::hasChanged ( ) const
```

hasChanged

Returns

17.472.2.12 insert()

```
bool ModelComponentManager::insert (
            ModelComponent * comp )
```

insert

Parameters

comp	

Returns

True when insertion succeeds.

17.472.2.13 next()

```
ModelComponent * ModelComponentManager::next ( )
```

next

Returns

Next component in iteration order.

17.472.2.14 remove()

```
void ModelComponentManager::remove (
            ModelComponent * comp )
```

remove

Parameters

comp	

17.472.2.15 setHasChanged()

```
void ModelComponentManager::setHasChanged (
            bool _hasChanged )
```

setHasChanged

Parameters

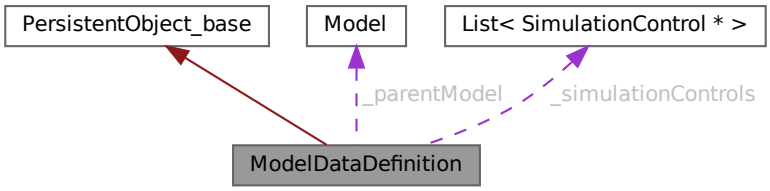
_hasChanged	

The documentation for this class was generated from the following files:

- kernel/simulator/model/ModelComponentManager.h
- kernel/simulator/model/ModelComponentManager.cpp

17.473 ModelDataDefinition Class Reference

`#include <kernel/simulator/model/ModelDataDefinition.h>`
Inherits PersistentObject_base.
Inherited by AISupport, Attribute, BacteriaSignalGrid, BioCompartment, BioNetwork, BioParameter, BioReaction, BioSimulatorRunner, BioSpecies, CPNArc, Conveyor, Counter, CppCompiler, DefaultNetwork, DefaultNode, DiffusionField, Distance, DummyElement, Entity, EntityGroup, EntityType, Failure, File, Formula, GeneticCircuit, GeneticCircuitPart, GeneticRegulation, GraphEdge, GroProgram, Label, MetabolicNetwork, MetabolicReaction, ModelComponent, MolecularSpecies, ODESolver, OLD_ODEelement, OctaveRunner, PythonRuntime, Queue, RSimulatorRunner, Resource, SPICERunner, Schedule, Segment, Sequence, Set, SignalData, Station, StatisticsCollector, StochasticReactionRule, Storage, Transporter, and WholeCellState.
Collaboration diagram for ModelDataDefinition:



Public Member Functions

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.
- virtual std::string show ()

  Returns a textual representation of this data definition.

Static Public Member Functions

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- bool _getSaveDefaultsOption ()
- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes

- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes

- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()

- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Protected Attributes

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

Friends

- class **ModelDataDefinitionAssociations**

17.473.1 Detailed Description

This class is the basis for any modeldatum of the model (such as Queue, Resource, Variable, etc.) and also for any component of the model. It has the infrastructure to read and write on file and to verify symbols.

17.473.2 Constructor & Destructor Documentation

17.473.2.1 ModelDataDefinition()

```
ModelDataDefinition::ModelDataDefinition (
            Model * model,
            std::string datadefinitionTypename,
            std::string name = "",
            bool insertIntoModel = true )
```

Creates a model data definition and optionally inserts it into the model.

Parameters

model	Parent model that owns this data definition.
datadefinitionTypename	Runtime type name for the concrete data definition.
name	Initial object name.
insertIntoModel	If `true`, the new instance is registered in the model immediately.

17.473.3 Member Function Documentation

17.473.3.1 _addSimulationControl()

```
void ModelDataDefinition::_addSimulationControl (
            SimulationControl * control )  [protected], [virtual]
```

Reimplemented in Buffer, PickStation, and Schedule.

17.473.3.2 _check()

```
bool ModelDataDefinition::_check (
            std::string & errorMessage )  [protected], [virtual]
```

could be overriden by derived classes

Reimplemented in Attribute, Counter, Entity, StatisticsCollector, SinkModelComponent, SourceModelComponent, AIAssistant, MarkovChain, BacteriaColony, BioRunnerCommand, BioSimulate, BioSteadyState, GeneticCircuitSimulate, GeneticExpressionStep, MetabolicFluxBalance, ContinuousSystemComponent, DiffEquations, DiffusionSimulate, LSODE, OLD_ODEelement, Decide, DropOff, Remove, Search, Buffer, Clone, Delay, Process, Release, Seize, CppForG, PythonForG, RSimulator, Batch, Separate, Record, Write, Assign, Create, Dispose, Access, Enter, Exit, Leave, Move, PickStation, Route, Start, Stop, Store, Unstore, ModalModelDefault, Submodel, Match, Signal, Wait, BioStateProjectionComponent, CellCycleCheckpointComponent, CellDivisionEvent, CellFateDecisionComponent, CellGrowthComponent, CompartmentExchangeComponent, EukaryoticCellCycleComponent, FtsZPolymerizationComponent, MetabolicStateProjectionComponent, MetabolicSubmodelComponent, PathwayStressResponseComponent, ResourceAllocationComponent, StochasticReactionComponent, StochasticTranscription, StochasticTranslation, AISupport, BacteriaSignalGrid, BioNetwork, BioParameter, BioReaction, BioSimulatorRunner, BioSpecies, GeneticCircuit, GeneticCircuitPart, GeneticRegulation, GroProgram, MetabolicNetwork, MetabolicReaction, DiffusionField, ODESolver, Failure, Queue, Resource, Schedule, CppCompiler, OctaveRunner, PythonRuntime, RSimulatorRunner, SPICERunner, EntityGroup, File, Formula, Label, Set, Variable, Conveyor, Distance, Segment, Sequence, Station, Storage, Transporter, ColoredPetriNetNetwork, CPNArc, CPNTransition, DefaultNetwork, DirectedAcyclicGraphNetwork, EFSMNetwork, GraphEdge, GraphNetwork, MarkovChainNetwork, SignalData, DummyElement, BioCompartment, MolecularSpecies, StochasticReactionRule, and WholeCellState.

17.473.3.3 _createAttachedAttributes()

```
void ModelDataDefinition::_createAttachedAttributes ( )  [protected], [virtual]
```

Reimplemented in Formula.

17.473.3.4 _createEditableDataDefinitions()

```
void ModelDataDefinition::_createEditableDataDefinitions ( )  [protected], [virtual]
```

Reimplemented in DiffEquations, Remove, Search, RSimulator, Assign, and BioNetwork.

17.473.3.5 _createInternalAndAttachedData()

```
void ModelDataDefinition::_createInternalAndAttachedData ( )  [protected], [virtual]
```

Legacy compatibility hook. Prefer the granular create∗ methods and the composed association manager.

17.473.3.6 _createNonEditableDataDefinitions()

```
void ModelDataDefinition::_createNonEditableDataDefinitions ( )  [protected], [virtual]
```

Reimplemented in SPICECircuit, and CppForG.

17.473.3.7 _getParserChangesInformation()

```
ParserChangesInformation * ModelDataDefinition::_getParserChangesInformation ( )  [protected],
[virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes

Reimplemented in Buffer, PickStation, Queue, Schedule, File, Set, and Storage.

17.473.3.8 _getSaveDefaultsOption()

```
bool ModelDataDefinition::_getSaveDefaultsOption ( )  [protected]
```

Returns a list of keys (names) of internal ModelDatas, cuch as Counters, StatisticsCollectors and others. Children↩ Elements are ModelDatas used by this ModelDataDefinition thar are needed before model checking

17.473.3.9 _initBetweenReplications()

```
void ModelDataDefinition::_initBetweenReplications ( )  [protected], [virtual]
```

Reimplemented in CppForG, and CellularAutomataComp.

17.473.3.10 _loadInstance()

```
bool ModelDataDefinition::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Implements PersistentObject_base.
Reimplemented in Attribute, Counter, Entity, EntityType, StatisticsCollector, ModelComponent, SinkModelComponent, SourceModelComponent, AIAssistant, MarkovChain, BacteriaColony, BioRunnerCommand, BioSimulate, BioSteadyState, GeneticCircuitSimulate, GeneticExpressionStep, MetabolicFluxBalance, ContinuousSystemComponent, DiffEquations, DiffusionSimulate, LSODE, OLD_ODEelement, Decide, DropOff, Remove, Search, Buffer, Clone, Delay, Process, Release, Seize, SPICECircuit, Resistor, Vsource, Vpulse, Vsine, Capacitor, Diode, PMOS, NMOS, NOT, NAND, AND, NOR, OR, XOR, XNOR, SPICENode, CppForG, PythonForG, RSimulator, Batch, Separate, Record, Write, Assign, Create, Dispose, Access, Enter, Exit, Leave, Move, PickStation, Route, Start, Stop, Store, Unstore, CellularAutomataComp, DefaultNode, FSMState, ModalModelDefault, PetriPlace, Submodel, Match, Signal, Wait, DummyComponent, BioStateProjectionComponent, CellCycleCheckpointComponent, CellDivisionEvent, CellFateDecisionComponent, CellGrowthComponent, CompartmentExchangeComponent, EukaryoticCellCycleComponent, FtsZPolymerizationComponent, MetabolicStateProjectionComponent, MetabolicSubmodelCompone PathwayStressResponseComponent, ResourceAllocationComponent, StochasticReactionComponent, StochasticTranscription, StochasticTranslation, AISupport, BacteriaSignalGrid, BioNetwork, BioParameter, BioReaction, BioSimulatorRunner, BioSpecies, GeneticCircuit, GeneticCircuitPart, GeneticRegulation, GroProgram, MetabolicNetwork, MetabolicReaction, DiffusionField, ODESolver, Failure, Queue, Resource, Schedule, CppCompiler, OctaveRunner, PythonRuntime, RSimulatorRunner, SPICERunner, EntityGroup, File, Formula, Label, Set, Variable, Conveyor, Distance, Segment, Sequence, Station, Storage, Transporter, ColoredPetriNetNetwork, CPNArc, CPNTransition, DefaultNetwork, EFSMNetwork, GraphEdge, GraphNetwork, MarkovChainNetwork, SignalData, DummyElement, BioCompartment, MolecularSpecies, StochasticReactionRule, and WholeCellState.

17.473.3.11 _saveInstance()

```
void ModelDataDefinition::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Implements PersistentObject_base.
Reimplemented in Attribute, Counter, Entity, EntityType, StatisticsCollector, ModelComponent, SinkModelComponent, SourceModelComponent, AIAssistant, MarkovChain, BacteriaColony, BioRunnerCommand, BioSimulate, BioSteadyState, GeneticCircuitSimulate, GeneticExpressionStep, MetabolicFluxBalance, ContinuousSystemComponent, DiffEquations, DiffusionSimulate, LSODE, OLD_ODEelement, Decide, DropOff, Remove, Search, Buffer, Clone, Delay, Process, Release, Seize, SPICECircuit, Resistor, Vsource, Vpulse, Vsine, Capacitor, Diode, PMOS, NMOS, NOT, NAND, AND, NOR, OR, XOR, XNOR, SPICENode, CppForG, PythonForG, RSimulator, Batch, Separate, Record, Write, Assign, Create, Dispose, Access, Enter, Exit, Leave, Move, PickStation, Route, Start, Stop, Store, Unstore, CellularAutomataComp, DefaultNode, FSMState, ModalModelDefault, PetriPlace, Submodel, Match, Signal, Wait, DummyComponent, BioStateProjectionComponent, CellCycleCheckpointComponent,

CellDivisionEvent, CellFateDecisionComponent, CellGrowthComponent, CompartmentExchangeComponent, EukaryoticCellCycleComponent, FtsZPolymerizationComponent, MetabolicStateProjectionComponent, MetabolicSubmodelCompone PathwayStressResponseComponent, ResourceAllocationComponent, StochasticReactionComponent, StochasticTranscription, StochasticTranslation, AISupport, BacteriaSignalGrid, BioNetwork, BioParameter, BioReaction, BioSimulatorRunner, BioSpecies, GeneticCircuit, GeneticCircuitPart, GeneticRegulation, GroProgram, MetabolicNetwork, MetabolicReaction, DiffusionField, ODESolver, Failure, Queue, Resource, Schedule, CppCompiler, OctaveRunner, PythonRuntime, RSimulatorRunner, SPICERunner, EntityGroup, File, Formula, Label, Set, Variable, Conveyor, Distance, Segment, Sequence, Station, Storage, Transporter, ColoredPetriNetNetwork, CPNArc, CPNTransition, DefaultNetwork, EFSMNetwork, GraphEdge, GraphNetwork, MarkovChainNetwork, SignalData, DummyElement, BioCompartment, MolecularSpecies, StochasticReactionRule, and WholeCellState.

17.473.3.12 Check()

```
bool ModelDataDefinition::Check (
            ModelDataDefinition * modeldatum,
            std::string & errorMessage )  [static]
```

Validates a data definition instance.
This helper invokes the instance _check() method to validate the object and any data it creates or references.

Parameters

modeldatum	Instance to validate.
errorMessage	Output error message when validation fails.

Returns

`true` when the instance is valid.

17.473.3.13 CreateInternalData()

```
void ModelDataDefinition::CreateInternalData (
            ModelDataDefinition * modeldatum )  [static]
```

Creates internal and attached data required by this instance.
This helper invokes the protected _check() path of the instance, which creates internal ModelDataDefinitions and other related data such as attributes or variables.

Parameters

modeldatum	Instance whose associated data should be created.

17.473.3.14 CreateRelatedDataElements()

```
bool ModelDataDefinition::CreateRelatedDataElements (
            ModelDataDefinition * modeldatum,
            std::string & errorMessage )  [static]
```

Creates related data and validates the instance in a single call.
Materializes related data and runs local checks for the instance in a single step.

Parameters

modeldatum	Instance whose related data should be materialized.
errorMessage	Output error message when validation fails.

Returns

`true` when creation and validation succeed.

17.473.3.15 defineTraceLevelSpecific()

```
void ModelDataDefinition::defineTraceLevelSpecific (
            TraceManager::Level traceLevelspecific,
            bool traceLevelSpecificEnabled = true )
```

Defines the specific trace level and optionally enables it immediately.

Parameters

traceLevelspecific	New trace level.
traceLevelSpecificEnabled	Whether the specific trace level starts enabled.

17.473.3.16 getAttachedData()

```
std::map< std::string, ModelDataDefinition * > * ModelDataDefinition::getAttachedData ( )
const
```

Returns the full map of attached data definitions.

Returns

Map of attached data indexed by name.

17.473.3.17 getClassname()

```
const std::string & ModelDataDefinition::getClassname ( ) const
```

Returns the runtime class name used by the plugin system.

Returns

Concrete class name.

17.473.3.18 getId()

```
Util::identification ModelDataDefinition::getId ( ) const
```

Returns the unique identifier of this instance.

Returns

Instance identifier.

17.473.3.19 getInternalData() [1/2]

```
std::map< std::string, ModelDataDefinition * > * ModelDataDefinition::getInternalData ( )
const
```

Returns the full map of internal child data definitions.

Returns

Map of internal children indexed by name.

17.473.3.20 getInternalData() [2/2]

```
ModelDataDefinition * ModelDataDefinition::getInternalData (
            const std::string & name ) const
```

Looks up a named internal child data definition.

Parameters

name	Child name to search for.

Returns

Matching internal child, or `nullptr` when not found.

17.473.3.21 getLabel()

`const std::string & ModelDataDefinition::getLabel ( ) const`
Returns the user-visible label of this instance.

Returns

Instance label.

17.473.3.22 getLevel()

`unsigned int ModelDataDefinition::getLevel ( ) const`
Returns the hierarchical editing context containing this element.
Zero denotes the root model view. A nonzero value identifies the ModelDataDefinition that owns the editing context; a ModelComponent may identify a context because it derives from ModelDataDefinition. Context membership does not transfer object ownership or imply Entity flow.

Returns

Current editing-context identifier (not an ordinal depth).

17.473.3.23 getName()

`const std::string & ModelDataDefinition::getName ( ) const`
Returns the logical name of this instance.

Returns

Instance name.

17.473.3.24 getParentModel()

`Model * ModelDataDefinition::getParentModel ( ) const`
Returns the parent model that owns this definition.

Returns

Parent model pointer, or `nullptr` when detached.

17.473.3.25 getSimulationControls()

`List< SimulationControl * > * ModelDataDefinition::getSimulationControls ( ) const`
Returns the simulation controls owned by this data definition.
Preferred API for writable controls owned by this model data definition.

Returns

Writable control list managed by the definition.

17.473.3.26 getTraceLevelSpecific()

```
TraceManager::Level ModelDataDefinition::getTraceLevelSpecific ( ) const
```

Returns the trace level that applies specifically to this definition.

Returns

Explicit trace level for this instance.

17.473.3.27 hasChanged()

```
bool ModelDataDefinition::hasChanged ( ) const
```

Indicates whether this instance has been modified since the last reset.

Returns

`true` when the object has pending changes.

17.473.3.28 InitBetweenReplications()

```
void ModelDataDefinition::InitBetweenReplications (
            ModelDataDefinition * modeldatum )  [static]
```

Resets simulation-time state between replications.

Parameters

modeldatum	Instance to reset.

17.473.3.29 isReportStatistics()

```
bool ModelDataDefinition::isReportStatistics ( ) const
```

Indicates whether this data definition contributes report statistics.

Returns

`true` when this definition emits simulation statistics.

17.473.3.30 isTraceLevelSpecificEnabled()

```
bool ModelDataDefinition::isTraceLevelSpecificEnabled ( ) const
```

Indicates whether the specific trace level is enabled.

Returns

`true` when the instance-specific trace level is active.

17.473.3.31 LoadInstance()

```
ModelDataDefinition * ModelDataDefinition::LoadInstance (
            Model * model,
            PersistenceRecord * fields,
            bool insertIntoModel )  [static]
```

Loads a model data definition instance from serialized fields.

This helper receives a map of fields read from a file or other source, creates the instance, and invokes its protected _loadInstance() method. The instance can be automatically inserted into the simulation model when requested.

Parameters

model	Parent model that will own the loaded instance.
fields	Serialized fields to deserialize.
insertIntoModel	If `true`, registers the loaded instance in the model.

Returns

Newly created instance when loading succeeds.

17.473.3.32 NewInstance()

```
ModelDataDefinition * ModelDataDefinition::NewInstance (
            Model * model,
            std::string name = "" )  [static]
```

Creates a new instance of a data definition using the plugin registry.
This helper invokes the constructor and returns a new instance that usually requires a cast to the correct subclass. It is used when plugins are connected through dynamically loaded libraries.

Parameters

model	Parent model that will own the instance.
name	Initial object name.

Returns

Newly created data definition instance.

17.473.3.33 SaveInstance()

```
void ModelDataDefinition::SaveInstance (
            PersistenceRecord * fields,
            ModelDataDefinition * modeldatum )  [static]
```

Serializes a data definition instance into a persistence record.
This helper calls the instance _saveInstance() method and returns a field map that can be saved to a file or other destination.

Parameters

fields	Output record that receives the serialized fields.
modeldatum	Instance to serialize.

17.473.3.34 setHasChanged()

```
void ModelDataDefinition::setHasChanged (
            bool hasChanged )
```

Updates the changed flag for this data definition.

Parameters

hasChanged	New changed-state value.

17.473.3.35 setLabel()

```
void ModelDataDefinition::setLabel (
            const std::string & label )
```

Updates the user-visible label of this instance.

Parameters

label	New label.

17.473.3.36 setModelLevel()

```
void ModelDataDefinition::setModelLevel (
            unsigned int _modelLevel )
```

Sets this element's hierarchical editing context.
Use zero for the root context, or the ID of the containing ModelDataDefinition / ModelComponent. This API only updates the context key: the caller remains responsible for domain ownership and persistence.

Parameters

_modelLevel	Editing-context identifier.

17.473.3.37 setName()

```
void ModelDataDefinition::setName (
            const std::string & name )
```

Updates the logical name of this instance.

Parameters

name	New name.

17.473.3.38 setReportStatistics()

```
void ModelDataDefinition::setReportStatistics (
            bool reportStatistics )
```

Enables or disables report-statistics generation for this definition.

Parameters

reportStatistics	New report-statistics flag.

17.473.3.39 setTraceLevelSpecific()

```
void ModelDataDefinition::setTraceLevelSpecific (
            TraceManager::Level level )
```

Updates the trace level used by this definition.

Parameters

level	New trace level.

17.473.3.40 setTraceLevelSpecificEnabled()

```
void ModelDataDefinition::setTraceLevelSpecificEnabled (
            bool traceLevelSpecificEnabled )
```

Enables or disables the specific trace level.

Parameters

traceLevelSpecificEnabled	New enabled-state value.

17.473.3.41 show()

```
std::string ModelDataDefinition::show ( )  [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented in Attribute, Counter, Entity, EntityType, StatisticsCollector, ModelComponent, SourceModelComponent, AIAssistant, MarkovChain, BacteriaColony, BioRunnerCommand, BioSimulate, BioSteadyState, GeneticCircuitSimulate, GeneticExpressionStep, MetabolicFluxBalance, ContinuousSystemComponent, DiffEquations, DiffusionSimulate, LSODE, OLD_ODEelement, Decide, DropOff, Remove, Search, Buffer, Clone, Delay, Process, Release, Seize, SPICECircuit, SPICENode, CppForG, PythonForG, RSimulator, Batch, Separate, Record, Write, Assign, Create, Dispose, Access, Enter, Exit, Leave, Move, PickStation, Route, Start, Stop, Store, Unstore, CellularAutomataComp, DefaultNode, ModalModelDefault, Submodel, Match, Signal, Wait, DummyComponent, BioStateProjectionComponent, CellCycleCheckpointComponent, CellDivisionEvent, CellFateDecisionComponent, CellGrowthComponent, CompartmentExchangeComponent, EukaryoticCellCycleComponent, FtsZPolymerizationComponent, MetabolicStateProjectionComponent, MetabolicSubmodelComponent, PathwayStressResponseComponent, ResourceAllocationComponent, StochasticReactionComponent, StochasticTranscription, StochasticTranslation, AISupport, BacteriaSignalGrid, BioNetwork, BioParameter, BioReaction, BioSimulatorRunner, BioSpecies, GeneticCircuit, GeneticCircuitPart, GeneticRegulation, GroProgram, MetabolicNetwork, MetabolicReaction, DiffusionField, ODESolver, Failure, Queue, Resource, Schedule, CppCompiler, OctaveRunner, PythonRuntime, RSimulatorRunner, SPICERunner, EntityGroup, File, Formula, Label, Set, Variable, Conveyor, Distance, Segment, Sequence, Station, Storage, Transporter, ColoredPetriNetNetwork, CPNArc, CPNTransition, DefaultNetwork, EFSMNetwork, GraphEdge, GraphNetwork, MarkovChainNetwork, SignalData, DummyElement, BioCompartment, MolecularSpecies, StochasticReactionRule, and WholeCellState.

17.473.3.42 traceSimulation()

```
void ModelDataDefinition::traceSimulation (
            void * thisobject,
            const std::string & text,
            TraceManager::Level level = TraceManager::Level::L8_detailed )  [protected]
```

Traces text to the simulation output.
Used only while a simulation is running, for example when components or data elements need to inform something.

Parameters

thisobject	Caller object used for trace attribution.
text	Trace text to emit.
level	Trace severity level.

The documentation for this class was generated from the following files:

- kernel/simulator/model/ModelDataDefinition.h
- kernel/simulator/model/ModelDataDefinition.cpp
- tests/unit/test_simulator_support.cpp

17.474 ModelDataDefinitionAssociations Class Reference

```
#include <kernel/simulator/model/ModelDataDefinitionAssociations.h>
```

Public Member Functions

- **ModelDataDefinitionAssociations** (ModelDataDefinition *owner)
- std::map< std::string, ModelDataDefinition * > * **getInternalData** () const

- std::map< std::string, ModelDataDefinition * > * **getAttachedData** () const
- ModelDataDefinition * **getInternalData** (const std::string &key) const
- void **clearAll** ()
- void **clearInternalData** ()
- void **clearAttachedData** ()
- void **internalDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **internalDataRemove** (const std::string &key)
- void **attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **attachedDataRemove** (const std::string &key)
- void **attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **checkCreateAttachedReferencedDataDefinition** (const std::string &expression)
- void **createInternalAndAttachedData** ()
- void **createInternalStatisticReporters** ()
- void **createAttachedAttributes** ()
- void **createNonEditableDataDefinitions** ()
- void **createEditableDataDefinitions** ()
- void **insertStatisticReporter** (const std::string &key, ModelDataDefinition *data)
- void **removeStatisticReporter** (const std::string &key)
- void **clearStatisticReporters** ()
- void **insertMandatoryAttachedAttribute** (const std::string &key, ModelDataDefinition *data)
- void **removeMandatoryAttachedAttribute** (const std::string &key)
- void **clearMandatoryAttachedAttributes** ()
- void **insertMandatoryEditableDataDefinition** (const std::string &key, ModelDataDefinition *data)
- void **removeMandatoryEditableDataDefinition** (const std::string &key)
- void **clearMandatoryEditableDataDefinitions** ()
- void **insertOptionalEditableDataDefinition** (const std::string &key, ModelDataDefinition *data)
- void **removeOptionalEditableDataDefinition** (const std::string &key)
- void **clearOptionalEditableDataDefinitions** ()
- void **insertMandatoryNonEditableDataDefinition** (const std::string &key, ModelDataDefinition *data)
- void **removeMandatoryNonEditableDataDefinition** (const std::string &key)
- void **clearMandatoryNonEditableDataDefinitions** ()
- void **insertOptionalNonEditableDataDefinition** (const std::string &key, ModelDataDefinition *data)
- void **removeOptionalNonEditableDataDefinition** (const std::string &key)
- void **clearOptionalNonEditableDataDefinitions** ()

17.474.1 Detailed Description

Owns the internal/attached registries and the new lifecycle buckets for a ModelDataDefinition.
The class starts as a compatibility layer around the current internal/attached storage so the kernel can move bucket by bucket without breaking direct callers that still inspect the legacy aggregate maps.
The documentation for this class was generated from the following files:

- kernel/simulator/model/ModelDataDefinitionAssociations.h
- kernel/simulator/model/ModelDataDefinitionAssociations.cpp

17.475 ModelDataManager Class Reference

Central registry for `ModelDataDefinition` instances owned by a model.
`#include <kernel/simulator/model/ModelDataManager.h>`

Public Member Functions

- **ModelDataManager** (Model *model)

  Creates a data manager attached to `model`.
- bool insert (ModelDataDefinition *anElement)

  insert
- void remove (ModelDataDefinition *anElement)

  remove
- bool insert (std::string datadefinitionTypename, ModelDataDefinition *anElement)

  insert
- void remove (std::string datadefinitionTypename, ModelDataDefinition *anElement)

  remove
- bool check (std::string datadefinitionTypename, ModelDataDefinition *anElement, std::string expressionName, std::string &errorMessage)

  check
- bool check (std::string datadefinitionTypename, std::string elementName, std::string expressionName, bool mandatory, std::string &errorMessage)

  check
- void **clear** ()

  clear
- ModelDataDefinition * getDataDefinition (std::string datadefinitionTypename, Util::identification id)

  getDataDefinition
- ModelDataDefinition * getDataDefinition (std::string datadefinitionTypename, std::string name)

  getDataDefinition
- unsigned int getNumberOfDataDefinitions (std::string datadefinitionTypename)

  getNumberOfDataDefinitions
- unsigned int getNumberOfDataDefinitions ()

  getNumberOfDataDefinitions
- int getRankOf (std::string datadefinitionTypename, std::string name)

  getRankOf
- std::list< std::string > getDataDefinitionClassnames () const

  getDataDefinitionClassnames
- List< ModelDataDefinition * > * **getDataDefinitionList** (std::string datadefinitionTypename) const
- void **show** ()

  Prints a textual summary of registered model data definitions.
- Model * **getParentModel** () const

  Returns the model that owns this data manager.
- bool **hasChanged** () const

  Indicates whether managed data collections were modified.
- void **setHasChanged** (bool _hasChanged)

  Updates the internal changed flag.

17.475.1 Detailed Description

Central registry for `ModelDataDefinition` instances owned by a model.
This manager groups model data by typename, supports insertion/removal/lookup operations and offers consistency checks used by components and model validation.

17.475.2 Member Function Documentation

17.475.2.1 check() [1/2]

```
bool ModelDataManager::check (
            std::string datadefinitionTypename,
            ModelDataDefinition * anElement,
            std::string expressionName,
            std::string & errorMessage )
```

check

Parameters

datadefinitionTypename	
anElement	
expressionName	
errorMessage	

Returns

17.475.2.2 check() [2/2]

```
bool ModelDataManager::check (
            std::string datadefinitionTypename,
            std::string elementName,
            std::string expressionName,
            bool mandatory,
            std::string & errorMessage )
```

check

Parameters

datadefinitionTypename	
elementName	
expressionName	
mandatory	
errorMessage	

Returns

17.475.2.3 getDataDefinition() [1/2]

```
ModelDataDefinition * ModelDataManager::getDataDefinition (
            std::string datadefinitionTypename,
            std::string name )
```

getDataDefinition

Parameters

datadefinitionTypename	
name	

Returns

17.475.2.4 getDataDefinition() [2/2]

```
ModelDataDefinition * ModelDataManager::getDataDefinition (
            std::string datadefinitionTypename,
            Util::identification id )
```

getDataDefinition

Parameters

datadefinitionTypename	
id	

Returns

17.475.2.5 getDataDefinitionClassnames()

```
std::list< std::string > ModelDataManager::getDataDefinitionClassnames ( ) const
```

getDataDefinitionClassnames

Returns

Snapshot list of current data-definition class names returned by value.

17.475.2.6 getNumberOfDataDefinitions() [1/2]

```
unsigned int ModelDataManager::getNumberOfDataDefinitions ( )
```

getNumberOfDataDefinitions

Returns

17.475.2.7 getNumberOfDataDefinitions() [2/2]

```
unsigned int ModelDataManager::getNumberOfDataDefinitions (
            std::string datadefinitionTypename )
```

getNumberOfDataDefinitions

Parameters

datadefinitionTypename	

Returns

17.475.2.8 getRankOf()

```
int ModelDataManager::getRankOf (
            std::string datadefinitionTypename,
            std::string name )
```

getRankOf

Parameters

datadefinitionTypename	
name	

Returns

returns the position (1st position=0) of the modeldatum if found, or negative value if not found

17.475.2.9 insert() [1/2]

```
bool ModelDataManager::insert (
            ModelDataDefinition * anElement )
```

insert

Parameters

anElement	

Returns

17.475.2.10 insert() [2/2]

```
bool ModelDataManager::insert (
            std::string datadefinitionTypename,
            ModelDataDefinition * anElement )
```

insert

Parameters

datadefinitionTypename	
anElement	

Returns

Deprected

17.475.2.11 remove() [1/2]

```
void ModelDataManager::remove (
            ModelDataDefinition * anElement )
```

remove

Parameters

anElement	Deprected

17.475.2.12 remove() [2/2]

```
void ModelDataManager::remove (
            std::string datadefinitionTypename,
            ModelDataDefinition * anElement )
```

remove

Parameters

datadefinitionTypename	
anElement	

The documentation for this class was generated from the following files:

- kernel/simulator/model/ModelDataManager.h
- kernel/simulator/model/ModelDataManager.cpp

17.476 ModelEvent Class Reference

`#include <kernel/simulator/OnEventManager.h>`

Public Member Functions

- **ModelEvent** (Model *model)
- const Model * **getModel** ()

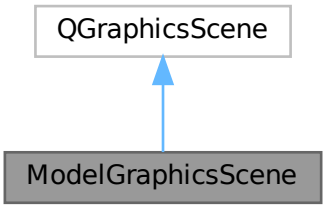
17.476.1 Detailed Description

Stores an event that happened on a specific model
The documentation for this class was generated from the following file:

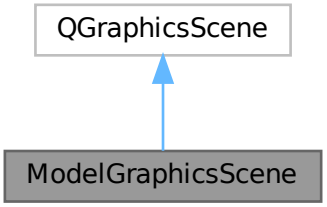
- kernel/simulator/OnEventManager.h

17.477 ModelGraphicsScene Class Reference

Scene that owns and manipulates the graphical representation of a model.
`#include <applications/gui/genesys/graphicals/ModelGraphicsScene.h>`
Inheritance diagram for ModelGraphicsScene:



Collaboration diagram for ModelGraphicsScene:



Classes

- struct GRID

Public Types

- enum **DrawingMode** {
  NONE , **LINE** , **TEXT** , **RECTANGLE** ,
  ELLIPSE , **POLYGON** , **POLYGON_POINTS** , **POLYGON_FINISHED** ,
  COUNTER , **VARIABLE** , **TIMER** , **ATTRIBUTE** ,
  ENTITY , **EVENT** , **EXPRESSION** , **PLOT** ,
  QUEUE_PLACEHOLDER , **RESOURCE** , **STATION** , **STATISTICS** }

Public Member Functions

- **ModelGraphicsScene** (qreal x, qreal y, qreal width, qreal height, QObject *parent=nullptr)
- **ModelGraphicsScene** (const ModelGraphicsScene &orig)=delete
- ModelGraphicsScene & **operator=** (const ModelGraphicsScene &other)=delete
- void **clear** ()

  Clears the scene and all formal-network projection bookkeeping.
- GraphicalModelComponent * **addGraphicalModelComponent** (Plugin *plugin, ModelComponent *component, QPointF position, QColor color=Qt::blue, bool notify=false, GraphicalModelComponent *autoConnectSource=nullptr)

  Creates and inserts one graphical component into the scene.
- GraphicalConnection * **addGraphicalConnection** (GraphicalComponentPort *sourcePort, GraphicalComponentPort *destinationPort, unsigned int portSourceConnection, unsigned int portDestinationConnection, bool notify=false)

  Creates and inserts one graphical connection between two ports.
- GraphicalModelDataDefinition * **addGraphicalModelDataDefinition** (Plugin *plugin, ModelDataDefinition *element, QPointF position, QColor color=Qt::blue)

  Creates and inserts one graphical data definition into the scene.
- GraphicalDiagramConnection * **addGraphicalDiagramConnection** (QGraphicsItem *dataDefinition, QGraphicsItem *linkedTo, GraphicalDiagramConnection::ConnectionType type)

  Creates or updates a diagram-level connection between data definitions.
- void **initializeAnimationDrawing** (QGraphicsSceneMouseEvent *mouseEvent)

  Starts the interactive drawing workflow for animation-related items.
- void **continueAnimationDrawing** (QGraphicsSceneMouseEvent *mouseEvent)

  Continues the interactive drawing workflow.
- void **finishAnimationDrawing** (QGraphicsSceneMouseEvent *mouseEvent)

Finalizes the interactive drawing workflow.

- void **startTextEditing** ()

  Opens inline text editing for the current text item.

- void **removeComponent** (GraphicalModelComponent ∗gmc, bool notify=false)

  Removes one component and optionally emits the graphical change notification.

- void **clearConnectionsComponent** (GraphicalModelComponent ∗gmc)

  Clears input and output connections attached to one component.

- void **clearInputConnectionsComponent** (GraphicalModelComponent ∗graphicalComponent)

  Removes all incoming connections from one component.

- void **clearOutputConnectionsComponent** (GraphicalModelComponent ∗graphicalComponent)

  Removes all outgoing connections from one component.

- void **clearPorts** (GraphicalConnection ∗connection, GraphicalModelComponent ∗source, GraphicalModelComponent ∗destination)

  Clears the connection ports associated with one connection before removal.

- void **connectComponents** (GraphicalConnection ∗connection, GraphicalModelComponent ∗source=nullptr, GraphicalModelComponent ∗destination=nullptr, bool notify=false)

  Connects one prepared connection to its source and destination items.

- bool **connectSource** (GraphicalConnection ∗connection, GraphicalModelComponent ∗source=nullptr)

  Binds a source component to a connection while checking port constraints.

- bool **connectDestination** (GraphicalConnection ∗connection, GraphicalModelComponent ∗destination=nullptr)

  Binds a destination component to a connection while checking port constraints.

- void **redoConnections** (GraphicalModelComponent ∗graphicalComponent, QList< GraphicalConnection ∗ > ∗inputConnections, QList< GraphicalConnection ∗ > ∗outputConnections)

  Recreates all connection links for one graphical component.

- void **removeComponentInModel** (GraphicalModelComponent ∗gmc)

  Removes one component from the kernel model as part of scene deletion.

- void **insertComponent** (GraphicalModelComponent ∗gmc, QList< GraphicalConnection ∗ > ∗input↩ Connections, QList< GraphicalConnection ∗ > ∗outputConnections, bool addGMC=true, bool addAll↩ GMC=true, bool notify=false)

  Inserts an existing component back into the scene and restores its connections.

- void **removeGraphicalConnection** (GraphicalConnection ∗graphicalConnection, GraphicalModelComponent ∗source, GraphicalModelComponent ∗destination, bool notify=false)

  Removes a graphical connection and updates the source/destination items.

- void **removeConnectionInModel** (GraphicalConnection ∗graphicalConnection, GraphicalModelComponent ∗source)

  Removes the connection from the kernel model while keeping scene state consistent.

- void **addGeometry** (QPointF endPoint, bool moving)

  Adds one freeform drawing geometry item to the scene.

- void **addDrawing** (QGraphicsItem ∗item, bool notify=false)

  Adds one drawing item to the scene and optionally notifies listeners.

- bool **addDrawingGeometry** (QGraphicsItem ∗item)

  Attempts to register one geometry drawing item.

- bool **addDrawingAnimation** (QGraphicsItem ∗item)

  Attempts to register one animation drawing item.

- void **removeGraphicalModelDataDefinition** (GraphicalModelDataDefinition ∗gmdd)

  Removes one graphical data definition from the scene.

- void **detachGraphicalModelDataDefinition** (GraphicalModelDataDefinition ∗gmdd)

  Detaches one graphical data definition without deleting the underlying model data.

- void **restoreGraphicalModelDataDefinition** (GraphicalModelDataDefinition ∗gmdd)

  Restores one previously detached graphical data definition.

- void **removeGraphicalDiagramConnection** (GraphicalDiagramConnection ∗connection)

  Removes one diagram connection from the scene.

- void **clearGraphicalModelDataDefinitions** ()

  Clears all graphical data definitions from the scene.
- void **clearGraphicalDiagramConnections** ()

  Clears all diagram connections from the scene.
- void **sanitizeGraphicalDataDefinitionsBookkeeping** ()

  Repairs bookkeeping used to track graphical data definitions.
- void **setDiagramLayerState** (bool diagramCreated, bool visible)

  Shows or hides the diagram layer and its creation state.
- QList< QGraphicsItem * > **userOperableItems** (const QList< QGraphicsItem * > &items) const

  Filters out scene items that the user cannot directly manipulate.
- QList< QGraphicsItem * > **userDeletableItems** (const QList< QGraphicsItem * > &items) const

  Filters the selected items down to items that can be deleted.
- void **ensureInitialInternalDataDefinitionGrouping** (GraphicalModelDataDefinition *dataDefinition, GraphicalModelComponent *component)

  Optionally groups newly created internal data definitions under their component.
- void **setPersistedGuiRestoreInProgress** (bool restoring)

  Marks that persisted GUI state is being restored.
- bool **isPersistedGuiRestoreInProgress** () const

  Returns whether persisted GUI state restoration is in progress.
- void **removeDrawing** (QGraphicsItem *item, bool notify=false)

  Removes one freeform drawing item from the scene.
- bool **removeDrawingGeometry** (QGraphicsItem *item)
- bool **removeDrawingAnimation** (QGraphicsItem *item)
- void **removeGroup** (QGraphicsItemGroup *group, bool notify=false)

  Removes one graphical group item from the scene.
- void **clearGraphicalModelComponents** ()

  Clears all graphical component items from the scene.
- void **clearGraphicalModelConnections** ()

  Clears all graphical connection items from the scene.
- void **groupComponents** (bool notify=false)

  Groups the current selection into a scene group when possible.
- void **groupModelComponents** (QList< GraphicalModelComponent * > *graphicalComponents, QGraphicsItemGroup *group)

  Groups the supplied graphical components into the provided group item.
- void **ungroupComponents** (bool notify=false)

  Ungroups the current selection when it contains a valid group item.
- void **ungroupModelComponents** (QGraphicsItemGroup *group)

  Ungroups one group item and restores its child components.
- void notifyGraphicalModelChange (GraphicalModelEvent::EventType eventType, GraphicalModelEvent::↩ EventObjectType eventObjectType, QGraphicsItem *item)

  Emits a scene-level change notification used by higher GUI layers.
- QList< GraphicalModelComponent * > **graphicalModelComponentItems** ()

  Returns the list of graphical component items currently in the scene.
- GraphicalModelComponent * **findGraphicalModelComponent** (Util::identification id)

  Finds one graphical component by kernel identification.
- GraphicalModelDataDefinition * **findGraphicalModelDataDefinition** (ModelDataDefinition *dataDefinition)

  Finds one graphical data definition by kernel pointer.
- void **clearEFSMNetworkDiagramItems** ()

  Removes all read-only EFSM transition graphics from this scene.
- void **addEFSMNetworkDiagramItem** (QGraphicsItem *item)

  Adds one scene-owned read-only EFSM transition graphic.

- QList< QGraphicsItem * > * **getEFSMNetworkDiagramItems** () const

  Returns the scene-owned EFSM transition graphics.
- GraphicalModelComponent * **resolveSourceComponent** (GraphicalConnection *connection) const

  Resolves the source component associated with one graphical connection.
- GraphicalModelComponent * **resolveDestinationComponent** (GraphicalConnection *connection) const

  Resolves the destination component associated with one graphical connection.
- GRID * **grid** ()

  Returns the current grid state owned by the scene.
- void **showGrid** ()

  Shows the grid using the current grid configuration.
- void **setGridVisible** (bool visible)

  Sets grid visibility without toggling the state implicitly.
- bool **isGridVisible** () const

  Returns whether the grid is currently visible.
- void **snapItemsToGrid** ()

  Snaps movable items to the active grid spacing.
- void **actualizeDiagramArrows** ()

  Recomputes the direction arrows used by diagram connections.
- void **showDiagrams** ()

  Shows all diagram-layer items.
- void **hideDiagrams** ()

  Hides all diagram-layer items.
- bool **existDiagram** ()

  Returns whether diagram-layer items exist.
- bool **visibleDiagram** ()

  Returns whether diagram-layer items are currently visible.
- void **destroyDiagram** ()

  Destroys the diagram layer and its cached items.
- void **createDiagrams** ()

  Creates the diagram layer and its cached items.
- QUndoStack * **getUndoStack** ()

  Returns the undo stack used by the scene.
- Simulator * **getSimulator** ()

  Returns the simulator bound to this scene.
- void **setUndoStack** (QUndoStack *undo)

  Replaces the undo stack used by the scene.
- void **beginConnection** ()

  Starts a new component connection gesture.
- void **setSimulator** (Simulator *simulator)

  Sets the simulator dependency used by scene logic.
- void **setPropertyEditor** (PropertyEditorGenesys *propEditor)

  Sets the property editor dependency used by scene logic.
- void **setPropertyList** (std::map< SimulationControl *, DataComponentProperty * > *propList)

  Sets the property list dependency used by scene logic.
- void **setPropertyEditorUI** (std::map< SimulationControl *, DataComponentEditor * > *propEditorUI)

  Sets the property-editor widget map used by scene logic.
- void **setComboBox** (std::map< SimulationControl *, ComboBoxEnum * > *propCombo)

  Sets the enum combo-box map used by scene logic.
- void **setObjectBeingDragged** (QTreeWidgetItem *objectBeingDragged)

  Records which tree item is currently being dragged into the scene.
- void **setRestoringPersistedGuiLayout** (bool restoring)

Marks whether the scene is restoring persisted GUI layout.

- bool **isRestoringPersistedGuiLayout** () const

  Returns whether the scene is restoring persisted GUI layout.

- void **setParentWidget** (QWidget *parentWidget)

  Sets the parent widget used by transient UI interactions.

- unsigned short **connectingStep** () const

  Returns the current step in the connection gesture.

- void **setConnectingStep** (unsigned short connectingStep)

  Updates the current step in the connection gesture.

- void **setSnapToGrid** (bool activated)

  Enables or disables grid snapping.

- bool **getSnapToGrid** ()

  Returns whether grid snapping is enabled.

- void **arranjeModels** (int direction)

  Rearranges model items using the specified layout direction.

- void **setDrawingMode** (DrawingMode drawingMode)

  Sets the current drawing mode used by mouse gestures.

- void **clearDrawingMode** ()

  Clears the current drawing mode.

- void **setGraphicalComponentPort** (GraphicalComponentPort *in)

  Sets the current port used by connection gestures.

- DrawingMode **getDrawingMode** ()

  Returns the current drawing mode.

- void **setAction** (QAction *action)

  Sets the QAction that tracks the current scene tool.

- QList< GraphicalModelComponent * > * **getAllComponents** ()

  Returns all component items owned by the scene.

- QList< GraphicalConnection * > * **getAllConnections** ()

  Returns all connection items owned by the scene.

- QList< GraphicalModelDataDefinition * > * **getAllDataDefinitions** ()

  Returns all data-definition items owned by the scene.

- QList< GraphicalDiagramConnection * > * **getAllGraphicalDiagramsConnections** ()

  Returns all diagram-connection items owned by the scene.

- QMap< QGraphicsItemGroup *, QList< GraphicalModelComponent * > > **getListComponentsGroup** ()

  Returns the mapping used to restore grouped components.

- void **insertComponentGroup** (QGraphicsItemGroup *group, QList< GraphicalModelComponent * > componentsGroup)

  Stores the component list associated with one graphical group.

- void **insertOldPositionItem** (QGraphicsItem *item, QPointF position)

  Stores the previous scene position for one item.

- QPointF **getOldPositionItem** (QGraphicsItem *item) const

  Returns the previous scene position stored for one item.

- QList< AnimationTransition * > * **getAnimationsTransition** ()

  Returns the active transition animations tracked by the scene.

- QList< AnimationCounter * > * **getAnimationsCounter** ()

  Returns the active counter animations tracked by the scene.

- QList< AnimationVariable * > * **getAnimationsVariable** ()

  Returns the active variable animations tracked by the scene.

- QList< AnimationTimer * > * **getAnimationsTimer** ()

  Returns the active timer animations tracked by the scene.

- QList< AnimationPlaceholder * > * **getAnimationsPlaceholder** ()

Returns the placeholder animation items tracked by the scene.

- QMap< Event *, QList< AnimationTransition * > * > * **getAnimationPaused** ()

  Returns paused animations indexed by their simulation event.
- bool **checkIgnoreEvent** ()

  Returns true when the scene should suppress animation-driven event handling.
- void **clearAnimations** ()

  Clears all animation categories tracked by the scene.
- void **clearAnimationsTransition** ()

  Clears transition animations tracked by the scene.
- void **clearAnimationsCounter** ()

  Clears counter animations tracked by the scene.
- void **clearAnimationsVariable** ()

  Clears variable animations tracked by the scene.
- void **clearAnimationsTimer** ()

  Clears timer animations tracked by the scene.
- void **clearAnimationsPlaceholder** ()

  Clears placeholder animations tracked by the scene.
- void **clearAnimationsQueue** ()

  Clears queue animation overlays tracked by the scene.
- void **animatePlot** ()

  Refreshes plot animation overlays for the current scene state.
- QList< QString > * **getImagesAnimation** ()

  Returns the list of animation image identifiers used by the scene.
- void **drawingCounter** ()

  Creates counter overlay geometry.
- void **drawingVariable** ()

  Creates variable overlay geometry.
- void **drawingTimer** ()

  Creates timer overlay geometry.
- void **drawingAttribute** ()

  Creates attribute overlay geometry.
- void **drawingEntity** ()

  Creates entity overlay geometry.
- void **drawingEvent** ()

  Creates event overlay geometry.
- void **drawingExpression** ()

  Creates expression overlay geometry.
- void **drawingPlot** ()

  Creates plot overlay geometry.
- void **drawingQueue** ()

  Creates queue overlay geometry.
- void **drawingResource** ()

  Creates resource overlay geometry.
- void **drawingStation** ()

  Creates station overlay geometry.
- void **drawingStatistics** ()

  Creates statistics overlay geometry.
- void **clearAnimationsValues** ()

  Clears animation-specific cached values.
- void **setCounters** ()

  Synchronizes the component counters used by animation overlays.

- void **setVariables** ()

  Synchronizes the variable counters used by animation overlays.
- void **insertRestoredDataDefinitions** (bool loaded)

  Restores data definitions removed by previous delete/check cycles.
- void **saveDataDefinitions** ()

  Caches data definitions before destructive scene operations.
- void **requestGraphicalDataDefinitionsSync** ()

  Requests an immediate data-definition synchronization pass.
- void **scheduleGraphicalDataDefinitionsSync** ()

  Schedules a deferred data-definition synchronization pass.
- bool **isGraphicalDataDefinitionsSyncInProgress** () const

  Returns whether a data-definition sync is currently in progress.
- void **setShowStatisticsDataDefinitions** (bool show)

  Shows or hides statistics-related data definitions.
- bool **showStatisticsDataDefinitions** () const

  Returns whether statistics-related data definitions are visible.
- void **setShowEditableDataDefinitions** (bool show)

  Shows or hides editable data definitions.
- bool **showEditableDataDefinitions** () const

  Returns whether editable data definitions are visible.
- void **setShowSharedDataDefinitions** (bool show)

  Shows or hides shared data definitions.
- bool **showSharedDataDefinitions** () const

  Returns whether shared data definitions are visible.
- void **setShowRecursiveDataDefinitions** (bool show)

  Shows or hides recursive data-definition expansion.
- bool **showRecursiveDataDefinitions** () const

  Returns whether recursive data-definition expansion is enabled.
- void setModelLevelFilter (unsigned int modelLevel)

  Restricts graphical rebuild/synchronization services to one kernel model level.
- void **clearModelLevelFilter** ()

  Clears any level restriction and lets rebuild services render all model levels.
- bool **hasModelLevelFilter** () const

  Returns true when this scene is scoped to a specific ModelDataDefinition level.
- unsigned int **modelLevelFilter** () const

  Returns the active level used by rebuild services when the filter is enabled.
- void **setShowInternalDataDefinitions** (bool show)

  Shows or hides internal data definitions.
- bool **showInternalDataDefinitions** () const

  Returns whether internal data definitions are visible.
- void **setShowAttachedDataDefinitions** (bool show)

  Shows or hides attached data definitions.
- bool **showAttachedDataDefinitions** () const

  Returns whether attached data definitions are visible.
- void **setConnectionGeometryUpdatesBlocked** (bool blocked)

  Temporarily blocks connection-geometry updates.
- bool **areConnectionGeometryUpdatesBlocked** () const

  Returns whether connection-geometry updates are blocked.
- QList< QGraphicsItem * > * **getGraphicalModelDataDefinitions** () const

  Returns the scene items classified as model data definitions.
- QList< QGraphicsItem * > * **getGraphicalModelComponents** () const

Returns the scene items classified as model components.

- QList< QGraphicsItem * > * **getGraphicalConnections** () const

  Returns the scene items classified as connections.

- QList< QGraphicsItem * > * **getGraphicalGeometries** () const

  Returns the scene items classified as geometry drawings.

- QList< QGraphicsItem * > * **getGraphicalAnimations** () const

  Returns the scene items classified as animations.

- QList< QGraphicsItem * > * **getGraphicalEntities** () const

  Returns the scene items classified as entities.

- QList< QGraphicsItem * > * **getGraphicalDiagramsConnections** () const

  Returns the scene items classified as diagram connections.

- QList< QGraphicsItemGroup * > * **getGraphicalGroups** () const

  Returns the groups currently tracked by the scene.

- void **animateTransition** (ModelComponent *source, ModelComponent *destination, bool viewSimulation, Event *event)

  Starts a transition animation between two model components.

- void **runAnimateTransition** (AnimationTransition *animationTransition, Event *event, bool restart=false)

  Runs one transition animation instance until completion.

- void **handleAnimationStateChanged** (QAbstractAnimation::State newState, QEventLoop *loop, Event *event, AnimationTransition *animationTransition)

  Handles animation state changes for transition-driven event loops.

- void **animateQueueInsert** (ModelComponent *component, bool visivible)

  Creates the queue-insert animation overlay for one component.

- void **animateQueueRemove** (ModelComponent *component)

  Creates the queue-remove animation overlay for one component.

- void **animateCounter** ()

  Updates the counter animation overlay.

- void **animateVariable** (Entity *entity=nullptr)

  Updates the variable animation overlay.

- void **animateTimer** (double time)

  Updates the timer animation overlay.

Protected Member Functions

- virtual void **contextMenuEvent** (QGraphicsSceneContextMenuEvent *contextMenuEvent)
- virtual void **dragEnterEvent** (QGraphicsSceneDragDropEvent *event)
- virtual void **dragLeaveEvent** (QGraphicsSceneDragDropEvent *event)
- virtual void **dragMoveEvent** (QGraphicsSceneDragDropEvent *event)
- virtual void **dropEvent** (QGraphicsSceneDragDropEvent *event)
- virtual void **focusInEvent** (QFocusEvent *focusEvent)
- virtual void **focusOutEvent** (QFocusEvent *focusEvent)
- virtual void **keyPressEvent** (QKeyEvent *keyEvent)
- virtual void **keyReleaseEvent** (QKeyEvent *keyEvent)
- virtual void **mouseDoubleClickEvent** (QGraphicsSceneMouseEvent *mouseEvent)
- virtual void **mouseMoveEvent** (QGraphicsSceneMouseEvent *mouseEvent)
- virtual void **mousePressEvent** (QGraphicsSceneMouseEvent *mouseEvent)
- virtual void **mouseReleaseEvent** (QGraphicsSceneMouseEvent *mouseEvent)
- virtual void **wheelEvent** (QGraphicsSceneWheelEvent *wheelEvent)

17.477.1 Detailed Description

Scene that owns and manipulates the graphical representation of a model.
The scene manages components, ports, connections, drawings, animations, and the auxiliary data-definition structures used by the GUI. It is the central editing surface for the graphical model and the main consumer of the view/controller services documented elsewhere.

17.477.2 Member Function Documentation

17.477.2.1 notifyGraphicalModelChange()

```
void ModelGraphicsScene::notifyGraphicalModelChange (
            GraphicalModelEvent::EventType eventType,
            GraphicalModelEvent::EventObjectType eventObjectType,
            QGraphicsItem * item )
```

Emits a scene-level change notification used by higher GUI layers.
Notifies view-level handlers about changes in graphical model.
Event object is stack-allocated and forwarded by const-reference to avoid heap ownership ambiguity.

Todo Consider broadcasting to all attached views when multi-view editing is supported.

17.477.2.2 setModelLevelFilter()

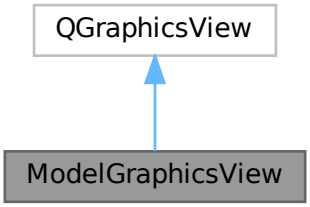
```
void ModelGraphicsScene::setModelLevelFilter (
            unsigned int modelLevel )
```

Restricts graphical rebuild/synchronization services to one kernel model level.
ModelGraphicsScene keeps this rendering context because some synchronization entry points are static services called from the property editor. Storing the active level in the scene lets those calls preserve the same root/submodel scope as the visible tab.
The documentation for this class was generated from the following files:

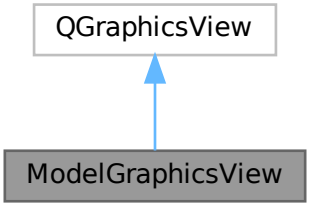
- applications/gui/genesys/graphicals/ModelGraphicsScene.h
- applications/gui/genesys/graphicals/ModelGraphicsScene.cpp

17.478 ModelGraphicsView Class Reference

Inheritance diagram for ModelGraphicsView:



Collaboration diagram for ModelGraphicsView:



Public Member Functions

- ModelGraphicsView (QWidget *parent=nullptr)

  View wrapper that hosts the model graphics scene and delegates GUI events.
- **ModelGraphicsView** (const ModelGraphicsView &orig)

  Copy constructor used by legacy compatibility paths.
- virtual ~**ModelGraphicsView** ()

  Releases the Qt view and its compatibility state.
- ModelGraphicsScene * **getScene** ()

  Returns the scene currently owned by the view.
- void **showGrid** ()

  Requests grid visibility from the underlying scene.
- void **clear** ()

  Clears the current scene contents and compatibility state.
- void **beginConnection** ()

  Starts the connection-creation workflow in the scene.
- void **selectModelComponent** (ModelComponent *component)

  Highlights the requested component in the scene and centers it when possible.
- void **setSimulator** (Simulator *simulator)

  Injects the simulator dependency used by scene callbacks.
- void **setPropertyEditor** (PropertyEditorGenesys *propEditor)

  Injects the property editor dependency used by scene callbacks.
- void **setPropertyList** (std::map< SimulationControl *, DataComponentProperty * > *propList)

  Injects the property list map used by scene/editor synchronization.
- void **setPropertyEditorUI** (std::map< SimulationControl *, DataComponentEditor * > *propEditorUI)

  Injects the property-editor widget map used by scene/editor synchronization.
- void **setComboBox** (std::map< SimulationControl *, ComboBoxEnum * > *propBox)

  Injects the enum combo-box map used by scene/editor synchronization.
- void **setEnabled** (bool enabled)

  Enables or disables interaction while keeping the scene wiring intact.
- QList< QGraphicsItem * > **selectedItems** ()

  Returns the current selected items from the underlying scene.
- template<typename Class >
  void **setSceneMouseEventHandler** (Class *object, void(Class::*function)(QGraphicsSceneMouseEvent *))
- template<typename Class >
  void **setGraphicalModelEventHandler** (Class *object, void(Class::*function)(const GraphicalModelEvent &))

- template<typename Class >
  void **setSceneWheelInEventHandler** (Class *object, void(Class::*function)())
- template<typename Class >
  void **setSceneWheelOutEventHandler** (Class *object, void(Class::*function)())
- template<typename Class >
  void **setContextMenuEventHandler** (Class *object, void(Class::*function)(QContextMenuEvent *))
- void **notifySceneMouseEventHandler** (QGraphicsSceneMouseEvent *mouseEvent)

  Notifies the registered mouse-event handler, if any.
- void **notifySceneWheelInEventHandler** ()

  Notifies the registered wheel-in handler, if any.
- void **notifySceneWheelOutEventHandler** ()

  Notifies the registered wheel-out handler, if any.
- void notifySceneGraphicalModelEventHandler (const GraphicalModelEvent &modelGraphicsEvent)

  Notifies the registered graphical-model-event handler, if any.
- void **setCanNotifyGraphicalModelEventHandlers** (bool can)

  Enables or disables graphical-model event emission from the view.
- void **clearEventHandlers** ()

  Clears all custom event handlers installed by the composition root.
- void **setParentWidget** (QWidget *parentWidget)

  Sets the widget used as parent for dialogs and transient controls.
- void **setRuleVisible** (bool visible)

  Enables or disables ruler rendering over the graphics viewport.
- bool **isRuleVisible** () const

  Returns whether ruler rendering is active.
- void **setGuidesVisible** (bool visible)

  Enables or disables guide line rendering over the graphics viewport.
- bool **isGuidesVisible** () const

  Returns whether guide line rendering is active.

Protected Member Functions

- void **changed** (const QList< QRectF > &region)
- void **focusItemChanged** (QGraphicsItem *newFocusItem, QGraphicsItem *oldFocusItem, Qt::FocusReason reason)
- void **sceneRectChanged** (const QRectF &rect)
- void **selectionChanged** ()
- virtual void **contextMenuEvent** (QContextMenuEvent *event) override
- virtual void **dragEnterEvent** (QDragEnterEvent *event) override
- virtual void **keyPressEvent** (QKeyEvent *event) override
- virtual void **keyReleaseEvent** (QKeyEvent *event) override
- virtual void **wheelEvent** (QWheelEvent *event) override
- virtual void **drawForeground** (QPainter *painter, const QRectF &rect) override

17.478.1 Constructor & Destructor Documentation

17.478.1.1 ModelGraphicsView()

```
ModelGraphicsView::ModelGraphicsView (
            QWidget * parent = nullptr )
```

View wrapper that hosts the model graphics scene and delegates GUI events.
The view keeps the scene-specific input handlers outside the raw Qt widget layer so the MainWindow composition root can wire context menus, wheel behavior, and selection changes through injected callbacks.

17.478.2 Member Function Documentation

17.478.2.1 notifySceneGraphicalModelEventHandler()

```
void ModelGraphicsView::notifySceneGraphicalModelEventHandler (
            const GraphicalModelEvent & modelGraphicsEvent )
```

Notifies the registered graphical-model-event handler, if any.
Dispatches graphical model change event to registered callback.

Parameters

modelGraphicsEvent	Event payload from scene.

Todo Replace callback with signal/slot to improve composability.

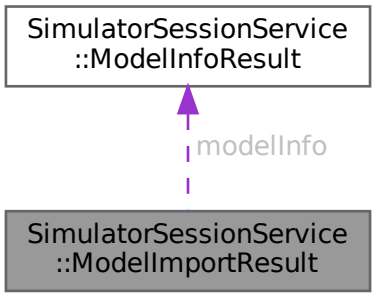
Todo actualize property editor?

The documentation for this class was generated from the following files:

- applications/gui/genesys/graphicals/ModelGraphicsView.h
- applications/gui/genesys/graphicals/ModelGraphicsView.cpp

17.479 SimulatorSessionService::ModelImportResult Struct Reference

Contains model import output and error state.
`#include <applications/worker/service/SimulatorSessionService.h>`
Collaboration diagram for SimulatorSessionService::ModelImportResult:



Public Attributes

- bool **success** = false
- ModelImportError **error** = ModelImportError::None
- ModelInfoResult **modelInfo**

17.479.1 Detailed Description

Contains model import output and error state.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.480 ModelInfo Class Reference

`#include <kernel/simulator/model/ModelInfo.h>`

Public Member Functions

- std::string **show** ()
- void setName (std::string _name)

  setName
- std::string getName () const

  getName
- void setAnalystName (std::string _analystName)

  setAnalystName
- std::string getAnalystName () const

  getAnalystName
- void setDescription (std::string _description)

  setDescription
- std::string getDescription () const

  getDescription
- void setProjectTitle (std::string _projectTitle)

  setProjectTitle
- std::string getProjectTitle () const

  getProjectTitle
- void setVersion (std::string _version)

  setVersion
- std::string getVersion () const

  getVersion
- void loadInstance (PersistenceRecord *fields)

  loadInstance
- void saveInstance (PersistenceRecord *fields)

  saveInstance
- bool hasChanged () const

  hasChanged
- void **setHasChanged** (bool hasChanged)

  Updates the changed flag for model metadata.

17.480.1 Detailed Description

ModelInfo stores basic model project information, such as the project and analyst names.

17.480.2 Member Function Documentation

17.480.2.1 getAnalystName()

`std::string ModelInfo::getAnalystName ( ) const`

getAnalystName

Returns

17.480.2.2 getDescription()

```
std::string ModelInfo::getDescription ( ) const
```

getDescription

Returns

17.480.2.3 getName()

```
std::string ModelInfo::getName ( ) const
```

getName

Returns

17.480.2.4 getProjectTitle()

```
std::string ModelInfo::getProjectTitle ( ) const
```

getProjectTitle

Returns

17.480.2.5 getVersion()

```
std::string ModelInfo::getVersion ( ) const
```

getVersion

Returns

17.480.2.6 hasChanged()

```
bool ModelInfo::hasChanged ( ) const
```

hasChanged

Returns

17.480.2.7 loadInstance()

```
void ModelInfo::loadInstance (
            PersistenceRecord * fields )
```

loadInstance

Parameters

fields	

17.480.2.8 saveInstance()

```
void ModelInfo::saveInstance (
            PersistenceRecord * fields )
```

saveInstance

Parameters

fields	

17.480.2.9 setAnalystName()

```
void ModelInfo::setAnalystName (
            std::string _analystName )
```

setAnalystName

Parameters

_analystName	

17.480.2.10 setDescription()

```
void ModelInfo::setDescription (
            std::string _description )
```

setDescription

Parameters

_description	

17.480.2.11 setName()

```
void ModelInfo::setName (
            std::string _name )
```

setName

Parameters

_name	

17.480.2.12 setProjectTitle()

```
void ModelInfo::setProjectTitle (
            std::string _projectTitle )
```

setProjectTitle

Parameters

_projectTitle	

17.480.2.13 setVersion()

```
void ModelInfo::setVersion (
            std::string _version )
```

setVersion

Parameters

_version	

The documentation for this class was generated from the following files:

- kernel/simulator/model/ModelInfo.h
- kernel/simulator/model/ModelInfo.cpp

17.481 SimulatorSessionService::ModelInfoResult Struct Reference

Describes the currently active model information for a session.
`#include <applications/worker/service/SimulatorSessionService.h>`

Public Attributes

- bool **exists** = false
- unsigned int **modelId** = 0
- bool **hasChanged** = false
- unsigned int **level** = 0
- std::string **name**
- std::string **analystName**
- std::string **projectTitle**
- std::string **version**
- std::string **description**
- unsigned int **componentCount** = 0

17.481.1 Detailed Description

Describes the currently active model information for a session.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.482 ModelInspectorController Class Reference

Controller for model inspection views and scene-selection synchronization.
`#include <applications/gui/genesys/controllers/ModelInspectorController.h>`

Classes

- struct DataDefinitionSelection

Public Member Functions

- **ModelInspectorController** (Simulator ∗simulator, QTreeWidget ∗componentsTree, QTreeWidget ∗data↩ DefinitionsTree, ModelGraphicsView ∗graphicsView)

  Creates the model-inspector delegated controller.
- void **actualizeModelComponents** (bool force) const

  Synchronizes the Components tree from the current model representation.
- void **actualizeModelDataDefinitions** (bool force) const

  Synchronizes the Data Definitions tree from current model data definitions.
- void **beginDataDefinitionNameEdit** (QTreeWidgetItem ∗item, int column) const

  Starts in-place rename editing for editable data-definition entries.
- void **applyDataDefinitionNameChange** (QTreeWidgetItem ∗item, int column) const

  Commits edited data-definition names back to the kernel model.

- void **syncSelectedComponentTreeItemToScene** () const

  Synchronizes tree selection with scene selection and viewport focus.
- DataDefinitionSelection **syncSelectedDataDefinitionTreeItemToScene** () const

  Synchronizes Data Definitions tree selection with visible GMDDs, when any.

17.482.1 Detailed Description

Controller for model inspection views and scene-selection synchronization.
In the refactored GUI architecture, this controller receives delegated callbacks from MainWindow to keep model-inspector responsibilities outside the composition root while preserving the existing UI contract. It synchronizes Components/Data Definitions trees with the current kernel model and keeps tree-to-scene selection behavior consistent.
Responsibilities:

- refresh model component and data-definition trees from Simulator state;
- manage in-place rename flow for editable data-definition names;
- propagate current tree selection to the graphical scene and viewport.

Boundaries:

- it does not create/destroy models or files (model lifecycle remains elsewhere);
- it does not own scene command logic, undo/redo, or simulation commands;
- it does not act as a persistence or export service.

The documentation for this class was generated from the following files:

- applications/gui/genesys/controllers/ModelInspectorController.h
- applications/gui/genesys/controllers/ModelInspectorController.cpp

17.483 ModelLanguageSynchronizer Class Reference

Service layer that synchronizes textual model language with kernel model state.
`#include <applications/gui/genesys/services/ModelLanguageSynchronizer.h>`

Public Member Functions

- ModelLanguageSynchronizer (Simulator *simulator, QPlainTextEdit *modelTextEditor, bool *textModelHas↩ ChangedFlag, QWidget *ownerWidget, std::function< void()> onModelCreatedOrLoaded)

  Creates the synchronization service used by MainWindow compatibility wrappers.
- void actualizeModelSimLanguage () const

  Rebuilds textual model representation from the current kernel model.
- bool setSimulationModelBasedOnText () const

  Parses text editor content and applies it to the simulator model.

17.483.1 Detailed Description

Service layer that synchronizes textual model language with kernel model state.
This service was extracted from MainWindow representation code to isolate text/model synchronization concerns. MainWindow remains the composition root and delegates wrapper methods to this class.
Responsibilities:

- generate/update the textual model representation from current Simulator model;
- parse/apply textual edits back into simulator model state;
- notify MainWindow via callback when model creation/loading side effects are required.

Boundaries:

- it does not own simulator or editor widgets;
- it does not handle graphical persistence, Graphviz export, or C++ code export;
- it is a synchronization service, not a controller.

17.483.2 Constructor & Destructor Documentation

17.483.2.1 ModelLanguageSynchronizer()

```
ModelLanguageSynchronizer::ModelLanguageSynchronizer (
            Simulator * simulator,
            QPlainTextEdit * modelTextEditor,
            bool * textModelHasChangedFlag,
            QWidget * ownerWidget,
            std::function< void()> onModelCreatedOrLoaded )
```

Creates the synchronization service used by MainWindow compatibility wrappers.
Stores explicit dependencies once to keep wrapper calls thin.

17.483.3 Member Function Documentation

17.483.3.1 actualizeModelSimLanguage()

```
void ModelLanguageSynchronizer::actualizeModelSimLanguage ( ) const
```

Rebuilds textual model representation from the current kernel model.
Regenerates the text editor content from the current kernel model.
This is part of the synchronization bridge used after lifecycle, property, and scene updates.

17.483.3.2 setSimulationModelBasedOnText()

```
bool ModelLanguageSynchronizer::setSimulationModelBasedOnText ( ) const
```

Parses text editor content and applies it to the simulator model.
Applies the text editor content to the current kernel model.

Returns

true when synchronization succeeds and model state is valid for downstream workflows.

The documentation for this class was generated from the following files:

- applications/gui/genesys/services/ModelLanguageSynchronizer.h
- applications/gui/genesys/services/ModelLanguageSynchronizer.cpp

17.484 ModelLifecycleController Class Reference

Controller for model/application lifecycle flows delegated from MainWindow.
`#include <applications/gui/genesys/controllers/ModelLifecycleController.h>`

Classes

- struct Callbacks

Public Member Functions

- **ModelLifecycleController** (QWidget ∗ownerWidget, Simulator ∗simulator, Ui::MainWindow ∗ui, QString ∗modelFilename, bool ∗textModelHasChanged, bool ∗graphicalModelHasChanged, bool ∗closingApproved, bool ∗loaded, bool &parallelizationEnabled, int &parallelizationThreads, int &parallelizationBatchSize, Callbacks callbacks)

  Creates the lifecycle orchestration controller used by MainWindow wrappers.

- void **onActionModelNewTriggered** () const

  Starts the new-model lifecycle flow and reinitializes GUI/model state.
- void **onActionModelOpenTriggered** () const

  Starts model-open flow with persistence loading and UI synchronization.
- bool **openModelFile** (const QString &fileName) const

  Opens a model from an explicit file path using the same lifecycle flow as the open action.
- void **onActionModelSaveTriggered** () const

  Starts model-save flow for graphical/text representations.
- void **onActionModelCloseTriggered** () const

  Closes current model after pending-change checks and cleanup orchestration.
- void **onActionModelInformationTriggered** () const

  Opens model information dialog for the current model context.
- void **onActionModelCheckTriggered** () const

  Runs model-check workflow as a delegated lifecycle operation.
- void **onActionSimulationConfigureTriggered** () const

  Opens simulation configuration workflow anchored to lifecycle façade.
- void **onActionSimulatorExitTriggered** () const

  Executes application-exit flow with pending-change confirmation semantics.
- bool **hasPendingModelChanges** () const

  Indicates whether current model/text state has unsaved changes.
- bool **confirmApplicationExit** () const

  Requests exit confirmation preserving compatibility prompt behavior.

17.484.1 Detailed Description

Controller for model/application lifecycle flows delegated from MainWindow.
This controller centralizes lifecycle actions (new/open/save/close/check/configure/exit) while MainWindow remains the composition root and compatibility façade. It uses explicit callback injection to invoke existing UI update and persistence routines without changing signatures or ownership.
Responsibilities:

- orchestrate user-triggered lifecycle actions and guard pending-change confirmations;
- coordinate model load/save paths through injected serializer callbacks;
- trigger UI/tab/action refreshes required after lifecycle transitions.

Boundaries:

- it does not directly implement file formats (delegated to services);
- it does not own simulator/model objects;
- it does not manage scene tools, property editor internals, or simulation traces.

The documentation for this class was generated from the following files:

- applications/gui/genesys/controllers/ModelLifecycleController.h
- applications/gui/genesys/controllers/ModelLifecycleController.cpp

17.485 ModelManager Class Reference

The ModelManager class.
`#include <kernel/simulator/model/ModelManager.h>`

Public Member Functions

- ModelManager (Simulator *simulator)

  Creates a model manager bound to a simulator instance.
- virtual ~**ModelManager** ()

  Releases the model manager and any owned bookkeeping structures.
- Model * newModel ()

  Creates a new model, inserts it into the open-model list and makes it current.
- void insert (Model *model)

  Inserts an existing model into the open-model list.
- void remove (Model *model)

  Removes a model from the open-model list.
- bool setCurrent (Model *model)

  Makes an open model the current one.
- bool saveModel (std::string filename)

  Saves the current model to a file.
- Model * loadModel (std::string filename)

  Loads a model from a file and opens it in the manager.
- Model * createFromLanguage (std::string modelSpecification)

  Creates a model from a textual model specification.
- unsigned int size ()

  Returns the number of open models.
- std::vector< Model * > models () const

  Returns a stable snapshot of the currently open models in tab/navigation order.
- bool hasModel (Model *model) const

  Checks whether a model is owned by this manager.
- int indexOf (Model *model) const

  Returns the open-model index of a model.
- Model * modelAt (unsigned int index) const

  Returns the model at a given open-model index.
- int currentIndex () const

  Returns the index of the current model.
- Model * front ()

  Returns the first open model.
- Model * last ()

  Returns the last open model.
- Model * current ()

  Returns the current model.
- Model * next ()

  Advances to the next open model.
- Model * previous ()

  Moves to the previous open model.
- bool canGoNext () const

  Indicates whether a next model exists.
- bool canGoPrevious () const

  Indicates whether a previous model exists.

17.485.1 Detailed Description

The ModelManager class.

17.485.2 Constructor & Destructor Documentation

17.485.2.1 ModelManager()

```
ModelManager::ModelManager (
            Simulator * simulator )
```

Creates a model manager bound to a simulator instance.

Parameters

simulator	Parent simulator that owns this manager.

17.485.3 Member Function Documentation

17.485.3.1 canGoNext()

```
bool ModelManager::canGoNext ( ) const
```

Indicates whether a next model exists.

Returns

`true` when a call to `next()` can advance.

17.485.3.2 canGoPrevious()

```
bool ModelManager::canGoPrevious ( ) const
```

Indicates whether a previous model exists.

Returns

`true` when a call to `previous()` can retreat.

17.485.3.3 createFromLanguage()

```
Model * ModelManager::createFromLanguage (
            std::string modelSpecification )
```

Creates a model from a textual model specification.

Parameters

modelSpecification	Source text in the model language.

Returns

Newly created model instance, or `nullptr` on failure.

17.485.3.4 current()

```
Model * ModelManager::current ( )
```

Returns the current model.

Returns

Currently selected model, or `nullptr` when none is selected.

17.485.3.5 currentIndex()

```
int ModelManager::currentIndex ( ) const
```

Returns the index of the current model.

Returns

Zero-based current model index, or -1 when there is no current model.

17.485.3.6 front()

```
Model * ModelManager::front ( )
```

Returns the first open model.

Returns

First model in open-model order, or `nullptr` when empty.

17.485.3.7 hasModel()

```
bool ModelManager::hasModel (
            Model * model ) const
```

Checks whether a model is owned by this manager.

Parameters

model	Model pointer to test.

Returns

True when model is in the open-model list.

17.485.3.8 indexOf()

```
int ModelManager::indexOf (
            Model * model ) const
```

Returns the open-model index of a model.

Parameters

model	Model pointer to locate.

Returns

Zero-based index, or -1 when not open.

17.485.3.9 insert()

```
void ModelManager::insert (
            Model * model )
```

Inserts an existing model into the open-model list.

Parameters

model	Model to register.

17.485.3.10 last()

```
Model * ModelManager::last ( )
```

Returns the last open model.

Returns

Last model in open-model order, or `nullptr` when empty.

17.485.3.11 loadModel()

```
Model * ModelManager::loadModel (
            std::string filename )
```

Loads a model from a file and opens it in the manager.

Parameters

filename	Input filename.

Returns

Loaded model instance, or `nullptr` on failure.

17.485.3.12 modelAt()

```
Model * ModelManager::modelAt (
            unsigned int index ) const
```

Returns the model at a given open-model index.

Parameters

index	Zero-based model index.

Returns

Model pointer, or nullptr when index is out of range.

17.485.3.13 models()

```
std::vector< Model * > ModelManager::models ( ) const
```

Returns a stable snapshot of the currently open models in tab/navigation order.

Returns

Vector with the open model pointers owned by this manager.

17.485.3.14 newModel()

```
Model * ModelManager::newModel ( )
```

Creates a new model, inserts it into the open-model list and makes it current.

Returns

The newly opened current model.

17.485.3.15 next()

```
Model * ModelManager::next ( )
```

Advances to the next open model.

Returns

Next model in order, or `nullptr` when already at the end.

17.485.3.16 previous()

```
Model * ModelManager::previous ( )
```

Moves to the previous open model.

Returns

Previous model in order, or `nullptr` when already at the beginning.

17.485.3.17 remove()

```
void ModelManager::remove (
            Model * model )
```

Removes a model from the open-model list.

Parameters

model	Model to remove.

17.485.3.18 saveModel()

```
bool ModelManager::saveModel (
            std::string filename )
```

Saves the current model to a file.

Parameters

filename	Output filename.

Returns

`true` when saving succeeds.

17.485.3.19 setCurrent()

```
bool ModelManager::setCurrent (
            Model * model )
```

Makes an open model the current one.

Parameters

model	Model to activate.

Returns

`true` when the model was found and selected.

17.485.3.20 size()

```
unsigned int ModelManager::size ( )
```

Returns the number of open models.

Returns

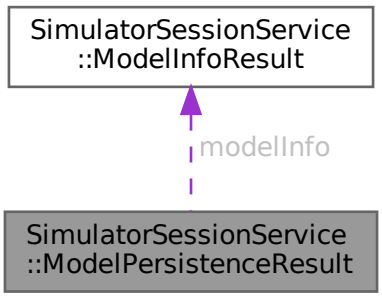
Open-model count.

The documentation for this class was generated from the following files:

- kernel/simulator/model/ModelManager.h
- kernel/simulator/model/ModelManager.cpp

17.486 SimulatorSessionService::ModelPersistenceResult Struct Reference

Contains save/load model operation output and error state.
`#include <applications/worker/service/SimulatorSessionService.h>`
Collaboration diagram for SimulatorSessionService::ModelPersistenceResult:



Public Attributes

- bool **success** = false
- PersistenceError **error** = PersistenceError::None
- std::string **filename**
- ModelInfoResult **modelInfo**

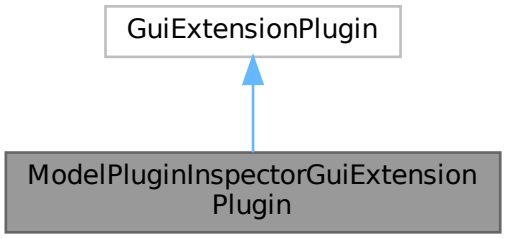
17.486.1 Detailed Description

Contains save/load model operation output and error state.
The documentation for this struct was generated from the following file:

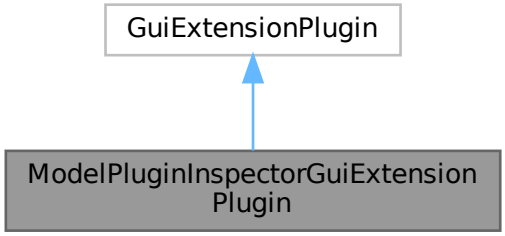
- applications/worker/service/SimulatorSessionService.h

17.487 ModelPluginInspectorGuiExtensionPlugin Class Reference

Inheritance diagram for ModelPluginInspectorGuiExtensionPlugin:



Collaboration diagram for ModelPluginInspectorGuiExtensionPlugin:



Public Member Functions

- std::string extensionId () const override
- std::string displayName () const override
- std::string version () const override
- std::vector< std::string > requiredModelPlugins () const override
- void registerContributions (GuiExtensionRegistry *registry) const override

17.487.1 Member Function Documentation

17.487.1.1 displayName()

```
std::string ModelPluginInspectorGuiExtensionPlugin::displayName ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.487.1.2 extensionId()

```
std::string ModelPluginInspectorGuiExtensionPlugin::extensionId ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

17.487.1.3 registerContributions()

```
void ModelPluginInspectorGuiExtensionPlugin::registerContributions (
            GuiExtensionRegistry * registry ) const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.487.1.4 requiredModelPlugins()

```
std::vector< std::string > ModelPluginInspectorGuiExtensionPlugin::requiredModelPlugins ( )
const  [inline], [override], [virtual]
```

Implements GuiExtensionPlugin.

17.487.1.5 version()

```
std::string ModelPluginInspectorGuiExtensionPlugin::version ( ) const  [inline], [override],
[virtual]
```

Implements GuiExtensionPlugin.

The documentation for this class was generated from the following file:

- applications/gui/genesys/extensions/ModelPluginInspectorGuiExtensionPlugin.cpp

17.488 ModelSimulation Class Reference

Controls the mature, currently functional simulation-execution layer of a model.
`#include <kernel/simulator/model/ModelSimulation.h>`

Public Member Functions

- **ModelSimulation** (Model *model)
- std::string **show** ()
- void start ()

  Starts a sequential execution of a simulation, ie, a set of replications of this model.
- void **pause** ()
- void **step** ()

  Executes the processing of a single event, the next one in the future events list.
- void **stop** ()
- void **setNumberOfReplications** (unsigned int _numberOfReplications)
- unsigned int **getNumberOfReplications** () const
- void **setReplicationLength** (double _replicationLength, Util::TimeUnit _replicationLengthTimeUnit=Util::TimeUnit::unknown)
- double **getReplicationLength** () const
- void **setReplicationLengthTimeUnit** (Util::TimeUnit _replicationLengthTimeUnit)
- Util::TimeUnit **getReplicationLengthTimeUnit** () const
- void **setReplicationReportBaseTimeUnit** (Util::TimeUnit _replicationReportBaseTimeUnit)
- Util::TimeUnit **getReplicationBaseTimeUnit** () const
- void **setWarmUpPeriod** (double warmUpPeriod, Util::TimeUnit warmUpPeriodTimeUnit=Util::TimeUnit::unknown)
- double **getWarmUpPeriod** () const
- void **setWarmUpPeriodTimeUnit** (Util::TimeUnit _warmUpPeriodTimeUnit)
- Util::TimeUnit **getWarmUpPeriodTimeUnit** () const
- void **setTerminatingCondition** (std::string _terminatingCondition)
- std::string **getTerminatingCondition** () const
- void **setPauseOnEvent** (bool _pauseOnEvent)
- bool **isPauseOnEvent** () const
- void **setStepByStep** (bool _stepByStep)
- bool **isStepByStep** () const
- void **setInitializeStatistics** (bool _initializeStatistics)
- bool **isInitializeStatistics** () const
- void **setInitializeSystem** (bool _initializeSystem)
- bool **isInitializeSystem** () const
- void **setPauseOnReplication** (bool _pauseBetweenReplications)
- bool **isPauseOnReplication** () const
- void **setReporter** (SimulationReporter_if *_simulationReporter)
- SimulationReporter_if * **getReporter** () const
- double **getSimulatedTime** () const
- bool isRunning () const
- bool **isPaused** () const
- unsigned int **getCurrentReplicationNumber** () const
- void **setShowReportsAfterReplication** (bool showReportsAfterReplication)
- bool **isShowReportsAfterReplication** () const
- void **setShowReportsAfterSimulation** (bool showReportsAfterSimulation)
- bool **isShowReportsAfterSimulation** () const
- List< double > * **getBreakpointsOnTime** () const
- List< Entity * > * **getBreakpointsOnEntity** () const
- List< ModelComponent * > * **getBreakpointsOnComponent** () const
- const List< ModelDataDefinition * > * **getSimulationStatisticsAggregates** () const

- void **loadInstance** (PersistenceRecord *fields)
- void **saveInstance** (PersistenceRecord *fields, bool saveDefaults)
- Event * **getCurrentEvent** () const
- void **setShowSimulationResposesInReport** (bool _showSimulationResposesInReport)
- bool **isShowSimulationResposesInReport** () const
- void **setShowSimulationControlsInReport** (bool _showSimulationControlsInReport)
- bool **isShowSimulationControlsInReport** () const

Friends

- class **Model**

17.488.1 Detailed Description

Controls the mature, currently functional simulation-execution layer of a model.
The ModelSimulation controls the simulation of a model, alowing to start, pause, resume e stop a simulation, composed by a set of replications.
Historically, this class has been the original and operational experiment mechanism in GenESyS. It already supports the core execution workflow based on replications, replication length, warm-up configuration and event processing. Later experiment-oriented classes such as SimulationExperiment and SimulationScenario were introduced as an attempt to move richer design-of-experiments concerns into the kernel. Those newer classes are still under development, while ModelSimulation remains the stable execution core.

17.488.2 Member Function Documentation

17.488.2.1 isRunning()

```
bool ModelSimulation::isRunning ( ) const
```

The current time in the model being simulated, i.e., the instant when the current event was triggered

17.488.2.2 start()

```
void ModelSimulation::start ( )
```

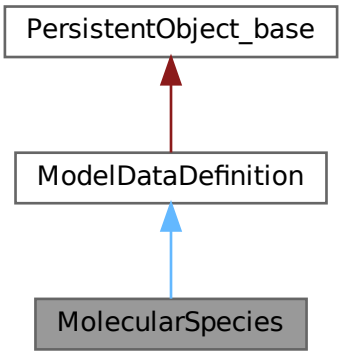
Starts a sequential execution of a simulation, ie, a set of replications of this model.
Checks the model and if ok then initialize the simulation, execute repeatedly each replication and then show simulation statistics
The documentation for this class was generated from the following files:

- kernel/simulator/model/ModelSimulation.h
- kernel/simulator/model/ModelSimulation.cpp

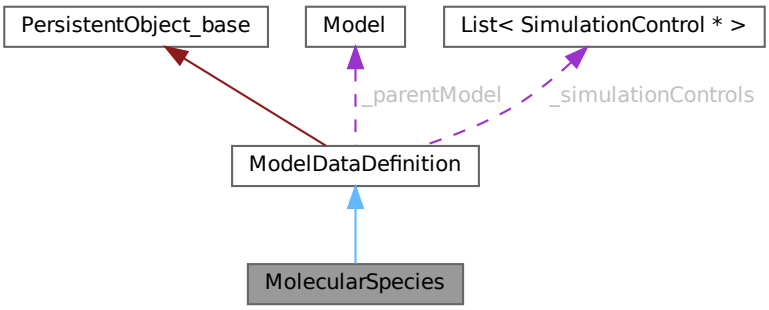
17.489 MolecularSpecies Class Reference

```
#include <plugins/data/WholeCellModeling/MolecularSpecies.h>
```

Inheritance diagram for MolecularSpecies:



Collaboration diagram for MolecularSpecies:



Public Member Functions

- **MolecularSpecies** (Model *model, std::string name="")
- void **setInitialCount** (int initialCount)
- int **getInitialCount** () const
- void **setCount** (int count)
- int **getCount** () const
- void **setCompartment** (std::string compartment)
- std::string **getCompartment** () const
- void **setUnit** (std::string unit)
- std::string **getUnit** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)

- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.489.1 Detailed Description

Discrete molecular species with integer copy count for Gillespie SSA.
Unlike BioSpecies (continuous double amounts for ODE), MolecularSpecies tracks integer molecule counts. Compartment and unit annotate the biological context without affecting simulation logic.

17.489.2 Member Function Documentation

17.489.2.1 _check()

```
bool MolecularSpecies::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.489.2.2 _initBetweenReplications()

```
void MolecularSpecies::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.489.2.3 _loadInstance()

```
bool MolecularSpecies::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.489.2.4 _saveInstance()

```
void MolecularSpecies::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.489.2.5 show()

```
std::string MolecularSpecies::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/WholeCellModeling/MolecularSpecies.h
- plugins/data/WholeCellModeling/MolecularSpecies.cpp

17.490 GroProgramRuntime::MotionMutation Struct Reference

Public Attributes

- MotionMutationType **type** = MotionMutationType::Run
- double **value** = 0.0
- double **previousDirection** = 0.0
- double **resultingDirection** = 0.0

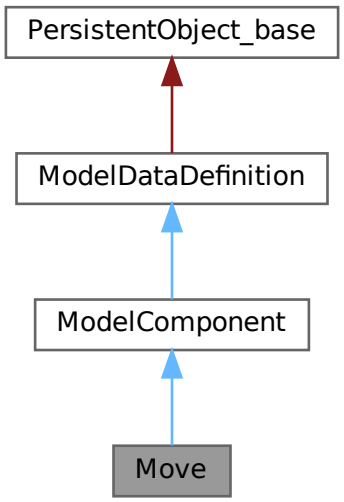
- double **previousSpeed** = 0.0
- double **resultingSpeed** = 0.0

The documentation for this struct was generated from the following file:

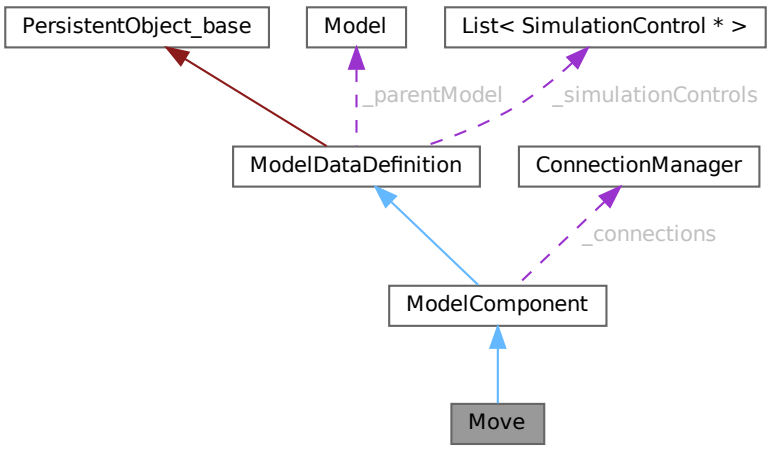
- plugins/data/BiochemicalSimulation/GroProgramRuntime.h

17.491 Move Class Reference

Minimal transporter movement component.
`#include <plugins/components/MaterialHandling/Move.h>`
Inheritance diagram for Move:



Collaboration diagram for Move:



Public Member Functions

- **Move** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setTransporter** (Transporter *transporter)
- Transporter * **getTransporter** () const
- void **setStation** (Station *station)
- Station * **getStation** () const
- void **setStationExpression** (std::string stationExpression)
- std::string **getStationExpression** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createInternalStatisticReporters () override
- virtual void _createEditableDataDefinitions () override
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.491.1 Detailed Description

Minimal transporter movement component.
This component intentionally collapses Arena's request/transport/free chain into one atomic operation suitable for the current closeout: reserve one transporter, move the entity to a destination station using the attached Distance set, and free the transporter on arrival.

17.491.2 Member Function Documentation

17.491.2.1 _check()

```
bool Move::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.491.2.2 _createAttachedAttributes()

```
void Move::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.491.2.3 _createEditableDataDefinitions()

```
void Move::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.491.2.4 _createInternalStatisticReporters()

```
void Move::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.491.2.5 _loadInstance()

```
bool Move::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.491.2.6 _onDispatchEvent()

```
void Move::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.491.2.7 _saveInstance()

```
void Move::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.491.2.8 show()

```
std::string Move::show ( )  [override], [virtual]
```

virtual public methods

Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

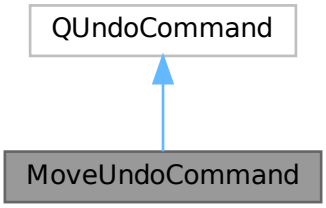
Reimplemented from ModelComponent.

The documentation for this class was generated from the following files:

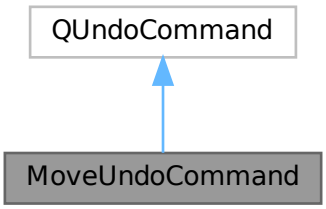
- plugins/components/MaterialHandling/Move.h
- plugins/components/MaterialHandling/Move.cpp

17.492 MoveUndoCommand Class Reference

Inheritance diagram for MoveUndoCommand:



Collaboration diagram for MoveUndoCommand:



Public Member Functions

- **MoveUndoCommand** (QList< QGraphicsItem * > item, ModelGraphicsScene *scene, QList< QPointF > &oldPos, QList< QPointF > &newPos, QUndoCommand *parent=nullptr)
- void **undo** () override
- void **redo** () override

The documentation for this class was generated from the following files:

- applications/gui/genesys/actions/MoveUndoCommand.h
- applications/gui/genesys/actions/MoveUndoCommand.cpp

17.493 testing::internal::Mutex Class Reference

Public Member Functions

- void **Lock** ()
- void **Unlock** ()
- void **AssertHeld** () const

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-port.h

17.494 dialogBreakpoint::MVCResult Class Reference

Public Member Functions

- **MVCResult** (std::string type, std::string on)

Public Attributes

- std::string **type**
- std::string **on**

The documentation for this class was generated from the following file:

- applications/gui/genesys/dialogs/dialogBreakpoint.h

17.495 GroProgramAst::NamedProgram Struct Reference

Public Attributes

- std::string **name** = ""
- std::vector< std::string > **parameters**
- std::string **bodySource** = ""
- std::vector< Statement > **statements**

The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/GroProgramAst.h

17.496 GroProgramIr::NamedProgramDefinition Struct Reference

Public Attributes

- std::vector< std::string > **parameters**
- std::string **sourceText** = ""
- std::vector< Command > **commands**

The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/GroProgramIr.h

17.497 testing::internal::NameGeneratorSelector< Provided > Struct Template Reference

Public Types

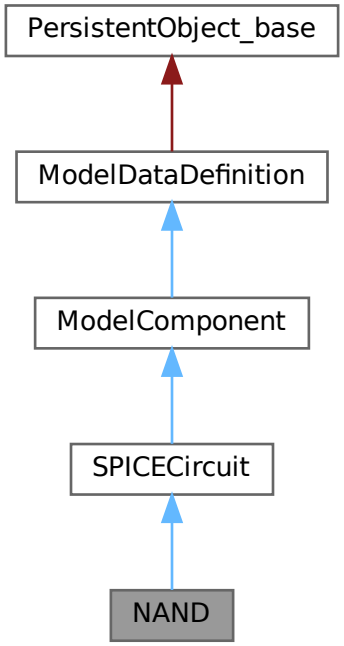
- typedef Provided **type**

The documentation for this struct was generated from the following file:

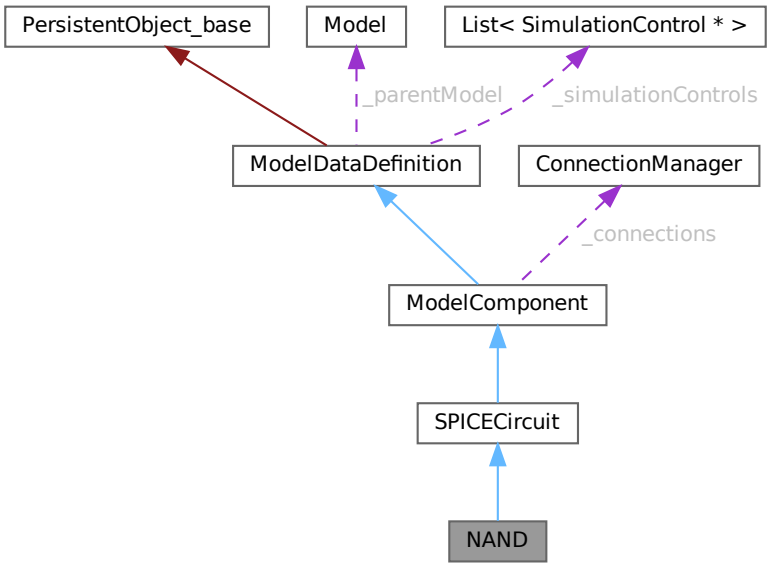
- gtest/gtest/internal/gtest-internal.h

17.498 NAND Class Reference

Inheritance diagram for NAND:



Collaboration diagram for NAND:



Public Member Functions

- **NAND** (Model *model, std::string name="")
- void Build () override
- void **setPMOSWidth** (float PMOSWidth)
- void **setPMOSLength** (float length)
- void **setNMOSWidth** (float NMOSWidth)
- void **setNMOSLength** (float NMOSLength)
- void **setElectricalModel** (std::string electricalModel)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.498.1 Member Function Documentation

17.498.1.1 _loadInstance()

```
bool NAND::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.498.1.2 _saveInstance()

```
void NAND::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.498.1.3 Build()

```
void NAND::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.
The documentation for this class was generated from the following files:

- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.499 testing::internal::NativeArray< Element > Class Template Reference

Public Types

- typedef Element **value_type**
- typedef Element * **iterator**
- typedef const Element * **const_iterator**

Public Member Functions

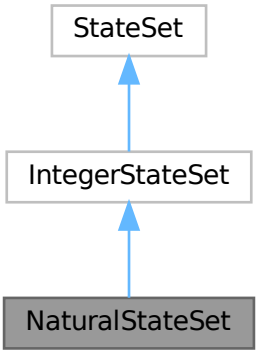
- **NativeArray** (const Element *array, size_t count, RelationToSourceReference)
- **NativeArray** (const Element *array, size_t count, RelationToSourceCopy)
- **NativeArray** (const NativeArray &rhs)
- size_t **size** () const
- const_iterator **begin** () const
- const_iterator **end** () const
- bool **operator==** (const NativeArray &rhs) const

The documentation for this class was generated from the following file:

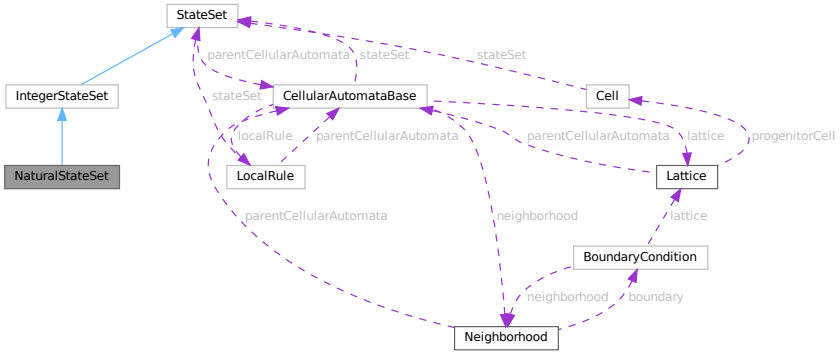
- gtest/gtest/internal/gtest-internal.h

17.500 NaturalStateSet Class Reference

Inheritance diagram for NaturalStateSet:



Collaboration diagram for NaturalStateSet:



Public Member Functions

- **NaturalStateSet** (CellularAutomataBase *parentCellularAutomata, std::optional< long > maximum=std::nullopt)
- std::string name () const override

Public Member Functions inherited from IntegerStateSet

- **IntegerStateSet** (CellularAutomataBase *parentCellularAutomata, std::optional< long > minimum=std::nullopt, std::optional< long > maximum=std::nullopt)
- bool contains (const State &state) const override
- std::unique_ptr< State > createDefaultState () const override
- std::unique_ptr< State > parseState (const std::string &text) const override
- bool isFinite () const override
- bool isDiscrete () const override
- std::vector< std::unique_ptr< State > > enumerateStates () const override

Public Member Functions inherited from StateSet

- **StateSet** (CellularAutomataBase *parentCellularAutomata)
- **StateSet** (const StateSet &orig)
- virtual bool **isContinuous** () const
- virtual bool **isComposite** () const

Additional Inherited Members

Protected Attributes inherited from StateSet

- CellularAutomataBase * **parentCellularAutomata**

17.500.1 Member Function Documentation

17.500.1.1 name()

```
std::string NaturalStateSet::name ( ) const  [inline], [override], [virtual]
```

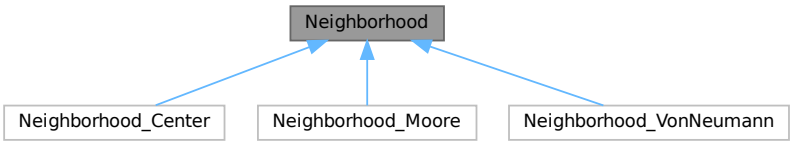
Reimplemented from IntegerStateSet.

The documentation for this class was generated from the following file:

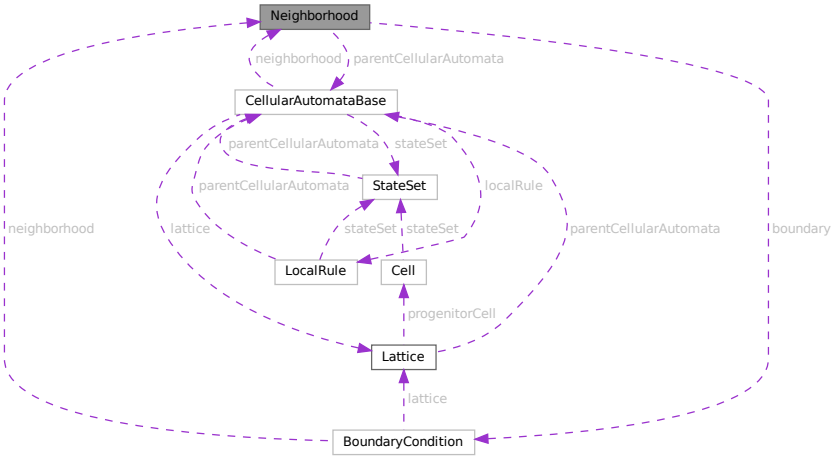
- plugins/components/ModalModel/CellularAutomata/StateSet.h

17.501 Neighborhood Class Reference

Defines the complete neighborhood of a cell.
`#include <plugins/components/ModalModel/CellularAutomata/Neighborhood.h>`
Inheritance diagram for Neighborhood:



Collaboration diagram for Neighborhood:



Public Member Functions

- **Neighborhood** (CellularAutomataBase ∗parentCellularAutomata, unsigned short radius=1, BoundaryCondition ∗boundary=nullptr, bool includeCellItself=false)
- **Neighborhood** (const Neighborhood &orig)
- virtual std::string **show** ()
- virtual std::vector< Cell ∗ > getNeighbors (Cell ∗cell)=0

  pure virtual method that has to be overriden by derived class, responsible for defining the neighborhood itself.
- void **setRadius** (unsigned short radius)
- unsigned short **getRadius** () const
- void **setBoundary** (BoundaryCondition ∗boundary)
- BoundaryCondition ∗ **getBoundary** () const
- void **setIncludeCellItself** (bool includeCellItself)
- bool **getIncludeCellItself** () const

Protected Member Functions

- Cell ∗ **getNeighborCell** (std::vector< int > cellPosition, std::vector< std::pair< unsigned short, int > > dimensionChanges)

Protected Attributes

- std::string **name**
- CellularAutomataBase * **parentCellularAutomata**
- BoundaryCondition * **boundary**
- unsigned short **radius**
- bool **includeCellItself**

17.501.1 Detailed Description

Defines the complete neighborhood of a cell.

17.501.2 Member Function Documentation

17.501.2.1 getNeighbors()

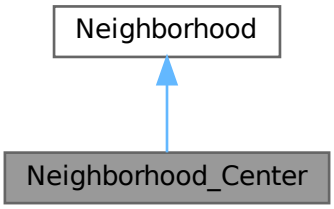
```
virtual std::vector< Cell * > Neighborhood::getNeighbors (
            Cell * cell )  [pure virtual]
```

pure virtual method that has to be overriden by derived class, responsible for defining the neighborhood itself.
Implemented in Neighborhood_Center, Neighborhood_Moore, and Neighborhood_VonNeumann.
The documentation for this class was generated from the following file:

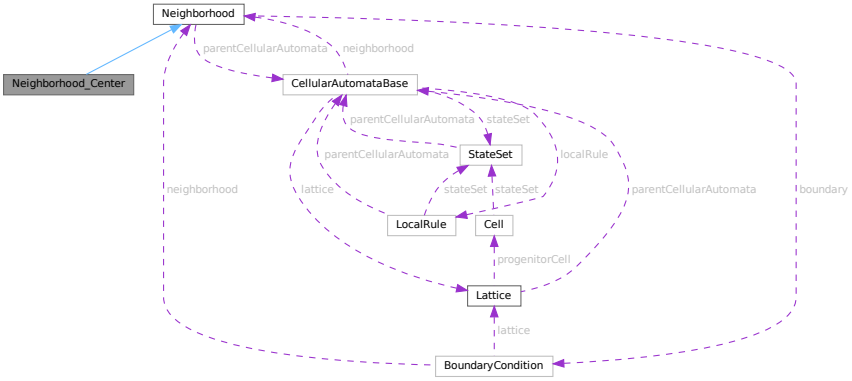
- plugins/components/ModalModel/CellularAutomata/Neighborhood.h

17.502 Neighborhood_Center Class Reference

Inheritance diagram for Neighborhood_Center:



Collaboration diagram for Neighborhood_Center:



Public Member Functions

- **Neighborhood_Center** (CellularAutomataBase ∗parentCellularAutomata, unsigned short radius=1, BoundaryCondition ∗boundary=nullptr, bool includeCellItself=false)
- **Neighborhood_Center** (const Neighborhood_Center &orig)
- virtual std::vector< Cell ∗ > getNeighbors (Cell ∗cell) override

  pure virtual method that has to be overriden by derived class, responsible for defining the neighborhood itself.

Public Member Functions inherited from Neighborhood

- **Neighborhood** (CellularAutomataBase ∗parentCellularAutomata, unsigned short radius=1, BoundaryCondition ∗boundary=nullptr, bool includeCellItself=false)
- **Neighborhood** (const Neighborhood &orig)
- virtual std::string **show** ()
- void **setRadius** (unsigned short radius)
- unsigned short **getRadius** () const
- void **setBoundary** (BoundaryCondition ∗boundary)
- BoundaryCondition ∗ **getBoundary** () const
- void **setIncludeCellItself** (bool includeCellItself)
- bool **getIncludeCellItself** () const

Additional Inherited Members

Protected Member Functions inherited from Neighborhood

- Cell ∗ **getNeighborCell** (std::vector< int > cellPosition, std::vector< std::pair< unsigned short, int > > dimensionChanges)

Protected Attributes inherited from Neighborhood

- std::string **name**
- CellularAutomataBase ∗ **parentCellularAutomata**
- BoundaryCondition ∗ **boundary**
- unsigned short **radius**
- bool **includeCellItself**

17.502.1 Member Function Documentation

17.502.1.1 getNeighbors()

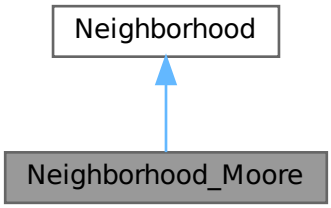
```
virtual std::vector< Cell * > Neighborhood_Center::getNeighbors (
            Cell * cell )  [inline], [override], [virtual]
```

pure virtual method that has to be overriden by derived class, responsible for defining the neighborhood itself.
Implements Neighborhood.
The documentation for this class was generated from the following file:

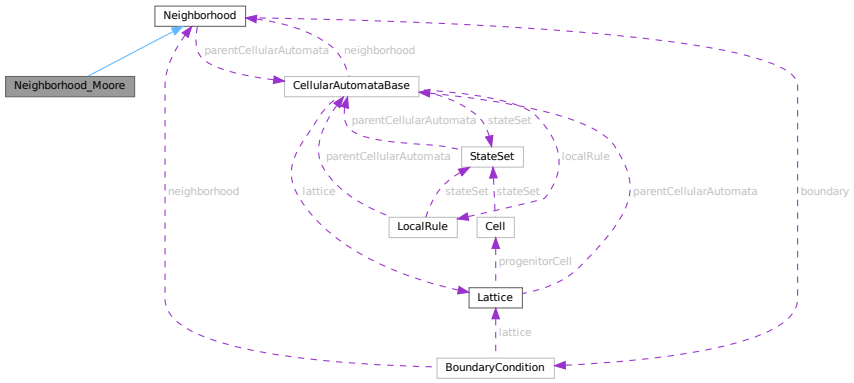
- plugins/components/ModalModel/CellularAutomata/Neighborhood_Center.h

17.503 Neighborhood_Moore Class Reference

Inheritance diagram for Neighborhood_Moore:



Collaboration diagram for Neighborhood_Moore:



Public Member Functions

- **Neighborhood_Moore** (CellularAutomataBase *parentCellularAutomata, unsigned short radius=1, BoundaryCondition *boundary=nullptr)
- **Neighborhood_Moore** (const Neighborhood_Moore &orig)
- virtual std::string show () override
- virtual std::vector< Cell * > getNeighbors (Cell *cell) override

  pure virtual method that has to be overriden by derived class, responsible for defining the neighborhood itself.

Public Member Functions inherited from Neighborhood

- **Neighborhood** (CellularAutomataBase *parentCellularAutomata, unsigned short radius=1, BoundaryCondition *boundary=nullptr, bool includeCellItself=false)
- **Neighborhood** (const Neighborhood &orig)
- void **setRadius** (unsigned short radius)
- unsigned short **getRadius** () const
- void **setBoundary** (BoundaryCondition *boundary)
- BoundaryCondition * **getBoundary** () const
- void **setIncludeCellItself** (bool includeCellItself)
- bool **getIncludeCellItself** () const

Additional Inherited Members

Protected Member Functions inherited from Neighborhood

- Cell * **getNeighborCell** (std::vector< int > cellPosition, std::vector< std::pair< unsigned short, int > > dimensionChanges)

Protected Attributes inherited from Neighborhood

- std::string **name**
- CellularAutomataBase * **parentCellularAutomata**
- BoundaryCondition * **boundary**
- unsigned short **radius**
- bool **includeCellItself**

17.503.1 Member Function Documentation

17.503.1.1 getNeighbors()

```
virtual std::vector< Cell * > Neighborhood_Moore::getNeighbors (
            Cell * cell )  [inline], [override], [virtual]
```

pure virtual method that has to be overriden by derived class, responsible for defining the neighborhood itself.
Implements Neighborhood.

17.503.1.2 show()

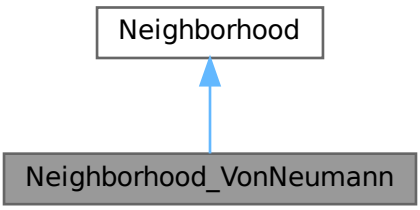
```
virtual std::string Neighborhood_Moore::show ( )  [inline], [override], [virtual]
```

Reimplemented from Neighborhood.
The documentation for this class was generated from the following file:

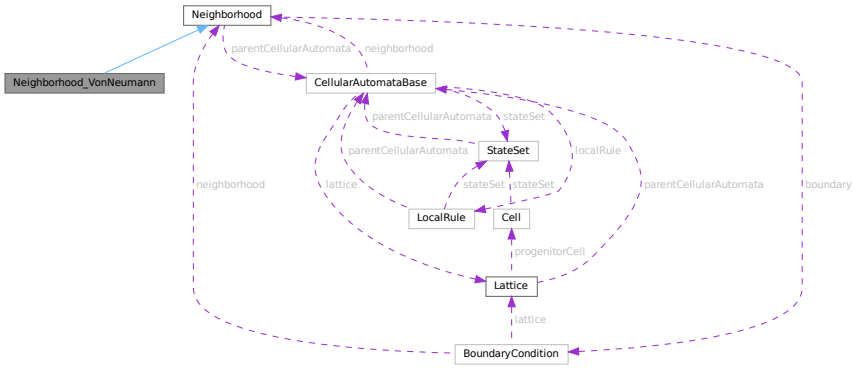
- plugins/components/ModalModel/CellularAutomata/Neighborhood_Moore.h

17.504 Neighborhood_VonNeumann Class Reference

Inheritance diagram for Neighborhood_VonNeumann:



Collaboration diagram for Neighborhood_VonNeumann:



Public Member Functions

- **Neighborhood_VonNeumann** (CellularAutomataBase ∗parentCellularAutomata, unsigned short radius=1, BoundaryCondition ∗boundary=nullptr)
- **Neighborhood_VonNeumann** (const Neighborhood_VonNeumann &orig)
- virtual std::string show () override
- virtual std::vector< Cell ∗ > getNeighbors (Cell ∗cell) override

  pure virtual method that has to be overriden by derived class, responsible for defining the neighborhood itself.

Public Member Functions inherited from Neighborhood

- **Neighborhood** (CellularAutomataBase ∗parentCellularAutomata, unsigned short radius=1, BoundaryCondition ∗boundary=nullptr, bool includeCellItself=false)
- **Neighborhood** (const Neighborhood &orig)
- void **setRadius** (unsigned short radius)
- unsigned short **getRadius** () const
- void **setBoundary** (BoundaryCondition ∗boundary)
- BoundaryCondition ∗ **getBoundary** () const
- void **setIncludeCellItself** (bool includeCellItself)
- bool **getIncludeCellItself** () const

Additional Inherited Members

Protected Member Functions inherited from Neighborhood

- Cell * **getNeighborCell** (std::vector< int > cellPosition, std::vector< std::pair< unsigned short, int > > dimensionChanges)

Protected Attributes inherited from Neighborhood

- std::string **name**
- CellularAutomataBase * **parentCellularAutomata**
- BoundaryCondition * **boundary**
- unsigned short **radius**
- bool **includeCellItself**

17.504.1 Member Function Documentation

17.504.1.1 getNeighbors()

```
virtual std::vector< Cell * > Neighborhood_VonNeumann::getNeighbors (
            Cell * cell )  [inline], [override], [virtual]
```

pure virtual method that has to be overriden by derived class, responsible for defining the neighborhood itself.
Implements Neighborhood.

17.504.1.2 show()

```
virtual std::string Neighborhood_VonNeumann::show ( )  [inline], [override], [virtual]
```

Reimplemented from Neighborhood.
The documentation for this class was generated from the following file:

- plugins/components/ModalModel/CellularAutomata/Neighborhood_VonNeumann.h

17.505 NetworkActivationFrame Class Reference

Input side of one `DefaultNetwork::activate()` call: the presence and value of every declared input port for this activation only.
`#include <plugins/data/ModalModel/NetworkActivation.h>`

Public Member Functions

- **NetworkActivationFrame** (unsigned int numInputs=0)
- void **reset** (unsigned int numInputs)

  Resizes the frame, clearing every input to absent.
- unsigned int **size** () const
- void **setPresent** (unsigned int index, double value)

  Marks input `index` present with `value`. No-op when `index` is out of range.
- void **setAbsent** (unsigned int index)

  Marks input `index` absent (value reads back as 0.0). No-op when `index` is out of range.
- bool **isPresent** (unsigned int index) const
- double **getValue** (unsigned int index) const

  Returns the input value, or 0.0 when absent or out of range.

17.505.1 Detailed Description

Input side of one `DefaultNetwork::activate()` call: the presence and value of every declared input port for this activation only.
A frame is a short-lived, per-activation value object. It is not persistent network state (see `DefaultNetwork::getLastInputValue()` for the separate, optional last-observed-value concept described in architecture reference §13).
The ordinary GenESyS `ModalModel` adapter normally marks only the input port that received the arriving entity as present (architecture reference §12); this class itself places no such restriction, so a future multi-input activation remains representable.
The documentation for this class was generated from the following files:

- plugins/data/ModalModel/NetworkActivation.h
- plugins/data/ModalModel/NetworkActivation.cpp

17.506 NetworkActivationResult Class Reference

Output side of one `DefaultNetwork::activate()` call: the presence and value of every declared output port produced by that activation.
`#include <plugins/data/ModalModel/NetworkActivation.h>`

Public Member Functions

- **NetworkActivationResult** (unsigned int numOutputs=0)
- void **reset** (unsigned int numOutputs)

  Resizes the result, clearing every output to absent.
- unsigned int **size** () const
- void **setPresent** (unsigned int index, double value)
- void **setAbsent** (unsigned int index)
- bool **isPresent** (unsigned int index) const
- double **getValue** (unsigned int index) const
- unsigned int **countPresent** () const

  Returns how many output ports are currently present.

17.506.1 Detailed Description

Output side of one `DefaultNetwork::activate()` call: the presence and value of every declared output port produced by that activation.
See architecture reference §15/§17: only present outputs cause the `ModalModel` adapter to route/consume/clone the process entity. An absent output is not interpreted as zero.
The documentation for this class was generated from the following files:

- plugins/data/ModalModel/NetworkActivation.h
- plugins/data/ModalModel/NetworkActivation.cpp

17.507 NetworkPortValue Struct Reference

Presence/value pair carried by one network port during one activation.
`#include <plugins/data/ModalModel/NetworkActivation.h>`

Public Attributes

- bool **present** = false
- double **value** = 0.0

17.507.1 Detailed Description

Presence/value pair carried by one network port during one activation.
See `docs/ai_assistants/reference/GENESYS_MODAL_MODEL_NETWORK_ARCHITECTURE.md` §8/§12: an absent port is not equivalent to a present port whose value is zero, and is not equivalent to reusing a previous value. `value` is only meaningful when `present` is `true`.
The value is numeric (`double`) in this first implementation, matching the current GenESyS parser/expression contract (§20 of the architecture reference); this does not prevent a future typed/vector/matrix port value, which would extend this struct rather than replace its presence semantics.
The documentation for this struct was generated from the following file:

- plugins/data/ModalModel/NetworkActivation.h

17.508 genesys::launcher::NetworkRequestOptions Struct Reference

Public Attributes

- qint64 **maximumBytes** = 1024 ∗ 1024
- int **timeoutMs** = 8000
- int **maximumRedirects** = 3
- bool **httpsOnly** = true

The documentation for this struct was generated from the following file:

- applications/launcher/NetworkTransport.h

17.509 genesys::launcher::NetworkResponse Struct Reference

Public Attributes

- bool **ok** = false
- bool **cancelled** = false
- int **httpStatus** = 0
- qint64 **bytesReceived** = 0
- QUrl **finalUrl**
- QString **error**

The documentation for this struct was generated from the following file:

- applications/launcher/NetworkTransport.h

17.510 ParserManager::NewParser Struct Reference

Paths associated with a generated parser artifact set.
`#include <kernel/simulator/ParserManager.h>`

Public Attributes

- std::string **bisonFilename**
- std::string **flexFilename**
- std::string **compiledParserFilename**

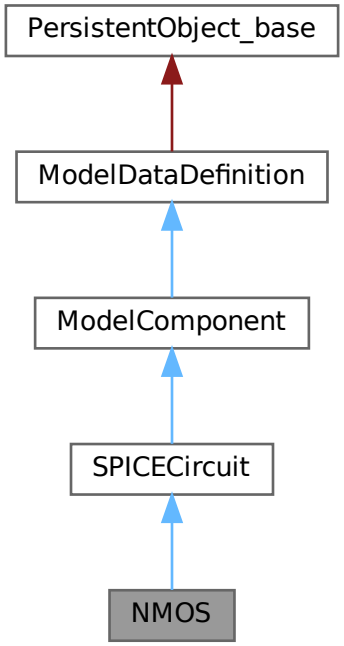
17.510.1 Detailed Description

Paths associated with a generated parser artifact set.
The documentation for this struct was generated from the following file:

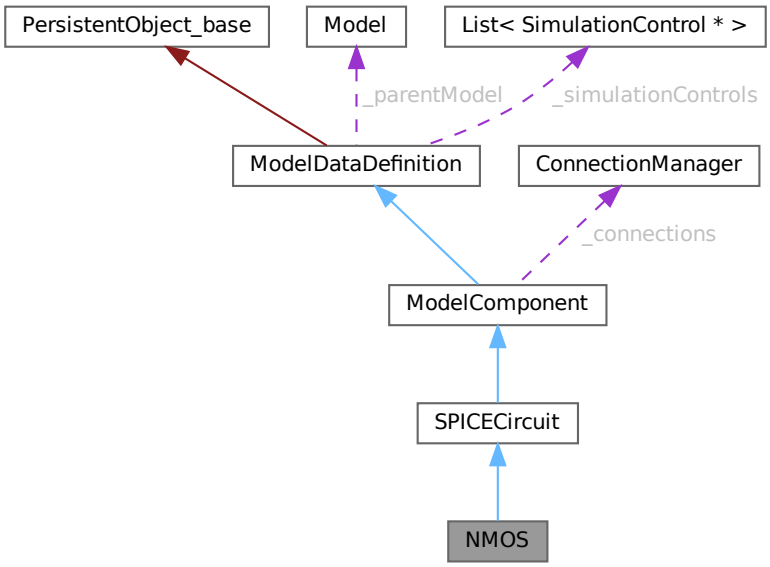
- kernel/simulator/ParserManager.h

17.511 NMOS Class Reference

Inheritance diagram for NMOS:



Collaboration diagram for NMOS:



Public Member Functions

- **NMOS** (Model *model, std::string name="")
- void Build () override
- void **setWidth** (float width)
- void **setLength** (float length)
- void **setElectricalModel** (std::string electricalModel)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component

- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.511.1 Member Function Documentation

17.511.1.1 _loadInstance()

```
bool NMOS::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.511.1.2 _saveInstance()

```
void NMOS::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.511.1.3 Build()

```
void NMOS::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.
The documentation for this class was generated from the following files:

- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

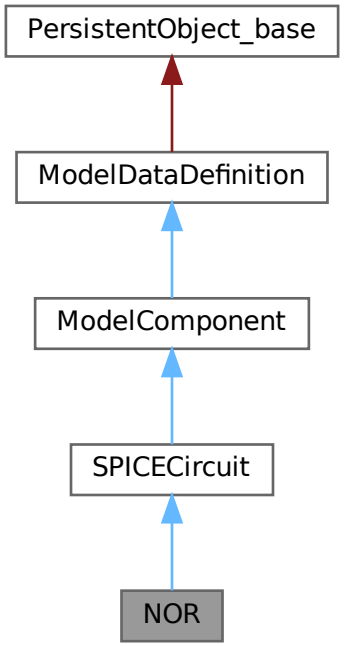
17.512 testing::internal::None Struct Reference

The documentation for this struct was generated from the following file:

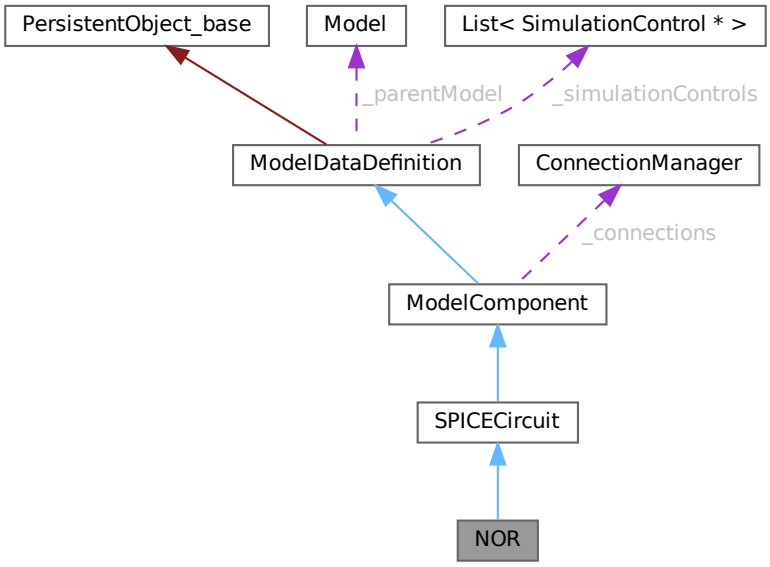
- gtest/gtest/internal/gtest-type-util.h

17.513 NOR Class Reference

Inheritance diagram for NOR:



Collaboration diagram for NOR:



Public Member Functions

- **NOR** (Model *model, std::string name="")
- void Build () override
- void **setPMOSWidth** (float PMOSWidth)
- void **setPMOSLength** (float length)
- void **setNMOSWidth** (float NMOSWidth)
- void **setNMOSLength** (float NMOSLength)
- void **setElectricalModel** (std::string electricalModel)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

Connects this component to another component.

- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.

- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden

- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden

- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.

- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes

- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)

- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.513.1 Member Function Documentation

17.513.1.1 _loadInstance()

```
bool NOR::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.513.1.2 _saveInstance()

```
void NOR::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.513.1.3 Build()

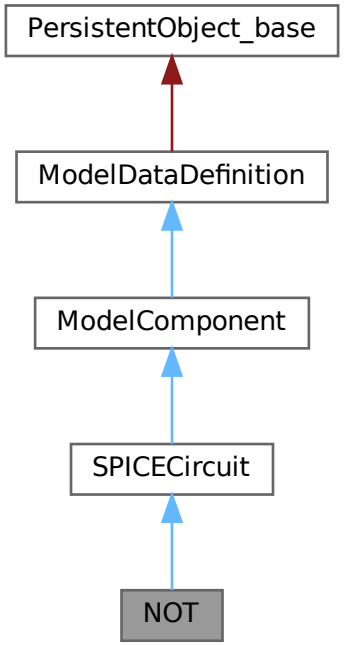
```
void NOR::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.
The documentation for this class was generated from the following files:

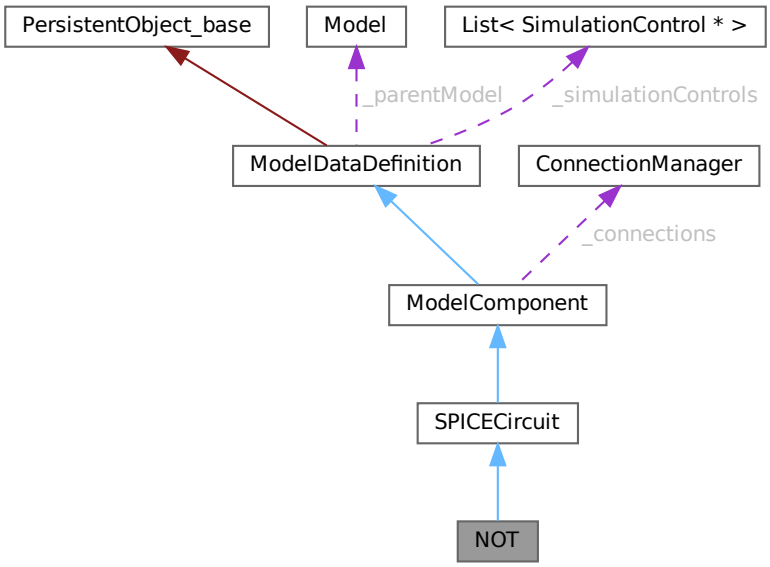
- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.514 NOT Class Reference

Inheritance diagram for NOT:



Collaboration diagram for NOT:



Public Member Functions

- **NOT** (Model *model, std::string name="")
- void Build () override
- void **setPMOSWidth** (float PMOSWidth)
- void **setPMOSLength** (float length)
- void **setNMOSWidth** (float NMOSWidth)
- void **setNMOSLength** (float NMOSLength)
- void **setElectricalModel** (std::string electricalModel)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.514.1 Member Function Documentation

17.514.1.1 _loadInstance()

```
bool NOT::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.514.1.2 _saveInstance()

```
void NOT::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.514.1.3 Build()

```
void NOT::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.
The documentation for this class was generated from the following files:

- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.515 obj_t Class Reference

Public Member Functions

- **obj_t** (double v, std::string t)
- **obj_t** (double v, std::string t, unsigned long uid)

Public Attributes

- double **valor**
- std::string **tipo**
- unsigned long **id**
- bool **isReference**
- bool **isAttributeReference**

The documentation for this class was generated from the following files:

- parser/obj_t.h
- parser/obj_t.cpp

17.516 Optimizer_if::ObjectiveDefinition Struct Reference

Configurable objective definition.
`#include <tools/Optimization/Optimizer_if.h>`

Public Attributes

- std::string **name**
- ObjectiveSense **sense**
- std::string **expression**
- double **weight** = 1.0
- bool **enabled** = true

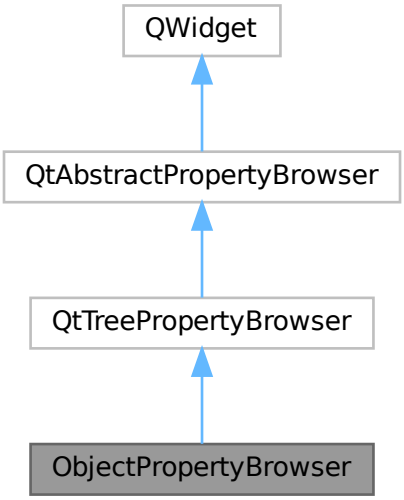
17.516.1 Detailed Description

Configurable objective definition.
The expression is parsed by the model/parser context and can reference selected responses, controls, constants or user-defined expressions.
The documentation for this struct was generated from the following file:

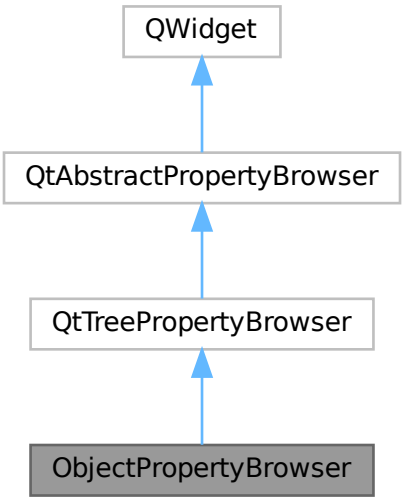
- tools/Optimization/Optimizer_if.h

17.517 ObjectPropertyBrowser Class Reference

Inheritance diagram for ObjectPropertyBrowser:



Collaboration diagram for ObjectPropertyBrowser:



Public Types

- using **ModelChangedCallback** = std::function< void()>

Public Types inherited from QtTreePropertyBrowser

- enum ResizeMode { **Interactive** , **Stretch** , **Fixed** , **ResizeToContents** }

Public Slots

- void **objectUpdated** ()

Public Slots inherited from QtAbstractPropertyBrowser

- QtBrowserItem * addProperty (QtProperty *property)
- QtBrowserItem * insertProperty (QtProperty *property, QtProperty *afterProperty)
- void removeProperty (QtProperty *property)

Public Member Functions

- **ObjectPropertyBrowser** (QWidget *parent=nullptr)
- void **setActiveObject** (QObject *obj, ModelDataDefinition *mdd=nullptr, const QSet< QString > &graphicallyRepresentedModelObjects={}, const QSet< QString > &editableModelObjects={}, PropertyEditorGenesys *peg=nullptr, std::map< SimulationControl *, DataComponentProperty * > *pl=nullptr, std::map< SimulationControl *, DataComponentEditor * > *peUI=nullptr, std::map< SimulationControl *, ComboBoxEnum * > *pb=nullptr)
- void **setActiveGraphicsItem** (QGraphicsItem *item)
- void **clearCurrentlyConnectedObject** ()
- void **setModelChangedCallback** (ModelChangedCallback callback)
- bool **isCommitPipelineBusy** () const

Public Member Functions inherited from QtTreePropertyBrowser

- QtTreePropertyBrowser (QWidget *parent=0)
- ~QtTreePropertyBrowser ()
- int **indentation** () const
- void **setIndentation** (int i)
- bool **rootIsDecorated** () const
- void **setRootIsDecorated** (bool show)
- bool **alternatingRowColors** () const
- void **setAlternatingRowColors** (bool enable)
- bool **isHeaderVisible** () const
- void **setHeaderVisible** (bool visible)
- ResizeMode **resizeMode** () const
- void **setResizeMode** (ResizeMode mode)
- int **splitterPosition** () const
- void **setSplitterPosition** (int position)
- void setExpanded (QtBrowserItem *item, bool expanded)
- bool isExpanded (QtBrowserItem *item) const
- bool isItemVisible (QtBrowserItem *item) const
- void setItemVisible (QtBrowserItem *item, bool visible)
- void setBackgroundColor (QtBrowserItem *item, const QColor &color)
- QColor backgroundColor (QtBrowserItem *item) const
- QColor calculatedBackgroundColor (QtBrowserItem *item) const
- void **setPropertiesWithoutValueMarked** (bool mark)
- bool **propertiesWithoutValueMarked** () const
- void editItem (QtBrowserItem *item)

Public Member Functions inherited from QtAbstractPropertyBrowser

- QtAbstractPropertyBrowser (QWidget *parent=0)
- ~QtAbstractPropertyBrowser ()
- QList< QtProperty * > properties () const
- QList< QtBrowserItem * > items (QtProperty *property) const
- QtBrowserItem * topLevelItem (QtProperty *property) const
- QList< QtBrowserItem * > topLevelItems () const
- void clear ()
- template<class PropertyManager >
  void setFactoryForManager (PropertyManager *manager, QtAbstractEditorFactory< PropertyManager > *factory)
- void unsetFactoryForManager (QtAbstractPropertyManager *manager)
- QtBrowserItem * currentItem () const
- void setCurrentItem (QtBrowserItem *)

Protected Member Functions

- void **keyPressEvent** (QKeyEvent *event) override
- void **contextMenuEvent** (QContextMenuEvent *event) override
- void **mouseDoubleClickEvent** (QMouseEvent *event) override

Protected Member Functions inherited from QtTreePropertyBrowser

- void itemInserted (QtBrowserItem *item, QtBrowserItem *afterItem) override
- void itemRemoved (QtBrowserItem *item) override
- void itemChanged (QtBrowserItem *item) override

Protected Member Functions inherited from QtAbstractPropertyBrowser

- virtual QWidget * createEditor (QtProperty *property, QWidget *parent)

Additional Inherited Members

Signals inherited from QtTreePropertyBrowser

- void collapsed (QtBrowserItem *item)
- void expanded (QtBrowserItem *item)

Signals inherited from QtAbstractPropertyBrowser

- void currentItemChanged (QtBrowserItem *)

Properties inherited from QtTreePropertyBrowser

- int **indentation**

  indentation of the items in the tree view.
- bool **rootIsDecorated**

  whether to show controls for expanding and collapsing root items.
- bool **alternatingRowColors**

  whether to draw the background using alternating colors. By default this property is set to true.
- bool **headerVisible**

  whether to show the header.
- ResizeMode **resizeMode**

  the resize mode of setions in the header.
- int **splitterPosition**

the position of the splitter between the columns.

- bool propertiesWithoutValueMarked

  whether to enable or disable marking properties without value.

The documentation for this class was generated from the following files:

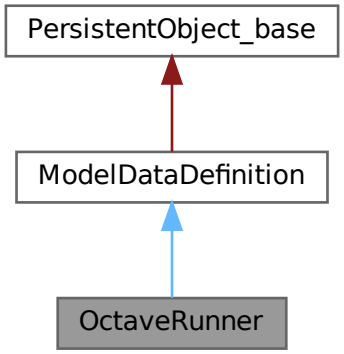
- applications/gui/genesys/propertyeditor/ObjectPropertyBrowser.h
- applications/gui/genesys/propertyeditor/ObjectPropertyBrowser.cpp

17.518 OctaveRunner Class Reference

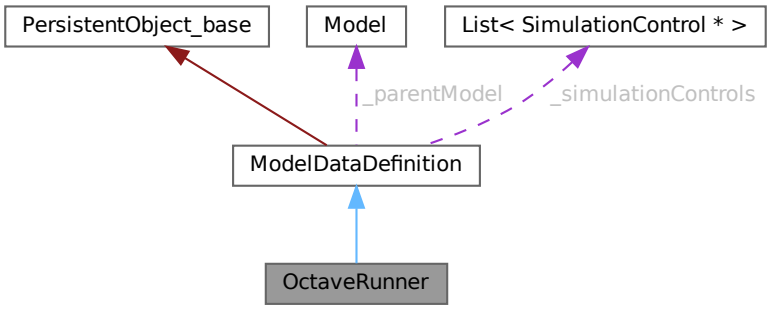
Model data definition that executes Octave code through Rscript.

`#include <plugins/data/ExternalIntegration/OctaveRunner.h>`

Inheritance diagram for OctaveRunner:



Collaboration diagram for OctaveRunner:



Public Member Functions

- **OctaveRunner** (Model *model, std::string name="")

  Creates an Octave Runner bound to a model.

- virtual ~**OctaveRunner** ()=default

Destroys the Octave Runner.

- void **setOctaveExecutable** (std::string octaveExecutable)

  Sets the Rscript executable name or path.
- std::string **getOctaveExecutable** () const

  Returns the Rscript executable name or path.
- void **setWorkingDirectory** (std::string workingDirectory)

  Sets the working directory used for generated scripts and command execution.
- std::string **getWorkingDirectory** () const

  Returns the working directory used for generated scripts and command execution.
- void **setPreludeScript** (std::string preludeScript)

  Sets optional Octave code prepended before the command block.
- std::string **getPreludeScript** () const

  Returns optional Octave code prepended before the command block.
- void **setCommand** (std::string command)

  Sets the Octave command/script body to execute.
- std::string **getCommand** () const

  Returns the Octave command/script body to execute.
- void **setLastStatus** (std::string lastStatus)

  Sets the last execution status.
- std::string **getLastStatus** () const

  Returns the last execution status.
- void **setLastExitCode** (int lastExitCode)

  Sets the last process exit code.
- int **getLastExitCode** () const

  Returns the last process exit code.
- void **setLastStdout** (std::string lastStdout)

  Sets stdout captured from the last Octave execution.
- std::string **getLastStdout** () const

  Returns stdout captured from the last Octave execution.
- void **setLastStderr** (std::string lastStderr)

  Sets stderr captured from the last Octave execution.
- std::string **getLastStderr** () const

  Returns stderr captured from the last Octave execution.
- void **setLastResponsePayload** (std::string lastResponsePayload)

  Sets the response payload exposed to other tools.
- std::string **getLastResponsePayload** () const

  Returns the response payload exposed to other tools.
- void **setLastScriptFilename** (std::string lastScriptFilename)

  Sets the generated Octave script filename from the last execution.
- std::string **getLastScriptFilename** () const

  Returns the generated Octave script filename from the last execution.
- void **setLastResponseFilename** (std::string lastResponseFilename)

  Sets the stdout file path from the last execution.
- std::string **getLastResponseFilename** () const

  Returns the stdout file path from the last execution.
- bool **executeCommand** (std::string &errorMessage)

  Executes the configured Octave code and updates last-execution fields atomically.
- virtual std::string show () override

  Returns a compact human-readable description of the runner.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)

  Loads an Octave Runner from a persistence record.
- static PluginInformation * **GetPluginInformation** ()

  Returns plugin metadata and system dependency declarations.
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

  Creates a new Octave Runner instance for plugin construction.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  Loads persisted Octave Runner configuration and last-execution state.
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves Octave Runner configuration and last-execution state.
- virtual bool _check (std::string &errorMessage) override

  Validates the runner configuration without executing Octave.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)

- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.518.1 Detailed Description

Model data definition that executes Octave code through Rscript.
The runner stores a compact Octave execution configuration, writes a temporary script, executes it with Rscript, and keeps status, exit code, stdout, stderr and response payload inside the model data definition for later inspection.

17.518.2 Member Function Documentation

17.518.2.1 _check()

```
bool OctaveRunner::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

Validates the runner configuration without executing Octave.
Reimplemented from ModelDataDefinition.

17.518.2.2 _loadInstance()

```
bool OctaveRunner::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

Loads persisted Octave Runner configuration and last-execution state.
Reimplemented from ModelDataDefinition.

17.518.2.3 _saveInstance()

```
void OctaveRunner::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves Octave Runner configuration and last-execution state.
Reimplemented from ModelDataDefinition.

17.518.2.4 show()

```
std::string OctaveRunner::show ( )  [override], [virtual]
```

Returns a compact human-readable description of the runner.
Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/ExternalIntegration/OctaveRunner.h
- plugins/data/ExternalIntegration/OctaveRunner.cpp

17.519 ODEfunction Class Reference

Public Member Functions

- **ODEfunction** (std::string expression, double initialPoint, double initialValue)

Public Attributes

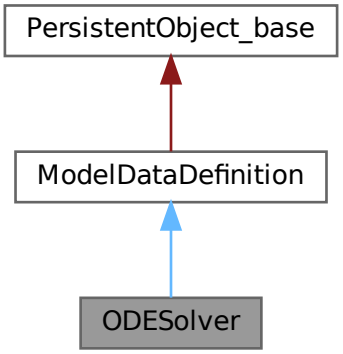
- std::string **expression**
- double **initialPoint**
- double **initialValue**

The documentation for this class was generated from the following file:

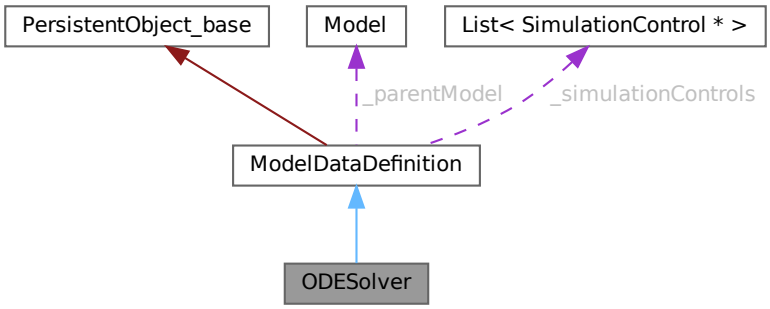
- plugins/components/Continuous/OLD_ODEelement.h

17.520 ODESolver Class Reference

Inheritance diagram for ODESolver:



Collaboration diagram for ODESolver:



Public Member Functions

- **ODESolver** (Model ∗model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **setTimeVariableName** (std::string timeVariableName)
- std::string **getTimeVariableName** () const
- void **setStateVariableNames** (std::vector< std::string > stateVariableNames)
- std::vector< std::string > **getStateVariableNames** () const
- void **setEquationExpressions** (std::vector< std::string > equationExpressions)
- std::vector< std::string > **getEquationExpressions** () const
- void **setStep** (double step)
- double **getStep** () const
- void **setPrecision** (double precision)
- double **getPrecision** () const

- void **setMaxSteps** (int maxSteps)
- int **getMaxSteps** () const
- void **setCurrentTime** (double currentTime)
- double **getCurrentTime** () const
- void **setStateValues** (std::vector< double > stateValues)
- std::vector< double > **getStateValues** () const
- void **setInitialStateValues** (std::vector< double > initialStateValues)
- std::vector< double > **getInitialStateValues** () const
- void **setAutoAdvance** (bool autoAdvance)
- bool **isAutoAdvance** () const
- void **setOutputFile** (std::string outputFile)
- std::string **getOutputFile** () const
- void **integrate** (double targetTime)
- void **resetState** ()
- double **getStateValue** (unsigned int index) const
- void **setStateValue** (unsigned int index, double value)
- void **synchronizeStateFromVariables** ()
- void **writeStateToVariables** () const

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static void **SaveInstance** (PersistenceRecord *fields, ModelDataDefinition *dataDef)
- static bool **Check** (ModelDataDefinition *dataDef, std::string &errorMessage)
- static void **InitBetweenReplications** (ModelDataDefinition *dataDef)
- static void **CreateInternalData** (ModelDataDefinition *dataDef)

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

not just an easy access to trace manager, but wrappers to check if specific trace level applies

- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.520.1 Member Function Documentation

17.520.1.1 _check()

```
bool ODESolver::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.520.1.2 _createEditableDataDefinitions()

```
void ODESolver::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.520.1.3 _initBetweenReplications()

```
void ODESolver::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.520.1.4 _loadInstance()

```
bool ODESolver::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.520.1.5 _saveInstance()

```
void ODESolver::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.520.1.6 show()

```
std::string ODESolver::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

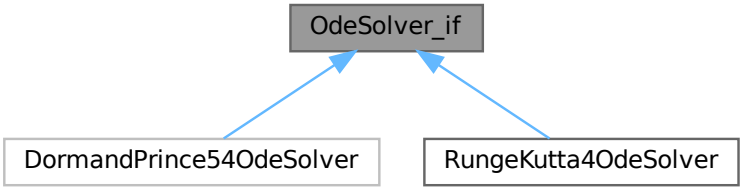
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/Continuous/ODESolver.h
- plugins/data/Continuous/ODESolver.cpp

17.521 OdeSolver_if Class Reference

ODE solver abstraction.
`#include <tools/Continuous/OdeSolver_if.h>`
Inheritance diagram for OdeSolver_if:



Public Member Functions

- virtual bool **advance** (const OdeSystem_if &system, double t0, double dt, const double *y0, double *y1) const =0

17.521.1 Detailed Description

ODE solver abstraction.
Architectural role:

- Planned evolution of the "derivate" responsibility currently exposed by legacy Solver_if.

The documentation for this class was generated from the following file:

- tools/Continuous/OdeSolver_if.h

17.522 OdeSolverFactory Class Reference

Public Types

- using **Creator** = std::function< std::unique_ptr< OdeSolver_if >()>

Static Public Member Functions

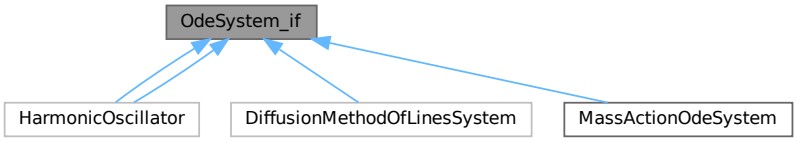
- static const std::string & **keyRungeKutta4** ()
- static const std::string & **keyDormandPrince54** ()
- static const std::string & **defaultKey** ()
- static std::unique_ptr< OdeSolver_if > **create** (OdeSolverType type)
- static std::unique_ptr< OdeSolver_if > **create** (const std::string &key)
- static bool **isRegistered** (const std::string &key)
- static std::vector< std::string > **availableKeys** ()
- static bool **registerCreator** (const std::string &key, Creator creator)
- static std::string **toKey** (OdeSolverType type)
- static bool **fromKey** (const std::string &key, OdeSolverType &out)

The documentation for this class was generated from the following file:

- tools/Continuous/OdeSolverFactory.h

17.523 OdeSystem_if Class Reference

Interface for first-order ODE systems dy/dt = f(t, y).
`#include <tools/Continuous/OdeSystem_if.h>`
Inheritance diagram for OdeSystem_if:



Public Member Functions

- virtual unsigned int **dimension** () const =0
- virtual void **evaluate** (double t, const double *y, double *dydt) const =0

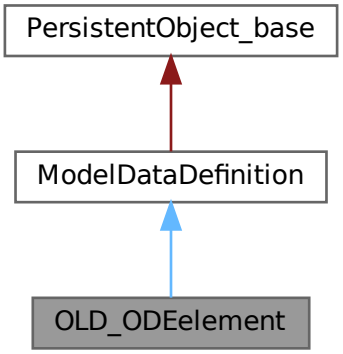
17.523.1 Detailed Description

Interface for first-order ODE systems dy/dt = f(t, y).
The documentation for this class was generated from the following file:

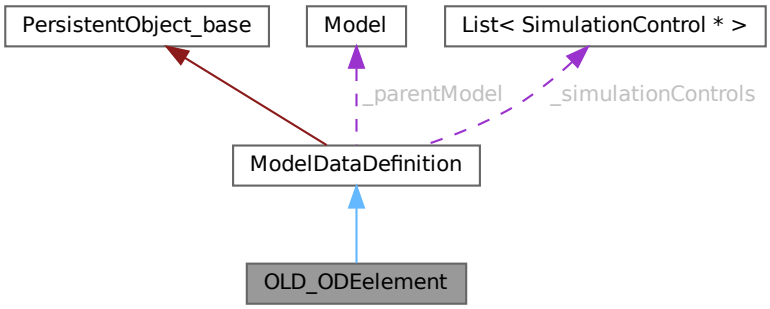
- tools/Continuous/OdeSystem_if.h

17.524 OLD_ODEelement Class Reference

Inheritance diagram for OLD_ODEelement:



Collaboration diagram for OLD_ODEelement:



Public Member Functions

- **OLD_ODEelement** (Model ∗model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- double **solve** ()
- void **setStepH** (double _h)
- double **getStepH** () const
- void **setEndTime** (double _endTime)
- double **getEndTime** () const
- List< ODEfunction ∗ > ∗ **getODEfunctions** () const
- void **addODEfunction** (ODEfunction ∗newFunction)
- void **removeODEfunction** (ODEfunction ∗function)

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)

- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.524.1 Member Function Documentation

17.524.1.1 _check()

```
bool OLD_ODEelement::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.524.1.2 _loadInstance()

```
bool OLD_ODEelement::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.524.1.3 _saveInstance()

```
void OLD_ODEelement::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.524.1.4 show()

```
std::string OLD_ODEelement::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/components/Continuous/OLD_ODEelement.h
- plugins/components/Continuous/OLD_ODEelement.cpp

17.525 OnEventManager Class Reference

```
#include <kernel/simulator/OnEventManager.h>
```

Public Member Functions

- void **addOnModelCheckSucessHandler** (modelEventHandler EventHandler)
- void **addOnModelLoadHandler** (modelEventHandler EventHandler)
- void **addOnModelSaveHandler** (modelEventHandler EventHandler)
- void **addOnReplicationStartHandler** (simulationEventHandler EventHandler)
- void **addOnReplicationStepHandler** (simulationEventHandler EventHandler)
- void **addOnReplicationEndHandler** (simulationEventHandler EventHandler)
- void **addOnProcessEventHandler** (simulationEventHandler EventHandler)
- void **addOnAfterProcessEventHandler** (simulationEventHandler EventHandler)
- void **addOnEntityCreateHandler** (simulationEventHandler EventHandler)
- void **addOnEntityMoveHandler** (simulationEventHandler EventHandler)
- void **addOnEntityRemoveHandler** (simulationEventHandler EventHandler)
- void **addOnSimulationStartHandler** (simulationEventHandler EventHandler)
- void **addOnSimulationPausedHandler** (simulationEventHandler EventHandler)
- void **addOnSimulationResumeHandler** (simulationEventHandler EventHandler)
- void **addOnSimulationEndHandler** (simulationEventHandler EventHandler)

- void **addOnBreakpointHandler** (simulationEventHandler EventHandler)
- template<typename Class >
  void **addOnModelCheckSuccessHandler** (Class *object, void(Class::*function)(ModelEvent *))
- template<typename Class >
  void **addOnModelLoadHandler** (Class *object, void(Class::*function)(ModelEvent *))
- template<typename Class >
  void **addOnModelSaveHandler** (Class *object, void(Class::*function)(ModelEvent *))
- template<typename Class >
  void **addOnReplicationStartHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnReplicationStepHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnReplicationEndHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnProcessEventHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnAfterProcessEventHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnEntityCreateHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnEntityMoveHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnEntityRemoveHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnSimulationStartHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnSimulationPausedHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnSimulationResumeHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnSimulationEndHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **addOnBreakpointHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- void **NotifyModelCheckSuccessHandlers** (ModelEvent *se)
- void **NotifyModelLoadHandlers** (ModelEvent *se)
- void **NotifyModelSaveHandlers** (ModelEvent *se)
- void **NotifyReplicationStartHandlers** (SimulationEvent *se)
- void **NotifyReplicationStepHandlers** (SimulationEvent *se)
- void **NotifyReplicationEndHandlers** (SimulationEvent *se)
- void **NotifyProcessEventHandlers** (SimulationEvent *se)
- void **NotifyAfterProcessEventHandlers** (SimulationEvent *se)
- void **NotifyEntityCreateHandlers** (SimulationEvent *se)
- void **NotifyEntityMoveHandlers** (SimulationEvent *se)
- void **NotifyEntityRemoveHandlers** (SimulationEvent *se)
- void **NotifySimulationStartHandlers** (SimulationEvent *se)
- void **NotifySimulationPausedHandlers** (SimulationEvent *se)
- void **NotifySimulationResumeHandlers** (SimulationEvent *se)
- void **NotifySimulationEndHandlers** (SimulationEvent *se)
- void **NotifyBreakpointHandlers** (SimulationEvent *se)

17.525.1 Detailed Description

OnEventManager allows external methods to hook interval simulation events as listeners (or observers) of pecific events. All methods added as listeners of an event will be invovked when that event is triggered.
The documentation for this class was generated from the following files:

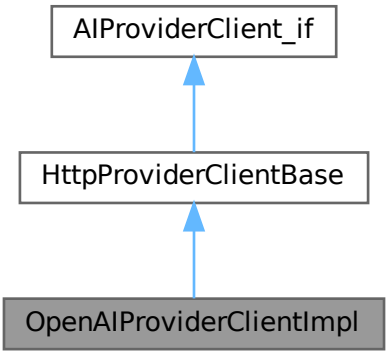
- kernel/simulator/OnEventManager.h
- kernel/simulator/OnEventManager.cpp

17.526 OpenAIProviderClientImpl Class Reference

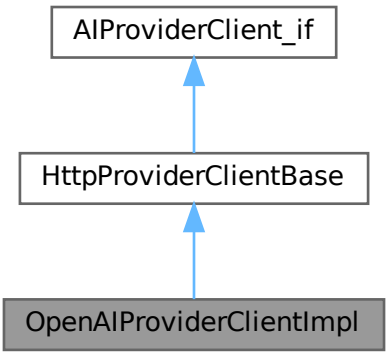
AI provider client for the OpenAI Chat Completions API.

`#include <tools/AIAssistant/OpenAIProviderClientImpl.h>`

Inheritance diagram for OpenAIProviderClientImpl:



Collaboration diagram for OpenAIProviderClientImpl:



Public Member Functions

- OpenAIProviderClientImpl ()

  Constructs the client with the standard OpenAI base URL.
- std::string getProviderName () const override

  Returns a human-readable provider name (e.g. "OpenAI", "Anthropic").
- bool isAvailable () const override

  Returns `true` when an API key is set and a base URL is configured.
- void **setOrganizationId** (const std::string &orgId)

  Sets the OpenAI organization identifier (sent as OpenAI-Organization header).

- std::string **getOrganizationId** () const

  Returns the configured organization identifier.
- void **setProjectId** (const std::string &projectId)

  Sets the OpenAI project identifier (sent as OpenAI-Project header).
- std::string **getProjectId** () const

  Returns the configured project identifier.

Public Member Functions inherited from HttpProviderClientBase

- **HttpProviderClientBase** (const HttpProviderClientBase &)=delete
- HttpProviderClientBase & **operator=** (const HttpProviderClientBase &)=delete
- void setBaseUrl (const std::string &url) override

  Sets the base URL for the provider API endpoint.
- std::string getBaseUrl () const override

  Returns the currently configured base URL.
- void setTimeoutSeconds (unsigned int seconds) override

  Sets the HTTP request timeout in seconds.
- unsigned int getTimeoutSeconds () const override

  Returns the configured HTTP request timeout.
- void setApiKey (const std::string &apiKey) override

  Stores the API key in memory.
- void clearApiKey () override

  Overwrites (best-effort) and clears the in-memory API key.
- bool hasApiKey () const override

  Returns `true` when a non-empty API key is held.
- ProviderResponse send (const ProviderRequest &request) override

  Executes the HTTP request via curl and returns the provider response.

Protected Member Functions

- std::string _buildRequestBody (const ProviderRequest &request) const override

  Serialises `request` to the provider's JSON wire format.
- ProviderResponse _parseResponseBody (const std::string &rawBody, int httpStatus) const override

  Deserialises `rawBody` (the full curl stdout) into a ProviderResponse.
- std::string _authorizationHeader () const override

  Returns the Authorization header line without the "header = " prefix.
- std::vector< std::string > _extraRequestHeaders () const override

  Returns additional HTTP headers to send with every request.

Additional Inherited Members

Static Public Member Functions inherited from HttpProviderClientBase

- static std::string **_escapeJsonString** (const std::string &s)

  Escapes a plain string for safe embedding inside a JSON string value.
- static std::string **_extractJsonString** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON string field by key. Returns an empty string when not found or value is not a quoted string.
- static int **_extractJsonInt** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON integer field by key. Returns 0 when not found.
- static double **_extractJsonDouble** (const std::string &json, const std::string &key)

  Extracts the value of a top-level JSON number field (int or float) by key. Returns 0.0 when not found.
- static std::string **_buildMessagesArray** (const std::vector< ChatMessage > &messages)

Serialises a vector of ChatMessage objects to a JSON array string.

- static std::string **_extractFirstArrayElement** (const std::string &json, const std::string &arrayKey)

  Extracts the raw substring of the first element of a named JSON array. Returns an empty string when not found.

Protected Attributes inherited from HttpProviderClientBase

- std::string **_baseUrl**
- unsigned int **_timeoutSeconds** = 30
- std::string _apiKey

  In-memory API key.

17.526.1 Detailed Description

AI provider client for the OpenAI Chat Completions API.
Wraps the OpenAI `/v1/chat/completions` endpoint. Default base URL: `https://api.openai.com/v1/chat/completions`
Supported features:

- Standard chat completions (gpt-4o, gpt-4-turbo, gpt-3.5-turbo, etc.).
- Reasoning models via highReasoningMode (o3, o4-mini): temperature is omitted and reasoning_effort is set to "high" in the request body.
- Organization and project headers when set via setOrganizationId() / setProjectId().

17.526.1.1 API key

Passed as `Authorization: Bearer <key>`. Stored in the parent's _apiKey member; never written to the command line or logs.

17.526.2 Constructor & Destructor Documentation

17.526.2.1 OpenAIProviderClientImpl()

```
OpenAIProviderClientImpl::OpenAIProviderClientImpl ( )
```

Constructs the client with the standard OpenAI base URL.
To use with Azure OpenAI or a compatible proxy, call setBaseUrl() after construction.

17.526.3 Member Function Documentation

17.526.3.1 _authorizationHeader()

```
std::string OpenAIProviderClientImpl::_authorizationHeader ( ) const  [override], [protected],
[virtual]
```

Returns the Authorization header line without the "header = " prefix.
Example return values:

- "Authorization: Bearer sk-..." (OpenAI, LocalProvider)
- "x-api-key: sk-ant-..." (Anthropic)

Called only from within send(); the value is written to a 0600 temp file and never logged.
Implements HttpProviderClientBase.

17.526.3.2 _buildRequestBody()

```
std::string OpenAIProviderClientImpl::_buildRequestBody (
            const ProviderRequest & request ) const  [override], [protected], [virtual]
```

Serialises `request` to the provider's JSON wire format.
Implements HttpProviderClientBase.

17.526.3.3 _extraRequestHeaders()

```
std::vector< std::string > OpenAIProviderClientImpl::_extraRequestHeaders ( ) const  [override], [protected], [virtual]
```

Returns additional HTTP headers to send with every request.
Default implementation returns an empty vector. Anthropic overrides to add "anthropic-version: 2023-06-01".
Each element must be a complete header line, e.g. "anthropic-version: 2023-06-01".
Reimplemented from HttpProviderClientBase.

17.526.3.4 _parseResponseBody()

```
ProviderResponse OpenAIProviderClientImpl::_parseResponseBody (
            const std::string & rawBody,
            int httpStatus ) const  [override], [protected], [virtual]
```

Deserialises `rawBody` (the full curl stdout) into a ProviderResponse.

Parameters

rawBody	Complete HTTP response body as received from curl.
httpStatus	HTTP status code captured from curl exit / response line, or 0 when unavailable.

Implements HttpProviderClientBase.

17.526.3.5 getProviderName()

```
std::string OpenAIProviderClientImpl::getProviderName ( ) const  [override], [virtual]
```

Returns a human-readable provider name (e.g. "OpenAI", "Anthropic").
Implements AIProviderClient_if.

17.526.3.6 isAvailable()

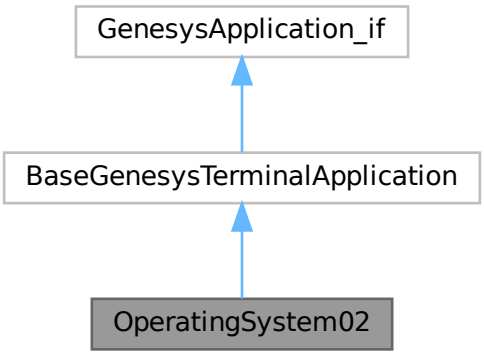
```
bool OpenAIProviderClientImpl::isAvailable ( ) const  [override], [virtual]
```

Returns `true` when an API key is set and a base URL is configured.
Reimplemented from HttpProviderClientBase.
The documentation for this class was generated from the following files:

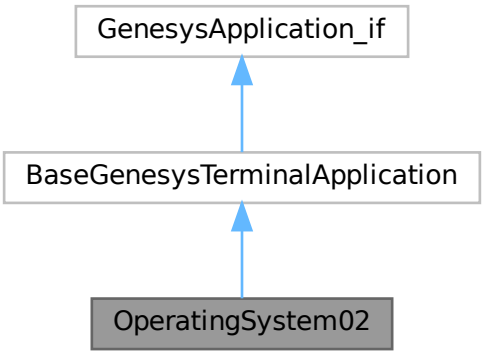
- tools/AIAssistant/OpenAIProviderClientImpl.h
- tools/AIAssistant/OpenAIProviderClientImpl.cpp

17.527 OperatingSystem02 Class Reference

Inheritance diagram for OperatingSystem02:



Collaboration diagram for OperatingSystem02:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.527.1 Member Function Documentation

17.527.1.1 main()

```
int OperatingSystem02::main (
            int argc,
            char ** argv )  [override], [virtual]
```

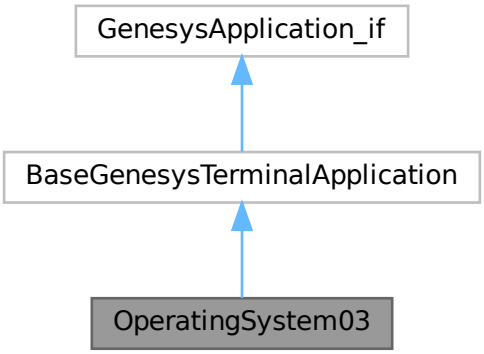
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

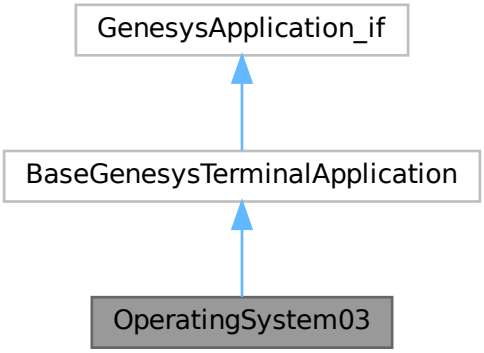
- applications/modelSpecific/teaching/OperatingSystem02.h
- applications/modelSpecific/teaching/OperatingSystem02.cpp

17.528 OperatingSystem03 Class Reference

Inheritance diagram for OperatingSystem03:



Collaboration diagram for OperatingSystem03:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.528.1 Member Function Documentation

17.528.1.1 main()

```
int OperatingSystem03::main (
            int argc,
            char ** argv )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/teaching/OperatingSystem03.h
- applications/modelSpecific/teaching/OperatingSystem03.cpp

17.529 Optimizer_if::OptimizationSettings Struct Reference

Generic optimization settings independent of a specific algorithm.
`#include <tools/Optimization/Optimizer_if.h>`

Public Attributes

- unsigned int **maxIterations** = 100
- unsigned int **maxSimulations** = 0
- unsigned int **replicationsPerSolution** = 1
- unsigned int **bestSolutionsToKeep** = 10
- unsigned int **randomSeed** = 0
- double **improvementTolerance** = 0.0
- double **timeLimitSeconds** = 0.0

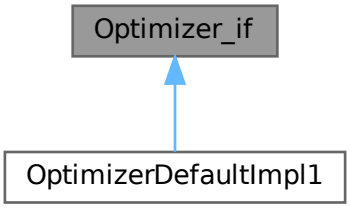
17.529.1 Detailed Description

Generic optimization settings independent of a specific algorithm.
The documentation for this struct was generated from the following file:

- tools/Optimization/Optimizer_if.h

17.530 Optimizer_if Class Reference

Interface for simulation-based optimization engines (OptQuest-like).
`#include <tools/Optimization/Optimizer_if.h>`
Inheritance diagram for Optimizer_if:



Classes

- struct ConstraintDefinition

  Feasibility restriction represented as an expression.
- struct ObjectiveDefinition

  Configurable objective definition.
- struct OptimizationSettings

  Generic optimization settings independent of a specific algorithm.
- struct SolutionSummary

  Captures one candidate or best-known optimization solution.

Public Types

- enum class ObjectiveSense { **MINIMIZE** , **MAXIMIZE** }

  Direction used by an optimization objective.
- enum class ExecutionState {
  NOT_READY , **READY** , **RUNNING** , **PAUSED** ,
  STOPPED , **FINISHED** , **ERROR** }

  Execution state of the optimization process.

Public Member Functions

- virtual void setModel (Model *model)=0

  Associates an already instantiated model with this optimizer.
- virtual bool setModelFilename (const std::string &filename)=0

  Loads/associates a model from a file supported by the simulator.
- virtual Model * getModel () const =0

  Returns the model currently associated with the optimizer.
- virtual std::string getModelFilename () const =0

  Returns the source filename previously configured for model loading.
- virtual List< SimulationControl * > * getAvailableControls () const =0

  Retrieves available model controls discoverable from Model.
- virtual List< SimulationResponse * > * getAvailableResponses () const =0

  Retrieves available model responses discoverable from Model.
- virtual void setSelectedControls (List< SimulationControl * > *controls)=0

  Replaces the set of controls selected for optimization.
- virtual void setSelectedResponses (List< SimulationResponse * > *responses)=0

  Replaces the set of responses selected for optimization.
- virtual List< SimulationControl * > * getSelectedControls () const =0

  Returns currently selected controls used as decision variables.
- virtual List< SimulationResponse * > * getSelectedResponses () const =0

  Returns currently selected responses monitored by optimization.
- virtual void addObjective (const ObjectiveDefinition &objective)=0

  Adds one objective to the optimization model.
- virtual void clearObjectives ()=0

  Clears all configured objectives.
- virtual List< ObjectiveDefinition > * getObjectives () const =0

  Returns all configured optimization objectives.
- virtual void addConstraint (const ConstraintDefinition &constraint)=0

  Adds one feasibility constraint expression.
- virtual void clearConstraints ()=0

  Clears all configured constraints.
- virtual List< ConstraintDefinition > * getConstraints () const =0

Returns all configured feasibility constraints.

- virtual void setSettings (const OptimizationSettings &settings)=0

  Applies algorithm settings such as iteration and retention limits.
- virtual OptimizationSettings getSettings () const =0

  Returns currently active optimization settings.
- virtual bool checkReady (std::string *message=nullptr) const =0

  Validates readiness before running optimization.
- virtual bool start ()=0

  Starts a new optimization run.
- virtual bool step ()=0

  Executes one optimization iteration (or one scenario decision step).
- virtual bool pause ()=0

  Pauses an active optimization execution.
- virtual bool resume ()=0

  Resumes optimization execution after a pause.
- virtual bool stop ()=0

  Stops optimization execution and finalizes current run.
- virtual ExecutionState getExecutionState () const =0

  Returns current lifecycle state of the optimizer.
- virtual unsigned int getCurrentIteration () const =0

  Returns the number of optimization iterations already performed.
- virtual unsigned int getTotalSimulations () const =0

  Returns total number of simulated scenarios evaluated so far.
- virtual List< SolutionSummary > * getBestSolutions () const =0

  Returns all feasible solutions retained as "best" by the algorithm.
- virtual const SolutionSummary * getCurrentBestSolution () const =0

  Returns the best-known feasible solution, or nullptr when absent.
- virtual void resetResults ()=0

  Clears results from previous runs while preserving configuration.

17.530.1 Detailed Description

Interface for simulation-based optimization engines (OptQuest-like).
This contract defines how an optimization tool can be attached to a model, configured, and executed while managing:

- decision variables (selected SimulationControl objects),
- observed KPIs (selected SimulationResponse objects),
- objectives (minimize/maximize expressions), and
- constraints (boolean expressions evaluated as false==0 and true!=0).

Typical flow: 1) Associate a model (from file or direct instance). 2) Select controls and responses participating in the search. 3) Register optimization objectives and feasibility constraints. 4) Configure search parameters (iterations, retained best solutions, etc.). 5) Run lifecycle commands (start/step/pause/resume/stop). 6) Query current status, best solutions and progress metrics.

17.530.2 Member Function Documentation

17.530.2.1 addConstraint()

```
virtual void Optimizer_if::addConstraint (
            const ConstraintDefinition & constraint )  [pure virtual]
```

Adds one feasibility constraint expression.
Implemented in OptimizerDefaultImpl1.

17.530.2.2 addObjective()

```
virtual void Optimizer_if::addObjective (
            const ObjectiveDefinition & objective )  [pure virtual]
```

Adds one objective to the optimization model.
Implemented in OptimizerDefaultImpl1.

17.530.2.3 checkReady()

```
virtual bool Optimizer_if::checkReady (
            std::string * message = nullptr ) const  [pure virtual]
```

Validates readiness before running optimization.

Parameters

message	Receives validation details when non-null.

Returns

true when optimizer has enough data/configuration to start.

Implemented in OptimizerDefaultImpl1.

17.530.2.4 clearConstraints()

```
virtual void Optimizer_if::clearConstraints ( )  [pure virtual]
```

Clears all configured constraints.
Implemented in OptimizerDefaultImpl1.

17.530.2.5 clearObjectives()

```
virtual void Optimizer_if::clearObjectives ( )  [pure virtual]
```

Clears all configured objectives.
Implemented in OptimizerDefaultImpl1.

17.530.2.6 getAvailableControls()

```
virtual List< SimulationControl * > * Optimizer_if::getAvailableControls ( ) const  [pure
virtual]
```

Retrieves available model controls discoverable from Model.
Implemented in OptimizerDefaultImpl1.

17.530.2.7 getAvailableResponses()

```
virtual List< SimulationResponse * > * Optimizer_if::getAvailableResponses ( ) const  [pure
virtual]
```

Retrieves available model responses discoverable from Model.
Implemented in OptimizerDefaultImpl1.

17.530.2.8 getBestSolutions()

```
virtual List< SolutionSummary > * Optimizer_if::getBestSolutions ( ) const  [pure virtual]
```

Returns all feasible solutions retained as "best" by the algorithm.
Implemented in OptimizerDefaultImpl1.

17.530.2.9 getConstraints()

```
virtual List< ConstraintDefinition > * Optimizer_if::getConstraints ( ) const  [pure virtual]
```

Returns all configured feasibility constraints.
Implemented in OptimizerDefaultImpl1.

17.530.2.10 getCurrentBestSolution()

```
virtual const SolutionSummary * Optimizer_if::getCurrentBestSolution ( ) const  [pure virtual]
```

Returns the best-known feasible solution, or nullptr when absent.
Implemented in OptimizerDefaultImpl1.

17.530.2.11 getCurrentIteration()

```
virtual unsigned int Optimizer_if::getCurrentIteration ( ) const  [pure virtual]
```

Returns the number of optimization iterations already performed.
Implemented in OptimizerDefaultImpl1.

17.530.2.12 getExecutionState()

```
virtual ExecutionState Optimizer_if::getExecutionState ( ) const  [pure virtual]
```

Returns current lifecycle state of the optimizer.
Implemented in OptimizerDefaultImpl1.

17.530.2.13 getModel()

```
virtual Model * Optimizer_if::getModel ( ) const  [pure virtual]
```

Returns the model currently associated with the optimizer.
Implemented in OptimizerDefaultImpl1.

17.530.2.14 getModelFilename()

```
virtual std::string Optimizer_if::getModelFilename ( ) const  [pure virtual]
```

Returns the source filename previously configured for model loading.
Implemented in OptimizerDefaultImpl1.

17.530.2.15 getObjectives()

```
virtual List< ObjectiveDefinition > * Optimizer_if::getObjectives ( ) const  [pure virtual]
```

Returns all configured optimization objectives.
Implemented in OptimizerDefaultImpl1.

17.530.2.16 getSelectedControls()

```
virtual List< SimulationControl * > * Optimizer_if::getSelectedControls ( ) const  [pure virtual]
```

Returns currently selected controls used as decision variables.
Implemented in OptimizerDefaultImpl1.

17.530.2.17 getSelectedResponses()

```
virtual List< SimulationResponse * > * Optimizer_if::getSelectedResponses ( ) const  [pure
virtual]
```

Returns currently selected responses monitored by optimization.
Implemented in OptimizerDefaultImpl1.

17.530.2.18 getSettings()

```
virtual OptimizationSettings Optimizer_if::getSettings ( ) const  [pure virtual]
```

Returns currently active optimization settings.
Implemented in OptimizerDefaultImpl1.

17.530.2.19 getTotalSimulations()

```
virtual unsigned int Optimizer_if::getTotalSimulations ( ) const  [pure virtual]
```

Returns total number of simulated scenarios evaluated so far.
Implemented in OptimizerDefaultImpl1.

17.530.2.20 pause()

```
virtual bool Optimizer_if::pause ( )  [pure virtual]
```

Pauses an active optimization execution.
Implemented in OptimizerDefaultImpl1.

17.530.2.21 resetResults()

```
virtual void Optimizer_if::resetResults ( )  [pure virtual]
```

Clears results from previous runs while preserving configuration.
Implemented in OptimizerDefaultImpl1.

17.530.2.22 resume()

```
virtual bool Optimizer_if::resume ( )  [pure virtual]
```

Resumes optimization execution after a pause.
Implemented in OptimizerDefaultImpl1.

17.530.2.23 setModel()

```
virtual void Optimizer_if::setModel (
            Model * model )  [pure virtual]
```

Associates an already instantiated model with this optimizer.
Implemented in OptimizerDefaultImpl1.

17.530.2.24 setModelFilename()

```
virtual bool Optimizer_if::setModelFilename (
            const std::string & filename )  [pure virtual]
```

Loads/associates a model from a file supported by the simulator.

Parameters

filename	Model file path.

Returns

true when the model file was successfully loaded and linked.

Implemented in OptimizerDefaultImpl1.

17.530.2.25 setSelectedControls()

```
virtual void Optimizer_if::setSelectedControls (
            List< SimulationControl * > * controls )  [pure virtual]
```

Replaces the set of controls selected for optimization.
Implemented in OptimizerDefaultImpl1.

17.530.2.26 setSelectedResponses()

```
virtual void Optimizer_if::setSelectedResponses (
            List< SimulationResponse * > * responses )  [pure virtual]
```

Replaces the set of responses selected for optimization.
Implemented in OptimizerDefaultImpl1.

17.530.2.27 setSettings()

```
virtual void Optimizer_if::setSettings (
            const OptimizationSettings & settings )  [pure virtual]
```

Applies algorithm settings such as iteration and retention limits.
Implemented in OptimizerDefaultImpl1.

17.530.2.28 start()

```
virtual bool Optimizer_if::start ( )  [pure virtual]
```

Starts a new optimization run.

Returns

true on successful transition to RUNNING state.

Implemented in OptimizerDefaultImpl1.

17.530.2.29 step()

```
virtual bool Optimizer_if::step ( )  [pure virtual]
```

Executes one optimization iteration (or one scenario decision step).

Returns

true when a step was executed.

Implemented in OptimizerDefaultImpl1.

17.530.2.30 stop()

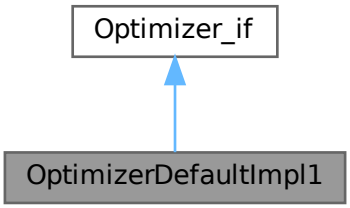
```
virtual bool Optimizer_if::stop ( )  [pure virtual]
```

Stops optimization execution and finalizes current run.
Implemented in OptimizerDefaultImpl1.
The documentation for this class was generated from the following file:

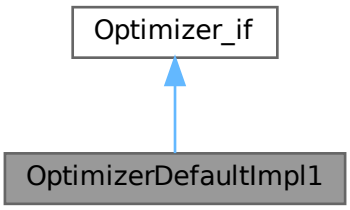
- tools/Optimization/Optimizer_if.h

17.531 OptimizerDefaultImpl1 Class Reference

Initial concrete scaffold for an OptQuest-like optimizer.
`#include <tools/Optimization/OptimizerDefaultImpl1.h>`
Inheritance diagram for OptimizerDefaultImpl1:



Collaboration diagram for OptimizerDefaultImpl1:



Public Member Functions

- **OptimizerDefaultImpl1** (const OptimizerDefaultImpl1 &)=delete
- OptimizerDefaultImpl1 & **operator=** (const OptimizerDefaultImpl1 &)=delete
- **OptimizerDefaultImpl1** (OptimizerDefaultImpl1 &&)=delete
- OptimizerDefaultImpl1 & **operator=** (OptimizerDefaultImpl1 &&)=delete
- virtual void setModel (Model *model) override

  Associates an already instantiated model with this optimizer.
- virtual bool setModelFilename (const std::string &filename) override

  Loads/associates a model from a file supported by the simulator.
- virtual Model * getModel () const override

  Returns the model currently associated with the optimizer.
- virtual std::string getModelFilename () const override

  Returns the source filename previously configured for model loading.
- virtual List< SimulationControl * > * getAvailableControls () const override

  Retrieves available model controls discoverable from Model.
- virtual List< SimulationResponse * > * getAvailableResponses () const override

  Retrieves available model responses discoverable from Model.
- virtual void setSelectedControls (List< SimulationControl * > *controls) override

  Replaces the set of controls selected for optimization.
- virtual void setSelectedResponses (List< SimulationResponse * > *responses) override

  Replaces the set of responses selected for optimization.
- virtual List< SimulationControl * > * getSelectedControls () const override

  Returns currently selected controls used as decision variables.
- virtual List< SimulationResponse * > * getSelectedResponses () const override

  Returns currently selected responses monitored by optimization.
- virtual void addObjective (const ObjectiveDefinition &objective) override

  Adds one objective to the optimization model.
- virtual void clearObjectives () override

  Clears all configured objectives.
- virtual List< ObjectiveDefinition > * getObjectives () const override

  Returns all configured optimization objectives.
- virtual void addConstraint (const ConstraintDefinition &constraint) override

  Adds one feasibility constraint expression.
- virtual void clearConstraints () override

  Clears all configured constraints.
- virtual List< ConstraintDefinition > * getConstraints () const override

Returns all configured feasibility constraints.

- virtual void setSettings (const OptimizationSettings &settings) override

  Applies algorithm settings such as iteration and retention limits.
- virtual OptimizationSettings getSettings () const override

  Returns currently active optimization settings.
- virtual bool checkReady (std::string *message=nullptr) const override

  Validates readiness before running optimization.
- virtual bool start () override

  Starts a new optimization run.
- virtual bool step () override

  Executes one optimization iteration (or one scenario decision step).
- virtual bool pause () override

  Pauses an active optimization execution.
- virtual bool resume () override

  Resumes optimization execution after a pause.
- virtual bool stop () override

  Stops optimization execution and finalizes current run.
- virtual ExecutionState getExecutionState () const override

  Returns current lifecycle state of the optimizer.
- virtual unsigned int getCurrentIteration () const override

  Returns the number of optimization iterations already performed.
- virtual unsigned int getTotalSimulations () const override

  Returns total number of simulated scenarios evaluated so far.
- virtual List< SolutionSummary > * getBestSolutions () const override

  Returns all feasible solutions retained as "best" by the algorithm.
- virtual const SolutionSummary * getCurrentBestSolution () const override

  Returns the best-known feasible solution, or nullptr when absent.
- virtual void resetResults () override

  Clears results from previous runs while preserving configuration.

Additional Inherited Members

Public Types inherited from Optimizer_if

- enum class ObjectiveSense { **MINIMIZE** , **MAXIMIZE** }

  Direction used by an optimization objective.
- enum class ExecutionState {
  NOT_READY , **READY** , **RUNNING** , **PAUSED** ,
  STOPPED , **FINISHED** , **ERROR** }

  Execution state of the optimization process.

17.531.1 Detailed Description

Initial concrete scaffold for an OptQuest-like optimizer.
This class intentionally provides only the public API declaration for the first default implementation.
Current status:

- interface contract declared;
- behavior/algorithm implementation intentionally deferred.

Planned behavior:

- discover controls/responses from an associated model,
- evaluate objective and constraint expressions through parser context,
- execute iterative search preserving top feasible solutions.

17.531.2 Member Function Documentation

17.531.2.1 addConstraint()

```
void OptimizerDefaultImpl1::addConstraint (
            const ConstraintDefinition & constraint )  [override], [virtual]
```

Adds one feasibility constraint expression.
Implements Optimizer_if.

17.531.2.2 addObjective()

```
void OptimizerDefaultImpl1::addObjective (
            const ObjectiveDefinition & objective )  [override], [virtual]
```

Adds one objective to the optimization model.
Implements Optimizer_if.

17.531.2.3 checkReady()

```
bool OptimizerDefaultImpl1::checkReady (
            std::string * message = nullptr ) const  [override], [virtual]
```

Validates readiness before running optimization.

Parameters

message	Receives validation details when non-null.

Returns

true when optimizer has enough data/configuration to start.

Implements Optimizer_if.

17.531.2.4 clearConstraints()

```
void OptimizerDefaultImpl1::clearConstraints ( )  [override], [virtual]
```

Clears all configured constraints.
Implements Optimizer_if.

17.531.2.5 clearObjectives()

```
void OptimizerDefaultImpl1::clearObjectives ( )  [override], [virtual]
```

Clears all configured objectives.
Implements Optimizer_if.

17.531.2.6 getAvailableControls()

```
List< SimulationControl * > * OptimizerDefaultImpl1::getAvailableControls ( ) const  [override],
[virtual]
```

Retrieves available model controls discoverable from Model.
Implements Optimizer_if.

17.531.2.7 getAvailableResponses()

```
List< SimulationResponse * > * OptimizerDefaultImpl1::getAvailableResponses ( ) const  [override],
[virtual]
```

Retrieves available model responses discoverable from Model.
Implements Optimizer_if.

17.531.2.8 getBestSolutions()

`List< Optimizer_if::SolutionSummary > * OptimizerDefaultImpl1::getBestSolutions ( ) const [override], [virtual]`
Returns all feasible solutions retained as "best" by the algorithm.
Implements Optimizer_if.

17.531.2.9 getConstraints()

`List< Optimizer_if::ConstraintDefinition > * OptimizerDefaultImpl1::getConstraints ( ) const [override], [virtual]`
Returns all configured feasibility constraints.
Implements Optimizer_if.

17.531.2.10 getCurrentBestSolution()

`const Optimizer_if::SolutionSummary * OptimizerDefaultImpl1::getCurrentBestSolution ( ) const [override], [virtual]`
Returns the best-known feasible solution, or nullptr when absent.
Implements Optimizer_if.

17.531.2.11 getCurrentIteration()

`unsigned int OptimizerDefaultImpl1::getCurrentIteration ( ) const [override], [virtual]`
Returns the number of optimization iterations already performed.
Implements Optimizer_if.

17.531.2.12 getExecutionState()

`Optimizer_if::ExecutionState OptimizerDefaultImpl1::getExecutionState ( ) const [override], [virtual]`
Returns current lifecycle state of the optimizer.
Implements Optimizer_if.

17.531.2.13 getModel()

`Model * OptimizerDefaultImpl1::getModel ( ) const [override], [virtual]`
Returns the model currently associated with the optimizer.
Implements Optimizer_if.

17.531.2.14 getModelFilename()

`std::string OptimizerDefaultImpl1::getModelFilename ( ) const [override], [virtual]`
Returns the source filename previously configured for model loading.
Implements Optimizer_if.

17.531.2.15 getObjectives()

`List< Optimizer_if::ObjectiveDefinition > * OptimizerDefaultImpl1::getObjectives ( ) const [override], [virtual]`
Returns all configured optimization objectives.
Implements Optimizer_if.

17.531.2.16 getSelectedControls()

`List< SimulationControl * > * OptimizerDefaultImpl1::getSelectedControls ( ) const [override], [virtual]`
Returns currently selected controls used as decision variables.
Implements Optimizer_if.

17.531.2.17 getSelectedResponses()

```
List< SimulationResponse * > * OptimizerDefaultImpl1::getSelectedResponses ( ) const  [override], [virtual]
```

Returns currently selected responses monitored by optimization.
Implements Optimizer_if.

17.531.2.18 getSettings()

```
Optimizer_if::OptimizationSettings OptimizerDefaultImpl1::getSettings ( ) const  [override], [virtual]
```

Returns currently active optimization settings.
Implements Optimizer_if.

17.531.2.19 getTotalSimulations()

```
unsigned int OptimizerDefaultImpl1::getTotalSimulations ( ) const  [override], [virtual]
```

Returns total number of simulated scenarios evaluated so far.
Implements Optimizer_if.

17.531.2.20 pause()

```
bool OptimizerDefaultImpl1::pause ( )  [override], [virtual]
```

Pauses an active optimization execution.
Implements Optimizer_if.

17.531.2.21 resetResults()

```
void OptimizerDefaultImpl1::resetResults ( )  [override], [virtual]
```

Clears results from previous runs while preserving configuration.
Implements Optimizer_if.

17.531.2.22 resume()

```
bool OptimizerDefaultImpl1::resume ( )  [override], [virtual]
```

Resumes optimization execution after a pause.
Implements Optimizer_if.

17.531.2.23 setModel()

```
void OptimizerDefaultImpl1::setModel (
            Model * model )  [override], [virtual]
```

Associates an already instantiated model with this optimizer.
Implements Optimizer_if.

17.531.2.24 setModelFilename()

```
bool OptimizerDefaultImpl1::setModelFilename (
            const std::string & filename )  [override], [virtual]
```

Loads/associates a model from a file supported by the simulator.

Parameters

filename	Model file path.

Returns

true when the model file was successfully loaded and linked.

Implements Optimizer_if.

17.531.2.25 setSelectedControls()

```
void OptimizerDefaultImpl1::setSelectedControls (
            List< SimulationControl * > * controls )  [override], [virtual]
```

Replaces the set of controls selected for optimization.
Implements Optimizer_if.

17.531.2.26 setSelectedResponses()

```
void OptimizerDefaultImpl1::setSelectedResponses (
            List< SimulationResponse * > * responses )  [override], [virtual]
```

Replaces the set of responses selected for optimization.
Implements Optimizer_if.

17.531.2.27 setSettings()

```
void OptimizerDefaultImpl1::setSettings (
            const OptimizationSettings & settings )  [override], [virtual]
```

Applies algorithm settings such as iteration and retention limits.
Implements Optimizer_if.

17.531.2.28 start()

```
bool OptimizerDefaultImpl1::start ( )  [override], [virtual]
```

Starts a new optimization run.

Returns

true on successful transition to RUNNING state.

Implements Optimizer_if.

17.531.2.29 step()

```
bool OptimizerDefaultImpl1::step ( )  [override], [virtual]
```

Executes one optimization iteration (or one scenario decision step).

Returns

true when a step was executed.

Implements Optimizer_if.

17.531.2.30 stop()

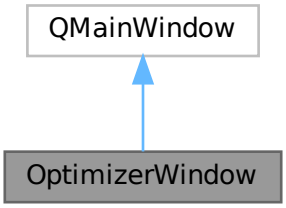
```
bool OptimizerDefaultImpl1::stop ( )  [override], [virtual]
```

Stops optimization execution and finalizes current run.
Implements Optimizer_if.
The documentation for this class was generated from the following files:

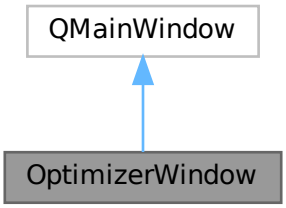
- tools/Optimization/OptimizerDefaultImpl1.h
- tools/Optimization/OptimizerDefaultImpl1.cpp

17.532 OptimizerWindow Class Reference

Inheritance diagram for OptimizerWindow:



Collaboration diagram for OptimizerWindow:



Classes

- class ProgressPlot

Public Member Functions

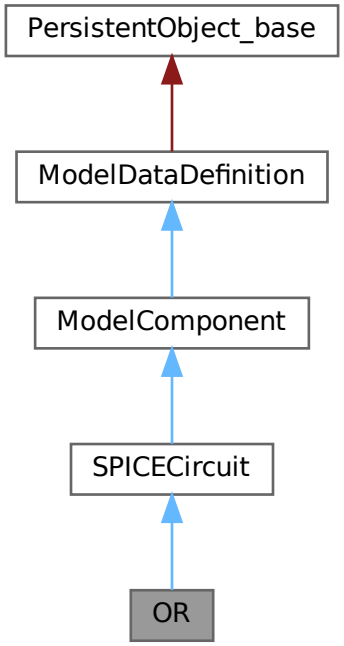
- **OptimizerWindow** (Simulator ∗simulator, QWidget ∗parent=nullptr)

The documentation for this class was generated from the following files:

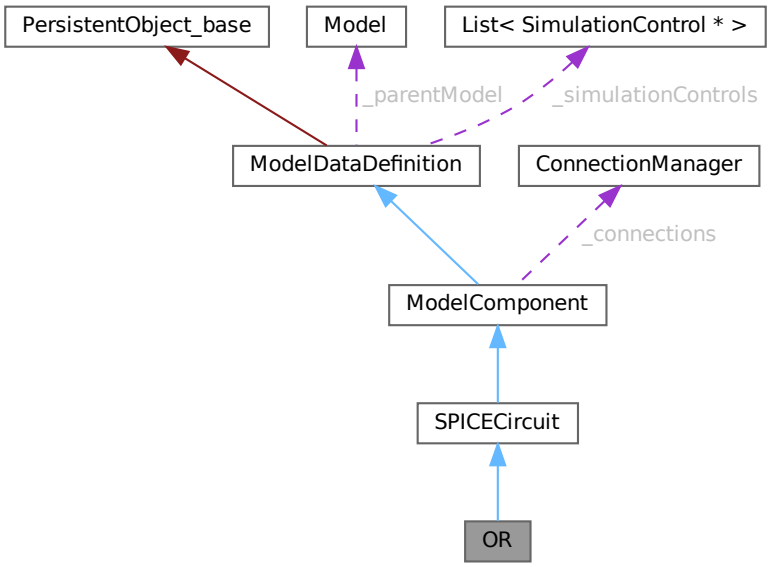
- applications/gui/genesys/tools/optimizer/OptimizerWindow.h
- applications/gui/genesys/tools/optimizer/OptimizerWindow.cpp

17.533 OR Class Reference

Inheritance diagram for OR:



Collaboration diagram for OR:



Public Member Functions

- **OR** (Model *model, std::string name="")
- void Build () override
- void **setPMOSWidth** (float PMOSWidth)
- void **setPMOSLength** (float length)
- void **setNMOSWidth** (float NMOSWidth)
- void **setNMOSLength** (float NMOSLength)
- void **setElectricalModel** (std::string electricalModel)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.533.1 Member Function Documentation

17.533.1.1 _loadInstance()

```
bool OR::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.533.1.2 _saveInstance()

```
void OR::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.533.1.3 Build()

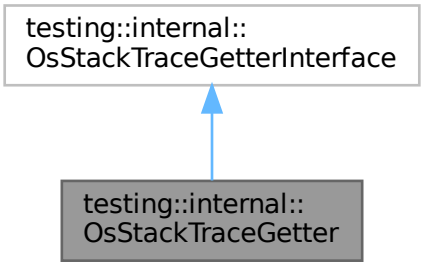
```
void OR::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.
The documentation for this class was generated from the following files:

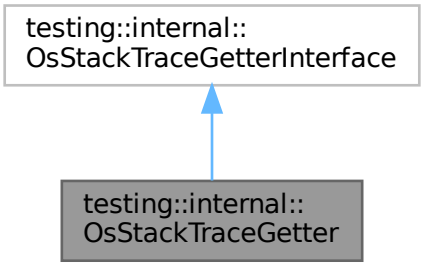
- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.534 testing::internal::OsStackTraceGetter Class Reference

Inheritance diagram for testing::internal::OsStackTraceGetter:



Collaboration diagram for testing::internal::OsStackTraceGetter:



Public Member Functions

- std::string CurrentStackTrace (int max_depth, int skip_count) override
- void UponLeavingGTest () override

Additional Inherited Members

Static Public Attributes inherited from testing::internal::OsStackTraceGetterInterface

- static const char ∗const kElidedFramesMarker

17.534.1 Member Function Documentation

17.534.1.1 CurrentStackTrace()

```
std::string testing::internal::OsStackTraceGetter::CurrentStackTrace (
            int max_depth,
            int skip_count )  [override], [virtual]
```

Implements testing::internal::OsStackTraceGetterInterface.

17.534.1.2 UponLeavingGTest()

```
void testing::internal::OsStackTraceGetter::UponLeavingGTest ( )  [override], [virtual]
```

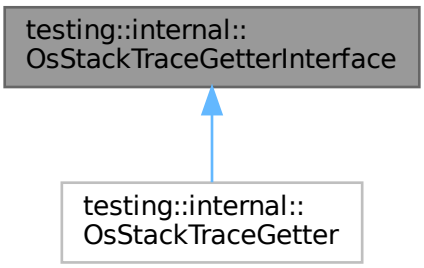
Implements testing::internal::OsStackTraceGetterInterface.

The documentation for this class was generated from the following files:

- gtest/src/gtest-internal-inl.h
- gtest/src/gtest.cc

17.535 testing::internal::OsStackTraceGetterInterface Class Reference

Inheritance diagram for testing::internal::OsStackTraceGetterInterface:



Public Member Functions

- virtual std::string **CurrentStackTrace** (int max_depth, int skip_count)=0
- virtual void **UponLeavingGTest** ()=0

Static Public Attributes

- static const char ∗const kElidedFramesMarker

17.535.1 Member Data Documentation

17.535.1.1 kElidedFramesMarker

```
const char *const testing::internal::OsStackTraceGetterInterface::kElidedFramesMarker  [static]
```

Initial value:

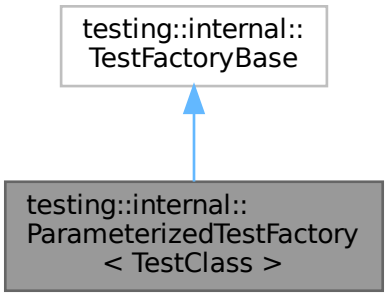
```
=
                "... " GTEST_NAME_ " internal frames ..."
```

The documentation for this class was generated from the following files:

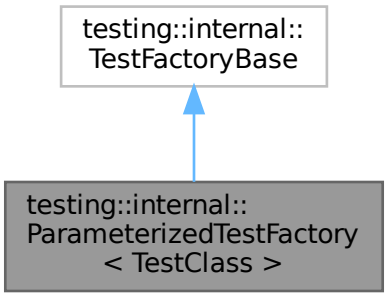
- gtest/src/gtest-internal-inl.h
- gtest/src/gtest.cc

17.536 testing::internal::ParameterizedTestFactory< TestClass > Class Template Reference

Inheritance diagram for testing::internal::ParameterizedTestFactory< TestClass >:



Collaboration diagram for testing::internal::ParameterizedTestFactory< TestClass >:



Public Types

- typedef TestClass::ParamType **ParamType**

Public Member Functions

- **ParameterizedTestFactory** (ParamType parameter)
- Test * CreateTest () override

17.536.1 Member Function Documentation

17.536.1.1 CreateTest()

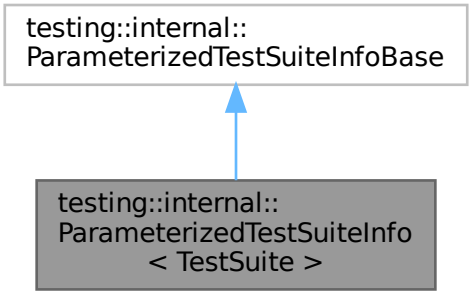
```
template<class TestClass >
Test * testing::internal::ParameterizedTestFactory< TestClass >::CreateTest ( )  [inline],
[override], [virtual]
```

Implements testing::internal::TestFactoryBase.
The documentation for this class was generated from the following file:

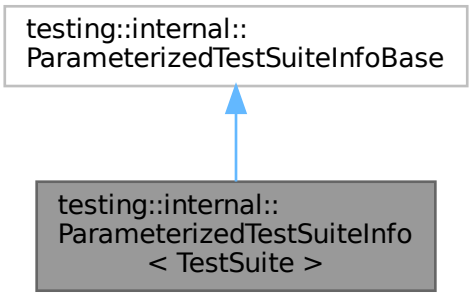
- gtest/gtest/internal/gtest-param-util.h

17.537 testing::internal::ParameterizedTestSuiteInfo< TestSuite > Class Template Reference

Inheritance diagram for testing::internal::ParameterizedTestSuiteInfo< TestSuite >:



Collaboration diagram for testing::internal::ParameterizedTestSuiteInfo< TestSuite >:



Public Types

- using **ParamType** = typename TestSuite::ParamType
- using **ParamNameGeneratorFunc** = std::string(const TestParamInfo< ParamType > &)

Public Member Functions

- typedef **ParamGenerator** (GeneratorCreationFunc)()
- **ParameterizedTestSuiteInfo** (const char *name, CodeLocation code_location)
- const std::string & GetTestSuiteName () const override

- TypeId GetTestSuiteTypeId () const override
- void **AddTestPattern** (const char *test_suite_name, const char *test_base_name, TestMetaFactoryBase< ParamType > *meta_factory, CodeLocation code_location)
- int **AddTestSuiteInstantiation** (const std::string &instantiation_name, GeneratorCreationFunc *func, ParamNameGeneratorFunc *name_func, const char *file, int line)
- void RegisterTests () override

17.537.1 Member Function Documentation

17.537.1.1 GetTestSuiteName()

```
template<class TestSuite >
const std::string & testing::internal::ParameterizedTestSuiteInfo< TestSuite >::GetTestSuite↩
Name ( ) const  [inline], [override], [virtual]
```

Implements testing::internal::ParameterizedTestSuiteInfoBase.

17.537.1.2 GetTestSuiteTypeId()

```
template<class TestSuite >
TypeId testing::internal::ParameterizedTestSuiteInfo< TestSuite >::GetTestSuiteTypeId ( )
const  [inline], [override], [virtual]
```

Implements testing::internal::ParameterizedTestSuiteInfoBase.

17.537.1.3 RegisterTests()

```
template<class TestSuite >
void testing::internal::ParameterizedTestSuiteInfo< TestSuite >::RegisterTests ( )  [inline],
[override], [virtual]
```

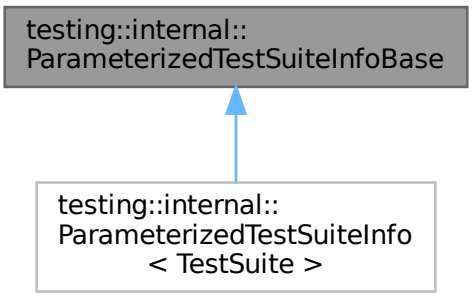
Implements testing::internal::ParameterizedTestSuiteInfoBase.

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.538 testing::internal::ParameterizedTestSuiteInfoBase Class Reference

Inheritance diagram for testing::internal::ParameterizedTestSuiteInfoBase:



Public Member Functions

- virtual const std::string & **GetTestSuiteName** () const =0
- virtual TypeId **GetTestSuiteTypeId** () const =0
- virtual void **RegisterTests** ()=0

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.539 testing::internal::ParameterizedTestSuiteRegistry Class Reference

Public Member Functions

- template<class TestSuite >
  ParameterizedTestSuiteInfo< TestSuite > * **GetTestSuitePatternHolder** (const char *test_suite_name, CodeLocation code_location)
- void **RegisterTests** ()
- template<class TestCase >
  ParameterizedTestCaseInfo< TestCase > * **GetTestCasePatternHolder** (const char *test_case_name, CodeLocation code_location)

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.540 testing::internal::ParamGenerator< T > Class Template Reference

Public Types

- typedef ParamIterator< T > **iterator**

Public Member Functions

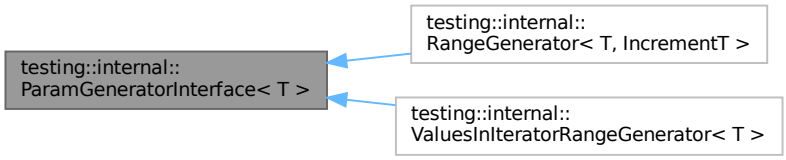
- **ParamGenerator** (ParamGeneratorInterface< T > *impl)
- **ParamGenerator** (const ParamGenerator &other)
- ParamGenerator & **operator=** (const ParamGenerator &other)
- iterator **begin** () const
- iterator **end** () const

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.541 testing::internal::ParamGeneratorInterface< T > Class Template Reference

Inheritance diagram for testing::internal::ParamGeneratorInterface< T >:



Public Types

- typedef T **ParamType**

Public Member Functions

- virtual ParamIteratorInterface< T > * **Begin** () const =0
- virtual ParamIteratorInterface< T > * **End** () const =0

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.542 testing::internal::ParamIterator< T > Class Template Reference

Public Types

- typedef T **value_type**
- typedef const T & **reference**
- typedef ptrdiff_t **difference_type**

Public Member Functions

- **ParamIterator** (const ParamIterator &other)
- ParamIterator & **operator=** (const ParamIterator &other)
- const T & **operator*** () const
- const T * **operator->** () const
- ParamIterator & **operator++** ()
- ParamIterator **operator++** (int)
- bool **operator==** (const ParamIterator &other) const
- bool **operator!=** (const ParamIterator &other) const

Friends

- class **ParamGenerator< T >**

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.543 testing::internal::ParamIteratorInterface< T > Class Template Reference

Public Member Functions

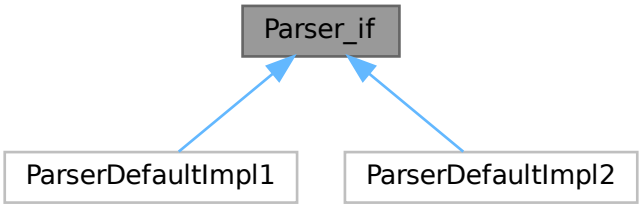
- virtual const ParamGeneratorInterface< T > * **BaseGenerator** () const =0
- virtual void **Advance** ()=0
- virtual ParamIteratorInterface * **Clone** () const =0
- virtual const T * **Current** () const =0
- virtual bool **Equals** (const ParamIteratorInterface &other) const =0

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.544 Parser_if Class Reference

Interface for evaluating model expressions and stochastic functions.
`#include <kernel/simulator/Parser_if.h>`
Inheritance diagram for Parser_if:



Public Member Functions

- virtual double parse (const std::string expression)=0

  parse
- virtual double parse (const std::string expression, bool &success, std::string &errorMessage)=0

  parse
- virtual std::string getErrorMessage ()=0

  getErrorMessage
- virtual void setSampler (Sampler_if *sampler)=0

  Sets the sampler used by stochastic language functions.
- virtual Sampler_if * getSampler () const =0

  Returns the sampler currently associated with the parser.
- virtual genesyspp_driver getParser () const =0

  getParser

17.544.1 Detailed Description

Interface for evaluating model expressions and stochastic functions.
Parser implementations are used by model components/data definitions to evaluate arithmetic/logical expressions, integrating deterministic symbols and stochastic calls backed by a `Sampler_if` instance.

17.544.2 Member Function Documentation

17.544.2.1 getErrorMessage()

```
virtual std::string Parser_if::getErrorMessage ( )  [pure virtual]
```

getErrorMessage

Returns

Returns the last error message produced by the parser.
Implemented in ParserDefaultImpl1, and ParserDefaultImpl2.

17.544.2.2 getParser()

```
virtual genesyspp_driver Parser_if::getParser ( ) const  [pure virtual]
```

getParser

Returns

Exposes the internal parser driver for advanced integration scenarios.
Implemented in ParserDefaultImpl2.

17.544.2.3 getSampler()

```
virtual Sampler_if * Parser_if::getSampler ( ) const  [pure virtual]
```

Returns the sampler currently associated with the parser.
The returned pointer is non-owning. Callers must consult concrete parser documentation for lifetime/ownership details of the underlying instance.
Implemented in ParserDefaultImpl2.

17.544.2.4 parse() [1/2]

```
virtual double Parser_if::parse (
            const std::string expression )  [pure virtual]
```

parse

Parameters

expression	

Returns

Evaluates an expression and returns its numeric value; may throw on syntax/semantic errors.
Implemented in ParserDefaultImpl1, and ParserDefaultImpl2.

17.544.2.5 parse() [2/2]

```
virtual double Parser_if::parse (
            const std::string expression,
            bool & success,
            std::string & errorMessage )  [pure virtual]
```

parse

Parameters

expression	

Parameters

success	
errorMessage	

Returns

Evaluates an expression without throwing and reports status via output parameters.
Implemented in ParserDefaultImpl1, and ParserDefaultImpl2.

17.544.2.6 setSampler()

```
virtual void Parser_if::setSampler (
            Sampler_if * sampler )  [pure virtual]
```

Sets the sampler used by stochastic language functions.

Parameters

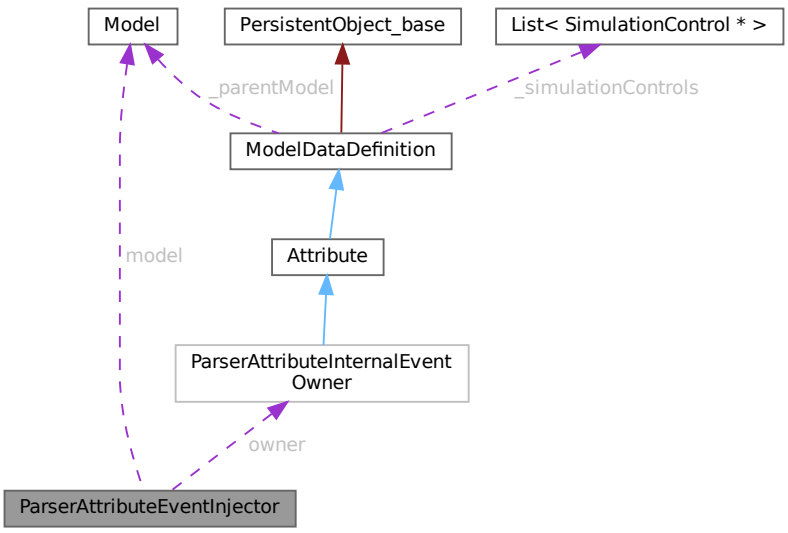
sampler	Pointer to sampler instance that must outlive parser usage unless an implementation-specific ownership transfer is explicitly documented.

This interface does not enforce ownership transfer. Concrete parser implementations may own a sampler created/provided at construction time, but samplers passed through `setSampler` are expected to be externally owned by default unless documented otherwise by the implementation.
Implemented in ParserDefaultImpl2.
The documentation for this class was generated from the following file:

- kernel/simulator/Parser_if.h

17.545 ParserAttributeEventInjector Struct Reference

Collaboration diagram for ParserAttributeEventInjector:



Public Member Functions

- void **OnReplicationStart** (SimulationEvent ∗)

Public Attributes

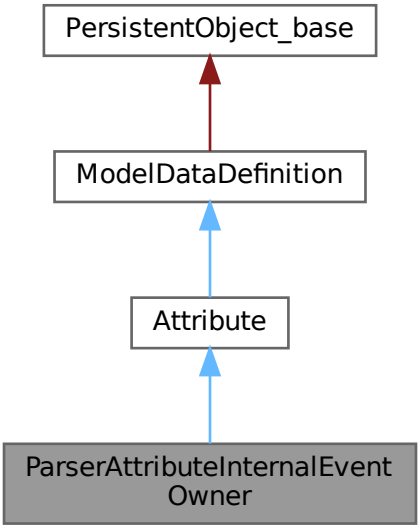
- Model ∗ **model** = nullptr
- ParserAttributeInternalEventOwner ∗ **owner** = nullptr
- std::string **attributeName** = "ParserAttrND"
- bool **inserted** = false

The documentation for this struct was generated from the following file:

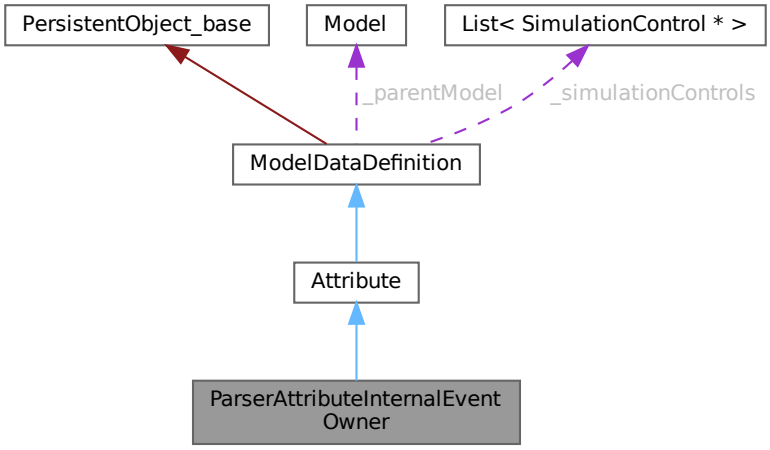
- tests/unit/test_parser_expressions.cpp

17.546 ParserAttributeInternalEventOwner Class Reference

Inheritance diagram for ParserAttributeInternalEventOwner:



Collaboration diagram for ParserAttributeInternalEventOwner:



Public Member Functions

- **ParserAttributeInternalEventOwner** (Model *model, const std::string &name="")
- void **Noop** (void *)

Public Member Functions inherited from Attribute

- **Attribute** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< Attribute >())

  Creates an entity attribute definition in the model.
- virtual ~**Attribute** () override

  Releases sparse initial values owned by this definition.
- virtual std::string show () override

  Returns a textual representation of this data definition.
- virtual std::string getInitialValuesText () const

  Returns the initial values serialized in bracket notation.
- virtual void setInitialValuesText (std::string valuesText)

  Parses bracket notation and replaces the initial value store.
- double **getInitialValue** (std::string index="")

  Reads the initial sparse value, returning 0.0 when the index is absent.
- void **setInitialValue** (double value, std::string index="")

  Writes the initial sparse value at the scalar or indexed position.
- void **setInitialValues** (const std::vector< std::pair< std::string, double > > values)

  Replaces/extends initial sparse values from textual index/value pairs.
- void **copyInitialValuesFrom** (const Attribute &source)

  Copies the full initial sparse store from another attribute.
- void **insertDimentionSize** (unsigned int size)

  Appends one dimension size to the attribute definition.
- std::list< unsigned int > * **getDimensionSizes** () const

  Returns dimension sizes for compatibility with existing callers.
- std::map< std::string, double > * **getInitialValues** () const

  Returns mutable initial sparse values for compatibility with existing callers.

- SparseValueStore * **getInitialValueStore** ()

  Returns the sparse initial value store used by this attribute.
- bool **isInitialValuesTextValid** () const

  Returns whether the last textual initial value is syntactically valid.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Additional Inherited Members

Static Public Member Functions inherited from Attribute

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions inherited from Attribute

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()

- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model ∗ **_parentModel**
- List< SimulationControl ∗ > ∗ **_simulationControls** = new List<SimulationControl∗>()

- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

The documentation for this class was generated from the following file:

- tests/unit/test_parser_expressions.cpp

17.547 ParserAttributeProcessObserver Struct Reference

Public Member Functions

- void **OnProcessEvent** (SimulationEvent *event)

Public Attributes

- std::function< void(Entity *)> **onProcess**

The documentation for this struct was generated from the following file:

- tests/unit/test_parser_expressions.cpp

17.548 ParserChangesInformation Class Reference

Public Member Functions

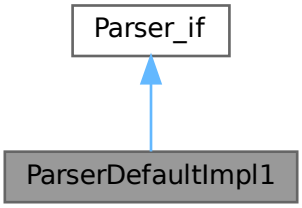
- std::string **getincludes** () const
- void **setIncludes** (const std::string &newIncludes)
- std::string **gettokens** () const
- void **setTokens** (const std::string &newTokens)
- std::string **gettypeObjs** () const
- void **setTypeObjs** (const std::string &newTypeObjs)
- std::string **getexpressions** () const
- void **setExpressions** (const std::string &newExpressions)
- std::string **getexpressionProductions** () const
- void **setExpressionProductions** (const std::string &newExpressionProductions)
- std::string **getassignments** () const
- void **setAssignments** (const std::string &newAssignments)
- std::string **getfunctionProdutions** () const
- void **setFunctionProdutions** (const std::string &newFunctionProdutions)

The documentation for this class was generated from the following files:

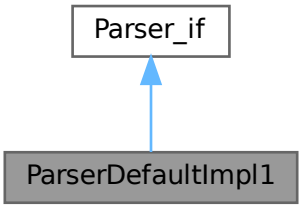
- kernel/simulator/ParserChangesInformation.h
- kernel/simulator/ParserChangesInformation.cpp

17.549 ParserDefaultImpl1 Class Reference

Inheritance diagram for ParserDefaultImpl1:



Collaboration diagram for ParserDefaultImpl1:



Public Member Functions

- **ParserDefaultImpl1** (Model ∗model)
- virtual double parse (const std::string expression) override

  parse
- virtual double parse (const std::string expression, bool &success, std::string &errorMessage) override

  parse
- virtual std::string getErrorMessage () override

  getErrorMessage

Public Member Functions inherited from Parser_if

- virtual void setSampler (Sampler_if ∗sampler)=0

  Sets the sampler used by stochastic language functions.
- virtual Sampler_if ∗ getSampler () const =0

  Returns the sampler currently associated with the parser.
- virtual genesyspp_driver getParser () const =0

  getParser

17.549.1 Member Function Documentation

17.549.1.1 getErrorMessage()

```
std::string ParserDefaultImpl1::getErrorMessage ( )  [override], [virtual]
```

getErrorMessage

Returns

Returns the last error message produced by the parser.
Implements Parser_if.

17.549.1.2 parse() [1/2]

```
double ParserDefaultImpl1::parse (
            const std::string expression )  [override], [virtual]
```

parse

Parameters

expression	

Returns

Evaluates an expression and returns its numeric value; may throw on syntax/semantic errors.
Implements Parser_if.

17.549.1.3 parse() [2/2]

```
double ParserDefaultImpl1::parse (
            const std::string expression,
            bool & success,
            std::string & errorMessage )  [override], [virtual]
```

parse

Parameters

expression	
success	
errorMessage	

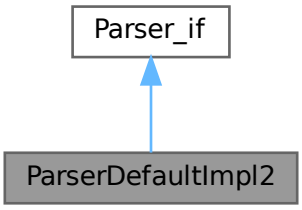
Returns

Evaluates an expression without throwing and reports status via output parameters.
Implements Parser_if.
The documentation for this class was generated from the following files:

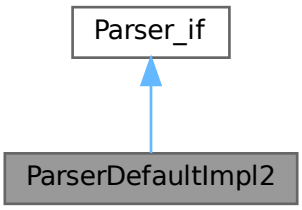
- kernel/simulator/ParserDefaultImpl1.h
- kernel/simulator/ParserDefaultImpl1.cpp

17.550 ParserDefaultImpl2 Class Reference

Inheritance diagram for ParserDefaultImpl2:



Collaboration diagram for ParserDefaultImpl2:



Public Member Functions

- ParserDefaultImpl2 (Model ∗model, Sampler_if ∗sampler, bool throws=false)

  Creates parser wrapper bound to a model and an initial sampler.
- virtual ∼ParserDefaultImpl2 ()

  Destroys parser wrapper and releases only parser-owned sampler.
- virtual double parse (const std::string expression) override

  parse
- virtual double parse (const std::string expression, bool &success, std::string &errorMessage) override

  parse
- virtual std::string getErrorMessage () override

getErrorMessage

- virtual void setSampler (Sampler_if ∗_sampler) override

  Replaces parser sampler with an externally-owned sampler.

- virtual Sampler_if ∗ getSampler () const override

  Returns the sampler currently associated with the parser.

- virtual genesyspp_driver getParser () const override

  getParser

17.550.1 Constructor & Destructor Documentation

17.550.1.1 ParserDefaultImpl2()

```
ParserDefaultImpl2::ParserDefaultImpl2 (
            Model * model,
            Sampler_if * sampler,
            bool throws = false )
```

Creates parser wrapper bound to a model and an initial sampler.
The constructor stores `sampler` in the internal driver and marks it as parser-owned when non-null. This preserves current model behavior where the parser starts with an internally managed sampler instance.

17.550.1.2 ∼ParserDefaultImpl2()

```
ParserDefaultImpl2::~ParserDefaultImpl2 ( )  [virtual]
```

Destroys parser wrapper and releases only parser-owned sampler.
The destructor deletes the current sampler only when ownership is flagged as internal. Samplers installed through setSampler(...) are non-owned by default and therefore not deleted at destruction time.

17.550.2 Member Function Documentation

17.550.2.1 getErrorMessage()

```
std::string ParserDefaultImpl2::getErrorMessage ( )  [override], [virtual]
```

getErrorMessage

Returns

Returns the last error message produced by the parser.
Implements Parser_if.

17.550.2.2 getParser()

```
genesyspp_driver ParserDefaultImpl2::getParser ( ) const  [override], [virtual]
```

getParser

Returns

Exposes the internal parser driver for advanced integration scenarios.
Implements Parser_if.

17.550.2.3 getSampler()

```
Sampler_if * ParserDefaultImpl2::getSampler ( ) const  [override], [virtual]
```

Returns the sampler currently associated with the parser.
The returned pointer is non-owning. Callers must consult concrete parser documentation for lifetime/ownership details of the underlying instance.
Implements Parser_if.

17.550.2.4 parse() [1/2]

```
double ParserDefaultImpl2::parse (
            const std::string expression )  [override], [virtual]
```

parse

Parameters

expression	

Returns

Evaluates an expression and returns its numeric value; may throw on syntax/semantic errors.
Implements Parser_if.

17.550.2.5 parse() [2/2]

```
double ParserDefaultImpl2::parse (
            const std::string expression,
            bool & success,
            std::string & errorMessage )  [override], [virtual]
```

parse

Parameters

expression	
success	
errorMessage	

Returns

Evaluates an expression without throwing and reports status via output parameters.
Implements Parser_if.

17.550.2.6 setSampler()

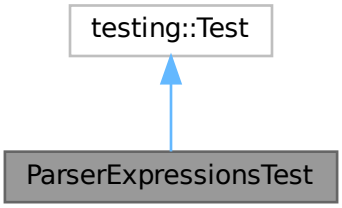
```
void ParserDefaultImpl2::setSampler (
            Sampler_if * _sampler )  [override], [virtual]
```

Replaces parser sampler with an externally-owned sampler.
If the parser currently owns a previous sampler, that previous sampler is released before binding the new pointer.
The provided sampler pointer is then tracked as non-owned.
Implements Parser_if.
The documentation for this class was generated from the following files:

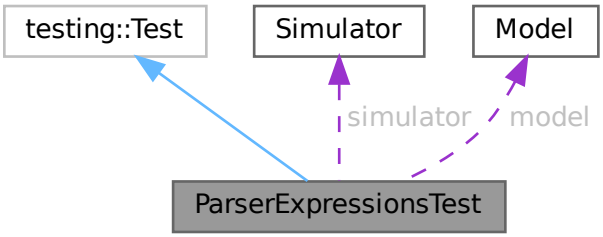
- kernel/simulator/ParserDefaultImpl2.h
- kernel/simulator/ParserDefaultImpl2.cpp

17.551 ParserExpressionsTest Class Reference

Inheritance diagram for ParserExpressionsTest:



Collaboration diagram for ParserExpressionsTest:



Protected Member Functions

- void SetUp () override

Protected Member Functions inherited from testing::Test

- virtual void **TearDown** ()

Protected Attributes

- Simulator **simulator**
- Model * **model** = nullptr

Additional Inherited Members

Static Public Member Functions inherited from testing::Test

- static void **SetUpTestSuite** ()
- static void **TearDownTestSuite** ()
- static void **TearDownTestCase** ()
- static void **SetUpTestCase** ()
- static bool **HasFatalFailure** ()

- static bool **HasNonfatalFailure** ()
- static bool **IsSkipped** ()
- static bool **HasFailure** ()
- static void **RecordProperty** (const std::string &key, const std::string &value)
- static void **RecordProperty** (const std::string &key, int value)

17.551.1 Member Function Documentation

17.551.1.1 SetUp()

```
void ParserExpressionsTest::SetUp ( )  [inline], [override], [protected], [virtual]
```

Reimplemented from testing::Test.

The documentation for this class was generated from the following file:

- tests/unit/test_parser_expressions.cpp

17.552 ParserManager Class Reference

Classes

- struct GenerateNewParserResult

  Result bundle for parser generation requests.
- struct NewParser

  Paths associated with a generated parser artifact set.

Public Member Functions

- ParserManager::GenerateNewParserResult generateNewParser (ParserChangesInformation *changes)

  generateNewParser
- bool connectNewParser (ParserManager::NewParser newParser)

  connectNewParser

17.552.1 Member Function Documentation

17.552.1.1 connectNewParser()

```
bool ParserManager::connectNewParser (
            ParserManager::NewParser newParser )
```

connectNewParser

Parameters

newParser	

Returns

17.552.1.2 generateNewParser()

```
ParserManager::GenerateNewParserResult ParserManager::generateNewParser (
            ParserChangesInformation * changes )
```

generateNewParser

Parameters

changes	

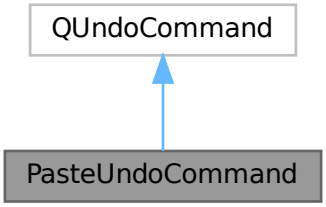
Returns

The documentation for this class was generated from the following files:

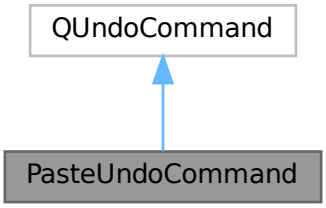
- kernel/simulator/ParserManager.h
- kernel/simulator/ParserManager.cpp

17.553 PasteUndoCommand Class Reference

Inheritance diagram for PasteUndoCommand:



Collaboration diagram for PasteUndoCommand:



Public Member Functions

- **PasteUndoCommand** (QList< GraphicalModelComponent * > *graphicalComponents, QList< GraphicalConnection * > *connections, QList< QGraphicsItemGroup * > *groups, QList< QGraphicsItem * > *drawing, ModelGraphicsScene *scene, QUndoCommand *parent=nullptr)
- void **undo** () override
- void **redo** () override

The documentation for this class was generated from the following files:

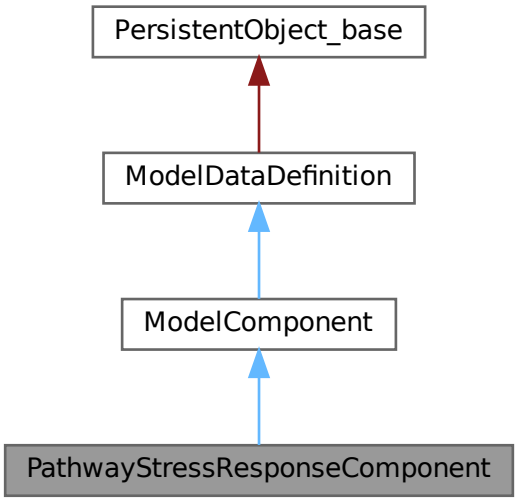
- applications/gui/genesys/actions/PasteUndoCommand.h
- applications/gui/genesys/actions/PasteUndoCommand.cpp

17.554 PathwayStressResponseComponent Class Reference

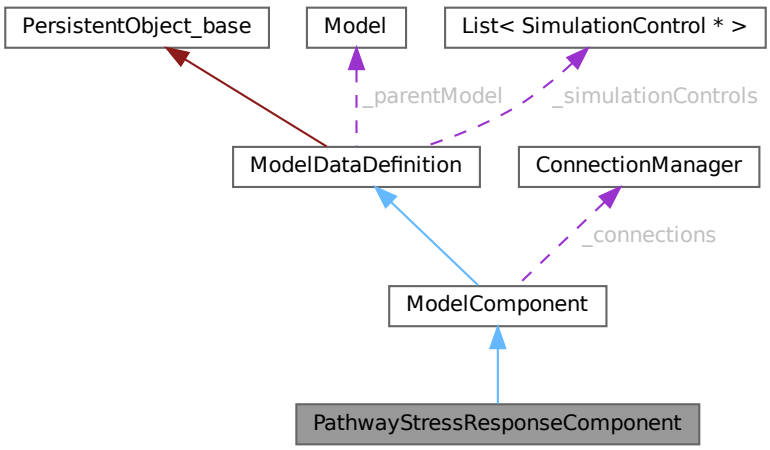
Applies sustained pathway-collapse responses to a whole-cell state.

`#include <plugins/components/WholeCellModeling/PathwayStressResponseComponent.h>`

Inheritance diagram for PathwayStressResponseComponent:



Collaboration diagram for PathwayStressResponseComponent:



Public Member Functions

- **PathwayStressResponseComponent** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)
- WholeCellState * **getWholeCellState** () const

- void **setMonitoredPathwayKey** (std::string key)
- std::string **getMonitoredPathwayKey** () const
- void **setStressThreshold** (double threshold)
- double **getStressThreshold** () const
- void **setArrestAfterSteps** (unsigned int steps)
- unsigned int **getArrestAfterSteps** () const
- void **setDeathAfterSteps** (unsigned int steps)
- unsigned int **getDeathAfterSteps** () const
- void **setArrestPhase** (std::string phase)
- std::string **getArrestPhase** () const
- void **setDeadPhase** (std::string phase)
- std::string **getDeadPhase** () const
- void **setRecoveryPhase** (std::string phase)
- std::string **getRecoveryPhase** () const
- void **setResetStreakOnRecovery** (bool reset)
- bool **getResetStreakOnRecovery** () const
- unsigned int **getStressStreak** () const
- double **getLastObservedActivity** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()

- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager ∗ _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.554.1 Detailed Description

Applies sustained pathway-collapse responses to a whole-cell state.
PathwayStressResponseComponent watches one pathway activity proxy already stored in WholeCellState and converts repeated low-activity windows into higher-level lifecycle responses such as arrest and death.
The component is intentionally simple:

- it does not solve metabolism;
- it does not schedule time;
- it does not route entities.

Instead, it acts as a dedicated biological stress-response layer that can be placed between checkpointing and fate-routing components in hybrid whole-cell examples.

17.554.2 Member Function Documentation

17.554.2.1 _check()

```
bool PathwayStressResponseComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.554.2.2 _createEditableDataDefinitions()

```
void PathwayStressResponseComponent::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.554.2.3 _initBetweenReplications()

```
void PathwayStressResponseComponent::_initBetweenReplications ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.554.2.4 _loadInstance()

```
bool PathwayStressResponseComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.554.2.5 _onDispatchEvent()

```
void PathwayStressResponseComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.554.2.6 _saveInstance()

```
void PathwayStressResponseComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.554.2.7 show()

```
std::string PathwayStressResponseComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

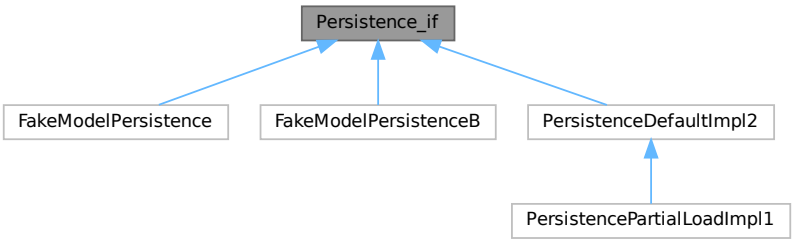
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/WholeCellModeling/PathwayStressResponseComponent.h
- plugins/components/WholeCellModeling/PathwayStressResponseComponent.cpp

17.555 Persistence_if Class Reference

Interface for model serialization/deserialization services.
`#include <kernel/simulator/persistence/Persistence_if.h>`

Inheritance diagram for Persistence_if:



Public Types

- enum class **Options** : int {
  SAVEDEFAULTS = 1 , **HIDEIDKEY** = 2 , **HIDETYPEKEY** = 4 , **HIDENAMEKEY** = 8 ,
  SORTALPHLY = 16 }

Public Member Functions

- virtual bool save (std::string filename)=0

  Serializes and saves the model to the provided filename.
- virtual bool load (std::string filename)=0

  Loads and deserializes a model from file.
- virtual bool hasChanged ()=0

  Indicates whether persistent state changed since the last save/load.
- virtual void setHasChanged (bool hasChanged)=0

  Updates the persistent-state changed flag.
- virtual bool getOption (Persistence_if::Options option)=0

  Returns whether a persistence option is currently enabled.
- virtual void setOption (Persistence_if::Options option, bool value)=0

  Enables or disables a persistence option.
- virtual std::string getFormatedField (PersistenceRecord *fields)=0

  Formats a set of fields as text according to the persistence format.

17.555.1 Detailed Description

Interface for model serialization/deserialization services.
Persistence implementations convert in-memory model structures to/from external representations (e.g., GEN/↩ XML/JSON-like formats) and expose formatting/options hooks used by the kernel persistence pipeline.

17.555.2 Member Function Documentation

17.555.2.1 getFormatedField()

```
virtual std::string Persistence_if::getFormatedField (
            PersistenceRecord * fields ) [pure virtual]
```

Formats a set of fields as text according to the persistence format.
Implemented in FakeModelPersistence, FakeModelPersistenceB, and PersistenceDefaultImpl2.

17.555.2.2 getOption()

```
virtual bool Persistence_if::getOption (
            Persistence_if::Options option )  [pure virtual]
```

Returns whether a persistence option is currently enabled.
Implemented in PersistenceDefaultImpl2, FakeModelPersistence, and FakeModelPersistenceB.

17.555.2.3 hasChanged()

```
virtual bool Persistence_if::hasChanged ( )  [pure virtual]
```

Indicates whether persistent state changed since the last save/load.
Implemented in PersistenceDefaultImpl2, FakeModelPersistence, and FakeModelPersistenceB.

17.555.2.4 load()

```
virtual bool Persistence_if::load (
            std::string filename )  [pure virtual]
```

Loads and deserializes a model from file.
Implemented in PersistenceDefaultImpl2, PersistencePartialLoadImpl1, FakeModelPersistence, and FakeModelPersistenceB.

17.555.2.5 save()

```
virtual bool Persistence_if::save (
            std::string filename )  [pure virtual]
```

Serializes and saves the model to the provided filename.
Implemented in PersistenceDefaultImpl2, FakeModelPersistence, and FakeModelPersistenceB.

17.555.2.6 setHasChanged()

```
virtual void Persistence_if::setHasChanged (
            bool hasChanged )  [pure virtual]
```

Updates the persistent-state changed flag.
Implemented in PersistenceDefaultImpl2, FakeModelPersistence, and FakeModelPersistenceB.

17.555.2.7 setOption()

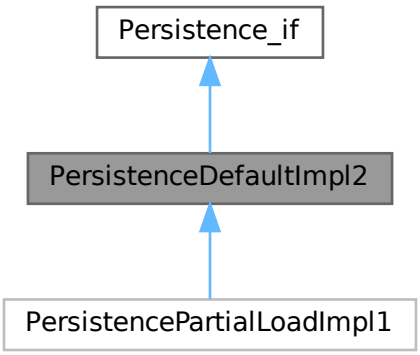
```
virtual void Persistence_if::setOption (
            Persistence_if::Options option,
            bool value )  [pure virtual]
```

Enables or disables a persistence option.
Implemented in PersistenceDefaultImpl2, FakeModelPersistence, and FakeModelPersistenceB.
The documentation for this class was generated from the following file:

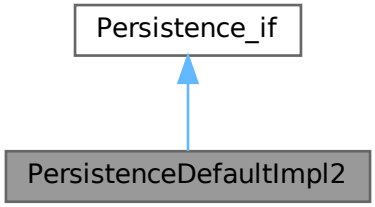
- kernel/simulator/persistence/Persistence_if.h

17.556 PersistenceDefaultImpl2 Class Reference

Inheritance diagram for PersistenceDefaultImpl2:



Collaboration diagram for PersistenceDefaultImpl2:



Public Member Functions

- **PersistenceDefaultImpl2** (Model *model)
- bool save (std::string filename) override

  Serializes and saves the model to the provided filename.
- bool load (std::string filename) override

  Loads and deserializes a model from file.
- bool hasChanged () override

  Indicates whether persistent state changed since the last save/load.
- void setHasChanged (bool hasChanged) override

  Updates the persistent-state changed flag.
- bool getOption (Persistence_if::Options option) override

  Returns whether a persistence option is currently enabled.
- void setOption (Persistence_if::Options option, bool value) override

  Enables or disables a persistence option.
- std::string getFormatedField (PersistenceRecord *fields) override

  Formats a set of fields as text according to the persistence format.

Additional Inherited Members

Public Types inherited from Persistence_if

- enum class **Options** : int {
  SAVEDEFAULTS = 1 , **HIDEIDKEY** = 2 , **HIDETYPEKEY** = 4 , **HIDENAMEKEY** = 8 ,
  SORTALPHLY = 16 }

17.556.1 Member Function Documentation

17.556.1.1 getFormatedField()

```
std::string PersistenceDefaultImpl2::getFormatedField (
            PersistenceRecord * fields )  [override], [virtual]
```

Formats a set of fields as text according to the persistence format.
Implements Persistence_if.

17.556.1.2 getOption()

```
bool PersistenceDefaultImpl2::getOption (
            Persistence_if::Options option )  [override], [virtual]
```

Returns whether a persistence option is currently enabled.
Implements Persistence_if.

17.556.1.3 hasChanged()

```
bool PersistenceDefaultImpl2::hasChanged ( )  [override], [virtual]
```

Indicates whether persistent state changed since the last save/load.
Implements Persistence_if.

17.556.1.4 load()

```
bool PersistenceDefaultImpl2::load (
            std::string filename )  [override], [virtual]
```

Loads and deserializes a model from file.
Implements Persistence_if.
Reimplemented in PersistencePartialLoadImpl1.

17.556.1.5 save()

```
bool PersistenceDefaultImpl2::save (
            std::string filename )  [override], [virtual]
```

Serializes and saves the model to the provided filename.
Implements Persistence_if.

17.556.1.6 setHasChanged()

```
void PersistenceDefaultImpl2::setHasChanged (
            bool hasChanged )  [override], [virtual]
```

Updates the persistent-state changed flag.
Implements Persistence_if.

17.556.1.7 setOption()

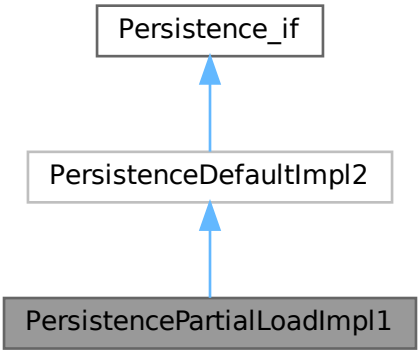
```
void PersistenceDefaultImpl2::setOption (
            Persistence_if::Options option,
            bool value )  [override], [virtual]
```

Enables or disables a persistence option.
Implements Persistence_if.
The documentation for this class was generated from the following files:

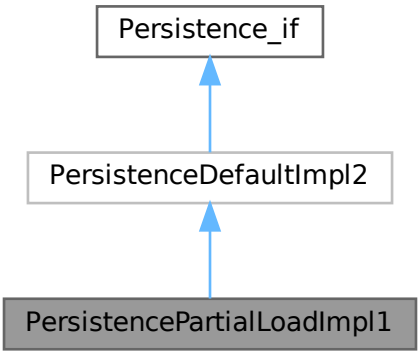
- kernel/simulator/persistence/PersistenceDefaultImpl2.h
- kernel/simulator/persistence/PersistenceDefaultImpl2.cpp

17.557 PersistencePartialLoadImpl1 Class Reference

Inheritance diagram for PersistencePartialLoadImpl1:



Collaboration diagram for PersistencePartialLoadImpl1:



Public Member Functions

- **PersistencePartialLoadImpl1** (Model *model)
- bool load (std::string filename) override

  Loads and deserializes a model from file.

Public Member Functions inherited from PersistenceDefaultImpl2

- **PersistenceDefaultImpl2** (Model *model)

- bool save (std::string filename) override

  Serializes and saves the model to the provided filename.
- bool hasChanged () override

  Indicates whether persistent state changed since the last save/load.
- void setHasChanged (bool hasChanged) override

  Updates the persistent-state changed flag.
- bool getOption (Persistence_if::Options option) override

  Returns whether a persistence option is currently enabled.
- void setOption (Persistence_if::Options option, bool value) override

  Enables or disables a persistence option.
- std::string getFormatedField (PersistenceRecord *fields) override

  Formats a set of fields as text according to the persistence format.

Additional Inherited Members

Public Types inherited from Persistence_if

- enum class **Options** : int {
  SAVEDEFAULTS = 1 , **HIDEIDKEY** = 2 , **HIDETYPEKEY** = 4 , **HIDENAMEKEY** = 8 ,
  SORTALPHLY = 16 }

17.557.1 Member Function Documentation

17.557.1.1 load()

```
bool PersistencePartialLoadImpl1::load (
            std::string filename )  [override], [virtual]
```

Loads and deserializes a model from file.
Reimplemented from PersistenceDefaultImpl2.
The documentation for this class was generated from the following files:

- kernel/simulator/persistence/PersistencePartialLoadImpl1.h
- kernel/simulator/persistence/PersistencePartialLoadImpl1.cpp

17.558 PersistenceRecord Class Reference

Serialization/Deserialization structure which behaves like an associative array.
`#include <kernel/simulator/Persistence.h>`

Classes

- struct Entry

  "Raw" entry, users should treat this as an opaque type since payloads could be in any format.

Public Types

- using **Payload** = std::string
- using **Iterator** = std::unordered_map< std::string, Entry >::iterator

Public Member Functions

- virtual PersistenceRecord * **newInstance** ()
- virtual std::string **loadField** (const std::string &key, const std::string &defaultValue="")
- virtual double **loadField** (const std::string &key, double defaultValue)
- virtual unsigned int **loadField** (const std::string &key, unsigned int defaultValue)
- virtual int **loadField** (const std::string &key, int defaultValue)
- virtual Util::TimeUnit **loadField** (const std::string &key, Util::TimeUnit defaultValue)

- virtual void **saveField** (const std::string &key, const std::string &value, const std::string &defaultValue, const bool saveIfDefault=false)
- virtual void **saveField** (const std::string &key, const std::string &value)
- virtual void **saveField** (const std::string &key, double value, const double defaultValue, const bool saveIf↩ Default=false)
- virtual void **saveField** (const std::string &key, unsigned int value, const unsigned int defaultValue, const bool saveIfDefault=false)
- virtual void **saveField** (const std::string &key, unsigned int value)
- virtual void **saveField** (const std::string &key, int value, const int defaultValue, const bool saveIfDefault=false)
- virtual void **saveField** (const std::string &key, Util::TimeUnit value, const Util::TimeUnit defaultValue, const bool saveIfDefault=false)
- **PersistenceRecord** (Persistence_if &config)
- std::size_t **size** () const

  Returns the number of distinct entries in the record.
- void **insert** (Entry entry)

  Inserts a raw entry in the record.
- void insert (Iterator it, Iterator end)

  Inserts multiple entries from the given iterators.
- void **erase** (const std::string &key)

  Removes a specific entry from the record.
- void **clear** ()

  Removes all entries from this record.
- Iterator **find** (const std::string &key)

  Returns an iterator over a specific entry, if found.
- Iterator **begin** ()

  Returns an iterator over raw entries.
- Iterator **end** ()

  Past-the-end sentinel iterator.

17.558.1 Detailed Description

Serialization/Deserialization structure which behaves like an associative array.
Simulation models/components/definitions use this as a "visitor", which is guided across their innards during ser/deserialization and filled with fields. See `PersistentObject_base`.
When this structure reaches the main model persistence interface, it finishes the serialization process in a format-specific manner. See `Persistence_if` and `Serializer`.

17.558.2 Member Function Documentation

17.558.2.1 insert()

```
void PersistenceRecord::insert (
            Iterator it,
            Iterator end )
```

Inserts multiple entries from the given iterators.
Insert the raw entry payload from the source iterator range. Skip entries that already exist to avoid overwriting without warning.
The documentation for this class was generated from the following files:

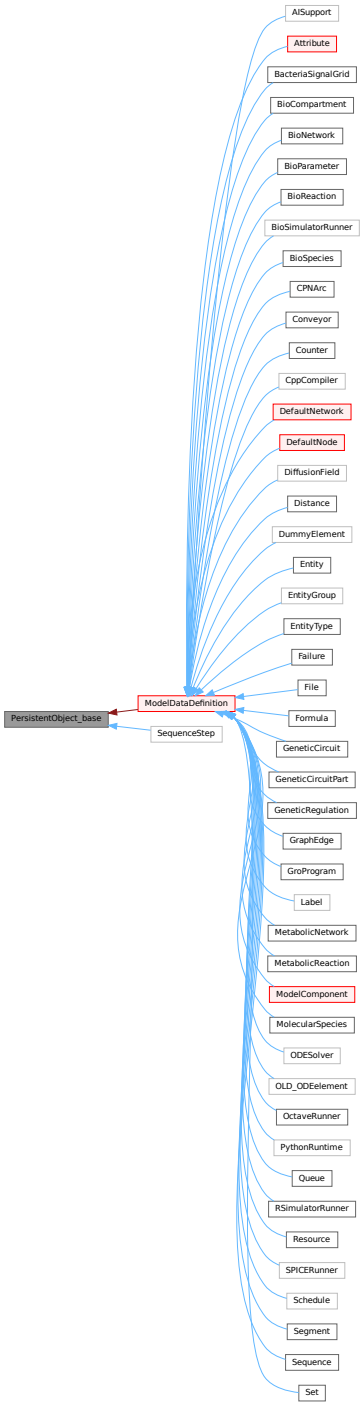
- kernel/simulator/Persistence.h
- kernel/simulator/Persistence.cpp

17.559 PersistentObject_base Class Reference

Dynamic serializable interface.

`#include <kernel/simulator/Persistence.h>`

Inheritance diagram for PersistentObject_base:



Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord ∗fields)=0

  Loads object fields from a deserialization structure.

- virtual void _saveInstance (PersistenceRecord ∗fields, bool saveDefaultValues=false)=0

Saves object fields into a serialization structure.

17.559.1 Detailed Description

Dynamic serializable interface.
Any object that's supposed to be saved to and/or loaded from a file must implement this interface.

17.559.2 Member Function Documentation

17.559.2.1 _loadInstance()

```
virtual bool PersistentObject_base::_loadInstance (
            PersistenceRecord * fields )  [protected], [pure virtual]
```

Loads object fields from a deserialization structure.

Parameters

fields	a table-like structure which can be queried for parsed fields.

Returns

a boolean which is true iff the load operation was successful.

Implemented in Attribute, Counter, Entity, EntityType, StatisticsCollector, ModelComponent, ModelDataDefinition, SinkModelComponent, SourceModelComponent, AIAssistant, MarkovChain, BacteriaColony, BioRunnerCommand, BioSimulate, BioSteadyState, GeneticCircuitSimulate, GeneticExpressionStep, MetabolicFluxBalance, ContinuousSystemComponent, DiffEquations, DiffusionSimulate, LSODE, OLD_ODEelement, Decide, DropOff, Remove, Search, Buffer, Clone, Delay, Process, Release, Seize, SPICECircuit, Resistor, Vsource, Vpulse, Vsine, Capacitor, Diode, PMOS, NMOS, NOT, NAND, AND, NOR, OR, XOR, XNOR, SPICENode, CppForG, PythonForG, RSimulator, Batch, Separate, Record, Write, Assign, Create, Dispose, Access, Enter, Exit, Leave, Move, PickStation, Route, Start, Stop, Store, Unstore, CellularAutomataComp, DefaultNode, FSMState, ModalModelDefault, PetriPlace, Submodel, Match, Signal, Wait, DummyComponent, BioStateProjectionComponent, CellCycleCheckpointComponent, CellDivisionEvent, CellFateDecisionComponent, CellGrowthComponent, CompartmentExchangeComponent, EukaryoticCellCycleComponent, FtsZPolymerizationComponent, MetabolicStateProjectionComponent, MetabolicSubmodelCompone PathwayStressResponseComponent, ResourceAllocationComponent, StochasticReactionComponent, StochasticTranscription, StochasticTranslation, AISupport, BacteriaSignalGrid, BioNetwork, BioParameter, BioReaction, BioSimulatorRunner, BioSpecies, GeneticCircuit, GeneticCircuitPart, GeneticRegulation, GroProgram, MetabolicNetwork, MetabolicReaction, DiffusionField, ODESolver, Failure, Queue, Resource, Schedule, CppCompiler, OctaveRunner, PythonRuntime, RSimulatorRunner, SPICERunner, EntityGroup, File, Formula, Label, Set, Variable, Conveyor, Distance, Segment, SequenceStep, Sequence, Station, Storage, Transporter, ColoredPetriNetNetwork, CPNArc, CPNTransition, DefaultNetwork, EFSMNetwork, GraphEdge, GraphNetwork, MarkovChainNetwork, SignalData, DummyElement, BioCompartment, MolecularSpecies, StochasticReactionRule, and WholeCellState.

17.559.2.2 _saveInstance()

```
virtual void PersistentObject_base::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues = false )  [protected], [pure virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

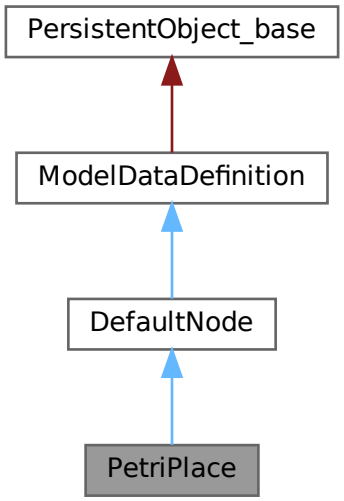
Implemented in Attribute, Counter, Entity, EntityType, StatisticsCollector, ModelComponent, ModelDataDefinition, SinkModelComponent, SourceModelComponent, AIAssistant, MarkovChain, BacteriaColony, BioRunnerCommand, BioSimulate, BioSteadyState, GeneticCircuitSimulate, GeneticExpressionStep, MetabolicFluxBalance, ContinuousSystemComponent

DiffEquations, DiffusionSimulate, LSODE, OLD_ODEelement, Decide, DropOff, Remove, Search, Buffer, Clone, Delay, Process, Release, Seize, SPICECircuit, Resistor, Vsource, Vpulse, Vsine, Capacitor, Diode, PMOS, NMOS, NOT, NAND, AND, NOR, OR, XOR, XNOR, SPICENode, CppForG, PythonForG, RSimulator, Batch, Separate, Record, Write, Assign, Create, Dispose, Access, Enter, Exit, Leave, Move, PickStation, Route, Start, Stop, Store, Unstore, CellularAutomataComp, DefaultNode, FSMState, ModalModelDefault, PetriPlace, Submodel, Match, Signal, Wait, DummyComponent, BioStateProjectionComponent, CellCycleCheckpointComponent, CellDivisionEvent, CellFateDecisionComponent, CellGrowthComponent, CompartmentExchangeComponent, EukaryoticCellCycleComponent, FtsZPolymerizationComponent, MetabolicStateProjectionComponent, MetabolicSubmodelCompone PathwayStressResponseComponent, ResourceAllocationComponent, StochasticReactionComponent, StochasticTranscription, StochasticTranslation, AISupport, BacteriaSignalGrid, BioNetwork, BioParameter, BioReaction, BioSimulatorRunner, BioSpecies, GeneticCircuit, GeneticCircuitPart, GeneticRegulation, GroProgram, MetabolicNetwork, MetabolicReaction, DiffusionField, ODESolver, Failure, Queue, Resource, Schedule, CppCompiler, OctaveRunner, PythonRuntime, RSimulatorRunner, SPICERunner, EntityGroup, File, Formula, Label, Set, Variable, Conveyor, Distance, Segment, SequenceStep, Sequence, Station, Storage, Transporter, ColoredPetriNetNetwork, CPNArc, CPNTransition, DefaultNetwork, EFSMNetwork, GraphEdge, GraphNetwork, MarkovChainNetwork, SignalData, DummyElement, BioCompartment, MolecularSpecies, StochasticReactionRule, and WholeCellState.
The documentation for this class was generated from the following file:

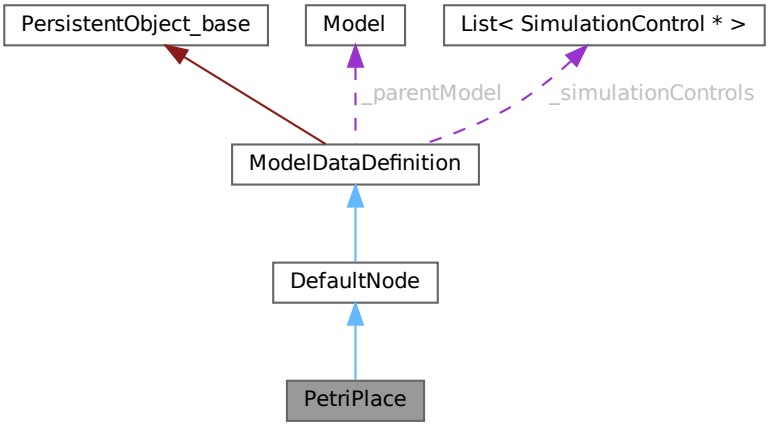
- kernel/simulator/Persistence.h

17.560 PetriPlace Class Reference

Inheritance diagram for PetriPlace:



Collaboration diagram for PetriPlace:



Public Member Functions

- **PetriPlace** (Model ∗model, std::string name="")
- unsigned int **getTokens** (std::string color="default") const
- void **setTokens** (unsigned int quantity, std::string color="default")
- void **addTokens** (unsigned int quantity, std::string color="default")
- bool **removeTokens** (unsigned int quantity, std::string color="default")
- void **clearTokens** ()
- std::map< std::string, unsigned int > ∗ **getAllTokens** ()

Public Member Functions inherited from DefaultNode

- DefaultNode (Model ∗model, std::string componentTypename, std::string name)

  constructors
- **DefaultNode** (Model ∗model, std::string name="")
- void **addTransition** (DefaultNodeTransition ∗transition)

  new public user methods for this component
- void **removeTransition** (DefaultNodeTransition ∗transition)
- List< DefaultNodeTransition ∗ > ∗ **getTransitions** () const
- void **setInitialNode** (bool initialNode)
- bool **isInitialNode** () const
- void **setFinalNode** (bool finalNode)
- bool **isFinalNode** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model ∗model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ∼**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from DefaultNode

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)

- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.560.1 Member Function Documentation

17.560.1.1 _loadInstance()

```
bool PetriPlace::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from DefaultNode.

17.560.1.2 _saveInstance()

```
void PetriPlace::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from DefaultNode.
The documentation for this class was generated from the following files:

- plugins/components/ModalModel/PetriPlace.h
- plugins/components/ModalModel/PetriPlace.cpp

17.561 PickableStationItem Class Reference

Public Member Functions

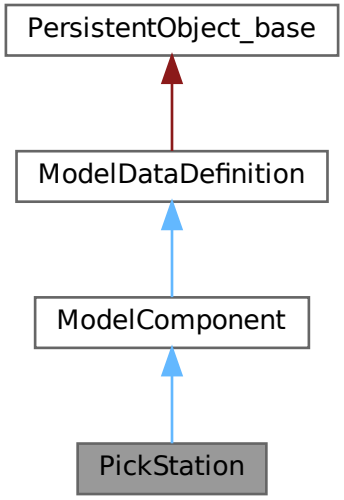
- **PickableStationItem** (Model *model, std::string stationName, std::string queueName="", std::string resourceName="")
- **PickableStationItem** (Model *model, std::string stationName, std::string expression)
- **PickableStationItem** (Station *station, Queue *queue)
- **PickableStationItem** (Station *station, Resource *resource)
- **PickableStationItem** (Station *station, std::string expression)
- **PickableStationItem** (Station *station)
- std::string **getExpression** () const
- Queue * **getQueue** () const
- Resource * **getResource** () const
- Station * **getStation** () const
- void **setExpression** (std::string _expression)
- void **setQueue** (Queue *_queue)
- void **setResource** (Resource *_resource)
- void **setStation** (Station *_station)
- std::string **getName** ()
- void **_addSimulationControl** (SimulationControl *control)
- List< SimulationControl * > * **getSimulationControls** () const
- std::string **getTypeDC** ()

The documentation for this class was generated from the following files:

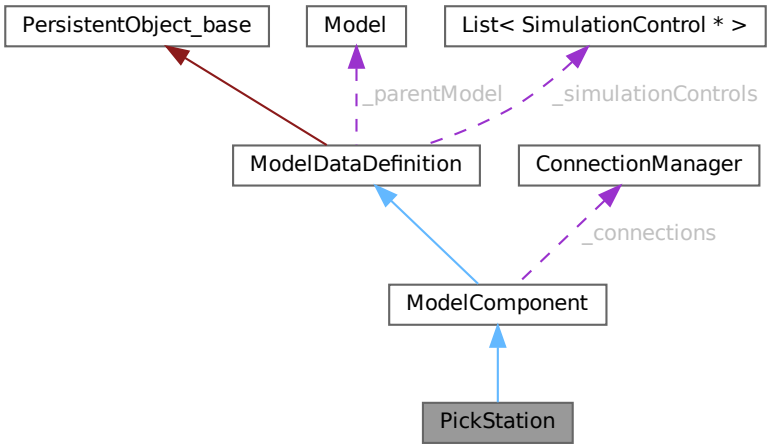
- plugins/components/MaterialHandling/PickableStationItem.h
- plugins/components/MaterialHandling/PickableStationItem.cpp

17.562 PickStation Class Reference

`#include <plugins/components/MaterialHandling/PickStation.h>`
Inheritance diagram for PickStation:



Collaboration diagram for PickStation:



Public Types

- enum class **TestCondition** : int { **MINIMUM** = 0 , **MAXIMUM** = 1 , **num_elements** = 2 }

Public Member Functions

- **PickStation** (Model ∗model, std::string name="")

- virtual std::string show () override

  virtual public methods
- void **setTestCondition** (PickStation::TestCondition _testCondition)
- PickStation::TestCondition **getTestCondition** () const
- void **setSaveAttribute** (std::string _saveAttribute)
- std::string **getSaveAttribute** () const
- List< PickableStationItem * > * **getPickableStationItens** () const
- void **addPickableStationItem** (PickableStationItem *newItem)
- void **removePickableStationItem** (PickableStationItem *item)
- void **setPickConditionExpression** (bool _pickConditionExpression)
- bool **isPickConditionExpression** () const
- void **setPickConditionNumberInQueue** (bool _pickConditionNumberInQueue)
- bool **isPickConditionNumberInQueue** () const
- void **setPickConditionNumberBusyResource** (bool _pickConditionNumberBusyResource)
- bool **isPickConditionNumberBusyResource** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static std::string **convertEnumToStr** (TestCondition condition)
- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation () override
- virtual void _initBetweenReplications () override
- virtual void _addSimulationControl (SimulationControl *property) override
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.562.1 Detailed Description

PickStation module DESCRIPTION The PickStation module allows an entity to select a particular station from the multiple stations specified. This module picks among the group of stations based on the selection logic defined with the module. The entity may then route, transport, convey, or connect to the station specified. If the method chosen is connect, the selected station is assigned to an entity attribute. The station selection process is based on the minimum or maximum value of a variety of system variables and expressions. TYPICAL USES A part sent to a processing station based on machine's availability at each station A loan application sent to a set of loan officers based on the number sent to each officer A customer selecting among cashier lines based on the least number waiting in each line PROMPTS Prompt Description Name Unique name of the module that will be displayed in the flowchart. Test Condition Test condition to use for the station selection process, either Minimum or Maximum. Number En Route to Station The number of entities transferring to the station is considered in the station selection process. Number in Queue The number of entities in the queue at the station is considered in the station selection process. Number of Resources Busy The number of busy resources at the station is considered in the station selection process. Expression Determines if an additional user-defined expression is considered in the station selection process. Transfer Type Determines how an entity will be transferred out of this module to its next destination station—either Route, Convey, Transport, or Connect. Save Attribute Defines the name of the attribute that will store the station name that is selected, visible when the transfer method is Connect. Route Time Move time of the entity from its current station to the station determined through this module. Units Time units for route-time parameters.

17.562.2 Member Function Documentation

17.562.2.1 _addSimulationControl()

```
void PickStation::_addSimulationControl (
            SimulationControl * property )  [override], [protected], [virtual]
```

This method is necessary only for those components that instantiate internal elements that must exist before simulation starts and even before model checking. That's the case of components that have internal StatisticsCollectors, since others components may refer to them as expressions (as in "TVAG(ThisCSTAT)") and therefore the modeldatum must exist before checking such expression
Reimplemented from ModelDataDefinition.

17.562.2.2 _check()

```
bool PickStation::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.562.2.3 _createEditableDataDefinitions()

```
void PickStation::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.562.2.4 _getParserChangesInformation()

```
ParserChangesInformation * PickStation::_getParserChangesInformation ( )  [override], [protected],
[virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
Reimplemented from ModelDataDefinition.

17.562.2.5 _initBetweenReplications()

```
void PickStation::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.562.2.6 _loadInstance()

```
bool PickStation::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.562.2.7 _onDispatchEvent()

```
void PickStation::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.562.2.8 _saveInstance()

```
void PickStation::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.562.2.9 show()

```
std::string PickStation::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

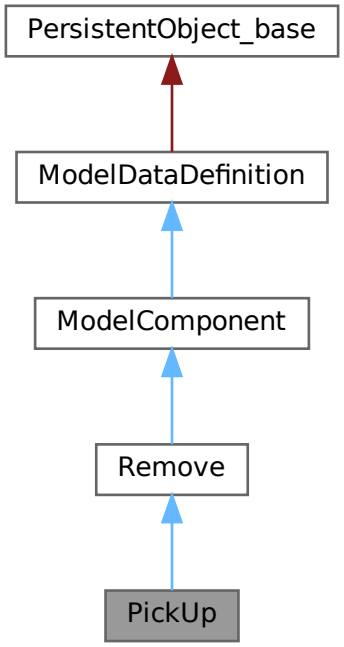
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/MaterialHandling/PickStation.h
- plugins/components/MaterialHandling/PickStation.cpp

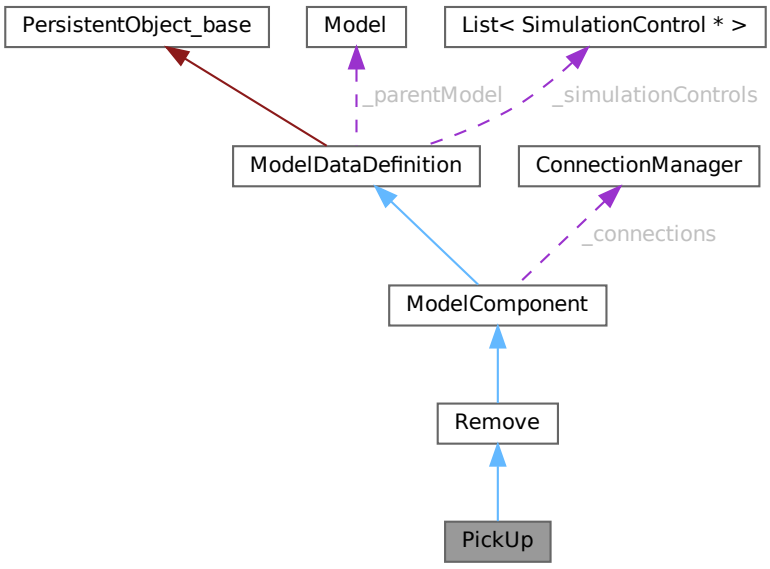
17.563 PickUp Class Reference

Removes a consecutive run of entities from a queue, appending them to the incoming entity's group.
`#include <plugins/components/Decisions/PickUp.h>`

Inheritance diagram for PickUp:



Collaboration diagram for PickUp:



Public Member Functions

- **PickUp** (Model *model, std::string name="")

Public Member Functions inherited from Remove

- **Remove** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setRemoveStartRank** (std::string _removeFromRank)
- std::string **getRemoveStartRank** () const
- void **setRemoveEndRank** (std::string _removeEndRank)
- std::string **getRemoveEndRank** () const
- void **setRemoveFromType** (Remove::RemoveFromType _removeFromType)
- Remove::RemoveFromType **getRemoveFromType** () const
- void **setRemoveFrom** (ModelDataDefinition *_removeFrom)
- ModelDataDefinition * **getRemoveFrom** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from Remove

- static std::string **convertEnumToStr** (RemoveFromType type)
- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Additional Inherited Members

Public Types inherited from Remove

- enum class **RemoveFromType** : int { **QUEUE** = 0 , **ENTITYGROUP** = 1 , **num_elements** = 2 }

Protected Member Functions inherited from Remove

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.563.1 Detailed Description

Removes a consecutive run of entities from a queue, appending them to the incoming entity's group.
Arena correspondence: the "Pickup module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 57). Implemented by inheriting from Remove and reusing its Queue Name/Starting Rank plumbing (a start/end rank range plays the role of Arena's Quantity).
The documentation for this class was generated from the following files:

- plugins/components/Decisions/PickUp.h
- plugins/components/Decisions/PickUp.cpp

17.564 genesys::launcher::PlatformInfo Struct Reference

Static Public Member Functions

- static PlatformInfo **detect** (const QString &osReleasePath=QStringLiteral("/etc/os-release"), const QString &architectureOverride={})

Public Attributes

- QString **osId**
- QString **osVersion**
- QString **architecture**
- QString **manifestKey**
- bool **valid** = false

The documentation for this struct was generated from the following files:

- applications/launcher/PlatformInfo.h
- applications/launcher/PlatformInfo.cpp

17.565 Plugin Class Reference

`#include <kernel/simulator/Plugin.h>`

Public Member Functions

- Plugin (std::string filename_so_dll)

  Creates a plugin wrapper for a dynamic library file.
- Plugin (StaticGetPluginInformation getInformation)

  Creates a plugin wrapper from a compiled-in metadata callback.
- virtual ~**Plugin** ()

  Releases metadata allocated for the wrapped plugin.
- std::string show ()

  Returns a human-readable description of the plugin state.
- bool isIsValidPlugin () const

  Indicates whether the plugin metadata was successfully initialized.
- PluginInformation * getPluginInfo () const

  Returns the plugin metadata object.
- ModelDataDefinition * loadNew (Model *model, PersistenceRecord *fields)

  Creates a new data definition instance from serialized fields.
- bool loadAndInsertNew (Model *model, PersistenceRecord *fields)

  Loads a new instance and inserts it into the model in one step.
- ModelDataDefinition * newInstance (Model *model, std::string name="")

  Creates a new in-memory instance of the plugin's primary type.

17.565.1 Detailed Description

A Plugin represents a dynamically linked component class (ModelComponent) or modeldatum class (ModelDataDefinition); It gives access to a ModelComponent so it can be used by the model. Classes like Create, Delay, and Dispose are examples of PlugIns. It corresponds directly to the "Expansible" part (the capitalized 'E') of the GenESyS acronymous PlugIns are NOT implemented yet

17.565.2 Constructor & Destructor Documentation

17.565.2.1 Plugin() [1/2]

```
Plugin::Plugin (
            std::string filename_so_dll )
```

Creates a plugin wrapper for a dynamic library file.

Parameters

filename_so_dll	Plugin library filename or path.

17.565.2.2 Plugin() [2/2]

```
Plugin::Plugin (
            StaticGetPluginInformation getInformation )
```

Creates a plugin wrapper from a compiled-in metadata callback.

Parameters

getInformation	Callback that returns the plugin metadata.

17.565.3 Member Function Documentation

17.565.3.1 getPluginInfo()

```
PluginInformation * Plugin::getPluginInfo ( ) const
```

Returns the plugin metadata object.

Returns

Plugin information block, or `nullptr` when invalid.

17.565.3.2 isIsValidPlugin()

```
bool Plugin::isIsValidPlugin ( ) const
```

Indicates whether the plugin metadata was successfully initialized.

Returns

`true` when this wrapper contains a valid plugin.

17.565.3.3 loadAndInsertNew()

```
bool Plugin::loadAndInsertNew (
            Model * model,
            PersistenceRecord * fields )
```

Loads a new instance and inserts it into the model in one step.

Parameters

model	Parent model that will own the new instance.
fields	Serialized fields to deserialize.

Returns

`true` when loading and insertion succeed.

17.565.3.4 loadNew()

```
ModelDataDefinition * Plugin::loadNew (
            Model * model,
            PersistenceRecord * fields )
```

Creates a new data definition instance from serialized fields.

Creates a new ModelDataDefinition from fields loaded from a file or another persistence source.

Parameters

model	Parent model that will own the new instance.
fields	Serialized fields to deserialize.

Returns

New data definition instance, or `nullptr` on failure.

17.565.3.5 newInstance()

```
ModelDataDefinition * Plugin::newInstance (
            Model * model,
            std::string name = "" )
```

Creates a new in-memory instance of the plugin's primary type.

Parameters

model	Parent model that will own the instance.
name	Initial object name.

Returns

Newly created instance, or `nullptr` on failure.

17.565.3.6 show()

`std::string Plugin::show ( )`
Returns a human-readable description of the plugin state.

Returns

Textual summary used in traces and diagnostics.

responsibilities should remain in PluginConnector/PluginManager layers.
The documentation for this class was generated from the following files:

- kernel/simulator/Plugin.h
- kernel/simulator/Plugin.cpp

17.566 PluginCatalogController Class Reference

Controller responsible for plugin catalog presentation and plugin palette behavior.
`#include <applications/gui/genesys/controllers/PluginCatalogController.h>`

Public Member Functions

- **PluginCatalogController** (Simulator *simulator, QTreeWidget *pluginsTree, QPlainTextEdit *modelText↩ Editor, std::map< std::string, QColor > *pluginCategoryColor)

  Creates the plugin-catalog controller used by MainWindow compatibility slots.
- void **insertPluginUI** (Plugin *plugin) const

  Inserts one plugin entry into the categorized plugin tree representation.
- void **reloadFromPluginManager** () const

  Rebuilds the categorized plugin tree from the current PluginManager state.
- void **applyCategoryExpansionPolicy** () const

  Applies the default expansion policy for plugin category roots.
- void **showContextMenu** (const QPoint &pos) const

  Shows the plugin-tree context menu used to reorder top-level categories.
- void **insertFakePlugins** () const

  Inserts compatibility fake plugins used by legacy GUI flows.
- void **handlePluginItemDoubleClicked** (QTreeWidgetItem *item, int column) const

  Handles plugin double-click by delegating insertion/selection behavior.
- void **handlePluginItemClicked** (QTreeWidgetItem *item, int column) const

  Handles plugin single-click behavior without mutating model lifecycle state.
- void **moveTopLevelCategory** (QTreeWidgetItem *item, int delta) const

  Moves a top-level category before or after its siblings.

17.566.1 Detailed Description

Controller responsible for plugin catalog presentation and plugin palette behavior.
This controller encapsulates the plugin-tree behavior used by MainWindow wrappers, keeping plugin browsing and insertion affordances consistent while reducing direct widget logic in the MainWindow compatibility façade.
Responsibilities:

- populate plugin entries grouped by category with legacy color conventions;
- provide compatibility helpers for fake-plugin insertion paths;
- react to plugin-tree click/double-click events delegated by MainWindow.

Boundaries:

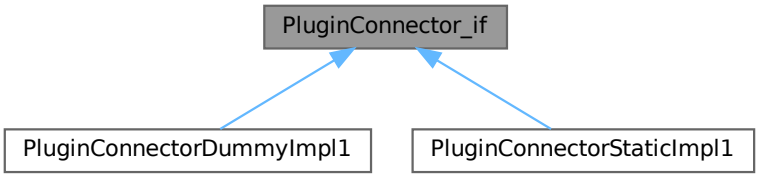
- it does not load plugin binaries or manage kernel plugin lifecycle;
- it does not own textual model synchronization beyond delegated editor insertions;
- it is a UI controller layer, not a persistence/export service.

The documentation for this class was generated from the following files:

- applications/gui/genesys/controllers/PluginCatalogController.h
- applications/gui/genesys/controllers/PluginCatalogController.cpp

17.567 PluginConnector_if Class Reference

Interface for dynamic discovery/loading/unloading of plugin libraries.
`#include <kernel/simulator/PluginConnector_if.h>`
Inheritance diagram for PluginConnector_if:



Public Member Functions

- virtual Plugin * check (const std::string dynamicLibraryFilename)=0
  check
- virtual Plugin * connect (const std::string dynamicLibraryFilename)=0
  connect
- virtual List< std::string > * find ()=0
  find dynamicLibraryFilenames
- virtual bool disconnect (const std::string dynamicLibraryFilename)=0
  disconnect
- virtual bool disconnect (Plugin *plugin)=0
  disconnect

17.567.1 Detailed Description

Interface for dynamic discovery/loading/unloading of plugin libraries.
Connector implementations bridge OS-level dynamic library handling and the kernel plugin abstraction, validating candidates and managing plugin lifecycles.

17.567.2 Member Function Documentation

17.567.2.1 check()

```
virtual Plugin * PluginConnector_if::check (
            const std::string dynamicLibraryFilename )  [pure virtual]
```

check

Parameters

dynamicLibraryFilename	

Returns

Validates whether a dynamic library contains a compatible plugin without permanently connecting it.
Implemented in PluginConnectorDummyImpl1, and PluginConnectorStaticImpl1.

17.567.2.2 connect()

```
virtual Plugin * PluginConnector_if::connect (
            const std::string dynamicLibraryFilename )  [pure virtual]
```

connect

Parameters

dynamicLibraryFilename	

Returns

Loads and connects a plugin from a dynamic library.
Implemented in PluginConnectorDummyImpl1, and PluginConnectorStaticImpl1.

17.567.2.3 disconnect() [1/2]

```
virtual bool PluginConnector_if::disconnect (
            const std::string dynamicLibraryFilename )  [pure virtual]
```

disconnect

Parameters

dynamicLibraryFilename	

Returns

List of all Plugins found and connected

Disconnects the plugin associated with the provided library file.
Implemented in PluginConnectorDummyImpl1, and PluginConnectorStaticImpl1.

17.567.2.4 disconnect() [2/2]

```
virtual bool PluginConnector_if::disconnect (
            Plugin * plugin )  [pure virtual]
```

disconnect

Parameters

plugin	

Returns

Disconnects a previously connected plugin instance.
Implemented in PluginConnectorDummyImpl1, and PluginConnectorStaticImpl1.

17.567.2.5 find()

```
virtual List< std::string > * PluginConnector_if::find ( )  [pure virtual]
```

find dynamicLibraryFilenames

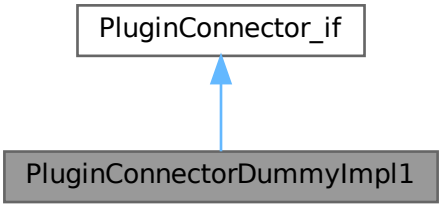
Parameters

return List of dynamicLibraryFilenames
Searches for candidate plugin libraries in the configured environment.
Implemented in PluginConnectorDummyImpl1, and PluginConnectorStaticImpl1.
The documentation for this class was generated from the following file:

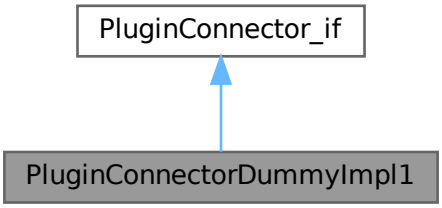
- kernel/simulator/PluginConnector_if.h

17.568 PluginConnectorDummyImpl1 Class Reference

In-memory plugin connector that resolves built-in plugin metadata.
`#include <plugins/PluginConnectorDummyImpl1.h>`
Inheritance diagram for PluginConnectorDummyImpl1:



Collaboration diagram for PluginConnectorDummyImpl1:



Public Member Functions

- Plugin * check (const std::string dynamicLibraryFilename) override
  Verifies whether a built-in plugin name can be resolved.
- Plugin * connect (const std::string dynamicLibraryFilename) override
  Resolves a built-in plugin by filename alias.
- List< std::string > * find () override
  Returns the list of built-in plugin filenames recognized by this connector.
- bool disconnect (const std::string dynamicLibraryFilename) override
  Disconnects a plugin identified by filename.
- bool disconnect (Plugin *plugin) override
  Disconnects a plugin instance.

17.568.1 Detailed Description

In-memory plugin connector that resolves built-in plugin metadata.
This connector does not load shared libraries. Instead, it maps known file names to compiled-in plugin classes and returns the corresponding metadata.

17.568.2 Member Function Documentation

17.568.2.1 check()

```
Plugin * PluginConnectorDummyImpl1::check (
            const std::string dynamicLibraryFilename )  [override], [virtual]
```

Verifies whether a built-in plugin name can be resolved.

Parameters

dynamicLibraryFilename	Plugin library filename or alias.

Returns

Metadata for the resolved plugin, or `nullptr` if unknown.

Implements PluginConnector_if.

17.568.2.2 connect()

```
Plugin * PluginConnectorDummyImpl1::connect (
            const std::string dynamicLibraryFilename )  [override], [virtual]
```

Resolves a built-in plugin by filename alias.

Parameters

dynamicLibraryFilename	Plugin library filename or alias.

Returns

Metadata for the resolved plugin, or `nullptr` if unknown.

Implements PluginConnector_if.

17.568.2.3 disconnect() [1/2]

```
bool PluginConnectorDummyImpl1::disconnect (
            const std::string dynamicLibraryFilename )  [override], [virtual]
```

Disconnects a plugin identified by filename.

Parameters

dynamicLibraryFilename	Plugin library filename or alias.

Returns

Always `true` in the current dummy implementation.

Implements PluginConnector_if.

17.568.2.4 disconnect() [2/2]

```
bool PluginConnectorDummyImpl1::disconnect (
            Plugin * plugin )  [override], [virtual]
```

Disconnects a plugin instance.

Parameters

plugin	Plugin instance to disconnect.

Returns

Always `true` in the current dummy implementation.

Implements PluginConnector_if.

17.568.2.5 find()

```
List< std::string > * PluginConnectorDummyImpl1::find ( )  [override], [virtual]
```

Returns the list of built-in plugin filenames recognized by this connector.

Returns

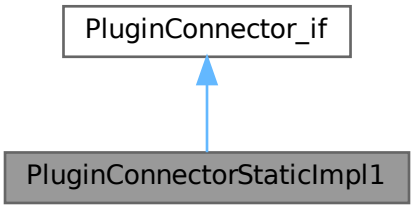
Newly allocated list with known plugin aliases.

Implements PluginConnector_if.
The documentation for this class was generated from the following files:

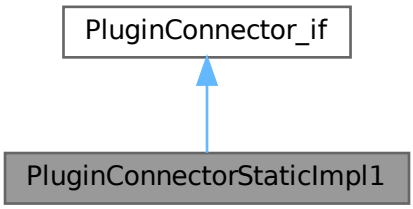
- plugins/PluginConnectorDummyImpl1.h
- kernel/simulator/PluginConnectorDummyBootstrap.cpp
- plugins/PluginConnectorDummyImpl1.cpp

17.569 PluginConnectorStaticImpl1 Class Reference

Static plugin connector stub used by the kernel interface.
`#include <plugins/PluginConnectorStaticImpl1.h>`
Inheritance diagram for PluginConnectorStaticImpl1:



Collaboration diagram for PluginConnectorStaticImpl1:



Public Member Functions

- Plugin * check (const std::string dynamicLibraryFilename) override
  Checks whether a plugin library can be connected.
- Plugin * connect (const std::string dynamicLibraryFilename) override
  Connects a plugin library.
- List< std::string > * find () override
  Lists the available plugin libraries.
- bool disconnect (const std::string dynamicLibraryFilename) override
  Disconnects a plugin by library filename.
- bool disconnect (Plugin *plugin) override
  Disconnects a plugin instance.

17.569.1 Detailed Description

Static plugin connector stub used by the kernel interface.
This implementation currently provides a no-op connector that satisfies the plugin connector contract without loading dynamic libraries.

17.569.2 Member Function Documentation

17.569.2.1 check()

```
Plugin * PluginConnectorStaticImpl1::check (
            const std::string dynamicLibraryFilename )  [override], [virtual]
```

Checks whether a plugin library can be connected.

Parameters

dynamicLibraryFilename	Path to the plugin library.

Returns

Always `nullptr` in the current static stub implementation.

Implements PluginConnector_if.

17.569.2.2 connect()

```
Plugin * PluginConnectorStaticImpl1::connect (
            const std::string dynamicLibraryFilename )  [override], [virtual]
```

Connects a plugin library.

Parameters

dynamicLibraryFilename	Path to the plugin library.

Returns

Always `nullptr` in the current static stub implementation.

Implements PluginConnector_if.

17.569.2.3 disconnect() [1/2]

```
bool PluginConnectorStaticImpl1::disconnect (
            const std::string dynamicLibraryFilename )  [override], [virtual]
```

Disconnects a plugin by library filename.

Parameters

dynamicLibraryFilename	Path to the plugin library.

Returns

Always `true` in the current static stub implementation.

Implements PluginConnector_if.

17.569.2.4 disconnect() [2/2]

```
bool PluginConnectorStaticImpl1::disconnect (
            Plugin * plugin )  [override], [virtual]
```

Disconnects a plugin instance.

Parameters

plugin	Plugin instance to disconnect.

Returns

Always `true` in the current static stub implementation.

Implements PluginConnector_if.

17.569.2.5 find()

`List< std::string > * PluginConnectorStaticImpl1::find ( ) [override], [virtual]`
Lists the available plugin libraries.

Returns

Always `nullptr` in the current static stub implementation.

Implements PluginConnector_if.
The documentation for this class was generated from the following files:

- plugins/PluginConnectorStaticImpl1.h
- plugins/PluginConnectorStaticImpl1.cpp

17.570 PluginInformation Class Reference

Public Member Functions

- **PluginInformation** (std::string pluginTypename, StaticLoaderComponentInstance componentloader, StaticConstructorDataDefinitionInstance elementConstructor)
- **PluginInformation** (std::string pluginTypename, StaticLoaderDataDefinitionInstance elementloader, Static↩ ConstructorDataDefinitionInstance elementConstructor)
- StaticLoaderDataDefinitionInstance **getDataDefinitionLoader** () const
- StaticLoaderComponentInstance **GetComponentLoader** () const
- StaticConstructorDataDefinitionInstance **getDataDefinitionConstructor** () const
- bool **isGenerateReport** () const
- bool **isComponent** () const
- bool **isSendTransfer** () const
- bool **isReceiveTransfer** () const
- bool **isSink** () const
- bool **isSource** () const
- std::string **getObservation** () const
- std::string **getVersion** () const
- std::string **getDate** () const
- std::string **getAuthor** () const
- std::string **getPluginTypename** () const
- void **insertDynamicLibFileDependence** (std::string filename)
- void **setDynamicLibFilenameDependencies** (std::list< std::string > *dynamicLibFilenameDependencies)
- std::list< std::string > * **getDynamicLibFilenameDependencies** () const
- void **insertSystemDependency** (const SystemDependency &dependency)
- const std::list< SystemDependency > * **getSystemDependencies** () const
- bool **hasSystemDependencies** () const
- void **setGenerateReport** (bool generateReport)
- void **setSendTransfer** (bool sendTransfer)
- void **setReceiveTransfer** (bool receiveTransfer)
- void **setSink** (bool Sink)
- void **setSource** (bool Source)
- void **setObservation** (std::string observation)
- void **setVersion** (std::string version)
- void **setDate** (std::string date)
- void **setAuthor** (std::string author)

- void **setMaximumOutputs** (unsigned short _maximumOutputs)
- unsigned short **getMaximumOutputs** () const
- void **setMinimumOutputs** (unsigned short _minimumOutputs)
- unsigned short **getMinimumOutputs** () const
- void **setMaximumInputs** (unsigned short _maximumInputs)
- unsigned short **getMaximumInputs** () const
- void **setMinimumInputs** (unsigned short _minimumInputs)
- unsigned short **getMinimumInputs** () const
- void **setDescriptionHelp** (std::string _descriptionHelp)
- std::string **getDescriptionHelp** () const
- void **setFields** (std::map< std::string, std::string > *_fiewlds)
- std::map< std::string, std::string > * **getFields** () const
- void **setLanguageTemplate** (std::string _languageTemplate)
- std::string **getLanguageTemplate** () const
- void **setCategory** (std::string _category)
- std::string **getCategory** () const

Static Public Member Functions

- static std::string **categoryFolderName** (const std::string &category)

The documentation for this class was generated from the following files:

- kernel/simulator/PluginInformation.h
- kernel/simulator/PluginInformation.cpp

17.571 PluginInsertionOptions Struct Reference

Optional policy hooks used while inserting a plugin.
`#include <kernel/simulator/PluginManager.h>`

Public Attributes

- std::function< bool(const SystemDependencyCheckResult &)> **confirmSystemDependencyInstallation**

  Called before running any install command for missing system dependencies.

17.571.1 Detailed Description

Optional policy hooks used while inserting a plugin.
The kernel reports dependency diagnostics through this structure and leaves user confirmation to the caller. An empty callback means non-interactive insertion: missing dependencies are reported through traces and no install command is executed.
The documentation for this struct was generated from the following file:

- kernel/simulator/PluginManager.h

17.572 PluginLoadIssue Class Reference

Diagnostic record for a plugin candidate that could not be loaded.
`#include <kernel/simulator/PluginManager.h>`

Public Types

- enum class Reason {
  InvalidPlugin , **MissingSystemDependency** , **DynamicDependencyFailure** , **ConnectionFailure** , **InsertionFailure** , **Exception** }

  High-level reason why a plugin candidate was not loaded.

Public Member Functions

- **PluginLoadIssue** ()

  Creates an empty diagnostic record.
- **PluginLoadIssue** (std::string filename, std::string pluginTypename, Reason reason, std::string message, SystemDependencyCheckResult systemDependencyResult={}) noexcept

  Creates a diagnostic record for a failed plugin load attempt.
- const std::string & **getFilename** () const

  Returns the filename passed to the plugin connector.
- const std::string & **getPluginTypename** () const

  Returns the plugin typename when it could be read from PluginInformation.
- Reason **getReason** () const

  Returns the high-level failure reason.
- const std::string & **getMessage** () const

  Returns the human-readable diagnostic message captured by the manager.
- const SystemDependencyCheckResult & **getSystemDependencyResult** () const

  Returns dependency diagnostics captured during system dependency preflight.
- bool **hasSystemDependencyResult** () const

  Returns true when this issue includes system dependency diagnostics.
- std::string **diagnosticText** () const

  Formats the issue for trace logs or GUI details.

Static Public Member Functions

- static std::string **reasonToString** (Reason reason)

  Returns a stable textual label for a failure reason.

17.572.1 Detailed Description

Diagnostic record for a plugin candidate that could not be loaded.
The manager keeps these records so interactive front-ends can explain startup autoload failures after the application UI is ready. The kernel stores only technical diagnostics; GUI code decides how to present and resolve them.
The documentation for this class was generated from the following files:

- kernel/simulator/PluginManager.h
- kernel/simulator/PluginManager.cpp

17.573 PluginManager Class Reference

Manages discovery, loading, lookup and instantiation of simulation plugins.
`#include <kernel/simulator/PluginManager.h>`

Public Member Functions

- **PluginManager** (Simulator *simulator)

  Creates a plugin manager bound to a simulator instance.
- PluginManager (Simulator *simulator, PluginConnector_if *pluginConnector, SystemCommandExecutor_if *systemCommandExecutor)

  Creates a plugin manager with injectable connector and command executor.
- virtual ~**PluginManager** ()

  Destroys owned plugin wrappers and connector resources.
- std::string **show** ()

  Returns a human-readable summary of currently connected plugins.
- List< Plugin * > * **completePluginsFieldsAndTemplates** ()

Completes plugin metadata (fields/templates) after loading.

- bool **check** (const std::string &dynamicLibraryFilename)

  Validates whether a dynamic library provides a compatible plugin.
- SystemDependencyCheckResult **checkSystemDependencies** (const std::string &dynamicLibraryFilename)

  Evaluates system dependencies declared by a dynamic-library plugin without inserting it.
- List< std::string > * **discoverPluginFilenames** () const

  Discovers plugin filenames through the configured connector without inserting them.
- List< PluginLoadIssue > * **getPluginLoadIssues** () const

  Returns plugin candidates that failed to load during check/connect/preflight.
- void **clearPluginLoadIssues** ()

  Clears all stored plugin load diagnostics.
- void **clearPluginLoadIssue** (const std::string &dynamicLibraryFilename)

  Removes stored diagnostics for one plugin candidate filename.
- Plugin * **insert** (const std::string &dynamicLibraryFilename)

  Loads and inserts a plugin from a dynamic library file.
- Plugin * **insert** (const std::string &dynamicLibraryFilename, const PluginInsertionOptions &options)

  Loads and inserts a plugin, optionally asking the caller to authorize dependency installation.
- bool **remove** (const std::string &dynamicLibraryFilename)

  Removes/disconnects a plugin by its dynamic library filename.
- bool **remove** (Plugin *plugin)

  Removes/disconnects a plugin by pointer.
- Plugin * **find** (const std::string &pluginTypeName)

  Finds a connected plugin by plugin type name.
- std::vector< std::string > getDataDefinitionPluginTypenames () const

  Returns connected plugin type names that can create ModelDataDefinition instances.
- std::string **sourceIncludePathFor** (const std::string &pluginTypeName)

  Returns the source include path for a connected plugin type.
- List< Plugin * > * **autoInsertPlugins** (const std::string &pluginsListFilename, const bool lookForPluginsIf↩ FilenameNotFound=true)

  Auto-loads plugins listed in file (or discovered automatically as fallback).
- List< Plugin * > * **autoInsertPlugins** (const std::string &pluginsListFilename, const bool lookForPluginsIf↩ FilenameNotFound, const PluginInsertionOptions &options)

  Auto-loads plugins listed in file using insertion policy hooks.
- List< Plugin * > * **autoInsertPlugins** ()

  Auto-loads plugins discovered automatically).
- List< Plugin * > * **autoInsertPlugins** (const PluginInsertionOptions &options)

  Auto-loads plugins discovered automatically using insertion policy hooks.
- Plugin * **front** ()

  Returns the first plugin in the internal plugin list.
- Plugin * **next** ()

  Returns the next plugin using the internal iterator cursor.
- Plugin * **last** ()

  Returns the last plugin in the internal plugin list.
- unsigned int **size** ()

  Returns the number of connected plugins.
- Plugin * **getAtRank** (unsigned int rank)

  Returns the plugin at a specific list rank.
- ModelDataDefinition * **newInstance** (const std::string &pluginTypename, Model *model, std::string name="")

  Creates a new model data/component instance using a plugin typename.
- template<typename T >
  T * **newInstance** (Model *model, std::string name="")

17.573.1 Detailed Description

Manages discovery, loading, lookup and instantiation of simulation plugins.
The plugin manager centralizes dynamic-library checks/connections through a `PluginConnector_if` and keeps the in-memory list of connected plugins used by the simulator to create model data/component instances.

17.573.2 Constructor & Destructor Documentation

17.573.2.1 PluginManager()

```
PluginManager::PluginManager (
            Simulator * simulator,
            PluginConnector_if * pluginConnector,
            SystemCommandExecutor_if * systemCommandExecutor )
```

Creates a plugin manager with injectable connector and command executor.
Ownership of both injected pointers is transferred to the manager. This constructor exists primarily for unit tests and narrow integration seams.

17.573.3 Member Function Documentation

17.573.3.1 getDataDefinitionPluginTypenames()

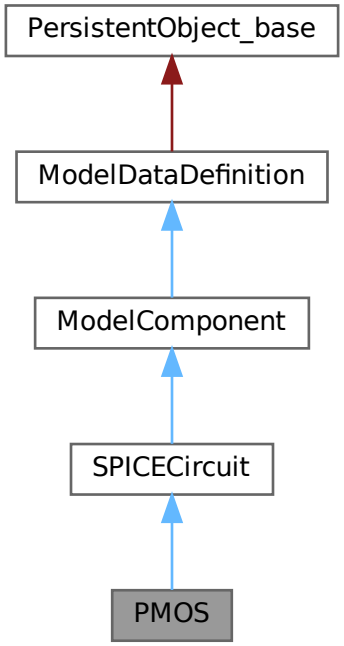
```
std::vector< std::string > PluginManager::getDataDefinitionPluginTypenames ( ) const
```

Returns connected plugin type names that can create ModelDataDefinition instances.
Polymorphic data structures such as Set use this snapshot to expose concrete, creatable element types to editors without hard-coding plugin names in the GUI.
The documentation for this class was generated from the following files:

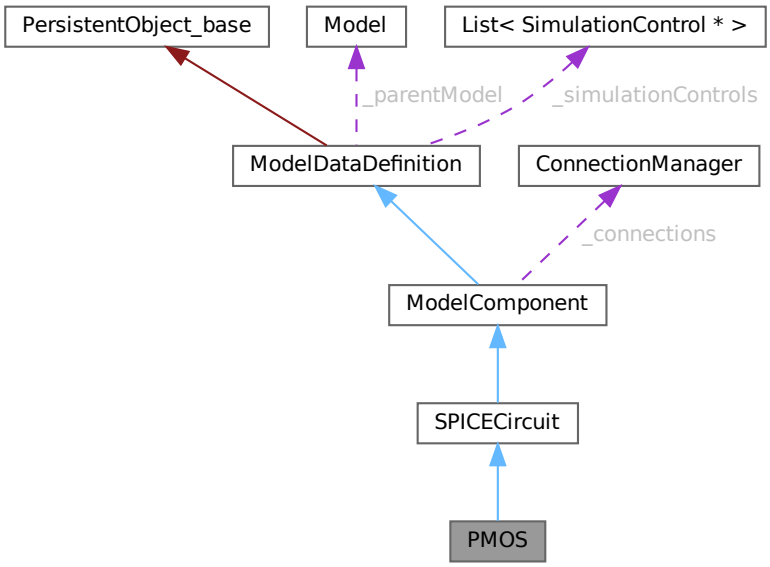
- kernel/simulator/PluginManager.h
- kernel/simulator/PluginManager.cpp

17.574 PMOS Class Reference

Inheritance diagram for PMOS:



Collaboration diagram for PMOS:



Public Member Functions

- **PMOS** (Model *model, std::string name="")
- void Build () override
- void **setWidth** (float width)
- void **setLength** (float length)
- void **setElectricalModel** (std::string electricalModel)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component

- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.574.1 Member Function Documentation

17.574.1.1 _loadInstance()

```
bool PMOS::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.574.1.2 _saveInstance()

```
void PMOS::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.574.1.3 Build()

```
void PMOS::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.
The documentation for this class was generated from the following files:

- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.575 testing::internal::PointerPrinter Struct Reference

Static Public Member Functions

- template<typename T >
  static void **PrintValue** (T *p, ::std::ostream *os)

The documentation for this struct was generated from the following file:

- gtest/gtest/gtest-printers.h

17.576 GroProgramRuntime::PopulationMutation Struct Reference

Public Attributes

- PopulationMutationType **type** = PopulationMutationType::Grow
- unsigned int **value** = 0
- unsigned int **previousPopulationSize** = 0
- unsigned int **resultingPopulationSize** = 0

The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/GroProgramRuntime.h

17.577 yy::position Class Reference

A point in a source file.
`#include <parser/location.hh>`

Public Types

- typedef const std::string **filename_type**

  Type for file name.
- typedef int **counter_type**

  Type for line and column numbers.

Public Member Functions

- **position** (filename_type *f=YY_NULLPTR, counter_type l=1, counter_type c=1)

  Construct a position.
- void **initialize** (filename_type *fn=YY_NULLPTR, counter_type l=1, counter_type c=1)

  Initialization.

Line and Column related manipulators

- void **lines** (counter_type count=1)

  (line related) Advance to the COUNT next lines.
- void **columns** (counter_type count=1)

  (column related) Advance to the COUNT next columns.

Public Attributes

- filename_type * **filename**

  File name to which this position refers.
- counter_type **line**

  Current line number.
- counter_type **column**

  Current column number.

17.577.1 Detailed Description

A point in a source file.
The documentation for this class was generated from the following file:

- parser/location.hh

17.578 testing::PrintToStringParamName Struct Reference

Public Member Functions

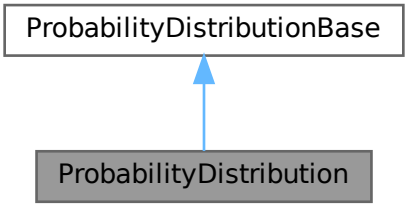
- template<class ParamType >
  std::string **operator()** (const TestParamInfo< ParamType > &info) const

The documentation for this struct was generated from the following file:

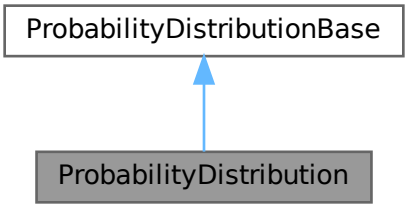
- gtest/gtest/internal/gtest-param-util.h

17.579 ProbabilityDistribution Class Reference

Static façade for inverse/CDF-related distribution utilities.
`#include <tools/Statistics/ProbabilityDistribution.h>`
Inheritance diagram for ProbabilityDistribution:



Collaboration diagram for ProbabilityDistribution:



Static Public Member Functions

- static double inverseChi2 (double cumulativeProbability, double degreeFreedom)

- static double inverseFFisherSnedecor (double cumulativeProbability, double d1, double d2)
- static double inverseNormal (double cumulativeProbability, double mean, double stddev)
- static double inverseTStudent (double cumulativeProbability, double mean, double stddev, double degree↩ Freedom)

Static Public Member Functions inherited from ProbabilityDistributionBase

- static double beta (double x, double alpha, double beta)
- static double chi2 (double x, double degreeFreedom)
- static double erlang (double x, double shape, double scale)
- static double exponential (double x, double mean)
- static double fisherSnedecor (double x, double d1, double d2)
- static double gamma (double x, double shape, double scale)
- static double logNormal (double x, double mean, double stddev)
- static double normal (double x, double mean, double stddev)
- static double poisson (double x, double mean)
- static double triangular (double x, double min, double mode, double max)
- static double tStudent (double x, double mean, double stddev, double degreeFreedom)
- static double uniform (double x, double min, double max)
- static double weibull (double x, double shape, double scale)

Additional Inherited Members

Static Protected Member Functions inherited from ProbabilityDistributionBase

- static double _gammaFunction (double x)
- static double _betaFunction (double x, double y)
- static double _factorial (double x)

17.579.1 Detailed Description

Static façade for inverse/CDF-related distribution utilities.
Architectural role:

- Exposes quantile/inverse computations used by hypothesis testing and confidence-interval routines.

Implementation notes:

- Uses numerical procedures and internal cache state.
- Current static design is legacy-oriented and not the final target model.

Planned evolution:

- Future versions may delegate responsibilities to Distribution_if, Quadrature_if and RootFinder_if abstractions.

17.579.2 Member Function Documentation

17.579.2.1 inverseChi2()

```
double ProbabilityDistribution::inverseChi2 (
            double cumulativeProbability,
            double degreeFreedom )  [static]
```

@TODO: Is there a way to better determine the upper limit?

17.579.2.2 inverseFFisherSnedecor()

```
double ProbabilityDistribution::inverseFFisherSnedecor (
            double cumulativeProbability,
            double d1,
            double d2 )  [static]
```

@TODO: Is there a way to better determine the upper limit?

17.579.2.3 inverseNormal()

```
double ProbabilityDistribution::inverseNormal (
            double cumulativeProbability,
            double mean,
            double stddev )  [static]
```

@TODO: Could be better (separate into 1,2,3xstddev)

17.579.2.4 inverseTStudent()

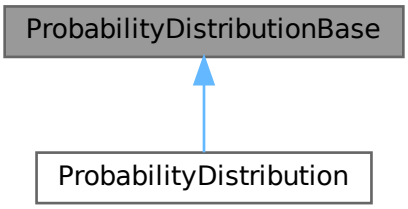
```
double ProbabilityDistribution::inverseTStudent (
            double cumulativeProbability,
            double mean,
            double stddev,
            double degreeFreedom )  [static]
```

@TODO: Could be better
The documentation for this class was generated from the following files:

- tools/Statistics/ProbabilityDistribution.h
- tools/Statistics/ProbabilityDistribution.cpp

17.580 ProbabilityDistributionBase Class Reference

Mathematical base library for distribution-related functions.
`#include <tools/Statistics/ProbabilityDistributionBase.h>`
Inheritance diagram for ProbabilityDistributionBase:



Static Public Member Functions

- static double beta (double x, double alpha, double beta)
- static double chi2 (double x, double degreeFreedom)
- static double erlang (double x, double shape, double scale)
- static double exponential (double x, double mean)
- static double fisherSnedecor (double x, double d1, double d2)
- static double gamma (double x, double shape, double scale)
- static double logNormal (double x, double mean, double stddev)
- static double normal (double x, double mean, double stddev)
- static double poisson (double x, double mean)
- static double triangular (double x, double min, double mode, double max)
- static double tStudent (double x, double mean, double stddev, double degreeFreedom)
- static double uniform (double x, double min, double max)
- static double weibull (double x, double shape, double scale)

Static Protected Member Functions

- static double _gammaFunction (double x)
- static double _betaFunction (double x, double y)
- static double _factorial (double x)

17.580.1 Detailed Description

Mathematical base library for distribution-related functions.
Architectural role:

- Provides static PDF/PMF and helper formulas used by higher-level tools.
- Not an OO hierarchy of distribution objects yet.

Mathematical meaning:

- Public methods represent density/mass functions or mathematically-related helpers for known distributions.

17.580.2 Member Function Documentation

17.580.2.1 _betaFunction()

```
double ProbabilityDistributionBase::_betaFunction (
            double x,
            double y )  [static], [protected]
```

Beta special function helper.

17.580.2.2 _factorial()

```
double ProbabilityDistributionBase::_factorial (
            double x )  [static], [protected]
```

Factorial helper, typically used by PMFs.

17.580.2.3 _gammaFunction()

```
double ProbabilityDistributionBase::_gammaFunction (
            double x )  [static], [protected]
```

Gamma special function approximation/helper. @TODO: Implement by myself

17.580.2.4 beta()

```
double ProbabilityDistributionBase::beta (
            double x,
            double alpha,
            double beta )  [static]
```

Beta distribution PDF.

17.580.2.5 chi2()

```
double ProbabilityDistributionBase::chi2 (
            double x,
            double degreeFreedom )  [static]
```

Chi-square distribution PDF.

17.580.2.6 erlang()

```
double ProbabilityDistributionBase::erlang (
            double x,
            double shape,
            double scale )  [static]
```

Erlang distribution PDF.

17.580.2.7 exponential()

```
double ProbabilityDistributionBase::exponential (
            double x,
            double mean )  [static]
```

Exponential distribution PDF.

17.580.2.8 fisherSnedecor()

```
double ProbabilityDistributionBase::fisherSnedecor (
            double x,
            double d1,
            double d2 )  [static]
```

Fisher-Snedecor F distribution PDF.

17.580.2.9 gamma()

```
double ProbabilityDistributionBase::gamma (
            double x,
            double shape,
            double scale )  [static]
```

Gamma distribution PDF.

17.580.2.10 logNormal()

```
double ProbabilityDistributionBase::logNormal (
            double x,
            double mean,
            double stddev )  [static]
```

Log-normal distribution PDF.

17.580.2.11 normal()

```
double ProbabilityDistributionBase::normal (
            double x,
            double mean,
            double stddev )  [static]
```

Normal distribution PDF.

17.580.2.12 poisson()

```
double ProbabilityDistributionBase::poisson (
            double x,
            double mean )  [static]
```

Poisson distribution PMF.

17.580.2.13 triangular()

```
double ProbabilityDistributionBase::triangular (
            double x,
            double min,
            double mode,
            double max )  [static]
```

Triangular distribution PDF.

17.580.2.14 tStudent()

```
double ProbabilityDistributionBase::tStudent (
            double x,
            double mean,
```

```
            double stddev,
            double degreeFreedom )  [static]
```

Student t distribution PDF.

17.580.2.15 uniform()

```
double ProbabilityDistributionBase::uniform (
            double x,
            double min,
            double max )  [static]
```

Uniform distribution PDF.

17.580.2.16 weibull()

```
double ProbabilityDistributionBase::weibull (
            double x,
            double shape,
            double scale )  [static]
```

Weibull distribution PDF.
The documentation for this class was generated from the following files:

- tools/Statistics/ProbabilityDistributionBase.h
- tools/Statistics/ProbabilityDistributionBase.cpp

17.581 MetabolicFluxBalanceSolver::Problem Struct Reference

Public Attributes

- std::vector< std::vector< double > > **stoichiometry**
- std::vector< double > **lowerBounds**
- std::vector< double > **upperBounds**
- std::vector< double > **objective**
- bool **maximize** = true

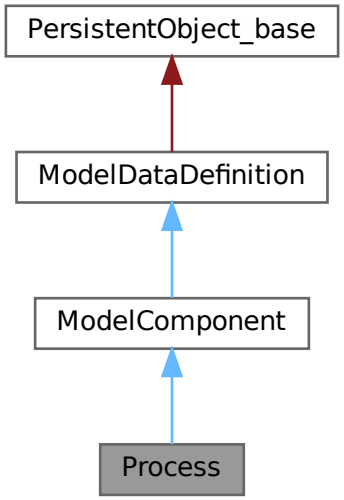
The documentation for this struct was generated from the following file:

- tools/Biochemical/MetabolicFluxBalanceSolver.h

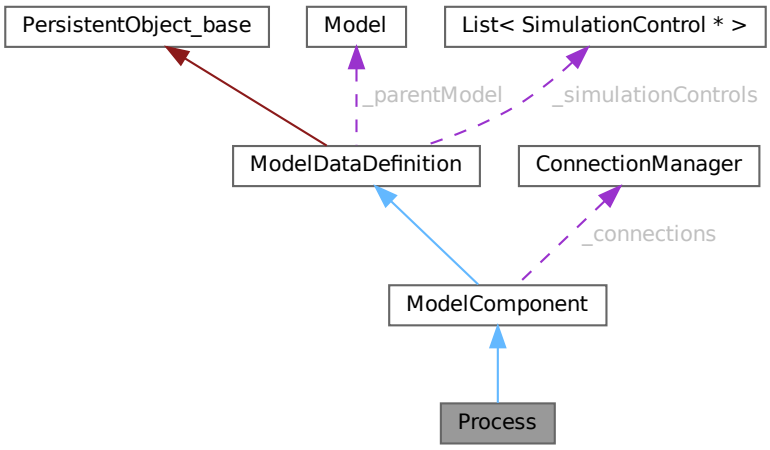
17.582 Process Class Reference

Composite component that always chains an internal Seize, Delay and Release to model "seize a resource, hold it for a while, release it".
`#include <plugins/components/DiscreteProcessing/Process.h>`

Inheritance diagram for Process:



Collaboration diagram for Process:



Public Member Functions

- **Process** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setPriority** (unsigned short _priority)
- unsigned short **getPriority** () const
- void **setPriorityExpression** (std::string priority)
- std::string **getPriorityExpression** () const

- void **setAllocationType** (Util::AllocationType _allocationType)
- Util::AllocationType **getAllocationType** () const
- List< SeizableItem * > * **getSeizeRequests** () const
- void **addSeizeRequest** (SeizableItem *newRequest)
- void **removeSeizeRequest** (SeizableItem *request)
- void **setQueueableItem** (QueueableItem *_queueableItem)
- QueueableItem * **getQueueableItem** () const
- void **setDelayExpression** (std::string _delayExpression)
- void **setDelayExpression** (std::string _delayExpression, Util::TimeUnit _delayTimeUnit)
- std::string **delayExpression** () const
- void **setDelayTimeUnit** (Util::TimeUnit _delayTimeUnit)
- Util::TimeUnit **delayTimeUnit** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createNonEditableDataDefinitions () override
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.582.1 Detailed Description

Composite component that always chains an internal Seize, Delay and Release to model "seize a resource, hold it for a while, release it".
Arena correspondence: the "Process module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 33-35, plus the "Process module — Resource dialog box", p. 35). Priority, the QueueableItem and the SeizeRequests list are forwarded to the internal Seize; the delay expression/time unit are forwarded to the internal Delay; see Seize, Delay and Release for their own accessors.
Known differences from Arena, see `docs/ai_assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.5 for full detail:

- there is no Action selector (Delay / Seize Delay Release / Seize Delay / Delay Release); a Process instance always builds all three sub-components — a standalone Delay is the closest substitute for Arena's resource-free Action variants;
- `setAllocationType()/getAllocationType()` control the internal Seize's allocation category, not the internal Delay's, so a Process component's own processing time stays permanently categorized as "Wait" (Delay's own default) rather than reaching Arena's typical "Value Added" default — flagged as a bug candidate pending maintainer confirmation, not fixed here.

17.582.2 Member Function Documentation

17.582.2.1 _check()

```
bool Process::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.582.2.2 _createEditableDataDefinitions()

```
void Process::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.582.2.3 _createNonEditableDataDefinitions()

```
void Process::_createNonEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.582.2.4 _loadInstance()

```
bool Process::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.582.2.5 _onDispatchEvent()

```
void Process::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.582.2.6 _saveInstance()

```
void Process::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.582.2.7 show()

```
std::string Process::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

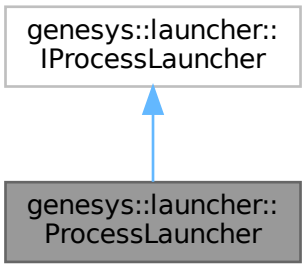
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

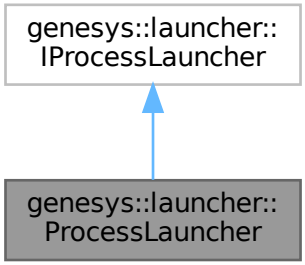
- plugins/components/DiscreteProcessing/Process.h
- plugins/components/DiscreteProcessing/Process.cpp

17.583 genesys::launcher::ProcessLauncher Class Reference

Inheritance diagram for genesys::launcher::ProcessLauncher:



Collaboration diagram for genesys::launcher::ProcessLauncher:



Public Member Functions

- **ProcessLauncher** (LauncherLogger *logger=nullptr)
- int launchReplacingProcess (const RuntimeSelection &selection, const QStringList &arguments, QString *error=nullptr) override

17.583.1 Member Function Documentation

17.583.1.1 launchReplacingProcess()

```
int genesys::launcher::ProcessLauncher::launchReplacingProcess (
            const RuntimeSelection & selection,
            const QStringList & arguments,
            QString * error = nullptr )  [override], [virtual]
```

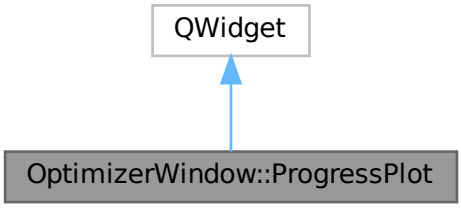
Implements genesys::launcher::IProcessLauncher.

The documentation for this class was generated from the following files:

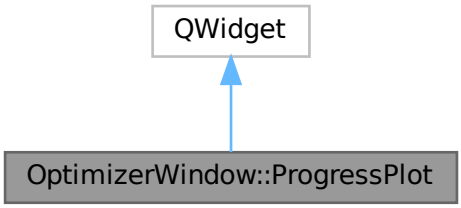
- applications/launcher/ProcessLauncher.h
- applications/launcher/ProcessLauncher.cpp

17.584 OptimizerWindow::ProgressPlot Class Reference

Inheritance diagram for OptimizerWindow::ProgressPlot:



Collaboration diagram for OptimizerWindow::ProgressPlot:



Public Member Functions

- **ProgressPlot** (QWidget ∗parent=nullptr)
- void **setValues** (const QVector< double > &values)

Protected Member Functions

- void **paintEvent** (QPaintEvent ∗) override

The documentation for this class was generated from the following file:

- applications/gui/genesys/tools/optimizer/OptimizerWindow.cpp

17.585 BioStateProjectionComponent::Projection Struct Reference

Public Attributes

- std::string **sourceSpeciesName**
- std::string **targetKey**
- ProjectionKind **kind** = ProjectionKind::MoleculeCount
- double **scale** = 1.0
- double **offset** = 0.0

The documentation for this struct was generated from the following file:

- plugins/components/WholeCellModeling/BioStateProjectionComponent.h

17.586 MetabolicStateProjectionComponent::Projection Struct Reference

Public Attributes

- std::string **reactionName**
- std::string **targetKey**
- ProjectionKind **kind** = ProjectionKind::MetaboliteAmount
- ProjectionUpdateMode **updateMode** = ProjectionUpdateMode::Overwrite
- double **scale** = 1.0
- double **offset** = 0.0
- double **turnoverFraction** = 0.0
- std::string **compartmentName**

The documentation for this struct was generated from the following file:

- plugins/components/WholeCellModeling/MetabolicStateProjectionComponent.h

17.587 EFSMStatePropertiesUndoCommand::Properties Struct Reference

Public Attributes

- std::string **name**
- std::string **entryActionExpression**
- std::string **exitActionExpression**
- bool **finalNode** = false

The documentation for this struct was generated from the following file:

- applications/gui/genesys/actions/EFSMStatePropertiesUndoCommand.h

17.588 EFSMTransitionPropertiesUndoCommand::Properties Struct Reference

Public Attributes

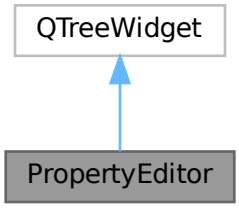
- std::string **name**
- std::string **guardExpression**
- std::string **outputExpression**
- std::string **inputEvent**
- unsigned int **priority** = 0
- std::string **triggerEvent**
- std::string **probabilityExpression**

The documentation for this struct was generated from the following file:

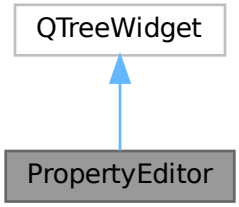
- applications/gui/genesys/actions/EFSMTransitionPropertiesUndoCommand.h

17.589 PropertyEditor Class Reference

Inheritance diagram for PropertyEditor:



Collaboration diagram for PropertyEditor:



Public Member Functions

- **PropertyEditor** (QWidget *parent=nullptr)
- void setModelBlock (ModelDataDefinition *modelblock)

17.589.1 Member Function Documentation

17.589.1.1 setModelBlock()

```
void PropertyEditor::setModelBlock (
            ModelDataDefinition * modelblock )
```

!@TODO Era para ser getClassName());
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/PropertyEditor.h
- applications/gui/genesys/propertyeditor/PropertyEditor.cpp

17.590 PropertyEditorController Class Reference

Controller that orchestrates property editor state and scene-selection coupling.
`#include <applications/gui/genesys/controllers/PropertyEditorController.h>`

Public Member Functions

- **PropertyEditorController** (ObjectPropertyBrowser *propertyBrowser, ModelGraphicsView *graphicsView, PropertyEditorGenesys *propertyGenesys, std::map< SimulationControl *, DataComponentProperty * > *propertyList, std::map< SimulationControl *, DataComponentEditor * > *propertyEditorUI, std::map< SimulationControl *, ComboBoxEnum * > *propertyBox, std::function< void()> actualizeModelSim↩ Language, std::function< void(bool)> actualizeModelComponents, std::function< void(bool)> actualize↩ ModelDataDefinitions, std::function< void()> actualizeModelCppCode, std::function< bool()> create↩ ModelImage, std::function< void()> actualizeTabPanes, std::function< void()> actualizeActions)

  Creates the property-editor controller and its synchronization callbacks.
- void **sceneSelectionChanged** () const

  Synchronizes current scene selection into property editor bindings.
- void **onPropertyEditorModelChanged** () const

  Executes post-commit synchronization across model representations and panes.
- void **clearPropertyEditorSelection** () const

  Clears property editor selection/bindings during lifecycle resets.
- void **bindDataDefinitionFromInspector** (ModelDataDefinition *dataDefinition, QObject *graphical↩ Object=nullptr) const

Binds a DataDefinition selected from the inspector tree to the property editor.

- bool **isPostCommitPipelineActive** () const

  Reports whether the deferred global refresh pipeline is active.

17.590.1 Detailed Description

Controller that orchestrates property editor state and scene-selection coupling.
This Phase-6 extraction isolates property editor wiring from MainWindow while preserving the existing compatibility façade methods. It coordinates selection-driven property binding and the deferred refresh cascade required after property commits.
Responsibilities:

- synchronize current scene selection into PropertyEditorGenesys/ObjectPropertyBrowser;
- execute post-edit refresh callbacks for model language, trees, C++ export, and tabs;
- coalesce heavy refresh requests to avoid re-entrant UI update storms.

Boundaries:

- it does not own property objects/editors lifetime (maps are injected);
- it does not alter undo/redo infrastructure or scene command orchestration;
- it does not persist models or execute simulation commands.

The documentation for this class was generated from the following files:

- applications/gui/genesys/controllers/PropertyEditorController.h
- applications/gui/genesys/controllers/PropertyEditorController.cpp

17.591 PropertyEditorGenesys Class Reference

Public Member Functions

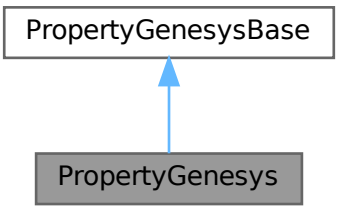
- std::list< ModelComponent * > **getElements** ()
- void **changeSimulationControl** (SimulationControl *control, const std::string &value, bool remove=false)
- SimulationControl * **findSimulationControl** (const std::string &id, const std::string &attribute)
- void **addElement** (ModelComponent *component)

The documentation for this class was generated from the following file:

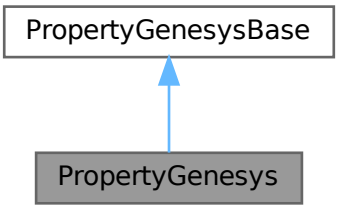
- kernel/simulator/PropertyGenesys.h

17.592 PropertyGenesys Class Reference

Inheritance diagram for PropertyGenesys:



Collaboration diagram for PropertyGenesys:



Public Member Functions

- **PropertyGenesys** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", std::string propertyId="", std::string toolTip="", std::string statusTip="")
- std::list< PropertyGenesys * > **subProperties** () const
- std::string **toolTip** () const
- std::string **statusTip** () const
- std::string **propertyName** () const
- std::string **propertyId** () const
- bool **isEnabled** () const
- bool **isModified** () const
- bool **isSubProperty** () const
- bool **hasValue** () const
- std::string **valueText** () const
- void **setToolTip** (const std::string text)
- void **setStatusTip** (const std::string text)
- void **setWhatsThis** (const std::string text)
- void **setPropertyName** (const std::string text)
- void **setPropertyId** (const std::string text)
- void **setEnabled** (bool enable)
- void **setModified** (bool modified)
- virtual bool **compare** (PropertyGenesys *otherProperty) const
- void **addSubProperty** (PropertyGenesys *property)
- void **insertSubProperty** (PropertyGenesys *property, PropertyGenesys *afterProperty)
- void **removeSubProperty** (PropertyGenesys *property)

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Protected Member Functions

- void **propertyChanged** ()

Protected Attributes

- std::string **_className**
- std::string **_propertyName**
- std::string **_whatsThis**
- std::string **_propertyId**
- std::string **_toolTip**
- std::string **_statusTip**
- std::list< PropertyGenesys * > **_subProperties**
- bool **_isEnabled**
- bool **_isModified**
- bool **_hasValue**
- bool **_isSubProperty**

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

The documentation for this class was generated from the following file:

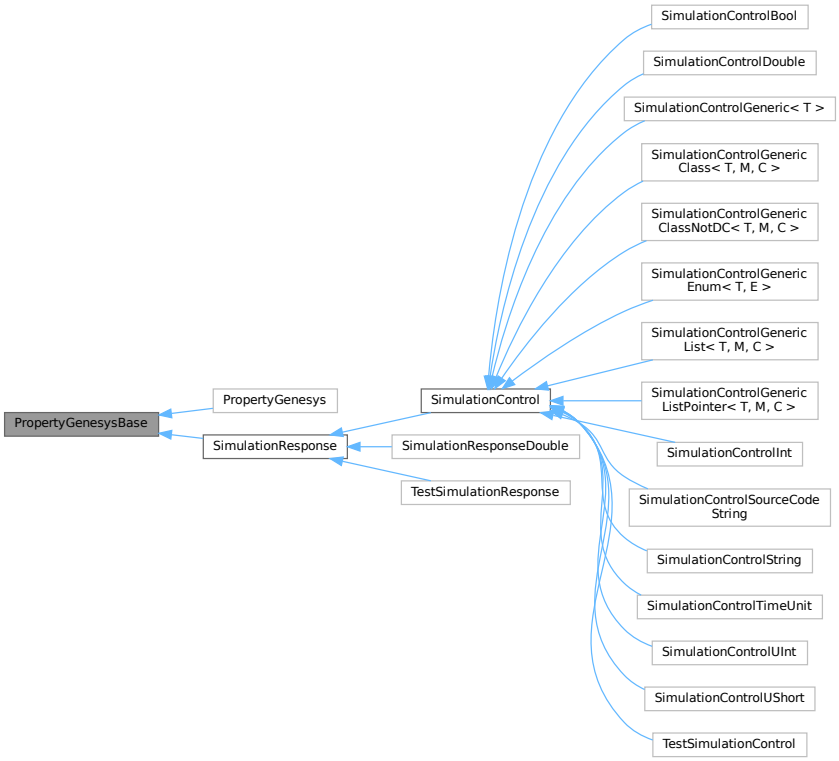
- kernel/simulator/SimulationControlAndResponse.h

17.593 PropertyGenesysBase Class Reference

Base metadata for kernel-side control/response abstractions.

`#include <kernel/simulator/SimulationControlAndResponse.h>`

Inheritance diagram for PropertyGenesysBase:



Public Member Functions

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std↩ ::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Protected Attributes

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.593.1 Detailed Description

Base metadata for kernel-side control/response abstractions.
Historically, the experiment layer in GenESyS evolved from the basic ModelSimulation controls toward a more generic mechanism capable of exposing arbitrary getters and setters from model-related classes.
Despite the legacy name, this base currently lives in the simulation kernel and supports kernel-side experiment/control abstractions rather than a user-interface property editor.
The documentation for this class was generated from the following file:

- kernel/simulator/SimulationControlAndResponse.h

17.594 testing::internal::ProtobufPrinter Struct Reference

Static Public Member Functions

- template<typename T , typename = typename std::enable_if< internal::HasDebugStringAndShortDebugString<T>::value>::type> static void **PrintValue** (const T &value, ::std::ostream *os)

Static Public Attributes

- static const size_t **kProtobufOneLinerMaxLength** = 50

The documentation for this struct was generated from the following file:

- gtest/gtest/gtest-printers.h

17.595 ProviderRequest Struct Reference

Input to a single LLM chat-completion call.
`#include <tools/AIAssistant/AIProviderClient_if.h>`

Public Attributes

- std::string **model**

  Provider model identifier.
- std::vector< ChatMessage > **messages**

  Conversation turns.
- double **temperature** = 0.2

  Sampling temperature [0,2].
- unsigned int **maxTokens** = 4096

  Maximum output tokens.
- bool **highReasoningMode** = false

  Request extended reasoning (o-series / Claude thinking).

17.595.1 Detailed Description

Input to a single LLM chat-completion call.
The documentation for this struct was generated from the following file:

- tools/AIAssistant/AIProviderClient_if.h

17.596 ProviderResponse Struct Reference

Output from a single LLM chat-completion call.
`#include <tools/AIAssistant/AIProviderClient_if.h>`

Public Attributes

- bool **success** = false

  `true` *when the provider returned a valid completion.*
- std::string **content**

  Model reply text. Empty on failure.
- std::string **errorMessage**

  Provider or transport error description.
- int **httpStatusCode** = 0

  Raw HTTP status (0 when transport not used).
- unsigned int **inputTokens** = 0

  Prompt tokens consumed (0 when unreported).
- unsigned int **outputTokens** = 0

  Completion tokens produced (0 when unreported).

17.596.1 Detailed Description

Output from a single LLM chat-completion call.
The documentation for this struct was generated from the following file:

- tools/AIAssistant/AIProviderClient_if.h

17.597 testing::internal::ProxyTypeList< Ts > Struct Template Reference

Public Types

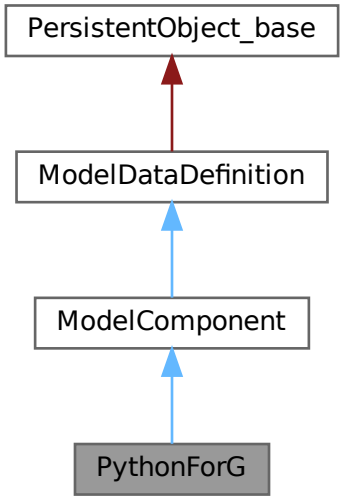
- using **type** = Types< Ts... >

The documentation for this struct was generated from the following file:

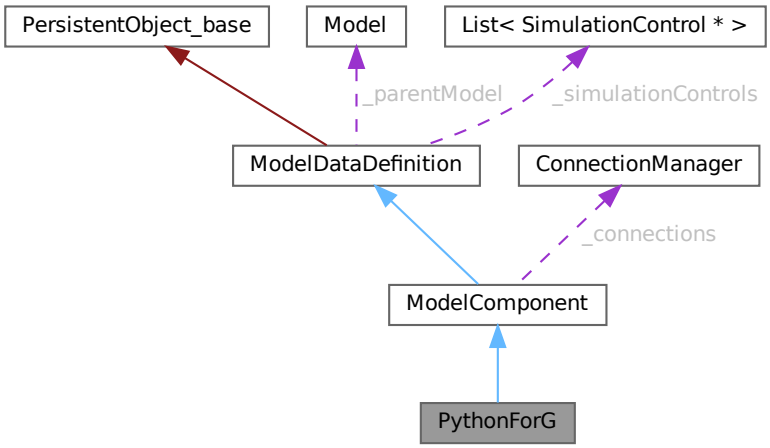
- gtest/gtest/internal/gtest-type-util.h

17.598 PythonForG Class Reference

Inheritance diagram for PythonForG:



Collaboration diagram for PythonForG:



Public Member Functions

- **PythonForG** (Model *model, std::string name="")
- void **setInitBetweenReplicationCode** (std::string initBetweenReplicationCode)
- std::string **getInitBetweenReplicationCode** () const
- void **setInitBetweenReplicationCodeProperty** (SourceCodeString initBetweenReplicationCode)
- SourceCodeString **getInitBetweenReplicationCodeProperty** () const

- void **setOnDispatchEventCode** (std::string onDispatchEventCode)
- std::string **getOnDispatchEventCode** () const
- void **setOnDispatchEventCodeProperty** (SourceCodeString onDispatchEventCode)
- SourceCodeString **getOnDispatchEventCodeProperty** () const
- void **setForwardEntityOnError** (bool forwardEntityOnError)
- bool **isForwardEntityOnError** () const
- PythonRuntime * **getPythonRuntime** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createNonEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.598.1 Member Function Documentation

17.598.1.1 _check()

```
bool PythonForG::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.598.1.2 _createNonEditableDataDefinitions()

```
void PythonForG::_createNonEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.598.1.3 _initBetweenReplications()

```
void PythonForG::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.598.1.4 _loadInstance()

```
bool PythonForG::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.598.1.5 _onDispatchEvent()

```
void PythonForG::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.598.1.6 _saveInstance()

```
void PythonForG::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.598.1.7 show()

```
std::string PythonForG::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

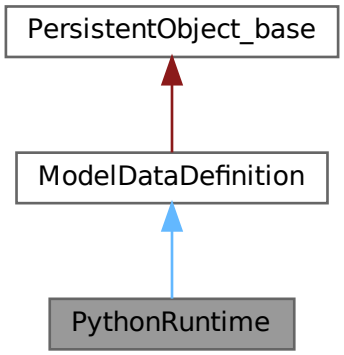
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

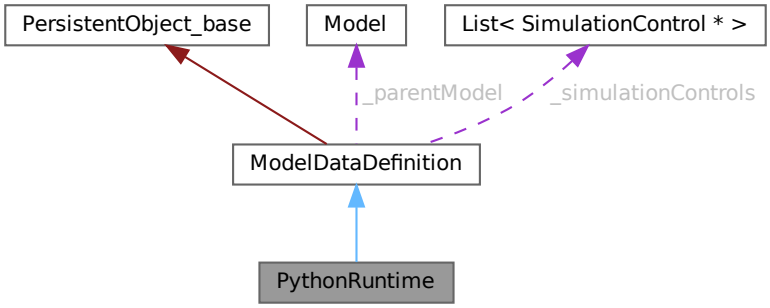
- plugins/components/ExternalIntegration/PythonForG.h
- plugins/components/ExternalIntegration/PythonForG.cpp

17.599 PythonRuntime Class Reference

Inheritance diagram for PythonRuntime:



Collaboration diagram for PythonRuntime:



Public Member Functions

- **PythonRuntime** (Model *model, std::string name="")
- void **setPythonExecutable** (std::string pythonExecutable)
- std::string **getPythonExecutable** () const
- void **setFacadeObjectName** (std::string facadeObjectName)
- std::string **getFacadeObjectName** () const
- void **setCaptureOutput** (bool captureOutput)
- bool **isCaptureOutput** () const
- void **setLastStatus** (std::string lastStatus)
- std::string **getLastStatus** () const
- void **setLastErrorMessage** (std::string lastErrorMessage)
- std::string **getLastErrorMessage** () const
- void **setLastStdout** (std::string lastStdout)
- std::string **getLastStdout** () const
- void **setLastStderr** (std::string lastStderr)

- std::string **getLastStderr** () const
- bool **validateHooks** (const std::string &initCode, const std::string &onDispatchCode, std::string &error↩ Message) const
- bool **executeInitHook** (ModelComponent ∗component, const std::string &initCode, const std::string &on↩ DispatchCode, std::string &errorMessage)
- bool **executeOnDispatchHook** (ModelComponent ∗component, Entity ∗entity, const std::string &initCode, const std::string &onDispatchCode, std::string &errorMessage)
- void **resetExecutionState** ()
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model ∗model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ∼**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model ∗ getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition ∗ getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition ∗ > ∗ getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition ∗ > ∗ getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl ∗ > ∗ getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition ∗ **LoadInstance** (Model ∗model, PersistenceRecord ∗fields)
- static PluginInformation ∗ **GetPluginInformation** ()
- static ModelDataDefinition ∗ **NewInstance** (Model ∗model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition ∗ LoadInstance (Model ∗model, PersistenceRecord ∗fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition ∗ NewInstance (Model ∗model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord ∗fields, ModelDataDefinition ∗modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition ∗modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition ∗modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition ∗modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition ∗modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord ∗fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord ∗fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation ∗ _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl ∗control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()

- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.599.1 Member Function Documentation

17.599.1.1 _check()

```
bool PythonRuntime::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.599.1.2 _loadInstance()

```
bool PythonRuntime::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.599.1.3 _saveInstance()

```
void PythonRuntime::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.599.1.4 show()

```
std::string PythonRuntime::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

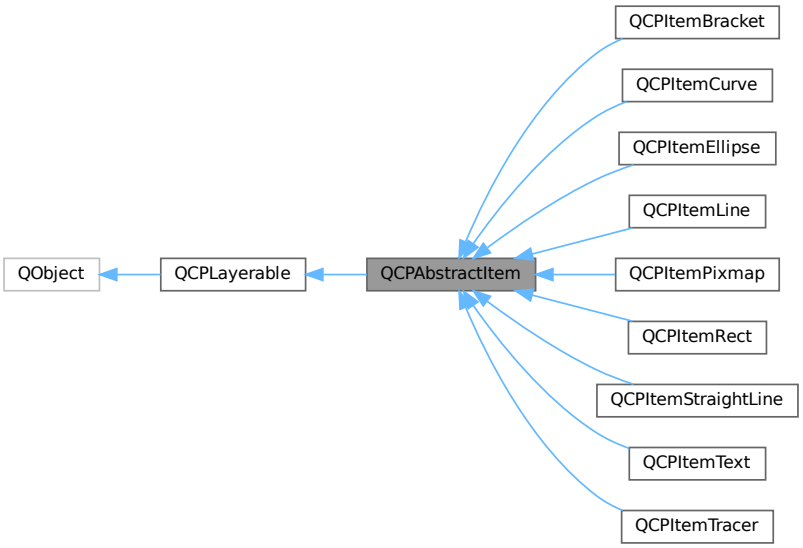
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

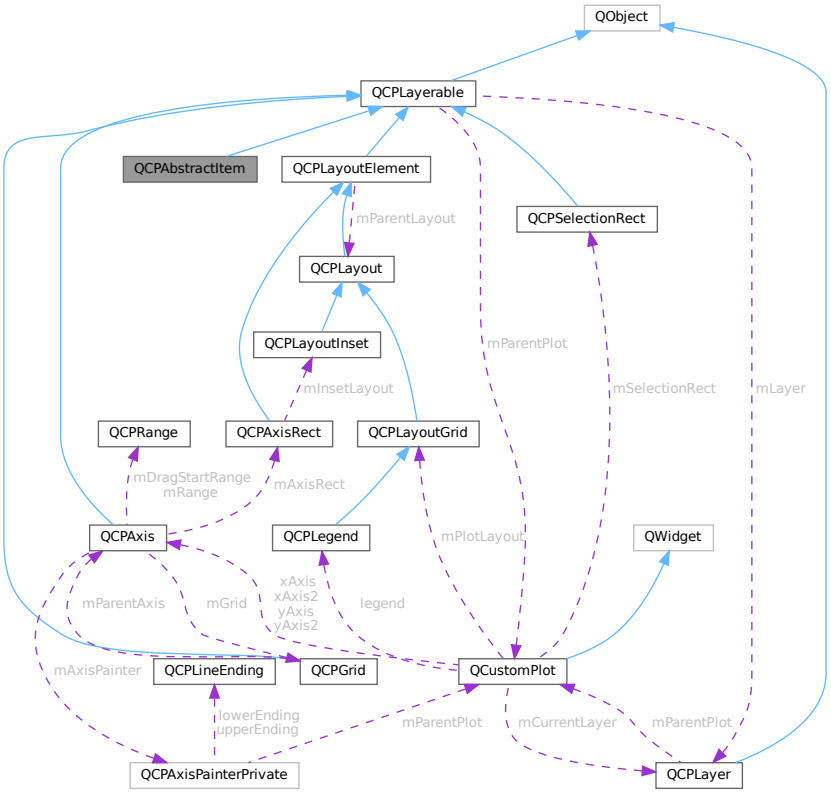
- plugins/data/ExternalIntegration/PythonRuntime.h
- plugins/data/ExternalIntegration/PythonRuntime.cpp

17.600 QCPAbstractItem Class Reference

The abstract base class for all items in a plot.
Inheritance diagram for QCPAbstractItem:



Collaboration diagram for QCPAbstractItem:



Signals

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPAbstractItem (QCustomPlot *parentPlot)
- bool **clipToAxisRect** () const
- QCPAxisRect * **clipAxisRect** () const
- bool **selectable** () const
- bool **selected** () const
- void setClipToAxisRect (bool clip)
- void setClipAxisRect (QCPAxisRect *rect)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE=0
- QList< QCPItemPosition * > positions () const
- QList< QCPItemAnchor * > anchors () const
- QCPItemPosition * position (const QString &name) const
- QCPItemAnchor * anchor (const QString &name) const
- bool hasAnchor (const QString &name) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE=0
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual QPointF **anchorPixelPosition** (int anchorId) const
- double **rectDistance** (const QRectF &rect, const QPointF &pos, bool filledRect) const
- QCPItemPosition * **createPosition** (const QString &name)
- QCPItemAnchor * **createAnchor** (const QString &name, int anchorId)

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot ∗parentPlot)
- virtual void mousePressEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent ∗event)
- void **initializeParentPlot** (QCustomPlot ∗parentPlot)
- void **setParentLayerable** (QCPLayerable ∗parentLayerable)
- bool **moveToLayer** (QCPLayer ∗layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter ∗painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- bool **mClipToAxisRect**
- QPointer< QCPAxisRect > **mClipAxisRect**
- QList< QCPItemPosition ∗ > **mPositions**
- QList< QCPItemAnchor ∗ > **mAnchors**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot ∗ **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer ∗ **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPItemAnchor**

17.600.1 Detailed Description

The abstract base class for all items in a plot.
In QCustomPlot, items are supplemental graphical elements that are neither plottables (QCPAbstractPlottable) nor axes (QCPAxis). While plottables are always tied to two axes and thus plot coordinates, items can also be placed in absolute coordinates independent of any axes. Each specific item has at least one QCPItemPosition member which controls the positioning. Some items are defined by more than one coordinate and thus have two or more QCPItemPosition members (For example, QCPItemRect has *topLeft* and *bottomRight*).
This abstract base class defines a very basic interface like visibility and clipping. Since this class is abstract, it can't be instantiated. Use one of the subclasses or create a subclass yourself to create new items.
The built-in items are:

QCPItemLine	A line defined by a start and an end point. May have different ending styles on each side (e.g. arrows).
QCPItemStraightLine	A straight line defined by a start and a direction point. Unlike QCPItemLine, the straight line is infinitely long and has no endings.
QCPItemCurve	A curve defined by start, end and two intermediate control points. May have different ending styles on each side (e.g. arrows).
QCPItemRect	A rectangle
QCPItemEllipse	An ellipse
QCPItemPixmap	An arbitrary pixmap

QCPItemText	A text label
QCPItemBracket	A bracket which may be used to reference/highlight certain parts in the plot.
QCPItemTracer	An item that can be attached to a QCPGraph and sticks to its data points, given a key coordinate.

17.600.2 Clipping

Items are by default clipped to the main axis rect (they are only visible inside the axis rect). To make an item visible outside that axis rect, disable clipping via setClipToAxisRect(false).
On the other hand if you want the item to be clipped to a different axis rect, specify it via setClipAxisRect. This clipAxisRect property of an item is only used for clipping behaviour, and in principle is independent of the coordinate axes the item might be tied to via its position members (QCPItemPosition::setAxes). However, it is common that the axis rect for clipping also contains the axes used for the item positions.

17.600.3 Using items

First you instantiate the item you want to use and add it to the plot:
by default, the positions of the item are bound to the x- and y-Axis of the plot. So we can just set the plot coordinates where the line should start/end:
If we don't want the line to be positioned in plot coordinates but a different coordinate system, e.g. absolute pixel positions on the QCustomPlot surface, we need to change the position type like this:
Then we can set the coordinates, this time in pixels:
and make the line visible on the entire QCustomPlot, by disabling clipping to the axis rect:
For more advanced plots, it is even possible to set different types and parent anchors per X/Y coordinate of an item position, using for example QCPItemPosition::setTypeX or QCPItemPosition::setParentAnchorX. For details, see the documentation of QCPItemPosition.

17.600.4 Creating own items

To create an own item, you implement a subclass of QCPAbstractItem. These are the pure virtual functions, you must implement:

- selectTest
- draw

See the documentation of those functions for what they need to do.

17.600.4.1 Allowing the item to be positioned

As mentioned, item positions are represented by QCPItemPosition members. Let's assume the new item shall have only one point as its position (as opposed to two like a rect or multiple like a polygon). You then add a public member of type QCPItemPosition like so:

```
QCPItemPosition * const myPosition;
```

the const makes sure the pointer itself can't be modified from the user of your new item (the QCPItemPosition instance it points to, can be modified, of course). The initialization of this pointer is made easy with the createPosition function. Just assign the return value of this function to each QCPItemPosition in the constructor of your item. createPosition takes a string which is the name of the position, typically this is identical to the variable name. For example, the constructor of QCPItemExample could look like this:

```
QCPItemExample::QCPItemExample(QCustomPlot *parentPlot) :
  QCPAbstractItem(parentPlot),
  myPosition(createPosition("myPosition"))
{
  // other constructor code
}
```

17.600.4.2 The draw function

To give your item a visual representation, reimplement the draw function and use the passed QCPPainter to draw the item. You can retrieve the item position in pixel coordinates from the position member(s) via QCPItemPosition::pixelPosition.

To optimize performance you should calculate a bounding rect first (don't forget to take the pen width into account), check whether it intersects the clipRect, and only draw the item at all if this is the case.

17.600.4.3 The selectTest function

Your implementation of the selectTest function may use the helpers QCPVector2D::distanceSquaredToLine and rectDistance. With these, the implementation of the selection test becomes significantly simpler for most items. See the documentation of selectTest for what the function parameters mean and what the function should return.

17.600.4.4 Providing anchors

Providing anchors (QCPItemAnchor) starts off like adding a position. First you create a public member, e.g.
`QCPItemAnchor * const bottom;`
and create it in the constructor with the createAnchor function, assigning it a name and an anchor id (an integer enumerating all anchors on the item, you may create an own enum for this). Since anchors can be placed anywhere, relative to the item's position(s), your item needs to provide the position of every anchor with the reimplementation of the anchorPixelPosition(int anchorId) function.
In essence the QCPItemAnchor is merely an intermediary that itself asks your item for the pixel position when anything attached to the anchor needs to know the coordinates.

17.600.5 Constructor & Destructor Documentation

17.600.5.1 QCPAbstractItem()

```
QCPAbstractItem::QCPAbstractItem (
            QCustomPlot * parentPlot )  [explicit]
```
Base class constructor which initializes base class members.

17.600.6 Member Function Documentation

17.600.6.1 anchor()

```
QCPItemAnchor * QCPAbstractItem::anchor (
            const QString & name ) const
```
Returns the QCPItemAnchor with the specified *name*. If this item doesn't have an anchor by that name, returns `nullptr`.
This function provides an alternative way to access item anchors. Normally, you access anchors direcly by their member pointers (which typically have the same variable name as *name*).

See also

anchors, position

17.600.6.2 anchors()

```
QList< QCPItemAnchor * > QCPAbstractItem::anchors ( ) const  [inline]
```
Returns all anchors of the item in a list. Note that since a position (QCPItemPosition) is always also an anchor, the list will also contain the positions of this item.

See also

positions, anchor

17.600.6.3 applyDefaultAntialiasingHint()

```
void QCPAbstractItem::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```
Implements QCPLayerable.

17.600.6.4 clipRect()

```
QRect QCPAbstractItem::clipRect ( ) const  [protected], [virtual]
```
Reimplemented from QCPLayerable.

17.600.6.5 deselectEvent()

```
void QCPAbstractItem::deselectEvent (
            bool * selectionStateChanged )  [protected], [virtual]
```
Reimplemented from QCPLayerable.

17.600.6.6 draw()

```
void QCPAbstractItem::draw (
            QCPPainter * painter )  [protected], [pure virtual]
```
Implements QCPLayerable.

17.600.6.7 hasAnchor()

```
bool QCPAbstractItem::hasAnchor (
            const QString & name ) const
```
Returns whether this item has an anchor with the specified *name*.
Note that you can check for positions with this function, too. This is because every position is also an anchor (QCPItemPosition inherits from QCPItemAnchor).

See also

anchor, position

17.600.6.8 position()

```
QCPItemPosition * QCPAbstractItem::position (
            const QString & name ) const
```
Returns the QCPItemPosition with the specified *name*. If this item doesn't have a position by that name, returns `nullptr`.
This function provides an alternative way to access item positions. Normally, you access positions direcly by their member pointers (which typically have the same variable name as *name*).

See also

positions, anchor

17.600.6.9 positions()

```
QList< QCPItemPosition * > QCPAbstractItem::positions ( ) const  [inline]
```
Returns all positions of the item in a list.

See also

anchors, position

17.600.6.10 selectEvent()

```
void QCPAbstractItem::selectEvent (
            QMouseEvent * event,
            bool additive,
            const QVariant & details,
            bool * selectionStateChanged )  [protected], [virtual]
```
Reimplemented from QCPLayerable.

17.600.6.11 selectionCategory()

```
QCP::Interaction QCPAbstractItem::selectionCategory ( ) const  [protected], [virtual]
```
Reimplemented from QCPLayerable.

17.600.6.12 selectionChanged

```
void QCPAbstractItem::selectionChanged (
            bool selected )  [signal]
```
This signal is emitted when the selection state of this item has changed, either by user interaction or by a direct call to setSelected.

17.600.6.13 selectTest()

```
virtual double QCPAbstractItem::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [pure virtual]
```
This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Reimplemented from QCPLayerable.
Implemented in QCPItemStraightLine, QCPItemLine, QCPItemCurve, QCPItemRect, QCPItemText, QCPItemEllipse, QCPItemPixmap, QCPItemTracer, and QCPItemBracket.

17.600.6.14 setClipAxisRect()

```
void QCPAbstractItem::setClipAxisRect (
            QCPAxisRect * rect )
```
Sets the clip axis rect. It defines the rect that will be used to clip the item when setClipToAxisRect is set to true.

See also

setClipToAxisRect

17.600.6.15 setClipToAxisRect()

```
void QCPAbstractItem::setClipToAxisRect (
            bool clip )
```

Sets whether the item shall be clipped to an axis rect or whether it shall be visible on the entire QCustomPlot. The axis rect can be set with setClipAxisRect.

See also

setClipAxisRect

17.600.6.16 setSelectable()

```
void QCPAbstractItem::setSelectable (
            bool selectable )
```

Sets whether the user can (de-)select this item by clicking on the QCustomPlot surface. (When QCustomPlot::setInteractions contains QCustomPlot::iSelectItems.)
However, even when *selectable* was set to false, it is possible to set the selection manually, by calling setSelected.

See also

QCustomPlot::setInteractions, setSelected

17.600.6.17 setSelected()

```
void QCPAbstractItem::setSelected (
            bool selected )
```

Sets whether this item is selected or not. When selected, it might use a different visual appearance (e.g. pen and brush), this depends on the specific item though.
The entire selection mechanism for items is handled automatically when QCustomPlot::setInteractions contains QCustomPlot::iSelectItems. You only need to call this function when you wish to change the selection state manually.
This function can change the selection state even when setSelectable was set to false.
emits the selectionChanged signal when *selected* is different from the previous selection state.

See also

setSelectable, selectTest

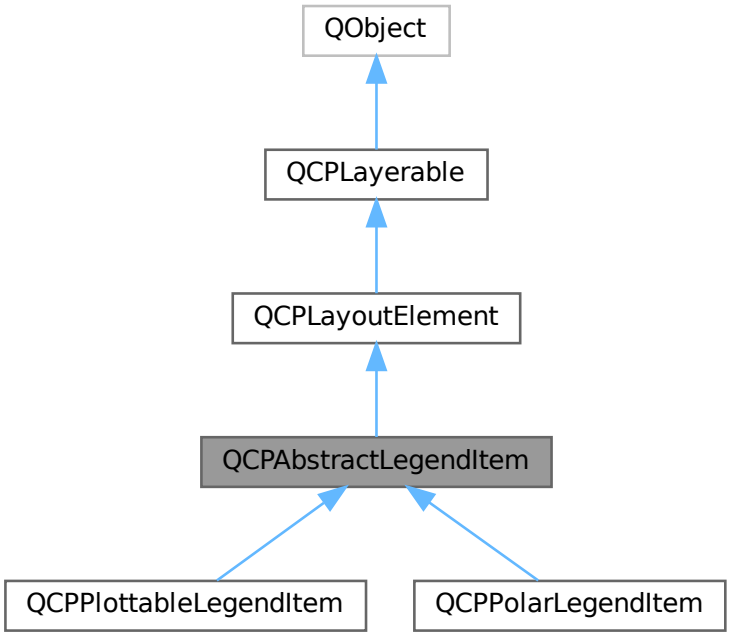
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

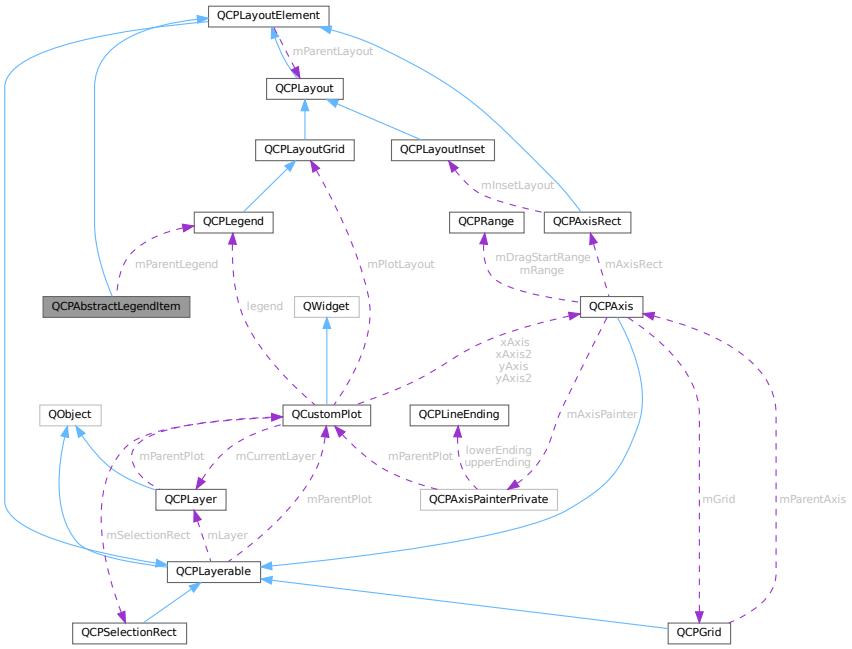
17.601 QCPAbstractLegendItem Class Reference

The abstract base class for all entries in a QCPLegend.

Inheritance diagram for QCPAbstractLegendItem:



Collaboration diagram for QCPAbstractLegendItem:



Signals

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPAbstractLegendItem (QCPLegend *parent)
- QCPLegend * **parentLegend** () const
- QFont **font** () const
- QColor **textColor** () const
- QFont **selectedFont** () const
- QColor **selectedTextColor** () const
- bool **selectable** () const
- bool **selected** () const
- void setFont (const QFont &font)
- void setTextColor (const QColor &color)
- void setSelectedFont (const QFont &font)
- void setSelectedTextColor (const QColor &color)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot *parentPlot=nullptr)
- QCPLayout * layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const
- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)
- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual void update (UpdatePhase phase)
- virtual QSize minimumOuterSizeHint () const
- virtual QSize maximumOuterSizeHint () const
- virtual QList< QCPLayoutElement * > elements (bool recursive) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE=0
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayoutElement

- virtual int **calculateAutoMargin** (QCP::MarginSide side)
- virtual void **layoutChanged** ()
- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QCPLegend * **mParentLegend**
- QFont **mFont**
- QColor **mTextColor**
- QFont **mSelectedFont**
- QColor **mSelectedTextColor**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayoutElement

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCPLegend**

Additional Inherited Members

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

17.601.1 Detailed Description

The abstract base class for all entries in a QCPLegend.
It defines a very basic interface for entries in a QCPLegend. For representing plottables in the legend, the subclass QCPPlottableLegendItem is more suitable.
Only derive directly from this class when you need absolute freedom (e.g. a custom legend entry that's not even associated with a plottable).
You must implement the following pure virtual functions:

- draw (from QCPLayerable)

You inherit the following members you may use:

QCPLegend *__mParentLegend__	A pointer to the parent QCPLegend.
QFont **mFont**	The generic font of the item. You should use this font for all or at least the most prominent text of the item.

17.601.2 Constructor & Destructor Documentation

17.601.2.1 QCPAbstractLegendItem()

```
QCPAbstractLegendItem::QCPAbstractLegendItem (
            QCPLegend * parent )  [explicit]
```

Constructs a QCPAbstractLegendItem and associates it with the QCPLegend *parent*. This does not cause the item to be added to *parent*, so QCPLegend::addItem must be called separately.

17.601.3 Member Function Documentation

17.601.3.1 applyDefaultAntialiasingHint()

```
void QCPAbstractLegendItem::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```

Reimplemented from QCPLayoutElement.

17.601.3.2 clipRect()

```
QRect QCPAbstractLegendItem::clipRect ( ) const  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.601.3.3 deselectEvent()

```
void QCPAbstractLegendItem::deselectEvent (
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.601.3.4 draw()

```
virtual void QCPAbstractLegendItem::draw (
            QCPPainter * painter )  [protected], [pure virtual]
```

Reimplemented from QCPLayoutElement.

17.601.3.5 selectEvent()

```
void QCPAbstractLegendItem::selectEvent (
            QMouseEvent * event,
            bool additive,
            const QVariant & details,
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.601.3.6 selectionCategory()

```
QCP::Interaction QCPAbstractLegendItem::selectionCategory ( ) const  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.601.3.7 selectionChanged

```
void QCPAbstractLegendItem::selectionChanged (
            bool selected )  [signal]
```

This signal is emitted when the selection state of this legend item has changed, either by user interaction or by a direct call to setSelected.

17.601.3.8 selectTest()

```
double QCPAbstractLegendItem::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

Layout elements are sensitive to events inside their outer rect. If *pos* is within the outer rect, this method returns a value corresponding to 0.99 times the parent plot's selection tolerance. However, layout elements are not selectable by default. So if *onlySelectable* is true, -1.0 is returned.
See QCPLayerable::selectTest for a general explanation of this virtual method.
QCPLayoutElement subclasses may reimplement this method to provide more specific selection test behaviour.
Reimplemented from QCPLayoutElement.

17.601.3.9 setFont()

```
void QCPAbstractLegendItem::setFont (
            const QFont & font )
```

Sets the default font of this specific legend item to *font*.

See also

setTextColor, QCPLegend::setFont

17.601.3.10 setSelectable()

```
void QCPAbstractLegendItem::setSelectable (
            bool selectable )
```

Sets whether this specific legend item is selectable.

See also

setSelectedParts, QCustomPlot::setInteractions

17.601.3.11 setSelected()

```
void QCPAbstractLegendItem::setSelected (
            bool selected )
```

Sets whether this specific legend item is selected.
It is possible to set the selection state of this item by calling this function directly, even if setSelectable is set to false.

See also

setSelectableParts, QCustomPlot::setInteractions

17.601.3.12 setSelectedFont()

```
void QCPAbstractLegendItem::setSelectedFont (
            const QFont & font )
```

When this legend item is selected, *font* is used to draw generic text, instead of the normal font set with setFont.

See also

setFont, QCPLegend::setSelectedFont

17.601.3.13 setSelectedTextColor()

```
void QCPAbstractLegendItem::setSelectedTextColor (
            const QColor & color )
```

When this legend item is selected, *color* is used to draw generic text, instead of the normal color set with setTextColor.

See also

setTextColor, QCPLegend::setSelectedTextColor

17.601.3.14 setTextColor()

```
void QCPAbstractLegendItem::setTextColor (
            const QColor & color )
```

Sets the default text color of this specific legend item to *color*.

See also

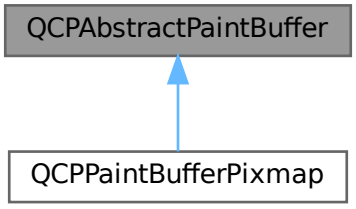
setFont, QCPLegend::setTextColor

The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.602 QCPAbstractPaintBuffer Class Reference

The abstract base class for paint buffers, which define the rendering backend.
Inheritance diagram for QCPAbstractPaintBuffer:



Public Member Functions

- QCPAbstractPaintBuffer (const QSize &size, double devicePixelRatio)
- QSize **size** () const
- bool **invalidated** () const
- double **devicePixelRatio** () const
- void setSize (const QSize &size)
- void setInvalidated (bool invalidated=true)
- void setDevicePixelRatio (double ratio)
- virtual QCPPainter * startPainting ()=0
- virtual void donePainting ()
- virtual void draw (QCPPainter *painter) const =0
- virtual void clear (const QColor &color)=0

Protected Member Functions

- virtual void reallocateBuffer ()=0

Protected Attributes

- QSize **mSize**
- double **mDevicePixelRatio**
- bool **mInvalidated**

17.602.1 Detailed Description

The abstract base class for paint buffers, which define the rendering backend.
This abstract base class defines the basic interface that a paint buffer needs to provide in order to be usable by QCustomPlot.
A paint buffer manages both a surface to draw onto, and the matching paint device. The size of the surface can be changed via setSize. External classes (QCustomPlot and QCPLayer) request a painter via startPainting and then perform the draw calls. Once the painting is complete, donePainting is called, so the paint buffer implementation can do clean up if necessary. Before rendering a frame, each paint buffer is usually filled with a color using clear (usually the color is `Qt::transparent`), to remove the contents of the previous frame.
The simplest paint buffer implementation is QCPPaintBufferPixmap which allows regular software rendering via the raster engine. Hardware accelerated rendering via pixel buffers and frame buffer objects is provided by QCPPaint↩ BufferGlPbuffer and QCPPaintBufferGlFbo. They are used automatically if QCustomPlot::setOpenGl is enabled.

17.602.2 Constructor & Destructor Documentation

17.602.2.1 QCPAbstractPaintBuffer()

```
QCPAbstractPaintBuffer::QCPAbstractPaintBuffer (
            const QSize & size,
            double devicePixelRatio )  [explicit]
```

Creates a paint buffer and initializes it with the provided *size* and *devicePixelRatio*.
Subclasses must call their reallocateBuffer implementation in their respective constructors.

17.602.3 Member Function Documentation

17.602.3.1 clear()

```
void QCPAbstractPaintBuffer::clear (
            const QColor & color )  [pure virtual]
```

Fills the entire buffer with the provided *color*. To have an empty transparent buffer, use the named color `Qt::transparent`.
This method must not be called if there is currently a painter (acquired with startPainting) active.
Implemented in QCPPaintBufferPixmap.

17.602.3.2 donePainting()

```
void QCPAbstractPaintBuffer::donePainting ( )  [inline], [virtual]
```

If you have acquired a QCPPainter to paint onto this paint buffer via startPainting, call this method as soon as you are done with the painting operations and have deleted the painter.
paint buffer subclasses may use this method to perform any type of cleanup that is necessary. The default implementation does nothing.

17.602.3.3 draw()

```
void QCPAbstractPaintBuffer::draw (
            QCPPainter * painter ) const  [pure virtual]
```

Draws the contents of this buffer with the provided *painter*. This is the method that is used to finally join all paint buffers and draw them onto the screen.
Implemented in QCPPaintBufferPixmap.

17.602.3.4 reallocateBuffer()

```
void QCPAbstractPaintBuffer::reallocateBuffer ( )  [protected], [pure virtual]
```

Reallocates the internal buffer with the currently configured size (setSize) and device pixel ratio, if applicable (setDevicePixelRatio). It is called as soon as any of those properties are changed on this paint buffer.

Note

> Subclasses of QCPAbstractPaintBuffer must call their reimplementation of this method in their constructor, to perform the first allocation (this can not be done by the base class because calling pure virtual methods in base class constructors is not possible).

Implemented in QCPPaintBufferPixmap.

17.602.3.5 setDevicePixelRatio()

```
void QCPAbstractPaintBuffer::setDevicePixelRatio (
            double ratio )
```

Sets the device pixel ratio to *ratio*. This is useful to render on high-DPI output devices. The ratio is automatically set to the device pixel ratio used by the parent QCustomPlot instance.
The buffer is reallocated (by calling reallocateBuffer), so any painters that were obtained by startPainting are invalidated and must not be used after calling this method.

Note

This method is only available for Qt versions 5.4 and higher.

17.602.3.6 setInvalidated()

```
void QCPAbstractPaintBuffer::setInvalidated (
            bool invalidated = true )
```

Sets the invalidated flag to *invalidated*.
This mechanism is used internally in conjunction with isolated replotting of QCPLayer instances (in QCPLayer::lmBuffered mode). If QCPLayer::replot is called on a buffered layer, i.e. an isolated repaint of only that layer (and its dedicated paint buffer) is requested, QCustomPlot will decide depending on the invalidated flags of other paint buffers whether it also replots them, instead of only the layer on which the replot was called.
The invalidated flag is set to true when QCPLayer association has changed, i.e. if layers were added or removed from this buffer, or if they were reordered. It is set to false as soon as all associated QCPLayer instances are drawn onto the buffer.
Under normal circumstances, it is not necessary to manually call this method.

17.602.3.7 setSize()

```
void QCPAbstractPaintBuffer::setSize (
            const QSize & size )
```

Sets the paint buffer size.
The buffer is reallocated (by calling reallocateBuffer), so any painters that were obtained by startPainting are invalidated and must not be used after calling this method.
If *size* is already the current buffer size, this method does nothing.

17.602.3.8 startPainting()

```
QCPPainter * QCPAbstractPaintBuffer::startPainting ( )  [pure virtual]
```

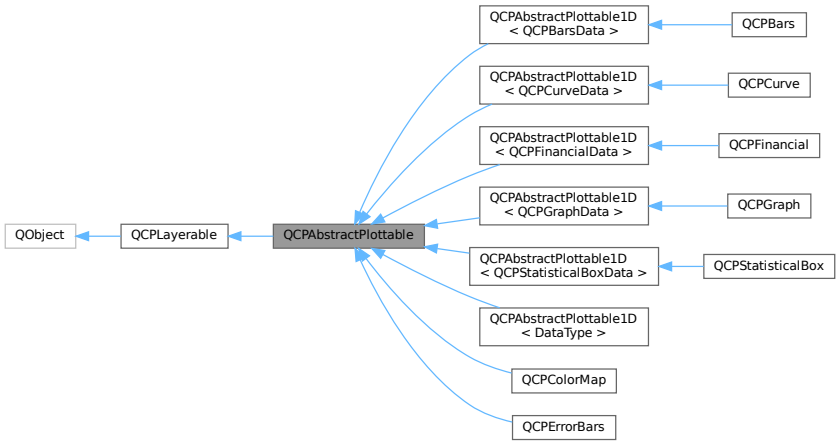
Returns a QCPPainter which is ready to draw to this buffer. The ownership and thus the responsibility to delete the painter after the painting operations are complete is given to the caller of this method.
Once you are done using the painter, delete the painter and call donePainting.
While a painter generated with this method is active, you must not call setSize, setDevicePixelRatio or clear.
This method may return 0, if a painter couldn't be activated on the buffer. This usually indicates a problem with the respective painting backend.
Implemented in QCPPaintBufferPixmap.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

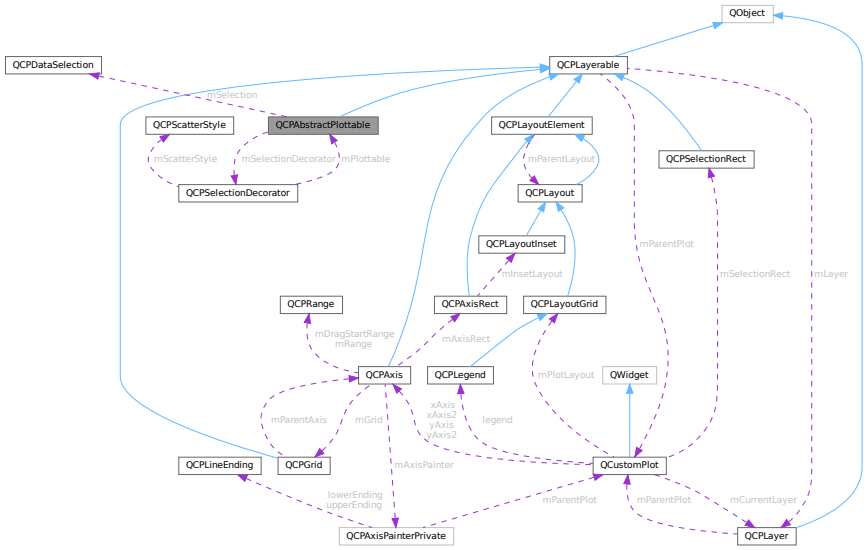
17.603 QCPAbstractPlottable Class Reference

The abstract base class for all data representing objects in a plot.

Inheritance diagram for QCPAbstractPlottable:



Collaboration diagram for QCPAbstractPlottable:



Signals

- void selectionChanged (bool selected)
- void selectionChanged (const QCPDataSelection &selection)
- void selectableChanged (QCP::SelectionType selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPAbstractPlottable (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QString **name** () const

- bool **antialiasedFill** () const
- bool **antialiasedScatters** () const
- QPen **pen** () const
- QBrush **brush** () const
- QCPAxis * **keyAxis** () const
- QCPAxis * **valueAxis** () const
- QCP::SelectionType **selectable** () const
- bool selected () const
- QCPDataSelection selection () const
- QCPSelectionDecorator * selectionDecorator () const
- void setName (const QString &name)
- void setAntialiasedFill (bool enabled)
- void setAntialiasedScatters (bool enabled)
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setKeyAxis (QCPAxis *axis)
- void setValueAxis (QCPAxis *axis)
- Q_SLOT void setSelectable (QCP::SelectionType selectable)
- Q_SLOT void setSelection (QCPDataSelection selection)
- void setSelectionDecorator (QCPSelectionDecorator *decorator)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE=0
- virtual QCPPlottableInterface1D * interface1D ()
- virtual QCPRange getKeyRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth) const =0
- virtual QCPRange getValueRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth, const QCPRange &inKeyRange=QCPRange()) const =0
- void coordsToPixels (double key, double value, double &x, double &y) const
- const QPointF coordsToPixels (double key, double value) const
- void pixelsToCoords (double x, double y, double &key, double &value) const
- void pixelsToCoords (const QPointF &pixelPos, double &key, double &value) const
- void rescaleAxes (bool onlyEnlarge=false) const
- void rescaleKeyAxis (bool onlyEnlarge=false) const
- void rescaleValueAxis (bool onlyEnlarge=false, bool inKeyRange=false) const
- bool addToLegend (QCPLegend *legend)
- bool addToLegend ()
- bool removeFromLegend (QCPLegend *legend) const
- bool removeFromLegend () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE=0
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void **drawLegendIcon** (QCPPainter *painter, const QRectF &rect) const =0
- void **applyFillAntialiasingHint** (QCPPainter *painter) const
- void **applyScattersAntialiasingHint** (QCPPainter *painter) const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QString **mName**
- bool **mAntialiasedFill**
- bool **mAntialiasedScatters**
- QPen **mPen**
- QBrush **mBrush**
- QPointer< QCPAxis > **mKeyAxis**
- QPointer< QCPAxis > **mValueAxis**
- QCP::SelectionType **mSelectable**
- QCPDataSelection **mSelection**
- QCPSelectionDecorator * **mSelectionDecorator**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPAxis**
- class **QCPPlottableLegendItem**

17.603.1 Detailed Description

The abstract base class for all data representing objects in a plot.
It defines a very basic interface like name, pen, brush, visibility etc. Since this class is abstract, it can't be instantiated. Use one of the subclasses or create a subclass yourself to create new ways of displaying data (see "Creating own plottables" below). Plottables that display one-dimensional data (i.e. data points have a single key dimension and one or multiple values at each key) are based off of the template subclass QCPAbstractPlottable1D, see details there.
All further specifics are in the subclasses, for example:

- A normal graph with possibly a line and/or scatter points QCPGraph (typically created with QCustomPlot::addGraph)
- A parametric curve: QCPCurve
- A bar chart: QCPBars
- A statistical box plot: QCPStatisticalBox
- A color encoded two-dimensional map: QCPColorMap
- An OHLC/Candlestick chart: QCPFinancial

17.603.2 Creating own plottables

Subclassing directly from QCPAbstractPlottable is only recommended if you wish to display two-dimensional data like QCPColorMap, i.e. two logical key dimensions and one (or more) data dimensions. If you want to display data with only one logical key dimension, you should rather derive from QCPAbstractPlottable1D.
If subclassing QCPAbstractPlottable directly, these are the pure virtual functions you must implement:

- selectTest
- draw
- drawLegendIcon
- getKeyRange
- getValueRange

See the documentation of those functions for what they need to do.
For drawing your plot, you can use the coordsToPixels functions to translate a point in plot coordinates to pixel coordinates. This function is quite convenient, because it takes the orientation of the key and value axes into account for you (x and y are swapped when the key axis is vertical and the value axis horizontal). If you are worried about performance (i.e. you need to translate many points in a loop like QCPGraph), you can directly use QCPAxis::coordToPixel. However, you must then take care about the orientation of the axis yourself.
Here are some important members you inherit from QCPAbstractPlottable:

QCustomPlot *__mParentPlot__	A pointer to the parent QCustomPlot instance. The parent plot is inferred from the axes that are passed in the constructor.
QString __mName__	The name of the plottable.
QPen __mPen__	The generic pen of the plottable. You should use this pen for the most prominent data representing lines in the plottable (e.g QCPGraph uses this pen for its graph lines and scatters)
QBrush __mBrush__	The generic brush of the plottable. You should use this brush for the most prominent fillable structures in the plottable (e.g. QCPGraph uses this brush to control filling under the graph)
QPointer<QCPAxis> __mKeyAxis__, __mValueAxis__	The key and value axes this plottable is attached to. Call their QCPAxis::coordToPixel functions to translate coordinates to pixels in either the key or value dimension. Make sure to check whether the pointer is `nullptr` before using it. If one of the axes is null, don't draw the plottable.

QCPSelectionDecorator **mSelectionDecorator**	The currently set selection decorator which specifies how selected data of the plottable shall be drawn and decorated. When drawing your data, you must consult this decorator for the appropriate pen/brush before drawing unselected/selected data segments. Finally, you should call its QCPSelectionDecorator::drawDecoration method at the end of your draw implementation.
QCP::SelectionType **mSelectable**	In which composition, if at all, this plottable's data may be selected. Enforcing this setting on the data selection is done by QCPAbstractPlottable automatically.
QCPDataSelection **mSelection**	Holds the current selection state of the plottable's data, i.e. the selected data ranges (QCPDataRange).

17.603.3 Constructor & Destructor Documentation

17.603.3.1 QCPAbstractPlottable()

```
QCPAbstractPlottable::QCPAbstractPlottable (
            QCPAxis * keyAxis,
            QCPAxis * valueAxis )
```

Constructs an abstract plottable which uses *keyAxis* as its key axis ("x") and *valueAxis* as its value axis ("y"). *key↩ Axis* and *valueAxis* must reside in the same QCustomPlot instance and have perpendicular orientations. If either of these restrictions is violated, a corresponding message is printed to the debug output (qDebug), the construction is not aborted, though.
Since QCPAbstractPlottable is an abstract class that defines the basic interface to plottables, it can't be directly instantiated.
You probably want one of the subclasses like QCPGraph or QCPCurve instead.

17.603.4 Member Function Documentation

17.603.4.1 addToLegend() [1/2]

```
bool QCPAbstractPlottable::addToLegend ( )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds this plottable to the legend of the parent QCustomPlot (QCustomPlot::legend).

See also

removeFromLegend

17.603.4.2 addToLegend() [2/2]

```
bool QCPAbstractPlottable::addToLegend (
            QCPLegend * legend )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds this plottable to the specified *legend*.
Creates a QCPPlottableLegendItem which is inserted into the legend. Returns true on success, i.e. when the legend exists and a legend item associated with this plottable isn't already in the legend.
If the plottable needs a more specialized representation in the legend, you can create a corresponding subclass of QCPPlottableLegendItem and add it to the legend manually instead of calling this method.

See also

removeFromLegend, QCPLegend::addItem

17.603.4.3 applyDefaultAntialiasingHint()

```
void QCPAbstractPlottable::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```

Implements QCPLayerable.

17.603.4.4 clipRect()

```
QRect QCPAbstractPlottable::clipRect ( ) const  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.603.4.5 coordsToPixels() [1/2]

```
const QPointF QCPAbstractPlottable::coordsToPixels (
            double key,
            double value ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Transforms the given *key* and *value* to pixel coordinates and returns them in a QPointF.

17.603.4.6 coordsToPixels() [2/2]

```
void QCPAbstractPlottable::coordsToPixels (
            double key,
            double value,
            double & x,
            double & y ) const
```

Convenience function for transforming a key/value pair to pixels on the QCustomPlot surface, taking the orientations of the axes associated with this plottable into account (e.g. whether key represents x or y).
key and *value* are transformed to the coodinates in pixels and are written to *x* and *y*.

See also

pixelsToCoords, QCPAxis::coordToPixel

17.603.4.7 deselectEvent()

```
void QCPAbstractPlottable::deselectEvent (
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.603.4.8 draw()

```
virtual void QCPAbstractPlottable::draw (
            QCPPainter * painter )  [protected], [pure virtual]
```

Implements QCPLayerable.

17.603.4.9 getKeyRange()

```
QCPRange QCPAbstractPlottable::getKeyRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth ) const  [pure virtual]
```

Returns the coordinate range that all data in this plottable span in the key axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSignDomain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getValueRange

Implemented in QCPGraph, QCPCurve, QCPBars, QCPStatisticalBox, QCPColorMap, QCPFinancial, and QCPErrorBars.

17.603.4.10 getValueRange()

```
QCPRange QCPAbstractPlottable::getValueRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth,
            const QCPRange & inKeyRange = QCPRange() ) const  [pure virtual]
```

Returns the coordinate range that the data points in the specified key range (*inKeyRange*) span in the value axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSign↩ Domain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
If *inKeyRange* has both lower and upper bound set to zero (is equal to `QCPRange()`), all data points are considered, without any restriction on the keys.
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getKeyRange

Implemented in QCPGraph, QCPCurve, QCPBars, QCPStatisticalBox, QCPColorMap, QCPFinancial, and QCPErrorBars.

17.603.4.11 interface1D()

```
QCPPlottableInterface1D * QCPAbstractPlottable::interface1D ( )  [inline], [virtual]
```

If this plottable is a one-dimensional plottable, i.e. it implements the QCPPlottableInterface1D, returns the *this* pointer with that type. Otherwise (e.g. in the case of a QCPColorMap) returns zero.
You can use this method to gain read access to data coordinates while holding a pointer to the abstract base class only.
Reimplemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< Q QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat and QCPErrorBars.

17.603.4.12 pixelsToCoords() [1/2]

```
void QCPAbstractPlottable::pixelsToCoords (
            const QPointF & pixelPos,
            double & key,
            double & value ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Returns the pixel input *pixelPos* as plot coordinates *key* and *value*.

17.603.4.13 pixelsToCoords() [2/2]

```
void QCPAbstractPlottable::pixelsToCoords (
            double x,
            double y,
            double & key,
            double & value ) const
```

Convenience function for transforming a x/y pixel pair on the QCustomPlot surface to plot coordinates, taking the orientations of the axes associated with this plottable into account (e.g. whether key represents x or y).
x and *y* are transformed to the plot coodinates and are written to *key* and *value*.

See also

coordsToPixels, QCPAxis::coordToPixel

17.603.4.14 removeFromLegend() [1/2]

```
bool QCPAbstractPlottable::removeFromLegend ( ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Removes the plottable from the legend of the parent QCustomPlot.

See also

addToLegend

17.603.4.15 removeFromLegend() [2/2]

```
bool QCPAbstractPlottable::removeFromLegend (
            QCPLegend * legend ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Removes the plottable from the specifed *legend*. This means the QCPPlottableLegendItem that is associated with this plottable is removed.
Returns true on success, i.e. if the legend exists and a legend item associated with this plottable was found and removed.

See also

addToLegend, QCPLegend::removeItem

17.603.4.16 rescaleAxes()

```
void QCPAbstractPlottable::rescaleAxes (
            bool onlyEnlarge = false ) const
```

Rescales the key and value axes associated with this plottable to contain all displayed data, so the whole plottable is visible. If the scaling of an axis is logarithmic, rescaleAxes will make sure not to rescale to an illegal range i.e. a range containing different signs and/or zero. Instead it will stay in the current sign domain and ignore all parts of the plottable that lie outside of that domain.
onlyEnlarge makes sure the ranges are only expanded, never reduced. So it's possible to show multiple plottables in their entirety by multiple calls to rescaleAxes where the first call has *onlyEnlarge* set to false (the default), and all subsequent set to true.

See also

rescaleKeyAxis, rescaleValueAxis, QCustomPlot::rescaleAxes, QCPAxis::rescale

17.603.4.17 rescaleKeyAxis()

```
void QCPAbstractPlottable::rescaleKeyAxis (
            bool onlyEnlarge = false ) const
```

Rescales the key axis of the plottable so the whole plottable is visible.
See rescaleAxes for detailed behaviour.

17.603.4.18 rescaleValueAxis()

```
void QCPAbstractPlottable::rescaleValueAxis (
            bool onlyEnlarge = false,
            bool inKeyRange = false ) const
```

Rescales the value axis of the plottable so the whole plottable is visible. If *inKeyRange* is set to true, only the data points which are in the currently visible key axis range are considered.
Returns true if the axis was actually scaled. This might not be the case if this plottable has an invalid range, e.g. because it has no data points.
See rescaleAxes for detailed behaviour.

17.603.4.19 selectableChanged

```
void QCPAbstractPlottable::selectableChanged (
            QCP::SelectionType selectable ) [signal]
```

This signal is emitted when the selectability of this plottable has changed.

See also

setSelectable

17.603.4.20 selected()

```
bool QCPAbstractPlottable::selected ( ) const [inline]
```

Returns true if there are any data points of the plottable currently selected. Use selection to retrieve the current QCPDataSelection.

17.603.4.21 selectEvent()

```
void QCPAbstractPlottable::selectEvent (
            QMouseEvent * event,
            bool additive,
            const QVariant & details,
            bool * selectionStateChanged ) [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.603.4.22 selection()

```
QCPDataSelection QCPAbstractPlottable::selection ( ) const [inline]
```

Returns a QCPDataSelection encompassing all the data points that are currently selected on this plottable.

See also

selected, setSelection, setSelectable

17.603.4.23 selectionCategory()

```
QCP::Interaction QCPAbstractPlottable::selectionCategory ( ) const [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.603.4.24 selectionChanged [1/2]

```
void QCPAbstractPlottable::selectionChanged (
            bool selected ) [signal]
```

This signal is emitted when the selection state of this plottable has changed, either by user interaction or by a direct call to setSelection. The parameter *selected* indicates whether there are any points selected or not.

See also

selectionChanged(const QCPDataSelection &selection)

17.603.4.25 selectionChanged [2/2]

```
void QCPAbstractPlottable::selectionChanged (
            const QCPDataSelection & selection )  [signal]
```

This signal is emitted when the selection state of this plottable has changed, either by user interaction or by a direct call to setSelection. The parameter *selection* holds the currently selected data ranges.

See also

selectionChanged(bool selected)

17.603.4.26 selectionDecorator()

```
QCPSelectionDecorator * QCPAbstractPlottable::selectionDecorator ( ) const  [inline]
```

Provides access to the selection decorator of this plottable. The selection decorator controls how selected data ranges are drawn (e.g. their pen color and fill), see QCPSelectionDecorator for details.
If you wish to use an own QCPSelectionDecorator subclass, pass an instance of it to setSelectionDecorator.

17.603.4.27 selectTest()

```
virtual double QCPAbstractPlottable::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [pure virtual]
```

This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Reimplemented from QCPLayerable.
Implemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCP
QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat
QCPGraph, QCPCurve, QCPBars, QCPStatisticalBox, QCPColorMap, QCPFinancial, and QCPErrorBars.

17.603.4.28 setAntialiasedFill()

```
void QCPAbstractPlottable::setAntialiasedFill (
            bool enabled )
```

Sets whether fills of this plottable are drawn antialiased or not.

Note that this setting may be overridden by QCustomPlot::setAntialiasedElements and QCustomPlot::setNotAntialiasedElements.

17.603.4.29 setAntialiasedScatters()

```
void QCPAbstractPlottable::setAntialiasedScatters (
            bool enabled )
```

Sets whether the scatter symbols of this plottable are drawn antialiased or not.
Note that this setting may be overridden by QCustomPlot::setAntialiasedElements and QCustomPlot::setNotAntialiasedElements.

17.603.4.30 setBrush()

```
void QCPAbstractPlottable::setBrush (
            const QBrush & brush )
```

The brush is used to draw basic fills of the plottable representation in the plot. The Fill can be a color, gradient or texture, see the usage of QBrush.
For example, the QCPGraph subclass draws the fill under the graph with this brush, when it's not set to Qt::NoBrush.

See also

setPen

17.603.4.31 setKeyAxis()

```
void QCPAbstractPlottable::setKeyAxis (
            QCPAxis * axis )
```

The key axis of a plottable can be set to any axis of a QCustomPlot, as long as it is orthogonal to the plottable's value axis. This function performs no checks to make sure this is the case. The typical mathematical choice is to use the x-axis (QCustomPlot::xAxis) as key axis and the y-axis (QCustomPlot::yAxis) as value axis.
Normally, the key and value axes are set in the constructor of the plottable (or QCustomPlot::addGraph when working with QCPGraphs through the dedicated graph interface).

See also

setValueAxis

17.603.4.32 setName()

```
void QCPAbstractPlottable::setName (
            const QString & name )
```

The name is the textual representation of this plottable as it is displayed in the legend (QCPLegend). It may contain any UTF-8 characters, including newlines.

17.603.4.33 setPen()

```
void QCPAbstractPlottable::setPen (
            const QPen & pen )
```

The pen is used to draw basic lines that make up the plottable representation in the plot.
For example, the QCPGraph subclass draws its graph lines with this pen.

See also

setBrush

17.603.4.34 setSelectable()

```
void QCPAbstractPlottable::setSelectable (
            QCP::SelectionType selectable )
```

Sets whether and to which granularity this plottable can be selected.
A selection can happen by clicking on the QCustomPlot surface (When QCustomPlot::setInteractions contains QCP::iSelectPlottables), by dragging a selection rect (When QCustomPlot::setSelectionRectMode is QCP::srmSelect), or programmatically by calling setSelection.

See also

setSelection, QCP::SelectionType

17.603.4.35 setSelection()

```
void QCPAbstractPlottable::setSelection (
            QCPDataSelection selection )
```

Sets which data ranges of this plottable are selected. Selected data ranges are drawn differently (e.g. color) in the plot. This can be controlled via the selection decorator (see selectionDecorator).
The entire selection mechanism for plottables is handled automatically when QCustomPlot::setInteractions contains iSelectPlottables. You only need to call this function when you wish to change the selection state programmatically.
Using setSelectable you can further specify for each plottable whether and to which granularity it is selectable. If *selection* is not compatible with the current QCP::SelectionType set via setSelectable, the resulting selection will be adjusted accordingly (see QCPDataSelection::enforceType).
emits the selectionChanged signal when *selected* is different from the previous selection state.

See also

setSelectable, selectTest

17.603.4.36 setSelectionDecorator()

```
void QCPAbstractPlottable::setSelectionDecorator (
            QCPSelectionDecorator * decorator )
```

Use this method to set an own QCPSelectionDecorator (subclass) instance. This allows you to customize the visual representation of selected data ranges further than by using the default QCPSelectionDecorator.
The plottable takes ownership of the *decorator*.
The currently set decorator can be accessed via selectionDecorator.

17.603.4.37 setValueAxis()

```
void QCPAbstractPlottable::setValueAxis (
            QCPAxis * axis )
```

The value axis of a plottable can be set to any axis of a QCustomPlot, as long as it is orthogonal to the plottable's key axis. This function performs no checks to make sure this is the case. The typical mathematical choice is to use the x-axis (QCustomPlot::xAxis) as key axis and the y-axis (QCustomPlot::yAxis) as value axis.
Normally, the key and value axes are set in the constructor of the plottable (or QCustomPlot::addGraph when working with QCPGraphs through the dedicated graph interface).

See also

setKeyAxis

The documentation for this class was generated from the following files:

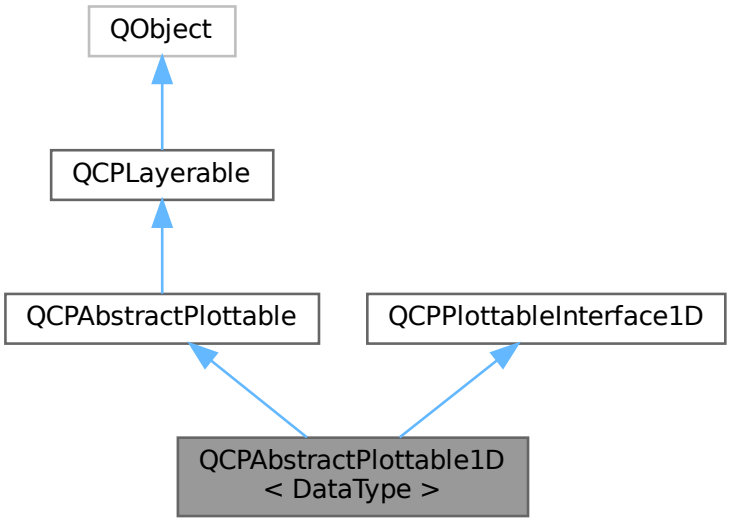
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.604 QCPAbstractPlottable1D< DataType > Class Template Reference

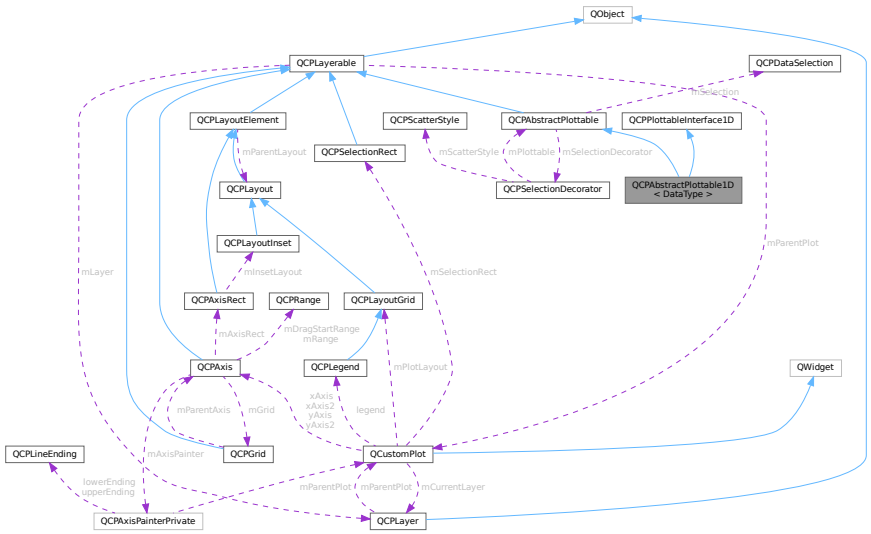
A template base class for plottables with one-dimensional data.

`#include <applications/gui/genesys/qcustomplot.h>`

Inheritance diagram for QCPAbstractPlottable1D< DataType >:



Collaboration diagram for QCPAbstractPlottable1D< DataType >:



Public Member Functions

- QCPAbstractPlottable1D (QCPAxis *keyAxis, QCPAxis *valueAxis)
- virtual int dataCount () const Q_DECL_OVERRIDE
- virtual double dataMainKey (int index) const Q_DECL_OVERRIDE

- virtual double dataSortKey (int index) const Q_DECL_OVERRIDE
- virtual double dataMainValue (int index) const Q_DECL_OVERRIDE
- virtual QCPRange dataValueRange (int index) const Q_DECL_OVERRIDE
- virtual QPointF dataPixelPosition (int index) const Q_DECL_OVERRIDE
- virtual bool sortKeyIsMainKey () const Q_DECL_OVERRIDE
- virtual QCPDataSelection selectTestRect (const QRectF &rect, bool onlySelectable) const Q_DECL_↩ OVERRIDE
- virtual int findBegin (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual int findEnd (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE
- virtual QCPPlottableInterface1D * interface1D () Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable

- QCPAbstractPlottable (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QString **name** () const
- bool **antialiasedFill** () const
- bool **antialiasedScatters** () const
- QPen **pen** () const
- QBrush **brush** () const
- QCPAxis * **keyAxis** () const
- QCPAxis * **valueAxis** () const
- QCP::SelectionType **selectable** () const
- bool selected () const
- QCPDataSelection selection () const
- QCPSelectionDecorator * selectionDecorator () const
- void setName (const QString &name)
- void setAntialiasedFill (bool enabled)
- void setAntialiasedScatters (bool enabled)
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setKeyAxis (QCPAxis *axis)
- void setValueAxis (QCPAxis *axis)
- Q_SLOT void setSelectable (QCP::SelectionType selectable)
- Q_SLOT void setSelection (QCPDataSelection selection)
- void setSelectionDecorator (QCPSelectionDecorator *decorator)
- virtual QCPRange getKeyRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth) const =0
- virtual QCPRange getValueRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth, const QCPRange &inKeyRange=QCPRange()) const =0
- void coordsToPixels (double key, double value, double &x, double &y) const
- const QPointF coordsToPixels (double key, double value) const
- void pixelsToCoords (double x, double y, double &key, double &value) const
- void pixelsToCoords (const QPointF &pixelPos, double &key, double &value) const
- void rescaleAxes (bool onlyEnlarge=false) const
- void rescaleKeyAxis (bool onlyEnlarge=false) const
- void rescaleValueAxis (bool onlyEnlarge=false, bool inKeyRange=false) const
- bool addToLegend (QCPLegend *legend)
- bool addToLegend ()
- bool removeFromLegend (QCPLegend *legend) const
- bool removeFromLegend () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- void getDataSegments (QList< QCPDataRange > &selectedSegments, QList< QCPDataRange > &unselectedSegments) const
- void drawPolyline (QCPPainter *painter, const QVector< QPointF > &lineData) const

Protected Member Functions inherited from QCPAbstractPlottable

- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE=0
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void **drawLegendIcon** (QCPPainter *painter, const QRectF &rect) const =0
- void **applyFillAntialiasingHint** (QCPPainter *painter) const
- void **applyScattersAntialiasingHint** (QCPPainter *painter) const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QSharedPointer< QCPDataContainer< DataType > > **mDataContainer**

Protected Attributes inherited from QCPAbstractPlottable

- QString **mName**
- bool **mAntialiasedFill**
- bool **mAntialiasedScatters**
- QPen **mPen**
- QBrush **mBrush**
- QPointer< QCPAxis > **mKeyAxis**
- QPointer< QCPAxis > **mValueAxis**
- QCP::SelectionType **mSelectable**
- QCPDataSelection **mSelection**
- QCPSelectionDecorator * **mSelectionDecorator**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPAbstractPlottable

- void selectionChanged (bool selected)
- void selectionChanged (const QCPDataSelection &selection)
- void selectableChanged (QCP::SelectionType selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.604.1 Detailed Description

template<class DataType>
class QCPAbstractPlottable1D< DataType >

A template base class for plottables with one-dimensional data.
This template class derives from QCPAbstractPlottable and from the abstract interface QCPPlottableInterface1D. It serves as a base class for all one-dimensional data (i.e. data with one key dimension), such as QCPGraph and QCPCurve.
The template parameter *DataType* is the type of the data points of this plottable (e.g. QCPGraphData or QCPCurveData). The main purpose of this base class is to provide the member *mDataContainer* (a shared pointer to a QCPDataContainer<DataType>) and implement the according virtual methods of the QCPPlottableInterface1D, such that most subclassed plottables don't need to worry about this anymore.
Further, it provides a convenience method for retrieving selected/unselected data segments via getDataSegments. This is useful when subclasses implement their draw method and need to draw selected segments with a different pen/brush than unselected segments (also see QCPSelectionDecorator).
This class implements basic functionality of QCPAbstractPlottable::selectTest and QCPPlottableInterface1D::selectTestRect, assuming point-like data points, based on the 1D data interface. In spite of that, most plottable subclasses will want to reimplement those methods again, to provide a more accurate hit test based on their specific data visualization geometry.

17.604.2 Constructor & Destructor Documentation

17.604.2.1 QCPAbstractPlottable1D()

```
template<class DataType >
QCPAbstractPlottable1D< DataType >::QCPAbstractPlottable1D (
            QCPAxis * keyAxis,
            QCPAxis * valueAxis )
```

Forwards *keyAxis* and *valueAxis* to the QCPAbstractPlottable constructor and allocates the *mDataContainer*.

17.604.3 Member Function Documentation

17.604.3.1 dataCount()

```
template<class DataType >
int QCPAbstractPlottable1D< DataType >::dataCount ( ) const  [virtual]
```

Returns the number of data points of the plottable.
Implements QCPPlottableInterface1D.

17.604.3.2 dataMainKey()

```
template<class DataType >
double QCPAbstractPlottable1D< DataType >::dataMainKey (
            int index ) const  [virtual]
```

Returns the main key of the data point at the given *index*.
What the main key is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.
Implements QCPPlottableInterface1D.

17.604.3.3 dataMainValue()

```
template<class DataType >
double QCPAbstractPlottable1D< DataType >::dataMainValue (
            int index ) const  [virtual]
```

Returns the main value of the data point at the given *index*.
What the main value is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.
Implements QCPPlottableInterface1D.

17.604.3.4 dataPixelPosition()

```
template<class DataType >
QPointF QCPAbstractPlottable1D< DataType >::dataPixelPosition (
            int index ) const  [virtual]
```

Returns the pixel position on the widget surface at which the data point at the given *index* appears.
Usually this corresponds to the point of dataMainKey/ dataMainValue, in pixel coordinates. However, depending on the plottable, this might be a different apparent position than just a coord-to-pixel transform of those values. For example, QCPBars apparent data values can be shifted depending on their stacking, bar grouping or configured base value.
Implements QCPPlottableInterface1D.
Reimplemented in QCPBars.

17.604.3.5 dataSortKey()

```
template<class DataType >
double QCPAbstractPlottable1D< DataType >::dataSortKey (
            int index ) const  [virtual]
```

Returns the sort key of the data point at the given *index*.
What the sort key is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.

Implements QCPPlottableInterface1D.

17.604.3.6 dataValueRange()

```
template<class DataType >
QCPRange QCPAbstractPlottable1D< DataType >::dataValueRange (
            int index ) const  [virtual]
```

Returns the value range of the data point at the given *index*.
What the value range is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.
Implements QCPPlottableInterface1D.

17.604.3.7 drawPolyline()

```
template<class DataType >
void QCPAbstractPlottable1D< DataType >::drawPolyline (
            QCPPainter * painter,
            const QVector< QPointF > & lineData ) const  [protected]
```

A helper method which draws a line with the passed *painter*, according to the pixel data in *lineData*. NaN points create gaps in the line, as expected from QCustomPlot's plottables (this is the main difference to QPainter's regular drawPolyline, which handles NaNs by lagging or crashing).
Further it uses a faster line drawing technique based on QCPPainter::drawLine rather than `QPainter::drawPolyline` if the configured QCustomPlot::setPlottingHints() and *painter* style allows.

17.604.3.8 findBegin()

```
template<class DataType >
int QCPAbstractPlottable1D< DataType >::findBegin (
            double sortKey,
            bool expandedRange = true ) const  [virtual]
```

Returns the index of the data point with a (sort-)key that is equal to, just below, or just above *sortKey*. If *expandedRange* is true, the data point just below *sortKey* will be considered, otherwise the one just above.
This can be used in conjunction with findEnd to iterate over data points within a given key range, including or excluding the bounding data points that are just beyond the specified range.
If *expandedRange* is true but there are no data points below *sortKey*, 0 is returned.
If the container is empty, returns 0 (in that case, findEnd will also return 0, so a loop using these methods will not iterate over the index 0).

See also

findEnd, QCPDataContainer::findBegin

Implements QCPPlottableInterface1D.

17.604.3.9 findEnd()

```
template<class DataType >
int QCPAbstractPlottable1D< DataType >::findEnd (
            double sortKey,
            bool expandedRange = true ) const  [virtual]
```

Returns the index one after the data point with a (sort-)key that is equal to, just above, or just below *sortKey*. If *expandedRange* is true, the data point just above *sortKey* will be considered, otherwise the one just below.
This can be used in conjunction with findBegin to iterate over data points within a given key range, including the bounding data points that are just below and above the specified range.
If *expandedRange* is true but there are no data points above *sortKey*, the index just above the highest data point is returned.
If the container is empty, returns 0.

See also

findBegin, QCPDataContainer::findEnd

Implements QCPPlottableInterface1D.

17.604.3.10 getDataSegments()

```
template<class DataType >
void QCPAbstractPlottable1D< DataType >::getDataSegments (
            QList< QCPDataRange > & selectedSegments,
            QList< QCPDataRange > & unselectedSegments ) const  [protected]
```

Splits all data into selected and unselected segments and outputs them via *selectedSegments* and *unselected↩ Segments*, respectively.
This is useful when subclasses implement their draw method and need to draw selected segments with a different pen/brush than unselected segments (also see QCPSelectionDecorator).

See also

setSelection

17.604.3.11 interface1D()

```
template<class DataType >
QCPPlottableInterface1D * QCPAbstractPlottable1D< DataType >::interface1D ( )  [inline], [virtual]
```

Returns a QCPPlottableInterface1D pointer to this plottable, providing access to its 1D interface.
\seebaseclassmethod
Reimplemented from QCPAbstractPlottable.

17.604.3.12 selectTest()

```
template<class DataType >
double QCPAbstractPlottable1D< DataType >::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

Implements a point-selection algorithm assuming the data (accessed via the 1D data interface) is point-like. Most subclasses will want to reimplement this method again, to provide a more accurate hit test based on the true data visualization geometry.
If *details* is not 0, it will be set to a QCPDataSelection, describing the closest data point to *pos*.
\seebaseclassmethod
Implements QCPAbstractPlottable.
Reimplemented in QCPGraph, QCPCurve, QCPBars, QCPStatisticalBox, and QCPFinancial.

17.604.3.13 selectTestRect()

```
template<class DataType >
QCPDataSelection QCPAbstractPlottable1D< DataType >::selectTestRect (
            const QRectF & rect,
            bool onlySelectable ) const  [virtual]
```

Implements a rect-selection algorithm assuming the data (accessed via the 1D data interface) is point-like. Most subclasses will want to reimplement this method again, to provide a more accurate hit test based on the true data visualization geometry.
\seebaseclassmethod
Implements QCPPlottableInterface1D.
Reimplemented in QCPBars, QCPStatisticalBox, and QCPFinancial.

17.604.3.14 sortKeyIsMainKey()

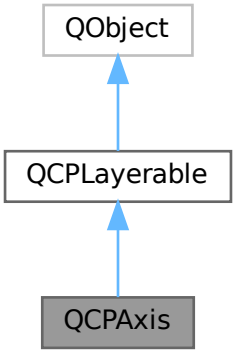
```
template<class DataType >
bool QCPAbstractPlottable1D< DataType >::sortKeyIsMainKey ( ) const  [virtual]
```

Returns whether the sort key (dataSortKey) is identical to the main key (dataMainKey).
What the sort and main keys are, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.
Implements QCPPlottableInterface1D.
The documentation for this class was generated from the following file:

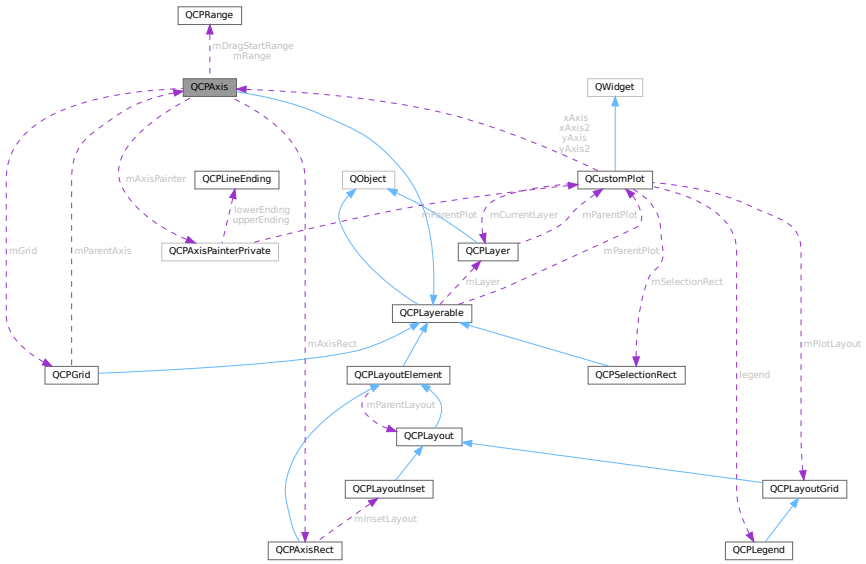
- applications/gui/genesys/qcustomplot.h

17.605 QCPAxis Class Reference

Manages a single axis inside a QCustomPlot.
Inheritance diagram for QCPAxis:



Collaboration diagram for QCPAxis:



Public Types

- enum AxisType { atLeft = 0x01 , atRight = 0x02 , atTop = 0x04 , atBottom = 0x08 }
- enum LabelSide { lsInside , lsOutside }
- enum ScaleType { stLinear , stLogarithmic }
- enum SelectablePart { spNone = 0 , spAxis = 0x001 , spTickLabels = 0x002 , spAxisLabel = 0x004 }

Signals

- void rangeChanged (const QCPRange &newRange)
- void rangeChanged (const QCPRange &newRange, const QCPRange &oldRange)
- void scaleTypeChanged (QCPAxis::ScaleType scaleType)
- void selectionChanged (const QCPAxis::SelectableParts &parts)
- void selectableChanged (const QCPAxis::SelectableParts &parts)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPAxis (QCPAxisRect *parent, AxisType type)
- AxisType **axisType** () const
- QCPAxisRect * **axisRect** () const
- ScaleType **scaleType** () const
- const QCPRange **range** () const
- bool **rangeReversed** () const
- QSharedPointer< QCPAxisTicker > ticker () const
- bool **ticks** () const
- bool **tickLabels** () const
- int **tickLabelPadding** () const
- QFont **tickLabelFont** () const
- QColor **tickLabelColor** () const
- double **tickLabelRotation** () const
- LabelSide **tickLabelSide** () const
- QString **numberFormat** () const
- int **numberPrecision** () const
- QVector< double > **tickVector** () const
- QVector< QString > **tickVectorLabels** () const
- int **tickLengthIn** () const
- int **tickLengthOut** () const
- bool **subTicks** () const
- int **subTickLengthIn** () const
- int **subTickLengthOut** () const
- QPen **basePen** () const
- QPen **tickPen** () const
- QPen **subTickPen** () const
- QFont **labelFont** () const
- QColor **labelColor** () const
- QString **label** () const
- int **labelPadding** () const
- int **padding** () const
- int **offset** () const
- SelectableParts **selectedParts** () const
- SelectableParts **selectableParts** () const
- QFont **selectedTickLabelFont** () const
- QFont **selectedLabelFont** () const
- QColor **selectedTickLabelColor** () const
- QColor **selectedLabelColor** () const
- QPen **selectedBasePen** () const
- QPen **selectedTickPen** () const
- QPen **selectedSubTickPen** () const
- QCPLineEnding **lowerEnding** () const

- QCPLineEnding **upperEnding** () const
- QCPGrid * grid () const
- Q_SLOT void setScaleType (QCPAxis::ScaleType type)
- Q_SLOT void setRange (const QCPRange &range)
- void setRange (double lower, double upper)
- void setRange (double position, double size, Qt::AlignmentFlag alignment)
- void setRangeLower (double lower)
- void setRangeUpper (double upper)
- void setRangeReversed (bool reversed)
- void setTicker (QSharedPointer< QCPAxisTicker > ticker)
- void setTicks (bool show)
- void setTickLabels (bool show)
- void setTickLabelPadding (int padding)
- void setTickLabelFont (const QFont &font)
- void setTickLabelColor (const QColor &color)
- void setTickLabelRotation (double degrees)
- void setTickLabelSide (LabelSide side)
- void setNumberFormat (const QString &formatCode)
- void setNumberPrecision (int precision)
- void setTickLength (int inside, int outside=0)
- void setTickLengthIn (int inside)
- void setTickLengthOut (int outside)
- void setSubTicks (bool show)
- void setSubTickLength (int inside, int outside=0)
- void setSubTickLengthIn (int inside)
- void setSubTickLengthOut (int outside)
- void setBasePen (const QPen &pen)
- void setTickPen (const QPen &pen)
- void setSubTickPen (const QPen &pen)
- void setLabelFont (const QFont &font)
- void setLabelColor (const QColor &color)
- void setLabel (const QString &str)
- void setLabelPadding (int padding)
- void setPadding (int padding)
- void setOffset (int offset)
- void setSelectedTickLabelFont (const QFont &font)
- void setSelectedLabelFont (const QFont &font)
- void setSelectedTickLabelColor (const QColor &color)
- void setSelectedLabelColor (const QColor &color)
- void setSelectedBasePen (const QPen &pen)
- void setSelectedTickPen (const QPen &pen)
- void setSelectedSubTickPen (const QPen &pen)
- Q_SLOT void setSelectableParts (const QCPAxis::SelectableParts &selectableParts)
- Q_SLOT void setSelectedParts (const QCPAxis::SelectableParts &selectedParts)
- void setLowerEnding (const QCPLineEnding &ending)
- void setUpperEnding (const QCPLineEnding &ending)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE
- Qt::Orientation orientation () const
- int pixelOrientation () const
- void moveRange (double diff)
- void scaleRange (double factor)
- void scaleRange (double factor, double center)
- void setScaleRatio (const QCPAxis *otherAxis, double ratio=1.0)
- void rescale (bool onlyVisiblePlottables=false)

- double pixelToCoord (double value) const
- double coordToPixel (double value) const
- SelectablePart getPartAt (const QPointF &pos) const
- QList< QCPAbstractPlottable * > plottables () const
- QList< QCPGraph * > graphs () const
- QList< QCPAbstractItem * > items () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Static Public Member Functions

- static AxisType marginSideToAxisType (QCP::MarginSide side)
- static Qt::Orientation orientation (AxisType type)
- static AxisType opposite (AxisType type)

Protected Member Functions

- virtual int **calculateMargin** ()
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void wheelEvent (QWheelEvent *event) Q_DECL_OVERRIDE
- void **setupTickVectors** ()
- QPen **getBasePen** () const
- QPen **getTickPen** () const
- QPen **getSubTickPen** () const
- QFont **getTickLabelFont** () const
- QFont **getLabelFont** () const
- QColor **getTickLabelColor** () const
- QColor **getLabelColor** () const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual QRect **clipRect** () const
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- AxisType **mAxisType**
- QCPAxisRect * **mAxisRect**
- int **mPadding**
- Qt::Orientation **mOrientation**
- SelectableParts **mSelectableParts**
- SelectableParts **mSelectedParts**
- QPen **mBasePen**
- QPen **mSelectedBasePen**
- QString **mLabel**
- QFont **mLabelFont**
- QFont **mSelectedLabelFont**
- QColor **mLabelColor**
- QColor **mSelectedLabelColor**
- bool **mTickLabels**
- QFont **mTickLabelFont**
- QFont **mSelectedTickLabelFont**
- QColor **mTickLabelColor**
- QColor **mSelectedTickLabelColor**
- int **mNumberPrecision**
- QLatin1Char **mNumberFormatChar**
- bool **mNumberBeautifulPowers**
- bool **mTicks**
- bool **mSubTicks**
- QPen **mTickPen**
- QPen **mSelectedTickPen**
- QPen **mSubTickPen**
- QPen **mSelectedSubTickPen**
- QCPRange **mRange**
- bool **mRangeReversed**
- ScaleType **mScaleType**
- QCPGrid * **mGrid**
- QCPAxisPainterPrivate * **mAxisPainter**
- QSharedPointer< QCPAxisTicker > **mTicker**
- QVector< double > **mTickVector**
- QVector< QString > **mTickVectorLabels**
- QVector< double > **mSubTickVector**
- bool **mCachedMarginValid**
- int **mCachedMargin**
- bool **mDragging**
- QCPRange **mDragStartRange**
- QCP::AntialiasedElements **mAADragBackup**
- QCP::AntialiasedElements **mNotAADragBackup**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPGrid**
- class **QCPAxisRect**

17.605.1 Detailed Description

Manages a single axis inside a QCustomPlot.
Usually doesn't need to be instantiated externally. Access QCustomPlot's default four axes via QCustomPlot::xAxis (bottom), QCustomPlot::yAxis (left), QCustomPlot::xAxis2 (top) and QCustomPlot::yAxis2 (right).
Axes are always part of an axis rect, see QCPAxisRect.

Naming convention of axis parts

Overview of the spacings and paddings that define the geometry of an axis. The dashed gray line on the left represents the QCustomPlot widget border.

Each axis holds an instance of QCPAxisTicker which is used to generate the tick coordinates and tick labels. You can access the currently installed ticker or set a new one (possibly one of the specialized subclasses, or your own subclass) via setTicker. For details, see the documentation of QCPAxisTicker.

17.605.2 Member Enumeration Documentation

17.605.2.1 AxisType

`enum QCPAxis::AxisType`
Defines at which side of the axis rect the axis will appear. This also affects how the tick marks are drawn, on which side the labels are placed etc.

Enumerator

atLeft	`0x01` Axis is vertical and on the left side of the axis rect
atRight	`0x02` Axis is vertical and on the right side of the axis rect
atTop	`0x04` Axis is horizontal and on the top side of the axis rect
atBottom	`0x08` Axis is horizontal and on the bottom side of the axis rect

17.605.2.2 LabelSide

`enum QCPAxis::LabelSide`
Defines on which side of the axis the tick labels (numbers) shall appear.

See also

setTickLabelSide

Enumerator

lsInside	Tick labels will be displayed inside the axis rect and clipped to the inner axis rect.
lsOutside	Tick labels will be displayed outside the axis rect.

17.605.2.3 ScaleType

`enum QCPAxis::ScaleType`
Defines the scale of an axis.

See also

setScaleType

Enumerator

stLinear	Linear scaling.
stLogarithmic	Logarithmic scaling with correspondingly transformed axis coordinates (possibly also setTicker to a QCPAxisTickerLog instance).

17.605.2.4 SelectablePart

```
enum QCPAxis::SelectablePart
```

Defines the selectable parts of an axis.

See also

setSelectableParts, setSelectedParts

Enumerator

spNone	None of the selectable parts.
spAxis	The axis backbone and tick marks.
spTickLabels	Tick labels (numbers) of this axis (as a whole, not individually)
spAxisLabel	The axis label.

17.605.3 Constructor & Destructor Documentation

17.605.3.1 QCPAxis()

```
QCPAxis::QCPAxis (
            QCPAxisRect * parent,
            AxisType type )  [explicit]
```

Constructs an Axis instance of Type *type* for the axis rect *parent*.
Usually it isn't necessary to instantiate axes directly, because you can let QCustomPlot create them for you with QCPAxisRect::addAxis. If you want to use own QCPAxis-subclasses however, create them manually and then inject them also via QCPAxisRect::addAxis.

17.605.4 Member Function Documentation

17.605.4.1 applyDefaultAntialiasingHint()

```
void QCPAxis::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```

Implements QCPLayerable.

17.605.4.2 coordToPixel()

```
double QCPAxis::coordToPixel (
            double value ) const
```

Transforms *value*, in coordinates of the axis, to pixel coordinates of the QCustomPlot widget.

17.605.4.3 deselectEvent()

```
void QCPAxis::deselectEvent (
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.605.4.4 draw()

```
void QCPAxis::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPLayerable.

17.605.4.5 getPartAt()

```
QCPAxis::SelectablePart QCPAxis::getPartAt (
            const QPointF & pos ) const
```

Returns the part of the axis that is hit by *pos* (in pixels). The return value of this function is independent of the user-selectable parts defined with setSelectableParts. Further, this function does not change the current selection state of the axis.
If the axis is not visible (setVisible), this function always returns spNone.

See also

setSelectedParts, setSelectableParts, QCustomPlot::setInteractions

17.605.4.6 graphs()

```
QList< QCPGraph * > QCPAxis::graphs ( ) const
```

Returns a list of all the graphs that have this axis as key or value axis.

See also

plottables, items

17.605.4.7 grid()

```
QCPGrid * QCPAxis::grid ( ) const  [inline]
```

Returns the QCPGrid instance belonging to this axis. Access it to set details about the way the grid is displayed.

17.605.4.8 items()

```
QList< QCPAbstractItem * > QCPAxis::items ( ) const
```

Returns a list of all the items that are associated with this axis. An item is considered associated with an axis if at least one of its positions uses the axis as key or value axis.

See also

plottables, graphs

17.605.4.9 marginSideToAxisType()

```
QCPAxis::AxisType QCPAxis::marginSideToAxisType (
            QCP::MarginSide side )  [static]
```

Transforms a margin side to the logically corresponding axis type. (QCP::msLeft to QCPAxis::atLeft, QCP::msRight to QCPAxis::atRight, etc.)

17.605.4.10 mouseMoveEvent()

```
void QCPAxis::mouseMoveEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user moves the mouse while holding a mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.605.4.11 mousePressEvent()

```
void QCPAxis::mousePressEvent (
            QMouseEvent * event,
            const QVariant & details )  [protected], [virtual]
```

This event gets called when the user presses a mouse button while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *details* contains layerable-specific details about the hit, which were generated in the previous call to selectTest. For example, One-dimensional plottables like QCPGraph or QCPBars convey the clicked data point in the *details* parameter, as QCPDataSelection packed as QVariant. Multi-part objects convey the specific `SelectablePart` that was hit (e.g. QCPAxis::SelectablePart in the case of axes).
QCustomPlot uses an event propagation system that works the same as Qt's system. If your layerable doesn't reimplement the mousePressEvent or explicitly calls `event->ignore()` in its reimplementation, the event will be propagated to the next layerable in the stacking order.
Once a layerable has accepted the mousePressEvent, it is considered the mouse grabber and will receive all following calls to mouseMoveEvent or mouseReleaseEvent for this mouse interaction (a "mouse interaction" in this context ends with the release).
The default implementation does nothing except explicitly ignoring the event with `event->ignore()`.

See also

mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.605.4.12 mouseReleaseEvent()

```
void QCPAxis::mouseReleaseEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user releases the mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseMoveEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.605.4.13 moveRange()

```
void QCPAxis::moveRange (
            double diff )
```

If the scale type (setScaleType) is stLinear, *diff* is added to the lower and upper bounds of the range. The range is simply moved by *diff*.
If the scale type is stLogarithmic, the range bounds are multiplied by *diff*. This corresponds to an apparent "linear" move in logarithmic scaling by a distance of log(diff).

17.605.4.14 opposite()

```
QCPAxis::AxisType QCPAxis::opposite (
            QCPAxis::AxisType type )  [static]
```

Returns the axis type that describes the opposite axis of an axis with the specified *type*.

17.605.4.15 orientation() [1/2]

```
Qt::Orientation QCPAxis::orientation ( ) const  [inline]
```
Returns the orientation of this axis. The axis orientation (horizontal or vertical) is deduced from the axis type (left, top, right or bottom).

See also

orientation(AxisType type), pixelOrientation

17.605.4.16 orientation() [2/2]

```
static Qt::Orientation QCPAxis::orientation (
            AxisType type )  [inline], [static]
```
Returns the orientation of the specified axis type

See also

orientation(), pixelOrientation

17.605.4.17 pixelOrientation()

```
int QCPAxis::pixelOrientation ( ) const  [inline]
```
Returns which direction points towards higher coordinate values/keys, in pixel space.
This method returns either 1 or -1. If it returns 1, then going in the positive direction along the orientation of the axis in pixels corresponds to going from lower to higher axis coordinates. On the other hand, if this method returns -1, going to smaller pixel values corresponds to going from lower to higher axis coordinates.
For example, this is useful to easily shift axis coordinates by a certain amount given in pixels, without having to care about reversed or vertically aligned axes:
```
double newKey = keyAxis->pixelToCoord(keyAxis->coordToPixel(oldKey)+10*keyAxis->pixelOrientation());
```
newKey will then contain a key that is ten pixels towards higher keys, starting from *oldKey*.

17.605.4.18 pixelToCoord()

```
double QCPAxis::pixelToCoord (
            double value ) const
```
Transforms *value*, in pixel coordinates of the QCustomPlot widget, to axis coordinates.

17.605.4.19 plottables()

```
QList< QCPAbstractPlottable * > QCPAxis::plottables ( ) const
```
Returns a list of all the plottables that have this axis as key or value axis.
If you are only interested in plottables of type QCPGraph, see graphs.

See also

graphs, items

17.605.4.20 rangeChanged [1/2]

```
void QCPAxis::rangeChanged (
            const QCPRange & newRange )  [signal]
```
This signal is emitted when the range of this axis has changed. You can connect it to the setRange slot of another axis to communicate the new range to the other axis, in order for it to be synchronized.
You may also manipulate/correct the range with setRange in a slot connected to this signal. This is useful if for example a maximum range span shall not be exceeded, or if the lower/upper range shouldn't go beyond certain values (see QCPRange::bounded). For example, the following slot would limit the x axis to ranges between 0 and 10:
```
customPlot->xAxis->setRange(newRange.bounded(0, 10))
```

17.605.4.21 rangeChanged [2/2]

```
void QCPAxis::rangeChanged (
            const QCPRange & newRange,
            const QCPRange & oldRange )  [signal]
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Additionally to the new range, this signal also provides the previous range held by the axis as *oldRange*.

17.605.4.22 rescale()

```
void QCPAxis::rescale (
            bool onlyVisiblePlottables = false )
```

Changes the axis range such that all plottables associated with this axis are fully visible in that dimension.

See also

QCPAbstractPlottable::rescaleAxes, QCustomPlot::rescaleAxes

17.605.4.23 scaleRange() [1/2]

```
void QCPAxis::scaleRange (
            double factor )
```

Scales the range of this axis by *factor* around the center of the current axis range. For example, if *factor* is 2.0, then the axis range will double its size, and the point at the axis range center won't have changed its position in the QCustomPlot widget (i.e. coordinates around the center will have moved symmetrically closer).
If you wish to scale around a different coordinate than the current axis range center, use the overload scaleRange(double factor, double center).

17.605.4.24 scaleRange() [2/2]

```
void QCPAxis::scaleRange (
            double factor,
            double center )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Scales the range of this axis by *factor* around the coordinate *center*. For example, if *factor* is 2.0, *center* is 1.0, then the axis range will double its size, and the point at coordinate 1.0 won't have changed its position in the QCustomPlot widget (i.e. coordinates around 1.0 will have moved symmetrically closer to 1.0).

See also

scaleRange(double factor)

17.605.4.25 scaleTypeChanged

```
void QCPAxis::scaleTypeChanged (
            QCPAxis::ScaleType scaleType )  [signal]
```

This signal is emitted when the scale type changes, by calls to setScaleType

17.605.4.26 selectableChanged

```
void QCPAxis::selectableChanged (
            const QCPAxis::SelectableParts & parts )  [signal]
```

This signal is emitted when the selectability changes, by calls to setSelectableParts

17.605.4.27 selectEvent()

```
void QCPAxis::selectEvent (
            QMouseEvent * event,
            bool additive,
            const QVariant & details,
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.605.4.28 selectionCategory()

```
QCP::Interaction QCPAxis::selectionCategory ( ) const  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.605.4.29 selectionChanged

```
void QCPAxis::selectionChanged (
            const QCPAxis::SelectableParts & parts )  [signal]
```

This signal is emitted when the selection state of this axis has changed, either by user interaction or by a direct call to setSelectedParts.

17.605.4.30 selectTest()

```
double QCPAxis::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Reimplemented from QCPLayerable.

17.605.4.31 setBasePen()

```
void QCPAxis::setBasePen (
            const QPen & pen )
```

Sets the pen, the axis base line is drawn with.

See also

setTickPen, setSubTickPen

17.605.4.32 setLabel()

```
void QCPAxis::setLabel (
            const QString & str )
```

Sets the text of the axis label that will be shown below/above or next to the axis, depending on its orientation. To disable axis labels, pass an empty string as *str*.

17.605.4.33 setLabelColor()

```
void QCPAxis::setLabelColor (
            const QColor & color )
```

Sets the color of the axis label.

See also

setLabelFont

17.605.4.34 setLabelFont()

```
void QCPAxis::setLabelFont (
            const QFont & font )
```

Sets the font of the axis label.

See also

setLabelColor

17.605.4.35 setLabelPadding()

```
void QCPAxis::setLabelPadding (
            int padding )
```

Sets the distance between the tick labels and the axis label.

See also

setTickLabelPadding, setPadding

17.605.4.36 setLowerEnding()

```
void QCPAxis::setLowerEnding (
            const QCPLineEnding & ending )
```

Sets the style for the lower axis ending. See the documentation of QCPLineEnding for available styles.
For horizontal axes, this method refers to the left ending, for vertical axes the bottom ending. Note that this meaning does not change when the axis range is reversed with setRangeReversed.

See also

setUpperEnding

17.605.4.37 setNumberFormat()

```
void QCPAxis::setNumberFormat (
            const QString & formatCode )
```

Sets the number format for the numbers in tick labels. This *formatCode* is an extended version of the format code used e.g. by QString::number() and QLocale::toString(). For reference about that, see the "Argument Formats" section in the detailed description of the QString class.

formatCode is a string of one, two or three characters.
The first character is identical to the normal format code used by Qt. In short, this means: 'e'/'E' scientific format, 'f' fixed format, 'g'/'G' scientific or fixed, whichever is shorter. For the 'e', 'E', and 'f' formats, the precision set by setNumberPrecision represents the number of digits after the decimal point. For the 'g' and 'G' formats, the precision represents the maximum number of significant digits, trailing zeroes are omitted.
The second and third characters are optional and specific to QCustomPlot:
If the first char was 'e' or 'g', numbers are/might be displayed in the scientific format, e.g. "5.5e9", which is ugly in a plot. So when the second char of *formatCode* is set to 'b' (for "beautiful"), those exponential numbers are formatted in a more natural way, i.e. "5.5 [multiplication sign] 10 [superscript] 9". By default, the multiplication sign is a centered dot. If instead a cross should be shown (as is usual in the USA), the third char of *formatCode* can be set to 'c'. The inserted multiplication signs are the UTF-8 characters 215 (0xD7) for the cross and 183 (0xB7) for the dot.
Examples for *formatCode:*

- `g` normal format code behaviour. If number is small, fixed format is used, if number is large, normal scientific format is used
- `gb` If number is small, fixed format is used, if number is large, scientific format is used with beautifully typeset decimal powers and a dot as multiplication sign
- `ebc` All numbers are in scientific format with beautifully typeset decimal power and a cross as multiplication sign
- `fb` illegal format code, since fixed format doesn't support (or need) beautifully typeset decimal powers. Format code will be reduced to 'f'.
- `hello` illegal format code, since first char is not 'e', 'E', 'f', 'g' or 'G'. Current format code will not be changed.

17.605.4.38 setNumberPrecision()

```
void QCPAxis::setNumberPrecision (
            int precision )
```

Sets the precision of the tick label numbers. See QLocale::toString(double i, char f, int prec) for details. The effect of precisions are most notably for number Formats starting with 'e', see setNumberFormat

17.605.4.39 setOffset()

```
void QCPAxis::setOffset (
            int offset )
```

Sets the offset the axis has to its axis rect side.
If an axis rect side has multiple axes and automatic margin calculation is enabled for that side, only the offset of the inner most axis has meaning (even if it is set to be invisible). The offset of the other, outer axes is controlled automatically, to place them at appropriate positions.

17.605.4.40 setPadding()

```
void QCPAxis::setPadding (
            int padding )
```

Sets the padding of the axis.
When QCPAxisRect::setAutoMargins is enabled, the padding is the additional outer most space, that is left blank.
The axis padding has no meaning if QCPAxisRect::setAutoMargins is disabled.

See also

setLabelPadding, setTickLabelPadding

17.605.4.41 setRange() [1/3]

```
void QCPAxis::setRange (
            const QCPRange & range )
```

Sets the range of the axis.
This slot may be connected with the rangeChanged signal of another axis so this axis is always synchronized with the other axis range, when it changes.
To invert the direction of an axis, use setRangeReversed.

17.605.4.42 setRange() [2/3]

```
void QCPAxis::setRange (
            double lower,
            double upper )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the lower and upper bound of the axis range.
To invert the direction of an axis, use setRangeReversed.
There is also a slot to set a range, see setRange(const QCPRange &range).

17.605.4.43 setRange() [3/3]

```
void QCPAxis::setRange (
            double position,
            double size,
            Qt::AlignmentFlag alignment )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the range of the axis.
The *position* coordinate indicates together with the *alignment* parameter, where the new range will be positioned. *size* defines the size of the new axis range. *alignment* may be Qt::AlignLeft, Qt::AlignRight or Qt::AlignCenter. This will cause the left border, right border, or center of the range to be aligned with *position*. Any other values of *alignment* will default to Qt::AlignCenter.

17.605.4.44 setRangeLower()

```
void QCPAxis::setRangeLower (
            double lower )
```

Sets the lower bound of the axis range. The upper bound is not changed.

See also

setRange

17.605.4.45 setRangeReversed()

```
void QCPAxis::setRangeReversed (
            bool reversed )
```

Sets whether the axis range (direction) is displayed reversed. Normally, the values on horizontal axes increase left to right, on vertical axes bottom to top. When *reversed* is set to true, the direction of increasing values is inverted.
Note that the range and data interface stays the same for reversed axes, e.g. the *lower* part of the setRange interface will still reference the mathematically smaller number than the *upper* part.

17.605.4.46 setRangeUpper()

```
void QCPAxis::setRangeUpper (
            double upper )
```

Sets the upper bound of the axis range. The lower bound is not changed.

See also

setRange

17.605.4.47 setScaleRatio()

```
void QCPAxis::setScaleRatio (
            const QCPAxis * otherAxis,
            double ratio = 1.0 )
```

Scales the range of this axis to have a certain scale *ratio* to *otherAxis*. The scaling will be done around the center of the current axis range.
For example, if *ratio* is 1, this axis is the *yAxis* and *otherAxis* is *xAxis*, graphs plotted with those axes will appear in a 1:1 aspect ratio, independent of the aspect ratio the axis rect has.
This is an operation that changes the range of this axis once, it doesn't fix the scale ratio indefinitely. Note that calling this function in the constructor of the QCustomPlot's parent won't have the desired effect, since the widget dimensions aren't defined yet, and a resizeEvent will follow.

17.605.4.48 setScaleType()

```
void QCPAxis::setScaleType (
            QCPAxis::ScaleType type )
```

Sets whether the axis uses a linear scale or a logarithmic scale.
Note that this method controls the coordinate transformation. For logarithmic scales, you will likely also want to use a logarithmic tick spacing and labeling, which can be achieved by setting the axis ticker to an instance of QCPAxisTickerLog :
See the documentation of QCPAxisTickerLog about the details of logarithmic axis tick creation.
setNumberPrecision

17.605.4.49 setSelectableParts()

```
void QCPAxis::setSelectableParts (
            const QCPAxis::SelectableParts & selectableParts )
```

Sets whether the user can (de-)select the parts in *selectable* by clicking on the QCustomPlot surface. (When QCustomPlot::setInteractions contains iSelectAxes.)
However, even when *selectable* is set to a value not allowing the selection of a specific part, it is still possible to set the selection of this part manually, by calling setSelectedParts directly.

See also

SelectablePart, setSelectedParts

17.605.4.50 setSelectedBasePen()

```
void QCPAxis::setSelectedBasePen (
            const QPen & pen )
```

Sets the pen that is used to draw the axis base line when selected.

See also

setBasePen, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.605.4.51 setSelectedLabelColor()

```
void QCPAxis::setSelectedLabelColor (
            const QColor & color )
```

Sets the color that is used for the axis label when it is selected.

See also

setLabelColor, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.605.4.52 setSelectedLabelFont()

```
void QCPAxis::setSelectedLabelFont (
            const QFont & font )
```

Sets the font that is used for the axis label when it is selected.

See also

setLabelFont, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.605.4.53 setSelectedParts()

```
void QCPAxis::setSelectedParts (
            const QCPAxis::SelectableParts & selectedParts )
```

Sets the selected state of the respective axis parts described by SelectablePart. When a part is selected, it uses a different pen/font.
The entire selection mechanism for axes is handled automatically when QCustomPlot::setInteractions contains iSelectAxes. You only need to call this function when you wish to change the selection state manually.
This function can change the selection state of a part, independent of the setSelectableParts setting.
emits the selectionChanged signal when *selected* is different from the previous selection state.

See also

SelectablePart, setSelectableParts, selectTest, setSelectedBasePen, setSelectedTickPen, setSelectedSubTickPen, setSelectedTickLabelFont, setSelectedLabelFont, setSelectedTickLabelColor, setSelectedLabelColor

17.605.4.54 setSelectedSubTickPen()

```
void QCPAxis::setSelectedSubTickPen (
            const QPen & pen )
```

Sets the pen that is used to draw the subticks when selected.

See also

setSubTickPen, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.605.4.55 setSelectedTickLabelColor()

```
void QCPAxis::setSelectedTickLabelColor (
            const QColor & color )
```

Sets the color that is used for tick labels when they are selected.

See also

setTickLabelColor, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.605.4.56 setSelectedTickLabelFont()

```
void QCPAxis::setSelectedTickLabelFont (
            const QFont & font )
```

Sets the font that is used for tick labels when they are selected.

See also

setTickLabelFont, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.605.4.57 setSelectedTickPen()

```
void QCPAxis::setSelectedTickPen (
            const QPen & pen )
```

Sets the pen that is used to draw the (major) ticks when selected.

See also

setTickPen, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.605.4.58 setSubTickLength()

```
void QCPAxis::setSubTickLength (
            int inside,
            int outside = 0 )
```

Sets the length of the subticks in pixels. *inside* is the length the subticks will reach inside the plot and *outside* is the length they will reach outside the plot. If *outside* is greater than zero, the tick labels and axis label will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setTickLength, setSubTickLengthIn, setSubTickLengthOut

17.605.4.59 setSubTickLengthIn()

```
void QCPAxis::setSubTickLengthIn (
            int inside )
```

Sets the length of the inward subticks in pixels. *inside* is the length the subticks will reach inside the plot.

See also

setSubTickLengthOut, setSubTickLength, setTickLength

17.605.4.60 setSubTickLengthOut()

```
void QCPAxis::setSubTickLengthOut (
            int outside )
```

Sets the length of the outward subticks in pixels. *outside* is the length the subticks will reach outside the plot. If *outside* is greater than zero, the tick labels will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setSubTickLengthIn, setSubTickLength, setTickLength

17.605.4.61 setSubTickPen()

```
void QCPAxis::setSubTickPen (
            const QPen & pen )
```

Sets the pen, subtick marks will be drawn with.

See also

setSubTickCount, setSubTickLength, setBasePen

17.605.4.62 setSubTicks()

```
void QCPAxis::setSubTicks (
            bool show )
```

Sets whether sub tick marks are displayed.
Sub ticks are only potentially visible if (major) ticks are also visible (see setTicks)

See also

setTicks

17.605.4.63 setTicker()

```
void QCPAxis::setTicker (
            QSharedPointer< QCPAxisTicker > ticker )
```

The axis ticker is responsible for generating the tick positions and tick labels. See the documentation of QCPAxisTicker for details on how to work with axis tickers.
You can change the tick positioning/labeling behaviour of this axis by setting a different QCPAxisTicker subclass using this method. If you only wish to modify the currently installed axis ticker, access it via ticker.
Since the ticker is stored in the axis as a shared pointer, multiple axes may share the same axis ticker simply by passing the same shared pointer to multiple axes.

See also

ticker

17.605.4.64 setTickLabelColor()

```
void QCPAxis::setTickLabelColor (
            const QColor & color )
```

Sets the color of the tick labels.

See also

setTickLabels, setTickLabelFont

17.605.4.65 setTickLabelFont()

```
void QCPAxis::setTickLabelFont (
            const QFont & font )
```

Sets the font of the tick labels.

See also

setTickLabels, setTickLabelColor

17.605.4.66 setTickLabelPadding()

```
void QCPAxis::setTickLabelPadding (
            int padding )
```

Sets the distance between the axis base line (including any outward ticks) and the tick labels.

See also

setLabelPadding, setPadding

17.605.4.67 setTickLabelRotation()

```
void QCPAxis::setTickLabelRotation (
            double degrees )
```

Sets the rotation of the tick labels. If *degrees* is zero, the labels are drawn normally. Else, the tick labels are drawn rotated by *degrees* clockwise. The specified angle is bound to values from -90 to 90 degrees.
If *degrees* is exactly -90, 0 or 90, the tick labels are centered on the tick coordinate. For other angles, the label is drawn with an offset such that it seems to point toward or away from the tick mark.

17.605.4.68 setTickLabels()

```
void QCPAxis::setTickLabels (
            bool show )
```

Sets whether tick labels are displayed. Tick labels are the numbers drawn next to tick marks.

17.605.4.69 setTickLabelSide()

```
void QCPAxis::setTickLabelSide (
            LabelSide side )
```

Sets whether the tick labels (numbers) shall appear inside or outside the axis rect.
The usual and default setting is lsOutside. Very compact plots sometimes require tick labels to be inside the axis rect, to save space. If *side* is set to lsInside, the tick labels appear on the inside are additionally clipped to the axis rect.

17.605.4.70 setTickLength()

```
void QCPAxis::setTickLength (
            int inside,
            int outside = 0 )
```

Sets the length of the ticks in pixels. *inside* is the length the ticks will reach inside the plot and *outside* is the length they will reach outside the plot. If *outside* is greater than zero, the tick labels and axis label will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setSubTickLength, setTickLengthIn, setTickLengthOut

17.605.4.71 setTickLengthIn()

```
void QCPAxis::setTickLengthIn (
            int inside )
```

Sets the length of the inward ticks in pixels. *inside* is the length the ticks will reach inside the plot.

See also

setTickLengthOut, setTickLength, setSubTickLength

17.605.4.72 setTickLengthOut()

```
void QCPAxis::setTickLengthOut (
            int outside )
```

Sets the length of the outward ticks in pixels. *outside* is the length the ticks will reach outside the plot. If *outside* is greater than zero, the tick labels and axis label will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setTickLengthIn, setTickLength, setSubTickLength

17.605.4.73 setTickPen()

```
void QCPAxis::setTickPen (
            const QPen & pen )
```

Sets the pen, tick marks will be drawn with.

See also

setTickLength, setBasePen

17.605.4.74 setTicks()

```
void QCPAxis::setTicks (
            bool show )
```

Sets whether tick marks are displayed.
Note that setting *show* to false does not imply that tick labels are invisible, too. To achieve that, see setTickLabels.

See also

setSubTicks

17.605.4.75 setUpperEnding()

```
void QCPAxis::setUpperEnding (
            const QCPLineEnding & ending )
```

Sets the style for the upper axis ending. See the documentation of QCPLineEnding for available styles.
For horizontal axes, this method refers to the right ending, for vertical axes the top ending. Note that this meaning does not change when the axis range is reversed with setRangeReversed.

See also

setLowerEnding

17.605.4.76 ticker()

```
QSharedPointer< QCPAxisTicker > QCPAxis::ticker ( ) const  [inline]
```

Returns a modifiable shared pointer to the currently installed axis ticker. The axis ticker is responsible for generating the tick positions and tick labels of this axis. You can access the QCPAxisTicker with this method and modify basic properties such as the approximate tick count (QCPAxisTicker::setTickCount).
You can gain more control over the axis ticks by setting a different QCPAxisTicker subclass, see the documentation there. A new axis ticker can be set with setTicker.
Since the ticker is stored in the axis as a shared pointer, multiple axes may share the same axis ticker simply by passing the same shared pointer to multiple axes.

See also

setTicker

17.605.4.77 wheelEvent()

```
void QCPAxis::wheelEvent (
            QWheelEvent * event )  [protected], [virtual]
```

This event gets called when the user turns the mouse scroll wheel while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`.
The `event->angleDelta()` indicates how far the mouse wheel was turned, which is usually +/- 120 for single rotation steps. However, if the mouse wheel is turned rapidly, multiple steps may accumulate to one event, making the delta larger. On the other hand, if the wheel has very smooth steps or none at all, the delta may be smaller.
The default implementation does nothing.

See also

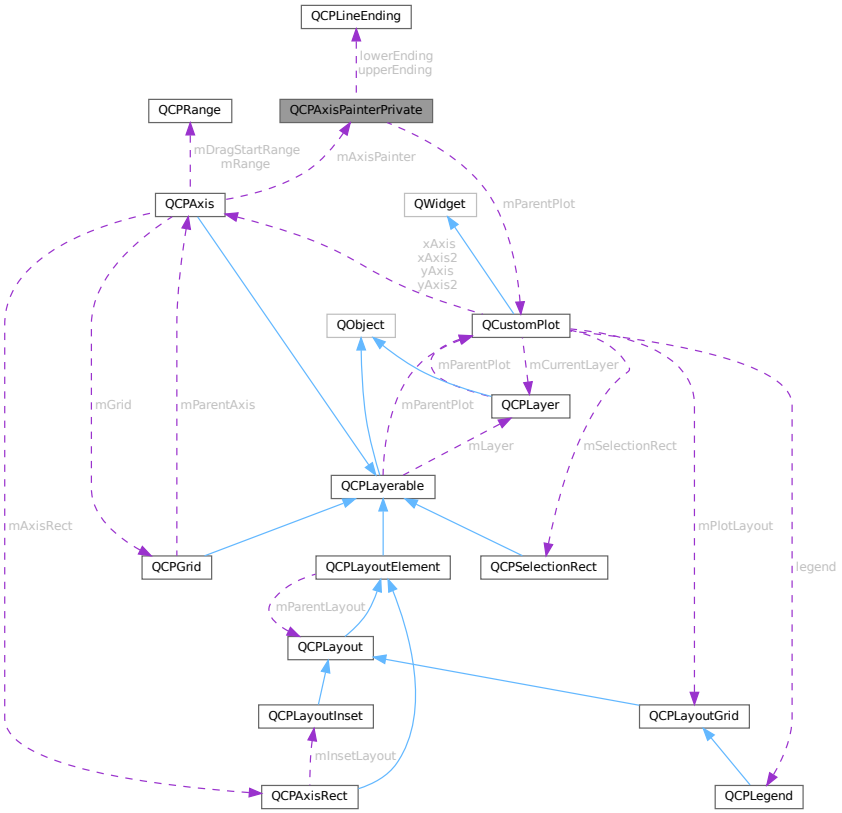
mousePressEvent, mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent

Reimplemented from QCPLayerable.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.606 QCPAxisPainterPrivate Class Reference

Collaboration diagram for QCPAxisPainterPrivate:



Classes

- struct CachedLabel
- struct TickLabelData

Public Member Functions

- QCPAxisPainterPrivate (QCustomPlot ∗parentPlot)
- virtual void **draw** (QCPPainter ∗painter)
- virtual int **size** ()
- void **clearCache** ()
- QRect **axisSelectionBox** () const
- QRect **tickLabelsSelectionBox** () const
- QRect **labelSelectionBox** () const

Public Attributes

- QCPAxis::AxisType **type**
- QPen **basePen**
- QCPLineEnding **lowerEnding**
- QCPLineEnding **upperEnding**
- int **labelPadding**
- QFont **labelFont**
- QColor **labelColor**

- QString **label**
- int **tickLabelPadding**
- double **tickLabelRotation**
- QCPAxis::LabelSide **tickLabelSide**
- bool **substituteExponent**
- bool **numberMultiplyCross**
- int **tickLengthIn**
- int **tickLengthOut**
- int **subTickLengthIn**
- int **subTickLengthOut**
- QPen **tickPen**
- QPen **subTickPen**
- QFont **tickLabelFont**
- QColor **tickLabelColor**
- QRect **axisRect**
- QRect **viewportRect**
- int **offset**
- bool **abbreviateDecimalPowers**
- bool **reversedEndings**
- QVector< double > **subTickPositions**
- QVector< double > **tickPositions**
- QVector< QString > **tickLabels**

Protected Member Functions

- virtual QByteArray **generateLabelParameterHash** () const
- virtual void **placeTickLabel** (QCPPainter *painter, double position, int distanceToAxis, const QString &text, QSize *tickLabelsSize)
- virtual void **drawTickLabel** (QCPPainter *painter, double x, double y, const TickLabelData &labelData) const
- virtual TickLabelData **getTickLabelData** (const QFont &font, const QString &text) const
- virtual QPointF **getTickLabelDrawOffset** (const TickLabelData &labelData) const
- virtual void **getMaxTickLabelSize** (const QFont &font, const QString &text, QSize *tickLabelsSize) const

Protected Attributes

- QCustomPlot * **mParentPlot**
- QByteArray **mLabelParameterHash**
- QCache< QString, CachedLabel > **mLabelCache**
- QRect **mAxisSelectionBox**
- QRect **mTickLabelsSelectionBox**
- QRect **mLabelSelectionBox**

17.606.1 Constructor & Destructor Documentation

17.606.1.1 QCPAxisPainterPrivate()

```
QCPAxisPainterPrivate::QCPAxisPainterPrivate (
            QCustomPlot * parentPlot )  [explicit]
```

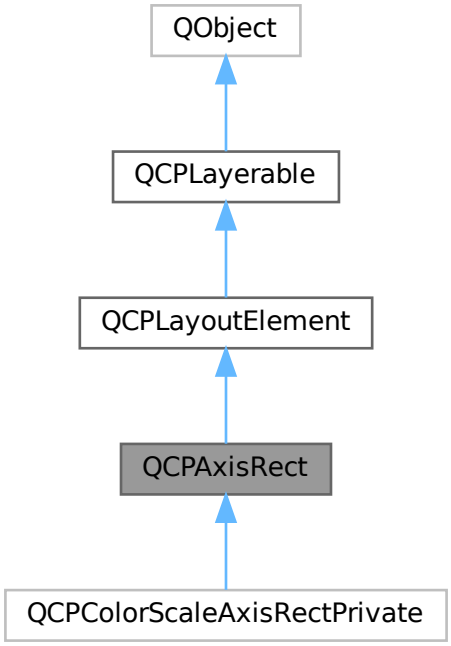
Constructs a QCPAxisPainterPrivate instance. Make sure to not create a new instance on every redraw, to utilize the caching mechanisms.

The documentation for this class was generated from the following files:

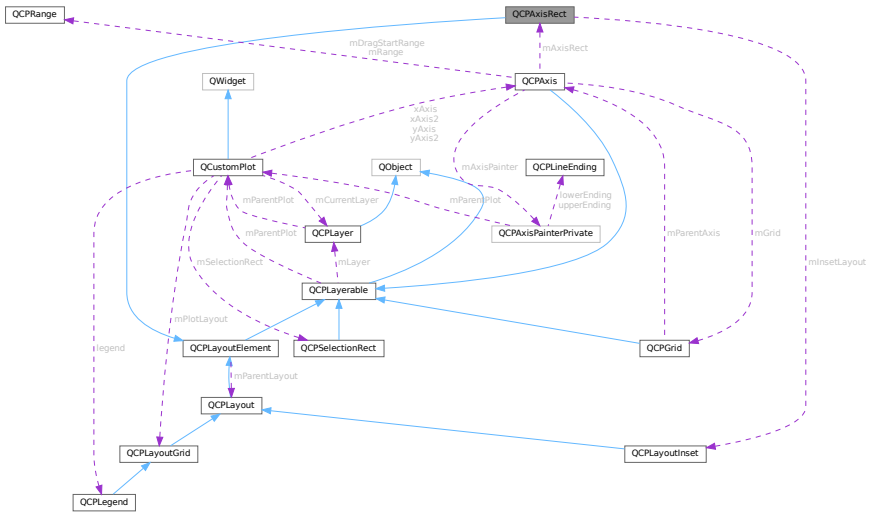
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.607 QCPAxisRect Class Reference

Holds multiple axes and arranges them in a rectangular shape.
Inheritance diagram for QCPAxisRect:



Collaboration diagram for QCPAxisRect:



Public Member Functions

- QCPAxisRect (QCustomPlot *parentPlot, bool setupDefaultAxes=true)

- QPixmap **background** () const
- QBrush **backgroundBrush** () const
- bool **backgroundScaled** () const
- Qt::AspectRatioMode **backgroundScaledMode** () const
- Qt::Orientations **rangeDrag** () const
- Qt::Orientations **rangeZoom** () const
- QCPAxis * rangeDragAxis (Qt::Orientation orientation)
- QCPAxis * rangeZoomAxis (Qt::Orientation orientation)
- QList< QCPAxis * > rangeDragAxes (Qt::Orientation orientation)
- QList< QCPAxis * > rangeZoomAxes (Qt::Orientation orientation)
- double rangeZoomFactor (Qt::Orientation orientation)
- void setBackground (const QPixmap &pm)
- void setBackground (const QPixmap &pm, bool scaled, Qt::AspectRatioMode mode=Qt::KeepAspectRatio↩ ByExpanding)
- void setBackground (const QBrush &brush)
- void setBackgroundScaled (bool scaled)
- void setBackgroundScaledMode (Qt::AspectRatioMode mode)
- void setRangeDrag (Qt::Orientations orientations)
- void setRangeZoom (Qt::Orientations orientations)
- void setRangeDragAxes (QCPAxis *horizontal, QCPAxis *vertical)
- void setRangeDragAxes (QList< QCPAxis * > axes)
- void setRangeDragAxes (QList< QCPAxis * > horizontal, QList< QCPAxis * > vertical)
- void setRangeZoomAxes (QCPAxis *horizontal, QCPAxis *vertical)
- void setRangeZoomAxes (QList< QCPAxis * > axes)
- void setRangeZoomAxes (QList< QCPAxis * > horizontal, QList< QCPAxis * > vertical)
- void setRangeZoomFactor (double horizontalFactor, double verticalFactor)
- void setRangeZoomFactor (double factor)
- int axisCount (QCPAxis::AxisType type) const
- QCPAxis * axis (QCPAxis::AxisType type, int index=0) const
- QList< QCPAxis * > axes (QCPAxis::AxisTypes types) const
- QList< QCPAxis * > axes () const
- QCPAxis * addAxis (QCPAxis::AxisType type, QCPAxis *axis=nullptr)
- QList< QCPAxis * > addAxes (QCPAxis::AxisTypes types)
- bool removeAxis (QCPAxis *axis)
- QCPLayoutInset * insetLayout () const
- void zoom (const QRectF &pixelRect)
- void zoom (const QRectF &pixelRect, const QList< QCPAxis * > &affectedAxes)
- void setupFullAxesBox (bool connectRanges=false)
- QList< QCPAbstractPlottable * > plottables () const
- QList< QCPGraph * > graphs () const
- QList< QCPAbstractItem * > items () const
- int left () const
- int right () const
- int top () const
- int bottom () const
- int width () const
- int height () const
- QSize size () const
- QPoint topLeft () const
- QPoint topRight () const
- QPoint bottomLeft () const
- QPoint bottomRight () const
- QPoint center () const
- virtual void update (UpdatePhase phase) Q_DECL_OVERRIDE
- virtual QList< QCPLayoutElement * > elements (bool recursive) const Q_DECL_OVERRIDE

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot *parentPlot=nullptr)
- QCPLayout * layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const
- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)
- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual QSize minimumOuterSizeHint () const
- virtual QSize maximumOuterSizeHint () const
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual int calculateAutoMargin (QCP::MarginSide side) Q_DECL_OVERRIDE
- virtual void layoutChanged () Q_DECL_OVERRIDE
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void wheelEvent (QWheelEvent *event) Q_DECL_OVERRIDE
- void **drawBackground** (QCPPainter *painter)
- void **updateAxesOffset** (QCPAxis::AxisType type)

Protected Member Functions inherited from QCPLayoutElement

- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void **selectEvent** (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged)
- virtual void **deselectEvent** (bool *selectionStateChanged)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QBrush **mBackgroundBrush**
- QPixmap **mBackgroundPixmap**
- QPixmap **mScaledBackgroundPixmap**
- bool **mBackgroundScaled**
- Qt::AspectRatioMode **mBackgroundScaledMode**
- QCPLayoutInset * **mInsetLayout**
- Qt::Orientations **mRangeDrag**
- Qt::Orientations **mRangeZoom**
- QList< QPointer< QCPAxis > > **mRangeDragHorzAxis**
- QList< QPointer< QCPAxis > > **mRangeDragVertAxis**
- QList< QPointer< QCPAxis > > **mRangeZoomHorzAxis**
- QList< QPointer< QCPAxis > > **mRangeZoomVertAxis**
- double **mRangeZoomFactorHorz**
- double **mRangeZoomFactorVert**
- QList< QCPRange > **mDragStartHorzRange**
- QList< QCPRange > **mDragStartVertRange**
- QCP::AntialiasedElements **mAADragBackup**
- QCP::AntialiasedElements **mNotAADragBackup**
- bool **mDragging**
- QHash< QCPAxis::AxisType, QList< QCPAxis * > > **mAxes**

Protected Attributes inherited from QCPLayoutElement

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**

Additional Inherited Members

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.607.1 Detailed Description

Holds multiple axes and arranges them in a rectangular shape.
This class represents an axis rect, a rectangular area that is bounded on all sides with an arbitrary number of axes. Initially QCustomPlot has one axis rect, accessible via QCustomPlot::axisRect(). However, the layout system allows to have multiple axis rects, e.g. arranged in a grid layout (QCustomPlot::plotLayout).
By default, QCPAxisRect comes with four axes, at bottom, top, left and right. They can be accessed via axis by providing the respective axis type (QCPAxis::AxisType) and index. If you need all axes in the axis rect, use axes. The top and right axes are set to be invisible initially (QCPAxis::setVisible). To add more axes to a side, use addAxis or addAxes. To remove an axis, use removeAxis.
The axis rect layerable itself only draws a background pixmap or color, if specified (setBackground). It is placed on the "background" layer initially (see QCPLayer for an explanation of the QCustomPlot layer system). The axes that are held by the axis rect can be placed on other layers, independently of the axis rect.
Every axis rect has a child layout of type QCPLayoutInset. It is accessible via insetLayout and can be used to have other layout elements (or even other layouts with multiple elements) hovering inside the axis rect.
If an axis rect is clicked and dragged, it processes this by moving certain axis ranges. The behaviour can be controlled with setRangeDrag and setRangeDragAxes. If the mouse wheel is scrolled while the cursor is on the axis rect, certain axes are scaled. This is controllable via setRangeZoom, setRangeZoomAxes and setRangeZoomFactor. These interactions are only enabled if QCustomPlot::setInteractions contains QCP::iRangeDrag and QCP::iRangeZoom.

Overview of the spacings and paddings that define the geometry of an axis. The dashed line on the far left indicates the viewport/widget border.

17.607.2 Constructor & Destructor Documentation

17.607.2.1 QCPAxisRect()

```
QCPAxisRect::QCPAxisRect (
            QCustomPlot * parentPlot,
            bool setupDefaultAxes = true )  [explicit]
```

Creates a QCPAxisRect instance and sets default values. An axis is added for each of the four sides, the top and right axes are set invisible initially.

17.607.3 Member Function Documentation

17.607.3.1 addAxes()

```
QList< QCPAxis * > QCPAxisRect::addAxes (
            QCPAxis::AxisTypes types )
```

Adds a new axis with addAxis to each axis rect side specified in *types*. This may be an `or`-combination of QCPAxis::AxisType, so axes can be added to multiple sides at once.
Returns a list of the added axes.

See also

addAxis, setupFullAxesBox

17.607.3.2 addAxis()

```
QCPAxis * QCPAxisRect::addAxis (
            QCPAxis::AxisType type,
            QCPAxis * axis = nullptr )
```

Adds a new axis to the axis rect side specified with *type*, and returns it. If *axis* is 0, a new QCPAxis instance is created internally. QCustomPlot owns the returned axis, so if you want to remove an axis, use removeAxis instead of deleting it manually.
You may inject QCPAxis instances (or subclasses of QCPAxis) by setting *axis* to an axis that was previously created outside QCustomPlot. It is important to note that QCustomPlot takes ownership of the axis, so you may not delete it afterwards. Further, the *axis* must have been created with this axis rect as parent and with the same axis type as specified in *type*. If this is not the case, a debug output is generated, the axis is not added, and the method returns `nullptr`.
This method can not be used to move *axis* between axis rects. The same *axis* instance must not be added multiple times to the same or different axis rects.
If an axis rect side already contains one or more axes, the lower and upper endings of the new axis (QCPAxis::setLowerEnding, QCPAxis::setUpperEnding) are set to QCPLineEnding::esHalfBar.

See also

addAxes, setupFullAxesBox

17.607.3.3 applyDefaultAntialiasingHint()

```
void QCPAxisRect::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```

Reimplemented from QCPLayoutElement.

17.607.3.4 axes() [1/2]

```
QList< QCPAxis * > QCPAxisRect::axes ( ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Returns all axes of this axis rect.

17.607.3.5 axes() [2/2]

```
QList< QCPAxis * > QCPAxisRect::axes (
            QCPAxis::AxisTypes types ) const
```

Returns all axes on the axis rect sides specified with *types*.
types may be a single QCPAxis::AxisType or an `or`-combination, to get the axes of multiple sides.

See also

axis

17.607.3.6 axis()

```
QCPAxis * QCPAxisRect::axis (
            QCPAxis::AxisType type,
            int index = 0 ) const
```

Returns the axis with the given *index* on the axis rect side specified with *type*.

See also

axisCount, axes

17.607.3.7 axisCount()

```
int QCPAxisRect::axisCount (
            QCPAxis::AxisType type ) const
```

Returns the number of axes on the axis rect side specified with *type*.

See also

axis

17.607.3.8 bottom()

```
int QCPAxisRect::bottom ( ) const  [inline]
```

Returns the pixel position of the bottom border of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.9 bottomLeft()

```
QPoint QCPAxisRect::bottomLeft ( ) const  [inline]
```

Returns the bottom left corner of this axis rect in pixels. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.10 bottomRight()

```
QPoint QCPAxisRect::bottomRight ( ) const  [inline]
```

Returns the bottom right corner of this axis rect in pixels. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.11 calculateAutoMargin()

```
int QCPAxisRect::calculateAutoMargin (
            QCP::MarginSide side )  [protected], [virtual]
```

Reimplemented from QCPLayoutElement.

17.607.3.12 center()

```
QPoint QCPAxisRect::center ( ) const  [inline]
```

Returns the center of this axis rect in pixels. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.13 draw()

```
void QCPAxisRect::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Reimplemented from QCPLayoutElement.

17.607.3.14 elements()

```
QList< QCPLayoutElement * > QCPAxisRect::elements (
            bool recursive ) const  [virtual]
```

Returns a list of all child elements in this layout element. If *recursive* is true, all sub-child elements are included in the list, too.

Warning

There may be `nullptr` entries in the returned list. For example, QCPLayoutGrid may have empty cells which yield `nullptr` at the respective index.

Reimplemented from QCPLayoutElement.

17.607.3.15 graphs()

```
QList< QCPGraph * > QCPAxisRect::graphs ( ) const
```

Returns a list of all the graphs that are associated with this axis rect.
A graph is considered associated with an axis rect if its key or value axis (or both) is in this axis rect.

See also

plottables, items

17.607.3.16 height()

```
int QCPAxisRect::height ( ) const  [inline]
```

Returns the pixel height of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.17 insetLayout()

```
QCPLayoutInset * QCPAxisRect::insetLayout ( ) const  [inline]
```

Returns the inset layout of this axis rect. It can be used to place other layout elements (or even layouts with multiple other elements) inside/on top of an axis rect.

See also

QCPLayoutInset

17.607.3.18 items()

```
QList< QCPAbstractItem * > QCPAxisRect::items ( ) const
```

Returns a list of all the items that are associated with this axis rect.
An item is considered associated with an axis rect if any of its positions has key or value axis set to an axis that is in this axis rect, or if any of its positions has QCPItemPosition::setAxisRect set to the axis rect, or if the clip axis rect (QCPAbstractItem::setClipAxisRect) is set to this axis rect.

See also

plottables, graphs

17.607.3.19 layoutChanged()

```
void QCPAxisRect::layoutChanged ( )  [protected], [virtual]
```

Reimplemented from QCPLayoutElement.

17.607.3.20 left()

```
int QCPAxisRect::left ( ) const  [inline]
```

Returns the pixel position of the left border of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.21 mouseMoveEvent()

```
void QCPAxisRect::mouseMoveEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user moves the mouse while holding a mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.
Reimplemented in QCPColorScaleAxisRectPrivate.

17.607.3.22 mousePressEvent()

```
void QCPAxisRect::mousePressEvent (
            QMouseEvent * event,
            const QVariant & details )  [protected], [virtual]
```

This event gets called when the user presses a mouse button while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *details* contains layerable-specific details about the hit, which were generated in the previous call to selectTest. For example, One-dimensional plottables like QCPGraph or QCPBars convey the clicked data point in the *details* parameter, as QCPDataSelection packed as QVariant. Multi-part objects convey the specific `SelectablePart` that was hit (e.g. QCPAxis::SelectablePart in the case of axes).
QCustomPlot uses an event propagation system that works the same as Qt's system. If your layerable doesn't reimplement the mousePressEvent or explicitly calls `event->ignore()` in its reimplementation, the event will be propagated to the next layerable in the stacking order.
Once a layerable has accepted the mousePressEvent, it is considered the mouse grabber and will receive all following calls to mouseMoveEvent or mouseReleaseEvent for this mouse interaction (a "mouse interaction" in this context ends with the release).
The default implementation does nothing except explicitly ignoring the event with `event->ignore()`.

See also

mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.
Reimplemented in QCPColorScaleAxisRectPrivate.

17.607.3.23 mouseReleaseEvent()

```
void QCPAxisRect::mouseReleaseEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user releases the mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseMoveEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.
Reimplemented in QCPColorScaleAxisRectPrivate.

17.607.3.24 plottables()

```
QList< QCPAbstractPlottable * > QCPAxisRect::plottables ( ) const
```

Returns a list of all the plottables that are associated with this axis rect.
A plottable is considered associated with an axis rect if its key or value axis (or both) is in this axis rect.

See also

graphs, items

17.607.3.25 rangeDragAxes()

```
QList< QCPAxis * > QCPAxisRect::rangeDragAxes (
            Qt::Orientation orientation )
```

Returns all range drag axes of the *orientation* provided.

See also

rangeZoomAxis, setRangeZoomAxes

17.607.3.26 rangeDragAxis()

```
QCPAxis * QCPAxisRect::rangeDragAxis (
            Qt::Orientation orientation )
```

Returns the range drag axis of the *orientation* provided. If multiple axes were set, returns the first one (use rangeDragAxes to retrieve a list with all set axes).

See also

setRangeDragAxes

17.607.3.27 rangeZoomAxes()

```
QList< QCPAxis * > QCPAxisRect::rangeZoomAxes (
            Qt::Orientation orientation )
```

Returns all range zoom axes of the *orientation* provided.

See also

rangeDragAxis, setRangeDragAxes

17.607.3.28 rangeZoomAxis()

```
QCPAxis * QCPAxisRect::rangeZoomAxis (
            Qt::Orientation orientation )
```

Returns the range zoom axis of the *orientation* provided. If multiple axes were set, returns the first one (use rangeZoomAxes to retrieve a list with all set axes).

See also

setRangeZoomAxes

17.607.3.29 rangeZoomFactor()

```
double QCPAxisRect::rangeZoomFactor (
            Qt::Orientation orientation )
```

Returns the range zoom factor of the *orientation* provided.

See also

setRangeZoomFactor

17.607.3.30 removeAxis()

```
bool QCPAxisRect::removeAxis (
            QCPAxis * axis )
```

Removes the specified *axis* from the axis rect and deletes it.
Returns true on success, i.e. if *axis* was a valid axis in this axis rect.

See also

addAxis

17.607.3.31 right()

```
int QCPAxisRect::right ( ) const  [inline]
```

Returns the pixel position of the right border of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.32 setBackground() [1/3]

```
void QCPAxisRect::setBackground (
            const QBrush & brush )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets *brush* as the background brush. The axis rect background will be filled with this brush. Since axis rects place themselves on the "background" layer by default, the axis rect backgrounds are usually drawn below everything else.
The brush will be drawn before (under) any background pixmap, which may be specified with setBackground(const QPixmap &pm).
To disable drawing of a background brush, set *brush* to Qt::NoBrush.

See also

setBackground(const QPixmap &pm)

17.607.3.33 setBackground() [2/3]

```
void QCPAxisRect::setBackground (
            const QPixmap & pm )
```

Sets *pm* as the axis background pixmap. The axis background pixmap will be drawn inside the axis rect. Since axis rects place themselves on the "background" layer by default, the axis rect backgrounds are usually drawn below everything else.
For cases where the provided pixmap doesn't have the same size as the axis rect, scaling can be enabled with setBackgroundScaled and the scaling mode (i.e. whether and how the aspect ratio is preserved) can be set with setBackgroundScaledMode. To set all these options in one call, consider using the overloaded version of this function.
Below the pixmap, the axis rect may be optionally filled with a brush, if specified with setBackground(const QBrush &brush).

See also

setBackgroundScaled, setBackgroundScaledMode, setBackground(const QBrush &brush)

17.607.3.34 setBackground() [3/3]

```
void QCPAxisRect::setBackground (
            const QPixmap & pm,
            bool scaled,
            Qt::AspectRatioMode mode = Qt::KeepAspectRatioByExpanding )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Allows setting the background pixmap of the axis rect, whether it shall be scaled and how it shall be scaled in one call.

See also

setBackground(const QPixmap &pm), setBackgroundScaled, setBackgroundScaledMode

17.607.3.35 setBackgroundScaled()

```
void QCPAxisRect::setBackgroundScaled (
            bool scaled )
```

Sets whether the axis background pixmap shall be scaled to fit the axis rect or not. If *scaled* is set to true, you may control whether and how the aspect ratio of the original pixmap is preserved with setBackgroundScaledMode.
Note that the scaled version of the original pixmap is buffered, so there is no performance penalty on replots. (Except when the axis rect dimensions are changed continuously.)

See also

setBackground, setBackgroundScaledMode

17.607.3.36 setBackgroundScaledMode()

```
void QCPAxisRect::setBackgroundScaledMode (
            Qt::AspectRatioMode mode )
```

If scaling of the axis background pixmap is enabled (setBackgroundScaled), use this function to define whether and how the aspect ratio of the original pixmap passed to setBackground is preserved.

See also

setBackground, setBackgroundScaled

17.607.3.37 setRangeDrag()

```
void QCPAxisRect::setRangeDrag (
            Qt::Orientations orientations )
```

Sets which axis orientation may be range dragged by the user with mouse interaction. What orientation corresponds to which specific axis can be set with setRangeDragAxes(QCPAxis *horizontal, QCPAxis *vertical). By default, the horizontal axis is the bottom axis (xAxis) and the vertical axis is the left axis (yAxis).
To disable range dragging entirely, pass `nullptr` as *orientations* or remove QCP::iRangeDrag from QCustomPlot::setInteractions. To enable range dragging for both directions, pass `Qt::Horizontal | Qt::Vertical` as *orientations*.
In addition to setting *orientations* to a non-zero value, make sure QCustomPlot::setInteractions contains QCP::iRangeDrag to enable the range dragging interaction.

See also

setRangeZoom, setRangeDragAxes, QCustomPlot::setNoAntialiasingOnDrag

17.607.3.38 setRangeDragAxes() [1/3]

```
void QCPAxisRect::setRangeDragAxes (
            QCPAxis * horizontal,
            QCPAxis * vertical )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the axes whose range will be dragged when setRangeDrag enables mouse range dragging on the QCustomPlot widget. Pass `nullptr` if no axis shall be dragged in the respective orientation.
Use the overload taking a list of axes, if multiple axes (more than one per orientation) shall react to dragging interactions.

See also

setRangeZoomAxes

17.607.3.39 setRangeDragAxes() [2/3]

```
void QCPAxisRect::setRangeDragAxes (
            QList< QCPAxis * > axes )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
This method allows to set up multiple axes to react to horizontal and vertical dragging. The drag orientation that the respective axis will react to is deduced from its orientation (QCPAxis::orientation).
In the unusual case that you wish to e.g. drag a vertically oriented axis with a horizontal drag motion, use the overload taking two separate lists for horizontal and vertical dragging.

17.607.3.40 setRangeDragAxes() [3/3]

```
void QCPAxisRect::setRangeDragAxes (
            QList< QCPAxis * > horizontal,
            QList< QCPAxis * > vertical )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
This method allows to set multiple axes up to react to horizontal and vertical dragging, and define specifically which axis reacts to which drag orientation (irrespective of the axis orientation).

17.607.3.41 setRangeZoom()

```
void QCPAxisRect::setRangeZoom (
            Qt::Orientations orientations )
```

Sets which axis orientation may be zoomed by the user with the mouse wheel. What orientation corresponds to which specific axis can be set with setRangeZoomAxes(QCPAxis *horizontal, QCPAxis *vertical). By default, the horizontal axis is the bottom axis (xAxis) and the vertical axis is the left axis (yAxis).
To disable range zooming entirely, pass `nullptr` as *orientations* or remove QCP::iRangeZoom from QCustomPlot::setInteractions. To enable range zooming for both directions, pass `Qt::Horizontal | Qt::Vertical` as *orientations*.
In addition to setting *orientations* to a non-zero value, make sure QCustomPlot::setInteractions contains QCP::iRangeZoom to enable the range zooming interaction.

See also

setRangeZoomFactor, setRangeZoomAxes, setRangeDrag

17.607.3.42 setRangeZoomAxes() [1/3]

```
void QCPAxisRect::setRangeZoomAxes (
            QCPAxis * horizontal,
            QCPAxis * vertical )
```

Sets the axes whose range will be zoomed when setRangeZoom enables mouse wheel zooming on the QCustomPlot widget. Pass `nullptr` if no axis shall be zoomed in the respective orientation.
The two axes can be zoomed with different strengths, when different factors are passed to setRangeZoomFactor(double horizontalFact
Use the overload taking a list of axes, if multiple axes (more than one per orientation) shall react to zooming interactions.

See also

setRangeDragAxes

17.607.3.43 setRangeZoomAxes() [2/3]

```
void QCPAxisRect::setRangeZoomAxes (
            QList< QCPAxis * > axes )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.

This method allows to set up multiple axes to react to horizontal and vertical range zooming. The zoom orientation that the respective axis will react to is deduced from its orientation (QCPAxis::orientation).
In the unusual case that you wish to e.g. zoom a vertically oriented axis with a horizontal zoom interaction, use the overload taking two separate lists for horizontal and vertical zooming.

17.607.3.44 setRangeZoomAxes() [3/3]

```
void QCPAxisRect::setRangeZoomAxes (
            QList< QCPAxis * > horizontal,
            QList< QCPAxis * > vertical )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
This method allows to set multiple axes up to react to horizontal and vertical zooming, and define specifically which axis reacts to which zoom orientation (irrespective of the axis orientation).

17.607.3.45 setRangeZoomFactor() [1/2]

```
void QCPAxisRect::setRangeZoomFactor (
            double factor )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets both the horizontal and vertical zoom *factor*.

17.607.3.46 setRangeZoomFactor() [2/2]

```
void QCPAxisRect::setRangeZoomFactor (
            double horizontalFactor,
            double verticalFactor )
```

Sets how strong one rotation step of the mouse wheel zooms, when range zoom was activated with setRangeZoom. The two parameters *horizontalFactor* and *verticalFactor* provide a way to let the horizontal axis zoom at different rates than the vertical axis. Which axis is horizontal and which is vertical, can be set with setRangeZoomAxes.
When the zoom factor is greater than one, scrolling the mouse wheel backwards (towards the user) will zoom in (make the currently visible range smaller). For zoom factors smaller than one, the same scrolling direction will zoom out.

17.607.3.47 setupFullAxesBox()

```
void QCPAxisRect::setupFullAxesBox (
            bool connectRanges = false )
```

Convenience function to create an axis on each side that doesn't have any axes yet and set their visibility to true. Further, the top/right axes are assigned the following properties of the bottom/left axes:

- range (QCPAxis::setRange)
- range reversed (QCPAxis::setRangeReversed)
- scale type (QCPAxis::setScaleType)
- tick visibility (QCPAxis::setTicks)
- number format (QCPAxis::setNumberFormat)
- number precision (QCPAxis::setNumberPrecision)
- tick count of ticker (QCPAxisTicker::setTickCount)
- tick origin of ticker (QCPAxisTicker::setTickOrigin)

Tick label visibility (QCPAxis::setTickLabels) of the right and top axes are set to false.
If *connectRanges* is true, the rangeChanged signals of the bottom and left axes are connected to the QCPAxis::setRange slots of the top and right axes.

17.607.3.48 size()

```
QSize QCPAxisRect::size ( ) const  [inline]
```
Returns the pixel size of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.49 top()

```
int QCPAxisRect::top ( ) const  [inline]
```
Returns the pixel position of the top border of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.50 topLeft()

```
QPoint QCPAxisRect::topLeft ( ) const  [inline]
```
Returns the top left corner of this axis rect in pixels. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.51 topRight()

```
QPoint QCPAxisRect::topRight ( ) const  [inline]
```
Returns the top right corner of this axis rect in pixels. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.52 update()

```
void QCPAxisRect::update (
            UpdatePhase phase )  [virtual]
```
This method is called automatically upon replot and doesn't need to be called by users of QCPAxisRect.
Calls the base class implementation to update the margins (see QCPLayoutElement::update), and finally passes the rect to the inset layout (insetLayout) and calls its QCPInsetLayout::update function.
\seebaseclassmethod
Reimplemented from QCPLayoutElement.
Reimplemented in QCPColorScaleAxisRectPrivate.

17.607.3.53 wheelEvent()

```
void QCPAxisRect::wheelEvent (
            QWheelEvent * event )  [protected], [virtual]
```
This event gets called when the user turns the mouse scroll wheel while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`.
The `event->angleDelta()` indicates how far the mouse wheel was turned, which is usually +/- 120 for single rotation steps. However, if the mouse wheel is turned rapidly, multiple steps may accumulate to one event, making the delta larger. On the other hand, if the wheel has very smooth steps or none at all, the delta may be smaller.
The default implementation does nothing.

See also

mousePressEvent, mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent

Reimplemented from QCPLayerable.
Reimplemented in QCPColorScaleAxisRectPrivate.

17.607.3.54 width()

```
int QCPAxisRect::width ( ) const  [inline]
```
Returns the pixel width of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.607.3.55 zoom() [1/2]

```
void QCPAxisRect::zoom (
            const QRectF & pixelRect )
```

Zooms in (or out) to the passed rectangular region *pixelRect*, given in pixel coordinates.
All axes of this axis rect will have their range zoomed accordingly. If you only wish to zoom specific axes, use the overloaded version of this method.

See also

QCustomPlot::setSelectionRectMode

17.607.3.56 zoom() [2/2]

```
void QCPAxisRect::zoom (
            const QRectF & pixelRect,
            const QList< QCPAxis * > & affectedAxes )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Zooms in (or out) to the passed rectangular region *pixelRect*, given in pixel coordinates.
Only the axes passed in *affectedAxes* will have their ranges zoomed accordingly.

See also

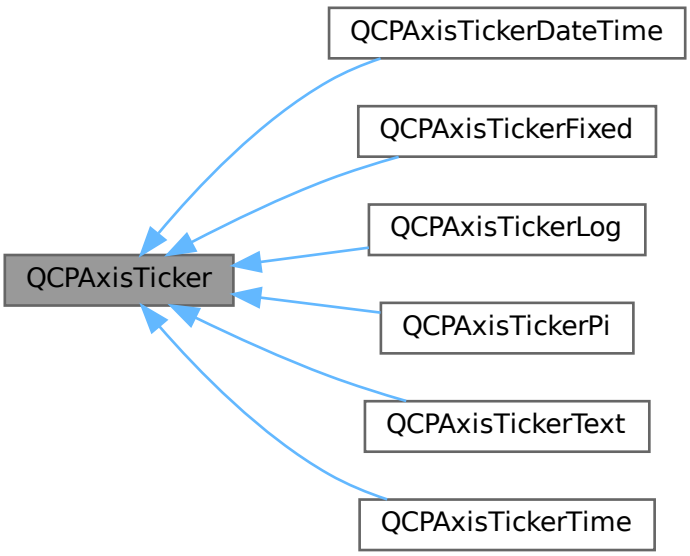
QCustomPlot::setSelectionRectMode

The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.608 QCPAxisTicker Class Reference

The base class tick generator used by QCPAxis to create tick positions and tick labels.
Inheritance diagram for QCPAxisTicker:



Public Types

- enum TickStepStrategy { tssReadability , tssMeetTickCount }

Public Member Functions

- QCPAxisTicker ()
- TickStepStrategy **tickStepStrategy** () const
- int **tickCount** () const
- double **tickOrigin** () const
- void setTickStepStrategy (TickStepStrategy strategy)
- void setTickCount (int count)
- void setTickOrigin (double origin)
- virtual void generate (const QCPRange &range, const QLocale &locale, QChar formatChar, int precision, QVector< double > &ticks, QVector< double > *subTicks, QVector< QString > *tickLabels)

Protected Member Functions

- virtual double getTickStep (const QCPRange &range)
- virtual int getSubTickCount (double tickStep)
- virtual QString getTickLabel (double tick, const QLocale &locale, QChar formatChar, int precision)
- virtual QVector< double > createTickVector (double tickStep, const QCPRange &range)
- virtual QVector< double > **createSubTickVector** (int subTickCount, const QVector< double > &ticks)
- virtual QVector< QString > **createLabelVector** (const QVector< double > &ticks, const QLocale &locale, QChar formatChar, int precision)
- void **trimTicks** (const QCPRange &range, QVector< double > &ticks, bool keepOneOutlier) const
- double **pickClosest** (double target, const QVector< double > &candidates) const
- double **getMantissa** (double input, double *magnitude=nullptr) const
- double **cleanMantissa** (double input) const

Protected Attributes

- TickStepStrategy **mTickStepStrategy**
- int **mTickCount**
- double **mTickOrigin**

17.608.1 Detailed Description

The base class tick generator used by QCPAxis to create tick positions and tick labels.
Each QCPAxis has an internal QCPAxisTicker (or a subclass) in order to generate tick positions and tick labels for the current axis range. The ticker of an axis can be set via QCPAxis::setTicker. Since that method takes a `QSharedPointer<QCPAxisTicker>`, multiple axes can share the same ticker instance.
This base class generates normal tick coordinates and numeric labels for linear axes. It picks a reasonable tick step (the separation between ticks) which results in readable tick labels. The number of ticks that should be approximately generated can be set via setTickCount. Depending on the current tick step strategy (setTickStepStrategy), the algorithm either sacrifices readability to better match the specified tick count (QCPAxisTicker::tssMeetTickCount) or relaxes the tick count in favor of better tick steps (QCPAxisTicker::tssReadability), which is the default.
The following more specialized axis ticker subclasses are available, see details in the respective class documentation:

QCPAxisTickerFixed	td
QCPAxisTickerLog	td
QCPAxisTickerPi	td
QCPAxisTickerText	td
QCPAxisTickerDateTime	td
QCPAxisTickerTime	td

17.608.2 Creating own axis tickers

Creating own axis tickers can be achieved very easily by sublassing QCPAxisTicker and reimplementing some or all of the available virtual methods.
In the simplest case you might wish to just generate different tick steps than the other tickers, so you only reimplement the method getTickStep. If you additionally want control over the string that will be shown as tick label, reimplement getTickLabel.
If you wish to have complete control, you can generate the tick vectors and tick label vectors yourself by reimplementing createTickVector and createLabelVector. The default implementations use the previously mentioned virtual methods getTickStep and getTickLabel, but your reimplementations don't necessarily need to do so. For example in the case of unequal tick steps, the method getTickStep loses its usefulness and can be ignored.
The sub tick count between major ticks can be controlled with getSubTickCount. Full sub tick placement control is obtained by reimplementing createSubTickVector.
See the documentation of all these virtual methods in QCPAxisTicker for detailed information about the parameters and expected return values.

17.608.3 Member Enumeration Documentation

17.608.3.1 TickStepStrategy

`enum QCPAxisTicker::TickStepStrategy`
Defines the strategies that the axis ticker may follow when choosing the size of the tick step.

See also

setTickStepStrategy

Enumerator

tssReadability	A nicely readable tick step is prioritized over matching the requested number of ticks (see setTickCount)
tssMeetTickCount	Less readable tick steps are allowed which in turn facilitates getting closer to the requested tick count.

17.608.4 Constructor & Destructor Documentation

17.608.4.1 QCPAxisTicker()

```
QCPAxisTicker::QCPAxisTicker ( )
```
Constructs the ticker and sets reasonable default values. Axis tickers are commonly created managed by a QSharedPointer, which then can be passed to QCPAxis::setTicker.

17.608.5 Member Function Documentation

17.608.5.1 createTickVector()

```
QVector< double > QCPAxisTicker::createTickVector (
            double tickStep,
            const QCPRange & range )  [protected], [virtual]
```
Reimplemented in QCPAxisTickerText.

17.608.5.2 generate()

```
void QCPAxisTicker::generate (
            const QCPRange & range,
            const QLocale & locale,
```

```
            QChar formatChar,
            int precision,
            QVector< double > & ticks,
            QVector< double > * subTicks,
            QVector< QString > * tickLabels )  [virtual]
```

This is the method called by QCPAxis in order to actually generate tick coordinates (*ticks*), tick label strings (*tick↩ Labels*) and sub tick coordinates (*subTicks*).
The ticks are generated for the specified *range*. The generated labels typically follow the specified *locale*, *formatChar* and number *precision*, however this might be different (or even irrelevant) for certain QCPAxisTicker subclasses.
The output parameter *ticks* is filled with the generated tick positions in axis coordinates. The output parameters *subTicks* and *tickLabels* are optional (set them to `nullptr` if not needed) and are respectively filled with sub tick coordinates, and tick label strings belonging to *ticks* by index.

17.608.5.3 getSubTickCount()

```
int QCPAxisTicker::getSubTickCount (
            double tickStep )  [protected], [virtual]
```

Reimplemented in QCPAxisTickerText.

17.608.5.4 getTickLabel()

```
QString QCPAxisTicker::getTickLabel (
            double tick,
            const QLocale & locale,
            QChar formatChar,
            int precision )  [protected], [virtual]
```

Reimplemented in QCPAxisTickerText.

17.608.5.5 getTickStep()

```
double QCPAxisTicker::getTickStep (
            const QCPRange & range )  [protected], [virtual]
```

Reimplemented in QCPAxisTickerText.

17.608.5.6 setTickCount()

```
void QCPAxisTicker::setTickCount (
            int count )
```

Sets how many ticks this ticker shall aim to generate across the axis range. Note that *count* is not guaranteed to be matched exactly, as generating readable tick intervals may conflict with the requested number of ticks.
Whether the readability has priority over meeting the requested *count* can be specified with setTickStepStrategy.

17.608.5.7 setTickOrigin()

```
void QCPAxisTicker::setTickOrigin (
            double origin )
```

Sets the mathematical coordinate (or "offset") of the zeroth tick. This tick coordinate is just a concept and doesn't need to be inside the currently visible axis range.
By default *origin* is zero, which for example yields ticks {-5, 0, 5, 10, 15,...} when the tick step is five. If *origin* is now set to 1 instead, the correspondingly generated ticks would be {-4, 1, 6, 11, 16,...}.

17.608.5.8 setTickStepStrategy()

```
void QCPAxisTicker::setTickStepStrategy (
            QCPAxisTicker::TickStepStrategy strategy )
```

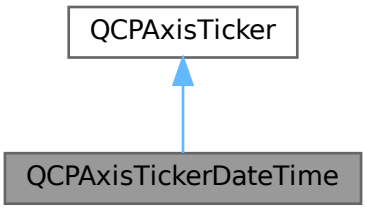
Sets which strategy the axis ticker follows when choosing the size of the tick step. For the available strategies, see TickStepStrategy.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h

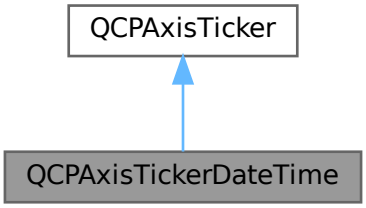
- applications/gui/genesys/qcustomplot.cpp

17.609 QCPAxisTickerDateTime Class Reference

Specialized axis ticker for calendar dates and times as axis ticks.
Inheritance diagram for QCPAxisTickerDateTime:



Collaboration diagram for QCPAxisTickerDateTime:



Public Member Functions

- QCPAxisTickerDateTime ()
- QString **dateTimeFormat** () const
- Qt::TimeSpec **dateTimeSpec** () const
- QTimeZone **timeZone** () const
- void setDateTimeFormat (const QString &format)
- void setDateTimeSpec (Qt::TimeSpec spec)
- void setTimeZone (const QTimeZone &zone)
- void setTickOrigin (double origin)
- void setTickOrigin (const QDateTime &origin)

Public Member Functions inherited from QCPAxisTicker

- QCPAxisTicker ()
- TickStepStrategy **tickStepStrategy** () const
- int **tickCount** () const
- double **tickOrigin** () const

- void setTickStepStrategy (TickStepStrategy strategy)
- void setTickCount (int count)
- void setTickOrigin (double origin)
- virtual void generate (const QCPRange &range, const QLocale &locale, QChar formatChar, int precision, QVector< double > &ticks, QVector< double > *subTicks, QVector< QString > *tickLabels)

Static Public Member Functions

- static QDateTime keyToDateTime (double key)
- static double dateTimeToKey (const QDateTime &dateTime)
- static double dateTimeToKey (const QDate &date, Qt::TimeSpec timeSpec=Qt::LocalTime)

Protected Types

- enum **DateStrategy** { **dsNone** , **dsUniformTimeInDay** , **dsUniformDayInMonth** }

Protected Member Functions

- virtual double getTickStep (const QCPRange &range) Q_DECL_OVERRIDE
- virtual int getSubTickCount (double tickStep) Q_DECL_OVERRIDE
- virtual QString getTickLabel (double tick, const QLocale &locale, QChar formatChar, int precision) Q_DECL↩ _OVERRIDE
- virtual QVector< double > createTickVector (double tickStep, const QCPRange &range) Q_DECL_↩ OVERRIDE

Protected Member Functions inherited from QCPAxisTicker

- virtual QVector< double > **createSubTickVector** (int subTickCount, const QVector< double > &ticks)
- virtual QVector< QString > **createLabelVector** (const QVector< double > &ticks, const QLocale &locale, QChar formatChar, int precision)
- void **trimTicks** (const QCPRange &range, QVector< double > &ticks, bool keepOneOutlier) const
- double **pickClosest** (double target, const QVector< double > &candidates) const
- double **getMantissa** (double input, double *magnitude=nullptr) const
- double **cleanMantissa** (double input) const

Protected Attributes

- QString **mDateTimeFormat**
- Qt::TimeSpec **mDateTimeSpec**
- QTimeZone **mTimeZone**
- enum QCPAxisTickerDateTime::DateStrategy **mDateStrategy**

Protected Attributes inherited from QCPAxisTicker

- TickStepStrategy **mTickStepStrategy**
- int **mTickCount**
- double **mTickOrigin**

Additional Inherited Members

Public Types inherited from QCPAxisTicker

- enum TickStepStrategy { tssReadability , tssMeetTickCount }

17.609.1 Detailed Description

Specialized axis ticker for calendar dates and times as axis ticks.
This QCPAxisTicker subclass generates ticks that correspond to real calendar dates and times. The plot axis coordinate is interpreted as Unix Time, so seconds since Epoch (January 1, 1970, 00:00 UTC). This is also used for example by QDateTime in the `toTime_t()/setTime_t()` methods with a precision of one second. Since Qt 4.7, millisecond accuracy can be obtained from QDateTime by using `QDateTime::fromMSecsSinceEpoch()/1000.0`. The static methods dateTimeToKey and keyToDateTime conveniently perform this conversion achieving a precision of one millisecond on all Qt versions.
The format of the date/time display in the tick labels is controlled with setDateTimeFormat. If a different time spec or time zone shall be used for the tick label appearance, see setDateTimeSpec or setTimeZone, respectively.
This ticker produces unequal tick spacing in order to provide intuitive date and time-of-day ticks. For example, if the axis range spans a few years such that there is one tick per year, ticks will be positioned on 1. January of every year. This is intuitive but, due to leap years, will result in slightly unequal tick intervals (visually unnoticeable). The same can be seen in the image above: even though the number of days varies month by month, this ticker generates ticks on the same day of each month.
If you would like to change the date/time that is used as a (mathematical) starting date for the ticks, use the setTickOrigin(const QDateTime &origin) method overload, which takes a QDateTime. If you pass 15. July, 9:45 to this method, the yearly ticks will end up on 15. July at 9:45 of every year.
The ticker can be created and assigned to an axis like this:

Note

> If you rather wish to display relative times in terms of days, hours, minutes, seconds and milliseconds, and are not interested in the intricacies of real calendar dates with months and (leap) years, have a look at QCPAxisTickerTime instead.

17.609.2 Constructor & Destructor Documentation

17.609.2.1 QCPAxisTickerDateTime()

```
QCPAxisTickerDateTime::QCPAxisTickerDateTime ( )
```

Constructs the ticker and sets reasonable default values. Axis tickers are commonly created managed by a QSharedPointer, which then can be passed to QCPAxis::setTicker.

17.609.3 Member Function Documentation

17.609.3.1 createTickVector()

```
QVector< double > QCPAxisTickerDateTime::createTickVector (
            double tickStep,
            const QCPRange & range )  [protected], [virtual]
```

Reimplemented from QCPAxisTicker.

17.609.3.2 dateTimeToKey() [1/2]

```
double QCPAxisTickerDateTime::dateTimeToKey (
            const QDate & date,
            Qt::TimeSpec timeSpec = Qt::LocalTime )  [static]
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
A convenience method which turns a QDate object into a double value that corresponds to seconds since Epoch (1. Jan 1970, 00:00 UTC). This is the format used as axis coordinates by QCPAxisTickerDateTime.
The returned value will be the start of the passed day of *date*, interpreted in the given *timeSpec*.

See also

> keyToDateTime

17.609.3.3 dateTimeToKey() [2/2]

```
double QCPAxisTickerDateTime::dateTimeToKey (
            const QDateTime & dateTime )  [static]
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
A convenience method which turns a QDateTime object into a double value that corresponds to seconds since Epoch (1. Jan 1970, 00:00 UTC). This is the format used as axis coordinates by QCPAxisTickerDateTime.
The accuracy achieved by this method is one millisecond, irrespective of the used Qt version (it works around the lack of a QDateTime::toMSecsSinceEpoch in Qt 4.6)

See also

keyToDateTime

17.609.3.4 getSubTickCount()

```
int QCPAxisTickerDateTime::getSubTickCount (
            double tickStep )  [protected], [virtual]
```

Reimplemented from QCPAxisTicker.

17.609.3.5 getTickLabel()

```
QString QCPAxisTickerDateTime::getTickLabel (
            double tick,
            const QLocale & locale,
            QChar formatChar,
            int precision )  [protected], [virtual]
```

Reimplemented from QCPAxisTicker.

17.609.3.6 getTickStep()

```
double QCPAxisTickerDateTime::getTickStep (
            const QCPRange & range )  [protected], [virtual]
```

Reimplemented from QCPAxisTicker.

17.609.3.7 keyToDateTime()

```
QDateTime QCPAxisTickerDateTime::keyToDateTime (
            double key )  [static]
```

A convenience method which turns *key* (in seconds since Epoch 1. Jan 1970, 00:00 UTC) into a QDateTime object. This can be used to turn axis coordinates to actual QDateTimes.
The accuracy achieved by this method is one millisecond, irrespective of the used Qt version (it works around the lack of a QDateTime::fromMSecsSinceEpoch in Qt 4.6)

See also

dateTimeToKey

17.609.3.8 setDateTimeFormat()

```
void QCPAxisTickerDateTime::setDateTimeFormat (
            const QString & format )
```

Sets the format in which dates and times are displayed as tick labels. For details about the *format* string, see the documentation of QDateTime::toString().
Typical expressions are

d	The day as a number without a leading zero (1 to 31)
dd	The day as a number with a leading zero (01 to 31)
ddd	The abbreviated localized day name (e.g. 'Mon' to 'Sun'). Uses the system locale to localize the name, i.e. QLocale::system().
dddd	The long localized day name (e.g. 'Monday' to 'Sunday'). Uses the system locale to localize the name, i.e. QLocale::system().
M	The month as a number without a leading zero (1 to 12)
MM	The month as a number with a leading zero (01 to 12)
MMM	The abbreviated localized month name (e.g. 'Jan' to 'Dec'). Uses the system locale to localize the name, i.e. QLocale::system().
MMMM	The long localized month name (e.g. 'January' to 'December'). Uses the system locale to localize the name, i.e. QLocale::system().
yy	The year as a two digit number (00 to 99)
yyyy	The year as a four digit number. If the year is negative, a minus sign is prepended, making five characters.
h	The hour without a leading zero (0 to 23 or 1 to 12 if AM/PM display)
hh	The hour with a leading zero (00 to 23 or 01 to 12 if AM/PM display)
H	The hour without a leading zero (0 to 23, even with AM/PM display)
HH	The hour with a leading zero (00 to 23, even with AM/PM display)
m	The minute without a leading zero (0 to 59)
mm	The minute with a leading zero (00 to 59)
s	The whole second, without any leading zero (0 to 59)
ss	The whole second, with a leading zero where applicable (00 to 59)
z	The fractional part of the second, to go after a decimal point, without trailing zeroes (0 to 999). Thus "s.z" reports the seconds to full available (millisecond) precision without trailing zeroes.
zzz	The fractional part of the second, to millisecond precision, including trailing zeroes where applicable (000 to 999).
AP or A	Use AM/PM display. A/AP will be replaced by an upper-case version of either QLocale::amText() or QLocale::pmText().
ap or a	Use am/pm display. a/ap will be replaced by a lower-case version of either QLocale::amText() or QLocale::pmText().
t	The timezone (for example "CEST")

Newlines can be inserted with "\n", literal strings (even when containing above expressions) by encapsulating them using single-quotes. A literal single quote can be generated by using two consecutive single quotes in the format.

See also

setDateTimeSpec, setTimeZone

17.609.3.9 setDateTimeSpec()

```
void QCPAxisTickerDateTime::setDateTimeSpec (
            Qt::TimeSpec spec )
```

Sets the time spec that is used for creating the tick labels from corresponding dates/times.
The default value of QDateTime objects (and also QCPAxisTickerDateTime) is Qt::LocalTime. However, if the displayed tick labels shall be given in UTC, set *spec* to Qt::UTC.
Tick labels corresponding to other time zones can be achieved with setTimeZone (which sets *spec* to Qt::TimeZone internally). Note that if *spec* is afterwards set to not be Qt::TimeZone again, the setTimeZone setting will be ignored accordingly.

See also

setDateTimeFormat, setTimeZone

17.609.3.10 setTickOrigin() [1/2]

```
void QCPAxisTickerDateTime::setTickOrigin (
            const QDateTime & origin )
```

Sets the tick origin (see QCPAxisTicker::setTickOrigin) as a QDateTime *origin*.
This is useful to define the month/day/time recurring at greater tick interval steps. For example, If you pass 15. July, 9:45 to this method and the tick interval happens to be one tick per year, the ticks will end up on 15. July at 9:45 of every year.

17.609.3.11 setTickOrigin() [2/2]

```
void QCPAxisTickerDateTime::setTickOrigin (
            double origin )
```

Sets the tick origin (see QCPAxisTicker::setTickOrigin) in seconds since Epoch (1. Jan 1970, 00:00 UTC). For the date time ticker it might be more intuitive to use the overload which directly takes a QDateTime, see setTickOrigin(const QDateTime &origin).
This is useful to define the month/day/time recurring at greater tick interval steps. For example, If you pass 15. July, 9:45 to this method and the tick interval happens to be one tick per year, the ticks will end up on 15. July at 9:45 of every year.

17.609.3.12 setTimeZone()

```
void QCPAxisTickerDateTime::setTimeZone (
            const QTimeZone & zone )
```

Sets the time zone that is used for creating the tick labels from corresponding dates/times. The time spec (setDateTimeSpec) is set to `Qt::TimeZone`.

See also

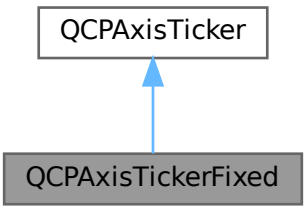
setDateTimeFormat, setTimeZone

The documentation for this class was generated from the following files:

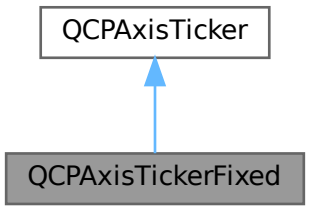
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.610 QCPAxisTickerFixed Class Reference

Specialized axis ticker with a fixed tick step.
Inheritance diagram for QCPAxisTickerFixed:



Collaboration diagram for QCPAxisTickerFixed:



Public Types

- enum ScaleStrategy { ssNone , ssMultiples , ssPowers }

Public Types inherited from QCPAxisTicker

- enum TickStepStrategy { tssReadability , tssMeetTickCount }

Public Member Functions

- QCPAxisTickerFixed ()
- double **tickStep** () const
- ScaleStrategy **scaleStrategy** () const
- void setTickStep (double step)
- void setScaleStrategy (ScaleStrategy strategy)

Public Member Functions inherited from QCPAxisTicker

- QCPAxisTicker ()
- TickStepStrategy **tickStepStrategy** () const
- int **tickCount** () const
- double **tickOrigin** () const
- void setTickStepStrategy (TickStepStrategy strategy)
- void setTickCount (int count)
- void setTickOrigin (double origin)
- virtual void generate (const QCPRange &range, const QLocale &locale, QChar formatChar, int precision, QVector< double > &ticks, QVector< double > *subTicks, QVector< QString > *tickLabels)

Protected Member Functions

- virtual double getTickStep (const QCPRange &range) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPAxisTicker

- virtual int getSubTickCount (double tickStep)
- virtual QString getTickLabel (double tick, const QLocale &locale, QChar formatChar, int precision)
- virtual QVector< double > createTickVector (double tickStep, const QCPRange &range)
- virtual QVector< double > **createSubTickVector** (int subTickCount, const QVector< double > &ticks)
- virtual QVector< QString > **createLabelVector** (const QVector< double > &ticks, const QLocale &locale, QChar formatChar, int precision)
- void **trimTicks** (const QCPRange &range, QVector< double > &ticks, bool keepOneOutlier) const

- double **pickClosest** (double target, const QVector< double > &candidates) const
- double **getMantissa** (double input, double *magnitude=nullptr) const
- double **cleanMantissa** (double input) const

Protected Attributes

- double **mTickStep**
- ScaleStrategy **mScaleStrategy**

Protected Attributes inherited from QCPAxisTicker

- TickStepStrategy **mTickStepStrategy**
- int **mTickCount**
- double **mTickOrigin**

17.610.1 Detailed Description

Specialized axis ticker with a fixed tick step.
This QCPAxisTicker subclass generates ticks with a fixed tick step set with setTickStep. It is also possible to allow integer multiples and integer powers of the specified tick step with setScaleStrategy.
A typical application of this ticker is to make an axis only display integers, by setting the tick step of the ticker to 1.0 and the scale strategy to ssMultiples.
Another case is when a certain number has a special meaning and axis ticks should only appear at multiples of that value. In this case you might also want to consider QCPAxisTickerPi because despite the name it is not limited to only pi symbols/values.
The ticker can be created and assigned to an axis like this:

17.610.2 Member Enumeration Documentation

17.610.2.1 ScaleStrategy

`enum QCPAxisTickerFixed::ScaleStrategy`
Defines how the axis ticker may modify the specified tick step (setTickStep) in order to control the number of ticks in the axis range.

See also

setScaleStrategy

Enumerator

ssNone	Modifications are not allowed, the specified tick step is absolutely fixed. This might cause a high tick density and overlapping labels if the axis range is zoomed out.
ssMultiples	An integer multiple of the specified tick step is allowed. The used factor follows the base class properties of setTickStepStrategy and setTickCount.
ssPowers	An integer power of the specified tick step is allowed.

17.610.3 Constructor & Destructor Documentation

17.610.3.1 QCPAxisTickerFixed()

`QCPAxisTickerFixed::QCPAxisTickerFixed ( )`
Constructs the ticker and sets reasonable default values. Axis tickers are commonly created managed by a QSharedPointer, which then can be passed to QCPAxis::setTicker.

17.610.4 Member Function Documentation

17.610.4.1 getTickStep()

```
double QCPAxisTickerFixed::getTickStep (
            const QCPRange & range )  [protected], [virtual]
```

Reimplemented from QCPAxisTicker.

17.610.4.2 setScaleStrategy()

```
void QCPAxisTickerFixed::setScaleStrategy (
            QCPAxisTickerFixed::ScaleStrategy strategy )
```

Sets whether the specified tick step (setTickStep) is absolutely fixed or whether modifications may be applied to it before calculating the finally used tick step, such as permitting multiples or powers. See ScaleStrategy for details.
The default strategy is ssNone, which means the tick step is absolutely fixed.

17.610.4.3 setTickStep()

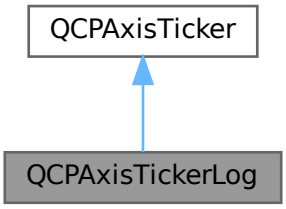
```
void QCPAxisTickerFixed::setTickStep (
            double step )
```

Sets the fixed tick interval to *step*.
The axis ticker will only use this tick step when generating axis ticks. This might cause a very high tick density and overlapping labels if the axis range is zoomed out. Using setScaleStrategy it is possible to relax the fixed step and also allow multiples or powers of *step*. This will enable the ticker to reduce the number of ticks to a reasonable amount (see setTickCount).
The documentation for this class was generated from the following files:

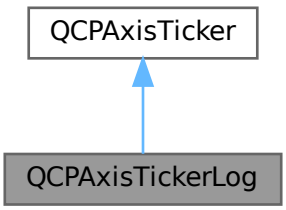
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.611 QCPAxisTickerLog Class Reference

Specialized axis ticker suited for logarithmic axes.
Inheritance diagram for QCPAxisTickerLog:



Collaboration diagram for QCPAxisTickerLog:



Public Member Functions

- QCPAxisTickerLog ()
- double **logBase** () const
- int **subTickCount** () const
- void setLogBase (double base)
- void setSubTickCount (int subTicks)

Public Member Functions inherited from QCPAxisTicker

- QCPAxisTicker ()
- TickStepStrategy **tickStepStrategy** () const
- int **tickCount** () const
- double **tickOrigin** () const
- void setTickStepStrategy (TickStepStrategy strategy)
- void setTickCount (int count)
- void setTickOrigin (double origin)
- virtual void generate (const QCPRange &range, const QLocale &locale, QChar formatChar, int precision, QVector< double > &ticks, QVector< double > *subTicks, QVector< QString > *tickLabels)

Protected Member Functions

- virtual int getSubTickCount (double tickStep) Q_DECL_OVERRIDE
- virtual QVector< double > createTickVector (double tickStep, const QCPRange &range) Q_DECL_↩ OVERRIDE

Protected Member Functions inherited from QCPAxisTicker

- virtual double getTickStep (const QCPRange &range)
- virtual QString getTickLabel (double tick, const QLocale &locale, QChar formatChar, int precision)
- virtual QVector< double > **createSubTickVector** (int subTickCount, const QVector< double > &ticks)
- virtual QVector< QString > **createLabelVector** (const QVector< double > &ticks, const QLocale &locale, QChar formatChar, int precision)
- void **trimTicks** (const QCPRange &range, QVector< double > &ticks, bool keepOneOutlier) const
- double **pickClosest** (double target, const QVector< double > &candidates) const
- double **getMantissa** (double input, double *magnitude=nullptr) const
- double **cleanMantissa** (double input) const

Protected Attributes

- double **mLogBase**
- int **mSubTickCount**
- double **mLogBaseLnInv**

Protected Attributes inherited from QCPAxisTicker

- TickStepStrategy **mTickStepStrategy**
- int **mTickCount**
- double **mTickOrigin**

Additional Inherited Members

Public Types inherited from QCPAxisTicker

- enum TickStepStrategy { tssReadability , tssMeetTickCount }

17.611.1 Detailed Description

Specialized axis ticker suited for logarithmic axes.
This QCPAxisTicker subclass generates ticks with unequal tick intervals suited for logarithmic axis scales. The ticks are placed at powers of the specified log base (setLogBase).
Especially in the case of a log base equal to 10 (the default), it might be desirable to have tick labels in the form of powers of ten without mantissa display. To achieve this, set the number precision (QCPAxis::setNumberPrecision) to zero and the number format (QCPAxis::setNumberFormat) to scientific (exponential) display with beautifully typeset decimal powers, so a format string of `"eb"`. This will result in the following axis tick labels:
The ticker can be created and assigned to an axis like this:
Note that the nature of logarithmic ticks imply that there exists a smallest possible tick step, corresponding to one multiplication by the log base. If the user zooms in further than that, no new ticks would appear, leading to very sparse or even no axis ticks on the axis. To prevent this situation, this ticker falls back to regular tick generation if the axis range would be covered by too few logarithmically placed ticks.

17.611.2 Constructor & Destructor Documentation

17.611.2.1 QCPAxisTickerLog()

```
QCPAxisTickerLog::QCPAxisTickerLog ( )
```

Constructs the ticker and sets reasonable default values. Axis tickers are commonly created managed by a QSharedPointer, which then can be passed to QCPAxis::setTicker.

17.611.3 Member Function Documentation

17.611.3.1 createTickVector()

```
QVector< double > QCPAxisTickerLog::createTickVector (
            double tickStep,
            const QCPRange & range )  [protected], [virtual]
```

Reimplemented from QCPAxisTicker.

17.611.3.2 getSubTickCount()

```
int QCPAxisTickerLog::getSubTickCount (
            double tickStep )  [protected], [virtual]
```

Reimplemented from QCPAxisTicker.

17.611.3.3 setLogBase()

```
void QCPAxisTickerLog::setLogBase (
            double base )
```

Sets the logarithm base used for tick coordinate generation. The ticks will be placed at integer powers of *base*.

17.611.3.4 setSubTickCount()

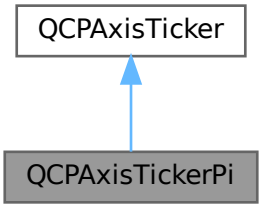
```
void QCPAxisTickerLog::setSubTickCount (
            int subTicks )
```

Sets the number of sub ticks in a tick interval. Within each interval, the sub ticks are spaced linearly to provide a better visual guide, so the sub tick density increases toward the higher tick.
Note that *subTicks* is the number of sub ticks (not sub intervals) in one tick interval. So in the case of logarithm base 10 an intuitive sub tick spacing would be achieved with eight sub ticks (the default). This means e.g. between the ticks 10 and 100 there will be eight ticks, namely at 20, 30, 40, 50, 60, 70, 80 and 90.
The documentation for this class was generated from the following files:

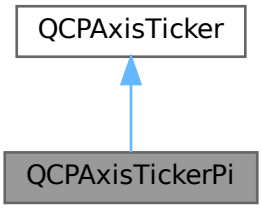
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.612 QCPAxisTickerPi Class Reference

Specialized axis ticker to display ticks in units of an arbitrary constant, for example pi.
Inheritance diagram for QCPAxisTickerPi:



Collaboration diagram for QCPAxisTickerPi:



Public Types

- enum FractionStyle { fsFloatingPoint , fsAsciiFractions , fsUnicodeFractions }

Public Types inherited from QCPAxisTicker

- enum TickStepStrategy { tssReadability , tssMeetTickCount }

Public Member Functions

- QCPAxisTickerPi ()
- QString **piSymbol** () const
- double **piValue** () const
- bool **periodicity** () const
- FractionStyle **fractionStyle** () const
- void setPiSymbol (QString symbol)
- void setPiValue (double pi)
- void setPeriodicity (int multiplesOfPi)
- void setFractionStyle (FractionStyle style)

Public Member Functions inherited from QCPAxisTicker

- QCPAxisTicker ()
- TickStepStrategy **tickStepStrategy** () const
- int **tickCount** () const
- double **tickOrigin** () const
- void setTickStepStrategy (TickStepStrategy strategy)
- void setTickCount (int count)
- void setTickOrigin (double origin)
- virtual void generate (const QCPRange &range, const QLocale &locale, QChar formatChar, int precision, QVector< double > &ticks, QVector< double > *subTicks, QVector< QString > *tickLabels)

Protected Member Functions

- virtual double getTickStep (const QCPRange &range) Q_DECL_OVERRIDE
- virtual int getSubTickCount (double tickStep) Q_DECL_OVERRIDE
- virtual QString getTickLabel (double tick, const QLocale &locale, QChar formatChar, int precision) Q_DECL↩ _OVERRIDE
- void **simplifyFraction** (int &numerator, int &denominator) const
- QString **fractionToString** (int numerator, int denominator) const
- QString **unicodeFraction** (int numerator, int denominator) const
- QString **unicodeSuperscript** (int number) const
- QString **unicodeSubscript** (int number) const

Protected Member Functions inherited from QCPAxisTicker

- virtual QVector< double > createTickVector (double tickStep, const QCPRange &range)
- virtual QVector< double > **createSubTickVector** (int subTickCount, const QVector< double > &ticks)
- virtual QVector< QString > **createLabelVector** (const QVector< double > &ticks, const QLocale &locale, QChar formatChar, int precision)
- void **trimTicks** (const QCPRange &range, QVector< double > &ticks, bool keepOneOutlier) const
- double **pickClosest** (double target, const QVector< double > &candidates) const
- double **getMantissa** (double input, double *magnitude=nullptr) const
- double **cleanMantissa** (double input) const

Protected Attributes

- QString **mPiSymbol**
- double **mPiValue**
- int **mPeriodicity**
- FractionStyle **mFractionStyle**
- double **mPiTickStep**

Protected Attributes inherited from QCPAxisTicker

- TickStepStrategy **mTickStepStrategy**
- int **mTickCount**
- double **mTickOrigin**

17.612.1 Detailed Description

Specialized axis ticker to display ticks in units of an arbitrary constant, for example pi.
This QCPAxisTicker subclass generates ticks that are expressed with respect to a given symbolic constant with a numerical value specified with setPiValue and an appearance in the tick labels specified with setPiSymbol.
Ticks may be generated at fractions of the symbolic constant. How these fractions appear in the tick label can be configured with setFractionStyle.
The ticker can be created and assigned to an axis like this:

17.612.2 Member Enumeration Documentation

17.612.2.1 FractionStyle

```
enum QCPAxisTickerPi::FractionStyle
```
Defines how fractions should be displayed in tick labels.

See also

setFractionStyle

Enumerator

fsFloatingPoint	Fractions are displayed as regular decimal floating point numbers, e.g. "0.25" or "0.125".
fsAsciiFractions	Fractions are written as rationals using ASCII characters only, e.g. "1/4" or "1/8".
fsUnicodeFractions	Fractions are written using sub- and superscript UTF-8 digits and the fraction symbol.

17.612.3 Constructor & Destructor Documentation

17.612.3.1 QCPAxisTickerPi()

```
QCPAxisTickerPi::QCPAxisTickerPi ( )
```
Constructs the ticker and sets reasonable default values. Axis tickers are commonly created managed by a QSharedPointer, which then can be passed to QCPAxis::setTicker.

17.612.4 Member Function Documentation

17.612.4.1 getSubTickCount()

```
int QCPAxisTickerPi::getSubTickCount (
            double tickStep )  [protected], [virtual]
```
Reimplemented from QCPAxisTicker.

17.612.4.2 getTickLabel()

```
QString QCPAxisTickerPi::getTickLabel (
            double tick,
            const QLocale & locale,
            QChar formatChar,
            int precision )  [protected], [virtual]
```
Reimplemented from QCPAxisTicker.

17.612.4.3 getTickStep()

```
double QCPAxisTickerPi::getTickStep (
            const QCPRange & range )  [protected], [virtual]
```

Reimplemented from QCPAxisTicker.

17.612.4.4 setFractionStyle()

```
void QCPAxisTickerPi::setFractionStyle (
            QCPAxisTickerPi::FractionStyle style )
```

Sets how the numerical/fractional part preceding the symbolic constant is displayed in tick labels. See FractionStyle for the various options.

17.612.4.5 setPeriodicity()

```
void QCPAxisTickerPi::setPeriodicity (
            int multiplesOfPi )
```

Sets whether the axis labels shall appear periodicly and if so, at which multiplicity of the symbolic constant.
To disable periodicity, set *multiplesOfPi* to zero.
For example, an axis that identifies 0 with 2pi would set *multiplesOfPi* to two.

17.612.4.6 setPiSymbol()

```
void QCPAxisTickerPi::setPiSymbol (
            QString symbol )
```

Sets how the symbol part (which is always a suffix to the number) shall appear in the axis tick label.
If a space shall appear between the number and the symbol, make sure the space is contained in *symbol*.

17.612.4.7 setPiValue()

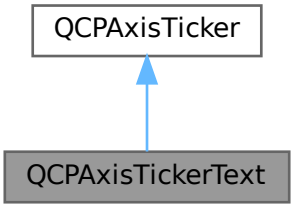
```
void QCPAxisTickerPi::setPiValue (
            double pi )
```

Sets the numerical value that the symbolic constant has.
This will be used to place the appropriate fractions of the symbol at the respective axis coordinates.
The documentation for this class was generated from the following files:

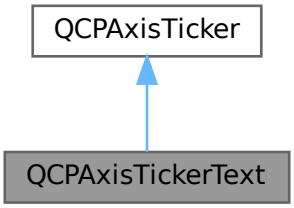
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.613 QCPAxisTickerText Class Reference

Specialized axis ticker which allows arbitrary labels at specified coordinates.
Inheritance diagram for QCPAxisTickerText:



Collaboration diagram for QCPAxisTickerText:



Public Member Functions

- QCPAxisTickerText ()
- QMap< double, QString > & ticks ()
- int **subTickCount** () const
- void setTicks (const QMap< double, QString > &ticks)
- void setTicks (const QVector< double > &positions, const QVector< QString > &labels)
- void setSubTickCount (int subTicks)
- void clear ()
- void addTick (double position, const QString &label)
- void addTicks (const QMap< double, QString > &ticks)
- void addTicks (const QVector< double > &positions, const QVector< QString > &labels)

Public Member Functions inherited from QCPAxisTicker

- QCPAxisTicker ()
- TickStepStrategy **tickStepStrategy** () const
- int **tickCount** () const
- double **tickOrigin** () const
- void setTickStepStrategy (TickStepStrategy strategy)
- void setTickCount (int count)
- void setTickOrigin (double origin)
- virtual void generate (const QCPRange &range, const QLocale &locale, QChar formatChar, int precision, QVector< double > &ticks, QVector< double > *subTicks, QVector< QString > *tickLabels)

Protected Member Functions

- virtual double getTickStep (const QCPRange &range) Q_DECL_OVERRIDE
- virtual int getSubTickCount (double tickStep) Q_DECL_OVERRIDE
- virtual QString getTickLabel (double tick, const QLocale &locale, QChar formatChar, int precision) Q_DECL↩ _OVERRIDE
- virtual QVector< double > createTickVector (double tickStep, const QCPRange &range) Q_DECL_↩ OVERRIDE

Protected Member Functions inherited from QCPAxisTicker

- virtual QVector< double > **createSubTickVector** (int subTickCount, const QVector< double > &ticks)
- virtual QVector< QString > **createLabelVector** (const QVector< double > &ticks, const QLocale &locale, QChar formatChar, int precision)
- void **trimTicks** (const QCPRange &range, QVector< double > &ticks, bool keepOneOutlier) const

- double **pickClosest** (double target, const QVector< double > &candidates) const
- double **getMantissa** (double input, double *magnitude=nullptr) const
- double **cleanMantissa** (double input) const

Protected Attributes

- QMap< double, QString > **mTicks**
- int **mSubTickCount**

Protected Attributes inherited from QCPAxisTicker

- TickStepStrategy **mTickStepStrategy**
- int **mTickCount**
- double **mTickOrigin**

Additional Inherited Members

Public Types inherited from QCPAxisTicker

- enum TickStepStrategy { tssReadability , tssMeetTickCount }

17.613.1 Detailed Description

Specialized axis ticker which allows arbitrary labels at specified coordinates.
This QCPAxisTicker subclass generates ticks which can be directly specified by the user as coordinates and associated strings. They can be passed as a whole with setTicks or one at a time with addTick. Alternatively you can directly access the internal storage via ticks and modify the tick/label data there.
This is useful for cases where the axis represents categories rather than numerical values.
If you are updating the ticks of this ticker regularly and in a dynamic fasion (e.g. dependent on the axis range), it is a sign that you should probably create an own ticker by subclassing QCPAxisTicker, instead of using this one.
The ticker can be created and assigned to an axis like this:

17.613.2 Constructor & Destructor Documentation

17.613.2.1 QCPAxisTickerText()

```
QCPAxisTickerText::QCPAxisTickerText ( )
```

Constructs the ticker and sets reasonable default values. Axis tickers are commonly created managed by a QSharedPointer, which then can be passed to QCPAxis::setTicker.

17.613.3 Member Function Documentation

17.613.3.1 addTick()

```
void QCPAxisTickerText::addTick (
            double position,
            const QString & label )
```

Adds a single tick to the axis at the given axis coordinate *position*, with the provided tick *label*.

See also

addTicks, setTicks, clear

17.613.3.2 addTicks() [1/2]

```
void QCPAxisTickerText::addTicks (
            const QMap< double, QString > & ticks )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.

Adds the provided *ticks* to the ones already existing. The map key of *ticks* corresponds to the axis coordinate, and the map value is the string that will appear as tick label.
An alternative to manipulate ticks is to directly access the internal storage with the ticks getter.

See also

addTick, setTicks, clear

17.613.3.3 addTicks() [2/2]

```
void QCPAxisTickerText::addTicks (
            const QVector< double > & positions,
            const QVector< QString > & labels )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds the provided ticks to the ones already existing. The entries of *positions* correspond to the axis coordinates, and the entries of *labels* are the respective strings that will appear as tick labels.
An alternative to manipulate ticks is to directly access the internal storage with the ticks getter.

See also

addTick, setTicks, clear

17.613.3.4 clear()

```
void QCPAxisTickerText::clear ( )
```

Clears all ticks.
An alternative to manipulate ticks is to directly access the internal storage with the ticks getter.

See also

setTicks, addTicks, addTick

17.613.3.5 createTickVector()

```
QVector< double > QCPAxisTickerText::createTickVector (
            double tickStep,
            const QCPRange & range )  [protected], [virtual]
```

Returns the externally provided tick coordinates which are in the specified *range*. If available, one tick above and below the range is provided in addition, to allow possible sub tick calculation. The parameter *tickStep* is ignored.
\seebaseclassmethod
Reimplemented from QCPAxisTicker.

17.613.3.6 getSubTickCount()

```
int QCPAxisTickerText::getSubTickCount (
            double tickStep )  [protected], [virtual]
```

Returns the sub tick count that was configured with setSubTickCount.
\seebaseclassmethod
Reimplemented from QCPAxisTicker.

17.613.3.7 getTickLabel()

```
QString QCPAxisTickerText::getTickLabel (
            double tick,
            const QLocale & locale,
            QChar formatChar,
            int precision )  [protected], [virtual]
```

Returns the tick label which corresponds to the key *tick* in the internal tick storage. Since the labels are provided externally, *locale*, *formatChar*, and *precision* are ignored.
\seebaseclassmethod
Reimplemented from QCPAxisTicker.

17.613.3.8 getTickStep()

```
double QCPAxisTickerText::getTickStep (
            const QCPRange & range )  [protected], [virtual]
```

Since the tick coordinates are provided externally, this method implementation does nothing.
\seebaseclassmethod
Reimplemented from QCPAxisTicker.

17.613.3.9 setSubTickCount()

```
void QCPAxisTickerText::setSubTickCount (
            int subTicks )
```

Sets the number of sub ticks that shall appear between ticks. For QCPAxisTickerText, there is no automatic sub tick count calculation. So if sub ticks are needed, they must be configured with this method.

17.613.3.10 setTicks() [1/2]

```
void QCPAxisTickerText::setTicks (
            const QMap< double, QString > & ticks )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the ticks that shall appear on the axis. The map key of *ticks* corresponds to the axis coordinate, and the map value is the string that will appear as tick label.
An alternative to manipulate ticks is to directly access the internal storage with the ticks getter.

See also

addTicks, addTick, clear

17.613.3.11 setTicks() [2/2]

```
void QCPAxisTickerText::setTicks (
            const QVector< double > & positions,
            const QVector< QString > & labels )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the ticks that shall appear on the axis. The entries of *positions* correspond to the axis coordinates, and the entries of *labels* are the respective strings that will appear as tick labels.

See also

addTicks, addTick, clear

17.613.3.12 ticks()

```
QMap< double, QString > & QCPAxisTickerText::ticks ( )  [inline]
```

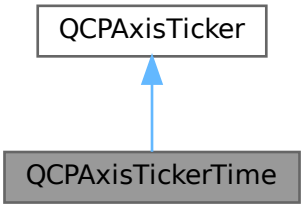
Returns a non-const reference to the internal map which stores the tick coordinates and their labels.
You can access the map directly in order to add, remove or manipulate ticks, as an alternative to using the methods provided by QCPAxisTickerText, such as setTicks and addTick.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

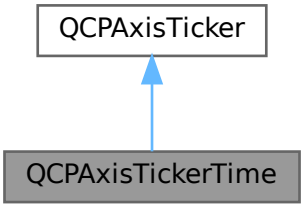
17.614 QCPAxisTickerTime Class Reference

Specialized axis ticker for time spans in units of milliseconds to days.

Inheritance diagram for QCPAxisTickerTime:



Collaboration diagram for QCPAxisTickerTime:



Public Types

- enum TimeUnit { tuMilliseconds , tuSeconds , tuMinutes , tuHours , tuDays }

Public Types inherited from QCPAxisTicker

- enum TickStepStrategy { tssReadability , tssMeetTickCount }

Public Member Functions

- QCPAxisTickerTime ()
- QString **timeFormat** () const
- int **fieldWidth** (TimeUnit unit) const
- void setTimeFormat (const QString &format)
- void setFieldWidth (TimeUnit unit, int width)

Public Member Functions inherited from QCPAxisTicker

- QCPAxisTicker ()
- TickStepStrategy **tickStepStrategy** () const
- int **tickCount** () const
- double **tickOrigin** () const

- void setTickStepStrategy (TickStepStrategy strategy)
- void setTickCount (int count)
- void setTickOrigin (double origin)
- virtual void generate (const QCPRange &range, const QLocale &locale, QChar formatChar, int precision, QVector< double > &ticks, QVector< double > *subTicks, QVector< QString > *tickLabels)

Protected Member Functions

- virtual double getTickStep (const QCPRange &range) Q_DECL_OVERRIDE
- virtual int getSubTickCount (double tickStep) Q_DECL_OVERRIDE
- virtual QString getTickLabel (double tick, const QLocale &locale, QChar formatChar, int precision) Q_DECL↩_OVERRIDE
- void **replaceUnit** (QString &text, TimeUnit unit, int value) const

Protected Member Functions inherited from QCPAxisTicker

- virtual QVector< double > createTickVector (double tickStep, const QCPRange &range)
- virtual QVector< double > **createSubTickVector** (int subTickCount, const QVector< double > &ticks)
- virtual QVector< QString > **createLabelVector** (const QVector< double > &ticks, const QLocale &locale, QChar formatChar, int precision)
- void **trimTicks** (const QCPRange &range, QVector< double > &ticks, bool keepOneOutlier) const
- double **pickClosest** (double target, const QVector< double > &candidates) const
- double **getMantissa** (double input, double *magnitude=nullptr) const
- double **cleanMantissa** (double input) const

Protected Attributes

- QString **mTimeFormat**
- QHash< TimeUnit, int > **mFieldWidth**
- TimeUnit **mSmallestUnit**
- TimeUnit **mBiggestUnit**
- QHash< TimeUnit, QString > **mFormatPattern**

Protected Attributes inherited from QCPAxisTicker

- TickStepStrategy **mTickStepStrategy**
- int **mTickCount**
- double **mTickOrigin**

17.614.1 Detailed Description

Specialized axis ticker for time spans in units of milliseconds to days.
This QCPAxisTicker subclass generates ticks that corresponds to time intervals.
The format of the time display in the tick labels is controlled with setTimeFormat and setFieldWidth. The time coordinate is in the unit of seconds with respect to the time coordinate zero. Unlike with QCPAxisTickerDateTime, the ticks don't correspond to a specific calendar date and time.
The time can be displayed in milliseconds, seconds, minutes, hours and days. Depending on the largest available unit in the format specified with setTimeFormat, any time spans above will be carried in that largest unit. So for example if the format string is "%m:%s" and a tick at coordinate value 7815 (being 2 hours, 10 minutes and 15 seconds) is created, the resulting tick label will show "130:15" (130 minutes, 15 seconds). If the format string is "%h:%m:%s", the hour unit will be used and the label will thus be "02:10:15". Negative times with respect to the axis zero will carry a leading minus sign.
The ticker can be created and assigned to an axis like this:
Here is an example of a time axis providing time information in days, hours and minutes. Due to the axis range spanning a few days and the wanted tick count (setTickCount), the ticker decided to use tick steps of 12 hours:
The format string for this example is

Note

If you rather wish to display calendar dates and times, have a look at QCPAxisTickerDateTime instead.

17.614.2 Member Enumeration Documentation

17.614.2.1 TimeUnit

`enum QCPAxisTickerTime::TimeUnit`

Defines the logical units in which fractions of time spans can be expressed.

See also

setFieldWidth, setTimeFormat

Enumerator

tuMilliseconds	Milliseconds, one thousandth of a second (%z in setTimeFormat)
tuSeconds	Seconds (%s in setTimeFormat)
tuMinutes	Minutes (%m in setTimeFormat)
tuHours	Hours (%h in setTimeFormat)
tuDays	Days (%d in setTimeFormat)

17.614.3 Constructor & Destructor Documentation

17.614.3.1 QCPAxisTickerTime()

```
QCPAxisTickerTime::QCPAxisTickerTime ( )
```

Constructs the ticker and sets reasonable default values. Axis tickers are commonly created managed by a QSharedPointer, which then can be passed to QCPAxis::setTicker.

17.614.4 Member Function Documentation

17.614.4.1 getSubTickCount()

```
int QCPAxisTickerTime::getSubTickCount (
            double tickStep )  [protected], [virtual]
```

Reimplemented from QCPAxisTicker.

17.614.4.2 getTickLabel()

```
QString QCPAxisTickerTime::getTickLabel (
            double tick,
            const QLocale & locale,
            QChar formatChar,
            int precision )  [protected], [virtual]
```

Reimplemented from QCPAxisTicker.

17.614.4.3 getTickStep()

```
double QCPAxisTickerTime::getTickStep (
            const QCPRange & range )  [protected], [virtual]
```

Reimplemented from QCPAxisTicker.

17.614.4.4 setFieldWidth()

```
void QCPAxisTickerTime::setFieldWidth (
            QCPAxisTickerTime::TimeUnit unit,
            int width )
```

Sets the field widh of the specified *unit* to be *width* digits, when displayed in the tick label. If the number for the specific unit is shorter than *width*, it will be padded with an according number of zeros to the left in order to reach the field width.

See also

setTimeFormat

17.614.4.5 setTimeFormat()

```
void QCPAxisTickerTime::setTimeFormat (
            const QString & format )
```

Sets the format that will be used to display time in the tick labels.
The available patterns are:

- %z for milliseconds
- %s for seconds
- %m for minutes
- %h for hours
- %d for days

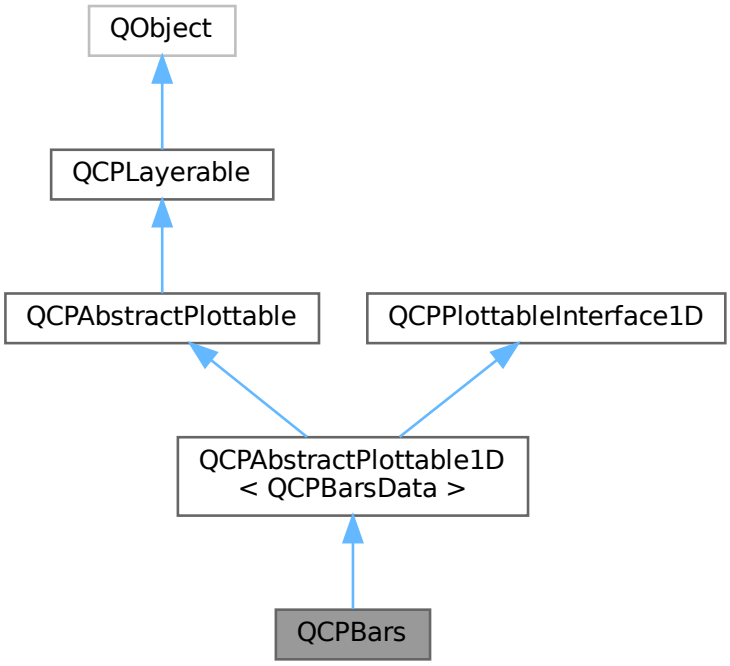
The field width (zero padding) can be controlled for each unit with setFieldWidth.
The largest unit that appears in *format* will carry all the remaining time of a certain tick coordinate, even if it overflows the natural limit of the unit. For example, if %m is the largest unit it might become larger than 59 in order to consume larger time values. If on the other hand %h is available, the minutes will wrap around to zero after 59 and the time will carry to the hour digit.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

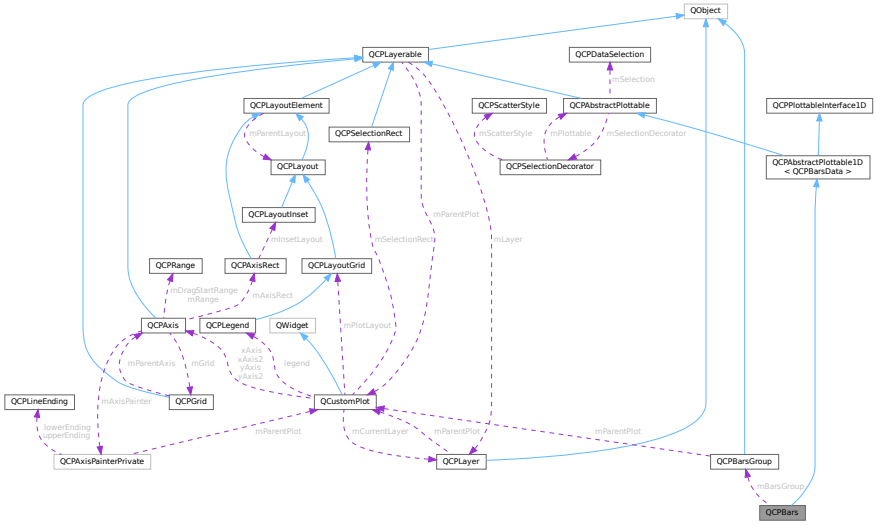
17.615 QCPBars Class Reference

A plottable representing a bar chart in a plot.

Inheritance diagram for QCPBars:



Collaboration diagram for QCPBars:



Public Types

- enum WidthType { wtAbsolute , wtAxisRectRatio , wtPlotCoords }

Public Member Functions

- QCPBars (QCPAxis *keyAxis, QCPAxis *valueAxis)
- double **width** () const
- WidthType **widthType** () const
- QCPBarsGroup * **barsGroup** () const
- double **baseValue** () const
- double **stackingGap** () const
- QCPBars * barBelow () const
- QCPBars * barAbove () const
- QSharedPointer< QCPBarsDataContainer > data () const
- void setData (QSharedPointer< QCPBarsDataContainer > data)
- void setData (const QVector< double > &keys, const QVector< double > &values, bool alreadySorted=false)
- void setWidth (double width)
- void setWidthType (WidthType widthType)
- void setBarsGroup (QCPBarsGroup *barsGroup)
- void setBaseValue (double baseValue)
- void setStackingGap (double pixels)
- void addData (const QVector< double > &keys, const QVector< double > &values, bool already↩ Sorted=false)
- void addData (double key, double value)
- void moveBelow (QCPBars *bars)
- void moveAbove (QCPBars *bars)
- virtual QCPDataSelection selectTestRect (const QRectF &rect, bool onlySelectable) const Q_DECL_↩ OVERRIDE
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE
- virtual QCPRange getKeyRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth) const Q_DECL_OVERRIDE
- virtual QCPRange getValueRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth, const QCPRange &inKeyRange=QCPRange()) const Q_DECL_OVERRIDE
- virtual QPointF dataPixelPosition (int index) const Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable1D< QCPBarsData >

- QCPAbstractPlottable1D (QCPAxis *keyAxis, QCPAxis *valueAxis)
- virtual int dataCount () const Q_DECL_OVERRIDE
- virtual double dataMainKey (int index) const Q_DECL_OVERRIDE
- virtual double dataSortKey (int index) const Q_DECL_OVERRIDE
- virtual double dataMainValue (int index) const Q_DECL_OVERRIDE
- virtual QCPRange dataValueRange (int index) const Q_DECL_OVERRIDE
- virtual bool sortKeyIsMainKey () const Q_DECL_OVERRIDE
- virtual int findBegin (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual int findEnd (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual QCPPlottableInterface1D * interface1D () Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable

- QCPAbstractPlottable (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QString **name** () const
- bool **antialiasedFill** () const
- bool **antialiasedScatters** () const
- QPen **pen** () const
- QBrush **brush** () const
- QCPAxis * **keyAxis** () const
- QCPAxis * **valueAxis** () const

- QCP::SelectionType **selectable** () const
- bool selected () const
- QCPDataSelection selection () const
- QCPSelectionDecorator * selectionDecorator () const
- void setName (const QString &name)
- void setAntialiasedFill (bool enabled)
- void setAntialiasedScatters (bool enabled)
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setKeyAxis (QCPAxis *axis)
- void setValueAxis (QCPAxis *axis)
- Q_SLOT void setSelectable (QCP::SelectionType selectable)
- Q_SLOT void setSelection (QCPDataSelection selection)
- void setSelectionDecorator (QCPSelectionDecorator *decorator)
- void coordsToPixels (double key, double value, double &x, double &y) const
- const QPointF coordsToPixels (double key, double value) const
- void pixelsToCoords (double x, double y, double &key, double &value) const
- void pixelsToCoords (const QPointF &pixelPos, double &key, double &value) const
- void rescaleAxes (bool onlyEnlarge=false) const
- void rescaleKeyAxis (bool onlyEnlarge=false) const
- void rescaleValueAxis (bool onlyEnlarge=false, bool inKeyRange=false) const
- bool addToLegend (QCPLegend *legend)
- bool addToLegend ()
- bool removeFromLegend (QCPLegend *legend) const
- bool removeFromLegend () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void drawLegendIcon (QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE
- void **getVisibleDataBounds** (QCPBarsDataContainer::const_iterator &begin, QCPBarsDataContainer::const_iterator &end) const
- QRectF **getBarRect** (double key, double value) const
- void **getPixelWidth** (double key, double &lower, double &upper) const
- double **getStackedBaseValue** (double key, bool positive) const

Protected Member Functions inherited from QCPAbstractPlottable1D< QCPBarsData >

- void getDataSegments (QList< QCPDataRange > &selectedSegments, QList< QCPDataRange > &unselectedSegments) const
- void drawPolyline (QCPPainter *painter, const QVector< QPointF > &lineData) const

Protected Member Functions inherited from QCPAbstractPlottable

- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- void **applyFillAntialiasingHint** (QCPPainter *painter) const
- void **applyScattersAntialiasingHint** (QCPPainter *painter) const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Static Protected Member Functions

- static void **connectBars** (QCPBars *lower, QCPBars *upper)

Protected Attributes

- double **mWidth**
- WidthType **mWidthType**
- QCPBarsGroup * **mBarsGroup**
- double **mBaseValue**
- double **mStackingGap**
- QPointer< QCPBars > **mBarBelow**
- QPointer< QCPBars > **mBarAbove**

Protected Attributes inherited from QCPAbstractPlottable1D< QCPBarsData >

- QSharedPointer< QCPDataContainer< QCPBarsData > > **mDataContainer**

Protected Attributes inherited from QCPAbstractPlottable

- QString **mName**
- bool **mAntialiasedFill**
- bool **mAntialiasedScatters**
- QPen **mPen**
- QBrush **mBrush**
- QPointer< QCPAxis > **mKeyAxis**
- QPointer< QCPAxis > **mValueAxis**
- QCP::SelectionType **mSelectable**
- QCPDataSelection **mSelection**
- QCPSelectionDecorator * **mSelectionDecorator**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPLegend**
- class **QCPBarsGroup**

Additional Inherited Members

Signals inherited from QCPAbstractPlottable

- void selectionChanged (bool selected)
- void selectionChanged (const QCPDataSelection &selection)
- void selectableChanged (QCP::SelectionType selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.615.1 Detailed Description

A plottable representing a bar chart in a plot.
To plot data, assign it with the setData or addData functions.

17.615.2 Changing the appearance

The appearance of the bars is determined by the pen and the brush (setPen, setBrush). The width of the individual bars can be controlled with setWidthType and setWidth.
Bar charts are stackable. This means, two QCPBars plottables can be placed on top of each other (see QCPBars::moveAbove). So when two bars are at the same key position, they will appear stacked.
If you would like to group multiple QCPBars plottables together so they appear side by side as shown below, use QCPBarsGroup.

17.615.3 Usage

Like all data representing objects in QCustomPlot, the QCPBars is a plottable (QCPAbstractPlottable). So the plottable-interface of QCustomPlot applies (QCustomPlot::plottable, QCustomPlot::removePlottable, etc.)
Usually, you first create an instance:
which registers it with the QCustomPlot instance of the passed axes. Note that this QCustomPlot instance takes ownership of the plottable, so do not delete it manually but use QCustomPlot::removePlottable() instead. The newly created plottable can be modified, e.g.:

17.615.4 Member Enumeration Documentation

17.615.4.1 WidthType

`enum QCPBars::WidthType`
Defines the ways the width of the bar can be specified. Thus it defines what the number passed to setWidth actually means.

See also

setWidthType, setWidth

Enumerator

wtAbsolute	Bar width is in absolute pixels.
wtAxisRectRatio	Bar width is given by a fraction of the axis rect size.
wtPlotCoords	Bar width is in key coordinates and thus scales with the key axis range.

17.615.5 Constructor & Destructor Documentation

17.615.5.1 QCPBars()

```
QCPBars::QCPBars (
            QCPAxis * keyAxis,
            QCPAxis * valueAxis )  [explicit]
```

Constructs a bar chart which uses *keyAxis* as its key axis ("x") and *valueAxis* as its value axis ("y"). *keyAxis* and *valueAxis* must reside in the same QCustomPlot instance and not have the same orientation. If either of these restrictions is violated, a corresponding message is printed to the debug output (qDebug), the construction is not aborted, though.
The created QCPBars is automatically registered with the QCustomPlot instance inferred from *keyAxis*. This QCustomPlot instance takes ownership of the QCPBars, so do not delete it manually but use QCustomPlot::removePlottable() instead.

17.615.6 Member Function Documentation

17.615.6.1 addData() [1/2]

```
void QCPBars::addData (
            const QVector< double > & keys,
            const QVector< double > & values,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds the provided points in *keys* and *values* to the current data. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.
If you can guarantee that the passed data points are sorted by *keys* in ascending order, you can set *alreadySorted* to true, to improve performance by saving a sorting run.
Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

17.615.6.2 addData() [2/2]

```
void QCPBars::addData (
            double key,
            double value )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts. Adds the provided data point as *key* and *value* to the current data.
Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

17.615.6.3 barAbove()

```
QCPBars * QCPBars::barAbove ( ) const  [inline]
```

Returns the bars plottable that is directly above this bars plottable. If there is no such plottable, returns `nullptr`.

See also

barBelow, moveBelow, moveAbove

17.615.6.4 barBelow()

```
QCPBars * QCPBars::barBelow ( ) const  [inline]
```
Returns the bars plottable that is directly below this bars plottable. If there is no such plottable, returns `nullptr`.

See also

barAbove, moveBelow, moveAbove

17.615.6.5 data()

```
QSharedPointer< QCPBarsDataContainer > QCPBars::data ( ) const  [inline]
```
Returns a shared pointer to the internal data storage of type QCPBarsDataContainer. You may use it to directly manipulate the data, which may be more convenient and faster than using the regular setData or addData methods.

17.615.6.6 dataPixelPosition()

```
QPointF QCPBars::dataPixelPosition (
            int index ) const  [virtual]
```
Returns the pixel position on the widget surface at which the data point at the given *index* appears.
Usually this corresponds to the point of dataMainKey/ dataMainValue, in pixel coordinates. However, depending on the plottable, this might be a different apparent position than just a coord-to-pixel transform of those values. For example, QCPBars apparent data values can be shifted depending on their stacking, bar grouping or configured base value.
Reimplemented from QCPAbstractPlottable1D< QCPBarsData >.

17.615.6.7 draw()

```
void QCPBars::draw (
            QCPPainter * painter )  [protected], [virtual]
```
Implements QCPAbstractPlottable.

17.615.6.8 drawLegendIcon()

```
void QCPBars::drawLegendIcon (
            QCPPainter * painter,
            const QRectF & rect ) const  [protected], [virtual]
```
Implements QCPAbstractPlottable.

17.615.6.9 getKeyRange()

```
QCPRange QCPBars::getKeyRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth ) const  [virtual]
```
Returns the coordinate range that all data in this plottable span in the key axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSignDomain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getValueRange

Implements QCPAbstractPlottable.

17.615.6.10 getValueRange()

```
QCPRange QCPBars::getValueRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth,
            const QCPRange & inKeyRange = QCPRange() ) const  [virtual]
```

Returns the coordinate range that the data points in the specified key range (*inKeyRange*) span in the value axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSign↩ Domain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
If *inKeyRange* has both lower and upper bound set to zero (is equal to `QCPRange()`), all data points are considered, without any restriction on the keys.
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getKeyRange

Implements QCPAbstractPlottable.

17.615.6.11 moveAbove()

```
void QCPBars::moveAbove (
            QCPBars * bars )
```

Moves this bars plottable above *bars*. In other words, the bars of this plottable will appear above the bars of *bars*. The move target *bars* must use the same key and value axis as this plottable.
Inserting into and removing from existing bar stacking is handled gracefully. If *bars* already has a bars object above itself, this bars object is inserted between the two. If this bars object is already between two other bars, the two other bars will be stacked on top of each other after the operation.
To remove this bars plottable from any stacking, set *bars* to `nullptr`.

See also

moveBelow, barBelow, barAbove

17.615.6.12 moveBelow()

```
void QCPBars::moveBelow (
            QCPBars * bars )
```

Moves this bars plottable below *bars*. In other words, the bars of this plottable will appear below the bars of *bars*. The move target *bars* must use the same key and value axis as this plottable.
Inserting into and removing from existing bar stacking is handled gracefully. If *bars* already has a bars object below itself, this bars object is inserted between the two. If this bars object is already between two other bars, the two other bars will be stacked on top of each other after the operation.
To remove this bars plottable from any stacking, set *bars* to `nullptr`.

See also

moveBelow, barAbove, barBelow

17.615.6.13 selectTest()

```
double QCPBars::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

Implements a selectTest specific to this plottable's point geometry.

If *details* is not 0, it will be set to a QCPDataSelection, describing the closest data point to *pos*.
\seebaseclassmethod QCPAbstractPlottable::selectTest
Reimplemented from QCPAbstractPlottable1D< QCPBarsData >.

17.615.6.14 selectTestRect()

```
QCPDataSelection QCPBars::selectTestRect (
            const QRectF & rect,
            bool onlySelectable ) const  [virtual]
```

Returns a data selection containing all the data points of this plottable which are contained (or hit by) *rect*. This is used mainly in the selection rect interaction for data selection (data selection mechanism).
If *onlySelectable* is true, an empty QCPDataSelection is returned if this plottable is not selectable (i.e. if QCPAbstractPlottable::setSelectable is QCP::stNone).

Note

> *rect* must be a normalized rect (positive or zero width and height). This is especially important when using the rect of QCPSelectionRect::accepted, which is not necessarily normalized. Use `QRect::normalized()` when passing a rect which might not be normalized.

Reimplemented from QCPAbstractPlottable1D< QCPBarsData >.

17.615.6.15 setBarsGroup()

```
void QCPBars::setBarsGroup (
            QCPBarsGroup * barsGroup )
```

Sets to which QCPBarsGroup this QCPBars instance belongs to. Alternatively, you can also use QCPBarsGroup::append.
To remove this QCPBars from any group, set *barsGroup* to `nullptr`.

17.615.6.16 setBaseValue()

```
void QCPBars::setBaseValue (
            double baseValue )
```

Sets the base value of this bars plottable.
The base value defines where on the value coordinate the bars start. How far the bars extend from the base value is given by their individual value data. For example, if the base value is set to 1, a bar with data value 2 will have its lowest point at value coordinate 1 and highest point at 3.
For stacked bars, only the base value of the bottom-most QCPBars has meaning.
The default base value is 0.

17.615.6.17 setData() [1/2]

```
void QCPBars::setData (
            const QVector< double > & keys,
            const QVector< double > & values,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data with the provided points in *keys* and *values*. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.
If you can guarantee that the passed data points are sorted by *keys* in ascending order, you can set *alreadySorted* to true, to improve performance by saving a sorting run.

See also

> addData

17.615.6.18 setData() [2/2]

```
void QCPBars::setData (
            QSharedPointer< QCPBarsDataContainer > data )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data container with the provided *data* container.
Since a QSharedPointer is used, multiple QCPBars may share the same data container safely. Modifying the data in the container will then affect all bars that share the container. Sharing can be achieved by simply exchanging the data containers wrapped in shared pointers:
If you do not wish to share containers, but create a copy from an existing container, rather use the QCPDataContainer<DataType>::set method on the bar's data container directly:

See also

addData

17.615.6.19 setStackingGap()

```
void QCPBars::setStackingGap (
            double pixels )
```

If this bars plottable is stacked on top of another bars plottable (moveAbove), this method allows specifying a distance in *pixels*, by which the drawn bar rectangles will be separated by the bars below it.

17.615.6.20 setWidth()

```
void QCPBars::setWidth (
            double width )
```

Sets the width of the bars.
How the number passed as *width* is interpreted (e.g. screen pixels, plot coordinates,...), depends on the currently set width type, see setWidthType and WidthType.

17.615.6.21 setWidthType()

```
void QCPBars::setWidthType (
            QCPBars::WidthType widthType )
```

Sets how the width of the bars is defined. See the documentation of WidthType for an explanation of the possible values for *widthType*.
The default value is wtPlotCoords.

See also

setWidth

The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.616 QCPBarsData Class Reference

Holds the data of one single data point (one bar) for QCPBars.

Public Member Functions

- QCPBarsData ()
- QCPBarsData (double key, double value)
- double sortKey () const
- double mainKey () const
- double mainValue () const
- QCPRange valueRange () const

Static Public Member Functions

- static QCPBarsData fromSortKey (double sortKey)
- static bool sortKeyIsMainKey ()

Public Attributes

- double **key**
- double **value**

17.616.1 Detailed Description

Holds the data of one single data point (one bar) for QCPBars.
The stored data is:

- *key:* coordinate on the key axis of this bar (this is the *mainKey* and the *sortKey*)
- *value:* height coordinate on the value axis of this bar (this is the *mainValue*)

The container for storing multiple data points is QCPBarsDataContainer. It is a typedef for QCPDataContainer with QCPBarsData as the DataType template parameter. See the documentation there for an explanation regarding the data type's generic methods.

See also

QCPBarsDataContainer

17.616.2 Constructor & Destructor Documentation

17.616.2.1 QCPBarsData() [1/2]

```
QCPBarsData::QCPBarsData ( )
```

Constructs a bar data point with key and value set to zero.

17.616.2.2 QCPBarsData() [2/2]

```
QCPBarsData::QCPBarsData (
            double key,
            double value )
```

Constructs a bar data point with the specified *key* and *value*.

17.616.3 Member Function Documentation

17.616.3.1 fromSortKey()

```
static QCPBarsData QCPBarsData::fromSortKey (
            double sortKey )  [inline], [static]
```

Returns a data point with the specified *sortKey*. All other members are set to zero.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.616.3.2 mainKey()

```
double QCPBarsData::mainKey ( ) const  [inline]
```

Returns the *key* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.616.3.3 mainValue()

```
double QCPBarsData::mainValue ( ) const  [inline]
```

Returns the *value* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.616.3.4 sortKey()

```
double QCPBarsData::sortKey ( ) const  [inline]
```

Returns the *key* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.616.3.5 sortKeyIsMainKey()

```
static static bool QCPBarsData::sortKeyIsMainKey ( )  [inline], [static]
```

Since the member *key* is both the data point key coordinate and the data ordering parameter, this method returns true.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.616.3.6 valueRange()

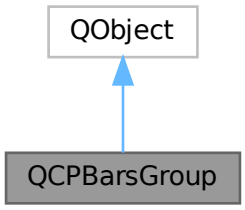
```
QCPRange QCPBarsData::valueRange ( ) const  [inline]
```

Returns a QCPRange with both lower and upper boundary set to *value* of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.
The documentation for this class was generated from the following files:

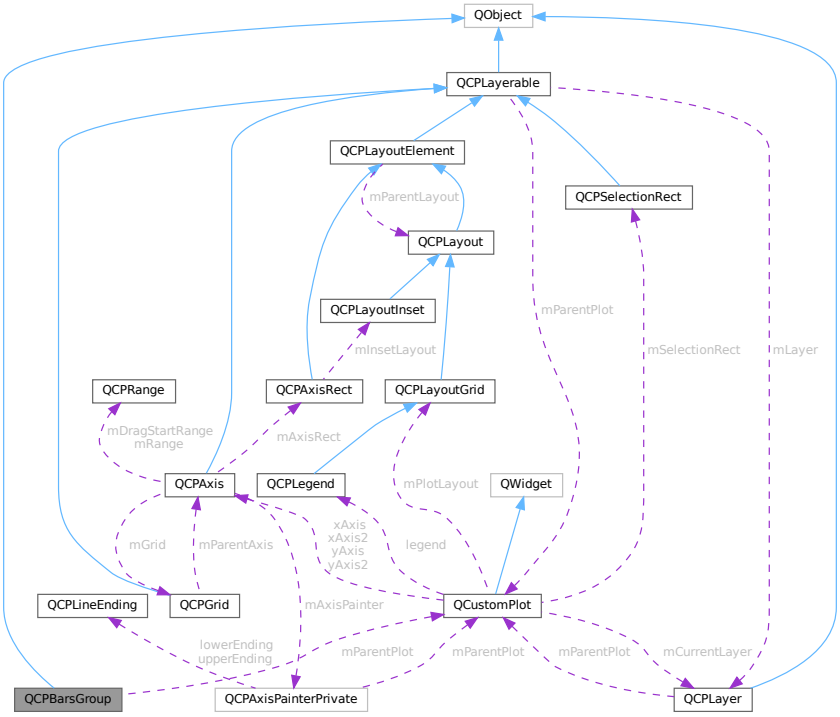
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.617 QCPBarsGroup Class Reference

Groups multiple QCPBars together so they appear side by side.
Inheritance diagram for QCPBarsGroup:



Collaboration diagram for QCPBarsGroup:



Public Types

- enum SpacingType { stAbsolute , stAxisRectRatio , stPlotCoords }

Public Member Functions

- QCPBarsGroup (QCustomPlot ∗parentPlot)
- SpacingType **spacingType** () const
- double **spacing** () const
- void setSpacingType (SpacingType spacingType)
- void setSpacing (double spacing)
- QList< QCPBars ∗ > bars () const
- QCPBars ∗ bars (int index) const
- int size () const
- bool isEmpty () const
- void clear ()
- bool contains (QCPBars ∗bars) const
- void append (QCPBars ∗bars)
- void insert (int i, QCPBars ∗bars)
- void remove (QCPBars ∗bars)

Protected Member Functions

- void **registerBars** (QCPBars ∗bars)
- void **unregisterBars** (QCPBars ∗bars)
- double **keyPixelOffset** (const QCPBars ∗bars, double keyCoord)
- double **getPixelSpacing** (const QCPBars ∗bars, double keyCoord)

Protected Attributes

- QCustomPlot * **mParentPlot**
- SpacingType **mSpacingType**
- double **mSpacing**
- QList< QCPBars * > **mBars**

Friends

- class **QCPBars**

17.617.1 Detailed Description

Groups multiple QCPBars together so they appear side by side.
When showing multiple QCPBars in one plot which have bars at identical keys, it may be desirable to have them appearing next to each other at each key. This is what adding the respective QCPBars plottables to a QCPBarsGroup achieves. (An alternative approach is to stack them on top of each other, see QCPBars::moveAbove.)

17.617.2 Usage

To add a QCPBars plottable to the group, create a new group and then add the respective bars intances:
Alternatively to appending to the group like shown above, you can also set the group on the QCPBars plottable via QCPBars::setBarsGroup.
The spacing between the bars can be configured via setSpacingType and setSpacing. The bars in this group appear in the plot in the order they were appended. To insert a bars plottable at a certain index position, or to reposition a bars plottable which is already in the group, use insert.
To remove specific bars from the group, use either remove or call QCPBars::setBarsGroup(0) on the respective bars plottable.
To clear the entire group, call clear, or simply delete the group.

17.617.3 Example

The image above is generated with the following code:

17.617.4 Member Enumeration Documentation

17.617.4.1 SpacingType

```
enum QCPBarsGroup::SpacingType
```

Defines the ways the spacing between bars in the group can be specified. Thus it defines what the number passed to setSpacing actually means.

See also

setSpacingType, setSpacing

Enumerator

stAbsolute	Bar spacing is in absolute pixels.
stAxisRectRatio	Bar spacing is given by a fraction of the axis rect size.
stPlotCoords	Bar spacing is in key coordinates and thus scales with the key axis range.

17.617.5 Constructor & Destructor Documentation

17.617.5.1 QCPBarsGroup()

```
QCPBarsGroup::QCPBarsGroup (
            QCustomPlot * parentPlot )  [explicit]
```

Constructs a new bars group for the specified QCustomPlot instance.

17.617.6 Member Function Documentation

17.617.6.1 append()

```
void QCPBarsGroup::append (
            QCPBars * bars )
```

Adds the specified *bars* plottable to this group. Alternatively, you can also use QCPBars::setBarsGroup on the *bars* instance.

See also

insert, remove

17.617.6.2 bars() [1/2]

```
QList< QCPBars * > QCPBarsGroup::bars ( ) const  [inline]
```

Returns all bars currently in this group.

See also

bars(int index)

17.617.6.3 bars() [2/2]

```
QCPBars * QCPBarsGroup::bars (
            int index ) const
```

Returns the QCPBars instance with the specified *index* in this group. If no such QCPBars exists, returns `nullptr`.

See also

bars(), size

17.617.6.4 clear()

```
void QCPBarsGroup::clear ( )
```

Removes all QCPBars plottables from this group.

See also

isEmpty

17.617.6.5 contains()

```
bool QCPBarsGroup::contains (
            QCPBars * bars ) const  [inline]
```

Returns whether the specified *bars* plottable is part of this group.

17.617.6.6 insert()

```
void QCPBarsGroup::insert (
            int i,
            QCPBars * bars )
```

Inserts the specified *bars* plottable into this group at the specified index position *i*. This gives you full control over the ordering of the bars.
bars may already be part of this group. In that case, *bars* is just moved to the new index position.

See also

append, remove

17.617.6.7 isEmpty()

```
bool QCPBarsGroup::isEmpty ( ) const  [inline]
```
Returns whether this bars group is empty.

See also

size

17.617.6.8 remove()

```
void QCPBarsGroup::remove (
            QCPBars * bars )
```
Removes the specified *bars* plottable from this group.

See also

contains, clear

17.617.6.9 setSpacing()

```
void QCPBarsGroup::setSpacing (
            double spacing )
```
Sets the spacing between adjacent bars. What the number passed as *spacing* actually means, is defined by the current SpacingType, which can be set with setSpacingType.

See also

setSpacingType

17.617.6.10 setSpacingType()

```
void QCPBarsGroup::setSpacingType (
            SpacingType spacingType )
```
Sets how the spacing between adjacent bars is interpreted. See SpacingType.
The actual spacing can then be specified with setSpacing.

See also

setSpacing

17.617.6.11 size()

```
int QCPBarsGroup::size ( ) const  [inline]
```
Returns the number of QCPBars plottables that are part of this group.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.618 QCPColorGradient Class Reference

Defines a color gradient for use with e.g. QCPColorMap.

Public Types

- enum ColorInterpolation { ciRGB , ciHSV }
- enum NanHandling {
  nhNone , nhLowestColor , nhHighestColor , nhTransparent ,
  nhNanColor }
- enum GradientPreset {
  gpGrayscale , gpHot , gpCold , gpNight ,
  gpCandy , gpGeography , gpIon , gpThermal ,
  gpPolar , gpSpectrum , gpJet , gpHues }

Public Member Functions

- QCPColorGradient ()
- QCPColorGradient (GradientPreset preset)
- bool **operator==** (const QCPColorGradient &other) const
- bool **operator!=** (const QCPColorGradient &other) const
- int **levelCount** () const
- QMap< double, QColor > **colorStops** () const
- ColorInterpolation **colorInterpolation** () const
- NanHandling **nanHandling** () const
- QColor **nanColor** () const
- bool **periodic** () const
- void setLevelCount (int n)
- void setColorStops (const QMap< double, QColor > &colorStops)
- void setColorStopAt (double position, const QColor &color)
- void setColorInterpolation (ColorInterpolation interpolation)
- void setNanHandling (NanHandling handling)
- void setNanColor (const QColor &color)
- void setPeriodic (bool enabled)
- void colorize (const double *data, const QCPRange &range, QRgb *scanLine, int n, int dataIndexFactor=1, bool logarithmic=false)
- void colorize (const double *data, const unsigned char *alpha, const QCPRange &range, QRgb *scanLine, int n, int dataIndexFactor=1, bool logarithmic=false)
- QRgb **color** (double position, const QCPRange &range, bool logarithmic=false)
- void loadPreset (GradientPreset preset)
- void clearColorStops ()
- QCPColorGradient inverted () const

Protected Member Functions

- bool **stopsUseAlpha** () const
- void **updateColorBuffer** ()

Protected Attributes

- int **mLevelCount**
- QMap< double, QColor > **mColorStops**
- ColorInterpolation **mColorInterpolation**
- NanHandling **mNanHandling**
- QColor **mNanColor**
- bool **mPeriodic**
- QVector< QRgb > **mColorBuffer**
- bool **mColorBufferInvalidated**

17.618.1 Detailed Description

Defines a color gradient for use with e.g. QCPColorMap.
This class describes a color gradient which can be used to encode data with color. For example, QCPColorMap and QCPColorScale have setGradient methods which take an instance of this class. Colors are set with setColorStopAt(double position, const QColor &color) with a *position* from 0 to 1. In between these defined color positions, the color will be interpolated linearly either in RGB or HSV space, see setColorInterpolation.
Alternatively, load one of the preset color gradients shown in the image below, with loadPreset, or by directly specifying the preset in the constructor.
Apart from red, green and blue components, the gradient also interpolates the alpha values of the configured color stops. This allows to display some portions of the data range as transparent in the plot.
How NaN values are interpreted can be configured with setNanHandling.
The constructor QCPColorGradient(GradientPreset preset) allows directly converting a GradientPreset to a QCPColorGradient. This means that you can directly pass GradientPreset to all the *setGradient* methods, e.g.:
The total number of levels used in the gradient can be set with setLevelCount. Whether the color gradient shall be applied periodically (wrapping around) to data values that lie outside the data range specified on the plottable instance can be controlled with setPeriodic.

17.618.2 Member Enumeration Documentation

17.618.2.1 ColorInterpolation

`enum QCPColorGradient::ColorInterpolation`
Defines the color spaces in which color interpolation between gradient stops can be performed.

See also

setColorInterpolation

Enumerator

ciRGB	Color channels red, green and blue are linearly interpolated.
ciHSV	Color channels hue, saturation and value are linearly interpolated (The hue is interpolated over the shortest angle distance)

17.618.2.2 GradientPreset

`enum QCPColorGradient::GradientPreset`
Defines the available presets that can be loaded with loadPreset. See the documentation there for an image of the presets.

Enumerator

gpGrayscale	Continuous lightness from black to white (suited for non-biased data representation)
gpHot	Continuous lightness from black over firey colors to white (suited for non-biased data representation)
gpCold	Continuous lightness from black over icey colors to white (suited for non-biased data representation)
gpNight	Continuous lightness from black over weak blueish colors to white (suited for non-biased data representation)
gpCandy	Blue over pink to white.
gpGeography	Colors suitable to represent different elevations on geographical maps.
gpIon	Half hue spectrum from black over purple to blue and finally green (creates banding illusion but allows more precise magnitude estimates)
gpThermal	Colors suitable for thermal imaging, ranging from dark blue over purple to orange, yellow and white.

Enumerator

gpPolar	Colors suitable to emphasize polarity around the center, with blue for negative, black in the middle and red for positive values.
gpSpectrum	An approximation of the visible light spectrum (creates banding illusion but allows more precise magnitude estimates)
gpJet	Hue variation similar to a spectrum, often used in numerical visualization (creates banding illusion but allows more precise magnitude estimates)
gpHues	Full hue cycle, with highest and lowest color red (suitable for periodic data, such as angles and phases, see setPeriodic)

17.618.2.3 NanHandling

`enum QCPColorGradient::NanHandling`

Defines how NaN data points shall appear in the plot.

See also

setNanHandling, setNanColor

Enumerator

nhNone	NaN data points are not explicitly handled and shouldn't occur in the data (this gives slight performance improvement)
nhLowestColor	NaN data points appear as the lowest color defined in this QCPColorGradient.
nhHighestColor	NaN data points appear as the highest color defined in this QCPColorGradient.
nhTransparent	NaN data points appear transparent.
nhNanColor	NaN data points appear as the color defined with setNanColor.

17.618.3 Constructor & Destructor Documentation

17.618.3.1 QCPColorGradient() [1/2]

```
QCPColorGradient::QCPColorGradient ( )
```

Constructs a new, empty QCPColorGradient with no predefined color stops. You can add own color stops with setColorStopAt.
The color level count is initialized to 350.

17.618.3.2 QCPColorGradient() [2/2]

```
QCPColorGradient::QCPColorGradient (
            GradientPreset preset )
```

Constructs a new QCPColorGradient initialized with the colors and color interpolation according to *preset*.
The color level count is initialized to 350.

17.618.4 Member Function Documentation

17.618.4.1 clearColorStops()

```
void QCPColorGradient::clearColorStops ( )
```

Clears all color stops.

See also

setColorStops, setColorStopAt

17.618.4.2 colorize() [1/2]

```
void QCPColorGradient::colorize (
            const double * data,
            const QCPRange & range,
            QRgb * scanLine,
            int n,
            int dataIndexFactor = 1,
            bool logarithmic = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
This method is used to quickly convert a *data* array to colors. The colors will be output in the array *scanLine*. Both *data* and *scanLine* must have the length *n* when passed to this function. The data range that shall be used for mapping the data value to the gradient is passed in *range*. *logarithmic* indicates whether the data values shall be mapped to colors logarithmically.
if *data* actually contains 2D-data linearized via `[row*columnCount + column]`, you can set *dataIndexFactor* to `columnCount` to convert a column instead of a row of the data array, in *scanLine*. *scanLine* will remain a regular (1D) array. This works because *data* is addressed `data[i*dataIndexFactor]`.
Use the overloaded method to additionally provide alpha map data.
The QRgb values that are placed in *scanLine* have their r, g, and b components premultiplied with alpha (see QImage::Format_ARGB32_Premultiplied).

17.618.4.3 colorize() [2/2]

```
void QCPColorGradient::colorize (
            const double * data,
            const unsigned char * alpha,
            const QCPRange & range,
            QRgb * scanLine,
            int n,
            int dataIndexFactor = 1,
            bool logarithmic = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Additionally to the other overload of colorize, this method takes the array *alpha*, which has the same size and structure as *data* and encodes the alpha information per data point.
The QRgb values that are placed in *scanLine* have their r, g and b components premultiplied with alpha (see QImage::Format_ARGB32_Premultiplied).

17.618.4.4 inverted()

```
QCPColorGradient QCPColorGradient::inverted ( ) const
```

Returns an inverted gradient. The inverted gradient has all properties as this QCPColorGradient, but the order of the color stops is inverted.

See also

setColorStops, setColorStopAt

17.618.4.5 loadPreset()

```
void QCPColorGradient::loadPreset (
            GradientPreset preset )
```

Clears the current color stops and loads the specified *preset*. A preset consists of predefined color stops and the corresponding color interpolation method.
The available presets are:

17.618.4.6 setColorInterpolation()

```
void QCPColorGradient::setColorInterpolation (
            QCPColorGradient::ColorInterpolation interpolation )
```

Sets whether the colors in between the configured color stops (see setColorStopAt) shall be interpolated linearly in RGB or in HSV color space.
For example, a sweep in RGB space from red to green will have a muddy brown intermediate color, whereas in HSV space the intermediate color is yellow.

17.618.4.7 setColorStopAt()

```
void QCPColorGradient::setColorStopAt (
            double position,
            const QColor & color )
```

Sets the *color* the gradient will have at the specified *position* (from 0 to 1). In between these color stops, the color is interpolated according to setColorInterpolation.

See also

setColorStops, clearColorStops

17.618.4.8 setColorStops()

```
void QCPColorGradient::setColorStops (
            const QMap< double, QColor > & colorStops )
```

Sets at which positions from 0 to 1 which color shall occur. The positions are the keys, the colors are the values of the passed QMap *colorStops*. In between these color stops, the color is interpolated according to setColorInterpolation.
A more convenient way to create a custom gradient may be to clear all color stops with clearColorStops (or creating a new, empty QCPColorGradient) and then adding them one by one with setColorStopAt.

See also

clearColorStops

17.618.4.9 setLevelCount()

```
void QCPColorGradient::setLevelCount (
            int n )
```

Sets the number of discretization levels of the color gradient to *n*. The default is 350 which is typically enough to create a smooth appearance. The minimum number of levels is 2.

17.618.4.10 setNanColor()

```
void QCPColorGradient::setNanColor (
            const QColor & color )
```

Sets the color that NaN data is represented by, if setNanHandling is set to ref nhNanColor.

See also

setNanHandling

17.618.4.11 setNanHandling()

```
void QCPColorGradient::setNanHandling (
            QCPColorGradient::NanHandling handling )
```

Sets how NaNs in the data are displayed in the plot.

See also

setNanColor

17.618.4.12 setPeriodic()

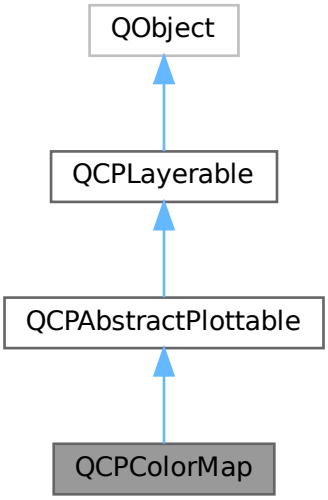
```
void QCPColorGradient::setPeriodic (
            bool enabled )
```

Sets whether data points that are outside the configured data range (e.g. QCPColorMap::setDataRange) are colored by periodically repeating the color gradient or whether they all have the same color, corresponding to the respective gradient boundary color.
As shown in the image above, gradients that have the same start and end color are especially suitable for a periodic gradient mapping, since they produce smooth color transitions throughout the color map. A preset that has this property is gpHues.
In practice, using periodic color gradients makes sense when the data corresponds to a periodic dimension, such as an angle or a phase. If this is not the case, the color encoding might become ambiguous, because multiple different data values are shown as the same color.
The documentation for this class was generated from the following files:

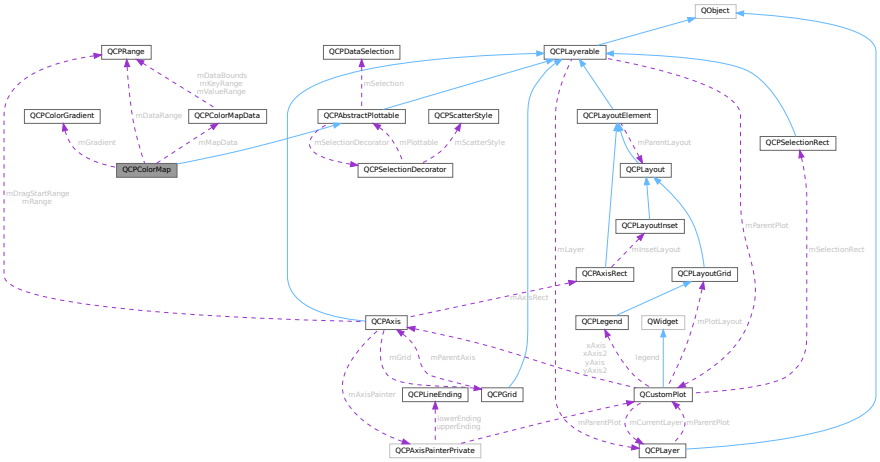
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.619 QCPColorMap Class Reference

A plottable representing a two-dimensional color map in a plot.
Inheritance diagram for QCPColorMap:



Collaboration diagram for QCPColorMap:



Signals

- void dataRangeChanged (const QCPRange &newRange)
- void dataScaleTypeChanged (QCPAxis::ScaleType scaleType)
- void gradientChanged (const QCPColorGradient &newGradient)

Signals inherited from QCPAbstractPlottable

- void selectionChanged (bool selected)
- void selectionChanged (const QCPDataSelection &selection)
- void selectableChanged (QCP::SelectionType selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPColorMap (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QCPColorMapData * data () const
- QCPRange **dataRange** () const
- QCPAxis::ScaleType **dataScaleType** () const
- bool **interpolate** () const
- bool **tightBoundary** () const
- QCPColorGradient **gradient** () const
- QCPColorScale * **colorScale** () const
- void setData (QCPColorMapData *data, bool copy=false)
- Q_SLOT void setDataRange (const QCPRange &dataRange)
- Q_SLOT void setDataScaleType (QCPAxis::ScaleType scaleType)
- Q_SLOT void setGradient (const QCPColorGradient &gradient)
- void setInterpolate (bool enabled)
- void setTightBoundary (bool enabled)
- void setColorScale (QCPColorScale *colorScale)
- void rescaleDataRange (bool recalculateDataBounds=false)
- Q_SLOT void updateLegendIcon (Qt::TransformationMode transformMode=Qt::SmoothTransformation, const QSize &thumbSize=QSize(32, 18))

- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE
- virtual QCPRange getKeyRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth) const Q_DECL_OVERRIDE
- virtual QCPRange getValueRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth, const QCPRange &inKeyRange=QCPRange()) const Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable

- QCPAbstractPlottable (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QString **name** () const
- bool **antialiasedFill** () const
- bool **antialiasedScatters** () const
- QPen **pen** () const
- QBrush **brush** () const
- QCPAxis * **keyAxis** () const
- QCPAxis * **valueAxis** () const
- QCP::SelectionType **selectable** () const
- bool selected () const
- QCPDataSelection selection () const
- QCPSelectionDecorator * selectionDecorator () const
- void setName (const QString &name)
- void setAntialiasedFill (bool enabled)
- void setAntialiasedScatters (bool enabled)
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setKeyAxis (QCPAxis *axis)
- void setValueAxis (QCPAxis *axis)
- Q_SLOT void setSelectable (QCP::SelectionType selectable)
- Q_SLOT void setSelection (QCPDataSelection selection)
- void setSelectionDecorator (QCPSelectionDecorator *decorator)
- virtual QCPPlottableInterface1D * interface1D ()
- void coordsToPixels (double key, double value, double &x, double &y) const
- const QPointF coordsToPixels (double key, double value) const
- void pixelsToCoords (double x, double y, double &key, double &value) const
- void pixelsToCoords (const QPointF &pixelPos, double &key, double &value) const
- void rescaleAxes (bool onlyEnlarge=false) const
- void rescaleKeyAxis (bool onlyEnlarge=false) const
- void rescaleValueAxis (bool onlyEnlarge=false, bool inKeyRange=false) const
- bool addToLegend (QCPLegend *legend)
- bool addToLegend ()
- bool removeFromLegend (QCPLegend *legend) const
- bool removeFromLegend () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void **updateMapImage** ()
- virtual void draw (QCPPainter ∗painter) Q_DECL_OVERRIDE
- virtual void drawLegendIcon (QCPPainter ∗painter, const QRectF &rect) const Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPAbstractPlottable

- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- void applyDefaultAntialiasingHint (QCPPainter ∗painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent ∗event, bool additive, const QVariant &details, bool ∗selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool ∗selectionStateChanged) Q_DECL_OVERRIDE
- void **applyFillAntialiasingHint** (QCPPainter ∗painter) const
- void **applyScattersAntialiasingHint** (QCPPainter ∗painter) const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot ∗parentPlot)
- virtual void mousePressEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent ∗event)
- void **initializeParentPlot** (QCustomPlot ∗parentPlot)
- void **setParentLayerable** (QCPLayerable ∗parentLayerable)
- bool **moveToLayer** (QCPLayer ∗layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter ∗painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QCPRange **mDataRange**
- QCPAxis::ScaleType **mDataScaleType**
- QCPColorMapData ∗ **mMapData**
- QCPColorGradient **mGradient**
- bool **mInterpolate**
- bool **mTightBoundary**
- QPointer< QCPColorScale > **mColorScale**
- QImage **mMapImage**
- QImage **mUndersampledMapImage**
- QPixmap **mLegendIcon**
- bool **mMapImageInvalidated**

Protected Attributes inherited from QCPAbstractPlottable

- QString **mName**
- bool **mAntialiasedFill**
- bool **mAntialiasedScatters**
- QPen **mPen**
- QBrush **mBrush**
- QPointer< QCPAxis > **mKeyAxis**
- QPointer< QCPAxis > **mValueAxis**
- QCP::SelectionType **mSelectable**
- QCPDataSelection **mSelection**
- QCPSelectionDecorator ∗ **mSelectionDecorator**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPLegend**

17.619.1 Detailed Description

A plottable representing a two-dimensional color map in a plot.
The data is stored in the class QCPColorMapData, which can be accessed via the data() method.
A color map has three dimensions to represent a data point: The *key* dimension, the *value* dimension and the *data* dimension. As with other plottables such as graphs, *key* and *value* correspond to two orthogonal axes on the QCustomPlot surface that you specify in the QCPColorMap constructor. The *data* dimension however is encoded as the color of the point at (*key*, *value*).
Set the number of points (or *cells*) in the key/value dimension via QCPColorMapData::setSize. The plot coordinate range over which these points will be displayed is specified via QCPColorMapData::setRange. The first cell will be centered on the lower range boundary and the last cell will be centered on the upper range boundary. The data can be set by either accessing the cells directly with QCPColorMapData::setCell or by addressing the cells via their plot coordinates with QCPColorMapData::setData. If possible, you should prefer setCell, since it doesn't need to do any coordinate transformation and thus performs a bit better.
The cell with index (0, 0) is at the bottom left, if the color map uses normal (i.e. not reversed) key and value axes.
To show the user which colors correspond to which *data* values, a QCPColorScale is typically placed to the right of the axis rect. See the documentation there for details on how to add and use a color scale.

17.619.2 Changing the appearance

Most important to the appearance is the color gradient, which can be specified via setGradient. See the documentation of QCPColorGradient for details on configuring a color gradient.
The *data* range that is mapped to the colors of the gradient can be specified with setDataRange. To make the data range encompass the whole data set minimum to maximum, call rescaleDataRange. If your data may contain NaN values, use QCPColorGradient::setNanHandling to define how they are displayed.

17.619.3 Transparency

Transparency in color maps can be achieved by two mechanisms. On one hand, you can specify alpha values for color stops of the QCPColorGradient, via the regular QColor interface. This will cause the color map data which gets mapped to colors around those color stops to appear with the accordingly interpolated transparency.
On the other hand you can also directly apply an alpha value to each cell independent of its data, by using the alpha map feature of QCPColorMapData. The relevant methods are QCPColorMapData::setAlpha, QCPColorMapData::fillAlpha and QCPColorMapData::clearAlpha().
The two transparencies will be joined together in the plot and otherwise not interfere with each other. They are mixed in a multiplicative matter, so an alpha of e.g. 50% (128/255) in both modes simultaneously, will result in a total transparency of 25% (64/255).

17.619.4 Usage

Like all data representing objects in QCustomPlot, the QCPColorMap is a plottable (QCPAbstractPlottable). So the plottable-interface of QCustomPlot applies (QCustomPlot::plottable, QCustomPlot::removePlottable, etc.)
Usually, you first create an instance:
which registers it with the QCustomPlot instance of the passed axes. Note that this QCustomPlot instance takes ownership of the plottable, so do not delete it manually but use QCustomPlot::removePlottable() instead. The newly created plottable can be modified, e.g.:

Note

The QCPColorMap always displays the data at equal key/value intervals, even if the key or value axis is set to a logarithmic scaling. If you want to use QCPColorMap with logarithmic axes, you shouldn't use the QCPColorMapData::setData method as it uses a linear transformation to determine the cell index. Rather directly access the cell index with QCPColorMapData::setCell.

17.619.5 Constructor & Destructor Documentation

17.619.5.1 QCPColorMap()

```
QCPColorMap::QCPColorMap (
            QCPAxis * keyAxis,
            QCPAxis * valueAxis )  [explicit]
```

Constructs a color map with the specified *keyAxis* and *valueAxis*.
The created QCPColorMap is automatically registered with the QCustomPlot instance inferred from *keyAxis*. This QCustomPlot instance takes ownership of the QCPColorMap, so do not delete it manually but use QCustomPlot::removePlottable() instead.

17.619.6 Member Function Documentation

17.619.6.1 data()

```
QCPColorMapData * QCPColorMap::data ( ) const  [inline]
```

Returns a pointer to the internal data storage of type QCPColorMapData. Access this to modify data points (cells) and the color map key/value range.

See also

setData

17.619.6.2 dataRangeChanged

```
void QCPColorMap::dataRangeChanged (
            const QCPRange & newRange )  [signal]
```

This signal is emitted when the data range changes.

See also

setDataRange

17.619.6.3 dataScaleTypeChanged

```
void QCPColorMap::dataScaleTypeChanged (
            QCPAxis::ScaleType scaleType )  [signal]
```

This signal is emitted when the data scale type changes.

See also

setDataScaleType

17.619.6.4 draw()

```
void QCPColorMap::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.619.6.5 drawLegendIcon()

```
void QCPColorMap::drawLegendIcon (
            QCPPainter * painter,
            const QRectF & rect ) const  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.619.6.6 getKeyRange()

```
QCPRange QCPColorMap::getKeyRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth ) const  [virtual]
```

Returns the coordinate range that all data in this plottable span in the key axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSignDomain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getValueRange

Implements QCPAbstractPlottable.

17.619.6.7 getValueRange()

```
QCPRange QCPColorMap::getValueRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth,
            const QCPRange & inKeyRange = QCPRange() ) const  [virtual]
```

Returns the coordinate range that the data points in the specified key range (*inKeyRange*) span in the value axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSign↩ Domain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
If *inKeyRange* has both lower and upper bound set to zero (is equal to `QCPRange()`), all data points are considered, without any restriction on the keys.
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getKeyRange

Implements QCPAbstractPlottable.

17.619.6.8 gradientChanged

```
void QCPColorMap::gradientChanged (
            const QCPColorGradient & newGradient )  [signal]
```

This signal is emitted when the gradient changes.

See also

setGradient

17.619.6.9 rescaleDataRange()

```
void QCPColorMap::rescaleDataRange (
            bool recalculateDataBounds = false )
```

Sets the data range (setDataRange) to span the minimum and maximum values that occur in the current data set. This corresponds to the rescaleKeyAxis or rescaleValueAxis methods, only for the third data dimension of the color map.

The minimum and maximum values of the data set are buffered in the internal QCPColorMapData instance (data). As data is updated via its QCPColorMapData::setCell or QCPColorMapData::setData, the buffered minimum and maximum values are updated, too. For performance reasons, however, they are only updated in an expanding fashion. So the buffered maximum can only increase and the buffered minimum can only decrease. In consequence, changes to the data that actually lower the maximum of the data set (by overwriting the cell holding the current maximum with a smaller value), aren't recognized and the buffered maximum overestimates the true maximum of the data set. The same happens for the buffered minimum. To recalculate the true minimum and maximum by explicitly looking at each cell, the method QCPColorMapData::recalculateDataBounds can be used. For convenience, setting the parameter *recalculateDataBounds* calls this method before setting the data range to the buffered minimum and maximum.

See also

setDataRange

17.619.6.10 selectTest()

```
double QCPColorMap::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.

pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.

If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).

The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.

details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.

In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.

You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Implements QCPAbstractPlottable.

17.619.6.11 setColorScale()

```
void QCPColorMap::setColorScale (
            QCPColorScale * colorScale )
```

Associates the color scale *colorScale* with this color map.

This means that both the color scale and the color map synchronize their gradient, data range and data scale type (setGradient, setDataRange, setDataScaleType). Multiple color maps can be associated with one single color scale. This causes the color maps to also synchronize those properties, via the mutual color scale.

This function causes the color map to adopt the current color gradient, data range and data scale type of *colorScale*. After this call, you may change these properties at either the color map or the color scale, and the setting will be applied to both.

Pass `nullptr` as *colorScale* to disconnect the color scale from this color map again.

17.619.6.12 setData()

```
void QCPColorMap::setData (
            QCPColorMapData * data,
            bool copy = false )
```

Replaces the current data with the provided *data*.
If *copy* is set to true, the *data* object will only be copied. if false, the color map takes ownership of the passed data and replaces the internal data pointer with it. This is significantly faster than copying for large datasets.

17.619.6.13 setDataRange()

```
void QCPColorMap::setDataRange (
            const QCPRange & dataRange )
```

Sets the data range of this color map to *dataRange*. The data range defines which data values are mapped to the color gradient.
To make the data range span the full range of the data set, use rescaleDataRange.

See also

QCPColorScale::setDataRange

17.619.6.14 setDataScaleType()

```
void QCPColorMap::setDataScaleType (
            QCPAxis::ScaleType scaleType )
```

Sets whether the data is correlated with the color gradient linearly or logarithmically.

See also

QCPColorScale::setDataScaleType

17.619.6.15 setGradient()

```
void QCPColorMap::setGradient (
            const QCPColorGradient & gradient )
```

Sets the color gradient that is used to represent the data. For more details on how to create an own gradient or use one of the preset gradients, see QCPColorGradient.
The colors defined by the gradient will be used to represent data values in the currently set data range, see setDataRange. Data points that are outside this data range will either be colored uniformly with the respective gradient boundary color, or the gradient will repeat, depending on QCPColorGradient::setPeriodic.

See also

QCPColorScale::setGradient

17.619.6.16 setInterpolate()

```
void QCPColorMap::setInterpolate (
            bool enabled )
```

Sets whether the color map image shall use bicubic interpolation when displaying the color map shrinked or expanded, and not at a 1:1 pixel-to-data scale.

17.619.6.17 setTightBoundary()

```
void QCPColorMap::setTightBoundary (
            bool enabled )
```

Sets whether the outer most data rows and columns are clipped to the specified key and value range (see QCPColorMapData::setKeyRange, QCPColorMapData::setValueRange).
if *enabled* is set to false, the data points at the border of the color map are drawn with the same width and height as all other data points. Since the data points are represented by rectangles of one color centered on the data coordinate, this means that the shown color map extends by half a data point over the specified key/value range in each direction.

17.619.6.18 updateLegendIcon()

```
void QCPColorMap::updateLegendIcon (
            Qt::TransformationMode transformMode = Qt::SmoothTransformation,
            const QSize & thumbSize = QSize(32, 18) )
```

Takes the current appearance of the color map and updates the legend icon, which is used to represent this color map in the legend (see QCPLegend).
The *transformMode* specifies whether the rescaling is done by a faster, low quality image scaling algorithm (Qt::FastTransformation) or by a slower, higher quality algorithm (Qt::SmoothTransformation).
The current color map appearance is scaled down to *thumbSize*. Ideally, this should be equal to the size of the legend icon (see QCPLegend::setIconSize). If it isn't exactly the configured legend icon size, the thumb will be rescaled during drawing of the legend item.

See also

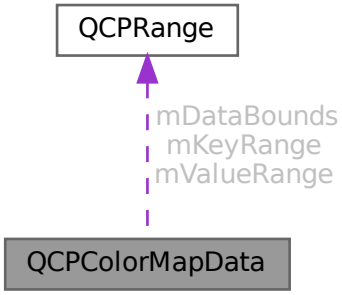
setDataRange

The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.620 QCPColorMapData Class Reference

Holds the two-dimensional data of a QCPColorMap plottable.
Collaboration diagram for QCPColorMapData:



Public Member Functions

- QCPColorMapData (int keySize, int valueSize, const QCPRange &keyRange, const QCPRange &valueRange)
- QCPColorMapData (const QCPColorMapData &other)
- QCPColorMapData & operator= (const QCPColorMapData &other)
- int **keySize** () const
- int **valueSize** () const
- QCPRange **keyRange** () const
- QCPRange **valueRange** () const
- QCPRange **dataBounds** () const
- double **data** (double key, double value)
- double **cell** (int keyIndex, int valueIndex)
- unsigned char alpha (int keyIndex, int valueIndex)

- void setSize (int keySize, int valueSize)
- void setKeySize (int keySize)
- void setValueSize (int valueSize)
- void setRange (const QCPRange &keyRange, const QCPRange &valueRange)
- void setKeyRange (const QCPRange &keyRange)
- void setValueRange (const QCPRange &valueRange)
- void setData (double key, double value, double z)
- void setCell (int keyIndex, int valueIndex, double z)
- void setAlpha (int keyIndex, int valueIndex, unsigned char alpha)
- void recalculateDataBounds ()
- void clear ()
- void clearAlpha ()
- void fill (double z)
- void fillAlpha (unsigned char alpha)
- bool isEmpty () const
- void coordToCell (double key, double value, int *keyIndex, int *valueIndex) const
- void cellToCoord (int keyIndex, int valueIndex, double *key, double *value) const

Protected Member Functions

- bool **createAlpha** (bool initializeOpaque=true)

Protected Attributes

- int **mKeySize**
- int **mValueSize**
- QCPRange **mKeyRange**
- QCPRange **mValueRange**
- bool **mIsEmpty**
- double * **mData**
- unsigned char * **mAlpha**
- QCPRange **mDataBounds**
- bool **mDataModified**

Friends

- class **QCPColorMap**

17.620.1 Detailed Description

Holds the two-dimensional data of a QCPColorMap plottable.
This class is a data storage for QCPColorMap. It holds a two-dimensional array, which QCPColorMap then displays as a 2D image in the plot, where the array values are represented by a color, depending on the value.
The size of the array can be controlled via setSize (or setKeySize, setValueSize). Which plot coordinates these cells correspond to can be configured with setRange (or setKeyRange, setValueRange).
The data cells can be accessed in two ways: They can be directly addressed by an integer index with setCell. This is the fastest method. Alternatively, they can be addressed by their plot coordinate with setData. plot coordinate to cell index transformations and vice versa are provided by the functions coordToCell and cellToCoord.
A QCPColorMapData also holds an on-demand two-dimensional array of alpha values which (if allocated) has the same size as the data map. It can be accessed via setAlpha, fillAlpha and clearAlpha. The memory for the alpha map is only allocated if needed, i.e. on the first call of setAlpha. clearAlpha restores full opacity and frees the alpha map.
This class also buffers the minimum and maximum values that are in the data set, to provide QCPColorMap::rescaleDataRange with the necessary information quickly. Setting a cell to a value that is greater than the current maximum increases this maximum to the new value. However, setting the cell that currently holds the maximum value to a smaller value doesn't decrease the maximum again, because finding the true new maximum would require going through the entire data array, which might be time consuming. The same holds for the data minimum. This functionality is

given by recalculateDataBounds, such that you can decide when it is sensible to find the true current minimum and maximum. The method QCPColorMap::rescaleDataRange offers a convenience parameter *recalculateDataBounds* which may be set to true to automatically call recalculateDataBounds internally.

17.620.2 Constructor & Destructor Documentation

17.620.2.1 QCPColorMapData() [1/2]

```
QCPColorMapData::QCPColorMapData (
            int keySize,
            int valueSize,
            const QCPRange & keyRange,
            const QCPRange & valueRange )
```

Constructs a new QCPColorMapData instance. The instance has *keySize* cells in the key direction and *valueSize* cells in the value direction. These cells will be displayed by the QCPColorMap at the coordinates *keyRange* and *valueRange*.

See also

setSize, setKeySize, setValueSize, setRange, setKeyRange, setValueRange

17.620.2.2 QCPColorMapData() [2/2]

```
QCPColorMapData::QCPColorMapData (
            const QCPColorMapData & other )
```

Constructs a new QCPColorMapData instance copying the data and range of *other*.

17.620.3 Member Function Documentation

17.620.3.1 alpha()

```
unsigned char QCPColorMapData::alpha (
            int keyIndex,
            int valueIndex )
```

Returns the alpha map value of the cell with the indices *keyIndex* and *valueIndex*.
If this color map data doesn't have an alpha map (because setAlpha was never called after creation or after a call to clearAlpha), returns 255, which corresponds to full opacity.

See also

setAlpha

17.620.3.2 cellToCoord()

```
void QCPColorMapData::cellToCoord (
            int keyIndex,
            int valueIndex,
            double * key,
            double * value ) const
```

Transforms cell indices given by *keyIndex* and *valueIndex* to cell indices of this QCPColorMapData instance. The resulting coordinates are returned via the output parameters *key* and *value*.
If you are only interested in a key or value coordinate, you may pass `nullptr` as *key* or *value*.

Note

The QCPColorMap always displays the data at equal key/value intervals, even if the key or value axis is set to a logarithmic scaling. If you want to use QCPColorMap with logarithmic axes, you shouldn't use the QCPColorMapData::cellToCoord method as it uses a linear transformation to determine the cell index.

See also

coordToCell, QCPAxis::pixelToCoord

17.620.3.3 clear()

```
void QCPColorMapData::clear ( )
```

Frees the internal data memory.
This is equivalent to calling setSize(0, 0).

17.620.3.4 clearAlpha()

```
void QCPColorMapData::clearAlpha ( )
```

Frees the internal alpha map. The color map will have full opacity again.

17.620.3.5 coordToCell()

```
void QCPColorMapData::coordToCell (
            double key,
            double value,
            int * keyIndex,
            int * valueIndex ) const
```

Transforms plot coordinates given by *key* and *value* to cell indices of this QCPColorMapData instance. The resulting cell indices are returned via the output parameters *keyIndex* and *valueIndex*.
The retrieved key/value cell indices can then be used for example with setCell.
If you are only interested in a key or value index, you may pass `nullptr` as *valueIndex* or *keyIndex*.

Note

The QCPColorMap always displays the data at equal key/value intervals, even if the key or value axis is set to a logarithmic scaling. If you want to use QCPColorMap with logarithmic axes, you shouldn't use the QCPColorMapData::coordToCell method as it uses a linear transformation to determine the cell index.

See also

cellToCoord, QCPAxis::coordToPixel

17.620.3.6 fill()

```
void QCPColorMapData::fill (
            double z )
```

Sets all cells to the value *z*.

17.620.3.7 fillAlpha()

```
void QCPColorMapData::fillAlpha (
            unsigned char alpha )
```

Sets the opacity of all color map cells to *alpha*. A value of 0 for *alpha* results in a fully transparent color map, and a value of 255 results in a fully opaque color map.
If you wish to restore opacity to 100% and free any used memory for the alpha map, rather use clearAlpha.

See also

setAlpha

17.620.3.8 isEmpty()

```
bool QCPColorMapData::isEmpty ( ) const  [inline]
```

Returns whether this instance carries no data. This is equivalent to having a size where at least one of the dimensions is 0 (see setSize).

17.620.3.9 operator=()

```
QCPColorMapData & QCPColorMapData::operator= (
            const QCPColorMapData & other )
```

Overwrites this color map data instance with the data stored in *other*. The alpha map state is transferred, too.

17.620.3.10 recalculateDataBounds()

```
void QCPColorMapData::recalculateDataBounds ( )
```

Goes through the data and updates the buffered minimum and maximum data values.
Calling this method is only advised if you are about to call QCPColorMap::rescaleDataRange and can not guarantee that the cells holding the maximum or minimum data haven't been overwritten with a smaller or larger value respectively, since the buffered maximum/minimum values have been updated the last time. Why this is the case is explained in the class description (QCPColorMapData).
Note that the method QCPColorMap::rescaleDataRange provides a parameter *recalculateDataBounds* for convenience. Setting this to true will call this method for you, before doing the rescale.

17.620.3.11 setAlpha()

```
void QCPColorMapData::setAlpha (
            int keyIndex,
            int valueIndex,
            unsigned char alpha )
```

Sets the alpha of the color map cell given by *keyIndex* and *valueIndex* to *alpha*. A value of 0 for *alpha* results in a fully transparent cell, and a value of 255 results in a fully opaque cell.
If an alpha map doesn't exist yet for this color map data, it will be created here. If you wish to restore full opacity and free any allocated memory of the alpha map, call clearAlpha.
Note that the cell-wise alpha which can be configured here is independent of any alpha configured in the color map's gradient (QCPColorGradient). If a cell is affected both by the cell-wise and gradient alpha, the alpha values will be blended accordingly during rendering of the color map.

See also

fillAlpha, clearAlpha

17.620.3.12 setCell()

```
void QCPColorMapData::setCell (
            int keyIndex,
            int valueIndex,
            double z )
```

Sets the data of the cell with indices *keyIndex* and *valueIndex* to *z*. The indices enumerate the cells starting from zero, up to the map's size-1 in the respective dimension (see setSize).
In the standard plot configuration (horizontal key axis and vertical value axis, both not range-reversed), the cell with indices (0, 0) is in the bottom left corner and the cell with indices (keySize-1, valueSize-1) is in the top right corner of the color map.

See also

setData, setSize

17.620.3.13 setData()

```
void QCPColorMapData::setData (
            double key,
            double value,
            double z )
```

Sets the data of the cell, which lies at the plot coordinates given by *key* and *value*, to *z*.

Note

The QCPColorMap always displays the data at equal key/value intervals, even if the key or value axis is set to a logarithmic scaling. If you want to use QCPColorMap with logarithmic axes, you shouldn't use the QCPColorMapData::setData method as it uses a linear transformation to determine the cell index. Rather directly access the cell index with QCPColorMapData::setCell.

See also

setCell, setRange

17.620.3.14 setKeyRange()

```
void QCPColorMapData::setKeyRange (
            const QCPRange & keyRange )
```

Sets the coordinate range the data shall be distributed over in the key dimension. Together with the value range, This defines the rectangular area covered by the color map in plot coordinates.
The outer cells will be centered on the range boundaries given to this function. For example, if the key size (setKeySize) is 3 and *keyRange* is set to `QCPRange(2, 3)` there will be cells centered on the key coordinates 2, 2.5 and 3.

See also

setRange, setValueRange, setSize

17.620.3.15 setKeySize()

```
void QCPColorMapData::setKeySize (
            int keySize )
```

Resizes the data array to have *keySize* cells in the key dimension.
The current data is discarded and the map cells are set to 0, unless the map had already the requested size.
Setting *keySize* to zero frees the internal data array and isEmpty returns true.

See also

setKeyRange, setSize, setValueSize

17.620.3.16 setRange()

```
void QCPColorMapData::setRange (
            const QCPRange & keyRange,
            const QCPRange & valueRange )
```

Sets the coordinate ranges the data shall be distributed over. This defines the rectangular area covered by the color map in plot coordinates.
The outer cells will be centered on the range boundaries given to this function. For example, if the key size (setKeySize) is 3 and *keyRange* is set to `QCPRange(2, 3)` there will be cells centered on the key coordinates 2, 2.5 and 3.

See also

setSize

17.620.3.17 setSize()

```
void QCPColorMapData::setSize (
            int keySize,
            int valueSize )
```

Resizes the data array to have *keySize* cells in the key dimension and *valueSize* cells in the value dimension.
The current data is discarded and the map cells are set to 0, unless the map had already the requested size.
Setting at least one of *keySize* or *valueSize* to zero frees the internal data array and isEmpty returns true.

See also

setRange, setKeySize, setValueSize

17.620.3.18 setValueRange()

```
void QCPColorMapData::setValueRange (
            const QCPRange & valueRange )
```

Sets the coordinate range the data shall be distributed over in the value dimension. Together with the key range, This defines the rectangular area covered by the color map in plot coordinates.
The outer cells will be centered on the range boundaries given to this function. For example, if the value size (setValueSize) is 3 and *valueRange* is set to `QCPRange(2, 3)` there will be cells centered on the value coordinates 2, 2.5 and 3.

See also

setRange, setKeyRange, setSize

17.620.3.19 setValueSize()

```
void QCPColorMapData::setValueSize (
            int valueSize )
```

Resizes the data array to have *valueSize* cells in the value dimension.
The current data is discarded and the map cells are set to 0, unless the map had already the requested size.
Setting *valueSize* to zero frees the internal data array and isEmpty returns true.

See also

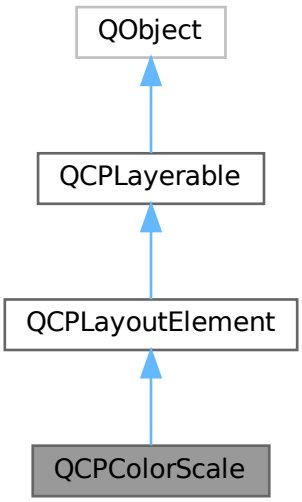
setValueRange, setSize, setKeySize

The documentation for this class was generated from the following files:

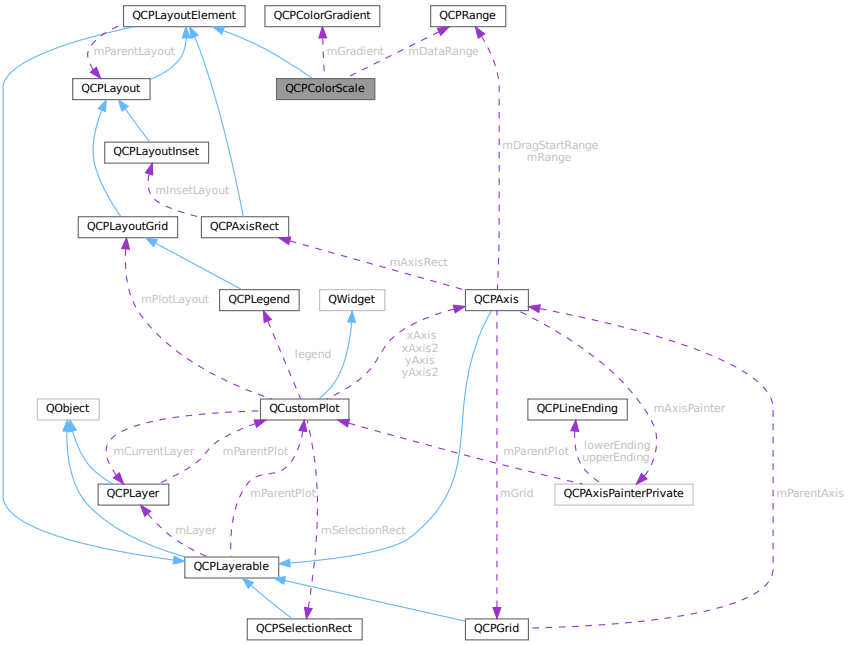
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.621 QCPColorScale Class Reference

A color scale for use with color coding data such as QCPColorMap.
Inheritance diagram for QCPColorScale:



Collaboration diagram for QCPColorScale:



Signals

- void dataRangeChanged (const QCPRange &newRange)
- void dataScaleTypeChanged (QCPAxis::ScaleType scaleType)
- void gradientChanged (const QCPColorGradient &newGradient)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPColorScale (QCustomPlot *parentPlot)
- QCPAxis * axis () const
- QCPAxis::AxisType **type** () const
- QCPRange **dataRange** () const
- QCPAxis::ScaleType **dataScaleType** () const
- QCPColorGradient **gradient** () const
- QString **label** () const
- int **barWidth** () const
- bool **rangeDrag** () const
- bool **rangeZoom** () const
- void setType (QCPAxis::AxisType type)
- Q_SLOT void setDataRange (const QCPRange &dataRange)
- Q_SLOT void setDataScaleType (QCPAxis::ScaleType scaleType)
- Q_SLOT void setGradient (const QCPColorGradient &gradient)
- void setLabel (const QString &str)
- void setBarWidth (int width)
- void setRangeDrag (bool enabled)
- void setRangeZoom (bool enabled)

- QList< QCPColorMap * > colorMaps () const
- void rescaleDataRange (bool onlyVisibleMaps)
- virtual void update (UpdatePhase phase) Q_DECL_OVERRIDE

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot *parentPlot=nullptr)
- QCPLayout * layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const
- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)
- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual QSize minimumOuterSizeHint () const
- virtual QSize maximumOuterSizeHint () const
- virtual QList< QCPLayoutElement * > elements (bool recursive) const
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void wheelEvent (QWheelEvent *event) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayoutElement

- virtual int **calculateAutoMargin** (QCP::MarginSide side)
- virtual void **layoutChanged** ()
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void **selectEvent** (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged)
- virtual void **deselectEvent** (bool *selectionStateChanged)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QCPAxis::AxisType **mType**
- QCPRange **mDataRange**
- QCPAxis::ScaleType **mDataScaleType**
- QCPColorGradient **mGradient**
- int **mBarWidth**
- QPointer< QCPColorScaleAxisRectPrivate > **mAxisRect**
- QPointer< QCPAxis > **mColorAxis**

Protected Attributes inherited from QCPLayoutElement

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCPColorScaleAxisRectPrivate**

Additional Inherited Members

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

17.621.1 Detailed Description

A color scale for use with color coding data such as QCPColorMap.
This layout element can be placed on the plot to correlate a color gradient with data values. It is usually used in combination with one or multiple QCPColorMaps.
The color scale can be either horizontal or vertical, as shown in the image above. The orientation and the side where the numbers appear is controlled with setType.
Use QCPColorMap::setColorScale to connect a color map with a color scale. Once they are connected, they share their gradient, data range and data scale type (setGradient, setDataRange, setDataScaleType). Multiple color maps may be associated with a single color scale, to make them all synchronize these properties.
To have finer control over the number display and axis behaviour, you can directly access the axis. See the documentation of QCPAxis for details about configuring axes. For example, if you want to change the number of automatically generated ticks, call
Placing a color scale next to the main axis rect works like with any other layout element:
In this case we have placed it to the right of the default axis rect, so it wasn't necessary to call setType, since QCPAxis::atRight is already the default. The text next to the color scale can be set with setLabel.
For optimum appearance (like in the image above), it may be desirable to line up the axis rect and the borders of the color scale. Use a QCPMarginGroup to achieve this:
Color scales are initialized with a non-zero minimum top and bottom margin (setMinimumMargins), because vertical color scales are most common and the minimum top/bottom margin makes sure it keeps some distance to the top/bottom widget border. So if you change to a horizontal color scale by setting setType to QCPAxis::atBottom or QCPAxis::atTop, you might want to also change the minimum margins accordingly, e.g. `setMinimum↩Margins(QMargins(6, 0, 6, 0))`.

17.621.2 Constructor & Destructor Documentation

17.621.2.1 QCPColorScale()

```
QCPColorScale::QCPColorScale (
            QCustomPlot * parentPlot )  [explicit]
```

Constructs a new QCPColorScale.

17.621.3 Member Function Documentation

17.621.3.1 applyDefaultAntialiasingHint()

```
void QCPColorScale::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```

Reimplemented from QCPLayoutElement.

17.621.3.2 axis()

```
QCPAxis * QCPColorScale::axis ( ) const  [inline]
```

Returns the internal QCPAxis instance of this color scale. You can access it to alter the appearance and behaviour of the axis. QCPColorScale duplicates some properties in its interface for convenience. Those are setDataRange (QCPAxis::setRange), setDataScaleType (QCPAxis::setScaleType), and the method setLabel (QCPAxis::setLabel). As they each are connected, it does not matter whether you use the method on the QCPColorScale or on its QCPAxis.
If the type of the color scale is changed with setType, the axis returned by this method will change, too, to either the left, right, bottom or top axis, depending on which type was set.

17.621.3.3 colorMaps()

```
QList< QCPColorMap * > QCPColorScale::colorMaps ( ) const
```
Returns a list of all the color maps associated with this color scale.

17.621.3.4 dataRangeChanged

```
void QCPColorScale::dataRangeChanged (
            const QCPRange & newRange )  [signal]
```
This signal is emitted when the data range changes.

See also

setDataRange

17.621.3.5 dataScaleTypeChanged

```
void QCPColorScale::dataScaleTypeChanged (
            QCPAxis::ScaleType scaleType )  [signal]
```
This signal is emitted when the data scale type changes.

See also

setDataScaleType

17.621.3.6 gradientChanged

```
void QCPColorScale::gradientChanged (
            const QCPColorGradient & newGradient )  [signal]
```
This signal is emitted when the gradient changes.

See also

setGradient

17.621.3.7 mouseMoveEvent()

```
void QCPColorScale::mouseMoveEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```
This event gets called when the user moves the mouse while holding a mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.621.3.8 mousePressEvent()

```
void QCPColorScale::mousePressEvent (
            QMouseEvent * event,
            const QVariant & details )  [protected], [virtual]
```
This event gets called when the user presses a mouse button while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *details* contains layerable-specific details about the hit, which were generated in the previous call to selectTest.

For example, One-dimensional plottables like QCPGraph or QCPBars convey the clicked data point in the *details* parameter, as QCPDataSelection packed as QVariant. Multi-part objects convey the specific `SelectablePart` that was hit (e.g. QCPAxis::SelectablePart in the case of axes).
QCustomPlot uses an event propagation system that works the same as Qt's system. If your layerable doesn't reimplement the mousePressEvent or explicitly calls `event->ignore()` in its reimplementation, the event will be propagated to the next layerable in the stacking order.
Once a layerable has accepted the mousePressEvent, it is considered the mouse grabber and will receive all following calls to mouseMoveEvent or mouseReleaseEvent for this mouse interaction (a "mouse interaction" in this context ends with the release).
The default implementation does nothing except explicitly ignoring the event with `event->ignore()`.

See also

mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.621.3.9 mouseReleaseEvent()

```
void QCPColorScale::mouseReleaseEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user releases the mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseMoveEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.621.3.10 rescaleDataRange()

```
void QCPColorScale::rescaleDataRange (
            bool onlyVisibleMaps )
```

Changes the data range such that all color maps associated with this color scale are fully mapped to the gradient in the data dimension.

See also

setDataRange

17.621.3.11 setBarWidth()

```
void QCPColorScale::setBarWidth (
            int width )
```

Sets the width (or height, for horizontal color scales) the bar where the gradient is displayed will have.

17.621.3.12 setDataRange()

```
void QCPColorScale::setDataRange (
            const QCPRange & dataRange )
```

Sets the range spanned by the color gradient and that is shown by the axis in the color scale.
It is equivalent to calling QCPColorMap::setDataRange on any of the connected color maps. It is also equivalent to directly accessing the axis and setting its range with QCPAxis::setRange.

See also

setDataScaleType, setGradient, rescaleDataRange

17.621.3.13 setDataScaleType()

```
void QCPColorScale::setDataScaleType (
            QCPAxis::ScaleType scaleType )
```

Sets the scale type of the color scale, i.e. whether values are associated with colors linearly or logarithmically.
It is equivalent to calling QCPColorMap::setDataScaleType on any of the connected color maps. It is also equivalent to directly accessing the axis and setting its scale type with QCPAxis::setScaleType.
Note that this method controls the coordinate transformation. For logarithmic scales, you will likely also want to use a logarithmic tick spacing and labeling, which can be achieved by setting the color scale's axis ticker to an instance of QCPAxisTickerLog :
See the documentation of QCPAxisTickerLog about the details of logarithmic axis tick creation.

See also

setDataRange, setGradient

17.621.3.14 setGradient()

```
void QCPColorScale::setGradient (
            const QCPColorGradient & gradient )
```

Sets the color gradient that will be used to represent data values.
It is equivalent to calling QCPColorMap::setGradient on any of the connected color maps.

See also

setDataRange, setDataScaleType

17.621.3.15 setLabel()

```
void QCPColorScale::setLabel (
            const QString & str )
```

Sets the axis label of the color scale. This is equivalent to calling QCPAxis::setLabel on the internal axis.

17.621.3.16 setRangeDrag()

```
void QCPColorScale::setRangeDrag (
            bool enabled )
```

Sets whether the user can drag the data range (setDataRange).
Note that QCP::iRangeDrag must be in the QCustomPlot's interactions (QCustomPlot::setInteractions) to allow range dragging.

17.621.3.17 setRangeZoom()

```
void QCPColorScale::setRangeZoom (
            bool enabled )
```

Sets whether the user can zoom the data range (setDataRange) by scrolling the mouse wheel.
Note that QCP::iRangeZoom must be in the QCustomPlot's interactions (QCustomPlot::setInteractions) to allow range dragging.

17.621.3.18 setType()

```
void QCPColorScale::setType (
            QCPAxis::AxisType type )
```

Sets at which side of the color scale the axis is placed, and thus also its orientation.
Note that after setting *type* to a different value, the axis returned by axis() will be a different one. The new axis will adopt the following properties from the previous axis: The range, scale type, label and ticker (the latter will be shared and not copied).

17.621.3.19 update()

```
void QCPColorScale::update (
            UpdatePhase phase )  [virtual]
```

Updates the layout element and sub-elements. This function is automatically called before every replot by the parent layout element. It is called multiple times, once for every UpdatePhase. The phases are run through in the order of the enum values. For details about what happens at the different phases, see the documentation of UpdatePhase.
Layout elements that have child elements should call the update method of their child elements, and pass the current *phase* unchanged.
The default implementation executes the automatic margin mechanism in the upMargins phase. Subclasses should make sure to call the base class implementation.
Reimplemented from QCPLayoutElement.

17.621.3.20 wheelEvent()

```
void QCPColorScale::wheelEvent (
            QWheelEvent * event )  [protected], [virtual]
```

This event gets called when the user turns the mouse scroll wheel while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`.
The `event->angleDelta()` indicates how far the mouse wheel was turned, which is usually +/- 120 for single rotation steps. However, if the mouse wheel is turned rapidly, multiple steps may accumulate to one event, making the delta larger. On the other hand, if the wheel has very smooth steps or none at all, the delta may be smaller.
The default implementation does nothing.

See also

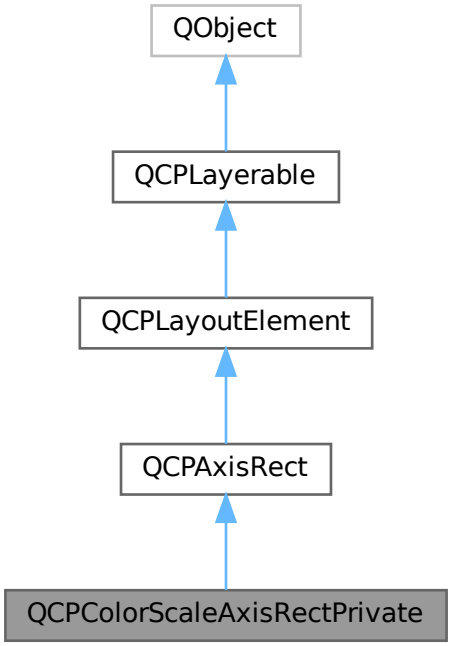
mousePressEvent, mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent

Reimplemented from QCPLayerable.
The documentation for this class was generated from the following files:

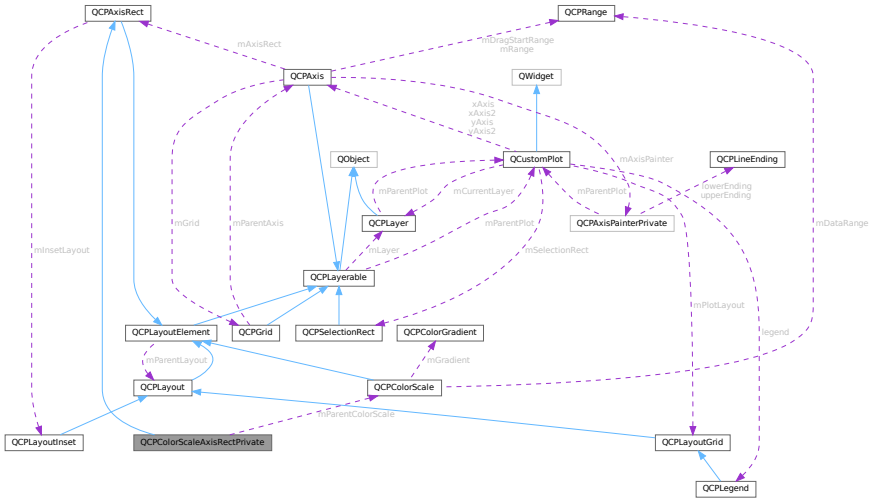
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.622 QCPColorScaleAxisRectPrivate Class Reference

Inheritance diagram for QCPColorScaleAxisRectPrivate:



Collaboration diagram for QCPColorScaleAxisRectPrivate:



Public Member Functions

- QCPColorScaleAxisRectPrivate (QCPColorScale *parentColorScale)

Public Member Functions inherited from QCPAxisRect

- QCPAxisRect (QCustomPlot *parentPlot, bool setupDefaultAxes=true)
- QPixmap **background** () const
- QBrush **backgroundBrush** () const
- bool **backgroundScaled** () const
- Qt::AspectRatioMode **backgroundScaledMode** () const
- Qt::Orientations **rangeDrag** () const
- Qt::Orientations **rangeZoom** () const
- QCPAxis * rangeDragAxis (Qt::Orientation orientation)
- QCPAxis * rangeZoomAxis (Qt::Orientation orientation)
- QList< QCPAxis * > rangeDragAxes (Qt::Orientation orientation)
- QList< QCPAxis * > rangeZoomAxes (Qt::Orientation orientation)
- double rangeZoomFactor (Qt::Orientation orientation)
- void setBackground (const QPixmap &pm)
- void setBackground (const QPixmap &pm, bool scaled, Qt::AspectRatioMode mode=Qt::KeepAspectRatio↩ ByExpanding)
- void setBackground (const QBrush &brush)
- void setBackgroundScaled (bool scaled)
- void setBackgroundScaledMode (Qt::AspectRatioMode mode)
- void setRangeDrag (Qt::Orientations orientations)
- void setRangeZoom (Qt::Orientations orientations)
- void setRangeDragAxes (QCPAxis *horizontal, QCPAxis *vertical)
- void setRangeDragAxes (QList< QCPAxis * > axes)
- void setRangeDragAxes (QList< QCPAxis * > horizontal, QList< QCPAxis * > vertical)
- void setRangeZoomAxes (QCPAxis *horizontal, QCPAxis *vertical)
- void setRangeZoomAxes (QList< QCPAxis * > axes)
- void setRangeZoomAxes (QList< QCPAxis * > horizontal, QList< QCPAxis * > vertical)
- void setRangeZoomFactor (double horizontalFactor, double verticalFactor)
- void setRangeZoomFactor (double factor)
- int axisCount (QCPAxis::AxisType type) const
- QCPAxis * axis (QCPAxis::AxisType type, int index=0) const
- QList< QCPAxis * > axes (QCPAxis::AxisTypes types) const
- QList< QCPAxis * > axes () const
- QCPAxis * addAxis (QCPAxis::AxisType type, QCPAxis *axis=nullptr)
- QList< QCPAxis * > addAxes (QCPAxis::AxisTypes types)
- bool removeAxis (QCPAxis *axis)
- QCPLayoutInset * insetLayout () const
- void zoom (const QRectF &pixelRect)
- void zoom (const QRectF &pixelRect, const QList< QCPAxis * > &affectedAxes)
- void setupFullAxesBox (bool connectRanges=false)
- QList< QCPAbstractPlottable * > plottables () const
- QList< QCPGraph * > graphs () const
- QList< QCPAbstractItem * > items () const
- int left () const
- int right () const
- int top () const
- int bottom () const
- int width () const
- int height () const
- QSize size () const
- QPoint topLeft () const
- QPoint topRight () const
- QPoint bottomLeft () const
- QPoint bottomRight () const
- QPoint center () const
- virtual QList< QCPLayoutElement * > elements (bool recursive) const Q_DECL_OVERRIDE

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot *parentPlot=nullptr)
- QCPLayout * layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const
- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)
- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual QSize minimumOuterSizeHint () const
- virtual QSize maximumOuterSizeHint () const
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- void **updateGradientImage** ()
- Q_SLOT void **axisSelectionChanged** (QCPAxis::SelectableParts selectedParts)
- Q_SLOT void **axisSelectableChanged** (QCPAxis::SelectableParts selectableParts)
- virtual int calculateAutoMargin (QCP::MarginSide side) Q_DECL_OVERRIDE
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void wheelEvent (QWheelEvent *event) Q_DECL_OVERRIDE
- virtual void update (UpdatePhase phase) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPAxisRect

- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void layoutChanged () Q_DECL_OVERRIDE
- void **drawBackground** (QCPPainter *painter)
- void **updateAxesOffset** (QCPAxis::AxisType type)

Protected Member Functions inherited from QCPLayoutElement

- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void **selectEvent** (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged)
- virtual void **deselectEvent** (bool *selectionStateChanged)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QCPColorScale * **mParentColorScale**
- QImage **mGradientImage**
- bool **mGradientImageInvalidated**

Protected Attributes inherited from QCPAxisRect

- QBrush **mBackgroundBrush**
- QPixmap **mBackgroundPixmap**
- QPixmap **mScaledBackgroundPixmap**
- bool **mBackgroundScaled**
- Qt::AspectRatioMode **mBackgroundScaledMode**
- QCPLayoutInset * **mInsetLayout**
- Qt::Orientations **mRangeDrag**
- Qt::Orientations **mRangeZoom**
- QList< QPointer< QCPAxis > > **mRangeDragHorzAxis**
- QList< QPointer< QCPAxis > > **mRangeDragVertAxis**
- QList< QPointer< QCPAxis > > **mRangeZoomHorzAxis**
- QList< QPointer< QCPAxis > > **mRangeZoomVertAxis**
- double **mRangeZoomFactorHorz**
- double **mRangeZoomFactorVert**
- QList< QCPRange > **mDragStartHorzRange**
- QList< QCPRange > **mDragStartVertRange**
- QCP::AntialiasedElements **mAADragBackup**
- QCP::AntialiasedElements **mNotAADragBackup**
- bool **mDragging**
- QHash< QCPAxis::AxisType, QList< QCPAxis * > > **mAxes**

Protected Attributes inherited from QCPLayoutElement

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCPColorScale**

Additional Inherited Members

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.622.1 Constructor & Destructor Documentation

17.622.1.1 QCPColorScaleAxisRectPrivate()

```
QCPColorScaleAxisRectPrivate::QCPColorScaleAxisRectPrivate (
            QCPColorScale * parentColorScale )  [explicit]
```

Creates a new instance, as a child of *parentColorScale*.

17.622.2 Member Function Documentation

17.622.2.1 calculateAutoMargin()

```
int QCPAxisRect::calculateAutoMargin (
            QCP::MarginSide side )  [protected], [virtual]
```

Reimplemented from QCPAxisRect.

17.622.2.2 draw()

```
void QCPColorScaleAxisRectPrivate::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Reimplemented from QCPAxisRect.

17.622.2.3 mouseMoveEvent()

```
void QCPAxisRect::mouseMoveEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user moves the mouse while holding a mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPAxisRect.

17.622.2.4 mousePressEvent()

```
void QCPAxisRect::mousePressEvent (
            QMouseEvent * event,
            const QVariant & details )  [protected], [virtual]
```

This event gets called when the user presses a mouse button while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *details* contains layerable-specific details about the hit, which were generated in the previous call to selectTest. For example, One-dimensional plottables like QCPGraph or QCPBars convey the clicked data point in the *details* parameter, as QCPDataSelection packed as QVariant. Multi-part objects convey the specific `SelectablePart` that was hit (e.g. QCPAxis::SelectablePart in the case of axes).
QCustomPlot uses an event propagation system that works the same as Qt's system. If your layerable doesn't reimplement the mousePressEvent or explicitly calls `event->ignore()` in its reimplementation, the event will be propagated to the next layerable in the stacking order.
Once a layerable has accepted the mousePressEvent, it is considered the mouse grabber and will receive all following calls to mouseMoveEvent or mouseReleaseEvent for this mouse interaction (a "mouse interaction" in this context ends with the release).
The default implementation does nothing except explicitly ignoring the event with `event->ignore()`.

See also

mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPAxisRect.

17.622.2.5 mouseReleaseEvent()

```
void QCPAxisRect::mouseReleaseEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user releases the mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseMoveEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPAxisRect.

17.622.2.6 update()

```
void QCPAxisRect::update (
            UpdatePhase phase )  [protected], [virtual]
```

This method is called automatically upon replot and doesn't need to be called by users of QCPAxisRect.
Calls the base class implementation to update the margins (see QCPLayoutElement::update), and finally passes the rect to the inset layout (insetLayout) and calls its QCPInsetLayout::update function.
\seebaseclassmethod
Reimplemented from QCPAxisRect.

17.622.2.7 wheelEvent()

```
void QCPAxisRect::wheelEvent (
            QWheelEvent * event )  [protected], [virtual]
```

This event gets called when the user turns the mouse scroll wheel while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`.
The `event->angleDelta()` indicates how far the mouse wheel was turned, which is usually +/- 120 for single rotation steps. However, if the mouse wheel is turned rapidly, multiple steps may accumulate to one event, making the delta larger. On the other hand, if the wheel has very smooth steps or none at all, the delta may be smaller.
The default implementation does nothing.

See also

mousePressEvent, mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent

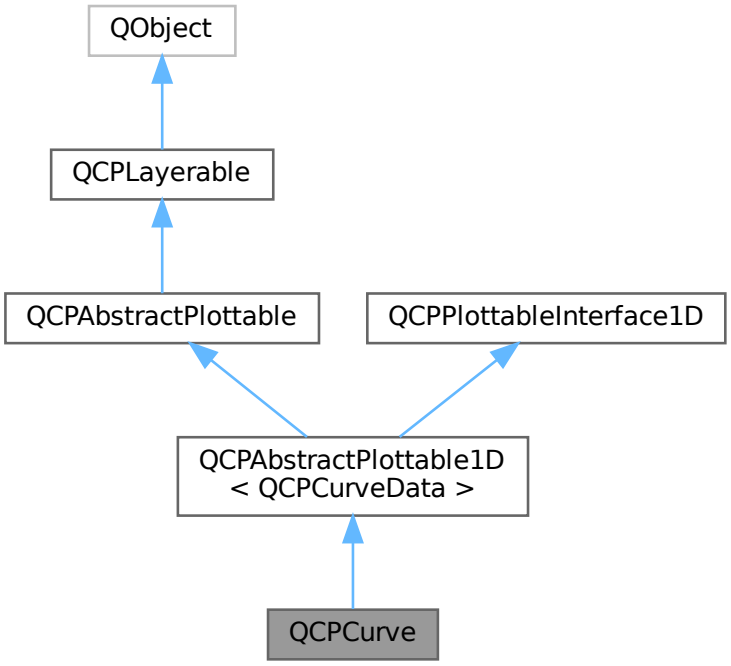
Reimplemented from QCPAxisRect.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

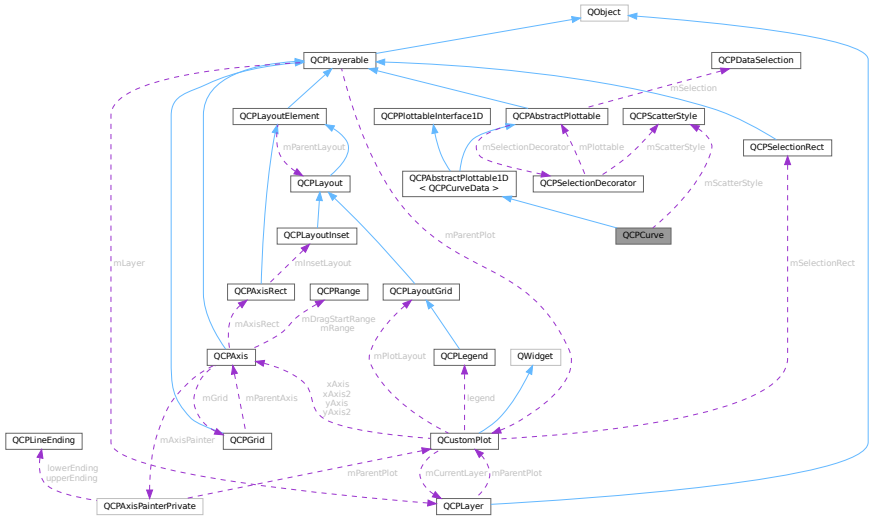
17.623 QCPCurve Class Reference

A plottable representing a parametric curve in a plot.

Inheritance diagram for QCPCurve:



Collaboration diagram for QCPCurve:



Public Types

- enum LineStyle { lsNone , lsLine }

Public Member Functions

- QCPCurve (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QSharedPointer< QCPCurveDataContainer > data () const
- QCPScatterStyle **scatterStyle** () const
- int **scatterSkip** () const
- LineStyle **lineStyle** () const
- void setData (QSharedPointer< QCPCurveDataContainer > data)
- void setData (const QVector< double > &t, const QVector< double > &keys, const QVector< double > &values, bool alreadySorted=false)
- void setData (const QVector< double > &keys, const QVector< double > &values)
- void setScatterStyle (const QCPScatterStyle &style)
- void setScatterSkip (int skip)
- void setLineStyle (LineStyle style)
- void addData (const QVector< double > &t, const QVector< double > &keys, const QVector< double > &values, bool alreadySorted=false)
- void addData (const QVector< double > &keys, const QVector< double > &values)
- void addData (double t, double key, double value)
- void addData (double key, double value)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_DECL_OVERRIDE
- virtual QCPRange getKeyRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth) const Q_DECL_OVERRIDE
- virtual QCPRange getValueRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth, const QCPRange &inKeyRange=QCPRange()) const Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable1D< QCPCurveData >

- QCPAbstractPlottable1D (QCPAxis *keyAxis, QCPAxis *valueAxis)
- virtual int dataCount () const Q_DECL_OVERRIDE
- virtual double dataMainKey (int index) const Q_DECL_OVERRIDE
- virtual double dataSortKey (int index) const Q_DECL_OVERRIDE
- virtual double dataMainValue (int index) const Q_DECL_OVERRIDE
- virtual QCPRange dataValueRange (int index) const Q_DECL_OVERRIDE
- virtual QPointF dataPixelPosition (int index) const Q_DECL_OVERRIDE
- virtual bool sortKeyIsMainKey () const Q_DECL_OVERRIDE
- virtual QCPDataSelection selectTestRect (const QRectF &rect, bool onlySelectable) const Q_DECL_OVERRIDE
- virtual int findBegin (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual int findEnd (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual QCPPlottableInterface1D * interface1D () Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable

- QCPAbstractPlottable (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QString **name** () const
- bool **antialiasedFill** () const
- bool **antialiasedScatters** () const
- QPen **pen** () const
- QBrush **brush** () const
- QCPAxis * **keyAxis** () const
- QCPAxis * **valueAxis** () const
- QCP::SelectionType **selectable** () const
- bool selected () const
- QCPDataSelection selection () const
- QCPSelectionDecorator * selectionDecorator () const

- void setName (const QString &name)
- void setAntialiasedFill (bool enabled)
- void setAntialiasedScatters (bool enabled)
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setKeyAxis (QCPAxis *axis)
- void setValueAxis (QCPAxis *axis)
- Q_SLOT void setSelectable (QCP::SelectionType selectable)
- Q_SLOT void setSelection (QCPDataSelection selection)
- void setSelectionDecorator (QCPSelectionDecorator *decorator)
- void coordsToPixels (double key, double value, double &x, double &y) const
- const QPointF coordsToPixels (double key, double value) const
- void pixelsToCoords (double x, double y, double &key, double &value) const
- void pixelsToCoords (const QPointF &pixelPos, double &key, double &value) const
- void rescaleAxes (bool onlyEnlarge=false) const
- void rescaleKeyAxis (bool onlyEnlarge=false) const
- void rescaleValueAxis (bool onlyEnlarge=false, bool inKeyRange=false) const
- bool addToLegend (QCPLegend *legend)
- bool addToLegend ()
- bool removeFromLegend (QCPLegend *legend) const
- bool removeFromLegend () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void drawLegendIcon (QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE
- virtual void **drawCurveLine** (QCPPainter *painter, const QVector< QPointF > &lines) const
- virtual void **drawScatterPlot** (QCPPainter *painter, const QVector< QPointF > &points, const QCPScatterStyle &style) const
- void **getCurveLines** (QVector< QPointF > *lines, const QCPDataRange &dataRange, double penWidth) const
- void **getScatters** (QVector< QPointF > *scatters, const QCPDataRange &dataRange, double scatterWidth) const
- int **getRegion** (double key, double value, double keyMin, double valueMax, double keyMax, double valueMin) const
- QPointF **getOptimizedPoint** (int otherRegion, double otherKey, double otherValue, double key, double value, double keyMin, double valueMax, double keyMax, double valueMin) const
- QVector< QPointF > **getOptimizedCornerPoints** (int prevRegion, int currentRegion, double prevKey, double prevValue, double key, double value, double keyMin, double valueMax, double keyMax, double valueMin) const
- bool **mayTraverse** (int prevRegion, int currentRegion) const

- bool **getTraverse** (double prevKey, double prevValue, double key, double value, double keyMin, double valueMax, double keyMax, double valueMin, QPointF &crossA, QPointF &crossB) const
- void **getTraverseCornerPoints** (int prevRegion, int currentRegion, double keyMin, double valueMax, double keyMax, double valueMin, QVector< QPointF > &beforeTraverse, QVector< QPointF > &afterTraverse) const
- double **pointDistance** (const QPointF &pixelPoint, QCPCurveDataContainer::const_iterator &closestData) const

Protected Member Functions inherited from QCPAbstractPlottable1D< QCPCurveData >

- void getDataSegments (QList< QCPDataRange > &selectedSegments, QList< QCPDataRange > &unselectedSegments) const
- void drawPolyline (QCPPainter *painter, const QVector< QPointF > &lineData) const

Protected Member Functions inherited from QCPAbstractPlottable

- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- void **applyFillAntialiasingHint** (QCPPainter *painter) const
- void **applyScattersAntialiasingHint** (QCPPainter *painter) const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QCPScatterStyle **mScatterStyle**
- int **mScatterSkip**
- LineStyle **mLineStyle**

Protected Attributes inherited from QCPAbstractPlottable1D< QCPCurveData >

- QSharedPointer< QCPDataContainer< QCPCurveData > > **mDataContainer**

Protected Attributes inherited from QCPAbstractPlottable

- QString **mName**
- bool **mAntialiasedFill**
- bool **mAntialiasedScatters**
- QPen **mPen**
- QBrush **mBrush**
- QPointer< QCPAxis > **mKeyAxis**

- QPointer< QCPAxis > **mValueAxis**
- QCP::SelectionType **mSelectable**
- QCPDataSelection **mSelection**
- QCPSelectionDecorator * **mSelectionDecorator**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPLegend**

Additional Inherited Members

Signals inherited from QCPAbstractPlottable

- void selectionChanged (bool selected)
- void selectionChanged (const QCPDataSelection &selection)
- void selectableChanged (QCP::SelectionType selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.623.1 Detailed Description

A plottable representing a parametric curve in a plot.
Unlike QCPGraph, plottables of this type may have multiple points with the same key coordinate, so their visual representation can have *loops*. This is realized by introducing a third coordinate *t*, which defines the order of the points described by the other two coordinates *x* and *y*.
To plot data, assign it with the setData or addData functions. Alternatively, you can also access and modify the curve's data via the data method, which returns a pointer to the internal QCPCurveDataContainer.
Gaps in the curve can be created by adding data points with NaN as key and value (`qQNaN()` or `std::numeric_limits<double>::quiet_NaN()`) in between the two data points that shall be separated.

17.623.2 Changing the appearance

The appearance of the curve is determined by the pen and the brush (setPen, setBrush).

17.623.3 Usage

Like all data representing objects in QCustomPlot, the QCPCurve is a plottable (QCPAbstractPlottable). So the plottable-interface of QCustomPlot applies (QCustomPlot::plottable, QCustomPlot::removePlottable, etc.)
Usually, you first create an instance:
which registers it with the QCustomPlot instance of the passed axes. Note that this QCustomPlot instance takes ownership of the plottable, so do not delete it manually but use QCustomPlot::removePlottable() instead. The newly created plottable can be modified, e.g.:

17.623.4 Member Enumeration Documentation

17.623.4.1 LineStyle

```
enum QCPCurve::LineStyle
```

Defines how the curve's line is represented visually in the plot. The line is drawn with the current pen of the curve (setPen).

See also

setLineStyle

Enumerator

lsNone	No line is drawn between data points (e.g. only scatters)
lsLine	Data points are connected with a straight line.

17.623.5 Constructor & Destructor Documentation

17.623.5.1 QCPCurve()

```
QCPCurve::QCPCurve (
            QCPAxis * keyAxis,
            QCPAxis * valueAxis )  [explicit]
```

Constructs a curve which uses *keyAxis* as its key axis ("x") and *valueAxis* as its value axis ("y"). *keyAxis* and *valueAxis* must reside in the same QCustomPlot instance and not have the same orientation. If either of these restrictions is violated, a corresponding message is printed to the debug output (qDebug), the construction is not aborted, though.

The created QCPCurve is automatically registered with the QCustomPlot instance inferred from *keyAxis*. This QCustomPlot instance takes ownership of the QCPCurve, so do not delete it manually but use QCustomPlot::removePlottable() instead.

17.623.6 Member Function Documentation

17.623.6.1 addData() [1/4]

```
void QCPCurve::addData (
            const QVector< double > & keys,
            const QVector< double > & values )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.

Adds the provided points in *keys* and *values* to the current data. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.

The t parameter of each data point will be set to the integer index of the respective key/value pair.

Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

17.623.6.2 addData() [2/4]

```
void QCPCurve::addData (
            const QVector< double > & t,
            const QVector< double > & keys,
            const QVector< double > & values,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.

Adds the provided points in *t*, *keys* and *values* to the current data. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.

If you can guarantee that the passed data points are sorted by *keys* in ascending order, you can set *alreadySorted* to true, to improve performance by saving a sorting run.
Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

17.623.6.3 addData() [3/4]

```
void QCPCurve::addData (
            double key,
            double value )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds the provided data point as *key* and *value* to the current data.
The t parameter is generated automatically by increments of 1 for each point, starting at the highest t of previously existing data or 0, if the curve data is empty.
Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

17.623.6.4 addData() [4/4]

```
void QCPCurve::addData (
            double t,
            double key,
            double value )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts. Adds the provided data point as *t*, *key* and *value* to the current data.
Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

17.623.6.5 data()

```
QSharedPointer< QCPCurveDataContainer > QCPCurve::data ( ) const  [inline]
```

Returns a shared pointer to the internal data storage of type QCPCurveDataContainer. You may use it to directly manipulate the data, which may be more convenient and faster than using the regular setData or addData methods.

17.623.6.6 draw()

```
void QCPCurve::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.623.6.7 drawLegendIcon()

```
void QCPCurve::drawLegendIcon (
            QCPPainter * painter,
            const QRectF & rect ) const  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.623.6.8 getKeyRange()

```
QCPRange QCPCurve::getKeyRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth ) const  [virtual]
```

Returns the coordinate range that all data in this plottable span in the key axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSignDomain* to QCP::sdBoth (default). *foundRange*

is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getValueRange

Implements QCPAbstractPlottable.

17.623.6.9 getValueRange()

```
QCPRange QCPCurve::getValueRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth,
            const QCPRange & inKeyRange = QCPRange() ) const  [virtual]
```

Returns the coordinate range that the data points in the specified key range (*inKeyRange*) span in the value axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSign↩ Domain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
If *inKeyRange* has both lower and upper bound set to zero (is equal to `QCPRange()`), all data points are considered, without any restriction on the keys.
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getKeyRange

Implements QCPAbstractPlottable.

17.623.6.10 selectTest()

```
double QCPCurve::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

Implements a selectTest specific to this plottable's point geometry.
If *details* is not 0, it will be set to a QCPDataSelection, describing the closest data point to *pos*.
\seebaseclassmethod QCPAbstractPlottable::selectTest
Reimplemented from QCPAbstractPlottable1D< QCPCurveData >.

17.623.6.11 setData() [1/3]

```
void QCPCurve::setData (
            const QVector< double > & keys,
            const QVector< double > & values )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data with the provided points in *keys* and *values*. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.
The t parameter of each data point will be set to the integer index of the respective key/value pair.

See also

addData

17.623.6.12 setData() [2/3]

```
void QCPCurve::setData (
            const QVector< double > & t,
            const QVector< double > & keys,
            const QVector< double > & values,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data with the provided points in *t*, *keys* and *values*. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.
If you can guarantee that the passed data points are sorted by *t* in ascending order, you can set *alreadySorted* to true, to improve performance by saving a sorting run.

See also

addData

17.623.6.13 setData() [3/3]

```
void QCPCurve::setData (
            QSharedPointer< QCPCurveDataContainer > data )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data container with the provided *data* container.
Since a QSharedPointer is used, multiple QCPCurves may share the same data container safely. Modifying the data in the container will then affect all curves that share the container. Sharing can be achieved by simply exchanging the data containers wrapped in shared pointers:
If you do not wish to share containers, but create a copy from an existing container, rather use the QCPDataContainer<DataType>::set method on the curve's data container directly:

See also

addData

17.623.6.14 setLineStyle()

```
void QCPCurve::setLineStyle (
            QCPCurve::LineStyle style )
```

Sets how the single data points are connected in the plot or how they are represented visually apart from the scatter symbol. For scatter-only plots, set *style* to lsNone and setScatterStyle to the desired scatter style.

See also

setScatterStyle

17.623.6.15 setScatterSkip()

```
void QCPCurve::setScatterSkip (
            int skip )
```

If scatters are displayed (scatter style not QCPScatterStyle::ssNone), *skip* number of scatter points are skipped/not drawn after every drawn scatter point.
This can be used to make the data appear sparser while for example still having a smooth line, and to improve performance for very high density plots.
If *skip* is set to 0 (default), all scatter points are drawn.

See also

setScatterStyle

17.623.6.16 setScatterStyle()

```
void QCPCurve::setScatterStyle (
            const QCPScatterStyle & style )
```

Sets the visual appearance of single data points in the plot. If set to QCPScatterStyle::ssNone, no scatter points are drawn (e.g. for line-only plots with appropriate line style).

See also

QCPScatterStyle, setLineStyle

The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.624 QCPCurveData Class Reference

Holds the data of one single data point for QCPCurve.

Public Member Functions

- QCPCurveData ()
- QCPCurveData (double t, double key, double value)
- double sortKey () const
- double mainKey () const
- double mainValue () const
- QCPRange valueRange () const

Static Public Member Functions

- static QCPCurveData fromSortKey (double sortKey)
- static bool sortKeyIsMainKey ()

Public Attributes

- double **t**
- double **key**
- double **value**

17.624.1 Detailed Description

Holds the data of one single data point for QCPCurve.
The stored data is:

- *t:* the free ordering parameter of this curve point, like in the mathematical vector *(x(t), y(t))*. (This is the *sortKey*)
- *key:* coordinate on the key axis of this curve point (this is the *mainKey*)
- *value:* coordinate on the value axis of this curve point (this is the *mainValue*)

The container for storing multiple data points is QCPCurveDataContainer. It is a typedef for QCPDataContainer with QCPCurveData as the DataType template parameter. See the documentation there for an explanation regarding the data type's generic methods.

See also

QCPCurveDataContainer

17.624.2 Constructor & Destructor Documentation

17.624.2.1 QCPCurveData() [1/2]

```
QCPCurveData::QCPCurveData ( )
```
Constructs a curve data point with t, key and value set to zero.

17.624.2.2 QCPCurveData() [2/2]

```
QCPCurveData::QCPCurveData (
            double t,
            double key,
            double value )
```
Constructs a curve data point with the specified *t*, *key* and *value*.

17.624.3 Member Function Documentation

17.624.3.1 fromSortKey()

```
static QCPCurveData QCPCurveData::fromSortKey (
            double sortKey )  [inline], [static]
```
Returns a data point with the specified *sortKey* (assigned to the data point's *t* member). All other members are set to zero.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.624.3.2 mainKey()

```
double QCPCurveData::mainKey ( ) const  [inline]
```
Returns the *key* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.624.3.3 mainValue()

```
double QCPCurveData::mainValue ( ) const  [inline]
```
Returns the *value* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.624.3.4 sortKey()

```
double QCPCurveData::sortKey ( ) const  [inline]
```
Returns the *t* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.624.3.5 sortKeyIsMainKey()

```
static static bool QCPCurveData::sortKeyIsMainKey ( )  [inline], [static]
```
Since the member *key* is the data point key coordinate and the member *t* is the data ordering parameter, this method returns false.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.624.3.6 valueRange()

```
QCPRange QCPCurveData::valueRange ( ) const  [inline]
```
Returns a QCPRange with both lower and upper boundary set to *value* of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.625 QCPDataContainer< DataType > Class Template Reference

The generic data container for one-dimensional plottables.
`#include <applications/gui/genesys/qcustomplot.h>`

Public Types

- typedef QVector< DataType >::const_iterator **const_iterator**
- typedef QVector< DataType >::iterator **iterator**

Public Member Functions

- QCPDataContainer ()
- int size () const
- bool isEmpty () const
- bool **autoSqueeze** () const
- void setAutoSqueeze (bool enabled)
- void set (const QCPDataContainer< DataType > &data)
- void set (const QVector< DataType > &data, bool alreadySorted=false)
- void add (const QCPDataContainer< DataType > &data)
- void add (const QVector< DataType > &data, bool alreadySorted=false)
- void add (const DataType &data)
- void removeBefore (double sortKey)
- void removeAfter (double sortKey)
- void remove (double sortKeyFrom, double sortKeyTo)
- void remove (double sortKey)
- void clear ()
- void sort ()
- void squeeze (bool preAllocation=true, bool postAllocation=true)
- const_iterator constBegin () const
- const_iterator constEnd () const
- iterator begin ()
- iterator end ()
- const_iterator findBegin (double sortKey, bool expandedRange=true) const
- const_iterator findEnd (double sortKey, bool expandedRange=true) const
- const_iterator at (int index) const
- QCPRange keyRange (bool &foundRange, QCP::SignDomain signDomain=QCP::sdBoth)
- QCPRange valueRange (bool &foundRange, QCP::SignDomain signDomain=QCP::sdBoth, const QCPRange &inKeyRange=QCPRange())
- QCPDataRange dataRange () const
- void limitIteratorsToDataRange (const_iterator &begin, const_iterator &end, const QCPDataRange &dataRange) const

Protected Member Functions

- void **preallocateGrow** (int minimumPreallocSize)
- void **performAutoSqueeze** ()

Protected Attributes

- bool **mAutoSqueeze**
- QVector< DataType > **mData**
- int **mPreallocSize**
- int **mPreallocIteration**

Related Symbols

(Note that these are not member symbols.)

- template<class DataType >
  bool qcpLessThanSortKey (const DataType &a, const DataType &b)

17.625.1 Detailed Description

template<class DataType>
class QCPDataContainer< DataType >

The generic data container for one-dimensional plottables.
This class template provides a fast container for data storage of one-dimensional data. The data type is specified as template parameter (called *DataType* in the following) and must provide some methods as described in the next section.
The data is stored in a sorted fashion, which allows very quick lookups by the sorted key as well as retrieval of ranges (see findBegin, findEnd, keyRange) using binary search. The container uses a preallocation and a postallocation scheme, such that appending and prepending data (with respect to the sort key) is very fast and minimizes reallocations. If data is added which needs to be inserted between existing keys, the merge usually can be done quickly too, using the fact that existing data is always sorted. The user can further improve performance by specifying that added data is already itself sorted by key, if he can guarantee that this is the case (see for example add(const QVector<DataType> &data, bool alreadySorted)).
The data can be accessed with the provided const iterators (constBegin, constEnd). If it is necessary to alter existing data in-place, the non-const iterators can be used (begin, end). Changing data members that are not the sort key (for most data types called *key*) is safe from the container's perspective.
Great care must be taken however if the sort key is modified through the non-const iterators. For performance reasons, the iterators don't automatically cause a re-sorting upon their manipulation. It is thus the responsibility of the user to leave the container in a sorted state when finished with the data manipulation, before calling any other methods on the container. A complete re-sort (e.g. after finishing all sort key manipulation) can be done by calling sort. Failing to do so can not be detected by the container efficiently and will cause both rendering artifacts and potential data loss.
Implementing one-dimensional plottables that make use of a QCPDataContainer<T> is usually done by subclassing from QCPAbstractPlottable1D<T>, which introduces an according *mDataContainer* member and some convenience methods.

17.625.2 Requirements for the DataType template parameter

The template parameter `DataType` is the type of the stored data points. It must be trivially copyable and have the following public methods, preferably inline:

- `double sortKey() const`
  Returns the member variable of this data point that is the sort key, defining the ordering in the container. Often this variable is simply called *key*.

- `static DataType fromSortKey(double sortKey)`
  Returns a new instance of the data type initialized with its sort key set to *sortKey*.

- `static bool sortKeyIsMainKey()`
  Returns true if the sort key is equal to the main key (see method `mainKey` below). For most plottables this is the case. It is not the case for example for QCPCurve, which uses *t* as sort key and *key* as main key. This is the reason why QCPCurve unlike QCPGraph can display parametric curves with loops.

- `double mainKey() const`
  Returns the variable of this data point considered the main key. This is commonly the variable that is used as the coordinate of this data point on the key axis of the plottable. This method is used for example when determining the automatic axis rescaling of key axes (QCPAxis::rescale).

- `double mainValue() const`
  Returns the variable of this data point considered the main value. This is commonly the variable that is used as the coordinate of this data point on the value axis of the plottable.

- `QCPRange valueRange() const`
  Returns the range this data point spans in the value axis coordinate. If the data is single-valued (e.g. QCPGraphData), this is simply a range with both lower and upper set to the main data point value. However if the data points can represent multiple values at once (e.g QCPFinancialData with its *high*, *low*, *open* and *close* values at each *key*) this method should return the range those values span. This method is used for example when determining the automatic axis rescaling of value axes (QCPAxis::rescale).

17.625.3 Constructor & Destructor Documentation

17.625.3.1 QCPDataContainer()

```
template<class DataType >
QCPDataContainer< DataType >::QCPDataContainer ( )
```

Constructs a QCPDataContainer used for plottable classes that represent a series of key-sorted data

17.625.4 Member Function Documentation

17.625.4.1 add() [1/3]

```
template<class DataType >
void QCPDataContainer< DataType >::add (
            const DataType & data )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds the provided single data point to the current data.

See also

remove

17.625.4.2 add() [2/3]

```
template<class DataType >
void QCPDataContainer< DataType >::add (
            const QCPDataContainer< DataType > & data )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds the provided *data* to the current data in this container.

See also

set, remove

17.625.4.3 add() [3/3]

```
template<class DataType >
void QCPDataContainer< DataType >::add (
            const QVector< DataType > & data,
            bool alreadySorted = false )
```

Adds the provided data points in *data* to the current data.
If you can guarantee that the data points in *data* have ascending order with respect to the DataType's sort key, set *alreadySorted* to true to avoid an unnecessary sorting run.

See also

set, remove

17.625.4.4 at()

```
template<class DataType >
QCPDataContainer::const_iterator QCPDataContainer< DataType >::at (
            int index ) const  [inline]
```

Returns a const iterator to the element with the specified *index*. If *index* points beyond the available elements in this container, returns constEnd, i.e. an iterator past the last valid element.
You can use this method to easily obtain iterators from a QCPDataRange, see the data selection page for an example.

17.625.4.5 begin()

```
template<class DataType >
QCPDataContainer::iterator QCPDataContainer< DataType >::begin ( )  [inline]
```

Returns a non-const iterator to the first data point in this container.
You can manipulate the data points in-place through the non-const iterators, but great care must be taken when manipulating the sort key of a data point, see sort, or the detailed description of this class.

17.625.4.6 clear()

```
template<class DataType >
void QCPDataContainer< DataType >::clear ( )
```

Removes all data points.

See also

remove, removeAfter, removeBefore

17.625.4.7 constBegin()

```
template<class DataType >
QCPDataContainer::const_iterator QCPDataContainer< DataType >::constBegin ( ) const  [inline]
```

Returns a const iterator to the first data point in this container.

17.625.4.8 constEnd()

```
template<class DataType >
QCPDataContainer::const_iterator QCPDataContainer< DataType >::constEnd ( ) const  [inline]
```

Returns a const iterator to the element past the last data point in this container.

17.625.4.9 dataRange()

```
template<class DataType >
QCPDataRange QCPDataContainer< DataType >::dataRange ( ) const  [inline]
```

Returns a QCPDataRange encompassing the entire data set of this container. This means the begin index of the returned range is 0, and the end index is size.

17.625.4.10 end()

```
template<class DataType >
QCPDataContainer::iterator QCPDataContainer< DataType >::end ( )  [inline]
```

Returns a non-const iterator to the element past the last data point in this container.
You can manipulate the data points in-place through the non-const iterators, but great care must be taken when manipulating the sort key of a data point, see sort, or the detailed description of this class.

17.625.4.11 findBegin()

```
template<class DataType >
QCPDataContainer< DataType >::const_iterator QCPDataContainer< DataType >::findBegin (
            double sortKey,
            bool expandedRange = true ) const
```

Returns an iterator to the data point with a (sort-)key that is equal to, just below, or just above *sortKey*. If *expandedRange* is true, the data point just below *sortKey* will be considered, otherwise the one just above.
This can be used in conjunction with findEnd to iterate over data points within a given key range, including or excluding the bounding data points that are just beyond the specified range.
If *expandedRange* is true but there are no data points below *sortKey*, constBegin is returned.
If the container is empty, returns constEnd.

See also

findEnd, QCPPlottableInterface1D::findBegin

17.625.4.12 findEnd()

```
template<class DataType >
QCPDataContainer< DataType >::const_iterator QCPDataContainer< DataType >::findEnd (
            double sortKey,
            bool expandedRange = true ) const
```

Returns an iterator to the element after the data point with a (sort-)key that is equal to, just above or just below *sortKey*. If *expandedRange* is true, the data point just above *sortKey* will be considered, otherwise the one just below.
This can be used in conjunction with findBegin to iterate over data points within a given key range, including the bounding data points that are just below and above the specified range.
If *expandedRange* is true but there are no data points above *sortKey*, constEnd is returned.
If the container is empty, constEnd is returned.

See also

findBegin, QCPPlottableInterface1D::findEnd

17.625.4.13 isEmpty()

```
template<class DataType >
bool QCPDataContainer< DataType >::isEmpty ( ) const  [inline]
```

Returns whether this container holds no data points.

17.625.4.14 keyRange()

```
template<class DataType >
QCPRange QCPDataContainer< DataType >::keyRange (
            bool & foundRange,
            QCP::SignDomain signDomain = QCP::sdBoth )
```

Returns the range encompassed by the (main-)key coordinate of all data points. The output parameter *foundRange* indicates whether a sensible range was found. If this is false, you should not use the returned QCPRange (e.g. the data container is empty or all points have the same key).
Use *signDomain* to control which sign of the key coordinates should be considered. This is relevant e.g. for logarithmic plots which can mathematically only display one sign domain at a time.
If the DataType reports that its main key is equal to the sort key (*sortKeyIsMainKey*), as is the case for most plottables, this method uses this fact and finds the range very quickly.

See also

valueRange

17.625.4.15 limitIteratorsToDataRange()

```
template<class DataType >
void QCPDataContainer< DataType >::limitIteratorsToDataRange (
            const_iterator & begin,
            const_iterator & end,
            const QCPDataRange & dataRange ) const
```

Makes sure *begin* and *end* mark a data range that is both within the bounds of this data container's data, as well as within the specified *dataRange*. The initial range described by the passed iterators *begin* and *end* is never expanded, only contracted if necessary.
This function doesn't require for *dataRange* to be within the bounds of this data container's valid range.

17.625.4.16 remove() [1/2]

```
template<class DataType >
void QCPDataContainer< DataType >::remove (
            double sortKey )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Removes a single data point at *sortKey*. If the position is not known with absolute (binary) precision, consider using remove(double sortKeyFrom, double sortKeyTo) with a small fuzziness interval around the suspected position, depeding on the precision with which the (sort-)key is known.

See also

removeBefore, removeAfter, clear

17.625.4.17 remove() [2/2]

```
template<class DataType >
void QCPDataContainer< DataType >::remove (
            double sortKeyFrom,
            double sortKeyTo )
```

Removes all data points with (sort-)keys between *sortKeyFrom* and *sortKeyTo*. if *sortKeyFrom* is greater or equal to *sortKeyTo*, the function does nothing. To remove a single data point with known (sort-)key, use remove(double sortKey).

See also

removeBefore, removeAfter, clear

17.625.4.18 removeAfter()

```
template<class DataType >
void QCPDataContainer< DataType >::removeAfter (
            double sortKey )
```

Removes all data points with (sort-)keys greater than or equal to *sortKey*.

See also

removeBefore, remove, clear

17.625.4.19 removeBefore()

```
template<class DataType >
void QCPDataContainer< DataType >::removeBefore (
            double sortKey )
```

Removes all data points with (sort-)keys smaller than or equal to *sortKey*.

See also

removeAfter, remove, clear

17.625.4.20 set() [1/2]

```
template<class DataType >
void QCPDataContainer< DataType >::set (
            const QCPDataContainer< DataType > & data )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data in this container with the provided *data*.

See also

add, remove

17.625.4.21 set() [2/2]

```
template<class DataType >
void QCPDataContainer< DataType >::set (
            const QVector< DataType > & data,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data in this container with the provided *data*
If you can guarantee that the data points in *data* have ascending order with respect to the DataType's sort key, set *alreadySorted* to true to avoid an unnecessary sorting run.

See also

add, remove

17.625.4.22 setAutoSqueeze()

```
template<class DataType >
void QCPDataContainer< DataType >::setAutoSqueeze (
            bool enabled )
```

Sets whether the container automatically decides when to release memory from its post- and preallocation pools when data points are removed. By default this is enabled and for typical applications shouldn't be changed.
If auto squeeze is disabled, you can manually decide when to release pre-/postallocation with squeeze.

17.625.4.23 size()

```
template<class DataType >
int QCPDataContainer< DataType >::size ( ) const  [inline]
```

Returns the number of data points in the container.

17.625.4.24 sort()

```
template<class DataType >
void QCPDataContainer< DataType >::sort ( )
```

Re-sorts all data points in the container by their sort key.
When setting, adding or removing points using the QCPDataContainer interface (set, add, remove, etc.), the container makes sure to always stay in a sorted state such that a full resort is never necessary. However, if you choose to directly manipulate the sort key on data points by accessing and modifying it through the non-const iterators (begin, end), it is your responsibility to bring the container back into a sorted state before any other methods are called on it. This can be achieved by calling this method immediately after finishing the sort key manipulation.

17.625.4.25 squeeze()

```
template<class DataType >
void QCPDataContainer< DataType >::squeeze (
```

```
            bool preAllocation = true,
            bool postAllocation = true )
```

Frees all unused memory that is currently in the preallocation and postallocation pools.
Note that QCPDataContainer automatically decides whether squeezing is necessary, if setAutoSqueeze is left enabled. It should thus not be necessary to use this method for typical applications.
The parameters *preAllocation* and *postAllocation* control whether pre- and/or post allocation should be freed, respectively.

17.625.4.26 valueRange()

```
template<class DataType >
QCPRange QCPDataContainer< DataType >::valueRange (
            bool & foundRange,
            QCP::SignDomain signDomain = QCP::sdBoth,
            const QCPRange & inKeyRange = QCPRange() )
```

Returns the range encompassed by the value coordinates of the data points in the specified key range (*inKeyRange*), using the full *DataType::valueRange* reported by the data points. The output parameter *foundRange* indicates whether a sensible range was found. If this is false, you should not use the returned QCPRange (e.g. the data container is empty or all points have the same value).
If *inKeyRange* has both lower and upper bound set to zero (is equal to `QCPRange()`), all data points are considered, without any restriction on the keys.
Use *signDomain* to control which sign of the value coordinates should be considered. This is relevant e.g. for logarithmic plots which can mathematically only display one sign domain at a time.

See also

keyRange

17.625.5 Friends And Related Symbol Documentation

17.625.5.1 qcpLessThanSortKey()

```
template<class DataType >
bool qcpLessThanSortKey (
            const DataType & a,
            const DataType & b )  [related]
```

Returns whether the sort key of *a* is less than the sort key of *b*.

See also

QCPDataContainer::sort

The documentation for this class was generated from the following file:

- applications/gui/genesys/qcustomplot.h

17.626 QCPDataRange Class Reference

Describes a data range given by begin and end index.

Public Member Functions

- QCPDataRange ()
- QCPDataRange (int begin, int end)
- bool **operator==** (const QCPDataRange &other) const
- bool **operator!=** (const QCPDataRange &other) const
- int **begin** () const
- int **end** () const
- int size () const
- int length () const

- void setBegin (int begin)
- void setEnd (int end)
- bool isValid () const
- bool isEmpty () const
- QCPDataRange bounded (const QCPDataRange &other) const
- QCPDataRange expanded (const QCPDataRange &other) const
- QCPDataRange intersection (const QCPDataRange &other) const
- QCPDataRange adjusted (int changeBegin, int changeEnd) const
- bool intersects (const QCPDataRange &other) const
- bool contains (const QCPDataRange &other) const

Related Symbols

(Note that these are not member symbols.)

- QDebug operator<< (QDebug d, const QCPDataRange &dataRange)

17.626.1 Detailed Description

Describes a data range given by begin and end index.
QCPDataRange holds two integers describing the begin (setBegin) and end (setEnd) index of a contiguous set of data points. The *end* index corresponds to the data point just after the last data point of the data range, like in standard iterators.
Data Ranges are not bound to a certain plottable, thus they can be freely exchanged, created and modified. If a non-contiguous data set shall be described, the class QCPDataSelection is used, which holds and manages multiple instances of QCPDataRange. In most situations, QCPDataSelection is thus used.
Both QCPDataRange and QCPDataSelection offer convenience methods to work with them, e.g. bounded, expanded, intersects, intersection, adjusted, contains. Further, addition and subtraction operators (defined in QCPDataSelection) can be used to join/subtract data ranges and data selections (or mixtures), to retrieve a corresponding QCPDataSelection.
QCustomPlot's data selection mechanism is based on QCPDataSelection and QCPDataRange.

Note

> Do not confuse QCPDataRange with QCPRange. A QCPRange describes an interval in floating point plot coordinates, e.g. the current axis range.

17.626.2 Constructor & Destructor Documentation

17.626.2.1 QCPDataRange() [1/2]

```
QCPDataRange::QCPDataRange ( )
```

Creates an empty QCPDataRange, with begin and end set to 0.

17.626.2.2 QCPDataRange() [2/2]

```
QCPDataRange::QCPDataRange (
            int begin,
            int end )
```

Creates a QCPDataRange, initialized with the specified *begin* and *end*.
No checks or corrections are made to ensure the resulting range is valid (isValid).

17.626.3 Member Function Documentation

17.626.3.1 adjusted()

```
QCPDataRange QCPDataRange::adjusted (
            int changeBegin,
            int changeEnd ) const  [inline]
```

Returns a data range where *changeBegin* and *changeEnd* were added to the begin and end indices, respectively.

17.626.3.2 bounded()

```
QCPDataRange QCPDataRange::bounded (
            const QCPDataRange & other ) const
```

Returns a data range that matches this data range, except that parts exceeding *other* are excluded.
This method is very similar to intersection, with one distinction: If this range and the *other* range share no intersection, the returned data range will be empty with begin and end set to the respective boundary side of *other*, at which this range is residing. (intersection would just return a range with begin and end set to 0.)

17.626.3.3 contains()

```
bool QCPDataRange::contains (
            const QCPDataRange & other ) const
```

Returns whether all data points of *other* are also contained inside this data range.

See also

intersects

17.626.3.4 expanded()

```
QCPDataRange QCPDataRange::expanded (
            const QCPDataRange & other ) const
```

Returns a data range that contains both this data range as well as *other*.

17.626.3.5 intersection()

```
QCPDataRange QCPDataRange::intersection (
            const QCPDataRange & other ) const
```

Returns the data range which is contained in both this data range and *other*.
This method is very similar to bounded, with one distinction: If this range and the *other* range share no intersection, the returned data range will be empty with begin and end set to 0. (bounded would return a range with begin and end set to one of the boundaries of *other*, depending on which side this range is on.)

See also

QCPDataSelection::intersection

17.626.3.6 intersects()

```
bool QCPDataRange::intersects (
            const QCPDataRange & other ) const
```

Returns whether this data range and *other* share common data points.

See also

intersection, contains

17.626.3.7 isEmpty()

```
bool QCPDataRange::isEmpty ( ) const  [inline]
```

Returns whether this range is empty, i.e. whether its begin index equals its end index.

See also

size, length

17.626.3.8 isValid()

```
bool QCPDataRange::isValid ( ) const  [inline]
```

Returns whether this range is valid. A valid range has a begin index greater or equal to 0, and an end index greater or equal to the begin index.

Note

Invalid ranges should be avoided and are never the result of any of QCustomPlot's methods (unless they are themselves fed with invalid ranges). Do not pass invalid ranges to QCustomPlot's methods. The invalid range is not inherently prevented in QCPDataRange, to allow temporary invalid begin/end values while manipulating the range. An invalid range is not necessarily empty (isEmpty), since its length can be negative and thus non-zero.

17.626.3.9 length()

```
int QCPDataRange::length ( ) const  [inline]
```

Returns the number of data points described by this data range. Equivalent to size.

17.626.3.10 setBegin()

```
void QCPDataRange::setBegin (
            int begin )  [inline]
```

Sets the begin of this data range. The *begin* index points to the first data point that is part of the data range.
No checks or corrections are made to ensure the resulting range is valid (isValid).

See also

setEnd

17.626.3.11 setEnd()

```
void QCPDataRange::setEnd (
            int end )  [inline]
```

Sets the end of this data range. The *end* index points to the data point just after the last data point that is part of the data range.
No checks or corrections are made to ensure the resulting range is valid (isValid).

See also

setBegin

17.626.3.12 size()

```
int QCPDataRange::size ( ) const  [inline]
```

Returns the number of data points described by this data range. This is equal to the end index minus the begin index.

See also

length

17.626.4 Friends And Related Symbol Documentation

17.626.4.1 operator<<()

```
QDebug operator<< (
            QDebug d,
            const QCPDataRange & dataRange )  [related]
```

Prints *dataRange* in a human readable format to the qDebug output.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.627 QCPDataSelection Class Reference

Describes a data set by holding multiple QCPDataRange instances.

Public Member Functions

- QCPDataSelection ()
- QCPDataSelection (const QCPDataRange &range)
- bool operator== (const QCPDataSelection &other) const
- bool **operator!=** (const QCPDataSelection &other) const
- QCPDataSelection & operator+= (const QCPDataSelection &other)
- QCPDataSelection & operator+= (const QCPDataRange &other)
- QCPDataSelection & operator-= (const QCPDataSelection &other)
- QCPDataSelection & operator-= (const QCPDataRange &other)
- int dataRangeCount () const
- int dataPointCount () const
- QCPDataRange dataRange (int index=0) const
- QList< QCPDataRange > dataRanges () const
- QCPDataRange span () const
- void addDataRange (const QCPDataRange &dataRange, bool simplify=true)
- void clear ()
- bool isEmpty () const
- void simplify ()
- void enforceType (QCP::SelectionType type)
- bool contains (const QCPDataSelection &other) const
- QCPDataSelection intersection (const QCPDataRange &other) const
- QCPDataSelection intersection (const QCPDataSelection &other) const
- QCPDataSelection inverse (const QCPDataRange &outerRange) const

Friends

- const QCPDataSelection operator+ (const QCPDataSelection &a, const QCPDataSelection &b)
- const QCPDataSelection operator+ (const QCPDataRange &a, const QCPDataSelection &b)
- const QCPDataSelection operator+ (const QCPDataSelection &a, const QCPDataRange &b)
- const QCPDataSelection operator+ (const QCPDataRange &a, const QCPDataRange &b)
- const QCPDataSelection operator- (const QCPDataSelection &a, const QCPDataSelection &b)
- const QCPDataSelection operator- (const QCPDataRange &a, const QCPDataSelection &b)
- const QCPDataSelection operator- (const QCPDataSelection &a, const QCPDataRange &b)
- const QCPDataSelection operator- (const QCPDataRange &a, const QCPDataRange &b)

Related Symbols

(Note that these are not member symbols.)

- QDebug operator<< (QDebug d, const QCPDataSelection &selection)

17.627.1 Detailed Description

Describes a data set by holding multiple QCPDataRange instances.
QCPDataSelection manages multiple instances of QCPDataRange in order to represent any (possibly disjoint) set of data selection.
The data selection can be modified with addition and subtraction operators which take QCPDataSelection and QCPDataRange instances, as well as methods such as addDataRange and clear. Read access is provided by dataRange, dataRanges, dataRangeCount, etc.
The method simplify is used to join directly adjacent or even overlapping QCPDataRange instances. QCPDataSelection automatically simplifies when using the addition/subtraction operators. The only case when simplify is left to the user, is when calling addDataRange, with the parameter *simplify* explicitly set to false. This is

useful if many data ranges will be added to the selection successively and the overhead for simplifying after each iteration shall be avoided. In this case, you should make sure to call simplify after completing the operation.
Use enforceType to bring the data selection into a state complying with the constraints for selections defined in QCP::SelectionType.
QCustomPlot's data selection mechanism is based on QCPDataSelection and QCPDataRange.

17.627.2 Iterating over a data selection

As an example, the following code snippet calculates the average value of a graph's data selection:

17.627.3 Constructor & Destructor Documentation

17.627.3.1 QCPDataSelection() [1/2]

```
QCPDataSelection::QCPDataSelection ( )  [explicit]
```

Creates an empty QCPDataSelection.

17.627.3.2 QCPDataSelection() [2/2]

```
QCPDataSelection::QCPDataSelection (
            const QCPDataRange & range )  [explicit]
```

Creates a QCPDataSelection containing the provided *range*.

17.627.4 Member Function Documentation

17.627.4.1 addDataRange()

```
void QCPDataSelection::addDataRange (
            const QCPDataRange & dataRange,
            bool simplify = true )
```

Adds the given *dataRange* to this data selection. This is equivalent to the += operator but allows disabling immediate simplification by setting *simplify* to false. This can improve performance if adding a very large amount of data ranges successively. In this case, make sure to call simplify manually, after the operation.

17.627.4.2 clear()

```
void QCPDataSelection::clear ( )
```

Removes all data ranges. The data selection then contains no data points.
isEmpty

17.627.4.3 contains()

```
bool QCPDataSelection::contains (
            const QCPDataSelection & other ) const
```

Returns true if the data selection *other* is contained entirely in this data selection, i.e. all data point indices that are in *other* are also in this data selection.

See also

QCPDataRange::contains

17.627.4.4 dataPointCount()

```
int QCPDataSelection::dataPointCount ( ) const
```

Returns the total number of data points contained in all data ranges that make up this data selection.

17.627.4.5 dataRange()

```
QCPDataRange QCPDataSelection::dataRange (
            int index = 0 ) const
```

Returns the data range with the specified *index*.
If the data selection is simplified (the usual state of the selection, see simplify), the ranges are sorted by ascending data point index.

See also

dataRangeCount

17.627.4.6 dataRangeCount()

```
int QCPDataSelection::dataRangeCount ( ) const  [inline]
```

Returns the number of ranges that make up the data selection. The ranges can be accessed by dataRange via their index.

See also

dataRange, dataPointCount

17.627.4.7 dataRanges()

```
QList< QCPDataRange > QCPDataSelection::dataRanges ( ) const  [inline]
```

Returns all data ranges that make up the data selection. If the data selection is simplified (the usual state of the selection, see simplify), the ranges are sorted by ascending data point index.

See also

dataRange

17.627.4.8 enforceType()

```
void QCPDataSelection::enforceType (
            QCP::SelectionType type )
```

Makes sure this data selection conforms to the specified *type* selection type. Before the type is enforced, simplify is called.
Depending on *type*, enforcing means adding new data points that were previously not part of the selection, or removing data points from the selection. If the current selection already conforms to *type*, the data selection is not changed.

See also

QCP::SelectionType

17.627.4.9 intersection() [1/2]

```
QCPDataSelection QCPDataSelection::intersection (
            const QCPDataRange & other ) const
```

Returns a data selection containing the points which are both in this data selection and in the data range *other*.
A common use case is to limit an unknown data selection to the valid range of a data container, using QCPDataContainer::dataRange as *other*. One can then safely iterate over the returned data selection without exceeding the data container's bounds.

17.627.4.10 intersection() [2/2]

```
QCPDataSelection QCPDataSelection::intersection (
            const QCPDataSelection & other ) const
```

Returns a data selection containing the points which are both in this data selection and in the data selection *other*.

17.627.4.11 inverse()

```
QCPDataSelection QCPDataSelection::inverse (
            const QCPDataRange & outerRange ) const
```

Returns a data selection which is the exact inverse of this data selection, with *outerRange* defining the base range on which to invert. If *outerRange* is smaller than the span of this data selection, it is expanded accordingly.
For example, this method can be used to retrieve all unselected segments by setting *outerRange* to the full data range of the plottable, and calling this method on a data selection holding the selected segments.

17.627.4.12 isEmpty()

```
bool QCPDataSelection::isEmpty ( ) const  [inline]
```

Returns true if there are no data ranges, and thus no data points, in this QCPDataSelection instance.

See also

dataRangeCount

17.627.4.13 operator+=() [1/2]

```
QCPDataSelection & QCPDataSelection::operator+= (
            const QCPDataRange & other )
```

Adds the data range *other* to this data selection, and then simplifies this data selection (see simplify).

17.627.4.14 operator+=() [2/2]

```
QCPDataSelection & QCPDataSelection::operator+= (
            const QCPDataSelection & other )
```

Adds the data selection of *other* to this data selection, and then simplifies this data selection (see simplify).

17.627.4.15 operator-=() [1/2]

```
QCPDataSelection & QCPDataSelection::operator-= (
            const QCPDataRange & other )
```

Removes all data point indices that are described by *other* from this data selection.

17.627.4.16 operator-=() [2/2]

```
QCPDataSelection & QCPDataSelection::operator-= (
            const QCPDataSelection & other )
```

Removes all data point indices that are described by *other* from this data selection.

17.627.4.17 operator==()

```
bool QCPDataSelection::operator== (
            const QCPDataSelection & other ) const
```

Returns true if this selection is identical (contains the same data ranges with the same begin and end indices) to *other*.
Note that both data selections must be in simplified state (the usual state of the selection, see simplify) for this operator to return correct results.

17.627.4.18 simplify()

```
void QCPDataSelection::simplify ( )
```

Sorts all data ranges by range begin index in ascending order, and then joins directly adjacent or overlapping ranges. This can reduce the number of individual data ranges in the selection, and prevents possible double-counting when iterating over the data points held by the data ranges.
This method is automatically called when using the addition/subtraction operators. The only case when simplify is left to the user, is when calling addDataRange, with the parameter *simplify* explicitly set to false.

17.627.4.19 span()

```
QCPDataRange QCPDataSelection::span ( ) const
```

Returns a QCPDataRange which spans the entire data selection, including possible intermediate segments which are not part of the original data selection.

17.627.5 Friends And Related Symbol Documentation

17.627.5.1 operator+ [1/4]

```
const QCPDataSelection operator+ (
            const QCPDataRange & a,
            const QCPDataRange & b )  [friend]
```

Return a QCPDataSelection with the data points in *a* joined with the data points in *b*. The resulting data selection is already simplified (see QCPDataSelection::simplify).

17.627.5.2 operator+ [2/4]

```
const QCPDataSelection operator+ (
            const QCPDataRange & a,
            const QCPDataSelection & b )  [friend]
```

Return a QCPDataSelection with the data points in *a* joined with the data points in *b*. The resulting data selection is already simplified (see QCPDataSelection::simplify).

17.627.5.3 operator+ [3/4]

```
const QCPDataSelection operator+ (
            const QCPDataSelection & a,
            const QCPDataRange & b )  [friend]
```

Return a QCPDataSelection with the data points in *a* joined with the data points in *b*. The resulting data selection is already simplified (see QCPDataSelection::simplify).

17.627.5.4 operator+ [4/4]

```
const QCPDataSelection operator+ (
            const QCPDataSelection & a,
            const QCPDataSelection & b )  [friend]
```

Return a QCPDataSelection with the data points in *a* joined with the data points in *b*. The resulting data selection is already simplified (see QCPDataSelection::simplify).

17.627.5.5 operator- [1/4]

```
const QCPDataSelection operator- (
            const QCPDataRange & a,
            const QCPDataRange & b )  [friend]
```

Return a QCPDataSelection with the data points which are in *a* but not in *b*.

17.627.5.6 operator- [2/4]

```
const QCPDataSelection operator- (
            const QCPDataRange & a,
            const QCPDataSelection & b )  [friend]
```

Return a QCPDataSelection with the data points which are in *a* but not in *b*.

17.627.5.7 operator- [3/4]

```
const QCPDataSelection operator- (
            const QCPDataSelection & a,
            const QCPDataRange & b )  [friend]
```

Return a QCPDataSelection with the data points which are in *a* but not in *b*.

17.627.5.8 operator- [4/4]

```
const QCPDataSelection operator- (
            const QCPDataSelection & a,
            const QCPDataSelection & b )  [friend]
```

Return a QCPDataSelection with the data points which are in *a* but not in *b*.

17.627.5.9 operator<<()

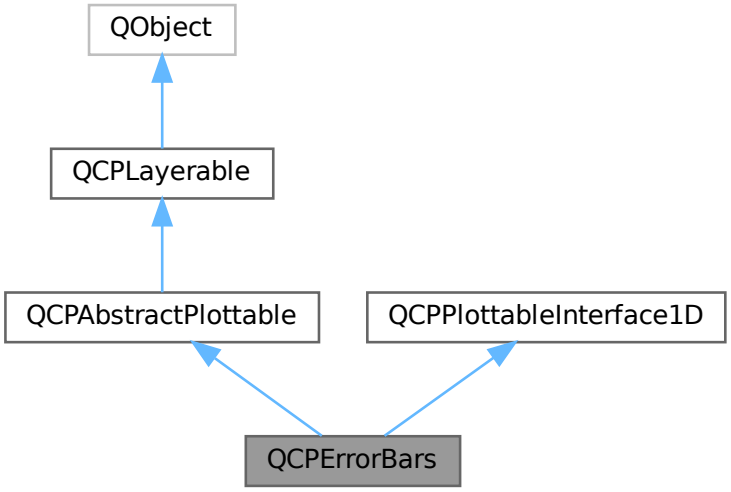
```
QDebug operator<< (
            QDebug d,
            const QCPDataSelection & selection )  [related]
```

Prints *selection* in a human readable format to the qDebug output.
The documentation for this class was generated from the following files:

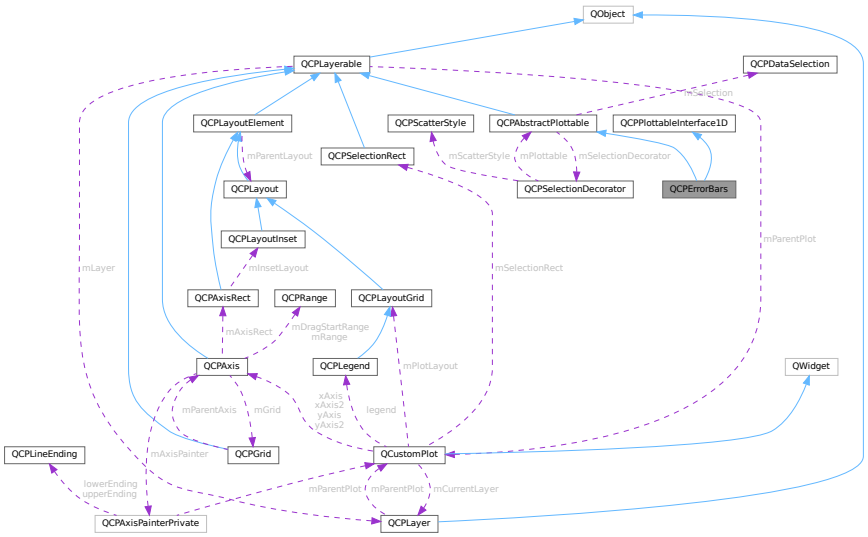
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.628 QCPErrorBars Class Reference

A plottable that adds a set of error bars to other plottables.
Inheritance diagram for QCPErrorBars:



Collaboration diagram for QCPErrorBars:



Public Types

- enum ErrorType { etKeyError , etValueError }

Public Member Functions

- QCPErrorBars (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QSharedPointer< QCPErrorBarsDataContainer > data () const
- QCPAbstractPlottable * **dataPlottable** () const
- ErrorType **errorType** () const
- double **whiskerWidth** () const
- double **symbolGap** () const
- void setData (QSharedPointer< QCPErrorBarsDataContainer > data)
- void setData (const QVector< double > &error)
- void setData (const QVector< double > &errorMinus, const QVector< double > &errorPlus)
- void setDataPlottable (QCPAbstractPlottable *plottable)
- void setErrorType (ErrorType type)
- void setWhiskerWidth (double pixels)
- void setSymbolGap (double pixels)
- void addData (const QVector< double > &error)
- void addData (const QVector< double > &errorMinus, const QVector< double > &errorPlus)
- void addData (double error)
- void addData (double errorMinus, double errorPlus)
- virtual int dataCount () const Q_DECL_OVERRIDE
- virtual double dataMainKey (int index) const Q_DECL_OVERRIDE
- virtual double dataSortKey (int index) const Q_DECL_OVERRIDE
- virtual double dataMainValue (int index) const Q_DECL_OVERRIDE
- virtual QCPRange dataValueRange (int index) const Q_DECL_OVERRIDE
- virtual QPointF dataPixelPosition (int index) const Q_DECL_OVERRIDE
- virtual bool sortKeyIsMainKey () const Q_DECL_OVERRIDE
- virtual QCPDataSelection selectTestRect (const QRectF &rect, bool onlySelectable) const Q_DECL_↩ OVERRIDE
- virtual int findBegin (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual int findEnd (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE

- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE
- virtual QCPPlottableInterface1D * interface1D () Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable

- QCPAbstractPlottable (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QString **name** () const
- bool **antialiasedFill** () const
- bool **antialiasedScatters** () const
- QPen **pen** () const
- QBrush **brush** () const
- QCPAxis * **keyAxis** () const
- QCPAxis * **valueAxis** () const
- QCP::SelectionType **selectable** () const
- bool selected () const
- QCPDataSelection selection () const
- QCPSelectionDecorator * selectionDecorator () const
- void setName (const QString &name)
- void setAntialiasedFill (bool enabled)
- void setAntialiasedScatters (bool enabled)
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setKeyAxis (QCPAxis *axis)
- void setValueAxis (QCPAxis *axis)
- Q_SLOT void setSelectable (QCP::SelectionType selectable)
- Q_SLOT void setSelection (QCPDataSelection selection)
- void setSelectionDecorator (QCPSelectionDecorator *decorator)
- void coordsToPixels (double key, double value, double &x, double &y) const
- const QPointF coordsToPixels (double key, double value) const
- void pixelsToCoords (double x, double y, double &key, double &value) const
- void pixelsToCoords (const QPointF &pixelPos, double &key, double &value) const
- void rescaleAxes (bool onlyEnlarge=false) const
- void rescaleKeyAxis (bool onlyEnlarge=false) const
- void rescaleValueAxis (bool onlyEnlarge=false, bool inKeyRange=false) const
- bool addToLegend (QCPLegend *legend)
- bool addToLegend ()
- bool removeFromLegend (QCPLegend *legend) const
- bool removeFromLegend () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void drawLegendIcon (QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE
- virtual QCPRange getKeyRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth) const Q_DECL_OVERRIDE
- virtual QCPRange getValueRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth, const QCPRange &inKeyRange=QCPRange()) const Q_DECL_OVERRIDE
- void **getErrorBarLines** (QCPErrorBarsDataContainer::const_iterator it, QVector< QLineF > &backbones, QVector< QLineF > &whiskers) const
- void **getVisibleDataBounds** (QCPErrorBarsDataContainer::const_iterator &begin, QCPErrorBarsDataContainer::const_iterator &end, const QCPDataRange &rangeRestriction) const
- double **pointDistance** (const QPointF &pixelPoint, QCPErrorBarsDataContainer::const_iterator &closestData) const
- void **getDataSegments** (QList< QCPDataRange > &selectedSegments, QList< QCPDataRange > &unselectedSegments) const
- bool **errorBarVisible** (int index) const
- bool **rectIntersectsLine** (const QRectF &pixelRect, const QLineF &line) const

Protected Member Functions inherited from QCPAbstractPlottable

- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- void **applyFillAntialiasingHint** (QCPPainter *painter) const
- void **applyScattersAntialiasingHint** (QCPPainter *painter) const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QSharedPointer< QCPErrorBarsDataContainer > **mDataContainer**
- QPointer< QCPAbstractPlottable > **mDataPlottable**
- ErrorType **mErrorType**
- double **mWhiskerWidth**
- double **mSymbolGap**

Protected Attributes inherited from QCPAbstractPlottable

- QString **mName**
- bool **mAntialiasedFill**
- bool **mAntialiasedScatters**
- QPen **mPen**
- QBrush **mBrush**
- QPointer< QCPAxis > **mKeyAxis**
- QPointer< QCPAxis > **mValueAxis**
- QCP::SelectionType **mSelectable**
- QCPDataSelection **mSelection**
- QCPSelectionDecorator * **mSelectionDecorator**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPLegend**

Additional Inherited Members

Signals inherited from QCPAbstractPlottable

- void selectionChanged (bool selected)
- void selectionChanged (const QCPDataSelection &selection)
- void selectableChanged (QCP::SelectionType selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.628.1 Detailed Description

A plottable that adds a set of error bars to other plottables.
The QCPErrorBars plottable can be attached to other one-dimensional plottables (e.g. QCPGraph, QCPCurve, QCPBars, etc.) and equips them with error bars.
Use setDataPlottable to define for which plottable the QCPErrorBars shall display the error bars. The orientation of the error bars can be controlled with setErrorType.
By using setData, you can supply the actual error data, either as symmetric error or plus/minus asymmetric errors. QCPErrorBars only stores the error data. The absolute key/value position of each error bar will be adopted from the configured data plottable. The error data of the QCPErrorBars are associated one-to-one via their index to the data points of the data plottable. You can directly access and manipulate the error bar data via data.
Set either of the plus/minus errors to NaN (`qQNaN()` or `std::numeric_limits<double>::quiet_NaN()`) to not show the respective error bar on the data point at that index.

17.628.2 Changing the appearance

The appearance of the error bars is defined by the pen (setPen), and the width of the whiskers (setWhiskerWidth). Further, the error bar backbones may leave a gap around the data point center to prevent that error bars are drawn too close to or even through scatter points. This gap size can be controlled via setSymbolGap.

17.628.3 Member Enumeration Documentation

17.628.3.1 ErrorType

```
enum QCPErrorBars::ErrorType
```

Defines in which orientation the error bars shall appear. If your data needs both error dimensions, create two QCPErrorBars with different ErrorType.

See also

setErrorType

Enumerator

etKeyError	The errors are for the key dimension (bars appear parallel to the key axis)
etValueError	The errors are for the value dimension (bars appear parallel to the value axis)

17.628.4 Constructor & Destructor Documentation

17.628.4.1 QCPErrorBars()

```
QCPErrorBars::QCPErrorBars (
            QCPAxis * keyAxis,
            QCPAxis * valueAxis )  [explicit]
```

Constructs an error bars plottable which uses *keyAxis* as its key axis ("x") and *valueAxis* as its value axis ("y"). *keyAxis* and *valueAxis* must reside in the same QCustomPlot instance and not have the same orientation. If either of these restrictions is violated, a corresponding message is printed to the debug output (qDebug), the construction is not aborted, though.
It is also important that the *keyAxis* and *valueAxis* are the same for the error bars plottable and the data plottable that the error bars shall be drawn on (setDataPlottable).
The created QCPErrorBars is automatically registered with the QCustomPlot instance inferred from *keyAxis*. This QCustomPlot instance takes ownership of the QCPErrorBars, so do not delete it manually but use QCustomPlot::removePlottable() instead.

17.628.5 Member Function Documentation

17.628.5.1 addData() [1/4]

```
void QCPErrorBars::addData (
            const QVector< double > & error )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds symmetrical error values as specified in *error*. The errors will be associated one-to-one by the data point index to the associated data plottable (setDataPlottable).
You can directly access and manipulate the error bar data via data.

See also

setData

17.628.5.2 addData() [2/4]

```
void QCPErrorBars::addData (
            const QVector< double > & errorMinus,
            const QVector< double > & errorPlus )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds asymmetrical errors as specified in *errorMinus* and *errorPlus*. The errors will be associated one-to-one by the data point index to the associated data plottable (setDataPlottable).
You can directly access and manipulate the error bar data via data.

See also

setData

17.628.5.3 addData() [3/4]

```
void QCPErrorBars::addData (
            double error )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds a single symmetrical error bar as specified in *error*. The errors will be associated one-to-one by the data point index to the associated data plottable (setDataPlottable).
You can directly access and manipulate the error bar data via data.

See also

setData

17.628.5.4 addData() [4/4]

```
void QCPErrorBars::addData (
            double errorMinus,
            double errorPlus )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds a single asymmetrical error bar as specified in *errorMinus* and *errorPlus*. The errors will be associated one-to-one by the data point index to the associated data plottable (setDataPlottable).
You can directly access and manipulate the error bar data via data.

See also

setData

17.628.5.5 data()

```
QSharedPointer< QCPErrorBarsDataContainer > QCPErrorBars::data ( ) const  [inline]
```

Returns a shared pointer to the internal data storage of type QCPErrorBarsDataContainer. You may use it to directly manipulate the error values, which may be more convenient and faster than using the regular setData methods.

17.628.5.6 dataCount()

```
int QCPErrorBars::dataCount ( ) const  [virtual]
```

Returns the number of data points of the plottable.
Implements QCPPlottableInterface1D.

17.628.5.7 dataMainKey()

```
double QCPErrorBars::dataMainKey (
            int index ) const  [virtual]
```

Returns the main key of the data point at the given *index*.
What the main key is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.
Implements QCPPlottableInterface1D.

17.628.5.8 dataMainValue()

```
double QCPErrorBars::dataMainValue (
            int index ) const  [virtual]
```

Returns the main value of the data point at the given *index*.
What the main value is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.
Implements QCPPlottableInterface1D.

17.628.5.9 dataPixelPosition()

```
QPointF QCPErrorBars::dataPixelPosition (
            int index ) const  [virtual]
```

Returns the pixel position on the widget surface at which the data point at the given *index* appears.
Usually this corresponds to the point of dataMainKey/ dataMainValue, in pixel coordinates. However, depending on the plottable, this might be a different apparent position than just a coord-to-pixel transform of those values. For example, QCPBars apparent data values can be shifted depending on their stacking, bar grouping or configured base value.
Implements QCPPlottableInterface1D.

17.628.5.10 dataSortKey()

```
double QCPErrorBars::dataSortKey (
            int index ) const  [virtual]
```

Returns the sort key of the data point at the given *index*.
What the sort key is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.
Implements QCPPlottableInterface1D.

17.628.5.11 dataValueRange()

```
QCPRange QCPErrorBars::dataValueRange (
            int index ) const  [virtual]
```

Returns the value range of the data point at the given *index*.
What the value range is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.
Implements QCPPlottableInterface1D.

17.628.5.12 draw()

```
void QCPErrorBars::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.628.5.13 drawLegendIcon()

```
void QCPErrorBars::drawLegendIcon (
            QCPPainter * painter,
            const QRectF & rect ) const  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.628.5.14 findBegin()

```
int QCPErrorBars::findBegin (
            double sortKey,
            bool expandedRange = true ) const  [virtual]
```

Returns the index of the data point with a (sort-)key that is equal to, just below, or just above *sortKey*. If *expanded↩ Range* is true, the data point just below *sortKey* will be considered, otherwise the one just above.
This can be used in conjunction with findEnd to iterate over data points within a given key range, including or excluding the bounding data points that are just beyond the specified range.
If *expandedRange* is true but there are no data points below *sortKey*, 0 is returned.
If the container is empty, returns 0 (in that case, findEnd will also return 0, so a loop using these methods will not iterate over the index 0).

See also

findEnd, QCPDataContainer::findBegin

Implements QCPPlottableInterface1D.

17.628.5.15 findEnd()

```
int QCPErrorBars::findEnd (
            double sortKey,
            bool expandedRange = true ) const  [virtual]
```

Returns the index one after the data point with a (sort-)key that is equal to, just above, or just below *sortKey*. If *expandedRange* is true, the data point just above *sortKey* will be considered, otherwise the one just below.
This can be used in conjunction with findBegin to iterate over data points within a given key range, including the bounding data points that are just below and above the specified range.
If *expandedRange* is true but there are no data points above *sortKey*, the index just above the highest data point is returned.
If the container is empty, returns 0.

See also

findBegin, QCPDataContainer::findEnd

Implements QCPPlottableInterface1D.

17.628.5.16 getKeyRange()

```
QCPRange QCPErrorBars::getKeyRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth ) const  [protected], [virtual]
```

Returns the coordinate range that all data in this plottable span in the key axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSignDomain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getValueRange

Implements QCPAbstractPlottable.

17.628.5.17 getValueRange()

```
QCPRange QCPErrorBars::getValueRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth,
            const QCPRange & inKeyRange = QCPRange() ) const  [protected], [virtual]
```

Returns the coordinate range that the data points in the specified key range (*inKeyRange*) span in the value axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSign↩ Domain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
If *inKeyRange* has both lower and upper bound set to zero (is equal to `QCPRange()`), all data points are considered, without any restriction on the keys.
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getKeyRange

Implements QCPAbstractPlottable.

17.628.5.18 interface1D()

```
virtual QCPPlottableInterface1D * QCPErrorBars::interface1D ( )  [inline], [virtual]
```

If this plottable is a one-dimensional plottable, i.e. it implements the QCPPlottableInterface1D, returns the *this* pointer with that type. Otherwise (e.g. in the case of a QCPColorMap) returns zero.
You can use this method to gain read access to data coordinates while holding a pointer to the abstract base class only.
Reimplemented from QCPAbstractPlottable.

17.628.5.19 selectTest()

```
double QCPErrorBars::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

Implements a selectTest specific to this plottable's point geometry.
If *details* is not 0, it will be set to a QCPDataSelection, describing the closest data point to *pos*.
\seebaseclassmethod QCPAbstractPlottable::selectTest
Implements QCPAbstractPlottable.

17.628.5.20 selectTestRect()

```
QCPDataSelection QCPErrorBars::selectTestRect (
            const QRectF & rect,
            bool onlySelectable ) const  [virtual]
```

Returns a data selection containing all the data points of this plottable which are contained (or hit by) *rect*. This is used mainly in the selection rect interaction for data selection (data selection mechanism).
If *onlySelectable* is true, an empty QCPDataSelection is returned if this plottable is not selectable (i.e. if QCPAbstractPlottable::setSelectable is QCP::stNone).

Note

> *rect* must be a normalized rect (positive or zero width and height). This is especially important when using the rect of QCPSelectionRect::accepted, which is not necessarily normalized. Use `QRect::normalized()` when passing a rect which might not be normalized.

Implements QCPPlottableInterface1D.

17.628.5.21 setData() [1/3]

```
void QCPErrorBars::setData (
            const QVector< double > & error )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets symmetrical error values as specified in *error*. The errors will be associated one-to-one by the data point index to the associated data plottable (setDataPlottable).
You can directly access and manipulate the error bar data via data.

See also

> addData

17.628.5.22 setData() [2/3]

```
void QCPErrorBars::setData (
            const QVector< double > & errorMinus,
            const QVector< double > & errorPlus )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets asymmetrical errors as specified in *errorMinus* and *errorPlus*. The errors will be associated one-to-one by the data point index to the associated data plottable (setDataPlottable).
You can directly access and manipulate the error bar data via data.

See also

addData

17.628.5.23 setData() [3/3]

```
void QCPErrorBars::setData (
            QSharedPointer< QCPErrorBarsDataContainer > data )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data container with the provided *data* container.
Since a QSharedPointer is used, multiple QCPErrorBars instances may share the same data container safely. Modifying the data in the container will then affect all QCPErrorBars instances that share the container. Sharing can be achieved by simply exchanging the data containers wrapped in shared pointers:
If you do not wish to share containers, but create a copy from an existing container, assign the data containers directly:
(This uses different notation compared with other plottables, because the QCPErrorBars uses a `QVector<QCPErrorBarsData>` as its data container, instead of a QCPDataContainer.)

See also

addData

17.628.5.24 setDataPlottable()

```
void QCPErrorBars::setDataPlottable (
            QCPAbstractPlottable * plottable )
```

Sets the data plottable to which the error bars will be applied. The error values specified e.g. via setData will be associated one-to-one by the data point index to the data points of *plottable*. This means that the error bars will adopt the key/value coordinates of the data point with the same index.
The passed *plottable* must be a one-dimensional plottable, i.e. it must implement the QCPPlottableInterface1D. Further, it must not be a QCPErrorBars instance itself. If either of these restrictions is violated, a corresponding qDebug output is generated, and the data plottable of this QCPErrorBars instance is set to zero.
For proper display, care must also be taken that the key and value axes of the *plottable* match those configured for this QCPErrorBars instance.

17.628.5.25 setErrorType()

```
void QCPErrorBars::setErrorType (
            ErrorType type )
```

Sets in which orientation the error bars shall appear on the data points. If your data needs both error dimensions, create two QCPErrorBars with different *type*.

17.628.5.26 setSymbolGap()

```
void QCPErrorBars::setSymbolGap (
            double pixels )
```

Sets the gap diameter around the data points that will be left out when drawing the error bar backbones. This gap prevents that error bars are drawn too close to or even through scatter points.

17.628.5.27 setWhiskerWidth()

```
void QCPErrorBars::setWhiskerWidth (
            double pixels )
```

Sets the width of the whiskers (the short bars at the end of the actual error bar backbones) to *pixels*.

17.628.5.28 sortKeyIsMainKey()

```
bool QCPErrorBars::sortKeyIsMainKey ( ) const  [virtual]
```

Returns whether the sort key (dataSortKey) is identical to the main key (dataMainKey).
What the sort and main keys are, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.
Implements QCPPlottableInterface1D.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.629 QCPErrorBarsData Class Reference

Holds the data of one single error bar for QCPErrorBars.

Public Member Functions

- QCPErrorBarsData ()
- QCPErrorBarsData (double error)
- QCPErrorBarsData (double errorMinus, double errorPlus)

Public Attributes

- double **errorMinus**
- double **errorPlus**

17.629.1 Detailed Description

Holds the data of one single error bar for QCPErrorBars.
The stored data is:

- *errorMinus:* how much the error bar extends towards negative coordinates from the data point position
- *errorPlus:* how much the error bar extends towards positive coordinates from the data point position

The container for storing the error bar information is QCPErrorBarsDataContainer. It is a typedef for `QVector<QCPErrorBarsData>`.

See also

QCPErrorBarsDataContainer

17.629.2 Constructor & Destructor Documentation

17.629.2.1 QCPErrorBarsData() [1/3]

```
QCPErrorBarsData::QCPErrorBarsData ( )
```

Constructs an error bar with errors set to zero.

17.629.2.2 QCPErrorBarsData() [2/3]

```
QCPErrorBarsData::QCPErrorBarsData (
            double error )  [explicit]
```

Constructs an error bar with equal *error* in both negative and positive direction.

17.629.2.3 QCPErrorBarsData() [3/3]

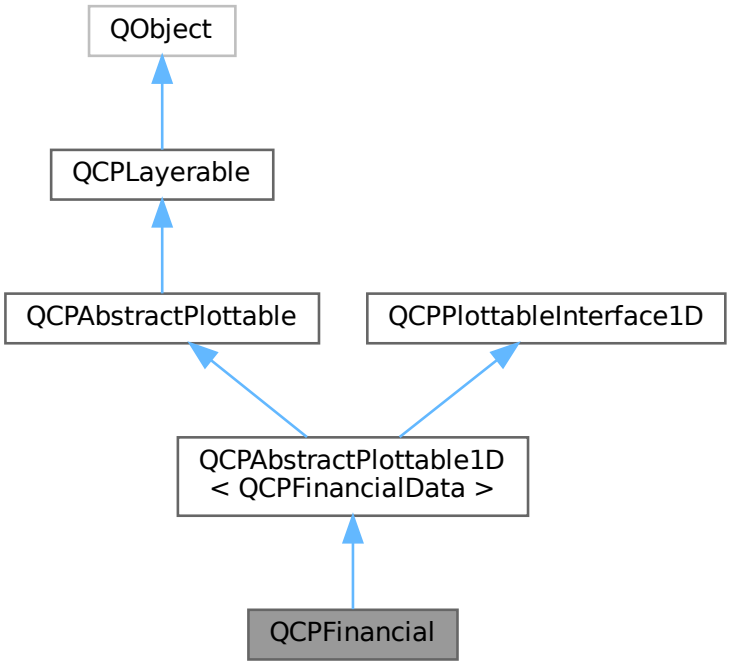
```
QCPErrorBarsData::QCPErrorBarsData (
            double errorMinus,
            double errorPlus )
```

Constructs an error bar with negative and positive errors set to *errorMinus* and *errorPlus*, respectively.
The documentation for this class was generated from the following files:

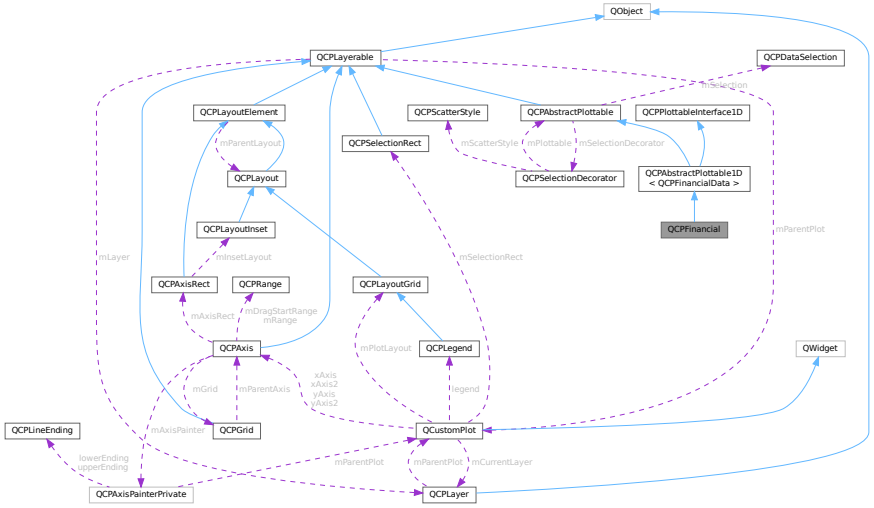
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.630 QCPFinancial Class Reference

A plottable representing a financial stock chart.
Inheritance diagram for QCPFinancial:



Collaboration diagram for QCPFinancial:



Public Types

- enum WidthType { wtAbsolute , wtAxisRectRatio , wtPlotCoords }
- enum ChartStyle { csOhlc , csCandlestick }

Public Member Functions

- QCPFinancial (QCPAxis ∗keyAxis, QCPAxis ∗valueAxis)
- QSharedPointer< QCPFinancialDataContainer > data () const
- ChartStyle **chartStyle** () const
- double **width** () const
- WidthType **widthType** () const
- bool **twoColored** () const
- QBrush **brushPositive** () const
- QBrush **brushNegative** () const
- QPen **penPositive** () const
- QPen **penNegative** () const
- void setData (QSharedPointer< QCPFinancialDataContainer > data)
- void setData (const QVector< double > &keys, const QVector< double > &open, const QVector< double > &high, const QVector< double > &low, const QVector< double > &close, bool alreadySorted=false)
- void setChartStyle (ChartStyle style)
- void setWidth (double width)
- void setWidthType (WidthType widthType)
- void setTwoColored (bool twoColored)
- void setBrushPositive (const QBrush &brush)
- void setBrushNegative (const QBrush &brush)
- void setPenPositive (const QPen &pen)
- void setPenNegative (const QPen &pen)
- void addData (const QVector< double > &keys, const QVector< double > &open, const QVector< double > &high, const QVector< double > &low, const QVector< double > &close, bool alreadySorted=false)
- void addData (double key, double open, double high, double low, double close)
- virtual QCPDataSelection selectTestRect (const QRectF &rect, bool onlySelectable) const Q_DECL_↩ OVERRIDE
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant ∗details=nullptr) const Q_↩ DECL_OVERRIDE

- virtual QCPRange getKeyRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth) const Q_DECL_OVERRIDE
- virtual QCPRange getValueRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth, const QCPRange &inKeyRange=QCPRange()) const Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable1D< QCPFinancialData >

- QCPAbstractPlottable1D (QCPAxis *keyAxis, QCPAxis *valueAxis)
- virtual int dataCount () const Q_DECL_OVERRIDE
- virtual double dataMainKey (int index) const Q_DECL_OVERRIDE
- virtual double dataSortKey (int index) const Q_DECL_OVERRIDE
- virtual double dataMainValue (int index) const Q_DECL_OVERRIDE
- virtual QCPRange dataValueRange (int index) const Q_DECL_OVERRIDE
- virtual QPointF dataPixelPosition (int index) const Q_DECL_OVERRIDE
- virtual bool sortKeyIsMainKey () const Q_DECL_OVERRIDE
- virtual int findBegin (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual int findEnd (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual QCPPlottableInterface1D * interface1D () Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable

- QCPAbstractPlottable (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QString **name** () const
- bool **antialiasedFill** () const
- bool **antialiasedScatters** () const
- QPen **pen** () const
- QBrush **brush** () const
- QCPAxis * **keyAxis** () const
- QCPAxis * **valueAxis** () const
- QCP::SelectionType **selectable** () const
- bool selected () const
- QCPDataSelection selection () const
- QCPSelectionDecorator * selectionDecorator () const
- void setName (const QString &name)
- void setAntialiasedFill (bool enabled)
- void setAntialiasedScatters (bool enabled)
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setKeyAxis (QCPAxis *axis)
- void setValueAxis (QCPAxis *axis)
- Q_SLOT void setSelectable (QCP::SelectionType selectable)
- Q_SLOT void setSelection (QCPDataSelection selection)
- void setSelectionDecorator (QCPSelectionDecorator *decorator)
- void coordsToPixels (double key, double value, double &x, double &y) const
- const QPointF coordsToPixels (double key, double value) const
- void pixelsToCoords (double x, double y, double &key, double &value) const
- void pixelsToCoords (const QPointF &pixelPos, double &key, double &value) const
- void rescaleAxes (bool onlyEnlarge=false) const
- void rescaleKeyAxis (bool onlyEnlarge=false) const
- void rescaleValueAxis (bool onlyEnlarge=false, bool inKeyRange=false) const
- bool addToLegend (QCPLegend *legend)
- bool addToLegend ()
- bool removeFromLegend (QCPLegend *legend) const
- bool removeFromLegend () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Static Public Member Functions

- static QCPFinancialDataContainer timeSeriesToOhlc (const QVector< double > &time, const QVector< double > &value, double timeBinSize, double timeBinOffset=0)

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void drawLegendIcon (QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE
- void **drawOhlcPlot** (QCPPainter *painter, const QCPFinancialDataContainer::const_iterator &begin, const QCPFinancialDataContainer::const_iterator &end, bool isSelected)
- void **drawCandlestickPlot** (QCPPainter *painter, const QCPFinancialDataContainer::const_iterator &begin, const QCPFinancialDataContainer::const_iterator &end, bool isSelected)
- double **getPixelWidth** (double key, double keyPixel) const
- double **ohlcSelectTest** (const QPointF &pos, const QCPFinancialDataContainer::const_iterator &begin, const QCPFinancialDataContainer::const_iterator &end, QCPFinancialDataContainer::const_iterator &closestDataPoint) const
- double **candlestickSelectTest** (const QPointF &pos, const QCPFinancialDataContainer::const_iterator &begin, const QCPFinancialDataContainer::const_iterator &end, QCPFinancialDataContainer::const_iterator &closestDataPoint) const
- void **getVisibleDataBounds** (QCPFinancialDataContainer::const_iterator &begin, QCPFinancialDataContainer::const_iterator &end) const
- QRectF **selectionHitBox** (QCPFinancialDataContainer::const_iterator it) const

Protected Member Functions inherited from QCPAbstractPlottable1D< QCPFinancialData >

- void getDataSegments (QList< QCPDataRange > &selectedSegments, QList< QCPDataRange > &unselectedSegments) const
- void drawPolyline (QCPPainter *painter, const QVector< QPointF > &lineData) const

Protected Member Functions inherited from QCPAbstractPlottable

- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- void **applyFillAntialiasingHint** (QCPPainter *painter) const
- void **applyScattersAntialiasingHint** (QCPPainter *painter) const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot ∗parentPlot)
- virtual void mousePressEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent ∗event)
- void **initializeParentPlot** (QCustomPlot ∗parentPlot)
- void **setParentLayerable** (QCPLayerable ∗parentLayerable)
- bool **moveToLayer** (QCPLayer ∗layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter ∗painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- ChartStyle **mChartStyle**
- double **mWidth**
- WidthType **mWidthType**
- bool **mTwoColored**
- QBrush **mBrushPositive**
- QBrush **mBrushNegative**
- QPen **mPenPositive**
- QPen **mPenNegative**

Protected Attributes inherited from QCPAbstractPlottable1D< QCPFinancialData >

- QSharedPointer< QCPDataContainer< QCPFinancialData > > **mDataContainer**

Protected Attributes inherited from QCPAbstractPlottable

- QString **mName**
- bool **mAntialiasedFill**
- bool **mAntialiasedScatters**
- QPen **mPen**
- QBrush **mBrush**
- QPointer< QCPAxis > **mKeyAxis**
- QPointer< QCPAxis > **mValueAxis**
- QCP::SelectionType **mSelectable**
- QCPDataSelection **mSelection**
- QCPSelectionDecorator ∗ **mSelectionDecorator**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot ∗ **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer ∗ **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPLegend**

Additional Inherited Members

Signals inherited from QCPAbstractPlottable

- void selectionChanged (bool selected)
- void selectionChanged (const QCPDataSelection &selection)
- void selectableChanged (QCP::SelectionType selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.630.1 Detailed Description

A plottable representing a financial stock chart.
This plottable represents time series data binned to certain intervals, mainly used for stock charts. The two common representations OHLC (Open-High-Low-Close) bars and Candlesticks can be set via setChartStyle.
The data is passed via setData as a set of open/high/low/close values at certain keys (typically times). This means the data must be already binned appropriately. If data is only available as a series of values (e.g. *price* against *time*), you can use the static convenience function timeSeriesToOhlc to generate binned OHLC-data which can then be passed to setData.
The width of the OHLC bars/candlesticks can be controlled with setWidth and setWidthType. A typical choice is to set the width type to wtPlotCoords (the default) and the width to (or slightly less than) one time bin interval width.

17.630.2 Changing the appearance

Charts can be either single- or two-colored (setTwoColored). If set to be single-colored, lines are drawn with the plottable's pen (setPen) and fills with the brush (setBrush).
If set to two-colored, positive changes of the value during an interval (*close* >= *open*) are represented with a different pen and brush than negative changes (*close* < *open*). These can be configured with setPenPositive, setPenNegative, setBrushPositive, and setBrushNegative. In two-colored mode, the normal plottable pen/brush is ignored. Upon selection however, the normal selected pen/brush (provided by the selectionDecorator) is used, irrespective of whether the chart is single- or two-colored.

17.630.3 Usage

Like all data representing objects in QCustomPlot, the QCPFinancial is a plottable (QCPAbstractPlottable). So the plottable-interface of QCustomPlot applies (QCustomPlot::plottable, QCustomPlot::removePlottable, etc.)
Usually, you first create an instance:
which registers it with the QCustomPlot instance of the passed axes. Note that this QCustomPlot instance takes ownership of the plottable, so do not delete it manually but use QCustomPlot::removePlottable() instead. The newly created plottable can be modified, e.g.:
Here we have used the static helper method timeSeriesToOhlc, to turn a time-price data series into a 24-hour binned open-high-low-close data series as QCPFinancial uses.

17.630.4 Member Enumeration Documentation

17.630.4.1 ChartStyle

`enum QCPFinancial::ChartStyle`
Defines the possible representations of OHLC data in the plot.

See also

setChartStyle

Enumerator

csOhlc	Open-High-Low-Close bar representation.
csCandlestick	Candlestick representation.

17.630.4.2 WidthType

```
enum QCPFinancial::WidthType
```

Defines the ways the width of the financial bar can be specified. Thus it defines what the number passed to setWidth actually means.

See also

setWidthType, setWidth

Enumerator

wtAbsolute	width is in absolute pixels
wtAxisRectRatio	width is given by a fraction of the axis rect size
wtPlotCoords	width is in key coordinates and thus scales with the key axis range

17.630.5 Constructor & Destructor Documentation

17.630.5.1 QCPFinancial()

```
QCPFinancial::QCPFinancial (
            QCPAxis * keyAxis,
            QCPAxis * valueAxis )  [explicit]
```

Constructs a financial chart which uses *keyAxis* as its key axis ("x") and *valueAxis* as its value axis ("y"). *keyAxis* and *valueAxis* must reside in the same QCustomPlot instance and not have the same orientation. If either of these restrictions is violated, a corresponding message is printed to the debug output (qDebug), the construction is not aborted, though.

The created QCPFinancial is automatically registered with the QCustomPlot instance inferred from *key*↩ *Axis*. This QCustomPlot instance takes ownership of the QCPFinancial, so do not delete it manually but use QCustomPlot::removePlottable() instead.

17.630.6 Member Function Documentation

17.630.6.1 addData() [1/2]

```
void QCPFinancial::addData (
            const QVector< double > & keys,
            const QVector< double > & open,
            const QVector< double > & high,
            const QVector< double > & low,
            const QVector< double > & close,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.

Adds the provided points in *keys*, *open*, *high*, *low* and *close* to the current data. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.

If you can guarantee that the passed data points are sorted by *keys* in ascending order, you can set *alreadySorted* to true, to improve performance by saving a sorting run.

Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

See also

timeSeriesToOhlc

17.630.6.2 addData() [2/2]

```
void QCPFinancial::addData (
            double key,
```

```
            double open,
            double high,
            double low,
            double close )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds the provided data point as *key*, *open*, *high*, *low* and *close* to the current data.
Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

See also

timeSeriesToOhlc

17.630.6.3 data()

```
QCPFinancialDataContainer * QCPFinancial::data ( ) const  [inline]
```

Returns a pointer to the internal data storage of type QCPFinancialDataContainer. You may use it to directly manipulate the data, which may be more convenient and faster than using the regular setData or addData methods, in certain situations.

17.630.6.4 draw()

```
void QCPFinancial::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.630.6.5 drawLegendIcon()

```
void QCPFinancial::drawLegendIcon (
            QCPPainter * painter,
            const QRectF & rect ) const  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.630.6.6 getKeyRange()

```
QCPRange QCPFinancial::getKeyRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth ) const  [virtual]
```

Returns the coordinate range that all data in this plottable span in the key axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSignDomain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getValueRange

Implements QCPAbstractPlottable.

17.630.6.7 getValueRange()

```
QCPRange QCPFinancial::getValueRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth,
            const QCPRange & inKeyRange = QCPRange() ) const  [virtual]
```

Returns the coordinate range that the data points in the specified key range (*inKeyRange*) span in the value axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSign↩ Domain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
If *inKeyRange* has both lower and upper bound set to zero (is equal to `QCPRange()`), all data points are considered, without any restriction on the keys.
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getKeyRange

Implements QCPAbstractPlottable.

17.630.6.8 selectTest()

```
double QCPFinancial::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

Implements a selectTest specific to this plottable's point geometry.
If *details* is not 0, it will be set to a QCPDataSelection, describing the closest data point to *pos*.
\seebaseclassmethod QCPAbstractPlottable::selectTest
Reimplemented from QCPAbstractPlottable1D< QCPFinancialData >.

17.630.6.9 selectTestRect()

```
QCPDataSelection QCPFinancial::selectTestRect (
            const QRectF & rect,
            bool onlySelectable ) const  [virtual]
```

Returns a data selection containing all the data points of this plottable which are contained (or hit by) *rect*. This is used mainly in the selection rect interaction for data selection (data selection mechanism).
If *onlySelectable* is true, an empty QCPDataSelection is returned if this plottable is not selectable (i.e. if QCPAbstractPlottable::setSelectable is QCP::stNone).

Note

rect must be a normalized rect (positive or zero width and height). This is especially important when using the rect of QCPSelectionRect::accepted, which is not necessarily normalized. Use `QRect::normalized()` when passing a rect which might not be normalized.

Reimplemented from QCPAbstractPlottable1D< QCPFinancialData >.

17.630.6.10 setBrushNegative()

```
void QCPFinancial::setBrushNegative (
            const QBrush & brush )
```

If setTwoColored is set to true, this function controls the brush that is used to draw fills of data points with a negative trend (i.e. bars/candlesticks with close < open).
If *twoColored* is false, the normal plottable's pen and brush are used (setPen, setBrush).

See also

setBrushPositive, setPenNegative, setPenPositive

17.630.6.11 setBrushPositive()

```
void QCPFinancial::setBrushPositive (
            const QBrush & brush )
```

If setTwoColored is set to true, this function controls the brush that is used to draw fills of data points with a positive trend (i.e. bars/candlesticks with close >= open).
If *twoColored* is false, the normal plottable's pen and brush are used (setPen, setBrush).

See also

setBrushNegative, setPenPositive, setPenNegative

17.630.6.12 setChartStyle()

```
void QCPFinancial::setChartStyle (
            QCPFinancial::ChartStyle style )
```

Sets which representation style shall be used to display the OHLC data.

17.630.6.13 setData() [1/2]

```
void QCPFinancial::setData (
            const QVector< double > & keys,
            const QVector< double > & open,
            const QVector< double > & high,
            const QVector< double > & low,
            const QVector< double > & close,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data with the provided points in *keys*, *open*, *high*, *low* and *close*. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.
If you can guarantee that the passed data points are sorted by *keys* in ascending order, you can set *alreadySorted* to true, to improve performance by saving a sorting run.

See also

addData, timeSeriesToOhlc

17.630.6.14 setData() [2/2]

```
void QCPFinancial::setData (
            QSharedPointer< QCPFinancialDataContainer > data )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data container with the provided *data* container.
Since a QSharedPointer is used, multiple QCPFinancials may share the same data container safely. Modifying the data in the container will then affect all financials that share the container. Sharing can be achieved by simply exchanging the data containers wrapped in shared pointers:
If you do not wish to share containers, but create a copy from an existing container, rather use the QCPDataContainer<DataType>::set method on the financial's data container directly:

See also

addData, timeSeriesToOhlc

17.630.6.15 setPenNegative()

```
void QCPFinancial::setPenNegative (
            const QPen & pen )
```

If setTwoColored is set to true, this function controls the pen that is used to draw outlines of data points with a negative trend (i.e. bars/candlesticks with close < open).
If *twoColored* is false, the normal plottable's pen and brush are used (setPen, setBrush).

See also

setPenPositive, setBrushNegative, setBrushPositive

17.630.6.16 setPenPositive()

```
void QCPFinancial::setPenPositive (
            const QPen & pen )
```

If setTwoColored is set to true, this function controls the pen that is used to draw outlines of data points with a positive trend (i.e. bars/candlesticks with close >= open).
If *twoColored* is false, the normal plottable's pen and brush are used (setPen, setBrush).

See also

setPenNegative, setBrushPositive, setBrushNegative

17.630.6.17 setTwoColored()

```
void QCPFinancial::setTwoColored (
            bool twoColored )
```

Sets whether this chart shall contrast positive from negative trends per data point by using two separate colors to draw the respective bars/candlesticks.
If *twoColored* is false, the normal plottable's pen and brush are used (setPen, setBrush).

See also

setPenPositive, setPenNegative, setBrushPositive, setBrushNegative

17.630.6.18 setWidth()

```
void QCPFinancial::setWidth (
            double width )
```

Sets the width of the individual bars/candlesticks to *width* in plot key coordinates.
A typical choice is to set it to (or slightly less than) one bin interval width.

17.630.6.19 setWidthType()

```
void QCPFinancial::setWidthType (
            QCPFinancial::WidthType widthType )
```

Sets how the width of the financial bars is defined. See the documentation of WidthType for an explanation of the possible values for *widthType*.
The default value is wtPlotCoords.

See also

setWidth

17.630.6.20 timeSeriesToOhlc()

```
QCPFinancialDataContainer QCPFinancial::timeSeriesToOhlc (
            const QVector< double > & time,
            const QVector< double > & value,
            double timeBinSize,
            double timeBinOffset = 0 )  [static]
```

A convenience function that converts time series data (*value* against *time*) to OHLC binned data points. The return value can then be passed on to QCPFinancialDataContainer::set(const QCPFinancialDataContainer&).
The size of the bins can be controlled with *timeBinSize* in the same units as *time* is given. For example, if the unit of *time* is seconds and single OHLC/Candlesticks should span an hour each, set *timeBinSize* to 3600.
timeBinOffset allows to control precisely at what *time* coordinate a bin should start. The value passed as *timeBinOffset* doesn't need to be in the range encompassed by the *time* keys. It merely defines the mathematical offset/phase of the bins that will be used to process the data.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.631 QCPFinancialData Class Reference

Holds the data of one single data point for QCPFinancial.

Public Member Functions

- QCPFinancialData ()
- QCPFinancialData (double key, double open, double high, double low, double close)
- double sortKey () const
- double mainKey () const
- double mainValue () const
- QCPRange valueRange () const

Static Public Member Functions

- static QCPFinancialData fromSortKey (double sortKey)
- static bool sortKeyIsMainKey ()

Public Attributes

- double **key**
- double **open**
- double **high**
- double **low**
- double **close**

17.631.1 Detailed Description

Holds the data of one single data point for QCPFinancial.
The stored data is:

- *key:* coordinate on the key axis of this data point (this is the *mainKey* and the *sortKey*)
- *open:* The opening value at the data point (this is the *mainValue*)
- *high:* The high/maximum value at the data point
- *low:* The low/minimum value at the data point
- *close:* The closing value at the data point

The container for storing multiple data points is QCPFinancialDataContainer. It is a typedef for QCPDataContainer with QCPFinancialData as the DataType template parameter. See the documentation there for an explanation regarding the data type's generic methods.

See also

QCPFinancialDataContainer

17.631.2 Constructor & Destructor Documentation

17.631.2.1 QCPFinancialData() [1/2]

`QCPFinancialData::QCPFinancialData ( )`
Constructs a data point with key and all values set to zero.

17.631.2.2 QCPFinancialData() [2/2]

```
QCPFinancialData::QCPFinancialData (
            double key,
            double open,
            double high,
            double low,
            double close )
```

Constructs a data point with the specified *key* and OHLC values.

17.631.3 Member Function Documentation

17.631.3.1 fromSortKey()

```
static QCPFinancialData QCPFinancialData::fromSortKey (
            double sortKey )  [inline], [static]
```

Returns a data point with the specified *sortKey*. All other members are set to zero.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.631.3.2 mainKey()

```
double QCPFinancialData::mainKey ( ) const  [inline]
```

Returns the *key* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.631.3.3 mainValue()

```
double QCPFinancialData::mainValue ( ) const  [inline]
```

Returns the *open* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.631.3.4 sortKey()

```
double QCPFinancialData::sortKey ( ) const  [inline]
```

Returns the *key* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.631.3.5 sortKeyIsMainKey()

```
static static bool QCPFinancialData::sortKeyIsMainKey ( )  [inline], [static]
```

Since the member *key* is both the data point key coordinate and the data ordering parameter, this method returns true.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.631.3.6 valueRange()

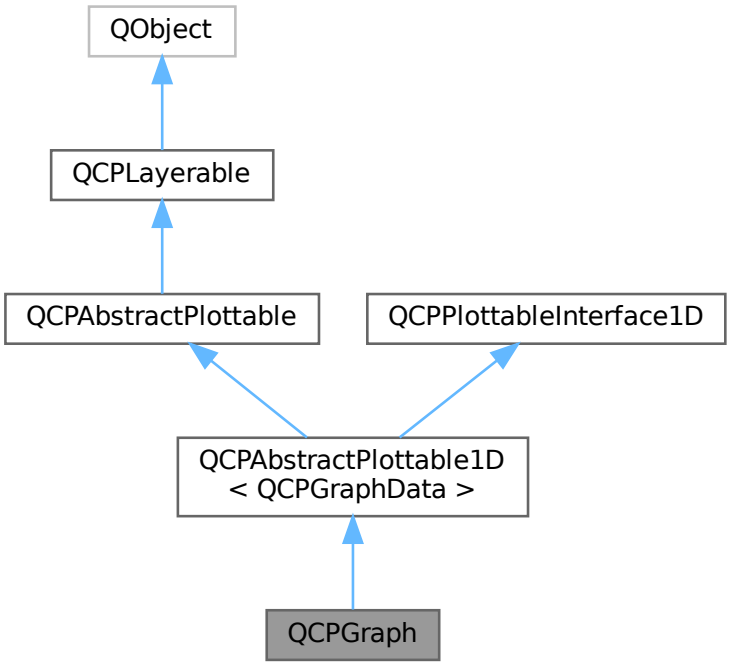
```
QCPRange QCPFinancialData::valueRange ( ) const  [inline]
```

Returns a QCPRange spanning from the *low* to the *high* value of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.
The documentation for this class was generated from the following files:

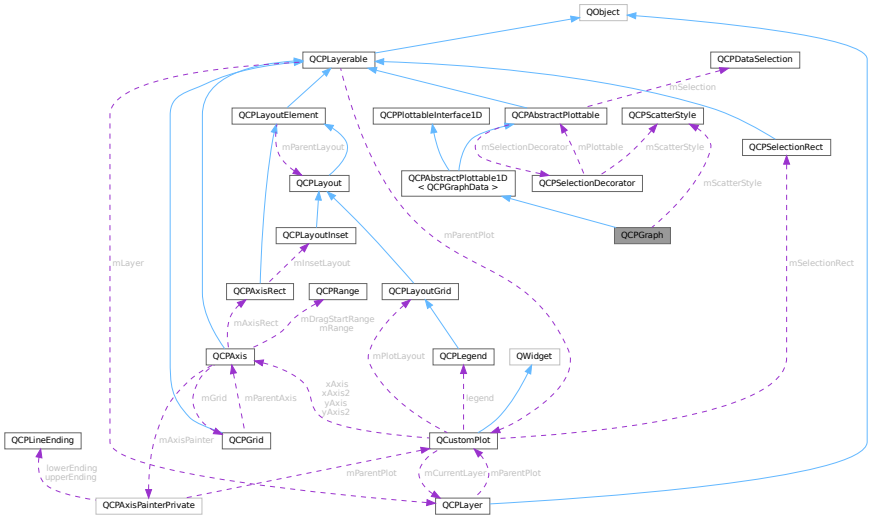
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.632 QCPGraph Class Reference

A plottable representing a graph in a plot.
Inheritance diagram for QCPGraph:



Collaboration diagram for QCPGraph:



Public Types

- enum LineStyle {
  lsNone , lsLine , lsStepLeft , lsStepRight ,
  lsStepCenter , lsImpulse }

Public Member Functions

- QCPGraph (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QSharedPointer< QCPGraphDataContainer > data () const
- LineStyle **lineStyle** () const
- QCPScatterStyle **scatterStyle** () const
- int **scatterSkip** () const
- QCPGraph * **channelFillGraph** () const
- bool **adaptiveSampling** () const
- void setData (QSharedPointer< QCPGraphDataContainer > data)
- void setData (const QVector< double > &keys, const QVector< double > &values, bool alreadySorted=false)
- void setLineStyle (LineStyle ls)
- void setScatterStyle (const QCPScatterStyle &style)
- void setScatterSkip (int skip)
- void setChannelFillGraph (QCPGraph *targetGraph)
- void setAdaptiveSampling (bool enabled)
- void addData (const QVector< double > &keys, const QVector< double > &values, bool already↩ Sorted=false)
- void addData (double key, double value)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE
- virtual QCPRange getKeyRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth) const Q_DECL_OVERRIDE
- virtual QCPRange getValueRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth, const QCPRange &inKeyRange=QCPRange()) const Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable1D< QCPGraphData >

- QCPAbstractPlottable1D (QCPAxis *keyAxis, QCPAxis *valueAxis)
- virtual int dataCount () const Q_DECL_OVERRIDE
- virtual double dataMainKey (int index) const Q_DECL_OVERRIDE
- virtual double dataSortKey (int index) const Q_DECL_OVERRIDE
- virtual double dataMainValue (int index) const Q_DECL_OVERRIDE
- virtual QCPRange dataValueRange (int index) const Q_DECL_OVERRIDE
- virtual QPointF dataPixelPosition (int index) const Q_DECL_OVERRIDE
- virtual bool sortKeyIsMainKey () const Q_DECL_OVERRIDE
- virtual QCPDataSelection selectTestRect (const QRectF &rect, bool onlySelectable) const Q_DECL_↩ OVERRIDE
- virtual int findBegin (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual int findEnd (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual QCPPlottableInterface1D * interface1D () Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable

- QCPAbstractPlottable (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QString **name** () const
- bool **antialiasedFill** () const
- bool **antialiasedScatters** () const
- QPen **pen** () const
- QBrush **brush** () const

- QCPAxis * **keyAxis** () const
- QCPAxis * **valueAxis** () const
- QCP::SelectionType **selectable** () const
- bool selected () const
- QCPDataSelection selection () const
- QCPSelectionDecorator * selectionDecorator () const
- void setName (const QString &name)
- void setAntialiasedFill (bool enabled)
- void setAntialiasedScatters (bool enabled)
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setKeyAxis (QCPAxis *axis)
- void setValueAxis (QCPAxis *axis)
- Q_SLOT void setSelectable (QCP::SelectionType selectable)
- Q_SLOT void setSelection (QCPDataSelection selection)
- void setSelectionDecorator (QCPSelectionDecorator *decorator)
- void coordsToPixels (double key, double value, double &x, double &y) const
- const QPointF coordsToPixels (double key, double value) const
- void pixelsToCoords (double x, double y, double &key, double &value) const
- void pixelsToCoords (const QPointF &pixelPos, double &key, double &value) const
- void rescaleAxes (bool onlyEnlarge=false) const
- void rescaleKeyAxis (bool onlyEnlarge=false) const
- void rescaleValueAxis (bool onlyEnlarge=false, bool inKeyRange=false) const
- bool addToLegend (QCPLegend *legend)
- bool addToLegend ()
- bool removeFromLegend (QCPLegend *legend) const
- bool removeFromLegend () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void drawLegendIcon (QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE
- virtual void **drawFill** (QCPPainter *painter, QVector< QPointF > *lines) const
- virtual void **drawScatterPlot** (QCPPainter *painter, const QVector< QPointF > &scatters, const QCPScatterStyle &style) const
- virtual void **drawLinePlot** (QCPPainter *painter, const QVector< QPointF > &lines) const
- virtual void **drawImpulsePlot** (QCPPainter *painter, const QVector< QPointF > &lines) const
- virtual void **getOptimizedLineData** (QVector< QCPGraphData > *lineData, const QCPGraphData↩ Container::const_iterator &begin, const QCPGraphDataContainer::const_iterator &end) const
- virtual void **getOptimizedScatterData** (QVector< QCPGraphData > *scatterData, QCPGraphData↩ Container::const_iterator begin, QCPGraphDataContainer::const_iterator end) const

- void getVisibleDataBounds (QCPGraphDataContainer::const_iterator &begin, QCPGraphDataContainer↩ ::const_iterator &end, const QCPDataRange &rangeRestriction) const
- void **getLines** (QVector< QPointF > *lines, const QCPDataRange &dataRange) const
- void **getScatters** (QVector< QPointF > *scatters, const QCPDataRange &dataRange) const
- QVector< QPointF > **dataToLines** (const QVector< QCPGraphData > &data) const
- QVector< QPointF > **dataToStepLeftLines** (const QVector< QCPGraphData > &data) const
- QVector< QPointF > **dataToStepRightLines** (const QVector< QCPGraphData > &data) const
- QVector< QPointF > **dataToStepCenterLines** (const QVector< QCPGraphData > &data) const
- QVector< QPointF > **dataToImpulseLines** (const QVector< QCPGraphData > &data) const
- QVector< QCPDataRange > **getNonNanSegments** (const QVector< QPointF > *lineData, Qt::Orientation keyOrientation) const
- QVector< QPair< QCPDataRange, QCPDataRange > > **getOverlappingSegments** (QVector< QCPDataRange > thisSegments, const QVector< QPointF > *thisData, QVector< QCPDataRange > otherSegments, const QVector< QPointF > *otherData) const
- bool **segmentsIntersect** (double aLower, double aUpper, double bLower, double bUpper, int &bPrecedence) const
- QPointF **getFillBasePoint** (QPointF matchingDataPoint) const
- const QPolygonF **getFillPolygon** (const QVector< QPointF > *lineData, QCPDataRange segment) const
- const QPolygonF **getChannelFillPolygon** (const QVector< QPointF > *thisData, QCPDataRange this↩ Segment, const QVector< QPointF > *otherData, QCPDataRange otherSegment) const
- int **findIndexBelowX** (const QVector< QPointF > *data, double x) const
- int **findIndexAboveX** (const QVector< QPointF > *data, double x) const
- int **findIndexBelowY** (const QVector< QPointF > *data, double y) const
- int **findIndexAboveY** (const QVector< QPointF > *data, double y) const
- double **pointDistance** (const QPointF &pixelPoint, QCPGraphDataContainer::const_iterator &closestData) const

Protected Member Functions inherited from QCPAbstractPlottable1D< QCPGraphData >

- void getDataSegments (QList< QCPDataRange > &selectedSegments, QList< QCPDataRange > &unselectedSegments) const
- void drawPolyline (QCPPainter *painter, const QVector< QPointF > &lineData) const

Protected Member Functions inherited from QCPAbstractPlottable

- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- void **applyFillAntialiasingHint** (QCPPainter *painter) const
- void **applyScattersAntialiasingHint** (QCPPainter *painter) const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- LineStyle **mLineStyle**
- QCPScatterStyle **mScatterStyle**
- int **mScatterSkip**
- QPointer< QCPGraph > **mChannelFillGraph**
- bool **mAdaptiveSampling**

Protected Attributes inherited from QCPAbstractPlottable1D< QCPGraphData >

- QSharedPointer< QCPDataContainer< QCPGraphData > > **mDataContainer**

Protected Attributes inherited from QCPAbstractPlottable

- QString **mName**
- bool **mAntialiasedFill**
- bool **mAntialiasedScatters**
- QPen **mPen**
- QBrush **mBrush**
- QPointer< QCPAxis > **mKeyAxis**
- QPointer< QCPAxis > **mValueAxis**
- QCP::SelectionType **mSelectable**
- QCPDataSelection **mSelection**
- QCPSelectionDecorator * **mSelectionDecorator**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPLegend**

Additional Inherited Members

Signals inherited from QCPAbstractPlottable

- void selectionChanged (bool selected)
- void selectionChanged (const QCPDataSelection &selection)
- void selectableChanged (QCP::SelectionType selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.632.1 Detailed Description

A plottable representing a graph in a plot.
Usually you create new graphs by calling QCustomPlot::addGraph. The resulting instance can be accessed via QCustomPlot::graph.
To plot data, assign it with the setData or addData functions. Alternatively, you can also access and modify the data via the data method, which returns a pointer to the internal QCPGraphDataContainer.
Graphs are used to display single-valued data. Single-valued means that there should only be one data point per unique key coordinate. In other words, the graph can't have *loops*. If you do want to plot non-single-valued curves, rather use the QCPCurve plottable.
Gaps in the graph line can be created by adding data points with NaN as value (`qQNaN()` or `std::numeric_limits<double>::quiet_NaN()`) in between the two data points that shall be separated.

17.632.2 Changing the appearance

The appearance of the graph is mainly determined by the line style, scatter style, brush and pen of the graph (setLineStyle, setScatterStyle, setBrush, setPen).

17.632.2.1 Filling under or between graphs

QCPGraph knows two types of fills: Normal graph fills towards the zero-value-line parallel to the key axis of the graph, and fills between two graphs, called channel fills. To enable a fill, just set a brush with setBrush which is neither Qt::NoBrush nor fully transparent.
By default, a normal fill towards the zero-value-line will be drawn. To set up a channel fill between this graph and another one, call setChannelFillGraph with the other graph as parameter.

See also

QCustomPlot::addGraph, QCustomPlot::graph

17.632.3 Member Enumeration Documentation

17.632.3.1 LineStyle

`enum QCPGraph::LineStyle`
Defines how the graph's line is represented visually in the plot. The line is drawn with the current pen of the graph (setPen).

See also

setLineStyle

Enumerator

lsNone	with symbols according to the scatter style, see setScatterStyle) data points are not connected with any lines (e.g. data only represented
lsLine	data points are connected by a straight line
lsStepLeft	line is drawn as steps where the step height is the value of the left data point
lsStepRight	line is drawn as steps where the step height is the value of the right data point
lsStepCenter	line is drawn as steps where the step is in between two data points
lsImpulse	each data point is represented by a line parallel to the value axis, which reaches from the data point to the zero-value-line

17.632.4 Constructor & Destructor Documentation

17.632.4.1 QCPGraph()

`QCPGraph::QCPGraph (`

```
            QCPAxis * keyAxis,
            QCPAxis * valueAxis )  [explicit]
```

Constructs a graph which uses *keyAxis* as its key axis ("x") and *valueAxis* as its value axis ("y"). *keyAxis* and *valueAxis* must reside in the same QCustomPlot instance and not have the same orientation. If either of these restrictions is violated, a corresponding message is printed to the debug output (qDebug), the construction is not aborted, though.
The created QCPGraph is automatically registered with the QCustomPlot instance inferred from *keyAxis*. This QCustomPlot instance takes ownership of the QCPGraph, so do not delete it manually but use QCustomPlot::removePlottable() instead.
To directly create a graph inside a plot, you can also use the simpler QCustomPlot::addGraph function.

17.632.5 Member Function Documentation

17.632.5.1 addData() [1/2]

```
void QCPGraph::addData (
            const QVector< double > & keys,
            const QVector< double > & values,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds the provided points in *keys* and *values* to the current data. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.
If you can guarantee that the passed data points are sorted by *keys* in ascending order, you can set *alreadySorted* to true, to improve performance by saving a sorting run.
Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

17.632.5.2 addData() [2/2]

```
void QCPGraph::addData (
            double key,
            double value )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Adds the provided data point as *key* and *value* to the current data.
Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

17.632.5.3 data()

```
QSharedPointer< QCPGraphDataContainer > QCPGraph::data ( ) const  [inline]
```

Returns a shared pointer to the internal data storage of type QCPGraphDataContainer. You may use it to directly manipulate the data, which may be more convenient and faster than using the regular setData or addData methods.

17.632.5.4 draw()

```
void QCPGraph::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.632.5.5 drawLegendIcon()

```
void QCPGraph::drawLegendIcon (
            QCPPainter * painter,
            const QRectF & rect ) const  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.632.5.6 getKeyRange()

```
QCPRange QCPGraph::getKeyRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth ) const  [virtual]
```

Returns the coordinate range that all data in this plottable span in the key axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSignDomain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getValueRange

Implements QCPAbstractPlottable.

17.632.5.7 getValueRange()

```
QCPRange QCPGraph::getValueRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth,
            const QCPRange & inKeyRange = QCPRange() ) const  [virtual]
```

Returns the coordinate range that the data points in the specified key range (*inKeyRange*) span in the value axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSign↩ Domain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
If *inKeyRange* has both lower and upper bound set to zero (is equal to `QCPRange()`), all data points are considered, without any restriction on the keys.
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getKeyRange

Implements QCPAbstractPlottable.

17.632.5.8 getVisibleDataBounds()

```
void QCPGraph::getVisibleDataBounds (
            QCPGraphDataContainer::const_iterator & begin,
            QCPGraphDataContainer::const_iterator & end,
            const QCPDataRange & rangeRestriction ) const  [protected]
```

This method outputs the currently visible data range via *begin* and *end*. The returned range will also never exceed *rangeRestriction*.
This method takes into account that the drawing of data lines at the axis rect border always requires the points just outside the visible axis range. So *begin* and *end* may actually indicate a range that contains one additional data point to the left and right of the visible axis range.

17.632.5.9 selectTest()

```
double QCPGraph::selectTest (
            const QPointF & pos,
```

```
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

Implements a selectTest specific to this plottable's point geometry.
If *details* is not 0, it will be set to a QCPDataSelection, describing the closest data point to *pos*.
\seebaseclassmethod QCPAbstractPlottable::selectTest
Reimplemented from QCPAbstractPlottable1D< QCPGraphData >.

17.632.5.10 setAdaptiveSampling()

```
void QCPGraph::setAdaptiveSampling (
            bool enabled )
```

Sets whether adaptive sampling shall be used when plotting this graph. QCustomPlot's adaptive sampling technique can drastically improve the replot performance for graphs with a larger number of points (e.g. above 10,000), without notably changing the appearance of the graph.
By default, adaptive sampling is enabled. Even if enabled, QCustomPlot decides whether adaptive sampling shall actually be used on a per-graph basis. So leaving adaptive sampling enabled has no disadvantage in almost all cases.
As can be seen, line plots experience no visual degradation from adaptive sampling. Outliers are reproduced reliably, as well as the overall shape of the data set. The replot time reduces dramatically though. This allows QCustomPlot to display large amounts of data in realtime.
Care must be taken when using high-density scatter plots in combination with adaptive sampling. The adaptive sampling algorithm treats scatter plots more carefully than line plots which still gives a significant reduction of replot times, but not quite as much as for line plots. This is because scatter plots inherently need more data points to be preserved in order to still resemble the original, non-adaptive-sampling plot. As shown above, the results still aren't quite identical, as banding occurs for the outer data points. This is in fact intentional, such that the boundaries of the data cloud stay visible to the viewer. How strong the banding appears, depends on the point density, i.e. the number of points in the plot.
For some situations with scatter plots it might thus be desirable to manually turn adaptive sampling off. For example, when saving the plot to disk. This can be achieved by setting *enabled* to false before issuing a command like QCustomPlot::savePng, and setting *enabled* back to true afterwards.

17.632.5.11 setChannelFillGraph()

```
void QCPGraph::setChannelFillGraph (
            QCPGraph * targetGraph )
```

Sets the target graph for filling the area between this graph and *targetGraph* with the current brush (setBrush).
When *targetGraph* is set to 0, a normal graph fill to the zero-value-line will be shown. To disable any filling, set the brush to Qt::NoBrush.

See also

setBrush

17.632.5.12 setData() [1/2]

```
void QCPGraph::setData (
            const QVector< double > & keys,
            const QVector< double > & values,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data with the provided points in *keys* and *values*. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.
If you can guarantee that the passed data points are sorted by *keys* in ascending order, you can set *alreadySorted* to true, to improve performance by saving a sorting run.

See also

addData

17.632.5.13 setData() [2/2]

```
void QCPGraph::setData (
            QSharedPointer< QCPGraphDataContainer > data )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data container with the provided *data* container.
Since a QSharedPointer is used, multiple QCPGraphs may share the same data container safely. Modifying the data in the container will then affect all graphs that share the container. Sharing can be achieved by simply exchanging the data containers wrapped in shared pointers:
If you do not wish to share containers, but create a copy from an existing container, rather use the QCPDataContainer<DataType>::set method on the graph's data container directly:

See also

addData

17.632.5.14 setLineStyle()

```
void QCPGraph::setLineStyle (
            LineStyle ls )
```

Sets how the single data points are connected in the plot. For scatter-only plots, set *ls* to lsNone and setScatterStyle to the desired scatter style.

See also

setScatterStyle

17.632.5.15 setScatterSkip()

```
void QCPGraph::setScatterSkip (
            int skip )
```

If scatters are displayed (scatter style not QCPScatterStyle::ssNone), *skip* number of scatter points are skipped/not drawn after every drawn scatter point.
This can be used to make the data appear sparser while for example still having a smooth line, and to improve performance for very high density plots.
If *skip* is set to 0 (default), all scatter points are drawn.

See also

setScatterStyle

17.632.5.16 setScatterStyle()

```
void QCPGraph::setScatterStyle (
            const QCPScatterStyle & style )
```

Sets the visual appearance of single data points in the plot. If set to QCPScatterStyle::ssNone, no scatter points are drawn (e.g. for line-only-plots with appropriate line style).

See also

QCPScatterStyle, setLineStyle

The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.633 QCPGraphData Class Reference

Holds the data of one single data point for QCPGraph.

Public Member Functions

- QCPGraphData ()
- QCPGraphData (double key, double value)
- double sortKey () const
- double mainKey () const
- double mainValue () const
- QCPRange valueRange () const

Static Public Member Functions

- static QCPGraphData fromSortKey (double sortKey)
- static bool sortKeyIsMainKey ()

Public Attributes

- double **key**
- double **value**

17.633.1 Detailed Description

Holds the data of one single data point for QCPGraph.
The stored data is:

- *key:* coordinate on the key axis of this data point (this is the *mainKey* and the *sortKey*)
- *value:* coordinate on the value axis of this data point (this is the *mainValue*)

The container for storing multiple data points is QCPGraphDataContainer. It is a typedef for QCPDataContainer with QCPGraphData as the DataType template parameter. See the documentation there for an explanation regarding the data type's generic methods.

See also

QCPGraphDataContainer

17.633.2 Constructor & Destructor Documentation

17.633.2.1 QCPGraphData() [1/2]

```
QCPGraphData::QCPGraphData ( )
```

Constructs a data point with key and value set to zero.

17.633.2.2 QCPGraphData() [2/2]

```
QCPGraphData::QCPGraphData (
            double key,
            double value )
```

Constructs a data point with the specified *key* and *value*.

17.633.3 Member Function Documentation

17.633.3.1 fromSortKey()

```
static QCPGraphData QCPGraphData::fromSortKey (
            double sortKey )  [inline], [static]
```

Returns a data point with the specified *sortKey*. All other members are set to zero.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.633.3.2 mainKey()

```
double QCPGraphData::mainKey ( ) const  [inline]
```

Returns the *key* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.633.3.3 mainValue()

```
double QCPGraphData::mainValue ( ) const  [inline]
```

Returns the *value* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.633.3.4 sortKey()

```
double QCPGraphData::sortKey ( ) const  [inline]
```

Returns the *key* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.633.3.5 sortKeyIsMainKey()

```
static static bool QCPGraphData::sortKeyIsMainKey ( )  [inline], [static]
```

Since the member *key* is both the data point key coordinate and the data ordering parameter, this method returns true.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.633.3.6 valueRange()

```
QCPRange QCPGraphData::valueRange ( ) const  [inline]
```

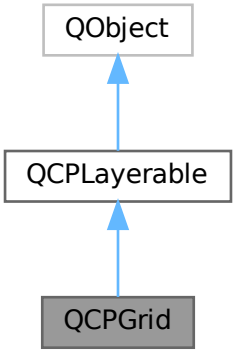
Returns a QCPRange with both lower and upper boundary set to *value* of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

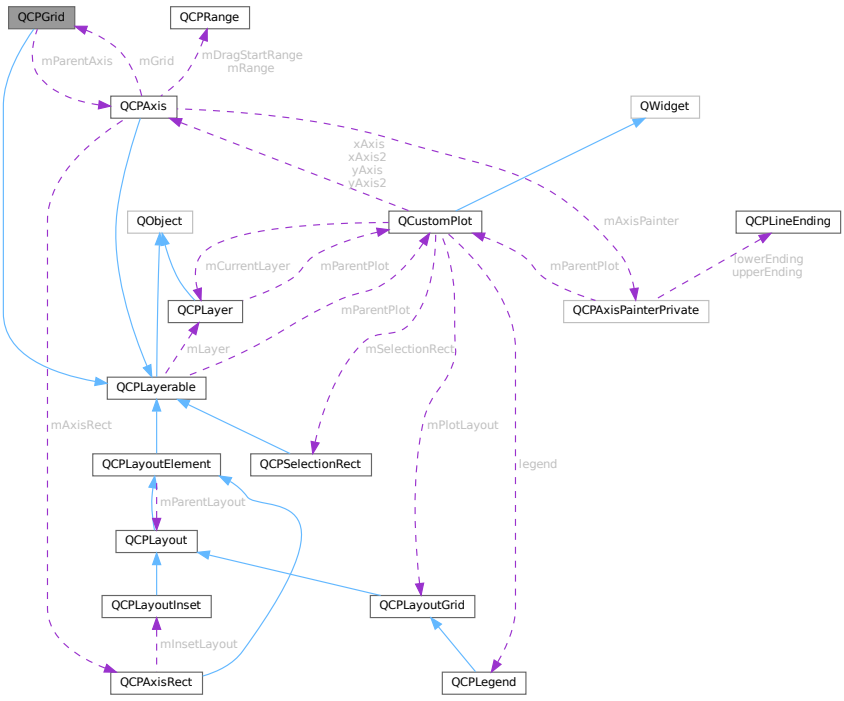
17.634 QCPGrid Class Reference

Responsible for drawing the grid of a QCPAxis.

Inheritance diagram for QCPGrid:



Collaboration diagram for QCPGrid:



Public Member Functions

- QCPGrid (QCPAxis *parentAxis)
- bool **subGridVisible** () const
- bool **antialiasedSubGrid** () const
- bool **antialiasedZeroLine** () const
- QPen **pen** () const
- QPen **subGridPen** () const

- QPen **zeroLinePen** () const
- void setSubGridVisible (bool visible)
- void setAntialiasedSubGrid (bool enabled)
- void setAntialiasedZeroLine (bool enabled)
- void setPen (const QPen &pen)
- void setSubGridPen (const QPen &pen)
- void setZeroLinePen (const QPen &pen)

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const
- bool realVisibility () const

Protected Member Functions

- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- void **drawGridLines** (QCPPainter *painter) const
- void **drawSubGridLines** (QCPPainter *painter) const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void **selectEvent** (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged)
- virtual void **deselectEvent** (bool *selectionStateChanged)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- bool **mSubGridVisible**
- bool **mAntialiasedSubGrid**
- bool **mAntialiasedZeroLine**
- QPen **mPen**
- QPen **mSubGridPen**
- QPen **mZeroLinePen**
- QCPAxis * **mParentAxis**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCPAxis**

Additional Inherited Members

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.634.1 Detailed Description

Responsible for drawing the grid of a QCPAxis.
This class is tightly bound to QCPAxis. Every axis owns a grid instance and uses it to draw the grid lines, sub grid lines and zero-line. You can interact with the grid of an axis via QCPAxis::grid. Normally, you don't need to create an instance of QCPGrid yourself.
The axis and grid drawing was split into two classes to allow them to be placed on different layers (both QCPAxis and QCPGrid inherit from QCPLayerable). Thus it is possible to have the grid in the background and the axes in the foreground, and any plottables/items in between. This described situation is the default setup, see the QCPLayer documentation.

17.634.2 Constructor & Destructor Documentation

17.634.2.1 QCPGrid()

```
QCPGrid::QCPGrid (
            QCPAxis * parentAxis )  [explicit]
```

Creates a QCPGrid instance and sets default values.
You shouldn't instantiate grids on their own, since every QCPAxis brings its own QCPGrid.

17.634.3 Member Function Documentation

17.634.3.1 applyDefaultAntialiasingHint()

```
void QCPGrid::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```

Implements QCPLayerable.

17.634.3.2 draw()

```
void QCPGrid::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPLayerable.

17.634.3.3 setAntialiasedSubGrid()

```
void QCPGrid::setAntialiasedSubGrid (
            bool enabled )
```

Sets whether sub grid lines are drawn antialiased.

17.634.3.4 setAntialiasedZeroLine()

```
void QCPGrid::setAntialiasedZeroLine (
            bool enabled )
```

Sets whether zero lines are drawn antialiased.

17.634.3.5 setPen()

```
void QCPGrid::setPen (
            const QPen & pen )
```

Sets the pen with which (major) grid lines are drawn.

17.634.3.6 setSubGridPen()

```
void QCPGrid::setSubGridPen (
            const QPen & pen )
```

Sets the pen with which sub grid lines are drawn.

17.634.3.7 setSubGridVisible()

```
void QCPGrid::setSubGridVisible (
            bool visible )
```

Sets whether grid lines at sub tick marks are drawn.

See also

setSubGridPen

17.634.3.8 setZeroLinePen()

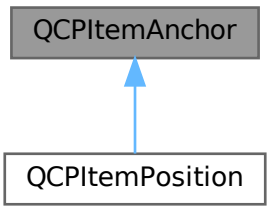
```
void QCPGrid::setZeroLinePen (
            const QPen & pen )
```

Sets the pen with which zero lines are drawn.
Zero lines are lines at value coordinate 0 which may be drawn with a different pen than other grid lines. To disable zero lines and just draw normal grid lines at zero, set *pen* to Qt::NoPen.
The documentation for this class was generated from the following files:

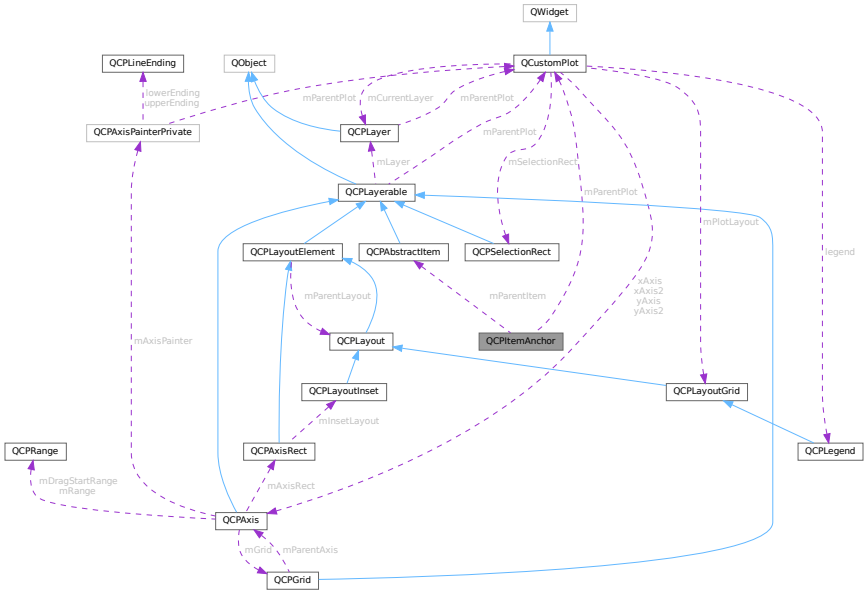
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.635 QCPItemAnchor Class Reference

An anchor of an item to which positions can be attached to.
Inheritance diagram for QCPItemAnchor:



Collaboration diagram for QCPItemAnchor:



Public Member Functions

- QCPItemAnchor (QCustomPlot *parentPlot, QCPAbstractItem *parentItem, const QString &name, int anchorId=-1)
- QString **name** () const
- virtual QPointF pixelPosition () const

Protected Member Functions

- virtual QCPItemPosition * toQCPItemPosition ()
- void **addChildX** (QCPItemPosition *pos)
- void **removeChildX** (QCPItemPosition *pos)
- void **addChildY** (QCPItemPosition *pos)
- void **removeChildY** (QCPItemPosition *pos)

Protected Attributes

- QString **mName**
- QCustomPlot * **mParentPlot**
- QCPAbstractItem * **mParentItem**
- int **mAnchorId**
- QSet< QCPItemPosition * > **mChildrenX**
- QSet< QCPItemPosition * > **mChildrenY**

Friends

- class **QCPItemPosition**

17.635.1 Detailed Description

An anchor of an item to which positions can be attached to.
An item (QCPAbstractItem) may have one or more anchors. Unlike QCPItemPosition, an anchor doesn't control anything on its item, but provides a way to tie other items via their positions to the anchor.

For example, a QCPItemRect is defined by its positions *topLeft* and *bottomRight*. Additionally it has various anchors like *top*, *topRight* or *bottomLeft* etc. So you can attach the *start* (which is a QCPItemPosition) of a QCPItemLine to one of the anchors by calling QCPItemPosition::setParentAnchor on *start*, passing the wanted anchor of the QCPItemRect. This way the start of the line will now always follow the respective anchor location on the rect item.
Note that QCPItemPosition derives from QCPItemAnchor, so every position can also serve as an anchor to other positions.
To learn how to provide anchors in your own item subclasses, see the subclassing section of the QCPAbstractItem documentation.

17.635.2 Constructor & Destructor Documentation

17.635.2.1 QCPItemAnchor()

```
QCPItemAnchor::QCPItemAnchor (
            QCustomPlot * parentPlot,
            QCPAbstractItem * parentItem,
            const QString & name,
            int anchorId = -1 )
```

Creates a new QCPItemAnchor. You shouldn't create QCPItemAnchor instances directly, even if you want to make a new item subclass. Use QCPAbstractItem::createAnchor instead, as explained in the subclassing section of the QCPAbstractItem documentation.

17.635.3 Member Function Documentation

17.635.3.1 pixelPosition()

```
QPointF QCPItemAnchor::pixelPosition ( ) const  [virtual]
```

Returns the final absolute pixel position of the QCPItemAnchor on the QCustomPlot surface.
The pixel information is internally retrieved via QCPAbstractItem::anchorPixelPosition of the parent item, QCPItemAnchor is just an intermediary.
Reimplemented in QCPItemPosition.

17.635.3.2 toQCPItemPosition()

```
QCPItemPosition * QCPItemAnchor::toQCPItemPosition ( )  [inline], [protected], [virtual]
```

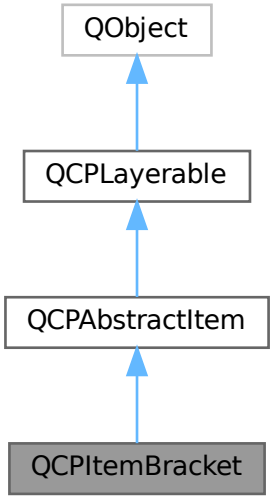
Returns `nullptr` if this instance is merely a QCPItemAnchor, and a valid pointer of type QCPItemPosition* if it actually is a QCPItemPosition (which is a subclass of QCPItemAnchor).
This safe downcast functionality could also be achieved with a dynamic_cast. However, QCustomPlot avoids dynamic_cast to work with projects that don't have RTTI support enabled (e.g. -fno-rtti flag with gcc compiler).
Reimplemented in QCPItemPosition.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

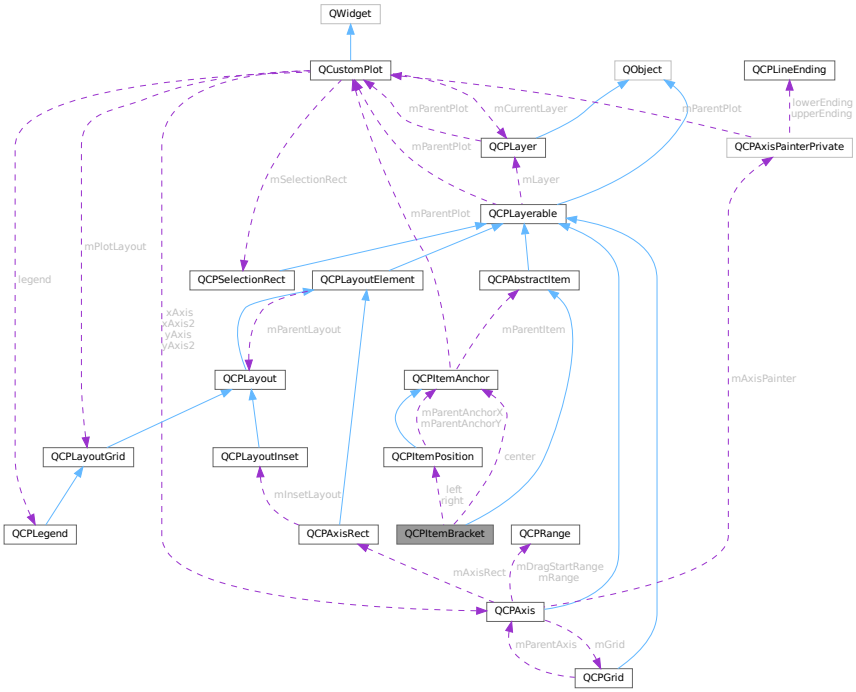
17.636 QCPItemBracket Class Reference

A bracket for referencing/highlighting certain parts in the plot.

Inheritance diagram for QCPItemBracket:



Collaboration diagram for QCPItemBracket:



Public Types

- enum BracketStyle { bsSquare , bsRound , bsCurly , bsCalligraphic }

Public Member Functions

- QCPItemBracket (QCustomPlot *parentPlot)
- QPen **pen** () const
- QPen **selectedPen** () const
- double **length** () const
- BracketStyle **style** () const
- void setPen (const QPen &pen)
- void setSelectedPen (const QPen &pen)
- void setLength (double length)
- void setStyle (BracketStyle style)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractItem

- QCPAbstractItem (QCustomPlot *parentPlot)
- bool **clipToAxisRect** () const
- QCPAxisRect * **clipAxisRect** () const
- bool **selectable** () const
- bool **selected** () const
- void setClipToAxisRect (bool clip)
- void setClipAxisRect (QCPAxisRect *rect)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- QList< QCPItemPosition * > positions () const
- QList< QCPItemAnchor * > anchors () const
- QCPItemPosition * position (const QString &name) const
- QCPItemAnchor * anchor (const QString &name) const
- bool hasAnchor (const QString &name) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Public Attributes

- QCPItemPosition *const **left**
- QCPItemPosition *const **right**
- QCPItemAnchor *const **center**

Protected Types

- enum **AnchorIndex** { **aiCenter** }

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual QPointF anchorPixelPosition (int anchorId) const Q_DECL_OVERRIDE
- QPen **mainPen** () const

Protected Member Functions inherited from QCPAbstractItem

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- double **rectDistance** (const QRectF &rect, const QPointF &pos, bool filledRect) const
- QCPItemPosition * **createPosition** (const QString &name)
- QCPItemAnchor * **createAnchor** (const QString &name, int anchorId)

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QPen **mPen**
- QPen **mSelectedPen**
- double **mLength**
- BracketStyle **mStyle**

Protected Attributes inherited from QCPAbstractItem

- bool **mClipToAxisRect**
- QPointer< QCPAxisRect > **mClipAxisRect**
- QList< QCPItemPosition * > **mPositions**
- QList< QCPItemAnchor * > **mAnchors**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPAbstractItem

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.636.1 Detailed Description

A bracket for referencing/highlighting certain parts in the plot.
It has two positions, *left* and *right*, which define the span of the bracket. If *left* is actually farther to the left than *right*, the bracket is opened to the bottom, as shown in the example image.
The bracket supports multiple styles via setStyle. The length, i.e. how far the bracket stretches away from the embraced span, can be controlled with setLength.

Demonstrating the effect of different values for setLength, for styles bsCalligraphic and bsSquare. Anchors and positions are displayed for reference.

It provides an anchor *center*, to allow connection of other items, e.g. an arrow (QCPItemLine or QCPItemCurve) or a text label (QCPItemText), to the bracket.

17.636.2 Member Enumeration Documentation

17.636.2.1 BracketStyle

```
enum QCPItemBracket::BracketStyle
```

Defines the various visual shapes of the bracket item. The appearance can be further modified by setLength and setPen.

See also

setStyle

Enumerator

bsSquare	A brace with angled edges.
bsRound	A brace with round edges.
bsCurly	A curly brace.
bsCalligraphic	A curly brace with varying stroke width giving a calligraphic impression.

17.636.3 Constructor & Destructor Documentation

17.636.3.1 QCPItemBracket()

```
QCPItemBracket::QCPItemBracket (
            QCustomPlot * parentPlot )  [explicit]
```

Creates a bracket item and sets default values.
The created item is automatically registered with *parentPlot*. This QCustomPlot instance takes ownership of the item, so do not delete it manually but use QCustomPlot::removeItem() instead.

17.636.4 Member Function Documentation

17.636.4.1 anchorPixelPosition()

```
QPointF QCPItemBracket::anchorPixelPosition (
```

```
            int anchorId ) const  [protected], [virtual]
```
Reimplemented from QCPAbstractItem.

17.636.4.2 draw()

```
void QCPItemBracket::draw (
            QCPPainter * painter )  [protected], [virtual]
```
Implements QCPAbstractItem.

17.636.4.3 selectTest()

```
double QCPItemBracket::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```
This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Implements QCPAbstractItem.

17.636.4.4 setLength()

```
void QCPItemBracket::setLength (
            double length )
```
Sets the *length* in pixels how far the bracket extends in the direction towards the embraced span of the bracket (i.e. perpendicular to the *left-right*-direction)

Demonstrating the effect of different values for setLength, for styles bsCalligraphic and bsSquare. Anchors and positions are displayed for reference.

17.636.4.5 setPen()

```
void QCPItemBracket::setPen (
            const QPen & pen )
```
Sets the pen that will be used to draw the bracket.
Note that when the style is bsCalligraphic, only the color will be taken from the pen, the stroke and width are ignored. To change the apparent stroke width of a calligraphic bracket, use setLength, which has a similar effect.

See also

setSelectedPen

17.636.4.6 setSelectedPen()

```
void QCPItemBracket::setSelectedPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the bracket when selected

See also

setPen, setSelected

17.636.4.7 setStyle()

```
void QCPItemBracket::setStyle (
            QCPItemBracket::BracketStyle style )
```

Sets the style of the bracket, i.e. the shape/visual appearance.

See also

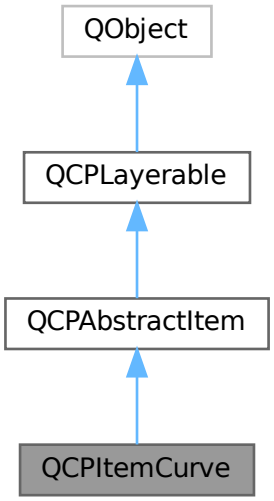
setPen

The documentation for this class was generated from the following files:

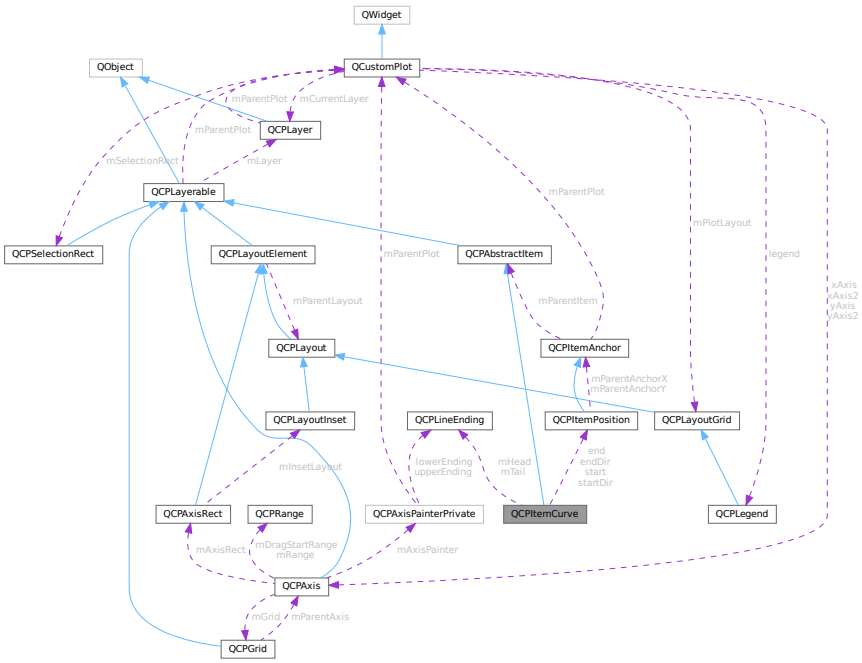
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.637 QCPItemCurve Class Reference

A curved line from one point to another.
Inheritance diagram for QCPItemCurve:



Collaboration diagram for QCPItemCurve:



Public Member Functions

- QCPItemCurve (QCustomPlot ∗parentPlot)
- QPen **pen** () const
- QPen **selectedPen** () const
- QCPLineEnding **head** () const
- QCPLineEnding **tail** () const
- void setPen (const QPen &pen)
- void setSelectedPen (const QPen &pen)
- void setHead (const QCPLineEnding &head)
- void setTail (const QCPLineEnding &tail)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant ∗details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractItem

- QCPAbstractItem (QCustomPlot ∗parentPlot)
- bool **clipToAxisRect** () const
- QCPAxisRect ∗ **clipAxisRect** () const
- bool **selectable** () const
- bool **selected** () const
- void setClipToAxisRect (bool clip)
- void setClipAxisRect (QCPAxisRect ∗rect)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- QList< QCPItemPosition ∗ > positions () const
- QList< QCPItemAnchor ∗ > anchors () const
- QCPItemPosition ∗ position (const QString &name) const
- QCPItemAnchor ∗ anchor (const QString &name) const
- bool hasAnchor (const QString &name) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Public Attributes

- QCPItemPosition *const **start**
- QCPItemPosition *const **startDir**
- QCPItemPosition *const **endDir**
- QCPItemPosition *const **end**

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- QPen **mainPen** () const

Protected Member Functions inherited from QCPAbstractItem

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual QPointF **anchorPixelPosition** (int anchorId) const
- double **rectDistance** (const QRectF &rect, const QPointF &pos, bool filledRect) const
- QCPItemPosition * **createPosition** (const QString &name)
- QCPItemAnchor * **createAnchor** (const QString &name, int anchorId)

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QPen **mPen**
- QPen **mSelectedPen**
- QCPLineEnding **mHead**
- QCPLineEnding **mTail**

Protected Attributes inherited from QCPAbstractItem

- bool **mClipToAxisRect**
- QPointer< QCPAxisRect > **mClipAxisRect**
- QList< QCPItemPosition * > **mPositions**
- QList< QCPItemAnchor * > **mAnchors**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPAbstractItem

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.637.1 Detailed Description

A curved line from one point to another.
It has four positions, *start* and *end*, which define the end points of the line, and two control points which define the direction the line exits from the start and the direction from which it approaches the end: *startDir* and *endDir*.
With setHead and setTail you may set different line ending styles, e.g. to create an arrow.
Often it is desirable for the control points to stay at fixed relative positions to the start/end point. This can be achieved by setting the parent anchor e.g. of *startDir* simply to *start*, and then specify the desired pixel offset with QCPItemPosition::setCoords on *startDir*.

17.637.2 Constructor & Destructor Documentation

17.637.2.1 QCPItemCurve()

```
QCPItemCurve::QCPItemCurve (
            QCustomPlot * parentPlot )  [explicit]
```

Creates a curve item and sets default values.
The created item is automatically registered with *parentPlot*. This QCustomPlot instance takes ownership of the item, so do not delete it manually but use QCustomPlot::removeItem() instead.

17.637.3 Member Function Documentation

17.637.3.1 draw()

```
void QCPItemCurve::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractItem.

17.637.3.2 selectTest()

```
double QCPItemCurve::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Implements QCPAbstractItem.

17.637.3.3 setHead()

```
void QCPItemCurve::setHead (
            const QCPLineEnding & head )
```

Sets the line ending style of the head. The head corresponds to the *end* position.
Note that due to the overloaded QCPLineEnding constructor, you may directly specify a QCPLineEnding::EndingStyle here, e.g.
`setHead(QCPLineEnding::esSpikeArrow)`

See also

setTail

17.637.3.4 setPen()

```
void QCPItemCurve::setPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line

See also

setSelectedPen

17.637.3.5 setSelectedPen()

```
void QCPItemCurve::setSelectedPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line when selected

See also

setPen, setSelected

17.637.3.6 setTail()

```
void QCPItemCurve::setTail (
            const QCPLineEnding & tail )
```

Sets the line ending style of the tail. The tail corresponds to the *start* position.
Note that due to the overloaded QCPLineEnding constructor, you may directly specify a QCPLineEnding::EndingStyle here, e.g.

```
setTail(QCPLineEnding::esSpikeArrow)
```

See also

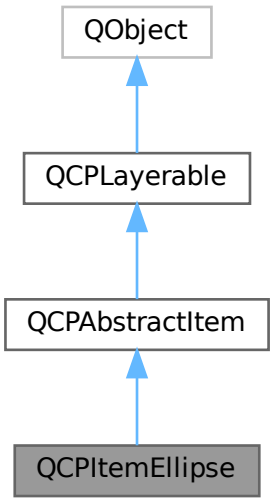
setHead

The documentation for this class was generated from the following files:

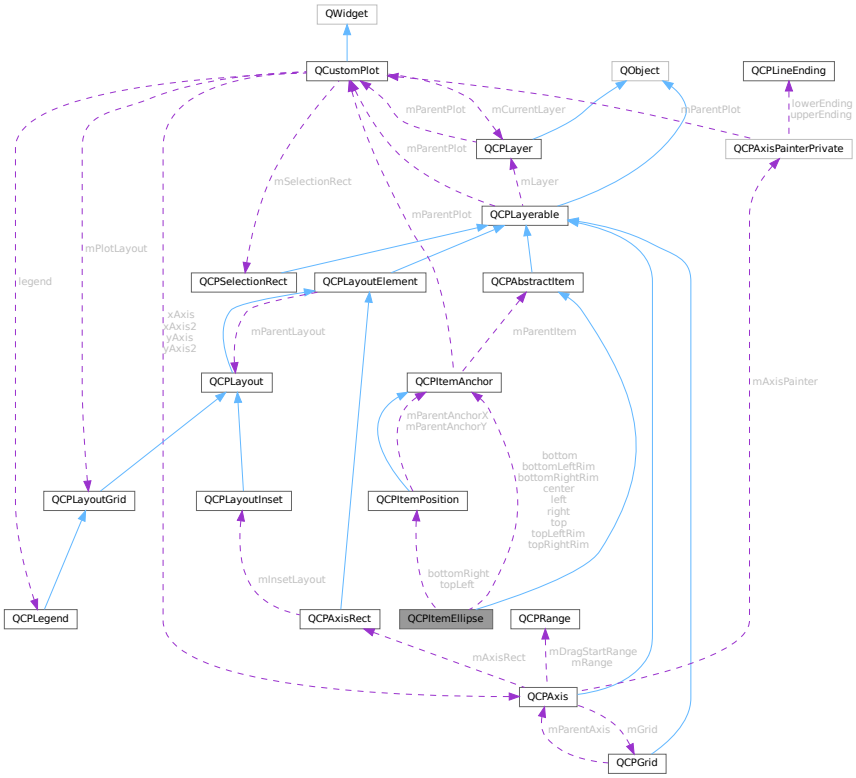
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.638 QCPItemEllipse Class Reference

An ellipse.
Inheritance diagram for QCPItemEllipse:



Collaboration diagram for QCPItemEllipse:



Public Member Functions

- QCPItemEllipse (QCustomPlot ∗parentPlot)
- QPen **pen** () const
- QPen **selectedPen** () const
- QBrush **brush** () const
- QBrush **selectedBrush** () const
- void setPen (const QPen &pen)
- void setSelectedPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setSelectedBrush (const QBrush &brush)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant ∗details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractItem

- QCPAbstractItem (QCustomPlot ∗parentPlot)
- bool **clipToAxisRect** () const
- QCPAxisRect ∗ **clipAxisRect** () const
- bool **selectable** () const
- bool **selected** () const
- void setClipToAxisRect (bool clip)
- void setClipAxisRect (QCPAxisRect ∗rect)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- QList< QCPItemPosition ∗ > positions () const

- QList< QCPItemAnchor * > anchors () const
- QCPItemPosition * position (const QString &name) const
- QCPItemAnchor * anchor (const QString &name) const
- bool hasAnchor (const QString &name) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Public Attributes

- QCPItemPosition *const **topLeft**
- QCPItemPosition *const **bottomRight**
- QCPItemAnchor *const **topLeftRim**
- QCPItemAnchor *const **top**
- QCPItemAnchor *const **topRightRim**
- QCPItemAnchor *const **right**
- QCPItemAnchor *const **bottomRightRim**
- QCPItemAnchor *const **bottom**
- QCPItemAnchor *const **bottomLeftRim**
- QCPItemAnchor *const **left**
- QCPItemAnchor *const **center**

Protected Types

- enum **AnchorIndex** {
  aiTopLeftRim , **aiTop** , **aiTopRightRim** , **aiRight** ,
  aiBottomRightRim , **aiBottom** , **aiBottomLeftRim** , **aiLeft** ,
  aiCenter }

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual QPointF anchorPixelPosition (int anchorId) const Q_DECL_OVERRIDE
- QPen **mainPen** () const
- QBrush **mainBrush** () const

Protected Member Functions inherited from QCPAbstractItem

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- double **rectDistance** (const QRectF &rect, const QPointF &pos, bool filledRect) const
- QCPItemPosition * **createPosition** (const QString &name)
- QCPItemAnchor * **createAnchor** (const QString &name, int anchorId)

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot ∗parentPlot)
- virtual void mousePressEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent ∗event)
- void **initializeParentPlot** (QCustomPlot ∗parentPlot)
- void **setParentLayerable** (QCPLayerable ∗parentLayerable)
- bool **moveToLayer** (QCPLayer ∗layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter ∗painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QPen **mPen**
- QPen **mSelectedPen**
- QBrush **mBrush**
- QBrush **mSelectedBrush**

Protected Attributes inherited from QCPAbstractItem

- bool **mClipToAxisRect**
- QPointer< QCPAxisRect > **mClipAxisRect**
- QList< QCPItemPosition ∗ > **mPositions**
- QList< QCPItemAnchor ∗ > **mAnchors**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot ∗ **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer ∗ **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPAbstractItem

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer ∗newLayer)

17.638.1 Detailed Description

An ellipse.
It has two positions, *topLeft* and *bottomRight*, which define the rect the ellipse will be drawn in.

17.638.2 Constructor & Destructor Documentation

17.638.2.1 QCPItemEllipse()

```
QCPItemEllipse::QCPItemEllipse (
            QCustomPlot * parentPlot )  [explicit]
```

Creates an ellipse item and sets default values.
The created item is automatically registered with *parentPlot*. This QCustomPlot instance takes ownership of the item, so do not delete it manually but use QCustomPlot::removeItem() instead.

17.638.3 Member Function Documentation

17.638.3.1 anchorPixelPosition()

```
QPointF QCPItemEllipse::anchorPixelPosition (
            int anchorId ) const  [protected], [virtual]
```

Reimplemented from QCPAbstractItem.

17.638.3.2 draw()

```
void QCPItemEllipse::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractItem.

17.638.3.3 selectTest()

```
double QCPItemEllipse::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Implements QCPAbstractItem.

17.638.3.4 setBrush()

```
void QCPItemEllipse::setBrush (
            const QBrush & brush )
```

Sets the brush that will be used to fill the ellipse. To disable filling, set *brush* to Qt::NoBrush.

See also

setSelectedBrush, setPen

17.638.3.5 setPen()

```
void QCPItemEllipse::setPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line of the ellipse

See also

setSelectedPen, setBrush

17.638.3.6 setSelectedBrush()

```
void QCPItemEllipse::setSelectedBrush (
            const QBrush & brush )
```

Sets the brush that will be used to fill the ellipse when selected. To disable filling, set *brush* to Qt::NoBrush.

See also

setBrush

17.638.3.7 setSelectedPen()

```
void QCPItemEllipse::setSelectedPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line of the ellipse when selected

See also

setPen, setSelected

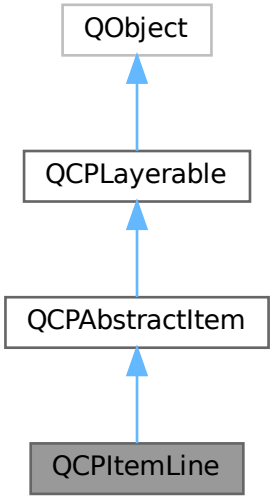
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

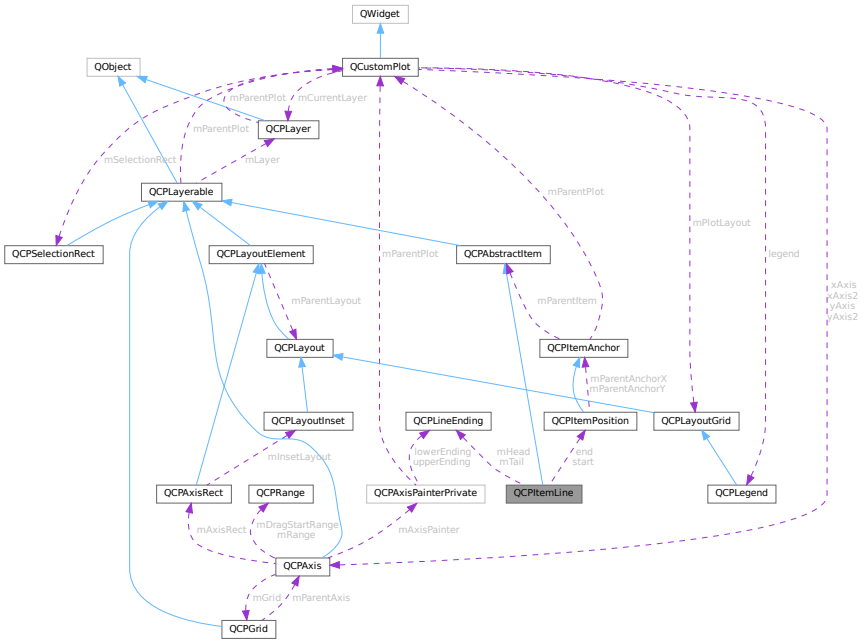
17.639 QCPItemLine Class Reference

A line from one point to another.

Inheritance diagram for QCPItemLine:



Collaboration diagram for QCPItemLine:



Public Member Functions

- QCPItemLine (QCustomPlot ∗parentPlot)
- QPen **pen** () const
- QPen **selectedPen** () const
- QCPLineEnding **head** () const

- QCPLineEnding **tail** () const
- void setPen (const QPen &pen)
- void setSelectedPen (const QPen &pen)
- void setHead (const QCPLineEnding &head)
- void setTail (const QCPLineEnding &tail)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractItem

- QCPAbstractItem (QCustomPlot *parentPlot)
- bool **clipToAxisRect** () const
- QCPAxisRect * **clipAxisRect** () const
- bool **selectable** () const
- bool **selected** () const
- void setClipToAxisRect (bool clip)
- void setClipAxisRect (QCPAxisRect *rect)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- QList< QCPItemPosition * > positions () const
- QList< QCPItemAnchor * > anchors () const
- QCPItemPosition * position (const QString &name) const
- QCPItemAnchor * anchor (const QString &name) const
- bool hasAnchor (const QString &name) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Public Attributes

- QCPItemPosition *const **start**
- QCPItemPosition *const **end**

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- QLineF **getRectClippedLine** (const QCPVector2D &start, const QCPVector2D &end, const QRect &rect) const
- QPen **mainPen** () const

Protected Member Functions inherited from QCPAbstractItem

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual QPointF **anchorPixelPosition** (int anchorId) const
- double **rectDistance** (const QRectF &rect, const QPointF &pos, bool filledRect) const
- QCPItemPosition * **createPosition** (const QString &name)
- QCPItemAnchor * **createAnchor** (const QString &name, int anchorId)

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QPen **mPen**
- QPen **mSelectedPen**
- QCPLineEnding **mHead**
- QCPLineEnding **mTail**

Protected Attributes inherited from QCPAbstractItem

- bool **mClipToAxisRect**
- QPointer< QCPAxisRect > **mClipAxisRect**
- QList< QCPItemPosition * > **mPositions**
- QList< QCPItemAnchor * > **mAnchors**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPAbstractItem

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.639.1 Detailed Description

A line from one point to another.
It has two positions, *start* and *end*, which define the end points of the line.
With setHead and setTail you may set different line ending styles, e.g. to create an arrow.

17.639.2 Constructor & Destructor Documentation

17.639.2.1 QCPItemLine()

```
QCPItemLine::QCPItemLine (
            QCustomPlot * parentPlot )  [explicit]
```

Creates a line item and sets default values.
The created item is automatically registered with *parentPlot*. This QCustomPlot instance takes ownership of the item, so do not delete it manually but use QCustomPlot::removeItem() instead.

17.639.3 Member Function Documentation

17.639.3.1 draw()

```
void QCPItemLine::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractItem.

17.639.3.2 selectTest()

```
double QCPItemLine::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99*selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Implements QCPAbstractItem.

17.639.3.3 setHead()

```
void QCPItemLine::setHead (
            const QCPLineEnding & head )
```

Sets the line ending style of the head. The head corresponds to the *end* position.
Note that due to the overloaded QCPLineEnding constructor, you may directly specify a QCPLineEnding::EndingStyle here, e.g.
`setHead(QCPLineEnding::esSpikeArrow)`

See also

setTail

17.639.3.4 setPen()

```
void QCPItemLine::setPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line

See also

setSelectedPen

17.639.3.5 setSelectedPen()

```
void QCPItemLine::setSelectedPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line when selected

See also

setPen, setSelected

17.639.3.6 setTail()

```
void QCPItemLine::setTail (
            const QCPLineEnding & tail )
```

Sets the line ending style of the tail. The tail corresponds to the *start* position.
Note that due to the overloaded QCPLineEnding constructor, you may directly specify a QCPLineEnding::EndingStyle here, e.g.
`setTail(QCPLineEnding::esSpikeArrow)`

See also

setHead

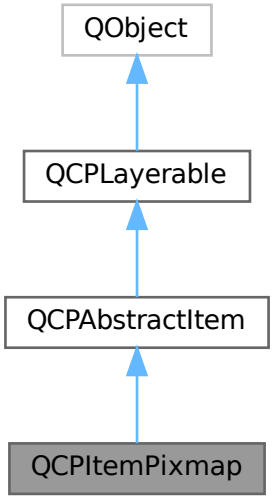
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

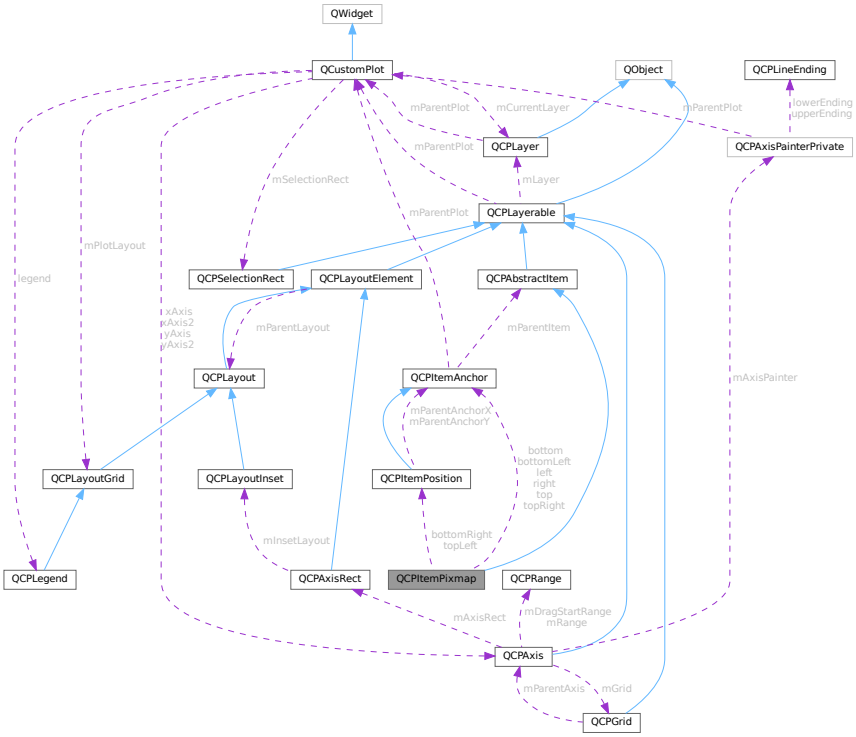
17.640 QCPItemPixmap Class Reference

An arbitrary pixmap.

Inheritance diagram for QCPItemPixmap:



Collaboration diagram for QCPItemPixmap:



Public Member Functions

- QCPItemPixmap (QCustomPlot *parentPlot)

- QPixmap **pixmap** () const
- bool **scaled** () const
- Qt::AspectRatioMode **aspectRatioMode** () const
- Qt::TransformationMode **transformationMode** () const
- QPen **pen** () const
- QPen **selectedPen** () const
- void setPixmap (const QPixmap &pixmap)
- void setScaled (bool scaled, Qt::AspectRatioMode aspectRatioMode=Qt::KeepAspectRatio, Qt::↩ TransformationMode transformationMode=Qt::SmoothTransformation)
- void setPen (const QPen &pen)
- void setSelectedPen (const QPen &pen)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractItem

- QCPAbstractItem (QCustomPlot *parentPlot)
- bool **clipToAxisRect** () const
- QCPAxisRect * **clipAxisRect** () const
- bool **selectable** () const
- bool **selected** () const
- void setClipToAxisRect (bool clip)
- void setClipAxisRect (QCPAxisRect *rect)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- QList< QCPItemPosition * > positions () const
- QList< QCPItemAnchor * > anchors () const
- QCPItemPosition * position (const QString &name) const
- QCPItemAnchor * anchor (const QString &name) const
- bool hasAnchor (const QString &name) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Public Attributes

- QCPItemPosition *const **topLeft**
- QCPItemPosition *const **bottomRight**
- QCPItemAnchor *const **top**
- QCPItemAnchor *const **topRight**
- QCPItemAnchor *const **right**
- QCPItemAnchor *const **bottom**
- QCPItemAnchor *const **bottomLeft**
- QCPItemAnchor *const **left**

Protected Types

- enum **AnchorIndex** {
  aiTop , **aiTopRight** , **aiRight** , **aiBottom** ,
  aiBottomLeft , **aiLeft** }

Protected Member Functions

- virtual void draw (QCPPainter ∗painter) Q_DECL_OVERRIDE
- virtual QPointF anchorPixelPosition (int anchorId) const Q_DECL_OVERRIDE
- void **updateScaledPixmap** (QRect finalRect=QRect(), bool flipHorz=false, bool flipVert=false)
- QRect **getFinalRect** (bool ∗flippedHorz=nullptr, bool ∗flippedVert=nullptr) const
- QPen **mainPen** () const

Protected Member Functions inherited from QCPAbstractItem

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter ∗painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent ∗event, bool additive, const QVariant &details, bool ∗selectionState↩ Changed) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool ∗selectionStateChanged) Q_DECL_OVERRIDE
- double **rectDistance** (const QRectF &rect, const QPointF &pos, bool filledRect) const
- QCPItemPosition ∗ **createPosition** (const QString &name)
- QCPItemAnchor ∗ **createAnchor** (const QString &name, int anchorId)

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot ∗parentPlot)
- virtual void mousePressEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent ∗event)
- void **initializeParentPlot** (QCustomPlot ∗parentPlot)
- void **setParentLayerable** (QCPLayerable ∗parentLayerable)
- bool **moveToLayer** (QCPLayer ∗layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter ∗painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QPixmap **mPixmap**
- QPixmap **mScaledPixmap**
- bool **mScaled**
- bool **mScaledPixmapInvalidated**
- Qt::AspectRatioMode **mAspectRatioMode**
- Qt::TransformationMode **mTransformationMode**
- QPen **mPen**
- QPen **mSelectedPen**

Protected Attributes inherited from QCPAbstractItem

- bool **mClipToAxisRect**
- QPointer< QCPAxisRect > **mClipAxisRect**
- QList< QCPItemPosition ∗ > **mPositions**
- QList< QCPItemAnchor ∗ > **mAnchors**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPAbstractItem

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.640.1 Detailed Description

An arbitrary pixmap.
It has two positions, *topLeft* and *bottomRight*, which define the rectangle the pixmap will be drawn in. Depending on the scale setting (setScaled), the pixmap will be either scaled to fit the rectangle or be drawn aligned to the topLeft position.
If scaling is enabled and *topLeft* is further to the bottom/right than *bottomRight* (as shown on the right side of the example image), the pixmap will be flipped in the respective orientations.

17.640.2 Constructor & Destructor Documentation

17.640.2.1 QCPItemPixmap()

```
QCPItemPixmap::QCPItemPixmap (
            QCustomPlot * parentPlot )  [explicit]
```

Creates a rectangle item and sets default values.
The created item is automatically registered with *parentPlot*. This QCustomPlot instance takes ownership of the item, so do not delete it manually but use QCustomPlot::removeItem() instead.

17.640.3 Member Function Documentation

17.640.3.1 anchorPixelPosition()

```
QPointF QCPItemPixmap::anchorPixelPosition (
            int anchorId ) const  [protected], [virtual]
```

Reimplemented from QCPAbstractItem.

17.640.3.2 draw()

```
void QCPItemPixmap::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractItem.

17.640.3.3 selectTest()

```
double QCPItemPixmap::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.

pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Implements QCPAbstractItem.

17.640.3.4 setPen()

```
void QCPItemPixmap::setPen (
            const QPen & pen )
```

Sets the pen that will be used to draw a border around the pixmap.

See also

setSelectedPen, setBrush

17.640.3.5 setPixmap()

```
void QCPItemPixmap::setPixmap (
            const QPixmap & pixmap )
```

Sets the pixmap that will be displayed.

17.640.3.6 setScaled()

```
void QCPItemPixmap::setScaled (
            bool scaled,
            Qt::AspectRatioMode aspectRatioMode = Qt::KeepAspectRatio,
            Qt::TransformationMode transformationMode = Qt::SmoothTransformation )
```

Sets whether the pixmap will be scaled to fit the rectangle defined by the *topLeft* and *bottomRight* positions.

17.640.3.7 setSelectedPen()

```
void QCPItemPixmap::setSelectedPen (
            const QPen & pen )
```

Sets the pen that will be used to draw a border around the pixmap when selected

See also

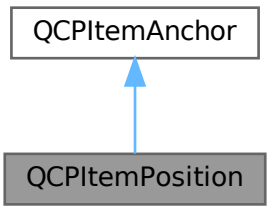
setPen, setSelected

The documentation for this class was generated from the following files:

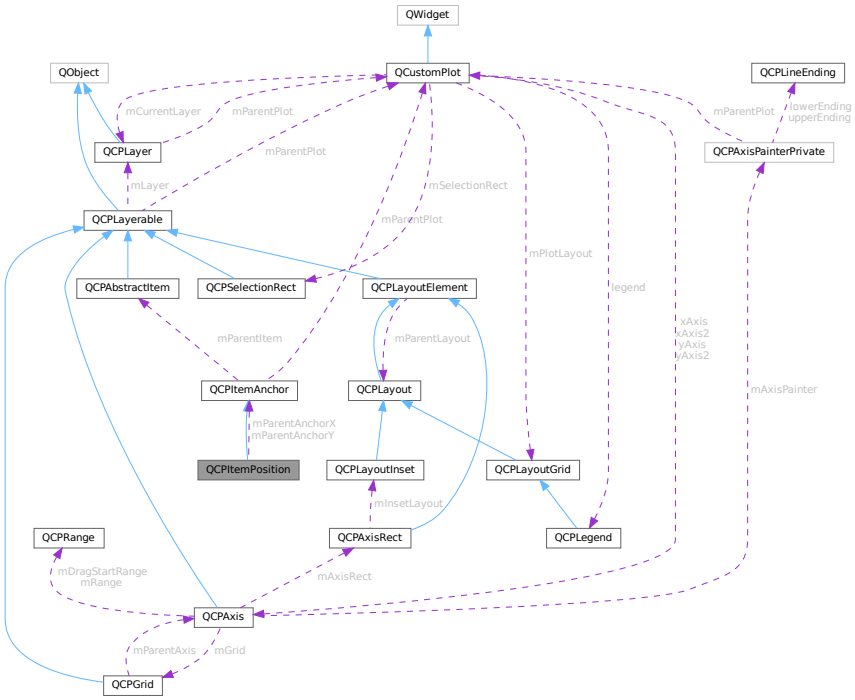
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.641 QCPItemPosition Class Reference

Manages the position of an item.
Inheritance diagram for QCPItemPosition:



Collaboration diagram for QCPItemPosition:



Public Types

- enum PositionType { ptAbsolute , ptViewportRatio , ptAxisRectRatio , ptPlotCoords }

Public Member Functions

- QCPItemPosition (QCustomPlot *parentPlot, QCPAbstractItem *parentItem, const QString &name)
- PositionType type () const
- PositionType **typeX** () const
- PositionType **typeY** () const
- QCPItemAnchor * parentAnchor () const
- QCPItemAnchor * **parentAnchorX** () const
- QCPItemAnchor * **parentAnchorY** () const
- double **key** () const
- double **value** () const
- QPointF **coords** () const
- QCPAxis * **keyAxis** () const
- QCPAxis * **valueAxis** () const
- QCPAxisRect * **axisRect** () const
- virtual QPointF pixelPosition () const Q_DECL_OVERRIDE
- void setType (PositionType type)
- void setTypeX (PositionType type)
- void setTypeY (PositionType type)
- bool setParentAnchor (QCPItemAnchor *parentAnchor, bool keepPixelPosition=false)
- bool setParentAnchorX (QCPItemAnchor *parentAnchor, bool keepPixelPosition=false)
- bool setParentAnchorY (QCPItemAnchor *parentAnchor, bool keepPixelPosition=false)
- void setCoords (double key, double value)
- void setCoords (const QPointF &pos)
- void setAxes (QCPAxis *keyAxis, QCPAxis *valueAxis)
- void setAxisRect (QCPAxisRect *axisRect)
- void setPixelPosition (const QPointF &pixelPosition)

Public Member Functions inherited from QCPItemAnchor

- QCPItemAnchor (QCustomPlot *parentPlot, QCPAbstractItem *parentItem, const QString &name, int anchorId=-1)
- QString **name** () const

Protected Member Functions

- virtual QCPItemPosition * toQCPItemPosition () Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPItemAnchor

- void **addChildX** (QCPItemPosition *pos)
- void **removeChildX** (QCPItemPosition *pos)
- void **addChildY** (QCPItemPosition *pos)
- void **removeChildY** (QCPItemPosition *pos)

Protected Attributes

- PositionType **mPositionTypeX**
- PositionType **mPositionTypeY**
- QPointer< QCPAxis > **mKeyAxis**
- QPointer< QCPAxis > **mValueAxis**
- QPointer< QCPAxisRect > **mAxisRect**
- double **mKey**
- double **mValue**
- QCPItemAnchor * **mParentAnchorX**
- QCPItemAnchor * **mParentAnchorY**

Protected Attributes inherited from QCPItemAnchor

- QString **mName**
- QCustomPlot * **mParentPlot**
- QCPAbstractItem * **mParentItem**
- int **mAnchorId**
- QSet< QCPItemPosition * > **mChildrenX**
- QSet< QCPItemPosition * > **mChildrenY**

17.641.1 Detailed Description

Manages the position of an item.
Every item has at least one public QCPItemPosition member pointer which provides ways to position the item on the QCustomPlot surface. Some items have multiple positions, for example QCPItemRect has two: *topLeft* and *bottomRight*.
QCPItemPosition has a type (PositionType) that can be set with setType. This type defines how coordinates passed to setCoords are to be interpreted, e.g. as absolute pixel coordinates, as plot coordinates of certain axes (QCPItemPosition::setAxes), as fractions of the axis rect (QCPItemPosition::setAxisRect), etc. For more advanced plots it is also possible to assign different types per X/Y coordinate of the position (see setTypeX, setTypeY). This way an item could be positioned for example at a fixed pixel distance from the top in the Y direction, while following a plot coordinate in the X direction.
A QCPItemPosition may have a parent QCPItemAnchor, see setParentAnchor. This way you can tie multiple items together. If the QCPItemPosition has a parent, its coordinates (setCoords) are considered to be absolute pixels in the reference frame of the parent anchor, where (0, 0) means directly ontop of the parent anchor. For example, You could attach the *start* position of a QCPItemLine to the *bottom* anchor of a QCPItemText to make the starting point of the line always be centered under the text label, no matter where the text is moved to. For more advanced plots, it is possible to assign different parent anchors per X/Y coordinate of the position, see setParentAnchorX, setParentAnchorY. This way an item could follow another item in the X direction but stay at a fixed position in the Y direction. Or even follow item A in X, and item B in Y.
Note that every QCPItemPosition inherits from QCPItemAnchor and thus can itself be used as parent anchor for other positions.
To set the apparent pixel position on the QCustomPlot surface directly, use setPixelPosition. This works no matter what type this QCPItemPosition is or what parent-child situation it is in, as setPixelPosition transforms the coordinates appropriately, to make the position appear at the specified pixel values.

17.641.2 Member Enumeration Documentation

17.641.2.1 PositionType

`enum QCPItemPosition::PositionType`
Defines the ways an item position can be specified. Thus it defines what the numbers passed to setCoords actually mean.

See also

setType

Enumerator

ptAbsolute	Static positioning in pixels, starting from the top left corner of the viewport/widget.
ptViewportRatio	Static positioning given by a fraction of the viewport size. For example, if you call setCoords(0, 0), the position will be at the top left corner of the viewport/widget. setCoords(1, 1) will be at the bottom right corner, setCoords(0.5, 0) will be horizontally centered and vertically at the top of the viewport/widget, etc.
ptAxisRectRatio	Static positioning given by a fraction of the axis rect size (see setAxisRect). For example, if you call setCoords(0, 0), the position will be at the top left corner of the axis rect. setCoords(1, 1) will be at the bottom right corner, setCoords(0.5, 0) will be horizontally centered and vertically at the top of the axis rect, etc. You can also go beyond the axis rect by providing negative coordinates or coordinates larger than 1.
ptPlotCoords	Dynamic positioning at a plot coordinate defined by two axes (see setAxes).

17.641.3 Constructor & Destructor Documentation

17.641.3.1 QCPItemPosition()

```
QCPItemPosition::QCPItemPosition (
            QCustomPlot * parentPlot,
            QCPAbstractItem * parentItem,
            const QString & name )
```

Creates a new QCPItemPosition. You shouldn't create QCPItemPosition instances directly, even if you want to make a new item subclass. Use QCPAbstractItem::createPosition instead, as explained in the subclassing section of the QCPAbstractItem documentation.

17.641.4 Member Function Documentation

17.641.4.1 parentAnchor()

```
QCPItemAnchor * QCPItemPosition::parentAnchor ( ) const  [inline]
```

Returns the current parent anchor.
If different parent anchors were set for X and Y (setParentAnchorX, setParentAnchorY), this method returns the parent anchor of the Y coordinate. In that case rather use *parentAnchorX()* and *parentAnchorY()*.

See also

setParentAnchor

17.641.4.2 pixelPosition()

```
QPointF QCPItemPosition::pixelPosition ( ) const  [virtual]
```

Returns the final absolute pixel position of the QCPItemPosition on the QCustomPlot surface. It includes all effects of type (setType) and possible parent anchors (setParentAnchor).

See also

setPixelPosition

Reimplemented from QCPItemAnchor.

17.641.4.3 setAxes()

```
void QCPItemPosition::setAxes (
            QCPAxis * keyAxis,
            QCPAxis * valueAxis )
```

When setType is ptPlotCoords, this function may be used to specify the axes the coordinates set with setCoords relate to. By default they are set to the initial xAxis and yAxis of the QCustomPlot.

17.641.4.4 setAxisRect()

```
void QCPItemPosition::setAxisRect (
            QCPAxisRect * axisRect )
```

When setType is ptAxisRectRatio, this function may be used to specify the axis rect the coordinates set with setCoords relate to. By default this is set to the main axis rect of the QCustomPlot.

17.641.4.5 setCoords() [1/2]

```
void QCPItemPosition::setCoords (
            const QPointF & pos )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the coordinates as a QPointF *pos* where pos.x has the meaning of *key* and pos.y the meaning of *value* of the setCoords(double key, double value) method.

17.641.4.6 setCoords() [2/2]

```
void QCPItemPosition::setCoords (
            double key,
            double value )
```

Sets the coordinates of this QCPItemPosition. What the coordinates mean, is defined by the type (setType, setTypeX, setTypeY).
For example, if the type is ptAbsolute, *key* and *value* mean the x and y pixel position on the QCustomPlot surface. In that case the origin (0, 0) is in the top left corner of the QCustomPlot viewport. If the type is ptPlotCoords, *key* and *value* mean a point in the plot coordinate system defined by the axes set by setAxes. By default those are the QCustomPlot's xAxis and yAxis. See the documentation of setType for other available coordinate types and their meaning.
If different types were configured for X and Y (setTypeX, setTypeY), *key* and *value* must also be provided in the different coordinate systems. Here, the X type refers to *key*, and the Y type refers to *value*.

See also

setPixelPosition

17.641.4.7 setParentAnchor()

```
bool QCPItemPosition::setParentAnchor (
            QCPItemAnchor * parentAnchor,
            bool keepPixelPosition = false )
```

Sets the parent of this QCPItemPosition to *parentAnchor*. This means the position will now follow any position changes of the anchor. The local coordinate system of positions with a parent anchor always is absolute pixels, with (0, 0) being exactly on top of the parent anchor. (Hence the type shouldn't be set to ptPlotCoords for positions with parent anchors.)
if *keepPixelPosition* is true, the current pixel position of the QCPItemPosition is preserved during reparenting. If it's set to false, the coordinates are set to (0, 0), i.e. the position will be exactly on top of the parent anchor.
To remove this QCPItemPosition from any parent anchor, set *parentAnchor* to `nullptr`.
If the QCPItemPosition previously had no parent and the type is ptPlotCoords, the type is set to ptAbsolute, to keep the position in a valid state.
This method sets the parent anchor for both X and Y directions. It is also possible to set different parents for X and Y, see setParentAnchorX, setParentAnchorY.

17.641.4.8 setParentAnchorX()

```
bool QCPItemPosition::setParentAnchorX (
            QCPItemAnchor * parentAnchor,
            bool keepPixelPosition = false )
```

This method sets the parent anchor of the X coordinate to *parentAnchor*.
For a detailed description of what a parent anchor is, see the documentation of setParentAnchor.

See also

setParentAnchor, setParentAnchorY

17.641.4.9 setParentAnchorY()

```
bool QCPItemPosition::setParentAnchorY (
            QCPItemAnchor * parentAnchor,
            bool keepPixelPosition = false )
```

This method sets the parent anchor of the Y coordinate to *parentAnchor*.
For a detailed description of what a parent anchor is, see the documentation of setParentAnchor.

See also

setParentAnchor, setParentAnchorX

17.641.4.10 setPixelPosition()

```
void QCPItemPosition::setPixelPosition (
            const QPointF & pixelPosition )
```

Sets the apparent pixel position. This works no matter what type (setType) this QCPItemPosition is or what parent-child situation it is in, as coordinates are transformed appropriately, to make the position finally appear at the specified pixel values.
Only if the type is ptAbsolute and no parent anchor is set, this function's effect is identical to that of setCoords.

See also

pixelPosition, setCoords

17.641.4.11 setType()

```
void QCPItemPosition::setType (
            QCPItemPosition::PositionType type )
```

Sets the type of the position. The type defines how the coordinates passed to setCoords should be handled and how the QCPItemPosition should behave in the plot.
The possible values for *type* can be separated in two main categories:

- The position is regarded as a point in plot coordinates. This corresponds to ptPlotCoords and requires two axes that define the plot coordinate system. They can be specified with setAxes. By default, the QCustomPlot's x- and yAxis are used.
- The position is fixed on the QCustomPlot surface, i.e. independent of axis ranges. This corresponds to all other types, i.e. ptAbsolute, ptViewportRatio and ptAxisRectRatio. They differ only in the way the absolute position is described, see the documentation of PositionType for details. For ptAxisRectRatio, note that you can specify the axis rect with setAxisRect. By default this is set to the main axis rect.

Note that the position type ptPlotCoords is only available (and sensible) when the position has no parent anchor (setParentAnchor).
If the type is changed, the apparent pixel position on the plot is preserved. This means the coordinates as retrieved with coords() and set with setCoords may change in the process.
This method sets the type for both X and Y directions. It is also possible to set different types for X and Y, see setTypeX, setTypeY.

17.641.4.12 setTypeX()

```
void QCPItemPosition::setTypeX (
            QCPItemPosition::PositionType type )
```

This method sets the position type of the X coordinate to *type*.
For a detailed description of what a position type is, see the documentation of setType.

See also

setType, setTypeY

17.641.4.13 setTypeY()

```
void QCPItemPosition::setTypeY (
            QCPItemPosition::PositionType type )
```

This method sets the position type of the Y coordinate to *type*.
For a detailed description of what a position type is, see the documentation of setType.

See also

setType, setTypeX

17.641.4.14 toQCPItemPosition()

`virtual QCPItemPosition * QCPItemPosition::toQCPItemPosition ( ) [inline], [protected], [virtual]`
Returns `nullptr` if this instance is merely a QCPItemAnchor, and a valid pointer of type QCPItemPosition* if it actually is a QCPItemPosition (which is a subclass of QCPItemAnchor).
This safe downcast functionality could also be achieved with a dynamic_cast. However, QCustomPlot avoids dynamic_cast to work with projects that don't have RTTI support enabled (e.g. -fno-rtti flag with gcc compiler).
Reimplemented from QCPItemAnchor.

17.641.4.15 type()

`QCPItemPosition::PositionType * QCPItemPosition::type ( ) const [inline]`
Returns the current position type.
If different types were set for X and Y (setTypeX, setTypeY), this method returns the type of the X coordinate. In that case rather use *typeX()* and *typeY()*.

See also

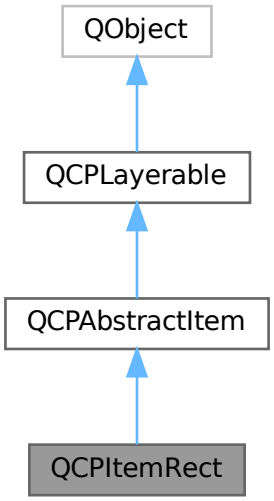
setType

The documentation for this class was generated from the following files:

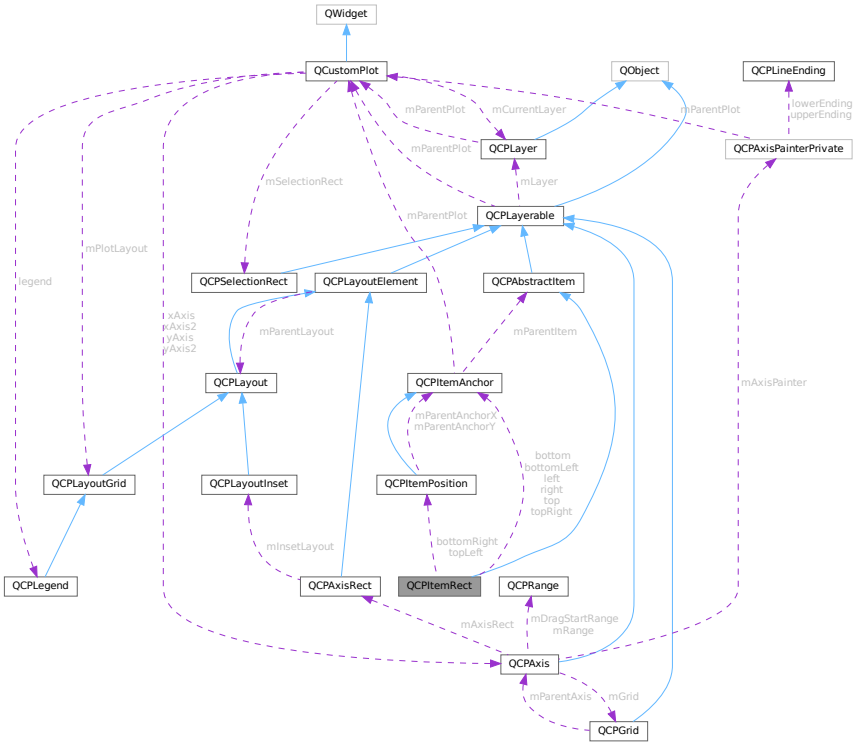
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.642 QCPItemRect Class Reference

A rectangle.
Inheritance diagram for QCPItemRect:



Collaboration diagram for QCPItemRect:



Public Member Functions

- QCPItemRect (QCustomPlot ∗parentPlot)
- QPen **pen** () const
- QPen **selectedPen** () const
- QBrush **brush** () const
- QBrush **selectedBrush** () const
- void setPen (const QPen &pen)
- void setSelectedPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setSelectedBrush (const QBrush &brush)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant ∗details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractItem

- QCPAbstractItem (QCustomPlot ∗parentPlot)
- bool **clipToAxisRect** () const
- QCPAxisRect ∗ **clipAxisRect** () const
- bool **selectable** () const
- bool **selected** () const
- void setClipToAxisRect (bool clip)
- void setClipAxisRect (QCPAxisRect ∗rect)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- QList< QCPItemPosition ∗ > positions () const
- QList< QCPItemAnchor ∗ > anchors () const

- QCPItemPosition * position (const QString &name) const
- QCPItemAnchor * anchor (const QString &name) const
- bool hasAnchor (const QString &name) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Public Attributes

- QCPItemPosition *const **topLeft**
- QCPItemPosition *const **bottomRight**
- QCPItemAnchor *const **top**
- QCPItemAnchor *const **topRight**
- QCPItemAnchor *const **right**
- QCPItemAnchor *const **bottom**
- QCPItemAnchor *const **bottomLeft**
- QCPItemAnchor *const **left**

Protected Types

- enum **AnchorIndex** {
  aiTop , **aiTopRight** , **aiRight** , **aiBottom** ,
  aiBottomLeft , **aiLeft** }

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual QPointF anchorPixelPosition (int anchorId) const Q_DECL_OVERRIDE
- QPen **mainPen** () const
- QBrush **mainBrush** () const

Protected Member Functions inherited from QCPAbstractItem

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- double **rectDistance** (const QRectF &rect, const QPointF &pos, bool filledRect) const
- QCPItemPosition * **createPosition** (const QString &name)
- QCPItemAnchor * **createAnchor** (const QString &name, int anchorId)

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot ∗parentPlot)
- virtual void mousePressEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent ∗event)
- void **initializeParentPlot** (QCustomPlot ∗parentPlot)
- void **setParentLayerable** (QCPLayerable ∗parentLayerable)
- bool **moveToLayer** (QCPLayer ∗layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter ∗painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QPen **mPen**
- QPen **mSelectedPen**
- QBrush **mBrush**
- QBrush **mSelectedBrush**

Protected Attributes inherited from QCPAbstractItem

- bool **mClipToAxisRect**
- QPointer< QCPAxisRect > **mClipAxisRect**
- QList< QCPItemPosition ∗ > **mPositions**
- QList< QCPItemAnchor ∗ > **mAnchors**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot ∗ **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer ∗ **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPAbstractItem

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer ∗newLayer)

17.642.1 Detailed Description

A rectangle.
It has two positions, *topLeft* and *bottomRight*, which define the rectangle.

17.642.2 Constructor & Destructor Documentation

17.642.2.1 QCPItemRect()

```
QCPItemRect::QCPItemRect (
            QCustomPlot * parentPlot )  [explicit]
```

Creates a rectangle item and sets default values.
The created item is automatically registered with *parentPlot*. This QCustomPlot instance takes ownership of the item, so do not delete it manually but use QCustomPlot::removeItem() instead.

17.642.3 Member Function Documentation

17.642.3.1 anchorPixelPosition()

```
QPointF QCPItemRect::anchorPixelPosition (
            int anchorId ) const  [protected], [virtual]
```

Reimplemented from QCPAbstractItem.

17.642.3.2 draw()

```
void QCPItemRect::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractItem.

17.642.3.3 selectTest()

```
double QCPItemRect::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Implements QCPAbstractItem.

17.642.3.4 setBrush()

```
void QCPItemRect::setBrush (
            const QBrush & brush )
```

Sets the brush that will be used to fill the rectangle. To disable filling, set *brush* to Qt::NoBrush.

See also

setSelectedBrush, setPen

17.642.3.5 setPen()

```
void QCPItemRect::setPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line of the rectangle

See also

setSelectedPen, setBrush

17.642.3.6 setSelectedBrush()

```
void QCPItemRect::setSelectedBrush (
            const QBrush & brush )
```

Sets the brush that will be used to fill the rectangle when selected. To disable filling, set *brush* to Qt::NoBrush.

See also

setBrush

17.642.3.7 setSelectedPen()

```
void QCPItemRect::setSelectedPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line of the rectangle when selected

See also

setPen, setSelected

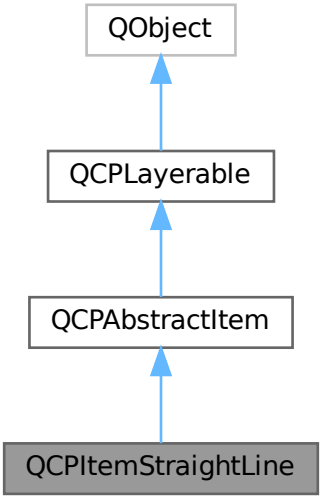
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

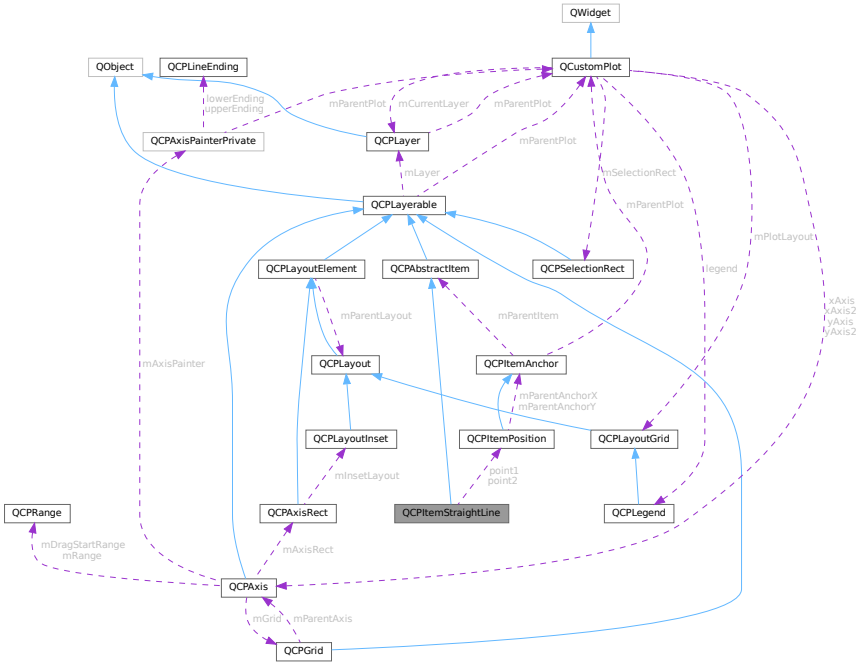
17.643 QCPItemStraightLine Class Reference

A straight line that spans infinitely in both directions.

Inheritance diagram for QCPItemStraightLine:



Collaboration diagram for QCPItemStraightLine:



Public Member Functions

- QCPItemStraightLine (QCustomPlot *parentPlot)
- QPen **pen** () const
- QPen **selectedPen** () const

- void setPen (const QPen &pen)
- void setSelectedPen (const QPen &pen)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractItem

- QCPAbstractItem (QCustomPlot *parentPlot)
- bool **clipToAxisRect** () const
- QCPAxisRect * **clipAxisRect** () const
- bool **selectable** () const
- bool **selected** () const
- void setClipToAxisRect (bool clip)
- void setClipAxisRect (QCPAxisRect *rect)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- QList< QCPItemPosition * > positions () const
- QList< QCPItemAnchor * > anchors () const
- QCPItemPosition * position (const QString &name) const
- QCPItemAnchor * anchor (const QString &name) const
- bool hasAnchor (const QString &name) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Public Attributes

- QCPItemPosition *const **point1**
- QCPItemPosition *const **point2**

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- QLineF **getRectClippedStraightLine** (const QCPVector2D &base, const QCPVector2D &vec, const QRect &rect) const
- QPen **mainPen** () const

Protected Member Functions inherited from QCPAbstractItem

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE

- virtual QPointF **anchorPixelPosition** (int anchorId) const
- double **rectDistance** (const QRectF &rect, const QPointF &pos, bool filledRect) const
- QCPItemPosition * **createPosition** (const QString &name)
- QCPItemAnchor * **createAnchor** (const QString &name, int anchorId)

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QPen **mPen**
- QPen **mSelectedPen**

Protected Attributes inherited from QCPAbstractItem

- bool **mClipToAxisRect**
- QPointer< QCPAxisRect > **mClipAxisRect**
- QList< QCPItemPosition * > **mPositions**
- QList< QCPItemAnchor * > **mAnchors**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPAbstractItem

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.643.1 Detailed Description

A straight line that spans infinitely in both directions.
It has two positions, *point1* and *point2*, which define the straight line.

17.643.2 Constructor & Destructor Documentation

17.643.2.1 QCPItemStraightLine()

```
QCPItemStraightLine::QCPItemStraightLine (
            QCustomPlot * parentPlot )  [explicit]
```

Creates a straight line item and sets default values.
The created item is automatically registered with *parentPlot*. This QCustomPlot instance takes ownership of the item, so do not delete it manually but use QCustomPlot::removeItem() instead.

17.643.3 Member Function Documentation

17.643.3.1 draw()

```
void QCPItemStraightLine::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractItem.

17.643.3.2 selectTest()

```
double QCPItemStraightLine::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Implements QCPAbstractItem.

17.643.3.3 setPen()

```
void QCPItemStraightLine::setPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line

See also

setSelectedPen

17.643.3.4 setSelectedPen()

```
void QCPItemStraightLine::setSelectedPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line when selected

See also

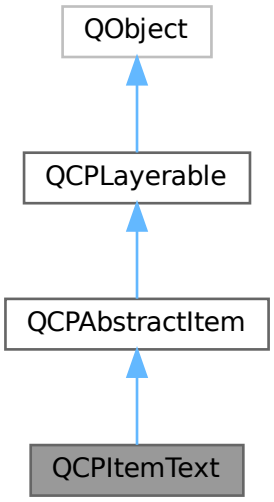
setPen, setSelected

The documentation for this class was generated from the following files:

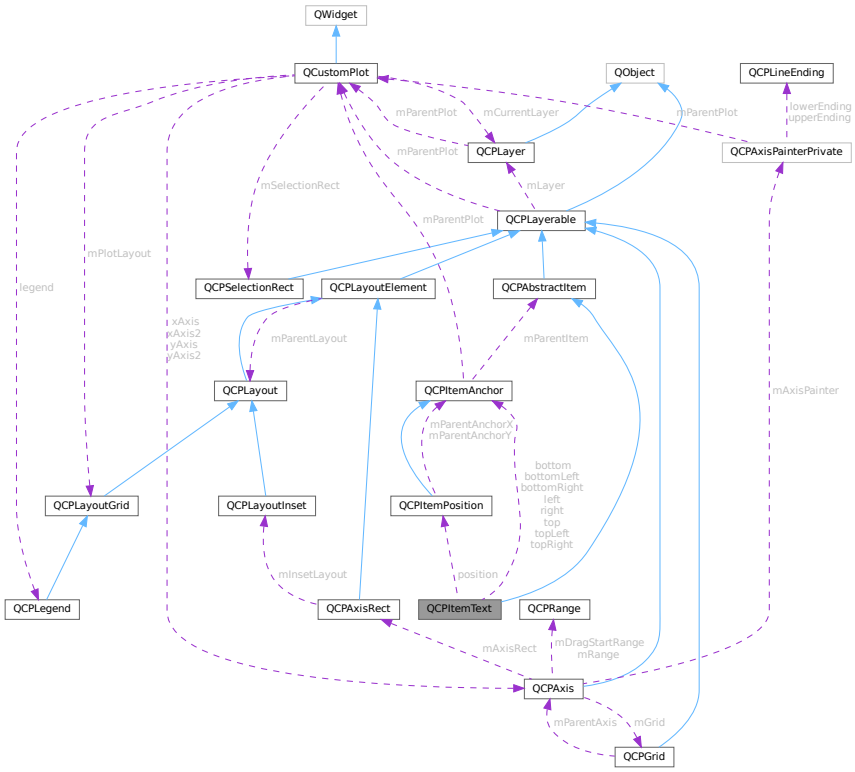
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.644 QCPItemText Class Reference

A text label.
Inheritance diagram for QCPItemText:



Collaboration diagram for QCPItemText:



Public Member Functions

- QCPItemText (QCustomPlot *parentPlot)
- QColor **color** () const
- QColor **selectedColor** () const
- QPen **pen** () const
- QPen **selectedPen** () const
- QBrush **brush** () const
- QBrush **selectedBrush** () const
- QFont **font** () const
- QFont **selectedFont** () const
- QString **text** () const
- Qt::Alignment **positionAlignment** () const
- Qt::Alignment **textAlignment** () const
- double **rotation** () const
- QMargins **padding** () const
- void setColor (const QColor &color)
- void setSelectedColor (const QColor &color)
- void setPen (const QPen &pen)
- void setSelectedPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setSelectedBrush (const QBrush &brush)
- void setFont (const QFont &font)
- void setSelectedFont (const QFont &font)
- void setText (const QString &text)
- void setPositionAlignment (Qt::Alignment alignment)

- void setTextAlignment (Qt::Alignment alignment)
- void setRotation (double degrees)
- void setPadding (const QMargins &padding)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractItem

- QCPAbstractItem (QCustomPlot *parentPlot)
- bool **clipToAxisRect** () const
- QCPAxisRect * **clipAxisRect** () const
- bool **selectable** () const
- bool **selected** () const
- void setClipToAxisRect (bool clip)
- void setClipAxisRect (QCPAxisRect *rect)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- QList< QCPItemPosition * > positions () const
- QList< QCPItemAnchor * > anchors () const
- QCPItemPosition * position (const QString &name) const
- QCPItemAnchor * anchor (const QString &name) const
- bool hasAnchor (const QString &name) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Public Attributes

- QCPItemPosition *const **position**
- QCPItemAnchor *const **topLeft**
- QCPItemAnchor *const **top**
- QCPItemAnchor *const **topRight**
- QCPItemAnchor *const **right**
- QCPItemAnchor *const **bottomRight**
- QCPItemAnchor *const **bottom**
- QCPItemAnchor *const **bottomLeft**
- QCPItemAnchor *const **left**

Protected Types

- enum **AnchorIndex** {
  aiTopLeft , **aiTop** , **aiTopRight** , **aiRight** ,
  aiBottomRight , **aiBottom** , **aiBottomLeft** , **aiLeft** }

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual QPointF anchorPixelPosition (int anchorId) const Q_DECL_OVERRIDE
- QPointF **getTextDrawPoint** (const QPointF &pos, const QRectF &rect, Qt::Alignment positionAlignment) const
- QFont **mainFont** () const
- QColor **mainColor** () const
- QPen **mainPen** () const
- QBrush **mainBrush** () const

Protected Member Functions inherited from QCPAbstractItem

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- double **rectDistance** (const QRectF &rect, const QPointF &pos, bool filledRect) const
- QCPItemPosition * **createPosition** (const QString &name)
- QCPItemAnchor * **createAnchor** (const QString &name, int anchorId)

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QColor **mColor**
- QColor **mSelectedColor**
- QPen **mPen**
- QPen **mSelectedPen**
- QBrush **mBrush**
- QBrush **mSelectedBrush**
- QFont **mFont**
- QFont **mSelectedFont**
- QString **mText**
- Qt::Alignment **mPositionAlignment**
- Qt::Alignment **mTextAlignment**
- double **mRotation**
- QMargins **mPadding**

Protected Attributes inherited from QCPAbstractItem

- bool **mClipToAxisRect**
- QPointer< QCPAxisRect > **mClipAxisRect**
- QList< QCPItemPosition * > **mPositions**
- QList< QCPItemAnchor * > **mAnchors**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPAbstractItem

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.644.1 Detailed Description

A text label.
Its position is defined by the member *position* and the setting of setPositionAlignment. The latter controls which part of the text rect shall be aligned with *position*.
The text alignment itself (i.e. left, center, right) can be controlled with setTextAlignment.
The text may be rotated around the *position* point with setRotation.

17.644.2 Constructor & Destructor Documentation

17.644.2.1 QCPItemText()

```
QCPItemText::QCPItemText (
            QCustomPlot * parentPlot )  [explicit]
```

Creates a text item and sets default values.
The created item is automatically registered with *parentPlot*. This QCustomPlot instance takes ownership of the item, so do not delete it manually but use QCustomPlot::removeItem() instead.

17.644.3 Member Function Documentation

17.644.3.1 anchorPixelPosition()

```
QPointF QCPItemText::anchorPixelPosition (
            int anchorId ) const  [protected], [virtual]
```

Reimplemented from QCPAbstractItem.

17.644.3.2 draw()

```
void QCPItemText::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractItem.

17.644.3.3 selectTest()

```
double QCPItemText::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Implements QCPAbstractItem.

17.644.3.4 setBrush()

```
void QCPItemText::setBrush (
            const QBrush & brush )
```

Sets the brush that will be used do fill the background of the text. To disable the background, set *brush* to Qt::No↩Brush.

See also

setSelectedBrush, setPen, setPadding

17.644.3.5 setColor()

```
void QCPItemText::setColor (
            const QColor & color )
```

Sets the color of the text.

17.644.3.6 setFont()

```
void QCPItemText::setFont (
            const QFont & font )
```

Sets the font of the text.

See also

setSelectedFont, setColor

17.644.3.7 setPadding()

```
void QCPItemText::setPadding (
            const QMargins & padding )
```

Sets the distance between the border of the text rectangle and the text. The appearance (and visibility) of the text rectangle can be controlled with setPen and setBrush.

17.644.3.8 setPen()

```
void QCPItemText::setPen (
            const QPen & pen )
```

Sets the pen that will be used do draw a rectangular border around the text. To disable the border, set *pen* to Qt::NoPen.

See also

setSelectedPen, setBrush, setPadding

17.644.3.9 setPositionAlignment()

```
void QCPItemText::setPositionAlignment (
            Qt::Alignment alignment )
```

Sets which point of the text rect shall be aligned with *position*.
Examples:

- If *alignment* is `Qt::AlignHCenter | Qt::AlignTop`, the text will be positioned such that the top of the text rect will be horizontally centered on *position*.
- If *alignment* is `Qt::AlignLeft | Qt::AlignBottom`, *position* will indicate the bottom left corner of the text rect.

If you want to control the alignment of (multi-lined) text within the text rect, use setTextAlignment.

17.644.3.10 setRotation()

```
void QCPItemText::setRotation (
            double degrees )
```

Sets the angle in degrees by which the text (and the text rectangle, if visible) will be rotated around *position*.

17.644.3.11 setSelectedBrush()

```
void QCPItemText::setSelectedBrush (
            const QBrush & brush )
```

Sets the brush that will be used do fill the background of the text, when the item is selected. To disable the background, set *brush* to Qt::NoBrush.

See also

setBrush

17.644.3.12 setSelectedColor()

```
void QCPItemText::setSelectedColor (
            const QColor & color )
```

Sets the color of the text that will be used when the item is selected.

17.644.3.13 setSelectedFont()

```
void QCPItemText::setSelectedFont (
            const QFont & font )
```

Sets the font of the text that will be used when the item is selected.

See also

setFont

17.644.3.14 setSelectedPen()

```
void QCPItemText::setSelectedPen (
            const QPen & pen )
```

Sets the pen that will be used do draw a rectangular border around the text, when the item is selected. To disable the border, set *pen* to Qt::NoPen.

See also

setPen

17.644.3.15 setText()

```
void QCPItemText::setText (
            const QString & text )
```

Sets the text that will be displayed. Multi-line texts are supported by inserting a line break character, e.g. '
'.

See also

setFont, setColor, setTextAlignment

17.644.3.16 setTextAlignment()

```
void QCPItemText::setTextAlignment (
            Qt::Alignment alignment )
```

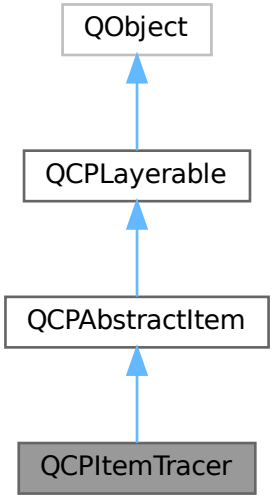
Controls how (multi-lined) text is aligned inside the text rect (typically Qt::AlignLeft, Qt::AlignCenter or Qt::Align↩ Right).
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

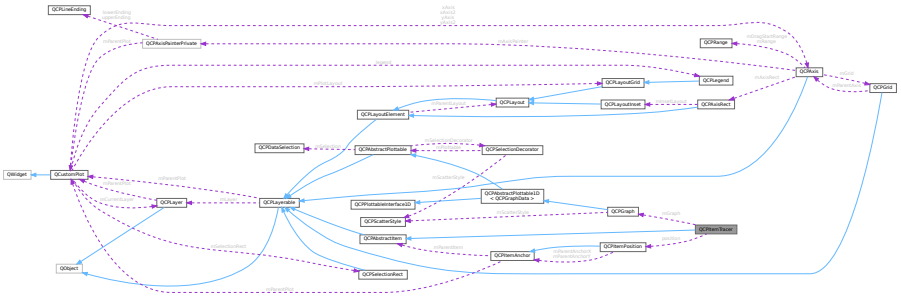
17.645 QCPItemTracer Class Reference

Item that sticks to QCPGraph data points.

Inheritance diagram for QCPItemTracer:



Collaboration diagram for QCPItemTracer:



Public Types

- enum TracerStyle {
  tsNone , tsPlus , tsCrosshair , tsCircle ,
  tsSquare }

Public Member Functions

- QCPItemTracer (QCustomPlot ∗parentPlot)
- QPen **pen** () const
- QPen **selectedPen** () const
- QBrush **brush** () const
- QBrush **selectedBrush** () const
- double **size** () const
- TracerStyle **style** () const
- QCPGraph ∗ **graph** () const
- double **graphKey** () const
- bool **interpolating** () const

- void setPen (const QPen &pen)
- void setSelectedPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setSelectedBrush (const QBrush &brush)
- void setSize (double size)
- void setStyle (TracerStyle style)
- void setGraph (QCPGraph *graph)
- void setGraphKey (double key)
- void setInterpolating (bool enabled)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE
- void updatePosition ()

Public Member Functions inherited from QCPAbstractItem

- QCPAbstractItem (QCustomPlot *parentPlot)
- bool **clipToAxisRect** () const
- QCPAxisRect * **clipAxisRect** () const
- bool **selectable** () const
- bool **selected** () const
- void setClipToAxisRect (bool clip)
- void setClipAxisRect (QCPAxisRect *rect)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- QList< QCPItemPosition * > positions () const
- QList< QCPItemAnchor * > anchors () const
- QCPItemPosition * position (const QString &name) const
- QCPItemAnchor * anchor (const QString &name) const
- bool hasAnchor (const QString &name) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Public Attributes

- QCPItemPosition *const **position**

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- QPen **mainPen** () const
- QBrush **mainBrush** () const

Protected Member Functions inherited from QCPAbstractItem

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual QPointF **anchorPixelPosition** (int anchorId) const
- double **rectDistance** (const QRectF &rect, const QPointF &pos, bool filledRect) const
- QCPItemPosition * **createPosition** (const QString &name)
- QCPItemAnchor * **createAnchor** (const QString &name, int anchorId)

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QPen **mPen**
- QPen **mSelectedPen**
- QBrush **mBrush**
- QBrush **mSelectedBrush**
- double **mSize**
- TracerStyle **mStyle**
- QCPGraph * **mGraph**
- double **mGraphKey**
- bool **mInterpolating**

Protected Attributes inherited from QCPAbstractItem

- bool **mClipToAxisRect**
- QPointer< QCPAxisRect > **mClipAxisRect**
- QList< QCPItemPosition * > **mPositions**
- QList< QCPItemAnchor * > **mAnchors**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPAbstractItem

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.645.1 Detailed Description

Item that sticks to QCPGraph data points.
The tracer can be connected with a QCPGraph via setGraph. Then it will automatically adopt the coordinate axes of the graph and update its *position* to be on the graph's data. This means the key stays controllable via setGraphKey, but the value will follow the graph data. If a QCPGraph is connected, note that setting the coordinates of the tracer item directly via *position* will have no effect because they will be overriden in the next redraw (this is when the coordinate update happens).
If the specified key in setGraphKey is outside the key bounds of the graph, the tracer will stay at the corresponding end of the graph.
With setInterpolating you may specify whether the tracer may only stay exactly on data points or whether it interpolates data points linearly, if given a key that lies between two data points of the graph.
The tracer has different visual styles, see setStyle. It is also possible to make the tracer have no own visual appearance (set the style to tsNone), and just connect other item positions to the tracer *position* (used as an anchor) via QCPItemPosition::setParentAnchor.

Note

The tracer position is only automatically updated upon redraws. So when the data of the graph changes and immediately afterwards (without a redraw) the position coordinates of the tracer are retrieved, they will not reflect the updated data of the graph. In this case updatePosition must be called manually, prior to reading the tracer coordinates.

17.645.2 Member Enumeration Documentation

17.645.2.1 TracerStyle

```
enum QCPItemTracer::TracerStyle
```

The different visual appearances a tracer item can have. Some styles size may be controlled with setSize.

See also

setStyle

Enumerator

tsNone	The tracer is not visible.
tsPlus	A plus shaped crosshair with limited size.
tsCrosshair	A plus shaped crosshair which spans the complete axis rect.
tsCircle	A circle.
tsSquare	A square.

17.645.3 Constructor & Destructor Documentation

17.645.3.1 QCPItemTracer()

```
QCPItemTracer::QCPItemTracer (
            QCustomPlot * parentPlot )  [explicit]
```

Creates a tracer item and sets default values.
The created item is automatically registered with *parentPlot*. This QCustomPlot instance takes ownership of the item, so do not delete it manually but use QCustomPlot::removeItem() instead.

17.645.4 Member Function Documentation

17.645.4.1 draw()

```
void QCPItemTracer::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractItem.

17.645.4.2 selectTest()

```
double QCPItemTracer::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Implements QCPAbstractItem.

17.645.4.3 setBrush()

```
void QCPItemTracer::setBrush (
            const QBrush & brush )
```

Sets the brush that will be used to draw any fills of the tracer

See also

setSelectedBrush, setPen

17.645.4.4 setGraph()

```
void QCPItemTracer::setGraph (
            QCPGraph * graph )
```

Sets the QCPGraph this tracer sticks to. The tracer *position* will be set to type QCPItemPosition::ptPlotCoords and the axes will be set to the axes of *graph*.
To free the tracer from any graph, set *graph* to `nullptr`. The tracer *position* can then be placed freely like any other item position. This is the state the tracer will assume when its graph gets deleted while still attached to it.

See also

setGraphKey

17.645.4.5 setGraphKey()

```
void QCPItemTracer::setGraphKey (
            double key )
```

Sets the key of the graph's data point the tracer will be positioned at. This is the only free coordinate of a tracer when attached to a graph.
Depending on setInterpolating, the tracer will be either positioned on the data point closest to *key*, or will stay exactly at *key* and interpolate the value linearly.

See also

setGraph, setInterpolating

17.645.4.6 setInterpolating()

```
void QCPItemTracer::setInterpolating (
            bool enabled )
```

Sets whether the value of the graph's data points shall be interpolated, when positioning the tracer.
If *enabled* is set to false and a key is given with setGraphKey, the tracer is placed on the data point of the graph which is closest to the key, but which is not necessarily exactly there. If *enabled* is true, the tracer will be positioned exactly at the specified key, and the appropriate value will be interpolated from the graph's data points linearly.

See also

setGraph, setGraphKey

17.645.4.7 setPen()

```
void QCPItemTracer::setPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line of the tracer

See also

setSelectedPen, setBrush

17.645.4.8 setSelectedBrush()

```
void QCPItemTracer::setSelectedBrush (
            const QBrush & brush )
```

Sets the brush that will be used to draw any fills of the tracer, when selected.

See also

setBrush, setSelected

17.645.4.9 setSelectedPen()

```
void QCPItemTracer::setSelectedPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the line of the tracer when selected

See also

setPen, setSelected

17.645.4.10 setSize()

```
void QCPItemTracer::setSize (
            double size )
```

Sets the size of the tracer in pixels, if the style supports setting a size (e.g. tsSquare does, tsCrosshair does not).

17.645.4.11 setStyle()

```
void QCPItemTracer::setStyle (
            QCPItemTracer::TracerStyle style )
```

Sets the style/visual appearance of the tracer.
If you only want to use the tracer *position* as an anchor for other items, set *style* to tsNone.

17.645.4.12 updatePosition()

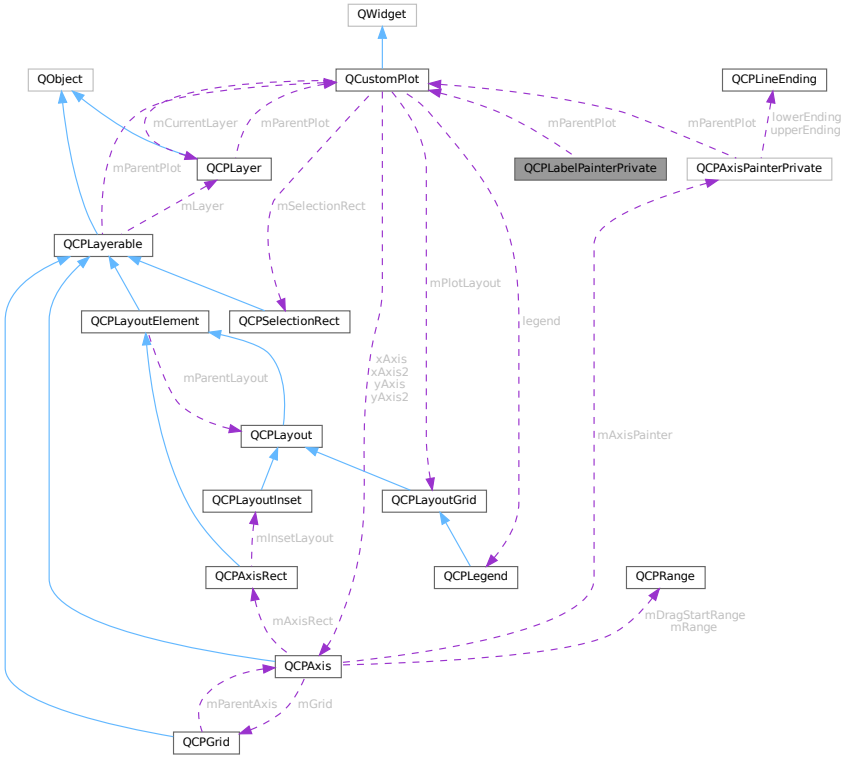
```
void QCPItemTracer::updatePosition ( )
```

If the tracer is connected with a graph (setGraph), this function updates the tracer's *position* to reside on the graph data, depending on the configured key (setGraphKey).
It is called automatically on every redraw and normally doesn't need to be called manually. One exception is when you want to read the tracer coordinates via *position* and are not sure that the graph's data (or the tracer key with setGraphKey) hasn't changed since the last redraw. In that situation, call this function before accessing *position*, to make sure you don't get out-of-date coordinates.
If there is no graph set on this tracer, this function does nothing.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.646 QCPLabelPainterPrivate Class Reference

Collaboration diagram for QCPLabelPainterPrivate:



Classes

- struct CachedLabel
- struct LabelData

Public Types

- enum AnchorMode { **amRectangular** , **amSkewedUpright** , **amSkewedRotated** }
- enum AnchorReferenceType { **artNormal** , **artTangent** }
- enum AnchorSide {
  asLeft , **asRight** , **asTop** , **asBottom** ,
  asTopLeft , **asTopRight** , **asBottomRight** , **asBottomLeft** }

Public Member Functions

- QCPLabelPainterPrivate (QCustomPlot *parentPlot)
- void **setAnchorSide** (AnchorSide side)
- void **setAnchorMode** (AnchorMode mode)
- void **setAnchorReference** (const QPointF &pixelPoint)
- void **setAnchorReferenceType** (AnchorReferenceType type)
- void **setFont** (const QFont &font)
- void **setColor** (const QColor &color)
- void **setPadding** (int padding)
- void **setRotation** (double rotation)
- void **setSubstituteExponent** (bool enabled)
- void **setMultiplicationSymbol** (QChar symbol)

- void **setAbbreviateDecimalPowers** (bool enabled)
- void **setCacheSize** (int labelCount)
- AnchorMode **anchorMode** () const
- AnchorSide **anchorSide** () const
- QPointF **anchorReference** () const
- AnchorReferenceType **anchorReferenceType** () const
- QFont **font** () const
- QColor **color** () const
- int **padding** () const
- double **rotation** () const
- bool **substituteExponent** () const
- QChar **multiplicationSymbol** () const
- bool **abbreviateDecimalPowers** () const
- int **cacheSize** () const
- void **drawTickLabel** (QCPPainter *painter, const QPointF &tickPos, const QString &text)
- void **clearCache** ()

Static Public Attributes

- static const QChar **SymbolDot**
- static const QChar **SymbolCross**

Protected Member Functions

- virtual void **drawLabelMaybeCached** (QCPPainter *painter, const QFont &font, const QColor &color, const QPointF &pos, AnchorSide side, double rotation, const QString &text)
- virtual QByteArray **generateLabelParameterHash** () const
- QPointF **getAnchorPos** (const QPointF &tickPos)
- void **drawText** (QCPPainter *painter, const QPointF &pos, const LabelData &labelData) const
- LabelData **getTickLabelData** (const QFont &font, const QColor &color, double rotation, AnchorSide side, const QString &text) const
- void **applyAnchorTransform** (LabelData &labelData) const
- CachedLabel * **createCachedLabel** (const LabelData &labelData) const
- QByteArray **cacheKey** (const QString &text, const QColor &color, double rotation, AnchorSide side) const
- AnchorSide **skewedAnchorSide** (const QPointF &tickPos, double sideExpandHorz, double sideExpandVert) const
- AnchorSide **rotationCorrectedSide** (AnchorSide side, double rotation) const
- void **analyzeFontMetrics** ()

Protected Attributes

- AnchorMode **mAnchorMode**
- AnchorSide **mAnchorSide**
- QPointF **mAnchorReference**
- AnchorReferenceType **mAnchorReferenceType**
- QFont **mFont**
- QColor **mColor**
- int **mPadding**
- double **mRotation**
- bool **mSubstituteExponent**
- QChar **mMultiplicationSymbol**
- bool **mAbbreviateDecimalPowers**
- QCustomPlot * **mParentPlot**
- QByteArray **mLabelParameterHash**
- QCache< QString, CachedLabel > **mLabelCache**
- QRect **mAxisSelectionBox**

- QRect **mTickLabelsSelectionBox**
- QRect **mLabelSelectionBox**
- int **mLetterCapHeight**
- int **mLetterDescent**

17.646.1 Member Enumeration Documentation

17.646.1.1 AnchorMode

```
enum QCPLabelPainterPrivate::AnchorMode
```
TODO

17.646.1.2 AnchorReferenceType

```
enum QCPLabelPainterPrivate::AnchorReferenceType
```
TODO

17.646.1.3 AnchorSide

```
enum QCPLabelPainterPrivate::AnchorSide
```
TODO

17.646.2 Constructor & Destructor Documentation

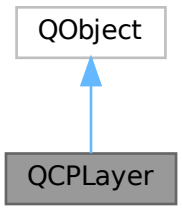
17.646.2.1 QCPLabelPainterPrivate()

```
QCPLabelPainterPrivate::QCPLabelPainterPrivate (
            QCustomPlot * parentPlot )  [explicit]
```
Constructs a QCPLabelPainterPrivate instance. Make sure to not create a new instance on every redraw, to utilize the caching mechanisms.
the *parentPlot* does not take ownership of the label painter. Make sure to delete it appropriately.
The documentation for this class was generated from the following files:

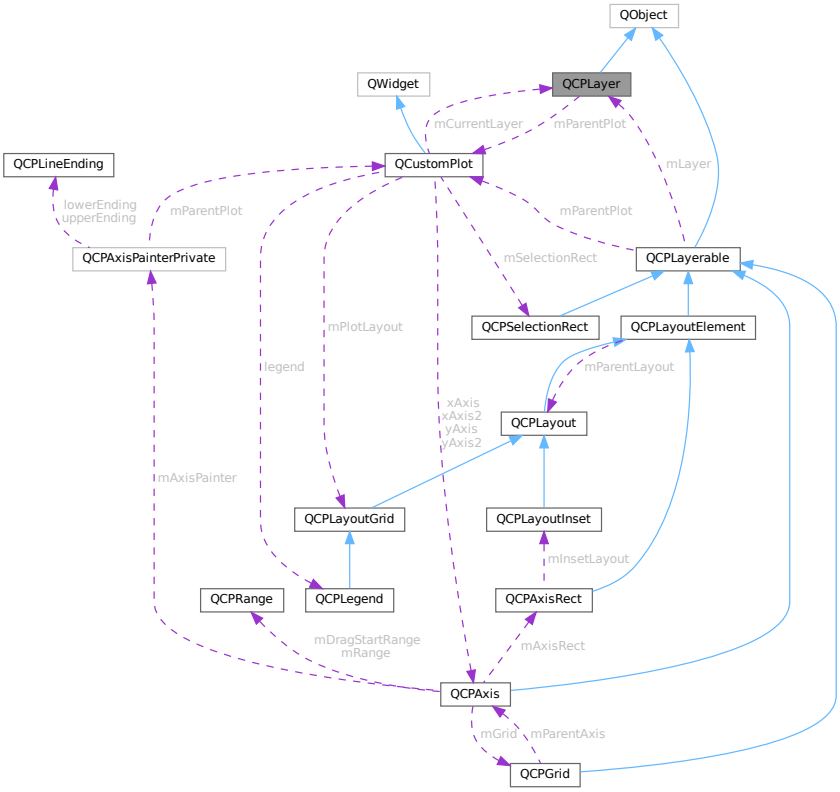
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.647 QCPLayer Class Reference

A layer that may contain objects, to control the rendering order.
Inheritance diagram for QCPLayer:



Collaboration diagram for QCPLayer:



Public Types

- enum LayerMode { lmLogical , lmBuffered }

Public Member Functions

- QCPLayer (QCustomPlot ∗parentPlot, const QString &layerName)
- QCustomPlot ∗ **parentPlot** () const
- QString **name** () const
- int index () const
- QList< QCPLayerable ∗ > children () const
- bool **visible** () const
- LayerMode **mode** () const
- void setVisible (bool visible)
- void setMode (LayerMode mode)
- void replot ()

Protected Member Functions

- void **draw** (QCPPainter ∗painter)
- void **drawToPaintBuffer** ()
- void **addChild** (QCPLayerable ∗layerable, bool prepend)
- void **removeChild** (QCPLayerable ∗layerable)

Protected Attributes

- QCustomPlot * **mParentPlot**
- QString **mName**
- int **mIndex**
- QList< QCPLayerable * > **mChildren**
- bool **mVisible**
- LayerMode **mMode**
- QWeakPointer< QCPAbstractPaintBuffer > **mPaintBuffer**

Friends

- class **QCustomPlot**
- class **QCPLayerable**

17.647.1 Detailed Description

A layer that may contain objects, to control the rendering order.
The Layering system of QCustomPlot is the mechanism to control the rendering order of the elements inside the plot.
It is based on the two classes QCPLayer and QCPLayerable. QCustomPlot holds an ordered list of one or more instances of QCPLayer (see QCustomPlot::addLayer, QCustomPlot::layer, QCustomPlot::moveLayer, etc.). When replotting, QCustomPlot goes through the list of layers bottom to top and successively draws the layerables of the layers into the paint buffer(s).
A QCPLayer contains an ordered list of QCPLayerable instances. QCPLayerable is an abstract base class from which almost all visible objects derive, like axes, grids, graphs, items, etc.

17.647.2 Default layers

Initially, QCustomPlot has six layers: "background", "grid", "main", "axes", "legend" and "overlay" (in that order). On top is the "overlay" layer, which only contains the QCustomPlot's selection rect (QCustomPlot::selectionRect). The next two layers "axes" and "legend" contain the default axes and legend, so they will be drawn above plottables. In the middle, there is the "main" layer. It is initially empty and set as the current layer (see QCustomPlot::setCurrentLayer). This means, all new plottables, items etc. are created on this layer by default. Then comes the "grid" layer which contains the QCPGrid instances (which belong tightly to QCPAxis, see QCPAxis::grid). The Axis rect background shall be drawn behind everything else, thus the default QCPAxisRect instance is placed on the "background" layer. Of course, the layer affiliation of the individual objects can be changed as required (QCPLayerable::setLayer).

17.647.3 Controlling the rendering order via layers

Controlling the ordering of layerables in the plot is easy: Create a new layer in the position you want the layerable to be in, e.g. above "main", with QCustomPlot::addLayer. Then set the current layer with QCustomPlot::setCurrentLayer to that new layer and finally create the objects normally. They will be placed on the new layer automatically, due to the current layer setting. Alternatively you could have also ignored the current layer setting and just moved the objects with QCPLayerable::setLayer to the desired layer after creating them.
It is also possible to move whole layers. For example, If you want the grid to be shown in front of all plottables/items on the "main" layer, just move it above "main" with QCustomPlot::moveLayer.
The rendering order within one layer is simply by order of creation or insertion. The item created last (or added last to the layer), is drawn on top of all other objects on that layer.
When a layer is deleted, the objects on it are not deleted with it, but fall on the layer below the deleted layer, see QCustomPlot::removeLayer.

17.647.4 Replotting only a specific layer

If the layer mode (setMode) is set to lmBuffered, you can replot only this specific layer by calling replot. In certain situations this can provide better replot performance, compared with a full replot of all layers. Upon creation of a new layer, the layer mode is initialized to lmLogical. The only layer that is set to lmBuffered in a new QCustomPlot instance is the "overlay" layer, containing the selection rect.

17.647.5 Member Enumeration Documentation

17.647.5.1 LayerMode

`enum QCPLayer::LayerMode`

Defines the different rendering modes of a layer. Depending on the mode, certain layers can be replotted individually, without the need to replot (possibly complex) layerables on other layers.

See also

setMode

Enumerator

lmLogical	Layer is used only for rendering order, and shares paint buffer with all other adjacent logical layers.
lmBuffered	Layer has its own paint buffer and may be replotted individually (see replot).

17.647.6 Constructor & Destructor Documentation

17.647.6.1 QCPLayer()

```
QCPLayer::QCPLayer (
            QCustomPlot * parentPlot,
            const QString & layerName )
```

Creates a new QCPLayer instance.
Normally you shouldn't directly instantiate layers, use QCustomPlot::addLayer instead.

Warning

It is not checked that *layerName* is actually a unique layer name in *parentPlot*. This check is only performed by QCustomPlot::addLayer.

17.647.7 Member Function Documentation

17.647.7.1 children()

`QList< QCPLayerable * > QCPLayer::children ( ) const [inline]`

Returns a list of all layerables on this layer. The order corresponds to the rendering order: layerables with higher indices are drawn above layerables with lower indices.

17.647.7.2 index()

`int QCPLayer::index ( ) const [inline]`

Returns the index this layer has in the QCustomPlot. The index is the integer number by which this layer can be accessed via QCustomPlot::layer.
Layers with higher indices will be drawn above layers with lower indices.

17.647.7.3 replot()

`void QCPLayer::replot ( )`

If the layer mode (setMode) is set to lmBuffered, this method allows replotting only the layerables on this specific layer, without the need to replot all other layers (as a call to QCustomPlot::replot would do).
QCustomPlot also makes sure to replot all layers instead of only this one, if the layer ordering or any layerable-layer-association has changed since the last full replot and any other paint buffers were thus invalidated.
If the layer mode is lmLogical however, this method simply calls QCustomPlot::replot on the parent QCustomPlot instance.

See also

draw

17.647.7.4 setMode()

```
void QCPLayer::setMode (
            QCPLayer::LayerMode mode )
```

Sets the rendering mode of this layer.
If *mode* is set to lmBuffered for a layer, it will be given a dedicated paint buffer by the parent QCustomPlot instance. This means it may be replotted individually by calling QCPLayer::replot, without needing to replot all other layers.
Layers which are set to lmLogical (the default) are used only to define the rendering order and can't be replotted individually.
Note that each layer which is set to lmBuffered requires additional paint buffers for the layers below, above and for the layer itself. This increases the memory consumption and (slightly) decreases the repainting speed because multiple paint buffers need to be joined. So you should carefully choose which layers benefit from having their own paint buffer. A typical example would be a layer which contains certain layerables (e.g. items) that need to be changed and thus replotted regularly, while all other layerables on other layers stay static. By default, only the topmost layer called "overlay" is in mode lmBuffered, and contains the selection rect.

See also

replot

17.647.7.5 setVisible()

```
void QCPLayer::setVisible (
            bool visible )
```

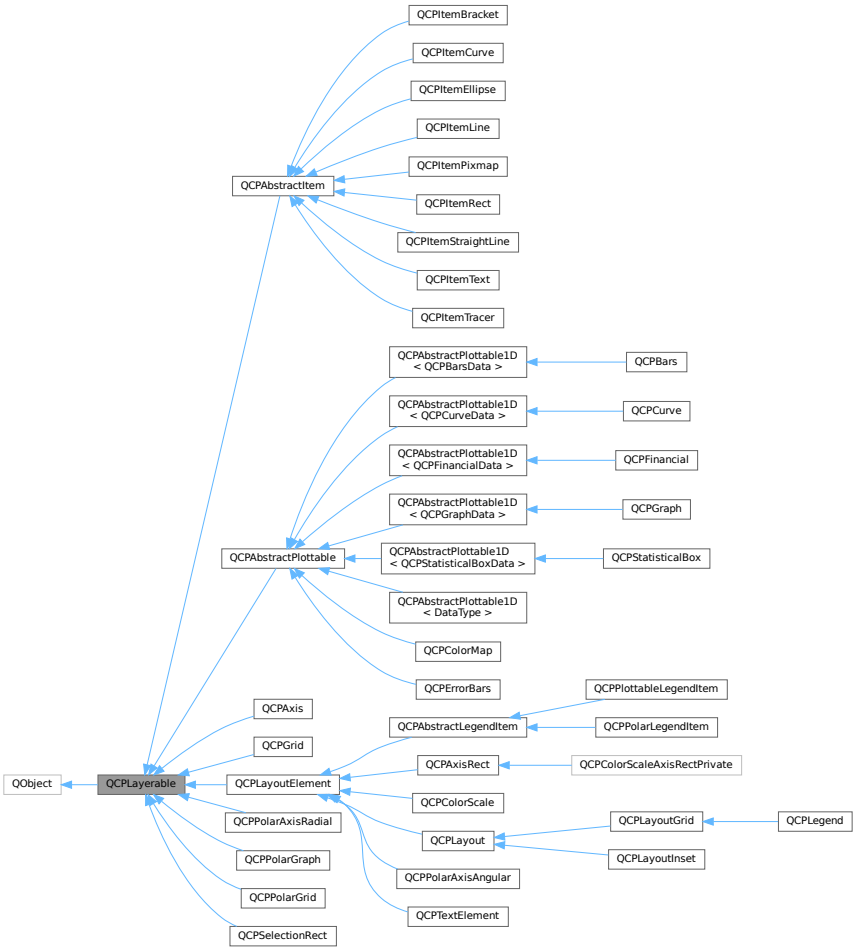
Sets whether this layer is visible or not. If *visible* is set to false, all layerables on this layer will be invisible.
This function doesn't change the visibility property of the layerables (QCPLayerable::setVisible), but the QCPLayerable::realVisibility of each layerable takes the visibility of the parent layer into account.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

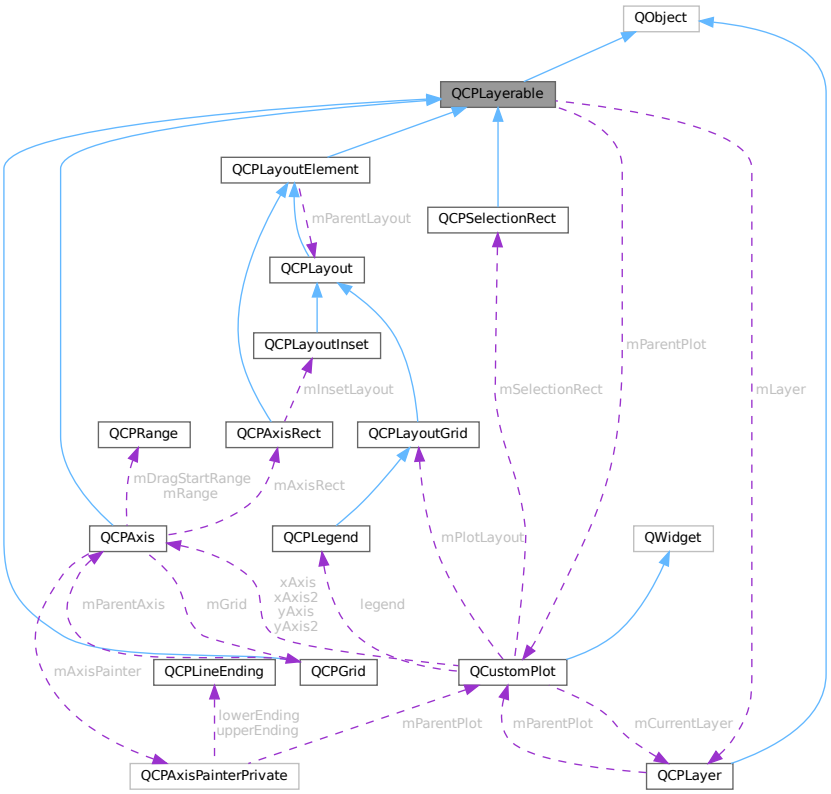
17.648 QCPLayerable Class Reference

Base class for all drawable objects.

Inheritance diagram for QCPLayerable:



Collaboration diagram for QCPLayerable:



Signals

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const
- bool realVisibility () const

Protected Member Functions

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void **applyDefaultAntialiasingHint** (QCPPainter *painter) const =0
- virtual void **draw** (QCPPainter *painter)=0

- virtual void **selectEvent** (QMouseEvent ∗event, bool additive, const QVariant &details, bool ∗selectionStateChanged)
- virtual void **deselectEvent** (bool ∗selectionStateChanged)
- virtual void mousePressEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent ∗event)
- void **initializeParentPlot** (QCustomPlot ∗parentPlot)
- void **setParentLayerable** (QCPLayerable ∗parentLayerable)
- bool **moveToLayer** (QCPLayer ∗layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter ∗painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- bool **mVisible**
- QCustomPlot ∗ **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer ∗ **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPLayer**
- class **QCPAxisRect**

17.648.1 Detailed Description

Base class for all drawable objects.
This is the abstract base class most visible objects derive from, e.g. plottables, axes, grid etc.
Every layerable is on a layer (QCPLayer) which allows controlling the rendering order by stacking the layers accordingly.
For details about the layering mechanism, see the QCPLayer documentation.

17.648.2 Constructor & Destructor Documentation

17.648.2.1 QCPLayerable()

```
QCPLayerable::QCPLayerable (
            QCustomPlot * plot,
            QString targetLayer = QString(),
            QCPLayerable * parentLayerable = nullptr )
```

Creates a new QCPLayerable instance.
Since QCPLayerable is an abstract base class, it can't be instantiated directly. Use one of the derived classes.
If *plot* is provided, it automatically places itself on the layer named *targetLayer*. If *targetLayer* is an empty string, it places itself on the current layer of the plot (see QCustomPlot::setCurrentLayer).
It is possible to provide `nullptr` as *plot*. In that case, you should assign a parent plot at a later time with initializeParentPlot.
The layerable's parent layerable is set to *parentLayerable*, if provided. Direct layerable parents are mainly used to control visibility in a hierarchy of layerables. This means a layerable is only drawn, if all its ancestor layerables are also visible. Note that *parentLayerable* does not become the QObject-parent (for memory management) of this layerable, *plot* does. It is not uncommon to set the QObject-parent to something else in the constructors of QCPLayerable subclasses, to guarantee a working destruction hierarchy.

17.648.3 Member Function Documentation

17.648.3.1 layerChanged

```
void QCPLayerable::layerChanged (
            QCPLayer * newLayer )  [signal]
```

This signal is emitted when the layer of this layerable changes, i.e. this layerable is moved to a different layer.

See also

setLayer

17.648.3.2 mouseDoubleClickEvent()

```
void QCPLayerable::mouseDoubleClickEvent (
            QMouseEvent * event,
            const QVariant & details )  [protected], [virtual]
```

This event gets called when the user presses the mouse button a second time in a double-click, while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The mouseDoubleClickEvent is called instead of the second mousePressEvent. So in the case of a double-click, the event succession is *pressEvent – releaseEvent – doubleClickEvent – releaseEvent*.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *details* contains layerable-specific details about the hit, which were generated in the previous call to selectTest. For example, One-dimensional plottables like QCPGraph or QCPBars convey the clicked data point in the *details* parameter, as QCPDataSelection packed as QVariant. Multi-part objects convey the specific `SelectablePart` that was hit (e.g. QCPAxis::SelectablePart in the case of axes).
Similarly to mousePressEvent, once a layerable has accepted the mouseDoubleClickEvent, it is considered the mouse grabber and will receive all following calls to mouseMoveEvent and mouseReleaseEvent for this mouse interaction (a "mouse interaction" in this context ends with the release).
The default implementation does nothing except explicitly ignoring the event with `event->ignore()`.

See also

mousePressEvent, mouseMoveEvent, mouseReleaseEvent, wheelEvent

Reimplemented in QCPTextElement.

17.648.3.3 mouseMoveEvent()

```
void QCPLayerable::mouseMoveEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user moves the mouse while holding a mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented in QCPAxis, QCPAxisRect, QCPColorScaleAxisRectPrivate, QCPColorScale, QCPPolarAxisRadial, and QCPPolarAxisAngular.

17.648.3.4 mousePressEvent()

```
void QCPLayerable::mousePressEvent (
            QMouseEvent * event,
            const QVariant & details )  [protected], [virtual]
```

This event gets called when the user presses a mouse button while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.

The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *details* contains layerable-specific details about the hit, which were generated in the previous call to selectTest. For example, One-dimensional plottables like QCPGraph or QCPBars convey the clicked data point in the *details* parameter, as QCPDataSelection packed as QVariant. Multi-part objects convey the specific `SelectablePart` that was hit (e.g. QCPAxis::SelectablePart in the case of axes).
QCustomPlot uses an event propagation system that works the same as Qt's system. If your layerable doesn't reimplement the mousePressEvent or explicitly calls `event->ignore()` in its reimplementation, the event will be propagated to the next layerable in the stacking order.
Once a layerable has accepted the mousePressEvent, it is considered the mouse grabber and will receive all following calls to mouseMoveEvent or mouseReleaseEvent for this mouse interaction (a "mouse interaction" in this context ends with the release).
The default implementation does nothing except explicitly ignoring the event with `event->ignore()`.

See also

mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented in QCPAxis, QCPAxisRect, QCPTextElement, QCPColorScaleAxisRectPrivate, QCPColorScale, QCPPolarAxisRadial, and QCPPolarAxisAngular.

17.648.3.5 mouseReleaseEvent()

```
void QCPLayerable::mouseReleaseEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user releases the mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseMoveEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented in QCPAxis, QCPAxisRect, QCPTextElement, QCPColorScaleAxisRectPrivate, QCPColorScale, QCPPolarAxisRadial, and QCPPolarAxisAngular.

17.648.3.6 parentLayerable()

```
QCPLayerable * QCPLayerable::parentLayerable ( ) const  [inline]
```

Returns the parent layerable of this layerable. The parent layerable is used to provide visibility hierarchies in conjunction with the method realVisibility. This way, layerables only get drawn if their parent layerables are visible, too.
Note that a parent layerable is not necessarily also the QObject parent for memory management. Further, a layerable doesn't always have a parent layerable, so this function may return `nullptr`.
A parent layerable is set implicitly when placed inside layout elements and doesn't need to be set manually by the user.

17.648.3.7 realVisibility()

```
bool QCPLayerable::realVisibility ( ) const
```

Returns whether this layerable is visible, taking the visibility of the layerable parent and the visibility of this layerable's layer into account. This is the method that is consulted to decide whether a layerable shall be drawn or not.
If this layerable has a direct layerable parent (usually set via hierarchies implemented in subclasses, like in the case of QCPLayoutElement), this function returns true only if this layerable has its visibility set to true and the parent layerable's realVisibility returns true.

17.648.3.8 selectTest()

```
double QCPLayerable::selectTest (
            const QPointF & pos,
```

```
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Reimplemented in QCPPolarGraph, QCPPolarAxisRadial, QCPPolarAxisAngular, QCPLayoutElement, QCPLayoutInset, QCPAxis, QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCPCurv QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat QCPAbstractLegendItem, QCPLegend, QCPTextElement, QCPGraph, QCPCurve, QCPBars, QCPStatisticalBox, QCPColorMap, QCPFinancial, QCPErrorBars, QCPItemStraightLine, QCPItemLine, QCPItemCurve, QCPItemRect, QCPItemText, QCPItemEllipse, QCPItemPixmap, QCPItemTracer, QCPItemBracket, QCPAbstractPlottable, and QCPAbstractItem.

17.648.3.9 setAntialiased()

```
void QCPLayerable::setAntialiased (
            bool enabled )
```

Sets whether this object will be drawn antialiased or not.
Note that antialiasing settings may be overridden by QCustomPlot::setAntialiasedElements and QCustomPlot::setNotAntialiasedEleme

17.648.3.10 setLayer() [1/2]

```
bool QCPLayerable::setLayer (
            const QString & layerName )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts. Sets the layer of this layerable object by name
Returns true on success, i.e. if *layerName* is a valid layer name.

17.648.3.11 setLayer() [2/2]

```
bool QCPLayerable::setLayer (
            QCPLayer * layer )
```

Sets the *layer* of this layerable object. The object will be placed on top of the other objects already on *layer*.
If *layer* is 0, this layerable will not be on any layer and thus not appear in the plot (or interact/receive events).
Returns true if the layer of this layerable was successfully changed to *layer*.

17.648.3.12 setVisible()

```
void QCPLayerable::setVisible (
            bool on )
```

Sets the visibility of this layerable object. If an object is not visible, it will not be drawn on the QCustomPlot surface, and user interaction with it (e.g. click and selection) is not possible.

17.648.3.13 wheelEvent()

```
void QCPLayerable::wheelEvent (
            QWheelEvent * event )  [protected], [virtual]
```

This event gets called when the user turns the mouse scroll wheel while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`.
The `event->angleDelta()` indicates how far the mouse wheel was turned, which is usually +/- 120 for single rotation steps. However, if the mouse wheel is turned rapidly, multiple steps may accumulate to one event, making the delta larger. On the other hand, if the wheel has very smooth steps or none at all, the delta may be smaller.
The default implementation does nothing.

See also

mousePressEvent, mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent

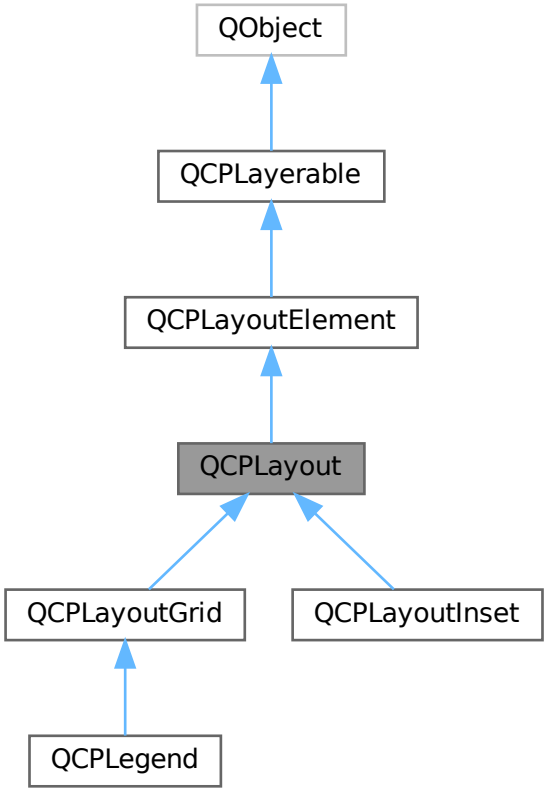
Reimplemented in QCPAxis, QCPAxisRect, QCPColorScaleAxisRectPrivate, QCPColorScale, QCPPolarAxisRadial, and QCPPolarAxisAngular.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

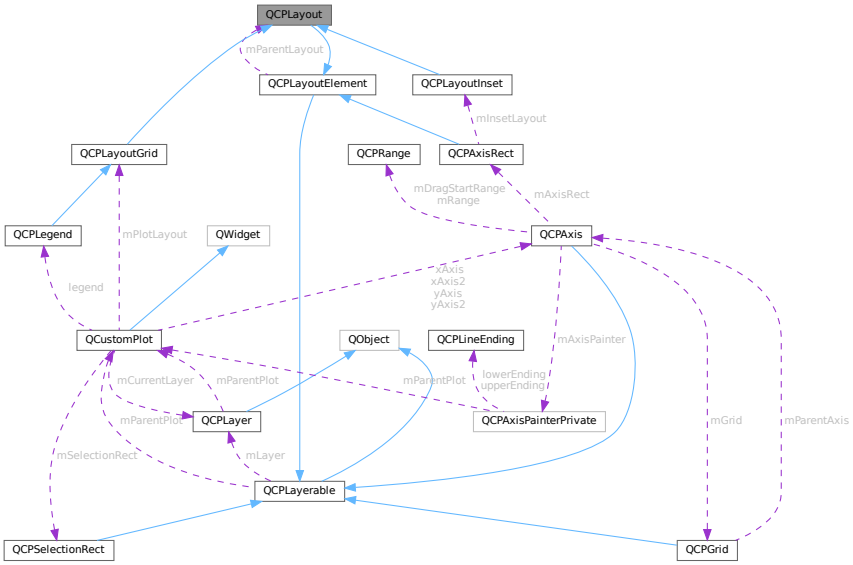
17.649 QCPLayout Class Reference

The abstract base class for layouts.

Inheritance diagram for QCPLayout:



Collaboration diagram for QCPLayout:



Public Member Functions

- QCPLayout ()
- virtual void update (UpdatePhase phase) Q_DECL_OVERRIDE
- virtual QList< QCPLayoutElement * > elements (bool recursive) const Q_DECL_OVERRIDE
- virtual int elementCount () const =0
- virtual QCPLayoutElement * elementAt (int index) const =0
- virtual QCPLayoutElement * takeAt (int index)=0
- virtual bool take (QCPLayoutElement *element)=0
- virtual void simplify ()
- bool removeAt (int index)
- bool remove (QCPLayoutElement *element)
- void clear ()

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot *parentPlot=nullptr)
- QCPLayout * layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const
- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)
- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual QSize minimumOuterSizeHint () const
- virtual QSize maximumOuterSizeHint () const
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void **updateLayout** ()
- void sizeConstraintsChanged () const
- void **adoptElement** (QCPLayoutElement *el)
- void **releaseElement** (QCPLayoutElement *el)
- QVector< int > **getSectionSizes** (QVector< int > maxSizes, QVector< int > minSizes, QVector< double > stretchFactors, int totalSize) const

Protected Member Functions inherited from QCPLayoutElement

- virtual int **calculateAutoMargin** (QCP::MarginSide side)
- virtual void **layoutChanged** ()
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void **selectEvent** (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged)
- virtual void **deselectEvent** (bool *selectionStateChanged)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Static Protected Member Functions

- static QSize **getFinalMinimumOuterSize** (const QCPLayoutElement *el)
- static QSize **getFinalMaximumOuterSize** (const QCPLayoutElement *el)

Friends

- class **QCPLayoutElement**

Additional Inherited Members

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Protected Attributes inherited from QCPLayoutElement

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

17.649.1 Detailed Description

The abstract base class for layouts.
This is an abstract base class for layout elements whose main purpose is to define the position and size of other child layout elements. In most cases, layouts don't draw anything themselves (but there are exceptions to this, e.g. QCPLegend).
QCPLayout derives from QCPLayoutElement, and thus can itself be nested in other layouts.
QCPLayout introduces a common interface for accessing and manipulating the child elements. Those functions are most notably elementCount, elementAt, takeAt, take, simplify, removeAt, remove and clear. Individual subclasses may add more functions to this interface which are more specialized to the form of the layout. For example, QCPLayoutGrid adds functions that take row and column indices to access cells of the layout grid more conveniently.
Since this is an abstract base class, you can't instantiate it directly. Rather use one of its subclasses like QCPLayoutGrid or QCPLayoutInset.
For a general introduction to the layout system, see the dedicated documentation page The Layout System.

17.649.2 Constructor & Destructor Documentation

17.649.2.1 QCPLayout()

```
QCPLayout::QCPLayout ( )  [explicit]
```

Creates an instance of QCPLayout and sets default values. Note that since QCPLayout is an abstract base class, it can't be instantiated directly.

17.649.3 Member Function Documentation

17.649.3.1 clear()

```
void QCPLayout::clear ( )
```

Removes and deletes all layout elements in this layout. Finally calls simplify to make sure all empty cells are collapsed.

See also

remove, removeAt

17.649.3.2 elementAt()

```
QCPLayoutElement * QCPLayout::elementAt (
            int index ) const  [pure virtual]
```

Returns the element in the cell with the given *index*. If *index* is invalid, returns `nullptr`.
Note that even if *index* is valid, the respective cell may be empty in some layouts (e.g. QCPLayoutGrid), so this function may return `nullptr` in those cases. You may use this function to check whether a cell is empty or not.

See also

elements, elementCount, takeAt

Implemented in QCPLayoutGrid, and QCPLayoutInset.

17.649.3.3 elementCount()

```
int QCPLayout::elementCount ( ) const  [pure virtual]
```

Returns the number of elements/cells in the layout.

See also

elements, elementAt

Implemented in QCPLayoutGrid, and QCPLayoutInset.

17.649.3.4 elements()

```
QList< QCPLayoutElement * > QCPLayout::elements (
            bool recursive ) const  [virtual]
```

Returns a list of all child elements in this layout element. If *recursive* is true, all sub-child elements are included in the list, too.

Warning

There may be `nullptr` entries in the returned list. For example, QCPLayoutGrid may have empty cells which yield `nullptr` at the respective index.

Reimplemented from QCPLayoutElement.
Reimplemented in QCPLayoutGrid.

17.649.3.5 remove()

```
bool QCPLayout::remove (
            QCPLayoutElement * element )
```

Removes and deletes the provided *element*. Returns true on success. If *element* is not in the layout, returns false.
This function internally uses takeAt to remove the element from the layout and then deletes the element. Note that some layouts don't remove the respective cell right away but leave an empty cell after successful removal of the layout element. To collapse empty cells, use simplify.

See also

removeAt, take

17.649.3.6 removeAt()

```
bool QCPLayout::removeAt (
            int index )
```

Removes and deletes the element at the provided *index*. Returns true on success. If *index* is invalid or points to an empty cell, returns false.
This function internally uses takeAt to remove the element from the layout and then deletes the returned element. Note that some layouts don't remove the respective cell right away but leave an empty cell after successful removal of the layout element. To collapse empty cells, use simplify.

See also

remove, takeAt

17.649.3.7 simplify()

```
void QCPLayout::simplify ( )  [virtual]
```

Simplifies the layout by collapsing empty cells. The exact behavior depends on subclasses, the default implementation does nothing.
Not all layouts need simplification. For example, QCPLayoutInset doesn't use explicit simplification while QCPLayoutGrid does.
Reimplemented in QCPLayoutGrid, and QCPLayoutInset.

17.649.3.8 sizeConstraintsChanged()

```
void QCPLayout::sizeConstraintsChanged ( ) const  [protected]
```

Subclasses call this method to report changed (minimum/maximum) size constraints.
If the parent of this layout is again a QCPLayout, forwards the call to the parent's sizeConstraintsChanged. If the parent is a QWidget (i.e. is the QCustomPlot::plotLayout of QCustomPlot), calls QWidget::updateGeometry, so if the QCustomPlot widget is inside a Qt QLayout, it may update itself and resize cells accordingly.

17.649.3.9 take()

```
bool QCPLayout::take (
            QCPLayoutElement * element )  [pure virtual]
```

Removes the specified *element* from the layout and returns true on success.
If the *element* isn't in this layout, returns false.
Note that some layouts don't remove the respective cell right away but leave an empty cell after successful removal of the layout element. To collapse empty cells, use simplify.

See also

takeAt

Implemented in QCPLayoutGrid, and QCPLayoutInset.

17.649.3.10 takeAt()

```
QCPLayoutElement * QCPLayout::takeAt (
            int index )  [pure virtual]
```

Removes the element with the given *index* from the layout and returns it.
If the *index* is invalid or the cell with that index is empty, returns `nullptr`.
Note that some layouts don't remove the respective cell right away but leave an empty cell after successful removal of the layout element. To collapse empty cells, use simplify.

See also

elementAt, take

Implemented in QCPLayoutGrid, and QCPLayoutInset.

17.649.3.11 update()

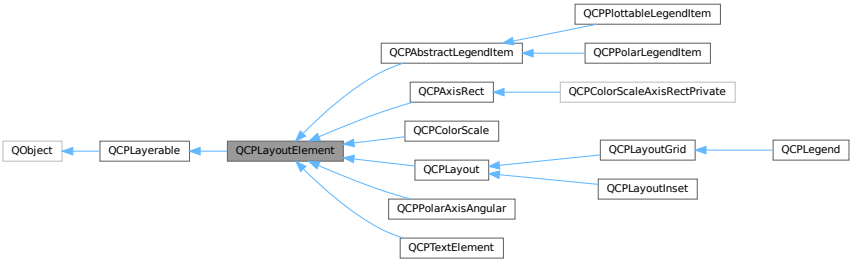
```
void QCPLayout::update (
            UpdatePhase phase )  [virtual]
```

If *phase* is upLayout, calls updateLayout, which subclasses may reimplement to reposition and resize their cells. Finally, the call is propagated down to all child QCPLayoutElements.
For details about this method and the update phases, see the documentation of QCPLayoutElement::update.
Reimplemented from QCPLayoutElement.
The documentation for this class was generated from the following files:

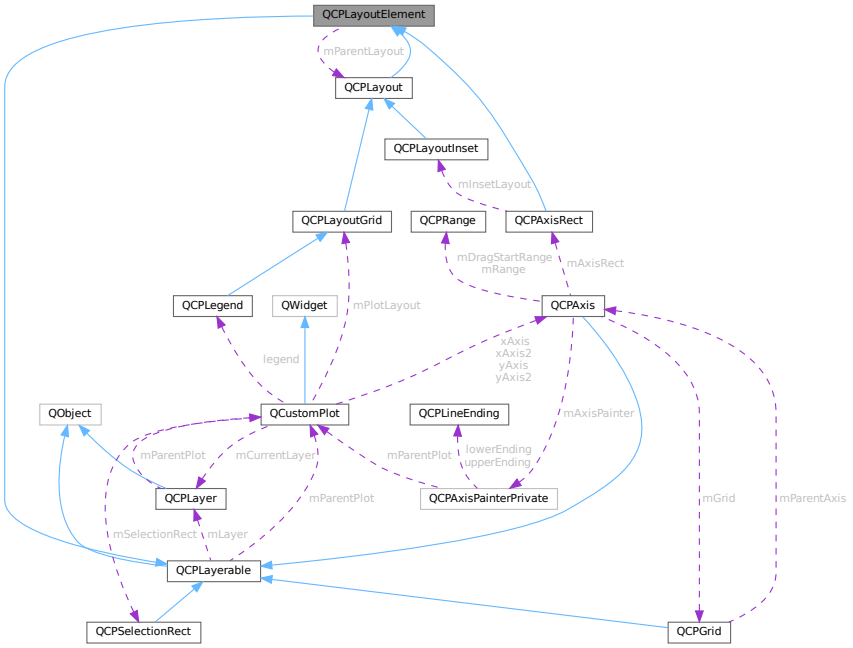
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.650 QCPLayoutElement Class Reference

The abstract base class for all objects that form the layout system.
Inheritance diagram for QCPLayoutElement:



Collaboration diagram for QCPLayoutElement:



Public Types

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

Public Member Functions

- QCPLayoutElement (QCustomPlot ∗parentPlot=nullptr)
- QCPLayout ∗ layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const

- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)
- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual void update (UpdatePhase phase)
- virtual QSize minimumOuterSizeHint () const
- virtual QSize maximumOuterSizeHint () const
- virtual QList< QCPLayoutElement * > elements (bool recursive) const
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual int **calculateAutoMargin** (QCP::MarginSide side)
- virtual void **layoutChanged** ()
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void **selectEvent** (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed)
- virtual void **deselectEvent** (bool *selectionStateChanged)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)

- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPLayout**
- class **QCPMarginGroup**

Additional Inherited Members

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.650.1 Detailed Description

The abstract base class for all objects that form the layout system.
This is an abstract base class. As such, it can't be instantiated directly, rather use one of its subclasses.
A Layout element is a rectangular object which can be placed in layouts. It has an outer rect (QCPLayoutElement::outerRect) and an inner rect (QCPLayoutElement::rect). The difference between outer and inner rect is called its margin. The margin can either be set to automatic or manual (setAutoMargins) on a per-side basis. If a side is set to manual, that margin can be set explicitly with setMargins and will stay fixed at that value. If it's set to automatic, the layout element subclass will control the value itself (via calculateAutoMargin).
Layout elements can be placed in layouts (base class QCPLayout) like QCPLayoutGrid. The top level layout is reachable via QCustomPlot::plotLayout, and is a QCPLayoutGrid. Since QCPLayout itself derives from QCPLayoutElement, layouts can be nested.
Thus in QCustomPlot one can divide layout elements into two categories: The ones that are invisible by themselves, because they don't draw anything. Their only purpose is to manage the position and size of other layout elements. This category of layout elements usually use QCPLayout as base class. Then there is the category of layout elements which actually draw something. For example, QCPAxisRect, QCPLegend and QCPTextElement are of this category. This does not necessarily mean that the latter category can't have child layout elements. QCPLegend for instance, actually derives from QCPLayoutGrid and the individual legend items are child layout elements in the grid layout.

17.650.2 Member Enumeration Documentation

17.650.2.1 SizeConstraintRect

`enum QCPLayoutElement::SizeConstraintRect`

Defines to which rect of a layout element the size constraints that can be set via setMinimumSize and setMaximumSize apply. The outer rect (outerRect) includes the margins (e.g. in the case of a QCPAxisRect the axis labels), whereas the inner rect (rect) does not.

See also

setSizeConstraintRect

Enumerator

scrInnerRect	Minimum/Maximum size constraints apply to inner rect.
scrOuterRect	Minimum/Maximum size constraints apply to outer rect, thus include layout element margins.

17.650.2.2 UpdatePhase

`enum QCPLayoutElement::UpdatePhase`

Defines the phases of the update process, that happens just before a replot. At each phase, update is called with the according UpdatePhase value.

Enumerator

upPreparation	Phase used for any type of preparation that needs to be done before margin calculation and layout.
upMargins	Phase in which the margins are calculated and set.
upLayout	Final phase in which the layout system places the rects of the elements.

17.650.3 Constructor & Destructor Documentation

17.650.3.1 QCPLayoutElement()

```
QCPLayoutElement::QCPLayoutElement (
            QCustomPlot * parentPlot = nullptr )  [explicit]
```

Creates an instance of QCPLayoutElement and sets default values.

17.650.4 Member Function Documentation

17.650.4.1 applyDefaultAntialiasingHint()

```
virtual void QCPLayoutElement::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [inline], [protected], [virtual]
```

Implements QCPLayerable.

17.650.4.2 draw()

```
virtual void QCPLayoutElement::draw (
            QCPPainter * painter )  [inline], [protected], [virtual]
```

Implements QCPLayerable.

17.650.4.3 elements()

```
QList< QCPLayoutElement * > QCPLayoutElement::elements (
            bool recursive ) const  [virtual]
```

Returns a list of all child elements in this layout element. If *recursive* is true, all sub-child elements are included in the list, too.

Warning

There may be `nullptr` entries in the returned list. For example, QCPLayoutGrid may have empty cells which yield `nullptr` at the respective index.

Reimplemented in QCPLayout, QCPLayoutGrid, QCPAxisRect, and QCPPolarAxisAngular.

17.650.4.4 layout()

```
QCPLayout * QCPLayoutElement::layout ( ) const  [inline]
```

Returns the parent layout of this layout element.

17.650.4.5 maximumOuterSizeHint()

```
QSize QCPLayoutElement::maximumOuterSizeHint ( ) const  [virtual]
```

Returns the suggested maximum size this layout element (the outerRect) may be expanded to, if no manual maximum size is set.
if a maximum size (setMaximumSize) was not set manually, parent layouts use the returned size (usually indirectly through QCPLayout::getFinalMaximumOuterSize) to determine the maximum allowed size of this layout element.
A manual maximum size is considered set if it is smaller than Qt's `QWIDGETSIZE_MAX`.
The default implementation simply returns `QWIDGETSIZE_MAX` for both width and height, implying no suggested maximum size. Reimplementations may use their detailed knowledge about the layout element's content to provide size hints.
Reimplemented in QCPLayoutGrid, and QCPTextElement.

17.650.4.6 minimumOuterSizeHint()

```
QSize QCPLayoutElement::minimumOuterSizeHint ( ) const  [virtual]
```

Returns the suggested minimum size this layout element (the outerRect) may be compressed to, if no manual minimum size is set.
if a minimum size (setMinimumSize) was not set manually, parent layouts use the returned size (usually indirectly through QCPLayout::getFinalMinimumOuterSize) to determine the minimum allowed size of this layout element.
A manual minimum size is considered set if it is non-zero.
The default implementation simply returns the sum of the horizontal margins for the width and the sum of the vertical margins for the height. Reimplementations may use their detailed knowledge about the layout element's content to provide size hints.
Reimplemented in QCPLayoutGrid, QCPPlottableLegendItem, QCPTextElement, and QCPPolarLegendItem.

17.650.4.7 outerRect()

```
QRect QCPLayoutElement::outerRect ( ) const  [inline]
```

Returns the outer rect of this layout element. The outer rect is the inner rect expanded by the margins (setMargins, setAutoMargins). The outer rect is used (and set via setOuterRect) by the parent QCPLayout to control the size of this layout element.

See also

rect

17.650.4.8 parentPlotInitialized()

```
void QCPLayoutElement::parentPlotInitialized (
            QCustomPlot * parentPlot )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.650.4.9 rect()

```
QRect QCPLayoutElement::rect ( ) const  [inline]
```

Returns the inner rect of this layout element. The inner rect is the outer rect (outerRect, setOuterRect) shrinked by the margins (setMargins, setAutoMargins).

In some cases, the area between outer and inner rect is left blank. In other cases the margin area is used to display peripheral graphics while the main content is in the inner rect. This is where automatic margin calculation becomes interesting because it allows the layout element to adapt the margins to the peripheral graphics it wants to draw. For example, QCPAxisRect draws the axis labels and tick labels in the margin area, thus needs to adjust the margins (if setAutoMargins is enabled) according to the space required by the labels of the axes.

See also

outerRect

17.650.4.10 selectTest()

```
double QCPLayoutElement::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

Layout elements are sensitive to events inside their outer rect. If *pos* is within the outer rect, this method returns a value corresponding to 0.99 times the parent plot's selection tolerance. However, layout elements are not selectable by default. So if *onlySelectable* is true, -1.0 is returned.

See QCPLayerable::selectTest for a general explanation of this virtual method.

QCPLayoutElement subclasses may reimplement this method to provide more specific selection test behaviour.

Reimplemented from QCPLayerable.

Reimplemented in QCPPolarAxisAngular, QCPLayoutInset, QCPAbstractLegendItem, QCPLegend, and QCPTextElement.

17.650.4.11 setAutoMargins()

```
void QCPLayoutElement::setAutoMargins (
            QCP::MarginSides sides )
```

Sets on which sides the margin shall be calculated automatically. If a side is calculated automatically, a minimum margin value may be provided with setMinimumMargins. If a side is set to be controlled manually, the value may be specified with setMargins.

Margin sides that are under automatic control may participate in a QCPMarginGroup (see setMarginGroup), to synchronize (align) it with other layout elements in the plot.

See also

setMinimumMargins, setMargins, QCP::MarginSide

17.650.4.12 setMarginGroup()

```
void QCPLayoutElement::setMarginGroup (
            QCP::MarginSides sides,
            QCPMarginGroup * group )
```

Sets the margin *group* of the specified margin *sides*.

Margin groups allow synchronizing specified margins across layout elements, see the documentation of QCPMarginGroup.

To unset the margin group of *sides*, set *group* to `nullptr`.

Note that margin groups only work for margin sides that are set to automatic (setAutoMargins).

See also

QCP::MarginSide

17.650.4.13 setMargins()

```
void QCPLayoutElement::setMargins (
            const QMargins & margins )
```

Sets the margins of this layout element. If setAutoMargins is disabled for some or all sides, this function is used to manually set the margin on those sides. Sides that are still set to be handled automatically are ignored and may have any value in *margins*.
The margin is the distance between the outer rect (controlled by the parent layout via setOuterRect) and the inner rect (which usually contains the main content of this layout element).

See also

setAutoMargins

17.650.4.14 setMaximumSize() [1/2]

```
void QCPLayoutElement::setMaximumSize (
            const QSize & size )
```

Sets the maximum size of this layout element. A parent layout tries to respect the *size* here by changing row/column sizes in the layout accordingly.
Whether this constraint applies to the inner or the outer rect can be specified with setSizeConstraintRect (see rect and outerRect).

17.650.4.15 setMaximumSize() [2/2]

```
void QCPLayoutElement::setMaximumSize (
            int width,
            int height )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the maximum size of this layout element.
Whether this constraint applies to the inner or the outer rect can be specified with setSizeConstraintRect (see rect and outerRect).

17.650.4.16 setMinimumMargins()

```
void QCPLayoutElement::setMinimumMargins (
            const QMargins & margins )
```

If setAutoMargins is enabled on some or all margins, this function is used to provide minimum values for those margins.
The minimum values are not enforced on margin sides that were set to be under manual control via setAutoMargins.

See also

setAutoMargins

17.650.4.17 setMinimumSize() [1/2]

```
void QCPLayoutElement::setMinimumSize (
            const QSize & size )
```

Sets the minimum size of this layout element. A parent layout tries to respect the *size* here by changing row/column sizes in the layout accordingly.
If the parent layout size is not sufficient to satisfy all minimum size constraints of its child layout elements, the layout may set a size that is actually smaller than *size*. QCustomPlot propagates the layout's size constraints to the outside by setting its own minimum QWidget size accordingly, so violations of *size* should be exceptions.
Whether this constraint applies to the inner or the outer rect can be specified with setSizeConstraintRect (see rect and outerRect).

17.650.4.18 setMinimumSize() [2/2]

```
void QCPLayoutElement::setMinimumSize (
            int width,
            int height )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the minimum size of this layout element.
Whether this constraint applies to the inner or the outer rect can be specified with setSizeConstraintRect (see rect and outerRect).

17.650.4.19 setOuterRect()

```
void QCPLayoutElement::setOuterRect (
            const QRect & rect )
```

Sets the outer rect of this layout element. If the layout element is inside a layout, the layout sets the position and size of this layout element using this function.
Calling this function externally has no effect, since the layout will overwrite any changes to the outer rect upon the next replot.
The layout element will adapt its inner rect by applying the margins inward to the outer rect.

See also

rect

17.650.4.20 setSizeConstraintRect()

```
void QCPLayoutElement::setSizeConstraintRect (
            SizeConstraintRect constraintRect )
```

Sets to which rect of a layout element the size constraints apply. Size constraints can be set via setMinimumSize and setMaximumSize.
The outer rect (outerRect) includes the margins (e.g. in the case of a QCPAxisRect the axis labels), whereas the inner rect (rect) does not.

See also

setMinimumSize, setMaximumSize

17.650.4.21 update()

```
void QCPLayoutElement::update (
            UpdatePhase phase ) [virtual]
```

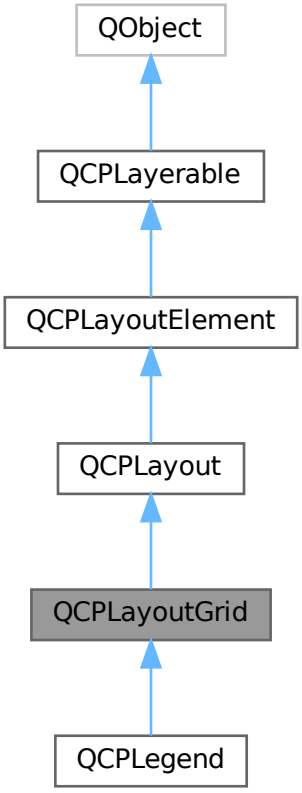
Updates the layout element and sub-elements. This function is automatically called before every replot by the parent layout element. It is called multiple times, once for every UpdatePhase. The phases are run through in the order of the enum values. For details about what happens at the different phases, see the documentation of UpdatePhase.
Layout elements that have child elements should call the update method of their child elements, and pass the current *phase* unchanged.
The default implementation executes the automatic margin mechanism in the upMargins phase. Subclasses should make sure to call the base class implementation.
Reimplemented in QCPLayout, QCPAxisRect, QCPColorScaleAxisRectPrivate, QCPColorScale, and QCPPolarAxisAngular.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

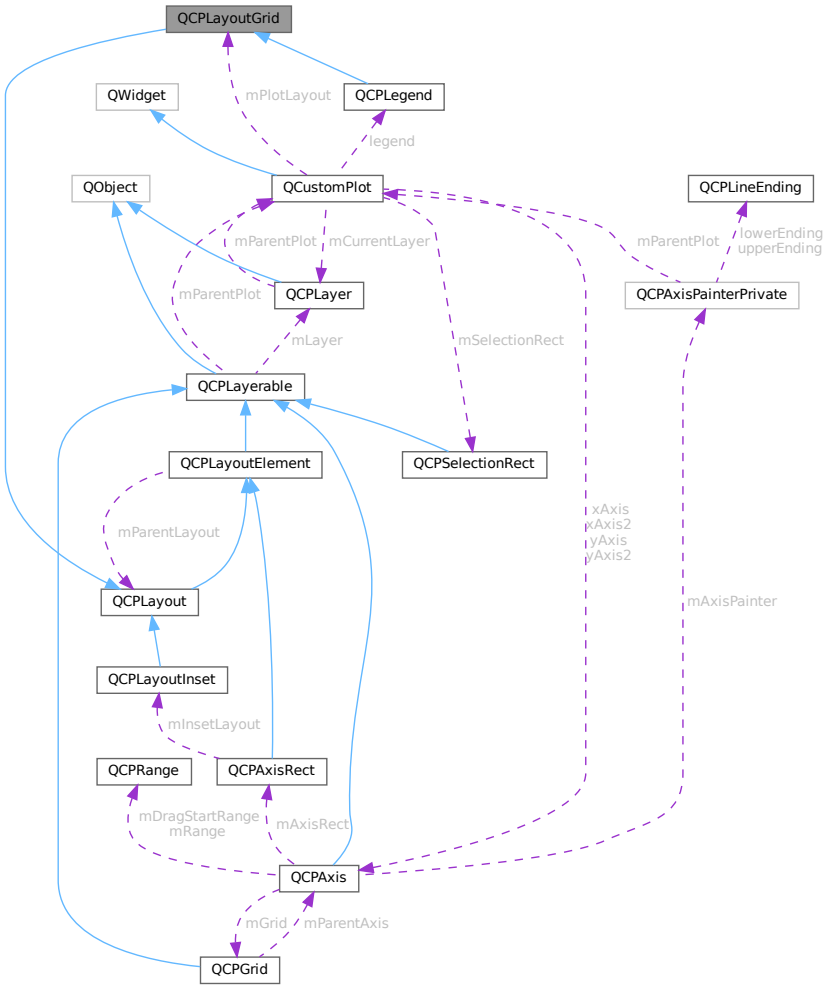
17.651 QCPLayoutGrid Class Reference

A layout that arranges child elements in a grid.

Inheritance diagram for QCPLayoutGrid:



Collaboration diagram for QCPLayoutGrid:



Public Types

- enum FillOrder { foRowsFirst , foColumnsFirst }

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

Public Member Functions

- QCPLayoutGrid ()
- int rowCount () const
- int columnCount () const
- QList< double > **columnStretchFactors** () const
- QList< double > **rowStretchFactors** () const
- int **columnSpacing** () const
- int **rowSpacing** () const
- int **wrap** () const
- FillOrder **fillOrder** () const

- void setColumnStretchFactor (int column, double factor)
- void setColumnStretchFactors (const QList< double > &factors)
- void setRowStretchFactor (int row, double factor)
- void setRowStretchFactors (const QList< double > &factors)
- void setColumnSpacing (int pixels)
- void setRowSpacing (int pixels)
- void setWrap (int count)
- void setFillOrder (FillOrder order, bool rearrange=true)
- virtual void updateLayout () Q_DECL_OVERRIDE
- virtual int elementCount () const Q_DECL_OVERRIDE
- virtual QCPLayoutElement * elementAt (int index) const Q_DECL_OVERRIDE
- virtual QCPLayoutElement * takeAt (int index) Q_DECL_OVERRIDE
- virtual bool take (QCPLayoutElement *element) Q_DECL_OVERRIDE
- virtual QList< QCPLayoutElement * > elements (bool recursive) const Q_DECL_OVERRIDE
- virtual void simplify () Q_DECL_OVERRIDE
- virtual QSize minimumOuterSizeHint () const Q_DECL_OVERRIDE
- virtual QSize maximumOuterSizeHint () const Q_DECL_OVERRIDE
- QCPLayoutElement * element (int row, int column) const
- bool addElement (int row, int column, QCPLayoutElement *element)
- bool addElement (QCPLayoutElement *element)
- bool hasElement (int row, int column)
- void expandTo (int newRowCount, int newColumnCount)
- void insertRow (int newIndex)
- void insertColumn (int newIndex)
- int rowColToIndex (int row, int column) const
- void indexToRowCol (int index, int &row, int &column) const

Public Member Functions inherited from QCPLayout

- QCPLayout ()
- virtual void update (UpdatePhase phase) Q_DECL_OVERRIDE
- bool removeAt (int index)
- bool remove (QCPLayoutElement *element)
- void clear ()

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot *parentPlot=nullptr)
- QCPLayout * layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const
- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)
- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)

- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- void **getMinimumRowColSizes** (QVector< int > *minColWidths, QVector< int > *minRowHeights) const
- void **getMaximumRowColSizes** (QVector< int > *maxColWidths, QVector< int > *maxRowHeights) const

Protected Member Functions inherited from QCPLayout

- void sizeConstraintsChanged () const
- void **adoptElement** (QCPLayoutElement *el)
- void **releaseElement** (QCPLayoutElement *el)
- QVector< int > **getSectionSizes** (QVector< int > maxSizes, QVector< int > minSizes, QVector< double > stretchFactors, int totalSize) const

Protected Member Functions inherited from QCPLayoutElement

- virtual int **calculateAutoMargin** (QCP::MarginSide side)
- virtual void **layoutChanged** ()
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void **selectEvent** (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed)
- virtual void **deselectEvent** (bool *selectionStateChanged)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QList< QList< QCPLayoutElement * > > **mElements**
- QList< double > **mColumnStretchFactors**
- QList< double > **mRowStretchFactors**
- int **mColumnSpacing**
- int **mRowSpacing**
- int **mWrap**
- FillOrder **mFillOrder**

Protected Attributes inherited from QCPLayoutElement

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Static Protected Member Functions inherited from QCPLayout

- static QSize **getFinalMinimumOuterSize** (const QCPLayoutElement *el)
- static QSize **getFinalMaximumOuterSize** (const QCPLayoutElement *el)

17.651.1 Detailed Description

A layout that arranges child elements in a grid.
Elements are laid out in a grid with configurable stretch factors (setColumnStretchFactor, setRowStretchFactor) and spacing (setColumnSpacing, setRowSpacing).
Elements can be added to cells via addElement. The grid is expanded if the specified row or column doesn't exist yet. Whether a cell contains a valid layout element can be checked with hasElement, that element can be retrieved with element. If rows and columns that only have empty cells shall be removed, call simplify. Removal of elements is either done by just adding the element to a different layout or by using the QCPLayout interface take or remove.
If you use addElement(QCPLayoutElement*) without explicit parameters for *row* and *column*, the grid layout will choose the position according to the current setFillOrder and the wrapping (setWrap).
Row and column insertion can be performed with insertRow and insertColumn.

17.651.2 Member Enumeration Documentation

17.651.2.1 FillOrder

enum QCPLayoutGrid::FillOrder

Defines in which direction the grid is filled when using addElement(QCPLayoutElement*). The column/row at which wrapping into the next row/column occurs can be specified with setWrap.

See also

setFillOrder

Enumerator

foRowsFirst	Rows are filled first, and a new element is wrapped to the next column if the row count would exceed setWrap.
foColumnsFirst	Columns are filled first, and a new element is wrapped to the next row if the column count would exceed setWrap.

17.651.3 Constructor & Destructor Documentation

17.651.3.1 QCPLayoutGrid()

QCPLayoutGrid::QCPLayoutGrid ( ) [explicit]

Creates an instance of QCPLayoutGrid and sets default values.

17.651.4 Member Function Documentation

17.651.4.1 addElement() [1/2]

```
bool QCPLayoutGrid::addElement (
            int row,
            int column,
            QCPLayoutElement * element )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.

Adds the *element* to cell with *row* and *column*. If *element* is already in a layout, it is first removed from there. If *row* or *column* don't exist yet, the layout is expanded accordingly.

Returns true if the element was added successfully, i.e. if the cell at *row* and *column* didn't already have an element.

Use the overload of this method without explicit row/column index to place the element according to the configured fill order and wrapping settings.

See also

element, hasElement, take, remove

17.651.4.2 addElement() [2/2]

```
bool QCPLayoutGrid::addElement (
            QCPLayoutElement * element )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.

Adds the *element* to the next empty cell according to the current fill order (setFillOrder) and wrapping (setWrap). If *element* is already in a layout, it is first removed from there. If necessary, the layout is expanded to hold the new element.

Returns true if the element was added successfully.

See also

setFillOrder, setWrap, element, hasElement, take, remove

17.651.4.3 columnCount()

```
int QCPLayoutGrid::columnCount ( ) const  [inline]
```
Returns the number of columns in the layout.

See also

rowCount

17.651.4.4 element()

```
QCPLayoutElement * QCPLayoutGrid::element (
            int row,
            int column ) const
```
Returns the element in the cell in *row* and *column*.
Returns `nullptr` if either the row/column is invalid or if the cell is empty. In those cases, a qDebug message is printed. To check whether a cell exists and isn't empty, use hasElement.

See also

addElement, hasElement

17.651.4.5 elementAt()

```
QCPLayoutElement * QCPLayoutGrid::elementAt (
            int index ) const  [virtual]
```
\seebaseclassmethod
Note that the association of the linear *index* to the row/column based cells depends on the current setting of setFillOrder.

See also

rowColToIndex

Implements QCPLayout.

17.651.4.6 elementCount()

```
virtual int QCPLayoutGrid::elementCount ( ) const  [inline], [virtual]
```
Returns the number of elements/cells in the layout.

See also

elements, elementAt

Implements QCPLayout.

17.651.4.7 elements()

```
QList< QCPLayoutElement * > QCPLayoutGrid::elements (
            bool recursive ) const  [virtual]
```
Returns a list of all child elements in this layout element. If *recursive* is true, all sub-child elements are included in the list, too.

Warning

There may be `nullptr` entries in the returned list. For example, QCPLayoutGrid may have empty cells which yield `nullptr` at the respective index.

Reimplemented from QCPLayout.

17.651.4.8 expandTo()

```
void QCPLayoutGrid::expandTo (
            int newRowCount,
            int newColumnCount )
```

Expands the layout to have *newRowCount* rows and *newColumnCount* columns. So the last valid row index will be *newRowCount-1*, the last valid column index will be *newColumnCount-1*.
If the current column/row count is already larger or equal to *newColumnCount/newRowCount,* this function does nothing in that dimension.
Newly created cells are empty, new rows and columns have the stretch factor 1.
Note that upon a call to addElement, the layout is expanded automatically to contain the specified row and column, using this function.

See also

simplify

17.651.4.9 hasElement()

```
bool QCPLayoutGrid::hasElement (
            int row,
            int column )
```

Returns whether the cell at *row* and *column* exists and contains a valid element, i.e. isn't empty.

See also

element

17.651.4.10 indexToRowCol()

```
void QCPLayoutGrid::indexToRowCol (
            int index,
            int & row,
            int & column ) const
```

Converts the linear index to row and column indices and writes the result to *row* and *column*.
The way the cells are indexed depends on setFillOrder. If it is foRowsFirst, the indices increase left to right and then top to bottom. If it is foColumnsFirst, the indices increase top to bottom and then left to right.
If there are no cells (i.e. column or row count is zero), sets *row* and *column* to -1.
For the retrieved *row* and *column* to be valid, the passed *index* must be valid itself, i.e. greater or equal to zero and smaller than the current elementCount.

See also

rowColToIndex

17.651.4.11 insertColumn()

```
void QCPLayoutGrid::insertColumn (
            int newIndex )
```

Inserts a new column with empty cells at the column index *newIndex*. Valid values for *newIndex* range from 0 (inserts a column at the left) to *columnCount* (appends a column at the right).

See also

insertRow

17.651.4.12 insertRow()

```
void QCPLayoutGrid::insertRow (
            int newIndex )
```

Inserts a new row with empty cells at the row index *newIndex*. Valid values for *newIndex* range from 0 (inserts a row at the top) to *rowCount* (appends a row at the bottom).

See also

insertColumn

17.651.4.13 maximumOuterSizeHint()

```
QSize QCPLayoutGrid::maximumOuterSizeHint ( ) const  [virtual]
```

Returns the suggested maximum size this layout element (the outerRect) may be expanded to, if no manual maximum size is set.
if a maximum size (setMaximumSize) was not set manually, parent layouts use the returned size (usually indirectly through QCPLayout::getFinalMaximumOuterSize) to determine the maximum allowed size of this layout element.
A manual maximum size is considered set if it is smaller than Qt's QWIDGETSIZE_MAX.
The default implementation simply returns QWIDGETSIZE_MAX for both width and height, implying no suggested maximum size. Reimplementations may use their detailed knowledge about the layout element's content to provide size hints.
Reimplemented from QCPLayoutElement.

17.651.4.14 minimumOuterSizeHint()

```
QSize QCPLayoutGrid::minimumOuterSizeHint ( ) const  [virtual]
```

Returns the suggested minimum size this layout element (the outerRect) may be compressed to, if no manual minimum size is set.
if a minimum size (setMinimumSize) was not set manually, parent layouts use the returned size (usually indirectly through QCPLayout::getFinalMinimumOuterSize) to determine the minimum allowed size of this layout element.
A manual minimum size is considered set if it is non-zero.
The default implementation simply returns the sum of the horizontal margins for the width and the sum of the vertical margins for the height. Reimplementations may use their detailed knowledge about the layout element's content to provide size hints.
Reimplemented from QCPLayoutElement.

17.651.4.15 rowColToIndex()

```
int QCPLayoutGrid::rowColToIndex (
            int row,
            int column ) const
```

Converts the given *row* and *column* to the linear index used by some methods of QCPLayoutGrid and QCPLayout.
The way the cells are indexed depends on setFillOrder. If it is foRowsFirst, the indices increase left to right and then top to bottom. If it is foColumnsFirst, the indices increase top to bottom and then left to right.
For the returned index to be valid, *row* and *column* must be valid indices themselves, i.e. greater or equal to zero and smaller than the current rowCount/ columnCount.

See also

indexToRowCol

17.651.4.16 rowCount()

```
int QCPLayoutGrid::rowCount ( ) const  [inline]
```

Returns the number of rows in the layout.

See also

columnCount

17.651.4.17 setColumnSpacing()

```
void QCPLayoutGrid::setColumnSpacing (
            int pixels )
```

Sets the gap that is left blank between columns to *pixels*.

See also

setRowSpacing

17.651.4.18 setColumnStretchFactor()

```
void QCPLayoutGrid::setColumnStretchFactor (
            int column,
            double factor )
```

Sets the stretch *factor* of *column*.
Stretch factors control the relative sizes of rows and columns. Cells will not be resized beyond their minimum and maximum widths/heights, regardless of the stretch factor. (see QCPLayoutElement::setMinimumSize, QCPLayoutElement::setMaximumSize, QCPLayoutElement::setSizeConstraintRect.)
The default stretch factor of newly created rows/columns is 1.

See also

setColumnStretchFactors, setRowStretchFactor

17.651.4.19 setColumnStretchFactors()

```
void QCPLayoutGrid::setColumnStretchFactors (
            const QList< double > & factors )
```

Sets the stretch *factors* of all columns. *factors* must have the size columnCount.
Stretch factors control the relative sizes of rows and columns. Cells will not be resized beyond their minimum and maximum widths/heights, regardless of the stretch factor. (see QCPLayoutElement::setMinimumSize, QCPLayoutElement::setMaximumSize, QCPLayoutElement::setSizeConstraintRect.)
The default stretch factor of newly created rows/columns is 1.

See also

setColumnStretchFactor, setRowStretchFactors

17.651.4.20 setFillOrder()

```
void QCPLayoutGrid::setFillOrder (
            FillOrder order,
            bool rearrange = true )
```

Sets the filling order and wrapping behaviour that is used when adding new elements with the method addElement(QCPLayoutElement*).
The specified *order* defines whether rows or columns are filled first. Using setWrap, you can control at which row/column count wrapping into the next column/row will occur. If you set it to zero, no wrapping will ever occur. Changing the fill order also changes the meaning of the linear index used e.g. in elementAt and takeAt. The default fill order for QCPLayoutGrid is foColumnsFirst.
If you want to have all current elements arranged in the new order, set *rearrange* to true. The elements will be rearranged in a way that tries to preserve their linear index. However, empty cells are skipped during build-up of the new cell order, which shifts the succeeding element's index. The rearranging is performed even if the specified *order* is already the current fill order. Thus this method can be used to re-wrap the current elements.
If *rearrange* is false, the current element arrangement is not changed, which means the linear indexes change (because the linear index is dependent on the fill order).
Note that the method addElement(int row, int column, QCPLayoutElement *element) with explicitly stated row and column is not subject to wrapping and can place elements even beyond the specified wrapping point.

See also

setWrap, addElement(QCPLayoutElement*)

17.651.4.21 setRowSpacing()

```
void QCPLayoutGrid::setRowSpacing (
            int pixels )
```

Sets the gap that is left blank between rows to *pixels*.

See also

setColumnSpacing

17.651.4.22 setRowStretchFactor()

```
void QCPLayoutGrid::setRowStretchFactor (
            int row,
            double factor )
```

Sets the stretch *factor* of *row*.
Stretch factors control the relative sizes of rows and columns. Cells will not be resized beyond their minimum and maximum widths/heights, regardless of the stretch factor. (see QCPLayoutElement::setMinimumSize, QCPLayoutElement::setMaximumSize, QCPLayoutElement::setSizeConstraintRect.)
The default stretch factor of newly created rows/columns is 1.

See also

setColumnStretchFactors, setRowStretchFactor

17.651.4.23 setRowStretchFactors()

```
void QCPLayoutGrid::setRowStretchFactors (
            const QList< double > & factors )
```

Sets the stretch *factors* of all rows. *factors* must have the size rowCount.
Stretch factors control the relative sizes of rows and columns. Cells will not be resized beyond their minimum and maximum widths/heights, regardless of the stretch factor. (see QCPLayoutElement::setMinimumSize, QCPLayoutElement::setMaximumSize, QCPLayoutElement::setSizeConstraintRect.)
The default stretch factor of newly created rows/columns is 1.

See also

setRowStretchFactor, setColumnStretchFactors

17.651.4.24 setWrap()

```
void QCPLayoutGrid::setWrap (
            int count )
```

Sets the maximum number of columns or rows that are used, before new elements added with addElement(QCPLayoutElement*) will start to fill the next row or column, respectively. It depends on setFillOrder, whether rows or columns are wrapped.
If *count* is set to zero, no wrapping will ever occur.
If you wish to re-wrap the elements currently in the layout, call setFillOrder with *rearrange* set to true (the actual fill order doesn't need to be changed for the rearranging to be done).
Note that the method addElement(int row, int column, QCPLayoutElement *element) with explicitly stated row and column is not subject to wrapping and can place elements even beyond the specified wrapping point.

See also

setFillOrder

17.651.4.25 simplify()

```
void QCPLayoutGrid::simplify ( )  [virtual]
```

Simplifies the layout by collapsing rows and columns which only contain empty cells.
Reimplemented from QCPLayout.

17.651.4.26 take()

```
bool QCPLayoutGrid::take (
            QCPLayoutElement * element )  [virtual]
```

Removes the specified *element* from the layout and returns true on success.
If the *element* isn't in this layout, returns false.
Note that some layouts don't remove the respective cell right away but leave an empty cell after successful removal of the layout element. To collapse empty cells, use simplify.

See also

takeAt

Implements QCPLayout.

17.651.4.27 takeAt()

```
QCPLayoutElement * QCPLayoutGrid::takeAt (
            int index )  [virtual]
```

\seebaseclassmethod
Note that the association of the linear *index* to the row/column based cells depends on the current setting of setFillOrder.

See also

rowColToIndex

Implements QCPLayout.

17.651.4.28 updateLayout()

```
void QCPLayoutGrid::updateLayout ( )  [virtual]
```

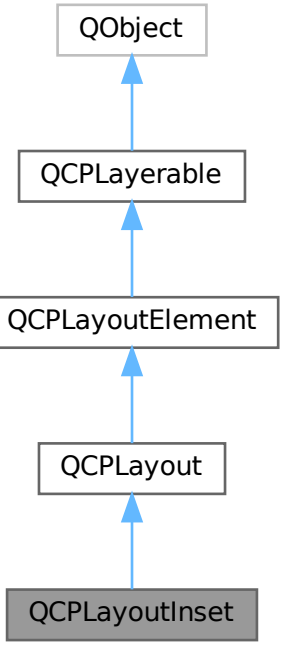
Reimplemented from QCPLayout.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

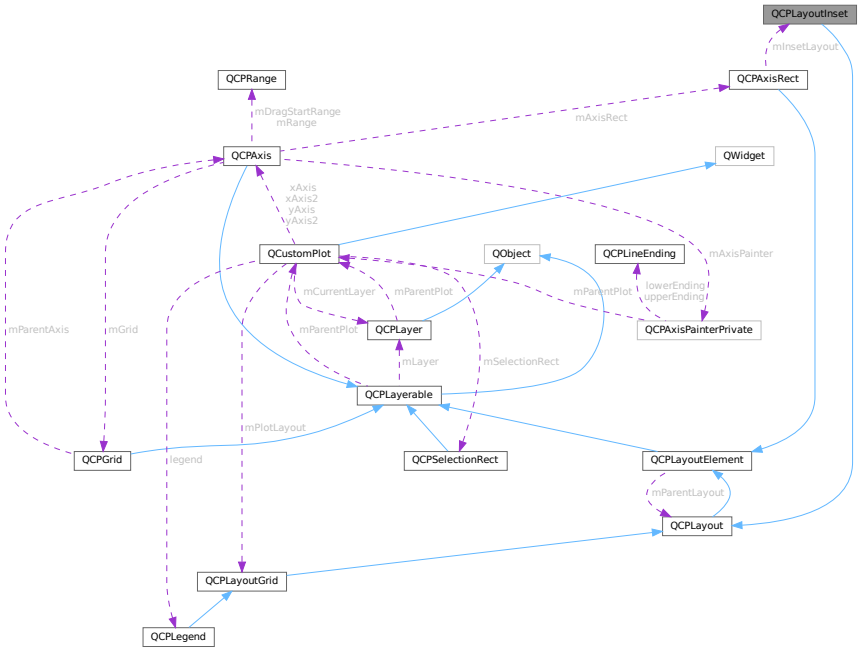
17.652 QCPLayoutInset Class Reference

A layout that places child elements aligned to the border or arbitrarily positioned.

Inheritance diagram for QCPLayoutInset:



Collaboration diagram for QCPLayoutInset:



Public Types

- enum InsetPlacement { ipFree , ipBorderAligned }

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

Public Member Functions

- QCPLayoutInset ()
- InsetPlacement insetPlacement (int index) const
- Qt::Alignment insetAlignment (int index) const
- QRectF insetRect (int index) const
- void setInsetPlacement (int index, InsetPlacement placement)
- void setInsetAlignment (int index, Qt::Alignment alignment)
- void setInsetRect (int index, const QRectF &rect)
- virtual void updateLayout () Q_DECL_OVERRIDE
- virtual int elementCount () const Q_DECL_OVERRIDE
- virtual QCPLayoutElement * elementAt (int index) const Q_DECL_OVERRIDE
- virtual QCPLayoutElement * takeAt (int index) Q_DECL_OVERRIDE
- virtual bool take (QCPLayoutElement *element) Q_DECL_OVERRIDE
- virtual void simplify () Q_DECL_OVERRIDE
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE
- void addElement (QCPLayoutElement *element, Qt::Alignment alignment)
- void addElement (QCPLayoutElement *element, const QRectF &rect)

Public Member Functions inherited from QCPLayout

- QCPLayout ()
- virtual void update (UpdatePhase phase) Q_DECL_OVERRIDE
- virtual QList< QCPLayoutElement * > elements (bool recursive) const Q_DECL_OVERRIDE
- bool removeAt (int index)
- bool remove (QCPLayoutElement *element)
- void clear ()

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot *parentPlot=nullptr)
- QCPLayout * layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const
- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)

- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual QSize minimumOuterSizeHint () const
- virtual QSize maximumOuterSizeHint () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Attributes

- QList< QCPLayoutElement * > **mElements**
- QList< InsetPlacement > **mInsetPlacement**
- QList< Qt::Alignment > **mInsetAlignment**
- QList< QRectF > **mInsetRect**

Protected Attributes inherited from QCPLayoutElement

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Protected Member Functions inherited from QCPLayout

- void sizeConstraintsChanged () const
- void **adoptElement** (QCPLayoutElement *el)
- void **releaseElement** (QCPLayoutElement *el)
- QVector< int > **getSectionSizes** (QVector< int > maxSizes, QVector< int > minSizes, QVector< double > stretchFactors, int totalSize) const

Protected Member Functions inherited from QCPLayoutElement

- virtual int **calculateAutoMargin** (QCP::MarginSide side)
- virtual void **layoutChanged** ()
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void **selectEvent** (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged)
- virtual void **deselectEvent** (bool *selectionStateChanged)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Static Protected Member Functions inherited from QCPLayout

- static QSize **getFinalMinimumOuterSize** (const QCPLayoutElement *el)
- static QSize **getFinalMaximumOuterSize** (const QCPLayoutElement *el)

17.652.1 Detailed Description

A layout that places child elements aligned to the border or arbitrarily positioned.
Elements are placed either aligned to the border or at arbitrary position in the area of the layout. Which placement applies is controlled with the InsetPlacement (setInsetPlacement).
Elements are added via addElement(QCPLayoutElement *element, Qt::Alignment alignment) or addElement(QCPLayoutElement *ele
If the first method is used, the inset placement will default to ipBorderAligned and the element will be aligned according to the *alignment* parameter. The second method defaults to ipFree and allows placing elements at arbitrary position and size, defined by *rect*.
The alignment or rect can be set via setInsetAlignment or setInsetRect, respectively.
This is the layout that every QCPAxisRect has as QCPAxisRect::insetLayout.

17.652.2 Member Enumeration Documentation

17.652.2.1 InsetPlacement

`enum QCPLayoutInset::InsetPlacement`
Defines how the placement and sizing is handled for a certain element in a QCPLayoutInset.

Enumerator

ipFree	The element may be positioned/sized arbitrarily, see setInsetRect.
ipBorderAligned	The element is aligned to one of the layout sides, see setInsetAlignment.

17.652.3 Constructor & Destructor Documentation

17.652.3.1 QCPLayoutInset()

```
QCPLayoutInset::QCPLayoutInset ( )  [explicit]
```

Creates an instance of QCPLayoutInset and sets default values.

17.652.4 Member Function Documentation

17.652.4.1 addElement() [1/2]

```
void QCPLayoutInset::addElement (
            QCPLayoutElement * element,
            const QRectF & rect )
```

Adds the specified *element* to the layout as an inset with free positioning/sizing (setInsetAlignment is initialized with ipFree). The position and size is set to *rect*.
rect is given in fractions of the whole inset layout rect. So an inset with rect (0, 0, 1, 1) will span the entire layout. An inset with rect (0.6, 0.1, 0.35, 0.35) will be in the top right corner of the layout, with 35% width and height of the parent layout.

See also

addElement(QCPLayoutElement *element, Qt::Alignment alignment)

17.652.4.2 addElement() [2/2]

```
void QCPLayoutInset::addElement (
            QCPLayoutElement * element,
            Qt::Alignment alignment )
```

Adds the specified *element* to the layout as an inset aligned at the border (setInsetAlignment is initialized with ipBorderAligned). The alignment is set to *alignment*.
alignment is an or combination of the following alignment flags: Qt::AlignLeft, Qt::AlignHCenter, Qt::AlighRight, Qt::AlignTop, Qt::AlignVCenter, Qt::AlignBottom. Any other alignment flags will be ignored.

See also

addElement(QCPLayoutElement *element, const QRectF &rect)

17.652.4.3 elementAt()

```
QCPLayoutElement * QCPLayoutInset::elementAt (
            int index ) const  [virtual]
```

Returns the element in the cell with the given *index*. If *index* is invalid, returns `nullptr`.
Note that even if *index* is valid, the respective cell may be empty in some layouts (e.g. QCPLayoutGrid), so this function may return `nullptr` in those cases. You may use this function to check whether a cell is empty or not.

See also

elements, elementCount, takeAt

Implements QCPLayout.

17.652.4.4 elementCount()

```
int QCPLayoutInset::elementCount ( ) const  [virtual]
```
Returns the number of elements/cells in the layout.

See also

elements, elementAt

Implements QCPLayout.

17.652.4.5 insetAlignment()

```
Qt::Alignment QCPLayoutInset::insetAlignment (
            int index ) const
```
Returns the alignment of the element with the specified *index*. The alignment only has a meaning, if the inset placement (setInsetPlacement) is ipBorderAligned.

17.652.4.6 insetPlacement()

```
QCPLayoutInset::InsetPlacement QCPLayoutInset::insetPlacement (
            int index ) const
```
Returns the placement type of the element with the specified *index*.

17.652.4.7 insetRect()

```
QRectF QCPLayoutInset::insetRect (
            int index ) const
```
Returns the rect of the element with the specified *index*. The rect only has a meaning, if the inset placement (setInsetPlacement) is ipFree.

17.652.4.8 selectTest()

```
double QCPLayoutInset::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```
The inset layout is sensitive to events only at areas where its (visible) child elements are sensitive. If the selectTest method of any of the child elements returns a positive number for *pos*, this method returns a value corresponding to 0.99 times the parent plot's selection tolerance. The inset layout is not selectable itself by default. So if *onlySelectable* is true, -1.0 is returned.
See QCPLayerable::selectTest for a general explanation of this virtual method.
Reimplemented from QCPLayoutElement.

17.652.4.9 setInsetAlignment()

```
void QCPLayoutInset::setInsetAlignment (
            int index,
            Qt::Alignment alignment )
```
If the inset placement (setInsetPlacement) is ipBorderAligned, this function is used to set the alignment of the element with the specified *index* to *alignment*.
alignment is an or combination of the following alignment flags: Qt::AlignLeft, Qt::AlignHCenter, Qt::AlighRight, Qt::AlignTop, Qt::AlignVCenter, Qt::AlignBottom. Any other alignment flags will be ignored.

17.652.4.10 setInsetPlacement()

```
void QCPLayoutInset::setInsetPlacement (
            int index,
            QCPLayoutInset::InsetPlacement placement )
```
Sets the inset placement type of the element with the specified *index* to *placement*.

See also

InsetPlacement

17.652.4.11 setInsetRect()

```
void QCPLayoutInset::setInsetRect (
            int index,
            const QRectF & rect )
```

If the inset placement (setInsetPlacement) is ipFree, this function is used to set the position and size of the element with the specified *index* to *rect*.
rect is given in fractions of the whole inset layout rect. So an inset with rect (0, 0, 1, 1) will span the entire layout. An inset with rect (0.6, 0.1, 0.35, 0.35) will be in the top right corner of the layout, with 35% width and height of the parent layout.
Note that the minimum and maximum sizes of the embedded element (QCPLayoutElement::setMinimumSize, QCPLayoutElement::setMaximumSize) are enforced.

17.652.4.12 simplify()

```
void QCPLayoutInset::simplify ( )  [inline], [virtual]
```

The QCPInsetLayout does not need simplification since it can never have empty cells due to its linear index structure. This method does nothing.
Reimplemented from QCPLayout.

17.652.4.13 take()

```
bool QCPLayoutInset::take (
            QCPLayoutElement * element )  [virtual]
```

Removes the specified *element* from the layout and returns true on success.
If the *element* isn't in this layout, returns false.
Note that some layouts don't remove the respective cell right away but leave an empty cell after successful removal of the layout element. To collapse empty cells, use simplify.

See also

takeAt

Implements QCPLayout.

17.652.4.14 takeAt()

```
QCPLayoutElement * QCPLayoutInset::takeAt (
            int index )  [virtual]
```

Removes the element with the given *index* from the layout and returns it.
If the *index* is invalid or the cell with that index is empty, returns `nullptr`.
Note that some layouts don't remove the respective cell right away but leave an empty cell after successful removal of the layout element. To collapse empty cells, use simplify.

See also

elementAt, take

Implements QCPLayout.

17.652.4.15 updateLayout()

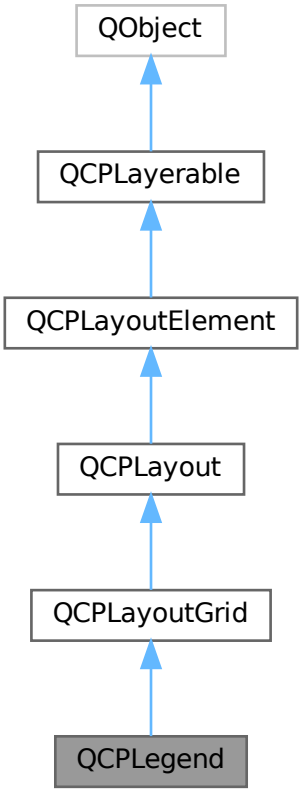
```
void QCPLayoutInset::updateLayout ( )  [virtual]
```

Reimplemented from QCPLayout.
The documentation for this class was generated from the following files:

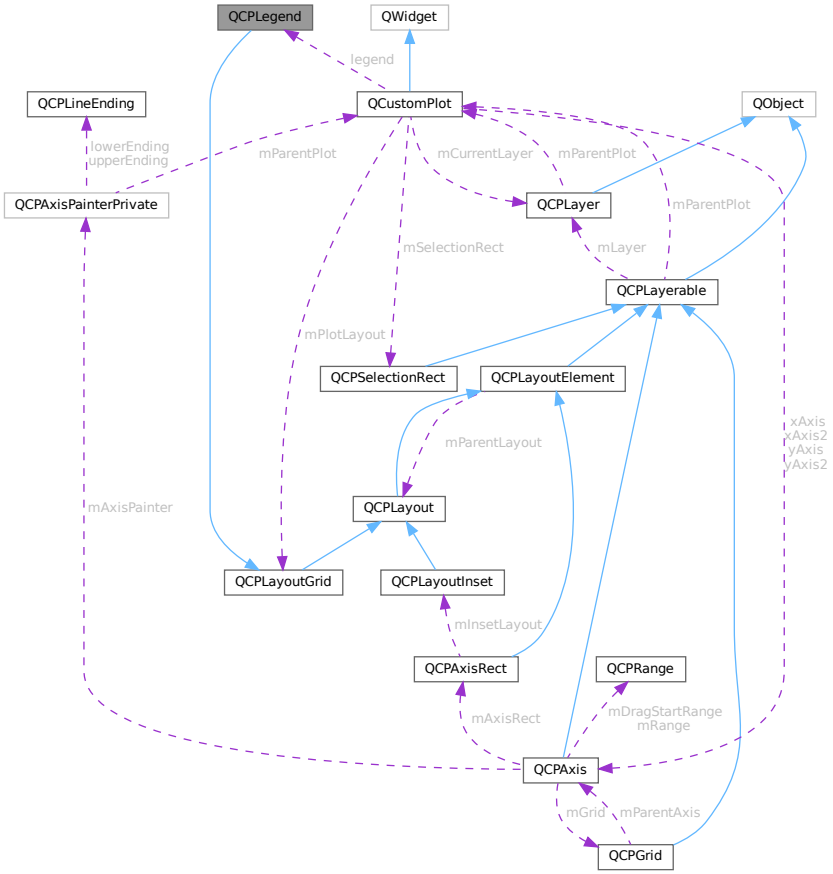
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.653 QCPLegend Class Reference

Manages a legend inside a QCustomPlot.
Inheritance diagram for QCPLegend:



Collaboration diagram for QCPLegend:



Public Types

- enum SelectablePart { spNone = 0x000 , spLegendBox = 0x001 , spItems = 0x002 }

Public Types inherited from QCPLayoutGrid

- enum FillOrder { foRowsFirst , foColumnsFirst }

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

Signals

- void selectionChanged (QCPLegend::SelectableParts parts)
- void **selectableChanged** (QCPLegend::SelectableParts parts)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPLegend ()
- QPen **borderPen** () const
- QBrush **brush** () const
- QFont **font** () const
- QColor **textColor** () const
- QSize **iconSize** () const
- int **iconTextPadding** () const
- QPen **iconBorderPen** () const
- SelectableParts **selectableParts** () const
- SelectableParts **selectedParts** () const
- QPen **selectedBorderPen** () const
- QPen **selectedIconBorderPen** () const
- QBrush **selectedBrush** () const
- QFont **selectedFont** () const
- QColor **selectedTextColor** () const
- void setBorderPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setFont (const QFont &font)
- void setTextColor (const QColor &color)
- void setIconSize (const QSize &size)
- void setIconSize (int width, int height)
- void setIconTextPadding (int padding)
- void setIconBorderPen (const QPen &pen)
- Q_SLOT void setSelectableParts (const SelectableParts &selectableParts)
- Q_SLOT void setSelectedParts (const SelectableParts &selectedParts)
- void setSelectedBorderPen (const QPen &pen)
- void setSelectedIconBorderPen (const QPen &pen)
- void setSelectedBrush (const QBrush &brush)
- void setSelectedFont (const QFont &font)
- void setSelectedTextColor (const QColor &color)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE
- QCPAbstractLegendItem * item (int index) const
- QCPPlottableLegendItem * itemWithPlottable (const QCPAbstractPlottable *plottable) const
- int itemCount () const
- bool hasItem (QCPAbstractLegendItem *item) const
- bool hasItemWithPlottable (const QCPAbstractPlottable *plottable) const
- bool addItem (QCPAbstractLegendItem *item)
- bool removeItem (int index)
- bool removeItem (QCPAbstractLegendItem *item)
- void clearItems ()
- QList< QCPAbstractLegendItem * > selectedItems () const

Public Member Functions inherited from QCPLayoutGrid

- QCPLayoutGrid ()
- int rowCount () const
- int columnCount () const
- QList< double > **columnStretchFactors** () const
- QList< double > **rowStretchFactors** () const
- int **columnSpacing** () const
- int **rowSpacing** () const
- int **wrap** () const

- FillOrder **fillOrder** () const
- void setColumnStretchFactor (int column, double factor)
- void setColumnStretchFactors (const QList< double > &factors)
- void setRowStretchFactor (int row, double factor)
- void setRowStretchFactors (const QList< double > &factors)
- void setColumnSpacing (int pixels)
- void setRowSpacing (int pixels)
- void setWrap (int count)
- void setFillOrder (FillOrder order, bool rearrange=true)
- virtual void updateLayout () Q_DECL_OVERRIDE
- virtual int elementCount () const Q_DECL_OVERRIDE
- virtual QCPLayoutElement * elementAt (int index) const Q_DECL_OVERRIDE
- virtual QCPLayoutElement * takeAt (int index) Q_DECL_OVERRIDE
- virtual bool take (QCPLayoutElement *element) Q_DECL_OVERRIDE
- virtual QList< QCPLayoutElement * > elements (bool recursive) const Q_DECL_OVERRIDE
- virtual void simplify () Q_DECL_OVERRIDE
- virtual QSize minimumOuterSizeHint () const Q_DECL_OVERRIDE
- virtual QSize maximumOuterSizeHint () const Q_DECL_OVERRIDE
- QCPLayoutElement * element (int row, int column) const
- bool addElement (int row, int column, QCPLayoutElement *element)
- bool addElement (QCPLayoutElement *element)
- bool hasElement (int row, int column)
- void expandTo (int newRowCount, int newColumnCount)
- void insertRow (int newIndex)
- void insertColumn (int newIndex)
- int rowColToIndex (int row, int column) const
- void indexToRowCol (int index, int &row, int &column) const

Public Member Functions inherited from QCPLayout

- QCPLayout ()
- virtual void update (UpdatePhase phase) Q_DECL_OVERRIDE
- bool removeAt (int index)
- bool remove (QCPLayoutElement *element)
- void clear ()

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot *parentPlot=nullptr)
- QCPLayout * layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const
- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)
- void setMinimumSize (const QSize &size)

- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- QPen **getBorderPen** () const
- QBrush **getBrush** () const

Protected Member Functions inherited from QCPLayoutGrid

- void **getMinimumRowColSizes** (QVector< int > *minColWidths, QVector< int > *minRowHeights) const
- void **getMaximumRowColSizes** (QVector< int > *maxColWidths, QVector< int > *maxRowHeights) const

Protected Member Functions inherited from QCPLayout

- void sizeConstraintsChanged () const
- void **adoptElement** (QCPLayoutElement *el)
- void **releaseElement** (QCPLayoutElement *el)
- QVector< int > **getSectionSizes** (QVector< int > maxSizes, QVector< int > minSizes, QVector< double > stretchFactors, int totalSize) const

Protected Member Functions inherited from QCPLayoutElement

- virtual int **calculateAutoMargin** (QCP::MarginSide side)
- virtual void **layoutChanged** ()

Protected Member Functions inherited from QCPLayerable

- virtual QRect **clipRect** () const
- virtual void mousePressEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent ∗event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent ∗event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent ∗event)
- void **initializeParentPlot** (QCustomPlot ∗parentPlot)
- void **setParentLayerable** (QCPLayerable ∗parentLayerable)
- bool **moveToLayer** (QCPLayer ∗layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter ∗painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QPen **mBorderPen**
- QPen **mIconBorderPen**
- QBrush **mBrush**
- QFont **mFont**
- QColor **mTextColor**
- QSize **mIconSize**
- int **mIconTextPadding**
- SelectableParts **mSelectedParts**
- SelectableParts **mSelectableParts**
- QPen **mSelectedBorderPen**
- QPen **mSelectedIconBorderPen**
- QBrush **mSelectedBrush**
- QFont **mSelectedFont**
- QColor **mSelectedTextColor**

Protected Attributes inherited from QCPLayoutGrid

- QList< QList< QCPLayoutElement ∗ > > **mElements**
- QList< double > **mColumnStretchFactors**
- QList< double > **mRowStretchFactors**
- int **mColumnSpacing**
- int **mRowSpacing**
- int **mWrap**
- FillOrder **mFillOrder**

Protected Attributes inherited from QCPLayoutElement

- QCPLayout ∗ **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup ∗ > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPAbstractLegendItem**

Additional Inherited Members

Static Protected Member Functions inherited from QCPLayout

- static QSize **getFinalMinimumOuterSize** (const QCPLayoutElement *el)
- static QSize **getFinalMaximumOuterSize** (const QCPLayoutElement *el)

17.653.1 Detailed Description

Manages a legend inside a QCustomPlot.
A legend is a small box somewhere in the plot which lists plottables with their name and icon.
A legend is populated with legend items by calling QCPAbstractPlottable::addToLegend on the plottable, for which a legend item shall be created. In the case of the main legend (QCustomPlot::legend), simply adding plottables to the plot while QCustomPlot::setAutoAddPlottableToLegend is set to true (the default) creates corresponding legend items. The legend item associated with a certain plottable can be removed with QCPAbstractPlottable::removeFromLegend. However, QCPLegend also offers an interface to add and manipulate legend items directly: item, itemWithPlottable, itemCount, addItem, removeItem, etc.
Since QCPLegend derives from QCPLayoutGrid, it can be placed in any position a QCPLayoutElement may be positioned. The legend items are themselves QCPLayoutElements which are placed in the grid layout of the legend. QCPLegend only adds an interface specialized for handling child elements of type QCPAbstractLegendItem, as mentioned above. In principle, any other layout elements may also be added to a legend via the normal QCPLayoutGrid interface. See the special page about The Layout System for examples on how to add other elements to the legend and move it outside the axis rect.
Use the methods setFillOrder and setWrap inherited from QCPLayoutGrid to control in which order (column first or row first) the legend is filled up when calling addItem, and at which column or row wrapping occurs. The default fill order for legends is foRowsFirst.
By default, every QCustomPlot has one legend (QCustomPlot::legend) which is placed in the inset layout of the main axis rect (QCPAxisRect::insetLayout). To move the legend to another position inside the axis rect, use the methods of the QCPLayoutInset. To move the legend outside of the axis rect, place it anywhere else with the QCPLayout/ QCPLayoutElement interface.

17.653.2 Member Enumeration Documentation

17.653.2.1 SelectablePart

`enum QCPLegend::SelectablePart`
Defines the selectable parts of a legend

See also

setSelectedParts, setSelectableParts

Enumerator

spNone	`0x000` None
spLegendBox	`0x001` The legend box (frame)
spItems	`0x002` Legend items individually (see selectedItems)

17.653.3 Constructor & Destructor Documentation

17.653.3.1 QCPLegend()

```
QCPLegend::QCPLegend ( )  [explicit]
```

Constructs a new QCPLegend instance with default values.
Note that by default, QCustomPlot already contains a legend ready to be used as QCustomPlot::legend

17.653.4 Member Function Documentation

17.653.4.1 addItem()

```
bool QCPLegend::addItem (
            QCPAbstractLegendItem * item )
```

Adds *item* to the legend, if it's not present already. The element is arranged according to the current fill order (setFillOrder) and wrapping (setWrap).
Returns true on sucess, i.e. if the item wasn't in the list already and has been successfuly added.
The legend takes ownership of the item.

See also

removeItem, item, hasItem

17.653.4.2 applyDefaultAntialiasingHint()

```
void QCPLegend::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```

Reimplemented from QCPLayoutElement.

17.653.4.3 clearItems()

```
void QCPLegend::clearItems ( )
```

Removes all items from the legend.

17.653.4.4 deselectEvent()

```
void QCPLegend::deselectEvent (
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.653.4.5 draw()

```
void QCPLegend::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Reimplemented from QCPLayoutElement.

17.653.4.6 hasItem()

```
bool QCPLegend::hasItem (
            QCPAbstractLegendItem * item ) const
```

Returns whether the legend contains *item*.

See also

hasItemWithPlottable

17.653.4.7 hasItemWithPlottable()

```
bool QCPLegend::hasItemWithPlottable (
            const QCPAbstractPlottable * plottable ) const
```

Returns whether the legend contains a QCPPlottableLegendItem which is associated with *plottable* (e.g. a QCPGraph*). If such an item isn't in the legend, returns false.

See also

itemWithPlottable

17.653.4.8 item()

```
QCPAbstractLegendItem * QCPLegend::item (
            int index ) const
```

Returns the item with index *i*. If non-legend items were added to the legend, and the element at the specified cell index is not a QCPAbstractLegendItem, returns `nullptr`.
Note that the linear index depends on the current fill order (setFillOrder).

See also

itemCount, addItem, itemWithPlottable

17.653.4.9 itemCount()

```
int QCPLegend::itemCount ( ) const
```

Returns the number of items currently in the legend. It is identical to the base class QCPLayoutGrid::elementCount(), and unlike the other "item" interface methods of QCPLegend, doesn't only address elements which can be cast to QCPAbstractLegendItem.
Note that if empty cells are in the legend (e.g. by calling methods of the QCPLayoutGrid base class which allows creating empty cells), they are included in the returned count.

See also

item

17.653.4.10 itemWithPlottable()

```
QCPPlottableLegendItem * QCPLegend::itemWithPlottable (
            const QCPAbstractPlottable * plottable ) const
```

Returns the QCPPlottableLegendItem which is associated with *plottable* (e.g. a QCPGraph*). If such an item isn't in the legend, returns `nullptr`.

See also

hasItemWithPlottable

17.653.4.11 parentPlotInitialized()

```
void QCPLegend::parentPlotInitialized (
            QCustomPlot * parentPlot )  [protected], [virtual]
```

Reimplemented from QCPLayoutElement.

17.653.4.12 removeItem() [1/2]

```
bool QCPLegend::removeItem (
            int index )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Removes the item with the specified *index* from the legend and deletes it.
After successful removal, the legend is reordered according to the current fill order (setFillOrder) and wrapping (setWrap), so no empty cell remains where the removed *item* was. If you don't want this, rather use the raw element interface of QCPLayoutGrid.
Returns true, if successful. Unlike QCPLayoutGrid::removeAt, this method only removes elements derived from QCPAbstractLegendItem.

See also

itemCount, clearItems

17.653.4.13 removeItem() [2/2]

```
bool QCPLegend::removeItem (
            QCPAbstractLegendItem * item )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Removes *item* from the legend and deletes it.
After successful removal, the legend is reordered according to the current fill order (setFillOrder) and wrapping (setWrap), so no empty cell remains where the removed *item* was. If you don't want this, rather use the raw element interface of QCPLayoutGrid.
Returns true, if successful.

See also

clearItems

17.653.4.14 selectedItems()

```
QList< QCPAbstractLegendItem * > QCPLegend::selectedItems ( ) const
```

Returns the legend items that are currently selected. If no items are selected, the list is empty.

See also

QCPAbstractLegendItem::setSelected, setSelectable

17.653.4.15 selectEvent()

```
void QCPLegend::selectEvent (
            QMouseEvent * event,
            bool additive,
            const QVariant & details,
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.653.4.16 selectionCategory()

```
QCP::Interaction QCPLegend::selectionCategory ( ) const  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.653.4.17 selectionChanged

```
void QCPLegend::selectionChanged (
            QCPLegend::SelectableParts selection )  [signal]
```

This signal is emitted when the selection state of this legend has changed.

See also

setSelectedParts, setSelectableParts

17.653.4.18 selectTest()

```
double QCPLegend::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

Layout elements are sensitive to events inside their outer rect. If *pos* is within the outer rect, this method returns a value corresponding to 0.99 times the parent plot's selection tolerance. However, layout elements are not selectable by default. So if *onlySelectable* is true, -1.0 is returned.
See QCPLayerable::selectTest for a general explanation of this virtual method.
QCPLayoutElement subclasses may reimplement this method to provide more specific selection test behaviour.
Reimplemented from QCPLayoutElement.

17.653.4.19 setBorderPen()

```
void QCPLegend::setBorderPen (
            const QPen & pen )
```

Sets the pen, the border of the entire legend is drawn with.

17.653.4.20 setBrush()

```
void QCPLegend::setBrush (
            const QBrush & brush )
```

Sets the brush of the legend background.

17.653.4.21 setFont()

```
void QCPLegend::setFont (
            const QFont & font )
```

Sets the default font of legend text. Legend items that draw text (e.g. the name of a graph) will use this font by default. However, a different font can be specified on a per-item-basis by accessing the specific legend item.
This function will also set *font* on all already existing legend items.

See also

QCPAbstractLegendItem::setFont

17.653.4.22 setIconBorderPen()

```
void QCPLegend::setIconBorderPen (
            const QPen & pen )
```

Sets the pen used to draw a border around each legend icon. Legend items that draw an icon (e.g. a visual representation of the graph) will use this pen by default.
If no border is wanted, set this to *Qt::NoPen*.

17.653.4.23 setIconSize() [1/2]

```
void QCPLegend::setIconSize (
            const QSize & size )
```

Sets the size of legend icons. Legend items that draw an icon (e.g. a visual representation of the graph) will use this size by default.

17.653.4.24 setIconSize() [2/2]

```
void QCPLegend::setIconSize (
            int width,
            int height )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.

17.653.4.25 setIconTextPadding()

```
void QCPLegend::setIconTextPadding (
            int padding )
```

Sets the horizontal space in pixels between the legend icon and the text next to it. Legend items that draw an icon (e.g. a visual representation of the graph) and text (e.g. the name of the graph) will use this space by default.

17.653.4.26 setSelectableParts()

```
void QCPLegend::setSelectableParts (
            const SelectableParts & selectable )
```

Sets whether the user can (de-)select the parts in *selectable* by clicking on the QCustomPlot surface. (When QCustomPlot::setInteractions contains QCP::iSelectLegend.)

However, even when *selectable* is set to a value not allowing the selection of a specific part, it is still possible to set the selection of this part manually, by calling setSelectedParts directly.

See also

SelectablePart, setSelectedParts

17.653.4.27 setSelectedBorderPen()

```
void QCPLegend::setSelectedBorderPen (
            const QPen & pen )
```

When the legend box is selected, this pen is used to draw the border instead of the normal pen set via setBorderPen.

See also

setSelectedParts, setSelectableParts, setSelectedBrush

17.653.4.28 setSelectedBrush()

```
void QCPLegend::setSelectedBrush (
            const QBrush & brush )
```

When the legend box is selected, this brush is used to draw the legend background instead of the normal brush set via setBrush.

See also

setSelectedParts, setSelectableParts, setSelectedBorderPen

17.653.4.29 setSelectedFont()

```
void QCPLegend::setSelectedFont (
            const QFont & font )
```

Sets the default font that is used by legend items when they are selected.
This function will also set *font* on all already existing legend items.

See also

setFont, QCPAbstractLegendItem::setSelectedFont

17.653.4.30 setSelectedIconBorderPen()

```
void QCPLegend::setSelectedIconBorderPen (
            const QPen & pen )
```

Sets the pen legend items will use to draw their icon borders, when they are selected.

See also

setSelectedParts, setSelectableParts, setSelectedFont

17.653.4.31 setSelectedParts()

```
void QCPLegend::setSelectedParts (
            const SelectableParts & selected )
```

Sets the selected state of the respective legend parts described by SelectablePart. When a part is selected, it uses a different pen/font and brush. If some legend items are selected and *selected* doesn't contain spItems, those items become deselected.
The entire selection mechanism is handled automatically when QCustomPlot::setInteractions contains iSelect↩ Legend. You only need to call this function when you wish to change the selection state manually.
This function can change the selection state of a part even when setSelectableParts was set to a value that actually excludes the part.
emits the selectionChanged signal when *selected* is different from the previous selection state.
Note that it doesn't make sense to set the selected state spItems here when it wasn't set before, because there's no way to specify which exact items to newly select. Do this by calling QCPAbstractLegendItem::setSelected directly on the legend item you wish to select.

See also

SelectablePart, setSelectableParts, selectTest, setSelectedBorderPen, setSelectedIconBorderPen, setSelectedBrush, setSelectedFont

17.653.4.32 setSelectedTextColor()

```
void QCPLegend::setSelectedTextColor (
            const QColor & color )
```

Sets the default text color that is used by legend items when they are selected.
This function will also set *color* on all already existing legend items.

See also

setTextColor, QCPAbstractLegendItem::setSelectedTextColor

17.653.4.33 setTextColor()

```
void QCPLegend::setTextColor (
            const QColor & color )
```

Sets the default color of legend text. Legend items that draw text (e.g. the name of a graph) will use this color by default. However, a different colors can be specified on a per-item-basis by accessing the specific legend item.
This function will also set *color* on all already existing legend items.

See also

QCPAbstractLegendItem::setTextColor

The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.654 QCPLineEnding Class Reference

Handles the different ending decorations for line-like items.

Public Types

- enum EndingStyle {
  esNone , esFlatArrow , esSpikeArrow , esLineArrow ,
  esDisc , esSquare , esDiamond , esBar ,
  esHalfBar , esSkewedBar }

Public Member Functions

- QCPLineEnding ()
- QCPLineEnding (EndingStyle style, double width=8, double length=10, bool inverted=false)
- EndingStyle **style** () const
- double **width** () const
- double **length** () const
- bool **inverted** () const
- void setStyle (EndingStyle style)
- void setWidth (double width)
- void setLength (double length)
- void setInverted (bool inverted)
- double **boundingDistance** () const
- double realLength () const
- void **draw** (QCPPainter *painter, const QCPVector2D &pos, const QCPVector2D &dir) const
- void **draw** (QCPPainter *painter, const QCPVector2D &pos, double angle) const

Protected Attributes

- EndingStyle **mStyle**
- double **mWidth**
- double **mLength**
- bool **mInverted**

17.654.1 Detailed Description

Handles the different ending decorations for line-like items.
For every ending a line-like item has, an instance of this class exists. For example, QCPItemLine has two endings which can be set with QCPItemLine::setHead and QCPItemLine::setTail.
The styles themselves are defined via the enum QCPLineEnding::EndingStyle. Most decorations can be modified regarding width and length, see setWidth and setLength. The direction of the ending decoration (e.g. direction an arrow is pointing) is controlled by the line-like item. For example, when both endings of a QCPItemLine are set to be arrows, they will point to opposite directions, e.g. "outward". This can be changed by setInverted, which would make the respective arrow point inward.
Note that due to the overloaded QCPLineEnding constructor, you may directly specify a QCPLineEnding::EndingStyle where actually a QCPLineEnding is expected, e.g.

17.654.2 Member Enumeration Documentation

17.654.2.1 EndingStyle

`enum QCPLineEnding::EndingStyle`
Defines the type of ending decoration for line-like items, e.g. an arrow.
The width and length of these decorations can be controlled with the functions setWidth and setLength. Some decorations like esDisc, esSquare, esDiamond and esBar only support a width, the length property is ignored.

See also

QCPItemLine::setHead, QCPItemLine::setTail, QCPItemCurve::setHead, QCPItemCurve::setTail, QCPAxis::setLowerEnding, QCPAxis::setUpperEnding

Enumerator

esNone	No ending decoration.
esFlatArrow	A filled arrow head with a straight/flat back (a triangle)
esSpikeArrow	A filled arrow head with an indented back.
esLineArrow	A non-filled arrow head with open back.
esDisc	A filled circle.
esSquare	A filled square.
esDiamond	A filled diamond (45 degrees rotated square)
esBar	A bar perpendicular to the line.
esHalfBar	A bar perpendicular to the line, pointing out to only one side (to which side can be changed with setInverted)
esSkewedBar	A bar that is skewed (skew controllable via setLength)

17.654.3 Constructor & Destructor Documentation

17.654.3.1 QCPLineEnding() [1/2]

`QCPLineEnding::QCPLineEnding ( )`
Creates a QCPLineEnding instance with default values (style esNone).

17.654.3.2 QCPLineEnding() [2/2]

```
QCPLineEnding::QCPLineEnding (
            QCPLineEnding::EndingStyle style,
            double width = 8,
            double length = 10,
            bool inverted = false )
```

Creates a QCPLineEnding instance with the specified values.

17.654.4 Member Function Documentation

17.654.4.1 realLength()

```
double QCPLineEnding::realLength ( ) const
```

Starting from the origin of this line ending (which is style specific), returns the length covered by the line ending symbol, in backward direction.
For example, the esSpikeArrow has a shorter real length than a esFlatArrow, even if both have the same setLength value, because the spike arrow has an inward curved back, which reduces the length along its center axis (the drawing origin for arrows is at the tip).
This function is used for precise, style specific placement of line endings, for example in QCPAxes.

17.654.4.2 setInverted()

```
void QCPLineEnding::setInverted (
            bool inverted )
```

Sets whether the ending decoration shall be inverted. For example, an arrow decoration will point inward when *inverted* is set to true.
Note that also the *width* direction is inverted. For symmetrical ending styles like arrows or discs, this doesn't make a difference. However, asymmetric styles like esHalfBar are affected by it, which can be used to control to which side the half bar points to.

17.654.4.3 setLength()

```
void QCPLineEnding::setLength (
            double length )
```

Sets the length of the ending decoration, if the style supports it. On arrows, for example, the length defines the size in pointing direction.

See also

setWidth

17.654.4.4 setStyle()

```
void QCPLineEnding::setStyle (
            QCPLineEnding::EndingStyle style )
```

Sets the style of the ending decoration.

17.654.4.5 setWidth()

```
void QCPLineEnding::setWidth (
            double width )
```

Sets the width of the ending decoration, if the style supports it. On arrows, for example, the width defines the size perpendicular to the arrow's pointing direction.

See also

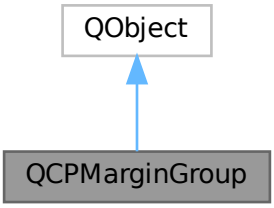
setLength

The documentation for this class was generated from the following files:

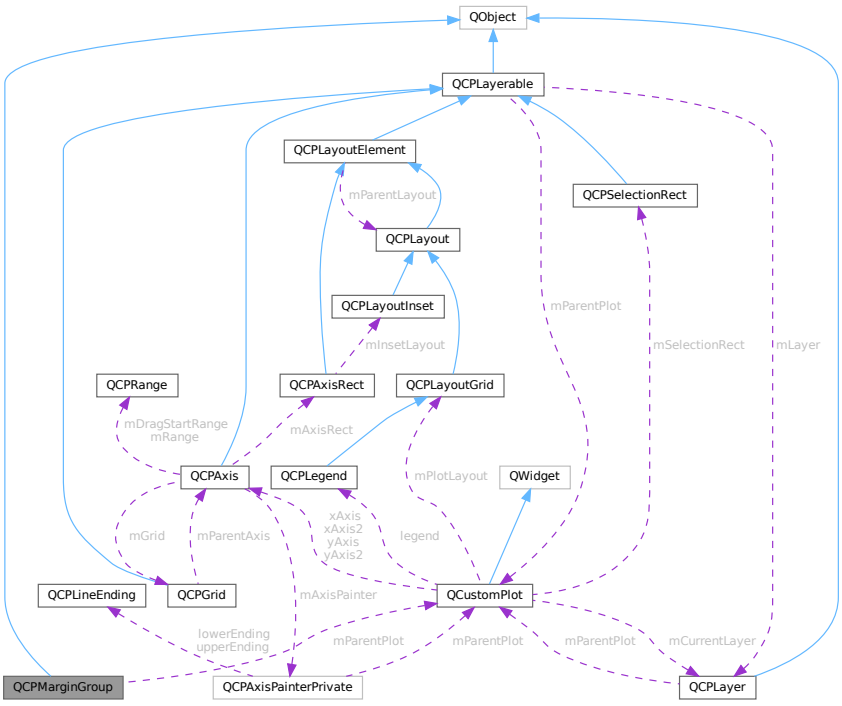
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.655 QCPMarginGroup Class Reference

A margin group allows synchronization of margin sides if working with multiple layout elements.
Inheritance diagram for QCPMarginGroup:



Collaboration diagram for QCPMarginGroup:



Public Member Functions

- QCPMarginGroup (QCustomPlot ∗parentPlot)
- QList< QCPLayoutElement ∗ > elements (QCP::MarginSide side) const
- bool isEmpty () const
- void clear ()

Protected Member Functions

- virtual int **commonMargin** (QCP::MarginSide side) const

- void **addChild** (QCP::MarginSide side, QCPLayoutElement *element)
- void **removeChild** (QCP::MarginSide side, QCPLayoutElement *element)

Protected Attributes

- QCustomPlot * **mParentPlot**
- QHash< QCP::MarginSide, QList< QCPLayoutElement * > > **mChildren**

Friends

- class **QCPLayoutElement**

17.655.1 Detailed Description

A margin group allows synchronization of margin sides if working with multiple layout elements.
QCPMarginGroup allows you to tie a margin side of two or more layout elements together, such that they will all have the same size, based on the largest required margin in the group.

In certain situations it is desirable that margins at specific sides are synchronized across layout elements. For example, if one QCPAxisRect is below another one in a grid layout, it will provide a cleaner look to the user if the left and right margins of the two axis rects are of the same size. The left axis of the top axis rect will then be at the same horizontal position as the left axis of the lower axis rect, making them appear aligned. The same applies for the right axes. This is what QCPMarginGroup makes possible.
To add/remove a specific side of a layout element to/from a margin group, use the QCPLayoutElement::setMarginGroup method. To completely break apart the margin group, either call clear, or just delete the margin group.

17.655.2 Example

First create a margin group:
Then set this group on the layout element sides:
Here, we've used the first two axis rects of the plot and synchronized their left margins with each other and their right margins with each other.

17.655.3 Constructor & Destructor Documentation

17.655.3.1 QCPMarginGroup()

```
QCPMarginGroup::QCPMarginGroup (
            QCustomPlot * parentPlot )  [explicit]
```

Creates a new QCPMarginGroup instance in *parentPlot*.

17.655.4 Member Function Documentation

17.655.4.1 clear()

```
void QCPMarginGroup::clear ( )
```

Clears this margin group. The synchronization of the margin sides that use this margin group is lifted and they will use their individual margin sizes again.

17.655.4.2 elements()

```
QList< QCPLayoutElement * > QCPMarginGroup::elements (
            QCP::MarginSide side ) const  [inline]
```

Returns a list of all layout elements that have their margin *side* associated with this margin group.

17.655.4.3 isEmpty()

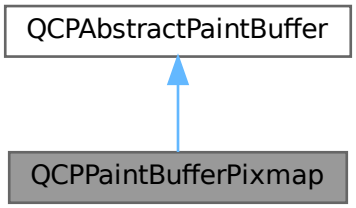
```
bool QCPMarginGroup::isEmpty ( ) const
```

Returns whether this margin group is empty. If this function returns true, no layout elements use this margin group to synchronize margin sides.
The documentation for this class was generated from the following files:

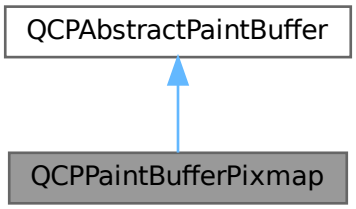
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.656 QCPPaintBufferPixmap Class Reference

A paint buffer based on QPixmap, using software raster rendering.
Inheritance diagram for QCPPaintBufferPixmap:



Collaboration diagram for QCPPaintBufferPixmap:



Public Member Functions

- QCPPaintBufferPixmap (const QSize &size, double devicePixelRatio)
- virtual QCPPainter * startPainting () Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) const Q_DECL_OVERRIDE
- void clear (const QColor &color) Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPaintBuffer

- QCPAbstractPaintBuffer (const QSize &size, double devicePixelRatio)
- QSize **size** () const

- bool **invalidated** () const
- double **devicePixelRatio** () const
- void setSize (const QSize &size)
- void setInvalidated (bool invalidated=true)
- void setDevicePixelRatio (double ratio)
- virtual void donePainting ()

Protected Member Functions

- virtual void reallocateBuffer () Q_DECL_OVERRIDE

Protected Attributes

- QPixmap **mBuffer**

Protected Attributes inherited from QCPAbstractPaintBuffer

- QSize **mSize**
- double **mDevicePixelRatio**
- bool **mInvalidated**

17.656.1 Detailed Description

A paint buffer based on QPixmap, using software raster rendering.
This paint buffer is the default and fall-back paint buffer which uses software rendering and QPixmap as internal buffer. It is used if QCustomPlot::setOpenGl is false.

17.656.2 Constructor & Destructor Documentation

17.656.2.1 QCPPaintBufferPixmap()

```
QCPPaintBufferPixmap::QCPPaintBufferPixmap (
            const QSize & size,
            double devicePixelRatio )  [explicit]
```

Creates a pixmap paint buffer instancen with the specified *size* and *devicePixelRatio*, if applicable.

17.656.3 Member Function Documentation

17.656.3.1 clear()

```
void QCPPaintBufferPixmap::clear (
            const QColor & color )  [virtual]
```

Fills the entire buffer with the provided *color*. To have an empty transparent buffer, use the named color `Qt::transparent`.
This method must not be called if there is currently a painter (acquired with startPainting) active.
Implements QCPAbstractPaintBuffer.

17.656.3.2 draw()

```
void QCPPaintBufferPixmap::draw (
            QCPPainter * painter ) const  [virtual]
```

Draws the contents of this buffer with the provided *painter*. This is the method that is used to finally join all paint buffers and draw them onto the screen.
Implements QCPAbstractPaintBuffer.

17.656.3.3 reallocateBuffer()

```
void QCPPaintBufferPixmap::reallocateBuffer ( )  [protected], [virtual]
```

Reallocates the internal buffer with the currently configured size (setSize) and device pixel ratio, if applicable (setDevicePixelRatio). It is called as soon as any of those properties are changed on this paint buffer.

Note

Subclasses of QCPAbstractPaintBuffer must call their reimplementation of this method in their constructor, to perform the first allocation (this can not be done by the base class because calling pure virtual methods in base class constructors is not possible).

Implements QCPAbstractPaintBuffer.

17.656.3.4 startPainting()

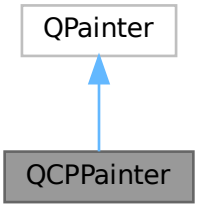
```
QCPPainter * QCPPaintBufferPixmap::startPainting ( )  [virtual]
```

Returns a QCPPainter which is ready to draw to this buffer. The ownership and thus the responsibility to delete the painter after the painting operations are complete is given to the caller of this method.
Once you are done using the painter, delete the painter and call donePainting.
While a painter generated with this method is active, you must not call setSize, setDevicePixelRatio or clear.
This method may return 0, if a painter couldn't be activated on the buffer. This usually indicates a problem with the respective painting backend.
Implements QCPAbstractPaintBuffer.
The documentation for this class was generated from the following files:

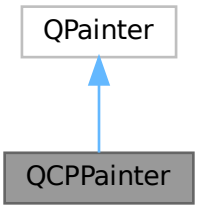
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.657 QCPPainter Class Reference

QPainter subclass used internally.
Inheritance diagram for QCPPainter:



Collaboration diagram for QCPPainter:



Public Types

- enum PainterMode { pmDefault = 0x00 , pmVectorized = 0x01 , pmNoCaching = 0x02 , pmNonCosmetic = 0x04 }

Public Member Functions

- QCPPainter ()
- QCPPainter (QPaintDevice ∗device)
- bool **antialiasing** () const
- PainterModes **modes** () const
- void setAntialiasing (bool enabled)
- void setMode (PainterMode mode, bool enabled=true)
- void setModes (PainterModes modes)
- bool begin (QPaintDevice ∗device)
- void setPen (const QPen &pen)
- void setPen (const QColor &color)
- void setPen (Qt::PenStyle penStyle)
- void drawLine (const QLineF &line)
- void **drawLine** (const QPointF &p1, const QPointF &p2)
- void save ()
- void restore ()
- void makeNonCosmetic ()

Protected Attributes

- PainterModes **mModes**
- bool **mIsAntialiasing**
- QStack< bool > **mAntialiasingStack**

17.657.1 Detailed Description

QPainter subclass used internally.
This QPainter subclass is used to provide some extended functionality e.g. for tweaking position consistency between antialiased and non-antialiased painting. Further it provides workarounds for QPainter quirks.

Warning

> This class intentionally hides non-virtual functions of QPainter, e.g. setPen, save and restore. So while it is possible to pass a QCPPainter instance to a function that expects a QPainter pointer, some of the workarounds and tweaks will be unavailable to the function (because it will call the base class implementations of the functions actually hidden by QCPPainter).

17.657.2 Member Enumeration Documentation

17.657.2.1 PainterMode

enum QCPPainter::PainterMode

Defines special modes the painter can operate in. They disable or enable certain subsets of features/fixes/workarounds, depending on whether they are wanted on the respective output device.

Enumerator

pmDefault	0x00 Default mode for painting on screen devices
pmVectorized	0x01 Mode for vectorized painting (e.g. PDF export). For example, this prevents some antialiasing fixes.
pmNoCaching	0x02 Mode for all sorts of exports (e.g. PNG, PDF,...). For example, this prevents using cached pixmap labels
pmNonCosmetic	0x04 Turns pen widths 0 to 1, i.e. disables cosmetic pens. (A cosmetic pen is always drawn with width 1 pixel in the vector image/pdf viewer, independent of zoom.)

17.657.3 Constructor & Destructor Documentation

17.657.3.1 QCPPainter() [1/2]

```
QCPPainter::QCPPainter ( )
```

Creates a new QCPPainter instance and sets default values

17.657.3.2 QCPPainter() [2/2]

```
QCPPainter::QCPPainter (
            QPaintDevice * device )  [explicit]
```

Creates a new QCPPainter instance on the specified paint *device* and sets default values. Just like the analogous QPainter constructor, begins painting on *device* immediately.
Like begin, this method sets QPainter::NonCosmeticDefaultPen in Qt versions before Qt5.

17.657.4 Member Function Documentation

17.657.4.1 begin()

```
bool QCPPainter::begin (
            QPaintDevice * device )
```

Sets the QPainter::NonCosmeticDefaultPen in Qt versions before Qt5 after beginning painting on *device*. This is necessary to get cosmetic pen consistency across Qt versions, because since Qt5, all pens are non-cosmetic by default, and in Qt4 this render hint must be set to get that behaviour.
The Constructor QCPPainter(QPaintDevice *device) which directly starts painting also sets the render hint as appropriate.

Note

this function hides the non-virtual base class implementation.

17.657.4.2 drawLine()

```
void QCPPainter::drawLine (
            const QLineF & line )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Works around a Qt bug introduced with Qt 4.8 which makes drawing QLineF unpredictable when antialiasing is disabled. Thus when antialiasing is disabled, it rounds the *line* to integer coordinates and then passes it to the original drawLine.

Note

this function hides the non-virtual base class implementation.

17.657.4.3 makeNonCosmetic()

```
void QCPPainter::makeNonCosmetic ( )
```

Changes the pen width to 1 if it currently is 0. This function is called in the setPen overrides when the pmNonCosmetic mode is set.

17.657.4.4 restore()

```
void QCPPainter::restore ( )
```

Restores the painter (see QPainter::restore). Since QCPPainter adds some new internal state to QPainter, the save/restore functions are reimplemented to also save/restore those members.

Note

this function hides the non-virtual base class implementation.

See also

save

17.657.4.5 save()

```
void QCPPainter::save ( )
```

Saves the painter (see QPainter::save). Since QCPPainter adds some new internal state to QPainter, the save/restore functions are reimplemented to also save/restore those members.

Note

this function hides the non-virtual base class implementation.

See also

restore

17.657.4.6 setAntialiasing()

```
void QCPPainter::setAntialiasing (
            bool enabled )
```

Sets whether painting uses antialiasing or not. Use this method instead of using setRenderHint with QPainter↩ ::Antialiasing directly, as it allows QCPPainter to regain pixel exactness between antialiased and non-antialiased painting (Since Qt $<$ 5.0 uses slightly different coordinate systems for AA/Non-AA painting).

17.657.4.7 setMode()

```
void QCPPainter::setMode (
            QCPPainter::PainterMode mode,
            bool enabled = true )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the mode of the painter. This controls whether the painter shall adjust its fixes/workarounds optimized for certain output devices.

17.657.4.8 setModes()

```
void QCPPainter::setModes (
            PainterModes modes )
```

Sets the mode of the painter. This controls whether the painter shall adjust its fixes/workarounds optimized for certain output devices.

17.657.4.9 setPen() [1/3]

```
void QCPPainter::setPen (
            const QColor & color )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the pen (by color) of the painter and applies certain fixes to it, depending on the mode of this QCPPainter.

Note

this function hides the non-virtual base class implementation.

17.657.4.10 setPen() [2/3]

```
void QCPPainter::setPen (
            const QPen & pen )
```

Sets the pen of the painter and applies certain fixes to it, depending on the mode of this QCPPainter.

Note

this function hides the non-virtual base class implementation.

17.657.4.11 setPen() [3/3]

```
void QCPPainter::setPen (
            Qt::PenStyle penStyle )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the pen (by style) of the painter and applies certain fixes to it, depending on the mode of this QCPPainter.

Note

this function hides the non-virtual base class implementation.

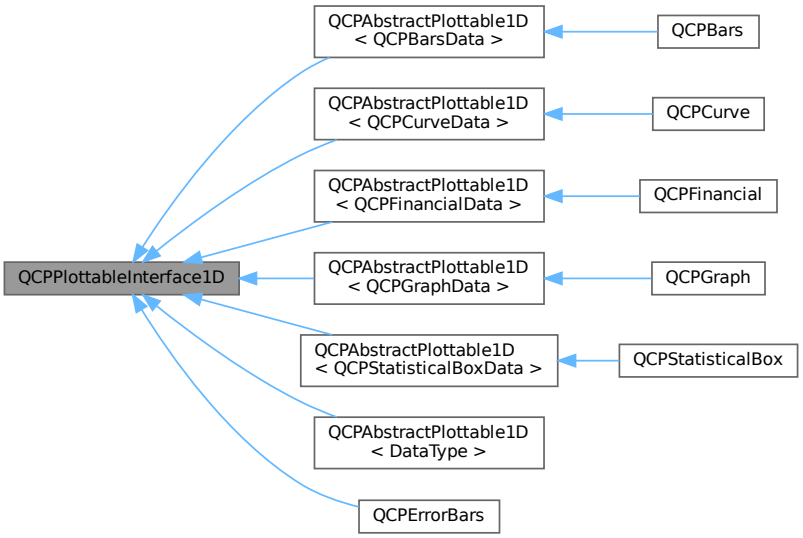
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.658 QCPPlottableInterface1D Class Reference

Defines an abstract interface for one-dimensional plottables.
`#include <applications/gui/genesys/qcustomplot.h>`

Inheritance diagram for QCPPlottableInterface1D:



Public Member Functions

- virtual int dataCount () const =0
- virtual double dataMainKey (int index) const =0
- virtual double dataSortKey (int index) const =0
- virtual double dataMainValue (int index) const =0
- virtual QCPRange dataValueRange (int index) const =0
- virtual QPointF dataPixelPosition (int index) const =0
- virtual bool sortKeyIsMainKey () const =0
- virtual QCPDataSelection selectTestRect (const QRectF &rect, bool onlySelectable) const =0
- virtual int findBegin (double sortKey, bool expandedRange=true) const =0
- virtual int findEnd (double sortKey, bool expandedRange=true) const =0

17.658.1 Detailed Description

Defines an abstract interface for one-dimensional plottables.
This class contains only pure virtual methods which define a common interface to the data of one-dimensional plottables.
For example, it is implemented by the template class QCPAbstractPlottable1D (the preferred base class for one-dimensional plottables). So if you use that template class as base class of your one-dimensional plottable, you won't have to care about implementing the 1d interface yourself.
If your plottable doesn't derive from QCPAbstractPlottable1D but still wants to provide a 1d interface (e.g. like QCPErrorBars does), you should inherit from both QCPAbstractPlottable and QCPPlottableInterface1D and accordingly reimplement the pure virtual methods of the 1d interface, matching your data container. Also, reimplement QCPAbstractPlottable::interface1D to return the `this` pointer.
If you have a QCPAbstractPlottable pointer, you can check whether it implements this interface by calling QCPAbstractPlottable::interface1D and testing it for a non-zero return value. If it indeed implements this interface, you may use it to access the plottable's data without needing to know the exact type of the plottable or its data point type.

17.658.2 Member Function Documentation

17.658.2.1 dataCount()

```
int QCPPlottableInterface1D::dataCount ( ) const  [pure virtual]
```

Returns the number of data points of the plottable.

Implemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCP QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat and QCPErrorBars.

17.658.2.2 dataMainKey()

```
double QCPPlottableInterface1D::dataMainKey (
            int index ) const  [pure virtual]
```

Returns the main key of the data point at the given *index*.

What the main key is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.

Implemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCP QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat and QCPErrorBars.

17.658.2.3 dataMainValue()

```
double QCPPlottableInterface1D::dataMainValue (
            int index ) const  [pure virtual]
```

Returns the main value of the data point at the given *index*.

What the main value is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.

Implemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCP QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat and QCPErrorBars.

17.658.2.4 dataPixelPosition()

```
QPointF QCPPlottableInterface1D::dataPixelPosition (
            int index ) const  [pure virtual]
```

Returns the pixel position on the widget surface at which the data point at the given *index* appears.

Usually this corresponds to the point of dataMainKey/ dataMainValue, in pixel coordinates. However, depending on the plottable, this might be a different apparent position than just a coord-to-pixel transform of those values. For example, QCPBars apparent data values can be shifted depending on their stacking, bar grouping or configured base value.

Implemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCP QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat QCPBars, and QCPErrorBars.

17.658.2.5 dataSortKey()

```
double QCPPlottableInterface1D::dataSortKey (
            int index ) const  [pure virtual]
```

Returns the sort key of the data point at the given *index*.

What the sort key is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.

Implemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCP QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat and QCPErrorBars.

17.658.2.6 dataValueRange()

```
QCPRange QCPPlottableInterface1D::dataValueRange (
            int index ) const  [pure virtual]
```

Returns the value range of the data point at the given *index*.
What the value range is, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.
Implemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCP QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat and QCPErrorBars.

17.658.2.7 findBegin()

```
int QCPPlottableInterface1D::findBegin (
            double sortKey,
            bool expandedRange = true ) const  [pure virtual]
```

Returns the index of the data point with a (sort-)key that is equal to, just below, or just above *sortKey*. If *expanded↩ Range* is true, the data point just below *sortKey* will be considered, otherwise the one just above.
This can be used in conjunction with findEnd to iterate over data points within a given key range, including or excluding the bounding data points that are just beyond the specified range.
If *expandedRange* is true but there are no data points below *sortKey*, 0 is returned.
If the container is empty, returns 0 (in that case, findEnd will also return 0, so a loop using these methods will not iterate over the index 0).

See also

findEnd, QCPDataContainer::findBegin

Implemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCP QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat and QCPErrorBars.

17.658.2.8 findEnd()

```
int QCPPlottableInterface1D::findEnd (
            double sortKey,
            bool expandedRange = true ) const  [pure virtual]
```

Returns the index one after the data point with a (sort-)key that is equal to, just above, or just below *sortKey*. If *expandedRange* is true, the data point just above *sortKey* will be considered, otherwise the one just below.
This can be used in conjunction with findBegin to iterate over data points within a given key range, including the bounding data points that are just below and above the specified range.
If *expandedRange* is true but there are no data points above *sortKey*, the index just above the highest data point is returned.
If the container is empty, returns 0.

See also

findBegin, QCPDataContainer::findEnd

Implemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCP QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat and QCPErrorBars.

17.658.2.9 selectTestRect()

```
QCPDataSelection QCPPlottableInterface1D::selectTestRect (
            const QRectF & rect,
            bool onlySelectable ) const  [pure virtual]
```

Returns a data selection containing all the data points of this plottable which are contained (or hit by) *rect*. This is used mainly in the selection rect interaction for data selection (data selection mechanism).
If *onlySelectable* is true, an empty QCPDataSelection is returned if this plottable is not selectable (i.e. if QCPAbstractPlottable::setSelectable is QCP::stNone).

Note

rect must be a normalized rect (positive or zero width and height). This is especially important when using the rect of QCPSelectionRect::accepted, which is not necessarily normalized. Use `QRect::normalized()` when passing a rect which might not be normalized.

Implemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCP
QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat
QCPBars, QCPStatisticalBox, QCPFinancial, and QCPErrorBars.

17.658.2.10 sortKeyIsMainKey()

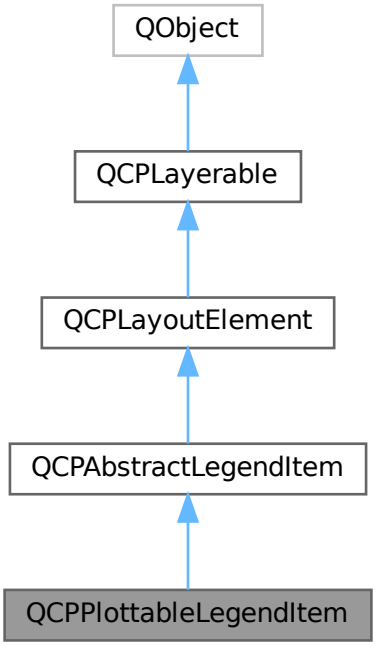
```
bool QCPPlottableInterface1D::sortKeyIsMainKey ( ) const  [pure virtual]
```

Returns whether the sort key (dataSortKey) is identical to the main key (dataMainKey).
What the sort and main keys are, is defined by the plottable's data type. See the QCPDataContainer DataType documentation for details about this naming convention.
Implemented in QCPAbstractPlottable1D< DataType >, QCPAbstractPlottable1D< QCPBarsData >, QCPAbstractPlottable1D< QCP
QCPAbstractPlottable1D< QCPFinancialData >, QCPAbstractPlottable1D< QCPGraphData >, QCPAbstractPlottable1D< QCPStat
and QCPErrorBars.
The documentation for this class was generated from the following file:

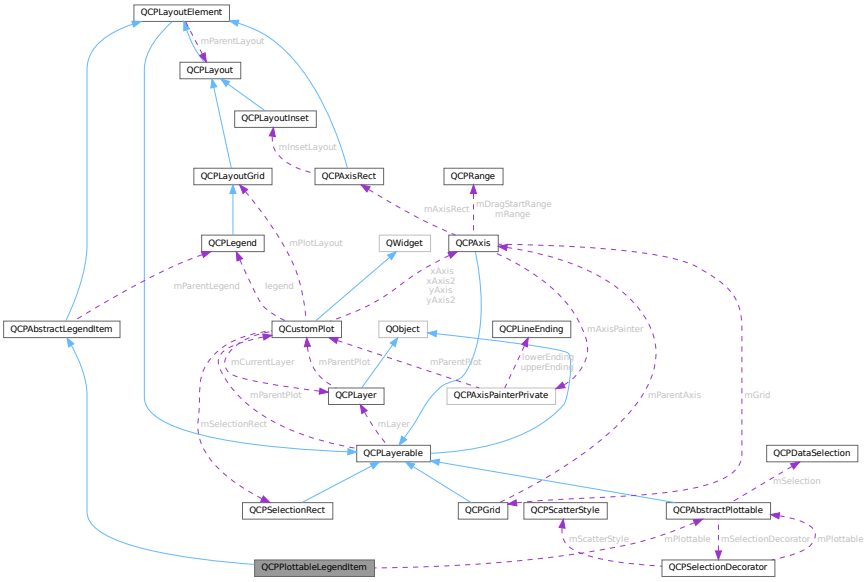
- applications/gui/genesys/qcustomplot.h

17.659 QCPPlottableLegendItem Class Reference

A legend item representing a plottable with an icon and the plottable name.
Inheritance diagram for QCPPlottableLegendItem:



Collaboration diagram for QCPPlottableLegendItem:



Public Member Functions

- QCPPlottableLegendItem (QCPLegend ∗parent, QCPAbstractPlottable ∗plottable)
- QCPAbstractPlottable ∗ **plottable** ()

Public Member Functions inherited from QCPAbstractLegendItem

- QCPAbstractLegendItem (QCPLegend ∗parent)
- QCPLegend ∗ **parentLegend** () const
- QFont **font** () const
- QColor **textColor** () const
- QFont **selectedFont** () const
- QColor **selectedTextColor** () const
- bool **selectable** () const
- bool **selected** () const
- void setFont (const QFont &font)
- void setTextColor (const QColor &color)
- void setSelectedFont (const QFont &font)
- void setSelectedTextColor (const QColor &color)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant ∗details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot ∗parentPlot=nullptr)
- QCPLayout ∗ layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const

- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)
- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual void update (UpdatePhase phase)
- virtual QSize maximumOuterSizeHint () const
- virtual QList< QCPLayoutElement * > elements (bool recursive) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual QSize minimumOuterSizeHint () const Q_DECL_OVERRIDE
- QPen **getIconBorderPen** () const
- QColor **getTextColor** () const
- QFont **getFont** () const

Protected Member Functions inherited from QCPAbstractLegendItem

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayoutElement

- virtual int **calculateAutoMargin** (QCP::MarginSide side)
- virtual void **layoutChanged** ()
- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QCPAbstractPlottable * **mPlottable**

Protected Attributes inherited from QCPAbstractLegendItem

- QCPLegend * **mParentLegend**
- QFont **mFont**
- QColor **mTextColor**
- QFont **mSelectedFont**
- QColor **mSelectedTextColor**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayoutElement

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

Signals inherited from QCPAbstractLegendItem

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.659.1 Detailed Description

A legend item representing a plottable with an icon and the plottable name.
This is the standard legend item for plottables. It displays an icon of the plottable next to the plottable name. The icon is drawn by the respective plottable itself (QCPAbstractPlottable::drawLegendIcon), and tries to give an intuitive symbol for the plottable. For example, the QCPGraph draws a centered horizontal line and/or a single scatter point in the middle.
Legend items of this type are always associated with one plottable (retrievable via the plottable() function and settable with the constructor). You may change the font of the plottable name with setFont. Icon padding and border pen is taken from the parent QCPLegend, see QCPLegend::setIconBorderPen and QCPLegend::setIconTextPadding.
The function QCPAbstractPlottable::addToLegend/ QCPAbstractPlottable::removeFromLegend creates/removes legend items of this type.
Since QCPLegend is based on QCPLayoutGrid, a legend item itself is just a subclass of QCPLayoutElement. While it could be added to a legend (or any other layout) via the normal layout interface, QCPLegend has specialized functions for handling legend items conveniently, see the documentation of QCPLegend.

17.659.2 Constructor & Destructor Documentation

17.659.2.1 QCPPlottableLegendItem()

```
QCPPlottableLegendItem::QCPPlottableLegendItem (
            QCPLegend * parent,
            QCPAbstractPlottable * plottable )
```

Creates a new legend item associated with *plottable*.
Once it's created, it can be added to the legend via QCPLegend::addItem.
A more convenient way of adding/removing a plottable to/from the legend is via the functions QCPAbstractPlottable::addToLegend and QCPAbstractPlottable::removeFromLegend.

17.659.3 Member Function Documentation

17.659.3.1 draw()

```
void QCPPlottableLegendItem::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractLegendItem.

17.659.3.2 minimumOuterSizeHint()

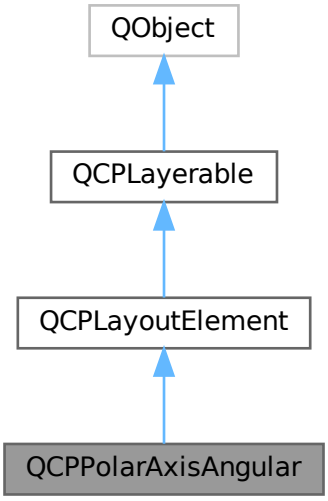
```
QSize QCPPlottableLegendItem::minimumOuterSizeHint ( ) const  [protected], [virtual]
```

Returns the suggested minimum size this layout element (the outerRect) may be compressed to, if no manual minimum size is set.
if a minimum size (setMinimumSize) was not set manually, parent layouts use the returned size (usually indirectly through QCPLayout::getFinalMinimumOuterSize) to determine the minimum allowed size of this layout element.
A manual minimum size is considered set if it is non-zero.
The default implementation simply returns the sum of the horizontal margins for the width and the sum of the vertical margins for the height. Reimplementations may use their detailed knowledge about the layout element's content to provide size hints.
Reimplemented from QCPLayoutElement.
The documentation for this class was generated from the following files:

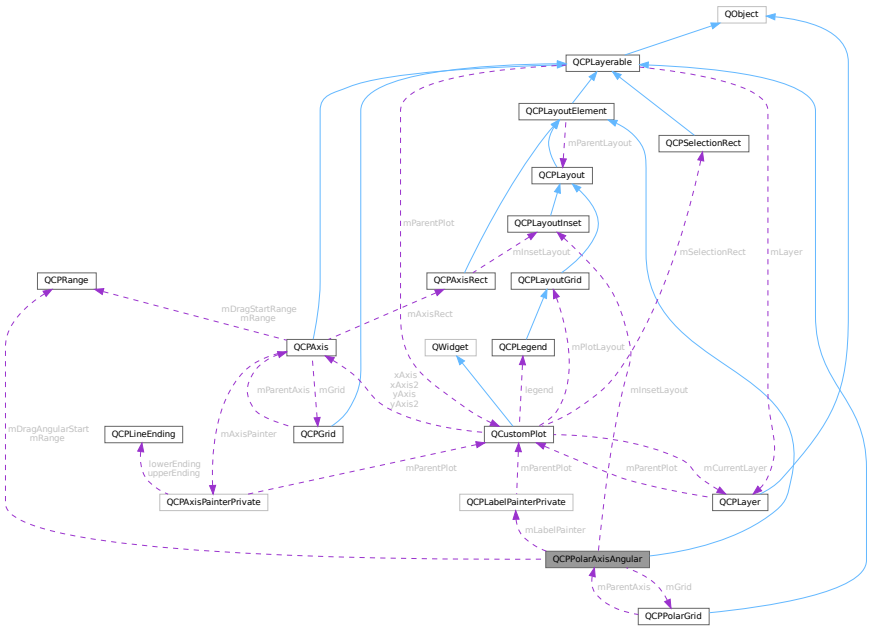
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.660 QCPPolarAxisAngular Class Reference

The main container for polar plots, representing the angular axis as a circle.
Inheritance diagram for QCPPolarAxisAngular:



Collaboration diagram for QCPPolarAxisAngular:



Public Types

- enum SelectablePart { spNone = 0 , spAxis = 0x001 , spTickLabels = 0x002 , spAxisLabel = 0x004 }
- enum LabelMode { **lmUpright** , **lmRotated** }

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

Signals

- void **rangeChanged** (const QCPRange &newRange)
- void **rangeChanged** (const QCPRange &newRange, const QCPRange &oldRange)
- void **selectionChanged** (const QCPPolarAxisAngular::SelectableParts &parts)
- void **selectableChanged** (const QCPPolarAxisAngular::SelectableParts &parts)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPPolarAxisAngular (QCustomPlot *parentPlot)
- QPixmap **background** () const
- QBrush **backgroundBrush** () const
- bool **backgroundScaled** () const
- Qt::AspectRatioMode **backgroundScaledMode** () const
- bool **rangeDrag** () const
- bool **rangeZoom** () const
- double **rangeZoomFactor** () const
- const QCPRange **range** () const
- bool **rangeReversed** () const
- double **angle** () const
- QSharedPointer< QCPAxisTicker > **ticker** () const
- bool **ticks** () const
- bool **tickLabels** () const
- int **tickLabelPadding** () const
- QFont **tickLabelFont** () const
- QColor **tickLabelColor** () const
- double **tickLabelRotation** () const
- LabelMode **tickLabelMode** () const
- QString **numberFormat** () const
- int **numberPrecision** () const
- QVector< double > **tickVector** () const
- QVector< QString > **tickVectorLabels** () const
- int **tickLengthIn** () const
- int **tickLengthOut** () const
- bool **subTicks** () const
- int **subTickLengthIn** () const
- int **subTickLengthOut** () const
- QPen **basePen** () const
- QPen **tickPen** () const
- QPen **subTickPen** () const
- QFont **labelFont** () const
- QColor **labelColor** () const
- QString **label** () const
- int **labelPadding** () const
- SelectableParts **selectedParts** () const
- SelectableParts **selectableParts** () const
- QFont **selectedTickLabelFont** () const

- QFont **selectedLabelFont** () const
- QColor **selectedTickLabelColor** () const
- QColor **selectedLabelColor** () const
- QPen **selectedBasePen** () const
- QPen **selectedTickPen** () const
- QPen **selectedSubTickPen** () const
- QCPPolarGrid * **grid** () const
- void setBackground (const QPixmap &pm)
- void setBackground (const QPixmap &pm, bool scaled, Qt::AspectRatioMode mode=Qt::KeepAspectRatio↩ ByExpanding)
- void setBackground (const QBrush &brush)
- void setBackgroundScaled (bool scaled)
- void setBackgroundScaledMode (Qt::AspectRatioMode mode)
- void **setRangeDrag** (bool enabled)
- void **setRangeZoom** (bool enabled)
- void **setRangeZoomFactor** (double factor)
- Q_SLOT void setRange (const QCPRange &range)
- void setRange (double lower, double upper)
- void setRange (double position, double size, Qt::AlignmentFlag alignment)
- void setRangeLower (double lower)
- void setRangeUpper (double upper)
- void setRangeReversed (bool reversed)
- void **setAngle** (double degrees)
- void setTicker (QSharedPointer< QCPAxisTicker > ticker)
- void setTicks (bool show)
- void setTickLabels (bool show)
- void setTickLabelPadding (int padding)
- void setTickLabelFont (const QFont &font)
- void setTickLabelColor (const QColor &color)
- void setTickLabelRotation (double degrees)
- void **setTickLabelMode** (LabelMode mode)
- void setNumberFormat (const QString &formatCode)
- void setNumberPrecision (int precision)
- void setTickLength (int inside, int outside=0)
- void setTickLengthIn (int inside)
- void setTickLengthOut (int outside)
- void setSubTicks (bool show)
- void setSubTickLength (int inside, int outside=0)
- void setSubTickLengthIn (int inside)
- void setSubTickLengthOut (int outside)
- void setBasePen (const QPen &pen)
- void setTickPen (const QPen &pen)
- void setSubTickPen (const QPen &pen)
- void setLabelFont (const QFont &font)
- void setLabelColor (const QColor &color)
- void setLabel (const QString &str)
- void setLabelPadding (int padding)
- void **setLabelPosition** (Qt::AlignmentFlag position)
- void setSelectedTickLabelFont (const QFont &font)
- void setSelectedLabelFont (const QFont &font)
- void setSelectedTickLabelColor (const QColor &color)
- void setSelectedLabelColor (const QColor &color)
- void setSelectedBasePen (const QPen &pen)
- void setSelectedTickPen (const QPen &pen)
- void setSelectedSubTickPen (const QPen &pen)

- Q_SLOT void setSelectableParts (const QCPPolarAxisAngular::SelectableParts &selectableParts)
- Q_SLOT void setSelectedParts (const QCPPolarAxisAngular::SelectableParts &selectedParts)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=0) const Q_DECL_OVERRIDE
- virtual void update (UpdatePhase phase) Q_DECL_OVERRIDE
- virtual QList< QCPLayoutElement * > elements (bool recursive) const Q_DECL_OVERRIDE
- bool **removeGraph** (QCPPolarGraph *graph)
- int radialAxisCount () const
- QCPPolarAxisRadial * radialAxis (int index=0) const
- QList< QCPPolarAxisRadial * > radialAxes () const
- QCPPolarAxisRadial * addRadialAxis (QCPPolarAxisRadial *axis=0)
- bool removeRadialAxis (QCPPolarAxisRadial *axis)
- QCPLayoutInset * insetLayout () const
- QRegion **exactClipRegion** () const
- void moveRange (double diff)
- void scaleRange (double factor)
- void scaleRange (double factor, double center)
- void rescale (bool onlyVisiblePlottables=false)
- double **coordToAngleRad** (double coord) const
- double **angleRadToCoord** (double angleRad) const
- void pixelToCoord (QPointF pixelPos, double &angleCoord, double &radiusCoord) const
- QPointF coordToPixel (double angleCoord, double radiusCoord) const
- SelectablePart getPartAt (const QPointF &pos) const
- int left () const
- int right () const
- int top () const
- int bottom () const
- int width () const
- int height () const
- QSize size () const
- QPoint topLeft () const
- QPoint topRight () const
- QPoint bottomLeft () const
- QPoint bottomRight () const
- QPointF center () const
- double **radius** () const

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot *parentPlot=nullptr)
- QCPLayout * layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const
- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)

- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual QSize minimumOuterSizeHint () const
- virtual QSize maximumOuterSizeHint () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void wheelEvent (QWheelEvent *event) Q_DECL_OVERRIDE
- bool **registerPolarGraph** (QCPPolarGraph *graph)
- void **drawBackground** (QCPPainter *painter, const QPointF &center, double radius)
- void **setupTickVectors** ()
- QPen **getBasePen** () const
- QPen **getTickPen** () const
- QPen **getSubTickPen** () const
- QFont **getTickLabelFont** () const
- QFont **getLabelFont** () const
- QColor **getTickLabelColor** () const
- QColor **getLabelColor** () const

Protected Member Functions inherited from QCPLayoutElement

- virtual int **calculateAutoMargin** (QCP::MarginSide side)
- virtual void **layoutChanged** ()
- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual QRect **clipRect** () const
- virtual void **selectEvent** (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged)
- virtual void **deselectEvent** (bool *selectionStateChanged)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QBrush **mBackgroundBrush**
- QPixmap **mBackgroundPixmap**
- QPixmap **mScaledBackgroundPixmap**
- bool **mBackgroundScaled**
- Qt::AspectRatioMode **mBackgroundScaledMode**
- QCPLayoutInset * **mInsetLayout**
- bool **mRangeDrag**
- bool **mRangeZoom**
- double **mRangeZoomFactor**
- double **mAngle**
- double **mAngleRad**
- SelectableParts **mSelectableParts**
- SelectableParts **mSelectedParts**
- QPen **mBasePen**
- QPen **mSelectedBasePen**
- int **mLabelPadding**
- QString **mLabel**
- QFont **mLabelFont**
- QFont **mSelectedLabelFont**
- QColor **mLabelColor**
- QColor **mSelectedLabelColor**
- bool **mTickLabels**
- QFont **mTickLabelFont**
- QFont **mSelectedTickLabelFont**
- QColor **mTickLabelColor**
- QColor **mSelectedTickLabelColor**
- int **mNumberPrecision**
- QLatin1Char **mNumberFormatChar**
- bool **mNumberBeautifulPowers**
- bool **mNumberMultiplyCross**
- bool **mTicks**
- bool **mSubTicks**
- int **mTickLengthIn**
- int **mTickLengthOut**
- int **mSubTickLengthIn**
- int **mSubTickLengthOut**
- QPen **mTickPen**
- QPen **mSelectedTickPen**
- QPen **mSubTickPen**
- QPen **mSelectedSubTickPen**
- QCPRange **mRange**

- bool **mRangeReversed**
- QPointF **mCenter**
- double **mRadius**
- QList< QCPPolarAxisRadial * > **mRadialAxes**
- QCPPolarGrid * **mGrid**
- QList< QCPPolarGraph * > **mGraphs**
- QSharedPointer< QCPAxisTicker > **mTicker**
- QVector< double > **mTickVector**
- QVector< QString > **mTickVectorLabels**
- QVector< QPointF > **mTickVectorCosSin**
- QVector< double > **mSubTickVector**
- QVector< QPointF > **mSubTickVectorCosSin**
- bool **mDragging**
- QCPRange **mDragAngularStart**
- QList< QCPRange > **mDragRadialStart**
- QCP::AntialiasedElements **mAADragBackup**
- QCP::AntialiasedElements **mNotAADragBackup**
- QCPLabelPainterPrivate **mLabelPainter**

Protected Attributes inherited from QCPLayoutElement

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPPolarGrid**
- class **QCPPolarGraph**

17.660.1 Detailed Description

The main container for polar plots, representing the angular axis as a circle.

Warning

> In this QCustomPlot version, polar plots are a tech preview. Expect documentation and functionality to be incomplete, as well as changing public interfaces in the future.

17.660.2 Member Enumeration Documentation

17.660.2.1 LabelMode

enum QCPPolarAxisAngular::LabelMode
TODO

17.660.2.2 SelectablePart

enum QCPPolarAxisAngular::SelectablePart
Defines the selectable parts of an axis.

See also

setSelectableParts, setSelectedParts

Enumerator

spNone	None of the selectable parts.
spAxis	The axis backbone and tick marks.
spTickLabels	Tick labels (numbers) of this axis (as a whole, not individually)
spAxisLabel	The axis label.

17.660.3 Constructor & Destructor Documentation

17.660.3.1 QCPPolarAxisAngular()

```
QCPPolarAxisAngular::QCPPolarAxisAngular (
            QCustomPlot * parentPlot )  [explicit]
```
Creates a QCPPolarAxis instance and sets default values. An axis is added for each of the four sides, the top and right axes are set invisible initially.

17.660.4 Member Function Documentation

17.660.4.1 addRadialAxis()

```
QCPPolarAxisRadial * QCPPolarAxisAngular::addRadialAxis (
            QCPPolarAxisRadial * axis = 0 )
```
Adds a new axis to the axis rect side specified with *type*, and returns it. If *axis* is 0, a new QCPAxis instance is created internally. QCustomPlot owns the returned axis, so if you want to remove an axis, use removeAxis instead of deleting it manually.
You may inject QCPAxis instances (or subclasses of QCPAxis) by setting *axis* to an axis that was previously created outside QCustomPlot. It is important to note that QCustomPlot takes ownership of the axis, so you may not delete it afterwards. Further, the *axis* must have been created with this axis rect as parent and with the same axis type as specified in *type*. If this is not the case, a debug output is generated, the axis is not added, and the method returns 0.
This method can not be used to move *axis* between axis rects. The same *axis* instance must not be added multiple times to the same or different axis rects.
If an axis rect side already contains one or more axes, the lower and upper endings of the new axis (QCPAxis::setLowerEnding, QCPAxis::setUpperEnding) are set to QCPLineEnding::esHalfBar.

See also

addAxes, setupFullAxesBox

17.660.4.2 applyDefaultAntialiasingHint()

```
void QCPPolarAxisAngular::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```
Reimplemented from QCPLayoutElement.

17.660.4.3 bottom()

```
int QCPPolarAxisAngular::bottom ( ) const  [inline]
```
Returns the pixel position of the bottom border of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.660.4.4 bottomLeft()

```
QPoint QCPPolarAxisAngular::bottomLeft ( ) const  [inline]
```
Returns the bottom left corner of this axis rect in pixels. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.660.4.5 bottomRight()

```
QPoint QCPPolarAxisAngular::bottomRight ( ) const  [inline]
```
Returns the bottom right corner of this axis rect in pixels. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.660.4.6 center()

```
QPoint QCPPolarAxisAngular::center ( ) const  [inline]
```
Returns the center of this axis rect in pixels. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.660.4.7 coordToPixel()

```
QPointF QCPPolarAxisAngular::coordToPixel (
            double angleCoord,
            double radiusCoord ) const
```
Transforms *value*, in coordinates of the axis, to pixel coordinates of the QCustomPlot widget.

17.660.4.8 draw()

```
void QCPPolarAxisAngular::draw (
            QCPPainter * painter )  [protected], [virtual]
```
Reimplemented from QCPLayoutElement.

17.660.4.9 elements()

```
QList< QCPLayoutElement * > QCPPolarAxisAngular::elements (
            bool recursive ) const  [virtual]
```
Returns a list of all child elements in this layout element. If *recursive* is true, all sub-child elements are included in the list, too.

Warning

> There may be `nullptr` entries in the returned list. For example, QCPLayoutGrid may have empty cells which yield `nullptr` at the respective index.

Reimplemented from QCPLayoutElement.

17.660.4.10 getPartAt()

```
QCPPolarAxisAngular::SelectablePart QCPPolarAxisAngular::getPartAt (
            const QPointF & pos ) const
```
Returns the part of the axis that is hit by *pos* (in pixels). The return value of this function is independent of the user-selectable parts defined with setSelectableParts. Further, this function does not change the current selection state of the axis.
If the axis is not visible (setVisible), this function always returns spNone.

See also

setSelectedParts, setSelectableParts, QCustomPlot::setInteractions

17.660.4.11 height()

```
int QCPPolarAxisAngular::height ( ) const  [inline]
```

Returns the pixel height of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.660.4.12 insetLayout()

```
QCPLayoutInset * QCPPolarAxisAngular::insetLayout ( ) const  [inline]
```

Returns the inset layout of this axis rect. It can be used to place other layout elements (or even layouts with multiple other elements) inside/on top of an axis rect.

See also

QCPLayoutInset

17.660.4.13 left()

```
int QCPPolarAxisAngular::left ( ) const  [inline]
```

Returns the pixel position of the left border of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.660.4.14 mouseMoveEvent()

```
void QCPPolarAxisAngular::mouseMoveEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user moves the mouse while holding a mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.660.4.15 mousePressEvent()

```
void QCPPolarAxisAngular::mousePressEvent (
            QMouseEvent * event,
            const QVariant & details )  [protected], [virtual]
```

This event gets called when the user presses a mouse button while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *details* contains layerable-specific details about the hit, which were generated in the previous call to selectTest. For example, One-dimensional plottables like QCPGraph or QCPBars convey the clicked data point in the *details* parameter, as QCPDataSelection packed as QVariant. Multi-part objects convey the specific `SelectablePart` that was hit (e.g. QCPAxis::SelectablePart in the case of axes).
QCustomPlot uses an event propagation system that works the same as Qt's system. If your layerable doesn't reimplement the mousePressEvent or explicitly calls `event->ignore()` in its reimplementation, the event will be propagated to the next layerable in the stacking order.
Once a layerable has accepted the mousePressEvent, it is considered the mouse grabber and will receive all following calls to mouseMoveEvent or mouseReleaseEvent for this mouse interaction (a "mouse interaction" in this context ends with the release).
The default implementation does nothing except explicitly ignoring the event with `event->ignore()`.

See also

mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.660.4.16 mouseReleaseEvent()

```
void QCPPolarAxisAngular::mouseReleaseEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user releases the mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseMoveEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.660.4.17 moveRange()

```
void QCPPolarAxisAngular::moveRange (
            double diff )
```

If the scale type (setScaleType) is stLinear, *diff* is added to the lower and upper bounds of the range. The range is simply moved by *diff*.
If the scale type is stLogarithmic, the range bounds are multiplied by *diff*. This corresponds to an apparent "linear" move in logarithmic scaling by a distance of log(diff).

17.660.4.18 pixelToCoord()

```
void QCPPolarAxisAngular::pixelToCoord (
            QPointF pixelPos,
            double & angleCoord,
            double & radiusCoord ) const
```

Transforms *value*, in pixel coordinates of the QCustomPlot widget, to axis coordinates.

17.660.4.19 radialAxes()

```
QList< QCPPolarAxisRadial * > QCPPolarAxisAngular::radialAxes ( ) const
```

Returns all axes on the axis rect sides specified with *types*.
types may be a single QCPAxis::AxisType or an `or`-combination, to get the axes of multiple sides.

See also

axis

17.660.4.20 radialAxis()

```
QCPPolarAxisRadial * QCPPolarAxisAngular::radialAxis (
            int index = 0 ) const
```

Returns the axis with the given *index* on the axis rect side specified with *type*.

See also

axisCount, axes

17.660.4.21 radialAxisCount()

```
int QCPPolarAxisAngular::radialAxisCount ( ) const
```

Returns the number of axes on the axis rect side specified with *type*.

See also

axis

17.660.4.22 removeRadialAxis()

```
bool QCPPolarAxisAngular::removeRadialAxis (
            QCPPolarAxisRadial * radialAxis )
```

Removes the specified *axis* from the axis rect and deletes it.
Returns true on success, i.e. if *axis* was a valid axis in this axis rect.

See also

addAxis

17.660.4.23 rescale()

```
void QCPPolarAxisAngular::rescale (
            bool onlyVisiblePlottables = false )
```

Changes the axis range such that all plottables associated with this axis are fully visible in that dimension.

See also

QCPAbstractPlottable::rescaleAxes, QCustomPlot::rescaleAxes

17.660.4.24 right()

```
int QCPPolarAxisAngular::right ( ) const  [inline]
```

Returns the pixel position of the right border of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.660.4.25 scaleRange() [1/2]

```
void QCPPolarAxisAngular::scaleRange (
            double factor )
```

Scales the range of this axis by *factor* around the center of the current axis range. For example, if *factor* is 2.0, then the axis range will double its size, and the point at the axis range center won't have changed its position in the QCustomPlot widget (i.e. coordinates around the center will have moved symmetrically closer).
If you wish to scale around a different coordinate than the current axis range center, use the overload scaleRange(double factor, double center).

17.660.4.26 scaleRange() [2/2]

```
void QCPPolarAxisAngular::scaleRange (
            double factor,
            double center )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Scales the range of this axis by *factor* around the coordinate *center*. For example, if *factor* is 2.0, *center* is 1.0, then the axis range will double its size, and the point at coordinate 1.0 won't have changed its position in the QCustomPlot widget (i.e. coordinates around 1.0 will have moved symmetrically closer to 1.0).

See also

scaleRange(double factor)

17.660.4.27 selectionCategory()

```
QCP::Interaction QCPPolarAxisAngular::selectionCategory ( ) const  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.660.4.28 selectTest()

```
double QCPPolarAxisAngular::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = 0 ) const  [virtual]
```

Layout elements are sensitive to events inside their outer rect. If *pos* is within the outer rect, this method returns a value corresponding to 0.99 times the parent plot's selection tolerance. However, layout elements are not selectable by default. So if *onlySelectable* is true, -1.0 is returned.
See QCPLayerable::selectTest for a general explanation of this virtual method.
QCPLayoutElement subclasses may reimplement this method to provide more specific selection test behaviour.
Reimplemented from QCPLayoutElement.

17.660.4.29 setBackground() [1/3]

```
void QCPPolarAxisAngular::setBackground (
            const QBrush & brush )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets *brush* as the background brush. The axis rect background will be filled with this brush. Since axis rects place themselves on the "background" layer by default, the axis rect backgrounds are usually drawn below everything else.
The brush will be drawn before (under) any background pixmap, which may be specified with setBackground(const QPixmap &pm).
To disable drawing of a background brush, set *brush* to Qt::NoBrush.

See also

setBackground(const QPixmap &pm)

17.660.4.30 setBackground() [2/3]

```
void QCPPolarAxisAngular::setBackground (
            const QPixmap & pm )
```

Sets *pm* as the axis background pixmap. The axis background pixmap will be drawn inside the axis rect. Since axis rects place themselves on the "background" layer by default, the axis rect backgrounds are usually drawn below everything else.
For cases where the provided pixmap doesn't have the same size as the axis rect, scaling can be enabled with setBackgroundScaled and the scaling mode (i.e. whether and how the aspect ratio is preserved) can be set with setBackgroundScaledMode. To set all these options in one call, consider using the overloaded version of this function.
Below the pixmap, the axis rect may be optionally filled with a brush, if specified with setBackground(const QBrush &brush).

See also

setBackgroundScaled, setBackgroundScaledMode, setBackground(const QBrush &brush)

17.660.4.31 setBackground() [3/3]

```
void QCPPolarAxisAngular::setBackground (
            const QPixmap & pm,
            bool scaled,
            Qt::AspectRatioMode mode = Qt::KeepAspectRatioByExpanding )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Allows setting the background pixmap of the axis rect, whether it shall be scaled and how it shall be scaled in one call.

See also

setBackground(const QPixmap &pm), setBackgroundScaled, setBackgroundScaledMode

17.660.4.32 setBackgroundScaled()

```
void QCPPolarAxisAngular::setBackgroundScaled (
            bool scaled )
```

Sets whether the axis background pixmap shall be scaled to fit the axis rect or not. If *scaled* is set to true, you may control whether and how the aspect ratio of the original pixmap is preserved with setBackgroundScaledMode.
Note that the scaled version of the original pixmap is buffered, so there is no performance penalty on replots. (Except when the axis rect dimensions are changed continuously.)

See also

setBackground, setBackgroundScaledMode

17.660.4.33 setBackgroundScaledMode()

```
void QCPPolarAxisAngular::setBackgroundScaledMode (
            Qt::AspectRatioMode mode )
```

If scaling of the axis background pixmap is enabled (setBackgroundScaled), use this function to define whether and how the aspect ratio of the original pixmap passed to setBackground is preserved.

See also

setBackground, setBackgroundScaled

17.660.4.34 setBasePen()

```
void QCPPolarAxisAngular::setBasePen (
            const QPen & pen )
```

Sets the pen, the axis base line is drawn with.

See also

setTickPen, setSubTickPen

17.660.4.35 setLabel()

```
void QCPPolarAxisAngular::setLabel (
            const QString & str )
```

Sets the text of the axis label that will be shown below/above or next to the axis, depending on its orientation. To disable axis labels, pass an empty string as *str*.

17.660.4.36 setLabelColor()

```
void QCPPolarAxisAngular::setLabelColor (
            const QColor & color )
```

Sets the color of the axis label.

See also

setLabelFont

17.660.4.37 setLabelFont()

```
void QCPPolarAxisAngular::setLabelFont (
            const QFont & font )
```

Sets the font of the axis label.

See also

setLabelColor

17.660.4.38 setLabelPadding()

```
void QCPPolarAxisAngular::setLabelPadding (
            int padding )
```

Sets the distance between the tick labels and the axis label.

See also

setTickLabelPadding, setPadding

17.660.4.39 setNumberFormat()

```
void QCPPolarAxisAngular::setNumberFormat (
            const QString & formatCode )
```

Sets the number format for the numbers in tick labels. This *formatCode* is an extended version of the format code used e.g. by QString::number() and QLocale::toString(). For reference about that, see the "Argument Formats" section in the detailed description of the QString class.
formatCode is a string of one, two or three characters. The first character is identical to the normal format code used by Qt. In short, this means: 'e'/'E' scientific format, 'f' fixed format, 'g'/'G' scientific or fixed, whichever is shorter.
The second and third characters are optional and specific to QCustomPlot:
If the first char was 'e' or 'g', numbers are/might be displayed in the scientific format, e.g. "5.5e9", which might be visually unappealing in a plot. So when the second char of *formatCode* is set to 'b' (for "beautiful"), those exponential numbers are formatted in a more natural way, i.e. "5.5 [multiplication sign] 10 [superscript] 9". By default, the multiplication sign is a centered dot. If instead a cross should be shown (as is usual in the USA), the third char of *formatCode* can be set to 'c'. The inserted multiplication signs are the UTF-8 characters 215 (0xD7) for the cross and 183 (0xB7) for the dot.
Examples for *formatCode:*

- `g` normal format code behaviour. If number is small, fixed format is used, if number is large, normal scientific format is used
- `gb` If number is small, fixed format is used, if number is large, scientific format is used with beautifully typeset decimal powers and a dot as multiplication sign
- `ebc` All numbers are in scientific format with beautifully typeset decimal power and a cross as multiplication sign
- `fb` illegal format code, since fixed format doesn't support (or need) beautifully typeset decimal powers. Format code will be reduced to 'f'.
- `hello` illegal format code, since first char is not 'e', 'E', 'f', 'g' or 'G'. Current format code will not be changed.

17.660.4.40 setNumberPrecision()

```
void QCPPolarAxisAngular::setNumberPrecision (
            int precision )
```

Sets the precision of the tick label numbers. See QLocale::toString(double i, char f, int prec) for details. The effect of precisions are most notably for number Formats starting with 'e', see setNumberFormat

17.660.4.41 setRange() [1/3]

```
void QCPPolarAxisAngular::setRange (
            const QCPRange & range )
```

Sets the range of the axis.
This slot may be connected with the rangeChanged signal of another axis so this axis is always synchronized with the other axis range, when it changes.
To invert the direction of an axis, use setRangeReversed.

17.660.4.42 setRange() [2/3]

```
void QCPPolarAxisAngular::setRange (
            double lower,
            double upper )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the lower and upper bound of the axis range.
To invert the direction of an axis, use setRangeReversed.
There is also a slot to set a range, see setRange(const QCPRange &range).

17.660.4.43 setRange() [3/3]

```
void QCPPolarAxisAngular::setRange (
            double position,
            double size,
            Qt::AlignmentFlag alignment )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the range of the axis.
The *position* coordinate indicates together with the *alignment* parameter, where the new range will be positioned. *size* defines the size of the new axis range. *alignment* may be Qt::AlignLeft, Qt::AlignRight or Qt::AlignCenter. This will cause the left border, right border, or center of the range to be aligned with *position*. Any other values of *alignment* will default to Qt::AlignCenter.

17.660.4.44 setRangeLower()

```
void QCPPolarAxisAngular::setRangeLower (
            double lower )
```

Sets the lower bound of the axis range. The upper bound is not changed.

See also

setRange

17.660.4.45 setRangeReversed()

```
void QCPPolarAxisAngular::setRangeReversed (
            bool reversed )
```

Sets whether the axis range (direction) is displayed reversed. Normally, the values on horizontal axes increase left to right, on vertical axes bottom to top. When *reversed* is set to true, the direction of increasing values is inverted.
Note that the range and data interface stays the same for reversed axes, e.g. the *lower* part of the setRange interface will still reference the mathematically smaller number than the *upper* part.

17.660.4.46 setRangeUpper()

```
void QCPPolarAxisAngular::setRangeUpper (
            double upper )
```

Sets the upper bound of the axis range. The lower bound is not changed.

See also

setRange

17.660.4.47 setSelectableParts()

```
void QCPPolarAxisAngular::setSelectableParts (
            const QCPPolarAxisAngular::SelectableParts & selectableParts )
```

Sets whether the user can (de-)select the parts in *selectable* by clicking on the QCustomPlot surface. (When QCustomPlot::setInteractions contains iSelectAxes.)
However, even when *selectable* is set to a value not allowing the selection of a specific part, it is still possible to set the selection of this part manually, by calling setSelectedParts directly.

See also

SelectablePart, setSelectedParts

17.660.4.48 setSelectedBasePen()

```
void QCPPolarAxisAngular::setSelectedBasePen (
            const QPen & pen )
```

Sets the pen that is used to draw the axis base line when selected.

See also

setBasePen, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.660.4.49 setSelectedLabelColor()

```
void QCPPolarAxisAngular::setSelectedLabelColor (
            const QColor & color )
```

Sets the color that is used for the axis label when it is selected.

See also

setLabelColor, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.660.4.50 setSelectedLabelFont()

```
void QCPPolarAxisAngular::setSelectedLabelFont (
            const QFont & font )
```

Sets the font that is used for the axis label when it is selected.

See also

setLabelFont, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.660.4.51 setSelectedParts()

```
void QCPPolarAxisAngular::setSelectedParts (
            const QCPPolarAxisAngular::SelectableParts & selectedParts )
```

Sets the selected state of the respective axis parts described by SelectablePart. When a part is selected, it uses a different pen/font.
The entire selection mechanism for axes is handled automatically when QCustomPlot::setInteractions contains iSelectAxes. You only need to call this function when you wish to change the selection state manually.
This function can change the selection state of a part, independent of the setSelectableParts setting.
emits the selectionChanged signal when *selected* is different from the previous selection state.

See also

SelectablePart, setSelectableParts, selectTest, setSelectedBasePen, setSelectedTickPen, setSelectedSubTickPen, setSelectedTickLabelFont, setSelectedLabelFont, setSelectedTickLabelColor, setSelectedLabelColor

17.660.4.52 setSelectedSubTickPen()

```
void QCPPolarAxisAngular::setSelectedSubTickPen (
            const QPen & pen )
```

Sets the pen that is used to draw the subticks when selected.

See also

setSubTickPen, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.660.4.53 setSelectedTickLabelColor()

```
void QCPPolarAxisAngular::setSelectedTickLabelColor (
            const QColor & color )
```

Sets the color that is used for tick labels when they are selected.

See also

setTickLabelColor, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.660.4.54 setSelectedTickLabelFont()

```
void QCPPolarAxisAngular::setSelectedTickLabelFont (
            const QFont & font )
```

Sets the font that is used for tick labels when they are selected.

See also

setTickLabelFont, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.660.4.55 setSelectedTickPen()

```
void QCPPolarAxisAngular::setSelectedTickPen (
            const QPen & pen )
```

Sets the pen that is used to draw the (major) ticks when selected.

See also

setTickPen, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.660.4.56 setSubTickLength()

```
void QCPPolarAxisAngular::setSubTickLength (
            int inside,
            int outside = 0 )
```

Sets the length of the subticks in pixels. *inside* is the length the subticks will reach inside the plot and *outside* is the length they will reach outside the plot. If *outside* is greater than zero, the tick labels and axis label will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setTickLength, setSubTickLengthIn, setSubTickLengthOut

17.660.4.57 setSubTickLengthIn()

```
void QCPPolarAxisAngular::setSubTickLengthIn (
            int inside )
```

Sets the length of the inward subticks in pixels. *inside* is the length the subticks will reach inside the plot.

See also

setSubTickLengthOut, setSubTickLength, setTickLength

17.660.4.58 setSubTickLengthOut()

```
void QCPPolarAxisAngular::setSubTickLengthOut (
            int outside )
```

Sets the length of the outward subticks in pixels. *outside* is the length the subticks will reach outside the plot. If *outside* is greater than zero, the tick labels will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setSubTickLengthIn, setSubTickLength, setTickLength

17.660.4.59 setSubTickPen()

```
void QCPPolarAxisAngular::setSubTickPen (
            const QPen & pen )
```

Sets the pen, subtick marks will be drawn with.

See also

setSubTickCount, setSubTickLength, setBasePen

17.660.4.60 setSubTicks()

```
void QCPPolarAxisAngular::setSubTicks (
            bool show )
```

Sets whether sub tick marks are displayed.
Sub ticks are only potentially visible if (major) ticks are also visible (see setTicks)

See also

setTicks

17.660.4.61 setTicker()

```
void QCPPolarAxisAngular::setTicker (
            QSharedPointer< QCPAxisTicker > ticker )
```

The axis ticker is responsible for generating the tick positions and tick labels. See the documentation of QCPAxisTicker for details on how to work with axis tickers.
You can change the tick positioning/labeling behaviour of this axis by setting a different QCPAxisTicker subclass using this method. If you only wish to modify the currently installed axis ticker, access it via ticker.
Since the ticker is stored in the axis as a shared pointer, multiple axes may share the same axis ticker simply by passing the same shared pointer to multiple axes.

See also

ticker

17.660.4.62 setTickLabelColor()

```
void QCPPolarAxisAngular::setTickLabelColor (
            const QColor & color )
```

Sets the color of the tick labels.

See also

setTickLabels, setTickLabelFont

17.660.4.63 setTickLabelFont()

```
void QCPPolarAxisAngular::setTickLabelFont (
            const QFont & font )
```

Sets the font of the tick labels.

See also

setTickLabels, setTickLabelColor

17.660.4.64 setTickLabelPadding()

```
void QCPPolarAxisAngular::setTickLabelPadding (
            int padding )
```

Sets the distance between the axis base line (including any outward ticks) and the tick labels.

See also

setLabelPadding, setPadding

17.660.4.65 setTickLabelRotation()

```
void QCPPolarAxisAngular::setTickLabelRotation (
            double degrees )
```

Sets the rotation of the tick labels. If *degrees* is zero, the labels are drawn normally. Else, the tick labels are drawn rotated by *degrees* clockwise. The specified angle is bound to values from -90 to 90 degrees.
If *degrees* is exactly -90, 0 or 90, the tick labels are centered on the tick coordinate. For other angles, the label is drawn with an offset such that it seems to point toward or away from the tick mark.

17.660.4.66 setTickLabels()

```
void QCPPolarAxisAngular::setTickLabels (
            bool show )
```

Sets whether tick labels are displayed. Tick labels are the numbers drawn next to tick marks.

17.660.4.67 setTickLength()

```
void QCPPolarAxisAngular::setTickLength (
            int inside,
            int outside = 0 )
```

Sets the length of the ticks in pixels. *inside* is the length the ticks will reach inside the plot and *outside* is the length they will reach outside the plot. If *outside* is greater than zero, the tick labels and axis label will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setSubTickLength, setTickLengthIn, setTickLengthOut

17.660.4.68 setTickLengthIn()

```
void QCPPolarAxisAngular::setTickLengthIn (
            int inside )
```

Sets the length of the inward ticks in pixels. *inside* is the length the ticks will reach inside the plot.

See also

setTickLengthOut, setTickLength, setSubTickLength

17.660.4.69 setTickLengthOut()

```
void QCPPolarAxisAngular::setTickLengthOut (
            int outside )
```

Sets the length of the outward ticks in pixels. *outside* is the length the ticks will reach outside the plot. If *outside* is greater than zero, the tick labels and axis label will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setTickLengthIn, setTickLength, setSubTickLength

17.660.4.70 setTickPen()

```
void QCPPolarAxisAngular::setTickPen (
            const QPen & pen )
```

Sets the pen, tick marks will be drawn with.

See also

setTickLength, setBasePen

17.660.4.71 setTicks()

```
void QCPPolarAxisAngular::setTicks (
            bool show )
```

Sets whether tick marks are displayed.
Note that setting *show* to false does not imply that tick labels are invisible, too. To achieve that, see setTickLabels.

See also

setSubTicks

17.660.4.72 size()

```
QSize QCPPolarAxisAngular::size ( ) const  [inline]
```

Returns the pixel size of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.660.4.73 top()

```
int QCPPolarAxisAngular::top ( ) const  [inline]
```

Returns the pixel position of the top border of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.660.4.74 topLeft()

```
QPoint QCPPolarAxisAngular::topLeft ( ) const  [inline]
```

Returns the top left corner of this axis rect in pixels. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.660.4.75 topRight()

```
QPoint QCPPolarAxisAngular::topRight ( ) const  [inline]
```

Returns the top right corner of this axis rect in pixels. Margins are not taken into account here, so the returned value is with respect to the inner rect.

17.660.4.76 update()

```
void QCPPolarAxisAngular::update (
            UpdatePhase phase )  [virtual]
```

This method is called automatically upon replot and doesn't need to be called by users of QCPPolarAxisAngular.
Calls the base class implementation to update the margins (see QCPLayoutElement::update), and finally passes the rect to the inset layout (insetLayout) and calls its QCPInsetLayout::update function.
\seebaseclassmethod
Reimplemented from QCPLayoutElement.

17.660.4.77 wheelEvent()

```
void QCPPolarAxisAngular::wheelEvent (
            QWheelEvent * event )  [protected], [virtual]
```

This event gets called when the user turns the mouse scroll wheel while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`.
The `event->angleDelta()` indicates how far the mouse wheel was turned, which is usually +/- 120 for single rotation steps. However, if the mouse wheel is turned rapidly, multiple steps may accumulate to one event, making the delta larger. On the other hand, if the wheel has very smooth steps or none at all, the delta may be smaller.
The default implementation does nothing.

See also

mousePressEvent, mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent

Reimplemented from QCPLayerable.

17.660.4.78 width()

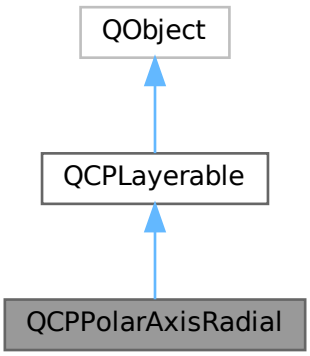
```
int QCPPolarAxisAngular::width ( ) const  [inline]
```

Returns the pixel width of this axis rect. Margins are not taken into account here, so the returned value is with respect to the inner rect.
The documentation for this class was generated from the following files:

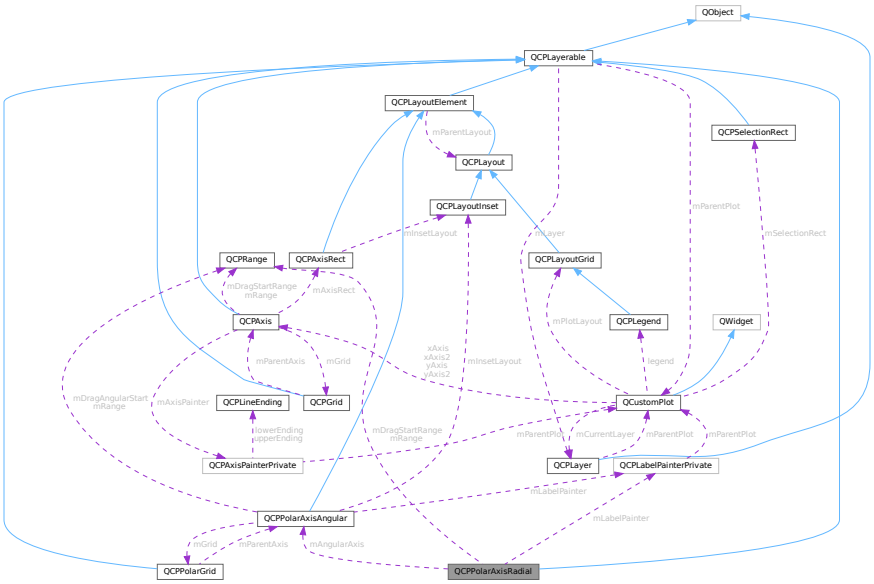
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.661 QCPPolarAxisRadial Class Reference

The radial axis inside a radial plot.
Inheritance diagram for QCPPolarAxisRadial:



Collaboration diagram for QCPPolarAxisRadial:



Public Types

- enum AngleReference { arAbsolute , arAngularAxis }
- enum ScaleType { stLinear , stLogarithmic }
- enum SelectablePart { spNone = 0 , spAxis = 0x001 , spTickLabels = 0x002 , spAxisLabel = 0x004 }
- enum **LabelMode** { **lmUpright** , **lmRotated** }

Signals

- void rangeChanged (const QCPRange &newRange)
- void rangeChanged (const QCPRange &newRange, const QCPRange &oldRange)
- void scaleTypeChanged (QCPPolarAxisRadial::ScaleType scaleType)
- void selectionChanged (const QCPPolarAxisRadial::SelectableParts &parts)
- void selectableChanged (const QCPPolarAxisRadial::SelectableParts &parts)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer ∗newLayer)

Public Member Functions

- QCPPolarAxisRadial (QCPPolarAxisAngular ∗parent)
- bool **rangeDrag** () const
- bool **rangeZoom** () const
- double **rangeZoomFactor** () const
- QCPPolarAxisAngular ∗ **angularAxis** () const
- ScaleType **scaleType** () const
- const QCPRange **range** () const
- bool **rangeReversed** () const
- double **angle** () const
- AngleReference **angleReference** () const
- QSharedPointer< QCPAxisTicker > ticker () const
- bool **ticks** () const

- bool **tickLabels** () const
- int **tickLabelPadding** () const
- QFont **tickLabelFont** () const
- QColor **tickLabelColor** () const
- double **tickLabelRotation** () const
- LabelMode **tickLabelMode** () const
- QString **numberFormat** () const
- int **numberPrecision** () const
- QVector< double > **tickVector** () const
- QVector< double > **subTickVector** () const
- QVector< QString > **tickVectorLabels** () const
- int **tickLengthIn** () const
- int **tickLengthOut** () const
- bool **subTicks** () const
- int **subTickLengthIn** () const
- int **subTickLengthOut** () const
- QPen **basePen** () const
- QPen **tickPen** () const
- QPen **subTickPen** () const
- QFont **labelFont** () const
- QColor **labelColor** () const
- QString **label** () const
- int **labelPadding** () const
- SelectableParts **selectedParts** () const
- SelectableParts **selectableParts** () const
- QFont **selectedTickLabelFont** () const
- QFont **selectedLabelFont** () const
- QColor **selectedTickLabelColor** () const
- QColor **selectedLabelColor** () const
- QPen **selectedBasePen** () const
- QPen **selectedTickPen** () const
- QPen **selectedSubTickPen** () const
- void **setRangeDrag** (bool enabled)
- void **setRangeZoom** (bool enabled)
- void **setRangeZoomFactor** (double factor)
- Q_SLOT void setScaleType (QCPPolarAxisRadial::ScaleType type)
- Q_SLOT void setRange (const QCPRange &range)
- void setRange (double lower, double upper)
- void setRange (double position, double size, Qt::AlignmentFlag alignment)
- void setRangeLower (double lower)
- void setRangeUpper (double upper)
- void setRangeReversed (bool reversed)
- void **setAngle** (double degrees)
- void **setAngleReference** (AngleReference reference)
- void setTicker (QSharedPointer< QCPAxisTicker > ticker)
- void setTicks (bool show)
- void setTickLabels (bool show)
- void setTickLabelPadding (int padding)
- void setTickLabelFont (const QFont &font)
- void setTickLabelColor (const QColor &color)
- void setTickLabelRotation (double degrees)
- void **setTickLabelMode** (LabelMode mode)
- void setNumberFormat (const QString &formatCode)
- void setNumberPrecision (int precision)
- void setTickLength (int inside, int outside=0)

- void setTickLengthIn (int inside)
- void setTickLengthOut (int outside)
- void setSubTicks (bool show)
- void setSubTickLength (int inside, int outside=0)
- void setSubTickLengthIn (int inside)
- void setSubTickLengthOut (int outside)
- void setBasePen (const QPen &pen)
- void setTickPen (const QPen &pen)
- void setSubTickPen (const QPen &pen)
- void setLabelFont (const QFont &font)
- void setLabelColor (const QColor &color)
- void setLabel (const QString &str)
- void setLabelPadding (int padding)
- void setSelectedTickLabelFont (const QFont &font)
- void setSelectedLabelFont (const QFont &font)
- void setSelectedTickLabelColor (const QColor &color)
- void setSelectedLabelColor (const QColor &color)
- void setSelectedBasePen (const QPen &pen)
- void setSelectedTickPen (const QPen &pen)
- void setSelectedSubTickPen (const QPen &pen)
- Q_SLOT void setSelectableParts (const QCPPolarAxisRadial::SelectableParts &selectableParts)
- Q_SLOT void setSelectedParts (const QCPPolarAxisRadial::SelectableParts &selectedParts)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=0) const Q_DECL_↩ OVERRIDE
- void moveRange (double diff)
- void scaleRange (double factor)
- void scaleRange (double factor, double center)
- void rescale (bool onlyVisiblePlottables=false)
- void pixelToCoord (QPointF pixelPos, double &angleCoord, double &radiusCoord) const
- QPointF coordToPixel (double angleCoord, double radiusCoord) const
- double **coordToRadius** (double coord) const
- double **radiusToCoord** (double radius) const
- SelectablePart getPartAt (const QPointF &pos) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void wheelEvent (QWheelEvent *event) Q_DECL_OVERRIDE
- void **updateGeometry** (const QPointF &center, double radius)
- void **setupTickVectors** ()
- QPen **getBasePen** () const
- QPen **getTickPen** () const
- QPen **getSubTickPen** () const
- QFont **getTickLabelFont** () const
- QFont **getLabelFont** () const
- QColor **getTickLabelColor** () const
- QColor **getLabelColor** () const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual QRect **clipRect** () const
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- bool **mRangeDrag**
- bool **mRangeZoom**
- double **mRangeZoomFactor**
- QCPPolarAxisAngular * **mAngularAxis**
- double **mAngle**
- AngleReference **mAngleReference**
- SelectableParts **mSelectableParts**
- SelectableParts **mSelectedParts**
- QPen **mBasePen**
- QPen **mSelectedBasePen**
- int **mLabelPadding**
- QString **mLabel**
- QFont **mLabelFont**
- QFont **mSelectedLabelFont**
- QColor **mLabelColor**
- QColor **mSelectedLabelColor**
- bool **mTickLabels**
- QFont **mTickLabelFont**
- QFont **mSelectedTickLabelFont**
- QColor **mTickLabelColor**
- QColor **mSelectedTickLabelColor**

- int **mNumberPrecision**
- QLatin1Char **mNumberFormatChar**
- bool **mNumberBeautifulPowers**
- bool **mNumberMultiplyCross**
- bool **mTicks**
- bool **mSubTicks**
- int **mTickLengthIn**
- int **mTickLengthOut**
- int **mSubTickLengthIn**
- int **mSubTickLengthOut**
- QPen **mTickPen**
- QPen **mSelectedTickPen**
- QPen **mSubTickPen**
- QPen **mSelectedSubTickPen**
- QCPRange **mRange**
- bool **mRangeReversed**
- ScaleType **mScaleType**
- QPointF **mCenter**
- double **mRadius**
- QSharedPointer< QCPAxisTicker > **mTicker**
- QVector< double > **mTickVector**
- QVector< QString > **mTickVectorLabels**
- QVector< double > **mSubTickVector**
- bool **mDragging**
- QCPRange **mDragStartRange**
- QCP::AntialiasedElements **mAADragBackup**
- QCP::AntialiasedElements **mNotAADragBackup**
- QCPLabelPainterPrivate **mLabelPainter**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPPolarAxisAngular**

17.661.1 Detailed Description

The radial axis inside a radial plot.

Warning

> In this QCustomPlot version, polar plots are a tech preview. Expect documentation and functionality to be incomplete, as well as changing public interfaces in the future.

Each axis holds an instance of QCPAxisTicker which is used to generate the tick coordinates and tick labels. You can access the currently installed ticker or set a new one (possibly one of the specialized subclasses, or your own subclass) via setTicker. For details, see the documentation of QCPAxisTicker.

17.661.2 Member Enumeration Documentation

17.661.2.1 AngleReference

enum QCPPolarAxisRadial::AngleReference

Defines the reference of the angle at which a radial axis is tilted (setAngle).

Enumerator

arAbsolute	The axis tilt is given in absolute degrees. The zero is to the right and positive angles are measured counter-clockwise.
arAngularAxis	The axis tilt is measured in the angular coordinate system given by the parent angular axis.

17.661.2.2 ScaleType

`enum QCPPolarAxisRadial::ScaleType`
Defines the scale of an axis.

See also

setScaleType

Enumerator

stLinear	Linear scaling.
stLogarithmic	Logarithmic scaling with correspondingly transformed axis coordinates (possibly also setTicker to a QCPAxisTickerLog instance).

17.661.2.3 SelectablePart

`enum QCPPolarAxisRadial::SelectablePart`
Defines the selectable parts of an axis.

See also

setSelectableParts, setSelectedParts

Enumerator

spNone	None of the selectable parts.
spAxis	The axis backbone and tick marks.
spTickLabels	Tick labels (numbers) of this axis (as a whole, not individually)
spAxisLabel	The axis label.

17.661.3 Constructor & Destructor Documentation

17.661.3.1 QCPPolarAxisRadial()

```
QCPPolarAxisRadial::QCPPolarAxisRadial (
            QCPPolarAxisAngular * parent )  [explicit]
```
Constructs an Axis instance of Type *type* for the axis rect *parent*.
Usually it isn't necessary to instantiate axes directly, because you can let QCustomPlot create them for you with QCPAxisRect::addAxis. If you want to use own QCPAxis-subclasses however, create them manually and then inject them also via QCPAxisRect::addAxis.

17.661.4 Member Function Documentation

17.661.4.1 applyDefaultAntialiasingHint()

```
void QCPPolarAxisRadial::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```
Implements QCPLayerable.

17.661.4.2 coordToPixel()

```
QPointF QCPPolarAxisRadial::coordToPixel (
            double angleCoord,
            double radiusCoord ) const
```

Transforms *value*, in coordinates of the axis, to pixel coordinates of the QCustomPlot widget.

17.661.4.3 deselectEvent()

```
void QCPPolarAxisRadial::deselectEvent (
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.661.4.4 draw()

```
void QCPPolarAxisRadial::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPLayerable.

17.661.4.5 getPartAt()

```
QCPPolarAxisRadial::SelectablePart QCPPolarAxisRadial::getPartAt (
            const QPointF & pos ) const
```

Returns the part of the axis that is hit by *pos* (in pixels). The return value of this function is independent of the user-selectable parts defined with setSelectableParts. Further, this function does not change the current selection state of the axis.
If the axis is not visible (setVisible), this function always returns spNone.

See also

setSelectedParts, setSelectableParts, QCustomPlot::setInteractions

17.661.4.6 mouseMoveEvent()

```
void QCPPolarAxisRadial::mouseMoveEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user moves the mouse while holding a mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.661.4.7 mousePressEvent()

```
void QCPPolarAxisRadial::mousePressEvent (
            QMouseEvent * event,
            const QVariant & details )  [protected], [virtual]
```

This event gets called when the user presses a mouse button while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *details* contains layerable-specific details about the hit, which were generated in the previous call to selectTest. For example, One-dimensional plottables like QCPGraph or QCPBars convey the clicked data point in the *details* parameter, as QCPDataSelection packed as QVariant. Multi-part objects convey the specific `SelectablePart` that was hit (e.g. QCPAxis::SelectablePart in the case of axes).

QCustomPlot uses an event propagation system that works the same as Qt's system. If your layerable doesn't reimplement the mousePressEvent or explicitly calls `event->ignore()` in its reimplementation, the event will be propagated to the next layerable in the stacking order.
Once a layerable has accepted the mousePressEvent, it is considered the mouse grabber and will receive all following calls to mouseMoveEvent or mouseReleaseEvent for this mouse interaction (a "mouse interaction" in this context ends with the release).
The default implementation does nothing except explicitly ignoring the event with `event->ignore()`.

See also

mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.661.4.8 mouseReleaseEvent()

```
void QCPPolarAxisRadial::mouseReleaseEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [protected], [virtual]
```

This event gets called when the user releases the mouse button, after this layerable has become the mouse grabber by accepting the preceding mousePressEvent.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`. The parameter *startPos* indicates the position where the initial mousePressEvent occurred, that started the mouse interaction.
The default implementation does nothing.

See also

mousePressEvent, mouseMoveEvent, mouseDoubleClickEvent, wheelEvent

Reimplemented from QCPLayerable.

17.661.4.9 moveRange()

```
void QCPPolarAxisRadial::moveRange (
            double diff )
```

If the scale type (setScaleType) is stLinear, *diff* is added to the lower and upper bounds of the range. The range is simply moved by *diff*.
If the scale type is stLogarithmic, the range bounds are multiplied by *diff*. This corresponds to an apparent "linear" move in logarithmic scaling by a distance of log(diff).

17.661.4.10 pixelToCoord()

```
void QCPPolarAxisRadial::pixelToCoord (
            QPointF pixelPos,
            double & angleCoord,
            double & radiusCoord ) const
```

Transforms *value*, in pixel coordinates of the QCustomPlot widget, to axis coordinates.

17.661.4.11 rangeChanged [1/2]

```
void QCPPolarAxisRadial::rangeChanged (
            const QCPRange & newRange )  [signal]
```

This signal is emitted when the range of this axis has changed. You can connect it to the setRange slot of another axis to communicate the new range to the other axis, in order for it to be synchronized.
You may also manipulate/correct the range with setRange in a slot connected to this signal. This is useful if for example a maximum range span shall not be exceeded, or if the lower/upper range shouldn't go beyond certain values (see QCPRange::bounded). For example, the following slot would limit the x axis to ranges between 0 and 10:

```
customPlot->xAxis->setRange(newRange.bounded(0, 10))
```

17.661.4.12 rangeChanged [2/2]

```
void QCPPolarAxisRadial::rangeChanged (
            const QCPRange & newRange,
            const QCPRange & oldRange )  [signal]
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Additionally to the new range, this signal also provides the previous range held by the axis as *oldRange*.

17.661.4.13 rescale()

```
void QCPPolarAxisRadial::rescale (
            bool onlyVisiblePlottables = false )
```

Changes the axis range such that all plottables associated with this axis are fully visible in that dimension.

See also

QCPAbstractPlottable::rescaleAxes, QCustomPlot::rescaleAxes

17.661.4.14 scaleRange() [1/2]

```
void QCPPolarAxisRadial::scaleRange (
            double factor )
```

Scales the range of this axis by *factor* around the center of the current axis range. For example, if *factor* is 2.0, then the axis range will double its size, and the point at the axis range center won't have changed its position in the QCustomPlot widget (i.e. coordinates around the center will have moved symmetrically closer).
If you wish to scale around a different coordinate than the current axis range center, use the overload scaleRange(double factor, double center).

17.661.4.15 scaleRange() [2/2]

```
void QCPPolarAxisRadial::scaleRange (
            double factor,
            double center )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Scales the range of this axis by *factor* around the coordinate *center*. For example, if *factor* is 2.0, *center* is 1.0, then the axis range will double its size, and the point at coordinate 1.0 won't have changed its position in the QCustomPlot widget (i.e. coordinates around 1.0 will have moved symmetrically closer to 1.0).

See also

scaleRange(double factor)

17.661.4.16 scaleTypeChanged

```
void QCPPolarAxisRadial::scaleTypeChanged (
            QCPPolarAxisRadial::ScaleType scaleType )  [signal]
```

This signal is emitted when the scale type changes, by calls to setScaleType

17.661.4.17 selectableChanged

```
void QCPPolarAxisRadial::selectableChanged (
            const QCPPolarAxisRadial::SelectableParts & parts )  [signal]
```

This signal is emitted when the selectability changes, by calls to setSelectableParts

17.661.4.18 selectEvent()

```
void QCPPolarAxisRadial::selectEvent (
            QMouseEvent * event,
            bool additive,
            const QVariant & details,
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.661.4.19 selectionCategory()

```
QCP::Interaction QCPPolarAxisRadial::selectionCategory ( ) const  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.661.4.20 selectionChanged

```
void QCPPolarAxisRadial::selectionChanged (
            const QCPPolarAxisRadial::SelectableParts & parts )  [signal]
```

This signal is emitted when the selection state of this axis has changed, either by user interaction or by a direct call to setSelectedParts.

17.661.4.21 selectTest()

```
double QCPPolarAxisRadial::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = 0 ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.
pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Reimplemented from QCPLayerable.

17.661.4.22 setBasePen()

```
void QCPPolarAxisRadial::setBasePen (
            const QPen & pen )
```

Sets the pen, the axis base line is drawn with.

See also

setTickPen, setSubTickPen

17.661.4.23 setLabel()

```
void QCPPolarAxisRadial::setLabel (
            const QString & str )
```

Sets the text of the axis label that will be shown below/above or next to the axis, depending on its orientation. To disable axis labels, pass an empty string as *str*.

17.661.4.24 setLabelColor()

```
void QCPPolarAxisRadial::setLabelColor (
            const QColor & color )
```

Sets the color of the axis label.

See also

setLabelFont

17.661.4.25 setLabelFont()

```
void QCPPolarAxisRadial::setLabelFont (
            const QFont & font )
```

Sets the font of the axis label.

See also

setLabelColor

17.661.4.26 setLabelPadding()

```
void QCPPolarAxisRadial::setLabelPadding (
            int padding )
```

Sets the distance between the tick labels and the axis label.

See also

setTickLabelPadding, setPadding

17.661.4.27 setNumberFormat()

```
void QCPPolarAxisRadial::setNumberFormat (
            const QString & formatCode )
```

Sets the number format for the numbers in tick labels. This *formatCode* is an extended version of the format code used e.g. by QString::number() and QLocale::toString(). For reference about that, see the "Argument Formats" section in the detailed description of the QString class.
formatCode is a string of one, two or three characters. The first character is identical to the normal format code used by Qt. In short, this means: 'e'/'E' scientific format, 'f' fixed format, 'g'/'G' scientific or fixed, whichever is shorter.
The second and third characters are optional and specific to QCustomPlot:
If the first char was 'e' or 'g', numbers are/might be displayed in the scientific format, e.g. "5.5e9", which is ugly in a plot. So when the second char of *formatCode* is set to 'b' (for "beautiful"), those exponential numbers are formatted in a more natural way, i.e. "5.5 [multiplication sign] 10 [superscript] 9". By default, the multiplication sign is a centered dot. If instead a cross should be shown (as is usual in the USA), the third char of *formatCode* can be set to 'c'. The inserted multiplication signs are the UTF-8 characters 215 (0xD7) for the cross and 183 (0xB7) for the dot.
Examples for *formatCode:*

- `g` normal format code behaviour. If number is small, fixed format is used, if number is large, normal scientific format is used

- `gb` If number is small, fixed format is used, if number is large, scientific format is used with beautifully typeset decimal powers and a dot as multiplication sign
- `ebc` All numbers are in scientific format with beautifully typeset decimal power and a cross as multiplication sign
- `fb` illegal format code, since fixed format doesn't support (or need) beautifully typeset decimal powers. Format code will be reduced to 'f'.
- `hello` illegal format code, since first char is not 'e', 'E', 'f', 'g' or 'G'. Current format code will not be changed.

17.661.4.28 setNumberPrecision()

```
void QCPPolarAxisRadial::setNumberPrecision (
            int precision )
```

Sets the precision of the tick label numbers. See QLocale::toString(double i, char f, int prec) for details. The effect of precisions are most notably for number Formats starting with 'e', see setNumberFormat

17.661.4.29 setRange() [1/3]

```
void QCPPolarAxisRadial::setRange (
            const QCPRange & range )
```

Sets the range of the axis.
This slot may be connected with the rangeChanged signal of another axis so this axis is always synchronized with the other axis range, when it changes.
To invert the direction of an axis, use setRangeReversed.

17.661.4.30 setRange() [2/3]

```
void QCPPolarAxisRadial::setRange (
            double lower,
            double upper )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the lower and upper bound of the axis range.
To invert the direction of an axis, use setRangeReversed.
There is also a slot to set a range, see setRange(const QCPRange &range).

17.661.4.31 setRange() [3/3]

```
void QCPPolarAxisRadial::setRange (
            double position,
            double size,
            Qt::AlignmentFlag alignment )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the range of the axis.
The *position* coordinate indicates together with the *alignment* parameter, where the new range will be positioned. *size* defines the size of the new axis range. *alignment* may be Qt::AlignLeft, Qt::AlignRight or Qt::AlignCenter. This will cause the left border, right border, or center of the range to be aligned with *position*. Any other values of *alignment* will default to Qt::AlignCenter.

17.661.4.32 setRangeLower()

```
void QCPPolarAxisRadial::setRangeLower (
            double lower )
```

Sets the lower bound of the axis range. The upper bound is not changed.

See also

setRange

17.661.4.33 setRangeReversed()

```
void QCPPolarAxisRadial::setRangeReversed (
            bool reversed )
```

Sets whether the axis range (direction) is displayed reversed. Normally, the values on horizontal axes increase left to right, on vertical axes bottom to top. When *reversed* is set to true, the direction of increasing values is inverted.
Note that the range and data interface stays the same for reversed axes, e.g. the *lower* part of the setRange interface will still reference the mathematically smaller number than the *upper* part.

17.661.4.34 setRangeUpper()

```
void QCPPolarAxisRadial::setRangeUpper (
            double upper )
```

Sets the upper bound of the axis range. The lower bound is not changed.

See also

setRange

17.661.4.35 setScaleType()

```
void QCPPolarAxisRadial::setScaleType (
            QCPPolarAxisRadial::ScaleType type )
```

Sets whether the axis uses a linear scale or a logarithmic scale.
Note that this method controls the coordinate transformation. For logarithmic scales, you will likely also want to use a logarithmic tick spacing and labeling, which can be achieved by setting the axis ticker to an instance of QCPAxisTickerLog :
See the documentation of QCPAxisTickerLog about the details of logarithmic axis tick creation.
setNumberPrecision

17.661.4.36 setSelectableParts()

```
void QCPPolarAxisRadial::setSelectableParts (
            const QCPPolarAxisRadial::SelectableParts & selectableParts )
```

Sets whether the user can (de-)select the parts in *selectable* by clicking on the QCustomPlot surface. (When QCustomPlot::setInteractions contains iSelectAxes.)
However, even when *selectable* is set to a value not allowing the selection of a specific part, it is still possible to set the selection of this part manually, by calling setSelectedParts directly.

See also

SelectablePart, setSelectedParts

17.661.4.37 setSelectedBasePen()

```
void QCPPolarAxisRadial::setSelectedBasePen (
            const QPen & pen )
```

Sets the pen that is used to draw the axis base line when selected.

See also

setBasePen, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.661.4.38 setSelectedLabelColor()

```
void QCPPolarAxisRadial::setSelectedLabelColor (
            const QColor & color )
```

Sets the color that is used for the axis label when it is selected.

See also

setLabelColor, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.661.4.39 setSelectedLabelFont()

```
void QCPPolarAxisRadial::setSelectedLabelFont (
            const QFont & font )
```

Sets the font that is used for the axis label when it is selected.

See also

setLabelFont, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.661.4.40 setSelectedParts()

```
void QCPPolarAxisRadial::setSelectedParts (
            const QCPPolarAxisRadial::SelectableParts & selectedParts )
```

Sets the selected state of the respective axis parts described by SelectablePart. When a part is selected, it uses a different pen/font.
The entire selection mechanism for axes is handled automatically when QCustomPlot::setInteractions contains iSelectAxes. You only need to call this function when you wish to change the selection state manually.
This function can change the selection state of a part, independent of the setSelectableParts setting.
emits the selectionChanged signal when *selected* is different from the previous selection state.

See also

SelectablePart, setSelectableParts, selectTest, setSelectedBasePen, setSelectedTickPen, setSelectedSubTickPen, setSelectedTickLabelFont, setSelectedLabelFont, setSelectedTickLabelColor, setSelectedLabelColor

17.661.4.41 setSelectedSubTickPen()

```
void QCPPolarAxisRadial::setSelectedSubTickPen (
            const QPen & pen )
```

Sets the pen that is used to draw the subticks when selected.

See also

setSubTickPen, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.661.4.42 setSelectedTickLabelColor()

```
void QCPPolarAxisRadial::setSelectedTickLabelColor (
            const QColor & color )
```

Sets the color that is used for tick labels when they are selected.

See also

setTickLabelColor, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.661.4.43 setSelectedTickLabelFont()

```
void QCPPolarAxisRadial::setSelectedTickLabelFont (
            const QFont & font )
```

Sets the font that is used for tick labels when they are selected.

See also

setTickLabelFont, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.661.4.44 setSelectedTickPen()

```
void QCPPolarAxisRadial::setSelectedTickPen (
            const QPen & pen )
```

Sets the pen that is used to draw the (major) ticks when selected.

See also

setTickPen, setSelectableParts, setSelectedParts, QCustomPlot::setInteractions

17.661.4.45 setSubTickLength()

```
void QCPPolarAxisRadial::setSubTickLength (
            int inside,
            int outside = 0 )
```

Sets the length of the subticks in pixels. *inside* is the length the subticks will reach inside the plot and *outside* is the length they will reach outside the plot. If *outside* is greater than zero, the tick labels and axis label will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setTickLength, setSubTickLengthIn, setSubTickLengthOut

17.661.4.46 setSubTickLengthIn()

```
void QCPPolarAxisRadial::setSubTickLengthIn (
            int inside )
```

Sets the length of the inward subticks in pixels. *inside* is the length the subticks will reach inside the plot.

See also

setSubTickLengthOut, setSubTickLength, setTickLength

17.661.4.47 setSubTickLengthOut()

```
void QCPPolarAxisRadial::setSubTickLengthOut (
            int outside )
```

Sets the length of the outward subticks in pixels. *outside* is the length the subticks will reach outside the plot. If *outside* is greater than zero, the tick labels will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setSubTickLengthIn, setSubTickLength, setTickLength

17.661.4.48 setSubTickPen()

```
void QCPPolarAxisRadial::setSubTickPen (
            const QPen & pen )
```

Sets the pen, subtick marks will be drawn with.

See also

setSubTickCount, setSubTickLength, setBasePen

17.661.4.49 setSubTicks()

```
void QCPPolarAxisRadial::setSubTicks (
            bool show )
```

Sets whether sub tick marks are displayed.
Sub ticks are only potentially visible if (major) ticks are also visible (see setTicks)

See also

setTicks

17.661.4.50 setTicker()

```
void QCPPolarAxisRadial::setTicker (
            QSharedPointer< QCPAxisTicker > ticker )
```

The axis ticker is responsible for generating the tick positions and tick labels. See the documentation of QCPAxisTicker for details on how to work with axis tickers.
You can change the tick positioning/labeling behaviour of this axis by setting a different QCPAxisTicker subclass using this method. If you only wish to modify the currently installed axis ticker, access it via ticker.
Since the ticker is stored in the axis as a shared pointer, multiple axes may share the same axis ticker simply by passing the same shared pointer to multiple axes.

See also

ticker

17.661.4.51 setTickLabelColor()

```
void QCPPolarAxisRadial::setTickLabelColor (
            const QColor & color )
```

Sets the color of the tick labels.

See also

setTickLabels, setTickLabelFont

17.661.4.52 setTickLabelFont()

```
void QCPPolarAxisRadial::setTickLabelFont (
            const QFont & font )
```

Sets the font of the tick labels.

See also

setTickLabels, setTickLabelColor

17.661.4.53 setTickLabelPadding()

```
void QCPPolarAxisRadial::setTickLabelPadding (
            int padding )
```

Sets the distance between the axis base line (including any outward ticks) and the tick labels.

See also

setLabelPadding, setPadding

17.661.4.54 setTickLabelRotation()

```
void QCPPolarAxisRadial::setTickLabelRotation (
            double degrees )
```

Sets the rotation of the tick labels. If *degrees* is zero, the labels are drawn normally. Else, the tick labels are drawn rotated by *degrees* clockwise. The specified angle is bound to values from -90 to 90 degrees.
If *degrees* is exactly -90, 0 or 90, the tick labels are centered on the tick coordinate. For other angles, the label is drawn with an offset such that it seems to point toward or away from the tick mark.

17.661.4.55 setTickLabels()

```
void QCPPolarAxisRadial::setTickLabels (
            bool show )
```

Sets whether tick labels are displayed. Tick labels are the numbers drawn next to tick marks.

17.661.4.56 setTickLength()

```
void QCPPolarAxisRadial::setTickLength (
            int inside,
            int outside = 0 )
```

Sets the length of the ticks in pixels. *inside* is the length the ticks will reach inside the plot and *outside* is the length they will reach outside the plot. If *outside* is greater than zero, the tick labels and axis label will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setSubTickLength, setTickLengthIn, setTickLengthOut

17.661.4.57 setTickLengthIn()

```
void QCPPolarAxisRadial::setTickLengthIn (
            int inside )
```

Sets the length of the inward ticks in pixels. *inside* is the length the ticks will reach inside the plot.

See also

setTickLengthOut, setTickLength, setSubTickLength

17.661.4.58 setTickLengthOut()

```
void QCPPolarAxisRadial::setTickLengthOut (
            int outside )
```

Sets the length of the outward ticks in pixels. *outside* is the length the ticks will reach outside the plot. If *outside* is greater than zero, the tick labels and axis label will increase their distance to the axis accordingly, so they won't collide with the ticks.

See also

setTickLengthIn, setTickLength, setSubTickLength

17.661.4.59 setTickPen()

```
void QCPPolarAxisRadial::setTickPen (
            const QPen & pen )
```

Sets the pen, tick marks will be drawn with.

See also

setTickLength, setBasePen

17.661.4.60 setTicks()

```
void QCPPolarAxisRadial::setTicks (
            bool show )
```

Sets whether tick marks are displayed.
Note that setting *show* to false does not imply that tick labels are invisible, too. To achieve that, see setTickLabels.

See also

setSubTicks

17.661.4.61 ticker()

```
QSharedPointer< QCPAxisTicker > QCPPolarAxisRadial::ticker ( ) const  [inline]
```

Returns a modifiable shared pointer to the currently installed axis ticker. The axis ticker is responsible for generating the tick positions and tick labels of this axis. You can access the QCPAxisTicker with this method and modify basic properties such as the approximate tick count (QCPAxisTicker::setTickCount).
You can gain more control over the axis ticks by setting a different QCPAxisTicker subclass, see the documentation there. A new axis ticker can be set with setTicker.
Since the ticker is stored in the axis as a shared pointer, multiple axes may share the same axis ticker simply by passing the same shared pointer to multiple axes.

See also

setTicker

17.661.4.62 wheelEvent()

```
void QCPPolarAxisRadial::wheelEvent (
            QWheelEvent * event )  [protected], [virtual]
```

This event gets called when the user turns the mouse scroll wheel while the cursor is over the layerable. Whether a cursor is over the layerable is decided by a preceding call to selectTest.
The current pixel position of the cursor on the QCustomPlot widget is accessible via `event->pos()`.
The `event->angleDelta()` indicates how far the mouse wheel was turned, which is usually +/- 120 for single rotation steps. However, if the mouse wheel is turned rapidly, multiple steps may accumulate to one event, making the delta larger. On the other hand, if the wheel has very smooth steps or none at all, the delta may be smaller.
The default implementation does nothing.

See also

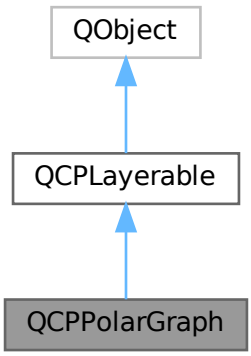
mousePressEvent, mouseMoveEvent, mouseReleaseEvent, mouseDoubleClickEvent

Reimplemented from QCPLayerable.
The documentation for this class was generated from the following files:

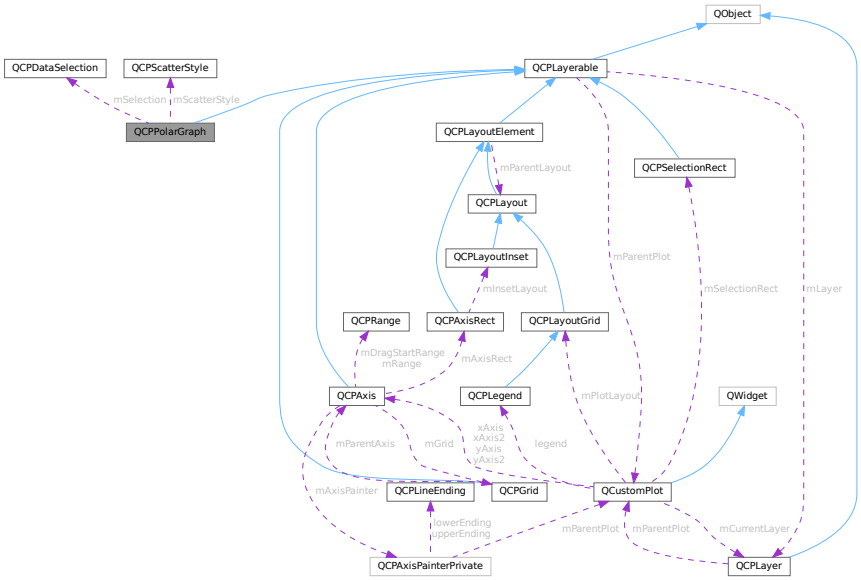
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.662 QCPPolarGraph Class Reference

A radial graph used to display data in polar plots.
Inheritance diagram for QCPPolarGraph:



Collaboration diagram for QCPPolarGraph:



Public Types

- enum LineStyle { lsNone , lsLine }

Signals

- void **selectionChanged** (bool selected)
- void **selectionChanged** (const QCPDataSelection &selection)
- void **selectableChanged** (QCP::SelectionType selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPPolarGraph (QCPPolarAxisAngular *keyAxis, QCPPolarAxisRadial *valueAxis)
- QString **name** () const
- bool **antialiasedFill** () const
- bool **antialiasedScatters** () const
- QPen **pen** () const
- QBrush **brush** () const
- bool **periodic** () const
- QCPPolarAxisAngular * **keyAxis** () const
- QCPPolarAxisRadial * **valueAxis** () const
- QCP::SelectionType **selectable** () const
- bool **selected** () const
- QCPDataSelection **selection** () const
- QSharedPointer< QCPGraphDataContainer > **data** () const
- LineStyle **lineStyle** () const
- QCPScatterStyle **scatterStyle** () const
- void setName (const QString &name)
- void setAntialiasedFill (bool enabled)

- void setAntialiasedScatters (bool enabled)
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void **setPeriodic** (bool enabled)
- void setKeyAxis (QCPPolarAxisAngular *axis)
- void setValueAxis (QCPPolarAxisRadial *axis)
- Q_SLOT void setSelectable (QCP::SelectionType selectable)
- Q_SLOT void setSelection (QCPDataSelection selection)
- void setData (QSharedPointer< QCPGraphDataContainer > data)
- void setData (const QVector< double > &keys, const QVector< double > &values, bool alreadySorted=false)
- void setLineStyle (LineStyle ls)
- void setScatterStyle (const QCPScatterStyle &style)
- void **addData** (const QVector< double > &keys, const QVector< double > &values, bool already↩ Sorted=false)
- void **addData** (double key, double value)
- void coordsToPixels (double key, double value, double &x, double &y) const
- const QPointF **coordsToPixels** (double key, double value) const
- void **pixelsToCoords** (double x, double y, double &key, double &value) const
- void **pixelsToCoords** (const QPointF &pixelPos, double &key, double &value) const
- void **rescaleAxes** (bool onlyEnlarge=false) const
- void **rescaleKeyAxis** (bool onlyEnlarge=false) const
- void **rescaleValueAxis** (bool onlyEnlarge=false, bool inKeyRange=false) const
- bool **addToLegend** (QCPLegend *legend)
- bool **addToLegend** ()
- bool **removeFromLegend** (QCPLegend *legend) const
- bool **removeFromLegend** () const
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=0) const
- virtual QCPPlottableInterface1D * **interface1D** ()
- virtual QCPRange **getKeyRange** (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth) const
- virtual QCPRange **getValueRange** (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth, const QCPRange &inKeyRange=QCPRange()) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual QRect clipRect () const
- virtual void draw (QCPPainter *painter)
- virtual QCP::Interaction selectionCategory () const
- void applyDefaultAntialiasingHint (QCPPainter *painter) const
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed)
- virtual void deselectEvent (bool *selectionStateChanged)

- virtual void **drawLinePlot** (QCPPainter *painter, const QVector< QPointF > &lines) const
- virtual void **drawFill** (QCPPainter *painter, QVector< QPointF > *lines) const
- virtual void **drawScatterPlot** (QCPPainter *painter, const QVector< QPointF > &scatters, const QCPScatterStyle &style) const
- virtual void **drawLegendIcon** (QCPPainter *painter, const QRectF &rect) const
- void **applyFillAntialiasingHint** (QCPPainter *painter) const
- void **applyScattersAntialiasingHint** (QCPPainter *painter) const
- double **pointDistance** (const QPointF &pixelPoint, QCPGraphDataContainer::const_iterator &closestData) const
- virtual int **dataCount** () const
- void **getDataSegments** (QList< QCPDataRange > &selectedSegments, QList< QCPDataRange > &unselectedSegments) const
- void **drawPolyline** (QCPPainter *painter, const QVector< QPointF > &lineData) const
- void **getVisibleDataBounds** (QCPGraphDataContainer::const_iterator &begin, QCPGraphDataContainer::const_iterator &end, const QCPDataRange &rangeRestriction) const
- void **getLines** (QVector< QPointF > *lines, const QCPDataRange &dataRange) const
- void **getScatters** (QVector< QPointF > *scatters, const QCPDataRange &dataRange) const
- void **getOptimizedLineData** (QVector< QCPGraphData > *lineData, const QCPGraphDataContainer::const_iterator &begin, const QCPGraphDataContainer::const_iterator &end) const
- void **getOptimizedScatterData** (QVector< QCPGraphData > *scatterData, QCPGraphDataContainer::const_iterator begin, QCPGraphDataContainer::const_iterator end) const
- QVector< QPointF > **dataToLines** (const QVector< QCPGraphData > &data) const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QSharedPointer< QCPGraphDataContainer > **mDataContainer**
- LineStyle **mLineStyle**
- QCPScatterStyle **mScatterStyle**
- QString **mName**
- bool **mAntialiasedFill**
- bool **mAntialiasedScatters**
- QPen **mPen**
- QBrush **mBrush**
- bool **mPeriodic**
- QPointer< QCPPolarAxisAngular > **mKeyAxis**
- QPointer< QCPPolarAxisRadial > **mValueAxis**
- QCP::SelectionType **mSelectable**
- QCPDataSelection **mSelection**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCPPolarLegendItem**

17.662.1 Detailed Description

A radial graph used to display data in polar plots.

Warning

> In this QCustomPlot version, polar plots are a tech preview. Expect documentation and functionality to be incomplete, as well as changing public interfaces in the future.

17.662.2 Member Enumeration Documentation

17.662.2.1 LineStyle

`enum QCPPolarGraph::LineStyle`

Defines how the graph's line is represented visually in the plot. The line is drawn with the current pen of the graph (setPen).

See also

> setLineStyle

Enumerator

lsNone	with symbols according to the scatter style, see setScatterStyle) data points are not connected with any lines (e.g. data only represented
lsLine	data points are connected by a straight line

17.662.3 Constructor & Destructor Documentation

17.662.3.1 QCPPolarGraph()

```
QCPPolarGraph::QCPPolarGraph (
            QCPPolarAxisAngular * keyAxis,
            QCPPolarAxisRadial * valueAxis )
```

Constructs a graph which uses *keyAxis* as its angular and *valueAxis* as its radial axis. *keyAxis* and *valueAxis* must reside in the same QCustomPlot, and the radial axis must be associated with the angular axis. If either of these restrictions is violated, a corresponding message is printed to the debug output (qDebug), the construction is not aborted, though.

The created QCPPolarGraph is automatically registered with the QCustomPlot instance inferred from *keyAxis*. This QCustomPlot instance takes ownership of the QCPPolarGraph, so do not delete it manually but use QCPPolar↩ AxisAngular::removeGraph() instead.

To directly create a QCPPolarGraph inside a plot, you shoud use the QCPPolarAxisAngular::addGraph method.

17.662.4 Member Function Documentation

17.662.4.1 applyDefaultAntialiasingHint()

```
void QCPPolarGraph::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```

Implements QCPLayerable.

17.662.4.2 clipRect()

```
QRect QCPPolarGraph::clipRect ( ) const  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.662.4.3 coordsToPixels()

```
void QCPPolarGraph::coordsToPixels (
            double key,
            double value,
            double & x,
            double & y ) const
```

Use this method to set an own QCPSelectionDecorator (subclass) instance. This allows you to customize the visual representation of selected data ranges further than by using the default QCPSelectionDecorator.
The plottable takes ownership of the *decorator*.
The currently set decorator can be accessed via selectionDecorator.

17.662.4.4 deselectEvent()

```
void QCPPolarGraph::deselectEvent (
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.662.4.5 draw()

```
void QCPPolarGraph::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPLayerable.

17.662.4.6 selectEvent()

```
void QCPPolarGraph::selectEvent (
            QMouseEvent * event,
            bool additive,
            const QVariant & details,
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.662.4.7 selectionCategory()

```
QCP::Interaction QCPPolarGraph::selectionCategory ( ) const  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.662.4.8 selectTest()

```
double QCPPolarGraph::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = 0 ) const  [virtual]
```

This function is used to decide whether a click hits a layerable object or not.

pos is a point in pixel coordinates on the QCustomPlot surface. This function returns the shortest pixel distance of this point to the object. If the object is either invisible or the distance couldn't be determined, -1.0 is returned. Further, if *onlySelectable* is true and the object is not selectable, -1.0 is returned, too.
If the object is represented not by single lines but by an area like a QCPItemText or the bars of a QCPBars plottable, a click inside the area should also be considered a hit. In these cases this function thus returns a constant value greater zero but still below the parent plot's selection tolerance. (typically the selectionTolerance multiplied by 0.99). Providing a constant value for area objects allows selecting line objects even when they are obscured by such area objects, by clicking close to the lines (i.e. closer than 0.99∗selectionTolerance).
The actual setting of the selection state is not done by this function. This is handled by the parent QCustomPlot when the mouseReleaseEvent occurs, and the finally selected object is notified via the selectEvent/ deselectEvent methods.
details is an optional output parameter. Every layerable subclass may place any information in *details*. This information will be passed to selectEvent when the parent QCustomPlot decides on the basis of this selectTest call, that the object was successfully selected. The subsequent call to selectEvent will carry the *details*. This is useful for multi-part objects (like QCPAxis). This way, a possibly complex calculation to decide which part was clicked is only done once in selectTest. The result (i.e. the actually clicked part) can then be placed in *details*. So in the subsequent selectEvent, the decision which part was selected doesn't have to be done a second time for a single selection operation.
In the case of 1D Plottables (QCPAbstractPlottable1D, like QCPGraph or QCPBars) *details* will be set to a QCPDataSelection, describing the closest data point to *pos*.
You may pass `nullptr` as *details* to indicate that you are not interested in those selection details.

See also

selectEvent, deselectEvent, mousePressEvent, wheelEvent, QCustomPlot::setInteractions, QCPAbstractPlottable1D::selectTest

Reimplemented from QCPLayerable.

17.662.4.9 setAntialiasedFill()

```
void QCPPolarGraph::setAntialiasedFill (
            bool enabled )
```

Sets whether fills of this plottable are drawn antialiased or not.
Note that this setting may be overridden by QCustomPlot::setAntialiasedElements and QCustomPlot::setNotAntialiasedElements.

17.662.4.10 setAntialiasedScatters()

```
void QCPPolarGraph::setAntialiasedScatters (
            bool enabled )
```

Sets whether the scatter symbols of this plottable are drawn antialiased or not.
Note that this setting may be overridden by QCustomPlot::setAntialiasedElements and QCustomPlot::setNotAntialiasedElements.

17.662.4.11 setBrush()

```
void QCPPolarGraph::setBrush (
            const QBrush & brush )
```

The brush is used to draw basic fills of the plottable representation in the plot. The Fill can be a color, gradient or texture, see the usage of QBrush.
For example, the QCPGraph subclass draws the fill under the graph with this brush, when it's not set to Qt::NoBrush.

See also

setPen

17.662.4.12 setData() [1/2]

```
void QCPPolarGraph::setData (
            const QVector< double > & keys,
```

```
            const QVector< double > & values,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data with the provided points in *keys* and *values*. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.
If you can guarantee that the passed data points are sorted by *keys* in ascending order, you can set *alreadySorted* to true, to improve performance by saving a sorting run.

See also

addData

17.662.4.13 setData() [2/2]

```
void QCPPolarGraph::setData (
            QSharedPointer< QCPGraphDataContainer > data )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data container with the provided *data* container.
Since a QSharedPointer is used, multiple QCPPolarGraphs may share the same data container safely. Modifying the data in the container will then affect all graphs that share the container. Sharing can be achieved by simply exchanging the data containers wrapped in shared pointers:
If you do not wish to share containers, but create a copy from an existing container, rather use the QCPDataContainer<DataType>::set method on the graph's data container directly:

See also

addData

17.662.4.14 setKeyAxis()

```
void QCPPolarGraph::setKeyAxis (
            QCPPolarAxisAngular * axis )
```

The key axis of a plottable can be set to any axis of a QCustomPlot, as long as it is orthogonal to the plottable's value axis. This function performs no checks to make sure this is the case. The typical mathematical choice is to use the x-axis (QCustomPlot::xAxis) as key axis and the y-axis (QCustomPlot::yAxis) as value axis.
Normally, the key and value axes are set in the constructor of the plottable (or QCustomPlot::addGraph when working with QCPGraphs through the dedicated graph interface).

See also

setValueAxis

17.662.4.15 setLineStyle()

```
void QCPPolarGraph::setLineStyle (
            LineStyle ls )
```

Sets how the single data points are connected in the plot. For scatter-only plots, set *ls* to lsNone and setScatterStyle to the desired scatter style.

See also

setScatterStyle

17.662.4.16 setName()

```
void QCPPolarGraph::setName (
            const QString & name )
```

The name is the textual representation of this plottable as it is displayed in the legend (QCPLegend). It may contain any UTF-8 characters, including newlines.

17.662.4.17 setPen()

```
void QCPPolarGraph::setPen (
            const QPen & pen )
```

The pen is used to draw basic lines that make up the plottable representation in the plot.
For example, the QCPGraph subclass draws its graph lines with this pen.

See also

setBrush

17.662.4.18 setScatterStyle()

```
void QCPPolarGraph::setScatterStyle (
            const QCPScatterStyle & style )
```

Sets the visual appearance of single data points in the plot. If set to QCPScatterStyle::ssNone, no scatter points are drawn (e.g. for line-only-plots with appropriate line style).

See also

QCPScatterStyle, setLineStyle

17.662.4.19 setSelectable()

```
void QCPPolarGraph::setSelectable (
            QCP::SelectionType selectable )
```

Sets whether and to which granularity this plottable can be selected.
A selection can happen by clicking on the QCustomPlot surface (When QCustomPlot::setInteractions contains QCP::iSelectPlottables), by dragging a selection rect (When QCustomPlot::setSelectionRectMode is QCP::srmSelect), or programmatically by calling setSelection.

See also

setSelection, QCP::SelectionType

17.662.4.20 setSelection()

```
void QCPPolarGraph::setSelection (
            QCPDataSelection selection )
```

Sets which data ranges of this plottable are selected. Selected data ranges are drawn differently (e.g. color) in the plot. This can be controlled via the selection decorator (see selectionDecorator).
The entire selection mechanism for plottables is handled automatically when QCustomPlot::setInteractions contains iSelectPlottables. You only need to call this function when you wish to change the selection state programmatically. Using setSelectable you can further specify for each plottable whether and to which granularity it is selectable. If *selection* is not compatible with the current QCP::SelectionType set via setSelectable, the resulting selection will be adjusted accordingly (see QCPDataSelection::enforceType).
emits the selectionChanged signal when *selected* is different from the previous selection state.

See also

setSelectable, selectTest

17.662.4.21 setValueAxis()

```
void QCPPolarGraph::setValueAxis (
            QCPPolarAxisRadial * axis )
```

The value axis of a plottable can be set to any axis of a QCustomPlot, as long as it is orthogonal to the plottable's key axis. This function performs no checks to make sure this is the case. The typical mathematical choice is to use the x-axis (QCustomPlot::xAxis) as key axis and the y-axis (QCustomPlot::yAxis) as value axis.
Normally, the key and value axes are set in the constructor of the plottable (or QCustomPlot::addGraph when working with QCPGraphs through the dedicated graph interface).

See also

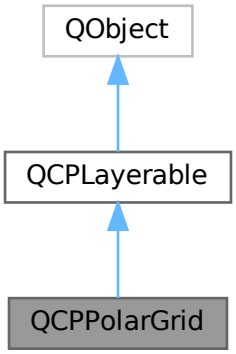
setKeyAxis

The documentation for this class was generated from the following files:

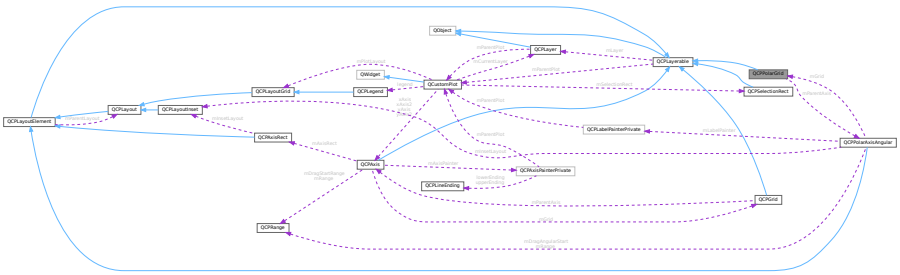
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.663 QCPPolarGrid Class Reference

The grid in both angular and radial dimensions for polar plots.
Inheritance diagram for QCPPolarGrid:



Collaboration diagram for QCPPolarGrid:



Public Types

- enum GridType { **gtAngular** = 0x01 , **gtRadial** = 0x02 , **gtAll** = 0xFF , **gtNone** = 0x00 }

Public Member Functions

- QCPPolarGrid (QCPPolarAxisAngular ∗parentAxis)
- QCPPolarAxisRadial ∗ **radialAxis** () const
- GridTypes **type** () const
- GridTypes **subGridType** () const
- bool **antialiasedSubGrid** () const
- bool **antialiasedZeroLine** () const

- QPen **angularPen** () const
- QPen **angularSubGridPen** () const
- QPen **radialPen** () const
- QPen **radialSubGridPen** () const
- QPen **radialZeroLinePen** () const
- void **setRadialAxis** (QCPPolarAxisRadial *axis)
- void **setType** (GridTypes type)
- void **setSubGridType** (GridTypes type)
- void setAntialiasedSubGrid (bool enabled)
- void setAntialiasedZeroLine (bool enabled)
- void setAngularPen (const QPen &pen)
- void setAngularSubGridPen (const QPen &pen)
- void **setRadialPen** (const QPen &pen)
- void **setRadialSubGridPen** (const QPen &pen)
- void **setRadialZeroLinePen** (const QPen &pen)

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const
- bool realVisibility () const

Protected Member Functions

- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- void **drawRadialGrid** (QCPPainter *painter, const QPointF &center, const QVector< double > &coords, const QPen &pen, const QPen &zeroPen=Qt::NoPen)
- void **drawAngularGrid** (QCPPainter *painter, const QPointF &center, double radius, const QVector< QPointF > &ticksCosSin, const QPen &pen)

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void **selectEvent** (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged)
- virtual void **deselectEvent** (bool *selectionStateChanged)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- GridTypes **mType**
- GridTypes **mSubGridType**
- bool **mAntialiasedSubGrid**
- bool **mAntialiasedZeroLine**
- QPen **mAngularPen**
- QPen **mAngularSubGridPen**
- QPen **mRadialPen**
- QPen **mRadialSubGridPen**
- QPen **mRadialZeroLinePen**
- QCPPolarAxisAngular * **mParentAxis**
- QPointer< QCPPolarAxisRadial > **mRadialAxis**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.663.1 Detailed Description

The grid in both angular and radial dimensions for polar plots.

Warning

> In this QCustomPlot version, polar plots are a tech preview. Expect documentation and functionality to be incomplete, as well as changing public interfaces in the future.

17.663.2 Member Enumeration Documentation

17.663.2.1 GridType

```
enum QCPPolarGrid::GridType
```

TODO

17.663.3 Constructor & Destructor Documentation

17.663.3.1 QCPPolarGrid()

```
QCPPolarGrid::QCPPolarGrid (
            QCPPolarAxisAngular * parentAxis )  [explicit]
```

Creates a QCPPolarGrid instance and sets default values.
You shouldn't instantiate grids on their own, since every axis brings its own grid.

17.663.4 Member Function Documentation

17.663.4.1 applyDefaultAntialiasingHint()

```
void QCPPolarGrid::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```

Implements QCPLayerable.

17.663.4.2 draw()

```
void QCPPolarGrid::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPLayerable.

17.663.4.3 setAngularPen()

```
void QCPPolarGrid::setAngularPen (
            const QPen & pen )
```

Sets the pen with which (major) grid lines are drawn.

17.663.4.4 setAngularSubGridPen()

```
void QCPPolarGrid::setAngularSubGridPen (
            const QPen & pen )
```

Sets the pen with which sub grid lines are drawn.

17.663.4.5 setAntialiasedSubGrid()

```
void QCPPolarGrid::setAntialiasedSubGrid (
            bool enabled )
```

Sets whether sub grid lines are drawn antialiased.

17.663.4.6 setAntialiasedZeroLine()

```
void QCPPolarGrid::setAntialiasedZeroLine (
            bool enabled )
```

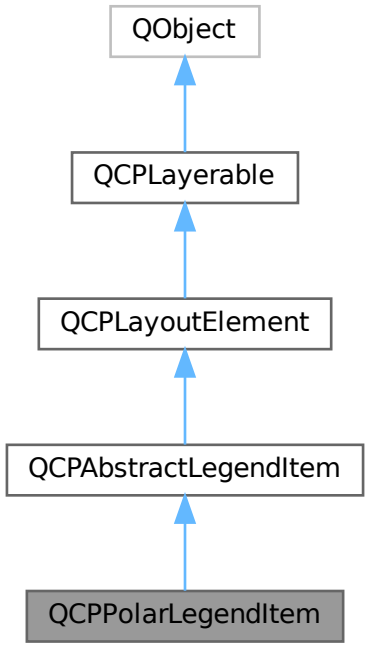
Sets whether zero lines are drawn antialiased.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

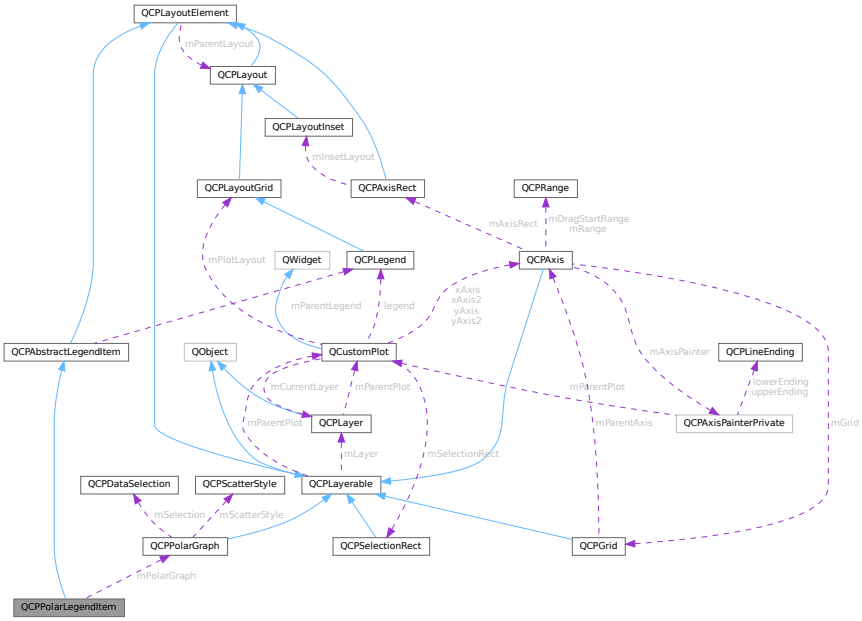
17.664 QCPPolarLegendItem Class Reference

A legend item for polar plots.

Inheritance diagram for QCPPolarLegendItem:



Collaboration diagram for QCPPolarLegendItem:



Public Member Functions

- **QCPPolarLegendItem** (QCPLegend ∗parent, QCPPolarGraph ∗graph)

- QCPPolarGraph * **polarGraph** ()

Public Member Functions inherited from QCPAbstractLegendItem

- QCPAbstractLegendItem (QCPLegend *parent)
- QCPLegend * **parentLegend** () const
- QFont **font** () const
- QColor **textColor** () const
- QFont **selectedFont** () const
- QColor **selectedTextColor** () const
- bool **selectable** () const
- bool **selected** () const
- void setFont (const QFont &font)
- void setTextColor (const QColor &color)
- void setSelectedFont (const QFont &font)
- void setSelectedTextColor (const QColor &color)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_↩ DECL_OVERRIDE

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot *parentPlot=nullptr)
- QCPLayout * layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const
- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)
- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)
- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual void update (UpdatePhase phase)
- virtual QSize maximumOuterSizeHint () const
- virtual QList< QCPLayoutElement * > elements (bool recursive) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual QSize minimumOuterSizeHint () const Q_DECL_OVERRIDE
- QPen **getIconBorderPen** () const
- QColor **getTextColor** () const
- QFont **getFont** () const

Protected Member Functions inherited from QCPAbstractLegendItem

- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayoutElement

- virtual int **calculateAutoMargin** (QCP::MarginSide side)
- virtual void **layoutChanged** ()
- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QCPPolarGraph * **mPolarGraph**

Protected Attributes inherited from QCPAbstractLegendItem

- QCPLegend * **mParentLegend**
- QFont **mFont**
- QColor **mTextColor**
- QFont **mSelectedFont**
- QColor **mSelectedTextColor**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayoutElement

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

Signals inherited from QCPAbstractLegendItem

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.664.1 Detailed Description

A legend item for polar plots.

Warning

> In this QCustomPlot version, polar plots are a tech preview. Expect documentation and functionality to be incomplete, as well as changing public interfaces in the future.

17.664.2 Member Function Documentation

17.664.2.1 draw()

```
void QCPPolarLegendItem::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractLegendItem.

17.664.2.2 minimumOuterSizeHint()

```
QSize QCPPolarLegendItem::minimumOuterSizeHint ( ) const  [protected], [virtual]
```

Returns the suggested minimum size this layout element (the outerRect) may be compressed to, if no manual minimum size is set.
if a minimum size (setMinimumSize) was not set manually, parent layouts use the returned size (usually indirectly through QCPLayout::getFinalMinimumOuterSize) to determine the minimum allowed size of this layout element.
A manual minimum size is considered set if it is non-zero.
The default implementation simply returns the sum of the horizontal margins for the width and the sum of the vertical margins for the height. Reimplementations may use their detailed knowledge about the layout element's content to provide size hints.
Reimplemented from QCPLayoutElement.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.665 QCPRange Class Reference

Represents the range an axis is encompassing.

Public Member Functions

- QCPRange ()
- QCPRange (double lower, double upper)
- bool **operator==** (const QCPRange &other) const
- bool **operator!=** (const QCPRange &other) const
- QCPRange & operator+= (const double &value)
- QCPRange & operator-= (const double &value)
- QCPRange & operator*= (const double &value)
- QCPRange & operator/= (const double &value)
- double size () const
- double center () const
- void normalize ()
- void expand (const QCPRange &otherRange)
- void expand (double includeCoord)
- QCPRange expanded (const QCPRange &otherRange) const
- QCPRange expanded (double includeCoord) const
- QCPRange bounded (double lowerBound, double upperBound) const
- QCPRange sanitizedForLogScale () const
- QCPRange sanitizedForLinScale () const
- bool contains (double value) const

Static Public Member Functions

- static bool validRange (double lower, double upper)
- static bool validRange (const QCPRange &range)

Public Attributes

- double **lower**
- double **upper**

Static Public Attributes

- static const double minRange = 1e-280
- static const double maxRange = 1e250

Friends

- const QCPRange operator+ (const QCPRange &, double)
- const QCPRange operator+ (double, const QCPRange &)
- const QCPRange operator- (const QCPRange &range, double value)
- const QCPRange operator* (const QCPRange &range, double value)
- const QCPRange operator* (double value, const QCPRange &range)
- const QCPRange operator/ (const QCPRange &range, double value)

Related Symbols

(Note that these are not member symbols.)

- QDebug operator<< (QDebug d, const QCPRange &range)

17.665.1 Detailed Description

Represents the range an axis is encompassing.
contains a *lower* and *upper* double value and provides convenience input, output and modification functions.

See also

QCPAxis::setRange

17.665.2 Constructor & Destructor Documentation

17.665.2.1 QCPRange() [1/2]

```
QCPRange::QCPRange ( )
```

Constructs a range with *lower* and *upper* set to zero.

17.665.2.2 QCPRange() [2/2]

```
QCPRange::QCPRange (
            double lower,
            double upper )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Constructs a range with the specified *lower* and *upper* values.
The resulting range will be normalized (see normalize), so if *lower* is not numerically smaller than *upper*, they will be swapped.

17.665.3 Member Function Documentation

17.665.3.1 bounded()

```
QCPRange QCPRange::bounded (
            double lowerBound,
            double upperBound ) const
```

Returns this range, possibly modified to not exceed the bounds provided as *lowerBound* and *upperBound*. If possible, the size of the current range is preserved in the process.

If the range shall only be bounded at the lower side, you can set *upperBound* to QCPRange::maxRange. If it shall only be bounded at the upper side, set *lowerBound* to -QCPRange::maxRange.

17.665.3.2 center()

```
double QCPRange::center ( ) const  [inline]
```

Returns the center of the range, i.e. (*upper+lower)∗0.5*

17.665.3.3 contains()

```
bool QCPRange::contains (
            double value ) const  [inline]
```

Returns true when *value* lies within or exactly on the borders of the range.

17.665.3.4 expand() [1/2]

```
void QCPRange::expand (
            const QCPRange & otherRange )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Expands this range such that *otherRange* is contained in the new range. It is assumed that both this range and *otherRange* are normalized (see normalize).
If this range contains NaN as lower or upper bound, it will be replaced by the respective bound of *otherRange*.
If *otherRange* is already inside the current range, this function does nothing.

See also

expanded

17.665.3.5 expand() [2/2]

```
void QCPRange::expand (
            double includeCoord )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Expands this range such that *includeCoord* is contained in the new range. It is assumed that this range is normalized (see normalize).
If this range contains NaN as lower or upper bound, the respective bound will be set to *includeCoord*.
If *includeCoord* is already inside the current range, this function does nothing.

See also

expand

17.665.3.6 expanded() [1/2]

```
QCPRange QCPRange::expanded (
            const QCPRange & otherRange ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Returns an expanded range that contains this and *otherRange*. It is assumed that both this range and *otherRange* are normalized (see normalize).
If this range contains NaN as lower or upper bound, the returned range's bound will be taken from *otherRange*.

See also

expand

17.665.3.7 expanded() [2/2]

```
QCPRange QCPRange::expanded (
            double includeCoord ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Returns an expanded range that includes the specified *includeCoord*. It is assumed that this range is normalized (see normalize).
If this range contains NaN as lower or upper bound, the returned range's bound will be set to *includeCoord*.

See also

expand

17.665.3.8 normalize()

```
void QCPRange::normalize ( )  [inline]
```

Makes sure *lower* is numerically smaller than *upper*. If this is not the case, the values are swapped.

17.665.3.9 operator*=()

```
QCPRange & QCPRange::operator*= (
            const double & value )  [inline]
```

Multiplies both boundaries of the range by *value*.

17.665.3.10 operator+=()

```
QCPRange & QCPRange::operator+= (
            const double & value )  [inline]
```

Adds *value* to both boundaries of the range.

17.665.3.11 operator-=()

```
QCPRange & QCPRange::operator-= (
            const double & value )  [inline]
```

Subtracts *value* from both boundaries of the range.

17.665.3.12 operator/=()

```
QCPRange & QCPRange::operator/= (
            const double & value )  [inline]
```

Divides both boundaries of the range by *value*.

17.665.3.13 sanitizedForLinScale()

```
QCPRange QCPRange::sanitizedForLinScale ( ) const
```

Returns a sanitized version of the range. Sanitized means for linear scales, that *lower* will always be numerically smaller (or equal) to *upper*.

17.665.3.14 sanitizedForLogScale()

```
QCPRange QCPRange::sanitizedForLogScale ( ) const
```

Returns a sanitized version of the range. Sanitized means for logarithmic scales, that the range won't span the positive and negative sign domain, i.e. contain zero. Further *lower* will always be numerically smaller (or equal) to *upper*.
If the original range does span positive and negative sign domains or contains zero, the returned range will try to approximate the original range as good as possible. If the positive interval of the original range is wider than the negative interval, the returned range will only contain the positive interval, with lower bound set to *rangeFac* or *rangeFac* ∗*upper*, whichever is closer to zero. Same procedure is used if the negative interval is wider than the positive interval, this time by changing the *upper* bound.

17.665.3.15 size()

```
double QCPRange::size ( ) const  [inline]
```
Returns the size of the range, i.e. *upper-lower*

17.665.3.16 validRange() [1/2]

```
bool QCPRange::validRange (
            const QCPRange & range )  [static]
```
This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts. Checks, whether the specified range is within valid bounds, which are defined as QCPRange::maxRange and QCPRange::minRange. A valid range means:

- range bounds within -maxRange and maxRange
- range size above minRange
- range size below maxRange

17.665.3.17 validRange() [2/2]

```
bool QCPRange::validRange (
            double lower,
            double upper )  [static]
```
Checks, whether the specified range is within valid bounds, which are defined as QCPRange::maxRange and QCPRange::minRange. A valid range means:

- range bounds within -maxRange and maxRange
- range size above minRange
- range size below maxRange

17.665.4 Friends And Related Symbol Documentation

17.665.4.1 operator* [1/2]

```
const QCPRange operator* (
            const QCPRange & range,
            double value )  [friend]
```
Multiplies both boundaries of the range by *value*.

17.665.4.2 operator* [2/2]

```
const QCPRange operator* (
            double value,
            const QCPRange & range )  [friend]
```
Multiplies both boundaries of the range by *value*.

17.665.4.3 operator+ [1/2]

```
const QCPRange operator+ (
            const QCPRange & range,
            double value )  [friend]
```
Adds *value* to both boundaries of the range.

17.665.4.4 operator+ [2/2]

```
const QCPRange operator+ (
            double value,
            const QCPRange & range )  [friend]
```
Adds *value* to both boundaries of the range.

17.665.4.5 operator-

```
const QCPRange operator- (
            const QCPRange & range,
            double value )  [friend]
```

Subtracts *value* from both boundaries of the range.

17.665.4.6 operator/

```
const QCPRange operator/ (
            const QCPRange & range,
            double value )  [friend]
```

Divides both boundaries of the range by *value*.

17.665.4.7 operator<<()

```
QDebug operator<< (
            QDebug d,
            const QCPRange & range )  [related]
```

Prints *range* in a human readable format to the qDebug output.

17.665.5 Member Data Documentation

17.665.5.1 maxRange

```
const double QCPRange::maxRange = 1e250  [static]
```

Maximum values (negative and positive) the range will accept in range-changing functions. Larger absolute values would cause errors due to the 11-bit exponent of double precision numbers, corresponding to a maximum magnitude of roughly 1e308.

Warning

Do not use this constant to indicate "arbitrarily large" values in plotting logic (as values that will appear in the plot)! It is intended only as a bound to compare against, e.g. to prevent axis ranges from obtaining overflowing ranges.

See also

validRange, minRange

17.665.5.2 minRange

```
const double QCPRange::minRange = 1e-280  [static]
```

Minimum range size (*upper* - *lower*) the range changing functions will accept. Smaller intervals would cause errors due to the 11-bit exponent of double precision numbers, corresponding to a minimum magnitude of roughly 1e-308.

Warning

Do not use this constant to indicate "arbitrarily small" values in plotting logic (as values that will appear in the plot)! It is intended only as a bound to compare against, e.g. to prevent axis ranges from obtaining underflowing ranges.

See also

validRange, maxRange

The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.666 QCPScatterStyle Class Reference

Represents the visual appearance of scatter points.

Public Types

- enum ScatterProperty {
  spNone = 0x00 , spPen = 0x01 , spBrush = 0x02 , spSize = 0x04 ,
  spShape = 0x08 , spAll = 0xFF }
- enum ScatterShape {
  ssNone , ssDot , ssCross , ssPlus ,
  ssCircle , ssDisc , ssSquare , ssDiamond ,
  ssStar , ssTriangle , ssTriangleInverted , ssCrossSquare ,
  ssPlusSquare , ssCrossCircle , ssPlusCircle , ssPeace ,
  ssPixmap , ssCustom }

Public Member Functions

- QCPScatterStyle ()
- QCPScatterStyle (ScatterShape shape, double size=6)
- QCPScatterStyle (ScatterShape shape, const QColor &color, double size)
- QCPScatterStyle (ScatterShape shape, const QColor &color, const QColor &fill, double size)
- QCPScatterStyle (ScatterShape shape, const QPen &pen, const QBrush &brush, double size)
- QCPScatterStyle (const QPixmap &pixmap)
- QCPScatterStyle (const QPainterPath &customPath, const QPen &pen, const QBrush &brush=Qt::NoBrush, double size=6)
- double **size** () const
- ScatterShape **shape** () const
- QPen **pen** () const
- QBrush **brush** () const
- QPixmap **pixmap** () const
- QPainterPath **customPath** () const
- void setFromOther (const QCPScatterStyle &other, ScatterProperties properties)
- void setSize (double size)
- void setShape (ScatterShape shape)
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setPixmap (const QPixmap &pixmap)
- void setCustomPath (const QPainterPath &customPath)
- bool isNone () const
- bool isPenDefined () const
- void undefinePen ()
- void applyTo (QCPPainter *painter, const QPen &defaultPen) const
- void drawShape (QCPPainter *painter, const QPointF &pos) const
- void drawShape (QCPPainter *painter, double x, double y) const

Protected Attributes

- double **mSize**
- ScatterShape **mShape**
- QPen **mPen**
- QBrush **mBrush**
- QPixmap **mPixmap**
- QPainterPath **mCustomPath**
- bool **mPenDefined**

17.666.1 Detailed Description

Represents the visual appearance of scatter points.
This class holds information about shape, color and size of scatter points. In plottables like QCPGraph it is used to store how scatter points shall be drawn. For example, QCPGraph::setScatterStyle takes a QCPScatterStyle instance.
A scatter style consists of a shape (setShape), a line color (setPen) and possibly a fill (setBrush), if the shape provides a fillable area. Further, the size of the shape can be controlled with setSize.

17.666.2 Specifying a scatter style

You can set all these configurations either by calling the respective functions on an instance:
Or you can use one of the various constructors that take different parameter combinations, making it easy to specify a scatter style in a single call, like so:

17.666.3 Leaving the color/pen up to the plottable

There are two constructors which leave the pen undefined: QCPScatterStyle() and QCPScatterStyle(ScatterShape shape, double size
If those constructors are used, a call to isPenDefined will return false. It leads to scatter points that inherit the pen from the plottable that uses the scatter style. Thus, if such a scatter style is passed to QCPGraph, the line color of the graph (QCPGraph::setPen) will be used by the scatter points. This makes it very convenient to set up typical scatter settings:
Notice that it wasn't even necessary to explicitly call a QCPScatterStyle constructor. This works because QCPScatterStyle provides a constructor that can transform a ScatterShape directly into a QCPScatterStyle instance (that's the QCPScatterStyle(ScatterShape shape, double size) constructor with a default for *size*). In those cases, C++ allows directly supplying a ScatterShape, where actually a QCPScatterStyle is expected.

17.666.4 Custom shapes and pixmaps

QCPScatterStyle supports drawing custom shapes and arbitrary pixmaps as scatter points.
For custom shapes, you can provide a QPainterPath with the desired shape to the setCustomPath function or call the constructor that takes a painter path. The scatter shape will automatically be set to ssCustom.
For pixmaps, you call setPixmap with the desired QPixmap. Alternatively you can use the constructor that takes a QPixmap. The scatter shape will automatically be set to ssPixmap. Note that setSize does not influence the appearance of the pixmap.

17.666.5 Member Enumeration Documentation

17.666.5.1 ScatterProperty

`enum QCPScatterStyle::ScatterProperty`
Represents the various properties of a scatter style instance. For example, this enum is used to specify which properties of QCPSelectionDecorator::setScatterStyle will be used when highlighting selected data points.
Specific scatter properties can be transferred between QCPScatterStyle instances via setFromOther.

Enumerator

spNone	`0x00` None
spPen	`0x01` The pen property, see setPen
spBrush	`0x02` The brush property, see setBrush
spSize	`0x04` The size property, see setSize
spShape	`0x08` The shape property, see setShape
spAll	`0xFF` All properties

17.666.5.2 ScatterShape

`enum QCPScatterStyle::ScatterShape`

Defines the shape used for scatter points.
On plottables/items that draw scatters, the sizes of these visualizations (with exception of ssDot and ssPixmap) can be controlled with the setSize function. Scatters are drawn with the pen and brush specified with setPen and setBrush.

Enumerator

ssNone	no scatter symbols are drawn (e.g. in QCPGraph, data only represented with lines)
ssDot	\enumimage{ssDot.png} a single pixel (use ssDisc or ssCircle if you want a round shape with a certain radius)
ssCross	\enumimage{ssCross.png} a cross
ssPlus	\enumimage{ssPlus.png} a plus
ssCircle	\enumimage{ssCircle.png} a circle
ssDisc	\enumimage{ssDisc.png} a circle which is filled with the pen's color (not the brush as with ssCircle)
ssSquare	\enumimage{ssSquare.png} a square
ssDiamond	\enumimage{ssDiamond.png} a diamond
ssStar	\enumimage{ssStar.png} a star with eight arms, i.e. a combination of cross and plus
ssTriangle	\enumimage{ssTriangle.png} an equilateral triangle, standing on baseline
ssTriangleInverted	\enumimage{ssTriangleInverted.png} an equilateral triangle, standing on corner
ssCrossSquare	\enumimage{ssCrossSquare.png} a square with a cross inside
ssPlusSquare	\enumimage{ssPlusSquare.png} a square with a plus inside
ssCrossCircle	\enumimage{ssCrossCircle.png} a circle with a cross inside
ssPlusCircle	\enumimage{ssPlusCircle.png} a circle with a plus inside
ssPeace	\enumimage{ssPeace.png} a circle, with one vertical and two downward diagonal lines
ssPixmap	a custom pixmap specified by setPixmap, centered on the data point coordinates
ssCustom	custom painter operations are performed per scatter (As QPainterPath, see setCustomPath)

17.666.6 Constructor & Destructor Documentation

17.666.6.1 QCPScatterStyle() [1/7]

```
QCPScatterStyle::QCPScatterStyle ( )
```

Creates a new QCPScatterStyle instance with size set to 6. No shape, pen or brush is defined.
Since the pen is undefined (isPenDefined returns false), the scatter color will be inherited from the plottable that uses this scatter style.

17.666.6.2 QCPScatterStyle() [2/7]

```
QCPScatterStyle::QCPScatterStyle (
            ScatterShape shape,
            double size = 6 )
```

Creates a new QCPScatterStyle instance with shape set to *shape* and size to *size*. No pen or brush is defined.
Since the pen is undefined (isPenDefined returns false), the scatter color will be inherited from the plottable that uses this scatter style.

17.666.6.3 QCPScatterStyle() [3/7]

```
QCPScatterStyle::QCPScatterStyle (
            ScatterShape shape,
            const QColor & color,
            double size )
```

Creates a new QCPScatterStyle instance with shape set to *shape*, the pen color set to *color*, and size to *size*. No brush is defined, i.e. the scatter point will not be filled.

17.666.6.4 QCPScatterStyle() [4/7]

```
QCPScatterStyle::QCPScatterStyle (
            ScatterShape shape,
            const QColor & color,
            const QColor & fill,
            double size )
```

Creates a new QCPScatterStyle instance with shape set to *shape*, the pen color set to *color*, the brush color to *fill* (with a solid pattern), and size to *size*.

17.666.6.5 QCPScatterStyle() [5/7]

```
QCPScatterStyle::QCPScatterStyle (
            ScatterShape shape,
            const QPen & pen,
            const QBrush & brush,
            double size )
```

Creates a new QCPScatterStyle instance with shape set to *shape*, the pen set to *pen*, the brush to *brush*, and size to *size*.

Warning

> In some cases it might be tempting to directly use a pen style like `Qt::NoPen` as *pen* and a color like `Qt::blue` as *brush*. Notice however, that the corresponding call `QCPScatterStyle(QCPScatterShape::ssCircle, Qt::NoPen, Qt::blue, 5)` doesn't necessarily lead C++ to use this constructor in some cases, but might mistake `Qt::NoPen` for a QColor and use the QCPScatterStyle(ScatterShape shape, const QColor &color, const QColor &fill, double size) constructor instead (which will lead to an unexpected look of the scatter points). To prevent this, be more explicit with the parameter types. For example, use `QBrush(Qt::blue)` instead of just `Qt::blue`, to clearly point out to the compiler that this constructor is wanted.

17.666.6.6 QCPScatterStyle() [6/7]

```
QCPScatterStyle::QCPScatterStyle (
            const QPixmap & pixmap )
```

Creates a new QCPScatterStyle instance which will show the specified *pixmap*. The scatter shape is set to ssPixmap.

17.666.6.7 QCPScatterStyle() [7/7]

```
QCPScatterStyle::QCPScatterStyle (
            const QPainterPath & customPath,
            const QPen & pen,
            const QBrush & brush = Qt::NoBrush,
            double size = 6 )
```

Creates a new QCPScatterStyle instance with a custom shape that is defined via *customPath*. The scatter shape is set to ssCustom.

The custom shape line will be drawn with *pen* and filled with *brush*. The size has a slightly different meaning than for built-in scatter points: The custom path will be drawn scaled by a factor of *size/6.0*. Since the default *size* is 6, the custom path will appear in its original size by default. To for example double the size of the path, set *size* to 12.

17.666.7 Member Function Documentation

17.666.7.1 applyTo()

```
void QCPScatterStyle::applyTo (
```

```
            QCPPainter * painter,
            const QPen & defaultPen ) const
```

Applies the pen and the brush of this scatter style to *painter*. If this scatter style has an undefined pen (isPenDefined), sets the pen of *painter* to *defaultPen* instead.
This function is used by plottables (or any class that wants to draw scatters) just before a number of scatters with this style shall be drawn with the *painter*.

See also

drawShape

17.666.7.2 drawShape() [1/2]

```
void QCPScatterStyle::drawShape (
            QCPPainter * painter,
            const QPointF & pos ) const
```

Draws the scatter shape with *painter* at position *pos*.
This function does not modify the pen or the brush on the painter, as applyTo is meant to be called before scatter points are drawn with drawShape.

See also

applyTo

17.666.7.3 drawShape() [2/2]

```
void QCPScatterStyle::drawShape (
            QCPPainter * painter,
            double x,
            double y ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts. Draws the scatter shape with *painter* at position *x* and *y*.

17.666.7.4 isNone()

```
bool QCPScatterStyle::isNone ( ) const  [inline]
```

Returns whether the scatter shape is ssNone.

See also

setShape

17.666.7.5 isPenDefined()

```
bool QCPScatterStyle::isPenDefined ( ) const  [inline]
```

Returns whether a pen has been defined for this scatter style.
The pen is undefined if a constructor is called that does not carry *pen* as parameter. Those are QCPScatterStyle() and QCPScatterStyle(ScatterShape shape, double size). If the pen is undefined, the pen of the respective plottable will be used for drawing scatters.
If a pen was defined for this scatter style instance, and you now wish to undefine the pen, call undefinePen.

See also

setPen

17.666.7.6 setBrush()

```
void QCPScatterStyle::setBrush (
            const QBrush & brush )
```

Sets the brush that will be used to fill scatter points to *brush*. Note that not all scatter shapes have fillable areas. For example, ssPlus does not while ssCircle does.

See also

setPen

17.666.7.7 setCustomPath()

```
void QCPScatterStyle::setCustomPath (
            const QPainterPath & customPath )
```

Sets the custom shape that will be drawn as scatter point to *customPath*.
The scatter shape is automatically set to ssCustom.

17.666.7.8 setFromOther()

```
void QCPScatterStyle::setFromOther (
            const QCPScatterStyle & other,
            ScatterProperties properties )
```

Copies the specified *properties* from the *other* scatter style to this scatter style.

17.666.7.9 setPen()

```
void QCPScatterStyle::setPen (
            const QPen & pen )
```

Sets the pen that will be used to draw scatter points to *pen*.
If the pen was previously undefined (see isPenDefined), the pen is considered defined after a call to this function, even if *pen* is `Qt::NoPen`. If you have defined a pen previously by calling this function and now wish to undefine the pen, call undefinePen.

See also

setBrush

17.666.7.10 setPixmap()

```
void QCPScatterStyle::setPixmap (
            const QPixmap & pixmap )
```

Sets the pixmap that will be drawn as scatter point to *pixmap*.
Note that setSize does not influence the appearance of the pixmap.
The scatter shape is automatically set to ssPixmap.

17.666.7.11 setShape()

```
void QCPScatterStyle::setShape (
            QCPScatterStyle::ScatterShape shape )
```

Sets the shape to *shape*.
Note that the calls setPixmap and setCustomPath automatically set the shape to ssPixmap and ssCustom, respectively.

See also

setSize

17.666.7.12 setSize()

```
void QCPScatterStyle::setSize (
            double size )
```

Sets the size (pixel diameter) of the drawn scatter points to *size*.

See also

setShape

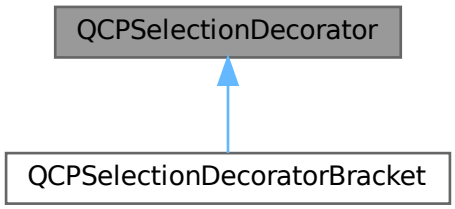
17.666.7.13 undefinePen()

`void QCPScatterStyle::undefinePen ( )`
Sets this scatter style to have an undefined pen (see isPenDefined for what an undefined pen implies).
A call to setPen will define a pen.
The documentation for this class was generated from the following files:

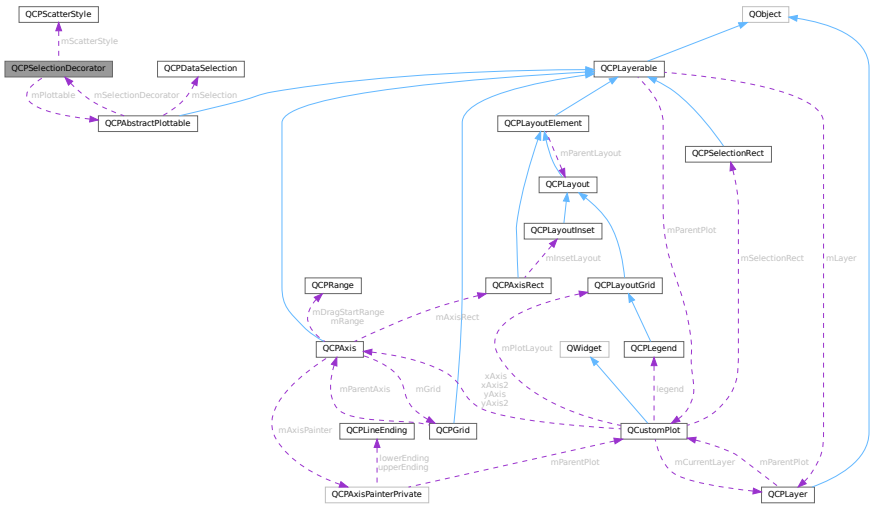
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.667 QCPSelectionDecorator Class Reference

Controls how a plottable's data selection is drawn.
Inheritance diagram for QCPSelectionDecorator:



Collaboration diagram for QCPSelectionDecorator:



Public Member Functions

- QCPSelectionDecorator ()
- QPen **pen** () const
- QBrush **brush** () const
- QCPScatterStyle **scatterStyle** () const
- QCPScatterStyle::ScatterProperties **usedScatterProperties** () const

- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setScatterStyle (const QCPScatterStyle &scatterStyle, QCPScatterStyle::ScatterProperties usedProperties=QCPScatterStyle::spPen)
- void setUsedScatterProperties (const QCPScatterStyle::ScatterProperties &properties)
- void applyPen (QCPPainter *painter) const
- void applyBrush (QCPPainter *painter) const
- QCPScatterStyle getFinalScatterStyle (const QCPScatterStyle &unselectedStyle) const
- virtual void copyFrom (const QCPSelectionDecorator *other)
- virtual void drawDecoration (QCPPainter *painter, QCPDataSelection selection)

Protected Member Functions

- virtual bool **registerWithPlottable** (QCPAbstractPlottable *plottable)

Protected Attributes

- QPen **mPen**
- QBrush **mBrush**
- QCPScatterStyle **mScatterStyle**
- QCPScatterStyle::ScatterProperties **mUsedScatterProperties**
- QCPAbstractPlottable * **mPlottable**

Friends

- class **QCPAbstractPlottable**

17.667.1 Detailed Description

Controls how a plottable's data selection is drawn.
Each QCPAbstractPlottable instance has one QCPSelectionDecorator (accessible via QCPAbstractPlottable::selectionDecorator) and uses it when drawing selected segments of its data.
The selection decorator controls both pen (setPen) and brush (setBrush), as well as the scatter style (setScatterStyle) if the plottable draws scatters. Since a QCPScatterStyle is itself composed of different properties such as color shape and size, the decorator allows specifying exactly which of those properties shall be used for the selected data point, via setUsedScatterProperties.
A QCPSelectionDecorator subclass instance can be passed to a plottable via QCPAbstractPlottable::setSelectionDecorator, allowing greater customizability of the appearance of selected segments.
Use copyFrom to easily transfer the settings of one decorator to another one. This is especially useful since plottables take ownership of the passed selection decorator, and thus the same decorator instance can not be passed to multiple plottables.
Selection decorators can also themselves perform drawing operations by reimplementing drawDecoration, which is called by the plottable's draw method. The base class QCPSelectionDecorator does not make use of this however. For example, QCPSelectionDecoratorBracket draws brackets around selected data segments.

17.667.2 Constructor & Destructor Documentation

17.667.2.1 QCPSelectionDecorator()

```
QCPSelectionDecorator::QCPSelectionDecorator ( )
```

Creates a new QCPSelectionDecorator instance with default values

17.667.3 Member Function Documentation

17.667.3.1 applyBrush()

```
void QCPSelectionDecorator::applyBrush (
            QCPPainter * painter ) const
```

Sets the brush of *painter* to the brush of this selection decorator.

See also

applyPen, getFinalScatterStyle

17.667.3.2 applyPen()

```
void QCPSelectionDecorator::applyPen (
            QCPPainter * painter ) const
```

Sets the pen of *painter* to the pen of this selection decorator.

See also

applyBrush, getFinalScatterStyle

17.667.3.3 copyFrom()

```
void QCPSelectionDecorator::copyFrom (
            const QCPSelectionDecorator * other )  [virtual]
```

Copies all properties (e.g. color, fill, scatter style) of the *other* selection decorator to this selection decorator.

17.667.3.4 drawDecoration()

```
void QCPSelectionDecorator::drawDecoration (
            QCPPainter * painter,
            QCPDataSelection selection )  [virtual]
```

This method is called by all plottables' draw methods to allow custom selection decorations to be drawn. Use the passed *painter* to perform the drawing operations. *selection* carries the data selection for which the decoration shall be drawn.
The default base class implementation of QCPSelectionDecorator has no special decoration, so this method does nothing.
Reimplemented in QCPSelectionDecoratorBracket.

17.667.3.5 getFinalScatterStyle()

```
QCPScatterStyle QCPSelectionDecorator::getFinalScatterStyle (
            const QCPScatterStyle & unselectedStyle ) const
```

Returns the scatter style that the parent plottable shall use for selected scatter points. The plottable's original (unselected) scatter style must be passed as *unselectedStyle*. Depending on the setting of setUsedScatterProperties, the returned scatter style is a mixture of this selecion decorator's scatter style (setScatterStyle), and *unselectedStyle*.

See also

applyPen, applyBrush, setScatterStyle

17.667.3.6 setBrush()

```
void QCPSelectionDecorator::setBrush (
            const QBrush & brush )
```

Sets the brush that will be used by the parent plottable to draw selected data segments.

17.667.3.7 setPen()

```
void QCPSelectionDecorator::setPen (
            const QPen & pen )
```

Sets the pen that will be used by the parent plottable to draw selected data segments.

17.667.3.8 setScatterStyle()

```
void QCPSelectionDecorator::setScatterStyle (
            const QCPScatterStyle & scatterStyle,
            QCPScatterStyle::ScatterProperties usedProperties = QCPScatterStyle::spPen )
```

Sets the scatter style that will be used by the parent plottable to draw scatters in selected data segments. *usedProperties* specifies which parts of the passed *scatterStyle* will be used by the plottable. The used properties can also be changed via setUsedScatterProperties.

17.667.3.9 setUsedScatterProperties()

```
void QCPSelectionDecorator::setUsedScatterProperties (
            const QCPScatterStyle::ScatterProperties & properties )
```

Use this method to define which properties of the scatter style (set via setScatterStyle) will be used for selected data segments. All properties of the scatter style that are not specified in *properties* will remain as specified in the plottable's original scatter style.

See also

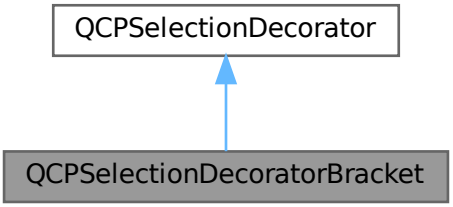
QCPScatterStyle::ScatterProperty

The documentation for this class was generated from the following files:

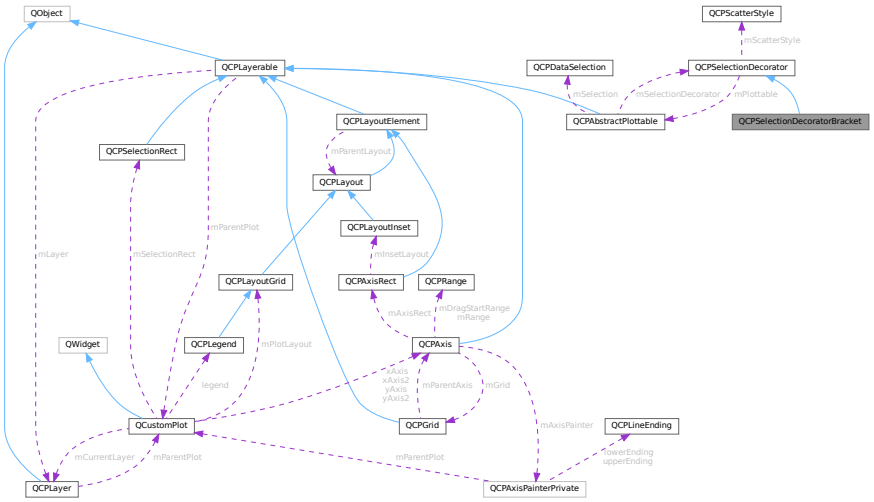
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.668 QCPSelectionDecoratorBracket Class Reference

A selection decorator which draws brackets around each selected data segment.
Inheritance diagram for QCPSelectionDecoratorBracket:



Collaboration diagram for QCPSelectionDecoratorBracket:



Public Types

- enum BracketStyle {
  bsSquareBracket , bsHalfEllipse , bsEllipse , bsPlus ,
  bsUserStyle }

Public Member Functions

- QCPSelectionDecoratorBracket ()
- QPen **bracketPen** () const
- QBrush **bracketBrush** () const
- int **bracketWidth** () const
- int **bracketHeight** () const
- BracketStyle **bracketStyle** () const
- bool **tangentToData** () const
- int **tangentAverage** () const
- void setBracketPen (const QPen &pen)
- void setBracketBrush (const QBrush &brush)
- void setBracketWidth (int width)
- void setBracketHeight (int height)
- void setBracketStyle (BracketStyle style)
- void setTangentToData (bool enabled)
- void setTangentAverage (int pointCount)
- virtual void drawBracket (QCPPainter *painter, int direction) const
- virtual void drawDecoration (QCPPainter *painter, QCPDataSelection selection) Q_DECL_OVERRIDE

Public Member Functions inherited from QCPSelectionDecorator

- QCPSelectionDecorator ()
- QPen **pen** () const
- QBrush **brush** () const
- QCPScatterStyle **scatterStyle** () const
- QCPScatterStyle::ScatterProperties **usedScatterProperties** () const
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)

- void setScatterStyle (const QCPScatterStyle &scatterStyle, QCPScatterStyle::ScatterProperties usedProperties=QCPScatterStyle::spPen)
- void setUsedScatterProperties (const QCPScatterStyle::ScatterProperties &properties)
- void applyPen (QCPPainter *painter) const
- void applyBrush (QCPPainter *painter) const
- QCPScatterStyle getFinalScatterStyle (const QCPScatterStyle &unselectedStyle) const
- virtual void copyFrom (const QCPSelectionDecorator *other)

Protected Member Functions

- double **getTangentAngle** (const QCPPlottableInterface1D *interface1d, int dataIndex, int direction) const
- QPointF **getPixelCoordinates** (const QCPPlottableInterface1D *interface1d, int dataIndex) const

Protected Member Functions inherited from QCPSelectionDecorator

- virtual bool **registerWithPlottable** (QCPAbstractPlottable *plottable)

Protected Attributes

- QPen **mBracketPen**
- QBrush **mBracketBrush**
- int **mBracketWidth**
- int **mBracketHeight**
- BracketStyle **mBracketStyle**
- bool **mTangentToData**
- int **mTangentAverage**

Protected Attributes inherited from QCPSelectionDecorator

- QPen **mPen**
- QBrush **mBrush**
- QCPScatterStyle **mScatterStyle**
- QCPScatterStyle::ScatterProperties **mUsedScatterProperties**
- QCPAbstractPlottable * **mPlottable**

17.668.1 Detailed Description

A selection decorator which draws brackets around each selected data segment.
Additionally to the regular highlighting of selected segments via color, fill and scatter style, this QCPSelectionDecorator subclass draws markers at the begin and end of each selected data segment of the plottable.
The shape of the markers can be controlled with setBracketStyle, setBracketWidth and setBracketHeight. The color/fill can be controlled with setBracketPen and setBracketBrush.
To introduce custom bracket styles, it is only necessary to sublcass QCPSelectionDecoratorBracket and reimplement drawBracket. The rest will be managed by the base class.

17.668.2 Member Enumeration Documentation

17.668.2.1 BracketStyle

`enum QCPSelectionDecoratorBracket::BracketStyle`
Defines which shape is drawn at the boundaries of selected data ranges.
Some of the bracket styles further allow specifying a height and/or width, see setBracketHeight and setBracketWidth.

Enumerator

bsSquareBracket	A square bracket is drawn.
bsHalfEllipse	A half ellipse is drawn. The size of the ellipse is given by the bracket width/height properties.
bsEllipse	An ellipse is drawn. The size of the ellipse is given by the bracket width/height properties.

Enumerator

bsPlus	A plus is drawn.
bsUserStyle	Start custom bracket styles at this index when subclassing and reimplementing drawBracket.

17.668.3 Constructor & Destructor Documentation

17.668.3.1 QCPSelectionDecoratorBracket()

```
QCPSelectionDecoratorBracket::QCPSelectionDecoratorBracket ( )
```

Creates a new QCPSelectionDecoratorBracket instance with default values.

17.668.4 Member Function Documentation

17.668.4.1 drawBracket()

```
void QCPSelectionDecoratorBracket::drawBracket (
            QCPPainter * painter,
            int direction ) const  [virtual]
```

Draws the bracket shape with *painter*. The parameter *direction* is either -1 or 1 and indicates whether the bracket shall point to the left or the right (i.e. is a closing or opening bracket, respectively).
The passed *painter* already contains all transformations that are necessary to position and rotate the bracket appropriately. Painting operations can be performed as if drawing upright brackets on flat data with horizontal key axis, with (0, 0) being the center of the bracket.
If you wish to sublcass QCPSelectionDecoratorBracket in order to provide custom bracket shapes (see QCPSelectionDecoratorBracket::bsUserStyle), this is the method you should reimplement.

17.668.4.2 drawDecoration()

```
void QCPSelectionDecoratorBracket::drawDecoration (
            QCPPainter * painter,
            QCPDataSelection selection )  [virtual]
```

Draws the bracket decoration on the data points at the begin and end of each selected data segment given in *seletion*.
It uses the method drawBracket to actually draw the shapes.
\seebaseclassmethod
Reimplemented from QCPSelectionDecorator.

17.668.4.3 setBracketBrush()

```
void QCPSelectionDecoratorBracket::setBracketBrush (
            const QBrush & brush )
```

Sets the brush that will be used to draw the brackets at the beginning and end of each selected data segment.

17.668.4.4 setBracketHeight()

```
void QCPSelectionDecoratorBracket::setBracketHeight (
            int height )
```

Sets the height of the drawn bracket. The height dimension is always perpendicular to the key axis of the data, or the tangent direction of the current data slope, if setTangentToData is enabled.

17.668.4.5 setBracketPen()

```
void QCPSelectionDecoratorBracket::setBracketPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the brackets at the beginning and end of each selected data segment.

17.668.4.6 setBracketStyle()

```
void QCPSelectionDecoratorBracket::setBracketStyle (
            QCPSelectionDecoratorBracket::BracketStyle style )
```

Sets the shape that the bracket/marker will have.

See also

setBracketWidth, setBracketHeight

17.668.4.7 setBracketWidth()

```
void QCPSelectionDecoratorBracket::setBracketWidth (
            int width )
```

Sets the width of the drawn bracket. The width dimension is always parallel to the key axis of the data, or the tangent direction of the current data slope, if setTangentToData is enabled.

17.668.4.8 setTangentAverage()

```
void QCPSelectionDecoratorBracket::setTangentAverage (
            int pointCount )
```

Controls over how many data points the slope shall be averaged, when brackets shall be aligned with the data (if setTangentToData is true).
From the position of the bracket, *pointCount* points towards the selected data range will be taken into account. The smallest value of *pointCount* is 1, which is effectively equivalent to disabling setTangentToData.

17.668.4.9 setTangentToData()

```
void QCPSelectionDecoratorBracket::setTangentToData (
            bool enabled )
```

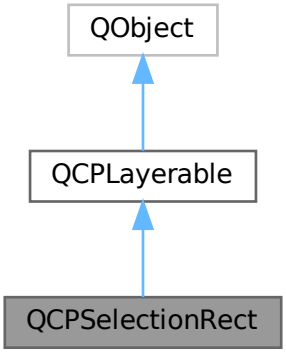
Sets whether the brackets will be rotated such that they align with the slope of the data at the position that they appear in.
For noisy data, it might be more visually appealing to average the slope over multiple data points. This can be configured via setTangentAverage.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

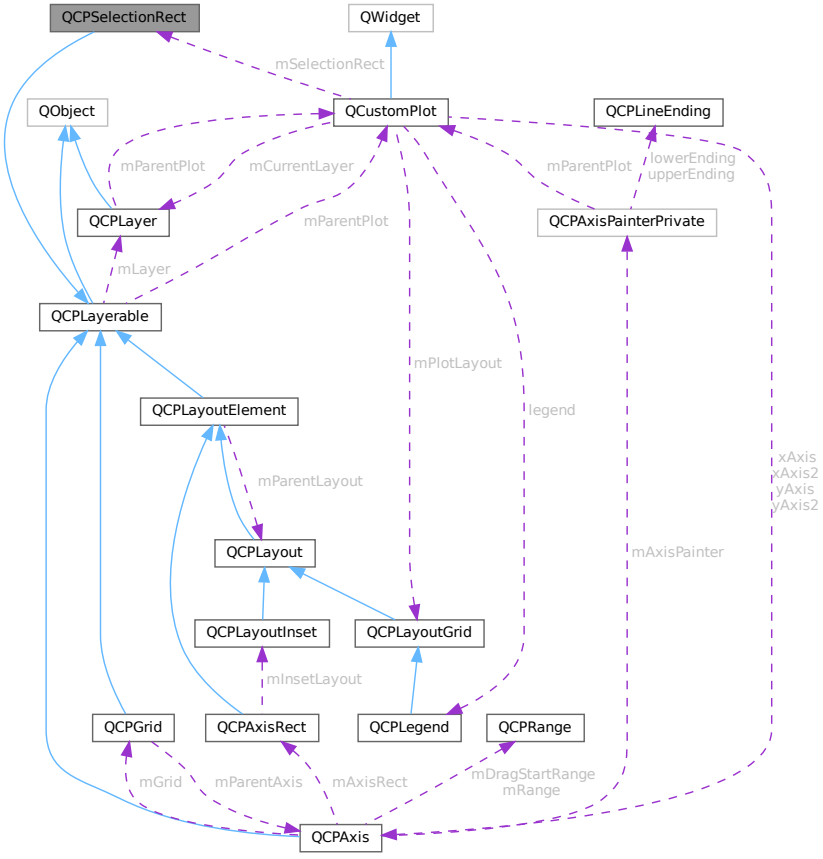
17.669 QCPSelectionRect Class Reference

Provides rect/rubber-band data selection and range zoom interaction.

Inheritance diagram for QCPSelectionRect:



Collaboration diagram for QCPSelectionRect:



Signals

- void started (QMouseEvent ∗event)

- void changed (const QRect &rect, QMouseEvent *event)
- void canceled (const QRect &rect, QInputEvent *event)
- void accepted (const QRect &rect, QMouseEvent *event)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPSelectionRect (QCustomPlot *parentPlot)
- QRect **rect** () const
- QCPRange range (const QCPAxis *axis) const
- QPen **pen** () const
- QBrush **brush** () const
- bool isActive () const
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- Q_SLOT void cancel ()

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const
- bool realVisibility () const

Protected Member Functions

- virtual void **startSelection** (QMouseEvent *event)
- virtual void **moveSelection** (QMouseEvent *event)
- virtual void **endSelection** (QMouseEvent *event)
- virtual void **keyPressEvent** (QKeyEvent *event)
- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void **selectEvent** (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed)
- virtual void **deselectEvent** (bool *selectionStateChanged)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)

- virtual void wheelEvent (QWheelEvent ∗event)
- void **initializeParentPlot** (QCustomPlot ∗parentPlot)
- void **setParentLayerable** (QCPLayerable ∗parentLayerable)
- bool **moveToLayer** (QCPLayer ∗layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter ∗painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QRect **mRect**
- QPen **mPen**
- QBrush **mBrush**
- bool **mActive**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot ∗ **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer ∗ **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**

17.669.1 Detailed Description

Provides rect/rubber-band data selection and range zoom interaction.
QCPSelectionRect is used by QCustomPlot when the QCustomPlot::setSelectionRectMode is not QCP::srmNone. When the user drags the mouse across the plot, the current selection rect instance (QCustomPlot::setSelectionRect) is forwarded these events and makes sure an according rect shape is drawn. At the begin, during, and after completion of the interaction, it emits the corresponding signals started, changed, canceled, and accepted.
The QCustomPlot instance connects own slots to the current selection rect instance, in order to react to an accepted selection rect interaction accordingly.
isActive can be used to check whether the selection rect is currently active. An ongoing selection interaction can be cancelled programmatically via calling cancel at any time.
The appearance of the selection rect can be controlled via setPen and setBrush.
If you wish to provide custom behaviour, e.g. a different visual representation of the selection rect (QCPSelectionRect::draw), you can subclass QCPSelectionRect and pass an instance of your subclass to QCustomPlot::setSelectionRect.

17.669.2 Constructor & Destructor Documentation

17.669.2.1 QCPSelectionRect()

```
QCPSelectionRect::QCPSelectionRect (
            QCustomPlot * parentPlot )  [explicit]
```

Creates a new QCPSelectionRect instance. To make QCustomPlot use the selection rect instance, pass it to QCustomPlot::setSelectionRect. *parentPlot* should be set to the same QCustomPlot widget.

17.669.3 Member Function Documentation

17.669.3.1 accepted

```
void QCPSelectionRect::accepted (
            const QRect & rect,
            QMouseEvent * event )  [signal]
```

This signal is emitted when the selection interaction was completed by the user releasing the mouse button.

Note that *rect* may have a negative width or height, if the selection is being dragged to the upper or left side of the selection rect origin.

17.669.3.2 applyDefaultAntialiasingHint()

```
void QCPSelectionRect::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```

Implements QCPLayerable.

17.669.3.3 cancel()

```
void QCPSelectionRect::cancel ( )
```

If there is currently a selection interaction going on (isActive), the interaction is canceled. The selection rect will emit the canceled signal.

17.669.3.4 canceled

```
void QCPSelectionRect::canceled (
            const QRect & rect,
            QInputEvent * event )  [signal]
```

This signal is emitted when the selection interaction was cancelled. Note that *event* is `nullptr` if the selection interaction was cancelled programmatically, by a call to cancel.
The user may cancel the selection interaction by pressing the escape key. In this case, *event* holds the respective input event.
Note that *rect* may have a negative width or height, if the selection is being dragged to the upper or left side of the selection rect origin.

17.669.3.5 changed

```
void QCPSelectionRect::changed (
            const QRect & rect,
            QMouseEvent * event )  [signal]
```

This signal is emitted while the selection rect interaction is ongoing and the *rect* has changed its size due to the user moving the mouse.
Note that *rect* may have a negative width or height, if the selection is being dragged to the upper or left side of the selection rect origin.

17.669.3.6 draw()

```
void QCPSelectionRect::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPLayerable.

17.669.3.7 isActive()

```
bool QCPSelectionRect::isActive ( ) const  [inline]
```

Returns true if there is currently a selection going on, i.e. the user has started dragging a selection rect, but hasn't released the mouse button yet.

See also

cancel

17.669.3.8 range()

```
QCPRange QCPSelectionRect::range (
            const QCPAxis * axis ) const
```

A convenience function which returns the coordinate range of the provided *axis*, that this selection rect currently encompasses.

17.669.3.9 setBrush()

```
void QCPSelectionRect::setBrush (
            const QBrush & brush )
```

Sets the brush that will be used to fill the selection rect. By default the selection rect is not filled, i.e. *brush* is `Qt::NoBrush`.

See also

setPen

17.669.3.10 setPen()

```
void QCPSelectionRect::setPen (
            const QPen & pen )
```

Sets the pen that will be used to draw the selection rect outline.

See also

setBrush

17.669.3.11 started

```
void QCPSelectionRect::started (
            QMouseEvent * event )  [signal]
```

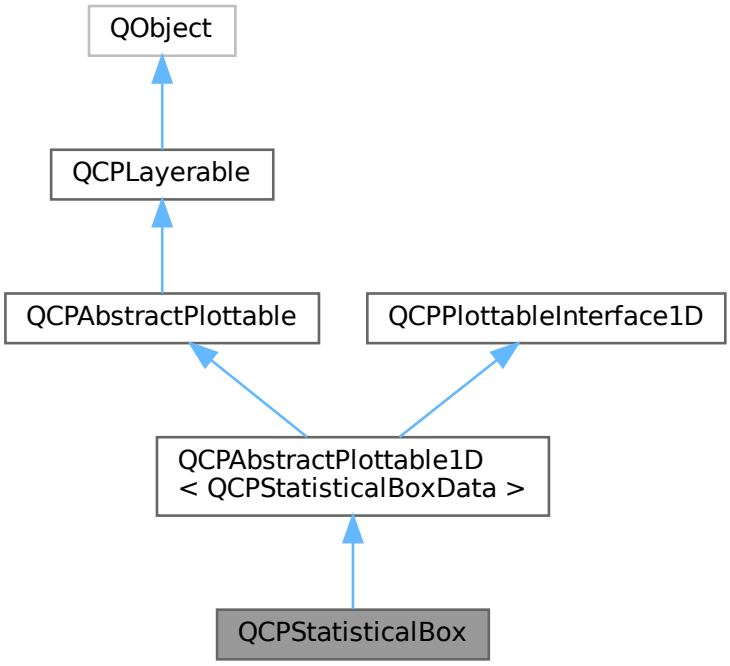
This signal is emitted when a selection rect interaction was initiated, i.e. the user just started dragging the selection rect with the mouse.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

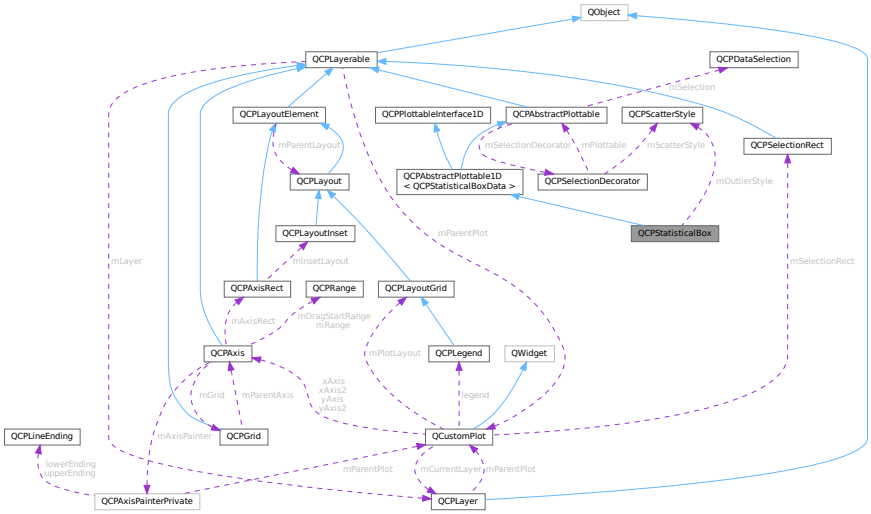
17.670 QCPStatisticalBox Class Reference

A plottable representing a single statistical box in a plot.

Inheritance diagram for QCPStatisticalBox:



Collaboration diagram for QCPStatisticalBox:



Public Member Functions

- QCPStatisticalBox (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QSharedPointer< QCPStatisticalBoxDataContainer > data () const
- double **width** () const

- double **whiskerWidth** () const
- QPen **whiskerPen** () const
- QPen **whiskerBarPen** () const
- bool **whiskerAntialiased** () const
- QPen **medianPen** () const
- QCPScatterStyle **outlierStyle** () const
- void setData (QSharedPointer< QCPStatisticalBoxDataContainer > data)
- void setData (const QVector< double > &keys, const QVector< double > &minimum, const QVector< double > &lowerQuartile, const QVector< double > &median, const QVector< double > &upperQuartile, const QVector< double > &maximum, bool alreadySorted=false)
- void setWidth (double width)
- void setWhiskerWidth (double width)
- void setWhiskerPen (const QPen &pen)
- void setWhiskerBarPen (const QPen &pen)
- void setWhiskerAntialiased (bool enabled)
- void setMedianPen (const QPen &pen)
- void setOutlierStyle (const QCPScatterStyle &style)
- void addData (const QVector< double > &keys, const QVector< double > &minimum, const QVector< double > &lowerQuartile, const QVector< double > &median, const QVector< double > &upperQuartile, const QVector< double > &maximum, bool alreadySorted=false)
- void addData (double key, double minimum, double lowerQuartile, double median, double upperQuartile, double maximum, const QVector< double > &outliers=QVector< double >())
- virtual QCPDataSelection selectTestRect (const QRectF &rect, bool onlySelectable) const Q_DECL_OVERRIDE
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_DECL_OVERRIDE
- virtual QCPRange getKeyRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth) const Q_DECL_OVERRIDE
- virtual QCPRange getValueRange (bool &foundRange, QCP::SignDomain inSignDomain=QCP::sdBoth, const QCPRange &inKeyRange=QCPRange()) const Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable1D< QCPStatisticalBoxData >

- QCPAbstractPlottable1D (QCPAxis *keyAxis, QCPAxis *valueAxis)
- virtual int dataCount () const Q_DECL_OVERRIDE
- virtual double dataMainKey (int index) const Q_DECL_OVERRIDE
- virtual double dataSortKey (int index) const Q_DECL_OVERRIDE
- virtual double dataMainValue (int index) const Q_DECL_OVERRIDE
- virtual QCPRange dataValueRange (int index) const Q_DECL_OVERRIDE
- virtual QPointF dataPixelPosition (int index) const Q_DECL_OVERRIDE
- virtual bool sortKeyIsMainKey () const Q_DECL_OVERRIDE
- virtual int findBegin (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual int findEnd (double sortKey, bool expandedRange=true) const Q_DECL_OVERRIDE
- virtual QCPPlottableInterface1D * interface1D () Q_DECL_OVERRIDE

Public Member Functions inherited from QCPAbstractPlottable

- QCPAbstractPlottable (QCPAxis *keyAxis, QCPAxis *valueAxis)
- QString **name** () const
- bool **antialiasedFill** () const
- bool **antialiasedScatters** () const
- QPen **pen** () const
- QBrush **brush** () const
- QCPAxis * **keyAxis** () const

- QCPAxis * **valueAxis** () const
- QCP::SelectionType **selectable** () const
- bool selected () const
- QCPDataSelection selection () const
- QCPSelectionDecorator * selectionDecorator () const
- void setName (const QString &name)
- void setAntialiasedFill (bool enabled)
- void setAntialiasedScatters (bool enabled)
- void setPen (const QPen &pen)
- void setBrush (const QBrush &brush)
- void setKeyAxis (QCPAxis *axis)
- void setValueAxis (QCPAxis *axis)
- Q_SLOT void setSelectable (QCP::SelectionType selectable)
- Q_SLOT void setSelection (QCPDataSelection selection)
- void setSelectionDecorator (QCPSelectionDecorator *decorator)
- void coordsToPixels (double key, double value, double &x, double &y) const
- const QPointF coordsToPixels (double key, double value) const
- void pixelsToCoords (double x, double y, double &key, double &value) const
- void pixelsToCoords (const QPointF &pixelPos, double &key, double &value) const
- void rescaleAxes (bool onlyEnlarge=false) const
- void rescaleKeyAxis (bool onlyEnlarge=false) const
- void rescaleValueAxis (bool onlyEnlarge=false, bool inKeyRange=false) const
- bool addToLegend (QCPLegend *legend)
- bool addToLegend ()
- bool removeFromLegend (QCPLegend *legend) const
- bool removeFromLegend () const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual void drawLegendIcon (QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE
- virtual void drawStatisticalBox (QCPPainter *painter, QCPStatisticalBoxDataContainer::const_iterator it, const QCPScatterStyle &outlierStyle) const
- void **getVisibleDataBounds** (QCPStatisticalBoxDataContainer::const_iterator &begin, QCPStatisticalBox↩DataContainer::const_iterator &end) const
- QRectF **getQuartileBox** (QCPStatisticalBoxDataContainer::const_iterator it) const
- QVector< QLineF > **getWhiskerBackboneLines** (QCPStatisticalBoxDataContainer::const_iterator it) const
- QVector< QLineF > **getWhiskerBarLines** (QCPStatisticalBoxDataContainer::const_iterator it) const

Protected Member Functions inherited from QCPAbstractPlottable1D< QCPStatisticalBoxData >

- void getDataSegments (QList< QCPDataRange > &selectedSegments, QList< QCPDataRange > &unselectedSegments) const
- void drawPolyline (QCPPainter *painter, const QVector< QPointF > &lineData) const

Protected Member Functions inherited from QCPAbstractPlottable

- virtual QRect clipRect () const Q_DECL_OVERRIDE
- virtual QCP::Interaction selectionCategory () const Q_DECL_OVERRIDE
- void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionStateChanged) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- void **applyFillAntialiasingHint** (QCPPainter *painter) const
- void **applyScattersAntialiasingHint** (QCPPainter *painter) const

Protected Member Functions inherited from QCPLayerable

- virtual void **parentPlotInitialized** (QCustomPlot *parentPlot)
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details)
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- double **mWidth**
- double **mWhiskerWidth**
- QPen **mWhiskerPen**
- QPen **mWhiskerBarPen**
- bool **mWhiskerAntialiased**
- QPen **mMedianPen**
- QCPScatterStyle **mOutlierStyle**

Protected Attributes inherited from QCPAbstractPlottable1D< QCPStatisticalBoxData >

- QSharedPointer< QCPDataContainer< QCPStatisticalBoxData > > **mDataContainer**

Protected Attributes inherited from QCPAbstractPlottable

- QString **mName**
- bool **mAntialiasedFill**
- bool **mAntialiasedScatters**
- QPen **mPen**
- QBrush **mBrush**
- QPointer< QCPAxis > **mKeyAxis**
- QPointer< QCPAxis > **mValueAxis**
- QCP::SelectionType **mSelectable**
- QCPDataSelection **mSelection**
- QCPSelectionDecorator * **mSelectionDecorator**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Friends

- class **QCustomPlot**
- class **QCPLegend**

Additional Inherited Members

Signals inherited from QCPAbstractPlottable

- void selectionChanged (bool selected)
- void selectionChanged (const QCPDataSelection &selection)
- void selectableChanged (QCP::SelectionType selectable)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

17.670.1 Detailed Description

A plottable representing a single statistical box in a plot.
To plot data, assign it with the setData or addData functions. Alternatively, you can also access and modify the data via the data method, which returns a pointer to the internal QCPStatisticalBoxDataContainer.
Additionally each data point can itself have a list of outliers, drawn as scatter points at the key coordinate of the respective statistical box data point. They can either be set by using the respective addData method or accessing the individual data points through data, and setting the `QVector<double> outliers` of the data points directly.

17.670.2 Changing the appearance

The appearance of each data point box, ranging from the lower to the upper quartile, is controlled via setPen and setBrush. You may change the width of the boxes with setWidth in plot coordinates.
Each data point's visual representation also consists of two whiskers. Whiskers are the lines which reach from the upper quartile to the maximum, and from the lower quartile to the minimum. The appearance of the whiskers can be modified with: setWhiskerPen, setWhiskerBarPen, setWhiskerWidth. The whisker width is the width of the bar perpendicular to the whisker at the top (for maximum) and bottom (for minimum). If the whisker pen is changed, make sure to set the `capStyle` to `Qt::FlatCap`. Otherwise the backbone line might exceed the whisker bars by a few pixels due to the pen cap being not perfectly flat.
The median indicator line inside the box has its own pen, setMedianPen.
The outlier data points are drawn as normal scatter points. Their look can be controlled with setOutlierStyle

17.670.3 Usage

Like all data representing objects in QCustomPlot, the QCPStatisticalBox is a plottable (QCPAbstractPlottable). So the plottable-interface of QCustomPlot applies (QCustomPlot::plottable, QCustomPlot::removePlottable, etc.)
Usually, you first create an instance:
which registers it with the QCustomPlot instance of the passed axes. Note that this QCustomPlot instance takes ownership of the plottable, so do not delete it manually but use QCustomPlot::removePlottable() instead. The newly created plottable can be modified, e.g.:

17.670.4 Constructor & Destructor Documentation

17.670.4.1 QCPStatisticalBox()

```
QCPStatisticalBox::QCPStatisticalBox (
            QCPAxis * keyAxis,
            QCPAxis * valueAxis )  [explicit]
```

Constructs a statistical box which uses *keyAxis* as its key axis ("x") and *valueAxis* as its value axis ("y"). *keyAxis* and *valueAxis* must reside in the same QCustomPlot instance and not have the same orientation. If either of these restrictions is violated, a corresponding message is printed to the debug output (qDebug), the construction is not aborted, though.

The created QCPStatisticalBox is automatically registered with the QCustomPlot instance inferred from *keyAxis*. This QCustomPlot instance takes ownership of the QCPStatisticalBox, so do not delete it manually but use QCustomPlot::removePlottable() instead.

17.670.5 Member Function Documentation

17.670.5.1 addData() [1/2]

```
void QCPStatisticalBox::addData (
            const QVector< double > & keys,
            const QVector< double > & minimum,
            const QVector< double > & lowerQuartile,
            const QVector< double > & median,
            const QVector< double > & upperQuartile,
            const QVector< double > & maximum,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.

Adds the provided points in *keys*, *minimum*, *lowerQuartile*, *median*, *upperQuartile* and *maximum* to the current data. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.

If you can guarantee that the passed data points are sorted by *keys* in ascending order, you can set *alreadySorted* to true, to improve performance by saving a sorting run.

Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

17.670.5.2 addData() [2/2]

```
void QCPStatisticalBox::addData (
            double key,
            double minimum,
            double lowerQuartile,
            double median,
            double upperQuartile,
            double maximum,
            const QVector< double > & outliers = QVector<double>() )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.

Adds the provided data point as *key*, *minimum*, *lowerQuartile*, *median*, *upperQuartile* and *maximum* to the current data.

Alternatively, you can also access and modify the data directly via the data method, which returns a pointer to the internal data container.

17.670.5.3 data()

```
QSharedPointer< QCPStatisticalBoxDataContainer > QCPStatisticalBox::data ( ) const  [inline]
```

Returns a shared pointer to the internal data storage of type QCPStatisticalBoxDataContainer. You may use it to directly manipulate the data, which may be more convenient and faster than using the regular setData or addData methods.

17.670.5.4 draw()

```
void QCPStatisticalBox::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.670.5.5 drawLegendIcon()

```
void QCPStatisticalBox::drawLegendIcon (
            QCPPainter * painter,
            const QRectF & rect ) const  [protected], [virtual]
```

Implements QCPAbstractPlottable.

17.670.5.6 drawStatisticalBox()

```
void QCPStatisticalBox::drawStatisticalBox (
            QCPPainter * painter,
            QCPStatisticalBoxDataContainer::const_iterator it,
            const QCPScatterStyle & outlierStyle ) const  [protected], [virtual]
```

Draws the graphical representation of a single statistical box with the data given by the iterator *it* with the provided *painter*.
If the statistical box has a set of outlier data points, they are drawn with *outlierStyle*.

See also

getQuartileBox, getWhiskerBackboneLines, getWhiskerBarLines

17.670.5.7 getKeyRange()

```
QCPRange QCPStatisticalBox::getKeyRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth ) const  [virtual]
```

Returns the coordinate range that all data in this plottable span in the key axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSignDomain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getValueRange

Implements QCPAbstractPlottable.

17.670.5.8 getValueRange()

```
QCPRange QCPStatisticalBox::getValueRange (
            bool & foundRange,
            QCP::SignDomain inSignDomain = QCP::sdBoth,
            const QCPRange & inKeyRange = QCPRange() ) const  [virtual]
```

Returns the coordinate range that the data points in the specified key range (*inKeyRange*) span in the value axis dimension. For logarithmic plots, one can set *inSignDomain* to either QCP::sdNegative or QCP::sdPositive in order to restrict the returned range to that sign domain. E.g. when only negative range is wanted, set *inSignDomain* to QCP::sdNegative and all positive points will be ignored for range calculation. For no restriction, just set *inSign↩ Domain* to QCP::sdBoth (default). *foundRange* is an output parameter that indicates whether a range could be found or not. If this is false, you shouldn't use the returned range (e.g. no points in data).

If *inKeyRange* has both lower and upper bound set to zero (is equal to `QCPRange()`), all data points are considered, without any restriction on the keys.
Note that *foundRange* is not the same as QCPRange::validRange, since the range returned by this function may have size zero (e.g. when there is only one data point). In this case *foundRange* would return true, but the returned range is not a valid range in terms of QCPRange::validRange.

See also

rescaleAxes, getKeyRange

Implements QCPAbstractPlottable.

17.670.5.9 selectTest()

```
double QCPStatisticalBox::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

Implements a selectTest specific to this plottable's point geometry.
If *details* is not 0, it will be set to a QCPDataSelection, describing the closest data point to *pos*.
\seebaseclassmethod QCPAbstractPlottable::selectTest
Reimplemented from QCPAbstractPlottable1D< QCPStatisticalBoxData >.

17.670.5.10 selectTestRect()

```
QCPDataSelection QCPStatisticalBox::selectTestRect (
            const QRectF & rect,
            bool onlySelectable ) const  [virtual]
```

Returns a data selection containing all the data points of this plottable which are contained (or hit by) *rect*. This is used mainly in the selection rect interaction for data selection (data selection mechanism).
If *onlySelectable* is true, an empty QCPDataSelection is returned if this plottable is not selectable (i.e. if QCPAbstractPlottable::setSelectable is QCP::stNone).

Note

rect must be a normalized rect (positive or zero width and height). This is especially important when using the rect of QCPSelectionRect::accepted, which is not necessarily normalized. Use `QRect::normalized()` when passing a rect which might not be normalized.

Reimplemented from QCPAbstractPlottable1D< QCPStatisticalBoxData >.

17.670.5.11 setData() [1/2]

```
void QCPStatisticalBox::setData (
            const QVector< double > & keys,
            const QVector< double > & minimum,
            const QVector< double > & lowerQuartile,
            const QVector< double > & median,
            const QVector< double > & upperQuartile,
            const QVector< double > & maximum,
            bool alreadySorted = false )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data with the provided points in *keys*, *minimum*, *lowerQuartile*, *median*, *upperQuartile* and *maximum*. The provided vectors should have equal length. Else, the number of added points will be the size of the smallest vector.
If you can guarantee that the passed data points are sorted by *keys* in ascending order, you can set *alreadySorted* to true, to improve performance by saving a sorting run.

See also

addData

17.670.5.12 setData() [2/2]

```
void QCPStatisticalBox::setData (
            QSharedPointer< QCPStatisticalBoxDataContainer > data )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Replaces the current data container with the provided *data* container.
Since a QSharedPointer is used, multiple QCPStatisticalBoxes may share the same data container safely. Modifying the data in the container will then affect all statistical boxes that share the container. Sharing can be achieved by simply exchanging the data containers wrapped in shared pointers:
If you do not wish to share containers, but create a copy from an existing container, rather use the QCPDataContainer<DataType>::set method on the statistical box data container directly:

See also

addData

17.670.5.13 setMedianPen()

```
void QCPStatisticalBox::setMedianPen (
            const QPen & pen )
```

Sets the pen used for drawing the median indicator line inside the statistical boxes.

17.670.5.14 setOutlierStyle()

```
void QCPStatisticalBox::setOutlierStyle (
            const QCPScatterStyle & style )
```

Sets the appearance of the outlier data points.
Outliers can be specified with the method addData(double key, double minimum, double lowerQuartile, double median, double upperQ

17.670.5.15 setWhiskerAntialiased()

```
void QCPStatisticalBox::setWhiskerAntialiased (
            bool enabled )
```

Sets whether the statistical boxes whiskers are drawn with antialiasing or not.
Note that antialiasing settings may be overridden by QCustomPlot::setAntialiasedElements and QCustomPlot::setNotAntialiasedEleme

17.670.5.16 setWhiskerBarPen()

```
void QCPStatisticalBox::setWhiskerBarPen (
            const QPen & pen )
```

Sets the pen used for drawing the whisker bars. Those are the lines parallel to the key axis at each end of the whisker backbone.
Whiskers are the lines which reach from the upper quartile to the maximum, and from the lower quartile to the minimum.

See also

setWhiskerPen

17.670.5.17 setWhiskerPen()

```
void QCPStatisticalBox::setWhiskerPen (
            const QPen & pen )
```

Sets the pen used for drawing the whisker backbone.
Whiskers are the lines which reach from the upper quartile to the maximum, and from the lower quartile to the minimum.
Make sure to set the `capStyle` of the passed *pen* to `Qt::FlatCap`. Otherwise the backbone line might exceed the whisker bars by a few pixels due to the pen cap being not perfectly flat.

See also

setWhiskerBarPen

17.670.5.18 setWhiskerWidth()

```
void QCPStatisticalBox::setWhiskerWidth (
            double width )
```

Sets the width of the whiskers in key coordinates.
Whiskers are the lines which reach from the upper quartile to the maximum, and from the lower quartile to the minimum.

See also

setWidth

17.670.5.19 setWidth()

```
void QCPStatisticalBox::setWidth (
            double width )
```

Sets the width of the boxes in key coordinates.

See also

setWhiskerWidth

The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.671 QCPStatisticalBoxData Class Reference

Holds the data of one single data point for QCPStatisticalBox.

Public Member Functions

- QCPStatisticalBoxData ()
- QCPStatisticalBoxData (double key, double minimum, double lowerQuartile, double median, double upper↩ Quartile, double maximum, const QVector< double > &outliers=QVector< double >())
- double sortKey () const
- double mainKey () const
- double mainValue () const
- QCPRange valueRange () const

Static Public Member Functions

- static QCPStatisticalBoxData fromSortKey (double sortKey)
- static bool sortKeyIsMainKey ()

Public Attributes

- double **key**
- double **minimum**
- double **lowerQuartile**
- double **median**
- double **upperQuartile**
- double **maximum**
- QVector< double > **outliers**

17.671.1 Detailed Description

Holds the data of one single data point for QCPStatisticalBox.
The stored data is:

- *key:* coordinate on the key axis of this data point (this is the *mainKey* and the *sortKey*)
- *minimum:* the position of the lower whisker, typically the minimum measurement of the sample that's not considered an outlier.
- *lowerQuartile:* the lower end of the box. The lower and the upper quartiles are the two statistical quartiles around the median of the sample, they should contain 50% of the sample data.
- *median:* the value of the median mark inside the quartile box. The median separates the sample data in half (50% of the sample data is below/above the median). (This is the *mainValue*)
- *upperQuartile:* the upper end of the box. The lower and the upper quartiles are the two statistical quartiles around the median of the sample, they should contain 50% of the sample data.
- *maximum:* the position of the upper whisker, typically the maximum measurement of the sample that's not considered an outlier.
- *outliers:* a QVector of outlier values that will be drawn as scatter points at the *key* coordinate of this data point (see QCPStatisticalBox::setOutlierStyle)

The container for storing multiple data points is QCPStatisticalBoxDataContainer. It is a typedef for QCPDataContainer with QCPStatisticalBoxData as the DataType template parameter. See the documentation there for an explanation regarding the data type's generic methods.

See also

QCPStatisticalBoxDataContainer

17.671.2 Constructor & Destructor Documentation

17.671.2.1 QCPStatisticalBoxData() [1/2]

```
QCPStatisticalBoxData::QCPStatisticalBoxData ( )
```

Constructs a data point with key and all values set to zero.

17.671.2.2 QCPStatisticalBoxData() [2/2]

```
QCPStatisticalBoxData::QCPStatisticalBoxData (
            double key,
            double minimum,
            double lowerQuartile,
            double median,
            double upperQuartile,
            double maximum,
            const QVector< double > & outliers = QVector<double>() )
```

Constructs a data point with the specified *key*, *minimum*, *lowerQuartile*, *median*, *upperQuartile*, *maximum* and optionally a number of *outliers*.

17.671.3 Member Function Documentation

17.671.3.1 fromSortKey()

```
static QCPStatisticalBoxData QCPStatisticalBoxData::fromSortKey (
            double sortKey )  [inline], [static]
```

Returns a data point with the specified *sortKey*. All other members are set to zero.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.671.3.2 mainKey()

```
double QCPStatisticalBoxData::mainKey ( ) const  [inline]
```

Returns the *key* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.671.3.3 mainValue()

```
double QCPStatisticalBoxData::mainValue ( ) const  [inline]
```

Returns the *median* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.671.3.4 sortKey()

```
double QCPStatisticalBoxData::sortKey ( ) const  [inline]
```

Returns the *key* member of this data point.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.671.3.5 sortKeyIsMainKey()

```
static static bool QCPStatisticalBoxData::sortKeyIsMainKey ( )  [inline], [static]
```

Since the member *key* is both the data point key coordinate and the data ordering parameter, this method returns true.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.

17.671.3.6 valueRange()

```
QCPRange QCPStatisticalBoxData::valueRange ( ) const  [inline]
```

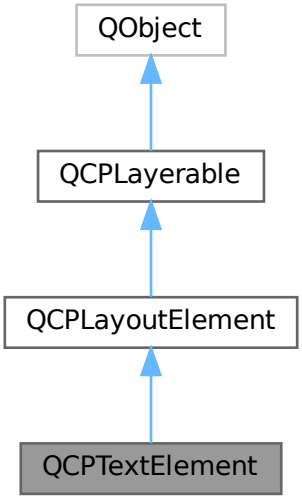
Returns a QCPRange spanning from the *minimum* to the *maximum* member of this statistical box data point, possibly further expanded by outliers.
For a general explanation of what this method is good for in the context of the data container, see the documentation of QCPDataContainer.
The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

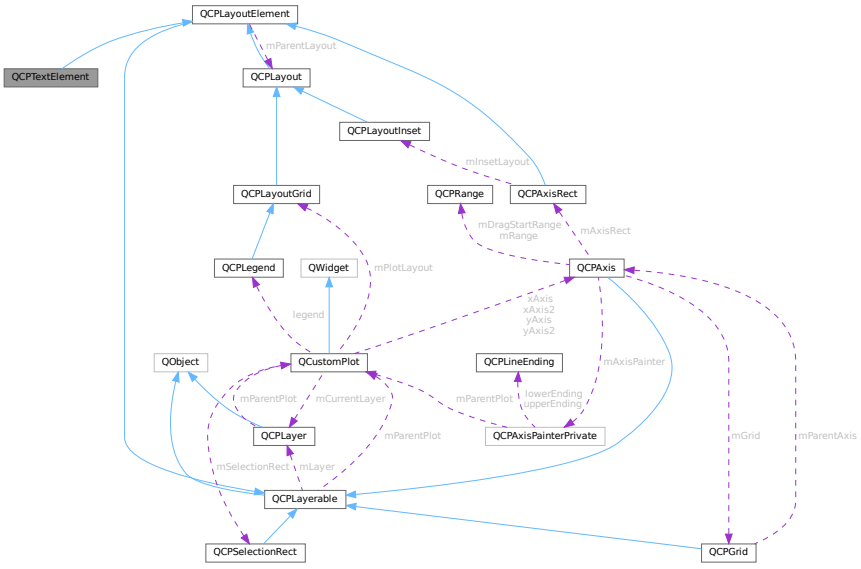
17.672 QCPTextElement Class Reference

A layout element displaying a text.

Inheritance diagram for QCPTextElement:



Collaboration diagram for QCPTextElement:



Signals

- void selectionChanged (bool selected)
- void **selectableChanged** (bool selectable)
- void clicked (QMouseEvent ∗event)
- void doubleClicked (QMouseEvent ∗event)

Signals inherited from QCPLayerable

- void layerChanged (QCPLayer *newLayer)

Public Member Functions

- QCPTextElement (QCustomPlot *parentPlot)
- QCPTextElement (QCustomPlot *parentPlot, const QString &text)
- QCPTextElement (QCustomPlot *parentPlot, const QString &text, double pointSize)
- QCPTextElement (QCustomPlot *parentPlot, const QString &text, const QString &fontFamily, double pointSize)
- QCPTextElement (QCustomPlot *parentPlot, const QString &text, const QFont &font)
- QString **text** () const
- int **textFlags** () const
- QFont **font** () const
- QColor **textColor** () const
- QFont **selectedFont** () const
- QColor **selectedTextColor** () const
- bool **selectable** () const
- bool **selected** () const
- void setText (const QString &text)
- void setTextFlags (int flags)
- void setFont (const QFont &font)
- void setTextColor (const QColor &color)
- void setSelectedFont (const QFont &font)
- void setSelectedTextColor (const QColor &color)
- Q_SLOT void setSelectable (bool selectable)
- Q_SLOT void setSelected (bool selected)
- virtual double selectTest (const QPointF &pos, bool onlySelectable, QVariant *details=nullptr) const Q_DECL_OVERRIDE
- virtual void mousePressEvent (QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE
- virtual void mouseReleaseEvent (QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE
- virtual void mouseDoubleClickEvent (QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE

Public Member Functions inherited from QCPLayoutElement

- QCPLayoutElement (QCustomPlot *parentPlot=nullptr)
- QCPLayout * layout () const
- QRect rect () const
- QRect outerRect () const
- QMargins **margins** () const
- QMargins **minimumMargins** () const
- QCP::MarginSides **autoMargins** () const
- QSize **minimumSize** () const
- QSize **maximumSize** () const
- SizeConstraintRect **sizeConstraintRect** () const
- QCPMarginGroup * **marginGroup** (QCP::MarginSide side) const
- QHash< QCP::MarginSide, QCPMarginGroup * > **marginGroups** () const
- void setOuterRect (const QRect &rect)
- void setMargins (const QMargins &margins)
- void setMinimumMargins (const QMargins &margins)
- void setAutoMargins (QCP::MarginSides sides)
- void setMinimumSize (const QSize &size)
- void setMinimumSize (int width, int height)
- void setMaximumSize (const QSize &size)
- void setMaximumSize (int width, int height)

- void setSizeConstraintRect (SizeConstraintRect constraintRect)
- void setMarginGroup (QCP::MarginSides sides, QCPMarginGroup *group)
- virtual void update (UpdatePhase phase)
- virtual QList< QCPLayoutElement * > elements (bool recursive) const

Public Member Functions inherited from QCPLayerable

- QCPLayerable (QCustomPlot *plot, QString targetLayer=QString(), QCPLayerable *parentLayerable=nullptr)
- bool **visible** () const
- QCustomPlot * **parentPlot** () const
- QCPLayerable * parentLayerable () const
- QCPLayer * **layer** () const
- bool **antialiased** () const
- void setVisible (bool on)
- Q_SLOT bool setLayer (QCPLayer *layer)
- bool setLayer (const QString &layerName)
- void setAntialiased (bool enabled)
- bool realVisibility () const

Protected Member Functions

- virtual void applyDefaultAntialiasingHint (QCPPainter *painter) const Q_DECL_OVERRIDE
- virtual void draw (QCPPainter *painter) Q_DECL_OVERRIDE
- virtual QSize minimumOuterSizeHint () const Q_DECL_OVERRIDE
- virtual QSize maximumOuterSizeHint () const Q_DECL_OVERRIDE
- virtual void selectEvent (QMouseEvent *event, bool additive, const QVariant &details, bool *selectionState↩ Changed) Q_DECL_OVERRIDE
- virtual void deselectEvent (bool *selectionStateChanged) Q_DECL_OVERRIDE
- QFont **mainFont** () const
- QColor **mainTextColor** () const

Protected Member Functions inherited from QCPLayoutElement

- virtual int **calculateAutoMargin** (QCP::MarginSide side)
- virtual void **layoutChanged** ()
- virtual void parentPlotInitialized (QCustomPlot *parentPlot) Q_DECL_OVERRIDE

Protected Member Functions inherited from QCPLayerable

- virtual QCP::Interaction **selectionCategory** () const
- virtual QRect **clipRect** () const
- virtual void mouseMoveEvent (QMouseEvent *event, const QPointF &startPos)
- virtual void wheelEvent (QWheelEvent *event)
- void **initializeParentPlot** (QCustomPlot *parentPlot)
- void **setParentLayerable** (QCPLayerable *parentLayerable)
- bool **moveToLayer** (QCPLayer *layer, bool prepend)
- void **applyAntialiasingHint** (QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement overrideElement) const

Protected Attributes

- QString **mText**
- int **mTextFlags**
- QFont **mFont**
- QColor **mTextColor**
- QFont **mSelectedFont**
- QColor **mSelectedTextColor**
- QRect **mTextBoundingRect**
- bool **mSelectable**
- bool **mSelected**

Protected Attributes inherited from QCPLayoutElement

- QCPLayout * **mParentLayout**
- QSize **mMinimumSize**
- QSize **mMaximumSize**
- SizeConstraintRect **mSizeConstraintRect**
- QRect **mRect**
- QRect **mOuterRect**
- QMargins **mMargins**
- QMargins **mMinimumMargins**
- QCP::MarginSides **mAutoMargins**
- QHash< QCP::MarginSide, QCPMarginGroup * > **mMarginGroups**

Protected Attributes inherited from QCPLayerable

- bool **mVisible**
- QCustomPlot * **mParentPlot**
- QPointer< QCPLayerable > **mParentLayerable**
- QCPLayer * **mLayer**
- bool **mAntialiased**

Additional Inherited Members

Public Types inherited from QCPLayoutElement

- enum UpdatePhase { upPreparation , upMargins , upLayout }
- enum SizeConstraintRect { scrInnerRect , scrOuterRect }

17.672.1 Detailed Description

A layout element displaying a text.
The text may be specified with setText, the formatting can be controlled with setFont, setTextColor, and setTextFlags.
A text element can be added as follows:

17.672.2 Constructor & Destructor Documentation

17.672.2.1 QCPTextElement() [1/5]

```
QCPTextElement::QCPTextElement (
            QCustomPlot * parentPlot )  [explicit]
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Creates a new QCPTextElement instance and sets default values. The initial text is empty (setText).

17.672.2.2 QCPTextElement() [2/5]

```
QCPTextElement::QCPTextElement (
            QCustomPlot * parentPlot,
            const QString & text )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Creates a new QCPTextElement instance and sets default values.
The initial text is set to *text*.

17.672.2.3 QCPTextElement() [3/5]

```
QCPTextElement::QCPTextElement (
            QCustomPlot * parentPlot,
            const QString & text,
            double pointSize )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Creates a new QCPTextElement instance and sets default values.
The initial text is set to *text* with *pointSize*.

17.672.2.4 QCPTextElement() [4/5]

```
QCPTextElement::QCPTextElement (
            QCustomPlot * parentPlot,
            const QString & text,
            const QString & fontFamily,
            double pointSize )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Creates a new QCPTextElement instance and sets default values.
The initial text is set to *text* with *pointSize* and the specified *fontFamily*.

17.672.2.5 QCPTextElement() [5/5]

```
QCPTextElement::QCPTextElement (
            QCustomPlot * parentPlot,
            const QString & text,
            const QFont & font )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Creates a new QCPTextElement instance and sets default values.
The initial text is set to *text* with the specified *font*.

17.672.3 Member Function Documentation

17.672.3.1 applyDefaultAntialiasingHint()

```
void QCPTextElement::applyDefaultAntialiasingHint (
            QCPPainter * painter ) const  [protected], [virtual]
```

Reimplemented from QCPLayoutElement.

17.672.3.2 clicked

```
void QCPTextElement::clicked (
            QMouseEvent * event )  [signal]
```

This signal is emitted when the text element is clicked.

See also

doubleClicked, selectTest

17.672.3.3 deselectEvent()

```
void QCPTextElement::deselectEvent (
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.672.3.4 doubleClicked

```
void QCPTextElement::doubleClicked (
            QMouseEvent * event )  [signal]
```

This signal is emitted when the text element is double clicked.

See also

clicked, selectTest

17.672.3.5 draw()

```
void QCPTextElement::draw (
            QCPPainter * painter )  [protected], [virtual]
```

Reimplemented from QCPLayoutElement.

17.672.3.6 maximumOuterSizeHint()

```
QSize QCPTextElement::maximumOuterSizeHint ( ) const  [protected], [virtual]
```

Returns the suggested maximum size this layout element (the outerRect) may be expanded to, if no manual maximum size is set.
if a maximum size (setMaximumSize) was not set manually, parent layouts use the returned size (usually indirectly through QCPLayout::getFinalMaximumOuterSize) to determine the maximum allowed size of this layout element.
A manual maximum size is considered set if it is smaller than Qt's QWIDGETSIZE_MAX.
The default implementation simply returns QWIDGETSIZE_MAX for both width and height, implying no suggested maximum size. Reimplementations may use their detailed knowledge about the layout element's content to provide size hints.
Reimplemented from QCPLayoutElement.

17.672.3.7 minimumOuterSizeHint()

```
QSize QCPTextElement::minimumOuterSizeHint ( ) const  [protected], [virtual]
```

Returns the suggested minimum size this layout element (the outerRect) may be compressed to, if no manual minimum size is set.
if a minimum size (setMinimumSize) was not set manually, parent layouts use the returned size (usually indirectly through QCPLayout::getFinalMinimumOuterSize) to determine the minimum allowed size of this layout element.
A manual minimum size is considered set if it is non-zero.
The default implementation simply returns the sum of the horizontal margins for the width and the sum of the vertical margins for the height. Reimplementations may use their detailed knowledge about the layout element's content to provide size hints.
Reimplemented from QCPLayoutElement.

17.672.3.8 mouseDoubleClickEvent()

```
void QCPTextElement::mouseDoubleClickEvent (
            QMouseEvent * event,
            const QVariant & details )  [virtual]
```

Emits the doubleClicked signal.
\seebaseclassmethod
Reimplemented from QCPLayerable.

17.672.3.9 mousePressEvent()

```
void QCPTextElement::mousePressEvent (
            QMouseEvent * event,
            const QVariant & details )  [virtual]
```

Accepts the mouse event in order to emit the according click signal in the mouseReleaseEvent.
\seebaseclassmethod
Reimplemented from QCPLayerable.

17.672.3.10 mouseReleaseEvent()

```
void QCPTextElement::mouseReleaseEvent (
            QMouseEvent * event,
            const QPointF & startPos )  [virtual]
```

Emits the clicked signal if the cursor hasn't moved by more than a few pixels since the mousePressEvent.
\seebaseclassmethod
Reimplemented from QCPLayerable.

17.672.3.11 selectEvent()

```
void QCPTextElement::selectEvent (
            QMouseEvent * event,
            bool additive,
            const QVariant & details,
            bool * selectionStateChanged )  [protected], [virtual]
```

Reimplemented from QCPLayerable.

17.672.3.12 selectionChanged

```
void QCPTextElement::selectionChanged (
            bool selected )  [signal]
```

This signal is emitted when the selection state has changed to *selected*, either by user interaction or by a direct call to setSelected.

See also

setSelected, setSelectable

17.672.3.13 selectTest()

```
double QCPTextElement::selectTest (
            const QPointF & pos,
            bool onlySelectable,
            QVariant * details = nullptr ) const  [virtual]
```

Returns 0.99*selectionTolerance (see QCustomPlot::setSelectionTolerance) when *pos* is within the bounding box of the text element's text. Note that this bounding box is updated in the draw call.
If *pos* is outside the text's bounding box or if *onlySelectable* is true and this text element is not selectable (setSelectable), returns -1.
\seebaseclassmethod
Reimplemented from QCPLayoutElement.

17.672.3.14 setFont()

```
void QCPTextElement::setFont (
            const QFont & font )
```

Sets the *font* of the text.

See also

setTextColor, setSelectedFont

17.672.3.15 setSelectable()

```
void QCPTextElement::setSelectable (
            bool selectable )
```

Sets whether the user may select this text element.
Note that even when *selectable* is set to `false`, the selection state may be changed programmatically via setSelected.

17.672.3.16 setSelected()

```
void QCPTextElement::setSelected (
            bool selected )
```

Sets the selection state of this text element to *selected*. If the selection has changed, selectionChanged is emitted.
Note that this function can change the selection state independently of the current setSelectable state.

17.672.3.17 setSelectedFont()

```
void QCPTextElement::setSelectedFont (
            const QFont & font )
```

Sets the *font* of the text that will be used if the text element is selected (setSelected).

See also

setFont

17.672.3.18 setSelectedTextColor()

```
void QCPTextElement::setSelectedTextColor (
            const QColor & color )
```

Sets the *color* of the text that will be used if the text element is selected (setSelected).

See also

setTextColor

17.672.3.19 setText()

```
void QCPTextElement::setText (
            const QString & text )
```

Sets the text that will be displayed to *text*. Multiple lines can be created by insertion of "\n".

See also

setFont, setTextColor, setTextFlags

17.672.3.20 setTextColor()

```
void QCPTextElement::setTextColor (
            const QColor & color )
```

Sets the *color* of the text.

See also

setFont, setSelectedTextColor

17.672.3.21 setTextFlags()

```
void QCPTextElement::setTextFlags (
            int flags )
```

Sets options for text alignment and wrapping behaviour. *flags* is a bitwise OR-combination of `Qt::AlignmentFlag` and `Qt::TextFlag` enums.
Possible enums are:

- Qt::AlignLeft
- Qt::AlignRight
- Qt::AlignHCenter
- Qt::AlignJustify
- Qt::AlignTop
- Qt::AlignBottom
- Qt::AlignVCenter
- Qt::AlignCenter
- Qt::TextDontClip
- Qt::TextSingleLine
- Qt::TextExpandTabs
- Qt::TextShowMnemonic
- Qt::TextWordWrap
- Qt::TextIncludeTrailingSpaces

The documentation for this class was generated from the following files:

- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.673 QCPVector2D Class Reference

Represents two doubles as a mathematical 2D vector.

Public Member Functions

- QCPVector2D ()
- QCPVector2D (double x, double y)
- QCPVector2D (const QPoint &point)
- QCPVector2D (const QPointF &point)
- double **x** () const
- double **y** () const
- double & **rx** ()
- double & **ry** ()
- void setX (double x)
- void setY (double y)
- double length () const
- double lengthSquared () const
- double angle () const
- QPoint toPoint () const
- QPointF toPointF () const
- bool isNull () const

- void normalize ()
- QCPVector2D normalized () const
- QCPVector2D perpendicular () const
- double dot (const QCPVector2D &vec) const
- double distanceSquaredToLine (const QCPVector2D &start, const QCPVector2D &end) const
- double distanceSquaredToLine (const QLineF &line) const
- double distanceToStraightLine (const QCPVector2D &base, const QCPVector2D &direction) const
- QCPVector2D & operator*= (double factor)
- QCPVector2D & operator/= (double divisor)
- QCPVector2D & operator+= (const QCPVector2D &vector)
- QCPVector2D & operator-= (const QCPVector2D &vector)

Friends

- const QCPVector2D **operator*** (double factor, const QCPVector2D &vec)
- const QCPVector2D **operator*** (const QCPVector2D &vec, double factor)
- const QCPVector2D **operator/** (const QCPVector2D &vec, double divisor)
- const QCPVector2D **operator+** (const QCPVector2D &vec1, const QCPVector2D &vec2)
- const QCPVector2D **operator-** (const QCPVector2D &vec1, const QCPVector2D &vec2)
- const QCPVector2D **operator-** (const QCPVector2D &vec)

Related Symbols

(Note that these are not member symbols.)

- QDebug operator<< (QDebug d, const QCPVector2D &vec)

17.673.1 Detailed Description

Represents two doubles as a mathematical 2D vector.
This class acts as a replacement for QVector2D with the advantage of double precision instead of single, and some convenience methods tailored for the QCustomPlot library.

17.673.2 Constructor & Destructor Documentation

17.673.2.1 QCPVector2D() [1/4]

```
QCPVector2D::QCPVector2D ( )
```

Creates a QCPVector2D object and initializes the x and y coordinates to 0.

17.673.2.2 QCPVector2D() [2/4]

```
QCPVector2D::QCPVector2D (
            double x,
            double y )
```

Creates a QCPVector2D object and initializes the *x* and *y* coordinates with the specified values.

17.673.2.3 QCPVector2D() [3/4]

```
QCPVector2D::QCPVector2D (
            const QPoint & point )
```

Creates a QCPVector2D object and initializes the x and y coordinates respective coordinates of the specified *point*.

17.673.2.4 QCPVector2D() [4/4]

```
QCPVector2D::QCPVector2D (
            const QPointF & point )
```

Creates a QCPVector2D object and initializes the x and y coordinates respective coordinates of the specified *point*.

17.673.3 Member Function Documentation

17.673.3.1 angle()

```
double QCPVector2D::angle ( ) const  [inline]
```

Returns the angle of the vector in radians. The angle is measured between the positive x line and the vector, counter-clockwise in a mathematical coordinate system (y axis upwards positive). In screen/widget coordinates where the y axis is inverted, the angle appears clockwise.

17.673.3.2 distanceSquaredToLine() [1/2]

```
double QCPVector2D::distanceSquaredToLine (
            const QCPVector2D & start,
            const QCPVector2D & end ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Returns the squared shortest distance of this vector (interpreted as a point) to the finite line segment given by *start* and *end*.

See also

distanceToStraightLine

17.673.3.3 distanceSquaredToLine() [2/2]

```
double QCPVector2D::distanceSquaredToLine (
            const QLineF & line ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Returns the squared shortest distance of this vector (interpreted as a point) to the finite line segment given by *line*.

See also

distanceToStraightLine

17.673.3.4 distanceToStraightLine()

```
double QCPVector2D::distanceToStraightLine (
            const QCPVector2D & base,
            const QCPVector2D & direction ) const
```

Returns the shortest distance of this vector (interpreted as a point) to the infinite straight line given by a *base* point and a *direction* vector.

See also

distanceSquaredToLine

17.673.3.5 dot()

```
double QCPVector2D::dot (
            const QCPVector2D & vec ) const  [inline]
```

Returns the dot/scalar product of this vector with the specified vector *vec*.

17.673.3.6 isNull()

```
bool QCPVector2D::isNull ( ) const  [inline]
```

Returns whether this vector is null. A vector is null if `qIsNull` returns true for both x and y coordinates, i.e. if both are binary equal to 0.

17.673.3.7 length()

```
double QCPVector2D::length ( ) const  [inline]
```
Returns the length of this vector.

See also

lengthSquared

17.673.3.8 lengthSquared()

```
double QCPVector2D::lengthSquared ( ) const  [inline]
```
Returns the squared length of this vector. In some situations, e.g. when just trying to find the shortest vector of a group, this is faster than calculating length, because it avoids calculation of a square root.

See also

length

17.673.3.9 normalize()

```
void QCPVector2D::normalize ( )
```
Normalizes this vector. After this operation, the length of the vector is equal to 1.
If the vector has both entries set to zero, this method does nothing.

See also

normalized, length, lengthSquared

17.673.3.10 normalized()

```
QCPVector2D QCPVector2D::normalized ( ) const
```
Returns a normalized version of this vector. The length of the returned vector is equal to 1.
If the vector has both entries set to zero, this method returns the vector unmodified.

See also

normalize, length, lengthSquared

17.673.3.11 operator*=()

```
QCPVector2D & QCPVector2D::operator*= (
            double factor )
```
Scales this vector by the given *factor*, i.e. the x and y components are multiplied by *factor*.

17.673.3.12 operator+=()

```
QCPVector2D & QCPVector2D::operator+= (
            const QCPVector2D & vector )
```
Adds the given *vector* to this vector component-wise.

17.673.3.13 operator-=()

```
QCPVector2D & QCPVector2D::operator-= (
            const QCPVector2D & vector )
```
subtracts the given *vector* from this vector component-wise.

17.673.3.14 operator/=()

```
QCPVector2D & QCPVector2D::operator/= (
            double divisor )
```
Scales this vector by the given *divisor*, i.e. the x and y components are divided by *divisor*.

17.673.3.15 perpendicular()

```
QCPVector2D QCPVector2D::perpendicular ( ) const  [inline]
```
Returns a vector perpendicular to this vector, with the same length.

17.673.3.16 setX()

```
void QCPVector2D::setX (
            double x )  [inline]
```
Sets the x coordinate of this vector to *x*.

See also

setY

17.673.3.17 setY()

```
void QCPVector2D::setY (
            double y )  [inline]
```
Sets the y coordinate of this vector to *y*.

See also

setX

17.673.3.18 toPoint()

```
QPoint QCPVector2D::toPoint ( ) const  [inline]
```
Returns a QPoint which has the x and y coordinates of this vector, truncating any floating point information.

See also

toPointF

17.673.3.19 toPointF()

```
QPointF QCPVector2D::toPointF ( ) const  [inline]
```
Returns a QPointF which has the x and y coordinates of this vector.

See also

toPoint

17.673.4 Friends And Related Symbol Documentation

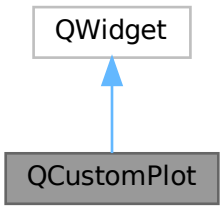
17.673.4.1 operator<<()

```
QDebug operator<< (
            QDebug d,
            const QCPVector2D & vec )  [related]
```
Prints *vec* in a human readable format to the qDebug output.
The documentation for this class was generated from the following files:

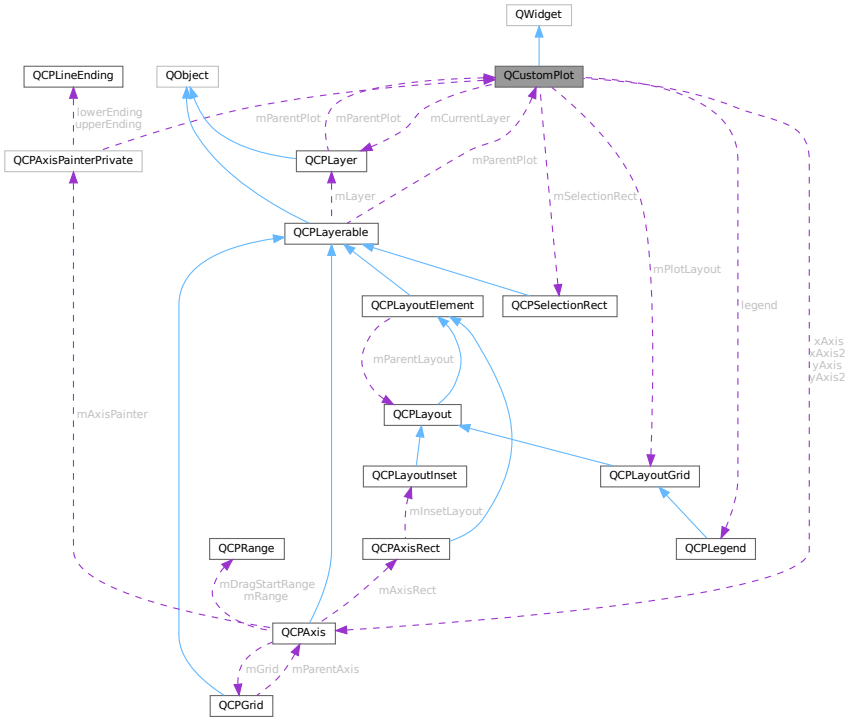
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.674 QCustomPlot Class Reference

The central class of the library. This is the QWidget which displays the plot and interacts with the user.
Inheritance diagram for QCustomPlot:



Collaboration diagram for QCustomPlot:



Public Types

- enum LayerInsertMode { limBelow , limAbove }
- enum RefreshPriority { rpImmediateRefresh , rpQueuedRefresh , rpRefreshHint , rpQueuedReplot }

Signals

- void mouseDoubleClick (QMouseEvent ∗event)
- void mousePress (QMouseEvent ∗event)
- void mouseMove (QMouseEvent ∗event)

- void mouseRelease (QMouseEvent *event)
- void mouseWheel (QWheelEvent *event)
- void plottableClick (QCPAbstractPlottable *plottable, int dataIndex, QMouseEvent *event)
- void plottableDoubleClick (QCPAbstractPlottable *plottable, int dataIndex, QMouseEvent *event)
- void itemClick (QCPAbstractItem *item, QMouseEvent *event)
- void itemDoubleClick (QCPAbstractItem *item, QMouseEvent *event)
- void axisClick (QCPAxis *axis, QCPAxis::SelectablePart part, QMouseEvent *event)
- void axisDoubleClick (QCPAxis *axis, QCPAxis::SelectablePart part, QMouseEvent *event)
- void legendClick (QCPLegend *legend, QCPAbstractLegendItem *item, QMouseEvent *event)
- void legendDoubleClick (QCPLegend *legend, QCPAbstractLegendItem *item, QMouseEvent *event)
- void selectionChangedByUser ()
- void beforeReplot ()
- void afterLayout ()
- void afterReplot ()

Public Member Functions

- QCustomPlot (QWidget *parent=nullptr)
- QRect **viewport** () const
- double **bufferDevicePixelRatio** () const
- QPixmap **background** () const
- bool **backgroundScaled** () const
- Qt::AspectRatioMode **backgroundScaledMode** () const
- QCPLayoutGrid * plotLayout () const
- QCP::AntialiasedElements **antialiasedElements** () const
- QCP::AntialiasedElements **notAntialiasedElements** () const
- bool **autoAddPlottableToLegend** () const
- const QCP::Interactions **interactions** () const
- int **selectionTolerance** () const
- bool **noAntialiasingOnDrag** () const
- QCP::PlottingHints **plottingHints** () const
- Qt::KeyboardModifier **multiSelectModifier** () const
- QCP::SelectionRectMode **selectionRectMode** () const
- QCPSelectionRect * selectionRect () const
- bool **openGl** () const
- void setViewport (const QRect &rect)
- void setBufferDevicePixelRatio (double ratio)
- void setBackground (const QPixmap &pm)
- void setBackground (const QPixmap &pm, bool scaled, Qt::AspectRatioMode mode=Qt::KeepAspectRatio↩ ByExpanding)
- void setBackground (const QBrush &brush)
- void setBackgroundScaled (bool scaled)
- void setBackgroundScaledMode (Qt::AspectRatioMode mode)
- void setAntialiasedElements (const QCP::AntialiasedElements &antialiasedElements)
- void setAntialiasedElement (QCP::AntialiasedElement antialiasedElement, bool enabled=true)
- void setNotAntialiasedElements (const QCP::AntialiasedElements &notAntialiasedElements)
- void setNotAntialiasedElement (QCP::AntialiasedElement notAntialiasedElement, bool enabled=true)
- void setAutoAddPlottableToLegend (bool on)
- void setInteractions (const QCP::Interactions &interactions)
- void setInteraction (const QCP::Interaction &interaction, bool enabled=true)
- void setSelectionTolerance (int pixels)
- void setNoAntialiasingOnDrag (bool enabled)
- void setPlottingHints (const QCP::PlottingHints &hints)
- void setPlottingHint (QCP::PlottingHint hint, bool enabled=true)
- void setMultiSelectModifier (Qt::KeyboardModifier modifier)

- void setSelectionRectMode (QCP::SelectionRectMode mode)
- void setSelectionRect (QCPSelectionRect *selectionRect)
- void setOpenGl (bool enabled, int multisampling=16)
- QCPAbstractPlottable * plottable (int index)
- QCPAbstractPlottable * plottable ()
- bool removePlottable (QCPAbstractPlottable *plottable)
- bool removePlottable (int index)
- int clearPlottables ()
- int plottableCount () const
- QList< QCPAbstractPlottable * > selectedPlottables () const
- template<class PlottableType >
  PlottableType * plottableAt (const QPointF &pos, bool onlySelectable=false, int *dataIndex=nullptr) const
- QCPAbstractPlottable * plottableAt (const QPointF &pos, bool onlySelectable=false, int *dataIndex=nullptr) const
- bool hasPlottable (QCPAbstractPlottable *plottable) const
- QCPGraph * graph (int index) const
- QCPGraph * graph () const
- QCPGraph * addGraph (QCPAxis *keyAxis=nullptr, QCPAxis *valueAxis=nullptr)
- bool removeGraph (QCPGraph *graph)
- bool removeGraph (int index)
- int clearGraphs ()
- int graphCount () const
- QList< QCPGraph * > selectedGraphs () const
- QCPAbstractItem * item (int index) const
- QCPAbstractItem * item () const
- bool removeItem (QCPAbstractItem *item)
- bool removeItem (int index)
- int clearItems ()
- int itemCount () const
- QList< QCPAbstractItem * > selectedItems () const
- template<class ItemType >
  ItemType * itemAt (const QPointF &pos, bool onlySelectable=false) const
- QCPAbstractItem * itemAt (const QPointF &pos, bool onlySelectable=false) const
- bool hasItem (QCPAbstractItem *item) const
- QCPLayer * layer (const QString &name) const
- QCPLayer * layer (int index) const
- QCPLayer * currentLayer () const
- bool setCurrentLayer (const QString &name)
- bool setCurrentLayer (QCPLayer *layer)
- int layerCount () const
- bool addLayer (const QString &name, QCPLayer *otherLayer=nullptr, LayerInsertMode insert↩ Mode=limAbove)
- bool removeLayer (QCPLayer *layer)
- bool moveLayer (QCPLayer *layer, QCPLayer *otherLayer, LayerInsertMode insertMode=limAbove)
- int axisRectCount () const
- QCPAxisRect * axisRect (int index=0) const
- QList< QCPAxisRect * > axisRects () const
- QCPLayoutElement * layoutElementAt (const QPointF &pos) const
- QCPAxisRect * axisRectAt (const QPointF &pos) const
- Q_SLOT void rescaleAxes (bool onlyVisiblePlottables=false)
- QList< QCPAxis * > selectedAxes () const
- QList< QCPLegend * > selectedLegends () const
- Q_SLOT void deselectAll ()
- bool savePdf (const QString &fileName, int width=0, int height=0, QCP::ExportPen exportPen=QCP::epAllowCosmetic, const QString &pdfCreator=QString(), const QString &pdfTitle=QString())

- bool savePng (const QString &fileName, int width=0, int height=0, double scale=1.0, int quality=-1, int resolution=96, QCP::ResolutionUnit resolutionUnit=QCP::ruDotsPerInch)
- bool saveJpg (const QString &fileName, int width=0, int height=0, double scale=1.0, int quality=-1, int resolution=96, QCP::ResolutionUnit resolutionUnit=QCP::ruDotsPerInch)
- bool saveBmp (const QString &fileName, int width=0, int height=0, double scale=1.0, int resolution=96, QCP::ResolutionUnit resolutionUnit=QCP::ruDotsPerInch)
- bool saveRastered (const QString &fileName, int width, int height, double scale, const char *format, int quality=-1, int resolution=96, QCP::ResolutionUnit resolutionUnit=QCP::ruDotsPerInch)
- QPixmap toPixmap (int width=0, int height=0, double scale=1.0)
- void toPainter (QCPPainter *painter, int width=0, int height=0)
- Q_SLOT void replot (QCustomPlot::RefreshPriority refreshPriority=QCustomPlot::rpRefreshHint)
- double replotTime (bool average=false) const

Public Attributes

- QCPAxis * xAxis
- QCPAxis * yAxis
- QCPAxis * xAxis2
- QCPAxis * yAxis2
- QCPLegend * legend

Protected Member Functions

- virtual QSize **minimumSizeHint** () const Q_DECL_OVERRIDE
- virtual QSize **sizeHint** () const Q_DECL_OVERRIDE
- virtual void **paintEvent** (QPaintEvent *event) Q_DECL_OVERRIDE
- virtual void **resizeEvent** (QResizeEvent *event) Q_DECL_OVERRIDE
- virtual void **mouseDoubleClickEvent** (QMouseEvent *event) Q_DECL_OVERRIDE
- virtual void **mousePressEvent** (QMouseEvent *event) Q_DECL_OVERRIDE
- virtual void **mouseMoveEvent** (QMouseEvent *event) Q_DECL_OVERRIDE
- virtual void **mouseReleaseEvent** (QMouseEvent *event) Q_DECL_OVERRIDE
- virtual void **wheelEvent** (QWheelEvent *event) Q_DECL_OVERRIDE
- virtual void **draw** (QCPPainter *painter)
- virtual void **updateLayout** ()
- virtual void **axisRemoved** (QCPAxis *axis)
- virtual void **legendRemoved** (QCPLegend *legend)
- virtual Q_SLOT void **processRectSelection** (QRect rect, QMouseEvent *event)
- virtual Q_SLOT void **processRectZoom** (QRect rect, QMouseEvent *event)
- virtual Q_SLOT void **processPointSelection** (QMouseEvent *event)
- bool **registerPlottable** (QCPAbstractPlottable *plottable)
- bool **registerGraph** (QCPGraph *graph)
- bool **registerItem** (QCPAbstractItem *item)
- void **updateLayerIndices** () const
- QCPLayerable * **layerableAt** (const QPointF &pos, bool onlySelectable, QVariant *selectionDetails=nullptr) const
- QList< QCPLayerable * > **layerableListAt** (const QPointF &pos, bool onlySelectable, QList< QVariant > *selectionDetails=nullptr) const
- void **drawBackground** (QCPPainter *painter)
- void **setupPaintBuffers** ()
- QCPAbstractPaintBuffer * **createPaintBuffer** ()
- bool hasInvalidatedPaintBuffers ()
- bool **setupOpenGl** ()
- void **freeOpenGl** ()

Protected Attributes

- QRect **mViewport**
- double **mBufferDevicePixelRatio**
- QCPLayoutGrid * **mPlotLayout**
- bool **mAutoAddPlottableToLegend**
- QList< QCPAbstractPlottable * > **mPlottables**
- QList< QCPGraph * > **mGraphs**
- QList< QCPAbstractItem * > **mItems**
- QList< QCPLayer * > **mLayers**
- QCP::AntialiasedElements **mAntialiasedElements**
- QCP::AntialiasedElements **mNotAntialiasedElements**
- QCP::Interactions **mInteractions**
- int **mSelectionTolerance**
- bool **mNoAntialiasingOnDrag**
- QBrush **mBackgroundBrush**
- QPixmap **mBackgroundPixmap**
- QPixmap **mScaledBackgroundPixmap**
- bool **mBackgroundScaled**
- Qt::AspectRatioMode **mBackgroundScaledMode**
- QCPLayer * **mCurrentLayer**
- QCP::PlottingHints **mPlottingHints**
- Qt::KeyboardModifier **mMultiSelectModifier**
- QCP::SelectionRectMode **mSelectionRectMode**
- QCPSelectionRect * **mSelectionRect**
- bool **mOpenGl**
- QList< QSharedPointer< QCPAbstractPaintBuffer > > **mPaintBuffers**
- QPoint **mMousePressPos**
- bool **mMouseHasMoved**
- QPointer< QCPLayerable > **mMouseEventLayerable**
- QPointer< QCPLayerable > **mMouseSignalLayerable**
- QVariant **mMouseEventLayerableDetails**
- QVariant **mMouseSignalLayerableDetails**
- bool **mReplotting**
- bool **mReplotQueued**
- double **mReplotTime**
- double **mReplotTimeAverage**
- int **mOpenGlMultisamples**
- QCP::AntialiasedElements **mOpenGlAntialiasedElementsBackup**
- bool **mOpenGlCacheLabelsBackup**

Friends

- class **QCPLegend**
- class **QCPAxis**
- class **QCPLayer**
- class **QCPAxisRect**
- class **QCPAbstractPlottable**
- class **QCPGraph**
- class **QCPAbstractItem**

17.674.1 Detailed Description

The central class of the library. This is the QWidget which displays the plot and interacts with the user.
For tutorials on how to use QCustomPlot, see the website
http://www.qcustomplot.com/

17.674.2 Member Enumeration Documentation

17.674.2.1 LayerInsertMode

`enum QCustomPlot::LayerInsertMode`
Defines how a layer should be inserted relative to an other layer.

See also

addLayer, moveLayer

Enumerator

limBelow	Layer is inserted below other layer.
limAbove	Layer is inserted above other layer.

17.674.2.2 RefreshPriority

`enum QCustomPlot::RefreshPriority`
Defines with what timing the QCustomPlot surface is refreshed after a replot.

See also

replot

Enumerator

rpImmediateRefresh	Replots immediately and repaints the widget immediately by calling QWidget::repaint() after the replot.
rpQueuedRefresh	Replots immediately, but queues the widget repaint, by calling QWidget::update() after the replot. This way multiple redundant widget repaints can be avoided.
rpRefreshHint	Whether to use immediate or queued refresh depends on whether the plotting hint QCP::phImmediateRefresh is set, see setPlottingHints.
rpQueuedReplot	Queues the entire replot for the next event loop iteration. This way multiple redundant replots can be avoided. The actual replot is then done with rpRefreshHint priority.

17.674.3 Constructor & Destructor Documentation

17.674.3.1 QCustomPlot()

```
QCustomPlot::QCustomPlot (
            QWidget * parent = nullptr )  [explicit]
```
Constructs a QCustomPlot and sets reasonable default values.

17.674.4 Member Function Documentation

17.674.4.1 addGraph()

```
QCPGraph * QCustomPlot::addGraph (
            QCPAxis * keyAxis = nullptr,
            QCPAxis * valueAxis = nullptr )
```
Creates a new graph inside the plot. If *keyAxis* and *valueAxis* are left unspecified (0), the bottom (xAxis) is used as key and the left (yAxis) is used as value axis. If specified, *keyAxis* and *valueAxis* must reside in this QCustomPlot.
keyAxis will be used as key axis (typically "x") and *valueAxis* as value axis (typically "y") for the graph.
Returns a pointer to the newly created graph, or `nullptr` if adding the graph failed.

See also

graph, graphCount, removeGraph, clearGraphs

17.674.4.2 addLayer()

```
bool QCustomPlot::addLayer (
            const QString & name,
            QCPLayer * otherLayer = nullptr,
            QCustomPlot::LayerInsertMode insertMode = limAbove )
```

Adds a new layer to this QCustomPlot instance. The new layer will have the name *name*, which must be unique. Depending on *insertMode*, it is positioned either below or above *otherLayer*.
Returns true on success, i.e. if there is no other layer named *name* and *otherLayer* is a valid layer inside this QCustomPlot.
If *otherLayer* is 0, the highest layer in the QCustomPlot will be used.
For an explanation of what layers are in QCustomPlot, see the documentation of QCPLayer.

See also

layer, moveLayer, removeLayer

17.674.4.3 afterLayout

```
void QCustomPlot::afterLayout ( )  [signal]
```

This signal is emitted immediately after the layout step has been completed, which occurs right before drawing the plot. This is typically during a call to replot, and in such cases this signal is emitted in between the signals beforeReplot and afterReplot. Unlike those signals however, this signal is also emitted during off-screen painting, such as when calling toPixmap or savePdf.
The layout step queries all layouts and layout elements in the plot for their proposed size and arranges the objects accordingly as preparation for the subsequent drawing step. Through this signal, you have the opportunity to update certain things in your plot that depend crucially on the exact dimensions/positioning of layout elements such as axes and axis rects.

Warning

However, changing any parameters of this QCustomPlot instance which would normally affect the layouting (e.g. axis range order of magnitudes, tick label sizes, etc.) will not issue a second run of the layout step. It will propagate directly to the draw step and may cause graphical inconsistencies such as overlapping objects, if sizes or positions have changed.

See also

updateLayout, beforeReplot, afterReplot

17.674.4.4 afterReplot

```
void QCustomPlot::afterReplot ( )  [signal]
```

This signal is emitted immediately after a replot has taken place (caused by a call to the slot replot).
It is safe to mutually connect the replot slot with this signal on two QCustomPlots to make them replot synchronously, it won't cause an infinite recursion.

See also

replot, beforeReplot, afterLayout

17.674.4.5 axisClick

```
void QCustomPlot::axisClick (
            QCPAxis * axis,
            QCPAxis::SelectablePart part,
            QMouseEvent * event )  [signal]
```

This signal is emitted when an axis is clicked.
event is the mouse event that caused the click, *axis* is the axis that received the click and *part* indicates the part of the axis that was clicked.

See also

axisDoubleClick

17.674.4.6 axisDoubleClick

```
void QCustomPlot::axisDoubleClick (
            QCPAxis * axis,
            QCPAxis::SelectablePart part,
            QMouseEvent * event )  [signal]
```

This signal is emitted when an axis is double clicked.
event is the mouse event that caused the click, *axis* is the axis that received the click and *part* indicates the part of the axis that was clicked.

See also

axisClick

17.674.4.7 axisRect()

```
QCPAxisRect * QCustomPlot::axisRect (
            int index = 0 ) const
```

Returns the axis rect with *index*.
Initially, only one axis rect (with index 0) exists in the plot. If multiple axis rects were added, all of them may be accessed with this function in a linear fashion (even when they are nested in a layout hierarchy or inside other axis rects via QCPAxisRect::insetLayout).
The order of the axis rects is given by the fill order of the QCPLayout that is holding them. For example, if the axis rects are in the top level grid layout (accessible via QCustomPlot::plotLayout), they are ordered from left to right, top to bottom, if the layout's default setFillOrder of foColumnsFirst wasn't changed.
If you want to access axis rects by their row and column index, use the layout interface. For example, use QCPLayoutGrid::element of the top level grid layout, and `qobject_cast` the returned layout element to QCPAxisRect. (See also thelayoutsystem.)

See also

axisRectCount, axisRects, QCPLayoutGrid::setFillOrder

17.674.4.8 axisRectAt()

```
QCPAxisRect * QCustomPlot::axisRectAt (
            const QPointF & pos ) const
```

Returns the layout element of type QCPAxisRect at pixel position *pos*. This method ignores other layout elements even if they are visually in front of the axis rect (e.g. a QCPLegend). If there is no axis rect at that position, returns `nullptr`.
Only visible axis rects are used. If QCPLayoutElement::setVisible on the axis rect itself or on any of its parent elements is set to false, it will not be considered.

See also

layoutElementAt

17.674.4.9 axisRectCount()

```
int QCustomPlot::axisRectCount ( ) const
```

Returns the number of axis rects in the plot.
All axis rects can be accessed via QCustomPlot::axisRect().
Initially, only one axis rect exists in the plot.

See also

axisRect, axisRects

17.674.4.10 axisRects()

```
QList< QCPAxisRect * > QCustomPlot::axisRects ( ) const
```

Returns all axis rects in the plot.
The order of the axis rects is given by the fill order of the QCPLayout that is holding them. For example, if the axis rects are in the top level grid layout (accessible via QCustomPlot::plotLayout), they are ordered from left to right, top to bottom, if the layout's default setFillOrder of foColumnsFirst wasn't changed.

See also

axisRectCount, axisRect, QCPLayoutGrid::setFillOrder

17.674.4.11 beforeReplot

```
void QCustomPlot::beforeReplot ( )  [signal]
```

This signal is emitted immediately before a replot takes place (caused by a call to the slot replot).
It is safe to mutually connect the replot slot with this signal on two QCustomPlots to make them replot synchronously, it won't cause an infinite recursion.

See also

replot, afterReplot, afterLayout

17.674.4.12 clearGraphs()

```
int QCustomPlot::clearGraphs ( )
```

Removes all graphs from the plot and deletes them. Corresponding legend items are also removed from the default legend (QCustomPlot::legend).
Returns the number of graphs removed.

See also

removeGraph

17.674.4.13 clearItems()

```
int QCustomPlot::clearItems ( )
```

Removes all items from the plot and deletes them.
Returns the number of items removed.

See also

removeItem

17.674.4.14 clearPlottables()

```
int QCustomPlot::clearPlottables ( )
```

Removes all plottables from the plot and deletes them. Corresponding legend items are also removed from the default legend (QCustomPlot::legend).
Returns the number of plottables removed.

See also

removePlottable

17.674.4.15 currentLayer()

```
QCPLayer * QCustomPlot::currentLayer ( ) const
```

Returns the layer that is set as current layer (see setCurrentLayer).

17.674.4.16 deselectAll()

```
void QCustomPlot::deselectAll ( )
```

Deselects all layerables (plottables, items, axes, legends,...) of the QCustomPlot.
Since calling this function is not a user interaction, this does not emit the selectionChangedByUser signal. The individual selectionChanged signals are emitted though, if the objects were previously selected.

See also

setInteractions, selectedPlottables, selectedItems, selectedAxes, selectedLegends

17.674.4.17 graph() [1/2]

```
QCPGraph * QCustomPlot::graph ( ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Returns the last graph, that was created with addGraph. If there are no graphs in the plot, returns `nullptr`.

See also

graphCount, addGraph

17.674.4.18 graph() [2/2]

```
QCPGraph * QCustomPlot::graph (
            int index ) const
```

Returns the graph with *index*. If the index is invalid, returns `nullptr`.
There is an overloaded version of this function with no parameter which returns the last created graph, see QCustomPlot::graph()

See also

graphCount, addGraph

17.674.4.19 graphCount()

```
int QCustomPlot::graphCount ( ) const
```

Returns the number of currently existing graphs in the plot

See also

graph, addGraph

17.674.4.20 hasInvalidatedPaintBuffers()

```
bool QCustomPlot::hasInvalidatedPaintBuffers ( )  [protected]
```

This method returns whether any of the paint buffers held by this QCustomPlot instance are invalidated.
If any buffer is invalidated, a partial replot (QCPLayer::replot) is not allowed and always causes a full replot (QCustomPlot::replot) of all layers. This is the case when for example the layer order has changed, new layers were added or removed, layer modes were changed (QCPLayer::setMode), or layerables were added or removed.

See also

QCPAbstractPaintBuffer::setInvalidated

17.674.4.21 hasItem()

```
bool QCustomPlot::hasItem (
            QCPAbstractItem * item ) const
```

Returns whether this QCustomPlot contains the *item*.

See also

item

17.674.4.22 hasPlottable()

```
bool QCustomPlot::hasPlottable (
            QCPAbstractPlottable * plottable ) const
```

Returns whether this QCustomPlot instance contains the *plottable*.

17.674.4.23 item() [1/2]

```
QCPAbstractItem * QCustomPlot::item ( ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Returns the last item that was added to this plot. If there are no items in the plot, returns `nullptr`.

See also

itemCount

17.674.4.24 item() [2/2]

```
QCPAbstractItem * QCustomPlot::item (
            int index ) const
```

Returns the item with *index*. If the index is invalid, returns `nullptr`.
There is an overloaded version of this function with no parameter which returns the last added item, see QCustomPlot::item()

See also

itemCount

17.674.4.25 itemAt() [1/2]

```
template<class ItemType >
ItemType * QCustomPlot::itemAt (
            const QPointF & pos,
            bool onlySelectable = false ) const
```

Returns the item at the pixel position *pos*. The item type (a QCPAbstractItem subclass) that shall be taken into consideration can be specified via the template parameter. Items that only consist of single lines (e.g. QCPItemLine or QCPItemCurve) have a tolerance band around them, see setSelectionTolerance. If multiple items come into consideration, the one closest to *pos* is returned.
If *onlySelectable* is true, only items that are selectable (QCPAbstractItem::setSelectable) are considered.
If there is no item at *pos*, returns `nullptr`.

See also

plottableAt, layoutElementAt

17.674.4.26 itemAt() [2/2]

```
QCPAbstractItem * QCustomPlot::itemAt (
            const QPointF & pos,
            bool onlySelectable = false ) const
```

Returns the item at the pixel position *pos*. Since it can capture all items, the return type is the abstract base class of all items, QCPAbstractItem.
For details, and if you wish to specify a certain item type (e.g. QCPItemLine), see the template method itemAt<ItemType>()

See also

itemAt<ItemType>(), plottableAt, layoutElementAt

17.674.4.27 itemClick

```
void QCustomPlot::itemClick (
            QCPAbstractItem * item,
            QMouseEvent * event )  [signal]
```

This signal is emitted when an item is clicked.
event is the mouse event that caused the click and *item* is the item that received the click.

See also

itemDoubleClick

17.674.4.28 itemCount()

```
int QCustomPlot::itemCount ( ) const
```

Returns the number of currently existing items in the plot

See also

item

17.674.4.29 itemDoubleClick

```
void QCustomPlot::itemDoubleClick (
            QCPAbstractItem * item,
            QMouseEvent * event )  [signal]
```

This signal is emitted when an item is double clicked.
event is the mouse event that caused the click and *item* is the item that received the click.

See also

itemClick

17.674.4.30 layer() [1/2]

```
QCPLayer * QCustomPlot::layer (
            const QString & name ) const
```

Returns the layer with the specified *name*. If there is no layer with the specified name, `nullptr` is returned.
Layer names are case-sensitive.

See also

addLayer, moveLayer, removeLayer

17.674.4.31 layer() [2/2]

```
QCPLayer * QCustomPlot::layer (
            int index ) const
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Returns the layer by *index*. If the index is invalid, `nullptr` is returned.

See also

addLayer, moveLayer, removeLayer

17.674.4.32 layerCount()

```
int QCustomPlot::layerCount ( ) const
```

Returns the number of currently existing layers in the plot

See also

layer, addLayer

17.674.4.33 layoutElementAt()

```
QCPLayoutElement * QCustomPlot::layoutElementAt (
            const QPointF & pos ) const
```

Returns the layout element at pixel position *pos*. If there is no element at that position, returns `nullptr`.
Only visible elements are used. If QCPLayoutElement::setVisible on the element itself or on any of its parent elements is set to false, it will not be considered.

See also

itemAt, plottableAt

17.674.4.34 legendClick

```
void QCustomPlot::legendClick (
            QCPLegend * legend,
            QCPAbstractLegendItem * item,
            QMouseEvent * event )  [signal]
```

This signal is emitted when a legend (item) is clicked.
event is the mouse event that caused the click, *legend* is the legend that received the click and *item* is the legend item that received the click. If only the legend and no item is clicked, *item* is `nullptr`. This happens for a click inside the legend padding or the space between two items.

See also

legendDoubleClick

17.674.4.35 legendDoubleClick

```
void QCustomPlot::legendDoubleClick (
            QCPLegend * legend,
            QCPAbstractLegendItem * item,
            QMouseEvent * event )  [signal]
```

This signal is emitted when a legend (item) is double clicked.
event is the mouse event that caused the click, *legend* is the legend that received the click and *item* is the legend item that received the click. If only the legend and no item is clicked, *item* is `nullptr`. This happens for a click inside the legend padding or the space between two items.

See also

legendClick

17.674.4.36 mouseDoubleClick

```
void QCustomPlot::mouseDoubleClick (
            QMouseEvent * event )  [signal]
```

This signal is emitted when the QCustomPlot receives a mouse double click event.

17.674.4.37 mouseMove

```
void QCustomPlot::mouseMove (
            QMouseEvent * event )  [signal]
```

This signal is emitted when the QCustomPlot receives a mouse move event.
It is emitted before QCustomPlot handles any other mechanism like range dragging. So a slot connected to this signal can still influence the behaviour e.g. with QCPAxisRect::setRangeDrag or QCPAxisRect::setRangeDragAxes.

Warning

> It is discouraged to change the drag-axes with QCPAxisRect::setRangeDragAxes here, because the dragging starting point was saved the moment the mouse was pressed. Thus it only has a meaning for the range drag axes that were set at that moment. If you want to change the drag axes, consider doing this in the mousePress signal instead.

17.674.4.38 mousePress

```
void QCustomPlot::mousePress (
            QMouseEvent * event )  [signal]
```

This signal is emitted when the QCustomPlot receives a mouse press event.
It is emitted before QCustomPlot handles any other mechanism like range dragging. So a slot connected to this signal can still influence the behaviour e.g. with QCPAxisRect::setRangeDrag or QCPAxisRect::setRangeDragAxes.

17.674.4.39 mouseRelease

```
void QCustomPlot::mouseRelease (
            QMouseEvent * event )  [signal]
```

This signal is emitted when the QCustomPlot receives a mouse release event.
It is emitted before QCustomPlot handles any other mechanisms like object selection. So a slot connected to this signal can still influence the behaviour e.g. with setInteractions or QCPAbstractPlottable::setSelectable.

17.674.4.40 mouseWheel

```
void QCustomPlot::mouseWheel (
            QWheelEvent * event )  [signal]
```

This signal is emitted when the QCustomPlot receives a mouse wheel event.
It is emitted before QCustomPlot handles any other mechanisms like range zooming. So a slot connected to this signal can still influence the behaviour e.g. with QCPAxisRect::setRangeZoom, QCPAxisRect::setRangeZoomAxes or QCPAxisRect::setRangeZoomFactor.

17.674.4.41 moveLayer()

```
bool QCustomPlot::moveLayer (
            QCPLayer * layer,
            QCPLayer * otherLayer,
            QCustomPlot::LayerInsertMode insertMode = limAbove )
```

Moves the specified *layer* either above or below *otherLayer*. Whether it's placed above or below is controlled with *insertMode*.
Returns true on success, i.e. when both *layer* and *otherLayer* are valid layers in the QCustomPlot.

See also

> layer, addLayer, moveLayer

17.674.4.42 plotLayout()

```
QCPLayoutGrid * QCustomPlot::plotLayout ( ) const  [inline]
```

Returns the top level layout of this QCustomPlot instance. It is a QCPLayoutGrid, initially containing just one cell with the main QCPAxisRect inside.

17.674.4.43 plottable() [1/2]

```
QCPAbstractPlottable * QCustomPlot::plottable ( )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Returns the last plottable that was added to the plot. If there are no plottables in the plot, returns `nullptr`.

See also

plottableCount

17.674.4.44 plottable() [2/2]

```
QCPAbstractPlottable * QCustomPlot::plottable (
            int index )
```

Returns the plottable with *index*. If the index is invalid, returns `nullptr`.
There is an overloaded version of this function with no parameter which returns the last added plottable, see QCustomPlot::plottable()

See also

plottableCount

17.674.4.45 plottableAt() [1/2]

```
template<class PlottableType >
PlottableType * QCustomPlot::plottableAt (
            const QPointF & pos,
            bool onlySelectable = false,
            int * dataIndex = nullptr ) const
```

Returns the plottable at the pixel position *pos*. The plottable type (a QCPAbstractPlottable subclass) that shall be taken into consideration can be specified via the template parameter.
Plottables that only consist of single lines (like graphs) have a tolerance band around them, see setSelectionTolerance. If multiple plottables come into consideration, the one closest to *pos* is returned.
If *onlySelectable* is true, only plottables that are selectable (QCPAbstractPlottable::setSelectable) are considered.
if *dataIndex* is non-null, it is set to the index of the plottable's data point that is closest to *pos*.
If there is no plottable of the specified type at *pos*, returns `nullptr`.

See also

itemAt, layoutElementAt

17.674.4.46 plottableAt() [2/2]

```
QCPAbstractPlottable * QCustomPlot::plottableAt (
            const QPointF & pos,
            bool onlySelectable = false,
            int * dataIndex = nullptr ) const
```

Returns any plottable at the pixel position *pos*. Since it can capture all plottables, the return type is the abstract base class of all plottables, QCPAbstractPlottable.
For details, and if you wish to specify a certain plottable type (e.g. QCPGraph), see the template method plottableAt<PlottableType>()

See also

plottableAt<PlottableType>(), itemAt, layoutElementAt

17.674.4.47 plottableClick

```
void QCustomPlot::plottableClick (
            QCPAbstractPlottable * plottable,
            int dataIndex,
            QMouseEvent * event )  [signal]
```

This signal is emitted when a plottable is clicked.
event is the mouse event that caused the click and *plottable* is the plottable that received the click. The parameter *dataIndex* indicates the data point that was closest to the click position.

See also

plottableDoubleClick

17.674.4.48 plottableCount()

```
int QCustomPlot::plottableCount ( ) const
```

Returns the number of currently existing plottables in the plot

See also

plottable

17.674.4.49 plottableDoubleClick

```
void QCustomPlot::plottableDoubleClick (
            QCPAbstractPlottable * plottable,
            int dataIndex,
            QMouseEvent * event )  [signal]
```

This signal is emitted when a plottable is double clicked.
event is the mouse event that caused the click and *plottable* is the plottable that received the click. The parameter *dataIndex* indicates the data point that was closest to the click position.

See also

plottableClick

17.674.4.50 removeGraph() [1/2]

```
bool QCustomPlot::removeGraph (
            int index )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Removes and deletes the graph by its *index*.

17.674.4.51 removeGraph() [2/2]

```
bool QCustomPlot::removeGraph (
            QCPGraph * graph )
```

Removes the specified *graph* from the plot and deletes it. If necessary, the corresponding legend item is also removed from the default legend (QCustomPlot::legend). If any other graphs in the plot have a channel fill set towards the removed graph, the channel fill property of those graphs is reset to `nullptr` (no channel fill).
Returns true on success.

See also

clearGraphs

17.674.4.52 removeItem() [1/2]

```
bool QCustomPlot::removeItem (
            int index )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Removes and deletes the item by its *index*.

17.674.4.53 removeItem() [2/2]

```
bool QCustomPlot::removeItem (
            QCPAbstractItem * item )
```

Removes the specified item from the plot and deletes it.
Returns true on success.

See also

clearItems

17.674.4.54 removeLayer()

```
bool QCustomPlot::removeLayer (
            QCPLayer * layer )
```

Removes the specified *layer* and returns true on success.
All layerables (e.g. plottables and items) on the removed layer will be moved to the layer below *layer*. If *layer* is the bottom layer, the layerables are moved to the layer above. In both cases, the total rendering order of all layerables in the QCustomPlot is preserved.
If *layer* is the current layer (setCurrentLayer), the layer below (or above, if bottom layer) becomes the new current layer.
It is not possible to remove the last layer of the plot.

See also

layer, addLayer, moveLayer

17.674.4.55 removePlottable() [1/2]

```
bool QCustomPlot::removePlottable (
            int index )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Removes and deletes the plottable by its *index*.

17.674.4.56 removePlottable() [2/2]

```
bool QCustomPlot::removePlottable (
            QCPAbstractPlottable * plottable )
```

Removes the specified plottable from the plot and deletes it. If necessary, the corresponding legend item is also removed from the default legend (QCustomPlot::legend).
Returns true on success.

See also

clearPlottables

17.674.4.57 replot()

```
void QCustomPlot::replot (
            QCustomPlot::RefreshPriority refreshPriority = QCustomPlot::rpRefreshHint )
```

Causes a complete replot into the internal paint buffer(s). Finally, the widget surface is refreshed with the new buffer contents. This is the method that must be called to make changes to the plot, e.g. on the axis ranges or data points of graphs, visible.
The parameter *refreshPriority* can be used to fine-tune the timing of the replot. For example if your application calls replot very quickly in succession (e.g. multiple independent functions change some aspects of the plot and each wants to make sure the change gets replotted), it is advisable to set *refreshPriority* to QCustomPlot::rpQueuedReplot. This way, the actual replotting is deferred to the next event loop iteration. Multiple successive calls of replot with this priority will only cause a single replot, avoiding redundant replots and improving performance.
Under a few circumstances, QCustomPlot causes a replot by itself. Those are resize events of the QCustomPlot widget and user interactions (object selection and range dragging/zooming).
Before the replot happens, the signal beforeReplot is emitted. After the replot, afterReplot is emitted. It is safe to mutually connect the replot slot with any of those two signals on two QCustomPlots to make them replot synchronously, it won't cause an infinite recursion.
If a layer is in mode QCPLayer::lmBuffered (QCPLayer::setMode), it is also possible to replot only that specific layer via QCPLayer::replot. See the documentation there for details.

See also

replotTime

17.674.4.58 replotTime()

```
double QCustomPlot::replotTime (
            bool average = false ) const
```

Returns the time in milliseconds that the last replot took. If *average* is set to true, an exponential moving average over the last couple of replots is returned.

See also

replot

17.674.4.59 rescaleAxes()

```
void QCustomPlot::rescaleAxes (
            bool onlyVisiblePlottables = false )
```

Rescales the axes such that all plottables (like graphs) in the plot are fully visible.
if *onlyVisiblePlottables* is set to true, only the plottables that have their visibility set to true (QCPLayerable::setVisible), will be used to rescale the axes.

See also

QCPAbstractPlottable::rescaleAxes, QCPAxis::rescale

17.674.4.60 saveBmp()

```
bool QCustomPlot::saveBmp (
            const QString & fileName,
            int width = 0,
            int height = 0,
            double scale = 1.0,
            int resolution = 96,
            QCP::ResolutionUnit resolutionUnit = QCP::ruDotsPerInch )
```

Saves a BMP image file to *fileName* on disc. The output plot will have the dimensions *width* and *height* in pixels, multiplied by *scale*. If either *width* or *height* is zero, the current width and height of the QCustomPlot widget is used instead. Line widths and texts etc. are not scaled up when larger widths/heights are used. If you want that effect, use the *scale* parameter.
For example, if you set both *width* and *height* to 100 and *scale* to 2, you will end up with an image file of size 200∗200 in which all graphical elements are scaled up by factor 2 (line widths, texts, etc.). This scaling is not done by stretching a 100∗100 image, the result will have full 200∗200 pixel resolution.

If you use a high scaling factor, it is recommended to enable antialiasing for all elements by temporarily setting QCustomPlot::setAntialiasedElements to QCP::aeAll as this allows QCustomPlot to place objects with sub-pixel accuracy.
The *resolution* will be written to the image file header and has no direct consequence for the quality or the pixel size. However, if opening the image with a tool which respects the metadata, it will be able to scale the image to match either a given size in real units of length (inch, centimeters, etc.), or the target display DPI. You can specify in which units *resolution* is given, by setting *resolutionUnit*. The *resolution* is converted to the format's expected resolution unit internally.
Returns true on success. If this function fails, most likely the BMP format isn't supported by the system, see Qt docs about QImageWriter::supportedImageFormats().
The objects of the plot will appear in the current selection state. If you don't want any selected objects to be painted in their selected look, deselect everything with deselectAll before calling this function.

Warning

> If calling this function inside the constructor of the parent of the QCustomPlot widget (i.e. the MainWindow constructor, if QCustomPlot is inside the MainWindow), always provide explicit non-zero widths and heights. If you leave *width* or *height* as 0 (default), this function uses the current width and height of the QCustomPlot widget. However, in Qt, these aren't defined yet inside the constructor, so you would get an image that has strange widths/heights.

See also

savePdf, savePng, saveJpg, saveRastered

17.674.4.61 saveJpg()

```
bool QCustomPlot::saveJpg (
              const QString & fileName,
              int width = 0,
              int height = 0,
              double scale = 1.0,
              int quality = -1,
              int resolution = 96,
              QCP::ResolutionUnit resolutionUnit = QCP::ruDotsPerInch )
```

Saves a JPEG image file to *fileName* on disc. The output plot will have the dimensions *width* and *height* in pixels, multiplied by *scale*. If either *width* or *height* is zero, the current width and height of the QCustomPlot widget is used instead. Line widths and texts etc. are not scaled up when larger widths/heights are used. If you want that effect, use the *scale* parameter.
For example, if you set both *width* and *height* to 100 and *scale* to 2, you will end up with an image file of size 200∗200 in which all graphical elements are scaled up by factor 2 (line widths, texts, etc.). This scaling is not done by stretching a 100∗100 image, the result will have full 200∗200 pixel resolution.
If you use a high scaling factor, it is recommended to enable antialiasing for all elements by temporarily setting QCustomPlot::setAntialiasedElements to QCP::aeAll as this allows QCustomPlot to place objects with sub-pixel accuracy.
image compression can be controlled with the *quality* parameter which must be between 0 and 100 or -1 to use the default setting.
The *resolution* will be written to the image file header and has no direct consequence for the quality or the pixel size. However, if opening the image with a tool which respects the metadata, it will be able to scale the image to match either a given size in real units of length (inch, centimeters, etc.), or the target display DPI. You can specify in which units *resolution* is given, by setting *resolutionUnit*. The *resolution* is converted to the format's expected resolution unit internally.
Returns true on success. If this function fails, most likely the JPEG format isn't supported by the system, see Qt docs about QImageWriter::supportedImageFormats().
The objects of the plot will appear in the current selection state. If you don't want any selected objects to be painted in their selected look, deselect everything with deselectAll before calling this function.

Warning

If calling this function inside the constructor of the parent of the QCustomPlot widget (i.e. the MainWindow constructor, if QCustomPlot is inside the MainWindow), always provide explicit non-zero widths and heights. If you leave *width* or *height* as 0 (default), this function uses the current width and height of the QCustomPlot widget. However, in Qt, these aren't defined yet inside the constructor, so you would get an image that has strange widths/heights.

See also

savePdf, savePng, saveBmp, saveRastered

17.674.4.62 savePdf()

```
bool QCustomPlot::savePdf (
            const QString & fileName,
            int width = 0,
            int height = 0,
            QCP::ExportPen exportPen = QCP::epAllowCosmetic,
            const QString & pdfCreator = QString(),
            const QString & pdfTitle = QString() )
```

Saves a PDF with the vectorized plot to the file *fileName*. The axis ratio as well as the scale of texts and lines will be derived from the specified *width* and *height*. This means, the output will look like the normal on-screen output of a QCustomPlot widget with the corresponding pixel width and height. If either *width* or *height* is zero, the exported image will have the same dimensions as the QCustomPlot widget currently has.
Setting *exportPen* to QCP::epNoCosmetic allows to disable the use of cosmetic pens when drawing to the PDF file. Cosmetic pens are pens with numerical width 0, which are always drawn as a one pixel wide line, no matter what zoom factor is set in the PDF-Viewer. For more information about cosmetic pens, see the QPainter and QPen documentation.
The objects of the plot will appear in the current selection state. If you don't want any selected objects to be painted in their selected look, deselect everything with deselectAll before calling this function.
Returns true on success.

Warning

- If you plan on editing the exported PDF file with a vector graphics editor like Inkscape, it is advised to set *exportPen* to QCP::epNoCosmetic to avoid losing those cosmetic lines (which might be quite many, because cosmetic pens are the default for e.g. axes and tick marks).
- If calling this function inside the constructor of the parent of the QCustomPlot widget (i.e. the MainWindow constructor, if QCustomPlot is inside the MainWindow), always provide explicit non-zero widths and heights. If you leave *width* or *height* as 0 (default), this function uses the current width and height of the QCustomPlot widget. However, in Qt, these aren't defined yet inside the constructor, so you would get an image that has strange widths/heights.

pdfCreator and *pdfTitle* may be used to set the according metadata fields in the resulting PDF file.

Note

On Android systems, this method does nothing and issues an according qDebug warning message. This is also the case if for other reasons the define flag `QT_NO_PRINTER` is set.

See also

savePng, saveBmp, saveJpg, saveRastered

17.674.4.63 savePng()

```
bool QCustomPlot::savePng (
            const QString & fileName,
```

```
int width = 0,
int height = 0,
double scale = 1.0,
int quality = -1,
int resolution = 96,
QCP::ResolutionUnit resolutionUnit = QCP::ruDotsPerInch )
```

Saves a PNG image file to *fileName* on disc. The output plot will have the dimensions *width* and *height* in pixels, multiplied by *scale*. If either *width* or *height* is zero, the current width and height of the QCustomPlot widget is used instead. Line widths and texts etc. are not scaled up when larger widths/heights are used. If you want that effect, use the *scale* parameter.
For example, if you set both *width* and *height* to 100 and *scale* to 2, you will end up with an image file of size 200∗200 in which all graphical elements are scaled up by factor 2 (line widths, texts, etc.). This scaling is not done by stretching a 100∗100 image, the result will have full 200∗200 pixel resolution.
If you use a high scaling factor, it is recommended to enable antialiasing for all elements by temporarily setting QCustomPlot::setAntialiasedElements to QCP::aeAll as this allows QCustomPlot to place objects with sub-pixel accuracy.
image compression can be controlled with the *quality* parameter which must be between 0 and 100 or -1 to use the default setting.
The *resolution* will be written to the image file header and has no direct consequence for the quality or the pixel size. However, if opening the image with a tool which respects the metadata, it will be able to scale the image to match either a given size in real units of length (inch, centimeters, etc.), or the target display DPI. You can specify in which units *resolution* is given, by setting *resolutionUnit*. The *resolution* is converted to the format's expected resolution unit internally.
Returns true on success. If this function fails, most likely the PNG format isn't supported by the system, see Qt docs about QImageWriter::supportedImageFormats().
The objects of the plot will appear in the current selection state. If you don't want any selected objects to be painted in their selected look, deselect everything with deselectAll before calling this function.
If you want the PNG to have a transparent background, call setBackground(const QBrush &brush) with no brush (Qt::NoBrush) or a transparent color (Qt::transparent), before saving.

Warning

> If calling this function inside the constructor of the parent of the QCustomPlot widget (i.e. the MainWindow constructor, if QCustomPlot is inside the MainWindow), always provide explicit non-zero widths and heights. If you leave *width* or *height* as 0 (default), this function uses the current width and height of the QCustomPlot widget. However, in Qt, these aren't defined yet inside the constructor, so you would get an image that has strange widths/heights.

See also

savePdf, saveBmp, saveJpg, saveRastered

17.674.4.64 saveRastered()

```
bool QCustomPlot::saveRastered (
const QString & fileName,
int width,
int height,
double scale,
const char * format,
int quality = -1,
int resolution = 96,
QCP::ResolutionUnit resolutionUnit = QCP::ruDotsPerInch )
```

Saves the plot to a rastered image file *fileName* in the image format *format*. The plot is sized to *width* and *height* in pixels and scaled with *scale*. (width 100 and scale 2.0 lead to a full resolution file with width 200.) If the *format* supports compression, *quality* may be between 0 and 100 to control it.
Returns true on success. If this function fails, most likely the given *format* isn't supported by the system, see Qt docs about QImageWriter::supportedImageFormats().

The *resolution* will be written to the image file header (if the file format supports this) and has no direct consequence for the quality or the pixel size. However, if opening the image with a tool which respects the metadata, it will be able to scale the image to match either a given size in real units of length (inch, centimeters, etc.), or the target display DPI. You can specify in which units *resolution* is given, by setting *resolutionUnit*. The *resolution* is converted to the format's expected resolution unit internally.

See also

saveBmp, saveJpg, savePng, savePdf

17.674.4.65 selectedAxes()

`QList< QCPAxis * > QCustomPlot::selectedAxes ( ) const`
Returns the axes that currently have selected parts, i.e. whose selection state is not QCPAxis::spNone.

See also

selectedPlottables, selectedLegends, setInteractions, QCPAxis::setSelectedParts, QCPAxis::setSelectableParts

17.674.4.66 selectedGraphs()

`QList< QCPGraph * > QCustomPlot::selectedGraphs ( ) const`
Returns a list of the selected graphs. If no graphs are currently selected, the list is empty.
If you are not only interested in selected graphs but other plottables like QCPCurve, QCPBars, etc., use selectedPlottables.

See also

setInteractions, selectedPlottables, QCPAbstractPlottable::setSelectable, QCPAbstractPlottable::setSelection

17.674.4.67 selectedItems()

`QList< QCPAbstractItem * > QCustomPlot::selectedItems ( ) const`
Returns a list of the selected items. If no items are currently selected, the list is empty.

See also

setInteractions, QCPAbstractItem::setSelectable, QCPAbstractItem::setSelected

17.674.4.68 selectedLegends()

`QList< QCPLegend * > QCustomPlot::selectedLegends ( ) const`
Returns the legends that currently have selected parts, i.e. whose selection state is not QCPLegend::spNone.

See also

selectedPlottables, selectedAxes, setInteractions, QCPLegend::setSelectedParts, QCPLegend::setSelectableParts, QCPLegend::selectedItems

17.674.4.69 selectedPlottables()

`QList< QCPAbstractPlottable * > QCustomPlot::selectedPlottables ( ) const`
Returns a list of the selected plottables. If no plottables are currently selected, the list is empty.
There is a convenience function if you're only interested in selected graphs, see selectedGraphs.

See also

setInteractions, QCPAbstractPlottable::setSelectable, QCPAbstractPlottable::setSelection

17.674.4.70 selectionChangedByUser

```
void QCustomPlot::selectionChangedByUser ( )  [signal]
```

This signal is emitted after the user has changed the selection in the QCustomPlot, e.g. by clicking. It is not emitted when the selection state of an object has changed programmatically by a direct call to `setSelected()`/`setSelection()` on an object or by calling deselectAll.

In addition to this signal, selectable objects also provide individual signals, for example QCPAxis::selectionChanged or QCPAbstractPlottable::selectionChanged. Note that those signals are emitted even if the selection state is changed programmatically.

See the documentation of setInteractions for details about the selection mechanism.

See also

selectedPlottables, selectedGraphs, selectedItems, selectedAxes, selectedLegends

17.674.4.71 selectionRect()

```
QCPSelectionRect * QCustomPlot::selectionRect ( ) const  [inline]
```

Allows access to the currently used QCPSelectionRect instance (or subclass thereof), that is used to handle and draw selection rect interactions (see setSelectionRectMode).

See also

setSelectionRect

17.674.4.72 setAntialiasedElement()

```
void QCustomPlot::setAntialiasedElement (
            QCP::AntialiasedElement antialiasedElement,
            bool enabled = true )
```

Sets whether the specified *antialiasedElement* is forcibly drawn antialiased.

See setAntialiasedElements for details.

See also

setNotAntialiasedElement

17.674.4.73 setAntialiasedElements()

```
void QCustomPlot::setAntialiasedElements (
            const QCP::AntialiasedElements & antialiasedElements )
```

Sets which elements are forcibly drawn antialiased as an *or* combination of QCP::AntialiasedElement.

This overrides the antialiasing settings for whole element groups, normally controlled with the *setAntialiasing* function on the individual elements. If an element is neither specified in setAntialiasedElements nor in setNotAntialiasedElements, the antialiasing setting on each individual element instance is used.

For example, if *antialiasedElements* contains QCP::aePlottables, all plottables will be drawn antialiased, no matter what the specific QCPAbstractPlottable::setAntialiased value was set to.

if an element in *antialiasedElements* is already set in setNotAntialiasedElements, it is removed from there.

See also

setNotAntialiasedElements

17.674.4.74 setAutoAddPlottableToLegend()

```
void QCustomPlot::setAutoAddPlottableToLegend (
            bool on )
```

If set to true, adding a plottable (e.g. a graph) to the QCustomPlot automatically also adds the plottable to the legend (QCustomPlot::legend).

See also

addGraph, QCPLegend::addItem

17.674.4.75 setBackground() [1/3]

```
void QCustomPlot::setBackground (
            const QBrush & brush )
```

Sets the background brush of the viewport (see setViewport).
Before drawing everything else, the background is filled with *brush*. If a background pixmap was set with setBackground(const QPixmap &pm), this brush will be used to fill the viewport before the background pixmap is drawn. This can be useful for background pixmaps with translucent areas.
Set *brush* to Qt::NoBrush or Qt::Transparent to leave background transparent. This can be useful for exporting to image formats which support transparency, e.g. savePng.

See also

setBackgroundScaled, setBackgroundScaledMode

17.674.4.76 setBackground() [2/3]

```
void QCustomPlot::setBackground (
            const QPixmap & pm )
```

Sets *pm* as the viewport background pixmap (see setViewport). The pixmap is always drawn below all other objects in the plot.
For cases where the provided pixmap doesn't have the same size as the viewport, scaling can be enabled with setBackgroundScaled and the scaling mode (whether and how the aspect ratio is preserved) can be set with setBackgroundScaledMode. To set all these options in one call, consider using the overloaded version of this function.
If a background brush was set with setBackground(const QBrush &brush), the viewport will first be filled with that brush, before drawing the background pixmap. This can be useful for background pixmaps with translucent areas.

See also

setBackgroundScaled, setBackgroundScaledMode

17.674.4.77 setBackground() [3/3]

```
void QCustomPlot::setBackground (
            const QPixmap & pm,
            bool scaled,
            Qt::AspectRatioMode mode = Qt::KeepAspectRatioByExpanding )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Allows setting the background pixmap of the viewport, whether it shall be scaled and how it shall be scaled in one call.

See also

setBackground(const QPixmap &pm), setBackgroundScaled, setBackgroundScaledMode

17.674.4.78 setBackgroundScaled()

```
void QCustomPlot::setBackgroundScaled (
            bool scaled )
```

Sets whether the viewport background pixmap shall be scaled to fit the viewport. If *scaled* is set to true, control whether and how the aspect ratio of the original pixmap is preserved with setBackgroundScaledMode.
Note that the scaled version of the original pixmap is buffered, so there is no performance penalty on replots. (Except when the viewport dimensions are changed continuously.)

See also

setBackground, setBackgroundScaledMode

17.674.4.79 setBackgroundScaledMode()

```
void QCustomPlot::setBackgroundScaledMode (
            Qt::AspectRatioMode mode )
```

If scaling of the viewport background pixmap is enabled (setBackgroundScaled), use this function to define whether and how the aspect ratio of the original pixmap is preserved.

See also

setBackground, setBackgroundScaled

17.674.4.80 setBufferDevicePixelRatio()

```
void QCustomPlot::setBufferDevicePixelRatio (
            double ratio )
```

Sets the device pixel ratio used by the paint buffers of this QCustomPlot instance.
Normally, this doesn't need to be set manually, because it is initialized with the regular *QWidget::devicePixelRatio* which is configured by Qt to fit the display device (e.g. 1 for normal displays, 2 for High-DPI displays).
Device pixel ratios are supported by Qt only for Qt versions since 5.4. If this method is called when QCustomPlot is being used with older Qt versions, outputs an according qDebug message and leaves the internal buffer device pixel ratio at 1.0.

17.674.4.81 setCurrentLayer() [1/2]

```
bool QCustomPlot::setCurrentLayer (
            const QString & name )
```

Sets the layer with the specified *name* to be the current layer. All layerables (QCPLayerable), e.g. plottables and items, are created on the current layer.
Returns true on success, i.e. if there is a layer with the specified *name* in the QCustomPlot.
Layer names are case-sensitive.

See also

addLayer, moveLayer, removeLayer, QCPLayerable::setLayer

17.674.4.82 setCurrentLayer() [2/2]

```
bool QCustomPlot::setCurrentLayer (
            QCPLayer * layer )
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Sets the provided *layer* to be the current layer.
Returns true on success, i.e. when *layer* is a valid layer in the QCustomPlot.

See also

addLayer, moveLayer, removeLayer

17.674.4.83 setInteraction()

```
void QCustomPlot::setInteraction (
            const QCP::Interaction & interaction,
            bool enabled = true )
```

Sets the single *interaction* of this QCustomPlot to *enabled*.
For details about the interaction system, see setInteractions.

See also

setInteractions

17.674.4.84 setInteractions()

```
void QCustomPlot::setInteractions (
            const QCP::Interactions & interactions )
```

Sets the possible interactions of this QCustomPlot as an or-combination of QCP::Interaction enums. There are the following types of interactions:

Axis range manipulation is controlled via QCP::iRangeDrag and QCP::iRangeZoom. When the respective interaction is enabled, the user may drag axes ranges and zoom with the mouse wheel. For details how to control which axes the user may drag/zoom and in what orientations, see QCPAxisRect::setRangeDrag, QCPAxisRect::setRangeZoom, QCPAxisRect::setRangeDragAxes, QCPAxisRect::setRangeZoomAxes.

Plottable data selection is controlled by QCP::iSelectPlottables. If QCP::iSelectPlottables is set, the user may select plottables (graphs, curves, bars,...) and their data by clicking on them or in their vicinity (setSelectionTolerance). Whether the user can actually select a plottable and its data can further be restricted with the QCPAbstractPlottable::setSelectable method on the specific plottable. For details, see the special page about the data selection mechanism. To retrieve a list of all currently selected plottables, call selectedPlottables. If you're only interested in QCPGraphs, you may use the convenience function selectedGraphs.

Item selection is controlled by QCP::iSelectItems. If QCP::iSelectItems is set, the user may select items (QCPItemLine, QCPItemText,...) by clicking on them or in their vicinity. To find out whether a specific item is selected, call QCPAbstractItem::selected(). To retrieve a list of all currently selected items, call selectedItems.

Axis selection is controlled with QCP::iSelectAxes. If QCP::iSelectAxes is set, the user may select parts of the axes by clicking on them. What parts exactly (e.g. Axis base line, tick labels, axis label) are selectable can be controlled via QCPAxis::setSelectableParts for each axis. To retrieve a list of all axes that currently contain selected parts, call selectedAxes. Which parts of an axis are selected, can be retrieved with QCPAxis::selectedParts().

Legend selection is controlled with QCP::iSelectLegend. If this is set, the user may select the legend itself or individual items by clicking on them. What parts exactly are selectable can be controlled via QCPLegend::setSelectableParts. To find out whether the legend or any of its child items are selected, check the value of QCPLegend::selectedParts. To find out which child items are selected, call QCPLegend::selectedItems.

All other selectable elements The selection of all other selectable objects (e.g. QCPTextElement, or your own layerable subclasses) is controlled with QCP::iSelectOther. If set, the user may select those objects by clicking on them. To find out which are currently selected, you need to check their selected state explicitly.

If the selection state has changed by user interaction, the selectionChangedByUser signal is emitted. Each selectable object additionally emits an individual selectionChanged signal whenever their selection state has changed, i.e. not only by user interaction.

To allow multiple objects to be selected by holding the selection modifier (setMultiSelectModifier), set the flag QCP::iMultiSelect.

Note

In addition to the selection mechanism presented here, QCustomPlot always emits corresponding signals, when an object is clicked or double clicked. see plottableClick and plottableDoubleClick for example.

See also

setInteraction, setSelectionTolerance

17.674.4.85 setMultiSelectModifier()

```
void QCustomPlot::setMultiSelectModifier (
            Qt::KeyboardModifier modifier )
```

Sets the keyboard modifier that will be recognized as multi-select-modifier.

If QCP::iMultiSelect is specified in setInteractions, the user may select multiple objects (or data points) by clicking on them one after the other while holding down *modifier*.

By default the multi-select-modifier is set to Qt::ControlModifier.

See also

setInteractions

17.674.4.86 setNoAntialiasingOnDrag()

```
void QCustomPlot::setNoAntialiasingOnDrag (
            bool enabled )
```

Sets whether antialiasing is disabled for this QCustomPlot while the user is dragging axes ranges. If many objects, especially plottables, are drawn antialiased, this greatly improves performance during dragging. Thus it creates a more responsive user experience. As soon as the user stops dragging, the last replot is done with normal antialiasing, to restore high image quality.

See also

setAntialiasedElements, setNotAntialiasedElements

17.674.4.87 setNotAntialiasedElement()

```
void QCustomPlot::setNotAntialiasedElement (
            QCP::AntialiasedElement notAntialiasedElement,
            bool enabled = true )
```

Sets whether the specified *notAntialiasedElement* is forcibly drawn not antialiased.
See setNotAntialiasedElements for details.

See also

setAntialiasedElement

17.674.4.88 setNotAntialiasedElements()

```
void QCustomPlot::setNotAntialiasedElements (
            const QCP::AntialiasedElements & notAntialiasedElements )
```

Sets which elements are forcibly drawn not antialiased as an *or* combination of QCP::AntialiasedElement.
This overrides the antialiasing settings for whole element groups, normally controlled with the *setAntialiasing* function on the individual elements. If an element is neither specified in setAntialiasedElements nor in setNotAntialiasedElements, the antialiasing setting on each individual element instance is used.
For example, if *notAntialiasedElements* contains QCP::aePlottables, no plottables will be drawn antialiased, no matter what the specific QCPAbstractPlottable::setAntialiased value was set to.
if an element in *notAntialiasedElements* is already set in setAntialiasedElements, it is removed from there.

See also

setAntialiasedElements

17.674.4.89 setOpenGl()

```
void QCustomPlot::setOpenGl (
            bool enabled,
            int multisampling = 16 )
```

Warning

This is still an experimental feature and its performance depends on the system that it runs on. Having multiple QCustomPlot widgets in one application with enabled OpenGL rendering might cause context conflicts on some systems.

This method allows to enable OpenGL plot rendering, for increased plotting performance of graphically demanding plots (thick lines, translucent fills, etc.).
If *enabled* is set to true, QCustomPlot will try to initialize OpenGL and, if successful, continue plotting with hardware acceleration. The parameter *multisampling* controls how many samples will be used per pixel, it essentially controls the antialiasing quality. If *multisampling* is set too high for the current graphics hardware, the maximum allowed value will be used.
You can test whether switching to OpenGL rendering was successful by checking whether the according getter *QCustomPlot::openGl()* returns true. If the OpenGL initialization fails, rendering continues with the regular software rasterizer, and an according qDebug output is generated.

If switching to OpenGL was successful, this method disables label caching (setPlottingHint(QCP::phCacheLabels, false)) and turns on QCustomPlot's antialiasing override for all elements (setAntialiasedElements(QCP::aeAll)), leading to a higher quality output. The antialiasing override allows for pixel-grid aligned drawing in the OpenGL paint device. As stated before, in OpenGL rendering the actual antialiasing of the plot is controlled with *multisampling*. If *enabled* is set to false, the antialiasing/label caching settings are restored to what they were before OpenGL was enabled, if they weren't altered in the meantime.

Note

OpenGL support is only enabled if QCustomPlot is compiled with the macro `QCUSTOMPLOT_USE_OPENGL` defined. This define must be set before including the QCustomPlot header both during compilation of the QCustomPlot library as well as when compiling your application. It is best to just include the line `DEFINES += QCUSTOMPLOT_USE_OPENGL` in the respective qmake project files.

If you are using a Qt version before 5.0, you must also add the module "opengl" to your `QT` variable in the qmake project files. For Qt versions 5.0 and higher, QCustomPlot switches to a newer OpenGL interface which is already in the "gui" module.

17.674.4.90 setPlottingHint()

```
void QCustomPlot::setPlottingHint (
            QCP::PlottingHint hint,
            bool enabled = true )
```

Sets the specified plotting *hint* to *enabled*.

See also

setPlottingHints

17.674.4.91 setPlottingHints()

```
void QCustomPlot::setPlottingHints (
            const QCP::PlottingHints & hints )
```

Sets the plotting hints for this QCustomPlot instance as an *or* combination of QCP::PlottingHint.

See also

setPlottingHint

17.674.4.92 setSelectionRect()

```
void QCustomPlot::setSelectionRect (
            QCPSelectionRect * selectionRect )
```

Sets the QCPSelectionRect instance that QCustomPlot will use if *mode* is not QCP::srmNone and the user performs a click-and-drag interaction. QCustomPlot takes ownership of the passed *selectionRect*. It can be accessed later via selectionRect.

This method is useful if you wish to replace the default QCPSelectionRect instance with an instance of a QCPSelectionRect subclass, to introduce custom behaviour of the selection rect.

See also

setSelectionRectMode

17.674.4.93 setSelectionRectMode()

```
void QCustomPlot::setSelectionRectMode (
            QCP::SelectionRectMode mode )
```

Sets how QCustomPlot processes mouse click-and-drag interactions by the user.

If *mode* is QCP::srmNone, the mouse drag is forwarded to the underlying objects. For example, QCPAxisRect may process a mouse drag by dragging axis ranges, see QCPAxisRect::setRangeDrag. If *mode* is not QCP::srmNone, the current selection rect (selectionRect) becomes activated and allows e.g. rect zooming and data point selection.

If you wish to provide your user both with axis range dragging and data selection/range zooming, use this method to switch between the modes just before the interaction is processed, e.g. in reaction to the mousePress or mouseMove signals. For example you could check whether the user is holding a certain keyboard modifier, and then decide which *mode* shall be set.
If a selection rect interaction is currently active, and *mode* is set to QCP::srmNone, the interaction is canceled (QCPSelectionRect::cancel). Switching between any of the other modes will keep the selection rect active. Upon completion of the interaction, the behaviour is as defined by the currently set *mode*, not the mode that was set when the interaction started.

See also

setInteractions, setSelectionRect, QCPSelectionRect

17.674.4.94 setSelectionTolerance()

```
void QCustomPlot::setSelectionTolerance (
            int pixels )
```

Sets the tolerance that is used to decide whether a click selects an object (e.g. a plottable) or not.
If the user clicks in the vicinity of the line of e.g. a QCPGraph, it's only regarded as a potential selection when the minimum distance between the click position and the graph line is smaller than *pixels*. Objects that are defined by an area (e.g. QCPBars) only react to clicks directly inside the area and ignore this selection tolerance. In other words, it only has meaning for parts of objects that are too thin to exactly hit with a click and thus need such a tolerance.

See also

setInteractions, QCPLayerable::selectTest

17.674.4.95 setViewport()

```
void QCustomPlot::setViewport (
            const QRect & rect )
```

Sets the viewport of this QCustomPlot. Usually users of QCustomPlot don't need to change the viewport manually.
The viewport is the area in which the plot is drawn. All mechanisms, e.g. margin calculation take the viewport to be the outer border of the plot. The viewport normally is the rect() of the QCustomPlot widget, i.e. a rect with top left (0, 0) and size of the QCustomPlot widget.
Don't confuse the viewport with the axis rect (QCustomPlot::axisRect). An axis rect is typically an area enclosed by four axes, where the graphs/plottables are drawn in. The viewport is larger and contains also the axes themselves, their tick numbers, their labels, or even additional axis rects, color scales and other layout elements.
This function is used to allow arbitrary size exports with toPixmap, savePng, savePdf, etc. by temporarily changing the viewport size.

17.674.4.96 toPainter()

```
void QCustomPlot::toPainter (
            QCPPainter * painter,
            int width = 0,
            int height = 0 )
```

Renders the plot using the passed *painter*.
The plot is sized to *width* and *height* in pixels. If the *painter's* scale is not 1.0, the resulting plot will appear scaled accordingly.

Note

If you are restricted to using a QPainter (instead of QCPPainter), create a temporary QPicture and open a QCPPainter on it. Then call toPainter with this QCPPainter. After ending the paint operation on the picture, draw it with the QPainter. This will reproduce the painter actions the QCPPainter took, with a QPainter.

See also

toPixmap

17.674.4.97 toPixmap()

```
QPixmap QCustomPlot::toPixmap (
            int width = 0,
            int height = 0,
            double scale = 1.0 )
```

Renders the plot to a pixmap and returns it.
The plot is sized to *width* and *height* in pixels and scaled with *scale*. (width 100 and scale 2.0 lead to a full resolution pixmap with width 200.)

See also

toPainter, saveRastered, saveBmp, savePng, saveJpg, savePdf

17.674.5 Member Data Documentation

17.674.5.1 legend

```
QCPLegend * QCustomPlot::legend
```

A pointer to the default legend of the main axis rect. The legend is invisible by default. Use QCPLegend::setVisible to change this.
QCustomPlot offers convenient pointers to the axes (xAxis, yAxis, xAxis2, yAxis2) and the legend. They make it very easy working with plots that only have a single axis rect and at most one axis at each axis rect side. If you use the layout system to add multiple legends to the plot, use the layout system interface to access the new legend. For example, legends can be placed inside an axis rect's inset layout, and must then also be accessed via the inset layout. If the default legend is removed due to manipulation of the layout system (e.g. by removing the main axis rect), the corresponding pointer becomes `nullptr`.
If an axis convenience pointer is currently `nullptr` and a new axis rect or a corresponding axis is added in the place of the main axis rect, QCustomPlot resets the convenience pointers to the according new axes. Similarly the legend convenience pointer will be reset if a legend is added after the main legend was removed before.

17.674.5.2 xAxis

```
QCPAxis * QCustomPlot::xAxis
```

A pointer to the primary x Axis (bottom) of the main axis rect of the plot.
QCustomPlot offers convenient pointers to the axes (xAxis, yAxis, xAxis2, yAxis2) and the legend. They make it very easy working with plots that only have a single axis rect and at most one axis at each axis rect side. If you use the layout system to add multiple axis rects or multiple axes to one side, use the QCPAxisRect::axis interface to access the new axes. If one of the four default axes or the default legend is removed due to manipulation of the layout system (e.g. by removing the main axis rect), the corresponding pointers become `nullptr`.
If an axis convenience pointer is currently `nullptr` and a new axis rect or a corresponding axis is added in the place of the main axis rect, QCustomPlot resets the convenience pointers to the according new axes. Similarly the legend convenience pointer will be reset if a legend is added after the main legend was removed before.

17.674.5.3 xAxis2

```
QCPAxis * QCustomPlot::xAxis2
```

A pointer to the secondary x Axis (top) of the main axis rect of the plot. Secondary axes are invisible by default. Use QCPAxis::setVisible to change this (or use QCPAxisRect::setupFullAxesBox).
QCustomPlot offers convenient pointers to the axes (xAxis, yAxis, xAxis2, yAxis2) and the legend. They make it very easy working with plots that only have a single axis rect and at most one axis at each axis rect side. If you use the layout system to add multiple axis rects or multiple axes to one side, use the QCPAxisRect::axis interface to access the new axes. If one of the four default axes or the default legend is removed due to manipulation of the layout system (e.g. by removing the main axis rect), the corresponding pointers become `nullptr`.
If an axis convenience pointer is currently `nullptr` and a new axis rect or a corresponding axis is added in the place of the main axis rect, QCustomPlot resets the convenience pointers to the according new axes. Similarly the legend convenience pointer will be reset if a legend is added after the main legend was removed before.

17.674.5.4 yAxis

```
QCPAxis * QCustomPlot::yAxis
```

A pointer to the primary y Axis (left) of the main axis rect of the plot.
QCustomPlot offers convenient pointers to the axes (xAxis, yAxis, xAxis2, yAxis2) and the legend. They make it very easy working with plots that only have a single axis rect and at most one axis at each axis rect side. If you use the layout system to add multiple axis rects or multiple axes to one side, use the QCPAxisRect::axis interface to access the new axes. If one of the four default axes or the default legend is removed due to manipulation of the layout system (e.g. by removing the main axis rect), the corresponding pointers become `nullptr`.
If an axis convenience pointer is currently `nullptr` and a new axis rect or a corresponding axis is added in the place of the main axis rect, QCustomPlot resets the convenience pointers to the according new axes. Similarly the legend convenience pointer will be reset if a legend is added after the main legend was removed before.

17.674.5.5 yAxis2

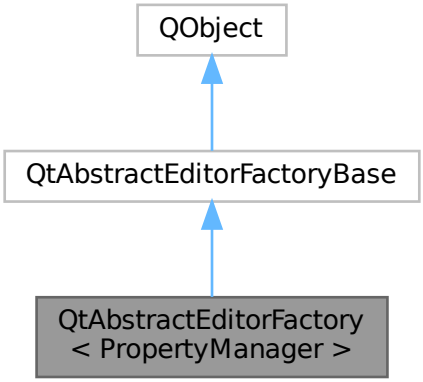
```
QCPAxis * QCustomPlot::yAxis2
```

A pointer to the secondary y Axis (right) of the main axis rect of the plot. Secondary axes are invisible by default. Use QCPAxis::setVisible to change this (or use QCPAxisRect::setupFullAxesBox).
QCustomPlot offers convenient pointers to the axes (xAxis, yAxis, xAxis2, yAxis2) and the legend. They make it very easy working with plots that only have a single axis rect and at most one axis at each axis rect side. If you use the layout system to add multiple axis rects or multiple axes to one side, use the QCPAxisRect::axis interface to access the new axes. If one of the four default axes or the default legend is removed due to manipulation of the layout system (e.g. by removing the main axis rect), the corresponding pointers become `nullptr`.
If an axis convenience pointer is currently `nullptr` and a new axis rect or a corresponding axis is added in the place of the main axis rect, QCustomPlot resets the convenience pointers to the according new axes. Similarly the legend convenience pointer will be reset if a legend is added after the main legend was removed before.
The documentation for this class was generated from the following files:

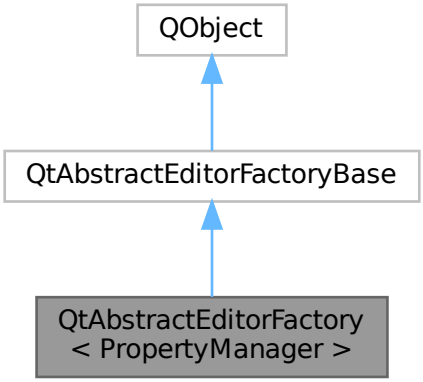
- applications/gui/genesys/qcustomplot.h
- applications/gui/genesys/qcustomplot.cpp

17.675 QtAbstractEditorFactory< PropertyManager > Class Template Reference

Inheritance diagram for QtAbstractEditorFactory< PropertyManager >:



Collaboration diagram for QtAbstractEditorFactory< PropertyManager >:



Public Member Functions

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (PropertyManager *manager)
- void removePropertyManager (PropertyManager *manager)
- QSet< PropertyManager * > propertyManagers () const
- PropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- virtual void connectPropertyManager (PropertyManager *manager)=0
- virtual QWidget * createEditor (PropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void disconnectPropertyManager (PropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Friends

- class **QtAbstractPropertyEditor**

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.675.1 Constructor & Destructor Documentation

17.675.1.1 QtAbstractEditorFactory()

```
template<class PropertyManager >
QtAbstractEditorFactory< PropertyManager >::QtAbstractEditorFactory (
            QObject * parent = 0 )  [inline], [explicit]
```

Creates an editor factory with the given *parent*.

See also

addPropertyManager()

17.675.2 Member Function Documentation

17.675.2.1 addPropertyManager()

```
template<class PropertyManager >
void QtAbstractEditorFactory< PropertyManager >::addPropertyManager (
            PropertyManager * manager )  [inline]
```

Adds the given *manager* to this factory's set of managers, making this factory produce editing widgets for properties created by the given manager.

The PropertyManager type is a template argument class, and represents the chosen QtAbstractPropertyManager subclass.

See also

propertyManagers(), removePropertyManager()

17.675.2.2 connectPropertyManager()

```
template<class PropertyManager >
void QtAbstractEditorFactory< PropertyManager >::connectPropertyManager (
            PropertyManager * manager )  [protected], [pure virtual]
```

Connects this factory to the given *manager's* signals. The PropertyManager type is a template argument class, and represents the chosen QtAbstractPropertyManager subclass.

This function is used internally by the addPropertyManager() function, and makes it possible to update an editing widget when the associated property's data changes. This is typically done in custom slots responding to the signals emitted by the property's manager, e.g. QtIntPropertyManager::valueChanged() and QtIntPropertyManager::rangeChanged().

See also

propertyManagers(), disconnectPropertyManager()

17.675.2.3 createEditor() [1/2]

```
template<class PropertyManager >
QWidget * QtAbstractEditorFactory< PropertyManager >::createEditor (
            PropertyManager * manager,
            QtProperty * property,
            QWidget * parent )  [protected], [pure virtual]
```

Creates an editing widget with the given *parent* for the specified *property* created by the given *manager*. The PropertyManager type is a template argument class, and represents the chosen QtAbstractPropertyManager subclass.

This function must be implemented in derived classes: It is recommended to store a pointer to the widget and map it to the given *property*, since the widget must be updated whenever the associated property's data changes. This is typically done in custom slots responding to the signals emitted by the property's manager, e.g. QtIntPropertyManager::valueChanged() and QtIntPropertyManager::rangeChanged().

See also

connectPropertyManager()

17.675.2.4 createEditor() [2/2]

```
template<class PropertyManager >
QWidget * QtAbstractEditorFactory< PropertyManager >::createEditor (
            QtProperty * property,
            QWidget * parent )  [inline], [override], [virtual]
```

Creates an editing widget (with the given *parent*) for the given *property*.

Implements QtAbstractEditorFactoryBase.

17.675.2.5 disconnectPropertyManager()

```
template<class PropertyManager >
void QtAbstractEditorFactory< PropertyManager >::disconnectPropertyManager (
            PropertyManager * manager )  [protected], [pure virtual]
```

Disconnects this factory from the given *manager's* signals. The PropertyManager type is a template argument class, and represents the chosen QtAbstractPropertyManager subclass.
This function is used internally by the removePropertyManager() function.

See also

propertyManagers(), connectPropertyManager()

17.675.2.6 managerDestroyed()

```
template<class PropertyManager >
void QtAbstractEditorFactory< PropertyManager >::managerDestroyed (
            QObject * manager )  [inline], [override], [protected], [virtual]
```

Implements QtAbstractEditorFactoryBase.

17.675.2.7 propertyManager()

```
template<class PropertyManager >
PropertyManager * QtAbstractEditorFactory< PropertyManager >::propertyManager (
            QtProperty * property ) const  [inline]
```

Returns the property manager for the given *property*, or 0 if the given *property* doesn't belong to any of this factory's registered managers.
The PropertyManager type is a template argument class, and represents the chosen QtAbstractPropertyManager subclass.

See also

propertyManagers()

17.675.2.8 propertyManagers()

```
template<class PropertyManager >
QSet< PropertyManager * > QtAbstractEditorFactory< PropertyManager >::propertyManagers ( )
const  [inline]
```

Returns the factory's set of associated managers. The PropertyManager type is a template argument class, and represents the chosen QtAbstractPropertyManager subclass.

See also

addPropertyManager(), removePropertyManager()

17.675.2.9 removePropertyManager()

```
template<class PropertyManager >
void QtAbstractEditorFactory< PropertyManager >::removePropertyManager (
            PropertyManager * manager )  [inline]
```

Removes the given *manager* from this factory's set of managers. The PropertyManager type is a template argument class, and may be any QtAbstractPropertyManager subclass.

See also

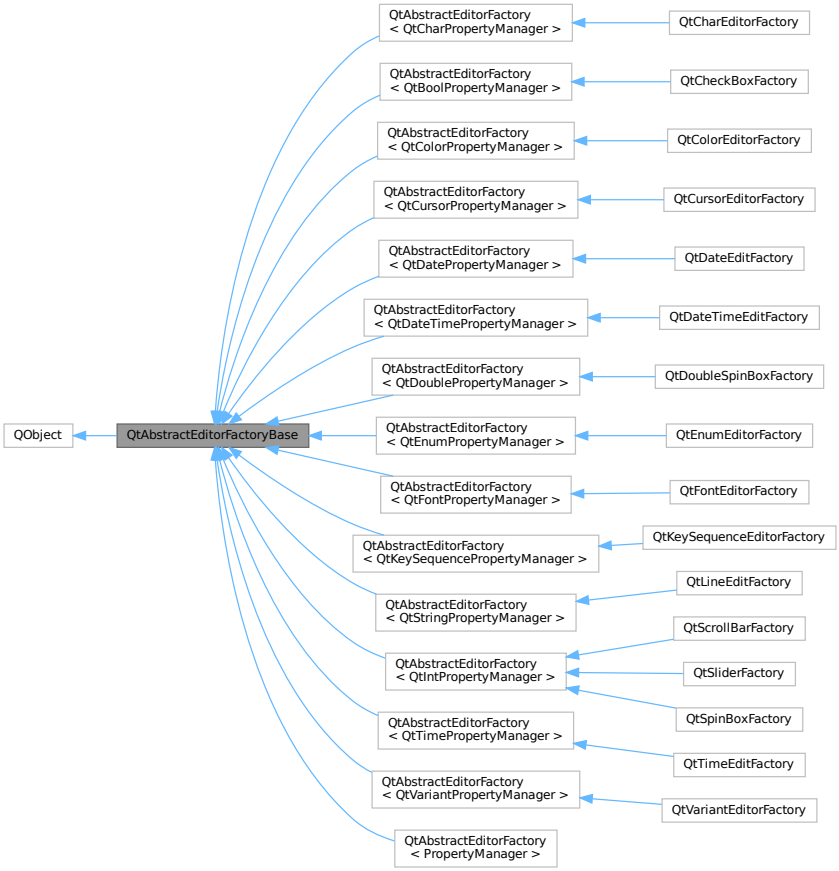
propertyManagers(), addPropertyManager()

The documentation for this class was generated from the following files:

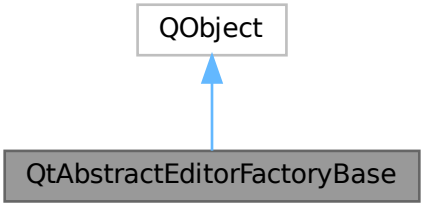
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.cpp

17.676 QtAbstractEditorFactoryBase Class Reference

Inheritance diagram for QtAbstractEditorFactoryBase:



Collaboration diagram for QtAbstractEditorFactoryBase:



Public Member Functions

- virtual QWidget ∗ createEditor (QtProperty ∗property, QWidget ∗parent)=0

Protected Slots

- virtual void **managerDestroyed** (QObject ∗manager)=0

Protected Member Functions

- QtAbstractEditorFactoryBase (QObject ∗parent=0)
- virtual void **breakConnection** (QtAbstractPropertyManager ∗manager)=0

Friends

- class **QtAbstractPropertyBrowser**

17.676.1 Constructor & Destructor Documentation

17.676.1.1 QtAbstractEditorFactoryBase()

```
QtAbstractEditorFactoryBase::QtAbstractEditorFactoryBase (
            QObject * parent = 0 )  [inline], [explicit], [protected]
```

Creates an abstract editor factory with the given *parent*.

17.676.2 Member Function Documentation

17.676.2.1 createEditor()

```
QWidget * QtAbstractEditorFactoryBase::createEditor (
            QtProperty * property,
            QWidget * parent )  [pure virtual]
```

Creates an editing widget (with the given *parent*) for the given *property*.
This function is reimplemented in QtAbstractEditorFactory template class which also provides a pure virtual convenience overload of this function enabling access to the property's manager.

See also

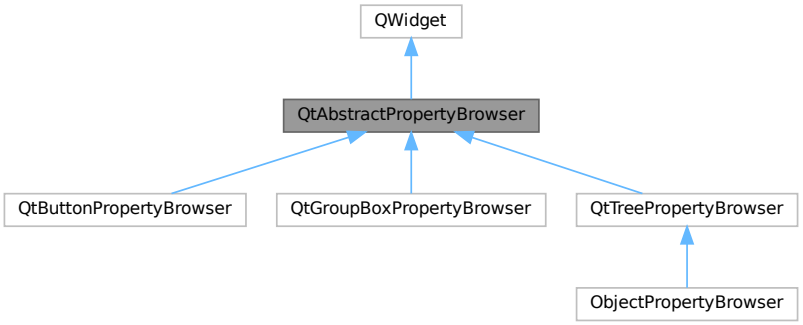
QtAbstractEditorFactory::createEditor()

Implemented in QtAbstractEditorFactory< PropertyManager >, QtAbstractEditorFactory< QtBoolPropertyManager >, QtAbstractEditorFactory< QtCharPropertyManager >, QtAbstractEditorFactory< QtColorPropertyManager >, QtAbstractEditorFactory< QtCursorPropertyManager >, QtAbstractEditorFactory< QtDatePropertyManager >, QtAbstractEditorFactory< QtDateTimePropertyManager >, QtAbstractEditorFactory< QtDoublePropertyManager >, QtAbstractEditorFactory< QtEnumPropertyManager >, QtAbstractEditorFactory< QtFontPropertyManager >, QtAbstractEditorFactory< QtIntPropertyManager >, QtAbstractEditorFactory< QtKeySequencePropertyManager >, QtAbstractEditorFactory< QtStringPropertyManager >, QtAbstractEditorFactory< QtTimePropertyManager >, and QtAbstractEditorFactory< QtVariantPropertyManager >.
The documentation for this class was generated from the following files:

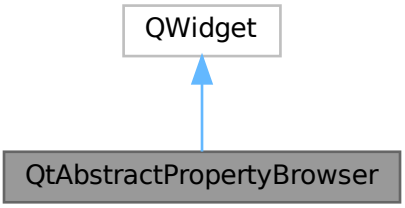
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.cpp

17.677 QtAbstractPropertyBrowser Class Reference

Inheritance diagram for QtAbstractPropertyBrowser:



Collaboration diagram for QtAbstractPropertyBrowser:



Public Slots

- QtBrowserItem * addProperty (QtProperty *property)
- QtBrowserItem * insertProperty (QtProperty *property, QtProperty *afterProperty)
- void removeProperty (QtProperty *property)

Signals

- void currentItemChanged (QtBrowserItem *)

Public Member Functions

- QtAbstractPropertyBrowser (QWidget *parent=0)
- ~QtAbstractPropertyBrowser ()
- QList< QtProperty * > properties () const
- QList< QtBrowserItem * > items (QtProperty *property) const
- QtBrowserItem * topLevelItem (QtProperty *property) const
- QList< QtBrowserItem * > topLevelItems () const
- void clear ()
- template<class PropertyManager >
  void setFactoryForManager (PropertyManager *manager, QtAbstractEditorFactory< PropertyManager > *factory)

- void unsetFactoryForManager (QtAbstractPropertyManager *manager)
- QtBrowserItem * currentItem () const
- void setCurrentItem (QtBrowserItem *)

Protected Member Functions

- virtual void itemInserted (QtBrowserItem *item, QtBrowserItem *afterItem)=0
- virtual void itemRemoved (QtBrowserItem *item)=0
- virtual void itemChanged (QtBrowserItem *item)=0
- virtual QWidget * createEditor (QtProperty *property, QWidget *parent)

17.677.1 Constructor & Destructor Documentation

17.677.1.1 QtAbstractPropertyBrowser()

```
QtAbstractPropertyBrowser::QtAbstractPropertyBrowser (
            QWidget * parent = 0 )  [explicit]
```

Creates an abstract property browser with the given *parent*.

17.677.1.2 ~QtAbstractPropertyBrowser()

```
QtAbstractPropertyBrowser::~QtAbstractPropertyBrowser ( )
```

Destroys the property browser, and destroys all the items that were created by this property browser.
Note that the properties that were displayed in the editor are not deleted since they still can be used in other editors. Neither does the destructor delete the property managers and editor factories that were used by this property browser widget unless this widget was their parent.

See also

QtAbstractPropertyManager::~QtAbstractPropertyManager()

17.677.2 Member Function Documentation

17.677.2.1 addProperty

```
QtBrowserItem * QtAbstractPropertyBrowser::addProperty (
            QtProperty * property )  [slot]
```

Appends the given *property* (and its subproperties) to the property browser's list of top level properties. Returns the item created by property browser which is associated with the *property*. In order to get all children items created by the property browser in this call, the returned item should be traversed.
If the specified *property* is already added, this function does nothing and returns 0.

See also

insertProperty(), QtProperty::addSubProperty(), properties()

17.677.2.2 clear()

```
void QtAbstractPropertyBrowser::clear ( )
```

Removes all the properties from the editor, but does not delete them since they can still be used in other editors.

See also

removeProperty(), QtAbstractPropertyManager::clear()

17.677.2.3 createEditor()

```
QWidget * QtAbstractPropertyBrowser::createEditor (
            QtProperty * property,
            QWidget * parent )  [protected], [virtual]
```

Creates an editing widget (with the given *parent*) for the given *property* according to the previously established associations between property managers and editor factories.
If the property is created by a property manager which was not associated with any of the existing factories in *this* property editor, the function returns 0.
To make a property editable in the property browser, the createEditor() function must be called to provide the property with a suitable editing widget.
Reimplement this function to provide additional decoration for the editing widgets created by the installed factories.

See also

setFactoryForManager()

17.677.2.4 currentItem()

```
QtBrowserItem * QtAbstractPropertyBrowser::currentItem ( ) const
```

Returns the current item in the property browser.

See also

setCurrentItem()

17.677.2.5 currentItemChanged

```
void QtAbstractPropertyBrowser::currentItemChanged (
            QtBrowserItem * current )  [signal]
```

This signal is emitted when the current item changes. The current item is specified by *current*.

See also

QtAbstractPropertyBrowser::setCurrentItem()

17.677.2.6 insertProperty

```
QtBrowserItem * QtAbstractPropertyBrowser::insertProperty (
            QtProperty * property,
            QtProperty * afterProperty )  [slot]
```

Inserts the given *property* (and its subproperties) after the specified *afterProperty* in the browser's list of top level properties. Returns item created by property browser which is associated with the *property*. In order to get all children items created by the property browser in this call returned item should be traversed.
If the specified *afterProperty* is 0, the given *property* is inserted at the beginning of the list. If *property* is already inserted, this function does nothing and returns 0.

See also

addProperty(), QtProperty::insertSubProperty(), properties()

17.677.2.7 itemChanged()

```
void QtAbstractPropertyBrowser::itemChanged (
            QtBrowserItem * item )  [protected], [pure virtual]
```

This function is called whenever a property's data changes, passing a pointer to the *item* of property as parameter.
This function must be reimplemented in derived classes in order to update the property browser widget whenever a property's name, tool tip, status tip, "what's this" text, value text or value icon changes.
Note that if the property browser contains several occurrences of the same property, this method will be called once for each occurrence (with a different item each time).

See also

QtProperty, items()

Implemented in QtButtonPropertyBrowser, QtGroupBoxPropertyBrowser, and QtTreePropertyBrowser.

17.677.2.8 itemInserted()

```
void QtAbstractPropertyBrowser::itemInserted (
            QtBrowserItem * insertedItem,
            QtBrowserItem * precedingItem )  [protected], [pure virtual]
```

This function is called to update the widget whenever a property is inserted or added to the property browser, passing pointers to the *insertedItem* of property and the specified *precedingItem* as parameters.
If *precedingItem* is 0, the *insertedItem* was put at the beginning of its parent item's list of subproperties. If the parent of *insertedItem* is 0, the *insertedItem* was added as a top level property of *this* property browser.
This function must be reimplemented in derived classes. Note that if the *insertedItem's* property has subproperties, this method will be called for those properties as soon as the current call is finished.

See also

insertProperty(), addProperty()

Implemented in QtButtonPropertyBrowser, QtGroupBoxPropertyBrowser, and QtTreePropertyBrowser.

17.677.2.9 itemRemoved()

```
void QtAbstractPropertyBrowser::itemRemoved (
            QtBrowserItem * item )  [protected], [pure virtual]
```

This function is called to update the widget whenever a property is removed from the property browser, passing the pointer to the *item* of the property as parameters. The passed *item* is deleted just after this call is finished.
If the the parent of *item* is 0, the removed *item* was a top level property in this editor.
This function must be reimplemented in derived classes. Note that if the removed *item's* property has subproperties, this method will be called for those properties just before the current call is started.

See also

removeProperty()

Implemented in QtButtonPropertyBrowser, QtGroupBoxPropertyBrowser, and QtTreePropertyBrowser.

17.677.2.10 items()

```
QList< QtBrowserItem * > QtAbstractPropertyBrowser::items (
            QtProperty * property ) const
```

Returns the property browser's list of all items associated with the given *property*.
There is one item per instance of the property in the browser.

See also

topLevelItem()

17.677.2.11 properties()

```
QList< QtProperty * > QtAbstractPropertyBrowser::properties ( ) const
```

Returns the property browser's list of top level properties.
To traverse the subproperties, use the QtProperty::subProperties() function.

See also

addProperty(), insertProperty(), removeProperty()

17.677.2.12 removeProperty

```
void QtAbstractPropertyBrowser::removeProperty (
            QtProperty * property )  [slot]
```

Removes the specified *property* (and its subproperties) from the property browser's list of top level properties. All items that were associated with the given *property* and its children are deleted.
Note that the properties are *not* deleted since they can still be used in other editors.

See also

clear(), QtProperty::removeSubProperty(), properties()

17.677.2.13 setCurrentItem()

```
void QtAbstractPropertyBrowser::setCurrentItem (
            QtBrowserItem * item )
```

Sets the current item in the property browser to *item*.

See also

currentItem(), currentItemChanged()

17.677.2.14 setFactoryForManager()

```
template<class PropertyManager >
void QtAbstractPropertyBrowser::setFactoryForManager (
            PropertyManager * manager,
            QtAbstractEditorFactory< PropertyManager > * factory )  [inline]
```

Connects the given *manager* to the given *factory*, ensuring that properties of the *manager's* type will be displayed with an editing widget suitable for their value.
For example:
In this example the `myInteger` property's value is displayed with a QSpinBox widget, while the `myDouble` property's value is displayed with a QDoubleSpinBox widget.
Note that a factory can be associated with many managers, but a manager can only be associated with one single factory. If the given *manager* already is associated with another factory, the old association is broken before the new one established.
This function ensures that the given *manager* and the given *factory* are compatible, and it automatically calls the QtAbstractEditorFactory::addPropertyManager() function if necessary.

See also

unsetFactoryForManager()

17.677.2.15 topLevelItem()

```
QtBrowserItem * QtAbstractPropertyBrowser::topLevelItem (
            QtProperty * property ) const
```

Returns the top-level items associated with the given *property*.
Returns 0 if *property* wasn't inserted into this property browser or isn't a top-level one.

See also

topLevelItems(), items()

17.677.2.16 topLevelItems()

```
QList< QtBrowserItem * > QtAbstractPropertyBrowser::topLevelItems ( ) const
```

Returns the list of top-level items.

See also

topLevelItem()

17.677.2.17 unsetFactoryForManager()

```
void QtAbstractPropertyBrowser::unsetFactoryForManager (
            QtAbstractPropertyManager * manager )
```

Removes the association between the given *manager* and the factory bound to it, automatically calling the QtAbstractEditorFactory::removePropertyManager() function if necessary.

See also

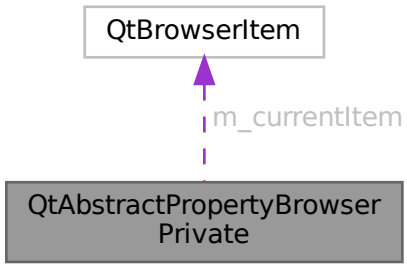
setFactoryForManager()

The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.cpp

17.678 QtAbstractPropertyBrowserPrivate Class Reference

Collaboration diagram for QtAbstractPropertyBrowserPrivate:



Public Member Functions

- void **insertSubTree** (QtProperty *property, QtProperty *parentProperty)
- void **removeSubTree** (QtProperty *property, QtProperty *parentProperty)
- void **createBrowserIndexes** (QtProperty *property, QtProperty *parentProperty, QtProperty *afterProperty)
- void **removeBrowserIndexes** (QtProperty *property, QtProperty *parentProperty)
- QtBrowserItem * **createBrowserIndex** (QtProperty *property, QtBrowserItem *parentIndex, QtBrowserItem *afterIndex)
- void **removeBrowserIndex** (QtBrowserItem *index)
- void **clearIndex** (QtBrowserItem *index)
- void **slotPropertyInserted** (QtProperty *property, QtProperty *parentProperty, QtProperty *afterProperty)
- void **slotPropertyRemoved** (QtProperty *property, QtProperty *parentProperty)
- void **slotPropertyDestroyed** (QtProperty *property)
- void **slotPropertyDataChanged** (QtProperty *property)

Public Attributes

- QList< QtProperty * > **m_subItems**
- QMap< QtAbstractPropertyManager *, QList< QtProperty * > > **m_managerToProperties**
- QMap< QtProperty *, QList< QtProperty * > > **m_propertyToParents**
- QMap< QtProperty *, QtBrowserItem * > **m_topLevelPropertyToIndex**
- QList< QtBrowserItem * > **m_topLevelIndexes**

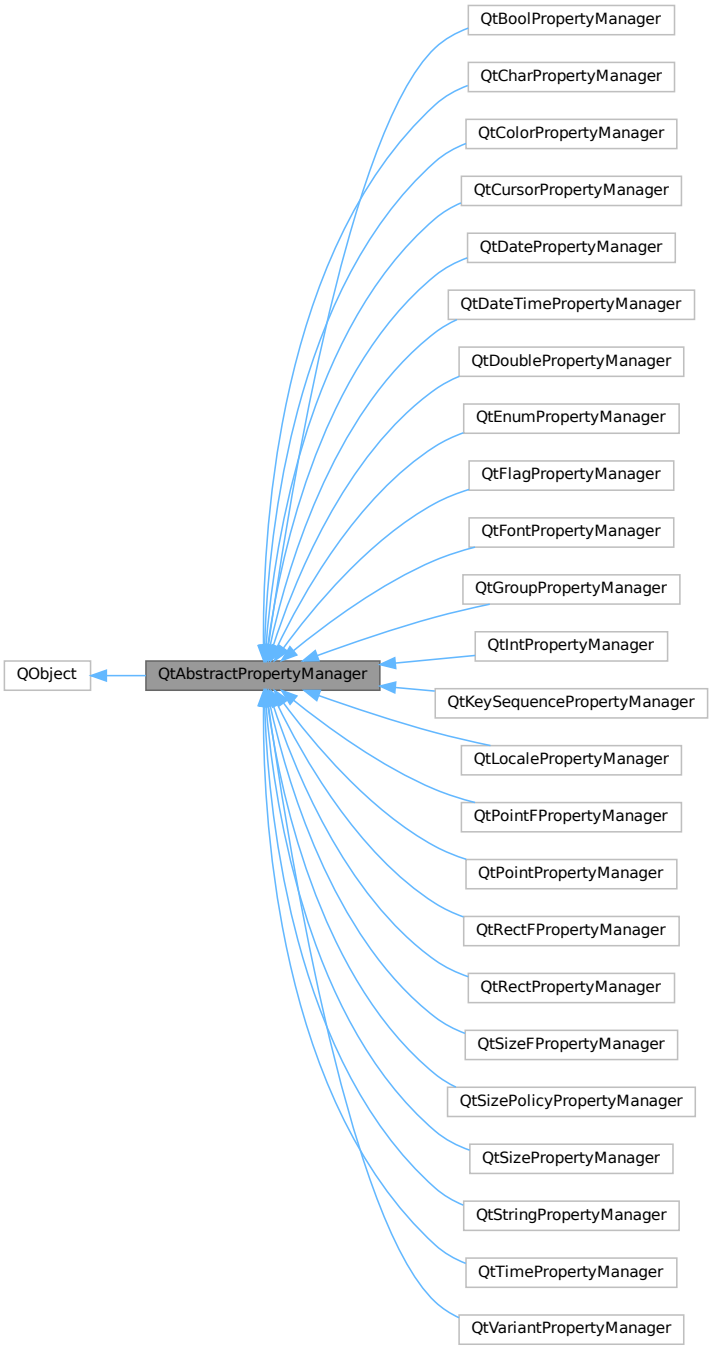
- QMap< QtProperty *, QList< QtBrowserItem * > > **m_propertyToIndexes**
- QtBrowserItem * **m_currentItem**

The documentation for this class was generated from the following file:

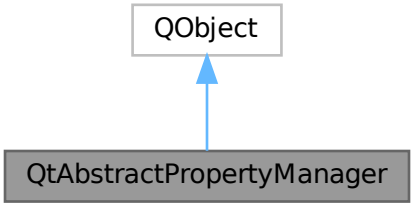
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.cpp

17.679 QtAbstractPropertyManager Class Reference

Inheritance diagram for QtAbstractPropertyManager:



Collaboration diagram for QtAbstractPropertyManager:



Signals

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QString valueText (const QtProperty *property) const
- virtual void initializeProperty (QtProperty *property)=0
- virtual void uninitializeProperty (QtProperty *property)
- virtual QtProperty * createProperty ()

Friends

- class **QtProperty**

17.679.1 Constructor & Destructor Documentation

17.679.1.1 QtAbstractPropertyManager()

```
QtAbstractPropertyManager::QtAbstractPropertyManager (
            QObject * parent = 0 )  [explicit]
```

Creates an abstract property manager with the given *parent*.

17.679.1.2 ~QtAbstractPropertyManager()

```
QtAbstractPropertyManager::~QtAbstractPropertyManager ( )
```

Destroys the manager. All properties created by the manager are destroyed.

17.679.2 Member Function Documentation

17.679.2.1 addProperty()

```
QtProperty * QtAbstractPropertyManager::addProperty (
            const QString & name = QString() )
```

Creates a property with the given *name* which then is owned by this manager.
Internally, this function calls the createProperty() and initializeProperty() functions.

See also

initializeProperty(), properties()

17.679.2.2 clear()

```
void QtAbstractPropertyManager::clear ( ) const
```

Destroys all the properties that this manager has created.

See also

propertyDestroyed(), uninitializeProperty()

17.679.2.3 createProperty()

```
QtProperty * QtAbstractPropertyManager::createProperty ( )  [protected], [virtual]
```

Creates a property.
The base implementation produce QtProperty instances; Reimplement this function to make this manager produce objects of a QtProperty subclass.

See also

addProperty(), initializeProperty()

Reimplemented in QtVariantPropertyManager.

17.679.2.4 hasValue()

```
bool QtAbstractPropertyManager::hasValue (
            const QtProperty * property ) const  [protected], [virtual]
```

Returns whether the given *property* has a value.
The default implementation of this function returns true.

See also

QtProperty::hasValue()

Reimplemented in QtGroupPropertyManager, and QtVariantPropertyManager.

17.679.2.5 initializeProperty()

```
void QtAbstractPropertyManager::initializeProperty (
            QtProperty * property )  [protected], [pure virtual]
```

This function is called whenever a new valid property pointer has been created, passing the pointer as parameter.
The purpose is to let the manager know that the *property* has been created so that it can provide additional attributes for the new property, e.g. QtIntPropertyManager adds \l {QtIntPropertyManager::value()}{value}, \l {QtIntPropertyManager::minimum()}{minimum} and \l {QtIntPropertyManager::maximum()}{maximum} attributes.
Since each manager subclass adds type specific attributes, this function is pure virtual and must be reimplemented when deriving from the QtAbstractPropertyManager class.

See also

addProperty(), createProperty()

Implemented in QtGroupPropertyManager, QtIntPropertyManager, QtBoolPropertyManager, QtDoublePropertyManager, QtStringPropertyManager, QtDatePropertyManager, QtTimePropertyManager, QtDateTimePropertyManager, QtKeySequencePropertyManager, QtCharPropertyManager, QtLocalePropertyManager, QtPointPropertyManager, QtPointFPropertyManager, QtSizePropertyManager, QtSizeFPropertyManager, QtRectPropertyManager, QtRectFPropertyManager, QtEnumPropertyManager, QtFlagPropertyManager, QtSizePolicyPropertyManager, QtFontPropertyManager, QtColorPropertyManager, QtCursorPropertyManager, and QtVariantPropertyManager.

17.679.2.6 properties()

```
QSet< QtProperty * > QtAbstractPropertyManager::properties ( ) const
```

Returns the set of properties created by this manager.

See also

addProperty()

17.679.2.7 propertyChanged

```
void QtAbstractPropertyManager::propertyChanged (
            QtProperty * property )  [signal]
```

This signal is emitted whenever a property's data changes, passing a pointer to the *property* as parameter.
Note that signal is only emitted for properties that are created by this manager.

See also

QtAbstractPropertyBrowser::itemChanged()

17.679.2.8 propertyDestroyed

```
void QtAbstractPropertyManager::propertyDestroyed (
            QtProperty * property )  [signal]
```

This signal is emitted when the specified *property* is about to be destroyed.
Note that signal is only emitted for properties that are created by this manager.

See also

clear(), uninitializeProperty()

17.679.2.9 propertyInserted

```
void QtAbstractPropertyManager::propertyInserted (
            QtProperty * newProperty,
            QtProperty * parentProperty,
            QtProperty * precedingProperty )  [signal]
```

This signal is emitted when a new subproperty is inserted into an existing property, passing pointers to the *newProperty*, *parentProperty* and *precedingProperty* as parameters.
If *precedingProperty* is 0, the *newProperty* was inserted at the beginning of the *parentProperty's* subproperties list.
Note that signal is emitted only if the *parentProperty* is created by this manager.

See also

QtAbstractPropertyBrowser::itemInserted()

17.679.2.10 propertyRemoved

```
void QtAbstractPropertyManager::propertyRemoved (
            QtProperty * property,
            QtProperty * parent )  [signal]
```

This signal is emitted when a subproperty is removed, passing pointers to the removed *property* and the *parent* property as parameters.
Note that signal is emitted only when the *parent* property is created by this manager.

See also

QtAbstractPropertyBrowser::itemRemoved()

17.679.2.11 uninitializeProperty()

```
void QtAbstractPropertyManager::uninitializeProperty (
            QtProperty * property )  [protected], [virtual]
```

This function is called just before the specified *property* is destroyed.
The purpose is to let the property manager know that the *property* is being destroyed so that it can remove the property's additional attributes.

See also

clear(), propertyDestroyed()

Reimplemented in QtGroupPropertyManager, QtIntPropertyManager, QtBoolPropertyManager, QtDoublePropertyManager, QtStringPropertyManager, QtDatePropertyManager, QtTimePropertyManager, QtDateTimePropertyManager, QtKeySequencePropertyManager, QtCharPropertyManager, QtLocalePropertyManager, QtPointPropertyManager, QtPointFPropertyManager, QtSizePropertyManager, QtSizeFPropertyManager, QtRectPropertyManager, QtRectFPropertyManager, QtEnumPropertyManager, QtFlagPropertyManager, QtSizePolicyPropertyManager, QtFontPropertyManager, QtColorPropertyManager, QtCursorPropertyManager, and QtVariantPropertyManager.

17.679.2.12 valueIcon()

```
QIcon QtAbstractPropertyManager::valueIcon (
            const QtProperty * property ) const  [protected], [virtual]
```

Returns an icon representing the current state of the given *property*.
The default implementation of this function returns an invalid icon.

See also

QtProperty::valueIcon()

Reimplemented in QtBoolPropertyManager, QtEnumPropertyManager, QtFontPropertyManager, QtColorPropertyManager, QtCursorPropertyManager, and QtVariantPropertyManager.

17.679.2.13 valueText()

```
QString QtAbstractPropertyManager::valueText (
            const QtProperty * property ) const  [protected], [virtual]
```

Returns a string representing the current state of the given *property*.
The default implementation of this function returns an empty string.

See also

QtProperty::valueText()

Reimplemented in QtIntPropertyManager, QtBoolPropertyManager, QtDoublePropertyManager, QtStringPropertyManager, QtDatePropertyManager, QtTimePropertyManager, QtDateTimePropertyManager, QtKeySequencePropertyManager, QtCharPropertyManager, QtLocalePropertyManager, QtPointPropertyManager, QtPointFPropertyManager, QtSizePropertyManager, QtSizeFPropertyManager, QtRectPropertyManager, QtRectFPropertyManager, QtEnumPropertyManager, QtFlagPropertyManager, QtSizePolicyPropertyManager, QtFontPropertyManager, QtColorPropertyManager, QtCursorPropertyManager, and QtVariantPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.cpp

17.680 QtAbstractPropertyManagerPrivate Class Reference

Public Member Functions

- void **propertyDestroyed** (QtProperty ∗property)
- void **propertyChanged** (QtProperty ∗property) const
- void **propertyRemoved** (QtProperty ∗property, QtProperty ∗parentProperty) const
- void **propertyInserted** (QtProperty ∗property, QtProperty ∗parentProperty, QtProperty ∗afterProperty) const

Public Attributes

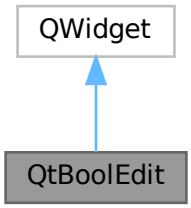
- QSet< QtProperty ∗ > **m_properties**

The documentation for this class was generated from the following file:

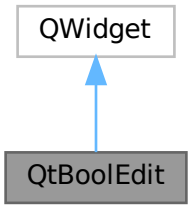
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.cpp

17.681 QtBoolEdit Class Reference

Inheritance diagram for QtBoolEdit:



Collaboration diagram for QtBoolEdit:



Signals

- void **toggled** (bool)

Public Member Functions

- **QtBoolEdit** (QWidget ∗parent=0)
- bool **textVisible** () const
- void **setTextVisible** (bool textVisible)
- Qt::CheckState **checkState** () const
- void **setCheckState** (Qt::CheckState state)
- bool **isChecked** () const
- void **setChecked** (bool c)
- void **setInitialChecked** (bool c)
- bool **blockCheckBoxSignals** (bool block)

Protected Member Functions

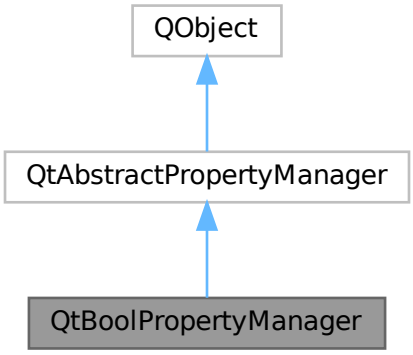
- void **mousePressEvent** (QMouseEvent ∗event) override

The documentation for this class was generated from the following files:

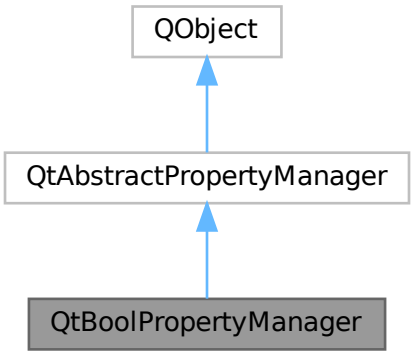
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils_p.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils.cpp

17.682 QtBoolPropertyManager Class Reference

Inheritance diagram for QtBoolPropertyManager:



Collaboration diagram for QtBoolPropertyManager:



Public Slots

- void setValue (QtProperty *property, bool val)
- void **setValueOnly** (QtProperty *property, bool val)

Signals

- void valueChanged (QtProperty *property, bool val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtBoolPropertyManager (QObject *parent=0)
- ~QtBoolPropertyManager ()
- bool value (const QtProperty *property) const
- bool **initialValue** (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- QIcon valueIcon (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.682.1 Constructor & Destructor Documentation

17.682.1.1 QtBoolPropertyManager()

```
QtBoolPropertyManager::QtBoolPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.682.1.2 ~QtBoolPropertyManager()

```
QtBoolPropertyManager::~QtBoolPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.682.2 Member Function Documentation

17.682.2.1 initializeProperty()

```
void QtBoolPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.682.2.2 setValue

```
void QtBoolPropertyManager::setValue (
            QtProperty * property,
            bool value )  [slot]
```

Sets the value of the given *property* to *value*.

See also

value()

17.682.2.3 uninitializeProperty()

```
void QtBoolPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.682.2.4 value()

```
bool QtBoolPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by *this* manager, this function returns false.

See also

setValue()

17.682.2.5 valueChanged

```
void QtBoolPropertyManager::valueChanged (
            QtProperty * property,
            bool value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

17.682.2.6 valueIcon()

```
QIcon QtBoolPropertyManager::valueIcon (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.682.2.7 valueText()

```
QString QtBoolPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.683 QtBoolPropertyManagerPrivate Class Reference

Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Attributes

- QMap< const QtProperty *, Data > **m_values**
- const QIcon **m_checkedIcon**
- const QIcon **m_uncheckedIcon**

The documentation for this class was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.684 QtBrowserItem Class Reference

Public Member Functions

- QtProperty * property () const
- QtBrowserItem * parent () const
- QList< QtBrowserItem * > children () const
- QtAbstractPropertyBrowser * browser () const

Friends

- class **QtAbstractPropertyBrowserPrivate**

17.684.1 Member Function Documentation

17.684.1.1 browser()

```
QtAbstractPropertyBrowser * QtBrowserItem::browser ( ) const
```

Returns the property browser which owns *this* item.

17.684.1.2 children()

```
QList< QtBrowserItem * > QtBrowserItem::children ( ) const
```

Returns the children items of *this* item. The properties reproduced from children items are always the same as reproduced from associated property' children, for example:
The *childrenItems* list represents the same list as *childrenProperties*.

17.684.1.3 parent()

```
QtBrowserItem * QtBrowserItem::parent ( ) const
```

Returns the parent item of *this* item. Returns 0 if *this* item is associated with top-level property in item's property browser.

See also

children()

17.684.1.4 property()

```
QtProperty * QtBrowserItem::property ( ) const
```

Returns the property which is accosiated with this item. Note that several items can be associated with the same property instance in the same property browser.

See also

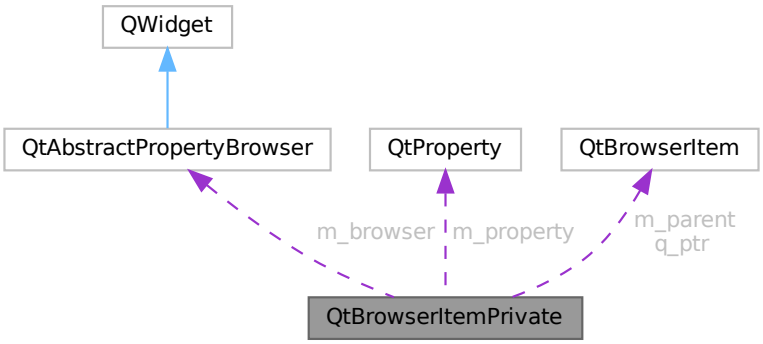
QtAbstractPropertyBrowser::items()

The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.cpp

17.685 QtBrowserItemPrivate Class Reference

Collaboration diagram for QtBrowserItemPrivate:



Public Member Functions

- **QtBrowserItemPrivate** (QtAbstractPropertyBrowser ∗browser, QtProperty ∗property, QtBrowserItem ∗parent)
- void **addChild** (QtBrowserItem ∗index, QtBrowserItem ∗after)
- void **removeChild** (QtBrowserItem ∗index)

Public Attributes

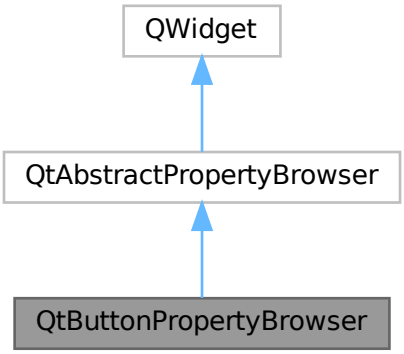
- QtAbstractPropertyBrowser ∗const **m_browser**
- QtProperty ∗ **m_property**
- QtBrowserItem ∗ **m_parent**
- QtBrowserItem ∗ **q_ptr**
- QList< QtBrowserItem ∗ > **m_children**

The documentation for this class was generated from the following file:

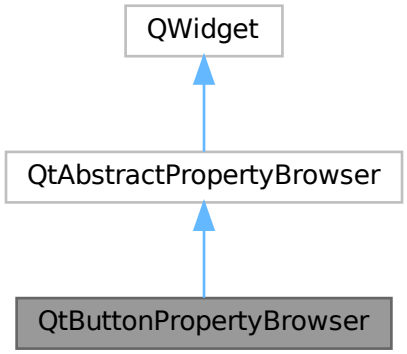
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.cpp

17.686 QtButtonPropertyBrowser Class Reference

Inheritance diagram for QtButtonPropertyBrowser:



Collaboration diagram for QtButtonPropertyBrowser:



Signals

- void collapsed (QtBrowserItem *item)
- void expanded (QtBrowserItem *item)

Signals inherited from QtAbstractPropertyBrowser

- void currentItemChanged (QtBrowserItem *)

Public Member Functions

- QtButtonPropertyBrowser (QWidget *parent=0)
- ~QtButtonPropertyBrowser ()
- void setExpanded (QtBrowserItem *item, bool expanded)
- bool isExpanded (QtBrowserItem *item) const

Public Member Functions inherited from QtAbstractPropertyBrowser

- QtAbstractPropertyBrowser (QWidget *parent=0)
- ~QtAbstractPropertyBrowser ()
- QList< QtProperty * > properties () const
- QList< QtBrowserItem * > items (QtProperty *property) const
- QtBrowserItem * topLevelItem (QtProperty *property) const
- QList< QtBrowserItem * > topLevelItems () const
- void clear ()
- template<class PropertyManager >
  void setFactoryForManager (PropertyManager *manager, QtAbstractEditorFactory< PropertyManager > *factory)
- void unsetFactoryForManager (QtAbstractPropertyManager *manager)
- QtBrowserItem * currentItem () const
- void setCurrentItem (QtBrowserItem *)

Protected Member Functions

- void itemInserted (QtBrowserItem *item, QtBrowserItem *afterItem) override
- void itemRemoved (QtBrowserItem *item) override
- void itemChanged (QtBrowserItem *item) override

Protected Member Functions inherited from QtAbstractPropertyBrowser

- virtual QWidget * createEditor (QtProperty *property, QWidget *parent)

Additional Inherited Members

Public Slots inherited from QtAbstractPropertyBrowser

- QtBrowserItem * addProperty (QtProperty *property)
- QtBrowserItem * insertProperty (QtProperty *property, QtProperty *afterProperty)
- void removeProperty (QtProperty *property)

17.686.1 Constructor & Destructor Documentation

17.686.1.1 QtButtonPropertyBrowser()

```
QtButtonPropertyBrowser::QtButtonPropertyBrowser (
            QWidget * parent = 0 )
```

Creates a property browser with the given *parent*.

17.686.1.2 ~QtButtonPropertyBrowser()

```
QtButtonPropertyBrowser::~QtButtonPropertyBrowser ( )
```

Destroys this property browser.

Note that the properties that were inserted into this browser are *not* destroyed since they may still be used in other browsers. The properties are owned by the manager that created them.

See also

QtProperty, QtAbstractPropertyManager

17.686.2 Member Function Documentation

17.686.2.1 collapsed

```
void QtButtonPropertyBrowser::collapsed (
            QtBrowserItem * item )  [signal]
```

This signal is emitted when the *item* is collapsed.

See also

expanded(), setExpanded()

17.686.2.2 expanded

```
void QtButtonPropertyBrowser::expanded (
            QtBrowserItem * item )  [signal]
```

This signal is emitted when the *item* is expanded.

See also

collapsed(), setExpanded()

17.686.2.3 isExpanded()

```
bool QtButtonPropertyBrowser::isExpanded (
            QtBrowserItem * item ) const
```

Returns true if the *item* is expanded; otherwise returns false.

See also

setExpanded()

17.686.2.4 itemChanged()

```
void QtButtonPropertyBrowser::itemChanged (
            QtBrowserItem * item )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyBrowser.

17.686.2.5 itemInserted()

```
void QtButtonPropertyBrowser::itemInserted (
            QtBrowserItem * item,
            QtBrowserItem * afterItem )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyBrowser.

17.686.2.6 itemRemoved()

```
void QtButtonPropertyBrowser::itemRemoved (
            QtBrowserItem * item )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyBrowser.

17.686.2.7 setExpanded()

```
void QtButtonPropertyBrowser::setExpanded (
            QtBrowserItem * item,
            bool expanded )
```

Sets the *item* to either collapse or expanded, depending on the value of *expanded*.

See also

isExpanded(), expanded(), collapsed()

The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtbuttonpropertybrowser.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtbuttonpropertybrowser.cpp

17.687 QtButtonPropertyBrowserPrivate Class Reference

Classes

- struct WidgetItem

Public Member Functions

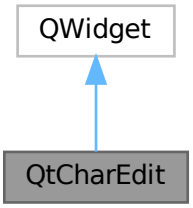
- void **init** (QWidget *parent)
- void **propertyInserted** (QtBrowserItem *index, QtBrowserItem *afterIndex)
- void **propertyRemoved** (QtBrowserItem *index)
- void **propertyChanged** (QtBrowserItem *index)
- QWidget * **createEditor** (QtProperty *property, QWidget *parent) const
- void **slotEditorDestroyed** ()
- void **slotUpdate** ()
- void **slotToggled** (bool checked)

The documentation for this class was generated from the following file:

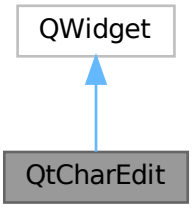
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtbuttonpropertybrowser.cpp

17.688 QtCharEdit Class Reference

Inheritance diagram for QtCharEdit:



Collaboration diagram for QtCharEdit:



Public Slots

- void **setValue** (const QChar &value)

Signals

- void **valueChanged** (const QChar &value)

Public Member Functions

- **QtCharEdit** (QWidget ∗parent=0)
- QChar **value** () const
- bool **eventFilter** (QObject ∗o, QEvent ∗e) override

Protected Member Functions

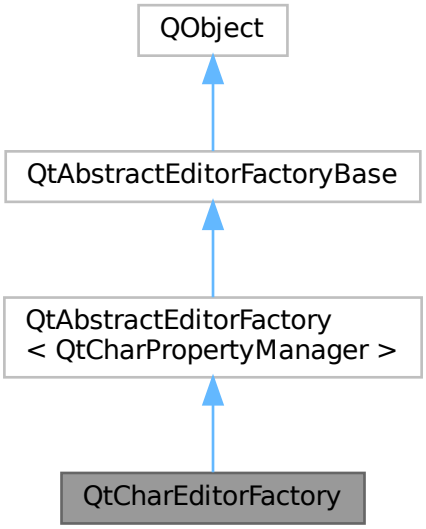
- void **focusInEvent** (QFocusEvent ∗e) override
- void **focusOutEvent** (QFocusEvent ∗e) override
- void **keyPressEvent** (QKeyEvent ∗e) override
- void **keyReleaseEvent** (QKeyEvent ∗e) override
- bool **event** (QEvent ∗e) override

The documentation for this class was generated from the following file:

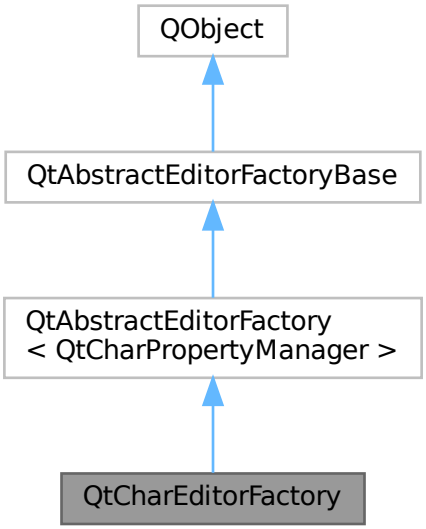
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.689 QtCharEditorFactory Class Reference

Inheritance diagram for QtCharEditorFactory:



Collaboration diagram for QtCharEditorFactory:



Public Member Functions

- QtCharEditorFactory (QObject *parent=0)

- ~QtCharEditorFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtCharPropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtCharPropertyManager *manager)
- void removePropertyManager (QtCharPropertyManager *manager)
- QSet< QtCharPropertyManager * > propertyManagers () const
- QtCharPropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtCharPropertyManager *manager) override
- QWidget * **createEditor** (QtCharPropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtCharPropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtCharPropertyManager >

- virtual QWidget * createEditor (QtCharPropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtCharPropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtCharPropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.689.1 Constructor & Destructor Documentation

17.689.1.1 QtCharEditorFactory()

```
QtCharEditorFactory::QtCharEditorFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.689.1.2 ~QtCharEditorFactory()

```
QtCharEditorFactory::~QtCharEditorFactory ( )
```

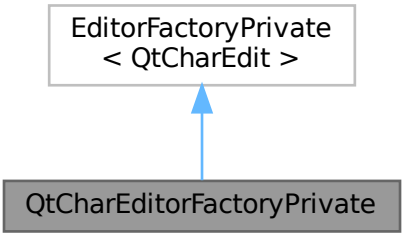
Destroys this factory, and all the widgets it has created.

The documentation for this class was generated from the following files:

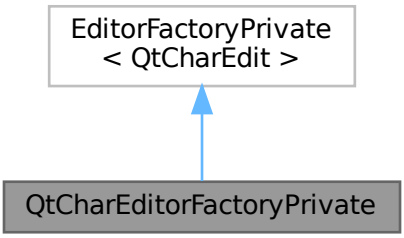
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.690 QtCharEditorFactoryPrivate Class Reference

Inheritance diagram for QtCharEditorFactoryPrivate:



Collaboration diagram for QtCharEditorFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, const QChar &value)
- void **slotSetValue** (const QChar &value)

Public Member Functions inherited from EditorFactoryPrivate< QtCharEdit >

- QtCharEdit * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QtCharEdit *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QtCharEdit >

- typedef QList< QtCharEdit * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QtCharEdit *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QtCharEdit >

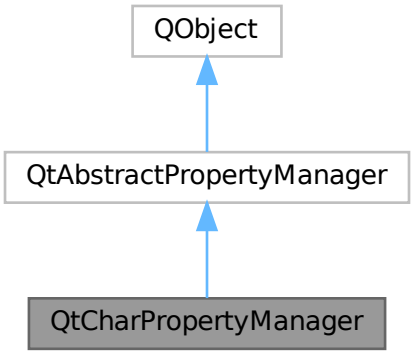
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

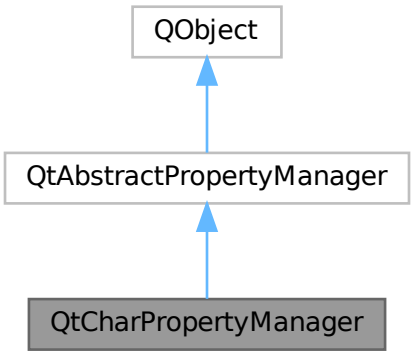
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.691 QtCharPropertyManager Class Reference

Inheritance diagram for QtCharPropertyManager:



Collaboration diagram for QtCharPropertyManager:



Public Slots

- void setValue (QtProperty *property, const QChar &val)

Signals

- void valueChanged (QtProperty *property, const QChar &val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtCharPropertyManager (QObject *parent=0)
- ~QtCharPropertyManager ()
- QChar value (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.691.1 Constructor & Destructor Documentation

17.691.1.1 QtCharPropertyManager()

```
QtCharPropertyManager::QtCharPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.691.1.2 ~QtCharPropertyManager()

```
QtCharPropertyManager::~QtCharPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.691.2 Member Function Documentation

17.691.2.1 initializeProperty()

```
void QtCharPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.691.2.2 setValue

```
void QtCharPropertyManager::setValue (
            QtProperty * property,
            const QChar & value )  [slot]
```

Sets the value of the given *property* to *value*.

See also

value(), valueChanged()

17.691.2.3 uninitializeProperty()

```
void QtCharPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.691.2.4 value()

```
QChar QtCharPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by this manager, this function returns an null QChar object.

See also

setValue()

17.691.2.5 valueChanged

```
void QtCharPropertyManager::valueChanged (
            QtProperty * property,
            const QChar & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

17.691.2.6 valueText()

```
QString QtCharPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.692 QtCharPropertyManagerPrivate Class Reference

Public Types

- typedef QMap< const QtProperty *, QChar > **PropertyValueMap**

Public Attributes

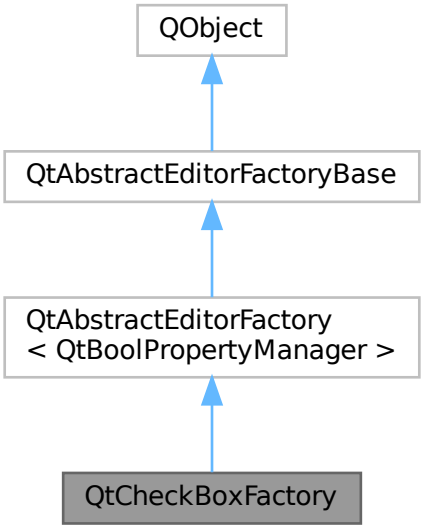
- PropertyValueMap **m_values**

The documentation for this class was generated from the following file:

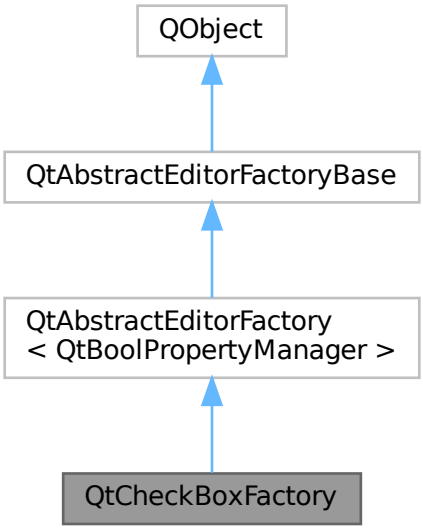
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.693 QtCheckBoxFactory Class Reference

Inheritance diagram for QtCheckBoxFactory:



Collaboration diagram for QtCheckBoxFactory:



Public Member Functions

- QtCheckBoxFactory (QObject ∗parent=0)

- ~QtCheckBoxFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtBoolPropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtBoolPropertyManager *manager)
- void removePropertyManager (QtBoolPropertyManager *manager)
- QSet< QtBoolPropertyManager * > propertyManagers () const
- QtBoolPropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtBoolPropertyManager *manager) override
- QWidget * **createEditor** (QtBoolPropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtBoolPropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtBoolPropertyManager >

- virtual QWidget * createEditor (QtBoolPropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtBoolPropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtBoolPropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.693.1 Constructor & Destructor Documentation

17.693.1.1 QtCheckBoxFactory()

```
QtCheckBoxFactory::QtCheckBoxFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.693.1.2 ~QtCheckBoxFactory()

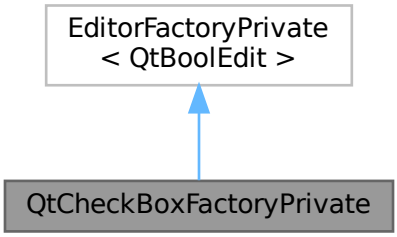
```
QtCheckBoxFactory::~QtCheckBoxFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

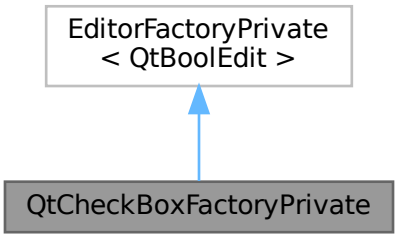
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.694 QtCheckBoxFactoryPrivate Class Reference

Inheritance diagram for QtCheckBoxFactoryPrivate:



Collaboration diagram for QtCheckBoxFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, bool value)
- void **slotSetValue** (bool value)

Public Member Functions inherited from EditorFactoryPrivate< QtBoolEdit >

- QtBoolEdit * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QtBoolEdit *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QtBoolEdit >

- typedef QList< QtBoolEdit * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QtBoolEdit *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QtBoolEdit >

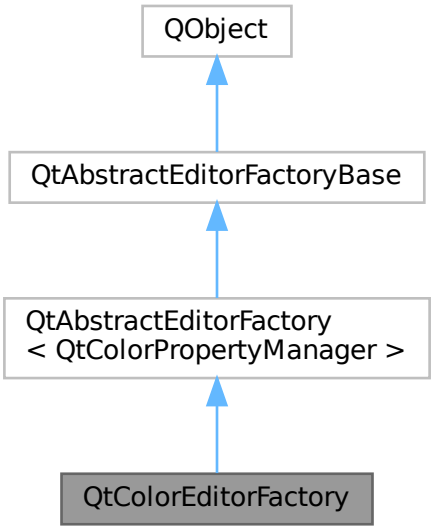
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

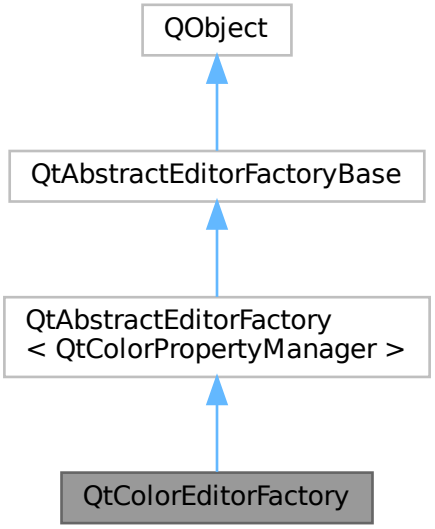
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.695 QtColorEditorFactory Class Reference

Inheritance diagram for QtColorEditorFactory:



Collaboration diagram for QtColorEditorFactory:



Public Member Functions

- QtColorEditorFactory (QObject ∗parent=0)
- ∼QtColorEditorFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtColorPropertyManager >

- QtAbstractEditorFactory (QObject ∗parent)
- QWidget ∗ createEditor (QtProperty ∗property, QWidget ∗parent) override
- void addPropertyManager (QtColorPropertyManager ∗manager)
- void removePropertyManager (QtColorPropertyManager ∗manager)
- QSet< QtColorPropertyManager ∗ > propertyManagers () const
- QtColorPropertyManager ∗ propertyManager (QtProperty ∗property) const

Protected Member Functions

- void **connectPropertyManager** (QtColorPropertyManager ∗manager) override
- QWidget ∗ **createEditor** (QtColorPropertyManager ∗manager, QtProperty ∗property, QWidget ∗parent) override
- void **disconnectPropertyManager** (QtColorPropertyManager ∗manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtColorPropertyManager >

- virtual QWidget ∗ createEditor (QtColorPropertyManager ∗manager, QtProperty ∗property, QWidget ∗parent)=0
- virtual void connectPropertyManager (QtColorPropertyManager ∗manager)=0
- virtual void disconnectPropertyManager (QtColorPropertyManager ∗manager)=0
- void managerDestroyed (QObject ∗manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject ∗parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.695.1 Constructor & Destructor Documentation

17.695.1.1 QtColorEditorFactory()

```
QtColorEditorFactory::QtColorEditorFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.695.1.2 ∼QtColorEditorFactory()

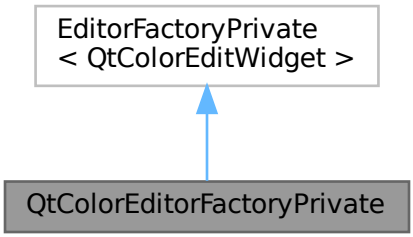
```
QtColorEditorFactory::~QtColorEditorFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

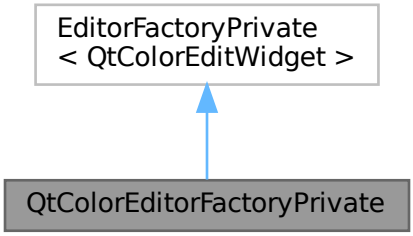
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.696 QtColorEditorFactoryPrivate Class Reference

Inheritance diagram for QtColorEditorFactoryPrivate:



Collaboration diagram for QtColorEditorFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, const QColor &value)
- void **slotSetValue** (const QColor &value)

Public Member Functions inherited from EditorFactoryPrivate< QtColorEditWidget >

- QtColorEditWidget * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QtColorEditWidget *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QtColorEditWidget >

- typedef QList< QtColorEditWidget * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QtColorEditWidget *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QtColorEditWidget >

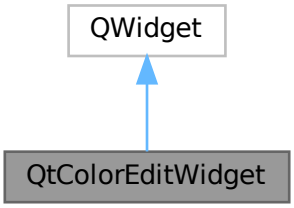
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

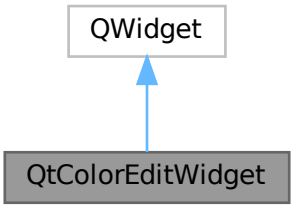
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.697 QtColorEditWidget Class Reference

Inheritance diagram for QtColorEditWidget:



Collaboration diagram for QtColorEditWidget:



Public Slots

- void **setValue** (const QColor &value)

Signals

- void **valueChanged** (const QColor &value)

Public Member Functions

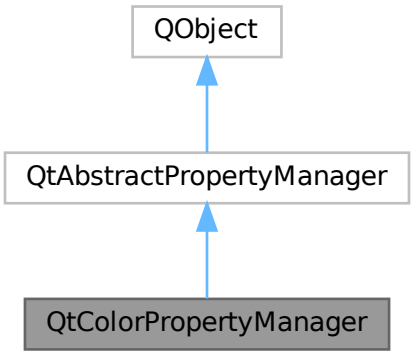
- **QtColorEditWidget** (QWidget ∗parent)
- bool **eventFilter** (QObject ∗obj, QEvent ∗ev) override

The documentation for this class was generated from the following file:

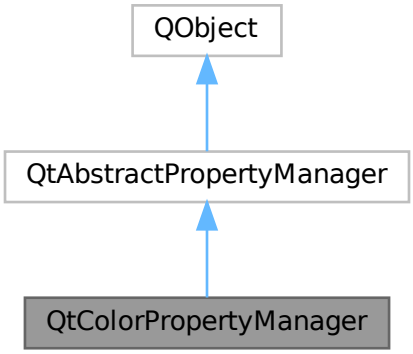
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.698 QtColorPropertyManager Class Reference

Inheritance diagram for QtColorPropertyManager:



Collaboration diagram for QtColorPropertyManager:



Public Slots

- void setValue (QtProperty *property, const QColor &val)

Signals

- void valueChanged (QtProperty *property, const QColor &val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtColorPropertyManager (QObject ∗parent=0)
- ∼QtColorPropertyManager ()
- QtIntPropertyManager ∗ subIntPropertyManager () const
- QColor value (const QtProperty ∗property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject ∗parent=0)
- ∼QtAbstractPropertyManager ()
- QSet< QtProperty ∗ > properties () const
- void clear () const
- QtProperty ∗ addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty ∗property) const override
- QIcon valueIcon (const QtProperty ∗property) const override
- void initializeProperty (QtProperty ∗property) override
- void uninitializeProperty (QtProperty ∗property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty ∗property) const
- virtual QtProperty ∗ createProperty ()

17.698.1 Constructor & Destructor Documentation

17.698.1.1 QtColorPropertyManager()

```
QtColorPropertyManager::QtColorPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.698.1.2 ∼QtColorPropertyManager()

```
QtColorPropertyManager::~QtColorPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.698.2 Member Function Documentation

17.698.2.1 initializeProperty()

```
void QtColorPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.698.2.2 setValue

```
void QtColorPropertyManager::setValue (
            QtProperty * property,
            const QColor & value )  [slot]
```

Sets the value of the given *property* to *value*. Nested properties are updated automatically.

See also

value(), valueChanged()

17.698.2.3 subIntPropertyManager()

```
QtIntPropertyManager * QtColorPropertyManager::subIntPropertyManager ( ) const
```

Returns the manager that produces the nested *red*, *green* and *blue* subproperties.
In order to provide editing widgets for the subproperties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.698.2.4 uninitializeProperty()

```
void QtColorPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.698.2.5 value()

```
QColor QtColorPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by *this* manager, this function returns an invalid color.

See also

setValue()

17.698.2.6 valueChanged

```
void QtColorPropertyManager::valueChanged (
            QtProperty * property,
            const QColor & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.698.2.7 valueIcon()

```
QIcon QtColorPropertyManager::valueIcon (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.698.2.8 valueText()

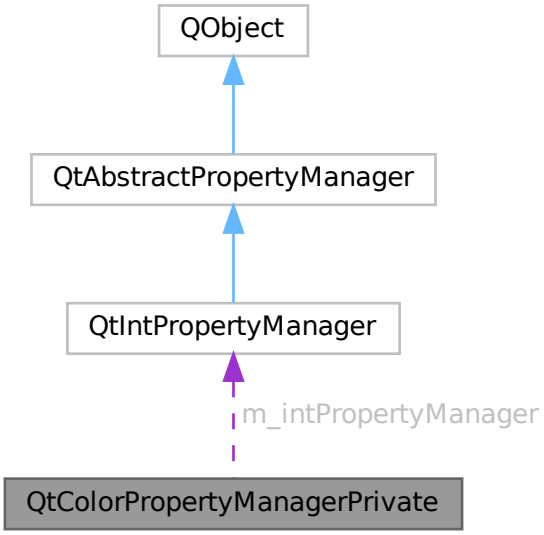
```
QString QtColorPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.699 QtColorPropertyManagerPrivate Class Reference

Collaboration diagram for QtColorPropertyManagerPrivate:



Public Types

- typedef QMap< const QtProperty *, QColor > **PropertyValueMap**

Public Member Functions

- void **slotIntChanged** (QtProperty *property, int value)
- void **slotPropertyDestroyed** (QtProperty *property)

Public Attributes

- PropertyValueMap **m_values**
- QtIntPropertyManager * **m_intPropertyManager**
- QMap< const QtProperty *, QtProperty * > **m_propertyToR**
- QMap< const QtProperty *, QtProperty * > **m_propertyToG**
- QMap< const QtProperty *, QtProperty * > **m_propertyToB**
- QMap< const QtProperty *, QtProperty * > **m_propertyToA**
- QMap< const QtProperty *, QtProperty * > **m_rToProperty**
- QMap< const QtProperty *, QtProperty * > **m_gToProperty**
- QMap< const QtProperty *, QtProperty * > **m_bToProperty**
- QMap< const QtProperty *, QtProperty * > **m_aToProperty**

The documentation for this class was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.700 QtCursorDatabase Class Reference

Public Member Functions

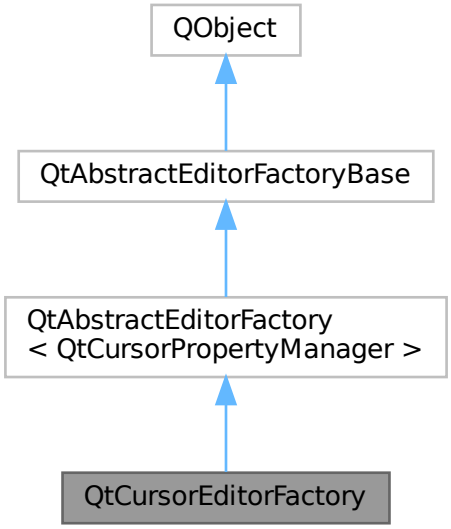
- void **clear** ()
- QStringList **cursorShapeNames** () const
- QMap< int, QIcon > **cursorShapeIcons** () const
- QString **cursorToShapeName** (const QCursor &cursor) const
- QIcon **cursorToShapeIcon** (const QCursor &cursor) const
- int **cursorToValue** (const QCursor &cursor) const
- QCursor **valueToCursor** (int value) const

The documentation for this class was generated from the following files:

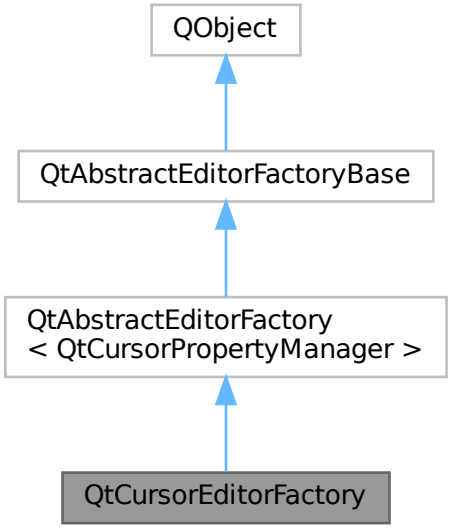
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils_p.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils.cpp

17.701 QtCursorEditorFactory Class Reference

Inheritance diagram for QtCursorEditorFactory:



Collaboration diagram for QtCursorEditorFactory:



Public Member Functions

- QtCursorEditorFactory (QObject *parent=0)
- ~QtCursorEditorFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtCursorPropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtCursorPropertyManager *manager)
- void removePropertyManager (QtCursorPropertyManager *manager)
- QSet< QtCursorPropertyManager * > propertyManagers () const
- QtCursorPropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtCursorPropertyManager *manager) override
- QWidget * **createEditor** (QtCursorPropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtCursorPropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtCursorPropertyManager >

- virtual QWidget * createEditor (QtCursorPropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtCursorPropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtCursorPropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject ∗parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.701.1 Constructor & Destructor Documentation

17.701.1.1 QtCursorEditorFactory()

```
QtCursorEditorFactory::QtCursorEditorFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.701.1.2 ∼QtCursorEditorFactory()

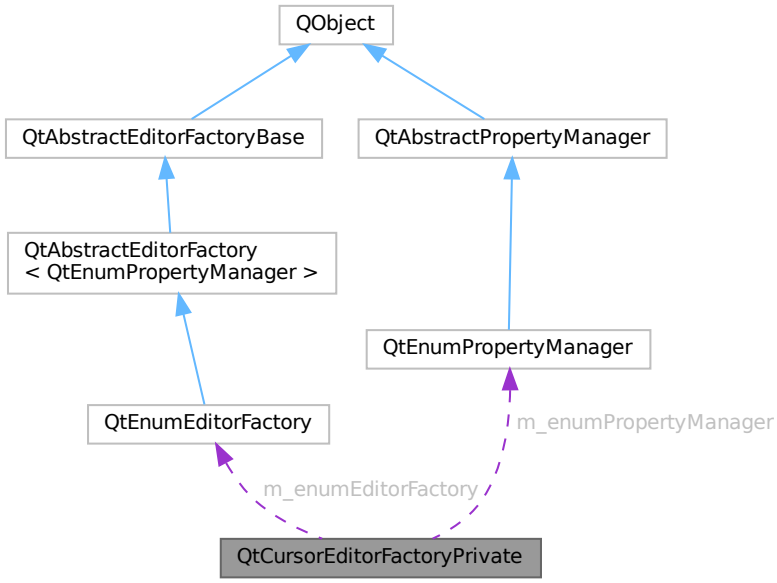
```
QtCursorEditorFactory::~QtCursorEditorFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.702 QtCursorEditorFactoryPrivate Class Reference

Collaboration diagram for QtCursorEditorFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty ∗property, const QCursor &cursor)
- void **slotEnumChanged** (QtProperty ∗property, int value)
- void **slotEditorDestroyed** (QObject ∗object)

Public Attributes

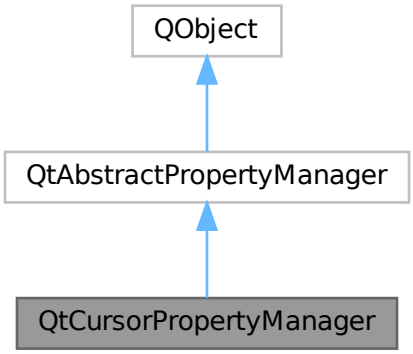
- QtEnumEditorFactory * **m_enumEditorFactory**
- QtEnumPropertyManager * **m_enumPropertyManager**
- QMap< QtProperty *, QtProperty * > **m_propertyToEnum**
- QMap< QtProperty *, QtProperty * > **m_enumToProperty**
- QMap< QtProperty *, QWidgetList > **m_enumToEditors**
- QMap< QWidget *, QtProperty * > **m_editorToEnum**
- bool **m_updatingEnum**

The documentation for this class was generated from the following file:

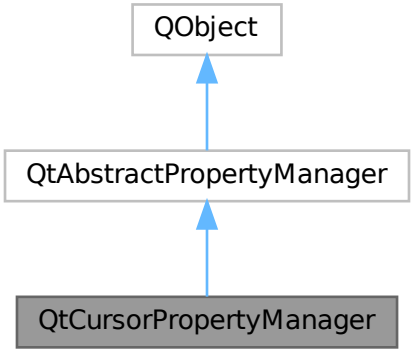
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.703 QtCursorPropertyManager Class Reference

Inheritance diagram for QtCursorPropertyManager:



Collaboration diagram for QtCursorPropertyManager:



Public Slots

- void setValue (QtProperty *property, const QCursor &val)

Signals

- void valueChanged (QtProperty *property, const QCursor &val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtCursorPropertyManager (QObject *parent=0)
- ~QtCursorPropertyManager ()
- QCursor value (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- QIcon valueIcon (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.703.1 Constructor & Destructor Documentation

17.703.1.1 QtCursorPropertyManager()

```
QtCursorPropertyManager::QtCursorPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.703.1.2 ~QtCursorPropertyManager()

```
QtCursorPropertyManager::~QtCursorPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.703.2 Member Function Documentation

17.703.2.1 initializeProperty()

```
void QtCursorPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.703.2.2 setValue

```
void QtCursorPropertyManager::setValue (
            QtProperty * property,
            const QCursor & value )  [slot]
```

Sets the value of the given *property* to *value*.

See also

value(), valueChanged()

17.703.2.3 uninitializeProperty()

```
void QtCursorPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.703.2.4 value()

```
QCursor QtCursorPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by this manager, this function returns a default QCursor object.

See also

setValue()

17.703.2.5 valueChanged

```
void QtCursorPropertyManager::valueChanged (
            QtProperty * property,
            const QCursor & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.703.2.6 valueIcon()

```
QIcon QtCursorPropertyManager::valueIcon (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.703.2.7 valueText()

```
QString QtCursorPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.704 QtCursorPropertyManagerPrivate Class Reference

Public Types

- typedef QMap< const QtProperty *, QCursor > **PropertyValueMap**

Public Attributes

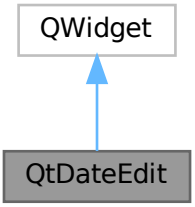
- PropertyValueMap **m_values**

The documentation for this class was generated from the following file:

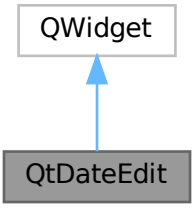
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.705 QtDateEdit Class Reference

Inheritance diagram for QtDateEdit:



Collaboration diagram for QtDateEdit:



Signals

- void **dateChanged** (QDate)

Public Member Functions

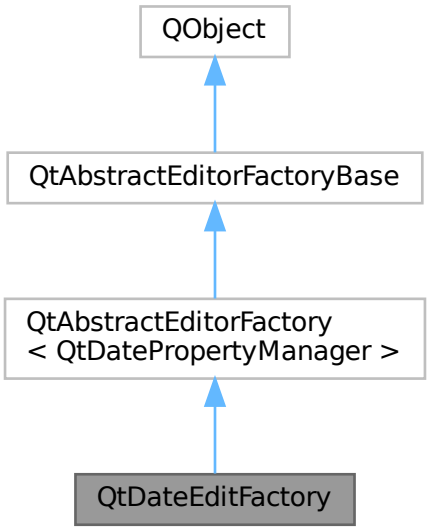
- **QtDateEdit** (QWidget ∗parent=Q_NULLPTR)
- QDate **date** ()
- void **setDate** (QDate)
- void **setInitialDate** (QDate)
- void **setDateRange** (QDate, QDate)
- void **setDisplayFormat** (QString)
- void **setCalendarPopup** (bool)

The documentation for this class was generated from the following files:

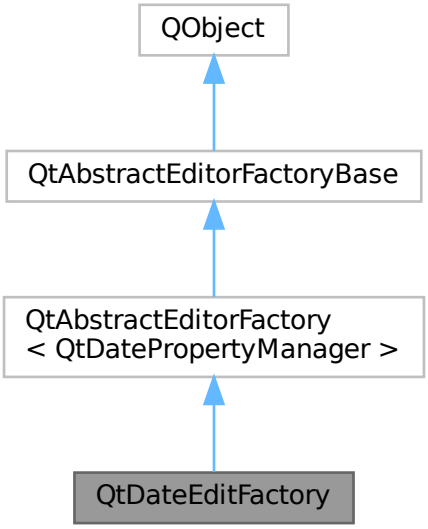
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils_p.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils.cpp

17.706 QtDateEditFactory Class Reference

Inheritance diagram for QtDateEditFactory:



Collaboration diagram for QtDateEditFactory:



Public Member Functions

- QtDateEditFactory (QObject *parent=0)
- ~QtDateEditFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtDatePropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtDatePropertyManager *manager)
- void removePropertyManager (QtDatePropertyManager *manager)
- QSet< QtDatePropertyManager * > propertyManagers () const
- QtDatePropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtDatePropertyManager *manager) override
- QWidget * **createEditor** (QtDatePropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtDatePropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtDatePropertyManager >

- virtual QWidget * createEditor (QtDatePropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtDatePropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtDatePropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject ∗parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.706.1 Constructor & Destructor Documentation

17.706.1.1 QtDateEditFactory()

```
QtDateEditFactory::QtDateEditFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.706.1.2 ∼QtDateEditFactory()

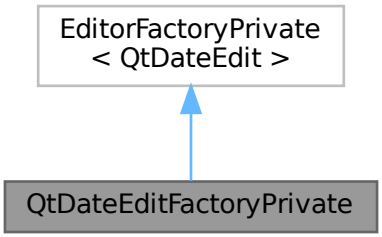
```
QtDateEditFactory::~QtDateEditFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

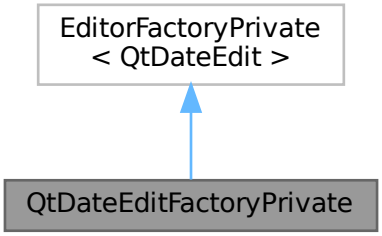
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.707 QtDateEditFactoryPrivate Class Reference

Inheritance diagram for QtDateEditFactoryPrivate:



Collaboration diagram for QtDateEditFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, QDate value)
- void **slotRangeChanged** (QtProperty *property, QDate min, QDate max)
- void **slotSetValue** (QDate value)

Public Member Functions inherited from EditorFactoryPrivate< QtDateEdit >

- QtDateEdit * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QtDateEdit *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QtDateEdit >

- typedef QList< QtDateEdit * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QtDateEdit *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QtDateEdit >

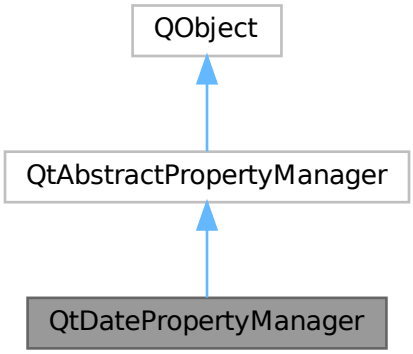
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

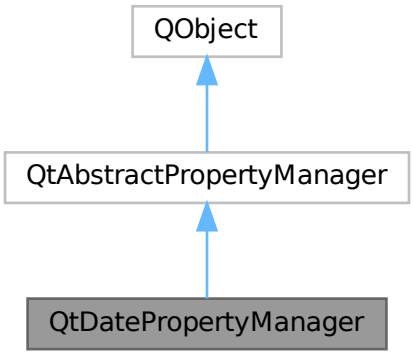
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.708 QtDatePropertyManager Class Reference

Inheritance diagram for QtDatePropertyManager:



Collaboration diagram for QtDatePropertyManager:



Public Slots

- void setValue (QtProperty *property, QDate val)
- void **setValueOnly** (QtProperty *property, QDate val)
- void setMinimum (QtProperty *property, QDate minVal)
- void setMaximum (QtProperty *property, QDate maxVal)
- void setRange (QtProperty *property, QDate minVal, QDate maxVal)

Signals

- void valueChanged (QtProperty *property, QDate val)
- void rangeChanged (QtProperty *property, QDate minVal, QDate maxVal)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty ∗property, QtProperty ∗parent, QtProperty ∗after)
- void propertyChanged (QtProperty ∗property)
- void propertyRemoved (QtProperty ∗property, QtProperty ∗parent)
- void propertyDestroyed (QtProperty ∗property)

Public Member Functions

- QtDatePropertyManager (QObject ∗parent=0)
- ∼QtDatePropertyManager ()
- QDate value (const QtProperty ∗property) const
- QDate **initialValue** (const QtProperty ∗property) const
- QDate minimum (const QtProperty ∗property) const
- QDate maximum (const QtProperty ∗property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject ∗parent=0)
- ∼QtAbstractPropertyManager ()
- QSet< QtProperty ∗ > properties () const
- void clear () const
- QtProperty ∗ addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty ∗property) const override
- void initializeProperty (QtProperty ∗property) override
- void uninitializeProperty (QtProperty ∗property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty ∗property) const
- virtual QIcon valueIcon (const QtProperty ∗property) const
- virtual QtProperty ∗ createProperty ()

17.708.1 Constructor & Destructor Documentation

17.708.1.1 QtDatePropertyManager()

```
QtDatePropertyManager::QtDatePropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.708.1.2 ∼QtDatePropertyManager()

```
QtDatePropertyManager::~QtDatePropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.708.2 Member Function Documentation

17.708.2.1 initializeProperty()

```
void QtDatePropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp

Implements QtAbstractPropertyManager.

17.708.2.2 maximum()

```
QDate QtDatePropertyManager::maximum (
            const QtProperty * property ) const
```

Returns the given *property's* maximum date.

See also

minimum(), setRange()

17.708.2.3 minimum()

```
QDate QtDatePropertyManager::minimum (
            const QtProperty * property ) const
```

Returns the given *property's* minimum date.

See also

maximum(), setRange()

17.708.2.4 rangeChanged

```
void QtDatePropertyManager::rangeChanged (
            QtProperty * property,
            QDate minimum,
            QDate maximum )  [signal]
```

This signal is emitted whenever a property created by this manager changes its range of valid dates, passing a pointer to the *property* and the new *minimum* and *maximum* dates.

See also

setRange()

17.708.2.5 setMaximum

```
void QtDatePropertyManager::setMaximum (
            QtProperty * property,
            QDate maxVal )  [slot]
```

Sets the maximum value for the given *property* to *maxVal*.
When setting the maximum value, the minimum and current values are adjusted if necessary (ensuring that the range remains valid and that the current value is within in the range).

See also

maximum(), setRange()

17.708.2.6 setMinimum

```
void QtDatePropertyManager::setMinimum (
            QtProperty * property,
            QDate minVal )  [slot]
```

Sets the minimum value for the given *property* to *minVal*.
When setting the minimum value, the maximum and current values are adjusted if necessary (ensuring that the range remains valid and that the current value is within in the range).

See also

minimum(), setRange()

17.708.2.7 setRange

```
void QtDatePropertyManager::setRange (
            QtProperty * property,
            QDate minimum,
            QDate maximum )  [slot]
```

Sets the range of valid dates.
This is a convenience function defining the range of valid dates in one go; setting the *minimum* and *maximum* values for the given *property* with a single function call.
When setting a new date range, the current value is adjusted if necessary (ensuring that the value remains in date range).

See also

setMinimum(), setMaximum(), rangeChanged()

17.708.2.8 setValue

```
void QtDatePropertyManager::setValue (
            QtProperty * property,
            QDate value )  [slot]
```

Sets the value of the given *property* to *value*.
If the specified *value* is not a valid date according to the given *property's* range, the value is adjusted to the nearest valid value within the range.

See also

value(), setRange(), valueChanged()

17.708.2.9 uninitializeProperty()

```
void QtDatePropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.708.2.10 value()

```
QDate QtDatePropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by *this* manager, this function returns an invalid date.

See also

setValue()

17.708.2.11 valueChanged

```
void QtDatePropertyManager::valueChanged (
            QtProperty * property,
            QDate value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.708.2.12 valueText()

```
QString QtDatePropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.709 QtDatePropertyManagerPrivate Class Reference

Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Member Functions

- **QtDatePropertyManagerPrivate** (QtDatePropertyManager *q)

Public Attributes

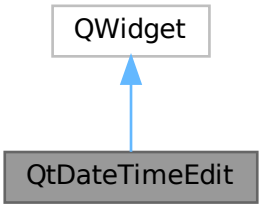
- QString **m_format**
- QMap< const QtProperty *, Data > **m_values**

The documentation for this class was generated from the following file:

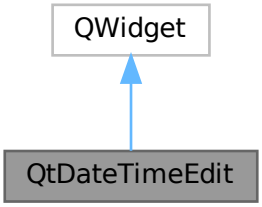
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.710 QtDateTimeEdit Class Reference

Inheritance diagram for QtDateTimeEdit:



Collaboration diagram for QtDateTimeEdit:



Signals

- void **dateTimeChanged** (const QDateTime &)

Public Member Functions

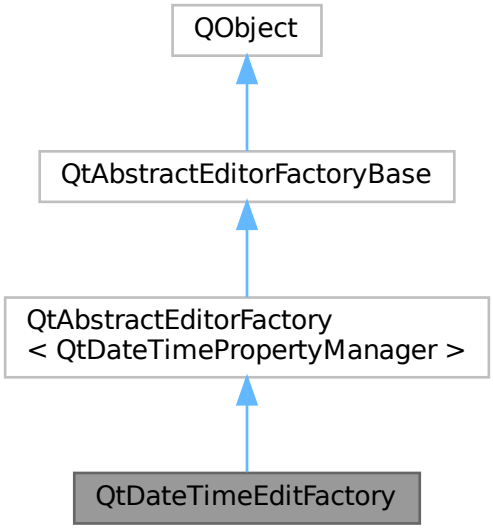
- **QtDateTimeEdit** (QWidget ∗parent=Q_NULLPTR)
- QDateTime **dateTime** ()
- void **setDateTime** (const QDateTime &)
- void **setInitialDateTime** (const QDateTime &)
- void **setDateTimeRange** (const QDateTime &, const QDateTime &)
- void **setDisplayFormat** (const QString &)
- void **setCalendarPopup** (bool)

The documentation for this class was generated from the following files:

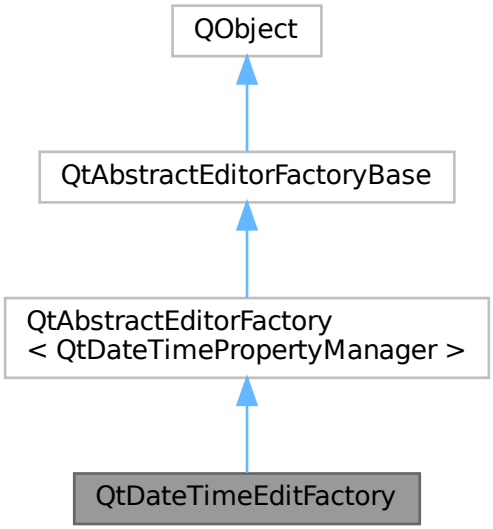
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils_p.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils.cpp

17.711 QtDateTimeEditFactory Class Reference

Inheritance diagram for QtDateTimeEditFactory:



Collaboration diagram for QtDateTimeEditFactory:



Public Member Functions

- QtDateTimeEditFactory (QObject *parent=0)

- ~QtDateTimeEditFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtDateTimePropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtDateTimePropertyManager *manager)
- void removePropertyManager (QtDateTimePropertyManager *manager)
- QSet< QtDateTimePropertyManager * > propertyManagers () const
- QtDateTimePropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtDateTimePropertyManager *manager) override
- QWidget * **createEditor** (QtDateTimePropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtDateTimePropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtDateTimePropertyManager >

- virtual QWidget * createEditor (QtDateTimePropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtDateTimePropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtDateTimePropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.711.1 Constructor & Destructor Documentation

17.711.1.1 QtDateTimeEditFactory()

```
QtDateTimeEditFactory::QtDateTimeEditFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.711.1.2 ~QtDateTimeEditFactory()

```
QtDateTimeEditFactory::~QtDateTimeEditFactory ( )
```

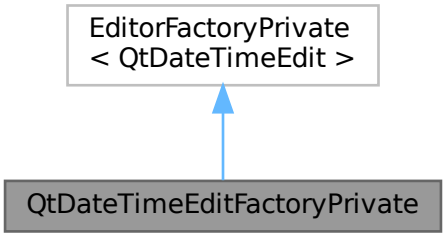
Destroys this factory, and all the widgets it has created.

The documentation for this class was generated from the following files:

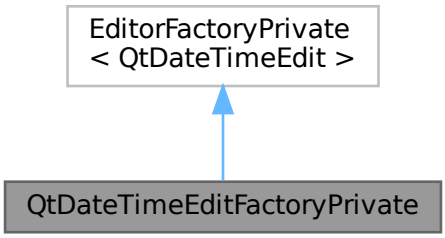
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.712 QtDateTimeEditFactoryPrivate Class Reference

Inheritance diagram for QtDateTimeEditFactoryPrivate:



Collaboration diagram for QtDateTimeEditFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, const QDateTime &value)
- void **slotSetValue** (const QDateTime &value)

Public Member Functions inherited from EditorFactoryPrivate< QtDateTimeEdit >

- QtDateTimeEdit * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QtDateTimeEdit *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QtDateTimeEdit >

- typedef QList< QtDateTimeEdit * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QtDateTimeEdit *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QtDateTimeEdit >

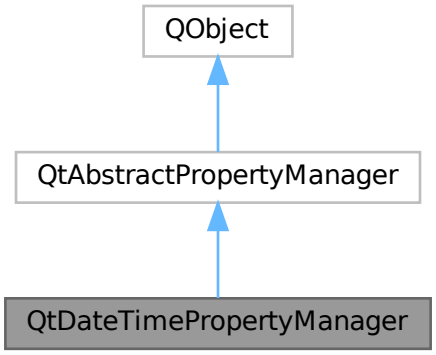
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

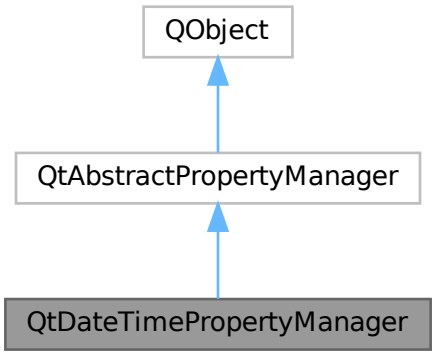
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.713 QtDateTimePropertyManager Class Reference

Inheritance diagram for QtDateTimePropertyManager:



Collaboration diagram for QtDateTimePropertyManager:



Public Slots

- void setValue (QtProperty *property, const QDateTime &val)
- void **setValueOnly** (QtProperty *property, const QDateTime &val)

Signals

- void valueChanged (QtProperty *property, const QDateTime &val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtDateTimePropertyManager (QObject *parent=0)
- ~QtDateTimePropertyManager ()
- QDateTime value (const QtProperty *property) const
- QDateTime **initialValue** (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.713.1 Constructor & Destructor Documentation

17.713.1.1 QtDateTimePropertyManager()

```
QtDateTimePropertyManager::QtDateTimePropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.713.1.2 ~QtDateTimePropertyManager()

```
QtDateTimePropertyManager::~QtDateTimePropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.713.2 Member Function Documentation

17.713.2.1 initializeProperty()

```
void QtDateTimePropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.713.2.2 setValue

```
void QtDateTimePropertyManager::setValue (
            QtProperty * property,
            const QDateTime & value )  [slot]
```

Sets the value of the given *property* to *value*.

See also

value(), valueChanged()

17.713.2.3 uninitializeProperty()

```
void QtDateTimePropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.713.2.4 value()

```
QDateTime QtDateTimePropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by this manager, this function returns an invalid QDateTime object.

See also

setValue()

17.713.2.5 valueChanged

```
void QtDateTimePropertyManager::valueChanged (
            QtProperty * property,
            const QDateTime & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

17.713.2.6 valueText()

```
QString QtDateTimePropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.714 QtDateTimePropertyManagerPrivate Class Reference

Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Member Functions

- **QtDateTimePropertyManagerPrivate** (QtDateTimePropertyManager *q)

Public Attributes

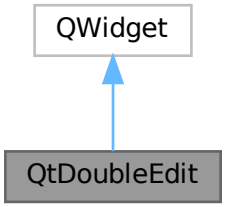
- const QString **m_format**
- PropertyValueMap **m_values**

The documentation for this class was generated from the following file:

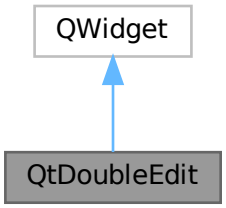
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.715 QtDoubleEdit Class Reference

Inheritance diagram for QtDoubleEdit:



Collaboration diagram for QtDoubleEdit:



Signals

- void **valueChanged** (double)

Public Member Functions

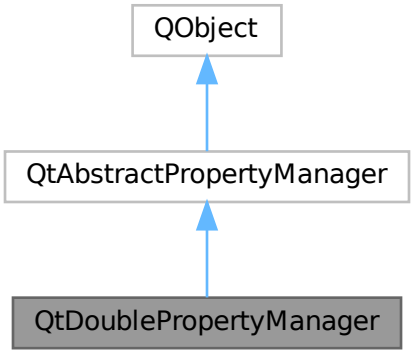
- **QtDoubleEdit** (QWidget ∗parent=Q_NULLPTR)
- int **value** ()
- void **setValue** (double)
- void **setInitialValue** (double)
- void **setDecimals** (double)
- void **setSingleStep** (double)
- void **setRange** (double, double)
- void **setKeyboardTracking** (bool)

The documentation for this class was generated from the following files:

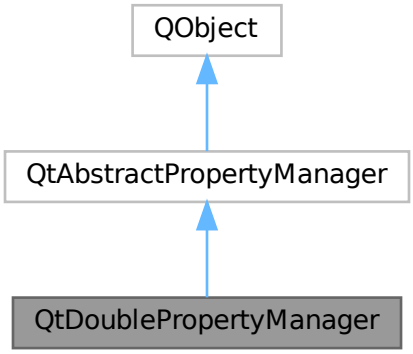
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils_p.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils.cpp

17.716 QtDoublePropertyManager Class Reference

Inheritance diagram for QtDoublePropertyManager:



Collaboration diagram for QtDoublePropertyManager:



Public Slots

- void setValue (QtProperty *property, double val)
- void **setValueOnly** (QtProperty *property, double val)
- void setMinimum (QtProperty *property, double minVal)
- void setMaximum (QtProperty *property, double maxVal)
- void setRange (QtProperty *property, double minVal, double maxVal)
- void setSingleStep (QtProperty *property, double step)
- void setDecimals (QtProperty *property, int prec)

Signals

- void valueChanged (QtProperty *property, double val)
- void rangeChanged (QtProperty *property, double minVal, double maxVal)
- void singleStepChanged (QtProperty *property, double step)
- void decimalsChanged (QtProperty *property, int prec)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtDoublePropertyManager (QObject *parent=0)
- ~QtDoublePropertyManager ()
- double value (const QtProperty *property) const
- double **initialValue** (const QtProperty *property) const
- double minimum (const QtProperty *property) const
- double maximum (const QtProperty *property) const
- double singleStep (const QtProperty *property) const
- int decimals (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.716.1 Constructor & Destructor Documentation

17.716.1.1 QtDoublePropertyManager()

```
QtDoublePropertyManager::QtDoublePropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.716.1.2 ~QtDoublePropertyManager()

```
QtDoublePropertyManager::~QtDoublePropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.716.2 Member Function Documentation

17.716.2.1 decimals()

```
int QtDoublePropertyManager::decimals (
            const QtProperty * property ) const
```

Returns the given *property's* precision, in decimals.

See also

setDecimals()

17.716.2.2 decimalsChanged

```
void QtDoublePropertyManager::decimalsChanged (
            QtProperty * property,
            int prec )  [signal]
```

This signal is emitted whenever a property created by this manager changes its precision of value, passing a pointer to the *property* and the new *prec* value

See also

setDecimals()

17.716.2.3 initializeProperty()

```
void QtDoublePropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.716.2.4 maximum()

```
double QtDoublePropertyManager::maximum (
            const QtProperty * property ) const
```

Returns the given *property's* maximum value.

See also

minimum(), setRange()

17.716.2.5 minimum()

```
double QtDoublePropertyManager::minimum (
            const QtProperty * property ) const
```

Returns the given *property's* minimum value.

See also

maximum(), setRange()

17.716.2.6 rangeChanged

```
void QtDoublePropertyManager::rangeChanged (
            QtProperty * property,
            double minimum,
            double maximum )  [signal]
```

This signal is emitted whenever a property created by this manager changes its range of valid values, passing a pointer to the *property* and the new *minimum* and *maximum* values

See also

setRange()

17.716.2.7 setDecimals

```
void QtDoublePropertyManager::setDecimals (
            QtProperty * property,
            int prec )  [slot]
```

Sets the precision of the given *property* to *prec*.
The valid decimal range is 0-13. The default is 2.

See also

decimals()

17.716.2.8 setMaximum

```
void QtDoublePropertyManager::setMaximum (
            QtProperty * property,
            double maxVal )  [slot]
```

Sets the maximum value for the given *property* to *maxVal*.
When setting the maximum value, the minimum and current values are adjusted if necessary (ensuring that the range remains valid and that the current value is within in the range).

See also

maximum(), setRange(), rangeChanged()

17.716.2.9 setMinimum

```
void QtDoublePropertyManager::setMinimum (
            QtProperty * property,
            double minVal )  [slot]
```

Sets the minimum value for the given *property* to *minVal*.
When setting the minimum value, the maximum and current values are adjusted if necessary (ensuring that the range remains valid and that the current value is within in the range).

See also

minimum(), setRange(), rangeChanged()

17.716.2.10 setRange

```
void QtDoublePropertyManager::setRange (
            QtProperty * property,
            double minimum,
            double maximum )  [slot]
```

Sets the range of valid values.
This is a convenience function defining the range of valid values in one go; setting the *minimum* and *maximum* values for the given *property* with a single function call.
When setting a new range, the current value is adjusted if necessary (ensuring that the value remains within range).

See also

setMinimum(), setMaximum(), rangeChanged()

17.716.2.11 setSingleStep

```
void QtDoublePropertyManager::setSingleStep (
            QtProperty * property,
            double step )  [slot]
```

Sets the step value for the given *property* to *step*.
The step is typically used to increment or decrement a property value while pressing an arrow key.

See also

singleStep()

17.716.2.12 setValue

```
void QtDoublePropertyManager::setValue (
            QtProperty * property,
            double value )  [slot]
```

Sets the value of the given *property* to *value*.
If the specified *value* is not valid according to the given *property's* range, the *value* is adjusted to the nearest valid value within the range.

See also

value(), setRange(), valueChanged()

17.716.2.13 singleStep()

```
double QtDoublePropertyManager::singleStep (
            const QtProperty * property ) const
```

Returns the given *property's* step value.
The step is typically used to increment or decrement a property value while pressing an arrow key.

See also

setSingleStep()

17.716.2.14 singleStepChanged

```
void QtDoublePropertyManager::singleStepChanged (
            QtProperty * property,
            double step )  [signal]
```

This signal is emitted whenever a property created by this manager changes its single step property, passing a pointer to the *property* and the new *step* value

See also

setSingleStep()

17.716.2.15 uninitializeProperty()

```
void QtDoublePropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.716.2.16 value()

```
double QtDoublePropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given property is not managed by this manager, this function returns 0.

See also

setValue()

17.716.2.17 valueChanged

```
void QtDoublePropertyManager::valueChanged (
            QtProperty * property,
            double value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.716.2.18 valueText()

```
QString QtDoublePropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.717 QtDoublePropertyManagerPrivate Class Reference

Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Attributes

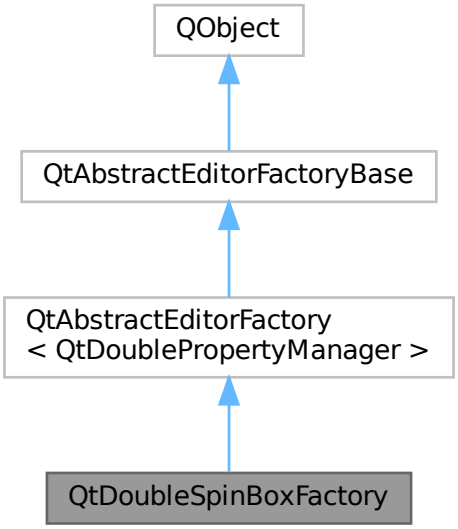
- PropertyValueMap **m_values**

The documentation for this class was generated from the following file:

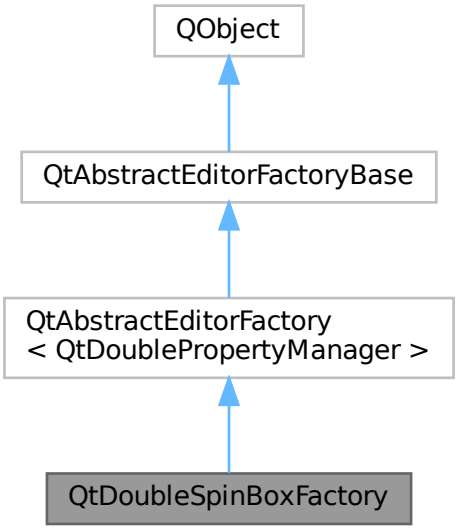
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.718 QtDoubleSpinBoxFactory Class Reference

Inheritance diagram for QtDoubleSpinBoxFactory:



Collaboration diagram for QtDoubleSpinBoxFactory:



Public Member Functions

- QtDoubleSpinBoxFactory (QObject ∗parent=0)

- ~QtDoubleSpinBoxFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtDoublePropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtDoublePropertyManager *manager)
- void removePropertyManager (QtDoublePropertyManager *manager)
- QSet< QtDoublePropertyManager * > propertyManagers () const
- QtDoublePropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtDoublePropertyManager *manager) override
- QWidget * **createEditor** (QtDoublePropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtDoublePropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtDoublePropertyManager >

- virtual QWidget * createEditor (QtDoublePropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtDoublePropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtDoublePropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.718.1 Constructor & Destructor Documentation

17.718.1.1 QtDoubleSpinBoxFactory()

```
QtDoubleSpinBoxFactory::QtDoubleSpinBoxFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.718.1.2 ~QtDoubleSpinBoxFactory()

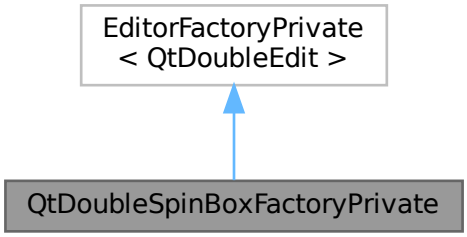
```
QtDoubleSpinBoxFactory::~QtDoubleSpinBoxFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

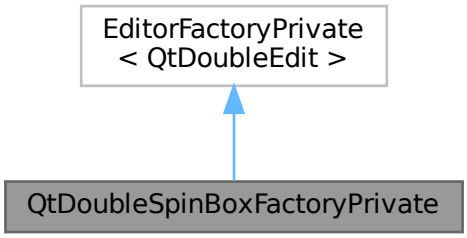
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.719 QtDoubleSpinBoxFactoryPrivate Class Reference

Inheritance diagram for QtDoubleSpinBoxFactoryPrivate:



Collaboration diagram for QtDoubleSpinBoxFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, double value)
- void **slotRangeChanged** (QtProperty *property, double min, double max)
- void **slotSingleStepChanged** (QtProperty *property, double step)
- void **slotDecimalsChanged** (QtProperty *property, int prec)
- void **slotSetValue** (double value)

Public Member Functions inherited from EditorFactoryPrivate< QtDoubleEdit >

- QtDoubleEdit * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QtDoubleEdit *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QtDoubleEdit >

- typedef QList< QtDoubleEdit * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QtDoubleEdit *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QtDoubleEdit >

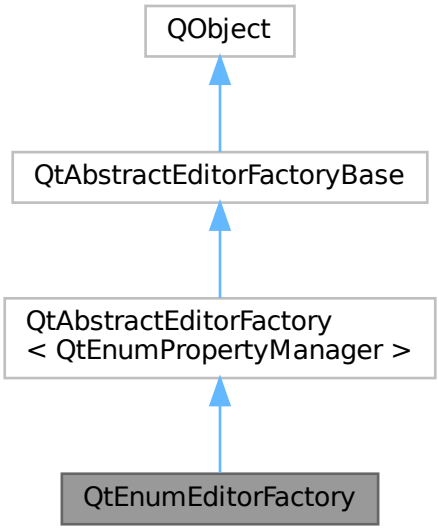
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

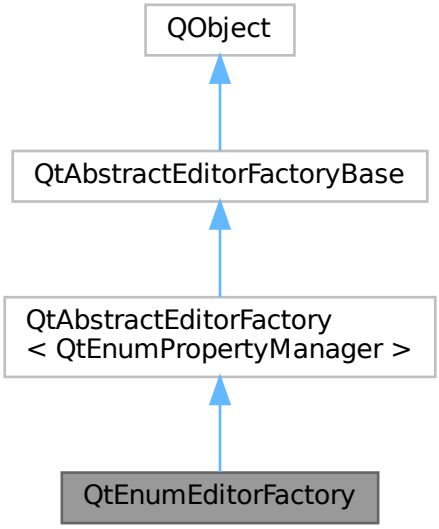
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.720 QtEnumEditorFactory Class Reference

Inheritance diagram for QtEnumEditorFactory:



Collaboration diagram for QtEnumEditorFactory:



Public Member Functions

- QtEnumEditorFactory (QObject *parent=0)
- ~QtEnumEditorFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtEnumPropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtEnumPropertyManager *manager)
- void removePropertyManager (QtEnumPropertyManager *manager)
- QSet< QtEnumPropertyManager * > propertyManagers () const
- QtEnumPropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtEnumPropertyManager *manager) override
- QWidget * **createEditor** (QtEnumPropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtEnumPropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtEnumPropertyManager >

- virtual QWidget * createEditor (QtEnumPropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtEnumPropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtEnumPropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.720.1 Constructor & Destructor Documentation

17.720.1.1 QtEnumEditorFactory()

```
QtEnumEditorFactory::QtEnumEditorFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.720.1.2 ~QtEnumEditorFactory()

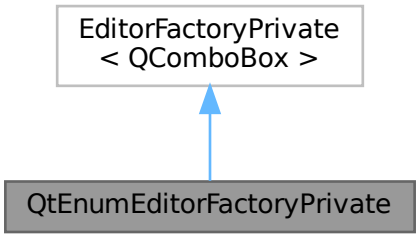
```
QtEnumEditorFactory::~QtEnumEditorFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

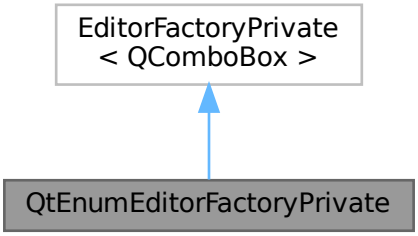
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.721 QtEnumEditorFactoryPrivate Class Reference

Inheritance diagram for QtEnumEditorFactoryPrivate:



Collaboration diagram for QtEnumEditorFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, int value)
- void **slotEnumNamesChanged** (QtProperty *property, const QStringList &)
- void **slotEnumIconsChanged** (QtProperty *property, const QMap< int, QIcon > &)
- void **slotSetValue** (int value)

Public Member Functions inherited from EditorFactoryPrivate< QComboBox >

- QComboBox * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QComboBox *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QComboBox >

- typedef QList< QComboBox * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QComboBox *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QComboBox >

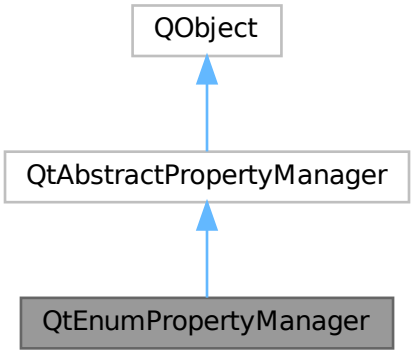
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

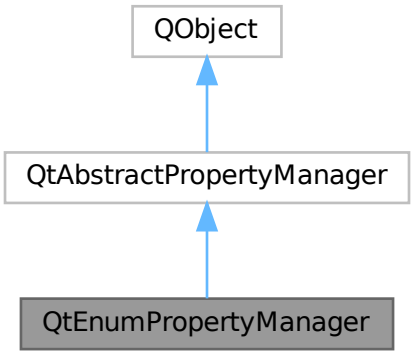
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.722 QtEnumPropertyManager Class Reference

Inheritance diagram for QtEnumPropertyManager:



Collaboration diagram for QtEnumPropertyManager:



Public Slots

- void setValue (QtProperty *property, int val)
- void setEnumNames (QtProperty *property, const QStringList &names)
- void setEnumIcons (QtProperty *property, const QMap< int, QIcon > &icons)

Signals

- void valueChanged (QtProperty *property, int val)
- void enumNamesChanged (QtProperty *property, const QStringList &names)
- void enumIconsChanged (QtProperty *property, const QMap< int, QIcon > &icons)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtEnumPropertyManager (QObject *parent=0)
- ~QtEnumPropertyManager ()
- int value (const QtProperty *property) const
- QStringList enumNames (const QtProperty *property) const
- QMap< int, QIcon > enumIcons (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- QIcon valueIcon (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.722.1 Constructor & Destructor Documentation

17.722.1.1 QtEnumPropertyManager()

```
QtEnumPropertyManager::QtEnumPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.722.1.2 ~QtEnumPropertyManager()

```
QtEnumPropertyManager::~QtEnumPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.722.2 Member Function Documentation

17.722.2.1 enumIcons()

```
QMap< int, QIcon > QtEnumPropertyManager::enumIcons (
            const QtProperty * property ) const
```

Returns the given *property's* map of enum values to their icons.

See also

value(), setEnumIcons()

17.722.2.2 enumIconsChanged

```
void QtEnumPropertyManager::enumIconsChanged (
            QtProperty * property,
            const QMap< int, QIcon > & icons )  [signal]
```

This signal is emitted whenever a property created by this manager changes its enum icons, passing a pointer to the *property* and the new mapping of values to *icons* as parameters.

See also

setEnumIcons()

17.722.2.3 enumNames()

```
QStringList QtEnumPropertyManager::enumNames (
            const QtProperty * property ) const
```

Returns the given *property's* list of enum names.

See also

value(), setEnumNames()

17.722.2.4 enumNamesChanged

```
void QtEnumPropertyManager::enumNamesChanged (
            QtProperty * property,
            const QStringList & names )  [signal]
```

This signal is emitted whenever a property created by this manager changes its enum names, passing a pointer to the *property* and the new *names* as parameters.

See also

setEnumNames()

17.722.2.5 initializeProperty()

```
void QtEnumPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.722.2.6 setEnumIcons

```
void QtEnumPropertyManager::setEnumIcons (
            QtProperty * property,
            const QMap< int, QIcon > & enumIcons )  [slot]
```

Sets the given *property's* map of enum values to their icons to *enumIcons*.
Each enum value can have associated icon. This association is represented with passed *enumIcons* map.

See also

enumNames(), enumNamesChanged()

17.722.2.7 setEnumNames

```
void QtEnumPropertyManager::setEnumNames (
            QtProperty * property,
            const QStringList & enumNames )  [slot]
```

Sets the given *property's* list of enum names to *enumNames*. The *property's* current value is reset to 0 indicating the first item of the list.
If the specified *enumNames* list is empty, the *property's* current value is set to -1.

See also

enumNames(), enumNamesChanged()

17.722.2.8 setValue

```
void QtEnumPropertyManager::setValue (
            QtProperty * property,
            int value )  [slot]
```

Sets the value of the given *property* to *value*.
The specified *value* must be less than the size of the given *property's* enumNames() list. Use -1 to leave the editor without a current selection.

See also

value(), valueChanged()

17.722.2.9 uninitializeProperty()

```
void QtEnumPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.722.2.10 value()

```
int QtEnumPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value which is an index in the list returned by enumNames()
If the given property is not managed by this manager, this function returns -1.

See also

enumNames(), setValue()

17.722.2.11 valueChanged

```
void QtEnumPropertyManager::valueChanged (
            QtProperty * property,
            int value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.722.2.12 valueIcon()

```
QIcon QtEnumPropertyManager::valueIcon (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.722.2.13 valueText()

```
QString QtEnumPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.723 QtEnumPropertyManagerPrivate Class Reference

Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Attributes

- PropertyValueMap **m_values**

The documentation for this class was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

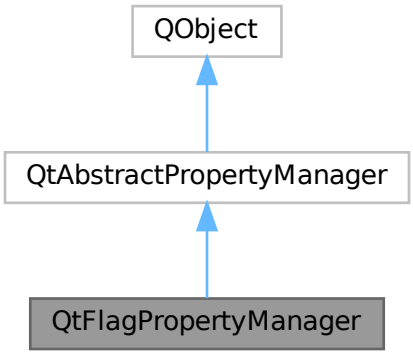
17.724 QtEnumPropertyType Class Reference

The documentation for this class was generated from the following file:

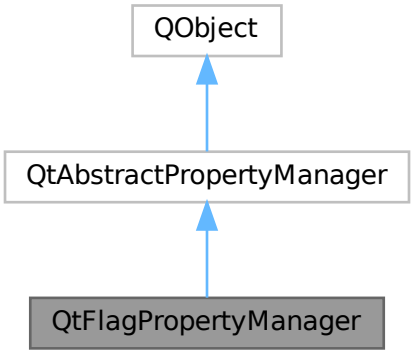
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.cpp

17.725 QtFlagPropertyManager Class Reference

Inheritance diagram for QtFlagPropertyManager:



Collaboration diagram for QtFlagPropertyManager:



Public Slots

- void setValue (QtProperty *property, int val)
- void setFlagNames (QtProperty *property, const QStringList &names)

Signals

- void valueChanged (QtProperty *property, int val)
- void flagNamesChanged (QtProperty *property, const QStringList &names)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtFlagPropertyManager (QObject *parent=0)
- ~QtFlagPropertyManager ()
- QtBoolPropertyManager * subBoolPropertyManager () const
- int value (const QtProperty *property) const
- QStringList flagNames (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.725.1 Constructor & Destructor Documentation

17.725.1.1 QtFlagPropertyManager()

```
QtFlagPropertyManager::QtFlagPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.725.1.2 ~QtFlagPropertyManager()

```
QtFlagPropertyManager::~QtFlagPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.725.2 Member Function Documentation

17.725.2.1 flagNames()

```
QStringList QtFlagPropertyManager::flagNames (
            const QtProperty * property ) const
```

Returns the given *property's* list of flag names.

See also

value(), setFlagNames()

17.725.2.2 flagNamesChanged

```
void QtFlagPropertyManager::flagNamesChanged (
            QtProperty * property,
            const QStringList & names )  [signal]
```

This signal is emitted whenever a property created by this manager changes its flag names, passing a pointer to the *property* and the new *names* as parameters.

See also

setFlagNames()

17.725.2.3 initializeProperty()

```
void QtFlagPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.725.2.4 setFlagNames

```
void QtFlagPropertyManager::setFlagNames (
            QtProperty * property,
            const QStringList & flagNames )  [slot]
```

Sets the given *property's* list of flag names to *flagNames*. The property's current value is reset to 0 indicating the first item of the list.

See also

flagNames(), flagNamesChanged()

17.725.2.5 setValue

```
void QtFlagPropertyManager::setValue (
            QtProperty * property,
            int value )  [slot]
```

Sets the value of the given *property* to *value*. Nested properties are updated automatically.
The specified *value* must be less than the binary combination of the property's flagNames() list size (i.e. less than 2\sup n, where n is the size of the list) and larger than (or equal to) 0.

See also

value(), valueChanged()

17.725.2.6 subBoolPropertyManager()

```
QtBoolPropertyManager * QtFlagPropertyManager::subBoolPropertyManager ( ) const
```

Returns the manager that produces the nested boolean subproperties representing each flag.
In order to provide editing widgets for the subproperties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.725.2.7 uninitializeProperty()

```
void QtFlagPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.725.2.8 value()

```
int QtFlagPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given property is not managed by this manager, this function returns 0.

See also

flagNames(), setValue()

17.725.2.9 valueChanged

```
void QtFlagPropertyManager::valueChanged (
            QtProperty * property,
            int value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.725.2.10 valueText()

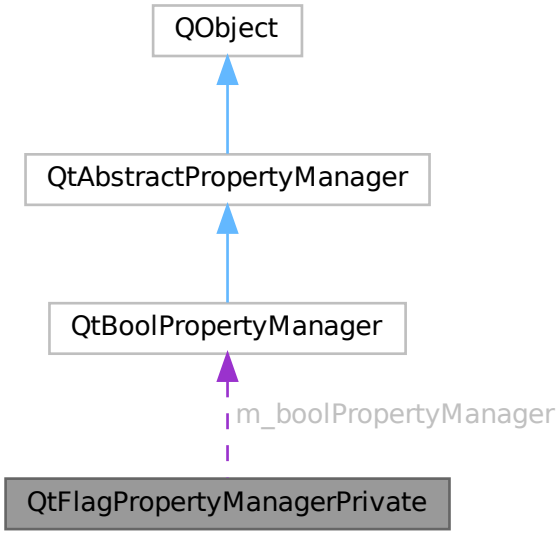
```
QString QtFlagPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.726 QtFlagPropertyManagerPrivate Class Reference

Collaboration diagram for QtFlagPropertyManagerPrivate:



Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Member Functions

- void **slotBoolChanged** (QtProperty *property, bool value)
- void **slotPropertyDestroyed** (QtProperty *property)

Public Attributes

- PropertyValueMap **m_values**
- QtBoolPropertyManager * **m_boolPropertyManager**
- QMap< const QtProperty *, QList< QtProperty * > > **m_propertyToFlags**
- QMap< const QtProperty *, QtProperty * > **m_flagToProperty**

The documentation for this class was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

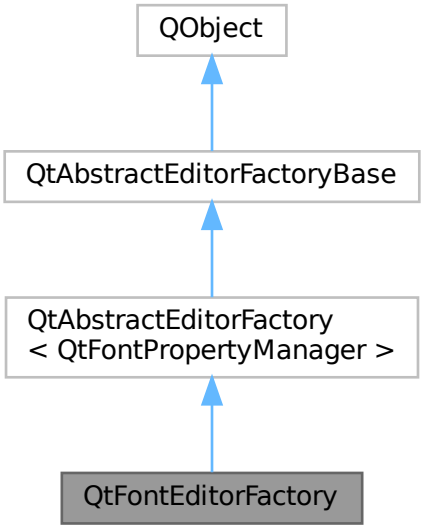
17.727 QtFlagPropertyType Class Reference

The documentation for this class was generated from the following file:

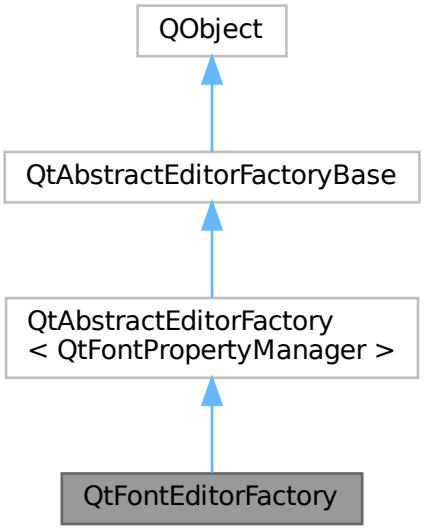
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.cpp

17.728 QtFontEditorFactory Class Reference

Inheritance diagram for QtFontEditorFactory:



Collaboration diagram for QtFontEditorFactory:



Public Member Functions

- QtFontEditorFactory (QObject *parent=0)

- ~QtFontEditorFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtFontPropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtFontPropertyManager *manager)
- void removePropertyManager (QtFontPropertyManager *manager)
- QSet< QtFontPropertyManager * > propertyManagers () const
- QtFontPropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtFontPropertyManager *manager) override
- QWidget * **createEditor** (QtFontPropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtFontPropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtFontPropertyManager >

- virtual QWidget * createEditor (QtFontPropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtFontPropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtFontPropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.728.1 Constructor & Destructor Documentation

17.728.1.1 QtFontEditorFactory()

```
QtFontEditorFactory::QtFontEditorFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.728.1.2 ~QtFontEditorFactory()

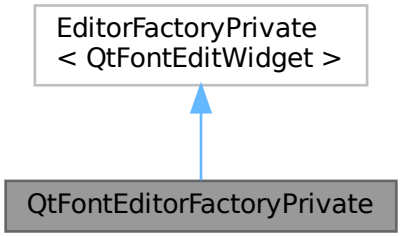
```
QtFontEditorFactory::~QtFontEditorFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

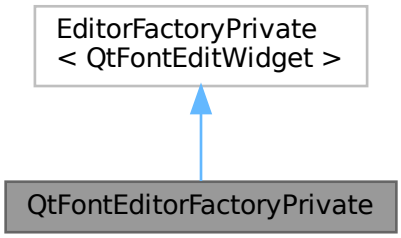
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.729 QtFontEditorFactoryPrivate Class Reference

Inheritance diagram for QtFontEditorFactoryPrivate:



Collaboration diagram for QtFontEditorFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, const QFont &value)
- void **slotSetValue** (const QFont &value)

Public Member Functions inherited from EditorFactoryPrivate< QtFontEditWidget >

- QtFontEditWidget * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QtFontEditWidget *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QtFontEditWidget >

- typedef QList< QtFontEditWidget * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QtFontEditWidget *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QtFontEditWidget >

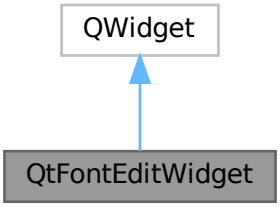
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

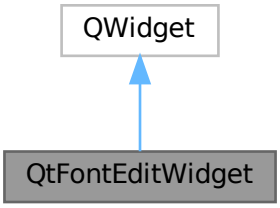
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.730 QtFontEditWidget Class Reference

Inheritance diagram for QtFontEditWidget:



Collaboration diagram for QtFontEditWidget:



Public Slots

- void **setValue** (const QFont &value)

Signals

- void **valueChanged** (const QFont &value)

Public Member Functions

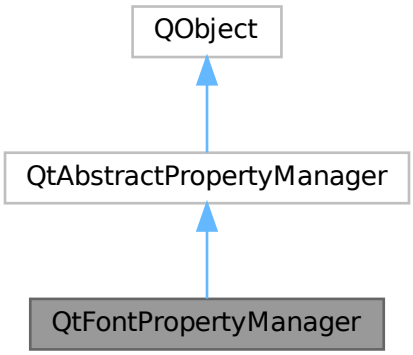
- **QtFontEditWidget** (QWidget ∗parent)
- bool **eventFilter** (QObject ∗obj, QEvent ∗ev) override

The documentation for this class was generated from the following file:

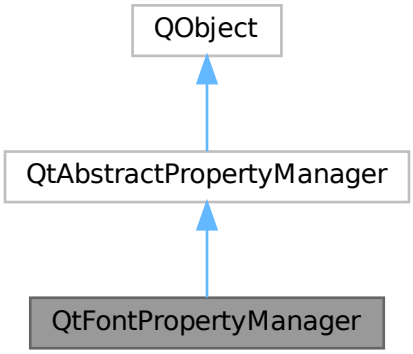
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.731 QtFontPropertyManager Class Reference

Inheritance diagram for QtFontPropertyManager:



Collaboration diagram for QtFontPropertyManager:



Public Slots

- void setValue (QtProperty *property, const QFont &val)

Signals

- void valueChanged (QtProperty *property, const QFont &val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtFontPropertyManager (QObject ∗parent=0)
- ∼QtFontPropertyManager ()
- QtIntPropertyManager ∗ subIntPropertyManager () const
- QtEnumPropertyManager ∗ subEnumPropertyManager () const
- QtBoolPropertyManager ∗ subBoolPropertyManager () const
- QFont value (const QtProperty ∗property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject ∗parent=0)
- ∼QtAbstractPropertyManager ()
- QSet< QtProperty ∗ > properties () const
- void clear () const
- QtProperty ∗ addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty ∗property) const override
- QIcon valueIcon (const QtProperty ∗property) const override
- void initializeProperty (QtProperty ∗property) override
- void uninitializeProperty (QtProperty ∗property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty ∗property) const
- virtual QtProperty ∗ createProperty ()

17.731.1 Constructor & Destructor Documentation

17.731.1.1 QtFontPropertyManager()

```
QtFontPropertyManager::QtFontPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.731.1.2 ∼QtFontPropertyManager()

```
QtFontPropertyManager::~QtFontPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.731.2 Member Function Documentation

17.731.2.1 initializeProperty()

```
void QtFontPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.731.2.2 setValue

```
void QtFontPropertyManager::setValue (
            QtProperty * property,
            const QFont & value )  [slot]
```

Sets the value of the given *property* to *value*. Nested properties are updated automatically.

See also

value(), valueChanged()

17.731.2.3 subBoolPropertyManager()

```
QtBoolPropertyManager * QtFontPropertyManager::subBoolPropertyManager ( ) const
```

Returns the manager that creates the *bold*, *italic*, *underline*, *strikeOut* and *kerning* subproperties.
In order to provide editing widgets for the mentioned properties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.731.2.4 subEnumPropertyManager()

```
QtEnumPropertyManager * QtFontPropertyManager::subEnumPropertyManager ( ) const
```

Returns the manager that create the *family* subproperty.
In order to provide editing widgets for the *family* property in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.731.2.5 subIntPropertyManager()

```
QtIntPropertyManager * QtFontPropertyManager::subIntPropertyManager ( ) const
```

Returns the manager that creates the *pointSize* subproperty.
In order to provide editing widgets for the *pointSize* property in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.731.2.6 uninitializeProperty()

```
void QtFontPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.731.2.7 value()

```
QFont QtFontPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given property is not managed by this manager, this function returns a font object that uses the application's default font.

See also

setValue()

17.731.2.8 valueChanged

```
void QtFontPropertyManager::valueChanged (
            QtProperty * property,
            const QFont & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.731.2.9 valueIcon()

```
QIcon QtFontPropertyManager::valueIcon (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.731.2.10 valueText()

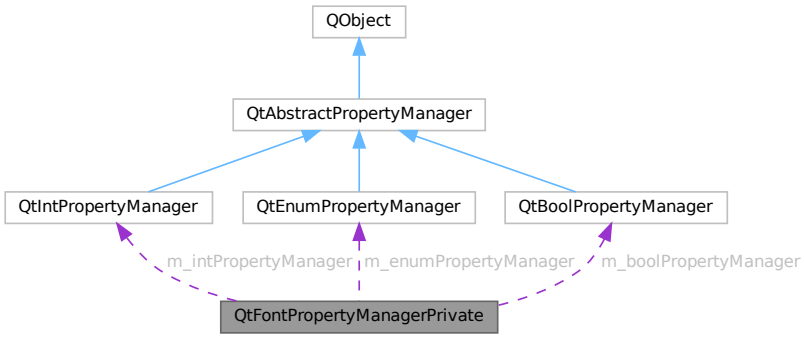
```
QString QtFontPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.732 QtFontPropertyManagerPrivate Class Reference

Collaboration diagram for QtFontPropertyManagerPrivate:



Public Types

- typedef QMap< const QtProperty *, QFont > **PropertyValueMap**

Public Member Functions

- void **slotIntChanged** (QtProperty *property, int value)
- void **slotEnumChanged** (QtProperty *property, int value)
- void **slotBoolChanged** (QtProperty *property, bool value)

- void **slotPropertyDestroyed** (QtProperty *property)
- void **slotFontDatabaseChanged** ()
- void **slotFontDatabaseDelayedChange** ()

Public Attributes

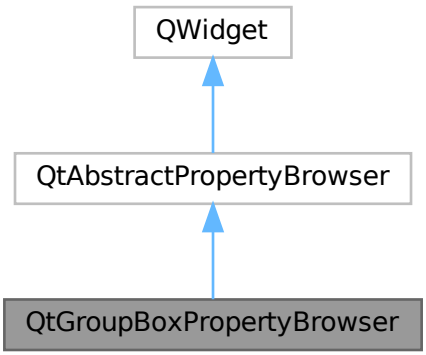
- QStringList **m_familyNames**
- PropertyValueMap **m_values**
- QtIntPropertyManager * **m_intPropertyManager**
- QtEnumPropertyManager * **m_enumPropertyManager**
- QtBoolPropertyManager * **m_boolPropertyManager**
- QMap< const QtProperty *, QtProperty * > **m_propertyToFamily**
- QMap< const QtProperty *, QtProperty * > **m_propertyToPointSize**
- QMap< const QtProperty *, QtProperty * > **m_propertyToBold**
- QMap< const QtProperty *, QtProperty * > **m_propertyToItalic**
- QMap< const QtProperty *, QtProperty * > **m_propertyToUnderline**
- QMap< const QtProperty *, QtProperty * > **m_propertyToStrikeOut**
- QMap< const QtProperty *, QtProperty * > **m_propertyToKerning**
- QMap< const QtProperty *, QtProperty * > **m_familyToProperty**
- QMap< const QtProperty *, QtProperty * > **m_pointSizeToProperty**
- QMap< const QtProperty *, QtProperty * > **m_boldToProperty**
- QMap< const QtProperty *, QtProperty * > **m_italicToProperty**
- QMap< const QtProperty *, QtProperty * > **m_underlineToProperty**
- QMap< const QtProperty *, QtProperty * > **m_strikeOutToProperty**
- QMap< const QtProperty *, QtProperty * > **m_kerningToProperty**
- bool **m_settingValue**
- QTimer * **m_fontDatabaseChangeTimer**

The documentation for this class was generated from the following file:

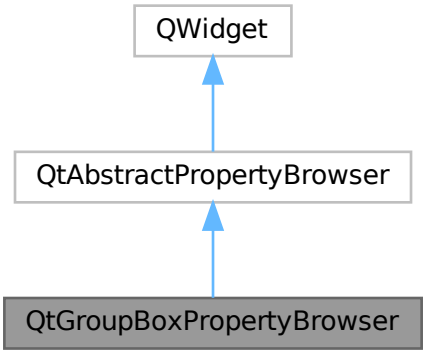
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.733 QtGroupBoxPropertyBrowser Class Reference

Inheritance diagram for QtGroupBoxPropertyBrowser:



Collaboration diagram for QtGroupBoxPropertyBrowser:



Public Member Functions

- QtGroupBoxPropertyBrowser (QWidget ∗parent=0)
- ∼QtGroupBoxPropertyBrowser ()

Public Member Functions inherited from QtAbstractPropertyBrowser

- QtAbstractPropertyBrowser (QWidget ∗parent=0)
- ∼QtAbstractPropertyBrowser ()
- QList< QtProperty ∗ > properties () const
- QList< QtBrowserItem ∗ > items (QtProperty ∗property) const
- QtBrowserItem ∗ topLevelItem (QtProperty ∗property) const
- QList< QtBrowserItem ∗ > topLevelItems () const
- void clear ()
- template<class PropertyManager >
  void setFactoryForManager (PropertyManager ∗manager, QtAbstractEditorFactory< PropertyManager > ∗factory)
- void unsetFactoryForManager (QtAbstractPropertyManager ∗manager)
- QtBrowserItem ∗ currentItem () const
- void setCurrentItem (QtBrowserItem ∗)

Protected Member Functions

- void itemInserted (QtBrowserItem ∗item, QtBrowserItem ∗afterItem) override
- void itemRemoved (QtBrowserItem ∗item) override
- void itemChanged (QtBrowserItem ∗item) override

Protected Member Functions inherited from QtAbstractPropertyBrowser

- virtual QWidget ∗ createEditor (QtProperty ∗property, QWidget ∗parent)

Additional Inherited Members

Public Slots inherited from QtAbstractPropertyBrowser

- QtBrowserItem ∗ addProperty (QtProperty ∗property)
- QtBrowserItem ∗ insertProperty (QtProperty ∗property, QtProperty ∗afterProperty)
- void removeProperty (QtProperty ∗property)

Signals inherited from QtAbstractPropertyBrowser

- void currentItemChanged (QtBrowserItem *)

17.733.1 Constructor & Destructor Documentation

17.733.1.1 QtGroupBoxPropertyBrowser()

```
QtGroupBoxPropertyBrowser::QtGroupBoxPropertyBrowser (
            QWidget * parent = 0 )
```

Creates a property browser with the given *parent*.

17.733.1.2 ~QtGroupBoxPropertyBrowser()

```
QtGroupBoxPropertyBrowser::~QtGroupBoxPropertyBrowser ( )
```

Destroys this property browser.
Note that the properties that were inserted into this browser are *not* destroyed since they may still be used in other browsers. The properties are owned by the manager that created them.

See also

QtProperty, QtAbstractPropertyManager

17.733.2 Member Function Documentation

17.733.2.1 itemChanged()

```
void QtGroupBoxPropertyBrowser::itemChanged (
            QtBrowserItem * item )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyBrowser.

17.733.2.2 itemInserted()

```
void QtGroupBoxPropertyBrowser::itemInserted (
            QtBrowserItem * item,
            QtBrowserItem * afterItem )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyBrowser.

17.733.2.3 itemRemoved()

```
void QtGroupBoxPropertyBrowser::itemRemoved (
            QtBrowserItem * item )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyBrowser.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtgroupboxpropertybrowser.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtgroupboxpropertybrowser.cpp

17.734 QtGroupBoxPropertyBrowserPrivate Class Reference

Classes

- struct WidgetItem

Public Member Functions

- void **init** (QWidget *parent)
- void **propertyInserted** (QtBrowserItem *index, QtBrowserItem *afterIndex)

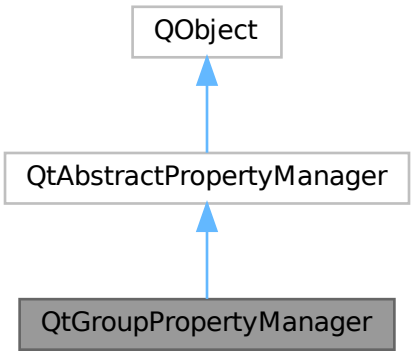
- void **propertyRemoved** (QtBrowserItem ∗index)
- void **propertyChanged** (QtBrowserItem ∗index)
- QWidget ∗ **createEditor** (QtProperty ∗property, QWidget ∗parent) const
- void **slotEditorDestroyed** ()
- void **slotUpdate** ()

The documentation for this class was generated from the following file:

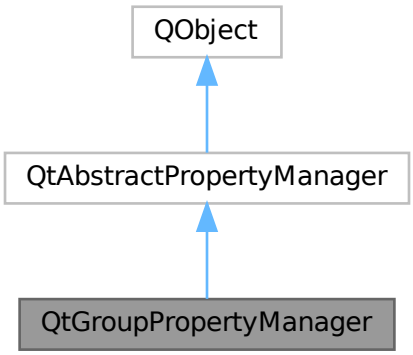
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtgroupboxpropertybrowser.cpp

17.735 QtGroupPropertyManager Class Reference

Inheritance diagram for QtGroupPropertyManager:



Collaboration diagram for QtGroupPropertyManager:



Public Member Functions

- QtGroupPropertyManager (QObject ∗parent=0)
- ∼QtGroupPropertyManager ()

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- bool hasValue (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QString valueText (const QtProperty *property) const
- virtual QtProperty * createProperty ()

Additional Inherited Members

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

17.735.1 Constructor & Destructor Documentation

17.735.1.1 QtGroupPropertyManager()

```
QtGroupPropertyManager::QtGroupPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.735.1.2 ~QtGroupPropertyManager()

```
QtGroupPropertyManager::~QtGroupPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.735.2 Member Function Documentation

17.735.2.1 hasValue()

```
bool QtGroupPropertyManager::hasValue (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp

Reimplemented from QtAbstractPropertyManager.

17.735.2.2 initializeProperty()

```
void QtGroupPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp

Implements QtAbstractPropertyManager.

17.735.2.3 uninitializeProperty()

```
void QtGroupPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

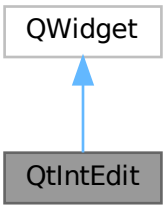
17.736 QtGroupPropertyType Class Reference

The documentation for this class was generated from the following file:

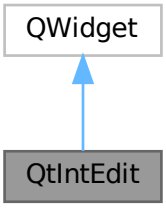
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.cpp

17.737 QtIntEdit Class Reference

Inheritance diagram for QtIntEdit:



Collaboration diagram for QtIntEdit:



Signals

- void **valueChanged** (int)

Public Member Functions

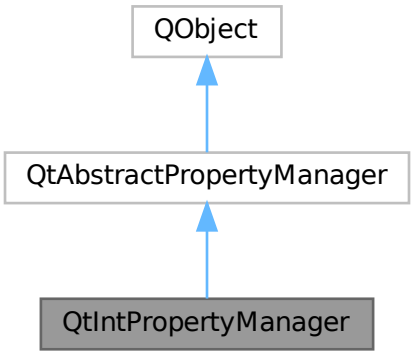
- **QtIntEdit** (QWidget ∗parent=Q_NULLPTR)
- int **value** ()
- void **setValue** (int)
- void **setInitialValue** (int)
- void **setSingleStep** (int)
- void **setRange** (int, int)
- void **setKeyboardTracking** (bool)

The documentation for this class was generated from the following files:

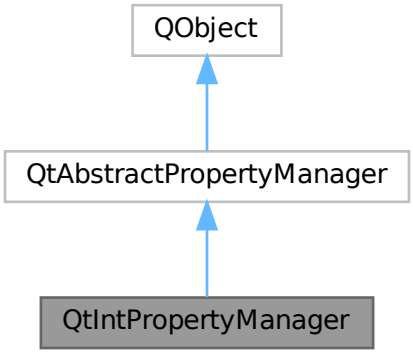
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils_p.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils.cpp

17.738 QtIntPropertyManager Class Reference

Inheritance diagram for QtIntPropertyManager:



Collaboration diagram for QtIntPropertyManager:



Public Slots

- void setValue (QtProperty *property, int val)
- void **setValueOnly** (QtProperty *property, int val)
- void setMinimum (QtProperty *property, int minVal)
- void setMaximum (QtProperty *property, int maxVal)
- void setRange (QtProperty *property, int minVal, int maxVal)
- void setSingleStep (QtProperty *property, int step)

Signals

- void valueChanged (QtProperty *property, int val)
- void rangeChanged (QtProperty *property, int minVal, int maxVal)
- void singleStepChanged (QtProperty *property, int step)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtIntPropertyManager (QObject *parent=0)
- ~QtIntPropertyManager ()
- int value (const QtProperty *property) const
- int **initialValue** (const QtProperty *property) const
- int minimum (const QtProperty *property) const
- int maximum (const QtProperty *property) const
- int singleStep (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.738.1 Constructor & Destructor Documentation

17.738.1.1 QtIntPropertyManager()

```
QtIntPropertyManager::QtIntPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.738.1.2 ~QtIntPropertyManager()

```
QtIntPropertyManager::~QtIntPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.738.2 Member Function Documentation

17.738.2.1 initializeProperty()

```
void QtIntPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.738.2.2 maximum()

```
int QtIntPropertyManager::maximum (
            const QtProperty * property ) const
```

Returns the given *property's* maximum value.

See also

setMaximum(), minimum(), setRange()

17.738.2.3 minimum()

```
int QtIntPropertyManager::minimum (
            const QtProperty * property ) const
```

Returns the given *property's* minimum value.

See also

setMinimum(), maximum(), setRange()

17.738.2.4 rangeChanged

```
void QtIntPropertyManager::rangeChanged (
            QtProperty * property,
            int minimum,
            int maximum )  [signal]
```

This signal is emitted whenever a property created by this manager changes its range of valid values, passing a pointer to the *property* and the new *minimum* and *maximum* values.

See also

setRange()

17.738.2.5 setMaximum

```
void QtIntPropertyManager::setMaximum (
            QtProperty * property,
            int maxVal )  [slot]
```

Sets the maximum value for the given *property* to *maxVal*.
When setting maximum value, the minimum and current values are adjusted if necessary (ensuring that the range remains valid and that the current value is within the range).

See also

maximum(), setRange(), rangeChanged()

17.738.2.6 setMinimum

```
void QtIntPropertyManager::setMinimum (
            QtProperty * property,
            int minVal )  [slot]
```

Sets the minimum value for the given *property* to *minVal*.
When setting the minimum value, the maximum and current values are adjusted if necessary (ensuring that the range remains valid and that the current value is within the range).

See also

minimum(), setRange(), rangeChanged()

17.738.2.7 setRange

```
void QtIntPropertyManager::setRange (
            QtProperty * property,
            int minimum,
            int maximum )  [slot]
```

Sets the range of valid values.
This is a convenience function defining the range of valid values in one go; setting the *minimum* and *maximum* values for the given *property* with a single function call.
When setting a new range, the current value is adjusted if necessary (ensuring that the value remains within range).

See also

setMinimum(), setMaximum(), rangeChanged()

17.738.2.8 setSingleStep

```
void QtIntPropertyManager::setSingleStep (
            QtProperty * property,
            int step )  [slot]
```

Sets the step value for the given *property* to *step*.
The step is typically used to increment or decrement a property value while pressing an arrow key.

See also

singleStep()

17.738.2.9 setValue

```
void QtIntPropertyManager::setValue (
            QtProperty * property,
            int value )  [slot]
```

Sets the value of the given *property* to *value*.
If the specified *value* is not valid according to the given *property's* range, the *value* is adjusted to the nearest valid value within the range.

See also

value(), setRange(), valueChanged()

17.738.2.10 singleStep()

```
int QtIntPropertyManager::singleStep (
            const QtProperty * property ) const
```

Returns the given *property's* step value.
The step is typically used to increment or decrement a property value while pressing an arrow key.

See also

setSingleStep()

17.738.2.11 singleStepChanged

```
void QtIntPropertyManager::singleStepChanged (
            QtProperty * property,
            int step )  [signal]
```

This signal is emitted whenever a property created by this manager changes its single step property, passing a pointer to the *property* and the new *step* value

See also

setSingleStep()

17.738.2.12 uninitializeProperty()

```
void QtIntPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.738.2.13 value()

```
int QtIntPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given property is not managed by this manager, this function returns 0.

See also

setValue()

17.738.2.14 valueChanged

```
void QtIntPropertyManager::valueChanged (
            QtProperty * property,
            int value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.738.2.15 valueText()

```
QString QtIntPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.739 QtIntPropertyManagerPrivate Class Reference

Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Attributes

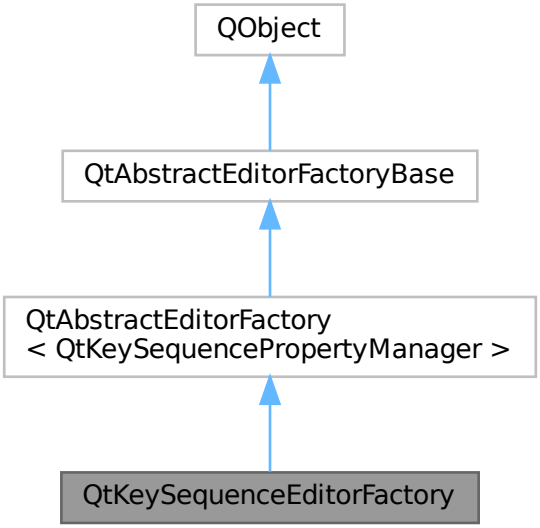
- PropertyValueMap **m_values**

The documentation for this class was generated from the following file:

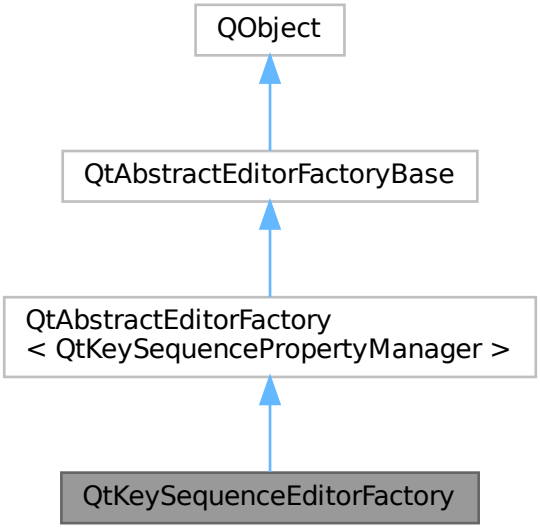
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.740 QtKeySequenceEditorFactory Class Reference

Inheritance diagram for QtKeySequenceEditorFactory:



Collaboration diagram for QtKeySequenceEditorFactory:



Public Member Functions

- QtKeySequenceEditorFactory (QObject ∗parent=0)
- ∼QtKeySequenceEditorFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtKeySequencePropertyManager >

- QtAbstractEditorFactory (QObject ∗parent)
- QWidget ∗ createEditor (QtProperty ∗property, QWidget ∗parent) override
- void addPropertyManager (QtKeySequencePropertyManager ∗manager)
- void removePropertyManager (QtKeySequencePropertyManager ∗manager)
- QSet< QtKeySequencePropertyManager ∗ > propertyManagers () const
- QtKeySequencePropertyManager ∗ propertyManager (QtProperty ∗property) const

Protected Member Functions

- void **connectPropertyManager** (QtKeySequencePropertyManager ∗manager) override
- QWidget ∗ **createEditor** (QtKeySequencePropertyManager ∗manager, QtProperty ∗property, QWidget ∗parent) override
- void **disconnectPropertyManager** (QtKeySequencePropertyManager ∗manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtKeySequencePropertyManager >

- virtual QWidget ∗ createEditor (QtKeySequencePropertyManager ∗manager, QtProperty ∗property, QWidget ∗parent)=0
- virtual void connectPropertyManager (QtKeySequencePropertyManager ∗manager)=0
- virtual void disconnectPropertyManager (QtKeySequencePropertyManager ∗manager)=0
- void managerDestroyed (QObject ∗manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject ∗parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.740.1 Constructor & Destructor Documentation

17.740.1.1 QtKeySequenceEditorFactory()

```
QtKeySequenceEditorFactory::QtKeySequenceEditorFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.740.1.2 ∼QtKeySequenceEditorFactory()

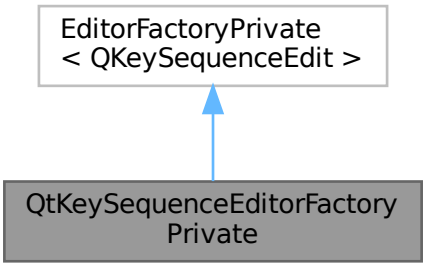
```
QtKeySequenceEditorFactory::~QtKeySequenceEditorFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

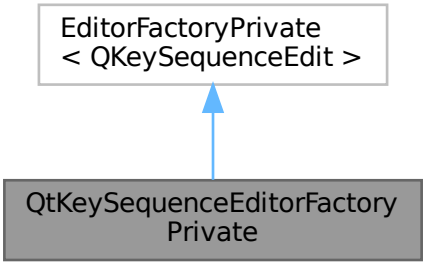
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.741 QtKeySequenceEditorFactoryPrivate Class Reference

Inheritance diagram for QtKeySequenceEditorFactoryPrivate:



Collaboration diagram for QtKeySequenceEditorFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, const QKeySequence &value)
- void **slotSetValue** (const QKeySequence &value)

Public Member Functions inherited from EditorFactoryPrivate< QKeySequenceEdit >

- QKeySequenceEdit * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QKeySequenceEdit *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QKeySequenceEdit >

- typedef QList< QKeySequenceEdit * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QKeySequenceEdit *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QKeySequenceEdit >

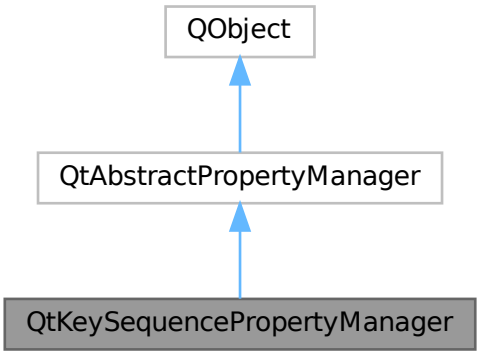
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

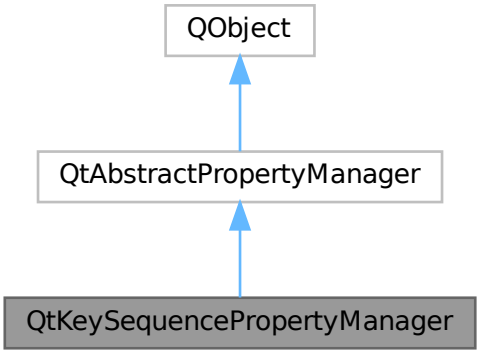
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.742 QtKeySequencePropertyManager Class Reference

Inheritance diagram for QtKeySequencePropertyManager:



Collaboration diagram for QtKeySequencePropertyManager:



Public Slots

- void setValue (QtProperty *property, const QKeySequence &val)

Signals

- void valueChanged (QtProperty *property, const QKeySequence &val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtKeySequencePropertyManager (QObject ∗parent=0)
- ∼QtKeySequencePropertyManager ()
- QKeySequence value (const QtProperty ∗property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject ∗parent=0)
- ∼QtAbstractPropertyManager ()
- QSet< QtProperty ∗ > properties () const
- void clear () const
- QtProperty ∗ addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty ∗property) const override
- void initializeProperty (QtProperty ∗property) override
- void uninitializeProperty (QtProperty ∗property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty ∗property) const
- virtual QIcon valueIcon (const QtProperty ∗property) const
- virtual QtProperty ∗ createProperty ()

17.742.1 Constructor & Destructor Documentation

17.742.1.1 QtKeySequencePropertyManager()

```
QtKeySequencePropertyManager::QtKeySequencePropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.742.1.2 ∼QtKeySequencePropertyManager()

```
QtKeySequencePropertyManager::~QtKeySequencePropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.742.2 Member Function Documentation

17.742.2.1 initializeProperty()

```
void QtKeySequencePropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.742.2.2 setValue

```
void QtKeySequencePropertyManager::setValue (
            QtProperty * property,
            const QKeySequence & value )  [slot]
```

Sets the value of the given *property* to *value*.

See also

value(), valueChanged()

17.742.2.3 uninitializeProperty()

```
void QtKeySequencePropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.742.2.4 value()

```
QKeySequence QtKeySequencePropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by this manager, this function returns an empty QKeySequence object.

See also

setValue()

17.742.2.5 valueChanged

```
void QtKeySequencePropertyManager::valueChanged (
            QtProperty * property,
            const QKeySequence & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

17.742.2.6 valueText()

```
QString QtKeySequencePropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.743 QtKeySequencePropertyManagerPrivate Class Reference

Public Types

- typedef QMap< const QtProperty *, QKeySequence > **PropertyValueMap**

Public Attributes

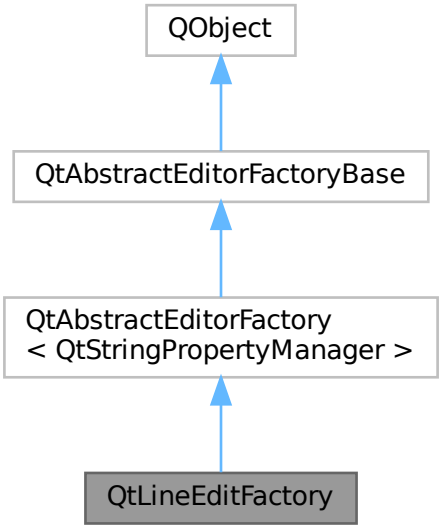
- QString **m_format**
- PropertyValueMap **m_values**

The documentation for this class was generated from the following file:

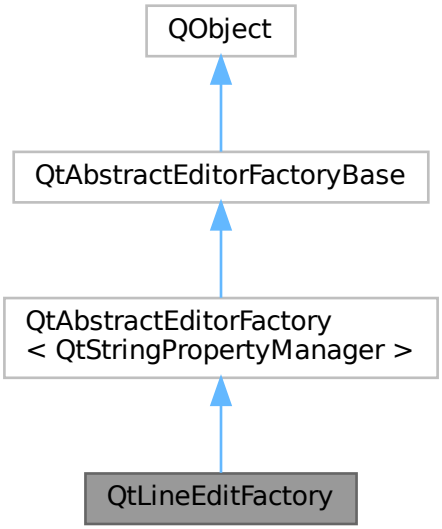
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.744 QtLineEditFactory Class Reference

Inheritance diagram for QtLineEditFactory:



Collaboration diagram for QtLineEditFactory:



Public Member Functions

- QtLineEditFactory (QObject *parent=0)

- ~QtLineEditFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtStringPropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtStringPropertyManager *manager)
- void removePropertyManager (QtStringPropertyManager *manager)
- QSet< QtStringPropertyManager * > propertyManagers () const
- QtStringPropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtStringPropertyManager *manager) override
- QWidget * **createEditor** (QtStringPropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtStringPropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtStringPropertyManager >

- virtual QWidget * createEditor (QtStringPropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtStringPropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtStringPropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.744.1 Constructor & Destructor Documentation

17.744.1.1 QtLineEditFactory()

```
QtLineEditFactory::QtLineEditFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.744.1.2 ~QtLineEditFactory()

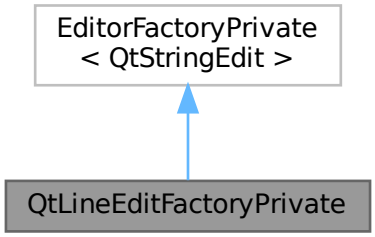
```
QtLineEditFactory::~QtLineEditFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

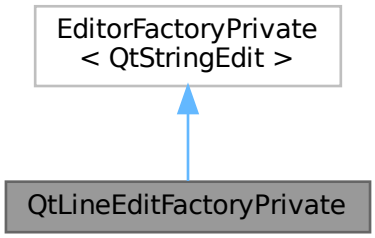
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.745 QtLineEditFactoryPrivate Class Reference

Inheritance diagram for QtLineEditFactoryPrivate:



Collaboration diagram for QtLineEditFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, const QString &value)
- void **slotRegExpChanged** (QtProperty *property, const QRegularExpression &regExp)
- void **slotSetValue** (const QString &value)

Public Member Functions inherited from EditorFactoryPrivate< QtStringEdit >

- QtStringEdit * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QtStringEdit *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QtStringEdit >

- typedef QList< QtStringEdit * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QtStringEdit *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QtStringEdit >

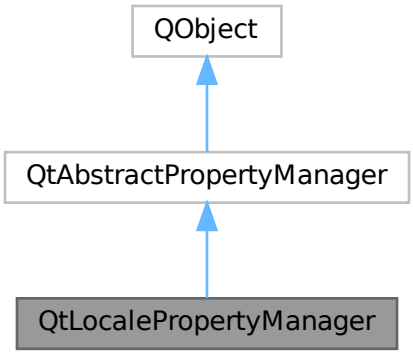
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

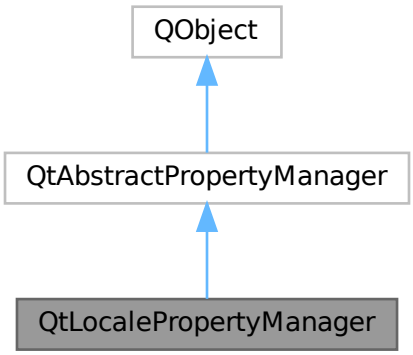
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.746 QtLocalePropertyManager Class Reference

Inheritance diagram for QtLocalePropertyManager:



Collaboration diagram for QtLocalePropertyManager:



Public Slots

- void setValue (QtProperty *property, const QLocale &val)

Signals

- void valueChanged (QtProperty *property, const QLocale &val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtLocalePropertyManager (QObject *parent=0)
- ~QtLocalePropertyManager ()
- QtEnumPropertyManager * subEnumPropertyManager () const
- QLocale value (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.746.1 Constructor & Destructor Documentation

17.746.1.1 QtLocalePropertyManager()

```
QtLocalePropertyManager::QtLocalePropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.746.1.2 ~QtLocalePropertyManager()

```
QtLocalePropertyManager::~QtLocalePropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.746.2 Member Function Documentation

17.746.2.1 initializeProperty()

```
void QtLocalePropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp

Implements QtAbstractPropertyManager.

17.746.2.2 setValue

```
void QtLocalePropertyManager::setValue (
            QtProperty * property,
            const QLocale & value )  [slot]
```

Sets the value of the given *property* to *value*. Nested properties are updated automatically.

See also

value(), valueChanged()

17.746.2.3 subEnumPropertyManager()

```
QtEnumPropertyManager * QtLocalePropertyManager::subEnumPropertyManager ( ) const
```

Returns the manager that creates the nested *language* and *country* subproperties.
In order to provide editing widgets for the mentioned subproperties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.746.2.4 uninitializeProperty()

```
void QtLocalePropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.746.2.5 value()

```
QLocale QtLocalePropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given property is not managed by this manager, this function returns the default locale.

See also

setValue()

17.746.2.6 valueChanged

```
void QtLocalePropertyManager::valueChanged (
            QtProperty * property,
            const QLocale & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.746.2.7 valueText()

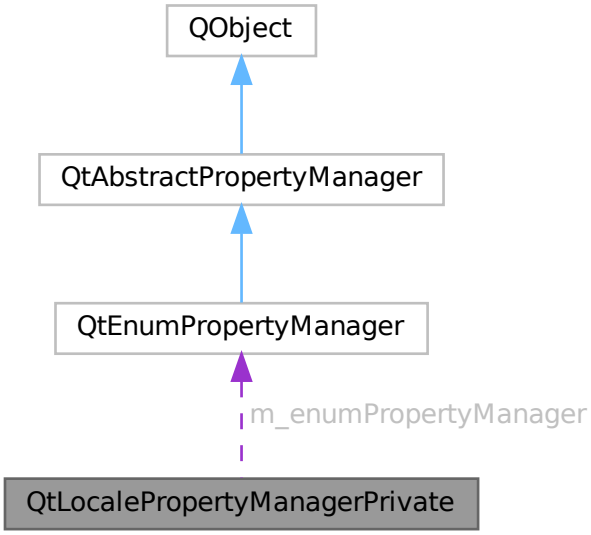
```
QString QtLocalePropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.747 QtLocalePropertyManagerPrivate Class Reference

Collaboration diagram for QtLocalePropertyManagerPrivate:



Public Types

- typedef QMap< const QtProperty *, QLocale > **PropertyValueMap**

Public Member Functions

- void **slotEnumChanged** (QtProperty *property, int value)
- void **slotPropertyDestroyed** (QtProperty *property)

Public Attributes

- PropertyValueMap **m_values**
- QtEnumPropertyManager * **m_enumPropertyManager**
- QMap< const QtProperty *, QtProperty * > **m_propertyToLanguage**
- QMap< const QtProperty *, QtProperty * > **m_propertyToCountry**
- QMap< const QtProperty *, QtProperty * > **m_languageToProperty**
- QMap< const QtProperty *, QtProperty * > **m_countryToProperty**

The documentation for this class was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.748 QtMetaEnumProvider Class Reference

Public Member Functions

- QStringList **policyEnumNames** () const
- QStringList **languageEnumNames** () const
- QStringList **countryEnumNames** (QLocale::Language language) const

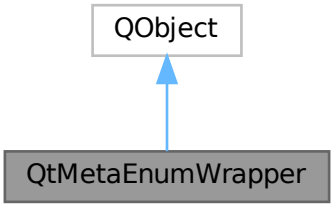
- QSizePolicy::Policy **indexToSizePolicy** (int index) const
- int **sizePolicyToIndex** (QSizePolicy::Policy policy) const
- void **indexToLocale** (int languageIndex, int countryIndex, QLocale::Language ∗language, QLocale::Country ∗country) const
- void **localeToIndex** (QLocale::Language language, QLocale::Country country, int ∗languageIndex, int ∗countryIndex) const

The documentation for this class was generated from the following file:

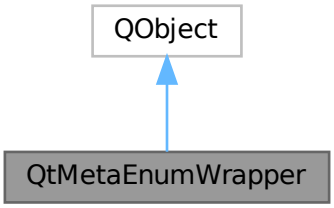
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.749 QtMetaEnumWrapper Class Reference

Inheritance diagram for QtMetaEnumWrapper:



Collaboration diagram for QtMetaEnumWrapper:



Public Member Functions

- QSizePolicy::Policy **policy** () const

Properties

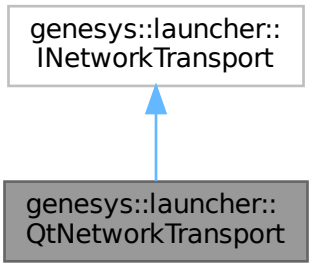
- QSizePolicy::Policy **policy**

The documentation for this class was generated from the following file:

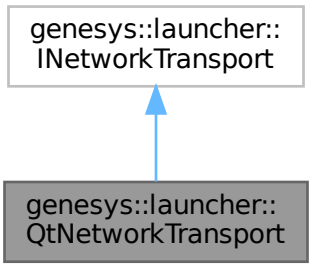
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.750 genesys::launcher::QtNetworkTransport Class Reference

Inheritance diagram for genesys::launcher::QtNetworkTransport:



Collaboration diagram for genesys::launcher::QtNetworkTransport:



Public Member Functions

- NetworkResponse download (const QUrl &url, QIODevice &destination, const NetworkRequestOptions &options, ProgressCallback progress={}, CancellationCallback cancelled={}) override

Public Member Functions inherited from genesys::launcher::INetworkTransport

- virtual std::pair< NetworkResponse, QByteArray > **get** (const QUrl &url, const NetworkRequestOptions &options, ProgressCallback progress={}, CancellationCallback cancelled={})

17.750.1 Member Function Documentation

17.750.1.1 download()

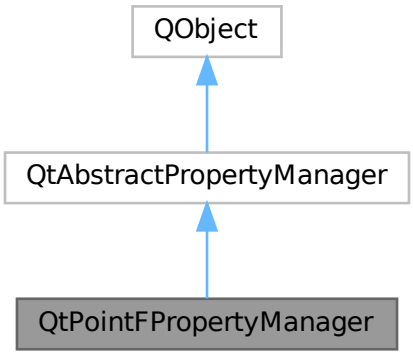
```
NetworkResponse genesys::launcher::QtNetworkTransport::download (
            const QUrl & url,
            QIODevice & destination,
            const NetworkRequestOptions & options,
            ProgressCallback progress = {},
            CancellationCallback cancelled = {} )  [override], [virtual]
```

Implements genesys::launcher::INetworkTransport.
The documentation for this class was generated from the following files:

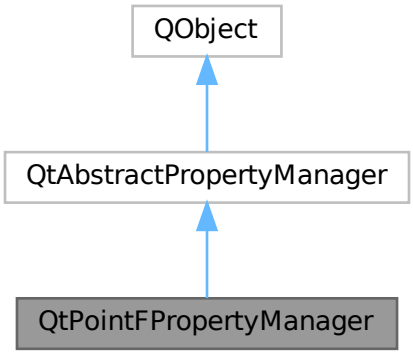
- applications/launcher/NetworkTransport.h
- applications/launcher/NetworkTransport.cpp

17.751 QtPointFPropertyManager Class Reference

Inheritance diagram for QtPointFPropertyManager:



Collaboration diagram for QtPointFPropertyManager:



Public Slots

- void setValue (QtProperty *property, const QPointF &val)
- void setDecimals (QtProperty *property, int prec)

Signals

- void valueChanged (QtProperty *property, const QPointF &val)

- void decimalsChanged (QtProperty *property, int prec)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtPointFPropertyManager (QObject *parent=0)
- ~QtPointFPropertyManager ()
- QtDoublePropertyManager * subDoublePropertyManager () const
- QPointF value (const QtProperty *property) const
- int decimals (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.751.1 Constructor & Destructor Documentation

17.751.1.1 QtPointFPropertyManager()

```
QtPointFPropertyManager::QtPointFPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.751.1.2 ~QtPointFPropertyManager()

```
QtPointFPropertyManager::~QtPointFPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.751.2 Member Function Documentation

17.751.2.1 decimals()

```
int QtPointFPropertyManager::decimals (
            const QtProperty * property ) const
```

Returns the given *property's* precision, in decimals.

See also

setDecimals()

17.751.2.2 decimalsChanged

```
void QtPointFPropertyManager::decimalsChanged (
            QtProperty * property,
            int prec )  [signal]
```

This signal is emitted whenever a property created by this manager changes its precision of value, passing a pointer to the *property* and the new *prec* value

See also

setDecimals()

17.751.2.3 initializeProperty()

```
void QtPointFPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.751.2.4 setDecimals

```
void QtPointFPropertyManager::setDecimals (
            QtProperty * property,
            int prec )  [slot]
```

Sets the precision of the given *property* to *prec*.
The valid decimal range is 0-13. The default is 2.

See also

decimals()

17.751.2.5 setValue

```
void QtPointFPropertyManager::setValue (
            QtProperty * property,
            const QPointF & value )  [slot]
```

Sets the value of the given *property* to *value*. Nested properties are updated automatically.

See also

value(), valueChanged()

17.751.2.6 subDoublePropertyManager()

```
QtDoublePropertyManager * QtPointFPropertyManager::subDoublePropertyManager ( ) const
```

Returns the manager that creates the nested *x* and *y* subproperties.
In order to provide editing widgets for the subproperties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.751.2.7 uninitializeProperty()

```
void QtPointFPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.751.2.8 value()

```
QPointF QtPointFPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by this manager, this function returns a point with coordinates (0, 0).

See also

setValue()

17.751.2.9 valueChanged

```
void QtPointFPropertyManager::valueChanged (
            QtProperty * property,
            const QPointF & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.751.2.10 valueText()

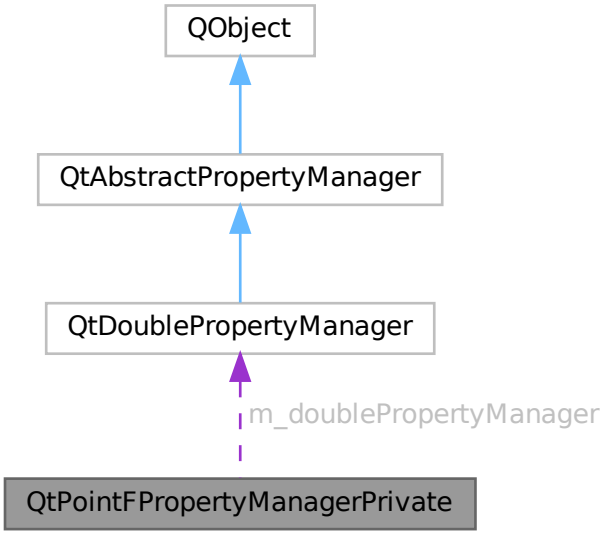
```
QString QtPointFPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.752 QtPointFPropertyManagerPrivate Class Reference

Collaboration diagram for QtPointFPropertyManagerPrivate:



Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Member Functions

- void **slotDoubleChanged** (QtProperty *property, double value)
- void **slotPropertyDestroyed** (QtProperty *property)

Public Attributes

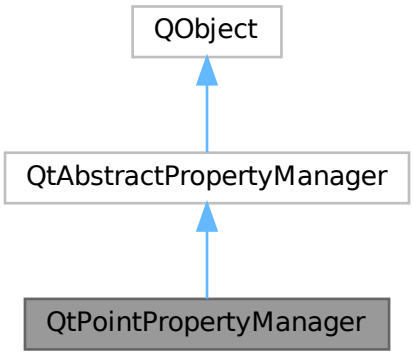
- PropertyValueMap **m_values**
- QtDoublePropertyManager * **m_doublePropertyManager**
- QMap< const QtProperty *, QtProperty * > **m_propertyToX**
- QMap< const QtProperty *, QtProperty * > **m_propertyToY**
- QMap< const QtProperty *, QtProperty * > **m_xToProperty**
- QMap< const QtProperty *, QtProperty * > **m_yToProperty**

The documentation for this class was generated from the following file:

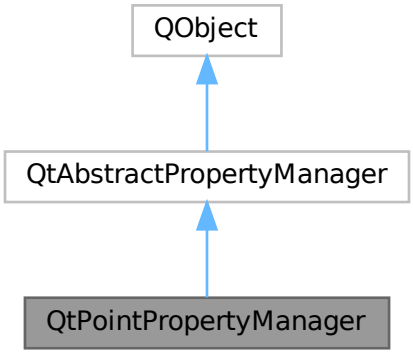
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.753 QtPointPropertyManager Class Reference

Inheritance diagram for QtPointPropertyManager:



Collaboration diagram for QtPointPropertyManager:



Public Slots

- void setValue (QtProperty *property, const QPoint &val)

Signals

- void valueChanged (QtProperty *property, const QPoint &val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtPointPropertyManager (QObject ∗parent=0)
- ∼QtPointPropertyManager ()
- QtIntPropertyManager ∗ subIntPropertyManager () const
- QPoint value (const QtProperty ∗property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject ∗parent=0)
- ∼QtAbstractPropertyManager ()
- QSet< QtProperty ∗ > properties () const
- void clear () const
- QtProperty ∗ addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty ∗property) const override
- void initializeProperty (QtProperty ∗property) override
- void uninitializeProperty (QtProperty ∗property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty ∗property) const
- virtual QIcon valueIcon (const QtProperty ∗property) const
- virtual QtProperty ∗ createProperty ()

17.753.1 Constructor & Destructor Documentation

17.753.1.1 QtPointPropertyManager()

```
QtPointPropertyManager::QtPointPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.753.1.2 ∼QtPointPropertyManager()

```
QtPointPropertyManager::~QtPointPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.753.2 Member Function Documentation

17.753.2.1 initializeProperty()

```
void QtPointPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.753.2.2 setValue

```
void QtPointPropertyManager::setValue (
            QtProperty * property,
            const QPoint & value )  [slot]
```

Sets the value of the given *property* to *value*. Nested properties are updated automatically.

See also

value(), valueChanged()

17.753.2.3 subIntPropertyManager()

```
QtIntPropertyManager * QtPointPropertyManager::subIntPropertyManager ( ) const
```

Returns the manager that creates the nested *x* and *y* subproperties.
In order to provide editing widgets for the subproperties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.753.2.4 uninitializeProperty()

```
void QtPointPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.753.2.5 value()

```
QPoint QtPointPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by this manager, this function returns a point with coordinates (0, 0).

See also

setValue()

17.753.2.6 valueChanged

```
void QtPointPropertyManager::valueChanged (
            QtProperty * property,
            const QPoint & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.753.2.7 valueText()

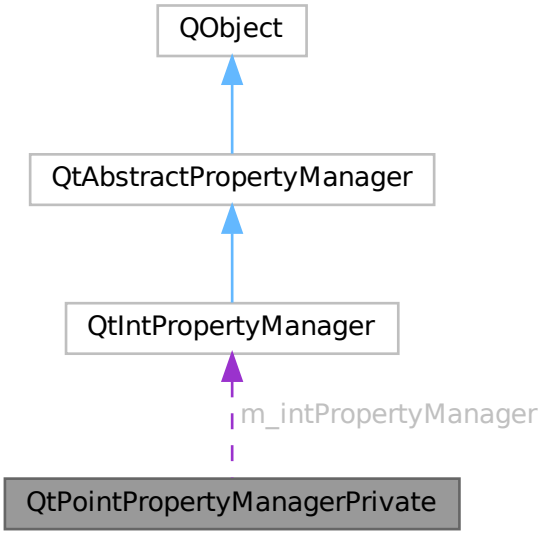
```
QString QtPointPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.754 QtPointPropertyManagerPrivate Class Reference

Collaboration diagram for QtPointPropertyManagerPrivate:



Public Types

- typedef QMap< const QtProperty *, QPoint > **PropertyValueMap**

Public Member Functions

- void **slotIntChanged** (QtProperty *property, int value)
- void **slotPropertyDestroyed** (QtProperty *property)

Public Attributes

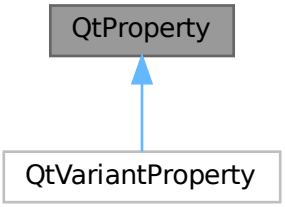
- PropertyValueMap **m_values**
- QtIntPropertyManager * **m_intPropertyManager**
- QMap< const QtProperty *, QtProperty * > **m_propertyToX**
- QMap< const QtProperty *, QtProperty * > **m_propertyToY**
- QMap< const QtProperty *, QtProperty * > **m_xToProperty**
- QMap< const QtProperty *, QtProperty * > **m_yToProperty**

The documentation for this class was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.755 QtProperty Class Reference

Inheritance diagram for QtProperty:



Public Member Functions

- virtual ~QtProperty ()
- QList< QtProperty * > subProperties () const
- QtAbstractPropertyManager * propertyManager () const
- QString **toolTip** () const
- QString valueToolTip () const
- QString descriptionToolTip () const
- QString statusTip () const
- QString whatsThis () const
- QString propertyName () const
- bool isEnabled () const
- bool isModified () const
- bool hasValue () const
- QIcon valueIcon () const
- QString valueText () const
- void **setToolTip** (const QString &text)
- void setValueToolTip (const QString &text)
- void setDescriptionToolTip (const QString &text)
- void setStatusTip (const QString &text)
- void setWhatsThis (const QString &text)
- void setPropertyName (const QString &text)
- void setEnabled (bool enable)
- void setModified (bool modified)
- void addSubProperty (QtProperty *property)
- void insertSubProperty (QtProperty *property, QtProperty *afterProperty)
- void removeSubProperty (QtProperty *property)

Protected Member Functions

- QtProperty (QtAbstractPropertyManager *manager)
- void **propertyChanged** ()

Friends

- class **QtAbstractPropertyManager**

17.755.1 Constructor & Destructor Documentation

17.755.1.1 ~QtProperty()

```
QtProperty::~QtProperty ( )  [virtual]
```

Destroys this property.
Note that subproperties are detached but not destroyed, i.e. they can still be used in another context.

See also

QtAbstractPropertyManager::clear()

17.755.1.2 QtProperty()

```
QtProperty::QtProperty (
            QtAbstractPropertyManager * manager )  [explicit], [protected]
```

Creates a property with the given *manager*.
This constructor is only useful when creating a custom QtProperty subclass (e.g. QtVariantProperty). To create a regular QtProperty object, use the QtAbstractPropertyManager::addProperty() function instead.

See also

QtAbstractPropertyManager::addProperty()

17.755.2 Member Function Documentation

17.755.2.1 addSubProperty()

```
void QtProperty::addSubProperty (
            QtProperty * property )
```

Appends the given *property* to this property's subproperties.
If the given *property* already is added, this function does nothing.

See also

insertSubProperty(), removeSubProperty()

17.755.2.2 descriptionToolTip()

```
QString QtProperty::descriptionToolTip ( ) const
```

Returns the property description's tool tip.
This is suitable for tool tips over the description (label).

Since

5.6

See also

setDescriptionToolTip()

17.755.2.3 hasValue()

```
bool QtProperty::hasValue ( ) const
```

Returns whether the property has a value.

See also

QtAbstractPropertyManager::hasValue()

17.755.2.4 insertSubProperty()

```
void QtProperty::insertSubProperty (
            QtProperty * property,
            QtProperty * precedingProperty )
```

Inserts the given *property* after the specified *precedingProperty* into this property's list of subproperties. If *precedingProperty* is 0, the specified *property* is inserted at the beginning of the list.
If the given *property* already is inserted, this function does nothing.

See also

addSubProperty(), removeSubProperty()

17.755.2.5 isEnabled()

```
bool QtProperty::isEnabled ( ) const
```

Returns whether the property is enabled.

See also

setEnabled()

17.755.2.6 isModified()

```
bool QtProperty::isModified ( ) const
```

Returns whether the property is modified.

See also

setModified()

17.755.2.7 propertyManager()

```
QtAbstractPropertyManager * QtProperty::propertyManager ( ) const
```

Returns a pointer to the manager that owns this property.

17.755.2.8 propertyName()

```
QString QtProperty::propertyName ( ) const
```

Returns the property's name.

See also

setPropertyName()

17.755.2.9 removeSubProperty()

```
void QtProperty::removeSubProperty (
            QtProperty * property )
```

Removes the given *property* from the list of subproperties without deleting it.

See also

addSubProperty(), insertSubProperty()

17.755.2.10 setDescriptionToolTip()

```
void QtProperty::setDescriptionToolTip (
            const QString & text )
```

Sets the property description's tool tip to the given *text*.

Since

5.6

See also

descriptionToolTip()

17.755.2.11 setEnabled()

```
void QtProperty::setEnabled (
            bool enable )
```

Enables or disables the property according to the passed *enable* value.

See also

isEnabled()

17.755.2.12 setModified()

```
void QtProperty::setModified (
            bool modified )
```

Sets the property's modified state according to the passed *modified* value.

See also

isModified()

17.755.2.13 setPropertyName()

```
void QtProperty::setPropertyName (
            const QString & name )
```

Sets the property's name to the given *name*.

See also

propertyName()

17.755.2.14 setStatusTip()

```
void QtProperty::setStatusTip (
            const QString & text )
```

Sets the property's status tip to the given *text*.

See also

statusTip()

17.755.2.15 setValueToolTip()

```
void QtProperty::setValueToolTip (
            const QString & text )
```

Sets the property value's tool tip to the given *text*.

Since

5.6

See also

valueToolTip()

17.755.2.16 setWhatsThis()

```
void QtProperty::setWhatsThis (
            const QString & text )
```

Sets the property's "What's This" help text to the given *text*.

See also

whatsThis()

17.755.2.17 statusTip()

```
QString QtProperty::statusTip ( ) const
```

Returns the property's status tip.

See also

setStatusTip()

17.755.2.18 subProperties()

```
QList< QtProperty * > QtProperty::subProperties ( ) const
```

Returns the set of subproperties.
Note that subproperties are not owned by *this* property, but by the manager that created them.

See also

insertSubProperty(), removeSubProperty()

17.755.2.19 valueIcon()

```
QIcon QtProperty::valueIcon ( ) const
```

Returns an icon representing the current state of this property.
If the given property type can not generate such an icon, this function returns an invalid icon.

See also

QtAbstractPropertyManager::valueIcon()

17.755.2.20 valueText()

```
QString QtProperty::valueText ( ) const
```

Returns a string representing the current state of this property.
If the given property type can not generate such a string, this function returns an empty string.

See also

QtAbstractPropertyManager::valueText()

17.755.2.21 valueToolTip()

```
QString QtProperty::valueToolTip ( ) const
```

Returns the property value's tool tip.
This is suitable for tool tips over the value (item delegate).

Since

5.6

See also

setValueToolTip()

17.755.2.22 whatsThis()

```
QString QtProperty::whatsThis ( ) const
```

Returns the property's "What's This" help text.

See also

setWhatsThis()

The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.cpp

17.756 QtPropertyBrowserUtils Class Reference

Static Public Member Functions

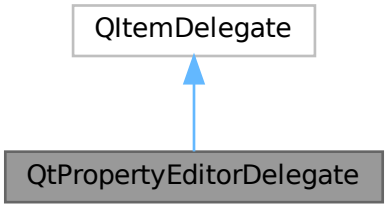
- static QPixmap **brushValuePixmap** (const QBrush &b)
- static QIcon **brushValueIcon** (const QBrush &b)
- static QString **colorValueText** (const QColor &c)
- static QPixmap **fontValuePixmap** (const QFont &f)
- static QIcon **fontValueIcon** (const QFont &f)
- static QString **fontValueText** (const QFont &f)
- static QString **dateFormat** ()
- static QString **timeFormat** ()
- static QString **dateTimeFormat** ()

The documentation for this class was generated from the following files:

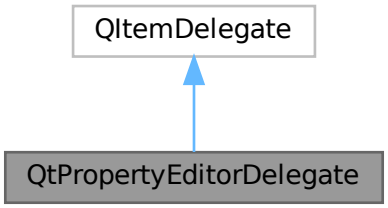
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils_p.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils.cpp

17.757 QtPropertyEditorDelegate Class Reference

Inheritance diagram for QtPropertyEditorDelegate:



Collaboration diagram for QtPropertyEditorDelegate:



Public Member Functions

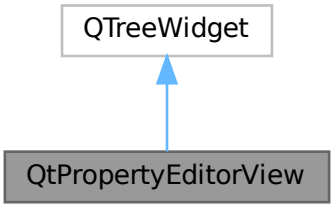
- **QtPropertyEditorDelegate** (QObject ∗parent=0)
- void **setEditorPrivate** (QtTreePropertyBrowserPrivate ∗editorPrivate)
- QWidget ∗ **createEditor** (QWidget ∗parent, const QStyleOptionViewItem &option, const QModelIndex &index) const override
- void **updateEditorGeometry** (QWidget ∗editor, const QStyleOptionViewItem &option, const QModelIndex &index) const override
- void **paint** (QPainter ∗painter, const QStyleOptionViewItem &option, const QModelIndex &index) const override
- QSize **sizeHint** (const QStyleOptionViewItem &option, const QModelIndex &index) const override
- void **setModelData** (QWidget ∗, QAbstractItemModel ∗, const QModelIndex &) const override
- void **setEditorData** (QWidget ∗, const QModelIndex &) const override
- bool **eventFilter** (QObject ∗object, QEvent ∗event) override
- void **closeEditor** (QtProperty ∗property)
- QTreeWidgetItem ∗ **editedItem** () const

The documentation for this class was generated from the following file:

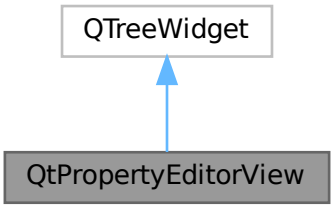
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qttreepropertybrowser.cpp

17.758 QtPropertyEditorView Class Reference

Inheritance diagram for QtPropertyEditorView:



Collaboration diagram for QtPropertyEditorView:



Public Member Functions

- **QtPropertyEditorView** (QWidget ∗parent=0)
- void **setEditorPrivate** (QtTreePropertyBrowserPrivate ∗editorPrivate)
- QTreeWidgetItem ∗ **indexToItem** (const QModelIndex &index) const

Protected Member Functions

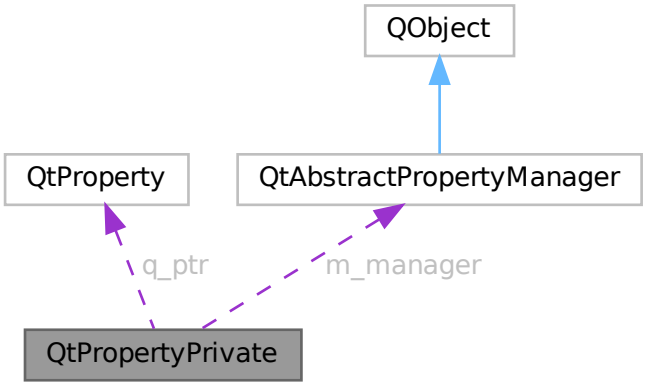
- void **keyPressEvent** (QKeyEvent ∗event) override
- void **mousePressEvent** (QMouseEvent ∗event) override
- void **drawRow** (QPainter ∗painter, const QStyleOptionViewItem &option, const QModelIndex &index) const override

The documentation for this class was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qttreepropertybrowser.cpp

17.759 QtPropertyPrivate Class Reference

Collaboration diagram for QtPropertyPrivate:



Public Member Functions

- **QtPropertyPrivate** (QtAbstractPropertyManager ∗manager)

Public Attributes

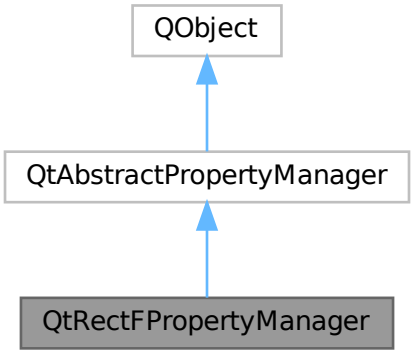
- QtProperty ∗ **q_ptr**
- QSet< QtProperty ∗ > **m_parentItems**
- QList< QtProperty ∗ > **m_subItems**
- QString **m_valueToolTip**
- QString **m_descriptionToolTip**
- QString **m_statusTip**
- QString **m_whatsThis**
- QString **m_name**
- bool **m_enabled**
- bool **m_modified**
- QtAbstractPropertyManager ∗const **m_manager**

The documentation for this class was generated from the following file:

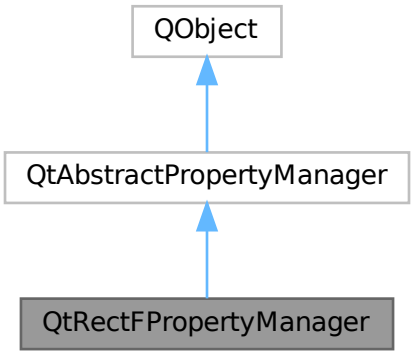
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.cpp

17.760 QtRectFPropertyManager Class Reference

Inheritance diagram for QtRectFPropertyManager:



Collaboration diagram for QtRectFPropertyManager:



Public Slots

- void setValue (QtProperty *property, const QRectF &val)
- void setConstraint (QtProperty *property, const QRectF &constraint)
- void setDecimals (QtProperty *property, int prec)

Signals

- void valueChanged (QtProperty *property, const QRectF &val)
- void constraintChanged (QtProperty *property, const QRectF &constraint)
- void decimalsChanged (QtProperty *property, int prec)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtRectFPropertyManager (QObject *parent=0)
- ~QtRectFPropertyManager ()
- QtDoublePropertyManager * subDoublePropertyManager () const
- QRectF value (const QtProperty *property) const
- QRectF constraint (const QtProperty *property) const
- int decimals (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.760.1 Constructor & Destructor Documentation

17.760.1.1 QtRectFPropertyManager()

```
QtRectFPropertyManager::QtRectFPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.760.1.2 ~QtRectFPropertyManager()

```
QtRectFPropertyManager::~QtRectFPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.760.2 Member Function Documentation

17.760.2.1 constraint()

```
QRectF QtRectFPropertyManager::constraint (
            const QtProperty * property ) const
```

Returns the given *property's* constraining rectangle. If returned value is null QRectF it means there is no constraint applied.

See also

value(), setConstraint()

17.760.2.2 constraintChanged

```
void QtRectFPropertyManager::constraintChanged (
            QtProperty * property,
            const QRectF & constraint )  [signal]
```

This signal is emitted whenever property changes its constraint rectangle, passing a pointer to the *property* and the new *constraint* rectangle as parameters.

See also

setConstraint()

17.760.2.3 decimals()

```
int QtRectFPropertyManager::decimals (
            const QtProperty * property ) const
```

Returns the given *property's* precision, in decimals.

See also

setDecimals()

17.760.2.4 decimalsChanged

```
void QtRectFPropertyManager::decimalsChanged (
            QtProperty * property,
            int prec )  [signal]
```

This signal is emitted whenever a property created by this manager changes its precision of value, passing a pointer to the *property* and the new *prec* value

See also

setDecimals()

17.760.2.5 initializeProperty()

```
void QtRectFPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.760.2.6 setConstraint

```
void QtRectFPropertyManager::setConstraint (
            QtProperty * property,
            const QRectF & constraint )  [slot]
```

Sets the given *property's* constraining rectangle to *constraint*.
When setting the constraint, the current value is adjusted if necessary (ensuring that the current rectangle value is inside the constraint). In order to reset the constraint pass a null QRectF value.

See also

setValue(), constraint(), constraintChanged()

17.760.2.7 setDecimals

```
void QtRectFPropertyManager::setDecimals (
            QtProperty * property,
            int prec )  [slot]
```

Sets the precision of the given *property* to *prec*.
The valid decimal range is 0-13. The default is 2.

See also

decimals()

17.760.2.8 setValue

```
void QtRectFPropertyManager::setValue (
            QtProperty * property,
            const QRectF & value )  [slot]
```

Sets the value of the given *property* to *value*. Nested properties are updated automatically.
If the specified *value* is not inside the given *property's* constraining rectangle, the value is adjusted accordingly to fit within the constraint.

See also

value(), setConstraint(), valueChanged()

17.760.2.9 subDoublePropertyManager()

```
QtDoublePropertyManager * QtRectFPropertyManager::subDoublePropertyManager ( ) const
```

Returns the manager that creates the nested *x*, *y*, *width* and *height* subproperties.
In order to provide editing widgets for the mentioned subproperties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.760.2.10 uninitializeProperty()

```
void QtRectFPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.760.2.11 value()

```
QRectF QtRectFPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by this manager, this function returns an invalid rectangle.

See also

setValue(), constraint()

17.760.2.12 valueChanged

```
void QtRectFPropertyManager::valueChanged (
            QtProperty * property,
            const QRectF & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.760.2.13 valueText()

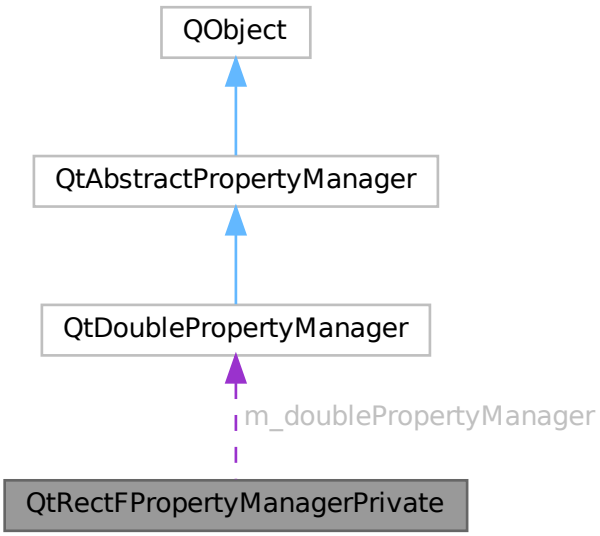
```
QString QtRectFPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.761 QtRectFPropertyManagerPrivate Class Reference

Collaboration diagram for QtRectFPropertyManagerPrivate:



Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Member Functions

- void **slotDoubleChanged** (QtProperty *property, double value)
- void **slotPropertyDestroyed** (QtProperty *property)
- void **setConstraint** (QtProperty *property, const QRectF &constraint, const QRectF &val)

Public Attributes

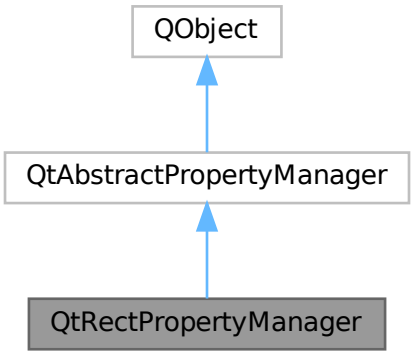
- PropertyValueMap **m_values**
- QtDoublePropertyManager * **m_doublePropertyManager**
- QMap< const QtProperty *, QtProperty * > **m_propertyToX**
- QMap< const QtProperty *, QtProperty * > **m_propertyToY**
- QMap< const QtProperty *, QtProperty * > **m_propertyToW**
- QMap< const QtProperty *, QtProperty * > **m_propertyToH**
- QMap< const QtProperty *, QtProperty * > **m_xToProperty**
- QMap< const QtProperty *, QtProperty * > **m_yToProperty**
- QMap< const QtProperty *, QtProperty * > **m_wToProperty**
- QMap< const QtProperty *, QtProperty * > **m_hToProperty**

The documentation for this class was generated from the following file:

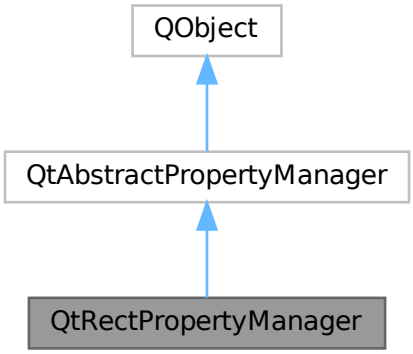
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.762 QtRectPropertyManager Class Reference

Inheritance diagram for QtRectPropertyManager:



Collaboration diagram for QtRectPropertyManager:



Public Slots

- void setValue (QtProperty *property, const QRect &val)
- void setConstraint (QtProperty *property, const QRect &constraint)

Signals

- void valueChanged (QtProperty *property, const QRect &val)
- void constraintChanged (QtProperty *property, const QRect &constraint)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtRectPropertyManager (QObject *parent=0)
- ~QtRectPropertyManager ()
- QtIntPropertyManager * subIntPropertyManager () const
- QRect value (const QtProperty *property) const
- QRect constraint (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.762.1 Constructor & Destructor Documentation

17.762.1.1 QtRectPropertyManager()

```
QtRectPropertyManager::QtRectPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.762.1.2 ~QtRectPropertyManager()

```
QtRectPropertyManager::~QtRectPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.762.2 Member Function Documentation

17.762.2.1 constraint()

```
QRect QtRectPropertyManager::constraint (
            const QtProperty * property ) const
```

Returns the given *property's* constraining rectangle. If returned value is null QRect it means there is no constraint applied.

See also

value(), setConstraint()

17.762.2.2 constraintChanged

```
void QtRectPropertyManager::constraintChanged (
            QtProperty * property,
            const QRect & constraint )  [signal]
```

This signal is emitted whenever property changes its constraint rectangle, passing a pointer to the *property* and the new *constraint* rectangle as parameters.

See also

setConstraint()

17.762.2.3 initializeProperty()

```
void QtRectPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.762.2.4 setConstraint

```
void QtRectPropertyManager::setConstraint (
            QtProperty * property,
            const QRect & constraint )  [slot]
```

Sets the given *property's* constraining rectangle to *constraint*.
When setting the constraint, the current value is adjusted if necessary (ensuring that the current rectangle value is inside the constraint). In order to reset the constraint pass a null QRect value.

See also

setValue(), constraint(), constraintChanged()

17.762.2.5 setValue

```
void QtRectPropertyManager::setValue (
            QtProperty * property,
            const QRect & value )  [slot]
```

Sets the value of the given *property* to *value*. Nested properties are updated automatically.
If the specified *value* is not inside the given *property's* constraining rectangle, the value is adjusted accordingly to fit within the constraint.

See also

value(), setConstraint(), valueChanged()

17.762.2.6 subIntPropertyManager()

```
QtIntPropertyManager * QtRectPropertyManager::subIntPropertyManager ( ) const
```

Returns the manager that creates the nested *x*, *y*, *width* and *height* subproperties.
In order to provide editing widgets for the mentioned subproperties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.762.2.7 uninitializeProperty()

```
void QtRectPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.762.2.8 value()

```
QRect QtRectPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by this manager, this function returns an invalid rectangle.

See also

setValue(), constraint()

17.762.2.9 valueChanged

```
void QtRectPropertyManager::valueChanged (
            QtProperty * property,
            const QRect & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.762.2.10 valueText()

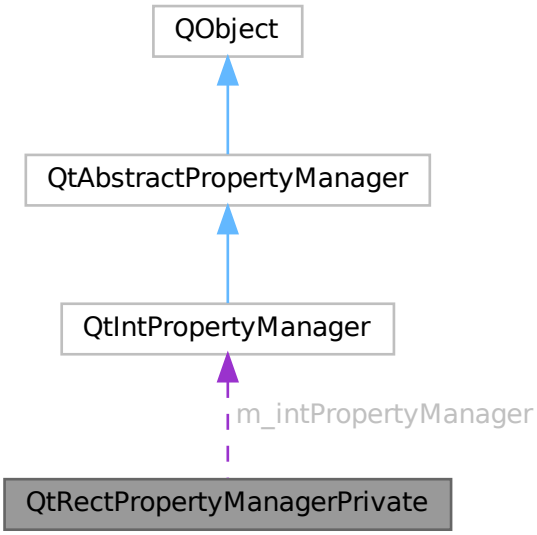
```
QString QtRectPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.763 QtRectPropertyManagerPrivate Class Reference

Collaboration diagram for QtRectPropertyManagerPrivate:



Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Member Functions

- void **slotIntChanged** (QtProperty *property, int value)
- void **slotPropertyDestroyed** (QtProperty *property)
- void **setConstraint** (QtProperty *property, const QRect &constraint, const QRect &val)

Public Attributes

- PropertyValueMap **m_values**
- QtIntPropertyManager * **m_intPropertyManager**

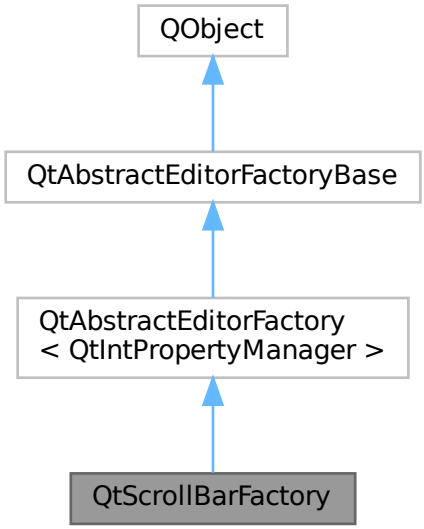
- QMap< const QtProperty *, QtProperty * > **m_propertyToX**
- QMap< const QtProperty *, QtProperty * > **m_propertyToY**
- QMap< const QtProperty *, QtProperty * > **m_propertyToW**
- QMap< const QtProperty *, QtProperty * > **m_propertyToH**
- QMap< const QtProperty *, QtProperty * > **m_xToProperty**
- QMap< const QtProperty *, QtProperty * > **m_yToProperty**
- QMap< const QtProperty *, QtProperty * > **m_wToProperty**
- QMap< const QtProperty *, QtProperty * > **m_hToProperty**

The documentation for this class was generated from the following file:

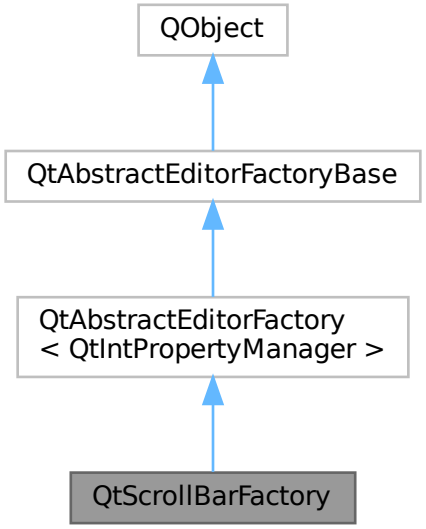
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.764 QtScrollBarFactory Class Reference

Inheritance diagram for QtScrollBarFactory:



Collaboration diagram for QtScrollBarFactory:



Public Member Functions

- QtScrollBarFactory (QObject ∗parent=0)
- ∼QtScrollBarFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtIntPropertyManager >

- QtAbstractEditorFactory (QObject ∗parent)
- QWidget ∗ createEditor (QtProperty ∗property, QWidget ∗parent) override
- void addPropertyManager (QtIntPropertyManager ∗manager)
- void removePropertyManager (QtIntPropertyManager ∗manager)
- QSet< QtIntPropertyManager ∗ > propertyManagers () const
- QtIntPropertyManager ∗ propertyManager (QtProperty ∗property) const

Protected Member Functions

- void **connectPropertyManager** (QtIntPropertyManager ∗manager) override
- QWidget ∗ **createEditor** (QtIntPropertyManager ∗manager, QtProperty ∗property, QWidget ∗parent) override
- void **disconnectPropertyManager** (QtIntPropertyManager ∗manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtIntPropertyManager >

- virtual QWidget ∗ createEditor (QtIntPropertyManager ∗manager, QtProperty ∗property, QWidget ∗parent)=0
- virtual void connectPropertyManager (QtIntPropertyManager ∗manager)=0
- virtual void disconnectPropertyManager (QtIntPropertyManager ∗manager)=0
- void managerDestroyed (QObject ∗manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.764.1 Constructor & Destructor Documentation

17.764.1.1 QtScrollBarFactory()

```
QtScrollBarFactory::QtScrollBarFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.764.1.2 ~QtScrollBarFactory()

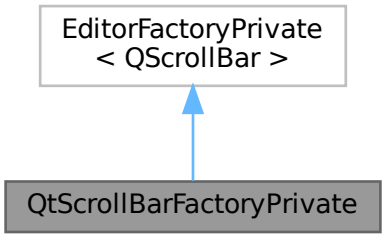
```
QtScrollBarFactory::~QtScrollBarFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

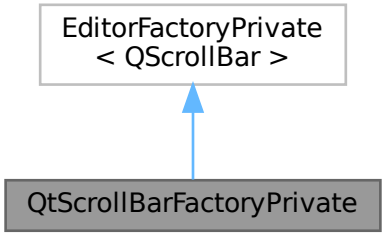
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.765 QtScrollBarFactoryPrivate Class Reference

Inheritance diagram for QtScrollBarFactoryPrivate:



Collaboration diagram for QtScrollBarFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, int value)
- void **slotRangeChanged** (QtProperty *property, int min, int max)
- void **slotSingleStepChanged** (QtProperty *property, int step)
- void **slotSetValue** (int value)

Public Member Functions inherited from EditorFactoryPrivate< QScrollBar >

- QScrollBar * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QScrollBar *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QScrollBar >

- typedef QList< QScrollBar * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QScrollBar *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QScrollBar >

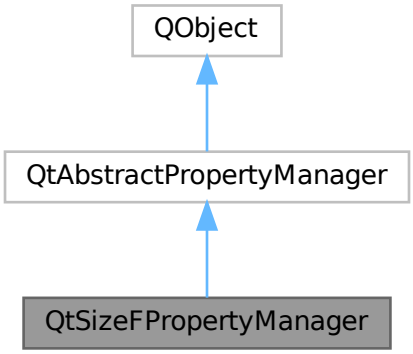
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

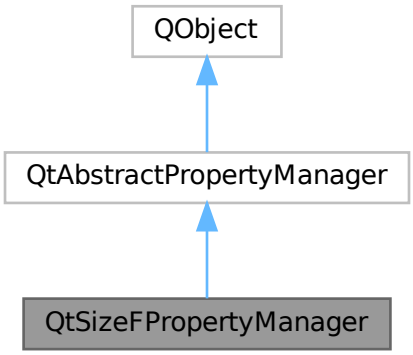
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.766 QtSizeFPropertyManager Class Reference

Inheritance diagram for QtSizeFPropertyManager:



Collaboration diagram for QtSizeFPropertyManager:



Public Slots

- void setValue (QtProperty *property, const QSizeF &val)
- void setMinimum (QtProperty *property, const QSizeF &minVal)
- void setMaximum (QtProperty *property, const QSizeF &maxVal)
- void setRange (QtProperty *property, const QSizeF &minVal, const QSizeF &maxVal)
- void setDecimals (QtProperty *property, int prec)

Signals

- void valueChanged (QtProperty *property, const QSizeF &val)
- void rangeChanged (QtProperty *property, const QSizeF &minVal, const QSizeF &maxVal)
- void decimalsChanged (QtProperty *property, int prec)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtSizeFPropertyManager (QObject *parent=0)
- ~QtSizeFPropertyManager ()
- QtDoublePropertyManager * subDoublePropertyManager () const
- QSizeF value (const QtProperty *property) const
- QSizeF minimum (const QtProperty *property) const
- QSizeF maximum (const QtProperty *property) const
- int decimals (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.766.1 Constructor & Destructor Documentation

17.766.1.1 QtSizeFPropertyManager()

```
QtSizeFPropertyManager::QtSizeFPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.766.1.2 ~QtSizeFPropertyManager()

```
QtSizeFPropertyManager::~QtSizeFPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.766.2 Member Function Documentation

17.766.2.1 decimals()

```
int QtSizeFPropertyManager::decimals (
            const QtProperty * property ) const
```

Returns the given *property's* precision, in decimals.

See also

setDecimals()

17.766.2.2 decimalsChanged

```
void QtSizeFPropertyManager::decimalsChanged (
            QtProperty * property,
            int prec )  [signal]
```

This signal is emitted whenever a property created by this manager changes its precision of value, passing a pointer to the *property* and the new *prec* value

See also

setDecimals()

17.766.2.3 initializeProperty()

```
void QtSizeFPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.766.2.4 maximum()

```
QSizeF QtSizeFPropertyManager::maximum (
            const QtProperty * property ) const
```

Returns the given *property's* maximum size value.

See also

setMaximum(), minimum(), setRange()

17.766.2.5 minimum()

```
QSizeF QtSizeFPropertyManager::minimum (
            const QtProperty * property ) const
```

Returns the given *property's* minimum size value.

See also

setMinimum(), maximum(), setRange()

17.766.2.6 rangeChanged

```
void QtSizeFPropertyManager::rangeChanged (
            QtProperty * property,
            const QSizeF & minimum,
            const QSizeF & maximum )  [signal]
```

This signal is emitted whenever a property created by this manager changes its range of valid sizes, passing a pointer to the *property* and the new *minimum* and *maximum* sizes.

See also

setRange()

17.766.2.7 setDecimals

```
void QtSizeFPropertyManager::setDecimals (
            QtProperty * property,
            int prec )  [slot]
```

Sets the precision of the given *property* to *prec*.
The valid decimal range is 0-13. The default is 2.

See also

decimals()

17.766.2.8 setMaximum

```
void QtSizeFPropertyManager::setMaximum (
            QtProperty * property,
            const QSizeF & maxVal )  [slot]
```

Sets the maximum size value for the given *property* to *maxVal*.
When setting the maximum size value, the minimum and current values are adjusted if necessary (ensuring that the size range remains valid and that the current value is within the range).

See also

maximum(), setRange(), rangeChanged()

17.766.2.9 setMinimum

```
void QtSizeFPropertyManager::setMinimum (
            QtProperty * property,
            const QSizeF & minVal )  [slot]
```

Sets the minimum size value for the given *property* to *minVal*.
When setting the minimum size value, the maximum and current values are adjusted if necessary (ensuring that the size range remains valid and that the current value is within the range).

See also

minimum(), setRange(), rangeChanged()

17.766.2.10 setRange

```
void QtSizeFPropertyManager::setRange (
            QtProperty * property,
            const QSizeF & minimum,
            const QSizeF & maximum )  [slot]
```

Sets the range of valid values.
This is a convenience function defining the range of valid values in one go; setting the *minimum* and *maximum* values for the given *property* with a single function call.
When setting a new range, the current value is adjusted if necessary (ensuring that the value remains within the range).

See also

setMinimum(), setMaximum(), rangeChanged()

17.766.2.11 setValue

```
void QtSizeFPropertyManager::setValue (
            QtProperty * property,
            const QSizeF & value )  [slot]
```

Sets the value of the given *property* to *value*.
If the specified *value* is not valid according to the given *property's* size range, the *value* is adjusted to the nearest valid value within the size range.

See also

value(), setRange(), valueChanged()

17.766.2.12 subDoublePropertyManager()

```
QtDoublePropertyManager * QtSizeFPropertyManager::subDoublePropertyManager ( ) const
```

Returns the manager that creates the nested *width* and *height* subproperties.
In order to provide editing widgets for the *width* and *height* properties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.766.2.13 uninitializeProperty()

```
void QtSizeFPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.766.2.14 value()

```
QSizeF QtSizeFPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by this manager, this function returns an invalid size

See also

setValue()

17.766.2.15 valueChanged

```
void QtSizeFPropertyManager::valueChanged (
            QtProperty * property,
            const QSizeF & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.766.2.16 valueText()

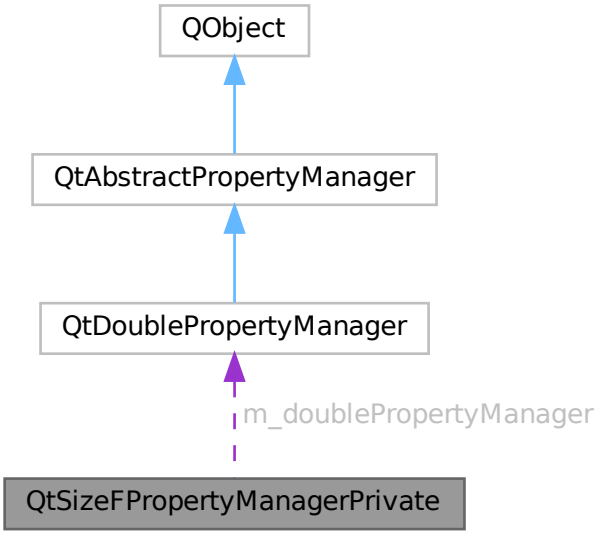
```
QString QtSizeFPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.767 QtSizeFPropertyManagerPrivate Class Reference

Collaboration diagram for QtSizeFPropertyManagerPrivate:



Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Member Functions

- void **slotDoubleChanged** (QtProperty *property, double value)
- void **slotPropertyDestroyed** (QtProperty *property)
- void **setValue** (QtProperty *property, const QSizeF &val)
- void **setRange** (QtProperty *property, const QSizeF &minVal, const QSizeF &maxVal, const QSizeF &val)

Public Attributes

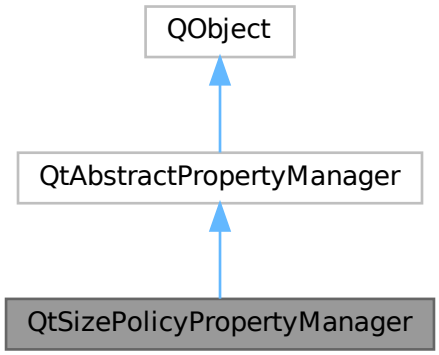
- PropertyValueMap **m_values**
- QtDoublePropertyManager * **m_doublePropertyManager**
- QMap< const QtProperty *, QtProperty * > **m_propertyToW**
- QMap< const QtProperty *, QtProperty * > **m_propertyToH**
- QMap< const QtProperty *, QtProperty * > **m_wToProperty**
- QMap< const QtProperty *, QtProperty * > **m_hToProperty**

The documentation for this class was generated from the following file:

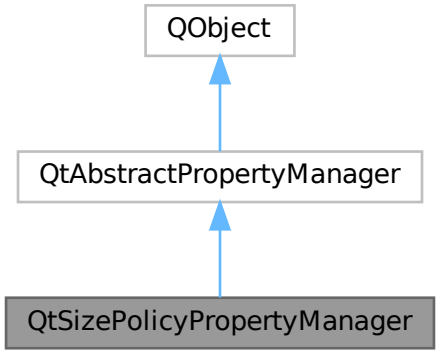
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.768 QtSizePolicyPropertyManager Class Reference

Inheritance diagram for QtSizePolicyPropertyManager:



Collaboration diagram for QtSizePolicyPropertyManager:



Public Slots

- void setValue (QtProperty *property, const QSizePolicy &val)

Signals

- void valueChanged (QtProperty *property, const QSizePolicy &val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtSizePolicyPropertyManager (QObject ∗parent=0)
- ∼QtSizePolicyPropertyManager ()
- QtIntPropertyManager ∗ subIntPropertyManager () const
- QtEnumPropertyManager ∗ subEnumPropertyManager () const
- QSizePolicy value (const QtProperty ∗property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject ∗parent=0)
- ∼QtAbstractPropertyManager ()
- QSet< QtProperty ∗ > properties () const
- void clear () const
- QtProperty ∗ addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty ∗property) const override
- void initializeProperty (QtProperty ∗property) override
- void uninitializeProperty (QtProperty ∗property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty ∗property) const
- virtual QIcon valueIcon (const QtProperty ∗property) const
- virtual QtProperty ∗ createProperty ()

17.768.1 Constructor & Destructor Documentation

17.768.1.1 QtSizePolicyPropertyManager()

```
QtSizePolicyPropertyManager::QtSizePolicyPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.768.1.2 ∼QtSizePolicyPropertyManager()

```
QtSizePolicyPropertyManager::~QtSizePolicyPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.768.2 Member Function Documentation

17.768.2.1 initializeProperty()

```
void QtSizePolicyPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.768.2.2 setValue

```
void QtSizePolicyPropertyManager::setValue (
            QtProperty * property,
            const QSizePolicy & value )  [slot]
```

Sets the value of the given *property* to *value*. Nested properties are updated automatically.

See also

value(), valueChanged()

17.768.2.3 subEnumPropertyManager()

```
QtEnumPropertyManager * QtSizePolicyPropertyManager::subEnumPropertyManager ( ) const
```

Returns the manager that creates the nested *horizontalPolicy* and *verticalPolicy* subproperties.
In order to provide editing widgets for the mentioned subproperties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.768.2.4 subIntPropertyManager()

```
QtIntPropertyManager * QtSizePolicyPropertyManager::subIntPropertyManager ( ) const
```

Returns the manager that creates the nested *horizontalStretch* and *verticalStretch* subproperties.
In order to provide editing widgets for the mentioned subproperties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.768.2.5 uninitializeProperty()

```
void QtSizePolicyPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.768.2.6 value()

```
QSizePolicy QtSizePolicyPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given property is not managed by this manager, this function returns the default size policy.

See also

setValue()

17.768.2.7 valueChanged

```
void QtSizePolicyPropertyManager::valueChanged (
            QtProperty * property,
            const QSizePolicy & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.768.2.8 valueText()

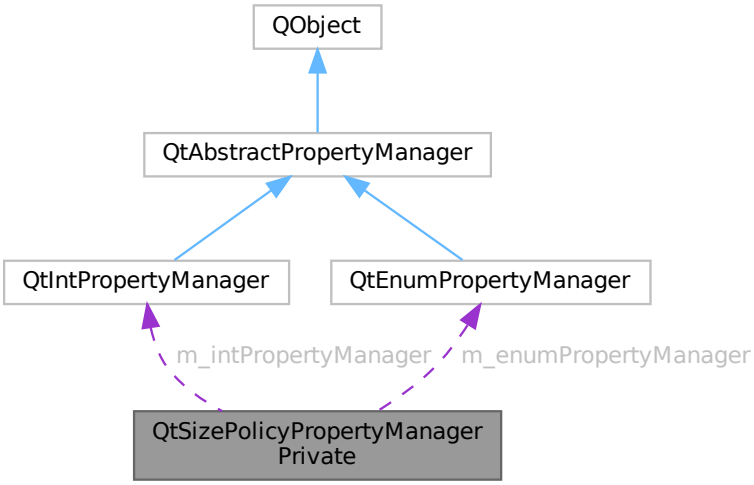
```
QString QtSizePolicyPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp Unknown size policy on reading invalid uic3 files
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.769 QtSizePolicyPropertyManagerPrivate Class Reference

Collaboration diagram for QtSizePolicyPropertyManagerPrivate:



Public Types

- typedef QMap< const QtProperty *, QSizePolicy > **PropertyValueMap**

Public Member Functions

- void **slotIntChanged** (QtProperty *property, int value)
- void **slotEnumChanged** (QtProperty *property, int value)
- void **slotPropertyDestroyed** (QtProperty *property)

Public Attributes

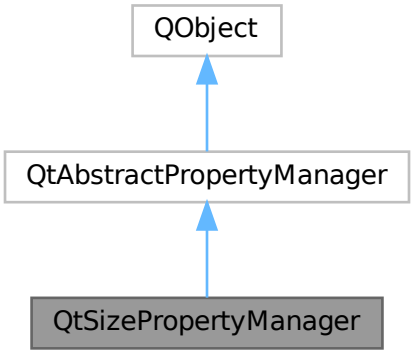
- PropertyValueMap **m_values**
- QtIntPropertyManager * **m_intPropertyManager**
- QtEnumPropertyManager * **m_enumPropertyManager**
- QMap< const QtProperty *, QtProperty * > **m_propertyToHPolicy**
- QMap< const QtProperty *, QtProperty * > **m_propertyToVPolicy**
- QMap< const QtProperty *, QtProperty * > **m_propertyToHStretch**
- QMap< const QtProperty *, QtProperty * > **m_propertyToVStretch**
- QMap< const QtProperty *, QtProperty * > **m_hPolicyToProperty**
- QMap< const QtProperty *, QtProperty * > **m_vPolicyToProperty**
- QMap< const QtProperty *, QtProperty * > **m_hStretchToProperty**
- QMap< const QtProperty *, QtProperty * > **m_vStretchToProperty**

The documentation for this class was generated from the following file:

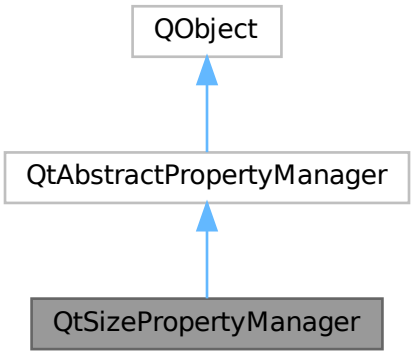
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.770 QtSizePropertyManager Class Reference

Inheritance diagram for QtSizePropertyManager:



Collaboration diagram for QtSizePropertyManager:



Public Slots

- void setValue (QtProperty *property, const QSize &val)
- void setMinimum (QtProperty *property, const QSize &minVal)
- void setMaximum (QtProperty *property, const QSize &maxVal)
- void setRange (QtProperty *property, const QSize &minVal, const QSize &maxVal)

Signals

- void valueChanged (QtProperty *property, const QSize &val)
- void rangeChanged (QtProperty *property, const QSize &minVal, const QSize &maxVal)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtSizePropertyManager (QObject *parent=0)
- ~QtSizePropertyManager ()
- QtIntPropertyManager * subIntPropertyManager () const
- QSize value (const QtProperty *property) const
- QSize minimum (const QtProperty *property) const
- QSize maximum (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.770.1 Constructor & Destructor Documentation

17.770.1.1 QtSizePropertyManager()

```
QtSizePropertyManager::QtSizePropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.770.1.2 ~QtSizePropertyManager()

```
QtSizePropertyManager::~QtSizePropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.770.2 Member Function Documentation

17.770.2.1 initializeProperty()

```
void QtSizePropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp

Implements QtAbstractPropertyManager.

17.770.2.2 maximum()

```
QSize QtSizePropertyManager::maximum (
            const QtProperty * property ) const
```

Returns the given *property's* maximum size value.

See also

setMaximum(), minimum(), setRange()

17.770.2.3 minimum()

```
QSize QtSizePropertyManager::minimum (
            const QtProperty * property ) const
```

Returns the given *property's* minimum size value.

See also

setMinimum(), maximum(), setRange()

17.770.2.4 rangeChanged

```
void QtSizePropertyManager::rangeChanged (
            QtProperty * property,
            const QSize & minimum,
            const QSize & maximum )  [signal]
```

This signal is emitted whenever a property created by this manager changes its range of valid sizes, passing a pointer to the *property* and the new *minimum* and *maximum* sizes.

See also

setRange()

17.770.2.5 setMaximum

```
void QtSizePropertyManager::setMaximum (
            QtProperty * property,
            const QSize & maxVal )  [slot]
```

Sets the maximum size value for the given *property* to *maxVal*.
When setting the maximum size value, the minimum and current values are adjusted if necessary (ensuring that the size range remains valid and that the current value is within the range).

See also

maximum(), setRange(), rangeChanged()

17.770.2.6 setMinimum

```
void QtSizePropertyManager::setMinimum (
            QtProperty * property,
            const QSize & minVal )  [slot]
```

Sets the minimum size value for the given *property* to *minVal*.
When setting the minimum size value, the maximum and current values are adjusted if necessary (ensuring that the size range remains valid and that the current value is within the range).

See also

minimum(), setRange(), rangeChanged()

17.770.2.7 setRange

```
void QtSizePropertyManager::setRange (
            QtProperty * property,
            const QSize & minimum,
            const QSize & maximum )  [slot]
```

Sets the range of valid values.
This is a convenience function defining the range of valid values in one go; setting the *minimum* and *maximum* values for the given *property* with a single function call.
When setting a new range, the current value is adjusted if necessary (ensuring that the value remains within the range).

See also

setMinimum(), setMaximum(), rangeChanged()

17.770.2.8 setValue

```
void QtSizePropertyManager::setValue (
            QtProperty * property,
            const QSize & value )  [slot]
```

Sets the value of the given *property* to *value*.
If the specified *value* is not valid according to the given *property's* size range, the *value* is adjusted to the nearest valid value within the size range.

See also

value(), setRange(), valueChanged()

17.770.2.9 subIntPropertyManager()

```
QtIntPropertyManager * QtSizePropertyManager::subIntPropertyManager ( ) const
```

Returns the manager that creates the nested *width* and *height* subproperties.
In order to provide editing widgets for the *width* and *height* properties in a property browser widget, this manager must be associated with an editor factory.

See also

QtAbstractPropertyBrowser::setFactoryForManager()

17.770.2.10 uninitializeProperty()

```
void QtSizePropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.770.2.11 value()

```
QSize QtSizePropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given *property* is not managed by this manager, this function returns an invalid size

See also

setValue()

17.770.2.12 valueChanged

```
void QtSizePropertyManager::valueChanged (
            QtProperty * property,
            const QSize & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.770.2.13 valueText()

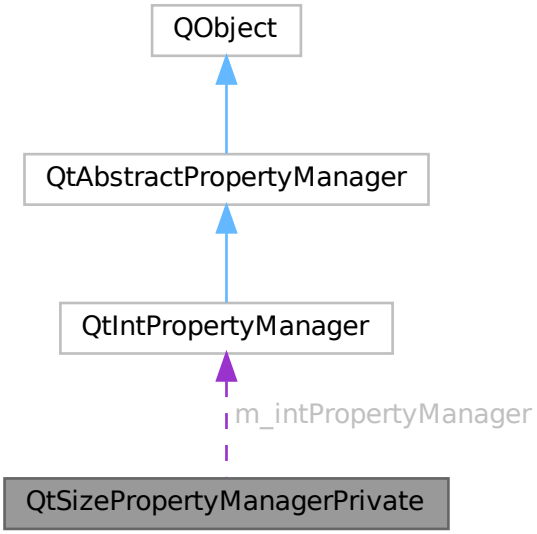
```
QString QtSizePropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.771 QtSizePropertyManagerPrivate Class Reference

Collaboration diagram for QtSizePropertyManagerPrivate:



Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Member Functions

- void **slotIntChanged** (QtProperty ∗property, int value)
- void **slotPropertyDestroyed** (QtProperty ∗property)
- void **setValue** (QtProperty ∗property, const QSize &val)
- void **setRange** (QtProperty ∗property, const QSize &minVal, const QSize &maxVal, const QSize &val)

Public Attributes

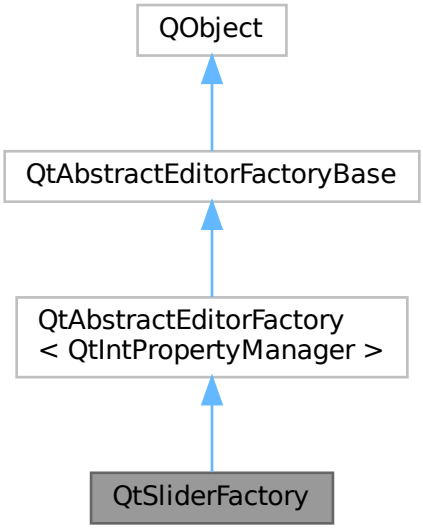
- PropertyValueMap **m_values**
- QtIntPropertyManager ∗ **m_intPropertyManager**
- QMap< const QtProperty ∗, QtProperty ∗ > **m_propertyToW**
- QMap< const QtProperty ∗, QtProperty ∗ > **m_propertyToH**
- QMap< const QtProperty ∗, QtProperty ∗ > **m_wToProperty**
- QMap< const QtProperty ∗, QtProperty ∗ > **m_hToProperty**

The documentation for this class was generated from the following file:

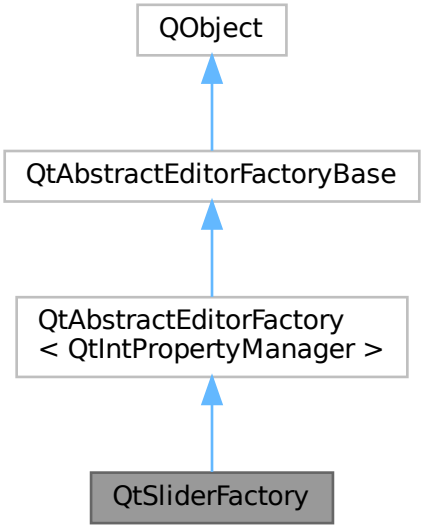
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.772 QtSliderFactory Class Reference

Inheritance diagram for QtSliderFactory:



Collaboration diagram for QtSliderFactory:



Public Member Functions

- QtSliderFactory (QObject *parent=0)
- ~QtSliderFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtIntPropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtIntPropertyManager *manager)
- void removePropertyManager (QtIntPropertyManager *manager)
- QSet< QtIntPropertyManager * > propertyManagers () const
- QtIntPropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtIntPropertyManager *manager) override
- QWidget * **createEditor** (QtIntPropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtIntPropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtIntPropertyManager >

- virtual QWidget * createEditor (QtIntPropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtIntPropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtIntPropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject ∗parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.772.1 Constructor & Destructor Documentation

17.772.1.1 QtSliderFactory()

```
QtSliderFactory::QtSliderFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.772.1.2 ∼QtSliderFactory()

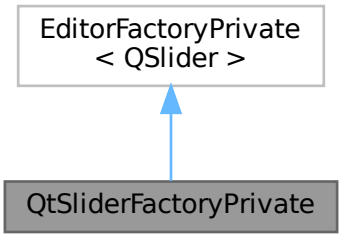
```
QtSliderFactory::~QtSliderFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

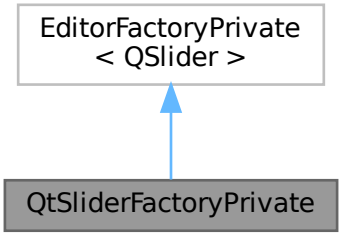
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.773 QtSliderFactoryPrivate Class Reference

Inheritance diagram for QtSliderFactoryPrivate:



Collaboration diagram for QtSliderFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, int value)
- void **slotRangeChanged** (QtProperty *property, int min, int max)
- void **slotSingleStepChanged** (QtProperty *property, int step)
- void **slotSetValue** (int value)

Public Member Functions inherited from EditorFactoryPrivate< QSlider >

- QSlider * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QSlider *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QSlider >

- typedef QList< QSlider * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QSlider *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QSlider >

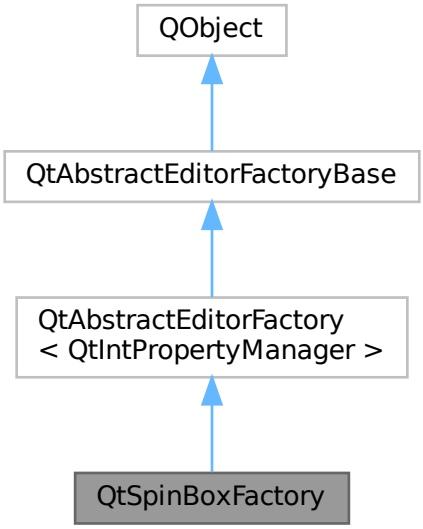
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

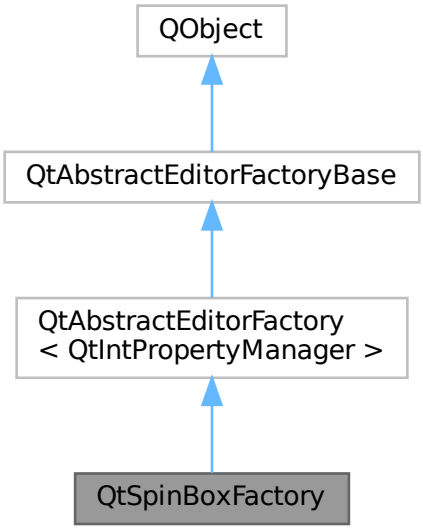
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.774 QtSpinBoxFactory Class Reference

Inheritance diagram for QtSpinBoxFactory:



Collaboration diagram for QtSpinBoxFactory:



Public Member Functions

- QtSpinBoxFactory (QObject *parent=0)

- ~QtSpinBoxFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtIntPropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtIntPropertyManager *manager)
- void removePropertyManager (QtIntPropertyManager *manager)
- QSet< QtIntPropertyManager * > propertyManagers () const
- QtIntPropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtIntPropertyManager *manager) override
- QWidget * **createEditor** (QtIntPropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtIntPropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtIntPropertyManager >

- virtual QWidget * createEditor (QtIntPropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtIntPropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtIntPropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.774.1 Constructor & Destructor Documentation

17.774.1.1 QtSpinBoxFactory()

```
QtSpinBoxFactory::QtSpinBoxFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.774.1.2 ~QtSpinBoxFactory()

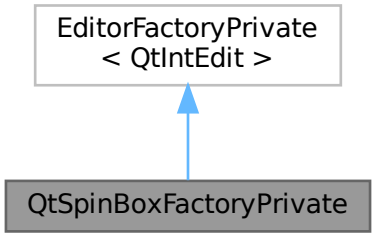
```
QtSpinBoxFactory::~QtSpinBoxFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

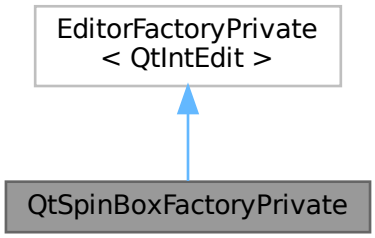
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.775 QtSpinBoxFactoryPrivate Class Reference

Inheritance diagram for QtSpinBoxFactoryPrivate:



Collaboration diagram for QtSpinBoxFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, int value)
- void **slotRangeChanged** (QtProperty *property, int min, int max)
- void **slotSingleStepChanged** (QtProperty *property, int step)
- void **slotSetValue** (int value)

Public Member Functions inherited from EditorFactoryPrivate< QtIntEdit >

- QtIntEdit * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QtIntEdit *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QtIntEdit >

- typedef QList< QtIntEdit * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QtIntEdit *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QtIntEdit >

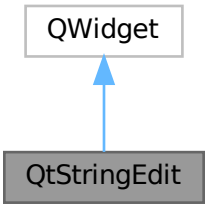
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

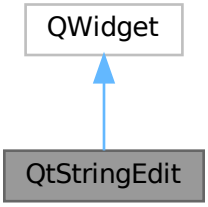
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.776 QtStringEdit Class Reference

Inheritance diagram for QtStringEdit:



Collaboration diagram for QtStringEdit:



Signals

- void **textChanged** (QString)
- void **textEdited** (QString)

Public Member Functions

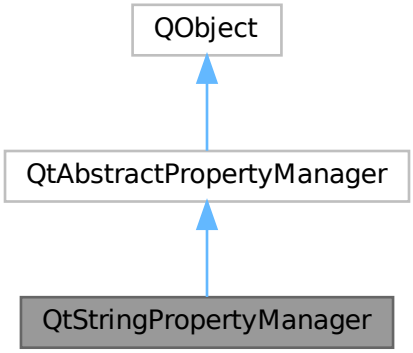
- **QtStringEdit** (QWidget *parent=Q_NULLPTR)
- QString **text** ()
- void **setText** (QString)
- void **setInitialText** (QString)
- const QValidator * **validator** ()
- void **setValidator** (QValidator *)

The documentation for this class was generated from the following files:

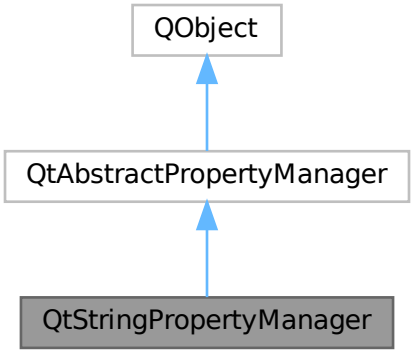
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils_p.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils.cpp

17.777 QtStringPropertyManager Class Reference

Inheritance diagram for QtStringPropertyManager:



Collaboration diagram for QtStringPropertyManager:



Public Slots

- void setValue (QtProperty *property, const QString &val)
- void **setValueOnly** (QtProperty *property, const QString &val)
- void setRegExp (QtProperty *property, const QRegularExpression &regExp)

Signals

- void valueChanged (QtProperty *property, const QString &val)

- void regExpChanged (QtProperty *property, const QRegularExpression &regExp)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtStringPropertyManager (QObject *parent=0)
- ~QtStringPropertyManager ()
- QString value (const QtProperty *property) const
- QString **initialValue** (const QtProperty *property) const
- QRegularExpression regExp (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.777.1 Constructor & Destructor Documentation

17.777.1.1 QtStringPropertyManager()

```
QtStringPropertyManager::QtStringPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.777.1.2 ~QtStringPropertyManager()

```
QtStringPropertyManager::~QtStringPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.777.2 Member Function Documentation

17.777.2.1 initializeProperty()

```
void QtStringPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.777.2.2 regExp()

```
QRegularExpression QtStringPropertyManager::regExp (
            const QtProperty * property ) const
```

Returns the given *property's* currently set regular expression.
If the given *property* is not managed by this manager, this function returns an empty expression.

See also

setRegExp()

17.777.2.3 regExpChanged

```
void QtStringPropertyManager::regExpChanged (
            QtProperty * property,
            const QRegularExpression & regExp ) [signal]
```

This signal is emitted whenever a property created by this manager changes its currenlty set regular expression, passing a pointer to the *property* and the new *regExp* as parameters.

See also

setRegExp()

17.777.2.4 setRegExp

```
void QtStringPropertyManager::setRegExp (
            QtProperty * property,
            const QRegularExpression & regExp ) [slot]
```

Sets the regular expression of the given *property* to *regExp*.

See also

regExp(), setValue(), regExpChanged()

17.777.2.5 setValue

```
void QtStringPropertyManager::setValue (
            QtProperty * property,
            const QString & value ) [slot]
```

Sets the value of the given *property* to *value*.
If the specified *value* doesn't match the given *property's* regular expression, this function does nothing.

See also

value(), setRegExp(), valueChanged()

17.777.2.6 uninitializeProperty()

```
void QtStringPropertyManager::uninitializeProperty (
            QtProperty * property ) [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.777.2.7 value()

```
QString QtStringPropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given property is not managed by this manager, this function returns an empty string.

See also

setValue()

17.777.2.8 valueChanged

```
void QtStringPropertyManager::valueChanged (
            QtProperty * property,
            const QString & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.777.2.9 valueText()

```
QString QtStringPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.778 QtStringPropertyManagerPrivate Class Reference

Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Attributes

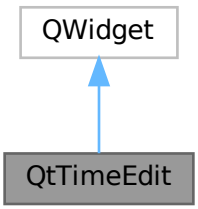
- QMap< const QtProperty *, Data > **m_values**

The documentation for this class was generated from the following file:

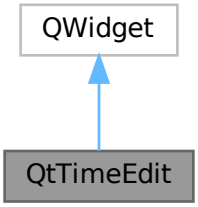
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.779 QtTimeEdit Class Reference

Inheritance diagram for QtTimeEdit:



Collaboration diagram for QtTimeEdit:



Signals

- void **timeChanged** (QTime)

Public Member Functions

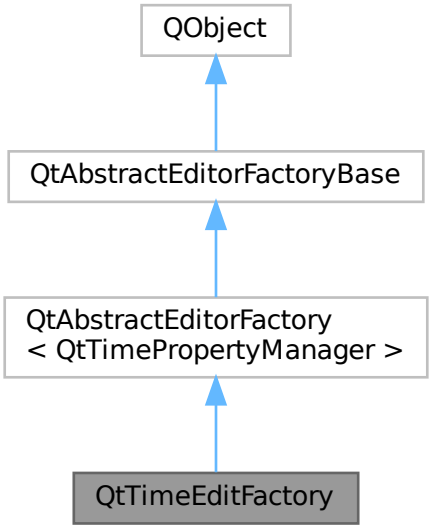
- **QtTimeEdit** (QWidget ∗parent=Q_NULLPTR)
- QTime **time** ()
- void **setTime** (QTime)
- void **setInitialTime** (QTime)
- void **setTimeRange** (QTime, QTime)
- void **setDisplayFormat** (QString)
- void **setCalendarPopup** (bool)

The documentation for this class was generated from the following files:

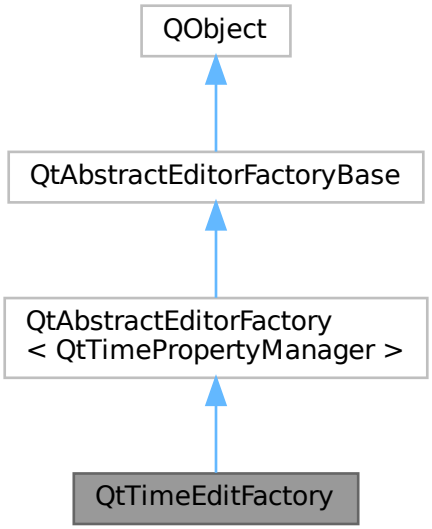
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils_p.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils.cpp

17.780 QtTimeEditFactory Class Reference

Inheritance diagram for QtTimeEditFactory:



Collaboration diagram for QtTimeEditFactory:



Public Member Functions

- QtTimeEditFactory (QObject *parent=0)

- ~QtTimeEditFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtTimePropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtTimePropertyManager *manager)
- void removePropertyManager (QtTimePropertyManager *manager)
- QSet< QtTimePropertyManager * > propertyManagers () const
- QtTimePropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtTimePropertyManager *manager) override
- QWidget * **createEditor** (QtTimePropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtTimePropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtTimePropertyManager >

- virtual QWidget * createEditor (QtTimePropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtTimePropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtTimePropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.780.1 Constructor & Destructor Documentation

17.780.1.1 QtTimeEditFactory()

```
QtTimeEditFactory::QtTimeEditFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.780.1.2 ~QtTimeEditFactory()

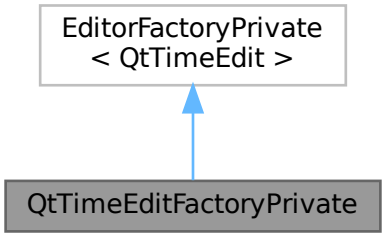
```
QtTimeEditFactory::~QtTimeEditFactory ( )
```

Destroys this factory, and all the widgets it has created.
The documentation for this class was generated from the following files:

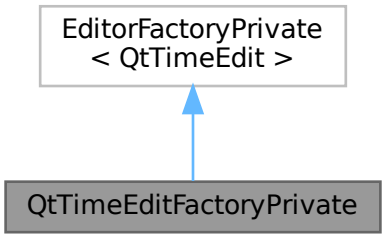
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.781 QtTimeEditFactoryPrivate Class Reference

Inheritance diagram for QtTimeEditFactoryPrivate:



Collaboration diagram for QtTimeEditFactoryPrivate:



Public Member Functions

- void **slotPropertyChanged** (QtProperty *property, QTime value)
- void **slotSetValue** (QTime value)

Public Member Functions inherited from EditorFactoryPrivate< QtTimeEdit >

- QtTimeEdit * **createEditor** (QtProperty *property, QWidget *parent)
- void **initializeEditor** (QtProperty *property, QtTimeEdit *e)
- void **slotEditorDestroyed** (QObject *object)

Additional Inherited Members

Public Types inherited from EditorFactoryPrivate< QtTimeEdit >

- typedef QList< QtTimeEdit * > **EditorList**
- typedef QMap< QtProperty *, EditorList > **PropertyToEditorListMap**
- typedef QMap< QtTimeEdit *, QtProperty * > **EditorToPropertyMap**

Public Attributes inherited from EditorFactoryPrivate< QtTimeEdit >

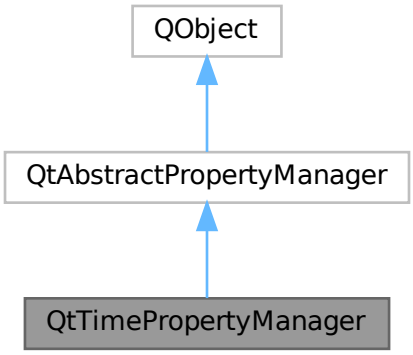
- PropertyToEditorListMap **m_createdEditors**
- EditorToPropertyMap **m_editorToProperty**

The documentation for this class was generated from the following file:

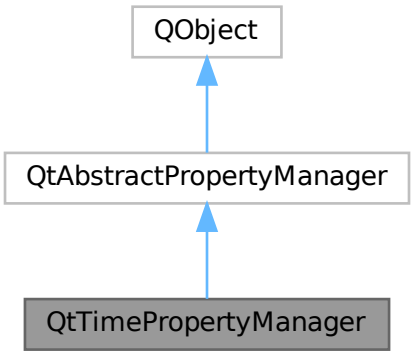
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.cpp

17.782 QtTimePropertyManager Class Reference

Inheritance diagram for QtTimePropertyManager:



Collaboration diagram for QtTimePropertyManager:



Public Slots

- void setValue (QtProperty *property, QTime val)
- void **setValueOnly** (QtProperty *property, QTime val)

Signals

- void valueChanged (QtProperty *property, QTime val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)
- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtTimePropertyManager (QObject *parent=0)
- ~QtTimePropertyManager ()
- QTime value (const QtProperty *property) const
- QTime **initialValue** (const QtProperty *property) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Protected Member Functions

- QString valueText (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override

Protected Member Functions inherited from QtAbstractPropertyManager

- virtual bool hasValue (const QtProperty *property) const
- virtual QIcon valueIcon (const QtProperty *property) const
- virtual QtProperty * createProperty ()

17.782.1 Constructor & Destructor Documentation

17.782.1.1 QtTimePropertyManager()

```
QtTimePropertyManager::QtTimePropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.782.1.2 ~QtTimePropertyManager()

```
QtTimePropertyManager::~QtTimePropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.782.2 Member Function Documentation

17.782.2.1 initializeProperty()

```
void QtTimePropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyManager.

17.782.2.2 setValue

```
void QtTimePropertyManager::setValue (
            QtProperty * property,
            QTime value )  [slot]
```

Sets the value of the given *property* to *value*.

See also

value(), valueChanged()

17.782.2.3 uninitializeProperty()

```
void QtTimePropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.

17.782.2.4 value()

```
QTime QtTimePropertyManager::value (
            const QtProperty * property ) const
```

Returns the given *property's* value.
If the given property is not managed by this manager, this function returns an invalid time object.

See also

setValue()

17.782.2.5 valueChanged

```
void QtTimePropertyManager::valueChanged (
            QtProperty * property,
            QTime value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.782.2.6 valueText()

```
QString QtTimePropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

\reimp
Reimplemented from QtAbstractPropertyManager.
The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.783 QtTimePropertyManagerPrivate Class Reference

Classes

- struct Data

Public Types

- typedef QMap< const QtProperty *, Data > **PropertyValueMap**

Public Member Functions

- **QtTimePropertyManagerPrivate** (QtTimePropertyManager *q)

Public Attributes

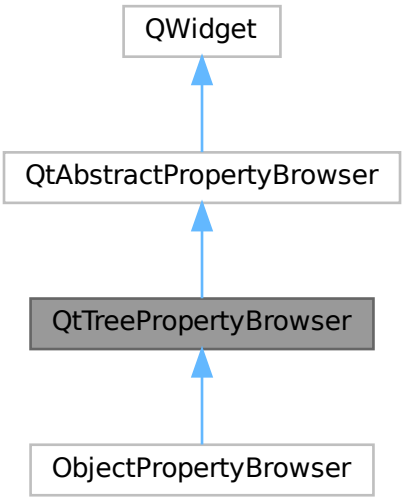
- const QString **m_format**
- PropertyValueMap **m_values**

The documentation for this class was generated from the following file:

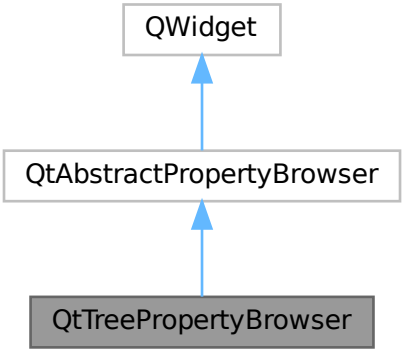
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.cpp

17.784 QtTreePropertyBrowser Class Reference

Inheritance diagram for QtTreePropertyBrowser:



Collaboration diagram for QtTreePropertyBrowser:



Public Types

- enum ResizeMode { **Interactive** , **Stretch** , **Fixed** , **ResizeToContents** }

Signals

- void collapsed (QtBrowserItem ∗item)
- void expanded (QtBrowserItem ∗item)

Signals inherited from QtAbstractPropertyBrowser

- void currentItemChanged (QtBrowserItem ∗)

Public Member Functions

- QtTreePropertyBrowser (QWidget ∗parent=0)
- ∼QtTreePropertyBrowser ()
- int **indentation** () const
- void **setIndentation** (int i)
- bool **rootIsDecorated** () const
- void **setRootIsDecorated** (bool show)
- bool **alternatingRowColors** () const
- void **setAlternatingRowColors** (bool enable)
- bool **isHeaderVisible** () const
- void **setHeaderVisible** (bool visible)
- ResizeMode **resizeMode** () const
- void **setResizeMode** (ResizeMode mode)
- int **splitterPosition** () const
- void **setSplitterPosition** (int position)
- void setExpanded (QtBrowserItem ∗item, bool expanded)
- bool isExpanded (QtBrowserItem ∗item) const
- bool isItemVisible (QtBrowserItem ∗item) const
- void setItemVisible (QtBrowserItem ∗item, bool visible)
- void setBackgroundColor (QtBrowserItem ∗item, const QColor &color)
- QColor backgroundColor (QtBrowserItem ∗item) const
- QColor calculatedBackgroundColor (QtBrowserItem ∗item) const

- void **setPropertiesWithoutValueMarked** (bool mark)
- bool **propertiesWithoutValueMarked** () const
- void editItem (QtBrowserItem *item)

Public Member Functions inherited from QtAbstractPropertyBrowser

- QtAbstractPropertyBrowser (QWidget *parent=0)
- ~QtAbstractPropertyBrowser ()
- QList< QtProperty * > properties () const
- QList< QtBrowserItem * > items (QtProperty *property) const
- QtBrowserItem * topLevelItem (QtProperty *property) const
- QList< QtBrowserItem * > topLevelItems () const
- void clear ()
- template<class PropertyManager >
  void setFactoryForManager (PropertyManager *manager, QtAbstractEditorFactory< PropertyManager > *factory)
- void unsetFactoryForManager (QtAbstractPropertyManager *manager)
- QtBrowserItem * currentItem () const
- void setCurrentItem (QtBrowserItem *)

Protected Member Functions

- void itemInserted (QtBrowserItem *item, QtBrowserItem *afterItem) override
- void itemRemoved (QtBrowserItem *item) override
- void itemChanged (QtBrowserItem *item) override

Protected Member Functions inherited from QtAbstractPropertyBrowser

- virtual QWidget * createEditor (QtProperty *property, QWidget *parent)

Properties

- int **indentation**

  indentation of the items in the tree view.
- bool **rootIsDecorated**

  whether to show controls for expanding and collapsing root items.
- bool **alternatingRowColors**

  whether to draw the background using alternating colors. By default this property is set to true.
- bool **headerVisible**

  whether to show the header.
- ResizeMode **resizeMode**

  the resize mode of setions in the header.
- int **splitterPosition**

  the position of the splitter between the colunms.
- bool propertiesWithoutValueMarked

  whether to enable or disable marking properties without value.

Additional Inherited Members

Public Slots inherited from QtAbstractPropertyBrowser

- QtBrowserItem * addProperty (QtProperty *property)
- QtBrowserItem * insertProperty (QtProperty *property, QtProperty *afterProperty)
- void removeProperty (QtProperty *property)

17.784.1 Member Enumeration Documentation

17.784.1.1 ResizeMode

enum QtTreePropertyBrowser::ResizeMode
The resize mode specifies the behavior of the header sections.
\value Interactive The user can resize the sections. The sections can also be resized programmatically using setSplitterPosition().
\value Fixed The user cannot resize the section. The section can only be resized programmatically using setSplitterPosition().
\value Stretch QHeaderView will automatically resize the section to fill the available space. The size cannot be changed by the user or programmatically.
\value ResizeToContents QHeaderView will automatically resize the section to its optimal size based on the contents of the entire column. The size cannot be changed by the user or programmatically.

See also

setResizeMode()

17.784.2 Constructor & Destructor Documentation

17.784.2.1 QtTreePropertyBrowser()

```
QtTreePropertyBrowser::QtTreePropertyBrowser (
            QWidget * parent = 0 )
```

Creates a property browser with the given *parent*.

17.784.2.2 ~QtTreePropertyBrowser()

```
QtTreePropertyBrowser::~QtTreePropertyBrowser ( )
```

Destroys this property browser.
Note that the properties that were inserted into this browser are *not* destroyed since they may still be used in other browsers. The properties are owned by the manager that created them.

See also

QtProperty, QtAbstractPropertyManager

17.784.3 Member Function Documentation

17.784.3.1 backgroundColor()

```
QColor QtTreePropertyBrowser::backgroundColor (
            QtBrowserItem * item ) const
```

Returns the *item's* color. If there is no color set for item it returns invalid color.

See also

calculatedBackgroundColor(), setBackgroundColor()

17.784.3.2 calculatedBackgroundColor()

```
QColor QtTreePropertyBrowser::calculatedBackgroundColor (
            QtBrowserItem * item ) const
```

Returns the *item's* color. If there is no color set for item it returns parent *item's* color (if there is no color set for parent it returns grandparent's color and so on). In case the color is not set for *item* and it's top level item it returns invalid color.

See also

backgroundColor(), setBackgroundColor()

17.784.3.3 collapsed

```
void QtTreePropertyBrowser::collapsed (
            QtBrowserItem * item )  [signal]
```

This signal is emitted when the *item* is collapsed.

See also

expanded(), setExpanded()

17.784.3.4 editItem()

```
void QtTreePropertyBrowser::editItem (
            QtBrowserItem * item )
```

Sets the current item to *item* and opens the relevant editor for it.

17.784.3.5 expanded

```
void QtTreePropertyBrowser::expanded (
            QtBrowserItem * item )  [signal]
```

This signal is emitted when the *item* is expanded.

See also

collapsed(), setExpanded()

17.784.3.6 isExpanded()

```
bool QtTreePropertyBrowser::isExpanded (
            QtBrowserItem * item ) const
```

Returns true if the *item* is expanded; otherwise returns false.

See also

setExpanded()

17.784.3.7 isItemVisible()

```
bool QtTreePropertyBrowser::isItemVisible (
            QtBrowserItem * item ) const
```

Returns true if the *item* is visible; otherwise returns false.

See also

setItemVisible()

Since

4.5

17.784.3.8 itemChanged()

```
void QtTreePropertyBrowser::itemChanged (
            QtBrowserItem * item )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyBrowser.

17.784.3.9 itemInserted()

```
void QtTreePropertyBrowser::itemInserted (
            QtBrowserItem * item,
            QtBrowserItem * afterItem )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyBrowser.

17.784.3.10 itemRemoved()

```
void QtTreePropertyBrowser::itemRemoved (
            QtBrowserItem * item )  [override], [protected], [virtual]
```

\reimp
Implements QtAbstractPropertyBrowser.

17.784.3.11 setBackgroundColor()

```
void QtTreePropertyBrowser::setBackgroundColor (
            QtBrowserItem * item,
            const QColor & color )
```

Sets the *item's* background color to *color*. Note that while item's background is rendered every second row is being drawn with alternate color (which is a bit lighter than items *color*)

See also

backgroundColor(), calculatedBackgroundColor()

17.784.3.12 setExpanded()

```
void QtTreePropertyBrowser::setExpanded (
            QtBrowserItem * item,
            bool expanded )
```

Sets the *item* to either collapse or expanded, depending on the value of *expanded*.

See also

isExpanded(), expanded(), collapsed()

17.784.3.13 setItemVisible()

```
void QtTreePropertyBrowser::setItemVisible (
            QtBrowserItem * item,
            bool visible )
```

Sets the *item* to be visible, depending on the value of *visible*.

See also

isItemVisible()

Since

4.5

17.784.4 Property Documentation

17.784.4.1 propertiesWithoutValueMarked

```
QtTreePropertyBrowser::propertiesWithoutValueMarked  [read], [write]
```

whether to enable or disable marking properties without value.
When marking is enabled the item's background is rendered in dark color and item's foreground is rendered with light color.

See also

propertiesWithoutValueMarked()

The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qttreepropertybrowser.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qttreepropertybrowser.cpp

17.785 QtTreePropertyBrowserPrivate Class Reference

Public Member Functions

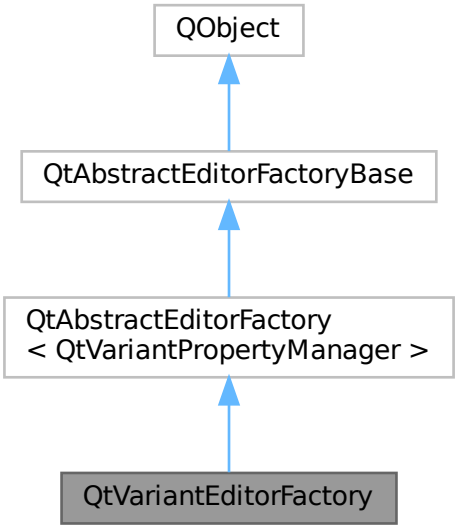
- void **init** (QWidget *parent)
- void **propertyInserted** (QtBrowserItem *index, QtBrowserItem *afterIndex)
- void **propertyRemoved** (QtBrowserItem *index)
- void **propertyChanged** (QtBrowserItem *index)
- QWidget * **createEditor** (QtProperty *property, QWidget *parent) const
- QtProperty * **indexToProperty** (const QModelIndex &index) const
- QTreeWidgetItem * **indexToItem** (const QModelIndex &index) const
- QtBrowserItem * **indexToBrowserItem** (const QModelIndex &index) const
- bool **lastColumn** (int column) const
- void **disableItem** (QTreeWidgetItem *item) const
- void **enableItem** (QTreeWidgetItem *item) const
- bool **hasValue** (QTreeWidgetItem *item) const
- void **slotCollapsed** (const QModelIndex &index)
- void **slotExpanded** (const QModelIndex &index)
- QColor **calculatedBackgroundColor** (QtBrowserItem *item) const
- QtPropertyEditorView * **treeWidget** () const
- bool **markPropertiesWithoutValue** () const
- QtBrowserItem * **currentItem** () const
- void **setCurrentItem** (QtBrowserItem *browserItem, bool block)
- void **editItem** (QtBrowserItem *browserItem)
- void **slotCurrentBrowserItemChanged** (QtBrowserItem *item)
- void **slotCurrentTreeItemChanged** (QTreeWidgetItem *newItem, QTreeWidgetItem *)
- QTreeWidgetItem * **editedItem** () const

The documentation for this class was generated from the following file:

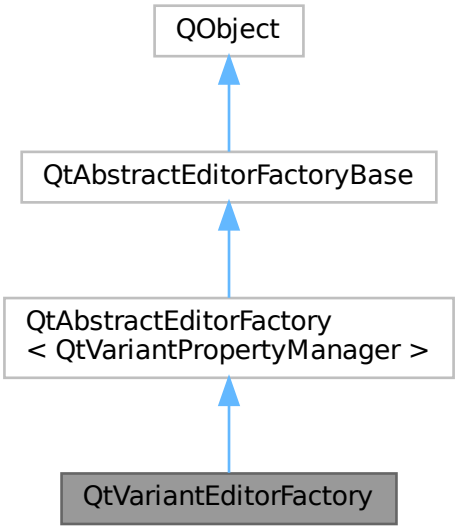
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qttreepropertybrowser.cpp

17.786 QtVariantEditorFactory Class Reference

Inheritance diagram for QtVariantEditorFactory:



Collaboration diagram for QtVariantEditorFactory:



Public Member Functions

- QtVariantEditorFactory (QObject *parent=0)

- ~QtVariantEditorFactory ()

Public Member Functions inherited from QtAbstractEditorFactory< QtVariantPropertyManager >

- QtAbstractEditorFactory (QObject *parent)
- QWidget * createEditor (QtProperty *property, QWidget *parent) override
- void addPropertyManager (QtVariantPropertyManager *manager)
- void removePropertyManager (QtVariantPropertyManager *manager)
- QSet< QtVariantPropertyManager * > propertyManagers () const
- QtVariantPropertyManager * propertyManager (QtProperty *property) const

Protected Member Functions

- void **connectPropertyManager** (QtVariantPropertyManager *manager) override
- QWidget * **createEditor** (QtVariantPropertyManager *manager, QtProperty *property, QWidget *parent) override
- void **disconnectPropertyManager** (QtVariantPropertyManager *manager) override

Protected Member Functions inherited from QtAbstractEditorFactory< QtVariantPropertyManager >

- virtual QWidget * createEditor (QtVariantPropertyManager *manager, QtProperty *property, QWidget *parent)=0
- virtual void connectPropertyManager (QtVariantPropertyManager *manager)=0
- virtual void disconnectPropertyManager (QtVariantPropertyManager *manager)=0
- void managerDestroyed (QObject *manager) override

Protected Member Functions inherited from QtAbstractEditorFactoryBase

- QtAbstractEditorFactoryBase (QObject *parent=0)

Additional Inherited Members

Protected Slots inherited from QtAbstractEditorFactoryBase

17.786.1 Constructor & Destructor Documentation

17.786.1.1 QtVariantEditorFactory()

```
QtVariantEditorFactory::QtVariantEditorFactory (
            QObject * parent = 0 )
```

Creates a factory with the given *parent*.

17.786.1.2 ~QtVariantEditorFactory()

```
QtVariantEditorFactory::~QtVariantEditorFactory ( )
```

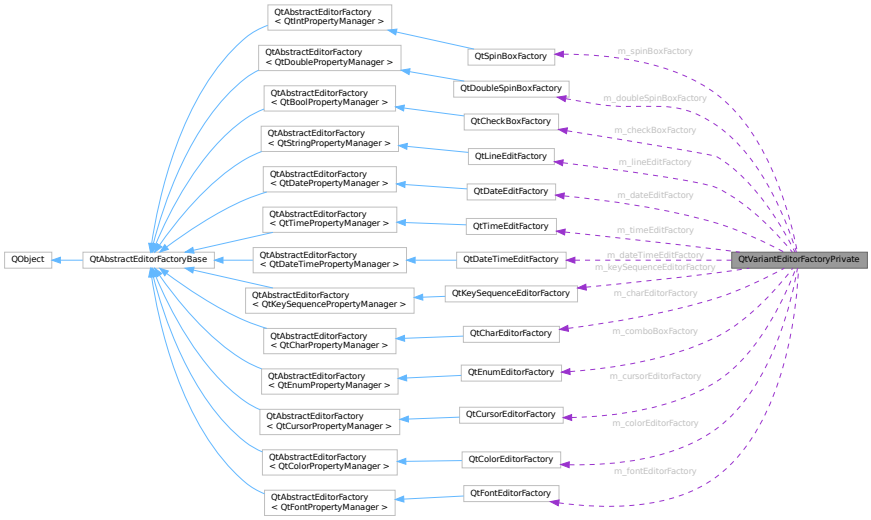
Destroys this factory, and all the widgets it has created.

The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.cpp

17.787 QtVariantEditorFactoryPrivate Class Reference

Collaboration diagram for QtVariantEditorFactoryPrivate:



Public Attributes

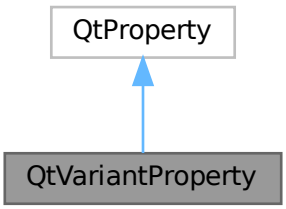
- QtSpinBoxFactory * **m_spinBoxFactory**
- QtDoubleSpinBoxFactory * **m_doubleSpinBoxFactory**
- QtCheckBoxFactory * **m_checkBoxFactory**
- QtLineEditFactory * **m_lineEditFactory**
- QtDateEditFactory * **m_dateEditFactory**
- QtTimeEditFactory * **m_timeEditFactory**
- QtDateTimeEditFactory * **m_dateTimeEditFactory**
- QtKeySequenceEditorFactory * **m_keySequenceEditorFactory**
- QtCharEditorFactory * **m_charEditorFactory**
- QtEnumEditorFactory * **m_comboBoxFactory**
- QtCursorEditorFactory * **m_cursorEditorFactory**
- QtColorEditorFactory * **m_colorEditorFactory**
- QtFontEditorFactory * **m_fontEditorFactory**
- QMap< QtAbstractEditorFactoryBase *, int > **m_factoryToType**
- QMap< int, QtAbstractEditorFactoryBase * > **m_typeToFactory**

The documentation for this class was generated from the following file:

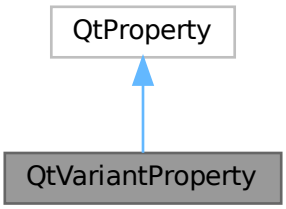
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.cpp

17.788 QtVariantProperty Class Reference

Inheritance diagram for QtVariantProperty:



Collaboration diagram for QtVariantProperty:



Public Member Functions

- ~QtVariantProperty ()
- QVariant value () const
- QVariant attributeValue (const QString &attribute) const
- int valueType () const
- int propertyType () const
- void setValue (const QVariant &value)
- void setAttribute (const QString &attribute, const QVariant &value)

Public Member Functions inherited from QtProperty

- virtual ~QtProperty ()
- QList< QtProperty * > subProperties () const
- QtAbstractPropertyManager * propertyManager () const
- QString **toolTip** () const
- QString valueToolTip () const
- QString descriptionToolTip () const
- QString statusTip () const
- QString whatsThis () const
- QString propertyName () const
- bool isEnabled () const

- bool isModified () const
- bool hasValue () const
- QIcon valueIcon () const
- QString valueText () const
- void **setToolTip** (const QString &text)
- void setValueToolTip (const QString &text)
- void setDescriptionToolTip (const QString &text)
- void setStatusTip (const QString &text)
- void setWhatsThis (const QString &text)
- void setPropertyName (const QString &text)
- void setEnabled (bool enable)
- void setModified (bool modified)
- void addSubProperty (QtProperty *property)
- void insertSubProperty (QtProperty *property, QtProperty *afterProperty)
- void removeSubProperty (QtProperty *property)

Protected Member Functions

- QtVariantProperty (QtVariantPropertyManager *manager)

Protected Member Functions inherited from QtProperty

- QtProperty (QtAbstractPropertyManager *manager)
- void **propertyChanged** ()

Friends

- class **QtVariantPropertyManager**

17.788.1 Constructor & Destructor Documentation

17.788.1.1 ~QtVariantProperty()

```
QtVariantProperty::~QtVariantProperty ( )
```

Destroys this property.

See also

QtProperty::~QtProperty()

17.788.1.2 QtVariantProperty()

```
QtVariantProperty::QtVariantProperty (
            QtVariantPropertyManager * manager )  [protected]
```

Creates a variant property using the given *manager*.

Do not use this constructor to create variant property instances; use the QtVariantPropertyManager::addProperty() function instead. This constructor is used internally by the QtVariantPropertyManager::createProperty() function.

See also

QtVariantPropertyManager

17.788.2 Member Function Documentation

17.788.2.1 attributeValue()

```
QVariant QtVariantProperty::attributeValue (
            const QString & attribute ) const
```

Returns this property's value for the specified *attribute*.

QtVariantPropertyManager provides a couple of related functions: \l{QtVariantPropertyManager::attributes()}{attributes()} and \l{QtVariantPropertyManager::attributeType()}{attributeType()}.

See also

setAttribute()

17.788.2.2 propertyType()

```
int QtVariantProperty::propertyType ( ) const
```

Returns this property's type.
QtVariantPropertyManager provides several related functions: \l{QtVariantPropertyManager::enumTypeId()}{enumTypeId()}, \l{QtVariantPropertyManager::flagTypeId()}{flagTypeId()} and \l{QtVariantPropertyManager::groupTypeId()}{groupTypeId()}.

See also

valueType()

17.788.2.3 setAttribute()

```
void QtVariantProperty::setAttribute (
            const QString & attribute,
            const QVariant & value )
```

Sets the *attribute* of property to *value*.
QtVariantPropertyManager provides the related \l{QtVariantPropertyManager::setAttribute()}{setAttribute()} function.

See also

attributeValue()

17.788.2.4 setValue()

```
void QtVariantProperty::setValue (
            const QVariant & value )
```

Sets the value of this property to *value*.
The specified *value* must be of the type returned by valueType(), or of a type that can be converted to valueType() using the QVariant::canConvert() function; otherwise this function does nothing.

See also

value()

17.788.2.5 value()

```
QVariant QtVariantProperty::value ( ) const
```

Returns the property's current value.

See also

valueType(), setValue()

17.788.2.6 valueType()

```
int QtVariantProperty::valueType ( ) const
```

Returns the type of this property's value.

See also

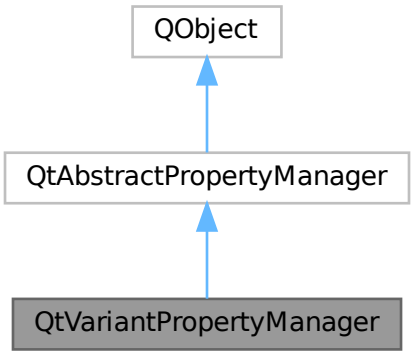
propertyType()

The documentation for this class was generated from the following files:

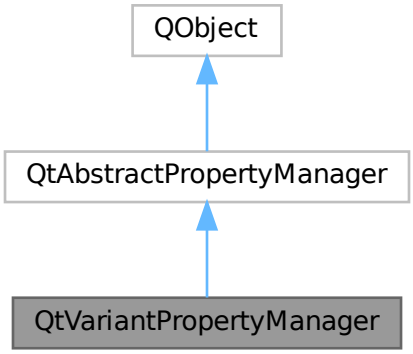
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.cpp

17.789 QtVariantPropertyManager Class Reference

Inheritance diagram for QtVariantPropertyManager:



Collaboration diagram for QtVariantPropertyManager:



Public Slots

- virtual void setValue (QtProperty *property, const QVariant &val)
- virtual void setAttribute (QtProperty *property, const QString &attribute, const QVariant &value)

Signals

- void valueChanged (QtProperty *property, const QVariant &val)
- void attributeChanged (QtProperty *property, const QString &attribute, const QVariant &val)

Signals inherited from QtAbstractPropertyManager

- void propertyInserted (QtProperty *property, QtProperty *parent, QtProperty *after)
- void propertyChanged (QtProperty *property)

- void propertyRemoved (QtProperty *property, QtProperty *parent)
- void propertyDestroyed (QtProperty *property)

Public Member Functions

- QtVariantPropertyManager (QObject *parent=0)
- ~QtVariantPropertyManager ()
- virtual QtVariantProperty * addProperty (int propertyType, const QString &name=QString())
- int propertyType (const QtProperty *property) const
- int valueType (const QtProperty *property) const
- QtVariantProperty * variantProperty (const QtProperty *property) const
- virtual bool isPropertyTypeSupported (int propertyType) const
- virtual int valueType (int propertyType) const
- virtual QStringList attributes (int propertyType) const
- virtual int attributeType (int propertyType, const QString &attribute) const
- virtual QVariant value (const QtProperty *property) const
- virtual QVariant attributeValue (const QtProperty *property, const QString &attribute) const

Public Member Functions inherited from QtAbstractPropertyManager

- QtAbstractPropertyManager (QObject *parent=0)
- ~QtAbstractPropertyManager ()
- QSet< QtProperty * > properties () const
- void clear () const
- QtProperty * addProperty (const QString &name=QString())

Static Public Member Functions

- static int enumTypeId ()
- static int flagTypeId ()
- static int groupTypeId ()
- static int iconMapTypeId ()

Protected Member Functions

- bool hasValue (const QtProperty *property) const override
- QString valueText (const QtProperty *property) const override
- QIcon valueIcon (const QtProperty *property) const override
- void initializeProperty (QtProperty *property) override
- void uninitializeProperty (QtProperty *property) override
- QtProperty * createProperty () override

17.789.1 Constructor & Destructor Documentation

17.789.1.1 QtVariantPropertyManager()

```
QtVariantPropertyManager::QtVariantPropertyManager (
            QObject * parent = 0 )
```

Creates a manager with the given *parent*.

17.789.1.2 ~QtVariantPropertyManager()

```
QtVariantPropertyManager::~QtVariantPropertyManager ( )
```

Destroys this manager, and all the properties it has created.

17.789.2 Member Function Documentation

17.789.2.1 addProperty()

```
QtVariantProperty * QtVariantPropertyManager::addProperty (
            int propertyType,
            const QString & name = QString() )  [virtual]
```

Creates and returns a variant property of the given *propertyType* with the given *name*.
If the specified *propertyType* is not supported by this variant manager, this function returns 0.
Do not use the inherited QtAbstractPropertyManager::addProperty() function to create a variant property (that function will always return 0 since it will not be clear what type the property should have).

See also

isPropertyTypeSupported()

17.789.2.2 attributeChanged

```
void QtVariantPropertyManager::attributeChanged (
            QtProperty * property,
            const QString & attribute,
            const QVariant & value )  [signal]
```

This signal is emitted whenever an attribute of a property created by this manager changes its value, passing a pointer to the *property*, the *attribute* and the new *value* as parameters.

See also

setAttribute()

17.789.2.3 attributes()

```
QStringList QtVariantPropertyManager::attributes (
            int propertyType ) const  [virtual]
```

Returns a list of the given *propertyType* 's attributes.

See also

attributeValue(), attributeType()

17.789.2.4 attributeType()

```
int QtVariantPropertyManager::attributeType (
            int propertyType,
            const QString & attribute ) const  [virtual]
```

Returns the type of the specified *attribute* of the given *propertyType*.
If the given *propertyType* is not supported by *this* manager, or if the given *propertyType* does not possess the specified *attribute*, this function returns QMetaType::UnknownType.

See also

attributes(), valueType()

17.789.2.5 attributeValue()

```
QVariant QtVariantPropertyManager::attributeValue (
            const QtProperty * property,
            const QString & attribute ) const  [virtual]
```

Returns the given *property's* value for the specified *attribute*
If the given *property* was not created by *this* manager, or if the specified *attribute* does not exist, this function returns an invalid variant.

See also

attributes(), attributeType(), setAttribute()

17.789.2.6 createProperty()

```
QtProperty * QtVariantPropertyManager::createProperty ( )  [override], [protected], [virtual]
```

Creates a property.
The base implementation produce QtProperty instances; Reimplement this function to make this manager produce objects of a QtProperty subclass.

See also

addProperty(), initializeProperty()

Reimplemented from QtAbstractPropertyManager.

17.789.2.7 enumTypeId()

```
QT_END_NAMESPACE QT_BEGIN_NAMESPACE int QtVariantPropertyManager::enumTypeId ( )  [static]
```

Returns the type id for an enum property.
Note that the property's value type can be retrieved using the valueType() function (which is QMetaType::Int for the enum property type).

See also

propertyType(), valueType()

17.789.2.8 flagTypeId()

```
int QtVariantPropertyManager::flagTypeId ( )  [static]
```

Returns the type id for a flag property.
Note that the property's value type can be retrieved using the valueType() function (which is QMetaType::Int for the flag property type).

See also

propertyType(), valueType()

17.789.2.9 groupTypeId()

```
int QtVariantPropertyManager::groupTypeId ( )  [static]
```

Returns the type id for a group property.
Note that the property's value type can be retrieved using the valueType() function (which is QMetaType::Unknown↩ Type for the group property type, since it doesn't provide any value).

See also

propertyType(), valueType()

17.789.2.10 hasValue()

```
bool QtVariantPropertyManager::hasValue (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

Returns whether the given *property* has a value.
The default implementation of this function returns true.

See also

QtProperty::hasValue()

Reimplemented from QtAbstractPropertyManager.

17.789.2.11 iconMapTypeId()

```
int QtVariantPropertyManager::iconMapTypeId ( )  [static]
```

Returns the type id for a icon map attribute.
Note that the property's attribute type can be retrieved using the attributeType() function.

See also

attributeType(), QtEnumPropertyManager::enumIcons()

17.789.2.12 initializeProperty()

```
void QtVariantPropertyManager::initializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

This function is called whenever a new valid property pointer has been created, passing the pointer as parameter.
The purpose is to let the manager know that the *property* has been created so that it can provide additional attributes for the new property, e.g. QtIntPropertyManager adds \l {QtIntPropertyManager::value()}{value}, \l {QtIntPropertyManager::minimum()}{minimum} and \l {QtIntPropertyManager::maximum()}{maximum} attributes. Since each manager subclass adds type specific attributes, this function is pure virtual and must be reimplemented when deriving from the QtAbstractPropertyManager class.

See also

addProperty(), createProperty()

Implements QtAbstractPropertyManager.

17.789.2.13 isPropertyTypeSupported()

```
bool QtVariantPropertyManager::isPropertyTypeSupported (
            int propertyType ) const  [virtual]
```

Returns true if the given *propertyType* is supported by this variant manager; otherwise false.

See also

propertyType()

17.789.2.14 propertyType()

```
int QtVariantPropertyManager::propertyType (
            const QtProperty * property ) const
```

Returns the given *property's* type.

See also

valueType()

17.789.2.15 setAttribute

```
void QtVariantPropertyManager::setAttribute (
            QtProperty * property,
            const QString & attribute,
            const QVariant & value )  [virtual], [slot]
```

Sets the value of the specified *attribute* of the given *property*, to *value*.
The new *value's* type must be of the type returned by attributeType(), or of a type that can be converted to attributeType() using the QVariant::canConvert() function, otherwise this function does nothing.

See also

attributeValue(), QtVariantProperty::setAttribute(), attributeChanged()

17.789.2.16 setValue

```
void QtVariantPropertyManager::setValue (
            QtProperty * property,
            const QVariant & value )  [virtual], [slot]
```

Sets the value of the given *property* to *value*.
The specified *value* must be of a type returned by valueType(), or of type that can be converted to valueType() using the QVariant::canConvert() function, otherwise this function does nothing.

See also

value(), QtVariantProperty::setValue(), valueChanged()

17.789.2.17 uninitializeProperty()

```
void QtVariantPropertyManager::uninitializeProperty (
            QtProperty * property )  [override], [protected], [virtual]
```

This function is called just before the specified *property* is destroyed.
The purpose is to let the property manager know that the *property* is being destroyed so that it can remove the property's additional attributes.

See also

clear(), propertyDestroyed()

Reimplemented from QtAbstractPropertyManager.

17.789.2.18 value()

```
QVariant QtVariantPropertyManager::value (
            const QtProperty * property ) const  [virtual]
```

Returns the given *property's* value.
If the given *property* is not managed by this manager, this function returns an invalid variant.

See also

setValue()

17.789.2.19 valueChanged

```
void QtVariantPropertyManager::valueChanged (
            QtProperty * property,
            const QVariant & value )  [signal]
```

This signal is emitted whenever a property created by this manager changes its value, passing a pointer to the *property* and the new *value* as parameters.

See also

setValue()

17.789.2.20 valueIcon()

```
QIcon QtVariantPropertyManager::valueIcon (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

Returns an icon representing the current state of the given *property*.
The default implementation of this function returns an invalid icon.

See also

QtProperty::valueIcon()

Reimplemented from QtAbstractPropertyManager.

17.789.2.21 valueText()

```
QString QtVariantPropertyManager::valueText (
            const QtProperty * property ) const  [override], [protected], [virtual]
```

Returns a string representing the current state of the given *property*.
The default implementation of this function returns an empty string.

See also

QtProperty::valueText()

Reimplemented from QtAbstractPropertyManager.

17.789.2.22 valueType() [1/2]

```
int QtVariantPropertyManager::valueType (
            const QtProperty * property ) const
```

Returns the given *property's* value type.

See also

propertyType()

17.789.2.23 valueType() [2/2]

```
int QtVariantPropertyManager::valueType (
            int propertyType ) const  [virtual]
```

This is an overloaded member function, provided for convenience. It differs from the above function only in what argument(s) it accepts.
Returns the value type associated with the given *propertyType*.

17.789.2.24 variantProperty()

```
QtVariantProperty * QtVariantPropertyManager::variantProperty (
            const QtProperty * property ) const
```

Returns the given *property* converted into a QtVariantProperty.
If the *property* was not created by this variant manager, the function returns 0.

See also

createProperty()

The documentation for this class was generated from the following files:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.h
- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.cpp

17.790 QtVariantPropertyManagerPrivate Class Reference

Public Member Functions

- void **slotValueChanged** (QtProperty *property, int val)
- void **slotRangeChanged** (QtProperty *property, int min, int max)
- void **slotSingleStepChanged** (QtProperty *property, int step)
- void **slotValueChanged** (QtProperty *property, double val)
- void **slotRangeChanged** (QtProperty *property, double min, double max)
- void **slotSingleStepChanged** (QtProperty *property, double step)
- void **slotDecimalsChanged** (QtProperty *property, int prec)
- void **slotValueChanged** (QtProperty *property, bool val)
- void **slotValueChanged** (QtProperty *property, const QString &val)
- void **slotRegExpChanged** (QtProperty *property, const QRegularExpression &regExp)

- void **slotValueChanged** (QtProperty *property, QDate val)
- void **slotRangeChanged** (QtProperty *property, QDate min, QDate max)
- void **slotValueChanged** (QtProperty *property, QTime val)
- void **slotValueChanged** (QtProperty *property, const QDateTime &val)
- void **slotValueChanged** (QtProperty *property, const QKeySequence &val)
- void **slotValueChanged** (QtProperty *property, const QChar &val)
- void **slotValueChanged** (QtProperty *property, const QLocale &val)
- void **slotValueChanged** (QtProperty *property, const QPoint &val)
- void **slotValueChanged** (QtProperty *property, const QPointF &val)
- void **slotValueChanged** (QtProperty *property, const QSize &val)
- void **slotRangeChanged** (QtProperty *property, const QSize &min, const QSize &max)
- void **slotValueChanged** (QtProperty *property, const QSizeF &val)
- void **slotRangeChanged** (QtProperty *property, const QSizeF &min, const QSizeF &max)
- void **slotValueChanged** (QtProperty *property, const QRect &val)
- void **slotConstraintChanged** (QtProperty *property, const QRect &val)
- void **slotValueChanged** (QtProperty *property, const QRectF &val)
- void **slotConstraintChanged** (QtProperty *property, const QRectF &val)
- void **slotValueChanged** (QtProperty *property, const QColor &val)
- void **slotEnumChanged** (QtProperty *property, int val)
- void **slotEnumNamesChanged** (QtProperty *property, const QStringList &enumNames)
- void **slotEnumIconsChanged** (QtProperty *property, const QMap< int, QIcon > &enumIcons)
- void **slotValueChanged** (QtProperty *property, const QSizePolicy &val)
- void **slotValueChanged** (QtProperty *property, const QFont &val)
- void **slotValueChanged** (QtProperty *property, const QCursor &val)
- void **slotFlagChanged** (QtProperty *property, int val)
- void **slotFlagNamesChanged** (QtProperty *property, const QStringList &flagNames)
- void **slotPropertyInserted** (QtProperty *property, QtProperty *parent, QtProperty *after)
- void **slotPropertyRemoved** (QtProperty *property, QtProperty *parent)
- void **valueChanged** (QtProperty *property, const QVariant &val)
- int **internalPropertyToType** (QtProperty *property) const
- QtVariantProperty * **createSubProperty** (QtVariantProperty *parent, QtVariantProperty *after, QtProperty *internal)
- void **removeSubProperty** (QtVariantProperty *property)

Public Attributes

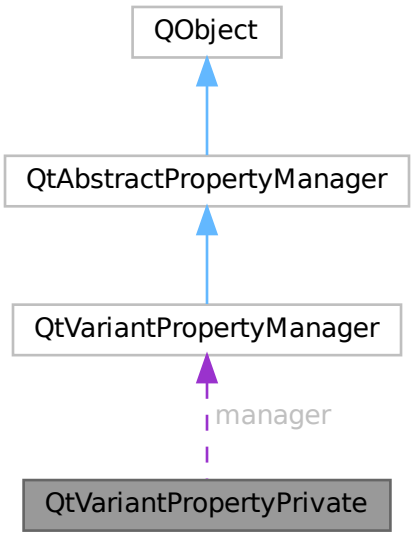
- bool **m_creatingProperty**
- bool **m_creatingSubProperties**
- bool **m_destroyingSubProperties**
- int **m_propertyType**
- QMap< int, QtAbstractPropertyManager * > **m_typeToPropertyManager**
- QMap< int, QMap< QString, int > > **m_typeToAttributeToAttributeType**
- QMap< const QtProperty *, QPair< QtVariantProperty *, int > > **m_propertyToType**
- QMap< int, int > **m_typeToValueType**
- QMap< QtProperty *, QtVariantProperty * > **m_internalToProperty**
- const QString **m_constraintAttribute**
- const QString **m_singleStepAttribute**
- const QString **m_decimalsAttribute**
- const QString **m_enumIconsAttribute**
- const QString **m_enumNamesAttribute**
- const QString **m_flagNamesAttribute**
- const QString **m_maximumAttribute**
- const QString **m_minimumAttribute**
- const QString **m_regExpAttribute**

The documentation for this class was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.cpp

17.791 QtVariantPropertyPrivate Class Reference

Collaboration diagram for QtVariantPropertyPrivate:



Public Member Functions

- **QtVariantPropertyPrivate** (QtVariantPropertyManager ∗m)

Public Attributes

- QtVariantPropertyManager ∗ **manager**

The documentation for this class was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.cpp

17.792 Quadrature_if Class Reference

One-dimensional numerical quadrature abstraction.
`#include <tools/Continuous/Quadrature_if.h>`

Public Member Functions

- virtual double **integrate** (double min, double max, double(∗f)(double), double precision) const =0

17.792.1 Detailed Description

One-dimensional numerical quadrature abstraction.
Architectural role:

- Planned extraction of integration responsibility from Solver_if.

The documentation for this class was generated from the following file:

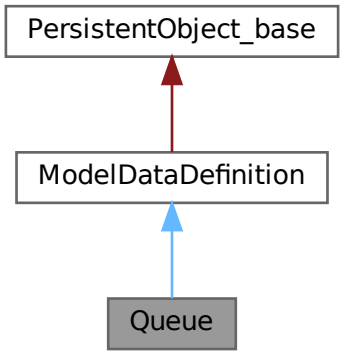
- tools/Continuous/Quadrature_if.h

17.793 Queue Class Reference

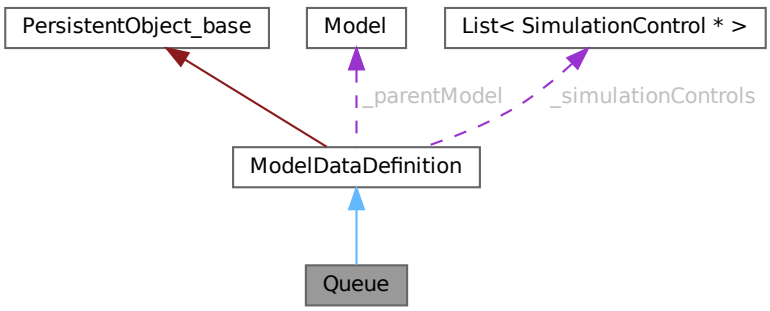
Data definition holding waiting entities for a component (a Seize-family component, Process, Batch, etc.) and defining their ranking rule.

`#include <plugins/data/DiscreteProcessing/Queue.h>`

Inheritance diagram for Queue:



Collaboration diagram for Queue:



Public Types

- enum class **OrderRule** : int {
  FIFO = 0 , **LIFO** = 1 , **HIGHESTVALUE** = 2 , **SMALLESTVALUE** = 3 ,
  num_elements = 4 }

Public Member Functions

- **Queue** (Model ∗model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **insertElement** (Waiting ∗modeldatum)
- void **removeElement** (Waiting ∗modeldatum)

- unsigned int **size** ()
- Waiting * **first** ()
- Waiting * **getAtRank** (unsigned int rank)
- void **setAttributeName** (std::string _attributeName)
- std::string **getAttributeName** () const
- void **setOrderRule** (OrderRule _orderRule)
- Queue::OrderRule **getOrderRule** () const
- void **setOrderRuleInt** (int orderRule)
- int **getOrderRuleInt** () const
- double **sumAttributesFromWaiting** (Util::identification attributeID)
- double **getAttributeFromWaitingRank** (unsigned int rank, Util::identification attributeID)

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static std::string **convertEnumToStr** (OrderRule rule)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.

- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.

- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.

- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.

- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes

- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

- virtual void _initBetweenReplications () override
- virtual ParserChangesInformation * _getParserChangesInformation () override
- virtual void _createInternalStatisticReporters () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.793.1 Detailed Description

Data definition holding waiting entities for a component (a Seize-family component, Process, Batch, etc.) and defining their ranking rule.
Arena correspondence: the "Queue module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 45-46). `OrderRule` mirrors Arena's four ranking rules (First In First Out, Last In First Out, lowest/highest attribute value first); `_attributeName` is the evaluated attribute for the two attribute-based rules. Known difference from Arena: this class has no explicit "Shared" flag. In GenESyS a Queue is simply a referenceable ModelDataDefinition, so any number of components may already reference the same instance; Arena needs an explicit opt-in because its queues are otherwise private to the module that created them.
`isReportStatistics()`/`setReportStatistics`() (inherited from ModelDataDefinition) correspond to Arena's per-module "Report Statistics" checkbox and gate the internal `StatisticsCollector` instances created by `_createInternalStatisticReporters()`.

17.793.2 Member Function Documentation

17.793.2.1 _check()

```
bool Queue::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.793.2.2 _createInternalStatisticReporters()

```
void Queue::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.793.2.3 _getParserChangesInformation()

```
ParserChangesInformation * Queue::_getParserChangesInformation ( )  [override], [protected],
[virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
Reimplemented from ModelDataDefinition.

17.793.2.4 _initBetweenReplications()

```
void Queue::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.793.2.5 _loadInstance()

```
bool Queue::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.793.2.6 _saveInstance()

```
void Queue::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.793.2.7 show()

```
std::string Queue::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/DiscreteProcessing/Queue.h
- plugins/data/DiscreteProcessing/Queue.cpp

17.794 QueueableItem Class Reference

Public Types

- enum class **QueueableType** : int { **QUEUE** = 0 , **SET** = 1 , **num_elements** = 2 }

Public Member Functions

- **QueueableItem** (ModelDataDefinition *queueOrSet, QueueableItem::QueueableType queueable↩ Type=QueueableItem::QueueableType::QUEUE, std::string index="0")
- **QueueableItem** (Model *model, std::string queueName)
- bool **loadInstance** (PersistenceRecord *fields)
- void **saveInstance** (PersistenceRecord *fields, bool saveDefaultValues)
- std::string **show** ()
- void **setIndex** (std::string index)
- std::string **getIndex** () const
- std::string **getQueueableName** () const
- std::string **getName** () const
- void **setQueue** (Queue *resource)
- Queue * **getQueue** () const
- void **setSet** (Set *set)

- Set * **getSet** () const
- void **setQueueableType** (QueueableType queueableType)
- QueueableType **getQueueableType** () const
- ModelDataDefinition * **getQueueable** () const
- void **setElementManager** (ModelDataManager *_modeldataManager)
- List< SimulationControl * > * **getSimulationControls** () const
- void **_addSimulationControl** (SimulationControl *control)
- std::string **getTypeDC** ()

Static Public Member Functions

- static std::string **convertEnumToStr** (QueueableType type)

The documentation for this class was generated from the following files:

- plugins/components/DiscreteProcessing/auxiliar/QueueableItem.h
- plugins/components/DiscreteProcessing/auxiliar/QueueableItem.cpp

17.795 testing::internal::Random Class Reference

Public Member Functions

- **Random** (uint32_t seed)
- void **Reseed** (uint32_t seed)
- uint32_t **Generate** (uint32_t range)

Static Public Attributes

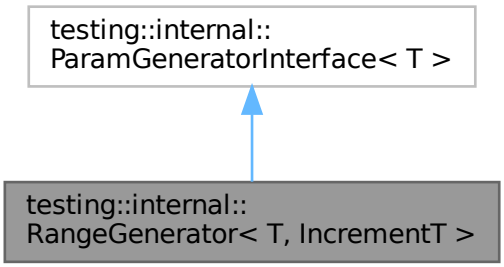
- static const uint32_t **kMaxRange** = 1u << 31

The documentation for this class was generated from the following files:

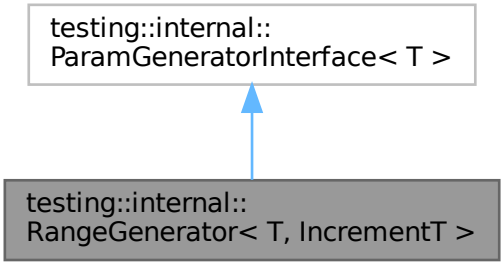
- gtest/gtest/internal/gtest-internal.h
- gtest/src/gtest.cc

17.796 testing::internal::RangeGenerator< T, IncrementT > Class Template Reference

Inheritance diagram for testing::internal::RangeGenerator< T, IncrementT >:



Collaboration diagram for testing::internal::RangeGenerator< T, IncrementT >:



Public Member Functions

- **RangeGenerator** (T begin, T end, IncrementT step)
- ParamIteratorInterface< T > * Begin () const override
- ParamIteratorInterface< T > * End () const override

Additional Inherited Members

Public Types inherited from testing::internal::ParamGeneratorInterface< T >

- typedef T **ParamType**

17.796.1 Member Function Documentation

17.796.1.1 Begin()

```
template<typename T , typename IncrementT >
ParamIteratorInterface< T > * testing::internal::RangeGenerator< T, IncrementT >::Begin ( )
const  [inline], [override], [virtual]
```

Implements testing::internal::ParamGeneratorInterface< T >.

17.796.1.2 End()

```
template<typename T , typename IncrementT >
ParamIteratorInterface< T > * testing::internal::RangeGenerator< T, IncrementT >::End ( )
const  [inline], [override], [virtual]
```

Implements testing::internal::ParamGeneratorInterface< T >.

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.797 testing::internal::RawBytesPrinter Struct Reference

Static Public Member Functions

- template<typename T , size_t = sizeof (T)>
  static void **PrintValue** (const T &value, ::std::ostream *os)

The documentation for this struct was generated from the following file:

- gtest/gtest/gtest-printers.h

17.798 testing::internal::RE Class Reference

Public Member Functions

- **RE** (const RE &other)
- **RE** (const ::std::string &regex)
- **RE** (const char *regex)
- const char * **pattern** () const

Static Public Member Functions

- static bool **FullMatch** (const ::std::string &str, const RE &re)
- static bool **PartialMatch** (const ::std::string &str, const RE &re)
- static bool **FullMatch** (const char *str, const RE &re)
- static bool **PartialMatch** (const char *str, const RE &re)

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-port.h

17.799 MassActionOdeSystem::Reaction Struct Reference

Public Attributes

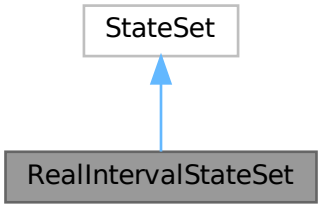
- std::string **name**
- double **rateConstant** = 0.0
- std::vector< StoichiometricTerm > **reactants**
- std::vector< StoichiometricTerm > **products**
- std::string **kineticLawExpression**
- std::vector< std::pair< std::string, double > > **parameters**
- std::string **reverseKineticLawExpression**
- double **reverseRateConstant** = 0.0
- bool **reversible** = false

The documentation for this struct was generated from the following file:

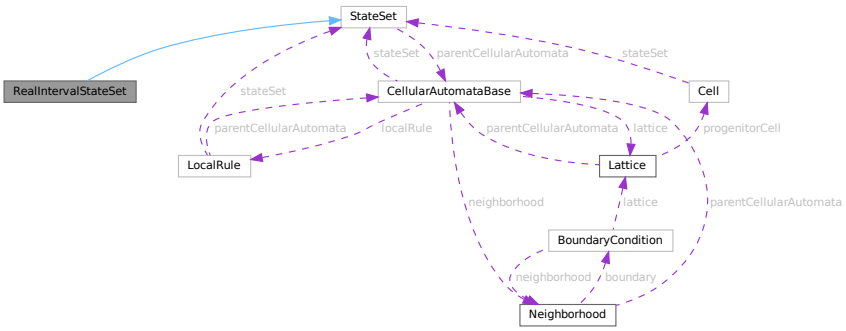
- tools/Biochemical/MassActionOdeSystem.h

17.800 RealIntervalStateSet Class Reference

Inheritance diagram for RealIntervalStateSet:



Collaboration diagram for RealIntervalStateSet:



Public Member Functions

- **RealIntervalStateSet** (CellularAutomataBase ∗parentCellularAutomata, std::optional< double > minimum=std::nullopt, std::optional< double > maximum=std::nullopt)
- bool contains (const State &state) const override
- std::unique_ptr< State > createDefaultState () const override
- std::unique_ptr< State > parseState (const std::string &text) const override
- bool isContinuous () const override
- std::string name () const override

Public Member Functions inherited from StateSet

- **StateSet** (CellularAutomataBase ∗parentCellularAutomata)
- **StateSet** (const StateSet &orig)
- virtual bool **isFinite** () const
- virtual bool **isDiscrete** () const
- virtual bool **isComposite** () const
- virtual std::vector< std::unique_ptr< State > > **enumerateStates** () const

Additional Inherited Members

Protected Attributes inherited from StateSet

- CellularAutomataBase ∗ **parentCellularAutomata**

17.800.1 Member Function Documentation

17.800.1.1 contains()

```
bool RealIntervalStateSet::contains (
            const State & state ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.800.1.2 createDefaultState()

```
std::unique_ptr< State > RealIntervalStateSet::createDefaultState ( ) const  [inline], [override],
[virtual]
```

Reimplemented from StateSet.

17.800.1.3 isContinuous()

```
bool RealIntervalStateSet::isContinuous ( ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.800.1.4 name()

```
std::string RealIntervalStateSet::name ( ) const  [inline], [override], [virtual]
```

Reimplemented from StateSet.

17.800.1.5 parseState()

```
std::unique_ptr< State > RealIntervalStateSet::parseState (
            const std::string & text ) const  [inline], [override], [virtual]
```

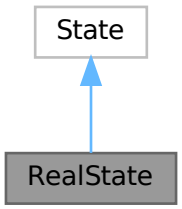
Reimplemented from StateSet.

The documentation for this class was generated from the following file:

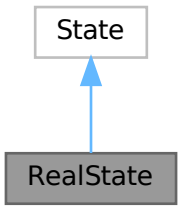
- plugins/components/ModalModel/CellularAutomata/StateSet.h

17.801 RealState Class Reference

Inheritance diagram for RealState:



Collaboration diagram for RealState:



Public Member Functions

- **RealState** (double value=0.0)
- std::unique_ptr< State > clone () const override
- bool equals (const State &other) const override
- std::string toString () const override
- std::string typeName () const override
- long getValue () const override
- void setValue (long value) override

- double **realValue** () const
- void **setRealValue** (double value)

Public Member Functions inherited from State

- **State** (long value)
- **State** (const State &orig)
- virtual std::string **show** () const
- template<typename T >
  const T * **as** () const noexcept
- template<typename T >
  T * **as** () noexcept

Additional Inherited Members

Protected Attributes inherited from State

- long **_value**

17.801.1 Member Function Documentation

17.801.1.1 clone()

```
std::unique_ptr< State > RealState::clone ( ) const  [inline], [override], [virtual]
```

Reimplemented from State.

17.801.1.2 equals()

```
bool RealState::equals (
            const State & other ) const  [inline], [override], [virtual]
```

Reimplemented from State.

17.801.1.3 getValue()

```
long RealState::getValue ( ) const  [inline], [override], [virtual]
```

Reimplemented from State.

17.801.1.4 setValue()

```
void RealState::setValue (
            long value )  [inline], [override], [virtual]
```

Reimplemented from State.

17.801.1.5 toString()

```
std::string RealState::toString ( ) const  [inline], [override], [virtual]
```

Reimplemented from State.

17.801.1.6 typeName()

```
std::string RealState::typeName ( ) const  [inline], [override], [virtual]
```

Reimplemented from State.

The documentation for this class was generated from the following file:

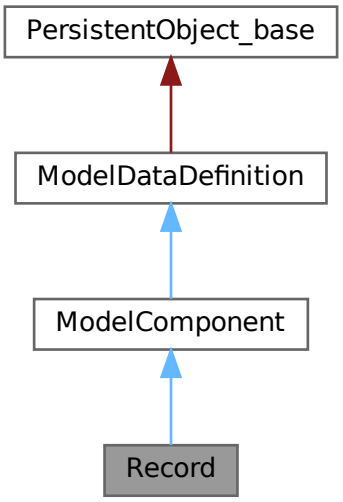
- plugins/components/ModalModel/CellularAutomata/State.h

17.802 Record Class Reference

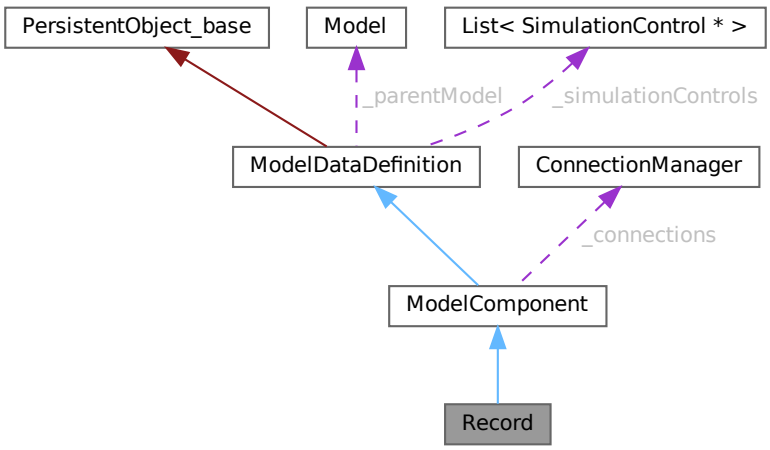
Evaluates a parser expression each time an entity passes through and records the value into a StatisticsCollector and/or an external file.

`#include <plugins/components/InputOutput/Record.h>`

Inheritance diagram for Record:



Collaboration diagram for Record:



Public Member Functions

- **Record** (Model *model, std::string name="")
- void **setFilename** (std::string filename)
- std::string **getFileName** () const

- void **setExpression** (const std::string expression)
- std::string **getExpression** () const
- void setExpressionName (std::string expressionName)
- std::string **getExpressionName** () const
- void setDatasetName (std::string datasetName)
- std::string **getDatasetName** () const
- void setRandomVariableName (std::string randomVariableName)
- std::string **getRandomVariableName** () const
- void setVariableType (std::string variableType)
- std::string **getVariableType** () const
- void setDatasetDescription (std::string description)
- std::string **getDatasetDescription** () const
- StatisticsCollector * **getCstatExpression** () const
- bool **getTimeDependent** () const
- void **setTimeDependent** (bool timeDependent)
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component

- static bool Check (ModelComponent *component)

  Validates a component instance.

- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.

- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _initBetweenReplications () override
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createInternalStatisticReporters () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)

- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.802.1 Detailed Description

Evaluates a parser expression each time an entity passes through and records the value into a StatisticsCollector and/or an external file.
Arena correspondence: the "Record module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 41-42), which offers five Types: Count, Entity Statistics, Time Interval, Time Between, and Expression, plus optional Tally/Counter Sets.
Only the Expression Type is implemented — there is no `Type` field at all. GenESyS extends this single mode with dataset-export metadata that has no Arena counterpart (`_datasetName`, `_randomVariableName`, `_variableType`, `_description`) and direct file recording (`_filename`, `_timeDependent`) intended for external tooling such as the Data Analyser.
Known differences from Arena: Count, Entity Statistics, Time Interval and Time Between have no equivalent Type here, and there is no Tally/Counter Set support ("Record into Set"). This does not mean statistics collection itself is unavailable in GenESyS — `Counter` and `StatisticsCollector` are used directly by several other components/data definitions — only that this specific component only reaches the Expression case.

17.802.2 Member Function Documentation

17.802.2.1 _check()

```
bool Record::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.802.2.2 _createInternalStatisticReporters()

```
void Record::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.802.2.3 _initBetweenReplications()

```
void Record::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.802.2.4 _loadInstance()

```
bool Record::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.802.2.5 _onDispatchEvent()

```
void Record::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.802.2.6 _saveInstance()

```
void Record::_saveInstance (
```

```
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.802.2.7 setDatasetDescription()

```
void Record::setDatasetDescription (
            std::string description )
```

Sets the free-text dataset description emitted in enriched Record file headers.

17.802.2.8 setDatasetName()

```
void Record::setDatasetName (
            std::string datasetName )
```

Sets the analytical dataset name emitted in enriched Record file headers.

17.802.2.9 setExpressionName()

```
void Record::setExpressionName (
            std::string expressionName )
```

Sets the legacy expression-name metadata kept for old Record datasets.

17.802.2.10 setRandomVariableName()

```
void Record::setRandomVariableName (
            std::string randomVariableName )
```

Sets the random variable name emitted in enriched Record file headers.

17.802.2.11 setVariableType()

```
void Record::setVariableType (
            std::string variableType )
```

Sets the canonical random variable type for enriched Record file headers.

17.802.2.12 show()

```
std::string Record::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

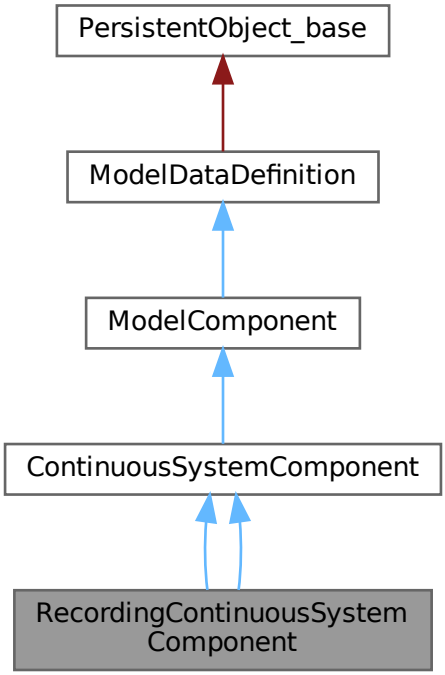
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

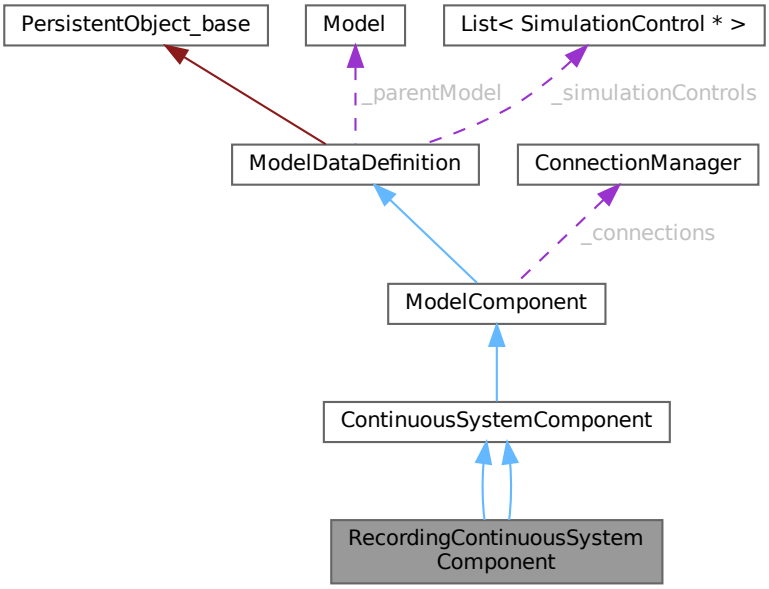
- plugins/components/InputOutput/Record.h
- plugins/components/InputOutput/Record.cpp

17.803 RecordingContinuousSystemComponent Class Reference

Inheritance diagram for RecordingContinuousSystemComponent:



Collaboration diagram for RecordingContinuousSystemComponent:



Public Member Functions

- **RecordingContinuousSystemComponent** (Model ∗model, ODESolver ∗solver, std::vector< Sample > ∗samples, std::string name="")
- **RecordingContinuousSystemComponent** (Model ∗model, ODESolver ∗solver, std::vector< Sample > ∗samples, std::string name="")

Public Member Functions inherited from ContinuousSystemComponent

- **ContinuousSystemComponent** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setOdeSolver** (ODESolver ∗solver)
- ODESolver ∗ **getOdeSolver** () const
- void **setOdeSolverName** (std::string name)
- std::string **getOdeSolverName** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.

- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Protected Member Functions

- void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.

Protected Member Functions inherited from ContinuousSystemComponent

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)

- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Static Public Member Functions inherited from ContinuousSystemComponent

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static void **SaveInstance** (PersistenceRecord *fields, ModelComponent *component)
- static bool **Check** (ModelComponent *component)
- static void **CreateInternalData** (ModelComponent *component)
- static void **DispatchEvent** (Event *event)

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.803.1 Member Function Documentation

17.803.1.1 _onDispatchEvent() [1/2]

```
void RecordingContinuousSystemComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [inline], [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Reimplemented from ContinuousSystemComponent.

17.803.1.2 _onDispatchEvent() [2/2]

```
void RecordingContinuousSystemComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [inline], [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

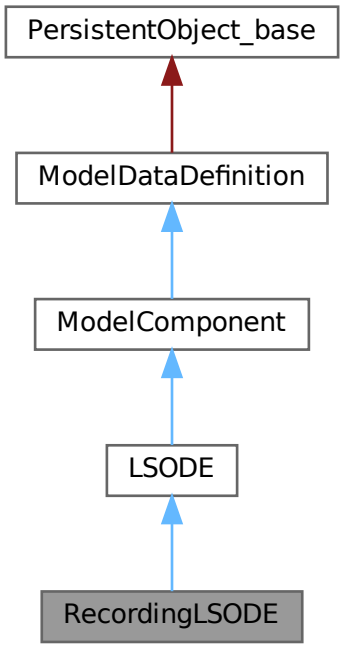
entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Reimplemented from ContinuousSystemComponent.
The documentation for this class was generated from the following files:

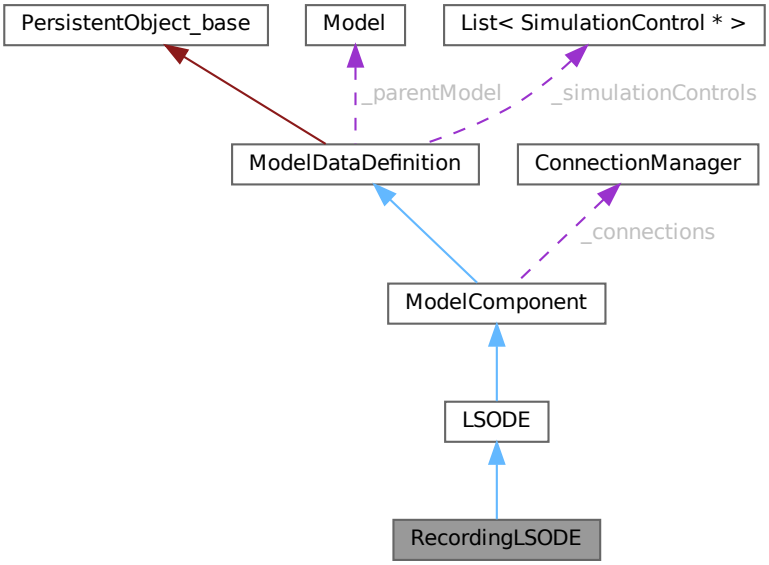
- tests/demo_continuous_trace.cpp
- tests/test_continuous_system.cpp

17.804 RecordingLSODE Class Reference

Inheritance diagram for RecordingLSODE:



Collaboration diagram for RecordingLSODE:



Public Member Functions

- **RecordingLSODE** (Model *model, Variable *timeVariable, Variable *stateVariable, std::vector< LSODESample > *samples, std::string name="")

Public Member Functions inherited from LSODE

- **LSODE** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setTimeVariable** (Variable *_timeVariable)
- Variable * **getTimeVariable** () const
- void **setStep** (double _step)
- double **getStep** () const
- void **setVariable** (Variable *_variables)
- Variable * **getVariable** () const
- List< std::string > * **getDiffEquations** () const
- void **addDiffEquation** (std::string newDiffEquation)
- void **removeDiffEquation** (std::string diffEquation)
- void **setFilename** (std::string filename)
- std::string **getFileName** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component

- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Protected Member Functions

- void _onDispatchEvent (Entity ∗entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.

Protected Member Functions inherited from LSODE

- virtual bool _loadInstance (PersistenceRecord ∗fields) override

  virtual protected method that must be overriden

- virtual void _saveInstance (PersistenceRecord ∗fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation ∗ _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl ∗control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()

- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Static Public Member Functions inherited from LSODE

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.804.1 Member Function Documentation

17.804.1.1 _onDispatchEvent()

```
void RecordingLSODE::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [inline], [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

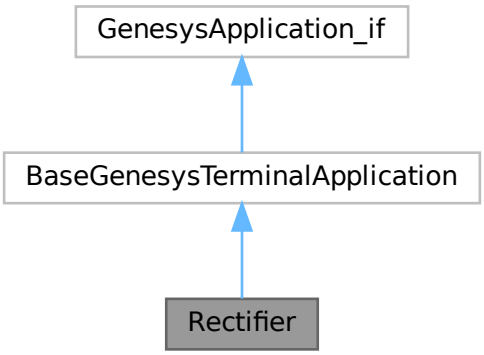
entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Reimplemented from LSODE.
The documentation for this class was generated from the following file:

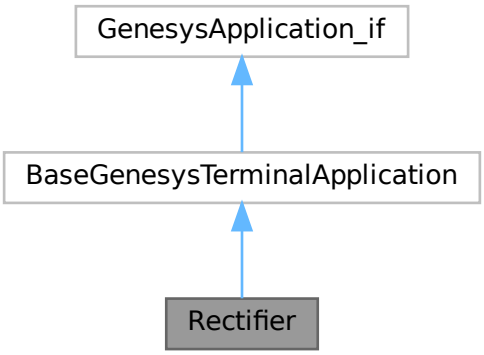
- tests/test_lsode.cpp

17.805 Rectifier Class Reference

Inheritance diagram for Rectifier:



Collaboration diagram for Rectifier:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.805.1 Member Function Documentation

17.805.1.1 main()

```
int Rectifier::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/teaching/Rectifier.h
- applications/modelSpecific/teaching/Rectifier.cpp

17.806 testing::internal::RelationToSourceCopy Struct Reference

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.807 testing::internal::RelationToSourceReference Struct Reference

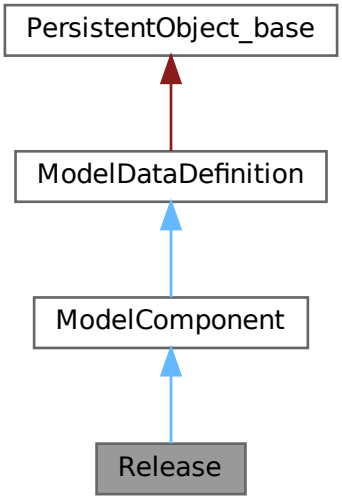
The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

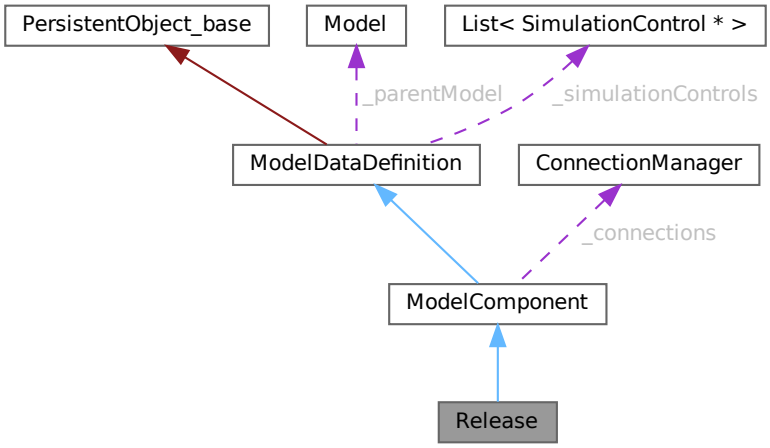
17.808 Release Class Reference

Gives up control of one or more previously seized resources, immediately unblocking any entity waiting for them.
`#include <plugins/components/DiscreteProcessing/Release.h>`

Inheritance diagram for Release:



Collaboration diagram for Release:



Public Member Functions

- **Release** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setPriority** (unsigned short _priority)
- unsigned short **priority** () const
- List< SeizableItem ∗ > ∗ **getReleaseRequests** () const
- void **addReleaseRequests** (SeizableItem ∗newRequest)
- void **removeReleaseRequests** (SeizableItem ∗request)

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _initBetweenReplications () override
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.808.1 Detailed Description

Gives up control of one or more previously seized resources, immediately unblocking any entity waiting for them. Arena correspondence: the "Release module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 60). `_releaseRequests` (a list of `SeizableItem`, reused from Seize) matches Arena's Type/Resource Name/Set Name/Quantity/Release Rule fields.
Confirmed behavior: reads back `Entity.Allocation.<ResourceName>` (set by Seize) and credits the resource's held time (`resource->getLastTimeSeized()`) to that category, both into a `<EntityType>.<Category>Time` StatisticsCollector and a running `Entity.Total<Category>Time` attribute — see `docs/ai_assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.10.

17.808.2 Member Function Documentation

17.808.2.1 _check()

```
bool Release::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.808.2.2 _createEditableDataDefinitions()

```
void Release::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.808.2.3 _initBetweenReplications()

```
void Release::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.808.2.4 _loadInstance()

```
bool Release::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.808.2.5 _onDispatchEvent()

```
void Release::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.808.2.6 _saveInstance()

```
void Release::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.808.2.7 show()

```
std::string Release::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

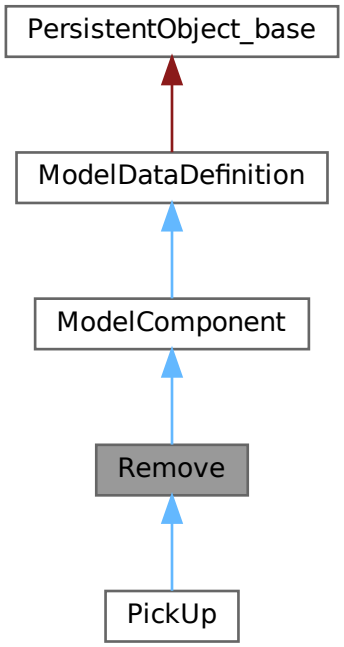
- plugins/components/DiscreteProcessing/Release.h
- plugins/components/DiscreteProcessing/Release.cpp

17.809 Remove Class Reference

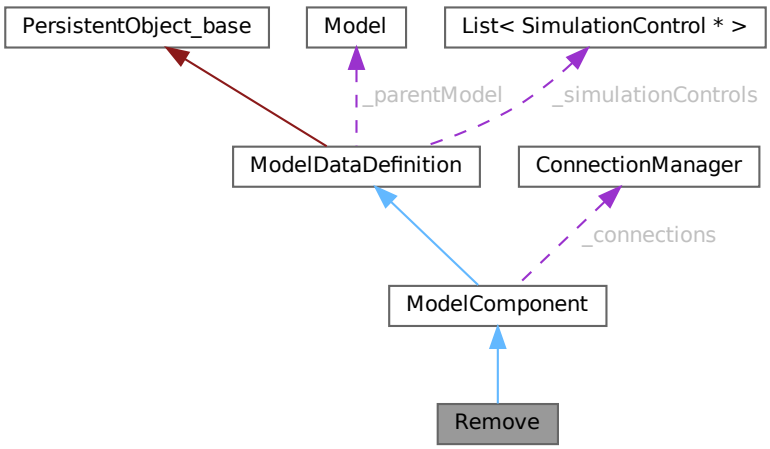
Removes one or more entities at a given rank (or rank range) from a Queue or an EntityGroup and forwards them.

`#include <plugins/components/Decisions/Remove.h>`

Inheritance diagram for Remove:



Collaboration diagram for Remove:



Public Types

- enum class **RemoveFromType** : int { **QUEUE** = 0 , **ENTITYGROUP** = 1 , **num_elements** = 2 }

Public Member Functions

- **Remove** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setRemoveStartRank** (std::string _removeFromRank)
- std::string **getRemoveStartRank** () const
- void **setRemoveEndRank** (std::string _removeEndRank)
- std::string **getRemoveEndRank** () const
- void **setRemoveFromType** (Remove::RemoveFromType _removeFromType)
- Remove::RemoveFromType **getRemoveFromType** () const
- void **setRemoveFrom** (ModelDataDefinition *_removeFrom)
- ModelDataDefinition * **getRemoveFrom** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static std::string **convertEnumToStr** (RemoveFromType type)
- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

Creates internal data required by a component instance.

- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)

- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.809.1 Detailed Description

Removes one or more entities at a given rank (or rank range) from a Queue or an EntityGroup and forwards them. Arena correspondence: the "Remove module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 61), which removes a single entity at one rank from a queue.
Known difference from Arena (broader, not narrower): `RemoveFromType` (`QUEUE/ENTITYGROUP`) also supports removing from an EntityGroup, and `_removeStartRank`/`_removeEndRank` support a rank *range* rather than Arena's single-rank-only removal. PickUp (in this same directory) reuses this class for Arena's "Pickup module".

17.809.2 Member Function Documentation

17.809.2.1 _check()

```
bool Remove::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.809.2.2 _createEditableDataDefinitions()

```
void Remove::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
Reimplemented from ModelDataDefinition.

17.809.2.3 _loadInstance()

```
bool Remove::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.809.2.4 _onDispatchEvent()

```
void Remove::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.809.2.5 _saveInstance()

```
void Remove::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.809.2.6 show()

```
std::string Remove::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

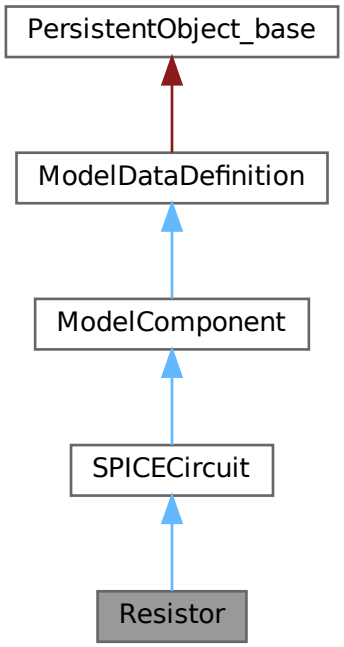
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

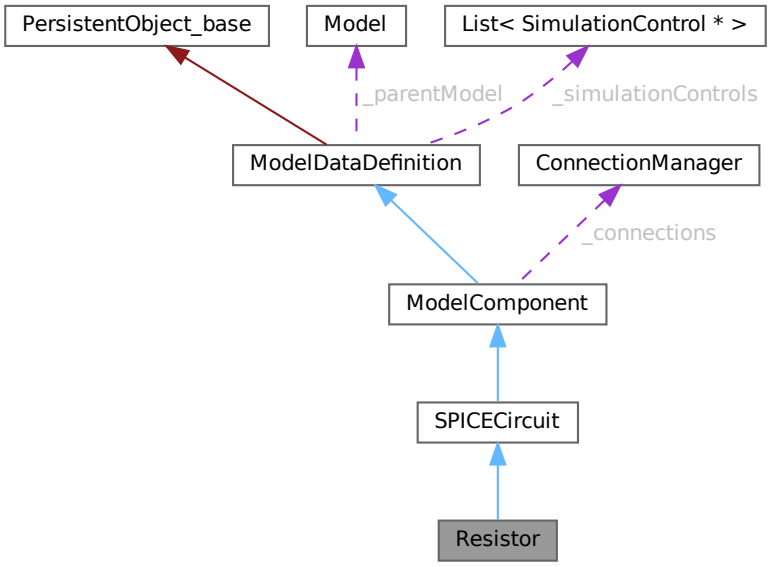
- plugins/components/Decisions/Remove.h
- plugins/components/Decisions/Remove.cpp

17.810 Resistor Class Reference

Inheritance diagram for Resistor:



Collaboration diagram for Resistor:



Public Member Functions

- **Resistor** (Model *model, std::string name="")
- void Build () override
- void **setResistance** (float resistance)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)

- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager ∗ _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model ∗ **_parentModel**
- List< SimulationControl ∗ > ∗ **_simulationControls** = new List<SimulationControl∗>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.810.1 Member Function Documentation

17.810.1.1 _loadInstance()

```
bool Resistor::_loadInstance (
            PersistenceRecord * fields ) [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.810.1.2 _saveInstance()

```
void Resistor::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.810.1.3 Build()

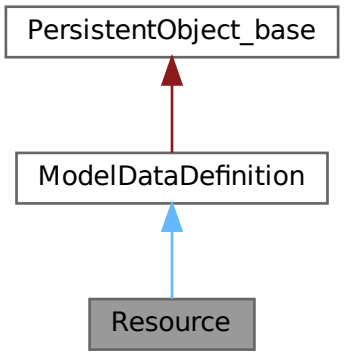
```
void Resistor::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.
The documentation for this class was generated from the following files:

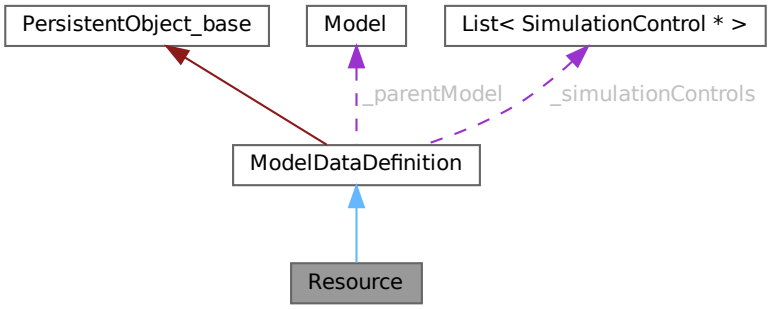
- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.811 Resource Class Reference

Data definition representing a seizable capacity pool (equipment, staff, etc.), its costing, its optional capacity schedule, and its associated Failure definitions.
`#include <plugins/data/DiscreteProcessing/Resource.h>`
Inheritance diagram for Resource:



Collaboration diagram for Resource:



Public Types

- enum class **ResourceState** : int {
  IDLE = 0 , **BUSY** = 1 , **FAILED** = 2 , **INACTIVE** = 3 ,
  OTHER = 4 , **num_elements** = 5 }
- typedef std::function< void(Resource *) > **ResourceEventHandler**
- typedef std::pair< std::pair< ResourceEventHandler, ModelComponent * >, unsigned int > **Sorted↩ ResourceEventHandler**

Public Member Functions

- Resource (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- bool **seize** (unsigned int quantity, double priority=0)
- void **release** (unsigned int quantity)
- void **insertFailure** (Failure *failure)
- void **removeFailure** (Failure *failure)
- double **getInstantCapacityUtilization** () const
- double **getCapacityUtilization** () const
- double **getSeizedUtilization** () const
- double **getLastTimeSeized** () const
- void **addReleaseResourceEventHandler** (ResourceEventHandler eventHandler, ModelComponent *component, unsigned int priority)
- void **setResourceState** (ResourceState _resourceState)
- Resource::ResourceState **getResourceState** () const
- void **setCapacity** (unsigned int capacity)
- unsigned int **getCapacity** () const
- void **setCostBusyTimeUnit** (double _costBusyTimeUnit)
- double **getCostBusyTimeUnit** () const
- void **setCostIdleTimeUnit** (double _costIdleTimeUnit)
- double **getCostIdleTimeUnit** () const
- void **setCostPerUse** (double _costPerUse)
- double **getCostPerUse** () const
- void **setCapacitySchedule** (Schedule *_capacitySchedule)
- Schedule * **getCapacitySchedule** () const
- unsigned int **getNumberBusy** () const

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- template<typename Class >
  static ResourceEventHandler **SetResourceEventHandler** (void(Class::*function)(Resource *), Class *object)
- static std::string **convertEnumToStr** (ResourceState state)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()

- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Friends

- class **Failure**
- class **ResourceTestProbe**

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.811.1 Detailed Description

Data definition representing a seizable capacity pool (equipment, staff, etc.), its costing, its optional capacity schedule, and its associated Failure definitions.
Arena correspondence: the "Resource module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 46-48). `_capacity` is Arena's fixed "Capacity"; `_capacitySchedule` (a Schedule*) is Arena's "Based on Schedule" type; `_costBusyTimeUnit`, `_costIdleTimeUnit` and `_costPerUse` are Arena's Busy/Idle-per-time-unit and Per Use costs (GenESyS generalizes Arena's "per hour" to the model's base time unit); `_failures` is Arena's Failures list (see Failure).
`ResourceState` (IDLE/BUSY/FAILED/INACTIVE/OTHER) is the closest correspondence to Arena's resource states, but it is a fixed enumeration.
Known differences from Arena, to validate against `Resource.cpp` and `Schedule.cpp` before treating them as closed:

- Arena's "StateSet" data module lets a modeler define arbitrary named states mapped onto autostates or failures; this class only exposes the fixed `ResourceState` enumeration, so a fully custom named-state catalog has no GenESyS equivalent;
- Arena's per-resource "Schedule Rule" (how a capacity decrease behaves against a busy unit: ignore/preempt/wait) has no field on this class; the closest candidate is the per-item `Schedulable↩ Item::Rule` on Schedule, which is a different granularity and needs confirmation.

17.811.2 Constructor & Destructor Documentation

17.811.2.1 Resource()

```
Resource::Resource (
            Model * model,
            std::string name = "" )
```

Handlers are sorted by priority

17.811.3 Member Function Documentation

17.811.3.1 _check()

```
bool Resource::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Validate resource basic semantic constraints.
Reimplemented from ModelDataDefinition.

17.811.3.2 _createAttachedAttributes()

```
void Resource::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.811.3.3 _initBetweenReplications()

```
void Resource::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.811.3.4 _loadInstance()

```
bool Resource::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.811.3.5 _saveInstance()

```
void Resource::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.811.3.6 show()

```
std::string Resource::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

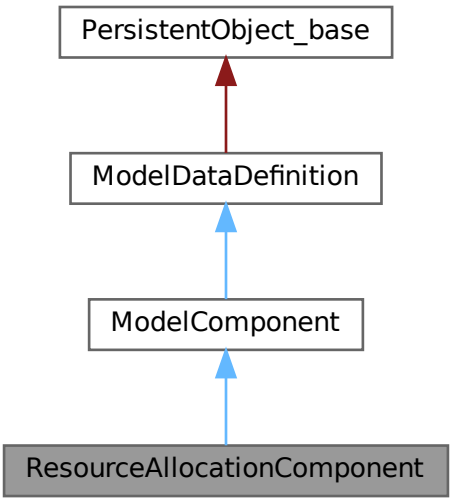
Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/DiscreteProcessing/Resource.h
- plugins/data/DiscreteProcessing/Resource.cpp

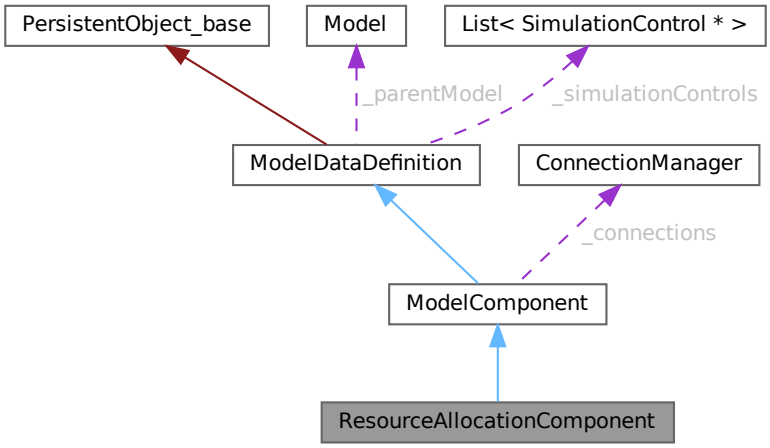
17.812 ResourceAllocationComponent Class Reference

```
#include <plugins/components/WholeCellModeling/ResourceAllocationComponent.↩
h>
```

Inheritance diagram for ResourceAllocationComponent:



Collaboration diagram for ResourceAllocationComponent:



Public Member Functions

- **ResourceAllocationComponent** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)
- WholeCellState * **getWholeCellState** () const
- void **setRnapCountKey** (std::string key)
- std::string **getRnapCountKey** () const
- void **setRibosomeCountKey** (std::string key)
- std::string **getRibosomeCountKey** () const
- void **setMRNASpeciesPrefix** (std::string prefix)
- std::string **getMRNASpeciesPrefix** () const
- void **setProteinSpeciesPrefix** (std::string prefix)
- std::string **getProteinSpeciesPrefix** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.812.1 Detailed Description

Fair allocation of molecular machines (RNAP, ribosomes) before each SSA step.
Implements the "ResourceAllocation" contract from Karr et al. (2012): available RNAP and ribosome counts are divided proportionally across active molecular species, then written to WholeCellState::setResourceBudget() so StochasticTranscription and StochasticTranslation can honour the budget.
Allocation logic:

- Count N_mRNA = number of keys in WholeCellState molecule counts whose name starts with m↩ RNASpeciesPrefix (default "mRNA_").
- Count N_protein = number of keys starting with proteinSpeciesPrefix (default "protein_").
- RNAP budget per mRNA gene = floor(RNAP_free / max(N_mRNA, 1))
- Ribosome budget per protein = floor(ribosome_free / max(N_mRNA, 1)) (per active mRNA transcript, not per protein species)
- Write per-species budget to WholeCellState::setResourceBudget().

This component must fire BEFORE StochasticTranscription and StochasticTranslation in the simulation pipeline.

17.812.2 Member Function Documentation

17.812.2.1 _check()

```
bool ResourceAllocationComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.812.2.2 _createEditableDataDefinitions()

```
void ResourceAllocationComponent::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.812.2.3 _initBetweenReplications()

```
void ResourceAllocationComponent::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.812.2.4 _loadInstance()

```
bool ResourceAllocationComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.812.2.5 _onDispatchEvent()

```
void ResourceAllocationComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.812.2.6 _saveInstance()

```
void ResourceAllocationComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.812.2.7 show()

```
std::string ResourceAllocationComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/WholeCellModeling/ResourceAllocationComponent.h
- plugins/components/WholeCellModeling/ResourceAllocationComponent.cpp

17.813 ResourceTestProbe Class Reference

Static Public Member Functions

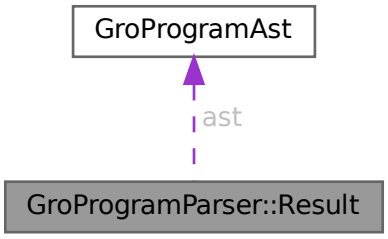
- static bool **HasStatisticsInternals** (const Resource &resource)
- static bool **HasFailure** (const Resource &resource, const Failure *failure)
- static size_t **FailureCount** (const Resource &resource)

The documentation for this class was generated from the following file:

- tests/unit/test_simulator_runtime.cpp

17.814 GroProgramParser::Result Struct Reference

Collaboration diagram for GroProgramParser::Result:



Public Attributes

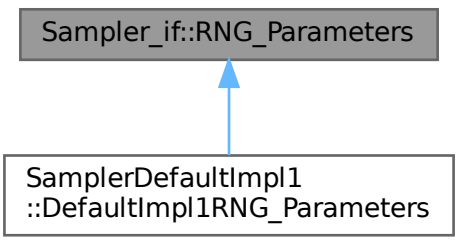
- bool **accepted** = false
- std::string **errorMessage** = ""
- GroProgramAst **ast**

The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/GroProgramParser.h

17.815 Sampler_if::RNG_Parameters Struct Reference

Encapsulates generator-specific configuration/state parameters.
`#include <kernel/statistics/Sampler_if.h>`
Inheritance diagram for Sampler_if::RNG_Parameters:



17.815.1 Detailed Description

Encapsulates generator-specific configuration/state parameters.
Implementations may derive this struct to expose seed, multiplier and other algorithm-specific controls required by a concrete RNG strategy.
The documentation for this struct was generated from the following file:

- kernel/statistics/Sampler_if.h

17.816 RootFinder_if Class Reference

Root-finding abstraction for scalar equations.
`#include <tools/Continuous/RootFinder_if.h>`

Public Member Functions

- virtual double **findRoot** (double min, double max, double(∗f)(double), double tolerance) const =0

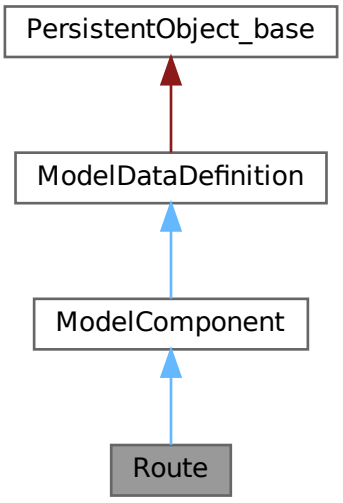
17.816.1 Detailed Description

Root-finding abstraction for scalar equations.
The documentation for this class was generated from the following file:

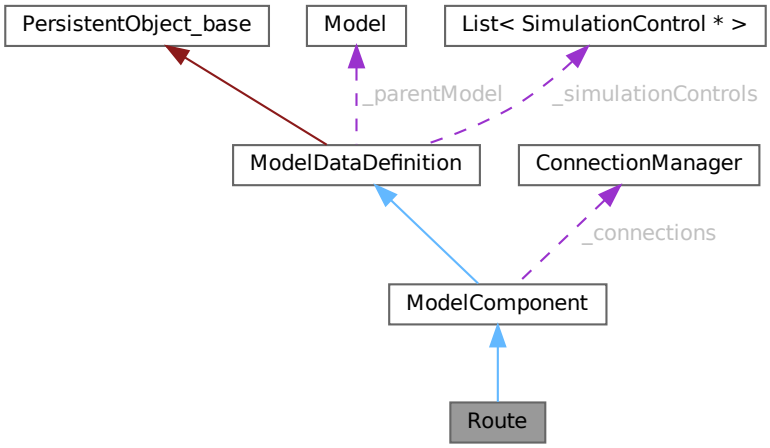
- tools/Continuous/RootFinder_if.h

17.817 Route Class Reference

`#include <plugins/components/MaterialHandling/Route.h>`
Inheritance diagram for Route:



Collaboration diagram for Route:



Public Types

- enum class **DestinationType** : int { **Station** = 0 , **Sequence** = 1 , **Label** = 2 , **num_elements** = 3 }

Public Member Functions

- **Route** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setStationName** (std::string stationName)
- void **setStation** (Station *_station)
- Station * **getStation** () const
- void **setStationExpression** (std::string _stationExpression)
- std::string **getStationExpression** () const
- void **setRouteTimeExpression** (std::string _routeTimeExpression)
- void **setRouteTimeExpression** (std::string _routeTimeExpression, Util::TimeUnit _routeTimeTimeUnit)
- std::string **getRouteTimeExpression** () const
- void **setRouteTimeTimeUnit** (Util::TimeUnit _routeTimeTimeUnit)
- Util::TimeUnit **getRouteTimeTimeUnit** () const
- void **setRouteDestinationType** (DestinationType _routeDestinationType)
- Route::DestinationType **getRouteDestinationType** () const
- void **setLabel** (Label *_label)
- Label * **getLabel** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

Connects this component to another component.

- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.

- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static std::string **convertEnumToStr** (DestinationType type)
- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createInternalStatisticReporters () override
- virtual void _createEditableDataDefinitions () override
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions

- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.817.1 Detailed Description

Route module DESCRIPTION The Route module transfers an entity to a specified station or the next station in the station visitation sequence defined for the entity. A delay time to transfer to the next station may be defined. When an entity enters the Route module, its Station attribute (Entity.Station) is set to the destination station. The entity is then sent to the destination station, using the route time specified. If the station destination is entered as By Sequence, the next station is determined by the entity's sequence and step within the set (defined by special-purpose attributes Entity.Sequence and Entity.Jobstep, respectively). TYPICAL USES Send a part to its next processing station based on its routing slip Send an account balance call to an account agent Send restaurant customers to a specific table PROMPTS Prompt Description Name Unique name of the module that will be displayed in the flowchart. Route Time Travel time from the entity's current location to the destination station. Units Time units for route-time parameters. Destination Type Method for determining the entity destination location. Selection of By Sequence requires that the entity has been assigned a sequence name and that the sequence itself has been defined. Station Name Name of the individual destination station. Attribute Name Name of the attribute that stores the station name to which entities will route. Expression Expression that is evaluated to the station name where entities will route.

17.817.2 Member Function Documentation

17.817.2.1 _check()

```
bool Route::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.817.2.2 _createAttachedAttributes()

```
void Route::_createAttachedAttributes ( )  [override], [protected], [virtual]
```
Reimplemented from ModelDataDefinition.

17.817.2.3 _createEditableDataDefinitions()

```
void Route::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```
Reimplemented from ModelDataDefinition.

17.817.2.4 _createInternalStatisticReporters()

```
void Route::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```
Reimplemented from ModelDataDefinition.

17.817.2.5 _loadInstance()

```
bool Route::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```
virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.817.2.6 _onDispatchEvent()

```
void Route::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```
Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.817.2.7 _saveInstance()

```
void Route::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```
Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.817.2.8 show()

```
std::string Route::show ( )  [override], [virtual]
```
virtual public methods
Returns a textual representation of this component.

Returns

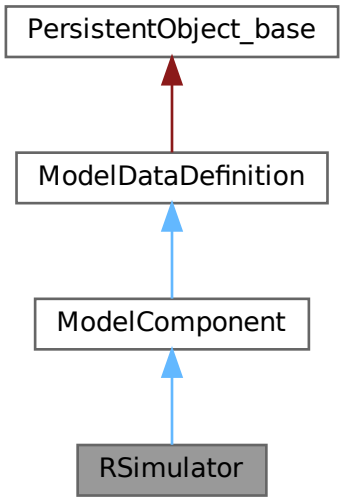
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

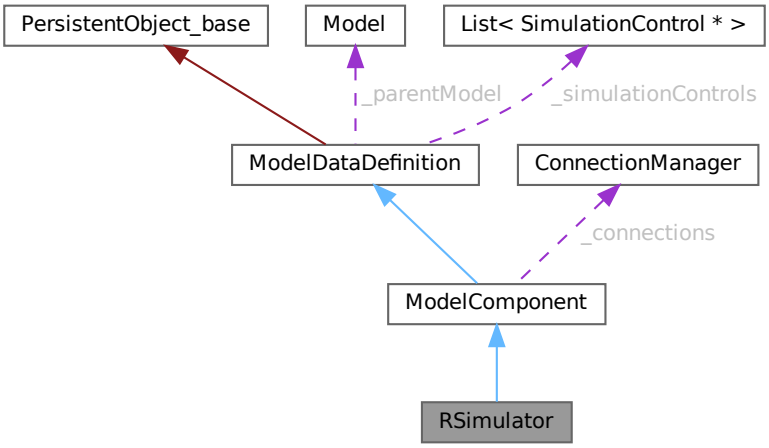
- plugins/components/MaterialHandling/Route.h
- plugins/components/MaterialHandling/Route.cpp

17.818 RSimulator Class Reference

Component that sends configured R commands to an internal RSimulatorRunner.
`#include <plugins/components/ExternalIntegration/RSimulator.h>`
Inheritance diagram for RSimulator:



Collaboration diagram for RSimulator:



Public Member Functions

- **RSimulator** (Model *model, std::string name="")

  Creates an RSimulator component attached to a model.
- virtual ~**RSimulator** ()=default

  Destroys the RSimulator component.
- void **setRExecutable** (std::string rExecutable)

  Sets the Rscript executable name or path used by the internal runner.
- std::string **getRExecutable** () const

  Returns the Rscript executable name or path used by the internal runner.
- void **setWorkingDirectory** (std::string workingDirectory)

  Sets the working directory used by generated R scripts.
- std::string **getWorkingDirectory** () const

  Returns the working directory used by generated R scripts.
- void **setPreludeScript** (std::string preludeScript)

  Sets optional R code prepended to every command.
- std::string **getPreludeScript** () const

  Returns optional R code prepended to every command.
- void **insertCommand** (std::string command)

  Appends one R command to the execution list.
- void **setCommands** (std::list< std::string > commands)

  Replaces the command list with the provided commands.
- List< std::string > * **getCommands** () const

  Returns the command list used when entities arrive.
- void **setCommandList** (std::string commandList)

  Replaces the command list using one command per non-empty line.
- std::string **getCommandList** () const

  Returns commands serialized as one command per line for GUI editing.
- RSimulatorRunner * **getRunner** ()

  Returns the internal runner, creating it first if necessary.
- virtual std::string show () override

  Returns a compact human-readable description of the component.

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()

  Returns plugin metadata for the RSimulator component.
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)

  Loads an RSimulator component from persistence.
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

  Creates a new RSimulator component instance.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

Validates a data definition instance.

- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Executes R commands for the arriving entity and forwards it.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  Loads persisted RSimulator configuration.
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves RSimulator configuration.
- virtual bool _check (std::string &errorMessage) override

  Validates the executable, working directory and command list.
- virtual void _createEditableDataDefinitions () override

  Creates or updates the internal RSimulatorRunner.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)

- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.818.1 Detailed Description

Component that sends configured R commands to an internal RSimulatorRunner.
Each arriving entity triggers the configured command list. Commands are executed sequentially by the internal RSimulatorRunner, and the first version exposes results through simulation trace messages before forwarding the entity.

17.818.2 Member Function Documentation

17.818.2.1 _check()

```
bool RSimulator::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

Validates the executable, working directory and command list.
Reimplemented from ModelDataDefinition.

17.818.2.2 _createEditableDataDefinitions()

```
void RSimulator::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Creates or updates the internal RSimulatorRunner.
Reimplemented from ModelDataDefinition.

17.818.2.3 _loadInstance()

```
bool RSimulator::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

Loads persisted RSimulator configuration.
Reimplemented from ModelComponent.

17.818.2.4 _onDispatchEvent()

```
void RSimulator::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Executes R commands for the arriving entity and forwards it.
Implements ModelComponent.

17.818.2.5 _saveInstance()

```
void RSimulator::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves RSimulator configuration.
Reimplemented from ModelComponent.

17.818.2.6 show()

```
std::string RSimulator::show ( )  [override], [virtual]
```

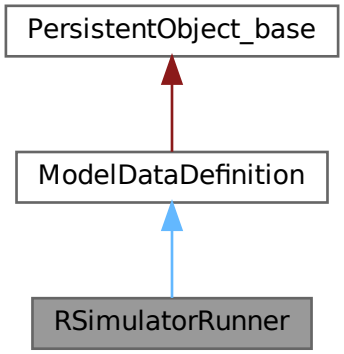
Returns a compact human-readable description of the component.
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/ExternalIntegration/RSimulator.h
- plugins/components/ExternalIntegration/RSimulator.cpp

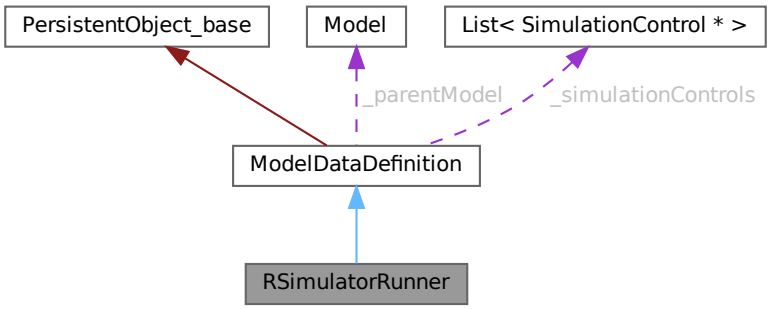
17.819 RSimulatorRunner Class Reference

Model data definition that executes R code through Rscript.
`#include <plugins/data/ExternalIntegration/RSimulatorRunner.h>`

Inheritance diagram for RSimulatorRunner:



Collaboration diagram for RSimulatorRunner:



Public Member Functions

- **RSimulatorRunner** (Model *model, std::string name="")

  Creates an R runner bound to a model.
- virtual ~**RSimulatorRunner** ()=default

  Destroys the R runner.
- void **setRExecutable** (std::string rExecutable)

  Sets the Rscript executable name or path.
- std::string **getRExecutable** () const

  Returns the Rscript executable name or path.
- void **setWorkingDirectory** (std::string workingDirectory)

  Sets the working directory used for generated scripts and command execution.
- std::string **getWorkingDirectory** () const

  Returns the working directory used for generated scripts and command execution.
- void **setPreludeScript** (std::string preludeScript)

  Sets optional R code prepended before the command block.

- std::string **getPreludeScript** () const

  Returns optional R code prepended before the command block.
- void **setCommand** (std::string command)

  Sets the R command/script body to execute.
- std::string **getCommand** () const

  Returns the R command/script body to execute.
- void **setLastStatus** (std::string lastStatus)

  Sets the last execution status.
- std::string **getLastStatus** () const

  Returns the last execution status.
- void **setLastExitCode** (int lastExitCode)

  Sets the last process exit code.
- int **getLastExitCode** () const

  Returns the last process exit code.
- void **setLastStdout** (std::string lastStdout)

  Sets stdout captured from the last R execution.
- std::string **getLastStdout** () const

  Returns stdout captured from the last R execution.
- void **setLastStderr** (std::string lastStderr)

  Sets stderr captured from the last R execution.
- std::string **getLastStderr** () const

  Returns stderr captured from the last R execution.
- void **setLastResponsePayload** (std::string lastResponsePayload)

  Sets the response payload exposed to other tools.
- std::string **getLastResponsePayload** () const

  Returns the response payload exposed to other tools.
- void **setLastScriptFilename** (std::string lastScriptFilename)

  Sets the generated R script filename from the last execution.
- std::string **getLastScriptFilename** () const

  Returns the generated R script filename from the last execution.
- void **setLastResponseFilename** (std::string lastResponseFilename)

  Sets the stdout file path from the last execution.
- std::string **getLastResponseFilename** () const

  Returns the stdout file path from the last execution.
- bool **executeCommand** (std::string &errorMessage)

  Executes the configured R code and updates last-execution fields atomically.
- virtual std::string show () override

  Returns a compact human-readable description of the runner.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)

  Loads an R runner from a persistence record.
- static PluginInformation * **GetPluginInformation** ()

  Returns plugin metadata and system dependency declarations.
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

  Creates a new R runner instance for plugin construction.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  Loads persisted R runner configuration and last-execution state.
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves R runner configuration and last-execution state.
- virtual bool _check (std::string &errorMessage) override

  Validates the runner configuration without executing R.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)

- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.819.1 Detailed Description

Model data definition that executes R code through Rscript.
The runner stores a compact R execution configuration, writes a temporary script, executes it with Rscript, and keeps status, exit code, stdout, stderr and response payload inside the model data definition for later inspection.

17.819.2 Member Function Documentation

17.819.2.1 _check()

```
bool RSimulatorRunner::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

Validates the runner configuration without executing R.
Reimplemented from ModelDataDefinition.

17.819.2.2 _loadInstance()

```
bool RSimulatorRunner::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

Loads persisted R runner configuration and last-execution state.
Reimplemented from ModelDataDefinition.

17.819.2.3 _saveInstance()

```
void RSimulatorRunner::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves R runner configuration and last-execution state.
Reimplemented from ModelDataDefinition.

17.819.2.4 show()

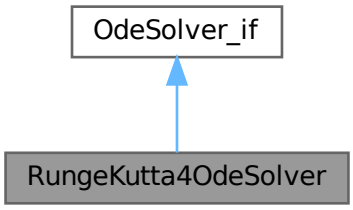
```
std::string RSimulatorRunner::show ( )  [override], [virtual]
```

Returns a compact human-readable description of the runner.
Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

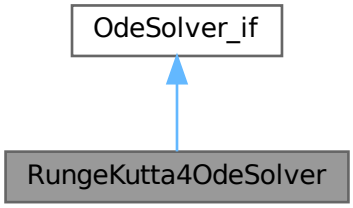
- plugins/data/ExternalIntegration/RSimulatorRunner.h
- plugins/data/ExternalIntegration/RSimulatorRunner.cpp

17.820 RungeKutta4OdeSolver Class Reference

Fixed-step fourth-order Runge-Kutta solver for first-order ODE systems.
`#include <tools/Continuous/RungeKutta4OdeSolver.h>`
Inheritance diagram for RungeKutta4OdeSolver:



Collaboration diagram for RungeKutta4OdeSolver:



Public Member Functions

- bool advance (const OdeSystem_if &system, double t0, double dt, const double *y0, double *y1) const override

17.820.1 Detailed Description

Fixed-step fourth-order Runge-Kutta solver for first-order ODE systems.

17.820.2 Member Function Documentation

17.820.2.1 advance()

```
bool RungeKutta4OdeSolver::advance (
            const OdeSystem_if & system,
            double t0,
            double dt,
            const double * y0,
            double * y1 ) const  [inline], [override], [virtual]
```

Implements OdeSolver_if.

The documentation for this class was generated from the following file:

- tools/Continuous/RungeKutta4OdeSolver.h

17.821 genesys::launcher::RuntimeInstallResult Struct Reference

Public Attributes

- bool **ok** = false
- QString **runtimeRoot**
- QString **previousRuntimeRoot**
- QString **error**

The documentation for this struct was generated from the following file:

- applications/launcher/UserRuntimeManager.h

17.822 genesys::launcher::RuntimePaths Class Reference

Public Member Functions

- **RuntimePaths** (const QProcessEnvironment &environment, QString systemRuntimeRoot={}, QString systemConfigPath={}, QString updateKeyringPath={})

- bool **hasUsableHome** () const noexcept
- const QString & **configRoot** () const noexcept
- const QString & **cacheRoot** () const noexcept
- const QString & **dataRoot** () const noexcept
- QString **userConfigPath** () const
- QString **systemConfigPath** () const
- QString **appsRoot** () const
- QString **currentRuntimeLink** () const
- QString **downloadsRoot** () const
- QString **logsRoot** () const
- QString **launcherLogPath** () const
- QString **updateStatePath** () const
- QString **updateLockPath** () const
- QString **systemRuntimeRoot** () const
- QString **systemBinRoot** () const
- QString **updateKeyringPath** () const

The documentation for this class was generated from the following files:

- applications/launcher/RuntimePaths.h
- applications/launcher/RuntimePaths.cpp

17.823 genesys::launcher::RuntimeSelection Struct Reference

Public Attributes

- bool **ok** = false
- QString **requestedApplication**
- QString **resolvedApplication**
- QString **primaryExecutable**
- QString **fallbackExecutable**
- RuntimeSource **source** = RuntimeSource::None
- QString **reason**

The documentation for this struct was generated from the following file:

- applications/launcher/RuntimeSelector.h

17.824 genesys::launcher::RuntimeSelector Class Reference

Public Member Functions

- **RuntimeSelector** (const RuntimePaths &paths, LauncherLogger *logger=nullptr)
- RuntimeSelection **select** (const QString &requestedApplication, const LauncherConfig &config) const
- Version **currentUserVersion** () const

Static Public Member Functions

- static QStringList **applicationCandidates** (const QString &requestedApplication)
- static bool **isSupportedApplicationName** (const QString &application)

The documentation for this class was generated from the following files:

- applications/launcher/RuntimeSelector.h
- applications/launcher/RuntimeSelector.cpp

17.825 Sample Struct Reference

Public Attributes

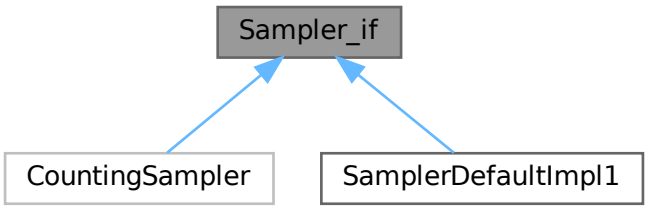
- double **t**
- double **x**
- double **v**
- double **energy**

The documentation for this struct was generated from the following files:

- tests/demo_continuous_trace.cpp
- tests/test_continuous_system.cpp

17.826 Sampler_if Class Reference

Abstraction for random-number generation and probability-distribution sampling.
`#include <kernel/statistics/Sampler_if.h>`
Inheritance diagram for Sampler_if:



Classes

- struct RNG_Parameters

  Encapsulates generator-specific configuration/state parameters.

Public Member Functions

- virtual double random ()=0

  Generates a pseudo-random value uniformly distributed in the range [0,1).
- virtual double sampleBeta (double alpha, double beta, double infLimit, double supLimit)=0

  Samples a Beta-distributed value scaled to an arbitrary output interval.
- virtual double sampleBeta (double alpha, double beta)=0

  Samples a standard Beta-distributed value in [0,1].
- virtual double sampleErlang (double mean, int M, double offset=0.0)=0

  Samples from an Erlang distribution using the provided mean and number of phases.
- virtual double sampleExponential (double mean, double offset=0.0)=0

  Samples from an Exponential distribution with the provided mean.
- virtual double sampleGamma (double alpha, double beta, double offset=0.0)=0

  Samples from a Gamma distribution using the provided shape/scale parameters.
- virtual double sampleGumbell (double mode, double scale)=0

  Samples from a Gumbel distribution using mode and scale.

- virtual double sampleLogNormal (double mean, double stddev, double offset=0.0)=0

  Samples from a Lognormal distribution parameterized by mean and standard deviation.
- virtual double sampleNormal (double mean, double stddev)=0

  Samples from a Normal distribution.
- virtual double sampleTriangular (double min, double mode, double max)=0

  Samples from a Triangular distribution.
- virtual double sampleUniform (double min, double max)=0

  Samples from a continuous Uniform distribution in [min, max].
- virtual double sampleWeibull (double alpha, double scale)=0

  Samples from a Weibull distribution.
- virtual double sampleBinomial (int trials, double p)=0

  Samples from a Binomial distribution with `trials` and success probability `p`.
- virtual double sampleBernoulli (double p)=0

  Samples from a Bernoulli distribution (0/1).
- virtual double sampleDiscrete (double prob, double value,...)=0

  Samples from a discrete distribution using a variadic list of probability/value pairs.
- virtual double sampleDiscrete (double *prob, double *value, int size)=0

  Samples from a discrete distribution using probability/value arrays.
- virtual double sampleGeometric (double p)=0

  Samples from a Geometric distribution with success probability `p`.
- virtual void setRNGparameters (RNG_Parameters *param)=0

  Updates the pseudo-random generator internal parameters.
- virtual RNG_Parameters * getRNGparameters () const =0

  Returns the current pseudo-random generator parameters.

17.826.1 Detailed Description

Abstraction for random-number generation and probability-distribution sampling.
Concrete implementations provide a base uniform pseudo-random generator and helper methods to sample supported continuous and discrete probability distributions used across the simulation kernel (e.g., parser functions and stochastic model logic).

17.826.2 Member Function Documentation

17.826.2.1 getRNGparameters()

```
virtual RNG_Parameters * Sampler_if::getRNGparameters ( ) const  [pure virtual]
```

Returns the current pseudo-random generator parameters.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.2 random()

```
virtual double Sampler_if::random ( )  [pure virtual]
```

Generates a pseudo-random value uniformly distributed in the range [0,1).

Returns

Base pseudo-random value used by the other distribution samplers.

Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.3 sampleBernoulli()

```
virtual double Sampler_if::sampleBernoulli (
            double p )  [pure virtual]
```

Samples from a Bernoulli distribution (0/1).
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.4 sampleBeta() [1/2]

```
virtual double Sampler_if::sampleBeta (
            double alpha,
            double beta )  [pure virtual]
```

Samples a standard Beta-distributed value in [0,1].
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.5 sampleBeta() [2/2]

```
virtual double Sampler_if::sampleBeta (
            double alpha,
            double beta,
            double infLimit,
            double supLimit )  [pure virtual]
```

Samples a Beta-distributed value scaled to an arbitrary output interval.

Parameters

alpha	Alpha shape parameter (>0).
beta	Beta shape parameter (>0).
infLimit	Lower bound of the output interval.
supLimit	Upper bound of the output interval.

Returns

Sampled value within [infLimit, supLimit].

Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.6 sampleBinomial()

```
virtual double Sampler_if::sampleBinomial (
            int trials,
            double p )  [pure virtual]
```

Samples from a Binomial distribution with `trials` and success probability `p`.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.7 sampleDiscrete() [1/2]

```
virtual double Sampler_if::sampleDiscrete (
            double * prob,
            double * value,
            int size )  [pure virtual]
```

Samples from a discrete distribution using probability/value arrays.
Implemented in CountingSampler, and SamplerDefaultImpl1.

17.826.2.8 sampleDiscrete() [2/2]

```
virtual double Sampler_if::sampleDiscrete (
            double prob,
            double value,
             ...  )  [pure virtual]
```

Samples from a discrete distribution using a variadic list of probability/value pairs.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.9 sampleErlang()

```
virtual double Sampler_if::sampleErlang (
            double mean,
```

```
            int M,
            double offset = 0.0 )  [pure virtual]
```

Samples from an Erlang distribution using the provided mean and number of phases.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.10 sampleExponential()

```
virtual double Sampler_if::sampleExponential (
            double mean,
            double offset = 0.0 )  [pure virtual]
```

Samples from an Exponential distribution with the provided mean.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.11 sampleGamma()

```
virtual double Sampler_if::sampleGamma (
            double alpha,
            double beta,
            double offset = 0.0 )  [pure virtual]
```

Samples from a Gamma distribution using the provided shape/scale parameters.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.12 sampleGeometric()

```
virtual double Sampler_if::sampleGeometric (
            double p )  [pure virtual]
```

Samples from a Geometric distribution with success probability p.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.13 sampleGumbell()

```
virtual double Sampler_if::sampleGumbell (
            double mode,
            double scale )  [pure virtual]
```

Samples from a Gumbel distribution using mode and scale.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.14 sampleLogNormal()

```
virtual double Sampler_if::sampleLogNormal (
            double mean,
            double stddev,
            double offset = 0.0 )  [pure virtual]
```

Samples from a Lognormal distribution parameterized by mean and standard deviation.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.15 sampleNormal()

```
virtual double Sampler_if::sampleNormal (
            double mean,
            double stddev )  [pure virtual]
```

Samples from a Normal distribution.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.16 sampleTriangular()

```
virtual double Sampler_if::sampleTriangular (
            double min,
```

```
            double mode,
            double max )  [pure virtual]
```

Samples from a Triangular distribution.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.17 sampleUniform()

```
virtual double Sampler_if::sampleUniform (
            double min,
            double max )  [pure virtual]
```

Samples from a continuous Uniform distribution in [min, max].
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.18 sampleWeibull()

```
virtual double Sampler_if::sampleWeibull (
            double alpha,
            double scale )  [pure virtual]
```

Samples from a Weibull distribution.
Implemented in SamplerDefaultImpl1, and CountingSampler.

17.826.2.19 setRNGparameters()

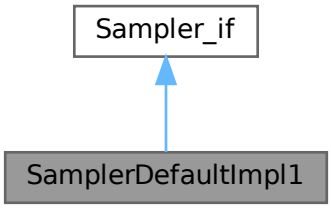
```
virtual void Sampler_if::setRNGparameters (
            RNG_Parameters * param )  [pure virtual]
```

Updates the pseudo-random generator internal parameters.
Implemented in CountingSampler, and SamplerDefaultImpl1.
The documentation for this class was generated from the following file:

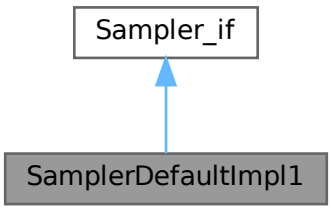
- kernel/statistics/Sampler_if.h

17.827 SamplerDefaultImpl1 Class Reference

Default pseudo-random sampler implementation used by the kernel.
`#include <kernel/statistics/SamplerDefaultImpl1.h>`
Inheritance diagram for SamplerDefaultImpl1:



Collaboration diagram for SamplerDefaultImpl1:



Classes

- struct DefaultImpl1RNG_Parameters

  Parameter bundle for the default RNG algorithm.

Public Member Functions

- **SamplerDefaultImpl1** ()

  Creates the sampler with default RNG parameters.
- virtual ~**SamplerDefaultImpl1** ()

  Destroys sampler and owned parameter objects when applicable.
- virtual double random () override

  Generates one base uniform pseudo-random number in [0,1).
- virtual double sampleBeta (double alpha, double beta, double infLimit, double supLimit) override

  Samples Beta in arbitrary interval [infLimit, supLimit].
- virtual double sampleBeta (double alpha, double beta) override

  Samples standard Beta in [0,1].
- virtual double sampleErlang (double mean, int M, double offset=0.0) override

  Samples Erlang distribution.
- virtual double sampleExponential (double mean, double offset=0.0) override

  Samples Exponential distribution.
- virtual double sampleGamma (double alpha, double beta, double offset=0.0) override

  Samples Gamma distribution.
- virtual double sampleGumbell (double mode, double scale) override

  Samples Gumbel distribution.
- virtual double sampleLogNormal (double mean, double stddev, double offset=0.0) override

  Samples Lognormal distribution.
- virtual double sampleNormal (double mean, double stddev) override

  Samples Normal distribution.
- virtual double sampleTriangular (double min, double mode, double max) override

  Samples Triangular distribution.
- virtual double sampleUniform (double min, double max) override

  Samples Uniform distribution in [min,max].
- virtual double sampleWeibull (double alpha, double scale) override

  Samples Weibull distribution.
- virtual double sampleBinomial (int trials, double p) override

  Samples Binomial distribution.

- virtual double sampleBernoulli (double p) override

  Samples Bernoulli distribution.
- virtual double sampleDiscrete (double prob, double value,...) override

  Samples arbitrary discrete distribution (variadic interface).
- virtual double sampleDiscrete (double *prob, double *value, int size) override

  Samples arbitrary discrete distribution (array interface).
- virtual double sampleGeometric (double p) override

  Samples Geometric distribution.
- void **reset** ()

  Reinitializes RNG state so the pseudo-random sequence restarts.
- virtual void setRNGparameters (RNG_Parameters *param) override

  Updates the pseudo-random generator internal parameters.
- virtual RNG_Parameters * getRNGparameters () const override

  Returns the current pseudo-random generator parameters.

17.827.1 Detailed Description

Default pseudo-random sampler implementation used by the kernel.
Implements a base RNG and multiple continuous/discrete distributions commonly used by simulation components and expression parser stochastic functions.

17.827.2 Member Function Documentation

17.827.2.1 getRNGparameters()

```
Sampler_if::RNG_Parameters * SamplerDefaultImpl1::getRNGparameters ( ) const  [override], [virtual]
```

Returns the current pseudo-random generator parameters.
Implements Sampler_if.

17.827.2.2 random()

```
double SamplerDefaultImpl1::random ( )  [override], [virtual]
```

Generates one base uniform pseudo-random number in [0,1).
Implements Sampler_if.

17.827.2.3 sampleBernoulli()

```
double SamplerDefaultImpl1::sampleBernoulli (
            double p )  [override], [virtual]
```

Samples Bernoulli distribution.
Implements Sampler_if.

17.827.2.4 sampleBeta() [1/2]

```
double SamplerDefaultImpl1::sampleBeta (
            double alpha,
            double beta )  [override], [virtual]
```

Samples standard Beta in [0,1].
Implements Sampler_if.

17.827.2.5 sampleBeta() [2/2]

```
double SamplerDefaultImpl1::sampleBeta (
            double alpha,
            double beta,
            double infLimit,
            double supLimit )  [override], [virtual]
```

Samples Beta in arbitrary interval [infLimit, supLimit].
Implements Sampler_if.

17.827.2.6 sampleBinomial()

```
double SamplerDefaultImpl1::sampleBinomial (
            int trials,
            double p )  [override], [virtual]
```

Samples Binomial distribution.
Implements Sampler_if.

17.827.2.7 sampleDiscrete() [1/2]

```
double SamplerDefaultImpl1::sampleDiscrete (
            double * prob,
            double * value,
            int size )  [override], [virtual]
```

Samples arbitrary discrete distribution (array interface).
Implements Sampler_if.

17.827.2.8 sampleDiscrete() [2/2]

```
double SamplerDefaultImpl1::sampleDiscrete (
            double prob,
            double value,
             ...  )  [override], [virtual]
```

Samples arbitrary discrete distribution (variadic interface).
Implements Sampler_if.

17.827.2.9 sampleErlang()

```
double SamplerDefaultImpl1::sampleErlang (
            double mean,
            int M,
            double offset = 0.0 )  [override], [virtual]
```

Samples Erlang distribution.
Implements Sampler_if.

17.827.2.10 sampleExponential()

```
double SamplerDefaultImpl1::sampleExponential (
            double mean,
            double offset = 0.0 )  [override], [virtual]
```

Samples Exponential distribution.
Implements Sampler_if.

17.827.2.11 sampleGamma()

```
double SamplerDefaultImpl1::sampleGamma (
            double alpha,
            double beta,
            double offset = 0.0 )  [override], [virtual]
```

Samples Gamma distribution.
Implements Sampler_if.

17.827.2.12 sampleGeometric()

```
double SamplerDefaultImpl1::sampleGeometric (
            double p )  [override], [virtual]
```

Samples Geometric distribution.
Implements Sampler_if.

17.827.2.13 sampleGumbell()

```
double SamplerDefaultImpl1::sampleGumbell (
            double mode,
            double scale )  [override], [virtual]
```

Samples Gumbel distribution.
Implements Sampler_if.

17.827.2.14 sampleLogNormal()

```
double SamplerDefaultImpl1::sampleLogNormal (
            double mean,
            double stddev,
            double offset = 0.0 )  [override], [virtual]
```

Samples Lognormal distribution.
Implements Sampler_if.

17.827.2.15 sampleNormal()

```
double SamplerDefaultImpl1::sampleNormal (
            double mean,
            double stddev )  [override], [virtual]
```

Samples Normal distribution.
Implements Sampler_if.

17.827.2.16 sampleTriangular()

```
double SamplerDefaultImpl1::sampleTriangular (
            double min,
            double mode,
            double max )  [override], [virtual]
```

Samples Triangular distribution.
Implements Sampler_if.

17.827.2.17 sampleUniform()

```
double SamplerDefaultImpl1::sampleUniform (
            double min,
            double max )  [override], [virtual]
```

Samples Uniform distribution in [min,max].
Implements Sampler_if.

17.827.2.18 sampleWeibull()

```
double SamplerDefaultImpl1::sampleWeibull (
            double alpha,
            double scale )  [override], [virtual]
```

Samples Weibull distribution.
Implements Sampler_if.

17.827.2.19 setRNGparameters()

```
void SamplerDefaultImpl1::setRNGparameters (
            RNG_Parameters * param )  [override], [virtual]
```

Updates the pseudo-random generator internal parameters.
Implements Sampler_if.
The documentation for this class was generated from the following files:

- kernel/statistics/SamplerDefaultImpl1.h
- kernel/statistics/SamplerDefaultImpl1.cpp

17.828 ScenarioExperiment_if Class Reference

The documentation for this class was generated from the following file:

- kernel/simulator/ScenarioExperiment_if.h

17.829 SceneToolController Class Reference

Controller for scene visualization, drawing tools, alignment, and zoom orchestration.
`#include <applications/gui/genesys/controllers/SceneToolController.h>`

Public Member Functions

- SceneToolController (ModelGraphicsView *graphicsView, Ui::MainWindow *ui, std::function< ModelGraphicsScene *()> currentScene, std::function< bool()> createModelImage, std::function< void()> unselectDrawIcons, std::function< bool()> checkSelectedDrawIcons, std::function< void(double)> gentleZoom, std::function< void()> actualizeActions, std::function< void()> actualizeTabPanes, int &zoomValue, bool &firstClickShow↩ Connection)

  Builds the scene-tool delegated controller used by MainWindow compatibility slots.
- void onActionShowGridTriggered ()

  Toggles grid visibility and requests scene redraw updates.
- void **onActionShowRuleTriggered** ()

  Toggles ruler visibility through the delegated scene callback path.
- void **onActionShowGuidesTriggered** ()

  Toggles guide visibility and keeps tab/action synchronization.
- void **onActionShowSnapTriggered** ()

  Toggles snap behavior on the current scene without changing tool wiring.
- void **onActionZoomInTriggered** ()

  Applies incremental zoom-in and updates shared zoom state.
- void **onActionZoomOutTriggered** ()

  Applies incremental zoom-out and updates shared zoom state.
- void onActionZoomAllTriggered ()

  Fits the current graphical model in view and re-synchronizes zoom widgets.
- void onActionZoomActualSizeTriggered ()

  Restores the graphical view to its default 1:1 zoom.
- void **syncZoomWidgetsForCurrentScale** ()

  Synchronizes the zoom slider and label with the current view scale.
- void onHorizontalSliderZoomGraphicalValueChanged (int value)

  Synchronizes slider-originated zoom changes with scene/view scale.
- void onActionDrawLineTriggered ()

  Activates line drawing mode as a compatibility façade wrapper.
- void **onActionDrawRectangleTriggered** ()

  Activates rectangle drawing mode as a compatibility façade wrapper.
- void **onActionDrawEllipseTriggered** ()

  Activates ellipse drawing mode as a compatibility façade wrapper.
- void **onActionDrawTextTriggered** ()

  Activates text drawing mode as a compatibility façade wrapper.
- void **onActionDrawPoligonTriggered** ()

  Activates polygon drawing mode as a compatibility façade wrapper.

- void **onActionAnimateSimulatedTimeTriggered** ()

  Toggles simulated-time animation overlays in the current scene.
- void **onActionAnimateVariableTriggered** ()

  Toggles variable animation overlays in the current scene.
- void **onActionAnimateCounterTriggered** ()

  Toggles counter animation overlays in the current scene.
- void **onActionAnimateAttributeTriggered** ()

  Activates attribute animation placeholder drawing.
- void **onActionAnimateEntityTriggered** ()

  Activates entity animation placeholder drawing.
- void **onActionAnimateEventTriggered** ()

  Activates event animation placeholder drawing.
- void **onActionAnimateExpressionTriggered** ()

  Activates expression animation placeholder drawing.
- void **onActionAnimatePlotTriggered** ()

  Activates plot animation placeholder drawing.
- void **onActionAnimateQueueTriggered** ()

  Activates queue animation placeholder drawing.
- void **onActionAnimateResourceTriggered** ()

  Activates resource animation placeholder drawing.
- void **onActionAnimateStationTriggered** ()

  Activates station animation placeholder drawing.
- void **onActionAnimateStatisticsTriggered** ()

  Activates statistics animation placeholder drawing.
- void **onActionConnectTriggered** ()

  Activates connection-creation mode and resets first-click compatibility state.
- void **onActionArranjeLeftTriggered** ()

  Aligns selected graphical items to the left boundary.
- void **onActionArranjeRightTriggered** ()

  Aligns selected graphical items to the right boundary.
- void **onActionArranjeTopTriggered** ()

  Aligns selected graphical items to the top boundary.
- void **onActionArranjeBototmTriggered** ()

  Aligns selected graphical items to the bottom boundary.
- void **onActionArranjeCenterTriggered** ()

  Aligns selected graphical items to horizontal center.
- void **onActionArranjeMiddleTriggered** ()

  Aligns selected graphical items to vertical middle.
- void **onActionActivateGraphicalSimulationTriggered** ()

  Toggles graphical simulation visualization mode and action state.
- void **onHorizontalSliderAnimationSpeedValueChanged** (int value)

  Applies animation-speed slider values to scene animation pacing.
- void **onActionSelectAllTriggered** ()

  Selects all graphical items in the active scene context.
- void **onActionShowInternalElementsTriggered** ()

  Toggles rendering of statistics data definitions and triggers representation refresh.
- void **onActionShowEditableElementsTriggered** ()

  Toggles rendering of editable data definitions and triggers representation refresh.
- void **onActionShowAttachedElementsTriggered** ()

  Toggles rendering of shared data definitions and triggers representation refresh.
- void **onCheckBoxShowElementsStateChanged** (int arg1)

Bridges checkbox state changes to the show-elements action wrapper.

- void **onCheckBoxShowInternalsStateChanged** (int arg1)

  Bridges checkbox state changes to the show-internals action wrapper.
- void **onCheckBoxShowEditableElementsStateChanged** (int arg1)

  Bridges checkbox state changes to the show-editable-elements action wrapper.
- void **onActionShowRecursiveElementsTriggered** ()

  Toggles recursive data-definition expansion and triggers representation refresh.
- void **onCheckBoxShowRecursiveStateChanged** (int arg1)

  Bridges recursive-render checkbox changes to model-image refresh workflow.
- void **onCheckBoxShowLevelsStateChanged** (int arg1)

  Bridges level-annotation checkbox changes to model-image refresh workflow.
- void **onActionGModelShowConnectTriggered** ()

  Toggles interactive connection-hint visibility in scene callback wiring.

17.829.1 Detailed Description

Controller for scene visualization, drawing tools, alignment, and zoom orchestration.
MainWindow delegates many menu/toolbar wrappers to this Phase-10 controller to keep the compatibility façade thin while concentrating scene-tool behavior in a dedicated component.
Responsibilities:

- handle grid/ruler/guides/snap toggles and zoom commands;
- orchestrate drawing/connection modes and alignment actions;
- coordinate animation toggles and scene-display filters through injected callbacks.

Boundaries:

- it does not own MainWindow or scene/view lifetime;
- it does not perform model persistence/export/simulation lifecycle work;
- it relies on injected references for shared UI state (zoom, first-click connection state).

17.829.2 Constructor & Destructor Documentation

17.829.2.1 SceneToolController()

```
SceneToolController::SceneToolController (
            ModelGraphicsView * graphicsView,
            Ui::MainWindow * ui,
            std::function< ModelGraphicsScene *()> currentScene,
            std::function< bool()> createModelImage,
            std::function< void()> unselectDrawIcons,
            std::function< bool()> checkSelectedDrawIcons,
            std::function< void(double)> gentleZoom,
            std::function< void()> actualizeActions,
            std::function< void()> actualizeTabPanes,
            int & zoomValue,
            bool & firstClickShowConnection )
```

Builds the scene-tool delegated controller used by MainWindow compatibility slots.
Stores only the narrow collaborators needed for scene-tool orchestration.

17.829.3 Member Function Documentation

17.829.3.1 onActionDrawLineTriggered()

```
void SceneToolController::onActionDrawLineTriggered ( )
```

Activates line drawing mode as a compatibility façade wrapper.
Activates the line drawing tool while preserving cursor semantics.

17.829.3.2 onActionShowGridTriggered()

```
void SceneToolController::onActionShowGridTriggered ( )
```

Toggles grid visibility and requests scene redraw updates.
Synchronizes the grid action with the current scene state.

17.829.3.3 onActionZoomActualSizeTriggered()

```
void SceneToolController::onActionZoomActualSizeTriggered ( )
```

Restores the graphical view to its default 1:1 zoom.
Restores the graphics view to the 1:1 zoom level and re-synchronizes the slider.

17.829.3.4 onActionZoomAllTriggered()

```
void SceneToolController::onActionZoomAllTriggered ( )
```

Fits the current graphical model in view and re-synchronizes zoom widgets.
Fits the scene to view and resets the zoom slider baseline.

17.829.3.5 onHorizontalSliderZoomGraphicalValueChanged()

```
void SceneToolController::onHorizontalSliderZoomGraphicalValueChanged (
            int value )
```

Synchronizes slider-originated zoom changes with scene/view scale.
Applies the zoom slider value to the graphics view transform.
The documentation for this class was generated from the following files:

- applications/gui/genesys/controllers/SceneToolController.h
- applications/gui/genesys/controllers/SceneToolController.cpp

17.830 SchedulableItem Class Reference

One (expression, duration, rule) triple inside a Schedule.
`#include <plugins/data/DiscreteProcessing/Schedule.h>`

Public Types

- enum class **Rule** : int { **IGNORE** = 1 , **PREEMPT** = 2 , **WAIT** = 3 }

Public Member Functions

- **SchedulableItem** (std::string expression, double duration, SchedulableItem::Rule rule=SchedulableItem::↩ Rule::IGNORE)
- void **setDuration** (double duration)
- double **getDuration** () const
- void **setExpression** (std::string expression)
- std::string **getExpression** () const
- void **setRule** (SchedulableItem::Rule rule)
- SchedulableItem::Rule **getRule** () const

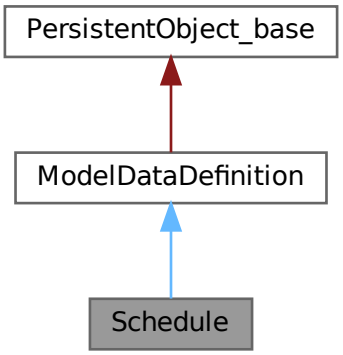
17.830.1 Detailed Description

One (expression, duration, rule) triple inside a Schedule.
`expression` is evaluated to produce the value that applies for `duration` time units before moving to the next item; this is the closest correspondence to one row of Arena's Schedule module "Durations" repeat group (Value/↩ Duration pairs), generalized from a static value to a parsed expression. `rule` mirrors Arena's per-resource "↩ Schedule Rule" (ignore/preempt/wait a busy unit on a capacity decrease), attached here per schedule item rather than per Resource; see Resource for the open question of how/whether the two map onto each other.
The documentation for this class was generated from the following file:

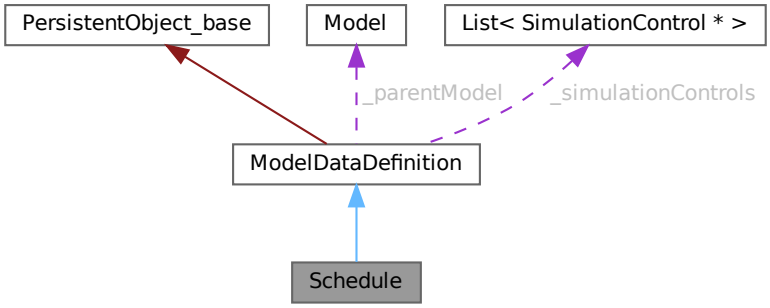
- plugins/data/DiscreteProcessing/Schedule.h

17.831 Schedule Class Reference

Inheritance diagram for Schedule:



Collaboration diagram for Schedule:



Public Member Functions

- **Schedule** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- std::string **getExpression** ()
- List< SchedulableItem * > * **getSchedulableItems** () const
- void **setRepeatAfterLast** (bool _repeatAfterLast)
- bool **isRepeatAfterLast** () const

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation () override
- virtual void _initBetweenReplications () override
- virtual void _addSimulationControl (SimulationControl *property) override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)

- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model ∗ **_parentModel**
- List< SimulationControl ∗ > ∗ **_simulationControls** = new List<SimulationControl∗>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.831.1 Member Function Documentation

17.831.1.1 _addSimulationControl()

```
void Schedule::_addSimulationControl (
            SimulationControl * property )  [override], [protected], [virtual]
```

Keep local control registration aligned with the base model-data contract.
Reimplemented from ModelDataDefinition.

17.831.1.2 _check()

```
bool Schedule::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Validate that schedule has items and non-negative durations.
Reimplemented from ModelDataDefinition.

17.831.1.3 _getParserChangesInformation()

```
ParserChangesInformation * Schedule::_getParserChangesInformation ( )  [override], [protected],
[virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes Return parser changes required by Schedule.
No parser customization is currently necessary.
Reimplemented from ModelDataDefinition.

17.831.1.4 _initBetweenReplications()

```
void Schedule::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.831.1.5 _loadInstance()

```
bool Schedule::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Load schedulable items and repeat behavior.
Reimplemented from ModelDataDefinition.

17.831.1.6 _saveInstance()

```
void Schedule::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Persist schedule items and repeat behavior.
Reimplemented from ModelDataDefinition.

17.831.1.7 show()

```
std::string Schedule::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/DiscreteProcessing/Schedule.h
- plugins/data/DiscreteProcessing/Schedule.cpp

17.832 testing::ScopedTrace Class Reference

Public Member Functions

- template<typename T >
  ScopedTrace (const char *file, int line, const T &message)
- **ScopedTrace** (const char *file, int line, const char *message)
- **ScopedTrace** (const char *file, int line, const std::string &message)

The documentation for this class was generated from the following files:

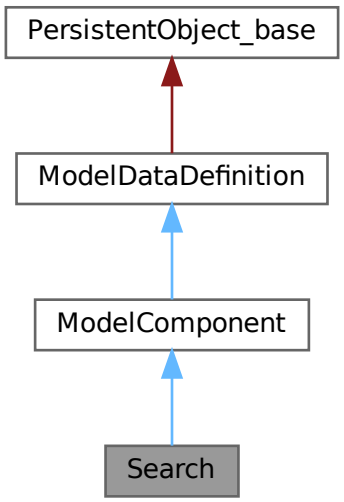
- gtest/gtest/gtest.h
- gtest/src/gtest.cc

17.833 Search Class Reference

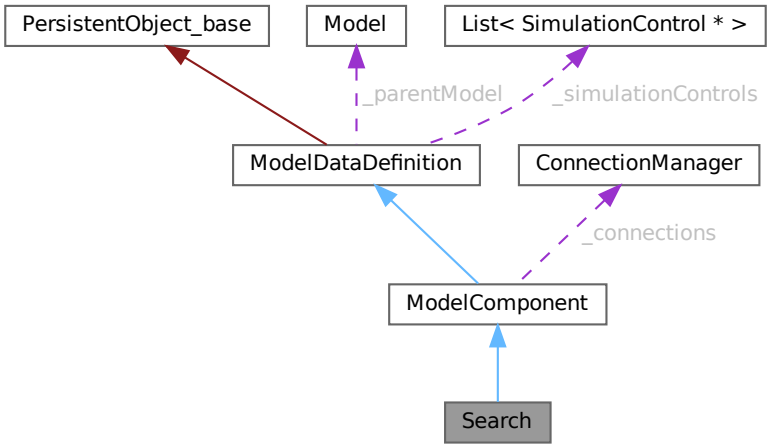
Scans a Queue or an EntityGroup rank range for the first entity satisfying a condition, saving the found rank (or 0) to an attribute.

`#include <plugins/components/Decisions/Search.h>`

Inheritance diagram for Search:



Collaboration diagram for Search:



Public Types

- enum class **SearchInType** : int { **QUEUE** = 0 , **ENTITYGROUP** = 1 , **num_elements** = 2 }

Public Member Functions

- **Search** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setSaveFounRankAttribute** (std::string _saveFounRankAttribute)
- std::string **getSaveFounRankAttribute** () const
- void **setSearchCondition** (std::string _searchCondition)
- std::string **getSearchCondition** () const
- void **setEndRank** (std::string _endRank)
- std::string **getEndRank** () const
- void **setStartRank** (std::string _startRank)
- std::string **getStartRank** () const
- void **setSearchInName** (std::string searchInName)
- std::string **getSearchInName** () const
- void **setSearchIn** (ModelDataDefinition *_searchIn)
- ModelDataDefinition * **getSearchIn** () const
- void **setSearchInType** (Search::SearchInType _searchInType)
- Search::SearchInType **getSearchInType** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

Connects this component to another component.

- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.

- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static std::string **convertEnumToStr** (SearchInType type)
- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions

- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.833.1 Detailed Description

Scans a Queue or an EntityGroup rank range for the first entity satisfying a condition, saving the found rank (or 0) to an attribute.
Arena correspondence: the "Search module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 64), which also supports a third Type: searching a free-standing index range via an expression (setting the global system variable `J`), with no queue or group involved at all.
`SearchInType` (`QUEUE/ENTITYGROUP`) matches two of Arena's three Types; `_startRank/_endRank/_searchCondition/_saveFounRankAttribute` match Starting/Ending Value and Search Condition.
Known difference from Arena: the free-standing expression-only search Type has no corresponding `SearchInType` enumerator.

17.833.2 Member Function Documentation

17.833.2.1 _check()

```
bool Search::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.833.2.2 _createEditableDataDefinitions()

```
void Search::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
Reimplemented from ModelDataDefinition.

17.833.2.3 _loadInstance()

```
bool Search::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.833.2.4 _onDispatchEvent()

```
void Search::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.833.2.5 _saveInstance()

```
void Search::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.833.2.6 show()

```
std::string Search::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

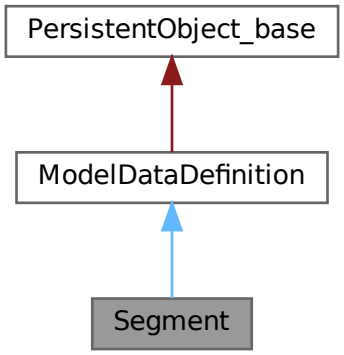
- plugins/components/Decisions/Search.h
- plugins/components/Decisions/Search.cpp

17.834 Segment Class Reference

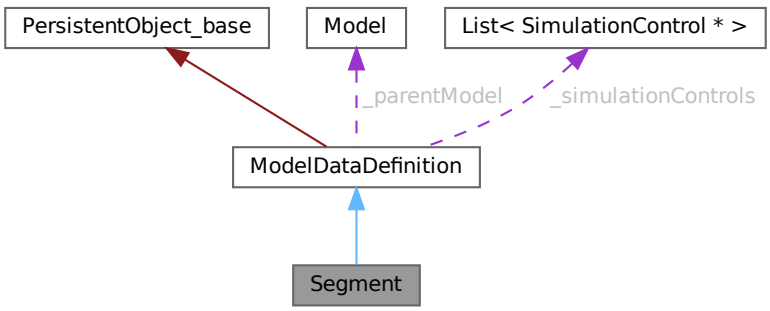
Data definition holding the ordered sequence of stations (and the length between consecutive stations) that make up one Conveyor's physical path.

`#include <plugins/data/MaterialHandling/Segment.h>`

Inheritance diagram for Segment:



Collaboration diagram for Segment:



Public Member Functions

- **Segment** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- List< SegmentStep * > * **getSteps** () const
- void **insertStep** (SegmentStep *step)
- double **getDistanceBetween** (const std::string &fromStationName, const std::string &toStationName) const

  Sums lengthToNext from fromStationName up to (not including) toStationName; returns -1 when either station is absent or out of order.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition ∗ **LoadInstance** (Model ∗model, PersistenceRecord ∗fields)
- static PluginInformation ∗ **GetPluginInformation** ()
- static ModelDataDefinition ∗ **NewInstance** (Model ∗model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition ∗ LoadInstance (Model ∗model, PersistenceRecord ∗fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition ∗ NewInstance (Model ∗model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord ∗fields, ModelDataDefinition ∗modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition ∗modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition ∗modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition ∗modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition ∗modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord ∗fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord ∗fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation ∗ _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl ∗control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)

- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.834.1 Detailed Description

Data definition holding the ordered sequence of stations (and the length between consecutive stations) that make up one Conveyor's physical path.
Arena correspondence: the "Segment module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Transfer Panel", p. 103), which defines a Segment Set (an ordered list of stations along a conveyor with the length of each segment).
Known limitation of this initial implementation: distance lookup only walks the path forward in declaration order (`getDistanceBetween()` sums `lengthToNext` from the "from" station up to the "to" station); there is no support for a conveyor whose stations are not visited in a single consistent order.

17.834.2 Member Function Documentation

17.834.2.1 _check()

```
bool Segment::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.834.2.2 _loadInstance()

```
bool Segment::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.834.2.3 _saveInstance()

```
void Segment::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.834.2.4 show()

```
std::string Segment::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/MaterialHandling/Segment.h
- plugins/data/MaterialHandling/Segment.cpp

17.835 SegmentStep Class Reference

One (station, length-to-next-station) step of a Segment's ordered conveyor path.

`#include <plugins/data/MaterialHandling/Segment.h>`

Public Member Functions

- **SegmentStep** (std::string stationName, double lengthToNext)

Public Attributes

- std::string **stationName**
- double **lengthToNext**

17.835.1 Detailed Description

One (station, length-to-next-station) step of a Segment's ordered conveyor path.
The documentation for this class was generated from the following file:

- plugins/data/MaterialHandling/Segment.h

17.836 SeizableItem Class Reference

Public Types

- enum class **SelectionRule** : int {
  CYCLICAL = 0 , **RANDOM** = 1 , **SPECIFICMEMBER** = 2 , **LARGESTREMAININGCAPACITY** = 3 ,
  SMALLESTNUMBERBUSY = 4 , **PREFEREDORDER** = 5 , **num_elements** = 6 }
- enum class **SeizableType** : int { **RESOURCE** = 0 , **SET** = 1 , **num_elements** = 2 }

Public Member Functions

- **SeizableItem** (ModelDataDefinition *resourceOrSet, std::string quantityExpression="1", SeizableItem↩
  ::SelectionRule selectionRule=SeizableItem::SelectionRule::LARGESTREMAININGCAPACITY, std::string
  saveAttribute="", std::string index="0")
- **SeizableItem** (Model *model, std::string resourceName, std::string quantityExpression="1", SeizableItem↩
  ::SelectionRule selectionRule=SeizableItem::SelectionRule::LARGESTREMAININGCAPACITY, std::string
  saveAttribute="", std::string index="0")
- **SeizableItem** (SeizableItem *original)
- bool **loadInstance** (PersistenceRecord *fields, unsigned int parentIndex)
- void **saveInstance** (PersistenceRecord *fields, unsigned int parentIndex, bool saveDefault)
- bool **loadInstance** (PersistenceRecord *fields)
- void **saveInstance** (PersistenceRecord *fields, bool saveDefaults)
- std::string **show** ()
- void **setIndex** (std::string index)
- std::string **getIndex** () const
- void **setSaveAttribute** (std::string saveAttribute)
- std::string **getSaveAttribute** () const
- void **setSelectionRule** (SelectionRule selectionRule)
- SelectionRule **getSelectionRule** () const
- void **setQuantityExpression** (std::string quantityExpression)
- std::string **getQuantityExpression** () const
- std::string **getResourceName** () const
- std::string **getName** () const
- void **setResource** (Resource *resource)
- Resource * **getResource** () const
- void **setSet** (Set *set)
- Set * **getSet** () const
- SeizableType **getSeizableType** () const
- void **setLastMemberSeized** (unsigned int lastMemberSeized)

- unsigned int **getLastMemberSeized** () const
- ModelDataDefinition * **getSeizable** () const
- void **setElementManager** (ModelDataManager *modeldataManager)
- void **setLastPreferedOrder** (unsigned int lastPreferedOrder)
- unsigned int **getLastPreferedOrder** () const
- List< SimulationControl * > * **getSimulationControls** () const
- void **_addSimulationControl** (SimulationControl *control)
- std::string **getTypeDC** ()

Static Public Member Functions

- static std::string **convertEnumToStr** (SelectionRule rule)
- static std::string **convertEnumToStr** (SeizableType type)

The documentation for this class was generated from the following files:

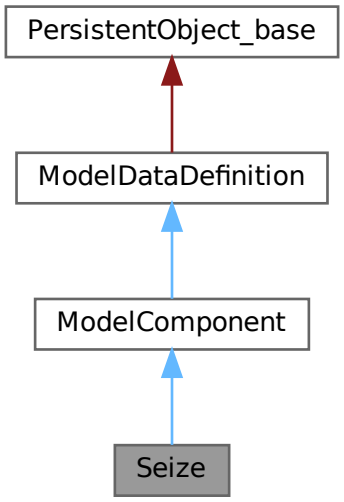
- plugins/components/DiscreteProcessing/auxiliar/SeizableItem.h
- plugins/components/DiscreteProcessing/auxiliar/SeizableItem.cpp

17.837 Seize Class Reference

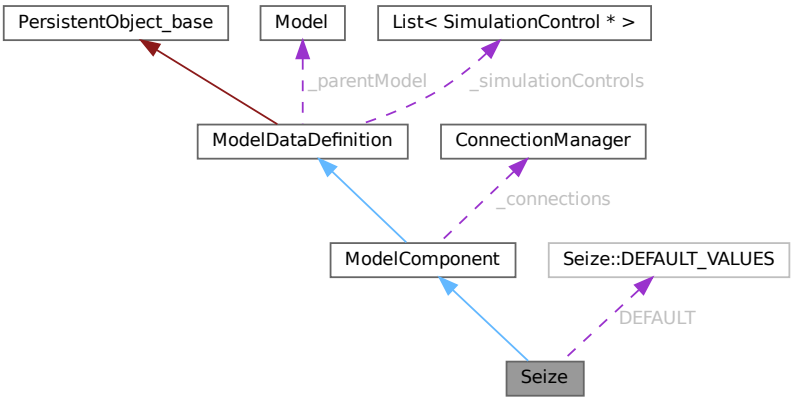
Allocates one or more resources (or resource-set members) to an entity, queueing it if they are not simultaneously available.

`#include <plugins/components/DiscreteProcessing/Seize.h>`

Inheritance diagram for Seize:



Collaboration diagram for Seize:



Classes

- struct DEFAULT_VALUES

Public Member Functions

- **Seize** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setPriority** (unsigned short _priority)
- unsigned short **getPriority** () const
- void **setAllocationType** (Util::AllocationType _allocationType)
- Util::AllocationType **getAllocationType** () const
- void **setQueue** (Queue *queue)

  Deprected.
- List< SeizableItem * > * **getSeizeRequests** () const
- void **addRequest** (SeizableItem *newRequest)
- void **removeRequest** (SeizableItem *request)
- void **setQueueableItem** (QueueableItem *_queueableItem)
- QueueableItem * **getQueueableItem** () const
- void **setPriorityExpression** (std::string _priorityExpression)
- std::string **getPriorityExpression** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.

- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Public Attributes

- const struct Seize::DEFAULT_VALUES **DEFAULT**

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _initBetweenReplications () override
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions

- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.837.1 Detailed Description

Allocates one or more resources (or resource-set members) to an entity, queueing it if they are not simultaneously available.
Arena correspondence: the "Seize module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", pp. 62-63). `_seizeRequests` (a list of `SeizableItem`) matches Arena's Type/Resource Name/Set Name/Quantity/Selection Rule/Save Attribute fields; `_queueableItem` matches Queue Type; `_priority`/`_priorityExpression` match Priority; `_allocationType` matches Allocation.
Confirmed behavior: on a successful seize, stamps `Entity.Allocation.<ResourceName>` with `_allocationType` — this is read back by Release, which credits the resource's held time to that category (see `docs/ai_assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.10 and §6.3 for the full category-time-allocation mechanism shared with Delay).
Known difference from Arena: no "Resource State" field (post-seize state assignment) — GenESyS's `ResourceState` is a fixed enum, not a user-defined StateSet (see Resource, §5.4).

17.837.2 Member Function Documentation

17.837.2.1 _check()

```
bool Seize::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.837.2.2 _createEditableDataDefinitions()

```
void Seize::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```
Reimplemented from ModelDataDefinition.

17.837.2.3 _initBetweenReplications()

```
void Seize::_initBetweenReplications ( )  [override], [protected], [virtual]
```
Reimplemented from ModelDataDefinition.

17.837.2.4 _loadInstance()

```
bool Seize::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```
virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.837.2.5 _onDispatchEvent()

```
void Seize::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```
Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.837.2.6 _saveInstance()

```
void Seize::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```
Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.837.2.7 show()

```
std::string Seize::show ( )  [override], [virtual]
```
virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

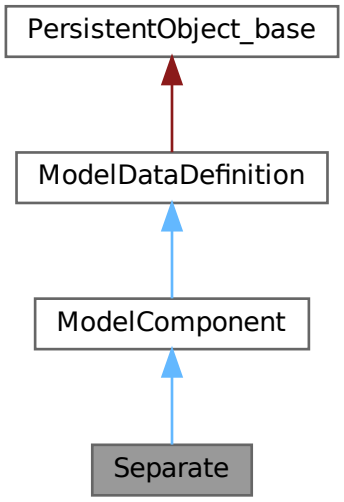
- plugins/components/DiscreteProcessing/Seize.h
- plugins/components/DiscreteProcessing/Seize.cpp

17.838 Separate Class Reference

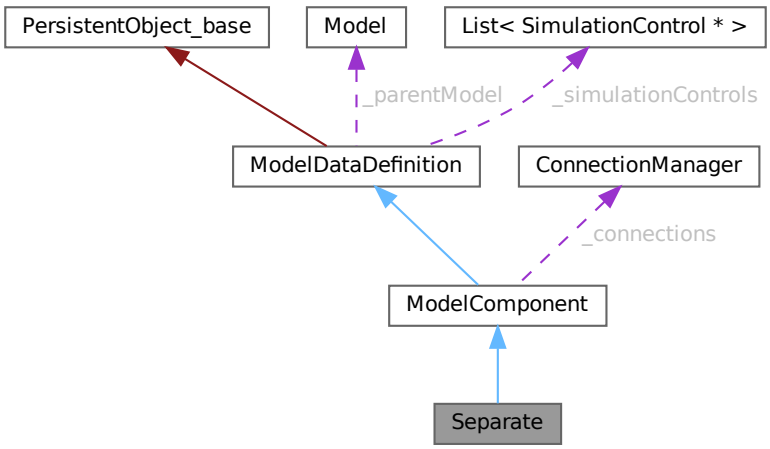
Releases the original member entities of a temporary Batch group back into the model.

`#include <plugins/components/Grouping/Separate.h>`

Inheritance diagram for Separate:



Collaboration diagram for Separate:



Public Member Functions

- **Separate** (Model *model, std::string name="")
- virtual std::string show () override

virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.838.1 Detailed Description

Releases the original member entities of a temporary Batch group back into the model.
Arena correspondence: the "Separate module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", p. 39), which has two Types: Duplicate Original (clone the incoming entity N times, optionally splitting VA/NVA/wait/transfer/other cost and time between the copies) and Split Existing Batch (recover the original entities grouped by Batch).
Only Split Existing Batch is implemented here: `_onDispatchEvent()` reads the `Entity.Group` marker attribute, looks up the matching EntityGroup (created by Batch), and releases every original member with its group marker cleared before removing the temporary representative entity.
Known difference from Arena: there is no Duplicate Original mode — no entity-cloning code path exists in this component at all. See `docs/ai_assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.8; Clone (`plugins/components/DiscreteProcessing/Clone.h`) may be the intended substitute but has not been audited yet.

17.838.2 Member Function Documentation

17.838.2.1 _check()

```
bool Separate::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.838.2.2 _loadInstance()

```
bool Separate::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.838.2.3 _onDispatchEvent()

```
void Separate::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.838.2.4 _saveInstance()

```
void Separate::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.838.2.5 show()

```
std::string Separate::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

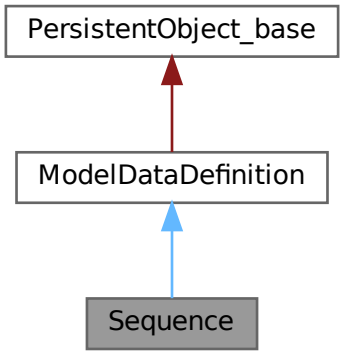
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

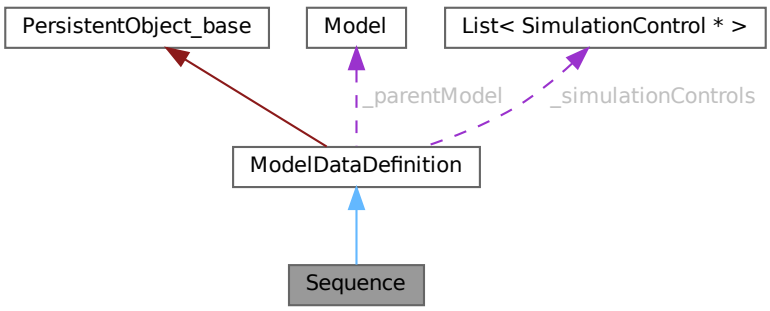
- plugins/components/Grouping/Separate.h
- plugins/components/Grouping/Separate.cpp

17.839 Sequence Class Reference

Data definition holding an ordered list of visitation steps (SequenceStep), each naming a Station or Label an entity should reach next, with optional attribute/variable assignments per step.
`#include <plugins/data/MaterialHandling/Sequence.h>`
Inheritance diagram for Sequence:



Collaboration diagram for Sequence:



Public Member Functions

- **Sequence** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- List< SequenceStep * > * **getSteps** () const

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)

- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.839.1 Detailed Description

Data definition holding an ordered list of visitation steps (`SequenceStep`), each naming a Station or Label an entity should reach next, with optional attribute/variable assignments per step.
Arena correspondence: the "Sequence module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Transfer Panel", pp. 100-101), which defines an ordered visitation list of stations (jobsteps) plus per-step attribute/variable/picture assignments, tracked at runtime through the special-purpose Entity.Sequence/↩ Entity.Jobstep/ Entity.PlannedStation attributes.
GenESyS extends the concept by letting a SequenceStep target a Label instead of a Station (useful outside pure station-to-station material handling flows).
Known difference from Arena, to confirm in `Sequence.cpp`: steps are stored as a plain ordered `List`, so Arena's named, out-of-order "Step Name"/"Next Step" jump targets have no obvious equivalent here.

17.839.2 Member Function Documentation

17.839.2.1 _check()

```
bool Sequence::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.839.2.2 _createAttachedAttributes()

```
void Sequence::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.839.2.3 _loadInstance()

```
bool Sequence::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.839.2.4 _saveInstance()

```
void Sequence::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.839.2.5 show()

```
std::string Sequence::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

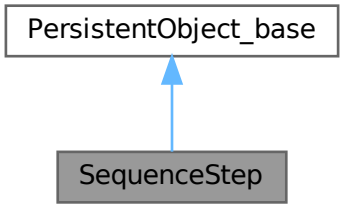
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

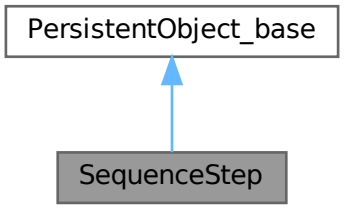
- plugins/data/MaterialHandling/Sequence.h
- plugins/data/MaterialHandling/Sequence.cpp

17.840 SequenceStep Class Reference

Inheritance diagram for SequenceStep:



Collaboration diagram for SequenceStep:



Public Member Functions

- **SequenceStep** (Station *station, std::list< Assignment * > *assignments=nullptr)
- **SequenceStep** (Label *label, std::list< Assignment * > *assignments=nullptr)
- **SequenceStep** (Model *model, std::string stationOrLabelName, bool isStation=true, std::list< Assignment * > *assignments=nullptr)
- virtual bool **_loadInstance** (PersistenceRecord *fields, unsigned int parentIndex)
- virtual void **_saveInstance** (PersistenceRecord *fields, unsigned int parentIndex, bool saveDefaultValues)
- virtual bool _loadInstance (PersistenceRecord *fields) override

  Loads object fields from a deserialization structure.
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

- std::list< Assignment * > * **getAssignments** () const
- void **setStation** (Station *_station)
- Station * **getStation** () const
- void **setElementManager** (ModelDataManager *_modeldataManager)
- void **setLabel** (Label *_label)
- Label * **getLabel** () const

Additional Inherited Members

17.840.1 Member Function Documentation

17.840.1.1 _loadInstance()

```
bool SequenceStep::_loadInstance (
            PersistenceRecord * fields )  [override], [virtual]
```

Loads object fields from a deserialization structure.

Parameters

fields	a table-like structure which can be queried for parsed fields.

Returns

a boolean which is true iff the load operation was successful.

Implements PersistentObject_base.

17.840.1.2 _saveInstance()

```
void SequenceStep::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

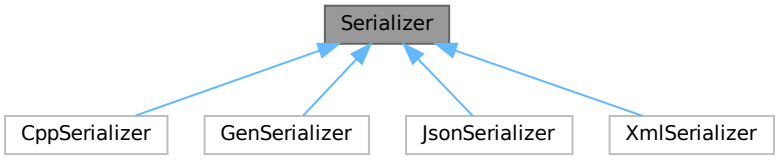
Implements PersistentObject_base.
The documentation for this class was generated from the following files:

- plugins/data/MaterialHandling/Sequence.h
- plugins/data/MaterialHandling/Sequence.cpp

17.841 Serializer Class Reference

Serialization/Deserialization interface for simulation models.
`#include <kernel/simulator/persistence/Serializer.h>`

Inheritance diagram for Serializer:



Public Member Functions

- virtual PersistenceRecord * newPersistenceRecord ()=0

  Creates an empty persistence record.
- virtual bool dump (std::ostream &output)=0

  Writes formatted contents to a given output stream.
- virtual bool load (std::istream &input)=0

  Attempts to load contents from a given input stream.
- virtual bool get (const std::string &name, PersistenceRecord *entry)=0

  Tries to fetch an entry with the given label as a key.
- virtual bool put (const std::string name, const std::string type, const Util::identification id, PersistenceRecord *fields)=0

  Upserts an entry into this structure.
- virtual int for_each (std::function< int(const std::string &) > delegate)=0

  Iterates over this structure's entries, ordered by ID.

17.841.1 Detailed Description

Serialization/Deserialization interface for simulation models.
This can be seen as an associative array which knows how to ser/deserialize itself. The "contents" of a serializer are the set of entries it contains.
Its entries are identified by strings, which correspond to unique component/datadef names. Each entry has its own name, a type, an id, and an associated set of fields. All of these are contained within a `PersistenceRecord` instance.
Serializers also act as factories for `PersistenceRecord`s, which can then only be used with the serializer which originated them.

17.841.2 Member Function Documentation

17.841.2.1 dump()

```
virtual bool Serializer::dump (
            std::ostream & output )  [pure virtual]
```

Writes formatted contents to a given output stream.

Parameters

output	output stream to write to.

Returns

whether the current state of this structure's contents was deemed valid for serialization.

Implemented in CppSerializer, GenSerializer, JsonSerializer, and XmlSerializer.

17.841.2.2 for_each()

```
virtual int Serializer::for_each (
            std::function< int(const std::string &) > delegate )  [pure virtual]
```

Iterates over this structure's entries, ordered by ID.

Parameters

delegate	delegate which is applied to every entry-identifying key, until it returns a non-zero value.

Returns

either zero, or the first non-zero value returned by the delegate.

Implemented in CppSerializer, GenSerializer, JsonSerializer, and XmlSerializer.

17.841.2.3 get()

```
virtual bool Serializer::get (
            const std::string & name,
            PersistenceRecord * entry )  [pure virtual]
```

Tries to fetch an entry with the given label as a key.

Parameters

name	a label which uniquely identifies an entry.
entry	(out-param) a pointer to a caller-owned entry, which is filled on success.

Returns

whether the entry was found and the out-param was actually written to.

Implemented in CppSerializer, GenSerializer, JsonSerializer, and XmlSerializer.

17.841.2.4 load()

```
virtual bool Serializer::load (
            std::istream & input )  [pure virtual]
```

Attempts to load contents from a given input stream.

Returns

whether the input was well-formatted and loading was successful.

Implemented in CppSerializer, GenSerializer, JsonSerializer, and XmlSerializer.

17.841.2.5 newPersistenceRecord()

```
virtual PersistenceRecord * Serializer::newPersistenceRecord ( )  [pure virtual]
```

Creates an empty persistence record.
Although the returned record can be freed normally with `delete`, care must be taken to only use records which come from a implementation, when configuring that implementation's contents.
Implemented in CppSerializer, GenSerializer, JsonSerializer, and XmlSerializer.

17.841.2.6 put()

```
virtual bool Serializer::put (
            const std::string name,
            const std::string type,
            const Util::identification id,
            PersistenceRecord * fields )  [pure virtual]
```

Upserts an entry into this structure.

Parameters

name	a label which uniquely identifies the entry.
type	a type name.
id	a component/datadef ID number.
fields	a persistence record for the entry, whose contents will be copied if valid.

Returns

whether the entry was valid and was actually copied into this structure.

Implemented in CppSerializer, GenSerializer, JsonSerializer, and XmlSerializer.
The documentation for this class was generated from the following files:

- kernel/simulator/persistence/Serializer.h
- kernel/simulator/persistence/Serializer.cpp

17.842 SessionContext Struct Reference

Owns the state that belongs to one authenticated web session.
`#include <applications/worker/session/SessionContext.h>`

Public Attributes

- std::string **sessionId**

  Stable session identifier returned to the client.
- std::string **accessToken**

  Bearer token used to authenticate API calls.
- std::filesystem::path **workspacePath**

  Filesystem workspace reserved for the session.
- std::unique_ptr< Simulator > **simulator**

  Simulator instance scoped to this session.
- std::mutex **mutex**

  Guards session-local mutable state.

17.842.1 Detailed Description

Owns the state that belongs to one authenticated web session.
Each session binds one access token to one workspace directory and one simulator instance. The mutex protects session-local state during API calls and background job execution.
The documentation for this struct was generated from the following file:

- applications/worker/session/SessionContext.h

17.843 SessionManager Class Reference

Creates, stores and resolves authenticated web sessions.
`#include <applications/worker/session/SessionManager.h>`

Public Types

- using **SimulatorFactory** = std::function< std::unique_ptr< Simulator >() >

  Factory used to build one simulator per session.

Public Member Functions

- SessionManager (TokenService &tokenService, SimulatorFactory simulatorFactory)

  Builds a manager with the token service and simulator factory to use.
- SessionContext * createSession ()

  Creates a new session, workspace and simulator instance.
- SessionContext * getSessionByToken (const std::string &token)

  Resolves the session associated with a bearer token.

17.843.1 Detailed Description

Creates, stores and resolves authenticated web sessions.
SessionManager owns the token-to-session map used by the web API. It also creates per-session simulator instances and assigns each session an isolated workspace directory.

17.843.2 Constructor & Destructor Documentation

17.843.2.1 SessionManager()

```
SessionManager::SessionManager (
            TokenService & tokenService,
            SimulatorFactory simulatorFactory )
```

Builds a manager with the token service and simulator factory to use.
Builds the session registry and session workspace root.

Parameters

tokenService	Service responsible for generating and validating tokens.
simulatorFactory	Factory used to create a fresh simulator per session.

17.843.3 Member Function Documentation

17.843.3.1 createSession()

```
SessionContext * SessionManager::createSession ( )
```

Creates a new session, workspace and simulator instance.
Creates a session, allocates token, simulator instance and workspace.

Returns

Pointer to the stored session context.

17.843.3.2 getSessionByToken()

```
SessionContext * SessionManager::getSessionByToken (
            const std::string & token )
```

Resolves the session associated with a bearer token.
Resolves a session by bearer token.

Parameters

token	Bearer token presented by the client.

Returns

Pointer to the matching session or nullptr when not found.

The documentation for this class was generated from the following files:

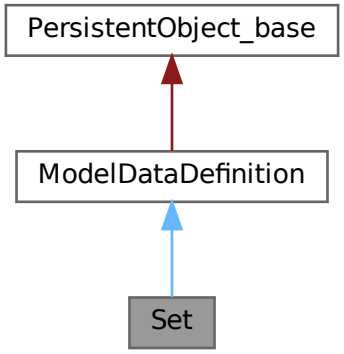
- applications/worker/session/SessionManager.h
- applications/worker/session/SessionManager.cpp

17.844 Set Class Reference

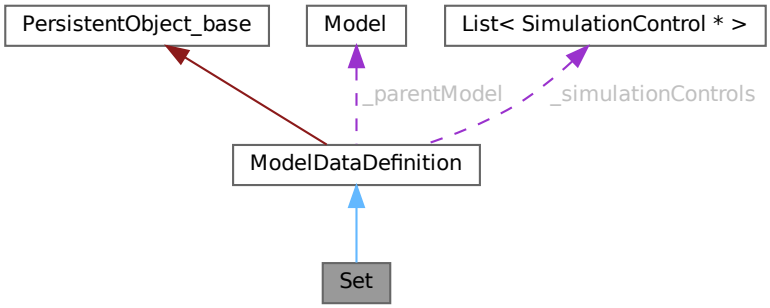
Data definition grouping an ordered list of ModelDataDefinition members of one concrete type, used by selection-rule-based components (e.g. Resource sets consumed by Seize-family components).

`#include <plugins/data/Logic/Set.h>`

Inheritance diagram for Set:



Collaboration diagram for Set:



Public Member Functions

- **Set** (Model *model, std::string name="", std::vector< std::string > allowedElementTypes={})
- virtual std::string show () override

  Returns a textual representation of this data definition.
- std::string **getSetOfType** () const
- bool setSetOfType (const std::string &setOfType)

  Sets the concrete member type for this Set while it is still structurally compatible.
- bool **canChangeSetOfType** () const

Returns true while the selected concrete member type can still be changed.

- void setAllowedElementTypes (const std::vector< std::string > &allowedElementTypes)

  Restricts which ModelDataDefinition subclasses may be created through this Set.
- void **addAllowedElementType** (const std::string &allowedElementType)

  Adds one accepted ModelDataDefinition subclass name to this Set creation contract.
- std::vector< std::string > **getAllowedElementTypes** () const

  Returns the currently accepted concrete member types for GUI and tooling contracts.
- List< ModelDataDefinition * > * **getElementSet** () const
- void **addElementSet** (ModelDataDefinition *newElement)
- void **removeElementSet** (ModelDataDefinition *element)
- ModelDataDefinition * createElementSetOfType (const std::string &typeName, const std::string &name="")

  Creates a new model data definition of the selected Set member type.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation () override
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.844.1 Detailed Description

Data definition grouping an ordered list of ModelDataDefinition members of one concrete type, used by selection-rule-based components (e.g. Resource sets consumed by Seize-family components).
Arena correspondence: Arena's "Set module"/"Advanced Set module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel" p. 51 and "The Advanced Process Panel" p. 69), which define closed, named set kinds: Resource, Counter, Tally, Entity (type), Entity Picture, Queue, Storage and Other.
Known difference from Arena (broader, not narrower, than the Arena concept): this class is polymorphic rather than a fixed enumeration of kinds. `setSetOfType()`/`getSetOfType()` record the concrete ModelDataDefinition subclass name of the first inserted member, and `setAllowedElementTypes()`/`addAllowedElement↩Type()` let a component owner (e.g. SeizableItem, QueueableItem) restrict a Set instance to one or more accepted subclasses. This means a GenESyS Set can, in principle, hold any registered ModelDataDefinition plugin type, not only the five kinds Arena hard-codes. There is no dedicated "picture" data definition, so Arena's Entity Picture sets have no GenESyS equivalent.

17.844.2 Member Function Documentation

17.844.2.1 _check()

```
bool Set::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.844.2.2 _createAttachedAttributes()

```
void Set::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.844.2.3 _getParserChangesInformation()

```
ParserChangesInformation * Set::_getParserChangesInformation ( )  [override], [protected],
[virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
Reimplemented from ModelDataDefinition.

17.844.2.4 _loadInstance()

```
bool Set::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.844.2.5 _saveInstance()

```
void Set::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.844.2.6 createElementSetOfType()

```
ModelDataDefinition * Set::createElementSetOfType (
            const std::string & typeName,
            const std::string & name = "" )
```

Creates a new model data definition of the selected Set member type.
This method is the kernel-side factory used by generic property editors. The editor provides the intended type; Set validates the current type/allowed-type contract and delegates actual object construction to PluginManager so plugin registration rules stay centralized.

17.844.2.7 setAllowedElementTypes()

```
void Set::setAllowedElementTypes (
            const std::vector< std::string > & allowedElementTypes )
```

Restricts which ModelDataDefinition subclasses may be created through this Set.
Empty means "use every creatable ModelDataDefinition plugin known by the simulator". Context owners such as SeizableItem and QueueableItem use this to expose Resource-only or Queue-only Set editing in the GUI without hard-coding those rules in the editor.

17.844.2.8 setSetOfType()

```
bool Set::setSetOfType (
            const std::string & setOfType )
```

Sets the concrete member type for this Set while it is still structurally compatible.
A Set is polymorphic only before its first member is inserted. Once members exist, all subsequent members must match this selected ModelDataDefinition subclass name.

17.844.2.9 show()

```
std::string Set::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

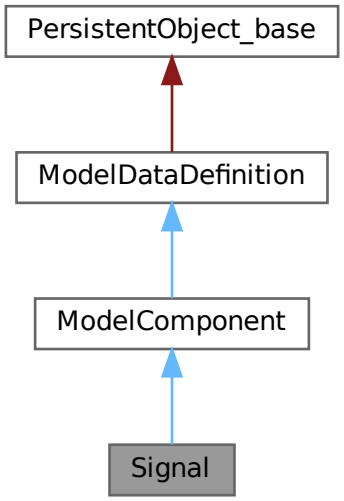
- plugins/data/Logic/Set.h
- plugins/data/Logic/Set.cpp

17.845 Signal Class Reference

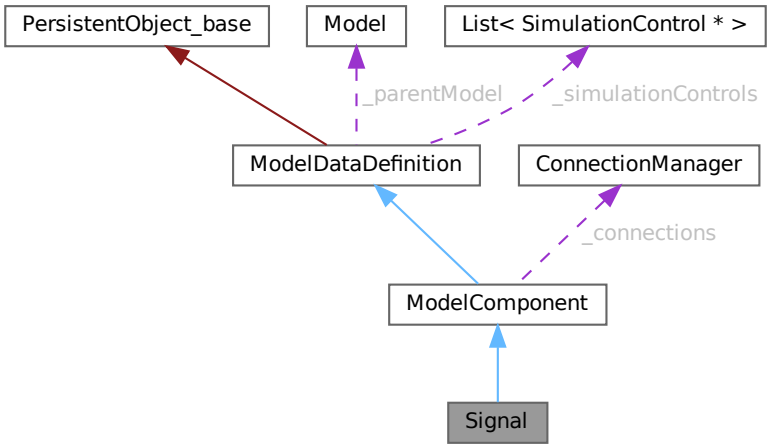
Broadcasts a signal value, releasing up to `_limitExpression` entities waiting at any `Wait` (Arena Hold) component in `WaitForSignal` mode with a matching signal.

`#include <plugins/components/Synchronization/Signal.h>`

Inheritance diagram for Signal:



Collaboration diagram for Signal:



Public Member Functions

- **Signal** (Model *model, std::string name="")
- virtual std::string show () override

virtual public methods

- void **setSignalData** (SignalData *signal)
- const std::string & **limitExpression** () const
- void **setLimitExpression** (const std::string &newLimitExpression)

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component

- static bool Check (ModelComponent *component)

  Validates a component instance.

- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.

- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.

- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.

- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.

- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.

- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)

- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.845.1 Detailed Description

Broadcasts a signal value, releasing up to `_limitExpression` entities waiting at any `Wait` (Arena Hold) component in `WaitForSignal` mode with a matching signal.
Arena correspondence: the "Signal module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 65). The attached `SignalData` matches Signal Value; `_limitExpression` matches Limit.

17.845.2 Member Function Documentation

17.845.2.1 _check()

```
bool Signal::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.845.2.2 _createEditableDataDefinitions()

```
void Signal::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```
Reimplemented from ModelDataDefinition.

17.845.2.3 _initBetweenReplications()

```
void Signal::_initBetweenReplications ( )  [override], [protected], [virtual]
```
Reimplemented from ModelDataDefinition.

17.845.2.4 _loadInstance()

```
bool Signal::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```
virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.845.2.5 _onDispatchEvent()

```
void Signal::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```
Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.845.2.6 _saveInstance()

```
void Signal::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```
Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.845.2.7 show()

```
std::string Signal::show ( )  [override], [virtual]
```
virtual public methods
Returns a textual representation of this component.

Returns

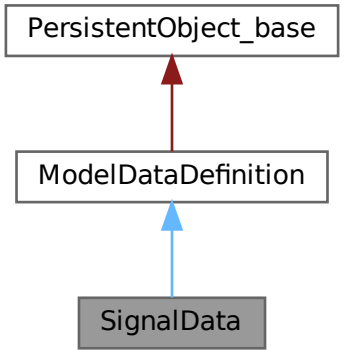
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

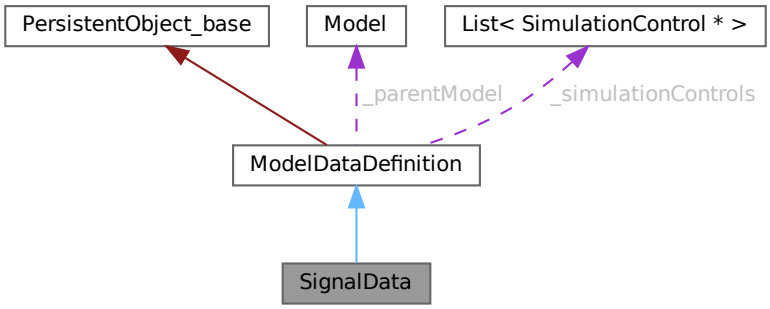
- plugins/components/Synchronization/Signal.h
- plugins/components/Synchronization/Signal.cpp

17.846 SignalData Class Reference

Inheritance diagram for SignalData:



Collaboration diagram for SignalData:



Public Types

- typedef std::function< unsigned int(SignalData *) > **SignalDataEventHandler**
- typedef std::pair< SignalDataEventHandler, ModelComponent * > **PairSignalDataEventHandler**

Public Member Functions

- **SignalData** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- unsigned int **generateSignal** (double signalValue, unsigned int limit)
- void **addSignalDataEventHandler** (SignalDataEventHandler eventHandler, ModelComponent *component)
- void **removeSignalDataEventHandler** (ModelComponent *component)
- bool **hasSignalDataEventHandler** (ModelComponent *component) const
- unsigned int **remainsToLimit** () const
- void **decreaseRemainLimit** ()

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- template<typename Class >
  static SignalDataEventHandler **SetSignalDataEventHandler** (unsigned int(Class::*function)(SignalData *), Class *object)
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)

- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.846.1 Member Function Documentation

17.846.1.1 _check()

```
bool SignalData::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.846.1.2 _initBetweenReplications()

```
void SignalData::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.846.1.3 _loadInstance()

```
bool SignalData::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.846.1.4 _saveInstance()

```
void SignalData::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.846.1.5 show()

```
std::string SignalData::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/Synchronization/SignalData.h
- plugins/data/Synchronization/SignalData.cpp

17.847 GroProgramRuntime::SignalMutation Struct Reference

Public Attributes

- SignalMutationType **type** = SignalMutationType::Emit
- double **value** = 0.0
- unsigned int **channel** = 0

The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/GroProgramRuntime.h

17.848 SimpleHttpServer Class Reference

Minimal blocking HTTP/1.1 server for single-request-per-connection handling.
`#include <applications/worker/http/SimpleHttpServer.h>`

Public Types

- using **RequestHandler** = std::function< HttpResponse(const HttpRequest &)>

Public Member Functions

- SimpleHttpServer (unsigned short port)

  Construct a server bound to a TCP port.
- bool serve (const RequestHandler &handler, unsigned long maxRequests=0)

  Serve HTTP requests until the optional request limit is reached.
- void **requestStop** ()

  Requests the serving loop to stop as soon as it reaches a safe point.

17.848.1 Detailed Description

Minimal blocking HTTP/1.1 server for single-request-per-connection handling.
This server intentionally keeps the transport model simple (no keep-alive, no chunked transfer decoding) and delegates application routing to a request handler callback.

17.848.2 Constructor & Destructor Documentation

17.848.2.1 SimpleHttpServer()

```
SimpleHttpServer::SimpleHttpServer (
            unsigned short port )  [explicit]
```

Construct a server bound to a TCP port.

Parameters

port	TCP port that the server will bind/listen on.

17.848.3 Member Function Documentation

17.848.3.1 serve()

```
bool SimpleHttpServer::serve (
            const RequestHandler & handler,
            unsigned long maxRequests = 0 )
```

Serve HTTP requests until the optional request limit is reached.
Serves one request at a time over a blocking TCP listener.

Parameters

handler	Callback used to transform a parsed request into a response.
maxRequests	Maximum number of client requests to process (0 means unlimited).

Returns

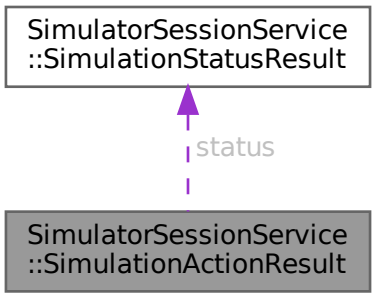
true when the serving loop ends normally; false on socket setup/runtime failure.

The loop accepts a connection, reads a bounded HTTP/1.1 request, delegates to the provided handler, serializes the response and closes the connection.
The documentation for this class was generated from the following files:

- applications/worker/http/SimpleHttpServer.h
- applications/worker/http/SimpleHttpServer.cpp

17.849 SimulatorSessionService::SimulationActionResult Struct Reference

Contains the result of a simulation action such as run or step.
`#include <applications/worker/service/SimulatorSessionService.h>`
Collaboration diagram for SimulatorSessionService::SimulationActionResult:



Public Attributes

- bool **success** = false
- bool **invalidToken** = false
- bool **missingCurrentModel** = false
- SimulationStatusResult **status**

17.849.1 Detailed Description

Contains the result of a simulation action such as run or step.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.850 SimulationCommandController Class Reference

Controller that orchestrates UI simulation commands over SimulationController guards.
`#include <applications/gui/genesys/controllers/SimulationCommandController.h>`

Public Member Functions

- **SimulationCommandController** (SimulationController *simulationController, const std::function< void(const std::string &)> &insertCommandInConsole, const std::function< void()> &actualizeActions, const std::function< bool()> &checkModel, const std::function< bool()> &setSimulationModelBasedOnText, const std::function< void()> &prepareNewSimulationOutputs)

  Creates a command controller with narrow callbacks from MainWindow.
- void **onActionSimulationStartTriggered** (bool modelChecked) const

  Delegates the start simulation command flow.

- void **onActionSimulationStepTriggered** (bool modelChecked) const

  Delegates the step simulation command flow.
- void **onActionSimulationPauseTriggered** () const

  Delegates the pause simulation command flow.
- void **onActionSimulationResumeTriggered** (bool modelChecked) const

  Delegates the resume simulation command flow.
- void **onActionSimulationStopTriggered** () const

  Delegates the stop simulation command flow.

17.850.1 Detailed Description

Controller that orchestrates UI simulation commands over SimulationController guards.
This class sits between MainWindow compatibility slots and the low-level simulation readiness controller. It coordinates command intent (start/step/pause/resume/stop), command logging, and action refresh callbacks while preserving the existing UI trigger surface.
Responsibilities:

- execute simulation command flows delegated from MainWindow;
- use SimulationController precondition checks before unsafe commands;
- invoke injected callbacks for console command logging and action-state refresh.

Boundaries:

- it does not own simulation state or kernel event wiring;
- it does not render trace/event widgets;
- it is a controller/orchestration layer, not a persistence or synchronization service.

The documentation for this class was generated from the following files:

- applications/gui/genesys/controllers/SimulationCommandController.h
- applications/gui/genesys/controllers/SimulationCommandController.cpp

17.851 SimulatorSessionService::SimulationConfigInput Struct Reference

Defines user-provided simulation parameters to apply to the current model.
`#include <applications/worker/service/SimulatorSessionService.h>`

Public Attributes

- unsigned int **numberOfReplications** = 0
- double **replicationLength** = 0.0
- double **warmUpPeriod** = 0.0
- bool **pauseOnEvent** = false
- bool **pauseOnReplication** = false
- bool **initializeStatistics** = false
- bool **initializeSystem** = false

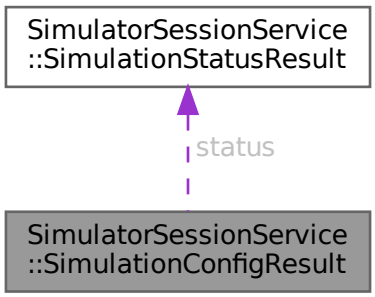
17.851.1 Detailed Description

Defines user-provided simulation parameters to apply to the current model.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.852 SimulatorSessionService::SimulationConfigResult Struct Reference

Contains the result of applying simulation configuration.
`#include <applications/worker/service/SimulatorSessionService.h>`
Collaboration diagram for SimulatorSessionService::SimulationConfigResult:



Public Attributes

- bool **success** = false
- bool **invalidToken** = false
- bool **missingCurrentModel** = false
- SimulationStatusResult **status**

17.852.1 Detailed Description

Contains the result of applying simulation configuration.
The documentation for this struct was generated from the following file:

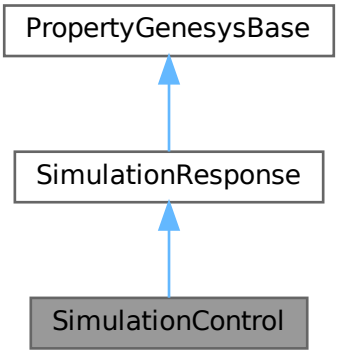
- applications/worker/service/SimulatorSessionService.h

17.853 SimulationControl Class Reference

Read/write kernel-side simulation control abstraction.
`#include <kernel/simulator/SimulationControlAndResponse.h>`

Inheritance diagram for SimulationControl:

Collaboration diagram for SimulationControl:



Public Member Functions

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const

- bool **isReadOnly** () const
- virtual void **setValue** (std::string value, bool remove=false)=0
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual std::string **getValue** () const =0
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Protected Member Functions

- void **_ensureWritable** (const char *operation) const

Protected Attributes

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.853.1 Detailed Description

Read/write kernel-side simulation control abstraction.
SimulationControl extends SimulationResponse with setter-based mutation and therefore represents writable kernel-side experiment controls.

17.853.2 Member Function Documentation

17.853.2.1 getChildSimulationControls()

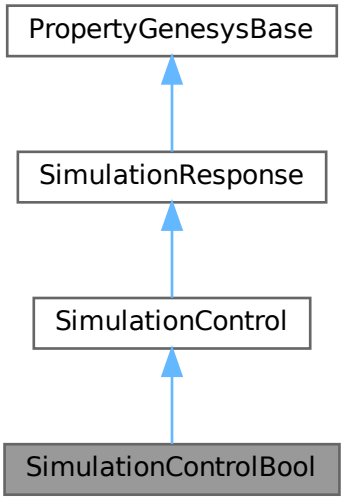
```
virtual List< SimulationControl * > * SimulationControl::getChildSimulationControls (
            int index = 0 )  [inline], [virtual]
```

Returns nested SimulationControls exposed by this control.
This is the kernel-facing API for traversing child controls belonging to inline objects, references, or list elements represented by a control.
Reimplemented in SimulationControlGenericClass< T, M, C >, SimulationControlGenericClassNotDC< T, M, C >, and SimulationControlGenericListPointer< T, M, C >.
The documentation for this class was generated from the following file:

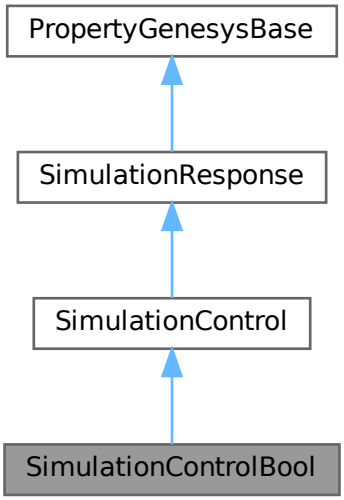
- kernel/simulator/SimulationControlAndResponse.h

17.854 SimulationControlBool Class Reference

Inheritance diagram for SimulationControlBool:



Collaboration diagram for SimulationControlBool:



Public Member Functions

- **SimulationControlBool** (GetterBool getter, SetterBool setter, std::string className, std::string element↩ Name, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool is↩ Enum=false)

- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char ∗operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.854.1 Member Function Documentation

17.854.1.1 getValue()

```
virtual std::string SimulationControlBool::getValue ( ) const  [inline], [override], [virtual]
```

Implements SimulationResponse.

17.854.1.2 setValue()

```
virtual void SimulationControlBool::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

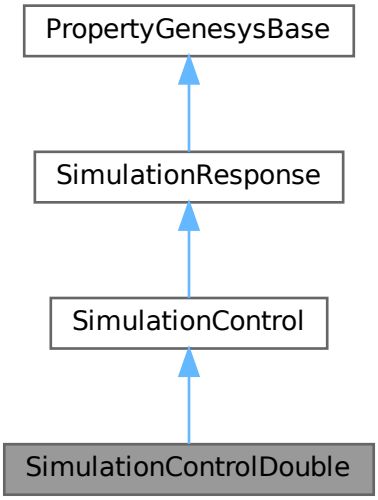
Implements SimulationControl.

The documentation for this class was generated from the following file:

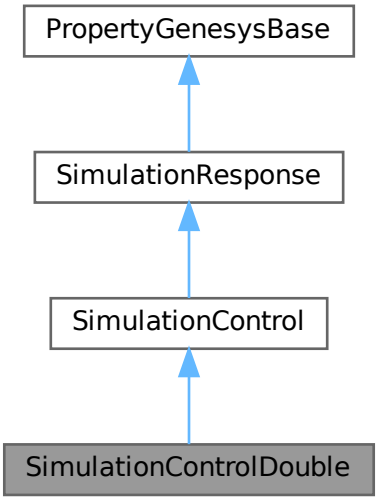
- kernel/simulator/SimulationControlAndResponse.h

17.855 SimulationControlDouble Class Reference

Inheritance diagram for SimulationControlDouble:



Collaboration diagram for SimulationControlDouble:



Public Member Functions

- **SimulationControlDouble** (GetterDouble getter, SetterDouble setter, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)

- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.855.1 Member Function Documentation

17.855.1.1 getValue()

```
virtual std::string SimulationControlDouble::getValue ( ) const  [inline], [override], [virtual]
```

Implements SimulationResponse.

17.855.1.2 setValue()

```
virtual void SimulationControlDouble::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

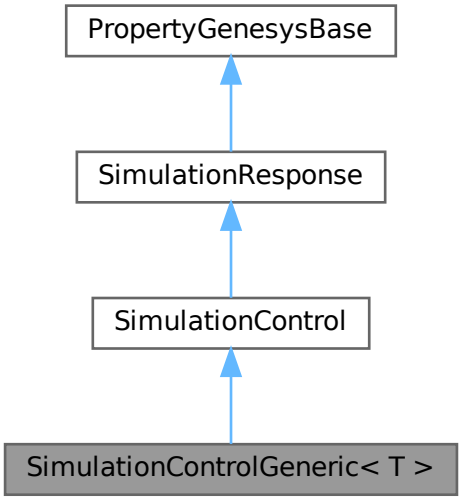
Implements SimulationControl.

The documentation for this class was generated from the following file:

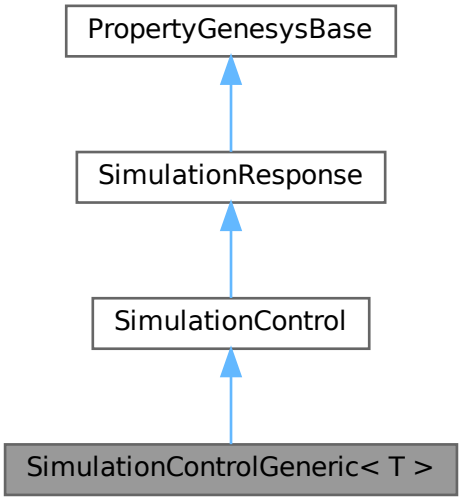
- kernel/simulator/SimulationControlAndResponse.h

17.856 SimulationControlGeneric< T > Class Template Reference

Inheritance diagram for SimulationControlGeneric< T >:



Collaboration diagram for SimulationControlGeneric< T >:



Public Member Functions

- **SimulationControlGeneric** (GetterGeneric< T > getter, SetterGeneric< T > setter, std::string class↩ Name, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)

- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.856.1 Member Function Documentation

17.856.1.1 getValue()

```
template<typename T >
virtual std::string SimulationControlGeneric< T >::getValue ( ) const  [inline], [override],
[virtual]
```

Implements SimulationResponse.

17.856.1.2 setValue()

```
template<typename T >
virtual void SimulationControlGeneric< T >::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

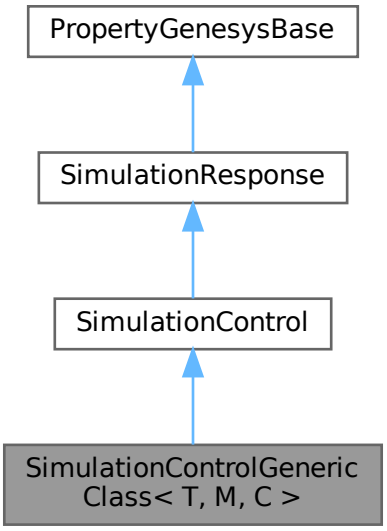
Implements SimulationControl.

The documentation for this class was generated from the following file:

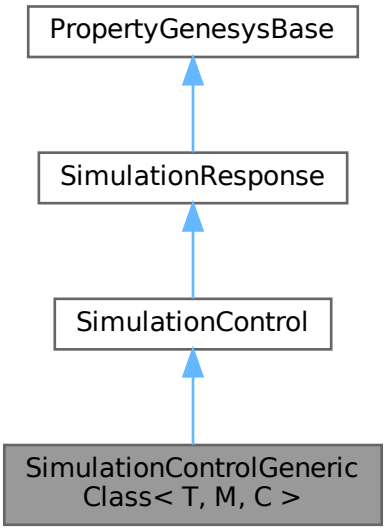
- kernel/simulator/SimulationControlAndResponse.h

17.857 SimulationControlGenericClass< T, M, C > Class Template Reference

Inheritance diagram for SimulationControlGenericClass< T, M, C >:



Collaboration diagram for SimulationControlGenericClass< T, M, C >:



Public Types

- using **Creator** = std::function< T(M, const std::string &)>

Public Member Functions

- **SimulationControlGenericClass** (M model, GetterGeneric< T > getter, SetterGeneric< T > setter, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=true, bool isEnum=false, Creator creator=nullptr)
- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override
- virtual bool isModelDataDefinitionReference () const override
- virtual ModelDataDefinition * getReferencedModelDataDefinition () const override
- virtual bool supportsInlineExpansion () const override
- virtual bool supportsExistingObjectSelection () const override
- virtual bool supportsObjectCreation () const override
- virtual bool isInlineObjectProperty () const override
- virtual bool hasObjectInstance () const override
- virtual bool ensureObjectInstance () override
- virtual bool createObjectInstance (const std::string &value="") override
- virtual List< std::string > * getStrValues () override
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0) override

  Returns nested SimulationControls exposed by this control.

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.857.1 Member Function Documentation

17.857.1.1 createObjectInstance()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClass< T, M, C >::createObjectInstance (
            const std::string & value = "" )  [inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.857.1.2 ensureObjectInstance()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClass< T, M, C >::ensureObjectInstance ( )  [inline],
[override], [virtual]
```

Reimplemented from SimulationControl.

17.857.1.3 getChildSimulationControls()

```
template<typename T , typename M , typename C >
virtual List< SimulationControl * > * SimulationControlGenericClass< T, M, C >::getChild↩
SimulationControls (
            int index = 0 )  [inline], [override], [virtual]
```

Returns nested SimulationControls exposed by this control.
This is the kernel-facing API for traversing child controls belonging to inline objects, references, or list elements represented by a control.
Reimplemented from SimulationControl.

17.857.1.4 getReferencedModelDataDefinition()

```
template<typename T , typename M , typename C >
virtual ModelDataDefinition * SimulationControlGenericClass< T, M, C >::getReferencedModel↩
DataDefinition ( ) const  [inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.857.1.5 getStrValues()

```
template<typename T , typename M , typename C >
virtual List< std::string > * SimulationControlGenericClass< T, M, C >::getStrValues ( )
[inline], [override], [virtual]
```

Reimplemented from SimulationResponse.

17.857.1.6 getValue()

```
template<typename T , typename M , typename C >
virtual std::string SimulationControlGenericClass< T, M, C >::getValue ( ) const  [inline],
[override], [virtual]
```

Implements SimulationResponse.

17.857.1.7 hasObjectInstance()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClass< T, M, C >::hasObjectInstance ( ) const  [inline],
[override], [virtual]
```

Reimplemented from SimulationControl.

17.857.1.8 isInlineObjectProperty()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClass< T, M, C >::isInlineObjectProperty ( ) const  [inline],
[override], [virtual]
```

Reimplemented from SimulationControl.

17.857.1.9 isModelDataDefinitionReference()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClass< T, M, C >::isModelDataDefinitionReference ( )
const  [inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.857.1.10 setValue()

```
template<typename T , typename M , typename C >
virtual void SimulationControlGenericClass< T, M, C >::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

Implements SimulationControl.

17.857.1.11 supportsExistingObjectSelection()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClass< T, M, C >::supportsExistingObjectSelection ( )
const  [inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.857.1.12 supportsInlineExpansion()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClass< T, M, C >::supportsInlineExpansion ( ) const
[inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.857.1.13 supportsObjectCreation()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClass< T, M, C >::supportsObjectCreation ( ) const  [inline],
[override], [virtual]
```

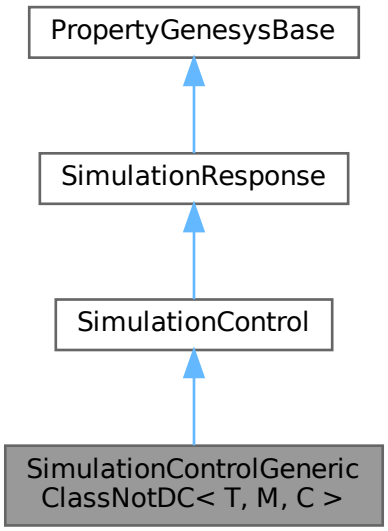
Reimplemented from SimulationControl.

The documentation for this class was generated from the following file:

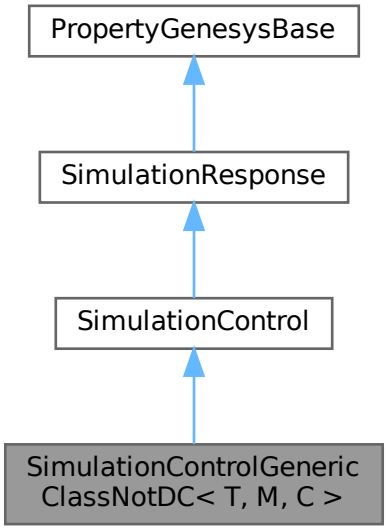
- kernel/simulator/SimulationControlAndResponse.h

17.858 SimulationControlGenericClassNotDC< T, M, C > Class Template Reference

Inheritance diagram for SimulationControlGenericClassNotDC< T, M, C >:



Collaboration diagram for SimulationControlGenericClassNotDC< T, M, C >:



Public Types

- using **Creator** = std::function< T(M)>
- using **NamedCreator** = std::function< T(M, const std::string &)>

Public Member Functions

- **SimulationControlGenericClassNotDC** (M model, GetterGeneric< T > getter, SetterGeneric< T > setter, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=true, bool isEnum=false, Creator creator=nullptr, NamedCreator namedCreator=nullptr)
- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override
- virtual bool hasObjectInstance () const override
- virtual bool supportsInlineExpansion () const override
- virtual bool supportsObjectCreation () const override
- virtual bool isInlineObjectProperty () const override
- virtual bool ensureObjectInstance () override
- virtual bool createObjectInstance (const std::string &value="") override
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0) override

  Returns nested SimulationControls exposed by this control.

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const

- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.858.1 Member Function Documentation

17.858.1.1 createObjectInstance()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClassNotDC< T, M, C >::createObjectInstance (
            const std::string & value = "" )  [inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.858.1.2 ensureObjectInstance()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClassNotDC< T, M, C >::ensureObjectInstance ( )  [inline],
[override], [virtual]
```

Reimplemented from SimulationControl.

17.858.1.3 getChildSimulationControls()

```
template<typename T , typename M , typename C >
virtual List< SimulationControl * > * SimulationControlGenericClassNotDC< T, M, C >::get↩
ChildSimulationControls (
            int index = 0 )  [inline], [override], [virtual]
```

Returns nested SimulationControls exposed by this control.
This is the kernel-facing API for traversing child controls belonging to inline objects, references, or list elements represented by a control.
Reimplemented from SimulationControl.

17.858.1.4 getValue()

```
template<typename T , typename M , typename C >
virtual std::string SimulationControlGenericClassNotDC< T, M, C >::getValue ( ) const  [inline],
[override], [virtual]
```

Implements SimulationResponse.

17.858.1.5 hasObjectInstance()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClassNotDC< T, M, C >::hasObjectInstance ( ) const  [inline],
[override], [virtual]
```

Reimplemented from SimulationControl.

17.858.1.6 isInlineObjectProperty()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClassNotDC< T, M, C >::isInlineObjectProperty ( ) const
[inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.858.1.7 setValue()

```
template<typename T , typename M , typename C >
virtual void SimulationControlGenericClassNotDC< T, M, C >::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

Implements SimulationControl.

17.858.1.8 supportsInlineExpansion()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClassNotDC< T, M, C >::supportsInlineExpansion ( ) const
[inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.858.1.9 supportsObjectCreation()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericClassNotDC< T, M, C >::supportsObjectCreation ( ) const
[inline], [override], [virtual]
```

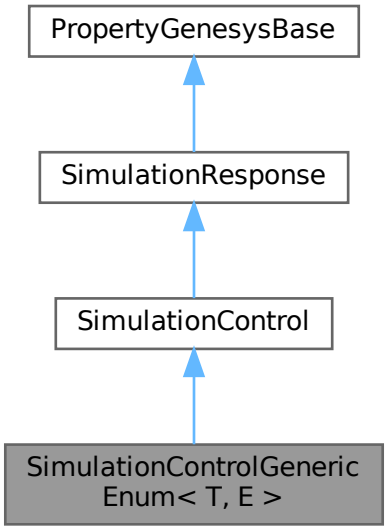
Reimplemented from SimulationControl.

The documentation for this class was generated from the following file:

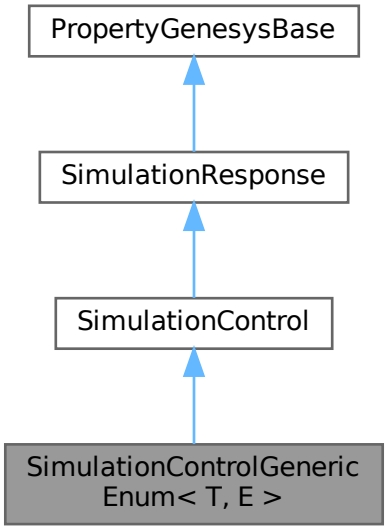
- kernel/simulator/SimulationControlAndResponse.h

17.859 SimulationControlGenericEnum< T, E > Class Template Reference

Inheritance diagram for SimulationControlGenericEnum< T, E >:



Collaboration diagram for SimulationControlGenericEnum< T, E >:



Public Member Functions

- **SimulationControlGenericEnum** (GetterGeneric< T > getter, SetterGeneric< T > setter, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=true)
- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override
- virtual List< std::string > * getStrValues () override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const

- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.859.1 Member Function Documentation

17.859.1.1 getStrValues()

```
template<typename T , typename E >
virtual List< std::string > * SimulationControlGenericEnum< T, E >::getStrValues ( )  [inline],
[override], [virtual]
```

Reimplemented from SimulationResponse.

17.859.1.2 getValue()

```
template<typename T , typename E >
virtual std::string SimulationControlGenericEnum< T, E >::getValue ( ) const  [inline], [override],
[virtual]
```

Implements SimulationResponse.

17.859.1.3 setValue()

```
template<typename T , typename E >
virtual void SimulationControlGenericEnum< T, E >::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

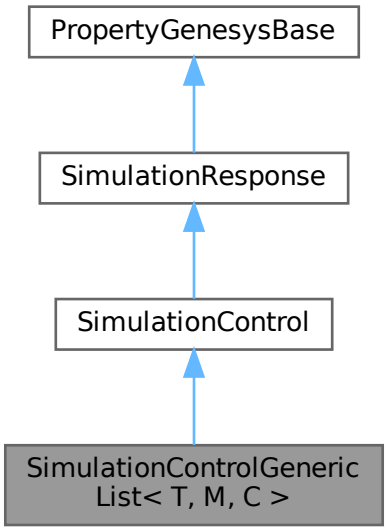
Implements SimulationControl.

The documentation for this class was generated from the following file:

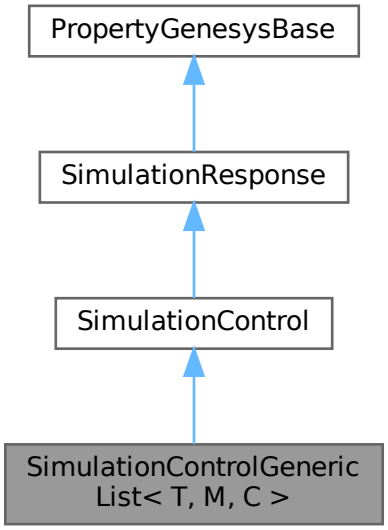
- kernel/simulator/SimulationControlAndResponse.h

17.860 SimulationControlGenericList< T, M, C > Class Template Reference

Inheritance diagram for SimulationControlGenericList< T, M, C >:



Collaboration diagram for SimulationControlGenericList< T, M, C >:



Public Member Functions

- **SimulationControlGenericList** (M model, GetterGeneric< List< T > * > getter, AdderGeneric< T > adder, RemoverGeneric< T > remover, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=true, bool isClass=false, bool isEnum=false)
- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override
- virtual List< std::string > * getStrValues () override
- virtual bool supportsListEditor () const override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const

- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.860.1 Member Function Documentation

17.860.1.1 getStrValues()

```
template<typename T , typename M , typename C >
virtual List< std::string > * SimulationControlGenericList< T, M, C >::getStrValues ( )  [inline],
[override], [virtual]
```

Reimplemented from SimulationResponse.

17.860.1.2 getValue()

```
template<typename T , typename M , typename C >
virtual std::string SimulationControlGenericList< T, M, C >::getValue ( ) const  [inline],
[override], [virtual]
```

Implements SimulationResponse.

17.860.1.3 setValue()

```
template<typename T , typename M , typename C >
virtual void SimulationControlGenericList< T, M, C >::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

Implements SimulationControl.

17.860.1.4 supportsListEditor()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericList< T, M, C >::supportsListEditor ( ) const  [inline],
[override], [virtual]
```

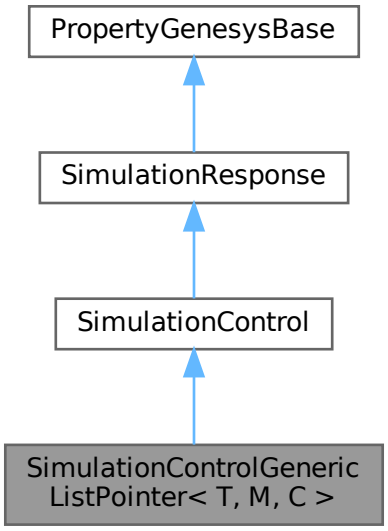
Reimplemented from SimulationControl.

The documentation for this class was generated from the following file:

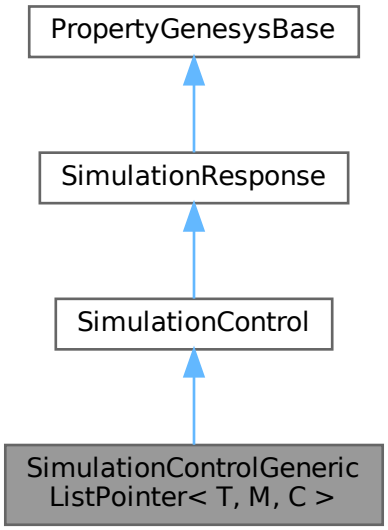
- kernel/simulator/SimulationControlAndResponse.h

17.861 SimulationControlGenericListPointer< T, M, C > Class Template Reference

Inheritance diagram for SimulationControlGenericListPointer< T, M, C >:



Collaboration diagram for SimulationControlGenericListPointer< T, M, C >:



Public Types

- using **Creator** = std::function< T(M, const std::string &)>
- using **TypedCreator** = std::function< T(M)>
- using **ElementTypeNamesGetter** = std::function< std::vector< std::string >()>
- using **CurrentElementTypeGetter** = std::function< std::string()>
- using **CurrentElementTypeSetter** = std::function< bool(const std::string &)>
- using **TypedNamedCreator** = std::function< T(M, const std::string &, const std::string &)>
- using **ListElementDataDefinitionGetter** = std::function< ModelDataDefinition *(int)>

Public Member Functions

- **SimulationControlGenericListPointer** (M model, GetterGeneric< List< T > * > getter, AdderGeneric< T > adder, RemoverGeneric< T > remover, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=true, bool isClass=true, bool isEnum=false, Creator creator=nullptr, TypedCreator typedCreator=nullptr, ElementTypeNamesGetter elementTypeNamesGetter=nullptr, CurrentElementTypeGetter currentElementTypeGetter=nullptr, CurrentElementTypeSetter currentElementTypeSetter=nullptr, TypedNamedCreator typedNamedCreator=nullptr, ListElementDataDefinitionGetter listElementDataDefinitionGetter=nullptr)
- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override
- virtual bool supportsListEditor () const override
- virtual bool supportsNewListElementCreation () const override
- virtual List< std::string > * getCreatableListElementTypes () override
- virtual std::string getCurrentListElementType () const override
- virtual bool setCurrentListElementType (const std::string &typeName) override
- virtual bool createNewListElement (const std::string &value="") override
- virtual bool createNewListElementOfType (const std::string &typeName, const std::string &value="") override
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0) override

  Returns nested SimulationControls exposed by this control.
- virtual ModelDataDefinition * getListElementModelDataDefinition (int index) const override
- virtual List< std::string > * getStrValues () override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.861.1 Member Function Documentation

17.861.1.1 createNewListElement()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericListPointer< T, M, C >::createNewListElement (
            const std::string & value = "" )  [inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.861.1.2 createNewListElementOfType()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericListPointer< T, M, C >::createNewListElementOfType (
            const std::string & typeName,
            const std::string & value = "" )  [inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.861.1.3 getChildSimulationControls()

```
template<typename T , typename M , typename C >
virtual List< SimulationControl * > * SimulationControlGenericListPointer< T, M, C >::get↩
ChildSimulationControls (
            int index = 0 )  [inline], [override], [virtual]
```

Returns nested SimulationControls exposed by this control.
This is the kernel-facing API for traversing child controls belonging to inline objects, references, or list elements represented by a control.
Reimplemented from SimulationControl.

17.861.1.4 getCreatableListElementTypes()

```
template<typename T , typename M , typename C >
virtual List< std::string > * SimulationControlGenericListPointer< T, M, C >::getCreatable↩
ListElementTypes ( )  [inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.861.1.5 getCurrentListElementType()

```
template<typename T , typename M , typename C >
virtual std::string SimulationControlGenericListPointer< T, M, C >::getCurrentListElementType
( ) const  [inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.861.1.6 getListElementModelDataDefinition()

```
template<typename T , typename M , typename C >
virtual ModelDataDefinition * SimulationControlGenericListPointer< T, M, C >::getListElement↩
ModelDataDefinition (
            int index ) const  [inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.861.1.7 getStrValues()

```
template<typename T , typename M , typename C >
virtual List< std::string > * SimulationControlGenericListPointer< T, M, C >::getStrValues (
)  [inline], [override], [virtual]
```

Reimplemented from SimulationResponse.

17.861.1.8 getValue()

```
template<typename T , typename M , typename C >
virtual std::string SimulationControlGenericListPointer< T, M, C >::getValue ( ) const  [inline],
[override], [virtual]
```

Implements SimulationResponse.

17.861.1.9 setCurrentListElementType()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericListPointer< T, M, C >::setCurrentListElementType (
            const std::string & typeName )  [inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.861.1.10 setValue()

```
template<typename T , typename M , typename C >
virtual void SimulationControlGenericListPointer< T, M, C >::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

Implements SimulationControl.

17.861.1.11 supportsListEditor()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericListPointer< T, M, C >::supportsListEditor ( ) const
[inline], [override], [virtual]
```

Reimplemented from SimulationControl.

17.861.1.12 supportsNewListElementCreation()

```
template<typename T , typename M , typename C >
virtual bool SimulationControlGenericListPointer< T, M, C >::supportsNewListElementCreation (
) const  [inline], [override], [virtual]
```

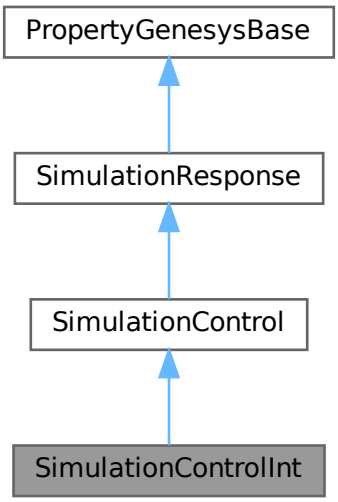
Reimplemented from SimulationControl.

The documentation for this class was generated from the following file:

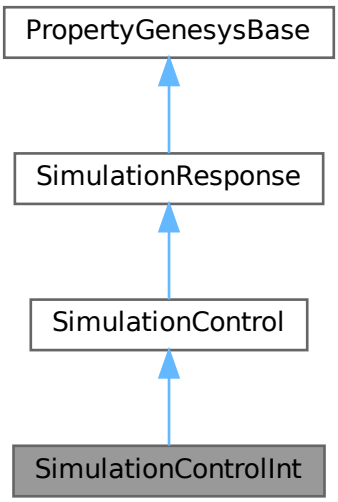
- kernel/simulator/SimulationControlAndResponse.h

17.862 SimulationControlInt Class Reference

Inheritance diagram for SimulationControlInt:



Collaboration diagram for SimulationControlInt:



Public Member Functions

- **SimulationControlInt** (GetterInt getter, SetterInt setter, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)

- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.862.1 Member Function Documentation

17.862.1.1 getValue()

```
virtual std::string SimulationControlInt::getValue ( ) const  [inline], [override], [virtual]
```

Implements SimulationResponse.

17.862.1.2 setValue()

```
virtual void SimulationControlInt::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

Implements SimulationControl.

The documentation for this class was generated from the following file:

- kernel/simulator/SimulationControlAndResponse.h

17.863 SimulationController Class Reference

Controller that validates simulation command readiness for MainWindow wrappers.

`#include <applications/gui/genesys/controllers/SimulationController.h>`

Public Member Functions

- SimulationController (QWidget *ownerWidget, Simulator *simulator)

  Creates a controller bound to GUI owner and simulator.
- bool hasCurrentModelSimulation () const

  Checks whether there is a valid current model with simulation object.
- ModelSimulation * currentSimulation () const

  Returns current simulation object when available.
- bool ensureReady (bool checkModel, bool modelChecked, const std::function< bool()> &runModelCheck, const std::function< bool()> &syncModelFromText) const

  Validates whether GUI can execute a simulation command now.

17.863.1 Detailed Description

Controller that validates simulation command readiness for MainWindow wrappers.

This controller is part of the incremental extraction that keeps MainWindow as a compatibility façade while moving simulation-command orchestration behind smaller collaborators. It centralizes precondition checks shared by start/step/pause/resume/stop flows, including model availability and model-text synchronization guards.

Responsibilities:

- verify that the current model and its ModelSimulation are available;
- execute callback-based readiness checks delegated by MainWindow;
- return a single command-ready decision used by higher-level command controllers.

Boundaries:

- it does not own simulation state or lifecycle (owned by Simulator);
- it does not execute GUI commands directly (handled by SimulationCommandController);
- it does not update widgets, tabs, or trace outputs.

Dependencies are injected narrowly (owner widget + simulator + callbacks) to preserve existing ownership and enable incremental compatibility without changing slot signatures.

17.863.2 Constructor & Destructor Documentation

17.863.2.1 SimulationController()

```
SimulationController::SimulationController (
            QWidget * ownerWidget,
            Simulator * simulator )
```

Creates a controller bound to GUI owner and simulator.

Parameters

ownerWidget	Widget used to show warning dialogs.
simulator	Simulator facade used by GUI.

17.863.3 Member Function Documentation

17.863.3.1 currentSimulation()

```
ModelSimulation * SimulationController::currentSimulation ( ) const
```

Returns current simulation object when available.

Returns

Pointer to current ModelSimulation or nullptr when unavailable.

17.863.3.2 ensureReady()

```
bool SimulationController::ensureReady (
            bool checkModel,
            bool modelChecked,
            const std::function< bool()> & runModelCheck,
            const std::function< bool()> & syncModelFromText ) const
```

Validates whether GUI can execute a simulation command now.

Validation pipeline: 1) Ensure current model/simulation exists; 2) Optionally execute model check callback when model is not checked yet; 3) Execute model synchronization callback (text -> model) and ensure simulation still exists.

Parameters

checkModel	If true, model-check callback may be executed.
modelChecked	Indicates current model check state from MainWindow.
runModelCheck	Callback that executes model check when needed.
syncModelFromText	Callback that synchronizes textual model representation.

Returns

true if command can proceed safely.

Todo Replace callback-based contract by dedicated interfaces to improve testability.

17.863.3.3 hasCurrentModelSimulation()

```
bool SimulationController::hasCurrentModelSimulation ( ) const
```

Checks whether there is a valid current model with simulation object.

Returns

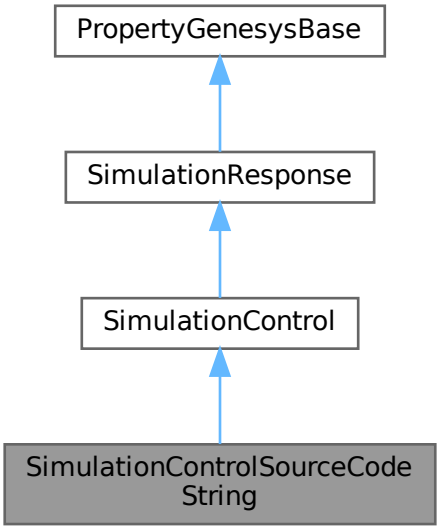
true when simulation commands can access `current()->getSimulation()` safely.

The documentation for this class was generated from the following files:

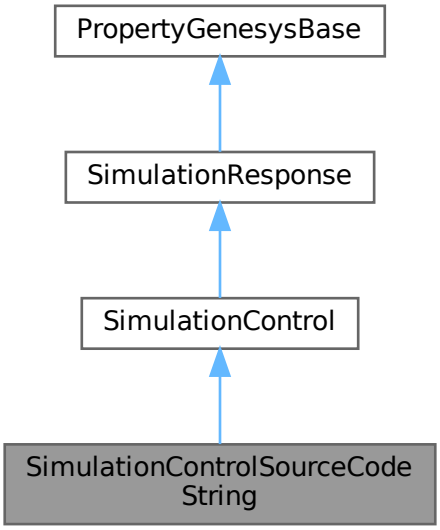
- applications/gui/genesys/controllers/SimulationController.h
- applications/gui/genesys/controllers/SimulationController.cpp

17.864 SimulationControlSourceCodeString Class Reference

Inheritance diagram for SimulationControlSourceCodeString:



Collaboration diagram for SimulationControlSourceCodeString:



Public Member Functions

- **SimulationControlSourceCodeString** (GetterSourceCodeString getter, SetterSourceCodeString setter, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false, ValidatorString validator=nullptr)
- virtual std::string getValue () const override
- virtual bool validateProposedValue (const std::string &value, std::string &errorMessage) const override
- virtual void setValue (std::string value, bool remove=false) override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()

- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.864.1 Member Function Documentation

17.864.1.1 getValue()

```
virtual std::string SimulationControlSourceCodeString::getValue ( ) const  [inline], [override],
[virtual]
```

Implements SimulationResponse.

17.864.1.2 setValue()

```
virtual void SimulationControlSourceCodeString::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

Implements SimulationControl.

17.864.1.3 validateProposedValue()

```
virtual bool SimulationControlSourceCodeString::validateProposedValue (
            const std::string & value,
            std::string & errorMessage ) const  [inline], [override], [virtual]
```

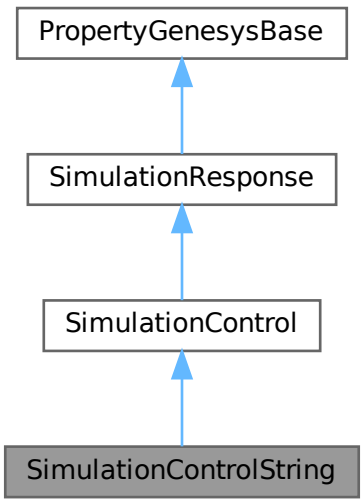
Reimplemented from SimulationControl.

The documentation for this class was generated from the following file:

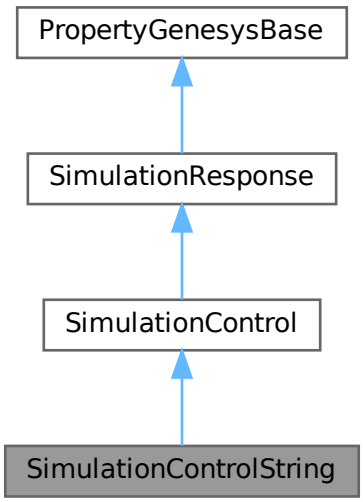
- kernel/simulator/SimulationControlAndResponse.h

17.865 SimulationControlString Class Reference

Inheritance diagram for SimulationControlString:



Collaboration diagram for SimulationControlString:



Public Member Functions

- **SimulationControlString** (GetterString getter, SetterString setter, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false, ValidatorString validator=nullptr)
- virtual std::string getValue () const override
- virtual bool validateProposedValue (const std::string &value, std::string &errorMessage) const override
- virtual void setValue (std::string value, bool remove=false) override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const

- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.865.1 Member Function Documentation

17.865.1.1 getValue()

```
virtual std::string SimulationControlString::getValue ( ) const  [inline], [override], [virtual]
```

Implements SimulationResponse.

17.865.1.2 setValue()

```
virtual void SimulationControlString::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

Implements SimulationControl.

17.865.1.3 validateProposedValue()

```
virtual bool SimulationControlString::validateProposedValue (
            const std::string & value,
            std::string & errorMessage ) const  [inline], [override], [virtual]
```

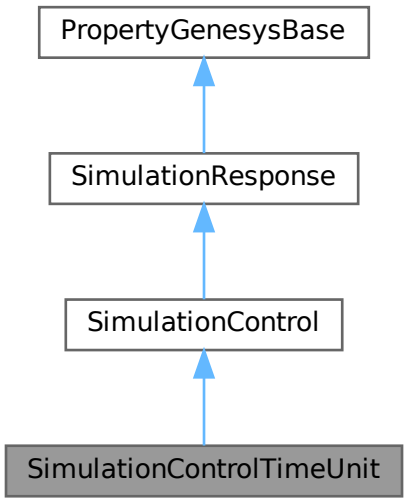
Reimplemented from SimulationControl.

The documentation for this class was generated from the following file:

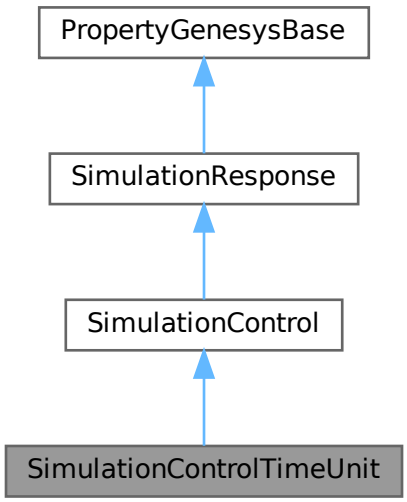
- kernel/simulator/SimulationControlAndResponse.h

17.866 SimulationControlTimeUnit Class Reference

Inheritance diagram for SimulationControlTimeUnit:



Collaboration diagram for SimulationControlTimeUnit:



Public Member Functions

- **SimulationControlTimeUnit** (GetterTimeUnit getter, SetterTimeUnit setter, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const

- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.866.1 Member Function Documentation

17.866.1.1 getValue()

```
virtual std::string SimulationControlTimeUnit::getValue ( ) const  [inline], [override], [virtual]
```

Implements SimulationResponse.

17.866.1.2 setValue()

```
virtual void SimulationControlTimeUnit::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

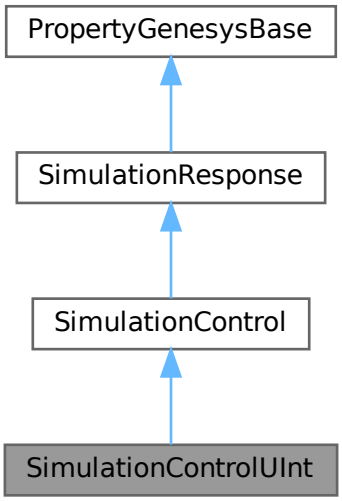
Implements SimulationControl.

The documentation for this class was generated from the following file:

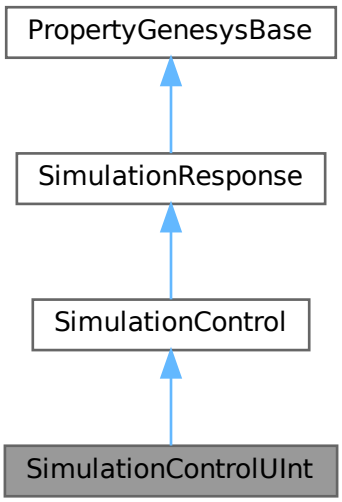
- kernel/simulator/SimulationControlAndResponse.h

17.867 SimulationControlUInt Class Reference

Inheritance diagram for SimulationControlUInt:



Collaboration diagram for SimulationControlUInt:



Public Member Functions

- **SimulationControlUInt** (GetterUInt getter, SetterUInt setter, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)

- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.867.1 Member Function Documentation

17.867.1.1 getValue()

```
virtual std::string SimulationControlUInt::getValue ( ) const  [inline], [override], [virtual]
```

Implements SimulationResponse.

17.867.1.2 setValue()

```
virtual void SimulationControlUInt::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

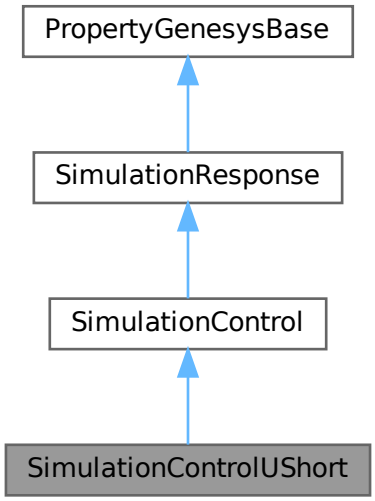
Implements SimulationControl.

The documentation for this class was generated from the following file:

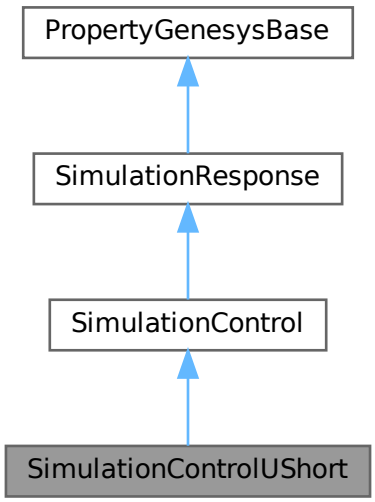
- kernel/simulator/SimulationControlAndResponse.h

17.868 SimulationControlUShort Class Reference

Inheritance diagram for SimulationControlUShort:



Collaboration diagram for SimulationControlUShort:



Public Member Functions

- **SimulationControlUShort** (GetterUShort getter, SetterUShort setter, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- virtual std::string getValue () const override
- virtual void setValue (std::string value, bool remove=false) override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const

- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.868.1 Member Function Documentation

17.868.1.1 getValue()

```
virtual std::string SimulationControlUShort::getValue ( ) const  [inline], [override], [virtual]
```

Implements SimulationResponse.

17.868.1.2 setValue()

```
virtual void SimulationControlUShort::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

Implements SimulationControl.

The documentation for this class was generated from the following file:

- kernel/simulator/SimulationControlAndResponse.h

17.869 SimulationEvent Class Reference

```
#include <kernel/simulator/OnEventManager.h>
```

Public Member Functions

- void **setSimulatedTime** (double simulatedTime)
- double **getSimulatedTime** () const
- void **setStopRequested** (bool stopRequested)
- bool **isStopRequested** () const
- void **setPauseRequested** (bool pauseRequested)
- bool **isPauseRequested** () const
- void **setPaused** (bool Paused)
- bool **isPaused** () const
- void **setRunning** (bool Running)
- bool **isRunning** () const
- void **setCustomObject** (void *customObject)
- void * **getCustomObject** () const
- void **setCurrentReplicationNumber** (unsigned int currentReplicationNumber)
- unsigned int **getCurrentReplicationNumber** () const
- void **setCurrentEvent** (Event *currentEvent)
- Event * **getCurrentEvent** () const
- void **setEntityCreated** (Entity *entityCreated)
- Entity * **getEntityCreated** () const
- void **setDestinationComponent** (ModelComponent *destinationComponent)
- ModelComponent * **getDestinationComponent** () const
- void **setEntityMoveTimeDelay** (double entityMoveTimeDelay)
- double **getEntityMoveTimeDelay** () const

Friends

- class **ModelSimulation**

17.869.1 Detailed Description

Stores an event that happened on a specific replication

The documentation for this class was generated from the following file:

- kernel/simulator/OnEventManager.h

17.870 SimulationEventController Class Reference

Controller for simulation event callbacks, UI updates, and handler wiring.
`#include <applications/gui/genesys/controllers/SimulationEventController.h>`

Classes

- struct Callbacks

Public Member Functions

- **SimulationEventController** (Simulator *simulator, ModelGraphicsScene *scene, ModelGraphicsView *graphicsView, QLabel *replicationLabel, QProgressBar *simulationProgressBar, QTableWidget *simulationEventsTable, QTableWidget *entitiesTable, QTableWidget *variablesTable, QTextEdit *simulationText, QTextEdit *reportsText, QTabWidget *centralTabWidget, QAction *activateGraphicalSimulation, QAction *animationEnabledAction, bool *modelChecked, int tabCentralReportsIndex, Callbacks callbacks)

  Creates the simulation-event bridge used by MainWindow compatibility handlers.
- void **onModelCheckSuccessHandler** (ModelEvent *re) const

  Handles model-check success and updates delegated GUI state.
- void **onReplicationStartHandler** (SimulationEvent *re) const

  Handles replication-start event and updates simulation/report panes.
- void **onSimulationStartHandler** (SimulationEvent *re) const

  Handles simulation-start event and resets run-scoped UI state.
- void **onSimulationPausedHandler** (SimulationEvent *re) const

  Handles simulation pause event and refreshes action availability.
- void **onSimulationResumeHandler** (SimulationEvent *re) const

  Handles simulation resume event and resumes graphical simulation synchronization.
- void **onSimulationEndHandler** (SimulationEvent *re) const

  Handles simulation end event and performs delegated cleanup/refresh.
- void **onReplicationEndHandler** (SimulationEvent *re) const

  Handles replication end event and forces progress display completion.
- void **onProcessEventHandler** (SimulationEvent *re) const

  Handles process-event updates for progress tables and scene refresh hooks.
- void **onEntityCreateHandler** (SimulationEvent *re) const

  Handles entity-create events for compatibility callback wiring.
- void **onEntityRemoveHandler** (SimulationEvent *re) const

  Handles entity-remove events for compatibility callback wiring.
- void **onMoveEntityEvent** (SimulationEvent *re) const

  Handles entity-move event updates used by graphical simulation animation.
- void **onAfterProcessEvent** (SimulationEvent *re) const

  Handles after-process event updates used by animation/frame synchronization.
- void **setOnEventHandlers** (MainWindow *owner) const

  Registers kernel on-event callbacks through MainWindow compatibility façade wrappers.

17.870.1 Detailed Description

Controller for simulation event callbacks, UI updates, and handler wiring.
This controller concentrates event-reaction logic previously embedded in MainWindow and keeps compatibility by exposing methods invoked from MainWindow wrappers. It also provides the event-registration entry point used to wire kernel OnEventManager callbacks to those wrappers without changing the existing signal/slot surface.
Responsibilities:

- react to model-check, replication, simulation, process, and entity events;
- update progress/status widgets and debug tables through injected UI dependencies;

- trigger delegated refresh callbacks for actions, debug panes, and scene synchronization;
- register handlers through MainWindow compatibility façade ownership.

Boundaries:

- it does not own MainWindow or widget lifetime;
- it does not execute simulation commands (handled by command controllers);
- it does not perform persistence/export/model-language synchronization.

The documentation for this class was generated from the following files:

- applications/gui/genesys/controllers/SimulationEventController.h
- applications/gui/genesys/controllers/SimulationEventController.cpp

17.871 SimulationExperiment Class Reference

Represents a higher-level simulation experiment definition.
`#include <kernel/simulator/SimulationExperiment.h>`

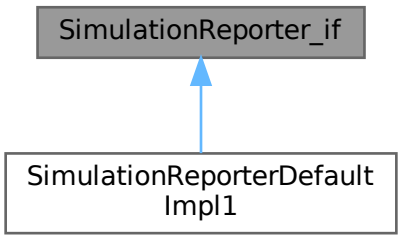
17.871.1 Detailed Description

Represents a higher-level simulation experiment definition.
Historically, GenESyS started with experiment control concentrated in ModelSimulation, mainly through replication count and replication length.
SimulationExperiment was introduced later as part of an unfinished effort to move richer experiment-design concepts into the simulation kernel, including experiment sets, scenarios and Process Analyzer-like workflows.
This class is still under development and should currently be treated as a placeholder for future experiment-layer behavior.
The documentation for this class was generated from the following files:

- kernel/simulator/SimulationExperiment.h
- kernel/simulator/SimulationExperiment.cpp

17.872 SimulationReporter_if Class Reference

Interface for reporting simulation results and experiment-facing outputs.
`#include <kernel/simulator/SimulationReporter_if.h>`
Inheritance diagram for SimulationReporter_if:



Public Member Functions

- virtual void showReplicationStatistics ()=0

  showReplicationStatistics
- virtual void showSimulationStatistics ()=0

  showSimulationStatistics
- virtual void showSimulationResponses ()=0

  showSimulationResponses
- virtual void showSimulationControls ()=0

  showSimulationControls

17.872.1 Detailed Description

Interface for reporting simulation results and experiment-facing outputs.
Reporter implementations define how replication/simulation summaries, responses and controls are presented to users or external consumers.

17.872.2 Member Function Documentation

17.872.2.1 showReplicationStatistics()

```
virtual void SimulationReporter_if::showReplicationStatistics ( )  [pure virtual]
```

showReplicationStatistics
Shows detailed statistics for the current/finished replication.
Implemented in SimulationReporterDefaultImpl1.

17.872.2.2 showSimulationControls()

```
virtual void SimulationReporter_if::showSimulationControls ( )  [pure virtual]
```

showSimulationControls
Shows monitored/configurable simulation controls (inputs).
Implemented in SimulationReporterDefaultImpl1.

17.872.2.3 showSimulationResponses()

```
virtual void SimulationReporter_if::showSimulationResponses ( )  [pure virtual]
```

showSimulationResponses
Shows configured simulation responses (outputs) for experimental analysis.
Implemented in SimulationReporterDefaultImpl1.

17.872.2.4 showSimulationStatistics()

```
virtual void SimulationReporter_if::showSimulationStatistics ( )  [pure virtual]
```

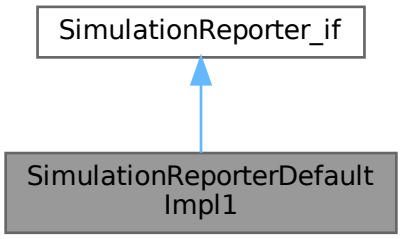
showSimulationStatistics
Shows aggregated statistics for the full simulation.
Implemented in SimulationReporterDefaultImpl1.
The documentation for this class was generated from the following file:

- kernel/simulator/SimulationReporter_if.h

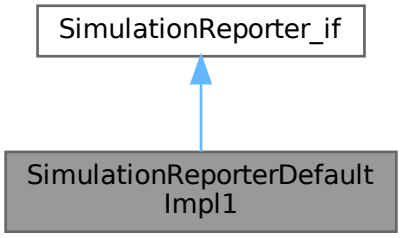
17.873 SimulationReporterDefaultImpl1 Class Reference

```
#include <kernel/simulator/SimulationReporterDefaultImpl1.h>
```

Inheritance diagram for SimulationReporterDefaultImpl1:



Collaboration diagram for SimulationReporterDefaultImpl1:



Public Member Functions

- **SimulationReporterDefaultImpl1** (ModelSimulation ∗simulation, Model ∗model, List< ModelDataDefinition ∗ > ∗statsCountersSimulation)
- virtual void showReplicationStatistics () override

  showReplicationStatistics
- virtual void showSimulationStatistics () override

  showSimulationStatistics
- virtual void showSimulationResponses () override

  showSimulationResponses
- virtual void showSimulationControls () override

  showSimulationControls

17.873.1 Detailed Description

Class that implements SimulationReporter_if interface and is responsible for building and showing replication and simulation reports

17.873.2 Member Function Documentation

17.873.2.1 showReplicationStatistics()

```
void SimulationReporterDefaultImpl1::showReplicationStatistics ( )  [override], [virtual]
```

showReplicationStatistics
Shows detailed statistics for the current/finished replication.
Implements SimulationReporter_if.

17.873.2.2 showSimulationControls()

```
void SimulationReporterDefaultImpl1::showSimulationControls ( )  [override], [virtual]
```

showSimulationControls
Shows monitored/configurable simulation controls (inputs).
Implements SimulationReporter_if.

17.873.2.3 showSimulationResponses()

```
void SimulationReporterDefaultImpl1::showSimulationResponses ( )  [override], [virtual]
```

showSimulationResponses
Shows configured simulation responses (outputs) for experimental analysis.
Implements SimulationReporter_if.

17.873.2.4 showSimulationStatistics()

```
void SimulationReporterDefaultImpl1::showSimulationStatistics ( )  [override], [virtual]
```

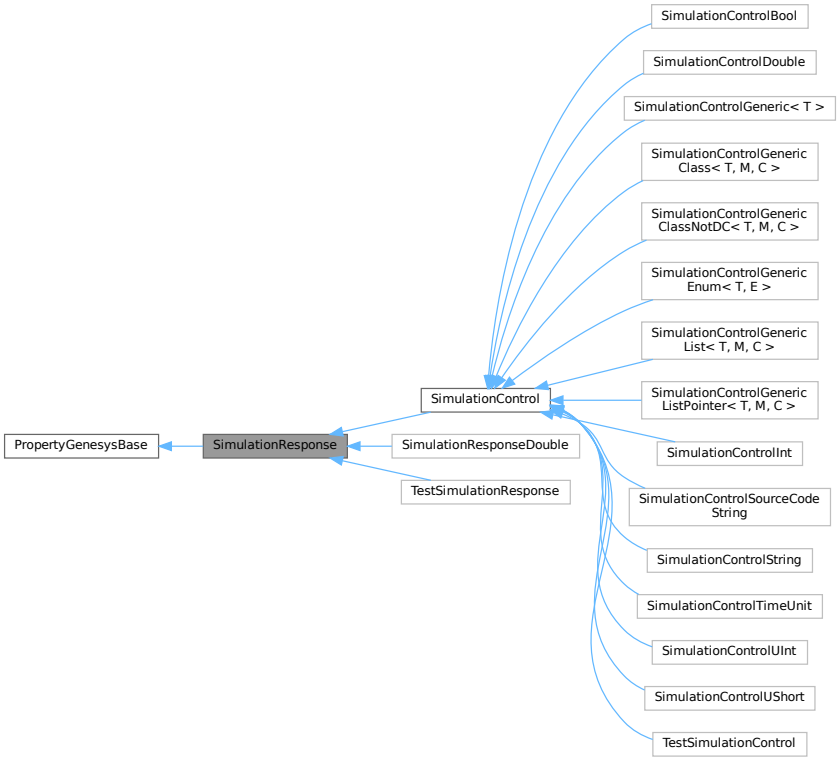
showSimulationStatistics
Shows aggregated statistics for the full simulation.
Implements SimulationReporter_if.
The documentation for this class was generated from the following files:

- kernel/simulator/SimulationReporterDefaultImpl1.h
- kernel/simulator/SimulationReporterDefaultImpl1.cpp

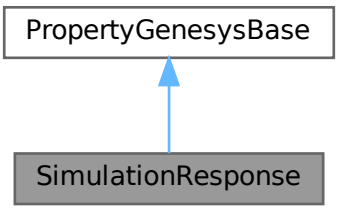
17.874 SimulationResponse Class Reference

Read-only kernel-side simulation response abstraction.
`#include <kernel/simulator/SimulationControlAndResponse.h>`

Inheritance diagram for SimulationResponse:



Collaboration diagram for SimulationResponse:



Public Member Functions

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual std::string **getValue** () const =0
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std↩

::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

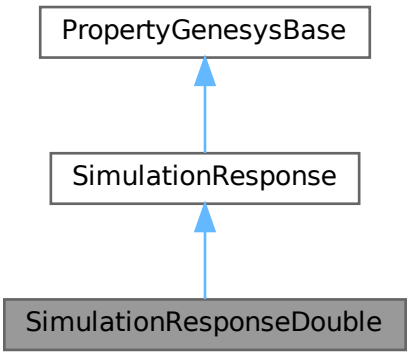
17.874.1 Detailed Description

Read-only kernel-side simulation response abstraction.
This is the intended kernel-level base for generic getter-based access to observable model attributes, statistics and experiment responses.
The documentation for this class was generated from the following file:

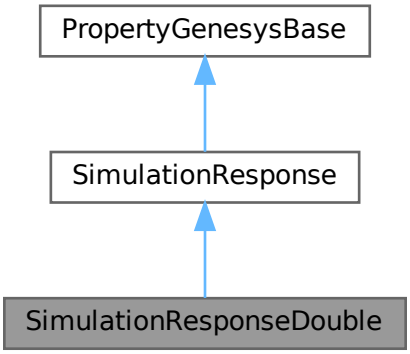
- kernel/simulator/SimulationControlAndResponse.h

17.875 SimulationResponseDouble Class Reference

Inheritance diagram for SimulationResponseDouble:



Collaboration diagram for SimulationResponseDouble:



Public Member Functions

- **SimulationResponseDouble** (GetterDouble getter, std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- virtual std::string getValue () const override

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**

- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.875.1 Member Function Documentation

17.875.1.1 getValue()

```
virtual std::string SimulationResponseDouble::getValue ( ) const  [inline], [override], [virtual]
```

Implements SimulationResponse.

The documentation for this class was generated from the following file:

- kernel/simulator/SimulationControlAndResponse.h

17.876 SimulationResultsDataset Struct Reference

Public Member Functions

- std::string formatKindName () const
- bool isRecordFormat () const
- std::vector< double > values () const
- std::vector< unsigned int > replications () const
- std::vector< double > valuesForReplication (unsigned int replication) const

Public Attributes

- std::string **datasetName**
- std::string **randomVariableName**
- std::string **description**
- std::string **variableType** = "Continuous numeric"
- std::string **source**
- std::string **sourceDescription**
- std::string **expression**
- std::string **expressionName**
- std::string **formatVersion**
- SimulationResultsDatasetFormat **formatKind** = SimulationResultsDatasetFormat::Unknown
- bool **recordFile** = false
- bool **timeDependent** = false
- std::vector< SimulationResultsObservation > **observations**
- std::vector< std::string > **previewLines**

17.876.1 Member Function Documentation

17.876.1.1 formatKindName()

```
std::string SimulationResultsDataset::formatKindName ( ) const
```

Returns the display name for the detected dataset format.

17.876.1.2 isRecordFormat()

```
bool SimulationResultsDataset::isRecordFormat ( ) const
```

Returns true when this dataset was imported from a Genesys Record file.

17.876.1.3 replications()

```
std::vector< unsigned int > SimulationResultsDataset::replications ( ) const
```

Returns the distinct replication identifiers present in the observations.

17.876.1.4 values()

```
std::vector< double > SimulationResultsDataset::values ( ) const
```

Returns the observation values without replication or time metadata.

17.876.1.5 valuesForReplication()

```
std::vector< double > SimulationResultsDataset::valuesForReplication (
            unsigned int replication ) const
```

Returns all observation values for a single replication.
The documentation for this struct was generated from the following files:

- tools/Statistics/SimulationResultsDataset.h
- tools/Statistics/SimulationResultsDataset.cpp

17.877 SimulationResultsDatasetParser Class Reference

Static Public Member Functions

- static bool loadFromTextFile (const std::string &fileName, SimulationResultsDataset *dataset, std::string *errorMessage=nullptr)

17.877.1 Member Function Documentation

17.877.1.1 loadFromTextFile()

```
bool SimulationResultsDatasetParser::loadFromTextFile (
            const std::string & fileName,
            SimulationResultsDataset * dataset,
            std::string * errorMessage = nullptr )  [static]
```

Loads a simulation-results dataset from a text file.
The parser intentionally detects formats in this order: GUI tabular, Genesys Record, then raw numeric. Once a format is detected, no residual heuristic reinterprets the same file as another format.
The documentation for this class was generated from the following files:

- tools/Statistics/SimulationResultsDataset.h
- tools/Statistics/SimulationResultsDataset.cpp

17.878 SimulationResultsObservation Struct Reference

Public Attributes

- unsigned int **replication** = 1
- double **time** = 0.0
- bool **hasTime** = false
- double **value** = 0.0
- unsigned int **sourceLine** = 0

The documentation for this struct was generated from the following file:

- tools/Statistics/SimulationResultsDataset.h

17.879 SimulationScenario Class Reference

Represents a candidate simulation scenario within the unfinished experiment layer.
`#include <kernel/simulator/SimulationScenario.h>`

Public Member Functions

- bool **startSimulation** (Simulator *sim, std::string &errorMessage)
- std::list< std::pair< std::string, double > * > * getResponseValues () const
- std::list< std::pair< std::string, double > * > * **getControlValues** () const
- double **getResponseValue** (const std::string &responseName)
- void **setModelFilename** (std::string _modelFilename)
- std::string **getModelFilename** () const
- void **setScenarioName** (std::string _name)
- std::string **getScenarioName** () const
- void **setScenarioDescription** (std::string _scenarioDescription)
- std::string **getScenarioDescription** () const
- std::list< std::string > * **getSelectedControls** () const
- double **getControlValue** (const std::string &controlName)
- std::list< std::string > * **getSelectedResponses** () const
- void **setSelectedControls** (std::list< std::string > *selectedControls)
- void **setControl** (std::string name, double value) const

17.879.1 Detailed Description

Represents a candidate simulation scenario within the unfinished experiment layer.
Represents a scenario where a specific model (defined my ModelFilename) will be simulated. To each scenario will be associated a set of SimulationControl and SimulationResponse, and their values are set to the scenario by the ProcessAnalyser.
The original GenESyS execution flow relied on ModelSimulation for the basic experimental controls already supported by the kernel.
SimulationScenario was introduced later to support richer scenario-based experimentation, potentially including alternative models, modified controls and future Process Analyzer-like workflows.
This class is still under development and should currently be treated as an evolving kernel-side abstraction for experiment scenarios.

17.879.2 Member Function Documentation

17.879.2.1 getResponseValues()

```
std::list< std::pair< std::string, double > * > * SimulationScenario::getResponseValues ( )
const
```

The final result of the simulationScenario
The documentation for this class was generated from the following files:

- kernel/simulator/SimulationScenario.h
- kernel/simulator/SimulationScenario.cpp

17.880 SimulatorSessionService::SimulationStatusResult Struct Reference

Captures the current simulation execution state.
`#include <applications/worker/service/SimulatorSessionService.h>`

Public Attributes

- bool **success** = false
- bool **invalidToken** = false
- bool **hasCurrentModel** = false
- bool **isRunning** = false
- bool **isPaused** = false
- double **simulatedTime** = 0.0

- unsigned int **currentReplicationNumber** = 0
- unsigned int **numberOfReplications** = 0
- double **replicationLength** = 0.0
- double **warmUpPeriod** = 0.0
- bool **pauseOnEvent** = false
- bool **pauseOnReplication** = false
- bool **initializeStatistics** = false
- bool **initializeSystem** = false

17.880.1 Detailed Description

Captures the current simulation execution state.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.881 Simulator Class Reference

`#include <kernel/simulator/Simulator.h>`

Public Member Functions

- Simulator ()
- std::string getVersion () const

  getVersion
- unsigned int getVersionNumber () const

  getVersionNumber
- std::string getName () const

  getName
- LicenceManager * getLicenceManager () const

  getLicence
- PluginManager * getPluginManager () const

  getPlugins
- ModelManager * getModelManager () const

  getModelManager
- TraceManager * getTraceManager () const

  getTraceManager
- ParserManager * getParserManager () const

  getParserManager
- ExperimentManager * getExperimentManager () const

  getExperimentManager

Friends

- class **PluginManager**

17.881.1 Detailed Description

The main class of the Genesys KERNEL simulation. It gives access to simulation models and tools. Simulation is the top level class and is supossed to be available to application as a dynamic linked library.

17.881.2 Constructor & Destructor Documentation

17.881.2.1 Simulator()

```
Simulator::Simulator ( )
```

Simulator constructor basically constructs inner classes (managers) and shows basic information on the console. This is the only method on the entire simulator thats directlly plots something to the output

17.881.3 Member Function Documentation

17.881.3.1 getExperimentManager()

```
ExperimentManager * Simulator::getExperimentManager ( ) const
```

getExperimentManager

Returns

17.881.3.2 getLicenceManager()

```
LicenceManager * Simulator::getLicenceManager ( ) const
```

getLicence

Returns

17.881.3.3 getModelManager()

```
ModelManager * Simulator::getModelManager ( ) const
```

getModelManager

Returns

17.881.3.4 getName()

```
std::string Simulator::getName ( ) const
```

getName

Returns

17.881.3.5 getParserManager()

```
ParserManager * Simulator::getParserManager ( ) const
```

getParserManager

Returns

17.881.3.6 getPluginManager()

```
PluginManager * Simulator::getPluginManager ( ) const
```

getPlugins

Returns

17.881.3.7 getTraceManager()

```
TraceManager * Simulator::getTraceManager ( ) const
```
getTraceManager

Returns

17.881.3.8 getVersion()

```
std::string Simulator::getVersion ( ) const
```
getVersion

Returns

17.881.3.9 getVersionNumber()

```
unsigned int Simulator::getVersionNumber ( ) const
```
getVersionNumber

Returns

The documentation for this class was generated from the following files:

- kernel/simulator/Simulator.h
- kernel/simulator/Simulator.cpp
- tests/unit/test_simulator_support.cpp
- tests/unit/test_support_experimentmanager.cpp
- tests/unit/test_support_modelmanager.cpp

17.882 SimulatorFacade Class Reference

High-level non-owning facade over Simulator and its main composed services.
`#include <kernel/simulator/SimulatorFacade.h>`

Public Member Functions

- **SimulatorFacade** (Simulator *simulator)
- std::string **getVersion** () const
- unsigned int **getVersionNumber** () const
- std::string **getName** () const
- const std::string **licenceShow** () const
- const std::string **licenceShowLimits** () const
- const std::string **licenceShowActivationCode** () const
- bool **licenceLookforActivationCode** ()
- bool **licenceInsertActivationCode** ()
- void **licenceRemoveActivationCode** ()
- unsigned int **licenceGetModelComponentsLimit** ()
- unsigned int **licenceGetModelDatasLimit** ()
- unsigned int **licenceGetEntityLimit** ()
- unsigned int **licenceGetHostsLimit** ()
- unsigned int **licenceGetThreadsLimit** ()
- std::string **showPlugins** ()
- List< Plugin * > * **completePluginsFieldsAndTemplates** ()

- bool **pluginCheck** (const std::string &dynamicLibraryFilename)
- SystemDependencyCheckResult **pluginCheckSystemDependencies** (const std::string &dynamicLibrary↩ Filename)
- List< std::string > * **discoverPluginFilenames** () const
- List< PluginLoadIssue > * **getPluginLoadIssues** () const
- void **clearPluginLoadIssues** ()
- void **clearPluginLoadIssue** (const std::string &dynamicLibraryFilename)
- Plugin * **insertPlugin** (const std::string &dynamicLibraryFilename)
- Plugin * **insertPlugin** (const std::string &dynamicLibraryFilename, const PluginInsertionOptions &options)
- bool **removePlugin** (const std::string &dynamicLibraryFilename)
- bool **removePlugin** (Plugin *plugin)
- Plugin * **findPlugin** (const std::string &pluginTypeName)
- std::vector< std::string > **getDataDefinitionPluginTypenames** () const
- std::string **pluginSourceIncludePathFor** (const std::string &pluginTypeName)
- List< Plugin * > * **autoInsertPlugins** (const std::string &pluginsListFilename, const bool lookForPluginsIf↩ FilenameNotFound=true)
- List< Plugin * > * **autoInsertPlugins** (const std::string &pluginsListFilename, const bool lookForPluginsIf↩ FilenameNotFound, const PluginInsertionOptions &options)
- List< Plugin * > * **autoInsertPlugins** ()
- List< Plugin * > * **autoInsertPlugins** (const PluginInsertionOptions &options)
- Plugin * **firstPlugin** ()
- Plugin * **nextPlugin** ()
- Plugin * **lastPlugin** ()
- unsigned int **pluginCount** ()
- Plugin * **getPluginAtRank** (unsigned int rank)
- ModelDataDefinition * **newInstance** (const std::string &pluginTypename, Model *model, std::string name="")
- template<typename T >
  T * **newInstance** (Model *model, std::string name="")
- Model * **newModel** ()
- void **insertModel** (Model *model)
- void **removeModel** (Model *model)
- bool **setCurrentModel** (Model *model)
- bool **saveModel** (std::string filename)
- Model * **loadModel** (std::string filename)
- Model * **createModelFromLanguage** (std::string modelSpecification)
- unsigned int **modelCount** ()
- std::vector< Model * > **models** () const
- bool **hasModel** (Model *model) const
- int **indexOfModel** (Model *model) const
- Model * **modelAt** (unsigned int index) const
- int **currentModelIndex** () const
- Model * **firstModel** ()
- Model * **lastModel** ()
- Model * **currentModel** ()
- Model * **nextModel** ()
- Model * **previousModel** ()
- bool **canGoNextModel** () const
- bool **canGoPreviousModel** () const
- void **addTraceHandler** (traceListener traceListener)
- void **addTraceReportHandler** (traceListener traceReportListener)
- void **addTraceSimulationHandler** (traceSimulationListener traceSimulationListener)
- void **addTraceErrorHandler** (traceErrorListener traceErrorListener)
- template<typename Class >
  void **addTraceHandler** (Class *object, void(Class::*function)(TraceEvent))

- template<typename Class >
  void **addTraceErrorHandler** (Class *object, void(Class::*function)(TraceErrorEvent))
- template<typename Class >
  void **addTraceReportHandler** (Class *object, void(Class::*function)(TraceEvent))
- template<typename Class >
  void **addTraceSimulationHandler** (Class *object, void(Class::*function)(TraceSimulationEvent))
- void **addTraceSimulationExceptionRuleModelData** (void *thisobject)
- void **beginTraceShutdown** ()
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed, bool showAnyway=false)
- void **traceSimulation** (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed, bool showAnyway=false)
- List< std::string > * **traceGetErrorMessages** () const
- void **setTraceLevel** (TraceManager::Level traceLevel)
- TraceManager::Level **getTraceLevel** () const
- void **setTraceSimulationRuleAllAllowed** (bool traceSimulationRuleAllAllowed)
- bool **isTraceSimulationRuleAllAllowed** () const
- ParserManager::GenerateNewParserResult **parserGenerateNewParser** (ParserChangesInformation *changes)
- bool **parserConnectNewParser** (ParserManager::NewParser newParser)
- SimulationExperiment * **newSimulationExperiment** ()
- void **insertSimulationExperiment** (SimulationExperiment *experiment)
- void **removeSimulationExperiment** (SimulationExperiment *experiment)
- void **setCurrentSimulationExperiment** (SimulationExperiment *experiment)
- bool **saveSimulationExperiment** (std::string filename)
- bool **loadSimulationExperiment** (std::string filename)
- unsigned int **simulationExperimentCount** ()
- SimulationExperiment * **firstSimulationExperiment** ()
- SimulationExperiment * **currentSimulationExperiment** ()
- SimulationExperiment * **nextSimulationExperiment** ()
- List< SimulationExperiment * > * **getSimulationExperiments** () const
- bool **modelSave** (std::string filename)
- bool **modelLoad** (std::string filename)
- bool **modelCheck** ()
- void **modelClear** ()
- std::string **modelShowLanguage** ()
- void **modelShow** ()
- bool **modelInsert** (ModelDataDefinition *elemOrComp)
- void **modelRemove** (ModelDataDefinition *elemOrComp)
- std::list< ModelDataDefinition * > **modelCollectDataDefinitionsRemovedWith** (const std::list< ModelDataDefinition * > &roots) const
- Entity * **modelCreateEntity** (std::string name, bool insertIntoModel=true)
- void **modelRemoveEntity** (Entity *entity)
- void **modelSendEntityToComponent** (Entity *entity, Connection *connection, double timeDelay=0.0)
- void **modelSendEntityToComponent** (Entity *entity, ModelComponent *component, double timeDelay=0.0, unsigned int componentinputPortNumber=0)
- double **modelParseExpression** (const std::string expression)
- double **modelParseExpression** (const std::string expression, bool &success, std::string &errorMessage)
- bool **modelCheckExpression** (const std::string expression, const std::string expressionName, std::string &errorMessage)

- void **modelCheckReferencesToDataDefinitions** (std::string expression, std::map< std::string, std::list< std::string > * > *referencedDataDefinitions)
- Util::identification **modelGetId** () const
- bool **modelHasChanged** () const
- void **modelSetHasChanged** (bool hasChanged)
- List< Event * > * **modelGetFutureEvents** () const
- List< SimulationControl * > * **modelGetControls** () const
- List< SimulationResponse * > * **modelGetResponses** () const
- Persistence_if * **modelGetPersistence** () const
- unsigned int **modelGetLevel** () const
- std::string **infoShow** ()
- void **infoSetName** (std::string name)
- std::string **infoGetName** () const
- void **infoSetAnalystName** (std::string analystName)
- std::string **infoGetAnalystName** () const
- void **infoSetDescription** (std::string description)
- std::string **infoGetDescription** () const
- void **infoSetProjectTitle** (std::string projectTitle)
- std::string **infoGetProjectTitle** () const
- void **infoSetVersion** (std::string version)
- std::string **infoGetVersion** () const
- void **infoLoadInstance** (PersistenceRecord *fields)
- void **infoSaveInstance** (PersistenceRecord *fields)
- bool **infoHasChanged** () const
- void **infoSetHasChanged** (bool hasChanged)
- std::string **simShow** ()
- void **simStart** ()
- void **simPause** ()
- void **simStep** ()
- void **simStop** ()
- void **simSetNumberOfReplications** (unsigned int numberOfReplications)
- unsigned int **simGetNumberOfReplications** () const
- void **simSetReplicationLength** (double replicationLength, Util::TimeUnit replicationLengthTimeUnit=Util::↩ TimeUnit::unknown)
- double **simGetReplicationLength** () const
- void **simSetReplicationLengthTimeUnit** (Util::TimeUnit replicationLengthTimeUnit)
- Util::TimeUnit **simGetReplicationLengthTimeUnit** () const
- void **simSetReplicationReportBaseTimeUnit** (Util::TimeUnit replicationReportBaseTimeUnit)
- Util::TimeUnit **simGetReplicationBaseTimeUnit** () const
- void **simSetWarmUpPeriod** (double warmUpPeriod, Util::TimeUnit warmUpPeriodTimeUnit=Util::TimeUnit↩ ::unknown)
- double **simGetWarmUpPeriod** () const
- void **simSetWarmUpPeriodTimeUnit** (Util::TimeUnit warmUpPeriodTimeUnit)
- Util::TimeUnit **simGetWarmUpPeriodTimeUnit** () const
- void **simSetTerminatingCondition** (std::string terminatingCondition)
- std::string **simGetTerminatingCondition** () const
- void **simSetPauseOnEvent** (bool pauseOnEvent)
- bool **simIsPauseOnEvent** () const
- void **simSetStepByStep** (bool stepByStep)
- bool **simIsStepByStep** () const
- void **simSetInitializeStatistics** (bool initializeStatistics)
- bool **simIsInitializeStatistics** () const
- void **simSetInitializeSystem** (bool initializeSystem)
- bool **simIsInitializeSystem** () const
- void **simSetPauseOnReplication** (bool pauseBetweenReplications)

- bool **simIsPauseOnReplication** () const
- void **simSetReporter** (SimulationReporter_if *simulationReporter)
- SimulationReporter_if * **simGetReporter** () const
- double **simGetSimulatedTime** () const
- bool **simIsRunning** () const
- bool **simIsPaused** () const
- unsigned int **simGetCurrentReplicationNumber** () const
- void **simSetShowReportsAfterReplication** (bool showReportsAfterReplication)
- bool **simIsShowReportsAfterReplication** () const
- void **simSetShowReportsAfterSimulation** (bool showReportsAfterSimulation)
- bool **simIsShowReportsAfterSimulation** () const
- List< double > * **simGetBreakpointsOnTime** () const
- List< Entity * > * **simGetBreakpointsOnEntity** () const
- List< ModelComponent * > * **simGetBreakpointsOnComponent** () const
- const List< ModelDataDefinition * > * **simGetSimulationStatisticsAggregates** () const
- void **simLoadInstance** (PersistenceRecord *fields)
- void **simSaveInstance** (PersistenceRecord *fields, bool saveDefaults)
- Event * **simGetCurrentEvent** () const
- void **simSetShowSimulationResposesInReport** (bool showSimulationResposesInReport)
- bool **simIsShowSimulationResposesInReport** () const
- void **simSetShowSimulationControlsInReport** (bool showSimulationControlsInReport)
- bool **simIsShowSimulationControlsInReport** () const
- bool **dataInsert** (ModelDataDefinition *anElement)
- void **dataRemove** (ModelDataDefinition *anElement)
- bool **dataInsert** (std::string datadefinitionTypename, ModelDataDefinition *anElement)
- void **dataRemove** (std::string datadefinitionTypename, ModelDataDefinition *anElement)
- bool **dataCheck** (std::string datadefinitionTypename, ModelDataDefinition *anElement, std::string expressionName, std::string &errorMessage)
- bool **dataCheck** (std::string datadefinitionTypename, std::string elementName, std::string expressionName, bool mandatory, std::string &errorMessage)
- void **dataClear** ()
- ModelDataDefinition * **dataGetDataDefinition** (std::string datadefinitionTypename, Util::identification id)
- ModelDataDefinition * **dataGetDataDefinition** (std::string datadefinitionTypename, std::string name)
- unsigned int **dataGetNumberOfDataDefinitions** (std::string datadefinitionTypename)
- unsigned int **dataGetNumberOfDataDefinitions** ()
- int **dataGetRankOf** (std::string datadefinitionTypename, std::string name)
- std::list< std::string > **dataGetDataDefinitionClassnames** () const
- List< ModelDataDefinition * > * **dataGetDataDefinitionList** (std::string datadefinitionTypename) const
- void **dataShow** ()
- bool **dataHasChanged** () const
- void **dataSetHasChanged** (bool hasChanged)
- bool **componentInsert** (ModelComponent *comp)
- void **componentRemove** (ModelComponent *comp)
- ModelComponent * **componentFind** (std::string name)
- ModelComponent * **componentFind** (Util::identification id)
- void **componentClear** ()
- unsigned int **componentGetNumberOfComponents** ()
- std::list< ModelComponent * >::iterator **componentBegin** ()
- std::list< ModelComponent * >::iterator **componentEnd** ()
- ModelComponent * **componentFront** ()
- ModelComponent * **componentNext** ()
- bool **componentHasChanged** () const
- void **componentSetHasChanged** (bool hasChanged)
- std::list< SourceModelComponent * > * **componentGetSourceComponents** ()
- std::list< ModelComponent * > * **componentGetTransferInComponents** ()

- std::list< ModelComponent * > * **componentGetAllComponents** () const
- void **eventAddOnModelCheckSucessHandler** (modelEventHandler EventHandler)
- void **eventAddOnModelLoadHandler** (modelEventHandler EventHandler)
- void **eventAddOnModelSaveHandler** (modelEventHandler EventHandler)
- void **eventAddOnReplicationStartHandler** (simulationEventHandler EventHandler)
- void **eventAddOnReplicationStepHandler** (simulationEventHandler EventHandler)
- void **eventAddOnReplicationEndHandler** (simulationEventHandler EventHandler)
- void **eventAddOnProcessEventHandler** (simulationEventHandler EventHandler)
- void **eventAddOnAfterProcessEventHandler** (simulationEventHandler EventHandler)
- void **eventAddOnEntityCreateHandler** (simulationEventHandler EventHandler)
- void **eventAddOnEntityMoveHandler** (simulationEventHandler EventHandler)
- void **eventAddOnEntityRemoveHandler** (simulationEventHandler EventHandler)
- void **eventAddOnSimulationStartHandler** (simulationEventHandler EventHandler)
- void **eventAddOnSimulationPausedHandler** (simulationEventHandler EventHandler)
- void **eventAddOnSimulationResumeHandler** (simulationEventHandler EventHandler)
- void **eventAddOnSimulationEndHandler** (simulationEventHandler EventHandler)
- void **eventAddOnBreakpointHandler** (simulationEventHandler EventHandler)
- template<typename Class >
  void **eventAddOnModelCheckSuccessHandler** (Class *object, void(Class::*function)(ModelEvent *))
- template<typename Class >
  void **eventAddOnModelLoadHandler** (Class *object, void(Class::*function)(ModelEvent *))
- template<typename Class >
  void **eventAddOnModelSaveHandler** (Class *object, void(Class::*function)(ModelEvent *))
- template<typename Class >
  void **eventAddOnReplicationStartHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnReplicationStepHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnReplicationEndHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnProcessEventHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnAfterProcessEventHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnEntityCreateHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnEntityMoveHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnEntityRemoveHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnSimulationStartHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnSimulationPausedHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnSimulationResumeHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnSimulationEndHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- template<typename Class >
  void **eventAddOnBreakpointHandler** (Class *object, void(Class::*function)(SimulationEvent *))
- void **eventNotifyModelCheckSuccessHandlers** (ModelEvent *se)
- void **eventNotifyModelLoadHandlers** (ModelEvent *se)
- void **eventNotifyModelSaveHandlers** (ModelEvent *se)
- void **eventNotifyReplicationStartHandlers** (SimulationEvent *se)
- void **eventNotifyReplicationStepHandlers** (SimulationEvent *se)
- void **eventNotifyReplicationEndHandlers** (SimulationEvent *se)
- void **eventNotifyProcessEventHandlers** (SimulationEvent *se)

- void **eventNotifyAfterProcessEventHandlers** (SimulationEvent ∗se)
- void **eventNotifyEntityCreateHandlers** (SimulationEvent ∗se)
- void **eventNotifyEntityMoveHandlers** (SimulationEvent ∗se)
- void **eventNotifyEntityRemoveHandlers** (SimulationEvent ∗se)
- void **eventNotifySimulationStartHandlers** (SimulationEvent ∗se)
- void **eventNotifySimulationPausedHandlers** (SimulationEvent ∗se)
- void **eventNotifySimulationResumeHandlers** (SimulationEvent ∗se)
- void **eventNotifySimulationEndHandlers** (SimulationEvent ∗se)
- void **eventNotifyBreakpointHandlers** (SimulationEvent ∗se)

Static Public Member Functions

- static std::string **traceConvertEnumToStr** (TraceManager::Level level)

17.882.1 Detailed Description

High-level non-owning facade over Simulator and its main composed services.
Construct with a non-owning pointer to an existing Simulator instance. All public methods delegate through the private helpers; none of those helpers are exposed publicly.
The documentation for this class was generated from the following file:

- kernel/simulator/SimulatorFacade.h

17.883 SimulatorSessionService::SimulatorInfoResult Struct Reference

Describes simulator identity metadata for a token-scoped session.
`#include <applications/worker/service/SimulatorSessionService.h>`

Public Attributes

- std::string **name**
- std::string **versionName**
- unsigned int **versionNumber**

17.883.1 Detailed Description

Describes simulator identity metadata for a token-scoped session.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.884 SimulatorSessionService Class Reference

Encapsulates session-scoped simulator operations exposed by the Web API.
`#include <applications/worker/service/SimulatorSessionService.h>`

Classes

- struct CreateSessionResult

  Contains session creation identifiers returned to API clients.
- struct ModelImportResult

  Contains model import output and error state.
- struct ModelInfoResult

  Describes the currently active model information for a session.
- struct ModelPersistenceResult

  Contains save/load model operation output and error state.

- struct SimulationActionResult

  Contains the result of a simulation action such as run or step.
- struct SimulationConfigInput

  Defines user-provided simulation parameters to apply to the current model.
- struct SimulationConfigResult

  Contains the result of applying simulation configuration.
- struct SimulationStatusResult

  Captures the current simulation execution state.
- struct SimulatorInfoResult

  Describes simulator identity metadata for a token-scoped session.
- struct WorkerCapabilitiesResult

  Describes currently supported worker features.
- struct WorkerInfoResult

  Describes worker identity metadata for distributed discovery.
- struct WorkerJobCreationResult

  Contains worker job creation output and error state.
- struct WorkerJobInfoResult

  Describes worker job metadata exposed by worker stage 3 endpoints.
- struct WorkerJobQueryResult

  Contains worker job lookup output and error state.
- struct WorkerJobResultInfo

  Describes terminal worker job result data exposed by worker stage 5 endpoint.
- struct WorkerJobResultQueryResult

  Contains worker job terminal result lookup output and error state.
- struct WorkerJobRunResult

  Contains worker job run output and error state.

Public Types

- enum class PersistenceError {
  None , **InvalidToken** , **InvalidFilename** , **MissingCurrentModel** ,
  FileNotFound , **OperationFailed** }

  Enumerates persistence-specific errors for save/load model operations.
- enum class ModelImportError {
  None , **InvalidToken** , **EmptySpecification** , **InvalidSpecification** ,
  OperationFailed }

  Enumerates model import errors for worker language-ingress operations.
- enum class WorkerJobError {
  None , **InvalidToken** , **MissingCurrentModel** , **JobNotFound** ,
  AccessDenied , **OperationFailed** , **ResultNotReady** }

  Enumerates worker job operation errors.

Public Member Functions

- SimulatorSessionService (SessionManager &sessionManager, WorkerJobManager &workerJobManager)

  Builds a service bound to the provided session manager.
- CreateSessionResult createSession ()

  Creates a new authenticated session.
- bool tryGetSimulatorInfo (const std::string &accessToken, SimulatorInfoResult &outInfo)

  Fetches simulator metadata using a session token.
- WorkerInfoResult getWorkerInfo () const

  Fetches public worker identity metadata.
- WorkerCapabilitiesResult getWorkerCapabilities () const

Fetches worker capability flags for the current implementation state.

- bool tryCreateModel (const std::string &accessToken, ModelInfoResult &outInfo)

  Creates a new model in the token-scoped simulator session.
- bool tryGetCurrentModelInfo (const std::string &accessToken, ModelInfoResult &outInfo)

  Returns metadata for the current model in a session.
- SimulationStatusResult getSimulationStatus (const std::string &accessToken)

  Returns simulation status for the session's current model.
- SimulationConfigResult configureSimulation (const std::string &accessToken, const SimulationConfigInput &input)

  Applies simulation configuration to the current model.
- SimulationActionResult runSimulation (const std::string &accessToken)

  Executes a synchronous simulation run.
- SimulationActionResult stepSimulation (const std::string &accessToken)

  Executes one synchronous simulation step.
- ModelPersistenceResult saveCurrentModel (const std::string &accessToken, const std::string &filename)

  Saves the current model into the session workspace.
- ModelPersistenceResult loadModel (const std::string &accessToken, const std::string &filename)

  Loads a model from the session workspace by filename.
- ModelImportResult importModelFromLanguage (const std::string &accessToken, const std::string &model↩ Specification)

  Imports a model from plain text model language into the token-scoped session.
- WorkerJobCreationResult createWorkerJob (const std::string &accessToken)

  Creates a queued worker job for the token-scoped session model snapshot.
- WorkerJobQueryResult getWorkerJob (const std::string &accessToken, const std::string &jobId)

  Returns worker job metadata for a token-scoped session.
- WorkerJobRunResult runWorkerJob (const std::string &accessToken, const std::string &jobId)

  Executes a worker job synchronously in the token-scoped simulator session.
- WorkerJobResultQueryResult getWorkerJobResult (const std::string &accessToken, const std::string &jobId)

  Returns persisted terminal result data for a token-scoped worker job.

17.884.1 Detailed Description

Encapsulates session-scoped simulator operations exposed by the Web API.

17.884.2 Constructor & Destructor Documentation

17.884.2.1 SimulatorSessionService()

```
SimulatorSessionService::SimulatorSessionService (
            SessionManager & sessionManager,
            WorkerJobManager & workerJobManager )  [explicit]
```

Builds a service bound to the provided session manager.
Wires session lookup with per-session worker job storage.

Parameters

sessionManager	Session manager used to resolve active sessions.
workerJobManager	In-memory worker job storage manager.

17.884.3 Member Function Documentation

17.884.3.1 configureSimulation()

```
SimulatorSessionService::SimulationConfigResult SimulatorSessionService::configureSimulation (
            const std::string & accessToken,
            const SimulationConfigInput & input )
```

Applies simulation configuration to the current model.
Applies user-provided simulation parameters to the current model.

Parameters

accessToken	Bearer token associated with a session.
input	Configuration values to apply.

Returns

Structured configuration result and resulting status.

17.884.3.2 createSession()

```
SimulatorSessionService::CreateSessionResult SimulatorSessionService::createSession ( )
```

Creates a new authenticated session.
Creates one authenticated session and returns its public identifiers.

Returns

Session identifiers and token for API usage.

17.884.3.3 createWorkerJob()

```
SimulatorSessionService::WorkerJobCreationResult SimulatorSessionService::createWorkerJob (
            const std::string & accessToken )
```

Creates a queued worker job for the token-scoped session model snapshot.
Creates a queued worker job by snapshotting the current model.

Parameters

accessToken	Bearer token associated with a session.

Returns

Worker job creation output and error state.

17.884.3.4 getSimulationStatus()

```
SimulatorSessionService::SimulationStatusResult SimulatorSessionService::getSimulationStatus (
            const std::string & accessToken )
```

Returns simulation status for the session's current model.
Returns the current simulation status for a session token.

Parameters

accessToken	Bearer token associated with a session.

Returns

Structured simulation status query result.

17.884.3.5 getWorkerCapabilities()

```
SimulatorSessionService::WorkerCapabilitiesResult SimulatorSessionService::getWorkerCapabilities
( ) const
```

Fetches worker capability flags for the current implementation state.
Advertises the runtime capabilities supported by this worker flavor.

Returns

Worker capability flags.

17.884.3.6 getWorkerInfo()

```
SimulatorSessionService::WorkerInfoResult SimulatorSessionService::getWorkerInfo ( ) const
```

Fetches public worker identity metadata.
Returns the fixed worker identity metadata used for discovery.

Returns

Worker identity information for discovery endpoints.

17.884.3.7 getWorkerJob()

```
SimulatorSessionService::WorkerJobQueryResult SimulatorSessionService::getWorkerJob (
            const std::string & accessToken,
            const std::string & jobId )
```

Returns worker job metadata for a token-scoped session.
Retrieves worker-job metadata when the job belongs to the session.

Parameters

accessToken	Bearer token associated with a session.
jobId	Worker job identifier to query.

Returns

Worker job lookup output and error state.

17.884.3.8 getWorkerJobResult()

```
SimulatorSessionService::WorkerJobResultQueryResult SimulatorSessionService::getWorkerJob↩
Result (
            const std::string & accessToken,
            const std::string & jobId )
```

Returns persisted terminal result data for a token-scoped worker job.
Retrieves the terminal result for a completed worker job.

Parameters

accessToken	Bearer token associated with a session.
jobId	Worker job identifier to query.

Returns

Worker job terminal result lookup output and error state.

17.884.3.9 importModelFromLanguage()

```
SimulatorSessionService::ModelImportResult SimulatorSessionService::importModelFromLanguage (
            const std::string & accessToken,
            const std::string & modelSpecification )
```

Imports a model from plain text model language into the token-scoped session.
Imports a model from textual Genesys language.

Parameters

accessToken	Bearer token associated with a session.
modelSpecification	Plain text model language content.

Returns

Import output and error state.

17.884.3.10 loadModel()

```
SimulatorSessionService::ModelPersistenceResult SimulatorSessionService::loadModel (
            const std::string & accessToken,
            const std::string & filename )
```

Loads a model from the session workspace by filename.
Loads a saved model file from the session workspace.

Parameters

accessToken	Bearer token associated with a session.
filename	Safe basename to load.

Returns

Persistence output and error state.

17.884.3.11 runSimulation()

```
SimulatorSessionService::SimulationActionResult SimulatorSessionService::runSimulation (
            const std::string & accessToken )
```

Executes a synchronous simulation run.
Starts the current simulation for the token-scoped session.

Parameters

accessToken	Bearer token associated with a session.

Returns

Structured action result and updated simulation status.

17.884.3.12 runWorkerJob()

```
SimulatorSessionService::WorkerJobRunResult SimulatorSessionService::runWorkerJob (
```

```
            const std::string & accessToken,
            const std::string & jobId )
```

Executes a worker job synchronously in the token-scoped simulator session.
Executes a queued worker job synchronously from its stored snapshot.

Parameters

accessToken	Bearer token associated with a session.
jobId	Worker job identifier to execute.

Returns

Worker job run output and error state.

17.884.3.13 saveCurrentModel()

```
SimulatorSessionService::ModelPersistenceResult SimulatorSessionService::saveCurrentModel (
            const std::string & accessToken,
            const std::string & filename )
```

Saves the current model into the session workspace.
Persists the current model into the session workspace.

Parameters

accessToken	Bearer token associated with a session.
filename	Safe basename to use for persistence.

Returns

Persistence output and error state.

17.884.3.14 stepSimulation()

```
SimulatorSessionService::SimulationActionResult SimulatorSessionService::stepSimulation (
            const std::string & accessToken )
```

Executes one synchronous simulation step.
Advances the current simulation by one step.

Parameters

accessToken	Bearer token associated with a session.

Returns

Structured action result and updated simulation status.

17.884.3.15 tryCreateModel()

```
bool SimulatorSessionService::tryCreateModel (
            const std::string & accessToken,
            ModelInfoResult & outInfo )
```

Creates a new model in the token-scoped simulator session.
Creates a new kernel model in the current session.

Parameters

accessToken	Bearer token associated with a session.
outInfo	Receives model information on success.

Returns

True when the model was created and output was populated.

17.884.3.16 tryGetCurrentModelInfo()

```
bool SimulatorSessionService::tryGetCurrentModelInfo (
            const std::string & accessToken,
            ModelInfoResult & outInfo )
```

Returns metadata for the current model in a session.
Returns metadata for the current model if one exists.

Parameters

accessToken	Bearer token associated with a session.
outInfo	Receives current model data.

Returns

True when token is valid and query completed.

17.884.3.17 tryGetSimulatorInfo()

```
bool SimulatorSessionService::tryGetSimulatorInfo (
            const std::string & accessToken,
            SimulatorInfoResult & outInfo )
```

Fetches simulator metadata using a session token.
Resolves the simulator metadata visible to a session token.

Parameters

accessToken	Bearer token associated with a session.
outInfo	Receives simulator identity information on success.

Returns

True when token is valid and metadata was populated.

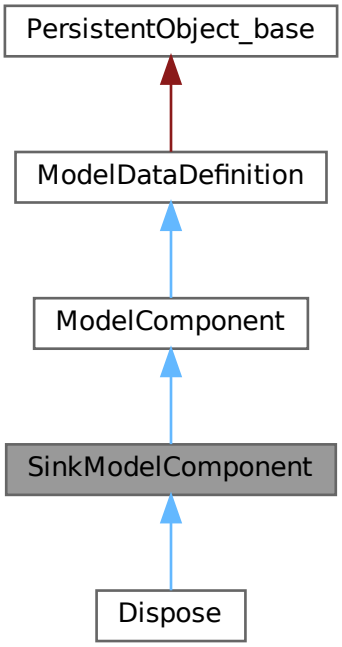
The documentation for this class was generated from the following files:

- applications/worker/service/SimulatorSessionService.h
- applications/worker/service/SimulatorSessionService.cpp

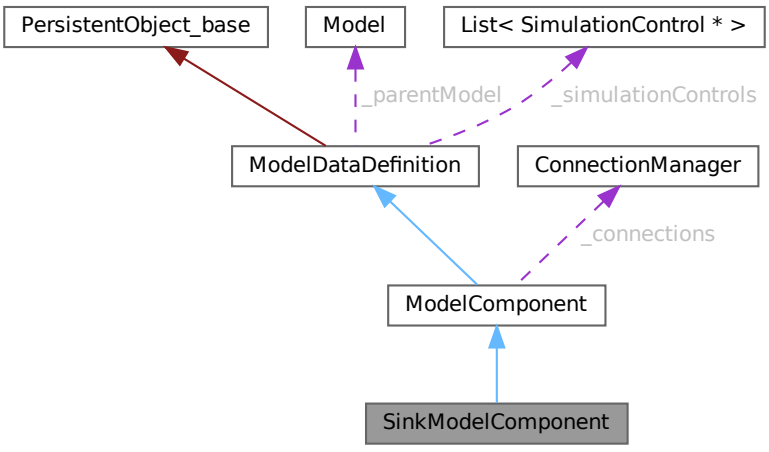
17.885 SinkModelComponent Class Reference

```
#include <kernel/simulator/SinkModelComponent.h>
```

Inheritance diagram for SinkModelComponent:



Collaboration diagram for SinkModelComponent:



Public Member Functions

- **SinkModelComponent** (Model *model, std::string componentTypename, std::string name="")

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelComponent

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber)=0

  Handles a dispatched event for this component type.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.885.1 Detailed Description

This class is the basis for any component representing the end of a process flow, such as a Dispose. It can remove entities from the system and collect statistics.

17.885.2 Member Function Documentation

17.885.2.1 _check()

```
bool SinkModelComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.
Reimplemented in Dispose.

17.885.2.2 _loadInstance()

```
bool SinkModelComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.
Reimplemented in Dispose.

17.885.2.3 _saveInstance()

```
void SinkModelComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.
Reimplemented in Dispose.
The documentation for this class was generated from the following files:

- kernel/simulator/SinkModelComponent.h
- kernel/simulator/SinkModelComponent.cpp

17.886 yy::genesyspp_parser::stack< T, S >::slice Class Reference

Present a slice of the top of a stack.
`#include <parser/GenesysParser.h>`

Public Member Functions

- **slice** (const stack &stack, index_type range) YY_NOEXCEPT
- const T & **operator[ ]** (index_type i) const

17.886.1 Detailed Description

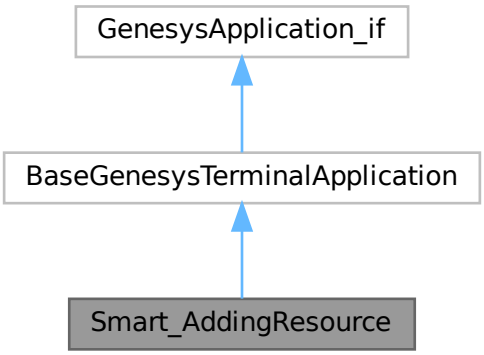
template<typename T, typename S = std::vector<T>>
class yy::genesyspp_parser::stack< T, S >::slice

Present a slice of the top of a stack.
The documentation for this class was generated from the following file:

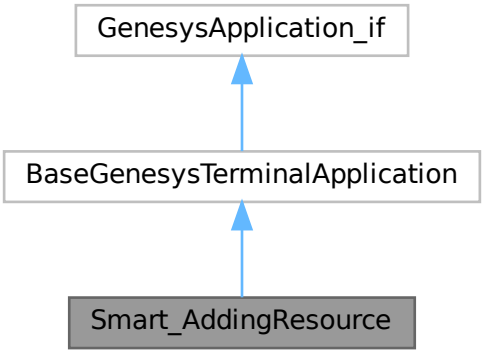
- parser/GenesysParser.h

17.887 Smart_AddingResource Class Reference

Inheritance diagram for Smart_AddingResource:



Collaboration diagram for Smart_AddingResource:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.887.1 Member Function Documentation

17.887.1.1 main()

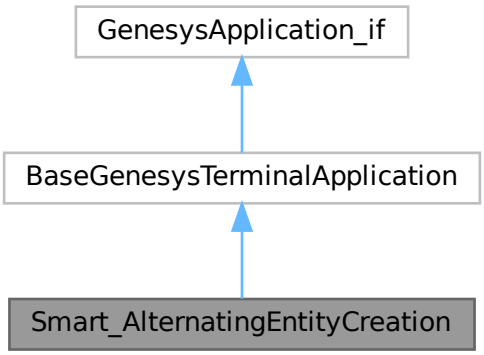
```
int Smart_AddingResource::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

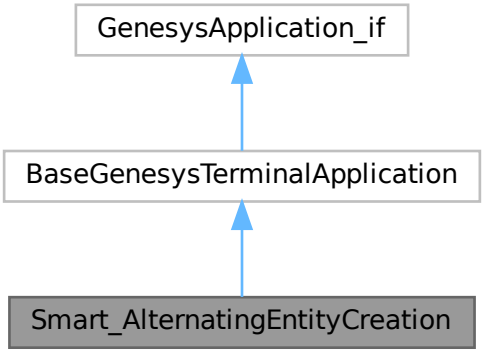
- applications/modelSpecific/arenaSmarts/Smart_AddingResource.h
- applications/modelSpecific/arenaSmarts/Smart_AddingResource.cpp

17.888 Smart_AlternatingEntityCreation Class Reference

Inheritance diagram for Smart_AlternatingEntityCreation:



Collaboration diagram for Smart_AlternatingEntityCreation:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.888.1 Member Function Documentation

17.888.1.1 main()

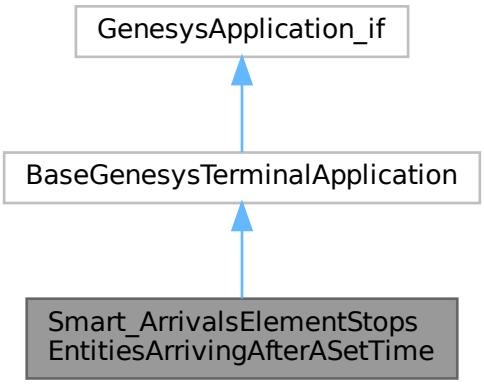
```
int Smart_AlternatingEntityCreation::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

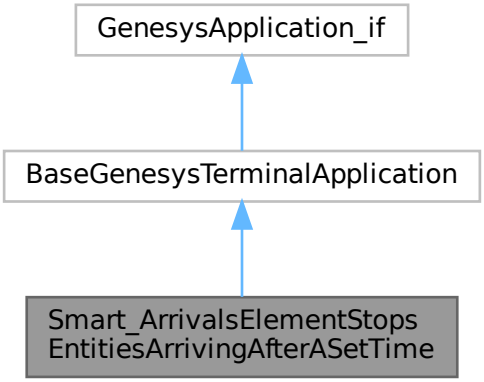
- applications/modelSpecific/arenaSmarts/Smart_AlternatingEntityCreation.h
- applications/modelSpecific/arenaSmarts/Smart_AlternatingEntityCreation.cpp

17.889 Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime Class Reference

Inheritance diagram for Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime:



Collaboration diagram for Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.889.1 Member Function Documentation

17.889.1.1 main()

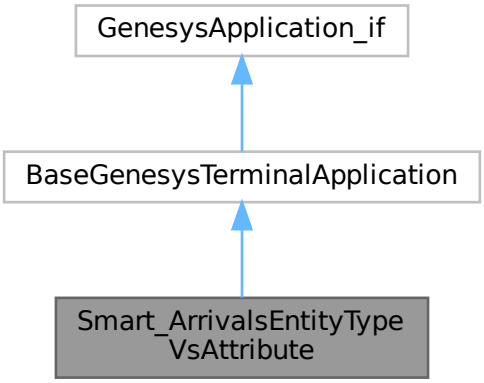
```
int Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

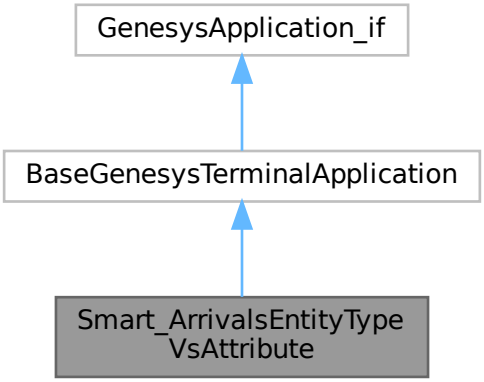
- applications/modelSpecific/arenaSmarts/Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime.h
- applications/modelSpecific/arenaSmarts/Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime.cpp

17.890 Smart_ArrivalsEntityTypeVsAttribute Class Reference

Inheritance diagram for Smart_ArrivalsEntityTypeVsAttribute:



Collaboration diagram for Smart_ArrivalsEntityTypeVsAttribute:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.890.1 Member Function Documentation

17.890.1.1 main()

```
int Smart_ArrivalsEntityTypeVsAttribute::main (
            int argc,
            char ** argv )  [override], [virtual]
```

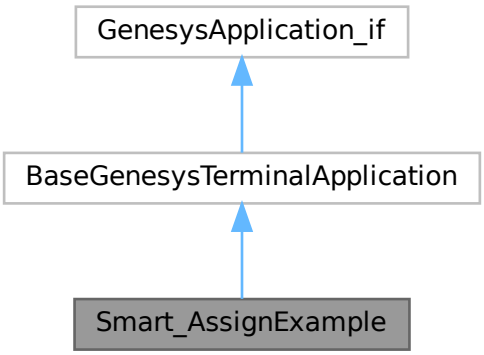
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

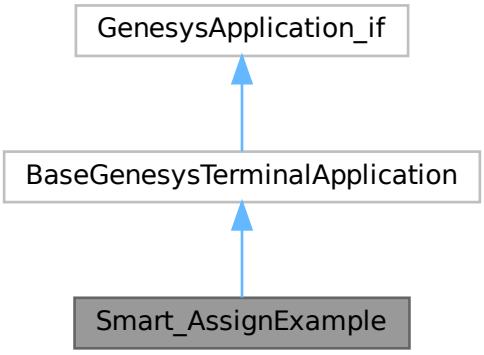
- applications/modelSpecific/arenaSmarts/Smart_ArrivalsEntityTypeVsAttribute.h
- applications/modelSpecific/arenaSmarts/Smart_ArrivalsEntityTypeVsAttribute.cpp

17.891 Smart_AssignExample Class Reference

Inheritance diagram for Smart_AssignExample:



Collaboration diagram for Smart_AssignExample:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.891.1 Member Function Documentation

17.891.1.1 main()

```
int Smart_AssignExample::main (
            int argc,
            char ** argv )  [override], [virtual]
```

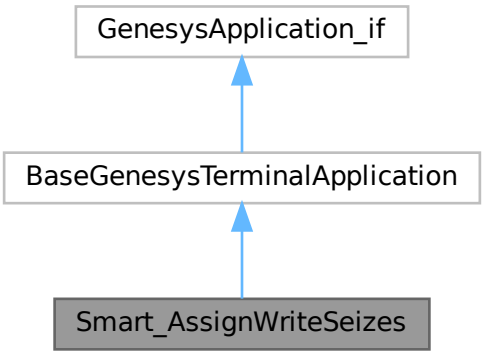
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

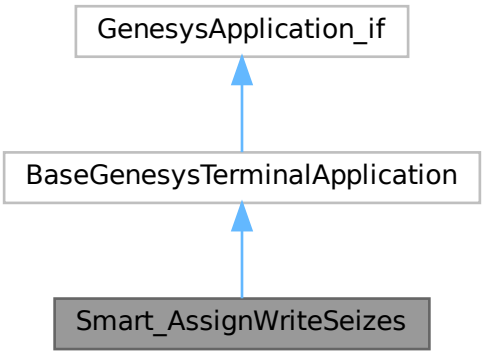
- applications/modelSpecific/arenaSmarts/Smart_AssignExample.h
- applications/modelSpecific/arenaSmarts/Smart_AssignExample.cpp

17.892 Smart_AssignWriteSeizes Class Reference

Inheritance diagram for Smart_AssignWriteSeizes:



Collaboration diagram for Smart_AssignWriteSeizes:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.892.1 Member Function Documentation

17.892.1.1 main()

```
int Smart_AssignWriteSeizes::main (
            int argc,
            char ** argv )  [override], [virtual]
```

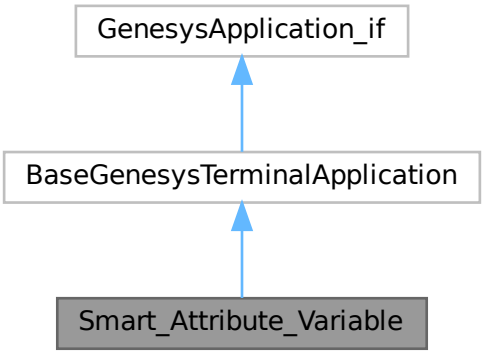
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

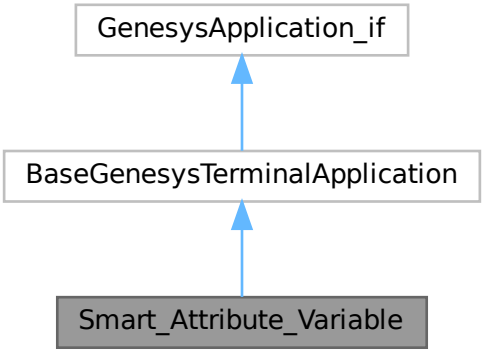
- applications/modelSpecific/smarts/Logic/Smart_AssignWriteSeizes.h
- applications/modelSpecific/smarts/Logic/Smart_AssignWriteSeizes.cpp

17.893 Smart_Attribute_Variable Class Reference

Inheritance diagram for Smart_Attribute_Variable:



Collaboration diagram for Smart_Attribute_Variable:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.893.1 Member Function Documentation

17.893.1.1 main()

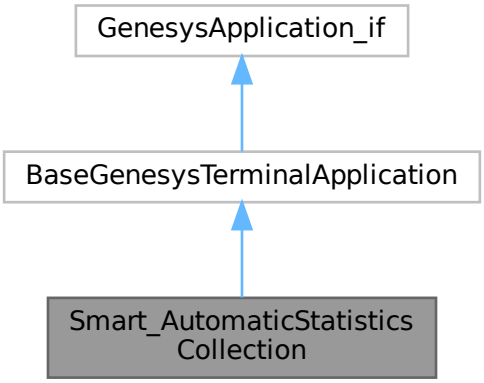
```
int Smart_Attribute_Variable::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

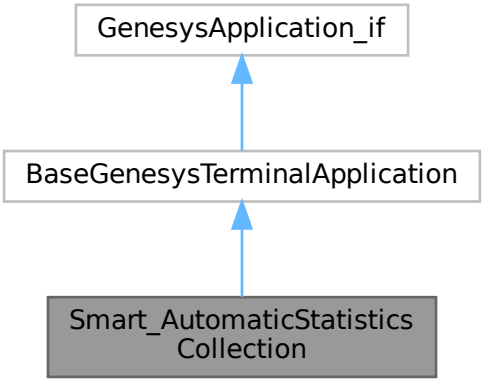
- applications/modelSpecific/smarts/Logic/Smart_Attribute_Variable.h
- applications/modelSpecific/smarts/Logic/Smart_Attribute_Variable.cpp

17.894 Smart_AutomaticStatisticsCollection Class Reference

Inheritance diagram for Smart_AutomaticStatisticsCollection:



Collaboration diagram for Smart_AutomaticStatisticsCollection:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.894.1 Member Function Documentation

17.894.1.1 main()

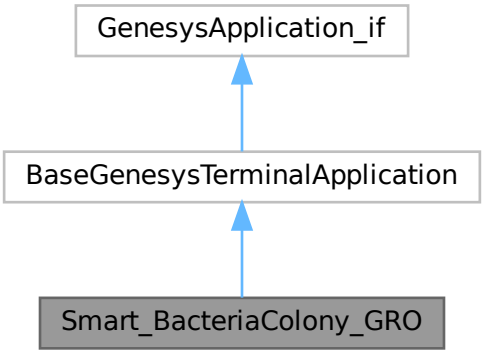
```
int Smart_AutomaticStatisticsCollection::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

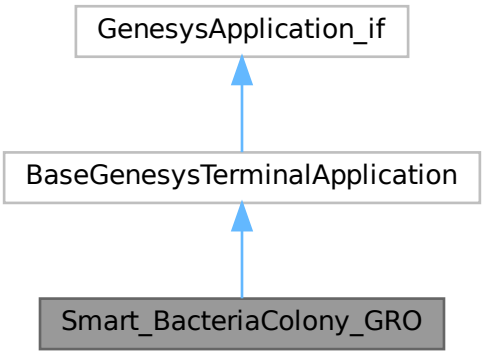
- applications/modelSpecific/arenaSmarts/Smart_AutomaticStatisticsCollection.h
- applications/modelSpecific/arenaSmarts/Smart_AutomaticStatisticsCollection.cpp

17.895 Smart_BacteriaColony_GRO Class Reference

Inheritance diagram for Smart_BacteriaColony_GRO:



Collaboration diagram for Smart_BacteriaColony_GRO:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.895.1 Member Function Documentation

17.895.1.1 main()

```
int Smart_BacteriaColony_GRO::main (
            int argc,
            char ** argv )  [override], [virtual]
```

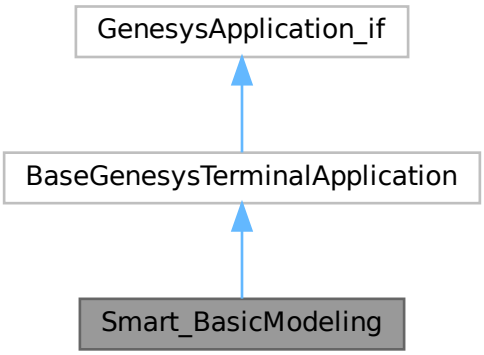
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

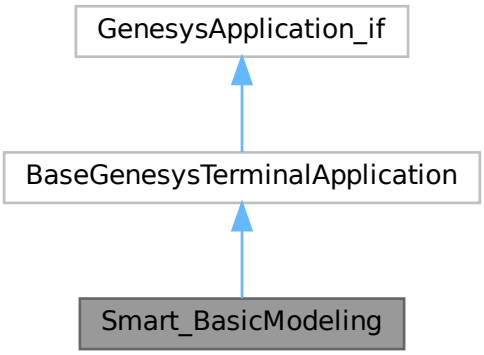
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BacteriaColony_GRO.h
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BacteriaColony_GRO.cpp

17.896 Smart_BasicModeling Class Reference

Inheritance diagram for Smart_BasicModeling:



Collaboration diagram for Smart_BasicModeling:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.896.1 Member Function Documentation

17.896.1.1 main()

```
int Smart_BasicModeling::main (
            int argc,
            char ** argv )  [override], [virtual]
```

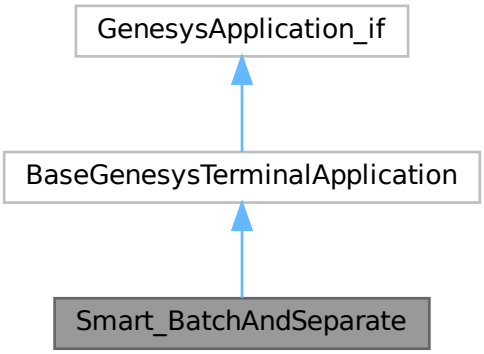
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

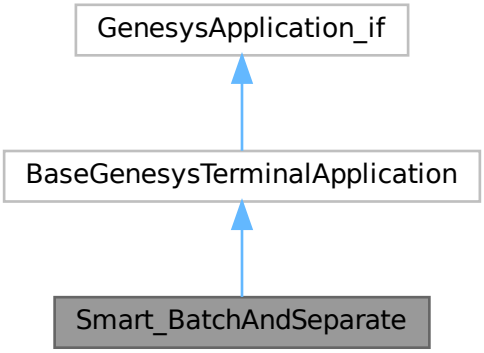
- applications/modelSpecific/arenaSmarts/Smart_BasicModeling.h
- applications/modelSpecific/arenaSmarts/Smart_BasicModeling.cpp

17.897 Smart_BatchAndSeparate Class Reference

Inheritance diagram for Smart_BatchAndSeparate:



Collaboration diagram for Smart_BatchAndSeparate:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.897.1 Member Function Documentation

17.897.1.1 main()

```
int Smart_BatchAndSeparate::main (
            int argc,
            char ** argv )  [override], [virtual]
```

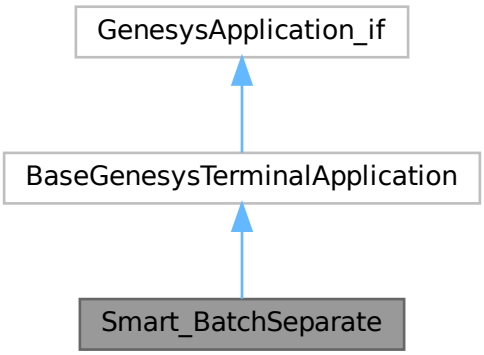
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

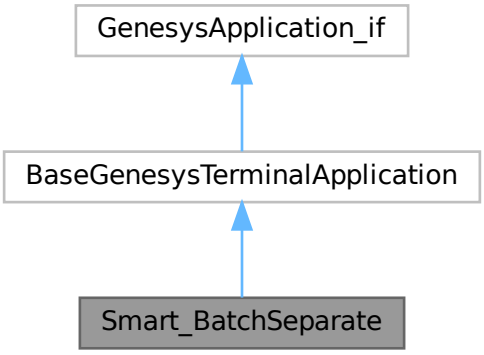
- applications/modelSpecific/arenaSmarts/Smart_BatchAndSeparate.h
- applications/modelSpecific/arenaSmarts/Smart_BatchAndSeparate.cpp

17.898 Smart_BatchSeparate Class Reference

Inheritance diagram for Smart_BatchSeparate:



Collaboration diagram for Smart_BatchSeparate:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.898.1 Member Function Documentation

17.898.1.1 main()

```
int Smart_BatchSeparate::main (
            int argc,
            char ** argv )  [override], [virtual]
```

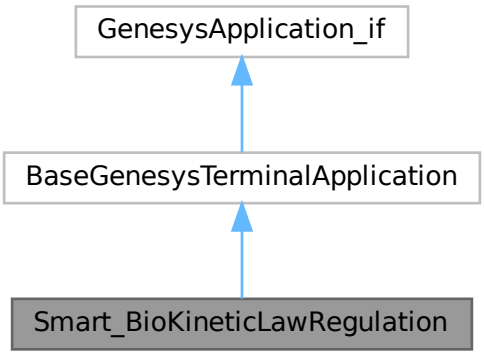
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

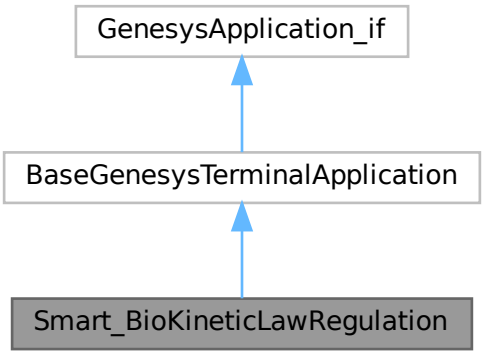
- applications/modelSpecific/smarts/Grouping/Smart_BatchSeparate.h
- applications/modelSpecific/smarts/Grouping/Smart_BatchSeparate.cpp

17.899 Smart_BioKineticLawRegulation Class Reference

Inheritance diagram for Smart_BioKineticLawRegulation:



Collaboration diagram for Smart_BioKineticLawRegulation:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.899.1 Member Function Documentation

17.899.1.1 main()

```
int Smart_BioKineticLawRegulation::main (
            int argc,
            char ** argv )  [override], [virtual]
```

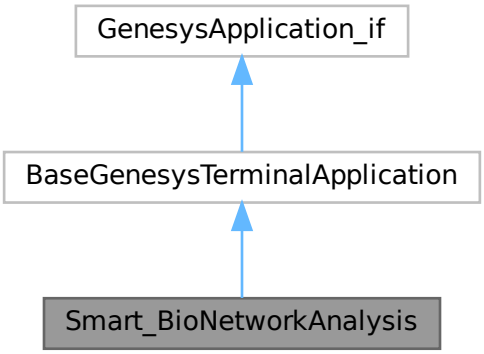
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

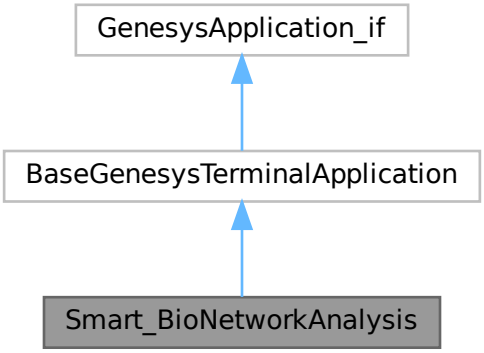
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioKineticLawRegulation.h
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioKineticLawRegulation.cpp

17.900 Smart_BioNetworkAnalysis Class Reference

Inheritance diagram for Smart_BioNetworkAnalysis:



Collaboration diagram for Smart_BioNetworkAnalysis:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.900.1 Member Function Documentation

17.900.1.1 main()

```
int Smart_BioNetworkAnalysis::main (
            int argc,
            char ** argv )  [override], [virtual]
```

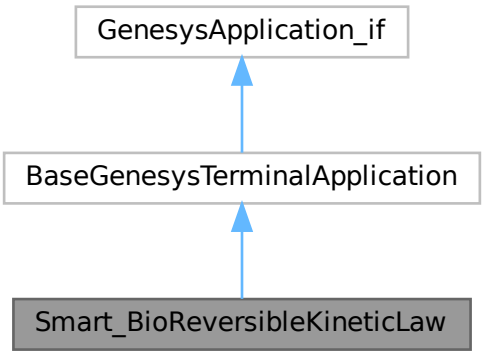
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

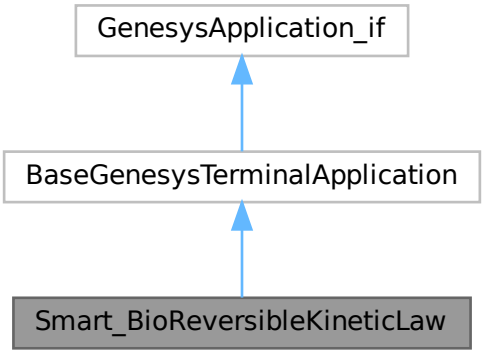
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioNetworkAnalysis.h
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioNetworkAnalysis.cpp

17.901 Smart_BioReversibleKineticLaw Class Reference

Inheritance diagram for Smart_BioReversibleKineticLaw:



Collaboration diagram for Smart_BioReversibleKineticLaw:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.901.1 Member Function Documentation

17.901.1.1 main()

```
int Smart_BioReversibleKineticLaw::main (
            int argc,
            char ** argv )  [override], [virtual]
```

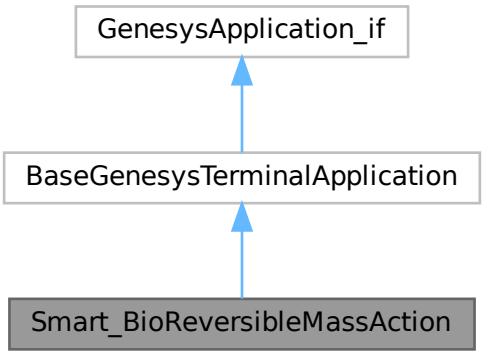
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

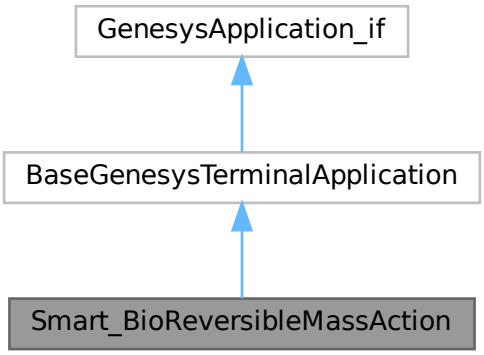
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioReversibleKineticLaw.h
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioReversibleKineticLaw.cpp

17.902 Smart_BioReversibleMassAction Class Reference

Inheritance diagram for Smart_BioReversibleMassAction:



Collaboration diagram for Smart_BioReversibleMassAction:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.902.1 Member Function Documentation

17.902.1.1 main()

```
int Smart_BioReversibleMassAction::main (
            int argc,
            char ** argv )  [override], [virtual]
```

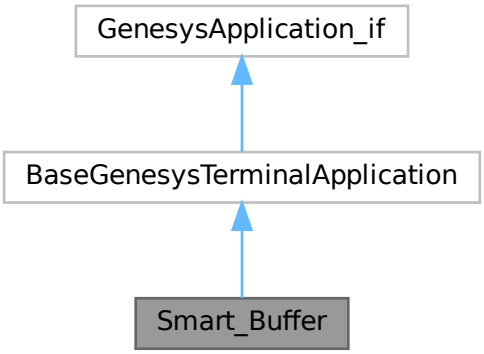
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

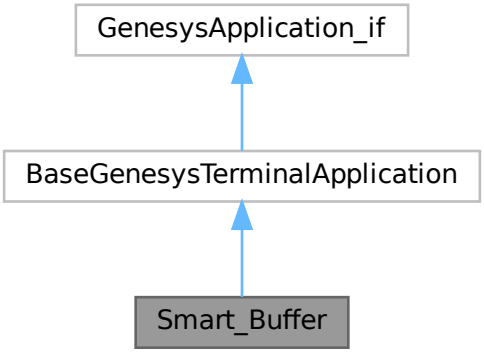
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioReversibleMassAction.h
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioReversibleMassAction.cpp

17.903 Smart_Buffer Class Reference

Inheritance diagram for Smart_Buffer:



Collaboration diagram for Smart_Buffer:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.903.1 Member Function Documentation

17.903.1.1 main()

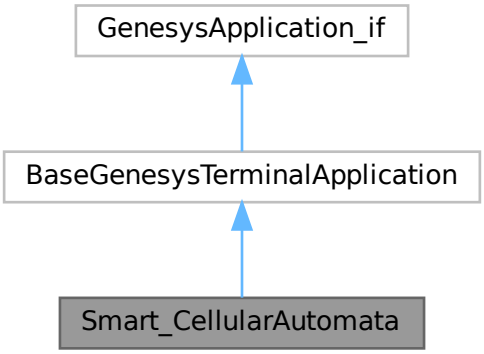
```
int Smart_Buffer::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

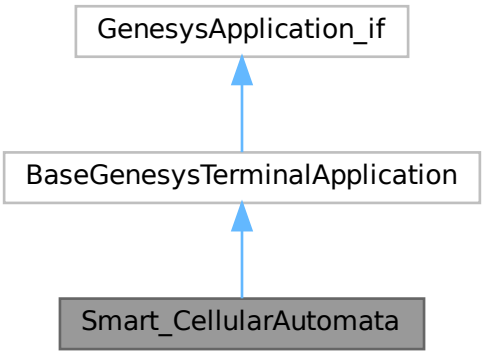
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_Buffer.h
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_Buffer.cpp

17.904 Smart_CellularAutomata Class Reference

Inheritance diagram for Smart_CellularAutomata:



Collaboration diagram for Smart_CellularAutomata:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.904.1 Member Function Documentation

17.904.1.1 main()

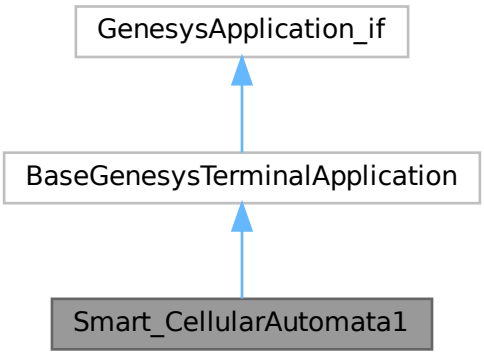
```
int Smart_CellularAutomata::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

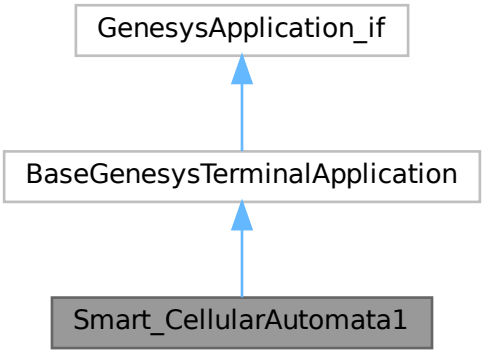
- applications/modelSpecific/smarts/ModalModel/Smart_CellularAutomata.h
- applications/modelSpecific/smarts/ModalModel/Smart_CellularAutomata.cpp

17.905 Smart_CellularAutomata1 Class Reference

Inheritance diagram for Smart_CellularAutomata1:



Collaboration diagram for Smart_CellularAutomata1:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.905.1 Member Function Documentation

17.905.1.1 main()

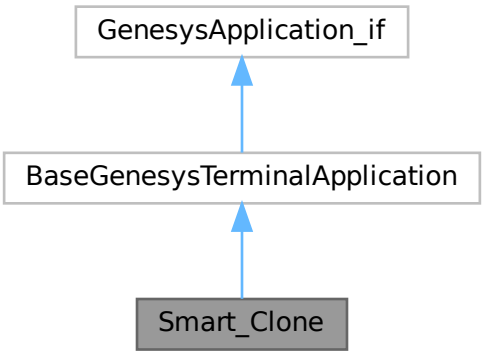
```
int Smart_CellularAutomata1::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

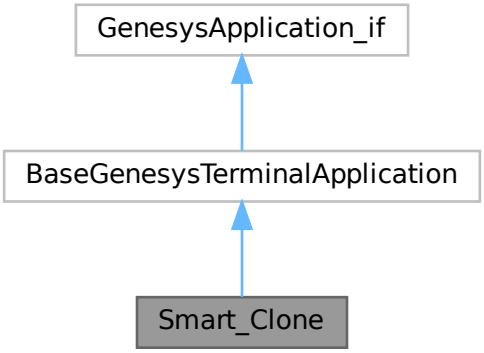
- applications/modelSpecific/smarts/ModalModel/Smart_CellularAutomata1.h
- applications/modelSpecific/smarts/ModalModel/Smart_CellularAutomata1.cpp

17.906 Smart_Clone Class Reference

Inheritance diagram for Smart_Clone:



Collaboration diagram for Smart_Clone:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.906.1 Member Function Documentation

17.906.1.1 main()

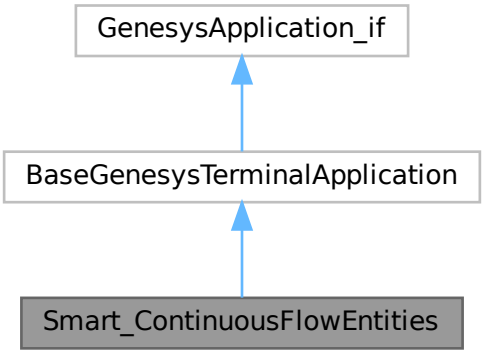
```
int Smart_Clone::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

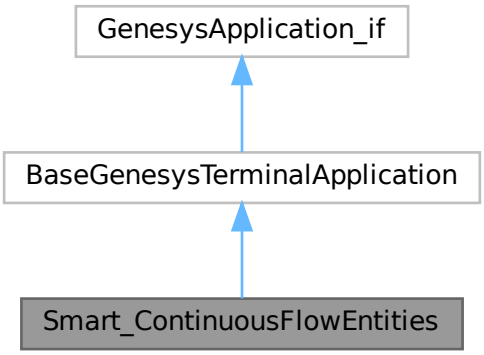
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_Clone.h
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_Clone.cpp

17.907 Smart_ContinuousFlowEntities Class Reference

Inheritance diagram for Smart_ContinuousFlowEntities:



Collaboration diagram for Smart_ContinuousFlowEntities:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.907.1 Member Function Documentation

17.907.1.1 main()

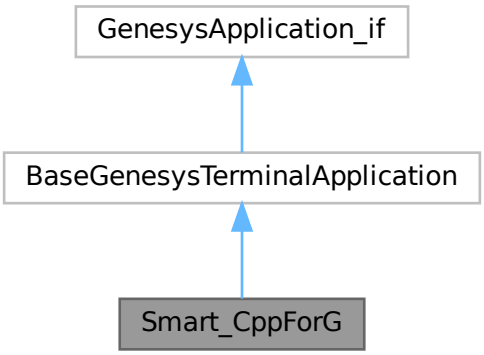
```
int Smart_ContinuousFlowEntities::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

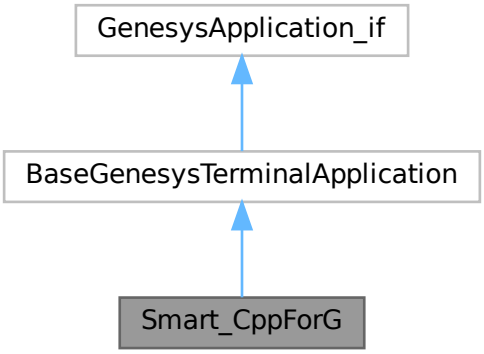
- applications/modelSpecific/arenaSmarts/Smart_ContinuousFlowEntities.h
- applications/modelSpecific/arenaSmarts/Smart_ContinuousFlowEntities.cpp

17.908 Smart_CppForG Class Reference

Inheritance diagram for Smart_CppForG:



Collaboration diagram for Smart_CppForG:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.908.1 Member Function Documentation

17.908.1.1 main()

```
int Smart_CppForG::main (
            int argc,
            char ** argv )  [override], [virtual]
```

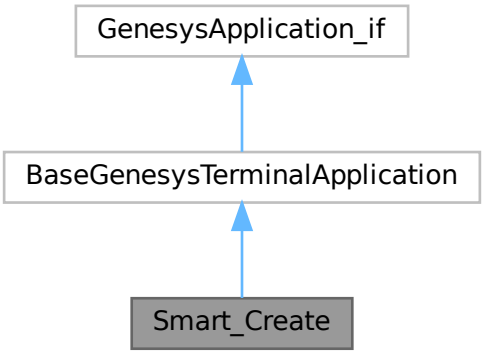
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

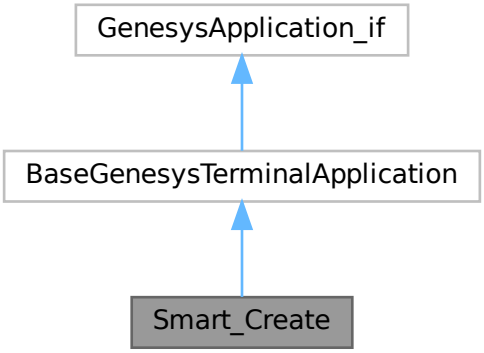
- applications/modelSpecific/smarts/ExternalIntegration/Smart_CppForG.h
- applications/modelSpecific/smarts/ExternalIntegration/Smart_CppForG.cpp

17.909 Smart_Create Class Reference

Inheritance diagram for Smart_Create:



Collaboration diagram for Smart_Create:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.909.1 Member Function Documentation

17.909.1.1 main()

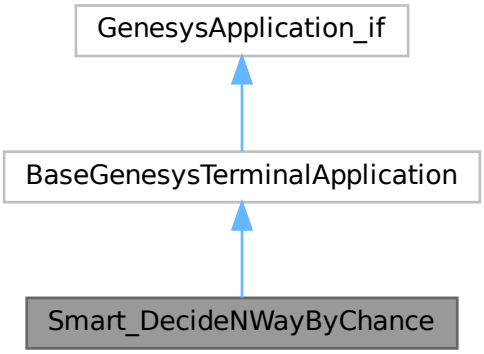
```
int Smart_Create::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

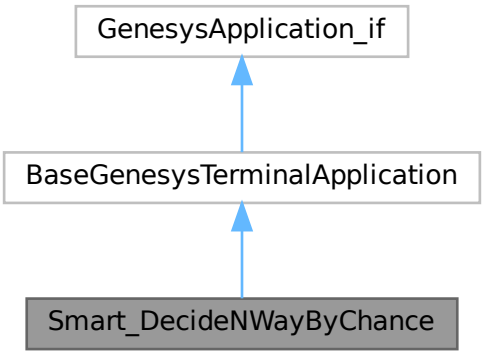
- applications/modelSpecific/arenaSmarts/Smart_Create.h
- applications/modelSpecific/arenaSmarts/Smart_Create.cpp

17.910 Smart_DecideNWayByChance Class Reference

Inheritance diagram for Smart_DecideNWayByChance:



Collaboration diagram for Smart_DecideNWayByChance:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.910.1 Member Function Documentation

17.910.1.1 main()

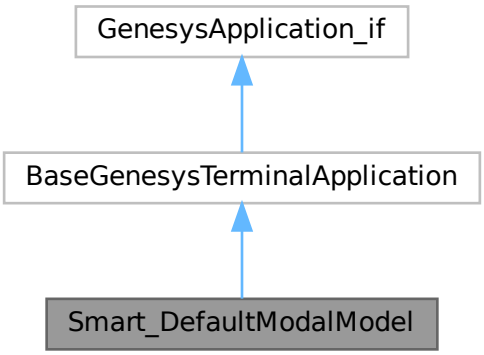
```
int Smart_DecideNWayByChance::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It create a new simulator instance, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

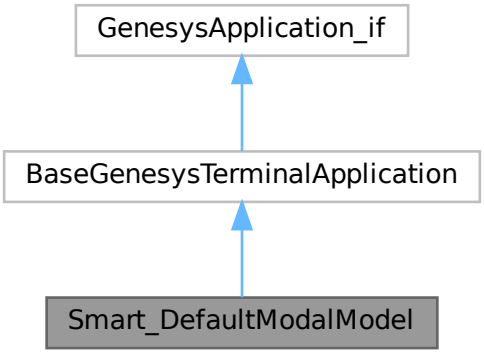
- applications/modelSpecific/arenaSmarts/Smart_DecideNWayByChance.h
- applications/modelSpecific/arenaSmarts/Smart_DecideNWayByChance.cpp

17.911 Smart_DefaultModalModel Class Reference

Inheritance diagram for Smart_DefaultModalModel:



Collaboration diagram for Smart_DefaultModalModel:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.911.1 Member Function Documentation

17.911.1.1 main()

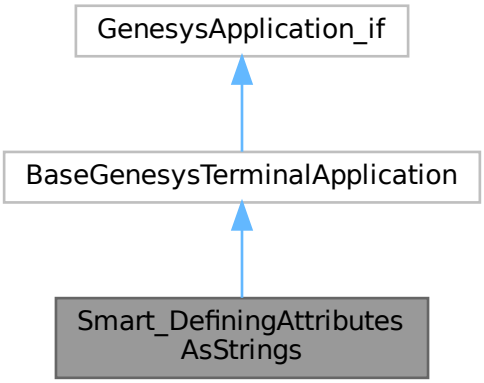
```
int Smart_DefaultModalModel::main (
            int argc,
            char ** argv )  [override], [virtual]
```

Demonstrates ModalModelDefault attached to an EFSMNetwork.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

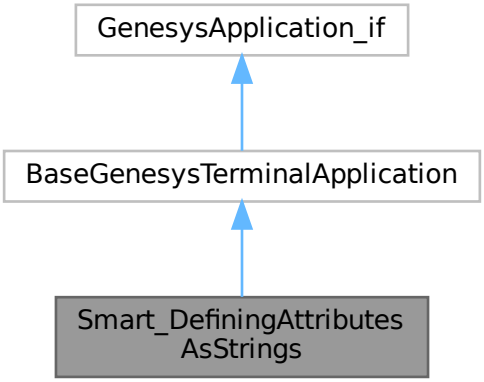
- applications/modelSpecific/smarts/ModalModel/Smart_DefaultModalModel.h
- applications/modelSpecific/smarts/ModalModel/Smaty_DefaultModalModel.cpp

17.912 Smart_DefiningAttributesAsStrings Class Reference

Inheritance diagram for Smart_DefiningAttributesAsStrings:



Collaboration diagram for Smart_DefiningAttributesAsStrings:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.912.1 Member Function Documentation

17.912.1.1 main()

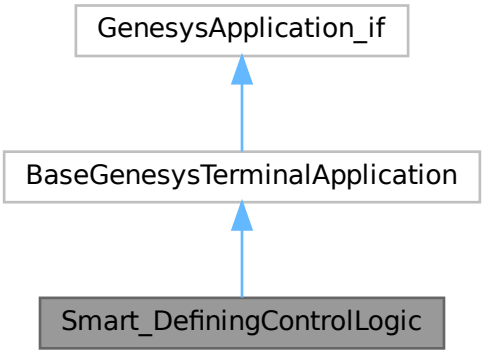
```
int Smart_DefiningAttributesAsStrings::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

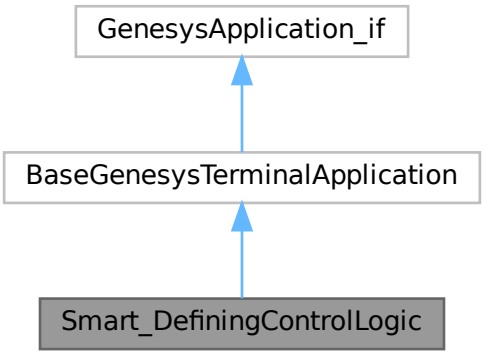
- applications/modelSpecific/arenaSmarts/Smart_DefiningAttributesAsStrings.h
- applications/modelSpecific/arenaSmarts/Smart_DefiningAttributesAsStrings.cpp

17.913 Smart_DefiningControlLogic Class Reference

Inheritance diagram for Smart_DefiningControlLogic:



Collaboration diagram for Smart_DefiningControlLogic:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.913.1 Member Function Documentation

17.913.1.1 main()

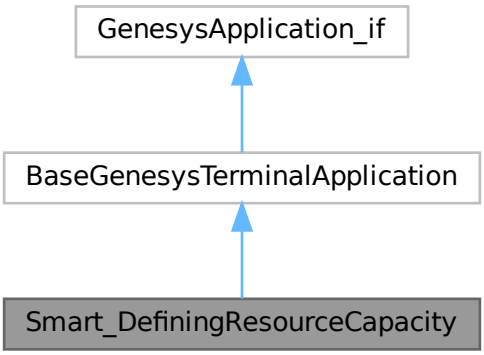
```
int Smart_DefiningControlLogic::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

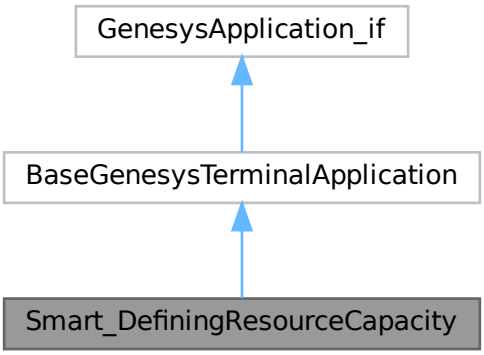
- applications/modelSpecific/arenaSmarts/Smart_DefiningControlLogic.h
- applications/modelSpecific/arenaSmarts/Smart_DefiningControlLogic.cpp

17.914 Smart_DefiningResourceCapacity Class Reference

Inheritance diagram for Smart_DefiningResourceCapacity:



Collaboration diagram for Smart_DefiningResourceCapacity:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.914.1 Member Function Documentation

17.914.1.1 main()

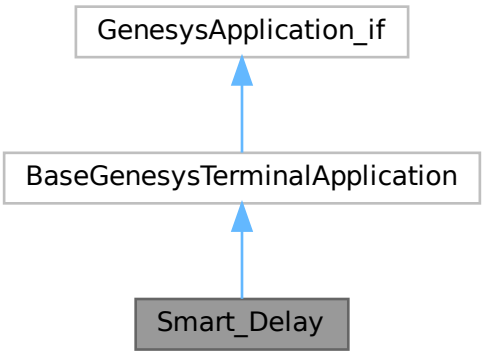
```
int Smart_DefiningResourceCapacity::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

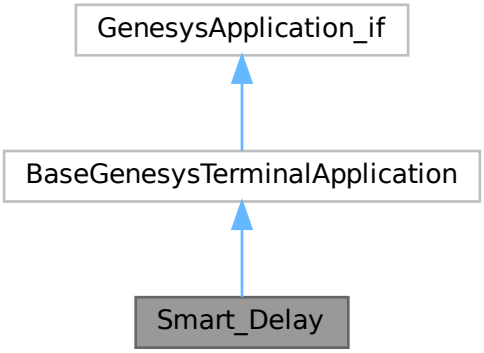
- applications/modelSpecific/arenaSmarts/Smart_DefiningResourceCapacity.h
- applications/modelSpecific/arenaSmarts/Smart_DefiningResourceCapacity.cpp

17.915 Smart_Delay Class Reference

Inheritance diagram for Smart_Delay:



Collaboration diagram for Smart_Delay:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.915.1 Member Function Documentation

17.915.1.1 main()

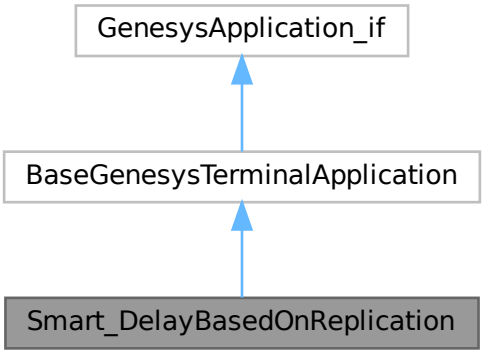
```
int Smart_Delay::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

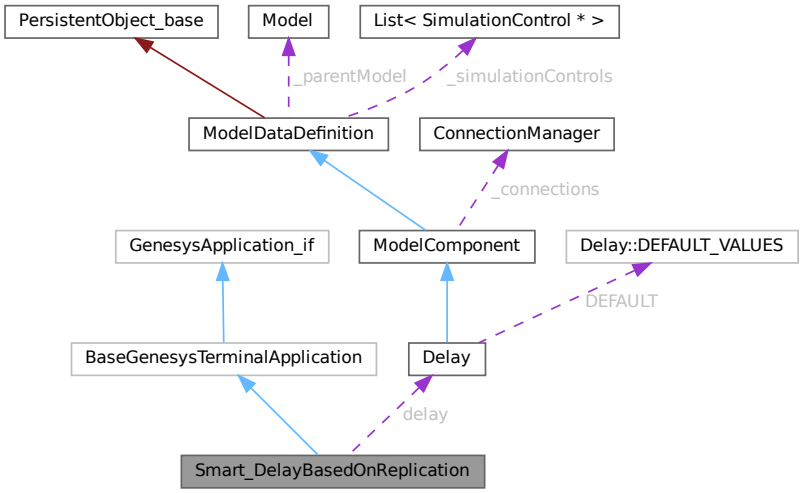
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_Delay.h
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_Delay.cpp

17.916 Smart_DelayBasedOnReplication Class Reference

Inheritance diagram for Smart_DelayBasedOnReplication:



Collaboration diagram for Smart_DelayBasedOnReplication:



Public Member Functions

- void TestEventHandler (SimulationEvent *re)
- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Public Attributes

- Delay * **delay**
- int * **proctimes**

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)

- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.916.1 Member Function Documentation

17.916.1.1 main()

```
int Smart_DelayBasedOnReplication::main (
            int argc,
            char ** argv )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.916.1.2 TestEventHandler()

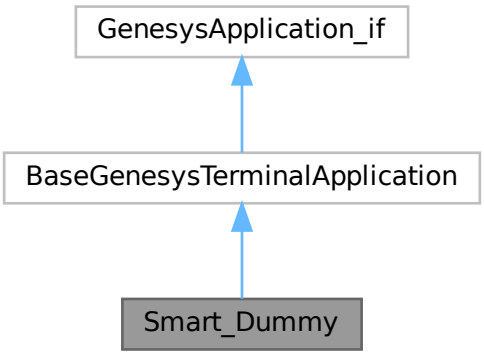
```
void Smart_DelayBasedOnReplication::TestEventHandler (
            SimulationEvent * re )
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
The documentation for this class was generated from the following files:

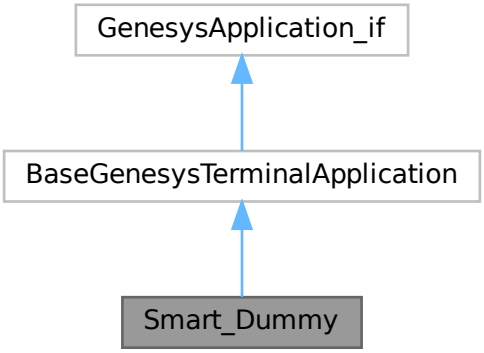
- applications/modelSpecific/arenaSmarts/Smart_DelayBasedOnReplication.h
- applications/modelSpecific/arenaSmarts/Smart_DelayBasedOnReplication.cpp

17.917 Smart_Dummy Class Reference

Inheritance diagram for Smart_Dummy:



Collaboration diagram for Smart_Dummy:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.917.1 Member Function Documentation

17.917.1.1 main()

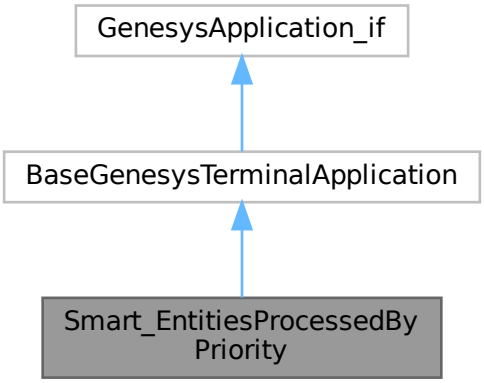
```
int Smart_Dummy::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

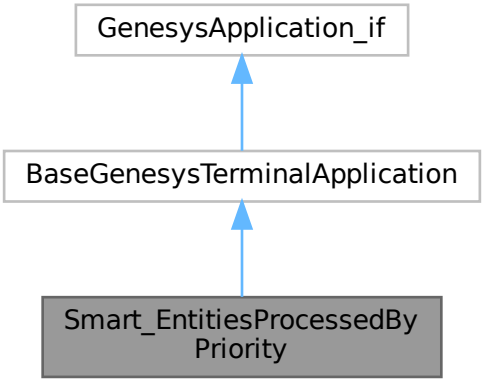
- applications/modelSpecific/smarts/ModalModel/Smart_Dummy.h
- applications/modelSpecific/smarts/ModalModel/Smart_Dummy.cpp

17.918 Smart_EntitiesProcessedByPriority Class Reference

Inheritance diagram for Smart_EntitiesProcessedByPriority:



Collaboration diagram for Smart_EntitiesProcessedByPriority:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.918.1 Member Function Documentation

17.918.1.1 main()

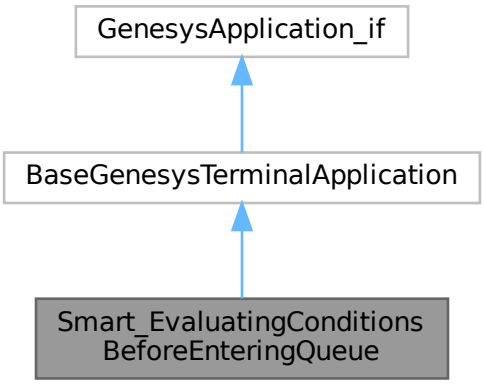
```
int Smart_EntitiesProcessedByPriority::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

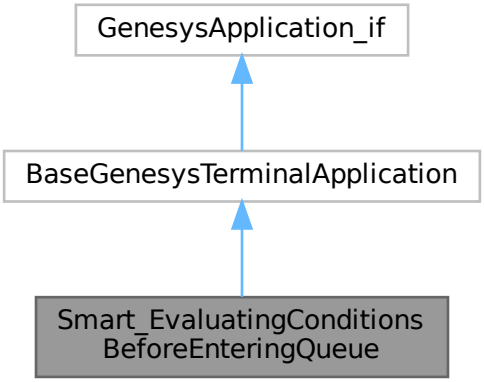
- applications/modelSpecific/arenaSmarts/Smart_EntitiesProcessedByPriority.h
- applications/modelSpecific/arenaSmarts/Smart_EntitiesProcessedByPriority.cpp

17.919 Smart_EvaluatingConditionsBeforeEnteringQueue Class Reference

Inheritance diagram for Smart_EvaluatingConditionsBeforeEnteringQueue:



Collaboration diagram for Smart_EvaluatingConditionsBeforeEnteringQueue:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.919.1 Member Function Documentation

17.919.1.1 main()

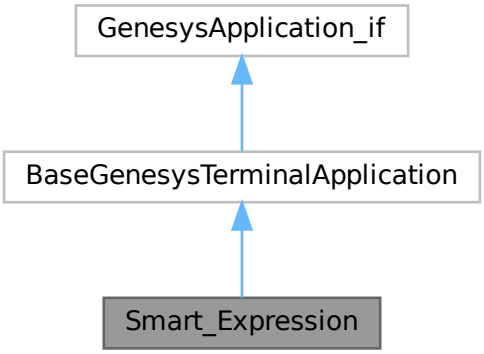
```
int Smart_EvaluatingConditionsBeforeEnteringQueue::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

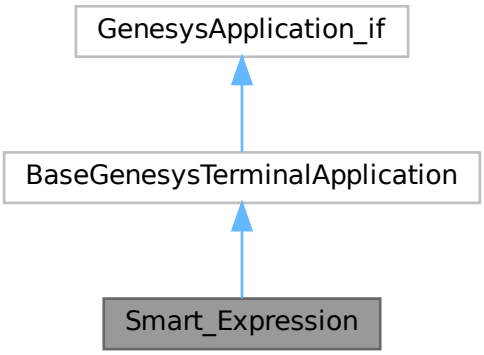
- applications/modelSpecific/arenaSmarts/Smart_EvaluatingConditionsBeforeEnteringQueue.h
- applications/modelSpecific/arenaSmarts/Smart_EvaluatingConditionsBeforeEnteringQueue.cpp

17.920 Smart_Expression Class Reference

Inheritance diagram for Smart_Expression:



Collaboration diagram for Smart_Expression:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.920.1 Member Function Documentation

17.920.1.1 main()

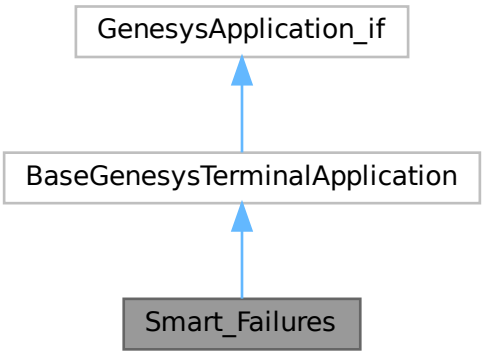
```
int Smart_Expression::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

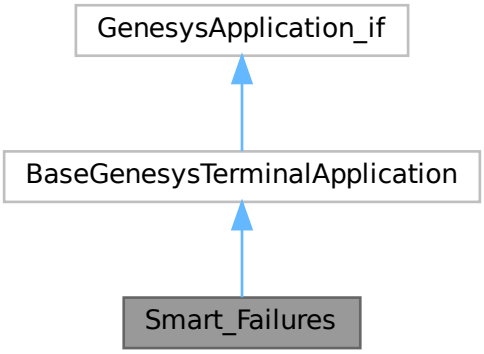
- applications/modelSpecific/arenaSmarts/Smart_Expression.h
- applications/modelSpecific/arenaSmarts/Smart_Expression.cpp

17.921 Smart_Failures Class Reference

Inheritance diagram for Smart_Failures:



Collaboration diagram for Smart_Failures:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.921.1 Member Function Documentation

17.921.1.1 main()

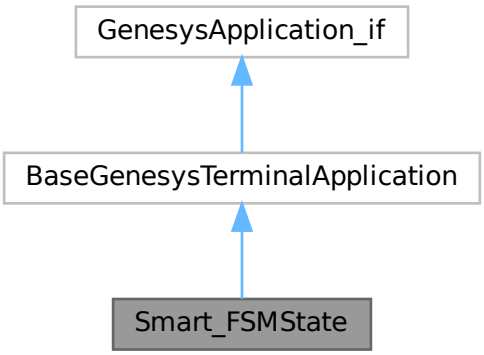
```
int Smart_Failures::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

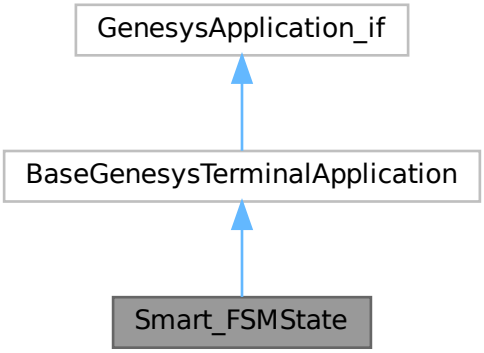
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_Failures.h
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_Failures.cpp

17.922 Smart_FSMState Class Reference

Inheritance diagram for Smart_FSMState:



Collaboration diagram for Smart_FSMState:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.922.1 Member Function Documentation

17.922.1.1 main()

```
int Smart_FSMState::main (
            int argc,
            char ** argv )  [override], [virtual]
```

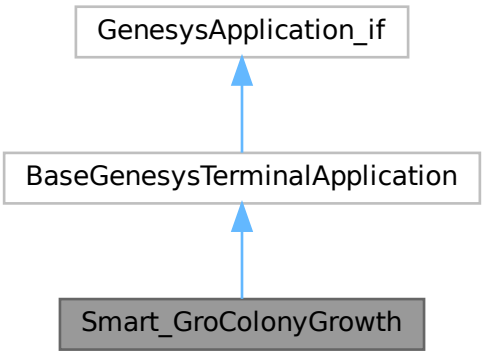
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

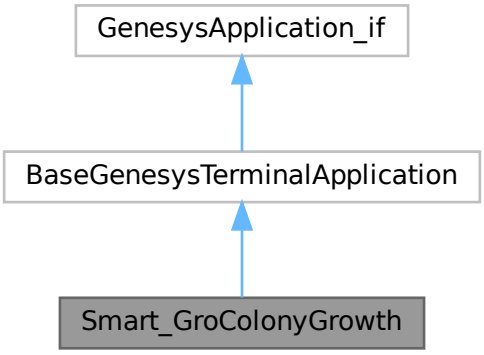
- applications/modelSpecific/smarts/ModalModel/Smart_FSMState.h
- applications/modelSpecific/smarts/ModalModel/Smart_FSMState.cpp

17.923 Smart_GroColonyGrowth Class Reference

Inheritance diagram for Smart_GroColonyGrowth:



Collaboration diagram for Smart_GroColonyGrowth:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.923.1 Member Function Documentation

17.923.1.1 main()

```
int Smart_GroColonyGrowth::main (
            int argc,
            char ** argv )  [override], [virtual]
```

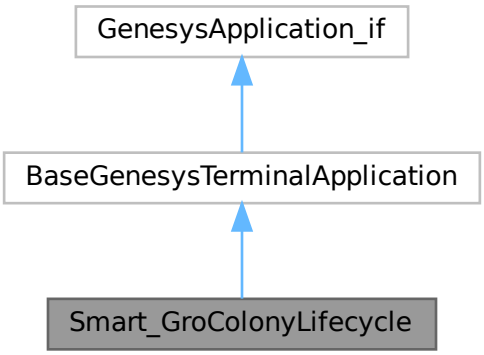
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

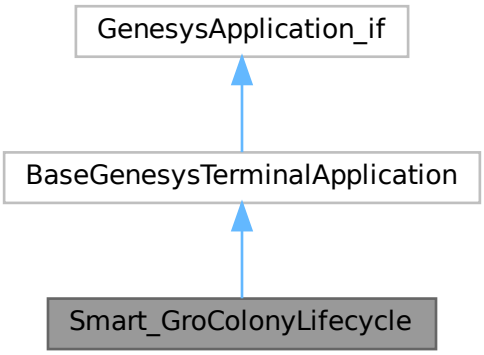
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_GroColonyGrowth.h
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_GroColonyGrowth.cpp

17.924 Smart_GroColonyLifecycle Class Reference

Inheritance diagram for Smart_GroColonyLifecycle:



Collaboration diagram for Smart_GroColonyLifecycle:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.924.1 Member Function Documentation

17.924.1.1 main()

```
int Smart_GroColonyLifecycle::main (
            int argc,
            char ** argv )  [override], [virtual]
```

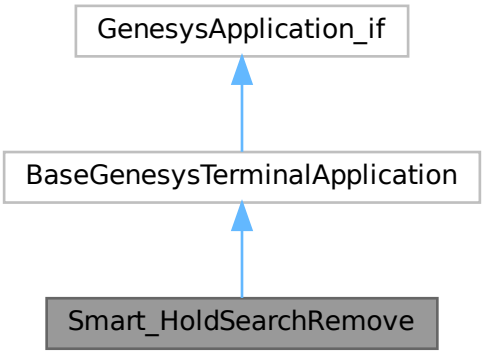
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

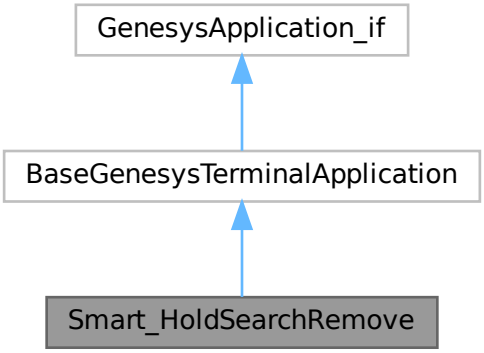
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_GroColonyLifecycle.h
- applications/modelSpecific/smarts/BiochemicalSimulation/Smart_GroColonyLifecycle.cpp

17.925 Smart_HoldSearchRemove Class Reference

Inheritance diagram for Smart_HoldSearchRemove:



Collaboration diagram for Smart_HoldSearchRemove:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.925.1 Member Function Documentation

17.925.1.1 main()

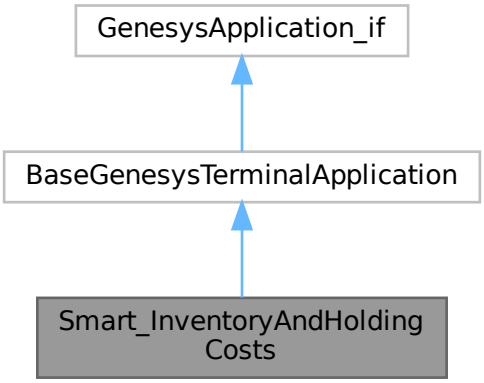
```
int Smart_HoldSearchRemove::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

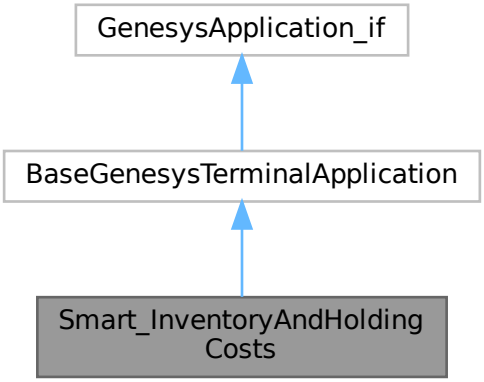
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_HoldSearchRemove.h
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_HoldSearchRemove.cpp

17.926 Smart_InventoryAndHoldingCosts Class Reference

Inheritance diagram for Smart_InventoryAndHoldingCosts:



Collaboration diagram for Smart_InventoryAndHoldingCosts:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.926.1 Member Function Documentation

17.926.1.1 main()

```
int Smart_InventoryAndHoldingCosts::main (
            int argc,
            char ** argv )  [override], [virtual]
```

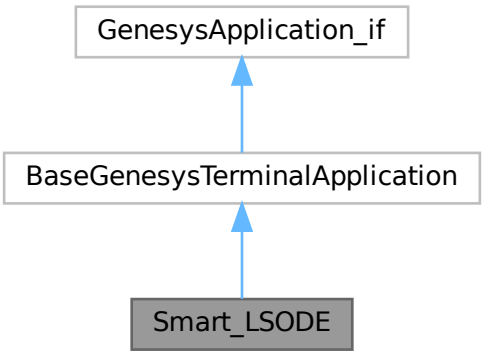
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

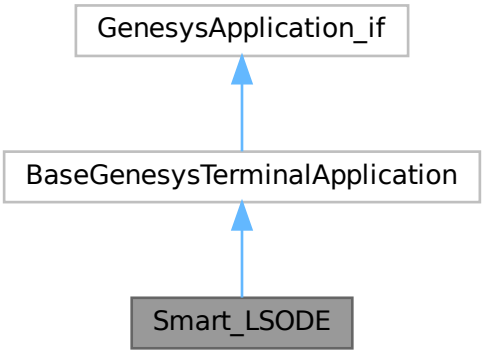
- applications/modelSpecific/arenaSmarts/Smart_InventoryAndHoldingCosts.h
- applications/modelSpecific/arenaSmarts/Smart_InventoryAndHoldingCosts.cpp

17.927 Smart_LSODE Class Reference

Inheritance diagram for Smart_LSODE:



Collaboration diagram for Smart_LSODE:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.927.1 Member Function Documentation

17.927.1.1 main()

```
int Smart_LSODE::main (
            int argc,
            char ** argv )  [override], [virtual]
```

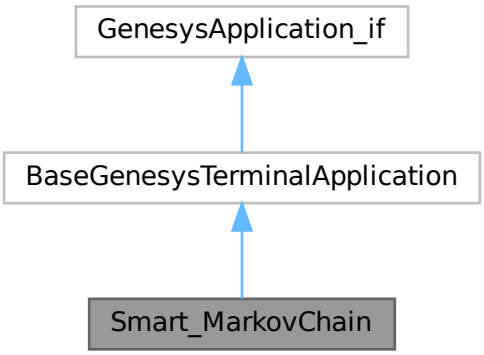
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

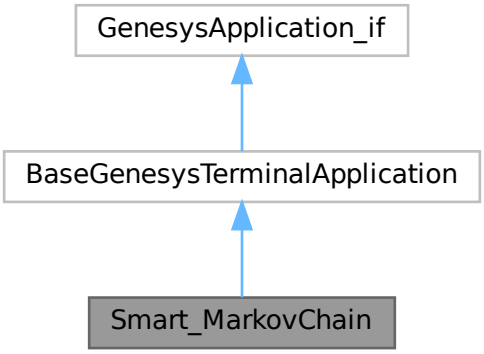
- applications/modelSpecific/smarts/Continuous/Smart_LSODE.h
- applications/modelSpecific/smarts/Continuous/Smart_LSODE.cpp

17.928 Smart_MarkovChain Class Reference

Inheritance diagram for Smart_MarkovChain:



Collaboration diagram for Smart_MarkovChain:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.928.1 Member Function Documentation

17.928.1.1 main()

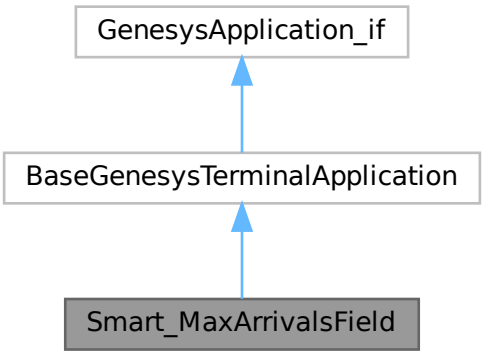
```
int Smart_MarkovChain::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

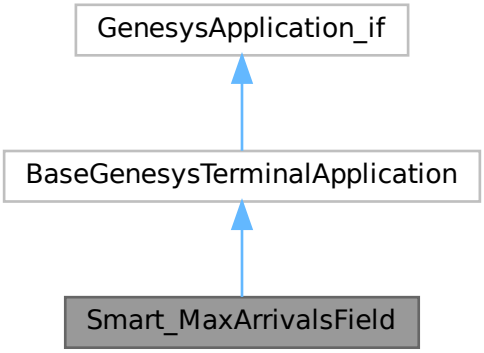
- applications/modelSpecific/smarts/AnalyticalModeling/Smart_MarkovChain.h
- applications/modelSpecific/smarts/AnalyticalModeling/Smart_MarkovChain.cpp

17.929 Smart_MaxArrivalsField Class Reference

Inheritance diagram for Smart_MaxArrivalsField:



Collaboration diagram for Smart_MaxArrivalsField:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.929.1 Member Function Documentation

17.929.1.1 main()

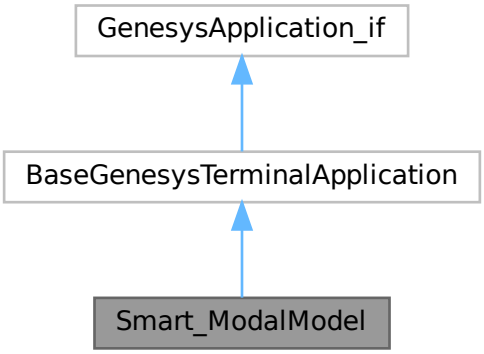
```
int Smart_MaxArrivalsField::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

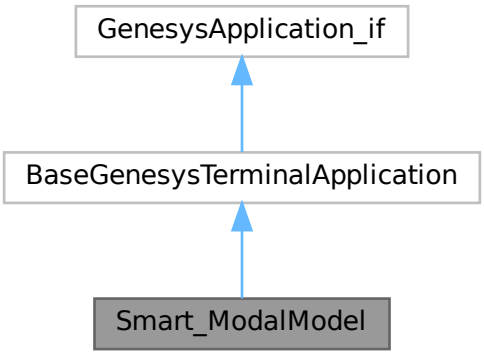
- applications/modelSpecific/arenaSmarts/Smart_MaxArrivalsField.h
- applications/modelSpecific/arenaSmarts/Smart_MaxArrivalsField.cpp

17.930 Smart_ModalModel Class Reference

Inheritance diagram for Smart_ModalModel:



Collaboration diagram for Smart_ModalModel:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.930.1 Member Function Documentation

17.930.1.1 main()

```
int Smart_ModalModel::main (
            int argc,
            char ** argv )  [override], [virtual]
```

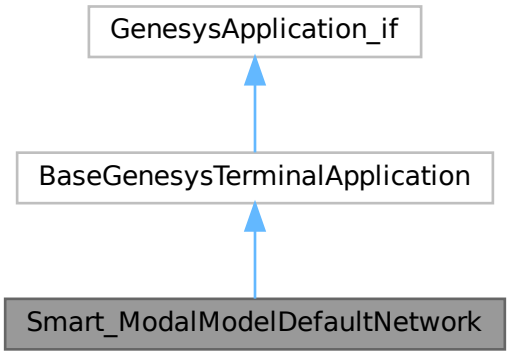
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

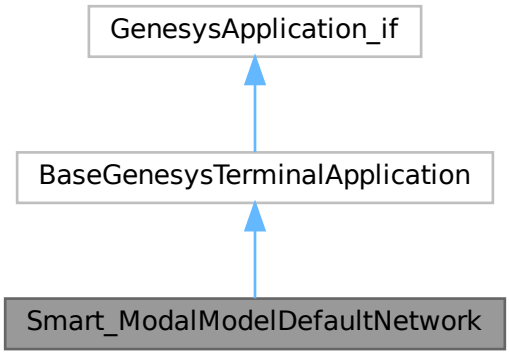
- applications/modelSpecific/smarts/ModalModel/Smart_Modalmodel.h
- applications/modelSpecific/smarts/ModalModel/Smart_Modalmodel.cpp

17.931 Smart_ModalModelDefaultNetwork Class Reference

Inheritance diagram for Smart_ModalModelDefaultNetwork:



Collaboration diagram for Smart_ModalModelDefaultNetwork:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.931.1 Member Function Documentation

17.931.1.1 main()

```
int Smart_ModalModelDefaultNetwork::main (
            int argc,
            char ** argv )  [override], [virtual]
```

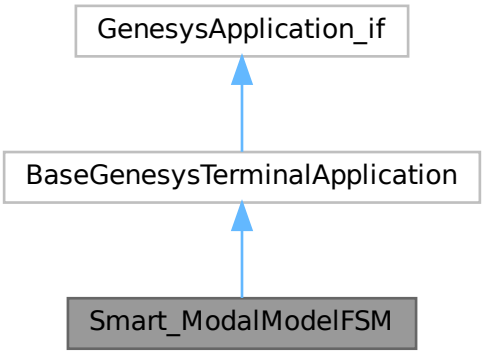
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

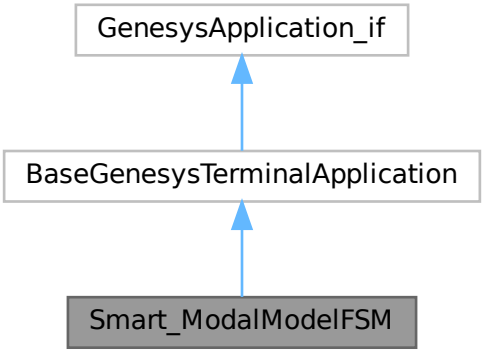
- applications/modelSpecific/smarts/ModalModel/Smart_ModalModelDefaultNetwork.h
- applications/modelSpecific/smarts/ModalModel/Smart_ModalModelDefaultNetwork.cpp

17.932 Smart_ModalModelFSM Class Reference

Inheritance diagram for Smart_ModalModelFSM:



Collaboration diagram for Smart_ModalModelFSM:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.932.1 Member Function Documentation

17.932.1.1 main()

```
int Smart_ModalModelFSM::main (
            int argc,
            char ** argv )  [override], [virtual]
```

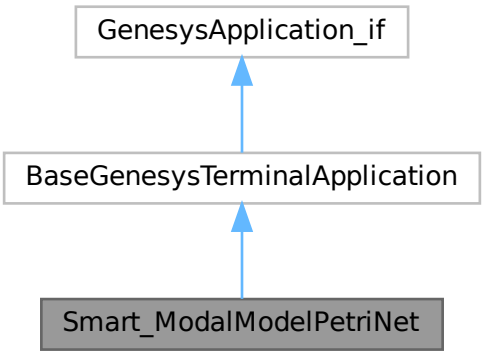
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

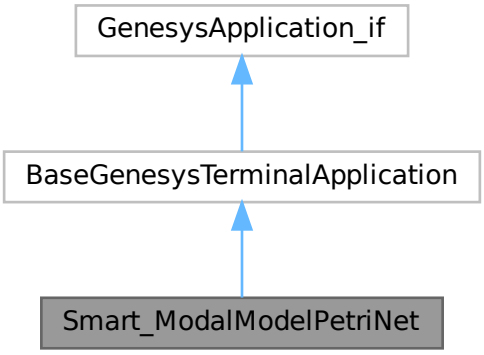
- applications/modelSpecific/smarts/ModalModel/Smart_ModalModelFSM.h
- applications/modelSpecific/smarts/ModalModel/Smart_ModalModelFSM.cpp

17.933 Smart_ModalModelPetriNet Class Reference

Inheritance diagram for Smart_ModalModelPetriNet:



Collaboration diagram for Smart_ModalModelPetriNet:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.933.1 Member Function Documentation

17.933.1.1 main()

```
int Smart_ModalModelPetriNet::main (
            int argc,
            char ** argv )  [override], [virtual]
```

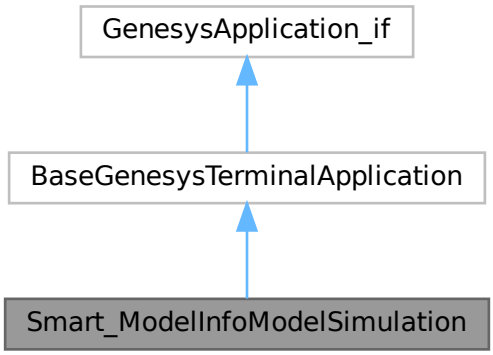
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

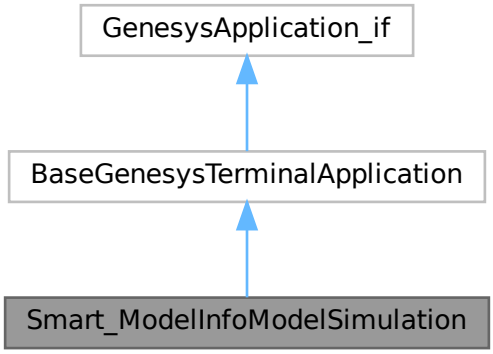
- applications/modelSpecific/smarts/ModalModel/Smart_ModalModelPetriNet.h
- applications/modelSpecific/smarts/ModalModel/Smart_ModalModelPetriNet.cpp

17.934 Smart_ModelInfoModelSimulation Class Reference

Inheritance diagram for Smart_ModelInfoModelSimulation:



Collaboration diagram for Smart_ModelInfoModelSimulation:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.934.1 Member Function Documentation

17.934.1.1 main()

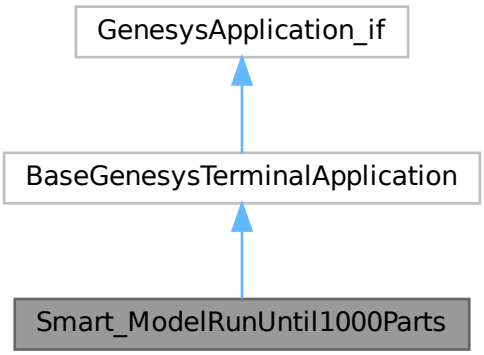
```
int Smart_ModelInfoModelSimulation::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

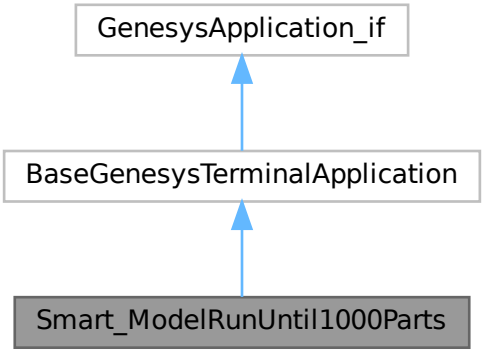
- applications/modelSpecific/smarts/Logic/Smart_ModelInfoModelSimulation.h
- applications/modelSpecific/smarts/Logic/Smart_ModelInfoModelSimulation.cpp

17.935 Smart_ModelRunUntil1000Parts Class Reference

Inheritance diagram for Smart_ModelRunUntil1000Parts:



Collaboration diagram for Smart_ModelRunUntil1000Parts:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.935.1 Member Function Documentation

17.935.1.1 main()

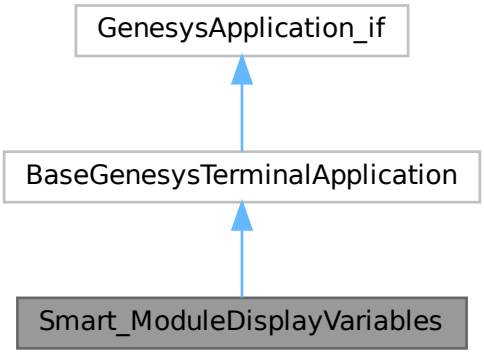
```
int Smart_ModelRunUntil1000Parts::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It create a new simulator instance, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

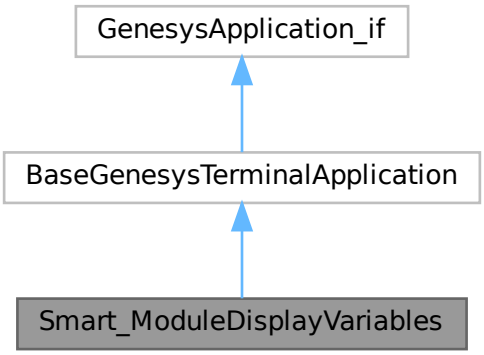
- applications/modelSpecific/arenaSmarts/Smart_ModelRunUntil1000Parts.h
- applications/modelSpecific/arenaSmarts/Smart_ModelRunUntil1000Parts.cpp

17.936 Smart_ModuleDisplayVariables Class Reference

Inheritance diagram for Smart_ModuleDisplayVariables:



Collaboration diagram for Smart_ModuleDisplayVariables:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.936.1 Member Function Documentation

17.936.1.1 main()

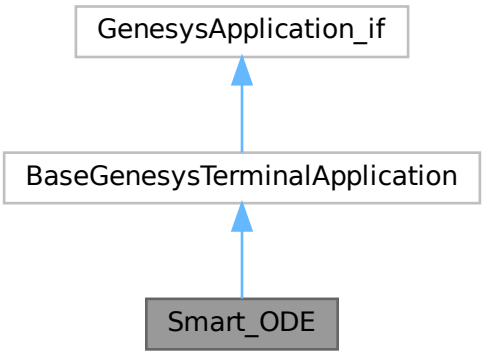
```
int Smart_ModuleDisplayVariables::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

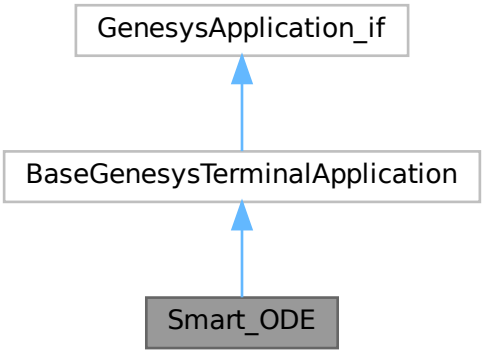
- applications/modelSpecific/arenaSmarts/Smart_ModuleDisplayVariables.h
- applications/modelSpecific/arenaSmarts/Smart_ModuleDisplayVariables.cpp

17.937 Smart_ODE Class Reference

Inheritance diagram for Smart_ODE:



Collaboration diagram for Smart_ODE:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.937.1 Member Function Documentation

17.937.1.1 main()

```
int Smart_ODE::main (
            int argc,
            char ** argv )  [override], [virtual]
```

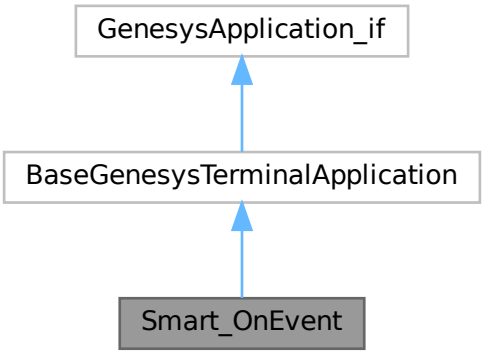
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

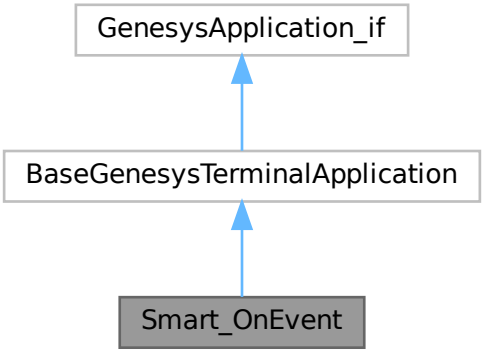
- applications/modelSpecific/smarts/Continuous/Smart_ODE.h
- applications/modelSpecific/smarts/Continuous/Smart_ODE.cpp

17.938 Smart_OnEvent Class Reference

Inheritance diagram for Smart_OnEvent:



Collaboration diagram for Smart_OnEvent:



Public Member Functions

- virtual int main (int argc, char **argv) override
- virtual void onBreakpointHandler (SimulationEvent *re) override
- virtual void onEntityCreateHandler (SimulationEvent *re) override
- virtual void onEntityMoveHandler (SimulationEvent *re) override
- virtual void onSimulationStartHandler (SimulationEvent *re) override
- virtual void onReplicationStartHandler (SimulationEvent *re) override
- virtual void onReplicationStepHandler (SimulationEvent *re) override
- virtual void onProcessEventHandler (SimulationEvent *re) override
- virtual void onReplicationEndHandler (SimulationEvent *re) override
- virtual void onSimulationEndHandler (SimulationEvent *re) override
- virtual void onSimulationPausedHandler (SimulationEvent *re) override
- virtual void onSimulationResumeHandler (SimulationEvent *re) override
- virtual void onEntityRemoveHandler (SimulationEvent *re) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)

17.938.1 Member Function Documentation

17.938.1.1 main()

```
int Smart_OnEvent::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.

17.938.1.2 onBreakpointHandler()

```
void Smart_OnEvent::onBreakpointHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.938.1.3 onEntityCreateHandler()

```
void Smart_OnEvent::onEntityCreateHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.938.1.4 onEntityMoveHandler()

```
void Smart_OnEvent::onEntityMoveHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.938.1.5 onEntityRemoveHandler()

```
void Smart_OnEvent::onEntityRemoveHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.938.1.6 onProcessEventHandler()

```
void Smart_OnEvent::onProcessEventHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.938.1.7 onReplicationEndHandler()

```
void Smart_OnEvent::onReplicationEndHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.938.1.8 onReplicationStartHandler()

```
void Smart_OnEvent::onReplicationStartHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.938.1.9 onReplicationStepHandler()

```
void Smart_OnEvent::onReplicationStepHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.938.1.10 onSimulationEndHandler()

```
void Smart_OnEvent::onSimulationEndHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.938.1.11 onSimulationPausedHandler()

```
void Smart_OnEvent::onSimulationPausedHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.938.1.12 onSimulationResumeHandler()

```
void Smart_OnEvent::onSimulationResumeHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.

17.938.1.13 onSimulationStartHandler()

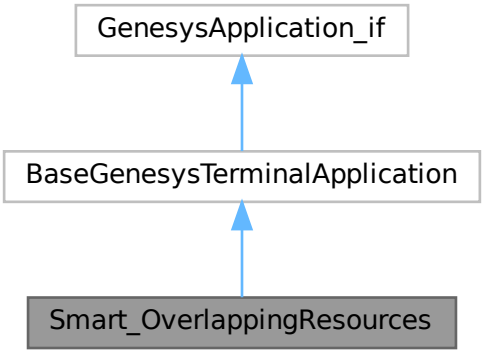
```
void Smart_OnEvent::onSimulationStartHandler (
            SimulationEvent * re )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

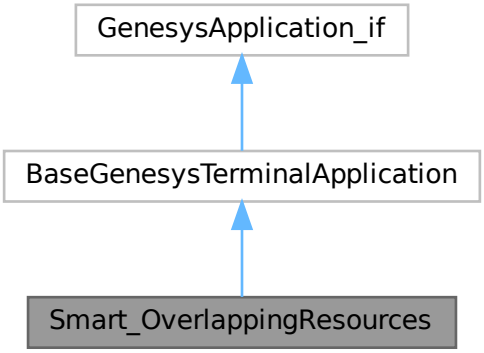
- applications/modelSpecific/smarts/Synchronization/Smart_OnEvent.h
- applications/modelSpecific/smarts/Synchronization/Smart_OnEvent.cpp

17.939 Smart_OverlappingResources Class Reference

Inheritance diagram for Smart_OverlappingResources:



Collaboration diagram for Smart_OverlappingResources:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.939.1 Member Function Documentation

17.939.1.1 main()

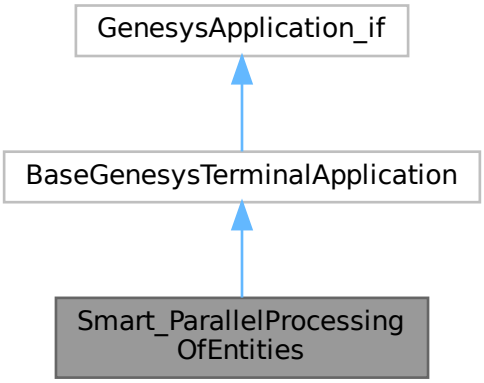
```
int Smart_OverlappingResources::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

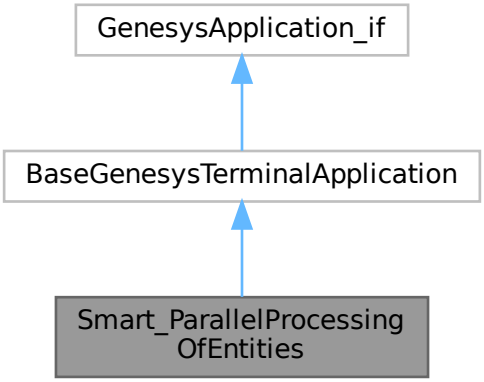
- applications/modelSpecific/arenaSmarts/Smart_OverlappingResources.h
- applications/modelSpecific/arenaSmarts/Smart_OverlappingResources.cpp

17.940 Smart_ParallelProcessingOfEntities Class Reference

Inheritance diagram for Smart_ParallelProcessingOfEntities:



Collaboration diagram for Smart_ParallelProcessingOfEntities:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.940.1 Member Function Documentation

17.940.1.1 main()

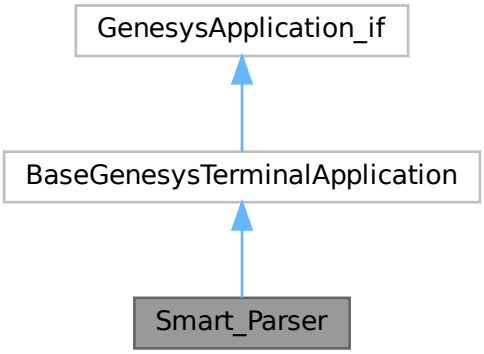
```
int Smart_ParallelProcessingOfEntities::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

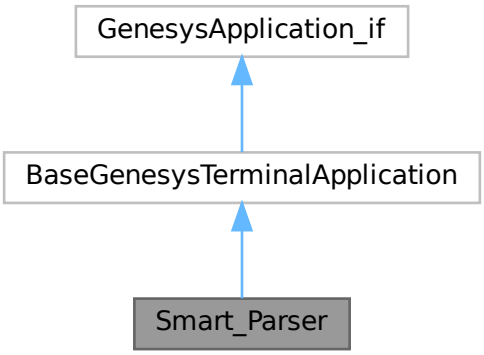
- applications/modelSpecific/arenaSmarts/Smart_ParallelProcessingOfEntities.h
- applications/modelSpecific/arenaSmarts/Smart_ParallelProcessingOfEntities.cpp

17.941 Smart_Parser Class Reference

Inheritance diagram for Smart_Parser:



Collaboration diagram for Smart_Parser:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.941.1 Member Function Documentation

17.941.1.1 main()

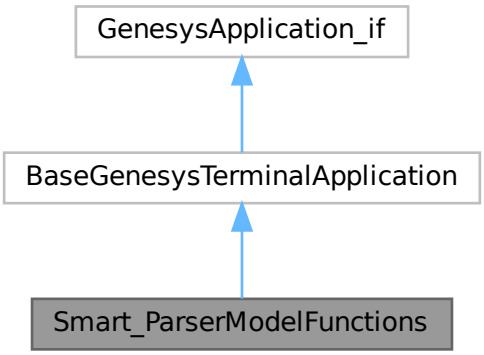
```
int Smart_Parser::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

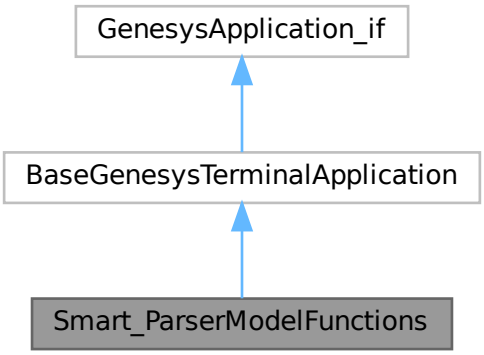
- applications/modelSpecific/smarts/Logic/Smart_Parser.h
- applications/modelSpecific/smarts/Logic/Smart_Parser.cpp

17.942 Smart_ParserModelFunctions Class Reference

Inheritance diagram for Smart_ParserModelFunctions:



Collaboration diagram for Smart_ParserModelFunctions:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.942.1 Member Function Documentation

17.942.1.1 main()

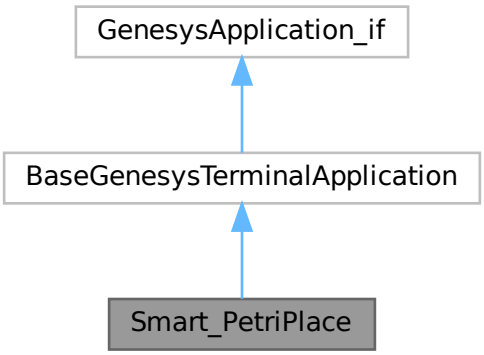
```
int Smart_ParserModelFunctions::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

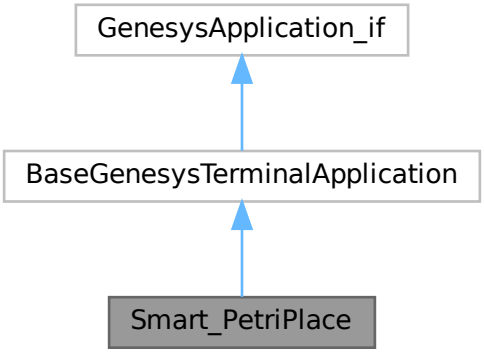
- applications/modelSpecific/smarts/Logic/Smart_ParserModelFunctions.h
- applications/modelSpecific/smarts/Logic/Smart_ParserModelFunctions.cpp

17.943 Smart_PetriPlace Class Reference

Inheritance diagram for Smart_PetriPlace:



Collaboration diagram for Smart_PetriPlace:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.943.1 Member Function Documentation

17.943.1.1 main()

```
int Smart_PetriPlace::main (
            int argc,
            char ** argv )  [override], [virtual]
```

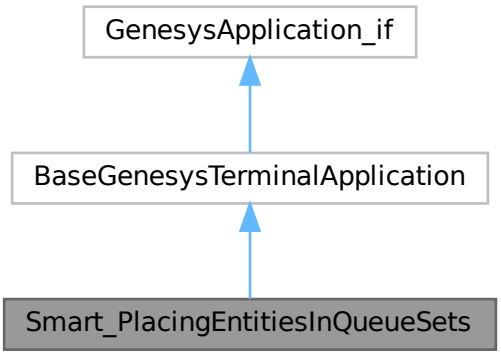
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

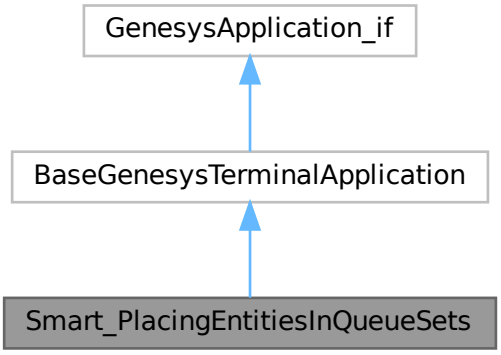
- applications/modelSpecific/smarts/ModalModel/Smart_PetriPlace.h
- applications/modelSpecific/smarts/ModalModel/Smart_PetriPlace.cpp

17.944 Smart_PlacingEntitiesInQueueSets Class Reference

Inheritance diagram for Smart_PlacingEntitiesInQueueSets:



Collaboration diagram for Smart_PlacingEntitiesInQueueSets:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.944.1 Member Function Documentation

17.944.1.1 main()

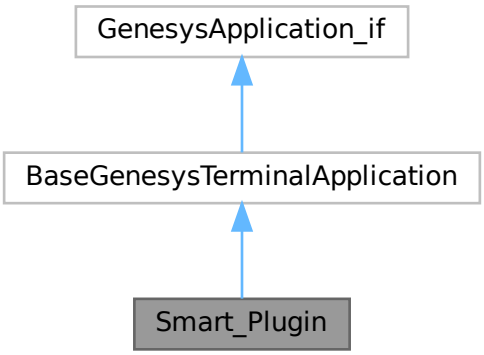
```
int Smart_PlacingEntitiesInQueueSets::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

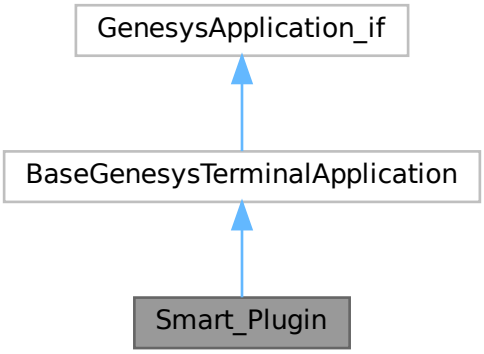
- applications/modelSpecific/arenaSmarts/Smart_PlacingEntitiesInQueueSets.h
- applications/modelSpecific/arenaSmarts/Smart_PlacingEntitiesInQueueSets.cpp

17.945 Smart_Plugin Class Reference

Inheritance diagram for Smart_Plugin:



Collaboration diagram for Smart_Plugin:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.945.1 Member Function Documentation

17.945.1.1 main()

```
int Smart_Plugin::main (
            int argc,
            char ** argv )  [override], [virtual]
```

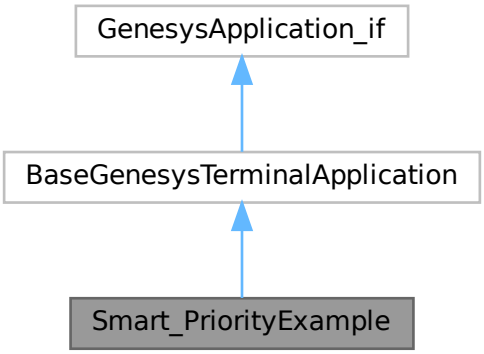
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

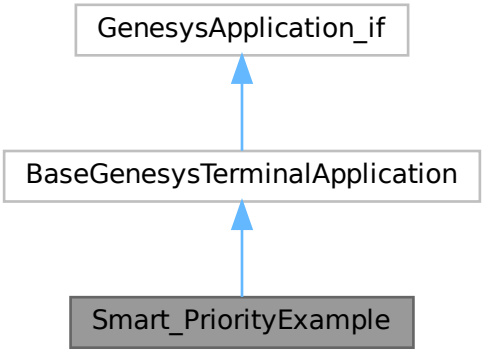
- applications/modelSpecific/smarts/Logic/Smart_Plugin.h
- applications/modelSpecific/smarts/Logic/Smart_Plugin.cpp

17.946 Smart_PriorityExample Class Reference

Inheritance diagram for Smart_PriorityExample:



Collaboration diagram for Smart_PriorityExample:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.946.1 Member Function Documentation

17.946.1.1 main()

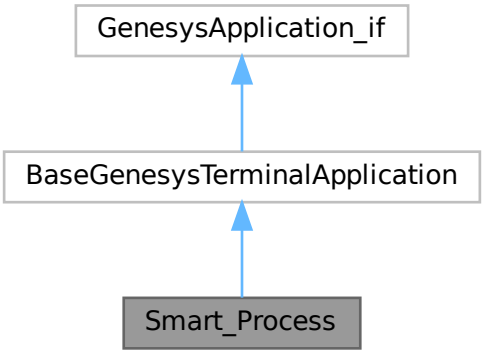
```
int Smart_PriorityExample::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

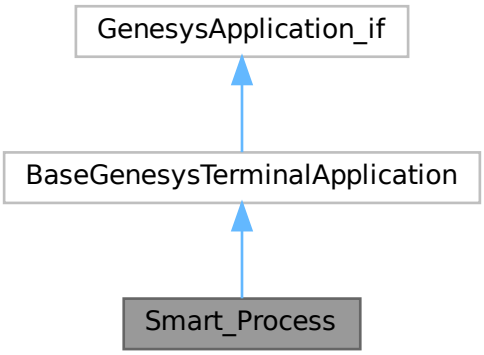
- applications/modelSpecific/arenaSmarts/Smart_PriorityExample.h
- applications/modelSpecific/arenaSmarts/Smart_PriorityExample.cpp

17.947 Smart_Process Class Reference

Inheritance diagram for Smart_Process:



Collaboration diagram for Smart_Process:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.947.1 Member Function Documentation

17.947.1.1 main()

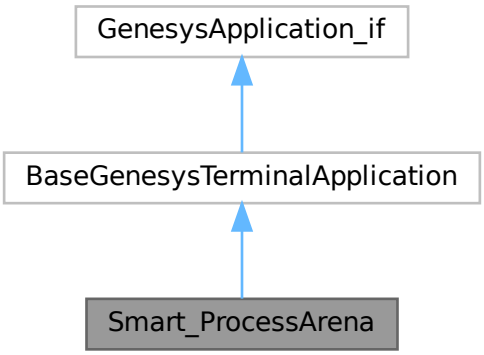
```
int Smart_Process::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

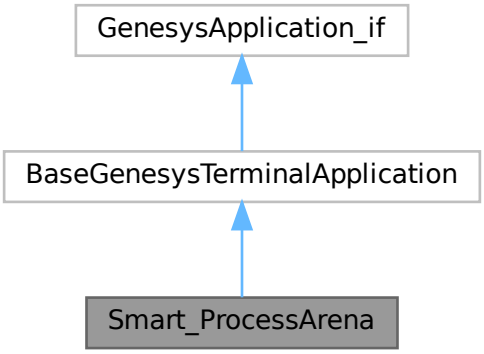
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_Process.h
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_Process.cpp

17.948 Smart_ProcessArena Class Reference

Inheritance diagram for Smart_ProcessArena:



Collaboration diagram for Smart_ProcessArena:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.948.1 Member Function Documentation

17.948.1.1 main()

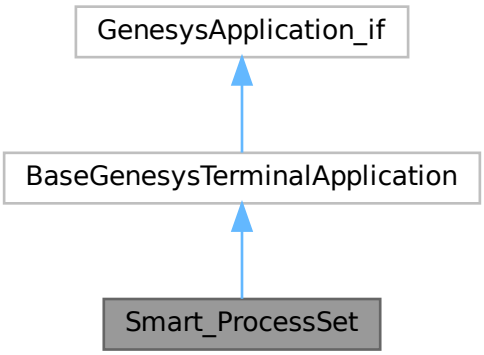
```
int Smart_ProcessArena::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

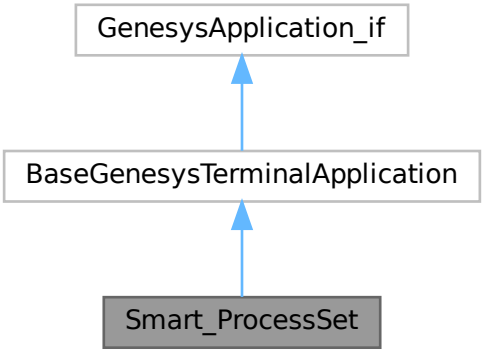
- applications/modelSpecific/arenaSmarts/Smart_ProcessArena.h
- applications/modelSpecific/arenaSmarts/Smart_ProcessArena.cpp

17.949 Smart_ProcessSet Class Reference

Inheritance diagram for Smart_ProcessSet:



Collaboration diagram for Smart_ProcessSet:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.949.1 Member Function Documentation

17.949.1.1 main()

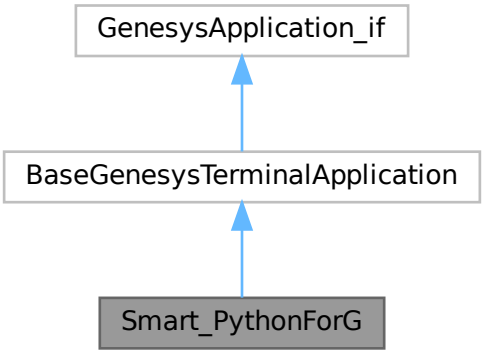
```
int Smart_ProcessSet::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

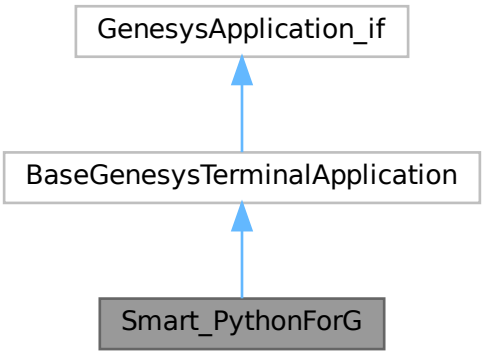
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_ProcessSet.h
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_ProcessSet.cpp

17.950 Smart_PythonForG Class Reference

Inheritance diagram for Smart_PythonForG:



Collaboration diagram for Smart_PythonForG:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.950.1 Member Function Documentation

17.950.1.1 main()

```
int Smart_PythonForG::main (
            int argc,
            char ** argv )  [override], [virtual]
```

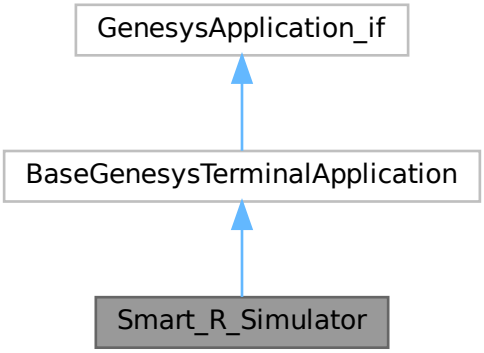
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/smarts/ExternalIntegration/Smart_PythonForG.h
- applications/modelSpecific/smarts/ExternalIntegration/Smart_PythonForG.cpp

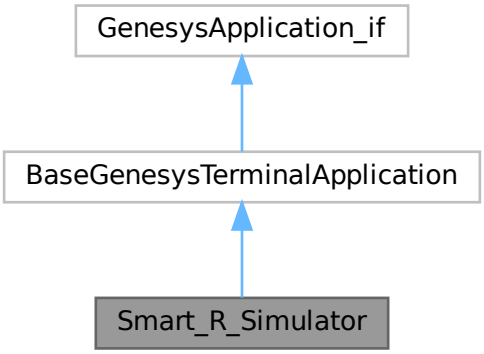
17.951 Smart_R_Simulator Class Reference

Builds a minimal Create -> RSimulator -> Dispose model.
`#include <applications/modelSpecific/smarts/ExternalIntegration/Smart_R_↩ Simulator.h>`

Inheritance diagram for Smart_R_Simulator:



Collaboration diagram for Smart_R_Simulator:



Public Member Functions

- **Smart_R_Simulator** ()

  Creates the smart R simulator example.
- virtual int main (int argc, char **argv) override

  Builds, saves and simulates the example model.

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.951.1 Detailed Description

Builds a minimal Create -> RSimulator -> Dispose model.
The example exercises the component-level R integration and is intended as a compact smoke model for RSimulator and its internal RSimulatorRunner.

17.951.2 Member Function Documentation

17.951.2.1 main()

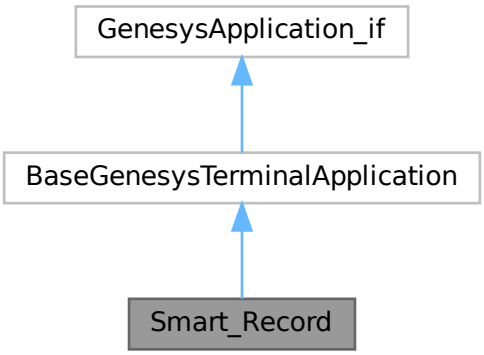
```
int Smart_R_Simulator::main (
            int argc,
            char ** argv )  [override], [virtual]
```

Builds, saves and simulates the example model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

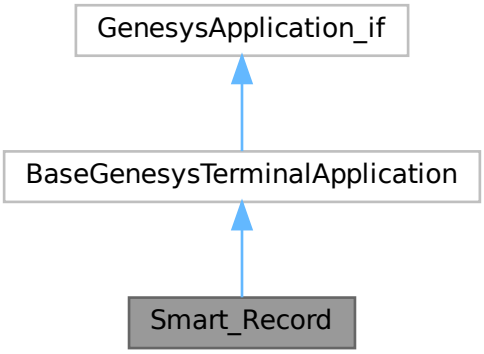
- applications/modelSpecific/smarts/ExternalIntegration/Smart_R_Simulator.h
- applications/modelSpecific/smarts/ExternalIntegration/Smart_R_Simulator.cpp

17.952 Smart_Record Class Reference

Inheritance diagram for Smart_Record:



Collaboration diagram for Smart_Record:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.952.1 Member Function Documentation

17.952.1.1 main()

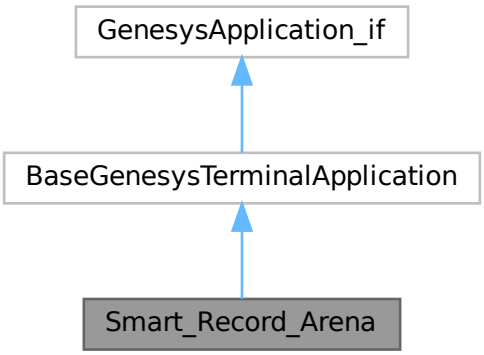
```
int Smart_Record::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

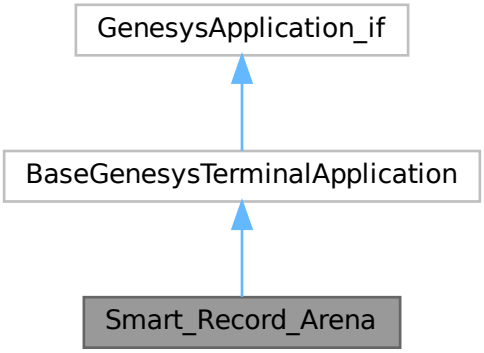
- applications/modelSpecific/smarts/InputOutput/Smart_Record.h
- applications/modelSpecific/smarts/InputOutput/Smart_Record.cpp

17.953 Smart_Record_Arena Class Reference

Inheritance diagram for Smart_Record_Arena:



Collaboration diagram for Smart_Record_Arena:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.953.1 Member Function Documentation

17.953.1.1 main()

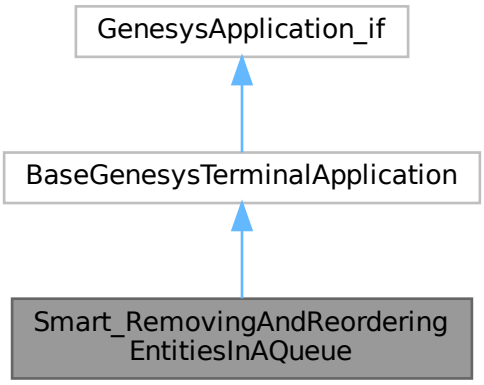
```
int Smart_Record_Arena::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

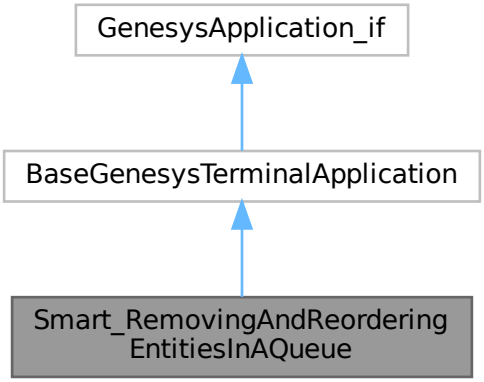
- applications/modelSpecific/arenaSmarts/Smart_Record_Arena.h
- applications/modelSpecific/arenaSmarts/Smart_Record_Arena.cpp

17.954 Smart_RemovingAndReorderingEntitiesInAQueue Class Reference

Inheritance diagram for Smart_RemovingAndReorderingEntitiesInAQueue:



Collaboration diagram for Smart_RemovingAndReorderingEntitiesInAQueue:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.954.1 Member Function Documentation

17.954.1.1 main()

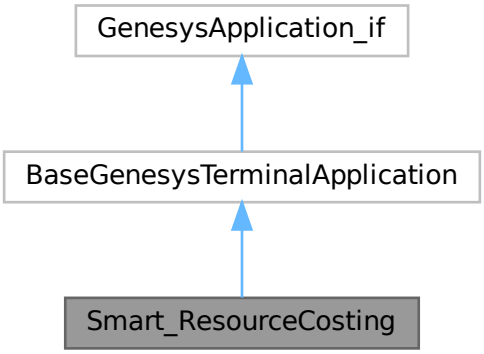
```
int Smart_RemovingAndReorderingEntitiesInAQueue::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

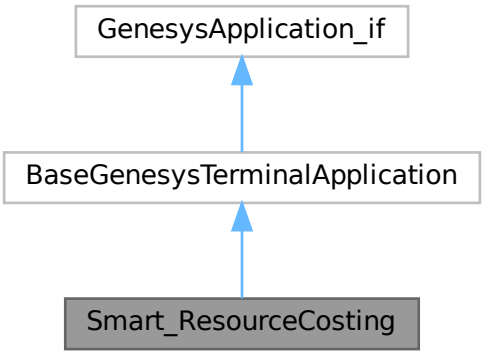
- applications/modelSpecific/arenaSmarts/Smart_RemovingAndReorderingEntitiesInAQueue.h
- applications/modelSpecific/arenaSmarts/Smart_RemovingAndReorderingEntitiesInAQueue.cpp

17.955 Smart_ResourceCosting Class Reference

Inheritance diagram for Smart_ResourceCosting:



Collaboration diagram for Smart_ResourceCosting:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.955.1 Member Function Documentation

17.955.1.1 main()

```
int Smart_ResourceCosting::main (
            int argc,
            char ** argv )  [override], [virtual]
```

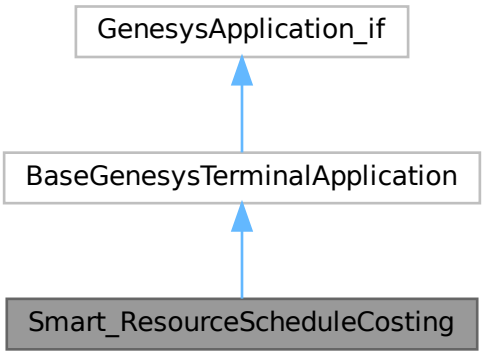
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

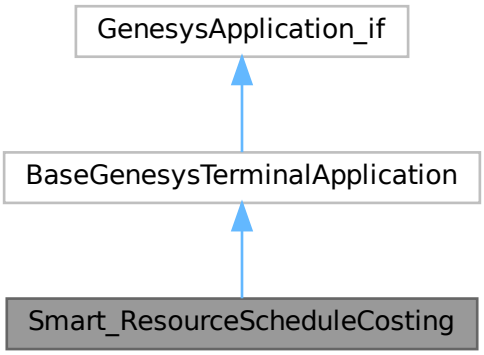
- applications/modelSpecific/arenaSmarts/Smart_ResourceCosting.h
- applications/modelSpecific/arenaSmarts/Smart_ResourceCosting.cpp

17.956 Smart_ResourceScheduleCosting Class Reference

Inheritance diagram for Smart_ResourceScheduleCosting:



Collaboration diagram for Smart_ResourceScheduleCosting:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.956.1 Member Function Documentation

17.956.1.1 main()

```
int Smart_ResourceScheduleCosting::main (
            int argc,
            char ** argv )  [override], [virtual]
```

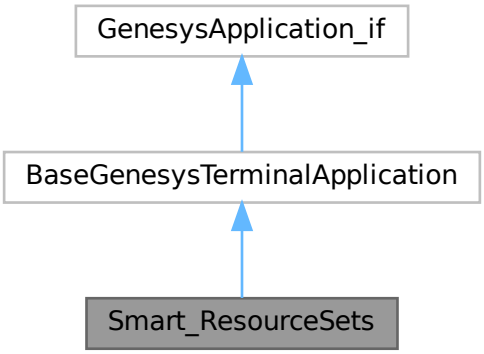
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

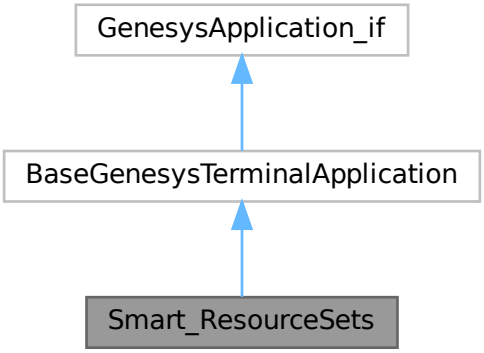
- applications/modelSpecific/arenaSmarts/Smart_ResourceScheduleCosting.h
- applications/modelSpecific/arenaSmarts/Smart_ResourceScheduleCosting.cpp

17.957 Smart_ResourceSets Class Reference

Inheritance diagram for Smart_ResourceSets:



Collaboration diagram for Smart_ResourceSets:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.957.1 Member Function Documentation

17.957.1.1 main()

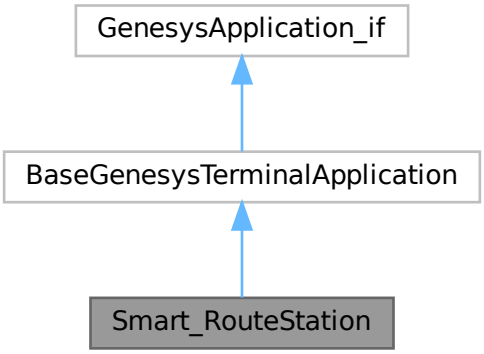
```
int Smart_ResourceSets::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It create a new simulator instance, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

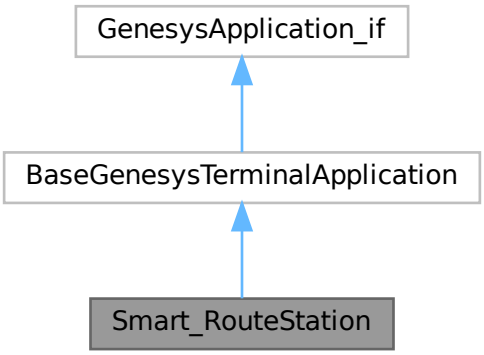
- applications/modelSpecific/arenaSmarts/Smart_ResourceSets.h
- applications/modelSpecific/arenaSmarts/Smart_ResourceSets.cpp

17.958 Smart_RouteStation Class Reference

Inheritance diagram for Smart_RouteStation:



Collaboration diagram for Smart_RouteStation:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.958.1 Member Function Documentation

17.958.1.1 main()

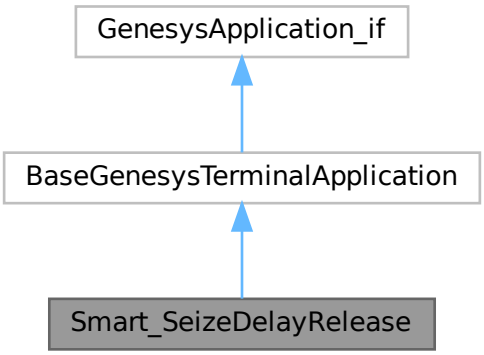
```
int Smart_RouteStation::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

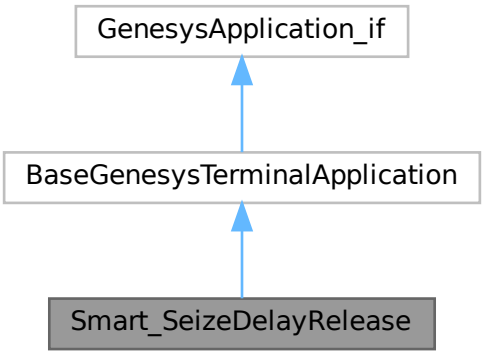
- applications/modelSpecific/smarts/MaterialHandling/Smart_RouteStation.h
- applications/modelSpecific/smarts/MaterialHandling/Smart_RouteStation.cpp

17.959 Smart_SeizeDelayRelease Class Reference

Inheritance diagram for Smart_SeizeDelayRelease:



Collaboration diagram for Smart_SeizeDelayRelease:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.959.1 Member Function Documentation

17.959.1.1 main()

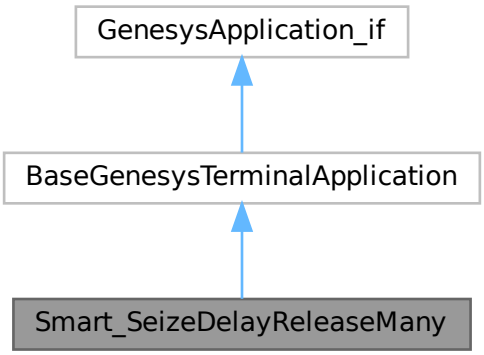
```
int Smart_SeizeDelayRelease::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

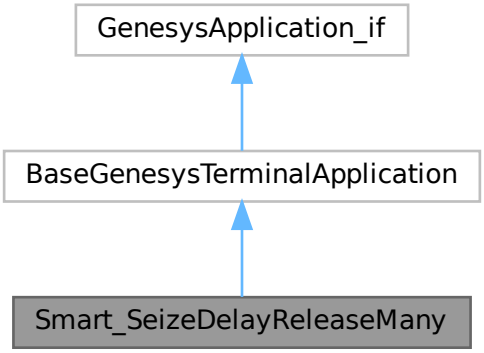
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayRelease.h
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayRelease.cpp

17.960 Smart_SeizeDelayReleaseMany Class Reference

Inheritance diagram for Smart_SeizeDelayReleaseMany:



Collaboration diagram for Smart_SeizeDelayReleaseMany:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.960.1 Member Function Documentation

17.960.1.1 main()

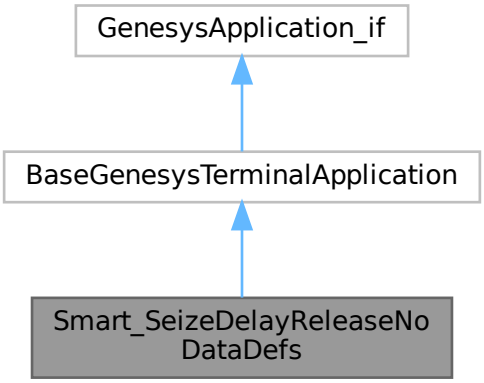
```
int Smart_SeizeDelayReleaseMany::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

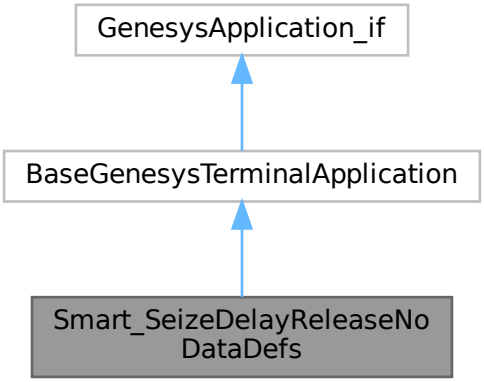
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayReleaseMany.h
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayReleaseMany.cpp

17.961 Smart_SeizeDelayReleaseNoDataDefs Class Reference

Inheritance diagram for Smart_SeizeDelayReleaseNoDataDefs:



Collaboration diagram for Smart_SeizeDelayReleaseNoDataDefs:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.961.1 Member Function Documentation

17.961.1.1 main()

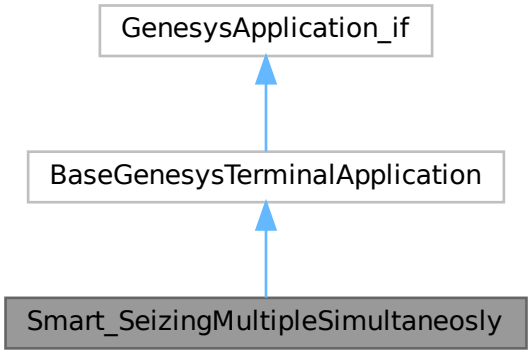
```
int Smart_SeizeDelayReleaseNoDataDefs::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

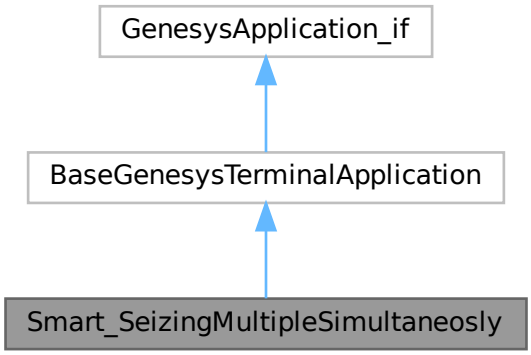
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayReleaseNoDataDefs.h
- applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayReleaseNoDataDefs.cpp

17.962 Smart_SeizingMultipleSimultaneosly Class Reference

Inheritance diagram for Smart_SeizingMultipleSimultaneosly:



Collaboration diagram for Smart_SeizingMultipleSimultaneosly:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.962.1 Member Function Documentation

17.962.1.1 main()

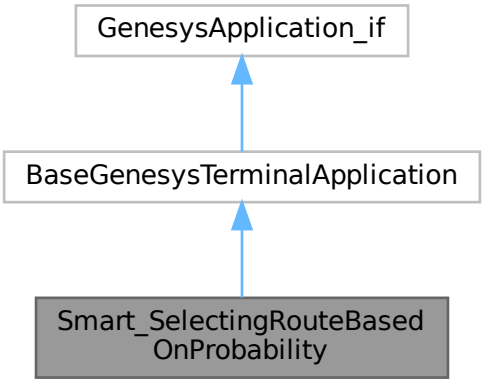
```
int Smart_SeizingMultipleSimultaneosly::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

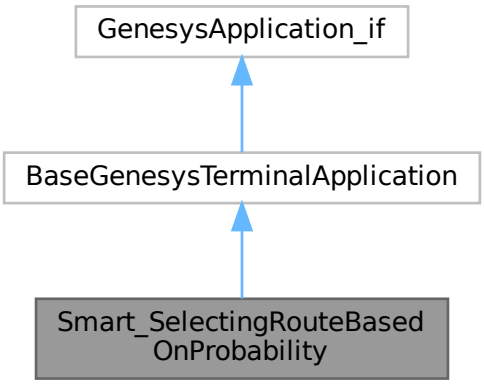
- applications/modelSpecific/arenaSmarts/Smart_SeizingMultipleSimultaneosly.h
- applications/modelSpecific/arenaSmarts/Smart_SeizingMultipleSimultaneosly.cpp

17.963 Smart_SelectingRouteBasedOnProbability Class Reference

Inheritance diagram for Smart_SelectingRouteBasedOnProbability:



Collaboration diagram for Smart_SelectingRouteBasedOnProbability:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.963.1 Member Function Documentation

17.963.1.1 main()

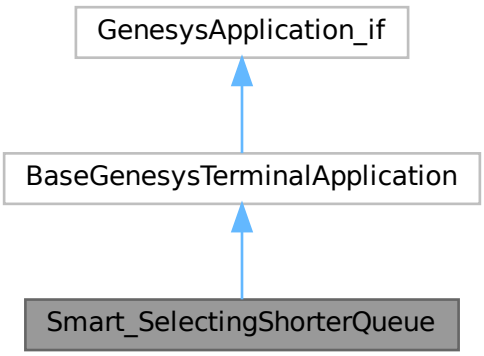
```
int Smart_SelectingRouteBasedOnProbability::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

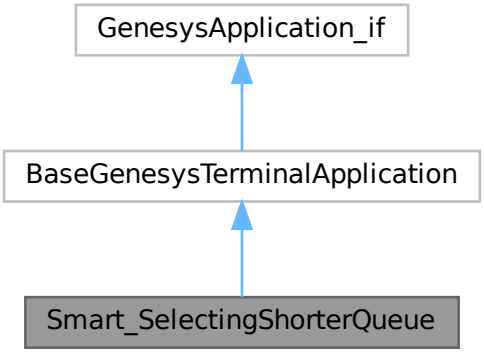
- applications/modelSpecific/arenaSmarts/Smart_SelectingRouteBasedOnProbability.h
- applications/modelSpecific/arenaSmarts/Smart_SelectingRouteBasedOnProbability.cpp

17.964 Smart_SelectingShorterQueue Class Reference

Inheritance diagram for Smart_SelectingShorterQueue:



Collaboration diagram for Smart_SelectingShorterQueue:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.964.1 Member Function Documentation

17.964.1.1 main()

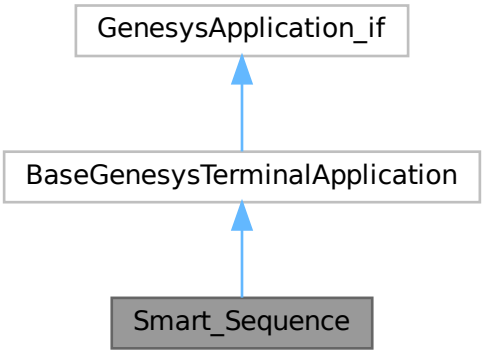
```
int Smart_SelectingShorterQueue::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

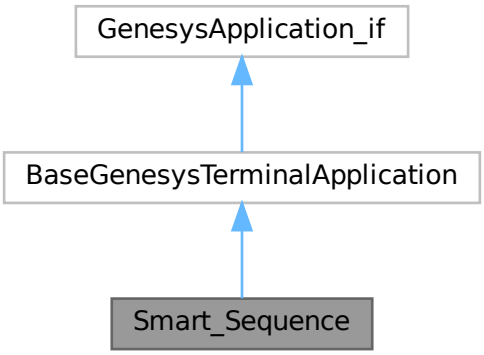
- applications/modelSpecific/arenaSmarts/Smart_SelectingShorterQueue.h
- applications/modelSpecific/arenaSmarts/Smart_SelectingShorterQueue.cpp

17.965 Smart_Sequence Class Reference

Inheritance diagram for Smart_Sequence:



Collaboration diagram for Smart_Sequence:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.965.1 Member Function Documentation

17.965.1.1 main()

```
int Smart_Sequence::main (
            int argc,
            char ** argv )  [override], [virtual]
```

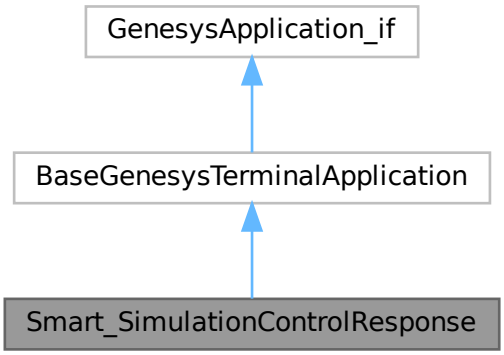
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

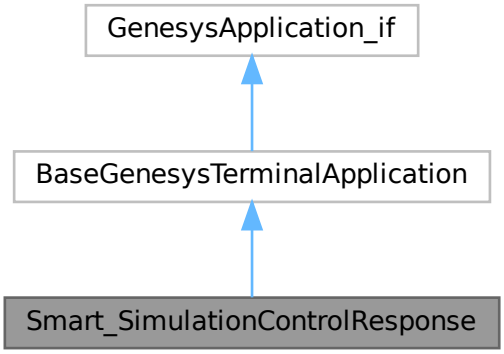
- applications/modelSpecific/smarts/MaterialHandling/Smart_Sequence.h
- applications/modelSpecific/smarts/MaterialHandling/Smart_Sequence.cpp

17.966 Smart_SimulationControlResponse Class Reference

Inheritance diagram for Smart_SimulationControlResponse:



Collaboration diagram for Smart_SimulationControlResponse:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.966.1 Member Function Documentation

17.966.1.1 main()

```
int Smart_SimulationControlResponse::main (
            int argc,
            char ** argv )  [override], [virtual]
```

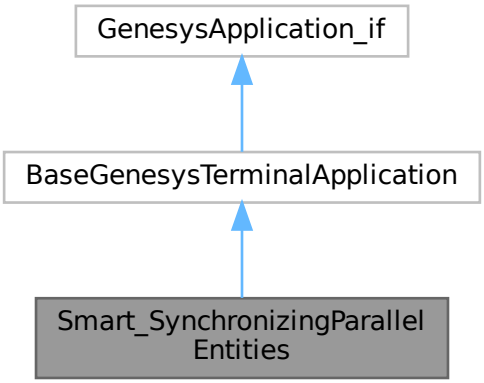
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

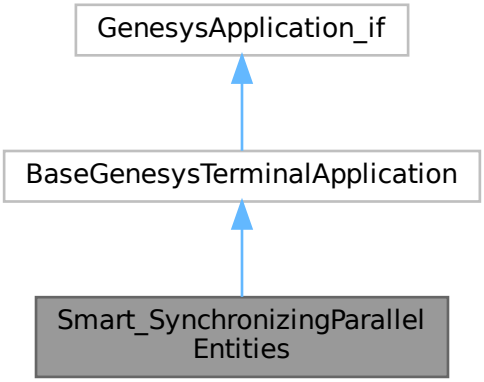
- applications/modelSpecific/smarts/Synchronization/Smart_SimulationControlResponse.h
- applications/modelSpecific/smarts/Synchronization/Smart_SimulationControlResponse.cpp

17.967 Smart_SynchronizingParallelEntities Class Reference

Inheritance diagram for Smart_SynchronizingParallelEntities:



Collaboration diagram for Smart_SynchronizingParallelEntities:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.967.1 Member Function Documentation

17.967.1.1 main()

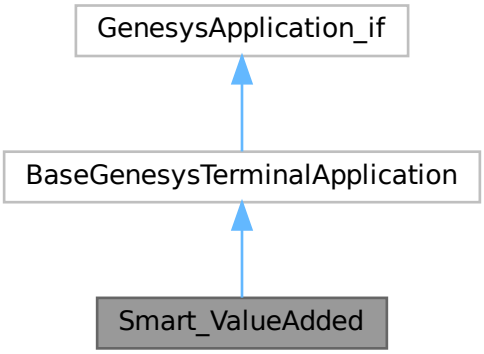
```
int Smart_SynchronizingParallelEntities::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

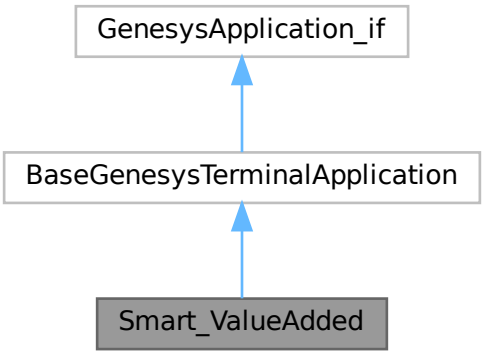
- applications/modelSpecific/arenaSmarts/Smart_SynchronizingParallelEntities.h
- applications/modelSpecific/arenaSmarts/Smart_SynchronizingParallelEntities.cpp

17.968 Smart_ValueAdded Class Reference

Inheritance diagram for Smart_ValueAdded:



Collaboration diagram for Smart_ValueAdded:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.968.1 Member Function Documentation

17.968.1.1 main()

```
int Smart_ValueAdded::main (
            int argc,
            char ** argv )  [override], [virtual]
```

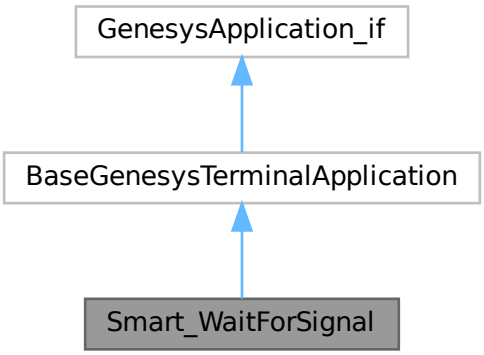
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

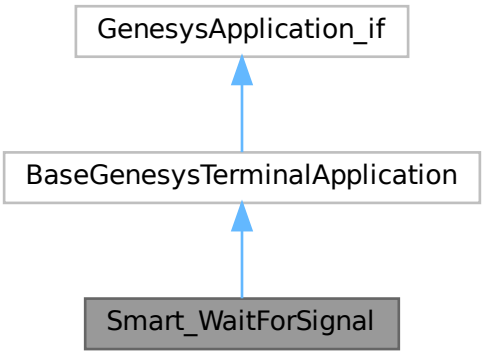
- applications/modelSpecific/arenaSmarts/Smart_ValueAdded.h
- applications/modelSpecific/arenaSmarts/Smart_ValueAdded.cpp

17.969 Smart_WaitForSignal Class Reference

Inheritance diagram for Smart_WaitForSignal:



Collaboration diagram for Smart_WaitForSignal:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.969.1 Member Function Documentation

17.969.1.1 main()

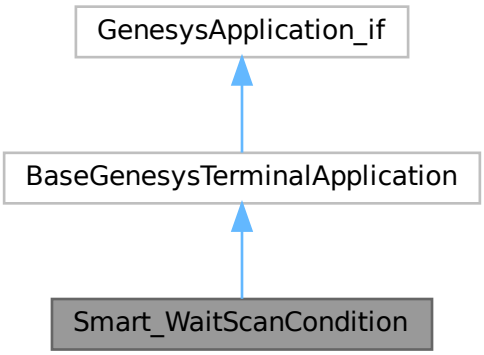
```
int Smart_WaitForSignal::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

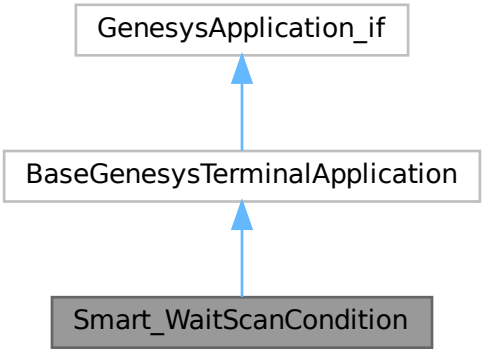
- applications/modelSpecific/arenaSmarts/Smart_WaitForSignal.h
- applications/modelSpecific/arenaSmarts/Smart_WaitForSignal.cpp

17.970 Smart_WaitScanCondition Class Reference

Inheritance diagram for Smart_WaitScanCondition:



Collaboration diagram for Smart_WaitScanCondition:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.970.1 Member Function Documentation

17.970.1.1 main()

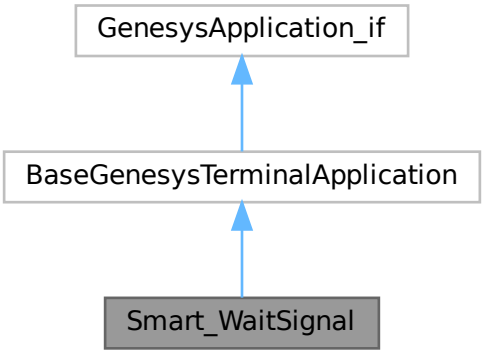
```
int Smart_WaitScanCondition::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

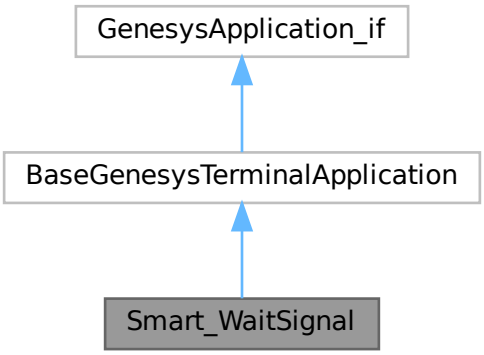
- applications/modelSpecific/smarts/Synchronization/Smart_WaitScanCondition.h
- applications/modelSpecific/smarts/Synchronization/Smart_WaitScanCondition.cpp

17.971 Smart_WaitSignal Class Reference

Inheritance diagram for Smart_WaitSignal:



Collaboration diagram for Smart_WaitSignal:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.971.1 Member Function Documentation

17.971.1.1 main()

```
int Smart_WaitSignal::main (
            int argc,
            char ** argv )  [override], [virtual]
```

This is the main function of the application. It instanciates the simulator, builds a simulation model and then simulate that model.
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/smarts/Synchronization/Smart_WaitSignal.h
- applications/modelSpecific/smarts/Synchronization/Smart_WaitSignal.cpp

17.972 WebWorkerRuntime::Snapshot Struct Reference

Captures a thread-safe snapshot of the worker state.
`#include <applications/worker/service/WebWorkerRuntime.h>`

Public Attributes

- bool **isRunning** = false
- bool **isStarting** = false
- bool **isStopping** = false
- bool **hasFailed** = false
- unsigned short **port** = 8080
- unsigned long **maxRequests** = 0
- unsigned long **servedRequests** = 0
- unsigned long **activeRequests** = 0
- std::string **statusMessage**
- std::string **lastError**
- std::vector< std::string > **recentEvents**
- std::vector< std::string > **recentErrors**

17.972.1 Detailed Description

Captures a thread-safe snapshot of the worker state.
The documentation for this struct was generated from the following file:

- applications/worker/service/WebWorkerRuntime.h

17.973 MetabolicFluxBalanceSolver::Solution Struct Reference

Public Attributes

- bool **feasible** = false
- double **objectiveValue** = 0.0
- std::vector< double > **fluxes**
- std::string **errorMessage**

The documentation for this struct was generated from the following file:

- tools/Biochemical/MetabolicFluxBalanceSolver.h

17.974 Optimizer_if::SolutionSummary Struct Reference

Captures one candidate or best-known optimization solution.
`#include <tools/Optimization/Optimizer_if.h>`

Public Attributes

- unsigned int **iteration** = 0
- unsigned int **simulationNumber** = 0
- bool **feasible** = false
- double **aggregatedObjectiveValue** = 0.0
- std::string **description**

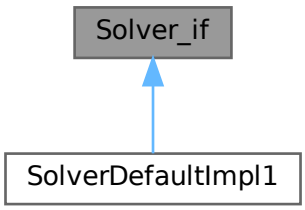
17.974.1 Detailed Description

Captures one candidate or best-known optimization solution.
A solution stores the controls/responses snapshot produced by one simulation scenario and whether all constraints were satisfied.
The documentation for this struct was generated from the following file:

- tools/Optimization/Optimizer_if.h

17.975 Solver_if Class Reference

Legacy compatibility interface for numerical routines.
`#include <tools/Continuous/Solver_if.h>`
Inheritance diagram for Solver_if:



Public Member Functions

- virtual void **setPrecision** (double e)=0
- virtual double **getPrecision** ()=0
- virtual void **setMaxSteps** (double steps)=0
- virtual double **getMaxSteps** ()=0
- virtual double **integrate** (double min, double max, double(∗f)(double, double), double p2)=0
- virtual double **integrate** (double min, double max, double(∗f)(double, double, double), double p2, double p3)=0
- virtual double **integrate** (double min, double max, double(∗f)(double, double, double, double), double p2, double p3, double p4)=0
- virtual double **integrate** (double min, double max, double(∗f)(double, double, double, double, double), double p2, double p3, double p4, double p5)=0
- virtual double **derivate** (double initPoint, double initValue, double(∗f)(double, double), double p2)=0
- virtual double **derivate** (double initPoint, double initValue, double(∗f)(double, double, double), double p2, double p3)=0
- virtual double **derivate** (double initPoint, double initValue, double(∗f)(double, double, double, double), double p2, double p3, double p4)=0
- virtual double **derivate** (double initPoint, double initValue, double(∗f)(double, double, double, double, double), double p2, double p3, double p4, double p5)=0

17.975.1 Detailed Description

Legacy compatibility interface for numerical routines.
Architectural role:

- Historically groups two responsibilities: 1) Numerical quadrature/integration. 2) Numerical advancement/derivation for ODE-like flows.

Mathematical meaning:

- Function signatures use the first functional parameter as time `t`.
- Parameters p2..p5 represent model parameters passed through old-style function pointers.

Limitations:

- The p2/p3/p4/p5 model reflects a historical constraint and is not the preferred long-term abstraction.

Planned evolution:

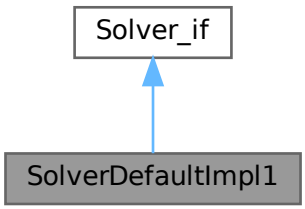
- Split into focused interfaces (quadrature, root finding, ODE systems and ODE solvers) while preserving this API for backward compatibility.

The documentation for this class was generated from the following file:

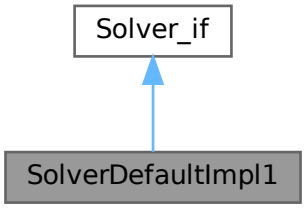
- tools/Continuous/Solver_if.h

17.976 SolverDefaultImpl1 Class Reference

Legacy default implementation of Solver_if.
`#include <tools/Continuous/SolverDefaultImpl1.h>`
Inheritance diagram for SolverDefaultImpl1:



Collaboration diagram for SolverDefaultImpl1:



Public Member Functions

- **SolverDefaultImpl1** (double precision=1e-6, unsigned int steps=1e3)
- virtual void setPrecision (double e) override
- virtual double getPrecision () override
- virtual void setMaxSteps (double steps) override
- virtual double getMaxSteps () override
- virtual double integrate (double min, double max, double(∗f)(double, double), double p2) override
- virtual double integrate (double min, double max, double(∗f)(double, double, double), double p2, double p3) override
- virtual double integrate (double min, double max, double(∗f)(double, double, double, double), double p2, double p3, double p4) override
- virtual double integrate (double min, double max, double(∗f)(double, double, double, double, double), double p2, double p3, double p4, double p5) override

- virtual double derivate (double initPoint, double initValue, double(∗f)(double, double), double p2) override
- virtual double derivate (double initPoint, double initValue, double(∗f)(double, double, double), double p2, double p3) override
- virtual double derivate (double initPoint, double initValue, double(∗f)(double, double, double, double), double p2, double p3, double p4) override
- virtual double derivate (double initPoint, double initValue, double(∗f)(double, double, double, double, double), double p2, double p3, double p4, double p5) override

17.976.1 Detailed Description

Legacy default implementation of Solver_if.
Supported behavior:

- Numerical quadrature through composite Simpson 1/3 with a positive even number of subintervals.

Unsupported behavior:

- The historical derivate() signatures do not expose a defined step-size contract and are not used by current repository clients. They fail explicitly instead of returning silent or undefined numerical results.

Planned evolution:

- Split the historical interface into focused quadrature and ODE abstractions while preserving compatibility where a valid contract exists.

17.976.2 Member Function Documentation

17.976.2.1 derivate() [1/4]

```
double SolverDefaultImpl1::derivate (
            double initPoint,
            double initValue,
            double(*)(double, double) f,
            double p2 )  [override], [virtual]
```

Implements Solver_if.

17.976.2.2 derivate() [2/4]

```
double SolverDefaultImpl1::derivate (
            double initPoint,
            double initValue,
            double(*)(double, double, double) f,
            double p2,
            double p3 )  [override], [virtual]
```

Implements Solver_if.

17.976.2.3 derivate() [3/4]

```
double SolverDefaultImpl1::derivate (
            double initPoint,
            double initValue,
            double(*)(double, double, double, double) f,
            double p2,
            double p3,
            double p4 )  [override], [virtual]
```

Implements Solver_if.

17.976.2.4 derivate() [4/4]

```
double SolverDefaultImpl1::derivate (
            double initPoint,
```

```
            double initValue,
            double(*)(double, double, double, double, double) f,
            double p2,
            double p3,
            double p4,
            double p5 )  [override], [virtual]
```

Implements Solver_if.

17.976.2.5 getMaxSteps()

```
double SolverDefaultImpl1::getMaxSteps ( )  [override], [virtual]
```

Implements Solver_if.

17.976.2.6 getPrecision()

```
double SolverDefaultImpl1::getPrecision ( )  [override], [virtual]
```

Implements Solver_if.

17.976.2.7 integrate() [1/4]

```
double SolverDefaultImpl1::integrate (
            double min,
            double max,
            double(*)(double, double) f,
            double p2 )  [override], [virtual]
```

Implements Solver_if.

17.976.2.8 integrate() [2/4]

```
double SolverDefaultImpl1::integrate (
            double min,
            double max,
            double(*)(double, double, double) f,
            double p2,
            double p3 )  [override], [virtual]
```

Implements Solver_if.

17.976.2.9 integrate() [3/4]

```
double SolverDefaultImpl1::integrate (
            double min,
            double max,
            double(*)(double, double, double, double) f,
            double p2,
            double p3,
            double p4 )  [override], [virtual]
```

Implements Solver_if.

17.976.2.10 integrate() [4/4]

```
double SolverDefaultImpl1::integrate (
            double min,
            double max,
            double(*)(double, double, double, double, double) f,
            double p2,
            double p3,
            double p4,
            double p5 )  [override], [virtual]
```

Implements Solver_if.

17.976.2.11 setMaxSteps()

```
void SolverDefaultImpl1::setMaxSteps (
            double steps )  [override], [virtual]
```

Implements Solver_if.

17.976.2.12 setPrecision()

```
void SolverDefaultImpl1::setPrecision (
            double e )  [override], [virtual]
```

Implements Solver_if.
The documentation for this class was generated from the following files:

- tools/Continuous/SolverDefaultImpl1.h
- tools/Continuous/SolverDefaultImpl1.cpp

17.977 SortFile Class Reference

Utility class that sorts numeric values persisted in a data file.
`#include <kernel/statistics/SorttFile.h>`

Public Member Functions

- **SortFile** ()

  Creates an unbound sorter (filename must be configured before use).
- bool sort ()

  Executes in-file sorting for the configured dataset.
- void setDataFilename (std::string filename)

  Sets the target data filename to be sorted.

17.977.1 Detailed Description

Utility class that sorts numeric values persisted in a data file.
Used by file-based statistics implementations to support quantile/median computations that require ordered samples.

17.977.2 Member Function Documentation

17.977.2.1 setDataFilename()

```
void SortFile::setDataFilename (
            std::string filename )
```

Sets the target data filename to be sorted.

Parameters

filename	Path to the backing data file.

17.977.2.2 sort()

```
bool SortFile::sort ( )
```

Executes in-file sorting for the configured dataset.

Returns

True if sorting operation completes successfully.

The documentation for this class was generated from the following files:

- kernel/statistics/SorttFile.h
- kernel/statistics/SorttFile.cpp

17.978 SourceCodeString Class Reference

Public Member Functions

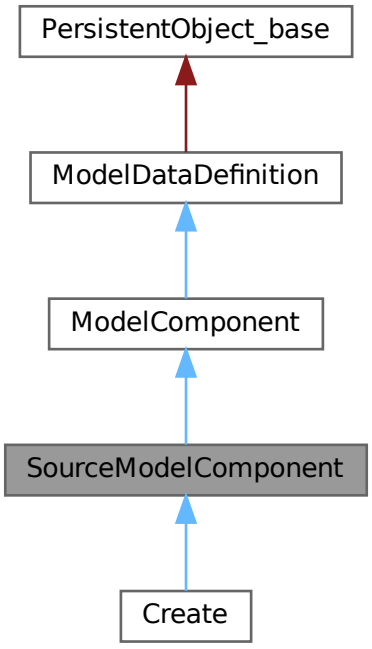
- **SourceCodeString** (const std::string &value)
- **SourceCodeString** (std::string &&value)
- const std::string & **str** () const
- **operator std::string** () const

The documentation for this class was generated from the following file:

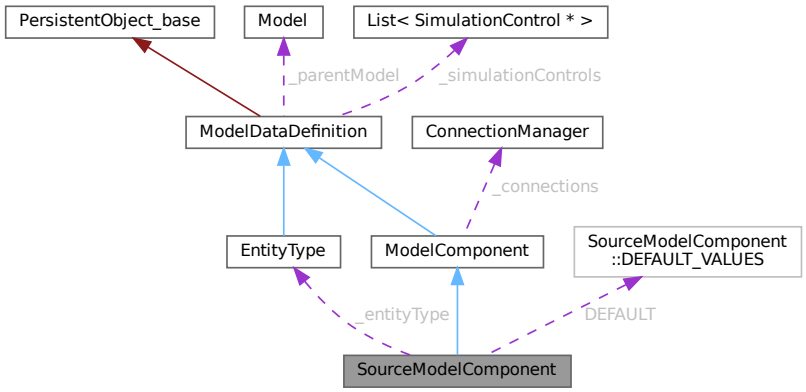
- kernel/simulator/SimulationControlAndResponse.h

17.979 SourceModelComponent Class Reference

Kernel base class for components that inject entities into the model (only Create derives from it so far), generalizing the fields Arena exposes on its "Create module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 31-32): Entity Type (`_entityType`), First Creation (`_firstCreation`), Entities per Arrival (`_entitiesPerCreation`), Max Arrivals (`_maxCreationsExpression`, a parsed expression rather than a static integer) and the time-between-creations expression/time unit.
`#include <kernel/simulator/SourceModelComponent.h>`
Inheritance diagram for SourceModelComponent:



Collaboration diagram for SourceModelComponent:



Classes

- struct DEFAULT_VALUES

Public Member Functions

- **SourceModelComponent** (Model *model, std::string componentTypename, std::string name="")
- void setFirstCreation (double _firstCreation)

  setFirstCreation
- double getFirstCreation () const

  getFirstCreation
- void setEntityType (EntityType *_entityType)

  setEntityType
- void setEntityTypeName (std::string entityTypeName)

  setEntityTypeName
- EntityType * getEntityType () const

  getEntityType
- void setTimeUnit (Util::TimeUnit _timeUnit)

  setTimeUnit
- Util::TimeUnit getTimeUnit () const

  getTimeUnit
- void setTimeBetweenCreationsExpression (std::string _timeBetweenCreations, Util::TimeUnit _time↩ Unit=Util::TimeUnit::unknown)

  setTimeBetweenCreationsExpression
- std::string getTimeBetweenCreationsExpression () const

  getTimeBetweenCreationsExpression
- void setMaxCreations (unsigned long _maxCreations)

  setMaxCreations
- void setMaxCreations (std::string _maxCreationsExpression)

  setMaxCreations
- std::string getMaxCreations () const

  getMaxCreations
- unsigned int getEntitiesCreated () const

  getEntitiesCreated

- void setEntitiesCreated (unsigned int _entitiesCreated)

  setEntitiesCreated
- void setEntitiesPerCreation (unsigned int _entitiesPerCreation)

  setEntitiesPerCreation
- unsigned int getEntitiesPerCreation () const

  getEntitiesPerCreation
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _initBetweenReplications () override
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _createEditableDataDefinitions ()
- virtual void _createAttachedAttributes ()

Protected Member Functions inherited from ModelComponent

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber)=0

  Handles a dispatched event for this component type.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Protected Attributes

- EntityType * **_entityType** = nullptr
- const struct SourceModelComponent::DEFAULT_VALUES **DEFAULT**
- double **_firstCreation** = DEFAULT.firstCreation
- unsigned int **_entitiesPerCreation** = DEFAULT.entitiesPerCreation
- std::string **_maxCreationsExpression** = DEFAULT.maxCreationsExpression
- std::string **_timeBetweenCreationsExpression** = DEFAULT.timeBetweenCreationsExpression
- Util::TimeUnit **_timeBetweenCreationsTimeUnit** = DEFAULT.timeBetweenCreationsTimeUnit
- unsigned int **_entitiesCreatedSoFar** = 0

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

Additional Inherited Members

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

17.979.1 Detailed Description

Kernel base class for components that inject entities into the model (only Create derives from it so far), generalizing the fields Arena exposes on its "Create module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 31-32): Entity Type (`_entityType`), First Creation (`_firstCreation`), Entities per Arrival (`_entitiesPerCreation`), Max Arrivals (`_maxCreationsExpression`, a parsed expression rather than a static integer) and the time-between-creations expression/time unit.
A source component populates the initially empty future events list with the first creation events for every source component in the model.

17.979.2 Member Function Documentation

17.979.2.1 _check()

```
bool SourceModelComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.
Reimplemented in Create.

17.979.2.2 _createAttachedAttributes()

```
void SourceModelComponent::_createAttachedAttributes ( )  [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.979.2.3 _createEditableDataDefinitions()

```
void SourceModelComponent::_createEditableDataDefinitions ( )  [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.979.2.4 _initBetweenReplications()

```
void SourceModelComponent::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.979.2.5 _loadInstance()

```
bool SourceModelComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.
Reimplemented in Create.

17.979.2.6 _saveInstance()

```
void SourceModelComponent::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.
Reimplemented in Create.

17.979.2.7 getEntitiesCreated()

```
unsigned int SourceModelComponent::getEntitiesCreated ( ) const
```
getEntitiesCreated

Returns

17.979.2.8 getEntitiesPerCreation()

```
unsigned int SourceModelComponent::getEntitiesPerCreation ( ) const
```
getEntitiesPerCreation

Returns

17.979.2.9 getEntityType()

```
EntityType * SourceModelComponent::getEntityType ( ) const
```
getEntityType

Returns

17.979.2.10 getFirstCreation()

```
double SourceModelComponent::getFirstCreation ( ) const
```
getFirstCreation

Returns

17.979.2.11 getMaxCreations()

```
std::string SourceModelComponent::getMaxCreations ( ) const
```
getMaxCreations

Returns

17.979.2.12 getTimeBetweenCreationsExpression()

```
std::string SourceModelComponent::getTimeBetweenCreationsExpression ( ) const
```
getTimeBetweenCreationsExpression

Returns

17.979.2.13 getTimeUnit()

```
Util::TimeUnit SourceModelComponent::getTimeUnit ( ) const
```

getTimeUnit

Returns

17.979.2.14 setEntitiesCreated()

```
void SourceModelComponent::setEntitiesCreated (
            unsigned int _entitiesCreated )
```

setEntitiesCreated

Parameters

_entitiesCreated	

17.979.2.15 setEntitiesPerCreation()

```
void SourceModelComponent::setEntitiesPerCreation (
            unsigned int _entitiesPerCreation )
```

setEntitiesPerCreation

Parameters

_entitiesPerCreation	

17.979.2.16 setEntityType()

```
void SourceModelComponent::setEntityType (
            EntityType * _entityType )
```

setEntityType

Parameters

_entityType	

17.979.2.17 setEntityTypeName()

```
void SourceModelComponent::setEntityTypeName (
            std::string entityTypeName )
```

setEntityTypeName

Parameters

entityTypeName	

17.979.2.18 setFirstCreation()

```
void SourceModelComponent::setFirstCreation (
            double _firstCreation )
```

setFirstCreation

Parameters

_firstCreation	

17.979.2.19 setMaxCreations() [1/2]

```
void SourceModelComponent::setMaxCreations (
            std::string _maxCreationsExpression )
```

setMaxCreations

Parameters

_maxCreationsExpression	

17.979.2.20 setMaxCreations() [2/2]

```
void SourceModelComponent::setMaxCreations (
            unsigned long _maxCreations )
```

setMaxCreations

Parameters

_maxCreations	

17.979.2.21 setTimeBetweenCreationsExpression()

```
void SourceModelComponent::setTimeBetweenCreationsExpression (
            std::string _timeBetweenCreations,
            Util::TimeUnit _timeUnit = Util::TimeUnit::unknown )
```

setTimeBetweenCreationsExpression

Parameters

_timeBetweenCreations	
_timeUnit	

17.979.2.22 setTimeUnit()

```
void SourceModelComponent::setTimeUnit (
            Util::TimeUnit _timeUnit )
```

setTimeUnit

Parameters

_timeUnit	

17.979.2.23 show()

```
std::string SourceModelComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
Reimplemented in Create.
The documentation for this class was generated from the following files:

- kernel/simulator/SourceModelComponent.h
- kernel/simulator/SourceModelComponent.cpp

17.980 SparseValueStore Class Reference

Sparse N-dimensional numeric store using GenESyS textual index keys.
`#include <kernel/simulator/essentialPlugins/SparseValueStore.h>`

Public Member Functions

- double **value** (const std::string &index="") const

  Reads a sparse position, returning 0.0 when the position is absent.
- void **setValue** (double value, const std::string &index="")

  Writes a sparse position, creating it when needed.
- void **setValues** (const std::map< std::string, double > &values)

  Replaces all sparse values with a copy of the provided values.
- void **copyFrom** (const SparseValueStore &other)

  Replaces this store with a deep copy of another store.
- void **insertDimensionSize** (unsigned int size)

  Inserts one dimension size at the end of the dimension list.
- std::list< unsigned int > * **dimensionSizes** ()

  Returns mutable dimension sizes for compatibility with legacy callers.
- const std::list< unsigned int > * **dimensionSizes** () const

  Returns immutable dimension sizes.
- std::map< std::string, double > * **values** ()

  Returns mutable sparse values for compatibility with legacy callers.
- const std::map< std::string, double > * **values** () const

  Returns immutable sparse values.
- void **clear** ()

  Clears dimensions and values.
- void **loadDimensions** (PersistenceRecord *fields)

  Loads dimensions using the legacy fields dimensions/dimension[i].
- void **saveDimensions** (PersistenceRecord *fields, bool saveDefaultValues) const

  Saves dimensions using the legacy fields dimensions/dimension[i].
- void **loadValues** (PersistenceRecord *fields, const std::string &countField, const std::string &positionPrefix, const std::string &valuePrefix)

  Loads sparse values using the provided count/position/value field prefixes.
- void **saveValues** (PersistenceRecord *fields, const std::string &countField, const std::string &positionPrefix, const std::string &valuePrefix, bool saveDefaultValues) const

  Saves sparse values using the provided count/position/value field prefixes.
- std::string **showValues** () const

  Returns a compact textual representation used by show/debug paths.

Static Public Member Functions

- static std::string **makeIndexKey** (const std::vector< unsigned int > &indexes)

  Builds the textual key used by legacy sparse matrices from index values.
- static std::string **makeIndexKeyFromDoubles** (const std::vector< double > &indexes)

  Builds the textual key used by legacy sparse matrices from expression values.
- static std::string **appendIndexKey** (const std::string &currentKey, unsigned int index)

  Appends one index to an existing textual sparse key.
- static std::string **appendIndexKeyFromDouble** (const std::string &currentKey, double index)

  Appends one expression value to an existing textual sparse key.
- static std::string **normalizeIndexKey** (const std::string &index)

  Returns a normalized copy of an already textual key.

17.980.1 Detailed Description

Sparse N-dimensional numeric store using GenESyS textual index keys.
The store preserves the existing Variable semantics: the scalar position is represented by an empty key, multidimensional indexes are represented as comma-separated unsigned integers such as "1,2,3", and any missing position reads back as 0.0. It is intentionally small so Variable, Attribute and Entity can share the same behavior without changing the persistence format.
The documentation for this class was generated from the following file:

- kernel/simulator/essentialPlugins/SparseValueStore.h

17.981 MassActionOdeSystem::Species Struct Reference

Public Attributes

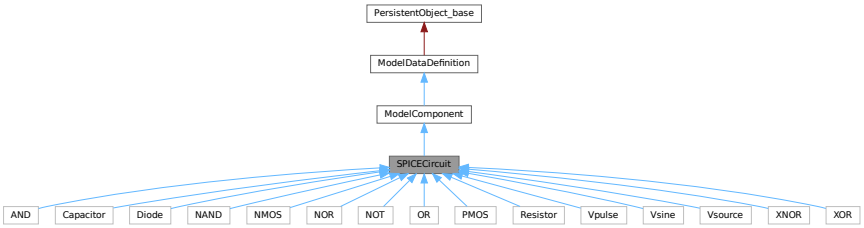
- std::string **name**
- double **initialAmount** = 0.0
- bool **boundaryCondition** = false
- bool **constant** = false

The documentation for this struct was generated from the following file:

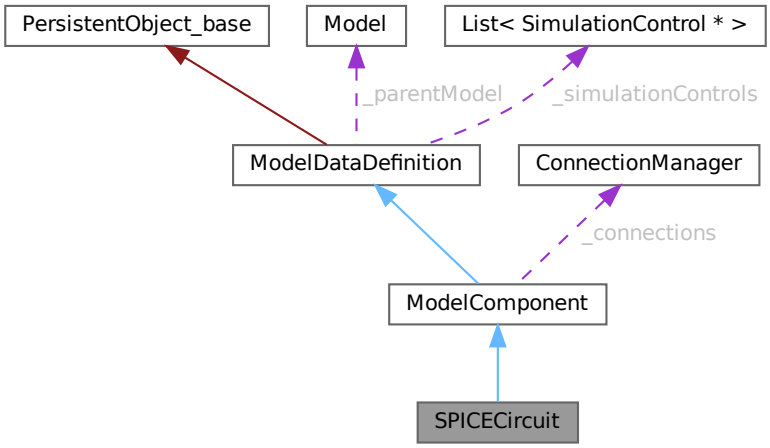
- tools/Biochemical/MassActionOdeSystem.h

17.982 SPICECircuit Class Reference

`#include <plugins/components/ElectronicsSimulation/SPICECircuit.h>`
Inheritance diagram for SPICECircuit:



Collaboration diagram for SPICECircuit:



Public Member Functions

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void **Build** ()
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Protected Attributes

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.982.1 Detailed Description

This component ...

17.982.2 Member Function Documentation

17.982.2.1 _createNonEditableDataDefinitions()

```
void SPICECircuit::_createNonEditableDataDefinitions ( )  [override], [protected], [virtual]
```

virtual protected methods that could be overriden by derived classes, if needed
This method is called by ModelChecker during model check. The component should check itself to verify if user parameters are ok (ex: correct syntax for the parser) and everithing in its parameters allow the model too run without errors in this component
This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
This method is called by ModelSimulation when initianting the replication. The model should set all value for a new replication (Ex: setting back to 0 any internal counter, clearing lists, etc.
Reimplemented from ModelDataDefinition.

17.982.2.2 _loadInstance()

```
bool SPICECircuit::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.
Reimplemented in Resistor, Vsource, Vpulse, Vsine, Capacitor, Diode, PMOS, NMOS, NOT, NAND, AND, NOR, OR, XOR, and XNOR.

17.982.2.3 _onDispatchEvent()

```
void SPICECircuit::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

This method is only for ModelComponents, not ModelDataElements.
Implements ModelComponent.

17.982.2.4 _saveInstance()

```
void SPICECircuit::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.
Reimplemented in Resistor, Vsource, Vpulse, Vsine, Capacitor, Diode, PMOS, NMOS, NOT, NAND, AND, NOR, OR, XOR, and XNOR.

17.982.2.5 show()

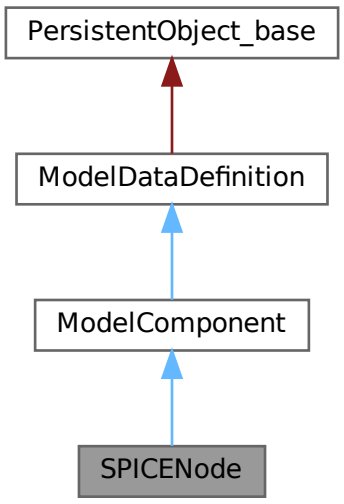
```
std::string SPICECircuit::show ( )  [override], [virtual]
```

virtual public methods
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

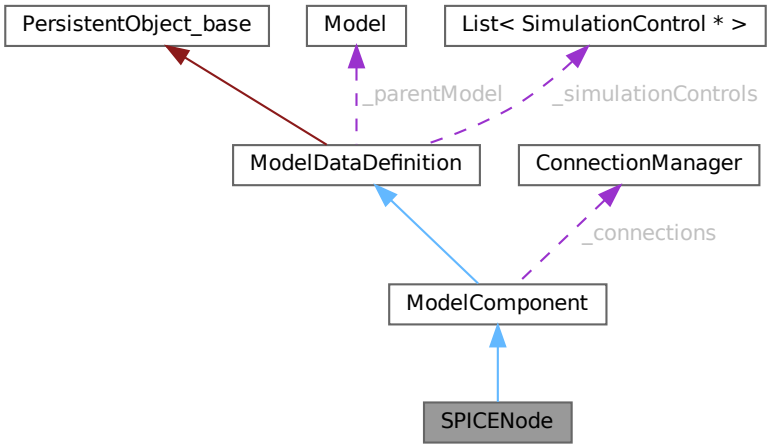
- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.983 SPICENode Class Reference

`#include <plugins/components/ElectronicsSimulation/SPICENode.h>`
Inheritance diagram for SPICENode:



Collaboration diagram for SPICENode:



Public Member Functions

- **SPICENode** (Model *model, std::string name="")

  constructors
- std::string **getNodeName** ()

  new public user methods for this component
- virtual std::string show () override

  virtual public methods
- **operator std::string** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.983.1 Detailed Description

This component ...

17.983.2 Member Function Documentation

17.983.2.1 _loadInstance()

```
bool SPICENode::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.983.2.2 _onDispatchEvent()

```
void SPICENode::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

This method is only for ModelComponents, not ModelDataElements.
Implements ModelComponent.

17.983.2.3 _saveInstance()

```
void SPICENode::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.983.2.4 show()

```
std::string SPICENode::show ( )  [override], [virtual]
```

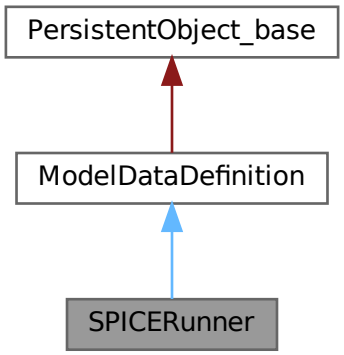
virtual public methods
Reimplemented from ModelComponent.

The documentation for this class was generated from the following files:

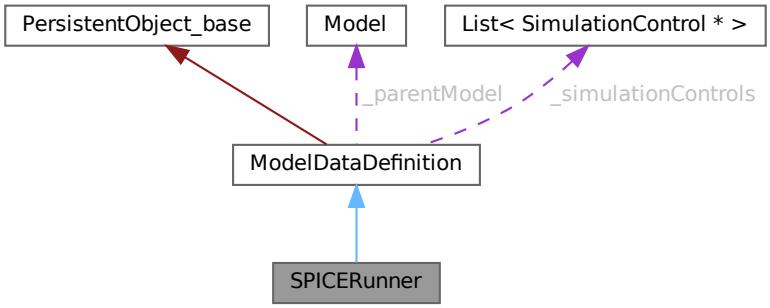
- plugins/components/ElectronicsSimulation/SPICENode.h
- plugins/components/ElectronicsSimulation/SPICENode.cpp

17.984 SPICERunner Class Reference

Inheritance diagram for SPICERunner:



Collaboration diagram for SPICERunner:



Public Member Functions

- **SPICERunner** (Model ∗model, std::string name="")

  constructors
- std::string **CompileSpiceFile** ()

  new public user methods for this component
- void **SendComponent** (std::string ∗instance, std::string subcircuit="", std::string model="")
- void **SendCallback** (std::function< void()> callback)
- void **ConfigSim** (double duration, double step)
- void **Run** ()

- void **PlotV** (std::string net)
- template<typename... Args>
  void **PlotV** (std::string net, Args... args)
- void **PlotVRelative** (std::string comparison_base, std::string net)
- template<typename... Args>
  void **PlotVRelative** (std::string comparison_base, std::string net, Args... args)
- void **PlotI** (std::string net)
- template<typename... Args>
  void **PlotI** (std::string net, Args... args)
- void **PlotIRelative** (std::string comparison_base, std::string net)
- template<typename... Args>
  void **PlotIRelative** (std::string comparison_base, std::string net, Args... args)
- double * **MeasurePeak** (std::string label, std::string peak, std::string quantity, std::string node, float start, float finish)
- double * **MeasureTrigTarg** (std::string label, std::string trig, float trig_value, std::string trig_inclin, std::string targ, float targ_value, std::string targ_inclin)
- void **ParseOutput** ()
- void **setRunnerCommand** (std::string runnerCommand)
- std::string **getRunnerCommand** () const
- void **setModelsPath** (std::string modelsPath)
- std::string **getModelsPath** () const
- void **setWorkingInputFilename** (std::string workingInputFilename)
- std::string **getWorkingInputFilename** () const
- void **setWorkingOutputFilename** (std::string workingOutputFilename)
- std::string **getWorkingOutputFilename** () const
- void **setWorkingDirectory** (std::string workingDirectory)
- std::string **getWorkingDirectory** () const
- std::string **getLastRunCommand** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)

- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.984.1 Member Function Documentation

17.984.1.1 _check()

```
bool SPICERunner::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

virtual protected methods that could be overriden by derived classes, if needed
This method is called by ModelChecker during model check. The component should check itself to verify if user parameters are ok (ex: correct syntax for the parser) and everithing in its parameters allow the model too run without errors in this component
Reimplemented from ModelDataDefinition.

17.984.1.2 _loadInstance()

```
bool SPICERunner::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelDataDefinition.

17.984.1.3 _saveInstance()

```
void SPICERunner::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.984.1.4 show()

```
std::string SPICERunner::show ( )  [override], [virtual]
```

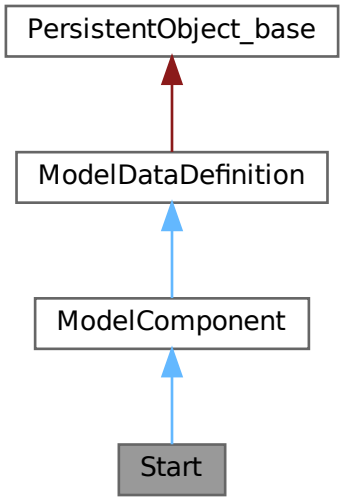
virtual public methods
Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/ExternalIntegration/SPICERunner.h
- plugins/data/ExternalIntegration/SPICERunner.cpp

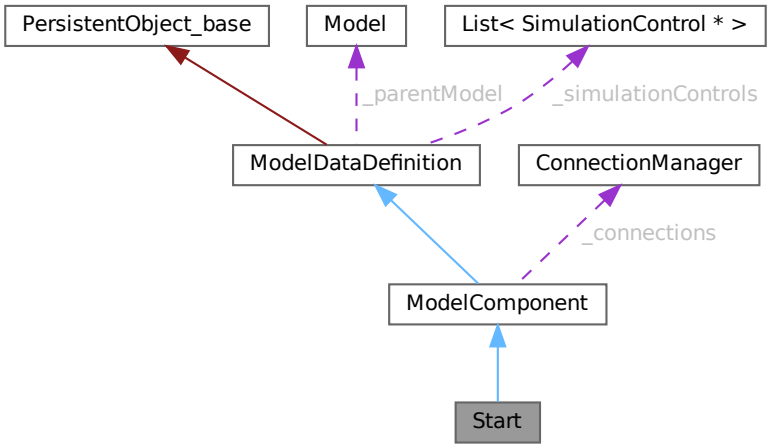
17.985 Start Class Reference

```
#include <plugins/components/MaterialHandling/Start.h>
```

Inheritance diagram for Start:



Collaboration diagram for Start:



Public Member Functions

- **Start** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setConveyor** (Conveyor *conveyor)
- Conveyor * **getConveyor** () const
- void **setVelocityExpression** (std::string velocityExpression)
- std::string **getVelocityExpression** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.985.1 Detailed Description

Start module DESCRIPTION The Start module changes the status of a conveyor from inactive to active. The conveyor may have been deactivated from either the Stop module or by initially being set to inactive at the start of the simulation. The velocity of the conveyor may be changed permanently when the conveyor is started. TYPICAL USES Start a bottling conveyor after scheduled maintenance Start a baggage claim conveyor when bags have arrived PROMPTS Prompt Description Name Unique name of the module that will be displayed in the flowchart. Conveyor Name Name of the conveyor to start. Velocity Speed of the conveyor once it begins to operate. This value will change the speed of the conveyor permanently, until it is changed in another module. Units Velocity time units.

17.985.2 Member Function Documentation

17.985.2.1 _check()

```
bool Start::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.985.2.2 _loadInstance()

```
bool Start::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.985.2.3 _onDispatchEvent()

```
void Start::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.985.2.4 _saveInstance()

```
void Start::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.985.2.5 show()

```
std::string Start::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

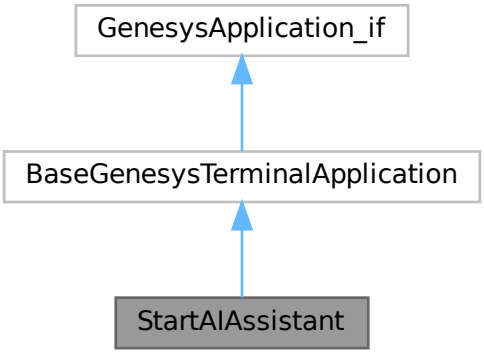
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

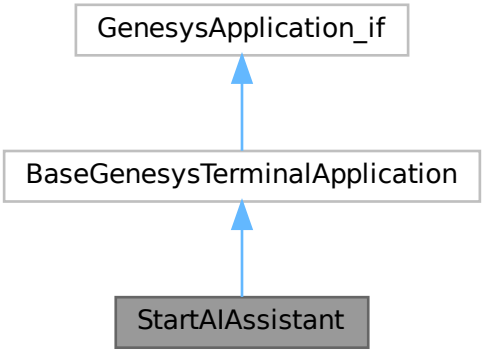
- plugins/components/MaterialHandling/Start.h
- plugins/components/MaterialHandling/Start.cpp

17.986 StartAIAssistant Class Reference

Inheritance diagram for StartAIAssistant:



Collaboration diagram for StartAIAssistant:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.986.1 Member Function Documentation

17.986.1.1 main()

```
int StartAIAssistant::main (
            int argc,
            char ** argv )  [override], [virtual]
```

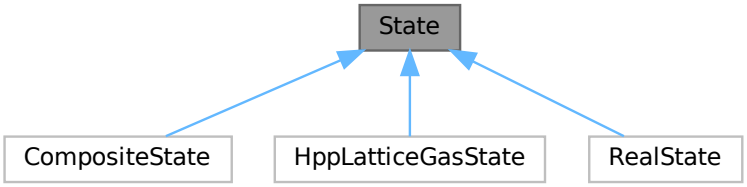
Reimplemented from BaseGenesysTerminalApplication.

The documentation for this class was generated from the following files:

- applications/modelSpecific/smarts/AI/StartAIAssistant.h
- applications/modelSpecific/smarts/AI/StartAIAssistant.cpp

17.987 State Class Reference

Inheritance diagram for State:



Public Member Functions

- **State** (long value)
- **State** (const State &orig)
- virtual std::unique_ptr< State > **clone** () const
- virtual bool **equals** (const State &other) const
- virtual std::string **toString** () const
- virtual std::string **show** () const
- virtual std::string **typeName** () const
- virtual long **getValue** () const
- virtual void **setValue** (long value)
- template<typename T >
  const T * **as** () const noexcept
- template<typename T >
  T * **as** () noexcept

Protected Attributes

- long **_value**

The documentation for this class was generated from the following file:

- plugins/components/ModalModel/CellularAutomata/State.h

17.988 GroProgramAst::Statement Struct Reference

Public Attributes

- std::string **sourceText** = ""

The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/GroProgramAst.h

17.989 EFSMNetworkDiagramLayout::StatePosition Struct Reference

Public Attributes

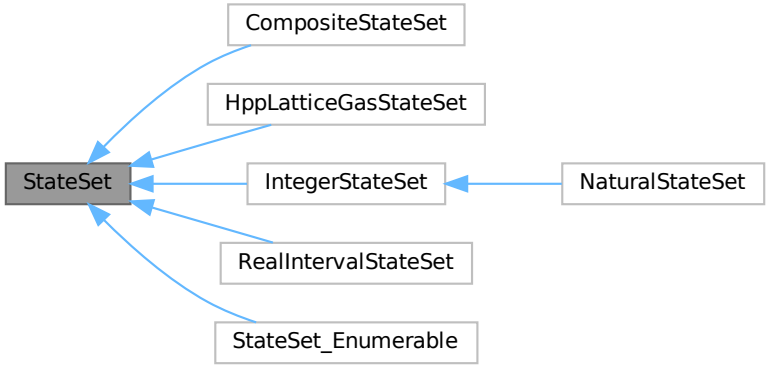
- unsigned long **stateId** = 0
- QString **stateName**
- QPointF **position**

The documentation for this struct was generated from the following file:

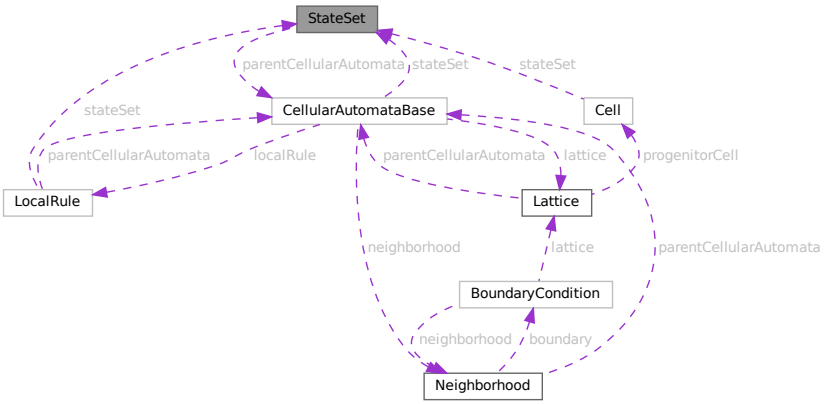
- applications/gui/genesys/services/EFSMNetworkDiagramLayout.h

17.990 StateSet Class Reference

Inheritance diagram for StateSet:



Collaboration diagram for StateSet:



Public Member Functions

- **StateSet** (CellularAutomataBase *parentCellularAutomata)
- **StateSet** (const StateSet &orig)
- virtual bool **contains** (const State &state) const
- virtual std::unique_ptr< State > **createDefaultState** () const
- virtual std::unique_ptr< State > **parseState** (const std::string &text) const
- virtual bool **isFinite** () const
- virtual bool **isDiscrete** () const
- virtual bool **isContinuous** () const
- virtual bool **isComposite** () const
- virtual std::vector< std::unique_ptr< State > > **enumerateStates** () const
- virtual std::string **name** () const

Protected Attributes

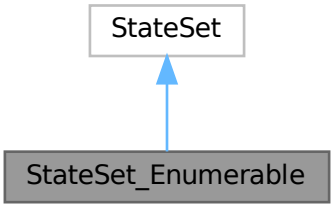
- CellularAutomataBase * **parentCellularAutomata**

The documentation for this class was generated from the following files:

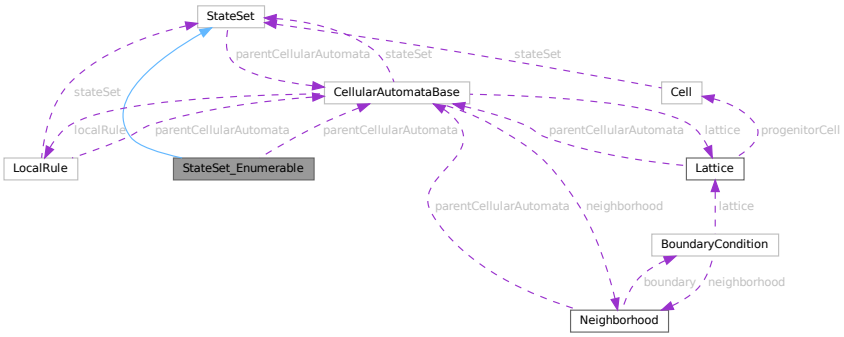
- plugins/components/ModalModel/CellularAutomata/StateSet.h
- plugins/components/ModalModel/CellularAutomata/StateSet.cpp

17.991 StateSet_Enumerable Class Reference

Inheritance diagram for StateSet_Enumerable:



Collaboration diagram for StateSet_Enumerable:



Public Member Functions

- **StateSet_Enumerable** (CellularAutomataBase *parentCellularAutomata, std::vector< State * > states={})
- **StateSet_Enumerable** (const StateSet_Enumerable &orig)
- virtual std::string **show** ()
- bool contains (const State &state) const override
- std::unique_ptr< State > createDefaultState () const override
- std::unique_ptr< State > parseState (const std::string &text) const override
- bool isFinite () const override
- bool isDiscrete () const override
- std::vector< std::unique_ptr< State > > enumerateStates () const override
- std::string name () const override
- unsigned int **size** ()
- void **addState** (State *state)
- void **addState** (const State &state)
- unsigned int **getStatesSize** ()
- State * **getState** (unsigned int rank)
- const State * **getState** (unsigned int rank) const
- State * **getState** (std::string name)
- void **setStates** (std::vector< State * > states)
- std::vector< State * > **getStates** () const

Public Member Functions inherited from StateSet

- **StateSet** (CellularAutomataBase *parentCellularAutomata)
- **StateSet** (const StateSet &orig)
- virtual bool **isContinuous** () const
- virtual bool **isComposite** () const

Protected Attributes

- CellularAutomataBase * **parentCellularAutomata**
- std::vector< std::unique_ptr< State > > **ownedStates**
- std::vector< State * > **states**

Protected Attributes inherited from StateSet

- CellularAutomataBase * **parentCellularAutomata**

17.991.1 Member Function Documentation

17.991.1.1 contains()

```
bool StateSet_Enumerable::contains (
            const State & state ) const  [override], [virtual]
```

Reimplemented from StateSet.

17.991.1.2 createDefaultState()

```
std::unique_ptr< State > StateSet_Enumerable::createDefaultState ( ) const  [override], [virtual]
```

Reimplemented from StateSet.

17.991.1.3 enumerateStates()

```
std::vector< std::unique_ptr< State > > StateSet_Enumerable::enumerateStates ( ) const  [override],
[virtual]
```

Reimplemented from StateSet.

17.991.1.4 isDiscrete()

```
bool StateSet_Enumerable::isDiscrete ( ) const  [override], [virtual]
```

Reimplemented from StateSet.

17.991.1.5 isFinite()

```
bool StateSet_Enumerable::isFinite ( ) const  [override], [virtual]
```

Reimplemented from StateSet.

17.991.1.6 name()

```
std::string StateSet_Enumerable::name ( ) const  [override], [virtual]
```

Reimplemented from StateSet.

17.991.1.7 parseState()

```
std::unique_ptr< State > StateSet_Enumerable::parseState (
            const std::string & text ) const  [override], [virtual]
```

Reimplemented from StateSet.

The documentation for this class was generated from the following files:

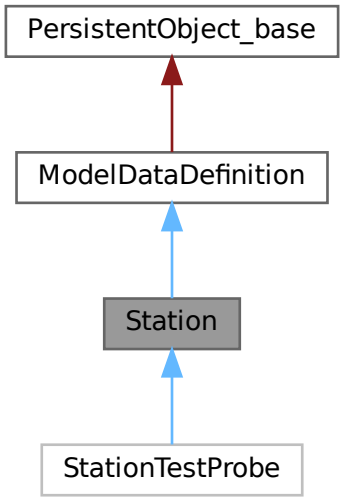
- plugins/components/ModalModel/CellularAutomata/StateSet_Enumerable.h
- plugins/components/ModalModel/CellularAutomata/StateSet_Enumerable.cpp

17.992 Station Class Reference

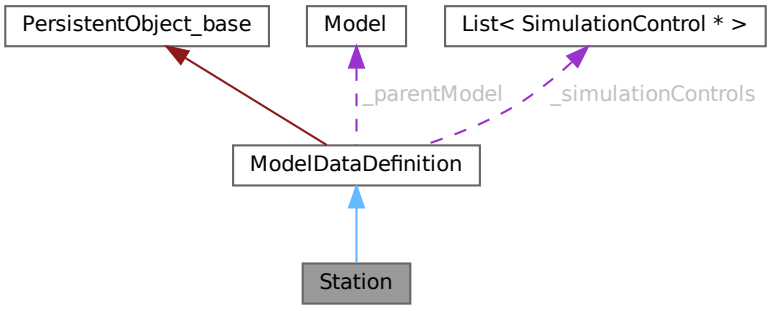
Data definition representing one physical/logical processing location, tracking the entities currently in it and their sojourn time.

`#include <plugins/data/MaterialHandling/Station.h>`

Inheritance diagram for Station:



Collaboration diagram for Station:



Public Member Functions

- **Station** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **initBetweenReplications** ()
- void **enter** (Entity *entity)
- void **leave** (Entity *entity)

- void **setEnterIntoStationComponent** (ModelComponent *_enterIntoStationComponent)
- ModelComponent * **getEnterIntoStationComponent** () const

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createInternalStatisticReporters () override
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Friends

- class **StationTestProbe**

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.992.1 Detailed Description

Data definition representing one physical/logical processing location, tracking the entities currently in it and their sojourn time.
Arena correspondence: Arena's "Station module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Transfer Panel", pp. 86-87). Note that in Arena, Station is a flowchart module placed on the diagram (with a matching Activity Area for cost/time roll-up), whereas in GenESyS this class is purely a data definition referenced by Material↩ Handling components (Enter, Route, PickStation, etc.); `enter()` / `leave()` and the `NumberInStation`/↩ `TimeInStation` statistics collectors are the closest correspondence to Arena's per-station/Activity-Area counts and times.
Known differences from Arena, still to verify against the MaterialHandling components in Phase B: no station-set concept (Arena's "Station Type = Set" with member list and Save Attribute), no parent Activity Area for hierarchical cost roll-up, and no associated-intersection field for a guided transporter network — GenESyS currently has no Transporter/Network data definitions at all (see the compatibility matrix).

17.992.2 Member Function Documentation

17.992.2.1 _check()

```
bool Station::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.992.2.2 _createAttachedAttributes()

```
void Station::_createAttachedAttributes ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.992.2.3 _createInternalStatisticReporters()

```
void Station::_createInternalStatisticReporters ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.992.2.4 _initBetweenReplications()

```
void Station::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.992.2.5 _loadInstance()

```
bool Station::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.992.2.6 _saveInstance()

```
void Station::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.992.2.7 show()

`std::string Station::show ( ) [override], [virtual]`

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

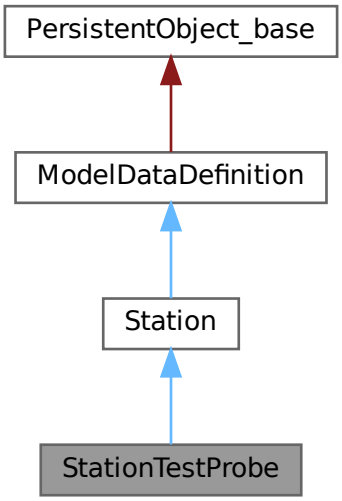
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

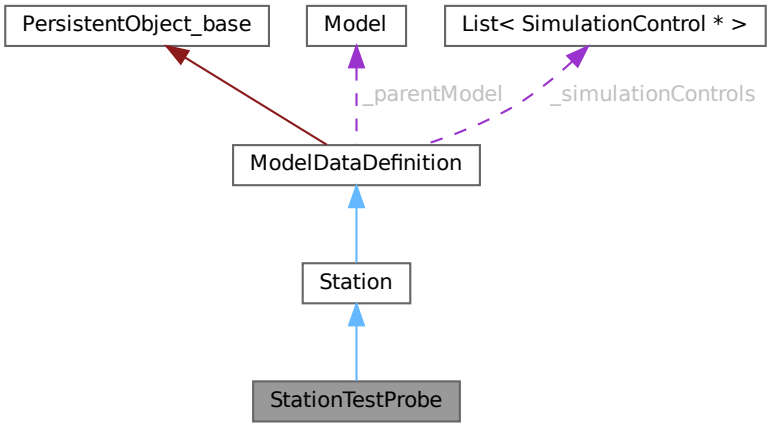
- plugins/data/MaterialHandling/Station.h
- plugins/data/MaterialHandling/Station.cpp

17.993 StationTestProbe Class Reference

Inheritance diagram for StationTestProbe:



Collaboration diagram for StationTestProbe:



Public Member Functions

- **StationTestProbe** (Model *model, const std::string &name="")
- void **CreateInternalAndAttachedDataProbe** ()
- void **InitBetweenReplicationsProbe** ()
- StatisticsCollector * **NumberInStationCollectorProbe** () const
- StatisticsCollector * **TimeInStationCollectorProbe** () const
- unsigned int **NumberInStationProbe** () const

Public Member Functions inherited from Station

- **Station** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **initBetweenReplications** ()
- void **enter** (Entity *entity)
- void **leave** (Entity *entity)
- void **setEnterIntoStationComponent** (ModelComponent *_enterIntoStationComponent)
- ModelComponent * **getEnterIntoStationComponent** () const

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Additional Inherited Members

Static Public Member Functions inherited from Station

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions inherited from Station

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createInternalStatisticReporters () override
- virtual void _createAttachedAttributes () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)

- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

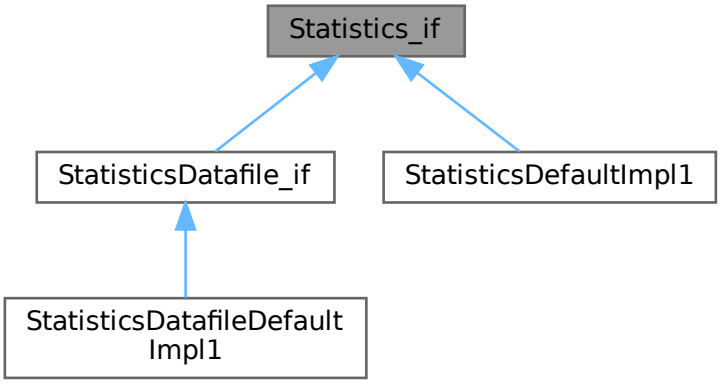
The documentation for this class was generated from the following file:

- tests/unit/test_simulator_runtime.cpp

17.994 Statistics_if Class Reference

Interface for statistical synthesis over values provided by a `Collector_if`.
`#include <kernel/statistics/Statistics_if.h>`

Inheritance diagram for Statistics_if:



Public Member Functions

- virtual Collector_if * getCollector () const =0
  Returns the data collector associated with these statistics.
- virtual void setCollector (Collector_if *collector)=0
  Sets the collector used as source for statistical calculations.
- virtual unsigned int numElements ()=0
  numElements
- virtual double min ()=0
  min
- virtual double max ()=0
  max
- virtual double average ()=0
  average
- virtual double variance ()=0
  variance
- virtual double stddeviation ()=0
  stddeviation
- virtual double variationCoef ()=0
  variationCoef
- virtual double halfWidthConfidenceInterval ()=0
  halfWidthConfidenceInterval
- virtual double confidenceLevel ()=0
  confidenceLevel
- virtual unsigned int newSampleSize (double halfWidth)=0
  newSampleSize
- virtual void setConfidenceLevel (double confidencelevel)=0
  setConfidenceLevel

17.994.1 Detailed Description

Interface for statistical synthesis over values provided by a `Collector_if`.
Implementations may compute metrics incrementally (online) or based on persisted datasets, but must expose a consistent API for descriptive and inferential measures.

17.994.2 Member Function Documentation

17.994.2.1 average()

```
virtual double Statistics_if::average ( )  [pure virtual]
```

average

Returns

Returns the sample mean of observations.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.2 confidenceLevel()

```
virtual double Statistics_if::confidenceLevel ( )  [pure virtual]
```

confidenceLevel

Returns

Returns the confidence level used for interval estimates.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.3 getCollector()

```
virtual Collector_if * Statistics_if::getCollector ( ) const  [pure virtual]
```

Returns the data collector associated with these statistics.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.4 halfWidthConfidenceInterval()

```
virtual double Statistics_if::halfWidthConfidenceInterval ( )  [pure virtual]
```

halfWidthConfidenceInterval

Returns

Returns the half-width of the current confidence interval.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.5 max()

```
virtual double Statistics_if::max ( )  [pure virtual]
```

max

Returns

Returns the maximum observed value.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.6 min()

```
virtual double Statistics_if::min ( )  [pure virtual]
```

min

Returns

Returns the minimum observed value.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.7 newSampleSize()

```
virtual unsigned int Statistics_if::newSampleSize (
            double halfWidth )  [pure virtual]
```

newSampleSize

Parameters

halfWidth	

Returns

Estimates the sample size needed to reach a target half-width.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.8 numElements()

```
virtual unsigned int Statistics_if::numElements ( )  [pure virtual]
```

numElements

Returns

Returns the number of observations considered in calculations.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.9 setCollector()

```
virtual void Statistics_if::setCollector (
            Collector_if * collector )  [pure virtual]
```

Sets the collector used as source for statistical calculations.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.10 setConfidenceLevel()

```
virtual void Statistics_if::setConfidenceLevel (
            double confidencelevel )  [pure virtual]
```

setConfidenceLevel

Parameters

confidencelevel	

Sets the confidence level used in inferential calculations.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.11 stddeviation()

```
virtual double Statistics_if::stddeviation ( )  [pure virtual]
```

stddeviation

Returns

Returns the sample standard deviation.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.12 variance()

```
virtual double Statistics_if::variance ( )  [pure virtual]
```

variance

Returns

Returns the sample variance of observations.
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.

17.994.2.13 variationCoef()

```
virtual double Statistics_if::variationCoef ( )  [pure virtual]
```

variationCoef

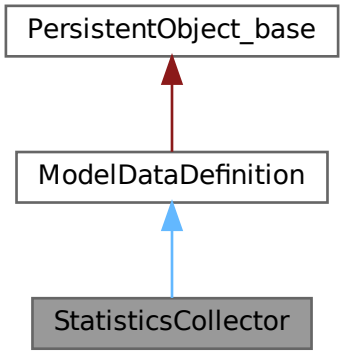
Returns

Returns the coefficient of variation (standard deviation / mean).
Implemented in StatisticsDatafileDefaultImpl1, and StatisticsDefaultImpl1.
The documentation for this class was generated from the following file:

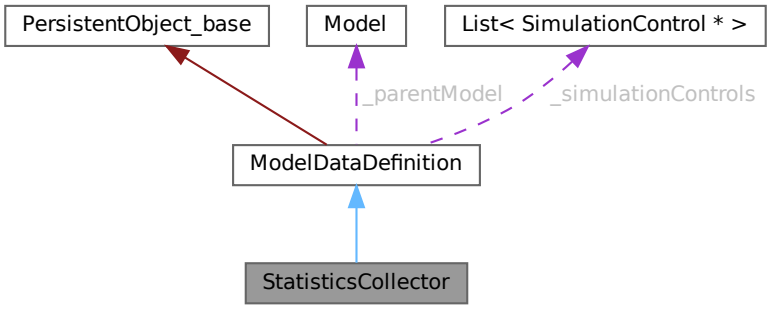
- kernel/statistics/Statistics_if.h

17.995 StatisticsCollector Class Reference

`#include <kernel/simulator/essentialPlugins/StatisticsCollector.h>`
Inheritance diagram for StatisticsCollector:



Collaboration diagram for StatisticsCollector:



Public Member Functions

- **StatisticsCollector** (Model *model, std::string name="", ModelDataDefinition *parent=nullptr, bool insert↩ IntoModel=true)
- virtual std::string show () override

  Returns a textual representation of this data definition.
- ModelDataDefinition * getParent () const

  getParent
- Statistics_if * getStatistics () const

  getStatistics

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

not just an easy access to trace manager, but wrappers to check if specific trace level applies

- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.995.1 Detailed Description

The StatisticsCollector is the ModelDataDefinition responsible for collecting data from the model (using the Collector) and simultaneouly keeping statistics updated (using the Statistics)

17.995.2 Member Function Documentation

17.995.2.1 _check()

```
bool StatisticsCollector::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.995.2.2 _loadInstance()

```
bool StatisticsCollector::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.995.2.3 _saveInstance()

```
void StatisticsCollector::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.995.2.4 getParent()

```
ModelDataDefinition * StatisticsCollector::getParent ( ) const
```

getParent

Returns

17.995.2.5 getStatistics()

```
Statistics_if * StatisticsCollector::getStatistics ( ) const
```

getStatistics

Returns

17.995.2.6 show()

```
std::string StatisticsCollector::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

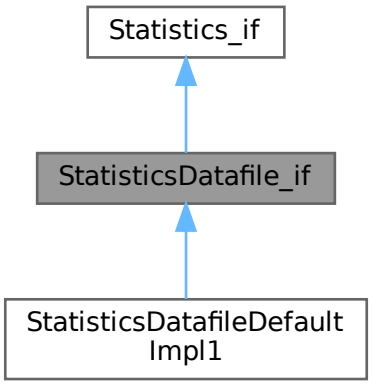
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

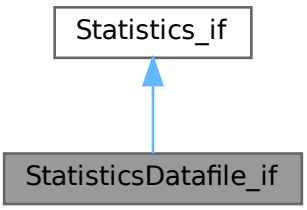
- kernel/simulator/essentialPlugins/StatisticsCollector.h
- kernel/simulator/essentialPlugins/StatisticsCollector.cpp

17.996 StatisticsDatafile_if Class Reference

Extension of `Statistics_if` for metrics that require full data access.
`#include <kernel/statistics/StatisticsDataFile_if.h>`
Inheritance diagram for StatisticsDatafile_if:



Collaboration diagram for StatisticsDatafile_if:



Public Member Functions

- virtual double mode ()=0
  mode
- virtual double mediane ()=0
  mediane
- virtual double quartil (unsigned short num)=0
  quartil
- virtual double decil (unsigned short num)=0
  decil
- virtual double centil (unsigned short num)=0
  centil
- virtual void setHistogramNumClasses (unsigned short num)=0
  setHistogramNumClasses
- virtual unsigned short histogramNumClasses ()=0
  histogramNumClasses
- virtual double histogramClassLowerLimit (unsigned short classNum)=0
  histogramClassLowerLimit
- virtual unsigned int histogramClassFrequency (unsigned short classNum)=0
  histogramClassFrequency

Public Member Functions inherited from Statistics_if

- virtual Collector_if * getCollector () const =0
  Returns the data collector associated with these statistics.
- virtual void setCollector (Collector_if *collector)=0
  Sets the collector used as source for statistical calculations.
- virtual unsigned int numElements ()=0
  numElements
- virtual double min ()=0
  min
- virtual double max ()=0
  max
- virtual double average ()=0
  average
- virtual double variance ()=0
  variance

- virtual double stddeviation ()=0

  stddeviation
- virtual double variationCoef ()=0

  variationCoef
- virtual double halfWidthConfidenceInterval ()=0

  halfWidthConfidenceInterval
- virtual double confidenceLevel ()=0

  confidenceLevel
- virtual unsigned int newSampleSize (double halfWidth)=0

  newSampleSize
- virtual void setConfidenceLevel (double confidencelevel)=0

  setConfidenceLevel

17.996.1 Detailed Description

Extension of `Statistics_if` for metrics that require full data access.
This interface targets collector implementations backed by data files, enabling order-based measures (median, quantiles) and histogram-related operations.

17.996.2 Member Function Documentation

17.996.2.1 centil()

```
virtual double StatisticsDatafile_if::centil (
            unsigned short num )  [pure virtual]
```

centil

Parameters

num	

Returns

Returns the requested percentile (1 to 99).
Implemented in StatisticsDatafileDefaultImpl1.

17.996.2.2 decil()

```
virtual double StatisticsDatafile_if::decil (
            unsigned short num )  [pure virtual]
```

decil

Parameters

num	

Returns

Returns the requested decile (1 to 9).
Implemented in StatisticsDatafileDefaultImpl1.

17.996.2.3 histogramClassFrequency()

```
virtual unsigned int StatisticsDatafile_if::histogramClassFrequency (
            unsigned short classNum )  [pure virtual]
```

histogramClassFrequency

Parameters

classNum	

Returns

Returns the absolute frequency of the requested histogram class.
Implemented in StatisticsDatafileDefaultImpl1.

17.996.2.4 histogramClassLowerLimit()

```
virtual double StatisticsDatafile_if::histogramClassLowerLimit (
            unsigned short classNum )  [pure virtual]
```

histogramClassLowerLimit

Parameters

classNum	

Returns

Returns the lower bound of the histogram class at index `classNum`.
Implemented in StatisticsDatafileDefaultImpl1.

17.996.2.5 histogramNumClasses()

```
virtual unsigned short StatisticsDatafile_if::histogramNumClasses ( )  [pure virtual]
```

histogramNumClasses

Returns

Returns the configured number of histogram classes.
Implemented in StatisticsDatafileDefaultImpl1.

17.996.2.6 mediane()

```
virtual double StatisticsDatafile_if::mediane ( )  [pure virtual]
```

mediane

Returns

Returns the sample median.
Implemented in StatisticsDatafileDefaultImpl1.

17.996.2.7 mode()

```
virtual double StatisticsDatafile_if::mode ( )  [pure virtual]
```

mode

Returns

Returns the mode of the sample stored in file.
Implemented in StatisticsDatafileDefaultImpl1.

17.996.2.8 quartil()

```
virtual double StatisticsDatafile_if::quartil (
            unsigned short num )  [pure virtual]
```

quartil

Parameters

num	

Returns

Returns the requested quartile (1 to 3).
Implemented in StatisticsDatafileDefaultImpl1.

17.996.2.9 setHistogramNumClasses()

```
virtual void StatisticsDatafile_if::setHistogramNumClasses (
            unsigned short num )  [pure virtual]
```

setHistogramNumClasses

Parameters

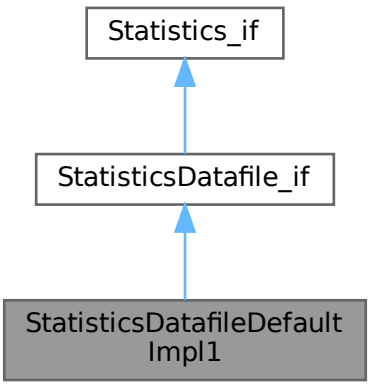
num	

Sets the number of classes used in the histogram.
Implemented in StatisticsDatafileDefaultImpl1.
The documentation for this class was generated from the following file:

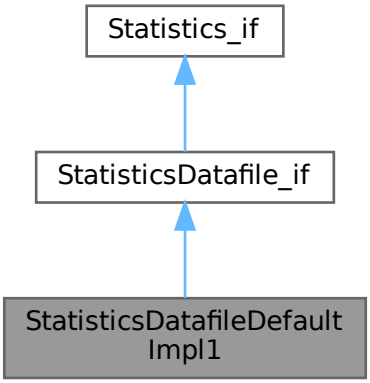
- kernel/statistics/StatisticsDataFile_if.h

17.997 StatisticsDatafileDefaultImpl1 Class Reference

Default implementation of `StatisticsDatafile_if` for file-backed samples.
`#include <kernel/statistics/StatisticsDataFileDefaultImpl.h>`
Inheritance diagram for StatisticsDatafileDefaultImpl1:



Collaboration diagram for StatisticsDatafileDefaultImpl1:



Public Member Functions

- **StatisticsDatafileDefaultImpl1** ()

  Creates statistics object with default collector implementation.
- virtual ~**StatisticsDatafileDefaultImpl1** ()

  Releases owned collector/sorter resources when configured as owner.
- virtual Collector_if * getCollector () const override

  Returns the collector currently used as data source.
- virtual void setCollector (Collector_if *collector) override

  Sets a collector to be used as data source for calculations.
- virtual unsigned int numElements () override

  Returns number of observations available in the collector.
- virtual double min () override

  Returns minimum observed value.
- virtual double max () override

  Returns maximum observed value.
- virtual double average () override

  Returns arithmetic mean.
- virtual double variance () override

  Returns sample variance.
- virtual double stddeviation () override

  Returns sample standard deviation.
- virtual double variationCoef () override

  Returns coefficient of variation.
- virtual double halfWidthConfidenceInterval () override

  Returns current confidence interval half-width.
- virtual double confidenceLevel () override

  Returns configured confidence level.
- virtual unsigned int newSampleSize (double halfWidth) override

  Estimates sample size needed for a desired half-width.
- virtual void setConfidenceLevel (double confidencelevel) override

Sets confidence level used in inferential calculations.

- virtual double mode () override

  Returns sample mode.
- virtual double mediane () override

  Returns sample median.
- virtual double quartil (unsigned short num) override

  Returns requested quartile.
- virtual double decil (unsigned short num) override

  Returns requested decile.
- virtual double centil (unsigned short num) override

  Returns requested percentile.
- virtual void setHistogramNumClasses (unsigned short num) override

  Sets number of histogram classes.
- virtual unsigned short histogramNumClasses () override

  Returns current number of histogram classes.
- virtual double histogramClassLowerLimit (unsigned short classNum) override

  Returns lower limit of histogram class `classNum`.
- virtual unsigned int histogramClassFrequency (unsigned short classNum) override

  Returns absolute frequency of histogram class `classNum`.

17.997.1 Detailed Description

Default implementation of `StatisticsDatafile_if` for file-backed samples.
Computes descriptive/inferential statistics from a `CollectorDatafile_if` and caches results to avoid recomputation until source data changes.

17.997.2 Member Function Documentation

17.997.2.1 average()

```
double StatisticsDatafileDefaultImpl1::average ( )  [override], [virtual]
```

Returns arithmetic mean.
Implements Statistics_if.

17.997.2.2 centil()

```
double StatisticsDatafileDefaultImpl1::centil (
            unsigned short num )  [override], [virtual]
```

Returns requested percentile.
Implements StatisticsDatafile_if.

17.997.2.3 confidenceLevel()

```
double StatisticsDatafileDefaultImpl1::confidenceLevel ( )  [override], [virtual]
```

Returns configured confidence level.
Implements Statistics_if.

17.997.2.4 decil()

```
double StatisticsDatafileDefaultImpl1::decil (
            unsigned short num )  [override], [virtual]
```

Returns requested decile.
Implements StatisticsDatafile_if.

17.997.2.5 getCollector()

```
Collector_if * StatisticsDatafileDefaultImpl1::getCollector ( ) const  [override], [virtual]
```

Returns the collector currently used as data source.
Implements Statistics_if.

17.997.2.6 halfWidthConfidenceInterval()

```
double StatisticsDatafileDefaultImpl1::halfWidthConfidenceInterval ( )  [override], [virtual]
```

Returns current confidence interval half-width.
Implements Statistics_if.

17.997.2.7 histogramClassFrequency()

```
unsigned int StatisticsDatafileDefaultImpl1::histogramClassFrequency (
            unsigned short classNum )  [override], [virtual]
```

Returns absolute frequency of histogram class `classNum`.
Implements StatisticsDatafile_if.

17.997.2.8 histogramClassLowerLimit()

```
double StatisticsDatafileDefaultImpl1::histogramClassLowerLimit (
            unsigned short classNum )  [override], [virtual]
```

Returns lower limit of histogram class `classNum`.
Implements StatisticsDatafile_if.

17.997.2.9 histogramNumClasses()

```
unsigned short StatisticsDatafileDefaultImpl1::histogramNumClasses ( )  [override], [virtual]
```

Returns current number of histogram classes.
Implements StatisticsDatafile_if.

17.997.2.10 max()

```
double StatisticsDatafileDefaultImpl1::max ( )  [override], [virtual]
```

Returns maximum observed value.
Implements Statistics_if.

17.997.2.11 mediane()

```
double StatisticsDatafileDefaultImpl1::mediane ( )  [override], [virtual]
```

Returns sample median.
Implements StatisticsDatafile_if.

17.997.2.12 min()

```
double StatisticsDatafileDefaultImpl1::min ( )  [override], [virtual]
```

Returns minimum observed value.
Implements Statistics_if.

17.997.2.13 mode()

```
double StatisticsDatafileDefaultImpl1::mode ( )  [override], [virtual]
```

Returns sample mode.
Implements StatisticsDatafile_if.

17.997.2.14 newSampleSize()

```
unsigned int StatisticsDatafileDefaultImpl1::newSampleSize (
            double halfWidth ) [override], [virtual]
```

Estimates sample size needed for a desired half-width.
Implements Statistics_if.

17.997.2.15 numElements()

```
unsigned int StatisticsDatafileDefaultImpl1::numElements ( ) [override], [virtual]
```

Returns number of observations available in the collector.
Implements Statistics_if.

17.997.2.16 quartil()

```
double StatisticsDatafileDefaultImpl1::quartil (
            unsigned short num ) [override], [virtual]
```

Returns requested quartile.
Implements StatisticsDatafile_if.

17.997.2.17 setCollector()

```
void StatisticsDatafileDefaultImpl1::setCollector (
            Collector_if * collector ) [override], [virtual]
```

Sets a collector to be used as data source for calculations.
Implements Statistics_if.

17.997.2.18 setConfidenceLevel()

```
void StatisticsDatafileDefaultImpl1::setConfidenceLevel (
            double confidencelevel ) [override], [virtual]
```

Sets confidence level used in inferential calculations.
Implements Statistics_if.

17.997.2.19 setHistogramNumClasses()

```
void StatisticsDatafileDefaultImpl1::setHistogramNumClasses (
            unsigned short num ) [override], [virtual]
```

Sets number of histogram classes.
Implements StatisticsDatafile_if.

17.997.2.20 stddeviation()

```
double StatisticsDatafileDefaultImpl1::stddeviation ( ) [override], [virtual]
```

Returns sample standard deviation.
Implements Statistics_if.

17.997.2.21 variance()

```
double StatisticsDatafileDefaultImpl1::variance ( ) [override], [virtual]
```

Returns sample variance.
Implements Statistics_if.

17.997.2.22 variationCoef()

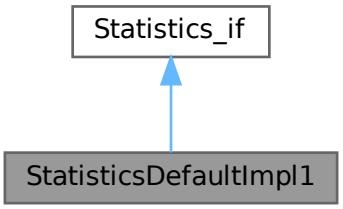
```
double StatisticsDatafileDefaultImpl1::variationCoef ( ) [override], [virtual]
```

Returns coefficient of variation.
Implements Statistics_if.
The documentation for this class was generated from the following files:

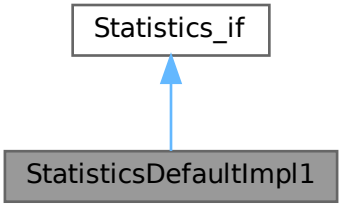
- kernel/statistics/StatisticsDataFileDefaultImpl.h
- kernel/statistics/StatisticsDataFileDefaultImpl.cpp

17.998 StatisticsDefaultImpl1 Class Reference

Default online statistics implementation over a `Collector_if` source.
`#include <kernel/statistics/StatisticsDefaultImpl1.h>`
Inheritance diagram for StatisticsDefaultImpl1:



Collaboration diagram for StatisticsDefaultImpl1:



Public Member Functions

- StatisticsDefaultImpl1 ()

  Creates statistics object using configured default collector implementation.
- **StatisticsDefaultImpl1** (Collector_if ∗collector)

  Creates statistics object bound to an external collector.
- virtual Collector_if ∗ getCollector () const override

  Returns current collector data source.
- virtual void setCollector (Collector_if ∗collector) override

  Sets collector source and reconnects internal callbacks.
- virtual unsigned int numElements () override

  Returns number of observed samples.
- virtual double min () override

  Returns minimum observed value.
- virtual double max () override

Returns maximum observed value.

- virtual double average () override

  Returns arithmetic mean.
- virtual double variance () override

  Returns sample variance.
- virtual double stddeviation () override

  Returns sample standard deviation.
- virtual double variationCoef () override

  Returns coefficient of variation.
- virtual double halfWidthConfidenceInterval () override

  Returns confidence interval half-width.
- virtual double confidenceLevel () override

  Returns configured confidence level.
- virtual unsigned int newSampleSize (double halfWidth) override

  Estimates sample size required for desired half-width.
- virtual void setConfidenceLevel (double confidencelevel) override

  Sets confidence level used in inferential formulas.

17.998.1 Detailed Description

Default online statistics implementation over a `Collector_if` source.
Updates core metrics incrementally through collector callbacks, avoiding complete rescans for every query.

17.998.2 Constructor & Destructor Documentation

17.998.2.1 StatisticsDefaultImpl1()

```
StatisticsDefaultImpl1::StatisticsDefaultImpl1 ( )
```

Creates statistics object using configured default collector implementation.
When constructor is invoked without a Collector, it is taken from Traits<Statistics_if>::CollectorImplementation configuration

17.998.3 Member Function Documentation

17.998.3.1 average()

```
double StatisticsDefaultImpl1::average ( )  [override], [virtual]
```

Returns arithmetic mean.
Implements Statistics_if.

17.998.3.2 confidenceLevel()

```
double StatisticsDefaultImpl1::confidenceLevel ( )  [override], [virtual]
```

Returns configured confidence level.
Implements Statistics_if.

17.998.3.3 getCollector()

```
Collector_if * StatisticsDefaultImpl1::getCollector ( ) const  [override], [virtual]
```

Returns current collector data source.
Implements Statistics_if.

17.998.3.4 halfWidthConfidenceInterval()

```
double StatisticsDefaultImpl1::halfWidthConfidenceInterval ( )  [override], [virtual]
```

Returns confidence interval half-width.
Implements Statistics_if.

17.998.3.5 max()

```
double StatisticsDefaultImpl1::max ( )  [override], [virtual]
```

Returns maximum observed value.
Implements Statistics_if.

17.998.3.6 min()

```
double StatisticsDefaultImpl1::min ( )  [override], [virtual]
```

Returns minimum observed value.
Implements Statistics_if.

17.998.3.7 newSampleSize()

```
unsigned int StatisticsDefaultImpl1::newSampleSize (
            double halfWidth )  [override], [virtual]
```

Estimates sample size required for desired half-width.
Implements Statistics_if.

17.998.3.8 numElements()

```
unsigned int StatisticsDefaultImpl1::numElements ( )  [override], [virtual]
```

Returns number of observed samples.
Implements Statistics_if.

17.998.3.9 setCollector()

```
void StatisticsDefaultImpl1::setCollector (
            Collector_if * collector )  [override], [virtual]
```

Sets collector source and reconnects internal callbacks.
Implements Statistics_if.

17.998.3.10 setConfidenceLevel()

```
void StatisticsDefaultImpl1::setConfidenceLevel (
            double confidencelevel )  [override], [virtual]
```

Sets confidence level used in inferential formulas.
Implements Statistics_if.

17.998.3.11 stddeviation()

```
double StatisticsDefaultImpl1::stddeviation ( )  [override], [virtual]
```

Returns sample standard deviation.
Implements Statistics_if.

17.998.3.12 variance()

```
double StatisticsDefaultImpl1::variance ( )  [override], [virtual]
```

Returns sample variance.
Implements Statistics_if.

17.998.3.13 variationCoef()

```
double StatisticsDefaultImpl1::variationCoef ( )  [override], [virtual]
```

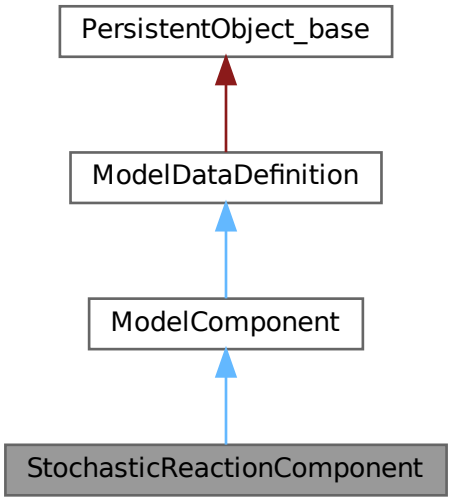
Returns coefficient of variation.
Implements Statistics_if.
The documentation for this class was generated from the following files:

- kernel/statistics/StatisticsDefaultImpl1.h
- kernel/statistics/StatisticsDefaultImpl1.cpp

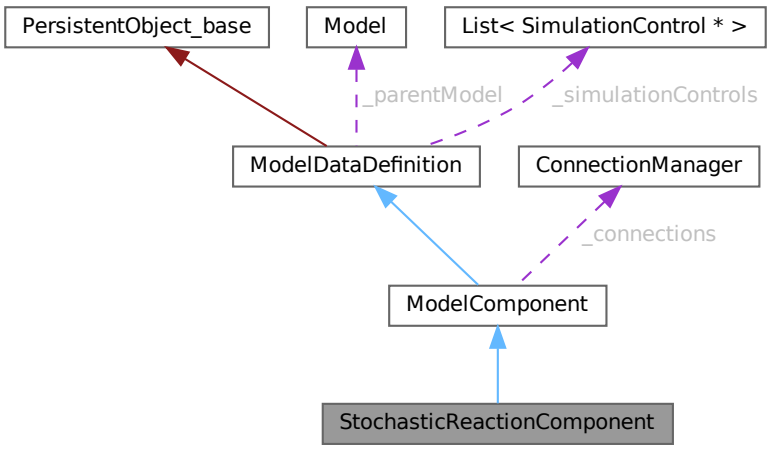
17.999 StochasticReactionComponent Class Reference

`#include <plugins/components/WholeCellModeling/StochasticReactionComponent.h>`

Inheritance diagram for StochasticReactionComponent:



Collaboration diagram for StochasticReactionComponent:



Public Member Functions

- **StochasticReactionComponent** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)
- WholeCellState * **getWholeCellState** () const

- void **setTimeWindow** (double timeWindow)
- double **getTimeWindow** () const
- void **setAdvanceWholeCellClock** (bool advance)
- bool **getAdvanceWholeCellClock** () const
- void **setRandomSeed** (unsigned int seed)
- unsigned int **getRandomSeed** () const
- int **getLastReactionCount** () const
- double **getLastSimulatedTime** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)

- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.999.1 Detailed Description

Gillespie Direct Method SSA runner for whole-cell stochastic simulations.
On each entity arrival the component runs the Direct Method (Gillespie 1977) for a configurable time window [0, timeWindow] using all StochasticReactionRule definitions registered in the model. Species counts are read from and written back to the referenced WholeCellState (if set) or directly to MolecularSpecies objects in the model.
Contract (Q7b): Input: current molecule counts from WholeCellState (or MolecularSpecies)

- resource budget from WholeCellState Output: updated counts written back to WholeCellState; entity forwarded

Resource cap: reactions whose firing would deplete a species below its resource budget floor are skipped, matching the fair-allocation contract of the WholeCell M. genitalium model (Karr et al. 2012).

17.999.2 Member Function Documentation

17.999.2.1 _check()

```
bool StochasticReactionComponent::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.999.2.2 _createEditableDataDefinitions()

```
void StochasticReactionComponent::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.999.2.3 _initBetweenReplications()

```
void StochasticReactionComponent::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.999.2.4 _loadInstance()

```
bool StochasticReactionComponent::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.999.2.5 _onDispatchEvent()

```
void StochasticReactionComponent::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.999.2.6 _saveInstance()

```
void StochasticReactionComponent::_saveInstance (
            PersistenceRecord * fields,
```

```
bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.999.2.7 show()

```
std::string StochasticReactionComponent::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

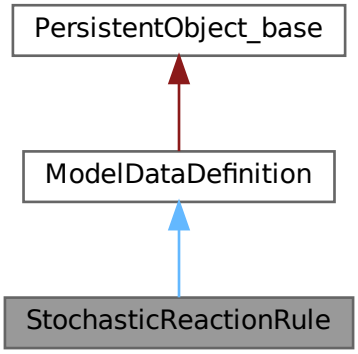
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/WholeCellModeling/StochasticReactionComponent.h
- plugins/components/WholeCellModeling/StochasticReactionComponent.cpp

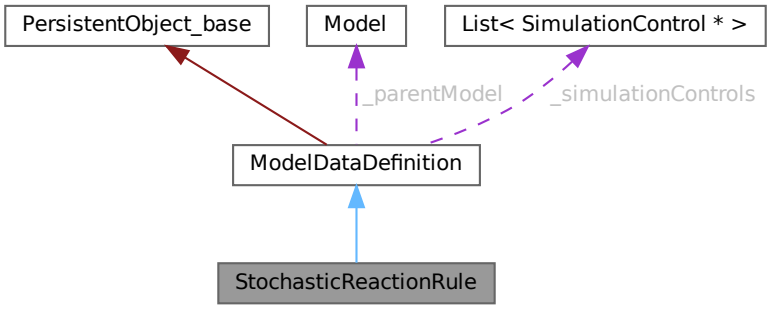
17.1000 StochasticReactionRule Class Reference

```
#include <plugins/data/WholeCellModeling/StochasticReactionRule.h>
```

Inheritance diagram for StochasticReactionRule:



Collaboration diagram for StochasticReactionRule:



Classes

- struct StoichiometricTerm

Public Member Functions

- **StochasticReactionRule** (Model *model, std::string name="")
- void **addReactant** (std::string speciesName, int stoichiometry=1)
- void **addProduct** (std::string speciesName, int stoichiometry=1)
- void **clearReactants** ()
- void **clearProducts** ()
- const std::vector< StoichiometricTerm > & **getReactants** () const
- const std::vector< StoichiometricTerm > & **getProducts** () const
- void **setRateConstant** (double rateConstant)
- double **getRateConstant** () const
- void **setCompartment** (std::string compartment)
- std::string **getCompartment** () const
- double **computePropensity** (const std::map< std::string, int > &counts) const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

Validates a data definition instance.

- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)

- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1000.1 Detailed Description

Stochastic reaction rule for Gillespie SSA whole-cell simulations.
Reactants and products are named by MolecularSpecies with integer stoichiometry. The propensity is computed as k * combinatorial(counts), where the combinatorial term is the product of C(x_i, n_i) for each reactant species i with stoichiometry n_i.
The rate constant k uses stochastic (microscopic) units: s^-1 for unimolecular, molecules^-1 * s^-1 for bimolecular reactions.

17.1000.2 Member Function Documentation

17.1000.2.1 _check()

```
bool StochasticReactionRule::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1000.2.2 _loadInstance()

```
bool StochasticReactionRule::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1000.2.3 _saveInstance()

```
void StochasticReactionRule::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.1000.2.4 show()

```
std::string StochasticReactionRule::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

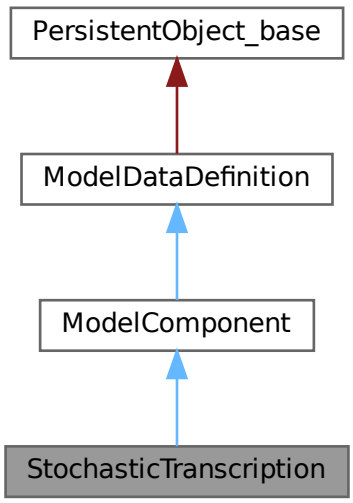
Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/WholeCellModeling/StochasticReactionRule.h
- plugins/data/WholeCellModeling/StochasticReactionRule.cpp

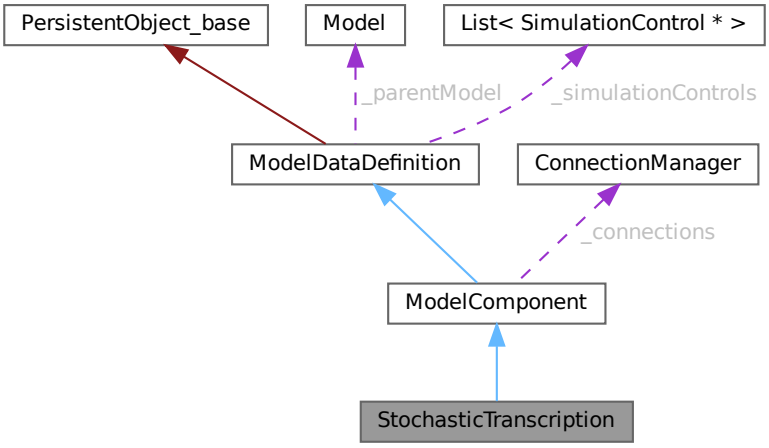
17.1001 StochasticTranscription Class Reference

```
#include <plugins/components/WholeCellModeling/StochasticTranscription.h>
```

Inheritance diagram for StochasticTranscription:



Collaboration diagram for StochasticTranscription:



Public Member Functions

- **StochasticTranscription** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)
- WholeCellState * **getWholeCellState** () const
- void **setElongationRate** (double rate)
- double **getElongationRate** () const
- void **setMeanGeneLength** (double length)
- double **getMeanGeneLength** () const
- void **setBindingProbability** (double prob)
- double **getBindingProbability** () const
- void **setTimeWindow** (double window)
- double **getTimeWindow** () const
- void **setMRNASpeciesPrefix** (std::string prefix)
- std::string **getMRNASpeciesPrefix** () const
- void **setRnapCountKey** (std::string key)
- std::string **getRnapCountKey** () const
- void **setRandomSeed** (unsigned int seed)
- unsigned int **getRandomSeed** () const
- int **getLastSynthesizedCount** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.

- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions

- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1001.1 Detailed Description

Stochastic mRNA synthesis via Poisson tau-leaping for whole-cell simulations.
Models the M. genitalium Transcription process (Karr et al. 2012): λ_i = (elongationRate / meanGeneLength) $\times$ bindingProbability $\times$ RNAP_free $\times$ Δt N_synth_i $\sim$ Poisson(λ_i)
This is a tau-leaping approximation (not full Gillespie) appropriate for transcription because RNAP count ($\sim$200) acting over many genes ($\sim$473) with a 1–10 s window produces moderate propensities where the Poisson approximation is valid. Full Gillespie is implemented in StochasticReactionComponent for individual reactions with low copy numbers.
Parameters loaded from WholeCellState (set by loadParameters()): param.rnaPolymeraseElongationRate — 50 nt/s (from parameters.json)
mRNA species in WholeCellState are identified by the prefix set in mRNASpeciesPrefix (default "mRNA_"). RNAP free count is read from the WholeCellState molecule count with key rnapCountKey (default "RNAP_free").

17.1001.2 Member Function Documentation

17.1001.2.1 _check()

```
bool StochasticTranscription::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1001.2.2 _createEditableDataDefinitions()

```
void StochasticTranscription::_createEditableDataDefinitions ( )  [override], [protected],
[virtual]
```

Reimplemented from ModelDataDefinition.

17.1001.2.3 _initBetweenReplications()

```
void StochasticTranscription::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1001.2.4 _loadInstance()

```
bool StochasticTranscription::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.1001.2.5 _onDispatchEvent()

```
void StochasticTranscription::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.1001.2.6 _saveInstance()

```
void StochasticTranscription::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.1001.2.7 show()

```
std::string StochasticTranscription::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

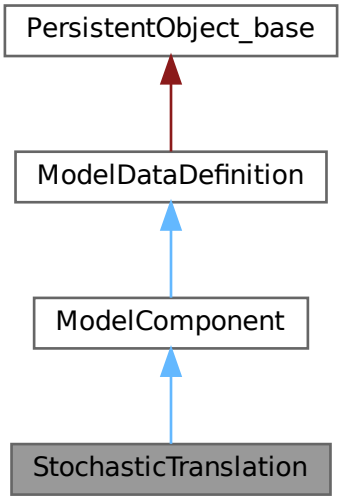
Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/WholeCellModeling/StochasticTranscription.h

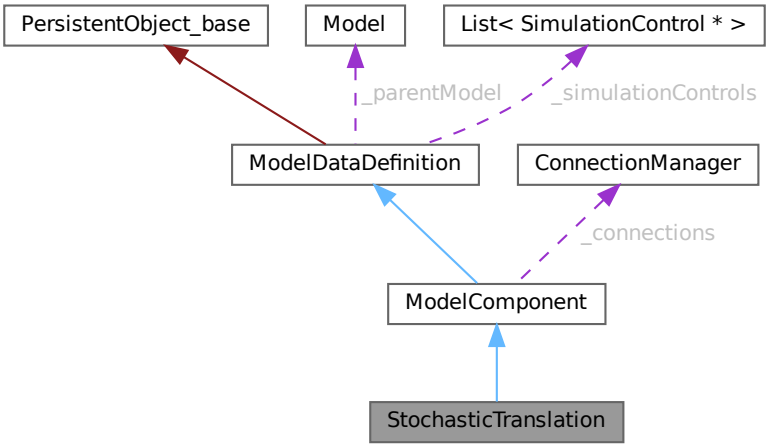
- plugins/components/WholeCellModeling/StochasticTranscription.cpp

17.1002 StochasticTranslation Class Reference

`#include <plugins/components/WholeCellModeling/StochasticTranslation.h>`
Inheritance diagram for StochasticTranslation:



Collaboration diagram for StochasticTranslation:



Public Member Functions

- **StochasticTranslation** (Model *model, std::string name="")
- void **setWholeCellState** (WholeCellState *state)

- WholeCellState * **getWholeCellState** () const
- void **setElongationRate** (double rate)
- double **getElongationRate** () const
- void **setMeanProteinLength** (double length)
- double **getMeanProteinLength** () const
- void **setTimeWindow** (double window)
- double **getTimeWindow** () const
- void **setMRNASpeciesPrefix** (std::string prefix)
- std::string **getMRNASpeciesPrefix** () const
- void **setProteinSpeciesPrefix** (std::string prefix)
- std::string **getProteinSpeciesPrefix** () const
- void **setRibosomeCountKey** (std::string key)
- std::string **getRibosomeCountKey** () const
- void **setRandomSeed** (unsigned int seed)
- unsigned int **getRandomSeed** () const
- int **getLastSynthesizedCount** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

Creates internal data required by a component instance.

- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)

- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1002.1 Detailed Description

Stochastic protein synthesis via Poisson tau-leaping for whole-cell simulations.
Models the M. genitalium Translation process (Karr et al. 2012): λ_i = (ribosomeElongationRate / meanProtein↩ Length) $\times$ mRNA_count_i $\times$ ribosome_free / max(1, total_mRNA) $\times$ Δt N_protein_i $\sim$ Poisson(λ_i)
Each mRNA template (prefix proteinSpeciesPrefix) is matched to its corresponding mRNA species (prefix m↩ RNASpeciesPrefix) in WholeCellState. Free ribosome count is read from WholeCellState with key ribosome↩ CountKey.
Parameters loaded from WholeCellState (set by loadParameters()): param.ribosomeElongationRate — 16 AA/s (from parameters.json)
mRNA degradation is not modeled here; use a separate StochasticReactionRule for mRNA decay (first-order, rate $\sim$1/mRNA_halflife).

17.1002.2 Member Function Documentation

17.1002.2.1 _check()

```
bool StochasticTranslation::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1002.2.2 _createEditableDataDefinitions()

```
void StochasticTranslation::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1002.2.3 _initBetweenReplications()

```
void StochasticTranslation::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1002.2.4 _loadInstance()

```
bool StochasticTranslation::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.1002.2.5 _onDispatchEvent()

```
void StochasticTranslation::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.1002.2.6 _saveInstance()

```
void StochasticTranslation::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.1002.2.7 show()

`std::string StochasticTranslation::show ( ) [override], [virtual]`
virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/WholeCellModeling/StochasticTranslation.h
- plugins/components/WholeCellModeling/StochasticTranslation.cpp

17.1003 BioReaction::StoichiometricTerm Struct Reference

Public Attributes

- std::string **speciesName**
- double **stoichiometry** = 1.0

The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/BioReaction.h

17.1004 MassActionOdeSystem::StoichiometricTerm Struct Reference

Public Attributes

- unsigned int **speciesIndex** = 0
- double **stoichiometry** = 1.0

The documentation for this struct was generated from the following file:

- tools/Biochemical/MassActionOdeSystem.h

17.1005 MetabolicReaction::StoichiometricTerm Struct Reference

Public Attributes

- std::string **speciesName**
- double **stoichiometry** = 1.0

The documentation for this struct was generated from the following file:

- plugins/data/BiochemicalSimulation/MetabolicReaction.h

17.1006 StochasticReactionRule::StoichiometricTerm Struct Reference

Public Attributes

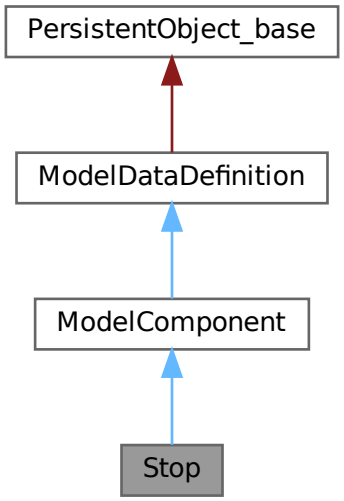
- std::string **speciesName**
- int **stoichiometry** = 1

The documentation for this struct was generated from the following file:

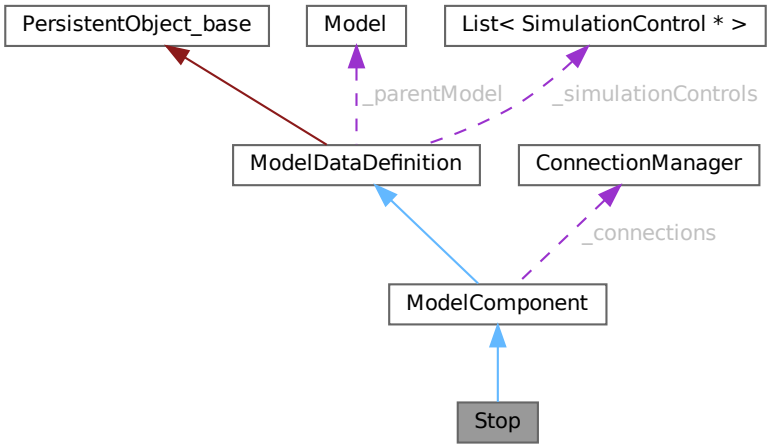
- plugins/data/WholeCellModeling/StochasticReactionRule.h

17.1007 Stop Class Reference

`#include <plugins/components/MaterialHandling/Stop.h>`
Inheritance diagram for Stop:



Collaboration diagram for Stop:



Public Member Functions

- **Stop** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setConveyor** (Conveyor *conveyor)
- Conveyor * **getConveyor** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()

- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create∗ hooks
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition ∗child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition ∗data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void ∗thisobject, double time, Entity ∗entity, ModelComponent ∗component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void ∗thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void ∗thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager ∗ _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1007.1 Detailed Description

Stop module DESCRIPTION The Stop module sets the operational status of a conveyor to inactive. The conveyor may have been activated from either the Start module or by initially being set to active at the start of the simulation. When the entity enters the Stop module, the conveyor will stop immediately, regardless of the type of conveyor or the number of entities currently on the conveyor. TYPICAL USES Stop a baggage conveyor after a pre-determined amount of time Stop a conveyor for scheduled maintenance PROMPTS Prompt Description Name Unique name of the module that will be displayed in the flowchart. Conveyor Name Name of the conveyor to stop.

17.1007.2 Member Function Documentation

17.1007.2.1 _check()

```
bool Stop::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1007.2.2 _loadInstance()

```
bool Stop::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.1007.2.3 _onDispatchEvent()

```
void Stop::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.1007.2.4 _saveInstance()

```
void Stop::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.1007.2.5 show()

```
std::string Stop::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

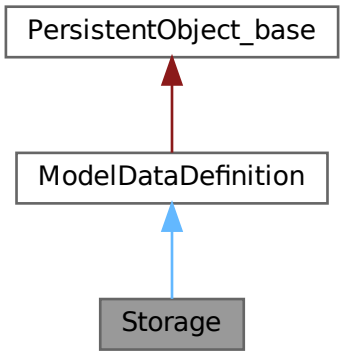
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

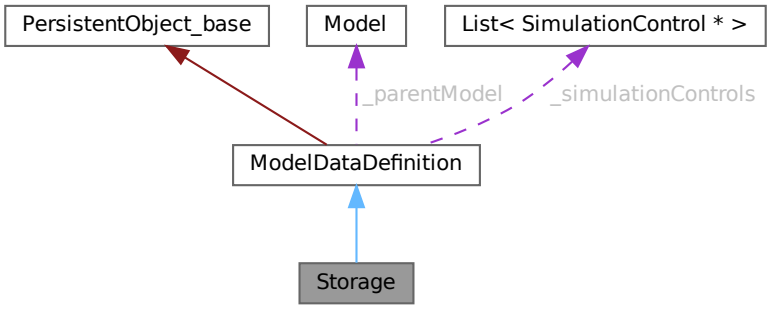
- plugins/components/MaterialHandling/Stop.h
- plugins/components/MaterialHandling/Stop.cpp

17.1008 Storage Class Reference

Data definition representing a named storage location with explicit area/capacity/density accounting.
`#include <plugins/data/MaterialHandling/Storage.h>`
Inheritance diagram for Storage:



Collaboration diagram for Storage:



Public Member Functions

- **Storage** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **setTotalArea** (double _totalArea)
- double **getTotalArea** () const
- void **setCapacity** (unsigned int _capacity)
- unsigned int **getCapacity** () const
- void **setUnitsPerArea** (double _unitsPerArea)
- double **getUnitsPerArea** () const
- unsigned int **getCurrentOccupation** () const
- bool **store** (unsigned int quantity=1)
- bool **unstore** (unsigned int quantity=1)

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation () override
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

not just an easy access to trace manager, but wrappers to check if specific trace level applies

- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1008.1 Detailed Description

Data definition representing a named storage location with explicit area/capacity/density accounting.
Arena correspondence: Arena's "Storage module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 78), which is a near-empty placeholder object (only a Name) auto-created by the Store/Unstore modules and only needed explicitly as a Storage-Set member.
Known difference from Arena (GenESyS extends the concept rather than narrowing it): this class carries real capacity semantics — `_totalArea`, `_capacity` and `_unitsPerArea` — that Arena's Storage module does not model at all; Arena tracks storage occupancy purely through animation, not through a declared capacity/area contract.

17.1008.2 Member Function Documentation

17.1008.2.1 _check()

```
bool Storage::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1008.2.2 _getParserChangesInformation()

```
ParserChangesInformation * Storage::_getParserChangesInformation ( )  [override], [protected],
[virtual]
```

This method returns all changes in the parser that are needed by plugins of this ModelDatas. When connecting a new plugin, ParserChangesInformation are used to change parser source code, whch is after compiled and dinamically linked to to simulator kernel to reflect the changes
Reimplemented from ModelDataDefinition.

17.1008.2.3 _initBetweenReplications()

```
void Storage::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1008.2.4 _loadInstance()

```
bool Storage::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1008.2.5 _saveInstance()

```
void Storage::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Persist Storage parameters using symmetric keys from _loadInstance.
Reimplemented from ModelDataDefinition.

17.1008.2.6 show()

```
std::string Storage::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

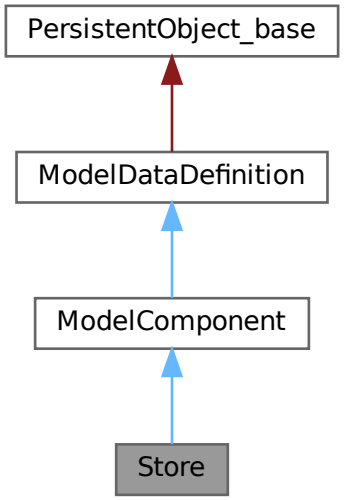
Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/MaterialHandling/Storage.h
- plugins/data/MaterialHandling/Storage.cpp

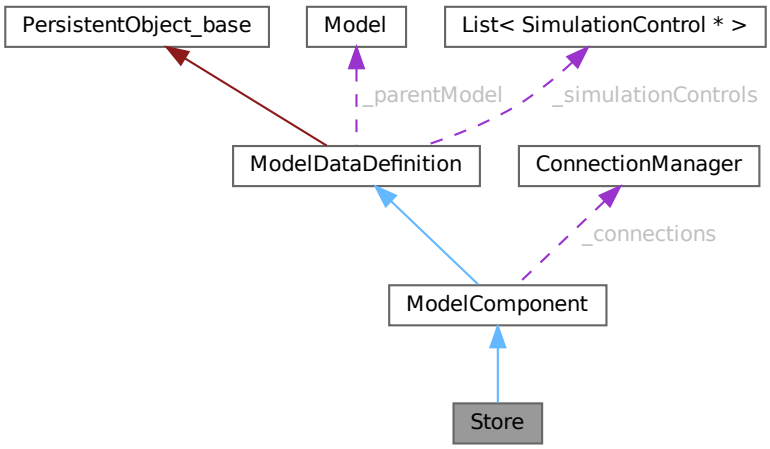
17.1009 Store Class Reference

Placeholder for adding an entity to a Storage (§5.11).
`#include <plugins/components/MaterialHandling/Store.h>`

Inheritance diagram for Store:



Collaboration diagram for Store:



Public Member Functions

- **Store** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setStorage** (Storage *storage)
- Storage * **getStorage** () const
- void **setQuantityExpression** (std::string quantityExpression)
- std::string **getQuantityExpression** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _createEditableDataDefinitions () override
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1009.1 Detailed Description

Placeholder for adding an entity to a Storage (§5.11).
Arena correspondence: the "Store module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 66).

Warning

> Incomplete stub template: this class declares no fields at all, and `_onDispatchEvent()` unconditionally forwards the entity without touching any Storage. `_check()`/`_loadInstance()`/`_saveInstance()` are not implemented either. See `docs/ai_assistants/reference/ARENA_GENESYS_↩ COMPATIBILITY.md` §6.17.

17.1009.2 Member Function Documentation

17.1009.2.1 _check()

```
bool Store::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1009.2.2 _createEditableDataDefinitions()

```
void Store::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1009.2.3 _loadInstance()

```
bool Store::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.1009.2.4 _onDispatchEvent()

```
void Store::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.1009.2.5 _saveInstance()

```
void Store::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.1009.2.6 show()

```
std::string Store::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/MaterialHandling/Store.h
- plugins/components/MaterialHandling/Store.cpp

17.1010 testing::internal::internal_stream_operator_without_lexical_↩ name_lookup::StreamPrinter Struct Reference

Static Public Member Functions

- template<typename T , typename = typename std::enable_if< !std::is_member_pointer<T>::value>::type, typename = decltype(std↩ ::declval<std::ostream&>() << std::declval<const T&>())>
  static void **PrintValue** (const T &value, ::std::ostream *os)

The documentation for this struct was generated from the following file:

- gtest/gtest/gtest-printers.h

17.1011 testing::internal::String Class Reference

Static Public Member Functions

- static const char * **CloneCString** (const char *c_str)
- static bool **CStringEquals** (const char *lhs, const char *rhs)
- static std::string **ShowWideCString** (const wchar_t *wide_c_str)
- static bool **WideCStringEquals** (const wchar_t *lhs, const wchar_t *rhs)
- static bool **CaseInsensitiveCStringEquals** (const char *lhs, const char *rhs)
- static bool **CaseInsensitiveWideCStringEquals** (const wchar_t *lhs, const wchar_t *rhs)
- static bool **EndsWithCaseInsensitive** (const std::string &str, const std::string &suffix)
- static std::string **FormatIntWidth2** (int value)
- static std::string **FormatIntWidthN** (int value, int width)
- static std::string **FormatHexInt** (int value)
- static std::string **FormatHexUInt32** (uint32_t value)
- static std::string **FormatByte** (unsigned char value)

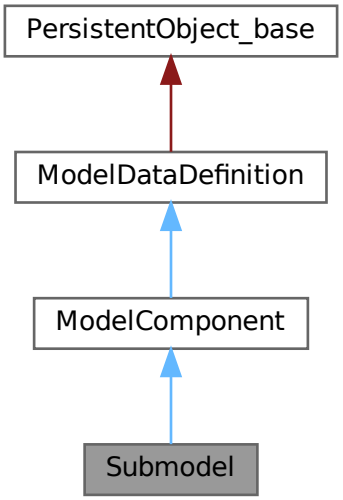
The documentation for this class was generated from the following files:

- gtest/gtest/internal/gtest-string.h
- gtest/src/gtest.cc

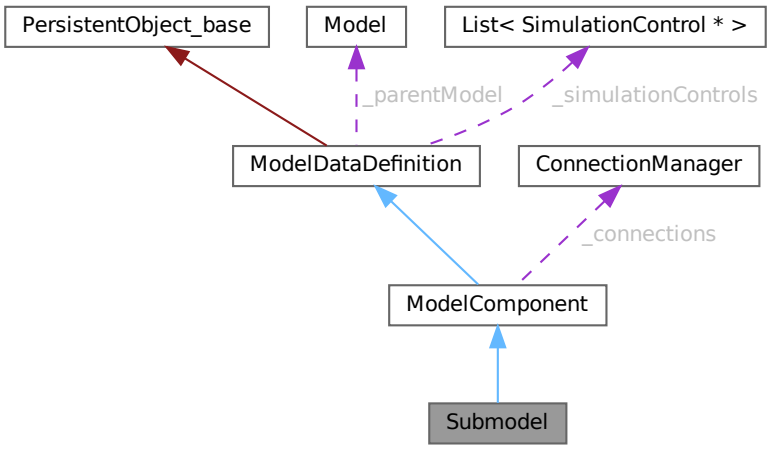
17.1012 Submodel Class Reference

`#include <plugins/components/ModalModel/Submodel.h>`

Inheritance diagram for Submodel:



Collaboration diagram for Submodel:



Public Member Functions

- **Submodel** (Model *model, std::string name="")
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

constructors

- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions

- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1012.1 Detailed Description

This component ...

17.1012.2 Member Function Documentation

17.1012.2.1 _check()

```
bool Submodel::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1012.2.2 _loadInstance()

```
bool Submodel::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.1012.2.3 _onDispatchEvent()

```
void Submodel::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.1012.2.4 _saveInstance()

```
void Submodel::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.1012.2.5 show()

```
std::string Submodel::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

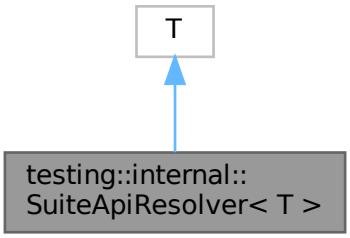
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

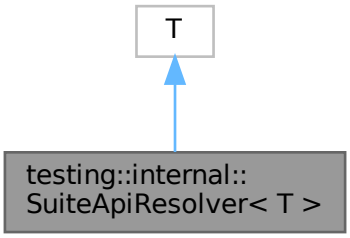
- plugins/components/ModalModel/Submodel.h
- plugins/components/ModalModel/Submodel.cpp

17.1013 testing::internal::SuiteApiResolver< T > Struct Template Reference

Inheritance diagram for testing::internal::SuiteApiResolver< T >:



Collaboration diagram for testing::internal::SuiteApiResolver< T >:



Public Types

- using **Test** = typename std::conditional< sizeof(T) !=0, ::testing::Test, void >::type

Static Public Member Functions

- static SetUpTearDownSuiteFuncType **GetSetUpCaseOrSuite** (const char *filename, int line_num)
- static SetUpTearDownSuiteFuncType **GetTearDownCaseOrSuite** (const char *filename, int line_num)

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.1014 yy::genesyspp_parser::symbol_kind Struct Reference

Symbol kinds.

`#include <parser/GenesysParser.h>`

Public Types

- enum symbol_kind_type {
  YYNTOKENS = 95 , **S_YYEMPTY** = -2 , **S_YYEOF** = 0 , **S_YYerror** = 1 ,
  S_YYUNDEF = 2 , **S_NUMD** = 3 , **S_NUMH** = 4 , **S_CTEZERO** = 5 ,
  S_oLE = 6 , **S_oGE** = 7 , **S_oEQ** = 8 , **S_oNE** = 9 ,
  S_oAND = 10 , **S_oOR** = 11 , **S_oNAND** = 12 , **S_oXOR** = 13 ,
  S_oNOT = 14 , **S_fSIN** = 15 , **S_fCOS** = 16 , **S_fROUND** = 17 ,
  S_fMOD = 18 , **S_fTRUNC** = 19 , **S_fFRAC** = 20 , **S_fEXP** = 21 ,
  S_fSQRT = 22 , **S_fLOG** = 23 , **S_fLN** = 24 , **S_mathMIN** = 25 ,
  S_mathMAX = 26 , **S_fVAL** = 27 , **S_fEVAL** = 28 , **S_fLENG** = 29 ,
  S_fRND1 = 30 , **S_fEXPO** = 31 , **S_fNORM** = 32 , **S_fUNIF** = 33 ,
  S_fWEIB = 34 , **S_fLOGN** = 35 , **S_fGAMM** = 36 , **S_fERLA** = 37 ,
  S_fTRIA = 38 , **S_fBETA** = 39 , **S_fDISC** = 40 , **S_fTNOW** = 41 ,
  S_fTFIN = 42 , **S_fMAXREP** = 43 , **S_fNUMREP** = 44 , **S_fIDENT** = 45 ,
  S_simulEntitiesWIP = 46 , **S_cIF** = 47 , **S_cELSE** = 48 , **S_cFOR** = 49 ,
  S_cTO = 50 , **S_cDO** = 51 , **S_ATRIB** = 52 , **S_CSTAT** = 53 ,
  S_COUNTER = 54 , **S_SIMRESP** = 55 , **S_SIMCTRL** = 56 , **S_fTAVG** = 57 ,
  S_fCOUNT = 58 , **S_ILLEGAL** = 59 , **S_RESOURCE** = 60 , **S_fNR** = 61 ,
  S_fMR = 62 , **S_fIRF** = 63 , **S_fRESSEIZES** = 64 , **S_fSTATE** = 65 ,
  S_fSETSUM = 66 , **S_fRESUTIL** = 67 , **S_QUEUE** = 68 , **S_fNQ** = 69 ,
  S_fFIRSTINQ = 70 , **S_fLASTINQ** = 71 , **S_fSAQUE** = 72 , **S_fAQUE** = 73 ,
  S_fENTATRANK = 74 , **S_SET** = 75 , **S_fNUMSET** = 76 , **S_VARI** = 77 ,
  S_FORM = 78 , **S_fNUMGR** = 79 , **S_fATRGR** = 80 , **S_LPAREN** = 81 ,
  S_RPAREN = 82 , **S_LBRACKET** = 83 , **S_RBRACKET** = 84 , **S_PLUS** = 85 ,
  S_MINUS = 86 , **S_STAR** = 87 , **S_POWER** = 88 , **S_SLASH** = 89 ,
  S_LESS = 90 , **S_GREATER** = 91 , **S_ASSIGN** = 92 , **S_COMMA** = 93 ,
  S_94_USER_ = 94 , **S_YYACCEPT** = 95 , **S_input** = 96 , **S_expression** = 97 ,
  S_logicalOr = 98 , **S_logicalXor** = 99 , **S_logicalAnd** = 100 , **S_logicalNot** = 101 ,
  S_relational = 102 , **S_additive** = 103 , **S_multiplicative** = 104 , **S_power** = 105 ,
  S_unary = 106 , **S_primary** = 107 , **S_number** = 108 , **S_command** = 109 ,
  S_commandIF = 110 , **S_commandFOR** = 111 , **S_function** = 112 , **S_kernelFunction** = 113 ,
  S_elementFunction = 114 , **S_trigonFunction** = 115 , **S_mathFunction** = 116 , **S_probFunction** = 117 ,
  S_userFunction = 118 , **S_listaparm** = 119 , **S_indexList** = 120 , **S_illegal** = 121 ,
  S_attribute = 122 , **S_simulationResponse** = 123 , **S_simulationControl** = 124 , **S_variable** = 125 ,
  S_formula = 126 , **S_assigment** = 127 , **S_pluginFunction** = 128 }

17.1014.1 Detailed Description

Symbol kinds.

17.1014.2 Member Enumeration Documentation

17.1014.2.1 symbol_kind_type

enum yy::genesyspp_parser::symbol_kind::symbol_kind_type

Enumerator

YYNTOKENS	Number of tokens.

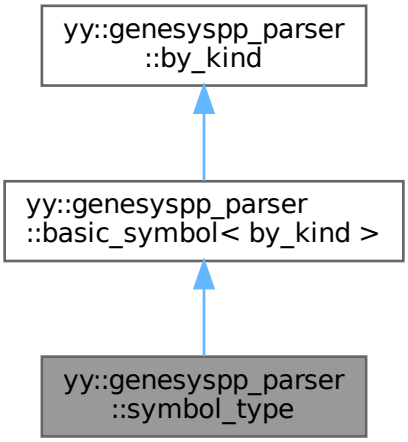
The documentation for this struct was generated from the following file:

- parser/GenesysParser.h

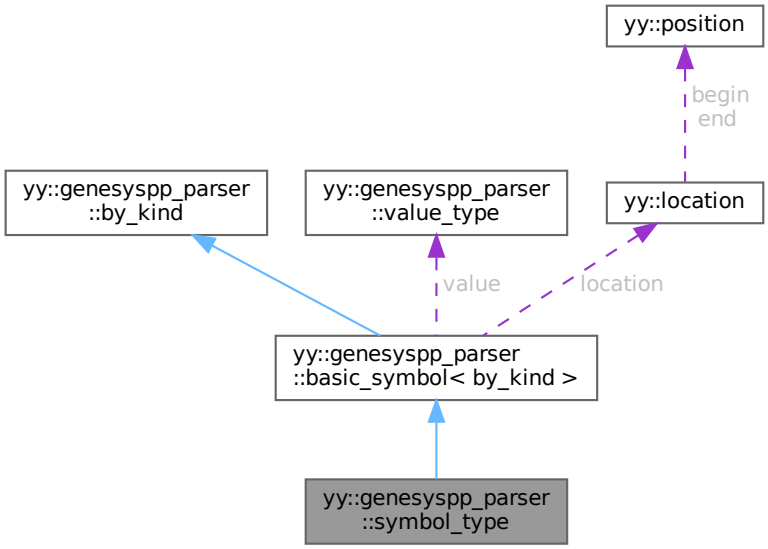
17.1015 yy::genesyspp_parser::symbol_type Struct Reference

"External" symbols: returned by the scanner.
`#include <parser/GenesysParser.h>`

Inheritance diagram for yy::genesyspp_parser::symbol_type:



Collaboration diagram for yy::genesyspp_parser::symbol_type:



Public Types

- typedef basic_symbol< by_kind > **super_type**

  Superclass.

Public Types inherited from yy::genesyspp_parser::basic_symbol< by_kind >

- typedef by_kind **super_type**

Alias to Base.

Public Types inherited from yy::genesyspp_parser::by_kind

- typedef token_kind_type **kind_type**

  The symbol kind as needed by the constructor.

Public Member Functions

- **symbol_type** () YY_NOEXCEPT

  Empty symbol.
- **symbol_type** (int tok, const location_type &l)

  Constructor for valueless symbols, and symbols from each type.
- **symbol_type** (int tok, const obj_t &v, const location_type &l)

Public Member Functions inherited from yy::genesyspp_parser::basic_symbol< by_kind >

- **basic_symbol** () YY_NOEXCEPT

  Default constructor.
- **basic_symbol** (const basic_symbol &that)

  Copy constructor.
- **basic_symbol** (typename Base::kind_type t, const location_type &l)

  Constructors for typed symbols.
- **basic_symbol** (typename Base::kind_type t, const obj_t &v, const location_type &l)
- **basic_symbol** (typename Base::kind_type t, const std::string &v, const location_type &l)
- ~**basic_symbol** ()

  Destroy the symbol.
- void **clear** () YY_NOEXCEPT

  Destroy contents, and record that is empty.
- std::string **name** () const YY_NOEXCEPT

  The user-facing name of this symbol.
- symbol_kind_type **type_get** () const YY_NOEXCEPT

  Backward compatibility (Bison 3.6).
- bool **empty** () const YY_NOEXCEPT

  Whether empty.
- void **move** (basic_symbol &s)

  Destructive move, s is emptied into this.

Public Member Functions inherited from yy::genesyspp_parser::by_kind

- **by_kind** () YY_NOEXCEPT

  Default constructor.
- **by_kind** (const by_kind &that) YY_NOEXCEPT

  Copy constructor.
- **by_kind** (kind_type t) YY_NOEXCEPT

  Constructor from (external) token numbers.
- void **clear** () YY_NOEXCEPT

  Record that this symbol is empty.
- void **move** (by_kind &that)

  Steal the symbol kind from that.
- symbol_kind_type kind () const YY_NOEXCEPT
- symbol_kind_type **type_get** () const YY_NOEXCEPT

  Backward compatibility (Bison 3.6).

Additional Inherited Members

Public Attributes inherited from yy::genesyspp_parser::basic_symbol< by_kind >

- value_type **value**

  The semantic value.
- location_type **location**

  The location.

Public Attributes inherited from yy::genesyspp_parser::by_kind

- symbol_kind_type kind_

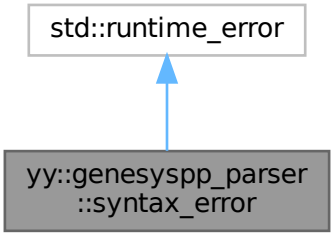
17.1015.1 Detailed Description

"External" symbols: returned by the scanner.
The documentation for this struct was generated from the following file:

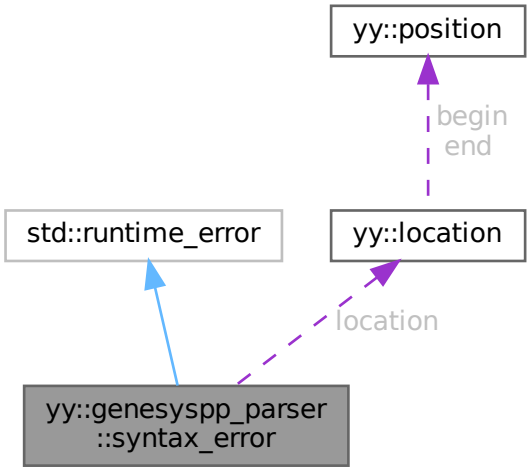
- parser/GenesysParser.h

17.1016 yy::genesyspp_parser::syntax_error Struct Reference

Syntax errors thrown from user actions.
`#include <parser/GenesysParser.h>`
Inheritance diagram for yy::genesyspp_parser::syntax_error:



Collaboration diagram for yy::genesyspp_parser::syntax_error:



Public Member Functions

- **syntax_error** (const location_type &l, const std::string &m)
- **syntax_error** (const syntax_error &s)

Public Attributes

- location_type **location**

17.1016.1 Detailed Description

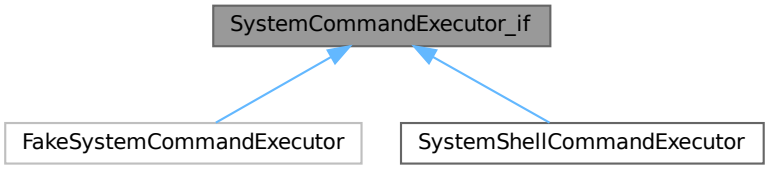
Syntax errors thrown from user actions.
The documentation for this struct was generated from the following files:

- parser/GenesysParser.h
- parser/GenesysParser.cpp

17.1017 SystemCommandExecutor_if Class Reference

Minimal command execution interface used to make system dependency checks testable.
`#include <kernel/simulator/SystemDependencyResolver.h>`

Inheritance diagram for SystemCommandExecutor_if:



Public Member Functions

- virtual ~**SystemCommandExecutor_if** ()=default

  Destroys the command executor abstraction.
- virtual SystemCommandResult run (const std::string &command)=0

  Runs a shell command and returns its observable result.

17.1017.1 Detailed Description

Minimal command execution interface used to make system dependency checks testable.

17.1017.2 Member Function Documentation

17.1017.2.1 run()

```
virtual SystemCommandResult SystemCommandExecutor_if::run (
            const std::string & command )  [pure virtual]
```

Runs a shell command and returns its observable result.
Implemented in SystemShellCommandExecutor, and FakeSystemCommandExecutor.
The documentation for this class was generated from the following file:

- kernel/simulator/SystemDependencyResolver.h

17.1018 SystemCommandResult Struct Reference

Captures the result of executing a system command used by dependency checks.
`#include <kernel/simulator/SystemDependencyResolver.h>`

Public Member Functions

- bool **succeeded** () const

  Returns true when the command started and exited successfully.

Public Attributes

- bool **started** = false

  True when the command process was started by the executor.
- int **exitCode** = -1

  Process exit code, or -1 when the command could not be started.
- std::string **output** = ""

  Combined standard output and diagnostic text produced by the command.
- std::string **errorMessage** = ""

  Internal executor error message, when starting or collecting the command failed.

17.1018.1 Detailed Description

Captures the result of executing a system command used by dependency checks.
The documentation for this struct was generated from the following files:

- kernel/simulator/SystemDependencyResolver.h
- kernel/simulator/SystemDependencyResolver.cpp

17.1019 SystemDependency Class Reference

Public Types

- enum class **OS** {
  Any , **Linux** , **Windows** , **MacOS** ,
  Unknown }

Public Member Functions

- **SystemDependency** (OS os, std::string name, std::string installCommand="", std::string checkCommand="")
- OS **getOS** () const
- std::string **getName** () const
- std::string **getInstallCommand** () const
- std::string **getCheckCommand** () const
- std::string **show** () const

Static Public Member Functions

- static std::string **osToString** (OS os)

The documentation for this class was generated from the following files:

- kernel/simulator/PluginInformation.h
- kernel/simulator/PluginInformation.cpp

17.1020 SystemDependencyCheckEntry Class Reference

One dependency entry produced by a system dependency preflight.
`#include <kernel/simulator/SystemDependencyResolver.h>`

Public Types

- enum class Status { **Satisfied** , **Missing** , **NotVerifiable** , **IgnoredDifferentOS** }

  Classification assigned to one declared dependency after preflight.

Public Member Functions

- **SystemDependencyCheckEntry** (SystemDependency dependency, Status status, SystemCommandResult checkResult={}, std::string message="")

  Builds a preflight entry for one declared dependency.
- const SystemDependency & **dependency** () const

  Returns the dependency declaration evaluated by this entry.
- Status **status** () const

  Returns the preflight status for this dependency.
- const SystemCommandResult & **checkResult** () const

  Returns the result of the check command, when one was executed.
- const std::string & **message** () const

  Returns human-readable diagnostic context for this entry.

- bool **blocksInsertion** () const

  Returns true when this entry should prevent plugin insertion.
- bool **canAttemptInstall** () const

  Returns true when this missing dependency has an installation command.
- std::string **diagnosticText** () const

  Returns a multi-line diagnostic with check/install commands and command results.

Static Public Member Functions

- static std::string **statusToString** (Status status)

  Converts a status value into a stable textual representation.

17.1020.1 Detailed Description

One dependency entry produced by a system dependency preflight.
The documentation for this class was generated from the following files:

- kernel/simulator/SystemDependencyResolver.h
- kernel/simulator/SystemDependencyResolver.cpp

17.1021 SystemDependencyCheckResult Class Reference

Full preflight result for the system dependencies declared by one plugin.
`#include <kernel/simulator/SystemDependencyResolver.h>`

Public Member Functions

- void **add** (SystemDependencyCheckEntry entry)

  Adds one dependency evaluation entry.
- const std::list< SystemDependencyCheckEntry > & **entries** () const

  Returns every evaluated dependency entry, including ignored entries.
- bool **canInsertPlugin** () const

  Returns true when no applicable dependency blocks insertion.
- bool **hasBlockingEntries** () const

  Returns true when at least one dependency blocks insertion.
- bool **canAttemptInstallForAllMissing** () const

  Returns true when every blocking missing dependency has an installation command.
- std::string **summary** () const

  Returns a compact diagnostic summary suitable for trace logs.
- std::string **diagnosticText** (bool includeSatisfied=false) const

  Returns a multi-line diagnostic suitable for terminal, GUI and trace details.

17.1021.1 Detailed Description

Full preflight result for the system dependencies declared by one plugin.
The documentation for this class was generated from the following files:

- kernel/simulator/SystemDependencyResolver.h
- kernel/simulator/SystemDependencyResolver.cpp

17.1022 SystemDependencyInstallEntry Class Reference

One command result from an attempted system dependency installation.
`#include <kernel/simulator/SystemDependencyResolver.h>`

Public Member Functions

- **SystemDependencyInstallEntry** (SystemDependency dependency, SystemCommandResult installResult)

  Builds an installation result for a single dependency.
- const SystemDependency & **dependency** () const

  Returns the dependency whose install command was executed.
- const SystemCommandResult & **installResult** () const

  Returns the result produced by the install command.
- std::string **diagnosticText** () const

  Returns command, exit code and captured output for trace or GUI feedback.

17.1022.1 Detailed Description

One command result from an attempted system dependency installation.
The documentation for this class was generated from the following files:

- kernel/simulator/SystemDependencyResolver.h
- kernel/simulator/SystemDependencyResolver.cpp

17.1023 SystemDependencyInstallResult Class Reference

Aggregates commands executed while trying to satisfy missing system dependencies.
`#include <kernel/simulator/SystemDependencyResolver.h>`

Public Member Functions

- void **add** (SystemDependencyInstallEntry entry)

  Adds one installation command result.
- const std::list< SystemDependencyInstallEntry > & **entries** () const

  Returns all attempted installation command results.
- bool **succeeded** () const

  Returns true when every attempted installation command succeeded.
- std::string **summary** () const

  Returns a compact diagnostic summary suitable for trace logs.
- std::string **diagnosticText** () const

  Returns detailed diagnostic text for every executed install command.

17.1023.1 Detailed Description

Aggregates commands executed while trying to satisfy missing system dependencies.
The documentation for this class was generated from the following files:

- kernel/simulator/SystemDependencyResolver.h
- kernel/simulator/SystemDependencyResolver.cpp

17.1024 SystemDependencyResolver Class Reference

Evaluates and optionally installs plugin system dependencies.
`#include <kernel/simulator/SystemDependencyResolver.h>`

Static Public Member Functions

- static SystemDependency::OS **currentOS** ()

  Returns the operating system represented by the current executable.
- static bool **appliesToCurrentOS** (const SystemDependency &dependency)

  Returns true when a dependency declaration applies to the current OS.
- static SystemDependencyCheckResult **evaluate** (const std::list< SystemDependency > *dependencies, SystemCommandExecutor_if &executor)

  Evaluates all dependency declarations using check commands when available.
- static SystemDependencyInstallResult **installMissingDependencies** (const SystemDependencyCheckResult &checkResult, SystemCommandExecutor_if &executor)

  Executes install commands for missing dependencies from a preflight result.

17.1024.1 Detailed Description

Evaluates and optionally installs plugin system dependencies.
The resolver deliberately contains no user-interface code. It only decides whether dependencies apply to the current OS, runs declared check/install commands through an executor, and reports diagnostics to the caller.
The documentation for this class was generated from the following files:

- kernel/simulator/SystemDependencyResolver.h
- kernel/simulator/SystemDependencyResolver.cpp

17.1025 SystemPreferences Class Reference

Persistent application preferences used by the Qt GUI.
`#include <applications/gui/genesys/systempreferences.h>`

Public Types

- enum class StartupModelMode : unsigned short { **NoModel** = 0 , **NewModel** = 1 , **OpenSpecificModel** = 2 , **OpenLastModel** = 3 }

  Startup model selection remembered in the preferences file.
- enum class VisualTheme { **Light** , **Dark** }

  Global visual style used by the GUI.
- enum class InterfaceStyle { **Classic** , **Modern** }

  Interface density/style preset used by the Qt application.
- enum class AutomaticPositioningStrategy { **Legacy** = 0 , **Centered** = 1 }

  Automatic positioning policy for graphical elements.

Static Public Member Functions

- static bool load ()

  Loads preferences from the configuration file.
- static bool save ()

  Saves the current preferences to disk.
- static void **resetToDefaults** ()

  Restores every preference to its built-in default value.
- static QString **configFilePath** ()

  Returns the full path to the active preferences file.
- static bool **startMaximized** ()

  Returns whether the application should start maximized.
- static void **setStartMaximized** (bool newStartMaximized)

  Updates the start-maximized preference.

- static bool **autoLoadPlugins** ()

  Returns whether plug-ins are loaded automatically at startup.
- static void **setAutoLoadPlugins** (bool newAutoLoadPlugins)

  Updates the automatic plug-in loading preference.
- static StartupModelMode **startupModelMode** ()

  Returns the startup model mode.
- static void **setStartupModelMode** (StartupModelMode newStartupModelMode)

  Updates the startup model mode.
- static unsigned short **modelAtStart** ()

  Returns the configured model index for the startup flow.
- static void **setModelAtStart** (unsigned short newModelAtStart)

  Updates the configured model index for the startup flow.
- static std::string **modelfilename** ()

  Returns the filename requested for startup loading.
- static void **setModelfilename** (const std::string &newModelfilename)

  Updates the filename requested for startup loading.
- static std::string **lastModelFilename** ()

  Returns the last model filename successfully opened by the GUI.
- static void **setLastModelFilename** (const std::string &newLastModelFilename)

  Updates the last model filename successfully opened by the GUI.
- static std::vector< std::string > **recentModelFiles** ()

  Returns the list of recently opened model files.
- static void **setRecentModelFiles** (const std::vector< std::string > &recentModelFiles)

  Replaces the recent model file list.
- static void pushRecentModelFile (const std::string &modelFilename)

  Pushes a file to the recent model list with limit handling.
- static unsigned int **recentModelFilesLimit** ()

  Returns the maximum number of recent model entries to keep.
- static void **setRecentModelFilesLimit** (unsigned int recentModelFilesLimit)

  Updates the maximum number of recent model entries.
- static bool **checkSystemPackagesAtStart** ()

  Returns whether system packages should be checked at startup.
- static void **setCheckSystemPackagesAtStart** (bool newCheckSystemPackagesAtStart)

  Updates the system-package check preference.
- static TraceManager::Level **traceLevel** ()

  Returns the current trace verbosity level.
- static void **setTraceLevel** (TraceManager::Level newTraceLevel)

  Updates the trace verbosity level.
- static VisualTheme **visualTheme** ()

  Returns the selected visual theme.
- static void **setVisualTheme** (VisualTheme newVisualTheme)

  Updates the selected visual theme.
- static std::string **visualThemeName** ()

  Returns the persisted name of the selected visual theme.
- static InterfaceStyle **interfaceStyle** ()

  Returns the selected interface style preset.
- static void **setInterfaceStyle** (InterfaceStyle newInterfaceStyle)

  Updates the selected interface style preset.
- static std::string **interfaceStyleName** ()

  Returns the persisted name of the selected interface style.
- static int **applicationFontPointSize** ()

Returns the application font size used by the GUI.

- static void **setApplicationFontPointSize** (int newApplicationFontPointSize)

  Updates the application font size used by the GUI.

- static bool **diagramUsesThemeColors** ()

  Returns whether diagram widgets should derive colors from the theme.

- static void **setDiagramUsesThemeColors** (bool newDiagramUsesThemeColors)

  Updates whether diagram widgets should derive colors from the theme.

- static AutomaticPositioningStrategy **automaticPositioningStrategy** ()

  Returns the automatic placement strategy used by the diagram canvas.

- static void **setAutomaticPositioningStrategy** (AutomaticPositioningStrategy strategy)

  Updates the automatic placement strategy used by the diagram canvas.

- static unsigned int **canvasBackgroundColor** ()

  Returns the RGB color used for the main canvas background.

- static unsigned int **canvasDisabledBackgroundColor** ()

  Returns the RGB color used for the disabled canvas background.

- static unsigned int **gridColor** ()

  Returns the RGB color used for the diagram grid.

17.1025.1 Detailed Description

Persistent application preferences used by the Qt GUI.
The class behaves like a singleton configuration store backed by a local preferences file. All access is static so the GUI can query and update the current runtime settings without manually passing a preferences object around.

17.1025.2 Member Function Documentation

17.1025.2.1 load()

```
bool SystemPreferences::load ( )  [static]
```

Loads preferences from the configuration file.
Loads GUI preferences from disk and normalizes invalid data.

17.1025.2.2 pushRecentModelFile()

```
void SystemPreferences::pushRecentModelFile (
            const std::string & modelFilename )  [static]
```

Pushes a file to the recent model list with limit handling.
Pushes a model file to the top of the recent-file list.

17.1025.2.3 save()

```
bool SystemPreferences::save ( )  [static]
```

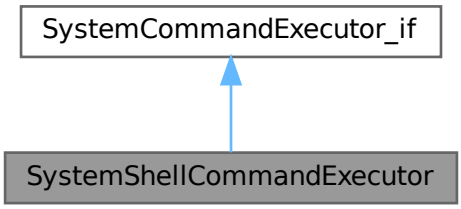
Saves the current preferences to disk.
Serializes the current GUI preferences to the preferences file.
The documentation for this class was generated from the following files:

- applications/gui/genesys/systempreferences.h
- applications/gui/genesys/systempreferences.cpp

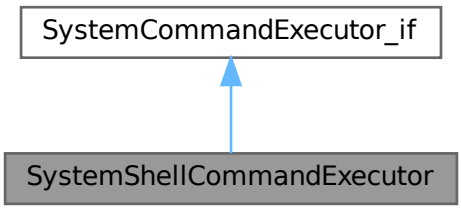
17.1026 SystemShellCommandExecutor Class Reference

Shell-based executor used by production plugin dependency preflight.
`#include <kernel/simulator/SystemDependencyResolver.h>`

Inheritance diagram for SystemShellCommandExecutor:



Collaboration diagram for SystemShellCommandExecutor:



Public Member Functions

- SystemCommandResult run (const std::string &command) override

  Executes a command through the host shell and captures combined output.

Public Member Functions inherited from SystemCommandExecutor_if

- virtual ~**SystemCommandExecutor_if** ()=default

  Destroys the command executor abstraction.

17.1026.1 Detailed Description

Shell-based executor used by production plugin dependency preflight.

17.1026.2 Member Function Documentation

17.1026.2.1 run()

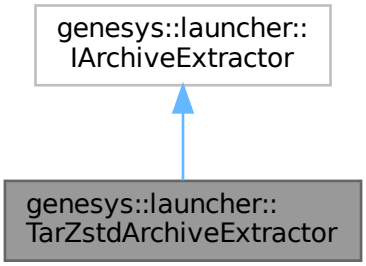
```
SystemCommandResult SystemShellCommandExecutor::run (
            const std::string & command )  [override], [virtual]
```

Executes a command through the host shell and captures combined output.
Implements SystemCommandExecutor_if.
The documentation for this class was generated from the following files:

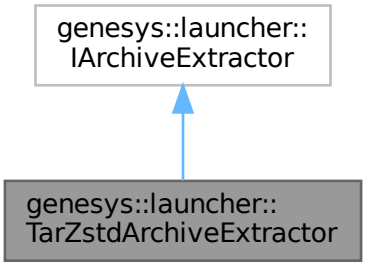
- kernel/simulator/SystemDependencyResolver.h
- kernel/simulator/SystemDependencyResolver.cpp

17.1027 genesys::launcher::TarZstdArchiveExtractor Class Reference

Inheritance diagram for genesys::launcher::TarZstdArchiveExtractor:



Collaboration diagram for genesys::launcher::TarZstdArchiveExtractor:



Public Member Functions

- **TarZstdArchiveExtractor** (int timeoutMs=60000)
- ExtractionResult extract (const QString &archivePath, const QString &destinationRoot) override

Static Public Member Functions

- static ExtractionResult **validatePathListing** (const QByteArray &listing)
- static ExtractionResult **validateVerboseListing** (const QByteArray &listing)

17.1027.1 Member Function Documentation

17.1027.1.1 extract()

```
ExtractionResult genesys::launcher::TarZstdArchiveExtractor::extract (
            const QString & archivePath,
            const QString & destinationRoot )  [override], [virtual]
```

Implements genesys::launcher::IArchiveExtractor.

The documentation for this class was generated from the following files:

- applications/launcher/ArchiveExtractor.h
- applications/launcher/ArchiveExtractor.cpp

17.1028 testing::internal::Templates< Head_, Tail_ > Struct Template Reference

Public Types

- using **Head** = TemplateSel< Head_ >
- using **Tail** = Templates< Tail_... >

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-type-util.h

17.1029 testing::internal::Templates< Head_ > Struct Template Reference

Public Types

- using **Head** = TemplateSel< Head_ >
- using **Tail** = None

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-type-util.h

17.1030 testing::internal::TemplateSel< Tmpl > Struct Template Reference

Classes

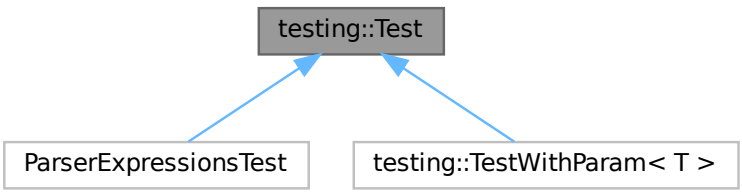
- struct Bind

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-type-util.h

17.1031 testing::Test Class Reference

Inheritance diagram for testing::Test:



Static Public Member Functions

- static void **SetUpTestSuite** ()
- static void **TearDownTestSuite** ()
- static void **TearDownTestCase** ()
- static void **SetUpTestCase** ()
- static bool **HasFatalFailure** ()
- static bool **HasNonfatalFailure** ()
- static bool **IsSkipped** ()
- static bool **HasFailure** ()
- static void **RecordProperty** (const std::string &key, const std::string &value)
- static void **RecordProperty** (const std::string &key, int value)

Protected Member Functions

- virtual void **SetUp** ()
- virtual void **TearDown** ()

Friends

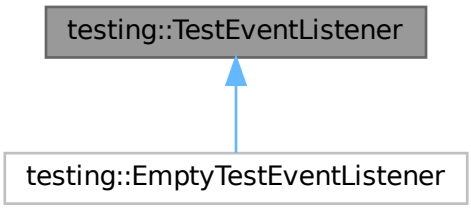
- class **TestInfo**

The documentation for this class was generated from the following files:

- gtest/gtest/gtest.h
- gtest/src/gtest.cc

17.1032 testing::TestEventListener Class Reference

Inheritance diagram for testing::TestEventListener:



Public Member Functions

- virtual void **OnTestProgramStart** (const UnitTest &unit_test)=0
- virtual void **OnTestIterationStart** (const UnitTest &unit_test, int iteration)=0
- virtual void **OnEnvironmentsSetUpStart** (const UnitTest &unit_test)=0
- virtual void **OnEnvironmentsSetUpEnd** (const UnitTest &unit_test)=0
- virtual void **OnTestSuiteStart** (const TestSuite &)
- virtual void **OnTestCaseStart** (const TestCase &)
- virtual void **OnTestStart** (const TestInfo &test_info)=0
- virtual void **OnTestDisabled** (const TestInfo &)
- virtual void **OnTestPartResult** (const TestPartResult &test_part_result)=0
- virtual void **OnTestEnd** (const TestInfo &test_info)=0

- virtual void **OnTestSuiteEnd** (const TestSuite &)
- virtual void **OnTestCaseEnd** (const TestCase &)
- virtual void **OnEnvironmentsTearDownStart** (const UnitTest &unit_test)=0
- virtual void **OnEnvironmentsTearDownEnd** (const UnitTest &unit_test)=0
- virtual void **OnTestIterationEnd** (const UnitTest &unit_test, int iteration)=0
- virtual void **OnTestProgramEnd** (const UnitTest &unit_test)=0

The documentation for this class was generated from the following file:

- gtest/gtest/gtest.h

17.1033 testing::TestEventListeners Class Reference

Public Member Functions

- void **Append** (TestEventListener *listener)
- TestEventListener * **Release** (TestEventListener *listener)
- TestEventListener * **default_result_printer** () const
- TestEventListener * **default_xml_generator** () const

Friends

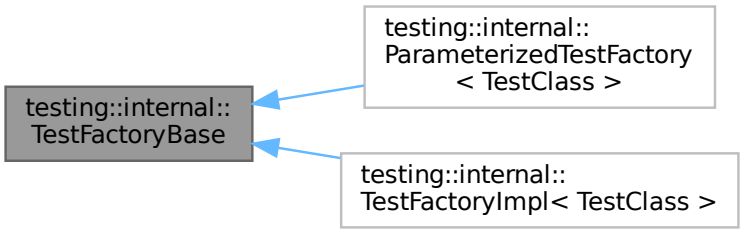
- class **TestSuite**
- class **TestInfo**
- class **internal::DefaultGlobalTestPartResultReporter**
- class **internal::NoExecDeathTest**
- class **internal::TestEventListenersAccessor**
- class **internal::UnitTestImpl**

The documentation for this class was generated from the following files:

- gtest/gtest/gtest.h
- gtest/src/gtest.cc

17.1034 testing::internal::TestFactoryBase Class Reference

Inheritance diagram for testing::internal::TestFactoryBase:



Public Member Functions

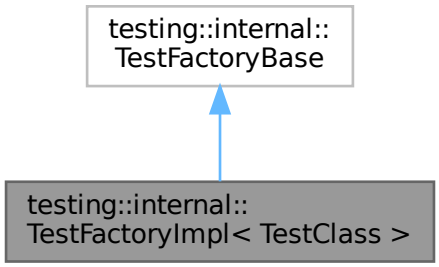
- virtual Test * **CreateTest** ()=0

The documentation for this class was generated from the following file:

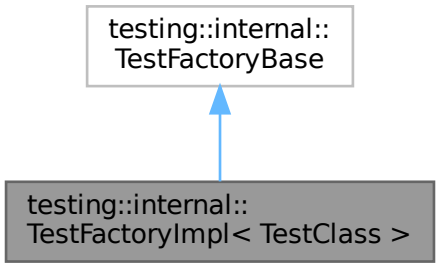
- gtest/gtest/internal/gtest-internal.h

17.1035 testing::internal::TestFactoryImpl< TestClass > Class Template Reference

Inheritance diagram for testing::internal::TestFactoryImpl< TestClass >:



Collaboration diagram for testing::internal::TestFactoryImpl< TestClass >:



Public Member Functions

- Test * CreateTest () override

17.1035.1 Member Function Documentation

17.1035.1.1 CreateTest()

```
template<class TestClass >
Test * testing::internal::TestFactoryImpl< TestClass >::CreateTest ( )  [inline], [override],
[virtual]
```

Implements testing::internal::TestFactoryBase.

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.1036 testing::TestInfo Class Reference

Public Member Functions

- const char * **test_suite_name** () const
- const char * **test_case_name** () const
- const char * **name** () const
- const char * **type_param** () const
- const char * **value_param** () const
- const char * **file** () const
- int **line** () const
- bool **is_in_another_shard** () const
- bool **should_run** () const
- bool **is_reportable** () const
- const TestResult * **result** () const

Friends

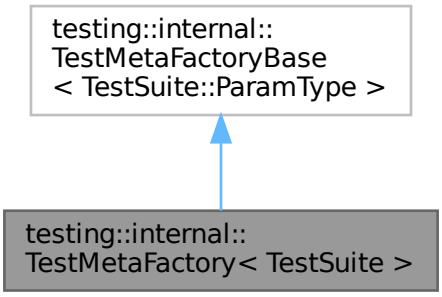
- class **Test**
- class **TestSuite**
- class **internal::UnitTestImpl**
- class **internal::StreamingListenerTest**
- TestInfo * **internal::MakeAndRegisterTestInfo** (const char *test_suite_name, const char *name, const char *type_param, const char *value_param, internal::CodeLocation code_location, internal::TypeId fixture_class_id, internal::SetUpTestSuiteFunc set_up_tc, internal::TearDownTestSuiteFunc tear_down_tc, internal::TestFactoryBase *factory)

The documentation for this class was generated from the following files:

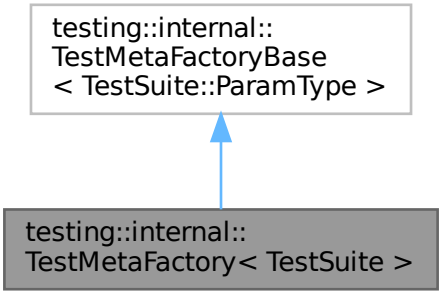
- gtest/gtest/gtest.h
- gtest/src/gtest.cc

17.1037 testing::internal::TestMetaFactory< TestSuite > Class Template Reference

Inheritance diagram for testing::internal::TestMetaFactory< TestSuite >:



Collaboration diagram for testing::internal::TestMetaFactory< TestSuite >:



Public Types

- using **ParamType** = typename TestSuite::ParamType

Public Member Functions

- TestFactoryBase * CreateTestFactory (ParamType parameter) override

17.1037.1 Member Function Documentation

17.1037.1.1 CreateTestFactory()

```
template<class TestSuite >
TestFactoryBase * testing::internal::TestMetaFactory< TestSuite >::CreateTestFactory (
            ParamType parameter )  [inline], [override], [virtual]
```

Implements testing::internal::TestMetaFactoryBase< TestSuite::ParamType >.
The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.1038 testing::internal::TestMetaFactoryBase< ParamType > Class Template Reference

Public Member Functions

- virtual TestFactoryBase * **CreateTestFactory** (ParamType parameter)=0

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.1039 testing::TestParamInfo< ParamType > Struct Template Reference

Public Member Functions

- **TestParamInfo** (const ParamType &a_param, size_t an_index)

Public Attributes

- ParamType **param**
- size_t **index**

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-param-util.h

17.1040 testing::TestProperty Class Reference

Public Member Functions

- **TestProperty** (const std::string &a_key, const std::string &a_value)
- const char * **key** () const
- const char * **value** () const
- void **SetValue** (const std::string &new_value)

The documentation for this class was generated from the following file:

- gtest/gtest/gtest.h

17.1041 testing::internal::TestPropertyKeyIs Class Reference

Public Member Functions

- **TestPropertyKeyIs** (const std::string &key)
- bool **operator()** (const TestProperty &test_property) const

The documentation for this class was generated from the following file:

- gtest/src/gtest-internal-inl.h

17.1042 HypothesisTester_if::TestResult Class Reference

Result of a parametric hypothesis test.
`#include <tools/Statistics/HypothesisTester_if.h>`

Public Member Functions

- **TestResult** (double pvalue, bool rejectH0, double acceptanceInferiorLimit, double acceptanceSuperiorLimit, double testStat)
- bool **rejectH0** () const
- bool **acceptH0** () const
- double **pValue** () const
- double **testStat** () const
- double **acceptanceInferiorLimit** () const
- double **acceptanceSuperiorLimit** () const

17.1042.1 Detailed Description

Result of a parametric hypothesis test.
Contains p-value, reject/accept decision for H0, acceptance bounds for the test statistic and observed statistic value.
The documentation for this class was generated from the following file:

- tools/Statistics/HypothesisTester_if.h

17.1043 testing::TestResult Class Reference

Public Member Functions

- int **total_part_count** () const
- int **test_property_count** () const
- bool **Passed** () const
- bool **Skipped** () const
- bool **Failed** () const
- bool **HasFatalFailure** () const
- bool **HasNonfatalFailure** () const
- TimeInMillis **elapsed_time** () const
- TimeInMillis **start_timestamp** () const
- const TestPartResult & **GetTestPartResult** (int i) const
- const TestProperty & **GetTestProperty** (int i) const

Friends

- class **TestInfo**
- class **TestSuite**
- class **UnitTest**
- class **internal::DefaultGlobalTestPartResultReporter**
- class **internal::ExecDeathTest**
- class **internal::TestResultAccessor**
- class **internal::UnitTestImpl**
- class **internal::WindowsDeathTest**
- class **internal::FuchsiaDeathTest**

The documentation for this class was generated from the following files:

- gtest/gtest/gtest.h
- gtest/src/gtest.cc

17.1044 testing::internal::TestResultAccessor Class Reference

Static Public Member Functions

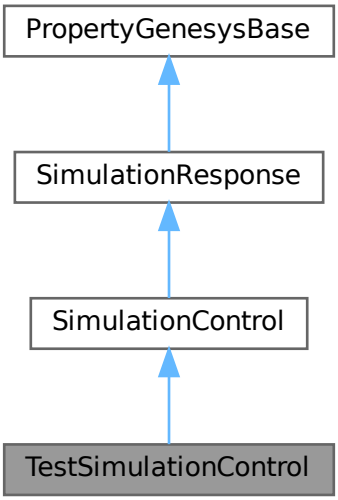
- static void **RecordProperty** (TestResult *test_result, const std::string &xml_element, const TestProperty &property)
- static void **ClearTestPartResults** (TestResult *test_result)
- static const std::vector< testing::TestPartResult > & **test_part_results** (const TestResult &test_result)

The documentation for this class was generated from the following file:

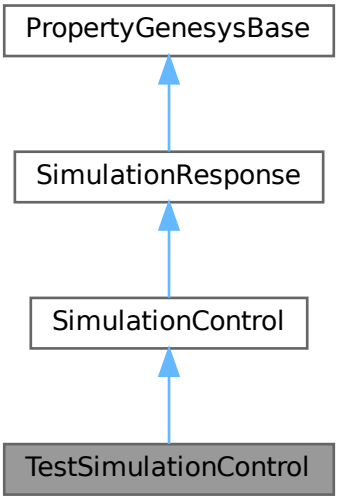
- gtest/src/gtest-internal-inl.h

17.1045 TestSimulationControl Class Reference

Inheritance diagram for TestSimulationControl:



Collaboration diagram for TestSimulationControl:



Public Member Functions

- **TestSimulationControl** (std::function< std::string()> getter, std::function< void(std::string)> setter)
- std::string getValue () const override
- void setValue (std::string value, bool remove=false) override

Public Member Functions inherited from SimulationControl

- **SimulationControl** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- bool **isReadOnly** () const
- virtual List< SimulationControl * > * getChildSimulationControls (int index=0)

  Returns nested SimulationControls exposed by this control.
- virtual bool **hasObjectInstance** () const
- virtual bool **ensureObjectInstance** ()
- virtual bool **isModelDataDefinitionReference** () const
- virtual ModelDataDefinition * **getReferencedModelDataDefinition** () const
- virtual bool **validateProposedValue** (const std::string &value, std::string &errorMessage) const
- virtual SimulationControlEditorHint **preferredEditorHint** () const
- void **setPreferredEditorHint** (SimulationControlEditorHint hint)
- virtual ModelDataDefinition * **getListElementModelDataDefinition** (int index) const
- virtual bool **supportsInlineExpansion** () const
- virtual bool **supportsListEditor** () const
- virtual bool **supportsExistingObjectSelection** () const
- virtual bool **supportsObjectCreation** () const
- virtual bool **supportsNewListElementCreation** () const
- virtual List< std::string > * **getCreatableListElementTypes** ()
- virtual std::string **getCurrentListElementType** () const
- virtual bool **setCurrentListElementType** (const std::string &typeName)
- virtual bool **isInlineObjectProperty** () const
- virtual bool **createObjectInstance** (const std::string &value="")
- virtual List< std::string > * **getCreatableReferenceTypes** ()
- virtual std::string **getCurrentReferenceType** () const
- virtual bool **setCurrentReferenceType** (const std::string &typeName)
- virtual bool **createObjectInstanceOfType** (const std::string &typeName, const std::string &value="")
- virtual bool **createNewListElement** (const std::string &value="")
- virtual bool **createNewListElementOfType** (const std::string &typeName, const std::string &value="")
- virtual List< SimulationControl * > * **getEditableChildSimulationControls** (int index=0)

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Member Functions inherited from SimulationControl

- void **_ensureWritable** (const char *operation) const

Protected Attributes inherited from SimulationControl

- bool **_readonly** = false
- SimulationControlEditorHint **_preferredEditorHint** = SimulationControlEditorHint::Default

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.1045.1 Member Function Documentation

17.1045.1.1 getValue()

```
std::string TestSimulationControl::getValue ( ) const  [inline], [override], [virtual]
```

Implements SimulationResponse.

17.1045.1.2 setValue()

```
void TestSimulationControl::setValue (
            std::string value,
            bool remove = false )  [inline], [override], [virtual]
```

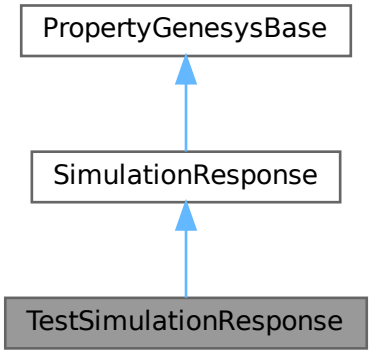
Implements SimulationControl.

The documentation for this class was generated from the following file:

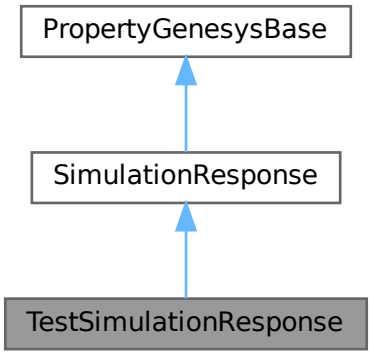
- tests/unit/test_simulator_support.cpp

17.1046 TestSimulationResponse Class Reference

Inheritance diagram for TestSimulationResponse:



Collaboration diagram for TestSimulationResponse:



Public Member Functions

- **TestSimulationResponse** (std::function< std::string()> getter, bool isEnum=false)
- std::string getValue () const override

Public Member Functions inherited from SimulationResponse

- **SimulationResponse** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **show** () const
- virtual List< std::string > * **getStrValues** ()

Public Member Functions inherited from PropertyGenesysBase

- **PropertyGenesysBase** (std::string className, std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
- std::string **whatsThis** () const
- std::string **getClassname** () const
- std::string **getElementName** () const
- std::string **getName** () const
- std::string **propertyType** () const
- bool **getIsList** () const
- bool **getIsClass** () const
- bool **getIsEnum** () const
- void **setWhatsThis** (const std::string text)
- void **setName** (std::string name)

Additional Inherited Members

Protected Attributes inherited from PropertyGenesysBase

- std::string **_elementName**
- std::string **_className**
- std::string **_propertyName**
- std::string **_propertyType**
- std::string **_whatsThis**
- bool **_isList**
- bool **_isClass**
- bool **_isEnum**

17.1046.1 Member Function Documentation

17.1046.1.1 getValue()

`std::string TestSimulationResponse::getValue ( ) const [inline], [override], [virtual]`
Implements SimulationResponse.
The documentation for this class was generated from the following file:

- tests/unit/test_simulator_support.cpp

17.1047 testing::TestSuite Class Reference

Public Member Functions

- **TestSuite** (const char *name, const char *a_type_param, internal::SetUpTestSuiteFunc set_up_tc, internal::TearDownTestSuiteFunc tear_down_tc)
- const char * **name** () const
- const char * **type_param** () const
- bool **should_run** () const
- int **successful_test_count** () const
- int **skipped_test_count** () const
- int **failed_test_count** () const
- int **reportable_disabled_test_count** () const
- int **disabled_test_count** () const
- int **reportable_test_count** () const
- int **test_to_run_count** () const
- int **total_test_count** () const
- bool **Passed** () const
- bool **Failed** () const

- TimeInMillis **elapsed_time** () const
- TimeInMillis **start_timestamp** () const
- const TestInfo * **GetTestInfo** (int i) const
- const TestResult & **ad_hoc_test_result** () const

Friends

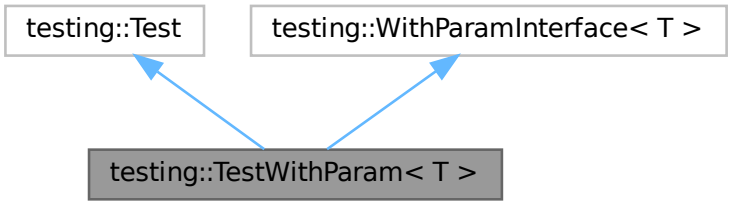
- class **Test**
- class **internal::UnitTestImpl**

The documentation for this class was generated from the following files:

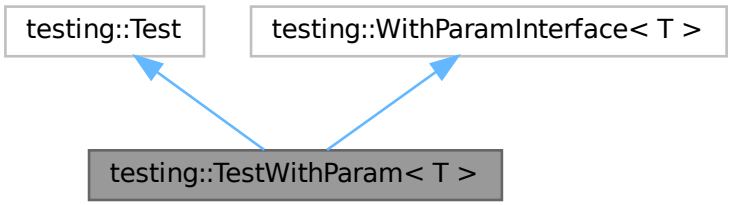
- gtest/gtest/gtest.h
- gtest/src/gtest.cc

17.1048 testing::TestWithParam< T > Class Template Reference

Inheritance diagram for testing::TestWithParam< T >:



Collaboration diagram for testing::TestWithParam< T >:



Additional Inherited Members

Public Types inherited from testing::WithParamInterface< T >

- typedef T **ParamType**

Static Public Member Functions inherited from testing::Test

- static void **SetUpTestSuite** ()
- static void **TearDownTestSuite** ()
- static void **TearDownTestCase** ()
- static void **SetUpTestCase** ()
- static bool **HasFatalFailure** ()
- static bool **HasNonfatalFailure** ()
- static bool **IsSkipped** ()
- static bool **HasFailure** ()
- static void **RecordProperty** (const std::string &key, const std::string &value)
- static void **RecordProperty** (const std::string &key, int value)

Static Public Member Functions inherited from testing::WithParamInterface< T >

- static const ParamType & **GetParam** ()

Protected Member Functions inherited from testing::Test

- virtual void **SetUp** ()
- virtual void **TearDown** ()

The documentation for this class was generated from the following file:

- gtest/gtest/gtest.h

17.1049 testing::internal::ThreadLocal< T > Class Template Reference

Public Member Functions

- **ThreadLocal** (const T &value)
- T * **pointer** ()
- const T * **pointer** () const
- const T & **get** () const
- void **set** (const T &value)

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-port.h

17.1050 QCPAxisPainterPrivate::TickLabelData Struct Reference

Public Attributes

- QString **basePart**
- QString **expPart**
- QString **suffixPart**
- QRect **baseBounds**
- QRect **expBounds**
- QRect **suffixBounds**
- QRect **totalBounds**
- QRect **rotatedTotalBounds**
- QFont **baseFont**
- QFont **expFont**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/qcustomplot.h

17.1051 Token Struct Reference

Public Types

- enum **Kind** {
  eof , **null** , **f** , **t** ,
  number , **string** , **lbracket** , **rbracket** ,
  lbrace , **rbrace** , **comma** , **colon** }

Public Attributes

- enum Token::Kind **kind**
- std::string **str** {}

The documentation for this struct was generated from the following file:

- kernel/simulator/persistence/JsonSerializer.cpp

17.1052 yy::genesyspp_parser::token Struct Reference

Token kinds.
`#include <parser/GenesysParser.h>`

Public Types

- enum **token_kind_type** {
  YYEMPTY = -2 , **END** = 0 , **YYerror** = 256 , **YYUNDEF** = 257 ,
  NUMD = 258 , **NUMH** = 259 , **CTEZERO** = 260 , **oLE** = 261 ,
  oGE = 262 , **oEQ** = 263 , **oNE** = 264 , **oAND** = 265 ,
  oOR = 266 , **oNAND** = 267 , **oXOR** = 268 , **oNOT** = 269 ,
  fSIN = 270 , **fCOS** = 271 , **fROUND** = 272 , **fMOD** = 273 ,
  fTRUNC = 274 , **fFRAC** = 275 , **fEXP** = 276 , **fSQRT** = 277 ,
  fLOG = 278 , **fLN** = 279 , **mathMIN** = 280 , **mathMAX** = 281 ,
  fVAL = 282 , **fEVAL** = 283 , **fLENG** = 284 , **fRND1** = 285 ,
  fEXPO = 286 , **fNORM** = 287 , **fUNIF** = 288 , **fWEIB** = 289 ,
  fLOGN = 290 , **fGAMM** = 291 , **fERLA** = 292 , **fTRIA** = 293 ,
  fBETA = 294 , **fDISC** = 295 , **fTNOW** = 296 , **fTFIN** = 297 ,
  fMAXREP = 298 , **fNUMREP** = 299 , **fIDENT** = 300 , **simulEntitiesWIP** = 301 ,
  cIF = 302 , **cELSE** = 303 , **cFOR** = 304 , **cTO** = 305 ,
  cDO = 306 , **ATRIB** = 307 , **CSTAT** = 308 , **COUNTER** = 309 ,
  SIMRESP = 310 , **SIMCTRL** = 311 , **fTAVG** = 312 , **fCOUNT** = 313 ,
  ILLEGAL = 314 , **RESOURCE** = 315 , **fNR** = 316 , **fMR** = 317 ,
  fIRF = 318 , **fRESSEIZES** = 319 , **fSTATE** = 320 , **fSETSUM** = 321 ,
  fRESUTIL = 322 , **QUEUE** = 323 , **fNQ** = 324 , **fFIRSTINQ** = 325 ,
  fLASTINQ = 326 , **fSAQUE** = 327 , **fAQUE** = 328 , **fENTATRANK** = 329 ,
  SET = 330 , **fNUMSET** = 331 , **VARI** = 332 , **FORM** = 333 ,
  fNUMGR = 334 , **fATRGR** = 335 , **LPAREN** = 336 , **RPAREN** = 337 ,
  LBRACKET = 338 , **RBRACKET** = 339 , **PLUS** = 340 , **MINUS** = 341 ,
  STAR = 342 , **POWER** = 343 , **SLASH** = 344 , **LESS** = 345 ,
  GREATER = 346 , **ASSIGN** = 347 , **COMMA** = 348 }
- typedef token_kind_type **yytokentype**

  Backward compatibility alias (Bison 3.6).

17.1052.1 Detailed Description

Token kinds.
The documentation for this struct was generated from the following file:

- parser/GenesysParser.h

17.1053 TokenService Class Reference

Public Member Functions

- std::string **generateSessionId** ()
- std::string **generateAccessToken** ()

The documentation for this class was generated from the following files:

- applications/worker/auth/TokenService.h
- applications/worker/auth/TokenService.cpp

17.1054 TraceConsoleController Class Reference

Controller that routes simulator trace events to GUI text widgets.
`#include <applications/gui/genesys/controllers/TraceConsoleController.h>`

Public Member Functions

- **TraceConsoleController** (QTextEdit *console, QTextEdit *simulationText, QTextEdit *reportsText, std::function< bool()> traceWidgetsEnabled={})

  Creates the trace-output bridge used by MainWindow trace wrappers.
- void **simulatorTraceHandler** (TraceEvent e) const

  Renders generic traces to the main console pane.
- void **simulatorTraceErrorHandler** (TraceErrorEvent e) const

  Renders error traces to the main console pane.
- void **simulatorTraceSimulationHandler** (TraceSimulationEvent e) const

  Renders simulation traces to the simulation output pane.
- void **simulatorTraceReportsHandler** (TraceEvent e) const

  Renders reports traces to the reports output pane.

17.1054.1 Detailed Description

Controller that routes simulator trace events to GUI text widgets.
This controller is the Phase-4 extraction for trace presentation. MainWindow keeps compatibility handlers registered in the kernel trace manager and delegates rendering to this class, reducing direct UI formatting logic in the façade.
Responsibilities:

- render regular/error traces to the main console pane;
- render simulation traces to the simulation output pane;
- render report traces to the reports pane.

Boundaries:

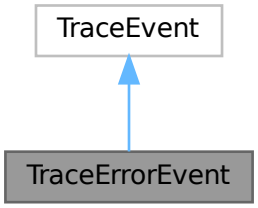
- it does not subscribe handlers by itself (registration stays in MainWindow/event layer);
- it does not change simulation flow, model state, or controller orchestration;
- it acts as a UI bridge, not as a domain service.

The documentation for this class was generated from the following files:

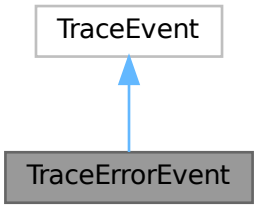
- applications/gui/genesys/controllers/TraceConsoleController.h
- applications/gui/genesys/controllers/TraceConsoleController.cpp

17.1055 TraceErrorEvent Class Reference

Inheritance diagram for TraceErrorEvent:



Collaboration diagram for TraceErrorEvent:



Public Member Functions

- **TraceErrorEvent** (const std::string &text, const std::exception &e)
- **TraceErrorEvent** (const std::string &text, TraceManager::Level level)
- const std::string & **getExceptionMessage** () const

Public Member Functions inherited from TraceEvent

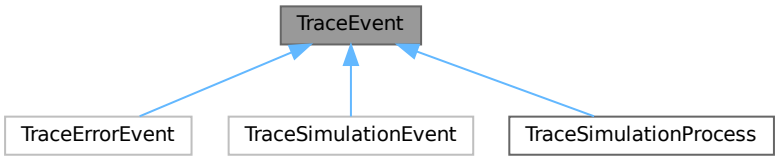
- **TraceEvent** (TraceManager::Level level, const std::string &text)
- **TraceEvent** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- TraceManager::Level **getTracelevel** () const
- const std::string & **getText** () const

The documentation for this class was generated from the following file:

- kernel/simulator/TraceManager.h

17.1056 TraceEvent Class Reference

Inheritance diagram for TraceEvent:



Public Member Functions

- **TraceEvent** (TraceManager::Level level, const std::string &text)
- **TraceEvent** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- TraceManager::Level **getTracelevel** () const
- const std::string & **getText** () const

The documentation for this class was generated from the following file:

- kernel/simulator/TraceManager.h

17.1057 testing::internal::TraceInfo Struct Reference

Public Attributes

- const char ∗ **file**
- int **line**
- std::string **message**

The documentation for this struct was generated from the following file:

- gtest/src/gtest-internal-inl.h

17.1058 TraceManager Class Reference

`#include <kernel/simulator/TraceManager.h>`

Public Types

- enum class **Level** : int {
  L0_noTraces = 0 , **L1_errorFatal** = 1 , **L2_results** = 2 , **L3_errorRecover** = 3 ,
  L4_warning = 4 , **L5_event** = 5 , **L6_arrival** = 6 , **L7_internal** = 7 ,
  L8_detailed = 8 , **L9_mostDetailed** = 9 , **num_elements** = 10 }

Public Member Functions

- **TraceManager** (Simulator ∗simulator)
- void addTraceHandler (traceListener traceListener)

  addTraceHandler
- void addTraceReportHandler (traceListener traceReportListener)

  addTraceReportHandler
- void addTraceSimulationHandler (traceSimulationListener traceSimulationListener)

addTraceSimulationHandler

- void addTraceErrorHandler (traceErrorListener traceErrorListener)

  addTraceErrorHandler

- template<typename Class >
  void addTraceHandler (Class *object, void(Class::*function)(TraceEvent))

  addTraceHandler

- template<typename Class >
  void addTraceErrorHandler (Class *object, void(Class::*function)(TraceErrorEvent))

  addTraceErrorHandler

- template<typename Class >
  void addTraceReportHandler (Class *object, void(Class::*function)(TraceEvent))

  addTraceReportHandler

- template<typename Class >
  void addTraceSimulationHandler (Class *object, void(Class::*function)(TraceSimulationEvent))

  addTraceSimulationHandler

- void **addTraceSimulationExceptionRuleModelData** (void *thisobject)
- void **trace** (TraceManager::Level level, const std::string &text)

  Deprecated.

- void **traceSimulation** (void *thisobject, TraceManager::Level level, double time, Entity *entity, ModelComponent *component, const std::string &text)

  Deprecated.

- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

  Deprecated.

- void **beginShutdown** ()
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  Trace to a general output, used when not simulating (eg: adding plugins, checking the model, etc)

- void **traceError** (const std::string &text, const std::exception &e)

  Trace to the error output, used in every situation an error happens (simulating, report, general)

- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)

  Trace to the error output, used in every situation an error happens (simulating, report, general)

- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)

  Trace to the report output, used only when generating the simulation report.

- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed, bool showAnyway=false)

  Trace to the simulation output, used only when simulation is running (eg: components or dataElements inform something)

- void **traceSimulation** (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩
  Level::L8_detailed, bool showAnyway=false)

  Trace to the simulation output, used only when simulation is running (eg: components or dataElements inform something)

- List< std::string > * errorMessages () const

  errorMessages

- void setTraceLevel (TraceManager::Level _traceLevel)

  setTraceLevel

- TraceManager::Level getTraceLevel () const

  getTraceLevel

- Simulator * getParentSimulator () const

  getParentSimulator

- void setTraceSimulationRuleAllAllowed (bool _traceSimulationRuleAllAllowed)

  setTraceSimulationRuleAllAllowed

- bool isTraceSimulationRuleAllAllowed () const

  isTraceSimulationRuleAllAllowed

Static Public Member Functions

- static std::string **convertEnumToStr** (Level level)

17.1058.1 Detailed Description

The TraceManager is used to trace back model simulation information and track/debug the simulation. It works as the model simulation output (cout) and allows external methods to hook up such output as listeners.

17.1058.2 Member Function Documentation

17.1058.2.1 addTraceErrorHandler() [1/2]

```
template<typename Class >
void TraceManager::addTraceErrorHandler (
            Class * object,
            void(Class::*)(TraceErrorEvent) function )
```

addTraceErrorHandler

Parameters

object	Add a method to by notified/invoked then a traceError is called

17.1058.2.2 addTraceErrorHandler() [2/2]

```
void TraceManager::addTraceErrorHandler (
            traceErrorListener traceErrorListener )
```

addTraceErrorHandler

Parameters

traceErrorListener	Add a function to by notified/invoked then a traceError is called

17.1058.2.3 addTraceHandler() [1/2]

```
template<typename Class >
void TraceManager::addTraceHandler (
            Class * object,
            void(Class::*)(TraceEvent) function )
```

addTraceHandler

Parameters

object	Add a method to by notified/invoked then a trace is called

17.1058.2.4 addTraceHandler() [2/2]

```
void TraceManager::addTraceHandler (
            traceListener traceListener )
```

addTraceHandler

Parameters

traceListener	Add a function to by notified/invoked then a trace is called

17.1058.2.5 addTraceReportHandler() [1/2]

```
template<typename Class >
void TraceManager::addTraceReportHandler (
            Class * object,
            void(Class::*)(TraceEvent) function )
```

addTraceReportHandler

Parameters

object	Add a method to by notified/invoked then a traceReport is called

17.1058.2.6 addTraceReportHandler() [2/2]

```
void TraceManager::addTraceReportHandler (
            traceListener traceReportListener )
```

addTraceReportHandler

Parameters

traceReportListener	Add a function to by notified/invoked then a traceReport is called

17.1058.2.7 addTraceSimulationHandler() [1/2]

```
template<typename Class >
void TraceManager::addTraceSimulationHandler (
            Class * object,
            void(Class::*)(TraceSimulationEvent) function )
```

addTraceSimulationHandler

Parameters

object	Add a method to by notified/invoked then a traceSimulation is called

17.1058.2.8 addTraceSimulationHandler() [2/2]

```
void TraceManager::addTraceSimulationHandler (
            traceSimulationListener traceSimulationListener )
```

addTraceSimulationHandler

Parameters

traceSimulationListener	Add a function to by notified/invoked then a traceSimulation is called

17.1058.2.9 errorMessages()

```
List< std::string > * TraceManager::errorMessages ( ) const
```

errorMessages

Returns

17.1058.2.10 getParentSimulator()

```
Simulator * TraceManager::getParentSimulator ( ) const
```

getParentSimulator

Returns

17.1058.2.11 getTraceLevel()

```
TraceManager::Level TraceManager::getTraceLevel ( ) const
```

getTraceLevel

Returns

17.1058.2.12 isTraceSimulationRuleAllAllowed()

```
bool TraceManager::isTraceSimulationRuleAllAllowed ( ) const
```

isTraceSimulationRuleAllAllowed

Returns

17.1058.2.13 setTraceLevel()

```
void TraceManager::setTraceLevel (
            TraceManager::Level _traceLevel )
```

setTraceLevel

Parameters

_traceLevel	

17.1058.2.14 setTraceSimulationRuleAllAllowed()

```
void TraceManager::setTraceSimulationRuleAllAllowed (
            bool _traceSimulationRuleAllAllowed )
```

setTraceSimulationRuleAllAllowed

Parameters

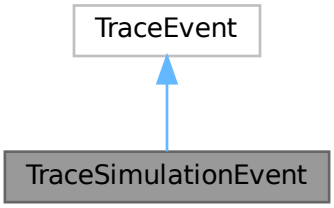
_traceSimulationRuleAllAllowed	

The documentation for this class was generated from the following files:

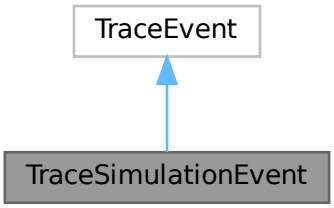
- kernel/simulator/TraceManager.h
- kernel/simulator/TraceManager.cpp

17.1059 TraceSimulationEvent Class Reference

Inheritance diagram for TraceSimulationEvent:



Collaboration diagram for TraceSimulationEvent:



Public Member Functions

- **TraceSimulationEvent** (TraceManager::Level level, double time, Entity *entity, ModelComponent *component, const std::string &text)
- ModelComponent * **getComponent** () const
- Entity * **getEntity** () const
- double **getTime** () const

Public Member Functions inherited from TraceEvent

- **TraceEvent** (TraceManager::Level level, const std::string &text)
- **TraceEvent** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- TraceManager::Level **getTracelevel** () const
- const std::string & **getText** () const

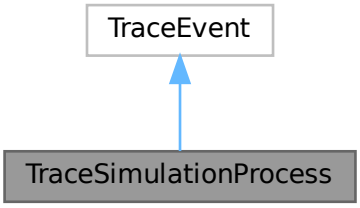
The documentation for this class was generated from the following file:

- kernel/simulator/TraceManager.h

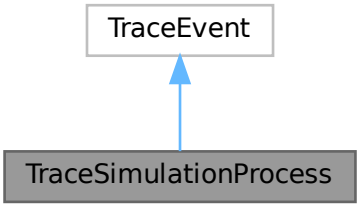
17.1060 TraceSimulationProcess Class Reference

`#include <kernel/simulator/TraceManager.h>`

Inheritance diagram for TraceSimulationProcess:



Collaboration diagram for TraceSimulationProcess:



Public Member Functions

- **TraceSimulationProcess** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_↩ detailed)

Public Member Functions inherited from TraceEvent

- **TraceEvent** (TraceManager::Level level, const std::string &text)
- **TraceEvent** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- TraceManager::Level **getTracelevel** () const
- const std::string & **getText** () const

17.1060.1 Detailed Description

Events related to simulation "process" (usually process analyser), associated to entire replication or simulation events (begin/end/pause of replication/simulation) @ToDo: (importante): CLASS NOT FULLY IMPLEMENTED (to be implemented for process analyser)
The documentation for this class was generated from the following file:

- kernel/simulator/TraceManager.h

17.1061 TraitsGUI< T > Struct Template Reference

GUI-specific trait constants used to tune default rendering behavior.
`#include <applications/gui/genesys/TraitsGUI.h>`

17.1061.1 Detailed Description

template<class T>
struct TraitsGUI< T >

GUI-specific trait constants used to tune default rendering behavior.
These specializations keep visual defaults close to the legacy Qt interface without scattering magic numbers across the graphics and editor code.
The documentation for this struct was generated from the following file:

- applications/gui/genesys/TraitsGUI.h

17.1062 TraitsGUI< GAssociation > Struct Reference

Static Public Attributes

- static const uint64_t **foreColor** = 0x808080FF
- static constexpr const int **margin** = 2
- static constexpr const int **selectionWidth** = 8

The documentation for this struct was generated from the following file:

- applications/gui/genesys/TraitsGUI.h

17.1063 TraitsGUI< GComponentPort > Struct Reference

Static Public Attributes

- static constexpr const int **width** = 20
- static constexpr const int **penWidth** = 1
- static constexpr const int **raise** = 3
- static constexpr const int **margin** = 2
- static const uint64_t **connectedBackgroundColor** = 0x2020FF80
- static const uint64_t **unconnectedBackgroundColor** = 0xFF202040
- static const uint64_t **borderColor** = 0x000000FF
- static const uint64_t **pathRaised** = 0x808080FF
- static const uint64_t **pathStunken** = 0xFFFFFFFF
- static const uint64_t **textColor** = 0xFFFFFFFF
- static const uint64_t **textShadowColor** = 0x000000FF
- static const uint64_t **selectionSquaresColor** = 0x000000FF

The documentation for this struct was generated from the following file:

- applications/gui/genesys/TraitsGUI.h

17.1064 TraitsGUI< GConnection > Struct Reference

Static Public Attributes

- static const uint64_t **foreColor** = 0x000000FF
- static constexpr const int **margin** = 2
- static constexpr const int **selectionWidth** = 8

The documentation for this struct was generated from the following file:

- applications/gui/genesys/TraitsGUI.h

17.1065 TraitsGUI< GMainWindow > Struct Reference

Static Public Attributes

- static const bool **startMaximized** = true
- static const int **zoomButtonChange** = 30
- static const uint64_t **consoleTextColor** = 0x008000FF

The documentation for this struct was generated from the following file:

- applications/gui/genesys/TraitsGUI.h

17.1066 TraitsGUI< GModelComponent > Struct Reference

Static Public Attributes

- static constexpr const int **opacity** = 256.0 ∗ 0.67
- static constexpr const int **colorIncrease** = 256.0 ∗ 0.2
- static constexpr const int **lighter** = 133
- static constexpr const int **width** = 150
- static constexpr const int **penWidth** = 1
- static constexpr const int **raise** = 5
- static constexpr const int **margin** = 8
- static constexpr const int **selectionWidth** = 8
- static constexpr const float **stretchPos** = 0.5
- static constexpr const float **stretch** = 0.0
- static constexpr const float **heightProportion** = 0.67
- static const uint64_t **breakpointColor** = 0xFF0000FF
- static const uint64_t **borderColor** = 0x000000FF
- static const uint64_t **pathRaised** = 0xFFFFFFFF
- static const uint64_t **pathStunken** = 0x808080FF
- static const uint64_t **textColor** = 0xFFFFFFFF
- static const uint64_t **textShadowColor** = 0x000000FF
- static const uint64_t **selectionSquaresColor** = 0x000000FF

The documentation for this struct was generated from the following file:

- applications/gui/genesys/TraitsGUI.h

17.1067 TraitsGUI< GModelDataDefinition > Struct Reference

Static Public Attributes

- static constexpr const int **opacity** = 256 ∗ 0.33
- static constexpr const int **nonEditableOpacity** = 58
- static constexpr const int **selectedNonEditableOpacity** = 72
- static constexpr const int **width** = 100
- static constexpr const int **penWidth** = 1
- static constexpr const int **raise** = 3
- static constexpr const int **margin** = 5
- static constexpr const int **selectionWidth** = 8
- static constexpr const float **stretchPos** = 0.5
- static constexpr const float **stretch** = 0.0
- static constexpr const float **heightProportion** = 0.50
- static const uint64_t **breakpointColor** = 0xFF0000FF
- static const uint64_t **borderColor** = 0x000000FF
- static const uint64_t **pathRaised** = 0x808080FF

- static const uint64_t **pathStunken** = 0xFFFFFFFF
- static const uint64_t **textColor** = 0x0
- static const uint64_t **textShadowColor** = 0x000000FF
- static const uint64_t **selectionSquaresColor** = 0x000000FF

The documentation for this struct was generated from the following file:

- applications/gui/genesys/TraitsGUI.h

17.1068 TraitsGUI< GModelGraphicPic > Struct Reference

Static Public Attributes

- static const int **nodeComponentFontSize** = 12
- static const uint64_t **nodeComponentFontColor** = 0x000000FF
- static const uint64_t **nodeComponentFillColor** = 0x80A040FF
- static const int **edgeComponentFontSize** = 7
- static const uint64_t **edgeComponentEdgeColor** = 0x000000FF
- static const uint64_t **edgeComponentFontColor** = 0x000000FF
- static const int **nodeDatadefInternalFontSize** = 8
- static const uint64_t **nodeDatadefInternalColor** = 0xA0A0A0FF
- static const uint64_t **nodeDatadefInternalFontColor** = 0xA0A0A0FF
- static const int **nodeDatadefAttachedFontSize** = 8
- static const uint64_t **nodeDatadefAttachedFontColor** = 0x808080FF

The documentation for this struct was generated from the following file:

- applications/gui/genesys/TraitsGUI.h

17.1069 TraitsGUI< GScene > Struct Reference

Static Public Attributes

- static constexpr const int **gridInterval** = 20
- static constexpr const int **gridPenWidth** = 1
- static const uint64_t **gridColor** = 0xC0C0C0E0

The documentation for this struct was generated from the following file:

- applications/gui/genesys/TraitsGUI.h

17.1070 TraitsGUI< GView > Struct Reference

Static Public Attributes

- static constexpr const int **sceneCenter** = 25000
- static constexpr const int **sceneDistanceCenter** = 25000
- static const uint64_t **backgroundEnabledColor** = 0xFFFF8040
- static const uint64_t **backgroundDisabledColor** = 0xC0C0C020

The documentation for this struct was generated from the following file:

- applications/gui/genesys/TraitsGUI.h

17.1071 TraitsKernel< T > Struct Template Reference

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1072 TraitsKernel< Collector_if > Struct Reference

Public Types

- typedef CollectorDatafileDefaultImpl1 **Implementation**
- typedef double **DataType**

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1073 TraitsKernel< Model > Struct Reference

Public Types

- typedef StatisticsDefaultImpl1 **StatisticsCollector_StatisticsImplementation**
- typedef CollectorDefaultImpl1 **StatisticsCollector_CollectorImplementation**

Static Public Attributes

- static const TraceManager::Level **traceLevel** = TraceManager::Level::L5_event

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1074 TraitsKernel< ModelChecker_if > Struct Reference

Public Types

- typedef ModelCheckerDefaultImpl1 **Implementation**

Static Public Attributes

- static const TraceManager::Level **traceLevel** = TraceManager::Level::L2_results

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1075 TraitsKernel< ModelComponent > Struct Reference

Static Public Attributes

- static constexpr bool **reportStatistics** = true
- static const TraceManager::Level **traceLevel** = TraceManager::Level::L2_results

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1076 TraitsKernel< ModelDataDefinition > Struct Reference

Static Public Attributes

- static constexpr bool **reportStatistics** = true
- static const TraceManager::Level **traceLevel** = TraceManager::Level::L2_results

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1077 TraitsKernel< Parser_if > Struct Reference

Public Types

- typedef ParserDefaultImpl2 **Implementation**

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1078 TraitsKernel< Persistence_if > Struct Reference

Public Types

- typedef PersistencePartialLoadImpl1 **Implementation**

Static Public Attributes

- static const TraceManager::Level **traceLevel** = TraceManager::Level::L2_results

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1079 TraitsKernel< PluginConnector_if > Struct Reference

Public Types

- typedef PluginConnectorDummyImpl1 **Implementation**

Static Public Attributes

- static const TraceManager::Level **traceLevel** = TraceManager::Level::L4_warning

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1080 TraitsKernel< Sampler_if > Struct Reference

Public Types

- typedef SamplerDefaultImpl1 **Implementation**
- typedef SamplerDefaultImpl1::DefaultImpl1RNG_Parameters **Parameters**

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1081 TraitsKernel< SimulationReporter_if > Struct Reference

Public Types

- typedef SimulationReporterDefaultImpl1 **Implementation**
- typedef Counter **CounterImplementation**

Static Public Attributes

- static const TraceManager::Level **traceLevel** = TraceManager::Level::L2_results

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1082 TraitsKernel< Simulator > Struct Reference

Static Public Attributes

- static const TraceManager::Level **traceLevel** = TraceManager::Level::L9_mostDetailed

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1083 TraitsKernel< Statistics_if > Struct Reference

Public Types

- typedef double **DataType**

Static Public Attributes

- static constexpr double **SignificanceLevel** = 0.05

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1084 TraitsKernel< StatisticsDatafile_if > Struct Reference

Public Types

- typedef StatisticsDatafileDefaultImpl1 **Implementation**
- typedef CollectorDatafileDefaultImpl1 **CollectorImplementation**

Static Public Attributes

- static constexpr double **SignificanceLevel** = 0.05

The documentation for this struct was generated from the following file:

- kernel/TraitsKernel.h

17.1085 TraitsTools< T > Struct Template Reference

Registry-like traits to bind abstractions to concrete tools classes.
`#include <tools/TraitsTools.h>`

17.1085.1 Detailed Description

template<typename T>
struct TraitsTools< T >

Registry-like traits to bind abstractions to concrete tools classes.
Architectural role:

- Central selection point for concrete implementations used by this package.

Implementation status:

- Currently covers only a subset of tool abstractions.

Planned evolution:

- Extend bindings as new stable interfaces receive concrete implementations.

- New abstractions without safe concrete implementations should remain documented-only for now to avoid build instability.

The documentation for this struct was generated from the following file:

- tools/TraitsTools.h

17.1086 TraitsTools< AIAssistant_if > Struct Reference

`#include <tools/TraitsTools.h>`

Public Types

- typedef AIAssistantDefaultImpl **Implementation**

17.1086.1 Detailed Description

Configure the AI Assistant to be used
The documentation for this struct was generated from the following file:

- tools/TraitsTools.h

17.1087 TraitsTools< Fitter_if > Struct Reference

`#include <tools/TraitsTools.h>`

Public Types

- typedef FitterDefaultImpl **Implementation**

17.1087.1 Detailed Description

Configure the Fitter to be used
The documentation for this struct was generated from the following file:

- tools/TraitsTools.h

17.1088 TraitsTools< HypothesisTester_if > Struct Reference

`#include <tools/TraitsTools.h>`

Public Types

- typedef HypothesisTesterDefaultImpl1 **Implementation**

Static Public Attributes

- static constexpr unsigned int **ConfidenceLevel** = 95

17.1088.1 Detailed Description

Configure the Hypothesis Tester to be used
The documentation for this struct was generated from the following file:

- tools/TraitsTools.h

17.1089 TraitsTools< Solver_if > Struct Reference

`#include <tools/TraitsTools.h>`

Public Types

- typedef SolverDefaultImpl1 **Implementation**

Static Public Attributes

- static constexpr double **Precision** = 1e-5
- static constexpr unsigned int **MaxSteps** = 1e2

17.1089.1 Detailed Description

Configure the Solver to be used
The documentation for this struct was generated from the following file:

- tools/TraitsTools.h

17.1090 EFSMNetworkDiagramLayout::TransitionLayout Struct Reference

Public Attributes

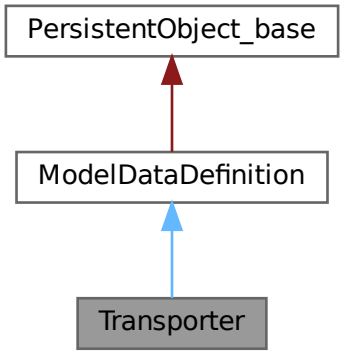
- QString **sourceName**
- QString **destinationName**
- QString **transitionName**
- int **ordinal** = 0
- QVector< QPointF > **controlPoints**
- bool **hasLabelPosition** = false
- QPointF **labelPosition**

The documentation for this struct was generated from the following file:

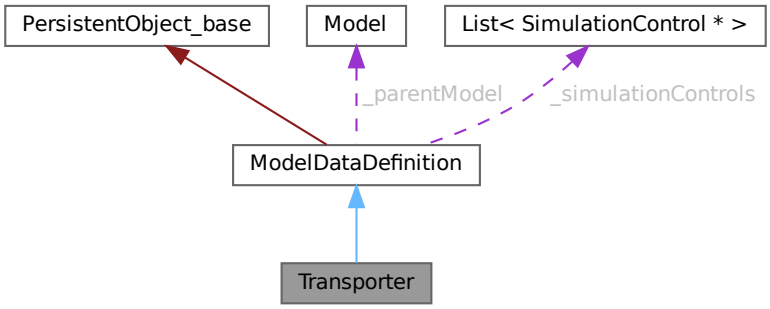
- applications/gui/genesys/services/EFSMNetworkDiagramLayout.h

17.1091 Transporter Class Reference

Inheritance diagram for Transporter:



Collaboration diagram for Transporter:



Classes

- struct TravelCompletion

Public Member Functions

- **Transporter** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- void **setDistance** (Distance *distance)
- Distance * **getDistance** () const
- void **setInitialStation** (Station *station)
- Station * **getInitialStation** () const
- Station * **getCurrentStation** () const
- void **setSpeed** (double speed)
- double **getSpeed** () const
- void **setInitiallyActive** (bool initiallyActive)
- bool **isInitiallyActive** () const
- void **setActive** (bool active)
- bool **isActive** () const
- bool **isBusy** () const
- bool **reserve** ()
- void **releaseAt** (Station *station)
- double **getTravelTime** (const Station *fromStation, const Station *toStation) const
- void **handleTravelCompletion** (void *parameter)

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)

- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1091.1 Member Function Documentation

17.1091.1.1 _check()

```
bool Transporter::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1091.1.2 _createEditableDataDefinitions()

```
void Transporter::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1091.1.3 _initBetweenReplications()

```
void Transporter::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1091.1.4 _loadInstance()

```
bool Transporter::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1091.1.5 _saveInstance()

```
void Transporter::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.1091.1.6 show()

```
std::string Transporter::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

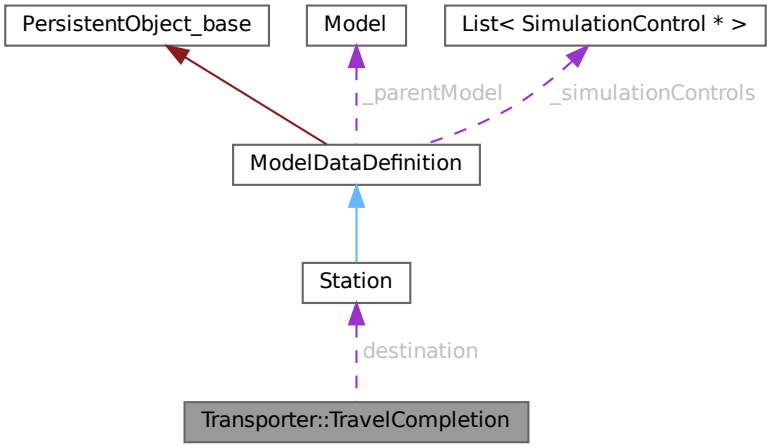
Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/MaterialHandling/Transporter.h
- plugins/data/MaterialHandling/Transporter.cpp

17.1092 Transporter::TravelCompletion Struct Reference

Collaboration diagram for Transporter::TravelCompletion:



Public Attributes

- Station * **destination** = nullptr

The documentation for this struct was generated from the following file:

- plugins/data/MaterialHandling/Transporter.h

17.1093 testing::internal::TrueWithString Struct Reference

Public Member Functions

- **TrueWithString** (const char *str)
- **TrueWithString** (const std::string &str)
- **operator bool** () const

Public Attributes

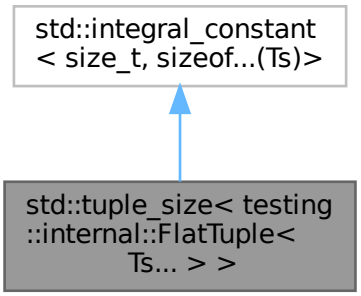
- std::string **value**

The documentation for this struct was generated from the following file:

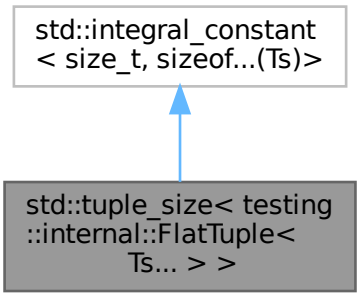
- gtest/gtest/internal/gtest-internal.h

17.1094 std::tuple_size< testing::internal::FlatTuple< Ts... > > Struct Template Reference

Inheritance diagram for std::tuple_size< testing::internal::FlatTuple< Ts... > >:



Collaboration diagram for std::tuple_size< testing::internal::FlatTuple< Ts... > >:



The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.1095 testing::internal::TypeIdHelper< T > Class Template Reference

Static Public Attributes

- static bool **dummy_** = false

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.1096 testing::internal::TypeParameterizedTest< Fixture, TestSel, Types > Class Template Reference

Static Public Member Functions

- static bool **Register** (const char *prefix, const CodeLocation &code_location, const char *case_name, const char *test_names, int index, const std::vector< std::string > &type_names=GenerateNames< DefaultNameGenerator, Types >())

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.1097 testing::internal::TypeParameterizedTest< Fixture, TestSel, internal::None > Class Template Reference

Static Public Member Functions

- static bool **Register** (const char *, const CodeLocation &, const char *, const char *, int, const std::vector< std::string > &=std::vector< std::string >())

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.1098 testing::internal::TypeParameterizedTestSuite< Fixture, Tests, Types > Class Template Reference

Static Public Member Functions

- static bool **Register** (const char *prefix, CodeLocation code_location, const TypedTestSuitePState *state, const char *case_name, const char *test_names, const std::vector< std::string > &type_names=Generate↩ Names< DefaultNameGenerator, Types >())

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.1099 testing::internal::TypeParameterizedTestSuite< Fixture, internal::None, Types > Class Template Reference

Static Public Member Functions

- static bool **Register** (const char *, const CodeLocation &, const TypedTestSuitePState *, const char *, const char *, const std::vector< std::string > &=std::vector< std::string >())

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-internal.h

17.1100 testing::internal::TypeParameterizedTestSuiteRegistry Class Reference

Public Member Functions

- void **RegisterTestSuite** (const char *test_suite_name, CodeLocation code_location)
- void **RegisterInstantiation** (const char *test_suite_name)

- void **CheckForInstantiations** ()

The documentation for this class was generated from the following files:

- gtest/gtest/internal/gtest-param-util.h
- gtest/src/gtest.cc

17.1101 testing::internal::Types< Head_, Tail_ > Struct Template Reference

Public Types

- using **Head** = Head_
- using **Tail** = Types< Tail_... >

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-type-util.h

17.1102 testing::internal::Types< Head_ > Struct Template Reference

Public Types

- using **Head** = Head_
- using **Tail** = None

The documentation for this struct was generated from the following file:

- gtest/gtest/internal/gtest-type-util.h

17.1103 testing::internal::TypeWithSize< size > Class Template Reference

Public Types

- using **UInt** = void

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-port.h

17.1104 testing::internal::TypeWithSize< 4 > Class Reference

Public Types

- using **Int** = std::int32_t
- using **UInt** = std::uint32_t

The documentation for this class was generated from the following file:

- gtest/gtest/internal/gtest-port.h

17.1105 testing::internal::TypeWithSize< 8 > Class Reference

Public Types

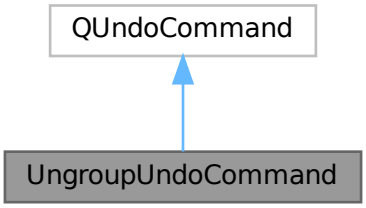
- using **Int** = std::int64_t
- using **UInt** = std::uint64_t

The documentation for this class was generated from the following file:

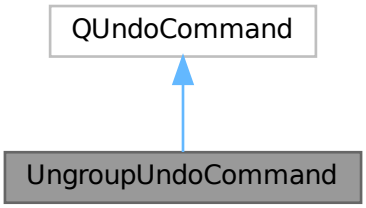
- gtest/gtest/internal/gtest-port.h

17.1106 UngroupUndoCommand Class Reference

Inheritance diagram for UngroupUndoCommand:



Collaboration diagram for UngroupUndoCommand:



Public Member Functions

- **UngroupUndoCommand** (QGraphicsItemGroup *group, ModelGraphicsScene *scene, QUndoCommand *parent=nullptr)
- void **undo** () override
- void **redo** () override

The documentation for this class was generated from the following files:

- applications/gui/genesys/actions/UngroupUndoCommand.h
- applications/gui/genesys/actions/UngroupUndoCommand.cpp

17.1107 testing::UnitTest Class Reference

Public Member Functions

- int **Run** () GTEST_MUST_USE_RESULT_
- const char * **original_working_dir** () const
- const TestSuite * **current_test_suite** () const GTEST_LOCK_EXCLUDED_(mutex_)
- const TestCase * **current_test_case** () const GTEST_LOCK_EXCLUDED_(mutex_)
- const TestInfo * **current_test_info** () const GTEST_LOCK_EXCLUDED_(mutex_)
- int **random_seed** () const

- internal::ParameterizedTestSuiteRegistry & **parameterized_test_registry** () GTEST_LOCK_EXCLUDED↩ _(mutex_)
- int **successful_test_suite_count** () const
- int **failed_test_suite_count** () const
- int **total_test_suite_count** () const
- int **test_suite_to_run_count** () const
- int **successful_test_case_count** () const
- int **failed_test_case_count** () const
- int **total_test_case_count** () const
- int **test_case_to_run_count** () const
- int **successful_test_count** () const
- int **skipped_test_count** () const
- int **failed_test_count** () const
- int **reportable_disabled_test_count** () const
- int **disabled_test_count** () const
- int **reportable_test_count** () const
- int **total_test_count** () const
- int **test_to_run_count** () const
- TimeInMillis **start_timestamp** () const
- TimeInMillis **elapsed_time** () const
- bool **Passed** () const
- bool **Failed** () const
- const TestSuite * **GetTestSuite** (int i) const
- const TestCase * **GetTestCase** (int i) const
- const TestResult & **ad_hoc_test_result** () const
- TestEventListeners & **listeners** ()

Static Public Member Functions

- static UnitTest * **GetInstance** ()

Friends

- class **ScopedTrace**
- class **Test**
- class **internal::AssertHelper**
- class **internal::StreamingListenerTest**
- class **internal::UnitTestRecordPropertyTestHelper**
- Environment * **AddGlobalTestEnvironment** (Environment *env)
- std::set< std::string > * **internal::GetIgnoredParameterizedTestSuites** ()
- internal::UnitTestImpl * **internal::GetUnitTestImpl** ()
- void **internal::ReportFailureInUnknownLocation** (TestPartResult::Type result_type, const std::string &message)

The documentation for this class was generated from the following files:

- gtest/gtest/gtest.h
- gtest/src/gtest.cc

17.1108 testing::internal::UnitTestImpl Class Reference

Public Types

- enum **ReactionToSharding** { **HONOR_SHARDING_PROTOCOL** , **IGNORE_SHARDING_PROTOCOL** }

Public Member Functions

- **UnitTestImpl** (UnitTest *parent)
- TestPartResultReporterInterface * **GetGlobalTestPartResultReporter** ()
- void **SetGlobalTestPartResultReporter** (TestPartResultReporterInterface *reporter)
- TestPartResultReporterInterface * **GetTestPartResultReporterForCurrentThread** ()
- void **SetTestPartResultReporterForCurrentThread** (TestPartResultReporterInterface *reporter)
- int **successful_test_suite_count** () const
- int **failed_test_suite_count** () const
- int **total_test_suite_count** () const
- int **test_suite_to_run_count** () const
- int **successful_test_count** () const
- int **skipped_test_count** () const
- int **failed_test_count** () const
- int **reportable_disabled_test_count** () const
- int **disabled_test_count** () const
- int **reportable_test_count** () const
- int **total_test_count** () const
- int **test_to_run_count** () const
- TimeInMillis **start_timestamp** () const
- TimeInMillis **elapsed_time** () const
- bool **Passed** () const
- bool **Failed** () const
- const TestSuite * **GetTestSuite** (int i) const
- const TestCase * **GetTestCase** (int i) const
- TestSuite * **GetMutableSuiteCase** (int i)
- TestEventListeners * **listeners** ()
- TestResult * **current_test_result** ()
- const TestResult * **ad_hoc_test_result** () const
- void **set_os_stack_trace_getter** (OsStackTraceGetterInterface *getter)
- OsStackTraceGetterInterface * **os_stack_trace_getter** ()
- std::string **CurrentOsStackTraceExceptTop** (int skip_count) GTEST_NO_INLINE_ GTEST_NO_TAIL_↩ CALL_
- TestSuite * **GetTestSuite** (const char *test_suite_name, const char *type_param, internal::SetUpTestSuite↩ Func set_up_tc, internal::TearDownTestSuiteFunc tear_down_tc)
- TestCase * **GetTestCase** (const char *test_case_name, const char *type_param, internal::SetUpTestSuite↩ Func set_up_tc, internal::TearDownTestSuiteFunc tear_down_tc)
- void **AddTestInfo** (internal::SetUpTestSuiteFunc set_up_tc, internal::TearDownTestSuiteFunc tear_down_tc, TestInfo *test_info)
- internal::ParameterizedTestSuiteRegistry & **parameterized_test_registry** ()
- std::set< std::string > * **ignored_parameterized_test_suites** ()
- internal::TypeParameterizedTestSuiteRegistry & **type_parameterized_test_registry** ()
- void **set_current_test_suite** (TestSuite *a_current_test_suite)
- void **set_current_test_info** (TestInfo *a_current_test_info)
- void **RegisterParameterizedTests** ()
- bool **RunAllTests** ()
- void **ClearNonAdHocTestResult** ()
- void **ClearAdHocTestResult** ()
- void **RecordProperty** (const TestProperty &test_property)
- int **FilterTests** (ReactionToSharding shard_tests)
- void **ListTestsMatchingFilter** ()
- const TestSuite * **current_test_suite** () const
- TestInfo * **current_test_info** ()
- const TestInfo * **current_test_info** () const
- std::vector< Environment * > & **environments** ()

- std::vector< TraceInfo > & **gtest_trace_stack** ()
- const std::vector< TraceInfo > & **gtest_trace_stack** () const
- void **ConfigureXmlOutput** ()
- void **PostFlagParsingInit** ()
- int **random_seed** () const
- internal::Random * **random** ()
- void **ShuffleTests** ()
- void **UnshuffleTests** ()
- bool **catch_exceptions** () const

Friends

- class **::testing::UnitTest**

The documentation for this class was generated from the following files:

- gtest/src/gtest-internal-inl.h
- gtest/src/gtest.cc

17.1109 testing::internal::UnitTestOptions Class Reference

Static Public Member Functions

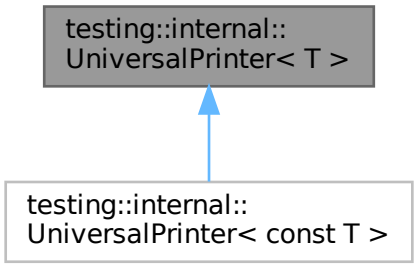
- static std::string **GetOutputFormat** ()
- static std::string **GetAbsolutePathToOutputFile** ()
- static bool **FilterMatchesTest** (const std::string &test_suite_name, const std::string &test_name)
- static bool **MatchesFilter** (const std::string &name, const char *filter)

The documentation for this class was generated from the following files:

- gtest/src/gtest-internal-inl.h
- gtest/src/gtest.cc

17.1110 testing::internal::UniversalPrinter< T > Class Template Reference

Inheritance diagram for testing::internal::UniversalPrinter< T >:



Static Public Member Functions

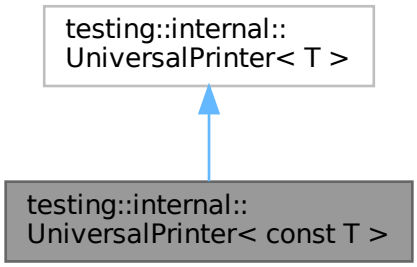
- static void **Print** (const T &value, ::std::ostream *os)

The documentation for this class was generated from the following file:

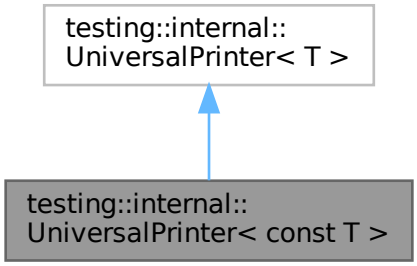
- gtest/gtest/gtest-printers.h

17.1111 testing::internal::UniversalPrinter< const T > Class Template Reference

Inheritance diagram for testing::internal::UniversalPrinter< const T >:



Collaboration diagram for testing::internal::UniversalPrinter< const T >:



Additional Inherited Members

Static Public Member Functions inherited from testing::internal::UniversalPrinter< T >

- static void **Print** (const T &value, ::std::ostream *os)

The documentation for this class was generated from the following file:

- gtest/gtest/gtest-printers.h

17.1112 testing::internal::UniversalPrinter< T & > Class Template Reference

Static Public Member Functions

- static void **Print** (const T &value, ::std::ostream *os)

The documentation for this class was generated from the following file:

- gtest/gtest/gtest-printers.h

17.1113 testing::internal::UniversalPrinter< T[N]> Class Template Reference

Static Public Member Functions

- static void **Print** (const T(&a)[N], ::std::ostream *os)

The documentation for this class was generated from the following file:

- gtest/gtest/gtest-printers.h

17.1114 testing::internal::UniversalTersePrinter< T > Class Template Reference

Static Public Member Functions

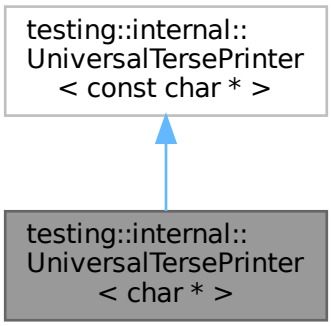
- static void **Print** (const T &value, ::std::ostream *os)

The documentation for this class was generated from the following file:

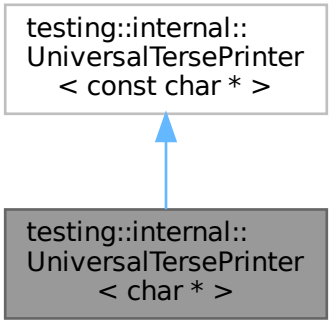
- gtest/gtest/gtest-printers.h

17.1115 testing::internal::UniversalTersePrinter< char * > Class Reference

Inheritance diagram for testing::internal::UniversalTersePrinter< char * >:



Collaboration diagram for testing::internal::UniversalTersePrinter< char * >:



Additional Inherited Members

Static Public Member Functions inherited from testing::internal::UniversalTersePrinter< const char * >

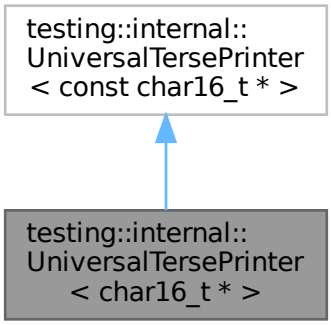
- static void **Print** (const char *str, ::std::ostream *os)

The documentation for this class was generated from the following file:

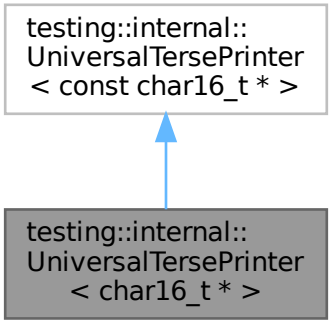
- gtest/gtest/gtest-printers.h

17.1116 testing::internal::UniversalTersePrinter< char16_t * > Class Reference

Inheritance diagram for testing::internal::UniversalTersePrinter< char16_t * >:



Collaboration diagram for testing::internal::UniversalTersePrinter< char16_t * >:



Additional Inherited Members

Static Public Member Functions inherited from testing::internal::UniversalTersePrinter< const char16_t * >

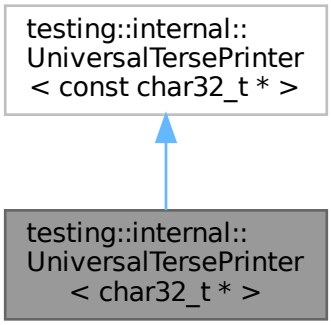
- static void **Print** (const char16_t *str, ::std::ostream *os)

The documentation for this class was generated from the following file:

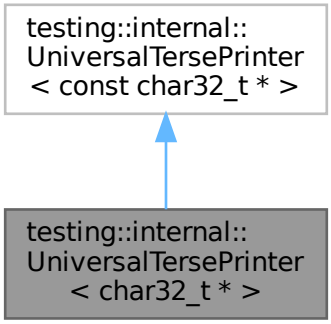
- gtest/gtest/gtest-printers.h

17.1117 testing::internal::UniversalTersePrinter< char32_t * > Class Reference

Inheritance diagram for testing::internal::UniversalTersePrinter< char32_t * >:



Collaboration diagram for testing::internal::UniversalTersePrinter< char32_t * >:



Additional Inherited Members

Static Public Member Functions inherited from testing::internal::UniversalTersePrinter< const char32_t * >

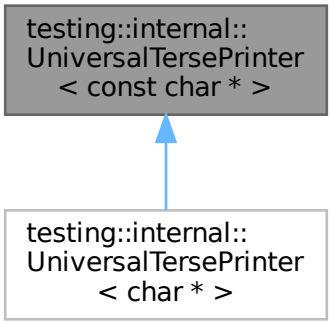
- static void **Print** (const char32_t *str, ::std::ostream *os)

The documentation for this class was generated from the following file:

- gtest/gtest/gtest-printers.h

17.1118 testing::internal::UniversalTersePrinter< const char * > Class Reference

Inheritance diagram for testing::internal::UniversalTersePrinter< const char * >:



Static Public Member Functions

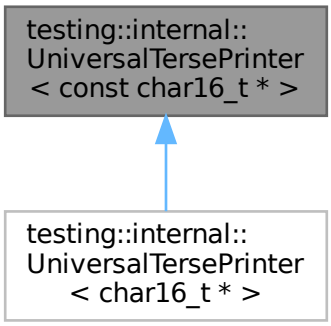
- static void **Print** (const char *str, ::std::ostream *os)

The documentation for this class was generated from the following file:

- gtest/gtest/gtest-printers.h

17.1119 testing::internal::UniversalTersePrinter< const char16_t * > Class Reference

Inheritance diagram for testing::internal::UniversalTersePrinter< const char16_t * >:



Static Public Member Functions

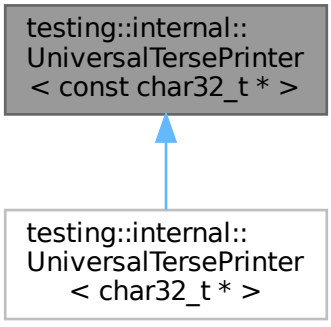
- static void **Print** (const char16_t *str, ::std::ostream *os)

The documentation for this class was generated from the following file:

- gtest/gtest/gtest-printers.h

17.1120 testing::internal::UniversalTersePrinter< const char32_t * > Class Reference

Inheritance diagram for testing::internal::UniversalTersePrinter< const char32_t * >:



Static Public Member Functions

- static void **Print** (const char32_t *str, ::std::ostream *os)

The documentation for this class was generated from the following file:

- gtest/gtest/gtest-printers.h

17.1121 testing::internal::UniversalTersePrinter< T & > Class Template Reference

Static Public Member Functions

- static void **Print** (const T &value, ::std::ostream *os)

The documentation for this class was generated from the following file:

- gtest/gtest/gtest-printers.h

17.1122 testing::internal::UniversalTersePrinter< T[N]> Class Template Reference

Static Public Member Functions

- static void **Print** (const T(&value)[N], ::std::ostream *os)

The documentation for this class was generated from the following file:

- gtest/gtest/gtest-printers.h

17.1123 testing::internal::UniversalTersePrinter< wchar_t * > Class Reference

Static Public Member Functions

- static void **Print** (wchar_t *str, ::std::ostream *os)

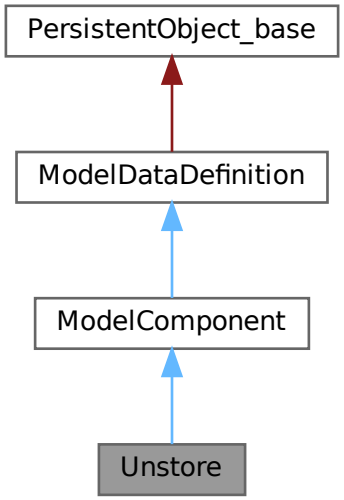
The documentation for this class was generated from the following file:

- gtest/gtest/gtest-printers.h

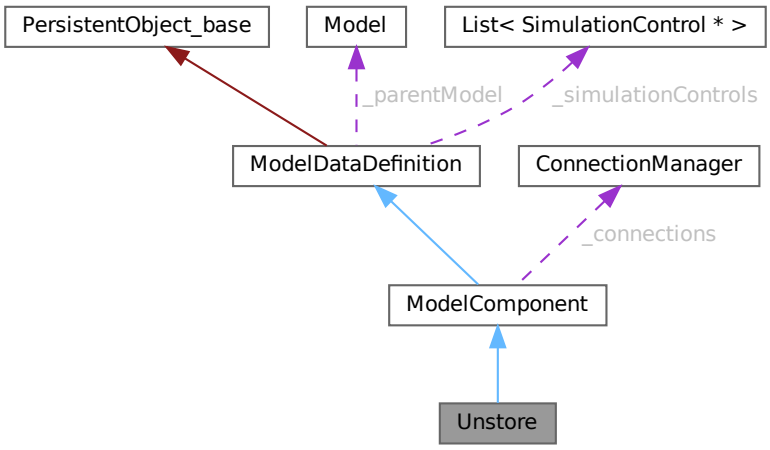
17.1124 Unstore Class Reference

Placeholder for removing an entity from a Storage (§5.11).
`#include <plugins/components/MaterialHandling/Unstore.h>`

Inheritance diagram for Unstore:



Collaboration diagram for Unstore:



Public Member Functions

- **Unstore** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setStorage** (Storage ∗storage)
- Storage ∗ **getStorage** () const
- void **setQuantityExpression** (std::string quantityExpression)
- std::string **getQuantityExpression** () const

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _createEditableDataDefinitions () override
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()

- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1124.1 Detailed Description

Placeholder for removing an entity from a Storage (§5.11).
Arena correspondence: the "Unstore module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 67).

Warning

Incomplete stub template: this class declares no fields at all, and `_onDispatchEvent()` unconditionally forwards the entity without touching any Storage. See `docs/ai_assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.17.

17.1124.2 Member Function Documentation

17.1124.2.1 _check()

```
bool Unstore::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1124.2.2 _createEditableDataDefinitions()

```
void Unstore::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1124.2.3 _loadInstance()

```
bool Unstore::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.1124.2.4 _onDispatchEvent()

```
void Unstore::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.1124.2.5 _saveInstance()

```
void Unstore::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.1124.2.6 show()

```
std::string Unstore::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/MaterialHandling/Unstore.h
- plugins/components/MaterialHandling/Unstore.cpp

17.1125 genesys::launcher::UpdateChecker Class Reference

Public Types

- using **Clock** = std::function< QDateTime()>

Public Member Functions

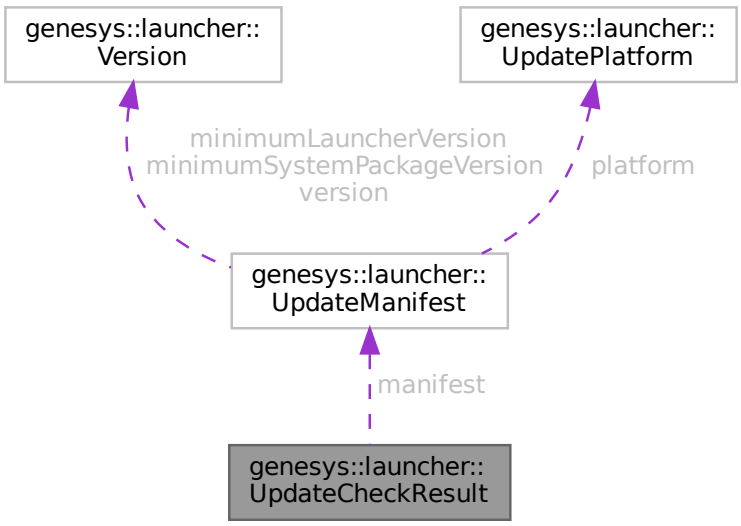
- **UpdateChecker** (RuntimePaths paths, LauncherConfig config, PlatformInfo platform, Version current↩ RuntimeVersion, Version launcherVersion, Version systemPackageVersion, INetworkTransport &transport, LauncherLogger *logger=nullptr, Clock clock={})
- UpdateCheckResult **check** (bool force=false)

The documentation for this class was generated from the following files:

- applications/launcher/UpdateChecker.h
- applications/launcher/UpdateChecker.cpp

17.1126 genesys::launcher::UpdateCheckResult Struct Reference

Collaboration diagram for genesys::launcher::UpdateCheckResult:



Public Attributes

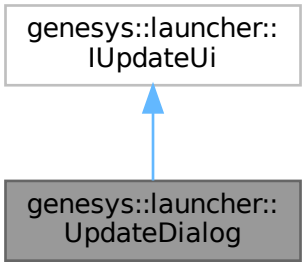
- UpdateCheckStatus **status** = UpdateCheckStatus::Disabled
- bool **ok** = false
- bool **updateAvailable** = false
- UpdateManifest **manifest**
- QString **message**

The documentation for this struct was generated from the following file:

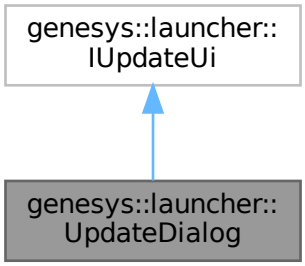
- applications/launcher/UpdateChecker.h

17.1127 genesys::launcher::UpdateDialog Class Reference

Inheritance diagram for genesys::launcher::UpdateDialog:



Collaboration diagram for genesys::launcher::UpdateDialog:



Public Member Functions

- **UpdateDialog** (QWidget *parent=nullptr)
- bool confirmUpdate (const Version &currentVersion, const Version &candidateVersion, const QString &releaseNotes) override
- void beginDownload (qint64 totalBytes) override
- bool updateDownloadProgress (qint64 receivedBytes, qint64 totalBytes) override
- void finishDownload () override
- void showError (const QString &message) override
- void showInfo (const QString &message) override

17.1127.1 Member Function Documentation

17.1127.1.1 beginDownload()

```
void genesys::launcher::UpdateDialog::beginDownload (
            qint64 totalBytes )  [override], [virtual]
```

Implements genesys::launcher::IUpdateUi.

17.1127.1.2 confirmUpdate()

```
bool genesys::launcher::UpdateDialog::confirmUpdate (
            const Version & currentVersion,
            const Version & candidateVersion,
            const QString & releaseNotes )  [override], [virtual]
```

Implements genesys::launcher::IUpdateUi.

17.1127.1.3 finishDownload()

```
void genesys::launcher::UpdateDialog::finishDownload ( )  [override], [virtual]
```

Implements genesys::launcher::IUpdateUi.

17.1127.1.4 showError()

```
void genesys::launcher::UpdateDialog::showError (
            const QString & message )  [override], [virtual]
```

Implements genesys::launcher::IUpdateUi.

17.1127.1.5 showInfo()

```
void genesys::launcher::UpdateDialog::showInfo (
            const QString & message )  [override], [virtual]
```

Implements genesys::launcher::IUpdateUi.

17.1127.1.6 updateDownloadProgress()

```
bool genesys::launcher::UpdateDialog::updateDownloadProgress (
            qint64 receivedBytes,
            qint64 totalBytes )  [override], [virtual]
```

Implements genesys::launcher::IUpdateUi.
The documentation for this class was generated from the following files:

- applications/launcher/UpdateDialog.h
- applications/launcher/UpdateDialog.cpp

17.1128 genesys::launcher::UpdateDownloader Class Reference

Public Member Functions

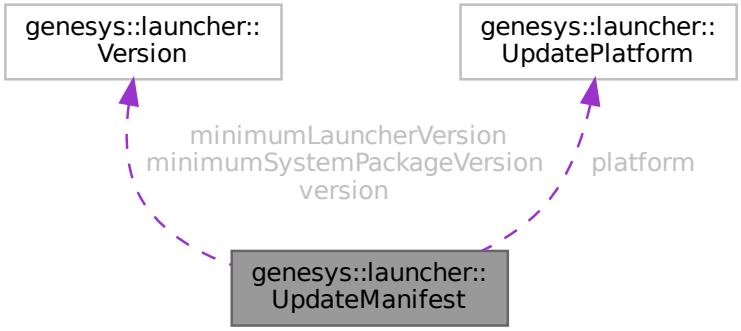
- **UpdateDownloader** (RuntimePaths paths, INetworkTransport &transport, LauncherLogger *logger=nullptr)
- DownloadResult **download** (const UpdateManifest &manifest, bool downloadSignature, ProgressCallback progress={}, CancellationCallback cancelled={})
- bool **promoteVerified** (const UpdateManifest &manifest, const DownloadResult &download, QString *verifiedBundlePath, QString *verifiedSignaturePath, QString *error=nullptr) const
- void **cleanup** (const DownloadResult &download) const

The documentation for this class was generated from the following files:

- applications/launcher/UpdateDownloader.h
- applications/launcher/UpdateDownloader.cpp

17.1129 genesys::launcher::UpdateManifest Struct Reference

Collaboration diagram for genesys::launcher::UpdateManifest:



Public Attributes

- int **schemaVersion** = 0
- QString **product**
- QString **channel**
- Version **version**
- QString **releaseTag**
- QString **publishedAt**
- Version **minimumLauncherVersion**
- Version **minimumSystemPackageVersion**
- UpdatePlatform **platform**
- QString **notesSummary**
- QUrl **notesUrl**

The documentation for this struct was generated from the following file:

- applications/launcher/UpdateManifest.h

17.1130 genesys::launcher::UpdateManifestParser Class Reference

Static Public Member Functions

- static ManifestParseResult **parse** (const QByteArray &json, const QString &expectedChannel, const QString &platformKey, const Version &currentRuntimeVersion, const Version &launcherVersion, const Version &systemPackageVersion)

Static Public Attributes

- static constexpr qint64 **MaximumBundleSize** = 2LL ∗ 1024 ∗ 1024 ∗ 1024

The documentation for this class was generated from the following files:

- applications/launcher/UpdateManifest.h
- applications/launcher/UpdateManifest.cpp

17.1131 genesys::launcher::UpdatePlatform Struct Reference

Public Attributes

- QString **key**
- QString **kind**
- QUrl **bundleUrl**
- QString **sha256**
- QUrl **signatureUrl**
- qint64 **size** = 0
- QStringList **applications**

The documentation for this struct was generated from the following file:

- applications/launcher/UpdateManifest.h

17.1132 genesys::launcher::UpdateVerifier Class Reference

Static Public Member Functions

- static VerificationResult **verifySha256** (const QString &filePath, const QString &expectedHexDigest)

The documentation for this class was generated from the following files:

- applications/launcher/UpdateVerifier.h
- applications/launcher/UpdateVerifier.cpp

17.1133 genesys::launcher::UserRuntimeManager Class Reference

Public Member Functions

- **UserRuntimeManager** (RuntimePaths paths, IArchiveExtractor &extractor, LauncherLogger ∗logger=nullptr)
- **UserRuntimeManager** (const UserRuntimeManager &)=delete
- UserRuntimeManager & **operator=** (const UserRuntimeManager &)=delete
- bool **beginUpdate** (QString ∗error=nullptr, int lockTimeoutMs=0)
- void **endUpdate** ()
- bool **hasUpdateLock** () const noexcept
- RuntimeInstallResult **installAndActivate** (const QString &verifiedBundlePath, const UpdateManifest &manifest, int maxUserVersions)
- bool **validateRuntime** (const QString &runtimeRoot, const UpdateManifest &manifest, QString ∗error=nullptr) const

The documentation for this class was generated from the following files:

- applications/launcher/UserRuntimeManager.h
- applications/launcher/UserRuntimeManager.cpp

17.1134 Util Class Reference

Classes

- class CommandResult

Public Types

- enum class TimeUnit : int {
  unknown = 0 , **picosecond** = 1 , **nanosecond** = 2 , **microsecond** = 3 ,
  milisecond = 4 , **second** = 5 , **minute** = 6 , **hour** = 7 ,
  day = 8 , **week** = 9 , **num_elements** = 10 }

  The TimeUnit enum.
- enum class **AllocationType** : int {
  ValueAdded = 0 , **NonValueAdded** = 1 , **Transfer** = 2 , **Wait** = 3 ,
  Others = 4 , **num_elements** = 5 }
- enum class **TimeFormat** : unsigned int { **twelve** = 12 , **twentyFour** = 24 }
- typedef unsigned long **identification**
- typedef unsigned int **rank**

Static Public Member Functions

- static std::string StrTimeUnitShort (Util::TimeUnit timeUnit)

  Returns the short textual symbol for a time unit.
- static std::string StrTimeUnitLong (Util::TimeUnit timeUnit)

  Returns the long textual label for a time unit.
- static std::string convertEnumToStr (Util::TimeUnit timeUnit)

  Converts a time unit enum value to its canonical string representation.
- static std::string StrAllocation (Util::AllocationType allocation)

  Returns the string representation for an allocation category.
- static std::string convertEnumToStr (Util::AllocationType allocation)

  Converts an allocation category enum value to a string.
- static void SetIndent (const unsigned short indent)

  Sets the indentation level used by `Indent()`.
- static void **IncIndent** ()

  Increments the current indentation level.
- static void **DecIndent** ()

  Decrements the current indentation level.
- static void SepKeyVal (std::string str, std::string &key, std::string &value)

  Splits a key-value string around the first = character.
- static std::string Indent ()

  Returns the current indentation prefix.
- static std::string SetW (std::string text, unsigned short width)

  Pads or truncates text to a fixed width.
- static std::string StrTruncIfInt (double value)

  Formats a numeric value without losing integer semantics.
- static std::string StrTruncIfInt (std::string strValue)

  Removes a trailing .000000 suffix when present.
- static std::string Trim (std::string str)

  Trims leading and trailing whitespace.
- static std::string StrReplace (std::string text, std::string searchFor, std::string replaceBy)

  Replaces all occurrences of one substring with another.
- static std::string StrReplaceSpecialChars (std::string text)

  Sanitizes a text for identifier-like usage.
- static std::string StrIndex (int index)

  Formats an integer index in bracket notation.
- static void Trimwithin (std::string &str)

  Removes all spaces from a string in place.
- static std::string Map2str (std::map< std::string, std::string > *mapss)

Serializes a string-to-string map to a compact text form.

- static std::string Map2str (std::map< std::string, double > *mapss)

  Serializes a string-to-double map to a compact text form.
- static std::string List2str (std::list< unsigned int > *list)

  Serializes a list of unsigned integers to a compact text form.
- static Util::identification GenerateNewId ()

  Generates a globally unique identifier.
- static Util::identification GenerateNewIdOfType (std::string objtype)

  Generates the next identifier for a given runtime type name.
- static Util::identification GetLastIdOfType (std::string objtype)

  Returns the most recently generated identifier for a type.
- static void ResetIdOfType (std::string objtype)

  Resets the identifier counter for a specific type.
- static void **ResetAllIds** ()

  Resets all identifier counters maintained by the utility class.
- static double TimeUnitConvert (Util::TimeUnit timeUnit1, Util::TimeUnit timeUnit2)

  Computes the multiplicative factor between two time units.
- static char DirSeparator ()

  Returns the platform directory separator character.
- static std::string FilenameFromFullFilename (const std::string &s)

  Extracts the file name from a full path.
- static void FileDelete (const std::string &filename)

  Deletes a file if it exists.
- static std::string PathFromFullFilename (const std::string &s)

  Extracts the directory path from a full filename.
- static std::string RunningPath ()

  Returns the directory that contains the running binary.
- static std::vector< std::string > ListFiles (std::string dir, std::string fileFilter="", mode_t attribFilter=S_IFREG &S_IFDIR)

  Lists files in a directory using an optional substring filter.
- static bool FileExists (const std::string &name)

  Checks whether a file can be opened for reading.
- static Util::CommandResult ExecuteCommand (std::string command)

  Executes a shell command and captures the result.
- template<typename T >
  static std::string TypeOf ()

  Returns the compiler type name used for `T`.
- template<class T >
  static Util::identification GenerateNewIdOfType ()

  Generates the next identifier in the namespace of `T`.

17.1134.1 Member Function Documentation

17.1134.1.1 convertEnumToStr() [1/2]

```
std::string Util::convertEnumToStr (
            Util::AllocationType allocation ) [static]
```

Converts an allocation category enum value to a string.

Parameters

allocation	Allocation category to convert.

Returns

Canonical string representation for the enum value.

17.1134.1.2 convertEnumToStr() [2/2]

```
std::string Util::convertEnumToStr (
            Util::TimeUnit timeUnit )  [static]
```

Converts a time unit enum value to its canonical string representation.

Parameters

timeUnit	Time unit to convert.

Returns

Canonical string representation for the enum value.

17.1134.1.3 DirSeparator()

```
char Util::DirSeparator ( )  [static]
```

Returns the platform directory separator character.

Returns

Directory separator for the current platform.

17.1134.1.4 ExecuteCommand()

```
Util::CommandResult Util::ExecuteCommand (
            std::string command )  [static]
```

Executes a shell command and captures the result.

Parameters

command	Command line to run.

Returns

Command execution outcome and captured output buffers.

17.1134.1.5 FileDelete()

```
void Util::FileDelete (
            const std::string & filename )  [static]
```

Deletes a file if it exists.

Parameters

filename	Path to the file to delete.

17.1134.1.6 FileExists()

```
bool Util::FileExists (
            const std::string & name )  [static]
```

Checks whether a file can be opened for reading.

Parameters

name	File path to test.

Returns

`true` if the file exists and can be opened.

17.1134.1.7 FilenameFromFullFilename()

```
std::string Util::FilenameFromFullFilename (
            const std::string & s )  [static]
```

Extracts the file name from a full path.

Parameters

s	Full file path.

Returns

Trailing file name portion.

17.1134.1.8 GenerateNewId()

```
Util::identification Util::GenerateNewId ( )  [static]
```

Generates a globally unique identifier.

Returns

Next process-wide identifier value.

17.1134.1.9 GenerateNewIdOfType() [1/2]

```
template<class T >
static Util::identification Util::GenerateNewIdOfType ( )  [inline], [static]
```

Generates the next identifier in the namespace of `T`.

Template Parameters

T	Type whose namespace should receive a new identifier.

Returns

Next sequential identifier for the type.

17.1134.1.10 GenerateNewIdOfType() [2/2]

```
Util::identification Util::GenerateNewIdOfType (
            std::string objtype )  [static]
```

Generates the next identifier for a given runtime type name.

Parameters

objtype	Type name used as identifier namespace.

Returns

Next identifier within the requested type namespace.

17.1134.1.11 GetLastIdOfType()

```
Util::identification Util::GetLastIdOfType (
            std::string objtype )  [static]
```

Returns the most recently generated identifier for a type.

Parameters

objtype	Type name used as identifier namespace.

Returns

Last identifier value for the requested type.

17.1134.1.12 Indent()

```
std::string Util::Indent ( )  [static]
```

Returns the current indentation prefix.

Returns

A string made of repeated indentation markers.

17.1134.1.13 List2str()

```
std::string Util::List2str (
            std::list< unsigned int > * list )  [static]
```

Serializes a list of unsigned integers to a compact text form.

Parameters

list	List to serialize.

Returns

One-line representation of the list contents.

17.1134.1.14 ListFiles()

```
std::vector< std::string > Util::ListFiles (
            std::string dir,
            std::string fileFilter = "",
            mode_t attribFilter = S_IFREG & S_IFDIR )  [static]
```

Lists files in a directory using an optional substring filter.

Parameters

dir	Directory to scan.
fileFilter	Optional substring filter for file names.
attribFilter	Attribute mask applied to `stat` results.

Returns

Matching file names found in the directory.

17.1134.1.15 Map2str() [1/2]

```
std::string Util::Map2str (
            std::map< std::string, double > * mapss )  [static]
```

Serializes a string-to-double map to a compact text form.

Parameters

mapss	Map to serialize.

Returns

One-line representation of the map contents.

17.1134.1.16 Map2str() [2/2]

```
std::string Util::Map2str (
            std::map< std::string, std::string > * mapss )  [static]
```

Serializes a string-to-string map to a compact text form.

Parameters

mapss	Map to serialize.

Returns

One-line representation of the map contents.

17.1134.1.17 PathFromFullFilename()

```
std::string Util::PathFromFullFilename (
            const std::string & s )  [static]
```

Extracts the directory path from a full filename.

Parameters

s	Full file path.

Returns

Directory portion of the path.

17.1134.1.18 ResetIdOfType()

```
void Util::ResetIdOfType (
            std::string objtype )  [static]
```

Resets the identifier counter for a specific type.

Parameters

objtype	Type name to reset.

17.1134.1.19 RunningPath()

```
std::string Util::RunningPath ( )  [static]
```

Returns the directory that contains the running binary.

Returns

Current executable directory.

17.1134.1.20 SepKeyVal()

```
void Util::SepKeyVal (
            std::string str,
            std::string & key,
            std::string & value )  [static]
```

Splits a key-value string around the first = character.

Parameters

str	Source text in the form `key=value`.
key	Output key portion.
value	Output value portion.

17.1134.1.21 SetIndent()

```
void Util::SetIndent (
            const unsigned short indent )  [static]
```

Sets the indentation level used by `Indent()`.

Parameters

indent	New indentation depth.

17.1134.1.22 SetW()

```
std::string Util::SetW (
            std::string text,
            unsigned short width )  [static]
```

Pads or truncates text to a fixed width.

Parameters

text	Text to adjust.
width	Target width.

Returns

Text resized to the requested width.

17.1134.1.23 StrAllocation()

```
std::string Util::StrAllocation (
            Util::AllocationType allocation )  [static]
```

Returns the string representation for an allocation category.

Parameters

allocation	Allocation category to convert.

Returns

Allocation label used in reports and logs.

17.1134.1.24 StrIndex()

```
std::string Util::StrIndex (
            int index )  [static]
```

Formats an integer index in bracket notation.

Parameters

index	Index to format.

Returns

Text such as [4].

17.1134.1.25 StrReplace()

```
std::string Util::StrReplace (
            std::string text,
            std::string searchFor,
            std::string replaceBy )  [static]
```

Replaces all occurrences of one substring with another.

Parameters

text	Input text.
searchFor	Substring to replace.
replaceBy	Replacement text.

Returns

Updated text.

17.1134.1.26 StrReplaceSpecialChars()

```
std::string Util::StrReplaceSpecialChars (
            std::string text )  [static]
```

Sanitizes a text for identifier-like usage.

Parameters

text	Input text.

Returns

Text with spaces and selected punctuation normalized.

17.1134.1.27 StrTimeUnitLong()

```
std::string Util::StrTimeUnitLong (
            Util::TimeUnit timeUnit )  [static]
```

Returns the long textual label for a time unit.

Parameters

timeUnit	Time unit to convert.

Returns

Long label such as `millisecond` or `hour`.

17.1134.1.28 StrTimeUnitShort()

```
std::string Util::StrTimeUnitShort (
            Util::TimeUnit timeUnit )  [static]
```

Returns the short textual symbol for a time unit.

Parameters

timeUnit	Time unit to convert.

Returns

Short symbol such as `ms` or `h`.

17.1134.1.29 StrTruncIfInt() [1/2]

```
std::string Util::StrTruncIfInt (
            double value )  [static]
```

Formats a numeric value without losing integer semantics.

Parameters

value	Value to format.

Returns

A string that preserves the decimal point for exact integers.

17.1134.1.30 StrTruncIfInt() [2/2]

```
std::string Util::StrTruncIfInt (
            std::string strValue )  [static]
```

Removes a trailing .000000 suffix when present.

Parameters

strValue	String representation to normalize.

Returns

Normalized string value.

17.1134.1.31 TimeUnitConvert()

```
double Util::TimeUnitConvert (
            Util::TimeUnit timeUnit1,
            Util::TimeUnit timeUnit2 )  [static]
```

Computes the multiplicative factor between two time units.

Parameters

timeUnit1	Source unit.
timeUnit2	Target unit.

Returns

Conversion factor from `timeUnit1` to `timeUnit2`.

17.1134.1.32 Trim()

```
std::string Util::Trim (
            std::string str )  [static]
```

Trims leading and trailing whitespace.

Parameters

str	Input string.

Returns

Trimmed string.

17.1134.1.33 Trimwithin()

```
void Util::Trimwithin (
            std::string & str )  [static]
```

Removes all spaces from a string in place.

Parameters

str	String to normalize.

17.1134.1.34 TypeOf()

```
template<typename T >
static std::string Util::TypeOf ( )  [inline], [static]
```

Returns the compiler type name used for `T`.

Template Parameters

T	Type to inspect.

Returns

Compiler-provided type name, normalized when possible.

The documentation for this class was generated from the following files:

- kernel/util/Util.h
- kernel/util/Util.cpp

17.1135 yy::genesyspp_parser::value_type Class Reference

`#include <parser/GenesysParser.h>`

Public Types

- typedef value_type **self_type**

  Type of ∗this.

Public Member Functions

- **value_type** () YY_NOEXCEPT

  Empty construction.
- template<typename T >
  value_type (YY_RVREF(T) t)

  Construct and fill.
- ∼**value_type** () YY_NOEXCEPT

  Destruction, allowed only if empty.
- template<typename T >
  T & **emplace** ()

  Instantiate an empty T in here.
- template<typename T >
  T & **emplace** (const T &t)

  Instantiate a T in here from t.
- template<typename T >
  T & build ()
- template<typename T >
  T & build (const T &t)
- template<typename T >
  T & **as** () YY_NOEXCEPT

  Accessor to a built T.
- template<typename T >
  const T & **as** () const YY_NOEXCEPT

  Const accessor to a built T (for printer).
- template<typename T >
  void swap (self_type &that) YY_NOEXCEPT
- template<typename T >
  void move (self_type &that)
- template<typename T >
  void **copy** (const self_type &that)

  Copy the content of that to this.
- template<typename T >
  void **destroy** ()

  Destroy the stored T.

17.1135.1 Detailed Description

A buffer to store and retrieve objects.
Sort of a variant, but does not keep track of the nature of the stored data, since that knowledge is available via the current parser state.

17.1135.2 Member Function Documentation

17.1135.2.1 build() [1/2]

```
template<typename T >
T & yy::genesyspp_parser::value_type::build ( )  [inline]
```

Instantiate an empty *T* in here. Obsolete, use emplace.

17.1135.2.2 build() [2/2]

```
template<typename T >
T & yy::genesyspp_parser::value_type::build (
            const T & t )  [inline]
```

Instantiate a *T* in here from *t*. Obsolete, use emplace.

17.1135.2.3 move()

```
template<typename T >
void yy::genesyspp_parser::value_type::move (
            self_type & that )  [inline]
```

Move the content of *that* to this.
Destroys *that*.

17.1135.2.4 swap()

```
template<typename T >
void yy::genesyspp_parser::value_type::swap (
            self_type & that )  [inline]
```

Swap the content with *that*, of same type.
Both variants must be built beforehand, because swapping the actual data requires reading it (with as()), and this is not possible on unconstructed variants: it would require some dynamic testing, which should not be the variant's responsibility. Swapping between built and (possibly) non-built is done with self_type::move ().
The documentation for this class was generated from the following file:

- parser/GenesysParser.h

17.1136 testing::internal::ValueArray< Ts > Class Template Reference

Public Member Functions

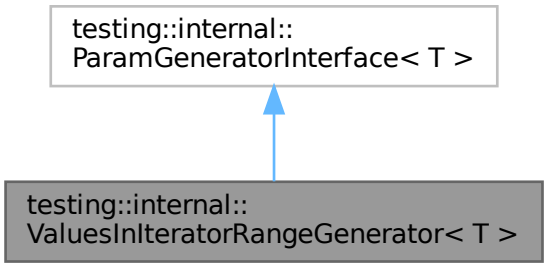
- **ValueArray** (Ts... v)
- template<typename T >
  operator ParamGenerator< T > () const

The documentation for this class was generated from the following file:

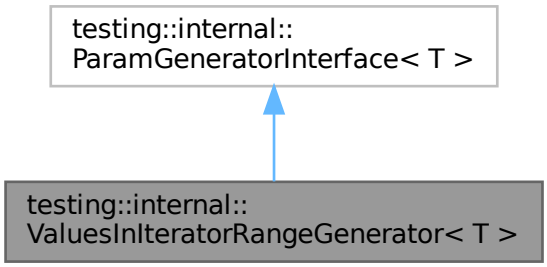
- gtest/gtest/internal/gtest-param-util.h

17.1137 testing::internal::ValuesInIteratorRangeGenerator< T > Class Template Reference

Inheritance diagram for testing::internal::ValuesInIteratorRangeGenerator< T >:



Collaboration diagram for testing::internal::ValuesInIteratorRangeGenerator< T >:



Public Member Functions

- template<typename ForwardIterator >
  ValuesInIteratorRangeGenerator (ForwardIterator begin, ForwardIterator end)
- ParamIteratorInterface< T > * Begin () const override
- ParamIteratorInterface< T > * End () const override

Additional Inherited Members

Public Types inherited from testing::internal::ParamGeneratorInterface< T >

- typedef T **ParamType**

17.1137.1 Member Function Documentation

17.1137.1.1 Begin()

```
template<typename T >
```

```
ParamIteratorInterface< T > * testing::internal::ValuesInIteratorRangeGenerator< T >::Begin (
) const  [inline], [override], [virtual]
```

Implements testing::internal::ParamGeneratorInterface< T >.

17.1137.1.2 End()

```
template<typename T >
ParamIteratorInterface< T > * testing::internal::ValuesInIteratorRangeGenerator< T >::End ( )
const  [inline], [override], [virtual]
```

Implements testing::internal::ParamGeneratorInterface< T >.

The documentation for this class was generated from the following file:

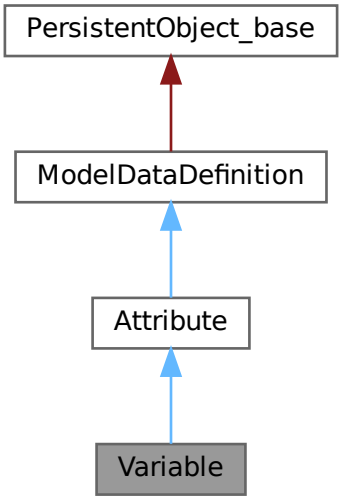
- gtest/gtest/internal/gtest-param-util.h

17.1138 Variable Class Reference

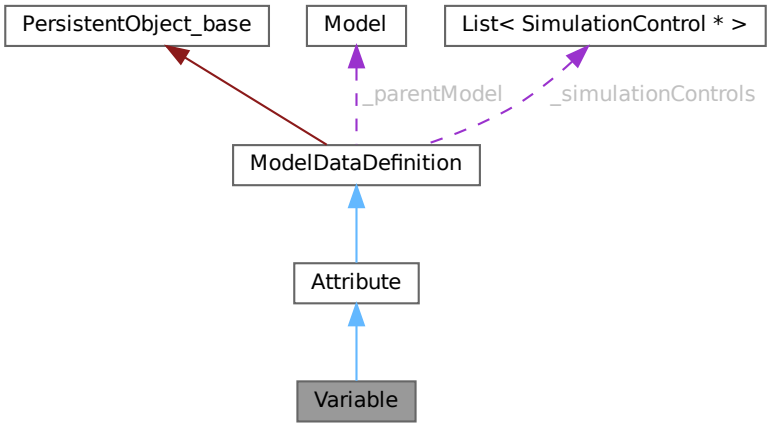
Data definition for a named value (scalar or indexed array) that is global to the whole model, as opposed to Attribute's per-entity values.

```
#include <plugins/data/Logic/Variable.h>
```

Inheritance diagram for Variable:



Collaboration diagram for Variable:



Public Member Functions

- **Variable** (Model *model, std::string name="")
- virtual std::string show () override

  Returns a textual representation of this data definition.
- double **getValue** (std::string index="")

  Reads the current sparse value, returning 0.0 when the index is absent.
- void **setValue** (double value, std::string index="")

  Writes the current sparse value at the scalar or indexed position.
- void **copyValuesFrom** (const Variable &source)

  Copies the full runtime sparse store from another variable.
- std::string getInitialValuesText () const override

  Returns the initial values serialized in bracket notation.
- void setInitialValuesText (std::string valuesText) override

  Parses bracket notation and replaces both initial and runtime stores.
- void **insertDimentionSize** (unsigned int size)

  Appends one dimension size to the variable definition.
- std::map< std::string, double > * **getValues** () const

  Returns mutable current sparse values for compatibility with existing callers.
- SparseValueStore * **getValueStore** ()

  Returns current value store used by this variable.
- ModelDataDefinition * **get_scope** ()
- void **set_scope** (ModelDataDefinition *const scope)

Public Member Functions inherited from Attribute

- **Attribute** (Model *model, std::string name="", std::string dataDefinitionTypename=Util::TypeOf< Attribute >())

  Creates an entity attribute definition in the model.
- virtual ~**Attribute** () override

  Releases sparse initial values owned by this definition.
- double **getInitialValue** (std::string index="")

Reads the initial sparse value, returning 0.0 when the index is absent.

- void **setInitialValue** (double value, std::string index="")

  Writes the initial sparse value at the scalar or indexed position.
- void **setInitialValues** (const std::vector< std::pair< std::string, double > > values)

  Replaces/extends initial sparse values from textual index/value pairs.
- void **copyInitialValuesFrom** (const Attribute &source)

  Copies the full initial sparse store from another attribute.
- void **insertDimentionSize** (unsigned int size)

  Appends one dimension size to the attribute definition.
- std::list< unsigned int > * **getDimensionSizes** () const

  Returns dimension sizes for compatibility with existing callers.
- std::map< std::string, double > * **getInitialValues** () const

  Returns mutable initial sparse values for compatibility with existing callers.
- SparseValueStore * **getInitialValueStore** ()

  Returns the sparse initial value store used by this attribute.
- bool **isInitialValuesTextValid** () const

  Returns whether the last textual initial value is syntactically valid.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from Attribute

- static PluginInformation * **GetPluginInformation** ()
- static ModelDataDefinition * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  must be overriden by derived classes
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

not just an easy access to trace manager, but wrappers to check if specific trace level applies

- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1138.1 Detailed Description

Data definition for a named value (scalar or indexed array) that is global to the whole model, as opposed to Attribute's per-entity values.
Arena correspondence: the "Variable module" (Rockwell Automation, *Getting Started with Arena*, "The Basic Process Panel", pp. 48-49).
GenESyS models Variable as a subclass of Attribute, reusing its dimension/rows/columns model and sparse initial-value store (`SparseValueStore`) instead of duplicating that infrastructure; `getValue()`/`setValue`() operate on a second, independent runtime store (`_values`) that starts from the inherited initial values but is not per-entity.
Known differences from Arena: there is no "Clear Option" (Statistics / System / None reset timing) and no external-file linkage (File Name, Recordset, File Read Time); a Variable's runtime values are only reset through `_init↩ BetweenReplications()`.

17.1138.2 Member Function Documentation

17.1138.2.1 _check()

```
bool Variable::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from Attribute.

17.1138.2.2 _initBetweenReplications()

```
void Variable::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1138.2.3 _loadInstance()

```
bool Variable::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from Attribute.

17.1138.2.4 _saveInstance()

```
void Variable::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from Attribute.

17.1138.2.5 getInitialValuesText()

```
std::string Variable::getInitialValuesText ( ) const  [override], [virtual]
```

Returns the initial values serialized in bracket notation.
Reimplemented from Attribute.

17.1138.2.6 setInitialValuesText()

```
void Variable::setInitialValuesText (
            std::string valuesText )  [override], [virtual]
```

Parses bracket notation and replaces both initial and runtime stores.
Reimplemented from Attribute.

17.1138.2.7 show()

```
std::string Variable::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from Attribute.
The documentation for this class was generated from the following files:

- plugins/data/Logic/Variable.h
- plugins/data/Logic/Variable.cpp

17.1139 genesys::launcher::VerificationResult Struct Reference

Public Attributes

- bool **ok** = false
- QString **error**

The documentation for this struct was generated from the following file:

- applications/launcher/UpdateVerifier.h

17.1140 genesys::launcher::Version Class Reference

Public Member Functions

- bool **isValid** () const noexcept
- QString **toString** () const
- const QVector< quint64 > & **components** () const noexcept

Static Public Member Functions

- static Version **parse** (const QString &text)

Friends

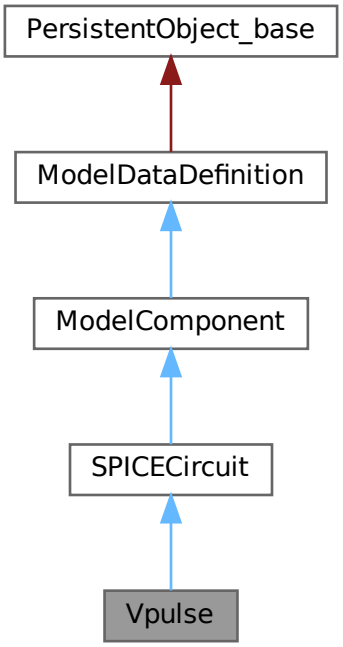
- bool **operator==** (const Version &lhs, const Version &rhs) noexcept
- bool **operator!=** (const Version &lhs, const Version &rhs) noexcept
- bool **operator**< (const Version &lhs, const Version &rhs) noexcept
- bool **operator**> (const Version &lhs, const Version &rhs) noexcept
- bool **operator**<**=** (const Version &lhs, const Version &rhs) noexcept
- bool **operator**>**=** (const Version &lhs, const Version &rhs) noexcept

The documentation for this class was generated from the following files:

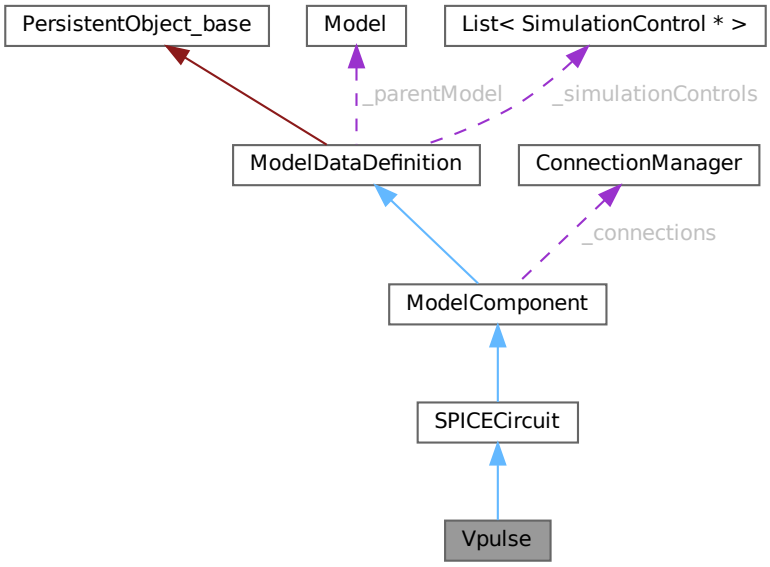
- applications/launcher/Version.h
- applications/launcher/Version.cpp

17.1141 Vpulse Class Reference

Inheritance diagram for Vpulse:



Collaboration diagram for Vpulse:



Public Member Functions

- **Vpulse** (Model *model, std::string name="")
- void Build () override
- void **setVoltage** (float voltage)
- void **setFreq** (float freq)
- void **setSlope** (float slope)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

Indicates whether a simulation breakpoint is set on this component.

- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)

- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1141.1 Member Function Documentation

17.1141.1.1 _loadInstance()

```
bool Vpulse::_loadInstance (
            PersistenceRecord * fields ) [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.1141.1.2 _saveInstance()

```
void Vpulse::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.1141.1.3 Build()

```
void Vpulse::Build ( )  [override], [virtual]
```

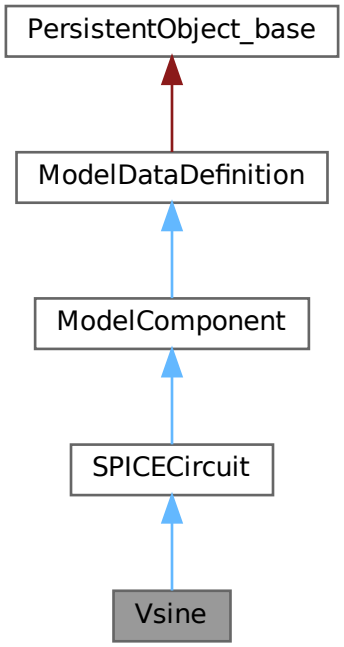
Reimplemented from SPICECircuit.

The documentation for this class was generated from the following files:

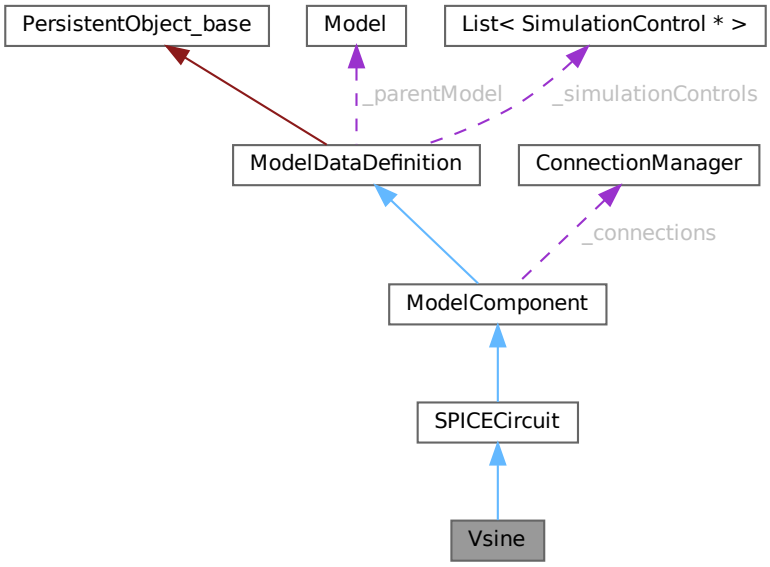
- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.1142 Vsine Class Reference

Inheritance diagram for Vsine:



Collaboration diagram for Vsine:



Public Member Functions

- **Vsine** (Model ∗model, std::string name="")
- void Build () override
- void **setVoltage** (float voltage)
- void **setFreq** (float freq)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model ∗model, std::string name="")

  constructors
- void **setRunner** (SPICERunner ∗runner)

  new public user methods for this components
- SPICERunner ∗ **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.

- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)

- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1142.1 Member Function Documentation

17.1142.1.1 _loadInstance()

```
bool Vsine::_loadInstance (
            PersistenceRecord * fields ) [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.1142.1.2 _saveInstance()

```
void Vsine::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.1142.1.3 Build()

```
void Vsine::Build ( )  [override], [virtual]
```

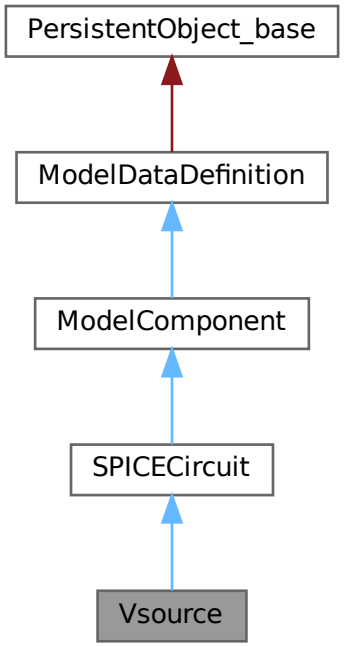
Reimplemented from SPICECircuit.

The documentation for this class was generated from the following files:

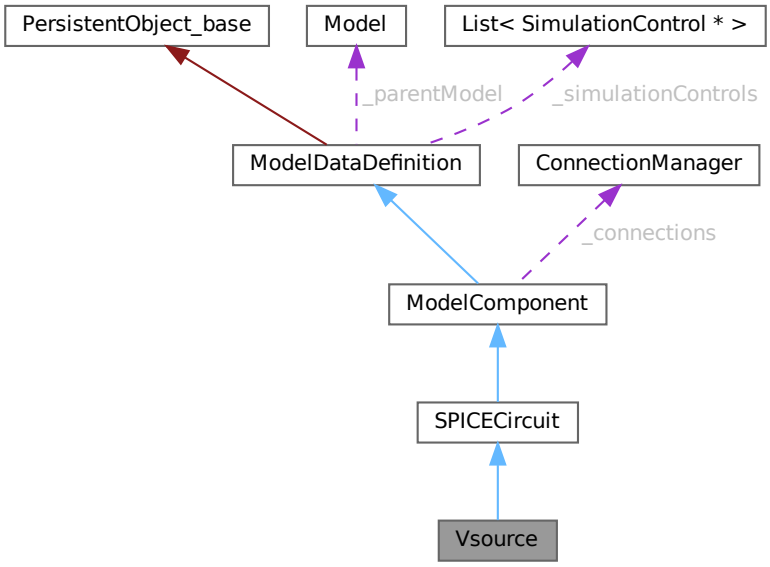
- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.1143 Vsource Class Reference

Inheritance diagram for Vsource:



Collaboration diagram for Vsource:



Public Member Functions

- **Vsource** (Model ∗model, std::string name="")
- void Build () override
- void **setVoltage** (float voltage)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model ∗model, std::string name="")

  constructors
- void **setRunner** (SPICERunner ∗runner)

  new public user methods for this components
- SPICERunner ∗ **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)

- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1143.1 Member Function Documentation

17.1143.1.1 _loadInstance()

```
bool Vsource::_loadInstance (
            PersistenceRecord * fields ) [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.1143.1.2 _saveInstance()

```
void Vsource::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.1143.1.3 Build()

```
void Vsource::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.

The documentation for this class was generated from the following files:

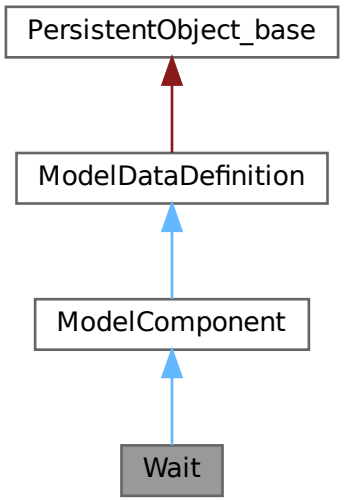
- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.1144 Wait Class Reference

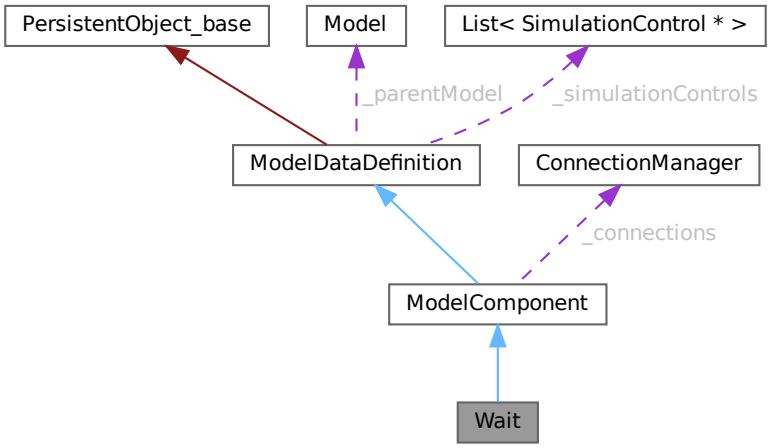
Holds an entity in a queue until a signal arrives, a condition becomes true, or it is explicitly removed elsewhere.

`#include <plugins/components/Synchronization/Wait.h>`

Inheritance diagram for Wait:



Collaboration diagram for Wait:



Public Types

- enum class **WaitType** : int { **WaitForSignal** = 0 , **InfiniteHold** = 1 , **ScanForCondition** = 2 , **num_elements** = 3 }

Public Member Functions

- **Wait** (Model ∗model, std::string name="")
- virtual std::string show () override

  virtual public methods
- void **setSignalData** (SignalData ∗signal)
- void **setWaitType** (WaitType _watType)
- Wait::WaitType **getWaitType** () const
- void **setCondition** (std::string _condition)
- std::string **getCondition** () const
- Queue ∗ **getQueue** () const
- void **setQueue** (Queue ∗queue)
- std::string **getlimitExpression** () const
- void **setLimitExpression** (const std::string &newLimitExpression)

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static std::string **convertEnumToStr** (WaitType type)
- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

Handles a dispatched event for this component type.

- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override
- virtual void _createNonEditableDataDefinitions () override
- virtual void _createEditableDataDefinitions () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)

- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1144.1 Detailed Description

Holds an entity in a queue until a signal arrives, a condition becomes true, or it is explicitly removed elsewhere.
Arena correspondence: this is Arena's "Hold module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 54), renamed `Wait` in GenESyS. `WaitType` (`WaitForSignal/InfiniteHold/ScanForCondition`) matches Arena's Type exactly; `_condition` and `_limitExpression` match Condition and Limit; the attached `SignalData` matches Wait for Value; an entity held with `InfiniteHold` is later released by Remove, exactly as in Arena.
Known difference from Arena: `_queue` is a plain `Queue*` rather than a full `QueueableItem` (Queue/Set/↩ Attribute/Expression), a gap already flagged with a `@TODO` in the source.

17.1144.2 Member Function Documentation

17.1144.2.1 _check()

```
bool Wait::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1144.2.2 _createEditableDataDefinitions()

```
void Wait::_createEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1144.2.3 _createNonEditableDataDefinitions()

```
void Wait::_createNonEditableDataDefinitions ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1144.2.4 _initBetweenReplications()

```
void Wait::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1144.2.5 _loadInstance()

```
bool Wait::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.1144.2.6 _onDispatchEvent()

```
void Wait::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.1144.2.7 _saveInstance()

```
void Wait::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.1144.2.8 show()

```
std::string Wait::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

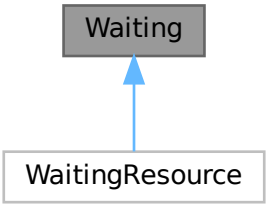
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

- plugins/components/Synchronization/Wait.h
- plugins/components/Synchronization/Wait.cpp

17.1145 Waiting Class Reference

Inheritance diagram for Waiting:



Public Member Functions

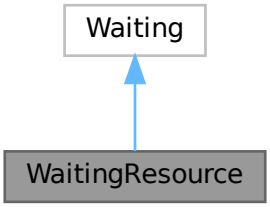
- **Waiting** (Entity *entity, double timeStartedWaiting, ModelComponent *thisComponent, unsigned int this↩ ComponentOutputPort=0)
- virtual std::string **show** ()
- double **getTimeStartedWaiting** () const
- ModelComponent * **geComponent** () const
- Entity * **getEntity** () const
- unsigned int **geComponentOutputPort** () const
- unsigned long long **getArrivalOrder** () const
- void **setArrivalOrder** (unsigned long long arrivalOrder)

The documentation for this class was generated from the following file:

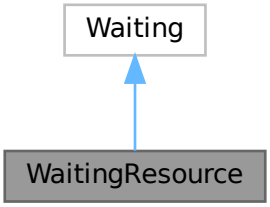
- plugins/data/DiscreteProcessing/Queue.h

17.1146 WaitingResource Class Reference

Inheritance diagram for WaitingResource:



Collaboration diagram for WaitingResource:



Public Member Functions

- **WaitingResource** (Entity *entity, double timeStartedWaiting, unsigned int quantity, ModelComponent *this↩ Component, unsigned int thisComponentOutputPort=0)
- **WaitingResource** (const WaitingResource &orig)
- virtual std::string show () override
- unsigned int **getQuantity** () const

Public Member Functions inherited from Waiting

- **Waiting** (Entity *entity, double timeStartedWaiting, ModelComponent *thisComponent, unsigned int this↩ ComponentOutputPort=0)
- double **getTimeStartedWaiting** () const
- ModelComponent * **geComponent** () const
- Entity * **getEntity** () const
- unsigned int **geComponentOutputPort** () const
- unsigned long long **getArrivalOrder** () const
- void **setArrivalOrder** (unsigned long long arrivalOrder)

17.1146.1 Member Function Documentation

17.1146.1.1 show()

```
virtual std::string WaitingResource::show ( )  [inline], [override], [virtual]
```

Reimplemented from Waiting.

The documentation for this class was generated from the following file:

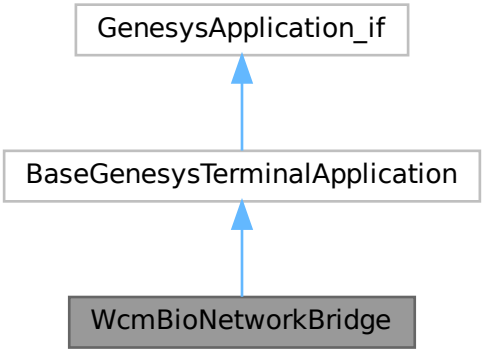
- plugins/components/DiscreteProcessing/Seize.h

17.1147 WcmBioNetworkBridge Class Reference

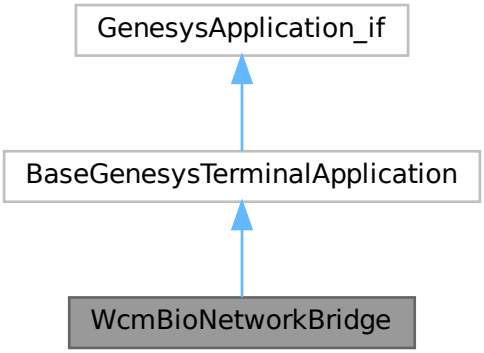
Demonstrates a minimal deterministic/stochastic whole-cell bridge.

```
#include <applications/modelSpecific/wcm/WcmBioNetworkBridge.h>
```

Inheritance diagram for WcmBioNetworkBridge:



Collaboration diagram for WcmBioNetworkBridge:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)

- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.1147.1 Detailed Description

Demonstrates a minimal deterministic/stochastic whole-cell bridge.
The example builds a small deterministic BioNetwork for nutrient conversion, projects its final state into a WholeCellState, and then runs stochastic transcription, translation, and degradation components. It is a didactic integration test for GenESyS abstractions, not a complete organism model.

17.1147.2 Member Function Documentation

17.1147.2.1 main()

```
int WcmBioNetworkBridge::main (
            int argc,
            char ** argv )  [override], [virtual]
```

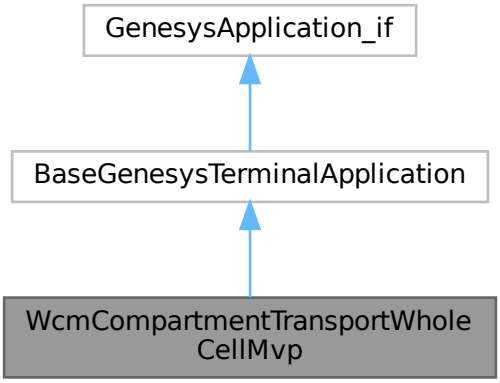
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/wcm/WcmBioNetworkBridge.h
- applications/modelSpecific/wcm/WcmBioNetworkBridge.cpp

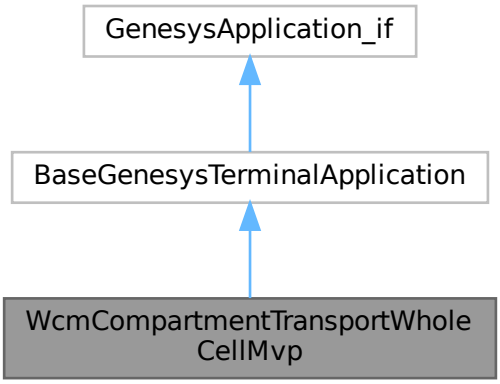
17.1148 WcmCompartmentTransportWholeCellMvp Class Reference

Whole-cell MVP with compartment-aware metabolic transport.
`#include <applications/modelSpecific/wcm/WcmCompartmentTransportWholeCell↩ Mvp.h>`

Inheritance diagram for WcmCompartmentTransportWholeCellMvp:



Collaboration diagram for WcmCompartmentTransportWholeCellMvp:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)

- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.1148.1 Detailed Description

Whole-cell MVP with compartment-aware metabolic transport.
This example extends the flux-balance whole-cell path with a tiny extracellular-to-cytosol transport chain. It demonstrates how projected metabolic fluxes can feed:

- global energy proxies;
- compartment-scoped metabolite pools; and
- pathway activity checkpoints.

The biological scope is still didactic. The purpose is to validate the architecture for multi-compartment hybrid whole-cell simulations.

17.1148.2 Member Function Documentation

17.1148.2.1 main()

```
int WcmCompartmentTransportWholeCellMvp::main (
            int argc,
            char ** argv )  [override], [virtual]
```

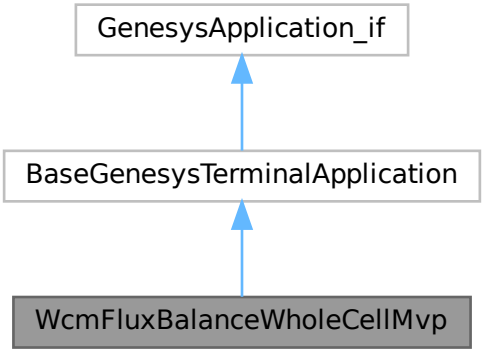
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/wcm/WcmCompartmentTransportWholeCellMvp.h
- applications/modelSpecific/wcm/WcmCompartmentTransportWholeCellMvp.cpp

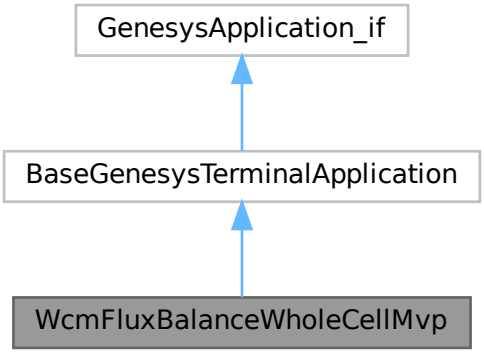
17.1149 WcmFluxBalanceWholeCellMvp Class Reference

Whole-cell MVP driven by flux-balance metabolism.
`#include <applications/modelSpecific/wcm/WcmFluxBalanceWholeCellMvp.h>`

Inheritance diagram for WcmFluxBalanceWholeCellMvp:



Collaboration diagram for WcmFluxBalanceWholeCellMvp:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)

- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.1149.1 Detailed Description

Whole-cell MVP driven by flux-balance metabolism.
This example demonstrates a second metabolic integration path for GenESyS whole-cell models. Instead of using the deterministic BioNetwork runner, it executes a small FBA problem through MetabolicFluxBalance, projects selected fluxes into WholeCellState, and then runs expression, growth, lifecycle checkpointing, fate routing, and division.

17.1149.2 Member Function Documentation

17.1149.2.1 main()

```
int WcmFluxBalanceWholeCellMvp::main (
            int argc,
            char ** argv )  [override], [virtual]
```

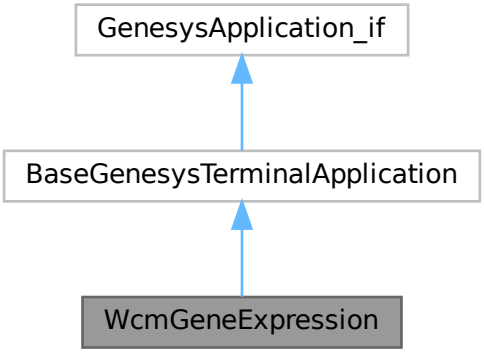
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/wcm/WcmFluxBalanceWholeCellMvp.h
- applications/modelSpecific/wcm/WcmFluxBalanceWholeCellMvp.cpp

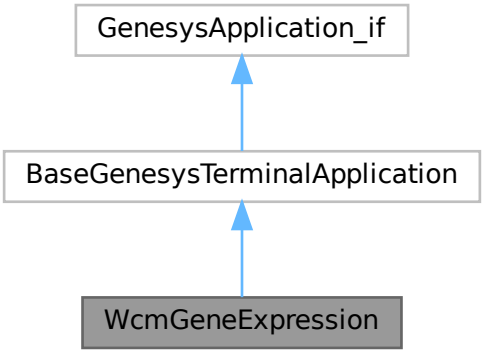
17.1150 WcmGeneExpression Class Reference

```
#include <applications/modelSpecific/wcm/WcmGeneExpression.h>
```

Inheritance diagram for WcmGeneExpression:



Collaboration diagram for WcmGeneExpression:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)

- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.1150.1 Detailed Description

Central dogma simulation: stochastic transcription + translation + SSA degradation.
Models 4 genes using two complementary approximations:

- Poisson tau-leaping for transcription (StochasticTranscription) and translation (StochasticTranslation) — valid for high-copy-number RNAP/ribosome
- Gillespie Direct Method SSA for low-copy-number mRNA and protein degradation

Biological parameters (M. genitalium ballpark): RNAP_free = 200 molecules ribosome_free = 400 molecules 4 genes: geneA (metabolic), geneB (regulatory), geneC (structural), geneD (transport) Transcription: elongRate=50 nt/s, meanGeneLen=900 nt, bindProb=0.25, Δt=60 s Translation: elongRate=16 AA/s, meanProtLen=300 AA, Δt=60 s mRNA decay: k = 0.01/s (mean lifetime $\approx$ 100 s) Protein decay: k = 0.001/s (mean lifetime $\approx$ 1000 s $\approx$ 16 min)
Simulation: 10 replications $\times$ 3600 s, 60 s time steps.
Run via GENESYS_TERMINAL_EXAMPLE: cmake -DGENESYS_TERMINAL_EXAMPLE=wcm/WcmGene↩ Expression.cpp ...

17.1150.2 Member Function Documentation

17.1150.2.1 main()

```
int WcmGeneExpression::main (
            int argc,
            char ** argv )  [override], [virtual]
```

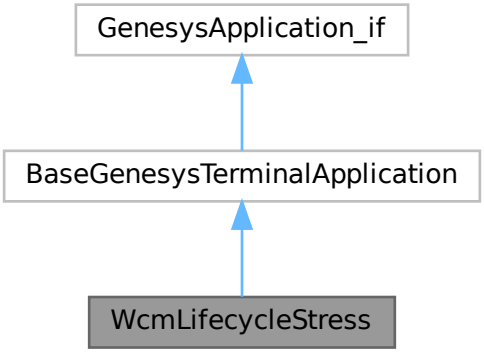
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/wcm/WcmGeneExpression.h
- applications/modelSpecific/wcm/WcmGeneExpression.cpp

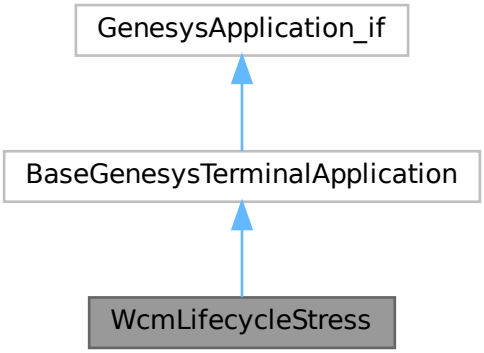
17.1151 WcmLifecycleStress Class Reference

Demonstrates starvation and death routing in a whole-cell stress scenario.
`#include <applications/modelSpecific/wcm/WcmLifecycleStress.h>`

Inheritance diagram for WcmLifecycleStress:



Collaboration diagram for WcmLifecycleStress:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)

- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.1151.1 Detailed Description

Demonstrates starvation and death routing in a whole-cell stress scenario.
This example focuses on lifecycle checkpointing rather than biosynthesis. It uses a low-energy whole-cell state plus the CellCycleCheckpointComponent and CellFateDecisionComponent to show the transition: newborn -> starved -> dead.

17.1151.2 Member Function Documentation

17.1151.2.1 main()

```
int WcmLifecycleStress::main (
            int argc,
            char ** argv )  [override], [virtual]
```

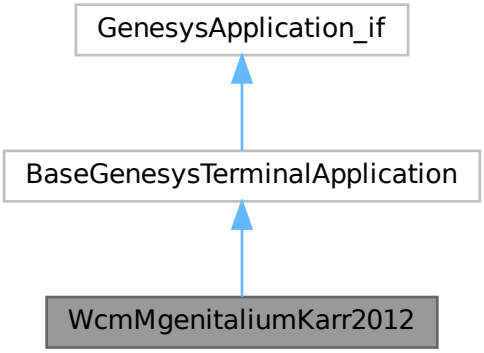
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/wcm/WcmLifecycleStress.h
- applications/modelSpecific/wcm/WcmLifecycleStress.cpp

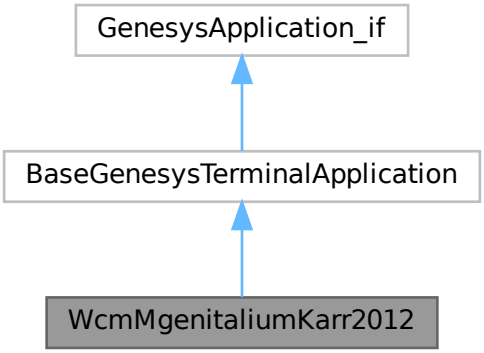
17.1152 WcmMgenitaliumKarr2012 Class Reference

```
#include <applications/modelSpecific/wcm/WcmMgenitaliumKarr2012.h>
```

Inheritance diagram for WcmMgenitaliumKarr2012:



Collaboration diagram for WcmMgenitaliumKarr2012:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)

- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.1152.1 Detailed Description

Whole-cell simulation analog of the M. genitalium model (Karr et al. 2012).
Reference: Karr JR, Sanghvi JC, Macklin DN, et al. "A Whole-Cell Computational Model Predicts Phenotype from Genotype." Cell 150(2):389–401, 2012. doi:10.1016/j.cell.2012.05.044
Architecture: This model uses all four GenESyS WholeCellModeling plugins in sequence:
SimClock → Transcription → Translation → MetabolicSSA → CellDivision |- port 0 (no division) → Dispose |- port 1 (division) → RecordDivision → Dispose
StochasticTranscription: Poisson tau-leaping for 10 representative M. genitalium genes StochasticTranslation↩ : Poisson tau-leaping mapping mRNA_ → prot_ species StochasticReactionComponent: Gillespie SSA for ATP/↩ GTP metabolism + FtsZ ring polymerization CellDivisionEvent: Triggers when FtsZ ring reaches 35% completion (350/1000 units) (mass-only trigger disabled; FtsZ-driven as in Karr et al.)
Biologically realistic parameters (from CovertLab/WholeCell parameters.json, MIT): RNAP_free = 200 molecules (Karr et al. 2012, Supplementary Table S2) ribosome_free = 400 molecules Transcription rate = 50 nt/s Translation rate = 16 AA/s Mean gene length = 900 nt Mean protein len = 300 AA Cell cycle ≈ 600 min (modeled as 600 × 60 s steps)
10 representative genes (selected from M. genitalium 473-gene model): MG_001 — metabolic (glycolytic enzyme) MG_006 — ATP synthase subunit (energy metabolism) MG_055 — DNA polymerase III subunit alpha (DNA replication) MG_139 — RNAP beta subunit (transcription machinery) MG_175 — 30S ribosomal protein S1 (translation machinery) MG_196 — FtsZ (cell division; drives the division trigger) MG_203 — RNAP sigma factor (transcription regulation) MG_339 — phosphoglycerate mutase (central metabolism) MG_398 — pyruvate kinase (central metabolism) MG_462 — ribosomal protein L7/L12 (translation machinery)
Simulation: 3 replications × 36000 s (10 h ≈ 1 M. genitalium cell cycle), 60 s steps.
Run via GENESYS_TERMINAL_EXAMPLE: cmake -DGENESYS_TERMINAL_EXAMPLE=wcm/Wcm↩ MgenitaliumKarr2012.cpp ...

17.1152.2 Member Function Documentation

17.1152.2.1 main()

```
int WcmMgenitaliumKarr2012::main (
            int argc,
            char ** argv )  [override], [virtual]
```

Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

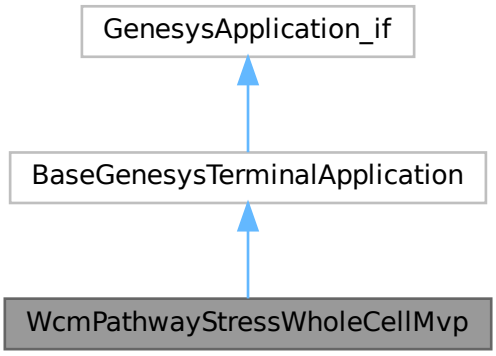
- applications/modelSpecific/wcm/WcmMgenitaliumKarr2012.h
- applications/modelSpecific/wcm/WcmMgenitaliumKarr2012.cpp

17.1153 WcmPathwayStressWholeCellMvp Class Reference

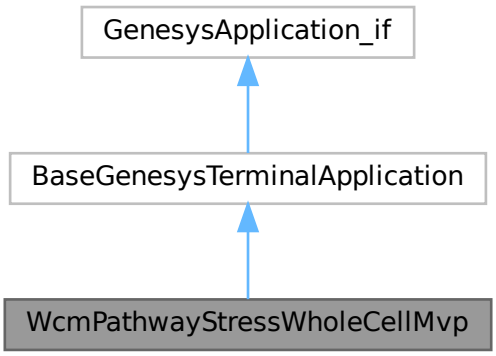
Whole-cell stress example driven by sustained low pathway activity.

`#include <applications/modelSpecific/wcm/WcmPathwayStressWholeCellMvp.h>`

Inheritance diagram for WcmPathwayStressWholeCellMvp:



Collaboration diagram for WcmPathwayStressWholeCellMvp:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)
- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.1153.1 Detailed Description

Whole-cell stress example driven by sustained low pathway activity.
This example keeps metabolism active but configures a monitored pathway threshold above the achievable FBA objective. The result is an arrest-then- death sequence driven by PathwayStressResponseComponent.

17.1153.2 Member Function Documentation

17.1153.2.1 main()

```
int WcmPathwayStressWholeCellMvp::main (
            int argc,
            char ** argv )  [override], [virtual]
```

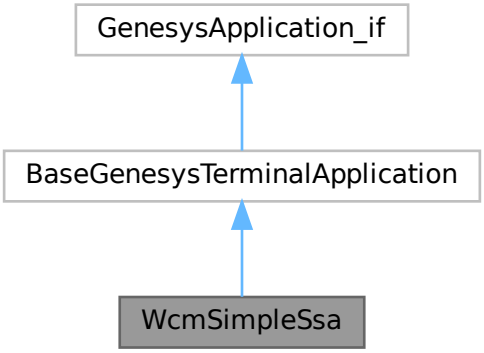
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/wcm/WcmPathwayStressWholeCellMvp.h
- applications/modelSpecific/wcm/WcmPathwayStressWholeCellMvp.cpp

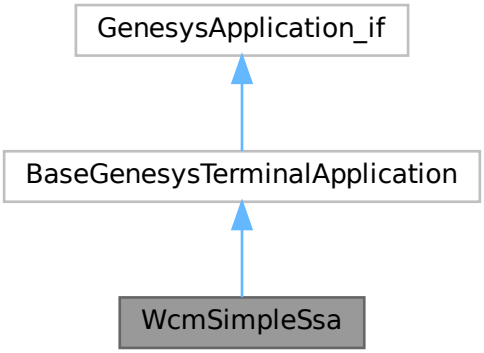
17.1154 WcmSimpleSsa Class Reference

```
#include <applications/modelSpecific/wcm/WcmSimpleSsa.h>
```

Inheritance diagram for WcmSimpleSsa:



Collaboration diagram for WcmSimpleSsa:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)

- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent *re)
- virtual void **onBreakpointHandler** (SimulationEvent *re)
- virtual void **onEntityCreateHandler** (SimulationEvent *re)
- virtual void **onEntityMoveHandler** (SimulationEvent *re)
- virtual void **onSimulationStartHandler** (SimulationEvent *re)
- virtual void **onReplicationStepHandler** (SimulationEvent *re)
- virtual void **onReplicationStartHandler** (SimulationEvent *re)
- virtual void **onProcessEventHandler** (SimulationEvent *re)
- virtual void **onReplicationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationEndHandler** (SimulationEvent *re)
- virtual void **onSimulationPausedHandler** (SimulationEvent *re)
- virtual void **onSimulationResumeHandler** (SimulationEvent *re)
- virtual void **onEntityRemoveHandler** (SimulationEvent *re)

17.1154.1 Detailed Description

Minimal whole-cell modeling example using the Gillespie Direct Method SSA.
Models stochastic gene expression of a single gene via 4 reactions: R1: transcription — DNA + RNAP $\rightarrow$ DNA + RNAP + mRNA_geneA (k = 0.02/s) R2: mRNA degradation — mRNA_geneA $\rightarrow \emptyset$ (k = 0.01/s) R3: translation — mRNA_geneA $\rightarrow$ mRNA_geneA + prot_geneA (k = 0.10/s) R4: protein degradation — prot_geneA $\rightarrow \emptyset$ (k = 0.001/s)
Steady-state analytic predictions: mRNA_geneA $\approx$ 2 molecules (k_tx / k_deg_mRNA = 0.02 / 0.01) prot_geneA $\approx$ 200 molecules (k_tl $*$ mRNA_ss / k_deg_prot = 0.1 $*$ 2 / 0.001)
Simulation runs 30 replications $\times$ 3600 s each (1 hour of biological time) in 60 s Gillespie windows triggered by periodic clock entities.
Run via GENESYS_TERMINAL_EXAMPLE: cmake -DGENESYS_TERMINAL_EXAMPLE=wcm/WcmSimple↩ Ssa.cpp ...

17.1154.2 Member Function Documentation

17.1154.2.1 main()

```
int WcmSimpleSsa::main (
            int argc,
            char ** argv )  [override], [virtual]
```

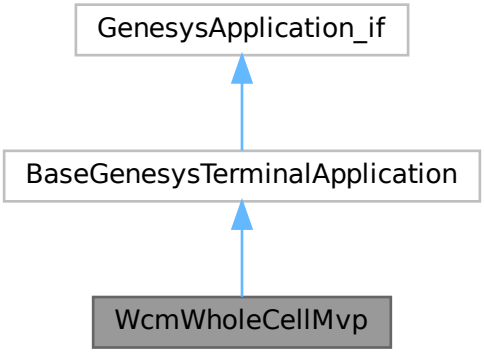
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/wcm/WcmSimpleSsa.h
- applications/modelSpecific/wcm/WcmSimpleSsa.cpp

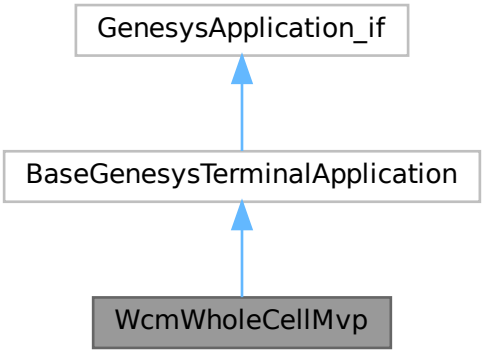
17.1155 WcmWholeCellMvp Class Reference

Minimal whole-cell MVP combining metabolism, expression, growth, and division.
`#include <applications/modelSpecific/wcm/WcmWholeCellMvp.h>`

Inheritance diagram for WcmWholeCellMvp:



Collaboration diagram for WcmWholeCellMvp:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)

- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.1155.1 Detailed Description

Minimal whole-cell MVP combining metabolism, expression, growth, and division.
This example is intentionally small and didactic. It combines:

- deterministic biochemical metabolism through BioNetwork;
- runtime projection into WholeCellState;
- stochastic transcription and translation;
- energy-gated mass growth;
- mass-triggered cell division.

The result is not a calibrated organism model. It is a compact whole-cell analog intended to validate the GenESyS architecture for hybrid deterministic/stochastic cell simulations.

17.1155.2 Member Function Documentation

17.1155.2.1 main()

```
int WcmWholeCellMvp::main (
            int argc,
            char ** argv )  [override], [virtual]
```

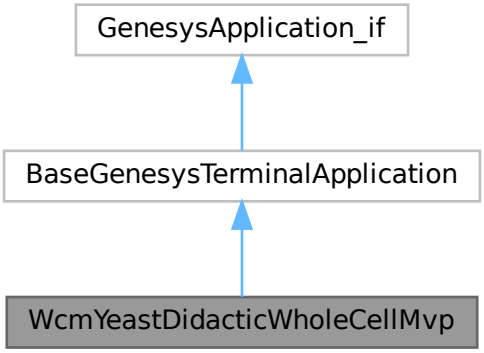
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/wcm/WcmWholeCellMvp.h
- applications/modelSpecific/wcm/WcmWholeCellMvp.cpp

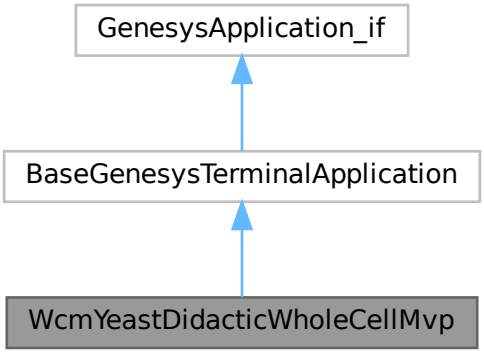
17.1156 WcmYeastDidacticWholeCellMvp Class Reference

Didactic yeast-oriented whole-cell MVP built on GEM-inspired metabolism.
`#include <applications/modelSpecific/wcm/WcmYeastDidacticWholeCellMvp.h>`

Inheritance diagram for WcmYeastDidacticWholeCellMvp:



Collaboration diagram for WcmYeastDidacticWholeCellMvp:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)

- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.1156.1 Detailed Description

Didactic yeast-oriented whole-cell MVP built on GEM-inspired metabolism.
This example is intentionally small and pedagogical. It combines a compact FBA network inspired by Saccharomyces cerevisiae carbon metabolism with whole-cell state projection, compartment exchange, stochastic expression, growth, lifecycle checkpointing, fate routing, and division.
It is not a canonical or validated yeast whole-cell model. It should be interpreted as a GenESyS architecture example informed by yeast-GEM style metabolic concepts, not as a public operational WCM for S. cerevisiae.

17.1156.2 Member Function Documentation

17.1156.2.1 main()

```
int WcmYeastDidacticWholeCellMvp::main (
            int argc,
            char ** argv )  [override], [virtual]
```

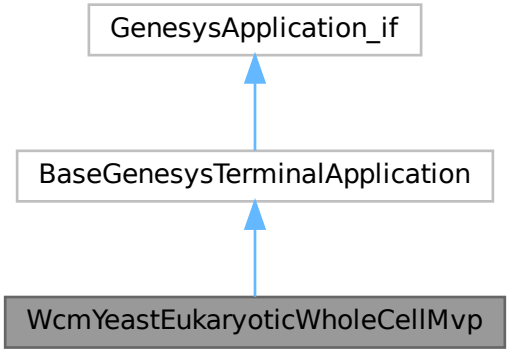
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/wcm/WcmYeastDidacticWholeCellMvp.h
- applications/modelSpecific/wcm/WcmYeastDidacticWholeCellMvp.cpp

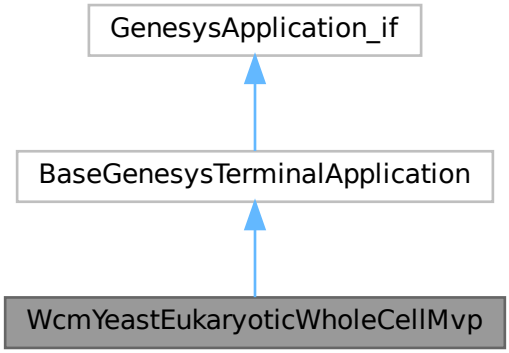
17.1157 WcmYeastEukaryoticWholeCellMvp Class Reference

Didactic eukaryotic whole-cell MVP inspired by budding yeast.
`#include <applications/modelSpecific/wcm/WcmYeastEukaryoticWholeCellMvp.h>`

Inheritance diagram for WcmYeastEukaryoticWholeCellMvp:



Collaboration diagram for WcmYeastEukaryoticWholeCellMvp:



Public Member Functions

- virtual int main (int argc, char **argv) override

Public Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **Simulate** (Simulator *genesys, Model *model, PluginManager *plugins)
- void **setDefaultTraceHandlers** (TraceManager *tm)
- void setDefaultEventHandlers (OnEventManager *oem)

Additional Inherited Members

Protected Member Functions inherited from BaseGenesysTerminalApplication

- virtual void **traceHandler** (TraceEvent e)
- virtual void **traceErrorHandler** (TraceErrorEvent e)

- virtual void **traceReportHandler** (TraceEvent e)
- virtual void **traceSimulationHandler** (TraceSimulationEvent e)
- virtual void **onAfterProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onBreakpointHandler** (SimulationEvent ∗re)
- virtual void **onEntityCreateHandler** (SimulationEvent ∗re)
- virtual void **onEntityMoveHandler** (SimulationEvent ∗re)
- virtual void **onSimulationStartHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStepHandler** (SimulationEvent ∗re)
- virtual void **onReplicationStartHandler** (SimulationEvent ∗re)
- virtual void **onProcessEventHandler** (SimulationEvent ∗re)
- virtual void **onReplicationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationEndHandler** (SimulationEvent ∗re)
- virtual void **onSimulationPausedHandler** (SimulationEvent ∗re)
- virtual void **onSimulationResumeHandler** (SimulationEvent ∗re)
- virtual void **onEntityRemoveHandler** (SimulationEvent ∗re)

17.1157.1 Detailed Description

Didactic eukaryotic whole-cell MVP inspired by budding yeast.
This example combines:

- a compact GEM-inspired FBA core;
- explicit biological compartments;
- compartment exchange and organelle energy proxies;
- stochastic expression and growth; and
- a simplified eukaryotic cell-cycle coordinator.

It is a didactic whole-cell example for Saccharomyces cerevisiae-inspired architecture. It is not a canonical public yeast whole-cell model.

17.1157.2 Member Function Documentation

17.1157.2.1 main()

```
int WcmYeastEukaryoticWholeCellMvp::main (
            int argc,
            char ** argv )  [override], [virtual]
```

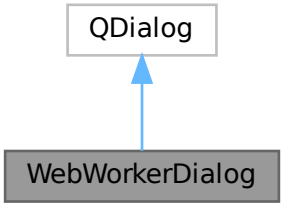
Reimplemented from BaseGenesysTerminalApplication.
The documentation for this class was generated from the following files:

- applications/modelSpecific/wcm/WcmYeastEukaryoticWholeCellMvp.h
- applications/modelSpecific/wcm/WcmYeastEukaryoticWholeCellMvp.cpp

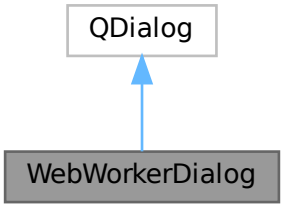
17.1158 WebWorkerDialog Class Reference

Lightweight control window for the embedded GenESyS web worker.
`#include <applications/gui/httpworker/WebWorkerDialog.h>`

Inheritance diagram for WebWorkerDialog:



Collaboration diagram for WebWorkerDialog:



Public Member Functions

- **WebWorkerDialog** (QWidget ∗parent=nullptr, bool allowClose=false)

Protected Member Functions

- void **closeEvent** (QCloseEvent ∗event) override
- void **showEvent** (QShowEvent ∗event) override

17.1158.1 Detailed Description

Lightweight control window for the embedded GenESyS web worker.
The documentation for this class was generated from the following files:

- applications/gui/httpworker/WebWorkerDialog.h
- applications/gui/httpworker/WebWorkerDialog.cpp

17.1159 WebWorkerRuntime Class Reference

Owns the embedded worker runtime used by the GUI and the standalone worker app.
`#include <applications/worker/service/WebWorkerRuntime.h>`

Classes

- struct Snapshot

  Captures a thread-safe snapshot of the worker state.

Public Member Functions

- **WebWorkerRuntime** (unsigned short defaultPort=8080, unsigned long maxRequests=0)

  Builds the reusable worker runtime with a default port and request cap.
- ~**WebWorkerRuntime** ()

  Stops the worker and releases background resources.
- bool start ()

  Starts the worker if it is not already running.
- void **stop** ()

  Requests a graceful shutdown and waits for the worker thread to finish.
- void **wait** ()

  Waits for the worker thread to complete if it is running.
- bool restart ()

  Restarts the worker using the currently configured settings.
- bool **isRunning** () const

  Returns whether the worker thread is currently serving requests.
- Snapshot **snapshot** () const

  Returns a snapshot of the current worker lifecycle and counters.
- void **setConfiguredPort** (unsigned short port)

  Updates the configured listening port used by the next start.
- unsigned short **configuredPort** () const

  Returns the listening port configured for the next start.
- void **setConfiguredMaxRequests** (unsigned long maxRequests)

  Updates the configured request cap used by the next start.
- unsigned long **configuredMaxRequests** () const

  Returns the configured request cap used by the next start.

17.1159.1 Detailed Description

Owns the embedded worker runtime used by the GUI and the standalone worker app.
The runtime wires the HTTP server, routing stack, and simulator-backed session services into a reusable lifecycle object. GUI code can start, stop, and inspect the worker without embedding HTTP details in the main window.

17.1159.2 Member Function Documentation

17.1159.2.1 restart()

```
bool WebWorkerRuntime::restart ( )
```

Restarts the worker using the currently configured settings.

Returns

True when the worker was scheduled to start again.

17.1159.2.2 start()

```
bool WebWorkerRuntime::start ( )
```

Starts the worker if it is not already running.

Returns

True when a new background worker thread was launched.

The documentation for this class was generated from the following files:

- applications/worker/service/WebWorkerRuntime.h
- applications/worker/service/WebWorkerRuntime.cpp

17.1160 WholeCellParameterReader Class Reference

```
#include <tools/Biochemical/WholeCellParameterReader.h>
```

Static Public Member Functions

- static std::map< std::string, double > **read** (const std::string &filePath)

17.1160.1 Detailed Description

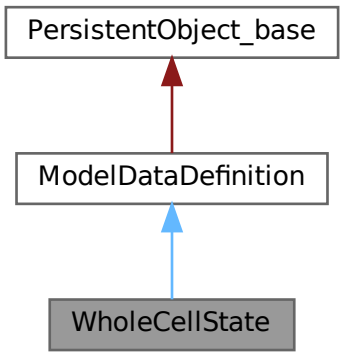
Minimal JSON reader for the CovertLab/WholeCell M. genitalium parameter files.
Reads parameters.json and fixedConstants.json (MIT license) which have the two-level structure: {"processes": {"ProcessName": {"key": value, ...}, ...}, "states": {"StateName": {"key": value, ...}, ...}}
Scalar double values are extracted into a flat map keyed by "processes.ProcessName.key" or "states.StateName.↩ key".
Arrays and nested objects are skipped. Non-numeric leaf values are ignored. No external JSON library is required. Usage: auto params = WholeCellParameterReader::read("data/parameters.json"); double rate = params.↩ count("processes.Transcription.rnaPolymeraseElongationRate") ? params.at("processes.Transcription.rna↩ PolymeraseElongationRate") : 50.0;
The documentation for this class was generated from the following file:

- tools/Biochemical/WholeCellParameterReader.h

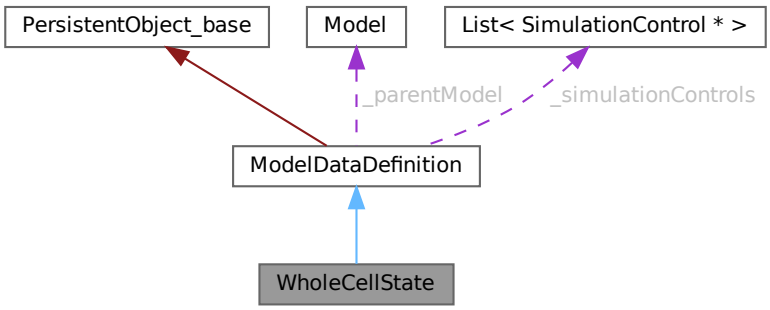
17.1161 WholeCellState Class Reference

```
#include <plugins/data/WholeCellModeling/WholeCellState.h>
```

Inheritance diagram for WholeCellState:



Collaboration diagram for WholeCellState:



Public Member Functions

- **WholeCellState** (Model *model, std::string name="")
- void **setMoleculeCount** (const std::string &speciesName, int count)
- int **getMoleculeCount** (const std::string &speciesName) const
- bool **hasMoleculeCount** (const std::string &speciesName) const
- const std::map< std::string, int > & **getMoleculeCounts** () const
- void **setMetaboliteAmount** (const std::string &metaboliteName, double amount)
- double **getMetaboliteAmount** (const std::string &metaboliteName) const
- const std::map< std::string, double > & **getMetabolitePool** () const
- void **setCompartmentMetaboliteAmount** (const std::string &compartmentName, const std::string &metaboliteName, double amount)
- double **getCompartmentMetaboliteAmount** (const std::string &compartmentName, const std::string &metaboliteName) const
- bool **hasCompartmentMetaboliteAmount** (const std::string &compartmentName, const std::string &metaboliteName) const
- const std::map< std::string, double > & **getCompartmentMetabolitePool** () const
- void **setPathwayActivity** (const std::string &pathwayName, double activity)
- double **getPathwayActivity** (const std::string &pathwayName) const
- bool **hasPathwayActivity** (const std::string &pathwayName) const
- const std::map< std::string, double > & **getPathwayActivities** () const
- void **setResourceBudget** (const std::string &speciesName, int budget)
- int **getResourceBudget** (const std::string &speciesName) const
- void **clearResourceBudget** ()
- void **setCellVolume** (double volume)
- double **getCellVolume** () const
- void **setCellMass** (double mass)
- double **getCellMass** () const
- void **setCurrentTime** (double time)
- double **getCurrentTime** () const
- void **setStepCount** (int stepCount)
- int **getStepCount** () const
- void **incrementStep** ()
- void **setLifecyclePhase** (const std::string &phase)
- std::string **getLifecyclePhase** () const
- void **setGenerationCount** (int generationCount)
- int **getGenerationCount** () const
- void **setViable** (bool viable)

- bool **isViable** () const
- void **setLastDivisionTime** (double time)
- double **getLastDivisionTime** () const
- bool **loadFixedConstants** (const std::string &jsonPath)
- bool **loadParameters** (const std::string &jsonPath)
- void **setFixedConstantsPath** (std::string path)
- std::string **getFixedConstantsPath** () const
- void **setParametersPath** (std::string path)
- std::string **getParametersPath** () const
- virtual std::string show () override

  Returns a textual representation of this data definition.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.

- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.

- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.

- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.

- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static ModelDataDefinition ∗ **LoadInstance** (Model ∗model, PersistenceRecord ∗fields)
- static PluginInformation ∗ **GetPluginInformation** ()
- static ModelDataDefinition ∗ **NewInstance** (Model ∗model, std::string name="")

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition ∗ LoadInstance (Model ∗model, PersistenceRecord ∗fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.

- static ModelDataDefinition ∗ NewInstance (Model ∗model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.

- static void SaveInstance (PersistenceRecord ∗fields, ModelDataDefinition ∗modeldatum)

  Serializes a data definition instance into a persistence record.

- static bool Check (ModelDataDefinition ∗modeldatum, std::string &errorMessage)

  Validates a data definition instance.

- static void InitBetweenReplications (ModelDataDefinition ∗modeldatum)

  Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition ∗modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition ∗modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord ∗fields) override

  must be overriden by derived classes

- virtual void _saveInstance (PersistenceRecord ∗fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes

- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1161.1 Detailed Description

Shared state container for whole-cell model simulations.
WholeCellState is the GenESyS analog to the Vivarium Store: a single ModelDataDefinition instance shared by all WCM component plugins in a simulation step. It holds:

- discrete molecule counts (mRNA, proteins) updated by SSA
- continuous metabolite pool (ATP, amino acids) updated by FBA
- resource budget allocated by fair-allocation before each step
- cell geometry (volume, mass) for propensity scaling

JSON loading (Q5a) reads fixedConstants.json and parameters.json from the CovertLab/WholeCell M. genitalium repository (MIT license).

17.1161.2 Member Function Documentation

17.1161.2.1 _check()

```
bool WholeCellState::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1161.2.2 _initBetweenReplications()

```
void WholeCellState::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1161.2.3 _loadInstance()

```
bool WholeCellState::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

must be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1161.2.4 _saveInstance()

```
void WholeCellState::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelDataDefinition.

17.1161.2.5 show()

```
std::string WholeCellState::show ( )  [override], [virtual]
```

Returns a textual representation of this data definition.
Returns the keys of internal ModelDataDefinitions such as counters, statistics collectors and other children needed before model checking.

Returns

Human-readable summary used by traces and debug output.

Reimplemented from ModelDataDefinition.
The documentation for this class was generated from the following files:

- plugins/data/WholeCellModeling/WholeCellState.h
- plugins/data/WholeCellModeling/WholeCellState.cpp

17.1162 QtButtonPropertyBrowserPrivate::WidgetItem Struct Reference

Collaboration diagram for QtButtonPropertyBrowserPrivate::WidgetItem:

Public Attributes

- QWidget * **widget** {nullptr}
- QLabel * **label** {nullptr}
- QLabel * **widgetLabel** {nullptr}
- QToolButton * **button** {nullptr}
- QWidget * **container** {nullptr}
- QGridLayout * **layout** {nullptr}
- WidgetItem * **parent** {nullptr}
- QList< WidgetItem * > **children**
- bool **expanded** {false}

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtbuttonpropertybrowser.cpp

17.1163 QtGroupBoxPropertyBrowserPrivate::WidgetItem Struct Reference

Collaboration diagram for QtGroupBoxPropertyBrowserPrivate::WidgetItem:

Public Attributes

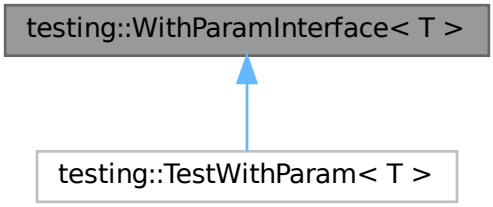
- QWidget * **widget** {nullptr}
- QLabel * **label** {nullptr}
- QLabel * **widgetLabel** {nullptr}
- QGroupBox * **groupBox** {nullptr}
- QGridLayout * **layout** {nullptr}
- QFrame * **line** {nullptr}
- WidgetItem * **parent** {nullptr}
- QList< WidgetItem * > **children**

The documentation for this struct was generated from the following file:

- applications/gui/genesys/propertyeditor/qtpropertybrowser/qtgroupboxpropertybrowser.cpp

17.1164 testing::WithParamInterface< T > Class Template Reference

Inheritance diagram for testing::WithParamInterface< T >:



Public Types

- typedef T **ParamType**

Static Public Member Functions

- static const ParamType & **GetParam** ()

Friends

- template<class TestClass >
  class **internal::ParameterizedTestFactory**

The documentation for this class was generated from the following file:

- gtest/gtest/gtest.h

17.1165 SimulatorSessionService::WorkerCapabilitiesResult Struct Reference

Describes currently supported worker features.
`#include <applications/worker/service/SimulatorSessionService.h>`

Public Attributes

- bool **supportsSessionApi** = false
- bool **supportsSessionScopedSimulator** = false
- bool **supportsModelCreation** = false
- bool **supportsModelPersistence** = false
- bool **supportsSimulationStatus** = false
- bool **supportsSimulationConfig** = false
- bool **supportsSynchronousRun** = false
- bool **supportsSynchronousStep** = false
- bool **supportsDistributedJobs** = false
- bool **supportsJobPolling** = false
- bool **supportsBackgroundExecution** = false
- bool **supportsModelUpload** = false
- bool **supportsStreamingEvents** = false
- bool **supportsJobResultRetrieval** = false

17.1165.1 Detailed Description

Describes currently supported worker features.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.1166 SimulatorSessionService::WorkerInfoResult Struct Reference

Describes worker identity metadata for distributed discovery.
`#include <applications/worker/service/SimulatorSessionService.h>`

Public Attributes

- std::string **role**
- std::string **application**
- std::string **apiFamily**
- std::string **apiVersion**
- std::string **simulatorName**
- std::string **simulatorVersionName**
- unsigned int **simulatorVersionNumber** = 0

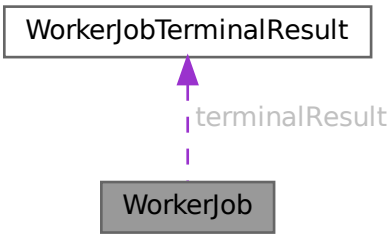
17.1166.1 Detailed Description

Describes worker identity metadata for distributed discovery.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.1167 WorkerJob Struct Reference

Stores metadata for a worker job snapshot owned by the web worker.
`#include <applications/worker/worker/WorkerJob.h>`
Collaboration diagram for WorkerJob:



Public Attributes

- std::string **jobId**

  Stable job identifier returned to API clients.
- std::string **sessionId**

  Session that owns the job snapshot and execution context.
- WorkerJobState **state** = WorkerJobState::Queued

  Current lifecycle state of the job.
- std::string **snapshotFilename**

  Snapshot filename stored in the session workspace.
- std::string **createdMarker**

  Monotonic creation marker used to preserve job ordering.
- std::string **message**

  Human-readable status or error text.
- bool **hasTerminalResult** = false

  True when terminalResult contains valid completion data.
- WorkerJobTerminalResult **terminalResult**

  Final simulation summary written when the job finishes.

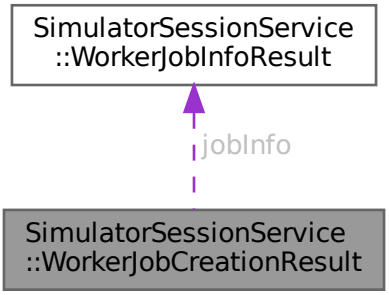
17.1167.1 Detailed Description

Stores metadata for a worker job snapshot owned by the web worker.
The documentation for this struct was generated from the following file:

- applications/worker/worker/WorkerJob.h

17.1168 SimulatorSessionService::WorkerJobCreationResult Struct Reference

Contains worker job creation output and error state.
`#include <applications/worker/service/SimulatorSessionService.h>`
Collaboration diagram for SimulatorSessionService::WorkerJobCreationResult:



Public Attributes

- bool **success** = false
- WorkerJobError **error** = WorkerJobError::None
- WorkerJobInfoResult **jobInfo**

17.1168.1 Detailed Description

Contains worker job creation output and error state.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.1169 SimulatorSessionService::WorkerJobInfoResult Struct Reference

Describes worker job metadata exposed by worker stage 3 endpoints.
`#include <applications/worker/service/SimulatorSessionService.h>`

Public Attributes

- std::string **jobId**
- std::string **sessionId**
- WorkerJobState **state** = WorkerJobState::Queued
- std::string **snapshotFilename**
- std::string **createdMarker**
- std::string **message**

17.1169.1 Detailed Description

Describes worker job metadata exposed by worker stage 3 endpoints.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.1170 WorkerJobManager Class Reference

Owns worker job records in memory and assigns unique identifiers.
`#include <applications/worker/worker/WorkerJobManager.h>`

Public Member Functions

- WorkerJob createQueuedJob (const std::string &sessionId)

  Creates and stores a queued worker job with generated identifiers.
- bool setSnapshotFilename (const std::string &jobId, const std::string &snapshotFilename)

  Replaces the snapshot filename for an existing job.
- bool setState (const std::string &jobId, WorkerJobState state)

  Updates the lifecycle state for an existing job.
- bool setMessage (const std::string &jobId, const std::string &message)

  Replaces the message text for an existing job.
- bool setTerminalResult (const std::string &jobId, const WorkerJobTerminalResult &terminalResult)

  Persists the terminal simulation summary for an existing job.
- std::optional< WorkerJob > getJob (const std::string &jobId) const

  Fetches a previously stored job record.

17.1170.1 Detailed Description

Owns worker job records in memory and assigns unique identifiers.
The manager is intentionally simple: it keeps all job metadata in memory and provides synchronized read/write helpers for the web API and background jobs.

17.1170.2 Member Function Documentation

17.1170.2.1 createQueuedJob()

```
WorkerJob WorkerJobManager::createQueuedJob (
            const std::string & sessionId )
```

Creates and stores a queued worker job with generated identifiers.
Creates and stores a queued worker job with a monotonic identifier.

Parameters

sessionId	Identifier of the owning session.

Returns

Newly stored worker job metadata.

17.1170.2.2 getJob()

```
std::optional< WorkerJob > WorkerJobManager::getJob (
            const std::string & jobId ) const
```

Fetches a previously stored job record.
Retrieves a copy of a job record when it exists.

Parameters

jobId	Job identifier.

Returns

Copy of the job when found.

17.1170.2.3 setMessage()

```
bool WorkerJobManager::setMessage (
            const std::string & jobId,
            const std::string & message )
```

Replaces the message text for an existing job.
Replaces the human-readable message for a stored job.

Parameters

jobId	Job identifier to update.
message	Human-readable status or error message.

Returns

True when the job exists and was updated.

17.1170.2.4 setSnapshotFilename()

```
bool WorkerJobManager::setSnapshotFilename (
            const std::string & jobId,
            const std::string & snapshotFilename )
```

Replaces the snapshot filename for an existing job.
Updates the snapshot filename for a stored job.

Parameters

jobId	Job identifier to update.
snapshotFilename	Snapshot filename persisted in session workspace.

Returns

True when the job exists and was updated.

17.1170.2.5 setState()

```
bool WorkerJobManager::setState (
            const std::string & jobId,
            WorkerJobState state )
```

Updates the lifecycle state for an existing job.
Updates the lifecycle state for a stored job.

Parameters

jobId	Job identifier to update.
state	New worker job state.

Returns

True when the job exists and was updated.

17.1170.2.6 setTerminalResult()

```
bool WorkerJobManager::setTerminalResult (
            const std::string & jobId,
            const WorkerJobTerminalResult & terminalResult )
```

Persists the terminal simulation summary for an existing job.
Stores the terminal simulation result and marks it available.

Parameters

jobId	Job identifier to update.
terminalResult	Terminal simulation summary values.

Returns

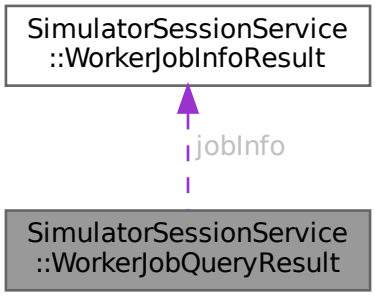
True when the job exists and was updated.

The documentation for this class was generated from the following files:

- applications/worker/worker/WorkerJobManager.h
- applications/worker/worker/WorkerJobManager.cpp

17.1171 SimulatorSessionService::WorkerJobQueryResult Struct Reference

Contains worker job lookup output and error state.
`#include <applications/worker/service/SimulatorSessionService.h>`
Collaboration diagram for SimulatorSessionService::WorkerJobQueryResult:



Public Attributes

- bool **success** = false
- WorkerJobError **error** = WorkerJobError::None
- WorkerJobInfoResult **jobInfo**

17.1171.1 Detailed Description

Contains worker job lookup output and error state.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.1172 SimulatorSessionService::WorkerJobResultInfo Struct Reference

Describes terminal worker job result data exposed by worker stage 5 endpoint.
`#include <applications/worker/service/SimulatorSessionService.h>`

Public Attributes

- std::string **jobId**
- WorkerJobState **state** = WorkerJobState::Queued
- std::string **message**
- double **simulatedTime** = 0.0
- unsigned int **currentReplicationNumber** = 0
- unsigned int **numberOfReplications** = 0
- double **replicationLength** = 0.0
- double **warmUpPeriod** = 0.0
- bool **hasIsPaused** = false
- bool **isPaused** = false

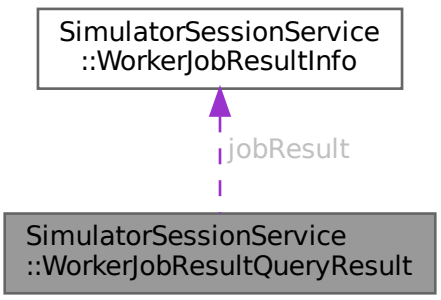
17.1172.1 Detailed Description

Describes terminal worker job result data exposed by worker stage 5 endpoint.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.1173 SimulatorSessionService::WorkerJobResultQueryResult Struct Reference

Contains worker job terminal result lookup output and error state.
`#include <applications/worker/service/SimulatorSessionService.h>`
Collaboration diagram for SimulatorSessionService::WorkerJobResultQueryResult:



Public Attributes

- bool **success** = false
- WorkerJobError **error** = WorkerJobError::None
- WorkerJobResultInfo **jobResult**

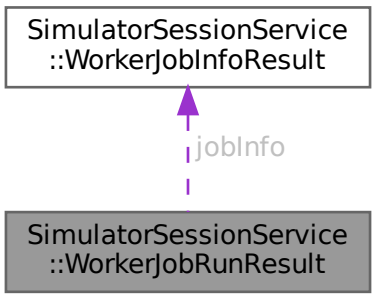
17.1173.1 Detailed Description

Contains worker job terminal result lookup output and error state.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.1174 SimulatorSessionService::WorkerJobRunResult Struct Reference

Contains worker job run output and error state.
`#include <applications/worker/service/SimulatorSessionService.h>`
Collaboration diagram for SimulatorSessionService::WorkerJobRunResult:



Public Attributes

- bool **success** = false
- WorkerJobError **error** = WorkerJobError::None
- WorkerJobInfoResult **jobInfo**

17.1174.1 Detailed Description

Contains worker job run output and error state.
The documentation for this struct was generated from the following file:

- applications/worker/service/SimulatorSessionService.h

17.1175 WorkerJobTerminalResult Struct Reference

Stores the terminal simulation summary persisted for a finished or failed worker job.
`#include <applications/worker/worker/WorkerJob.h>`

Public Attributes

- double **simulatedTime** = 0.0

  Simulated time reached when the job completed.
- unsigned int **currentReplicationNumber** = 0

  Current replication number at completion.
- unsigned int **numberOfReplications** = 0

Total number of replications configured for the run.

- double **replicationLength** = 0.0

  Replication length used by the run.

- double **warmUpPeriod** = 0.0

  Warm-up period used by the run.

- std::optional< bool > **isPaused**

  Pause state when the simulator explicitly reported it.

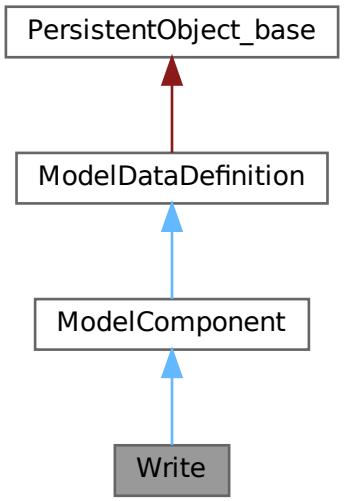
17.1175.1 Detailed Description

Stores the terminal simulation summary persisted for a finished or failed worker job.
The documentation for this struct was generated from the following file:

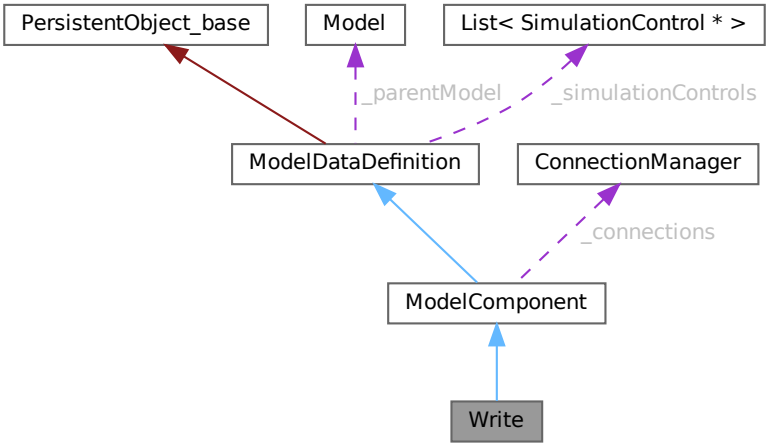
- applications/worker/worker/WorkerJob.h

17.1176 Write Class Reference

Writes one or more literal/expression text lines to the screen (trace output) or to a file.
`#include <plugins/components/InputOutput/Write.h>`
Inheritance diagram for Write:



Collaboration diagram for Write:



Public Types

- enum class **WriteToType** : int { **SCREEN** = 0 , **FILE** = 1 , **num_elements** = 2 }

Public Member Functions

- **Write** (Model ∗model, std::string name="")
- void **insertText** (std::list< std::string > texts)
- void **setFilename** (std::string _filename)
- std::string **filename** () const
- void **setWriteToType** (WriteToType _writeToType)
- Write::WriteToType **writeToType** () const
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent ∗comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insertIntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.
- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecificEnabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static std::string **convertEnumToStr** (WriteToType type)
- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  Handles a dispatched event for this component type.
- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.
- virtual bool _check (std::string &errorMessage) override

  could be overriden by derived classes
- virtual void _initBetweenReplications () override

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createNonEditableDataDefinitions ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1176.1 Detailed Description

Writes one or more literal/expression text lines to the screen (trace output) or to a file.
Arena correspondence: the write-only slice of Arena's "ReadWrite module" (Rockwell Automation, *Getting Started with Arena*, "The Advanced Process Panel", p. 58). `WriteToType` (`SCREEN/FILE`) matches Arena's corresponding output destinations.
Known difference from Arena: there is no read direction at all (no component reads a value from Screen/File/↩ Attribute/Variable back into the model), and no structured recordset integration with File (§5.10) — see `docs/ai↩ _assistants/reference/ARENA_GENESYS_COMPATIBILITY.md` §6.16.

17.1176.2 Member Function Documentation

17.1176.2.1 _check()

```
bool Write::_check (
            std::string & errorMessage )  [override], [protected], [virtual]
```

could be overriden by derived classes
Reimplemented from ModelDataDefinition.

17.1176.2.2 _initBetweenReplications()

```
void Write::_initBetweenReplications ( )  [override], [protected], [virtual]
```

Reimplemented from ModelDataDefinition.

17.1176.2.3 _loadInstance()

```
bool Write::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from ModelComponent.

17.1176.2.4 _onDispatchEvent()

```
void Write::_onDispatchEvent (
            Entity * entity,
            unsigned int inputPortNumber )  [override], [protected], [virtual]
```

Handles a dispatched event for this component type.
This method is only for ModelComponents, not for ModelDataElements.

Parameters

entity	Entity being processed.
inputPortNumber	Input port where the entity arrived.

Implements ModelComponent.

17.1176.2.5 _saveInstance()

```
void Write::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from ModelComponent.

17.1176.2.6 show()

```
std::string Write::show ( )  [override], [virtual]
```

virtual public methods
Returns a textual representation of this component.

Returns

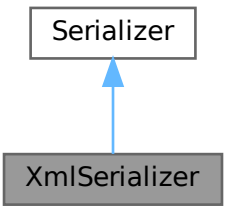
Human-readable description used for tracing and debugging.

Reimplemented from ModelComponent.
The documentation for this class was generated from the following files:

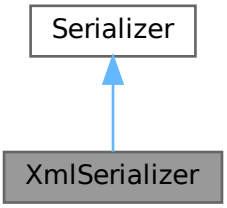
- plugins/components/InputOutput/Write.h
- plugins/components/InputOutput/Write.cpp

17.1177 XmlSerializer Class Reference

Inheritance diagram for XmlSerializer:



Collaboration diagram for XmlSerializer:



Public Member Functions

- **XmlSerializer** (Model *model)
- PersistenceRecord * newPersistenceRecord () override

  Creates an empty persistence record.
- bool dump (std::ostream &output) override

  Writes formatted contents to a given output stream.
- bool load (std::istream &input) override

  Attempts to load contents from a given input stream.
- bool get (const std::string &name, PersistenceRecord *entry) override

  Tries to fetch an entry with the given label as a key.
- bool put (const std::string name, const std::string type, const Util::identification id, PersistenceRecord *fields) override

  Upserts an entry into this structure.
- int for_each (std::function< int(const std::string &) > delegate) override

  Iterates over this structure's entries, ordered by ID.

17.1177.1 Member Function Documentation

17.1177.1.1 dump()

```
bool XmlSerializer::dump (
            std::ostream & output )  [override], [virtual]
```

Writes formatted contents to a given output stream.

Parameters

output	output stream to write to.

Returns

whether the current state of this structure's contents was deemed valid for serialization.

Implements Serializer.

17.1177.1.2 for_each()

```
int XmlSerializer::for_each (
            std::function< int(const std::string &) > delegate )  [override], [virtual]
```

Iterates over this structure's entries, ordered by ID.

Parameters

delegate	delegate which is applied to every entry-identifying key, until it returns a non-zero value.

Returns

either zero, or the first non-zero value returned by the delegate.

Implements Serializer.

17.1177.1.3 get()

```
bool XmlSerializer::get (
            const std::string & name,
            PersistenceRecord * entry )  [override], [virtual]
```

Tries to fetch an entry with the given label as a key.

Parameters

name	a label which uniquely identifies an entry.
entry	(out-param) a pointer to a caller-owned entry, which is filled on success.

Returns

whether the entry was found and the out-param was actually written to.

Implements Serializer.

17.1177.1.4 load()

```
bool XmlSerializer::load (
            std::istream & input )  [override], [virtual]
```

Attempts to load contents from a given input stream.

Returns

whether the input was well-formatted and loading was successful.

Implements Serializer.

17.1177.1.5 newPersistenceRecord()

```
PersistenceRecord * XmlSerializer::newPersistenceRecord ( )  [override], [virtual]
```

Creates an empty persistence record.
Although the returned record can be freed normally with `delete`, care must be taken to only use records which come from a implementation, when configuring that implementation's contents.
Implements Serializer.

17.1177.1.6 put()

```
bool XmlSerializer::put (
            const std::string name,
            const std::string type,
            const Util::identification id,
            PersistenceRecord * fields )  [override], [virtual]
```

Upserts an entry into this structure.

Parameters

name	a label which uniquely identifies the entry.

Parameters

type	a type name.
id	a component/datadef ID number.
fields	a persistence record for the entry, whose contents will be copied if valid.

Returns

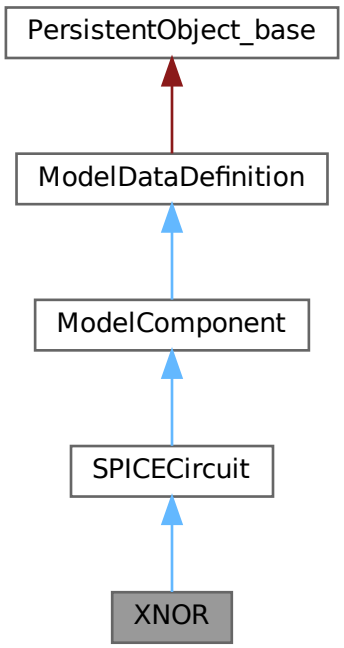
whether the entry was valid and was actually copied into this structure.

Implements Serializer.
The documentation for this class was generated from the following files:

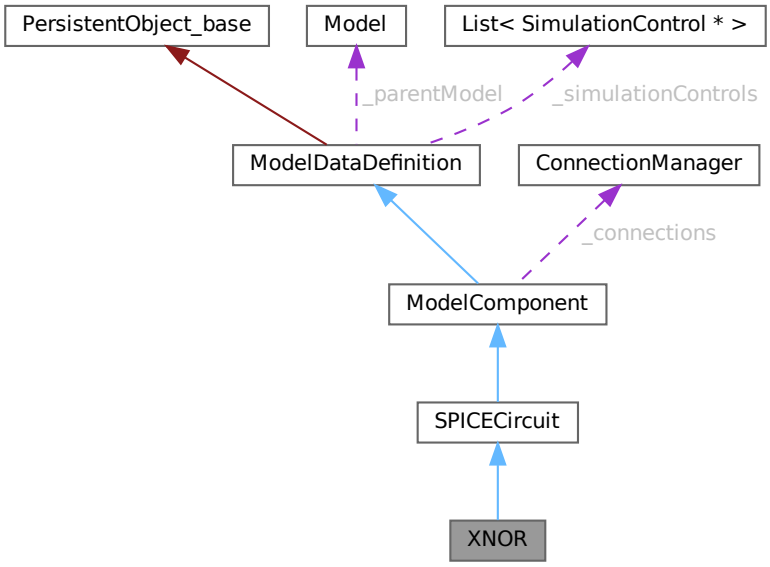
- kernel/simulator/persistence/XmlSerializer.h
- kernel/simulator/persistence/XmlSerializer.cpp

17.1178 XNOR Class Reference

Inheritance diagram for XNOR:



Collaboration diagram for XNOR:



Public Member Functions

- **XNOR** (Model ∗model, std::string name="")
- void Build () override
- void **setPMOSWidth** (float PMOSWidth)
- void **setPMOSLength** (float length)
- void **setNMOSWidth** (float NMOSWidth)
- void **setNMOSLength** (float NMOSLength)
- void **setElectricalModel** (std::string electricalModel)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model ∗model, std::string name="")

  constructors
- void **setRunner** (SPICERunner ∗runner)

  new public user methods for this components
- SPICERunner ∗ **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model ∗model, std::string componentTypename, std::string name="")

  constructors
- virtual ∼**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager ∗ getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent ∗component, unsigned int inputPortNumber=0)

Connects this component to another component.

- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.

- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.

- std::string getDescription () const

  Returns the free-form description associated with this component.

- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.

- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.

- Util::identification getId () const

  Returns the unique identifier of this instance.

- void setName (const std::string &name)

  Updates the logical name of this instance.

- const std::string & getName () const

  Returns the logical name of this instance.

- void setLabel (const std::string &label)

  Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.

- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.

- Model * getParentModel () const

  Returns the parent model that owns this definition.

- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.

- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.

- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.

- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.

- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.

- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.

- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.

- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.

- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.

- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.

- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

Resets simulation-time state between replications.

- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.

- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden

- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden

- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.

- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes

- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*

- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)

- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1178.1 Member Function Documentation

17.1178.1.1 _loadInstance()

```
bool XNOR::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.1178.1.2 _saveInstance()

```
void XNOR::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.1178.1.3 Build()

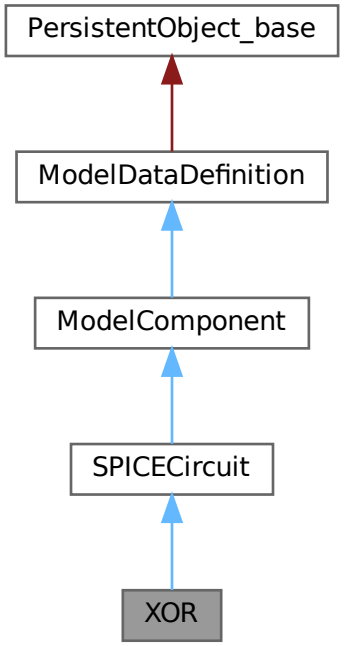
```
void XNOR::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.
The documentation for this class was generated from the following files:

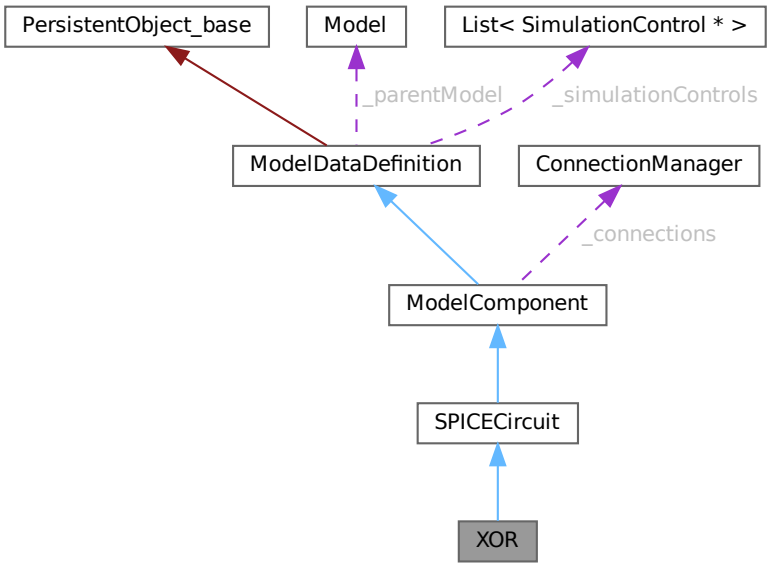
- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.1179 XOR Class Reference

Inheritance diagram for XOR:



Collaboration diagram for XOR:



Public Member Functions

- **XOR** (Model *model, std::string name="")
- void Build () override
- void **setPMOSWidth** (float PMOSWidth)
- void **setPMOSLength** (float length)
- void **setNMOSWidth** (float NMOSWidth)
- void **setNMOSLength** (float NMOSLength)
- void **setElectricalModel** (std::string electricalModel)

Public Member Functions inherited from SPICECircuit

- **SPICECircuit** (Model *model, std::string name="")

  constructors
- void **setRunner** (SPICERunner *runner)

  new public user methods for this components
- SPICERunner * **getRunner** ()
- virtual std::string show () override

  virtual public methods

Public Member Functions inherited from ModelComponent

- ModelComponent (Model *model, std::string componentTypename, std::string name="")

  constructors
- virtual ~**ModelComponent** ()

  Releases component-owned resources and unregisters it from the model.
- ConnectionManager * getConnectionManager () const

  new public user methods for this component
- void connectTo (ModelComponent *component, unsigned int inputPortNumber=0)

  Connects this component to another component.
- bool hasBreakpointAt ()

  Indicates whether a simulation breakpoint is set on this component.
- void setDescription (std::string _description)

  Updates the free-form description shown by the GUI.
- std::string getDescription () const

  Returns the free-form description associated with this component.
- void remove (ModelComponent *comp)

  Removes a connected component from this component.

Public Member Functions inherited from ModelDataDefinition

- ModelDataDefinition (Model *model, std::string datadefinitionTypename, std::string name="", bool insert↩ IntoModel=true)

  Creates a model data definition and optionally inserts it into the model.
- virtual ~**ModelDataDefinition** ()

  Releases resources owned by this data definition.
- Util::identification getId () const

  Returns the unique identifier of this instance.
- void setName (const std::string &name)

  Updates the logical name of this instance.
- const std::string & getName () const

  Returns the logical name of this instance.
- void setLabel (const std::string &label)

Updates the user-visible label of this instance.

- const std::string & getLabel () const

  Returns the user-visible label of this instance.
- const std::string & getClassname () const

  Returns the runtime class name used by the plugin system.
- Model * getParentModel () const

  Returns the parent model that owns this definition.
- bool isReportStatistics () const

  Indicates whether this data definition contributes report statistics.
- void setReportStatistics (bool reportStatistics)

  Enables or disables report-statistics generation for this definition.
- ModelDataDefinition * getInternalData (const std::string &name) const

  Looks up a named internal child data definition.
- std::map< std::string, ModelDataDefinition * > * getInternalData () const

  Returns the full map of internal child data definitions.
- std::map< std::string, ModelDataDefinition * > * getAttachedData () const

  Returns the full map of attached data definitions.
- bool hasChanged () const

  Indicates whether this instance has been modified since the last reset.
- void setHasChanged (bool hasChanged)

  Updates the changed flag for this data definition.
- unsigned int getLevel () const

  Returns the hierarchical editing context containing this element.
- void setModelLevel (unsigned int _modelLevel)

  Sets this element's hierarchical editing context.
- List< SimulationControl * > * getSimulationControls () const

  Returns the simulation controls owned by this data definition.
- TraceManager::Level getTraceLevelSpecific () const

  Returns the trace level that applies specifically to this definition.
- void setTraceLevelSpecific (TraceManager::Level level)

  Updates the trace level used by this definition.
- void defineTraceLevelSpecific (TraceManager::Level traceLevelspecific, bool traceLevelSpecific↩ Enabled=true)

  Defines the specific trace level and optionally enables it immediately.
- bool isTraceLevelSpecificEnabled () const

  Indicates whether the specific trace level is enabled.
- void setTraceLevelSpecificEnabled (bool traceLevelSpecificEnabled)

  Enables or disables the specific trace level.

Static Public Member Functions

- static PluginInformation * **GetPluginInformation** ()
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")

Static Public Member Functions inherited from SPICECircuit

- static PluginInformation * **GetPluginInformation** ()

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static ModelComponent * **LoadInstance** (Model *model, PersistenceRecord *fields)
- static ModelDataDefinition * **NewInstance** (Model *model, std::string name="")
- static std::vector< std::string > **split** (std::string word, char split_char)

  Externalize function GetPluginInformation to be accessible throught dynamic linked library.

Static Public Member Functions inherited from ModelComponent

- static ModelComponent * LoadInstance (Model *model, PersistenceRecord *fields)

  static public methods that must have implementations (Load and New just the same. GetInformation must provide specific infos for the new component
- static bool Check (ModelComponent *component)

  Validates a component instance.
- static void CreateInternalData (ModelComponent *component)

  Creates internal data required by a component instance.
- static void SaveInstance (PersistenceRecord *fields, ModelComponent *component)

  Serializes a component instance into a persistence record.
- static void DispatchEvent (Event *event)

  Dispatches a simulation event to the target component.

Static Public Member Functions inherited from ModelDataDefinition

- static ModelDataDefinition * LoadInstance (Model *model, PersistenceRecord *fields, bool insertIntoModel)

  Loads a model data definition instance from serialized fields.
- static ModelDataDefinition * NewInstance (Model *model, std::string name="")

  Creates a new instance of a data definition using the plugin registry.
- static void SaveInstance (PersistenceRecord *fields, ModelDataDefinition *modeldatum)

  Serializes a data definition instance into a persistence record.
- static bool Check (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Validates a data definition instance.
- static void InitBetweenReplications (ModelDataDefinition *modeldatum)

  Resets simulation-time state between replications.
- static void CreateInternalData (ModelDataDefinition *modeldatum)

  Creates internal and attached data required by this instance.
- static bool CreateRelatedDataElements (ModelDataDefinition *modeldatum, std::string &errorMessage)

  Creates related data and validates the instance in a single call.

Protected Member Functions

- virtual bool _loadInstance (PersistenceRecord *fields) override

  virtual protected method that must be overriden
- virtual void _saveInstance (PersistenceRecord *fields, bool saveDefaultValues) override

  Saves object fields into a serialization structure.

Protected Member Functions inherited from SPICECircuit

- void **BuildCircuit** (std::string description, unsigned int id, std::vector< std::string > the_params={""}, std::string the_model="", std::string the_model_file="")

  virtual protected method that must be overriden
- void **UpdateConnections** ()
- virtual void _onDispatchEvent (Entity *entity, unsigned int inputPortNumber) override

  This method is only for ModelComponents, not ModelDataElements.
- virtual void _createNonEditableDataDefinitions () override

  virtual protected methods that could be overriden by derived classes, if needed

Protected Member Functions inherited from ModelDataDefinition

- bool _getSaveDefaultsOption ()
- virtual bool _check (std::string &errorMessage)

  could be overriden by derived classes
- virtual ParserChangesInformation * _getParserChangesInformation ()
- virtual void _initBetweenReplications ()
- virtual void _addSimulationControl (SimulationControl *control)
- virtual void _createInternalAndAttachedData ()
- virtual void **_createInternalStatisticReporters** ()
- virtual void _createAttachedAttributes ()
- virtual void _createEditableDataDefinitions ()
- void **_templateCreateInternalStatisticReporters** ()
- void **_templateCreateAttachedAttributes** ()
- void **_templateCreateNonEditableDataDefinitions** ()
- void **_templateCreateEditableDataDefinitions** ()
- void **_internaStatisticReportersClear** ()

  legacy helper methods used by the granular create hooks*
- void **_internalDataClear** ()
- void **_internalDataInsert** (const std::string &key, ModelDataDefinition *child)
- void **_internalDataRemove** (const std::string &key)
- void **_attachedDataClear** ()
- void **_attachedDataInsert** (const std::string &key, ModelDataDefinition *data)
- void **_attachedDataRemove** (const std::string &key)
- void **_attachedAttributesInsert** (const std::vector< std::string > &neededNames)
- void **_statisticReporterInsert** (const std::string &key, ModelDataDefinition *data)
- void **_statisticReporterRemove** (const std::string &key)
- void **_statisticReportersClear** ()
- void **_mandatoryAttachedAttributeInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryAttachedAttributeRemove** (const std::string &key)
- void **_mandatoryAttachedAttributesClear** ()
- void **_mandatoryEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryEditableDataDefinitionsClear** ()
- void **_optionalEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalEditableDataDefinitionsClear** ()
- void **_mandatoryNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_mandatoryNonEditableDataDefinitionRemove** (const std::string &key)
- void **_mandatoryNonEditableDataDefinitionsClear** ()
- void **_optionalNonEditableDataDefinitionInsert** (const std::string &key, ModelDataDefinition *data)
- void **_optionalNonEditableDataDefinitionRemove** (const std::string &key)
- void **_optionalNonEditableDataDefinitionsClear** ()
- void **_checkCreateAttachedReferencedDataDefinition** (const std::string &expression)

  method to be called to insert attached dataelements that are referenced by string expressions (detected by the parser), to avoid orphaned data definitions
- void **trace** (const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)

  not just an easy access to trace manager, but wrappers to check if specific trace level applies
- void **traceError** (const std::string &text, const std::exception &e)
- void **traceError** (const std::string &text, TraceManager::Level level=TraceManager::Level::L1_errorFatal)
- void **traceReport** (const std::string &text, TraceManager::Level level=TraceManager::Level::L2_results)
- void **traceSimulation** (void *thisobject, double time, Entity *entity, ModelComponent *component, const std::string &text, TraceManager::Level level=TraceManager::Level::L8_detailed)
- void traceSimulation (void *thisobject, const std::string &text, TraceManager::Level level=TraceManager::↩ Level::L8_detailed)

  Traces text to the simulation output.
- void **traceSimulation** (void *thisobject, TraceManager::Level level, const std::string &text)

Additional Inherited Members

Protected Attributes inherited from SPICECircuit

- unsigned int **id**
- std::function< void()> **build**

Protected Attributes inherited from ModelComponent

- ConnectionManager * _connections = new ConnectionManager()

  virtual protected methods that could be overriden by derived classes, if needed

Protected Attributes inherited from ModelDataDefinition

- Util::identification **_id**
- std::string **_typename**
- bool **_reportStatistics**
- bool **_hasChanged**
- unsigned int **_modelLevel** = 0
- Model * **_parentModel**
- List< SimulationControl * > * **_simulationControls** = new List<SimulationControl*>()
- TraceManager::Level **_traceLevelSpecific**
- bool **_traceLevelSpecificEnabled** = false

17.1179.1 Member Function Documentation

17.1179.1.1 _loadInstance()

```
bool XOR::_loadInstance (
            PersistenceRecord * fields )  [override], [protected], [virtual]
```

virtual protected method that must be overriden
Reimplemented from SPICECircuit.

17.1179.1.2 _saveInstance()

```
void XOR::_saveInstance (
            PersistenceRecord * fields,
            bool saveDefaultValues )  [override], [protected], [virtual]
```

Saves object fields into a serialization structure.

Parameters

fields	a table-like structure used for serialization.
saveDefaultValues	indicates whether or not to save fields which are set to their default value.

Reimplemented from SPICECircuit.

17.1179.1.3 Build()

```
void XOR::Build ( )  [override], [virtual]
```

Reimplemented from SPICECircuit.
The documentation for this class was generated from the following files:

- plugins/components/ElectronicsSimulation/SPICECircuit.h
- plugins/components/ElectronicsSimulation/SPICECircuit.cpp

17.1180 yy_buffer_state Struct Reference

Public Attributes

- FILE * **yy_input_file**
- char * **yy_ch_buf**
- char * **yy_buf_pos**
- int **yy_buf_size**
- int **yy_n_chars**
- int **yy_is_our_buffer**
- int **yy_is_interactive**
- int **yy_at_bol**
- int yy_bs_lineno
- int yy_bs_column
- int **yy_fill_buffer**
- int **yy_buffer_status**

17.1180.1 Member Data Documentation

17.1180.1.1 yy_bs_column

`int yy_buffer_state::yy_bs_column`
The column count.

17.1180.1.2 yy_bs_lineno

`int yy_buffer_state::yy_bs_lineno`
The line count.
The documentation for this struct was generated from the following file:

- parser/Genesys++-scanner.cpp

17.1181 yy_trans_info Struct Reference

Public Attributes

- flex_int32_t **yy_verify**
- flex_int32_t **yy_nxt**

The documentation for this struct was generated from the following file:

- parser/Genesys++-scanner.cpp

Chapter 18

File Documentation

18.1 BaseGenesysTerminalApplication.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   BaseGenesysTerminalApplication.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 16:25
00012  */
00013
00014 #ifndef BaseGenesysTerminalApplication_H
00015 #define BaseGenesysTerminalApplication_H
00016
00017 #include "GenesysApplication_if.h"
00018 #include "kernel/simulator/TraceManager.h"
00019 #include "kernel/simulator/OnEventManager.h"
00020 #include "kernel/simulator/PluginManager.h"
00021 #include "../kernel/simulator/model/Model.h"
00022
00023 #include <thread>
00024
00025 class BaseGenesysTerminalApplication : public GenesysApplication_if {
00026 public:
00027     BaseGenesysTerminalApplication();
00028     virtual ~BaseGenesysTerminalApplication() = default;
00029 public:
00030     virtual int main(int argc, char** argv) override;
00031     virtual void Simulate(Simulator* genesys, Model* model, PluginManager *plugins);
00032 public:
00033     void setDefaultTraceHandlers(TraceManager* tm);
00034     void setDefaultEventHandlers(OnEventManager* oem);
00035 protected:
00036     // default Trace Handlers
00037     virtual void traceHandler(TraceEvent e);
00038     virtual void traceErrorHandler(TraceErrorEvent e);
00039     virtual void traceReportHandler(TraceEvent e);
00040     virtual void traceSimulationHandler(TraceSimulationEvent e);
00041     // default Event Handlers
00042     virtual void onAfterProcessEventHandler(SimulationEvent* re);
00043     virtual void onBreakpointHandler(SimulationEvent* re);
00044     virtual void onEntityCreateHandler(SimulationEvent* re);
00045     virtual void onEntityMoveHandler(SimulationEvent* re);
00046     virtual void onSimulationStartHandler(SimulationEvent* re);
00047     virtual void onReplicationStepHandler(SimulationEvent* re);
00048     virtual void onReplicationStartHandler(SimulationEvent* re);
00049     virtual void onProcessEventHandler(SimulationEvent* re);
00050     virtual void onReplicationEndHandler(SimulationEvent* re);
00051     virtual void onSimulationEndHandler(SimulationEvent* re);
00052     virtual void onSimulationPausedHandler(SimulationEvent* re);
00053     virtual void onSimulationResumeHandler(SimulationEvent* re);
00054     virtual void onEntityRemoveHandler(SimulationEvent* re);
00055 private:
00056
00057 };
00058
00059 #endif /* BaseGenesysTerminalApplication_H */
00060
```

18.2 GenesysApplication_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   GenesysApplication_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 2 de Outubro de 2018, 19:14
00012  */
00013
00014 #ifndef GENESYSAPPLICATION_IF_H
00015 #define GENESYSAPPLICATION_IF_H
00016
00017 class GenesysApplication_if {
00018 public:
00019     virtual ~GenesysApplication_if() = default;
00020     virtual int main(int argc, char** argv) = 0;
00021 };
00022
00023 #endif /* GENESYSAPPLICATION_IF_H */
```

18.3 AIAssistantWindow.h

```
00001 #ifndef AIASSISTANTWINDOW_H
00002 #define AIASSISTANTWINDOW_H
00003
00004 #include "../../../tools/AIAssistant/AIAssistantDefaultImpl.h"
00005 #include "../../../tools/AIAssistant/AISecretStore.h"
00006 #include "../../../kernel/simulator/SimulatorFacade.h"
00007
00008 #include <QMainWindow>
00009
00010 class QAction;
00011 class QCheckBox;
00012 class QComboBox;
00013 class QDoubleSpinBox;
00014 class QLabel;
00015 class QLineEdit;
00016 class QPlainTextEdit;
00017 class QPushButton;
00018 class QSpinBox;
00019 class QTabWidget;
00020 class QTableWidget;
00021 class QTextBrowser;
00022 class Simulator;
00023
00024 class AIAssistantWindow : public QMainWindow {
00025     Q_OBJECT
00026 public:
00027     explicit AIAssistantWindow(Simulator* simulator, QWidget* parent = nullptr);
00028     ~AIAssistantWindow() override;
00029
00030 private:
00031     void buildMenus();
00032     void buildWorkspace();
00033     void connectActions();
00034
00035     QWidget* buildConfigTab();
00036     QWidget* buildPromptTab();
00037     QWidget* buildExecutionTab();
00038     QWidget* buildAuditTab();
00039
00040     void applyConfigurationToBackend();
00041     void runAnalyzePrompt();
00042     void runCreateModelPlan();
00043     void runBuildModel();
00044     void runConfigureSimulation();
00045     void runSimulation();
00046     void runCollectResults();
00047     void runFullPipeline();
00048     void saveApiKeyToKeyring();
00049     void loadApiKeyFromKeyring();
00050     void refreshAuditTable();
00051     void exportAuditLog();
00052     void appendLog(const QString& text);
00053     void setStatus(const QString& text);
00054     void setRunning(bool running);
00055     AIAssistantRequest buildRequest() const;
00056     QString providerKeyringAccount() const;
```

```
00057
00058 private:
00059     Simulator*          _simulator = nullptr;
00060     SimulatorFacade     _facade;
00061     AIAssistantDefaultImpl _assistant;
00062
00063     // Configuration tab
00064     QComboBox*          _providerCombo       = nullptr;
00065     QLineEdit*          _modelNameEdit       = nullptr;
00066     QLineEdit*          _apiKeyEdit          = nullptr;
00067     QPushButton*        _apiKeyToggleBtn     = nullptr;
00068     QPushButton*        _saveKeyringBtn      = nullptr;
00069     QPushButton*        _loadKeyringBtn      = nullptr;
00070     QLineEdit*          _apiKeyEnvVarEdit    = nullptr;
00071     QLineEdit*          _baseUrlEdit         = nullptr;
00072     QSpinBox*           _timeoutSpin         = nullptr;
00073     QDoubleSpinBox*     _temperatureSpin     = nullptr;
00074     QCheckBox*          _allowNetworkChk     = nullptr;
00075     QCheckBox*          _allowMutationChk    = nullptr;
00076     QCheckBox*          _allowExecutionChk   = nullptr;
00077     QCheckBox*          _allowFilesystemChk  = nullptr;
00078     QCheckBox*          _sandboxChk          = nullptr;
00079     QCheckBox*          _dryRunChk           = nullptr;
00080
00081     // Prompt & Plan tab
00082     QPlainTextEdit*     _modelDescEdit       = nullptr;
00083     QPlainTextEdit*     _experimentDescEdit  = nullptr;
00084     QPlainTextEdit*     _userPromptEdit      = nullptr;
00085     QPushButton*        _analyzeBtn          = nullptr;
00086     QPushButton*        _planBtn             = nullptr;
00087     QTextBrowser*       _planOutput          = nullptr;
00088
00089     // Execution tab
00090     QPushButton*        _buildModelBtn       = nullptr;
00091     QPushButton*        _configureSimBtn     = nullptr;
00092     QPushButton*        _runSimBtn           = nullptr;
00093     QPushButton*        _collectResultsBtn   = nullptr;
00094     QPushButton*        _fullPipelineBtn     = nullptr;
00095     QTextBrowser*       _executionLog        = nullptr;
00096     QLabel*             _statusLabel         = nullptr;
00097
00098     // Audit log tab
00099     QTableWidget*       _auditTable          = nullptr;
00100     QPushButton*        _refreshAuditBtn     = nullptr;
00101     QPushButton*        _exportAuditBtn      = nullptr;
00102     QLabel*             _auditLogPathLabel   = nullptr;
00103
00104     QTabWidget*         _tabs                = nullptr;
00105
00106     // Actions
00107     QAction*            _applyConfigAction   = nullptr;
00108     QAction*            _clearLogAction      = nullptr;
00109     QAction*            _refreshAuditAction  = nullptr;
00110 };
00111
00112 #endif // AIASSISTANTWINDOW_H
```

18.4 AddUndoCommand.h

```
00001 #ifndef ADDUNDOCOMMAND_H
00002 #define ADDUNDOCOMMAND_H
00003
00004 #include <QUndoCommand>
00005 #include "graphicals/ModelGraphicsScene.h"
00006 #include "graphicals/GraphicalConnection.h"
00007 #include "graphicals/GraphicalModelComponent.h"
00008
00009 struct ComponentItem {
00010     GraphicalModelComponent *graphicalComponent;
00011     QPointF initialPosition;
00012     QList<GraphicalConnection *> inputConnections;
00013     QList<GraphicalConnection *> outputConnections;
00014 };
00015
00016 struct GroupItem {
00017     QGraphicsItemGroup *group;
00018     QList<ComponentItem> myComponentItems;
00019     QPointF initialPosition;
00020 };
00021
00022 struct DrawingItem {
00023     QGraphicsItem *myDrawingItem;
00024     QPointF initialPosition;
00025 };
00026
```

```
00027
00028 class AddUndoCommand: public QUndoCommand {
00029 public:
00030     AddUndoCommand(QGraphicsItem *item, ModelGraphicsScene *scene, QUndoCommand *parent = nullptr);
00031     ~AddUndoCommand();
00032
00033     void undo() override;
00034     void redo() override;
00035
00036 private:
00037     ComponentItem _myComponentItem;
00038     GraphicalConnection *_myConnectionItem;
00039     QGraphicsItem *_myDrawingItem;
00040     ModelGraphicsScene *_myGraphicsScene;
00041     bool _firstExecution;
00042 };
00043
00044 #endif // ADDUNDOCOMMAND_H
```

18.5 DeleteUndoCommand.h

```
00001 #ifndef DELETEUNDOCOMMAND_H
00002 #define DELETEUNDOCOMMAND_H
00003
00004 #include <QUndoCommand>
00005 #include <string>
00006 #include "AddUndoCommand.h"
00007 #include "graphicals/ModelGraphicsScene.h"
00008 #include "graphicals/GraphicalConnection.h"
00009 #include "graphicals/GraphicalModelDataDefinition.h"
00010
00011 struct DataDefinitionItem {
00012     GraphicalModelDataDefinition *graphicalDataDefinition;
00013     QPointF initialPosition;
00014     bool initiallySelected;
00015 };
00016
00017 struct DataDefinitionRelationItem {
00018     ModelDataDefinition *owner;
00019     ModelDataDefinition *dataDefinition;
00020     std::string key;
00021     bool internal;
00022 };
00023
00024 class DeleteUndoCommand : public QUndoCommand
00025 {
00026 public:
00027     explicit DeleteUndoCommand(QList<QGraphicsItem *> items, ModelGraphicsScene *scene, QUndoCommand
      *parent = nullptr);
00028     ~DeleteUndoCommand();
00029
00030     void undo() override;
00031     void redo() override;
00032
00033 private:
00034     void captureDataDefinitionsRemovedWithSelectedItems();
00035     void captureDataDefinitionRelations();
00036     void removeDataDefinitionRelations();
00037     void restoreDataDefinitionRelations();
00038     void removeDataDefinitionsFromModel();
00039     void restoreDataDefinitionsToModel();
00040     void detachDataDefinitionItems();
00041     void restoreDataDefinitionItems();
00042
00043     QList<ComponentItem> *_myComponentItems;
00044     QList<GraphicalConnection *> *_myConnectionItems;
00045     QList<DrawingItem> *_myDrawingItems;
00046     QList<GroupItem> *_myGroupItems;
00047     QList<DataDefinitionItem> *_myDataDefinitionItems;
00048     QList<DataDefinitionRelationItem> *_myDataDefinitionRelationItems;
00049     QList<ModelDataDefinition *> *_myDataDefinitions;
00050     ModelGraphicsScene *_myGraphicsScene;
00051 };
00052
00053 #endif // DELETEUNDOCOMMAND_H
```

18.6 EFSMInitialStateUndoCommand.h

```
00001 #ifndef EFSMINITIALSTATEUNDOCOMMAND_H
00002 #define EFSMINITIALSTATEUNDOCOMMAND_H
00003
00004 #include <QUndoCommand>
```

```
00005 #include <vector>
00006
00007 class EFSMNetwork;
00008 class FSMState;
00009
00010 class EFSMInitialStateUndoCommand final : public QUndoCommand {
00011 public:
00012     EFSMInitialStateUndoCommand(EFSMNetwork* network, FSMState* state, QUndoCommand* parent =
      nullptr);
00013     void undo() override;
00014     void redo() override;
00015
00016 private:
00017     bool _stateBelongsToNetwork(FSMState* state) const;
00018     bool _select(FSMState* state);
00019
00020     EFSMNetwork* _network = nullptr;
00021     FSMState* _before = nullptr;
00022     FSMState* _after = nullptr;
00023     std::vector<std::pair<FSMState*, bool» _beforeFlags;
00024 };
00025
00026 #endif
```

18.7 EFSMStateMembershipUndoCommand.h

```
00001 #ifndef EFSMSTATEMEMBERSHIPUNDOCOMMAND_H
00002 #define EFSMSTATEMEMBERSHIPUNDOCOMMAND_H
00003
00004 #include <QUndoCommand>
00005 #include <memory>
00006 #include <utility>
00007 #include <vector>
00008
00009 class EFSMNetwork;
00010 class EFSMTransition;
00011 class FSMState;
00012 class Model;
00013
00014 class EFSMStateMembershipUndoCommand final : public QUndoCommand {
00015 public:
00016     EFSMStateMembershipUndoCommand(Model* model, EFSMNetwork* network,
00017                                    std::unique_ptr<FSMState> state, QUndoCommand* parent = nullptr);
00018     EFSMStateMembershipUndoCommand(Model* model, EFSMNetwork* network,
00019                                    FSMState* state, QUndoCommand* parent = nullptr);
00020     ~EFSMStateMembershipUndoCommand() override = default;
00021     void undo() override;
00022     void redo() override;
00023
00024 private:
00025     struct DetachedTransition {
00026         std::unique_ptr<EFSMTransition> transition;
00027         unsigned int ordinal = 0;
00028     };
00029
00030     bool _belongs(FSMState* state) const;
00031     bool _insert();
00032     bool _remove();
00033     bool _restoreRemovedState();
00034     void _captureFlags();
00035     void _restoreFlags();
00036
00037     Model* _model = nullptr;
00038     EFSMNetwork* _network = nullptr;
00039     FSMState* _state = nullptr;
00040     FSMState* _beforeInitial = nullptr;
00041     FSMState* _beforeCurrent = nullptr;
00042     std::vector<std::pair<FSMState*, bool» _initialFlags;
00043     std::unique_ptr<FSMState> _ownedState;
00044     std::vector<DetachedTransition> _detachedTransitions;
00045     bool _insertionCommand = false;
00046 };
00047
00048 #endif
```

18.8 EFSMStatePropertiesUndoCommand.h

```
00001 #ifndef EFSMSTATEPROPERTIESUNDOCOMMAND_H
00002 #define EFSMSTATEPROPERTIESUNDOCOMMAND_H
00003
00004 #include <QUndoCommand>
00005 #include <string>
```

```
00006
00007 class EFSMNetwork;
00008 class FSMState;
00009
00010 class EFSMStatePropertiesUndoCommand final : public QUndoCommand {
00011 public:
00012     struct Properties {
00013         std::string name;
00014         std::string entryActionExpression;
00015         std::string exitActionExpression;
00016         bool finalNode = false;
00017     };
00018
00019     EFSMStatePropertiesUndoCommand(EFSMNetwork* network, FSMState* state,
00020                                    Properties updated, QUndoCommand* parent = nullptr);
00021     void undo() override;
00022     void redo() override;
00023
00024 private:
00025     static Properties _read(const FSMState* state);
00026     bool _apply(const Properties& properties);
00027
00028     EFSMNetwork* _network = nullptr;
00029     FSMState* _state = nullptr;
00030     Properties _before;
00031     Properties _after;
00032 };
00033
00034 #endif
```

18.9 EFSMTransitionPropertiesUndoCommand.h

```
00001 #ifndef EFSMTRANSITIONPROPERTIESUNDOCOMMAND_H
00002 #define EFSMTRANSITIONPROPERTIESUNDOCOMMAND_H
00003
00004 #include <QUndoCommand>
00005 #include <string>
00006
00007 class EFSMNetwork;
00008 class EFSMTransition;
00009
00010 class EFSMTransitionPropertiesUndoCommand final : public QUndoCommand {
00011 public:
00012     struct Properties {
00013         std::string name;
00014         std::string guardExpression;
00015         std::string outputExpression;
00016         std::string inputEvent;
00017         unsigned int priority = 0;
00018         std::string triggerEvent;
00019         std::string probabilityExpression;
00020     };
00021
00022     EFSMTransitionPropertiesUndoCommand(EFSMNetwork* network, EFSMTransition* transition,
00023                                         Properties updated, QUndoCommand* parent = nullptr);
00024     void undo() override;
00025     void redo() override;
00026
00027 private:
00028     bool _apply(const Properties& properties);
00029     static Properties _read(const EFSMTransition* transition);
00030
00031     EFSMNetwork* _network = nullptr;
00032     EFSMTransition* _transition = nullptr;
00033     Properties _before;
00034     Properties _after;
00035 };
00036
00037 #endif
```

18.10 EFSMTransitionUndoCommand.h

```
00001 #ifndef EFSMTRANSITIONUNDOCOMMAND_H
00002 #define EFSMTRANSITIONUNDOCOMMAND_H
00003
00004 #include <QUndoCommand>
00005 #include <memory>
00006
00007 class EFSMNetwork;
00008 class EFSMTransition;
00009
00017 class EFSMTransitionUndoCommand final : public QUndoCommand {
```

```
00018 public:
00019     EFSMTransitionUndoCommand(EFSMNetwork* network, std::unique_ptr<EFSMTransition> transition,
00020                               QUndoCommand* parent = nullptr);
00021     EFSMTransitionUndoCommand(EFSMNetwork* network, EFSMTransition* transition,
00022                               QUndoCommand* parent = nullptr);
00023     ~EFSMTransitionUndoCommand() override = default;
00024 
00025     void undo() override;
00026     void redo() override;
00027 
00028 private:
00029     bool _attach();
00030     bool _detach();
00031 
00032     EFSMNetwork* _network = nullptr;
00033     EFSMTransition* _transition = nullptr;
00034     std::unique_ptr<EFSMTransition> _detachedTransition;
00035     unsigned int _ordinal = 0;
00036     bool _insertOnRedo = false;
00037 };
00038 
00039 #endif
```

18.11 GroupUndoCommand.h

```
00001 #ifndef GROUPUNDOCOMMAND_H
00002 #define GROUPUNDOCOMMAND_H
00003 
00004 #include <QUndoCommand>
00005 #include "graphicals/ModelGraphicsScene.h"
00006 #include "graphicals/GraphicalModelComponent.h"
00007 
00008 class GroupUndoCommand: public QUndoCommand {
00009 public:
00010     GroupUndoCommand(QList<GraphicalModelComponent *> graphicalComponents, ModelGraphicsScene *scene,
      QUndoCommand *parent = nullptr);
00011     ~GroupUndoCommand();
00012 
00013     void undo() override;
00014     void redo() override;
00015 
00016 private:
00017     QList<GraphicalModelComponent *> _myGraphicalComponents;
00018     QGraphicsItemGroup *_group;
00019     ModelGraphicsScene *_myGraphicsScene;
00020     QPointF _initialPosition;
00021 };
00022 
00023 #endif // GROUPUNDOCOMMAND_H
```

18.12 ModalModelBindingsUndoCommand.h

```
00001 #ifndef MODALMODELBINDINGSUNDOCOMMAND_H
00002 #define MODALMODELBINDINGSUNDOCOMMAND_H
00003 
00004 #include <QUndoCommand>
00005 #include <string>
00006 #include <vector>
00007 
00008 class ModalModelDefault;
00009 class DefaultNetwork;
00010 
00011 class ModalModelBindingsUndoCommand final : public QUndoCommand {
00012 public:
00013     struct Bindings {
00014         std::vector<std::string> inputs;
00015         std::vector<std::string> outputs;
00016     };
00017 
00018     ModalModelBindingsUndoCommand(ModalModelDefault* adapter, DefaultNetwork* network,
00019                                   Bindings updated, QUndoCommand* parent = nullptr);
00020     void undo() override;
00021     void redo() override;
00022 
00023 private:
00024     bool _apply(const Bindings& bindings);
00025     ModalModelDefault* _adapter = nullptr;
00026     DefaultNetwork* _network = nullptr;
00027     Bindings _before;
00028     Bindings _after;
00029 };
00030 
```

```
00031 #endif
```

18.13 MoveUndoCommand.h

```
00001 #ifndef MOVEUNDOCOMMAND_H
00002 #define MOVEUNDOCOMMAND_H
00003
00004 #include <QUndoCommand>
00005 #include "graphicals/ModelGraphicsScene.h"
00006
00007 class MoveUndoCommand : public QUndoCommand
00008 {
00009 public:
00010     MoveUndoCommand(QList<QGraphicsItem *> item, ModelGraphicsScene *scene, QList<QPointF> &oldPos,
      QList<QPointF> &newPos, QUndoCommand *parent = nullptr);
00011     ~MoveUndoCommand();
00012
00013     void undo() override;
00014     void redo() override;
00015
00016     //bool mergeWith(const QUndoCommand *command) override;
00017
00018 private:
00019     QList<QGraphicsItem*> _myGraphicalItem;
00020     ModelGraphicsScene *_myGraphicsScene;
00021     QList<QPointF> _myOldPos;
00022     QList<QPointF> _myNewPos;
00023 };
00024
00025 #endif // MOVEUNDOCOMMAND_H
```

18.14 PasteUndoCommand.h

```
00001 #ifndef PASTEUNDOCOMMAND_H
00002 #define PASTEUNDOCOMMAND_H
00003
00004 #include <QUndoCommand>
00005 #include "AddUndoCommand.h"
00006 #include "graphicals/ModelGraphicsScene.h"
00007 #include "graphicals/GraphicalConnection.h"
00008
00009
00010 class PasteUndoCommand : public QUndoCommand
00011 {
00012 public:
00013     explicit PasteUndoCommand(QList<GraphicalModelComponent *> *graphicalComponents,
      QList<GraphicalConnection *> *connections, QList<QGraphicsItemGroup *> *groups, QList<QGraphicsItem *>
      *drawing, ModelGraphicsScene *scene, QUndoCommand *parent = nullptr);
00014     ~PasteUndoCommand();
00015
00016     void undo() override;
00017     void redo() override;
00018
00019 private:
00020     QList<ComponentItem> *_myComponentItems;
00021     QList<GraphicalConnection *> *_myConnectionItems;
00022     QList<DrawingItem> *_myDrawingItems;
00023     QList<GroupItem> *_myGroupItems;
00024     ModelGraphicsScene *_myGraphicsScene;
00025 };
00026
00027 #endif // PASTEUNDOCOMMAND_H
```

18.15 UngroupUndoCommand.h

```
00001 #ifndef UNGROUPUNDOCOMMAND_H
00002 #define UNGROUPUNDOCOMMAND_H
00003
00004
00005 #include <QUndoCommand>
00006 #include "graphicals/ModelGraphicsScene.h"
00007
00008 class UngroupUndoCommand: public QUndoCommand {
00009 public:
00010     UngroupUndoCommand(QGraphicsItemGroup* group, ModelGraphicsScene *scene, QUndoCommand *parent =
      nullptr);
00011     ~UngroupUndoCommand();
00012
00013     void undo() override;
```

```
00014     void redo() override;
00015
00016 private:
00017     QGraphicsItemGroup *_group;
00018     QList<GraphicalModelComponent *> *_myComponentsGroup;
00019     ModelGraphicsScene *_myGraphicsScene;
00020     QPointF _initialPosition;
00021 };
00022
00023 #endif // UNGROUPUNDOCOMMAND_H
```

18.16 AnimationCounter.h

```
00001 #ifndef ANIMATIONCOUNTER_H
00002 #define ANIMATIONCOUNTER_H
00003
00004 #include <QGraphicsRectItem>
00005 #include <QPainter>
00006 #include <QGraphicsSceneMouseEvent>
00007
00008 #include "../../../../kernel/simulator/essentialPlugins/Counter.h"
00009 class AnimationCounter : public QGraphicsRectItem {
00010 public:
00011     // Construtor
00012     AnimationCounter();
00013
00014     // Sobrecarga do método paint
00015     void paint(QPainter *painter, const QStyleOptionGraphicsItem *option, QWidget *widget = nullptr)
      override;
00016
00017     // Getters
00018     double getValue();
00019     Counter *getCounter();
00020
00021     // Setters
00022     void setValue(double value);
00023     void setCounter(Counter *myCounter);
00024     void setIdCounter(int id);
00025     void setNameCounter(std::string name);
00026
00027     // Outros
00028     void startDrawing(QGraphicsSceneMouseEvent *event); // Inicia o desenho da tela
00029     void continueDrawing(QGraphicsSceneMouseEvent *event); // Continua o desenho na tela
00030     void stopDrawing(QGraphicsSceneMouseEvent *event); // Finaliza o desenho na tela
00031     void adjustSizeAndPosition(QGraphicsSceneMouseEvent *event); // Ajusta posição e dimensão do
      retângulo no final
00032     bool isDrawingInicialized(); // Diz se o contador começou a ser desenhada
00033     bool isDrawingFinalized(); // Diz se o contador terminou de ser desenhada
00034     void setWhenLoaded(QList<Counter *> *counters); // seta o contador quando o modelo é carregado
00035
00036 private:
00037     double _value = 0.0;
00038     QPointF _startPoint = QPointF(0, 0);
00039     bool _isResizing = false;
00040     bool _isDrawingInicialized = false;
00041     bool _isDrawingFinalized = false;
00042     Counter *_counter = nullptr;
00043     int _idCounter = -1;
00044     std::string _counterName = "None";
00045 };
00046
00047 #endif // ANIMATIONCOUNTER_H
```

18.17 AnimationPlaceholder.h

```
00001 #ifndef ANIMATIONPLACEHOLDER_H
00002 #define ANIMATIONPLACEHOLDER_H
00003
00004 #include <QGraphicsRectItem>
00005 #include <QGraphicsSceneMouseEvent>
00006 #include <QPainter>
00007 #include <QVector>
00008 #include <QString>
00009 #include <limits>
00010
00011 class Model;
00012
00013 class AnimationPlaceholder : public QGraphicsRectItem {
00014 public:
00015     explicit AnimationPlaceholder(const QString& animationType);
00016
```

```
00017     void paint(QPainter* painter, const QStyleOptionGraphicsItem* option, QWidget* widget = nullptr)
      override;
00018
00019     QString getAnimationType() const;
00020     QString getTargetName() const;
00021     void setTargetName(const QString& targetName);
00022
00023     void startDrawing(QGraphicsSceneMouseEvent* event);
00024     void continueDrawing(QGraphicsSceneMouseEvent* event);
00025     void stopDrawing(QGraphicsSceneMouseEvent* event);
00026     void adjustSizeAndPosition(QGraphicsSceneMouseEvent* event);
00027     bool isDrawingInicialized() const;
00028     bool isDrawingFinalized() const;
00029
00030 private:
00031     QString _animationType;
00032     QString _targetName;
00033     QPointF _startPoint = QPointF(0, 0);
00034     bool _isResizing = false;
00035     bool _isDrawingInicialized = false;
00036     bool _isDrawingFinalized = false;
00037 };
00038
00039 class AnimationAttribute : public AnimationPlaceholder {
00040 public:
00041     AnimationAttribute();
00042 };
00043
00044 class AnimationEntity : public AnimationPlaceholder {
00045 public:
00046     AnimationEntity();
00047 };
00048
00049 class AnimationEvent : public AnimationPlaceholder {
00050 public:
00051     AnimationEvent();
00052 };
00053
00054 class AnimationExpression : public AnimationPlaceholder {
00055 public:
00056     AnimationExpression();
00057 };
00058
00059 class AnimationPlot : public AnimationPlaceholder {
00060 public:
00061     enum class PlotType {
00062         Line = 0,
00063         Bar = 1
00064     };
00065
00066     struct DataSeries {
00067         QString label;
00068         QString expression;
00069         QColor color = QColor(31, 119, 180);
00070         qreal lineWidth = 2.0;
00071         QVector<double> values;
00072     };
00073
00074     AnimationPlot();
00075
00076     QString getTitle() const;
00077     void setTitle(const QString& title);
00078     QString getYAxisTitle() const;
00079     void setYAxisTitle(const QString& title);
00080     QString getXAxisTitle() const;
00081     void setXAxisTitle(const QString& title);
00082     PlotType getPlotType() const;
00083     void setPlotType(PlotType type);
00084     QString getDatasetsText() const;
00085     void setDatasetsText(const QString& datasetsText);
00086     bool getShowGridLines() const;
00087     void setShowGridLines(bool show);
00088     bool getShowTicks() const;
00089     void setShowTicks(bool show);
00090     double getXAxisMin() const;
00091     void setXAxisMin(double value);
00092     double getXAxisMax() const;
00093     void setXAxisMax(double value);
00094     double getYAxisMin() const;
00095     void setYAxisMin(double value);
00096     double getYAxisMax() const;
00097     void setYAxisMax(double value);
00098     void clearValues();
00099     void appendSample(Model* model);
00100
00101     void paint(QPainter* painter, const QStyleOptionGraphicsItem* option, QWidget* widget = nullptr)
      override;
```

```
00102
00103 private:
00104     static QVector<DataSeries> parseDatasetsText(const QString& datasetsText);
00105     static QString datasetsTextFromSeries(const QVector<DataSeries>& series);
00106
00107     QString _title = "Plot";
00108     QString _xAxisTitle = "Time";
00109     QString _yAxisTitle = "Value";
00110     PlotType _plotType = PlotType::Line;
00111     bool _showGridLines = true;
00112     bool _showTicks = true;
00113     double _xAxisMin = std::numeric_limits<double>::quiet_NaN();
00114     double _xAxisMax = std::numeric_limits<double>::quiet_NaN();
00115     double _yAxisMin = std::numeric_limits<double>::quiet_NaN();
00116     double _yAxisMax = std::numeric_limits<double>::quiet_NaN();
00117     QVector<DataSeries> _series;
00118 };
00119
00120 class AnimationQueueDisplay : public AnimationPlaceholder {
00121 public:
00122     AnimationQueueDisplay();
00123 };
00124
00125 class AnimationResource : public AnimationPlaceholder {
00126 public:
00127     AnimationResource();
00128 };
00129
00130 class AnimationStation : public AnimationPlaceholder {
00131 public:
00132     AnimationStation();
00133 };
00134
00135 class AnimationStatistics : public AnimationPlaceholder {
00136 public:
00137     AnimationStatistics();
00138 };
00139
00140 #endif // ANIMATIONPLACEHOLDER_H
```

18.18 AnimationQueue.h

```
00001 #ifndef ANIMATIONQUEUE_H
00002 #define ANIMATIONQUEUE_H
00003
00004 #include "graphicals/GraphicalModelComponent.h"
00005
00006 class ModelGraphicsScene;
00007
00008 class AnimationQueue
00009 {
00010 public:
00011     // Construtor
00012     AnimationQueue(ModelGraphicsScene* myScene, ModelComponent* component);
00013     // Destrutor
00014     ~AnimationQueue();
00015
00016     // Getters
00017     GraphicalModelComponent* getGraphicalComponent() const;
00018
00019     // quando uma entidade esta movendo em relacao a um componente que tem fila, adiciona a animação
00020     void addAnimationQueue(bool visivible);
00021     // quando uma entidade esta movendo de um componente que tem fila, remove a animação
00022     void removeAnimationQueue();
00023
00024     // faz a animação de acordo com o tamanho da fila, mas não ficou muito condizente com a
      visualização da animação
00025     // verifica se precisa adiconar animação e adiciona se for o caso (observando valores da fila)
00026     void verifyAddAnimationQueue(bool visivible);
00027     // verifica se precisa remover animação e remove se for o caso (observando valores da fila)
00028     void verifyRemoveAnimationQueue();
00029
00030     // retorna a posição em que a animação da fila será desenhada
00031     QPointF calculatePositionImageQueue(unsigned int indexQueue, unsigned int sizeQueue, unsigned int
      width, unsigned int height);
00032
00033 private:
00034     ModelGraphicsScene* _myScene;
00035     GraphicalModelComponent* _graphicalComponent;
00036 };
00037
00038 #endif // ANIMATIONQUEUE_H
```

18.19 AnimationTimer.h

```
00001 #ifndef ANIMATIONTIMER_H
00002 #define ANIMATIONTIMER_H
00003
00004 #include <QGraphicsRectItem>
00005 #include <QPainter>
00006 #include <QGraphicsSceneMouseEvent>
00007 #include "kernel/util/Util.h"
00008
00009 class ModelGraphicsScene;
00010
00011 class AnimationTimer : public QGraphicsRectItem {
00012 public:
00013     // Construtor
00014     AnimationTimer(ModelGraphicsScene* myScene);
00015
00016     // Sobrecarga do método paint
00017     void paint(QPainter* painter, const QStyleOptionGraphicsItem* option, QWidget* widget = nullptr)
      override;
00018
00019     // Getters
00020     double getTime();
00021     unsigned int getInitialHours();
00022     unsigned int getInitialMinutes();
00023     unsigned int getInitialSeconds();
00024     unsigned int getHours();
00025     unsigned int getMinutes();
00026     unsigned int getSeconds();
00027     Util::TimeFormat getTimeFormat();
00028
00029     // Setters
00030     void setTime(double value);
00031     static void setConversionFactorToSeconds(double factor);
00032     void setInitialHours(unsigned int initialHours);
00033     void setInitialMinutes(unsigned int initialMinutes);
00034     void setInitialSeconds(unsigned int initialSeconds);
00035     void setHours(unsigned int hours);
00036     void setMinutes(unsigned int minutes);
00037     void setSeconds(unsigned int seconds);
00038     void setTimeFormat(Util::TimeFormat timeFormat);
00039
00040     // Outros
00041     void startDrawing(QGraphicsSceneMouseEvent* event); // Inicia o desenho da tela
00042     void continueDrawing(QGraphicsSceneMouseEvent* event); // Continua o desenho na tela
00043     void stopDrawing(QGraphicsSceneMouseEvent* event); // Finaliza o desenho na tela
00044     void adjustSizeAndPosition(QGraphicsSceneMouseEvent* event); // Ajusta posição e dimensão do
      retângulo no final
00045     void convertSeconds(); // converte o tempo recebido em segundos para horas, dias e segundos
00046     bool isDrawingInicialized(); // Diz se o relógio começou a ser desenhado
00047     bool isDrawingFinalized(); // Diz se o relógio terminou de ser desenhado
00048
00049 private:
00050     ModelGraphicsScene* _myScene;
00051     double _time = 0.0;
00052     static double _conversionFactorToSeconds;
00053     QPointF _startPoint = QPointF(0, 0);
00054     unsigned int _initialHours = 0;
00055     unsigned int _initialMinutes = 0;
00056     unsigned int _initialSeconds = 0;
00057     unsigned int _hours = 0;
00058     unsigned int _minutes = 0;
00059     unsigned int _seconds = 0;
00060     unsigned int _microsseconds = 0;
00061     Util::TimeFormat _timeFormat = Util::TimeFormat(24);
00062     bool _showMicrosseconds = true;
00063     bool _isResizing = false;
00064     bool _isDrawingInicialized = false;
00065     bool _isDrawingFinalized = false;
00066 };
00067
00068 #endif // ANIMATIONTIMER_H
```

18.20 AnimationTransition.h

```
00001 #ifndef ANIMATIONTRANSITION_H
00002 #define ANIMATIONTRANSITION_H
00003
00004 #include <QList>
00005 #include <QPainterPath>
00006 #include <QPointF>
00007 #include <QTimer>
00008 #include <QVariantAnimation>
00009 #include <QElapsedTimer>
```

```
00010
00011 #include "graphicals/GraphicalImageAnimation.h"
00012 #include "graphicals/GraphicalModelComponent.h"
00013 #include "graphicals/GraphicalConnection.h"
00014
00015 // Em segundos
00016 #define TEMPO_EXECUCAO_ANIMACAO 2.5
00017
00018 class ModelGraphicsScene;
00019
00020 class AnimationTransition : public QVariantAnimation
00021 {
00022 public:
00023     // Construtor
00024     AnimationTransition(ModelGraphicsScene* myScene, ModelComponent* graphicalStartComponent,
      ModelComponent* graphicalEndComponent, bool viewSimulation);
00025     // Destrutor
00026     ~AnimationTransition();
00027
00028     // Getters
00029     GraphicalModelComponent* getGraphicalStartComponent() const;
00030     GraphicalModelComponent* getGraphicalEndComponent() const;
00031     GraphicalConnection* getGraphicalConnection() const;
00032     static double getTimeExecution();
00033     QList<QPointF> getPointsForAnimation() const;
00034     GraphicalImageAnimation* getImageAnimation() const;
00035     unsigned int getPortNumber() const;
00036     // Expose whether transition has all runtime prerequisites to safely animate.
00037     bool isReadyToRun() const;
00038
00039     // Setters
00040     void setImageAnimation(GraphicalImageAnimation* imageAnimation);
00041     static void setTimeExecution(double timeExecution);
00042     static void setPause(bool pause);
00043     static void setRunning(bool running);
00044
00045     // Inicia a animação
00046     void startAnimation();
00047     // Finaliza a animação
00048     void stopAnimation();
00049     // Reinicia a animação
00050     void restartAnimation();
00051     // Configuração informações para a animação
00052     void configureAnimation();
00053     // Atualiza a duração da animação se ela for alterada em tempo de execução
00054     void updateDurationIfNeeded();
00055     // Conecta o sinal valueChanged da animação ao slot onAnimationValueChanged
00056     void connectValueChangedSignal();
00057     // Conecta o sinal finished da animação ao slot onAnimationFinished
00058     void connectFinishedSignal();
00059
00060 public slots:
00061     // Slot para ser conectado ao sinal valueChanged
00062     void onAnimationValueChanged(const QVariant& value);
00063     // Slot que será conectado a um sinal
00064     void onAnimationFinished();
00065
00066 private:
00067     ModelGraphicsScene* _myScene;
00068     GraphicalModelComponent* _graphicalStartComponent;
00069     GraphicalModelComponent* _graphicalEndComponent;
00070     GraphicalConnection* _graphicalConnection;
00071     static double _timeExecution;
00072     static double _oldTimeExecution;
00073     static bool _pause;
00074     static bool _running;
00075     QList<QPointF> _pointsForAnimation;
00076     QPainterPath _pathForAnimation;
00077     GraphicalImageAnimation* _imageAnimation;
00078     unsigned int _portNumber;
00079     qreal _currentProgress;
00080     bool _viewSimulation;
00081     // Track terminal lifecycle paths to keep cleanup idempotent.
00082     bool _isStopping;
00083     // Track whether finished callback was already handled.
00084     bool _isFinishedHandled;
00085     bool _usesPathAnimation;
00086 };
00087
00088 #endif // ANIMATIONTRANSITION_H
```

18.21 AnimationVariable.h

```
00001 #ifndef ANIMATIONVARIABLE_H
00002 #define ANIMATIONVARIABLE_H
```

```
00003
00004 #include <QString>
00005 #include <QList>
00006 #include <QVector>
00007 #include <QGraphicsRectItem>
00008 #include <QGraphicsSceneMouseEvent>
00009 #include <QPainter>
00010 #include <string>
00011
00012 #include "../../../../kernel/simulator/essentialPlugins/Attribute.h"
00013 #include "../../../../kernel/simulator/essentialPlugins/Entity.h"
00014 #include "../../../../kernel/simulator/model/Model.h"
00015 #include "../../../../kernel/simulator/model/ModelDataDefinition.h"
00016 #include "../../../../kernel/simulator/model/ModelDataManager.h"
00017 #include "../../../../plugins/data/Logic/Variable.h"
00018
00019 class AnimationVariable : public QGraphicsRectItem {
00020 public:
00021     AnimationVariable();
00022
00023     void paint(QPainter *painter, const QStyleOptionGraphicsItem *option, QWidget *widget = nullptr)
      override;
00024
00025     double getValue();
00026     ModelDataDefinition *getDataDefinition() const;
00027     Variable *getVariable();
00028     Attribute *getAttribute();
00029     QString getTargetName() const;
00030     int getTargetId() const;
00031     unsigned int getDimensionCount() const;
00032     unsigned int getDimensionSize(unsigned int dimensionIndex) const;
00033     unsigned int getSliceIndex(unsigned int dimensionIndex) const;
00034
00035     void setValue(double value);
00036     void setDataDefinition(ModelDataDefinition *definition);
00037     void setVariable(Variable *variable);
00038     void setAttribute(Attribute *attribute);
00039     void setModel(Model *model);
00040     void setTargetName(std::string name);
00041     void setIdVariable(int id);
00042     void setSliceIndex(unsigned int dimensionIndex, unsigned int value);
00043     void setWhenLoaded(QList<ModelDataDefinition *> *definitions);
00044     void refreshValue(Entity *entity = nullptr);
00045
00046     void startDrawing(QGraphicsSceneMouseEvent *event);
00047     void continueDrawing(QGraphicsSceneMouseEvent *event);
00048     void stopDrawing(QGraphicsSceneMouseEvent *event);
00049     void adjustSizeAndPosition(QGraphicsSceneMouseEvent *event);
00050     bool isDrawingInicialized();
00051     bool isDrawingFinalized();
00052
00053 private:
00054     double _value = 0.0;
00055     Model *_model = nullptr;
00056     ModelDataDefinition *_dataDefinition = nullptr;
00057     QString _targetName = "None";
00058     int _targetId = -1;
00059     QVector<unsigned int> _sliceIndexes;
00060     QVector<double> _cachedValues;
00061     QVector<unsigned int> _cachedShape;
00062     QString _cachedTypeName;
00063     QString _cachedTitle;
00064     QPointF _startPoint = QPointF(0, 0);
00065     bool _isResizing = false;
00066     bool _isDrawingInicialized = false;
00067     bool _isDrawingFinalized = false;
00068 };
00069
00070 #endif // ANIMATIONVARIABLE_H
```

18.22 BaseMVController.h

```
00001 #pragma once
00002
00003 #include <QMainWindow>
00004 #include <QObject>
00005 #include <QWidget>
00006 #include "kernel/simulator/Simulator.h"
00007 #include "ui_mainwindow.h"
00008
00016 class BaseMVController
00017 {
00018 public:
00019     BaseMVController();
00020 protected: // View (UI) and Model (Simulator) main elements to join to the Controller
```

```
00022     Ui::MainWindow *ui;
00024     Simulator* simulator;
00025 };
```

18.23 CellularAutomataDemoBuilder.h

```
00001 #pragma once
00002
00003 /*
00004
00005 • Para que a GUI use um novo autômato celular com nova regra local,
00006   você mexe em 3 camadas:
00007
00008   1) Implementar a regra no plugin
00009
00010   - Crie a classe da regra em source/plugins/components/ModalModel/
00011     CellularAutomata/.
00012   - Normalmente isso significa algo como:
00013       - LocalRule_MyRule.h
00014       - opcionalmente LocalRule_MyRule.cpp
00015   - A regra precisa herdar de LocalRule e sobrescrever
00016     applyRule(Cell* cell).
00017   - Se a regra precisar de vizinhança específica, ela vai ler
00018     cell->getNeighbors().
00019
00020   2) Registrar essa regra no builder da GUI
00021
00022   - O ponto de montagem do demo fica em:
00023       - source/applications/gui/genesys/cellular_automata/
00024         CellularAutomataDemoBuilder.h
00025       - source/applications/gui/genesys/cellular_automata/
00026         CellularAutomataDemoBuilder.cpp
00027   - É aqui que a GUI escolhe qual regra concreta instanciar.
00028   - Você precisa:
00029       - incluir o header da nova regra;
00030       - adicionar um valor novo no enum CellularAutomataRulePreset;
00031       - estender cellularAutomataRulePresetText(...);
00032       - atualizar makeLocalRule(...) para retornar sua classe.
00033
00034   3) Expor a escolha na janela
00035
00036   - A UI de seleção fica em:
00037       - source/applications/gui/genesys/cellular_automata/
00038         CellularAutomataViewerWindow.cpp
00039       - source/applications/gui/genesys/cellular_automata/
00040         CellularAutomataViewerWindow.h
00041   - Você precisa adicionar o novo preset ao combo de regra em
00042     _syncControlsFromController().
00043   - Se quiser salvar/restaurar essa escolha no JSON, ajuste também:
00044       - source/applications/gui/genesys/cellular_automata/
00045         CellularAutomataViewerController.cpp
00046
00047   Se a nova regra exigir outro tipo de vizinhança ou estado
00048
00049   - O ajuste continua na mesma área do builder:
00050       - CellularAutomataDemoBuilder.h/.cpp
00051   - Para vizinhança, hoje a GUI já suporta:
00052       - Moore
00053       - VonNeumann
00054   - Para tipo de estado, hoje ela expõe:
00055       - Binary
00056       - Enumerated
00057       - Numeric
00058   - Se sua regra precisar de um estado diferente, você precisa
00059     ampliar o preset de estado e o mapeamento de cores/menu.
00060
00061   Fluxo mínimo para fazer funcionar
00062
00063   1. Criar LocalRule_MyRule.
00064   2. Incluir essa classe em CellularAutomataDemoBuilder.h.
00065   3. Adicionar o preset no enum CellularAutomataRulePreset.
00066   4. Implementar a criação da regra em makeLocalRule(...).
00067   5. Adicionar o item no combo da janela em
00068      CellularAutomataViewerWindow.cpp.
00069   6. Atualizar serialização em CellularAutomataViewerController.cpp
00070      se quiser persistência.
00071
00072   Arquivos que você provavelmente vai mexer
00073
00074   - source/plugins/components/ModalModel/CellularAutomata/
00075     LocalRule_MyRule.h
00076   - source/plugins/components/ModalModel/CellularAutomata/
00077     LocalRule_MyRule.cpp se necessário
00078   - source/applications/gui/genesys/cellular_automata/
00079     CellularAutomataDemoBuilder.h
```

```
00080   - source/applications/gui/genesys/cellular_automata/
00081     CellularAutomataDemoBuilder.cpp
00082   - source/applications/gui/genesys/cellular_automata/
00083     CellularAutomataViewerWindow.cpp
00084   - source/applications/gui/genesys/cellular_automata/
00085     CellularAutomataViewerController.cpp
00086
00087 */
00088
00089
00090
00091
00092 #include <memory>
00093 #include <vector>
00094
00095 #include <QSize>
00096
00097 #include "plugins/components/ModalModel/CellularAutomata/Boundary_Closed.h"
00098 #include "plugins/components/ModalModel/CellularAutomata/Boundary_Fixed.h"
00099 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00100 #include "plugins/components/ModalModel/CellularAutomata/CellularAutomata_Classic.h"
00101 #include "plugins/components/ModalModel/CellularAutomata/Lattice.h"
00102 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00103 #include "plugins/components/ModalModel/CellularAutomata/LocalRule_GameOfLife.h"
00104 #include "plugins/components/ModalModel/CellularAutomata/LocalRule_FlorestalFire.h"
00105 #include "plugins/components/ModalModel/CellularAutomata/LocalRule_Growty.h"
00106 #include "plugins/components/ModalModel/CellularAutomata/LocalRule_HppLatticeGas.h"
00107 #include "plugins/components/ModalModel/CellularAutomata/Neighborhood_Moore.h"
00108 #include "plugins/components/ModalModel/CellularAutomata/Neighborhood_VonNeumann.h"
00109 #include "plugins/components/ModalModel/CellularAutomata/StateSet_Enumerable.h"
00110
00111 enum class CellularAutomataRulePreset {
00112     GameOfLife = 0,
00113     Growty = 1,
00114     Identity = 2,
00115     ForestFire = 3,
00116     HppLatticeGas = 4,
00117     SandRock = 5
00118 };
00119
00120 enum class CellularAutomataNeighborhoodPreset {
00121     Moore = 0,
00122     VonNeumann = 1
00123 };
00124
00125 enum class CellularAutomataBoundaryPreset {
00126     Closed = 0,
00127     Fixed = 1
00128 };
00129
00130 enum class CellularAutomataStatePreset {
00131     Binary = 0,
00132     Enumerated = 1,
00133     Numeric = 2,
00134     HppLatticeGas = 3
00135 };
00136
00137 struct CellularAutomataDemoSettings {
00138     QSize latticeSize = QSize{100, 100};
00139     CellularAutomataRulePreset rulePreset = CellularAutomataRulePreset::SandRock;
00140     CellularAutomataNeighborhoodPreset neighborhoodPreset = CellularAutomataNeighborhoodPreset::Moore;
00141     CellularAutomataBoundaryPreset boundaryPreset = CellularAutomataBoundaryPreset::Fixed;
00142     CellularAutomataStatePreset statePreset = CellularAutomataStatePreset::Enumerated;
00143 };
00144
00145 QString cellularAutomataRulePresetText(CellularAutomataRulePreset preset);
00146 QString cellularAutomataNeighborhoodPresetText(CellularAutomataNeighborhoodPreset preset);
00147 QString cellularAutomataBoundaryPresetText(CellularAutomataBoundaryPreset preset);
00148 QString cellularAutomataStatePresetText(CellularAutomataStatePreset preset);
00149
00150 class LocalRule_CopyCurrentState final : public LocalRule {
00151 public:
00152     explicit LocalRule_CopyCurrentState(CellularAutomataBase* parentCellularAutomata)
00153         : LocalRule(parentCellularAutomata) {
00154     }
00155
00156     void applyRule(Cell* cell) override {
00157         if (cell != nullptr) {
00158             cell->setNextState(cell->getCurrentState());
00159         }
00160     }
00161 };
00162
00163 struct CellularAutomataDemoModel {
00164     explicit CellularAutomataDemoModel(CellularAutomataDemoSettings settings = {});
00165
00166     CellularAutomataDemoSettings settings;
```

```
00167     std::vector<State> ownedStates;
00168     std::vector<State*> statePointers;
00169     std::unique_ptr<CellularAutomata_Classic> automaton;
00170     std::unique_ptr<Lattice> lattice;
00171     std::unique_ptr<StateSet> stateSet;
00172     std::unique_ptr<BoundaryCondition> boundary;
00173     std::unique_ptr<Neighborhood> neighborhood;
00174     std::unique_ptr<LocalRule> localRule;
00175     std::unique_ptr<LocalRule_CopyCurrentState> identityLocalRule;
00176 
00177     int columns() const;
00178     int rows() const;
00179     bool isValidPosition(int x, int y) const;
00180     Cell* cellAt(int x, int y) const;
00181     long stateAt(int x, int y) const;
00182     QString stateTextAt(int x, int y) const;
00183     bool setStateAt(int x, int y, long value);
00184     void fill(long value);
00185     void seedDefaultPattern();
00186     std::vector<long> availablePaintStates() const;
00187 };
00188 
00189 class CellularAutomataDemoBuilder {
00190 public:
00191     static std::unique_ptr<CellularAutomataDemoModel> buildDemo(CellularAutomataDemoSettings settings
    = {});
00192 };
```

18.24 CellularAutomataLatticeWidget.h

```
00001 #pragma once
00002 
00003 #include <QPoint>
00004 #include <QRectF>
00005 #include <QWidget>
00006 
00007 class CellularAutomataViewerController;
00008 
00009 class CellularAutomataLatticeWidget : public QWidget {
00010 public:
00011     explicit CellularAutomataLatticeWidget(QWidget* parent = nullptr);
00012 
00013     void setController(CellularAutomataViewerController* controller);
00014 
00015 protected:
00016     void paintEvent(QPaintEvent* event) override;
00017     void mousePressEvent(QMouseEvent* event) override;
00018     void mouseMoveEvent(QMouseEvent* event) override;
00019     void mouseReleaseEvent(QMouseEvent* event) override;
00020     void contextMenuEvent(QContextMenuEvent* event) override;
00021     QSize minimumSizeHint() const override;
00022 
00023 private:
00024     bool _ensureSelectedPaintState();
00025     QPoint _cellAtPosition(const QPoint& widgetPosition) const;
00026     QRectF _gridRect() const;
00027     void _paintCellState(const QPoint& cellPosition);
00028     void _showStateContextMenu(const QPoint& globalPosition, const QPoint& cellPosition);
00029 
00030 private:
00031     CellularAutomataViewerController* _controller = nullptr;
00032     bool _dragging = false;
00033     QPoint _lastPaintedCell = QPoint{-1, -1};
00034 };
```

18.25 CellularAutomataStateColorMap.h

```
00001 #pragma once
00002 
00003 #include <QColor>
00004 #include <QString>
00005 #include <functional>
00006 
00007 class CellularAutomataStateColorMap {
00008 public:
00009     static QColor colorForState(long state) {
00010         if (state <= 0) {
00011             return QColor(248, 248, 248);
00012         }
00013         if (state == 1) {
00014             return QColor(28, 30, 36);
00015         }
```

```
00016
00017         const int hue = static_cast<int>((state * 47) % 360);
00018         return QColor::fromHsv(hue, 180, 215);
00019     }
00020
00021     static QColor colorForStateText(const QString& stateText, long fallbackState) {
00022         if (stateText.isEmpty() || stateText == QString::number(fallbackState)) {
00023             return colorForState(fallbackState);
00024         }
00025         const size_t hash = std::hash<std::string>{}(stateText.toStdString());
00026         return QColor::fromHsv(static_cast<int>(hash % 360), 170, 220);
00027     }
00028
00029     static QColor backgroundColor() {
00030         return QColor(238, 239, 242);
00031     }
00032
00033     static QColor gridLineColor() {
00034         return QColor(190, 194, 201);
00035     }
00036
00037     static QColor emptyTextColor() {
00038         return QColor(110, 115, 124);
00039     }
00040 };
```

18.26 CellularAutomataViewerController.h

```
00001 #pragma once
00002
00003 #include <memory>
00004 #include <optional>
00005 #include <vector>
00006
00007 #include <QSize>
00008 #include <QString>
00009 #include <QStringList>
00010 #include <QTimer>
00011 #include <QObject>
00012
00013 #include "CellularAutomataDemoBuilder.h"
00014
00015 class CellularAutomataViewerController : public QObject {
00016     Q_OBJECT
00017 public:
00018     explicit CellularAutomataViewerController(QObject* parent = nullptr);
00019
00020     CellularAutomataDemoModel* model() const;
00021     CellularAutomataDemoSettings settings() const;
00022     QSize latticeSize() const;
00023     bool hasDemo() const;
00024     unsigned int simulatedTime() const;
00025     bool isRunning() const;
00026     QString configurationSummary() const;
00027     std::vector<long> availablePaintStates() const;
00028
00029     long cellState(int x, int y) const;
00030     bool setCellState(int x, int y, long value);
00031     void fill(long value);
00032
00033     bool hasSelectedPaintState() const;
00034     std::optional<long> selectedPaintState() const;
00035     QString selectedPaintStateText() const;
00036     void setSelectedPaintState(long state);
00037     void clearSelectedPaintState();
00038
00039     int autoStepIntervalMs() const;
00040     void setAutoStepIntervalMs(int intervalMs);
00041
00042     void rebuildDemo(CellularAutomataDemoSettings settings = {});
00043     void setSettings(CellularAutomataDemoSettings settings);
00044     void setRulePreset(CellularAutomataRulePreset preset);
00045     void setNeighborhoodPreset(CellularAutomataNeighborhoodPreset preset);
00046     void setBoundaryPreset(CellularAutomataBoundaryPreset preset);
00047     void setStatePreset(CellularAutomataStatePreset preset);
00048     void resetToInitialState();
00049     void step();
00050     void start();
00051     void pause();
00052     void stop();
00053
00054     bool saveConfiguration(const QString& filePath, QString* errorMessage = nullptr) const;
00055     bool loadConfiguration(const QString& filePath, QString* errorMessage = nullptr);
00056
00057 signals:
```

```
00058     void modelChanged();
00059     void settingsChanged();
00060     void timeChanged(unsigned int time);
00061     void runningChanged(bool running);
00062     void paintStateChanged();
00063     void statusMessage(const QString& message);
00064
00065 private slots:
00066     void _onTimerTimeout();
00067
00068 private:
00069     void _resetRuntimeState();
00070
00071 private:
00072     std::unique_ptr<CellularAutomataDemoModel> _demo;
00073     CellularAutomataDemoSettings _settings;
00074     QTimer _timer;
00075     unsigned int _simulatedTime = 0;
00076     bool _running = false;
00077     int _autoStepIntervalMs = 0;
00078     std::optional<long> _selectedPaintState;
00079 };
```

18.27 CellularAutomataViewerWindow.h

```
00001 #pragma once
00002
00003 #include <QMainWindow>
00004 #include <QString>
00005
00006 class QAction;
00007 class QComboBox;
00008 class QLabel;
00009 class CellularAutomataLatticeWidget;
00010 class CellularAutomataViewerController;
00011
00012 class CellularAutomataViewerWindow : public QMainWindow {
00013 public:
00014     explicit CellularAutomataViewerWindow(QWidget* parent = nullptr);
00015
00016 private:
00017     void _updateTimeLabel(unsigned int time);
00018     void _updateRunningState(bool running);
00019     void _updatePaintStateLabel();
00020     void _updateConfigurationLabel();
00021     void _updateStatusMessage(const QString& message);
00022     void _refreshUi();
00023     void _syncControlsFromController();
00024     void _onRulePresetChanged(int index);
00025     void _onNeighborhoodPresetChanged(int index);
00026     void _onBoundaryPresetChanged(int index);
00027     void _onStatePresetChanged(int index);
00028     bool _promptSavePath(QString* path) const;
00029     bool _promptLoadPath(QString* path) const;
00030
00031 private:
00032     CellularAutomataViewerController* _controller = nullptr;
00033     CellularAutomataLatticeWidget* _latticeWidget = nullptr;
00034     QComboBox* _ruleCombo = nullptr;
00035     QComboBox* _neighborhoodCombo = nullptr;
00036     QComboBox* _boundaryCombo = nullptr;
00037     QComboBox* _stateCombo = nullptr;
00038     QLabel* _timeLabel = nullptr;
00039     QLabel* _runningLabel = nullptr;
00040     QLabel* _paintStateLabel = nullptr;
00041     QLabel* _configurationLabel = nullptr;
00042     QLabel* _statusLabel = nullptr;
00043     QAction* _startAction = nullptr;
00044     QAction* _stepAction = nullptr;
00045     QAction* _continueAction = nullptr;
00046     QAction* _pauseAction = nullptr;
00047     QAction* _stopAction = nullptr;
00048     QAction* _restartAction = nullptr;
00049     QAction* _saveAction = nullptr;
00050     QAction* _loadAction = nullptr;
00051 };
```

18.28 CodeEditor.h

```
00001 #ifndef CODEEDITOR_H
00002 #define CODEEDITOR_H
00003
```

```
00004 #include <QPlainTextEdit>
00005
00006 class CodeEditor : public QPlainTextEdit {
00007     Q_OBJECT
00008
00009 public:
00010     CodeEditor(QWidget *parent = nullptr);
00011
00012     void lineNumberAreaPaintEvent(QPaintEvent *event);
00013     int lineNumberAreaWidth();
00014
00015 protected:
00016     void resizeEvent(QResizeEvent *event) override;
00017
00018 private slots:
00019     void updateLineNumberAreaWidth(int newBlockCount);
00020     void highlightCurrentLine();
00021     void updateLineNumberArea(const QRect &rect, int dy);
00022
00023 private:
00024     QWidget *lineNumberArea;
00025 };
00026
00027 #endif // CODEEDITOR_H
```

18.29 LineNumberArea.h

```
00001 #ifndef LINENUMBERAREA_H
00002 #define LINENUMBERAREA_H
00003
00004 #include <QWidget>
00005 #include "CodeEditor.h"
00006
00007 class LineNumberArea : public QWidget {
00008 public:
00009
00010     LineNumberArea(CodeEditor *editor) : QWidget(editor), codeEditor(editor) {
00011
00012     }
00013
00014     QSize sizeHint() const override {
00015         return QSize(codeEditor->lineNumberAreaWidth(), 0);
00016     }
00017
00018 protected:
00019
00020     void paintEvent(QPaintEvent *event) override {
00021         codeEditor->lineNumberAreaPaintEvent(event);
00022     }
00023
00024 private:
00025     CodeEditor *codeEditor;
00026 };
00027 #endif // LINENUMBERAREA_H
```

18.30 DialogUtilityController.h

```
00001 #ifndef DIALOGUTILITYCONTROLLER_H
00002 #define DIALOGUTILITYCONTROLLER_H
00003
00004 #include <functional>
00005
00006 class MainWindow;
00007 class Simulator;
00008 class ModelGraphicsView;
00009 class ModelGraphicsScene;
00010 class QString;
00011
00012 namespace Ui {
00013 class MainWindow;
00014 }
00015
00016 // Document the dialog and utility controller extracted from remaining MainWindow actions.
00034 class DialogUtilityController {
00035 public:
00037     DialogUtilityController(MainWindow* ownerWidget,
00038                             Simulator* simulator,
00039                             Ui::MainWindow* ui,
00040                             ModelGraphicsView* graphicsView,
00041                             std::function<void()> showMessageNotImplemented,
00042                             std::function<void(bool)> actualizeDebugBreakpoints,
00043                             std::function<bool()> createModelImage,
```

```
00044                             std::function<void()> actualizeActions,
00045                             std::function<void()> actualizeTabPanes,
00046                             std::function<void()> reloadPluginCatalog,
00047                             std::function<ModelGraphicsScene*()> currentScene,
00048                             double& optimizerPrecision,
00049                             unsigned int& optimizerMaxSteps,
00050                             bool& parallelizationEnabled,
00051                             int& parallelizationThreads,
00052                             int& parallelizationBatchSize,
00053                             QString& lastDataAnalyzerPath);
00054
00056     void onActionAboutAboutTriggered();
00058     void onActionAboutLicenceTriggered();
00060     void onActionAboutGetInvolvedTriggered();
00062     void onActionAboutReportIssueTriggered();
00064     void onActionEditFindTriggered();
00066     void onActionEditReplaceTriggered();
00068     void onActionToolsParserGrammarCheckerTriggered();
00070     void onActionToolsOptimizatorTriggered();
00072     void onActionToolsAIAssistantTriggered();
00074     void onActionToolsDataAnalyzerTriggered();
00076     void onActionViewConfigureTriggered();
00078     void onActionSimulatorPreferencesTriggered();
00080     void onActionSimulatorsPluginManagerTriggered(bool showProblemPlugins = false);
00082     void onPushButtonBreakpointInsertClicked();
00084     void onPushButtonBreakpointRemoveClicked();
00086     void onPushButtonExportClicked();
00088     void onActionParallelizationTriggered();
00089
00090 private:
00091     MainWindow* _ownerWidget;
00092     Simulator* _simulator;
00093     Ui::MainWindow* _ui;
00094     ModelGraphicsView* _graphicsView;
00095     std::function<void()> _showMessageNotImplemented;
00096     std::function<void(bool)> _actualizeDebugBreakpoints;
00097     std::function<bool()> _createModelImage;
00098     std::function<void()> _actualizeActions;
00099     std::function<void()> _actualizeTabPanes;
00100     std::function<void()> _reloadPluginCatalog;
00101     std::function<ModelGraphicsScene*()> _currentScene;
00102     double& _optimizerPrecision;
00103     unsigned int& _optimizerMaxSteps;
00104     bool& _parallelizationEnabled;
00105     int& _parallelizationThreads;
00106     int& _parallelizationBatchSize;
00107     QString& _lastDataAnalyzerPath;
00108 };
00109
00110 #endif // DIALOGUTILITYCONTROLLER_H
```

18.31 EditCommandController.h

```
00001 #ifndef EDITCOMMANDCONTROLLER_H
00002 #define EDITCOMMANDCONTROLLER_H
00003
00004 #include <QList>
00005 #include <QGraphicsItem>
00006 #include <functional>
00007
00008 class Simulator;
00009 class ModelGraphicsView;
00010 class GraphicalModelComponent;
00011 class GraphicalConnection;
00012 class QGraphicsItemGroup;
00013
00014 // Document edit command orchestration extracted from MainWindow wrappers.
00032 class EditCommandController {
00033 public:
00035     EditCommandController(Simulator* simulator,
00036                           ModelGraphicsView* graphicsView,
00037                           std::function<void()> actualizeActions,
00038                           bool* cut,
00039                           QList<GraphicalModelComponent*>** gmcCopies,
00040                           QList<GraphicalConnection*>** portsCopies,
00041                           QList<QGraphicsItem*>** drawCopy,
00042                           QList<QGraphicsItemGroup*>** groupCopy);
00043
00045     void onActionEditUndoTriggered() const;
00047     void onActionEditRedoTriggered() const;
00049     void onActionEditCutTriggered() const;
00051     void onActionEditCopyTriggered() const;
00053     void onActionEditPasteTriggered() const;
00055     void onActionEditDeleteTriggered() const;
00057     void onActionEditGroupTriggered() const;
```

```
00059     void onActionEditUngroupTriggered() const;
00061     void onActionViewGroupTriggered() const;
00063     void onActionViewUngroupTriggered() const;
00064
00066     void saveItemForCopy(QList<GraphicalModelComponent*>* gmcList, QList<GraphicalConnection*>*
      connList) const;
00068     void helpCopy() const;
00069
00070 private:
00071     // Resolve current scene from the injected graphics view.
00072     class ModelGraphicsScene* scene() const;
00073
00074 private:
00075     Simulator* _simulator;
00076     ModelGraphicsView* _graphicsView;
00077     std::function<void()> _actualizeActions;
00078     bool* _cut;
00079     QList<GraphicalModelComponent*>** _gmcCopies;
00080     QList<GraphicalConnection*>** _portsCopies;
00081     QList<QGraphicsItem*>** _drawCopy;
00082     QList<QGraphicsItemGroup*>** _groupCopy;
00083 };
00084
00085 #endif // EDITCOMMANDCONTROLLER_H
```

18.32 GraphicalContextMenuController.h

```
00001 #ifndef GRAPHICALCONTEXTMENUCONTROLLER_H
00002 #define GRAPHICALCONTEXTMENUCONTROLLER_H
00003
00004 #include <functional>
00005
00006 class QContextMenuEvent;
00007 class QGraphicsItem;
00008 class QMenu;
00009 class QAction;
00010 class ModelGraphicsView;
00011 class ModelGraphicsScene;
00012 class ModelDataDefinition;
00013
00014 namespace Ui {
00015 class MainWindow;
00016 }
00017
00024 class GraphicalContextMenuController {
00025 public:
00029     GraphicalContextMenuController(ModelGraphicsView* graphicsView,
00030                                    Ui::MainWindow* ui,
00031                                    QAction* openSubmodelAction,
00032                                    std::function<ModelGraphicsScene*()> currentScene,
00033                                    std::function<void()> actualizeActions);
00034
00038     void handleGraphicsViewContextMenu(QContextMenuEvent* event);
00039
00040 private:
00041     enum class ContextKind {
00042         Background,
00043         SingleComponent,
00044         SingleDataDefinition,
00045         SingleConnection,
00046         SingleDiagramConnection,
00047         SingleGroup,
00048         MultiSelection,
00049         GenericItem
00050     };
00051
00055     QGraphicsItem* semanticItemFor(QGraphicsItem* item) const;
00056
00060     void synchronizeSelectionForContext(QGraphicsItem* item) const;
00061
00065     ContextKind classifyContext(QGraphicsItem* item) const;
00066
00070     ModelDataDefinition* modelDataDefinitionFor(QGraphicsItem* item) const;
00071
00075     void populateBackgroundMenu(QMenu* menu) const;
00076
00080     void populateItemMenu(QMenu* menu, ContextKind contextKind, QGraphicsItem* clickedItem) const;
00081
00085     void addEditActions(QMenu* menu) const;
00086
00090     void addShowMenu(QMenu* menu) const;
00091
00095     void addZoomMenu(QMenu* menu) const;
00096
00100     void addTraceLevelMenu(QMenu* menu, QGraphicsItem* clickedItem) const;
```

```
00101
00105     void addBioReactionLinkMenu(QMenu* menu) const;
00106
00110     void addDrawMenu(QMenu* menu) const;
00111
00115     void addAnimateMenu(QMenu* menu) const;
00116
00120     void addAlignMenu(QMenu* menu) const;
00121
00122     ModelGraphicsView* _graphicsView;
00123     Ui::MainWindow* _ui;
00124     QAction* _openSubmodelAction;
00125     std::function<ModelGraphicsScene*()> _currentScene;
00126     std::function<void()> _actualizeActions;
00127 };
00128
00129 #endif // GRAPHICALCONTEXTMENUCONTROLLER_H
```

18.33 ModelInspectorController.h

```
00001 #ifndef MODELINSPECTORCONTROLLER_H
00002 #define MODELINSPECTORCONTROLLER_H
00003
00004 #include <QTreeWidget>
00005
00006 class Simulator;
00007 class ModelGraphicsView;
00008 class ModelDataDefinition;
00009 class GraphicalModelDataDefinition;
00010
00011 // Document the controller boundary for model-inspection and tree/scene synchronization.
00030 class ModelInspectorController {
00031 public:
00032     struct DataDefinitionSelection {
00033         bool handled = false;
00034         ModelDataDefinition* dataDefinition = nullptr;
00035         GraphicalModelDataDefinition* graphicalDataDefinition = nullptr;
00036     };
00037
00039     ModelInspectorController(Simulator* simulator,
00040                              QTreeWidget* componentsTree,
00041                              QTreeWidget* dataDefinitionsTree,
00042                              ModelGraphicsView* graphicsView);
00043
00045     void actualizeModelComponents(bool force) const;
00047     void actualizeModelDataDefinitions(bool force) const;
00049     void beginDataDefinitionNameEdit(QTreeWidgetItem* item, int column) const;
00051     void applyDataDefinitionNameChange(QTreeWidgetItem* item, int column) const;
00053     void syncSelectedComponentTreeItemToScene() const;
00055     DataDefinitionSelection syncSelectedDataDefinitionTreeItemToScene() const;
00056
00057 private:
00058     // Keep simulator access scoped to model-inspection responsibilities.
00059     Simulator* _simulator;
00060     // Keep direct access to the Components tree used by this controller.
00061     QTreeWidget* _componentsTree;
00062     // Keep direct access to the Data Definitions tree used by this controller.
00063     QTreeWidget* _dataDefinitionsTree;
00064     // Keep direct access to the graphics view for selection synchronization.
00065     ModelGraphicsView* _graphicsView;
00066 };
00067
00068 #endif // MODELINSPECTORCONTROLLER_H
```

18.34 ModelLifecycleController.h

```
00001 #ifndef MODELLIFECYCLECONTROLLER_H
00002 #define MODELLIFECYCLECONTROLLER_H
00003
00004 #include <functional>
00005 #include <string>
00006
00007 class QFile;
00008 class QString;
00009 class QWidget;
00010 class Simulator;
00011 class Model;
00012
00013 namespace Ui {
00014 class MainWindow;
00015 }
00016
```

```
00017 // Document model lifecycle orchestration delegated by MainWindow compatibility wrappers.
00036 class ModelLifecycleController {
00037 public:
00038     // Group callback dependencies to keep constructor arguments focused and explicit.
00039     struct Callbacks {
00040         std::function<void(const std::string&)> insertCommandInConsole;
00041         std::function<void(Model*)> initUiForNewModel;
00042         std::function<void()> actualizeActions;
00043         std::function<void()> actualizeTabPanes;
00044         std::function<void(bool)> actualizeModelTextHasChanged;
00045         std::function<bool()> checkModel;
00046         std::function<bool()> setSimulationModelBasedOnText;
00047         std::function<void()> clearModelEditors;
00048         std::function<bool(QString)> saveGraphicalModel;
00049         std::function<bool(QFile*, QString)> saveTextModel;
00050         std::function<Model*(std::string)> loadGraphicalModel;
00051         std::function<void()> connectSceneSignals;
00052         std::function<void(const char*)> disconnectSceneSignals;
00053     };
00054
00056     ModelLifecycleController(QWidget* ownerWidget,
00057                              Simulator* simulator,
00058                              Ui::MainWindow* ui,
00059                              QString* modelFilename,
00060                              bool* textModelHasChanged,
00061                              bool* graphicalModelHasChanged,
00062                              bool* closingApproved,
00063                              bool* loaded,
00064                              bool& parallelizationEnabled,
00065                              int& parallelizationThreads,
00066                              int& parallelizationBatchSize,
00067                              Callbacks callbacks);
00068
00070     void onActionModelNewTriggered() const;
00072     void onActionModelOpenTriggered() const;
00074     bool openModelFile(const QString& fileName) const;
00076     void onActionModelSaveTriggered() const;
00078     void onActionModelCloseTriggered() const;
00080     void onActionModelInformationTriggered() const;
00082     void onActionModelCheckTriggered() const;
00084     void onActionSimulationConfigureTriggered() const;
00086     void onActionSimulatorExitTriggered() const;
00088     bool hasPendingModelChanges() const;
00090     bool confirmApplicationExit() const;
00091
00092 private:
00093     bool openModelFileInternal(const QString& fileName, bool showDialogs) const;
00094
00095     // Keep QObject ownership separate while allowing dialog parenting.
00096     QWidget* _ownerWidget;
00097     Simulator* _simulator;
00098     Ui::MainWindow* _ui;
00099     QString* _modelFilename;
00100     bool* _textModelHasChanged;
00101     bool* _graphicalModelHasChanged;
00102     bool* _closingApproved;
00103     bool* _loaded;
00104     bool& _parallelizationEnabled;
00105     int& _parallelizationThreads;
00106     int& _parallelizationBatchSize;
00107     Callbacks _callbacks;
00108 };
00109
00110 #endif // MODELLIFECYCLECONTROLLER_H
```

18.35 PluginCatalogController.h

```
00001 #ifndef PLUGINCATALOGCONTROLLER_H
00002 #define PLUGINCATALOGCONTROLLER_H
00003
00004 #include <map>
00005 #include <string>
00006 #include <vector>
00007
00008 #include <QColor>
00009
00010 class Plugin;
00011 class Simulator;
00012 class QPlainTextEdit;
00013 class QPoint;
00014 class QString;
00015 class QTreeWidget;
00016 class QTreeWidgetItem;
00017
00018 // Document the plugin-catalog controller extracted from MainWindow UI code.
```

```
00036 class PluginCatalogController {
00037 public:
00039     PluginCatalogController(Simulator* simulator,
00040                             QTreeWidget* pluginsTree,
00041                             QPlainTextEdit* modelTextEditor,
00042                             std::map<std::string, QColor>* pluginCategoryColor);
00043
00045     void insertPluginUI(Plugin* plugin) const;
00047     void reloadFromPluginManager() const;
00049     void applyCategoryExpansionPolicy() const;
00051     void showContextMenu(const QPoint& pos) const;
00053     void insertFakePlugins() const;
00055     void handlePluginItemDoubleClicked(QTreeWidgetItem* item, int column) const;
00057     void handlePluginItemClicked(QTreeWidgetItem* item, int column) const;
00059     void moveTopLevelCategory(QTreeWidgetItem* item, int delta) const;
00060
00061 private:
00062     static QString topLevelCategory(const QString& categoryPath);
00063     QTreeWidgetItem* ensureCategoryPath(const QString& categoryPath) const;
00064     QTreeWidgetItem* ensureCategoryChild(QTreeWidgetItem* parent, const QString& segment) const;
00065     QTreeWidgetItem* findDirectChild(QTreeWidgetItem* parent, const QString& text) const;
00066     QColor backgroundColorForCategory(const QString& categoryPath) const;
00067     void rememberTopLevelCategory(const QString& topLevelCategory) const;
00068     void applyTopLevelCategoryOrder() const;
00069     void applyExpansionPolicyRecursive(QTreeWidgetItem* item) const;
00070     void registerCategoryColorPrefixes(const QString& categoryPath, const QColor& categoryColor)
      const;
00071
00072 private:
00073     Simulator* _simulator;
00074     QTreeWidget* _pluginsTree;
00075     QPlainTextEdit* _modelTextEditor;
00076     std::map<std::string, QColor>* _pluginCategoryColor;
00077     mutable std::vector<std::string> _topLevelCategoryOrder;
00078 };
00079
00080 #endif /* PLUGINCATALOGCONTROLLER_H */
```

18.36 PropertyEditorController.h

```
00001 #ifndef PROPERTYEDITORCONTROLLER_H
00002 #define PROPERTYEDITORCONTROLLER_H
00003
00004 #include <functional>
00005 #include <map>
00006
00007 class ObjectPropertyBrowser;
00008 class ModelGraphicsView;
00009 class PropertyEditorGenesys;
00010 class SimulationControl;
00011 class DataComponentProperty;
00012 class DataComponentEditor;
00013 class ComboBoxEnum;
00014 class ModelDataDefinition;
00015 class QObject;
00016
00017 // Document the property-editor controller and post-commit synchronization pipeline.
00035 class PropertyEditorController {
00036 public:
00038     PropertyEditorController(ObjectPropertyBrowser* propertyBrowser,
00039                              ModelGraphicsView* graphicsView,
00040                              PropertyEditorGenesys* propertyGenesys,
00041                              std::map<SimulationControl*, DataComponentProperty*>* propertyList,
00042                              std::map<SimulationControl*, DataComponentEditor*>* propertyEditorUI,
00043                              std::map<SimulationControl*, ComboBoxEnum*>* propertyBox,
00044                              std::function<void()> actualizeModelSimLanguage,
00045                              std::function<void(bool)> actualizeModelComponents,
00046                              std::function<void(bool)> actualizeModelDataDefinitions,
00047                              std::function<void()> actualizeModelCppCode,
00048                              std::function<bool()> createModelImage,
00049                              std::function<void()> actualizeTabPanes,
00050                              std::function<void()> actualizeActions);
00051
00053     void sceneSelectionChanged() const;
00055     void onPropertyEditorModelChanged() const;
00057     void clearPropertyEditorSelection() const;
00059     void bindDataDefinitionFromInspector(ModelDataDefinition* dataDefinition, QObject* graphicalObject
      = nullptr) const;
00061     bool isPostCommitPipelineActive() const;
00062
00063 private:
00064     // Keep direct access to the property browser used by this controller.
00065     ObjectPropertyBrowser* _propertyBrowser;
00066     // Keep direct access to the graphical view used for selection inspection.
00067     ModelGraphicsView* _graphicsView;
```

```
00068     // Keep direct access to property-editor dependencies required for activation.
00069     PropertyEditorGenesys* _propertyGenesys;
00070     std::map<SimulationControl*, DataComponentProperty*>* _propertyList;
00071     std::map<SimulationControl*, DataComponentEditor*>* _propertyEditorUI;
00072     std::map<SimulationControl*, ComboBoxEnum*>* _propertyBox;
00073
00074     // Keep narrow callbacks for MainWindow-owned update methods.
00075     std::function<void()> _actualizeModelSimLanguage;
00076     std::function<void(bool)> _actualizeModelComponents;
00077     std::function<void(bool)> _actualizeModelDataDefinitions;
00078     std::function<void()> _actualizeModelCppCode;
00079     std::function<bool()> _createModelImage;
00080     std::function<void()> _actualizeTabPanes;
00081     std::function<void()> _actualizeActions;
00082
00083     // Coalesce and serialize heavy refresh requests after property commits.
00084     mutable bool _isGlobalRefreshQueued = false;
00085     mutable bool _isGlobalRefreshRunning = false;
00086     mutable bool _pendingGlobalRefresh = false;
00087     // Coalesce deferred selection synchronization while commit/refresh is active.
00088     mutable bool _isDeferredSelectionSyncScheduled = false;
00089
00090     void _runGlobalRefresh() const;
00091 };
00092
00093 #endif // PROPERTYEDITORCONTROLLER_H
```

18.37 SceneToolController.h

```
00001 #ifndef SCENETOOLCONTROLLER_H
00002 #define SCENETOOLCONTROLLER_H
00003
00004 #include <functional>
00005 #include <QRectF>
00006
00007 class ModelGraphicsView;
00008 class ModelGraphicsScene;
00009 class QAction;
00010
00011 namespace Ui {
00012 class MainWindow;
00013 }
00014
00015 // Document the scene tooling controller that receives delegated UI commands.
00032 class SceneToolController {
00033 public:
00037     SceneToolController(ModelGraphicsView* graphicsView,
00038                         Ui::MainWindow* ui,
00039                         std::function<ModelGraphicsScene*()> currentScene,
00040                         std::function<bool()> createModelImage,
00041                         std::function<void()> unselectDrawIcons,
00042                         std::function<bool()> checkSelectedDrawIcons,
00043                         std::function<void(double)> gentleZoom,
00044                         std::function<void()> actualizeActions,
00045                         std::function<void()> actualizeTabPanes,
00046                         int& zoomValue,
00047                         bool& firstClickShowConnection);
00048
00050     void onActionShowGridTriggered();
00052     void onActionShowRuleTriggered();
00054     void onActionShowGuidesTriggered();
00056     void onActionShowSnapTriggered();
00058     void onActionZoomInTriggered();
00060     void onActionZoomOutTriggered();
00062     void onActionZoomAllTriggered();
00064     void onActionZoomActualSizeTriggered();
00066     void syncZoomWidgetsForCurrentScale();
00068     void onHorizontalSliderZoomGraphicalValueChanged(int value);
00070     void onActionDrawLineTriggered();
00072     void onActionDrawRectangleTriggered();
00074     void onActionDrawEllipseTriggered();
00076     void onActionDrawTextTriggered();
00078     void onActionDrawPoligonTriggered();
00080     void onActionAnimateSimulatedTimeTriggered();
00082     void onActionAnimateVariableTriggered();
00084     void onActionAnimateCounterTriggered();
00086     void onActionAnimateAttributeTriggered();
00088     void onActionAnimateEntityTriggered();
00090     void onActionAnimateEventTriggered();
00092     void onActionAnimateExpressionTriggered();
00094     void onActionAnimatePlotTriggered();
00096     void onActionAnimateQueueTriggered();
00098     void onActionAnimateResourceTriggered();
00100     void onActionAnimateStationTriggered();
00102     void onActionAnimateStatisticsTriggered();
```

```
00104     void onActionConnectTriggered();
00106     void onActionArranjeLeftTriggered();
00108     void onActionArranjeRightTriggered();
00110     void onActionArranjeTopTriggered();
00112     void onActionArranjeBototmTriggered();
00114     void onActionArranjeCenterTriggered();
00116     void onActionArranjeMiddleTriggered();
00118     void onActionActivateGraphicalSimulationTriggered();
00120     void onHorizontalSliderAnimationSpeedValueChanged(int value);
00122     void onActionSelectAllTriggered();
00124     void onActionShowInternalElementsTriggered();
00126     void onActionShowEditableElementsTriggered();
00128     void onActionShowAttachedElementsTriggered();
00130     void onCheckBoxShowElementsStateChanged(int arg1);
00132     void onCheckBoxShowInternalsStateChanged(int arg1);
00134     void onCheckBoxShowEditableElementsStateChanged(int arg1);
00136     void onActionShowRecursiveElementsTriggered();
00138     void onCheckBoxShowRecursiveStateChanged(int arg1);
00140     void onCheckBoxShowLevelsStateChanged(int arg1);
00142     void onActionGModelShowConnectTriggered();
00143
00144 private:
00145     QRectF _zoomableModelBounds() const;
00146     double _currentViewScale() const;
00147     double _minimumZoomScale() const;
00148     double _maximumZoomScale() const;
00149     double _zoomScaleFromSliderValue(int value) const;
00150     int _sliderValueForZoomScale(double scale) const;
00151     void _syncZoomWidgetsForCurrentScale();
00152
00153     void activateAnimationDrawingTool(QAction* action, void (ModelGraphicsScene::*drawingFunction)());
00154
00155     ModelGraphicsView* _graphicsView;
00156     Ui::MainWindow* _ui;
00157     std::function<ModelGraphicsScene*()> _currentScene;
00158     std::function<bool()> _createModelImage;
00159     std::function<void()> _unselectDrawIcons;
00160     std::function<bool()> _checkSelectedDrawIcons;
00161     std::function<void(double)> _gentleZoom;
00162     std::function<void()> _actualizeActions;
00163     std::function<void()> _actualizeTabPanes;
00164     int& _zoomValue;
00165     bool& _firstClickShowConnection;
00166 };
00167
00168 #endif // SCENETOOLCONTROLLER_H
```

18.38 SimulationCommandController.h

```
00001 #ifndef SIMULATIONCOMMANDCONTROLLER_H
00002 #define SIMULATIONCOMMANDCONTROLLER_H
00003
00004 #include <functional>
00005 #include <string>
00006
00007 class SimulationController;
00008
00009 // Document simulation command orchestration delegated by MainWindow slots.
00027 class SimulationCommandController {
00028 public:
00032     SimulationCommandController(
00033         SimulationController* simulationController,
00034         const std::function<void(const std::string&)>& insertCommandInConsole,
00035         const std::function<void()>& actualizeActions,
00036         const std::function<bool()>& checkModel,
00037         const std::function<bool()>& setSimulationModelBasedOnText,
00038         const std::function<void()>& prepareNewSimulationOutputs);
00039
00041     void onActionSimulationStartTriggered(bool modelChecked) const;
00043     void onActionSimulationStepTriggered(bool modelChecked) const;
00045     void onActionSimulationPauseTriggered() const;
00047     void onActionSimulationResumeTriggered(bool modelChecked) const;
00049     void onActionSimulationStopTriggered() const;
00050
00051 private:
00052     SimulationController* _simulationController;
00053     std::function<void(const std::string&)> _insertCommandInConsole;
00054     std::function<void()> _actualizeActions;
00055     std::function<bool()> _checkModel;
00056     std::function<bool()> _setSimulationModelBasedOnText;
00057     std::function<void()> _prepareNewSimulationOutputs;
00058 };
00059
00060 #endif // SIMULATIONCOMMANDCONTROLLER_H
```

18.39 SimulationController.h

```
00001 #ifndef SIMULATIONCONTROLLER_H
00002 #define SIMULATIONCONTROLLER_H
00003
00004 #include <functional>
00005
00006 class QWidget;
00007 class Simulator;
00008 class ModelSimulation;
00009
00010 // Document the simulation precondition controller role in the refactored GUI architecture.
00032 class SimulationController {
00033 public:
00039     SimulationController(QWidget* ownerWidget, Simulator* simulator);
00040
00045     bool hasCurrentModelSimulation() const;
00046
00051     ModelSimulation* currentSimulation() const;
00052
00069     bool ensureReady(bool checkModel,
00070                      bool modelChecked,
00071                      const std::function<bool()>& runModelCheck,
00072                      const std::function<bool()>& syncModelFromText) const;
00073
00074 private:
00075     QWidget* _ownerWidget;
00076     Simulator* _simulator;
00077 };
00078
00079 #endif // SIMULATIONCONTROLLER_H
```

18.40 SimulationEventController.h

```
00001 #ifndef SIMULATIONEVENTCONTROLLER_H
00002 #define SIMULATIONEVENTCONTROLLER_H
00003
00004 #include <functional>
00005
00006 #include "../../../../kernel/simulator/model/ModelSimulation.h"
00007 #include "kernel/simulator/OnEventManager.h"
00008 #include "kernel/simulator/Simulator.h"
00009
00010 class MainWindow;
00011 class ModelGraphicsScene;
00012 class ModelGraphicsView;
00013 class QLabel;
00014 class QProgressBar;
00015 class QTableWidget;
00016 class QTextEdit;
00017 class QTabWidget;
00018 class QAction;
00019
00020 // Document the controller that bridges kernel simulation events to GUI updates.
00040 class SimulationEventController {
00041 public:
00042     // Group callback dependencies to keep constructor explicit and narrow.
00043     struct Callbacks {
00044         std::function<void()> actualizeActions;
00045         std::function<void(SimulationEvent*)> actualizeSimulationEvents;
00046         std::function<void(bool)> actualizeDebugEntities;
00047         std::function<void(bool)> actualizeDebugVariables;
00048         std::function<void(SimulationEvent*)> actualizeGraphicalModel;
00049     };
00050
00052     SimulationEventController(Simulator* simulator,
00053                               ModelGraphicsScene* scene,
00054                               ModelGraphicsView* graphicsView,
00055                               QLabel* replicationLabel,
00056                               QProgressBar* simulationProgressBar,
00057                               QTableWidget* simulationEventsTable,
00058                               QTableWidget* entitiesTable,
00059                               QTableWidget* variablesTable,
00060                               QTextEdit* simulationText,
00061                               QTextEdit* reportsText,
00062                               QTabWidget* centralTabWidget,
00063                               QAction* activateGraphicalSimulation,
00064                               QAction* animationEnabledAction,
00065                               bool* modelChecked,
00066                               int tabCentralReportsIndex,
00067                               Callbacks callbacks);
00068
00070     void onModelCheckSuccessHandler(ModelEvent* re) const;
00072     void onReplicationStartHandler(SimulationEvent* re) const;
```

```
00074     void onSimulationStartHandler(SimulationEvent* re) const;
00076     void onSimulationPausedHandler(SimulationEvent* re) const;
00078     void onSimulationResumeHandler(SimulationEvent* re) const;
00080     void onSimulationEndHandler(SimulationEvent* re) const;
00082     void onReplicationEndHandler(SimulationEvent* re) const;
00084     void onProcessEventHandler(SimulationEvent* re) const;
00086     void onEntityCreateHandler(SimulationEvent* re) const;
00088     void onEntityRemoveHandler(SimulationEvent* re) const;
00090     void onMoveEntityEvent(SimulationEvent* re) const;
00092     void onAfterProcessEvent(SimulationEvent* re) const;
00094     void setOnEventHandlers(MainWindow* owner) const;
00095
00096 private:
00097     // Keep simulator access scoped to simulation-event responsibilities.
00098     Simulator* _simulator;
00099     // Keep scene dependency for animation-related handlers.
00100     ModelGraphicsScene* _scene;
00101     // Keep graphics-view dependency for legacy compatibility and potential scene access.
00102     ModelGraphicsView* _graphicsView;
00103     // Keep replication label dependency for replication-start updates.
00104     QLabel* _replicationLabel;
00105     // Keep simulation progress bar dependency for process-event updates.
00106     QProgressBar* _simulationProgressBar;
00107     // Keep simulation-events table dependency for replication-start updates.
00108     QTableWidget* _simulationEventsTable;
00109     // Keep entities table dependency for simulation-start resets.
00110     QTableWidget* _entitiesTable;
00111     // Keep variables table dependency for simulation-start resets.
00112     QTableWidget* _variablesTable;
00113     // Keep simulation text dependency for simulation-start resets.
00114     QTextEdit* _simulationText;
00115     // Keep reports text dependency for simulation-start resets.
00116     QTextEdit* _reportsText;
00117     // Keep central-tab dependency for simulation-end tab switch.
00118     QTabWidget* _centralTabWidget;
00119     // Keep action dependency used by graphical simulation toggle.
00120     QAction* _activateGraphicalSimulation;
00121     // Keep action dependency used by the global Simulation/Animation runtime gate.
00122     QAction* _animationEnabledAction;
00123     // Keep mutable model-check flag dependency used at simulation end.
00124     bool* _modelChecked;
00125     // Keep tab index dependency used for report-tab activation.
00126     int _tabCentralReportsIndex;
00127     // Keep callback dependencies for delegated MainWindow operations.
00128     Callbacks _callbacks;
00129     // Throttle GUI event processing when animation/output updates are suppressed.
00130     mutable unsigned int _suppressedUiProgressEvents = 0;
00131
00132     bool animationsEnabled() const;
00133     void processSuppressedUiProgressEvents() const;
00134 };
00135
00136 #endif // SIMULATIONEVENTCONTROLLER_H
```

18.41 TraceConsoleController.h

```
00001 #ifndef TRACECONSOLECONTROLLER_H
00002 #define TRACECONSOLECONTROLLER_H
00003
00004 #include "kernel/simulator/TraceManager.h"
00005
00006 #include <functional>
00007
00008 class QTextEdit;
00009
00010 // Document the trace rendering bridge used by MainWindow simulator callbacks.
00028 class TraceConsoleController {
00029 public:
00031     TraceConsoleController(QTextEdit* console,
00032                            QTextEdit* simulationText,
00033                            QTextEdit* reportsText,
00034                            std::function<bool()> traceWidgetsEnabled = {});
00035
00037     void simulatorTraceHandler(TraceEvent e) const;
00039     void simulatorTraceErrorHandler(TraceErrorEvent e) const;
00041     void simulatorTraceSimulationHandler(TraceSimulationEvent e) const;
00043     void simulatorTraceReportsHandler(TraceEvent e) const;
00044
00045 private:
00046     // Keep the console trace sink used by normal/error traces.
00047     QTextEdit* _console;
00048     // Keep the simulation trace sink used by simulation traces.
00049     QTextEdit* _simulationText;
00050     // Keep the reports trace sink used by reports traces.
00051     QTextEdit* _reportsText;
```

```
00052     // Runtime gate used to avoid expensive trace-widget churn during non-animated simulations.
00053     std::function<bool()> _traceWidgetsEnabled;
00054
00055     bool shouldRenderTraceWidgets() const;
00056 };
00057
00058 #endif // TRACECONSOLECONTROLLER_H
```

18.42 dialogBreakpoint.h

```
00001 /*
00002  * The MIT License
00003  *
00004  * Copyright 2022 rlcancian.
00005  *
00006  * Permission is hereby granted, free of charge, to any person obtaining a copy
00007  * of this software and associated documentation files (the "Software"), to deal
00008  * in the Software without restriction, including without limitation the rights
00009  * to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
00010  * copies of the Software, and to permit persons to whom the Software is
00011  * furnished to do so, subject to the following conditions:
00012  *
00013  * The above copyright notice and this permission notice shall be included in
00014  * all copies or substantial portions of the Software.
00015  *
00016  * THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR
00017  * IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
00018  * FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
00019  * AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
00020  * LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
00021  * OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN
00022  * THE SOFTWARE.
00023  */
00024
00025 /*
00026  * File:   dialogBreakpoint.h
00027  * Author: rlcancian
00028  *
00029  * Created on 6 de fevereiro de 2022, 14:24
00030  */
00031
00032 #ifndef _DIALOGBREAKPOINT_H
00033 #define _DIALOGBREAKPOINT_H
00034
00035 #include "ui_dialogBreakpoint.h"
00036 #include "kernel/simulator/Simulator.h"
00037
00038 class dialogBreakpoint : public QDialog {
00039
00040     Q_OBJECT
00041 public:
00042     class MVCResult {
00043     public:
00044
00045         MVCResult(std::string type, std::string on) {
00046             this->type = type;
00047             this->on = on;
00048         }
00049         std::string type;
00050         std::string on;
00051     };
00052 public:
00053     dialogBreakpoint();
00054     virtual ~dialogBreakpoint();
00055 public:
00056     void setMVCModel(Simulator* simulator);
00057     dialogBreakpoint::MVCResult* getMVCResult();
00058 private slots:
00059     void on_comboBox_Type_activated(const QString &arg1);
00060     void on_comboBox_On_activated(const QString &arg1);
00061     void on_buttonBox_accepted();
00062 private:
00063     Ui::dialogBreakpoint widget;
00064     Simulator* simulator;
00065 };
00066
00067 #endif /* _DIALOGBREAKPOINT_H */
```

18.43 DialogFind.h

```
00001 #ifndef DIALOGFIND_H
00002 #define DIALOGFIND_H
```

```
00003
00004 #include <QDialog>
00005 #include <QLabel>
00006 #include <QHBoxLayout>
00007 #include <QLineEdit>
00008 #include <QPushButton>
00009 #include <QMessageBox>
00010 #include "../graphicals/GraphicalModelComponent.h"
00011 #include "graphicals/ModelGraphicsScene.h"
00012 #include <QGraphicsItem>
00013 #include <QList>
00014
00015
00016 class QLineEdit;
00017 class QPushButton;
00018
00019 class FindLineEdit : public QLineEdit {
00020     Q_OBJECT
00021
00022 public:
00023     FindLineEdit(QWidget *parent = nullptr) : QLineEdit(parent) {
00024         connect(this, &QLineEdit::textChanged, this, &FindLineEdit::onTextChanged);
00025     }
00026
00027 private slots:
00028     void onTextChanged(const QString &text) {
00029 //        qDebug() << "Texto atual: " << text;
00030
00031         if (text.length() < previousText.length()) {
00032             this->setStyleSheet("color: black");
00033         }
00034
00035         previousText = text;
00036     }
00037
00038 private:
00039     QString previousText;
00040 };
00041
00042
00043 class DialogFind : public QDialog
00044 {
00045     Q_OBJECT
00046
00047 public:
00048     DialogFind(QWidget *parent = nullptr, ModelGraphicsScene * scene = nullptr);
00049     QString getFindText();
00050
00051 public slots:
00052     void findClicked();
00053     void nextClicked();
00054     void previousClicked();
00055
00056 private:
00057     QPushButton *findButton;
00058     QPushButton *nextButton;
00059     QPushButton *previousButton;
00060     FindLineEdit *lineEdit;
00061     bool clicked;
00062     int counter;
00063     QList<QGraphicsItem*>* finded = new QList<QGraphicsItem*>();
00064     ModelGraphicsScene * scene;
00065
00066 };
00067
00068 #endif // DIALOGFIND_H
00069
00070
```

18.44 Dialogmodelinformation.h

```
00001 #ifndef DIALOGMODELINFORMATION_H
00002 #define DIALOGMODELINFORMATION_H
00003
00004 #include "../../../../kernel/simulator/model/ModelInfo.h"
00005 #include <QDialog>
00006 #include <string>
00007
00008 namespace Ui {
00009     class DialogModelInformation;
00010 }
00011
00012 class DialogModelInformation : public QDialog
00013 {
00014     Q_OBJECT
```

```
00015
00016 public:
00017     explicit DialogModelInformation(QWidget *parent = nullptr);
00018     ~DialogModelInformation();
00019     void setModelInfo(ModelInfo* modelInfo);
00020
00021 public slots:
00022     void accept() override;
00023     void reject() override;
00024
00025 private:
00026     void _loadModelInfo();
00027     bool _hasPendingChanges() const;
00028     std::string _projectTitleFromUi() const;
00029     std::string _nameFromUi() const;
00030     std::string _versionFromUi() const;
00031     std::string _descriptionFromUi() const;
00032     std::string _analystNameFromUi() const;
00033
00034     Ui::DialogModelInformation *ui;
00035     ModelInfo* _modelInfo = nullptr;
00036     std::string _originalProjectTitle;
00037     std::string _originalName;
00038     std::string _originalVersion;
00039     std::string _originalDescription;
00040     std::string _originalAnalystName;
00041 };
00042
00043 #endif // DIALOGMODELINFORMATION_H
```

18.45 dialogpluginmanager.h

```
00001 #ifndef DIALOGPLUGINMANAGER_H
00002 #define DIALOGPLUGINMANAGER_H
00003
00004 #include <QDialog>
00005
00006 #include <functional>
00007 #include <string>
00008 #include <vector>
00009
00010 class Plugin;
00011 class PluginInformation;
00012 class PluginLoadIssue;
00013 class SystemDependencyCheckResult;
00014 class Simulator;
00015 class QTableWidgetItem;
00016 class QTableWidget;
00017
00018 namespace Ui {
00019     class DialogPluginManager;
00020 }
00021
00022 class DialogPluginManager : public QDialog {
00023     Q_OBJECT
00024
00025 public:
00026     explicit DialogPluginManager(QWidget* parent = nullptr);
00027     ~DialogPluginManager();
00028     void setSimulator(Simulator* simulator);
00029     void setPluginCatalogRefreshCallback(std::function<void()> refreshCallback);
00031     void showProblemPluginsTab();
00032
00033 private slots:
00034     void on_pushButtonBrowseAutoload_clicked();
00035     void on_pushButtonAutoLoadNow_clicked();
00036     void on_pushButtonCheck_clicked();
00037     void on_pushButtonInsert_clicked();
00038     void on_pushButtonResolveSelected_clicked();
00039     void on_pushButtonRemove_clicked();
00040     void on_pushButtonRefresh_clicked();
00041
00042 private:
00044     void _refreshPluginTable();
00046     void _configureLoadedPluginTable();
00048     void _configurePluginIssuesTable();
00050     void _refreshPluginCatalog();
00052     void _appendPluginIssueRow(const PluginLoadIssue& issue);
00054     Plugin* _selectedPlugin() const;
00056     std::vector<PluginLoadIssue> _selectedPluginLoadIssues() const;
00058     bool _isKernelPlugin(const Plugin* plugin) const;
00060     bool _isPluginUsedByCurrentModel(const Plugin* plugin) const;
00062     bool _choosePluginFilename(std::string& filename) const;
00064     QString _formatDynamicDependencyTableText(const PluginInformation* info) const;
00066     QString _formatSystemDependencyTableText(const PluginInformation* info) const;
```

```
00068     QString _formatSystemDependencyPreflight(const SystemDependencyCheckResult& result) const;
00070     QString _formatFields(const PluginInformation* info) const;
00072     bool _confirmSystemDependencyInstallation(const SystemDependencyCheckResult& result) const;
00074     bool _runInstallCommandsForIssue(const PluginLoadIssue& issue, QString* feedback);
00076     bool _runInstallCommandInteractive(const QString& command, QString* feedback);
00078     bool _terminalCommandForScript(const QString& script, QString* program, QStringList* arguments)
      const;
00080     void _configureCommonTable(QTableWidget* table) const;
00081     void _showOperationResult(const QString& title, const QString& message) const;
00082
00083     Ui::DialogPluginManager* ui;
00084     Simulator* _simulator = nullptr;
00085     std::function<void()> _refreshPluginCatalogCallback;
00086 };
00087
00088 #endif // DIALOGPLUGINMANAGER_H
```

18.46 DialogReportIssue.h

```
00001 #ifndef DIALOGREPORTISSUE_H
00002 #define DIALOGREPORTISSUE_H
00003
00004 #include <QDialog>
00005
00006 #include "../services/IssueReportRelayService.h"
00007
00008 class QCheckBox;
00009 class QComboBox;
00010 class QLabel;
00011 class QLineEdit;
00012 class QPlainTextEdit;
00013 class QPushButton;
00014
00028 class DialogReportIssue : public QDialog {
00029 public:
00030     explicit DialogReportIssue(QWidget* parent = nullptr);
00031
00033     IssueReportDraft draft() const;
00035     void setRelayConfigurationStatus(const QString& endpoint, bool configured, const QString& detail);
00036
00037 private:
00039     void _updateSendButton();
00040
00041     QComboBox* _categoryCombo = nullptr;
00042     QLineEdit* _titleEdit = nullptr;
00043     QPlainTextEdit* _summaryEdit = nullptr;
00044     QPlainTextEdit* _stepsEdit = nullptr;
00045     QPlainTextEdit* _expectedEdit = nullptr;
00046     QPlainTextEdit* _observedEdit = nullptr;
00047     QPlainTextEdit* _additionalContextEdit = nullptr;
00048     QLineEdit* _contactEdit = nullptr;
00049     QCheckBox* _includeScreenshotCheck = nullptr;
00050     QCheckBox* _includeLogTailCheck = nullptr;
00051     QCheckBox* _includeModelTextCheck = nullptr;
00052     QLabel* _relayStatusLabel = nullptr;
00053     QPushButton* _sendButton = nullptr;
00054     bool _relayConfigured = true;
00055 };
00056
00057 #endif // DIALOGREPORTISSUE_H
```

18.47 DialogSelectCounter.h

```
00001 #ifndef DIALOGSELECTCOUNTER_H
00002 #define DIALOGSELECTCOUNTER_H
00003
00004 #include <QDialog>
00005 #include <QComboBox>
00006 #include <QPushButton>
00007
00008 #include "../../../../kernel/simulator/essentialPlugins/Counter.h"
00009 class DialogSelectCounter : public QDialog {
00010     Q_OBJECT
00011
00012 public:
00013     explicit DialogSelectCounter(QWidget *parent = nullptr);
00014
00015     Counter* selectedIndex() const;
00016     void setCounterItems(QList<Counter *> *counters, Counter *counter);
00017
00018 private slots:
00019     void onOkClicked();
```

```
00020     void onCancelClicked();
00021
00022 private:
00023     QComboBox *comboBox;
00024     QPushButton *okButton;
00025     QPushButton *cancelButton;
00026     QMap<int, Counter*> indexToCounter;
00027 };
00028
00029 #endif // DIALOGSELECTCOUNTER_H
```

18.48 DialogSelectVariable.h

```
00001 #ifndef DIALOGSELECTVARIABLE_H
00002 #define DIALOGSELECTVARIABLE_H
00003
00004 #include <QDialog>
00005 #include <QComboBox>
00006 #include <QPushButton>
00007 #include <QList>
00008 #include <QMap>
00009
00010 class ModelDataDefinition;
00011
00012 class DialogSelectVariable : public QDialog {
00013     Q_OBJECT
00014
00015 public:
00016     explicit DialogSelectVariable(QWidget *parent = nullptr);
00017
00018     ModelDataDefinition* selectedIndex() const;
00019     void setVariableItems(QList<ModelDataDefinition *> *definitions, ModelDataDefinition *definition);
00020
00021 private slots:
00022     void onOkClicked();
00023     void onCancelClicked();
00024
00025 private:
00026     QComboBox *comboBox;
00027     QPushButton *okButton;
00028     QPushButton *cancelButton;
00029     QMap<int, ModelDataDefinition*> indexToVariable;
00030 };
00031
00032 #endif // DIALOGSELECTVARIABLE_H
```

18.49 dialogsimulationconfigure.h

```
00001 #ifndef DIALOGSIMULATIONCONFIGURE_H
00002 #define DIALOGSIMULATIONCONFIGURE_H
00003
00004 #include "../../../../kernel/simulator/model/ModelSimulation.h"
00005 #include <QDialog>
00006 #include <string>
00007
00008 class QComboBox;
00009 class ExperimentManager;
00010 class SimulationReporter_if;
00011
00012 namespace Ui {
00013     class DialogSimulationConfigure;
00014 }
00015
00016 class DialogSimulationConfigure : public QDialog
00017 {
00018     Q_OBJECT
00019
00020 public:
00021     explicit DialogSimulationConfigure(QWidget *parent = nullptr);
00022     ~DialogSimulationConfigure();
00023     void setModelSimulation(ModelSimulation* modelSimulation);
00024     void setExperimentManager(ExperimentManager* experimentManager);
00025     void setParallelizationSettings(bool* enabled, int* localThreads, int* batchSize);
00026
00027 public slots:
00028     void accept() override;
00029     void reject() override;
00030
00031 private:
00032     struct SimulationConfiguration {
00033         unsigned int numberOfReplications = 1;
00034         Util::TimeUnit replicationBaseTimeUnit = Util::TimeUnit::unknown;
```

```
00035         double replicationLength = 0.0;
00036         Util::TimeUnit replicationLengthTimeUnit = Util::TimeUnit::unknown;
00037         double warmUpPeriod = 0.0;
00038         Util::TimeUnit warmUpPeriodTimeUnit = Util::TimeUnit::unknown;
00039         std::string terminatingCondition;
00040         bool initializeSystem = true;
00041         bool initializeStatistics = true;
00042         bool stepByStep = false;
00043         bool pauseOnEvent = false;
00044         bool pauseOnReplication = false;
00045         bool parallelizationEnabled = false;
00046         int parallelizationThreads = 1;
00047         int parallelizationBatchSize = 100;
00048         bool distributedParallelizationEnabled = false;
00049         std::string distributedCoordinatorUrl;
00050         std::string distributedToken;
00051     };
00052
00053     void _populateTimeUnitComboBoxes();
00054     void _loadModelSimulation();
00055     void _loadSimulationReporter();
00056     void _loadExperimentManager();
00057     void _loadParallelizationSettings();
00058     SimulationConfiguration _configurationFromUi() const;
00059     bool _hasPendingChanges() const;
00060     Util::TimeUnit _timeUnitFromComboBox(const QComboBox* comboBox) const;
00061     void _setTimeUnitComboBoxIndex(QComboBox* comboBox, Util::TimeUnit timeUnit);
00062     void _applyIfChanged(const SimulationConfiguration& edited);
00063
00064     Ui::DialogSimulationConfigure *ui;
00065     ModelSimulation* _modelSimulation = nullptr;
00066     SimulationReporter_if* _simulationReporter = nullptr;
00067     ExperimentManager* _experimentManager = nullptr;
00068     bool* _parallelizationEnabled = nullptr;
00069     int* _parallelizationThreads = nullptr;
00070     int* _parallelizationBatchSize = nullptr;
00071     SimulationConfiguration _originalConfiguration;
00072 };
00073
00074 #endif // DIALOGSIMULATIONCONFIGURE_H
```

18.50 dialogsystempreferences.h

```
00001 #ifndef DIALOGSYSTEMPREFERENCES_H
00002 #define DIALOGSYSTEMPREFERENCES_H
00003
00004 #include <QDialog>
00005
00006 class Simulator;
00007
00008 namespace Ui {
00009     class DialogSystemPreferences;
00010 }
00011
00012 class DialogSystemPreferences : public QDialog
00013 {
00014     Q_OBJECT
00015
00016 public:
00017     explicit DialogSystemPreferences(Simulator* simulator = nullptr, QWidget *parent = nullptr);
00018     ~DialogSystemPreferences();
00019
00020 private:
00021     void loadPreferencesIntoUi();
00022     void savePreferencesFromUi();
00023     void populateTraceLevels();
00024     void browseSpecificModelFile();
00025
00026     Ui::DialogSystemPreferences *ui;
00027     Simulator* _simulator = nullptr;
00028 };
00029
00030 #endif // DIALOGSYSTEMPREFERENCES_H
```

18.51 DialogTimerConfigure.h

```
00001 #ifndef DIALOGTIMERCONFIGURE_H
00002 #define DIALOGTIMERCONFIGURE_H
00003
00004 #include <QDialog>
00005 #include <QPushButton>
00006
```

```
00007 #include "../animations/AnimationTimer.h"
00008
00009 namespace Ui {
00010     class DialogTimerConfigure;
00011 }
00012
00013 class DialogTimerConfigure : public QDialog
00014 {
00015     Q_OBJECT
00016
00017 public:
00018     explicit DialogTimerConfigure(QWidget *parent = nullptr);
00019     ~DialogTimerConfigure();
00020
00021     // recebe o Timer
00022     void setAnimationTimer(AnimationTimer *animationTimer);
00023     // atualiza os valores no Timer
00024     void updateAnimationTimer();
00025
00026 private slots:
00027     // slot para tratar o clique no botão "OK"
00028     void on_buttonBox_accepted();
00029
00030     // slot para tratar mudança no formato de hora para 12 horas
00031     void on_radioButtonTwelve_clicked(bool checked);
00032
00033     // slot para tratar mudança no formato de hora para 24 horas
00034     void on_radioButtonTwentyFour_clicked(bool checked);
00035
00036 private:
00037     Ui::DialogTimerConfigure *ui;
00038     AnimationTimer *_animationTimer;
00039 };
00040
00041 #endif // DIALOGTIMERCONFIGURE_H
```

18.52 BacteriaColonyViewerCoordinates.h

```
00001 #pragma once
00002
00003 #include <cstddef>
00004
00005 namespace BacteriaColonyViewerCoordinates {
00006
00007 // Returns the cell-space offset from the left/top grid edge for a centered
00008 // Gro coordinate. Cell centers range from 0.5 to gridSize - 0.5.
00009 inline double centeredCoordinateToCellOffset(double coordinate, std::size_t gridSize) {
00010     return coordinate + 0.5 * static_cast<double>(gridSize);
00011 }
00012
00013 } // namespace BacteriaColonyViewerCoordinates
```

18.53 GuiExtensionConfigurationDialog.h

```
00001 #ifndef GUIEXTENSIONCONFIGURATIONDIALOG_H
00002 #define GUIEXTENSIONCONFIGURATIONDIALOG_H
00003
00004 #include "GuiExtensionContracts.h"
00005
00006 #include <QDialog>
00007 #include <QJsonObject>
00008
00009 #include <functional>
00010 #include <vector>
00011
00012 class QTableWidget;
00013 class QPushButton;
00014
00015 class GuiExtensionConfigurationDialog : public QDialog {
00016 public:
00017     GuiExtensionConfigurationDialog(
00018         QWidget* parent,
00019         std::vector<const GuiExtensionPlugin*> plugins,
00020         std::function<void()> onSaved = {});
00021
00022 private:
00023     void reloadFromDisk();
00024     void saveToDisk();
00025     QString _configPath() const;
00026     QJsonObject _loadConfigRoot() const;
00027     bool _writeConfigRoot(const QJsonObject& root) const;
00028     void _populateFromConfig(const QJsonObject& root);
```

```
00029     void _configureTable();
00030     static std::string _normalizeId(const std::string& text);
00031
00032 private:
00033     std::vector<const GuiExtensionPlugin*> _plugins;
00034     std::function<void()> _onSaved;
00035     QTableWidget* _table = nullptr;
00036     QPushButton* _reloadButton = nullptr;
00037     QPushButton* _saveButton = nullptr;
00038     QPushButton* _closeButton = nullptr;
00039 };
00040
00041 #endif // GUIEXTENSIONCONFIGURATIONDIALOG_H
```

18.54 GuiExtensionContracts.h

```
00001 #ifndef GUIEXTENSIONCONTRACTS_H
00002 #define GUIEXTENSIONCONTRACTS_H
00003
00004 #include <functional>
00005 #include <string>
00006 #include <vector>
00007
00008 class Simulator;
00009 class QMainWindow;
00010 class QWidget;
00011 class ModelGraphicsView;
00012 class ModelGraphicsScene;
00013
00014 namespace Ui {
00015 class MainWindow;
00016 }
00017
00018 struct GuiExtensionRuntimeContext {
00019     Simulator* simulator = nullptr;
00020     QMainWindow* mainWindow = nullptr;
00021     Ui::MainWindow* ui = nullptr;
00022     ModelGraphicsView* graphicsView = nullptr;
00023     ModelGraphicsScene* graphicsScene = nullptr;
00024 };
00025
00026 struct GuiActionContribution {
00027     std::string id;
00028     std::string text;
00029     std::string menuPath;
00030     std::string toolBarId;
00031     std::string statusTip;
00032     std::string shortcut;
00033     std::function<bool(const GuiExtensionRuntimeContext&)> isVisible;
00034     std::function<bool(const GuiExtensionRuntimeContext&)> isEnabled;
00035     std::function<void(const GuiExtensionRuntimeContext&)> trigger;
00036 };
00037
00038 struct GuiDockContribution {
00039     std::string id;
00040     std::string title;
00041     std::string areaHint;
00042     std::function<bool(const GuiExtensionRuntimeContext&)> isVisible;
00043     std::function<QWidget*(const GuiExtensionRuntimeContext&)> createWidget;
00044 };
00045
00046 struct GuiWindowContribution {
00047     std::string id;
00048     std::string menuPath;
00049     std::string text;
00050     std::function<bool(const GuiExtensionRuntimeContext&)> isVisible;
00051     std::function<void(const GuiExtensionRuntimeContext&)> open;
00052 };
00053
00054 class GuiExtensionRegistry {
00055 public:
00056     void addAction(GuiActionContribution contribution) {
00057         _actions.push_back(std::move(contribution));
00058     }
00059
00060     void addDock(GuiDockContribution contribution) {
00061         _docks.push_back(std::move(contribution));
00062     }
00063
00064     void addWindow(GuiWindowContribution contribution) {
00065         _windows.push_back(std::move(contribution));
00066     }
00067
00068     const std::vector<GuiActionContribution>& actions() const {
00069         return _actions;
```

```
00070     }
00071
00072     const std::vector<GuiDockContribution>& docks() const {
00073         return _docks;
00074     }
00075
00076     const std::vector<GuiWindowContribution>& windows() const {
00077         return _windows;
00078     }
00079
00080     void clear() {
00081         _actions.clear();
00082         _docks.clear();
00083         _windows.clear();
00084     }
00085
00086 private:
00087     std::vector<GuiActionContribution> _actions;
00088     std::vector<GuiDockContribution> _docks;
00089     std::vector<GuiWindowContribution> _windows;
00090 };
00091
00092 class GuiExtensionPlugin {
00093 public:
00094     virtual ~GuiExtensionPlugin() = default;
00095
00096     virtual std::string extensionId() const = 0;
00097     virtual std::string displayName() const = 0;
00098     virtual std::string version() const = 0;
00099     virtual std::vector<std::string> requiredModelPlugins() const = 0;
00100     virtual void registerContributions(GuiExtensionRegistry* registry) const = 0;
00101 };
00102
00103 #endif /* GUIEXTENSIONCONTRACTS_H */
```

18.55 GuiExtensionManager.h

```
00001 #ifndef GUIEXTENSIONMANAGER_H
00002 #define GUIEXTENSIONMANAGER_H
00003
00004 #include "GuiExtensionContracts.h"
00005
00006 #include <string>
00007 #include <vector>
00008
00009 class QAction;
00010 class QDockWidget;
00011 class QMainWindow;
00012 class QMenu;
00013 class QToolBar;
00014
00015 class GuiExtensionManager {
00016 public:
00017     explicit GuiExtensionManager(QMainWindow* mainWindow);
00018     ~GuiExtensionManager();
00019
00020     void setPlugins(std::vector<const GuiExtensionPlugin*> plugins);
00021     void setLoadedModelPluginIds(std::vector<std::string> modelPluginIds);
00022     void rebuild(const GuiExtensionRuntimeContext& context);
00023     void clear();
00024
00025 private:
00026     QMenu* _resolveMenuPath(const std::string& menuPath);
00027     QToolBar* _resolveToolBar(const std::string& toolBarId) const;
00028     void _applyActionContribution(const GuiActionContribution& contribution, const
      GuiExtensionRuntimeContext& context);
00029     void _applyWindowContribution(const GuiWindowContribution& contribution, const
      GuiExtensionRuntimeContext& context);
00030     void _applyDockContribution(const GuiDockContribution& contribution, const
      GuiExtensionRuntimeContext& context);
00031     bool _isPluginDependenciesSatisfied(const GuiExtensionPlugin* plugin) const;
00032
00033 private:
00034     QMainWindow* _mainWindow = nullptr;
00035     std::vector<const GuiExtensionPlugin*> _plugins;
00036     std::vector<std::string> _loadedModelPluginIds;
00037     std::vector<QAction*> _createdActions;
00038     std::vector<QAction*> _createdMenuActions;
00039     std::vector<QDockWidget*> _createdDocks;
00040 };
00041
00042 #endif /* GUIEXTENSIONMANAGER_H */
```

18.56 GuiExtensionPluginCatalog.h

```
00001 #ifndef GUIEXTENSIONPLUGINCATALOG_H
00002 #define GUIEXTENSIONPLUGINCATALOG_H
00003
00004 #include "GuiExtensionContracts.h"
00005
00006 #include <string>
00007 #include <vector>
00008
00009 struct GuiExtensionPresentationMetadata {
00010     std::string category;
00011     std::string group;
00012     int priority = 0;
00013 };
00014
00015 class GuiExtensionPluginCatalog {
00016 public:
00017     static const std::vector<const GuiExtensionPlugin*>& plugins();
00018     static std::vector<const GuiExtensionPlugin*> resolvedPlugins();
00019     static GuiExtensionPresentationMetadata presentationMetadata(const std::string& extensionId);
00020     static void registerPlugin(const GuiExtensionPlugin* plugin);
00021 };
00022
00023 class GuiExtensionPluginAutoRegistrar {
00024 public:
00025     explicit GuiExtensionPluginAutoRegistrar(const GuiExtensionPlugin* plugin);
00026 };
00027
00028 #define REGISTER_GUI_EXTENSION_PLUGIN(PluginType)                \
00029     namespace {                                                    \
00030         static const PluginType _instance_##PluginType;           \
00031         static const GuiExtensionPluginAutoRegistrar              \
00032             _registrar_##PluginType(&_instance_##PluginType);     \
00033     }
00034
00035 #endif /* GUIEXTENSIONPLUGINCATALOG_H */
```

18.57 EFSMTransitionGraphicsItem.h

```
00001 #ifndef EFSMTRANSITIONGRAPHICSITEM_H
00002 #define EFSMTRANSITIONGRAPHICSITEM_H
00003
00004 #include <QGraphicsPathItem>
00005
00006 class EFSMTransition;
00007
00015 class EFSMTransitionGraphicsItem final : public QGraphicsPathItem {
00016 public:
00017     EFSMTransitionGraphicsItem(EFSMTransition* transition, int ordinal, QGraphicsItem* parent =
      nullptr);
00018
00019     EFSMTransition* transition() const;
00020     int ordinal() const;
00021     QString transitionLabel() const;
00022     void setGeometry(const QPointF& sourceCenter, const QPointF& destinationCenter, int parallelRank,
00023                      int parallelCount);
00024
00025 protected:
00026     void paint(QPainter* painter, const QStyleOptionGraphicsItem* option, QWidget* widget = nullptr)
      override;
00027
00028 private:
00029     EFSMTransition* _transition = nullptr;
00030     int _ordinal = -1;
00031 };
00032
00033 #endif
```

18.58 GraphicalAnimateExpression.h

```
00001 #ifndef GRAPHICALANIMATEEXPRESSION_H
00002 #define GRAPHICALANIMATEEXPRESSION_H
00003
00004 #include <QGraphicsObject>
00005
00006 class GraphicalAnimateExpression : public QGraphicsObject {
00007 public:
00008     GraphicalAnimateExpression();
00009 };
00010
00011 #endif // GRAPHICALANIMATEEXPRESSION_H
```

18.59 GraphicalAssociation.h

```
00001 #ifndef GRAPHICALASSOCIATION_H
00002 #define GRAPHICALASSOCIATION_H
00003
00004
00005 class GraphicalAssociation
00006 {
00007 public:
00008     GraphicalAssociation();
00009 };
00010
00011 #endif // GRAPHICALASSOCIATION_H
```

18.60 GraphicalComponentPort.h

```
00001 #ifndef GRAPHICALCOMPONENTPORT_H
00002 #define GRAPHICALCOMPONENTPORT_H
00003
00004 #include <QGraphicsItem>
00005 #include <QGraphicsObject>
00006 #include <QPen>
00007 #include <QBrush>
00008 #include "../../../../kernel/simulator/model/ModelComponent.h"
00009 #include "TraitsGUI.h"
00010 //#include "GraphicalConnection.h"
00011
00012 class GraphicalModelComponent;
00013 class GraphicalConnection;
00014
00015 class GraphicalComponentPort : public QGraphicsObject {
00016 public:
00024     GraphicalComponentPort(GraphicalModelComponent* componentGraph, bool isInputPort, unsigned int
      portNum = 0, QGraphicsItem *parent = nullptr);
00025
00026 protected: // virtual functions
00027     //virtual void mousePressEvent(QGraphicsSceneMouseEvent *mouseEvent);
00028     //virtual void mouseReleaseEvent(QGraphicsSceneMouseEvent *mouseEvent);
00029     //virtual void mouseMoveEvent(QGraphicsSceneMouseEvent *mouseEvent);
00031     virtual QVariant itemChange(QGraphicsItem::GraphicsItemChange change, const QVariant &value)
      override;
00032 public: //overrides
00034     QRectF boundingRect() const override;
00036     void paint(QPainter *painter, const QStyleOptionGraphicsItem *option, QWidget *widget) override;
00038     qreal width() const;
00040     qreal height() const;
00041 public:
00043     void addGraphicalConnection(GraphicalConnection* connection);
00045     void removeGraphicalConnection(GraphicalConnection* connection);
00046
00047 public: // sets and gets
00049     unsigned int portNum() const;
00051     bool isInputPort() const;
00053     GraphicalModelComponent *graphicalComponent() const;
00055     QList<GraphicalConnection*>*getConnections() const;
00056 private:
00057     QColor myrgba(uint64_t color); // TODO: Should NOT be here, but in UtilGUI.h, but then it
      generates multiple definitions error
00058 private:
00059     qreal _width = TraitsGUI<GComponentPort>::width;//20;
00060     qreal _height = _width;
00061     qreal _margin = TraitsGUI<GComponentPort>::margin; //2
00062     int _raise = TraitsGUI<GComponentPort>::raise; //3;
00063     bool _isInputPort;
00064     unsigned int _portNum;
00065     GraphicalModelComponent* _componentGraph;
00066     QList<GraphicalConnection*>* _connections = new QList<GraphicalConnection*>();
00067 };
00068
00069 #endif // GRAPHICALCOMPONENTPORT_H
```

18.61 GraphicalConnection.h

```
00001 #ifndef GRAPHICALCONNECTION_H
00002 #define GRAPHICALCONNECTION_H
00003
00004 #include <QGraphicsObject>
00005
00006 #include <QGraphicsItem>
00007 #include <QGraphicsObject>
00008 #include <QPointer>
00009 #include <QPainterPath>
```

```
00010 #include <QPen>
00011 #include <QBrush>
00012 #include "GraphicalConnectionStyle.h"
00013 #include "GraphicalComponentPort.h"
00014 //#include "kernel/simulator/ModelComponent.h"
00015 //#include "kernel/simulator/Plugin.h"
00016 #include "kernel/simulator/ConnectionManager.h"
00017
00029 class GraphicalConnection : public QGraphicsObject {
00030 public:
00031     enum class ConnectionType : int {
00032         HORIZONTAL = 1, VERTICAL = 2, DIRECT = 3, USERDEFINED = 4
00033     };
00034
00035 public:
00037     GraphicalConnection(GraphicalComponentPort* sourceGraphicalPort, GraphicalComponentPort*
      destinationGraphicalPort, unsigned int portSourceConnection = 0, unsigned int
      portDestinationConnection = 0, QColor color = Qt::black, QGraphicsItem *parent = nullptr);
00039     GraphicalConnection(const GraphicalConnection& orig);
00041     virtual ~GraphicalConnection();
00042 public:
00044     QRectF boundingRect() const override;
00046     QPainterPath shape() const override;
00048     void paint(QPainter *painter, const QStyleOptionGraphicsItem *option, QWidget *widget) override;
00050     Connection* getSource() const;
00052     Connection* getDestination() const;
00054     GraphicalComponentPort* getSourceGraphicalPort();
00056     GraphicalComponentPort* getDestinationGraphicalPort();
00058     GraphicalModelComponent* getSourceGraphicalComponent() const;
00060     GraphicalModelComponent* getDestinationGraphicalComponent() const;
00062     void updateDimensionsAndPosition();
00064     GraphicalConnection::ConnectionType connectionType() const;
00066     unsigned int getPortSourceConnection() const;
00068     unsigned int getPortDestinationConnection() const;
00070     void setConnectionType(GraphicalConnection::ConnectionType newConnectionType);
00072     QList<QPointF> getPoints() const;
00074     bool usesCurvedStyle() const;
00076     QPainterPath animationPathForImage(qreal imageWidth, qreal imageHeight) const;
00077
00078 protected: // virtual
00079     virtual bool sceneEvent(QEvent *event) override;
00080     //virtual void  hoverEnterEvent(QGraphicsSceneHoverEvent * event)
00081     //virtual void  hoverLeaveEvent(QGraphicsSceneHoverEvent * event)
00082     //virtual void  hoverMoveEvent(QGraphicsSceneHoverEvent * event)
00083     //virtual void  inputMethodEvent(QInputMethodEvent * event)
00084     //virtual QVariant  inputMethodQuery(Qt::InputMethodQuery query) const
00085     //virtual QVariant  itemChange(GraphicsItemChange change, const QVariant & value)
00086     //virtual void  keyPressEvent(QKeyEvent * event)
00087     //virtual void  keyReleaseEvent(QKeyEvent * event)
00088     //virtual void mouseDoubleClickEvent(QGraphicsSceneMouseEvent * event);
00089     //virtual void mouseMoveEvent(QGraphicsSceneMouseEvent * event);
00090     //virtual void mousePressEvent(QGraphicsSceneMouseEvent * event);
00091     //virtual void  mouseReleaseEvent(QGraphicsSceneMouseEvent * event)
00092 private:
00093     bool canRefreshGeometry() const;
00094     QPainterPath connectionPath() const;
00095     GraphicalConnectionStyle::RouteType routeTypeForStyle() const;
00096     qreal _width = 0.0;
00097     qreal _height = 0.0;
00098     QPointF _sourcePointLocal;
00099     QPointF _destinationPointLocal;
00100     unsigned int _margin = TraitsGUI<GConnection>::margin;//2;
00101     unsigned int _selWidth = TraitsGUI<GConnection>::selectionWidth;//8;
00102     ConnectionType _connectionType = ConnectionType::HORIZONTAL;
00103     Connection* _sourceConnection;
00104     Connection* _destinationConnection;
00105     QPointer<GraphicalComponentPort> _sourceGraphicalPort;
00106     QPointer<GraphicalComponentPort> _destinationGraphicalPort;
00107     unsigned int _portSourceConnection;
00108     unsigned int _portDestinationConnection;
00109     QColor _color;
00110     QList<QPointF> _points;
00111 };
00112 #endif // GRAPHICALCONNECTION_H
```

18.62 GraphicalConnectionStyle.h

```
00001 #ifndef GRAPHICALCONNECTIONSTYLE_H
00002 #define GRAPHICALCONNECTIONSTYLE_H
00003
00004 #include <QPainterPath>
00005 #include <QPointF>
00006
00007 class GraphicalConnectionStyle {
00008 public:
```

```
00009     enum class RouteType {
00010         Horizontal,
00011         Vertical,
00012         Direct,
00013         UserDefined
00014     };
00015
00016     static bool usesModernCurves();
00017     static QPainterPath modelConnectionPath(const QPointF& source,
00018                                             const QPointF& destination,
00019                                             RouteType routeType,
00020                                             bool modernCurves);
00021     static QPainterPath diagramConnectionPath(const QPointF& source,
00022                                               const QPointF& destination,
00023                                               bool modernCurves);
00024     static QPointF pointAtProgress(const QPainterPath& path, qreal progress);
00025
00026 private:
00027     GraphicalConnectionStyle() = default;
00028 };
00029
00030 #endif // GRAPHICALCONNECTIONSTYLE_H
```

18.63 GraphicalDiagramConnection.h

```
00001 #ifndef GRAPHICALDIAGRAMCONNECTION_H
00002 #define GRAPHICALDIAGRAMCONNECTION_H
00003
00004 #include <QGraphicsItem>
00005 #include <QGraphicsObject>
00006 #include <QPen>
00007 #include <QBrush>
00008 #include <QPainter>
00009 #include <QPainterPath>
00010 #include <cmath>
00011 #include <QGraphicsLineItem>
00012 #include "GraphicalConnectionStyle.h"
00013 #include "../../../../kernel/simulator/model/ModelComponent.h"
00014 #include "../../../../kernel/simulator/model/ModelDataDefinition.h"
00015 #include "GraphicalModelComponent.h"
00016 #include "GraphicalModelDataDefinition.h"
00017 #include "TraitsGUI.h"
00018 #include "UtilGUI.h"
00019
00020 class GraphicalDiagramConnection : public QGraphicsLineItem {
00021 public:
00022     enum class ConnectionType{
00023         ATTACHED, INTERNAL
00024     };
00025
00026 public:
00027     GraphicalDiagramConnection(QGraphicsItem* dataDefinition, QGraphicsItem* linkedTo, ConnectionType
      type);
00028     void refreshGeometry();
00029
00030     QRectF boundingRect() const override;
00031     QPainterPath shape() const override;
00032     void paint(QPainter* painter, const QStyleOptionGraphicsItem* option, QWidget* widget = nullptr)
      override;
00033
00034     QGraphicsItem* getDataDefinition();
00035     QGraphicsItem* getLinkedDataDefinition();
00036     ConnectionType getConnectionType();
00037
00038 private:
00039     ConnectionType _type;
00040     QGraphicsItem* _item1;
00041     QGraphicsItem* _item2;
00042     QPainterPath connectionPath() const;
00043 };
00044
00045
00046 #endif // GRAPHICALDIAGRAMCONNECTION_H
```

18.64 GraphicalImageAnimation.h

```
00001 #ifndef GRAPHICALIMAGEANIMATION_H
00002 #define GRAPHICALIMAGEANIMATION_H
00003
00004 #include <QGraphicsPixmapItem>
00005 #include <QPixmap>
00006
```

```
00007 class GraphicalImageAnimation : public QGraphicsPixmapItem {
00008 public:
00009     // Contrutor
00010     GraphicalImageAnimation(const QPointF startPoint, unsigned int width, unsigned int height, const
      QString imageName);
00011     // Destrutor
00012     ~GraphicalImageAnimation();
00013 
00014     // Getters
00015     QString getImagePath() const;
00016     QString getImageName() const;
00017     QPointF getStartPoint() const;
00018     unsigned int getWidth() const;
00019     unsigned int getHeight() const;
00020 
00021     // Setters
00022     void setImageName(const QString &imageName);
00023     void setStartPoint(const QPointF &startPoint);
00024     void setWidth(unsigned int imageWidth);
00025     void setHeight(unsigned int imageHeight);
00026 
00027     // Atualiza informações da imagem
00028     void updateImage();
00029 
00030 private:
00031     // Builds an in-memory marker image when the file-based image cannot be loaded.
00032     QPixmap buildFallbackPixmap() const;
00033 
00034     // Use a Qt resource base path to avoid working-directory dependent file paths.
00035     QString _resourceBasePath = ":/animations/";
00036 
00037     // Atributos
00038     QString _imagePath;
00039     QPointF _startPoint;
00040     unsigned int _width;
00041     unsigned int _height;
00042     QString _imageName;
00043 };
00044 
00045 #endif // GRAPHICALIMAGEANIMATION_H
```

18.65 GraphicalModelComponent.h

```
00001 #ifndef GRAPHICALMODELCOMPONENT_H
00002 #define GRAPHICALMODELCOMPONENT_H
00003 
00004 /*
00005  * File:   ModelComponentGraphicItem.h
00006  * Author: rlcancian
00007  *
00008  * Created on 16 de fevereiro de 2022, 11:41
00009  */
00010 
00011 #include <QGraphicsItem>
00012 #include <QGraphicsObject>
00013 #include <QPen>
00014 #include <QBrush>
00015 #include "../../../../kernel/simulator/model/ModelComponent.h"
00016 #include "../../../../kernel/simulator/model/ModelDataDefinition.h"
00017 #include "../../../../kernel/simulator/essentialPlugins/Attribute.h"
00018 #include "../../../../kernel/simulator/essentialPlugins/Counter.h"
00019 #include "../../../../kernel/simulator/PropertyGenesys.h"
00020 #include "../../../../kernel/simulator/Simulator.h"
00021 #include "../../../../plugins/data/Logic/Variable.h"
00022 #include "kernel/simulator/Plugin.h"
00023 #include "plugins/data/DiscreteProcessing/Queue.h"
00024 #include "graphicals/GraphicalImageAnimation.h"
00025 #include "GraphicalComponentPort.h"
00026 #include "GraphicalModelDataDefinition.h"
00027 #include "TraitsGUI.h"
00028 
00035 class GraphicalModelComponent : public GraphicalModelDataDefinition {
00036 public:
00038     GraphicalModelComponent(Plugin* plugin, ModelComponent* component, QPointF position, QColor color
      = Qt::blue, QGraphicsItem *parent = nullptr);
00040     GraphicalModelComponent(const GraphicalModelComponent& orig);
00042     virtual ~GraphicalModelComponent();
00043 public:
00045     QRectF boundingRect() const override;
00047     void paint(QPainter *painter, const QStyleOptionGraphicsItem *option, QWidget *widget) override;
00049     ModelComponent* getComponent() const;
00051     qreal getHeight() const;
00053     QPointF getOldPosition() const;
00055     void setOldPosition(QPointF oldPosition);
00057     QList<GraphicalComponentPort *> getGraphicalInputPorts() const;
```

```
00059     QList<GraphicalComponentPort *> getGraphicalOutputPorts() const;
00061     QColor getColor() const;
00063     void setColor(const QColor& color);
00065     QList<ModelDataDefinition *> *getInternalData() const;
00067     QList<ModelDataDefinition *> *getAttachedData() const;
00069     EntityType* getEntityType() const;
00071     void setEntityType(EntityType *entityType);
00073     unsigned int getOcupiedInputPorts() const;
00075     unsigned int getOcupiedOutputPorts() const;
00077     void setOcupiedInputPorts(unsigned int value);
00079     void setOcupiedOutputPorts(unsigned int value);
00080 private:
00081     QColor myrgba(uint64_t color); // TODO: Should NOT be here, but in UtilGUI.h, but then it
      generates multiple definitions error
00082 protected: // virtual
00083     virtual bool sceneEvent(QEvent *event) override;
00084     QVariant itemChange(GraphicsItemChange change, const QVariant& value) override;
00085     GraphicalModelItemRenderContext renderContext() const override;
00086     //virtual void  hoverEnterEvent(QGraphicsSceneHoverEvent * event)
00087     //virtual void  hoverLeaveEvent(QGraphicsSceneHoverEvent * event)
00088     //virtual void  hoverMoveEvent(QGraphicsSceneHoverEvent * event)
00089     //virtual void  inputMethodEvent(QInputMethodEvent * event)
00090     //virtual QVariant  inputMethodQuery(Qt::InputMethodQuery query) const
00091     //virtual QVariant  itemChange(GraphicsItemChange change, const QVariant & value)
00092     //virtual void  keyPressEvent(QKeyEvent * event)
00093     //virtual void  keyReleaseEvent(QKeyEvent * event)
00094     //   virtual void mouseDoubleClickEvent(QGraphicsSceneMouseEvent * event);
00095     //   virtual void mouseMoveEvent(QGraphicsSceneMouseEvent * event);
00096     //   virtual void mousePressEvent(QGraphicsSceneMouseEvent * event);
00097     //virtual void  mouseReleaseEvent(QGraphicsSceneMouseEvent * event)
00098 protected:
00099     qreal _width = TraitsGUI<GModelComponent>::width; //150;
00100     qreal _height = _width * TraitsGUI<GModelComponent>::heightProportion; //0.67;
00101     unsigned int _margin = TraitsGUI<GModelComponent>::margin;//8;
00102     unsigned int _selWidth = TraitsGUI<GModelComponent>::selectionWidth;//8;
00103     ModelComponent* _component;
00104     QColor _color;
00105     QPointF _oldPosition;
00106     qreal _stretchPosTop = TraitsGUI<GModelComponent>::stretchPos;//0.5;
00107     qreal _stretchPosBottom = TraitsGUI<GModelComponent>::stretchPos;//0.5;
00108     qreal _stretchPosLeft = TraitsGUI<GModelComponent>::stretchPos;//0.5;
00109     qreal _stretchPosRigth = TraitsGUI<GModelComponent>::stretchPos;//0.5;
00110     qreal _stretchRigth = TraitsGUI<GModelComponent>::stretch;//0;
00111     qreal _stretchLeft = TraitsGUI<GModelComponent>::stretch;//0;
00112     qreal _stretchRigthMidle = TraitsGUI<GModelComponent>::stretch;//0;
00113     qreal _stretchLeftMidle = TraitsGUI<GModelComponent>::stretch;//0;
00114     qreal _stretchTop = TraitsGUI<GModelComponent>::stretch;//0;
00115     qreal _stretchBottom = TraitsGUI<GModelComponent>::stretch;//0;
00116     qreal _stretchTopMidle = TraitsGUI<GModelComponent>::stretch;//0;
00117     qreal _stretchBottomMidle = TraitsGUI<GModelComponent>::stretch;//0;
00118 private:
00119     QList<GraphicalComponentPort*> _graphicalInputPorts = QList<GraphicalComponentPort*>();
00120     QList<GraphicalComponentPort*> _graphicalOutputPorts = QList<GraphicalComponentPort*>();
00121     QList<ModelDataDefinition *> *_internalData = new QList<ModelDataDefinition*>();
00122     QList<ModelDataDefinition *> *_attachedData = new QList<ModelDataDefinition*>();
00123     EntityType* _entityType = nullptr;
00124     unsigned int _ocupiedInputPorts = 0;
00125     unsigned int _ocupiedOutputPorts = 0;
00126
00127 private:
00128     QString _animationImageName = "default.png";
00129     // Map que mapeia o ponteiro da fila do componente para seu índice (caso tiver mais de uma) e
      tamanho (usado para animação)
00130     QMap<Queue *, QPair<unsigned int, unsigned int» *_mapQueue = new QMap<Queue *, QPair<unsigned int,
      unsigned int»();
00131     // Irá conter referência para as imagens usadas para mostrar a fila do componente
00132     QList<QList<GraphicalImageAnimation *> *> *_imagesQueue = new QList<QList<GraphicalImageAnimation
      *> *>;
00133     bool _hasQueue = false;
00134
00135 public:
00137     QString getAnimationImageName();
00139     void setAnimationImageName(QString name);
00141     QMap<Queue *, QPair<unsigned int, unsigned int»* getMapQueue();
00143     QList<QList<GraphicalImageAnimation *> *>* getImagesQueue();
00144
00146     void verifyQueue();
00148     bool hasQueue();
00150     static bool compareQueuesById(const Queue* a, const Queue* b);
00152     unsigned int getIndexQueue(Queue *queue);
00154     unsigned int getSizeQueue(Queue *queue);
00156     void populateMapQueue(QList<Queue *> queues);
00158     bool insertImageQueue(Queue *queue, GraphicalImageAnimation *image);
00160     QList<GraphicalImageAnimation *> *removeImageQueue(Queue *queue, unsigned int quantityRemoved);
00162     bool insertImageQueue(GraphicalImageAnimation *image);
00164     GraphicalImageAnimation *removeImageQueue();
00166     void actualizeMapQueue(Queue *queue);
```

```
00168     void visivibleImageQueue(bool visivible);
00170     void clearQueues();
00171 };
00172
00173 #endif /* MODELCOMPONENTGRAPHICITEM_H */
```

18.66 GraphicalModelDataDefinition.h

```
00001 #ifndef GRAPHICALMODELDATADEFINITION_H
00002 #define GRAPHICALMODELDATADEFINITION_H
00003
00004 /*
00005  * File:   ModelDataDefinitionGraphicItem.h
00006  * Author: rlcancian
00007  *
00008  * Created on 16 de fevereiro de 2022, 11:41
00009  */
00010
00011 #include <QGraphicsItem>
00012 #include <QGraphicsObject>
00013 #include <QPainterPath>
00014 #include <QPen>
00015 #include <QBrush>
00016 #include "kernel/simulator/Plugin.h"
00017 #include "graphicals/GraphicalModelItemRenderStrategy.h"
00018 #include "TraitsGUI.h"
00019
00026 class GraphicalModelDataDefinition : public QGraphicsObject {
00027 public:
00029     GraphicalModelDataDefinition(Plugin* plugin, ModelDataDefinition* element, QPointF position,
      QColor color = Qt::blue, QGraphicsItem *parent = nullptr);
00031     GraphicalModelDataDefinition(const GraphicalModelDataDefinition& orig);
00033     virtual ~GraphicalModelDataDefinition();
00034 public:
00036     QRectF boundingRect() const override;
00038     QPainterPath shape() const override;
00040     void paint(QPainter *painter, const QStyleOptionGraphicsItem *option, QWidget *widget) override;
00042     ModelDataDefinition* getDataDefinition() const;
00044     bool isEditableInPropertyEditor() const;
00046     void setEditableInPropertyEditor(bool editable);
00048     QPointF getOldPosition() const;
00050     void setOldPosition(qreal x, qreal y);
00052     QColor getColor() const;
00054     void setColor(QColor newColor);
00056     qreal getHeight() const;
00057
00058 protected:
00059     QColor myrgba(uint64_t color); // TODO: Should NOT be here, but in UtilGUI.h, but then it
      generates multiple definitions error
00060     virtual GraphicalModelItemRenderContext renderContext() const;
00061 protected: // virtual
00062     virtual bool sceneEvent(QEvent *event) override;
00063     //virtual void  hoverEnterEvent(QGraphicsSceneHoverEvent * event)
00064     //virtual void  hoverLeaveEvent(QGraphicsSceneHoverEvent * event)
00065     //virtual void  hoverMoveEvent(QGraphicsSceneHoverEvent * event)
00066     //virtual void  inputMethodEvent(QInputMethodEvent * event)
00067     //virtual QVariant  inputMethodQuery(Qt::InputMethodQuery query) const
00068     //virtual QVariant  itemChange(GraphicsItemChange change, const QVariant & value)
00069     //virtual void  keyPressEvent(QKeyEvent * event)
00070     //virtual void  keyReleaseEvent(QKeyEvent * event)
00071     //virtual void mouseDoubleClickEvent(QGraphicsSceneMouseEvent * event);
00072     //virtual void mouseMoveEvent(QGraphicsSceneMouseEvent * event);
00073     //virtual void mousePressEvent(QGraphicsSceneMouseEvent * event);
00074     //virtual void  mouseReleaseEvent(QGraphicsSceneMouseEvent * event)
00075 protected:
00076     qreal _width = TraitsGUI<GModelDataDefinition>::width; //150;
00077     qreal _height = _width * TraitsGUI<GModelDataDefinition>::heightProportion; //0.67;
00078     unsigned int _margin = TraitsGUI<GModelDataDefinition>::margin;//8;
00079     unsigned int _selWidth = TraitsGUI<GModelDataDefinition>::selectionWidth;//8;
00080     ModelDataDefinition* _element;
00081     bool _editableInPropertyEditor = false;
00082     QColor _color;
00083     QPointF _oldPosition;
00084     qreal _stretchPosTop = TraitsGUI<GModelDataDefinition>::stretchPos;//0.5;
00085     qreal _stretchPosBottom = TraitsGUI<GModelDataDefinition>::stretchPos;//0.5;
00086     qreal _stretchPosLeft = TraitsGUI<GModelDataDefinition>::stretchPos;//0.5;
00087     qreal _stretchPosRigth = TraitsGUI<GModelDataDefinition>::stretchPos;//0.5;
00088     qreal _stretchRigth = TraitsGUI<GModelDataDefinition>::stretch;//0;
00089     qreal _stretchLeft = TraitsGUI<GModelDataDefinition>::stretch;//0;
00090     qreal _stretchRigthMidle = TraitsGUI<GModelDataDefinition>::stretch;//0;
00091     qreal _stretchLeftMidle = TraitsGUI<GModelDataDefinition>::stretch;//0;
00092     qreal _stretchTop = TraitsGUI<GModelDataDefinition>::stretch;//0;
00093     qreal _stretchBottom = TraitsGUI<GModelDataDefinition>::stretch;//0;
00094     qreal _stretchTopMidle = TraitsGUI<GModelDataDefinition>::stretch;//0;
00095     qreal _stretchBottomMidle = TraitsGUI<GModelDataDefinition>::stretch;//0;
```

```
00096 private:
00097 };
00098
00099 #endif /* GRAPHICALMODELDATADEFINITION_H */
```

18.67 GraphicalModelItemRenderStrategy.h

```
00001 #ifndef GRAPHICALMODELITEMRENDERSTRATEGY_H
00002 #define GRAPHICALMODELITEMRENDERSTRATEGY_H
00003
00004 #include <cstdint>
00005
00006 #include <QColor>
00007 #include <QPainterPath>
00008 #include <QRectF>
00009 #include <QString>
00010
00011 class QPainter;
00012
00013 struct GraphicalModelItemRenderContext {
00014     enum class ItemKind {
00015         DataDefinition,
00016         Component
00017     };
00018
00019     ItemKind kind = ItemKind::DataDefinition;
00020     QRectF bounds;
00021     QColor fillColor;
00022     QString primaryText;
00023     QString secondaryText;
00024     QString tertiaryText;
00025     QString semanticClassName;
00026     QString semanticSubtype;
00027     bool selected = false;
00028     bool breakpoint = false;
00029
00030     qreal width = 0.0;
00031     qreal height = 0.0;
00032     int penWidth = 1;
00033     int raise = 0;
00034     int margin = 0;
00035     int selectionWidth = 0;
00036
00037     qreal stretchPosTop = 0.5;
00038     qreal stretchPosBottom = 0.5;
00039     qreal stretchPosLeft = 0.5;
00040     qreal stretchPosRight = 0.5;
00041     qreal stretchRight = 0.0;
00042     qreal stretchLeft = 0.0;
00043     qreal stretchRightMiddle = 0.0;
00044     qreal stretchLeftMiddle = 0.0;
00045     qreal stretchTop = 0.0;
00046     qreal stretchBottom = 0.0;
00047     qreal stretchTopMiddle = 0.0;
00048     qreal stretchBottomMiddle = 0.0;
00049
00050     uint64_t borderColor = 0;
00051     uint64_t raisedColor = 0;
00052     uint64_t sunkenColor = 0;
00053     uint64_t textColor = 0;
00054     uint64_t textShadowColor = 0;
00055     uint64_t selectionColor = 0;
00056     uint64_t breakpointColor = 0;
00057 };
00058
00059 class GraphicalModelItemRenderStrategy {
00060 public:
00061     virtual ~GraphicalModelItemRenderStrategy() = default;
00062
00063     virtual const char* name() const = 0;
00064     virtual void paint(QPainter* painter, const GraphicalModelItemRenderContext& context) const = 0;
00065     virtual QPainterPath shape(const GraphicalModelItemRenderContext& context) const = 0;
00066 };
00067
00068 class GraphicalModelItemRenderer {
00069 public:
00070     static void paint(QPainter* painter, const GraphicalModelItemRenderContext& context);
00071     static QPainterPath shape(const GraphicalModelItemRenderContext& context);
00072     static const GraphicalModelItemRenderStrategy& currentStrategy();
00073 };
00074
00075 #endif // GRAPHICALMODELITEMRENDERSTRATEGY_H
```

18.68 ModelGraphicsScene.h

```
00001 /*
00002  * The MIT License
00003  *
00004  * Copyright 2022 rlcancian.
00005  *
00006  * Permission is hereby granted, free of charge, to any person obtaining a copy
00007  * of this software and associated documentation files (the "Software"), to deal
00008  * in the Software without restriction, including without limitation the rights
00009  * to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
00010  * copies of the Software, and to permit persons to whom the Software is
00011  * furnished to do so, subject to the following conditions:
00012  *
00013  * The above copyright notice and this permission notice shall be included in
00014  * all copies or substantial portions of the Software.
00015  *
00016  * THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR
00017  * IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
00018  * FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
00019  * AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
00020  * LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
00021  * OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN
00022  * THE SOFTWARE.
00023  */
00024
00025 /*
00026  * File:   ModelGraphicsScene.h
00027  * Author: rlcancian
00028  *
00029  * Created on 16 de fevereiro de 2022, 09:52
00030  */
00031
00032 #ifndef MODELGRAPHICSSCENE_H
00033 #define MODELGRAPHICSSCENE_H
00034
00035 #include <QGraphicsScene>
00036 #include <QGraphicsProxyWidget>
00037 #include <QGraphicsRectItem>
00038 #include <QTreeWidgetItem>
00039 #include <QUndoStack>
00040 #include <QAction>
00041 #include <QEventLoop>
00042 #include "graphicals/GraphicalModelComponent.h"
00043 #include "graphicals/GraphicalComponentPort.h"
00044 #include "graphicals/GraphicalDiagramConnection.h"
00045 #include "propertyeditor/DataComponentProperty.h"
00046 #include "propertyeditor/DataComponentEditor.h"
00047 #include "propertyeditor/ComboBoxEnum.h"
00048 #include "../../../../kernel/simulator/model/ModelComponent.h"
00049 #include "kernel/simulator/Simulator.h"
00050 #include "kernel/simulator/Plugin.h"
00051 #include "animations/AnimationTransition.h"
00052 #include "animations/AnimationVariable.h"
00053 #include "animations/AnimationCounter.h"
00054 #include "animations/AnimationTimer.h"
00055 #include "animations/AnimationPlaceholder.h"
00056 #include "../../../../kernel/simulator/essentialPlugins/Counter.h"
00057 #include "../../../../kernel/simulator/essentialPlugins/Attribute.h"
00058 #include "kernel/simulator/PropertyGenesys.h"
00059 #include "../../../../kernel/simulator/model/ModelDataDefinition.h"
00060 #include "../../../../plugins/data/Logic/Variable.h"
00064 class GraphicalModelEvent {
00065 public:
00066
00067     enum class EventType : int {
00068         CREATE = 1,
00069         REMOVE = 2,
00070         EDIT = 3,
00071         CLONE = 4,
00072         OTHER = 9
00073     };
00074
00075     enum class EventObjectType : int {
00076         COMPONENT = 1,
00077         DATADEFINITION = 2,
00078         CONNECTION = 3,
00079         DRAWING = 4,
00080         ANIMATION = 5,
00081         PLOT = 6,
00082         OTHER = 9
00083     };
00084
00085 public:
00086
00088     GraphicalModelEvent(GraphicalModelEvent::EventType eventType, GraphicalModelEvent::EventObjectType
      eventObjectType, QGraphicsItem* item) {
```

```
00089         this->eventType = eventType;
00090         this->eventObjectType = eventObjectType;
00091         this->item = item;
00092     }
00093     GraphicalModelEvent::EventType eventType;
00094     GraphicalModelEvent::EventObjectType eventObjectType;
00095     QGraphicsItem* item;
00096 };
00097
00105 class ModelGraphicsScene : public QGraphicsScene {
00106 public:
00107     ModelGraphicsScene(qreal x, qreal y, qreal width, qreal height, QObject *parent = nullptr);
00108     // Disable copy construction to keep scene ownership state unique.
00109     ModelGraphicsScene(const ModelGraphicsScene& orig) = delete;
00110     // Disable copy assignment to prevent shallow copies of GUI-owned resources.
00111     ModelGraphicsScene& operator=(const ModelGraphicsScene& other) = delete;
00112     virtual ~ModelGraphicsScene();
00114     void clear();
00115 public: // editing graphic model
00116     enum DrawingMode{
00117         NONE, LINE, TEXT, RECTANGLE, ELLIPSE, POLYGON,  POLYGON_POINTS, POLYGON_FINISHED,
00118         COUNTER, VARIABLE, TIMER,
00119         ATTRIBUTE, ENTITY, EVENT, EXPRESSION, PLOT, QUEUE_PLACEHOLDER, RESOURCE, STATION, STATISTICS
00120     };
00122     GraphicalModelComponent* addGraphicalModelComponent(Plugin* plugin, ModelComponent* component,
      QPointF position, QColor color = Qt::blue, bool notify = false, GraphicalModelComponent*
      autoConnectSource = nullptr);
00124     GraphicalConnection* addGraphicalConnection(GraphicalComponentPort* sourcePort,
      GraphicalComponentPort* destinationPort, unsigned int portSourceConnection, unsigned int
      portDestinationConnection, bool notify = false);
00126     GraphicalModelDataDefinition* addGraphicalModelDataDefinition(Plugin* plugin, ModelDataDefinition*
      element, QPointF position, QColor color = Qt::blue);
00128     GraphicalDiagramConnection* addGraphicalDiagramConnection(QGraphicsItem* dataDefinition,
      QGraphicsItem* linkedTo, GraphicalDiagramConnection::ConnectionType type);
00130     void initializeAnimationDrawing(QGraphicsSceneMouseEvent *mouseEvent);
00132     void continueAnimationDrawing(QGraphicsSceneMouseEvent *mouseEvent);
00134     void finishAnimationDrawing(QGraphicsSceneMouseEvent *mouseEvent);
00136     void startTextEditing();
00138     void removeComponent(GraphicalModelComponent* gmc, bool notify = false);
00140     void clearConnectionsComponent(GraphicalModelComponent* gmc);
00142     void clearInputConnectionsComponent(GraphicalModelComponent* graphicalComponent);
00144     void clearOutputConnectionsComponent(GraphicalModelComponent* graphicalComponent);
00146     void clearPorts(GraphicalConnection* connection, GraphicalModelComponent *source,
      GraphicalModelComponent *destination);
00148     void connectComponents(GraphicalConnection* connection, GraphicalModelComponent *source = nullptr,
      GraphicalModelComponent *destination = nullptr, bool notify = false);
00150     bool connectSource(GraphicalConnection* connection, GraphicalModelComponent *source = nullptr);
00152     bool connectDestination(GraphicalConnection* connection, GraphicalModelComponent *destination =
      nullptr);
00154     void redoConnections(GraphicalModelComponent *graphicalComponent, QList<GraphicalConnection *>
      *inputConnections, QList<GraphicalConnection *> *outputConnections);
00156     void removeComponentInModel(GraphicalModelComponent* gmc);
00158     void insertComponent(GraphicalModelComponent* gmc, QList<GraphicalConnection *> *inputConnections,
      QList<GraphicalConnection *> *outputConnections, bool addGMC = true, bool addAllGMC = true, bool
      notify = false);
00160     void removeGraphicalConnection(GraphicalConnection* graphicalConnection, GraphicalModelComponent
      *source, GraphicalModelComponent *destination, bool notify = false);
00162     void removeConnectionInModel(GraphicalConnection* graphicalConnection, GraphicalModelComponent
      *source);
00164     void addGeometry(QPointF endPoint, bool moving);
00166     void addDrawing(QGraphicsItem * item, bool notify = false);
00168     bool addDrawingGeometry(QGraphicsItem * item);
00170     bool addDrawingAnimation(QGraphicsItem * item);
00172     void removeGraphicalModelDataDefinition(GraphicalModelDataDefinition* gmdd);
00174     void detachGraphicalModelDataDefinition(GraphicalModelDataDefinition* gmdd);
00176     void restoreGraphicalModelDataDefinition(GraphicalModelDataDefinition* gmdd);
00178     void removeGraphicalDiagramConnection(GraphicalDiagramConnection* connection);
00180     void clearGraphicalModelDataDefinitions();
00182     void clearGraphicalDiagramConnections();
00184     void sanitizeGraphicalDataDefinitionsBookkeeping();
00186     void setDiagramLayerState(bool diagramCreated, bool visible);
00187     // Return only items that can be directly manipulated by user edit commands.
00189     QList<QGraphicsItem*> userOperableItems(const QList<QGraphicsItem*>& items) const;
00190     // Filter out non-deletable items from user-triggered delete flows.
00192     QList<QGraphicsItem*> userDeletableItems(const QList<QGraphicsItem*>& items) const;
00193     // Keep internal data-definition initial grouping opt-in during model rebuild only.
00195     void ensureInitialInternalDataDefinitionGrouping(GraphicalModelDataDefinition* dataDefinition,
      GraphicalModelComponent* component);
00196     // Allow serializer to disable automatic initial grouping while persisted GUI state is being
      restored.
00198     void setPersistedGuiRestoreInProgress(bool restoring);
00200     bool isPersistedGuiRestoreInProgress() const;
00202     void removeDrawing(QGraphicsItem * item, bool notify = false);
00203     bool removeDrawingGeometry(QGraphicsItem * item);
00204     bool removeDrawingAnimation(QGraphicsItem * item);
00206     void removeGroup(QGraphicsItemGroup* group, bool notify = false);
00208     void clearGraphicalModelComponents();
```

```
00210     void clearGraphicalModelConnections();
00212     void groupComponents(bool notify = false); // tenta agrupar (verifica se sao
      ModelGraphicalComponents)
00214     void groupModelComponents(QList<GraphicalModelComponent *> *graphicalComponents,
      QGraphicsItemGroup *group); // agrupa componentes
00216     void ungroupComponents(bool notify = false);
00218     void ungroupModelComponents(QGraphicsItemGroup *group);
00220     void notifyGraphicalModelChange(GraphicalModelEvent::EventType eventType,
      GraphicalModelEvent::EventObjectType eventObjectType, QGraphicsItem *item);
00221     // Return model component items by value to avoid heap ownership transfer.
00223     QList<GraphicalModelComponent*> graphicalModelComponentItems();
00225     GraphicalModelComponent* findGraphicalModelComponent(Util::identification id);
00227     GraphicalModelDataDefinition* findGraphicalModelDataDefinition(ModelDataDefinition*
      dataDefinition);
00229     void clearEFSMNetworkDiagramItems();
00231     void addEFSMNetworkDiagramItem(QGraphicsItem* item);
00233     QList<QGraphicsItem*>* getEFSMNetworkDiagramItems() const;
00235     GraphicalModelComponent* resolveSourceComponent(GraphicalConnection* connection) const;
00237     GraphicalModelComponent* resolveDestinationComponent(GraphicalConnection* connection) const;
00238 public:
00239     struct GRID {
00240         unsigned int interval;
00241         QPen pen;
00242         std::list<QGraphicsLineItem *> *lines;
00243         bool visible;
00244         void clear();
00245     };
00247     GRID *grid();
00249     void showGrid();
00250     // Aplica o estado visual do grid de forma determinística sem alternância implícita.
00252     void setGridVisible(bool visible);
00253     // Informa o estado visual atual do grid para sincronização com QAction.
00255     bool isGridVisible() const;
00257     void snapItemsToGrid();
00259     void actualizeDiagramArrows();
00261     void showDiagrams();
00263     void hideDiagrams();
00265     bool existDiagram();
00267     bool visibleDiagram();
00269     void destroyDiagram();
00271     void createDiagrams();
00273     QUndoStack* getUndoStack();
00275     Simulator* getSimulator();
00277     void setUndoStack(QUndoStack* undo);
00279     void beginConnection();
00281     void setSimulator(Simulator *simulator);
00283     void setPropertyEditor(PropertyEditorGenesys *propEditor);
00285     void setPropertyList(std::map<SimulationControl*, DataComponentProperty*>* propList);
00287     void setPropertyEditorUI(std::map<SimulationControl*, DataComponentEditor*>* propEditorUI);
00289     void setComboBox(std::map<SimulationControl*, ComboBoxEnum*>* propCombo);
00291     void setObjectBeingDragged(QTreeWidgetItem* objectBeingDragged);
00293     void setRestoringPersistedGuiLayout(bool restoring);
00295     bool isRestoringPersistedGuiLayout() const;
00297     void setParentWidget(QWidget *parentWidget);
00299     unsigned short connectingStep() const;
00301     void setConnectingStep(unsigned short connectingStep);
00303     void setSnapToGrid(bool activated);
00305     bool getSnapToGrid();
00307     void arranjeModels(int direction);
00309     void setDrawingMode(DrawingMode drawingMode);
00311     void clearDrawingMode();
00313     void setGraphicalComponentPort(GraphicalComponentPort * in);
00315     DrawingMode getDrawingMode();
00317     void setAction(QAction* action);
00319     QList<GraphicalModelComponent*> *getAllComponents();
00321     QList<GraphicalConnection*> *getAllConnections();
00323     QList<GraphicalModelDataDefinition*> *getAllDataDefinitions();
00325     QList<GraphicalDiagramConnection*> *getAllGraphicalDiagramsConnections();
00327     QMap<QGraphicsItemGroup *, QList<GraphicalModelComponent *» getListComponentsGroup();
00329     void insertComponentGroup(QGraphicsItemGroup *group, QList<GraphicalModelComponent *>
      componentsGroup);
00331     void insertOldPositionItem(QGraphicsItem *item, QPointF position);
00333     QPointF getOldPositionItem(QGraphicsItem *item) const;
00335     QList<AnimationTransition *> *getAnimationsTransition();
00337     QList<AnimationCounter *> *getAnimationsCounter();
00339     QList<AnimationVariable *> *getAnimationsVariable();
00341     QList<AnimationTimer *> *getAnimationsTimer();
00343     QList<AnimationPlaceholder *> *getAnimationsPlaceholder();
00345     QMap<Event *, QList<AnimationTransition *> *> *getAnimationPaused();
00347     bool checkIgnoreEvent();
00349     void clearAnimations();
00351     void clearAnimationsTransition();
00353     void clearAnimationsCounter();
00355     void clearAnimationsVariable();
00357     void clearAnimationsTimer();
00359     void clearAnimationsPlaceholder();
00361     void clearAnimationsQueue();
```

```
00363     void animatePlot();
00365     QList<QString> *getImagesAnimation();
00367     void drawingCounter();
00369     void drawingVariable();
00371     void drawingTimer();
00373     void drawingAttribute();
00375     void drawingEntity();
00377     void drawingEvent();
00379     void drawingExpression();
00381     void drawingPlot();
00383     void drawingQueue();
00385     void drawingResource();
00387     void drawingStation();
00389     void drawingStatistics();
00391     void clearAnimationsValues();
00393     void setCounters();
00395     void setVariables();
00396     // TODO: Funções abaixo são usadas para reaver os dataDefinitions do componente nos
      dataDefinitions do modelo dos componentes deletados, "checados" e reinseridos (Control Z de um delete,
      por exemplo).
00397     // O kernel não trata este caso, ale acusa erro, pois não encontra os dataDefinitions do
      componente nos dataDefinitions do modelo, pois ele os remove como "órfãos" e não os reinsere quando
      voltados ao modelo.
00399     void insertRestoredDataDefinitions(bool loaded);
00401     void saveDataDefinitions();
00402     // -------------------------------- //
00404     void requestGraphicalDataDefinitionsSync();
00406     void scheduleGraphicalDataDefinitionsSync();
00408     bool isGraphicalDataDefinitionsSyncInProgress() const;
00410     void setShowStatisticsDataDefinitions(bool show);
00412     bool showStatisticsDataDefinitions() const;
00414     void setShowEditableDataDefinitions(bool show);
00416     bool showEditableDataDefinitions() const;
00418     void setShowSharedDataDefinitions(bool show);
00420     bool showSharedDataDefinitions() const;
00422     void setShowRecursiveDataDefinitions(bool show);
00424     bool showRecursiveDataDefinitions() const;
00432     void setModelLevelFilter(unsigned int modelLevel);
00434     void clearModelLevelFilter();
00436     bool hasModelLevelFilter() const;
00438     unsigned int modelLevelFilter() const;
00440     void setShowInternalDataDefinitions(bool show);
00442     bool showInternalDataDefinitions() const;
00444     void setShowAttachedDataDefinitions(bool show);
00446     bool showAttachedDataDefinitions() const;
00448     void setConnectionGeometryUpdatesBlocked(bool blocked);
00450     bool areConnectionGeometryUpdatesBlocked() const;
00451
00452 public:
00454     QList<QGraphicsItem*>*getGraphicalModelDataDefinitions() const;
00456     QList<QGraphicsItem*>*getGraphicalModelComponents() const;
00458     QList<QGraphicsItem*>*getGraphicalConnections() const;
00460     QList<QGraphicsItem*>*getGraphicalGeometries() const;
00462     QList<QGraphicsItem*>*getGraphicalAnimations() const;
00464     QList<QGraphicsItem*>*getGraphicalEntities() const;
00466     QList<QGraphicsItem*>*getGraphicalDiagramsConnections() const;
00468     QList<QGraphicsItemGroup*>*getGraphicalGroups() const;
00469
00470 public:
00472     void animateTransition(ModelComponent *source, ModelComponent *destination, bool viewSimulation,
      Event *event);
00474     void runAnimateTransition(AnimationTransition *animationTransition, Event *event, bool restart =
      false);
00476     void handleAnimationStateChanged(QAbstractAnimation::State newState, QEventLoop* loop, Event*
      event, AnimationTransition* animationTransition);
00478     void animateQueueInsert(ModelComponent *component, bool visivible);
00480     void animateQueueRemove(ModelComponent *component);
00482     void animateCounter();
00484     void animateVariable(Entity* entity = nullptr);
00486     void animateTimer(double time);
00487
00488 protected: // virtual functions
00489     virtual void contextMenuEvent(QGraphicsSceneContextMenuEvent *contextMenuEvent);
00490     virtual void dragEnterEvent(QGraphicsSceneDragDropEvent *event);
00491     virtual void dragLeaveEvent(QGraphicsSceneDragDropEvent *event);
00492     virtual void dragMoveEvent(QGraphicsSceneDragDropEvent *event);
00493     //virtual void  drawBackground(QPainter *painter, const QRectF &rect);
00494     //virtual void  drawForeground(QPainter *painter, const QRectF &rect);
00495     virtual void dropEvent(QGraphicsSceneDragDropEvent *event);
00496     virtual void focusInEvent(QFocusEvent *focusEvent);
00497     virtual void focusOutEvent(QFocusEvent *focusEvent);
00498     //virtual void  helpEvent(QGraphicsSceneHelpEvent *helpEvent);
00499     //virtual void  inputMethodEvent(QInputMethodEvent *event);
00500     virtual void keyPressEvent(QKeyEvent *keyEvent);
00501     virtual void keyReleaseEvent(QKeyEvent *keyEvent);
00502     virtual void mouseDoubleClickEvent(QGraphicsSceneMouseEvent *mouseEvent);
00503     virtual void mouseMoveEvent(QGraphicsSceneMouseEvent *mouseEvent);
```

```
00504     virtual void mousePressEvent(QGraphicsSceneMouseEvent *mouseEvent);
00505     virtual void mouseReleaseEvent(QGraphicsSceneMouseEvent *mouseEvent);
00506     virtual void wheelEvent(QGraphicsSceneWheelEvent *wheelEvent);
00507
00508 private:
00509     bool tryCreateConnection(GraphicalComponentPort* source, GraphicalComponentPort* destination, bool
      notify = true);
00510     GraphicalComponentPort* firstAvailableOutputPort(GraphicalModelComponent* component) const;
00511     GraphicalComponentPort* firstInputPort(GraphicalModelComponent* component) const;
00512     void resetConnectingState();
00513 private:
00514     GRID _grid;
00515     Simulator* _simulator = nullptr;
00516     PropertyEditorGenesys* _propertyEditor = nullptr;
00517     std::map<SimulationControl*, DataComponentProperty*>* _propertyList = nullptr;
00518     std::map<SimulationControl*, DataComponentEditor*>* _propertyEditorUI = nullptr;
00519     std::map<SimulationControl*, ComboBoxEnum*>* _propertyBox = nullptr;
00520     QTreeWidgetItem* _objectBeingDragged = nullptr;
00521     // Initialize the parent widget pointer to a known null state.
00522     QWidget* _parentWidget = nullptr;
00523     QList<GraphicalModelComponent*> _allGraphicalModelComponents;
00524     QList<GraphicalConnection*> _allGraphicalConnections;
00525     QList<GraphicalModelDataDefinition*> _allGraphicalModelDataDefinitions;
00526     QList<GraphicalDiagramConnection*> _allGraphicalDiagramConnections;
00527     QUndoStack *_undoStack = nullptr;
00528     QMap<QGraphicsItemGroup *, QList<GraphicalModelComponent *> > _listComponentsGroup;
00529     QMap<QGraphicsItem *, QPointF> _oldPositionsItems;
00530     QList<Counter *> *_counters = new QList<Counter *>();
00531     QList<ModelDataDefinition *> *_variables = new QList<ModelDataDefinition *>();
00532 private:
00533     DrawingMode _drawingMode = NONE;
00534     QGraphicsRectItem* _currentRectangle = nullptr;
00535     QGraphicsLineItem* _currentLine = nullptr;
00536     QGraphicsPolygonItem* _currentPolygon = nullptr;
00537     QGraphicsEllipseItem* _currentEllipse = nullptr;
00538     QPolygonF _currentPolygonPoints;
00539     QPointF _drawingStartPoint;
00540     QAction* _currentAction = nullptr;
00541     bool _drawing = false;
00542     bool _diagram = false;
00543     bool _visibleDiagram = false;
00544     bool _persistedGuiRestoreInProgress = false;
00545     unsigned short _connectingStep = 0; //0:nothing, 1:waiting click on source or destination, 2:
      click on source, 3: click on destination
00546     bool _controlIsPressed = false;
00547     bool _shiftSelectionInProgress = false;
00548     QGraphicsItem* _shiftClickedSelectableItem = nullptr;
00549     QList<QGraphicsItem*> _shiftSelectionBeforeClick;
00550     bool _snapToGrid = false;
00551     // Initialize the source port pointer before connection drawing starts.
00552     GraphicalComponentPort* _sourceGraphicalComponentPort = nullptr;
00553     // Initialize the destination port pointer before connection drawing starts.
00554     GraphicalComponentPort* _destinationGraphicalComponentPort = nullptr;
00555     // Initialize the current counter drawing pointer to avoid indeterminate access.
00556     AnimationCounter *_currentCounter = nullptr;
00557     // Initialize the current variable drawing pointer to avoid indeterminate access.
00558     AnimationVariable *_currentVariable = nullptr;
00559     // Initialize the current timer drawing pointer to avoid indeterminate access.
00560     AnimationTimer *_currentTimer = nullptr;
00561     AnimationPlaceholder *_currentPlaceholderAnimation = nullptr;
00562     QMap<Event *, QList<AnimationTransition *> *> *_animationPaused = new QMap<Event *,
      QList<AnimationTransition *> *>();
00563
00564 private:
00565     QList<QString> *_imagesAnimation = new QList<QString>;
00566     QList<AnimationTransition *> *_animationsTransition = new QList<AnimationTransition*>();
00567     QList<AnimationCounter *> *_animationsCounter = new QList<AnimationCounter*>();
00568     QList<AnimationVariable *> *_animationsVariable = new QList<AnimationVariable*>();
00569     QList<AnimationTimer *> *_animationsTimer = new QList<AnimationTimer*>();
00570     QList<AnimationPlaceholder *> *_animationsPlaceholder = new QList<AnimationPlaceholder*>();
00571 private:
00572     // IMPORTANT. MUST BE CONSISTENT WITH SIMULATOR->MODEL
00573     QList<QGraphicsItem*>* _graphicalModelComponents = new QList<QGraphicsItem*>();
00574     QList<QGraphicsItem*>* _graphicalModelDataDefinitions = new QList<QGraphicsItem*>();
00575     QList<QGraphicsItem*>* _graphicalConnections = new QList<QGraphicsItem*>();
00576     QList<QGraphicsItem*>* _graphicalDiagramConnections = new QList<QGraphicsItem*>();
00577     QList<QGraphicsItem*>* _efsmNetworkDiagramItems = new QList<QGraphicsItem*>();
00578     QList<QGraphicsItem*>* _graphicalAssociations = new QList<QGraphicsItem*>();
00579     QList<QGraphicsItem*>* _graphicalGeometries = new QList<QGraphicsItem*>();
00580     QList<QGraphicsItem*>* _graphicalAnimations = new QList<QGraphicsItem*>();
00581     QList<QGraphicsItem*>* _graphicalEntities = new QList<QGraphicsItem*>();
00582     QList<QGraphicsItemGroup*>* _graphicalGroups = new QList<QGraphicsItemGroup*>();
00583     bool _restoringPersistedGuiLayout = false;
00584     bool _graphicalDataDefinitionsSyncPending = false;
00585     bool _graphicalDataDefinitionsSyncInProgress = false;
00586     bool _graphicalDataDefinitionsSyncDeferredDuringRestore = false;
00587     bool _connectionGeometryUpdatesBlocked = false;
```

```
00588     bool _showStatisticsDataDefinitions = true;
00589     bool _showEditableDataDefinitions = true;
00590     bool _showSharedDataDefinitions = true;
00591     bool _showRecursiveDataDefinitions = true;
00592     // Rendering scope used by root/submodel tabs. The filter is intentionally scene-local so
00593     // static graphical synchronization calls can keep the same scope as the active view.
00594     bool _hasModelLevelFilter = false;
00595     unsigned int _modelLevelFilter = 0;
00596 };
00597
00598 #endif /* MODELGRAPHICSSCENE_H */
```

18.69 ModelGraphicsView.h

```
00001 /*
00002  * The MIT License
00003  *
00004  * Copyright 2022 rlcancian.
00005  *
00006  * Permission is hereby granted, free of charge, to any person obtaining a copy
00007  * of this software and associated documentation files (the "Software"), to deal
00008  * in the Software without restriction, including without limitation the rights
00009  * to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
00010  * copies of the Software, and to permit persons to whom the Software is
00011  * furnished to do so, subject to the following conditions:
00012  *
00013  * The above copyright notice and this permission notice shall be included in
00014  * all copies or substantial portions of the Software.
00015  *
00016  * THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR
00017  * IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
00018  * FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
00019  * AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
00020  * LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
00021  * OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN
00022  * THE SOFTWARE.
00023  */
00024
00025 /*
00026  * File:   QModelGraphicView.h
00027  * Author: rlcancian
00028  *
00029  * Created on 15 de fevereiro de 2022, 21:12
00030  */
00031
00032 #ifndef QMODELGRAPHICVIEW_H
00033 #define QMODELGRAPHICVIEW_H
00034
00035 #include <QGraphicsView>
00036 #include <Qt>
00037 #include <QColor>
00038 #include <QStyle>
00039 #include <QGraphicsSceneMouseEvent>
00040 #include <QContextMenuEvent>
00041 #include <QPainter>
00042 #include "graphicals/ModelGraphicsScene.h"
00043 #include "propertyeditor/DataComponentProperty.h"
00044 #include "propertyeditor/DataComponentEditor.h"
00045 #include "propertyeditor/ComboBoxEnum.h"
00046 #include "../../../../kernel/simulator/model/ModelComponent.h"
00047 #include "kernel/simulator/Simulator.h"
00048 #include "kernel/simulator/PropertyGenesys.h"
00049 //#include "kernel/simulator/Plugin.h"
00050
00051 class ModelGraphicsView : public QGraphicsView {
00052 public:
00060     ModelGraphicsView(QWidget *parent = nullptr);
00062     ModelGraphicsView(const ModelGraphicsView& orig);
00064     virtual ~ModelGraphicsView();
00065 public: // editing graphic model
00066     // TODO: AddGraphicalModelComponent should be only on scene
00067     //GraphicalModelComponent* addGraphicalModelComponent(Plugin* plugin, ModelComponent* component,
      QPointF position);
00068     //bool removeGraphicalModelComponent(GraphicalModelComponent* gmc);
00069     //bool addGraphicalConnection(GraphicalComponentPort* sourcePort, GraphicalComponentPort*
      destinationPort);
00070     //bool removeGraphicalConnection(GraphicalConnection* gc);
00071     //bool addDrawing();
00072     //bool removeDrawing();
00073     //bool addAnimation();
00074     //bool removeAnimation();
00076     ModelGraphicsScene* getScene();
00077 public:
00079     void showGrid();
00081     void clear();
```

```
00083     void beginConnection();
00085     void selectModelComponent(ModelComponent* component);
00087     void setSimulator(Simulator* simulator);
00089     void setPropertyEditor(PropertyEditorGenesys* propEditor);
00091     void setPropertyList(std::map<SimulationControl*, DataComponentProperty*>* propList);
00093     void setPropertyEditorUI(std::map<SimulationControl*, DataComponentEditor*>* propEditorUI);
00095     void setComboBox(std::map<SimulationControl*, ComboBoxEnum*>* propBox);
00097     void setEnabled(bool enabled);
00099     QList<QGraphicsItem*> selectedItems();
00100 public: // events and notifications
00101
00102     template<typename Class> void setSceneMouseEventHandler(Class * object, void
      (Class::*function)(QGraphicsSceneMouseEvent*)) {
00103         sceneMouseEventHandlerMethod handlerMethod = std::bind(function, object,
      std::placeholders::_1);
00104         this->_sceneMouseEventHandler = handlerMethod;
00105     }
00106
00107     template<typename Class> void setGraphicalModelEventHandler(Class * object, void
      (Class::*function)(const GraphicalModelEvent&)) {
00108         sceneGraphicalModelEventHandlerMethod handlerMethod = std::bind(function, object,
      std::placeholders::_1);
00109         this->_sceneGraphicalModelEventHandler = handlerMethod;
00110     }
00111
00112     template <typename Class> void setSceneWheelInEventHandler(Class *object, void
      (Class::*function)()) {
00113         sceneWheelEventHandlerMethod handlerMethod = std::bind(function, object);
00114         this->_sceneWheelInEventHandler = handlerMethod;
00115     }
00116
00117     template <typename Class> void setSceneWheelOutEventHandler(Class *object, void
      (Class::*function)()) {
00118         sceneWheelEventHandlerMethod handlerMethod = std::bind(function, object);
00119         this->_sceneWheelOutEventHandler = handlerMethod;
00120     }
00121
00122     template <typename Class> void setContextMenuEventHandler(Class* object, void
      (Class::*function)(QContextMenuEvent*)) {
00123         // Keep context-menu construction outside the view so menus can reuse MainWindow actions.
00124         contextMenuEventHandlerMethod handlerMethod = std::bind(function, object,
      std::placeholders::_1);
00125         this->_contextMenuEventHandler = handlerMethod;
00126     }
00127
00129     void notifySceneMouseEventHandler(QGraphicsSceneMouseEvent* mouseEvent);
00131     void notifySceneWheelInEventHandler();
00133     void notifySceneWheelOutEventHandler();
00135     void notifySceneGraphicalModelEventHandler(const GraphicalModelEvent& modelGraphicsEvent);
00137     void setCanNotifyGraphicalModelEventHandlers(bool can);
00139     void clearEventHandlers();
00141     void setParentWidget(QWidget *parentWidget);
00142     // Enables or disables ruler rendering over the graphics viewport.
00144     void setRuleVisible(bool visible);
00145     // Informs if ruler rendering is active for action synchronization.
00147     bool isRuleVisible() const;
00148     // Enables or disables guide line rendering over the graphics viewport.
00150     void setGuidesVisible(bool visible);
00151     // Informs if guide line rendering is active for action synchronization.
00153     bool isGuidesVisible() const;
00154 protected:// slots:
00155     void changed(const QList<QRectF> &region);
00156     void focusItemChanged(QGraphicsItem *newFocusItem, QGraphicsItem *oldFocusItem, Qt::FocusReason
      reason);
00157     void sceneRectChanged(const QRectF &rect);
00158     void selectionChanged();
00159 protected: // virtual functions
00160     virtual void contextMenuEvent(QContextMenuEvent *event) override;
00161     virtual void dragEnterEvent(QDragEnterEvent *event) override;
00162     //virtual void dragLeaveEvent(QDragLeaveEvent *event) override;
00163     //virtual void dragMoveEvent(QDragMoveEvent *event) override;
00164     //virtual void dropEvent(QDropEvent *event) override;
00165     //virtual bool event(QEvent *event) override;
00166     //virtual void focusInEvent(QFocusEvent *event) override;
00167     //virtual bool focusNextPrevChild(bool next) override;
00168     //virtual void focusOutEvent(QFocusEvent *event) override;
00169     //virtual void inputMethodEvent(QInputMethodEvent *event) override;
00170     virtual void keyPressEvent(QKeyEvent *event) override;
00171     virtual void keyReleaseEvent(QKeyEvent *event) override;
00172     //virtual void mouseDoubleClickEvent(QMouseEvent *event) override;
00173     //virtual void mouseMoveEvent(QMouseEvent *event) override;
00174     //virtual void mousePressEvent(QMouseEvent *event) override;
00175     //virtual void mouseReleaseEvent(QMouseEvent *event) override;
00176     //virtual void paintEvent(QPaintEvent *event) override;
00177     //virtual void resizeEvent(QResizeEvent *event) override;
00178     //virtual void scrollContentsBy(int dx, int dy) override;
00179     //virtual void showEvent(QShowEvent *event) override;
```

```
00180     //virtual bool viewportEvent(QEvent *event) override;
00181     virtual void wheelEvent(QWheelEvent *event) override;
00182     // Draws optional rulers and guides using the current visible scene rectangle.
00183     virtual void drawForeground(QPainter *painter, const QRectF &rect) override;
00184 private:
00185     typedef std::function<void(QGraphicsSceneMouseEvent*) > sceneMouseEventHandlerMethod;
00186     typedef std::function<void()> sceneWheelEventHandlerMethod;
00187     typedef std::function<void(const GraphicalModelEvent&)> sceneGraphicalModelEventHandlerMethod;
00188     typedef std::function<void(QContextMenuEvent*)> contextMenuEventHandlerMethod;
00189     sceneMouseEventHandlerMethod _sceneMouseEventHandler;
00190     sceneWheelEventHandlerMethod _sceneWheelInEventHandler;
00191     sceneWheelEventHandlerMethod _sceneWheelOutEventHandler;
00192     sceneGraphicalModelEventHandlerMethod _sceneGraphicalModelEventHandler;
00193     contextMenuEventHandlerMethod _contextMenuEventHandler;
00194     Simulator* _simulator = nullptr;
00195     PropertyEditorGenesys* _propertyEditor = nullptr;
00196     std::map<SimulationControl*, DataComponentProperty*>* _propertyList = nullptr;
00197     std::map<SimulationControl*, DataComponentEditor*>* _propertyEditorUI = nullptr;
00198     std::map<SimulationControl*, ComboBoxEnum*>* _propertyBox = nullptr;
00199     QWidget* _parentWidget;
00200     bool _notifyGraphicalModelEventHandlers = true;
00201     bool _ruleVisible = false;
00202     bool _guidesVisible = false;
00203 };
00204
00205 #endif /* QMODELGRAPHICVIEW_H */
```

18.70 GuiCrashDiagnostics.h

```
00001 #ifndef GUICRASHDIAGNOSTICS_H
00002 #define GUICRASHDIAGNOSTICS_H
00003
00011 namespace GuiCrashDiagnostics {
00012
00016 void installFatalSignalHandlers();
00017
00018 } // namespace GuiCrashDiagnostics
00019
00020 #endif // GUICRASHDIAGNOSTICS_H
```

18.71 GuiScopeTrace.h

```
00001 #ifndef GUISCOPETRACE_H
00002 #define GUISCOPETRACE_H
00003
00004 #include <chrono>
00005
00013 class GuiScopeTrace {
00014 public:
00020     GuiScopeTrace(const char* methodName, const void* thisPtr);
00024     ~GuiScopeTrace();
00025
00026 private:
00028     const char* _methodName;
00030     const void* _thisPtr;
00032     int _depth;
00034     std::chrono::steady_clock::time_point _start;
00035 };
00036
00037 #endif // GUISCOPETRACE_H
```

18.72 guithememanager.h

```
00001 #ifndef GUITHEMEMANAGER_H
00002 #define GUITHEMEMANAGER_H
00003
00004 class QApplication;
00005 class ModelGraphicsView;
00006
00013 class GuiThemeManager {
00014 public:
00019     static void applyApplicationTheme(QApplication* application);
00024     static void applyModelGraphicsTheme(ModelGraphicsView* graphicsView);
00025
00026 private:
00027     GuiThemeManager() = default;
00028 };
00029
00030 #endif // GUITHEMEMANAGER_H
```

18.73 applications/gui/genesys/mainwindow.cpp File Reference

Composition-root partition of MainWindow implementation.

```
#include "mainwindow.h"
#include "ui_mainwindow.h"
#include "kernel/simulator/SinkModelComponent.h"
#include "../../../kernel/simulator/essentialPlugins/Attribute.h"
#include "../../../kernel/simulator/essentialPlugins/Counter.h"
#include "../../../kernel/simulator/essentialPlugins/StatisticsCollector.h"
#include "kernel/simulator/PluginManager.h"
#include "kernel/simulator/Plugin.h"
#include "../../../kernel/simulator/model/ModelComponent.h"
#include "../../../kernel/simulator/model/ModelComponentManager.h"
#include "../../../kernel/simulator/model/ModelDataManager.h"
#include "plugins/components/ModalModel/ModalModelDefault.h"
#include "plugins/components/ModalModel/FSMState.h"
#include "plugins/data/ModalModel/DefaultNetwork.h"
#include "plugins/data/ModalModel/EFSMNetwork.h"
#include "graphicals/ModelGraphicsScene.h"
#include "TraitsGUI.h"
#include "graphicals/GraphicalConnection.h"
#include "graphicals/GraphicalModelDataDefinition.h"
#include "controllers/SimulationController.h"
#include "controllers/ModelInspectorController.h"
#include "controllers/TraceConsoleController.h"
#include "controllers/SimulationEventController.h"
#include "controllers/PluginCatalogController.h"
#include "controllers/PropertyEditorController.h"
#include "controllers/ModelLifecycleController.h"
#include "controllers/SimulationCommandController.h"
#include "controllers/EditCommandController.h"
#include "controllers/SceneToolController.h"
#include "controllers/GraphicalContextMenuController.h"
#include "controllers/DialogUtilityController.h"
#include "actions/EFSMTransitionUndoCommand.h"
#include "actions/EFSMTransitionPropertiesUndoCommand.h"
#include "actions/EFSMInitialStateUndoCommand.h"
#include "actions/EFSMStateMembershipUndoCommand.h"
#include "actions/EFSMStatePropertiesUndoCommand.h"
#include "actions/ModalModelBindingsUndoCommand.h"
#include "graphicals/EFSMTransitionGraphicsItem.h"
#include "services/EFSMNetworkDiagramProjection.h"
#include "extensions/GuiExtensionContracts.h"
#include "extensions/GuiExtensionManager.h"
#include "extensions/GuiExtensionPluginCatalog.h"
#include "services/ModelLanguageSynchronizer.h"
#include "services/GraphvizModelExporter.h"
#include "services/CppModelExporter.h"
#include "services/GraphicalModelSerializer.h"
#include "services/GraphicalModelBuilder.h"
#include "UtilGUI.h"
#include "guithememanager.h"
#include "propertyeditor/qtpropertybrowser/qttreepropertybrowser.h"
#include "animations/AnimationVariable.h"
#include "systempreferences.h"
#include "../../../plugins/data/Logic/Variable.h"
#include <string>
#include <fstream>
```

```
#include <sstream>
#include <cstdlib>
#include <algorithm>
#include <cmath>
#include <limits>
#include <QMessageBox>
#include <QInputDialog>
#include <QDialog>
#include <QFormLayout>
#include <QDialogButtonBox>
#include <QLineEdit>
#include <QSpinBox>
#include <QCheckBox>
#include <QComboBox>
#include <QTextStream>
#include <QFileDialog>
#include <QFileInfo>
#include <QDateTime>
#include <QEventLoop>
#include <QTemporaryFile>
#include <Qt>
#include <QGraphicsPixmapItem>
#include <QKeySequence>
#include <QPropertyAnimation>
#include <QtWidgets/qgraphicsitem.h>
#include <QScreen>
#include <QDebug>
#include <QRegularExpression>
#include <QRandomGenerator>
#include <QAction>
#include <QFrame>
#include <QLabel>
#include <QMap>
#include <QMenu>
#include <QPainter>
#include <QScrollArea>
#include <QSignalBlocker>
#include <QSizePolicy>
#include <QSplitter>
#include <QTabBar>
#include <QTabWidget>
#include <QTreeWidget>
#include <QTimer>
#include <QToolBar>
#include <QVBoxLayout>
```

18.73.1 Detailed Description

Composition-root partition of MainWindow implementation.
This file contains central MainWindow construction, initialization, and wiring logic, including creation of extracted controllers/services and baseline UI setup. It does not define a new class; it is a physical partition of the same MainWindow implementation used as a compatibility façade in the incremental refactoring.

18.74 mainwindow.h

```
00001 #ifndef MAINWINDOW_H
00002 #define MAINWINDOW_H
00003
```

```
00004 #include <QMainWindow>
00005 #include <QListWidget>
00006 #include <QTreeWidgetItem>
00007 #include <QGraphicsItem>
00008 #include <QUndoView>
00009 #include <QMetaObject>
00010 #include <memory>
00011 #include <map>
00012
00013 #include "propertyeditor/DataComponentProperty.h"
00014 #include "propertyeditor/DataComponentEditor.h"
00015 #include "propertyeditor/ComboBoxEnum.h"
00016
00017 #include "kernel/simulator/Simulator.h"
00018 #include "kernel/simulator/PropertyGenesys.h"
00019 #include "kernel/simulator/TraceManager.h"
00020 #include "graphicals/ModelGraphicsScene.h"
00021 #include "services/EFSMNetworkDiagramLayout.h"
00022
00023 QT_BEGIN_NAMESPACE
00024         namespace Ui {
00025     class MainWindow;
00026 }
00027
00028 QT_END_NAMESPACE
00029
00030 class ModelLanguageSynchronizer;
00031 class GraphvizModelExporter;
00032 class CppModelExporter;
00033 class GraphicalModelSerializer;
00034 class GraphicalModelBuilder;
00035 class ModelInspectorController;
00036 class TraceConsoleController;
00037 class SimulationEventController;
00038 class PluginCatalogController;
00039 class PropertyEditorController;
00040 class ModelLifecycleController;
00041 class SimulationCommandController;
00042 class EditCommandController;
00043 class SceneToolController;
00044 class GraphicalContextMenuController;
00045 class DialogUtilityController;
00046 class GuiExtensionManager;
00047 class QAction;
00048 class QMenu;
00049 class QTabWidget;
00050 class ModelGraphicsView;
00051 class ModelComponent;
00052 class ModelDataDefinition;
00053 class ModalModelDefault;
00054
00055 // Document MainWindow as composition root and incremental compatibility façade.
00075 class MainWindow : public QMainWindow {
00076     Q_OBJECT
00077
00078     // Allow the Phase 4 event controller to register private compatibility handlers.
00079     friend class SimulationEventController;
00080
00081 public:
00083     MainWindow(QWidget *parent = nullptr);
00085     ~MainWindow();
00087     ModelGraphicsScene* myScene() const;
00088
00089 public: // to notify changes
00091     bool graphicalModelHasChanged() const;
00093     void setGraphicalModelHasChanged(bool graphicalModelHasChanged);
00095     void unselectDrawIcons();
00097     bool checkSelectedDrawIcons();
00099     void refreshGuiExtensions();
00101     void refreshPluginCatalog();
00102
00103 private slots:
00104     // actions
00105     void on_actionEditUndo_triggered();
00106     void on_actionEditRedo_triggered();
00107     void on_actionEditFind_triggered();
00108     // void on_actionReplace_triggered(); // old name (without menu namespace)
00109     void on_actionEditReplace_triggered();
00110     void on_actionEditCut_triggered();
00111     void on_actionEditCopy_triggered();
00112     void on_actionEditPaste_triggered();
00113     void on_actionEditDelete_triggered();
00114     void on_actionEditGroup_triggered();
00115     void on_actionEditUngroup_triggered();
00116     void on_actionViewGroup_triggered();
00117     void on_actionViewUngroup_triggered();
00118
```

```
00119     void on_actionShowGrid_triggered();
00120     void on_actionShowRule_triggered();
00121     void on_actionShowGuides_triggered();
00122     void on_actionViewConfigure_triggered();
00123
00124     void on_actionZoom_In_triggered();
00125     void on_actionZoom_Out_triggered();
00126     void on_actionZoom_All_triggered();
00127     void on_actionZoom_OneToOne_triggered();
00128
00129     void on_actionDrawLine_triggered();
00130     void on_actionDrawRectangle_triggered();
00131     void on_actionDrawEllipse_triggered();
00132     void on_actionDrawText_triggered();
00133     void on_actionDrawPoligon_triggered();
00134
00135     void on_actionAnimateSimulatedTime_triggered();
00136     void on_actionAnimateVariable_triggered();
00137     void on_actionAnimateExpression_triggered();
00138     void on_actionAnimateResource_triggered();
00139     void on_actionAnimateQueue_triggered();
00140     void on_actionAnimateStation_triggered();
00141     void on_actionAnimateCounter_triggered();
00142     void on_actionAnimateEntity_triggered();
00143     void on_actionAnimateEvent_triggered();
00144     void on_actionAnimateAttribute_triggered();
00145     void on_actionAnimateStatistics_triggered();
00146     void on_actionAnimatePlot_triggered();
00147
00148     void on_actionSimulationStop_triggered();
00150     void on_actionSimulationStart_triggered();
00152     void on_actionSimulationStep_triggered();
00154     void on_actionSimulationPause_triggered();
00156     void on_actionSimulationResume_triggered();
00158     void on_actionSimulationConfigure_triggered();
00159
00160     void on_actionAboutAbout_triggered();
00161     void on_actionAboutLicence_triggered();
00162     void on_actionAboutGetInvolved_triggered();
00163
00164     void on_actionAlignMiddle_triggered();
00165     void on_actionAlignTop_triggered();
00166     void on_actionAlignRight_triggered();
00167     void on_actionAlignCenter_triggered();
00168     void on_actionAlignLeft_triggered();
00169
00170     void on_actionToolsParserGrammarChecker_triggered();
00171     void on_actionToolsWebWorker_triggered();
00172     // Legacy slot: there is no matching QAction in mainwindow.ui at this moment.
00173     void on_actionToolsExperimentation_triggered();
00174     void on_actionToolsOptimizator_triggered();
00175     void on_actionToolsAIAssistant_triggered();
00176     void on_actionToolsDataAnalyzer_triggered();
00177
00178     void on_actionSimulatorsPluginManager_triggered();
00179     void on_actionSimulatorExit_triggered();
00180     void on_actionSimulatorPreferences_triggered();
00181
00183     void on_actionModelNew_triggered();
00185     void on_actionModelOpen_triggered();
00187     void on_actionModelSave_triggered();
00189     void on_actionModelClose_triggered();
00191     void on_actionAnimation_triggered();
00193     void on_actionModelInformation_triggered();
00195     void on_actionModelCheck_triggered();
00196
00197     void on_actionGModelComponentBreakpoint_triggered();
00198     void on_actionShowInternalElements_triggered();
00199     void on_actionShowEditableElements_triggered();
00200     void on_actionShowAttachedElements_triggered();
00201     void on_actionShowRecursiveElements_triggered();
00202
00203
00204     // widget events
00205     //void on_textCodeEdit_Model_textChanged();
00206     void on_tabWidget_Model_tabBarClicked(int index);
00207     void on_tabWidget_Debug_currentChanged(int index);
00208     void on_tabWidgetCentral_currentChanged(int index);
00209     void on_tabWidgetCentral_tabBarClicked(int index);
00210     void on_tabWidgetModel_currentChanged(int index);
00211     void on_tabWidgetSimulation_currentChanged(int index);
00212     void on_tabWidgetReports_currentChanged(int index);
00213     void on_tabWidgetModelLanguages_currentChanged(int index);
00214
00215     void on_treeWidget_Plugins_itemDoubleClicked(QTreeWidgetItem *item, int column);
00216     void on_treeWidgetComponents_itemSelectionChanged();
00217     void on_treeWidgetDataDefnitions_itemSelectionChanged();
```

```
00218     void on_treeWidget_Plugins_itemClicked(QTreeWidgetItem *item, int column);
00219
00220     void on_horizontalSlider_Zoom_valueChanged(int value);
00221     void on_horizontalSlider_ZoomGraphical_valueChanged(int value);
00222
00223     void on_checkBox_ShowElements_stateChanged(int arg1);
00224     void on_checkBox_ShowInternals_stateChanged(int arg1);
00225     void on_checkBox_ShowEditableElements_stateChanged(int arg1);
00226     void on_checkBox_ShowRecursive_stateChanged(int arg1);
00227     void on_checkBox_ShowLevels_stateChanged(int arg1);
00228
00229     void on_pushButton_Breakpoint_Insert_clicked();
00230     void on_pushButton_Breakpoint_Remove_clicked();
00231     void on_pushButton_Export_clicked();
00232
00234     void on_graphicsView_rubberBandChanged(const QRect &viewportRect, const QPointF &fromScenePoint,
      const QPointF &toScenePoint);
00236     void on_TextCodeEditor_textChanged();
00237
00238     void on_treeWidgetDataDefnitions_itemDoubleClicked(QTreeWidgetItem *item, int column);
00239     void on_treeWidgetDataDefnitions_itemChanged(QTreeWidgetItem *item, int column);
00240     void on_actionArranjeLeft_triggered();
00241     void on_actionArranjeRight_triggered();
00242     void on_actionArranjeTop_triggered();
00243     void on_actionArranjeBototm_triggered();
00244     void on_actionArranjeCenter_triggered();
00245     void on_actionArranjeMiddle_triggered();
00246     void on_actionShowSnap_triggered();
00247     void on_actionGModelShowConnect_triggered(bool checked = false);
00248
00249     void on_actionActivateGraphicalSimulation_triggered();
00250     void on_horizontalSliderAnimationSpeed_valueChanged(int value);
00251     void on_actionSelectAll_triggered();
00252
00253     void on_actionParallelization_triggered();
00254
00255     void on_horizontalSlider_ZoomGraphical_actionTriggered(int action);
00257     void _onPropertyEditorModelChanged();
00258
00259 protected:
00260     void closeEvent(QCloseEvent *event) override;
00261
00262 private: // VIEW
00263
00264 private: // trace handlers
00266     void _simulatorTraceHandler(TraceEvent e);
00268     void _simulatorTraceErrorHandler(TraceErrorEvent e);
00270     void _simulatorTraceSimulationHandler(TraceSimulationEvent e);
00272     void _simulatorTraceReportsHandler(TraceEvent e);
00273 private: // simulator event handlers
00275     void _onModelCheckSuccessHandler(ModelEvent* re);
00277     void _onReplicationStartHandler(SimulationEvent* re);
00279     void _onReplicationEndHandler(SimulationEvent* re);
00281     void _onSimulationStartHandler(SimulationEvent* re);
00283     void _onSimulationPausedHandler(SimulationEvent* re);
00285     void _onSimulationResumeHandler(SimulationEvent* re);
00287     void _onSimulationEndHandler(SimulationEvent* re);
00289     void _onProcessEventHandler(SimulationEvent* re);
00291     void _onEntityCreateHandler(SimulationEvent* re);
00293     void _onEntityRemoveHandler(SimulationEvent* re);
00295     void _onMoveEntityEvent(SimulationEvent * re);
00297     void _onAfterProcessEvent(SimulationEvent * re);
00298 private: // model Graphics View handlers
00300     void _onSceneMouseEvent(QGraphicsSceneMouseEvent* mouseEvent);
00302     void _onSceneWheelInEvent();
00304     void _onSceneWheelOutEvent();
00306     void _onSceneGraphicalModelEvent(const GraphicalModelEvent& event);
00307 private: // QGraphicsScene Slots
00309     void sceneChanged(const QList<QRectF> &region);
00311     void sceneFocusItemChanged(QGraphicsItem *newFocusItem, QGraphicsItem *oldFocusItem,
      Qt::FocusReason reason);
00312     //void sceneRectChanged(const QRectF &rect);
00314     void sceneSelectionChanged();
00315 private: // Similar to QGraphicsScene Slots
00317     void sceneGraphicalModelChanged();
00318 private: // simulator related
00320     void _setOnEventHandlers();
00322     bool _ensureSimulationReady(bool checkModel = true);
00324     bool _hasCurrentModelSimulation() const;
00326     void _insertPluginUI(Plugin* plugin);
00328     void _insertFakePlugins();
00330     bool _setSimulationModelBasedOnText();
00332     bool _createModelImage();
00334     std::string _adjustDotName(std::string name);
00336     void _insertTextInDot(std::string text, unsigned int compLevel, unsigned int compRank,
      std::map<unsigned int, std::map<unsigned int, std::list<std::string>*>*>* dotmap, bool isNode =
      false);
```

```
00338     void _recursiveCreateModelGraphicPicture(ModelDataDefinition* componentOrData,
      std::list<ModelDataDefinition*>* visited, std::map<unsigned int, std::map<unsigned int,
      std::list<std::string>*>*>* dotmap);
00340     void _actualizeModelCppCode();
00342     std::string _addCppCodeLine(std::string line, unsigned int indent = 0);
00343 private: // view
00345     void _initModelGraphicsView();
00347     void _connectSceneSignals();
00349     void _disconnectSceneSignals(const char* context);
00351     void _initializeModelTabs();
00353     void _configureModelGraphicsView(ModelGraphicsView* graphicsView);
00355     ModelGraphicsView* _createModelGraphicsView(QWidget* parent);
00357     void _activateModelGraphicsView(ModelGraphicsView* graphicsView, bool rebuildControllers = true);
00359     ModelGraphicsView* _ensureModelTab(Model* model);
00361     void _removeModelTab(Model* model);
00363     void _updateModelTabs();
00365     void _activateModelTab(int index);
00367     void _activatePreviousModel();
00369     void _activateNextModel();
00371     void _rebuildViewDependentControllers();
00373     void _refreshGuiExtensions();
00375     void _initUiForNewModel(Model* m);
00377     void _actualizeActions();
00379     void _actualizeUndo();
00381     void _actualizeTabPanes();
00383     void _actualizeModelSimLanguage();
00385     void _actualizeModelTextHasChanged(bool hasChanged);
00387     void _actualizeSimulationEvents(SimulationEvent * re);
00389     void _actualizeDebugVariables(bool force);
00391     void _actualizeDebugEntities(bool force);
00393     void _actualizeDebugBreakpoints(bool force);
00395     void _prepareReportsResultsTable();
00397     void _clearReportsResultsTable();
00399     void _actualizeReportsResultsTable();
00401     void _prepareReportsPlots();
00403     void _clearReportsPlots();
00405     void _actualizeReportsPlots();
00407     void _showReplicationReportIfSuppressed();
00409     void _showSimulationReportIfSuppressed();
00411     void _actualizeModelComponents(bool force);
00413     void _actualizeModelDataDefinitions(bool force);
00415     void _actualizeGraphicalModel(SimulationEvent * re);
00417     void _insertCommandInConsole(std::string text);
00419     void _clearModelEditors();
00421     void _gentle_zoom(double factor);
00423     void _showMessageNotImplemented();
00425     void _recursivalyGenerateGraphicalModelFromModel(ModelComponent* component, List<ModelComponent*>*
      visited, std::map<ModelComponent*,GraphicalModelComponent*>* map, int *x, int *y, int *ymax, int
      sequenceInline);
00427     void _generateGraphicalModelFromModel();
00429     void saveItemForCopy(QList<GraphicalModelComponent*> * gmcList, QList<GraphicalConnection*> *
      connList);
00430     //bool _checkStartSimulation();
00431 private: // opened-model document state
00433     enum class GraphicalModelViewContextKind {
00434         RootModel,
00435         Submodel
00436     };
00437
00446     struct GraphicalModelViewContext {
00447         GraphicalModelViewContextKind kind = GraphicalModelViewContextKind::RootModel;
00448         Model* rootModel = nullptr;
00449         unsigned int modelLevel = 0;
00450         ModelComponent* ownerComponent = nullptr;
00451         ModelDataDefinition* ownerDataDefinition = nullptr;
00452         ModelGraphicsView* graphicsView = nullptr;
00453         QString explicitTitle;
00454     };
00455
00457     GraphicalModelViewContext* _activeViewContext() const;
00459     GraphicalModelViewContext* _contextForView(ModelGraphicsView* graphicsView) const;
00461     GraphicalModelViewContext* _rootContextForModel(Model* model) const;
00463     void _captureEFSMNetworkLayoutForContext(GraphicalModelViewContext* context);
00465     GraphicalModelViewContext* _ensureRootModelViewContext(Model* model, ModelGraphicsView*
      graphicsView);
00467     ModelGraphicsView* _ensureSubmodelViewContext(ModelDataDefinition* owner);
00469     void _closeSubmodelViewContext(ModelGraphicsView* graphicsView);
00471     void _removeSubmodelViewContextsForModel(Model* model);
00473     void _removeViewContext(ModelGraphicsView* graphicsView);
00475     QString _viewContextTitle(const GraphicalModelViewContext& context) const;
00477     bool _viewContextBelongsToOpenModel(const GraphicalModelViewContext& context) const;
00479     ModelDataDefinition* _selectedSubmodelOwner() const;
00481     bool _canOpenSubmodelFor(ModelDataDefinition* owner) const;
00483     void _openSelectedSubmodel();
00485     ModalModelDefault* _selectedModalModelDefault() const;
00487     void _createNetworkForSelectedModalModel();
00489     void _editModalModelBindings();
```

```
00491     void _selectNetworkForSelectedModalModel();
00493     void _createEfsmState();
00495     void _createEfsmTransition();
00496     void _editEfsmTransition();
00497     void _setEfsmInitialState();
00498     void _editEfsmState();
00499     void _refreshRecentModelsMenu();
00500     bool _openRecentModelFile(const QString& fileName);
00502     void _syncCurrentModelDocumentState();
00504     void _restoreModelDocumentState(Model* model);
00506     QString _modelDisplayName(Model* model) const;
00508     QString _modelSaveBaseFilename(QString filename) const;
00510     bool _modelHasPendingChanges(Model* model) const;
00512     bool _saveCurrentModel(bool promptForFilename = true);
00514     bool _saveModelWithActivation(Model* model);
00516     bool _confirmCloseModel(Model* model);
00518     bool _closeModel(Model* model);
00520     void _clearModelGraphicsViewForClose(ModelGraphicsView* graphicsView);
00521 private: // graphical model persistence
00523     bool _saveGraphicalModel(QString filename);
00525     bool _saveTextModel(QFile *saveFile, QString data);
00527     Model* _loadGraphicalModel(std::string filename);
00528 private: //???
00530     void _helpCopy();
00531 private: // interface and model main elements to join
00532     Ui::MainWindow *ui;
00533     Simulator* simulator;
00534     // Hosts one graphical scene/view per opened kernel model.
00535     QTabWidget* _modelGraphicsTabs = nullptr;
00536     // Maps opened kernel models to their graphical view so tab switching preserves each scene.
00537     std::map<Model*, ModelGraphicsView*> _modelGraphicsViews;
00538     // Reverse lookup used by tab activation without relying on QVariant pointer casts.
00539     std::map<ModelGraphicsView*, Model*> _modelsByGraphicsView;
00540     // Owns graphical contexts by view; currently every context is root-model, future ones may be
      submodels.
00541     std::map<ModelGraphicsView*, std::unique_ptr<GraphicalModelViewContext»
      _viewContextsByGraphicsView;
00542     // Fast lookup for the root graphical context of each opened kernel model.
00543     std::map<Model*, GraphicalModelViewContext*> _rootViewContextsByModel;
00544     // Keeps each opened model's last graphical/text filename independent while switching tabs.
00545     std::map<Model*, QString> _modelFilenames;
00546     // Caches the editable Simulang text per opened model so tab switches do not discard unsaved text.
00547     std::map<Model*, QString> _modelTextContents;
00548     // Tracks text-editor dirty state per opened model while legacy flags mirror only the active
      model.
00549     std::map<Model*, bool> _modelTextHasChanged;
00550     // Tracks graphical dirty state per opened model while legacy flags mirror only the active model.
00551     std::map<Model*, bool> _modelGraphicalHasChanged;
00552     // Serializer instances are rebuilt when the active graphical view changes.
00553     std::shared_ptr<EFSMNetworkDiagramLayoutsByModel> _efsmNetworkDiagramLayouts =
00554         std::make_shared<EFSMNetworkDiagramLayoutsByModel>();
00555     // Tracks tabs being changed programmatically to avoid re-entrant model switching.
00556     bool _changingModelTabProgrammatically = false;
00557     // Runtime menu actions for navigating the opened-model list.
00558     QAction* _actionModelPrevious = nullptr;
00559     QAction* _actionModelNext = nullptr;
00560     // Runtime action that opens a level-scoped graphical tab from the selected owner item.
00561     QAction* _actionOpenSelectedSubmodel = nullptr;
00562     QAction* _actionCreateModalNetwork = nullptr;
00563     QAction* _actionSelectModalNetwork = nullptr;
00564     QAction* _actionEditModalModelBindings = nullptr;
00565     QAction* _actionCreateEfsmState = nullptr;
00566     QAction* _actionCreateEfsmTransition = nullptr;
00567     QAction* _actionEditEfsmTransition = nullptr;
00568     QAction* _actionSetEfsmInitialState = nullptr;
00569     QAction* _actionEditEfsmState = nullptr;
00570     QMenu* _menuOpenRecent = nullptr;
00571     QList<QAction*> _recentModelActions;
00572     // Keep core simulation command gateway owned by MainWindow composition root.
00573     std::unique_ptr<class SimulationController> _simulationController;
00574     // Keep phase-ordered services to make composition dependencies explicit in Phase 12.
00575     // Synchronize textual model language with the kernel model manager.
00576     std::unique_ptr<ModelLanguageSynchronizer> _modelLanguageSynchronizer;
00577     // Generate Graphviz DOT and rendered model images for the GUI pane.
00578     std::unique_ptr<GraphvizModelExporter> _graphvizModelExporter;
00579     // Generate C++ model representation text for the code viewer pane.
00580     std::unique_ptr<CppModelExporter> _cppModelExporter;
00581     // Persist and restore graphical .gui state through the Phase 2 serializer service.
00582     std::unique_ptr<GraphicalModelSerializer> _graphicalModelSerializer;
00583     // Rebuild graphical components and links through the Phase 2 builder service.
00584     std::unique_ptr<GraphicalModelBuilder> _graphicalModelBuilder;
00585     // Keep phase-ordered controllers to preserve the final compatibility façade surface.
00586     // Add the Phase 3 model-inspector controller owned by MainWindow.
00587     std::unique_ptr<ModelInspectorController> _modelInspectorController;
00588     // Add the Phase 4 trace controller owned by MainWindow.
00589     std::unique_ptr<TraceConsoleController> _traceConsoleController;
00590     // Add the Phase 4 simulation-event controller owned by MainWindow.
```

```
00591     std::unique_ptr<SimulationEventController> _simulationEventController;
00592     // Add the Phase 5 plugin-catalog controller owned by MainWindow.
00593     std::unique_ptr<PluginCatalogController> _pluginCatalogController;
00594     // Add the Phase 6 property-editor controller owned by MainWindow.
00595     std::unique_ptr<PropertyEditorController> _propertyEditorController;
00596     // Add the Phase 7 model-lifecycle controller owned by MainWindow.
00597     std::unique_ptr<ModelLifecycleController> _modelLifecycleController;
00598     // Add the Phase 8 simulation-command controller owned by MainWindow.
00599     std::unique_ptr<SimulationCommandController> _simulationCommandController;
00600     // Add the Phase 9 edit-command controller owned by MainWindow.
00601     std::unique_ptr<EditCommandController> _editCommandController;
00602     // Add the Phase 10 scene-tool controller owned by MainWindow.
00603     std::unique_ptr<SceneToolController> _sceneToolController;
00604     // Add the graphical context-menu controller owned by MainWindow.
00605     std::unique_ptr<GraphicalContextMenuController> _graphicalContextMenuController;
00606     // Add the Phase 11 dialog-utility controller owned by MainWindow.
00607     std::unique_ptr<DialogUtilityController> _dialogUtilityController;
00608     // Keeps runtime GUI extension contributions isolated from core static menus/toolbars.
00609     std::unique_ptr<GuiExtensionManager> _guiExtensionManager;
00610     PropertyEditorGenesys* propertyGenesys;
00611     std::map<SimulationControl*, DataComponentProperty*>* propertyList;
00612     std::map<SimulationControl*, DataComponentEditor*>* propertyEditorUI;
00613     std::map<SimulationControl*, ComboBoxEnum*>* propertyBox;
00614 private: // attributes to be saved and loaded withing the graphical model
00615     int _zoomValue; // todo should be set for each open graphical model, such as view rect, etc
00616 private: // misc useful
00618     bool _check(bool success = true);
00620     bool _hasPendingModelChanges() const;
00622     bool _confirmApplicationExit();
00623     bool _textModelHasChanged = false;
00624     bool _graphicalModelHasChanged = false;
00625     bool _modelWasOpened = false;
00626     // Avoids duplicated prompts when exit is approved and Qt emits close events during shutdown.
00627     bool _closingApproved = false;
00628     QString _autoLoadPluginsFilename = "autoloadplugins.txt";
00630     bool _checkItemsScene();
00631     QString _modelfilename;
00632     std::map<std::string /*category*/,QColor>* _pluginCategoryColor = new
      std::map<std::string,QColor>();
00633     // TODO 1: Faz parte do mecanismo de restaurar dataDefinitions deletados do modelo e que são
      restaurados com um Control Z
00634     // Caso: Ao adicionar um Create no modelo e dar um check() o EntityType será criado,
00635     // mas ao deletar o Create, dar outro check() e em sequida dar um Control Z (voltando o Create no
      modelo) e checar novamente, o EntityType não é restaurado
00636     // na lista de dataDefinitions do modelo, apesar de ainda existir no Create.
00637     // Neste caso, ocorre um erro acusando que o EntityType do Create não está no modelo.
00638     // Isso se dá ao fato de que no Kernel ele verifica se _entityType = nullptr para criar um novo e
      reinserir no modelo, mas não trata o caso dele não ser nullptr
00639     // e não estar nos dataDefinitions do modelo, que é o que ocorre na situação que foi descrita.
00640     // Esta função foi feita para tratar esse caso, assim como é feito com
      insertRestoredDataDefinitions e saveDataDefinitions em ModelGraphicScene
00641     void setStatisticsCollector();
00642
00643     bool _cut;
00644     QList<GraphicalModelComponent*> * _gmc_copies  = new QList<GraphicalModelComponent*>();
00645     QList<GraphicalConnection*> * _ports_copies = new QList<GraphicalConnection*>();
00646     QList<QGraphicsItem *>  *_draw_copy = new QList<QGraphicsItem *>();
00647     QList<QGraphicsItemGroup *>  *_group_copy = new QList<QGraphicsItemGroup *>();
00648
00649     struct COPY {
00650         GraphicalModelComponent * old;
00651         GraphicalModelComponent * copy;
00652     };
00653
00654     bool _firstClickShowConnection = true;
00655
00656 private:
00657
00658     const struct TABINDEXES_STRUC {
00659         const int TabCentralModelIndex = 0;
00660         const int TabCentralSimulationIndex = 1;
00661         const int TabCentralReportsIndex = 2;
00662         //
00663         const int TabModelSimLangIndex = 0;
00664         const int TabModelCppCodeIndex = 1;
00665         const int TabModelDiagramIndex = 2;
00666         const int TabModelComponentsIndex = 3;
00667         //
00668         const int TabModelDataDefinitionsIndex = 4;
00669         const int TabSimulationBreakpointsIndex = 0;
00670         const int TabSimulationVariablesIndex = 1;
00671         const int TabSimulationEntitiesIndex = 2;
00672         const int TabSimulationTracesIndex = 3;
00673         const int TabSimulationEventsIndex = 4;
00674         //
00675         const int TabReportReportIndex = 0;
00676         const int TabReportResultIndex = 1;
```

```
00677         const int TabReportPlotIndex = 2;
00678     } CONST;
00679
00680     QUndoView *undoView = nullptr;
00681     bool _graphicalSimulation = false;
00682     bool _modelCheked = false;
00683     bool _loaded = false;
00684     bool _shuttingDown = false;
00685     // Persists lightweight tool settings introduced in Phase 5 dialogs.
00686     double _optimizerPrecision = 1e-6;
00687     unsigned int _optimizerMaxSteps = 1000;
00688     // Persists minimal parallel execution preferences configured in GUI.
00689     bool _parallelizationEnabled = false;
00690     int _parallelizationThreads = 1;
00691     int _parallelizationBatchSize = 100;
00692     // Remembers the last dataset location used by the data analyzer workflow.
00693     QString _lastDataAnalyzerPath;
00694     QMetaObject::Connection _sceneChangedConnection;
00695     QMetaObject::Connection _sceneFocusItemChangedConnection;
00696     QMetaObject::Connection _sceneSelectionChangedConnection;
00697     bool _isDeferredPropertyEditorModelChangedScheduled = false;
00698     //CodeEditor* textCodeEdit_Model;
00699 };
00700 #endif // MAINWINDOW_H
```

18.75 applications/gui/genesys/mainwindow_controller.cpp File Reference

Controller-delegation partition of MainWindow implementation.

```
#include "mainwindow.h"
#include "ui_mainwindow.h"
#include "controllers/ModelInspectorController.h"
#include "controllers/PropertyEditorController.h"
#include "controllers/PluginCatalogController.h"
#include "controllers/ModelLifecycleController.h"
#include "controllers/SimulationCommandController.h"
#include "controllers/EditCommandController.h"
#include "controllers/SceneToolController.h"
#include "controllers/DialogUtilityController.h"
#include "dialogs/dialogBreakpoint.h"
#include "dialogs/Dialogmodelinformation.h"
#include "dialogs/dialogsimulationconfigure.h"
#include "dialogs/dialogpluginmanager.h"
#include "dialogs/dialogsystempreferences.h"
#include "dialogs/DialogFind.h"
#include "controllers/SimulationController.h"
#include "services/ToolLauncher.h"
#include <string>
#include <fstream>
#include <memory>
#include <algorithm>
#include <cmath>
#include <cstdlib>
#include <QCoreApplication>
#include <QMessageBox>
#include <QTextStream>
#include <QFileDialog>
#include <QGraphicsScene>
#include <QDateTime>
#include <QEventLoop>
#include <QTemporaryFile>
#include <Qt>
#include <QGraphicsPixmapItem>
#include <QPropertyAnimation>
#include <QtWidgets/qgraphicsitem.h>
```

```
#include <QScreen>
#include <QDebug>
#include <QRegularExpression>
#include <QRandomGenerator>
#include <QSignalBlocker>
#include <QDialog>
#include <QVBoxLayout>
#include <QHBoxLayout>
#include <QFormLayout>
#include <QLineEdit>
#include <QPushButton>
#include <QDialogButtonBox>
#include <QLabel>
#include <QCheckBox>
#include <QSpinBox>
#include <QDoubleSpinBox>
#include <QPlainTextEdit>
#include <QTextEdit>
#include <QStatusBar>
#include <QImage>
#include <QPainter>
#include <QFileInfo>
#include "../../../kernel/simulator/model/ModelSimulation.h"
#include "../../../tools/Continuous/SolverDefaultImpl1.h"
```

18.75.1 Detailed Description

Controller-delegation partition of MainWindow implementation.
This file groups menu/toolbar/widget slot wrappers that delegate behavior to extracted controllers (for example lifecycle, simulation command, edit, scene tool, and dialog utility controllers). It does not define a new class; it is a thematic partition of MainWindow.

18.76 applications/gui/genesys/mainwindow_modelrepresentations.cpp File Reference

Model representation/persistence wrapper partition of MainWindow implementation.

```
#include "mainwindow.h"
#include "ui_mainwindow.h"
#include "controllers/ModelInspectorController.h"
#include "services/ModelLanguageSynchronizer.h"
#include "services/GraphvizModelExporter.h"
#include "services/CppModelExporter.h"
#include "services/GraphicalModelSerializer.h"
#include "services/GraphicalModelBuilder.h"
#include "services/EFSMNetworkDiagramProjection.h"
#include "kernel/simulator/SinkModelComponent.h"
#include "kernel/simulator/model/ModelDataManager.h"
#include "kernel/util/Util.h"
#include "plugins/data/ModalModel/EFSMNetwork.h"
#include <string>
#include <fstream>
#include <sstream>
#include <cstdlib>
#include <cctype>
#include <QMessageBox>
#include <QTextStream>
```

```
#include <QFileDialog>
#include <QDateTime>
#include <QEventLoop>
#include <QTemporaryFile>
#include <Qt>
#include <QGraphicsPixmapItem>
#include <QPropertyAnimation>
#include <QtWidgets/qgraphicsitem.h>
#include <QScreen>
#include <QDebug>
#include <QRegularExpression>
#include <QRandomGenerator>
#include <QScrollBar>
#include <QSignalBlocker>
#include <QTimer>
#include <QUrl>
```

Include dependency graph for mainwindow_modelrepresentations.cpp:

18.76.1 Detailed Description

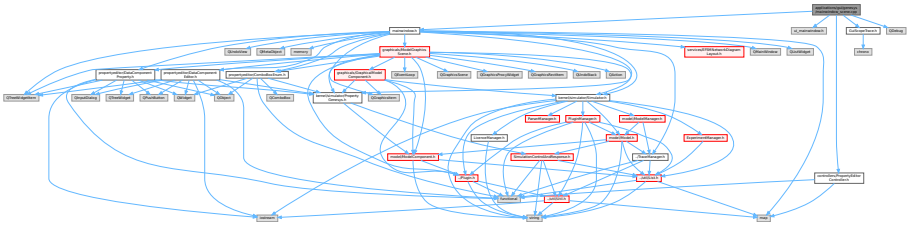
Model representation/persistence wrapper partition of MainWindow implementation.
This file groups MainWindow wrappers related to textual synchronization, Graphviz and C++ exports, graphical persistence/serialization, and graphical model reconstruction through extracted services/controllers. It does not define a new class; it is a thematic partition of MainWindow.

18.77 applications/gui/genesys/mainwindow_scene.cpp File Reference

Scene callback and scene synchronization partition of MainWindow implementation.

```
#include "mainwindow.h"
#include "ui_mainwindow.h"
#include "controllers/PropertyEditorController.h"
#include "GuiScopeTrace.h"
#include <QDebug>
```

Include dependency graph for mainwindow_scene.cpp:



18.77.1 Detailed Description

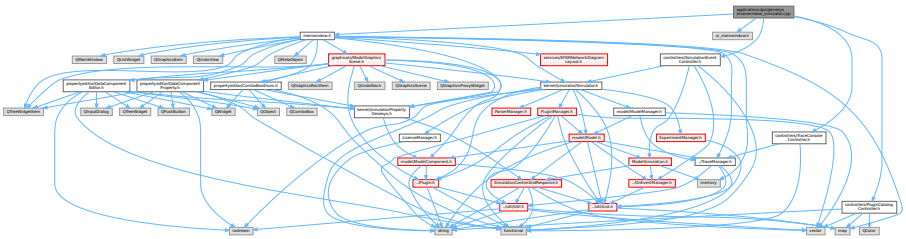
Scene callback and scene synchronization partition of MainWindow implementation.
This file concentrates scene-related callbacks, selection/focus/zoom bridging, and wrappers that synchronize scene state with UI elements such as actions and the property editor controller. It does not define a new class; it is a thematic partition of MainWindow.

18.78 applications/gui/genesys/mainwindow_simulator.cpp File Reference

Simulator trace/event wrapper partition of MainWindow implementation.

```
#include "mainwindow.h"
#include "ui_mainwindow.h"
#include "controllers/TraceConsoleController.h"
#include "controllers/SimulationEventController.h"
#include "controllers/PluginCatalogController.h"
```

Include dependency graph for mainwindow_simulator.cpp:



18.78.1 Detailed Description

Simulator trace/event wrapper partition of MainWindow implementation.

This file centralizes MainWindow compatibility wrappers for simulator traces and simulation event callbacks, delegating rendering and reactions to TraceConsoleController and SimulationEventController (plus plugin-catalog related delegation where applicable). It does not define a new class; it is a thematic partition of MainWindow.

18.79 MovedGuiCompatibilityIncludes.h

```
00001 #ifndef MOVED_GUI_COMPATIBILITY_INCLUDES_H
00002 #define MOVED_GUI_COMPATIBILITY_INCLUDES_H
00003
00004 // Temporary compatibility header for the physical GUI source-tree move.
00005 // Some legacy source files still contain multiple includes on one line after
00006 // the automated path adjustment used during the salvage move. The first include
00007 // on those lines is still seen by the preprocessor, but subsequent includes are
00008 // ignored as extra tokens. This header keeps the moved GUI buildable while those
00009 // source files are repaired incrementally.
00010
00011 #include "../../../kernel/simulator/Simulator.h"
00012 #include "../../../kernel/simulator/Plugin.h"
00013 #include "../../../kernel/simulator/PropertyGenesys.h"
00014 #include "../../../kernel/simulator/SimulationControlAndResponse.h"
00015 #include "../../../kernel/simulator/LicenceManager.h"
00016 #include "../../../kernel/simulator/SimulatorFacade.h"
00017
00018 #include "../../../kernel/simulator/model/Model.h"
00019 #include "../../../kernel/simulator/model/ModelManager.h"
00020 #include "../../../kernel/simulator/model/ModelSimulation.h"
00021 #include "../../../kernel/simulator/model/ModelComponent.h"
00022 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00023 #include "../../../kernel/simulator/model/ModelDataManager.h"
00024
00025 #include "../../../kernel/simulator/essentialPlugins/Attribute.h"
00026 #include "../../../kernel/simulator/essentialPlugins/Counter.h"
00027 #include "../../../kernel/simulator/essentialPlugins/Entity.h"
00028
00029 #include "../../../plugins/data/Logic/Variable.h"
00030 #include "../../../plugins/data/DiscreteProcessing/Queue.h"
00031
00032 #include "../../../tools/Statistics/FitterDefaultImpl.h"
00033 #include "../../../tools/Statistics/HypothesisTesterDefaultImpl1.h"
00034 #include "../../../tools/Optimization/OptimizerDefaultImpl1.h"
00035 #include "../../../tools/Continuous/SolverDefaultImpl1.h"
00036
00037 #endif // MOVED_GUI_COMPATIBILITY_INCLUDES_H
```

18.80 ComboBoxEnum.h

```
00001 #ifndef COMBOBOXENUM_H
00002 #define COMBOBOXENUM_H
00003
00004 #include <functional>
00005
00006 #include <QObject>
00007 #include <QWidget>
00008 #include <QComboBox>
00009
00010 #include "kernel/simulator/PropertyGenesys.h"
00011
00012 class ComboBoxEnum : public QObject {
00013 public:
00014     using AfterChange = std::function<void()>;
00015
00016     ComboBoxEnum(
00017         PropertyEditorGenesys* editor,
00018         SimulationControl* property,
00019         AfterChange afterChange = {}
00020         );
00021
00022     ~ComboBoxEnum() override = default;
00023
00024 public:
00025     void open_box();
00026     void configure_options(SimulationControl* property);
00027
00028 private Q_SLOTS:
00029     void changeValue();
00030
00031 private:
00032     QWidget* _window = nullptr;
00033     QComboBox* _comboBox = nullptr;
00034
00035     PropertyEditorGenesys* _editor = nullptr;
00036     SimulationControl* _property = nullptr;
00037     AfterChange _afterChange;
00038 };
00039
00040 #endif // COMBOBOXENUM_H
```

18.81 DataComponentEditor.h

```
00001 #ifndef DATACOMPONENTEDITOR_H
00002 #define DATACOMPONENTEDITOR_H
00003
00004 #include <functional>
00005 #include <iostream>
00006
00007 #include <QObject>
00008 #include <QWidget>
00009 #include <QTreeWidget>
00010 #include <QTreeWidgetItem>
00011 #include <QPushButton>
00012 #include <QInputDialog>
00013
00014 #include "kernel/simulator/PropertyGenesys.h"
00015
00016 class DataComponentProperty;
00017 class ComboBoxEnum;
00018
00019 class DataComponentEditor : public QObject {
00020 public:
00021     using AfterChange = std::function<void()>;
00022
00023     DataComponentEditor(
00024         PropertyEditorGenesys* editor,
00025         SimulationControl* property,
00026         AfterChange afterChange = {}
00027         );
00028
00029     DataComponentEditor(
00030         PropertyEditorGenesys* editor,
00031         List<SimulationControl*>* properties,
00032         AfterChange afterChange = {}
00033         );
00034
00035     ~DataComponentEditor() override = default;
00036
00037 public:
00038     void open_window(SimulationControl* property);
00039     void open_window(List<SimulationControl*>* properties);
```

```
00040
00041     void configure_properties(SimulationControl* property);
00042     void configure_properties(List<SimulationControl*>* properties);
00043
00044 private:
00045     void _notifyChanged();
00046
00047     void editProperty(SimulationControl* property);
00048     void editProperty(List<SimulationControl*>* properties);
00049
00050 private:
00051     QWidget* _window = nullptr;
00052     QTreeWidget* _view = nullptr;
00053     QPushButton* _edit = nullptr;
00054     QInputDialog* _newValue = nullptr;
00055
00056     PropertyEditorGenesys* _editor = nullptr;
00057     AfterChange _afterChange;
00058 };
00059
00060 #endif // DATACOMPONENTEDITOR_H
```

18.82 DataComponentProperty.h

```
00001 #ifndef DATACOMPONENTPROPERTY_H
00002 #define DATACOMPONENTPROPERTY_H
00003
00004 #include <functional>
00005 #include <iostream>
00006
00007 #include <QObject>
00008 #include <QWidget>
00009 #include <QTreeWidget>
00010 #include <QTreeWidgetItem>
00011 #include <QPushButton>
00012 #include <QInputDialog>
00013
00014 #include "kernel/simulator/PropertyGenesys.h"
00015
00016 class DataComponentEditor;
00017 class ComboBoxEnum;
00018
00019 class DataComponentProperty : public QObject {
00020 public:
00021     using AfterChange = std::function<void()>;
00022
00023     DataComponentProperty(
00024         PropertyEditorGenesys* editor,
00025         SimulationControl* property,
00026         bool necessaryConfig,
00027         AfterChange afterChange = {}
00028         );
00029
00030     ~DataComponentProperty() override = default;
00031
00032 public:
00033     void open_window();
00034     void config_values();
00035     bool isInList(const std::string& value) const;
00036
00037 private:
00038     void _notifyChanged();
00039
00040 private Q_SLOTS:
00041     void addElement();
00042     void removeElement();
00043     void editProperty();
00044
00045 private:
00046     QWidget* _window = nullptr;
00047     QTreeWidget* _view = nullptr;
00048     QPushButton* _add = nullptr;
00049     QPushButton* _remove = nullptr;
00050     QPushButton* _edit = nullptr;
00051     QInputDialog* _confirmation = nullptr;
00052
00053     PropertyEditorGenesys* _editor = nullptr;
00054     SimulationControl* _property = nullptr;
00055     AfterChange _afterChange;
00056 };
00057
00058 #endif // DATACOMPONENTPROPERTY_H
```

18.83 ObjectPropertyBrowser.h

```
00001 #ifndef OBJECTPROPERTYBROWSER_H
00002 #define OBJECTPROPERTYBROWSER_H
00003
00004 #include <map>
00005 #include <vector>
00006 #include <set>
00007 #include <functional>
00008
00009 #include <QObject>
00010 #include <QBrush>
00011 #include <QPen>
00012 #include <QPointer>
00013 #include <QMap>
00014 #include <QSet>
00015 #include <QStringList>
00016 #include <QContextMenuEvent>
00017 #include <QGraphicsItem>
00018 #include <QKeyEvent>
00019 #include <QMouseEvent>
00020 #include <QVariant>
00021
00022 #include "qtpropertybrowser/qttreepropertybrowser.h"
00023 #include "qtpropertybrowser/qtvariantproperty.h"
00024 #include "qtpropertybrowser/qtpropertymanager.h"
00025 #include "qtpropertybrowser/qteditorfactory.h"
00026
00027 #include "DataComponentProperty.h"
00028 #include "DataComponentEditor.h"
00029 #include "ComboBoxEnum.h"
00030
00031 #include "../../../../kernel/simulator/model/ModelDataDefinition.h"
00032 #include "kernel/simulator/PropertyGenesys.h"
00033 #include "kernel/simulator/GenesysPropertyIntrospection.h"
00034
00035 class ObjectPropertyBrowser : public QtTreePropertyBrowser {
00036     Q_OBJECT
00037
00038 public:
00039     explicit ObjectPropertyBrowser(QWidget* parent = nullptr);
00040     using ModelChangedCallback = std::function<void()>;
00041
00042     void setActiveObject(
00043         QObject *obj,
00044         ModelDataDefinition* mdd = nullptr,
00045         const QSet<QString>& graphicallyRepresentedModelObjects = {},
00046         const QSet<QString>& editableModelObjects = {},
00047         PropertyEditorGenesys* peg = nullptr,
00048         std::map<SimulationControl*, DataComponentProperty*>* pl = nullptr,
00049         std::map<SimulationControl*, DataComponentEditor*>* peUI = nullptr,
00050         std::map<SimulationControl*, ComboBoxEnum*>* pb = nullptr
00051         );
00052     void setActiveGraphicsItem(QGraphicsItem* item);
00053
00054     void clearCurrentlyConnectedObject();
00055     void setModelChangedCallback(ModelChangedCallback callback);
00056     bool isCommitPipelineBusy() const;
00057
00058 private:
00059     enum class ActiveMode {
00060         None,
00061         KernelObject,
00062         GraphicsItem
00063     };
00064
00065     enum class BindingKind {
00066         Kernel,
00067         GraphicsVariant,
00068         GraphicsEnum
00069     };
00070
00071     struct Binding {
00072         BindingKind kind = BindingKind::Kernel;
00073         ModelDataDefinition* owner = nullptr;
00074         SimulationControl* control = nullptr;
00075         GenesysPropertyDescriptor descriptor;
00076         bool isObjectSelector = false;
00077         bool isModelObjectAction = false;
00078         bool isObjectListAction = false;
00079         bool isMultilineTextAction = false;
00080         QGraphicsItem* graphicsItem = nullptr;
00081         bool graphicsRequiresCommit = false;
00082         std::function<void(const QVariant&)> graphicsVariantSetter;
00083         std::function<void(int)> graphicsEnumSetter;
00084     };
00085
```

```
00086     struct ModelObjectRelation {
00087         bool exists = false;
00088         bool internal = false;
00089         std::string key;
00090         ModelDataDefinition* dataDefinition = nullptr;
00091     };
00092
00093     struct ObjectListAction {
00094         enum class Kind {
00095             None,
00096             CreateNew,
00097             SetCurrentType
00098         };
00099
00100         Kind kind = Kind::None;
00101         std::string typeName;
00102     };
00103
00104     struct ModelObjectAction {
00105         enum class Kind {
00106             None,
00107             SetReference,
00108             CreateNew,
00109             SelectAndEdit,
00110             RemoveReference
00111         };
00112
00113         Kind kind = Kind::None;
00114         std::string objectValue;
00115         std::string typeName;
00116     };
00117
00118 private:
00119     // Ensure the property browser infrastructure is created exactly once.
00120     void _ensureBrowserInfrastructure();
00121     void _notifyModelChangeApplied();
00122     void _clearAll();
00123     void _rebuildProperties();
00124     void _populateKernelProperties(ModelDataDefinition* mdd);
00125     void _populateGraphicsProperties(QGraphicsItem* item);
00126     void _appendDescriptorRecursively(
00127         QtProperty* parent,
00128         SimulationControl* control,
00129         std::set<const SimulationControl*>& recursionPath,
00130         int depth = 0
00131         );
00132
00133     QtProperty* _createLeafProperty(const GenesysPropertyDescriptor& desc);
00134     QtProperty* _createObjectListProperty(
00135         SimulationControl* control,
00136         const GenesysPropertyDescriptor& desc,
00137         std::set<const SimulationControl*>& recursionPath,
00138         int depth
00139         );
00140     QtProperty* _createModelObjectActionProperty(const GenesysPropertyDescriptor& desc);
00141     bool _isObjectListProperty(const GenesysPropertyDescriptor& desc) const;
00142     bool _isKernelEditingEnabled(const GenesysPropertyDescriptor& desc) const;
00143     bool _hasGraphicalRepresentation(const GenesysPropertyDescriptor& desc) const;
00144     bool _isModelObjectActionProperty(const GenesysPropertyDescriptor& desc, SimulationControl*
      control) const;
00145     ModelObjectRelation _relationForDataDefinition(ModelDataDefinition* dataDefinition) const;
00146     ModelDataDefinition* _referencedModelDataDefinition(const Binding& binding) const;
00147     QString _modelObjectTypeName(const GenesysPropertyDescriptor& desc) const;
00148     QString _defaultModelObjectName(const GenesysPropertyDescriptor& desc, const std::string&
      concreteTypeName = "") const;
00149     bool _isRegisteredModelDataDefinition(ModelDataDefinition* dataDefinition) const;
00150     void _materializeAffectedModelDataDefinitions(ModelDataDefinition* referencedDataDefinition =
      nullptr) const;
00151     bool _createNewListElementForProperty(QtProperty* property, const std::string& typeName = "");
00152     bool _setCurrentListElementTypeForProperty(QtProperty* property, const std::string& typeName);
00153     bool _createModelObjectForProperty(QtProperty* property, const std::string& typeName = "");
00154     bool _setModelObjectReferenceForProperty(QtProperty* property, const QString& objectName);
00155     bool _selectModelObjectForProperty(QtProperty* property);
00156     bool _removeModelObjectReferenceForProperty(QtProperty* property);
00157     bool _deleteModelObjectForProperty(QtProperty* property);
00158     bool _applyObjectListSelection(QtProperty* property, int value);
00159     bool _applyModelObjectReferenceSelection(QtProperty* property, int value);
00160     void _synchronizeGraphicalModelDataDefinitionsNow() const;
00161     QtVariantProperty* _createGraphicsVariantProperty(
00162         const QString& name,
00163         int variantType,
00164         const QVariant& value,
00165         std::function<void(const QVariant&)> setter,
00166         bool requiresCommit = false,
00167         bool enabled = true
00168         );
00169     QtProperty* _createGraphicsEnumProperty(
```

```
00170         const QString& name,
00171         const QStringList& choices,
00172         int value,
00173         std::function<void(int)> setter,
00174         bool enabled = true
00175         );
00176     void _appendCommonGraphicsProperties(QtProperty* group, QGraphicsItem* item);
00177     void _appendPenProperties(QtProperty* group, const QPen& pen, std::function<void(const QPen&)>
      setter);
00178     void _appendBrushProperties(QtProperty* group, const QBrush& brush, std::function<void(const
      QBrush&)> setter);
00179     void _notifyGraphicsChangeApplied(QGraphicsItem* item);
00180     QVariant _toVariant(const GenesysPropertyDescriptor& desc) const;
00181     std::string _fromVariant(const GenesysPropertyDescriptor& desc, const QVariant& value) const;
00182     int _enumIndexFor(const GenesysPropertyDescriptor& desc) const;
00183     QStringList _toQStringList(const std::vector<std::string>& values) const;
00184     bool _hasSpecializedEditor(const GenesysPropertyDescriptor& desc) const;
00185     QString _specializedEditorActionText(const GenesysPropertyDescriptor& desc) const;
00186
00187     bool _openSpecializedEditorForCurrentItem();
00188     bool _openSpecializedEditor(QtProperty* property);
00189     bool _openTextDialogEditor(const Binding& binding, bool compactEditor);
00190     bool _openMultilineTextDialogEditor(const Binding& binding);
00191     bool _createObjectForProperty(QtProperty* property);
00192
00193 protected:
00194     void keyPressEvent(QKeyEvent* event) override;
00195     void contextMenuEvent(QContextMenuEvent* event) override;
00196     void mouseDoubleClickEvent(QMouseEvent* event) override;
00197
00198 private:
00199     // Guard against reentrant callback chains during property rebuild and notification.
00200     bool _isRebuildingProperties = false;
00201     // Guard against nested model-change notifications triggered by editor callbacks.
00202     bool _isNotifyingModelChange = false;
00203     // Keep track of pending rebuild requests raised during guarded execution.
00204     bool _pendingRebuild = false;
00205     // Coalesce queued rebuild dispatches while one deferred rebuild request is already scheduled.
00206     bool _isDeferredRebuildScheduled = false;
00207     // Coalesce queued model-change callbacks while one deferred dispatch is already scheduled.
00208     bool _isDeferredModelChangedScheduled = false;
00209     // Prevent recursive handling when a multiline text action resets its combo-box selection.
00210     bool _isHandlingMultilineTextAction = false;
00211
00212 private:
00213     // Execute property rebuild with reentrancy suppression and deferred retry support.
00214     void _rebuildPropertiesGuarded();
00215     // Request a deferred rebuild after current signal/slot stack finishes.
00216     void _scheduleDeferredRebuild();
00217     // Request a deferred model-changed callback after current signal/slot stack finishes.
00218     void _scheduleDeferredModelChangedCallback();
00219     // Check whether active editor bindings are currently valid before mutating model state.
00220     bool _hasValidActiveBindingContext(QtProperty* property = nullptr) const;
00221
00222 private:
00223     // Track the selected graphical object safely in case Qt destroys it during callbacks.
00224     QPointer<QObject> _graphicalObject = nullptr;
00225     QGraphicsItem* _graphicalItem = nullptr;
00226     ModelDataDefinition* _modelObject = nullptr;
00227     ActiveMode _activeMode = ActiveMode::None;
00228     QSet<QString> _graphicallyRepresentedModelObjects;
00229     QSet<QString> _editableModelObjects;
00230     bool _activeKernelObjectReadOnly = false;
00231     PropertyEditorGenesys* _propertyEditor = nullptr;
00232
00233     std::map<SimulationControl*, DataComponentProperty*>* _propertyList = nullptr;
00234     std::map<SimulationControl*, DataComponentEditor*>* _propertyEditorUI = nullptr;
00235     std::map<SimulationControl*, ComboBoxEnum*>* _propertyBox = nullptr;
00236
00237     QtVariantPropertyManager* _variantManager = nullptr;
00238     QtGroupPropertyManager* _groupManager = nullptr;
00239     QtEnumPropertyManager* _enumManager = nullptr;
00240
00241     QtVariantEditorFactory* _variantFactory = nullptr;
00242     QtEnumEditorFactory* _enumFactory = nullptr;
00243     // Track whether factories were already bound to managers in the browser.
00244     bool _browserInfrastructureBound = false;
00245
00246     QMap<QtProperty*, Binding> _bindings;
00247     QMap<QtProperty*, QStringList> _enumNames;
00248     QMap<QtProperty*, std::vector<ObjectListAction>> _objectListActions;
00249     QMap<QtProperty*, std::vector<ModelObjectAction>> _modelObjectActions;
00250     ModelChangedCallback _modelChangedCallback;
00251
00252 private:
00253     bool _requiresCommitConfirmation(const Binding& binding) const;
00254     bool _applyVariantChange(QtProperty* property, const QVariant& value, bool committed);
```

```
00255     bool _applyGraphicsVariantChange(QtProperty* property, const QVariant& value, bool committed);
00256     bool _applyGraphicsEnumChange(QtProperty* property, int value);
00257
00258 private slots:
00259     void valueChanged(QtProperty *property, const QVariant &value);
00260     void enumValueChanged(QtProperty *property, int value);
00261     void onVariantEditorCommitted(QtProperty* property, const QVariant& committedValue = QVariant());
00262
00263 public slots:
00264     void objectUpdated();
00265 };
00266
00267 #endif // OBJECTPROPERTYBROWSER_H
```

18.84 PropertyEditor.h

```
00001 #ifndef PROPERTYEDITOR_H
00002 #define PROPERTYEDITOR_H
00003
00004 #include <QTreeWidget>
00005 #include <QObject>
00006
00007 class ModelDataDefinition;
00008
00009 class PropertyEditor : public QTreeWidget {
00010 public:
00011     PropertyEditor(QWidget *parent = nullptr);
00012 public:
00013     void setModelBlock(ModelDataDefinition* modelblock);
00014 };
00015
00016 #endif // PROPERTYEDITOR_H
```

18.85 qtbuttonpropertybrowser.h

```
00001 /****************************************************************************
00002 **
00003 ** Copyright (C) 2016 The Qt Company Ltd.
00004 ** Contact: https://www.qt.io/licensing/
00005 **
00006 ** This file is part of the tools applications of the Qt Toolkit.
00007 **
00008 ** $QT_BEGIN_LICENSE:LGPL$
00009 ** Commercial License Usage
00010 ** Licensees holding valid commercial Qt licenses may use this file in
00011 ** accordance with the commercial license agreement provided with the
00012 ** Software or, alternatively, in accordance with the terms contained in
00013 ** a written agreement between you and The Qt Company. For licensing terms
00014 ** and conditions see https://www.qt.io/terms-conditions. For further
00015 ** information use the contact form at https://www.qt.io/contact-us.
00016 **
00017 ** GNU Lesser General Public License Usage
00018 ** Alternatively, this file may be used under the terms of the GNU Lesser
00019 ** General Public License version 3 as published by the Free Software
00020 ** Foundation and appearing in the file LICENSE.LGPL3 included in the
00021 ** packaging of this file. Please review the following information to
00022 ** ensure the GNU Lesser General Public License version 3 requirements
00023 ** will be met: https://www.gnu.org/licenses/lgpl-3.0.html.
00024 **
00025 ** GNU General Public License Usage
00026 ** Alternatively, this file may be used under the terms of the GNU
00027 ** General Public License version 2.0 or (at your option) the GNU General
00028 ** Public license version 3 or any later version approved by the KDE Free
00029 ** Qt Foundation. The licenses are as published by the Free Software
00030 ** Foundation and appearing in the file LICENSE.GPL2 and LICENSE.GPL3
00031 ** included in the packaging of this file. Please review the following
00032 ** information to ensure the GNU General Public License requirements will
00033 ** be met: https://www.gnu.org/licenses/gpl-2.0.html and
00034 ** https://www.gnu.org/licenses/gpl-3.0.html.
00035 **
00036 ** $QT_END_LICENSE$
00037 **
00038 ****************************************************************************/
00039
00040 #ifndef QTBUTTONPROPERTYBROWSER_H
00041 #define QTBUTTONPROPERTYBROWSER_H
00042
00043 #include "qtpropertybrowser.h"
00044
00045 QT_BEGIN_NAMESPACE
00046
00047 class QtButtonPropertyBrowserPrivate;
```

```
00048
00049 class QT_QTPROPERTYBROWSER_EXPORT QtButtonPropertyBrowser : public QtAbstractPropertyBrowser
00050 {
00051     Q_OBJECT
00052 public:
00053
00054     QtButtonPropertyBrowser(QWidget *parent = 0);
00055     ~QtButtonPropertyBrowser();
00056
00057     void setExpanded(QtBrowserItem *item, bool expanded);
00058     bool isExpanded(QtBrowserItem *item) const;
00059
00060 Q_SIGNALS:
00061
00062     void collapsed(QtBrowserItem *item);
00063     void expanded(QtBrowserItem *item);
00064
00065 protected:
00066     void itemInserted(QtBrowserItem *item, QtBrowserItem *afterItem) override;
00067     void itemRemoved(QtBrowserItem *item) override;
00068     void itemChanged(QtBrowserItem *item) override;
00069
00070 private:
00071
00072     QScopedPointer<QtButtonPropertyBrowserPrivate> d_ptr;
00073     Q_DECLARE_PRIVATE(QtButtonPropertyBrowser)
00074     Q_DISABLE_COPY_MOVE(QtButtonPropertyBrowser)
00075     Q_PRIVATE_SLOT(d_func(), void slotUpdate())
00076     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed())
00077     Q_PRIVATE_SLOT(d_func(), void slotToggled(bool))
00078
00079 };
00080
00081 QT_END_NAMESPACE
00082
00083 #endif
```

18.86 qteditorfactory.h

```
00001 /****************************************************************************
00002 **
00003 ** Copyright (C) 2016 The Qt Company Ltd.
00004 ** Contact: https://www.qt.io/licensing/
00005 **
00006 ** This file is part of the tools applications of the Qt Toolkit.
00007 **
00008 ** $QT_BEGIN_LICENSE:LGPL$
00009 ** Commercial License Usage
00010 ** Licensees holding valid commercial Qt licenses may use this file in
00011 ** accordance with the commercial license agreement provided with the
00012 ** Software or, alternatively, in accordance with the terms contained in
00013 ** a written agreement between you and The Qt Company. For licensing terms
00014 ** and conditions see https://www.qt.io/terms-conditions. For further
00015 ** information use the contact form at https://www.qt.io/contact-us.
00016 **
00017 ** GNU Lesser General Public License Usage
00018 ** Alternatively, this file may be used under the terms of the GNU Lesser
00019 ** General Public License version 3 as published by the Free Software
00020 ** Foundation and appearing in the file LICENSE.LGPL3 included in the
00021 ** packaging of this file. Please review the following information to
00022 ** ensure the GNU Lesser General Public License version 3 requirements
00023 ** will be met: https://www.gnu.org/licenses/lgpl-3.0.html.
00024 **
00025 ** GNU General Public License Usage
00026 ** Alternatively, this file may be used under the terms of the GNU
00027 ** General Public License version 2.0 or (at your option) the GNU General
00028 ** Public license version 3 or any later version approved by the KDE Free
00029 ** Qt Foundation. The licenses are as published by the Free Software
00030 ** Foundation and appearing in the file LICENSE.GPL2 and LICENSE.GPL3
00031 ** included in the packaging of this file. Please review the following
00032 ** information to ensure the GNU General Public License requirements will
00033 ** be met: https://www.gnu.org/licenses/gpl-2.0.html and
00034 ** https://www.gnu.org/licenses/gpl-3.0.html.
00035 **
00036 ** $QT_END_LICENSE$
00037 **
00038 ****************************************************************************/
00039
00040 #ifndef QTEDITORFACTORY_H
00041 #define QTEDITORFACTORY_H
00042
00043 #include "qtpropertymanager.h"
00044
00045 QT_BEGIN_NAMESPACE
00046
```

```
00047 class QRegularExpression;
00048
00049 class QtSpinBoxFactoryPrivate;
00050
00051 class QT_QTPROPERTYBROWSER_EXPORT QtSpinBoxFactory : public
      QtAbstractEditorFactory<QtIntPropertyManager>
00052 {
00053     Q_OBJECT
00054 public:
00055     QtSpinBoxFactory(QObject *parent = 0);
00056     ~QtSpinBoxFactory();
00057 protected:
00058     void connectPropertyManager(QtIntPropertyManager *manager) override;
00059     QWidget *createEditor(QtIntPropertyManager *manager, QtProperty *property,
00060                 QWidget *parent) override;
00061     void disconnectPropertyManager(QtIntPropertyManager *manager) override;
00062 private:
00063     QScopedPointer<QtSpinBoxFactoryPrivate> d_ptr;
00064     Q_DECLARE_PRIVATE(QtSpinBoxFactory)
00065     Q_DISABLE_COPY_MOVE(QtSpinBoxFactory)
00066     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, int))
00067     Q_PRIVATE_SLOT(d_func(), void slotRangeChanged(QtProperty *, int, int))
00068     Q_PRIVATE_SLOT(d_func(), void slotSingleStepChanged(QtProperty *, int))
00069     Q_PRIVATE_SLOT(d_func(), void slotSetValue(int))
00070     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00071 };
00072
00073 class QtSliderFactoryPrivate;
00074
00075 class QT_QTPROPERTYBROWSER_EXPORT QtSliderFactory : public
      QtAbstractEditorFactory<QtIntPropertyManager>
00076 {
00077     Q_OBJECT
00078 public:
00079     QtSliderFactory(QObject *parent = 0);
00080     ~QtSliderFactory();
00081 protected:
00082     void connectPropertyManager(QtIntPropertyManager *manager) override;
00083     QWidget *createEditor(QtIntPropertyManager *manager, QtProperty *property,
00084                 QWidget *parent) override;
00085     void disconnectPropertyManager(QtIntPropertyManager *manager) override;
00086 private:
00087     QScopedPointer<QtSliderFactoryPrivate> d_ptr;
00088     Q_DECLARE_PRIVATE(QtSliderFactory)
00089     Q_DISABLE_COPY_MOVE(QtSliderFactory)
00090     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, int))
00091     Q_PRIVATE_SLOT(d_func(), void slotRangeChanged(QtProperty *, int, int))
00092     Q_PRIVATE_SLOT(d_func(), void slotSingleStepChanged(QtProperty *, int))
00093     Q_PRIVATE_SLOT(d_func(), void slotSetValue(int))
00094     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00095 };
00096
00097 class QtScrollBarFactoryPrivate;
00098
00099 class QT_QTPROPERTYBROWSER_EXPORT QtScrollBarFactory : public
      QtAbstractEditorFactory<QtIntPropertyManager>
00100 {
00101     Q_OBJECT
00102 public:
00103     QtScrollBarFactory(QObject *parent = 0);
00104     ~QtScrollBarFactory();
00105 protected:
00106     void connectPropertyManager(QtIntPropertyManager *manager) override;
00107     QWidget *createEditor(QtIntPropertyManager *manager, QtProperty *property,
00108                 QWidget *parent) override;
00109     void disconnectPropertyManager(QtIntPropertyManager *manager) override;
00110 private:
00111     QScopedPointer<QtScrollBarFactoryPrivate> d_ptr;
00112     Q_DECLARE_PRIVATE(QtScrollBarFactory)
00113     Q_DISABLE_COPY_MOVE(QtScrollBarFactory)
00114     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, int))
00115     Q_PRIVATE_SLOT(d_func(), void slotRangeChanged(QtProperty *, int, int))
00116     Q_PRIVATE_SLOT(d_func(), void slotSingleStepChanged(QtProperty *, int))
00117     Q_PRIVATE_SLOT(d_func(), void slotSetValue(int))
00118     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00119 };
00120
00121 class QtCheckBoxFactoryPrivate;
00122
00123 class QT_QTPROPERTYBROWSER_EXPORT QtCheckBoxFactory : public
      QtAbstractEditorFactory<QtBoolPropertyManager>
00124 {
00125     Q_OBJECT
00126 public:
00127     QtCheckBoxFactory(QObject *parent = 0);
00128     ~QtCheckBoxFactory();
00129 protected:
```

```
00130     void connectPropertyManager(QtBoolPropertyManager *manager) override;
00131     QWidget *createEditor(QtBoolPropertyManager *manager, QtProperty *property,
00132                 QWidget *parent) override;
00133     void disconnectPropertyManager(QtBoolPropertyManager *manager) override;
00134 private:
00135     QScopedPointer<QtCheckBoxFactoryPrivate> d_ptr;
00136     Q_DECLARE_PRIVATE(QtCheckBoxFactory)
00137     Q_DISABLE_COPY_MOVE(QtCheckBoxFactory)
00138     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, bool))
00139     Q_PRIVATE_SLOT(d_func(), void slotSetValue(bool))
00140     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00141 };
00142
00143 class QtDoubleSpinBoxFactoryPrivate;
00144
00145 class QT_QTPROPERTYBROWSER_EXPORT QtDoubleSpinBoxFactory : public
    QtAbstractEditorFactory<QtDoublePropertyManager>
00146 {
00147     Q_OBJECT
00148 public:
00149     QtDoubleSpinBoxFactory(QObject *parent = 0);
00150     ~QtDoubleSpinBoxFactory();
00151 protected:
00152     void connectPropertyManager(QtDoublePropertyManager *manager) override;
00153     QWidget *createEditor(QtDoublePropertyManager *manager, QtProperty *property,
00154                 QWidget *parent) override;
00155     void disconnectPropertyManager(QtDoublePropertyManager *manager) override;
00156 private:
00157     QScopedPointer<QtDoubleSpinBoxFactoryPrivate> d_ptr;
00158     Q_DECLARE_PRIVATE(QtDoubleSpinBoxFactory)
00159     Q_DISABLE_COPY_MOVE(QtDoubleSpinBoxFactory)
00160     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, double))
00161     Q_PRIVATE_SLOT(d_func(), void slotRangeChanged(QtProperty *, double, double))
00162     Q_PRIVATE_SLOT(d_func(), void slotSingleStepChanged(QtProperty *, double))
00163     Q_PRIVATE_SLOT(d_func(), void slotDecimalsChanged(QtProperty *, int))
00164     Q_PRIVATE_SLOT(d_func(), void slotSetValue(double))
00165     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00166 };
00167
00168 class QtLineEditFactoryPrivate;
00169
00170 class QT_QTPROPERTYBROWSER_EXPORT QtLineEditFactory : public
    QtAbstractEditorFactory<QtStringPropertyManager>
00171 {
00172     Q_OBJECT
00173 public:
00174     QtLineEditFactory(QObject *parent = 0);
00175     ~QtLineEditFactory();
00176 protected:
00177     void connectPropertyManager(QtStringPropertyManager *manager) override;
00178     QWidget *createEditor(QtStringPropertyManager *manager, QtProperty *property,
00179                 QWidget *parent) override;
00180     void disconnectPropertyManager(QtStringPropertyManager *manager) override;
00181 private:
00182     QScopedPointer<QtLineEditFactoryPrivate> d_ptr;
00183     Q_DECLARE_PRIVATE(QtLineEditFactory)
00184     Q_DISABLE_COPY_MOVE(QtLineEditFactory)
00185     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, const QString &))
00186     Q_PRIVATE_SLOT(d_func(), void slotRegExpChanged(QtProperty *, const QRegularExpression &))
00187     Q_PRIVATE_SLOT(d_func(), void slotSetValue(const QString &))
00188     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00189 };
00190
00191 class QtDateEditFactoryPrivate;
00192
00193 class QT_QTPROPERTYBROWSER_EXPORT QtDateEditFactory : public
    QtAbstractEditorFactory<QtDatePropertyManager>
00194 {
00195     Q_OBJECT
00196 public:
00197     QtDateEditFactory(QObject *parent = 0);
00198     ~QtDateEditFactory();
00199 protected:
00200     void connectPropertyManager(QtDatePropertyManager *manager) override;
00201     QWidget *createEditor(QtDatePropertyManager *manager, QtProperty *property,
00202                 QWidget *parent) override;
00203     void disconnectPropertyManager(QtDatePropertyManager *manager) override;
00204 private:
00205     QScopedPointer<QtDateEditFactoryPrivate> d_ptr;
00206     Q_DECLARE_PRIVATE(QtDateEditFactory)
00207     Q_DISABLE_COPY_MOVE(QtDateEditFactory)
00208     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, QDate))
00209     Q_PRIVATE_SLOT(d_func(), void slotRangeChanged(QtProperty *, QDate, QDate))
00210     Q_PRIVATE_SLOT(d_func(), void slotSetValue(QDate))
00211     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00212 };
00213
```

```
00214 class QtTimeEditFactoryPrivate;
00215
00216 class QT_QTPROPERTYBROWSER_EXPORT QtTimeEditFactory : public
      QtAbstractEditorFactory<QtTimePropertyManager>
00217 {
00218     Q_OBJECT
00219 public:
00220     QtTimeEditFactory(QObject *parent = 0);
00221     ~QtTimeEditFactory();
00222 protected:
00223     void connectPropertyManager(QtTimePropertyManager *manager) override;
00224     QWidget *createEditor(QtTimePropertyManager *manager, QtProperty *property,
00225                 QWidget *parent) override;
00226     void disconnectPropertyManager(QtTimePropertyManager *manager) override;
00227 private:
00228     QScopedPointer<QtTimeEditFactoryPrivate> d_ptr;
00229     Q_DECLARE_PRIVATE(QtTimeEditFactory)
00230     Q_DISABLE_COPY_MOVE(QtTimeEditFactory)
00231     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, QTime))
00232     Q_PRIVATE_SLOT(d_func(), void slotSetValue(QTime))
00233     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00234 };
00235
00236 class QtDateTimeEditFactoryPrivate;
00237
00238 class QT_QTPROPERTYBROWSER_EXPORT QtDateTimeEditFactory : public
      QtAbstractEditorFactory<QtDateTimePropertyManager>
00239 {
00240     Q_OBJECT
00241 public:
00242     QtDateTimeEditFactory(QObject *parent = 0);
00243     ~QtDateTimeEditFactory();
00244 protected:
00245     void connectPropertyManager(QtDateTimePropertyManager *manager) override;
00246     QWidget *createEditor(QtDateTimePropertyManager *manager, QtProperty *property,
00247                 QWidget *parent) override;
00248     void disconnectPropertyManager(QtDateTimePropertyManager *manager) override;
00249 private:
00250     QScopedPointer<QtDateTimeEditFactoryPrivate> d_ptr;
00251     Q_DECLARE_PRIVATE(QtDateTimeEditFactory)
00252     Q_DISABLE_COPY_MOVE(QtDateTimeEditFactory)
00253     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, const QDateTime &))
00254     Q_PRIVATE_SLOT(d_func(), void slotSetValue(const QDateTime &))
00255     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00256 };
00257
00258 class QtKeySequenceEditorFactoryPrivate;
00259
00260 class QT_QTPROPERTYBROWSER_EXPORT QtKeySequenceEditorFactory : public
      QtAbstractEditorFactory<QtKeySequencePropertyManager>
00261 {
00262     Q_OBJECT
00263 public:
00264     QtKeySequenceEditorFactory(QObject *parent = 0);
00265     ~QtKeySequenceEditorFactory();
00266 protected:
00267     void connectPropertyManager(QtKeySequencePropertyManager *manager) override;
00268     QWidget *createEditor(QtKeySequencePropertyManager *manager, QtProperty *property,
00269                 QWidget *parent) override;
00270     void disconnectPropertyManager(QtKeySequencePropertyManager *manager) override;
00271 private:
00272     QScopedPointer<QtKeySequenceEditorFactoryPrivate> d_ptr;
00273     Q_DECLARE_PRIVATE(QtKeySequenceEditorFactory)
00274     Q_DISABLE_COPY_MOVE(QtKeySequenceEditorFactory)
00275     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, const QKeySequence &))
00276     Q_PRIVATE_SLOT(d_func(), void slotSetValue(const QKeySequence &))
00277     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00278 };
00279
00280 class QtCharEditorFactoryPrivate;
00281
00282 class QT_QTPROPERTYBROWSER_EXPORT QtCharEditorFactory : public
      QtAbstractEditorFactory<QtCharPropertyManager>
00283 {
00284     Q_OBJECT
00285 public:
00286     QtCharEditorFactory(QObject *parent = 0);
00287     ~QtCharEditorFactory();
00288 protected:
00289     void connectPropertyManager(QtCharPropertyManager *manager) override;
00290     QWidget *createEditor(QtCharPropertyManager *manager, QtProperty *property,
00291                 QWidget *parent) override;
00292     void disconnectPropertyManager(QtCharPropertyManager *manager) override;
00293 private:
00294     QScopedPointer<QtCharEditorFactoryPrivate> d_ptr;
00295     Q_DECLARE_PRIVATE(QtCharEditorFactory)
00296     Q_DISABLE_COPY_MOVE(QtCharEditorFactory)
```

```
00297     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, const QChar &))
00298     Q_PRIVATE_SLOT(d_func(), void slotSetValue(const QChar &))
00299     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00300 };
00301
00302 class QtEnumEditorFactoryPrivate;
00303
00304 class QT_QTPROPERTYBROWSER_EXPORT QtEnumEditorFactory : public
      QtAbstractEditorFactory<QtEnumPropertyManager>
00305 {
00306     Q_OBJECT
00307 public:
00308     QtEnumEditorFactory(QObject *parent = 0);
00309     ~QtEnumEditorFactory();
00310 protected:
00311     void connectPropertyManager(QtEnumPropertyManager *manager) override;
00312     QWidget *createEditor(QtEnumPropertyManager *manager, QtProperty *property,
00313                 QWidget *parent) override;
00314     void disconnectPropertyManager(QtEnumPropertyManager *manager) override;
00315 private:
00316     QScopedPointer<QtEnumEditorFactoryPrivate> d_ptr;
00317     Q_DECLARE_PRIVATE(QtEnumEditorFactory)
00318     Q_DISABLE_COPY_MOVE(QtEnumEditorFactory)
00319     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, int))
00320     Q_PRIVATE_SLOT(d_func(), void slotEnumNamesChanged(QtProperty *,
00321                         const QStringList &))
00322     Q_PRIVATE_SLOT(d_func(), void slotEnumIconsChanged(QtProperty *,
00323                         const QMap<int, QIcon> &))
00324     Q_PRIVATE_SLOT(d_func(), void slotSetValue(int))
00325     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00326 };
00327
00328 class QtCursorEditorFactoryPrivate;
00329
00330 class QT_QTPROPERTYBROWSER_EXPORT QtCursorEditorFactory : public
      QtAbstractEditorFactory<QtCursorPropertyManager>
00331 {
00332     Q_OBJECT
00333 public:
00334     QtCursorEditorFactory(QObject *parent = 0);
00335     ~QtCursorEditorFactory();
00336 protected:
00337     void connectPropertyManager(QtCursorPropertyManager *manager) override;
00338     QWidget *createEditor(QtCursorPropertyManager *manager, QtProperty *property,
00339                 QWidget *parent) override;
00340     void disconnectPropertyManager(QtCursorPropertyManager *manager) override;
00341 private:
00342     QScopedPointer<QtCursorEditorFactoryPrivate> d_ptr;
00343     Q_DECLARE_PRIVATE(QtCursorEditorFactory)
00344     Q_DISABLE_COPY_MOVE(QtCursorEditorFactory)
00345     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, const QCursor &))
00346     Q_PRIVATE_SLOT(d_func(), void slotEnumChanged(QtProperty *, int))
00347     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00348 };
00349
00350 class QtColorEditorFactoryPrivate;
00351
00352 class QT_QTPROPERTYBROWSER_EXPORT QtColorEditorFactory : public
      QtAbstractEditorFactory<QtColorPropertyManager>
00353 {
00354     Q_OBJECT
00355 public:
00356     QtColorEditorFactory(QObject *parent = 0);
00357     ~QtColorEditorFactory();
00358 protected:
00359     void connectPropertyManager(QtColorPropertyManager *manager) override;
00360     QWidget *createEditor(QtColorPropertyManager *manager, QtProperty *property,
00361                 QWidget *parent) override;
00362     void disconnectPropertyManager(QtColorPropertyManager *manager) override;
00363 private:
00364     QScopedPointer<QtColorEditorFactoryPrivate> d_ptr;
00365     Q_DECLARE_PRIVATE(QtColorEditorFactory)
00366     Q_DISABLE_COPY_MOVE(QtColorEditorFactory)
00367     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, const QColor &))
00368     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00369     Q_PRIVATE_SLOT(d_func(), void slotSetValue(const QColor &))
00370 };
00371
00372 class QtFontEditorFactoryPrivate;
00373
00374 class QT_QTPROPERTYBROWSER_EXPORT QtFontEditorFactory : public
      QtAbstractEditorFactory<QtFontPropertyManager>
00375 {
00376     Q_OBJECT
00377 public:
00378     QtFontEditorFactory(QObject *parent = 0);
00379     ~QtFontEditorFactory();
```

```
00380 protected:
00381     void connectPropertyManager(QtFontPropertyManager *manager) override;
00382     QWidget *createEditor(QtFontPropertyManager *manager, QtProperty *property,
00383                 QWidget *parent) override;
00384     void disconnectPropertyManager(QtFontPropertyManager *manager) override;
00385 private:
00386     QScopedPointer<QtFontEditorFactoryPrivate> d_ptr;
00387     Q_DECLARE_PRIVATE(QtFontEditorFactory)
00388     Q_DISABLE_COPY_MOVE(QtFontEditorFactory)
00389     Q_PRIVATE_SLOT(d_func(), void slotPropertyChanged(QtProperty *, const QFont &))
00390     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed(QObject *))
00391     Q_PRIVATE_SLOT(d_func(), void slotSetValue(const QFont &))
00392 };
00393
00394 QT_END_NAMESPACE
00395
00396 #endif
```

18.87 qtgroupboxpropertybrowser.h

```
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00034 ** https://www.gnu.org/licenses/gpl-3.0.html.
00035 **
00036 ** $QT_END_LICENSE$
00037 **
00038 ****************************************************************************/
00039
00040 #ifndef QTGROUPBOXPROPERTYBROWSER_H
00041 #define QTGROUPBOXPROPERTYBROWSER_H
00042
00043 #include "qtpropertybrowser.h"
00044
00045 QT_BEGIN_NAMESPACE
00046
00047 class QtGroupBoxPropertyBrowserPrivate;
00048
00049 class QT_QTPROPERTYBROWSER_EXPORT QtGroupBoxPropertyBrowser : public QtAbstractPropertyBrowser
00050 {
00051     Q_OBJECT
00052 public:
00053
00054     QtGroupBoxPropertyBrowser(QWidget *parent = 0);
00055     ~QtGroupBoxPropertyBrowser();
00056
00057 protected:
00058     void itemInserted(QtBrowserItem *item, QtBrowserItem *afterItem) override;
00059     void itemRemoved(QtBrowserItem *item) override;
00060     void itemChanged(QtBrowserItem *item) override;
00061
00062 private:
00063
00064     QScopedPointer<QtGroupBoxPropertyBrowserPrivate> d_ptr;
00065     Q_DECLARE_PRIVATE(QtGroupBoxPropertyBrowser)
```

```
00066     Q_DISABLE_COPY_MOVE(QtGroupBoxPropertyBrowser)
00067     Q_PRIVATE_SLOT(d_func(), void slotUpdate())
00068     Q_PRIVATE_SLOT(d_func(), void slotEditorDestroyed())
00069
00070 };
00071
00072 QT_END_NAMESPACE
00073
00074 #endif
```

18.88 qtpropertybrowser.h

```
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00034 ** https://www.gnu.org/licenses/gpl-3.0.html.
00035 **
00036 ** $QT_END_LICENSE$
00037 **
00038 ****************************************************************************/
00039
00040 #ifndef QTPROPERTYBROWSER_H
00041 #define QTPROPERTYBROWSER_H
00042
00043 #include <QtWidgets/QWidget>
00044 #include <QtCore/QSet>
00045
00046 QT_BEGIN_NAMESPACE
00047
00048 #if defined(Q_OS_WIN)
00049 #  if !defined(QT_QTPROPERTYBROWSER_EXPORT) && !defined(QT_QTPROPERTYBROWSER_IMPORT)
00050 #    define QT_QTPROPERTYBROWSER_EXPORT
00051 #  elif defined(QT_QTPROPERTYBROWSER_IMPORT)
00052 #    if defined(QT_QTPROPERTYBROWSER_EXPORT)
00053 #      undef QT_QTPROPERTYBROWSER_EXPORT
00054 #    endif
00055 #    define QT_QTPROPERTYBROWSER_EXPORT __declspec(dllimport)
00056 #  elif defined(QT_QTPROPERTYBROWSER_EXPORT)
00057 #    undef QT_QTPROPERTYBROWSER_EXPORT
00058 #    define QT_QTPROPERTYBROWSER_EXPORT __declspec(dllexport)
00059 #  endif
00060 #else
00061 #  define QT_QTPROPERTYBROWSER_EXPORT
00062 #endif
00063
00064 class QtAbstractPropertyManager;
00065 class QtPropertyPrivate;
00066
00067 class QT_QTPROPERTYBROWSER_EXPORT QtProperty
00068 {
00069 public:
00070     virtual ~QtProperty();
00071
00072     QList<QtProperty *> subProperties() const;
00073
```

```
00074     QtAbstractPropertyManager *propertyManager() const;
00075
00076     QString toolTip() const { return valueToolTip(); } // Compatibility
00077     QString valueToolTip() const;
00078     QString descriptionToolTip() const;
00079     QString statusTip() const;
00080     QString whatsThis() const;
00081     QString propertyName() const;
00082     bool isEnabled() const;
00083     bool isModified() const;
00084
00085     bool hasValue() const;
00086     QIcon valueIcon() const;
00087     QString valueText() const;
00088
00089     void setToolTip(const QString &text) { setValueToolTip(text); }  // Compatibility
00090     void setValueToolTip(const QString &text);
00091     void setDescriptionToolTip(const QString &text);
00092     void setStatusTip(const QString &text);
00093     void setWhatsThis(const QString &text);
00094     void setPropertyName(const QString &text);
00095     void setEnabled(bool enable);
00096     void setModified(bool modified);
00097
00098     void addSubProperty(QtProperty *property);
00099     void insertSubProperty(QtProperty *property, QtProperty *afterProperty);
00100     void removeSubProperty(QtProperty *property);
00101 protected:
00102     explicit QtProperty(QtAbstractPropertyManager *manager);
00103     void propertyChanged();
00104 private:
00105     friend class QtAbstractPropertyManager;
00106     QScopedPointer<QtPropertyPrivate> d_ptr;
00107 };
00108
00109 class QtAbstractPropertyManagerPrivate;
00110
00111 class QT_QTPROPERTYBROWSER_EXPORT QtAbstractPropertyManager : public QObject
00112 {
00113     Q_OBJECT
00114 public:
00115
00116     explicit QtAbstractPropertyManager(QObject *parent = 0);
00117     ~QtAbstractPropertyManager();
00118
00119     QSet<QtProperty *> properties() const;
00120     void clear() const;
00121
00122     QtProperty *addProperty(const QString &name = QString());
00123 Q_SIGNALS:
00124
00125     void propertyInserted(QtProperty *property,
00126                 QtProperty *parent, QtProperty *after);
00127     void propertyChanged(QtProperty *property);
00128     void propertyRemoved(QtProperty *property, QtProperty *parent);
00129     void propertyDestroyed(QtProperty *property);
00130 protected:
00131     virtual bool hasValue(const QtProperty *property) const;
00132     virtual QIcon valueIcon(const QtProperty *property) const;
00133     virtual QString valueText(const QtProperty *property) const;
00134     virtual void initializeProperty(QtProperty *property) = 0;
00135     virtual void uninitializeProperty(QtProperty *property);
00136     virtual QtProperty *createProperty();
00137 private:
00138     friend class QtProperty;
00139     QScopedPointer<QtAbstractPropertyManagerPrivate> d_ptr;
00140     Q_DECLARE_PRIVATE(QtAbstractPropertyManager)
00141     Q_DISABLE_COPY_MOVE(QtAbstractPropertyManager)
00142 };
00143
00144 class QT_QTPROPERTYBROWSER_EXPORT QtAbstractEditorFactoryBase : public QObject
00145 {
00146     Q_OBJECT
00147 public:
00148     virtual QWidget *createEditor(QtProperty *property, QWidget *parent) = 0;
00149 protected:
00150     explicit QtAbstractEditorFactoryBase(QObject *parent = 0)
00151         : QObject(parent) {}
00152
00153     virtual void breakConnection(QtAbstractPropertyManager *manager) = 0;
00154 protected Q_SLOTS:
00155     virtual void managerDestroyed(QObject *manager) = 0;
00156
00157     friend class QtAbstractPropertyBrowser;
00158 };
00159
00160 template <class PropertyManager>
```

```
00161 class QtAbstractEditorFactory : public QtAbstractEditorFactoryBase
00162 {
00163 public:
00164     explicit QtAbstractEditorFactory(QObject *parent) : QtAbstractEditorFactoryBase(parent) {}
00165     QWidget *createEditor(QtProperty *property, QWidget *parent) override
00166     {
00167         for (PropertyManager *manager : qAsConst(m_managers)) {
00168             if (manager == property->propertyManager()) {
00169                 return createEditor(manager, property, parent);
00170             }
00171         }
00172         return 0;
00173     }
00174     void addPropertyManager(PropertyManager *manager)
00175     {
00176         if (m_managers.contains(manager))
00177             return;
00178         m_managers.insert(manager);
00179         connectPropertyManager(manager);
00180         connect(manager, SIGNAL(destroyed(QObject *)),
00181                     this, SLOT(managerDestroyed(QObject *)));
00182     }
00183     void removePropertyManager(PropertyManager *manager)
00184     {
00185         if (!m_managers.contains(manager))
00186             return;
00187         disconnect(manager, SIGNAL(destroyed(QObject *)),
00188                     this, SLOT(managerDestroyed(QObject *)));
00189         disconnectPropertyManager(manager);
00190         m_managers.remove(manager);
00191     }
00192     QSet<PropertyManager *> propertyManagers() const
00193     {
00194         return m_managers;
00195     }
00196     PropertyManager *propertyManager(QtProperty *property) const
00197     {
00198         QtAbstractPropertyManager *manager = property->propertyManager();
00199         for (PropertyManager *m : qAsConst(m_managers)) {
00200             if (m == manager) {
00201                 return m;
00202             }
00203         }
00204         return 0;
00205     }
00206 protected:
00207     virtual void connectPropertyManager(PropertyManager *manager) = 0;
00208     virtual QWidget *createEditor(PropertyManager *manager, QtProperty *property,
00209                 QWidget *parent) = 0;
00210     virtual void disconnectPropertyManager(PropertyManager *manager) = 0;
00211     void managerDestroyed(QObject *manager) override
00212     {
00213         for (PropertyManager *m : qAsConst(m_managers)) {
00214             if (m == manager) {
00215                 m_managers.remove(m);
00216                 return;
00217             }
00218         }
00219     }
00220 private:
00221     void breakConnection(QtAbstractPropertyManager *manager) override
00222     {
00223         for (PropertyManager *m : qAsConst(m_managers)) {
00224             if (m == manager) {
00225                 removePropertyManager(m);
00226                 return;
00227             }
00228         }
00229     }
00230 private:
00231     QSet<PropertyManager *> m_managers;
00232     friend class QtAbstractPropertyEditor;
00233 };
00234
00235 class QtAbstractPropertyBrowser;
00236 class QtBrowserItemPrivate;
00237
00238 class QT_QTPROPERTYBROWSER_EXPORT QtBrowserItem
00239 {
00240 public:
00241     QtProperty *property() const;
00242     QtBrowserItem *parent() const;
00243     QList<QtBrowserItem *> children() const;
00244     QtAbstractPropertyBrowser *browser() const;
00245 private:
00246     explicit QtBrowserItem(QtAbstractPropertyBrowser *browser, QtProperty *property, QtBrowserItem
      *parent);
```

```
00247     ~QtBrowserItem();
00248     QScopedPointer<QtBrowserItemPrivate> d_ptr;
00249     friend class QtAbstractPropertyBrowserPrivate;
00250 };
00251
00252 class QtAbstractPropertyBrowserPrivate;
00253
00254 class QT_QTPROPERTYBROWSER_EXPORT QtAbstractPropertyBrowser : public QWidget
00255 {
00256     Q_OBJECT
00257 public:
00258
00259     explicit QtAbstractPropertyBrowser(QWidget *parent = 0);
00260     ~QtAbstractPropertyBrowser();
00261
00262     QList<QtProperty *> properties() const;
00263     QList<QtBrowserItem *> items(QtProperty *property) const;
00264     QtBrowserItem *topLevelItem(QtProperty *property) const;
00265     QList<QtBrowserItem *> topLevelItems() const;
00266     void clear();
00267
00268     template <class PropertyManager>
00269     void setFactoryForManager(PropertyManager *manager,
00270                               QtAbstractEditorFactory<PropertyManager> *factory) {
00271         QtAbstractPropertyManager *abstractManager = manager;
00272         QtAbstractEditorFactoryBase *abstractFactory = factory;
00273
00274         if (addFactory(abstractManager, abstractFactory))
00275             factory->addPropertyManager(manager);
00276     }
00277
00278     void unsetFactoryForManager(QtAbstractPropertyManager *manager);
00279
00280     QtBrowserItem *currentItem() const;
00281     void setCurrentItem(QtBrowserItem *);
00282
00283 Q_SIGNALS:
00284     void currentItemChanged(QtBrowserItem *);
00285
00286 public Q_SLOTS:
00287
00288     QtBrowserItem *addProperty(QtProperty *property);
00289     QtBrowserItem *insertProperty(QtProperty *property, QtProperty *afterProperty);
00290     void removeProperty(QtProperty *property);
00291
00292 protected:
00293
00294     virtual void itemInserted(QtBrowserItem *item, QtBrowserItem *afterItem) = 0;
00295     virtual void itemRemoved(QtBrowserItem *item) = 0;
00296     // can be tooltip, statustip, whatsthis, name, icon, text.
00297     virtual void itemChanged(QtBrowserItem *item) = 0;
00298
00299     virtual QWidget *createEditor(QtProperty *property, QWidget *parent);
00300 private:
00301
00302     bool addFactory(QtAbstractPropertyManager *abstractManager,
00303                     QtAbstractEditorFactoryBase *abstractFactory);
00304
00305     QScopedPointer<QtAbstractPropertyBrowserPrivate> d_ptr;
00306     Q_DECLARE_PRIVATE(QtAbstractPropertyBrowser)
00307     Q_DISABLE_COPY_MOVE(QtAbstractPropertyBrowser)
00308     Q_PRIVATE_SLOT(d_func(), void slotPropertyInserted(QtProperty *,
00309                                 QtProperty *, QtProperty *))
00310     Q_PRIVATE_SLOT(d_func(), void slotPropertyRemoved(QtProperty *,
00311                                 QtProperty *))
00312     Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty *))
00313     Q_PRIVATE_SLOT(d_func(), void slotPropertyDataChanged(QtProperty *))
00314
00315 };
00316
00317 QT_END_NAMESPACE
00318
00319 #endif // QTPROPERTYBROWSER_H
```

18.89 qtpropertybrowserutils_p.h

```
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00037 **
00038 ****************************************************************************/
00039
00040 //
00041 //  W A R N I N G
00042 //  -------------
00043 //
00044 // This file is not part of the Qt API.  It exists for the convenience
00045 // of Qt Designer.  This header
00046 // file may change from version to version without notice, or even be removed.
00047 //
00048 // We mean it.
00049 //
00050
00051 #ifndef QTPROPERTYBROWSERUTILS_H
00052 #define QTPROPERTYBROWSERUTILS_H
00053
00054 #include <QtCore/QMap>
00055 #include <QtGui/QIcon>
00056 #include <QtWidgets/QWidget>
00057 #include <QtWidgets/QPushButton>
00058 #include <QtWidgets/QDateEdit>
00059 #include <QtWidgets/QSpinBox>
00060 #include <QtCore/QDateTime>
00061 #include <QtCore/QStringList>
00062
00063 QT_BEGIN_NAMESPACE
00064
00065 class QMouseEvent;
00066 class QCheckBox;
00067 class QLineEdit;
00068
00069 class QtCursorDatabase
00070 {
00071 public:
00072     QtCursorDatabase();
00073     void clear();
00074
00075     QStringList cursorShapeNames() const;
00076     QMap<int, QIcon> cursorShapeIcons() const;
00077     QString cursorToShapeName(const QCursor& cursor) const;
00078     QIcon cursorToShapeIcon(const QCursor& cursor) const;
00079     int cursorToValue(const QCursor& cursor) const;
00080 #ifndef QT_NO_CURSOR
00081     QCursor valueToCursor(int value) const;
00082 #endif
00083 private:
00084     void appendCursor(Qt::CursorShape shape, const QString& name, const QIcon& icon);
00085     QStringList m_cursorNames;
00086     QMap<int, QIcon> m_cursorIcons;
00087     QMap<int, Qt::CursorShape> m_valueToCursorShape;
00088     QMap<Qt::CursorShape, int> m_cursorShapeToValue;
00089 };
00090
00091 class QtPropertyBrowserUtils
00092 {
00093 public:
00094     static QPixmap brushValuePixmap(const QBrush& b);
00095     static QIcon brushValueIcon(const QBrush& b);
00096     static QString colorValueText(const QColor& c);
```

```
00097     static QPixmap fontValuePixmap(const QFont& f);
00098     static QIcon fontValueIcon(const QFont& f);
00099     static QString fontValueText(const QFont& f);
00100     static QString dateFormat();
00101     static QString timeFormat();
00102     static QString dateTimeFormat();
00103 };
00104
00105 #pragma region QtBoolEdit
00106
00107 class QtBoolEdit : public QWidget {
00108     Q_OBJECT
00109 public:
00110     QtBoolEdit(QWidget* parent = 0);
00111
00112     bool textVisible() const { return m_textVisible; }
00113     void setTextVisible(bool textVisible);
00114
00115     Qt::CheckState checkState() const;
00116     void setCheckState(Qt::CheckState state);
00117
00118     bool isChecked() const;
00119     void setChecked(bool c);
00120     void setInitialChecked(bool c);
00121
00122     bool blockCheckBoxSignals(bool block);
00123
00124 Q_SIGNALS:
00125     void toggled(bool);
00126
00127 protected:
00128     void mousePressEvent(QMouseEvent* event) override;
00129
00130 private:
00131     QCheckBox* m_checkBox;
00132     QPushButton* m_button;
00133     bool m_textVisible;
00134     bool m_initialChecked;
00135 };
00136 #pragma endregion
00137
00138 #pragma region QtIntEdit
00139
00140 class QtIntEdit : public QWidget {
00141     Q_OBJECT
00142 public:
00143     QtIntEdit(QWidget* parent = Q_NULLPTR);
00144
00145     int value();
00146     void setValue(int);
00147     void setInitialValue(int);
00148     void setSingleStep(int);
00149     void setRange(int, int);
00150     void setKeyboardTracking(bool);
00151
00152 Q_SIGNALS:
00153     void valueChanged(int);
00154
00155 private:
00156     QSpinBox* m_spinbox;
00157     QPushButton* m_button;
00158     int m_initialvalue;
00159 };
00160
00161 #pragma endregion
00162
00163 #pragma region QtDoubleEdit
00164
00165 class QtDoubleEdit : public QWidget {
00166     Q_OBJECT
00167 public:
00168     QtDoubleEdit(QWidget* parent = Q_NULLPTR);
00169
00170     int value();
00171     void setValue(double);
00172     void setInitialValue(double);
00173     void setDecimals(double);
00174     void setSingleStep(double);
00175     void setRange(double, double);
00176     void setKeyboardTracking(bool);
00177
00178 Q_SIGNALS:
00179     void valueChanged(double);
00180
00181 private:
00182     QDoubleSpinBox* m_doublespinbox;
00183     QPushButton* m_button;
```

```
00184     double m_initialvalue;
00185 };
00186
00187 #pragma endregion
00188
00189 #pragma region QtStringEdit
00190
00191 class QtStringEdit : public QWidget {
00192     Q_OBJECT
00193 public:
00194     QtStringEdit(QWidget* parent = Q_NULLPTR);
00195
00196     QString text();
00197     void setText(QString);
00198     void setInitialText(QString);
00199     const QValidator* validator();
00200     void setValidator(QValidator*);
00201
00202 Q_SIGNALS:
00203     void textChanged(QString);
00204     void textEdited(QString);
00205
00206 private:
00207     QLineEdit* m_lineedit;
00208     QPushButton* m_button;
00209     QString m_initialvalue;
00210 };
00211
00212 #pragma endregion
00213
00214 #pragma region QtDateEdit
00215
00216 class QtDateEdit : public QWidget {
00217     Q_OBJECT
00218 public:
00219     QtDateEdit(QWidget* parent = Q_NULLPTR);
00220
00221     QDate date();
00222     void setDate(QDate);
00223     void setInitialDate(QDate);
00224     void setDateRange(QDate, QDate);
00225     void setDisplayFormat(QString);
00226     void setCalendarPopup(bool);
00227
00228 Q_SIGNALS:
00229     void dateChanged(QDate);
00230
00231 private:
00232     QDateEdit* m_dateedit;
00233     QPushButton* m_button;
00234     QDate m_initialvalue;
00235 };
00236
00237 #pragma endregion
00238
00239 #pragma region QtTimeEdit
00240
00241 class QtTimeEdit : public QWidget {
00242     Q_OBJECT
00243 public:
00244     QtTimeEdit(QWidget* parent = Q_NULLPTR);
00245
00246     QTime time();
00247     void setTime(QTime);
00248     void setInitialTime(QTime);
00249     void setTimeRange(QTime, QTime);
00250     void setDisplayFormat(QString);
00251     void setCalendarPopup(bool);
00252
00253 Q_SIGNALS:
00254     void timeChanged(QTime);
00255
00256 private:
00257     QTimeEdit* m_timeedit;
00258     QPushButton* m_button;
00259     QTime m_initialvalue;
00260 };
00261
00262 #pragma endregion
00263
00264 #pragma region QDateTimeEdit
00265
00266 class QtDateTimeEdit : public QWidget {
00267     Q_OBJECT
00268 public:
00269     QtDateTimeEdit(QWidget* parent = Q_NULLPTR);
00270
```

```
00271     QDateTime dateTime();
00272     void setDateTime(const QDateTime&);
00273     void setInitialDateTime(const QDateTime&);
00274     void setDateTimeRange(const QDateTime&, const QDateTime&);
00275     void setDisplayFormat(const QString&);
00276     void setCalendarPopup(bool);
00277
00278 Q_SIGNALS:
00279     void dateTimeChanged(const QDateTime&);
00280
00281 private:
00282     QDateTimeEdit* m_timeedit;
00283     QPushButton* m_button;
00284     QDateTime m_initialvalue;
00285 };
00286
00287 #pragma endregion
00288
00289
00290 QT_END_NAMESPACE
00291
00292 #endif
```

18.90 qtpropertymanager.h

```
00001 /****************************************************************************
00002 **
00003 ** Copyright (C) 2016 The Qt Company Ltd.
00004 ** Contact: https://www.qt.io/licensing/
00005 **
00006 ** This file is part of the tools applications of the Qt Toolkit.
00007 **
00008 ** $QT_BEGIN_LICENSE:LGPL$
00009 ** Commercial License Usage
00010 ** Licensees holding valid commercial Qt licenses may use this file in
00011 ** accordance with the commercial license agreement provided with the
00012 ** Software or, alternatively, in accordance with the terms contained in
00013 ** a written agreement between you and The Qt Company. For licensing terms
00014 ** and conditions see https://www.qt.io/terms-conditions. For further
00015 ** information use the contact form at https://www.qt.io/contact-us.
00016 **
00017 ** GNU Lesser General Public License Usage
00018 ** Alternatively, this file may be used under the terms of the GNU Lesser
00019 ** General Public License version 3 as published by the Free Software
00020 ** Foundation and appearing in the file LICENSE.LGPL3 included in the
00021 ** packaging of this file. Please review the following information to
00022 ** ensure the GNU Lesser General Public License version 3 requirements
00023 ** will be met: https://www.gnu.org/licenses/lgpl-3.0.html.
00024 **
00025 ** GNU General Public License Usage
00026 ** Alternatively, this file may be used under the terms of the GNU
00027 ** General Public License version 2.0 or (at your option) the GNU General
00028 ** Public license version 3 or any later version approved by the KDE Free
00029 ** Qt Foundation. The licenses are as published by the Free Software
00030 ** Foundation and appearing in the file LICENSE.GPL2 and LICENSE.GPL3
00031 ** included in the packaging of this file. Please review the following
00032 ** information to ensure the GNU General Public License requirements will
00033 ** be met: https://www.gnu.org/licenses/gpl-2.0.html and
00034 ** https://www.gnu.org/licenses/gpl-3.0.html.
00035 **
00036 ** $QT_END_LICENSE$
00037 **
00038 ****************************************************************************/
00039
00040 #ifndef QTPROPERTYMANAGER_H
00041 #define QTPROPERTYMANAGER_H
00042
00043 #include "qtpropertybrowser.h"
00044
00045 QT_BEGIN_NAMESPACE
00046
00047 class QDate;
00048 class QTime;
00049 class QDateTime;
00050 class QLocale;
00051 class QRegularExpression;
00052
00053 #pragma region QtGroupPropertyManager
00054
00055 class QT_QTPROPERTYBROWSER_EXPORT QtGroupPropertyManager : public QtAbstractPropertyManager
00056 {
00057     Q_OBJECT
00058 public:
00059     QtGroupPropertyManager(QObject* parent = 0);
00060     ~QtGroupPropertyManager();
```

```
00061
00062 protected:
00063     bool hasValue(const QtProperty* property) const override;
00064
00065     void initializeProperty(QtProperty* property) override;
00066     void uninitializeProperty(QtProperty* property) override;
00067 };
00068
00069 #pragma endregion
00070
00071 #pragma region QtIntPropertyManager
00072
00073 class QtIntPropertyManagerPrivate;
00074
00075 class QT_QTPROPERTYBROWSER_EXPORT QtIntPropertyManager : public QtAbstractPropertyManager
00076 {
00077     Q_OBJECT
00078 public:
00079     QtIntPropertyManager(QObject* parent = 0);
00080     ~QtIntPropertyManager();
00081
00082     int value(const QtProperty* property) const;
00083     int initialValue(const QtProperty* property) const;
00084     int minimum(const QtProperty* property) const;
00085     int maximum(const QtProperty* property) const;
00086     int singleStep(const QtProperty* property) const;
00087
00088 public Q_SLOTS:
00089     void setValue(QtProperty* property, int val);
00090     void setValueOnly(QtProperty* property, int val);
00091     void setMinimum(QtProperty* property, int minVal);
00092     void setMaximum(QtProperty* property, int maxVal);
00093     void setRange(QtProperty* property, int minVal, int maxVal);
00094     void setSingleStep(QtProperty* property, int step);
00095 Q_SIGNALS:
00096     void valueChanged(QtProperty* property, int val);
00097     void rangeChanged(QtProperty* property, int minVal, int maxVal);
00098     void singleStepChanged(QtProperty* property, int step);
00099 protected:
00100     QString valueText(const QtProperty* property) const override;
00101     void initializeProperty(QtProperty* property) override;
00102     void uninitializeProperty(QtProperty* property) override;
00103 private:
00104     QScopedPointer<QtIntPropertyManagerPrivate> d_ptr;
00105     Q_DECLARE_PRIVATE(QtIntPropertyManager)
00106         Q_DISABLE_COPY_MOVE(QtIntPropertyManager)
00107 };
00108 #pragma endregion
00109
00110 #pragma region QtBoolPropertyManager
00111
00112 class QtBoolPropertyManagerPrivate;
00113
00114 class QT_QTPROPERTYBROWSER_EXPORT QtBoolPropertyManager : public QtAbstractPropertyManager
00115 {
00116     Q_OBJECT
00117 public:
00118     QtBoolPropertyManager(QObject* parent = 0);
00119     ~QtBoolPropertyManager();
00120
00121     bool value(const QtProperty* property) const;
00122     bool initialValue(const QtProperty* property) const;
00123
00124 public Q_SLOTS:
00125     void setValue(QtProperty* property, bool val);
00126     void setValueOnly(QtProperty* property, bool val);
00127 Q_SIGNALS:
00128     void valueChanged(QtProperty* property, bool val);
00129 protected:
00130     QString valueText(const QtProperty* property) const override;
00131     QIcon valueIcon(const QtProperty* property) const override;
00132     void initializeProperty(QtProperty* property) override;
00133     void uninitializeProperty(QtProperty* property) override;
00134 private:
00135     QScopedPointer<QtBoolPropertyManagerPrivate> d_ptr;
00136     Q_DECLARE_PRIVATE(QtBoolPropertyManager)
00137         Q_DISABLE_COPY_MOVE(QtBoolPropertyManager)
00138 };
00139
00140 #pragma endregion
00141
00142 #pragma region QtDoublePropertyManager
00143
00144 class QtDoublePropertyManagerPrivate;
00145
00146 class QT_QTPROPERTYBROWSER_EXPORT QtDoublePropertyManager : public QtAbstractPropertyManager
00147 {
```

```
00148     Q_OBJECT
00149 public:
00150     QtDoublePropertyManager(QObject* parent = 0);
00151     ~QtDoublePropertyManager();
00152 
00153     double value(const QtProperty* property) const;
00154     double initialValue(const QtProperty* property) const;
00155     double minimum(const QtProperty* property) const;
00156     double maximum(const QtProperty* property) const;
00157     double singleStep(const QtProperty* property) const;
00158     int decimals(const QtProperty* property) const;
00159 
00160 public Q_SLOTS:
00161     void setValue(QtProperty* property, double val);
00162     void setValueOnly(QtProperty* property, double val);
00163     void setMinimum(QtProperty* property, double minVal);
00164     void setMaximum(QtProperty* property, double maxVal);
00165     void setRange(QtProperty* property, double minVal, double maxVal);
00166     void setSingleStep(QtProperty* property, double step);
00167     void setDecimals(QtProperty* property, int prec);
00168 Q_SIGNALS:
00169     void valueChanged(QtProperty* property, double val);
00170     void rangeChanged(QtProperty* property, double minVal, double maxVal);
00171     void singleStepChanged(QtProperty* property, double step);
00172     void decimalsChanged(QtProperty* property, int prec);
00173 protected:
00174     QString valueText(const QtProperty* property) const override;
00175     void initializeProperty(QtProperty* property) override;
00176     void uninitializeProperty(QtProperty* property) override;
00177 private:
00178     QScopedPointer<QtDoublePropertyManagerPrivate> d_ptr;
00179     Q_DECLARE_PRIVATE(QtDoublePropertyManager)
00180         Q_DISABLE_COPY_MOVE(QtDoublePropertyManager)
00181 };
00182 
00183 #pragma endregion
00184 
00185 #pragma region QtStringPropertyManager
00186 
00187 class QtStringPropertyManagerPrivate;
00188 
00189 class QT_QTPROPERTYBROWSER_EXPORT QtStringPropertyManager : public QtAbstractPropertyManager
00190 {
00191     Q_OBJECT
00192 public:
00193     QtStringPropertyManager(QObject* parent = 0);
00194     ~QtStringPropertyManager();
00195 
00196     QString value(const QtProperty* property) const;
00197     QString initialValue(const QtProperty* property) const;
00198     QRegularExpression regExp(const QtProperty* property) const;
00199 
00200 public Q_SLOTS:
00201     void setValue(QtProperty* property, const QString& val);
00202     void setValueOnly(QtProperty* property, const QString& val);
00203     void setRegExp(QtProperty* property, const QRegularExpression& regExp);
00204 Q_SIGNALS:
00205     void valueChanged(QtProperty* property, const QString& val);
00206     void regExpChanged(QtProperty* property, const QRegularExpression& regExp);
00207 protected:
00208     QString valueText(const QtProperty* property) const override;
00209     void initializeProperty(QtProperty* property) override;
00210     void uninitializeProperty(QtProperty* property) override;
00211 private:
00212     QScopedPointer<QtStringPropertyManagerPrivate> d_ptr;
00213     Q_DECLARE_PRIVATE(QtStringPropertyManager)
00214         Q_DISABLE_COPY_MOVE(QtStringPropertyManager)
00215 };
00216 
00217 #pragma endregion
00218 
00219 #pragma region QtDatePropertyManager
00220 
00221 class QtDatePropertyManagerPrivate;
00222 
00223 class QT_QTPROPERTYBROWSER_EXPORT QtDatePropertyManager : public QtAbstractPropertyManager
00224 {
00225     Q_OBJECT
00226 public:
00227     QtDatePropertyManager(QObject* parent = 0);
00228     ~QtDatePropertyManager();
00229 
00230     QDate value(const QtProperty* property) const;
00231     QDate initialValue(const QtProperty* property) const;
00232     QDate minimum(const QtProperty* property) const;
00233     QDate maximum(const QtProperty* property) const;
00234 
```

```
00235 public Q_SLOTS:
00236     void setValue(QtProperty* property, QDate val);
00237     void setValueOnly(QtProperty* property, QDate val);
00238     void setMinimum(QtProperty* property, QDate minVal);
00239     void setMaximum(QtProperty* property, QDate maxVal);
00240     void setRange(QtProperty* property, QDate minVal, QDate maxVal);
00241 Q_SIGNALS:
00242     void valueChanged(QtProperty* property, QDate val);
00243     void rangeChanged(QtProperty* property, QDate minVal, QDate maxVal);
00244 protected:
00245     QString valueText(const QtProperty* property) const override;
00246     void initializeProperty(QtProperty* property) override;
00247     void uninitializeProperty(QtProperty* property) override;
00248 private:
00249     QScopedPointer<QtDatePropertyManagerPrivate> d_ptr;
00250     Q_DECLARE_PRIVATE(QtDatePropertyManager)
00251         Q_DISABLE_COPY_MOVE(QtDatePropertyManager)
00252 };
00253
00254 #pragma endregion
00255
00256 #pragma region QtTimePropertyManager
00257
00258 class QtTimePropertyManagerPrivate;
00259
00260 class QT_QTPROPERTYBROWSER_EXPORT QtTimePropertyManager : public QtAbstractPropertyManager
00261 {
00262     Q_OBJECT
00263 public:
00264     QtTimePropertyManager(QObject* parent = 0);
00265     ~QtTimePropertyManager();
00266
00267     QTime value(const QtProperty* property) const;
00268     QTime initialValue(const QtProperty* property) const;
00269
00270 public Q_SLOTS:
00271     void setValue(QtProperty* property, QTime val);
00272     void setValueOnly(QtProperty* property, QTime val);
00273 Q_SIGNALS:
00274     void valueChanged(QtProperty* property, QTime val);
00275 protected:
00276     QString valueText(const QtProperty* property) const override;
00277     void initializeProperty(QtProperty* property) override;
00278     void uninitializeProperty(QtProperty* property) override;
00279 private:
00280     QScopedPointer<QtTimePropertyManagerPrivate> d_ptr;
00281     Q_DECLARE_PRIVATE(QtTimePropertyManager)
00282         Q_DISABLE_COPY_MOVE(QtTimePropertyManager)
00283 };
00284
00285 #pragma endregion
00286
00287 #pragma region QtDateTimePropertyManager
00288
00289 class QtDateTimePropertyManagerPrivate;
00290
00291 class QT_QTPROPERTYBROWSER_EXPORT QtDateTimePropertyManager : public QtAbstractPropertyManager
00292 {
00293     Q_OBJECT
00294 public:
00295     QtDateTimePropertyManager(QObject* parent = 0);
00296     ~QtDateTimePropertyManager();
00297
00298     QDateTime value(const QtProperty* property) const;
00299     QDateTime initialValue(const QtProperty* property) const;
00300
00301 public Q_SLOTS:
00302     void setValue(QtProperty* property, const QDateTime& val);
00303     void setValueOnly(QtProperty* property, const QDateTime& val);
00304 Q_SIGNALS:
00305     void valueChanged(QtProperty* property, const QDateTime& val);
00306 protected:
00307     QString valueText(const QtProperty* property) const override;
00308     void initializeProperty(QtProperty* property) override;
00309     void uninitializeProperty(QtProperty* property) override;
00310 private:
00311     QScopedPointer<QtDateTimePropertyManagerPrivate> d_ptr;
00312     Q_DECLARE_PRIVATE(QtDateTimePropertyManager)
00313         Q_DISABLE_COPY_MOVE(QtDateTimePropertyManager)
00314 };
00315
00316 #pragma endregion
00317
00318 #pragma region QtKeySequencePropertyManager
00319
00320 class QtKeySequencePropertyManagerPrivate;
00321
```

```
00322 class QT_QTPROPERTYBROWSER_EXPORT QtKeySequencePropertyManager : public QtAbstractPropertyManager
00323 {
00324     Q_OBJECT
00325 public:
00326     QtKeySequencePropertyManager(QObject* parent = 0);
00327     ~QtKeySequencePropertyManager();
00328 
00329     QKeySequence value(const QtProperty* property) const;
00330 
00331 public Q_SLOTS:
00332     void setValue(QtProperty* property, const QKeySequence& val);
00333 Q_SIGNALS:
00334     void valueChanged(QtProperty* property, const QKeySequence& val);
00335 protected:
00336     QString valueText(const QtProperty* property) const override;
00337     void initializeProperty(QtProperty* property) override;
00338     void uninitializeProperty(QtProperty* property) override;
00339 private:
00340     QScopedPointer<QtKeySequencePropertyManagerPrivate> d_ptr;
00341     Q_DECLARE_PRIVATE(QtKeySequencePropertyManager)
00342         Q_DISABLE_COPY_MOVE(QtKeySequencePropertyManager)
00343 };
00344 
00345 #pragma endregion
00346 
00347 #pragma region QtCharPropertyManager
00348 
00349 class QtCharPropertyManagerPrivate;
00350 
00351 class QT_QTPROPERTYBROWSER_EXPORT QtCharPropertyManager : public QtAbstractPropertyManager
00352 {
00353     Q_OBJECT
00354 public:
00355     QtCharPropertyManager(QObject* parent = 0);
00356     ~QtCharPropertyManager();
00357 
00358     QChar value(const QtProperty* property) const;
00359 
00360 public Q_SLOTS:
00361     void setValue(QtProperty* property, const QChar& val);
00362 Q_SIGNALS:
00363     void valueChanged(QtProperty* property, const QChar& val);
00364 protected:
00365     QString valueText(const QtProperty* property) const override;
00366     void initializeProperty(QtProperty* property) override;
00367     void uninitializeProperty(QtProperty* property) override;
00368 private:
00369     QScopedPointer<QtCharPropertyManagerPrivate> d_ptr;
00370     Q_DECLARE_PRIVATE(QtCharPropertyManager)
00371         Q_DISABLE_COPY_MOVE(QtCharPropertyManager)
00372 };
00373 
00374 #pragma endregion
00375 
00376 #pragma region QtLocalePropertyManager
00377 
00378 class QtEnumPropertyManager;
00379 class QtLocalePropertyManagerPrivate;
00380 
00381 class QT_QTPROPERTYBROWSER_EXPORT QtLocalePropertyManager : public QtAbstractPropertyManager
00382 {
00383     Q_OBJECT
00384 public:
00385     QtLocalePropertyManager(QObject* parent = 0);
00386     ~QtLocalePropertyManager();
00387 
00388     QtEnumPropertyManager* subEnumPropertyManager() const;
00389 
00390     QLocale value(const QtProperty* property) const;
00391 
00392 public Q_SLOTS:
00393     void setValue(QtProperty* property, const QLocale& val);
00394 Q_SIGNALS:
00395     void valueChanged(QtProperty* property, const QLocale& val);
00396 protected:
00397     QString valueText(const QtProperty* property) const override;
00398     void initializeProperty(QtProperty* property) override;
00399     void uninitializeProperty(QtProperty* property) override;
00400 private:
00401     QScopedPointer<QtLocalePropertyManagerPrivate> d_ptr;
00402     Q_DECLARE_PRIVATE(QtLocalePropertyManager)
00403         Q_DISABLE_COPY_MOVE(QtLocalePropertyManager)
00404         Q_PRIVATE_SLOT(d_func(), void slotEnumChanged(QtProperty*, int))
00405         Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty*))
00406 };
00407 
00408 #pragma endregion
```

```
00409
00410 #pragma region QtPointPropertyManager
00411
00412 class QtPointPropertyManagerPrivate;
00413
00414 class QT_QTPROPERTYBROWSER_EXPORT QtPointPropertyManager : public QtAbstractPropertyManager
00415 {
00416     Q_OBJECT
00417 public:
00418     QtPointPropertyManager(QObject* parent = 0);
00419     ~QtPointPropertyManager();
00420
00421     QtIntPropertyManager* subIntPropertyManager() const;
00422
00423     QPoint value(const QtProperty* property) const;
00424
00425 public Q_SLOTS:
00426     void setValue(QtProperty* property, const QPoint& val);
00427 Q_SIGNALS:
00428     void valueChanged(QtProperty* property, const QPoint& val);
00429 protected:
00430     QString valueText(const QtProperty* property) const override;
00431     void initializeProperty(QtProperty* property) override;
00432     void uninitializeProperty(QtProperty* property) override;
00433 private:
00434     QScopedPointer<QtPointPropertyManagerPrivate> d_ptr;
00435     Q_DECLARE_PRIVATE(QtPointPropertyManager)
00436         Q_DISABLE_COPY_MOVE(QtPointPropertyManager)
00437         Q_PRIVATE_SLOT(d_func(), void slotIntChanged(QtProperty*, int))
00438         Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty*))
00439 };
00440
00441 #pragma endregion
00442
00443 #pragma region QtPointFPropertyManager
00444
00445 class QtPointFPropertyManagerPrivate;
00446
00447 class QT_QTPROPERTYBROWSER_EXPORT QtPointFPropertyManager : public QtAbstractPropertyManager
00448 {
00449     Q_OBJECT
00450 public:
00451     QtPointFPropertyManager(QObject* parent = 0);
00452     ~QtPointFPropertyManager();
00453
00454     QtDoublePropertyManager* subDoublePropertyManager() const;
00455
00456     QPointF value(const QtProperty* property) const;
00457     int decimals(const QtProperty* property) const;
00458
00459 public Q_SLOTS:
00460     void setValue(QtProperty* property, const QPointF& val);
00461     void setDecimals(QtProperty* property, int prec);
00462 Q_SIGNALS:
00463     void valueChanged(QtProperty* property, const QPointF& val);
00464     void decimalsChanged(QtProperty* property, int prec);
00465 protected:
00466     QString valueText(const QtProperty* property) const override;
00467     void initializeProperty(QtProperty* property) override;
00468     void uninitializeProperty(QtProperty* property) override;
00469 private:
00470     QScopedPointer<QtPointFPropertyManagerPrivate> d_ptr;
00471     Q_DECLARE_PRIVATE(QtPointFPropertyManager)
00472         Q_DISABLE_COPY_MOVE(QtPointFPropertyManager)
00473         Q_PRIVATE_SLOT(d_func(), void slotDoubleChanged(QtProperty*, double))
00474         Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty*))
00475 };
00476
00477 #pragma endregion
00478
00479 #pragma region QtSizePropertyManager
00480
00481 class QtSizePropertyManagerPrivate;
00482
00483 class QT_QTPROPERTYBROWSER_EXPORT QtSizePropertyManager : public QtAbstractPropertyManager
00484 {
00485     Q_OBJECT
00486 public:
00487     QtSizePropertyManager(QObject* parent = 0);
00488     ~QtSizePropertyManager();
00489
00490     QtIntPropertyManager* subIntPropertyManager() const;
00491
00492     QSize value(const QtProperty* property) const;
00493     QSize minimum(const QtProperty* property) const;
00494     QSize maximum(const QtProperty* property) const;
00495
```

```
00496 public Q_SLOTS:
00497     void setValue(QtProperty* property, const QSize& val);
00498     void setMinimum(QtProperty* property, const QSize& minVal);
00499     void setMaximum(QtProperty* property, const QSize& maxVal);
00500     void setRange(QtProperty* property, const QSize& minVal, const QSize& maxVal);
00501 Q_SIGNALS:
00502     void valueChanged(QtProperty* property, const QSize& val);
00503     void rangeChanged(QtProperty* property, const QSize& minVal, const QSize& maxVal);
00504 protected:
00505     QString valueText(const QtProperty* property) const override;
00506     void initializeProperty(QtProperty* property) override;
00507     void uninitializeProperty(QtProperty* property) override;
00508 private:
00509     QScopedPointer<QtSizePropertyManagerPrivate> d_ptr;
00510     Q_DECLARE_PRIVATE(QtSizePropertyManager)
00511         Q_DISABLE_COPY_MOVE(QtSizePropertyManager)
00512         Q_PRIVATE_SLOT(d_func(), void slotIntChanged(QtProperty*, int))
00513         Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty*))
00514 };
00515
00516 #pragma endregion
00517
00518 #pragma region QtSizeFPropertyManager
00519
00520 class QtSizeFPropertyManagerPrivate;
00521
00522 class QT_QTPROPERTYBROWSER_EXPORT QtSizeFPropertyManager : public QtAbstractPropertyManager
00523 {
00524     Q_OBJECT
00525 public:
00526     QtSizeFPropertyManager(QObject* parent = 0);
00527     ~QtSizeFPropertyManager();
00528
00529     QtDoublePropertyManager* subDoublePropertyManager() const;
00530
00531     QSizeF value(const QtProperty* property) const;
00532     QSizeF minimum(const QtProperty* property) const;
00533     QSizeF maximum(const QtProperty* property) const;
00534     int decimals(const QtProperty* property) const;
00535
00536 public Q_SLOTS:
00537     void setValue(QtProperty* property, const QSizeF& val);
00538     void setMinimum(QtProperty* property, const QSizeF& minVal);
00539     void setMaximum(QtProperty* property, const QSizeF& maxVal);
00540     void setRange(QtProperty* property, const QSizeF& minVal, const QSizeF& maxVal);
00541     void setDecimals(QtProperty* property, int prec);
00542 Q_SIGNALS:
00543     void valueChanged(QtProperty* property, const QSizeF& val);
00544     void rangeChanged(QtProperty* property, const QSizeF& minVal, const QSizeF& maxVal);
00545     void decimalsChanged(QtProperty* property, int prec);
00546 protected:
00547     QString valueText(const QtProperty* property) const override;
00548     void initializeProperty(QtProperty* property) override;
00549     void uninitializeProperty(QtProperty* property) override;
00550 private:
00551     QScopedPointer<QtSizeFPropertyManagerPrivate> d_ptr;
00552     Q_DECLARE_PRIVATE(QtSizeFPropertyManager)
00553         Q_DISABLE_COPY_MOVE(QtSizeFPropertyManager)
00554         Q_PRIVATE_SLOT(d_func(), void slotDoubleChanged(QtProperty*, double))
00555         Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty*))
00556 };
00557
00558 #pragma endregion
00559
00560 #pragma region QtRectPropertyManager
00561
00562 class QtRectPropertyManagerPrivate;
00563
00564 class QT_QTPROPERTYBROWSER_EXPORT QtRectPropertyManager : public QtAbstractPropertyManager
00565 {
00566     Q_OBJECT
00567 public:
00568     QtRectPropertyManager(QObject* parent = 0);
00569     ~QtRectPropertyManager();
00570
00571     QtIntPropertyManager* subIntPropertyManager() const;
00572
00573     QRect value(const QtProperty* property) const;
00574     QRect constraint(const QtProperty* property) const;
00575
00576 public Q_SLOTS:
00577     void setValue(QtProperty* property, const QRect& val);
00578     void setConstraint(QtProperty* property, const QRect& constraint);
00579 Q_SIGNALS:
00580     void valueChanged(QtProperty* property, const QRect& val);
00581     void constraintChanged(QtProperty* property, const QRect& constraint);
00582 protected:
```

```
00583     QString valueText(const QtProperty* property) const override;
00584     void initializeProperty(QtProperty* property) override;
00585     void uninitializeProperty(QtProperty* property) override;
00586 private:
00587     QScopedPointer<QtRectPropertyManagerPrivate> d_ptr;
00588     Q_DECLARE_PRIVATE(QtRectPropertyManager)
00589         Q_DISABLE_COPY_MOVE(QtRectPropertyManager)
00590         Q_PRIVATE_SLOT(d_func(), void slotIntChanged(QtProperty*, int))
00591         Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty*))
00592 };
00593
00594 #pragma endregion
00595
00596 #pragma region QtRectFPropertyManager
00597
00598 class QtRectFPropertyManagerPrivate;
00599
00600 class QT_QTPROPERTYBROWSER_EXPORT QtRectFPropertyManager : public QtAbstractPropertyManager
00601 {
00602     Q_OBJECT
00603 public:
00604     QtRectFPropertyManager(QObject* parent = 0);
00605     ~QtRectFPropertyManager();
00606
00607     QtDoublePropertyManager* subDoublePropertyManager() const;
00608
00609     QRectF value(const QtProperty* property) const;
00610     QRectF constraint(const QtProperty* property) const;
00611     int decimals(const QtProperty* property) const;
00612
00613 public Q_SLOTS:
00614     void setValue(QtProperty* property, const QRectF& val);
00615     void setConstraint(QtProperty* property, const QRectF& constraint);
00616     void setDecimals(QtProperty* property, int prec);
00617 Q_SIGNALS:
00618     void valueChanged(QtProperty* property, const QRectF& val);
00619     void constraintChanged(QtProperty* property, const QRectF& constraint);
00620     void decimalsChanged(QtProperty* property, int prec);
00621 protected:
00622     QString valueText(const QtProperty* property) const override;
00623     void initializeProperty(QtProperty* property) override;
00624     void uninitializeProperty(QtProperty* property) override;
00625 private:
00626     QScopedPointer<QtRectFPropertyManagerPrivate> d_ptr;
00627     Q_DECLARE_PRIVATE(QtRectFPropertyManager)
00628         Q_DISABLE_COPY_MOVE(QtRectFPropertyManager)
00629         Q_PRIVATE_SLOT(d_func(), void slotDoubleChanged(QtProperty*, double))
00630         Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty*))
00631 };
00632
00633 #pragma endregion
00634
00635 #pragma region QtEnumPropertyManager
00636
00637 class QtEnumPropertyManagerPrivate;
00638
00639 class QT_QTPROPERTYBROWSER_EXPORT QtEnumPropertyManager : public QtAbstractPropertyManager
00640 {
00641     Q_OBJECT
00642 public:
00643     QtEnumPropertyManager(QObject* parent = 0);
00644     ~QtEnumPropertyManager();
00645
00646     int value(const QtProperty* property) const;
00647     QStringList enumNames(const QtProperty* property) const;
00648     QMap<int, QIcon> enumIcons(const QtProperty* property) const;
00649
00650 public Q_SLOTS:
00651     void setValue(QtProperty* property, int val);
00652     void setEnumNames(QtProperty* property, const QStringList& names);
00653     void setEnumIcons(QtProperty* property, const QMap<int, QIcon>& icons);
00654 Q_SIGNALS:
00655     void valueChanged(QtProperty* property, int val);
00656     void enumNamesChanged(QtProperty* property, const QStringList& names);
00657     void enumIconsChanged(QtProperty* property, const QMap<int, QIcon>& icons);
00658 protected:
00659     QString valueText(const QtProperty* property) const override;
00660     QIcon valueIcon(const QtProperty* property) const override;
00661     void initializeProperty(QtProperty* property) override;
00662     void uninitializeProperty(QtProperty* property) override;
00663 private:
00664     QScopedPointer<QtEnumPropertyManagerPrivate> d_ptr;
00665     Q_DECLARE_PRIVATE(QtEnumPropertyManager)
00666         Q_DISABLE_COPY_MOVE(QtEnumPropertyManager)
00667 };
00668
00669 #pragma endregion
```

```
00670
00671 #pragma region QtFlagPropertyManager
00672
00673 class QtFlagPropertyManagerPrivate;
00674
00675 class QT_QTPROPERTYBROWSER_EXPORT QtFlagPropertyManager : public QtAbstractPropertyManager
00676 {
00677     Q_OBJECT
00678 public:
00679     QtFlagPropertyManager(QObject* parent = 0);
00680     ~QtFlagPropertyManager();
00681
00682     QtBoolPropertyManager* subBoolPropertyManager() const;
00683
00684     int value(const QtProperty* property) const;
00685     QStringList flagNames(const QtProperty* property) const;
00686
00687 public Q_SLOTS:
00688     void setValue(QtProperty* property, int val);
00689     void setFlagNames(QtProperty* property, const QStringList& names);
00690 Q_SIGNALS:
00691     void valueChanged(QtProperty* property, int val);
00692     void flagNamesChanged(QtProperty* property, const QStringList& names);
00693 protected:
00694     QString valueText(const QtProperty* property) const override;
00695     void initializeProperty(QtProperty* property) override;
00696     void uninitializeProperty(QtProperty* property) override;
00697 private:
00698     QScopedPointer<QtFlagPropertyManagerPrivate> d_ptr;
00699     Q_DECLARE_PRIVATE(QtFlagPropertyManager)
00700         Q_DISABLE_COPY_MOVE(QtFlagPropertyManager)
00701         Q_PRIVATE_SLOT(d_func(), void slotBoolChanged(QtProperty*, bool))
00702         Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty*))
00703 };
00704
00705 #pragma endregion
00706
00707 #pragma region QtSizePolicyPropertyManager
00708
00709 class QtSizePolicyPropertyManagerPrivate;
00710
00711 class QT_QTPROPERTYBROWSER_EXPORT QtSizePolicyPropertyManager : public QtAbstractPropertyManager
00712 {
00713     Q_OBJECT
00714 public:
00715     QtSizePolicyPropertyManager(QObject* parent = 0);
00716     ~QtSizePolicyPropertyManager();
00717
00718     QtIntPropertyManager* subIntPropertyManager() const;
00719     QtEnumPropertyManager* subEnumPropertyManager() const;
00720
00721     QSizePolicy value(const QtProperty* property) const;
00722
00723 public Q_SLOTS:
00724     void setValue(QtProperty* property, const QSizePolicy& val);
00725 Q_SIGNALS:
00726     void valueChanged(QtProperty* property, const QSizePolicy& val);
00727 protected:
00728     QString valueText(const QtProperty* property) const override;
00729     void initializeProperty(QtProperty* property) override;
00730     void uninitializeProperty(QtProperty* property) override;
00731 private:
00732     QScopedPointer<QtSizePolicyPropertyManagerPrivate> d_ptr;
00733     Q_DECLARE_PRIVATE(QtSizePolicyPropertyManager)
00734         Q_DISABLE_COPY_MOVE(QtSizePolicyPropertyManager)
00735         Q_PRIVATE_SLOT(d_func(), void slotIntChanged(QtProperty*, int))
00736         Q_PRIVATE_SLOT(d_func(), void slotEnumChanged(QtProperty*, int))
00737         Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty*))
00738 };
00739
00740 #pragma endregion
00741
00742 #pragma region QtFontPropertyManager
00743
00744 class QtFontPropertyManagerPrivate;
00745
00746 class QT_QTPROPERTYBROWSER_EXPORT QtFontPropertyManager : public QtAbstractPropertyManager
00747 {
00748     Q_OBJECT
00749 public:
00750     QtFontPropertyManager(QObject* parent = 0);
00751     ~QtFontPropertyManager();
00752
00753     QtIntPropertyManager* subIntPropertyManager() const;
00754     QtEnumPropertyManager* subEnumPropertyManager() const;
00755     QtBoolPropertyManager* subBoolPropertyManager() const;
00756
```

```
00757     QFont value(const QtProperty* property) const;
00758
00759 public Q_SLOTS:
00760     void setValue(QtProperty* property, const QFont& val);
00761 Q_SIGNALS:
00762     void valueChanged(QtProperty* property, const QFont& val);
00763 protected:
00764     QString valueText(const QtProperty* property) const override;
00765     QIcon valueIcon(const QtProperty* property) const override;
00766     void initializeProperty(QtProperty* property) override;
00767     void uninitializeProperty(QtProperty* property) override;
00768 private:
00769     QScopedPointer<QtFontPropertyManagerPrivate> d_ptr;
00770     Q_DECLARE_PRIVATE(QtFontPropertyManager)
00771         Q_DISABLE_COPY_MOVE(QtFontPropertyManager)
00772         Q_PRIVATE_SLOT(d_func(), void slotIntChanged(QtProperty*, int))
00773         Q_PRIVATE_SLOT(d_func(), void slotEnumChanged(QtProperty*, int))
00774         Q_PRIVATE_SLOT(d_func(), void slotBoolChanged(QtProperty*, bool))
00775         Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty*))
00776         Q_PRIVATE_SLOT(d_func(), void slotFontDatabaseChanged())
00777         Q_PRIVATE_SLOT(d_func(), void slotFontDatabaseDelayedChange())
00778 };
00779
00780 #pragma endregion
00781
00782 #pragma region QtColorPropertyManager
00783
00784 class QtColorPropertyManagerPrivate;
00785
00786 class QT_QTPROPERTYBROWSER_EXPORT QtColorPropertyManager : public QtAbstractPropertyManager
00787 {
00788     Q_OBJECT
00789 public:
00790     QtColorPropertyManager(QObject* parent = 0);
00791     ~QtColorPropertyManager();
00792
00793     QtIntPropertyManager* subIntPropertyManager() const;
00794
00795     QColor value(const QtProperty* property) const;
00796
00797 public Q_SLOTS:
00798     void setValue(QtProperty* property, const QColor& val);
00799 Q_SIGNALS:
00800     void valueChanged(QtProperty* property, const QColor& val);
00801 protected:
00802     QString valueText(const QtProperty* property) const override;
00803     QIcon valueIcon(const QtProperty* property) const override;
00804     void initializeProperty(QtProperty* property) override;
00805     void uninitializeProperty(QtProperty* property) override;
00806 private:
00807     QScopedPointer<QtColorPropertyManagerPrivate> d_ptr;
00808     Q_DECLARE_PRIVATE(QtColorPropertyManager)
00809         Q_DISABLE_COPY_MOVE(QtColorPropertyManager)
00810         Q_PRIVATE_SLOT(d_func(), void slotIntChanged(QtProperty*, int))
00811         Q_PRIVATE_SLOT(d_func(), void slotPropertyDestroyed(QtProperty*))
00812 };
00813
00814 #pragma endregion
00815
00816 #pragma region QtCursorPropertyManager
00817
00818 class QtCursorPropertyManagerPrivate;
00819
00820 class QT_QTPROPERTYBROWSER_EXPORT QtCursorPropertyManager : public QtAbstractPropertyManager
00821 {
00822     Q_OBJECT
00823 public:
00824     QtCursorPropertyManager(QObject* parent = 0);
00825     ~QtCursorPropertyManager();
00826
00827 #ifndef QT_NO_CURSOR
00828     QCursor value(const QtProperty* property) const;
00829 #endif
00830
00831 public Q_SLOTS:
00832     void setValue(QtProperty* property, const QCursor& val);
00833 Q_SIGNALS:
00834     void valueChanged(QtProperty* property, const QCursor& val);
00835 protected:
00836     QString valueText(const QtProperty* property) const override;
00837     QIcon valueIcon(const QtProperty* property) const override;
00838     void initializeProperty(QtProperty* property) override;
00839     void uninitializeProperty(QtProperty* property) override;
00840 private:
00841     QScopedPointer<QtCursorPropertyManagerPrivate> d_ptr;
00842     Q_DECLARE_PRIVATE(QtCursorPropertyManager)
00843         Q_DISABLE_COPY_MOVE(QtCursorPropertyManager)
```

```
00844 };
00845
00846 #pragma endregion
00847
00848 QT_END_NAMESPACE
00849
00850 #endif
```

18.91 qttreepropertybrowser.h

```
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00034 ** https://www.gnu.org/licenses/gpl-3.0.html.
00035 **
00036 ** $QT_END_LICENSE$
00037 **
00038 ****************************************************************************/
00039
00040 #ifndef QTTREEPROPERTYBROWSER_H
00041 #define QTTREEPROPERTYBROWSER_H
00042
00043 #include "qtpropertybrowser.h"
00044
00045 QT_BEGIN_NAMESPACE
00046
00047 class QTreeWidgetItem;
00048 class QtTreePropertyBrowserPrivate;
00049
00050 class QT_QTPROPERTYBROWSER_EXPORT QtTreePropertyBrowser : public QtAbstractPropertyBrowser
00051 {
00052     Q_OBJECT
00053     Q_PROPERTY(int indentation READ indentation WRITE setIndentation)
00054     Q_PROPERTY(bool rootIsDecorated READ rootIsDecorated WRITE setRootIsDecorated)
00055     Q_PROPERTY(bool alternatingRowColors READ alternatingRowColors WRITE setAlternatingRowColors)
00056     Q_PROPERTY(bool headerVisible READ isHeaderVisible WRITE setHeaderVisible)
00057     Q_PROPERTY(ResizeMode resizeMode READ resizeMode WRITE setResizeMode)
00058     Q_PROPERTY(int splitterPosition READ splitterPosition WRITE setSplitterPosition)
00059     Q_PROPERTY(bool propertiesWithoutValueMarked READ propertiesWithoutValueMarked WRITE
      setPropertiesWithoutValueMarked)
00060 public:
00061
00062     enum ResizeMode
00063     {
00064         Interactive,
00065         Stretch,
00066         Fixed,
00067         ResizeToContents
00068     };
00069     Q_ENUM(ResizeMode)
00070
00071     QtTreePropertyBrowser(QWidget *parent = 0);
00072     ~QtTreePropertyBrowser();
00073
00074     int indentation() const;
```

```
00075     void setIndentation(int i);
00076
00077     bool rootIsDecorated() const;
00078     void setRootIsDecorated(bool show);
00079
00080     bool alternatingRowColors() const;
00081     void setAlternatingRowColors(bool enable);
00082
00083     bool isHeaderVisible() const;
00084     void setHeaderVisible(bool visible);
00085
00086     ResizeMode resizeMode() const;
00087     void setResizeMode(ResizeMode mode);
00088
00089     int splitterPosition() const;
00090     void setSplitterPosition(int position);
00091
00092     void setExpanded(QtBrowserItem *item, bool expanded);
00093     bool isExpanded(QtBrowserItem *item) const;
00094
00095     bool isItemVisible(QtBrowserItem *item) const;
00096     void setItemVisible(QtBrowserItem *item, bool visible);
00097
00098     void setBackgroundColor(QtBrowserItem *item, const QColor &color);
00099     QColor backgroundColor(QtBrowserItem *item) const;
00100     QColor calculatedBackgroundColor(QtBrowserItem *item) const;
00101
00102     void setPropertiesWithoutValueMarked(bool mark);
00103     bool propertiesWithoutValueMarked() const;
00104
00105     void editItem(QtBrowserItem *item);
00106
00107 Q_SIGNALS:
00108
00109     void collapsed(QtBrowserItem *item);
00110     void expanded(QtBrowserItem *item);
00111
00112 protected:
00113     void itemInserted(QtBrowserItem *item, QtBrowserItem *afterItem) override;
00114     void itemRemoved(QtBrowserItem *item) override;
00115     void itemChanged(QtBrowserItem *item) override;
00116
00117 private:
00118
00119     QScopedPointer<QtTreePropertyBrowserPrivate> d_ptr;
00120     Q_DECLARE_PRIVATE(QtTreePropertyBrowser)
00121     Q_DISABLE_COPY_MOVE(QtTreePropertyBrowser)
00122
00123     Q_PRIVATE_SLOT(d_func(), void slotCollapsed(const QModelIndex &))
00124     Q_PRIVATE_SLOT(d_func(), void slotExpanded(const QModelIndex &))
00125     Q_PRIVATE_SLOT(d_func(), void slotCurrentBrowserItemChanged(QtBrowserItem *))
00126     Q_PRIVATE_SLOT(d_func(), void slotCurrentTreeItemChanged(QTreeWidgetItem *, QTreeWidgetItem *))
00127
00128 };
00129
00130 QT_END_NAMESPACE
00131
00132 #endif
```

18.92 qtvariantproperty.h

```
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00002 **
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```

```
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00034 ** https://www.gnu.org/licenses/gpl-3.0.html.
00035 **
00036 ** $QT_END_LICENSE$
00037 **
00038 ****************************************************************************/
00039
00040 #ifndef QTVARIANTPROPERTY_H
00041 #define QTVARIANTPROPERTY_H
00042
00043 #include "qtpropertybrowser.h"
00044 #include <QtCore/QVariant>
00045 #include <QtGui/QIcon>
00046
00047 QT_BEGIN_NAMESPACE
00048
00049 class QRegularExpression;
00050
00051 typedef QMap<int, QIcon> QtIconMap;
00052
00053 class QtVariantPropertyManager;
00054
00055 class QT_QTPROPERTYBROWSER_EXPORT QtVariantProperty : public QtProperty
00056 {
00057 public:
00058     ~QtVariantProperty();
00059     QVariant value() const;
00060     QVariant attributeValue(const QString &attribute) const;
00061     int valueType() const;
00062     int propertyType() const;
00063
00064     void setValue(const QVariant &value);
00065     void setAttribute(const QString &attribute, const QVariant &value);
00066 protected:
00067     QtVariantProperty(QtVariantPropertyManager *manager);
00068 private:
00069     friend class QtVariantPropertyManager;
00070     QScopedPointer<class QtVariantPropertyPrivate> d_ptr;
00071 };
00072
00073 class QT_QTPROPERTYBROWSER_EXPORT QtVariantPropertyManager : public QtAbstractPropertyManager
00074 {
00075     Q_OBJECT
00076 public:
00077     QtVariantPropertyManager(QObject *parent = 0);
00078     ~QtVariantPropertyManager();
00079
00080     virtual QtVariantProperty *addProperty(int propertyType, const QString &name = QString());
00081
00082     int propertyType(const QtProperty *property) const;
00083     int valueType(const QtProperty *property) const;
00084     QtVariantProperty *variantProperty(const QtProperty *property) const;
00085
00086     virtual bool isPropertyTypeSupported(int propertyType) const;
00087     virtual int valueType(int propertyType) const;
00088     virtual QStringList attributes(int propertyType) const;
00089     virtual int attributeType(int propertyType, const QString &attribute) const;
00090
00091     virtual QVariant value(const QtProperty *property) const;
00092     virtual QVariant attributeValue(const QtProperty *property, const QString &attribute) const;
00093
00094     static int enumTypeId();
00095     static int flagTypeId();
00096     static int groupTypeId();
00097     static int iconMapTypeId();
00098 public Q_SLOTS:
00099     virtual void setValue(QtProperty *property, const QVariant &val);
00100     virtual void setAttribute(QtProperty *property,
00101                 const QString &attribute, const QVariant &value);
00102 Q_SIGNALS:
00103     void valueChanged(QtProperty *property, const QVariant &val);
00104     void attributeChanged(QtProperty *property,
00105                 const QString &attribute, const QVariant &val);
00106 protected:
00107     bool hasValue(const QtProperty *property) const override;
00108     QString valueText(const QtProperty *property) const override;
00109     QIcon valueIcon(const QtProperty *property) const override;
00110     void initializeProperty(QtProperty *property) override;
00111     void uninitializeProperty(QtProperty *property) override;
```

```
00112     QtProperty *createProperty() override;
00113 private:
00114     QScopedPointer<class QtVariantPropertyManagerPrivate> d_ptr;
00115     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, int))
00116     Q_PRIVATE_SLOT(d_func(), void slotRangeChanged(QtProperty *, int, int))
00117     Q_PRIVATE_SLOT(d_func(), void slotSingleStepChanged(QtProperty *, int))
00118     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, double))
00119     Q_PRIVATE_SLOT(d_func(), void slotRangeChanged(QtProperty *, double, double))
00120     Q_PRIVATE_SLOT(d_func(), void slotSingleStepChanged(QtProperty *, double))
00121     Q_PRIVATE_SLOT(d_func(), void slotDecimalsChanged(QtProperty *, int))
00122     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, bool))
00123     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QString &))
00124     Q_PRIVATE_SLOT(d_func(), void slotRegExpChanged(QtProperty *, const QRegularExpression &))
00125     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, QDate))
00126     Q_PRIVATE_SLOT(d_func(), void slotRangeChanged(QtProperty *, QDate, QDate))
00127     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, QTime))
00128     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QDateTime &))
00129     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QKeySequence &))
00130     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QChar &))
00131     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QLocale &))
00132     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QPoint &))
00133     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QPointF &))
00134     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QSize &))
00135     Q_PRIVATE_SLOT(d_func(), void slotRangeChanged(QtProperty *, const QSize &, const QSize &))
00136     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QSizeF &))
00137     Q_PRIVATE_SLOT(d_func(), void slotRangeChanged(QtProperty *, const QSizeF &, const QSizeF &))
00138     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QRect &))
00139     Q_PRIVATE_SLOT(d_func(), void slotConstraintChanged(QtProperty *, const QRect &))
00140     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QRectF &))
00141     Q_PRIVATE_SLOT(d_func(), void slotConstraintChanged(QtProperty *, const QRectF &))
00142     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QColor &))
00143     Q_PRIVATE_SLOT(d_func(), void slotEnumNamesChanged(QtProperty *, const QStringList &))
00144     Q_PRIVATE_SLOT(d_func(), void slotEnumIconsChanged(QtProperty *, const QMap<int, QIcon> &))
00145     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QSizePolicy &))
00146     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QFont &))
00147     Q_PRIVATE_SLOT(d_func(), void slotValueChanged(QtProperty *, const QCursor &))
00148     Q_PRIVATE_SLOT(d_func(), void slotFlagNamesChanged(QtProperty *, const QStringList &))
00149
00150     Q_PRIVATE_SLOT(d_func(), void slotPropertyInserted(QtProperty *, QtProperty *, QtProperty *))
00151     Q_PRIVATE_SLOT(d_func(), void slotPropertyRemoved(QtProperty *, QtProperty *))
00152     Q_DECLARE_PRIVATE(QtVariantPropertyManager)
00153     Q_DISABLE_COPY_MOVE(QtVariantPropertyManager)
00154 };
00155
00156 class QT_QTPROPERTYBROWSER_EXPORT QtVariantEditorFactory : public
      QtAbstractEditorFactory<QtVariantPropertyManager>
00157 {
00158     Q_OBJECT
00159 public:
00160     QtVariantEditorFactory(QObject *parent = 0);
00161     ~QtVariantEditorFactory();
00162 protected:
00163     void connectPropertyManager(QtVariantPropertyManager *manager) override;
00164     QWidget *createEditor(QtVariantPropertyManager *manager, QtProperty *property,
00165                 QWidget *parent) override;
00166     void disconnectPropertyManager(QtVariantPropertyManager *manager) override;
00167 private:
00168     QScopedPointer<class QtVariantEditorFactoryPrivate> d_ptr;
00169     Q_DECLARE_PRIVATE(QtVariantEditorFactory)
00170     Q_DISABLE_COPY_MOVE(QtVariantEditorFactory)
00171 };
00172
00173 QT_END_NAMESPACE
00174
00175 Q_DECLARE_METATYPE(QIcon)
00176 Q_DECLARE_METATYPE(QtIconMap)
00177 #endif
```

18.93 qcustomplot.h

```
00001 /**************************************************************************
00002  **                                                                      **
00003  **  QCustomPlot, an easy to use, modern plotting widget for Qt          **
00004  **  Copyright (C) 2011-2021 Emanuel Eichhammer                          **
00005  **                                                                      **
00006  **  This program is free software: you can redistribute it and/or modify  **
00007  **  it under the terms of the GNU General Public License as published by  **
00008  **  the Free Software Foundation, either version 3 of the License, or     **
00009  **  (at your option) any later version.                                 **
00010  **                                                                      **
00011  **  This program is distributed in the hope that it will be useful,       **
00012  **  but WITHOUT ANY WARRANTY; without even the implied warranty of        **
00013  **  MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the         **
00014  **  GNU General Public License for more details.                        **
00015  **                                                                      **
```

```
00016  **  You should have received a copy of the GNU General Public License     **
00017  **  along with this program.  If not, see http://www.gnu.org/licenses/.   **
00018  **                                                                         **
00019  *****************************************************************************
00020  **           Author: Emanuel Eichhammer                                    **
00021  **  Website/Contact: http://www.qcustomplot.com/                           **
00022  **             Date: 29.03.21                                              **
00023  **          Version: 2.1.0                                                 **
00024  *****************************************************************************/
00025
00026 // https://www.qcustomplot.com/documentation/index.html
00027
00028 #ifndef QCUSTOMPLOT_H
00029 #define QCUSTOMPLOT_H
00030
00031 #include <QtCore/qglobal.h>
00032
00033 // some Qt version/configuration dependent macros to include or exclude certain code paths:
00034 #ifdef QCUSTOMPLOT_USE_OPENGL
00035 #if QT_VERSION < QT_VERSION_CHECK(5, 0, 0)
00036 #define QCP_OPENGL_PBUFFER
00037 #else
00038 #define QCP_OPENGL_FBO
00039 #endif
00040 #if QT_VERSION >= QT_VERSION_CHECK(5, 3, 0)
00041 #define QCP_OPENGL_OFFSCREENSURFACE
00042 #endif
00043 #endif
00044
00045 #if QT_VERSION >= QT_VERSION_CHECK(5, 4, 0)
00046 #define QCP_DEVICEPIXELRATIO_SUPPORTED
00047 #if QT_VERSION >= QT_VERSION_CHECK(5, 6, 0)
00048 #define QCP_DEVICEPIXELRATIO_FLOAT
00049 #endif
00050 #endif
00051
00052 #include <QtCore/QObject>
00053 #include <QtCore/QPointer>
00054 #include <QtCore/QSharedPointer>
00055 #include <QtCore/QTimer>
00056 #include <QtGui/QPainter>
00057 #include <QtGui/QPainterPath>
00058 #include <QtGui/QPaintEvent>
00059 #include <QtGui/QMouseEvent>
00060 #include <QtGui/QWheelEvent>
00061 #include <QtGui/QPixmap>
00062 #include <QtCore/QVector>
00063 #include <QtCore/QString>
00064 #include <QtCore/QDateTime>
00065 #include <QtCore/QMultiMap>
00066 #include <QtCore/QFlags>
00067 #include <QtCore/QDebug>
00068 #include <QtCore/QStack>
00069 #include <QtCore/QCache>
00070 #include <QtCore/QMargins>
00071 #include <qmath.h>
00072 #include <limits>
00073 #include <algorithm>
00074 #ifdef QCP_OPENGL_FBO
00075 #include <QtGui/QOpenGLContext>
00076 #if QT_VERSION < QT_VERSION_CHECK(6, 0, 0)
00077 #include <QtGui/QOpenGLFramebufferObject>
00078 #else
00079 #include <QOpenGLFramebufferObject>
00080 #include <QOpenGLPaintDevice>
00081 #endif
00082 #ifdef QCP_OPENGL_OFFSCREENSURFACE
00083 #include <QtGui/QOffscreenSurface>
00084 #else
00085 #include <QtGui/QWindow>
00086 #endif
00087 #endif
00088 #ifdef QCP_OPENGL_PBUFFER
00089 #include <QtOpenGL/QGLPixelBuffer>
00090 #endif
00091 #if QT_VERSION < QT_VERSION_CHECK(5, 0, 0)
00092 #include <qnumeric.h>
00093 #include <QtGui/QWidget>
00094 #include <QtGui/QPrinter>
00095 #include <QtGui/QPrintEngine>
00096 #else
00097 #include <QtNumeric>
00098 #include <QtWidgets/QWidget>
00099 #include <QtPrintSupport/QtPrintSupport>
00100 #endif
00101 #if QT_VERSION >= QT_VERSION_CHECK(4, 8, 0)
00102 #include <QtCore/QElapsedTimer>
```

```
00103 #endif
00104 #if QT_VERSION >= QT_VERSION_CHECK(5, 2, 0)
00105 #include <QtCore/QTimeZone>
00106 #endif
00107
00108 class QCPPainter;
00109 class QCustomPlot;
00110 class QCPLayerable;
00111 class QCPLayoutElement;
00112 class QCPLayout;
00113 class QCPAxis;
00114 class QCPAxisRect;
00115 class QCPAxisPainterPrivate;
00116 class QCPAbstractPlottable;
00117 class QCPGraph;
00118 class QCPAbstractItem;
00119 class QCPPlottableInterface1D;
00120 class QCPLegend;
00121 class QCPItemPosition;
00122 class QCPLayer;
00123 class QCPAbstractLegendItem;
00124 class QCPSelectionRect;
00125 class QCPColorMap;
00126 class QCPColorScale;
00127 class QCPBars;
00128 class QCPPolarAxisRadial;
00129 class QCPPolarAxisAngular;
00130 class QCPPolarGrid;
00131 class QCPPolarGraph;
00132
00133 /* including file 'src/global.h'             */
00134 /* modified 2021-03-29T02:30:44, size 16981 */
00135
00136 #define QCUSTOMPLOT_VERSION_STR "2.1.0"
00137 #define QCUSTOMPLOT_VERSION 0x020100
00138
00139 // decl definitions for shared library compilation/usage:
00140 #if defined(QT_STATIC_BUILD)
00141 #define QCP_LIB_DECL
00142 #elif defined(QCUSTOMPLOT_COMPILE_LIBRARY)
00143 #define QCP_LIB_DECL Q_DECL_EXPORT
00144 #elif defined(QCUSTOMPLOT_USE_LIBRARY)
00145 #define QCP_LIB_DECL Q_DECL_IMPORT
00146 #else
00147 #define QCP_LIB_DECL
00148 #endif
00149
00150 // define empty macro for Q_DECL_OVERRIDE if it doesn't exist (Qt < 5)
00151 #ifndef Q_DECL_OVERRIDE
00152 #define Q_DECL_OVERRIDE
00153 #endif
00154
00161 #ifndef Q_MOC_RUN
00162 namespace QCP {
00163 #else
00164
00165 class QCP { // when in moc-run, make it look like a class, so we get Q_GADGET, Q_ENUMS/Q_FLAGS
      features in namespace
00166
00167     Q_GADGET
00168     Q_ENUMS(ExportPen)
00169     Q_ENUMS(ResolutionUnit)
00170     Q_ENUMS(SignDomain)
00171     Q_ENUMS(MarginSide)
00172     Q_FLAGS(MarginSides)
00173     Q_ENUMS(AntialiasedElement)
00174     Q_FLAGS(AntialiasedElements)
00175     Q_ENUMS(PlottingHint)
00176     Q_FLAGS(PlottingHints)
00177     Q_ENUMS(Interaction)
00178     Q_FLAGS(Interactions)
00179     Q_ENUMS(SelectionRectMode)
00180     Q_ENUMS(SelectionType)
00181 public:
00182 #endif
00183
00190     enum ResolutionUnit {
00191         ruDotsPerMeter
00192         , ruDotsPerCentimeter
00193         , ruDotsPerInch
00194     };
00195
00201     enum ExportPen {
00202         epNoCosmetic
00203         , epAllowCosmetic
00204     };
00205
```

```
00213     enum SignDomain {
00214         sdNegative
00215         , sdBoth
00216         , sdPositive
00217     };
00218
00224     enum MarginSide {
00225         msLeft = 0x01
00226         , msRight = 0x02
00227         , msTop = 0x04
00228         , msBottom = 0x08
00229         , msAll = 0xFF
00230         , msNone = 0x00
00231     };
00232
00233     Q_DECLARE_FLAGS(MarginSides, MarginSide)
00234
00235
00244     enum AntialiasedElement {
00245         aeAxes = 0x0001
00246         , aeGrid = 0x0002
00247         , aeSubGrid = 0x0004
00248         , aeLegend = 0x0008
00249         , aeLegendItems = 0x0010
00250         , aePlottables = 0x0020
00251         , aeItems = 0x0040
00252         , aeScatters = 0x0080
00253         , aeFills = 0x0100
00254         , aeZeroLine = 0x0200
00255         , aeOther = 0x8000
00256         , aeAll = 0xFFFF
00257         , aeNone = 0x0000
00258     };
00259
00260     Q_DECLARE_FLAGS(AntialiasedElements, AntialiasedElement)
00261
00262
00267     enum PlottingHint {
00268         phNone = 0x000
00269         , phFastPolylines = 0x001
00271         , phImmediateRefresh = 0x002
00273         , phCacheLabels = 0x004
00274     };
00275
00276     Q_DECLARE_FLAGS(PlottingHints, PlottingHint)
00277
00278
00285     enum Interaction {
00286         iNone = 0x000
00287         , iRangeDrag = 0x001
00288         , iRangeZoom = 0x002
00289         , iMultiSelect = 0x004
00290         , iSelectPlottables = 0x008
00291         , iSelectAxes = 0x010
00292         , iSelectLegend = 0x020
00293         , iSelectItems = 0x040
00294         , iSelectOther = 0x080
00295         , iSelectPlottablesBeyondAxisRect = 0x100
00296     };
00297
00298     Q_DECLARE_FLAGS(Interactions, Interaction)
00299
00300
00305     enum SelectionRectMode {
00306         srmNone
00307         , srmZoom
00308         , srmSelect
00309         , srmCustom
00310     };
00311
00330     enum SelectionType {
00331         stNone
00332         , stWhole
00333         , stSingleData
00334         , stDataRange
00335         , stMultipleDataRanges
00336     };
00337
00344     inline bool isInvalidData(double value) {
00345         return qIsNaN(value) || qIsInf(value);
00346     }
00347
00353     inline bool isInvalidData(double value1, double value2) {
00354         return isInvalidData(value1) || isInvalidData(value2);
00355     }
00356
00363     inline void setMarginValue(QMargins &margins, QCP::MarginSide side, int value) {
```

```
00364         switch (side) {
00365             case QCP::msLeft: margins.setLeft(value);
00366                 break;
00367             case QCP::msRight: margins.setRight(value);
00368                 break;
00369             case QCP::msTop: margins.setTop(value);
00370                 break;
00371             case QCP::msBottom: margins.setBottom(value);
00372                 break;
00373             case QCP::msAll: margins = QMargins(value, value, value, value);
00374                 break;
00375             default: break;
00376         }
00377     }
00378
00386     inline int getMarginValue(const QMargins &margins, QCP::MarginSide side) {
00387         switch (side) {
00388             case QCP::msLeft: return margins.left();
00389             case QCP::msRight: return margins.right();
00390             case QCP::msTop: return margins.top();
00391             case QCP::msBottom: return margins.bottom();
00392             default: break;
00393         }
00394         return 0;
00395     }
00396
00397
00398     extern const QMetaObject staticMetaObject; // in moc-run we create a static meta object for QCP
      "fake" object. This line is the link to it via QCP::staticMetaObject in normal operation as namespace
00399
00400 } // end of namespace QCP
00401
00402 Q_DECLARE_OPERATORS_FOR_FLAGS(QCP::AntialiasedElements)
00403 Q_DECLARE_OPERATORS_FOR_FLAGS(QCP::PlottingHints)
00404 Q_DECLARE_OPERATORS_FOR_FLAGS(QCP::MarginSides)
00405 Q_DECLARE_OPERATORS_FOR_FLAGS(QCP::Interactions)
00406 Q_DECLARE_METATYPE(QCP::ExportPen)
00407 Q_DECLARE_METATYPE(QCP::ResolutionUnit)
00408 Q_DECLARE_METATYPE(QCP::SignDomain)
00409 Q_DECLARE_METATYPE(QCP::MarginSide)
00410 Q_DECLARE_METATYPE(QCP::AntialiasedElement)
00411 Q_DECLARE_METATYPE(QCP::PlottingHint)
00412 Q_DECLARE_METATYPE(QCP::Interaction)
00413 Q_DECLARE_METATYPE(QCP::SelectionRectMode)
00414 Q_DECLARE_METATYPE(QCP::SelectionType)
00415
00416 /* end of 'src/global.h' */
00417
00418
00419 /* including file 'src/vector2d.h'         */
00420 /* modified 2021-03-29T02:30:44, size 4988 */
00421
00422 class QCP_LIB_DECL QCPVector2D {
00423 public:
00424     QCPVector2D();
00425     QCPVector2D(double x, double y);
00426     QCPVector2D(const QPoint &point);
00427     QCPVector2D(const QPointF &point);
00428
00429     // getters:
00430
00431     double x() const {
00432         return mX;
00433     }
00434
00435     double y() const {
00436         return mY;
00437     }
00438
00439     double &rx() {
00440         return mX;
00441     }
00442
00443     double &ry() {
00444         return mY;
00445     }
00446
00447     // setters:
00448
00449     void setX(double x) {
00450         mX = x;
00451     }
00452
00453     void setY(double y) {
00454         mY = y;
00455     }
00456
```

```
00457     // non-virtual methods:
00458
00459     double length() const {
00460         return qSqrt(mX * mX + mY * mY);
00461     }
00462
00463     double lengthSquared() const {
00464         return mX * mX + mY*mY;
00465     }
00466
00467     double angle() const {
00468         return qAtan2(mY, mX);
00469     }
00470
00471     QPoint toPoint() const {
00472         return QPoint(int(mX), int(mY));
00473     }
00474
00475     QPointF toPointF() const {
00476         return QPointF(mX, mY);
00477     }
00478
00479     bool isNull() const {
00480         return qIsNull(mX) && qIsNull(mY);
00481     }
00482     void normalize();
00483     QCPVector2D normalized() const;
00484
00485     QCPVector2D perpendicular() const {
00486         return QCPVector2D(-mY, mX);
00487     }
00488
00489     double dot(const QCPVector2D &vec) const {
00490         return mX * vec.mX + mY * vec.mY;
00491     }
00492     double distanceSquaredToLine(const QCPVector2D &start, const QCPVector2D &end) const;
00493     double distanceSquaredToLine(const QLineF &line) const;
00494     double distanceToStraightLine(const QCPVector2D &base, const QCPVector2D &direction) const;
00495
00496     QCPVector2D &operator*=(double factor);
00497     QCPVector2D &operator/=(double divisor);
00498     QCPVector2D &operator+=(const QCPVector2D &vector);
00499     QCPVector2D &operator-=(const QCPVector2D &vector);
00500
00501 private:
00502     // property members:
00503     double mX, mY;
00504
00505     friend inline const QCPVector2D operator*(double factor, const QCPVector2D &vec);
00506     friend inline const QCPVector2D operator*(const QCPVector2D &vec, double factor);
00507     friend inline const QCPVector2D operator/(const QCPVector2D &vec, double divisor);
00508     friend inline const QCPVector2D operator+(const QCPVector2D &vec1, const QCPVector2D &vec2);
00509     friend inline const QCPVector2D operator-(const QCPVector2D &vec1, const QCPVector2D &vec2);
00510     friend inline const QCPVector2D operator-(const QCPVector2D &vec);
00511 };
00512 Q_DECLARE_TYPEINFO(QCPVector2D, Q_MOVABLE_TYPE);
00513
00514 inline const QCPVector2D operator*(double factor, const QCPVector2D &vec) {
00515     return QCPVector2D(vec.mX*factor, vec.mY * factor);
00516 }
00517
00518 inline const QCPVector2D operator*(const QCPVector2D &vec, double factor) {
00519     return QCPVector2D(vec.mX*factor, vec.mY * factor);
00520 }
00521
00522 inline const QCPVector2D operator/(const QCPVector2D &vec, double divisor) {
00523     return QCPVector2D(vec.mX / divisor, vec.mY / divisor);
00524 }
00525
00526 inline const QCPVector2D operator+(const QCPVector2D &vec1, const QCPVector2D &vec2) {
00527     return QCPVector2D(vec1.mX + vec2.mX, vec1.mY + vec2.mY);
00528 }
00529
00530 inline const QCPVector2D operator-(const QCPVector2D &vec1, const QCPVector2D &vec2) {
00531     return QCPVector2D(vec1.mX - vec2.mX, vec1.mY - vec2.mY);
00532 }
00533
00534 inline const QCPVector2D operator-(const QCPVector2D &vec) {
00535     return QCPVector2D(-vec.mX, -vec.mY);
00536 }
00537
00542 inline QDebug operator«(QDebug d, const QCPVector2D &vec) {
00543     d.nospace() « "QCPVector2D(" « vec.x() « ", " « vec.y() « ")";
00544     return d.space();
00545 }
00546
00547 /* end of 'src/vector2d.h' */
```

```
00548
00549
00550 /* including file 'src/painter.h'          */
00551
00552 /* modified 2021-03-29T02:30:44, size 4035 */
00553
00554 class QCP_LIB_DECL QCPPainter : public QPainter {
00555
00556     Q_GADGET
00557 public:
00562     enum PainterMode {
00563         pmDefault = 0x00
00564         , pmVectorized = 0x01
00565         , pmNoCaching = 0x02
00566         , pmNonCosmetic = 0x04
00567     };
00568     Q_ENUMS(PainterMode)
00569     Q_FLAGS(PainterModes)
00570     Q_DECLARE_FLAGS(PainterModes, PainterMode)
00571
00572     QCPPainter();
00573     explicit QCPPainter(QPaintDevice *device);
00574
00575     // getters:
00576
00577     bool antialiasing() const {
00578         return testRenderHint(QPainter::Antialiasing);
00579     }
00580
00581     PainterModes modes() const {
00582         return mModes;
00583     }
00584
00585     // setters:
00586     void setAntialiasing(bool enabled);
00587     void setMode(PainterMode mode, bool enabled = true);
00588     void setModes(PainterModes modes);
00589
00590     // methods hiding non-virtual base class functions (QPainter bug workarounds):
00591     bool begin(QPaintDevice *device);
00592     void setPen(const QPen &pen);
00593     void setPen(const QColor &color);
00594     void setPen(Qt::PenStyle penStyle);
00595     void drawLine(const QLineF &line);
00596
00597     void drawLine(const QPointF &p1, const QPointF &p2) {
00598         drawLine(QLineF(p1, p2));
00599     }
00600     void save();
00601     void restore();
00602
00603     // non-virtual methods:
00604     void makeNonCosmetic();
00605
00606 protected:
00607     // property members:
00608     PainterModes mModes;
00609     bool mIsAntialiasing;
00610
00611     // non-property members:
00612     QStack<bool> mAntialiasingStack;
00613 };
00614
00615 Q_DECLARE_OPERATORS_FOR_FLAGS(QCPPainter::PainterModes)
00616 Q_DECLARE_METATYPE(QCPPainter::PainterMode)
00617
00618 /* end of 'src/painter.h' */
00619
00620
00621 /* including file 'src/paintbuffer.h'      */
00622 /* modified 2021-03-29T02:30:44, size 5006 */
00623
00624 class QCP_LIB_DECL QCPAbstractPaintBuffer {
00625 public:
00626     explicit QCPAbstractPaintBuffer(const QSize &size, double devicePixelRatio);
00627     virtual ~QCPAbstractPaintBuffer();
00628
00629     // getters:
00630
00631     QSize size() const {
00632         return mSize;
00633     }
00634
00635     bool invalidated() const {
00636         return mInvalidated;
00637     }
00638
```

```
00639     double devicePixelRatio() const {
00640         return mDevicePixelRatio;
00641     }
00642 
00643     // setters:
00644     void setSize(const QSize &size);
00645     void setInvalidated(bool invalidated = true);
00646     void setDevicePixelRatio(double ratio);
00647 
00648     // introduced virtual methods:
00649     virtual QCPPainter *startPainting() = 0;
00650 
00651     virtual void donePainting() {
00652     }
00653     virtual void draw(QCPPainter *painter) const = 0;
00654     virtual void clear(const QColor &color) = 0;
00655 
00656 protected:
00657     // property members:
00658     QSize mSize;
00659     double mDevicePixelRatio;
00660 
00661     // non-property members:
00662     bool mInvalidated;
00663 
00664     // introduced virtual methods:
00665     virtual void reallocateBuffer() = 0;
00666 };
00667 
00668 class QCP_LIB_DECL QCPPaintBufferPixmap : public QCPAbstractPaintBuffer {
00669 public:
00670     explicit QCPPaintBufferPixmap(const QSize &size, double devicePixelRatio);
00671     virtual ~QCPPaintBufferPixmap() Q_DECL_OVERRIDE;
00672 
00673     // reimplemented virtual methods:
00674     virtual QCPPainter *startPainting() Q_DECL_OVERRIDE;
00675     virtual void draw(QCPPainter *painter) const Q_DECL_OVERRIDE;
00676     void clear(const QColor &color) Q_DECL_OVERRIDE;
00677 
00678 protected:
00679     // non-property members:
00680     QPixmap mBuffer;
00681 
00682     // reimplemented virtual methods:
00683     virtual void reallocateBuffer() Q_DECL_OVERRIDE;
00684 };
00685 
00686 
00687 #ifdef QCP_OPENGL_PBUFFER
00688 
00689 class QCP_LIB_DECL QCPPaintBufferGlPbuffer : public QCPAbstractPaintBuffer {
00690 public:
00691     explicit QCPPaintBufferGlPbuffer(const QSize &size, double devicePixelRatio, int multisamples);
00692     virtual ~QCPPaintBufferGlPbuffer() Q_DECL_OVERRIDE;
00693 
00694     // reimplemented virtual methods:
00695     virtual QCPPainter *startPainting() Q_DECL_OVERRIDE;
00696     virtual void draw(QCPPainter *painter) const Q_DECL_OVERRIDE;
00697     void clear(const QColor &color) Q_DECL_OVERRIDE;
00698 
00699 protected:
00700     // non-property members:
00701     QGLPixelBuffer *mGlPBuffer;
00702     int mMultisamples;
00703 
00704     // reimplemented virtual methods:
00705     virtual void reallocateBuffer() Q_DECL_OVERRIDE;
00706 };
00707 #endif // QCP_OPENGL_PBUFFER
00708 
00709 
00710 #ifdef QCP_OPENGL_FBO
00711 
00712 class QCP_LIB_DECL QCPPaintBufferGlFbo : public QCPAbstractPaintBuffer {
00713 public:
00714     explicit QCPPaintBufferGlFbo(const QSize &size, double devicePixelRatio,
      QWeakPointer<QOpenGLContext> glContext, QWeakPointer<QOpenGLPaintDevice> glPaintDevice);
00715     virtual ~QCPPaintBufferGlFbo() Q_DECL_OVERRIDE;
00716 
00717     // reimplemented virtual methods:
00718     virtual QCPPainter *startPainting() Q_DECL_OVERRIDE;
00719     virtual void donePainting() Q_DECL_OVERRIDE;
00720     virtual void draw(QCPPainter *painter) const Q_DECL_OVERRIDE;
00721     void clear(const QColor &color) Q_DECL_OVERRIDE;
00722 
00723 protected:
00724     // non-property members:
```

```
00725     QWeakPointer<QOpenGLContext> mGlContext;
00726     QWeakPointer<QOpenGLPaintDevice> mGlPaintDevice;
00727     QOpenGLFramebufferObject *mGlFrameBuffer;
00728 
00729     // reimplemented virtual methods:
00730     virtual void reallocateBuffer() Q_DECL_OVERRIDE;
00731 };
00732 #endif // QCP_OPENGL_FBO
00733 
00734 /* end of 'src/paintbuffer.h' */
00735 
00736 
00737 /* including file 'src/layer.h'            */
00738 
00739 /* modified 2021-03-29T02:30:44, size 7038 */
00740 
00741 class QCP_LIB_DECL QCPLayer : public QObject {
00742 
00743     Q_OBJECT
00745     Q_PROPERTY(QCustomPlot* parentPlot READ parentPlot)
00746     Q_PROPERTY(QString name READ name)
00747     Q_PROPERTY(int index READ index)
00748     Q_PROPERTY(QList<QCPLayerable*> children READ children)
00749     Q_PROPERTY(bool visible READ visible WRITE setVisible)
00750     Q_PROPERTY(LayerMode mode READ mode WRITE setMode)
00752 public:
00753 
00761     enum LayerMode {
00762         lmLogical
00763         , lmBuffered
00764     };
00765     Q_ENUMS(LayerMode)
00766 
00767     QCPLayer(QCustomPlot* parentPlot, const QString &layerName);
00768     virtual ~QCPLayer();
00769 
00770     // getters:
00771 
00772     QCustomPlot *parentPlot() const {
00773         return mParentPlot;
00774     }
00775 
00776     QString name() const {
00777         return mName;
00778     }
00779 
00780     int index() const {
00781         return mIndex;
00782     }
00783 
00784     QList<QCPLayerable*> children() const {
00785         return mChildren;
00786     }
00787 
00788     bool visible() const {
00789         return mVisible;
00790     }
00791 
00792     LayerMode mode() const {
00793         return mMode;
00794     }
00795 
00796     // setters:
00797     void setVisible(bool visible);
00798     void setMode(LayerMode mode);
00799 
00800     // non-virtual methods:
00801     void replot();
00802 
00803 protected:
00804     // property members:
00805     QCustomPlot *mParentPlot;
00806     QString mName;
00807     int mIndex;
00808     QList<QCPLayerable*> mChildren;
00809     bool mVisible;
00810     LayerMode mMode;
00811 
00812     // non-property members:
00813     QWeakPointer<QCPAbstractPaintBuffer> mPaintBuffer;
00814 
00815     // non-virtual methods:
00816     void draw(QCPPainter *painter);
00817     void drawToPaintBuffer();
00818     void addChild(QCPLayerable *layerable, bool prepend);
00819     void removeChild(QCPLayerable *layerable);
00820 
```

```
00821 private:
00822     Q_DISABLE_COPY(QCPLayer)
00823 
00824     friend class QCustomPlot;
00825     friend class QCPLayerable;
00826 };
00827 
00828 Q_DECLARE_METATYPE(QCPLayer::LayerMode)
00829 
00830 class QCP_LIB_DECL QCPLayerable : public QObject {
00831     Q_OBJECT
00833     Q_PROPERTY(bool visible READ visible WRITE setVisible)
00834     Q_PROPERTY(QCustomPlot* parentPlot READ parentPlot)
00835     Q_PROPERTY(QCPLayerable* parentLayerable READ parentLayerable)
00836     Q_PROPERTY(QCPLayer* layer READ layer WRITE setLayer NOTIFY layerChanged)
00837     Q_PROPERTY(bool antialiased READ antialiased WRITE setAntialiased)
00839 public:
00840     QCPLayerable(QCustomPlot *plot, QString targetLayer = QString(), QCPLayerable *parentLayerable =
      nullptr);
00841     virtual ~QCPLayerable();
00842 
00843     // getters:
00844 
00845     bool visible() const {
00846         return mVisible;
00847     }
00848 
00849     QCustomPlot *parentPlot() const {
00850         return mParentPlot;
00851     }
00852 
00853     QCPLayerable *parentLayerable() const {
00854         return mParentLayerable.data();
00855     }
00856 
00857     QCPLayer *layer() const {
00858         return mLayer;
00859     }
00860 
00861     bool antialiased() const {
00862         return mAntialiased;
00863     }
00864 
00865     // setters:
00866     void setVisible(bool on);
00867     Q_SLOT bool setLayer(QCPLayer *layer);
00868     bool setLayer(const QString &layerName);
00869     void setAntialiased(bool enabled);
00870 
00871     // introduced virtual methods:
00872     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const;
00873 
00874     // non-property methods:
00875     bool realVisibility() const;
00876 
00877 signals:
00878     void layerChanged(QCPLayer *newLayer);
00879 
00880 protected:
00881     // property members:
00882     bool mVisible;
00883     QCustomPlot *mParentPlot;
00884     QPointer<QCPLayerable> mParentLayerable;
00885     QCPLayer *mLayer;
00886     bool mAntialiased;
00887 
00888     // introduced virtual methods:
00889     virtual void parentPlotInitialized(QCustomPlot *parentPlot);
00890     virtual QCP::Interaction selectionCategory() const;
00891     virtual QRect clipRect() const;
00892     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const = 0;
00893     virtual void draw(QCPPainter *painter) = 0;
00894     // selection events:
00895     virtual void selectEvent(QMouseEvent *event, bool additive, const QVariant &details, bool
      *selectionStateChanged);
00896     virtual void deselectEvent(bool *selectionStateChanged);
00897     // low-level mouse events:
00898     virtual void mousePressEvent(QMouseEvent *event, const QVariant &details);
00899     virtual void mouseMoveEvent(QMouseEvent *event, const QPointF &startPos);
00900     virtual void mouseReleaseEvent(QMouseEvent *event, const QPointF &startPos);
00901     virtual void mouseDoubleClickEvent(QMouseEvent *event, const QVariant &details);
00902     virtual void wheelEvent(QWheelEvent *event);
00903 
00904     // non-property methods:
00905     void initializeParentPlot(QCustomPlot *parentPlot);
00906     void setParentLayerable(QCPLayerable* parentLayerable);
```

```
00907     bool moveToLayer(QCPLayer *layer, bool prepend);
00908     void applyAntialiasingHint(QCPPainter *painter, bool localAntialiased, QCP::AntialiasedElement
      overrideElement) const;
00909
00910 private:
00911     Q_DISABLE_COPY(QCPLayerable)
00912
00913     friend class QCustomPlot;
00914     friend class QCPLayer;
00915     friend class QCPAxisRect;
00916 };
00917
00918 /* end of 'src/layer.h' */
00919
00920
00921 /* including file 'src/axis/range.h'       */
00922
00923 /* modified 2021-03-29T02:30:44, size 5280 */
00924
00925 class QCP_LIB_DECL QCPRange {
00926 public:
00927     double lower, upper;
00928
00929     QCPRange();
00930     QCPRange(double lower, double upper);
00931
00932     bool operator==(const QCPRange& other) const {
00933         return lower == other.lower && upper == other.upper;
00934     }
00935
00936     bool operator!=(const QCPRange& other) const {
00937         return !(*this == other);
00938     }
00939
00940     QCPRange &operator+=(const double& value) {
00941         lower += value;
00942         upper += value;
00943         return *this;
00944     }
00945
00946     QCPRange &operator-=(const double& value) {
00947         lower -= value;
00948         upper -= value;
00949         return *this;
00950     }
00951
00952     QCPRange &operator*=(const double& value) {
00953         lower *= value;
00954         upper *= value;
00955         return *this;
00956     }
00957
00958     QCPRange &operator/=(const double& value) {
00959         lower /= value;
00960         upper /= value;
00961         return *this;
00962     }
00963     friend inline const QCPRange operator+(const QCPRange&, double);
00964     friend inline const QCPRange operator+(double, const QCPRange&);
00965     friend inline const QCPRange operator-(const QCPRange& range, double value);
00966     friend inline const QCPRange operator*(const QCPRange& range, double value);
00967     friend inline const QCPRange operator*(double value, const QCPRange& range);
00968     friend inline const QCPRange operator/(const QCPRange& range, double value);
00969
00970     double size() const {
00971         return upper - lower;
00972     }
00973
00974     double center() const {
00975         return (upper + lower)*0.5;
00976     }
00977
00978     void normalize() {
00979         if (lower > upper) qSwap(lower, upper);
00980     }
00981     void expand(const QCPRange &otherRange);
00982     void expand(double includeCoord);
00983     QCPRange expanded(const QCPRange &otherRange) const;
00984     QCPRange expanded(double includeCoord) const;
00985     QCPRange bounded(double lowerBound, double upperBound) const;
00986     QCPRange sanitizedForLogScale() const;
00987     QCPRange sanitizedForLinScale() const;
00988
00989     bool contains(double value) const {
00990         return value >= lower && value <= upper;
00991     }
00992
```

```
00993     static bool validRange(double lower, double upper);
00994     static bool validRange(const QCPRange &range);
00995     static const double minRange;
00996     static const double maxRange;
00997
00998 };
00999 Q_DECLARE_TYPEINFO(QCPRange, Q_MOVABLE_TYPE);
01000
01005 inline QDebug operator<<(QDebug d, const QCPRange &range) {
01006     d.nospace() << "QCPRange(" << range.lower << ", " << range.upper << ")";
01007     return d.space();
01008 }
01009
01013 inline const QCPRange operator+(const QCPRange& range, double value) {
01014     QCPRange result(range);
01015     result += value;
01016     return result;
01017 }
01018
01022 inline const QCPRange operator+(double value, const QCPRange& range) {
01023     QCPRange result(range);
01024     result += value;
01025     return result;
01026 }
01027
01031 inline const QCPRange operator-(const QCPRange& range, double value) {
01032     QCPRange result(range);
01033     result -= value;
01034     return result;
01035 }
01036
01040 inline const QCPRange operator*(const QCPRange& range, double value) {
01041     QCPRange result(range);
01042     result *= value;
01043     return result;
01044 }
01045
01049 inline const QCPRange operator*(double value, const QCPRange& range) {
01050     QCPRange result(range);
01051     result *= value;
01052     return result;
01053 }
01054
01058 inline const QCPRange operator/(const QCPRange& range, double value) {
01059     QCPRange result(range);
01060     result /= value;
01061     return result;
01062 }
01063
01064 /* end of 'src/axis/range.h' */
01065
01066
01067 /* including file 'src/selection.h'        */
01068
01069 /* modified 2021-03-29T02:30:44, size 8569 */
01070
01071 class QCP_LIB_DECL QCPDataRange {
01072 public:
01073     QCPDataRange();
01074     QCPDataRange(int begin, int end);
01075
01076     bool operator==(const QCPDataRange& other) const {
01077         return mBegin == other.mBegin && mEnd == other.mEnd;
01078     }
01079
01080     bool operator!=(const QCPDataRange& other) const {
01081         return !(*this == other);
01082     }
01083
01084     // getters:
01085
01086     int begin() const {
01087         return mBegin;
01088     }
01089
01090     int end() const {
01091         return mEnd;
01092     }
01093
01094     int size() const {
01095         return mEnd - mBegin;
01096     }
01097
01098     int length() const {
01099         return size();
01100     }
01101
```

```
01102     // setters:
01103
01104     void setBegin(int begin) {
01105         mBegin = begin;
01106     }
01107
01108     void setEnd(int end) {
01109         mEnd = end;
01110     }
01111
01112     // non-property methods:
01113
01114     bool isValid() const {
01115         return (mEnd >= mBegin) && (mBegin >= 0);
01116     }
01117
01118     bool isEmpty() const {
01119         return length() == 0;
01120     }
01121     QCPDataRange bounded(const QCPDataRange &other) const;
01122     QCPDataRange expanded(const QCPDataRange &other) const;
01123     QCPDataRange intersection(const QCPDataRange &other) const;
01124
01125     QCPDataRange adjusted(int changeBegin, int changeEnd) const {
01126         return QCPDataRange(mBegin + changeBegin, mEnd + changeEnd);
01127     }
01128     bool intersects(const QCPDataRange &other) const;
01129     bool contains(const QCPDataRange &other) const;
01130
01131 private:
01132     // property members:
01133     int mBegin, mEnd;
01134
01135 };
01136 Q_DECLARE_TYPEINFO(QCPDataRange, Q_MOVABLE_TYPE);
01137
01138 class QCP_LIB_DECL QCPDataSelection {
01139 public:
01140     explicit QCPDataSelection();
01141     explicit QCPDataSelection(const QCPDataRange &range);
01142
01143     bool operator==(const QCPDataSelection& other) const;
01144
01145     bool operator!=(const QCPDataSelection& other) const {
01146         return !(*this == other);
01147     }
01148     QCPDataSelection &operator+=(const QCPDataSelection& other);
01149     QCPDataSelection &operator+=(const QCPDataRange& other);
01150     QCPDataSelection &operator-=(const QCPDataSelection& other);
01151     QCPDataSelection &operator-=(const QCPDataRange& other);
01152     friend inline const QCPDataSelection operator+(const QCPDataSelection& a, const QCPDataSelection&
      b);
01153     friend inline const QCPDataSelection operator+(const QCPDataRange& a, const QCPDataSelection& b);
01154     friend inline const QCPDataSelection operator+(const QCPDataSelection& a, const QCPDataRange& b);
01155     friend inline const QCPDataSelection operator+(const QCPDataRange& a, const QCPDataRange& b);
01156     friend inline const QCPDataSelection operator-(const QCPDataSelection& a, const QCPDataSelection&
      b);
01157     friend inline const QCPDataSelection operator-(const QCPDataRange& a, const QCPDataSelection& b);
01158     friend inline const QCPDataSelection operator-(const QCPDataSelection& a, const QCPDataRange& b);
01159     friend inline const QCPDataSelection operator-(const QCPDataRange& a, const QCPDataRange& b);
01160
01161     // getters:
01162
01163     int dataRangeCount() const {
01164         return mDataRanges.size();
01165     }
01166     int dataPointCount() const;
01167     QCPDataRange dataRange(int index = 0) const;
01168
01169     QList<QCPDataRange> dataRanges() const {
01170         return mDataRanges;
01171     }
01172     QCPDataRange span() const;
01173
01174     // non-property methods:
01175     void addDataRange(const QCPDataRange &dataRange, bool simplify = true);
01176     void clear();
01177
01178     bool isEmpty() const {
01179         return mDataRanges.isEmpty();
01180     }
01181     void simplify();
01182     void enforceType(QCP::SelectionType type);
01183     bool contains(const QCPDataSelection &other) const;
01184     QCPDataSelection intersection(const QCPDataRange &other) const;
01185     QCPDataSelection intersection(const QCPDataSelection &other) const;
01186     QCPDataSelection inverse(const QCPDataRange &outerRange) const;
```

```
01187
01188 private:
01189     // property members:
01190     QList<QCPDataRange> mDataRanges;
01191
01192     inline static bool lessThanDataRangeBegin(const QCPDataRange &a, const QCPDataRange &b) {
01193         return a.begin() < b.begin();
01194     }
01195 };
01196
01197 Q_DECLARE_METATYPE(QCPDataSelection)
01198
01199
01200
01204 inline const QCPDataSelection operator+(const QCPDataSelection& a, const QCPDataSelection& b) {
01205     QCPDataSelection result(a);
01206     result += b;
01207     return result;
01208 }
01209
01214 inline const QCPDataSelection operator+(const QCPDataRange& a, const QCPDataSelection& b) {
01215     QCPDataSelection result(a);
01216     result += b;
01217     return result;
01218 }
01219
01224 inline const QCPDataSelection operator+(const QCPDataSelection& a, const QCPDataRange& b) {
01225     QCPDataSelection result(a);
01226     result += b;
01227     return result;
01228 }
01229
01234 inline const QCPDataSelection operator+(const QCPDataRange& a, const QCPDataRange& b) {
01235     QCPDataSelection result(a);
01236     result += b;
01237     return result;
01238 }
01239
01243 inline const QCPDataSelection operator-(const QCPDataSelection& a, const QCPDataSelection& b) {
01244     QCPDataSelection result(a);
01245     result -= b;
01246     return result;
01247 }
01248
01252 inline const QCPDataSelection operator-(const QCPDataRange& a, const QCPDataSelection& b) {
01253     QCPDataSelection result(a);
01254     result -= b;
01255     return result;
01256 }
01257
01261 inline const QCPDataSelection operator-(const QCPDataSelection& a, const QCPDataRange& b) {
01262     QCPDataSelection result(a);
01263     result -= b;
01264     return result;
01265 }
01266
01270 inline const QCPDataSelection operator-(const QCPDataRange& a, const QCPDataRange& b) {
01271     QCPDataSelection result(a);
01272     result -= b;
01273     return result;
01274 }
01275
01280 inline QDebug operator<<(QDebug d, const QCPDataRange &dataRange) {
01281     d.nospace() << "QCPDataRange(" << dataRange.begin() << ", " << dataRange.end() << ")";
01282     return d;
01283 }
01284
01289 inline QDebug operator<<(QDebug d, const QCPDataSelection &selection) {
01290     d.nospace() << "QCPDataSelection(";
01291     for (int i = 0; i < selection.dataRangeCount(); ++i) {
01292         if (i != 0)
01293             d << ", ";
01294         d << selection.dataRange(i);
01295     }
01296     d << ")";
01297     return d;
01298 }
01299
01300
01301
01302 /* end of 'src/selection.h' */
01303
01304
01305 /* including file 'src/selectionrect.h'     */
01306
01307 /* modified 2021-03-29T02:30:44, size 3354 */
01308
```

```
01309 class QCP_LIB_DECL QCPSelectionRect : public QCPLayerable {
01310     Q_OBJECT
01311 public:
01312     explicit QCPSelectionRect(QCustomPlot *parentPlot);
01313     virtual ~QCPSelectionRect() Q_DECL_OVERRIDE;
01314
01315     // getters:
01316
01317     QRect rect() const {
01318         return mRect;
01319     }
01320     QCPRange range(const QCPAxis *axis) const;
01321
01322     QPen pen() const {
01323         return mPen;
01324     }
01325
01326     QBrush brush() const {
01327         return mBrush;
01328     }
01329
01330     bool isActive() const {
01331         return mActive;
01332     }
01333
01334     // setters:
01335     void setPen(const QPen &pen);
01336     void setBrush(const QBrush &brush);
01337
01338     // non-property methods:
01339     Q_SLOT void cancel();
01340
01341 signals:
01342     void started(QMouseEvent *event);
01343     void changed(const QRect &rect, QMouseEvent *event);
01344     void canceled(const QRect &rect, QInputEvent *event);
01345     void accepted(const QRect &rect, QMouseEvent *event);
01346
01347 protected:
01348     // property members:
01349     QRect mRect;
01350     QPen mPen;
01351     QBrush mBrush;
01352     // non-property members:
01353     bool mActive;
01354
01355     // introduced virtual methods:
01356     virtual void startSelection(QMouseEvent *event);
01357     virtual void moveSelection(QMouseEvent *event);
01358     virtual void endSelection(QMouseEvent *event);
01359     virtual void keyPressEvent(QKeyEvent *event);
01360
01361     // reimplemented virtual methods
01362     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
01363     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
01364
01365     friend class QCustomPlot;
01366 };
01367
01368 /* end of 'src/selectionrect.h' */
01369
01370
01371 /* including file 'src/layout.h'            */
01372
01373 /* modified 2021-03-29T02:30:44, size 14279 */
01374
01375 class QCP_LIB_DECL QCPMarginGroup : public QObject {
01376     Q_OBJECT
01377 public:
01378     explicit QCPMarginGroup(QCustomPlot *parentPlot);
01379     virtual ~QCPMarginGroup();
01380
01381     // non-virtual methods:
01382
01383     QList<QCPLayoutElement*> elements(QCP::MarginSide side) const {
01384         return mChildren.value(side);
01385     }
01386     bool isEmpty() const;
01387     void clear();
01388
01389 protected:
01390     // non-property members:
01391     QCustomPlot *mParentPlot;
01392     QHash<QCP::MarginSide, QList<QCPLayoutElement*> > mChildren;
01393
01394     // introduced virtual methods:
01395     virtual int commonMargin(QCP::MarginSide side) const;
```

```
01396
01397     // non-virtual methods:
01398     void addChild(QCP::MarginSide side, QCPLayoutElement *element);
01399     void removeChild(QCP::MarginSide side, QCPLayoutElement *element);
01400
01401 private:
01402     Q_DISABLE_COPY(QCPMarginGroup)
01403
01404     friend class QCPLayoutElement;
01405 };
01406
01407 class QCP_LIB_DECL QCPLayoutElement : public QCPLayerable {
01408
01409     Q_OBJECT
01411     Q_PROPERTY(QCPLayout* layout READ layout)
01412     Q_PROPERTY(QRect rect READ rect)
01413     Q_PROPERTY(QRect outerRect READ outerRect WRITE setOuterRect)
01414     Q_PROPERTY(QMargins margins READ margins WRITE setMargins)
01415     Q_PROPERTY(QMargins minimumMargins READ minimumMargins WRITE setMinimumMargins)
01416     Q_PROPERTY(QSize minimumSize READ minimumSize WRITE setMinimumSize)
01417     Q_PROPERTY(QSize maximumSize READ maximumSize WRITE setMaximumSize)
01418     Q_PROPERTY(SizeConstraintRect sizeConstraintRect READ sizeConstraintRect WRITE
      setSizeConstraintRect)
01420 public:
01425     enum UpdatePhase {
01426         upPreparation
01427         , upMargins
01428         , upLayout
01429     };
01430
01431     Q_ENUMS(UpdatePhase)
01432
01433
01441     enum SizeConstraintRect {
01442         scrInnerRect
01443         , scrOuterRect
01444     };
01445     Q_ENUMS(SizeConstraintRect)
01446
01447     explicit QCPLayoutElement(QCustomPlot *parentPlot = nullptr);
01448     virtual ~QCPLayoutElement() Q_DECL_OVERRIDE;
01449
01450     // getters:
01451
01452     QCPLayout *layout() const {
01453         return mParentLayout;
01454     }
01455
01456     QRect rect() const {
01457         return mRect;
01458     }
01459
01460     QRect outerRect() const {
01461         return mOuterRect;
01462     }
01463
01464     QMargins margins() const {
01465         return mMargins;
01466     }
01467
01468     QMargins minimumMargins() const {
01469         return mMinimumMargins;
01470     }
01471
01472     QCP::MarginSides autoMargins() const {
01473         return mAutoMargins;
01474     }
01475
01476     QSize minimumSize() const {
01477         return mMinimumSize;
01478     }
01479
01480     QSize maximumSize() const {
01481         return mMaximumSize;
01482     }
01483
01484     SizeConstraintRect sizeConstraintRect() const {
01485         return mSizeConstraintRect;
01486     }
01487
01488     QCPMarginGroup *marginGroup(QCP::MarginSide side) const {
01489         return mMarginGroups.value(side, nullptr);
01490     }
01491
01492     QHash<QCP::MarginSide, QCPMarginGroup*> marginGroups() const {
01493         return mMarginGroups;
01494     }
```

```
01495
01496     // setters:
01497     void setOuterRect(const QRect &rect);
01498     void setMargins(const QMargins &margins);
01499     void setMinimumMargins(const QMargins &margins);
01500     void setAutoMargins(QCP::MarginSides sides);
01501     void setMinimumSize(const QSize &size);
01502     void setMinimumSize(int width, int height);
01503     void setMaximumSize(const QSize &size);
01504     void setMaximumSize(int width, int height);
01505     void setSizeConstraintRect(SizeConstraintRect constraintRect);
01506     void setMarginGroup(QCP::MarginSides sides, QCPMarginGroup *group);
01507
01508     // introduced virtual methods:
01509     virtual void update(UpdatePhase phase);
01510     virtual QSize minimumOuterSizeHint() const;
01511     virtual QSize maximumOuterSizeHint() const;
01512     virtual QList<QCPLayoutElement*> elements(bool recursive) const;
01513
01514     // reimplemented virtual methods:
01515     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
01516
01517 protected:
01518     // property members:
01519     QCPLayout *mParentLayout;
01520     QSize mMinimumSize, mMaximumSize;
01521     SizeConstraintRect mSizeConstraintRect;
01522     QRect mRect, mOuterRect;
01523     QMargins mMargins, mMinimumMargins;
01524     QCP::MarginSides mAutoMargins;
01525     QHash<QCP::MarginSide, QCPMarginGroup*> mMarginGroups;
01526
01527     // introduced virtual methods:
01528     virtual int calculateAutoMargin(QCP::MarginSide side);
01529     virtual void layoutChanged();
01530
01531     // reimplemented virtual methods:
01532
01533     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE {
01534         Q_UNUSED(painter)
01535     }
01536
01537     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE {
01538         Q_UNUSED(painter)
01539     }
01540     virtual void parentPlotInitialized(QCustomPlot *parentPlot) Q_DECL_OVERRIDE;
01541
01542 private:
01543     Q_DISABLE_COPY(QCPLayoutElement)
01544
01545     friend class QCustomPlot;
01546     friend class QCPLayout;
01547     friend class QCPMarginGroup;
01548 };
01549
01550 Q_DECLARE_METATYPE(QCPLayoutElement::UpdatePhase)
01551
01552
01553 class QCP_LIB_DECL QCPLayout : public QCPLayoutElement {
01554     Q_OBJECT
01555 public:
01556     explicit QCPLayout();
01557
01558     // reimplemented virtual methods:
01559     virtual void update(UpdatePhase phase) Q_DECL_OVERRIDE;
01560     virtual QList<QCPLayoutElement*> elements(bool recursive) const Q_DECL_OVERRIDE;
01561
01562     // introduced virtual methods:
01563     virtual int elementCount() const = 0;
01564     virtual QCPLayoutElement* elementAt(int index) const = 0;
01565     virtual QCPLayoutElement* takeAt(int index) = 0;
01566     virtual bool take(QCPLayoutElement* element) = 0;
01567     virtual void simplify();
01568
01569     // non-virtual methods:
01570     bool removeAt(int index);
01571     bool remove(QCPLayoutElement* element);
01572     void clear();
01573
01574 protected:
01575     // introduced virtual methods:
01576     virtual void updateLayout();
01577
01578     // non-virtual methods:
01579     void sizeConstraintsChanged() const;
01580     void adoptElement(QCPLayoutElement *el);
```

```
01581     void releaseElement(QCPLayoutElement *el);
01582     QVector<int> getSectionSizes(QVector<int> maxSizes, QVector<int> minSizes, QVector<double>
      stretchFactors, int totalSize) const;
01583     static QSize getFinalMinimumOuterSize(const QCPLayoutElement *el);
01584     static QSize getFinalMaximumOuterSize(const QCPLayoutElement *el);
01585
01586 private:
01587     Q_DISABLE_COPY(QCPLayout)
01588     friend class QCPLayoutElement;
01589 };
01590
01591 class QCP_LIB_DECL QCPLayoutGrid : public QCPLayout {
01592
01593     Q_OBJECT
01595     Q_PROPERTY(int rowCount READ rowCount)
01596     Q_PROPERTY(int columnCount READ columnCount)
01597     Q_PROPERTY(QList<double> columnStretchFactors READ columnStretchFactors WRITE
      setColumnStretchFactors)
01598     Q_PROPERTY(QList<double> rowStretchFactors READ rowStretchFactors WRITE setRowStretchFactors)
01599     Q_PROPERTY(int columnSpacing READ columnSpacing WRITE setColumnSpacing)
01600     Q_PROPERTY(int rowSpacing READ rowSpacing WRITE setRowSpacing)
01601     Q_PROPERTY(FillOrder fillOrder READ fillOrder WRITE setFillOrder)
01602     Q_PROPERTY(int wrap READ wrap WRITE setWrap)
01604 public:
01605
01613     enum FillOrder {
01614         foRowsFirst
01615         , foColumnsFirst
01616     };
01617     Q_ENUMS(FillOrder)
01618
01619     explicit QCPLayoutGrid();
01620     virtual ~QCPLayoutGrid() Q_DECL_OVERRIDE;
01621
01622     // getters:
01623
01624     int rowCount() const {
01625         return mElements.size();
01626     }
01627
01628     int columnCount() const {
01629         return mElements.size() > 0 ? mElements.first().size() : 0;
01630     }
01631
01632     QList<double> columnStretchFactors() const {
01633         return mColumnStretchFactors;
01634     }
01635
01636     QList<double> rowStretchFactors() const {
01637         return mRowStretchFactors;
01638     }
01639
01640     int columnSpacing() const {
01641         return mColumnSpacing;
01642     }
01643
01644     int rowSpacing() const {
01645         return mRowSpacing;
01646     }
01647
01648     int wrap() const {
01649         return mWrap;
01650     }
01651
01652     FillOrder fillOrder() const {
01653         return mFillOrder;
01654     }
01655
01656     // setters:
01657     void setColumnStretchFactor(int column, double factor);
01658     void setColumnStretchFactors(const QList<double> &factors);
01659     void setRowStretchFactor(int row, double factor);
01660     void setRowStretchFactors(const QList<double> &factors);
01661     void setColumnSpacing(int pixels);
01662     void setRowSpacing(int pixels);
01663     void setWrap(int count);
01664     void setFillOrder(FillOrder order, bool rearrange = true);
01665
01666     // reimplemented virtual methods:
01667     virtual void updateLayout() Q_DECL_OVERRIDE;
01668
01669     virtual int elementCount() const Q_DECL_OVERRIDE {
01670         return rowCount() * columnCount();
01671     }
01672     virtual QCPLayoutElement* elementAt(int index) const Q_DECL_OVERRIDE;
01673     virtual QCPLayoutElement* takeAt(int index) Q_DECL_OVERRIDE;
01674     virtual bool take(QCPLayoutElement* element) Q_DECL_OVERRIDE;
```

```
01675     virtual QList<QCPLayoutElement*> elements(bool recursive) const Q_DECL_OVERRIDE;
01676     virtual void simplify() Q_DECL_OVERRIDE;
01677     virtual QSize minimumOuterSizeHint() const Q_DECL_OVERRIDE;
01678     virtual QSize maximumOuterSizeHint() const Q_DECL_OVERRIDE;
01679
01680     // non-virtual methods:
01681     QCPLayoutElement *element(int row, int column) const;
01682     bool addElement(int row, int column, QCPLayoutElement *element);
01683     bool addElement(QCPLayoutElement *element);
01684     bool hasElement(int row, int column);
01685     void expandTo(int newRowCount, int newColumnCount);
01686     void insertRow(int newIndex);
01687     void insertColumn(int newIndex);
01688     int rowColToIndex(int row, int column) const;
01689     void indexToRowCol(int index, int &row, int &column) const;
01690
01691 protected:
01692     // property members:
01693     QList<QList<QCPLayoutElement*> > mElements;
01694     QList<double> mColumnStretchFactors;
01695     QList<double> mRowStretchFactors;
01696     int mColumnSpacing, mRowSpacing;
01697     int mWrap;
01698     FillOrder mFillOrder;
01699
01700     // non-virtual methods:
01701     void getMinimumRowColSizes(QVector<int> *minColWidths, QVector<int> *minRowHeights) const;
01702     void getMaximumRowColSizes(QVector<int> *maxColWidths, QVector<int> *maxRowHeights) const;
01703
01704 private:
01705     Q_DISABLE_COPY(QCPLayoutGrid)
01706 };
01707
01708 Q_DECLARE_METATYPE(QCPLayoutGrid::FillOrder)
01709
01710
01711 class QCP_LIB_DECL QCPLayoutInset : public QCPLayout {
01712
01713     Q_OBJECT
01714 public:
01718     enum InsetPlacement {
01719         ipFree
01720         , ipBorderAligned
01721     };
01722     Q_ENUMS(InsetPlacement)
01723
01724     explicit QCPLayoutInset();
01725     virtual ~QCPLayoutInset() Q_DECL_OVERRIDE;
01726
01727     // getters:
01728     InsetPlacement insetPlacement(int index) const;
01729     Qt::Alignment insetAlignment(int index) const;
01730     QRectF insetRect(int index) const;
01731
01732     // setters:
01733     void setInsetPlacement(int index, InsetPlacement placement);
01734     void setInsetAlignment(int index, Qt::Alignment alignment);
01735     void setInsetRect(int index, const QRectF &rect);
01736
01737     // reimplemented virtual methods:
01738     virtual void updateLayout() Q_DECL_OVERRIDE;
01739     virtual int elementCount() const Q_DECL_OVERRIDE;
01740     virtual QCPLayoutElement* elementAt(int index) const Q_DECL_OVERRIDE;
01741     virtual QCPLayoutElement* takeAt(int index) Q_DECL_OVERRIDE;
01742     virtual bool take(QCPLayoutElement* element) Q_DECL_OVERRIDE;
01743
01744     virtual void simplify() Q_DECL_OVERRIDE {
01745     }
01746     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
    const Q_DECL_OVERRIDE;
01747
01748     // non-virtual methods:
01749     void addElement(QCPLayoutElement *element, Qt::Alignment alignment);
01750     void addElement(QCPLayoutElement *element, const QRectF &rect);
01751
01752 protected:
01753     // property members:
01754     QList<QCPLayoutElement*> mElements;
01755     QList<InsetPlacement> mInsetPlacement;
01756     QList<Qt::Alignment> mInsetAlignment;
01757     QList<QRectF> mInsetRect;
01758
01759 private:
01760     Q_DISABLE_COPY(QCPLayoutInset)
01761 };
01762
01763 Q_DECLARE_METATYPE(QCPLayoutInset::InsetPlacement)
```

```
01764
01765 /* end of 'src/layout.h' */
01766
01767
01768 /* including file 'src/lineending.h'       */
01769 /* modified 2021-03-29T02:30:44, size 4426 */
01770
01771 class QCP_LIB_DECL QCPLineEnding {
01772
01773     Q_GADGET
01774 public:
01786     enum EndingStyle {
01787         esNone
01788         , esFlatArrow
01789         , esSpikeArrow
01790         , esLineArrow
01791         , esDisc
01792         , esSquare
01793         , esDiamond
01794         , esBar
01795         , esHalfBar
01796         , esSkewedBar
01797     };
01798     Q_ENUMS(EndingStyle)
01799
01800     QCPLineEnding();
01801     QCPLineEnding(EndingStyle style, double width = 8, double length = 10, bool inverted = false);
01802
01803     // getters:
01804
01805     EndingStyle style() const {
01806         return mStyle;
01807     }
01808
01809     double width() const {
01810         return mWidth;
01811     }
01812
01813     double length() const {
01814         return mLength;
01815     }
01816
01817     bool inverted() const {
01818         return mInverted;
01819     }
01820
01821     // setters:
01822     void setStyle(EndingStyle style);
01823     void setWidth(double width);
01824     void setLength(double length);
01825     void setInverted(bool inverted);
01826
01827     // non-property methods:
01828     double boundingDistance() const;
01829     double realLength() const;
01830     void draw(QCPPainter *painter, const QCPVector2D &pos, const QCPVector2D &dir) const;
01831     void draw(QCPPainter *painter, const QCPVector2D &pos, double angle) const;
01832
01833 protected:
01834     // property members:
01835     EndingStyle mStyle;
01836     double mWidth, mLength;
01837     bool mInverted;
01838 };
01839 Q_DECLARE_TYPEINFO(QCPLineEnding, Q_MOVABLE_TYPE);
01840
01841 Q_DECLARE_METATYPE(QCPLineEnding::EndingStyle)
01842
01843 /* end of 'src/lineending.h' */
01844
01845
01846 /* including file 'src/axis/labelpainter.h' */
01847 /* modified 2021-03-29T02:30:44, size 7086  */
01848
01849 class QCPLabelPainterPrivate {
01850
01851     Q_GADGET
01852 public:
01856     enum AnchorMode {
01857         amRectangular
01858         , amSkewedUpright
01859         , amSkewedRotated
01860     };
01861
01862     Q_ENUMS(AnchorMode)
01863
01864
```

```
01867     enum AnchorReferenceType {
01868         artNormal
01869         , artTangent
01870     };
01871
01872     Q_ENUMS(AnchorReferenceType)
01873
01874
01877     enum AnchorSide {
01878         asLeft
01879         , asRight
01880         , asTop
01881         , asBottom
01882         , asTopLeft
01883         , asTopRight
01884         , asBottomRight
01885         , asBottomLeft
01886     };
01887     Q_ENUMS(AnchorSide)
01888
01889     explicit QCPLabelPainterPrivate(QCustomPlot *parentPlot);
01890     virtual ~QCPLabelPainterPrivate();
01891
01892     // setters:
01893     void setAnchorSide(AnchorSide side);
01894     void setAnchorMode(AnchorMode mode);
01895     void setAnchorReference(const QPointF &pixelPoint);
01896     void setAnchorReferenceType(AnchorReferenceType type);
01897     void setFont(const QFont &font);
01898     void setColor(const QColor &color);
01899     void setPadding(int padding);
01900     void setRotation(double rotation);
01901     void setSubstituteExponent(bool enabled);
01902     void setMultiplicationSymbol(QChar symbol);
01903     void setAbbreviateDecimalPowers(bool enabled);
01904     void setCacheSize(int labelCount);
01905
01906     // getters:
01907
01908     AnchorMode anchorMode() const {
01909         return mAnchorMode;
01910     }
01911
01912     AnchorSide anchorSide() const {
01913         return mAnchorSide;
01914     }
01915
01916     QPointF anchorReference() const {
01917         return mAnchorReference;
01918     }
01919
01920     AnchorReferenceType anchorReferenceType() const {
01921         return mAnchorReferenceType;
01922     }
01923
01924     QFont font() const {
01925         return mFont;
01926     }
01927
01928     QColor color() const {
01929         return mColor;
01930     }
01931
01932     int padding() const {
01933         return mPadding;
01934     }
01935
01936     double rotation() const {
01937         return mRotation;
01938     }
01939
01940     bool substituteExponent() const {
01941         return mSubstituteExponent;
01942     }
01943
01944     QChar multiplicationSymbol() const {
01945         return mMultiplicationSymbol;
01946     }
01947
01948     bool abbreviateDecimalPowers() const {
01949         return mAbbreviateDecimalPowers;
01950     }
01951     int cacheSize() const;
01952
01953     //virtual int size() const;
01954
01955     // non-property methods:
```

```
01956     void drawTickLabel(QCPPainter *painter, const QPointF &tickPos, const QString &text);
01957     void clearCache();
01958 
01959     // constants that may be used with setMultiplicationSymbol:
01960     static const QChar SymbolDot;
01961     static const QChar SymbolCross;
01962 
01963 protected:
01964 
01965     struct CachedLabel {
01966         QPoint offset;
01967         QPixmap pixmap;
01968     };
01969 
01970     struct LabelData {
01971         AnchorSide side;
01972         double rotation; // angle in degrees
01973         QTransform transform; // the transform about the label anchor which is at (0, 0). Does not
      contain final absolute x/y positioning on the plot/axis
01974         QString basePart, expPart, suffixPart;
01975         QRect baseBounds, expBounds, suffixBounds;
01976         QRect totalBounds; // is in a coordinate system where label top left is at (0, 0)
01977         QRect rotatedTotalBounds; // is in a coordinate system where the label anchor is at (0, 0)
01978         QFont baseFont, expFont;
01979         QColor color;
01980     };
01981 
01982     // property members:
01983     AnchorMode mAnchorMode;
01984     AnchorSide mAnchorSide;
01985     QPointF mAnchorReference;
01986     AnchorReferenceType mAnchorReferenceType;
01987     QFont mFont;
01988     QColor mColor;
01989     int mPadding;
01990     double mRotation; // this is the rotation applied uniformly to all labels, not the heterogeneous
      rotation in amCircularRotated mode
01991     bool mSubstituteExponent;
01992     QChar mMultiplicationSymbol;
01993     bool mAbbreviateDecimalPowers;
01994     // non-property members:
01995     QCustomPlot *mParentPlot;
01996     QByteArray mLabelParameterHash; // to determine whether mLabelCache needs to be cleared due to
      changed parameters
01997     QCache<QString, CachedLabel> mLabelCache;
01998     QRect mAxisSelectionBox, mTickLabelsSelectionBox, mLabelSelectionBox;
01999     int mLetterCapHeight, mLetterDescent;
02000 
02001     // introduced virtual methods:
02002     virtual void drawLabelMaybeCached(QCPPainter *painter, const QFont &font, const QColor &color,
      const QPointF &pos, AnchorSide side, double rotation, const QString &text);
02003     virtual QByteArray generateLabelParameterHash() const; // TODO: get rid of this in favor of
      invalidation flag upon setters?
02004 
02005     // non-virtual methods:
02006     QPointF getAnchorPos(const QPointF &tickPos);
02007     void drawText(QCPPainter *painter, const QPointF &pos, const LabelData &labelData) const;
02008     LabelData getTickLabelData(const QFont &font, const QColor &color, double rotation, AnchorSide
      side, const QString &text) const;
02009     void applyAnchorTransform(LabelData &labelData) const;
02010     //void getMaxTickLabelSize(const QFont &font, const QString &text, QSize *tickLabelsSize) const;
02011     CachedLabel *createCachedLabel(const LabelData &labelData) const;
02012     QByteArray cacheKey(const QString &text, const QColor &color, double rotation, AnchorSide side)
      const;
02013     AnchorSide skewedAnchorSide(const QPointF &tickPos, double sideExpandHorz, double sideExpandVert)
      const;
02014     AnchorSide rotationCorrectedSide(AnchorSide side, double rotation) const;
02015     void analyzeFontMetrics();
02016 };
02017 
02018 Q_DECLARE_METATYPE(QCPLabelPainterPrivate::AnchorMode)
02019 Q_DECLARE_METATYPE(QCPLabelPainterPrivate::AnchorSide)
02020 
02021 
02022 /* end of 'src/axis/labelpainter.h' */
02023 
02024 
02025 /* including file 'src/axis/axisticker.h'  */
02026 /* modified 2021-03-29T02:30:44, size 4230 */
02027 
02028 class QCP_LIB_DECL QCPAxisTicker {
02029 
02030     Q_GADGET
02031 public:
02037     enum TickStepStrategy {
02038         tssReadability
02039         , tssMeetTickCount
```

```
02040     };
02041     Q_ENUMS(TickStepStrategy)
02042
02043     QCPAxisTicker();
02044     virtual ~QCPAxisTicker();
02045
02046     // getters:
02047
02048     TickStepStrategy tickStepStrategy() const {
02049         return mTickStepStrategy;
02050     }
02051
02052     int tickCount() const {
02053         return mTickCount;
02054     }
02055
02056     double tickOrigin() const {
02057         return mTickOrigin;
02058     }
02059
02060     // setters:
02061     void setTickStepStrategy(TickStepStrategy strategy);
02062     void setTickCount(int count);
02063     void setTickOrigin(double origin);
02064
02065     // introduced virtual methods:
02066     virtual void generate(const QCPRange &range, const QLocale &locale, QChar formatChar, int
      precision, QVector<double> &ticks, QVector<double> *subTicks, QVector<QString> *tickLabels);
02067
02068 protected:
02069     // property members:
02070     TickStepStrategy mTickStepStrategy;
02071     int mTickCount;
02072     double mTickOrigin;
02073
02074     // introduced virtual methods:
02075     virtual double getTickStep(const QCPRange &range);
02076     virtual int getSubTickCount(double tickStep);
02077     virtual QString getTickLabel(double tick, const QLocale &locale, QChar formatChar, int precision);
02078     virtual QVector<double> createTickVector(double tickStep, const QCPRange &range);
02079     virtual QVector<double> createSubTickVector(int subTickCount, const QVector<double> &ticks);
02080     virtual QVector<QString> createLabelVector(const QVector<double> &ticks, const QLocale &locale,
      QChar formatChar, int precision);
02081
02082     // non-virtual methods:
02083     void trimTicks(const QCPRange &range, QVector<double> &ticks, bool keepOneOutlier) const;
02084     double pickClosest(double target, const QVector<double> &candidates) const;
02085     double getMantissa(double input, double *magnitude = nullptr) const;
02086     double cleanMantissa(double input) const;
02087
02088 private:
02089     Q_DISABLE_COPY(QCPAxisTicker)
02090
02091 };
02092
02093 Q_DECLARE_METATYPE(QCPAxisTicker::TickStepStrategy)
02094 Q_DECLARE_METATYPE(QSharedPointer<QCPAxisTicker>)
02095
02096 /* end of 'src/axis/axisticker.h' */
02097
02098
02099 /* including file 'src/axis/axistickerdatetime.h' */
02100 /* modified 2021-03-29T02:30:44, size 3600        */
02101
02102 class QCP_LIB_DECL QCPAxisTickerDateTime : public QCPAxisTicker {
02103 public:
02104     QCPAxisTickerDateTime();
02105
02106     // getters:
02107
02108     QString dateTimeFormat() const {
02109         return mDateTimeFormat;
02110     }
02111
02112     Qt::TimeSpec dateTimeSpec() const {
02113         return mDateTimeSpec;
02114     }
02115 #if QT_VERSION >= QT_VERSION_CHECK(5, 2, 0)
02116
02117     QTimeZone timeZone() const {
02118         return mTimeZone;
02119     }
02120 #endif
02121
02122     // setters:
02123     void setDateTimeFormat(const QString &format);
02124     void setDateTimeSpec(Qt::TimeSpec spec);
```

```
02125 #if QT_VERSION >= QT_VERSION_CHECK(5, 2, 0)
02126     void setTimeZone(const QTimeZone &zone);
02127 #endif
02128     void setTickOrigin(double origin); // hides base class method but calls baseclass implementation
      ("using" throws off IDEs and doxygen)
02129     void setTickOrigin(const QDateTime &origin);
02130
02131     // static methods:
02132     static QDateTime keyToDateTime(double key);
02133     static double dateTimeToKey(const QDateTime &dateTime);
02134     static double dateTimeToKey(const QDate &date, Qt::TimeSpec timeSpec = Qt::LocalTime);
02135
02136 protected:
02137     // property members:
02138     QString mDateTimeFormat;
02139     Qt::TimeSpec mDateTimeSpec;
02140 #if QT_VERSION >= QT_VERSION_CHECK(5, 2, 0)
02141     QTimeZone mTimeZone;
02142 #endif
02143     // non-property members:
02144
02145     enum DateStrategy {
02146         dsNone, dsUniformTimeInDay, dsUniformDayInMonth
02147     } mDateStrategy;
02148
02149     // reimplemented virtual methods:
02150     virtual double getTickStep(const QCPRange &range) Q_DECL_OVERRIDE;
02151     virtual int getSubTickCount(double tickStep) Q_DECL_OVERRIDE;
02152     virtual QString getTickLabel(double tick, const QLocale &locale, QChar formatChar, int precision)
      Q_DECL_OVERRIDE;
02153     virtual QVector<double> createTickVector(double tickStep, const QCPRange &range) Q_DECL_OVERRIDE;
02154 };
02155
02156 /* end of 'src/axis/axistickerdatetime.h' */
02157
02158
02159 /* including file 'src/axis/axistickertime.h' */
02160
02161 /* modified 2021-03-29T02:30:44, size 3542    */
02162
02163 class QCP_LIB_DECL QCPAxisTickerTime : public QCPAxisTicker {
02164
02165     Q_GADGET
02166 public:
02172     enum TimeUnit {
02173         tuMilliseconds
02174         , tuSeconds
02175         , tuMinutes
02176         , tuHours
02177         , tuDays
02178     };
02179     Q_ENUMS(TimeUnit)
02180
02181     QCPAxisTickerTime();
02182
02183     // getters:
02184
02185     QString timeFormat() const {
02186         return mTimeFormat;
02187     }
02188
02189     int fieldWidth(TimeUnit unit) const {
02190         return mFieldWidth.value(unit);
02191     }
02192
02193     // setters:
02194     void setTimeFormat(const QString &format);
02195     void setFieldWidth(TimeUnit unit, int width);
02196
02197 protected:
02198     // property members:
02199     QString mTimeFormat;
02200     QHash<TimeUnit, int> mFieldWidth;
02201
02202     // non-property members:
02203     TimeUnit mSmallestUnit, mBiggestUnit;
02204     QHash<TimeUnit, QString> mFormatPattern;
02205
02206     // reimplemented virtual methods:
02207     virtual double getTickStep(const QCPRange &range) Q_DECL_OVERRIDE;
02208     virtual int getSubTickCount(double tickStep) Q_DECL_OVERRIDE;
02209     virtual QString getTickLabel(double tick, const QLocale &locale, QChar formatChar, int precision)
      Q_DECL_OVERRIDE;
02210
02211     // non-virtual methods:
02212     void replaceUnit(QString &text, TimeUnit unit, int value) const;
02213 };
```

```
02214
02215 Q_DECLARE_METATYPE(QCPAxisTickerTime::TimeUnit)
02216
02217 /* end of 'src/axis/axistickertime.h' */
02218
02219
02220 /* including file 'src/axis/axistickerfixed.h' */
02221 /* modified 2021-03-29T02:30:44, size 3308     */
02222
02223 class QCP_LIB_DECL QCPAxisTickerFixed : public QCPAxisTicker {
02224
02225     Q_GADGET
02226 public:
02233     enum ScaleStrategy {
02234         ssNone
02235         , ssMultiples
02236         , ssPowers
02237     };
02238     Q_ENUMS(ScaleStrategy)
02239
02240     QCPAxisTickerFixed();
02241
02242     // getters:
02243
02244     double tickStep() const {
02245         return mTickStep;
02246     }
02247
02248     ScaleStrategy scaleStrategy() const {
02249         return mScaleStrategy;
02250     }
02251
02252     // setters:
02253     void setTickStep(double step);
02254     void setScaleStrategy(ScaleStrategy strategy);
02255
02256 protected:
02257     // property members:
02258     double mTickStep;
02259     ScaleStrategy mScaleStrategy;
02260
02261     // reimplemented virtual methods:
02262     virtual double getTickStep(const QCPRange &range) Q_DECL_OVERRIDE;
02263 };
02264
02265 Q_DECLARE_METATYPE(QCPAxisTickerFixed::ScaleStrategy)
02266
02267 /* end of 'src/axis/axistickerfixed.h' */
02268
02269
02270 /* including file 'src/axis/axistickertext.h' */
02271 /* modified 2021-03-29T02:30:44, size 3090    */
02272
02273 class QCP_LIB_DECL QCPAxisTickerText : public QCPAxisTicker {
02274 public:
02275     QCPAxisTickerText();
02276
02277     // getters:
02278
02279     QMap<double, QString> &ticks() {
02280         return mTicks;
02281     }
02282
02283     int subTickCount() const {
02284         return mSubTickCount;
02285     }
02286
02287     // setters:
02288     void setTicks(const QMap<double, QString> &ticks);
02289     void setTicks(const QVector<double> &positions, const QVector<QString> &labels);
02290     void setSubTickCount(int subTicks);
02291
02292     // non-virtual methods:
02293     void clear();
02294     void addTick(double position, const QString &label);
02295     void addTicks(const QMap<double, QString> &ticks);
02296     void addTicks(const QVector<double> &positions, const QVector<QString> &labels);
02297
02298 protected:
02299     // property members:
02300     QMap<double, QString> mTicks;
02301     int mSubTickCount;
02302
02303     // reimplemented virtual methods:
02304     virtual double getTickStep(const QCPRange &range) Q_DECL_OVERRIDE;
02305     virtual int getSubTickCount(double tickStep) Q_DECL_OVERRIDE;
02306     virtual QString getTickLabel(double tick, const QLocale &locale, QChar formatChar, int precision)
```

```
    Q_DECL_OVERRIDE;
02307     virtual QVector<double> createTickVector(double tickStep, const QCPRange &range) Q_DECL_OVERRIDE;
02308 };
02309
02310 /* end of 'src/axis/axistickertext.h' */
02311
02312
02313 /* including file 'src/axis/axistickerpi.h' */
02314
02315 /* modified 2021-03-29T02:30:44, size 3911  */
02316
02317 class QCP_LIB_DECL QCPAxisTickerPi : public QCPAxisTicker {
02318
02319     Q_GADGET
02320 public:
02326     enum FractionStyle {
02327         fsFloatingPoint
02328         , fsAsciiFractions
02329         , fsUnicodeFractions
02330     };
02331     Q_ENUMS(FractionStyle)
02332
02333     QCPAxisTickerPi();
02334
02335     // getters:
02336
02337     QString piSymbol() const {
02338         return mPiSymbol;
02339     }
02340
02341     double piValue() const {
02342         return mPiValue;
02343     }
02344
02345     bool periodicity() const {
02346         return mPeriodicity;
02347     }
02348
02349     FractionStyle fractionStyle() const {
02350         return mFractionStyle;
02351     }
02352
02353     // setters:
02354     void setPiSymbol(QString symbol);
02355     void setPiValue(double pi);
02356     void setPeriodicity(int multiplesOfPi);
02357     void setFractionStyle(FractionStyle style);
02358
02359 protected:
02360     // property members:
02361     QString mPiSymbol;
02362     double mPiValue;
02363     int mPeriodicity;
02364     FractionStyle mFractionStyle;
02365
02366     // non-property members:
02367     double mPiTickStep; // size of one tick step in units of mPiValue
02368
02369     // reimplemented virtual methods:
02370     virtual double getTickStep(const QCPRange &range) Q_DECL_OVERRIDE;
02371     virtual int getSubTickCount(double tickStep) Q_DECL_OVERRIDE;
02372     virtual QString getTickLabel(double tick, const QLocale &locale, QChar formatChar, int precision)
    Q_DECL_OVERRIDE;
02373
02374     // non-virtual methods:
02375     void simplifyFraction(int &numerator, int &denominator) const;
02376     QString fractionToString(int numerator, int denominator) const;
02377     QString unicodeFraction(int numerator, int denominator) const;
02378     QString unicodeSuperscript(int number) const;
02379     QString unicodeSubscript(int number) const;
02380 };
02381
02382 Q_DECLARE_METATYPE(QCPAxisTickerPi::FractionStyle)
02383
02384 /* end of 'src/axis/axistickerpi.h' */
02385
02386
02387 /* including file 'src/axis/axistickerlog.h' */
02388 /* modified 2021-03-29T02:30:44, size 2594  */
02389
02390 class QCP_LIB_DECL QCPAxisTickerLog : public QCPAxisTicker {
02391 public:
02392     QCPAxisTickerLog();
02393
02394     // getters:
02395
02396     double logBase() const {
```

```
02397         return mLogBase;
02398     }
02399
02400     int subTickCount() const {
02401         return mSubTickCount;
02402     }
02403
02404     // setters:
02405     void setLogBase(double base);
02406     void setSubTickCount(int subTicks);
02407
02408 protected:
02409     // property members:
02410     double mLogBase;
02411     int mSubTickCount;
02412
02413     // non-property members:
02414     double mLogBaseLnInv;
02415
02416     // reimplemented virtual methods:
02417     virtual int getSubTickCount(double tickStep) Q_DECL_OVERRIDE;
02418     virtual QVector<double> createTickVector(double tickStep, const QCPRange &range) Q_DECL_OVERRIDE;
02419 };
02420
02421 /* end of 'src/axis/axistickerlog.h' */
02422
02423
02424 /* including file 'src/axis/axis.h'          */
02425
02426 /* modified 2021-03-29T02:30:44, size 20913 */
02427
02428 class QCP_LIB_DECL QCPGrid : public QCPLayerable {
02429     Q_OBJECT
02431     Q_PROPERTY(bool subGridVisible READ subGridVisible WRITE setSubGridVisible)
02432     Q_PROPERTY(bool antialiasedSubGrid READ antialiasedSubGrid WRITE setAntialiasedSubGrid)
02433     Q_PROPERTY(bool antialiasedZeroLine READ antialiasedZeroLine WRITE setAntialiasedZeroLine)
02434     Q_PROPERTY(QPen pen READ pen WRITE setPen)
02435     Q_PROPERTY(QPen subGridPen READ subGridPen WRITE setSubGridPen)
02436     Q_PROPERTY(QPen zeroLinePen READ zeroLinePen WRITE setZeroLinePen)
02438 public:
02439     explicit QCPGrid(QCPAxis *parentAxis);
02440
02441     // getters:
02442
02443     bool subGridVisible() const {
02444         return mSubGridVisible;
02445     }
02446
02447     bool antialiasedSubGrid() const {
02448         return mAntialiasedSubGrid;
02449     }
02450
02451     bool antialiasedZeroLine() const {
02452         return mAntialiasedZeroLine;
02453     }
02454
02455     QPen pen() const {
02456         return mPen;
02457     }
02458
02459     QPen subGridPen() const {
02460         return mSubGridPen;
02461     }
02462
02463     QPen zeroLinePen() const {
02464         return mZeroLinePen;
02465     }
02466
02467     // setters:
02468     void setSubGridVisible(bool visible);
02469     void setAntialiasedSubGrid(bool enabled);
02470     void setAntialiasedZeroLine(bool enabled);
02471     void setPen(const QPen &pen);
02472     void setSubGridPen(const QPen &pen);
02473     void setZeroLinePen(const QPen &pen);
02474
02475 protected:
02476     // property members:
02477     bool mSubGridVisible;
02478     bool mAntialiasedSubGrid, mAntialiasedZeroLine;
02479     QPen mPen, mSubGridPen, mZeroLinePen;
02480
02481     // non-property members:
02482     QCPAxis *mParentAxis;
02483
02484     // reimplemented virtual methods:
02485     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
```

```
02486     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
02487
02488     // non-virtual methods:
02489     void drawGridLines(QCPPainter *painter) const;
02490     void drawSubGridLines(QCPPainter *painter) const;
02491
02492     friend class QCPAxis;
02493 };
02494
02495 class QCP_LIB_DECL QCPAxis : public QCPLayerable {
02496
02497     Q_OBJECT
02499     Q_PROPERTY(AxisType axisType READ axisType)
02500     Q_PROPERTY(QCPAxisRect* axisRect READ axisRect)
02501     Q_PROPERTY(ScaleType scaleType READ scaleType WRITE setScaleType NOTIFY scaleTypeChanged)
02502     Q_PROPERTY(QCPRange range READ range WRITE setRange NOTIFY rangeChanged)
02503     Q_PROPERTY(bool rangeReversed READ rangeReversed WRITE setRangeReversed)
02504     Q_PROPERTY(QSharedPointer<QCPAxisTicker> ticker READ ticker WRITE setTicker)
02505     Q_PROPERTY(bool ticks READ ticks WRITE setTicks)
02506     Q_PROPERTY(bool tickLabels READ tickLabels WRITE setTickLabels)
02507     Q_PROPERTY(int tickLabelPadding READ tickLabelPadding WRITE setTickLabelPadding)
02508     Q_PROPERTY(QFont tickLabelFont READ tickLabelFont WRITE setTickLabelFont)
02509     Q_PROPERTY(QColor tickLabelColor READ tickLabelColor WRITE setTickLabelColor)
02510     Q_PROPERTY(double tickLabelRotation READ tickLabelRotation WRITE setTickLabelRotation)
02511     Q_PROPERTY(LabelSide tickLabelSide READ tickLabelSide WRITE setTickLabelSide)
02512     Q_PROPERTY(QString numberFormat READ numberFormat WRITE setNumberFormat)
02513     Q_PROPERTY(int numberPrecision READ numberPrecision WRITE setNumberPrecision)
02514     Q_PROPERTY(QVector<double> tickVector READ tickVector)
02515     Q_PROPERTY(QVector<QString> tickVectorLabels READ tickVectorLabels)
02516     Q_PROPERTY(int tickLengthIn READ tickLengthIn WRITE setTickLengthIn)
02517     Q_PROPERTY(int tickLengthOut READ tickLengthOut WRITE setTickLengthOut)
02518     Q_PROPERTY(bool subTicks READ subTicks WRITE setSubTicks)
02519     Q_PROPERTY(int subTickLengthIn READ subTickLengthIn WRITE setSubTickLengthIn)
02520     Q_PROPERTY(int subTickLengthOut READ subTickLengthOut WRITE setSubTickLengthOut)
02521     Q_PROPERTY(QPen basePen READ basePen WRITE setBasePen)
02522     Q_PROPERTY(QPen tickPen READ tickPen WRITE setTickPen)
02523     Q_PROPERTY(QPen subTickPen READ subTickPen WRITE setSubTickPen)
02524     Q_PROPERTY(QFont labelFont READ labelFont WRITE setLabelFont)
02525     Q_PROPERTY(QColor labelColor READ labelColor WRITE setLabelColor)
02526     Q_PROPERTY(QString label READ label WRITE setLabel)
02527     Q_PROPERTY(int labelPadding READ labelPadding WRITE setLabelPadding)
02528     Q_PROPERTY(int padding READ padding WRITE setPadding)
02529     Q_PROPERTY(int offset READ offset WRITE setOffset)
02530     Q_PROPERTY(SelectableParts selectedParts READ selectedParts WRITE setSelectedParts NOTIFY
      selectionChanged)
02531     Q_PROPERTY(SelectableParts selectableParts READ selectableParts WRITE setSelectableParts NOTIFY
      selectableChanged)
02532     Q_PROPERTY(QFont selectedTickLabelFont READ selectedTickLabelFont WRITE setSelectedTickLabelFont)
02533     Q_PROPERTY(QFont selectedLabelFont READ selectedLabelFont WRITE setSelectedLabelFont)
02534     Q_PROPERTY(QColor selectedTickLabelColor READ selectedTickLabelColor WRITE
      setSelectedTickLabelColor)
02535     Q_PROPERTY(QColor selectedLabelColor READ selectedLabelColor WRITE setSelectedLabelColor)
02536     Q_PROPERTY(QPen selectedBasePen READ selectedBasePen WRITE setSelectedBasePen)
02537     Q_PROPERTY(QPen selectedTickPen READ selectedTickPen WRITE setSelectedTickPen)
02538     Q_PROPERTY(QPen selectedSubTickPen READ selectedSubTickPen WRITE setSelectedSubTickPen)
02539     Q_PROPERTY(QCPLineEnding lowerEnding READ lowerEnding WRITE setLowerEnding)
02540     Q_PROPERTY(QCPLineEnding upperEnding READ upperEnding WRITE setUpperEnding)
02541     Q_PROPERTY(QCPGrid* grid READ grid)
02543 public:
02548     enum AxisType {
02549         atLeft = 0x01
02550         , atRight = 0x02
02551         , atTop = 0x04
02552         , atBottom = 0x08
02553     };
02554
02555     Q_ENUMS(AxisType)
02556     Q_FLAGS(AxisTypes)
02557     Q_DECLARE_FLAGS(AxisTypes, AxisType)
02563     enum LabelSide {
02564         lsInside
02565         , lsOutside
02566     };
02567
02568     Q_ENUMS(LabelSide)
02573     enum ScaleType {
02574         stLinear
02575         , stLogarithmic
02576     };
02577
02578     Q_ENUMS(ScaleType)
02583     enum SelectablePart {
02584         spNone = 0
02585         , spAxis = 0x001
02586         , spTickLabels = 0x002
02587         , spAxisLabel = 0x004
02588     };
```

```
02589     Q_ENUMS(SelectablePart)
02590     Q_FLAGS(SelectableParts)
02591     Q_DECLARE_FLAGS(SelectableParts, SelectablePart)
02592 
02593     explicit QCPAxis(QCPAxisRect *parent, AxisType type);
02594     virtual ~QCPAxis() Q_DECL_OVERRIDE;
02595 
02596     // getters:
02597 
02598     AxisType axisType() const {
02599         return mAxisType;
02600     }
02601 
02602     QCPAxisRect *axisRect() const {
02603         return mAxisRect;
02604     }
02605 
02606     ScaleType scaleType() const {
02607         return mScaleType;
02608     }
02609 
02610     const QCPRange range() const {
02611         return mRange;
02612     }
02613 
02614     bool rangeReversed() const {
02615         return mRangeReversed;
02616     }
02617 
02618     QSharedPointer<QCPAxisTicker> ticker() const {
02619         return mTicker;
02620     }
02621 
02622     bool ticks() const {
02623         return mTicks;
02624     }
02625 
02626     bool tickLabels() const {
02627         return mTickLabels;
02628     }
02629     int tickLabelPadding() const;
02630 
02631     QFont tickLabelFont() const {
02632         return mTickLabelFont;
02633     }
02634 
02635     QColor tickLabelColor() const {
02636         return mTickLabelColor;
02637     }
02638     double tickLabelRotation() const;
02639     LabelSide tickLabelSide() const;
02640     QString numberFormat() const;
02641 
02642     int numberPrecision() const {
02643         return mNumberPrecision;
02644     }
02645 
02646     QVector<double> tickVector() const {
02647         return mTickVector;
02648     }
02649 
02650     QVector<QString> tickVectorLabels() const {
02651         return mTickVectorLabels;
02652     }
02653     int tickLengthIn() const;
02654     int tickLengthOut() const;
02655 
02656     bool subTicks() const {
02657         return mSubTicks;
02658     }
02659     int subTickLengthIn() const;
02660     int subTickLengthOut() const;
02661 
02662     QPen basePen() const {
02663         return mBasePen;
02664     }
02665 
02666     QPen tickPen() const {
02667         return mTickPen;
02668     }
02669 
02670     QPen subTickPen() const {
02671         return mSubTickPen;
02672     }
02673 
02674     QFont labelFont() const {
02675         return mLabelFont;
```

```
02676     }
02677
02678     QColor labelColor() const {
02679         return mLabelColor;
02680     }
02681
02682     QString label() const {
02683         return mLabel;
02684     }
02685     int labelPadding() const;
02686
02687     int padding() const {
02688         return mPadding;
02689     }
02690     int offset() const;
02691
02692     SelectableParts selectedParts() const {
02693         return mSelectedParts;
02694     }
02695
02696     SelectableParts selectableParts() const {
02697         return mSelectableParts;
02698     }
02699
02700     QFont selectedTickLabelFont() const {
02701         return mSelectedTickLabelFont;
02702     }
02703
02704     QFont selectedLabelFont() const {
02705         return mSelectedLabelFont;
02706     }
02707
02708     QColor selectedTickLabelColor() const {
02709         return mSelectedTickLabelColor;
02710     }
02711
02712     QColor selectedLabelColor() const {
02713         return mSelectedLabelColor;
02714     }
02715
02716     QPen selectedBasePen() const {
02717         return mSelectedBasePen;
02718     }
02719
02720     QPen selectedTickPen() const {
02721         return mSelectedTickPen;
02722     }
02723
02724     QPen selectedSubTickPen() const {
02725         return mSelectedSubTickPen;
02726     }
02727     QCPLineEnding lowerEnding() const;
02728     QCPLineEnding upperEnding() const;
02729
02730     QCPGrid *grid() const {
02731         return mGrid;
02732     }
02733
02734     // setters:
02735     Q_SLOT void setScaleType(QCPAxis::ScaleType type);
02736     Q_SLOT void setRange(const QCPRange &range);
02737     void setRange(double lower, double upper);
02738     void setRange(double position, double size, Qt::AlignmentFlag alignment);
02739     void setRangeLower(double lower);
02740     void setRangeUpper(double upper);
02741     void setRangeReversed(bool reversed);
02742     void setTicker(QSharedPointer<QCPAxisTicker> ticker);
02743     void setTicks(bool show);
02744     void setTickLabels(bool show);
02745     void setTickLabelPadding(int padding);
02746     void setTickLabelFont(const QFont &font);
02747     void setTickLabelColor(const QColor &color);
02748     void setTickLabelRotation(double degrees);
02749     void setTickLabelSide(LabelSide side);
02750     void setNumberFormat(const QString &formatCode);
02751     void setNumberPrecision(int precision);
02752     void setTickLength(int inside, int outside = 0);
02753     void setTickLengthIn(int inside);
02754     void setTickLengthOut(int outside);
02755     void setSubTicks(bool show);
02756     void setSubTickLength(int inside, int outside = 0);
02757     void setSubTickLengthIn(int inside);
02758     void setSubTickLengthOut(int outside);
02759     void setBasePen(const QPen &pen);
02760     void setTickPen(const QPen &pen);
02761     void setSubTickPen(const QPen &pen);
02762     void setLabelFont(const QFont &font);
```

```
02763     void setLabelColor(const QColor &color);
02764     void setLabel(const QString &str);
02765     void setLabelPadding(int padding);
02766     void setPadding(int padding);
02767     void setOffset(int offset);
02768     void setSelectedTickLabelFont(const QFont &font);
02769     void setSelectedLabelFont(const QFont &font);
02770     void setSelectedTickLabelColor(const QColor &color);
02771     void setSelectedLabelColor(const QColor &color);
02772     void setSelectedBasePen(const QPen &pen);
02773     void setSelectedTickPen(const QPen &pen);
02774     void setSelectedSubTickPen(const QPen &pen);
02775     Q_SLOT void setSelectableParts(const QCPAxis::SelectableParts &selectableParts);
02776     Q_SLOT void setSelectedParts(const QCPAxis::SelectableParts &selectedParts);
02777     void setLowerEnding(const QCPLineEnding &ending);
02778     void setUpperEnding(const QCPLineEnding &ending);
02779 
02780     // reimplemented virtual methods:
02781     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
02782 
02783     // non-property methods:
02784 
02785     Qt::Orientation orientation() const {
02786         return mOrientation;
02787     }
02788 
02789     int pixelOrientation() const {
02790         return rangeReversed() != (orientation() == Qt::Vertical) ? -1 : 1;
02791     }
02792     void moveRange(double diff);
02793     void scaleRange(double factor);
02794     void scaleRange(double factor, double center);
02795     void setScaleRatio(const QCPAxis *otherAxis, double ratio = 1.0);
02796     void rescale(bool onlyVisiblePlottables = false);
02797     double pixelToCoord(double value) const;
02798     double coordToPixel(double value) const;
02799     SelectablePart getPartAt(const QPointF &pos) const;
02800     QList<QCPAbstractPlottable*> plottables() const;
02801     QList<QCPGraph*> graphs() const;
02802     QList<QCPAbstractItem*> items() const;
02803 
02804     static AxisType marginSideToAxisType(QCP::MarginSide side);
02805 
02806     static Qt::Orientation orientation(AxisType type) {
02807         return type == atBottom || type == atTop ? Qt::Horizontal : Qt::Vertical;
02808     }
02809     static AxisType opposite(AxisType type);
02810 
02811 signals:
02812     void rangeChanged(const QCPRange &newRange);
02813     void rangeChanged(const QCPRange &newRange, const QCPRange &oldRange);
02814     void scaleTypeChanged(QCPAxis::ScaleType scaleType);
02815     void selectionChanged(const QCPAxis::SelectableParts &parts);
02816     void selectableChanged(const QCPAxis::SelectableParts &parts);
02817 
02818 protected:
02819     // property members:
02820     // axis base:
02821     AxisType mAxisType;
02822     QCPAxisRect *mAxisRect;
02823     //int mOffset; // in QCPAxisPainter
02824     int mPadding;
02825     Qt::Orientation mOrientation;
02826     SelectableParts mSelectableParts, mSelectedParts;
02827     QPen mBasePen, mSelectedBasePen;
02828     //QCPLineEnding mLowerEnding, mUpperEnding; // in QCPAxisPainter
02829     // axis label:
02830     //int mLabelPadding; // in QCPAxisPainter
02831     QString mLabel;
02832     QFont mLabelFont, mSelectedLabelFont;
02833     QColor mLabelColor, mSelectedLabelColor;
02834     // tick labels:
02835     //int mTickLabelPadding; // in QCPAxisPainter
02836     bool mTickLabels;
02837     //double mTickLabelRotation; // in QCPAxisPainter
02838     QFont mTickLabelFont, mSelectedTickLabelFont;
02839     QColor mTickLabelColor, mSelectedTickLabelColor;
02840     int mNumberPrecision;
02841     QLatin1Char mNumberFormatChar;
02842     bool mNumberBeautifulPowers;
02843     //bool mNumberMultiplyCross; // QCPAxisPainter
02844     // ticks and subticks:
02845     bool mTicks;
02846     bool mSubTicks;
02847     //int mTickLengthIn, mTickLengthOut, mSubTickLengthIn, mSubTickLengthOut; // QCPAxisPainter
02848     QPen mTickPen, mSelectedTickPen;
```

```
02849     QPen mSubTickPen, mSelectedSubTickPen;
02850     // scale and range:
02851     QCPRange mRange;
02852     bool mRangeReversed;
02853     ScaleType mScaleType;
02854 
02855     // non-property members:
02856     QCPGrid *mGrid;
02857     QCPAxisPainterPrivate *mAxisPainter;
02858     QSharedPointer<QCPAxisTicker> mTicker;
02859     QVector<double> mTickVector;
02860     QVector<QString> mTickVectorLabels;
02861     QVector<double> mSubTickVector;
02862     bool mCachedMarginValid;
02863     int mCachedMargin;
02864     bool mDragging;
02865     QCPRange mDragStartRange;
02866     QCP::AntialiasedElements mAADragBackup, mNotAADragBackup;
02867 
02868     // introduced virtual methods:
02869     virtual int calculateMargin();
02870 
02871     // reimplemented virtual methods:
02872     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
02873     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
02874     virtual QCP::Interaction selectionCategory() const Q_DECL_OVERRIDE;
02875     // events:
02876     virtual void selectEvent(QMouseEvent *event, bool additive, const QVariant &details, bool
      *selectionStateChanged) Q_DECL_OVERRIDE;
02877     virtual void deselectEvent(bool *selectionStateChanged) Q_DECL_OVERRIDE;
02878     // mouse events:
02879     virtual void mousePressEvent(QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE;
02880     virtual void mouseMoveEvent(QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE;
02881     virtual void mouseReleaseEvent(QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE;
02882     virtual void wheelEvent(QWheelEvent *event) Q_DECL_OVERRIDE;
02883 
02884     // non-virtual methods:
02885     void setupTickVectors();
02886     QPen getBasePen() const;
02887     QPen getTickPen() const;
02888     QPen getSubTickPen() const;
02889     QFont getTickLabelFont() const;
02890     QFont getLabelFont() const;
02891     QColor getTickLabelColor() const;
02892     QColor getLabelColor() const;
02893 
02894 private:
02895     Q_DISABLE_COPY(QCPAxis)
02896 
02897     friend class QCustomPlot;
02898     friend class QCPGrid;
02899     friend class QCPAxisRect;
02900 };
02901 
02902 Q_DECLARE_OPERATORS_FOR_FLAGS(QCPAxis::SelectableParts)
02903 Q_DECLARE_OPERATORS_FOR_FLAGS(QCPAxis::AxisTypes)
02904 Q_DECLARE_METATYPE(QCPAxis::AxisType)
02905 Q_DECLARE_METATYPE(QCPAxis::LabelSide)
02906 Q_DECLARE_METATYPE(QCPAxis::ScaleType)
02907 Q_DECLARE_METATYPE(QCPAxis::SelectablePart)
02908 
02909 
02910 class QCPAxisPainterPrivate {
02911 public:
02912     explicit QCPAxisPainterPrivate(QCustomPlot *parentPlot);
02913     virtual ~QCPAxisPainterPrivate();
02914 
02915     virtual void draw(QCPPainter *painter);
02916     virtual int size();
02917     void clearCache();
02918 
02919     QRect axisSelectionBox() const {
02920         return mAxisSelectionBox;
02921     }
02922 
02923     QRect tickLabelsSelectionBox() const {
02924         return mTickLabelsSelectionBox;
02925     }
02926 
02927     QRect labelSelectionBox() const {
02928         return mLabelSelectionBox;
02929     }
02930 
02931     // public property members:
02932     QCPAxis::AxisType type;
02933     QPen basePen;
02934     QCPLineEnding lowerEnding, upperEnding; // directly accessed by QCPAxis setters/getters
```

```
02935     int labelPadding; // directly accessed by QCPAxis setters/getters
02936     QFont labelFont;
02937     QColor labelColor;
02938     QString label;
02939     int tickLabelPadding; // directly accessed by QCPAxis setters/getters
02940     double tickLabelRotation; // directly accessed by QCPAxis setters/getters
02941     QCPAxis::LabelSide tickLabelSide; // directly accessed by QCPAxis setters/getters
02942     bool substituteExponent;
02943     bool numberMultiplyCross; // directly accessed by QCPAxis setters/getters
02944     int tickLengthIn, tickLengthOut, subTickLengthIn, subTickLengthOut; // directly accessed by
      QCPAxis setters/getters
02945     QPen tickPen, subTickPen;
02946     QFont tickLabelFont;
02947     QColor tickLabelColor;
02948     QRect axisRect, viewportRect;
02949     int offset; // directly accessed by QCPAxis setters/getters
02950     bool abbreviateDecimalPowers;
02951     bool reversedEndings;
02952
02953     QVector<double> subTickPositions;
02954     QVector<double> tickPositions;
02955     QVector<QString> tickLabels;
02956
02957 protected:
02958
02959     struct CachedLabel {
02960         QPointF offset;
02961         QPixmap pixmap;
02962     };
02963
02964     struct TickLabelData {
02965         QString basePart, expPart, suffixPart;
02966         QRect baseBounds, expBounds, suffixBounds, totalBounds, rotatedTotalBounds;
02967         QFont baseFont, expFont;
02968     };
02969     QCustomPlot *mParentPlot;
02970     QByteArray mLabelParameterHash; // to determine whether mLabelCache needs to be cleared due to
      changed parameters
02971     QCache<QString, CachedLabel> mLabelCache;
02972     QRect mAxisSelectionBox, mTickLabelsSelectionBox, mLabelSelectionBox;
02973
02974     virtual QByteArray generateLabelParameterHash() const;
02975
02976     virtual void placeTickLabel(QCPPainter *painter, double position, int distanceToAxis, const
      QString &text, QSize *tickLabelsSize);
02977     virtual void drawTickLabel(QCPPainter *painter, double x, double y, const TickLabelData
      &labelData) const;
02978     virtual TickLabelData getTickLabelData(const QFont &font, const QString &text) const;
02979     virtual QPointF getTickLabelDrawOffset(const TickLabelData &labelData) const;
02980     virtual void getMaxTickLabelSize(const QFont &font, const QString &text, QSize *tickLabelsSize)
      const;
02981 };
02982
02983 /* end of 'src/axis/axis.h' */
02984
02985
02986 /* including file 'src/scatterstyle.h'     */
02987
02988 /* modified 2021-03-29T02:30:44, size 7275 */
02989
02990 class QCP_LIB_DECL QCPScatterStyle {
02991
02992     Q_GADGET
02993 public:
03002     enum ScatterProperty {
03003         spNone = 0x00
03004         , spPen = 0x01
03005         , spBrush = 0x02
03006         , spSize = 0x04
03007         , spShape = 0x08
03008         , spAll = 0xFF
03009     };
03010
03011     Q_ENUMS(ScatterProperty)
03012     Q_FLAGS(ScatterProperties)
03013     Q_DECLARE_FLAGS(ScatterProperties, ScatterProperty)
03014
03015
03022     enum ScatterShape {
03023         ssNone
03024         , ssDot
03025         , ssCross
03026         , ssPlus
03027         , ssCircle
03028         , ssDisc
03029         , ssSquare
03030         , ssDiamond
```

```
03031         , ssStar
03032         , ssTriangle
03033         , ssTriangleInverted
03034         , ssCrossSquare
03035         , ssPlusSquare
03036         , ssCrossCircle
03037         , ssPlusCircle
03038         , ssPeace
03039         , ssPixmap
03040         , ssCustom
03041     };
03042     Q_ENUMS(ScatterShape)
03043 
03044     QCPScatterStyle();
03045     QCPScatterStyle(ScatterShape shape, double size = 6);
03046     QCPScatterStyle(ScatterShape shape, const QColor &color, double size);
03047     QCPScatterStyle(ScatterShape shape, const QColor &color, const QColor &fill, double size);
03048     QCPScatterStyle(ScatterShape shape, const QPen &pen, const QBrush &brush, double size);
03049     QCPScatterStyle(const QPixmap &pixmap);
03050     QCPScatterStyle(const QPainterPath &customPath, const QPen &pen, const QBrush &brush =
      Qt::NoBrush, double size = 6);
03051 
03052     // getters:
03053 
03054     double size() const {
03055         return mSize;
03056     }
03057 
03058     ScatterShape shape() const {
03059         return mShape;
03060     }
03061 
03062     QPen pen() const {
03063         return mPen;
03064     }
03065 
03066     QBrush brush() const {
03067         return mBrush;
03068     }
03069 
03070     QPixmap pixmap() const {
03071         return mPixmap;
03072     }
03073 
03074     QPainterPath customPath() const {
03075         return mCustomPath;
03076     }
03077 
03078     // setters:
03079     void setFromOther(const QCPScatterStyle &other, ScatterProperties properties);
03080     void setSize(double size);
03081     void setShape(ScatterShape shape);
03082     void setPen(const QPen &pen);
03083     void setBrush(const QBrush &brush);
03084     void setPixmap(const QPixmap &pixmap);
03085     void setCustomPath(const QPainterPath &customPath);
03086 
03087     // non-property methods:
03088 
03089     bool isNone() const {
03090         return mShape == ssNone;
03091     }
03092 
03093     bool isPenDefined() const {
03094         return mPenDefined;
03095     }
03096     void undefinePen();
03097     void applyTo(QCPPainter *painter, const QPen &defaultPen) const;
03098     void drawShape(QCPPainter *painter, const QPointF &pos) const;
03099     void drawShape(QCPPainter *painter, double x, double y) const;
03100 
03101 protected:
03102     // property members:
03103     double mSize;
03104     ScatterShape mShape;
03105     QPen mPen;
03106     QBrush mBrush;
03107     QPixmap mPixmap;
03108     QPainterPath mCustomPath;
03109 
03110     // non-property members:
03111     bool mPenDefined;
03112 };
03113 Q_DECLARE_TYPEINFO(QCPScatterStyle, Q_MOVABLE_TYPE);
03114 
03115 Q_DECLARE_OPERATORS_FOR_FLAGS(QCPScatterStyle::ScatterProperties)
03116 Q_DECLARE_METATYPE(QCPScatterStyle::ScatterProperty)
```

```
03117 Q_DECLARE_METATYPE(QCPScatterStyle::ScatterShape)
03118
03119 /* end of 'src/scatterstyle.h' */
03120
03121
03122 /* including file 'src/datacontainer.h'     */
03123 /* modified 2021-03-29T02:30:44, size 34070 */
03124
03125
03130 template <class DataType>
03131 inline bool qcpLessThanSortKey(const DataType &a, const DataType &b) {
03132     return a.sortKey() < b.sortKey();
03133 }
03134
03135 template <class DataType>
03136 class QCPDataContainer // no QCP_LIB_DECL, template class ends up in header (cpp included below)
03137 {
03138 public:
03139     typedef typename QVector<DataType>::const_iterator const_iterator;
03140     typedef typename QVector<DataType>::iterator iterator;
03141
03142     QCPDataContainer();
03143
03144     // getters:
03145
03146     int size() const {
03147         return mData.size() - mPreallocSize;
03148     }
03149
03150     bool isEmpty() const {
03151         return size() == 0;
03152     }
03153
03154     bool autoSqueeze() const {
03155         return mAutoSqueeze;
03156     }
03157
03158     // setters:
03159     void setAutoSqueeze(bool enabled);
03160
03161     // non-virtual methods:
03162     void set(const QCPDataContainer<DataType> &data);
03163     void set(const QVector<DataType> &data, bool alreadySorted = false);
03164     void add(const QCPDataContainer<DataType> &data);
03165     void add(const QVector<DataType> &data, bool alreadySorted = false);
03166     void add(const DataType &data);
03167     void removeBefore(double sortKey);
03168     void removeAfter(double sortKey);
03169     void remove(double sortKeyFrom, double sortKeyTo);
03170     void remove(double sortKey);
03171     void clear();
03172     void sort();
03173     void squeeze(bool preAllocation = true, bool postAllocation = true);
03174
03175     const_iterator constBegin() const {
03176         return mData.constBegin() + mPreallocSize;
03177     }
03178
03179     const_iterator constEnd() const {
03180         return mData.constEnd();
03181     }
03182
03183     iterator begin() {
03184         return mData.begin() + mPreallocSize;
03185     }
03186
03187     iterator end() {
03188         return mData.end();
03189     }
03190     const_iterator findBegin(double sortKey, bool expandedRange = true) const;
03191     const_iterator findEnd(double sortKey, bool expandedRange = true) const;
03192
03193     const_iterator at(int index) const {
03194         return constBegin() + qBound(0, index, size());
03195     }
03196     QCPRange keyRange(bool &foundRange, QCP::SignDomain signDomain = QCP::sdBoth);
03197     QCPRange valueRange(bool &foundRange, QCP::SignDomain signDomain = QCP::sdBoth, const QCPRange
    &inKeyRange = QCPRange());
03198
03199     QCPDataRange dataRange() const {
03200         return QCPDataRange(0, size());
03201     }
03202     void limitIteratorsToDataRange(const_iterator &begin, const_iterator &end, const QCPDataRange
    &dataRange) const;
03203
03204 protected:
03205     // property members:
```

```
03206     bool mAutoSqueeze;
03207
03208     // non-property memebers:
03209     QVector<DataType> mData;
03210     int mPreallocSize;
03211     int mPreallocIteration;
03212
03213     // non-virtual methods:
03214     void preallocateGrow(int minimumPreallocSize);
03215     void performAutoSqueeze();
03216 };
03217
03218
03219
03220 // include implementation in header since it is a class template:
03224
03292 /* start documentation of inline functions */
03293
03348 /* end documentation of inline functions */
03349
03354 template <class DataType>
03355 QCPDataContainer<DataType>::QCPDataContainer() :
03356 mAutoSqueeze(true),
03357 mPreallocSize(0),
03358 mPreallocIteration(0) {
03359 }
03360
03369 template <class DataType>
03370 void QCPDataContainer<DataType>::setAutoSqueeze(bool enabled) {
03371     if (mAutoSqueeze != enabled) {
03372         mAutoSqueeze = enabled;
03373         if (mAutoSqueeze)
03374             performAutoSqueeze();
03375     }
03376 }
03377
03384 template <class DataType>
03385 void QCPDataContainer<DataType>::set(const QCPDataContainer<DataType> &data) {
03386     clear();
03387     add(data);
03388 }
03389
03399 template <class DataType>
03400 void QCPDataContainer<DataType>::set(const QVector<DataType> &data, bool alreadySorted) {
03401     mData = data;
03402     mPreallocSize = 0;
03403     mPreallocIteration = 0;
03404     if (!alreadySorted)
03405         sort();
03406 }
03407
03414 template <class DataType>
03415 void QCPDataContainer<DataType>::add(const QCPDataContainer<DataType> &data) {
03416     if (data.isEmpty())
03417         return;
03418
03419     const int n = data.size();
03420     const int oldSize = size();
03421
03422     if (oldSize > 0 && !qcpLessThanSortKey<DataType>(*constBegin(), *(data.constEnd() - 1))) //
    prepend if new data keys are all smaller than or equal to existing ones
03423     {
03424         if (mPreallocSize < n)
03425             preallocateGrow(n);
03426         mPreallocSize -= n;
03427         std::copy(data.constBegin(), data.constEnd(), begin());
03428     } else // don't need to prepend, so append and merge if necessary
03429     {
03430         mData.resize(mData.size() + n);
03431         std::copy(data.constBegin(), data.constEnd(), end() - n);
03432         if (oldSize > 0 && !qcpLessThanSortKey<DataType>(*(constEnd() - n - 1), *(constEnd() - n))) //
    if appended range keys aren't all greater than existing ones, merge the two partitions
03433             std::inplace_merge(begin(), end() - n, end(), qcpLessThanSortKey<DataType>);
03434     }
03435 }
03436
03445 template <class DataType>
03446 void QCPDataContainer<DataType>::add(const QVector<DataType> &data, bool alreadySorted) {
03447     if (data.isEmpty())
03448         return;
03449     if (isEmpty()) {
03450         set(data, alreadySorted);
03451         return;
03452     }
03453
03454     const int n = data.size();
03455     const int oldSize = size();
```

```
03456
03457     if (alreadySorted && oldSize > 0 && !qcpLessThanSortKey<DataType>(*constBegin(), *(data.constEnd()
      - 1))) // prepend if new data is sorted and keys are all smaller than or equal to existing ones
03458     {
03459         if (mPreallocSize < n)
03460             preallocateGrow(n);
03461         mPreallocSize -= n;
03462         std::copy(data.constBegin(), data.constEnd(), begin());
03463     } else // don't need to prepend, so append and then sort and merge if necessary
03464     {
03465         mData.resize(mData.size() + n);
03466         std::copy(data.constBegin(), data.constEnd(), end() - n);
03467         if (!alreadySorted) // sort appended subrange if it wasn't already sorted
03468             std::sort(end() - n, end(), qcpLessThanSortKey<DataType>);
03469         if (oldSize > 0 && !qcpLessThanSortKey<DataType>(*(constEnd() - n - 1), *(constEnd() - n))) //
      if appended range keys aren't all greater than existing ones, merge the two partitions
03470             std::inplace_merge(begin(), end() - n, end(), qcpLessThanSortKey<DataType>);
03471     }
03472 }
03473
03480 template <class DataType>
03481 void QCPDataContainer<DataType>::add(const DataType &data) {
03482     if (isEmpty() || !qcpLessThanSortKey<DataType>(data, *(constEnd() - 1))) // quickly handle appends
      if new data key is greater or equal to existing ones
03483     {
03484         mData.append(data);
03485     } else if (qcpLessThanSortKey<DataType>(data, *constBegin())) // quickly handle prepends using
      preallocated space
03486     {
03487         if (mPreallocSize < 1)
03488             preallocateGrow(1);
03489         --mPreallocSize;
03490         *begin() = data;
03491     } else // handle inserts, maintaining sorted keys
03492     {
03493         QCPDataContainer<DataType>::iterator insertionPoint = std::lower_bound(begin(), end(), data,
      qcpLessThanSortKey<DataType>);
03494         mData.insert(insertionPoint, data);
03495     }
03496 }
03497
03503 template <class DataType>
03504 void QCPDataContainer<DataType>::removeBefore(double sortKey) {
03505     QCPDataContainer<DataType>::iterator it = begin();
03506     QCPDataContainer<DataType>::iterator itEnd = std::lower_bound(begin(), end(),
      DataType::fromSortKey(sortKey), qcpLessThanSortKey<DataType>);
03507     mPreallocSize += int(itEnd - it); // don't actually delete, just add it to the preallocated block
      (if it gets too large, squeeze will take care of it)
03508     if (mAutoSqueeze)
03509         performAutoSqueeze();
03510 }
03511
03517 template <class DataType>
03518 void QCPDataContainer<DataType>::removeAfter(double sortKey) {
03519     QCPDataContainer<DataType>::iterator it = std::upper_bound(begin(), end(),
      DataType::fromSortKey(sortKey), qcpLessThanSortKey<DataType>);
03520     QCPDataContainer<DataType>::iterator itEnd = end();
03521     mData.erase(it, itEnd); // typically adds it to the postallocated block
03522     if (mAutoSqueeze)
03523         performAutoSqueeze();
03524 }
03525
03533 template <class DataType>
03534 void QCPDataContainer<DataType>::remove(double sortKeyFrom, double sortKeyTo) {
03535     if (sortKeyFrom >= sortKeyTo || isEmpty())
03536         return;
03537
03538     QCPDataContainer<DataType>::iterator it = std::lower_bound(begin(), end(),
      DataType::fromSortKey(sortKeyFrom), qcpLessThanSortKey<DataType>);
03539     QCPDataContainer<DataType>::iterator itEnd = std::upper_bound(it, end(),
      DataType::fromSortKey(sortKeyTo), qcpLessThanSortKey<DataType>);
03540     mData.erase(it, itEnd);
03541     if (mAutoSqueeze)
03542         performAutoSqueeze();
03543 }
03544
03554 template <class DataType>
03555 void QCPDataContainer<DataType>::remove(double sortKey) {
03556     QCPDataContainer::iterator it = std::lower_bound(begin(), end(), DataType::fromSortKey(sortKey),
      qcpLessThanSortKey<DataType>);
03557     if (it != end() && it->sortKey() == sortKey) {
03558         if (it == begin())
03559             ++mPreallocSize; // don't actually delete, just add it to the preallocated block (if it
      gets too large, squeeze will take care of it)
03560         else
03561             mData.erase(it);
03562     }
```

```
03563     if (mAutoSqueeze)
03564         performAutoSqueeze();
03565 }
03566
03572 template <class DataType>
03573 void QCPDataContainer<DataType>::clear() {
03574     mData.clear();
03575     mPreallocIteration = 0;
03576     mPreallocSize = 0;
03577 }
03578
03590 template <class DataType>
03591 void QCPDataContainer<DataType>::sort() {
03592     std::sort(begin(), end(), qcpLessThanSortKey<DataType>);
03593 }
03594
03605 template <class DataType>
03606 void QCPDataContainer<DataType>::squeeze(bool preAllocation, bool postAllocation) {
03607     if (preAllocation) {
03608         if (mPreallocSize > 0) {
03609             std::copy(begin(), end(), mData.begin());
03610             mData.resize(size());
03611             mPreallocSize = 0;
03612         }
03613         mPreallocIteration = 0;
03614     }
03615     if (postAllocation)
03616         mData.squeeze();
03617 }
03618
03634 template <class DataType>
03635 typename QCPDataContainer<DataType>::const_iterator QCPDataContainer<DataType>::findBegin(double
      sortKey, bool expandedRange) const {
03636     if (isEmpty())
03637         return constEnd();
03638
03639     QCPDataContainer<DataType>::const_iterator it = std::lower_bound(constBegin(), constEnd(),
      DataType::fromSortKey(sortKey), qcpLessThanSortKey<DataType>);
03640     if (expandedRange && it != constBegin()) // also covers it == constEnd case, and we know
      --constEnd is valid because mData isn't empty
03641         --it;
03642     return it;
03643 }
03644
03660 template <class DataType>
03661 typename QCPDataContainer<DataType>::const_iterator QCPDataContainer<DataType>::findEnd(double
      sortKey, bool expandedRange) const {
03662     if (isEmpty())
03663         return constEnd();
03664
03665     QCPDataContainer<DataType>::const_iterator it = std::upper_bound(constBegin(), constEnd(),
      DataType::fromSortKey(sortKey), qcpLessThanSortKey<DataType>);
03666     if (expandedRange && it != constEnd())
03667         ++it;
03668     return it;
03669 }
03670
03686 template <class DataType>
03687 QCPRange QCPDataContainer<DataType>::keyRange(bool &foundRange, QCP::SignDomain signDomain) {
03688     if (isEmpty()) {
03689         foundRange = false;
03690         return QCPRange();
03691     }
03692     QCPRange range;
03693     bool haveLower = false;
03694     bool haveUpper = false;
03695     double current;
03696
03697     QCPDataContainer<DataType>::const_iterator it = constBegin();
03698     QCPDataContainer<DataType>::const_iterator itEnd = constEnd();
03699     if (signDomain == QCP::sdBoth) // range may be anywhere
03700     {
03701         if (DataType::sortKeyIsMainKey()) // if DataType is sorted by main key (e.g. QCPGraph, but not
      QCPCurve), use faster algorithm by finding just first and last key with non-NaN value
03702         {
03703             while (it != itEnd) // find first non-nan going up from left
03704             {
03705                 if (!qIsNaN(it->mainValue())) {
03706                     range.lower = it->mainKey();
03707                     haveLower = true;
03708                     break;
03709                 }
03710                 ++it;
03711             }
03712             it = itEnd;
03713             while (it != constBegin()) // find first non-nan going down from right
03714             {
```

```
03715                 --it;
03716                 if (!qIsNaN(it->mainValue())) {
03717                     range.upper = it->mainKey();
03718                     haveUpper = true;
03719                     break;
03720                 }
03721             }
03722         } else // DataType is not sorted by main key, go through all data points and accordingly expand range
03723         {
03724             while (it != itEnd) {
03725                 if (!qIsNaN(it->mainValue())) {
03726                     current = it->mainKey();
03727                     if (current < range.lower || !haveLower) {
03728                         range.lower = current;
03729                         haveLower = true;
03730                     }
03731                     if (current > range.upper || !haveUpper) {
03732                         range.upper = current;
03733                         haveUpper = true;
03734                     }
03735                 }
03736                 ++it;
03737             }
03738         }
03739     } else if (signDomain == QCP::sdNegative) // range may only be in the negative sign domain
03740     {
03741         while (it != itEnd) {
03742             if (!qIsNaN(it->mainValue())) {
03743                 current = it->mainKey();
03744                 if ((current < range.lower || !haveLower) && current < 0) {
03745                     range.lower = current;
03746                     haveLower = true;
03747                 }
03748                 if ((current > range.upper || !haveUpper) && current < 0) {
03749                     range.upper = current;
03750                     haveUpper = true;
03751                 }
03752             }
03753             ++it;
03754         }
03755     } else if (signDomain == QCP::sdPositive) // range may only be in the positive sign domain
03756     {
03757         while (it != itEnd) {
03758             if (!qIsNaN(it->mainValue())) {
03759                 current = it->mainKey();
03760                 if ((current < range.lower || !haveLower) && current > 0) {
03761                     range.lower = current;
03762                     haveLower = true;
03763                 }
03764                 if ((current > range.upper || !haveUpper) && current > 0) {
03765                     range.upper = current;
03766                     haveUpper = true;
03767                 }
03768             }
03769             ++it;
03770         }
03771     }
03772
03773     foundRange = haveLower && haveUpper;
03774     return range;
03775 }
03776
03793 template <class DataType>
03794 QCPRange QCPDataContainer<DataType>::valueRange(bool &foundRange, QCP::SignDomain signDomain, const QCPRange &inKeyRange) {
03795     if (isEmpty()) {
03796         foundRange = false;
03797         return QCPRange();
03798     }
03799     QCPRange range;
03800     const bool restrictKeyRange = inKeyRange != QCPRange();
03801     bool haveLower = false;
03802     bool haveUpper = false;
03803     QCPRange current;
03804     QCPDataContainer<DataType>::const_iterator itBegin = constBegin();
03805     QCPDataContainer<DataType>::const_iterator itEnd = constEnd();
03806     if (DataType::sortKeyIsMainKey() && restrictKeyRange) {
03807         itBegin = findBegin(inKeyRange.lower, false);
03808         itEnd = findEnd(inKeyRange.upper, false);
03809     }
03810     if (signDomain == QCP::sdBoth) // range may be anywhere
03811     {
03812         for (QCPDataContainer<DataType>::const_iterator it = itBegin; it != itEnd; ++it) {
03813             if (restrictKeyRange && (it->mainKey() < inKeyRange.lower || it->mainKey() > inKeyRange.upper))
03814                 continue;
```

```
03815             current = it->valueRange();
03816             if ((current.lower < range.lower || !haveLower) && !qIsNaN(current.lower)) {
03817                 range.lower = current.lower;
03818                 haveLower = true;
03819             }
03820             if ((current.upper > range.upper || !haveUpper) && !qIsNaN(current.upper)) {
03821                 range.upper = current.upper;
03822                 haveUpper = true;
03823             }
03824         }
03825     } else if (signDomain == QCP::sdNegative) // range may only be in the negative sign domain
03826     {
03827         for (QCPDataContainer<DataType>::const_iterator it = itBegin; it != itEnd; ++it) {
03828             if (restrictKeyRange && (it->mainKey() < inKeyRange.lower || it->mainKey() >
      inKeyRange.upper))
03829                 continue;
03830             current = it->valueRange();
03831             if ((current.lower < range.lower || !haveLower) && current.lower < 0 &&
      !qIsNaN(current.lower)) {
03832                 range.lower = current.lower;
03833                 haveLower = true;
03834             }
03835             if ((current.upper > range.upper || !haveUpper) && current.upper < 0 &&
      !qIsNaN(current.upper)) {
03836                 range.upper = current.upper;
03837                 haveUpper = true;
03838             }
03839         }
03840     } else if (signDomain == QCP::sdPositive) // range may only be in the positive sign domain
03841     {
03842         for (QCPDataContainer<DataType>::const_iterator it = itBegin; it != itEnd; ++it) {
03843             if (restrictKeyRange && (it->mainKey() < inKeyRange.lower || it->mainKey() >
      inKeyRange.upper))
03844                 continue;
03845             current = it->valueRange();
03846             if ((current.lower < range.lower || !haveLower) && current.lower > 0 &&
      !qIsNaN(current.lower)) {
03847                 range.lower = current.lower;
03848                 haveLower = true;
03849             }
03850             if ((current.upper > range.upper || !haveUpper) && current.upper > 0 &&
      !qIsNaN(current.upper)) {
03851                 range.upper = current.upper;
03852                 haveUpper = true;
03853             }
03854         }
03855     }
03856
03857     foundRange = haveLower && haveUpper;
03858     return range;
03859 }
03860
03869 template <class DataType>
03870 void QCPDataContainer<DataType>::limitIteratorsToDataRange(const_iterator &begin, const_iterator &end,
      const QCPDataRange &dataRange) const {
03871     QCPDataRange iteratorRange(int(begin - constBegin()), int(end - constBegin()));
03872     iteratorRange = iteratorRange.bounded(dataRange.bounded(this->dataRange()));
03873     begin = constBegin() + iteratorRange.begin();
03874     end = constBegin() + iteratorRange.end();
03875 }
03876
03886 template <class DataType>
03887 void QCPDataContainer<DataType>::preallocateGrow(int minimumPreallocSize) {
03888     if (minimumPreallocSize <= mPreallocSize)
03889         return;
03890
03891     int newPreallocSize = minimumPreallocSize;
03892     newPreallocSize += (1u << qBound(4, mPreallocIteration + 4, 15)) - 12; // do 4 up to 32768-12
      preallocation, doubling in each intermediate iteration
03893     ++mPreallocIteration;
03894
03895     int sizeDifference = newPreallocSize - mPreallocSize;
03896     mData.resize(mData.size() + sizeDifference);
03897     std::copy_backward(mData.begin() + mPreallocSize, mData.end() - sizeDifference, mData.end());
03898     mPreallocSize = newPreallocSize;
03899 }
03900
03915 template <class DataType>
03916 void QCPDataContainer<DataType>::performAutoSqueeze() {
03917     const int totalAlloc = mData.capacity();
03918     const int postAllocSize = totalAlloc - mData.size();
03919     const int usedSize = size();
03920     bool shrinkPostAllocation = false;
03921     bool shrinkPreAllocation = false;
03922     if (totalAlloc > 650000) // if allocation is larger, shrink earlier with respect to total used
      size
03923     {
```

```
03924         shrinkPostAllocation = postAllocSize > usedSize * 1.5; // QVector grow strategy is 2^n for
      static data. Watch out not to oscillate!
03925         shrinkPreAllocation = mPreallocSize * 10 > usedSize;
03926     } else if (totalAlloc > 1000) // below 10 MiB raw data be generous with preallocated memory, below
      1k points don't even bother
03927     {
03928         shrinkPostAllocation = postAllocSize > usedSize * 5;
03929         shrinkPreAllocation = mPreallocSize > usedSize * 1.5; // preallocation can grow into
      postallocation, so can be smaller
03930     }
03931
03932     if (shrinkPreAllocation || shrinkPostAllocation)
03933         squeeze(shrinkPreAllocation, shrinkPostAllocation);
03934 }
03935
03936
03937 /* end of 'src/datacontainer.h' */
03938
03939
03940 /* including file 'src/plottable.h'        */
03941
03942 /* modified 2021-03-29T02:30:44, size 8461 */
03943
03944 class QCP_LIB_DECL QCPSelectionDecorator {
03945     Q_GADGET
03946 public:
03947     QCPSelectionDecorator();
03948     virtual ~QCPSelectionDecorator();
03949
03950     // getters:
03951
03952     QPen pen() const {
03953         return mPen;
03954     }
03955
03956     QBrush brush() const {
03957         return mBrush;
03958     }
03959
03960     QCPScatterStyle scatterStyle() const {
03961         return mScatterStyle;
03962     }
03963
03964     QCPScatterStyle::ScatterProperties usedScatterProperties() const {
03965         return mUsedScatterProperties;
03966     }
03967
03968     // setters:
03969     void setPen(const QPen &pen);
03970     void setBrush(const QBrush &brush);
03971     void setScatterStyle(const QCPScatterStyle &scatterStyle, QCPScatterStyle::ScatterProperties
      usedProperties = QCPScatterStyle::spPen);
03972     void setUsedScatterProperties(const QCPScatterStyle::ScatterProperties &properties);
03973
03974     // non-virtual methods:
03975     void applyPen(QCPPainter *painter) const;
03976     void applyBrush(QCPPainter *painter) const;
03977     QCPScatterStyle getFinalScatterStyle(const QCPScatterStyle &unselectedStyle) const;
03978
03979     // introduced virtual methods:
03980     virtual void copyFrom(const QCPSelectionDecorator *other);
03981     virtual void drawDecoration(QCPPainter *painter, QCPDataSelection selection);
03982
03983 protected:
03984     // property members:
03985     QPen mPen;
03986     QBrush mBrush;
03987     QCPScatterStyle mScatterStyle;
03988     QCPScatterStyle::ScatterProperties mUsedScatterProperties;
03989     // non-property members:
03990     QCPAbstractPlottable *mPlottable;
03991
03992     // introduced virtual methods:
03993     virtual bool registerWithPlottable(QCPAbstractPlottable *plottable);
03994
03995 private:
03996     Q_DISABLE_COPY(QCPSelectionDecorator)
03997     friend class QCPAbstractPlottable;
03998 };
03999
04000 Q_DECLARE_METATYPE(QCPSelectionDecorator*)
04001
04002
04003 class QCP_LIB_DECL QCPAbstractPlottable : public QCPLayerable {
04004     Q_OBJECT
04006     Q_PROPERTY(QString name READ name WRITE setName)
04007     Q_PROPERTY(bool antialiasedFill READ antialiasedFill WRITE setAntialiasedFill)
```

```
04008     Q_PROPERTY(bool antialiasedScatters READ antialiasedScatters WRITE setAntialiasedScatters)
04009     Q_PROPERTY(QPen pen READ pen WRITE setPen)
04010     Q_PROPERTY(QBrush brush READ brush WRITE setBrush)
04011     Q_PROPERTY(QCPAxis* keyAxis READ keyAxis WRITE setKeyAxis)
04012     Q_PROPERTY(QCPAxis* valueAxis READ valueAxis WRITE setValueAxis)
04013     Q_PROPERTY(QCP::SelectionType selectable READ selectable WRITE setSelectable NOTIFY
      selectableChanged)
04014     Q_PROPERTY(QCPDataSelection selection READ selection WRITE setSelection NOTIFY selectionChanged)
04015     Q_PROPERTY(QCPSelectionDecorator* selectionDecorator READ selectionDecorator WRITE
      setSelectionDecorator)
04017 public:
04018     QCPAbstractPlottable(QCPAxis *keyAxis, QCPAxis *valueAxis);
04019     virtual ~QCPAbstractPlottable() Q_DECL_OVERRIDE;
04020
04021     // getters:
04022
04023     QString name() const {
04024         return mName;
04025     }
04026
04027     bool antialiasedFill() const {
04028         return mAntialiasedFill;
04029     }
04030
04031     bool antialiasedScatters() const {
04032         return mAntialiasedScatters;
04033     }
04034
04035     QPen pen() const {
04036         return mPen;
04037     }
04038
04039     QBrush brush() const {
04040         return mBrush;
04041     }
04042
04043     QCPAxis *keyAxis() const {
04044         return mKeyAxis.data();
04045     }
04046
04047     QCPAxis *valueAxis() const {
04048         return mValueAxis.data();
04049     }
04050
04051     QCP::SelectionType selectable() const {
04052         return mSelectable;
04053     }
04054
04055     bool selected() const {
04056         return !mSelection.isEmpty();
04057     }
04058
04059     QCPDataSelection selection() const {
04060         return mSelection;
04061     }
04062
04063     QCPSelectionDecorator *selectionDecorator() const {
04064         return mSelectionDecorator;
04065     }
04066
04067     // setters:
04068     void setName(const QString &name);
04069     void setAntialiasedFill(bool enabled);
04070     void setAntialiasedScatters(bool enabled);
04071     void setPen(const QPen &pen);
04072     void setBrush(const QBrush &brush);
04073     void setKeyAxis(QCPAxis *axis);
04074     void setValueAxis(QCPAxis *axis);
04075     Q_SLOT void setSelectable(QCP::SelectionType selectable);
04076     Q_SLOT void setSelection(QCPDataSelection selection);
04077     void setSelectionDecorator(QCPSelectionDecorator *decorator);
04078
04079     // introduced virtual methods:
04080     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE = 0; // actually introduced in QCPLayerable as non-pure, but we want to force
      reimplementation for plottables
04081
04082     virtual QCPPlottableInterface1D *interface1D() {
04083         return nullptr;
04084     }
04085     virtual QCPRange getKeyRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth) const =
      0;
04086     virtual QCPRange getValueRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth, const
      QCPRange &inKeyRange = QCPRange()) const = 0;
04087
04088     // non-property methods:
04089     void coordsToPixels(double key, double value, double &x, double &y) const;
```

```
04090     const QPointF coordsToPixels(double key, double value) const;
04091     void pixelsToCoords(double x, double y, double &key, double &value) const;
04092     void pixelsToCoords(const QPointF &pixelPos, double &key, double &value) const;
04093     void rescaleAxes(bool onlyEnlarge = false) const;
04094     void rescaleKeyAxis(bool onlyEnlarge = false) const;
04095     void rescaleValueAxis(bool onlyEnlarge = false, bool inKeyRange = false) const;
04096     bool addToLegend(QCPLegend *legend);
04097     bool addToLegend();
04098     bool removeFromLegend(QCPLegend *legend) const;
04099     bool removeFromLegend() const;
04100
04101 signals:
04102     void selectionChanged(bool selected);
04103     void selectionChanged(const QCPDataSelection &selection);
04104     void selectableChanged(QCP::SelectionType selectable);
04105
04106 protected:
04107     // property members:
04108     QString mName;
04109     bool mAntialiasedFill, mAntialiasedScatters;
04110     QPen mPen;
04111     QBrush mBrush;
04112     QPointer<QCPAxis> mKeyAxis, mValueAxis;
04113     QCP::SelectionType mSelectable;
04114     QCPDataSelection mSelection;
04115     QCPSelectionDecorator *mSelectionDecorator;
04116
04117     // reimplemented virtual methods:
04118     virtual QRect clipRect() const Q_DECL_OVERRIDE;
04119     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE = 0;
04120     virtual QCP::Interaction selectionCategory() const Q_DECL_OVERRIDE;
04121     void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
04122     // events:
04123     virtual void selectEvent(QMouseEvent *event, bool additive, const QVariant &details, bool
      *selectionStateChanged) Q_DECL_OVERRIDE;
04124     virtual void deselectEvent(bool *selectionStateChanged) Q_DECL_OVERRIDE;
04125
04126     // introduced virtual methods:
04127     virtual void drawLegendIcon(QCPPainter *painter, const QRectF &rect) const = 0;
04128
04129     // non-virtual methods:
04130     void applyFillAntialiasingHint(QCPPainter *painter) const;
04131     void applyScattersAntialiasingHint(QCPPainter *painter) const;
04132
04133 private:
04134     Q_DISABLE_COPY(QCPAbstractPlottable)
04135
04136     friend class QCustomPlot;
04137     friend class QCPAxis;
04138     friend class QCPPlottableLegendItem;
04139 };
04140
04141
04142 /* end of 'src/plottable.h' */
04143
04144
04145 /* including file 'src/item.h'              */
04146
04147 /* modified 2021-03-29T02:30:44, size 9425 */
04148
04149 class QCP_LIB_DECL QCPItemAnchor {
04150     Q_GADGET
04151 public:
04152     QCPItemAnchor(QCustomPlot *parentPlot, QCPAbstractItem *parentItem, const QString &name, int
      anchorId = -1);
04153     virtual ~QCPItemAnchor();
04154
04155     // getters:
04156
04157     QString name() const {
04158         return mName;
04159     }
04160     virtual QPointF pixelPosition() const;
04161
04162 protected:
04163     // property members:
04164     QString mName;
04165
04166     // non-property members:
04167     QCustomPlot *mParentPlot;
04168     QCPAbstractItem *mParentItem;
04169     int mAnchorId;
04170     QSet<QCPItemPosition*> mChildrenX, mChildrenY;
04171
04172     // introduced virtual methods:
04173
04174     virtual QCPItemPosition *toQCPItemPosition() {
```

```
04175         return nullptr;
04176     }
04177
04178     // non-virtual methods:
04179     void addChildX(QCPItemPosition* pos); // called from pos when this anchor is set as parent
04180     void removeChildX(QCPItemPosition *pos); // called from pos when its parent anchor is reset or pos
      deleted
04181     void addChildY(QCPItemPosition* pos); // called from pos when this anchor is set as parent
04182     void removeChildY(QCPItemPosition *pos); // called from pos when its parent anchor is reset or pos
      deleted
04183
04184 private:
04185     Q_DISABLE_COPY(QCPItemAnchor)
04186
04187     friend class QCPItemPosition;
04188 };
04189
04190 class QCP_LIB_DECL QCPItemPosition : public QCPItemAnchor {
04191
04192     Q_GADGET
04193 public:
04200     enum PositionType {
04201         ptAbsolute
04202         , ptViewportRatio
04205         , ptAxisRectRatio
04208         , ptPlotCoords
04209     };
04210     Q_ENUMS(PositionType)
04211
04212     QCPItemPosition(QCustomPlot *parentPlot, QCPAbstractItem *parentItem, const QString &name);
04213     virtual ~QCPItemPosition() Q_DECL_OVERRIDE;
04214
04215     // getters:
04216
04217     PositionType type() const {
04218         return typeX();
04219     }
04220
04221     PositionType typeX() const {
04222         return mPositionTypeX;
04223     }
04224
04225     PositionType typeY() const {
04226         return mPositionTypeY;
04227     }
04228
04229     QCPItemAnchor *parentAnchor() const {
04230         return parentAnchorX();
04231     }
04232
04233     QCPItemAnchor *parentAnchorX() const {
04234         return mParentAnchorX;
04235     }
04236
04237     QCPItemAnchor *parentAnchorY() const {
04238         return mParentAnchorY;
04239     }
04240
04241     double key() const {
04242         return mKey;
04243     }
04244
04245     double value() const {
04246         return mValue;
04247     }
04248
04249     QPointF coords() const {
04250         return QPointF(mKey, mValue);
04251     }
04252
04253     QCPAxis *keyAxis() const {
04254         return mKeyAxis.data();
04255     }
04256
04257     QCPAxis *valueAxis() const {
04258         return mValueAxis.data();
04259     }
04260     QCPAxisRect *axisRect() const;
04261     virtual QPointF pixelPosition() const Q_DECL_OVERRIDE;
04262
04263     // setters:
04264     void setType(PositionType type);
04265     void setTypeX(PositionType type);
04266     void setTypeY(PositionType type);
04267     bool setParentAnchor(QCPItemAnchor *parentAnchor, bool keepPixelPosition = false);
04268     bool setParentAnchorX(QCPItemAnchor *parentAnchor, bool keepPixelPosition = false);
04269     bool setParentAnchorY(QCPItemAnchor *parentAnchor, bool keepPixelPosition = false);
```

```
04270     void setCoords(double key, double value);
04271     void setCoords(const QPointF &pos);
04272     void setAxes(QCPAxis* keyAxis, QCPAxis* valueAxis);
04273     void setAxisRect(QCPAxisRect *axisRect);
04274     void setPixelPosition(const QPointF &pixelPosition);
04275
04276 protected:
04277     // property members:
04278     PositionType mPositionTypeX, mPositionTypeY;
04279     QPointer<QCPAxis> mKeyAxis, mValueAxis;
04280     QPointer<QCPAxisRect> mAxisRect;
04281     double mKey, mValue;
04282     QCPItemAnchor *mParentAnchorX, *mParentAnchorY;
04283
04284     // reimplemented virtual methods:
04285
04286     virtual QCPItemPosition *toQCPItemPosition() Q_DECL_OVERRIDE {
04287         return this;
04288     }
04289
04290 private:
04291     Q_DISABLE_COPY(QCPItemPosition)
04292
04293 };
04294
04295 Q_DECLARE_METATYPE(QCPItemPosition::PositionType)
04296
04297
04298 class QCP_LIB_DECL QCPAbstractItem : public QCPLayerable {
04299     Q_OBJECT
04301     Q_PROPERTY(bool clipToAxisRect READ clipToAxisRect WRITE setClipToAxisRect)
04302     Q_PROPERTY(QCPAxisRect* clipAxisRect READ clipAxisRect WRITE setClipAxisRect)
04303     Q_PROPERTY(bool selectable READ selectable WRITE setSelectable NOTIFY selectableChanged)
04304     Q_PROPERTY(bool selected READ selected WRITE setSelected NOTIFY selectionChanged)
04306 public:
04307     explicit QCPAbstractItem(QCustomPlot *parentPlot);
04308     virtual ~QCPAbstractItem() Q_DECL_OVERRIDE;
04309
04310     // getters:
04311
04312     bool clipToAxisRect() const {
04313         return mClipToAxisRect;
04314     }
04315     QCPAxisRect *clipAxisRect() const;
04316
04317     bool selectable() const {
04318         return mSelectable;
04319     }
04320
04321     bool selected() const {
04322         return mSelected;
04323     }
04324
04325     // setters:
04326     void setClipToAxisRect(bool clip);
04327     void setClipAxisRect(QCPAxisRect *rect);
04328     Q_SLOT void setSelectable(bool selectable);
04329     Q_SLOT void setSelected(bool selected);
04330
04331     // reimplemented virtual methods:
04332     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE = 0;
04333
04334     // non-virtual methods:
04335
04336     QList<QCPItemPosition*> positions() const {
04337         return mPositions;
04338     }
04339
04340     QList<QCPItemAnchor*> anchors() const {
04341         return mAnchors;
04342     }
04343     QCPItemPosition *position(const QString &name) const;
04344     QCPItemAnchor *anchor(const QString &name) const;
04345     bool hasAnchor(const QString &name) const;
04346
04347 signals:
04348     void selectionChanged(bool selected);
04349     void selectableChanged(bool selectable);
04350
04351 protected:
04352     // property members:
04353     bool mClipToAxisRect;
04354     QPointer<QCPAxisRect> mClipAxisRect;
04355     QList<QCPItemPosition*> mPositions;
04356     QList<QCPItemAnchor*> mAnchors;
04357     bool mSelectable, mSelected;
```

```
04358
04359     // reimplemented virtual methods:
04360     virtual QCP::Interaction selectionCategory() const Q_DECL_OVERRIDE;
04361     virtual QRect clipRect() const Q_DECL_OVERRIDE;
04362     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
04363     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE = 0;
04364     // events:
04365     virtual void selectEvent(QMouseEvent *event, bool additive, const QVariant &details, bool
      *selectionStateChanged) Q_DECL_OVERRIDE;
04366     virtual void deselectEvent(bool *selectionStateChanged) Q_DECL_OVERRIDE;
04367
04368     // introduced virtual methods:
04369     virtual QPointF anchorPixelPosition(int anchorId) const;
04370
04371     // non-virtual methods:
04372     double rectDistance(const QRectF &rect, const QPointF &pos, bool filledRect) const;
04373     QCPItemPosition *createPosition(const QString &name);
04374     QCPItemAnchor *createAnchor(const QString &name, int anchorId);
04375
04376 private:
04377     Q_DISABLE_COPY(QCPAbstractItem)
04378
04379     friend class QCustomPlot;
04380     friend class QCPItemAnchor;
04381 };
04382
04383 /* end of 'src/item.h' */
04384
04385
04386 /* including file 'src/core.h'              */
04387
04388 /* modified 2021-03-29T02:30:44, size 19304 */
04389
04390 class QCP_LIB_DECL QCustomPlot : public QWidget {
04391
04392     Q_OBJECT
04394     Q_PROPERTY(QRect viewport READ viewport WRITE setViewport)
04395     Q_PROPERTY(QPixmap background READ background WRITE setBackground)
04396     Q_PROPERTY(bool backgroundScaled READ backgroundScaled WRITE setBackgroundScaled)
04397     Q_PROPERTY(Qt::AspectRatioMode backgroundScaledMode READ backgroundScaledMode WRITE
      setBackgroundScaledMode)
04398     Q_PROPERTY(QCPLayoutGrid* plotLayout READ plotLayout)
04399     Q_PROPERTY(bool autoAddPlottableToLegend READ autoAddPlottableToLegend WRITE
      setAutoAddPlottableToLegend)
04400     Q_PROPERTY(int selectionTolerance READ selectionTolerance WRITE setSelectionTolerance)
04401     Q_PROPERTY(bool noAntialiasingOnDrag READ noAntialiasingOnDrag WRITE setNoAntialiasingOnDrag)
04402     Q_PROPERTY(Qt::KeyboardModifier multiSelectModifier READ multiSelectModifier WRITE
      setMultiSelectModifier)
04403     Q_PROPERTY(bool openGl READ openGl WRITE setOpenGl)
04405 public:
04411     enum LayerInsertMode {
04412         limBelow
04413         , limAbove
04414     };
04415
04416     Q_ENUMS(LayerInsertMode)
04417
04418
04423     enum RefreshPriority {
04424         rpImmediateRefresh
04425         , rpQueuedRefresh
04426         , rpRefreshHint
04427         , rpQueuedReplot
04428     };
04429     Q_ENUMS(RefreshPriority)
04430
04431     explicit QCustomPlot(QWidget *parent = nullptr);
04432     virtual ~QCustomPlot() Q_DECL_OVERRIDE;
04433
04434     // getters:
04435
04436     QRect viewport() const {
04437         return mViewport;
04438     }
04439
04440     double bufferDevicePixelRatio() const {
04441         return mBufferDevicePixelRatio;
04442     }
04443
04444     QPixmap background() const {
04445         return mBackgroundPixmap;
04446     }
04447
04448     bool backgroundScaled() const {
04449         return mBackgroundScaled;
04450     }
04451
```

```
04452     Qt::AspectRatioMode backgroundScaledMode() const {
04453         return mBackgroundScaledMode;
04454     }
04455 
04456     QCPLayoutGrid *plotLayout() const {
04457         return mPlotLayout;
04458     }
04459 
04460     QCP::AntialiasedElements antialiasedElements() const {
04461         return mAntialiasedElements;
04462     }
04463 
04464     QCP::AntialiasedElements notAntialiasedElements() const {
04465         return mNotAntialiasedElements;
04466     }
04467 
04468     bool autoAddPlottableToLegend() const {
04469         return mAutoAddPlottableToLegend;
04470     }
04471 
04472     const QCP::Interactions interactions() const {
04473         return mInteractions;
04474     }
04475 
04476     int selectionTolerance() const {
04477         return mSelectionTolerance;
04478     }
04479 
04480     bool noAntialiasingOnDrag() const {
04481         return mNoAntialiasingOnDrag;
04482     }
04483 
04484     QCP::PlottingHints plottingHints() const {
04485         return mPlottingHints;
04486     }
04487 
04488     Qt::KeyboardModifier multiSelectModifier() const {
04489         return mMultiSelectModifier;
04490     }
04491 
04492     QCP::SelectionRectMode selectionRectMode() const {
04493         return mSelectionRectMode;
04494     }
04495 
04496     QCPSelectionRect *selectionRect() const {
04497         return mSelectionRect;
04498     }
04499 
04500     bool openGl() const {
04501         return mOpenGl;
04502     }
04503 
04504     // setters:
04505     void setViewport(const QRect &rect);
04506     void setBufferDevicePixelRatio(double ratio);
04507     void setBackground(const QPixmap &pm);
04508     void setBackground(const QPixmap &pm, bool scaled, Qt::AspectRatioMode mode =
      Qt::KeepAspectRatioByExpanding);
04509     void setBackground(const QBrush &brush);
04510     void setBackgroundScaled(bool scaled);
04511     void setBackgroundScaledMode(Qt::AspectRatioMode mode);
04512     void setAntialiasedElements(const QCP::AntialiasedElements &antialiasedElements);
04513     void setAntialiasedElement(QCP::AntialiasedElement antialiasedElement, bool enabled = true);
04514     void setNotAntialiasedElements(const QCP::AntialiasedElements &notAntialiasedElements);
04515     void setNotAntialiasedElement(QCP::AntialiasedElement notAntialiasedElement, bool enabled = true);
04516     void setAutoAddPlottableToLegend(bool on);
04517     void setInteractions(const QCP::Interactions &interactions);
04518     void setInteraction(const QCP::Interaction &interaction, bool enabled = true);
04519     void setSelectionTolerance(int pixels);
04520     void setNoAntialiasingOnDrag(bool enabled);
04521     void setPlottingHints(const QCP::PlottingHints &hints);
04522     void setPlottingHint(QCP::PlottingHint hint, bool enabled = true);
04523     void setMultiSelectModifier(Qt::KeyboardModifier modifier);
04524     void setSelectionRectMode(QCP::SelectionRectMode mode);
04525     void setSelectionRect(QCPSelectionRect *selectionRect);
04526     void setOpenGl(bool enabled, int multisampling = 16);
04527 
04528     // non-property methods:
04529     // plottable interface:
04530     QCPAbstractPlottable *plottable(int index);
04531     QCPAbstractPlottable *plottable();
04532     bool removePlottable(QCPAbstractPlottable *plottable);
04533     bool removePlottable(int index);
04534     int clearPlottables();
04535     int plottableCount() const;
04536     QList<QCPAbstractPlottable*> selectedPlottables() const;
04537     template<class PlottableType>
```

```
04538     PlottableType *plottableAt(const QPointF &pos, bool onlySelectable = false, int *dataIndex =
      nullptr) const;
04539     QCPAbstractPlottable *plottableAt(const QPointF &pos, bool onlySelectable = false, int *dataIndex
      = nullptr) const;
04540     bool hasPlottable(QCPAbstractPlottable *plottable) const;
04541
04542     // specialized interface for QCPGraph:
04543     QCPGraph *graph(int index) const;
04544     QCPGraph *graph() const;
04545     QCPGraph *addGraph(QCPAxis *keyAxis = nullptr, QCPAxis *valueAxis = nullptr);
04546     bool removeGraph(QCPGraph *graph);
04547     bool removeGraph(int index);
04548     int clearGraphs();
04549     int graphCount() const;
04550     QList<QCPGraph*> selectedGraphs() const;
04551
04552     // item interface:
04553     QCPAbstractItem *item(int index) const;
04554     QCPAbstractItem *item() const;
04555     bool removeItem(QCPAbstractItem *item);
04556     bool removeItem(int index);
04557     int clearItems();
04558     int itemCount() const;
04559     QList<QCPAbstractItem*> selectedItems() const;
04560     template<class ItemType>
04561     ItemType *itemAt(const QPointF &pos, bool onlySelectable = false) const;
04562     QCPAbstractItem *itemAt(const QPointF &pos, bool onlySelectable = false) const;
04563     bool hasItem(QCPAbstractItem *item) const;
04564
04565     // layer interface:
04566     QCPLayer *layer(const QString &name) const;
04567     QCPLayer *layer(int index) const;
04568     QCPLayer *currentLayer() const;
04569     bool setCurrentLayer(const QString &name);
04570     bool setCurrentLayer(QCPLayer *layer);
04571     int layerCount() const;
04572     bool addLayer(const QString &name, QCPLayer *otherLayer = nullptr, LayerInsertMode insertMode =
      limAbove);
04573     bool removeLayer(QCPLayer *layer);
04574     bool moveLayer(QCPLayer *layer, QCPLayer *otherLayer, LayerInsertMode insertMode = limAbove);
04575
04576     // axis rect/layout interface:
04577     int axisRectCount() const;
04578     QCPAxisRect* axisRect(int index = 0) const;
04579     QList<QCPAxisRect*> axisRects() const;
04580     QCPLayoutElement* layoutElementAt(const QPointF &pos) const;
04581     QCPAxisRect* axisRectAt(const QPointF &pos) const;
04582     Q_SLOT void rescaleAxes(bool onlyVisiblePlottables = false);
04583
04584     QList<QCPAxis*> selectedAxes() const;
04585     QList<QCPLegend*> selectedLegends() const;
04586     Q_SLOT void deselectAll();
04587
04588     bool savePdf(const QString &fileName, int width = 0, int height = 0, QCP::ExportPen exportPen =
      QCP::epAllowCosmetic, const QString &pdfCreator = QString(), const QString &pdfTitle = QString());
04589     bool savePng(const QString &fileName, int width = 0, int height = 0, double scale = 1.0, int
      quality = -1, int resolution = 96, QCP::ResolutionUnit resolutionUnit = QCP::ruDotsPerInch);
04590     bool saveJpg(const QString &fileName, int width = 0, int height = 0, double scale = 1.0, int
      quality = -1, int resolution = 96, QCP::ResolutionUnit resolutionUnit = QCP::ruDotsPerInch);
04591     bool saveBmp(const QString &fileName, int width = 0, int height = 0, double scale = 1.0, int
      resolution = 96, QCP::ResolutionUnit resolutionUnit = QCP::ruDotsPerInch);
04592     bool saveRastered(const QString &fileName, int width, int height, double scale, const char
      *format, int quality = -1, int resolution = 96, QCP::ResolutionUnit resolutionUnit =
      QCP::ruDotsPerInch);
04593     QPixmap toPixmap(int width = 0, int height = 0, double scale = 1.0);
04594     void toPainter(QCPPainter *painter, int width = 0, int height = 0);
04595     Q_SLOT void replot(QCustomPlot::RefreshPriority refreshPriority = QCustomPlot::rpRefreshHint);
04596     double replotTime(bool average = false) const;
04597
04598     QCPAxis *xAxis, *yAxis, *xAxis2, *yAxis2;
04599     QCPLegend *legend;
04600
04601 signals:
04602     void mouseDoubleClick(QMouseEvent *event);
04603     void mousePress(QMouseEvent *event);
04604     void mouseMove(QMouseEvent *event);
04605     void mouseRelease(QMouseEvent *event);
04606     void mouseWheel(QWheelEvent *event);
04607
04608     void plottableClick(QCPAbstractPlottable *plottable, int dataIndex, QMouseEvent *event);
04609     void plottableDoubleClick(QCPAbstractPlottable *plottable, int dataIndex, QMouseEvent *event);
04610     void itemClick(QCPAbstractItem *item, QMouseEvent *event);
04611     void itemDoubleClick(QCPAbstractItem *item, QMouseEvent *event);
04612     void axisClick(QCPAxis *axis, QCPAxis::SelectablePart part, QMouseEvent *event);
04613     void axisDoubleClick(QCPAxis *axis, QCPAxis::SelectablePart part, QMouseEvent *event);
04614     void legendClick(QCPLegend *legend, QCPAbstractLegendItem *item, QMouseEvent *event);
04615     void legendDoubleClick(QCPLegend *legend, QCPAbstractLegendItem *item, QMouseEvent *event);
```

```
04616
04617     void selectionChangedByUser();
04618     void beforeReplot();
04619     void afterLayout();
04620     void afterReplot();
04621
04622 protected:
04623     // property members:
04624     QRect mViewport;
04625     double mBufferDevicePixelRatio;
04626     QCPLayoutGrid *mPlotLayout;
04627     bool mAutoAddPlottableToLegend;
04628     QList<QCPAbstractPlottable*> mPlottables;
04629     QList<QCPGraph*> mGraphs; // extra list of plottables also in mPlottables that are of type
      QCPGraph
04630     QList<QCPAbstractItem*> mItems;
04631     QList<QCPLayer*> mLayers;
04632     QCP::AntialiasedElements mAntialiasedElements, mNotAntialiasedElements;
04633     QCP::Interactions mInteractions;
04634     int mSelectionTolerance;
04635     bool mNoAntialiasingOnDrag;
04636     QBrush mBackgroundBrush;
04637     QPixmap mBackgroundPixmap;
04638     QPixmap mScaledBackgroundPixmap;
04639     bool mBackgroundScaled;
04640     Qt::AspectRatioMode mBackgroundScaledMode;
04641     QCPLayer *mCurrentLayer;
04642     QCP::PlottingHints mPlottingHints;
04643     Qt::KeyboardModifier mMultiSelectModifier;
04644     QCP::SelectionRectMode mSelectionRectMode;
04645     QCPSelectionRect *mSelectionRect;
04646     bool mOpenGl;
04647
04648     // non-property members:
04649     QList<QSharedPointer<QCPAbstractPaintBuffer> > mPaintBuffers;
04650     QPoint mMousePressPos;
04651     bool mMouseHasMoved;
04652     QPointer<QCPLayerable> mMouseEventLayerable;
04653     QPointer<QCPLayerable> mMouseSignalLayerable;
04654     QVariant mMouseEventLayerableDetails;
04655     QVariant mMouseSignalLayerableDetails;
04656     bool mReplotting;
04657     bool mReplotQueued;
04658     double mReplotTime, mReplotTimeAverage;
04659     int mOpenGlMultisamples;
04660     QCP::AntialiasedElements mOpenGlAntialiasedElementsBackup;
04661     bool mOpenGlCacheLabelsBackup;
04662 #ifdef QCP_OPENGL_FBO
04663     QSharedPointer<QOpenGLContext> mGlContext;
04664     QSharedPointer<QSurface> mGlSurface;
04665     QSharedPointer<QOpenGLPaintDevice> mGlPaintDevice;
04666 #endif
04667
04668     // reimplemented virtual methods:
04669     virtual QSize minimumSizeHint() const Q_DECL_OVERRIDE;
04670     virtual QSize sizeHint() const Q_DECL_OVERRIDE;
04671     virtual void paintEvent(QPaintEvent *event) Q_DECL_OVERRIDE;
04672     virtual void resizeEvent(QResizeEvent *event) Q_DECL_OVERRIDE;
04673     virtual void mouseDoubleClickEvent(QMouseEvent *event) Q_DECL_OVERRIDE;
04674     virtual void mousePressEvent(QMouseEvent *event) Q_DECL_OVERRIDE;
04675     virtual void mouseMoveEvent(QMouseEvent *event) Q_DECL_OVERRIDE;
04676     virtual void mouseReleaseEvent(QMouseEvent *event) Q_DECL_OVERRIDE;
04677     virtual void wheelEvent(QWheelEvent *event) Q_DECL_OVERRIDE;
04678
04679     // introduced virtual methods:
04680     virtual void draw(QCPPainter *painter);
04681     virtual void updateLayout();
04682     virtual void axisRemoved(QCPAxis *axis);
04683     virtual void legendRemoved(QCPLegend *legend);
04684     Q_SLOT virtual void processRectSelection(QRect rect, QMouseEvent *event);
04685     Q_SLOT virtual void processRectZoom(QRect rect, QMouseEvent *event);
04686     Q_SLOT virtual void processPointSelection(QMouseEvent *event);
04687
04688     // non-virtual methods:
04689     bool registerPlottable(QCPAbstractPlottable *plottable);
04690     bool registerGraph(QCPGraph *graph);
04691     bool registerItem(QCPAbstractItem* item);
04692     void updateLayerIndices() const;
04693     QCPLayerable *layerableAt(const QPointF &pos, bool onlySelectable, QVariant *selectionDetails =
      nullptr) const;
04694     QList<QCPLayerable*> layerableListAt(const QPointF &pos, bool onlySelectable, QList<QVariant>
      *selectionDetails = nullptr) const;
04695     void drawBackground(QCPPainter *painter);
04696     void setupPaintBuffers();
04697     QCPAbstractPaintBuffer *createPaintBuffer();
04698     bool hasInvalidatedPaintBuffers();
04699     bool setupOpenGl();
```

```
04700     void freeOpenGl();
04701
04702     friend class QCPLegend;
04703     friend class QCPAxis;
04704     friend class QCPLayer;
04705     friend class QCPAxisRect;
04706     friend class QCPAbstractPlottable;
04707     friend class QCPGraph;
04708     friend class QCPAbstractItem;
04709 };
04710
04711 Q_DECLARE_METATYPE(QCustomPlot::LayerInsertMode)
04712 Q_DECLARE_METATYPE(QCustomPlot::RefreshPriority)
04713
04714
04715 // implementation of template functions:
04716
04735 template<class PlottableType>
04736 PlottableType *QCustomPlot::plottableAt(const QPointF &pos, bool onlySelectable, int *dataIndex) const
      {
04737     PlottableType *resultPlottable = 0;
04738     QVariant resultDetails;
04739     double resultDistance = mSelectionTolerance; // only regard clicks with distances smaller than
      mSelectionTolerance as selections, so initialize with that value
04740
04741     foreach(QCPAbstractPlottable *plottable, mPlottables) {
04742         PlottableType *currentPlottable = qobject_cast<PlottableType*>(plottable);
04743         if (!currentPlottable || (onlySelectable && !currentPlottable->selectable())) // we could have
      also passed onlySelectable to the selectTest function, but checking here is faster, because we have
      access to QCPAbstractPlottable::selectable
04744             continue;
04745         if (currentPlottable->clipRect().contains(pos.toPoint())) // only consider clicks where the
      plottable is actually visible
04746         {
04747             QVariant details;
04748             double currentDistance = currentPlottable->selectTest(pos, false, dataIndex ? &details :
      nullptr);
04749             if (currentDistance >= 0 && currentDistance < resultDistance) {
04750                 resultPlottable = currentPlottable;
04751                 resultDetails = details;
04752                 resultDistance = currentDistance;
04753             }
04754         }
04755     }
04756
04757     if (resultPlottable && dataIndex) {
04758         QCPDataSelection sel = resultDetails.value<QCPDataSelection>();
04759         if (!sel.isEmpty())
04760             *dataIndex = sel.dataRange(0).begin();
04761     }
04762     return resultPlottable;
04763 }
04764
04778 template<class ItemType>
04779 ItemType *QCustomPlot::itemAt(const QPointF &pos, bool onlySelectable) const {
04780     ItemType *resultItem = 0;
04781     double resultDistance = mSelectionTolerance; // only regard clicks with distances smaller than
      mSelectionTolerance as selections, so initialize with that value
04782
04783     foreach(QCPAbstractItem *item, mItems) {
04784         ItemType *currentItem = qobject_cast<ItemType*>(item);
04785         if (!currentItem || (onlySelectable && !currentItem->selectable())) // we could have also
      passed onlySelectable to the selectTest function, but checking here is faster, because we have access
      to QCPAbstractItem::selectable
04786             continue;
04787         if (!currentItem->clipToAxisRect() || currentItem->clipRect().contains(pos.toPoint())) // only
      consider clicks inside axis cliprect of the item if actually clipped to it
04788         {
04789             double currentDistance = currentItem->selectTest(pos, false);
04790             if (currentDistance >= 0 && currentDistance < resultDistance) {
04791                 resultItem = currentItem;
04792                 resultDistance = currentDistance;
04793             }
04794         }
04795     }
04796
04797     return resultItem;
04798 }
04799
04800
04801
04802 /* end of 'src/core.h' */
04803
04804
04805 /* including file 'src/plottableId.h'       */
04806
04807 /* modified 2021-03-29T02:30:44, size 25638 */
```

```
04808
04809 class QCPPlottableInterface1D {
04810 public:
04811     virtual ~QCPPlottableInterface1D() = default;
04812     // introduced pure virtual methods:
04813     virtual int dataCount() const = 0;
04814     virtual double dataMainKey(int index) const = 0;
04815     virtual double dataSortKey(int index) const = 0;
04816     virtual double dataMainValue(int index) const = 0;
04817     virtual QCPRange dataValueRange(int index) const = 0;
04818     virtual QPointF dataPixelPosition(int index) const = 0;
04819     virtual bool sortKeyIsMainKey() const = 0;
04820     virtual QCPDataSelection selectTestRect(const QRectF &rect, bool onlySelectable) const = 0;
04821     virtual int findBegin(double sortKey, bool expandedRange = true) const = 0;
04822     virtual int findEnd(double sortKey, bool expandedRange = true) const = 0;
04823 };
04824
04825 template <class DataType>
04826 class QCPAbstractPlottable1D : public QCPAbstractPlottable, public QCPPlottableInterface1D // no
      QCP_LIB_DECL, template class ends up in header (cpp included below)
04827 {
04828     // No Q_OBJECT macro due to template class
04829
04830 public:
04831     QCPAbstractPlottable1D(QCPAxis *keyAxis, QCPAxis *valueAxis);
04832     virtual ~QCPAbstractPlottable1D() Q_DECL_OVERRIDE;
04833
04834     // virtual methods of 1d plottable interface:
04835     virtual int dataCount() const Q_DECL_OVERRIDE;
04836     virtual double dataMainKey(int index) const Q_DECL_OVERRIDE;
04837     virtual double dataSortKey(int index) const Q_DECL_OVERRIDE;
04838     virtual double dataMainValue(int index) const Q_DECL_OVERRIDE;
04839     virtual QCPRange dataValueRange(int index) const Q_DECL_OVERRIDE;
04840     virtual QPointF dataPixelPosition(int index) const Q_DECL_OVERRIDE;
04841     virtual bool sortKeyIsMainKey() const Q_DECL_OVERRIDE;
04842     virtual QCPDataSelection selectTestRect(const QRectF &rect, bool onlySelectable) const
      Q_DECL_OVERRIDE;
04843     virtual int findBegin(double sortKey, bool expandedRange = true) const Q_DECL_OVERRIDE;
04844     virtual int findEnd(double sortKey, bool expandedRange = true) const Q_DECL_OVERRIDE;
04845
04846     // reimplemented virtual methods:
04847     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
04848
04849     virtual QCPPlottableInterface1D *interface1D() Q_DECL_OVERRIDE {
04850         return this;
04851     }
04852
04853 protected:
04854     // property members:
04855     QSharedPointer<QCPDataContainer<DataType> > mDataContainer;
04856
04857     // helpers for subclasses:
04858     void getDataSegments(QList<QCPDataRange> &selectedSegments, QList<QCPDataRange>
      &unselectedSegments) const;
04859     void drawPolyline(QCPPainter *painter, const QVector<QPointF> &lineData) const;
04860
04861 private:
04862     Q_DISABLE_COPY(QCPAbstractPlottable1D)
04863
04864 };
04865
04866
04867
04868 // include implementation in header since it is a class template:
04872
04896 /* start documentation of pure virtual functions */
04897
05007 /* end documentation of pure virtual functions */
05008
05009
05013
05038 /* start documentation of inline functions */
05039
05048 /* end documentation of inline functions */
05049
05054 template <class DataType>
05055 QCPAbstractPlottable1D<DataType>::QCPAbstractPlottable1D(QCPAxis *keyAxis, QCPAxis *valueAxis) :
05056 QCPAbstractPlottable(keyAxis, valueAxis),
05057 mDataContainer(new QCPDataContainer<DataType>) {
05058 }
05059
05060 template <class DataType>
05061 QCPAbstractPlottable1D<DataType>::~QCPAbstractPlottable1D() {
05062 }
05063
05067 template <class DataType>
```

```
05068 int QCPAbstractPlottable1D<DataType>::dataCount() const {
05069     return mDataContainer->size();
05070 }
05071
05075 template <class DataType>
05076 double QCPAbstractPlottable1D<DataType>::dataMainKey(int index) const {
05077     if (index >= 0 && index < mDataContainer->size()) {
05078         return (mDataContainer->constBegin() + index)->mainKey();
05079     } else {
05080         qDebug() « Q_FUNC_INFO « "Index out of bounds" « index;
05081         return 0;
05082     }
05083 }
05084
05088 template <class DataType>
05089 double QCPAbstractPlottable1D<DataType>::dataSortKey(int index) const {
05090     if (index >= 0 && index < mDataContainer->size()) {
05091         return (mDataContainer->constBegin() + index)->sortKey();
05092     } else {
05093         qDebug() « Q_FUNC_INFO « "Index out of bounds" « index;
05094         return 0;
05095     }
05096 }
05097
05101 template <class DataType>
05102 double QCPAbstractPlottable1D<DataType>::dataMainValue(int index) const {
05103     if (index >= 0 && index < mDataContainer->size()) {
05104         return (mDataContainer->constBegin() + index)->mainValue();
05105     } else {
05106         qDebug() « Q_FUNC_INFO « "Index out of bounds" « index;
05107         return 0;
05108     }
05109 }
05110
05114 template <class DataType>
05115 QCPRange QCPAbstractPlottable1D<DataType>::dataValueRange(int index) const {
05116     if (index >= 0 && index < mDataContainer->size()) {
05117         return (mDataContainer->constBegin() + index)->valueRange();
05118     } else {
05119         qDebug() « Q_FUNC_INFO « "Index out of bounds" « index;
05120         return QCPRange(0, 0);
05121     }
05122 }
05123
05127 template <class DataType>
05128 QPointF QCPAbstractPlottable1D<DataType>::dataPixelPosition(int index) const {
05129     if (index >= 0 && index < mDataContainer->size()) {
05130         const typename QCPDataContainer<DataType>::const_iterator it = mDataContainer->constBegin() +
      index;
05131         return coordsToPixels(it->mainKey(), it->mainValue());
05132     } else {
05133         qDebug() « Q_FUNC_INFO « "Index out of bounds" « index;
05134         return QPointF();
05135     }
05136 }
05137
05141 template <class DataType>
05142 bool QCPAbstractPlottable1D<DataType>::sortKeyIsMainKey() const {
05143     return DataType::sortKeyIsMainKey();
05144 }
05145
05153 template <class DataType>
05154 QCPDataSelection QCPAbstractPlottable1D<DataType>::selectTestRect(const QRectF &rect, bool
      onlySelectable) const {
05155     QCPDataSelection result;
05156     if ((onlySelectable && mSelectable == QCP::stNone) || mDataContainer->isEmpty())
05157         return result;
05158     if (!mKeyAxis || !mValueAxis)
05159         return result;
05160
05161     // convert rect given in pixels to ranges given in plot coordinates:
05162     double key1, value1, key2, value2;
05163     pixelsToCoords(rect.topLeft(), key1, value1);
05164     pixelsToCoords(rect.bottomRight(), key2, value2);
05165     QCPRange keyRange(key1, key2); // QCPRange normalizes internally so we don't have to care about
      whether key1 < key2
05166     QCPRange valueRange(value1, value2);
05167     typename QCPDataContainer<DataType>::const_iterator begin = mDataContainer->constBegin();
05168     typename QCPDataContainer<DataType>::const_iterator end = mDataContainer->constEnd();
05169     if (DataType::sortKeyIsMainKey()) // we can assume that data is sorted by main key, so can reduce
      the searched key interval:
05170     {
05171         begin = mDataContainer->findBegin(keyRange.lower, false);
05172         end = mDataContainer->findEnd(keyRange.upper, false);
05173     }
05174     if (begin == end)
05175         return result;
```

```
05176
05177     int currentSegmentBegin = -1; // -1 means we're currently not in a segment that's contained in
      rect
05178     for (typename QCPDataContainer<DataType>::const_iterator it = begin; it != end; ++it) {
05179         if (currentSegmentBegin == -1) {
05180             if (valueRange.contains(it->mainValue()) && keyRange.contains(it->mainKey())) // start
      segment
05181                 currentSegmentBegin = int(it - mDataContainer->constBegin());
05182         } else if (!valueRange.contains(it->mainValue()) || !keyRange.contains(it->mainKey())) //
      segment just ended
05183         {
05184             result.addDataRange(QCPDataRange(currentSegmentBegin, int(it -
      mDataContainer->constBegin())), false);
05185             currentSegmentBegin = -1;
05186         }
05187     }
05188     // process potential last segment:
05189     if (currentSegmentBegin != -1)
05190         result.addDataRange(QCPDataRange(currentSegmentBegin, int(end -
      mDataContainer->constBegin())), false);
05191
05192     result.simplify();
05193     return result;
05194 }
05195
05199 template <class DataType>
05200 int QCPAbstractPlottable1D<DataType>::findBegin(double sortKey, bool expandedRange) const {
05201     return int(mDataContainer->findBegin(sortKey, expandedRange) - mDataContainer->constBegin());
05202 }
05203
05207 template <class DataType>
05208 int QCPAbstractPlottable1D<DataType>::findEnd(double sortKey, bool expandedRange) const {
05209     return int(mDataContainer->findEnd(sortKey, expandedRange) - mDataContainer->constBegin());
05210 }
05211
05222 template <class DataType>
05223 double QCPAbstractPlottable1D<DataType>::selectTest(const QPointF &pos, bool onlySelectable, QVariant
      *details) const {
05224     if ((onlySelectable && mSelectable == QCP::stNone) || mDataContainer->isEmpty())
05225         return -1;
05226     if (!mKeyAxis || !mValueAxis)
05227         return -1;
05228
05229     QCPDataSelection selectionResult;
05230     double minDistSqr = (std::numeric_limits<double>::max)();
05231     int minDistIndex = mDataContainer->size();
05232
05233     typename QCPDataContainer<DataType>::const_iterator begin = mDataContainer->constBegin();
05234     typename QCPDataContainer<DataType>::const_iterator end = mDataContainer->constEnd();
05235     if (DataType::sortKeyIsMainKey()) // we can assume that data is sorted by main key, so can reduce
      the searched key interval:
05236     {
05237         // determine which key range comes into question, taking selection tolerance around pos into
      account:
05238         double posKeyMin, posKeyMax, dummy;
05239         pixelsToCoords(pos - QPointF(mParentPlot->selectionTolerance(),
      mParentPlot->selectionTolerance()), posKeyMin, dummy);
05240         pixelsToCoords(pos + QPointF(mParentPlot->selectionTolerance(),
      mParentPlot->selectionTolerance()), posKeyMax, dummy);
05241         if (posKeyMin > posKeyMax)
05242             qSwap(posKeyMin, posKeyMax);
05243         begin = mDataContainer->findBegin(posKeyMin, true);
05244         end = mDataContainer->findEnd(posKeyMax, true);
05245     }
05246     if (begin == end)
05247         return -1;
05248     QCPRange keyRange(mKeyAxis->range());
05249     QCPRange valueRange(mValueAxis->range());
05250     for (typename QCPDataContainer<DataType>::const_iterator it = begin; it != end; ++it) {
05251         const double mainKey = it->mainKey();
05252         const double mainValue = it->mainValue();
05253         if (keyRange.contains(mainKey) && valueRange.contains(mainValue)) // make sure data point is
      inside visible range, for speedup in cases where sort key isn't main key and we iterate over all
      points
05254         {
05255             const double currentDistSqr = QCPVector2D(coordsToPixels(mainKey, mainValue) -
      pos).lengthSquared();
05256             if (currentDistSqr < minDistSqr) {
05257                 minDistSqr = currentDistSqr;
05258                 minDistIndex = int(it - mDataContainer->constBegin());
05259             }
05260         }
05261     }
05262     if (minDistIndex != mDataContainer->size())
05263         selectionResult.addDataRange(QCPDataRange(minDistIndex, minDistIndex + 1), false);
05264
05265     selectionResult.simplify();
```

```
05266     if (details)
05267         details->setValue(selectionResult);
05268     return qSqrt(minDistSqr);
05269 }
05270
05281 template <class DataType>
05282 void QCPAbstractPlottable1D<DataType>::getDataSegments(QList<QCPDataRange> &selectedSegments,
      QList<QCPDataRange> &unselectedSegments) const {
05283     selectedSegments.clear();
05284     unselectedSegments.clear();
05285     if (mSelectable == QCP::stWhole) // stWhole selection type draws the entire plottable with
      selected style if mSelection isn't empty
05286     {
05287         if (selected())
05288             selectedSegments << QCPDataRange(0, dataCount());
05289         else
05290             unselectedSegments << QCPDataRange(0, dataCount());
05291     } else {
05292         QCPDataSelection sel(selection());
05293         sel.simplify();
05294         selectedSegments = sel.dataRanges();
05295         unselectedSegments = sel.inverse(QCPDataRange(0, dataCount())).dataRanges();
05296     }
05297 }
05298
05309 template <class DataType>
05310 void QCPAbstractPlottable1D<DataType>::drawPolyline(QCPPainter *painter, const QVector<QPointF>
      &lineData) const {
05311     // if drawing lines in plot (instead of PDF), reduce 1px lines to cosmetic, because at least in
05312     // Qt6 drawing of "1px" width lines is much slower even though it has same appearance apart from
05313     // High-DPI. In High-DPI cases people must set a pen width slightly larger than 1.0 to get
05314     // correct DPI scaling of width, but of course with performance penalty.
05315     if (!painter->modes().testFlag(QCPPainter::pmVectorized) &&
05316             qFuzzyCompare(painter->pen().widthF(), 1.0)) {
05317         QPen newPen = painter->pen();
05318         newPen.setWidth(0);
05319         painter->setPen(newPen);
05320     }
05321
05322     // if drawing solid line and not in PDF, use much faster line drawing instead of polyline:
05323     if (mParentPlot->plottingHints().testFlag(QCP::phFastPolylines) &&
05324             painter->pen().style() == Qt::SolidLine &&
05325             !painter->modes().testFlag(QCPPainter::pmVectorized) &&
05326             !painter->modes().testFlag(QCPPainter::pmNoCaching)) {
05327         int i = 0;
05328         bool lastIsNan = false;
05329         const int lineDataSize = lineData.size();
05330         while (i < lineDataSize && (qIsNaN(lineData.at(i).y()) || qIsNaN(lineData.at(i).x()))) // make
      sure first point is not NaN
05331             ++i;
05332         ++i; // because drawing works in 1 point retrospect
05333         while (i < lineDataSize) {
05334             if (!qIsNaN(lineData.at(i).y()) && !qIsNaN(lineData.at(i).x())) // NaNs create a gap in
      the line
05335             {
05336                 if (!lastIsNan)
05337                     painter->drawLine(lineData.at(i - 1), lineData.at(i));
05338                 else
05339                     lastIsNan = false;
05340             } else
05341                 lastIsNan = true;
05342             ++i;
05343         }
05344     } else {
05345         int segmentStart = 0;
05346         int i = 0;
05347         const int lineDataSize = lineData.size();
05348         while (i < lineDataSize) {
05349             if (qIsNaN(lineData.at(i).y()) || qIsNaN(lineData.at(i).x()) ||
      qIsInf(lineData.at(i).y())) // NaNs create a gap in the line. Also filter Infs which make drawPolyline
      block
05350             {
05351                 painter->drawPolyline(lineData.constData() + segmentStart, i - segmentStart); // i,
      because we don't want to include the current NaN point
05352                 segmentStart = i + 1;
05353             }
05354             ++i;
05355         }
05356         // draw last segment:
05357         painter->drawPolyline(lineData.constData() + segmentStart, lineDataSize - segmentStart);
05358     }
05359 }
05360
05361
05362 /* end of 'src/plottable1d.h' */
05363
05364
```

```
05365 /* including file 'src/colorgradient.h'     */
05366
05367 /* modified 2021-03-29T02:30:44, size 7262 */
05368
05369 class QCP_LIB_DECL QCPColorGradient {
05370
05371     Q_GADGET
05372 public:
05378     enum ColorInterpolation {
05379         ciRGB
05380         , ciHSV
05381     };
05382
05383     Q_ENUMS(ColorInterpolation)
05384
05385
05390     enum NanHandling {
05391         nhNone
05392         , nhLowestColor
05393         , nhHighestColor
05394         , nhTransparent
05395         , nhNanColor
05396     };
05397
05398     Q_ENUMS(NanHandling)
05399
05400
05404     enum GradientPreset {
05405         gpGrayscale
05406         , gpHot
05407         , gpCold
05408         , gpNight
05409         , gpCandy
05410         , gpGeography
05411         , gpIon
05412         , gpThermal
05413         , gpPolar
05414         , gpSpectrum
05415         , gpJet
05416         , gpHues
05417     };
05418     Q_ENUMS(GradientPreset)
05419
05420     QCPColorGradient();
05421     QCPColorGradient(GradientPreset preset);
05422     bool operator==(const QCPColorGradient &other) const;
05423
05424     bool operator!=(const QCPColorGradient &other) const {
05425         return !(*this == other);
05426     }
05427
05428     // getters:
05429
05430     int levelCount() const {
05431         return mLevelCount;
05432     }
05433
05434     QMap<double, QColor> colorStops() const {
05435         return mColorStops;
05436     }
05437
05438     ColorInterpolation colorInterpolation() const {
05439         return mColorInterpolation;
05440     }
05441
05442     NanHandling nanHandling() const {
05443         return mNanHandling;
05444     }
05445
05446     QColor nanColor() const {
05447         return mNanColor;
05448     }
05449
05450     bool periodic() const {
05451         return mPeriodic;
05452     }
05453
05454     // setters:
05455     void setLevelCount(int n);
05456     void setColorStops(const QMap<double, QColor> &colorStops);
05457     void setColorStopAt(double position, const QColor &color);
05458     void setColorInterpolation(ColorInterpolation interpolation);
05459     void setNanHandling(NanHandling handling);
05460     void setNanColor(const QColor &color);
05461     void setPeriodic(bool enabled);
05462
05463     // non-property methods:
```

```
05464     void colorize(const double *data, const QCPRange &range, QRgb *scanLine, int n, int
      dataIndexFactor = 1, bool logarithmic = false);
05465     void colorize(const double *data, const unsigned char *alpha, const QCPRange &range, QRgb
      *scanLine, int n, int dataIndexFactor = 1, bool logarithmic = false);
05466     QRgb color(double position, const QCPRange &range, bool logarithmic = false);
05467     void loadPreset(GradientPreset preset);
05468     void clearColorStops();
05469     QCPColorGradient inverted() const;
05470
05471 protected:
05472     // property members:
05473     int mLevelCount;
05474     QMap<double, QColor> mColorStops;
05475     ColorInterpolation mColorInterpolation;
05476     NanHandling mNanHandling;
05477     QColor mNanColor;
05478     bool mPeriodic;
05479
05480     // non-property members:
05481     QVector<QRgb> mColorBuffer; // have colors premultiplied with alpha (for usage with
      QImage::Format_ARGB32_Premultiplied)
05482     bool mColorBufferInvalidated;
05483
05484     // non-virtual methods:
05485     bool stopsUseAlpha() const;
05486     void updateColorBuffer();
05487 };
05488
05489 Q_DECLARE_METATYPE(QCPColorGradient::ColorInterpolation)
05490 Q_DECLARE_METATYPE(QCPColorGradient::NanHandling)
05491 Q_DECLARE_METATYPE(QCPColorGradient::GradientPreset)
05492
05493 /* end of 'src/colorgradient.h' */
05494
05495
05496 /* including file 'src/selectiondecorator-bracket.h' */
05497 /* modified 2021-03-29T02:30:44, size 4458           */
05498
05499 class QCP_LIB_DECL QCPSelectionDecoratorBracket : public QCPSelectionDecorator {
05500
05501     Q_GADGET
05502 public:
05503
05510     enum BracketStyle {
05511         bsSquareBracket
05512         , bsHalfEllipse
05513         , bsEllipse
05514         , bsPlus
05515         , bsUserStyle
05516     };
05517     Q_ENUMS(BracketStyle)
05518
05519     QCPSelectionDecoratorBracket();
05520     virtual ~QCPSelectionDecoratorBracket() Q_DECL_OVERRIDE;
05521
05522     // getters:
05523
05524     QPen bracketPen() const {
05525         return mBracketPen;
05526     }
05527
05528     QBrush bracketBrush() const {
05529         return mBracketBrush;
05530     }
05531
05532     int bracketWidth() const {
05533         return mBracketWidth;
05534     }
05535
05536     int bracketHeight() const {
05537         return mBracketHeight;
05538     }
05539
05540     BracketStyle bracketStyle() const {
05541         return mBracketStyle;
05542     }
05543
05544     bool tangentToData() const {
05545         return mTangentToData;
05546     }
05547
05548     int tangentAverage() const {
05549         return mTangentAverage;
05550     }
05551
05552     // setters:
05553     void setBracketPen(const QPen &pen);
```

```
05554     void setBracketBrush(const QBrush &brush);
05555     void setBracketWidth(int width);
05556     void setBracketHeight(int height);
05557     void setBracketStyle(BracketStyle style);
05558     void setTangentToData(bool enabled);
05559     void setTangentAverage(int pointCount);
05560 
05561     // introduced virtual methods:
05562     virtual void drawBracket(QCPPainter *painter, int direction) const;
05563 
05564     // virtual methods:
05565     virtual void drawDecoration(QCPPainter *painter, QCPDataSelection selection) Q_DECL_OVERRIDE;
05566 
05567 protected:
05568     // property members:
05569     QPen mBracketPen;
05570     QBrush mBracketBrush;
05571     int mBracketWidth;
05572     int mBracketHeight;
05573     BracketStyle mBracketStyle;
05574     bool mTangentToData;
05575     int mTangentAverage;
05576 
05577     // non-virtual methods:
05578     double getTangentAngle(const QCPPlottableInterface1D *interface1d, int dataIndex, int direction)
    const;
05579     QPointF getPixelCoordinates(const QCPPlottableInterface1D *interface1d, int dataIndex) const;
05580 
05581 };
05582 
05583 Q_DECLARE_METATYPE(QCPSelectionDecoratorBracket::BracketStyle)
05584 
05585 /* end of 'src/selectiondecorator-bracket.h' */
05586 
05587 
05588 /* including file 'src/layoutelements/layoutelement-axisrect.h' */
05589 /* modified 2021-03-29T02:30:44, size 7529                      */
05590 
05591 class QCP_LIB_DECL QCPAxisRect : public QCPLayoutElement {
05592     Q_OBJECT
05594     Q_PROPERTY(QPixmap background READ background WRITE setBackground)
05595     Q_PROPERTY(bool backgroundScaled READ backgroundScaled WRITE setBackgroundScaled)
05596     Q_PROPERTY(Qt::AspectRatioMode backgroundScaledMode READ backgroundScaledMode WRITE
    setBackgroundScaledMode)
05597     Q_PROPERTY(Qt::Orientations rangeDrag READ rangeDrag WRITE setRangeDrag)
05598     Q_PROPERTY(Qt::Orientations rangeZoom READ rangeZoom WRITE setRangeZoom)
05600 public:
05601     explicit QCPAxisRect(QCustomPlot *parentPlot, bool setupDefaultAxes = true);
05602     virtual ~QCPAxisRect() Q_DECL_OVERRIDE;
05603 
05604     // getters:
05605 
05606     QPixmap background() const {
05607         return mBackgroundPixmap;
05608     }
05609 
05610     QBrush backgroundBrush() const {
05611         return mBackgroundBrush;
05612     }
05613 
05614     bool backgroundScaled() const {
05615         return mBackgroundScaled;
05616     }
05617 
05618     Qt::AspectRatioMode backgroundScaledMode() const {
05619         return mBackgroundScaledMode;
05620     }
05621 
05622     Qt::Orientations rangeDrag() const {
05623         return mRangeDrag;
05624     }
05625 
05626     Qt::Orientations rangeZoom() const {
05627         return mRangeZoom;
05628     }
05629     QCPAxis *rangeDragAxis(Qt::Orientation orientation);
05630     QCPAxis *rangeZoomAxis(Qt::Orientation orientation);
05631     QList<QCPAxis*> rangeDragAxes(Qt::Orientation orientation);
05632     QList<QCPAxis*> rangeZoomAxes(Qt::Orientation orientation);
05633     double rangeZoomFactor(Qt::Orientation orientation);
05634 
05635     // setters:
05636     void setBackground(const QPixmap &pm);
05637     void setBackground(const QPixmap &pm, bool scaled, Qt::AspectRatioMode mode =
    Qt::KeepAspectRatioByExpanding);
05638     void setBackground(const QBrush &brush);
05639     void setBackgroundScaled(bool scaled);
```

```
05640     void setBackgroundScaledMode(Qt::AspectRatioMode mode);
05641     void setRangeDrag(Qt::Orientations orientations);
05642     void setRangeZoom(Qt::Orientations orientations);
05643     void setRangeDragAxes(QCPAxis *horizontal, QCPAxis *vertical);
05644     void setRangeDragAxes(QList<QCPAxis*> axes);
05645     void setRangeDragAxes(QList<QCPAxis*> horizontal, QList<QCPAxis*> vertical);
05646     void setRangeZoomAxes(QCPAxis *horizontal, QCPAxis *vertical);
05647     void setRangeZoomAxes(QList<QCPAxis*> axes);
05648     void setRangeZoomAxes(QList<QCPAxis*> horizontal, QList<QCPAxis*> vertical);
05649     void setRangeZoomFactor(double horizontalFactor, double verticalFactor);
05650     void setRangeZoomFactor(double factor);
05651
05652     // non-property methods:
05653     int axisCount(QCPAxis::AxisType type) const;
05654     QCPAxis *axis(QCPAxis::AxisType type, int index = 0) const;
05655     QList<QCPAxis*> axes(QCPAxis::AxisTypes types) const;
05656     QList<QCPAxis*> axes() const;
05657     QCPAxis *addAxis(QCPAxis::AxisType type, QCPAxis *axis = nullptr);
05658     QList<QCPAxis*> addAxes(QCPAxis::AxisTypes types);
05659     bool removeAxis(QCPAxis *axis);
05660
05661     QCPLayoutInset *insetLayout() const {
05662         return mInsetLayout;
05663     }
05664
05665     void zoom(const QRectF &pixelRect);
05666     void zoom(const QRectF &pixelRect, const QList<QCPAxis*> &affectedAxes);
05667     void setupFullAxesBox(bool connectRanges = false);
05668     QList<QCPAbstractPlottable*> plottables() const;
05669     QList<QCPGraph*> graphs() const;
05670     QList<QCPAbstractItem*> items() const;
05671
05672     // read-only interface imitating a QRect:
05673
05674     int left() const {
05675         return mRect.left();
05676     }
05677
05678     int right() const {
05679         return mRect.right();
05680     }
05681
05682     int top() const {
05683         return mRect.top();
05684     }
05685
05686     int bottom() const {
05687         return mRect.bottom();
05688     }
05689
05690     int width() const {
05691         return mRect.width();
05692     }
05693
05694     int height() const {
05695         return mRect.height();
05696     }
05697
05698     QSize size() const {
05699         return mRect.size();
05700     }
05701
05702     QPoint topLeft() const {
05703         return mRect.topLeft();
05704     }
05705
05706     QPoint topRight() const {
05707         return mRect.topRight();
05708     }
05709
05710     QPoint bottomLeft() const {
05711         return mRect.bottomLeft();
05712     }
05713
05714     QPoint bottomRight() const {
05715         return mRect.bottomRight();
05716     }
05717
05718     QPoint center() const {
05719         return mRect.center();
05720     }
05721
05722     // reimplemented virtual methods:
05723     virtual void update(UpdatePhase phase) Q_DECL_OVERRIDE;
05724     virtual QList<QCPLayoutElement*> elements(bool recursive) const Q_DECL_OVERRIDE;
05725
05726 protected:
```

```
05727     // property members:
05728     QBrush mBackgroundBrush;
05729     QPixmap mBackgroundPixmap;
05730     QPixmap mScaledBackgroundPixmap;
05731     bool mBackgroundScaled;
05732     Qt::AspectRatioMode mBackgroundScaledMode;
05733     QCPLayoutInset *mInsetLayout;
05734     Qt::Orientations mRangeDrag, mRangeZoom;
05735     QList<QPointer<QCPAxis> > mRangeDragHorzAxis, mRangeDragVertAxis;
05736     QList<QPointer<QCPAxis> > mRangeZoomHorzAxis, mRangeZoomVertAxis;
05737     double mRangeZoomFactorHorz, mRangeZoomFactorVert;
05738
05739     // non-property members:
05740     QList<QCPRange> mDragStartHorzRange, mDragStartVertRange;
05741     QCP::AntialiasedElements mAADragBackup, mNotAADragBackup;
05742     bool mDragging;
05743     QHash<QCPAxis::AxisType, QList<QCPAxis*> > mAxes;
05744
05745     // reimplemented virtual methods:
05746     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
05747     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
05748     virtual int calculateAutoMargin(QCP::MarginSide side) Q_DECL_OVERRIDE;
05749     virtual void layoutChanged() Q_DECL_OVERRIDE;
05750     // events:
05751     virtual void mousePressEvent(QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE;
05752     virtual void mouseMoveEvent(QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE;
05753     virtual void mouseReleaseEvent(QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE;
05754     virtual void wheelEvent(QWheelEvent *event) Q_DECL_OVERRIDE;
05755
05756     // non-property methods:
05757     void drawBackground(QCPPainter *painter);
05758     void updateAxesOffset(QCPAxis::AxisType type);
05759
05760 private:
05761     Q_DISABLE_COPY(QCPAxisRect)
05762
05763     friend class QCustomPlot;
05764 };
05765
05766
05767 /* end of 'src/layoutelements/layoutelement-axisrect.h' */
05768
05769
05770 /* including file 'src/layoutelements/layoutelement-legend.h' */
05771
05772 /* modified 2021-03-29T02:30:44, size 10425                   */
05773
05774 class QCP_LIB_DECL QCPAbstractLegendItem : public QCPLayoutElement {
05775     Q_OBJECT
05777     Q_PROPERTY(QCPLegend* parentLegend READ parentLegend)
05778     Q_PROPERTY(QFont font READ font WRITE setFont)
05779     Q_PROPERTY(QColor textColor READ textColor WRITE setTextColor)
05780     Q_PROPERTY(QFont selectedFont READ selectedFont WRITE setSelectedFont)
05781     Q_PROPERTY(QColor selectedTextColor READ selectedTextColor WRITE setSelectedTextColor)
05782     Q_PROPERTY(bool selectable READ selectable WRITE setSelectable NOTIFY selectionChanged)
05783     Q_PROPERTY(bool selected READ selected WRITE setSelected NOTIFY selectableChanged)
05785 public:
05786     explicit QCPAbstractLegendItem(QCPLegend *parent);
05787
05788     // getters:
05789
05790     QCPLegend *parentLegend() const {
05791         return mParentLegend;
05792     }
05793
05794     QFont font() const {
05795         return mFont;
05796     }
05797
05798     QColor textColor() const {
05799         return mTextColor;
05800     }
05801
05802     QFont selectedFont() const {
05803         return mSelectedFont;
05804     }
05805
05806     QColor selectedTextColor() const {
05807         return mSelectedTextColor;
05808     }
05809
05810     bool selectable() const {
05811         return mSelectable;
05812     }
05813
05814     bool selected() const {
05815         return mSelected;
```

```
05816     }
05817
05818     // setters:
05819     void setFont(const QFont &font);
05820     void setTextColor(const QColor &color);
05821     void setSelectedFont(const QFont &font);
05822     void setSelectedTextColor(const QColor &color);
05823     Q_SLOT void setSelectable(bool selectable);
05824     Q_SLOT void setSelected(bool selected);
05825
05826     // reimplemented virtual methods:
05827     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
05828
05829 signals:
05830     void selectionChanged(bool selected);
05831     void selectableChanged(bool selectable);
05832
05833 protected:
05834     // property members:
05835     QCPLegend *mParentLegend;
05836     QFont mFont;
05837     QColor mTextColor;
05838     QFont mSelectedFont;
05839     QColor mSelectedTextColor;
05840     bool mSelectable, mSelected;
05841
05842     // reimplemented virtual methods:
05843     virtual QCP::Interaction selectionCategory() const Q_DECL_OVERRIDE;
05844     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
05845     virtual QRect clipRect() const Q_DECL_OVERRIDE;
05846     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE = 0;
05847     // events:
05848     virtual void selectEvent(QMouseEvent *event, bool additive, const QVariant &details, bool
      *selectionStateChanged) Q_DECL_OVERRIDE;
05849     virtual void deselectEvent(bool *selectionStateChanged) Q_DECL_OVERRIDE;
05850
05851 private:
05852     Q_DISABLE_COPY(QCPAbstractLegendItem)
05853
05854     friend class QCPLegend;
05855 };
05856
05857 class QCP_LIB_DECL QCPPlottableLegendItem : public QCPAbstractLegendItem {
05858     Q_OBJECT
05859 public:
05860     QCPPlottableLegendItem(QCPLegend *parent, QCPAbstractPlottable *plottable);
05861
05862     // getters:
05863
05864     QCPAbstractPlottable *plottable() {
05865         return mPlottable;
05866     }
05867
05868 protected:
05869     // property members:
05870     QCPAbstractPlottable *mPlottable;
05871
05872     // reimplemented virtual methods:
05873     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
05874     virtual QSize minimumOuterSizeHint() const Q_DECL_OVERRIDE;
05875
05876     // non-virtual methods:
05877     QPen getIconBorderPen() const;
05878     QColor getTextColor() const;
05879     QFont getFont() const;
05880 };
05881
05882 class QCP_LIB_DECL QCPLegend : public QCPLayoutGrid {
05883
05884     Q_OBJECT
05886     Q_PROPERTY(QPen borderPen READ borderPen WRITE setBorderPen)
05887     Q_PROPERTY(QBrush brush READ brush WRITE setBrush)
05888     Q_PROPERTY(QFont font READ font WRITE setFont)
05889     Q_PROPERTY(QColor textColor READ textColor WRITE setTextColor)
05890     Q_PROPERTY(QSize iconSize READ iconSize WRITE setIconSize)
05891     Q_PROPERTY(int iconTextPadding READ iconTextPadding WRITE setIconTextPadding)
05892     Q_PROPERTY(QPen iconBorderPen READ iconBorderPen WRITE setIconBorderPen)
05893     Q_PROPERTY(SelectableParts selectableParts READ selectableParts WRITE setSelectableParts NOTIFY
      selectionChanged)
05894     Q_PROPERTY(SelectableParts selectedParts READ selectedParts WRITE setSelectedParts NOTIFY
      selectableChanged)
05895     Q_PROPERTY(QPen selectedBorderPen READ selectedBorderPen WRITE setSelectedBorderPen)
05896     Q_PROPERTY(QPen selectedIconBorderPen READ selectedIconBorderPen WRITE setSelectedIconBorderPen)
05897     Q_PROPERTY(QBrush selectedBrush READ selectedBrush WRITE setSelectedBrush)
05898     Q_PROPERTY(QFont selectedFont READ selectedFont WRITE setSelectedFont)
05899     Q_PROPERTY(QColor selectedTextColor READ selectedTextColor WRITE setSelectedTextColor)
```

```
05901 public:
05907     enum SelectablePart {
05908         spNone = 0x000
05909         , spLegendBox = 0x001
05910         , spItems = 0x002
05911     };
05912     Q_ENUMS(SelectablePart)
05913     Q_FLAGS(SelectableParts)
05914     Q_DECLARE_FLAGS(SelectableParts, SelectablePart)
05915
05916     explicit QCPLegend();
05917     virtual ~QCPLegend() Q_DECL_OVERRIDE;
05918
05919     // getters:
05920
05921     QPen borderPen() const {
05922         return mBorderPen;
05923     }
05924
05925     QBrush brush() const {
05926         return mBrush;
05927     }
05928
05929     QFont font() const {
05930         return mFont;
05931     }
05932
05933     QColor textColor() const {
05934         return mTextColor;
05935     }
05936
05937     QSize iconSize() const {
05938         return mIconSize;
05939     }
05940
05941     int iconTextPadding() const {
05942         return mIconTextPadding;
05943     }
05944
05945     QPen iconBorderPen() const {
05946         return mIconBorderPen;
05947     }
05948
05949     SelectableParts selectableParts() const {
05950         return mSelectableParts;
05951     }
05952     SelectableParts selectedParts() const;
05953
05954     QPen selectedBorderPen() const {
05955         return mSelectedBorderPen;
05956     }
05957
05958     QPen selectedIconBorderPen() const {
05959         return mSelectedIconBorderPen;
05960     }
05961
05962     QBrush selectedBrush() const {
05963         return mSelectedBrush;
05964     }
05965
05966     QFont selectedFont() const {
05967         return mSelectedFont;
05968     }
05969
05970     QColor selectedTextColor() const {
05971         return mSelectedTextColor;
05972     }
05973
05974     // setters:
05975     void setBorderPen(const QPen &pen);
05976     void setBrush(const QBrush &brush);
05977     void setFont(const QFont &font);
05978     void setTextColor(const QColor &color);
05979     void setIconSize(const QSize &size);
05980     void setIconSize(int width, int height);
05981     void setIconTextPadding(int padding);
05982     void setIconBorderPen(const QPen &pen);
05983     Q_SLOT void setSelectableParts(const SelectableParts &selectableParts);
05984     Q_SLOT void setSelectedParts(const SelectableParts &selectedParts);
05985     void setSelectedBorderPen(const QPen &pen);
05986     void setSelectedIconBorderPen(const QPen &pen);
05987     void setSelectedBrush(const QBrush &brush);
05988     void setSelectedFont(const QFont &font);
05989     void setSelectedTextColor(const QColor &color);
05990
05991     // reimplemented virtual methods:
05992     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
```

```
    const Q_DECL_OVERRIDE;
05993
05994     // non-virtual methods:
05995     QCPAbstractLegendItem *item(int index) const;
05996     QCPPlottableLegendItem *itemWithPlottable(const QCPAbstractPlottable *plottable) const;
05997     int itemCount() const;
05998     bool hasItem(QCPAbstractLegendItem *item) const;
05999     bool hasItemWithPlottable(const QCPAbstractPlottable *plottable) const;
06000     bool addItem(QCPAbstractLegendItem *item);
06001     bool removeItem(int index);
06002     bool removeItem(QCPAbstractLegendItem *item);
06003     void clearItems();
06004     QList<QCPAbstractLegendItem*> selectedItems() const;
06005
06006 signals:
06007     void selectionChanged(QCPLegend::SelectableParts parts);
06008     void selectableChanged(QCPLegend::SelectableParts parts);
06009
06010 protected:
06011     // property members:
06012     QPen mBorderPen, mIconBorderPen;
06013     QBrush mBrush;
06014     QFont mFont;
06015     QColor mTextColor;
06016     QSize mIconSize;
06017     int mIconTextPadding;
06018     SelectableParts mSelectedParts, mSelectableParts;
06019     QPen mSelectedBorderPen, mSelectedIconBorderPen;
06020     QBrush mSelectedBrush;
06021     QFont mSelectedFont;
06022     QColor mSelectedTextColor;
06023
06024     // reimplemented virtual methods:
06025     virtual void parentPlotInitialized(QCustomPlot *parentPlot) Q_DECL_OVERRIDE;
06026     virtual QCP::Interaction selectionCategory() const Q_DECL_OVERRIDE;
06027     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
06028     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
06029     // events:
06030     virtual void selectEvent(QMouseEvent *event, bool additive, const QVariant &details, bool
    *selectionStateChanged) Q_DECL_OVERRIDE;
06031     virtual void deselectEvent(bool *selectionStateChanged) Q_DECL_OVERRIDE;
06032
06033     // non-virtual methods:
06034     QPen getBorderPen() const;
06035     QBrush getBrush() const;
06036
06037 private:
06038     Q_DISABLE_COPY(QCPLegend)
06039
06040     friend class QCustomPlot;
06041     friend class QCPAbstractLegendItem;
06042 };
06043
06044 Q_DECLARE_OPERATORS_FOR_FLAGS(QCPLegend::SelectableParts)
06045 Q_DECLARE_METATYPE(QCPLegend::SelectablePart)
06046
06047 /* end of 'src/layoutelements/layoutelement-legend.h' */
06048
06049
06050 /* including file 'src/layoutelements/layoutelement-textelement.h' */
06051 /* modified 2021-03-29T02:30:44, size 5359                        */
06052
06053 class QCP_LIB_DECL QCPTextElement : public QCPLayoutElement {
06054     Q_OBJECT
06056     Q_PROPERTY(QString text READ text WRITE setText)
06057     Q_PROPERTY(QFont font READ font WRITE setFont)
06058     Q_PROPERTY(QColor textColor READ textColor WRITE setTextColor)
06059     Q_PROPERTY(QFont selectedFont READ selectedFont WRITE setSelectedFont)
06060     Q_PROPERTY(QColor selectedTextColor READ selectedTextColor WRITE setSelectedTextColor)
06061     Q_PROPERTY(bool selectable READ selectable WRITE setSelectable NOTIFY selectableChanged)
06062     Q_PROPERTY(bool selected READ selected WRITE setSelected NOTIFY selectionChanged)
06064 public:
06065     explicit QCPTextElement(QCustomPlot *parentPlot);
06066     QCPTextElement(QCustomPlot *parentPlot, const QString &text);
06067     QCPTextElement(QCustomPlot *parentPlot, const QString &text, double pointSize);
06068     QCPTextElement(QCustomPlot *parentPlot, const QString &text, const QString &fontFamily, double
    pointSize);
06069     QCPTextElement(QCustomPlot *parentPlot, const QString &text, const QFont &font);
06070
06071     // getters:
06072
06073     QString text() const {
06074         return mText;
06075     }
06076
06077     int textFlags() const {
06078         return mTextFlags;
```

```
06079     }
06080
06081     QFont font() const {
06082         return mFont;
06083     }
06084
06085     QColor textColor() const {
06086         return mTextColor;
06087     }
06088
06089     QFont selectedFont() const {
06090         return mSelectedFont;
06091     }
06092
06093     QColor selectedTextColor() const {
06094         return mSelectedTextColor;
06095     }
06096
06097     bool selectable() const {
06098         return mSelectable;
06099     }
06100
06101     bool selected() const {
06102         return mSelected;
06103     }
06104
06105     // setters:
06106     void setText(const QString &text);
06107     void setTextFlags(int flags);
06108     void setFont(const QFont &font);
06109     void setTextColor(const QColor &color);
06110     void setSelectedFont(const QFont &font);
06111     void setSelectedTextColor(const QColor &color);
06112     Q_SLOT void setSelectable(bool selectable);
06113     Q_SLOT void setSelected(bool selected);
06114
06115     // reimplemented virtual methods:
06116     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
06117     virtual void mousePressEvent(QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE;
06118     virtual void mouseReleaseEvent(QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE;
06119     virtual void mouseDoubleClickEvent(QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE;
06120
06121 signals:
06122     void selectionChanged(bool selected);
06123     void selectableChanged(bool selectable);
06124     void clicked(QMouseEvent *event);
06125     void doubleClicked(QMouseEvent *event);
06126
06127 protected:
06128     // property members:
06129     QString mText;
06130     int mTextFlags;
06131     QFont mFont;
06132     QColor mTextColor;
06133     QFont mSelectedFont;
06134     QColor mSelectedTextColor;
06135     QRect mTextBoundingRect;
06136     bool mSelectable, mSelected;
06137
06138     // reimplemented virtual methods:
06139     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
06140     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
06141     virtual QSize minimumOuterSizeHint() const Q_DECL_OVERRIDE;
06142     virtual QSize maximumOuterSizeHint() const Q_DECL_OVERRIDE;
06143     // events:
06144     virtual void selectEvent(QMouseEvent *event, bool additive, const QVariant &details, bool
      *selectionStateChanged) Q_DECL_OVERRIDE;
06145     virtual void deselectEvent(bool *selectionStateChanged) Q_DECL_OVERRIDE;
06146
06147     // non-virtual methods:
06148     QFont mainFont() const;
06149     QColor mainTextColor() const;
06150
06151 private:
06152     Q_DISABLE_COPY(QCPTextElement)
06153 };
06154
06155
06156
06157 /* end of 'src/layoutelements/layoutelement-textelement.h' */
06158
06159
06160 /* including file 'src/layoutelements/layoutelement-colorscale.h' */
06161
06162 /* modified 2021-03-29T02:30:44, size 5939                        */
06163
```

```
06164
06165 class QCPColorScaleAxisRectPrivate : public QCPAxisRect {
06166     Q_OBJECT
06167 public:
06168     explicit QCPColorScaleAxisRectPrivate(QCPColorScale *parentColorScale);
06169 protected:
06170     QCPColorScale *mParentColorScale;
06171     QImage mGradientImage;
06172     bool mGradientImageInvalidated;
06173     // re-using some methods of QCPAxisRect to make them available to friend class QCPColorScale
06174     using QCPAxisRect::calculateAutoMargin;
06175     using QCPAxisRect::mousePressEvent;
06176     using QCPAxisRect::mouseMoveEvent;
06177     using QCPAxisRect::mouseReleaseEvent;
06178     using QCPAxisRect::wheelEvent;
06179     using QCPAxisRect::update;
06180     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
06181     void updateGradientImage();
06182     Q_SLOT void axisSelectionChanged(QCPAxis::SelectableParts selectedParts);
06183     Q_SLOT void axisSelectableChanged(QCPAxis::SelectableParts selectableParts);
06184     friend class QCPColorScale;
06185 };
06186
06187 class QCP_LIB_DECL QCPColorScale : public QCPLayoutElement {
06188     Q_OBJECT
06190     Q_PROPERTY(QCPAxis::AxisType type READ type WRITE setType)
06191     Q_PROPERTY(QCPRange dataRange READ dataRange WRITE setDataRange NOTIFY dataRangeChanged)
06192     Q_PROPERTY(QCPAxis::ScaleType dataScaleType READ dataScaleType WRITE setDataScaleType NOTIFY
      dataScaleTypeChanged)
06193     Q_PROPERTY(QCPColorGradient gradient READ gradient WRITE setGradient NOTIFY gradientChanged)
06194     Q_PROPERTY(QString label READ label WRITE setLabel)
06195     Q_PROPERTY(int barWidth READ barWidth WRITE setBarWidth)
06196     Q_PROPERTY(bool rangeDrag READ rangeDrag WRITE setRangeDrag)
06197     Q_PROPERTY(bool rangeZoom READ rangeZoom WRITE setRangeZoom)
06199 public:
06200     explicit QCPColorScale(QCustomPlot *parentPlot);
06201     virtual ~QCPColorScale() Q_DECL_OVERRIDE;
06202
06203     // getters:
06204
06205     QCPAxis *axis() const {
06206         return mColorAxis.data();
06207     }
06208
06209     QCPAxis::AxisType type() const {
06210         return mType;
06211     }
06212
06213     QCPRange dataRange() const {
06214         return mDataRange;
06215     }
06216
06217     QCPAxis::ScaleType dataScaleType() const {
06218         return mDataScaleType;
06219     }
06220
06221     QCPColorGradient gradient() const {
06222         return mGradient;
06223     }
06224     QString label() const;
06225
06226     int barWidth() const {
06227         return mBarWidth;
06228     }
06229     bool rangeDrag() const;
06230     bool rangeZoom() const;
06231
06232     // setters:
06233     void setType(QCPAxis::AxisType type);
06234     Q_SLOT void setDataRange(const QCPRange &dataRange);
06235     Q_SLOT void setDataScaleType(QCPAxis::ScaleType scaleType);
06236     Q_SLOT void setGradient(const QCPColorGradient &gradient);
06237     void setLabel(const QString &str);
06238     void setBarWidth(int width);
06239     void setRangeDrag(bool enabled);
06240     void setRangeZoom(bool enabled);
06241
06242     // non-property methods:
06243     QList<QCPColorMap*> colorMaps() const;
06244     void rescaleDataRange(bool onlyVisibleMaps);
06245
06246     // reimplemented virtual methods:
06247     virtual void update(UpdatePhase phase) Q_DECL_OVERRIDE;
06248
06249 signals:
06250     void dataRangeChanged(const QCPRange &newRange);
06251     void dataScaleTypeChanged(QCPAxis::ScaleType scaleType);
```

```
06252     void gradientChanged(const QCPColorGradient &newGradient);
06253
06254 protected:
06255     // property members:
06256     QCPAxis::AxisType mType;
06257     QCPRange mDataRange;
06258     QCPAxis::ScaleType mDataScaleType;
06259     QCPColorGradient mGradient;
06260     int mBarWidth;
06261
06262     // non-property members:
06263     QPointer<QCPColorScaleAxisRectPrivate> mAxisRect;
06264     QPointer<QCPAxis> mColorAxis;
06265
06266     // reimplemented virtual methods:
06267     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
06268     // events:
06269     virtual void mousePressEvent(QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE;
06270     virtual void mouseMoveEvent(QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE;
06271     virtual void mouseReleaseEvent(QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE;
06272     virtual void wheelEvent(QWheelEvent *event) Q_DECL_OVERRIDE;
06273
06274 private:
06275     Q_DISABLE_COPY(QCPColorScale)
06276
06277     friend class QCPColorScaleAxisRectPrivate;
06278 };
06279
06280
06281 /* end of 'src/layoutelements/layoutelement-colorscale.h' */
06282
06283
06284 /* including file 'src/plottables/plottable-graph.h' */
06285
06286 /* modified 2021-03-29T02:30:44, size 9316           */
06287
06288 class QCP_LIB_DECL QCPGraphData {
06289 public:
06290     QCPGraphData();
06291     QCPGraphData(double key, double value);
06292
06293     inline double sortKey() const {
06294         return key;
06295     }
06296
06297     inline static QCPGraphData fromSortKey(double sortKey) {
06298         return QCPGraphData(sortKey, 0);
06299     }
06300
06301     inline static bool sortKeyIsMainKey() {
06302         return true;
06303     }
06304
06305     inline double mainKey() const {
06306         return key;
06307     }
06308
06309     inline double mainValue() const {
06310         return value;
06311     }
06312
06313     inline QCPRange valueRange() const {
06314         return QCPRange(value, value);
06315     }
06316
06317     double key, value;
06318 };
06319 Q_DECLARE_TYPEINFO(QCPGraphData, Q_PRIMITIVE_TYPE);
06320
06321
06331 typedef QCPDataContainer<QCPGraphData> QCPGraphDataContainer;
06332
06333 class QCP_LIB_DECL QCPGraph : public QCPAbstractPlottable1D<QCPGraphData> {
06334
06335     Q_OBJECT
06337     Q_PROPERTY(LineStyle lineStyle READ lineStyle WRITE setLineStyle)
06338     Q_PROPERTY(QCPScatterStyle scatterStyle READ scatterStyle WRITE setScatterStyle)
06339     Q_PROPERTY(int scatterSkip READ scatterSkip WRITE setScatterSkip)
06340     Q_PROPERTY(QCPGraph* channelFillGraph READ channelFillGraph WRITE setChannelFillGraph)
06341     Q_PROPERTY(bool adaptiveSampling READ adaptiveSampling WRITE setAdaptiveSampling)
06343 public:
06349     enum LineStyle {
06350         lsNone
06352         , lsLine
06353         , lsStepLeft
06354         , lsStepRight
06355         , lsStepCenter
```

```
06356         , lsImpulse
06357     };
06358     Q_ENUMS(LineStyle)
06359 
06360     explicit QCPGraph(QCPAxis *keyAxis, QCPAxis *valueAxis);
06361     virtual ~QCPGraph() Q_DECL_OVERRIDE;
06362 
06363     // getters:
06364 
06365     QSharedPointer<QCPGraphDataContainer> data() const {
06366         return mDataContainer;
06367     }
06368 
06369     LineStyle lineStyle() const {
06370         return mLineStyle;
06371     }
06372 
06373     QCPScatterStyle scatterStyle() const {
06374         return mScatterStyle;
06375     }
06376 
06377     int scatterSkip() const {
06378         return mScatterSkip;
06379     }
06380 
06381     QCPGraph *channelFillGraph() const {
06382         return mChannelFillGraph.data();
06383     }
06384 
06385     bool adaptiveSampling() const {
06386         return mAdaptiveSampling;
06387     }
06388 
06389     // setters:
06390     void setData(QSharedPointer<QCPGraphDataContainer> data);
06391     void setData(const QVector<double> &keys, const QVector<double> &values, bool alreadySorted =
    false);
06392     void setLineStyle(LineStyle ls);
06393     void setScatterStyle(const QCPScatterStyle &style);
06394     void setScatterSkip(int skip);
06395     void setChannelFillGraph(QCPGraph *targetGraph);
06396     void setAdaptiveSampling(bool enabled);
06397 
06398     // non-property methods:
06399     void addData(const QVector<double> &keys, const QVector<double> &values, bool alreadySorted =
    false);
06400     void addData(double key, double value);
06401 
06402     // reimplemented virtual methods:
06403     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
    const Q_DECL_OVERRIDE;
06404     virtual QCPRange getKeyRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth) const
    Q_DECL_OVERRIDE;
06405     virtual QCPRange getValueRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth, const
    QCPRange &inKeyRange = QCPRange()) const Q_DECL_OVERRIDE;
06406 
06407 protected:
06408     // property members:
06409     LineStyle mLineStyle;
06410     QCPScatterStyle mScatterStyle;
06411     int mScatterSkip;
06412     QPointer<QCPGraph> mChannelFillGraph;
06413     bool mAdaptiveSampling;
06414 
06415     // reimplemented virtual methods:
06416     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
06417     virtual void drawLegendIcon(QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE;
06418 
06419     // introduced virtual methods:
06420     virtual void drawFill(QCPPainter *painter, QVector<QPointF> *lines) const;
06421     virtual void drawScatterPlot(QCPPainter *painter, const QVector<QPointF> &scatters, const
    QCPScatterStyle &style) const;
06422     virtual void drawLinePlot(QCPPainter *painter, const QVector<QPointF> &lines) const;
06423     virtual void drawImpulsePlot(QCPPainter *painter, const QVector<QPointF> &lines) const;
06424 
06425     virtual void getOptimizedLineData(QVector<QCPGraphData> *lineData, const
    QCPGraphDataContainer::const_iterator &begin, const QCPGraphDataContainer::const_iterator &end) const;
06426     virtual void getOptimizedScatterData(QVector<QCPGraphData> *scatterData,
    QCPGraphDataContainer::const_iterator begin, QCPGraphDataContainer::const_iterator end) const;
06427 
06428     // non-virtual methods:
06429     void getVisibleDataBounds(QCPGraphDataContainer::const_iterator &begin,
    QCPGraphDataContainer::const_iterator &end, const QCPDataRange &rangeRestriction) const;
06430     void getLines(QVector<QPointF> *lines, const QCPDataRange &dataRange) const;
06431     void getScatters(QVector<QPointF> *scatters, const QCPDataRange &dataRange) const;
06432     QVector<QPointF> dataToLines(const QVector<QCPGraphData> &data) const;
06433     QVector<QPointF> dataToStepLeftLines(const QVector<QCPGraphData> &data) const;
```

```
06434     QVector<QPointF> dataToStepRightLines(const QVector<QCPGraphData> &data) const;
06435     QVector<QPointF> dataToStepCenterLines(const QVector<QCPGraphData> &data) const;
06436     QVector<QPointF> dataToImpulseLines(const QVector<QCPGraphData> &data) const;
06437     QVector<QCPDataRange> getNonNanSegments(const QVector<QPointF> *lineData, Qt::Orientation
      keyOrientation) const;
06438     QVector<QPair<QCPDataRange, QCPDataRange> > getOverlappingSegments(QVector<QCPDataRange>
      thisSegments, const QVector<QPointF> *thisData, QVector<QCPDataRange> otherSegments, const
      QVector<QPointF> *otherData) const;
06439     bool segmentsIntersect(double aLower, double aUpper, double bLower, double bUpper, int
      &bPrecedence) const;
06440     QPointF getFillBasePoint(QPointF matchingDataPoint) const;
06441     const QPolygonF getFillPolygon(const QVector<QPointF> *lineData, QCPDataRange segment) const;
06442     const QPolygonF getChannelFillPolygon(const QVector<QPointF> *thisData, QCPDataRange thisSegment,
      const QVector<QPointF> *otherData, QCPDataRange otherSegment) const;
06443     int findIndexBelowX(const QVector<QPointF> *data, double x) const;
06444     int findIndexAboveX(const QVector<QPointF> *data, double x) const;
06445     int findIndexBelowY(const QVector<QPointF> *data, double y) const;
06446     int findIndexAboveY(const QVector<QPointF> *data, double y) const;
06447     double pointDistance(const QPointF &pixelPoint, QCPGraphDataContainer::const_iterator
      &closestData) const;
06448
06449     friend class QCustomPlot;
06450     friend class QCPLegend;
06451 };
06452
06453 Q_DECLARE_METATYPE(QCPGraph::LineStyle)
06454
06455 /* end of 'src/plottables/plottable-graph.h' */
06456
06457
06458 /* including file 'src/plottables/plottable-curve.h' */
06459 /* modified 2021-03-29T02:30:44, size 7434           */
06460
06461 class QCP_LIB_DECL QCPCurveData {
06462 public:
06463     QCPCurveData();
06464     QCPCurveData(double t, double key, double value);
06465
06466     inline double sortKey() const {
06467         return t;
06468     }
06469
06470     inline static QCPCurveData fromSortKey(double sortKey) {
06471         return QCPCurveData(sortKey, 0, 0);
06472     }
06473
06474     inline static bool sortKeyIsMainKey() {
06475         return false;
06476     }
06477
06478     inline double mainKey() const {
06479         return key;
06480     }
06481
06482     inline double mainValue() const {
06483         return value;
06484     }
06485
06486     inline QCPRange valueRange() const {
06487         return QCPRange(value, value);
06488     }
06489
06490     double t, key, value;
06491 };
06492 Q_DECLARE_TYPEINFO(QCPCurveData, Q_PRIMITIVE_TYPE);
06493
06494
06505 typedef QCPDataContainer<QCPCurveData> QCPCurveDataContainer;
06506
06507 class QCP_LIB_DECL QCPCurve : public QCPAbstractPlottable1D<QCPCurveData> {
06508
06509     Q_OBJECT
06511     Q_PROPERTY(QCPScatterStyle scatterStyle READ scatterStyle WRITE setScatterStyle)
06512     Q_PROPERTY(int scatterSkip READ scatterSkip WRITE setScatterSkip)
06513     Q_PROPERTY(LineStyle lineStyle READ lineStyle WRITE setLineStyle)
06515 public:
06521     enum LineStyle {
06522         lsNone
06523         , lsLine
06524     };
06525     Q_ENUMS(LineStyle)
06526
06527     explicit QCPCurve(QCPAxis *keyAxis, QCPAxis *valueAxis);
06528     virtual ~QCPCurve() Q_DECL_OVERRIDE;
06529
06530     // getters:
06531
```

```
06532     QSharedPointer<QCPCurveDataContainer> data() const {
06533         return mDataContainer;
06534     }
06535
06536     QCPScatterStyle scatterStyle() const {
06537         return mScatterStyle;
06538     }
06539
06540     int scatterSkip() const {
06541         return mScatterSkip;
06542     }
06543
06544     LineStyle lineStyle() const {
06545         return mLineStyle;
06546     }
06547
06548     // setters:
06549     void setData(QSharedPointer<QCPCurveDataContainer> data);
06550     void setData(const QVector<double> &t, const QVector<double> &keys, const QVector<double> &values,
      bool alreadySorted = false);
06551     void setData(const QVector<double> &keys, const QVector<double> &values);
06552     void setScatterStyle(const QCPScatterStyle &style);
06553     void setScatterSkip(int skip);
06554     void setLineStyle(LineStyle style);
06555
06556     // non-property methods:
06557     void addData(const QVector<double> &t, const QVector<double> &keys, const QVector<double> &values,
      bool alreadySorted = false);
06558     void addData(const QVector<double> &keys, const QVector<double> &values);
06559     void addData(double t, double key, double value);
06560     void addData(double key, double value);
06561
06562     // reimplemented virtual methods:
06563     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
06564     virtual QCPRange getKeyRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth) const
      Q_DECL_OVERRIDE;
06565     virtual QCPRange getValueRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth, const
      QCPRange &inKeyRange = QCPRange()) const Q_DECL_OVERRIDE;
06566
06567 protected:
06568     // property members:
06569     QCPScatterStyle mScatterStyle;
06570     int mScatterSkip;
06571     LineStyle mLineStyle;
06572
06573     // reimplemented virtual methods:
06574     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
06575     virtual void drawLegendIcon(QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE;
06576
06577     // introduced virtual methods:
06578     virtual void drawCurveLine(QCPPainter *painter, const QVector<QPointF> &lines) const;
06579     virtual void drawScatterPlot(QCPPainter *painter, const QVector<QPointF> &points, const
      QCPScatterStyle &style) const;
06580
06581     // non-virtual methods:
06582     void getCurveLines(QVector<QPointF> *lines, const QCPDataRange &dataRange, double penWidth) const;
06583     void getScatters(QVector<QPointF> *scatters, const QCPDataRange &dataRange, double scatterWidth)
      const;
06584     int getRegion(double key, double value, double keyMin, double valueMax, double keyMax, double
      valueMin) const;
06585     QPointF getOptimizedPoint(int otherRegion, double otherKey, double otherValue, double key, double
      value, double keyMin, double valueMax, double keyMax, double valueMin) const;
06586     QVector<QPointF> getOptimizedCornerPoints(int prevRegion, int currentRegion, double prevKey,
      double prevValue, double key, double value, double keyMin, double valueMax, double keyMax, double
      valueMin) const;
06587     bool mayTraverse(int prevRegion, int currentRegion) const;
06588     bool getTraverse(double prevKey, double prevValue, double key, double value, double keyMin, double
      valueMax, double keyMax, double valueMin, QPointF &crossA, QPointF &crossB) const;
06589     void getTraverseCornerPoints(int prevRegion, int currentRegion, double keyMin, double valueMax,
      double keyMax, double valueMin, QVector<QPointF> &beforeTraverse, QVector<QPointF> &afterTraverse)
      const;
06590     double pointDistance(const QPointF &pixelPoint, QCPCurveDataContainer::const_iterator
      &closestData) const;
06591
06592     friend class QCustomPlot;
06593     friend class QCPLegend;
06594 };
06595
06596 Q_DECLARE_METATYPE(QCPCurve::LineStyle)
06597
06598 /* end of 'src/plottables/plottable-curve.h' */
06599
06600
06601 /* including file 'src/plottables/plottable-bars.h' */
06602 /* modified 2021-03-29T02:30:44, size 8955          */
06603
```

```
06604 class QCP_LIB_DECL QCPBarsGroup : public QObject {
06605
06606     Q_OBJECT
06608     Q_PROPERTY(SpacingType spacingType READ spacingType WRITE setSpacingType)
06609     Q_PROPERTY(double spacing READ spacing WRITE setSpacing)
06611 public:
06618     enum SpacingType {
06619         stAbsolute
06620         , stAxisRectRatio
06621         , stPlotCoords
06622     };
06623     Q_ENUMS(SpacingType)
06624
06625     explicit QCPBarsGroup(QCustomPlot *parentPlot);
06626     virtual ~QCPBarsGroup();
06627
06628     // getters:
06629
06630     SpacingType spacingType() const {
06631         return mSpacingType;
06632     }
06633
06634     double spacing() const {
06635         return mSpacing;
06636     }
06637
06638     // setters:
06639     void setSpacingType(SpacingType spacingType);
06640     void setSpacing(double spacing);
06641
06642     // non-virtual methods:
06643
06644     QList<QCPBars*> bars() const {
06645         return mBars;
06646     }
06647     QCPBars* bars(int index) const;
06648
06649     int size() const {
06650         return mBars.size();
06651     }
06652
06653     bool isEmpty() const {
06654         return mBars.isEmpty();
06655     }
06656     void clear();
06657
06658     bool contains(QCPBars *bars) const {
06659         return mBars.contains(bars);
06660     }
06661     void append(QCPBars *bars);
06662     void insert(int i, QCPBars *bars);
06663     void remove(QCPBars *bars);
06664
06665 protected:
06666     // non-property members:
06667     QCustomPlot *mParentPlot;
06668     SpacingType mSpacingType;
06669     double mSpacing;
06670     QList<QCPBars*> mBars;
06671
06672     // non-virtual methods:
06673     void registerBars(QCPBars *bars);
06674     void unregisterBars(QCPBars *bars);
06675
06676     // virtual methods:
06677     double keyPixelOffset(const QCPBars *bars, double keyCoord);
06678     double getPixelSpacing(const QCPBars *bars, double keyCoord);
06679
06680 private:
06681     Q_DISABLE_COPY(QCPBarsGroup)
06682
06683     friend class QCPBars;
06684 };
06685
06686 Q_DECLARE_METATYPE(QCPBarsGroup::SpacingType)
06687
06688
06689 class QCP_LIB_DECL QCPBarsData {
06690 public:
06691     QCPBarsData();
06692     QCPBarsData(double key, double value);
06693
06694     inline double sortKey() const {
06695         return key;
06696     }
06697
06698     inline static QCPBarsData fromSortKey(double sortKey) {
```

```
06699         return QCPBarsData(sortKey, 0);
06700     }
06701
06702     inline static bool sortKeyIsMainKey() {
06703         return true;
06704     }
06705
06706     inline double mainKey() const {
06707         return key;
06708     }
06709
06710     inline double mainValue() const {
06711         return value;
06712     }
06713
06714     inline QCPRange valueRange() const {
06715         return QCPRange(value, value);
06716     } // note that bar base value isn't held in each QCPBarsData and thus can't/shouldn't be returned
      here
06717
06718     double key, value;
06719 };
06720 Q_DECLARE_TYPEINFO(QCPBarsData, Q_PRIMITIVE_TYPE);
06721
06722
06732 typedef QCPDataContainer<QCPBarsData> QCPBarsDataContainer;
06733
06734 class QCP_LIB_DECL QCPBars : public QCPAbstractPlottable1D<QCPBarsData> {
06735
06736     Q_OBJECT
06738     Q_PROPERTY(double width READ width WRITE setWidth)
06739     Q_PROPERTY(WidthType widthType READ widthType WRITE setWidthType)
06740     Q_PROPERTY(QCPBarsGroup* barsGroup READ barsGroup WRITE setBarsGroup)
06741     Q_PROPERTY(double baseValue READ baseValue WRITE setBaseValue)
06742     Q_PROPERTY(double stackingGap READ stackingGap WRITE setStackingGap)
06743     Q_PROPERTY(QCPBars* barBelow READ barBelow)
06744     Q_PROPERTY(QCPBars* barAbove READ barAbove)
06746 public:
06753     enum WidthType {
06754         wtAbsolute
06755         , wtAxisRectRatio
06756         , wtPlotCoords
06757     };
06758     Q_ENUMS(WidthType)
06759
06760     explicit QCPBars(QCPAxis *keyAxis, QCPAxis *valueAxis);
06761     virtual ~QCPBars() Q_DECL_OVERRIDE;
06762
06763     // getters:
06764
06765     double width() const {
06766         return mWidth;
06767     }
06768
06769     WidthType widthType() const {
06770         return mWidthType;
06771     }
06772
06773     QCPBarsGroup *barsGroup() const {
06774         return mBarsGroup;
06775     }
06776
06777     double baseValue() const {
06778         return mBaseValue;
06779     }
06780
06781     double stackingGap() const {
06782         return mStackingGap;
06783     }
06784
06785     QCPBars *barBelow() const {
06786         return mBarBelow.data();
06787     }
06788
06789     QCPBars *barAbove() const {
06790         return mBarAbove.data();
06791     }
06792
06793     QSharedPointer<QCPBarsDataContainer> data() const {
06794         return mDataContainer;
06795     }
06796
06797     // setters:
06798     void setData(QSharedPointer<QCPBarsDataContainer> data);
06799     void setData(const QVector<double> &keys, const QVector<double> &values, bool alreadySorted =
      false);
06800     void setWidth(double width);
```

```
06801     void setWidthType(WidthType widthType);
06802     void setBarsGroup(QCPBarsGroup *barsGroup);
06803     void setBaseValue(double baseValue);
06804     void setStackingGap(double pixels);
06805 
06806     // non-property methods:
06807     void addData(const QVector<double> &keys, const QVector<double> &values, bool alreadySorted =
      false);
06808     void addData(double key, double value);
06809     void moveBelow(QCPBars *bars);
06810     void moveAbove(QCPBars *bars);
06811 
06812     // reimplemented virtual methods:
06813     virtual QCPDataSelection selectTestRect(const QRectF &rect, bool onlySelectable) const
      Q_DECL_OVERRIDE;
06814     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
06815     virtual QCPRange getKeyRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth) const
      Q_DECL_OVERRIDE;
06816     virtual QCPRange getValueRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth, const
      QCPRange &inKeyRange = QCPRange()) const Q_DECL_OVERRIDE;
06817     virtual QPointF dataPixelPosition(int index) const Q_DECL_OVERRIDE;
06818 
06819 protected:
06820     // property members:
06821     double mWidth;
06822     WidthType mWidthType;
06823     QCPBarsGroup *mBarsGroup;
06824     double mBaseValue;
06825     double mStackingGap;
06826     QPointer<QCPBars> mBarBelow, mBarAbove;
06827 
06828     // reimplemented virtual methods:
06829     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
06830     virtual void drawLegendIcon(QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE;
06831 
06832     // non-virtual methods:
06833     void getVisibleDataBounds(QCPBarsDataContainer::const_iterator &begin,
      QCPBarsDataContainer::const_iterator &end) const;
06834     QRectF getBarRect(double key, double value) const;
06835     void getPixelWidth(double key, double &lower, double &upper) const;
06836     double getStackedBaseValue(double key, bool positive) const;
06837     static void connectBars(QCPBars* lower, QCPBars* upper);
06838 
06839     friend class QCustomPlot;
06840     friend class QCPLegend;
06841     friend class QCPBarsGroup;
06842 };
06843 
06844 Q_DECLARE_METATYPE(QCPBars::WidthType)
06845 
06846 /* end of 'src/plottables/plottable-bars.h' */
06847 
06848 
06849 /* including file 'src/plottables/plottable-statisticalbox.h' */
06850 /* modified 2021-03-29T02:30:44, size 7522                    */
06851 
06852 class QCP_LIB_DECL QCPStatisticalBoxData {
06853 public:
06854     QCPStatisticalBoxData();
06855     QCPStatisticalBoxData(double key, double minimum, double lowerQuartile, double median, double
      upperQuartile, double maximum, const QVector<double>& outliers = QVector<double>());
06856 
06857     inline double sortKey() const {
06858         return key;
06859     }
06860 
06861     inline static QCPStatisticalBoxData fromSortKey(double sortKey) {
06862         return QCPStatisticalBoxData(sortKey, 0, 0, 0, 0, 0);
06863     }
06864 
06865     inline static bool sortKeyIsMainKey() {
06866         return true;
06867     }
06868 
06869     inline double mainKey() const {
06870         return key;
06871     }
06872 
06873     inline double mainValue() const {
06874         return median;
06875     }
06876 
06877     inline QCPRange valueRange() const {
06878         QCPRange result(minimum, maximum);
06879         for (QVector<double>::const_iterator it = outliers.constBegin(); it != outliers.constEnd();
      ++it)
```

```
06880             result.expand(*it);
06881         return result;
06882     }
06883
06884     double key, minimum, lowerQuartile, median, upperQuartile, maximum;
06885     QVector<double> outliers;
06886 };
06887 Q_DECLARE_TYPEINFO(QCPStatisticalBoxData, Q_MOVABLE_TYPE);
06888
06889
06900 typedef QCPDataContainer<QCPStatisticalBoxData> QCPStatisticalBoxDataContainer;
06901
06902 class QCP_LIB_DECL QCPStatisticalBox : public QCPAbstractPlottable1D<QCPStatisticalBoxData> {
06903     Q_OBJECT
06905     Q_PROPERTY(double width READ width WRITE setWidth)
06906     Q_PROPERTY(double whiskerWidth READ whiskerWidth WRITE setWhiskerWidth)
06907     Q_PROPERTY(QPen whiskerPen READ whiskerPen WRITE setWhiskerPen)
06908     Q_PROPERTY(QPen whiskerBarPen READ whiskerBarPen WRITE setWhiskerBarPen)
06909     Q_PROPERTY(bool whiskerAntialiased READ whiskerAntialiased WRITE setWhiskerAntialiased)
06910     Q_PROPERTY(QPen medianPen READ medianPen WRITE setMedianPen)
06911     Q_PROPERTY(QCPScatterStyle outlierStyle READ outlierStyle WRITE setOutlierStyle)
06913 public:
06914     explicit QCPStatisticalBox(QCPAxis *keyAxis, QCPAxis *valueAxis);
06915
06916     // getters:
06917
06918     QSharedPointer<QCPStatisticalBoxDataContainer> data() const {
06919         return mDataContainer;
06920     }
06921
06922     double width() const {
06923         return mWidth;
06924     }
06925
06926     double whiskerWidth() const {
06927         return mWhiskerWidth;
06928     }
06929
06930     QPen whiskerPen() const {
06931         return mWhiskerPen;
06932     }
06933
06934     QPen whiskerBarPen() const {
06935         return mWhiskerBarPen;
06936     }
06937
06938     bool whiskerAntialiased() const {
06939         return mWhiskerAntialiased;
06940     }
06941
06942     QPen medianPen() const {
06943         return mMedianPen;
06944     }
06945
06946     QCPScatterStyle outlierStyle() const {
06947         return mOutlierStyle;
06948     }
06949
06950     // setters:
06951     void setData(QSharedPointer<QCPStatisticalBoxDataContainer> data);
06952     void setData(const QVector<double> &keys, const QVector<double> &minimum, const QVector<double>
    &lowerQuartile, const QVector<double> &median, const QVector<double> &upperQuartile, const
    QVector<double> &maximum, bool alreadySorted = false);
06953     void setWidth(double width);
06954     void setWhiskerWidth(double width);
06955     void setWhiskerPen(const QPen &pen);
06956     void setWhiskerBarPen(const QPen &pen);
06957     void setWhiskerAntialiased(bool enabled);
06958     void setMedianPen(const QPen &pen);
06959     void setOutlierStyle(const QCPScatterStyle &style);
06960
06961     // non-property methods:
06962     void addData(const QVector<double> &keys, const QVector<double> &minimum, const QVector<double>
    &lowerQuartile, const QVector<double> &median, const QVector<double> &upperQuartile, const
    QVector<double> &maximum, bool alreadySorted = false);
06963     void addData(double key, double minimum, double lowerQuartile, double median, double
    upperQuartile, double maximum, const QVector<double> &outliers = QVector<double>());
06964
06965     // reimplemented virtual methods:
06966     virtual QCPDataSelection selectTestRect(const QRectF &rect, bool onlySelectable) const
    Q_DECL_OVERRIDE;
06967     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
    const Q_DECL_OVERRIDE;
06968     virtual QCPRange getKeyRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth) const
    Q_DECL_OVERRIDE;
06969     virtual QCPRange getValueRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth, const
    QCPRange &inKeyRange = QCPRange()) const Q_DECL_OVERRIDE;
```

```
06970
06971 protected:
06972     // property members:
06973     double mWidth;
06974     double mWhiskerWidth;
06975     QPen mWhiskerPen, mWhiskerBarPen;
06976     bool mWhiskerAntialiased;
06977     QPen mMedianPen;
06978     QCPScatterStyle mOutlierStyle;
06979
06980     // reimplemented virtual methods:
06981     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
06982     virtual void drawLegendIcon(QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE;
06983
06984     // introduced virtual methods:
06985     virtual void drawStatisticalBox(QCPPainter *painter,
      QCPStatisticalBoxDataContainer::const_iterator it, const QCPScatterStyle &outlierStyle) const;
06986
06987     // non-virtual methods:
06988     void getVisibleDataBounds(QCPStatisticalBoxDataContainer::const_iterator &begin,
      QCPStatisticalBoxDataContainer::const_iterator &end) const;
06989     QRectF getQuartileBox(QCPStatisticalBoxDataContainer::const_iterator it) const;
06990     QVector<QLineF> getWhiskerBackboneLines(QCPStatisticalBoxDataContainer::const_iterator it) const;
06991     QVector<QLineF> getWhiskerBarLines(QCPStatisticalBoxDataContainer::const_iterator it) const;
06992
06993     friend class QCustomPlot;
06994     friend class QCPLegend;
06995 };
06996
06997 /* end of 'src/plottables/plottable-statisticalbox.h' */
06998
06999
07000 /* including file 'src/plottables/plottable-colormap.h' */
07001
07002 /* modified 2021-03-29T02:30:44, size 7092              */
07003
07004 class QCP_LIB_DECL QCPColorMapData {
07005 public:
07006     QCPColorMapData(int keySize, int valueSize, const QCPRange &keyRange, const QCPRange &valueRange);
07007     ~QCPColorMapData();
07008     QCPColorMapData(const QCPColorMapData &other);
07009     QCPColorMapData &operator=(const QCPColorMapData &other);
07010
07011     // getters:
07012
07013     int keySize() const {
07014         return mKeySize;
07015     }
07016
07017     int valueSize() const {
07018         return mValueSize;
07019     }
07020
07021     QCPRange keyRange() const {
07022         return mKeyRange;
07023     }
07024
07025     QCPRange valueRange() const {
07026         return mValueRange;
07027     }
07028
07029     QCPRange dataBounds() const {
07030         return mDataBounds;
07031     }
07032     double data(double key, double value);
07033     double cell(int keyIndex, int valueIndex);
07034     unsigned char alpha(int keyIndex, int valueIndex);
07035
07036     // setters:
07037     void setSize(int keySize, int valueSize);
07038     void setKeySize(int keySize);
07039     void setValueSize(int valueSize);
07040     void setRange(const QCPRange &keyRange, const QCPRange &valueRange);
07041     void setKeyRange(const QCPRange &keyRange);
07042     void setValueRange(const QCPRange &valueRange);
07043     void setData(double key, double value, double z);
07044     void setCell(int keyIndex, int valueIndex, double z);
07045     void setAlpha(int keyIndex, int valueIndex, unsigned char alpha);
07046
07047     // non-property methods:
07048     void recalculateDataBounds();
07049     void clear();
07050     void clearAlpha();
07051     void fill(double z);
07052     void fillAlpha(unsigned char alpha);
07053
07054     bool isEmpty() const {
```

```
07055         return mIsEmpty;
07056     }
07057     void coordToCell(double key, double value, int *keyIndex, int *valueIndex) const;
07058     void cellToCoord(int keyIndex, int valueIndex, double *key, double *value) const;
07059
07060 protected:
07061     // property members:
07062     int mKeySize, mValueSize;
07063     QCPRange mKeyRange, mValueRange;
07064     bool mIsEmpty;
07065
07066     // non-property members:
07067     double *mData;
07068     unsigned char *mAlpha;
07069     QCPRange mDataBounds;
07070     bool mDataModified;
07071
07072     bool createAlpha(bool initializeOpaque = true);
07073
07074     friend class QCPColorMap;
07075 };
07076
07077 class QCP_LIB_DECL QCPColorMap : public QCPAbstractPlottable {
07078     Q_OBJECT
07080     Q_PROPERTY(QCPRange dataRange READ dataRange WRITE setDataRange NOTIFY dataRangeChanged)
07081     Q_PROPERTY(QCPAxis::ScaleType dataScaleType READ dataScaleType WRITE setDataScaleType NOTIFY
      dataScaleTypeChanged)
07082     Q_PROPERTY(QCPColorGradient gradient READ gradient WRITE setGradient NOTIFY gradientChanged)
07083     Q_PROPERTY(bool interpolate READ interpolate WRITE setInterpolate)
07084     Q_PROPERTY(bool tightBoundary READ tightBoundary WRITE setTightBoundary)
07085     Q_PROPERTY(QCPColorScale* colorScale READ colorScale WRITE setColorScale)
07087 public:
07088     explicit QCPColorMap(QCPAxis *keyAxis, QCPAxis *valueAxis);
07089     virtual ~QCPColorMap() Q_DECL_OVERRIDE;
07090
07091     // getters:
07092
07093     QCPColorMapData *data() const {
07094         return mMapData;
07095     }
07096
07097     QCPRange dataRange() const {
07098         return mDataRange;
07099     }
07100
07101     QCPAxis::ScaleType dataScaleType() const {
07102         return mDataScaleType;
07103     }
07104
07105     bool interpolate() const {
07106         return mInterpolate;
07107     }
07108
07109     bool tightBoundary() const {
07110         return mTightBoundary;
07111     }
07112
07113     QCPColorGradient gradient() const {
07114         return mGradient;
07115     }
07116
07117     QCPColorScale *colorScale() const {
07118         return mColorScale.data();
07119     }
07120
07121     // setters:
07122     void setData(QCPColorMapData *data, bool copy = false);
07123     Q_SLOT void setDataRange(const QCPRange &dataRange);
07124     Q_SLOT void setDataScaleType(QCPAxis::ScaleType scaleType);
07125     Q_SLOT void setGradient(const QCPColorGradient &gradient);
07126     void setInterpolate(bool enabled);
07127     void setTightBoundary(bool enabled);
07128     void setColorScale(QCPColorScale *colorScale);
07129
07130     // non-property methods:
07131     void rescaleDataRange(bool recalculateDataBounds = false);
07132     Q_SLOT void updateLegendIcon(Qt::TransformationMode transformMode = Qt::SmoothTransformation,
      const QSize &thumbSize = QSize(32, 18));
07133
07134     // reimplemented virtual methods:
07135     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
07136     virtual QCPRange getKeyRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth) const
      Q_DECL_OVERRIDE;
07137     virtual QCPRange getValueRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth, const
      QCPRange &inKeyRange = QCPRange()) const Q_DECL_OVERRIDE;
07138
```

```
07139 signals:
07140     void dataRangeChanged(const QCPRange &newRange);
07141     void dataScaleTypeChanged(QCPAxis::ScaleType scaleType);
07142     void gradientChanged(const QCPColorGradient &newGradient);
07143 
07144 protected:
07145     // property members:
07146     QCPRange mDataRange;
07147     QCPAxis::ScaleType mDataScaleType;
07148     QCPColorMapData *mMapData;
07149     QCPColorGradient mGradient;
07150     bool mInterpolate;
07151     bool mTightBoundary;
07152     QPointer<QCPColorScale> mColorScale;
07153 
07154     // non-property members:
07155     QImage mMapImage, mUndersampledMapImage;
07156     QPixmap mLegendIcon;
07157     bool mMapImageInvalidated;
07158 
07159     // introduced virtual methods:
07160     virtual void updateMapImage();
07161 
07162     // reimplemented virtual methods:
07163     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
07164     virtual void drawLegendIcon(QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE;
07165 
07166     friend class QCustomPlot;
07167     friend class QCPLegend;
07168 };
07169 
07170 /* end of 'src/plottables/plottable-colormap.h' */
07171 
07172 
07173 /* including file 'src/plottables/plottable-financial.h' */
07174 
07175 /* modified 2021-03-29T02:30:44, size 8644               */
07176 
07177 class QCP_LIB_DECL QCPFinancialData {
07178 public:
07179     QCPFinancialData();
07180     QCPFinancialData(double key, double open, double high, double low, double close);
07181 
07182     inline double sortKey() const {
07183         return key;
07184     }
07185 
07186     inline static QCPFinancialData fromSortKey(double sortKey) {
07187         return QCPFinancialData(sortKey, 0, 0, 0, 0);
07188     }
07189 
07190     inline static bool sortKeyIsMainKey() {
07191         return true;
07192     }
07193 
07194     inline double mainKey() const {
07195         return key;
07196     }
07197 
07198     inline double mainValue() const {
07199         return open;
07200     }
07201 
07202     inline QCPRange valueRange() const {
07203         return QCPRange(low, high);
07204     } // open and close must lie between low and high, so we don't need to check them
07205 
07206     double key, open, high, low, close;
07207 };
07208 Q_DECLARE_TYPEINFO(QCPFinancialData, Q_PRIMITIVE_TYPE);
07209 
07210 
07220 typedef QCPDataContainer<QCPFinancialData> QCPFinancialDataContainer;
07221 
07222 class QCP_LIB_DECL QCPFinancial : public QCPAbstractPlottable1D<QCPFinancialData> {
07223 
07224     Q_OBJECT
07226     Q_PROPERTY(ChartStyle chartStyle READ chartStyle WRITE setChartStyle)
07227     Q_PROPERTY(double width READ width WRITE setWidth)
07228     Q_PROPERTY(WidthType widthType READ widthType WRITE setWidthType)
07229     Q_PROPERTY(bool twoColored READ twoColored WRITE setTwoColored)
07230     Q_PROPERTY(QBrush brushPositive READ brushPositive WRITE setBrushPositive)
07231     Q_PROPERTY(QBrush brushNegative READ brushNegative WRITE setBrushNegative)
07232     Q_PROPERTY(QPen penPositive READ penPositive WRITE setPenPositive)
07233     Q_PROPERTY(QPen penNegative READ penNegative WRITE setPenNegative)
07235 public:
07242     enum WidthType {
```

```
07243         wtAbsolute
07244         , wtAxisRectRatio
07245         , wtPlotCoords
07246     };
07247 
07248     Q_ENUMS(WidthType)
07249 
07250 
07255     enum ChartStyle {
07256         csOhlc
07257         , csCandlestick
07258     };
07259     Q_ENUMS(ChartStyle)
07260 
07261     explicit QCPFinancial(QCPAxis *keyAxis, QCPAxis *valueAxis);
07262     virtual ~QCPFinancial() Q_DECL_OVERRIDE;
07263 
07264     // getters:
07265 
07266     QSharedPointer<QCPFinancialDataContainer> data() const {
07267         return mDataContainer;
07268     }
07269 
07270     ChartStyle chartStyle() const {
07271         return mChartStyle;
07272     }
07273 
07274     double width() const {
07275         return mWidth;
07276     }
07277 
07278     WidthType widthType() const {
07279         return mWidthType;
07280     }
07281 
07282     bool twoColored() const {
07283         return mTwoColored;
07284     }
07285 
07286     QBrush brushPositive() const {
07287         return mBrushPositive;
07288     }
07289 
07290     QBrush brushNegative() const {
07291         return mBrushNegative;
07292     }
07293 
07294     QPen penPositive() const {
07295         return mPenPositive;
07296     }
07297 
07298     QPen penNegative() const {
07299         return mPenNegative;
07300     }
07301 
07302     // setters:
07303     void setData(QSharedPointer<QCPFinancialDataContainer> data);
07304     void setData(const QVector<double> &keys, const QVector<double> &open, const QVector<double>
    &high, const QVector<double> &low, const QVector<double> &close, bool alreadySorted = false);
07305     void setChartStyle(ChartStyle style);
07306     void setWidth(double width);
07307     void setWidthType(WidthType widthType);
07308     void setTwoColored(bool twoColored);
07309     void setBrushPositive(const QBrush &brush);
07310     void setBrushNegative(const QBrush &brush);
07311     void setPenPositive(const QPen &pen);
07312     void setPenNegative(const QPen &pen);
07313 
07314     // non-property methods:
07315     void addData(const QVector<double> &keys, const QVector<double> &open, const QVector<double>
    &high, const QVector<double> &low, const QVector<double> &close, bool alreadySorted = false);
07316     void addData(double key, double open, double high, double low, double close);
07317 
07318     // reimplemented virtual methods:
07319     virtual QCPDataSelection selectTestRect(const QRectF &rect, bool onlySelectable) const
    Q_DECL_OVERRIDE;
07320     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
    const Q_DECL_OVERRIDE;
07321     virtual QCPRange getKeyRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth) const
    Q_DECL_OVERRIDE;
07322     virtual QCPRange getValueRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth, const
    QCPRange &inKeyRange = QCPRange()) const Q_DECL_OVERRIDE;
07323 
07324     // static methods:
07325     static QCPFinancialDataContainer timeSeriesToOhlc(const QVector<double> &time, const
    QVector<double> &value, double timeBinSize, double timeBinOffset = 0);
07326 
```

```
07327 protected:
07328     // property members:
07329     ChartStyle mChartStyle;
07330     double mWidth;
07331     WidthType mWidthType;
07332     bool mTwoColored;
07333     QBrush mBrushPositive, mBrushNegative;
07334     QPen mPenPositive, mPenNegative;
07335 
07336     // reimplemented virtual methods:
07337     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
07338     virtual void drawLegendIcon(QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE;
07339 
07340     // non-virtual methods:
07341     void drawOhlcPlot(QCPPainter *painter, const QCPFinancialDataContainer::const_iterator &begin,
      const QCPFinancialDataContainer::const_iterator &end, bool isSelected);
07342     void drawCandlestickPlot(QCPPainter *painter, const QCPFinancialDataContainer::const_iterator
      &begin, const QCPFinancialDataContainer::const_iterator &end, bool isSelected);
07343     double getPixelWidth(double key, double keyPixel) const;
07344     double ohlcSelectTest(const QPointF &pos, const QCPFinancialDataContainer::const_iterator &begin,
      const QCPFinancialDataContainer::const_iterator &end, QCPFinancialDataContainer::const_iterator
      &closestDataPoint) const;
07345     double candlestickSelectTest(const QPointF &pos, const QCPFinancialDataContainer::const_iterator
      &begin, const QCPFinancialDataContainer::const_iterator &end,
      QCPFinancialDataContainer::const_iterator &closestDataPoint) const;
07346     void getVisibleDataBounds(QCPFinancialDataContainer::const_iterator &begin,
      QCPFinancialDataContainer::const_iterator &end) const;
07347     QRectF selectionHitBox(QCPFinancialDataContainer::const_iterator it) const;
07348 
07349     friend class QCustomPlot;
07350     friend class QCPLegend;
07351 };
07352 
07353 Q_DECLARE_METATYPE(QCPFinancial::ChartStyle)
07354 
07355 /* end of 'src/plottables/plottable-financial.h' */
07356 
07357 
07358 /* including file 'src/plottables/plottable-errorbar.h' */
07359 /* modified 2021-03-29T02:30:44, size 7749               */
07360 
07361 class QCP_LIB_DECL QCPErrorBarsData {
07362 public:
07363     QCPErrorBarsData();
07364     explicit QCPErrorBarsData(double error);
07365     QCPErrorBarsData(double errorMinus, double errorPlus);
07366 
07367     double errorMinus, errorPlus;
07368 };
07369 Q_DECLARE_TYPEINFO(QCPErrorBarsData, Q_PRIMITIVE_TYPE);
07370 
07371 
07387 typedef QVector<QCPErrorBarsData> QCPErrorBarsDataContainer;
07388 
07389 class QCP_LIB_DECL QCPErrorBars : public QCPAbstractPlottable, public QCPPlottableInterface1D {
07390 
07391     Q_OBJECT
07393     Q_PROPERTY(QSharedPointer<QCPErrorBarsDataContainer> data READ data WRITE setData)
07394     Q_PROPERTY(QCPAbstractPlottable* dataPlottable READ dataPlottable WRITE setDataPlottable)
07395     Q_PROPERTY(ErrorType errorType READ errorType WRITE setErrorType)
07396     Q_PROPERTY(double whiskerWidth READ whiskerWidth WRITE setWhiskerWidth)
07397     Q_PROPERTY(double symbolGap READ symbolGap WRITE setSymbolGap)
07399 public:
07400 
07407     enum ErrorType {
07408         etKeyError
07409         , etValueError
07410     };
07411     Q_ENUMS(ErrorType)
07412 
07413     explicit QCPErrorBars(QCPAxis *keyAxis, QCPAxis *valueAxis);
07414     virtual ~QCPErrorBars() Q_DECL_OVERRIDE;
07415     // getters:
07416 
07417     QSharedPointer<QCPErrorBarsDataContainer> data() const {
07418         return mDataContainer;
07419     }
07420 
07421     QCPAbstractPlottable *dataPlottable() const {
07422         return mDataPlottable.data();
07423     }
07424 
07425     ErrorType errorType() const {
07426         return mErrorType;
07427     }
07428 
07429     double whiskerWidth() const {
```

```
07430         return mWhiskerWidth;
07431     }
07432
07433     double symbolGap() const {
07434         return mSymbolGap;
07435     }
07436
07437     // setters:
07438     void setData(QSharedPointer<QCPErrorBarsDataContainer> data);
07439     void setData(const QVector<double> &error);
07440     void setData(const QVector<double> &errorMinus, const QVector<double> &errorPlus);
07441     void setDataPlottable(QCPAbstractPlottable* plottable);
07442     void setErrorType(ErrorType type);
07443     void setWhiskerWidth(double pixels);
07444     void setSymbolGap(double pixels);
07445
07446     // non-property methods:
07447     void addData(const QVector<double> &error);
07448     void addData(const QVector<double> &errorMinus, const QVector<double> &errorPlus);
07449     void addData(double error);
07450     void addData(double errorMinus, double errorPlus);
07451
07452     // virtual methods of 1d plottable interface:
07453     virtual int dataCount() const Q_DECL_OVERRIDE;
07454     virtual double dataMainKey(int index) const Q_DECL_OVERRIDE;
07455     virtual double dataSortKey(int index) const Q_DECL_OVERRIDE;
07456     virtual double dataMainValue(int index) const Q_DECL_OVERRIDE;
07457     virtual QCPRange dataValueRange(int index) const Q_DECL_OVERRIDE;
07458     virtual QPointF dataPixelPosition(int index) const Q_DECL_OVERRIDE;
07459     virtual bool sortKeyIsMainKey() const Q_DECL_OVERRIDE;
07460     virtual QCPDataSelection selectTestRect(const QRectF &rect, bool onlySelectable) const
      Q_DECL_OVERRIDE;
07461     virtual int findBegin(double sortKey, bool expandedRange = true) const Q_DECL_OVERRIDE;
07462     virtual int findEnd(double sortKey, bool expandedRange = true) const Q_DECL_OVERRIDE;
07463
07464     // reimplemented virtual methods:
07465     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
07466
07467     virtual QCPPlottableInterface1D *interface1D() Q_DECL_OVERRIDE {
07468         return this;
07469     }
07470
07471 protected:
07472     // property members:
07473     QSharedPointer<QCPErrorBarsDataContainer> mDataContainer;
07474     QPointer<QCPAbstractPlottable> mDataPlottable;
07475     ErrorType mErrorType;
07476     double mWhiskerWidth;
07477     double mSymbolGap;
07478
07479     // reimplemented virtual methods:
07480     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
07481     virtual void drawLegendIcon(QCPPainter *painter, const QRectF &rect) const Q_DECL_OVERRIDE;
07482     virtual QCPRange getKeyRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth) const
      Q_DECL_OVERRIDE;
07483     virtual QCPRange getValueRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth, const
      QCPRange &inKeyRange = QCPRange()) const Q_DECL_OVERRIDE;
07484
07485     // non-virtual methods:
07486     void getErrorBarLines(QCPErrorBarsDataContainer::const_iterator it, QVector<QLineF> &backbones,
      QVector<QLineF> &whiskers) const;
07487     void getVisibleDataBounds(QCPErrorBarsDataContainer::const_iterator &begin,
      QCPErrorBarsDataContainer::const_iterator &end, const QCPDataRange &rangeRestriction) const;
07488     double pointDistance(const QPointF &pixelPoint, QCPErrorBarsDataContainer::const_iterator
      &closestData) const;
07489     // helpers:
07490     void getDataSegments(QList<QCPDataRange> &selectedSegments, QList<QCPDataRange>
      &unselectedSegments) const;
07491     bool errorBarVisible(int index) const;
07492     bool rectIntersectsLine(const QRectF &pixelRect, const QLineF &line) const;
07493
07494     friend class QCustomPlot;
07495     friend class QCPLegend;
07496 };
07497
07498 /* end of 'src/plottables/plottable-errorbar.h' */
07499
07500
07501 /* including file 'src/items/item-straightline.h' */
07502
07503 /* modified 2021-03-29T02:30:44, size 3137        */
07504
07505 class QCP_LIB_DECL QCPItemStraightLine : public QCPAbstractItem {
07506     Q_OBJECT
07508     Q_PROPERTY(QPen pen READ pen WRITE setPen)
07509     Q_PROPERTY(QPen selectedPen READ selectedPen WRITE setSelectedPen)
```

```
07511 public:
07512     explicit QCPItemStraightLine(QCustomPlot *parentPlot);
07513     virtual ~QCPItemStraightLine() Q_DECL_OVERRIDE;
07514 
07515     // getters:
07516 
07517     QPen pen() const {
07518         return mPen;
07519     }
07520 
07521     QPen selectedPen() const {
07522         return mSelectedPen;
07523     }
07524 
07525     // setters;
07526     void setPen(const QPen &pen);
07527     void setSelectedPen(const QPen &pen);
07528 
07529     // reimplemented virtual methods:
07530     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
07531 
07532     QCPItemPosition * const point1;
07533     QCPItemPosition * const point2;
07534 
07535 protected:
07536     // property members:
07537     QPen mPen, mSelectedPen;
07538 
07539     // reimplemented virtual methods:
07540     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
07541 
07542     // non-virtual methods:
07543     QLineF getRectClippedStraightLine(const QCPVector2D &base, const QCPVector2D &vec, const QRect
      &rect) const;
07544     QPen mainPen() const;
07545 };
07546 
07547 /* end of 'src/items/item-straightline.h' */
07548 
07549 
07550 /* including file 'src/items/item-line.h'  */
07551 
07552 /* modified 2021-03-29T02:30:44, size 3429 */
07553 
07554 class QCP_LIB_DECL QCPItemLine : public QCPAbstractItem {
07555     Q_OBJECT
07557     Q_PROPERTY(QPen pen READ pen WRITE setPen)
07558     Q_PROPERTY(QPen selectedPen READ selectedPen WRITE setSelectedPen)
07559     Q_PROPERTY(QCPLineEnding head READ head WRITE setHead)
07560     Q_PROPERTY(QCPLineEnding tail READ tail WRITE setTail)
07562 public:
07563     explicit QCPItemLine(QCustomPlot *parentPlot);
07564     virtual ~QCPItemLine() Q_DECL_OVERRIDE;
07565 
07566     // getters:
07567 
07568     QPen pen() const {
07569         return mPen;
07570     }
07571 
07572     QPen selectedPen() const {
07573         return mSelectedPen;
07574     }
07575 
07576     QCPLineEnding head() const {
07577         return mHead;
07578     }
07579 
07580     QCPLineEnding tail() const {
07581         return mTail;
07582     }
07583 
07584     // setters;
07585     void setPen(const QPen &pen);
07586     void setSelectedPen(const QPen &pen);
07587     void setHead(const QCPLineEnding &head);
07588     void setTail(const QCPLineEnding &tail);
07589 
07590     // reimplemented virtual methods:
07591     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
07592 
07593     QCPItemPosition * const start;
07594     QCPItemPosition * const end;
07595 
07596 protected:
```

```
07597     // property members:
07598     QPen mPen, mSelectedPen;
07599     QCPLineEnding mHead, mTail;
07600 
07601     // reimplemented virtual methods:
07602     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
07603 
07604     // non-virtual methods:
07605     QLineF getRectClippedLine(const QCPVector2D &start, const QCPVector2D &end, const QRect &rect)
      const;
07606     QPen mainPen() const;
07607 };
07608 
07609 /* end of 'src/items/item-line.h' */
07610 
07611 
07612 /* including file 'src/items/item-curve.h' */
07613 
07614 /* modified 2021-03-29T02:30:44, size 3401 */
07615 
07616 class QCP_LIB_DECL QCPItemCurve : public QCPAbstractItem {
07617     Q_OBJECT
07619     Q_PROPERTY(QPen pen READ pen WRITE setPen)
07620     Q_PROPERTY(QPen selectedPen READ selectedPen WRITE setSelectedPen)
07621     Q_PROPERTY(QCPLineEnding head READ head WRITE setHead)
07622     Q_PROPERTY(QCPLineEnding tail READ tail WRITE setTail)
07624 public:
07625     explicit QCPItemCurve(QCustomPlot *parentPlot);
07626     virtual ~QCPItemCurve() Q_DECL_OVERRIDE;
07627 
07628     // getters:
07629 
07630     QPen pen() const {
07631         return mPen;
07632     }
07633 
07634     QPen selectedPen() const {
07635         return mSelectedPen;
07636     }
07637 
07638     QCPLineEnding head() const {
07639         return mHead;
07640     }
07641 
07642     QCPLineEnding tail() const {
07643         return mTail;
07644     }
07645 
07646     // setters;
07647     void setPen(const QPen &pen);
07648     void setSelectedPen(const QPen &pen);
07649     void setHead(const QCPLineEnding &head);
07650     void setTail(const QCPLineEnding &tail);
07651 
07652     // reimplemented virtual methods:
07653     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
07654 
07655     QCPItemPosition * const start;
07656     QCPItemPosition * const startDir;
07657     QCPItemPosition * const endDir;
07658     QCPItemPosition * const end;
07659 
07660 protected:
07661     // property members:
07662     QPen mPen, mSelectedPen;
07663     QCPLineEnding mHead, mTail;
07664 
07665     // reimplemented virtual methods:
07666     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
07667 
07668     // non-virtual methods:
07669     QPen mainPen() const;
07670 };
07671 
07672 /* end of 'src/items/item-curve.h' */
07673 
07674 
07675 /* including file 'src/items/item-rect.h'  */
07676 
07677 /* modified 2021-03-29T02:30:44, size 3710 */
07678 
07679 class QCP_LIB_DECL QCPItemRect : public QCPAbstractItem {
07680     Q_OBJECT
07682     Q_PROPERTY(QPen pen READ pen WRITE setPen)
07683     Q_PROPERTY(QPen selectedPen READ selectedPen WRITE setSelectedPen)
07684     Q_PROPERTY(QBrush brush READ brush WRITE setBrush)
```

```
07685     Q_PROPERTY(QBrush selectedBrush READ selectedBrush WRITE setSelectedBrush)
07687 public:
07688     explicit QCPItemRect(QCustomPlot *parentPlot);
07689     virtual ~QCPItemRect() Q_DECL_OVERRIDE;
07690
07691     // getters:
07692
07693     QPen pen() const {
07694         return mPen;
07695     }
07696
07697     QPen selectedPen() const {
07698         return mSelectedPen;
07699     }
07700
07701     QBrush brush() const {
07702         return mBrush;
07703     }
07704
07705     QBrush selectedBrush() const {
07706         return mSelectedBrush;
07707     }
07708
07709     // setters;
07710     void setPen(const QPen &pen);
07711     void setSelectedPen(const QPen &pen);
07712     void setBrush(const QBrush &brush);
07713     void setSelectedBrush(const QBrush &brush);
07714
07715     // reimplemented virtual methods:
07716     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
07717
07718     QCPItemPosition * const topLeft;
07719     QCPItemPosition * const bottomRight;
07720     QCPItemAnchor * const top;
07721     QCPItemAnchor * const topRight;
07722     QCPItemAnchor * const right;
07723     QCPItemAnchor * const bottom;
07724     QCPItemAnchor * const bottomLeft;
07725     QCPItemAnchor * const left;
07726
07727 protected:
07728
07729     enum AnchorIndex {
07730         aiTop, aiTopRight, aiRight, aiBottom, aiBottomLeft, aiLeft
07731     };
07732
07733     // property members:
07734     QPen mPen, mSelectedPen;
07735     QBrush mBrush, mSelectedBrush;
07736
07737     // reimplemented virtual methods:
07738     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
07739     virtual QPointF anchorPixelPosition(int anchorId) const Q_DECL_OVERRIDE;
07740
07741     // non-virtual methods:
07742     QPen mainPen() const;
07743     QBrush mainBrush() const;
07744 };
07745
07746 /* end of 'src/items/item-rect.h' */
07747
07748
07749 /* including file 'src/items/item-text.h'  */
07750
07751 /* modified 2021-03-29T02:30:44, size 5576 */
07752
07753 class QCP_LIB_DECL QCPItemText : public QCPAbstractItem {
07754     Q_OBJECT
07756     Q_PROPERTY(QColor color READ color WRITE setColor)
07757     Q_PROPERTY(QColor selectedColor READ selectedColor WRITE setSelectedColor)
07758     Q_PROPERTY(QPen pen READ pen WRITE setPen)
07759     Q_PROPERTY(QPen selectedPen READ selectedPen WRITE setSelectedPen)
07760     Q_PROPERTY(QBrush brush READ brush WRITE setBrush)
07761     Q_PROPERTY(QBrush selectedBrush READ selectedBrush WRITE setSelectedBrush)
07762     Q_PROPERTY(QFont font READ font WRITE setFont)
07763     Q_PROPERTY(QFont selectedFont READ selectedFont WRITE setSelectedFont)
07764     Q_PROPERTY(QString text READ text WRITE setText)
07765     Q_PROPERTY(Qt::Alignment positionAlignment READ positionAlignment WRITE setPositionAlignment)
07766     Q_PROPERTY(Qt::Alignment textAlignment READ textAlignment WRITE setTextAlignment)
07767     Q_PROPERTY(double rotation READ rotation WRITE setRotation)
07768     Q_PROPERTY(QMargins padding READ padding WRITE setPadding)
07770 public:
07771     explicit QCPItemText(QCustomPlot *parentPlot);
07772     virtual ~QCPItemText() Q_DECL_OVERRIDE;
07773
```

```
07774     // getters:
07775
07776     QColor color() const {
07777         return mColor;
07778     }
07779
07780     QColor selectedColor() const {
07781         return mSelectedColor;
07782     }
07783
07784     QPen pen() const {
07785         return mPen;
07786     }
07787
07788     QPen selectedPen() const {
07789         return mSelectedPen;
07790     }
07791
07792     QBrush brush() const {
07793         return mBrush;
07794     }
07795
07796     QBrush selectedBrush() const {
07797         return mSelectedBrush;
07798     }
07799
07800     QFont font() const {
07801         return mFont;
07802     }
07803
07804     QFont selectedFont() const {
07805         return mSelectedFont;
07806     }
07807
07808     QString text() const {
07809         return mText;
07810     }
07811
07812     Qt::Alignment positionAlignment() const {
07813         return mPositionAlignment;
07814     }
07815
07816     Qt::Alignment textAlignment() const {
07817         return mTextAlignment;
07818     }
07819
07820     double rotation() const {
07821         return mRotation;
07822     }
07823
07824     QMargins padding() const {
07825         return mPadding;
07826     }
07827
07828     // setters;
07829     void setColor(const QColor &color);
07830     void setSelectedColor(const QColor &color);
07831     void setPen(const QPen &pen);
07832     void setSelectedPen(const QPen &pen);
07833     void setBrush(const QBrush &brush);
07834     void setSelectedBrush(const QBrush &brush);
07835     void setFont(const QFont &font);
07836     void setSelectedFont(const QFont &font);
07837     void setText(const QString &text);
07838     void setPositionAlignment(Qt::Alignment alignment);
07839     void setTextAlignment(Qt::Alignment alignment);
07840     void setRotation(double degrees);
07841     void setPadding(const QMargins &padding);
07842
07843     // reimplemented virtual methods:
07844     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
07845
07846     QCPItemPosition * const position;
07847     QCPItemAnchor * const topLeft;
07848     QCPItemAnchor * const top;
07849     QCPItemAnchor * const topRight;
07850     QCPItemAnchor * const right;
07851     QCPItemAnchor * const bottomRight;
07852     QCPItemAnchor * const bottom;
07853     QCPItemAnchor * const bottomLeft;
07854     QCPItemAnchor * const left;
07855
07856 protected:
07857
07858     enum AnchorIndex {
07859         aiTopLeft, aiTop, aiTopRight, aiRight, aiBottomRight, aiBottom, aiBottomLeft, aiLeft
```

```
07860     };
07861
07862     // property members:
07863     QColor mColor, mSelectedColor;
07864     QPen mPen, mSelectedPen;
07865     QBrush mBrush, mSelectedBrush;
07866     QFont mFont, mSelectedFont;
07867     QString mText;
07868     Qt::Alignment mPositionAlignment;
07869     Qt::Alignment mTextAlignment;
07870     double mRotation;
07871     QMargins mPadding;
07872
07873     // reimplemented virtual methods:
07874     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
07875     virtual QPointF anchorPixelPosition(int anchorId) const Q_DECL_OVERRIDE;
07876
07877     // non-virtual methods:
07878     QPointF getTextDrawPoint(const QPointF &pos, const QRectF &rect, Qt::Alignment positionAlignment)
      const;
07879     QFont mainFont() const;
07880     QColor mainColor() const;
07881     QPen mainPen() const;
07882     QBrush mainBrush() const;
07883 };
07884
07885 /* end of 'src/items/item-text.h' */
07886
07887
07888 /* including file 'src/items/item-ellipse.h' */
07889
07890 /* modified 2021-03-29T02:30:44, size 3890   */
07891
07892 class QCP_LIB_DECL QCPItemEllipse : public QCPAbstractItem {
07893     Q_OBJECT
07895     Q_PROPERTY(QPen pen READ pen WRITE setPen)
07896     Q_PROPERTY(QPen selectedPen READ selectedPen WRITE setSelectedPen)
07897     Q_PROPERTY(QBrush brush READ brush WRITE setBrush)
07898     Q_PROPERTY(QBrush selectedBrush READ selectedBrush WRITE setSelectedBrush)
07900 public:
07901     explicit QCPItemEllipse(QCustomPlot *parentPlot);
07902     virtual ~QCPItemEllipse() Q_DECL_OVERRIDE;
07903
07904     // getters:
07905
07906     QPen pen() const {
07907         return mPen;
07908     }
07909
07910     QPen selectedPen() const {
07911         return mSelectedPen;
07912     }
07913
07914     QBrush brush() const {
07915         return mBrush;
07916     }
07917
07918     QBrush selectedBrush() const {
07919         return mSelectedBrush;
07920     }
07921
07922     // setters;
07923     void setPen(const QPen &pen);
07924     void setSelectedPen(const QPen &pen);
07925     void setBrush(const QBrush &brush);
07926     void setSelectedBrush(const QBrush &brush);
07927
07928     // reimplemented virtual methods:
07929     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
07930
07931     QCPItemPosition * const topLeft;
07932     QCPItemPosition * const bottomRight;
07933     QCPItemAnchor * const topLeftRim;
07934     QCPItemAnchor * const top;
07935     QCPItemAnchor * const topRightRim;
07936     QCPItemAnchor * const right;
07937     QCPItemAnchor * const bottomRightRim;
07938     QCPItemAnchor * const bottom;
07939     QCPItemAnchor * const bottomLeftRim;
07940     QCPItemAnchor * const left;
07941     QCPItemAnchor * const center;
07942
07943 protected:
07944
07945     enum AnchorIndex {
07946         aiTopLeftRim, aiTop, aiTopRightRim, aiRight, aiBottomRightRim, aiBottom, aiBottomLeftRim,
```

```
    aiLeft, aiCenter
07947     };
07948
07949     // property members:
07950     QPen mPen, mSelectedPen;
07951     QBrush mBrush, mSelectedBrush;
07952
07953     // reimplemented virtual methods:
07954     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
07955     virtual QPointF anchorPixelPosition(int anchorId) const Q_DECL_OVERRIDE;
07956
07957     // non-virtual methods:
07958     QPen mainPen() const;
07959     QBrush mainBrush() const;
07960 };
07961
07962 /* end of 'src/items/item-ellipse.h' */
07963
07964
07965 /* including file 'src/items/item-pixmap.h' */
07966
07967 /* modified 2021-03-29T02:30:44, size 4407  */
07968
07969 class QCP_LIB_DECL QCPItemPixmap : public QCPAbstractItem {
07970     Q_OBJECT
07972     Q_PROPERTY(QPixmap pixmap READ pixmap WRITE setPixmap)
07973     Q_PROPERTY(bool scaled READ scaled WRITE setScaled)
07974     Q_PROPERTY(Qt::AspectRatioMode aspectRatioMode READ aspectRatioMode)
07975     Q_PROPERTY(Qt::TransformationMode transformationMode READ transformationMode)
07976     Q_PROPERTY(QPen pen READ pen WRITE setPen)
07977     Q_PROPERTY(QPen selectedPen READ selectedPen WRITE setSelectedPen)
07979 public:
07980     explicit QCPItemPixmap(QCustomPlot *parentPlot);
07981     virtual ~QCPItemPixmap() Q_DECL_OVERRIDE;
07982
07983     // getters:
07984
07985     QPixmap pixmap() const {
07986         return mPixmap;
07987     }
07988
07989     bool scaled() const {
07990         return mScaled;
07991     }
07992
07993     Qt::AspectRatioMode aspectRatioMode() const {
07994         return mAspectRatioMode;
07995     }
07996
07997     Qt::TransformationMode transformationMode() const {
07998         return mTransformationMode;
07999     }
08000
08001     QPen pen() const {
08002         return mPen;
08003     }
08004
08005     QPen selectedPen() const {
08006         return mSelectedPen;
08007     }
08008
08009     // setters;
08010     void setPixmap(const QPixmap &pixmap);
08011     void setScaled(bool scaled, Qt::AspectRatioMode aspectRatioMode = Qt::KeepAspectRatio,
    Qt::TransformationMode transformationMode = Qt::SmoothTransformation);
08012     void setPen(const QPen &pen);
08013     void setSelectedPen(const QPen &pen);
08014
08015     // reimplemented virtual methods:
08016     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
    const Q_DECL_OVERRIDE;
08017
08018     QCPItemPosition * const topLeft;
08019     QCPItemPosition * const bottomRight;
08020     QCPItemAnchor * const top;
08021     QCPItemAnchor * const topRight;
08022     QCPItemAnchor * const right;
08023     QCPItemAnchor * const bottom;
08024     QCPItemAnchor * const bottomLeft;
08025     QCPItemAnchor * const left;
08026
08027 protected:
08028
08029     enum AnchorIndex {
08030         aiTop, aiTopRight, aiRight, aiBottom, aiBottomLeft, aiLeft
08031     };
08032
```

```
08033     // property members:
08034     QPixmap mPixmap;
08035     QPixmap mScaledPixmap;
08036     bool mScaled;
08037     bool mScaledPixmapInvalidated;
08038     Qt::AspectRatioMode mAspectRatioMode;
08039     Qt::TransformationMode mTransformationMode;
08040     QPen mPen, mSelectedPen;
08041
08042     // reimplemented virtual methods:
08043     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
08044     virtual QPointF anchorPixelPosition(int anchorId) const Q_DECL_OVERRIDE;
08045
08046     // non-virtual methods:
08047     void updateScaledPixmap(QRect finalRect = QRect(), bool flipHorz = false, bool flipVert = false);
08048     QRect getFinalRect(bool *flippedHorz = nullptr, bool *flippedVert = nullptr) const;
08049     QPen mainPen() const;
08050 };
08051
08052 /* end of 'src/items/item-pixmap.h' */
08053
08054
08055 /* including file 'src/items/item-tracer.h' */
08056
08057 /* modified 2021-03-29T02:30:44, size 4811  */
08058
08059 class QCP_LIB_DECL QCPItemTracer : public QCPAbstractItem {
08060
08061     Q_OBJECT
08063     Q_PROPERTY(QPen pen READ pen WRITE setPen)
08064     Q_PROPERTY(QPen selectedPen READ selectedPen WRITE setSelectedPen)
08065     Q_PROPERTY(QBrush brush READ brush WRITE setBrush)
08066     Q_PROPERTY(QBrush selectedBrush READ selectedBrush WRITE setSelectedBrush)
08067     Q_PROPERTY(double size READ size WRITE setSize)
08068     Q_PROPERTY(TracerStyle style READ style WRITE setStyle)
08069     Q_PROPERTY(QCPGraph* graph READ graph WRITE setGraph)
08070     Q_PROPERTY(double graphKey READ graphKey WRITE setGraphKey)
08071     Q_PROPERTY(bool interpolating READ interpolating WRITE setInterpolating)
08073 public:
08079     enum TracerStyle {
08080         tsNone
08081         , tsPlus
08082         , tsCrosshair
08083         , tsCircle
08084         , tsSquare
08085     };
08086     Q_ENUMS(TracerStyle)
08087
08088     explicit QCPItemTracer(QCustomPlot *parentPlot);
08089     virtual ~QCPItemTracer() Q_DECL_OVERRIDE;
08090
08091     // getters:
08092
08093     QPen pen() const {
08094         return mPen;
08095     }
08096
08097     QPen selectedPen() const {
08098         return mSelectedPen;
08099     }
08100
08101     QBrush brush() const {
08102         return mBrush;
08103     }
08104
08105     QBrush selectedBrush() const {
08106         return mSelectedBrush;
08107     }
08108
08109     double size() const {
08110         return mSize;
08111     }
08112
08113     TracerStyle style() const {
08114         return mStyle;
08115     }
08116
08117     QCPGraph *graph() const {
08118         return mGraph;
08119     }
08120
08121     double graphKey() const {
08122         return mGraphKey;
08123     }
08124
08125     bool interpolating() const {
08126         return mInterpolating;
```

```
08127     }
08128
08129     // setters;
08130     void setPen(const QPen &pen);
08131     void setSelectedPen(const QPen &pen);
08132     void setBrush(const QBrush &brush);
08133     void setSelectedBrush(const QBrush &brush);
08134     void setSize(double size);
08135     void setStyle(TracerStyle style);
08136     void setGraph(QCPGraph *graph);
08137     void setGraphKey(double key);
08138     void setInterpolating(bool enabled);
08139
08140     // reimplemented virtual methods:
08141     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
08142
08143     // non-virtual methods:
08144     void updatePosition();
08145
08146     QCPItemPosition * const position;
08147
08148 protected:
08149     // property members:
08150     QPen mPen, mSelectedPen;
08151     QBrush mBrush, mSelectedBrush;
08152     double mSize;
08153     TracerStyle mStyle;
08154     QCPGraph *mGraph;
08155     double mGraphKey;
08156     bool mInterpolating;
08157
08158     // reimplemented virtual methods:
08159     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
08160
08161     // non-virtual methods:
08162     QPen mainPen() const;
08163     QBrush mainBrush() const;
08164 };
08165
08166 Q_DECLARE_METATYPE(QCPItemTracer::TracerStyle)
08167
08168 /* end of 'src/items/item-tracer.h' */
08169
08170
08171 /* including file 'src/items/item-bracket.h' */
08172 /* modified 2021-03-29T02:30:44, size 3991   */
08173
08174 class QCP_LIB_DECL QCPItemBracket : public QCPAbstractItem {
08175
08176     Q_OBJECT
08178     Q_PROPERTY(QPen pen READ pen WRITE setPen)
08179     Q_PROPERTY(QPen selectedPen READ selectedPen WRITE setSelectedPen)
08180     Q_PROPERTY(double length READ length WRITE setLength)
08181     Q_PROPERTY(BracketStyle style READ style WRITE setStyle)
08183 public:
08190     enum BracketStyle {
08191         bsSquare
08192         , bsRound
08193         , bsCurly
08194         , bsCalligraphic
08195     };
08196     Q_ENUMS(BracketStyle)
08197
08198     explicit QCPItemBracket(QCustomPlot *parentPlot);
08199     virtual ~QCPItemBracket() Q_DECL_OVERRIDE;
08200
08201     // getters:
08202
08203     QPen pen() const {
08204         return mPen;
08205     }
08206
08207     QPen selectedPen() const {
08208         return mSelectedPen;
08209     }
08210
08211     double length() const {
08212         return mLength;
08213     }
08214
08215     BracketStyle style() const {
08216         return mStyle;
08217     }
08218
08219     // setters;
08220     void setPen(const QPen &pen);
```

```
08221     void setSelectedPen(const QPen &pen);
08222     void setLength(double length);
08223     void setStyle(BracketStyle style);
08224 
08225     // reimplemented virtual methods:
08226     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = nullptr)
      const Q_DECL_OVERRIDE;
08227 
08228     QCPItemPosition * const left;
08229     QCPItemPosition * const right;
08230     QCPItemAnchor * const center;
08231 
08232 protected:
08233     // property members:
08234 
08235     enum AnchorIndex {
08236         aiCenter
08237     };
08238     QPen mPen, mSelectedPen;
08239     double mLength;
08240     BracketStyle mStyle;
08241 
08242     // reimplemented virtual methods:
08243     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
08244     virtual QPointF anchorPixelPosition(int anchorId) const Q_DECL_OVERRIDE;
08245 
08246     // non-virtual methods:
08247     QPen mainPen() const;
08248 };
08249 
08250 Q_DECLARE_METATYPE(QCPItemBracket::BracketStyle)
08251 
08252 /* end of 'src/items/item-bracket.h' */
08253 
08254 
08255 /* including file 'src/polar/radialaxis.h'  */
08256 /* modified 2021-03-29T02:30:44, size 12227 */
08257 
08258 
08259 class QCP_LIB_DECL QCPPolarAxisRadial : public QCPLayerable {
08260 
08261     Q_OBJECT
08263 
08265 public:
08269     enum AngleReference {
08270         arAbsolute
08271         , arAngularAxis
08272     };
08273 
08274     Q_ENUMS(AngleReference)
08279     enum ScaleType {
08280         stLinear
08281         , stLogarithmic
08282     };
08283 
08284     Q_ENUMS(ScaleType)
08289     enum SelectablePart {
08290         spNone = 0
08291         , spAxis = 0x001
08292         , spTickLabels = 0x002
08293         , spAxisLabel = 0x004
08294     };
08295 
08296     Q_ENUMS(SelectablePart)
08297     Q_FLAGS(SelectableParts)
08298     Q_DECLARE_FLAGS(SelectableParts, SelectablePart)
08299 
08300     enum LabelMode {
08301         lmUpright
08302         , lmRotated
08303     };
08304     Q_ENUMS(LabelMode)
08305 
08306     explicit QCPPolarAxisRadial(QCPPolarAxisAngular *parent);
08307     virtual ~QCPPolarAxisRadial();
08308 
08309     // getters:
08310 
08311     bool rangeDrag() const {
08312         return mRangeDrag;
08313     }
08314 
08315     bool rangeZoom() const {
08316         return mRangeZoom;
08317     }
08318 
08319     double rangeZoomFactor() const {
```

```
08320         return mRangeZoomFactor;
08321     }
08322
08323     QCPPolarAxisAngular *angularAxis() const {
08324         return mAngularAxis;
08325     }
08326
08327     ScaleType scaleType() const {
08328         return mScaleType;
08329     }
08330
08331     const QCPRange range() const {
08332         return mRange;
08333     }
08334
08335     bool rangeReversed() const {
08336         return mRangeReversed;
08337     }
08338
08339     double angle() const {
08340         return mAngle;
08341     }
08342
08343     AngleReference angleReference() const {
08344         return mAngleReference;
08345     }
08346
08347     QSharedPointer<QCPAxisTicker> ticker() const {
08348         return mTicker;
08349     }
08350
08351     bool ticks() const {
08352         return mTicks;
08353     }
08354
08355     bool tickLabels() const {
08356         return mTickLabels;
08357     }
08358
08359     int tickLabelPadding() const {
08360         return mLabelPainter.padding();
08361     }
08362
08363     QFont tickLabelFont() const {
08364         return mTickLabelFont;
08365     }
08366
08367     QColor tickLabelColor() const {
08368         return mTickLabelColor;
08369     }
08370
08371     double tickLabelRotation() const {
08372         return mLabelPainter.rotation();
08373     }
08374     LabelMode tickLabelMode() const;
08375     QString numberFormat() const;
08376
08377     int numberPrecision() const {
08378         return mNumberPrecision;
08379     }
08380
08381     QVector<double> tickVector() const {
08382         return mTickVector;
08383     }
08384
08385     QVector<double> subTickVector() const {
08386         return mSubTickVector;
08387     }
08388
08389     QVector<QString> tickVectorLabels() const {
08390         return mTickVectorLabels;
08391     }
08392     int tickLengthIn() const;
08393     int tickLengthOut() const;
08394
08395     bool subTicks() const {
08396         return mSubTicks;
08397     }
08398     int subTickLengthIn() const;
08399     int subTickLengthOut() const;
08400
08401     QPen basePen() const {
08402         return mBasePen;
08403     }
08404
08405     QPen tickPen() const {
08406         return mTickPen;
```

```
08407     }
08408
08409     QPen subTickPen() const {
08410         return mSubTickPen;
08411     }
08412
08413     QFont labelFont() const {
08414         return mLabelFont;
08415     }
08416
08417     QColor labelColor() const {
08418         return mLabelColor;
08419     }
08420
08421     QString label() const {
08422         return mLabel;
08423     }
08424     int labelPadding() const;
08425
08426     SelectableParts selectedParts() const {
08427         return mSelectedParts;
08428     }
08429
08430     SelectableParts selectableParts() const {
08431         return mSelectableParts;
08432     }
08433
08434     QFont selectedTickLabelFont() const {
08435         return mSelectedTickLabelFont;
08436     }
08437
08438     QFont selectedLabelFont() const {
08439         return mSelectedLabelFont;
08440     }
08441
08442     QColor selectedTickLabelColor() const {
08443         return mSelectedTickLabelColor;
08444     }
08445
08446     QColor selectedLabelColor() const {
08447         return mSelectedLabelColor;
08448     }
08449
08450     QPen selectedBasePen() const {
08451         return mSelectedBasePen;
08452     }
08453
08454     QPen selectedTickPen() const {
08455         return mSelectedTickPen;
08456     }
08457
08458     QPen selectedSubTickPen() const {
08459         return mSelectedSubTickPen;
08460     }
08461
08462     // setters:
08463     void setRangeDrag(bool enabled);
08464     void setRangeZoom(bool enabled);
08465     void setRangeZoomFactor(double factor);
08466
08467     Q_SLOT void setScaleType(QCPPolarAxisRadial::ScaleType type);
08468     Q_SLOT void setRange(const QCPRange &range);
08469     void setRange(double lower, double upper);
08470     void setRange(double position, double size, Qt::AlignmentFlag alignment);
08471     void setRangeLower(double lower);
08472     void setRangeUpper(double upper);
08473     void setRangeReversed(bool reversed);
08474     void setAngle(double degrees);
08475     void setAngleReference(AngleReference reference);
08476     void setTicker(QSharedPointer<QCPAxisTicker> ticker);
08477     void setTicks(bool show);
08478     void setTickLabels(bool show);
08479     void setTickLabelPadding(int padding);
08480     void setTickLabelFont(const QFont &font);
08481     void setTickLabelColor(const QColor &color);
08482     void setTickLabelRotation(double degrees);
08483     void setTickLabelMode(LabelMode mode);
08484     void setNumberFormat(const QString &formatCode);
08485     void setNumberPrecision(int precision);
08486     void setTickLength(int inside, int outside = 0);
08487     void setTickLengthIn(int inside);
08488     void setTickLengthOut(int outside);
08489     void setSubTicks(bool show);
08490     void setSubTickLength(int inside, int outside = 0);
08491     void setSubTickLengthIn(int inside);
08492     void setSubTickLengthOut(int outside);
08493     void setBasePen(const QPen &pen);
```

```
08494     void setTickPen(const QPen &pen);
08495     void setSubTickPen(const QPen &pen);
08496     void setLabelFont(const QFont &font);
08497     void setLabelColor(const QColor &color);
08498     void setLabel(const QString &str);
08499     void setLabelPadding(int padding);
08500     void setSelectedTickLabelFont(const QFont &font);
08501     void setSelectedLabelFont(const QFont &font);
08502     void setSelectedTickLabelColor(const QColor &color);
08503     void setSelectedLabelColor(const QColor &color);
08504     void setSelectedBasePen(const QPen &pen);
08505     void setSelectedTickPen(const QPen &pen);
08506     void setSelectedSubTickPen(const QPen &pen);
08507     Q_SLOT void setSelectableParts(const QCPPolarAxisRadial::SelectableParts &selectableParts);
08508     Q_SLOT void setSelectedParts(const QCPPolarAxisRadial::SelectableParts &selectedParts);
08509
08510     // reimplemented virtual methods:
08511     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = 0) const
      Q_DECL_OVERRIDE;
08512
08513     // non-property methods:
08514     void moveRange(double diff);
08515     void scaleRange(double factor);
08516     void scaleRange(double factor, double center);
08517     void rescale(bool onlyVisiblePlottables = false);
08518     void pixelToCoord(QPointF pixelPos, double &angleCoord, double &radiusCoord) const;
08519     QPointF coordToPixel(double angleCoord, double radiusCoord) const;
08520     double coordToRadius(double coord) const;
08521     double radiusToCoord(double radius) const;
08522     SelectablePart getPartAt(const QPointF &pos) const;
08523
08524 signals:
08525     void rangeChanged(const QCPRange &newRange);
08526     void rangeChanged(const QCPRange &newRange, const QCPRange &oldRange);
08527     void scaleTypeChanged(QCPPolarAxisRadial::ScaleType scaleType);
08528     void selectionChanged(const QCPPolarAxisRadial::SelectableParts &parts);
08529     void selectableChanged(const QCPPolarAxisRadial::SelectableParts &parts);
08530
08531 protected:
08532     // property members:
08533     bool mRangeDrag;
08534     bool mRangeZoom;
08535     double mRangeZoomFactor;
08536
08537     // axis base:
08538     QCPPolarAxisAngular *mAngularAxis;
08539     double mAngle;
08540     AngleReference mAngleReference;
08541     SelectableParts mSelectableParts, mSelectedParts;
08542     QPen mBasePen, mSelectedBasePen;
08543     // axis label:
08544     int mLabelPadding;
08545     QString mLabel;
08546     QFont mLabelFont, mSelectedLabelFont;
08547     QColor mLabelColor, mSelectedLabelColor;
08548     // tick labels:
08549     //int mTickLabelPadding; in label painter
08550     bool mTickLabels;
08551     //double mTickLabelRotation; in label painter
08552     QFont mTickLabelFont, mSelectedTickLabelFont;
08553     QColor mTickLabelColor, mSelectedTickLabelColor;
08554     int mNumberPrecision;
08555     QLatin1Char mNumberFormatChar;
08556     bool mNumberBeautifulPowers;
08557     bool mNumberMultiplyCross;
08558     // ticks and subticks:
08559     bool mTicks;
08560     bool mSubTicks;
08561     int mTickLengthIn, mTickLengthOut, mSubTickLengthIn, mSubTickLengthOut;
08562     QPen mTickPen, mSelectedTickPen;
08563     QPen mSubTickPen, mSelectedSubTickPen;
08564     // scale and range:
08565     QCPRange mRange;
08566     bool mRangeReversed;
08567     ScaleType mScaleType;
08568
08569     // non-property members:
08570     QPointF mCenter;
08571     double mRadius;
08572     QSharedPointer<QCPAxisTicker> mTicker;
08573     QVector<double> mTickVector;
08574     QVector<QString> mTickVectorLabels;
08575     QVector<double> mSubTickVector;
08576     bool mDragging;
08577     QCPRange mDragStartRange;
08578     QCP::AntialiasedElements mAADragBackup, mNotAADragBackup;
08579     QCPLabelPainterPrivate mLabelPainter;
```

```
08580
08581     // reimplemented virtual methods:
08582     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
08583     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
08584     virtual QCP::Interaction selectionCategory() const Q_DECL_OVERRIDE;
08585     // events:
08586     virtual void selectEvent(QMouseEvent *event, bool additive, const QVariant &details, bool
      *selectionStateChanged) Q_DECL_OVERRIDE;
08587     virtual void deselectEvent(bool *selectionStateChanged) Q_DECL_OVERRIDE;
08588     // mouse events:
08589     virtual void mousePressEvent(QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE;
08590     virtual void mouseMoveEvent(QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE;
08591     virtual void mouseReleaseEvent(QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE;
08592     virtual void wheelEvent(QWheelEvent *event) Q_DECL_OVERRIDE;
08593
08594     // non-virtual methods:
08595     void updateGeometry(const QPointF &center, double radius);
08596     void setupTickVectors();
08597     QPen getBasePen() const;
08598     QPen getTickPen() const;
08599     QPen getSubTickPen() const;
08600     QFont getTickLabelFont() const;
08601     QFont getLabelFont() const;
08602     QColor getTickLabelColor() const;
08603     QColor getLabelColor() const;
08604
08605 private:
08606     Q_DISABLE_COPY(QCPPolarAxisRadial)
08607
08608     friend class QCustomPlot;
08609     friend class QCPPolarAxisAngular;
08610 };
08611
08612 Q_DECLARE_OPERATORS_FOR_FLAGS(QCPPolarAxisRadial::SelectableParts)
08613 Q_DECLARE_METATYPE(QCPPolarAxisRadial::AngleReference)
08614 Q_DECLARE_METATYPE(QCPPolarAxisRadial::ScaleType)
08615 Q_DECLARE_METATYPE(QCPPolarAxisRadial::SelectablePart)
08616
08617
08618
08619 /* end of 'src/polar/radialaxis.h' */
08620
08621
08622 /* including file 'src/polar/layoutelement-angularaxis.h' */
08623 /* modified 2021-03-29T02:30:44, size 13461               */
08624
08625 class QCP_LIB_DECL QCPPolarAxisAngular : public QCPLayoutElement {
08626
08627     Q_OBJECT
08629
08631 public:
08636     enum SelectablePart {
08637         spNone = 0
08638         , spAxis = 0x001
08639         , spTickLabels = 0x002
08640         , spAxisLabel = 0x004
08641     };
08642
08643     Q_ENUMS(SelectablePart)
08644     Q_FLAGS(SelectableParts)
08645     Q_DECLARE_FLAGS(SelectableParts, SelectablePart)
08646
08647
08650     enum LabelMode {
08651         lmUpright
08652         , lmRotated
08653     };
08654     Q_ENUMS(LabelMode)
08655
08656     explicit QCPPolarAxisAngular(QCustomPlot *parentPlot);
08657     virtual ~QCPPolarAxisAngular();
08658
08659     // getters:
08660
08661     QPixmap background() const {
08662         return mBackgroundPixmap;
08663     }
08664
08665     QBrush backgroundBrush() const {
08666         return mBackgroundBrush;
08667     }
08668
08669     bool backgroundScaled() const {
08670         return mBackgroundScaled;
08671     }
08672
08673     Qt::AspectRatioMode backgroundScaledMode() const {
```

```
08674         return mBackgroundScaledMode;
08675     }
08676
08677     bool rangeDrag() const {
08678         return mRangeDrag;
08679     }
08680
08681     bool rangeZoom() const {
08682         return mRangeZoom;
08683     }
08684
08685     double rangeZoomFactor() const {
08686         return mRangeZoomFactor;
08687     }
08688
08689     const QCPRange range() const {
08690         return mRange;
08691     }
08692
08693     bool rangeReversed() const {
08694         return mRangeReversed;
08695     }
08696
08697     double angle() const {
08698         return mAngle;
08699     }
08700
08701     QSharedPointer<QCPAxisTicker> ticker() const {
08702         return mTicker;
08703     }
08704
08705     bool ticks() const {
08706         return mTicks;
08707     }
08708
08709     bool tickLabels() const {
08710         return mTickLabels;
08711     }
08712
08713     int tickLabelPadding() const {
08714         return mLabelPainter.padding();
08715     }
08716
08717     QFont tickLabelFont() const {
08718         return mTickLabelFont;
08719     }
08720
08721     QColor tickLabelColor() const {
08722         return mTickLabelColor;
08723     }
08724
08725     double tickLabelRotation() const {
08726         return mLabelPainter.rotation();
08727     }
08728     LabelMode tickLabelMode() const;
08729     QString numberFormat() const;
08730
08731     int numberPrecision() const {
08732         return mNumberPrecision;
08733     }
08734
08735     QVector<double> tickVector() const {
08736         return mTickVector;
08737     }
08738
08739     QVector<QString> tickVectorLabels() const {
08740         return mTickVectorLabels;
08741     }
08742
08743     int tickLengthIn() const {
08744         return mTickLengthIn;
08745     }
08746
08747     int tickLengthOut() const {
08748         return mTickLengthOut;
08749     }
08750
08751     bool subTicks() const {
08752         return mSubTicks;
08753     }
08754
08755     int subTickLengthIn() const {
08756         return mSubTickLengthIn;
08757     }
08758
08759     int subTickLengthOut() const {
08760         return mSubTickLengthOut;
```

```
08761     }
08762
08763     QPen basePen() const {
08764         return mBasePen;
08765     }
08766
08767     QPen tickPen() const {
08768         return mTickPen;
08769     }
08770
08771     QPen subTickPen() const {
08772         return mSubTickPen;
08773     }
08774
08775     QFont labelFont() const {
08776         return mLabelFont;
08777     }
08778
08779     QColor labelColor() const {
08780         return mLabelColor;
08781     }
08782
08783     QString label() const {
08784         return mLabel;
08785     }
08786
08787     int labelPadding() const {
08788         return mLabelPadding;
08789     }
08790
08791     SelectableParts selectedParts() const {
08792         return mSelectedParts;
08793     }
08794
08795     SelectableParts selectableParts() const {
08796         return mSelectableParts;
08797     }
08798
08799     QFont selectedTickLabelFont() const {
08800         return mSelectedTickLabelFont;
08801     }
08802
08803     QFont selectedLabelFont() const {
08804         return mSelectedLabelFont;
08805     }
08806
08807     QColor selectedTickLabelColor() const {
08808         return mSelectedTickLabelColor;
08809     }
08810
08811     QColor selectedLabelColor() const {
08812         return mSelectedLabelColor;
08813     }
08814
08815     QPen selectedBasePen() const {
08816         return mSelectedBasePen;
08817     }
08818
08819     QPen selectedTickPen() const {
08820         return mSelectedTickPen;
08821     }
08822
08823     QPen selectedSubTickPen() const {
08824         return mSelectedSubTickPen;
08825     }
08826
08827     QCPPolarGrid *grid() const {
08828         return mGrid;
08829     }
08830
08831     // setters:
08832     void setBackground(const QPixmap &pm);
08833     void setBackground(const QPixmap &pm, bool scaled, Qt::AspectRatioMode mode =
      Qt::KeepAspectRatioByExpanding);
08834     void setBackground(const QBrush &brush);
08835     void setBackgroundScaled(bool scaled);
08836     void setBackgroundScaledMode(Qt::AspectRatioMode mode);
08837     void setRangeDrag(bool enabled);
08838     void setRangeZoom(bool enabled);
08839     void setRangeZoomFactor(double factor);
08840
08841     Q_SLOT void setRange(const QCPRange &range);
08842     void setRange(double lower, double upper);
08843     void setRange(double position, double size, Qt::AlignmentFlag alignment);
08844     void setRangeLower(double lower);
08845     void setRangeUpper(double upper);
08846     void setRangeReversed(bool reversed);
```

```
08847     void setAngle(double degrees);
08848     void setTicker(QSharedPointer<QCPAxisTicker> ticker);
08849     void setTicks(bool show);
08850     void setTickLabels(bool show);
08851     void setTickLabelPadding(int padding);
08852     void setTickLabelFont(const QFont &font);
08853     void setTickLabelColor(const QColor &color);
08854     void setTickLabelRotation(double degrees);
08855     void setTickLabelMode(LabelMode mode);
08856     void setNumberFormat(const QString &formatCode);
08857     void setNumberPrecision(int precision);
08858     void setTickLength(int inside, int outside = 0);
08859     void setTickLengthIn(int inside);
08860     void setTickLengthOut(int outside);
08861     void setSubTicks(bool show);
08862     void setSubTickLength(int inside, int outside = 0);
08863     void setSubTickLengthIn(int inside);
08864     void setSubTickLengthOut(int outside);
08865     void setBasePen(const QPen &pen);
08866     void setTickPen(const QPen &pen);
08867     void setSubTickPen(const QPen &pen);
08868     void setLabelFont(const QFont &font);
08869     void setLabelColor(const QColor &color);
08870     void setLabel(const QString &str);
08871     void setLabelPadding(int padding);
08872     void setLabelPosition(Qt::AlignmentFlag position);
08873     void setSelectedTickLabelFont(const QFont &font);
08874     void setSelectedTickLabelColor(const QColor &color);
08875     void setSelectedLabelFont(const QFont &font);
08876     void setSelectedLabelColor(const QColor &color);
08877     void setSelectedBasePen(const QPen &pen);
08878     void setSelectedTickPen(const QPen &pen);
08879     void setSelectedSubTickPen(const QPen &pen);
08880     Q_SLOT void setSelectableParts(const QCPPolarAxisAngular::SelectableParts &selectableParts);
08881     Q_SLOT void setSelectedParts(const QCPPolarAxisAngular::SelectableParts &selectedParts);
08882
08883     // reimplemented virtual methods:
08884     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = 0) const
      Q_DECL_OVERRIDE;
08885     virtual void update(UpdatePhase phase) Q_DECL_OVERRIDE;
08886     virtual QList<QCPLayoutElement*> elements(bool recursive) const Q_DECL_OVERRIDE;
08887
08888     // non-property methods:
08889     bool removeGraph(QCPPolarGraph *graph);
08890     int radialAxisCount() const;
08891     QCPPolarAxisRadial *radialAxis(int index = 0) const;
08892     QList<QCPPolarAxisRadial*> radialAxes() const;
08893     QCPPolarAxisRadial *addRadialAxis(QCPPolarAxisRadial *axis = 0);
08894     bool removeRadialAxis(QCPPolarAxisRadial *axis);
08895
08896     QCPLayoutInset *insetLayout() const {
08897         return mInsetLayout;
08898     }
08899     QRegion exactClipRegion() const;
08900
08901     void moveRange(double diff);
08902     void scaleRange(double factor);
08903     void scaleRange(double factor, double center);
08904     void rescale(bool onlyVisiblePlottables = false);
08905
08906     double coordToAngleRad(double coord) const {
08907         return mAngleRad + (coord - mRange.lower) / mRange.size()*(mRangeReversed ? -2.0 * M_PI : 2.0
      * M_PI);
08908     } // mention in doc that return doesn't wrap
08909
08910     double angleRadToCoord(double angleRad) const {
08911         return mRange.lower + (angleRad - mAngleRad) / (mRangeReversed ? -2.0 * M_PI : 2.0 * M_PI) *
      mRange.size();
08912     }
08913     void pixelToCoord(QPointF pixelPos, double &angleCoord, double &radiusCoord) const;
08914     QPointF coordToPixel(double angleCoord, double radiusCoord) const;
08915     SelectablePart getPartAt(const QPointF &pos) const;
08916
08917     // read-only interface imitating a QRect:
08918
08919     int left() const {
08920         return mRect.left();
08921     }
08922
08923     int right() const {
08924         return mRect.right();
08925     }
08926
08927     int top() const {
08928         return mRect.top();
08929     }
08930
```

```
08931     int bottom() const {
08932         return mRect.bottom();
08933     }
08934 
08935     int width() const {
08936         return mRect.width();
08937     }
08938 
08939     int height() const {
08940         return mRect.height();
08941     }
08942 
08943     QSize size() const {
08944         return mRect.size();
08945     }
08946 
08947     QPoint topLeft() const {
08948         return mRect.topLeft();
08949     }
08950 
08951     QPoint topRight() const {
08952         return mRect.topRight();
08953     }
08954 
08955     QPoint bottomLeft() const {
08956         return mRect.bottomLeft();
08957     }
08958 
08959     QPoint bottomRight() const {
08960         return mRect.bottomRight();
08961     }
08962 
08963     QPointF center() const {
08964         return mCenter;
08965     }
08966 
08967     double radius() const {
08968         return mRadius;
08969     }
08970 
08971 signals:
08972     void rangeChanged(const QCPRange &newRange);
08973     void rangeChanged(const QCPRange &newRange, const QCPRange &oldRange);
08974     void selectionChanged(const QCPPolarAxisAngular::SelectableParts &parts);
08975     void selectableChanged(const QCPPolarAxisAngular::SelectableParts &parts);
08976 
08977 protected:
08978     // property members:
08979     QBrush mBackgroundBrush;
08980     QPixmap mBackgroundPixmap;
08981     QPixmap mScaledBackgroundPixmap;
08982     bool mBackgroundScaled;
08983     Qt::AspectRatioMode mBackgroundScaledMode;
08984     QCPLayoutInset *mInsetLayout;
08985     bool mRangeDrag;
08986     bool mRangeZoom;
08987     double mRangeZoomFactor;
08988 
08989     // axis base:
08990     double mAngle, mAngleRad;
08991     SelectableParts mSelectableParts, mSelectedParts;
08992     QPen mBasePen, mSelectedBasePen;
08993     // axis label:
08994     int mLabelPadding;
08995     QString mLabel;
08996     QFont mLabelFont, mSelectedLabelFont;
08997     QColor mLabelColor, mSelectedLabelColor;
08998     // tick labels:
08999     //int mTickLabelPadding; in label painter
09000     bool mTickLabels;
09001     //double mTickLabelRotation; in label painter
09002     QFont mTickLabelFont, mSelectedTickLabelFont;
09003     QColor mTickLabelColor, mSelectedTickLabelColor;
09004     int mNumberPrecision;
09005     QLatin1Char mNumberFormatChar;
09006     bool mNumberBeautifulPowers;
09007     bool mNumberMultiplyCross;
09008     // ticks and subticks:
09009     bool mTicks;
09010     bool mSubTicks;
09011     int mTickLengthIn, mTickLengthOut, mSubTickLengthIn, mSubTickLengthOut;
09012     QPen mTickPen, mSelectedTickPen;
09013     QPen mSubTickPen, mSelectedSubTickPen;
09014     // scale and range:
09015     QCPRange mRange;
09016     bool mRangeReversed;
09017 
```

```
09018     // non-property members:
09019     QPointF mCenter;
09020     double mRadius;
09021     QList<QCPPolarAxisRadial*> mRadialAxes;
09022     QCPPolarGrid *mGrid;
09023     QList<QCPPolarGraph*> mGraphs;
09024     QSharedPointer<QCPAxisTicker> mTicker;
09025     QVector<double> mTickVector;
09026     QVector<QString> mTickVectorLabels;
09027     QVector<QPointF> mTickVectorCosSin;
09028     QVector<double> mSubTickVector;
09029     QVector<QPointF> mSubTickVectorCosSin;
09030     bool mDragging;
09031     QCPRange mDragAngularStart;
09032     QList<QCPRange> mDragRadialStart;
09033     QCP::AntialiasedElements mAADragBackup, mNotAADragBackup;
09034     QCPLabelPainterPrivate mLabelPainter;
09035
09036     // reimplemented virtual methods:
09037     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
09038     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
09039     virtual QCP::Interaction selectionCategory() const Q_DECL_OVERRIDE;
09040     // events:
09041     virtual void mousePressEvent(QMouseEvent *event, const QVariant &details) Q_DECL_OVERRIDE;
09042     virtual void mouseMoveEvent(QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE;
09043     virtual void mouseReleaseEvent(QMouseEvent *event, const QPointF &startPos) Q_DECL_OVERRIDE;
09044     virtual void wheelEvent(QWheelEvent *event) Q_DECL_OVERRIDE;
09045
09046     // non-virtual methods:
09047     bool registerPolarGraph(QCPPolarGraph *graph);
09048     void drawBackground(QCPPainter *painter, const QPointF &center, double radius);
09049     void setupTickVectors();
09050     QPen getBasePen() const;
09051     QPen getTickPen() const;
09052     QPen getSubTickPen() const;
09053     QFont getTickLabelFont() const;
09054     QFont getLabelFont() const;
09055     QColor getTickLabelColor() const;
09056     QColor getLabelColor() const;
09057
09058 private:
09059     Q_DISABLE_COPY(QCPPolarAxisAngular)
09060
09061     friend class QCustomPlot;
09062     friend class QCPPolarGrid;
09063     friend class QCPPolarGraph;
09064 };
09065
09066 Q_DECLARE_OPERATORS_FOR_FLAGS(QCPPolarAxisAngular::SelectableParts)
09067 Q_DECLARE_METATYPE(QCPPolarAxisAngular::SelectablePart)
09068
09069 /* end of 'src/polar/layoutelement-angularaxis.h' */
09070
09071
09072 /* including file 'src/polar/polargrid.h'  */
09073 /* modified 2021-03-29T02:30:44, size 4506 */
09074
09075 class QCP_LIB_DECL QCPPolarGrid : public QCPLayerable {
09076
09077     Q_OBJECT
09079
09081 public:
09085     enum GridType {
09086         gtAngular = 0x01
09087         , gtRadial = 0x02
09088         , gtAll = 0xFF
09089         , gtNone = 0x00
09090     };
09091     Q_ENUMS(GridType)
09092     Q_FLAGS(GridTypes)
09093     Q_DECLARE_FLAGS(GridTypes, GridType)
09094
09095     explicit QCPPolarGrid(QCPPolarAxisAngular *parentAxis);
09096
09097     // getters:
09098
09099     QCPPolarAxisRadial *radialAxis() const {
09100         return mRadialAxis.data();
09101     }
09102
09103     GridTypes type() const {
09104         return mType;
09105     }
09106
09107     GridTypes subGridType() const {
09108         return mSubGridType;
09109     }
```

```
09110
09111     bool antialiasedSubGrid() const {
09112         return mAntialiasedSubGrid;
09113     }
09114
09115     bool antialiasedZeroLine() const {
09116         return mAntialiasedZeroLine;
09117     }
09118
09119     QPen angularPen() const {
09120         return mAngularPen;
09121     }
09122
09123     QPen angularSubGridPen() const {
09124         return mAngularSubGridPen;
09125     }
09126
09127     QPen radialPen() const {
09128         return mRadialPen;
09129     }
09130
09131     QPen radialSubGridPen() const {
09132         return mRadialSubGridPen;
09133     }
09134
09135     QPen radialZeroLinePen() const {
09136         return mRadialZeroLinePen;
09137     }
09138
09139     // setters:
09140     void setRadialAxis(QCPPolarAxisRadial *axis);
09141     void setType(GridTypes type);
09142     void setSubGridType(GridTypes type);
09143     void setAntialiasedSubGrid(bool enabled);
09144     void setAntialiasedZeroLine(bool enabled);
09145     void setAngularPen(const QPen &pen);
09146     void setAngularSubGridPen(const QPen &pen);
09147     void setRadialPen(const QPen &pen);
09148     void setRadialSubGridPen(const QPen &pen);
09149     void setRadialZeroLinePen(const QPen &pen);
09150
09151 protected:
09152     // property members:
09153     GridTypes mType;
09154     GridTypes mSubGridType;
09155     bool mAntialiasedSubGrid, mAntialiasedZeroLine;
09156     QPen mAngularPen, mAngularSubGridPen;
09157     QPen mRadialPen, mRadialSubGridPen, mRadialZeroLinePen;
09158
09159     // non-property members:
09160     QCPPolarAxisAngular *mParentAxis;
09161     QPointer<QCPPolarAxisRadial> mRadialAxis;
09162
09163     // reimplemented virtual methods:
09164     virtual void applyDefaultAntialiasingHint(QCPPainter *painter) const Q_DECL_OVERRIDE;
09165     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
09166
09167     // non-virtual methods:
09168     void drawRadialGrid(QCPPainter *painter, const QPointF &center, const QVector<double> &coords,
      const QPen &pen, const QPen &zeroPen = Qt::NoPen);
09169     void drawAngularGrid(QCPPainter *painter, const QPointF &center, double radius, const
      QVector<QPointF> &ticksCosSin, const QPen &pen);
09170
09171 private:
09172     Q_DISABLE_COPY(QCPPolarGrid)
09173
09174 };
09175
09176 Q_DECLARE_OPERATORS_FOR_FLAGS(QCPPolarGrid::GridTypes)
09177 Q_DECLARE_METATYPE(QCPPolarGrid::GridType)
09178
09179
09180 /* end of 'src/polar/polargrid.h' */
09181
09182
09183 /* including file 'src/polar/polargraph.h' */
09184 /* modified 2021-03-29T02:30:44, size 9606 */
09185
09186
09187 class QCP_LIB_DECL QCPPolarLegendItem : public QCPAbstractLegendItem {
09188     Q_OBJECT
09189 public:
09190     QCPPolarLegendItem(QCPLegend *parent, QCPPolarGraph *graph);
09191
09192     // getters:
09193
09194     QCPPolarGraph *polarGraph() {
```

```
09195         return mPolarGraph;
09196     }
09197
09198 protected:
09199     // property members:
09200     QCPPolarGraph *mPolarGraph;
09201
09202     // reimplemented virtual methods:
09203     virtual void draw(QCPPainter *painter) Q_DECL_OVERRIDE;
09204     virtual QSize minimumOuterSizeHint() const Q_DECL_OVERRIDE;
09205
09206     // non-virtual methods:
09207     QPen getIconBorderPen() const;
09208     QColor getTextColor() const;
09209     QFont getFont() const;
09210 };
09211
09212 class QCP_LIB_DECL QCPPolarGraph : public QCPLayerable {
09213
09214     Q_OBJECT
09216
09218 public:
09224     enum LineStyle {
09225         lsNone
09227         , lsLine
09228     };
09229     Q_ENUMS(LineStyle)
09230
09231     QCPPolarGraph(QCPPolarAxisAngular *keyAxis, QCPPolarAxisRadial *valueAxis);
09232     virtual ~QCPPolarGraph();
09233
09234     // getters:
09235
09236     QString name() const {
09237         return mName;
09238     }
09239
09240     bool antialiasedFill() const {
09241         return mAntialiasedFill;
09242     }
09243
09244     bool antialiasedScatters() const {
09245         return mAntialiasedScatters;
09246     }
09247
09248     QPen pen() const {
09249         return mPen;
09250     }
09251
09252     QBrush brush() const {
09253         return mBrush;
09254     }
09255
09256     bool periodic() const {
09257         return mPeriodic;
09258     }
09259
09260     QCPPolarAxisAngular *keyAxis() const {
09261         return mKeyAxis.data();
09262     }
09263
09264     QCPPolarAxisRadial *valueAxis() const {
09265         return mValueAxis.data();
09266     }
09267
09268     QCP::SelectionType selectable() const {
09269         return mSelectable;
09270     }
09271
09272     bool selected() const {
09273         return !mSelection.isEmpty();
09274     }
09275
09276     QCPDataSelection selection() const {
09277         return mSelection;
09278     }
09279     //QCPSelectionDecorator *selectionDecorator() const { return mSelectionDecorator; }
09280
09281     QSharedPointer<QCPGraphDataContainer> data() const {
09282         return mDataContainer;
09283     }
09284
09285     LineStyle lineStyle() const {
09286         return mLineStyle;
09287     }
09288
09289     QCPScatterStyle scatterStyle() const {
```

```
09290         return mScatterStyle;
09291     }
09292
09293     // setters:
09294     void setName(const QString &name);
09295     void setAntialiasedFill(bool enabled);
09296     void setAntialiasedScatters(bool enabled);
09297     void setPen(const QPen &pen);
09298     void setBrush(const QBrush &brush);
09299     void setPeriodic(bool enabled);
09300     void setKeyAxis(QCPPolarAxisAngular *axis);
09301     void setValueAxis(QCPPolarAxisRadial *axis);
09302     Q_SLOT void setSelectable(QCP::SelectionType selectable);
09303     Q_SLOT void setSelection(QCPDataSelection selection);
09304     //void setSelectionDecorator(QCPSelectionDecorator *decorator);
09305     void setData(QSharedPointer<QCPGraphDataContainer> data);
09306     void setData(const QVector<double> &keys, const QVector<double> &values, bool alreadySorted =
      false);
09307     void setLineStyle(LineStyle ls);
09308     void setScatterStyle(const QCPScatterStyle &style);
09309
09310     // non-property methods:
09311     void addData(const QVector<double> &keys, const QVector<double> &values, bool alreadySorted =
      false);
09312     void addData(double key, double value);
09313     void coordsToPixels(double key, double value, double &x, double &y) const;
09314     const QPointF coordsToPixels(double key, double value) const;
09315     void pixelsToCoords(double x, double y, double &key, double &value) const;
09316     void pixelsToCoords(const QPointF &pixelPos, double &key, double &value) const;
09317     void rescaleAxes(bool onlyEnlarge = false) const;
09318     void rescaleKeyAxis(bool onlyEnlarge = false) const;
09319     void rescaleValueAxis(bool onlyEnlarge = false, bool inKeyRange = false) const;
09320     bool addToLegend(QCPLegend *legend);
09321     bool addToLegend();
09322     bool removeFromLegend(QCPLegend *legend) const;
09323     bool removeFromLegend() const;
09324
09325     // introduced virtual methods:
09326     virtual double selectTest(const QPointF &pos, bool onlySelectable, QVariant *details = 0) const;
      // actually introduced in QCPLayerable as non-pure, but we want to force reimplementation for
      plottables
09327
09328     virtual QCPPlottableInterface1D *interface1D() {
09329         return 0;
09330     } // TODO: return this later, when QCPAbstractPolarPlottable is created
09331     virtual QCPRange getKeyRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth) const;
09332     virtual QCPRange getValueRange(bool &foundRange, QCP::SignDomain inSignDomain = QCP::sdBoth, const
      QCPRange &inKeyRange = QCPRange()) const;
09333
09334 signals:
09335     void selectionChanged(bool selected);
09336     void selectionChanged(const QCPDataSelection &selection);
09337     void selectableChanged(QCP::SelectionType selectable);
09338
09339 protected:
09340     // property members:
09341     QSharedPointer<QCPGraphDataContainer> mDataContainer;
09342     LineStyle mLineStyle;
09343     QCPScatterStyle mScatterStyle;
09344     QString mName;
09345     bool mAntialiasedFill, mAntialiasedScatters;
09346     QPen mPen;
09347     QBrush mBrush;
09348     bool mPeriodic;
09349     QPointer<QCPPolarAxisAngular> mKeyAxis;
09350     QPointer<QCPPolarAxisRadial> mValueAxis;
09351     QCP::SelectionType mSelectable;
09352     QCPDataSelection mSelection;
09353     //QCPSelectionDecorator *mSelectionDecorator;
09354
09355     // introduced virtual methods (later reimplemented TODO from QCPAbstractPolarPlottable):
09356     virtual QRect clipRect() const;
09357     virtual void draw(QCPPainter *painter);
09358     virtual QCP::Interaction selectionCategory() const;
09359     void applyDefaultAntialiasingHint(QCPPainter *painter) const;
09360     // events:
09361     virtual void selectEvent(QMouseEvent *event, bool additive, const QVariant &details, bool
      *selectionStateChanged);
09362     virtual void deselectEvent(bool *selectionStateChanged);
09363     // virtual drawing helpers:
09364     virtual void drawLinePlot(QCPPainter *painter, const QVector<QPointF> &lines) const;
09365     virtual void drawFill(QCPPainter *painter, QVector<QPointF> *lines) const;
09366     virtual void drawScatterPlot(QCPPainter *painter, const QVector<QPointF> &scatters, const
      QCPScatterStyle &style) const;
09367
09368     // introduced virtual methods:
09369     virtual void drawLegendIcon(QCPPainter *painter, const QRectF &rect) const;
```

```
09370
09371     // non-virtual methods:
09372     void applyFillAntialiasingHint(QCPPainter *painter) const;
09373     void applyScattersAntialiasingHint(QCPPainter *painter) const;
09374     double pointDistance(const QPointF &pixelPoint, QCPGraphDataContainer::const_iterator
      &closestData) const;
09375     // drawing helpers:
09376     virtual int dataCount() const;
09377     void getDataSegments(QList<QCPDataRange> &selectedSegments, QList<QCPDataRange>
      &unselectedSegments) const;
09378     void drawPolyline(QCPPainter *painter, const QVector<QPointF> &lineData) const;
09379     void getVisibleDataBounds(QCPGraphDataContainer::const_iterator &begin,
      QCPGraphDataContainer::const_iterator &end, const QCPDataRange &rangeRestriction) const;
09380     void getLines(QVector<QPointF> *lines, const QCPDataRange &dataRange) const;
09381     void getScatters(QVector<QPointF> *scatters, const QCPDataRange &dataRange) const;
09382     void getOptimizedLineData(QVector<QCPGraphData> *lineData, const
      QCPGraphDataContainer::const_iterator &begin, const QCPGraphDataContainer::const_iterator &end) const;
09383     void getOptimizedScatterData(QVector<QCPGraphData> *scatterData,
      QCPGraphDataContainer::const_iterator begin, QCPGraphDataContainer::const_iterator end) const;
09384     QVector<QPointF> dataToLines(const QVector<QCPGraphData> &data) const;
09385
09386 private:
09387     Q_DISABLE_COPY(QCPPolarGraph)
09388
09389     friend class QCPPolarLegendItem;
09390 };
09391
09392 /* end of 'src/polar/polargraph.h' */
09393
09394
09395 #endif // QCUSTOMPLOT_H
09396
```

18.94 CppModelExporter.h

```
00001 #ifndef CPPMODELEXPORTER_H
00002 #define CPPMODELEXPORTER_H
00003
00004 #include <string>
00005
00006 class QPlainTextEdit;
00007 class Simulator;
00008
00009 // Document the service that generates C++ model representation text.
00026 class CppModelExporter {
00027 public:
00029     CppModelExporter(Simulator* simulator, QPlainTextEdit* cppCodeEditor);
00030
00035     std::string addCppCodeLine(const std::string& line, unsigned int indent = 0) const;
00037     void actualizeModelCppCode() const;
00038
00039 private:
00040     Simulator* _simulator;
00041     QPlainTextEdit* _cppCodeEditor;
00042 };
00043
00044 #endif // CPPMODELEXPORTER_H
```

18.95 EFSMNetworkDiagramCodec.h

```
00001 #ifndef EFSMNETWORKDIAGRAMCODEC_H
00002 #define EFSMNETWORKDIAGRAMCODEC_H
00003
00004 #include "EFSMNetworkDiagramLayout.h"
00005
00006 #include <QHash>
00007 #include <QStringList>
00008
00009 class QTextStream;
00010 class Simulator;
00011 class Model;
00012
00027 namespace EFSMNetworkDiagramCodec {
00028
00030 constexpr int kSupportedSchemaVersion = 1;
00031
00033 constexpr const char* kSectionName = "Graphical EFSM Network Diagrams";
00034
00041 void writeSection(QTextStream& out, Simulator* simulator,
00042                   const QHash<unsigned long, EFSMNetworkDiagramLayout>& layouts);
00043
00051 QHash<unsigned long, EFSMNetworkDiagramLayout> parseSection(const QStringList& lines, Model* model);
```

```
00052
00053 } // namespace EFSMNetworkDiagramCodec
00054
00055 #endif // EFSMNETWORKDIAGRAMCODEC_H
```

18.96 EFSMNetworkDiagramLayout.h

```
00001 #ifndef EFSMNETWORKDIAGRAMLAYOUT_H
00002 #define EFSMNETWORKDIAGRAMLAYOUT_H
00003
00004 #include <QPointF>
00005 #include <QHash>
00006 #include <QString>
00007 #include <QVector>
00008
00009 class Model;
00010
00029 struct EFSMNetworkDiagramLayout {
00030     struct StatePosition {
00031         unsigned long stateId = 0;
00032         QString stateName;
00033         QPointF position;
00034     };
00035
00036     struct TransitionLayout {
00037         QString sourceName;
00038         QString destinationName;
00039         QString transitionName;
00040         int ordinal = 0; // global EFSMNetwork transition-list index
00041         QVector<QPointF> controlPoints;
00042         bool hasLabelPosition = false;
00043         QPointF labelPosition;
00044     };
00045
00046     unsigned long networkId = 0;
00047     QString networkName;
00048     QVector<StatePosition> states;
00049     QVector<TransitionLayout> transitions;
00050
00051     bool isEmpty() const { return states.isEmpty() && transitions.isEmpty(); }
00052 };
00053
00054 using EFSMNetworkDiagramLayoutsByModel =
00055     QHash<Model*, QHash<unsigned long, EFSMNetworkDiagramLayout»;
00056
00057 #endif // EFSMNETWORKDIAGRAMLAYOUT_H
```

18.97 EFSMNetworkDiagramProjection.h

```
00001 #ifndef EFSMNETWORKDIAGRAMPROJECTION_H
00002 #define EFSMNETWORKDIAGRAMPROJECTION_H
00003
00004 class ModelGraphicsScene;
00005 class Simulator;
00006 class EFSMNetwork;
00007 struct EFSMNetworkDiagramLayout;
00008
00009 namespace EFSMNetworkDiagramProjection {
00010
00018 void synchronize(Simulator* simulator, ModelGraphicsScene* scene);
00019
00021 bool captureLayout(EFSMNetwork* network, ModelGraphicsScene* scene, EFSMNetworkDiagramLayout* layout);
00022
00024 void restoreLayout(Simulator* simulator, EFSMNetwork* network, ModelGraphicsScene* scene,
00025                    const EFSMNetworkDiagramLayout& layout);
00026
00027 } // namespace EFSMNetworkDiagramProjection
00028
00029 #endif
```

18.98 GraphicalAutomaticPositioningStrategy.h

```
00001 #ifndef GRAPHICALAUTOMATICPOSITIONINGSTRATEGY_H
00002 #define GRAPHICALAUTOMATICPOSITIONINGSTRATEGY_H
00003
00004 #include <QPointF>
00005 #include <QRectF>
00006 #include <QSizeF>
00007
```

```
00008 #include <memory>
00009
00010 class GraphicalAutomaticPositioningStrategy {
00011 public:
00019     virtual ~GraphicalAutomaticPositioningStrategy() = default;
00020
00022     virtual QPointF initialComponentLayoutOrigin() const = 0;
00023
00035     virtual QPointF dataDefinitionPosition(const QRectF& anchorBounds,
00036                                            const QSizeF& childSize,
00037                                            int index,
00038                                            int count,
00039                                            bool upperArc,
00040                                            int radialLayer) const = 0;
00041 };
00042
00046 class GraphicalAutomaticPositioningStrategyFactory {
00047 public:
00049     static std::unique_ptr<GraphicalAutomaticPositioningStrategy> createFromPreferences();
00050 };
00051
00052 #endif // GRAPHICALAUTOMATICPOSITIONINGSTRATEGY_H
```

18.99 GraphicalDataDefinitionLayout.h

```
00001 #ifndef GRAPHICALDATADEFINITIONLAYOUT_H
00002 #define GRAPHICALDATADEFINITIONLAYOUT_H
00003
00004 #include <QPointF>
00005 #include <QRectF>
00006 #include <QSizeF>
00007
00014 class GraphicalDataDefinitionLayout {
00015 public:
00027     static QPointF arcPosition(const QRectF& anchorBounds,
00028                                const QSizeF& childSize,
00029                                int index,
00030                                int count,
00031                                bool upperArc,
00032                                int radialLayer = 0);
00033
00034 private:
00035     GraphicalDataDefinitionLayout() = default;
00036 };
00037
00038 #endif // GRAPHICALDATADEFINITIONLAYOUT_H
```

18.100 GraphicalModelBuilder.h

```
00001 #ifndef GRAPHICALMODELBUILDER_H
00002 #define GRAPHICALMODELBUILDER_H
00003
00004 #include <map>
00005 #include <string>
00006
00007 #include <QColor>
00008
00009 class Simulator;
00010 class QTextEdit;
00011 class ModelGraphicsView;
00012 class ModelGraphicsScene;
00013 class ModelComponent;
00014 class GraphicalModelComponent;
00015 class ModelDataDefinition;
00016 class GraphicalModelDataDefinition;
00017 template<typename T> class List;
00018
00019 // Document the service that rebuilds graphical scene artifacts from kernel models.
00037 class GraphicalModelBuilder {
00038 public:
00040     GraphicalModelBuilder(Simulator* simulator,
00041                           ModelGraphicsView* graphicsView,
00042                           ModelGraphicsScene* scene,
00043                           std::map<std::string, QColor>* pluginCategoryColor,
00044                           QTextEdit* console);
00045
00052     void setModelLevelFilter(unsigned int modelLevel);
00054     void clearModelLevelFilter();
00055
00059     void recursivalyGenerateGraphicalModelFromModel(ModelComponent* component,
00060                                                     List<ModelComponent*>* visited,
```

```
00061                                                   std::map<ModelComponent*,
      GraphicalModelComponent*>* map,
00062                                                   int* x,
00063                                                   int* y,
00064                                                   int* ymax,
00065                                                   int sequenceInLine);
00066
00068     void generateGraphicalModelFromModel();
00076     static void synchronizeGraphicalDataDefinitionsLayer(Simulator* simulator, ModelGraphicsScene*
      scene);
00077
00078 private:
00079     void rebuildGraphicalDataDefinitionsLayer(std::map<ModelComponent*, GraphicalModelComponent*>*
      componentMap);
00080     bool componentBelongsToActiveModelLevel(ModelComponent* component) const;
00081
00082     Simulator* _simulator;
00083     ModelGraphicsView* _graphicsView;
00084     ModelGraphicsScene* _scene;
00085     std::map<std::string, QColor>* _pluginCategoryColor;
00086     QTextEdit* _console;
00087 };
00088
00089 #endif // GRAPHICALMODELBUILDER_H
```

18.101 GraphicalModelSerializer.h

```
00001 #ifndef GRAPHICALMODELSERIALIZER_H
00002 #define GRAPHICALMODELSERIALIZER_H
00003
00004 #include "EFSMNetworkDiagramLayout.h"
00005
00006 #include <QHash>
00007
00008 #include <functional>
00009 #include <memory>
00010 #include <string>
00011
00012 class Simulator;
00013 class Model;
00014 class QWidget;
00015 class QFile;
00016 class QString;
00017 class QPlainTextEdit;
00018 class QTextEdit;
00019 class QSlider;
00020 class QAction;
00021 class ModelGraphicsView;
00022
00023 // Document persistence responsibilities for textual and graphical GUI model data.
00041 class GraphicalModelSerializer {
00042 public:
00044     GraphicalModelSerializer(Simulator* simulator,
00045                              QWidget* ownerWidget,
00046                              QPlainTextEdit* modelTextEditor,
00047                              ModelGraphicsView* graphicsView,
00048                              QSlider* zoomSlider,
00049                              QAction* actionShowGrid,
00050                              QAction* actionShowRule,
00051                              QAction* actionShowSnap,
00052                              QAction* actionShowGuides,
00053                              QAction* actionShowInternalElements,
00054                              QAction* actionShowEditableElements,
00055                              QAction* actionShowAttachedElements,
00056                              QAction* actionShowRecursiveElements,
00057                              QTextEdit* console,
00058                              QString* modelFilename,
00059                              std::function<void()> clearModelEditors,
00060                              std::function<void()> rebuildGraphicalModelFromModel,
00061                              std::function<void()> applyShowInternalElements,
00062                              std::function<void()> applyShowEditableElements,
00063                              std::function<void()> applyShowAttachedElements,
00064                              std::shared_ptr<EFSMNetworkDiagramLayoutsByModel>
      efsmNetworkDiagramLayouts = nullptr);
00065
00070     bool saveTextModel(QFile* saveFile, const QString& data) const;
00075     bool saveGraphicalModel(const QString& filename) const;
00080     Model* loadGraphicalModel(const std::string& filename) const;
00081
00088     void setEFSMNetworkDiagramLayout(Model* model, unsigned long networkId, const
      EFSMNetworkDiagramLayout& layout);
00093     EFSMNetworkDiagramLayout efsmNetworkDiagramLayout(Model* model, unsigned long networkId) const;
00095     void forgetEFSMNetworkLayouts(Model* model);
00096
00097 private:
```

```
00098     // Encode free text fields into a tab-safe persisted value.
00099     static QString encodeGuiText(const QString& text);
00100     // Decode tab-safe persisted text into original UTF-8 value.
00101     static QString decodeGuiText(const QString& text);
00102
00103 private:
00104     Simulator* _simulator;
00105     QWidget* _ownerWidget;
00106     QPlainTextEdit* _modelTextEditor;
00107     ModelGraphicsView* _graphicsView;
00108     QSlider* _zoomSlider;
00109     QAction* _actionShowGrid;
00110     QAction* _actionShowRule;
00111     QAction* _actionShowSnap;
00112     QAction* _actionShowGuides;
00113     QAction* _actionShowInternalElements;
00114     QAction* _actionShowEditableElements;
00115     QAction* _actionShowAttachedElements;
00116     QAction* _actionShowRecursiveElements;
00117     QTextEdit* _console;
00118     QString* _modelFilename;
00119     std::function<void()> _clearModelEditors;
00120     std::function<void()> _rebuildGraphicalModelFromModel;
00121     std::function<void()> _applyShowInternalElements;
00122     std::function<void()> _applyShowEditableElements;
00123     std::function<void()> _applyShowAttachedElements;
00124
00125     // GUI-only EFSM diagram layouts keyed by root model and EFSMNetwork id. Mutable because
00126     // Shared with MainWindow because serializer instances are rebuilt when the
00127     // active graphics view changes. loadGraphicalModel() refreshes this cache
00128     // from the persisted GUI section without retaining widget state.
00129     std::shared_ptr<EFSMNetworkDiagramLayoutsByModel> _efsmNetworkDiagramLayouts;
00130 };
00131
00132 #endif // GRAPHICALMODELSERIALIZER_H
```

18.102 GraphvizModelExporter.h

```
00001 #ifndef GRAPHVIZMODELEXPORTER_H
00002 #define GRAPHVIZMODELEXPORTER_H
00003
00004 #include <functional>
00005 #include <list>
00006 #include <map>
00007 #include <string>
00008
00009 class QLabel;
00010 class QCheckBox;
00011 class Simulator;
00012 class ModelDataDefinition;
00013
00014 // Document the service used to export graphical model representations via Graphviz.
00032 class GraphvizModelExporter {
00033 public:
00037     GraphvizModelExporter(Simulator* simulator,
00038                           QLabel* modelGraphicLabel,
00039                           QCheckBox* showStatistics,
00040                           QCheckBox* showEditable,
00041                           QCheckBox* showShared,
00042                           QCheckBox* showRecursive,
00043                           QCheckBox* showLevels,
00044                           std::function<bool()> ensureModelSynchronized);
00045
00051     std::string adjustDotName(std::string name) const;
00052
00056     void insertTextInDot(std::string text,
00057                          unsigned int compLevel,
00058                          unsigned int compRank,
00059                          std::map<unsigned int, std::map<unsigned int, std::list<std::string>*>*>*
      dotmap,
00060                          bool isNode = false) const;
00064     void recursiveCreateModelGraphicPicture(ModelDataDefinition* componentOrData,
00065                                             std::list<ModelDataDefinition*>* visited,
00066                                             std::map<unsigned int, std::map<unsigned int,
      std::list<std::string>*>*>* dotmap) const;
00067
00072     bool createModelImage() const;
00073
00074 private:
00075     Simulator* _simulator;
00076     QLabel* _modelGraphicLabel;
00077     QCheckBox* _showStatistics;
00078     QCheckBox* _showEditable;
00079     QCheckBox* _showShared;
00080     QCheckBox* _showRecursive;
```

```
00081     QCheckBox* _showLevels;
00082     // Keep synchronization dependency narrow and explicit.
00083     std::function<bool()> _ensureModelSynchronized;
00084 };
00085
00086 #endif // GRAPHVIZMODELEXPORTER_H
```

18.103 IssueReportRelayService.h

```
00001 #ifndef ISSUEREPORTRELAYSERVICE_H
00002 #define ISSUEREPORTRELAYSERVICE_H
00003
00004 #include <QByteArray>
00005 #include <QString>
00006
00010 struct IssueReportDraft {
00011     QString categoryKey;
00012     QString categoryLabel;
00013     QString title;
00014     QString summary;
00015     QString reproductionSteps;
00016     QString expectedBehavior;
00017     QString observedBehavior;
00018     QString additionalContext;
00019     QString contactEmail;
00020     bool includeScreenshot = false;
00021     bool includeLogTail = true;
00022     bool includeModelText = false;
00023 };
00024
00028 struct IssueReportRuntimeContext {
00029     QString applicationName;
00030     QString applicationVersion;
00031     QString qtVersion;
00032     QString operatingSystem;
00033     QString currentModelName;
00034     QString currentModelSimulanguage;
00035     QString consoleTail;
00036     QString simulationTail;
00037     QString reportsTail;
00038     QString processLogTail;
00039     QByteArray screenshotPngBase64;
00040 };
00041
00045 struct IssueReportSubmissionResult {
00046     bool success = false;
00047     int httpStatus = 0;
00048     int issueNumber = 0;
00049     QString issueUrl;
00050     QString errorMessage;
00051     QString responseBody;
00052 };
00053
00067 class IssueReportRelayService {
00068 public:
00069     static QString configuredEndpoint();
00070     static QString optionalClientKey();
00071     static int requestTimeoutMs();
00072     static bool isEndpointConfigured(QString* reason = nullptr);
00073     static IssueReportSubmissionResult submit(const IssueReportDraft& draft,
00074                                               const IssueReportRuntimeContext& runtimeContext);
00075 };
00076
00077 #endif // ISSUEREPORTRELAYSERVICE_H
```

18.104 ModelLanguageSynchronizer.h

```
00001 #ifndef MODELLANGUAGESYNCHRONIZER_H
00002 #define MODELLANGUAGESYNCHRONIZER_H
00003
00004 #include <functional>
00005
00006 class QWidget;
00007 class QPlainTextEdit;
00008 class Simulator;
00009
00010 // Document the service boundary for textual model representation synchronization.
00028 class ModelLanguageSynchronizer {
00029 public:
00033     ModelLanguageSynchronizer(Simulator* simulator,
00034                               QPlainTextEdit* modelTextEditor,
00035                               bool* textModelHasChangedFlag,
```

```
00036                               QWidget* ownerWidget,
00037                               std::function<void()> onModelCreatedOrLoaded);
00038
00044     void actualizeModelSimLanguage() const;
00045
00050     bool setSimulationModelBasedOnText() const;
00051
00052 private:
00053     Simulator* _simulator;
00054     QPlainTextEdit* _modelTextEditor;
00055     bool* _textModelHasChangedFlag;
00056     QWidget* _ownerWidget;
00057     std::function<void()> _onModelCreatedOrLoaded;
00058 };
00059
00060 #endif // MODELLANGUAGESYNCHRONIZER_H
```

18.105 ToolLauncher.h

```
00001 #pragma once
00002
00003 #include <QString>
00004 #include <QStringList>
00005
00006 class QWidget;
00007
00008 namespace ToolLauncher {
00009
00010 bool launchDetached(QWidget* parent,
00011                     const QString& executableBaseName,
00012                     const QString& displayName,
00013                     const QStringList& arguments = {},
00014                     const QStringList& relativeSearchDirs = {});
00015
00016 }  // namespace ToolLauncher
```

18.106 systempreferences.h

```
00001 #ifndef SYSTEMPREFERENCES_H
00002 #define SYSTEMPREFERENCES_H
00003
00004 #include "kernel/simulator/TraceManager.h"
00005
00006 #include <QString>
00007 #include <string>
00008 #include <vector>
00009
00017 class SystemPreferences {
00018 public:
00022     enum class StartupModelMode : unsigned short {
00023         NoModel = 0,
00024         NewModel = 1,
00025         OpenSpecificModel = 2,
00026         OpenLastModel = 3
00027     };
00028
00032     enum class VisualTheme {
00033         Light,
00034         Dark
00035     };
00036
00040     enum class InterfaceStyle {
00041         Classic,
00042         Modern
00043     };
00044
00048     enum class AutomaticPositioningStrategy {
00049         Legacy = 0,
00050         Centered = 1
00051     };
00052
00054     static bool load();
00056     static bool save();
00058     static void resetToDefaults();
00060     static QString configFilePath();
00061
00063     static bool startMaximized();
00065     static void setStartMaximized(bool newStartMaximized);
00066
00068     static bool autoLoadPlugins();
00070     static void setAutoLoadPlugins(bool newAutoLoadPlugins);
00071
```

```
00073     static StartupModelMode startupModelMode();
00075     static void setStartupModelMode(StartupModelMode newStartupModelMode);
00077     static unsigned short modelAtStart();
00079     static void setModelAtStart(unsigned short newModelAtStart);
00080
00082     static std::string modelfilename();
00084     static void setModelfilename(const std::string &newModelfilename);
00086     static std::string lastModelFilename();
00088     static void setLastModelFilename(const std::string& newLastModelFilename);
00090     static std::vector<std::string> recentModelFiles();
00092     static void setRecentModelFiles(const std::vector<std::string>& recentModelFiles);
00094     static void pushRecentModelFile(const std::string& modelFilename);
00096     static unsigned int recentModelFilesLimit();
00098     static void setRecentModelFilesLimit(unsigned int recentModelFilesLimit);
00099
00101     static bool checkSystemPackagesAtStart();
00103     static void setCheckSystemPackagesAtStart(bool newCheckSystemPackagesAtStart);
00104
00106     static TraceManager::Level traceLevel();
00108     static void setTraceLevel(TraceManager::Level newTraceLevel);
00109
00111     static VisualTheme visualTheme();
00113     static void setVisualTheme(VisualTheme newVisualTheme);
00115     static std::string visualThemeName();
00116
00118     static InterfaceStyle interfaceStyle();
00120     static void setInterfaceStyle(InterfaceStyle newInterfaceStyle);
00122     static std::string interfaceStyleName();
00123
00125     static int applicationFontPointSize();
00127     static void setApplicationFontPointSize(int newApplicationFontPointSize);
00128
00130     static bool diagramUsesThemeColors();
00132     static void setDiagramUsesThemeColors(bool newDiagramUsesThemeColors);
00134     static AutomaticPositioningStrategy automaticPositioningStrategy();
00136     static void setAutomaticPositioningStrategy(AutomaticPositioningStrategy strategy);
00137
00139     static unsigned int canvasBackgroundColor();
00141     static unsigned int canvasDisabledBackgroundColor();
00143     static unsigned int gridColor();
00144
00145 private:
00146     SystemPreferences(){};
00147
00148     static QString defaultConfigDirectory();
00149     static StartupModelMode startupModelModeFromString(const QString& value);
00150     static QString startupModelModeToString(StartupModelMode value);
00151     static VisualTheme visualThemeFromString(const QString& value);
00152     static QString visualThemeToString(VisualTheme value);
00153     static InterfaceStyle interfaceStyleFromString(const QString& value);
00154     static QString interfaceStyleToString(InterfaceStyle value);
00155     static AutomaticPositioningStrategy automaticPositioningStrategyFromString(const QString& value);
00156     static QString automaticPositioningStrategyToString(AutomaticPositioningStrategy value);
00157     static TraceManager::Level traceLevelFromInt(int value);
00158
00159     static bool _startMaximized;
00160     static bool _autoLoadPlugins;
00161     static bool _checkSystemPackagesAtStart;
00162     static StartupModelMode _startupModelMode;
00163     static std::string _modelfilenameToOpen;
00164     static std::string _lastModelFilename;
00165     static std::vector<std::string> _recentModelFiles;
00166     static unsigned int _recentModelFilesLimit;
00167     static TraceManager::Level _traceLevel;
00168     static VisualTheme _visualTheme;
00169     static InterfaceStyle _interfaceStyle;
00170     static int _applicationFontPointSize;
00171     static bool _diagramUsesThemeColors;
00172     static AutomaticPositioningStrategy _automaticPositioningStrategy;
00173 };
00174
00175 #endif // SYSTEMPREFERENCES_H
```

18.107 DataAnalyzerWindow.h

```
00001 #ifndef DATAANALYZERWINDOW_H
00002 #define DATAANALYZERWINDOW_H
00003
00004 #include <QList>
00005 #include <QMainWindow>
00006 #include <QString>
00007 #include <QStringList>
00008
00009 class QGroupBox;
00010 class QCheckBox;
```

```
00011 class QComboBox;
00012 class QDoubleSpinBox;
00013 class QLabel;
00014 class QListWidget;
00015 class QPlainTextEdit;
00016 class QSpinBox;
00017 class QSplitter;
00018 class QTableWidget;
00019 class QTabWidget;
00020 class QTextEdit;
00021 class QTextBrowser;
00022 class Simulator;
00023
00024 class DataAnalyzerWindow : public QMainWindow {
00025 public:
00026     explicit DataAnalyzerWindow(Simulator* simulator, QString& lastDataAnalyzerPath, QWidget* parent =
      nullptr);
00027
00028 private:
00029     struct DataSummary {
00030         int count = 0;
00031         double min = 0.0;
00032         double max = 0.0;
00033         double mean = 0.0;
00034         double median = 0.0;
00035         QString mode;
00036         double firstQuartile = 0.0;
00037         double thirdQuartile = 0.0;
00038         double variance = 0.0;
00039         double stddev = 0.0;
00040         double cv = 0.0;
00041         double kurtosis = 0.0;
00042     };
00043
00044     struct DatasetObservation {
00045         int replication = 1;
00046         double time = 0.0;
00047         bool hasTime = false;
00048         double value = 0.0;
00049     };
00050
00051     struct DatasetDescriptor {
00052         QString datasetName;
00053         QString randomVariableName;
00054         QString description;
00055         QString variableType;
00056         QString source;
00057         QStringList previewLines;
00058         QList<double> values;
00059         bool raw = true;
00060         QList<DatasetObservation> observations;
00061         bool recordFile = false;
00062         bool timeDependent = false;
00063         QString formatKind;
00064         QString expression;
00065         QString expressionName;
00066         QString formatVersion;
00067     };
00068
00069     class HistogramPreview;
00070     class DoePreviewPlot;
00071
00072     void buildMenus();
00073     void buildWorkspace();
00074     void connectActions();
00075     void refreshModelResponses();
00076     void openDataset();
00077     void importSimulationResponsesSnapshot();
00078     void loadDatasetFromFile(const QString& fileName);
00079     void addDataset(const DatasetDescriptor& dataset);
00080     void filterDataset();
00081     void refreshDatasetList();
00082     void refreshScopeSelector();
00083     void showAnalysisSetupDialog(const QString& analysisName, int targetTabIndex);
00084     void refreshAnalysisViews();
00085     QList<double> scopedValues() const;
00086     QList<DatasetObservation> scopedObservations() const;
00087     QString scopedDatasetLabel() const;
00088     QString scopedDatasetDescription() const;
00089     QStringList scopedPreviewLines() const;
00090     void refreshStudySummaryView();
00091     void refreshDescriptiveView(const DataSummary& summary);
00092     void refreshDistributionFitView(const DataSummary& summary);
00093     void refreshInferenceView(const DataSummary& summary);
00094     void refreshReportView(const DataSummary& summary, const QString& fitConclusion);
00095     void refreshDoeDemoViews();
00096     void refreshDoePlanSummary();
```

```
00097     void exportDataset();
00098     void saveReport();
00099     void showSkeletonMessage(const QString& featureName);
00100
00101     static DataSummary summarizeValues(const QList<double>& values);
00102     static QList<double> sortedValues(const QList<double>& values);
00103     static double percentile(const QList<double>& sorted, double probability);
00104     static QString exactMode(const QList<double>& values);
00105     static QString formatNumber(double value);
00106
00107 private:
00108     Simulator* _simulator = nullptr;
00109     QString& _lastDataAnalyzerPath;
00110     QString _studyName;
00111     QList<DatasetDescriptor> _datasets;
00112     QString _lastFitConclusion;
00113
00114     QAction* _openDatasetAction = nullptr;
00115     QAction* _saveReportAction = nullptr;
00116     QAction* _importSimulationResponsesAction = nullptr;
00117     QAction* _refreshModelResponsesAction = nullptr;
00118     QAction* _exportDatasetAction = nullptr;
00119
00120     QListWidget* _datasetsList = nullptr;
00121     QListWidget* _analysisNavigator = nullptr;
00122     QTableWidget* _modelResponsesTable = nullptr;
00123     QTabWidget* _workspaceTabs = nullptr;
00124     QComboBox* _scopeCombo = nullptr;
00125     QLabel* _datasetSourceLabel = nullptr;
00126     QPlainTextEdit* _datasetPreview = nullptr;
00127     QTableWidget* _studySummaryTable = nullptr;
00128     QTableWidget* _centralTendencyTable = nullptr;
00129     QTableWidget* _dispersionTable = nullptr;
00130     QTableWidget* _rawDataTable = nullptr;
00131     QTableWidget* _movingAverageTable = nullptr;
00132     HistogramPreview* _histogramPreview = nullptr;
00133     QTableWidget* _fitTable = nullptr;
00134     QLabel* _fitConclusionLabel = nullptr;
00135     QTableWidget* _inferenceTable = nullptr;
00136     QTableWidget* _doeFactorsTable = nullptr;
00137     QTableWidget* _doeResponsesTable = nullptr;
00138     QTableWidget* _doeDesignTable = nullptr;
00139     QTableWidget* _doeAnovaTable = nullptr;
00140     QTableWidget* _doeOptimizationTable = nullptr;
00141     QTableWidget* _doeAdviceTable = nullptr;
00142     QTableWidget* _doeDesignQualityTable = nullptr;
00143     QTableWidget* _doeDesirabilityTable = nullptr;
00144     QComboBox* _doeDesignFamilyCombo = nullptr;
00145     QComboBox* _doeModelOrderCombo = nullptr;
00146     QComboBox* _doePlotSelector = nullptr;
00147     QSpinBox* _doeReplicationsSpin = nullptr;
00148     QDoubleSpinBox* _doeAlphaSpin = nullptr;
00149     QCheckBox* _doeRandomizeCheck = nullptr;
00150     QLabel* _doeAdvisorLabel = nullptr;
00151     QLabel* _doePlanSummaryLabel = nullptr;
00152     DoePreviewPlot* _doePreviewPlot = nullptr;
00153     QTextBrowser* _startPage = nullptr;
00154     QTextEdit* _reportText = nullptr;
00155 };
00156
00157 #endif // DATAANALYZERWINDOW_H
```

18.108 ExpressionBuilder.h

```
00001 #ifndef EXPRESSIONBUILDER_H
00002 #define EXPRESSIONBUILDER_H
00003
00004 #include <QDialog>
00005 #include <QTreeWidget>
00006 #include <QListWidget>
00007 #include <QFormLayout>
00008 #include <QLineEdit>
00009 #include <QPushButton>
00010 #include <QClipboard>
00011 #include <QApplication>
00012 #include <QVBoxLayout>
00013 #include <QHBoxLayout>
00014 #include <QLabel>
00015 #include <QString>
00016 #include <QMap>
00017 #include <QList>
00018
00019 namespace Ui {
00020 class ExpressionBuilder;
00021 }
```

```
00022
00023 struct ExpressionParameter {
00024     QString name;
00025     QString defaultValue;
00026     QString description;
00027 };
00028
00029 struct ExpressionTemplate {
00030     QString name;
00031     QString pattern;
00032     QString description;
00033     QList<ExpressionParameter> parameters;
00034 };
00035
00036 struct ExpressionCategory {
00037     QString name;
00038     QList<ExpressionTemplate> expressions;
00039 };
00040
00041 class ExpressionBuilder : public QDialog
00042 {
00043     Q_OBJECT
00044
00045 public:
00046     explicit ExpressionBuilder(QWidget *parent = nullptr);
00047     ~ExpressionBuilder();
00048
00049 private slots:
00050     void on_categorySelected(QTreeWidgetItem *item, int column);
00051     void on_expressionSelected(QListWidgetItem *item);
00052     void on_copyToClipboard_clicked();
00053     void on_close_clicked();
00054
00055 private:
00056     Ui::ExpressionBuilder *ui;
00057
00058     // Expression data
00059     QList<ExpressionCategory> categories;
00060
00061     // UI Components
00062     QTreeWidget *treeCategories;
00063     QListWidget *listExpressions;
00064     QWidget *parameterWidget;
00065     QFormLayout *parameterLayout;
00066     QLabel *expressionPreview;
00067     QPushButton *btnCopyToClipboard;
00068     QPushButton *btnClose;
00069
00070     // Parameter inputs
00071     QMap<QString, QLineEdit*> parameterInputs;
00072
00073     // Current selection
00074     ExpressionTemplate *currentExpression;
00075
00076     // Initialize expression categories
00077     void initializeCategories();
00078     void updateParameterForm();
00079     void updateExpressionPreview();
00080     QString buildExpression();
00081
00082     // Extensibility: Register new categories and expressions
00083     void registerCategory(const ExpressionCategory &category);
00084     void registerExpression(const QString &categoryName, const ExpressionTemplate &expression);
00085 };
00086
00087 #endif // EXPRESSIONBUILDER_H
```

18.109 OptimizerWindow.h

```
00001 #ifndef OPTIMIZERWINDOW_H
00002 #define OPTIMIZERWINDOW_H
00003
00004 #include "tools/Optimization/OptimizerDefaultImpl1.h"
00005
00006 #include <QList>
00007 #include <QMainWindow>
00008 #include <QString>
00009 #include <QVector>
00010
00011 #include <vector>
00012
00013 class QAction;
00014 class QCheckBox;
00015 class QComboBox;
00016 class QDoubleSpinBox;
```

```
00017 class QLabel;
00018 class QPlainTextEdit;
00019 class QPushButton;
00020 class QSpinBox;
00021 class QTableWidget;
00022 class QTableWidgetItem;
00023 class QTabWidget;
00024 class QTextBrowser;
00025 class Simulator;
00026 class SimulationControl;
00027 class SimulationResponse;
00028
00029 class OptimizerWindow : public QMainWindow {
00030 public:
00031     explicit OptimizerWindow(Simulator* simulator, QWidget* parent = nullptr);
00032
00033 private:
00034     class ProgressPlot;
00035
00036     void buildMenus();
00037     void buildWorkspace();
00038     void connectActions();
00039     void refreshFromCurrentModel();
00040     void refreshControlsTable();
00041     void refreshResponsesTable();
00042     void refreshReport();
00043     void applyConfigurationToBackend();
00044     void refreshRunState();
00045     void addObjective();
00046     void removeSelectedObjective();
00047     void addConstraint();
00048     void removeSelectedConstraint();
00049     void configureTechnique();
00050     void checkReadiness();
00051     void startOptimization();
00052     void stepOptimization();
00053     void pauseOptimization();
00054     void resumeOptimization();
00055     void stopOptimization();
00056     void showSkeletonMessage(const QString& featureName);
00057
00058     static QString formatNumber(double value);
00059     static QString stateText(Optimizer_if::ExecutionState state);
00060     static QString techniqueDescription(const QString& technique);
00061     static bool stringIsNumeric(const QString& value);
00062     static QTableWidgetItem* readOnlyItem(const QString& text);
00063     static QTableWidgetItem* editableItem(const QString& text);
00064     static QTableWidgetItem* checkableItem(bool checked, bool enabled = true);
00065
00066 private:
00067     Simulator* _simulator = nullptr;
00068     OptimizerDefaultImpl1 _optimizer;
00069     std::vector<SimulationControl*> _controls;
00070     std::vector<SimulationResponse*> _responses;
00071     QVector<double> _progressValues;
00072
00073     QAction* _refreshModelAction = nullptr;
00074     QAction* _configureTechniqueAction = nullptr;
00075     QAction* _checkAction = nullptr;
00076     QAction* _startAction = nullptr;
00077     QAction* _stepAction = nullptr;
00078     QAction* _pauseAction = nullptr;
00079     QAction* _resumeAction = nullptr;
00080     QAction* _stopAction = nullptr;
00081
00082     QTabWidget* _tabs = nullptr;
00083     QLabel* _modelStatusLabel = nullptr;
00084     QLabel* _runStatusLabel = nullptr;
00085     QTableWidget* _controlsTable = nullptr;
00086     QTableWidget* _responsesTable = nullptr;
00087     QTableWidget* _objectivesTable = nullptr;
00088     QTableWidget* _constraintsTable = nullptr;
00089     QTableWidget* _bestSolutionsTable = nullptr;
00090     QComboBox* _techniqueCombo = nullptr;
00091     QSpinBox* _maxIterationsSpin = nullptr;
00092     QSpinBox* _maxSimulationsSpin = nullptr;
00093     QSpinBox* _replicationsSpin = nullptr;
00094     QSpinBox* _bestSolutionsSpin = nullptr;
00095     QSpinBox* _randomSeedSpin = nullptr;
00096     QSpinBox* _populationSizeSpin = nullptr;
00097     QDoubleSpinBox* _mutationRateSpin = nullptr;
00098     QDoubleSpinBox* _crossoverRateSpin = nullptr;
00099     QDoubleSpinBox* _improvementToleranceSpin = nullptr;
00100     QDoubleSpinBox* _timeLimitSpin = nullptr;
00101     QPlainTextEdit* _techniqueNotes = nullptr;
00102     QTextBrowser* _overviewText = nullptr;
00103     QTextBrowser* _reportText = nullptr;
```

```
00104     ProgressPlot* _progressPlot = nullptr;
00105 };
00106
00107 #endif // OPTIMIZERWINDOW_H
```

18.110 TraitsGUI.h

```
00001 #ifndef TRAITSGUI_H
00002 #define TRAITSGUI_H
00003
00004 #include <cstdint>
00005 #include <string>
00006
00013 template <class T>
00014 struct TraitsGUI {
00015 };
00016
00017 // include only features for this GUI, such as theme, default widgets, ...
00018
00019 class GMainWindow;
00020
00021 template <>
00022 struct TraitsGUI<GMainWindow> {
00023     static const bool startMaximized = true;
00024     static const int zoomButtonChange = 30;
00025     static const uint64_t consoleTextColor = 0x008000FF;
00026 };
00027
00028 class GModelGraphicPic;
00029
00030 template <>
00031 struct TraitsGUI<GModelGraphicPic> {
00032     static const int nodeComponentFontSize = 12;
00033     static const uint64_t nodeComponentFontColor = 0x000000FF;
00034     static const uint64_t nodeComponentFillColor = 0x80A040FF;
00035     static const int edgeComponentFontSize = 7;
00036     static const uint64_t edgeComponentEdgeColor = 0x000000FF;
00037     static const uint64_t edgeComponentFontColor = 0x000000FF;
00038     static const int nodeDatadefInternalFontSize = 8;
00039     static const uint64_t nodeDatadefInternalColor = 0xA0A0A0FF;
00040     static const uint64_t nodeDatadefInternalFontColor = 0xA0A0A0FF;
00041     static const int nodeDatadefAttachedFontSize = 8;
00042     static const uint64_t nodeDatadefAttachedFontColor = 0x808080FF;
00043 };
00044
00045 class GAssociation;
00046
00047 template <>
00048 struct TraitsGUI<GAssociation> {
00049     static const uint64_t foreColor = 0x808080FF;
00050     constexpr static const int margin = 2;
00051     constexpr static const int selectionWidth = 8;
00052 };
00053
00054 class GConnection;
00055
00056 template <>
00057 struct TraitsGUI<GConnection> {
00058     static const uint64_t foreColor = 0x000000FF;
00059     constexpr static const int margin = 2;
00060     constexpr static const int selectionWidth = 8;
00061 };
00062
00063 class GView;
00064
00065 template <>
00066 struct TraitsGUI<GView> {
00067     constexpr static const int sceneCenter = 25000;
00068     constexpr static const int sceneDistanceCenter = 25000;
00069     static const uint64_t backgroundEnabledColor = 0xFFFF8040;
00070     static const uint64_t backgroundDisabledColor = 0xC0C0C020;
00071 };
00072
00073 class GScene;
00074
00075 template <>
00076 struct TraitsGUI<GScene> {
00077     constexpr static const int gridInterval = 20;
00078     constexpr static const int gridPenWidth = 1;
00079     static const uint64_t gridColor = 0xC0C0C0E0;
00080 };
00081
00082 class GComponentPort;
00083
00084 template <>
```

```
00085 struct TraitsGUI<GComponentPort> {
00086     constexpr static const int width = 20;
00087     constexpr static const int penWidth = 1;
00088     constexpr static const int raise = 3;
00089     constexpr static const int margin = 2;
00090     static const uint64_t connectedBackgroundColor = 0x2020FF80;
00091     static const uint64_t unconnectedBackgroundColor = 0xFF202040;
00092     static const uint64_t borderColor = 0x000000FF;
00093     static const uint64_t pathRaised = 0x808080FF;
00094     static const uint64_t pathStunken = 0xFFFFFFFF;
00095     static const uint64_t textColor = 0xFFFFFFFF;
00096     static const uint64_t textShadowColor = 0x000000FF;
00097     static const uint64_t selectionSquaresColor = 0x000000FF;
00098 };
00099
00100 class GModelDataDefinition;
00101
00102 template <>
00103 struct TraitsGUI<GModelDataDefinition> {
00104     //<GraphicalModelComponent> {
00105     constexpr static const int opacity = 256 * 0.33;
00106     constexpr static const int nonEditableOpacity = 58;
00107     constexpr static const int selectedNonEditableOpacity = 72;
00108     constexpr static const int width = 100;
00109     constexpr static const int penWidth = 1;
00110     constexpr static const int raise = 3;
00111     constexpr static const int margin = 5;
00112     constexpr static const int selectionWidth = 8;
00113     constexpr static const float stretchPos = 0.5;
00114     constexpr static const float stretch = 0.0;
00115     constexpr static const float heightProportion = 0.50;
00116     static const uint64_t breakpointColor = 0xFF0000FF;
00117     static const uint64_t borderColor = 0x000000FF;
00118     static const uint64_t pathRaised = 0x808080FF;
00119     static const uint64_t pathStunken = 0xFFFFFFFF;
00120     static const uint64_t textColor = 0x0; // 0xFFFFFFFF;
00121     static const uint64_t textShadowColor = 0x000000FF;
00122     static const uint64_t selectionSquaresColor = 0x000000FF;
00123 };
00124
00125 class GModelComponent;
00126
00127 template <>
00128 struct TraitsGUI<GModelComponent> {
00129     //<GraphicalModelComponent> {
00130     constexpr static const int opacity = 256.0 * 0.67;
00131     constexpr static const int colorIncrease = 256.0 * 0.2;
00132     // 20% more some color component for sources, sinks, transfers
00133     constexpr static const int lighter = 133; // 33% lighter for some components
00134     constexpr static const int width = 150;
00135     constexpr static const int penWidth = 1;
00136     constexpr static const int raise = 5;
00137     constexpr static const int margin = 8;
00138     constexpr static const int selectionWidth = 8;
00139     constexpr static const float stretchPos = 0.5;
00140     constexpr static const float stretch = 0.0;
00141     constexpr static const float heightProportion = 0.67;
00142     static const uint64_t breakpointColor = 0xFF0000FF;
00143     static const uint64_t borderColor = 0x000000FF;
00144     static const uint64_t pathRaised = 0xFFFFFFFF;
00145     static const uint64_t pathStunken = 0x808080FF;
00146     static const uint64_t textColor = 0xFFFFFFFF;
00147     static const uint64_t textShadowColor = 0x000000FF;
00148     static const uint64_t selectionSquaresColor = 0x000000FF;
00149 };
00150
00151 #endif // TRAITSGUI_H
```

18.111 UtilGUI.h

```
00001 #ifndef UTILGUI_H
00002 #define UTILGUI_H
00003 #include <QColor>
00004 #include <cstdint>
00005 #include <sstream>
00006 #include <string>
00007
00008 namespace UtilGUI {
00009
00017 inline QColor rgbaFromPacked(uint64_t color) {
00018     uint8_t r = (color & 0xFF000000) >> 24;
00019     uint8_t g = (color & 0x00FF0000) >> 16;
00020     uint8_t b = (color & 0x0000FF00) >> 8;
00021     uint8_t a = (color & 0x000000FF);
00022     return QColor(r, g, b, a);
```

```
00023 }
00024
00032 inline std::string dotColorFromPacked(uint64_t color) {
00033     std::stringstream stream;
00034     stream « std::hex « "#" « color;
00035     return stream.str();
00036 }
00037
00038 }
00039
00040 #endif // UTILGUI_H
```

18.112 WebWorkerDialog.h

```
00001 #pragma once
00002
00003 #include <QDialog>
00004 #include <memory>
00005
00006 class QCloseEvent;
00007 class QShowEvent;
00008 class QPlainTextEdit;
00009 class QSpinBox;
00010 class QLabel;
00011 class WebWorkerRuntime;
00012
00016 class WebWorkerDialog : public QDialog {
00017     Q_OBJECT
00018
00019 public:
00020     explicit WebWorkerDialog(QWidget* parent = nullptr, bool allowClose = false);
00021     ~WebWorkerDialog() override;
00022
00023 protected:
00024     void closeEvent(QCloseEvent* event) override;
00025     void showEvent(QShowEvent* event) override;
00026
00027 private:
00028     void _refresh();
00029
00030     bool _allowClose = false;
00031     std::unique_ptr<WebWorkerRuntime> _runtime;
00032     QSpinBox* _portSpin = nullptr;
00033     QSpinBox* _maxRequestsSpin = nullptr;
00034     QLabel* _statusValue = nullptr;
00035     QLabel* _phaseValue = nullptr;
00036     QLabel* _urlValue = nullptr;
00037     QLabel* _servedValue = nullptr;
00038     QLabel* _activeValue = nullptr;
00039     QLabel* _stateValue = nullptr;
00040     QPlainTextEdit* _historyText = nullptr;
00041     QPlainTextEdit* _errorText = nullptr;
00042 };
```

18.113 ArchiveExtractor.h

```
00001 #pragma once
00002
00003 #include <QByteArray>
00004 #include <QString>
00005
00006 namespace genesys::launcher {
00007
00008 struct ExtractionResult {
00009     bool ok = false;
00010     QString error;
00011 };
00012
00013 class IArchiveExtractor {
00014 public:
00015     virtual ~IArchiveExtractor() = default;
00016
00017     [[nodiscard]] virtual ExtractionResult extract(const QString& archivePath,
00018                                                    const QString& destinationRoot) = 0;
00019 };
00020
00021 class TarZstdArchiveExtractor final : public IArchiveExtractor {
00022 public:
00023     explicit TarZstdArchiveExtractor(int timeoutMs = 60000);
00024
00025     [[nodiscard]] ExtractionResult extract(const QString& archivePath,
00026                                            const QString& destinationRoot) override;
```

```
00027
00028     [[nodiscard]] static ExtractionResult validatePathListing(const QByteArray& listing);
00029     [[nodiscard]] static ExtractionResult validateVerboseListing(const QByteArray& listing);
00030
00031 private:
00032     [[nodiscard]] ExtractionResult inspect(const QString& archivePath) const;
00033     [[nodiscard]] static ExtractionResult validateExtractedTree(const QString& destinationRoot);
00034
00035     int timeoutMs_ = 60000;
00036 };
00037
00038 } // namespace genesys::launcher
```

18.114 DispatchController.h

```
00001 #pragma once
00002
00003 #include "LauncherConfig.h"
00004 #include "ProcessLauncher.h"
00005 #include "RuntimeSelector.h"
00006
00007 #include <QString>
00008 #include <QStringList>
00009
00010 namespace genesys::launcher {
00011
00012 class DispatchController {
00013 public:
00014     DispatchController(const RuntimeSelector& selector,
00015                        IProcessLauncher& processLauncher);
00016
00017     [[nodiscard]] int dispatch(const QString& application,
00018                                const QStringList& arguments,
00019                                const LauncherConfig& config,
00020                                QString* error = nullptr) const;
00021
00022 private:
00023     const RuntimeSelector& selector_;
00024     IProcessLauncher& processLauncher_;
00025 };
00026
00027 } // namespace genesys::launcher
```

18.115 LauncherConfig.h

```
00001 #pragma once
00002
00003 #include <QUrl>
00004 #include <QString>
00005 #include <QStringList>
00006
00007 namespace genesys::launcher {
00008
00009 struct LauncherConfig {
00010     bool updatesEnabled = false;
00011     QString channel = QStringLiteral("stable");
00012     bool allowUserRuntime = true;
00013     bool requireSignature = true;
00014     bool checkOnGuiStartup = true;
00015     bool fallbackToSystem = true;
00016     int minimumCheckIntervalHours = 12;
00017     int maxUserVersions = 2;
00018     QUrl manifestUrl;
00019     bool hasExplicitPolicy = false;
00020     QStringList warnings;
00021
00022     [[nodiscard]] static LauncherConfig load(const QString& systemConfigPath,
00023                                              const QString& userConfigPath);
00024 };
00025
00026 } // namespace genesys::launcher
```

18.116 LauncherController.h

```
00001 #pragma once
00002
00003 #include "LauncherConfig.h"
00004 #include "UpdateChecker.h"
00005 #include "UpdateDialog.h"
```

```
00006 #include "UpdateDownloader.h"
00007 #include "UpdateVerifier.h"
00008 #include "UserRuntimeManager.h"
00009
00010 #include <QString>
00011
00012 namespace genesys::launcher {
00013
00014 class LauncherLogger;
00015
00016 struct LauncherUpdateResult {
00017     bool checkCompleted = false;
00018     bool updateAvailable = false;
00019     bool installed = false;
00020     Version installedVersion;
00021     QString message;
00022 };
00023
00024 class LauncherController {
00025 public:
00026     LauncherController(LauncherConfig config,
00027                        UpdateChecker& checker,
00028                        UpdateDownloader& downloader,
00029                        ISignatureVerifier* signatureVerifier,
00030                        UserRuntimeManager& runtimeManager,
00031                        IUpdateUi& ui,
00032                        LauncherLogger* logger = nullptr);
00033
00034     [[nodiscard]] LauncherUpdateResult checkAndMaybeUpdate(const Version& currentVersion,
00035                                                            bool interactive,
00036                                                            bool forceCheck,
00037                                                            bool checkOnly);
00038
00039 private:
00040     LauncherUpdateResult failUpdate(QString message, bool interactive);
00041     void log(const QString& event, const QString& detail = {}) const;
00042
00043     LauncherConfig config_;
00044     UpdateChecker& checker_;
00045     UpdateDownloader& downloader_;
00046     ISignatureVerifier* signatureVerifier_ = nullptr;
00047     UserRuntimeManager& runtimeManager_;
00048     IUpdateUi& ui_;
00049     LauncherLogger* logger_ = nullptr;
00050 };
00051
00052 } // namespace genesys::launcher
```

18.117 LauncherLogger.h

```
00001 #pragma once
00002
00003 #include <QMutex>
00004 #include <QString>
00005
00006 namespace genesys::launcher {
00007
00008 class LauncherLogger {
00009 public:
00010     explicit LauncherLogger(QString path, qint64 maxBytes = 1024 * 1024);
00011
00012     void log(const QString& event, const QString& detail = {});
00013
00014 private:
00015     void rotateIfNeeded();
00016     static QString sanitize(QString value);
00017
00018     QString path_;
00019     qint64 maxBytes_;
00020     QMutex mutex_;
00021 };
00022
00023 } // namespace genesys::launcher
```

18.118 NetworkTransport.h

```
00001 #pragma once
00002
00003 #include <QByteArray>
00004 #include <QIODevice>
00005 #include <QString>
00006 #include <QUrl>
```

```
00007
00008 #include <functional>
00009
00010 namespace genesys::launcher {
00011
00012 struct NetworkRequestOptions {
00013     qint64 maximumBytes = 1024 * 1024;
00014     int timeoutMs = 8000;
00015     int maximumRedirects = 3;
00016     bool httpsOnly = true;
00017 };
00018
00019 struct NetworkResponse {
00020     bool ok = false;
00021     bool cancelled = false;
00022     int httpStatus = 0;
00023     qint64 bytesReceived = 0;
00024     QUrl finalUrl;
00025     QString error;
00026 };
00027
00028 using ProgressCallback = std::function<void(qint64, qint64)>;
00029 using CancellationCallback = std::function<bool()>;
00030
00031 class INetworkTransport {
00032 public:
00033     virtual ~INetworkTransport() = default;
00034
00035     virtual NetworkResponse download(const QUrl& url,
00036                                      QIODevice& destination,
00037                                      const NetworkRequestOptions& options,
00038                                      ProgressCallback progress = {},
00039                                      CancellationCallback cancelled = {}) = 0;
00040
00041     [[nodiscard]] virtual std::pair<NetworkResponse, QByteArray> get(
00042         const QUrl& url,
00043         const NetworkRequestOptions& options,
00044         ProgressCallback progress = {},
00045         CancellationCallback cancelled = {});
00046 };
00047
00048 class QtNetworkTransport final : public INetworkTransport {
00049 public:
00050     NetworkResponse download(const QUrl& url,
00051                              QIODevice& destination,
00052                              const NetworkRequestOptions& options,
00053                              ProgressCallback progress = {},
00054                              CancellationCallback cancelled = {}) override;
00055 };
00056
00057 } // namespace genesys::launcher
```

18.119 PlatformInfo.h

```
00001 #pragma once
00002
00003 #include <QString>
00004
00005 namespace genesys::launcher {
00006
00007 struct PlatformInfo {
00008     QString osId;
00009     QString osVersion;
00010     QString architecture;
00011     QString manifestKey;
00012     bool valid = false;
00013
00014     [[nodiscard]] static PlatformInfo detect(const QString& osReleasePath =
      QStringLiteral("/etc/os-release"),
00015                                              const QString& architectureOverride = {});
00016 };
00017
00018 } // namespace genesys::launcher
```

18.120 ProcessLauncher.h

```
00001 #pragma once
00002
00003 #include "RuntimeSelector.h"
00004
00005 #include <QString>
00006 #include <QStringList>
```

```
00007
00008 namespace genesys::launcher {
00009
00010 class LauncherLogger;
00011
00012 class IProcessLauncher {
00013 public:
00014     virtual ~IProcessLauncher() = default;
00015
00016     [[nodiscard]] virtual int launchReplacingProcess(const RuntimeSelection& selection,
00017                                                      const QStringList& arguments,
00018                                                      QString* error = nullptr) = 0;
00019 };
00020
00021 class ProcessLauncher final : public IProcessLauncher {
00022 public:
00023     explicit ProcessLauncher(LauncherLogger* logger = nullptr);
00024
00025     [[nodiscard]] int launchReplacingProcess(const RuntimeSelection& selection,
00026                                              const QStringList& arguments,
00027                                              QString* error = nullptr) override;
00028
00029 private:
00030     LauncherLogger* logger_ = nullptr;
00031 };
00032
00033 } // namespace genesys::launcher
```

18.121 RuntimePaths.h

```
00001 #pragma once
00002
00003 #include <QProcessEnvironment>
00004 #include <QString>
00005
00006 namespace genesys::launcher {
00007
00008 class RuntimePaths {
00009 public:
00010     RuntimePaths();
00011     explicit RuntimePaths(const QProcessEnvironment& environment,
00012                           QString systemRuntimeRoot = {},
00013                           QString systemConfigPath = {},
00014                           QString updateKeyringPath = {});
00015
00016     [[nodiscard]] bool hasUsableHome() const noexcept { return hasUsableHome_; }
00017     [[nodiscard]] const QString& configRoot() const noexcept { return configRoot_; }
00018     [[nodiscard]] const QString& cacheRoot() const noexcept { return cacheRoot_; }
00019     [[nodiscard]] const QString& dataRoot() const noexcept { return dataRoot_; }
00020     [[nodiscard]] QString userConfigPath() const;
00021     [[nodiscard]] QString systemConfigPath() const { return systemConfigPath_; }
00022     [[nodiscard]] QString appsRoot() const;
00023     [[nodiscard]] QString currentRuntimeLink() const;
00024     [[nodiscard]] QString downloadsRoot() const;
00025     [[nodiscard]] QString logsRoot() const;
00026     [[nodiscard]] QString launcherLogPath() const;
00027     [[nodiscard]] QString updateStatePath() const;
00028     [[nodiscard]] QString updateLockPath() const;
00029     [[nodiscard]] QString systemRuntimeRoot() const { return systemRuntimeRoot_; }
00030     [[nodiscard]] QString systemBinRoot() const;
00031     [[nodiscard]] QString updateKeyringPath() const { return updateKeyringPath_; }
00032
00033 private:
00034     static QString resolveXdgRoot(const QProcessEnvironment& environment,
00035                                   const QString& variable,
00036                                   const QString& home,
00037                                   const QString& fallbackSuffix);
00038
00039     bool hasUsableHome_ = false;
00040     QString configRoot_;
00041     QString cacheRoot_;
00042     QString dataRoot_;
00043     QString systemRuntimeRoot_;
00044     QString systemConfigPath_;
00045     QString updateKeyringPath_;
00046 };
00047
00048 } // namespace genesys::launcher
```

18.122 RuntimeSelector.h

```
00001 #pragma once
```

```
00002
00003 #include "LauncherConfig.h"
00004 #include "RuntimePaths.h"
00005 #include "Version.h"
00006
00007 #include <QString>
00008 #include <QStringList>
00009
00010 namespace genesys::launcher {
00011
00012 class LauncherLogger;
00013
00014 enum class RuntimeSource {
00015     None,
00016     User,
00017     System
00018 };
00019
00020 struct RuntimeSelection {
00021     bool ok = false;
00022     QString requestedApplication;
00023     QString resolvedApplication;
00024     QString primaryExecutable;
00025     QString fallbackExecutable;
00026     RuntimeSource source = RuntimeSource::None;
00027     QString reason;
00028 };
00029
00030 class RuntimeSelector {
00031 public:
00032     RuntimeSelector(const RuntimePaths& paths, LauncherLogger* logger = nullptr);
00033
00034     [[nodiscard]] RuntimeSelection select(const QString& requestedApplication,
00035                                           const LauncherConfig& config) const;
00036     [[nodiscard]] Version currentUserVersion() const;
00037
00038     [[nodiscard]] static QStringList applicationCandidates(const QString& requestedApplication);
00039     [[nodiscard]] static bool isSupportedApplicationName(const QString& application);
00040
00041 private:
00042     struct UserRuntime {
00043         bool valid = false;
00044         QString root;
00045         Version version;
00046         QStringList declaredApplications;
00047         QString reason;
00048     };
00049
00050     [[nodiscard]] UserRuntime inspectCurrentRuntime() const;
00051     [[nodiscard]] QString findUserExecutable(const UserRuntime& runtime,
00052                                              const QString& requestedApplication,
00053                                              QString* resolvedApplication) const;
00054     [[nodiscard]] QString findSystemExecutable(const QString& requestedApplication,
00055                                                QString* resolvedApplication) const;
00056     void log(const QString& event, const QString& detail) const;
00057
00058     RuntimePaths paths_;
00059     LauncherLogger* logger_ = nullptr;
00060 };
00061
00062 } // namespace genesys::launcher
```

18.123 UpdateChecker.h

```
00001 #pragma once
00002
00003 #include "LauncherConfig.h"
00004 #include "NetworkTransport.h"
00005 #include "PlatformInfo.h"
00006 #include "RuntimePaths.h"
00007 #include "UpdateManifest.h"
00008 #include "Version.h"
00009
00010 #include <QDateTime>
00011
00012 #include <functional>
00013
00014 namespace genesys::launcher {
00015
00016 class LauncherLogger;
00017
00018 enum class UpdateCheckStatus {
00019     Disabled,
00020     RateLimited,
00021     UnsupportedPlatform,
```

```
00022     NetworkError,
00023     InvalidManifest,
00024     UpToDate,
00025     UpdateAvailable
00026 };
00027
00028 struct UpdateCheckResult {
00029     UpdateCheckStatus status = UpdateCheckStatus::Disabled;
00030     bool ok = false;
00031     bool updateAvailable = false;
00032     UpdateManifest manifest;
00033     QString message;
00034 };
00035
00036 class UpdateChecker {
00037 public:
00038     using Clock = std::function<QDateTime()>;
00039
00040     UpdateChecker(RuntimePaths paths,
00041                   LauncherConfig config,
00042                   PlatformInfo platform,
00043                   Version currentRuntimeVersion,
00044                   Version launcherVersion,
00045                   Version systemPackageVersion,
00046                   INetworkTransport& transport,
00047                   LauncherLogger* logger = nullptr,
00048                   Clock clock = {});
00049
00050     [[nodiscard]] UpdateCheckResult check(bool force = false);
00051
00052 private:
00053     [[nodiscard]] bool isRateLimited(const QDateTime& now) const;
00054     void recordAttempt(const QDateTime& now) const;
00055     void log(const QString& event, const QString& detail = {}) const;
00056
00057     RuntimePaths paths_;
00058     LauncherConfig config_;
00059     PlatformInfo platform_;
00060     Version currentRuntimeVersion_;
00061     Version launcherVersion_;
00062     Version systemPackageVersion_;
00063     INetworkTransport& transport_;
00064     LauncherLogger* logger_ = nullptr;
00065     Clock clock_;
00066 };
00067
00068 } // namespace genesys::launcher
```

18.124 UpdateDialog.h

```
00001 #pragma once
00002
00003 #include "Version.h"
00004
00005 #include <QString>
00006
00007 #include <memory>
00008
00009 class QProgressDialog;
00010 class QWidget;
00011
00012 namespace genesys::launcher {
00013
00014 class IUpdateUi {
00015 public:
00016     virtual ~IUpdateUi() = default;
00017
00018     [[nodiscard]] virtual bool confirmUpdate(const Version& currentVersion,
00019                                              const Version& candidateVersion,
00020                                              const QString& releaseNotes) = 0;
00021     virtual void beginDownload(qint64 totalBytes) = 0;
00022     [[nodiscard]] virtual bool updateDownloadProgress(qint64 receivedBytes,
00023                                                       qint64 totalBytes) = 0;
00024     virtual void finishDownload() = 0;
00025     virtual void showError(const QString& message) = 0;
00026     virtual void showInfo(const QString& message) = 0;
00027 };
00028
00029 class UpdateDialog final : public IUpdateUi {
00030 public:
00031     explicit UpdateDialog(QWidget* parent = nullptr);
00032     ~UpdateDialog() override;
00033
00034     [[nodiscard]] bool confirmUpdate(const Version& currentVersion,
00035                                      const Version& candidateVersion,
```

```
00036                                     const QString& releaseNotes) override;
00037     void beginDownload(qint64 totalBytes) override;
00038     [[nodiscard]] bool updateDownloadProgress(qint64 receivedBytes,
00039                                               qint64 totalBytes) override;
00040     void finishDownload() override;
00041     void showError(const QString& message) override;
00042     void showInfo(const QString& message) override;
00043
00044 private:
00045     QWidget* parent_ = nullptr;
00046     std::unique_ptr<QProgressDialog> progress_;
00047 };
00048
00049 } // namespace genesys::launcher
```

18.125 UpdateDownloader.h

```
00001 #pragma once
00002
00003 #include "NetworkTransport.h"
00004 #include "RuntimePaths.h"
00005 #include "UpdateManifest.h"
00006
00007 #include <QString>
00008
00009 namespace genesys::launcher {
00010
00011 class LauncherLogger;
00012
00013 struct DownloadResult {
00014     bool ok = false;
00015     bool cancelled = false;
00016     QString bundlePartPath;
00017     QString signaturePartPath;
00018     QString error;
00019 };
00020
00021 class UpdateDownloader {
00022 public:
00023     UpdateDownloader(RuntimePaths paths,
00024                      INetworkTransport& transport,
00025                      LauncherLogger* logger = nullptr);
00026
00027     [[nodiscard]] DownloadResult download(const UpdateManifest& manifest,
00028                                           bool downloadSignature,
00029                                           ProgressCallback progress = {},
00030                                           CancellationCallback cancelled = {});
00031
00032     [[nodiscard]] bool promoteVerified(const UpdateManifest& manifest,
00033                                        const DownloadResult& download,
00034                                        QString* verifiedBundlePath,
00035                                        QString* verifiedSignaturePath,
00036                                        QString* error = nullptr) const;
00037
00038     void cleanup(const DownloadResult& download) const;
00039
00040 private:
00041     [[nodiscard]] QString bundleBaseName(const UpdateManifest& manifest) const;
00042     void log(const QString& event, const QString& detail = {}) const;
00043
00044     RuntimePaths paths_;
00045     INetworkTransport& transport_;
00046     LauncherLogger* logger_ = nullptr;
00047 };
00048
00049 } // namespace genesys::launcher
```

18.126 UpdateManifest.h

```
00001 #pragma once
00002
00003 #include "Version.h"
00004
00005 #include <QByteArray>
00006 #include <QString>
00007 #include <QStringList>
00008 #include <QUrl>
00009
00010 namespace genesys::launcher {
00011
00012 struct UpdatePlatform {
00013     QString key;
```

```
00014     QString kind;
00015     QUrl bundleUrl;
00016     QString sha256;
00017     QUrl signatureUrl;
00018     qint64 size = 0;
00019     QStringList applications;
00020 };
00021
00022 struct UpdateManifest {
00023     int schemaVersion = 0;
00024     QString product;
00025     QString channel;
00026     Version version;
00027     QString releaseTag;
00028     QString publishedAt;
00029     Version minimumLauncherVersion;
00030     Version minimumSystemPackageVersion;
00031     UpdatePlatform platform;
00032     QString notesSummary;
00033     QUrl notesUrl;
00034 };
00035
00036 struct ManifestParseResult {
00037     bool ok = false;
00038     UpdateManifest manifest;
00039     QString error;
00040 };
00041
00042 class UpdateManifestParser {
00043 public:
00044     static constexpr qint64 MaximumBundleSize = 2LL * 1024 * 1024 * 1024;
00045
00046     [[nodiscard]] static ManifestParseResult parse(const QByteArray& json,
00047                                                    const QString& expectedChannel,
00048                                                    const QString& platformKey,
00049                                                    const Version& currentRuntimeVersion,
00050                                                    const Version& launcherVersion,
00051                                                    const Version& systemPackageVersion);
00052 };
00053
00054 } // namespace genesys::launcher
```

18.127 UpdateVerifier.h

```
00001 #pragma once
00002
00003 #include <QString>
00004
00005 namespace genesys::launcher {
00006
00007 class LauncherLogger;
00008
00009 struct VerificationResult {
00010     bool ok = false;
00011     QString error;
00012 };
00013
00014 class ISignatureVerifier {
00015 public:
00016     virtual ~ISignatureVerifier() = default;
00017
00018     [[nodiscard]] virtual VerificationResult verify(const QString& dataPath,
00019                                                     const QString& signaturePath) = 0;
00020 };
00021
00022 class GpgvSignatureVerifier final : public ISignatureVerifier {
00023 public:
00024     explicit GpgvSignatureVerifier(QString keyringPath,
00025                                    LauncherLogger* logger = nullptr,
00026                                    int timeoutMs = 15000);
00027
00028     [[nodiscard]] VerificationResult verify(const QString& dataPath,
00029                                             const QString& signaturePath) override;
00030
00031 private:
00032     QString keyringPath_;
00033     LauncherLogger* logger_ = nullptr;
00034     int timeoutMs_ = 15000;
00035 };
00036
00037 class UpdateVerifier {
00038 public:
00039     [[nodiscard]] static VerificationResult verifySha256(const QString& filePath,
00040                                                          const QString& expectedHexDigest);
00041 };
```

```
00042
00043 } // namespace genesys::launcher
```

18.128 UserRuntimeManager.h

```
00001 #pragma once
00002
00003 #include "ArchiveExtractor.h"
00004 #include "RuntimePaths.h"
00005 #include "UpdateManifest.h"
00006
00007 #include <QLockFile>
00008 #include <QString>
00009
00010 #include <memory>
00011
00012 namespace genesys::launcher {
00013
00014 class LauncherLogger;
00015
00016 struct RuntimeInstallResult {
00017     bool ok = false;
00018     QString runtimeRoot;
00019     QString previousRuntimeRoot;
00020     QString error;
00021 };
00022
00023 class UserRuntimeManager {
00024 public:
00025     UserRuntimeManager(RuntimePaths paths,
00026                        IArchiveExtractor& extractor,
00027                        LauncherLogger* logger = nullptr);
00028     ~UserRuntimeManager();
00029
00030     UserRuntimeManager(const UserRuntimeManager&) = delete;
00031     UserRuntimeManager& operator=(const UserRuntimeManager&) = delete;
00032
00033     [[nodiscard]] bool beginUpdate(QString* error = nullptr, int lockTimeoutMs = 0);
00034     void endUpdate();
00035     [[nodiscard]] bool hasUpdateLock() const noexcept { return lock_ && lock_->isLocked(); }
00036
00037     [[nodiscard]] RuntimeInstallResult installAndActivate(const QString& verifiedBundlePath,
00038                                                           const UpdateManifest& manifest,
00039                                                           int maxUserVersions);
00040
00041     [[nodiscard]] bool validateRuntime(const QString& runtimeRoot,
00042                                        const UpdateManifest& manifest,
00043                                        QString* error = nullptr) const;
00044
00045 private:
00046     [[nodiscard]] QString activeRuntimeRoot() const;
00047     [[nodiscard]] bool activateAtomically(const QString& runtimeRoot,
00048                                           QString* error) const;
00049     void cleanupOldVersions(int maxUserVersions,
00050                             const QString& activeRuntime,
00051                             const QString& rollbackRuntime) const;
00052     void log(const QString& event, const QString& detail = {}) const;
00053
00054     RuntimePaths paths_;
00055     IArchiveExtractor& extractor_;
00056     LauncherLogger* logger_ = nullptr;
00057     std::unique_ptr<QLockFile> lock_;
00058 };
00059
00060 } // namespace genesys::launcher
```

18.129 Version.h

```
00001 #pragma once
00002
00003 #include <QString>
00004 #include <QVector>
00005
00006 namespace genesys::launcher {
00007
00008 class Version {
00009 public:
00010     Version() = default;
00011
00012     [[nodiscard]] static Version parse(const QString& text);
00013
00014     [[nodiscard]] bool isValid() const noexcept { return valid_; }
```

```
00015     [[nodiscard]] QString toString() const;
00016     [[nodiscard]] const QVector<quint64>& components() const noexcept { return components_; }
00017
00018     friend bool operator==(const Version& lhs, const Version& rhs) noexcept;
00019     friend bool operator!=(const Version& lhs, const Version& rhs) noexcept { return !(lhs == rhs); }
00020     friend bool operator<(const Version& lhs, const Version& rhs) noexcept;
00021     friend bool operator>(const Version& lhs, const Version& rhs) noexcept { return rhs < lhs; }
00022     friend bool operator<=(const Version& lhs, const Version& rhs) noexcept { return !(rhs < lhs); }
00023     friend bool operator>=(const Version& lhs, const Version& rhs) noexcept { return !(lhs < rhs); }
00024
00025 private:
00026     QVector<quint64> components_;
00027     bool valid_ = false;
00028 };
00029
00030 } // namespace genesys::launcher
```

18.130 Airport_Extended1.h

```
00001 #ifndef AIRPORT_EXTENDED1_H
00002 #define AIRPORT_EXTENDED1_H
00003
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00006 class Airport_Extended1 : public BaseGenesysTerminalApplication {
00007 public:
00008     Airport_Extended1();
00009 public:
00010     virtual int main(int argc, char** argv) override;
00011 };
00012
00013 #endif // SMART_FAILURES_H
```

18.131 AirportSecurityExample.h

```
00001 #pragma once
00002
00003 #include "../../BaseGenesysTerminalApplication.h"
00004
00005 class AirportSecurityExample : public BaseGenesysTerminalApplication {
00006 public:
00007     AirportSecurityExample();
00008     virtual int main(int argc, char** argv) override;
00009 };
```

18.132 AirportSecurityExampleExtended.h

```
00001 #pragma once
00002
00003 #include "../../BaseGenesysTerminalApplication.h"
00004
00005 class AirportSecurityExampleExtended : public BaseGenesysTerminalApplication {
00006 public:
00007     AirportSecurityExampleExtended();
00008     virtual int main(int argc, char** argv) override;
00009 };
```

18.133 Assembly_Line.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Delay.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef ASSEMBLY_LINE_H
00015 #define ASSEMBLY_LINE_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
```

```
00019 class Assembly_Line : public BaseGenesysTerminalApplication {
00020 public:
00021     Assembly_Line();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_DELAY_H */
00027
```

18.134 Banking_Transactions.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Dummy.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef BANKING_TRANSACTIONS_H
00015 #define BANKING_TRANSACTIONS_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Banking_Transactions : public BaseGenesysTerminalApplication {
00020 public:
00021     Banking_Transactions();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* BANKING_TRANSACTIONS_H*/
00027
```

18.135 Example_BasicOrderShipping.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Example_PublicTransport_SARA.h
00009  * Author: Bruno & Henrique
00010  *
00011  * Created 2022-2 DS3
00012  * */
00013
00014 #ifndef EXAMPLE_BASIC_ORDER_SHIPPING_H
00015 #define EXAMPLE_BASIC_ORDER_SHIPPING_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Example_BasicOrderShipping : public BaseGenesysTerminalApplication {
00020 public:
00021     Example_BasicOrderShipping();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024
00025
00026 };
00027
00028 #endif /* EXAMPLE_BASIC_ORDER_SHIPPING_H */
00029
```

18.136 Example_PortModel.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
```

```
00006
00007
00008 #ifndef EXAMPLE_PORTMODEL_H
00009 #define EXAMPLE_PORTMODEL_H
00010
00011 #include "../../BaseGenesysTerminalApplication.h"
00012
00013 class Example_PortModel : public BaseGenesysTerminalApplication {
00014 public:
00015     Example_PortModel();
00016 public:
00017     virtual int main(int argc, char** argv) override;
00018 };
00019
00020 #endif /* EXAMPLE_PORTMODEL_H */
00021
```

18.137 Example_PublicTransport.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Example_PublicTransport.h
00009  * Author: Bruno & Henrique
00010  *
00011  * Created
00012  * */
00013
00014 #ifndef EXAMPLE_PUBLIC_TRANSPORT_H
00015 #define EXAMPLE_PUBLIC_TRANSPORT_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Example_PublicTransport : public BaseGenesysTerminalApplication {
00020 public:
00021     Example_PublicTransport();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024
00025
00026 };
00027
00028 #endif /* EXAMPLE_PUBLIC_TRANSPORT_H */
00029
```

18.138 Smart_AddingResource.h

```
00001 /*
00002  * File:   AddingResource.h
00003  * Author: genesys
00004  *
00005  * Created on 10 de dezembro de 2022, 17:57
00006  */
00007
00008 #ifndef ADDINGRESOURCE_H
00009 #define ADDINGRESOURCE_H
00010
00011 #include "../../BaseGenesysTerminalApplication.h"
00012
00013 class Smart_AddingResource : public BaseGenesysTerminalApplication {
00014 public:
00015     Smart_AddingResource();
00016 public:
00017     virtual int main(int argc, char** argv) override;
00018 };
00019
00020 #endif /* ADDINGRESOURCE_H */
00021
```

18.139 Smart_AlternatingEntityCreation.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
```

```
00005  */
00006
00007 /*
00008  * File:   Smart_AlternatingEntityCreation.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_ALTERNATINGENTITYCREATION_H
00015 #define SMART_ALTERNATINGENTITYCREATION_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_AlternatingEntityCreation : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_AlternatingEntityCreation();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_ALTERNATINGENTITYCREATION_H */
```

18.140 Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime_H
00015 #define Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime : public BaseGenesysTerminalApplication
    {
00020 public:
00021     Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime_H */
```

18.141 Smart_ArrivalsEntityTypeVsAttribute.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/file.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_ArrivalsEntityTypeVsAttribute.h
00008  * Author: genesys
00009  *
00010  * Created on 16 de dezembro de 2022, 15:22
00011  */
00012
00013 #ifndef SMART_ARRIVALSENTITYTYPEVSATTRIBUTE_H
00014 #define SMART_ARRIVALSENTITYTYPEVSATTRIBUTE_H
00015
00016 #include "../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_ArrivalsEntityTypeVsAttribute : public BaseGenesysTerminalApplication {
00019     public:
00020         Smart_ArrivalsEntityTypeVsAttribute();
00021         virtual int main(int argc, char** argv) override;
00022 };
00023
00024 #endif /* SMART_ARRIVALSENTITYTYPEVSATTRIBUTE_H */
00025
```

18.142 Smart_AssignExample.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/file.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_AssignExample.h
00008  * Author: genesys
00009  *
00010  * Created on 15 de dezembro de 2022, 23:46
00011  */
00012
00013 #ifndef SMART_ASSIGNEXAMPLE_H
00014 #define SMART_ASSIGNEXAMPLE_H
00015
00016 #include "../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_AssignExample : public BaseGenesysTerminalApplication {
00019     public:
00020         Smart_AssignExample();
00021         virtual int main(int argc, char** argv) override;
00022 };
00023
00024
00025 #endif /* SMART_ASSIGNEXAMPLE_H */
00026
```

18.143 Smart_AutomaticStatisticsCollection.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_AutomaticStatisticsCollection.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef Smart_AutomaticStatisticsCollection_H
00015 #define Smart_AutomaticStatisticsCollection_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_AutomaticStatisticsCollection : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_AutomaticStatisticsCollection();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* Smart_AutomaticStatisticsCollection_H */
```

18.144 Smart_BasicModeling.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   BasicExample.h
00008  * Author: genesys
00009  *
00010  * Created on 12 de dezembro de 2022, 12:38
00011  */
00012
00013 #ifndef SMART_BASICMODELING_H
00014 #define SMART_BASICMODELING_H
00015
00016 #include "../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_BasicModeling : public BaseGenesysTerminalApplication {
00019     public:
00020         Smart_BasicModeling();
00021         virtual int main(int argc, char** argv) override;
00022 };
```

```
00023
00024
00025 #endif /* SMART_BASICMODELING_H */
00026
```

18.145 Smart_BatchAndSeparate.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/file.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_BatchAndSeparate.h
00008  * Author: genesys
00009  *
00010  * Created on 16 de dezembro de 2022, 00:26
00011  */
00012
00013 #ifndef SMART_BATCHANDSEPARATE_H
00014 #define SMART_BATCHANDSEPARATE_H
00015
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_BatchAndSeparate : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_BatchAndSeparate();
00022     virtual int main(int argc, char** argv) override;
00023 };
00024
00025 #endif /* SMART_BATCHANDSEPARATE_H */
00026
```

18.146 Smart_ContinuousFlowEntities.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_ModellingLearningCurve.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_CONTINUOUSFLOWENTITIES_H
00015 #define SMART_CONTINUOUSFLOWENTITIES_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ContinuousFlowEntities : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ContinuousFlowEntities();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_DUMMY_H */
00027
```

18.147 Smart_Create.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Create.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
```

```
00014 #ifndef SMART_CREATE_H
00015 #define SMART_CREATE_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_Create : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_Create();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_CREATE_H */
00027
```

18.148 Smart_DecideNWayByChance.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_DecideNWayByChance.h
00009  * Author: guifrrs and LuizValdiero
00010  *
00011  * Created on December 3, 2022, 8:04 PM
00012  */
00013
00014 #ifndef SMART_DECIDENWAYBYCHANCE_H
00015 #define SMART_DECIDENWAYBYCHANCE_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_DecideNWayByChance : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_DecideNWayByChance();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_DECIDENWAYBYCHANCE_H */
00027
```

18.149 Smart_DefiningAttributesAsStrings.h

```
00001 /*
00002  * File:   Smart_MaxArrivalsField.h
00003  * Author: genesys
00004  *
00005  * Created on 10 de dezembro de 2022, 15:34
00006  */
00007
00008 #ifndef SMART_DEFININGATTRIBUTESASSTRINGS_H
00009 #define SMART_DEFININGATTRIBUTESASSTRINGS_H
00010
00011 #include "../../BaseGenesysTerminalApplication.h"
00012
00013 class Smart_DefiningAttributesAsStrings: public BaseGenesysTerminalApplication {
00014 public:
00015     Smart_DefiningAttributesAsStrings();
00016 public:
00017     virtual int main(int argc, char** argv) override;
00018 };
00019
00020 #endif /* SMART_DEFININGATTRIBUTESASSTRINGS_H */
00021
```

18.150 Smart_DefiningControlLogic.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_DefiningControlLogic.h
```

```
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef Smart_DEFININGCONTROLLOGIC_H
00015 #define Smart_DEFININGCONTROLLOGIC_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018 #include "plugins/components/DiscreteProcessing/Delay.h"
00019
00020
00021 class Smart_DefiningControlLogic : public BaseGenesysTerminalApplication {
00022 public:
00023     Smart_DefiningControlLogic();
00024 public:
00025     virtual int main(int argc, char** argv) override;
00026 };
00027
00028 #endif /* Smart_DEFININGCONTROLLOGIC_H */
```

18.151 Smart_DefiningResourceCapacity.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_DefiningResourceCapacity.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef Smart_DefiningResourceCapacity_H
00015 #define Smart_DefiningResourceCapacity_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_DefiningResourceCapacity : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_DefiningResourceCapacity();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* Smart_DefiningResourceCapacity_H */
```

18.152 Smart_DelayBasedOnReplication.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_DelayBasedOnReplication.h
00009  * Author: João Janini
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef Smart_DELAYBASEDONREPLICATION_H
00015 #define Smart_DELAYBASEDONREPLICATION_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018 #include "plugins/components/DiscreteProcessing/Delay.h"
00019
00020
00021 class Smart_DelayBasedOnReplication : public BaseGenesysTerminalApplication {
00022 public:
00023     Smart_DelayBasedOnReplication();
00024     void TestEventHandler(SimulationEvent *re);
00025 public:
00026     virtual int main(int argc, char** argv) override;
00027     Delay* delay;
00028     int *proctimes;
00029 };
00030
```

```
00031 #endif /* Smart_DelayBasedOnReplication_H */
```

18.153 Smart_EntitiesProcessedByPriority.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_EntitiesProcessedByPriority.h
00009  * Author: bepasar
00010  *
00011  * Created on 09/12/22
00012  */
00013
00014 #ifndef SMART_ENTITIESPROCESSEDBYPRIORITY_H
00015 #define SMART_ENTITIESPROCESSEDBYPRIORITY_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_EntitiesProcessedByPriority : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_EntitiesProcessedByPriority();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_ENTITIESPROCESSEDBYPRIORITY_H */
00027
```

18.154 Smart_EvaluatingConditionsBeforeEnteringQueue.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_EvaluatingConditionsBeforeEnteringQueue.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef Smart_EvaluatingConditionsBeforeEnteringQueue_H
00015 #define Smart_EvaluatingConditionsBeforeEnteringQueue_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_EvaluatingConditionsBeforeEnteringQueue : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_EvaluatingConditionsBeforeEnteringQueue();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* Smart_EvaluatingConditionsBeforeEnteringQueue_H */
```

18.155 Smart_Expression.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Expression.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_EXPRESSION_H
00015 #define SMART_EXPRESSION_H
00016
```

```
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_Expression : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_Expression();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_EXPRESSION_H */
00027
```

18.156 Smart_InventoryAndHoldingCosts.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Dummy.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_INVENTORYANDHOLDINGCOSTS_H
00015 #define SMART_INVENTORYANDHOLDINGCOSTS_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_InventoryAndHoldingCosts : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_InventoryAndHoldingCosts();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_INVENTORYANDHOLDINGCOSTS_H */
00027
```

18.157 Smart_MaxArrivalsField.h

```
00001 /*
00002  * File:   Smart_MaxArrivalsField.h
00003  * Author: genesys
00004  *
00005  * Created on 10 de dezembro de 2022, 15:34
00006  */
00007
00008 #ifndef SMART_MAXARRIVALSFIELD_H
00009 #define SMART_MAXARRIVALSFIELD_H
00010
00011 #include "../../BaseGenesysTerminalApplication.h"
00012
00013 class Smart_MaxArrivalsField : public BaseGenesysTerminalApplication {
00014 public:
00015     Smart_MaxArrivalsField();
00016 public:
00017     virtual int main(int argc, char** argv) override;
00018 };
00019
00020 #endif /* SMART_MAXARRIVALSFIELD_H */
00021
```

18.158 Smart_ModelRunUntil1000Parts.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_ModelRunUntil1000Parts.h
00009  * Author: guifrrs and LuizValdiero
00010  *
00011  * Created on December 3, 2022, 8:04 PM
```

```
00012  */
00013
00014 #ifndef SMART_MODELRUNUNTIL1000PARTS_H
00015 #define SMART_MODELRUNUNTIL1000PARTS_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ModelRunUntil1000Parts : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ModelRunUntil1000Parts();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_MODELRUNUNTIL1000PARTS_H */
```

18.159 Smart_ModuleDisplayVariables.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_ModuleDisplayVariables.h
00009  * Author: bepasar
00010  *
00011  * Created on 06/12/22, 21:27
00012  */
00013
00014 #ifndef SMART_MODULEDISPLAYVARIABLES_H
00015 #define SMART_MODULEDISPLAYVARIABLES_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ModuleDisplayVariables : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ModuleDisplayVariables();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_MODULEDISPLAYVARIABLES_H */
00027
```

18.160 Smart_OverlappingResources.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007
00008 #ifndef SMART_OVERLAPPINGRESOURCES_H
00009 #define SMART_OVERLAPPINGRESOURCES_H
00010
00011 #include "../../BaseGenesysTerminalApplication.h"
00012
00013 class Smart_OverlappingResources : public BaseGenesysTerminalApplication {
00014 public:
00015     Smart_OverlappingResources();
00016 public:
00017     virtual int main(int argc, char** argv) override;
00018 };
00019
00020 #endif /* SMART_OVERLAPPINGRESOURCES_H */
00021
```

18.161 Smart_ParallelProcessingOfEntities.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
```

```
00008  * File:   Smart_ParallelProcessingOfEntities.h
00009  * Author: Luiz e Guilherme
00010  *
00011  * Created on Dezembro de 2022
00012  */
00013
00014 #ifndef SMART_PARALLEL_PROCESSING_OF_ENTITIES_H
00015 #define SMART_PARALLEL_PROCESSING_OF_ENTITIES_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ParallelProcessingOfEntities : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ParallelProcessingOfEntities();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_PARALLEL_PROCESSING_OF_ENTITIES_H */
```

18.162 Smart_PlacingEntitiesInQueueSets.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_PlacingEntitiesInQueueSets.h
00009  * Author: Joçao Janini
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_PLACINGENTITIESINQUEUESETS_H
00015 #define SMART_PLACINGENTITIESINQUEUESETS_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_PlacingEntitiesInQueueSets : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_PlacingEntitiesInQueueSets();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_PLACINGENTITIESINQUEUESETS_H */
```

18.163 Smart_PriorityExample.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Dummy.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_PRIORITYEXAMPLE_H
00015 #define SMART_PRIORITYEXAMPLE_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_PriorityExample : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_PriorityExample();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_DUMMY_H */
00027
```

18.164 Smart_ProcessArena.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_ProcessArena.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_PROCESSARENA_H
00015 #define SMART_PROCESSARENA_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ProcessArena : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ProcessArena();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_PROCESSARENA_H */
00027
```

18.165 Smart_Record_Arena.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Record_Arena.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef Smart_Record_Arena_ARENA_H
00015 #define Smart_Record_Arena_ARENA_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_Record_Arena : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_Record_Arena();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* Smart_Record_Arena_ARENA_H */
00027
```

18.166 Smart_RemovingAndReorderingEntitiesInAQueue.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007
00008 #ifndef SMART_REMOVINGANDREORDERINGENTITIESINAQUEUE_H
00009 #define SMART_REMOVINGANDREORDERINGENTITIESINAQUEUE_H
00010
00011 #include "../../BaseGenesysTerminalApplication.h"
00012
00013 class Smart_RemovingAndReorderingEntitiesInAQueue : public BaseGenesysTerminalApplication {
00014 public:
00015     Smart_RemovingAndReorderingEntitiesInAQueue();
00016 public:
00017     virtual int main(int argc, char** argv) override;
00018 };
00019
00020 #endif /* SMART_REMOVINGANDREORDERINGENTITIESINAQUEUE_H */
```

```
00021
```

18.167 Smart_ResourceCosting.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Dummy.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_RESOURCECOSTING_H
00015 #define SMART_RESOURCECOSTING_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ResourceCosting : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ResourceCosting();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_RESOURCECOSTING_H */
00027
```

18.168 Smart_ResourceScheduleCosting.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Dummy.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_RESOURCESCHEDULECOSTING_H
00015 #define SMART_RESOURCESCHEDULECOSTING_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ResourceScheduleCosting : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ResourceScheduleCosting();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_RESOURCESCHEDULECOSTING_H */
00027
```

18.169 Smart_ResourceSets.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_ResourceSets.h
00009  * Author: guifrrs and LuizValdiero
00010  *
00011  * Created on December 3, 2022, 8:04 PM
00012  */
00013
00014 #ifndef SMART_RESOURCESETS_H
00015 #define SMART_RESOURCESETS_H
```

```
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ResourceSets : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ResourceSets();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_RESOURCESETS_H */
00027
```

18.170 Smart_SeizingMultipleSimultaneosly.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_SeizingMultipleSimultaneosly.h
00009  * Author: douglaspereira04
00010  *
00011  */
00012
00013 #ifndef SMART_SEIZINGMULTIPLESIMULTANEOSLY_H
00014 #define SMART_SEIZINGMULTIPLESIMULTANEOSLY_H
00015
00016 #include "../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_SeizingMultipleSimultaneosly : public BaseGenesysTerminalApplication {
00019 public:
00020     Smart_SeizingMultipleSimultaneosly();
00021 public:
00022     virtual int main(int argc, char** argv) override;
00023 };
00024 #endif /* SMART_SEIZINGMULTIPLESIMULTANEOSLY_H */
```

18.171 Smart_SelectingRouteBasedOnProbability.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_WaitSignal.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_SELECTINGROUTEBASEDONPROBABILITY_H
00015 #define SMART_SELECTINGROUTEBASEDONPROBABILITY_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_SelectingRouteBasedOnProbability : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_SelectingRouteBasedOnProbability();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_SELECTINGROUTEBASEDONPROBABILITY_H */
00027
```

18.172 Smart_SelectingShorterQueue.h

```
00001 /*
00002  * File:   Smart_SelectingShorterQueue.h
00003  * Author: genesys
00004  *
00005  * Created on 10 de dezembro de 2022, 17:59
00006  */
00007
```

```
00008 #ifndef SMART_SELECTINGSHORTERQUEUE_H
00009 #define SMART_SELECTINGSHORTERQUEUE_H
00010
00011 #include "../../BaseGenesysTerminalApplication.h"
00012
00013 class Smart_SelectingShorterQueue : public BaseGenesysTerminalApplication {
00014 public:
00015     Smart_SelectingShorterQueue();
00016 public:
00017     virtual int main(int argc, char** argv) override;
00018 };
00019
00020 #endif /* SMART_SELECTINGSHORTERQUEUE_H */
00021
```

18.173 Smart_SynchronizingParallelEntities.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_SynchronizingParallelEntities.h
00009  * Author:
00010  *
00011  */
00012
00013 #ifndef SMART_SYNCHRONIZINGPARALLELENTITIES_H
00014 #define SMART_SYNCHRONIZINGPARALLELENTITIES_H
00015
00016 #include "../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_SynchronizingParallelEntities : public BaseGenesysTerminalApplication {
00019 public:
00020     Smart_SynchronizingParallelEntities();
00021 public:
00022     virtual int main(int argc, char** argv) override;
00023 };
00024 #endif /* SMART_SYNCHRONIZINGPARALLELENTITIES_H */
00025
```

18.174 Smart_ValueAdded.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Dummy.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_VALUEADDED_H
00015 #define SMART_VALUEADDED_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ValueAdded : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ValueAdded();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_VALUEADDED_H */
00027
```

18.175 Smart_WaitForSignal.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
```

```
00005  */
00006
00007 /*
00008  * File:   Smart_WaitForSignal.h
00009  * Author: douglaspereira04
00010  *
00011  */
00012
00013 #ifndef SMART_WAITFORSIGNAL_H
00014 #define SMART_WAITFORSIGNAL_H
00015
00016 #include "../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_WaitForSignal : public BaseGenesysTerminalApplication {
00019 public:
00020     Smart_WaitForSignal();
00021 public:
00022     virtual int main(int argc, char** argv) override;
00023 };
00024 #endif /* SMART_WAITFORSIGNAL_H */
```

18.176 Book_Cap02_Example01.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Book_Cap02_Example01.h
00008  * Author: rlcancian
00009  *
00010  * Created on 19 de outubro de 2021, 16:23
00011  */
00012
00013 #ifndef BOOK_CAP02_EXAMPLE01_H
00014 #define BOOK_CAP02_EXAMPLE01_H
00015
00016 #include "../../BaseGenesysTerminalApplication.h"
00017
00018 class Book_Cap02_Example01 : public BaseGenesysTerminalApplication {
00019 public:
00020     Book_Cap02_Example01();
00021     virtual int main(int argc, char** argv) override;
00022 };
00023
00024 #endif /* BOOK_CAP02_EXAMPLE01_H */
00025
```

18.177 StartAIAssistant.h

```
00001 #pragma once
00002
00003 #include "../../../BaseGenesysTerminalApplication.h"
00004
00005 class StartAIAssistant : public BaseGenesysTerminalApplication {
00006 public:
00007     StartAIAssistant();
00008 public:
00009     virtual int main(int argc, char** argv) override;
00010 };
```

18.178 Smart_MarkovChain.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_MarkovChain.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_MARKOVCHAIN_H
00015 #define SMART_MARKOVCHAIN_H
```

```
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_MarkovChain : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_MarkovChain();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_MARKOVCHAIN_H */
00027
```

18.179 Smart_BacteriaColony_GRO.h

```
00001 /*
00002  * File:   Smart_BacteriaColony_GRO.h
00003  * Author: GRO
00004  *
00005  * Created on 1 de Maio de 2026
00006  */
00007
00008 #ifndef SMART_BACTERIACOLONY_GRO_H
00009 #define SMART_BACTERIACOLONY_GRO_H
00010
00011 #include "../../../BaseGenesysTerminalApplication.h"
00012
00013 class Smart_BacteriaColony_GRO : public BaseGenesysTerminalApplication {
00014 public:
00015     Smart_BacteriaColony_GRO();
00016
00017 public:
00018     virtual int main(int argc, char** argv) override;
00019 };
00020
00021 #endif /* SMART_BACTERIACOLONY_GRO_H */
```

18.180 Smart_BioKineticLawRegulation.h

```
00001 /*
00002  * File:   Smart_BioKineticLawRegulation.h
00003  */
00004
00005 #ifndef SMART_BIOKINETICLAWREGULATION_H
00006 #define SMART_BIOKINETICLAWREGULATION_H
00007
00008 #include "../../../BaseGenesysTerminalApplication.h"
00009
00010 class Smart_BioKineticLawRegulation : public BaseGenesysTerminalApplication {
00011 public:
00012     Smart_BioKineticLawRegulation();
00013 public:
00014     virtual int main(int argc, char** argv) override;
00015 };
00016
00017 #endif /* SMART_BIOKINETICLAWREGULATION_H */
```

18.181 Smart_BioNetworkAnalysis.h

```
00001 /*
00002  * File:   Smart_BioNetworkAnalysis.h
00003  */
00004
00005 #ifndef SMART_BIONETWORKANALYSIS_H
00006 #define SMART_BIONETWORKANALYSIS_H
00007
00008 #include "../../../BaseGenesysTerminalApplication.h"
00009
00010 class Smart_BioNetworkAnalysis : public BaseGenesysTerminalApplication {
00011 public:
00012     Smart_BioNetworkAnalysis();
00013 public:
00014     virtual int main(int argc, char** argv) override;
00015 };
00016
00017 #endif /* SMART_BIONETWORKANALYSIS_H */
```

18.182 Smart_BioReversibleKineticLaw.h

```
00001 /*
00002  * File:   Smart_BioReversibleKineticLaw.h
00003  */
00004
00005 #ifndef SMART_BIOREVERSIBLEKINETICLAW_H
00006 #define SMART_BIOREVERSIBLEKINETICLAW_H
00007
00008 #include "../../../BaseGenesysTerminalApplication.h"
00009
00010 class Smart_BioReversibleKineticLaw : public BaseGenesysTerminalApplication {
00011 public:
00012     Smart_BioReversibleKineticLaw();
00013 public:
00014     virtual int main(int argc, char** argv) override;
00015 };
00016
00017 #endif /* SMART_BIOREVERSIBLEKINETICLAW_H */
```

18.183 Smart_BioReversibleMassAction.h

```
00001 /*
00002  * File:   Smart_BioReversibleMassAction.h
00003  */
00004
00005 #ifndef SMART_BIOREVERSIBLEMASSACTION_H
00006 #define SMART_BIOREVERSIBLEMASSACTION_H
00007
00008 #include "../../../BaseGenesysTerminalApplication.h"
00009
00010 class Smart_BioReversibleMassAction : public BaseGenesysTerminalApplication {
00011 public:
00012     Smart_BioReversibleMassAction();
00013 public:
00014     virtual int main(int argc, char** argv) override;
00015 };
00016
00017 #endif /* SMART_BIOREVERSIBLEMASSACTION_H */
```

18.184 Smart_GroColonyGrowth.h

```
00001 /*
00002  * File:   Smart_GroColonyGrowth.h
00003  * Author: GRO
00004  *
00005  * Created on 17 de Abril de 2026
00006  */
00007
00008 #ifndef SMART_GROCOLONYGROWTH_H
00009 #define SMART_GROCOLONYGROWTH_H
00010
00011 #include "../../../BaseGenesysTerminalApplication.h"
00012
00013 class Smart_GroColonyGrowth : public BaseGenesysTerminalApplication {
00014 public:
00015     Smart_GroColonyGrowth();
00016
00017 public:
00018     virtual int main(int argc, char** argv) override;
00019 };
00020
00021 #endif /* SMART_GROCOLONYGROWTH_H */
```

18.185 Smart_GroColonyLifecycle.h

```
00001 /*
00002  * File:   Smart_GroColonyLifecycle.h
00003  * Author: GRO
00004  *
00005  * Created on 17 de Abril de 2026
00006  */
00007
00008 #ifndef SMART_GROCOLONYLIFECYCLE_H
00009 #define SMART_GROCOLONYLIFECYCLE_H
00010
00011 #include "../../../BaseGenesysTerminalApplication.h"
00012
00013 class Smart_GroColonyLifecycle : public BaseGenesysTerminalApplication {
```

```
00014 public:
00015     Smart_GroColonyLifecycle();
00016
00017 public:
00018     virtual int main(int argc, char** argv) override;
00019 };
00020
00021 #endif /* SMART_GROCOLONYLIFECYCLE_H */
```

18.186 Smart_LSODE.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_ODE.h
00008  * Author: rlcancian
00009  *
00010  * Created on 11 de janeiro de 2022, 19:29
00011  */
00012
00013 #ifndef SMART_LSODE_H
00014 #define SMART_LSODE_H
00015
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_LSODE : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_LSODE();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_LSODE_H */
00027
```

18.187 Smart_ODE.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_ODE.h
00008  * Author: rlcancian
00009  *
00010  * Created on 11 de janeiro de 2022, 19:29
00011  */
00012
00013 #ifndef SMART_ODE_H
00014 #define SMART_ODE_H
00015
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ODE : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ODE();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_ODE_H */
00027
```

18.188 Smart_Buffer.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Buffer.h
```

```
00009  * Author: rlcancian
00010  *
00011  * Created on
00012  */
00013
00014 #ifndef SMART_BUFFER_H
00015 #define SMART_BUFFER_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_Buffer : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_Buffer();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_BUFFER_H */
00027
```

18.189 Smart_Clone.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Clone.h
00009  * Author: rlcancian
00010  *
00011  * Created on 30 de Novembro de 2022, 19:25
00012  */
00013
00014 #ifndef SMART_CLONE_H
00015 #define SMART_CLONE_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_Clone : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_Clone();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_CLONE_H */
00027
```

18.190 Smart_Delay.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Delay.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_DELAY_H
00015 #define SMART_DELAY_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_Delay : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_Delay();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_DELAY_H */
00027
```

18.191 Smart_Failures.h

```
00001 #ifndef SMART_FAILURES_H
00002 #define SMART_FAILURES_H
00003
00004 #include "../../../BaseGenesysTerminalApplication.h"
00005
00006 class Smart_Failures : public BaseGenesysTerminalApplication {
00007 public:
00008     Smart_Failures();
00009 public:
00010     virtual int main(int argc, char** argv) override;
00011 };
00012
00013 #endif // SMART_FAILURES_H
```

18.192 Smart_HoldSearchRemove.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_HoldSearchRemove.h
00009  * Author: rlcancian
00010  *
00011  * Created on 01 de Dezembro de 2022, 09:31
00012  */
00013
00014 #ifndef SMART_HOLDSEARCHREMOVE_H
00015 #define SMART_HOLDSEARCHREMOVE_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_HoldSearchRemove : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_HoldSearchRemove();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_HOLDSEARCHREMOVE_H */
00027
```

18.193 Smart_Process.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Process.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_PROCESS_H
00015 #define SMART_PROCESS_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_Process : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_Process();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_PROCESS_H */
00027
```

18.194 Smart_ProcessSet.h

```
00001 /*
```

```
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_ProcessSet.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_PROCESSSET_H
00015 #define SMART_PROCESSSET_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ProcessSet : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ProcessSet();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_PROCESSSET_H */
00027
```

18.195 Smart_SeizeDelayRelease.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_SeizeDelayRelease.h
00009  * Author: rlcancian
00010  *
00011  * Created on 24 de Setembro de 2019, 16:43
00012  */
00013
00014 #ifndef SMART_SEIZEDELAYRELEASE_H
00015 #define SMART_SEIZEDELAYRELEASE_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_SeizeDelayRelease : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_SeizeDelayRelease();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025 #endif /* SMART_SEIZEDELAYRELEASE_H */
00026
```

18.196 Smart_SeizeDelayReleaseMany.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_SeizeDelayReleaseMany.h
00009  * Author: rlcancian
00010  *
00011  * Created on 9 de abril de 2021, 18:09
00012  */
00013
00014 #ifndef MODEL_SEIZEDELAYRELEASEMANY_H
00015 #define MODEL_SEIZEDELAYRELEASEMANY_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_SeizeDelayReleaseMany : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_SeizeDelayReleaseMany();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
```

```
00025
00026 #endif /* MODEL_SEIZEDELAYRELEASEMANY_H */
00027
```

18.197 Smart_SeizeDelayReleaseNoDataDefs.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_SeizeDelayReleaseNoDataDefs.h
00008  * Author: rlcancian
00009  *
00010  * Created on 23 de maio de 2022, 12:39
00011  */
00012
00013 #ifndef SMART_SEIZEDELAYRELEASENODATADEFS_H
00014 #define SMART_SEIZEDELAYRELEASENODATADEFS_H
00015
00016 #include "../../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_SeizeDelayReleaseNoDataDefs : public BaseGenesysTerminalApplication {
00019 public:
00020     Smart_SeizeDelayReleaseNoDataDefs();
00021 public:
00022     virtual int main(int argc, char** argv) override;
00023 };
00024
00025 #endif /* SMART_SEIZEDELAYRELEASENODATADEFS_H */
00026
```

18.198 Smart_CppForG.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_CppForG.h
00008  * Author: rlcancian
00009  *
00010  * Created on 11 de janeiro de 2022, 22:39
00011  */
00012
00013 #pragma once
00014
00015 #include "../../../BaseGenesysTerminalApplication.h"
00016
00017 class Smart_CppForG : public BaseGenesysTerminalApplication {
00018 public:
00019     Smart_CppForG();
00020 public:
00021     virtual int main(int argc, char** argv) override;
00022 };
00023
```

18.199 Smart_PythonForG.h

```
00001 #ifndef SMART_PYTHONFORG_H
00002 #define SMART_PYTHONFORG_H
00003
00004 #include "../../../BaseGenesysTerminalApplication.h"
00005
00006 class Smart_PythonForG : public BaseGenesysTerminalApplication {
00007 public:
00008     Smart_PythonForG();
00009     virtual int main(int argc, char** argv) override;
00010 };
00011
00012 #endif /* SMART_PYTHONFORG_H */
```

18.200 Smart_R_Simulator.h

```
00001 /*
```

```
00002  * File:   Smart_R_Simulator.h
00003  * Author: GenESyS
00004  *
00005  * Minimal smart model for the RSimulator component.
00006  */
00007
00008 #ifndef SMART_R_SIMULATOR_H
00009 #define SMART_R_SIMULATOR_H
00010
00011 #include "../../../BaseGenesysTerminalApplication.h"
00012
00019 class Smart_R_Simulator : public BaseGenesysTerminalApplication {
00020 public:
00022     Smart_R_Simulator();
00023
00024 public:
00026     virtual int main(int argc, char** argv) override;
00027 };
00028
00029 #endif /* SMART_R_SIMULATOR_H */
```

18.201 Smart_BatchSeparate.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_BatchSeparate.h
00008  * Author: rlcancian
00009  *
00010  * Created on 15 de janeiro de 2022, 09:51
00011  */
00012
00013 #ifndef SMART_BATCHSEPARATE_H
00014 #define SMART_BATCHSEPARATE_H
00015
00016 #include "../../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_BatchSeparate : public BaseGenesysTerminalApplication {
00019 public:
00020     Smart_BatchSeparate();
00021 public:
00022     virtual int main(int argc, char** argv) override;
00023 };
00024
00025 #endif /* SMART_BATCHSEPARATE_H */
00026
```

18.202 Smart_Record.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Record.h
00009  * Author: rlcancian
00010  *
00011  * Created on
00012  */
00013
00014 #pragma once
00015
00016 #include "../../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_Record : public BaseGenesysTerminalApplication {
00019 public:
00020     Smart_Record();
00021 public:
00022     virtual int main(int argc, char** argv) override;
00023 };
00024
00025
```

18.203 Smart_AssignWriteSeizes.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_AssignWriteSeizes.h
00009  * Author: rlcancian
00010  *
00011  * Created on 24 de Setembro de 2019, 20:56
00012  */
00013
00014 #ifndef SMART_ASSIGNWRITE3SEIZES_H
00015 #define SMART_ASSIGNWRITE3SEIZES_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_AssignWriteSeizes : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_AssignWriteSeizes();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_ASSIGNWRITE3SEIZES_H */
00027
```

18.204 Smart_Attribute_Variable.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Attribute_Variable.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_ATTRIBUTE_VARIABLE_H
00015 #define SMART_ATTRIBUTE_VARIABLE_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_Attribute_Variable : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_Attribute_Variable();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_ATTRIBUTE_VARIABLE_H */
```

18.205 Smart_ModelInfoModelSimulation.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_ModelInfoModelSimulation.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:59
00012  */
00013
00014 #ifndef SMART_MODELINFOMODELSIMULATION_H
00015 #define SMART_MODELINFOMODELSIMULATION_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ModelInfoModelSimulation : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ModelInfoModelSimulation();
```

```
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_MODELINFOMODELSIMULATION_H */
00027
```

18.206 Smart_Parser.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Parser.h
00009  * Author: rlcancian
00010  *
00011  * Created on 5 de Janeiro de 2022, 07:21
00012  */
00013
00014 #ifndef SMART_PARSER_H
00015 #define SMART_PARSER_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_Parser : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_Parser();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_PARSER_H */
00027
```

18.207 Smart_ParserModelFunctions.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_ParserModelFunctions.h
00009  * Author: rlcancian
00010  *
00011  * Created on 5 de Janeiro de 2022, 07:21
00012  */
00013
00014 #ifndef SMART_PARSERMODELFUNCTIONS_H
00015 #define SMART_PARSERMODELFUNCTIONS_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_ParserModelFunctions : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_ParserModelFunctions();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_PARSERMODELFUNCTIONS_H */
00027
```

18.208 Smart_Plugin.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_Plugin.h
00008  * Author: rlcancian
00009  *
00010  * Created on 13 de janeiro de 2022, 17:47
```

```
00011  */
00012
00013 #ifndef SMART_PLUGIN_H
00014 #define SMART_PLUGIN_H
00015
00016 #include "../../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_Plugin : public BaseGenesysTerminalApplication {
00019 public:
00020     Smart_Plugin();
00021 public:
00022     virtual int main(int argc, char** argv) override;
00023 };
00024
00025 #endif /* SMART_PLUGIN_H */
00026
```

18.209 Smart_RouteStation.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_RouteStation.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_ROUTESTATION_H
00015 #define SMART_ROUTESTATION_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_RouteStation : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_RouteStation();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_ROUTESTATION_H */
00027
```

18.210 Smart_Sequence.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Sequence.h
00009  * Author: rlcancian
00010  *
00011  * Created on 12 de março de 2021, 15:39
00012  */
00013
00014 #ifndef SMART_SEQUENCE_H
00015 #define SMART_SEQUENCE_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_Sequence : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_Sequence();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_SEQUENCE_H */
00027
```

18.211 Smart_CellularAutomata.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_CellularAutomata.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_CELLULARAUTOMATA_H
00015 #define SMART_CELLULARAUTOMATA_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_CellularAutomata : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_CellularAutomata();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_CELLULARAUTOMATA_H */
00027
```

18.212 Smart_CellularAutomata1.h

```
00001 /*
00002 * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_CellularAutomata.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2025, 18:34
00012  */
00013
00014 #ifndef SMART_CELLULARAUTOMATA1_H
00015 #define SMART_CELLULARAUTOMATA1_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_CellularAutomata1 : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_CellularAutomata1();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_CELLULARAUTOMATA_H */
```

18.213 Smart_DefaultModalModel.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007
00008 #pragma once
00009
00010 #include "../../../BaseGenesysTerminalApplication.h"
00011
00012 class Smart_DefaultModalModel : public BaseGenesysTerminalApplication {
00013 public:
00014     Smart_DefaultModalModel();
00015 public:
00016     virtual int main(int argc, char** argv) override;
00017 };
00018
00019
```

18.214 Smart_Dummy.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_Dummy.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_DUMMY_H
00015 #define SMART_DUMMY_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_Dummy : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_Dummy();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_DUMMY_H */
00027
```

18.215 Smart_FSMState.h

```
00001 #pragma once
00002
00003 #include "../../../BaseGenesysTerminalApplication.h"
00004
00005 class Smart_FSMState : public BaseGenesysTerminalApplication {
00006 public:
00007     Smart_FSMState();
00008 public:
00009     virtual int main(int argc, char** argv) override;
00010 };
```

18.216 Smart_Modalmodel.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_ModelModel.h
00009  * Author: rlcancian
00010  *
00011  * Created on 30 de Setembro de 2025, 17:46
00012  */
00013
00014
00015 #ifndef SMART_MODELMODEL_H
00016 #define SMART_MODELMODEL_H
00017
00018 #include "../../../BaseGenesysTerminalApplication.h"
00019
00020 class Smart_ModalModel : public BaseGenesysTerminalApplication {
00021 public:
00022     Smart_ModalModel();
00023 public:
00024     virtual int main(int argc, char** argv) override;
00025 };
00026
00027 #endif /* SMART_MODELMODEL_H */
00028
```

18.217 Smart_ModalModelDefaultNetwork.h

```
00001 #pragma once
00002
00003 #include "../../../BaseGenesysTerminalApplication.h"
```

```
00004
00005 class Smart_ModalModelDefaultNetwork : public BaseGenesysTerminalApplication {
00006 public:
00007     Smart_ModalModelDefaultNetwork();
00008 public:
00009     virtual int main(int argc, char** argv) override;
00010 };
```

18.218 Smart_ModalModelFSM.h

```
00001 #pragma once
00002
00003 #include "../../../BaseGenesysTerminalApplication.h"
00004
00005 class Smart_ModalModelFSM : public BaseGenesysTerminalApplication {
00006 public:
00007     Smart_ModalModelFSM();
00008 public:
00009     virtual int main(int argc, char** argv) override;
00010 };
```

18.219 Smart_ModalModelPetriNet.h

```
00001 #pragma once
00002
00003 #include "../../../BaseGenesysTerminalApplication.h"
00004
00005 class Smart_ModalModelPetriNet : public BaseGenesysTerminalApplication {
00006 public:
00007     Smart_ModalModelPetriNet();
00008 public:
00009     virtual int main(int argc, char** argv) override;
00010 };
```

18.220 Smart_PetriPlace.h

```
00001 #pragma once
00002
00003 #include "../../../BaseGenesysTerminalApplication.h"
00004
00005 class Smart_PetriPlace : public BaseGenesysTerminalApplication {
00006 public:
00007     Smart_PetriPlace();
00008 public:
00009     virtual int main(int argc, char** argv) override;
00010 };
```

18.221 Smart_OnEvent.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_OnEvent.h
00009  * Author: rlcancian
00010  *
00011  * Created on 5 de Janeiro de 2021, 18:00
00012  */
00013
00014 #ifndef SMART_ONEVENT_H
00015 #define SMART_ONEVENT_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_OnEvent : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_OnEvent();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 public:
00025     virtual void onBreakpointHandler(SimulationEvent* re) override;
00026     virtual void onEntityCreateHandler(SimulationEvent* re) override;
00027     virtual void onEntityMoveHandler(SimulationEvent* re) override;
00028     virtual void onSimulationStartHandler(SimulationEvent* re) override;
```

```
00029     virtual void onReplicationStartHandler(SimulationEvent* re) override;
00030     virtual void onReplicationStepHandler(SimulationEvent* re) override;
00031     virtual void onProcessEventHandler(SimulationEvent* re) override;
00032     virtual void onReplicationEndHandler(SimulationEvent* re) override;
00033     virtual void onSimulationEndHandler(SimulationEvent* re) override;
00034     virtual void onSimulationPausedHandler(SimulationEvent* re) override;
00035     virtual void onSimulationResumeHandler(SimulationEvent* re) override;
00036     virtual void onEntityRemoveHandler(SimulationEvent* re) override;
00037 };
00038
00039 #endif /* SMART_ONEVENT_H */
```

18.222 Smart_SimulationControlResponse.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_SimulationControlResponse.h
00008  * Author: rlcancian
00009  *
00010  * Created on 29 de novembro de 2022, 21:07
00011  */
00012
00013 #ifndef SMART_SIMULATIONCONTROLRESPONSE_H
00014 #define SMART_SIMULATIONCONTROLRESPONSE_H
00015
00016 #include "../../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_SimulationControlResponse : public BaseGenesysTerminalApplication {
00019 public:
00020     Smart_SimulationControlResponse();
00021 public:
00022     virtual int main(int argc, char** argv) override;
00023 };
00024
00025 #endif /* SMART_SIMULATIONCONTROLRESPONSE_H */
00026
```

18.223 Smart_WaitScanCondition.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_WaitScanCondition.h
00008  * Author: rlcancian
00009  *
00010  * Created on 14 de junho de 2022, 13:00
00011  */
00012
00013 #ifndef SMART_WAITSCANCONDITION_H
00014 #define SMART_WAITSCANCONDITION_H
00015
00016 #include "../../../BaseGenesysTerminalApplication.h"
00017
00018 class Smart_WaitScanCondition : public BaseGenesysTerminalApplication {
00019 public:
00020     Smart_WaitScanCondition();
00021 public:
00022     virtual int main(int argc, char** argv) override;
00023 };
00024
00025 #endif /* SMART_WAITSCANCONDITION_H */
00026
```

18.224 Smart_WaitSignal.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
```

```
00008  * File:   Smart_WaitSignal.h
00009  * Author: rlcancian
00010  *
00011  * Created on 3 de Setembro de 2019, 18:34
00012  */
00013
00014 #ifndef SMART_WAITSIGNAL_H
00015 #define SMART_WAITSIGNAL_H
00016
00017 #include "../../../BaseGenesysTerminalApplication.h"
00018
00019 class Smart_WaitSignal : public BaseGenesysTerminalApplication {
00020 public:
00021     Smart_WaitSignal();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* SMART_WAITSIGNAL_H */
00027
```

18.225 AnElectronicAssemblyAndTestSystem.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Smart_AnElectronicAssemblyAndTestSystem.h
00008  * Author: rlcancian
00009  *
00010  * Created on 6 de dezembro de 2021, 11:09
00011  */
00012
00013 #ifndef ANELECTRONICASSEMBLYANDTESTSYSTEM_H
00014 #define ANELECTRONICASSEMBLYANDTESTSYSTEM_H
00015
00016 #include "../../BaseGenesysTerminalApplication.h"
00017
00018 class AnElectronicAssemblyAndTestSystem : public BaseGenesysTerminalApplication {
00019 public:
00020     AnElectronicAssemblyAndTestSystem();
00021 public:
00022     virtual int main(int argc, char** argv) override;
00023
00024 };
00025
00026 #endif /* ANELECTRONICASSEMBLYANDTESTSYSTEM_H */
00027
```

18.226 BufferFIFO.h

```
00001 #pragma once
00002
00003 /*
00004  * File:   BufferFIFO.h
00005  * Author: rlcancian
00006  *
00007  * Created on 13 de setembro de 2023, 18:03
00008  */
00009
00010 #include "../../BaseGenesysTerminalApplication.h"
00011
00012 #include "kernel/simulator/Simulator.h"
00013 #include "plugins/components/Logic/Create.h"
00014 #include "plugins/components/DiscreteProcessing/Clone.h"
00015 #include "plugins/components/Synchronization/Wait.h"
00016 #include "plugins/components/Synchronization/Signal.h"
00017 #include "plugins/components/Logic/Dispose.h"
00018 #include "plugins/components/Decisions/Decide.h"
00019 #include "plugins/data/Synchronization/SignalData.h"
00020 #include "plugins/data/Logic/Variable.h"
00021
00022 class BufferFIFO : public BaseGenesysTerminalApplication {
00023 public:
00024     BufferFIFO() {}
00025 public:
00026     virtual int main(int argc, char** argv) {
00027         Simulator* genesys = new Simulator();
00028         this->setDefaultTraceHandlers(genesys->getTraceManager());
00029         genesys->getPluginManager()->autoInsertPlugins("autoloadplugins.txt");
```

```
00030         Model* model = genesys->getModelManager()->newModel();
00031         PluginManager* plugins = genesys->getPluginManager();
00032 
00033         Create* create1 = plugins->newInstance<Create>(model);
00034         create1->setTimeBetweenCreationsExpression("1", Util::TimeUnit::hour);
00035 
00036         Clone* clone1 = plugins->newInstance<Clone>(model);
00037         clone1->setNumClonesExpression("1");
00038 
00039         create1->getConnectionManager()->insert(clone1);
00040 
00041         SignalData* sigdata1 = plugins->newInstance<SignalData>(model);
00042         Queue* queue1 = plugins->newInstance<Queue>(model, "waitingQueue");
00043         Wait* wait1 = plugins->newInstance<Wait>(model);
00044         wait1->setWaitType(Wait::WaitType::WaitForSignal);
00045         wait1->setSignalData(sigdata1);
00046         wait1->setQueue(queue1);
00047 
00048         clone1->getConnectionManager()->insert(wait1);
00049 
00050         Dispose* dispose1 = plugins->newInstance<Dispose>(model);
00051 
00052         wait1->getConnectionManager()->insert(dispose1);
00053 
00054         Decide* decide1 =  plugins->newInstance<Decide>(model);
00055         decide1->getConditions()->insert("nq(waitingQueue)<tamBuffer");
00056         Variable* var1 = plugins->newInstance<Variable>(model, "tamBuffer");
00057         var1->setInitialValue(5);
00058 
00059         clone1->getConnectionManager()->insert(decide1);
00060 
00061         Dispose* dispose2 = plugins->newInstance<Dispose>(model);
00062         dispose2->setReportStatistics(false);
00063 
00064         decide1->getConnectionManager()->insert(dispose2);
00065 
00066         Signal* signal1 = plugins->newInstance<Signal>(model);
00067         signal1->setSignalData(sigdata1);
00068 
00069         decide1->getConnectionManager()->insert(signal1);
00070         signal1->getConnectionManager()->insert(dispose2);
00071 
00072         ModelSimulation* sim = model->getSimulation();
00073         sim->setReplicationLength(50, Util::TimeUnit::hour);
00074         sim->setReplicationReportBaseTimeUnit(Util::TimeUnit::hour);
00075 
00076         model->save("./model/teachingBufferFIFO.gen");
00077         model->getTracer()->setTraceLevel(TraceManager::Level::L9_mostDetailed);
00078 
00079         do {
00080             sim->step();
00081             std::cout « "Press ENTER to continue...";
00082             std::cin.ignore(std::numeric_limits <std::streamsize> ::max(), '\n');
00083         } while (sim->isPaused());
00084         delete genesys;
00085         return 0;
00086     }
00087 };
00088 
```

18.227 ContinuousModel.h

```
00001 #pragma once
00002 
00003 #include "../../BaseGenesysTerminalApplication.h"
00004 
00005 class ContinuousModel: public BaseGenesysTerminalApplication {
00006 public:
00007     ContinuousModel();
00008 public: // virtual
00009     virtual void Simulate(Simulator* genesys, Model* model, PluginManager *plugins) override;
00010 };
00011 
```

18.228 FullSimulationOfComplexModel.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
```

```
00007 /*
00008  * File:   buildSimpleModel1.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 2 de Outubro de 2018, 19:18
00012  */
00013
00014 #ifndef COMPLEXMODEL01_H
00015 #define COMPLEXMODEL01_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class FullSimulationOfComplexModel : public BaseGenesysTerminalApplication {
00020 public:
00021     FullSimulationOfComplexModel();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* COMPLEXMODEL01_H */
00027
```

18.229 Half_Adder.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 #ifndef HALF_ADDER_H
00008 #define HALF_ADDER_H
00009
00010 #include "../../BaseGenesysTerminalApplication.h"
00011
00012 class Half_Adder : public BaseGenesysTerminalApplication {
00013 public:
00014     Half_Adder();
00015 public:
00016     virtual int main(int argc, char** argv) override;
00017 };
00018
00019 #endif /* HALF_ADDER_H */
00020
```

18.230 Loja01.h

```
00001 #pragma once
00002
00003
00004 /*
00005  * File:   Loja01.h
00006  * Author: rlcancian
00007  *
00008  * Created on 13 de Novembro de 2024, 10:25
00009  */
00010 #include "../../../BaseGenesysTerminalApplication.h"
00011
00012 class Loja01 : public BaseGenesysTerminalApplication {
00013 public:
00014     Loja01();
00015 public:
00016     virtual int main(int argc, char** argv) override;
00017 };
00018
00019
00020
```

18.231 OperatingSystem02.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_SistemasOperacionais02.h
00009  * Author: rlcancian
```

```
00010  *
00011  * Created on 12 de abril de 2021, 21:54
00012  */
00013
00014 #ifndef OPERATINGSYSTEMSS02_H
00015 #define OPERATINGSYSTEMSS02_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class OperatingSystem02 : public BaseGenesysTerminalApplication {
00020 public:
00021     OperatingSystem02();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* OPERATINGSYSTEMSS02_H */
00027
```

18.232 OperatingSystem03.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Smart_SistemaOperacional03.h
00009  * Author: rlcancian
00010  *
00011  * Created on 23 de abril de 2021, 14:03
00012  */
00013
00014 #ifndef OPERATINGSYSTEMS03_H
00015 #define OPERATINGSYSTEMS03_H
00016
00017 #include "../../BaseGenesysTerminalApplication.h"
00018
00019 class OperatingSystem03 : public BaseGenesysTerminalApplication {
00020 public:
00021     OperatingSystem03();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 private:
00025
00026 };
00027
00028 #endif /* OPERATINGSYSTEMS03_H */
00029
```

18.233 Rectifier.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 #ifndef RECTIFIER_H
00008 #define RECTIFIER_H
00009
00010 #include "../../BaseGenesysTerminalApplication.h"
00011
00012 class Rectifier : public BaseGenesysTerminalApplication {
00013 public:
00014     Rectifier();
00015 public:
00016     virtual int main(int argc, char** argv) override;
00017 };
00018
00019 #endif /* HALF_ADDER_H */
00020
```

18.234 WcmBioNetworkBridge.h

```
00001 #ifndef WCMBIONETWORKBRIDGE_H
00002 #define WCMBIONETWORKBRIDGE_H
00003
```

```
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00014 class WcmBioNetworkBridge : public BaseGenesysTerminalApplication {
00015 public:
00016     WcmBioNetworkBridge();
00017     virtual int main(int argc, char** argv) override;
00018 };
00019
00020 #endif /* WCMBIONETWORKBRIDGE_H */
```

18.235 WcmCompartmentTransportWholeCellMvp.h

```
00001 #ifndef WCMCOMPARTMENTTRANSPORTWHOLECELLMVP_H
00002 #define WCMCOMPARTMENTTRANSPORTWHOLECELLMVP_H
00003
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00019 class WcmCompartmentTransportWholeCellMvp : public BaseGenesysTerminalApplication {
00020 public:
00021     WcmCompartmentTransportWholeCellMvp();
00022     virtual int main(int argc, char** argv) override;
00023 };
00024
00025 #endif /* WCMCOMPARTMENTTRANSPORTWHOLECELLMVP_H */
```

18.236 WcmFluxBalanceWholeCellMvp.h

```
00001 #ifndef WCMFLUXBALANCEWHOLECELLMVP_H
00002 #define WCMFLUXBALANCEWHOLECELLMVP_H
00003
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00015 class WcmFluxBalanceWholeCellMvp : public BaseGenesysTerminalApplication {
00016 public:
00017     WcmFluxBalanceWholeCellMvp();
00018     virtual int main(int argc, char** argv) override;
00019 };
00020
00021 #endif /* WCMFLUXBALANCEWHOLECELLMVP_H */
```

18.237 WcmGeneExpression.h

```
00001 #ifndef WCMGENEEXPRESSION_H
00002 #define WCMGENEEXPRESSION_H
00003
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00028 class WcmGeneExpression : public BaseGenesysTerminalApplication {
00029 public:
00030     WcmGeneExpression();
00031     virtual int main(int argc, char** argv) override;
00032 };
00033
00034 #endif /* WCMGENEEXPRESSION_H */
```

18.238 WcmLifecycleStress.h

```
00001 #ifndef WCMLIFECYCLESTRESS_H
00002 #define WCMLIFECYCLESTRESS_H
00003
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00014 class WcmLifecycleStress : public BaseGenesysTerminalApplication {
00015 public:
00016     WcmLifecycleStress();
00017     virtual int main(int argc, char** argv) override;
00018 };
00019
00020 #endif /* WCMLIFECYCLESTRESS_H */
```

18.239 WcmMgenitaliumKarr2012.h

```
00001 #ifndef WCMMGENITALIUMKARR2012_H
00002 #define WCMMGENITALIUMKARR2012_H
```

```
00003
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00053 class WcmMgenitaliumKarr2012 : public BaseGenesysTerminalApplication {
00054 public:
00055     WcmMgenitaliumKarr2012();
00056     virtual int main(int argc, char** argv) override;
00057 };
00058
00059 #endif /* WCMMGENITALIUMKARR2012_H */
```

18.240 WcmPathwayStressWholeCellMvp.h

```
00001 #ifndef WCMPATHWAYSTRESSWHOLECELLMVP_H
00002 #define WCMPATHWAYSTRESSWHOLECELLMVP_H
00003
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00013 class WcmPathwayStressWholeCellMvp : public BaseGenesysTerminalApplication {
00014 public:
00015     WcmPathwayStressWholeCellMvp();
00016     virtual int main(int argc, char** argv) override;
00017 };
00018
00019 #endif /* WCMPATHWAYSTRESSWHOLECELLMVP_H */
```

18.241 WcmSimpleSsa.h

```
00001 #ifndef WCMSIMPLESSA_H
00002 #define WCMSIMPLESSA_H
00003
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00025 class WcmSimpleSsa : public BaseGenesysTerminalApplication {
00026 public:
00027     WcmSimpleSsa();
00028     virtual int main(int argc, char** argv) override;
00029 };
00030
00031 #endif /* WCMSIMPLESSA_H */
```

18.242 WcmWholeCellMvp.h

```
00001 #ifndef WCMWHOLECELLMVP_H
00002 #define WCMWHOLECELLMVP_H
00003
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00020 class WcmWholeCellMvp : public BaseGenesysTerminalApplication {
00021 public:
00022     WcmWholeCellMvp();
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* WCMWHOLECELLMVP_H */
```

18.243 WcmYeastDidacticWholeCellMvp.h

```
00001 #ifndef WCMYEASTDIDACTICWHOLECELLMVP_H
00002 #define WCMYEASTDIDACTICWHOCELLMVP_H
00003
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00018 class WcmYeastDidacticWholeCellMvp : public BaseGenesysTerminalApplication {
00019 public:
00020     WcmYeastDidacticWholeCellMvp();
00021     virtual int main(int argc, char** argv) override;
00022 };
00023
00024 #endif /* WCMYEASTDIDACTICWHOCELLMVP_H */
```

18.244 WcmYeastEukaryoticWholeCellMvp.h

```
00001 #ifndef WCMYEASTEUKARYOTICWHOLECELLMVP_H
```

```
00002 #define WCMYEASTEUKARYOTICWHOLECELLMVP_H
00003
00004 #include "../../BaseGenesysTerminalApplication.h"
00005
00019 class WcmYeastEukaryoticWholeCellMvp : public BaseGenesysTerminalApplication {
00020 public:
00021     WcmYeastEukaryoticWholeCellMvp();
00022 public:
00023     virtual int main(int argc, char** argv) override;
00024 };
00025
00026 #endif /* WCMYEASTEUKARYOTICWHOLECELLMVP_H */
```

18.245 GenesysShell.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   GenesysTerminalAppApp.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 23 de Maio de 2019, 13:02
00012  */
00013
00014 #ifndef GENESYSSHELL_H
00015 #define GENESYSSHELL_H
00016
00017 #include "../BaseGenesysTerminalApplication.h"
00018 #include "kernel/simulator/SimulatorFacade.h"
00019 #include "kernel/simulator/Simulator.h"
00020 #include "kernel/util/List.h"
00021 #include <memory>
00022 #include <regex>
00023 #include <fstream>
00024 #include <assert.h>
00025
00026 class GenesysShell : public BaseGenesysTerminalApplication {
00027 private:
00028     typedef std::function<void() > ExecuterMember;
00029
00030     template<typename Class>
00031     ExecuterMember DefineExecuterMember(Class * object, void (Class::*function)()) {
00032         return std::bind(function, object); //, std::placeholders::_1
00033     }
00034
00035     class ShellCommand {
00036     public:
00037
00038         ShellCommand(std::string shortname, std::string longname, std::string parameters, std::string
      descrition, ExecuterMember executer) {
00039             this->descrition = descrition;
00040             this->longname = longname;
00041             this->parameters = parameters;
00042             this->shortname = shortname;
00043             this->executer = executer;
00044         }
00045         std::string shortname;
00046         std::string longname;
00047         std::string parameters;
00048         std::string descrition;
00049         ExecuterMember executer;
00050     };
00051 public:
00052     GenesysShell();
00053     GenesysShell(const GenesysShell& orig);
00054     virtual ~GenesysShell() = default;
00055 public:
00056     virtual int main(int argc, char** argv) override;
00057 public: // commands
00058     void cmdScript();
00059     void cmdExecuteScript();
00060     void cmdHelp();
00061     void cmdQuit();
00062     void cmdBash();
00063     void cmdFacade();
00064     void cmdLicence();
00065     void cmdInfo();
00066     void cmdSim();
00067     void cmdData();
00068     void cmdComponent();
00069     void cmdExperiment();
```

```
00070     void cmdParser();
00071     void cmdTraceLevel();
00072     void cmdPlugin();
00073     //void cmdPluginInfo();
00074     //void cmdPluginAdd();
00075     //void cmdPluginRemove();
00076     void cmdSimulation();
00077     //void cmdSimulationStep();
00078     //void cmdSimulationStop();
00079     //void cmdSimulationInfo();
00080     void cmdReplication();
00081     //void cmdShowReport();
00082     void cmdModel();
00083     //void cmdModelClose();
00084     void cmdModelLoad();
00085     //void cmdModelCheck();
00086     void cmdModelSave();
00087     //void cmdModelShow();
00088     //void cmdModelSetInfos();
00089 private:
00090     void defineCommands();
00091     void run(List<std::string>* arguments);
00092     void Trace(std::string message);
00093     //void tryExecuteCommand(std::string inputText, const char* shortPrefix, const char* longPrefix,
      std::string separator);
00094     void tryExecuteCommand(std::string inputText);
00095     std::vector<std::string> split(std::string text, std::string separatorRegex);
00096     std::string joinArguments(std::size_t firstArgumentIndex = 1) const;
00097     std::string lowercase(std::string text) const;
00098     bool requireCurrentModel(const std::string& actionDescription);
00099     bool parseUnsigned(const std::string& text, unsigned int& value) const;
00100     bool parseInt(const std::string& text, int& value) const;
00101     bool parseDouble(const std::string& text, double& value) const;
00102     bool parseBool(const std::string& text, bool& value) const;
00103     bool parseTraceLevel(const std::string& text, TraceManager::Level& level) const;
00104     bool parseTimeUnit(const std::string& text, Util::TimeUnit& timeUnit) const;
00105 private:
00106     std::unique_ptr<Simulator> simulator;
00107     SimulatorFacade facade;
00108     Model* model = nullptr;
00109     std::vector<std::string> *_typedWords = new std::vector<std::string>();
00110     std::vector<std::string> *_history = new std::vector<std::string>();
00111     List<ShellCommand*>* _commands = new List<ShellCommand*>();
00112     std::string _prompt = "$genesys> ";
00113     bool _exitRequested = false;
00114 };
00115 #endif /* GENESYSSHELL_H */
```

18.246 GenesysShell_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   GenesysShell_if.h
00009  * Author: rlcancian
00010  *
00011  * Created on 24 de Maio de 2019, 11:02
00012  */
00013
00014 #ifndef GENESYSSHELL_IF_H
00015 #define GENESYSSHELL_IF_H
00016
00017 #include "GenesysApplication_if.h"
00018 #include "Util.h"
00019
00020 class GenesysShell_if : public GenesysApplication_if {
00021 public:
00022     virtual ~GenesysShell_if() = default;
00023     virtual void openModel(std::string filename) = 0;
00024     virtual void saveModelAs(std::string filename) = 0;
00025     virtual void saveModel() = 0;
00026     virtual void listElements() = 0;
00027     virtual void listComponents() = 0;
00028     virtual void listHosts() = 0;
00029     virtual void listPlugins() = 0;
00030     virtual void deleteTraceFiles() = 0;
00031     virtual void traceLevel(TraceManager::TraceLevel tracelevel) = 0;
00032     virtual void addPlugin(std::string filename) = 0;
00033     virtual void addFromFile(std::string filename) = 0;
00034     virtual void readCommandsFromFile(std::string filename) = 0;
00035     virtual void redirectTrace(std::string trace, std::string dest, std::string filename) = 0;
```

```
00036     virtual void closeModel() = 0;
00037     virtual void createModel() = 0;
00038     virtual void execLinuxCommand(std::string command) = 0;
00039     virtual void verboseMode(bool on) = 0;
00040     virtual void check() = 0;
00041     virtual void getGenesysInfo() = 0;
00042     virtual void getCommandLine() = 0;
00043     virtual void sendFile(std::string filename, std::string hostname, std::string portname) = 0;
00044     virtual void setActivationCode(std::string code) = 0;
00045     virtual void receiveFile(std::string filename) = 0;
00046     virtual void startSimulation() = 0;
00047     virtual void stepSimulation() = 0;
00048     virtual void stopSimulation() = 0;
00049     virtual void showInit() = 0;
00050     virtual void showHelp() = 0;
00051     virtual void showHostName() = 0;
00052     //virtual void execute() = 0;
00053     //property TraceHandler : TProcTrace write aTraceHandler;
00054 };
00055
00056 #endif /* GENESYSSHELL_IF_H */
```

18.247 ApiRouter.h

```
00001 #pragma once
00002
00003 #include "../http/HttpRequest.h"
00004 #include "../http/HttpResponse.h"
00005 #include "../service/SimulatorSessionService.h"
00006
00007 #include <optional>
00008
00012 class ApiRouter {
00013 public:
00018     explicit ApiRouter(SimulatorSessionService& simulatorService);
00019
00025     HttpResponse handle(const HttpRequest& request) const;
00026
00027 private:
00028     SimulatorSessionService& _simulatorService;
00029
00037     static HttpResponse _jsonError(int status, const char* code, const char* message);
00043     static std::string _extractBearerToken(const HttpRequest& request);
00050     static bool _tryExtractFilenameFromBody(const std::string& body, std::string& outFilename);
00058     static bool _tryExtractUnsignedIntField(const std::string& body, const std::string& fieldName,
      unsigned int& outValue);
00066     static bool _tryExtractDoubleField(const std::string& body, const std::string& fieldName, double&
      outValue);
00074     static bool _tryExtractBooleanField(const std::string& body, const std::string& fieldName, bool&
      outValue);
00080     static std::optional<SimulatorSessionService::SimulationConfigInput>
      _parseSimulationConfigBody(const std::string& body);
00086     static std::string _simulationStatusDataJson(const
      SimulatorSessionService::SimulationStatusResult& status);
00092     static HttpResponse _mapPersistenceError(const SimulatorSessionService::ModelPersistenceResult&
      result);
00098     static HttpResponse _mapModelImportError(const SimulatorSessionService::ModelImportResult&
      result);
00104     static std::string _escapeJson(const std::string& value);
00110     static std::string _modelInfoDataJson(const SimulatorSessionService::ModelInfoResult& info);
00116     static std::string _workerInfoDataJson(const SimulatorSessionService::WorkerInfoResult& info);
00122     static std::string _workerCapabilitiesDataJson(const
      SimulatorSessionService::WorkerCapabilitiesResult& capabilities);
00129     static bool _tryExtractWorkerJobIdFromPath(const std::string& path, std::string& outJobId);
00136     static bool _tryExtractWorkerJobRunIdFromPath(const std::string& path, std::string& outJobId);
00143     static bool _tryExtractWorkerJobResultIdFromPath(const std::string& path, std::string& outJobId);
00149     static const char* _workerJobStateToString(WorkerJobState state);
00155     static std::string _workerJobDataJson(const SimulatorSessionService::WorkerJobInfoResult& job);
00161     static std::string _workerJobResultDataJson(const SimulatorSessionService::WorkerJobResultInfo&
      result);
00168     static HttpResponse _mapWorkerJobError(SimulatorSessionService::WorkerJobError error, bool
      includeMissingModelMessage);
00169 };
```

18.248 TokenService.h

```
00001 #pragma once
00002
00003 #include <cstddef>
00004 #include <string>
00005
00006 class TokenService {
```

```
00007 public:
00008     TokenService();
00009
00010     [[nodiscard]] std::string generateSessionId();
00011     [[nodiscard]] std::string generateAccessToken();
00012
00013 private:
00014     [[nodiscard]] std::string _generateHexToken(std::size_t bytes);
00015 };
```

18.249 BaseGenesysWebApplication.h

```
00001 #pragma once
00002
00003 #include "../GenesysApplication_if.h"
00004
00012 class BaseGenesysWebApplication : public GenesysApplication_if {
00013 public:
00020     int main(int argc, char** argv) override;
00021
00022 private:
00029     static unsigned short _readPort(int argc, char** argv);
00036     static unsigned long _readMaxRequests(int argc, char** argv);
00037 };
```

18.250 HttpRequest.h

```
00001 #pragma once
00002
00003 #include <algorithm>
00004 #include <cctype>
00005 #include <string>
00006 #include <unordered_map>
00007
00015 struct HttpRequest {
00017     std::string method;
00019     std::string path;
00021     std::string body;
00023     std::unordered_map<std::string, std::string> headers;
00024
00030     [[nodiscard]] std::string header(const std::string& name) const {
00031         std::string normalized = name;
00032         std::transform(normalized.begin(), normalized.end(), normalized.begin(), [](unsigned char c) {
00033             return static_cast<char>(std::tolower(c));
00034         });
00035         auto it = headers.find(normalized);
00036         if (it == headers.end()) {
00037             return {};
00038         }
00039         return it->second;
00040     }
00041 };
```

18.251 HttpResponse.h

```
00001 #pragma once
00002
00003 #include <string>
00004 #include <unordered_map>
00005
00012 struct HttpResponse {
00014     int status = 200;
00016     std::string contentType = "application/json";
00018     std::string body;
00020     std::unordered_map<std::string, std::string> headers;
00021 };
```

18.252 SimpleHttpServer.h

```
00001 #pragma once
00002
00003 #include "HttpRequest.h"
00004 #include "HttpResponse.h"
00005
00006 #include <atomic>
00007 #include <functional>
00008
```

```
00015 class SimpleHttpServer {
00016 public:
00017     using RequestHandler = std::function<HttpResponse(const HttpRequest&)>;
00018
00023     explicit SimpleHttpServer(unsigned short port);
00024
00031     bool serve(const RequestHandler& handler, unsigned long maxRequests = 0);
00032
00036     void requestStop();
00037
00038 private:
00039     unsigned short _port;
00040     std::atomic<bool> _stopRequested{false};
00041 };
```

18.253 SimulatorSessionService.h

```
00001 #pragma once
00002
00003 #include "../session/SessionManager.h"
00004 #include "../worker/WorkerJob.h"
00005 #include "../worker/WorkerJobManager.h"
00006
00007 #include <string>
00008
00012 class SimulatorSessionService {
00013 public:
00017     enum class PersistenceError {
00018         None,
00019         InvalidToken,
00020         InvalidFilename,
00021         MissingCurrentModel,
00022         FileNotFound,
00023         OperationFailed
00024     };
00025
00029     enum class ModelImportError {
00030         None,
00031         InvalidToken,
00032         EmptySpecification,
00033         InvalidSpecification,
00034         OperationFailed
00035     };
00036
00040     enum class WorkerJobError {
00041         None,
00042         InvalidToken,
00043         MissingCurrentModel,
00044         JobNotFound,
00045         AccessDenied,
00046         OperationFailed,
00047         ResultNotReady
00048     };
00049
00053     struct CreateSessionResult {
00054         std::string sessionId;
00055         std::string accessToken;
00056     };
00057
00061     struct SimulatorInfoResult {
00062         std::string name;
00063         std::string versionName;
00064         unsigned int versionNumber;
00065     };
00066
00070     struct WorkerInfoResult {
00071         std::string role;
00072         std::string application;
00073         std::string apiFamily;
00074         std::string apiVersion;
00075         std::string simulatorName;
00076         std::string simulatorVersionName;
00077         unsigned int simulatorVersionNumber = 0;
00078     };
00079
00083     struct WorkerCapabilitiesResult {
00084         bool supportsSessionApi = false;
00085         bool supportsSessionScopedSimulator = false;
00086         bool supportsModelCreation = false;
00087         bool supportsModelPersistence = false;
00088         bool supportsSimulationStatus = false;
00089         bool supportsSimulationConfig = false;
00090         bool supportsSynchronousRun = false;
00091         bool supportsSynchronousStep = false;
00092         bool supportsDistributedJobs = false;
```

```
00093         bool supportsJobPolling = false;
00094         bool supportsBackgroundExecution = false;
00095         bool supportsModelUpload = false;
00096         bool supportsStreamingEvents = false;
00097         bool supportsJobResultRetrieval = false;
00098     };
00099
00103     struct ModelInfoResult {
00104         bool exists = false;
00105         unsigned int modelId = 0;
00106         bool hasChanged = false;
00107         unsigned int level = 0;
00108         std::string name;
00109         std::string analystName;
00110         std::string projectTitle;
00111         std::string version;
00112         std::string description;
00113         unsigned int componentCount = 0;
00114     };
00115
00119     struct ModelPersistenceResult {
00120         bool success = false;
00121         PersistenceError error = PersistenceError::None;
00122         std::string filename;
00123         ModelInfoResult modelInfo;
00124     };
00125
00129     struct ModelImportResult {
00130         bool success = false;
00131         ModelImportError error = ModelImportError::None;
00132         ModelInfoResult modelInfo;
00133     };
00134
00138     struct WorkerJobInfoResult {
00139         std::string jobId;
00140         std::string sessionId;
00141         WorkerJobState state = WorkerJobState::Queued;
00142         std::string snapshotFilename;
00143         std::string createdMarker;
00144         std::string message;
00145     };
00146
00150     struct WorkerJobCreationResult {
00151         bool success = false;
00152         WorkerJobError error = WorkerJobError::None;
00153         WorkerJobInfoResult jobInfo;
00154     };
00155
00159     struct WorkerJobQueryResult {
00160         bool success = false;
00161         WorkerJobError error = WorkerJobError::None;
00162         WorkerJobInfoResult jobInfo;
00163     };
00164
00168     struct WorkerJobRunResult {
00169         bool success = false;
00170         WorkerJobError error = WorkerJobError::None;
00171         WorkerJobInfoResult jobInfo;
00172     };
00173
00177     struct WorkerJobResultInfo {
00178         std::string jobId;
00179         WorkerJobState state = WorkerJobState::Queued;
00180         std::string message;
00181         double simulatedTime = 0.0;
00182         unsigned int currentReplicationNumber = 0;
00183         unsigned int numberOfReplications = 0;
00184         double replicationLength = 0.0;
00185         double warmUpPeriod = 0.0;
00186         bool hasIsPaused = false;
00187         bool isPaused = false;
00188     };
00189
00193     struct WorkerJobResultQueryResult {
00194         bool success = false;
00195         WorkerJobError error = WorkerJobError::None;
00196         WorkerJobResultInfo jobResult;
00197     };
00198
00202     struct SimulationStatusResult {
00203         bool success = false;
00204         bool invalidToken = false;
00205         bool hasCurrentModel = false;
00206         bool isRunning = false;
00207         bool isPaused = false;
00208         double simulatedTime = 0.0;
00209         unsigned int currentReplicationNumber = 0;
```

```
00210         unsigned int numberOfReplications = 0;
00211         double replicationLength = 0.0;
00212         double warmUpPeriod = 0.0;
00213         bool pauseOnEvent = false;
00214         bool pauseOnReplication = false;
00215         bool initializeStatistics = false;
00216         bool initializeSystem = false;
00217     };
00218 
00222     struct SimulationConfigInput {
00223         unsigned int numberOfReplications = 0;
00224         double replicationLength = 0.0;
00225         double warmUpPeriod = 0.0;
00226         bool pauseOnEvent = false;
00227         bool pauseOnReplication = false;
00228         bool initializeStatistics = false;
00229         bool initializeSystem = false;
00230     };
00231 
00235     struct SimulationConfigResult {
00236         bool success = false;
00237         bool invalidToken = false;
00238         bool missingCurrentModel = false;
00239         SimulationStatusResult status;
00240     };
00241 
00245     struct SimulationActionResult {
00246         bool success = false;
00247         bool invalidToken = false;
00248         bool missingCurrentModel = false;
00249         SimulationStatusResult status;
00250     };
00251 
00257     explicit SimulatorSessionService(SessionManager& sessionManager, WorkerJobManager&
      workerJobManager);
00258 
00263     CreateSessionResult createSession();
00270     bool tryGetSimulatorInfo(const std::string& accessToken, SimulatorInfoResult& outInfo);
00275     WorkerInfoResult getWorkerInfo() const;
00280     WorkerCapabilitiesResult getWorkerCapabilities() const;
00287     bool tryCreateModel(const std::string& accessToken, ModelInfoResult& outInfo);
00294     bool tryGetCurrentModelInfo(const std::string& accessToken, ModelInfoResult& outInfo);
00300     SimulationStatusResult getSimulationStatus(const std::string& accessToken);
00307     SimulationConfigResult configureSimulation(const std::string& accessToken, const
      SimulationConfigInput& input);
00313     SimulationActionResult runSimulation(const std::string& accessToken);
00319     SimulationActionResult stepSimulation(const std::string& accessToken);
00326     ModelPersistenceResult saveCurrentModel(const std::string& accessToken, const std::string&
      filename);
00333     ModelPersistenceResult loadModel(const std::string& accessToken, const std::string& filename);
00340     ModelImportResult importModelFromLanguage(const std::string& accessToken, const std::string&
      modelSpecification);
00346     WorkerJobCreationResult createWorkerJob(const std::string& accessToken);
00353     WorkerJobQueryResult getWorkerJob(const std::string& accessToken, const std::string& jobId);
00360     WorkerJobRunResult runWorkerJob(const std::string& accessToken, const std::string& jobId);
00367     WorkerJobResultQueryResult getWorkerJobResult(const std::string& accessToken, const std::string&
      jobId);
00368 
00369 private:
00375     static bool _isSafeFilename(const std::string& filename);
00376 
00382     static WorkerJobInfoResult _toWorkerJobInfoResult(const WorkerJob& job);
00383 
00389     static WorkerJobResultInfo _toWorkerJobResultInfo(const WorkerJob& job);
00390 
00391     SessionManager& _sessionManager;
00392     WorkerJobManager& _workerJobManager;
00393 };
```

18.254 WebWorkerRuntime.h

```
00001 #pragma once
00002 
00003 #include "api/ApiRouter.h"
00004 #include "auth/TokenService.h"
00005 #include "http/SimpleHttpServer.h"
00006 #include "kernel/simulator/Simulator.h"
00007 #include "service/SimulatorSessionService.h"
00008 #include "session/SessionManager.h"
00009 #include "worker/WorkerJobManager.h"
00010 
00011 #include <atomic>
00012 #include <deque>
00013 #include <memory>
00014 #include <mutex>
```

```
00015 #include <string>
00016 #include <thread>
00017 #include <vector>
00018
00026 class WebWorkerRuntime {
00027 public:
00031     struct Snapshot {
00032         bool isRunning = false;
00033         bool isStarting = false;
00034         bool isStopping = false;
00035         bool hasFailed = false;
00036         unsigned short port = 8080;
00037         unsigned long maxRequests = 0;
00038         unsigned long servedRequests = 0;
00039         unsigned long activeRequests = 0;
00040         std::string statusMessage;
00041         std::string lastError;
00042         std::vector<std::string> recentEvents;
00043         std::vector<std::string> recentErrors;
00044     };
00045
00049     explicit WebWorkerRuntime(unsigned short defaultPort = 8080, unsigned long maxRequests = 0);
00053     ~WebWorkerRuntime();
00054
00059     bool start();
00063     void stop();
00067     void wait();
00072     bool restart();
00073
00077     bool isRunning() const;
00081     Snapshot snapshot() const;
00082
00086     void setConfiguredPort(unsigned short port);
00090     unsigned short configuredPort() const;
00091
00095     void setConfiguredMaxRequests(unsigned long maxRequests);
00099     unsigned long configuredMaxRequests() const;
00100
00101 private:
00105     void _run();
00109     void _fail(const std::string& errorMessage);
00113     void _recordEventLocked(const std::string& message);
00117     void _recordErrorLocked(const std::string& message);
00121     std::string _statusMessageLocked() const;
00125     static std::unique_ptr<Simulator> _createSimulator();
00126
00127     mutable std::mutex _mutex;
00128     std::atomic<bool> _stopRequested{false};
00129     std::atomic<bool> _threadRunning{false};
00130     std::atomic<bool> _starting{false};
00131     std::atomic<bool> _stopping{false};
00132     std::atomic<unsigned long> _servedRequests{0};
00133     std::atomic<unsigned long> _activeRequests{0};
00134     unsigned short _configuredPort = 8080;
00135     unsigned long _configuredMaxRequests = 0;
00136     std::string _lastError;
00137     std::string _statusMessage;
00138     std::deque<std::string> _recentEvents;
00139     std::deque<std::string> _recentErrors;
00140     std::thread _thread;
00141
00142     std::unique_ptr<TokenService> _tokenService;
00143     std::unique_ptr<SessionManager> _sessionManager;
00144     std::unique_ptr<WorkerJobManager> _workerJobManager;
00145     std::unique_ptr<SimulatorSessionService> _simulatorSessionService;
00146     std::unique_ptr<ApiRouter> _router;
00147     std::unique_ptr<SimpleHttpServer> _server;
00148 };
```

18.255 SessionContext.h

```
00001 #pragma once
00002
00003 #include "kernel/simulator/Simulator.h"
00004
00005 #include <filesystem>
00006 #include <memory>
00007 #include <mutex>
00008 #include <string>
00009
00017 struct SessionContext {
00019     std::string sessionId;
00021     std::string accessToken;
00023     std::filesystem::path workspacePath;
00025     std::unique_ptr<Simulator> simulator;
```

```
00027     mutable std::mutex mutex;
00028 };
```

18.256 SessionManager.h

```
00001 #pragma once
00002
00003 #include "SessionContext.h"
00004 #include "../auth/TokenService.h"
00005
00006 #include <functional>
00007 #include <filesystem>
00008 #include <memory>
00009 #include <mutex>
00010 #include <string>
00011 #include <unordered_map>
00012
00020 class SessionManager {
00021 public:
00023     using SimulatorFactory = std::function<std::unique_ptr<Simulator>()>;
00024
00030     SessionManager(TokenService& tokenService, SimulatorFactory simulatorFactory);
00031
00036     SessionContext* createSession();
00042     SessionContext* getSessionByToken(const std::string& token);
00043
00044 private:
00049     static std::filesystem::path _buildWorkspaceRoot();
00050
00051     TokenService& _tokenService;
00052     SimulatorFactory _simulatorFactory;
00053     std::filesystem::path _workspaceRoot;
00054     mutable std::mutex _sessionsMutex;
00055     std::unordered_map<std::string, std::unique_ptr<SessionContext» _sessionsByToken;
00056 };
```

18.257 WorkerJob.h

```
00001 #pragma once
00002
00003 #include <optional>
00004 #include <string>
00005
00009 enum class WorkerJobState {
00010     Queued,
00011     Running,
00012     Finished,
00013     Failed
00014 };
00015
00019 struct WorkerJobTerminalResult {
00021     double simulatedTime = 0.0;
00023     unsigned int currentReplicationNumber = 0;
00025     unsigned int numberOfReplications = 0;
00027     double replicationLength = 0.0;
00029     double warmUpPeriod = 0.0;
00031     std::optional<bool> isPaused;
00032 };
00033
00037 struct WorkerJob {
00039     std::string jobId;
00041     std::string sessionId;
00043     WorkerJobState state = WorkerJobState::Queued;
00045     std::string snapshotFilename;
00047     std::string createdMarker;
00049     std::string message;
00051     bool hasTerminalResult = false;
00053     WorkerJobTerminalResult terminalResult;
00054 };
```

18.258 WorkerJobManager.h

```
00001 #pragma once
00002
00003 #include "WorkerJob.h"
00004
00005 #include <mutex>
00006 #include <optional>
00007 #include <string>
```

```
00008 #include <unordered_map>
00009
00016 class WorkerJobManager {
00017 public:
00023     WorkerJob createQueuedJob(const std::string& sessionId);
00024
00031     bool setSnapshotFilename(const std::string& jobId, const std::string& snapshotFilename);
00038     bool setState(const std::string& jobId, WorkerJobState state);
00039
00046     bool setMessage(const std::string& jobId, const std::string& message);
00047
00054     bool setTerminalResult(const std::string& jobId, const WorkerJobTerminalResult& terminalResult);
00055
00061     std::optional<WorkerJob> getJob(const std::string& jobId) const;
00062
00063 private:
00069     static std::string _makeCreationMarker(unsigned long long numericId);
00070
00071     mutable std::mutex _mutex;
00072     std::unordered_map<std::string, WorkerJob> _jobs;
00073     unsigned long long _nextJobNumber = 1;
00074 };
```

18.259 gtest-assertion-result.h

```
00001 // Copyright 2005, Google Inc.
00002 // All rights reserved.
00003 //
00004 // Redistribution and use in source and binary forms, with or without
00005 // modification, are permitted provided that the following conditions are
00006 // met:
00007 //
00008 //     * Redistributions of source code must retain the above copyright
00009 // notice, this list of conditions and the following disclaimer.
00010 //     * Redistributions in binary form must reproduce the above
00011 // copyright notice, this list of conditions and the following disclaimer
00012 // in the documentation and/or other materials provided with the
00013 // distribution.
00014 //     * Neither the name of Google Inc. nor the names of its
00015 // contributors may be used to endorse or promote products derived from
00016 // this software without specific prior written permission.
00017 //
00018 // THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS
00019 // "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT
00020 // LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR
00021 // A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT
00022 // OWNER OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL,
00023 // SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT
00024 // LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE,
00025 // DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY
00026 // THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // The Google C++ Testing and Mocking Framework (Google Test)
00031 //
00032 // This file implements the AssertionResult type.
00033
00034 // IWYU pragma: private, include "gtest/gtest.h"
00035 // IWYU pragma: friend gtest/.*
00036 // IWYU pragma: friend gmock/.*
00037
00038 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_ASSERTION_RESULT_H_
00039 #define GOOGLETEST_INCLUDE_GTEST_GTEST_ASSERTION_RESULT_H_
00040
00041 #include <memory>
00042 #include <ostream>
00043 #include <string>
00044 #include <type_traits>
00045
00046 #include "gtest/gtest-message.h"
00047 #include "gtest/internal/gtest-port.h"
00048
00049 GTEST_DISABLE_MSC_WARNINGS_PUSH_(4251 \
00050 /* class A needs to have dll-interface to be used by clients of class B */)
00051
00052 namespace testing {
00053
00054     // A class for indicating whether an assertion was successful.  When
00055     // the assertion wasn't successful, the AssertionResult object
00056     // remembers a non-empty message that describes how it failed.
00057     //
00058     // To create an instance of this class, use one of the factory functions
00059     // (AssertionSuccess() and AssertionFailure()).
00060     //
```

```
00061     // This class is useful for two purposes:
00062     //   1. Defining predicate functions to be used with Boolean test assertions
00063     //      EXPECT_TRUE/EXPECT_FALSE and their ASSERT_ counterparts
00064     //   2. Defining predicate-format functions to be
00065     //      used with predicate assertions (ASSERT_PRED_FORMAT*, etc).
00066     //
00067     // For example, if you define IsEven predicate:
00068     //
00069     //   testing::AssertionResult IsEven(int n) {
00070     //     if ((n % 2) == 0)
00071     //       return testing::AssertionSuccess();
00072     //     else
00073     //       return testing::AssertionFailure() « n « " is odd";
00074     //   }
00075     //
00076     // Then the failed expectation EXPECT_TRUE(IsEven(Fib(5)))
00077     // will print the message
00078     //
00079     //   Value of: IsEven(Fib(5))
00080     //     Actual: false (5 is odd)
00081     //   Expected: true
00082     //
00083     // instead of a more opaque
00084     //
00085     //   Value of: IsEven(Fib(5))
00086     //     Actual: false
00087     //   Expected: true
00088     //
00089     // in case IsEven is a simple Boolean predicate.
00090     //
00091     // If you expect your predicate to be reused and want to support informative
00092     // messages in EXPECT_FALSE and ASSERT_FALSE (negative assertions show up
00093     // about half as often as positive ones in our tests), supply messages for
00094     // both success and failure cases:
00095     //
00096     //   testing::AssertionResult IsEven(int n) {
00097     //     if ((n % 2) == 0)
00098     //       return testing::AssertionSuccess() « n « " is even";
00099     //     else
00100     //       return testing::AssertionFailure() « n « " is odd";
00101     //   }
00102     //
00103     // Then a statement EXPECT_FALSE(IsEven(Fib(6))) will print
00104     //
00105     //   Value of: IsEven(Fib(6))
00106     //     Actual: true (8 is even)
00107     //   Expected: false
00108     //
00109     // NB: Predicates that support negative Boolean assertions have reduced
00110     // performance in positive ones so be careful not to use them in tests
00111     // that have lots (tens of thousands) of positive Boolean assertions.
00112     //
00113     // To use this class with EXPECT_PRED_FORMAT assertions such as:
00114     //
00115     //   // Verifies that Foo() returns an even number.
00116     //   EXPECT_PRED_FORMAT1(IsEven, Foo());
00117     //
00118     // you need to define:
00119     //
00120     //   testing::AssertionResult IsEven(const char* expr, int n) {
00121     //     if ((n % 2) == 0)
00122     //       return testing::AssertionSuccess();
00123     //     else
00124     //       return testing::AssertionFailure()
00125     //         « "Expected: " « expr « " is even\n  Actual: it's " « n;
00126     //   }
00127     //
00128     // If Foo() returns 5, you will see the following message:
00129     //
00130     //   Expected: Foo() is even
00131     //     Actual: it's 5
00132     //
00133
00134     class GTEST_API_ AssertionResult {
00135     public:
00136         // Copy constructor.
00137         // Used in EXPECT_TRUE/FALSE(assertion_result).
00138         AssertionResult(const AssertionResult& other);
00139
00140         // C4800 is a level 3 warning in Visual Studio 2015 and earlier.
00141         // This warning is not emitted in Visual Studio 2017.
00142         // This warning is off by default starting in Visual Studio 2019 but can be
00143         // enabled with command-line options.
00144 #if defined(_MSC_VER) && (_MSC_VER < 1910 || _MSC_VER >= 1920)
00145
00146         GTEST_DISABLE_MSC_WARNINGS_PUSH_(4800 /* forcing value to bool */)
00147 #endif
```

```
00148
00149         // Used in the EXPECT_TRUE/FALSE(bool_expression).
00150         //
00151         // T must be contextually convertible to bool.
00152         //
00153         // The second parameter prevents this overload from being considered if
00154         // the argument is implicitly convertible to AssertionResult. In that case
00155         // we want AssertionResult's copy constructor to be used.
00156         template <typename T>
00157         explicit AssertionResult(
00158                 const T& success,
00159                 typename std::enable_if<
00160                 !std::is_convertible<T, AssertionResult>::value>::type*
00161                 /*enabler*/
00162                 = nullptr)
00163         : success_(success) {
00164         }
00165
00166 #if defined(_MSC_VER) && (_MSC_VER < 1910 || _MSC_VER >= 1920)
00167
00168         GTEST_DISABLE_MSC_WARNINGS_POP_()
00169 #endif
00170
00171         // Assignment operator.
00172         AssertionResult& operator=(AssertionResult other) {
00173             swap(other);
00174             return *this;
00175         }
00176
00177         // Returns true if and only if the assertion succeeded.
00178
00179         operator bool() const {
00180             return success_;
00181         } // NOLINT
00182
00183         // Returns the assertion's negation. Used with EXPECT/ASSERT_FALSE.
00184         AssertionResult operator!() const;
00185
00186         // Returns the text streamed into this AssertionResult. Test assertions
00187         // use it when they fail (i.e., the predicate's outcome doesn't match the
00188         // assertion's expectation). When nothing has been streamed into the
00189         // object, returns an empty string.
00190
00191         const char* message() const {
00192             return message_.get() != nullptr ? message_->c_str() : "";
00193         }
00194         // Deprecated; please use message() instead.
00195
00196         const char* failure_message() const {
00197             return message();
00198         }
00199
00200         // Streams a custom failure message into this object.
00201
00202         template <typename T>
00203         AssertionResult& operator<<(const T& value) {
00204             AppendMessage(Message() << value);
00205             return *this;
00206         }
00207
00208         // Allows streaming basic output manipulators such as endl or flush into
00209         // this object.
00210
00211         AssertionResult& operator<<(
00212                 ::std::ostream& (*basic_manipulator)(::std::ostream& stream)) {
00213             AppendMessage(Message() << basic_manipulator);
00214             return *this;
00215         }
00216
00217     private:
00218         // Appends the contents of message to message_.
00219
00220         void AppendMessage(const Message& a_message) {
00221             if (message_.get() == nullptr) message_.reset(new ::std::string);
00222             message_->append(a_message.GetString().c_str());
00223         }
00224
00225         // Swap the contents of this AssertionResult with other.
00226         void swap(AssertionResult& other);
00227
00228         // Stores result of the assertion predicate.
00229         bool success_;
00230         // Stores the message describing the condition in case the expectation
00231         // construct is not satisfied with the predicate's outcome.
00232         // Referenced via a pointer to avoid taking too much stack frame space
00233         // with test assertions.
00234         std::unique_ptr< ::std::string> message_;
```

```
00235     };
00236
00237     // Makes a successful assertion result.
00238     GTEST_API_ AssertionResult AssertionSuccess();
00239
00240     // Makes a failed assertion result.
00241     GTEST_API_ AssertionResult AssertionFailure();
00242
00243     // Makes a failed assertion result with the given failure message.
00244     // Deprecated; use AssertionFailure() << msg.
00245     GTEST_API_ AssertionResult AssertionFailure(const Message& msg);
00246
00247 } // namespace testing
00248
00249 GTEST_DISABLE_MSC_WARNINGS_POP_() // 4251
00250
00251 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_ASSERTION_RESULT_H_
```

18.260 gtest-death-test.h

```
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00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // The Google C++ Testing and Mocking Framework (Google Test)
00031 //
00032 // This header file defines the public API for death tests.  It is
00033 // #included by gtest.h so a user doesn't need to include this
00034 // directly.
00035
00036 // IWYU pragma: private, include "gtest/gtest.h"
00037 // IWYU pragma: friend gtest/.*
00038 // IWYU pragma: friend gmock/.*
00039
00040 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_DEATH_TEST_H_
00041 #define GOOGLETEST_INCLUDE_GTEST_GTEST_DEATH_TEST_H_
00042
00043 #include "gtest/internal/gtest-death-test-internal.h"
00044
00045 // This flag controls the style of death tests.  Valid values are "threadsafe",
00046 // meaning that the death test child process will re-execute the test binary
00047 // from the start, running only a single death test, or "fast",
00048 // meaning that the child process will execute the test logic immediately
00049 // after forking.
00050 GTEST_DECLARE_string_(death_test_style);
00051
00052 namespace testing {
00053
00054 #if GTEST_HAS_DEATH_TEST
00055
00056     namespace internal {
00057
00058         // Returns a Boolean value indicating whether the caller is currently
00059         // executing in the context of the death test child process.  Tools such as
00060         // Valgrind heap checkers may need this to modify their behavior in death
00061         // tests.  IMPORTANT: This is an internal utility.  Using it may break the
00062         // implementation of death tests.  User code MUST NOT use it.
00063         GTEST_API_ bool InDeathTestChild();
00064
00065     } // namespace internal
```

```
00066
00067     // The following macros are useful for writing death tests.
00068
00069     // Here's what happens when an ASSERT_DEATH* or EXPECT_DEATH* is
00070     // executed:
00071     //
00072     //   1. It generates a warning if there is more than one active
00073     //   thread.  This is because it's safe to fork() or clone() only
00074     //   when there is a single thread.
00075     //
00076     //   2. The parent process clone()s a sub-process and runs the death
00077     //   test in it; the sub-process exits with code 0 at the end of the
00078     //   death test, if it hasn't exited already.
00079     //
00080     //   3. The parent process waits for the sub-process to terminate.
00081     //
00082     //   4. The parent process checks the exit code and error message of
00083     //   the sub-process.
00084     //
00085     // Examples:
00086     //
00087     //   ASSERT_DEATH(server.SendMessage(56, "Hello"), "Invalid port number");
00088     //   for (int i = 0; i < 5; i++) {
00089     //     EXPECT_DEATH(server.ProcessRequest(i),
00090     //                  "Invalid request .* in ProcessRequest()")
00091     //                  << "Failed to die on request " << i;
00092     //   }
00093     //
00094     //   ASSERT_EXIT(server.ExitNow(), ::testing::ExitedWithCode(0), "Exiting");
00095     //
00096     //   bool KilledBySIGHUP(int exit_code) {
00097     //     return WIFSIGNALED(exit_code) && WTERMSIG(exit_code) == SIGHUP;
00098     //   }
00099     //
00100     //   ASSERT_EXIT(client.HangUpServer(), KilledBySIGHUP, "Hanging up!");
00101     //
00102     // The final parameter to each of these macros is a matcher applied to any data
00103     // the sub-process wrote to stderr.  For compatibility with existing tests, a
00104     // bare string is interpreted as a regular expression matcher.
00105     //
00106     // On the regular expressions used in death tests:
00107     //
00108     //   On POSIX-compliant systems (*nix), we use the <regex.h> library,
00109     //   which uses the POSIX extended regex syntax.
00110     //
00111     //   On other platforms (e.g. Windows or Mac), we only support a simple regex
00112     //   syntax implemented as part of Google Test.  This limited
00113     //   implementation should be enough most of the time when writing
00114     //   death tests; though it lacks many features you can find in PCRE
00115     //   or POSIX extended regex syntax.  For example, we don't support
00116     //   union ("x|y"), grouping ("(xy)"), brackets ("[xy]"), and
00117     //   repetition count ("x{5,7}"), among others.
00118     //
00119     //   Below is the syntax that we do support.  We chose it to be a
00120     //   subset of both PCRE and POSIX extended regex, so it's easy to
00121     //   learn wherever you come from.  In the following: 'A' denotes a
00122     //   literal character, period (.), or a single \\ escape sequence;
00123     //   'x' and 'y' denote regular expressions; 'm' and 'n' are for
00124     //   natural numbers.
00125     //
00126     //     c     matches any literal character c
00127     //     \\d   matches any decimal digit
00128     //     \\D   matches any character that's not a decimal digit
00129     //     \\f   matches \f
00130     //     \\n   matches \n
00131     //     \\r   matches \r
00132     //     \\s   matches any ASCII whitespace, including \n
00133     //     \\S   matches any character that's not a whitespace
00134     //     \\t   matches \t
00135     //     \\v   matches \v
00136     //     \\w   matches any letter, _, or decimal digit
00137     //     \\W   matches any character that \\w doesn't match
00138     //     \\c   matches any literal character c, which must be a punctuation
00139     //     .     matches any single character except \n
00140     //     A?    matches 0 or 1 occurrences of A
00141     //     A*    matches 0 or many occurrences of A
00142     //     A+    matches 1 or many occurrences of A
00143     //     ^     matches the beginning of a string (not that of each line)
00144     //     $     matches the end of a string (not that of each line)
00145     //     xy    matches x followed by y
00146     //
00147     //   If you accidentally use PCRE or POSIX extended regex features
00148     //   not implemented by us, you will get a run-time failure.  In that
00149     //   case, please try to rewrite your regular expression within the
00150     //   above syntax.
00151     //
00152     //   This implementation is *not* meant to be as highly tuned or robust
```

```
00153     //   as a compiled regex library, but should perform well enough for a
00154     //   death test, which already incurs significant overhead by launching
00155     //   a child process.
00156     //
00157     // Known caveats:
00158     //
00159     //   A "threadsafe" style death test obtains the path to the test
00160     //   program from argv[0] and re-executes it in the sub-process.  For
00161     //   simplicity, the current implementation doesn't search the PATH
00162     //   when launching the sub-process.  This means that the user must
00163     //   invoke the test program via a path that contains at least one
00164     //   path separator (e.g. path/to/foo_test and
00165     //   /absolute/path/to/bar_test are fine, but foo_test is not).  This
00166     //   is rarely a problem as people usually don't put the test binary
00167     //   directory in PATH.
00168     //
00169
00170     // Asserts that a given `statement` causes the program to exit, with an
00171     // integer exit status that satisfies `predicate`, and emitting error output
00172     // that matches `matcher`.
00173 #define ASSERT_EXIT(statement, predicate, matcher) \
00174   GTEST_DEATH_TEST_(statement, predicate, matcher, GTEST_FATAL_FAILURE_)
00175
00176     // Like `ASSERT_EXIT`, but continues on to successive tests in the
00177     // test suite, if any:
00178 #define EXPECT_EXIT(statement, predicate, matcher) \
00179   GTEST_DEATH_TEST_(statement, predicate, matcher, GTEST_NONFATAL_FAILURE_)
00180
00181     // Asserts that a given `statement` causes the program to exit, either by
00182     // explicitly exiting with a nonzero exit code or being killed by a
00183     // signal, and emitting error output that matches `matcher`.
00184 #define ASSERT_DEATH(statement, matcher) \
00185   ASSERT_EXIT(statement, ::testing::internal::ExitedUnsuccessfully, matcher)
00186
00187     // Like `ASSERT_DEATH`, but continues on to successive tests in the
00188     // test suite, if any:
00189 #define EXPECT_DEATH(statement, matcher) \
00190   EXPECT_EXIT(statement, ::testing::internal::ExitedUnsuccessfully, matcher)
00191
00192     // Two predicate classes that can be used in {ASSERT,EXPECT}_EXIT*:
00193
00194     // Tests that an exit code describes a normal exit with a given exit code.
00195
00196     class GTEST_API_ ExitedWithCode {
00197     public:
00198         explicit ExitedWithCode(int exit_code);
00199         ExitedWithCode(const ExitedWithCode&) = default;
00200         void operator=(const ExitedWithCode& other) = delete;
00201         bool operator()(int exit_status) const;
00202
00203     private:
00204         const int exit_code_;
00205     };
00206
00207 #if !GTEST_OS_WINDOWS && !GTEST_OS_FUCHSIA
00208     // Tests that an exit code describes an exit due to termination by a
00209     // given signal.
00210
00211     class GTEST_API_ KilledBySignal {
00212     public:
00213         explicit KilledBySignal(int signum);
00214         bool operator()(int exit_status) const;
00215
00216     private:
00217         const int signum_;
00218     };
00219 #endif  // !GTEST_OS_WINDOWS
00220
00221     // EXPECT_DEBUG_DEATH asserts that the given statements die in debug mode.
00222     // The death testing framework causes this to have interesting semantics,
00223     // since the sideeffects of the call are only visible in opt mode, and not
00224     // in debug mode.
00225     //
00226     // In practice, this can be used to test functions that utilize the
00227     // LOG(DFATAL) macro using the following style:
00228     //
00229     // int DieInDebugOr12(int* sideeffect) {
00230     //   if (sideeffect) {
00231     //     *sideeffect = 12;
00232     //   }
00233     //   LOG(DFATAL) << "death";
00234     //   return 12;
00235     // }
00236     //
00237     // TEST(TestSuite, TestDieOr12WorksInDgbAndOpt) {
00238     //   int sideeffect = 0;
00239     //   // Only asserts in dbg.
```

```
00240     //   EXPECT_DEBUG_DEATH(DieInDebugOr12(&sideeffect), "death");
00241     //
00242     // #ifdef NDEBUG
00243     //   // opt-mode has sideeffect visible.
00244     //   EXPECT_EQ(12, sideeffect);
00245     // #else
00246     //   // dbg-mode no visible sideeffect.
00247     //   EXPECT_EQ(0, sideeffect);
00248     // #endif
00249     // }
00250     //
00251     // This will assert that DieInDebugReturn12InOpt() crashes in debug
00252     // mode, usually due to a DCHECK or LOG(DFATAL), but returns the
00253     // appropriate fallback value (12 in this case) in opt mode. If you
00254     // need to test that a function has appropriate side-effects in opt
00255     // mode, include assertions against the side-effects.  A general
00256     // pattern for this is:
00257     //
00258     // EXPECT_DEBUG_DEATH({
00259     //   // Side-effects here will have an effect after this statement in
00260     //   // opt mode, but none in debug mode.
00261     //   EXPECT_EQ(12, DieInDebugOr12(&sideeffect));
00262     // }, "death");
00263     //
00264 #ifdef NDEBUG
00265
00266 #define EXPECT_DEBUG_DEATH(statement, regex) \
00267   GTEST_EXECUTE_STATEMENT_(statement, regex)
00268
00269 #define ASSERT_DEBUG_DEATH(statement, regex) \
00270   GTEST_EXECUTE_STATEMENT_(statement, regex)
00271
00272 #else
00273
00274 #define EXPECT_DEBUG_DEATH(statement, regex) EXPECT_DEATH(statement, regex)
00275
00276 #define ASSERT_DEBUG_DEATH(statement, regex) ASSERT_DEATH(statement, regex)
00277
00278 #endif  // NDEBUG for EXPECT_DEBUG_DEATH
00279 #endif  // GTEST_HAS_DEATH_TEST
00280
00281     // This macro is used for implementing macros such as
00282     // EXPECT_DEATH_IF_SUPPORTED and ASSERT_DEATH_IF_SUPPORTED on systems where
00283     // death tests are not supported. Those macros must compile on such systems
00284     // if and only if EXPECT_DEATH and ASSERT_DEATH compile with the same parameters
00285     // on systems that support death tests. This allows one to write such a macro on
00286     // a system that does not support death tests and be sure that it will compile
00287     // on a death-test supporting system. It is exposed publicly so that systems
00288     // that have death-tests with stricter requirements than GTEST_HAS_DEATH_TEST
00289     // can write their own equivalent of EXPECT_DEATH_IF_SUPPORTED and
00290     // ASSERT_DEATH_IF_SUPPORTED.
00291     //
00292     // Parameters:
00293     //   statement -  A statement that a macro such as EXPECT_DEATH would test
00294     //                for program termination. This macro has to make sure this
00295     //                statement is compiled but not executed, to ensure that
00296     //                EXPECT_DEATH_IF_SUPPORTED compiles with a certain
00297     //                parameter if and only if EXPECT_DEATH compiles with it.
00298     //   regex     -  A regex that a macro such as EXPECT_DEATH would use to test
00299     //                the output of statement.  This parameter has to be
00300     //                compiled but not evaluated by this macro, to ensure that
00301     //                this macro only accepts expressions that a macro such as
00302     //                EXPECT_DEATH would accept.
00303     //   terminator - Must be an empty statement for EXPECT_DEATH_IF_SUPPORTED
00304     //                and a return statement for ASSERT_DEATH_IF_SUPPORTED.
00305     //                This ensures that ASSERT_DEATH_IF_SUPPORTED will not
00306     //                compile inside functions where ASSERT_DEATH doesn't
00307     //                compile.
00308     //
00309     //  The branch that has an always false condition is used to ensure that
00310     //  statement and regex are compiled (and thus syntactically correct) but
00311     //  never executed. The unreachable code macro protects the terminator
00312     //  statement from generating an 'unreachable code' warning in case
00313     //  statement unconditionally returns or throws. The Message constructor at
00314     //  the end allows the syntax of streaming additional messages into the
00315     //  macro, for compilational compatibility with EXPECT_DEATH/ASSERT_DEATH.
00316 #define GTEST_UNSUPPORTED_DEATH_TEST(statement, regex, terminator)             \
00317   GTEST_AMBIGUOUS_ELSE_BLOCKER_                                               \
00318   if (::testing::internal::AlwaysTrue()) {                                    \
00319     GTEST_LOG_(WARNING) << "Death tests are not supported on this platform.\n" \
00320                         << "Statement '" #statement "' cannot be verified.";  \
00321   } else if (::testing::internal::AlwaysFalse()) {                            \
00322     ::testing::internal::RE::PartialMatch(".*", (regex));                     \
00323     GTEST_SUPPRESS_UNREACHABLE_CODE_WARNING_BELOW_(statement);                \
00324     terminator;                                                               \
00325   } else                                                                      \
00326     ::testing::Message()
```

```
00327
00328     // EXPECT_DEATH_IF_SUPPORTED(statement, regex) and
00329     // ASSERT_DEATH_IF_SUPPORTED(statement, regex) expand to real death tests if
00330     // death tests are supported; otherwise they just issue a warning.  This is
00331     // useful when you are combining death test assertions with normal test
00332     // assertions in one test.
00333 #if GTEST_HAS_DEATH_TEST
00334 #define EXPECT_DEATH_IF_SUPPORTED(statement, regex) \
00335   EXPECT_DEATH(statement, regex)
00336 #define ASSERT_DEATH_IF_SUPPORTED(statement, regex) \
00337   ASSERT_DEATH(statement, regex)
00338 #else
00339 #define EXPECT_DEATH_IF_SUPPORTED(statement, regex) \
00340   GTEST_UNSUPPORTED_DEATH_TEST(statement, regex, )
00341 #define ASSERT_DEATH_IF_SUPPORTED(statement, regex) \
00342   GTEST_UNSUPPORTED_DEATH_TEST(statement, regex, return)
00343 #endif
00344
00345 } // namespace testing
00346
00347 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_DEATH_TEST_H_
```

18.261 gtest-matchers.h

```
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00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // The Google C++ Testing and Mocking Framework (Google Test)
00031 //
00032 // This file implements just enough of the matcher interface to allow
00033 // EXPECT_DEATH and friends to accept a matcher argument.
00034
00035 // IWYU pragma: private, include "gtest/gtest.h"
00036 // IWYU pragma: friend gtest/.*
00037 // IWYU pragma: friend gmock/.*
00038
00039 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_MATCHERS_H_
00040 #define GOOGLETEST_INCLUDE_GTEST_GTEST_MATCHERS_H_
00041
00042 #include <atomic>
00043 #include <memory>
00044 #include <ostream>
00045 #include <string>
00046 #include <type_traits>
00047
00048 #include "gtest/gtest-printers.h"
00049 #include "gtest/internal/gtest-internal.h"
00050 #include "gtest/internal/gtest-port.h"
00051
00052 // MSVC warning C5046 is new as of VS2017 version 15.8.
00053 #if defined(_MSC_VER) && _MSC_VER >= 1915
00054 #define GTEST_MAYBE_5046_ 5046
00055 #else
00056 #define GTEST_MAYBE_5046_
00057 #endif
00058
00059 GTEST_DISABLE_MSC_WARNINGS_PUSH_(
00060         4251 GTEST_MAYBE_5046_ /* class A needs to have dll-interface to be used by
00061                                   clients of class B */
```

```
00062         /* Symbol involving type with internal linkage not defined */)
00063
00064 namespace testing {
00065
00066     // To implement a matcher Foo for type T, define:
00067     //   1. a class FooMatcherMatcher that implements the matcher interface:
00068     //     using is_gtest_matcher = void;
00069     //     bool MatchAndExplain(const T&, std::ostream*);
00070     //       (MatchResultListener* can also be used instead of std::ostream*)
00071     //     void DescribeTo(std::ostream*);
00072     //     void DescribeNegationTo(std::ostream*);
00073     //
00074     //   2. a factory function that creates a Matcher<T> object from a
00075     //      FooMatcherMatcher.
00076
00077     class MatchResultListener {
00078     public:
00079         // Creates a listener object with the given underlying ostream.  The
00080         // listener does not own the ostream, and does not dereference it
00081         // in the constructor or destructor.
00082
00083         explicit MatchResultListener(::std::ostream* os) : stream_(os) {
00084         }
00085         virtual ~MatchResultListener() = 0; // Makes this class abstract.
00086
00087         // Streams x to the underlying ostream; does nothing if the ostream
00088         // is NULL.
00089
00090         template <typename T>
00091         MatchResultListener& operator<<(const T& x) {
00092             if (stream_ != nullptr) *stream_ << x;
00093             return *this;
00094         }
00095
00096         // Returns the underlying ostream.
00097
00098         ::std::ostream* stream() {
00099             return stream_;
00100         }
00101
00102         // Returns true if and only if the listener is interested in an explanation
00103         // of the match result.  A matcher's MatchAndExplain() method can use
00104         // this information to avoid generating the explanation when no one
00105         // intends to hear it.
00106
00107         bool IsInterested() const {
00108             return stream_ != nullptr;
00109         }
00110
00111     private:
00112         ::std::ostream * const stream_;
00113
00114         MatchResultListener(const MatchResultListener&) = delete;
00115         MatchResultListener& operator=(const MatchResultListener&) = delete;
00116     };
00117
00118     inline MatchResultListener::~MatchResultListener() {
00119     }
00120
00121     // An instance of a subclass of this knows how to describe itself as a
00122     // matcher.
00123
00124     class GTEST_API_ MatcherDescriberInterface {
00125     public:
00126
00127         virtual ~MatcherDescriberInterface() {
00128         }
00129
00130         // Describes this matcher to an ostream.  The function should print
00131         // a verb phrase that describes the property a value matching this
00132         // matcher should have.  The subject of the verb phrase is the value
00133         // being matched.  For example, the DescribeTo() method of the Gt(7)
00134         // matcher prints "is greater than 7".
00135         virtual void DescribeTo(::std::ostream* os) const = 0;
00136
00137         // Describes the negation of this matcher to an ostream.  For
00138         // example, if the description of this matcher is "is greater than
00139         // 7", the negated description could be "is not greater than 7".
00140         // You are not required to override this when implementing
00141         // MatcherInterface, but it is highly advised so that your matcher
00142         // can produce good error messages.
00143
00144         virtual void DescribeNegationTo(::std::ostream* os) const {
00145             *os << "not (";
00146             DescribeTo(os);
00147             *os << ")";
00148         }
```

```
00149     };
00150
00151     // The implementation of a matcher.
00152
00153     template <typename T>
00154     class MatcherInterface : public MatcherDescriberInterface {
00155     public:
00156         // Returns true if and only if the matcher matches x; also explains the
00157         // match result to 'listener' if necessary (see the next paragraph), in
00158         // the form of a non-restrictive relative clause ("which ...",
00159         // "whose ...", etc) that describes x.  For example, the
00160         // MatchAndExplain() method of the Pointee(...) matcher should
00161         // generate an explanation like "which points to ...".
00162         //
00163         // Implementations of MatchAndExplain() should add an explanation of
00164         // the match result *if and only if* they can provide additional
00165         // information that's not already present (or not obvious) in the
00166         // print-out of x and the matcher's description.  Whether the match
00167         // succeeds is not a factor in deciding whether an explanation is
00168         // needed, as sometimes the caller needs to print a failure message
00169         // when the match succeeds (e.g. when the matcher is used inside
00170         // Not()).
00171         //
00172         // For example, a "has at least 10 elements" matcher should explain
00173         // what the actual element count is, regardless of the match result,
00174         // as it is useful information to the reader; on the other hand, an
00175         // "is empty" matcher probably only needs to explain what the actual
00176         // size is when the match fails, as it's redundant to say that the
00177         // size is 0 when the value is already known to be empty.
00178         //
00179         // You should override this method when defining a new matcher.
00180         //
00181         // It's the responsibility of the caller (Google Test) to guarantee
00182         // that 'listener' is not NULL.  This helps to simplify a matcher's
00183         // implementation when it doesn't care about the performance, as it
00184         // can talk to 'listener' without checking its validity first.
00185         // However, in order to implement dummy listeners efficiently,
00186         // listener->stream() may be NULL.
00187         virtual bool MatchAndExplain(T x, MatchResultListener* listener) const = 0;
00188
00189         // Inherits these methods from MatcherDescriberInterface:
00190         //   virtual void DescribeTo(::std::ostream* os) const = 0;
00191         //   virtual void DescribeNegationTo(::std::ostream* os) const;
00192     };
00193
00194     namespace internal {
00195
00196         struct AnyEq {
00197
00198             template <typename A, typename B>
00199             bool operator()(const A& a, const B& b) const {
00200                 return a == b;
00201             }
00202         };
00203
00204         struct AnyNe {
00205
00206             template <typename A, typename B>
00207             bool operator()(const A& a, const B& b) const {
00208                 return a != b;
00209             }
00210         };
00211
00212         struct AnyLt {
00213
00214             template <typename A, typename B>
00215             bool operator()(const A& a, const B& b) const {
00216                 return a < b;
00217             }
00218         };
00219
00220         struct AnyGt {
00221
00222             template <typename A, typename B>
00223             bool operator()(const A& a, const B& b) const {
00224                 return a > b;
00225             }
00226         };
00227
00228         struct AnyLe {
00229
00230             template <typename A, typename B>
00231             bool operator()(const A& a, const B& b) const {
00232                 return a <= b;
00233             }
00234         };
00235
```

```
00236         struct AnyGe {
00237
00238             template <typename A, typename B>
00239             bool operator()(const A& a, const B& b) const {
00240                 return a >= b;
00241             }
00242         };
00243
00244         // A match result listener that ignores the explanation.
00245
00246         class DummyMatchResultListener : public MatchResultListener {
00247         public:
00248
00249             DummyMatchResultListener() : MatchResultListener(nullptr) {
00250             }
00251
00252         private:
00253             DummyMatchResultListener(const DummyMatchResultListener&) = delete;
00254             DummyMatchResultListener& operator=(const DummyMatchResultListener&) = delete;
00255         };
00256
00257         // A match result listener that forwards the explanation to a given
00258         // ostream.  The difference between this and MatchResultListener is
00259         // that the former is concrete.
00260
00261         class StreamMatchResultListener : public MatchResultListener {
00262         public:
00263
00264             explicit StreamMatchResultListener(::std::ostream* os)
00265             : MatchResultListener(os) {
00266             }
00267
00268         private:
00269             StreamMatchResultListener(const StreamMatchResultListener&) = delete;
00270             StreamMatchResultListener& operator=(const StreamMatchResultListener&) =
00271                     delete;
00272         };
00273
00274         struct SharedPayloadBase {
00275             std::atomic<int> ref{1};
00276
00277             void Ref() {
00278                 ref.fetch_add(1, std::memory_order_relaxed);
00279             }
00280
00281             bool Unref() {
00282                 return ref.fetch_sub(1, std::memory_order_acq_rel) == 1;
00283             }
00284         };
00285
00286         template <typename T>
00287         struct SharedPayload : SharedPayloadBase {
00288
00289             explicit SharedPayload(const T& v) : value(v) {
00290             }
00291
00292             explicit SharedPayload(T&& v) : value(std::move(v)) {
00293             }
00294
00295             static void Destroy(SharedPayloadBase* shared) {
00296                 delete static_cast<SharedPayload*> (shared);
00297             }
00298
00299             T value;
00300         };
00301
00302         // An internal class for implementing Matcher<T>, which will derive
00303         // from it.  We put functionalities common to all Matcher<T>
00304         // specializations here to avoid code duplication.
00305
00306         template <typename T>
00307         class MatcherBase : private MatcherDescriberInterface {
00308         public:
00309             // Returns true if and only if the matcher matches x; also explains the
00310             // match result to 'listener'.
00311
00312             bool MatchAndExplain(const T& x, MatchResultListener* listener) const {
00313                 GTEST_CHECK_(vtable_ != nullptr);
00314                 return vtable_->match_and_explain(*this, x, listener);
00315             }
00316
00317             // Returns true if and only if this matcher matches x.
00318
00319             bool Matches(const T& x) const {
00320                 DummyMatchResultListener dummy;
00321                 return MatchAndExplain(x, &dummy);
00322             }
```

```
00323
00324           // Describes this matcher to an ostream.
00325
00326           void DescribeTo(::std::ostream* os) const final {
00327               GTEST_CHECK_(vtable_ != nullptr);
00328               vtable_->describe(*this, os, false);
00329           }
00330
00331           // Describes the negation of this matcher to an ostream.
00332
00333           void DescribeNegationTo(::std::ostream* os) const final {
00334               GTEST_CHECK_(vtable_ != nullptr);
00335               vtable_->describe(*this, os, true);
00336           }
00337
00338           // Explains why x matches, or doesn't match, the matcher.
00339
00340           void ExplainMatchResultTo(const T& x, ::std::ostream* os) const {
00341               StreamMatchResultListener listener(os);
00342               MatchAndExplain(x, &listener);
00343           }
00344
00345           // Returns the describer for this matcher object; retains ownership
00346           // of the describer, which is only guaranteed to be alive when
00347           // this matcher object is alive.
00348
00349           const MatcherDescriberInterface* GetDescriber() const {
00350               if (vtable_ == nullptr) return nullptr;
00351               return vtable_->get_describer(*this);
00352           }
00353
00354       protected:
00355
00356           MatcherBase() : vtable_(nullptr), buffer_() {
00357           }
00358
00359           // Constructs a matcher from its implementation.
00360
00361           template <typename U>
00362           explicit MatcherBase(const MatcherInterface<U>* impl)
00363           : vtable_(nullptr), buffer_() {
00364               Init(impl);
00365           }
00366
00367           template <typename M, typename = typename std::remove_reference<
00368           M>::type::is_gtest_matcher>
00369           MatcherBase(M&& m) : vtable_(nullptr), buffer_() { // NOLINT
00370               Init(std::forward<M>(m));
00371           }
00372
00373           MatcherBase(const MatcherBase& other)
00374           : vtable_(other.vtable_), buffer_(other.buffer_) {
00375               if (IsShared()) buffer_.shared->Ref();
00376           }
00377
00378           MatcherBase& operator=(const MatcherBase& other) {
00379               if (this == &other) return *this;
00380               Destroy();
00381               vtable_ = other.vtable_;
00382               buffer_ = other.buffer_;
00383               if (IsShared()) buffer_.shared->Ref();
00384               return *this;
00385           }
00386
00387           MatcherBase(MatcherBase&& other)
00388           : vtable_(other.vtable_), buffer_(other.buffer_) {
00389               other.vtable_ = nullptr;
00390           }
00391
00392           MatcherBase& operator=(MatcherBase&& other) {
00393               if (this == &other) return *this;
00394               Destroy();
00395               vtable_ = other.vtable_;
00396               buffer_ = other.buffer_;
00397               other.vtable_ = nullptr;
00398               return *this;
00399           }
00400
00401           ~MatcherBase() override {
00402               Destroy();
00403           }
00404
00405       private:
00406
00407           struct VTable {
00408               bool (*match_and_explain)(const MatcherBase&, const T&,
00409                       MatchResultListener*);
```

```
00410               void (*describe)(const MatcherBase&, std::ostream*, bool negation);
00411               // Returns the captured object if it implements the interface, otherwise
00412               // returns the MatcherBase itself.
00413               const MatcherDescriberInterface* (*get_describer)(const MatcherBase&);
00414               // Called on shared instances when the reference count reaches 0.
00415               void (*shared_destroy)(SharedPayloadBase*);
00416             };
00417
00418             bool IsShared() const {
00419               return vtable_ != nullptr && vtable_->shared_destroy != nullptr;
00420             }
00421
00422             // If the implementation uses a listener, call that.
00423             template <typename P>
00424             static auto MatchAndExplainImpl(const MatcherBase& m, const T& value,
00425                     MatchResultListener* listener)
00426             -> decltype(P::Get(m).MatchAndExplain(value, listener->stream())) {
00427               return P::Get(m).MatchAndExplain(value, listener->stream());
00428             }
00429
00430             template <typename P>
00431             static auto MatchAndExplainImpl(const MatcherBase& m, const T& value,
00432                     MatchResultListener* listener)
00433             -> decltype(P::Get(m).MatchAndExplain(value, listener)) {
00434               return P::Get(m).MatchAndExplain(value, listener);
00435             }
00436
00437             template <typename P>
00438             static void DescribeImpl(const MatcherBase& m, std::ostream* os,
00439                     bool negation) {
00440               if (negation) {
00441                 P::Get(m).DescribeNegationTo(os);
00442               } else {
00443                 P::Get(m).DescribeTo(os);
00444               }
00445             }
00446
00447             template <typename P>
00448             static const MatcherDescriberInterface* GetDescriberImpl(
00449                     const MatcherBase& m) {
00450               // If the impl is a MatcherDescriberInterface, then return it.
00451               // Otherwise use MatcherBase itself.
00452               // This allows us to implement the GetDescriber() function without support
00453               // from the impl, but some users really want to get their impl back when
00454               // they call GetDescriber().
00455               // We use std::get on a tuple as a workaround of not having `if constexpr`.
00456               return std::get < (
00457                       std::is_convertible<decltype(&P::Get(m)),
00458                       const MatcherDescriberInterface*>::value
00459                       ? 1
00460                       : 0)>(std::make_tuple(&m, &P::Get(m)));
00461             }
00462
00463             template <typename P>
00464             const VTable* GetVTable() {
00465               static constexpr VTable kVTable = {&MatchAndExplainImpl<P>,
00466                   &DescribeImpl<P>, &GetDescriberImpl<P>,
00467                   P::shared_destroy};
00468               return &kVTable;
00469             }
00470
00471             union Buffer {
00472               // Add some types to give Buffer some common alignment/size use cases.
00473               void* ptr;
00474               double d;
00475               int64_t i;
00476               // And add one for the out-of-line cases.
00477               SharedPayloadBase* shared;
00478             };
00479
00480             void Destroy() {
00481               if (IsShared() && buffer_.shared->Unref()) {
00482                 vtable_->shared_destroy(buffer_.shared);
00483               }
00484             }
00485
00486             template <typename M>
00487             static constexpr bool IsInlined() {
00488               return sizeof (M) <= sizeof (Buffer) && alignof (M) <= alignof (Buffer) &&
00489                       std::is_trivially_copy_constructible<M>::value &&
00490                       std::is_trivially_destructible<M>::value;
00491             }
00492
00493             template <typename M, bool = MatcherBase::IsInlined<M>()>
00494             struct ValuePolicy {
00495
00496               static const M& Get(const MatcherBase& m) {
```

```
00497                   // When inlined along with Init, need to be explicit to avoid violating
00498                   // strict aliasing rules.
00499                   const M* ptr =
00500                           static_cast<const M*> (static_cast<const void*> (&m.buffer_));
00501                   return *ptr;
00502               }
00503
00504               static void Init(MatcherBase& m, M impl) {
00505                   ::new (static_cast<void*> (&m.buffer_)) M(impl);
00506               }
00507               static constexpr auto shared_destroy = nullptr;
00508           };
00509
00510           template <typename M>
00511           struct ValuePolicy<M, false> {
00512               using Shared = SharedPayload<M>;
00513
00514               static const M& Get(const MatcherBase& m) {
00515                   return static_cast<Shared*> (m.buffer_.shared)->value;
00516               }
00517
00518               template <typename Arg>
00519               static void Init(MatcherBase& m, Arg&& arg) {
00520                   m.buffer_.shared = new Shared(std::forward<Arg>(arg));
00521               }
00522               static constexpr auto shared_destroy = &Shared::Destroy;
00523           };
00524
00525           template <typename U, bool B>
00526           struct ValuePolicy<const MatcherInterface<U>*, B> {
00527               using M = const MatcherInterface<U>;
00528               using Shared = SharedPayload<std::unique_ptr<M»;
00529
00530               static const M& Get(const MatcherBase& m) {
00531                   return *static_cast<Shared*> (m.buffer_.shared)->value;
00532               }
00533
00534               static void Init(MatcherBase& m, M* impl) {
00535                   m.buffer_.shared = new Shared(std::unique_ptr<M>(impl));
00536               }
00537
00538               static constexpr auto shared_destroy = &Shared::Destroy;
00539           };
00540
00541           template <typename M>
00542           void Init(M&& m) {
00543               using MM = typename std::decay<M>::type;
00544               using Policy = ValuePolicy<MM>;
00545               vtable_ = GetVTable<Policy>();
00546               Policy::Init(*this, std::forward<M>(m));
00547           }
00548
00549           const VTable* vtable_;
00550           Buffer buffer_;
00551       };
00552
00553   } // namespace internal
00554
00555   // A Matcher<T> is a copyable and IMMUTABLE (except by assignment)
00556   // object that can check whether a value of type T matches.  The
00557   // implementation of Matcher<T> is just a std::shared_ptr to const
00558   // MatcherInterface<T>.  Don't inherit from Matcher!
00559
00560   template <typename T>
00561   class Matcher : public internal::MatcherBase<T> {
00562   public:
00563       // Constructs a null matcher.  Needed for storing Matcher objects in STL
00564       // containers.  A default-constructed matcher is not yet initialized.  You
00565       // cannot use it until a valid value has been assigned to it.
00566
00567       explicit Matcher() {
00568       } // NOLINT
00569
00570       // Constructs a matcher from its implementation.
00571
00572       explicit Matcher(const MatcherInterface<const T&>* impl)
00573       : internal::MatcherBase<T>(impl) {
00574       }
00575
00576       template <typename U>
00577       explicit Matcher(
00578               const MatcherInterface<U>* impl,
00579               typename std::enable_if<!std::is_same<U, const U&>::value>::type* =
00580               nullptr)
00581       : internal::MatcherBase<T>(impl) {
00582       }
00583
```

```
00584         template <typename M, typename = typename std::remove_reference<
00585         M>::type::is_gtest_matcher>
00586         Matcher(M&& m) : internal::MatcherBase<T>(std::forward<M>(m)) {
00587         } // NOLINT
00588 
00589         // Implicit constructor here allows people to write
00590         // EXPECT_CALL(foo, Bar(5)) instead of EXPECT_CALL(foo, Bar(Eq(5))) sometimes
00591         Matcher(T value); // NOLINT
00592     };
00593 
00594     // The following two specializations allow the user to write str
00595     // instead of Eq(str) and "foo" instead of Eq("foo") when a std::string
00596     // matcher is expected.
00597 
00598     template <>
00599     class GTEST_API_ Matcher<const std::string&>
00600     : public internal::MatcherBase<const std::string&> {
00601     public:
00602 
00603         Matcher() {
00604         }
00605 
00606         explicit Matcher(const MatcherInterface<const std::string&>* impl)
00607         : internal::MatcherBase<const std::string&>(impl) {
00608         }
00609 
00610         template <typename M, typename = typename std::remove_reference<
00611         M>::type::is_gtest_matcher>
00612         Matcher(M&& m) // NOLINT
00613         : internal::MatcherBase<const std::string&>(std::forward<M>(m)) {
00614         }
00615 
00616         // Allows the user to write str instead of Eq(str) sometimes, where
00617         // str is a std::string object.
00618         Matcher(const std::string& s); // NOLINT
00619 
00620         // Allows the user to write "foo" instead of Eq("foo") sometimes.
00621         Matcher(const char* s); // NOLINT
00622     };
00623 
00624     template <>
00625     class GTEST_API_ Matcher<std::string>
00626     : public internal::MatcherBase<std::string> {
00627     public:
00628 
00629         Matcher() {
00630         }
00631 
00632         explicit Matcher(const MatcherInterface<const std::string&>* impl)
00633         : internal::MatcherBase<std::string>(impl) {
00634         }
00635 
00636         explicit Matcher(const MatcherInterface<std::string>* impl)
00637         : internal::MatcherBase<std::string>(impl) {
00638         }
00639 
00640         template <typename M, typename = typename std::remove_reference<
00641         M>::type::is_gtest_matcher>
00642         Matcher(M&& m) // NOLINT
00643         : internal::MatcherBase<std::string>(std::forward<M>(m)) {
00644         }
00645 
00646         // Allows the user to write str instead of Eq(str) sometimes, where
00647         // str is a string object.
00648         Matcher(const std::string& s); // NOLINT
00649 
00650         // Allows the user to write "foo" instead of Eq("foo") sometimes.
00651         Matcher(const char* s); // NOLINT
00652     };
00653 
00654 #if GTEST_INTERNAL_HAS_STRING_VIEW
00655     // The following two specializations allow the user to write str
00656     // instead of Eq(str) and "foo" instead of Eq("foo") when a absl::string_view
00657     // matcher is expected.
00658 
00659     template <>
00660     class GTEST_API_ Matcher<const internal::StringView&>
00661     : public internal::MatcherBase<const internal::StringView&> {
00662     public:
00663 
00664         Matcher() {
00665         }
00666 
00667         explicit Matcher(const MatcherInterface<const internal::StringView&>* impl)
00668         : internal::MatcherBase<const internal::StringView&>(impl) {
00669         }
00670 
```

```
00671         template <typename M, typename = typename std::remove_reference<
00672         M>::type::is_gtest_matcher>
00673         Matcher(M&& m) // NOLINT
00674         : internal::MatcherBase<const internal::StringView&>(std::forward<M>(m)) {
00675         }
00676
00677         // Allows the user to write str instead of Eq(str) sometimes, where
00678         // str is a std::string object.
00679         Matcher(const std::string& s); // NOLINT
00680
00681         // Allows the user to write "foo" instead of Eq("foo") sometimes.
00682         Matcher(const char* s); // NOLINT
00683
00684         // Allows the user to pass absl::string_views or std::string_views directly.
00685         Matcher(internal::StringView s); // NOLINT
00686     };
00687
00688     template <>
00689     class GTEST_API_ Matcher<internal::StringView>
00690     : public internal::MatcherBase<internal::StringView> {
00691     public:
00692
00693         Matcher() {
00694         }
00695
00696         explicit Matcher(const MatcherInterface<const internal::StringView&>* impl)
00697         : internal::MatcherBase<internal::StringView>(impl) {
00698         }
00699
00700         explicit Matcher(const MatcherInterface<internal::StringView>* impl)
00701         : internal::MatcherBase<internal::StringView>(impl) {
00702         }
00703
00704         template <typename M, typename = typename std::remove_reference<
00705         M>::type::is_gtest_matcher>
00706         Matcher(M&& m) // NOLINT
00707         : internal::MatcherBase<internal::StringView>(std::forward<M>(m)) {
00708         }
00709
00710         // Allows the user to write str instead of Eq(str) sometimes, where
00711         // str is a std::string object.
00712         Matcher(const std::string& s); // NOLINT
00713
00714         // Allows the user to write "foo" instead of Eq("foo") sometimes.
00715         Matcher(const char* s); // NOLINT
00716
00717         // Allows the user to pass absl::string_views or std::string_views directly.
00718         Matcher(internal::StringView s); // NOLINT
00719     };
00720 #endif  // GTEST_INTERNAL_HAS_STRING_VIEW
00721
00722     // Prints a matcher in a human-readable format.
00723
00724     template <typename T>
00725     std::ostream& operator<<(std::ostream& os, const Matcher<T>& matcher) {
00726         matcher.DescribeTo(&os);
00727         return os;
00728     }
00729
00730     // The PolymorphicMatcher class template makes it easy to implement a
00731     // polymorphic matcher (i.e. a matcher that can match values of more
00732     // than one type, e.g. Eq(n) and NotNull()).
00733     //
00734     // To define a polymorphic matcher, a user should provide an Impl
00735     // class that has a DescribeTo() method and a DescribeNegationTo()
00736     // method, and define a member function (or member function template)
00737     //
00738     //   bool MatchAndExplain(const Value& value,
00739     //                        MatchResultListener* listener) const;
00740     //
00741     // See the definition of NotNull() for a complete example.
00742
00743     template <class Impl>
00744     class PolymorphicMatcher {
00745     public:
00746
00747         explicit PolymorphicMatcher(const Impl& an_impl) : impl_(an_impl) {
00748         }
00749
00750         // Returns a mutable reference to the underlying matcher
00751         // implementation object.
00752
00753         Impl& mutable_impl() {
00754             return impl_;
00755         }
00756
00757         // Returns an immutable reference to the underlying matcher
```

```
00758         // implementation object.
00759
00760         const Impl& impl() const {
00761             return impl_;
00762         }
00763
00764         template <typename T>
00765         operator Matcher<T>() const {
00766             return Matcher<T>(new MonomorphicImpl<const T&>(impl_));
00767         }
00768
00769     private:
00770
00771         template <typename T>
00772         class MonomorphicImpl : public MatcherInterface<T> {
00773         public:
00774
00775             explicit MonomorphicImpl(const Impl& impl) : impl_(impl) {
00776             }
00777
00778             void DescribeTo(::std::ostream* os) const override {
00779                 impl_.DescribeTo(os);
00780             }
00781
00782             void DescribeNegationTo(::std::ostream* os) const override {
00783                 impl_.DescribeNegationTo(os);
00784             }
00785
00786             bool MatchAndExplain(T x, MatchResultListener* listener) const override {
00787                 return impl_.MatchAndExplain(x, listener);
00788             }
00789
00790         private:
00791             const Impl impl_;
00792         };
00793
00794         Impl impl_;
00795     };
00796
00797     // Creates a matcher from its implementation.
00798     // DEPRECATED: Especially in the generic code, prefer:
00799     //   Matcher<T>(new MyMatcherImpl<const T&>(...));
00800     //
00801     // MakeMatcher may create a Matcher that accepts its argument by value, which
00802     // leads to unnecessary copies & lack of support for non-copyable types.
00803
00804     template <typename T>
00805     inline Matcher<T> MakeMatcher(const MatcherInterface<T>* impl) {
00806         return Matcher<T>(impl);
00807     }
00808
00809     // Creates a polymorphic matcher from its implementation.  This is
00810     // easier to use than the PolymorphicMatcher<Impl> constructor as it
00811     // doesn't require you to explicitly write the template argument, e.g.
00812     //
00813     //   MakePolymorphicMatcher(foo);
00814     // vs
00815     //   PolymorphicMatcher<TypeOfFoo>(foo);
00816
00817     template <class Impl>
00818     inline PolymorphicMatcher<Impl> MakePolymorphicMatcher(const Impl& impl) {
00819         return PolymorphicMatcher<Impl>(impl);
00820     }
00821
00822     namespace internal {
00823         // Implements a matcher that compares a given value with a
00824         // pre-supplied value using one of the ==, <=, <, etc, operators.  The
00825         // two values being compared don't have to have the same type.
00826         //
00827         // The matcher defined here is polymorphic (for example, Eq(5) can be
00828         // used to match an int, a short, a double, etc).  Therefore we use
00829         // a template type conversion operator in the implementation.
00830         //
00831         // The following template definition assumes that the Rhs parameter is
00832         // a "bare" type (i.e. neither 'const T' nor 'T&').
00833
00834         template <typename D, typename Rhs, typename Op>
00835         class ComparisonBase {
00836         public:
00837
00838             explicit ComparisonBase(const Rhs& rhs) : rhs_(rhs) {
00839             }
00840
00841             using is_gtest_matcher = void;
00842
00843             template <typename Lhs>
00844             bool MatchAndExplain(const Lhs& lhs, std::ostream*) const {
```

```
00845                 return Op()(lhs, Unwrap(rhs_));
00846             }
00847
00848             void DescribeTo(std::ostream* os) const {
00849                 *os « D::Desc() « " ";
00850                 UniversalPrint(Unwrap(rhs_), os);
00851             }
00852
00853             void DescribeNegationTo(std::ostream* os) const {
00854                 *os « D::NegatedDesc() « " ";
00855                 UniversalPrint(Unwrap(rhs_), os);
00856             }
00857
00858         private:
00859
00860             template <typename T>
00861             static const T& Unwrap(const T& v) {
00862                 return v;
00863             }
00864
00865             template <typename T>
00866             static const T& Unwrap(std::reference_wrapper<T> v) {
00867                 return v;
00868             }
00869
00870             Rhs rhs_;
00871         };
00872
00873         template <typename Rhs>
00874         class EqMatcher : public ComparisonBase<EqMatcher<Rhs>, Rhs, AnyEq> {
00875         public:
00876
00877             explicit EqMatcher(const Rhs& rhs)
00878             : ComparisonBase<EqMatcher<Rhs>, Rhs, AnyEq>(rhs) {
00879             }
00880
00881             static const char* Desc() {
00882                 return "is equal to";
00883             }
00884
00885             static const char* NegatedDesc() {
00886                 return "isn't equal to";
00887             }
00888         };
00889
00890         template <typename Rhs>
00891         class NeMatcher : public ComparisonBase<NeMatcher<Rhs>, Rhs, AnyNe> {
00892         public:
00893
00894             explicit NeMatcher(const Rhs& rhs)
00895             : ComparisonBase<NeMatcher<Rhs>, Rhs, AnyNe>(rhs) {
00896             }
00897
00898             static const char* Desc() {
00899                 return "isn't equal to";
00900             }
00901
00902             static const char* NegatedDesc() {
00903                 return "is equal to";
00904             }
00905         };
00906
00907         template <typename Rhs>
00908         class LtMatcher : public ComparisonBase<LtMatcher<Rhs>, Rhs, AnyLt> {
00909         public:
00910
00911             explicit LtMatcher(const Rhs& rhs)
00912             : ComparisonBase<LtMatcher<Rhs>, Rhs, AnyLt>(rhs) {
00913             }
00914
00915             static const char* Desc() {
00916                 return "is <";
00917             }
00918
00919             static const char* NegatedDesc() {
00920                 return "isn't <";
00921             }
00922         };
00923
00924         template <typename Rhs>
00925         class GtMatcher : public ComparisonBase<GtMatcher<Rhs>, Rhs, AnyGt> {
00926         public:
00927
00928             explicit GtMatcher(const Rhs& rhs)
00929             : ComparisonBase<GtMatcher<Rhs>, Rhs, AnyGt>(rhs) {
00930             }
00931
```

```
00932             static const char* Desc() {
00933                 return "is >";
00934             }
00935
00936             static const char* NegatedDesc() {
00937                 return "isn't >";
00938             }
00939         };
00940
00941         template <typename Rhs>
00942         class LeMatcher : public ComparisonBase<LeMatcher<Rhs>, Rhs, AnyLe> {
00943         public:
00944
00945             explicit LeMatcher(const Rhs& rhs)
00946             : ComparisonBase<LeMatcher<Rhs>, Rhs, AnyLe>(rhs) {
00947             }
00948
00949             static const char* Desc() {
00950                 return "is <=";
00951             }
00952
00953             static const char* NegatedDesc() {
00954                 return "isn't <=";
00955             }
00956         };
00957
00958         template <typename Rhs>
00959         class GeMatcher : public ComparisonBase<GeMatcher<Rhs>, Rhs, AnyGe> {
00960         public:
00961
00962             explicit GeMatcher(const Rhs& rhs)
00963             : ComparisonBase<GeMatcher<Rhs>, Rhs, AnyGe>(rhs) {
00964             }
00965
00966             static const char* Desc() {
00967                 return "is >=";
00968             }
00969
00970             static const char* NegatedDesc() {
00971                 return "isn't >=";
00972             }
00973         };
00974
00975         template <typename T, typename = typename std::enable_if<
00976         std::is_constructible<std::string, T>::value>::type>
00977         using StringLike = T;
00978
00979         // Implements polymorphic matchers MatchesRegex(regex) and
00980         // ContainsRegex(regex), which can be used as a Matcher<T> as long as
00981         // T can be converted to a string.
00982
00983         class MatchesRegexMatcher {
00984         public:
00985
00986             MatchesRegexMatcher(const RE* regex, bool full_match)
00987             : regex_(regex), full_match_(full_match) {
00988             }
00989
00990 #if GTEST_INTERNAL_HAS_STRING_VIEW
00991
00992             bool MatchAndExplain(const internal::StringView& s,
00993                    MatchResultListener* listener) const {
00994                 return MatchAndExplain(std::string(s), listener);
00995             }
00996 #endif  // GTEST_INTERNAL_HAS_STRING_VIEW
00997
00998             // Accepts pointer types, particularly:
00999             //   const char*
01000             //   char*
01001             //   const wchar_t*
01002             //   wchar_t*
01003
01004             template <typename CharType>
01005             bool MatchAndExplain(CharType* s, MatchResultListener* listener) const {
01006                 return s != nullptr && MatchAndExplain(std::string(s), listener);
01007             }
01008
01009             // Matches anything that can convert to std::string.
01010             //
01011             // This is a template, not just a plain function with const std::string&,
01012             // because absl::string_view has some interfering non-explicit constructors.
01013
01014             template <class MatcheeStringType>
01015             bool MatchAndExplain(const MatcheeStringType& s,
01016                    MatchResultListener* /* listener */) const {
01017                 const std::string & s2(s);
01018                 return full_match_ ? RE::FullMatch(s2, *regex_)
```

```
01019                       : RE::PartialMatch(s2, *regex_);
01020           }
01021
01022           void DescribeTo(::std::ostream* os) const {
01023               *os « (full_match_ ? "matches" : "contains") « " regular expression ";
01024               UniversalPrinter<std::string>::Print(regex_->pattern(), os);
01025           }
01026
01027           void DescribeNegationTo(::std::ostream* os) const {
01028               *os « "doesn't " « (full_match_ ? "match" : "contain")
01029                       « " regular expression ";
01030               UniversalPrinter<std::string>::Print(regex_->pattern(), os);
01031           }
01032
01033       private:
01034           const std::shared_ptr<const RE> regex_;
01035           const bool full_match_;
01036       };
01037   } // namespace internal
01038
01039   // Matches a string that fully matches regular expression 'regex'.
01040   // The matcher takes ownership of 'regex'.
01041
01042   inline PolymorphicMatcher<internal::MatchesRegexMatcher> MatchesRegex(
01043           const internal::RE* regex) {
01044       return MakePolymorphicMatcher(internal::MatchesRegexMatcher(regex, true));
01045   }
01046
01047   template <typename T = std::string>
01048   PolymorphicMatcher<internal::MatchesRegexMatcher> MatchesRegex(
01049           const internal::StringLike<T>& regex) {
01050       return MatchesRegex(new internal::RE(std::string(regex)));
01051   }
01052
01053   // Matches a string that contains regular expression 'regex'.
01054   // The matcher takes ownership of 'regex'.
01055
01056   inline PolymorphicMatcher<internal::MatchesRegexMatcher> ContainsRegex(
01057           const internal::RE* regex) {
01058       return MakePolymorphicMatcher(internal::MatchesRegexMatcher(regex, false));
01059   }
01060
01061   template <typename T = std::string>
01062   PolymorphicMatcher<internal::MatchesRegexMatcher> ContainsRegex(
01063           const internal::StringLike<T>& regex) {
01064       return ContainsRegex(new internal::RE(std::string(regex)));
01065   }
01066
01067   // Creates a polymorphic matcher that matches anything equal to x.
01068   // Note: if the parameter of Eq() were declared as const T&, Eq("foo")
01069   // wouldn't compile.
01070
01071   template <typename T>
01072   inline internal::EqMatcher<T> Eq(T x) {
01073       return internal::EqMatcher<T>(x);
01074   }
01075
01076   // Constructs a Matcher<T> from a 'value' of type T.  The constructed
01077   // matcher matches any value that's equal to 'value'.
01078
01079   template <typename T>
01080   Matcher<T>::Matcher(T value) {
01081       *this = Eq(value);
01082   }
01083
01084   // Creates a monomorphic matcher that matches anything with type Lhs
01085   // and equal to rhs.  A user may need to use this instead of Eq(...)
01086   // in order to resolve an overloading ambiguity.
01087   //
01088   // TypedEq<T>(x) is just a convenient short-hand for Matcher<T>(Eq(x))
01089   // or Matcher<T>(x), but more readable than the latter.
01090   //
01091   // We could define similar monomorphic matchers for other comparison
01092   // operations (e.g. TypedLt, TypedGe, and etc), but decided not to do
01093   // it yet as those are used much less than Eq() in practice.  A user
01094   // can always write Matcher<T>(Lt(5)) to be explicit about the type,
01095   // for example.
01096
01097   template <typename Lhs, typename Rhs>
01098   inline Matcher<Lhs> TypedEq(const Rhs& rhs) {
01099       return Eq(rhs);
01100   }
01101
01102   // Creates a polymorphic matcher that matches anything >= x.
01103
01104   template <typename Rhs>
01105   inline internal::GeMatcher<Rhs> Ge(Rhs x) {
```

```
01106         return internal::GeMatcher<Rhs>(x);
01107     }
01108
01109     // Creates a polymorphic matcher that matches anything > x.
01110
01111     template <typename Rhs>
01112     inline internal::GtMatcher<Rhs> Gt(Rhs x) {
01113         return internal::GtMatcher<Rhs>(x);
01114     }
01115
01116     // Creates a polymorphic matcher that matches anything <= x.
01117
01118     template <typename Rhs>
01119     inline internal::LeMatcher<Rhs> Le(Rhs x) {
01120         return internal::LeMatcher<Rhs>(x);
01121     }
01122
01123     // Creates a polymorphic matcher that matches anything < x.
01124
01125     template <typename Rhs>
01126     inline internal::LtMatcher<Rhs> Lt(Rhs x) {
01127         return internal::LtMatcher<Rhs>(x);
01128     }
01129
01130     // Creates a polymorphic matcher that matches anything != x.
01131
01132     template <typename Rhs>
01133     inline internal::NeMatcher<Rhs> Ne(Rhs x) {
01134         return internal::NeMatcher<Rhs>(x);
01135     }
01136 } // namespace testing
01137
01138 GTEST_DISABLE_MSC_WARNINGS_POP_() //  4251 5046
01139
01140 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_MATCHERS_H_
```

18.262 gtest-message.h

```
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00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // The Google C++ Testing and Mocking Framework (Google Test)
00031 //
00032 // This header file defines the Message class.
00033 //
00034 // IMPORTANT NOTE: Due to limitation of the C++ language, we have to
00035 // leave some internal implementation details in this header file.
00036 // They are clearly marked by comments like this:
00037 //
00038 //   // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
00039 //
00040 // Such code is NOT meant to be used by a user directly, and is subject
00041 // to CHANGE WITHOUT NOTICE.  Therefore DO NOT DEPEND ON IT in a user
00042 // program!
00043
00044 // IWYU pragma: private, include "gtest/gtest.h"
00045 // IWYU pragma: friend gtest/.*
00046 // IWYU pragma: friend gmock/.*
00047
```

```
00048 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_MESSAGE_H_
00049 #define GOOGLETEST_INCLUDE_GTEST_GTEST_MESSAGE_H_
00050
00051 #include <limits>
00052 #include <memory>
00053 #include <sstream>
00054
00055 #include "gtest/internal/gtest-port.h"
00056
00057 GTEST_DISABLE_MSC_WARNINGS_PUSH_(4251 \
00058 /* class A needs to have dll-interface to be used by clients of class B */)
00059
00060 // Ensures that there is at least one operator« in the global namespace.
00061 // See Message& operator«(...) below for why.
00062 void operator«(const testing::internal::Secret&, int);
00063
00064 namespace testing {
00065
00066     // The Message class works like an ostream repeater.
00067     //
00068     // Typical usage:
00069     //
00070     //   1. You stream a bunch of values to a Message object.
00071     //      It will remember the text in a stringstream.
00072     //   2. Then you stream the Message object to an ostream.
00073     //      This causes the text in the Message to be streamed
00074     //      to the ostream.
00075     //
00076     // For example;
00077     //
00078     //   testing::Message foo;
00079     //   foo « 1 « " != " « 2;
00080     //   std::cout « foo;
00081     //
00082     // will print "1 != 2".
00083     //
00084     // Message is not intended to be inherited from.  In particular, its
00085     // destructor is not virtual.
00086     //
00087     // Note that stringstream behaves differently in gcc and in MSVC.  You
00088     // can stream a NULL char pointer to it in the former, but not in the
00089     // latter (it causes an access violation if you do).  The Message
00090     // class hides this difference by treating a NULL char pointer as
00091     // "(null)".
00092
00093     class GTEST_API_ Message {
00094     private:
00095         // The type of basic IO manipulators (endl, ends, and flush) for
00096         // narrow streams.
00097         typedef std::ostream& (*BasicNarrowIoManip)(std::ostream&);
00098
00099     public:
00100         // Constructs an empty Message.
00101         Message();
00102
00103         // Copy constructor.
00104
00105         Message(const Message& msg) : ss_(new ::std::stringstream) { // NOLINT
00106             *ss_ « msg.GetString();
00107         }
00108
00109         // Constructs a Message from a C-string.
00110
00111         explicit Message(const char* str) : ss_(new ::std::stringstream) {
00112             *ss_ « str;
00113         }
00114
00115         // Streams a non-pointer value to this object.
00116
00117         template <typename T>
00118         inline Message& operator«(const T& val) {
00119             // Some libraries overload « for STL containers.  These
00120             // overloads are defined in the global namespace instead of ::std.
00121             //
00122             // C++'s symbol lookup rule (i.e. Koenig lookup) says that these
00123             // overloads are visible in either the std namespace or the global
00124             // namespace, but not other namespaces, including the testing
00125             // namespace which Google Test's Message class is in.
00126             //
00127             // To allow STL containers (and other types that has a « operator
00128             // defined in the global namespace) to be used in Google Test
00129             // assertions, testing::Message must access the custom « operator
00130             // from the global namespace.  With this using declaration,
00131             // overloads of « defined in the global namespace and those
00132             // visible via Koenig lookup are both exposed in this function.
00133             using ::operator«;
00134             *ss_ « val;
```

```
00135             return *this;
00136         }
00137
00138         // Streams a pointer value to this object.
00139         //
00140         // This function is an overload of the previous one.  When you
00141         // stream a pointer to a Message, this definition will be used as it
00142         // is more specialized.  (The C++ Standard, section
00143         // [temp.func.order].)  If you stream a non-pointer, then the
00144         // previous definition will be used.
00145         //
00146         // The reason for this overload is that streaming a NULL pointer to
00147         // ostream is undefined behavior.  Depending on the compiler, you
00148         // may get "0", "(nil)", "(null)", or an access violation.  To
00149         // ensure consistent result across compilers, we always treat NULL
00150         // as "(null)".
00151
00152         template <typename T>
00153         inline Message& operator<<(T * const& pointer) { // NOLINT
00154             if (pointer == nullptr) {
00155                 *ss_ << "(null)";
00156             } else {
00157                 *ss_ << pointer;
00158             }
00159             return *this;
00160         }
00161
00162         // Since the basic IO manipulators are overloaded for both narrow
00163         // and wide streams, we have to provide this specialized definition
00164         // of operator <<, even though its body is the same as the
00165         // templatized version above.  Without this definition, streaming
00166         // endl or other basic IO manipulators to Message will confuse the
00167         // compiler.
00168
00169         Message& operator<<(BasicNarrowIoManip val) {
00170             *ss_ << val;
00171             return *this;
00172         }
00173
00174         // Instead of 1/0, we want to see true/false for bool values.
00175
00176         Message& operator<<(bool b) {
00177             return *this << (b ? "true" : "false");
00178         }
00179
00180         // These two overloads allow streaming a wide C string to a Message
00181         // using the UTF-8 encoding.
00182         Message& operator<<(const wchar_t* wide_c_str);
00183         Message& operator<<(wchar_t* wide_c_str);
00184
00185 #if GTEST_HAS_STD_WSTRING
00186         // Converts the given wide string to a narrow string using the UTF-8
00187         // encoding, and streams the result to this Message object.
00188         Message& operator<<(const ::std::wstring& wstr);
00189 #endif  // GTEST_HAS_STD_WSTRING
00190
00191         // Gets the text streamed to this object so far as an std::string.
00192         // Each '\0' character in the buffer is replaced with "\\0".
00193         //
00194         // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
00195         std::string GetString() const;
00196
00197     private:
00198         // We'll hold the text streamed to this object here.
00199         const std::unique_ptr< ::std::stringstream> ss_;
00200
00201         // We declare (but don't implement) this to prevent the compiler
00202         // from implementing the assignment operator.
00203         void operator=(const Message&);
00204     };
00205
00206     // Streams a Message to an ostream.
00207
00208     inline std::ostream& operator<<(std::ostream& os, const Message& sb) {
00209         return os << sb.GetString();
00210     }
00211
00212     namespace internal {
00213
00214         // Converts a streamable value to an std::string.  A NULL pointer is
00215         // converted to "(null)".  When the input value is a ::string,
00216         // ::std::string, ::wstring, or ::std::wstring object, each NUL
00217         // character in it is replaced with "\\0".
00218
00219         template <typename T>
00220         std::string StreamableToString(const T& streamable) {
00221             return (Message() << streamable).GetString();
```

```
00222         }
00223
00224     } // namespace internal
00225 } // namespace testing
00226
00227 GTEST_DISABLE_MSC_WARNINGS_POP_() //  4251
00228
00229 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_MESSAGE_H_
```

18.263 gtest-param-test.h

```
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00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // Macros and functions for implementing parameterized tests
00031 // in Google C++ Testing and Mocking Framework (Google Test)
00032
00033 // IWYU pragma: private, include "gtest/gtest.h"
00034 // IWYU pragma: friend gtest/.*
00035 // IWYU pragma: friend gmock/.*
00036
00037 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_PARAM_TEST_H_
00038 #define GOOGLETEST_INCLUDE_GTEST_GTEST_PARAM_TEST_H_
00039
00040 // Value-parameterized tests allow you to test your code with different
00041 // parameters without writing multiple copies of the same test.
00042 //
00043 // Here is how you use value-parameterized tests:
00044
00045 #if 0
00046
00047 // To write value-parameterized tests, first you should define a fixture
00048 // class. It is usually derived from testing::TestWithParam<T> (see below for
00049 // another inheritance scheme that's sometimes useful in more complicated
00050 // class hierarchies), where the type of your parameter values.
00051 // TestWithParam<T> is itself derived from testing::Test. T can be any
00052 // copyable type. If it's a raw pointer, you are responsible for managing the
00053 // lifespan of the pointed values.
00054
00055 class FooTest : public ::testing::TestWithParam<const char*> {
00056   // You can implement all the usual class fixture members here.
00057 };
00058
00059 // Then, use the TEST_P macro to define as many parameterized tests
00060 // for this fixture as you want. The _P suffix is for "parameterized"
00061 // or "pattern", whichever you prefer to think.
00062
00063 TEST_P(FooTest, DoesBlah) {
00064   // Inside a test, access the test parameter with the GetParam() method
00065   // of the TestWithParam<T> class:
00066   EXPECT_TRUE(foo.Blah(GetParam()));
00067   ...
00068 }
00069
00070 TEST_P(FooTest, HasBlahBlah) {
00071   ...
00072 }
00073
00074 // Finally, you can use INSTANTIATE_TEST_SUITE_P to instantiate the test
```

```
00075 // case with any set of parameters you want. Google Test defines a number
00076 // of functions for generating test parameters. They return what we call
00077 // (surprise!) parameter generators. Here is a summary of them, which
00078 // are all in the testing namespace:
00079 //
00080 //
00081 //  Range(begin, end [, step]) - Yields values {begin, begin+step,
00082 //                               begin+step+step, ...}. The values do not
00083 //                               include end. step defaults to 1.
00084 //  Values(v1, v2, ..., vN)    - Yields values {v1, v2, ..., vN}.
00085 //  ValuesIn(container)        - Yields values from a C-style array, an STL
00086 //  ValuesIn(begin,end)          container, or an iterator range [begin, end).
00087 //  Bool()                     - Yields sequence {false, true}.
00088 //  Combine(g1, g2, ..., gN)   - Yields all combinations (the Cartesian product
00089 //                               for the math savvy) of the values generated
00090 //                               by the N generators.
00091 //
00092 // For more details, see comments at the definitions of these functions below
00093 // in this file.
00094 //
00095 // The following statement will instantiate tests from the FooTest test suite
00096 // each with parameter values "meeny", "miny", and "moe".
00097
00098 INSTANTIATE_TEST_SUITE_P(InstantiationName,
00099                          FooTest,
00100                          Values("meeny", "miny", "moe"));
00101
00102 // To distinguish different instances of the pattern, (yes, you
00103 // can instantiate it more than once) the first argument to the
00104 // INSTANTIATE_TEST_SUITE_P macro is a prefix that will be added to the
00105 // actual test suite name. Remember to pick unique prefixes for different
00106 // instantiations. The tests from the instantiation above will have
00107 // these names:
00108 //
00109 //    * InstantiationName/FooTest.DoesBlah/0 for "meeny"
00110 //    * InstantiationName/FooTest.DoesBlah/1 for "miny"
00111 //    * InstantiationName/FooTest.DoesBlah/2 for "moe"
00112 //    * InstantiationName/FooTest.HasBlahBlah/0 for "meeny"
00113 //    * InstantiationName/FooTest.HasBlahBlah/1 for "miny"
00114 //    * InstantiationName/FooTest.HasBlahBlah/2 for "moe"
00115 //
00116 // You can use these names in --gtest_filter.
00117 //
00118 // This statement will instantiate all tests from FooTest again, each
00119 // with parameter values "cat" and "dog":
00120
00121 const char* pets[] = {"cat", "dog"};
00122 INSTANTIATE_TEST_SUITE_P(AnotherInstantiationName, FooTest, ValuesIn(pets));
00123
00124 // The tests from the instantiation above will have these names:
00125 //
00126 //    * AnotherInstantiationName/FooTest.DoesBlah/0 for "cat"
00127 //    * AnotherInstantiationName/FooTest.DoesBlah/1 for "dog"
00128 //    * AnotherInstantiationName/FooTest.HasBlahBlah/0 for "cat"
00129 //    * AnotherInstantiationName/FooTest.HasBlahBlah/1 for "dog"
00130 //
00131 // Please note that INSTANTIATE_TEST_SUITE_P will instantiate all tests
00132 // in the given test suite, whether their definitions come before or
00133 // AFTER the INSTANTIATE_TEST_SUITE_P statement.
00134 //
00135 // Please also note that generator expressions (including parameters to the
00136 // generators) are evaluated in InitGoogleTest(), after main() has started.
00137 // This allows the user on one hand, to adjust generator parameters in order
00138 // to dynamically determine a set of tests to run and on the other hand,
00139 // give the user a chance to inspect the generated tests with Google Test
00140 // reflection API before RUN_ALL_TESTS() is executed.
00141 //
00142 // You can see samples/sample7_unittest.cc and samples/sample8_unittest.cc
00143 // for more examples.
00144 //
00145 // In the future, we plan to publish the API for defining new parameter
00146 // generators. But for now this interface remains part of the internal
00147 // implementation and is subject to change.
00148 //
00149 //
00150 // A parameterized test fixture must be derived from testing::Test and from
00151 // testing::WithParamInterface<T>, where T is the type of the parameter
00152 // values. Inheriting from TestWithParam<T> satisfies that requirement because
00153 // TestWithParam<T> inherits from both Test and WithParamInterface. In more
00154 // complicated hierarchies, however, it is occasionally useful to inherit
00155 // separately from Test and WithParamInterface. For example:
00156
00157 class BaseTest : public ::testing::Test {
00158   // You can inherit all the usual members for a non-parameterized test
00159   // fixture here.
00160 };
00161
```

```
00162 class DerivedTest : public BaseTest, public ::testing::WithParamInterface<int> {
00163   // The usual test fixture members go here too.
00164 };
00165
00166 TEST_F(BaseTest, HasFoo) {
00167   // This is an ordinary non-parameterized test.
00168 }
00169
00170 TEST_P(DerivedTest, DoesBlah) {
00171   // GetParam works just the same here as if you inherit from TestWithParam.
00172   EXPECT_TRUE(foo.Blah(GetParam()));
00173 }
00174
00175 #endif  // 0
00176
00177 #include <iterator>
00178 #include <utility>
00179
00180 #include "gtest/internal/gtest-internal.h"
00181 #include "gtest/internal/gtest-param-util.h"
00182 #include "gtest/internal/gtest-port.h"
00183
00184 namespace testing {
00185
00186 // Functions producing parameter generators.
00187 //
00188 // Google Test uses these generators to produce parameters for value-
00189 // parameterized tests. When a parameterized test suite is instantiated
00190 // with a particular generator, Google Test creates and runs tests
00191 // for each element in the sequence produced by the generator.
00192 //
00193 // In the following sample, tests from test suite FooTest are instantiated
00194 // each three times with parameter values 3, 5, and 8:
00195 //
00196 // class FooTest : public TestWithParam<int> { ... };
00197 //
00198 // TEST_P(FooTest, TestThis) {
00199 // }
00200 // TEST_P(FooTest, TestThat) {
00201 // }
00202 // INSTANTIATE_TEST_SUITE_P(TestSequence, FooTest, Values(3, 5, 8));
00203 //
00204
00205 // Range() returns generators providing sequences of values in a range.
00206 //
00207 // Synopsis:
00208 // Range(start, end)
00209 //   - returns a generator producing a sequence of values {start, start+1,
00210 //     start+2, ..., }.
00211 // Range(start, end, step)
00212 //   - returns a generator producing a sequence of values {start, start+step,
00213 //     start+step+step, ..., }.
00214 // Notes:
00215 //   * The generated sequences never include end. For example, Range(1, 5)
00216 //     returns a generator producing a sequence {1, 2, 3, 4}. Range(1, 9, 2)
00217 //     returns a generator producing {1, 3, 5, 7}.
00218 //   * start and end must have the same type. That type may be any integral or
00219 //     floating-point type or a user defined type satisfying these conditions:
00220 //     * It must be assignable (have operator=() defined).
00221 //     * It must have operator+() (operator+(int-compatible type) for
00222 //       two-operand version).
00223 //     * It must have operator<() defined.
00224 //     Elements in the resulting sequences will also have that type.
00225 //   * Condition start < end must be satisfied in order for resulting sequences
00226 //     to contain any elements.
00227 //
00228 template <typename T, typename IncrementT>
00229 internal::ParamGenerator<T> Range(T start, T end, IncrementT step) {
00230   return internal::ParamGenerator<T>(
00231       new internal::RangeGenerator<T, IncrementT>(start, end, step));
00232 }
00233
00234 template <typename T>
00235 internal::ParamGenerator<T> Range(T start, T end) {
00236   return Range(start, end, 1);
00237 }
00238
00239 // ValuesIn() function allows generation of tests with parameters coming from
00240 // a container.
00241 //
00242 // Synopsis:
00243 // ValuesIn(const T (&array)[N])
00244 //   - returns a generator producing sequences with elements from
00245 //     a C-style array.
00246 // ValuesIn(const Container& container)
00247 //   - returns a generator producing sequences with elements from
00248 //     an STL-style container.
```

```
00249 // ValuesIn(Iterator begin, Iterator end)
00250 //   - returns a generator producing sequences with elements from
00251 //     a range [begin, end) defined by a pair of STL-style iterators. These
00252 //     iterators can also be plain C pointers.
00253 //
00254 // Please note that ValuesIn copies the values from the containers
00255 // passed in and keeps them to generate tests in RUN_ALL_TESTS().
00256 //
00257 // Examples:
00258 //
00259 // This instantiates tests from test suite StringTest
00260 // each with C-string values of "foo", "bar", and "baz":
00261 //
00262 // const char* strings[] = {"foo", "bar", "baz"};
00263 // INSTANTIATE_TEST_SUITE_P(StringSequence, StringTest, ValuesIn(strings));
00264 //
00265 // This instantiates tests from test suite StlStringTest
00266 // each with STL strings with values "a" and "b":
00267 //
00268 // ::std::vector< ::std::string> GetParameterStrings() {
00269 //   ::std::vector< ::std::string> v;
00270 //   v.push_back("a");
00271 //   v.push_back("b");
00272 //   return v;
00273 // }
00274 //
00275 // INSTANTIATE_TEST_SUITE_P(CharSequence,
00276 //                          StlStringTest,
00277 //                          ValuesIn(GetParameterStrings()));
00278 //
00279 //
00280 // This will also instantiate tests from CharTest
00281 // each with parameter values 'a' and 'b':
00282 //
00283 // ::std::list<char> GetParameterChars() {
00284 //   ::std::list<char> list;
00285 //   list.push_back('a');
00286 //   list.push_back('b');
00287 //   return list;
00288 // }
00289 // ::std::list<char> l = GetParameterChars();
00290 // INSTANTIATE_TEST_SUITE_P(CharSequence2,
00291 //                          CharTest,
00292 //                          ValuesIn(l.begin(), l.end()));
00293 //
00294 template <typename ForwardIterator>
00295 internal::ParamGenerator<
00296     typename std::iterator_traits<ForwardIterator>::value_type>
00297 ValuesIn(ForwardIterator begin, ForwardIterator end) {
00298   typedef typename std::iterator_traits<ForwardIterator>::value_type ParamType;
00299   return internal::ParamGenerator<ParamType>(
00300       new internal::ValuesInIteratorRangeGenerator<ParamType>(begin, end));
00301 }
00302
00303 template <typename T, size_t N>
00304 internal::ParamGenerator<T> ValuesIn(const T (&array)[N]) {
00305   return ValuesIn(array, array + N);
00306 }
00307
00308 template <class Container>
00309 internal::ParamGenerator<typename Container::value_type> ValuesIn(
00310     const Container& container) {
00311   return ValuesIn(container.begin(), container.end());
00312 }
00313
00314 // Values() allows generating tests from explicitly specified list of
00315 // parameters.
00316 //
00317 // Synopsis:
00318 // Values(T v1, T v2, ..., T vN)
00319 //   - returns a generator producing sequences with elements v1, v2, ..., vN.
00320 //
00321 // For example, this instantiates tests from test suite BarTest each
00322 // with values "one", "two", and "three":
00323 //
00324 // INSTANTIATE_TEST_SUITE_P(NumSequence,
00325 //                          BarTest,
00326 //                          Values("one", "two", "three"));
00327 //
00328 // This instantiates tests from test suite BazTest each with values 1, 2, 3.5.
00329 // The exact type of values will depend on the type of parameter in BazTest.
00330 //
00331 // INSTANTIATE_TEST_SUITE_P(FloatingNumbers, BazTest, Values(1, 2, 3.5));
00332 //
00333 //
00334 template <typename... T>
00335 internal::ValueArray<T...> Values(T... v) {
```

```
00336   return internal::ValueArray<T...>(std::move(v)...);
00337 }
00338
00339 // Bool() allows generating tests with parameters in a set of (false, true).
00340 //
00341 // Synopsis:
00342 // Bool()
00343 //   - returns a generator producing sequences with elements {false, true}.
00344 //
00345 // It is useful when testing code that depends on Boolean flags. Combinations
00346 // of multiple flags can be tested when several Bool()'s are combined using
00347 // Combine() function.
00348 //
00349 // In the following example all tests in the test suite FlagDependentTest
00350 // will be instantiated twice with parameters false and true.
00351 //
00352 // class FlagDependentTest : public testing::TestWithParam<bool> {
00353 //   virtual void SetUp() {
00354 //     external_flag = GetParam();
00355 //   }
00356 // }
00357 // INSTANTIATE_TEST_SUITE_P(BoolSequence, FlagDependentTest, Bool());
00358 //
00359 inline internal::ParamGenerator<bool> Bool() { return Values(false, true); }
00360
00361 // Combine() allows the user to combine two or more sequences to produce
00362 // values of a Cartesian product of those sequences' elements.
00363 //
00364 // Synopsis:
00365 // Combine(gen1, gen2, ..., genN)
00366 //   - returns a generator producing sequences with elements coming from
00367 //     the Cartesian product of elements from the sequences generated by
00368 //     gen1, gen2, ..., genN. The sequence elements will have a type of
00369 //     std::tuple<T1, T2, ..., TN> where T1, T2, ..., TN are the types
00370 //     of elements from sequences produces by gen1, gen2, ..., genN.
00371 //
00372 // Example:
00373 //
00374 // This will instantiate tests in test suite AnimalTest each one with
00375 // the parameter values tuple("cat", BLACK), tuple("cat", WHITE),
00376 // tuple("dog", BLACK), and tuple("dog", WHITE):
00377 //
00378 // enum Color { BLACK, GRAY, WHITE };
00379 // class AnimalTest
00380 //     : public testing::TestWithParam<std::tuple<const char*, Color> > {...};
00381 //
00382 // TEST_P(AnimalTest, AnimalLooksNice) {...}
00383 //
00384 // INSTANTIATE_TEST_SUITE_P(AnimalVariations, AnimalTest,
00385 //                          Combine(Values("cat", "dog"),
00386 //                                  Values(BLACK, WHITE)));
00387 //
00388 // This will instantiate tests in FlagDependentTest with all variations of two
00389 // Boolean flags:
00390 //
00391 // class FlagDependentTest
00392 //     : public testing::TestWithParam<std::tuple<bool, bool> > {
00393 //   virtual void SetUp() {
00394 //     // Assigns external_flag_1 and external_flag_2 values from the tuple.
00395 //     std::tie(external_flag_1, external_flag_2) = GetParam();
00396 //   }
00397 // };
00398 //
00399 // TEST_P(FlagDependentTest, TestFeature1) {
00400 //   // Test your code using external_flag_1 and external_flag_2 here.
00401 // }
00402 // INSTANTIATE_TEST_SUITE_P(TwoBoolSequence, FlagDependentTest,
00403 //                          Combine(Bool(), Bool()));
00404 //
00405 template <typename... Generator>
00406 internal::CartesianProductHolder<Generator...> Combine(const Generator&... g) {
00407   return internal::CartesianProductHolder<Generator...>(g...);
00408 }
00409
00410 #define TEST_P(test_suite_name, test_name)                                \
00411   class GTEST_TEST_CLASS_NAME_(test_suite_name, test_name)                \
00412       : public test_suite_name {                                          \
00413    public:                                                                \
00414     GTEST_TEST_CLASS_NAME_(test_suite_name, test_name)() {}               \
00415     void TestBody() override;                                             \
00416                                                                           \
00417    private:                                                               \
00418     static int AddToRegistry() {                                          \
00419       ::testing::UnitTest::GetInstance()                                  \
00420           ->parameterized_test_registry()                                 \
00421           .GetTestSuitePatternHolder<test_suite_name>(                    \
00422               GTEST_STRINGIFY_(test_suite_name),                          \
```

```
00423              ::testing::internal::CodeLocation(__FILE__, __LINE__))           \
00424          ->AddTestPattern(                                                    \
00425              GTEST_STRINGIFY_(test_suite_name), GTEST_STRINGIFY_(test_name),  \
00426              new ::testing::internal::TestMetaFactory<GTEST_TEST_CLASS_NAME_( \
00427                  test_suite_name, test_name)>(),                              \
00428              ::testing::internal::CodeLocation(__FILE__, __LINE__));          \
00429      return 0;                                                                \
00430    }                                                                          \
00431    static int gtest_registering_dummy_ GTEST_ATTRIBUTE_UNUSED_;               \
00432    GTEST_TEST_CLASS_NAME_(test_suite_name, test_name)                         \
00433    (const GTEST_TEST_CLASS_NAME_(test_suite_name, test_name) &) = delete;     \
00434    GTEST_TEST_CLASS_NAME_(test_suite_name, test_name) & operator=(            \
00435        const GTEST_TEST_CLASS_NAME_(test_suite_name,                          \
00436                                     test_name) &) = delete; /* NOLINT */      \
00437  };                                                                           \
00438  int GTEST_TEST_CLASS_NAME_(test_suite_name,                                  \
00439                             test_name)::gtest_registering_dummy_ =            \
00440      GTEST_TEST_CLASS_NAME_(test_suite_name, test_name)::AddToRegistry();     \
00441  void GTEST_TEST_CLASS_NAME_(test_suite_name, test_name)::TestBody()
00442
00443    // The last argument to INSTANTIATE_TEST_SUITE_P allows the user to specify
00444    // generator and an optional function or functor that generates custom test name
00445    // suffixes based on the test parameters. Such a function or functor should
00446    // accept one argument of type testing::TestParamInfo<class ParamType>, and
00447    // return std::string.
00448    //
00449    // testing::PrintToStringParamName is a builtin test suffix generator that
00450    // returns the value of testing::PrintToString(GetParam()).
00451    //
00452    // Note: test names must be non-empty, unique, and may only contain ASCII
00453    // alphanumeric characters or underscore. Because PrintToString adds quotes
00454    // to std::string and C strings, it won't work for these types.
00455
00456 #define GTEST_EXPAND_(arg) arg
00457 #define GTEST_GET_FIRST_(first, ...) first
00458 #define GTEST_GET_SECOND_(first, second, ...) second
00459
00460 #define INSTANTIATE_TEST_SUITE_P(prefix, test_suite_name, ...)               \
00461   static ::testing::internal::ParamGenerator<test_suite_name::ParamType>     \
00462       gtest_##prefix##test_suite_name##_EvalGenerator_() {                   \
00463     return GTEST_EXPAND_(GTEST_GET_FIRST_(__VA_ARGS__, DUMMY_PARAM_));       \
00464   }                                                                          \
00465   static ::std::string gtest_##prefix##test_suite_name##_EvalGenerateName_(  \
00466       const ::testing::TestParamInfo<test_suite_name::ParamType>& info) {    \
00467     if (::testing::internal::AlwaysFalse()) {                                \
00468       ::testing::internal::TestNotEmpty(GTEST_EXPAND_(GTEST_GET_SECOND_(     \
00469           __VA_ARGS__,                                                       \
00470           ::testing::internal::DefaultParamName<test_suite_name::ParamType>, \
00471           DUMMY_PARAM_)));                                                   \
00472       auto t = std::make_tuple(__VA_ARGS__);                                 \
00473       static_assert(std::tuple_size<decltype(t)>::value <= 2,                \
00474                     "Too Many Args!");                                       \
00475     }                                                                        \
00476     return ((GTEST_EXPAND_(GTEST_GET_SECOND_(                                \
00477         __VA_ARGS__,                                                         \
00478         ::testing::internal::DefaultParamName<test_suite_name::ParamType>,   \
00479         DUMMY_PARAM_))))(info);                                              \
00480   }                                                                          \
00481   static int gtest_##prefix##test_suite_name##_dummy_                        \
00482       GTEST_ATTRIBUTE_UNUSED_ =                                              \
00483           ::testing::UnitTest::GetInstance()                                 \
00484               ->parameterized_test_registry()                                \
00485               .GetTestSuitePatternHolder<test_suite_name>(                   \
00486                   GTEST_STRINGIFY_(test_suite_name),                         \
00487                   ::testing::internal::CodeLocation(__FILE__, __LINE__))     \
00488               ->AddTestSuiteInstantiation(                                   \
00489                   GTEST_STRINGIFY_(prefix),                                  \
00490                   &gtest_##prefix##test_suite_name##_EvalGenerator_,         \
00491                   &gtest_##prefix##test_suite_name##_EvalGenerateName_,      \
00492                   __FILE__, __LINE__)
00493
00494    // Allow Marking a Parameterized test class as not needing to be instantiated.
00495 #define GTEST_ALLOW_UNINSTANTIATED_PARAMETERIZED_TEST(T)                     \
00496   namespace gtest_do_not_use_outside_namespace_scope {}                      \
00497   static const ::testing::internal::MarkAsIgnored gtest_allow_ignore_##T( \
00498       GTEST_STRINGIFY_(T))
00499
00500    // Legacy API is deprecated but still available
00501 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00502 #define INSTANTIATE_TEST_CASE_P                                              \
00503   static_assert(::testing::internal::InstantiateTestCase_P_IsDeprecated(), \
00504                 "");                                                         \
00505   INSTANTIATE_TEST_SUITE_P
00506 #endif  // GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00507
00508 } // namespace testing
00509
```

```
00510 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_PARAM_TEST_H_
```

18.264 gtest-printers.h

```
00001 // Copyright 2007, Google Inc.
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00003 //
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00005 // modification, are permitted provided that the following conditions are
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00026 // THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // Google Test - The Google C++ Testing and Mocking Framework
00031 //
00032 // This file implements a universal value printer that can print a
00033 // value of any type T:
00034 //
00035 //   void ::testing::internal::UniversalPrinter<T>::Print(value, ostream_ptr);
00036 //
00037 // A user can teach this function how to print a class type T by
00038 // defining either operator<<() or PrintTo() in the namespace that
00039 // defines T.  More specifically, the FIRST defined function in the
00040 // following list will be used (assuming T is defined in namespace
00041 // foo):
00042 //
00043 //   1. foo::PrintTo(const T&, ostream*)
00044 //   2. operator<<(ostream&, const T&) defined in either foo or the
00045 //      global namespace.
00046 //
00047 // However if T is an STL-style container then it is printed element-wise
00048 // unless foo::PrintTo(const T&, ostream*) is defined. Note that
00049 // operator<<() is ignored for container types.
00050 //
00051 // If none of the above is defined, it will print the debug string of
00052 // the value if it is a protocol buffer, or print the raw bytes in the
00053 // value otherwise.
00054 //
00055 // To aid debugging: when T is a reference type, the address of the
00056 // value is also printed; when T is a (const) char pointer, both the
00057 // pointer value and the NUL-terminated string it points to are
00058 // printed.
00059 //
00060 // We also provide some convenient wrappers:
00061 //
00062 //   // Prints a value to a string.  For a (const or not) char
00063 //   // pointer, the NUL-terminated string (but not the pointer) is
00064 //   // printed.
00065 //   std::string ::testing::PrintToString(const T& value);
00066 //
00067 //   // Prints a value tersely: for a reference type, the referenced
00068 //   // value (but not the address) is printed; for a (const or not) char
00069 //   // pointer, the NUL-terminated string (but not the pointer) is
00070 //   // printed.
00071 //   void ::testing::internal::UniversalTersePrint(const T& value, ostream*);
00072 //
00073 //   // Prints value using the type inferred by the compiler.  The difference
00074 //   // from UniversalTersePrint() is that this function prints both the
00075 //   // pointer and the NUL-terminated string for a (const or not) char pointer.
00076 //   void ::testing::internal::UniversalPrint(const T& value, ostream*);
00077 //
00078 //   // Prints the fields of a tuple tersely to a string vector, one
00079 //   // element for each field. Tuple support must be enabled in
00080 //   // gtest-port.h.
00081 //   std::vector<string> UniversalTersePrintTupleFieldsToStrings(
```

```
00082 //       const Tuple& value);
00083 //
00084 // Known limitation:
00085 //
00086 // The print primitives print the elements of an STL-style container
00087 // using the compiler-inferred type of *iter where iter is a
00088 // const_iterator of the container.  When const_iterator is an input
00089 // iterator but not a forward iterator, this inferred type may not
00090 // match value_type, and the print output may be incorrect.  In
00091 // practice, this is rarely a problem as for most containers
00092 // const_iterator is a forward iterator.  We'll fix this if there's an
00093 // actual need for it.  Note that this fix cannot rely on value_type
00094 // being defined as many user-defined container types don't have
00095 // value_type.
00096
00097 // IWYU pragma: private, include "gtest/gtest.h"
00098 // IWYU pragma: friend gtest/.*
00099 // IWYU pragma: friend gmock/.*
00100
00101 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_PRINTERS_H_
00102 #define GOOGLETEST_INCLUDE_GTEST_GTEST_PRINTERS_H_
00103
00104 #include <functional>
00105 #include <memory>
00106 #include <ostream>  // NOLINT
00107 #include <sstream>
00108 #include <string>
00109 #include <tuple>
00110 #include <type_traits>
00111 #include <utility>
00112 #include <vector>
00113
00114 #include "gtest/internal/gtest-internal.h"
00115 #include "gtest/internal/gtest-port.h"
00116
00117 namespace testing {
00118
00119     // Definitions in the internal* namespaces are subject to change without notice.
00120     // DO NOT USE THEM IN USER CODE!
00121     namespace internal {
00122
00123         template <typename T>
00124         void UniversalPrint(const T& value, ::std::ostream* os);
00125
00126         // Used to print an STL-style container when the user doesn't define
00127         // a PrintTo() for it.
00128
00129         struct ContainerPrinter {
00130
00131             template <typename T,
00132             typename = typename std::enable_if<
00133             (sizeof (IsContainerTest<T>(0)) == sizeof (IsContainer)) &&
00134             !IsRecursiveContainer<T>::value>::type>
00135             static void PrintValue(const T& container, std::ostream* os) {
00136                 const size_t kMaxCount = 32; // The maximum number of elements to print.
00137                 *os « '{';
00138                 size_t count = 0;
00139                 for (auto&& elem : container) {
00140                     if (count > 0) {
00141                         *os « ',';
00142                         if (count == kMaxCount) { // Enough has been printed.
00143                             *os « " ...";
00144                             break;
00145                         }
00146                     }
00147                     *os « ' ';
00148                     // We cannot call PrintTo(elem, os) here as PrintTo() doesn't
00149                     // handle `elem` being a native array.
00150                     internal::UniversalPrint(elem, os);
00151                     ++count;
00152                 }
00153
00154                 if (count > 0) {
00155                     *os « ' ';
00156                 }
00157                 *os « '}';
00158             }
00159         };
00160
00161         // Used to print a pointer that is neither a char pointer nor a member
00162         // pointer, when the user doesn't define PrintTo() for it.  (A member
00163         // variable pointer or member function pointer doesn't really point to
00164         // a location in the address space.  Their representation is
00165         // implementation-defined.  Therefore they will be printed as raw
00166         // bytes.)
00167
00168         struct FunctionPointerPrinter {
```

```
00169
00170             template <typename T, typename = typename std::enable_if<
00171             std::is_function<T>::value>::type>
00172             static void PrintValue(T* p, ::std::ostream* os) {
00173                 if (p == nullptr) {
00174                     *os « "NULL";
00175                 } else {
00176                     // T is a function type, so '*os « p' doesn't do what we want
00177                     // (it just prints p as bool).  We want to print p as a const
00178                     // void*.
00179                     *os « reinterpret_cast<const void*> (p);
00180                 }
00181             }
00182         };
00183
00184         struct PointerPrinter {
00185
00186             template <typename T>
00187             static void PrintValue(T* p, ::std::ostream* os) {
00188                 if (p == nullptr) {
00189                     *os « "NULL";
00190                 } else {
00191                     // T is not a function type.  We just call « to print p,
00192                     // relying on ADL to pick up user-defined « for their pointer
00193                     // types, if any.
00194                     *os « p;
00195                 }
00196             }
00197         };
00198
00199         namespace internal_stream_operator_without_lexical_name_lookup {
00200
00201             // The presence of an operator« here will terminate lexical scope lookup
00202             // straight away (even though it cannot be a match because of its argument
00203             // types). Thus, the two operator« calls in StreamPrinter will find only ADL
00204             // candidates.
00205
00206             struct LookupBlocker {
00207             };
00208             void operator«(LookupBlocker, LookupBlocker);
00209
00210             struct StreamPrinter {
00211
00212                 template <typename T,
00213                 // Don't accept member pointers here. We'd print them via implicit
00214                 // conversion to bool, which isn't useful.
00215                 typename = typename std::enable_if<
00216                 !std::is_member_pointer<T>::value>::type,
00217                 // Only accept types for which we can find a streaming operator via
00218                 // ADL (possibly involving implicit conversions).
00219                 typename = decltype(std::declval<std::ostream&>()
00220                         « std::declval<const T&>())>
00221                 static void PrintValue(const T& value, ::std::ostream* os) {
00222                     // Call streaming operator found by ADL, possibly with implicit conversions
00223                     // of the arguments.
00224                     *os « value;
00225                 }
00226             };
00227
00228         } // namespace internal_stream_operator_without_lexical_name_lookup
00229
00230         struct ProtobufPrinter {
00231             // We print a protobuf using its ShortDebugString() when the string
00232             // doesn't exceed this many characters; otherwise we print it using
00233             // DebugString() for better readability.
00234             static const size_t kProtobufOneLinerMaxLength = 50;
00235
00236             template <typename T,
00237             typename = typename std::enable_if<
00238             internal::HasDebugStringAndShortDebugString<T>::value>::type>
00239             static void PrintValue(const T& value, ::std::ostream* os) {
00240                 std::string pretty_str = value.ShortDebugString();
00241                 if (pretty_str.length() > kProtobufOneLinerMaxLength) {
00242                     pretty_str = "\n" + value.DebugString();
00243                 }
00244                 *os « ("<" + pretty_str + ">");
00245             }
00246         };
00247
00248         struct ConvertibleToIntegerPrinter {
00249             // Since T has no « operator or PrintTo() but can be implicitly
00250             // converted to BiggestInt, we print it as a BiggestInt.
00251             //
00252             // Most likely T is an enum type (either named or unnamed), in which
00253             // case printing it as an integer is the desired behavior.  In case
00254             // T is not an enum, printing it as an integer is the best we can do
00255             // given that it has no user-defined printer.
```

```
00256
00257             static void PrintValue(internal::BiggestInt value, ::std::ostream* os) {
00258                 *os << value;
00259             }
00260         };
00261
00262         struct ConvertibleToStringViewPrinter {
00263 #if GTEST_INTERNAL_HAS_STRING_VIEW
00264
00265             static void PrintValue(internal::StringView value, ::std::ostream* os) {
00266                 internal::UniversalPrint(value, os);
00267             }
00268 #endif
00269         };
00270
00271         // Prints the given number of bytes in the given object to the given
00272         // ostream.
00273         GTEST_API_ void PrintBytesInObjectTo(const unsigned char* obj_bytes,
00274                 size_t count, ::std::ostream* os);
00275
00276         struct RawBytesPrinter {
00277             // SFINAE on `sizeof` to make sure we have a complete type.
00278
00279             template <typename T, size_t = sizeof (T)>
00280             static void PrintValue(const T& value, ::std::ostream* os) {
00281                 PrintBytesInObjectTo(
00282                         static_cast<const unsigned char*> (
00283                         // Load bearing cast to void* to support iOS
00284                         reinterpret_cast<const void*> (std::addressof(value))),
00285                         sizeof (value), os);
00286             }
00287         };
00288
00289         struct FallbackPrinter {
00290
00291             template <typename T>
00292             static void PrintValue(const T&, ::std::ostream* os) {
00293                 *os << "(incomplete type)";
00294             }
00295         };
00296
00297         // Try every printer in order and return the first one that works.
00298
00299         template <typename T, typename E, typename Printer, typename... Printers>
00300         struct FindFirstPrinter : FindFirstPrinter<T, E, Printers...> {
00301         };
00302
00303         template <typename T, typename Printer, typename... Printers>
00304         struct FindFirstPrinter<
00305         T, decltype(Printer::PrintValue(std::declval<const T&>(), nullptr)),
00306         Printer, Printers...> {
00307             using type = Printer;
00308         };
00309
00310         // Select the best printer in the following order:
00311         //  - Print containers (they have begin/end/etc).
00312         //  - Print function pointers.
00313         //  - Print object pointers.
00314         //  - Use the stream operator, if available.
00315         //  - Print protocol buffers.
00316         //  - Print types convertible to BiggestInt.
00317         //  - Print types convertible to StringView, if available.
00318         //  - Fallback to printing the raw bytes of the object.
00319
00320         template <typename T>
00321         void PrintWithFallback(const T& value, ::std::ostream* os) {
00322             using Printer = typename FindFirstPrinter<
00323                     T, void, ContainerPrinter, FunctionPointerPrinter, PointerPrinter,
00324                     internal_stream_operator_without_lexical_name_lookup::StreamPrinter,
00325                     ProtobufPrinter, ConvertibleToIntegerPrinter,
00326                     ConvertibleToStringViewPrinter, RawBytesPrinter, FallbackPrinter>::type;
00327             Printer::PrintValue(value, os);
00328         }
00329
00330         // FormatForComparison<ToPrint, OtherOperand>::Format(value) formats a
00331         // value of type ToPrint that is an operand of a comparison assertion
00332         // (e.g. ASSERT_EQ).  OtherOperand is the type of the other operand in
00333         // the comparison, and is used to help determine the best way to
00334         // format the value.  In particular, when the value is a C string
00335         // (char pointer) and the other operand is an STL string object, we
00336         // want to format the C string as a string, since we know it is
00337         // compared by value with the string object.  If the value is a char
00338         // pointer but the other operand is not an STL string object, we don't
00339         // know whether the pointer is supposed to point to a NUL-terminated
00340         // string, and thus want to print it as a pointer to be safe.
00341         //
00342         // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
```

```
00343
00344         // The default case.
00345
00346         template <typename ToPrint, typename OtherOperand>
00347         class FormatForComparison {
00348         public:
00349
00350             static ::std::string Format(const ToPrint& value) {
00351                 return ::testing::PrintToString(value);
00352             }
00353         };
00354
00355         // Array.
00356
00357         template <typename ToPrint, size_t N, typename OtherOperand>
00358         class FormatForComparison<ToPrint[N], OtherOperand> {
00359         public:
00360
00361             static ::std::string Format(const ToPrint* value) {
00362                 return FormatForComparison<const ToPrint*, OtherOperand>::Format(value);
00363             }
00364         };
00365
00366         // By default, print C string as pointers to be safe, as we don't know
00367         // whether they actually point to a NUL-terminated string.
00368
00369 #define GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_(CharType)                 \
00370   template <typename OtherOperand>                                       \
00371   class FormatForComparison<CharType*, OtherOperand> {                  \
00372    public:                                                             \
00373     static ::std::string Format(CharType* value) {                     \
00374       return ::testing::PrintToString(static_cast<const void*>(value)); \
00375     }                                                                  \
00376   }
00377
00378         GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_(char);
00379         GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_(const char);
00380         GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_(wchar_t);
00381         GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_(const wchar_t);
00382 #ifdef __cpp_lib_char8_t
00383         GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_(char8_t);
00384         GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_(const char8_t);
00385 #endif
00386         GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_(char16_t);
00387         GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_(const char16_t);
00388         GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_(char32_t);
00389         GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_(const char32_t);
00390
00391 #undef GTEST_IMPL_FORMAT_C_STRING_AS_POINTER_
00392
00393         // If a C string is compared with an STL string object, we know it's meant
00394         // to point to a NUL-terminated string, and thus can print it as a string.
00395
00396 #define GTEST_IMPL_FORMAT_C_STRING_AS_STRING_(CharType, OtherStringType) \
00397   template <>                                                            \
00398   class FormatForComparison<CharType*, OtherStringType> {                \
00399    public:                                                             \
00400     static ::std::string Format(CharType* value) {                     \
00401       return ::testing::PrintToString(value);                          \
00402     }                                                                  \
00403   }
00404
00405         GTEST_IMPL_FORMAT_C_STRING_AS_STRING_(char, ::std::string);
00406         GTEST_IMPL_FORMAT_C_STRING_AS_STRING_(const char, ::std::string);
00407 #ifdef __cpp_lib_char8_t
00408         GTEST_IMPL_FORMAT_C_STRING_AS_STRING_(char8_t, ::std::u8string);
00409         GTEST_IMPL_FORMAT_C_STRING_AS_STRING_(const char8_t, ::std::u8string);
00410 #endif
00411         GTEST_IMPL_FORMAT_C_STRING_AS_STRING_(char16_t, ::std::u16string);
00412         GTEST_IMPL_FORMAT_C_STRING_AS_STRING_(const char16_t, ::std::u16string);
00413         GTEST_IMPL_FORMAT_C_STRING_AS_STRING_(char32_t, ::std::u32string);
00414         GTEST_IMPL_FORMAT_C_STRING_AS_STRING_(const char32_t, ::std::u32string);
00415
00416 #if GTEST_HAS_STD_WSTRING
00417         GTEST_IMPL_FORMAT_C_STRING_AS_STRING_(wchar_t, ::std::wstring);
00418         GTEST_IMPL_FORMAT_C_STRING_AS_STRING_(const wchar_t, ::std::wstring);
00419 #endif
00420
00421 #undef GTEST_IMPL_FORMAT_C_STRING_AS_STRING_
00422
00423         // Formats a comparison assertion (e.g. ASSERT_EQ, EXPECT_LT, and etc)
00424         // operand to be used in a failure message.  The type (but not value)
00425         // of the other operand may affect the format.  This allows us to
00426         // print a char* as a raw pointer when it is compared against another
00427         // char* or void*, and print it as a C string when it is compared
00428         // against an std::string object, for example.
00429         //
```

```
00430         // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
00431
00432         template <typename T1, typename T2>
00433         std::string FormatForComparisonFailureMessage(const T1& value,
00434                 const T2& /* other_operand */) {
00435             return FormatForComparison<T1, T2>::Format(value);
00436         }
00437
00438         // UniversalPrinter<T>::Print(value, ostream_ptr) prints the given
00439         // value to the given ostream.  The caller must ensure that
00440         // 'ostream_ptr' is not NULL, or the behavior is undefined.
00441         //
00442         // We define UniversalPrinter as a class template (as opposed to a
00443         // function template), as we need to partially specialize it for
00444         // reference types, which cannot be done with function templates.
00445         template <typename T>
00446         class UniversalPrinter;
00447
00448         // Prints the given value using the << operator if it has one;
00449         // otherwise prints the bytes in it.  This is what
00450         // UniversalPrinter<T>::Print() does when PrintTo() is not specialized
00451         // or overloaded for type T.
00452         //
00453         // A user can override this behavior for a class type Foo by defining
00454         // an overload of PrintTo() in the namespace where Foo is defined.  We
00455         // give the user this option as sometimes defining a << operator for
00456         // Foo is not desirable (e.g. the coding style may prevent doing it,
00457         // or there is already a << operator but it doesn't do what the user
00458         // wants).
00459
00460         template <typename T>
00461         void PrintTo(const T& value, ::std::ostream* os) {
00462             internal::PrintWithFallback(value, os);
00463         }
00464
00465         // The following list of PrintTo() overloads tells
00466         // UniversalPrinter<T>::Print() how to print standard types (built-in
00467         // types, strings, plain arrays, and pointers).
00468
00469         // Overloads for various char types.
00470         GTEST_API_ void PrintTo(unsigned char c, ::std::ostream* os);
00471         GTEST_API_ void PrintTo(signed char c, ::std::ostream* os);
00472
00473         inline void PrintTo(char c, ::std::ostream* os) {
00474             // When printing a plain char, we always treat it as unsigned.  This
00475             // way, the output won't be affected by whether the compiler thinks
00476             // char is signed or not.
00477             PrintTo(static_cast<unsigned char> (c), os);
00478         }
00479
00480         // Overloads for other simple built-in types.
00481
00482         inline void PrintTo(bool x, ::std::ostream* os) {
00483             *os << (x ? "true" : "false");
00484         }
00485
00486         // Overload for wchar_t type.
00487         // Prints a wchar_t as a symbol if it is printable or as its internal
00488         // code otherwise and also as its decimal code (except for L'\0').
00489         // The L'\0' char is printed as "L'\\0'". The decimal code is printed
00490         // as signed integer when wchar_t is implemented by the compiler
00491         // as a signed type and is printed as an unsigned integer when wchar_t
00492         // is implemented as an unsigned type.
00493         GTEST_API_ void PrintTo(wchar_t wc, ::std::ostream* os);
00494
00495         GTEST_API_ void PrintTo(char32_t c, ::std::ostream* os);
00496
00497         inline void PrintTo(char16_t c, ::std::ostream* os) {
00498             PrintTo(ImplicitCast_<char32_t>(c), os);
00499         }
00500 #ifdef __cpp_char8_t
00501
00502         inline void PrintTo(char8_t c, ::std::ostream* os) {
00503             PrintTo(ImplicitCast_<char32_t>(c), os);
00504         }
00505 #endif
00506
00507         // gcc/clang __{u,}int128_t
00508 #if defined(__SIZEOF_INT128__)
00509         GTEST_API_ void PrintTo(__uint128_t v, ::std::ostream* os);
00510         GTEST_API_ void PrintTo(__int128_t v, ::std::ostream* os);
00511 #endif  // __SIZEOF_INT128__
00512
00513         // Overloads for C strings.
00514         GTEST_API_ void PrintTo(const char* s, ::std::ostream* os);
00515
00516         inline void PrintTo(char* s, ::std::ostream* os) {
```

```
00517             PrintTo(ImplicitCast_<const char*>(s), os);
00518         }
00519
00520         // signed/unsigned char is often used for representing binary data, so
00521         // we print pointers to it as void* to be safe.
00522
00523         inline void PrintTo(const signed char* s, ::std::ostream* os) {
00524             PrintTo(ImplicitCast_<const void*>(s), os);
00525         }
00526
00527         inline void PrintTo(signed char* s, ::std::ostream* os) {
00528             PrintTo(ImplicitCast_<const void*>(s), os);
00529         }
00530
00531         inline void PrintTo(const unsigned char* s, ::std::ostream* os) {
00532             PrintTo(ImplicitCast_<const void*>(s), os);
00533         }
00534
00535         inline void PrintTo(unsigned char* s, ::std::ostream* os) {
00536             PrintTo(ImplicitCast_<const void*>(s), os);
00537         }
00538 #ifdef __cpp_char8_t
00539         // Overloads for u8 strings.
00540         GTEST_API_ void PrintTo(const char8_t* s, ::std::ostream* os);
00541
00542         inline void PrintTo(char8_t* s, ::std::ostream* os) {
00543             PrintTo(ImplicitCast_<const char8_t*>(s), os);
00544         }
00545 #endif
00546         // Overloads for u16 strings.
00547         GTEST_API_ void PrintTo(const char16_t* s, ::std::ostream* os);
00548
00549         inline void PrintTo(char16_t* s, ::std::ostream* os) {
00550             PrintTo(ImplicitCast_<const char16_t*>(s), os);
00551         }
00552         // Overloads for u32 strings.
00553         GTEST_API_ void PrintTo(const char32_t* s, ::std::ostream* os);
00554
00555         inline void PrintTo(char32_t* s, ::std::ostream* os) {
00556             PrintTo(ImplicitCast_<const char32_t*>(s), os);
00557         }
00558
00559         // MSVC can be configured to define wchar_t as a typedef of unsigned
00560         // short.  It defines _NATIVE_WCHAR_T_DEFINED when wchar_t is a native
00561         // type.  When wchar_t is a typedef, defining an overload for const
00562         // wchar_t* would cause unsigned short* be printed as a wide string,
00563         // possibly causing invalid memory accesses.
00564 #if !defined(_MSC_VER) || defined(_NATIVE_WCHAR_T_DEFINED)
00565         // Overloads for wide C strings
00566         GTEST_API_ void PrintTo(const wchar_t* s, ::std::ostream* os);
00567
00568         inline void PrintTo(wchar_t* s, ::std::ostream* os) {
00569             PrintTo(ImplicitCast_<const wchar_t*>(s), os);
00570         }
00571 #endif
00572
00573         // Overload for C arrays.  Multi-dimensional arrays are printed
00574         // properly.
00575
00576         // Prints the given number of elements in an array, without printing
00577         // the curly braces.
00578
00579         template <typename T>
00580         void PrintRawArrayTo(const T a[], size_t count, ::std::ostream* os) {
00581             UniversalPrint(a[0], os);
00582             for (size_t i = 1; i != count; i++) {
00583                 *os << ", ";
00584                 UniversalPrint(a[i], os);
00585             }
00586         }
00587
00588         // Overloads for ::std::string.
00589         GTEST_API_ void PrintStringTo(const ::std::string& s, ::std::ostream* os);
00590
00591         inline void PrintTo(const ::std::string& s, ::std::ostream* os) {
00592             PrintStringTo(s, os);
00593         }
00594
00595         // Overloads for ::std::u8string
00596 #ifdef __cpp_lib_char8_t
00597         GTEST_API_ void PrintU8StringTo(const ::std::u8string& s, ::std::ostream* os);
00598
00599         inline void PrintTo(const ::std::u8string& s, ::std::ostream* os) {
00600             PrintU8StringTo(s, os);
00601         }
00602 #endif
00603
```

```
00604         // Overloads for ::std::u16string
00605         GTEST_API_ void PrintU16StringTo(const ::std::u16string& s, ::std::ostream* os);
00606
00607         inline void PrintTo(const ::std::u16string& s, ::std::ostream* os) {
00608             PrintU16StringTo(s, os);
00609         }
00610
00611         // Overloads for ::std::u32string
00612         GTEST_API_ void PrintU32StringTo(const ::std::u32string& s, ::std::ostream* os);
00613
00614         inline void PrintTo(const ::std::u32string& s, ::std::ostream* os) {
00615             PrintU32StringTo(s, os);
00616         }
00617
00618         // Overloads for ::std::wstring.
00619 #if GTEST_HAS_STD_WSTRING
00620         GTEST_API_ void PrintWideStringTo(const ::std::wstring& s, ::std::ostream* os);
00621
00622         inline void PrintTo(const ::std::wstring& s, ::std::ostream* os) {
00623             PrintWideStringTo(s, os);
00624         }
00625 #endif  // GTEST_HAS_STD_WSTRING
00626
00627 #if GTEST_INTERNAL_HAS_STRING_VIEW
00628         // Overload for internal::StringView.
00629
00630         inline void PrintTo(internal::StringView sp, ::std::ostream* os) {
00631             PrintTo(::std::string(sp), os);
00632         }
00633 #endif  // GTEST_INTERNAL_HAS_STRING_VIEW
00634
00635         inline void PrintTo(std::nullptr_t, ::std::ostream* os) {
00636             *os « "(nullptr)";
00637         }
00638
00639 #if GTEST_HAS_RTTI
00640
00641         inline void PrintTo(const std::type_info& info, std::ostream* os) {
00642             *os « internal::GetTypeName(info);
00643         }
00644 #endif  // GTEST_HAS_RTTI
00645
00646         template <typename T>
00647         void PrintTo(std::reference_wrapper<T> ref, ::std::ostream* os) {
00648             UniversalPrinter<T&>::Print(ref.get(), os);
00649         }
00650
00651         inline const void* VoidifyPointer(const void* p) {
00652             return p;
00653         }
00654
00655         inline const void* VoidifyPointer(volatile const void* p) {
00656             return const_cast<const void*> (p);
00657         }
00658
00659         template <typename T, typename Ptr>
00660         void PrintSmartPointer(const Ptr& ptr, std::ostream* os, char) {
00661             if (ptr == nullptr) {
00662                 *os « "(nullptr)";
00663             } else {
00664                 // We can’t print the value. Just print the pointer..
00665                 *os « "(" « (VoidifyPointer) (ptr.get()) « ")";
00666             }
00667         }
00668
00669         template <typename T, typename Ptr,
00670         typename = typename std::enable_if<!std::is_void<T>::value &&
00671         !std::is_array<T>::value>::type>
00672         void PrintSmartPointer(const Ptr& ptr, std::ostream* os, int) {
00673             if (ptr == nullptr) {
00674                 *os « "(nullptr)";
00675             } else {
00676                 *os « "(ptr = " « (VoidifyPointer) (ptr.get()) « ", value = ";
00677                 UniversalPrinter<T>::Print(*ptr, os);
00678                 *os « ")";
00679             }
00680         }
00681
00682         template <typename T, typename D>
00683         void PrintTo(const std::unique_ptr<T, D>& ptr, std::ostream* os) {
00684             (PrintSmartPointer<T>)(ptr, os, 0);
00685         }
00686
00687         template <typename T>
00688         void PrintTo(const std::shared_ptr<T>& ptr, std::ostream* os) {
00689             (PrintSmartPointer<T>)(ptr, os, 0);
00690         }
```

```
00691
00692         // Helper function for printing a tuple.  T must be instantiated with
00693         // a tuple type.
00694
00695         template <typename T>
00696         void PrintTupleTo(const T&, std::integral_constant<size_t, 0>,
00697                 ::std::ostream*) {
00698         }
00699
00700         template <typename T, size_t I>
00701         void PrintTupleTo(const T& t, std::integral_constant<size_t, I>,
00702                 ::std::ostream* os) {
00703             PrintTupleTo(t, std::integral_constant<size_t, I - 1 > (), os);
00704             GTEST_INTENTIONAL_CONST_COND_PUSH_()
00705             if (I > 1) {
00706                 GTEST_INTENTIONAL_CONST_COND_POP_()
00707                         *os << ", ";
00708             }
00709             UniversalPrinter<typename std::tuple_element < I - 1, T>::type>::Print(
00710                     std::get < I - 1 > (t), os);
00711         }
00712
00713         template <typename... Types>
00714         void PrintTo(const ::std::tuple<Types...>& t, ::std::ostream* os) {
00715             *os << "(";
00716             PrintTupleTo(t, std::integral_constant<size_t, sizeof...(Types)>(), os);
00717             *os << ")";
00718         }
00719
00720         // Overload for std::pair.
00721
00722         template <typename T1, typename T2>
00723         void PrintTo(const ::std::pair<T1, T2>& value, ::std::ostream* os) {
00724             *os << '(';
00725             // We cannot use UniversalPrint(value.first, os) here, as T1 may be
00726             // a reference type.  The same for printing value.second.
00727             UniversalPrinter<T1>::Print(value.first, os);
00728             *os << ", ";
00729             UniversalPrinter<T2>::Print(value.second, os);
00730             *os << ')';
00731         }
00732
00733         // Implements printing a non-reference type T by letting the compiler
00734         // pick the right overload of PrintTo() for T.
00735
00736         template <typename T>
00737         class UniversalPrinter {
00738         public:
00739             // MSVC warns about adding const to a function type, so we want to
00740             // disable the warning.
00741
00742             GTEST_DISABLE_MSC_WARNINGS_PUSH_(4180)
00743
00744             // Note: we deliberately don't call this PrintTo(), as that name
00745             // conflicts with ::testing::internal::PrintTo in the body of the
00746             // function.
00747             static void Print(const T& value, ::std::ostream* os) {
00748                 // By default, ::testing::internal::PrintTo() is used for printing
00749                 // the value.
00750                 //
00751                 // Thanks to Koenig look-up, if T is a class and has its own
00752                 // PrintTo() function defined in its namespace, that function will
00753                 // be visible here.  Since it is more specific than the generic ones
00754                 // in ::testing::internal, it will be picked by the compiler in the
00755                 // following statement - exactly what we want.
00756                 PrintTo(value, os);
00757             }
00758
00759             GTEST_DISABLE_MSC_WARNINGS_POP_()
00760         };
00761
00762         // Remove any const-qualifiers before passing a type to UniversalPrinter.
00763
00764         template <typename T>
00765         class UniversalPrinter<const T> : public UniversalPrinter<T> {
00766         };
00767
00768 #if GTEST_INTERNAL_HAS_ANY
00769
00770         // Printer for std::any / absl::any
00771
00772         template <>
00773         class UniversalPrinter<Any> {
00774         public:
00775
00776             static void Print(const Any& value, ::std::ostream* os) {
00777                 if (value.has_value()) {
```

```
00778                     *os « "value of type " « GetTypeName(value);
00779                 } else {
00780                     *os « "no value";
00781                 }
00782             }
00783
00784         private:
00785
00786             static std::string GetTypeName(const Any& value) {
00787 #if GTEST_HAS_RTTI
00788                 return internal::GetTypeName(value.type());
00789 #else
00790                 static_cast<void> (value); // possibly unused
00791                 return "<unknown_type>";
00792 #endif  // GTEST_HAS_RTTI
00793             }
00794         };
00795
00796 #endif  // GTEST_INTERNAL_HAS_ANY
00797
00798 #if GTEST_INTERNAL_HAS_OPTIONAL
00799
00800         // Printer for std::optional / absl::optional
00801
00802         template <typename T>
00803         class UniversalPrinter<Optional<T»
00804         {
00805             public:
00806
00807             static void Print(const Optional<T>& value, ::std::ostream * os) {
00808                 *os « '(';
00809                 if (!value) {
00810                     *os « "nullopt";
00811                 } else {
00812                     UniversalPrint(*value, os);
00813                 }
00814                 *os « ')';
00815             }
00816         };
00817
00818         template <>
00819         class UniversalPrinter<decltype(Nullopt())> {
00820         public:
00821
00822             static void Print(decltype(Nullopt()), ::std::ostream* os) {
00823                 *os « "(nullopt)";
00824             }
00825         };
00826
00827 #endif  // GTEST_INTERNAL_HAS_OPTIONAL
00828
00829 #if GTEST_INTERNAL_HAS_VARIANT
00830
00831         // Printer for std::variant / absl::variant
00832
00833         template <typename... T>
00834         class UniversalPrinter<Variant<T...»
00835         {
00836             public:
00837
00838             static void Print(const Variant<T...>& value, ::std::ostream * os) {
00839                 *os « '(';
00840 #if GTEST_HAS_ABSL
00841                 absl::visit(Visitor{os, value.index()}, value);
00842 #else
00843                 std::visit(Visitor{os, value.index()}, value);
00844 #endif  // GTEST_HAS_ABSL
00845                 *os « ')';
00846             }
00847
00848             private:
00849
00850             struct Visitor {
00851
00852                 template <typename U>
00853                 void operator()(const U& u) const {
00854                     *os « "'" « GetTypeName<U>() « "(index = " « index
00855                             « ")' with value ";
00856                     UniversalPrint(u, os);
00857                 }
00858                 ::std::ostream* os;
00859                 std::size_t index;
00860             };
00861         };
00862
00863 #endif  // GTEST_INTERNAL_HAS_VARIANT
00864
```

```
00865         // UniversalPrintArray(begin, len, os) prints an array of 'len'
00866         // elements, starting at address 'begin'.
00867
00868         template <typename T>
00869         void UniversalPrintArray(const T* begin, size_t len, ::std::ostream* os) {
00870             if (len == 0) {
00871                 *os << "{}";
00872             } else {
00873                 *os << "{ ";
00874                 const size_t kThreshold = 18;
00875                 const size_t kChunkSize = 8;
00876                 // If the array has more than kThreshold elements, we'll have to
00877                 // omit some details by printing only the first and the last
00878                 // kChunkSize elements.
00879                 if (len <= kThreshold) {
00880                     PrintRawArrayTo(begin, len, os);
00881                 } else {
00882                     PrintRawArrayTo(begin, kChunkSize, os);
00883                     *os << ", ..., ";
00884                     PrintRawArrayTo(begin + len - kChunkSize, kChunkSize, os);
00885                 }
00886                 *os << " }";
00887             }
00888         }
00889         // This overload prints a (const) char array compactly.
00890         GTEST_API_ void UniversalPrintArray(const char* begin, size_t len,
00891                 ::std::ostream* os);
00892
00893 #ifdef __cpp_char8_t
00894         // This overload prints a (const) char8_t array compactly.
00895         GTEST_API_ void UniversalPrintArray(const char8_t* begin, size_t len,
00896                 ::std::ostream* os);
00897 #endif
00898
00899         // This overload prints a (const) char16_t array compactly.
00900         GTEST_API_ void UniversalPrintArray(const char16_t* begin, size_t len,
00901                 ::std::ostream* os);
00902
00903         // This overload prints a (const) char32_t array compactly.
00904         GTEST_API_ void UniversalPrintArray(const char32_t* begin, size_t len,
00905                 ::std::ostream* os);
00906
00907         // This overload prints a (const) wchar_t array compactly.
00908         GTEST_API_ void UniversalPrintArray(const wchar_t* begin, size_t len,
00909                 ::std::ostream* os);
00910
00911         // Implements printing an array type T[N].
00912
00913         template <typename T, size_t N>
00914         class UniversalPrinter<T[N]> {
00915         public:
00916             // Prints the given array, omitting some elements when there are too
00917             // many.
00918
00919             static void Print(const T(&a)[N], ::std::ostream* os) {
00920                 UniversalPrintArray(a, N, os);
00921             }
00922         };
00923
00924         // Implements printing a reference type T&.
00925
00926         template <typename T>
00927         class UniversalPrinter<T&> {
00928         public:
00929             // MSVC warns about adding const to a function type, so we want to
00930             // disable the warning.
00931
00932             GTEST_DISABLE_MSC_WARNINGS_PUSH_(4180)
00933
00934             static void Print(const T& value, ::std::ostream* os) {
00935                 // Prints the address of the value.  We use reinterpret_cast here
00936                 // as static_cast doesn't compile when T is a function type.
00937                 *os << "@" << reinterpret_cast<const void*> (&value) << " ";
00938
00939                 // Then prints the value itself.
00940                 UniversalPrint(value, os);
00941             }
00942
00943             GTEST_DISABLE_MSC_WARNINGS_POP_()
00944         };
00945
00946         // Prints a value tersely: for a reference type, the referenced value
00947         // (but not the address) is printed; for a (const) char pointer, the
00948         // NUL-terminated string (but not the pointer) is printed.
00949
00950         template <typename T>
00951         class UniversalTersePrinter {
```

```
00952         public:
00953
00954             static void Print(const T& value, ::std::ostream* os) {
00955                 UniversalPrint(value, os);
00956             }
00957         };
00958
00959         template <typename T>
00960         class UniversalTersePrinter<T&> {
00961         public:
00962
00963             static void Print(const T& value, ::std::ostream* os) {
00964                 UniversalPrint(value, os);
00965             }
00966         };
00967
00968         template <typename T, size_t N>
00969         class UniversalTersePrinter<T[N]> {
00970         public:
00971
00972             static void Print(const T(&value)[N], ::std::ostream* os) {
00973                 UniversalPrinter < T[N]>::Print(value, os);
00974             }
00975         };
00976
00977         template <>
00978         class UniversalTersePrinter<const char*> {
00979         public:
00980
00981             static void Print(const char* str, ::std::ostream* os) {
00982                 if (str == nullptr) {
00983                     *os « "NULL";
00984                 } else {
00985                     UniversalPrint(std::string(str), os);
00986                 }
00987             }
00988         };
00989
00990         template <>
00991         class UniversalTersePrinter<char*> : public UniversalTersePrinter<const char*> {
00992         };
00993
00994 #ifdef __cpp_char8_t
00995
00996         template <>
00997         class UniversalTersePrinter<const char8_t*> {
00998         public:
00999
01000             static void Print(const char8_t* str, ::std::ostream* os) {
01001                 if (str == nullptr) {
01002                     *os « "NULL";
01003                 } else {
01004                     UniversalPrint(::std::u8string(str), os);
01005                 }
01006             }
01007         };
01008
01009         template <>
01010         class UniversalTersePrinter<char8_t*>
01011         : public UniversalTersePrinter<const char8_t*> {
01012         };
01013 #endif
01014
01015         template <>
01016         class UniversalTersePrinter<const char16_t*> {
01017         public:
01018
01019             static void Print(const char16_t* str, ::std::ostream* os) {
01020                 if (str == nullptr) {
01021                     *os « "NULL";
01022                 } else {
01023                     UniversalPrint(::std::u16string(str), os);
01024                 }
01025             }
01026         };
01027
01028         template <>
01029         class UniversalTersePrinter<char16_t*>
01030         : public UniversalTersePrinter<const char16_t*> {
01031         };
01032
01033         template <>
01034         class UniversalTersePrinter<const char32_t*> {
01035         public:
01036
01037             static void Print(const char32_t* str, ::std::ostream* os) {
01038                 if (str == nullptr) {
```

```
01039                     *os « "NULL";
01040                 } else {
01041                     UniversalPrint(::std::u32string(str), os);
01042                 }
01043             }
01044         };
01045
01046         template <>
01047         class UniversalTersePrinter<char32_t*>
01048         : public UniversalTersePrinter<const char32_t*> {
01049         };
01050
01051 #if GTEST_HAS_STD_WSTRING
01052
01053         template <>
01054         class UniversalTersePrinter<const wchar_t*> {
01055         public:
01056
01057             static void Print(const wchar_t* str, ::std::ostream* os) {
01058                 if (str == nullptr) {
01059                     *os « "NULL";
01060                 } else {
01061                     UniversalPrint(::std::wstring(str), os);
01062                 }
01063             }
01064         };
01065 #endif
01066
01067         template <>
01068         class UniversalTersePrinter<wchar_t*> {
01069         public:
01070
01071             static void Print(wchar_t* str, ::std::ostream* os) {
01072                 UniversalTersePrinter<const wchar_t*>::Print(str, os);
01073             }
01074         };
01075
01076         template <typename T>
01077         void UniversalTersePrint(const T& value, ::std::ostream* os) {
01078             UniversalTersePrinter<T>::Print(value, os);
01079         }
01080
01081         // Prints a value using the type inferred by the compiler.  The
01082         // difference between this and UniversalTersePrint() is that for a
01083         // (const) char pointer, this prints both the pointer and the
01084         // NUL-terminated string.
01085
01086         template <typename T>
01087         void UniversalPrint(const T& value, ::std::ostream* os) {
01088             // A workarond for the bug in VC++ 7.1 that prevents us from instantiating
01089             // UniversalPrinter with T directly.
01090             typedef T T1;
01091             UniversalPrinter<T1>::Print(value, os);
01092         }
01093
01094         typedef ::std::vector<::std::string> Strings;
01095
01096         // Tersely prints the first N fields of a tuple to a string vector,
01097         // one element for each field.
01098
01099         template <typename Tuple>
01100         void TersePrintPrefixToStrings(const Tuple&, std::integral_constant<size_t, 0>,
01101                 Strings*) {
01102         }
01103
01104         template <typename Tuple, size_t I>
01105         void TersePrintPrefixToStrings(const Tuple& t,
01106                 std::integral_constant<size_t, I>,
01107                 Strings* strings) {
01108             TersePrintPrefixToStrings(t, std::integral_constant<size_t, I - 1 > (),
01109                     strings);
01110             ::std::stringstream ss;
01111             UniversalTersePrint(std::get < I - 1 > (t), &ss);
01112             strings->push_back(ss.str());
01113         }
01114
01115         // Prints the fields of a tuple tersely to a string vector, one
01116         // element for each field.  See the comment before
01117         // UniversalTersePrint() for how we define "tersely".
01118
01119         template <typename Tuple>
01120         Strings UniversalTersePrintTupleFieldsToStrings(const Tuple& value) {
01121             Strings result;
01122             TersePrintPrefixToStrings(
01123                     value, std::integral_constant<size_t, std::tuple_size<Tuple>::value>(),
01124                     &result);
01125             return result;
```

```
01126         }
01127
01128     } // namespace internal
01129
01130     template <typename T>
01131     ::std::string PrintToString(const T& value) {
01132         ::std::stringstream ss;
01133         internal::UniversalTersePrinter<T>::Print(value, &ss);
01134         return ss.str();
01135     }
01136
01137 } // namespace testing
01138
01139 // Include any custom printer added by the local installation.
01140 // We must include this header at the end to make sure it can use the
01141 // declarations from this file.
01142 #include "gtest/internal/custom/gtest-printers.h"
01143
01144 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_PRINTERS_H_
```

18.265 gtest-printers.h

```
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00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029 //
00030 // This file provides an injection point for custom printers in a local
00031 // installation of gTest.
00032 // It will be included from gtest-printers.h and the overrides in this file
00033 // will be visible to everyone.
00034 //
00035 // Injection point for custom user configurations. See README for details
00036 //
00037 // ** Custom implementation starts here **
00038
00039 #ifndef GOOGLETEST_INCLUDE_GTEST_INTERNAL_CUSTOM_GTEST_PRINTERS_H_
00040 #define GOOGLETEST_INCLUDE_GTEST_INTERNAL_CUSTOM_GTEST_PRINTERS_H_
00041
00042 #endif  // GOOGLETEST_INCLUDE_GTEST_INTERNAL_CUSTOM_GTEST_PRINTERS_H_
```

18.266 gtest-spi.h

```
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```
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00026 // THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // Utilities for testing Google Test itself and code that uses Google Test
00031 // (e.g. frameworks built on top of Google Test).
00032
00033 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_SPI_H_
00034 #define GOOGLETEST_INCLUDE_GTEST_GTEST_SPI_H_
00035
00036 #include "gtest/gtest.h"
00037
00038 GTEST_DISABLE_MSC_WARNINGS_PUSH_(4251 \
00039 /* class A needs to have dll-interface to be used by clients of class B */)
00040
00041 namespace testing {
00042
00043     // This helper class can be used to mock out Google Test failure reporting
00044     // so that we can test Google Test or code that builds on Google Test.
00045     //
00046     // An object of this class appends a TestPartResult object to the
00047     // TestPartResultArray object given in the constructor whenever a Google Test
00048     // failure is reported. It can either intercept only failures that are
00049     // generated in the same thread that created this object or it can intercept
00050     // all generated failures. The scope of this mock object can be controlled with
00051     // the second argument to the two arguments constructor.
00052
00053     class GTEST_API_ ScopedFakeTestPartResultReporter
00054     : public TestPartResultReporterInterface {
00055     public:
00056         // The two possible mocking modes of this object.
00057
00058         enum InterceptMode {
00059             INTERCEPT_ONLY_CURRENT_THREAD, // Intercepts only thread local failures.
00060             INTERCEPT_ALL_THREADS // Intercepts all failures.
00061         };
00062
00063         // The c'tor sets this object as the test part result reporter used
00064         // by Google Test.  The 'result' parameter specifies where to report the
00065         // results. This reporter will only catch failures generated in the current
00066         // thread. DEPRECATED
00067         explicit ScopedFakeTestPartResultReporter(TestPartResultArray* result);
00068
00069         // Same as above, but you can choose the interception scope of this object.
00070         ScopedFakeTestPartResultReporter(InterceptMode intercept_mode,
00071                 TestPartResultArray* result);
00072
00073         // The d'tor restores the previous test part result reporter.
00074         ~ScopedFakeTestPartResultReporter() override;
00075
00076         // Appends the TestPartResult object to the TestPartResultArray
00077         // received in the constructor.
00078         //
00079         // This method is from the TestPartResultReporterInterface
00080         // interface.
00081         void ReportTestPartResult(const TestPartResult& result) override;
00082
00083     private:
00084         void Init();
00085
00086         const InterceptMode intercept_mode_;
00087         TestPartResultReporterInterface* old_reporter_;
00088         TestPartResultArray * const result_;
00089
00090         ScopedFakeTestPartResultReporter(const ScopedFakeTestPartResultReporter&) =
00091         delete;
00092         ScopedFakeTestPartResultReporter& operator=(
00093                 const ScopedFakeTestPartResultReporter&) = delete;
00094     };
00095
00096     namespace internal {
00097
00098         // A helper class for implementing EXPECT_FATAL_FAILURE() and
00099         // EXPECT_NONFATAL_FAILURE().  Its destructor verifies that the given
00100         // TestPartResultArray contains exactly one failure that has the given
00101         // type and contains the given substring.  If that's not the case, a
00102         // non-fatal failure will be generated.
```

```
00103
00104         class GTEST_API_ SingleFailureChecker {
00105         public:
00106             // The constructor remembers the arguments.
00107             SingleFailureChecker(const TestPartResultArray* results,
00108                     TestPartResult::Type type, const std::string& substr);
00109             ~SingleFailureChecker();
00110
00111         private:
00112             const TestPartResultArray * const results_;
00113             const TestPartResult::Type type_;
00114             const std::string substr_;
00115
00116             SingleFailureChecker(const SingleFailureChecker&) = delete;
00117             SingleFailureChecker& operator=(const SingleFailureChecker&) = delete;
00118         };
00119
00120     } // namespace internal
00121
00122 } // namespace testing
00123
00124 GTEST_DISABLE_MSC_WARNINGS_POP_() //  4251
00125
00126 // A set of macros for testing Google Test assertions or code that's expected
00127 // to generate Google Test fatal failures (e.g. a failure from an ASSERT_EQ, but
00128 // not a non-fatal failure, as from EXPECT_EQ).  It verifies that the given
00129 // statement will cause exactly one fatal Google Test failure with 'substr'
00130 // being part of the failure message.
00131 //
00132 // There are two different versions of this macro. EXPECT_FATAL_FAILURE only
00133 // affects and considers failures generated in the current thread and
00134 // EXPECT_FATAL_FAILURE_ON_ALL_THREADS does the same but for all threads.
00135 //
00136 // The verification of the assertion is done correctly even when the statement
00137 // throws an exception or aborts the current function.
00138 //
00139 // Known restrictions:
00140 //   - 'statement' cannot reference local non-static variables or
00141 //     non-static members of the current object.
00142 //   - 'statement' cannot return a value.
00143 //   - You cannot stream a failure message to this macro.
00144 //
00145 // Note that even though the implementations of the following two
00146 // macros are much alike, we cannot refactor them to use a common
00147 // helper macro, due to some peculiarity in how the preprocessor
00148 // works.  The AcceptsMacroThatExpandsToUnprotectedComma test in
00149 // gtest_unittest.cc will fail to compile if we do that.
00150 #define EXPECT_FATAL_FAILURE(statement, substr)                                 \
00151   do {                                                                          \
00152     class GTestExpectFatalFailureHelper {                                       \
00153      public:                                                                    \
00154       static void Execute() { statement; }                                      \
00155     };                                                                          \
00156     ::testing::TestPartResultArray gtest_failures;                              \
00157     ::testing::internal::SingleFailureChecker gtest_checker(                    \
00158         &gtest_failures, ::testing::TestPartResult::kFatalFailure, (substr)); \
00159     {                                                                           \
00160       ::testing::ScopedFakeTestPartResultReporter gtest_reporter(               \
00161           ::testing::ScopedFakeTestPartResultReporter::                         \
00162               INTERCEPT_ONLY_CURRENT_THREAD,                                    \
00163           &gtest_failures);                                                     \
00164       GTestExpectFatalFailureHelper::Execute();                                 \
00165     }                                                                           \
00166   } while (::testing::internal::AlwaysFalse())
00167
00168 #define EXPECT_FATAL_FAILURE_ON_ALL_THREADS(statement, substr)                  \
00169   do {                                                                          \
00170     class GTestExpectFatalFailureHelper {                                       \
00171      public:                                                                    \
00172       static void Execute() { statement; }                                      \
00173     };                                                                          \
00174     ::testing::TestPartResultArray gtest_failures;                              \
00175     ::testing::internal::SingleFailureChecker gtest_checker(                    \
00176         &gtest_failures, ::testing::TestPartResult::kFatalFailure, (substr)); \
00177     {                                                                           \
00178       ::testing::ScopedFakeTestPartResultReporter gtest_reporter(               \
00179           ::testing::ScopedFakeTestPartResultReporter::INTERCEPT_ALL_THREADS, \
00180           &gtest_failures);                                                     \
00181       GTestExpectFatalFailureHelper::Execute();                                 \
00182     }                                                                           \
00183   } while (::testing::internal::AlwaysFalse())
00184
00185 // A macro for testing Google Test assertions or code that's expected to
00186 // generate Google Test non-fatal failures (e.g. a failure from an EXPECT_EQ,
00187 // but not from an ASSERT_EQ). It asserts that the given statement will cause
00188 // exactly one non-fatal Google Test failure with 'substr' being part of the
00189 // failure message.
```

```
00190 //
00191 // There are two different versions of this macro. EXPECT_NONFATAL_FAILURE only
00192 // affects and considers failures generated in the current thread and
00193 // EXPECT_NONFATAL_FAILURE_ON_ALL_THREADS does the same but for all threads.
00194 //
00195 // 'statement' is allowed to reference local variables and members of
00196 // the current object.
00197 //
00198 // The verification of the assertion is done correctly even when the statement
00199 // throws an exception or aborts the current function.
00200 //
00201 // Known restrictions:
00202 //   - You cannot stream a failure message to this macro.
00203 //
00204 // Note that even though the implementations of the following two
00205 // macros are much alike, we cannot refactor them to use a common
00206 // helper macro, due to some peculiarity in how the preprocessor
00207 // works.  If we do that, the code won't compile when the user gives
00208 // EXPECT_NONFATAL_FAILURE() a statement that contains a macro that
00209 // expands to code containing an unprotected comma.  The
00210 // AcceptsMacroThatExpandsToUnprotectedComma test in gtest_unittest.cc
00211 // catches that.
00212 //
00213 // For the same reason, we have to write
00214 //   if (::testing::internal::AlwaysTrue()) { statement; }
00215 // instead of
00216 //   GTEST_SUPPRESS_UNREACHABLE_CODE_WARNING_BELOW_(statement)
00217 // to avoid an MSVC warning on unreachable code.
00218 #define EXPECT_NONFATAL_FAILURE(statement, substr)                     \
00219   do {                                                                 \
00220     ::testing::TestPartResultArray gtest_failures;                     \
00221     ::testing::internal::SingleFailureChecker gtest_checker(           \
00222         &gtest_failures, ::testing::TestPartResult::kNonFatalFailure, \
00223         (substr));                                                     \
00224     {                                                                  \
00225       ::testing::ScopedFakeTestPartResultReporter gtest_reporter(      \
00226           ::testing::ScopedFakeTestPartResultReporter::                \
00227               INTERCEPT_ONLY_CURRENT_THREAD,                           \
00228           &gtest_failures);                                            \
00229       if (::testing::internal::AlwaysTrue()) {                         \
00230         statement;                                                     \
00231       }                                                                \
00232     }                                                                  \
00233   } while (::testing::internal::AlwaysFalse())
00234
00235 #define EXPECT_NONFATAL_FAILURE_ON_ALL_THREADS(statement, substr)             \
00236   do {                                                                        \
00237     ::testing::TestPartResultArray gtest_failures;                            \
00238     ::testing::internal::SingleFailureChecker gtest_checker(                  \
00239         &gtest_failures, ::testing::TestPartResult::kNonFatalFailure,         \
00240         (substr));                                                            \
00241     {                                                                         \
00242       ::testing::ScopedFakeTestPartResultReporter gtest_reporter(             \
00243           ::testing::ScopedFakeTestPartResultReporter::INTERCEPT_ALL_THREADS, \
00244           &gtest_failures);                                                   \
00245       if (::testing::internal::AlwaysTrue()) {                                \
00246         statement;                                                            \
00247       }                                                                       \
00248     }                                                                         \
00249   } while (::testing::internal::AlwaysFalse())
00250
00251 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_SPI_H_
```

18.267 gtest-test-part.h

```
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00003 //
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00005 // modification, are permitted provided that the following conditions are
00006 // met:
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00009 // notice, this list of conditions and the following disclaimer.
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00011 // copyright notice, this list of conditions and the following disclaimer
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00020 // LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR
```

```
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00026 // THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // IWYU pragma: private, include "gtest/gtest.h"
00031 // IWYU pragma: friend gtest/.*
00032 // IWYU pragma: friend gmock/.*
00033
00034 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_TEST_PART_H_
00035 #define GOOGLETEST_INCLUDE_GTEST_GTEST_TEST_PART_H_
00036
00037 #include <iosfwd>
00038 #include <vector>
00039
00040 #include "gtest/internal/gtest-internal.h"
00041 #include "gtest/internal/gtest-string.h"
00042
00043 GTEST_DISABLE_MSC_WARNINGS_PUSH_(4251 \
00044 /* class A needs to have dll-interface to be used by clients of class B */)
00045
00046 namespace testing {
00047
00048     // A copyable object representing the result of a test part (i.e. an
00049     // assertion or an explicit FAIL(), ADD_FAILURE(), or SUCCESS()).
00050     //
00051     // Don't inherit from TestPartResult as its destructor is not virtual.
00052
00053     class GTEST_API_ TestPartResult {
00054     public:
00055         // The possible outcomes of a test part (i.e. an assertion or an
00056         // explicit SUCCEED(), FAIL(), or ADD_FAILURE()).
00057
00058         enum Type {
00059             kSuccess, // Succeeded.
00060             kNonFatalFailure, // Failed but the test can continue.
00061             kFatalFailure, // Failed and the test should be terminated.
00062             kSkip // Skipped.
00063         };
00064
00065         // C'tor.  TestPartResult does NOT have a default constructor.
00066         // Always use this constructor (with parameters) to create a
00067         // TestPartResult object.
00068
00069         TestPartResult(Type a_type, const char* a_file_name, int a_line_number,
00070                 const char* a_message)
00071         : type_(a_type),
00072         file_name_(a_file_name == nullptr ? "" : a_file_name),
00073         line_number_(a_line_number),
00074         summary_(ExtractSummary(a_message)),
00075         message_(a_message) {
00076         }
00077
00078         // Gets the outcome of the test part.
00079
00080         Type type() const {
00081             return type_;
00082         }
00083
00084         // Gets the name of the source file where the test part took place, or
00085         // NULL if it's unknown.
00086
00087         const char* file_name() const {
00088             return file_name_.empty() ? nullptr : file_name_.c_str();
00089         }
00090
00091         // Gets the line in the source file where the test part took place,
00092         // or -1 if it's unknown.
00093
00094         int line_number() const {
00095             return line_number_;
00096         }
00097
00098         // Gets the summary of the failure message.
00099
00100         const char* summary() const {
00101             return summary_.c_str();
00102         }
00103
00104         // Gets the message associated with the test part.
00105
00106         const char* message() const {
00107             return message_.c_str();
```

```
00108         }
00109
00110         // Returns true if and only if the test part was skipped.
00111
00112         bool skipped() const {
00113             return type_ == kSkip;
00114         }
00115
00116         // Returns true if and only if the test part passed.
00117
00118         bool passed() const {
00119             return type_ == kSuccess;
00120         }
00121
00122         // Returns true if and only if the test part non-fatally failed.
00123
00124         bool nonfatally_failed() const {
00125             return type_ == kNonFatalFailure;
00126         }
00127
00128         // Returns true if and only if the test part fatally failed.
00129
00130         bool fatally_failed() const {
00131             return type_ == kFatalFailure;
00132         }
00133
00134         // Returns true if and only if the test part failed.
00135
00136         bool failed() const {
00137             return fatally_failed() || nonfatally_failed();
00138         }
00139
00140     private:
00141         Type type_;
00142
00143         // Gets the summary of the failure message by omitting the stack
00144         // trace in it.
00145         static std::string ExtractSummary(const char* message);
00146
00147         // The name of the source file where the test part took place, or
00148         // "" if the source file is unknown.
00149         std::string file_name_;
00150         // The line in the source file where the test part took place, or -1
00151         // if the line number is unknown.
00152         int line_number_;
00153         std::string summary_; // The test failure summary.
00154         std::string message_; // The test failure message.
00155     };
00156
00157     // Prints a TestPartResult object.
00158     std::ostream& operator<<(std::ostream& os, const TestPartResult& result);
00159
00160     // An array of TestPartResult objects.
00161     //
00162     // Don't inherit from TestPartResultArray as its destructor is not
00163     // virtual.
00164
00165     class GTEST_API_ TestPartResultArray {
00166     public:
00167
00168         TestPartResultArray() {
00169         }
00170
00171         // Appends the given TestPartResult to the array.
00172         void Append(const TestPartResult& result);
00173
00174         // Returns the TestPartResult at the given index (0-based).
00175         const TestPartResult& GetTestPartResult(int index) const;
00176
00177         // Returns the number of TestPartResult objects in the array.
00178         int size() const;
00179
00180     private:
00181         std::vector<TestPartResult> array_;
00182
00183         TestPartResultArray(const TestPartResultArray&) = delete;
00184         TestPartResultArray& operator=(const TestPartResultArray&) = delete;
00185     };
00186
00187     // This interface knows how to report a test part result.
00188
00189     class GTEST_API_ TestPartResultReporterInterface {
00190     public:
00191
00192         virtual ~TestPartResultReporterInterface() {
00193         }
00194
```

```
00195         virtual void ReportTestPartResult(const TestPartResult& result) = 0;
00196     };
00197
00198     namespace internal {
00199
00200         // This helper class is used by {ASSERT|EXPECT}_NO_FATAL_FAILURE to check if a
00201         // statement generates new fatal failures. To do so it registers itself as the
00202         // current test part result reporter. Besides checking if fatal failures were
00203         // reported, it only delegates the reporting to the former result reporter.
00204         // The original result reporter is restored in the destructor.
00205         // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
00206
00207         class GTEST_API_ HasNewFatalFailureHelper
00208         : public TestPartResultReporterInterface {
00209         public:
00210             HasNewFatalFailureHelper();
00211             ~HasNewFatalFailureHelper() override;
00212             void ReportTestPartResult(const TestPartResult& result) override;
00213
00214             bool has_new_fatal_failure() const {
00215                 return has_new_fatal_failure_;
00216             }
00217
00218         private:
00219             bool has_new_fatal_failure_;
00220             TestPartResultReporterInterface* original_reporter_;
00221
00222             HasNewFatalFailureHelper(const HasNewFatalFailureHelper&) = delete;
00223             HasNewFatalFailureHelper& operator=(const HasNewFatalFailureHelper&) = delete;
00224         };
00225
00226     } // namespace internal
00227
00228 } // namespace testing
00229
00230 GTEST_DISABLE_MSC_WARNINGS_POP_() //  4251
00231
00232 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_TEST_PART_H_
```

18.268 gtest-typed-test.h

```
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00026 // THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // IWYU pragma: private, include "gtest/gtest.h"
00031 // IWYU pragma: friend gtest/.*
00032 // IWYU pragma: friend gmock/.*
00033
00034 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_TYPED_TEST_H_
00035 #define GOOGLETEST_INCLUDE_GTEST_GTEST_TYPED_TEST_H_
00036
00037 // This header implements typed tests and type-parameterized tests.
00038
00039 // Typed (aka type-driven) tests repeat the same test for types in a
00040 // list.  You must know which types you want to test with when writing
00041 // typed tests. Here's how you do it:
00042
00043 #if 0
00044
```

```
00045 // First, define a fixture class template.  It should be parameterized
00046 // by a type.  Remember to derive it from testing::Test.
00047
00048 template <typename T>
00049 class FooTest : public testing::Test {
00050 public:
00051     ...
00052     typedef std::list<T> List;
00053     static T shared_;
00054     T value_;
00055 };
00056
00057 // Next, associate a list of types with the test suite, which will be
00058 // repeated for each type in the list.  The typedef is necessary for
00059 // the macro to parse correctly.
00060 typedef testing::Types<char, int, unsigned int> MyTypes;
00061 TYPED_TEST_SUITE(FooTest, MyTypes);
00062
00063 // If the type list contains only one type, you can write that type
00064 // directly without Types<...>:
00065 //   TYPED_TEST_SUITE(FooTest, int);
00066
00067 // Then, use TYPED_TEST() instead of TEST_F() to define as many typed
00068 // tests for this test suite as you want.
00069
00070 TYPED_TEST(FooTest, DoesBlah) {
00071     // Inside a test, refer to the special name TypeParam to get the type
00072     // parameter.  Since we are inside a derived class template, C++ requires
00073     // us to visit the members of FooTest via 'this'.
00074     TypeParam n = this->value_;
00075
00076     // To visit static members of the fixture, add the TestFixture::
00077     // prefix.
00078     n += TestFixture::shared_;
00079
00080     // To refer to typedefs in the fixture, add the "typename
00081     // TestFixture::" prefix.
00082     typename TestFixture::List values;
00083     values.push_back(n);
00084     ...
00085 }
00086
00087 TYPED_TEST(FooTest, HasPropertyA) {
00088     ...
00089 }
00090
00091 // TYPED_TEST_SUITE takes an optional third argument which allows to specify a
00092 // class that generates custom test name suffixes based on the type. This should
00093 // be a class which has a static template function GetName(int index) returning
00094 // a string for each type. The provided integer index equals the index of the
00095 // type in the provided type list. In many cases the index can be ignored.
00096 //
00097 // For example:
00098 //   class MyTypeNames {
00099 //    public:
00100 //     template <typename T>
00101 //     static std::string GetName(int) {
00102 //       if (std::is_same<T, char>()) return "char";
00103 //       if (std::is_same<T, int>()) return "int";
00104 //       if (std::is_same<T, unsigned int>()) return "unsignedInt";
00105 //     }
00106 //   };
00107 //   TYPED_TEST_SUITE(FooTest, MyTypes, MyTypeNames);
00108
00109 #endif  // 0
00110
00111 // Type-parameterized tests are abstract test patterns parameterized
00112 // by a type.  Compared with typed tests, type-parameterized tests
00113 // allow you to define the test pattern without knowing what the type
00114 // parameters are.  The defined pattern can be instantiated with
00115 // different types any number of times, in any number of translation
00116 // units.
00117 //
00118 // If you are designing an interface or concept, you can define a
00119 // suite of type-parameterized tests to verify properties that any
00120 // valid implementation of the interface/concept should have.  Then,
00121 // each implementation can easily instantiate the test suite to verify
00122 // that it conforms to the requirements, without having to write
00123 // similar tests repeatedly.  Here's an example:
00124
00125 #if 0
00126
00127 // First, define a fixture class template.  It should be parameterized
00128 // by a type.  Remember to derive it from testing::Test.
00129
00130 template <typename T>
00131 class FooTest : public testing::Test {
```

```
00132     ...
00133 };
00134
00135 // Next, declare that you will define a type-parameterized test suite
00136 // (the _P suffix is for "parameterized" or "pattern", whichever you
00137 // prefer):
00138 TYPED_TEST_SUITE_P(FooTest);
00139
00140 // Then, use TYPED_TEST_P() to define as many type-parameterized tests
00141 // for this type-parameterized test suite as you want.
00142
00143 TYPED_TEST_P(FooTest, DoesBlah) {
00144     // Inside a test, refer to TypeParam to get the type parameter.
00145     TypeParam n = 0;
00146     ...
00147 }
00148
00149 TYPED_TEST_P(FooTest, HasPropertyA) {
00150     ...
00151 }
00152
00153 // Now the tricky part: you need to register all test patterns before
00154 // you can instantiate them.  The first argument of the macro is the
00155 // test suite name; the rest are the names of the tests in this test
00156 // case.
00157 REGISTER_TYPED_TEST_SUITE_P(FooTest,
00158         DoesBlah, HasPropertyA);
00159
00160 // Finally, you are free to instantiate the pattern with the types you
00161 // want.  If you put the above code in a header file, you can #include
00162 // it in multiple C++ source files and instantiate it multiple times.
00163 //
00164 // To distinguish different instances of the pattern, the first
00165 // argument to the INSTANTIATE_* macro is a prefix that will be added
00166 // to the actual test suite name.  Remember to pick unique prefixes for
00167 // different instances.
00168 typedef testing::Types<char, int, unsigned int> MyTypes;
00169 INSTANTIATE_TYPED_TEST_SUITE_P(My, FooTest, MyTypes);
00170
00171 // If the type list contains only one type, you can write that type
00172 // directly without Types<...>:
00173 //   INSTANTIATE_TYPED_TEST_SUITE_P(My, FooTest, int);
00174 //
00175 // Similar to the optional argument of TYPED_TEST_SUITE above,
00176 // INSTANTIATE_TEST_SUITE_P takes an optional fourth argument which allows to
00177 // generate custom names.
00178 //   INSTANTIATE_TYPED_TEST_SUITE_P(My, FooTest, MyTypes, MyTypeNames);
00179
00180 #endif  // 0
00181
00182 #include "gtest/internal/gtest-internal.h"
00183 #include "gtest/internal/gtest-port.h"
00184 #include "gtest/internal/gtest-type-util.h"
00185
00186 // Implements typed tests.
00187
00188 // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00189 //
00190 // Expands to the name of the typedef for the type parameters of the
00191 // given test suite.
00192 #define GTEST_TYPE_PARAMS_(TestSuiteName) gtest_type_params_##TestSuiteName##_
00193
00194 // Expands to the name of the typedef for the NameGenerator, responsible for
00195 // creating the suffixes of the name.
00196 #define GTEST_NAME_GENERATOR_(TestSuiteName) \
00197   gtest_type_params_##TestSuiteName##_NameGenerator
00198
00199 #define TYPED_TEST_SUITE(CaseName, Types, ...)                            \
00200   typedef ::testing::internal::GenerateTypeList<Types>::type            \
00201       GTEST_TYPE_PARAMS_(CaseName);                                       \
00202   typedef ::testing::internal::NameGeneratorSelector<__VA_ARGS__>::type \
00203   GTEST_NAME_GENERATOR_(CaseName)
00204
00205 #define TYPED_TEST(CaseName, TestName)                                    \
00206   static_assert(sizeof(GTEST_STRINGIFY_(TestName)) > 1,                   \
00207                 "test-name must not be empty");                           \
00208   template <typename gtest_TypeParam_>                                    \
00209   class GTEST_TEST_CLASS_NAME_(CaseName, TestName)                        \
00210       : public CaseName<gtest_TypeParam_> {                               \
00211    private:                                                               \
00212     typedef CaseName<gtest_TypeParam_> TestFixture;                       \
00213     typedef gtest_TypeParam_ TypeParam;                                   \
00214     void TestBody() override;                                             \
00215   };                                                                      \
00216   static bool gtest_##CaseName##_##TestName##_registered_                 \
00217       GTEST_ATTRIBUTE_UNUSED_ = ::testing::internal::TypeParameterizedTest<   \
00218           CaseName,                                                       \
```

```
00219           ::testing::internal::TemplateSel<GTEST_TEST_CLASS_NAME_(CaseName,   \
00220                                                                   TestName)>, \
00221           GTEST_TYPE_PARAMS_(                                                 \
00222               CaseName)>::Register("",                                        \
00223                                    ::testing::internal::CodeLocation(         \
00224                                        __FILE__, __LINE__),                   \
00225                                    GTEST_STRINGIFY_(CaseName),                \
00226                                    GTEST_STRINGIFY_(TestName), 0,             \
00227                                    ::testing::internal::GenerateNames<        \
00228                                        GTEST_NAME_GENERATOR_(CaseName),       \
00229                                        GTEST_TYPE_PARAMS_(CaseName)>());      \
00230   template <typename gtest_TypeParam_>                                         \
00231   void GTEST_TEST_CLASS_NAME_(CaseName,                                        \
00232                               TestName)<gtest_TypeParam_>::TestBody()
00233
00234 // Legacy API is deprecated but still available
00235 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00236 #define TYPED_TEST_CASE                                                 \
00237   static_assert(::testing::internal::TypedTestCaseIsDeprecated(), ""); \
00238   TYPED_TEST_SUITE
00239 #endif  // GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00240
00241 // Implements type-parameterized tests.
00242
00243 // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00244 //
00245 // Expands to the namespace name that the type-parameterized tests for
00246 // the given type-parameterized test suite are defined in.  The exact
00247 // name of the namespace is subject to change without notice.
00248 #define GTEST_SUITE_NAMESPACE_(TestSuiteName) gtest_suite_##TestSuiteName##_
00249
00250 // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00251 //
00252 // Expands to the name of the variable used to remember the names of
00253 // the defined tests in the given test suite.
00254 #define GTEST_TYPED_TEST_SUITE_P_STATE_(TestSuiteName) \
00255   gtest_typed_test_suite_p_state_##TestSuiteName##_
00256
00257 // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE DIRECTLY.
00258 //
00259 // Expands to the name of the variable used to remember the names of
00260 // the registered tests in the given test suite.
00261 #define GTEST_REGISTERED_TEST_NAMES_(TestSuiteName) \
00262   gtest_registered_test_names_##TestSuiteName##_
00263
00264 // The variables defined in the type-parameterized test macros are
00265 // static as typically these macros are used in a .h file that can be
00266 // #included in multiple translation units linked together.
00267 #define TYPED_TEST_SUITE_P(SuiteName)              \
00268   static ::testing::internal::TypedTestSuitePState \
00269   GTEST_TYPED_TEST_SUITE_P_STATE_(SuiteName)
00270
00271 // Legacy API is deprecated but still available
00272 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00273 #define TYPED_TEST_CASE_P                                                 \
00274   static_assert(::testing::internal::TypedTestCase_P_IsDeprecated(), ""); \
00275   TYPED_TEST_SUITE_P
00276 #endif  // GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00277
00278 #define TYPED_TEST_P(SuiteName, TestName)                             \
00279   namespace GTEST_SUITE_NAMESPACE_(SuiteName) {                       \
00280     template <typename gtest_TypeParam_>                              \
00281     class TestName : public SuiteName<gtest_TypeParam_> {             \
00282      private:                                                         \
00283       typedef SuiteName<gtest_TypeParam_> TestFixture;                \
00284       typedef gtest_TypeParam_ TypeParam;                             \
00285       void TestBody() override;                                       \
00286     };                                                                \
00287     static bool gtest_##TestName##_defined_ GTEST_ATTRIBUTE_UNUSED_ = \
00288         GTEST_TYPED_TEST_SUITE_P_STATE_(SuiteName).AddTestName(       \
00289             __FILE__, __LINE__, GTEST_STRINGIFY_(SuiteName),          \
00290             GTEST_STRINGIFY_(TestName));                              \
00291   }                                                                   \
00292   template <typename gtest_TypeParam_>                                \
00293   void GTEST_SUITE_NAMESPACE_(                                        \
00294       SuiteName)::TestName<gtest_TypeParam_>::TestBody()
00295
00296 // Note: this won't work correctly if the trailing arguments are macros.
00297 #define REGISTER_TYPED_TEST_SUITE_P(SuiteName, ...)                          \
00298   namespace GTEST_SUITE_NAMESPACE_(SuiteName) {                              \
00299     typedef ::testing::internal::Templates<__VA_ARGS__> gtest_AllTests_;     \
00300   }                                                                          \
00301   static const char* const GTEST_REGISTERED_TEST_NAMES_(                     \
00302       SuiteName) GTEST_ATTRIBUTE_UNUSED_ =                                   \
00303       GTEST_TYPED_TEST_SUITE_P_STATE_(SuiteName).VerifyRegisteredTestNames( \
00304           GTEST_STRINGIFY_(SuiteName), __FILE__, __LINE__, #__VA_ARGS__)
00305
```

```
00306 // Legacy API is deprecated but still available
00307 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00308 #define REGISTER_TYPED_TEST_CASE_P                                          \
00309   static_assert(::testing::internal::RegisterTypedTestCase_P_IsDeprecated(), \
00310                 "");                                                        \
00311   REGISTER_TYPED_TEST_SUITE_P
00312 #endif  // GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00313
00314 #define INSTANTIATE_TYPED_TEST_SUITE_P(Prefix, SuiteName, Types, ...)     \
00315   static_assert(sizeof(GTEST_STRINGIFY_(Prefix)) > 1,                     \
00316                 "test-suit-prefix must not be empty");                    \
00317   static bool gtest_##Prefix##_##SuiteName GTEST_ATTRIBUTE_UNUSED_ =      \
00318       ::testing::internal::TypeParameterizedTestSuite<                    \
00319           SuiteName, GTEST_SUITE_NAMESPACE_(SuiteName)::gtest_AllTests_,  \
00320           ::testing::internal::GenerateTypeList<Types>::type>::           \
00321           Register(GTEST_STRINGIFY_(Prefix),                              \
00322                    ::testing::internal::CodeLocation(__FILE__, __LINE__), \
00323                    &GTEST_TYPED_TEST_SUITE_P_STATE_(SuiteName),           \
00324                    GTEST_STRINGIFY_(SuiteName),                           \
00325                    GTEST_REGISTERED_TEST_NAMES_(SuiteName),               \
00326                    ::testing::internal::GenerateNames<                    \
00327                        ::testing::internal::NameGeneratorSelector<        \
00328                            __VA_ARGS__>::type,                            \
00329                        ::testing::internal::GenerateTypeList<Types>::type>())
00330
00331 // Legacy API is deprecated but still available
00332 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00333 #define INSTANTIATE_TYPED_TEST_CASE_P                                       \
00334   static_assert(                                                            \
00335       ::testing::internal::InstantiateTypedTestCase_P_IsDeprecated(), ""); \
00336   INSTANTIATE_TYPED_TEST_SUITE_P
00337 #endif  // GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00338
00339 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_TYPED_TEST_H_
```

18.269 gtest.h

```
00001 // Copyright 2005, Google Inc.
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00003 //
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00005 // modification, are permitted provided that the following conditions are
00006 // met:
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00009 // notice, this list of conditions and the following disclaimer.
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00024 // LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE,
00025 // DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY
00026 // THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // The Google C++ Testing and Mocking Framework (Google Test)
00031 //
00032 // This header file defines the public API for Google Test.  It should be
00033 // included by any test program that uses Google Test.
00034 //
00035 // IMPORTANT NOTE: Due to limitation of the C++ language, we have to
00036 // leave some internal implementation details in this header file.
00037 // They are clearly marked by comments like this:
00038 //
00039 //   // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
00040 //
00041 // Such code is NOT meant to be used by a user directly, and is subject
00042 // to CHANGE WITHOUT NOTICE.  Therefore DO NOT DEPEND ON IT in a user
00043 // program!
00044 //
00045 // Acknowledgment: Google Test borrowed the idea of automatic test
00046 // registration from Barthelemy Dagenais' (barthelemy@prologique.com)
00047 // easyUnit framework.
00048
```

```
00049 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_H_
00050 #define GOOGLETEST_INCLUDE_GTEST_GTEST_H_
00051
00052 #include <cstddef>
00053 #include <limits>
00054 #include <memory>
00055 #include <ostream>
00056 #include <type_traits>
00057 #include <vector>
00058
00059 #include "gtest/gtest-assertion-result.h"
00060 #include "gtest/gtest-death-test.h"
00061 #include "gtest/gtest-matchers.h"
00062 #include "gtest/gtest-message.h"
00063 #include "gtest/gtest-param-test.h"
00064 #include "gtest/gtest-printers.h"
00065 #include "gtest/gtest-test-part.h"
00066 #include "gtest/gtest-typed-test.h"
00067 #include "gtest/gtest_pred_impl.h"
00068 #include "gtest/gtest_prod.h"
00069 #include "gtest/internal/gtest-internal.h"
00070 #include "gtest/internal/gtest-string.h"
00071
00072 GTEST_DISABLE_MSC_WARNINGS_PUSH_(4251 \
00073 /* class A needs to have dll-interface to be used by clients of class B */)
00074
00075 // Declares the flags.
00076
00077 // This flag temporary enables the disabled tests.
00078 GTEST_DECLARE_bool_(also_run_disabled_tests);
00079
00080 // This flag brings the debugger on an assertion failure.
00081 GTEST_DECLARE_bool_(break_on_failure);
00082
00083 // This flag controls whether Google Test catches all test-thrown exceptions
00084 // and logs them as failures.
00085 GTEST_DECLARE_bool_(catch_exceptions);
00086
00087 // This flag enables using colors in terminal output. Available values are
00088 // "yes" to enable colors, "no" (disable colors), or "auto" (the default)
00089 // to let Google Test decide.
00090 GTEST_DECLARE_string_(color);
00091
00092 // This flag controls whether the test runner should continue execution past
00093 // first failure.
00094 GTEST_DECLARE_bool_(fail_fast);
00095
00096 // This flag sets up the filter to select by name using a glob pattern
00097 // the tests to run. If the filter is not given all tests are executed.
00098 GTEST_DECLARE_string_(filter);
00099
00100 // This flag controls whether Google Test installs a signal handler that dumps
00101 // debugging information when fatal signals are raised.
00102 GTEST_DECLARE_bool_(install_failure_signal_handler);
00103
00104 // This flag causes the Google Test to list tests. None of the tests listed
00105 // are actually run if the flag is provided.
00106 GTEST_DECLARE_bool_(list_tests);
00107
00108 // This flag controls whether Google Test emits a detailed XML report to a file
00109 // in addition to its normal textual output.
00110 GTEST_DECLARE_string_(output);
00111
00112 // This flags control whether Google Test prints only test failures.
00113 GTEST_DECLARE_bool_(brief);
00114
00115 // This flags control whether Google Test prints the elapsed time for each
00116 // test.
00117 GTEST_DECLARE_bool_(print_time);
00118
00119 // This flags control whether Google Test prints UTF8 characters as text.
00120 GTEST_DECLARE_bool_(print_utf8);
00121
00122 // This flag specifies the random number seed.
00123 GTEST_DECLARE_int32_(random_seed);
00124
00125 // This flag sets how many times the tests are repeated. The default value
00126 // is 1. If the value is -1 the tests are repeating forever.
00127 GTEST_DECLARE_int32_(repeat);
00128
00129 // This flag controls whether Google Test Environments are recreated for each
00130 // repeat of the tests. The default value is true. If set to false the global
00131 // test Environment objects are only set up once, for the first iteration, and
00132 // only torn down once, for the last.
00133 GTEST_DECLARE_bool_(recreate_environments_when_repeating);
00134
00135 // This flag controls whether Google Test includes Google Test internal
```

```
00136 // stack frames in failure stack traces.
00137 GTEST_DECLARE_bool_(show_internal_stack_frames);
00138
00139 // When this flag is specified, tests' order is randomized on every iteration.
00140 GTEST_DECLARE_bool_(shuffle);
00141
00142 // This flag specifies the maximum number of stack frames to be
00143 // printed in a failure message.
00144 GTEST_DECLARE_int32_(stack_trace_depth);
00145
00146 // When this flag is specified, a failed assertion will throw an
00147 // exception if exceptions are enabled, or exit the program with a
00148 // non-zero code otherwise. For use with an external test framework.
00149 GTEST_DECLARE_bool_(throw_on_failure);
00150
00151 // When this flag is set with a "host:port" string, on supported
00152 // platforms test results are streamed to the specified port on
00153 // the specified host machine.
00154 GTEST_DECLARE_string_(stream_result_to);
00155
00156 #if GTEST_USE_OWN_FLAGFILE_FLAG_
00157 GTEST_DECLARE_string_(flagfile);
00158 #endif  // GTEST_USE_OWN_FLAGFILE_FLAG_
00159
00160 namespace testing {
00161
00162     // Silence C4100 (unreferenced formal parameter) and 4805
00163     // unsafe mix of type 'const int' and type 'const bool'
00164 #ifdef _MSC_VER
00165 #pragma warning(push)
00166 #pragma warning(disable : 4805)
00167 #pragma warning(disable : 4100)
00168 #endif
00169
00170     // The upper limit for valid stack trace depths.
00171     const int kMaxStackTraceDepth = 100;
00172
00173     namespace internal {
00174
00175         class AssertHelper;
00176         class DefaultGlobalTestPartResultReporter;
00177         class ExecDeathTest;
00178         class NoExecDeathTest;
00179         class FinalSuccessChecker;
00180         class GTestFlagSaver;
00181         class StreamingListenerTest;
00182         class TestResultAccessor;
00183         class TestEventListenersAccessor;
00184         class TestEventRepeater;
00185         class UnitTestRecordPropertyTestHelper;
00186         class WindowsDeathTest;
00187         class FuchsiaDeathTest;
00188         class UnitTestImpl* GetUnitTestImpl();
00189         void ReportFailureInUnknownLocation(TestPartResult::Type result_type,
00190                 const std::string& message);
00191         std::set<std::string>* GetIgnoredParameterizedTestSuites();
00192
00193     } // namespace internal
00194
00195     // The friend relationship of some of these classes is cyclic.
00196     // If we don't forward declare them the compiler might confuse the classes
00197     // in friendship clauses with same named classes on the scope.
00198     class Test;
00199     class TestSuite;
00200
00201     // Old API is still available but deprecated
00202 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00203     using TestCase = TestSuite;
00204 #endif
00205     class TestInfo;
00206     class UnitTest;
00207
00208     // The abstract class that all tests inherit from.
00209     //
00210     // In Google Test, a unit test program contains one or many TestSuites, and
00211     // each TestSuite contains one or many Tests.
00212     //
00213     // When you define a test using the TEST macro, you don't need to
00214     // explicitly derive from Test - the TEST macro automatically does
00215     // this for you.
00216     //
00217     // The only time you derive from Test is when defining a test fixture
00218     // to be used in a TEST_F.  For example:
00219     //
00220     //   class FooTest : public testing::Test {
00221     //    protected:
00222     //     void SetUp() override { ... }
```

```
00223     //     void TearDown() override { ... }
00224     //     ...
00225     //   };
00226     //
00227     //   TEST_F(FooTest, Bar) { ... }
00228     //   TEST_F(FooTest, Baz) { ... }
00229     //
00230     // Test is not copyable.
00231
00232     class GTEST_API_ Test {
00233     public:
00234         friend class TestInfo;
00235
00236         // The d'tor is virtual as we intend to inherit from Test.
00237         virtual ~Test();
00238
00239         // Sets up the stuff shared by all tests in this test suite.
00240         //
00241         // Google Test will call Foo::SetUpTestSuite() before running the first
00242         // test in test suite Foo.  Hence a sub-class can define its own
00243         // SetUpTestSuite() method to shadow the one defined in the super
00244         // class.
00245
00246         static void SetUpTestSuite() {
00247         }
00248
00249         // Tears down the stuff shared by all tests in this test suite.
00250         //
00251         // Google Test will call Foo::TearDownTestSuite() after running the last
00252         // test in test suite Foo.  Hence a sub-class can define its own
00253         // TearDownTestSuite() method to shadow the one defined in the super
00254         // class.
00255
00256         static void TearDownTestSuite() {
00257         }
00258
00259         // Legacy API is deprecated but still available. Use SetUpTestSuite and
00260         // TearDownTestSuite instead.
00261 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00262
00263         static void TearDownTestCase() {
00264         }
00265
00266         static void SetUpTestCase() {
00267         }
00268 #endif  // GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00269
00270         // Returns true if and only if the current test has a fatal failure.
00271         static bool HasFatalFailure();
00272
00273         // Returns true if and only if the current test has a non-fatal failure.
00274         static bool HasNonfatalFailure();
00275
00276         // Returns true if and only if the current test was skipped.
00277         static bool IsSkipped();
00278
00279         // Returns true if and only if the current test has a (either fatal or
00280         // non-fatal) failure.
00281
00282         static bool HasFailure() {
00283             return HasFatalFailure() || HasNonfatalFailure();
00284         }
00285
00286         // Logs a property for the current test, test suite, or for the entire
00287         // invocation of the test program when used outside of the context of a
00288         // test suite.  Only the last value for a given key is remembered.  These
00289         // are public static so they can be called from utility functions that are
00290         // not members of the test fixture.  Calls to RecordProperty made during
00291         // lifespan of the test (from the moment its constructor starts to the
00292         // moment its destructor finishes) will be output in XML as attributes of
00293         // the <testcase> element.  Properties recorded from fixture's
00294         // SetUpTestSuite or TearDownTestSuite are logged as attributes of the
00295         // corresponding <testsuite> element.  Calls to RecordProperty made in the
00296         // global context (before or after invocation of RUN_ALL_TESTS and from
00297         // SetUp/TearDown method of Environment objects registered with Google
00298         // Test) will be output as attributes of the <testsuites> element.
00299         static void RecordProperty(const std::string& key, const std::string& value);
00300         static void RecordProperty(const std::string& key, int value);
00301
00302     protected:
00303         // Creates a Test object.
00304         Test();
00305
00306         // Sets up the test fixture.
00307         virtual void SetUp();
00308
00309         // Tears down the test fixture.
```

```
00310         virtual void TearDown();
00311
00312     private:
00313         // Returns true if and only if the current test has the same fixture class
00314         // as the first test in the current test suite.
00315         static bool HasSameFixtureClass();
00316
00317         // Runs the test after the test fixture has been set up.
00318         //
00319         // A sub-class must implement this to define the test logic.
00320         //
00321         // DO NOT OVERRIDE THIS FUNCTION DIRECTLY IN A USER PROGRAM.
00322         // Instead, use the TEST or TEST_F macro.
00323         virtual void TestBody() = 0;
00324
00325         // Sets up, executes, and tears down the test.
00326         void Run();
00327
00328         // Deletes self.  We deliberately pick an unusual name for this
00329         // internal method to avoid clashing with names used in user TESTs.
00330
00331         void DeleteSelf_() {
00332             delete this;
00333         }
00334
00335         const std::unique_ptr<GTEST_FLAG_SAVER_> gtest_flag_saver_;
00336
00337         // Often a user misspells SetUp() as Setup() and spends a long time
00338         // wondering why it is never called by Google Test.  The declaration of
00339         // the following method is solely for catching such an error at
00340         // compile time:
00341         //
00342         //   - The return type is deliberately chosen to be not void, so it
00343         //   will be a conflict if void Setup() is declared in the user's
00344         //   test fixture.
00345         //
00346         //   - This method is private, so it will be another compiler error
00347         //   if the method is called from the user's test fixture.
00348         //
00349         // DO NOT OVERRIDE THIS FUNCTION.
00350         //
00351         // If you see an error about overriding the following function or
00352         // about it being private, you have mis-spelled SetUp() as Setup().
00353
00354         struct Setup_should_be_spelled_SetUp {
00355         };
00356
00357         virtual Setup_should_be_spelled_SetUp* Setup() {
00358             return nullptr;
00359         }
00360
00361         // We disallow copying Tests.
00362         Test(const Test&) = delete;
00363         Test& operator=(const Test&) = delete;
00364     };
00365
00366     typedef internal::TimeInMillis TimeInMillis;
00367
00368     // A copyable object representing a user specified test property which can be
00369     // output as a key/value string pair.
00370     //
00371     // Don't inherit from TestProperty as its destructor is not virtual.
00372
00373     class TestProperty {
00374     public:
00375         // C'tor.  TestProperty does NOT have a default constructor.
00376         // Always use this constructor (with parameters) to create a
00377         // TestProperty object.
00378
00379         TestProperty(const std::string& a_key, const std::string& a_value)
00380         : key_(a_key), value_(a_value) {
00381         }
00382
00383         // Gets the user supplied key.
00384
00385         const char* key() const {
00386             return key_.c_str();
00387         }
00388
00389         // Gets the user supplied value.
00390
00391         const char* value() const {
00392             return value_.c_str();
00393         }
00394
00395         // Sets a new value, overriding the one supplied in the constructor.
00396
```

```
00397         void SetValue(const std::string& new_value) {
00398             value_ = new_value;
00399         }
00400
00401     private:
00402         // The key supplied by the user.
00403         std::string key_;
00404         // The value supplied by the user.
00405         std::string value_;
00406     };
00407
00408     // The result of a single Test.  This includes a list of
00409     // TestPartResults, a list of TestProperties, a count of how many
00410     // death tests there are in the Test, and how much time it took to run
00411     // the Test.
00412     //
00413     // TestResult is not copyable.
00414
00415     class GTEST_API_ TestResult {
00416     public:
00417         // Creates an empty TestResult.
00418         TestResult();
00419
00420         // D'tor.  Do not inherit from TestResult.
00421         ~TestResult();
00422
00423         // Gets the number of all test parts.  This is the sum of the number
00424         // of successful test parts and the number of failed test parts.
00425         int total_part_count() const;
00426
00427         // Returns the number of the test properties.
00428         int test_property_count() const;
00429
00430         // Returns true if and only if the test passed (i.e. no test part failed).
00431
00432         bool Passed() const {
00433             return !Skipped() && !Failed();
00434         }
00435
00436         // Returns true if and only if the test was skipped.
00437         bool Skipped() const;
00438
00439         // Returns true if and only if the test failed.
00440         bool Failed() const;
00441
00442         // Returns true if and only if the test fatally failed.
00443         bool HasFatalFailure() const;
00444
00445         // Returns true if and only if the test has a non-fatal failure.
00446         bool HasNonfatalFailure() const;
00447
00448         // Returns the elapsed time, in milliseconds.
00449
00450         TimeInMillis elapsed_time() const {
00451             return elapsed_time_;
00452         }
00453
00454         // Gets the time of the test case start, in ms from the start of the
00455         // UNIX epoch.
00456
00457         TimeInMillis start_timestamp() const {
00458             return start_timestamp_;
00459         }
00460
00461         // Returns the i-th test part result among all the results. i can range from 0
00462         // to total_part_count() - 1. If i is not in that range, aborts the program.
00463         const TestPartResult& GetTestPartResult(int i) const;
00464
00465         // Returns the i-th test property. i can range from 0 to
00466         // test_property_count() - 1. If i is not in that range, aborts the
00467         // program.
00468         const TestProperty& GetTestProperty(int i) const;
00469
00470     private:
00471         friend class TestInfo;
00472         friend class TestSuite;
00473         friend class UnitTest;
00474         friend class internal::DefaultGlobalTestPartResultReporter;
00475         friend class internal::ExecDeathTest;
00476         friend class internal::TestResultAccessor;
00477         friend class internal::UnitTestImpl;
00478         friend class internal::WindowsDeathTest;
00479         friend class internal::FuchsiaDeathTest;
00480
00481         // Gets the vector of TestPartResults.
00482
00483         const std::vector<TestPartResult>& test_part_results() const {
```

```
00484             return test_part_results_;
00485         }
00486
00487         // Gets the vector of TestProperties.
00488
00489         const std::vector<TestProperty>& test_properties() const {
00490             return test_properties_;
00491         }
00492
00493         // Sets the start time.
00494
00495         void set_start_timestamp(TimeInMillis start) {
00496             start_timestamp_ = start;
00497         }
00498
00499         // Sets the elapsed time.
00500
00501         void set_elapsed_time(TimeInMillis elapsed) {
00502             elapsed_time_ = elapsed;
00503         }
00504
00505         // Adds a test property to the list. The property is validated and may add
00506         // a non-fatal failure if invalid (e.g., if it conflicts with reserved
00507         // key names). If a property is already recorded for the same key, the
00508         // value will be updated, rather than storing multiple values for the same
00509         // key.  xml_element specifies the element for which the property is being
00510         // recorded and is used for validation.
00511         void RecordProperty(const std::string& xml_element,
00512                 const TestProperty& test_property);
00513
00514         // Adds a failure if the key is a reserved attribute of Google Test
00515         // testsuite tags.  Returns true if the property is valid.
00516         // FIXME: Validate attribute names are legal and human readable.
00517         static bool ValidateTestProperty(const std::string& xml_element,
00518                 const TestProperty& test_property);
00519
00520         // Adds a test part result to the list.
00521         void AddTestPartResult(const TestPartResult& test_part_result);
00522
00523         // Returns the death test count.
00524
00525         int death_test_count() const {
00526             return death_test_count_;
00527         }
00528
00529         // Increments the death test count, returning the new count.
00530
00531         int increment_death_test_count() {
00532             return ++death_test_count_;
00533         }
00534
00535         // Clears the test part results.
00536         void ClearTestPartResults();
00537
00538         // Clears the object.
00539         void Clear();
00540
00541         // Protects mutable state of the property vector and of owned
00542         // properties, whose values may be updated.
00543         internal::Mutex test_properties_mutex_;
00544
00545         // The vector of TestPartResults
00546         std::vector<TestPartResult> test_part_results_;
00547         // The vector of TestProperties
00548         std::vector<TestProperty> test_properties_;
00549         // Running count of death tests.
00550         int death_test_count_;
00551         // The start time, in milliseconds since UNIX Epoch.
00552         TimeInMillis start_timestamp_;
00553         // The elapsed time, in milliseconds.
00554         TimeInMillis elapsed_time_;
00555
00556         // We disallow copying TestResult.
00557         TestResult(const TestResult&) = delete;
00558         TestResult& operator=(const TestResult&) = delete;
00559     }; // class TestResult
00560
00561     // A TestInfo object stores the following information about a test:
00562     //
00563     //   Test suite name
00564     //   Test name
00565     //   Whether the test should be run
00566     //   A function pointer that creates the test object when invoked
00567     //   Test result
00568     //
00569     // The constructor of TestInfo registers itself with the UnitTest
00570     // singleton such that the RUN_ALL_TESTS() macro knows which tests to
```

```
00571     // run.
00572
00573     class GTEST_API_ TestInfo {
00574     public:
00575         // Destructs a TestInfo object.  This function is not virtual, so
00576         // don't inherit from TestInfo.
00577         ~TestInfo();
00578
00579         // Returns the test suite name.
00580
00581         const char* test_suite_name() const {
00582             return test_suite_name_.c_str();
00583         }
00584
00585         // Legacy API is deprecated but still available
00586 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00587
00588         const char* test_case_name() const {
00589             return test_suite_name();
00590         }
00591 #endif  // GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00592
00593         // Returns the test name.
00594
00595         const char* name() const {
00596             return name_.c_str();
00597         }
00598
00599         // Returns the name of the parameter type, or NULL if this is not a typed
00600         // or a type-parameterized test.
00601
00602         const char* type_param() const {
00603             if (type_param_.get() != nullptr) return type_param_->c_str();
00604             return nullptr;
00605         }
00606
00607         // Returns the text representation of the value parameter, or NULL if this
00608         // is not a value-parameterized test.
00609
00610         const char* value_param() const {
00611             if (value_param_.get() != nullptr) return value_param_->c_str();
00612             return nullptr;
00613         }
00614
00615         // Returns the file name where this test is defined.
00616
00617         const char* file() const {
00618             return location_.file.c_str();
00619         }
00620
00621         // Returns the line where this test is defined.
00622
00623         int line() const {
00624             return location_.line;
00625         }
00626
00627         // Return true if this test should not be run because it's in another shard.
00628
00629         bool is_in_another_shard() const {
00630             return is_in_another_shard_;
00631         }
00632
00633         // Returns true if this test should run, that is if the test is not
00634         // disabled (or it is disabled but the also_run_disabled_tests flag has
00635         // been specified) and its full name matches the user-specified filter.
00636         //
00637         // Google Test allows the user to filter the tests by their full names.
00638         // The full name of a test Bar in test suite Foo is defined as
00639         // "Foo.Bar".  Only the tests that match the filter will run.
00640         //
00641         // A filter is a colon-separated list of glob (not regex) patterns,
00642         // optionally followed by a '-' and a colon-separated list of
00643         // negative patterns (tests to exclude).  A test is run if it
00644         // matches one of the positive patterns and does not match any of
00645         // the negative patterns.
00646         //
00647         // For example, *A*:Foo.* is a filter that matches any string that
00648         // contains the character 'A' or starts with "Foo.".
00649
00650         bool should_run() const {
00651             return should_run_;
00652         }
00653
00654         // Returns true if and only if this test will appear in the XML report.
00655
00656         bool is_reportable() const {
00657             // The XML report includes tests matching the filter, excluding those
```

```
00658             // run in other shards.
00659             return matches_filter_ && !is_in_another_shard_;
00660         }
00661
00662         // Returns the result of the test.
00663
00664         const TestResult* result() const {
00665             return &result_;
00666         }
00667
00668     private:
00669 #if GTEST_HAS_DEATH_TEST
00670         friend class internal::DefaultDeathTestFactory;
00671 #endif  // GTEST_HAS_DEATH_TEST
00672         friend class Test;
00673         friend class TestSuite;
00674         friend class internal::UnitTestImpl;
00675         friend class internal::StreamingListenerTest;
00676         friend TestInfo* internal::MakeAndRegisterTestInfo(
00677                 const char* test_suite_name, const char* name, const char* type_param,
00678                 const char* value_param, internal::CodeLocation code_location,
00679                 internal::TypeId fixture_class_id, internal::SetUpTestSuiteFunc set_up_tc,
00680                 internal::TearDownTestSuiteFunc tear_down_tc,
00681                 internal::TestFactoryBase* factory);
00682
00683         // Constructs a TestInfo object. The newly constructed instance assumes
00684         // ownership of the factory object.
00685         TestInfo(const std::string& test_suite_name, const std::string& name,
00686                 const char* a_type_param, // NULL if not a type-parameterized test
00687                 const char* a_value_param, // NULL if not a value-parameterized test
00688                 internal::CodeLocation a_code_location,
00689                 internal::TypeId fixture_class_id,
00690                 internal::TestFactoryBase* factory);
00691
00692         // Increments the number of death tests encountered in this test so
00693         // far.
00694
00695         int increment_death_test_count() {
00696             return result_.increment_death_test_count();
00697         }
00698
00699         // Creates the test object, runs it, records its result, and then
00700         // deletes it.
00701         void Run();
00702
00703         // Skip and records the test result for this object.
00704         void Skip();
00705
00706         static void ClearTestResult(TestInfo* test_info) {
00707             test_info->result_.Clear();
00708         }
00709
00710         // These fields are immutable properties of the test.
00711         const std::string test_suite_name_; // test suite name
00712         const std::string name_; // Test name
00713         // Name of the parameter type, or NULL if this is not a typed or a
00714         // type-parameterized test.
00715         const std::unique_ptr<const ::std::string> type_param_;
00716         // Text representation of the value parameter, or NULL if this is not a
00717         // value-parameterized test.
00718         const std::unique_ptr<const ::std::string> value_param_;
00719         internal::CodeLocation location_;
00720         const internal::TypeId fixture_class_id_; // ID of the test fixture class
00721         bool should_run_; // True if and only if this test should run
00722         bool is_disabled_; // True if and only if this test is disabled
00723         bool matches_filter_; // True if this test matches the
00724         // user-specified filter.
00725         bool is_in_another_shard_; // Will be run in another shard.
00726         internal::TestFactoryBase * const factory_; // The factory that creates
00727         // the test object
00728
00729         // This field is mutable and needs to be reset before running the
00730         // test for the second time.
00731         TestResult result_;
00732
00733         TestInfo(const TestInfo&) = delete;
00734         TestInfo& operator=(const TestInfo&) = delete;
00735     };
00736
00737     // A test suite, which consists of a vector of TestInfos.
00738     //
00739     // TestSuite is not copyable.
00740
00741     class GTEST_API_ TestSuite {
00742     public:
00743         // Creates a TestSuite with the given name.
00744         //
```

```
00745         // TestSuite does NOT have a default constructor.  Always use this
00746         // constructor to create a TestSuite object.
00747         //
00748         // Arguments:
00749         //
00750         //   name:         name of the test suite
00751         //   a_type_param: the name of the test's type parameter, or NULL if
00752         //                 this is not a type-parameterized test.
00753         //   set_up_tc:    pointer to the function that sets up the test suite
00754         //   tear_down_tc: pointer to the function that tears down the test suite
00755         TestSuite(const char* name, const char* a_type_param,
00756                 internal::SetUpTestSuiteFunc set_up_tc,
00757                 internal::TearDownTestSuiteFunc tear_down_tc);
00758
00759         // Destructor of TestSuite.
00760         virtual ~TestSuite();
00761
00762         // Gets the name of the TestSuite.
00763
00764         const char* name() const {
00765             return name_.c_str();
00766         }
00767
00768         // Returns the name of the parameter type, or NULL if this is not a
00769         // type-parameterized test suite.
00770
00771         const char* type_param() const {
00772             if (type_param_.get() != nullptr) return type_param_->c_str();
00773             return nullptr;
00774         }
00775
00776         // Returns true if any test in this test suite should run.
00777
00778         bool should_run() const {
00779             return should_run_;
00780         }
00781
00782         // Gets the number of successful tests in this test suite.
00783         int successful_test_count() const;
00784
00785         // Gets the number of skipped tests in this test suite.
00786         int skipped_test_count() const;
00787
00788         // Gets the number of failed tests in this test suite.
00789         int failed_test_count() const;
00790
00791         // Gets the number of disabled tests that will be reported in the XML report.
00792         int reportable_disabled_test_count() const;
00793
00794         // Gets the number of disabled tests in this test suite.
00795         int disabled_test_count() const;
00796
00797         // Gets the number of tests to be printed in the XML report.
00798         int reportable_test_count() const;
00799
00800         // Get the number of tests in this test suite that should run.
00801         int test_to_run_count() const;
00802
00803         // Gets the number of all tests in this test suite.
00804         int total_test_count() const;
00805
00806         // Returns true if and only if the test suite passed.
00807
00808         bool Passed() const {
00809             return !Failed();
00810         }
00811
00812         // Returns true if and only if the test suite failed.
00813
00814         bool Failed() const {
00815             return failed_test_count() > 0 || ad_hoc_test_result().Failed();
00816         }
00817
00818         // Returns the elapsed time, in milliseconds.
00819
00820         TimeInMillis elapsed_time() const {
00821             return elapsed_time_;
00822         }
00823
00824         // Gets the time of the test suite start, in ms from the start of the
00825         // UNIX epoch.
00826
00827         TimeInMillis start_timestamp() const {
00828             return start_timestamp_;
00829         }
00830
00831         // Returns the i-th test among all the tests. i can range from 0 to
```

```
00832         // total_test_count() - 1. If i is not in that range, returns NULL.
00833         const TestInfo* GetTestInfo(int i) const;
00834
00835         // Returns the TestResult that holds test properties recorded during
00836         // execution of SetUpTestSuite and TearDownTestSuite.
00837
00838         const TestResult& ad_hoc_test_result() const {
00839             return ad_hoc_test_result_;
00840         }
00841
00842     private:
00843         friend class Test;
00844         friend class internal::UnitTestImpl;
00845
00846         // Gets the (mutable) vector of TestInfos in this TestSuite.
00847
00848         std::vector<TestInfo*>& test_info_list() {
00849             return test_info_list_;
00850         }
00851
00852         // Gets the (immutable) vector of TestInfos in this TestSuite.
00853
00854         const std::vector<TestInfo*>& test_info_list() const {
00855             return test_info_list_;
00856         }
00857
00858         // Returns the i-th test among all the tests. i can range from 0 to
00859         // total_test_count() - 1. If i is not in that range, returns NULL.
00860         TestInfo* GetMutableTestInfo(int i);
00861
00862         // Sets the should_run member.
00863
00864         void set_should_run(bool should) {
00865             should_run_ = should;
00866         }
00867
00868         // Adds a TestInfo to this test suite.  Will delete the TestInfo upon
00869         // destruction of the TestSuite object.
00870         void AddTestInfo(TestInfo* test_info);
00871
00872         // Clears the results of all tests in this test suite.
00873         void ClearResult();
00874
00875         // Clears the results of all tests in the given test suite.
00876
00877         static void ClearTestSuiteResult(TestSuite* test_suite) {
00878             test_suite->ClearResult();
00879         }
00880
00881         // Runs every test in this TestSuite.
00882         void Run();
00883
00884         // Skips the execution of tests under this TestSuite
00885         void Skip();
00886
00887         // Runs SetUpTestSuite() for this TestSuite.  This wrapper is needed
00888         // for catching exceptions thrown from SetUpTestSuite().
00889
00890         void RunSetUpTestSuite() {
00891             if (set_up_tc_ != nullptr) {
00892                 (*set_up_tc_)();
00893             }
00894         }
00895
00896         // Runs TearDownTestSuite() for this TestSuite.  This wrapper is
00897         // needed for catching exceptions thrown from TearDownTestSuite().
00898
00899         void RunTearDownTestSuite() {
00900             if (tear_down_tc_ != nullptr) {
00901                 (*tear_down_tc_)();
00902             }
00903         }
00904
00905         // Returns true if and only if test passed.
00906
00907         static bool TestPassed(const TestInfo* test_info) {
00908             return test_info->should_run() && test_info->result()->Passed();
00909         }
00910
00911         // Returns true if and only if test skipped.
00912
00913         static bool TestSkipped(const TestInfo* test_info) {
00914             return test_info->should_run() && test_info->result()->Skipped();
00915         }
00916
00917         // Returns true if and only if test failed.
00918
```

```
00919         static bool TestFailed(const TestInfo* test_info) {
00920             return test_info->should_run() && test_info->result()->Failed();
00921         }
00922
00923         // Returns true if and only if the test is disabled and will be reported in
00924         // the XML report.
00925
00926         static bool TestReportableDisabled(const TestInfo* test_info) {
00927             return test_info->is_reportable() && test_info->is_disabled_;
00928         }
00929
00930         // Returns true if and only if test is disabled.
00931
00932         static bool TestDisabled(const TestInfo* test_info) {
00933             return test_info->is_disabled_;
00934         }
00935
00936         // Returns true if and only if this test will appear in the XML report.
00937
00938         static bool TestReportable(const TestInfo* test_info) {
00939             return test_info->is_reportable();
00940         }
00941
00942         // Returns true if the given test should run.
00943
00944         static bool ShouldRunTest(const TestInfo* test_info) {
00945             return test_info->should_run();
00946         }
00947
00948         // Shuffles the tests in this test suite.
00949         void ShuffleTests(internal::Random* random);
00950
00951         // Restores the test order to before the first shuffle.
00952         void UnshuffleTests();
00953
00954         // Name of the test suite.
00955         std::string name_;
00956         // Name of the parameter type, or NULL if this is not a typed or a
00957         // type-parameterized test.
00958         const std::unique_ptr<const ::std::string> type_param_;
00959         // The vector of TestInfos in their original order.  It owns the
00960         // elements in the vector.
00961         std::vector<TestInfo*> test_info_list_;
00962         // Provides a level of indirection for the test list to allow easy
00963         // shuffling and restoring the test order.  The i-th element in this
00964         // vector is the index of the i-th test in the shuffled test list.
00965         std::vector<int> test_indices_;
00966         // Pointer to the function that sets up the test suite.
00967         internal::SetUpTestSuiteFunc set_up_tc_;
00968         // Pointer to the function that tears down the test suite.
00969         internal::TearDownTestSuiteFunc tear_down_tc_;
00970         // True if and only if any test in this test suite should run.
00971         bool should_run_;
00972         // The start time, in milliseconds since UNIX Epoch.
00973         TimeInMillis start_timestamp_;
00974         // Elapsed time, in milliseconds.
00975         TimeInMillis elapsed_time_;
00976         // Holds test properties recorded during execution of SetUpTestSuite and
00977         // TearDownTestSuite.
00978         TestResult ad_hoc_test_result_;
00979
00980         // We disallow copying TestSuites.
00981         TestSuite(const TestSuite&) = delete;
00982         TestSuite& operator=(const TestSuite&) = delete;
00983     };
00984
00985     // An Environment object is capable of setting up and tearing down an
00986     // environment.  You should subclass this to define your own
00987     // environment(s).
00988     //
00989     // An Environment object does the set-up and tear-down in virtual
00990     // methods SetUp() and TearDown() instead of the constructor and the
00991     // destructor, as:
00992     //
00993     //   1. You cannot safely throw from a destructor.  This is a problem
00994     //      as in some cases Google Test is used where exceptions are enabled, and
00995     //      we may want to implement ASSERT_* using exceptions where they are
00996     //      available.
00997     //   2. You cannot use ASSERT_* directly in a constructor or
00998     //      destructor.
00999
01000     class Environment {
01001     public:
01002         // The d'tor is virtual as we need to subclass Environment.
01003
01004         virtual ~Environment() {
01005         }
```

```
01006
01007         // Override this to define how to set up the environment.
01008
01009         virtual void SetUp() {
01010         }
01011
01012         // Override this to define how to tear down the environment.
01013
01014         virtual void TearDown() {
01015         }
01016
01017     private:
01018         // If you see an error about overriding the following function or
01019         // about it being private, you have mis-spelled SetUp() as Setup().
01020
01021         struct Setup_should_be_spelled_SetUp {
01022         };
01023
01024         virtual Setup_should_be_spelled_SetUp* Setup() {
01025             return nullptr;
01026         }
01027     };
01028
01029 #if GTEST_HAS_EXCEPTIONS
01030
01031     // Exception which can be thrown from TestEventListener::OnTestPartResult.
01032
01033     class GTEST_API_ AssertionException
01034     : public internal::GoogleTestFailureException {
01035     public:
01036
01037         explicit AssertionException(const TestPartResult& result)
01038         : GoogleTestFailureException(result) {
01039         }
01040     };
01041
01042 #endif  // GTEST_HAS_EXCEPTIONS
01043
01044     // The interface for tracing execution of tests. The methods are organized in
01045     // the order the corresponding events are fired.
01046
01047     class TestEventListener {
01048     public:
01049
01050         virtual ~TestEventListener() {
01051         }
01052
01053         // Fired before any test activity starts.
01054         virtual void OnTestProgramStart(const UnitTest& unit_test) = 0;
01055
01056         // Fired before each iteration of tests starts.  There may be more than
01057         // one iteration if GTEST_FLAG(repeat) is set. iteration is the iteration
01058         // index, starting from 0.
01059         virtual void OnTestIterationStart(const UnitTest& unit_test,
01060                 int iteration) = 0;
01061
01062         // Fired before environment set-up for each iteration of tests starts.
01063         virtual void OnEnvironmentsSetUpStart(const UnitTest& unit_test) = 0;
01064
01065         // Fired after environment set-up for each iteration of tests ends.
01066         virtual void OnEnvironmentsSetUpEnd(const UnitTest& unit_test) = 0;
01067
01068         // Fired before the test suite starts.
01069
01070         virtual void OnTestSuiteStart(const TestSuite& /*test_suite*/) {
01071         }
01072
01073         //  Legacy API is deprecated but still available
01074 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01075
01076         virtual void OnTestCaseStart(const TestCase& /*test_case*/) {
01077         }
01078 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01079
01080         // Fired before the test starts.
01081         virtual void OnTestStart(const TestInfo& test_info) = 0;
01082
01083         // Fired when a test is disabled
01084
01085         virtual void OnTestDisabled(const TestInfo& /*test_info*/) {
01086         }
01087
01088         // Fired after a failed assertion or a SUCCEED() invocation.
01089         // If you want to throw an exception from this function to skip to the next
01090         // TEST, it must be AssertionException defined above, or inherited from it.
01091         virtual void OnTestPartResult(const TestPartResult& test_part_result) = 0;
01092
```

```
01093         // Fired after the test ends.
01094         virtual void OnTestEnd(const TestInfo& test_info) = 0;
01095
01096         // Fired after the test suite ends.
01097
01098         virtual void OnTestSuiteEnd(const TestSuite& /*test_suite*/) {
01099         }
01100
01101         //  Legacy API is deprecated but still available
01102 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01103
01104         virtual void OnTestCaseEnd(const TestCase& /*test_case*/) {
01105         }
01106 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01107
01108         // Fired before environment tear-down for each iteration of tests starts.
01109         virtual void OnEnvironmentsTearDownStart(const UnitTest& unit_test) = 0;
01110
01111         // Fired after environment tear-down for each iteration of tests ends.
01112         virtual void OnEnvironmentsTearDownEnd(const UnitTest& unit_test) = 0;
01113
01114         // Fired after each iteration of tests finishes.
01115         virtual void OnTestIterationEnd(const UnitTest& unit_test, int iteration) = 0;
01116
01117         // Fired after all test activities have ended.
01118         virtual void OnTestProgramEnd(const UnitTest& unit_test) = 0;
01119     };
01120
01121     // The convenience class for users who need to override just one or two
01122     // methods and are not concerned that a possible change to a signature of
01123     // the methods they override will not be caught during the build.  For
01124     // comments about each method please see the definition of TestEventListener
01125     // above.
01126
01127     class EmptyTestEventListener : public TestEventListener {
01128     public:
01129
01130         void OnTestProgramStart(const UnitTest& /*unit_test*/) override {
01131         }
01132
01133         void OnTestIterationStart(const UnitTest& /*unit_test*/,
01134                 int /*iteration*/) override {
01135         }
01136
01137         void OnEnvironmentsSetUpStart(const UnitTest& /*unit_test*/) override {
01138         }
01139
01140         void OnEnvironmentsSetUpEnd(const UnitTest& /*unit_test*/) override {
01141         }
01142
01143         void OnTestSuiteStart(const TestSuite& /*test_suite*/) override {
01144         }
01145         //  Legacy API is deprecated but still available
01146 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01147
01148         void OnTestCaseStart(const TestCase& /*test_case*/) override {
01149         }
01150 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01151
01152         void OnTestStart(const TestInfo& /*test_info*/) override {
01153         }
01154
01155         void OnTestDisabled(const TestInfo& /*test_info*/) override {
01156         }
01157
01158         void OnTestPartResult(const TestPartResult& /*test_part_result*/) override {
01159         }
01160
01161         void OnTestEnd(const TestInfo& /*test_info*/) override {
01162         }
01163
01164         void OnTestSuiteEnd(const TestSuite& /*test_suite*/) override {
01165         }
01166 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01167
01168         void OnTestCaseEnd(const TestCase& /*test_case*/) override {
01169         }
01170 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01171
01172         void OnEnvironmentsTearDownStart(const UnitTest& /*unit_test*/) override {
01173         }
01174
01175         void OnEnvironmentsTearDownEnd(const UnitTest& /*unit_test*/) override {
01176         }
01177
01178         void OnTestIterationEnd(const UnitTest& /*unit_test*/,
01179                 int /*iteration*/) override {
```

```
01180         }
01181
01182         void OnTestProgramEnd(const UnitTest& /*unit_test*/) override {
01183         }
01184     };
01185
01186     // TestEventListeners lets users add listeners to track events in Google Test.
01187
01188     class GTEST_API_ TestEventListeners {
01189     public:
01190         TestEventListeners();
01191         ~TestEventListeners();
01192
01193         // Appends an event listener to the end of the list. Google Test assumes
01194         // the ownership of the listener (i.e. it will delete the listener when
01195         // the test program finishes).
01196         void Append(TestEventListener* listener);
01197
01198         // Removes the given event listener from the list and returns it.  It then
01199         // becomes the caller's responsibility to delete the listener. Returns
01200         // NULL if the listener is not found in the list.
01201         TestEventListener* Release(TestEventListener* listener);
01202
01203         // Returns the standard listener responsible for the default console
01204         // output.  Can be removed from the listeners list to shut down default
01205         // console output.  Note that removing this object from the listener list
01206         // with Release transfers its ownership to the caller and makes this
01207         // function return NULL the next time.
01208
01209         TestEventListener* default_result_printer() const {
01210             return default_result_printer_;
01211         }
01212
01213         // Returns the standard listener responsible for the default XML output
01214         // controlled by the --gtest_output=xml flag.  Can be removed from the
01215         // listeners list by users who want to shut down the default XML output
01216         // controlled by this flag and substitute it with custom one.  Note that
01217         // removing this object from the listener list with Release transfers its
01218         // ownership to the caller and makes this function return NULL the next
01219         // time.
01220
01221         TestEventListener* default_xml_generator() const {
01222             return default_xml_generator_;
01223         }
01224
01225     private:
01226         friend class TestSuite;
01227         friend class TestInfo;
01228         friend class internal::DefaultGlobalTestPartResultReporter;
01229         friend class internal::NoExecDeathTest;
01230         friend class internal::TestEventListenersAccessor;
01231         friend class internal::UnitTestImpl;
01232
01233         // Returns repeater that broadcasts the TestEventListener events to all
01234         // subscribers.
01235         TestEventListener* repeater();
01236
01237         // Sets the default_result_printer attribute to the provided listener.
01238         // The listener is also added to the listener list and previous
01239         // default_result_printer is removed from it and deleted. The listener can
01240         // also be NULL in which case it will not be added to the list. Does
01241         // nothing if the previous and the current listener objects are the same.
01242         void SetDefaultResultPrinter(TestEventListener* listener);
01243
01244         // Sets the default_xml_generator attribute to the provided listener.  The
01245         // listener is also added to the listener list and previous
01246         // default_xml_generator is removed from it and deleted. The listener can
01247         // also be NULL in which case it will not be added to the list. Does
01248         // nothing if the previous and the current listener objects are the same.
01249         void SetDefaultXmlGenerator(TestEventListener* listener);
01250
01251         // Controls whether events will be forwarded by the repeater to the
01252         // listeners in the list.
01253         bool EventForwardingEnabled() const;
01254         void SuppressEventForwarding();
01255
01256         // The actual list of listeners.
01257         internal::TestEventRepeater* repeater_;
01258         // Listener responsible for the standard result output.
01259         TestEventListener* default_result_printer_;
01260         // Listener responsible for the creation of the XML output file.
01261         TestEventListener* default_xml_generator_;
01262
01263         // We disallow copying TestEventListeners.
01264         TestEventListeners(const TestEventListeners&) = delete;
01265         TestEventListeners& operator=(const TestEventListeners&) = delete;
01266     };
```

```
01267
01268     // A UnitTest consists of a vector of TestSuites.
01269     //
01270     // This is a singleton class.  The only instance of UnitTest is
01271     // created when UnitTest::GetInstance() is first called.  This
01272     // instance is never deleted.
01273     //
01274     // UnitTest is not copyable.
01275     //
01276     // This class is thread-safe as long as the methods are called
01277     // according to their specification.
01278
01279     class GTEST_API_ UnitTest {
01280     public:
01281         // Gets the singleton UnitTest object.  The first time this method
01282         // is called, a UnitTest object is constructed and returned.
01283         // Consecutive calls will return the same object.
01284         static UnitTest* GetInstance();
01285
01286         // Runs all tests in this UnitTest object and prints the result.
01287         // Returns 0 if successful, or 1 otherwise.
01288         //
01289         // This method can only be called from the main thread.
01290         //
01291         // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01292         int Run() GTEST_MUST_USE_RESULT_;
01293
01294         // Returns the working directory when the first TEST() or TEST_F()
01295         // was executed.  The UnitTest object owns the string.
01296         const char* original_working_dir() const;
01297
01298         // Returns the TestSuite object for the test that's currently running,
01299         // or NULL if no test is running.
01300         const TestSuite* current_test_suite() const GTEST_LOCK_EXCLUDED_(mutex_);
01301
01302         // Legacy API is still available but deprecated
01303 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01304         const TestCase* current_test_case() const GTEST_LOCK_EXCLUDED_(mutex_);
01305 #endif
01306
01307         // Returns the TestInfo object for the test that's currently running,
01308         // or NULL if no test is running.
01309         const TestInfo* current_test_info() const GTEST_LOCK_EXCLUDED_(mutex_);
01310
01311         // Returns the random seed used at the start of the current test run.
01312         int random_seed() const;
01313
01314         // Returns the ParameterizedTestSuiteRegistry object used to keep track of
01315         // value-parameterized tests and instantiate and register them.
01316         //
01317         // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01318         internal::ParameterizedTestSuiteRegistry& parameterized_test_registry()
01319         GTEST_LOCK_EXCLUDED_(mutex_);
01320
01321         // Gets the number of successful test suites.
01322         int successful_test_suite_count() const;
01323
01324         // Gets the number of failed test suites.
01325         int failed_test_suite_count() const;
01326
01327         // Gets the number of all test suites.
01328         int total_test_suite_count() const;
01329
01330         // Gets the number of all test suites that contain at least one test
01331         // that should run.
01332         int test_suite_to_run_count() const;
01333
01334         //  Legacy API is deprecated but still available
01335 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01336         int successful_test_case_count() const;
01337         int failed_test_case_count() const;
01338         int total_test_case_count() const;
01339         int test_case_to_run_count() const;
01340 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01341
01342         // Gets the number of successful tests.
01343         int successful_test_count() const;
01344
01345         // Gets the number of skipped tests.
01346         int skipped_test_count() const;
01347
01348         // Gets the number of failed tests.
01349         int failed_test_count() const;
01350
01351         // Gets the number of disabled tests that will be reported in the XML report.
01352         int reportable_disabled_test_count() const;
01353
```

```
01354         // Gets the number of disabled tests.
01355         int disabled_test_count() const;
01356
01357         // Gets the number of tests to be printed in the XML report.
01358         int reportable_test_count() const;
01359
01360         // Gets the number of all tests.
01361         int total_test_count() const;
01362
01363         // Gets the number of tests that should run.
01364         int test_to_run_count() const;
01365
01366         // Gets the time of the test program start, in ms from the start of the
01367         // UNIX epoch.
01368         TimeInMillis start_timestamp() const;
01369
01370         // Gets the elapsed time, in milliseconds.
01371         TimeInMillis elapsed_time() const;
01372
01373         // Returns true if and only if the unit test passed (i.e. all test suites
01374         // passed).
01375         bool Passed() const;
01376
01377         // Returns true if and only if the unit test failed (i.e. some test suite
01378         // failed or something outside of all tests failed).
01379         bool Failed() const;
01380
01381         // Gets the i-th test suite among all the test suites. i can range from 0 to
01382         // total_test_suite_count() - 1. If i is not in that range, returns NULL.
01383         const TestSuite* GetTestSuite(int i) const;
01384
01385         //  Legacy API is deprecated but still available
01386 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01387         const TestCase* GetTestCase(int i) const;
01388 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
01389
01390         // Returns the TestResult containing information on test failures and
01391         // properties logged outside of individual test suites.
01392         const TestResult& ad_hoc_test_result() const;
01393
01394         // Returns the list of event listeners that can be used to track events
01395         // inside Google Test.
01396         TestEventListeners& listeners();
01397
01398     private:
01399         // Registers and returns a global test environment.  When a test
01400         // program is run, all global test environments will be set-up in
01401         // the order they were registered.  After all tests in the program
01402         // have finished, all global test environments will be torn-down in
01403         // the *reverse* order they were registered.
01404         //
01405         // The UnitTest object takes ownership of the given environment.
01406         //
01407         // This method can only be called from the main thread.
01408         Environment* AddEnvironment(Environment* env);
01409
01410         // Adds a TestPartResult to the current TestResult object.  All
01411         // Google Test assertion macros (e.g. ASSERT_TRUE, EXPECT_EQ, etc)
01412         // eventually call this to report their results.  The user code
01413         // should use the assertion macros instead of calling this directly.
01414         void AddTestPartResult(TestPartResult::Type result_type,
01415                 const char* file_name, int line_number,
01416                 const std::string& message,
01417                 const std::string& os_stack_trace)
01418         GTEST_LOCK_EXCLUDED_(mutex_);
01419
01420         // Adds a TestProperty to the current TestResult object when invoked from
01421         // inside a test, to current TestSuite's ad_hoc_test_result_ when invoked
01422         // from SetUpTestSuite or TearDownTestSuite, or to the global property set
01423         // when invoked elsewhere.  If the result already contains a property with
01424         // the same key, the value will be updated.
01425         void RecordProperty(const std::string& key, const std::string& value);
01426
01427         // Gets the i-th test suite among all the test suites. i can range from 0 to
01428         // total_test_suite_count() - 1. If i is not in that range, returns NULL.
01429         TestSuite* GetMutableTestSuite(int i);
01430
01431         // Accessors for the implementation object.
01432
01433         internal::UnitTestImpl* impl() {
01434             return impl_;
01435         }
01436
01437         const internal::UnitTestImpl* impl() const {
01438             return impl_;
01439         }
01440
```

```
01441         // These classes and functions are friends as they need to access private
01442         // members of UnitTest.
01443         friend class ScopedTrace;
01444         friend class Test;
01445         friend class internal::AssertHelper;
01446         friend class internal::StreamingListenerTest;
01447         friend class internal::UnitTestRecordPropertyTestHelper;
01448         friend Environment* AddGlobalTestEnvironment(Environment* env);
01449         friend std::set<std::string>* internal::GetIgnoredParameterizedTestSuites();
01450         friend internal::UnitTestImpl* internal::GetUnitTestImpl();
01451         friend void internal::ReportFailureInUnknownLocation(
01452                 TestPartResult::Type result_type, const std::string& message);
01453 
01454         // Creates an empty UnitTest.
01455         UnitTest();
01456 
01457         // D'tor
01458         virtual ~UnitTest();
01459 
01460         // Pushes a trace defined by SCOPED_TRACE() on to the per-thread
01461         // Google Test trace stack.
01462         void PushGTestTrace(const internal::TraceInfo& trace)
01463         GTEST_LOCK_EXCLUDED_(mutex_);
01464 
01465         // Pops a trace from the per-thread Google Test trace stack.
01466         void PopGTestTrace() GTEST_LOCK_EXCLUDED_(mutex_);
01467 
01468         // Protects mutable state in *impl_.  This is mutable as some const
01469         // methods need to lock it too.
01470         mutable internal::Mutex mutex_;
01471 
01472         // Opaque implementation object.  This field is never changed once
01473         // the object is constructed.  We don't mark it as const here, as
01474         // doing so will cause a warning in the constructor of UnitTest.
01475         // Mutable state in *impl_ is protected by mutex_.
01476         internal::UnitTestImpl* impl_;
01477 
01478         // We disallow copying UnitTest.
01479         UnitTest(const UnitTest&) = delete;
01480         UnitTest& operator=(const UnitTest&) = delete;
01481     };
01482 
01483     // A convenient wrapper for adding an environment for the test
01484     // program.
01485     //
01486     // You should call this before RUN_ALL_TESTS() is called, probably in
01487     // main().  If you use gtest_main, you need to call this before main()
01488     // starts for it to take effect.  For example, you can define a global
01489     // variable like this:
01490     //
01491     //   testing::Environment* const foo_env =
01492     //       testing::AddGlobalTestEnvironment(new FooEnvironment);
01493     //
01494     // However, we strongly recommend you to write your own main() and
01495     // call AddGlobalTestEnvironment() there, as relying on initialization
01496     // of global variables makes the code harder to read and may cause
01497     // problems when you register multiple environments from different
01498     // translation units and the environments have dependencies among them
01499     // (remember that the compiler doesn't guarantee the order in which
01500     // global variables from different translation units are initialized).
01501 
01502     inline Environment* AddGlobalTestEnvironment(Environment* env) {
01503         return UnitTest::GetInstance()->AddEnvironment(env);
01504     }
01505 
01506     // Initializes Google Test.  This must be called before calling
01507     // RUN_ALL_TESTS().  In particular, it parses a command line for the
01508     // flags that Google Test recognizes.  Whenever a Google Test flag is
01509     // seen, it is removed from argv, and *argc is decremented.
01510     //
01511     // No value is returned.  Instead, the Google Test flag variables are
01512     // updated.
01513     //
01514     // Calling the function for the second time has no user-visible effect.
01515     GTEST_API_ void InitGoogleTest(int* argc, char** argv);
01516 
01517     // This overloaded version can be used in Windows programs compiled in
01518     // UNICODE mode.
01519     GTEST_API_ void InitGoogleTest(int* argc, wchar_t** argv);
01520 
01521     // This overloaded version can be used on Arduino/embedded platforms where
01522     // there is no argc/argv.
01523     GTEST_API_ void InitGoogleTest();
01524 
01525     namespace internal {
01526 
01527         // Separate the error generating code from the code path to reduce the stack
```

```
01528         // frame size of CmpHelperEQ. This helps reduce the overhead of some sanitizers
01529         // when calling EXPECT_* in a tight loop.
01530
01531         template <typename T1, typename T2>
01532         AssertionResult CmpHelperEQFailure(const char* lhs_expression,
01533                 const char* rhs_expression, const T1& lhs,
01534                 const T2& rhs) {
01535             return EqFailure(lhs_expression, rhs_expression,
01536                              FormatForComparisonFailureMessage(lhs, rhs),
01537                              FormatForComparisonFailureMessage(rhs, lhs), false);
01538         }
01539
01540         // This block of code defines operator==/!=
01541         // to block lexical scope lookup.
01542         // It prevents using invalid operator==/!= defined at namespace scope.
01543
01544         struct faketype {
01545         };
01546
01547         inline bool operator==(faketype, faketype) {
01548             return true;
01549         }
01550
01551         inline bool operator!=(faketype, faketype) {
01552             return false;
01553         }
01554
01555         // The helper function for {ASSERT|EXPECT}_EQ.
01556
01557         template <typename T1, typename T2>
01558         AssertionResult CmpHelperEQ(const char* lhs_expression,
01559                 const char* rhs_expression, const T1& lhs,
01560                 const T2& rhs) {
01561             if (lhs == rhs) {
01562                 return AssertionSuccess();
01563             }
01564
01565             return CmpHelperEQFailure(lhs_expression, rhs_expression, lhs, rhs);
01566         }
01567
01568         class EqHelper {
01569         public:
01570             // This templatized version is for the general case.
01571
01572             template <
01573             typename T1, typename T2,
01574             // Disable this overload for cases where one argument is a pointer
01575             // and the other is the null pointer constant.
01576             typename std::enable_if<!std::is_integral<T1>::value ||
01577             !std::is_pointer<T2>::value>::type* = nullptr>
01578             static AssertionResult Compare(const char* lhs_expression,
01579                     const char* rhs_expression, const T1& lhs,
01580                     const T2& rhs) {
01581                 return CmpHelperEQ(lhs_expression, rhs_expression, lhs, rhs);
01582             }
01583
01584             // With this overloaded version, we allow anonymous enums to be used
01585             // in {ASSERT|EXPECT}_EQ when compiled with gcc 4, as anonymous
01586             // enums can be implicitly cast to BiggestInt.
01587             //
01588             // Even though its body looks the same as the above version, we
01589             // cannot merge the two, as it will make anonymous enums unhappy.
01590
01591             static AssertionResult Compare(const char* lhs_expression,
01592                     const char* rhs_expression, BiggestInt lhs,
01593                     BiggestInt rhs) {
01594                 return CmpHelperEQ(lhs_expression, rhs_expression, lhs, rhs);
01595             }
01596
01597             template <typename T>
01598             static AssertionResult Compare(
01599                     const char* lhs_expression, const char* rhs_expression,
01600                     // Handle cases where '0' is used as a null pointer literal.
01601                     std::nullptr_t /* lhs */, T* rhs) {
01602                 // We already know that 'lhs' is a null pointer.
01603                 return CmpHelperEQ(lhs_expression, rhs_expression, static_cast<T*> (nullptr),
01604                         rhs);
01605             }
01606         };
01607
01608         // Separate the error generating code from the code path to reduce the stack
01609         // frame size of CmpHelperOP. This helps reduce the overhead of some sanitizers
01610         // when calling EXPECT_OP in a tight loop.
01611
01612         template <typename T1, typename T2>
01613         AssertionResult CmpHelperOpFailure(const char* expr1, const char* expr2,
01614                 const T1& val1, const T2& val2,
```

```
01615                const char* op) {
01616            return AssertionFailure()
01617                   << "Expected: (" << expr1 << ") " << op << " (" << expr2
01618                   << "), actual: " << FormatForComparisonFailureMessage(val1, val2)
01619                   << " vs " << FormatForComparisonFailureMessage(val2, val1);
01620        }
01621
01622        // A macro for implementing the helper functions needed to implement
01623        // ASSERT_?? and EXPECT_??.  It is here just to avoid copy-and-paste
01624        // of similar code.
01625        //
01626        // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01627
01628 #define GTEST_IMPL_CMP_HELPER_(op_name, op)                                \
01629   template <typename T1, typename T2>                                      \
01630   AssertionResult CmpHelper##op_name(const char* expr1, const char* expr2, \
01631                                      const T1& val1, const T2& val2) {     \
01632     if (val1 op val2) {                                                    \
01633       return AssertionSuccess();                                           \
01634     } else {                                                               \
01635       return CmpHelperOpFailure(expr1, expr2, val1, val2, #op);            \
01636     }                                                                      \
01637   }
01638
01639        // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01640
01641        // Implements the helper function for {ASSERT|EXPECT}_NE
01642        GTEST_IMPL_CMP_HELPER_(NE, !=)
01643        // Implements the helper function for {ASSERT|EXPECT}_LE
01644        GTEST_IMPL_CMP_HELPER_(LE, <=)
01645        // Implements the helper function for {ASSERT|EXPECT}_LT
01646        GTEST_IMPL_CMP_HELPER_(LT, <)
01647        // Implements the helper function for {ASSERT|EXPECT}_GE
01648        GTEST_IMPL_CMP_HELPER_(GE, >=)
01649        // Implements the helper function for {ASSERT|EXPECT}_GT
01650        GTEST_IMPL_CMP_HELPER_(GT, >)
01651
01652 #undef GTEST_IMPL_CMP_HELPER_
01653
01654        // The helper function for {ASSERT|EXPECT}_STREQ.
01655        //
01656        // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01657        GTEST_API_ AssertionResult CmpHelperSTREQ(const char* s1_expression,
01658                const char* s2_expression,
01659                const char* s1, const char* s2);
01660
01661        // The helper function for {ASSERT|EXPECT}_STRCASEEQ.
01662        //
01663        // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01664        GTEST_API_ AssertionResult CmpHelperSTRCASEEQ(const char* s1_expression,
01665                const char* s2_expression,
01666                const char* s1, const char* s2);
01667
01668        // The helper function for {ASSERT|EXPECT}_STRNE.
01669        //
01670        // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01671        GTEST_API_ AssertionResult CmpHelperSTRNE(const char* s1_expression,
01672                const char* s2_expression,
01673                const char* s1, const char* s2);
01674
01675        // The helper function for {ASSERT|EXPECT}_STRCASENE.
01676        //
01677        // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01678        GTEST_API_ AssertionResult CmpHelperSTRCASENE(const char* s1_expression,
01679                const char* s2_expression,
01680                const char* s1, const char* s2);
01681
01682        // Helper function for *_STREQ on wide strings.
01683        //
01684        // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01685        GTEST_API_ AssertionResult CmpHelperSTREQ(const char* s1_expression,
01686                const char* s2_expression,
01687                const wchar_t* s1, const wchar_t* s2);
01688
01689        // Helper function for *_STRNE on wide strings.
01690        //
01691        // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01692        GTEST_API_ AssertionResult CmpHelperSTRNE(const char* s1_expression,
01693                const char* s2_expression,
01694                const wchar_t* s1, const wchar_t* s2);
01695
01696    } // namespace internal
01697
01698    // IsSubstring() and IsNotSubstring() are intended to be used as the
01699    // first argument to {EXPECT,ASSERT}_PRED_FORMAT2(), not by
01700    // themselves.  They check whether needle is a substring of haystack
01701    // (NULL is considered a substring of itself only), and return an
```

```
01702     // appropriate error message when they fail.
01703     //
01704     // The {needle,haystack}_expr arguments are the stringified
01705     // expressions that generated the two real arguments.
01706     GTEST_API_ AssertionResult IsSubstring(const char* needle_expr,
01707             const char* haystack_expr,
01708             const char* needle,
01709             const char* haystack);
01710     GTEST_API_ AssertionResult IsSubstring(const char* needle_expr,
01711             const char* haystack_expr,
01712             const wchar_t* needle,
01713             const wchar_t* haystack);
01714     GTEST_API_ AssertionResult IsNotSubstring(const char* needle_expr,
01715             const char* haystack_expr,
01716             const char* needle,
01717             const char* haystack);
01718     GTEST_API_ AssertionResult IsNotSubstring(const char* needle_expr,
01719             const char* haystack_expr,
01720             const wchar_t* needle,
01721             const wchar_t* haystack);
01722     GTEST_API_ AssertionResult IsSubstring(const char* needle_expr,
01723             const char* haystack_expr,
01724             const ::std::string& needle,
01725             const ::std::string& haystack);
01726     GTEST_API_ AssertionResult IsNotSubstring(const char* needle_expr,
01727             const char* haystack_expr,
01728             const ::std::string& needle,
01729             const ::std::string& haystack);
01730
01731 #if GTEST_HAS_STD_WSTRING
01732     GTEST_API_ AssertionResult IsSubstring(const char* needle_expr,
01733             const char* haystack_expr,
01734             const ::std::wstring& needle,
01735             const ::std::wstring& haystack);
01736     GTEST_API_ AssertionResult IsNotSubstring(const char* needle_expr,
01737             const char* haystack_expr,
01738             const ::std::wstring& needle,
01739             const ::std::wstring& haystack);
01740 #endif  // GTEST_HAS_STD_WSTRING
01741
01742     namespace internal {
01743
01744         // Helper template function for comparing floating-points.
01745         //
01746         // Template parameter:
01747         //
01748         //   RawType: the raw floating-point type (either float or double)
01749         //
01750         // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01751
01752         template <typename RawType>
01753         AssertionResult CmpHelperFloatingPointEQ(const char* lhs_expression,
01754                 const char* rhs_expression,
01755                 RawType lhs_value, RawType rhs_value) {
01756             const FloatingPoint<RawType> lhs(lhs_value), rhs(rhs_value);
01757
01758             if (lhs.AlmostEquals(rhs)) {
01759                 return AssertionSuccess();
01760             }
01761
01762             ::std::stringstream lhs_ss;
01763             lhs_ss << std::setprecision(std::numeric_limits<RawType>::digits10 + 2)
01764                    << lhs_value;
01765
01766             ::std::stringstream rhs_ss;
01767             rhs_ss << std::setprecision(std::numeric_limits<RawType>::digits10 + 2)
01768                    << rhs_value;
01769
01770             return EqFailure(lhs_expression, rhs_expression,
01771                     StringStreamToString(&lhs_ss), StringStreamToString(&rhs_ss),
01772                     false);
01773         }
01774
01775         // Helper function for implementing ASSERT_NEAR.
01776         //
01777         // INTERNAL IMPLEMENTATION - DO NOT USE IN A USER PROGRAM.
01778         GTEST_API_ AssertionResult DoubleNearPredFormat(const char* expr1,
01779                 const char* expr2,
01780                 const char* abs_error_expr,
01781                 double val1, double val2,
01782                 double abs_error);
01783
01784         // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
01785         // A class that enables one to stream messages to assertion macros
01786
01787         class GTEST_API_ AssertHelper {
01788         public:
```

```
01789             // Constructor.
01790             AssertHelper(TestPartResult::Type type, const char* file, int line,
01791                          const char* message);
01792             ~AssertHelper();
01793
01794             // Message assignment is a semantic trick to enable assertion
01795             // streaming; see the GTEST_MESSAGE_ macro below.
01796             void operator=(const Message& message) const;
01797
01798         private:
01799             // We put our data in a struct so that the size of the AssertHelper class can
01800             // be as small as possible.  This is important because gcc is incapable of
01801             // re-using stack space even for temporary variables, so every EXPECT_EQ
01802             // reserves stack space for another AssertHelper.
01803
01804             struct AssertHelperData {
01805
01806                 AssertHelperData(TestPartResult::Type t, const char* srcfile, int line_num,
01807                                  const char* msg)
01808                 : type(t), file(srcfile), line(line_num), message(msg) {
01809                 }
01810
01811                 TestPartResult::Type const type;
01812                 const char* const file;
01813                 int const line;
01814                 std::string const message;
01815
01816             private:
01817                 AssertHelperData(const AssertHelperData&) = delete;
01818                 AssertHelperData& operator=(const AssertHelperData&) = delete;
01819             };
01820
01821             AssertHelperData * const data_;
01822
01823             AssertHelper(const AssertHelper&) = delete;
01824             AssertHelper& operator=(const AssertHelper&) = delete;
01825         };
01826
01827     } // namespace internal
01828
01829     // The pure interface class that all value-parameterized tests inherit from.
01830     // A value-parameterized class must inherit from both ::testing::Test and
01831     // ::testing::WithParamInterface. In most cases that just means inheriting
01832     // from ::testing::TestWithParam, but more complicated test hierarchies
01833     // may need to inherit from Test and WithParamInterface at different levels.
01834     //
01835     // This interface has support for accessing the test parameter value via
01836     // the GetParam() method.
01837     //
01838     // Use it with one of the parameter generator defining functions, like Range(),
01839     // Values(), ValuesIn(), Bool(), and Combine().
01840     //
01841     // class FooTest : public ::testing::TestWithParam<int> {
01842     //  protected:
01843     //   FooTest() {
01844     //     // Can use GetParam() here.
01845     //   }
01846     //   ~FooTest() override {
01847     //     // Can use GetParam() here.
01848     //   }
01849     //   void SetUp() override {
01850     //     // Can use GetParam() here.
01851     //   }
01852     //   void TearDown override {
01853     //     // Can use GetParam() here.
01854     //   }
01855     // };
01856     // TEST_P(FooTest, DoesBar) {
01857     //   // Can use GetParam() method here.
01858     //   Foo foo;
01859     //   ASSERT_TRUE(foo.DoesBar(GetParam()));
01860     // }
01861     // INSTANTIATE_TEST_SUITE_P(OneToTenRange, FooTest, ::testing::Range(1, 10));
01862
01863     template <typename T>
01864     class WithParamInterface {
01865     public:
01866         typedef T ParamType;
01867
01868         virtual ~WithParamInterface() {
01869         }
01870
01871         // The current parameter value. Is also available in the test fixture's
01872         // constructor.
01873
01874         static const ParamType& GetParam() {
01875             GTEST_CHECK_(parameter_ != nullptr)
```

```
01876                     « "GetParam() can only be called inside a value-parameterized test "
01877                     « "-- did you intend to write TEST_P instead of TEST_F?";
01878             return *parameter_;
01879         }
01880
01881     private:
01882         // Sets parameter value. The caller is responsible for making sure the value
01883         // remains alive and unchanged throughout the current test.
01884
01885         static void SetParam(const ParamType* parameter) {
01886             parameter_ = parameter;
01887         }
01888
01889         // Static value used for accessing parameter during a test lifetime.
01890         static const ParamType* parameter_;
01891
01892         // TestClass must be a subclass of WithParamInterface<T> and Test.
01893         template <class TestClass>
01894         friend class internal::ParameterizedTestFactory;
01895     };
01896
01897     template <typename T>
01898     const T* WithParamInterface<T>::parameter_ = nullptr;
01899
01900     // Most value-parameterized classes can ignore the existence of
01901     // WithParamInterface, and can just inherit from ::testing::TestWithParam.
01902
01903     template <typename T>
01904     class TestWithParam : public Test, public WithParamInterface<T> {
01905     };
01906
01907     // Macros for indicating success/failure in test code.
01908
01909     // Skips test in runtime.
01910     // Skipping test aborts current function.
01911     // Skipped tests are neither successful nor failed.
01912 #define GTEST_SKIP() GTEST_SKIP_("")
01913
01914     // ADD_FAILURE unconditionally adds a failure to the current test.
01915     // SUCCEED generates a success - it doesn't automatically make the
01916     // current test successful, as a test is only successful when it has
01917     // no failure.
01918     //
01919     // EXPECT_* verifies that a certain condition is satisfied.  If not,
01920     // it behaves like ADD_FAILURE.  In particular:
01921     //
01922     //   EXPECT_TRUE  verifies that a Boolean condition is true.
01923     //   EXPECT_FALSE verifies that a Boolean condition is false.
01924     //
01925     // FAIL and ASSERT_* are similar to ADD_FAILURE and EXPECT_*, except
01926     // that they will also abort the current function on failure.  People
01927     // usually want the fail-fast behavior of FAIL and ASSERT_*, but those
01928     // writing data-driven tests often find themselves using ADD_FAILURE
01929     // and EXPECT_* more.
01930
01931     // Generates a nonfatal failure with a generic message.
01932 #define ADD_FAILURE() GTEST_NONFATAL_FAILURE_("Failed")
01933
01934     // Generates a nonfatal failure at the given source file location with
01935     // a generic message.
01936 #define ADD_FAILURE_AT(file, line)        \
01937   GTEST_MESSAGE_AT_(file, line, "Failed", \
01938                     ::testing::TestPartResult::kNonFatalFailure)
01939
01940     // Generates a fatal failure with a generic message.
01941 #define GTEST_FAIL() GTEST_FATAL_FAILURE_("Failed")
01942
01943     // Like GTEST_FAIL(), but at the given source file location.
01944 #define GTEST_FAIL_AT(file, line)         \
01945   GTEST_MESSAGE_AT_(file, line, "Failed", \
01946                     ::testing::TestPartResult::kFatalFailure)
01947
01948     // Define this macro to 1 to omit the definition of FAIL(), which is a
01949     // generic name and clashes with some other libraries.
01950 #if !GTEST_DONT_DEFINE_FAIL
01951 #define FAIL() GTEST_FAIL()
01952 #endif
01953
01954     // Generates a success with a generic message.
01955 #define GTEST_SUCCEED() GTEST_SUCCESS_("Succeeded")
01956
01957     // Define this macro to 1 to omit the definition of SUCCEED(), which
01958     // is a generic name and clashes with some other libraries.
01959 #if !GTEST_DONT_DEFINE_SUCCEED
01960 #define SUCCEED() GTEST_SUCCEED()
01961 #endif
01962
```

```
01963     // Macros for testing exceptions.
01964     //
01965     //    * {ASSERT|EXPECT}_THROW(statement, expected_exception):
01966     //         Tests that the statement throws the expected exception.
01967     //    * {ASSERT|EXPECT}_NO_THROW(statement):
01968     //         Tests that the statement doesn't throw any exception.
01969     //    * {ASSERT|EXPECT}_ANY_THROW(statement):
01970     //         Tests that the statement throws an exception.
01971
01972 #define EXPECT_THROW(statement, expected_exception) \
01973   GTEST_TEST_THROW_(statement, expected_exception, GTEST_NONFATAL_FAILURE_)
01974 #define EXPECT_NO_THROW(statement) \
01975   GTEST_TEST_NO_THROW_(statement, GTEST_NONFATAL_FAILURE_)
01976 #define EXPECT_ANY_THROW(statement) \
01977   GTEST_TEST_ANY_THROW_(statement, GTEST_NONFATAL_FAILURE_)
01978 #define ASSERT_THROW(statement, expected_exception) \
01979   GTEST_TEST_THROW_(statement, expected_exception, GTEST_FATAL_FAILURE_)
01980 #define ASSERT_NO_THROW(statement) \
01981   GTEST_TEST_NO_THROW_(statement, GTEST_FATAL_FAILURE_)
01982 #define ASSERT_ANY_THROW(statement) \
01983   GTEST_TEST_ANY_THROW_(statement, GTEST_FATAL_FAILURE_)
01984
01985     // Boolean assertions. Condition can be either a Boolean expression or an
01986     // AssertionResult. For more information on how to use AssertionResult with
01987     // these macros see comments on that class.
01988 #define GTEST_EXPECT_TRUE(condition)                      \
01989   GTEST_TEST_BOOLEAN_(condition, #condition, false, true, \
01990                       GTEST_NONFATAL_FAILURE_)
01991 #define GTEST_EXPECT_FALSE(condition)                       \
01992   GTEST_TEST_BOOLEAN_(!(condition), #condition, true, false, \
01993                       GTEST_NONFATAL_FAILURE_)
01994 #define GTEST_ASSERT_TRUE(condition) \
01995   GTEST_TEST_BOOLEAN_(condition, #condition, false, true, GTEST_FATAL_FAILURE_)
01996 #define GTEST_ASSERT_FALSE(condition)                       \
01997   GTEST_TEST_BOOLEAN_(!(condition), #condition, true, false, \
01998                       GTEST_FATAL_FAILURE_)
01999
02000     // Define these macros to 1 to omit the definition of the corresponding
02001     // EXPECT or ASSERT, which clashes with some users' own code.
02002
02003 #if !GTEST_DONT_DEFINE_EXPECT_TRUE
02004 #define EXPECT_TRUE(condition) GTEST_EXPECT_TRUE(condition)
02005 #endif
02006
02007 #if !GTEST_DONT_DEFINE_EXPECT_FALSE
02008 #define EXPECT_FALSE(condition) GTEST_EXPECT_FALSE(condition)
02009 #endif
02010
02011 #if !GTEST_DONT_DEFINE_ASSERT_TRUE
02012 #define ASSERT_TRUE(condition) GTEST_ASSERT_TRUE(condition)
02013 #endif
02014
02015 #if !GTEST_DONT_DEFINE_ASSERT_FALSE
02016 #define ASSERT_FALSE(condition) GTEST_ASSERT_FALSE(condition)
02017 #endif
02018
02019     // Macros for testing equalities and inequalities.
02020     //
02021     //    * {ASSERT|EXPECT}_EQ(v1, v2): Tests that v1 == v2
02022     //    * {ASSERT|EXPECT}_NE(v1, v2): Tests that v1 != v2
02023     //    * {ASSERT|EXPECT}_LT(v1, v2): Tests that v1 < v2
02024     //    * {ASSERT|EXPECT}_LE(v1, v2): Tests that v1 <= v2
02025     //    * {ASSERT|EXPECT}_GT(v1, v2): Tests that v1 > v2
02026     //    * {ASSERT|EXPECT}_GE(v1, v2): Tests that v1 >= v2
02027     //
02028     // When they are not, Google Test prints both the tested expressions and
02029     // their actual values.  The values must be compatible built-in types,
02030     // or you will get a compiler error.  By "compatible" we mean that the
02031     // values can be compared by the respective operator.
02032     //
02033     // Note:
02034     //
02035     //   1. It is possible to make a user-defined type work with
02036     //   {ASSERT|EXPECT}_??(), but that requires overloading the
02037     //   comparison operators and is thus discouraged by the Google C++
02038     //   Usage Guide.  Therefore, you are advised to use the
02039     //   {ASSERT|EXPECT}_TRUE() macro to assert that two objects are
02040     //   equal.
02041     //
02042     //   2. The {ASSERT|EXPECT}_??() macros do pointer comparisons on
02043     //   pointers (in particular, C strings).  Therefore, if you use it
02044     //   with two C strings, you are testing how their locations in memory
02045     //   are related, not how their content is related.  To compare two C
02046     //   strings by content, use {ASSERT|EXPECT}_STR*().
02047     //
02048     //   3. {ASSERT|EXPECT}_EQ(v1, v2) is preferred to
02049     //   {ASSERT|EXPECT}_TRUE(v1 == v2), as the former tells you
```

```
02050     //   what the actual value is when it fails, and similarly for the
02051     //   other comparisons.
02052     //
02053     //   4. Do not depend on the order in which {ASSERT|EXPECT}_??()
02054     //   evaluate their arguments, which is undefined.
02055     //
02056     //   5. These macros evaluate their arguments exactly once.
02057     //
02058     // Examples:
02059     //
02060     //   EXPECT_NE(Foo(), 5);
02061     //   EXPECT_EQ(a_pointer, NULL);
02062     //   ASSERT_LT(i, array_size);
02063     //   ASSERT_GT(records.size(), 0) << "There is no record left.";
02064
02065 #define EXPECT_EQ(val1, val2) \
02066   EXPECT_PRED_FORMAT2(::testing::internal::EqHelper::Compare, val1, val2)
02067 #define EXPECT_NE(val1, val2) \
02068   EXPECT_PRED_FORMAT2(::testing::internal::CmpHelperNE, val1, val2)
02069 #define EXPECT_LE(val1, val2) \
02070   EXPECT_PRED_FORMAT2(::testing::internal::CmpHelperLE, val1, val2)
02071 #define EXPECT_LT(val1, val2) \
02072   EXPECT_PRED_FORMAT2(::testing::internal::CmpHelperLT, val1, val2)
02073 #define EXPECT_GE(val1, val2) \
02074   EXPECT_PRED_FORMAT2(::testing::internal::CmpHelperGE, val1, val2)
02075 #define EXPECT_GT(val1, val2) \
02076   EXPECT_PRED_FORMAT2(::testing::internal::CmpHelperGT, val1, val2)
02077
02078 #define GTEST_ASSERT_EQ(val1, val2) \
02079   ASSERT_PRED_FORMAT2(::testing::internal::EqHelper::Compare, val1, val2)
02080 #define GTEST_ASSERT_NE(val1, val2) \
02081   ASSERT_PRED_FORMAT2(::testing::internal::CmpHelperNE, val1, val2)
02082 #define GTEST_ASSERT_LE(val1, val2) \
02083   ASSERT_PRED_FORMAT2(::testing::internal::CmpHelperLE, val1, val2)
02084 #define GTEST_ASSERT_LT(val1, val2) \
02085   ASSERT_PRED_FORMAT2(::testing::internal::CmpHelperLT, val1, val2)
02086 #define GTEST_ASSERT_GE(val1, val2) \
02087   ASSERT_PRED_FORMAT2(::testing::internal::CmpHelperGE, val1, val2)
02088 #define GTEST_ASSERT_GT(val1, val2) \
02089   ASSERT_PRED_FORMAT2(::testing::internal::CmpHelperGT, val1, val2)
02090
02091     // Define macro GTEST_DONT_DEFINE_ASSERT_XY to 1 to omit the definition of
02092     // ASSERT_XY(), which clashes with some users' own code.
02093
02094 #if !GTEST_DONT_DEFINE_ASSERT_EQ
02095 #define ASSERT_EQ(val1, val2) GTEST_ASSERT_EQ(val1, val2)
02096 #endif
02097
02098 #if !GTEST_DONT_DEFINE_ASSERT_NE
02099 #define ASSERT_NE(val1, val2) GTEST_ASSERT_NE(val1, val2)
02100 #endif
02101
02102 #if !GTEST_DONT_DEFINE_ASSERT_LE
02103 #define ASSERT_LE(val1, val2) GTEST_ASSERT_LE(val1, val2)
02104 #endif
02105
02106 #if !GTEST_DONT_DEFINE_ASSERT_LT
02107 #define ASSERT_LT(val1, val2) GTEST_ASSERT_LT(val1, val2)
02108 #endif
02109
02110 #if !GTEST_DONT_DEFINE_ASSERT_GE
02111 #define ASSERT_GE(val1, val2) GTEST_ASSERT_GE(val1, val2)
02112 #endif
02113
02114 #if !GTEST_DONT_DEFINE_ASSERT_GT
02115 #define ASSERT_GT(val1, val2) GTEST_ASSERT_GT(val1, val2)
02116 #endif
02117
02118     // C-string Comparisons.  All tests treat NULL and any non-NULL string
02119     // as different.  Two NULLs are equal.
02120     //
02121     //    * {ASSERT|EXPECT}_STREQ(s1, s2):     Tests that s1 == s2
02122     //    * {ASSERT|EXPECT}_STRNE(s1, s2):     Tests that s1 != s2
02123     //    * {ASSERT|EXPECT}_STRCASEEQ(s1, s2): Tests that s1 == s2, ignoring case
02124     //    * {ASSERT|EXPECT}_STRCASENE(s1, s2): Tests that s1 != s2, ignoring case
02125     //
02126     // For wide or narrow string objects, you can use the
02127     // {ASSERT|EXPECT}_??() macros.
02128     //
02129     // Don't depend on the order in which the arguments are evaluated,
02130     // which is undefined.
02131     //
02132     // These macros evaluate their arguments exactly once.
02133
02134 #define EXPECT_STREQ(s1, s2) \
02135   EXPECT_PRED_FORMAT2(::testing::internal::CmpHelperSTREQ, s1, s2)
02136 #define EXPECT_STRNE(s1, s2) \
```

```
02137   EXPECT_PRED_FORMAT2(::testing::internal::CmpHelperSTRNE, s1, s2)
02138 #define EXPECT_STRCASEEQ(s1, s2) \
02139   EXPECT_PRED_FORMAT2(::testing::internal::CmpHelperSTRCASEEQ, s1, s2)
02140 #define EXPECT_STRCASENE(s1, s2) \
02141   EXPECT_PRED_FORMAT2(::testing::internal::CmpHelperSTRCASENE, s1, s2)
02142
02143 #define ASSERT_STREQ(s1, s2) \
02144   ASSERT_PRED_FORMAT2(::testing::internal::CmpHelperSTREQ, s1, s2)
02145 #define ASSERT_STRNE(s1, s2) \
02146   ASSERT_PRED_FORMAT2(::testing::internal::CmpHelperSTRNE, s1, s2)
02147 #define ASSERT_STRCASEEQ(s1, s2) \
02148   ASSERT_PRED_FORMAT2(::testing::internal::CmpHelperSTRCASEEQ, s1, s2)
02149 #define ASSERT_STRCASENE(s1, s2) \
02150   ASSERT_PRED_FORMAT2(::testing::internal::CmpHelperSTRCASENE, s1, s2)
02151
02152     // Macros for comparing floating-point numbers.
02153     //
02154     //    * {ASSERT|EXPECT}_FLOAT_EQ(val1, val2):
02155     //         Tests that two float values are almost equal.
02156     //    * {ASSERT|EXPECT}_DOUBLE_EQ(val1, val2):
02157     //         Tests that two double values are almost equal.
02158     //    * {ASSERT|EXPECT}_NEAR(v1, v2, abs_error):
02159     //         Tests that v1 and v2 are within the given distance to each other.
02160     //
02161     // Google Test uses ULP-based comparison to automatically pick a default
02162     // error bound that is appropriate for the operands.  See the
02163     // FloatingPoint template class in gtest-internal.h if you are
02164     // interested in the implementation details.
02165
02166 #define EXPECT_FLOAT_EQ(val1, val2)                                          \
02167   EXPECT_PRED_FORMAT2(::testing::internal::CmpHelperFloatingPointEQ<float>, \
02168                       val1, val2)
02169
02170 #define EXPECT_DOUBLE_EQ(val1, val2)                                          \
02171   EXPECT_PRED_FORMAT2(::testing::internal::CmpHelperFloatingPointEQ<double>, \
02172                       val1, val2)
02173
02174 #define ASSERT_FLOAT_EQ(val1, val2)                                          \
02175   ASSERT_PRED_FORMAT2(::testing::internal::CmpHelperFloatingPointEQ<float>, \
02176                       val1, val2)
02177
02178 #define ASSERT_DOUBLE_EQ(val1, val2)                                          \
02179   ASSERT_PRED_FORMAT2(::testing::internal::CmpHelperFloatingPointEQ<double>, \
02180                       val1, val2)
02181
02182 #define EXPECT_NEAR(val1, val2, abs_error)                                    \
02183   EXPECT_PRED_FORMAT3(::testing::internal::DoubleNearPredFormat, val1, val2, \
02184                       abs_error)
02185
02186 #define ASSERT_NEAR(val1, val2, abs_error)                                    \
02187   ASSERT_PRED_FORMAT3(::testing::internal::DoubleNearPredFormat, val1, val2, \
02188                       abs_error)
02189
02190     // These predicate format functions work on floating-point values, and
02191     // can be used in {ASSERT|EXPECT}_PRED_FORMAT2*(), e.g.
02192     //
02193     //   EXPECT_PRED_FORMAT2(testing::DoubleLE, Foo(), 5.0);
02194
02195     // Asserts that val1 is less than, or almost equal to, val2.  Fails
02196     // otherwise.  In particular, it fails if either val1 or val2 is NaN.
02197     GTEST_API_ AssertionResult FloatLE(const char* expr1, const char* expr2,
02198             float val1, float val2);
02199     GTEST_API_ AssertionResult DoubleLE(const char* expr1, const char* expr2,
02200             double val1, double val2);
02201
02202 #if GTEST_OS_WINDOWS
02203
02204     // Macros that test for HRESULT failure and success, these are only useful
02205     // on Windows, and rely on Windows SDK macros and APIs to compile.
02206     //
02207     //    * {ASSERT|EXPECT}_HRESULT_{SUCCEEDED|FAILED}(expr)
02208     //
02209     // When expr unexpectedly fails or succeeds, Google Test prints the
02210     // expected result and the actual result with both a human-readable
02211     // string representation of the error, if available, as well as the
02212     // hex result code.
02213 #define EXPECT_HRESULT_SUCCEEDED(expr) \
02214   EXPECT_PRED_FORMAT1(::testing::internal::IsHRESULTSuccess, (expr))
02215
02216 #define ASSERT_HRESULT_SUCCEEDED(expr) \
02217   ASSERT_PRED_FORMAT1(::testing::internal::IsHRESULTSuccess, (expr))
02218
02219 #define EXPECT_HRESULT_FAILED(expr) \
02220   EXPECT_PRED_FORMAT1(::testing::internal::IsHRESULTFailure, (expr))
02221
02222 #define ASSERT_HRESULT_FAILED(expr) \
02223   ASSERT_PRED_FORMAT1(::testing::internal::IsHRESULTFailure, (expr))
```

```
02224
02225 #endif  // GTEST_OS_WINDOWS
02226
02227     // Macros that execute statement and check that it doesn't generate new fatal
02228     // failures in the current thread.
02229     //
02230     //   * {ASSERT|EXPECT}_NO_FATAL_FAILURE(statement);
02231     //
02232     // Examples:
02233     //
02234     //   EXPECT_NO_FATAL_FAILURE(Process());
02235     //   ASSERT_NO_FATAL_FAILURE(Process()) << "Process() failed";
02236     //
02237 #define ASSERT_NO_FATAL_FAILURE(statement) \
02238   GTEST_TEST_NO_FATAL_FAILURE_(statement, GTEST_FATAL_FAILURE_)
02239 #define EXPECT_NO_FATAL_FAILURE(statement) \
02240   GTEST_TEST_NO_FATAL_FAILURE_(statement, GTEST_NONFATAL_FAILURE_)
02241
02242     // Causes a trace (including the given source file path and line number,
02243     // and the given message) to be included in every test failure message generated
02244     // by code in the scope of the lifetime of an instance of this class. The effect
02245     // is undone with the destruction of the instance.
02246     //
02247     // The message argument can be anything streamable to std::ostream.
02248     //
02249     // Example:
02250     //   testing::ScopedTrace trace("file.cc", 123, "message");
02251     //
02252
02253     class GTEST_API_ ScopedTrace {
02254     public:
02255         // The c'tor pushes the given source file location and message onto
02256         // a trace stack maintained by Google Test.
02257
02258         // Template version. Uses Message() to convert the values into strings.
02259         // Slow, but flexible.
02260
02261         template <typename T>
02262         ScopedTrace(const char* file, int line, const T& message) {
02263             PushTrace(file, line, (Message() << message).GetString());
02264         }
02265
02266         // Optimize for some known types.
02267
02268         ScopedTrace(const char* file, int line, const char* message) {
02269             PushTrace(file, line, message ? message : "(null)");
02270         }
02271
02272         ScopedTrace(const char* file, int line, const std::string& message) {
02273             PushTrace(file, line, message);
02274         }
02275
02276         // The d'tor pops the info pushed by the c'tor.
02277         //
02278         // Note that the d'tor is not virtual in order to be efficient.
02279         // Don't inherit from ScopedTrace!
02280         ~ScopedTrace();
02281
02282     private:
02283         void PushTrace(const char* file, int line, std::string message);
02284
02285         ScopedTrace(const ScopedTrace&) = delete;
02286         ScopedTrace& operator=(const ScopedTrace&) = delete;
02287     } GTEST_ATTRIBUTE_UNUSED_; // A ScopedTrace object does its job in its
02288     // c'tor and d'tor.  Therefore it doesn't
02289     // need to be used otherwise.
02290
02291     // Causes a trace (including the source file path, the current line
02292     // number, and the given message) to be included in every test failure
02293     // message generated by code in the current scope.  The effect is
02294     // undone when the control leaves the current scope.
02295     //
02296     // The message argument can be anything streamable to std::ostream.
02297     //
02298     // In the implementation, we include the current line number as part
02299     // of the dummy variable name, thus allowing multiple SCOPED_TRACE()s
02300     // to appear in the same block - as long as they are on different
02301     // lines.
02302     //
02303     // Assuming that each thread maintains its own stack of traces.
02304     // Therefore, a SCOPED_TRACE() would (correctly) only affect the
02305     // assertions in its own thread.
02306 #define SCOPED_TRACE(message)                                          \
02307   ::testing::ScopedTrace GTEST_CONCAT_TOKEN_(gtest_trace_, __LINE__)( \
02308       __FILE__, __LINE__, (message))
02309
02310     // Compile-time assertion for type equality.
```

```
02311     // StaticAssertTypeEq<type1, type2>() compiles if and only if type1 and type2
02312     // are the same type.  The value it returns is not interesting.
02313     //
02314     // Instead of making StaticAssertTypeEq a class template, we make it a
02315     // function template that invokes a helper class template.  This
02316     // prevents a user from misusing StaticAssertTypeEq<T1, T2> by
02317     // defining objects of that type.
02318     //
02319     // CAVEAT:
02320     //
02321     // When used inside a method of a class template,
02322     // StaticAssertTypeEq<T1, T2>() is effective ONLY IF the method is
02323     // instantiated.  For example, given:
02324     //
02325     //   template <typename T> class Foo {
02326     //    public:
02327     //     void Bar() { testing::StaticAssertTypeEq<int, T>(); }
02328     //   };
02329     //
02330     // the code:
02331     //
02332     //   void Test1() { Foo<bool> foo; }
02333     //
02334     // will NOT generate a compiler error, as Foo<bool>::Bar() is never
02335     // actually instantiated.  Instead, you need:
02336     //
02337     //   void Test2() { Foo<bool> foo; foo.Bar(); }
02338     //
02339     // to cause a compiler error.
02340
02341     template <typename T1, typename T2>
02342     constexpr bool StaticAssertTypeEq() noexcept {
02343         static_assert(std::is_same<T1, T2>::value, "T1 and T2 are not the same type");
02344         return true;
02345     }
02346
02347     // Defines a test.
02348     //
02349     // The first parameter is the name of the test suite, and the second
02350     // parameter is the name of the test within the test suite.
02351     //
02352     // The convention is to end the test suite name with "Test".  For
02353     // example, a test suite for the Foo class can be named FooTest.
02354     //
02355     // Test code should appear between braces after an invocation of
02356     // this macro.  Example:
02357     //
02358     //   TEST(FooTest, InitializesCorrectly) {
02359     //     Foo foo;
02360     //     EXPECT_TRUE(foo.StatusIsOK());
02361     //   }
02362
02363     // Note that we call GetTestTypeId() instead of GetTypeId<
02364     // ::testing::Test>() here to get the type ID of testing::Test.  This
02365     // is to work around a suspected linker bug when using Google Test as
02366     // a framework on Mac OS X.  The bug causes GetTypeId<
02367     // ::testing::Test>() to return different values depending on whether
02368     // the call is from the Google Test framework itself or from user test
02369     // code.  GetTestTypeId() is guaranteed to always return the same
02370     // value, as it always calls GetTypeId<>() from the Google Test
02371     // framework.
02372 #define GTEST_TEST(test_suite_name, test_name)             \
02373   GTEST_TEST_(test_suite_name, test_name, ::testing::Test, \
02374               ::testing::internal::GetTestTypeId())
02375
02376     // Define this macro to 1 to omit the definition of TEST(), which
02377     // is a generic name and clashes with some other libraries.
02378 #if !GTEST_DONT_DEFINE_TEST
02379 #define TEST(test_suite_name, test_name) GTEST_TEST(test_suite_name, test_name)
02380 #endif
02381
02382     // Defines a test that uses a test fixture.
02383     //
02384     // The first parameter is the name of the test fixture class, which
02385     // also doubles as the test suite name.  The second parameter is the
02386     // name of the test within the test suite.
02387     //
02388     // A test fixture class must be declared earlier.  The user should put
02389     // the test code between braces after using this macro.  Example:
02390     //
02391     //   class FooTest : public testing::Test {
02392     //    protected:
02393     //     void SetUp() override { b_.AddElement(3); }
02394     //
02395     //     Foo a_;
02396     //     Foo b_;
02397     //   };
```

```
02398     //
02399     //   TEST_F(FooTest, InitializesCorrectly) {
02400     //     EXPECT_TRUE(a_.StatusIsOK());
02401     //   }
02402     //
02403     //   TEST_F(FooTest, ReturnsElementCountCorrectly) {
02404     //     EXPECT_EQ(a_.size(), 0);
02405     //     EXPECT_EQ(b_.size(), 1);
02406     //   }
02407 #define GTEST_TEST_F(test_fixture, test_name)        \
02408   GTEST_TEST_(test_fixture, test_name, test_fixture, \
02409               ::testing::internal::GetTypeId<test_fixture>())
02410 #if !GTEST_DONT_DEFINE_TEST_F
02411 #define TEST_F(test_fixture, test_name) GTEST_TEST_F(test_fixture, test_name)
02412 #endif
02413
02414     // Returns a path to temporary directory.
02415     // Tries to determine an appropriate directory for the platform.
02416     GTEST_API_ std::string TempDir();
02417
02418 #ifdef _MSC_VER
02419 #pragma warning(pop)
02420 #endif
02421
02422     // Dynamically registers a test with the framework.
02423     //
02424     // This is an advanced API only to be used when the `TEST` macros are
02425     // insufficient. The macros should be preferred when possible, as they avoid
02426     // most of the complexity of calling this function.
02427     //
02428     // The `factory` argument is a factory callable (move-constructible) object or
02429     // function pointer that creates a new instance of the Test object. It
02430     // handles ownership to the caller. The signature of the callable is
02431     // `Fixture*()`, where `Fixture` is the test fixture class for the test. All
02432     // tests registered with the same `test_suite_name` must return the same
02433     // fixture type. This is checked at runtime.
02434     //
02435     // The framework will infer the fixture class from the factory and will call
02436     // the `SetUpTestSuite` and `TearDownTestSuite` for it.
02437     //
02438     // Must be called before `RUN_ALL_TESTS()` is invoked, otherwise behavior is
02439     // undefined.
02440     //
02441     // Use case example:
02442     //
02443     // class MyFixture : public ::testing::Test {
02444     //  public:
02445     //   // All of these optional, just like in regular macro usage.
02446     //   static void SetUpTestSuite() { ... }
02447     //   static void TearDownTestSuite() { ... }
02448     //   void SetUp() override { ... }
02449     //   void TearDown() override { ... }
02450     // };
02451     //
02452     // class MyTest : public MyFixture {
02453     //  public:
02454     //   explicit MyTest(int data) : data_(data) {}
02455     //   void TestBody() override { ... }
02456     //
02457     //  private:
02458     //   int data_;
02459     // };
02460     //
02461     // void RegisterMyTests(const std::vector<int>& values) {
02462     //   for (int v : values) {
02463     //     ::testing::RegisterTest(
02464     //         "MyFixture", ("Test" + std::to_string(v)).c_str(), nullptr,
02465     //         std::to_string(v).c_str(),
02466     //         __FILE__, __LINE__,
02467     //         // Important to use the fixture type as the return type here.
02468     //         [=]() -> MyFixture* { return new MyTest(v); });
02469     //   }
02470     // }
02471     // ...
02472     // int main(int argc, char** argv) {
02473     //   ::testing::InitGoogleTest(&argc, argv);
02474     //   std::vector<int> values_to_test = LoadValuesFromConfig();
02475     //   RegisterMyTests(values_to_test);
02476     //   ...
02477     //   return RUN_ALL_TESTS();
02478     // }
02479     //
02480
02481     template <int&... ExplicitParameterBarrier, typename Factory>
02482     TestInfo* RegisterTest(const char* test_suite_name, const char* test_name,
02483            const char* type_param, const char* value_param,
02484            const char* file, int line, Factory factory) {
```

```
02485         using TestT = typename std::remove_pointer<decltype(factory())>::type;
02486
02487         class FactoryImpl : public internal::TestFactoryBase {
02488         public:
02489
02490             explicit FactoryImpl(Factory f) : factory_(std::move(f)) {
02491             }
02492
02493             Test* CreateTest() override {
02494                 return factory_();
02495             }
02496
02497         private:
02498             Factory factory_;
02499         };
02500
02501         return internal::MakeAndRegisterTestInfo(
02502                 test_suite_name, test_name, type_param, value_param,
02503                 internal::CodeLocation(file, line), internal::GetTypeId<TestT>(),
02504                 internal::SuiteApiResolver<TestT>::GetSetUpCaseOrSuite(file, line),
02505                 internal::SuiteApiResolver<TestT>::GetTearDownCaseOrSuite(file, line),
02506                 new FactoryImpl{std::move(factory)});
02507     }
02508
02509 } // namespace testing
02510
02511 // Use this function in main() to run all tests.  It returns 0 if all
02512 // tests are successful, or 1 otherwise.
02513 //
02514 // RUN_ALL_TESTS() should be invoked after the command line has been
02515 // parsed by InitGoogleTest().
02516 //
02517 // This function was formerly a macro; thus, it is in the global
02518 // namespace and has an all-caps name.
02519 int RUN_ALL_TESTS() GTEST_MUST_USE_RESULT_;
02520
02521 inline int RUN_ALL_TESTS() {
02522     return ::testing::UnitTest::GetInstance()->Run();
02523 }
02524
02525 GTEST_DISABLE_MSC_WARNINGS_POP_() //  4251
02526
02527 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_H_
```

18.270 gtest.h

```
00001 // Copyright 2015, Google Inc.
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00026 // THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029 //
00030 // Injection point for custom user configurations. See README for details
00031 //
00032 // ** Custom implementation starts here **
00033
00034 #ifndef GOOGLETEST_INCLUDE_GTEST_INTERNAL_CUSTOM_GTEST_H_
00035 #define GOOGLETEST_INCLUDE_GTEST_INTERNAL_CUSTOM_GTEST_H_
00036
00037 #endif  // GOOGLETEST_INCLUDE_GTEST_INTERNAL_CUSTOM_GTEST_H_
```

18.271 gtest_pred_impl.h

```
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00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029 //
00030 // Implements a family of generic predicate assertion macros.
00031
00032 // IWYU pragma: private, include "gtest/gtest.h"
00033 // IWYU pragma: friend gtest/.*
00034 // IWYU pragma: friend gmock/.*
00035
00036 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_PRED_IMPL_H_
00037 #define GOOGLETEST_INCLUDE_GTEST_GTEST_PRED_IMPL_H_
00038
00039 #include "gtest/gtest-assertion-result.h"
00040 #include "gtest/internal/gtest-internal.h"
00041 #include "gtest/internal/gtest-port.h"
00042
00043 namespace testing {
00044
00045     // This header implements a family of generic predicate assertion
00046     // macros:
00047     //
00048     //   ASSERT_PRED_FORMAT1(pred_format, v1)
00049     //   ASSERT_PRED_FORMAT2(pred_format, v1, v2)
00050     //   ...
00051     //
00052     // where pred_format is a function or functor that takes n (in the
00053     // case of ASSERT_PRED_FORMATn) values and their source expression
00054     // text, and returns a testing::AssertionResult.  See the definition
00055     // of ASSERT_EQ in gtest.h for an example.
00056     //
00057     // If you don't care about formatting, you can use the more
00058     // restrictive version:
00059     //
00060     //   ASSERT_PRED1(pred, v1)
00061     //   ASSERT_PRED2(pred, v1, v2)
00062     //   ...
00063     //
00064     // where pred is an n-ary function or functor that returns bool,
00065     // and the values v1, v2, ..., must support the « operator for
00066     // streaming to std::ostream.
00067     //
00068     // We also define the EXPECT_* variations.
00069     //
00070     // For now we only support predicates whose arity is at most 5.
00071     // Please email googletestframework@googlegroups.com if you need
00072     // support for higher arities.
00073
00074     // GTEST_ASSERT_ is the basic statement to which all of the assertions
00075     // in this file reduce.  Don't use this in your code.
00076
00077 #define GTEST_ASSERT_(expression, on_failure)                  \
00078   GTEST_AMBIGUOUS_ELSE_BLOCKER_                               \
00079   if (const ::testing::AssertionResult gtest_ar = (expression)) \
00080     ;                                                         \
00081   else                                                        \
00082     on_failure(gtest_ar.failure_message())
00083
00084     // Helper function for implementing {EXPECT|ASSERT}_PRED1.  Don't use
00085     // this in your code.
```

```
00086
00087     template <typename Pred, typename T1>
00088     AssertionResult AssertPred1Helper(const char* pred_text, const char* e1,
00089             Pred pred, const T1& v1) {
00090         if (pred(v1)) return AssertionSuccess();
00091
00092         return AssertionFailure()
00093                 << pred_text << "(" << e1 << ") evaluates to false, where"
00094                 << "\n"
00095                 << e1 << " evaluates to " << ::testing::PrintToString(v1);
00096     }
00097
00098     // Internal macro for implementing {EXPECT|ASSERT}_PRED_FORMAT1.
00099     // Don't use this in your code.
00100 #define GTEST_PRED_FORMAT1_(pred_format, v1, on_failure) \
00101   GTEST_ASSERT_(pred_format(#v1, v1), on_failure)
00102
00103     // Internal macro for implementing {EXPECT|ASSERT}_PRED1.  Don't use
00104     // this in your code.
00105 #define GTEST_PRED1_(pred, v1, on_failure) \
00106   GTEST_ASSERT_(::testing::AssertPred1Helper(#pred, #v1, pred, v1), on_failure)
00107
00108     // Unary predicate assertion macros.
00109 #define EXPECT_PRED_FORMAT1(pred_format, v1) \
00110   GTEST_PRED_FORMAT1_(pred_format, v1, GTEST_NONFATAL_FAILURE_)
00111 #define EXPECT_PRED1(pred, v1) GTEST_PRED1_(pred, v1, GTEST_NONFATAL_FAILURE_)
00112 #define ASSERT_PRED_FORMAT1(pred_format, v1) \
00113   GTEST_PRED_FORMAT1_(pred_format, v1, GTEST_FATAL_FAILURE_)
00114 #define ASSERT_PRED1(pred, v1) GTEST_PRED1_(pred, v1, GTEST_FATAL_FAILURE_)
00115
00116     // Helper function for implementing {EXPECT|ASSERT}_PRED2.  Don't use
00117     // this in your code.
00118
00119     template <typename Pred, typename T1, typename T2>
00120     AssertionResult AssertPred2Helper(const char* pred_text, const char* e1,
00121             const char* e2, Pred pred, const T1& v1,
00122             const T2& v2) {
00123         if (pred(v1, v2)) return AssertionSuccess();
00124
00125         return AssertionFailure()
00126                 << pred_text << "(" << e1 << ", " << e2
00127                 << ") evaluates to false, where"
00128                 << "\n"
00129                 << e1 << " evaluates to " << ::testing::PrintToString(v1) << "\n"
00130                 << e2 << " evaluates to " << ::testing::PrintToString(v2);
00131     }
00132
00133     // Internal macro for implementing {EXPECT|ASSERT}_PRED_FORMAT2.
00134     // Don't use this in your code.
00135 #define GTEST_PRED_FORMAT2_(pred_format, v1, v2, on_failure) \
00136   GTEST_ASSERT_(pred_format(#v1, #v2, v1, v2), on_failure)
00137
00138     // Internal macro for implementing {EXPECT|ASSERT}_PRED2.  Don't use
00139     // this in your code.
00140 #define GTEST_PRED2_(pred, v1, v2, on_failure)                                 \
00141   GTEST_ASSERT_(::testing::AssertPred2Helper(#pred, #v1, #v2, pred, v1, v2), \
00142                 on_failure)
00143
00144     // Binary predicate assertion macros.
00145 #define EXPECT_PRED_FORMAT2(pred_format, v1, v2) \
00146   GTEST_PRED_FORMAT2_(pred_format, v1, v2, GTEST_NONFATAL_FAILURE_)
00147 #define EXPECT_PRED2(pred, v1, v2) \
00148   GTEST_PRED2_(pred, v1, v2, GTEST_NONFATAL_FAILURE_)
00149 #define ASSERT_PRED_FORMAT2(pred_format, v1, v2) \
00150   GTEST_PRED_FORMAT2_(pred_format, v1, v2, GTEST_FATAL_FAILURE_)
00151 #define ASSERT_PRED2(pred, v1, v2) \
00152   GTEST_PRED2_(pred, v1, v2, GTEST_FATAL_FAILURE_)
00153
00154     // Helper function for implementing {EXPECT|ASSERT}_PRED3.  Don't use
00155     // this in your code.
00156
00157     template <typename Pred, typename T1, typename T2, typename T3>
00158     AssertionResult AssertPred3Helper(const char* pred_text, const char* e1,
00159             const char* e2, const char* e3, Pred pred,
00160             const T1& v1, const T2& v2, const T3& v3) {
00161         if (pred(v1, v2, v3)) return AssertionSuccess();
00162
00163         return AssertionFailure()
00164                 << pred_text << "(" << e1 << ", " << e2 << ", " << e3
00165                 << ") evaluates to false, where"
00166                 << "\n"
00167                 << e1 << " evaluates to " << ::testing::PrintToString(v1) << "\n"
00168                 << e2 << " evaluates to " << ::testing::PrintToString(v2) << "\n"
00169                 << e3 << " evaluates to " << ::testing::PrintToString(v3);
00170     }
00171
00172     // Internal macro for implementing {EXPECT|ASSERT}_PRED_FORMAT3.
```

```
00173     // Don't use this in your code.
00174 #define GTEST_PRED_FORMAT3_(pred_format, v1, v2, v3, on_failure) \
00175   GTEST_ASSERT_(pred_format(#v1, #v2, #v3, v1, v2, v3), on_failure)
00176
00177     // Internal macro for implementing {EXPECT|ASSERT}_PRED3.  Don't use
00178     // this in your code.
00179 #define GTEST_PRED3_(pred, v1, v2, v3, on_failure)                            \
00180   GTEST_ASSERT_(                                                              \
00181       ::testing::AssertPred3Helper(#pred, #v1, #v2, #v3, pred, v1, v2, v3), \
00182       on_failure)
00183
00184     // Ternary predicate assertion macros.
00185 #define EXPECT_PRED_FORMAT3(pred_format, v1, v2, v3) \
00186   GTEST_PRED_FORMAT3_(pred_format, v1, v2, v3, GTEST_NONFATAL_FAILURE_)
00187 #define EXPECT_PRED3(pred, v1, v2, v3) \
00188   GTEST_PRED3_(pred, v1, v2, v3, GTEST_NONFATAL_FAILURE_)
00189 #define ASSERT_PRED_FORMAT3(pred_format, v1, v2, v3) \
00190   GTEST_PRED_FORMAT3_(pred_format, v1, v2, v3, GTEST_FATAL_FAILURE_)
00191 #define ASSERT_PRED3(pred, v1, v2, v3) \
00192   GTEST_PRED3_(pred, v1, v2, v3, GTEST_FATAL_FAILURE_)
00193
00194     // Helper function for implementing {EXPECT|ASSERT}_PRED4.  Don't use
00195     // this in your code.
00196
00197     template <typename Pred, typename T1, typename T2, typename T3, typename T4>
00198     AssertionResult AssertPred4Helper(const char* pred_text, const char* e1,
00199             const char* e2, const char* e3,
00200             const char* e4, Pred pred, const T1& v1,
00201             const T2& v2, const T3& v3, const T4& v4) {
00202         if (pred(v1, v2, v3, v4)) return AssertionSuccess();
00203
00204         return AssertionFailure()
00205                 << pred_text << "(" << e1 << ", " << e2 << ", " << e3 << ", " << e4
00206                 << ") evaluates to false, where"
00207                 << "\n"
00208                 << e1 << " evaluates to " << ::testing::PrintToString(v1) << "\n"
00209                 << e2 << " evaluates to " << ::testing::PrintToString(v2) << "\n"
00210                 << e3 << " evaluates to " << ::testing::PrintToString(v3) << "\n"
00211                 << e4 << " evaluates to " << ::testing::PrintToString(v4);
00212     }
00213
00214     // Internal macro for implementing {EXPECT|ASSERT}_PRED_FORMAT4.
00215     // Don't use this in your code.
00216 #define GTEST_PRED_FORMAT4_(pred_format, v1, v2, v3, v4, on_failure) \
00217   GTEST_ASSERT_(pred_format(#v1, #v2, #v3, #v4, v1, v2, v3, v4), on_failure)
00218
00219     // Internal macro for implementing {EXPECT|ASSERT}_PRED4.  Don't use
00220     // this in your code.
00221 #define GTEST_PRED4_(pred, v1, v2, v3, v4, on_failure)                        \
00222   GTEST_ASSERT_(::testing::AssertPred4Helper(#pred, #v1, #v2, #v3, #v4, pred, \
00223                                              v1, v2, v3, v4),                 \
00224                 on_failure)
00225
00226     // 4-ary predicate assertion macros.
00227 #define EXPECT_PRED_FORMAT4(pred_format, v1, v2, v3, v4) \
00228   GTEST_PRED_FORMAT4_(pred_format, v1, v2, v3, v4, GTEST_NONFATAL_FAILURE_)
00229 #define EXPECT_PRED4(pred, v1, v2, v3, v4) \
00230   GTEST_PRED4_(pred, v1, v2, v3, v4, GTEST_NONFATAL_FAILURE_)
00231 #define ASSERT_PRED_FORMAT4(pred_format, v1, v2, v3, v4) \
00232   GTEST_PRED_FORMAT4_(pred_format, v1, v2, v3, v4, GTEST_FATAL_FAILURE_)
00233 #define ASSERT_PRED4(pred, v1, v2, v3, v4) \
00234   GTEST_PRED4_(pred, v1, v2, v3, v4, GTEST_FATAL_FAILURE_)
00235
00236     // Helper function for implementing {EXPECT|ASSERT}_PRED5.  Don't use
00237     // this in your code.
00238
00239     template <typename Pred, typename T1, typename T2, typename T3, typename T4,
00240     typename T5>
00241     AssertionResult AssertPred5Helper(const char* pred_text, const char* e1,
00242             const char* e2, const char* e3,
00243             const char* e4, const char* e5, Pred pred,
00244             const T1& v1, const T2& v2, const T3& v3,
00245             const T4& v4, const T5& v5) {
00246         if (pred(v1, v2, v3, v4, v5)) return AssertionSuccess();
00247
00248         return AssertionFailure()
00249                 << pred_text << "(" << e1 << ", " << e2 << ", " << e3 << ", " << e4
00250                 << ", " << e5 << ") evaluates to false, where"
00251                 << "\n"
00252                 << e1 << " evaluates to " << ::testing::PrintToString(v1) << "\n"
00253                 << e2 << " evaluates to " << ::testing::PrintToString(v2) << "\n"
00254                 << e3 << " evaluates to " << ::testing::PrintToString(v3) << "\n"
00255                 << e4 << " evaluates to " << ::testing::PrintToString(v4) << "\n"
00256                 << e5 << " evaluates to " << ::testing::PrintToString(v5);
00257     }
00258
00259     // Internal macro for implementing {EXPECT|ASSERT}_PRED_FORMAT5.
```

```
00260     // Don't use this in your code.
00261 #define GTEST_PRED_FORMAT5_(pred_format, v1, v2, v3, v4, v5, on_failure)  \
00262   GTEST_ASSERT_(pred_format(#v1, #v2, #v3, #v4, #v5, v1, v2, v3, v4, v5), \
00263                 on_failure)
00264
00265     // Internal macro for implementing {EXPECT|ASSERT}_PRED5.  Don't use
00266     // this in your code.
00267 #define GTEST_PRED5_(pred, v1, v2, v3, v4, v5, on_failure)                    \
00268   GTEST_ASSERT_(::testing::AssertPred5Helper(#pred, #v1, #v2, #v3, #v4, #v5, \
00269                                              pred, v1, v2, v3, v4, v5),      \
00270                 on_failure)
00271
00272     // 5-ary predicate assertion macros.
00273 #define EXPECT_PRED_FORMAT5(pred_format, v1, v2, v3, v4, v5) \
00274   GTEST_PRED_FORMAT5_(pred_format, v1, v2, v3, v4, v5, GTEST_NONFATAL_FAILURE_)
00275 #define EXPECT_PRED5(pred, v1, v2, v3, v4, v5) \
00276   GTEST_PRED5_(pred, v1, v2, v3, v4, v5, GTEST_NONFATAL_FAILURE_)
00277 #define ASSERT_PRED_FORMAT5(pred_format, v1, v2, v3, v4, v5) \
00278   GTEST_PRED_FORMAT5_(pred_format, v1, v2, v3, v4, v5, GTEST_FATAL_FAILURE_)
00279 #define ASSERT_PRED5(pred, v1, v2, v3, v4, v5) \
00280   GTEST_PRED5_(pred, v1, v2, v3, v4, v5, GTEST_FATAL_FAILURE_)
00281
00282 } // namespace testing
00283
00284 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_PRED_IMPL_H_
```

18.272 gtest_prod.h

```
00001 // Copyright 2006, Google Inc.
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00003 //
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00005 // modification, are permitted provided that the following conditions are
00006 // met:
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00026 // THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // Google C++ Testing and Mocking Framework definitions useful in production
00031 // code.
00032
00033 #ifndef GOOGLETEST_INCLUDE_GTEST_GTEST_PROD_H_
00034 #define GOOGLETEST_INCLUDE_GTEST_GTEST_PROD_H_
00035
00036 // When you need to test the private or protected members of a class,
00037 // use the FRIEND_TEST macro to declare your tests as friends of the
00038 // class.  For example:
00039 //
00040 // class MyClass {
00041 //  private:
00042 //   void PrivateMethod();
00043 //   FRIEND_TEST(MyClassTest, PrivateMethodWorks);
00044 // };
00045 //
00046 // class MyClassTest : public testing::Test {
00047 //   // ...
00048 // };
00049 //
00050 // TEST_F(MyClassTest, PrivateMethodWorks) {
00051 //   // Can call MyClass::PrivateMethod() here.
00052 // }
00053 //
00054 // Note: The test class must be in the same namespace as the class being tested.
00055 // For example, putting MyClassTest in an anonymous namespace will not work.
00056
00057 #define FRIEND_TEST(test_case_name, test_name) \
```

```
00058   friend class test_case_name##_##test_name##_Test
00059
00060 #endif  // GOOGLETEST_INCLUDE_GTEST_GTEST_PROD_H_
```

18.273 gtest-port.h

```
00001 // Copyright 2015, Google Inc.
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00013 // distribution.
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00025 // DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY
00026 // THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029 //
00030 // Injection point for custom user configurations. See README for details
00031 //
00032 // ** Custom implementation starts here **
00033
00034 #ifndef GOOGLETEST_INCLUDE_GTEST_INTERNAL_CUSTOM_GTEST_PORT_H_
00035 #define GOOGLETEST_INCLUDE_GTEST_INTERNAL_CUSTOM_GTEST_PORT_H_
00036
00037 #endif  // GOOGLETEST_INCLUDE_GTEST_INTERNAL_CUSTOM_GTEST_PORT_H_
```

18.274 gtest-port.h

```
00001 // Copyright 2005, Google Inc.
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00026 // THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // Low-level types and utilities for porting Google Test to various
00031 // platforms.  All macros ending with _ and symbols defined in an
00032 // internal namespace are subject to change without notice.  Code
00033 // outside Google Test MUST NOT USE THEM DIRECTLY.  Macros that don't
00034 // end with _ are part of Google Test's public API and can be used by
00035 // code outside Google Test.
00036 //
```

```
00037 // This file is fundamental to Google Test.  All other Google Test source
00038 // files are expected to #include this.  Therefore, it cannot #include
00039 // any other Google Test header.
00040
00041 // IWYU pragma: private, include "gtest/gtest.h"
00042 // IWYU pragma: friend gtest/.*
00043 // IWYU pragma: friend gmock/.*
00044
00045 #ifndef GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_PORT_H_
00046 #define GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_PORT_H_
00047
00048 // Environment-describing macros
00049 // -----------------------------
00050 //
00051 // Google Test can be used in many different environments.  Macros in
00052 // this section tell Google Test what kind of environment it is being
00053 // used in, such that Google Test can provide environment-specific
00054 // features and implementations.
00055 //
00056 // Google Test tries to automatically detect the properties of its
00057 // environment, so users usually don't need to worry about these
00058 // macros.  However, the automatic detection is not perfect.
00059 // Sometimes it's necessary for a user to define some of the following
00060 // macros in the build script to override Google Test's decisions.
00061 //
00062 // If the user doesn't define a macro in the list, Google Test will
00063 // provide a default definition.  After this header is #included, all
00064 // macros in this list will be defined to either 1 or 0.
00065 //
00066 // Notes to maintainers:
00067 //   - Each macro here is a user-tweakable knob; do not grow the list
00068 //     lightly.
00069 //   - Use #if to key off these macros.  Don't use #ifdef or "#if
00070 //     defined(...)", which will not work as these macros are ALWAYS
00071 //     defined.
00072 //
00073 //   GTEST_HAS_CLONE          - Define it to 1/0 to indicate that clone(2)
00074 //                              is/isn't available.
00075 //   GTEST_HAS_EXCEPTIONS     - Define it to 1/0 to indicate that exceptions
00076 //                              are enabled.
00077 //   GTEST_HAS_POSIX_RE       - Define it to 1/0 to indicate that POSIX regular
00078 //                              expressions are/aren't available.
00079 //   GTEST_HAS_PTHREAD        - Define it to 1/0 to indicate that <pthread.h>
00080 //                              is/isn't available.
00081 //   GTEST_HAS_RTTI           - Define it to 1/0 to indicate that RTTI is/isn't
00082 //                              enabled.
00083 //   GTEST_HAS_STD_WSTRING    - Define it to 1/0 to indicate that
00084 //                              std::wstring does/doesn't work (Google Test can
00085 //                              be used where std::wstring is unavailable).
00086 //   GTEST_HAS_SEH            - Define it to 1/0 to indicate whether the
00087 //                              compiler supports Microsoft's "Structured
00088 //                              Exception Handling".
00089 //   GTEST_HAS_STREAM_REDIRECTION
00090 //                            - Define it to 1/0 to indicate whether the
00091 //                              platform supports I/O stream redirection using
00092 //                              dup() and dup2().
00093 //   GTEST_LINKED_AS_SHARED_LIBRARY
00094 //                            - Define to 1 when compiling tests that use
00095 //                              Google Test as a shared library (known as
00096 //                              DLL on Windows).
00097 //   GTEST_CREATE_SHARED_LIBRARY
00098 //                            - Define to 1 when compiling Google Test itself
00099 //                              as a shared library.
00100 //   GTEST_DEFAULT_DEATH_TEST_STYLE
00101 //                            - The default value of --gtest_death_test_style.
00102 //                              The legacy default has been "fast" in the open
00103 //                              source version since 2008. The recommended value
00104 //                              is "threadsafe", and can be set in
00105 //                              custom/gtest-port.h.
00106
00107 // Platform-indicating macros
00108 // --------------------------
00109 //
00110 // Macros indicating the platform on which Google Test is being used
00111 // (a macro is defined to 1 if compiled on the given platform;
00112 // otherwise UNDEFINED -- it's never defined to 0.).  Google Test
00113 // defines these macros automatically.  Code outside Google Test MUST
00114 // NOT define them.
00115 //
00116 //   GTEST_OS_AIX      - IBM AIX
00117 //   GTEST_OS_CYGWIN   - Cygwin
00118 //   GTEST_OS_DRAGONFLY - DragonFlyBSD
00119 //   GTEST_OS_FREEBSD  - FreeBSD
00120 //   GTEST_OS_FUCHSIA  - Fuchsia
00121 //   GTEST_OS_GNU_HURD - GNU/Hurd
00122 //   GTEST_OS_GNU_KFREEBSD - GNU/kFreeBSD
00123 //   GTEST_OS_HAIKU    - Haiku
```

```
00124 //   GTEST_OS_HPUX     - HP-UX
00125 //   GTEST_OS_LINUX    - Linux
00126 //     GTEST_OS_LINUX_ANDROID - Google Android
00127 //   GTEST_OS_MAC      - Mac OS X
00128 //     GTEST_OS_IOS    - iOS
00129 //   GTEST_OS_NACL     - Google Native Client (NaCl)
00130 //   GTEST_OS_NETBSD   - NetBSD
00131 //   GTEST_OS_OPENBSD  - OpenBSD
00132 //   GTEST_OS_OS2      - OS/2
00133 //   GTEST_OS_QNX      - QNX
00134 //   GTEST_OS_SOLARIS  - Sun Solaris
00135 //   GTEST_OS_WINDOWS  - Windows (Desktop, MinGW, or Mobile)
00136 //     GTEST_OS_WINDOWS_DESKTOP  - Windows Desktop
00137 //     GTEST_OS_WINDOWS_MINGW    - MinGW
00138 //     GTEST_OS_WINDOWS_MOBILE   - Windows Mobile
00139 //     GTEST_OS_WINDOWS_PHONE    - Windows Phone
00140 //     GTEST_OS_WINDOWS_RT       - Windows Store App/WinRT
00141 //   GTEST_OS_ZOS      - z/OS
00142 //
00143 // Among the platforms, Cygwin, Linux, Mac OS X, and Windows have the
00144 // most stable support.  Since core members of the Google Test project
00145 // don't have access to other platforms, support for them may be less
00146 // stable.  If you notice any problems on your platform, please notify
00147 // googletestframework@googlegroups.com (patches for fixing them are
00148 // even more welcome!).
00149 //
00150 // It is possible that none of the GTEST_OS_* macros are defined.
00151
00152 // Feature-indicating macros
00153 // -------------------------
00154 //
00155 // Macros indicating which Google Test features are available (a macro
00156 // is defined to 1 if the corresponding feature is supported;
00157 // otherwise UNDEFINED -- it's never defined to 0.).  Google Test
00158 // defines these macros automatically.  Code outside Google Test MUST
00159 // NOT define them.
00160 //
00161 // These macros are public so that portable tests can be written.
00162 // Such tests typically surround code using a feature with an #if
00163 // which controls that code.  For example:
00164 //
00165 // #if GTEST_HAS_DEATH_TEST
00166 //   EXPECT_DEATH(DoSomethingDeadly());
00167 // #endif
00168 //
00169 //   GTEST_HAS_DEATH_TEST   - death tests
00170 //   GTEST_HAS_TYPED_TEST   - typed tests
00171 //   GTEST_HAS_TYPED_TEST_P - type-parameterized tests
00172 //   GTEST_IS_THREADSAFE    - Google Test is thread-safe.
00173 //   GTEST_USES_RE2         - the RE2 regular expression library is used
00174 //   GTEST_USES_POSIX_RE    - enhanced POSIX regex is used. Do not confuse with
00175 //                            GTEST_HAS_POSIX_RE (see above) which users can
00176 //                            define themselves.
00177 //   GTEST_USES_SIMPLE_RE   - our own simple regex is used;
00178 //                            the above RE\b(s) are mutually exclusive.
00179
00180 // Misc public macros
00181 // ------------------
00182 //
00183 //   GTEST_FLAG(flag_name)  - references the variable corresponding to
00184 //                            the given Google Test flag.
00185
00186 // Internal utilities
00187 // ------------------
00188 //
00189 // The following macros and utilities are for Google Test's INTERNAL
00190 // use only.  Code outside Google Test MUST NOT USE THEM DIRECTLY.
00191 //
00192 // Macros for basic C++ coding:
00193 //   GTEST_AMBIGUOUS_ELSE_BLOCKER_ - for disabling a gcc warning.
00194 //   GTEST_ATTRIBUTE_UNUSED_  - declares that a class' instances or a
00195 //                              variable don't have to be used.
00196 //   GTEST_MUST_USE_RESULT_   - declares that a function's result must be used.
00197 //   GTEST_INTENTIONAL_CONST_COND_PUSH_ - start code section where MSVC C4127 is
00198 //                                       suppressed (constant conditional).
00199 //   GTEST_INTENTIONAL_CONST_COND_POP_  - finish code section where MSVC C4127
00200 //                                       is suppressed.
00201 //   GTEST_INTERNAL_HAS_ANY - for enabling UniversalPrinter<std::any> or
00202 //                            UniversalPrinter<absl::any> specializations.
00203 //   GTEST_INTERNAL_HAS_OPTIONAL - for enabling UniversalPrinter<std::optional>
00204 //   or
00205 //                                 UniversalPrinter<absl::optional>
00206 //                                 specializations.
00207 //   GTEST_INTERNAL_HAS_STRING_VIEW - for enabling Matcher<std::string_view> or
00208 //                                    Matcher<absl::string_view>
00209 //                                    specializations.
00210 //   GTEST_INTERNAL_HAS_VARIANT - for enabling UniversalPrinter<std::variant> or
```

```
00211 //                               UniversalPrinter<absl::variant>
00212 //                               specializations.
00213 //
00214 // Synchronization:
00215 //   Mutex, MutexLock, ThreadLocal, GetThreadCount()
00216 //                           - synchronization primitives.
00217 //
00218 // Regular expressions:
00219 //   RE             - a simple regular expression class using
00220 //                      1) the RE2 syntax on all platforms when built with RE2
00221 //                         and Abseil as dependencies
00222 //                      2) the POSIX Extended Regular Expression syntax on
00223 //                         UNIX-like platforms,
00224 //                      3) A reduced regular exception syntax on other platforms,
00225 //                         including Windows.
00226 // Logging:
00227 //   GTEST_LOG_()   - logs messages at the specified severity level.
00228 //   LogToStderr()  - directs all log messages to stderr.
00229 //   FlushInfoLog() - flushes informational log messages.
00230 //
00231 // Stdout and stderr capturing:
00232 //   CaptureStdout()     - starts capturing stdout.
00233 //   GetCapturedStdout() - stops capturing stdout and returns the captured
00234 //                         string.
00235 //   CaptureStderr()     - starts capturing stderr.
00236 //   GetCapturedStderr() - stops capturing stderr and returns the captured
00237 //                         string.
00238 //
00239 // Integer types:
00240 //   TypeWithSize   - maps an integer to a int type.
00241 //   TimeInMillis   - integers of known sizes.
00242 //   BiggestInt     - the biggest signed integer type.
00243 //
00244 // Command-line utilities:
00245 //   GetInjectableArgvs() - returns the command line as a vector of strings.
00246 //
00247 // Environment variable utilities:
00248 //   GetEnv()             - gets the value of an environment variable.
00249 //   BoolFromGTestEnv()   - parses a bool environment variable.
00250 //   Int32FromGTestEnv()  - parses an int32_t environment variable.
00251 //   StringFromGTestEnv() - parses a string environment variable.
00252 //
00253 // Deprecation warnings:
00254 //   GTEST_INTERNAL_DEPRECATED(message) - attribute marking a function as
00255 //                                        deprecated; calling a marked function
00256 //                                        should generate a compiler warning
00257
00258 #include <ctype.h>   // for isspace, etc
00259 #include <stddef.h>  // for ptrdiff_t
00260 #include <stdio.h>
00261 #include <stdlib.h>
00262 #include <string.h>
00263
00264 #include <cerrno>
00265 // #include <condition_variable>  // Guarded by GTEST_IS_THREADSAFE below
00266 #include <cstdint>
00267 #include <iostream>
00268 #include <limits>
00269 #include <locale>
00270 #include <memory>
00271 #include <string>
00272 // #include <mutex>  // Guarded by GTEST_IS_THREADSAFE below
00273 #include <tuple>
00274 #include <type_traits>
00275 #include <vector>
00276
00277 #ifndef _WIN32_WCE
00278 #include <sys/stat.h>
00279 #include <sys/types.h>
00280 #endif  // !_WIN32_WCE
00281
00282 #if defined __APPLE__
00283 #include <AvailabilityMacros.h>
00284 #include <TargetConditionals.h>
00285 #endif
00286
00287 #include "gtest/internal/custom/gtest-port.h"
00288 #include "gtest/internal/gtest-port-arch.h"
00289
00290 #if GTEST_HAS_ABSL
00291 #include "absl/flags/declare.h"
00292 #include "absl/flags/flag.h"
00293 #include "absl/flags/reflection.h"
00294 #endif
00295
00296 #if !defined(GTEST_DEV_EMAIL_)
00297 #define GTEST_DEV_EMAIL_ "googletestframework@@googlegroups.com"
```

```
00298 #define GTEST_FLAG_PREFIX_ "gtest_"
00299 #define GTEST_FLAG_PREFIX_DASH_ "gtest-"
00300 #define GTEST_FLAG_PREFIX_UPPER_ "GTEST_"
00301 #define GTEST_NAME_ "Google Test"
00302 #define GTEST_PROJECT_URL_ "https://github.com/google/googletest/"
00303 #endif  // !defined(GTEST_DEV_EMAIL_)
00304
00305 #if !defined(GTEST_INIT_GOOGLE_TEST_NAME_)
00306 #define GTEST_INIT_GOOGLE_TEST_NAME_ "testing::InitGoogleTest"
00307 #endif  // !defined(GTEST_INIT_GOOGLE_TEST_NAME_)
00308
00309 // Determines the version of gcc that is used to compile this.
00310 #ifdef __GNUC__
00311 // 40302 means version 4.3.2.
00312 #define GTEST_GCC_VER_ \
00313   (__GNUC__ * 10000 + __GNUC_MINOR__ * 100 + __GNUC_PATCHLEVEL__)
00314 #endif  // __GNUC__
00315
00316 // Macros for disabling Microsoft Visual C++ warnings.
00317 //
00318 //   GTEST_DISABLE_MSC_WARNINGS_PUSH_(4800 4385)
00319 //   /* code that triggers warnings C4800 and C4385 */
00320 //   GTEST_DISABLE_MSC_WARNINGS_POP_()
00321 #if defined(_MSC_VER)
00322 #define GTEST_DISABLE_MSC_WARNINGS_PUSH_(warnings) \
00323   __pragma(warning(push)) __pragma(warning(disable : warnings))
00324 #define GTEST_DISABLE_MSC_WARNINGS_POP_() __pragma(warning(pop))
00325 #else
00326 // Not all compilers are MSVC
00327 #define GTEST_DISABLE_MSC_WARNINGS_PUSH_(warnings)
00328 #define GTEST_DISABLE_MSC_WARNINGS_POP_()
00329 #endif
00330
00331 // Clang on Windows does not understand MSVC's pragma warning.
00332 // We need clang-specific way to disable function deprecation warning.
00333 #ifdef __clang__
00334 #define GTEST_DISABLE_MSC_DEPRECATED_PUSH_()                                \
00335   _Pragma("clang diagnostic push")                                          \
00336       _Pragma("clang diagnostic ignored \"-Wdeprecated-declarations\"") \
00337           _Pragma("clang diagnostic ignored \"-Wdeprecated-implementations\"")
00338 #define GTEST_DISABLE_MSC_DEPRECATED_POP_() _Pragma("clang diagnostic pop")
00339 #else
00340 #define GTEST_DISABLE_MSC_DEPRECATED_PUSH_() \
00341   GTEST_DISABLE_MSC_WARNINGS_PUSH_(4996)
00342 #define GTEST_DISABLE_MSC_DEPRECATED_POP_() GTEST_DISABLE_MSC_WARNINGS_POP_()
00343 #endif
00344
00345 // Brings in definitions for functions used in the testing::internal::posix
00346 // namespace (read, write, close, chdir, isatty, stat). We do not currently
00347 // use them on Windows Mobile.
00348 #if GTEST_OS_WINDOWS
00349 #if !GTEST_OS_WINDOWS_MOBILE
00350 #include <direct.h>
00351 #include <io.h>
00352 #endif
00353 // In order to avoid having to include <windows.h>, use forward declaration
00354 #if GTEST_OS_WINDOWS_MINGW && !defined(__MINGW64_VERSION_MAJOR)
00355 // MinGW defined _CRITICAL_SECTION and _RTL_CRITICAL_SECTION as two
00356 // separate (equivalent) structs, instead of using typedef
00357 typedef struct _CRITICAL_SECTION GTEST_CRITICAL_SECTION;
00358 #else
00359 // Assume CRITICAL_SECTION is a typedef of _RTL_CRITICAL_SECTION.
00360 // This assumption is verified by
00361 // WindowsTypesTest.CRITICAL_SECTIONIs_RTL_CRITICAL_SECTION.
00362 typedef struct _RTL_CRITICAL_SECTION GTEST_CRITICAL_SECTION;
00363 #endif
00364 #elif GTEST_OS_XTENSA
00365 #include <unistd.h>
00366 // Xtensa toolchains define strcasecmp in the string.h header instead of
00367 // strings.h. string.h is already included.
00368 #else
00369 // This assumes that non-Windows OSes provide unistd.h. For OSes where this
00370 // is not the case, we need to include headers that provide the functions
00371 // mentioned above.
00372 #include <strings.h>
00373 #include <unistd.h>
00374 #endif  // GTEST_OS_WINDOWS
00375
00376 #if GTEST_OS_LINUX_ANDROID
00377 // Used to define __ANDROID_API__ matching the target NDK API level.
00378 #include <android/api-level.h>  // NOLINT
00379 #endif
00380
00381 // Defines this to true if and only if Google Test can use POSIX regular
00382 // expressions.
00383 #ifndef GTEST_HAS_POSIX_RE
00384 #if GTEST_OS_LINUX_ANDROID
```

```
00385 // On Android, <regex.h> is only available starting with Gingerbread.
00386 #define GTEST_HAS_POSIX_RE (__ANDROID_API__ >= 9)
00387 #else
00388 #define GTEST_HAS_POSIX_RE (!GTEST_OS_WINDOWS && !GTEST_OS_XTENSA)
00389 #endif
00390 #endif
00391
00392 // Select the regular expression implementation.
00393 #if GTEST_HAS_ABSL
00394 // When using Abseil, RE2 is required.
00395 #include "absl/strings/string_view.h"
00396 #include "re2/re2.h"
00397 #define GTEST_USES_RE2 1
00398 #elif GTEST_HAS_POSIX_RE
00399 #include <regex.h>  // NOLINT
00400 #define GTEST_USES_POSIX_RE 1
00401 #else
00402 // Use our own simple regex implementation.
00403 #define GTEST_USES_SIMPLE_RE 1
00404 #endif
00405
00406 #ifndef GTEST_HAS_EXCEPTIONS
00407 // The user didn't tell us whether exceptions are enabled, so we need
00408 // to figure it out.
00409 #if defined(_MSC_VER) && defined(_CPPUNWIND)
00410 // MSVC defines _CPPUNWIND to 1 if and only if exceptions are enabled.
00411 #define GTEST_HAS_EXCEPTIONS 1
00412 #elif defined(__BORLANDC__)
00413 // C++Builder's implementation of the STL uses the _HAS_EXCEPTIONS
00414 // macro to enable exceptions, so we'll do the same.
00415 // Assumes that exceptions are enabled by default.
00416 #ifndef _HAS_EXCEPTIONS
00417 #define _HAS_EXCEPTIONS 1
00418 #endif  // _HAS_EXCEPTIONS
00419 #define GTEST_HAS_EXCEPTIONS _HAS_EXCEPTIONS
00420 #elif defined(__clang__)
00421 // clang defines __EXCEPTIONS if and only if exceptions are enabled before clang
00422 // 220714, but if and only if cleanups are enabled after that. In Obj-C++ files,
00423 // there can be cleanups for ObjC exceptions which also need cleanups, even if
00424 // C++ exceptions are disabled. clang has __has_feature(cxx_exceptions) which
00425 // checks for C++ exceptions starting at clang r206352, but which checked for
00426 // cleanups prior to that. To reliably check for C++ exception availability with
00427 // clang, check for
00428 // __EXCEPTIONS && __has_feature(cxx_exceptions).
00429 #define GTEST_HAS_EXCEPTIONS (__EXCEPTIONS && __has_feature(cxx_exceptions))
00430 #elif defined(__GNUC__) && __EXCEPTIONS
00431 // gcc defines __EXCEPTIONS to 1 if and only if exceptions are enabled.
00432 #define GTEST_HAS_EXCEPTIONS 1
00433 #elif defined(__SUNPRO_CC)
00434 // Sun Pro CC supports exceptions.  However, there is no compile-time way of
00435 // detecting whether they are enabled or not.  Therefore, we assume that
00436 // they are enabled unless the user tells us otherwise.
00437 #define GTEST_HAS_EXCEPTIONS 1
00438 #elif defined(__IBMCPP__) && __EXCEPTIONS
00439 // xlC defines __EXCEPTIONS to 1 if and only if exceptions are enabled.
00440 #define GTEST_HAS_EXCEPTIONS 1
00441 #elif defined(__HP_aCC)
00442 // Exception handling is in effect by default in HP aCC compiler. It has to
00443 // be turned of by +noeh compiler option if desired.
00444 #define GTEST_HAS_EXCEPTIONS 1
00445 #else
00446 // For other compilers, we assume exceptions are disabled to be
00447 // conservative.
00448 #define GTEST_HAS_EXCEPTIONS 0
00449 #endif  // defined(_MSC_VER) || defined(__BORLANDC__)
00450 #endif  // GTEST_HAS_EXCEPTIONS
00451
00452 #ifndef GTEST_HAS_STD_WSTRING
00453 // The user didn't tell us whether ::std::wstring is available, so we need
00454 // to figure it out.
00455 // Cygwin 1.7 and below doesn't support ::std::wstring.
00456 // Solaris' libc++ doesn't support it either.  Android has
00457 // no support for it at least as recent as Froyo (2.2).
00458 #define GTEST_HAS_STD_WSTRING                                     \
00459   (!(GTEST_OS_LINUX_ANDROID || GTEST_OS_CYGWIN || GTEST_OS_SOLARIS || \
00460      GTEST_OS_HAIKU || GTEST_OS_ESP32 || GTEST_OS_ESP8266 || GTEST_OS_XTENSA))
00461
00462 #endif  // GTEST_HAS_STD_WSTRING
00463
00464 // Determines whether RTTI is available.
00465 #ifndef GTEST_HAS_RTTI
00466 // The user didn't tell us whether RTTI is enabled, so we need to
00467 // figure it out.
00468
00469 #ifdef _MSC_VER
00470
00471 #ifdef _CPPRTTI  // MSVC defines this macro if and only if RTTI is enabled.
```

```
00472 #define GTEST_HAS_RTTI 1
00473 #else
00474 #define GTEST_HAS_RTTI 0
00475 #endif
00476
00477 // Starting with version 4.3.2, gcc defines __GXX_RTTI if and only if RTTI is
00478 // enabled.
00479 #elif defined(__GNUC__)
00480
00481 #ifdef __GXX_RTTI
00482 // When building against STLport with the Android NDK and with
00483 // -frtti -fno-exceptions, the build fails at link time with undefined
00484 // references to __cxa_bad_typeid. Note sure if STL or toolchain bug,
00485 // so disable RTTI when detected.
00486 #if GTEST_OS_LINUX_ANDROID && defined(_STLPORT_MAJOR) && !defined(__EXCEPTIONS)
00487 #define GTEST_HAS_RTTI 0
00488 #else
00489 #define GTEST_HAS_RTTI 1
00490 #endif  // GTEST_OS_LINUX_ANDROID && __STLPORT_MAJOR && !__EXCEPTIONS
00491 #else
00492 #define GTEST_HAS_RTTI 0
00493 #endif  // __GXX_RTTI
00494
00495 // Clang defines __GXX_RTTI starting with version 3.0, but its manual recommends
00496 // using has_feature instead. has_feature(cxx_rtti) is supported since 2.7, the
00497 // first version with C++ support.
00498 #elif defined(__clang__)
00499
00500 #define GTEST_HAS_RTTI __has_feature(cxx_rtti)
00501
00502 // Starting with version 9.0 IBM Visual Age defines __RTTI_ALL__ to 1 if
00503 // both the typeid and dynamic_cast features are present.
00504 #elif defined(__IBMCPP__) && (__IBMCPP__ >= 900)
00505
00506 #ifdef __RTTI_ALL__
00507 #define GTEST_HAS_RTTI 1
00508 #else
00509 #define GTEST_HAS_RTTI 0
00510 #endif
00511
00512 #else
00513
00514 // For all other compilers, we assume RTTI is enabled.
00515 #define GTEST_HAS_RTTI 1
00516
00517 #endif  // _MSC_VER
00518
00519 #endif  // GTEST_HAS_RTTI
00520
00521 // It's this header's responsibility to #include <typeinfo> when RTTI
00522 // is enabled.
00523 #if GTEST_HAS_RTTI
00524 #include <typeinfo>
00525 #endif
00526
00527 // Determines whether Google Test can use the pthreads library.
00528 #ifndef GTEST_HAS_PTHREAD
00529 // The user didn't tell us explicitly, so we make reasonable assumptions about
00530 // which platforms have pthreads support.
00531 //
00532 // To disable threading support in Google Test, add -DGTEST_HAS_PTHREAD=0
00533 // to your compiler flags.
00534 #define GTEST_HAS_PTHREAD                                                   \
00535   (GTEST_OS_LINUX || GTEST_OS_MAC || GTEST_OS_HPUX || GTEST_OS_QNX ||           \
00536    GTEST_OS_FREEBSD || GTEST_OS_NACL || GTEST_OS_NETBSD || GTEST_OS_FUCHSIA || \
00537    GTEST_OS_DRAGONFLY || GTEST_OS_GNU_KFREEBSD || GTEST_OS_OPENBSD ||           \
00538    GTEST_OS_HAIKU || GTEST_OS_GNU_HURD)
00539 #endif  // GTEST_HAS_PTHREAD
00540
00541 #if GTEST_HAS_PTHREAD
00542 // gtest-port.h guarantees to #include <pthread.h> when GTEST_HAS_PTHREAD is
00543 // true.
00544 #include <pthread.h>  // NOLINT
00545
00546 // For timespec and nanosleep, used below.
00547 #include <time.h>  // NOLINT
00548 #endif
00549
00550 // Determines whether clone(2) is supported.
00551 // Usually it will only be available on Linux, excluding
00552 // Linux on the Itanium architecture.
00553 // Also see http://linux.die.net/man/2/clone.
00554 #ifndef GTEST_HAS_CLONE
00555 // The user didn't tell us, so we need to figure it out.
00556
00557 #if GTEST_OS_LINUX && !defined(__ia64__)
00558 #if GTEST_OS_LINUX_ANDROID
```

```
00559 // On Android, clone() became available at different API levels for each 32-bit
00560 // architecture.
00561 #if defined(__LP64__) || (defined(__arm__) && __ANDROID_API__ >= 9) || \
00562     (defined(__mips__) && __ANDROID_API__ >= 12) ||                    \
00563     (defined(__i386__) && __ANDROID_API__ >= 17)
00564 #define GTEST_HAS_CLONE 1
00565 #else
00566 #define GTEST_HAS_CLONE 0
00567 #endif
00568 #else
00569 #define GTEST_HAS_CLONE 1
00570 #endif
00571 #else
00572 #define GTEST_HAS_CLONE 0
00573 #endif  // GTEST_OS_LINUX && !defined(__ia64__)
00574
00575 #endif  // GTEST_HAS_CLONE
00576
00577 // Determines whether to support stream redirection. This is used to test
00578 // output correctness and to implement death tests.
00579 #ifndef GTEST_HAS_STREAM_REDIRECTION
00580 // By default, we assume that stream redirection is supported on all
00581 // platforms except known mobile ones.
00582 #if GTEST_OS_WINDOWS_MOBILE || GTEST_OS_WINDOWS_PHONE || \
00583     GTEST_OS_WINDOWS_RT || GTEST_OS_ESP8266 || GTEST_OS_XTENSA
00584 #define GTEST_HAS_STREAM_REDIRECTION 0
00585 #else
00586 #define GTEST_HAS_STREAM_REDIRECTION 1
00587 #endif  // !GTEST_OS_WINDOWS_MOBILE
00588 #endif  // GTEST_HAS_STREAM_REDIRECTION
00589
00590 // Determines whether to support death tests.
00591 // pops up a dialog window that cannot be suppressed programmatically.
00592 #if (GTEST_OS_LINUX || GTEST_OS_CYGWIN || GTEST_OS_SOLARIS ||             \
00593      (GTEST_OS_MAC && !GTEST_OS_IOS) ||                                   \
00594      (GTEST_OS_WINDOWS_DESKTOP && _MSC_VER) || GTEST_OS_WINDOWS_MINGW ||  \
00595      GTEST_OS_AIX || GTEST_OS_HPUX || GTEST_OS_OPENBSD || GTEST_OS_QNX || \
00596      GTEST_OS_FREEBSD || GTEST_OS_NETBSD || GTEST_OS_FUCHSIA ||           \
00597      GTEST_OS_DRAGONFLY || GTEST_OS_GNU_KFREEBSD || GTEST_OS_HAIKU ||     \
00598      GTEST_OS_GNU_HURD)
00599 #define GTEST_HAS_DEATH_TEST 1
00600 #endif
00601
00602 // Determines whether to support type-driven tests.
00603
00604 // Typed tests need <typeinfo> and variadic macros, which GCC, VC++ 8.0,
00605 // Sun Pro CC, IBM Visual Age, and HP aCC support.
00606 #if defined(__GNUC__) || defined(_MSC_VER) || defined(__SUNPRO_CC) || \
00607     defined(__IBMCPP__) || defined(__HP_aCC)
00608 #define GTEST_HAS_TYPED_TEST 1
00609 #define GTEST_HAS_TYPED_TEST_P 1
00610 #endif
00611
00612 // Determines whether the system compiler uses UTF-16 for encoding wide strings.
00613 #define GTEST_WIDE_STRING_USES_UTF16_ \
00614   (GTEST_OS_WINDOWS || GTEST_OS_CYGWIN || GTEST_OS_AIX || GTEST_OS_OS2)
00615
00616 // Determines whether test results can be streamed to a socket.
00617 #if GTEST_OS_LINUX || GTEST_OS_GNU_KFREEBSD || GTEST_OS_DRAGONFLY || \
00618     GTEST_OS_FREEBSD || GTEST_OS_NETBSD || GTEST_OS_OPENBSD ||       \
00619     GTEST_OS_GNU_HURD
00620 #define GTEST_CAN_STREAM_RESULTS_ 1
00621 #endif
00622
00623 // Defines some utility macros.
00624
00625 // The GNU compiler emits a warning if nested "if" statements are followed by
00626 // an "else" statement and braces are not used to explicitly disambiguate the
00627 // "else" binding.  This leads to problems with code like:
00628 //
00629 //   if (gate)
00630 //     ASSERT_*(condition) « "Some message";
00631 //
00632 // The "switch (0) case 0:" idiom is used to suppress this.
00633 #ifdef __INTEL_COMPILER
00634 #define GTEST_AMBIGUOUS_ELSE_BLOCKER_
00635 #else
00636 #define GTEST_AMBIGUOUS_ELSE_BLOCKER_ \
00637   switch (0)                          \
00638   case 0:                             \
00639   default:  // NOLINT
00640 #endif
00641
00642 // Use this annotation at the end of a struct/class definition to
00643 // prevent the compiler from optimizing away instances that are never
00644 // used.  This is useful when all interesting logic happens inside the
00645 // c'tor and / or d'tor.  Example:
```

```
00646 //
00647 //   struct Foo {
00648 //     Foo() { ... }
00649 //   } GTEST_ATTRIBUTE_UNUSED_;
00650 //
00651 // Also use it after a variable or parameter declaration to tell the
00652 // compiler the variable/parameter does not have to be used.
00653 #if defined(__GNUC__) && !defined(COMPILER_ICC)
00654 #define GTEST_ATTRIBUTE_UNUSED_ __attribute__((unused))
00655 #elif defined(__clang__)
00656 #if __has_attribute(unused)
00657 #define GTEST_ATTRIBUTE_UNUSED_ __attribute__((unused))
00658 #endif
00659 #endif
00660 #ifndef GTEST_ATTRIBUTE_UNUSED_
00661 #define GTEST_ATTRIBUTE_UNUSED_
00662 #endif
00663
00664 // Use this annotation before a function that takes a printf format string.
00665 #if (defined(__GNUC__) || defined(__clang__)) && !defined(COMPILER_ICC)
00666 #if defined(__MINGW_PRINTF_FORMAT)
00667 // MinGW has two different printf implementations. Ensure the format macro
00668 // matches the selected implementation. See
00669 // https://sourceforge.net/p/mingw-w64/wiki2/gnu%20printf/.
00670 #define GTEST_ATTRIBUTE_PRINTF_(string_index, first_to_check) \
00671   __attribute__((                                             \
00672       __format__(__MINGW_PRINTF_FORMAT, string_index, first_to_check)))
00673 #else
00674 #define GTEST_ATTRIBUTE_PRINTF_(string_index, first_to_check) \
00675   __attribute__((__format__(__printf__, string_index, first_to_check)))
00676 #endif
00677 #else
00678 #define GTEST_ATTRIBUTE_PRINTF_(string_index, first_to_check)
00679 #endif
00680
00681 // Tell the compiler to warn about unused return values for functions declared
00682 // with this macro.  The macro should be used on function declarations
00683 // following the argument list:
00684 //
00685 //   Sprocket* AllocateSprocket() GTEST_MUST_USE_RESULT_;
00686 #if defined(__GNUC__) && !defined(COMPILER_ICC)
00687 #define GTEST_MUST_USE_RESULT_ __attribute__((warn_unused_result))
00688 #else
00689 #define GTEST_MUST_USE_RESULT_
00690 #endif  // __GNUC__ && !COMPILER_ICC
00691
00692 // MS C++ compiler emits warning when a conditional expression is compile time
00693 // constant. In some contexts this warning is false positive and needs to be
00694 // suppressed. Use the following two macros in such cases:
00695 //
00696 // GTEST_INTENTIONAL_CONST_COND_PUSH_()
00697 // while (true) {
00698 // GTEST_INTENTIONAL_CONST_COND_POP_()
00699 // }
00700 #define GTEST_INTENTIONAL_CONST_COND_PUSH_() \
00701   GTEST_DISABLE_MSC_WARNINGS_PUSH_(4127)
00702 #define GTEST_INTENTIONAL_CONST_COND_POP_() GTEST_DISABLE_MSC_WARNINGS_POP_()
00703
00704 // Determine whether the compiler supports Microsoft's Structured Exception
00705 // Handling.  This is supported by several Windows compilers but generally
00706 // does not exist on any other system.
00707 #ifndef GTEST_HAS_SEH
00708 // The user didn't tell us, so we need to figure it out.
00709
00710 #if defined(_MSC_VER) || defined(__BORLANDC__)
00711 // These two compilers are known to support SEH.
00712 #define GTEST_HAS_SEH 1
00713 #else
00714 // Assume no SEH.
00715 #define GTEST_HAS_SEH 0
00716 #endif
00717
00718 #endif  // GTEST_HAS_SEH
00719
00720 #ifndef GTEST_IS_THREADSAFE
00721
00722 #define GTEST_IS_THREADSAFE                                                \
00723   (GTEST_HAS_MUTEX_AND_THREAD_LOCAL_ ||                                   \
00724    (GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_PHONE && !GTEST_OS_WINDOWS_RT) || \
00725    GTEST_HAS_PTHREAD)
00726
00727 #endif  // GTEST_IS_THREADSAFE
00728
00729 #if GTEST_IS_THREADSAFE
00730 // Some platforms don't support including these threading related headers.
00731 #include <condition_variable>  // NOLINT
00732 #include <mutex>              // NOLINT
```

```
00733 #endif                          // GTEST_IS_THREADSAFE
00734
00735 // GTEST_API_ qualifies all symbols that must be exported. The definitions below
00736 // are guarded by #ifndef to give embedders a chance to define GTEST_API_ in
00737 // gtest/internal/custom/gtest-port.h
00738 #ifndef GTEST_API_
00739
00740 #ifdef _MSC_VER
00741 #if GTEST_LINKED_AS_SHARED_LIBRARY
00742 #define GTEST_API_ __declspec(dllimport)
00743 #elif GTEST_CREATE_SHARED_LIBRARY
00744 #define GTEST_API_ __declspec(dllexport)
00745 #endif
00746 #elif __GNUC__ >= 4 || defined(__clang__)
00747 #define GTEST_API_ __attribute__((visibility("default")))
00748 #endif  // _MSC_VER
00749
00750 #endif  // GTEST_API_
00751
00752 #ifndef GTEST_API_
00753 #define GTEST_API_
00754 #endif  // GTEST_API_
00755
00756 #ifndef GTEST_DEFAULT_DEATH_TEST_STYLE
00757 #define GTEST_DEFAULT_DEATH_TEST_STYLE "fast"
00758 #endif  // GTEST_DEFAULT_DEATH_TEST_STYLE
00759
00760 #ifdef __GNUC__
00761 // Ask the compiler to never inline a given function.
00762 #define GTEST_NO_INLINE_ __attribute__((noinline))
00763 #else
00764 #define GTEST_NO_INLINE_
00765 #endif
00766
00767 #if defined(__clang__)
00768 // Nested ifs to avoid triggering MSVC warning.
00769 #if __has_attribute(disable_tail_calls)
00770 // Ask the compiler not to perform tail call optimization inside
00771 // the marked function.
00772 #define GTEST_NO_TAIL_CALL_ __attribute__((disable_tail_calls))
00773 #endif
00774 #elif __GNUC__
00775 #define GTEST_NO_TAIL_CALL_ \
00776   __attribute__((optimize("no-optimize-sibling-calls")))
00777 #else
00778 #define GTEST_NO_TAIL_CALL_
00779 #endif
00780
00781 // _LIBCPP_VERSION is defined by the libc++ library from the LLVM project.
00782 #if !defined(GTEST_HAS_CXXABI_H_)
00783 #if defined(__GLIBCXX__) || (defined(_LIBCPP_VERSION) && !defined(_MSC_VER))
00784 #define GTEST_HAS_CXXABI_H_ 1
00785 #else
00786 #define GTEST_HAS_CXXABI_H_ 0
00787 #endif
00788 #endif
00789
00790 // A function level attribute to disable checking for use of uninitialized
00791 // memory when built with MemorySanitizer.
00792 #if defined(__clang__)
00793 #if __has_feature(memory_sanitizer)
00794 #define GTEST_ATTRIBUTE_NO_SANITIZE_MEMORY_ __attribute__((no_sanitize_memory))
00795 #else
00796 #define GTEST_ATTRIBUTE_NO_SANITIZE_MEMORY_
00797 #endif  // __has_feature(memory_sanitizer)
00798 #else
00799 #define GTEST_ATTRIBUTE_NO_SANITIZE_MEMORY_
00800 #endif  // __clang__
00801
00802 // A function level attribute to disable AddressSanitizer instrumentation.
00803 #if defined(__clang__)
00804 #if __has_feature(address_sanitizer)
00805 #define GTEST_ATTRIBUTE_NO_SANITIZE_ADDRESS_ \
00806   __attribute__((no_sanitize_address))
00807 #else
00808 #define GTEST_ATTRIBUTE_NO_SANITIZE_ADDRESS_
00809 #endif  // __has_feature(address_sanitizer)
00810 #else
00811 #define GTEST_ATTRIBUTE_NO_SANITIZE_ADDRESS_
00812 #endif  // __clang__
00813
00814 // A function level attribute to disable HWAddressSanitizer instrumentation.
00815 #if defined(__clang__)
00816 #if __has_feature(hwaddress_sanitizer)
00817 #define GTEST_ATTRIBUTE_NO_SANITIZE_HWADDRESS_ \
00818   __attribute__((no_sanitize("hwaddress")))
00819 #else
```

```
00820 #define GTEST_ATTRIBUTE_NO_SANITIZE_HWADDRESS_
00821 #endif  // __has_feature(hwaddress_sanitizer)
00822 #else
00823 #define GTEST_ATTRIBUTE_NO_SANITIZE_HWADDRESS_
00824 #endif  // __clang__
00825
00826 // A function level attribute to disable ThreadSanitizer instrumentation.
00827 #if defined(__clang__)
00828 #if __has_feature(thread_sanitizer)
00829 #define GTEST_ATTRIBUTE_NO_SANITIZE_THREAD_ __attribute__((no_sanitize_thread))
00830 #else
00831 #define GTEST_ATTRIBUTE_NO_SANITIZE_THREAD_
00832 #endif  // __has_feature(thread_sanitizer)
00833 #else
00834 #define GTEST_ATTRIBUTE_NO_SANITIZE_THREAD_
00835 #endif  // __clang__
00836
00837 namespace testing {
00838
00839     class Message;
00840
00841     // Legacy imports for backwards compatibility.
00842     // New code should use std:: names directly.
00843     using std::get;
00844     using std::make_tuple;
00845     using std::tuple;
00846     using std::tuple_element;
00847     using std::tuple_size;
00848
00849     namespace internal {
00850
00851         // A secret type that Google Test users don't know about.  It has no
00852         // definition on purpose.  Therefore it's impossible to create a
00853         // Secret object, which is what we want.
00854         class Secret;
00855
00856         // A helper for suppressing warnings on constant condition.  It just
00857         // returns 'condition'.
00858         GTEST_API_ bool IsTrue(bool condition);
00859
00860         // Defines RE.
00861
00862 #if GTEST_USES_RE2
00863
00864         // This is almost `using RE = ::RE2`, except it is copy-constructible, and it
00865         // needs to disambiguate the `std::string`, `absl::string_view`, and `const
00866         // char*` constructors.
00867
00868         class GTEST_API_ RE {
00869         public:
00870
00871             RE(absl::string_view regex) : regex_(regex) {
00872             } // NOLINT
00873
00874             RE(const char* regex) : RE(absl::string_view(regex)) {
00875             } // NOLINT
00876
00877             RE(const std::string& regex) : RE(absl::string_view(regex)) {
00878             } // NOLINT
00879
00880             RE(const RE& other) : RE(other.pattern()) {
00881             }
00882
00883             const std::string& pattern() const {
00884                 return regex_.pattern();
00885             }
00886
00887             static bool FullMatch(absl::string_view str, const RE& re) {
00888                 return RE2::FullMatch(str, re.regex_);
00889             }
00890
00891             static bool PartialMatch(absl::string_view str, const RE& re) {
00892                 return RE2::PartialMatch(str, re.regex_);
00893             }
00894
00895         private:
00896             RE2 regex_;
00897         };
00898
00899 #elif GTEST_USES_POSIX_RE || GTEST_USES_SIMPLE_RE
00900
00901         // A simple C++ wrapper for <regex.h>.  It uses the POSIX Extended
00902         // Regular Expression syntax.
00903
00904         class GTEST_API_ RE {
00905         public:
00906             // A copy constructor is required by the Standard to initialize object
```

```
00907             // references from r-values.
00908
00909             RE(const RE& other) {
00910                 Init(other.pattern());
00911             }
00912
00913             // Constructs an RE from a string.
00914
00915             RE(const ::std::string& regex) {
00916                 Init(regex.c_str());
00917             } // NOLINT
00918
00919             RE(const char* regex) {
00920                 Init(regex);
00921             } // NOLINT
00922             ~RE();
00923
00924             // Returns the string representation of the regex.
00925
00926             const char* pattern() const {
00927                 return pattern_;
00928             }
00929
00930             // FullMatch(str, re) returns true if and only if regular expression re
00931             // matches the entire str.
00932             // PartialMatch(str, re) returns true if and only if regular expression re
00933             // matches a substring of str (including str itself).
00934
00935             static bool FullMatch(const ::std::string& str, const RE& re) {
00936                 return FullMatch(str.c_str(), re);
00937             }
00938
00939             static bool PartialMatch(const ::std::string& str, const RE& re) {
00940                 return PartialMatch(str.c_str(), re);
00941             }
00942
00943             static bool FullMatch(const char* str, const RE& re);
00944             static bool PartialMatch(const char* str, const RE& re);
00945
00946         private:
00947             void Init(const char* regex);
00948             const char* pattern_;
00949             bool is_valid_;
00950
00951 #if GTEST_USES_POSIX_RE
00952
00953             regex_t full_regex_; // For FullMatch().
00954             regex_t partial_regex_; // For PartialMatch().
00955
00956 #else  // GTEST_USES_SIMPLE_RE
00957
00958             const char* full_pattern_; // For FullMatch();
00959
00960 #endif
00961         };
00962
00963 #endif  // ::testing::internal::RE implementation
00964
00965         // Formats a source file path and a line number as they would appear
00966         // in an error message from the compiler used to compile this code.
00967         GTEST_API_::std::string FormatFileLocation(const char* file, int line);
00968
00969         // Formats a file location for compiler-independent XML output.
00970         // Although this function is not platform dependent, we put it next to
00971         // FormatFileLocation in order to contrast the two functions.
00972         GTEST_API_::std::string FormatCompilerIndependentFileLocation(const char* file,
00973                 int line);
00974
00975         // Defines logging utilities:
00976         //   GTEST_LOG_(severity) - logs messages at the specified severity level. The
00977         //                          message itself is streamed into the macro.
00978         //   LogToStderr()  - directs all log messages to stderr.
00979         //   FlushInfoLog() - flushes informational log messages.
00980
00981         enum GTestLogSeverity {
00982             GTEST_INFO, GTEST_WARNING, GTEST_ERROR, GTEST_FATAL
00983         };
00984
00985         // Formats log entry severity, provides a stream object for streaming the
00986         // log message, and terminates the message with a newline when going out of
00987         // scope.
00988
00989         class GTEST_API_ GTestLog {
00990         public:
00991             GTestLog(GTestLogSeverity severity, const char* file, int line);
00992
00993             // Flushes the buffers and, if severity is GTEST_FATAL, aborts the program.
```

```
00994             ~GTestLog();
00995
00996             ::std::ostream& GetStream() {
00997                 return ::std::cerr;
00998             }
00999
01000         private:
01001             const GTestLogSeverity severity_;
01002
01003             GTestLog(const GTestLog&) = delete;
01004             GTestLog& operator=(const GTestLog&) = delete;
01005         };
01006
01007 #if !defined(GTEST_LOG_)
01008
01009 #define GTEST_LOG_(severity)                                                \
01010   ::testing::internal::GTestLog(::testing::internal::GTEST_##severity, \
01011                                 __FILE__, __LINE__)                         \
01012       .GetStream()
01013
01014         inline void LogToStderr() {
01015         }
01016
01017         inline void FlushInfoLog() {
01018             fflush(nullptr);
01019         }
01020
01021 #endif  // !defined(GTEST_LOG_)
01022
01023 #if !defined(GTEST_CHECK_)
01024         // INTERNAL IMPLEMENTATION - DO NOT USE.
01025         //
01026         // GTEST_CHECK_ is an all-mode assert. It aborts the program if the condition
01027         // is not satisfied.
01028         //  Synopsis:
01029         //    GTEST_CHECK_(boolean_condition);
01030         //     or
01031         //    GTEST_CHECK_(boolean_condition) << "Additional message";
01032         //
01033         //    This checks the condition and if the condition is not satisfied
01034         //    it prints message about the condition violation, including the
01035         //    condition itself, plus additional message streamed into it, if any,
01036         //    and then it aborts the program. It aborts the program irrespective of
01037         //    whether it is built in the debug mode or not.
01038 #define GTEST_CHECK_(condition)               \
01039   GTEST_AMBIGUOUS_ELSE_BLOCKER_              \
01040   if (::testing::internal::IsTrue(condition)) \
01041     ;                                        \
01042   else                                       \
01043     GTEST_LOG_(FATAL) << "Condition " #condition " failed. "
01044 #endif  // !defined(GTEST_CHECK_)
01045
01046         // An all-mode assert to verify that the given POSIX-style function
01047         // call returns 0 (indicating success).  Known limitation: this
01048         // doesn't expand to a balanced 'if' statement, so enclose the macro
01049         // in {} if you need to use it as the only statement in an 'if'
01050         // branch.
01051 #define GTEST_CHECK_POSIX_SUCCESS_(posix_call) \
01052   if (const int gtest_error = (posix_call))    \
01053   GTEST_LOG_(FATAL) << #posix_call << "failed with error " << gtest_error
01054
01055         // Transforms "T" into "const T&" according to standard reference collapsing
01056         // rules (this is only needed as a backport for C++98 compilers that do not
01057         // support reference collapsing). Specifically, it transforms:
01058         //
01059         //   char         ==> const char&
01060         //   const char   ==> const char&
01061         //   char&        ==> char&
01062         //   const char&  ==> const char&
01063         //
01064         // Note that the non-const reference will not have "const" added. This is
01065         // standard, and necessary so that "T" can always bind to "const T&".
01066
01067         template <typename T>
01068         struct ConstRef {
01069             typedef const T& type;
01070         };
01071
01072         template <typename T>
01073         struct ConstRef<T&> {
01074             typedef T& type;
01075         };
01076
01077         // The argument T must depend on some template parameters.
01078 #define GTEST_REFERENCE_TO_CONST_(T) \
01079   typename ::testing::internal::ConstRef<T>::type
01080
```

```
01081         // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
01082         //
01083         // Use ImplicitCast_ as a safe version of static_cast for upcasting in
01084         // the type hierarchy (e.g. casting a Foo* to a SuperclassOfFoo* or a
01085         // const Foo*).  When you use ImplicitCast_, the compiler checks that
01086         // the cast is safe.  Such explicit ImplicitCast_s are necessary in
01087         // surprisingly many situations where C++ demands an exact type match
01088         // instead of an argument type convertible to a target type.
01089         //
01090         // The syntax for using ImplicitCast_ is the same as for static_cast:
01091         //
01092         //   ImplicitCast_<ToType>(expr)
01093         //
01094         // ImplicitCast_ would have been part of the C++ standard library,
01095         // but the proposal was submitted too late.  It will probably make
01096         // its way into the language in the future.
01097         //
01098         // This relatively ugly name is intentional. It prevents clashes with
01099         // similar functions users may have (e.g., implicit_cast). The internal
01100         // namespace alone is not enough because the function can be found by ADL.
01101
01102         template <typename To>
01103         inline To ImplicitCast_(To x) {
01104             return x;
01105         }
01106
01107         // When you upcast (that is, cast a pointer from type Foo to type
01108         // SuperclassOfFoo), it's fine to use ImplicitCast_<>, since upcasts
01109         // always succeed.  When you downcast (that is, cast a pointer from
01110         // type Foo to type SubclassOfFoo), static_cast<> isn't safe, because
01111         // how do you know the pointer is really of type SubclassOfFoo?  It
01112         // could be a bare Foo, or of type DifferentSubclassOfFoo.  Thus,
01113         // when you downcast, you should use this macro.  In debug mode, we
01114         // use dynamic_cast<> to double-check the downcast is legal (we die
01115         // if it's not).  In normal mode, we do the efficient static_cast<>
01116         // instead.  Thus, it's important to test in debug mode to make sure
01117         // the cast is legal!
01118         //    This is the only place in the code we should use dynamic_cast<>.
01119         // In particular, you SHOULDN'T be using dynamic_cast<> in order to
01120         // do RTTI (eg code like this:
01121         //    if (dynamic_cast<Subclass1>(foo)) HandleASubclass1Object(foo);
01122         //    if (dynamic_cast<Subclass2>(foo)) HandleASubclass2Object(foo);
01123         // You should design the code some other way not to need this.
01124         //
01125         // This relatively ugly name is intentional. It prevents clashes with
01126         // similar functions users may have (e.g., down_cast). The internal
01127         // namespace alone is not enough because the function can be found by ADL.
01128
01129         template <typename To, typename From> // use like this: DownCast_<T*>(foo);
01130         inline To DownCast_(From* f) { // so we only accept pointers
01131             // Ensures that To is a sub-type of From *.  This test is here only
01132             // for compile-time type checking, and has no overhead in an
01133             // optimized build at run-time, as it will be optimized away
01134             // completely.
01135             GTEST_INTENTIONAL_CONST_COND_PUSH_()
01136             if (false) {
01137                 GTEST_INTENTIONAL_CONST_COND_POP_()
01138                         const To to = nullptr;
01139                 ::testing::internal::ImplicitCast_<From*>(to);
01140             }
01141
01142 #if GTEST_HAS_RTTI
01143             // RTTI: debug mode only!
01144             GTEST_CHECK_(f == nullptr || dynamic_cast<To> (f) != nullptr);
01145 #endif
01146             return static_cast<To> (f);
01147         }
01148
01149         // Downcasts the pointer of type Base to Derived.
01150         // Derived must be a subclass of Base. The parameter MUST
01151         // point to a class of type Derived, not any subclass of it.
01152         // When RTTI is available, the function performs a runtime
01153         // check to enforce this.
01154
01155         template <class Derived, class Base>
01156         Derived* CheckedDowncastToActualType(Base* base) {
01157 #if GTEST_HAS_RTTI
01158             GTEST_CHECK_(typeid (*base) == typeid (Derived));
01159 #endif
01160
01161 #if GTEST_HAS_DOWNCAST_
01162             return ::down_cast<Derived*>(base);
01163 #elif GTEST_HAS_RTTI
01164             return dynamic_cast<Derived*> (base); // NOLINT
01165 #else
01166             return static_cast<Derived*> (base); // Poor man's downcast.
01167 #endif
```

```
01168         }
01169
01170 #if GTEST_HAS_STREAM_REDIRECTION
01171
01172         // Defines the stderr capturer:
01173         //   CaptureStdout     - starts capturing stdout.
01174         //   GetCapturedStdout - stops capturing stdout and returns the captured string.
01175         //   CaptureStderr     - starts capturing stderr.
01176         //   GetCapturedStderr - stops capturing stderr and returns the captured string.
01177         //
01178         GTEST_API_ void CaptureStdout();
01179         GTEST_API_ std::string GetCapturedStdout();
01180         GTEST_API_ void CaptureStderr();
01181         GTEST_API_ std::string GetCapturedStderr();
01182
01183 #endif  // GTEST_HAS_STREAM_REDIRECTION
01184         // Returns the size (in bytes) of a file.
01185         GTEST_API_ size_t GetFileSize(FILE* file);
01186
01187         // Reads the entire content of a file as a string.
01188         GTEST_API_ std::string ReadEntireFile(FILE* file);
01189
01190         // All command line arguments.
01191         GTEST_API_ std::vector<std::string> GetArgvs();
01192
01193 #if GTEST_HAS_DEATH_TEST
01194
01195         std::vector<std::string> GetInjectableArgvs();
01196         // Deprecated: pass the args vector by value instead.
01197         void SetInjectableArgvs(const std::vector<std::string>* new_argvs);
01198         void SetInjectableArgvs(const std::vector<std::string>& new_argvs);
01199         void ClearInjectableArgvs();
01200
01201 #endif  // GTEST_HAS_DEATH_TEST
01202
01203         // Defines synchronization primitives.
01204 #if GTEST_IS_THREADSAFE
01205
01206 #if GTEST_OS_WINDOWS
01207         // Provides leak-safe Windows kernel handle ownership.
01208         // Used in death tests and in threading support.
01209
01210         class GTEST_API_ AutoHandle {
01211         public:
01212             // Assume that Win32 HANDLE type is equivalent to void*. Doing so allows us to
01213             // avoid including <windows.h> in this header file. Including <windows.h> is
01214             // undesirable because it defines a lot of symbols and macros that tend to
01215             // conflict with client code. This assumption is verified by
01216             // WindowsTypesTest.HANDLEIsVoidStar.
01217             typedef void* Handle;
01218             AutoHandle();
01219             explicit AutoHandle(Handle handle);
01220
01221             ~AutoHandle();
01222
01223             Handle Get() const;
01224             void Reset();
01225             void Reset(Handle handle);
01226
01227         private:
01228             // Returns true if and only if the handle is a valid handle object that can be
01229             // closed.
01230             bool IsCloseable() const;
01231
01232             Handle handle_;
01233
01234             AutoHandle(const AutoHandle&) = delete;
01235             AutoHandle& operator=(const AutoHandle&) = delete;
01236         };
01237 #endif
01238
01239 #if GTEST_HAS_NOTIFICATION_
01240         // Notification has already been imported into the namespace.
01241         // Nothing to do here.
01242
01243 #else
01244
01245         GTEST_DISABLE_MSC_WARNINGS_PUSH_(4251 \
01246 /* class A needs to have dll-interface to be used by clients of class B */)
01247
01248         // Allows a controller thread to pause execution of newly created
01249         // threads until notified.  Instances of this class must be created
01250         // and destroyed in the controller thread.
01251         //
01252         // This class is only for testing Google Test's own constructs. Do not
01253         // use it in user tests, either directly or indirectly.
01254         // TODO(b/203539622): Replace unconditionally with absl::Notification.
```

```
01255         class GTEST_API_ Notification {
01256         public:
01257
01258             Notification() : notified_(false) {
01259             }
01260             Notification(const Notification&) = delete;
01261             Notification& operator=(const Notification&) = delete;
01262
01263             // Notifies all threads created with this notification to start. Must
01264             // be called from the controller thread.
01265
01266             void Notify() {
01267                 std::lock_guard<std::mutex> lock(mu_);
01268                 notified_ = true;
01269                 cv_.notify_all();
01270             }
01271
01272             // Blocks until the controller thread notifies. Must be called from a test
01273             // thread.
01274
01275             void WaitForNotification() {
01276                 std::unique_lock<std::mutex> lock(mu_);
01277                 cv_.wait(lock, [this]() {
01278                     return notified_; });
01279             }
01280
01281         private:
01282             std::mutex mu_;
01283             std::condition_variable cv_;
01284             bool notified_;
01285         };
01286
01287         GTEST_DISABLE_MSC_WARNINGS_POP_() // 4251
01288 #endif  // GTEST_HAS_NOTIFICATION_
01289
01290         // On MinGW, we can have both GTEST_OS_WINDOWS and GTEST_HAS_PTHREAD
01291         // defined, but we don't want to use MinGW's pthreads implementation, which
01292         // has conformance problems with some versions of the POSIX standard.
01293 #if GTEST_HAS_PTHREAD && !GTEST_OS_WINDOWS_MINGW
01294
01295         // As a C-function, ThreadFuncWithCLinkage cannot be templated itself.
01296         // Consequently, it cannot select a correct instantiation of ThreadWithParam
01297         // in order to call its Run(). Introducing ThreadWithParamBase as a
01298         // non-templated base class for ThreadWithParam allows us to bypass this
01299         // problem.
01300         class ThreadWithParamBase {
01301         public:
01302
01303             virtual ~ThreadWithParamBase() {
01304             }
01305             virtual void Run() = 0;
01306         };
01307
01308         // pthread_create() accepts a pointer to a function type with the C linkage.
01309         // According to the Standard (7.5/1), function types with different linkages
01310         // are different even if they are otherwise identical.  Some compilers (for
01311         // example, SunStudio) treat them as different types.  Since class methods
01312         // cannot be defined with C-linkage we need to define a free C-function to
01313         // pass into pthread_create().
01314
01315         extern "C" inline void* ThreadFuncWithCLinkage(void* thread) {
01316             static_cast<ThreadWithParamBase*> (thread)->Run();
01317             return nullptr;
01318         }
01319
01320         // Helper class for testing Google Test's multi-threading constructs.
01321         // To use it, write:
01322         //
01323         //   void ThreadFunc(int param) { /* Do things with param */ }
01324         //   Notification thread_can_start;
01325         //   ...
01326         //   // The thread_can_start parameter is optional; you can supply NULL.
01327         //   ThreadWithParam<int> thread(&ThreadFunc, 5, &thread_can_start);
01328         //   thread_can_start.Notify();
01329         //
01330         // These classes are only for testing Google Test's own constructs. Do
01331         // not use them in user tests, either directly or indirectly.
01332
01333         template <typename T>
01334         class ThreadWithParam : public ThreadWithParamBase {
01335         public:
01336             typedef void UserThreadFunc(T);
01337
01338             ThreadWithParam(UserThreadFunc* func, T param, Notification* thread_can_start)
01339             : func_(func),
01340             param_(param),
01341             thread_can_start_(thread_can_start),
```

```
01342             finished_(false) {
01343                 ThreadWithParamBase * const base = this;
01344                 // The thread can be created only after all fields except thread_
01345                 // have been initialized.
01346                 GTEST_CHECK_POSIX_SUCCESS_(
01347                         pthread_create(&thread_, nullptr, &ThreadFuncWithCLinkage, base));
01348             }
01349
01350             ~ThreadWithParam() override {
01351                 Join();
01352             }
01353
01354             void Join() {
01355                 if (!finished_) {
01356                     GTEST_CHECK_POSIX_SUCCESS_(pthread_join(thread_, nullptr));
01357                     finished_ = true;
01358                 }
01359             }
01360
01361             void Run() override {
01362                 if (thread_can_start_ != nullptr) thread_can_start_->WaitForNotification();
01363                 func_(param_);
01364             }
01365
01366         private:
01367             UserThreadFunc * const func_; // User-supplied thread function.
01368             const T param_; // User-supplied parameter to the thread function.
01369             // When non-NULL, used to block execution until the controller thread
01370             // notifies.
01371             Notification * const thread_can_start_;
01372             bool finished_; // true if and only if we know that the thread function has
01373             // finished.
01374             pthread_t thread_; // The native thread object.
01375
01376             ThreadWithParam(const ThreadWithParam&) = delete;
01377             ThreadWithParam& operator=(const ThreadWithParam&) = delete;
01378         };
01379 #endif  // !GTEST_OS_WINDOWS && GTEST_HAS_PTHREAD ||
01380         // GTEST_HAS_MUTEX_AND_THREAD_LOCAL_
01381
01382 #if GTEST_HAS_MUTEX_AND_THREAD_LOCAL_
01383         // Mutex and ThreadLocal have already been imported into the namespace.
01384         // Nothing to do here.
01385
01386 #elif GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_PHONE && !GTEST_OS_WINDOWS_RT
01387
01388         // Mutex implements mutex on Windows platforms.  It is used in conjunction
01389         // with class MutexLock:
01390         //
01391         //   Mutex mutex;
01392         //   ...
01393         //   MutexLock lock(&mutex);  // Acquires the mutex and releases it at the
01394         //                            // end of the current scope.
01395         //
01396         // A static Mutex *must* be defined or declared using one of the following
01397         // macros:
01398         //   GTEST_DEFINE_STATIC_MUTEX_(g_some_mutex);
01399         //   GTEST_DECLARE_STATIC_MUTEX_(g_some_mutex);
01400         //
01401         // (A non-static Mutex is defined/declared in the usual way).
01402
01403         class GTEST_API_ Mutex {
01404         public:
01405
01406             enum MutexType {
01407                 kStatic = 0, kDynamic = 1
01408             };
01409             // We rely on kStaticMutex being 0 as it is to what the linker initializes
01410             // type_ in static mutexes.  critical_section_ will be initialized lazily
01411             // in ThreadSafeLazyInit().
01412
01413             enum StaticConstructorSelector {
01414                 kStaticMutex = 0
01415             };
01416
01417             // This constructor intentionally does nothing.  It relies on type_ being
01418             // statically initialized to 0 (effectively setting it to kStatic) and on
01419             // ThreadSafeLazyInit() to lazily initialize the rest of the members.
01420
01421             explicit Mutex(StaticConstructorSelector /*dummy*/) {
01422             }
01423
01424             Mutex();
01425             ~Mutex();
01426
01427             void Lock();
01428
```

```
01429             void Unlock();
01430
01431             // Does nothing if the current thread holds the mutex. Otherwise, crashes
01432             // with high probability.
01433             void AssertHeld();
01434
01435         private:
01436             // Initializes owner_thread_id_ and critical_section_ in static mutexes.
01437             void ThreadSafeLazyInit();
01438
01439             // Per https://blogs.msdn.microsoft.com/oldnewthing/20040223-00/?p=40503,
01440             // we assume that 0 is an invalid value for thread IDs.
01441             unsigned int owner_thread_id_;
01442
01443             // For static mutexes, we rely on these members being initialized to zeros
01444             // by the linker.
01445             MutexType type_;
01446             long critical_section_init_phase_; // NOLINT
01447             GTEST_CRITICAL_SECTION* critical_section_;
01448
01449             Mutex(const Mutex&) = delete;
01450             Mutex& operator=(const Mutex&) = delete;
01451         };
01452
01453 #define GTEST_DECLARE_STATIC_MUTEX_(mutex) \
01454   extern ::testing::internal::Mutex mutex
01455
01456 #define GTEST_DEFINE_STATIC_MUTEX_(mutex) \
01457   ::testing::internal::Mutex mutex(::testing::internal::Mutex::kStaticMutex)
01458
01459         // We cannot name this class MutexLock because the ctor declaration would
01460         // conflict with a macro named MutexLock, which is defined on some
01461         // platforms. That macro is used as a defensive measure to prevent against
01462         // inadvertent misuses of MutexLock like "MutexLock(&mu)" rather than
01463         // "MutexLock l(&mu)".  Hence the typedef trick below.
01464
01465         class GTestMutexLock {
01466         public:
01467
01468             explicit GTestMutexLock(Mutex* mutex) : mutex_(mutex) {
01469                 mutex_->Lock();
01470             }
01471
01472             ~GTestMutexLock() {
01473                 mutex_->Unlock();
01474             }
01475
01476         private:
01477             Mutex * const mutex_;
01478
01479             GTestMutexLock(const GTestMutexLock&) = delete;
01480             GTestMutexLock& operator=(const GTestMutexLock&) = delete;
01481         };
01482
01483         typedef GTestMutexLock MutexLock;
01484
01485         // Base class for ValueHolder<T>.  Allows a caller to hold and delete a value
01486         // without knowing its type.
01487
01488         class ThreadLocalValueHolderBase {
01489         public:
01490
01491             virtual ~ThreadLocalValueHolderBase() {
01492             }
01493         };
01494
01495         // Provides a way for a thread to send notifications to a ThreadLocal
01496         // regardless of its parameter type.
01497
01498         class ThreadLocalBase {
01499         public:
01500             // Creates a new ValueHolder<T> object holding a default value passed to
01501             // this ThreadLocal<T>'s constructor and returns it.  It is the caller's
01502             // responsibility not to call this when the ThreadLocal<T> instance already
01503             // has a value on the current thread.
01504             virtual ThreadLocalValueHolderBase* NewValueForCurrentThread() const = 0;
01505
01506         protected:
01507
01508             ThreadLocalBase() {
01509             }
01510
01511             virtual ~ThreadLocalBase() {
01512             }
01513
01514         private:
01515             ThreadLocalBase(const ThreadLocalBase&) = delete;
```

```
01516           ThreadLocalBase& operator=(const ThreadLocalBase&) = delete;
01517       };
01518
01519       // Maps a thread to a set of ThreadLocals that have values instantiated on that
01520       // thread and notifies them when the thread exits.  A ThreadLocal instance is
01521       // expected to persist until all threads it has values on have terminated.
01522
01523       class GTEST_API_ ThreadLocalRegistry {
01524       public:
01525           // Registers thread_local_instance as having value on the current thread.
01526           // Returns a value that can be used to identify the thread from other threads.
01527           static ThreadLocalValueHolderBase* GetValueOnCurrentThread(
01528                   const ThreadLocalBase* thread_local_instance);
01529
01530           // Invoked when a ThreadLocal instance is destroyed.
01531           static void OnThreadLocalDestroyed(
01532                   const ThreadLocalBase* thread_local_instance);
01533       };
01534
01535       class GTEST_API_ ThreadWithParamBase {
01536       public:
01537           void Join();
01538
01539       protected:
01540
01541           class Runnable {
01542           public:
01543
01544               virtual ~Runnable() {
01545               }
01546               virtual void Run() = 0;
01547           };
01548
01549           ThreadWithParamBase(Runnable* runnable, Notification* thread_can_start);
01550           virtual ~ThreadWithParamBase();
01551
01552       private:
01553           AutoHandle thread_;
01554       };
01555
01556       // Helper class for testing Google Test's multi-threading constructs.
01557
01558       template <typename T>
01559       class ThreadWithParam : public ThreadWithParamBase {
01560       public:
01561           typedef void UserThreadFunc(T);
01562
01563           ThreadWithParam(UserThreadFunc* func, T param, Notification* thread_can_start)
01564           : ThreadWithParamBase(new RunnableImpl(func, param), thread_can_start) {
01565           }
01566
01567           virtual ~ThreadWithParam() {
01568           }
01569
01570       private:
01571
01572           class RunnableImpl : public Runnable {
01573           public:
01574
01575               RunnableImpl(UserThreadFunc* func, T param) : func_(func), param_(param) {
01576               }
01577
01578               virtual ~RunnableImpl() {
01579               }
01580
01581               virtual void Run() {
01582                   func_(param_);
01583               }
01584
01585           private:
01586               UserThreadFunc * const func_;
01587               const T param_;
01588
01589               RunnableImpl(const RunnableImpl&) = delete;
01590               RunnableImpl& operator=(const RunnableImpl&) = delete;
01591           };
01592
01593           ThreadWithParam(const ThreadWithParam&) = delete;
01594           ThreadWithParam& operator=(const ThreadWithParam&) = delete;
01595       };
01596
01597       // Implements thread-local storage on Windows systems.
01598       //
01599       //   // Thread 1
01600       //   ThreadLocal<int> tl(100);  // 100 is the default value for each thread.
01601       //
01602       //   // Thread 2
```

```
01603         //   tl.set(150);  // Changes the value for thread 2 only.
01604         //   EXPECT_EQ(150, tl.get());
01605         //
01606         //   // Thread 1
01607         //   EXPECT_EQ(100, tl.get());  // In thread 1, tl has the original value.
01608         //   tl.set(200);
01609         //   EXPECT_EQ(200, tl.get());
01610         //
01611         // The template type argument T must have a public copy constructor.
01612         // In addition, the default ThreadLocal constructor requires T to have
01613         // a public default constructor.
01614         //
01615         // The users of a TheadLocal instance have to make sure that all but one
01616         // threads (including the main one) using that instance have exited before
01617         // destroying it. Otherwise, the per-thread objects managed for them by the
01618         // ThreadLocal instance are not guaranteed to be destroyed on all platforms.
01619         //
01620         // Google Test only uses global ThreadLocal objects.  That means they
01621         // will die after main() has returned.  Therefore, no per-thread
01622         // object managed by Google Test will be leaked as long as all threads
01623         // using Google Test have exited when main() returns.
01624
01625         template <typename T>
01626         class ThreadLocal : public ThreadLocalBase {
01627         public:
01628
01629             ThreadLocal() : default_factory_(new DefaultValueHolderFactory()) {
01630             }
01631
01632             explicit ThreadLocal(const T& value)
01633             : default_factory_(new InstanceValueHolderFactory(value)) {
01634             }
01635
01636             ~ThreadLocal() override {
01637                 ThreadLocalRegistry::OnThreadLocalDestroyed(this);
01638             }
01639
01640             T* pointer() {
01641                 return GetOrCreateValue();
01642             }
01643
01644             const T* pointer() const {
01645                 return GetOrCreateValue();
01646             }
01647
01648             const T& get() const {
01649                 return *pointer();
01650             }
01651
01652             void set(const T& value) {
01653                 *pointer() = value;
01654             }
01655
01656         private:
01657             // Holds a value of T.  Can be deleted via its base class without the caller
01658             // knowing the type of T.
01659
01660             class ValueHolder : public ThreadLocalValueHolderBase {
01661             public:
01662
01663                 ValueHolder() : value_() {
01664                 }
01665
01666                 explicit ValueHolder(const T& value) : value_(value) {
01667                 }
01668
01669                 T* pointer() {
01670                     return &value_;
01671                 }
01672
01673             private:
01674                 T value_;
01675                 ValueHolder(const ValueHolder&) = delete;
01676                 ValueHolder& operator=(const ValueHolder&) = delete;
01677             };
01678
01679             T* GetOrCreateValue() const {
01680                 return static_cast<ValueHolder*> (
01681                         ThreadLocalRegistry::GetValueOnCurrentThread(this))
01682                         ->pointer();
01683             }
01684
01685             ThreadLocalValueHolderBase* NewValueForCurrentThread() const override {
01686                 return default_factory_->MakeNewHolder();
01687             }
01688
01689             class ValueHolderFactory {
```

```
01690             public:
01691
01692                 ValueHolderFactory() {
01693                 }
01694
01695                 virtual ~ValueHolderFactory() {
01696                 }
01697                 virtual ValueHolder* MakeNewHolder() const = 0;
01698
01699             private:
01700                 ValueHolderFactory(const ValueHolderFactory&) = delete;
01701                 ValueHolderFactory& operator=(const ValueHolderFactory&) = delete;
01702             };
01703
01704             class DefaultValueHolderFactory : public ValueHolderFactory {
01705             public:
01706
01707                 DefaultValueHolderFactory() {
01708                 }
01709
01710                 ValueHolder* MakeNewHolder() const override {
01711                     return new ValueHolder();
01712                 }
01713
01714             private:
01715                 DefaultValueHolderFactory(const DefaultValueHolderFactory&) = delete;
01716                 DefaultValueHolderFactory& operator=(const DefaultValueHolderFactory&) =
01717                         delete;
01718             };
01719
01720             class InstanceValueHolderFactory : public ValueHolderFactory {
01721             public:
01722
01723                 explicit InstanceValueHolderFactory(const T& value) : value_(value) {
01724                 }
01725
01726                 ValueHolder* MakeNewHolder() const override {
01727                     return new ValueHolder(value_);
01728                 }
01729
01730             private:
01731                 const T value_; // The value for each thread.
01732
01733                 InstanceValueHolderFactory(const InstanceValueHolderFactory&) = delete;
01734                 InstanceValueHolderFactory& operator=(const InstanceValueHolderFactory&) =
01735                         delete;
01736             };
01737
01738             std::unique_ptr<ValueHolderFactory> default_factory_;
01739
01740             ThreadLocal(const ThreadLocal&) = delete;
01741             ThreadLocal& operator=(const ThreadLocal&) = delete;
01742         };
01743
01744 #elif GTEST_HAS_PTHREAD
01745
01746         // MutexBase and Mutex implement mutex on pthreads-based platforms.
01747
01748         class MutexBase {
01749         public:
01750             // Acquires this mutex.
01751
01752             void Lock() {
01753                 GTEST_CHECK_POSIX_SUCCESS_(pthread_mutex_lock(&mutex_));
01754                 owner_ = pthread_self();
01755                 has_owner_ = true;
01756             }
01757
01758             // Releases this mutex.
01759
01760             void Unlock() {
01761                 // Since the lock is being released the owner_ field should no longer be
01762                 // considered valid. We don't protect writing to has_owner_ here, as it's
01763                 // the caller's responsibility to ensure that the current thread holds the
01764                 // mutex when this is called.
01765                 has_owner_ = false;
01766                 GTEST_CHECK_POSIX_SUCCESS_(pthread_mutex_unlock(&mutex_));
01767             }
01768
01769             // Does nothing if the current thread holds the mutex. Otherwise, crashes
01770             // with high probability.
01771
01772             void AssertHeld() const {
01773                 GTEST_CHECK_(has_owner_ && pthread_equal(owner_, pthread_self()))
01774                         << "The current thread is not holding the mutex @" << this;
01775             }
01776
```

```
01777             // A static mutex may be used before main() is entered.  It may even
01778             // be used before the dynamic initialization stage.  Therefore we
01779             // must be able to initialize a static mutex object at link time.
01780             // This means MutexBase has to be a POD and its member variables
01781             // have to be public.
01782         public:
01783             pthread_mutex_t mutex_; // The underlying pthread mutex.
01784             // has_owner_ indicates whether the owner_ field below contains a valid thread
01785             // ID and is therefore safe to inspect (e.g., to use in pthread_equal()). All
01786             // accesses to the owner_ field should be protected by a check of this field.
01787             // An alternative might be to memset() owner_ to all zeros, but there's no
01788             // guarantee that a zero'd pthread_t is necessarily invalid or even different
01789             // from pthread_self().
01790             bool has_owner_;
01791             pthread_t owner_; // The thread holding the mutex.
01792         };
01793
01794         // Forward-declares a static mutex.
01795 #define GTEST_DECLARE_STATIC_MUTEX_(mutex) \
01796   extern ::testing::internal::MutexBase mutex
01797
01798         // Defines and statically (i.e. at link time) initializes a static mutex.
01799         // The initialization list here does not explicitly initialize each field,
01800         // instead relying on default initialization for the unspecified fields. In
01801         // particular, the owner_ field (a pthread_t) is not explicitly initialized.
01802         // This allows initialization to work whether pthread_t is a scalar or struct.
01803         // The flag -Wmissing-field-initializers must not be specified for this to work.
01804 #define GTEST_DEFINE_STATIC_MUTEX_(mutex) \
01805   ::testing::internal::MutexBase mutex = {PTHREAD_MUTEX_INITIALIZER, false, 0}
01806
01807         // The Mutex class can only be used for mutexes created at runtime. It
01808         // shares its API with MutexBase otherwise.
01809
01810         class Mutex : public MutexBase {
01811         public:
01812
01813             Mutex() {
01814                 GTEST_CHECK_POSIX_SUCCESS_(pthread_mutex_init(&mutex_, nullptr));
01815                 has_owner_ = false;
01816             }
01817
01818             ~Mutex() {
01819                 GTEST_CHECK_POSIX_SUCCESS_(pthread_mutex_destroy(&mutex_));
01820             }
01821
01822         private:
01823             Mutex(const Mutex&) = delete;
01824             Mutex& operator=(const Mutex&) = delete;
01825         };
01826
01827         // We cannot name this class MutexLock because the ctor declaration would
01828         // conflict with a macro named MutexLock, which is defined on some
01829         // platforms. That macro is used as a defensive measure to prevent against
01830         // inadvertent misuses of MutexLock like "MutexLock(&mu)" rather than
01831         // "MutexLock l(&mu)".  Hence the typedef trick below.
01832
01833         class GTestMutexLock {
01834         public:
01835
01836             explicit GTestMutexLock(MutexBase* mutex) : mutex_(mutex) {
01837                 mutex_->Lock();
01838             }
01839
01840             ~GTestMutexLock() {
01841                 mutex_->Unlock();
01842             }
01843
01844         private:
01845             MutexBase * const mutex_;
01846
01847             GTestMutexLock(const GTestMutexLock&) = delete;
01848             GTestMutexLock& operator=(const GTestMutexLock&) = delete;
01849         };
01850
01851         typedef GTestMutexLock MutexLock;
01852
01853         // Helpers for ThreadLocal.
01854
01855         // pthread_key_create() requires DeleteThreadLocalValue() to have
01856         // C-linkage.  Therefore it cannot be templatized to access
01857         // ThreadLocal<T>.  Hence the need for class
01858         // ThreadLocalValueHolderBase.
01859
01860         class GTEST_API_ ThreadLocalValueHolderBase {
01861         public:
01862
01863             virtual ~ThreadLocalValueHolderBase() {
```

```
01864             }
01865         };
01866
01867         // Called by pthread to delete thread-local data stored by
01868         // pthread_setspecific().
01869
01870         extern "C" inline void DeleteThreadLocalValue(void* value_holder) {
01871             delete static_cast<ThreadLocalValueHolderBase*> (value_holder);
01872         }
01873
01874         // Implements thread-local storage on pthreads-based systems.
01875
01876         template <typename T>
01877         class GTEST_API_ ThreadLocal {
01878         public:
01879
01880             ThreadLocal()
01881             : key_(CreateKey()), default_factory_(new DefaultValueHolderFactory()) {
01882             }
01883
01884             explicit ThreadLocal(const T& value)
01885             : key_(CreateKey()),
01886             default_factory_(new InstanceValueHolderFactory(value)) {
01887             }
01888
01889             ~ThreadLocal() {
01890                 // Destroys the managed object for the current thread, if any.
01891                 DeleteThreadLocalValue(pthread_getspecific(key_));
01892
01893                 // Releases resources associated with the key.  This will *not*
01894                 // delete managed objects for other threads.
01895                 GTEST_CHECK_POSIX_SUCCESS_(pthread_key_delete(key_));
01896             }
01897
01898             T* pointer() {
01899                 return GetOrCreateValue();
01900             }
01901
01902             const T* pointer() const {
01903                 return GetOrCreateValue();
01904             }
01905
01906             const T& get() const {
01907                 return *pointer();
01908             }
01909
01910             void set(const T& value) {
01911                 *pointer() = value;
01912             }
01913
01914         private:
01915             // Holds a value of type T.
01916
01917             class ValueHolder : public ThreadLocalValueHolderBase {
01918             public:
01919
01920                 ValueHolder() : value_() {
01921                 }
01922
01923                 explicit ValueHolder(const T& value) : value_(value) {
01924                 }
01925
01926                 T* pointer() {
01927                     return &value_;
01928                 }
01929
01930             private:
01931                 T value_;
01932                 ValueHolder(const ValueHolder&) = delete;
01933                 ValueHolder& operator=(const ValueHolder&) = delete;
01934             };
01935
01936             static pthread_key_t CreateKey() {
01937                 pthread_key_t key;
01938                 // When a thread exits, DeleteThreadLocalValue() will be called on
01939                 // the object managed for that thread.
01940                 GTEST_CHECK_POSIX_SUCCESS_(
01941                         pthread_key_create(&key, &DeleteThreadLocalValue));
01942                 return key;
01943             }
01944
01945             T* GetOrCreateValue() const {
01946                 ThreadLocalValueHolderBase * const holder =
01947                         static_cast<ThreadLocalValueHolderBase*> (pthread_getspecific(key_));
01948                 if (holder != nullptr) {
01949                     return CheckedDowncastToActualType<ValueHolder>(holder)->pointer();
01950                 }
```

```
01951
01952                 ValueHolder * const new_holder = default_factory_->MakeNewHolder();
01953                 ThreadLocalValueHolderBase * const holder_base = new_holder;
01954                 GTEST_CHECK_POSIX_SUCCESS_(pthread_setspecific(key_, holder_base));
01955                 return new_holder->pointer();
01956             }
01957
01958             class ValueHolderFactory {
01959             public:
01960
01961                 ValueHolderFactory() {
01962                 }
01963
01964                 virtual ~ValueHolderFactory() {
01965                 }
01966                 virtual ValueHolder* MakeNewHolder() const = 0;
01967
01968             private:
01969                 ValueHolderFactory(const ValueHolderFactory&) = delete;
01970                 ValueHolderFactory& operator=(const ValueHolderFactory&) = delete;
01971             };
01972
01973             class DefaultValueHolderFactory : public ValueHolderFactory {
01974             public:
01975
01976                 DefaultValueHolderFactory() {
01977                 }
01978
01979                 ValueHolder* MakeNewHolder() const override {
01980                     return new ValueHolder();
01981                 }
01982
01983             private:
01984                 DefaultValueHolderFactory(const DefaultValueHolderFactory&) = delete;
01985                 DefaultValueHolderFactory& operator=(const DefaultValueHolderFactory&) =
01986                         delete;
01987             };
01988
01989             class InstanceValueHolderFactory : public ValueHolderFactory {
01990             public:
01991
01992                 explicit InstanceValueHolderFactory(const T& value) : value_(value) {
01993                 }
01994
01995                 ValueHolder* MakeNewHolder() const override {
01996                     return new ValueHolder(value_);
01997                 }
01998
01999             private:
02000                 const T value_; // The value for each thread.
02001
02002                 InstanceValueHolderFactory(const InstanceValueHolderFactory&) = delete;
02003                 InstanceValueHolderFactory& operator=(const InstanceValueHolderFactory&) =
02004                         delete;
02005             };
02006
02007             // A key pthreads uses for looking up per-thread values.
02008             const pthread_key_t key_;
02009             std::unique_ptr<ValueHolderFactory> default_factory_;
02010
02011             ThreadLocal(const ThreadLocal&) = delete;
02012             ThreadLocal& operator=(const ThreadLocal&) = delete;
02013         };
02014
02015 #endif  // GTEST_HAS_MUTEX_AND_THREAD_LOCAL_
02016
02017 #else  // GTEST_IS_THREADSAFE
02018
02019         // A dummy implementation of synchronization primitives (mutex, lock,
02020         // and thread-local variable).  Necessary for compiling Google Test where
02021         // mutex is not supported - using Google Test in multiple threads is not
02022         // supported on such platforms.
02023
02024         class Mutex {
02025         public:
02026
02027             Mutex() {
02028             }
02029
02030             void Lock() {
02031             }
02032
02033             void Unlock() {
02034             }
02035
02036             void AssertHeld() const {
02037             }
```

```
02038         };
02039
02040 #define GTEST_DECLARE_STATIC_MUTEX_(mutex) \
02041   extern ::testing::internal::Mutex mutex
02042
02043 #define GTEST_DEFINE_STATIC_MUTEX_(mutex) ::testing::internal::Mutex mutex
02044
02045         // We cannot name this class MutexLock because the ctor declaration would
02046         // conflict with a macro named MutexLock, which is defined on some
02047         // platforms. That macro is used as a defensive measure to prevent against
02048         // inadvertent misuses of MutexLock like "MutexLock(&mu)" rather than
02049         // "MutexLock l(&mu)".  Hence the typedef trick below.
02050
02051         class GTestMutexLock {
02052         public:
02053
02054             explicit GTestMutexLock(Mutex*) {
02055             } // NOLINT
02056         };
02057
02058         typedef GTestMutexLock MutexLock;
02059
02060         template <typename T>
02061         class GTEST_API_ ThreadLocal {
02062         public:
02063
02064             ThreadLocal() : value_() {
02065             }
02066
02067             explicit ThreadLocal(const T& value) : value_(value) {
02068             }
02069
02070             T* pointer() {
02071                 return &value_;
02072             }
02073
02074             const T* pointer() const {
02075                 return &value_;
02076             }
02077
02078             const T& get() const {
02079                 return value_;
02080             }
02081
02082             void set(const T& value) {
02083                 value_ = value;
02084             }
02085
02086         private:
02087             T value_;
02088         };
02089
02090 #endif  // GTEST_IS_THREADSAFE
02091
02092         // Returns the number of threads running in the process, or 0 to indicate that
02093         // we cannot detect it.
02094         GTEST_API_ size_t GetThreadCount();
02095
02096 #if GTEST_OS_WINDOWS
02097 #define GTEST_PATH_SEP_ "\\"
02098 #define GTEST_HAS_ALT_PATH_SEP_ 1
02099 #else
02100 #define GTEST_PATH_SEP_ "/"
02101 #define GTEST_HAS_ALT_PATH_SEP_ 0
02102 #endif  // GTEST_OS_WINDOWS
02103
02104         // Utilities for char.
02105
02106         // isspace(int ch) and friends accept an unsigned char or EOF.  char
02107         // may be signed, depending on the compiler (or compiler flags).
02108         // Therefore we need to cast a char to unsigned char before calling
02109         // isspace(), etc.
02110
02111         inline bool IsAlpha(char ch) {
02112             return isalpha(static_cast<unsigned char> (ch)) != 0;
02113         }
02114
02115         inline bool IsAlNum(char ch) {
02116             return isalnum(static_cast<unsigned char> (ch)) != 0;
02117         }
02118
02119         inline bool IsDigit(char ch) {
02120             return isdigit(static_cast<unsigned char> (ch)) != 0;
02121         }
02122
02123         inline bool IsLower(char ch) {
02124             return islower(static_cast<unsigned char> (ch)) != 0;
```

```
02125         }
02126
02127         inline bool IsSpace(char ch) {
02128             return isspace(static_cast<unsigned char> (ch)) != 0;
02129         }
02130
02131         inline bool IsUpper(char ch) {
02132             return isupper(static_cast<unsigned char> (ch)) != 0;
02133         }
02134
02135         inline bool IsXDigit(char ch) {
02136             return isxdigit(static_cast<unsigned char> (ch)) != 0;
02137         }
02138 #ifdef __cpp_char8_t
02139
02140         inline bool IsXDigit(char8_t ch) {
02141             return isxdigit(static_cast<unsigned char> (ch)) != 0;
02142         }
02143 #endif
02144
02145         inline bool IsXDigit(char16_t ch) {
02146             const unsigned char low_byte = static_cast<unsigned char> (ch);
02147             return ch == low_byte && isxdigit(low_byte) != 0;
02148         }
02149
02150         inline bool IsXDigit(char32_t ch) {
02151             const unsigned char low_byte = static_cast<unsigned char> (ch);
02152             return ch == low_byte && isxdigit(low_byte) != 0;
02153         }
02154
02155         inline bool IsXDigit(wchar_t ch) {
02156             const unsigned char low_byte = static_cast<unsigned char> (ch);
02157             return ch == low_byte && isxdigit(low_byte) != 0;
02158         }
02159
02160         inline char ToLower(char ch) {
02161             return static_cast<char> (tolower(static_cast<unsigned char> (ch)));
02162         }
02163
02164         inline char ToUpper(char ch) {
02165             return static_cast<char> (toupper(static_cast<unsigned char> (ch)));
02166         }
02167
02168         inline std::string StripTrailingSpaces(std::string str) {
02169             std::string::iterator it = str.end();
02170             while (it != str.begin() && IsSpace(*--it)) it = str.erase(it);
02171             return str;
02172         }
02173
02174         // The testing::internal::posix namespace holds wrappers for common
02175         // POSIX functions.  These wrappers hide the differences between
02176         // Windows/MSVC and POSIX systems.  Since some compilers define these
02177         // standard functions as macros, the wrapper cannot have the same name
02178         // as the wrapped function.
02179
02180         namespace posix {
02181
02182             // Functions with a different name on Windows.
02183
02184 #if GTEST_OS_WINDOWS
02185
02186             typedef struct _stat StatStruct;
02187
02188 #ifdef __BORLANDC__
02189
02190             inline int DoIsATTY(int fd) {
02191                 return isatty(fd);
02192             }
02193
02194             inline int StrCaseCmp(const char* s1, const char* s2) {
02195                 return stricmp(s1, s2);
02196             }
02197
02198             inline char* StrDup(const char* src) {
02199                 return strdup(src);
02200             }
02201 #else  // !__BORLANDC__
02202 #if GTEST_OS_WINDOWS_MOBILE || GTEST_OS_ZOS || GTEST_OS_IOS || \
02203     GTEST_OS_WINDOWS_PHONE || GTEST_OS_WINDOWS_RT || defined(ESP_PLATFORM)
02204
02205             inline int DoIsATTY(int /* fd */) {
02206                 return 0;
02207             }
02208 #else
02209
02210             inline int DoIsATTY(int fd) {
02211                 return _isatty(fd);
```

```
02212             }
02213 #endif  // GTEST_OS_WINDOWS_MOBILE
02214
02215             inline int StrCaseCmp(const char* s1, const char* s2) {
02216                 return _stricmp(s1, s2);
02217             }
02218
02219             inline char* StrDup(const char* src) {
02220                 return _strdup(src);
02221             }
02222 #endif  // __BORLANDC__
02223
02224 #if GTEST_OS_WINDOWS_MOBILE
02225
02226             inline int FileNo(FILE* file) {
02227                 return reinterpret_cast<int> (_fileno(file));
02228             }
02229             // Stat(), RmDir(), and IsDir() are not needed on Windows CE at this
02230             // time and thus not defined there.
02231 #else
02232
02233             inline int FileNo(FILE* file) {
02234                 return _fileno(file);
02235             }
02236
02237             inline int Stat(const char* path, StatStruct* buf) {
02238                 return _stat(path, buf);
02239             }
02240
02241             inline int RmDir(const char* dir) {
02242                 return _rmdir(dir);
02243             }
02244
02245             inline bool IsDir(const StatStruct& st) {
02246                 return (_S_IFDIR & st.st_mode) != 0;
02247             }
02248 #endif  // GTEST_OS_WINDOWS_MOBILE
02249
02250 #elif GTEST_OS_ESP8266
02251             typedef struct stat StatStruct;
02252
02253             inline int FileNo(FILE* file) {
02254                 return fileno(file);
02255             }
02256
02257             inline int DoIsATTY(int fd) {
02258                 return isatty(fd);
02259             }
02260
02261             inline int Stat(const char* path, StatStruct* buf) {
02262                 // stat function not implemented on ESP8266
02263                 return 0;
02264             }
02265
02266             inline int StrCaseCmp(const char* s1, const char* s2) {
02267                 return strcasecmp(s1, s2);
02268             }
02269
02270             inline char* StrDup(const char* src) {
02271                 return strdup(src);
02272             }
02273
02274             inline int RmDir(const char* dir) {
02275                 return rmdir(dir);
02276             }
02277
02278             inline bool IsDir(const StatStruct& st) {
02279                 return S_ISDIR(st.st_mode);
02280             }
02281
02282 #else
02283
02284             typedef struct stat StatStruct;
02285
02286             inline int FileNo(FILE* file) {
02287                 return fileno(file);
02288             }
02289
02290             inline int DoIsATTY(int fd) {
02291                 return isatty(fd);
02292             }
02293
02294             inline int Stat(const char* path, StatStruct* buf) {
02295                 return stat(path, buf);
02296             }
02297
02298             inline int StrCaseCmp(const char* s1, const char* s2) {
```

```
02299                 return strcasecmp(s1, s2);
02300             }
02301
02302             inline char* StrDup(const char* src) {
02303                 return strdup(src);
02304             }
02305
02306             inline int RmDir(const char* dir) {
02307                 return rmdir(dir);
02308             }
02309
02310             inline bool IsDir(const StatStruct& st) {
02311                 return S_ISDIR(st.st_mode);
02312             }
02313
02314 #endif  // GTEST_OS_WINDOWS
02315
02316             inline int IsATTY(int fd) {
02317                 // DoIsATTY might change errno (for example ENOTTY in case you redirect stdout
02318                 // to a file on Linux), which is unexpected, so save the previous value, and
02319                 // restore it after the call.
02320                 int savedErrno = errno;
02321                 int isAttyValue = DoIsATTY(fd);
02322                 errno = savedErrno;
02323
02324                 return isAttyValue;
02325             }
02326
02327             // Functions deprecated by MSVC 8.0.
02328
02329             GTEST_DISABLE_MSC_DEPRECATED_PUSH_()
02330
02331             // ChDir(), FReopen(), FDOpen(), Read(), Write(), Close(), and
02332             // StrError() aren't needed on Windows CE at this time and thus not
02333             // defined there.
02334
02335 #if !GTEST_OS_WINDOWS_MOBILE && !GTEST_OS_WINDOWS_PHONE && \
02336     !GTEST_OS_WINDOWS_RT && !GTEST_OS_ESP8266 && !GTEST_OS_XTENSA
02337             inline int ChDir(const char* dir) {
02338                 return chdir(dir);
02339             }
02340 #endif
02341
02342             inline FILE* FOpen(const char* path, const char* mode) {
02343 #if GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_MINGW
02344
02345                 struct wchar_codecvt : public std::codecvt<wchar_t, char, std::mbstate_t> {
02346                 };
02347                 std::wstring_convert<wchar_codecvt> converter;
02348                 std::wstring wide_path = converter.from_bytes(path);
02349                 std::wstring wide_mode = converter.from_bytes(mode);
02350                 return _wfopen(wide_path.c_str(), wide_mode.c_str());
02351 #else   // GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_MINGW
02352                 return fopen(path, mode);
02353 #endif  // GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_MINGW
02354             }
02355 #if !GTEST_OS_WINDOWS_MOBILE
02356
02357             inline FILE* FReopen(const char* path, const char* mode, FILE* stream) {
02358                 return freopen(path, mode, stream);
02359             }
02360
02361             inline FILE* FDOpen(int fd, const char* mode) {
02362                 return fdopen(fd, mode);
02363             }
02364 #endif
02365
02366             inline int FClose(FILE* fp) {
02367                 return fclose(fp);
02368             }
02369 #if !GTEST_OS_WINDOWS_MOBILE
02370
02371             inline int Read(int fd, void* buf, unsigned int count) {
02372                 return static_cast<int> (read(fd, buf, count));
02373             }
02374
02375             inline int Write(int fd, const void* buf, unsigned int count) {
02376                 return static_cast<int> (write(fd, buf, count));
02377             }
02378
02379             inline int Close(int fd) {
02380                 return close(fd);
02381             }
02382
02383             inline const char* StrError(int errnum) {
02384                 return strerror(errnum);
02385             }
```

```
02386 #endif
02387
02388             inline const char* GetEnv(const char* name) {
02389 #if GTEST_OS_WINDOWS_MOBILE || GTEST_OS_WINDOWS_PHONE || \
02390     GTEST_OS_WINDOWS_RT || GTEST_OS_ESP8266 || GTEST_OS_XTENSA
02391                 // We are on an embedded platform, which has no environment variables.
02392                 static_cast<void> (name); // To prevent 'unused argument' warning.
02393                 return nullptr;
02394 #elif defined(__BORLANDC__) || defined(__SunOS_5_8) || defined(__SunOS_5_9)
02395                 // Environment variables which we programmatically clear will be set to the
02396                 // empty string rather than unset (NULL).  Handle that case.
02397                 const char* const env = getenv(name);
02398                 return (env != nullptr && env[0] != '\0') ? env : nullptr;
02399 #else
02400                 return getenv(name);
02401 #endif
02402             }
02403
02404             GTEST_DISABLE_MSC_DEPRECATED_POP_()
02405
02406 #if GTEST_OS_WINDOWS_MOBILE
02407             // Windows CE has no C library. The abort() function is used in
02408             // several places in Google Test. This implementation provides a reasonable
02409             // imitation of standard behaviour.
02410             [[noreturn]] void Abort();
02411 #else
02412             [[noreturn]] inline void Abort() {
02413                 abort();
02414             }
02415 #endif  // GTEST_OS_WINDOWS_MOBILE
02416
02417         } // namespace posix
02418
02419         // MSVC "deprecates" snprintf and issues warnings wherever it is used.  In
02420         // order to avoid these warnings, we need to use _snprintf or _snprintf_s on
02421         // MSVC-based platforms.  We map the GTEST_SNPRINTF_ macro to the appropriate
02422         // function in order to achieve that.  We use macro definition here because
02423         // snprintf is a variadic function.
02424 #if _MSC_VER && !GTEST_OS_WINDOWS_MOBILE
02425         // MSVC 2005 and above support variadic macros.
02426 #define GTEST_SNPRINTF_(buffer, size, format, ...) \
02427   _snprintf_s(buffer, size, size, format, __VA_ARGS__)
02428 #elif defined(_MSC_VER)
02429         // Windows CE does not define _snprintf_s
02430 #define GTEST_SNPRINTF_ _snprintf
02431 #else
02432 #define GTEST_SNPRINTF_ snprintf
02433 #endif
02434
02435         // The biggest signed integer type the compiler supports.
02436         //
02437         // long long is guaranteed to be at least 64-bits in C++11.
02438         using BiggestInt = long long; // NOLINT
02439
02440         // The maximum number a BiggestInt can represent.
02441         constexpr BiggestInt kMaxBiggestInt = (std::numeric_limits<BiggestInt>::max)();
02442
02443         // This template class serves as a compile-time function from size to
02444         // type.  It maps a size in bytes to a primitive type with that
02445         // size. e.g.
02446         //
02447         //   TypeWithSize<4>::UInt
02448         //
02449         // is typedef-ed to be unsigned int (unsigned integer made up of 4
02450         // bytes).
02451         //
02452         // Such functionality should belong to STL, but I cannot find it
02453         // there.
02454         //
02455         // Google Test uses this class in the implementation of floating-point
02456         // comparison.
02457         //
02458         // For now it only handles UInt (unsigned int) as that's all Google Test
02459         // needs.  Other types can be easily added in the future if need
02460         // arises.
02461
02462         template <size_t size>
02463         class TypeWithSize {
02464         public:
02465             // This prevents the user from using TypeWithSize<N> with incorrect
02466             // values of N.
02467             using UInt = void;
02468         };
02469
02470         // The specialization for size 4.
02471
02472         template <>
```

```
02473         class TypeWithSize<4> {
02474         public:
02475             using Int = std::int32_t;
02476             using UInt = std::uint32_t;
02477         };
02478 
02479         // The specialization for size 8.
02480 
02481         template <>
02482         class TypeWithSize<8> {
02483         public:
02484             using Int = std::int64_t;
02485             using UInt = std::uint64_t;
02486         };
02487 
02488         // Integer types of known sizes.
02489         using TimeInMillis = int64_t; // Represents time in milliseconds.
02490 
02491         // Utilities for command line flags and environment variables.
02492 
02493         // Macro for referencing flags.
02494 #if !defined(GTEST_FLAG)
02495 #define GTEST_FLAG_NAME_(name) gtest_##name
02496 #define GTEST_FLAG(name) FLAGS_gtest_##name
02497 #endif  // !defined(GTEST_FLAG)
02498 
02499         // Pick a command line flags implementation.
02500 #if GTEST_HAS_ABSL
02501 
02502         // Macros for defining flags.
02503 #define GTEST_DEFINE_bool_(name, default_val, doc) \
02504   ABSL_FLAG(bool, GTEST_FLAG_NAME_(name), default_val, doc)
02505 #define GTEST_DEFINE_int32_(name, default_val, doc) \
02506   ABSL_FLAG(int32_t, GTEST_FLAG_NAME_(name), default_val, doc)
02507 #define GTEST_DEFINE_string_(name, default_val, doc) \
02508   ABSL_FLAG(std::string, GTEST_FLAG_NAME_(name), default_val, doc)
02509 
02510         // Macros for declaring flags.
02511 #define GTEST_DECLARE_bool_(name) \
02512   ABSL_DECLARE_FLAG(bool, GTEST_FLAG_NAME_(name))
02513 #define GTEST_DECLARE_int32_(name) \
02514   ABSL_DECLARE_FLAG(int32_t, GTEST_FLAG_NAME_(name))
02515 #define GTEST_DECLARE_string_(name) \
02516   ABSL_DECLARE_FLAG(std::string, GTEST_FLAG_NAME_(name))
02517 
02518 #define GTEST_FLAG_SAVER_ ::absl::FlagSaver
02519 
02520 #define GTEST_FLAG_GET(name) ::absl::GetFlag(GTEST_FLAG(name))
02521 #define GTEST_FLAG_SET(name, value) \
02522   (void)(::absl::SetFlag(&GTEST_FLAG(name), value))
02523 #define GTEST_USE_OWN_FLAGFILE_FLAG_ 0
02524 
02525 #else  // GTEST_HAS_ABSL
02526 
02527         // Macros for defining flags.
02528 #define GTEST_DEFINE_bool_(name, default_val, doc)  \
02529   namespace testing {                               \
02530   GTEST_API_ bool GTEST_FLAG(name) = (default_val); \
02531   }                                                 \
02532   static_assert(true, "no-op to require trailing semicolon")
02533 #define GTEST_DEFINE_int32_(name, default_val, doc)         \
02534   namespace testing {                                       \
02535   GTEST_API_ std::int32_t GTEST_FLAG(name) = (default_val); \
02536   }                                                         \
02537   static_assert(true, "no-op to require trailing semicolon")
02538 #define GTEST_DEFINE_string_(name, default_val, doc)         \
02539   namespace testing {                                        \
02540   GTEST_API_ ::std::string GTEST_FLAG(name) = (default_val); \
02541   }                                                          \
02542   static_assert(true, "no-op to require trailing semicolon")
02543 
02544         // Macros for declaring flags.
02545 #define GTEST_DECLARE_bool_(name)          \
02546   namespace testing {                      \
02547   GTEST_API_ extern bool GTEST_FLAG(name); \
02548   }                                        \
02549   static_assert(true, "no-op to require trailing semicolon")
02550 #define GTEST_DECLARE_int32_(name)                \
02551   namespace testing {                             \
02552   GTEST_API_ extern std::int32_t GTEST_FLAG(name); \
02553   }                                               \
02554   static_assert(true, "no-op to require trailing semicolon")
02555 #define GTEST_DECLARE_string_(name)                \
02556   namespace testing {                              \
02557   GTEST_API_ extern ::std::string GTEST_FLAG(name); \
02558   }                                                \
02559   static_assert(true, "no-op to require trailing semicolon")
```

```
02560
02561 #define GTEST_FLAG_SAVER_ ::testing::internal::GTestFlagSaver
02562
02563 #define GTEST_FLAG_GET(name) ::testing::GTEST_FLAG(name)
02564 #define GTEST_FLAG_SET(name, value) (void)(::testing::GTEST_FLAG(name) = value)
02565 #define GTEST_USE_OWN_FLAGFILE_FLAG_ 1
02566
02567 #endif  // GTEST_HAS_ABSL
02568
02569         // Thread annotations
02570 #if !defined(GTEST_EXCLUSIVE_LOCK_REQUIRED_)
02571 #define GTEST_EXCLUSIVE_LOCK_REQUIRED_(locks)
02572 #define GTEST_LOCK_EXCLUDED_(locks)
02573 #endif  // !defined(GTEST_EXCLUSIVE_LOCK_REQUIRED_)
02574
02575         // Parses 'str' for a 32-bit signed integer.  If successful, writes the result
02576         // to *value and returns true; otherwise leaves *value unchanged and returns
02577         // false.
02578         GTEST_API_ bool ParseInt32(const Message& src_text, const char* str,
02579                 int32_t* value);
02580
02581         // Parses a bool/int32_t/string from the environment variable
02582         // corresponding to the given Google Test flag.
02583         bool BoolFromGTestEnv(const char* flag, bool default_val);
02584         GTEST_API_ int32_t Int32FromGTestEnv(const char* flag, int32_t default_val);
02585         std::string OutputFlagAlsoCheckEnvVar();
02586         const char* StringFromGTestEnv(const char* flag, const char* default_val);
02587
02588     } // namespace internal
02589 } // namespace testing
02590
02591 #if !defined(GTEST_INTERNAL_DEPRECATED)
02592
02593 // Internal Macro to mark an API deprecated, for googletest usage only
02594 // Usage: class GTEST_INTERNAL_DEPRECATED(message) MyClass or
02595 // GTEST_INTERNAL_DEPRECATED(message) <return_type> myFunction(); Every usage of
02596 // a deprecated entity will trigger a warning when compiled with
02597 // `-Wdeprecated-declarations` option (clang, gcc, any __GNUC__ compiler).
02598 // For msvc /W3 option will need to be used
02599 // Note that for 'other' compilers this macro evaluates to nothing to prevent
02600 // compilations errors.
02601 #if defined(_MSC_VER)
02602 #define GTEST_INTERNAL_DEPRECATED(message) __declspec(deprecated(message))
02603 #elif defined(__GNUC__)
02604 #define GTEST_INTERNAL_DEPRECATED(message) __attribute__((deprecated(message)))
02605 #else
02606 #define GTEST_INTERNAL_DEPRECATED(message)
02607 #endif
02608
02609 #endif  // !defined(GTEST_INTERNAL_DEPRECATED)
02610
02611 #if GTEST_HAS_ABSL
02612 // Always use absl::any for UniversalPrinter<> specializations if googletest
02613 // is built with absl support.
02614 #define GTEST_INTERNAL_HAS_ANY 1
02615 #include "absl/types/any.h"
02616 namespace testing {
02617     namespace internal {
02618         using Any = ::absl::any;
02619     } // namespace internal
02620 } // namespace testing
02621 #else
02622 #ifdef __has_include
02623 #if __has_include(<any>) && __cplusplus >= 201703L
02624 // Otherwise for C++17 and higher use std::any for UniversalPrinter<>
02625 // specializations.
02626 #define GTEST_INTERNAL_HAS_ANY 1
02627 #include <any>
02628 namespace testing {
02629     namespace internal {
02630         using Any = ::std::any;
02631     } // namespace internal
02632 } // namespace testing
02633 // The case where absl is configured NOT to alias std::any is not
02634 // supported.
02635 #endif  // __has_include(<any>) && __cplusplus >= 201703L
02636 #endif  // __has_include
02637 #endif  // GTEST_HAS_ABSL
02638
02639 #if GTEST_HAS_ABSL
02640 // Always use absl::optional for UniversalPrinter<> specializations if
02641 // googletest is built with absl support.
02642 #define GTEST_INTERNAL_HAS_OPTIONAL 1
02643 #include "absl/types/optional.h"
02644 namespace testing {
02645     namespace internal {
02646         template <typename T>
```

```
02647         using Optional = ::absl::optional<T>;
02648
02649         inline ::absl::nullopt_t Nullopt() {
02650             return ::absl::nullopt;
02651         }
02652     } // namespace internal
02653 } // namespace testing
02654 #else
02655 #ifdef __has_include
02656 #if __has_include(<optional>) && __cplusplus >= 201703L
02657 // Otherwise for C++17 and higher use std::optional for UniversalPrinter<>
02658 // specializations.
02659 #define GTEST_INTERNAL_HAS_OPTIONAL 1
02660 #include <optional>
02661 namespace testing {
02662     namespace internal {
02663         template <typename T>
02664         using Optional = ::std::optional<T>;
02665
02666         inline ::std::nullopt_t Nullopt() {
02667             return ::std::nullopt;
02668         }
02669     } // namespace internal
02670 } // namespace testing
02671 // The case where absl is configured NOT to alias std::optional is not
02672 // supported.
02673 #endif  // __has_include(<optional>) && __cplusplus >= 201703L
02674 #endif  // __has_include
02675 #endif  // GTEST_HAS_ABSL
02676
02677 #if GTEST_HAS_ABSL
02678 // Always use absl::string_view for Matcher<> specializations if googletest
02679 // is built with absl support.
02680 #define GTEST_INTERNAL_HAS_STRING_VIEW 1
02681 #include "absl/strings/string_view.h"
02682 namespace testing {
02683     namespace internal {
02684         using StringView = ::absl::string_view;
02685     } // namespace internal
02686 } // namespace testing
02687 #else
02688 #ifdef __has_include
02689 #if __has_include(<string_view>) && __cplusplus >= 201703L
02690 // Otherwise for C++17 and higher use std::string_view for Matcher<>
02691 // specializations.
02692 #define GTEST_INTERNAL_HAS_STRING_VIEW 1
02693 #include <string_view>
02694 namespace testing {
02695     namespace internal {
02696         using StringView = ::std::string_view;
02697     } // namespace internal
02698 } // namespace testing
02699 // The case where absl is configured NOT to alias std::string_view is not
02700 // supported.
02701 #endif  // __has_include(<string_view>) && __cplusplus >= 201703L
02702 #endif  // __has_include
02703 #endif  // GTEST_HAS_ABSL
02704
02705 #if GTEST_HAS_ABSL
02706 // Always use absl::variant for UniversalPrinter<> specializations if googletest
02707 // is built with absl support.
02708 #define GTEST_INTERNAL_HAS_VARIANT 1
02709 #include "absl/types/variant.h"
02710 namespace testing {
02711     namespace internal {
02712         template <typename... T>
02713         using Variant = ::absl::variant<T...>;
02714     } // namespace internal
02715 } // namespace testing
02716 #else
02717 #ifdef __has_include
02718 #if __has_include(<variant>) && __cplusplus >= 201703L
02719 // Otherwise for C++17 and higher use std::variant for UniversalPrinter<>
02720 // specializations.
02721 #define GTEST_INTERNAL_HAS_VARIANT 1
02722 #include <variant>
02723 namespace testing {
02724     namespace internal {
02725         template <typename... T>
02726         using Variant = ::std::variant<T...>;
02727     } // namespace internal
02728 } // namespace testing
02729 // The case where absl is configured NOT to alias std::variant is not supported.
02730 #endif  // __has_include(<variant>) && __cplusplus >= 201703L
02731 #endif  // __has_include
02732 #endif  // GTEST_HAS_ABSL
02733
```

```
02734 #endif  // GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_PORT_H_
```

18.275 gtest-death-test-internal.h

```
00001 // Copyright 2005, Google Inc.
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00003 //
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00026 // THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // The Google C++ Testing and Mocking Framework (Google Test)
00031 //
00032 // This header file defines internal utilities needed for implementing
00033 // death tests.  They are subject to change without notice.
00034
00035 // IWYU pragma: private, include "gtest/gtest.h"
00036 // IWYU pragma: friend gtest/.*
00037 // IWYU pragma: friend gmock/.*
00038
00039 #ifndef GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_DEATH_TEST_INTERNAL_H_
00040 #define GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_DEATH_TEST_INTERNAL_H_
00041
00042 #include <stdio.h>
00043
00044 #include <memory>
00045
00046 #include "gtest/gtest-matchers.h"
00047 #include "gtest/internal/gtest-internal.h"
00048
00049 GTEST_DECLARE_string_(internal_run_death_test);
00050
00051 namespace testing {
00052     namespace internal {
00053
00054         // Names of the flags (needed for parsing Google Test flags).
00055         const char kDeathTestStyleFlag[] = "death_test_style";
00056         const char kDeathTestUseFork[] = "death_test_use_fork";
00057         const char kInternalRunDeathTestFlag[] = "internal_run_death_test";
00058
00059 #if GTEST_HAS_DEATH_TEST
00060
00061         GTEST_DISABLE_MSC_WARNINGS_PUSH_(4251 \
00062 /* class A needs to have dll-interface to be used by clients of class B */)
00063
00064         // DeathTest is a class that hides much of the complexity of the
00065         // GTEST_DEATH_TEST_ macro.  It is abstract; its static Create method
00066         // returns a concrete class that depends on the prevailing death test
00067         // style, as defined by the --gtest_death_test_style and/or
00068         // --gtest_internal_run_death_test flags.
00069
00070         // In describing the results of death tests, these terms are used with
00071         // the corresponding definitions:
00072         //
00073         // exit status:  The integer exit information in the format specified
00074         //               by wait(2)
00075         // exit code:    The integer code passed to exit(3), _exit(2), or
00076         //               returned from main()
00077         class GTEST_API_ DeathTest {
00078         public:
00079             // Create returns false if there was an error determining the
00080             // appropriate action to take for the current death test; for example,
00081             // if the gtest_death_test_style flag is set to an invalid value.
```

```
00082             // The LastMessage method will return a more detailed message in that
00083             // case.  Otherwise, the DeathTest pointer pointed to by the "test"
00084             // argument is set.  If the death test should be skipped, the pointer
00085             // is set to NULL; otherwise, it is set to the address of a new concrete
00086             // DeathTest object that controls the execution of the current test.
00087             static bool Create(const char* statement, Matcher<const std::string&> matcher,
00088                    const char* file, int line, DeathTest** test);
00089             DeathTest();
00090
00091             virtual ~DeathTest() {
00092             }
00093
00094             // A helper class that aborts a death test when it's deleted.
00095
00096             class ReturnSentinel {
00097             public:
00098
00099                 explicit ReturnSentinel(DeathTest* test) : test_(test) {
00100                 }
00101
00102                 ~ReturnSentinel() {
00103                     test_->Abort(TEST_ENCOUNTERED_RETURN_STATEMENT);
00104                 }
00105
00106             private:
00107                 DeathTest * const test_;
00108                 ReturnSentinel(const ReturnSentinel&) = delete;
00109                 ReturnSentinel& operator=(const ReturnSentinel&) = delete;
00110             } GTEST_ATTRIBUTE_UNUSED_;
00111
00112             // An enumeration of possible roles that may be taken when a death
00113             // test is encountered.  EXECUTE means that the death test logic should
00114             // be executed immediately.  OVERSEE means that the program should prepare
00115             // the appropriate environment for a child process to execute the death
00116             // test, then wait for it to complete.
00117
00118             enum TestRole {
00119                 OVERSEE_TEST, EXECUTE_TEST
00120             };
00121
00122             // An enumeration of the three reasons that a test might be aborted.
00123
00124             enum AbortReason {
00125                 TEST_ENCOUNTERED_RETURN_STATEMENT,
00126                 TEST_THREW_EXCEPTION,
00127                 TEST_DID_NOT_DIE
00128             };
00129
00130             // Assumes one of the above roles.
00131             virtual TestRole AssumeRole() = 0;
00132
00133             // Waits for the death test to finish and returns its status.
00134             virtual int Wait() = 0;
00135
00136             // Returns true if the death test passed; that is, the test process
00137             // exited during the test, its exit status matches a user-supplied
00138             // predicate, and its stderr output matches a user-supplied regular
00139             // expression.
00140             // The user-supplied predicate may be a macro expression rather
00141             // than a function pointer or functor, or else Wait and Passed could
00142             // be combined.
00143             virtual bool Passed(bool exit_status_ok) = 0;
00144
00145             // Signals that the death test did not die as expected.
00146             virtual void Abort(AbortReason reason) = 0;
00147
00148             // Returns a human-readable outcome message regarding the outcome of
00149             // the last death test.
00150             static const char* LastMessage();
00151
00152             static void set_last_death_test_message(const std::string& message);
00153
00154         private:
00155             // A string containing a description of the outcome of the last death test.
00156             static std::string last_death_test_message_;
00157
00158             DeathTest(const DeathTest&) = delete;
00159             DeathTest& operator=(const DeathTest&) = delete;
00160         };
00161
00162         GTEST_DISABLE_MSC_WARNINGS_POP_() //  4251
00163
00164         // Factory interface for death tests.  May be mocked out for testing.
00165         class DeathTestFactory {
00166         public:
00167
00168             virtual ~DeathTestFactory() {
```

```
00169             }
00170             virtual bool Create(const char* statement,
00171                     Matcher<const std::string&> matcher, const char* file,
00172                     int line, DeathTest** test) = 0;
00173         };
00174
00175         // A concrete DeathTestFactory implementation for normal use.
00176
00177         class DefaultDeathTestFactory : public DeathTestFactory {
00178         public:
00179             bool Create(const char* statement, Matcher<const std::string&> matcher,
00180                     const char* file, int line, DeathTest** test) override;
00181         };
00182
00183         // Returns true if exit_status describes a process that was terminated
00184         // by a signal, or exited normally with a nonzero exit code.
00185         GTEST_API_ bool ExitedUnsuccessfully(int exit_status);
00186
00187         // A string passed to EXPECT_DEATH (etc.) is caught by one of these overloads
00188         // and interpreted as a regex (rather than an Eq matcher) for legacy
00189         // compatibility.
00190
00191         inline Matcher<const ::std::string&> MakeDeathTestMatcher(
00192                 ::testing::internal::RE regex) {
00193             return ContainsRegex(regex.pattern());
00194         }
00195
00196         inline Matcher<const ::std::string&> MakeDeathTestMatcher(const char* regex) {
00197             return ContainsRegex(regex);
00198         }
00199
00200         inline Matcher<const ::std::string&> MakeDeathTestMatcher(
00201                 const ::std::string& regex) {
00202             return ContainsRegex(regex);
00203         }
00204
00205         // If a Matcher<const ::std::string&> is passed to EXPECT_DEATH (etc.), it's
00206         // used directly.
00207
00208         inline Matcher<const ::std::string&> MakeDeathTestMatcher(
00209                 Matcher<const ::std::string&> matcher) {
00210             return matcher;
00211         }
00212
00213         // Traps C++ exceptions escaping statement and reports them as test
00214         // failures. Note that trapping SEH exceptions is not implemented here.
00215 #if GTEST_HAS_EXCEPTIONS
00216 #define GTEST_EXECUTE_DEATH_TEST_STATEMENT_(statement, death_test)           \
00217   try {                                                                      \
00218     GTEST_SUPPRESS_UNREACHABLE_CODE_WARNING_BELOW_(statement);               \
00219   } catch (const ::std::exception& gtest_exception) {                        \
00220     fprintf(                                                                 \
00221         stderr,                                                              \
00222         "\n%s: Caught std::exception-derived exception escaping the "        \
00223         "death test statement. Exception message: %s\n",                     \
00224         ::testing::internal::FormatFileLocation(__FILE__, __LINE__).c_str(), \
00225         gtest_exception.what());                                             \
00226     fflush(stderr);                                                          \
00227     death_test->Abort(::testing::internal::DeathTest::TEST_THREW_EXCEPTION); \
00228   } catch (...) {                                                            \
00229     death_test->Abort(::testing::internal::DeathTest::TEST_THREW_EXCEPTION); \
00230   }
00231
00232 #else
00233 #define GTEST_EXECUTE_DEATH_TEST_STATEMENT_(statement, death_test) \
00234   GTEST_SUPPRESS_UNREACHABLE_CODE_WARNING_BELOW_(statement)
00235
00236 #endif
00237
00238         // This macro is for implementing ASSERT_DEATH*, EXPECT_DEATH*,
00239         // ASSERT_EXIT*, and EXPECT_EXIT*.
00240 #define GTEST_DEATH_TEST_(statement, predicate, regex_or_matcher, fail)        \
00241   GTEST_AMBIGUOUS_ELSE_BLOCKER_                                                \
00242   if (::testing::internal::AlwaysTrue()) {                                     \
00243     ::testing::internal::DeathTest* gtest_dt;                                  \
00244     if (!::testing::internal::DeathTest::Create(                               \
00245             #statement,                                                        \
00246             ::testing::internal::MakeDeathTestMatcher(regex_or_matcher),       \
00247             __FILE__, __LINE__, &gtest_dt)) {                                  \
00248       goto GTEST_CONCAT_TOKEN_(gtest_label_, __LINE__);                        \
00249     }                                                                          \
00250     if (gtest_dt != nullptr) {                                                 \
00251       std::unique_ptr< ::testing::internal::DeathTest> gtest_dt_ptr(gtest_dt); \
00252       switch (gtest_dt->AssumeRole()) {                                        \
00253         case ::testing::internal::DeathTest::OVERSEE_TEST:                     \
00254           if (!gtest_dt->Passed(predicate(gtest_dt->Wait()))) {                \
00255             goto GTEST_CONCAT_TOKEN_(gtest_label_, __LINE__);                  \
```

```
00256           }                                                                \
00257           break;                                                           \
00258         case ::testing::internal::DeathTest::EXECUTE_TEST: {               \
00259           ::testing::internal::DeathTest::ReturnSentinel gtest_sentinel(   \
00260               gtest_dt);                                                   \
00261           GTEST_EXECUTE_DEATH_TEST_STATEMENT_(statement, gtest_dt);        \
00262           gtest_dt->Abort(::testing::internal::DeathTest::TEST_DID_NOT_DIE); \
00263           break;                                                           \
00264         }                                                                  \
00265       }                                                                    \
00266     }                                                                      \
00267   } else                                                                   \
00268     GTEST_CONCAT_TOKEN_(gtest_label_, __LINE__)                            \
00269         : fail(::testing::internal::DeathTest::LastMessage())
00270         // The symbol "fail" here expands to something into which a message
00271         // can be streamed.
00272
00273         // This macro is for implementing ASSERT/EXPECT_DEBUG_DEATH when compiled in
00274         // NDEBUG mode. In this case we need the statements to be executed and the macro
00275         // must accept a streamed message even though the message is never printed.
00276         // The regex object is not evaluated, but it is used to prevent "unused"
00277         // warnings and to avoid an expression that doesn't compile in debug mode.
00278 #define GTEST_EXECUTE_STATEMENT_(statement, regex_or_matcher)    \
00279   GTEST_AMBIGUOUS_ELSE_BLOCKER_                                  \
00280   if (::testing::internal::AlwaysTrue()) {                       \
00281     GTEST_SUPPRESS_UNREACHABLE_CODE_WARNING_BELOW_(statement);   \
00282   } else if (!::testing::internal::AlwaysTrue()) {               \
00283     ::testing::internal::MakeDeathTestMatcher(regex_or_matcher); \
00284   } else                                                         \
00285     ::testing::Message()
00286
00287         // A class representing the parsed contents of the
00288         // --gtest_internal_run_death_test flag, as it existed when
00289         // RUN_ALL_TESTS was called.
00290
00291         class InternalRunDeathTestFlag {
00292         public:
00293
00294             InternalRunDeathTestFlag(const std::string& a_file, int a_line, int an_index,
00295                    int a_write_fd)
00296             : file_(a_file), line_(a_line), index_(an_index), write_fd_(a_write_fd) {
00297             }
00298
00299             ~InternalRunDeathTestFlag() {
00300                 if (write_fd_ >= 0) posix::Close(write_fd_);
00301             }
00302
00303             const std::string& file() const {
00304                 return file_;
00305             }
00306
00307             int line() const {
00308                 return line_;
00309             }
00310
00311             int index() const {
00312                 return index_;
00313             }
00314
00315             int write_fd() const {
00316                 return write_fd_;
00317             }
00318
00319         private:
00320             std::string file_;
00321             int line_;
00322             int index_;
00323             int write_fd_;
00324
00325             InternalRunDeathTestFlag(const InternalRunDeathTestFlag&) = delete;
00326             InternalRunDeathTestFlag& operator=(const InternalRunDeathTestFlag&) = delete;
00327         };
00328
00329         // Returns a newly created InternalRunDeathTestFlag object with fields
00330         // initialized from the GTEST_FLAG(internal_run_death_test) flag if
00331         // the flag is specified; otherwise returns NULL.
00332         InternalRunDeathTestFlag* ParseInternalRunDeathTestFlag();
00333
00334 #endif  // GTEST_HAS_DEATH_TEST
00335
00336     } // namespace internal
00337 } // namespace testing
00338
00339 #endif  // GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_DEATH_TEST_INTERNAL_H_
```

18.276 gtest-filepath.h

```
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00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // Google Test filepath utilities
00031 //
00032 // This header file declares classes and functions used internally by
00033 // Google Test.  They are subject to change without notice.
00034 //
00035 // This file is #included in gtest/internal/gtest-internal.h.
00036 // Do not include this header file separately!
00037
00038 // IWYU pragma: private, include "gtest/gtest.h"
00039 // IWYU pragma: friend gtest/.*
00040 // IWYU pragma: friend gmock/.*
00041
00042 #ifndef GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_FILEPATH_H_
00043 #define GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_FILEPATH_H_
00044
00045 #include "gtest/internal/gtest-string.h"
00046
00047 GTEST_DISABLE_MSC_WARNINGS_PUSH_(4251 \
00048 /* class A needs to have dll-interface to be used by clients of class B */)
00049
00050 namespace testing {
00051     namespace internal {
00052
00053         // FilePath - a class for file and directory pathname manipulation which
00054         // handles platform-specific conventions (like the pathname separator).
00055         // Used for helper functions for naming files in a directory for xml output.
00056         // Except for Set methods, all methods are const or static, which provides an
00057         // "immutable value object" -- useful for peace of mind.
00058         // A FilePath with a value ending in a path separator ("like/this/") represents
00059         // a directory, otherwise it is assumed to represent a file. In either case,
00060         // it may or may not represent an actual file or directory in the file system.
00061         // Names are NOT checked for syntax correctness -- no checking for illegal
00062         // characters, malformed paths, etc.
00063
00064         class GTEST_API_ FilePath {
00065         public:
00066
00067             FilePath() : pathname_("") {
00068             }
00069
00070             FilePath(const FilePath& rhs) : pathname_(rhs.pathname_) {
00071             }
00072
00073             explicit FilePath(const std::string& pathname) : pathname_(pathname) {
00074                 Normalize();
00075             }
00076
00077             FilePath& operator=(const FilePath& rhs) {
00078                 Set(rhs);
00079                 return *this;
00080             }
00081
00082             void Set(const FilePath& rhs) {
00083                 pathname_ = rhs.pathname_;
00084             }
00085
```

```
00086             const std::string& string() const {
00087                 return pathname_;
00088             }
00089
00090             const char* c_str() const {
00091                 return pathname_.c_str();
00092             }
00093
00094             // Returns the current working directory, or "" if unsuccessful.
00095             static FilePath GetCurrentDir();
00096
00097             // Given directory = "dir", base_name = "test", number = 0,
00098             // extension = "xml", returns "dir/test.xml". If number is greater
00099             // than zero (e.g., 12), returns "dir/test_12.xml".
00100            // On Windows platform, uses \ as the separator rather than /.
00101            static FilePath MakeFileName(const FilePath& directory,
00102                    const FilePath& base_name, int number,
00103                    const char* extension);
00104
00105            // Given directory = "dir", relative_path = "test.xml",
00106            // returns "dir/test.xml".
00107            // On Windows, uses \ as the separator rather than /.
00108            static FilePath ConcatPaths(const FilePath& directory,
00109                    const FilePath& relative_path);
00110
00111            // Returns a pathname for a file that does not currently exist. The pathname
00112            // will be directory/base_name.extension or
00113            // directory/base_name_<number>.extension if directory/base_name.extension
00114            // already exists. The number will be incremented until a pathname is found
00115            // that does not already exist.
00116            // Examples: 'dir/foo_test.xml' or 'dir/foo_test_1.xml'.
00117            // There could be a race condition if two or more processes are calling this
00118            // function at the same time -- they could both pick the same filename.
00119            static FilePath GenerateUniqueFileName(const FilePath& directory,
00120                    const FilePath& base_name,
00121                    const char* extension);
00122
00123            // Returns true if and only if the path is "".
00124
00125            bool IsEmpty() const {
00126                return pathname_.empty();
00127            }
00128
00129            // If input name has a trailing separator character, removes it and returns
00130            // the name, otherwise return the name string unmodified.
00131            // On Windows platform, uses \ as the separator, other platforms use /.
00132            FilePath RemoveTrailingPathSeparator() const;
00133
00134            // Returns a copy of the FilePath with the directory part removed.
00135            // Example: FilePath("path/to/file").RemoveDirectoryName() returns
00136            // FilePath("file"). If there is no directory part ("just_a_file"), it returns
00137            // the FilePath unmodified. If there is no file part ("just_a_dir/") it
00138            // returns an empty FilePath ("").
00139            // On Windows platform, '\' is the path separator, otherwise it is '/'.
00140            FilePath RemoveDirectoryName() const;
00141
00142            // RemoveFileName returns the directory path with the filename removed.
00143            // Example: FilePath("path/to/file").RemoveFileName() returns "path/to/".
00144            // If the FilePath is "a_file" or "/a_file", RemoveFileName returns
00145            // FilePath("./") or, on Windows, FilePath(".\\"). If the filepath does
00146            // not have a file, like "just/a/dir/", it returns the FilePath unmodified.
00147            // On Windows platform, '\' is the path separator, otherwise it is '/'.
00148            FilePath RemoveFileName() const;
00149
00150            // Returns a copy of the FilePath with the case-insensitive extension removed.
00151            // Example: FilePath("dir/file.exe").RemoveExtension("EXE") returns
00152            // FilePath("dir/file"). If a case-insensitive extension is not
00153            // found, returns a copy of the original FilePath.
00154            FilePath RemoveExtension(const char* extension) const;
00155
00156            // Creates directories so that path exists. Returns true if successful or if
00157            // the directories already exist; returns false if unable to create
00158            // directories for any reason. Will also return false if the FilePath does
00159            // not represent a directory (that is, it doesn't end with a path separator).
00160            bool CreateDirectoriesRecursively() const;
00161
00162            // Create the directory so that path exists. Returns true if successful or
00163            // if the directory already exists; returns false if unable to create the
00164            // directory for any reason, including if the parent directory does not
00165            // exist. Not named "CreateDirectory" because that's a macro on Windows.
00166            bool CreateFolder() const;
00167
00168            // Returns true if FilePath describes something in the file-system,
00169            // either a file, directory, or whatever, and that something exists.
00170            bool FileOrDirectoryExists() const;
00171
00172            // Returns true if pathname describes a directory in the file-system
```

```
00173             // that exists.
00174             bool DirectoryExists() const;
00175
00176             // Returns true if FilePath ends with a path separator, which indicates that
00177             // it is intended to represent a directory. Returns false otherwise.
00178             // This does NOT check that a directory (or file) actually exists.
00179             bool IsDirectory() const;
00180
00181             // Returns true if pathname describes a root directory. (Windows has one
00182             // root directory per disk drive.)
00183             bool IsRootDirectory() const;
00184
00185             // Returns true if pathname describes an absolute path.
00186             bool IsAbsolutePath() const;
00187
00188         private:
00189             // Replaces multiple consecutive separators with a single separator.
00190             // For example, "bar///foo" becomes "bar/foo". Does not eliminate other
00191             // redundancies that might be in a pathname involving "." or "..".
00192             //
00193             // A pathname with multiple consecutive separators may occur either through
00194             // user error or as a result of some scripts or APIs that generate a pathname
00195             // with a trailing separator. On other platforms the same API or script
00196             // may NOT generate a pathname with a trailing "/". Then elsewhere that
00197             // pathname may have another "/" and pathname components added to it,
00198             // without checking for the separator already being there.
00199             // The script language and operating system may allow paths like "foo//bar"
00200             // but some of the functions in FilePath will not handle that correctly. In
00201             // particular, RemoveTrailingPathSeparator() only removes one separator, and
00202             // it is called in CreateDirectoriesRecursively() assuming that it will change
00203             // a pathname from directory syntax (trailing separator) to filename syntax.
00204             //
00205             // On Windows this method also replaces the alternate path separator '/' with
00206             // the primary path separator '\\', so that for example "bar\\/\\foo" becomes
00207             // "bar\\foo".
00208
00209             void Normalize();
00210
00211             // Returns a pointer to the last occurrence of a valid path separator in
00212             // the FilePath. On Windows, for example, both '/' and '\' are valid path
00213             // separators. Returns NULL if no path separator was found.
00214             const char* FindLastPathSeparator() const;
00215
00216             std::string pathname_;
00217         }; // class FilePath
00218
00219     } // namespace internal
00220 } // namespace testing
00221
00222 GTEST_DISABLE_MSC_WARNINGS_POP_() //  4251
00223
00224 #endif  // GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_FILEPATH_H_
```

18.277 gtest-internal.h

```
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00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // The Google C++ Testing and Mocking Framework (Google Test)
```

```
00031 //
00032 // This header file declares functions and macros used internally by
00033 // Google Test.  They are subject to change without notice.
00034
00035 // IWYU pragma: private, include "gtest/gtest.h"
00036 // IWYU pragma: friend gtest/.*
00037 // IWYU pragma: friend gmock/.*
00038
00039 #ifndef GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_INTERNAL_H_
00040 #define GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_INTERNAL_H_
00041
00042 #include "gtest/internal/gtest-port.h"
00043
00044 #if GTEST_OS_LINUX
00045 #include <stdlib.h>
00046 #include <sys/types.h>
00047 #include <sys/wait.h>
00048 #include <unistd.h>
00049 #endif  // GTEST_OS_LINUX
00050
00051 #if GTEST_HAS_EXCEPTIONS
00052 #include <stdexcept>
00053 #endif
00054
00055 #include <ctype.h>
00056 #include <float.h>
00057 #include <string.h>
00058
00059 #include <cstdint>
00060 #include <functional>
00061 #include <iomanip>
00062 #include <limits>
00063 #include <map>
00064 #include <set>
00065 #include <string>
00066 #include <type_traits>
00067 #include <vector>
00068
00069 #include "gtest/gtest-message.h"
00070 #include "gtest/internal/gtest-filepath.h"
00071 #include "gtest/internal/gtest-string.h"
00072 #include "gtest/internal/gtest-type-util.h"
00073
00074 // Due to C++ preprocessor weirdness, we need double indirection to
00075 // concatenate two tokens when one of them is __LINE__.  Writing
00076 //
00077 //   foo ## __LINE__
00078 //
00079 // will result in the token foo__LINE__, instead of foo followed by
00080 // the current line number.  For more details, see
00081 // http://www.parashift.com/c++-faq-lite/misc-technical-issues.html#faq-39.6
00082 #define GTEST_CONCAT_TOKEN_(foo, bar) GTEST_CONCAT_TOKEN_IMPL_(foo, bar)
00083 #define GTEST_CONCAT_TOKEN_IMPL_(foo, bar) foo##bar
00084
00085 // Stringifies its argument.
00086 // Work around a bug in visual studio which doesn't accept code like this:
00087 //
00088 //   #define GTEST_STRINGIFY_(name) #name
00089 //   #define MACRO(a, b, c) ... GTEST_STRINGIFY_(a) ...
00090 //   MACRO(, x, y)
00091 //
00092 // Complaining about the argument to GTEST_STRINGIFY_ being empty.
00093 // This is allowed by the spec.
00094 #define GTEST_STRINGIFY_HELPER_(name, ...) #name
00095 #define GTEST_STRINGIFY_(...) GTEST_STRINGIFY_HELPER_(__VA_ARGS__, )
00096
00097 namespace proto2 {
00098 class MessageLite;
00099 }
00100
00101 namespace testing {
00102
00103 // Forward declarations.
00104
00105 class AssertionResult;  // Result of an assertion.
00106 class Message;          // Represents a failure message.
00107 class Test;             // Represents a test.
00108 class TestInfo;         // Information about a test.
00109 class TestPartResult;   // Result of a test part.
00110 class UnitTest;         // A collection of test suites.
00111
00112 template <typename T>
00113 ::std::string PrintToString(const T& value);
00114
00115 namespace internal {
00116
00117 struct TraceInfo;    // Information about a trace point.
```

```
00118 class TestInfoImpl;  // Opaque implementation of TestInfo
00119 class UnitTestImpl;  // Opaque implementation of UnitTest
00120
00121 // The text used in failure messages to indicate the start of the
00122 // stack trace.
00123 GTEST_API_ extern const char kStackTraceMarker[];
00124
00125 // An IgnoredValue object can be implicitly constructed from ANY value.
00126 class IgnoredValue {
00127   struct Sink {};
00128
00129  public:
00130   // This constructor template allows any value to be implicitly
00131   // converted to IgnoredValue.  The object has no data member and
00132   // doesn't try to remember anything about the argument.  We
00133   // deliberately omit the 'explicit' keyword in order to allow the
00134   // conversion to be implicit.
00135   // Disable the conversion if T already has a magical conversion operator.
00136   // Otherwise we get ambiguity.
00137   template <typename T,
00138             typename std::enable_if<!std::is_convertible<T, Sink>::value,
00139                                     int>::type = 0>
00140   IgnoredValue(const T& /* ignored */) {}  // NOLINT(runtime/explicit)
00141 };
00142
00143 // Appends the user-supplied message to the Google-Test-generated message.
00144 GTEST_API_ std::string AppendUserMessage(const std::string& gtest_msg,
00145                                          const Message& user_msg);
00146
00147 #if GTEST_HAS_EXCEPTIONS
00148
00149 GTEST_DISABLE_MSC_WARNINGS_PUSH_(
00150     4275 /* an exported class was derived from a class that was not exported */)
00151
00152 // This exception is thrown by (and only by) a failed Google Test
00153 // assertion when GTEST_FLAG(throw_on_failure) is true (if exceptions
00154 // are enabled).  We derive it from std::runtime_error, which is for
00155 // errors presumably detectable only at run time.  Since
00156 // std::runtime_error inherits from std::exception, many testing
00157 // frameworks know how to extract and print the message inside it.
00158 class GTEST_API_ GoogleTestFailureException : public ::std::runtime_error {
00159  public:
00160   explicit GoogleTestFailureException(const TestPartResult& failure);
00161 };
00162
00163 GTEST_DISABLE_MSC_WARNINGS_POP_()  //  4275
00164
00165 #endif  // GTEST_HAS_EXCEPTIONS
00166
00167 namespace edit_distance {
00168 // Returns the optimal edits to go from 'left' to 'right'.
00169 // All edits cost the same, with replace having lower priority than
00170 // add/remove.
00171 // Simple implementation of the Wagner-Fischer algorithm.
00172 // See http://en.wikipedia.org/wiki/Wagner-Fischer_algorithm
00173 enum EditType { kMatch, kAdd, kRemove, kReplace };
00174 GTEST_API_ std::vector<EditType> CalculateOptimalEdits(
00175     const std::vector<size_t>& left, const std::vector<size_t>& right);
00176
00177 // Same as above, but the input is represented as strings.
00178 GTEST_API_ std::vector<EditType> CalculateOptimalEdits(
00179     const std::vector<std::string>& left,
00180     const std::vector<std::string>& right);
00181
00182 // Create a diff of the input strings in Unified diff format.
00183 GTEST_API_ std::string CreateUnifiedDiff(const std::vector<std::string>& left,
00184                                          const std::vector<std::string>& right,
00185                                          size_t context = 2);
00186
00187 }  // namespace edit_distance
00188
00189 // Constructs and returns the message for an equality assertion
00190 // (e.g. ASSERT_EQ, EXPECT_STREQ, etc) failure.
00191 //
00192 // The first four parameters are the expressions used in the assertion
00193 // and their values, as strings.  For example, for ASSERT_EQ(foo, bar)
00194 // where foo is 5 and bar is 6, we have:
00195 //
00196 //   expected_expression: "foo"
00197 //   actual_expression:   "bar"
00198 //   expected_value:      "5"
00199 //   actual_value:        "6"
00200 //
00201 // The ignoring_case parameter is true if and only if the assertion is a
00202 // *_STRCASEEQ*.  When it's true, the string " (ignoring case)" will
00203 // be inserted into the message.
00204 GTEST_API_ AssertionResult EqFailure(const char* expected_expression,
```

```
00205                                     const char* actual_expression,
00206                                     const std::string& expected_value,
00207                                     const std::string& actual_value,
00208                                     bool ignoring_case);
00209
00210 // Constructs a failure message for Boolean assertions such as EXPECT_TRUE.
00211 GTEST_API_ std::string GetBoolAssertionFailureMessage(
00212     const AssertionResult& assertion_result, const char* expression_text,
00213     const char* actual_predicate_value, const char* expected_predicate_value);
00214
00215 // This template class represents an IEEE floating-point number
00216 // (either single-precision or double-precision, depending on the
00217 // template parameters).
00218 //
00219 // The purpose of this class is to do more sophisticated number
00220 // comparison.  (Due to round-off error, etc, it's very unlikely that
00221 // two floating-points will be equal exactly.  Hence a naive
00222 // comparison by the == operation often doesn't work.)
00223 //
00224 // Format of IEEE floating-point:
00225 //
00226 //   The most-significant bit being the leftmost, an IEEE
00227 //   floating-point looks like
00228 //
00229 //     sign_bit exponent_bits fraction_bits
00230 //
00231 //   Here, sign_bit is a single bit that designates the sign of the
00232 //   number.
00233 //
00234 //   For float, there are 8 exponent bits and 23 fraction bits.
00235 //
00236 //   For double, there are 11 exponent bits and 52 fraction bits.
00237 //
00238 //   More details can be found at
00239 //   http://en.wikipedia.org/wiki/IEEE_floating-point_standard.
00240 //
00241 // Template parameter:
00242 //
00243 //   RawType: the raw floating-point type (either float or double)
00244 template <typename RawType>
00245 class FloatingPoint {
00246  public:
00247   // Defines the unsigned integer type that has the same size as the
00248   // floating point number.
00249   typedef typename TypeWithSize<sizeof(RawType)>::UInt Bits;
00250
00251   // Constants.
00252
00253   // # of bits in a number.
00254   static const size_t kBitCount = 8 * sizeof(RawType);
00255
00256   // # of fraction bits in a number.
00257   static const size_t kFractionBitCount =
00258       std::numeric_limits<RawType>::digits - 1;
00259
00260   // # of exponent bits in a number.
00261   static const size_t kExponentBitCount = kBitCount - 1 - kFractionBitCount;
00262
00263   // The mask for the sign bit.
00264   static const Bits kSignBitMask = static_cast<Bits>(1) << (kBitCount - 1);
00265
00266   // The mask for the fraction bits.
00267   static const Bits kFractionBitMask = ~static_cast<Bits>(0) >>
00268                                        (kExponentBitCount + 1);
00269
00270   // The mask for the exponent bits.
00271   static const Bits kExponentBitMask = ~(kSignBitMask | kFractionBitMask);
00272
00273   // How many ULP's (Units in the Last Place) we want to tolerate when
00274   // comparing two numbers.  The larger the value, the more error we
00275   // allow.  A 0 value means that two numbers must be exactly the same
00276   // to be considered equal.
00277   //
00278   // The maximum error of a single floating-point operation is 0.5
00279   // units in the last place.  On Intel CPU's, all floating-point
00280   // calculations are done with 80-bit precision, while double has 64
00281   // bits.  Therefore, 4 should be enough for ordinary use.
00282   //
00283   // See the following article for more details on ULP:
00284   // http://randomascii.wordpress.com/2012/02/25/comparing-floating-point-numbers-2012-edition/
00285   static const uint32_t kMaxUlps = 4;
00286
00287   // Constructs a FloatingPoint from a raw floating-point number.
00288   //
00289   // On an Intel CPU, passing a non-normalized NAN (Not a Number)
00290   // around may change its bits, although the new value is guaranteed
00291   // to be also a NAN.  Therefore, don't expect this constructor to
```

```
00292   // preserve the bits in x when x is a NAN.
00293   explicit FloatingPoint(const RawType& x) { u_.value_ = x; }
00294
00295   // Static methods
00296
00297   // Reinterprets a bit pattern as a floating-point number.
00298   //
00299   // This function is needed to test the AlmostEquals() method.
00300   static RawType ReinterpretBits(const Bits bits) {
00301     FloatingPoint fp(0);
00302     fp.u_.bits_ = bits;
00303     return fp.u_.value_;
00304   }
00305
00306   // Returns the floating-point number that represent positive infinity.
00307   static RawType Infinity() { return ReinterpretBits(kExponentBitMask); }
00308
00309   // Returns the maximum representable finite floating-point number.
00310   static RawType Max();
00311
00312   // Non-static methods
00313
00314   // Returns the bits that represents this number.
00315   const Bits& bits() const { return u_.bits_; }
00316
00317   // Returns the exponent bits of this number.
00318   Bits exponent_bits() const { return kExponentBitMask & u_.bits_; }
00319
00320   // Returns the fraction bits of this number.
00321   Bits fraction_bits() const { return kFractionBitMask & u_.bits_; }
00322
00323   // Returns the sign bit of this number.
00324   Bits sign_bit() const { return kSignBitMask & u_.bits_; }
00325
00326   // Returns true if and only if this is NAN (not a number).
00327   bool is_nan() const {
00328     // It's a NAN if the exponent bits are all ones and the fraction
00329     // bits are not entirely zeros.
00330     return (exponent_bits() == kExponentBitMask) && (fraction_bits() != 0);
00331   }
00332
00333   // Returns true if and only if this number is at most kMaxUlps ULP's away
00334   // from rhs.  In particular, this function:
00335   //
00336   //   - returns false if either number is (or both are) NAN.
00337   //   - treats really large numbers as almost equal to infinity.
00338   //   - thinks +0.0 and -0.0 are 0 DLP's apart.
00339   bool AlmostEquals(const FloatingPoint& rhs) const {
00340     // The IEEE standard says that any comparison operation involving
00341     // a NAN must return false.
00342     if (is_nan() || rhs.is_nan()) return false;
00343
00344     return DistanceBetweenSignAndMagnitudeNumbers(u_.bits_, rhs.u_.bits_) <=
00345            kMaxUlps;
00346   }
00347
00348  private:
00349   // The data type used to store the actual floating-point number.
00350   union FloatingPointUnion {
00351     RawType value_;  // The raw floating-point number.
00352     Bits bits_;      // The bits that represent the number.
00353   };
00354
00355   // Converts an integer from the sign-and-magnitude representation to
00356   // the biased representation.  More precisely, let N be 2 to the
00357   // power of (kBitCount - 1), an integer x is represented by the
00358   // unsigned number x + N.
00359   //
00360   // For instance,
00361   //
00362   //   -N + 1 (the most negative number representable using
00363   //          sign-and-magnitude) is represented by 1;
00364   //   0      is represented by N; and
00365   //   N - 1  (the biggest number representable using
00366   //          sign-and-magnitude) is represented by 2N - 1.
00367   //
00368   // Read http://en.wikipedia.org/wiki/Signed_number_representations
00369   // for more details on signed number representations.
00370   static Bits SignAndMagnitudeToBiased(const Bits& sam) {
00371     if (kSignBitMask & sam) {
00372       // sam represents a negative number.
00373       return ~sam + 1;
00374     } else {
00375       // sam represents a positive number.
00376       return kSignBitMask | sam;
00377     }
00378   }
```

```
00379
00380   // Given two numbers in the sign-and-magnitude representation,
00381   // returns the distance between them as an unsigned number.
00382   static Bits DistanceBetweenSignAndMagnitudeNumbers(const Bits& sam1,
00383                                                      const Bits& sam2) {
00384     const Bits biased1 = SignAndMagnitudeToBiased(sam1);
00385     const Bits biased2 = SignAndMagnitudeToBiased(sam2);
00386     return (biased1 >= biased2) ? (biased1 - biased2) : (biased2 - biased1);
00387   }
00388
00389   FloatingPointUnion u_;
00390 };
00391
00392 // We cannot use std::numeric_limits<T>::max() as it clashes with the max()
00393 // macro defined by <windows.h>.
00394 template <>
00395 inline float FloatingPoint<float>::Max() {
00396   return FLT_MAX;
00397 }
00398 template <>
00399 inline double FloatingPoint<double>::Max() {
00400   return DBL_MAX;
00401 }
00402
00403 // Typedefs the instances of the FloatingPoint template class that we
00404 // care to use.
00405 typedef FloatingPoint<float> Float;
00406 typedef FloatingPoint<double> Double;
00407
00408 // In order to catch the mistake of putting tests that use different
00409 // test fixture classes in the same test suite, we need to assign
00410 // unique IDs to fixture classes and compare them.  The TypeId type is
00411 // used to hold such IDs.  The user should treat TypeId as an opaque
00412 // type: the only operation allowed on TypeId values is to compare
00413 // them for equality using the == operator.
00414 typedef const void* TypeId;
00415
00416 template <typename T>
00417 class TypeIdHelper {
00418  public:
00419   // dummy_ must not have a const type.  Otherwise an overly eager
00420   // compiler (e.g. MSVC 7.1 & 8.0) may try to merge
00421   // TypeIdHelper<T>::dummy_ for different Ts as an "optimization".
00422   static bool dummy_;
00423 };
00424
00425 template <typename T>
00426 bool TypeIdHelper<T>::dummy_ = false;
00427
00428 // GetTypeId<T>() returns the ID of type T.  Different values will be
00429 // returned for different types.  Calling the function twice with the
00430 // same type argument is guaranteed to return the same ID.
00431 template <typename T>
00432 TypeId GetTypeId() {
00433   // The compiler is required to allocate a different
00434   // TypeIdHelper<T>::dummy_ variable for each T used to instantiate
00435   // the template.  Therefore, the address of dummy_ is guaranteed to
00436   // be unique.
00437   return &(TypeIdHelper<T>::dummy_);
00438 }
00439
00440 // Returns the type ID of ::testing::Test.  Always call this instead
00441 // of GetTypeId< ::testing::Test>() to get the type ID of
00442 // ::testing::Test, as the latter may give the wrong result due to a
00443 // suspected linker bug when compiling Google Test as a Mac OS X
00444 // framework.
00445 GTEST_API_ TypeId GetTestTypeId();
00446
00447 // Defines the abstract factory interface that creates instances
00448 // of a Test object.
00449 class TestFactoryBase {
00450  public:
00451   virtual ~TestFactoryBase() {}
00452
00453   // Creates a test instance to run. The instance is both created and destroyed
00454   // within TestInfoImpl::Run()
00455   virtual Test* CreateTest() = 0;
00456
00457  protected:
00458   TestFactoryBase() {}
00459
00460  private:
00461   TestFactoryBase(const TestFactoryBase&) = delete;
00462   TestFactoryBase& operator=(const TestFactoryBase&) = delete;
00463 };
00464
00465 // This class provides implementation of TestFactoryBase interface.
```

```
00466 // It is used in TEST and TEST_F macros.
00467 template <class TestClass>
00468 class TestFactoryImpl : public TestFactoryBase {
00469  public:
00470   Test* CreateTest() override { return new TestClass; }
00471 };
00472
00473 #if GTEST_OS_WINDOWS
00474
00475 // Predicate-formatters for implementing the HRESULT checking macros
00476 // {ASSERT|EXPECT}_HRESULT_{SUCCEEDED|FAILED}
00477 // We pass a long instead of HRESULT to avoid causing an
00478 // include dependency for the HRESULT type.
00479 GTEST_API_ AssertionResult IsHRESULTSuccess(const char* expr,
00480                                             long hr);  // NOLINT
00481 GTEST_API_ AssertionResult IsHRESULTFailure(const char* expr,
00482                                             long hr);  // NOLINT
00483
00484 #endif  // GTEST_OS_WINDOWS
00485
00486 // Types of SetUpTestSuite() and TearDownTestSuite() functions.
00487 using SetUpTestSuiteFunc = void (*)();
00488 using TearDownTestSuiteFunc = void (*)();
00489
00490 struct CodeLocation {
00491   CodeLocation(const std::string& a_file, int a_line)
00492       : file(a_file), line(a_line) {}
00493
00494   std::string file;
00495   int line;
00496 };
00497
00498 //  Helper to identify which setup function for TestCase / TestSuite to call.
00499 //  Only one function is allowed, either TestCase or TestSute but not both.
00500
00501 // Utility functions to help SuiteApiResolver
00502 using SetUpTearDownSuiteFuncType = void (*)();
00503
00504 inline SetUpTearDownSuiteFuncType GetNotDefaultOrNull(
00505     SetUpTearDownSuiteFuncType a, SetUpTearDownSuiteFuncType def) {
00506   return a == def ? nullptr : a;
00507 }
00508
00509 template <typename T>
00510 //  Note that SuiteApiResolver inherits from T because
00511 //  SetUpTestSuite()/TearDownTestSuite() could be protected. This way
00512 //  SuiteApiResolver can access them.
00513 struct SuiteApiResolver : T {
00514   // testing::Test is only forward declared at this point. So we make it a
00515   // dependent class for the compiler to be OK with it.
00516   using Test =
00517       typename std::conditional<sizeof(T) != 0, ::testing::Test, void>::type;
00518
00519   static SetUpTearDownSuiteFuncType GetSetUpCaseOrSuite(const char* filename,
00520                                                         int line_num) {
00521 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00522     SetUpTearDownSuiteFuncType test_case_fp =
00523         GetNotDefaultOrNull(&T::SetUpTestCase, &Test::SetUpTestCase);
00524     SetUpTearDownSuiteFuncType test_suite_fp =
00525         GetNotDefaultOrNull(&T::SetUpTestSuite, &Test::SetUpTestSuite);
00526
00527     GTEST_CHECK_(!test_case_fp || !test_suite_fp)
00528         << "Test can not provide both SetUpTestSuite and SetUpTestCase, please "
00529            "make sure there is only one present at "
00530         << filename << ":" << line_num;
00531
00532     return test_case_fp != nullptr ? test_case_fp : test_suite_fp;
00533 #else
00534     (void)(filename);
00535     (void)(line_num);
00536     return &T::SetUpTestSuite;
00537 #endif
00538   }
00539
00540   static SetUpTearDownSuiteFuncType GetTearDownCaseOrSuite(const char* filename,
00541                                                            int line_num) {
00542 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00543     SetUpTearDownSuiteFuncType test_case_fp =
00544         GetNotDefaultOrNull(&T::TearDownTestCase, &Test::TearDownTestCase);
00545     SetUpTearDownSuiteFuncType test_suite_fp =
00546         GetNotDefaultOrNull(&T::TearDownTestSuite, &Test::TearDownTestSuite);
00547
00548     GTEST_CHECK_(!test_case_fp || !test_suite_fp)
00549         << "Test can not provide both TearDownTestSuite and TearDownTestCase,"
00550            " please make sure there is only one present at"
00551         << filename << ":" << line_num;
00552
```

```
00553     return test_case_fp != nullptr ? test_case_fp : test_suite_fp;
00554 #else
00555     (void)(filename);
00556     (void)(line_num);
00557     return &T::TearDownTestSuite;
00558 #endif
00559   }
00560 };
00561
00562 // Creates a new TestInfo object and registers it with Google Test;
00563 // returns the created object.
00564 //
00565 // Arguments:
00566 //
00567 //   test_suite_name:  name of the test suite
00568 //   name:             name of the test
00569 //   type_param:       the name of the test's type parameter, or NULL if
00570 //                     this is not a typed or a type-parameterized test.
00571 //   value_param:      text representation of the test's value parameter,
00572 //                     or NULL if this is not a type-parameterized test.
00573 //   code_location:    code location where the test is defined
00574 //   fixture_class_id: ID of the test fixture class
00575 //   set_up_tc:        pointer to the function that sets up the test suite
00576 //   tear_down_tc:     pointer to the function that tears down the test suite
00577 //   factory:          pointer to the factory that creates a test object.
00578 //                     The newly created TestInfo instance will assume
00579 //                     ownership of the factory object.
00580 GTEST_API_ TestInfo* MakeAndRegisterTestInfo(
00581     const char* test_suite_name, const char* name, const char* type_param,
00582     const char* value_param, CodeLocation code_location,
00583     TypeId fixture_class_id, SetUpTestSuiteFunc set_up_tc,
00584     TearDownTestSuiteFunc tear_down_tc, TestFactoryBase* factory);
00585
00586 // If *pstr starts with the given prefix, modifies *pstr to be right
00587 // past the prefix and returns true; otherwise leaves *pstr unchanged
00588 // and returns false.  None of pstr, *pstr, and prefix can be NULL.
00589 GTEST_API_ bool SkipPrefix(const char* prefix, const char** pstr);
00590
00591 GTEST_DISABLE_MSC_WARNINGS_PUSH_(4251 \
00592 /* class A needs to have dll-interface to be used by clients of class B */)
00593
00594 // State of the definition of a type-parameterized test suite.
00595 class GTEST_API_ TypedTestSuitePState {
00596  public:
00597   TypedTestSuitePState() : registered_(false) {}
00598
00599   // Adds the given test name to defined_test_names_ and return true
00600   // if the test suite hasn't been registered; otherwise aborts the
00601   // program.
00602   bool AddTestName(const char* file, int line, const char* case_name,
00603                    const char* test_name) {
00604     if (registered_) {
00605       fprintf(stderr,
00606               "%s Test %s must be defined before "
00607               "REGISTER_TYPED_TEST_SUITE_P(%s, ...).\n",
00608               FormatFileLocation(file, line).c_str(), test_name, case_name);
00609       fflush(stderr);
00610       posix::Abort();
00611     }
00612     registered_tests_.insert(
00613         ::std::make_pair(test_name, CodeLocation(file, line)));
00614     return true;
00615   }
00616
00617   bool TestExists(const std::string& test_name) const {
00618     return registered_tests_.count(test_name) > 0;
00619   }
00620
00621   const CodeLocation& GetCodeLocation(const std::string& test_name) const {
00622     RegisteredTestsMap::const_iterator it = registered_tests_.find(test_name);
00623     GTEST_CHECK_(it != registered_tests_.end());
00624     return it->second;
00625   }
00626
00627   // Verifies that registered_tests match the test names in
00628   // defined_test_names_; returns registered_tests if successful, or
00629   // aborts the program otherwise.
00630   const char* VerifyRegisteredTestNames(const char* test_suite_name,
00631                                         const char* file, int line,
00632                                         const char* registered_tests);
00633
00634  private:
00635   typedef ::std::map<std::string, CodeLocation, std::less<» RegisteredTestsMap;
00636
00637   bool registered_;
00638   RegisteredTestsMap registered_tests_;
00639 };
```

```
00640
00641 //  Legacy API is deprecated but still available
00642 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00643 using TypedTestCasePState = TypedTestSuitePState;
00644 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00645
00646 GTEST_DISABLE_MSC_WARNINGS_POP_()  //  4251
00647
00648 // Skips to the first non-space char after the first comma in 'str';
00649 // returns NULL if no comma is found in 'str'.
00650 inline const char* SkipComma(const char* str) {
00651   const char* comma = strchr(str, ',');
00652   if (comma == nullptr) {
00653     return nullptr;
00654   }
00655   while (IsSpace(*(++comma))) {
00656   }
00657   return comma;
00658 }
00659
00660 // Returns the prefix of 'str' before the first comma in it; returns
00661 // the entire string if it contains no comma.
00662 inline std::string GetPrefixUntilComma(const char* str) {
00663   const char* comma = strchr(str, ',');
00664   return comma == nullptr ? str : std::string(str, comma);
00665 }
00666
00667 // Splits a given string on a given delimiter, populating a given
00668 // vector with the fields.
00669 void SplitString(const ::std::string& str, char delimiter,
00670                  ::std::vector<::std::string>* dest);
00671
00672 // The default argument to the template below for the case when the user does
00673 // not provide a name generator.
00674 struct DefaultNameGenerator {
00675   template <typename T>
00676   static std::string GetName(int i) {
00677     return StreamableToString(i);
00678   }
00679 };
00680
00681 template <typename Provided = DefaultNameGenerator>
00682 struct NameGeneratorSelector {
00683   typedef Provided type;
00684 };
00685
00686 template <typename NameGenerator>
00687 void GenerateNamesRecursively(internal::None, std::vector<std::string>*, int) {}
00688
00689 template <typename NameGenerator, typename Types>
00690 void GenerateNamesRecursively(Types, std::vector<std::string>* result, int i) {
00691   result->push_back(NameGenerator::template GetName<typename Types::Head>(i));
00692   GenerateNamesRecursively<NameGenerator>(typename Types::Tail(), result,
00693                                           i + 1);
00694 }
00695
00696 template <typename NameGenerator, typename Types>
00697 std::vector<std::string> GenerateNames() {
00698   std::vector<std::string> result;
00699   GenerateNamesRecursively<NameGenerator>(Types(), &result, 0);
00700   return result;
00701 }
00702
00703 // TypeParameterizedTest<Fixture, TestSel, Types>::Register()
00704 // registers a list of type-parameterized tests with Google Test.  The
00705 // return value is insignificant - we just need to return something
00706 // such that we can call this function in a namespace scope.
00707 //
00708 // Implementation note: The GTEST_TEMPLATE_ macro declares a template
00709 // template parameter.  It's defined in gtest-type-util.h.
00710 template <GTEST_TEMPLATE_ Fixture, class TestSel, typename Types>
00711 class TypeParameterizedTest {
00712  public:
00713   // 'index' is the index of the test in the type list 'Types'
00714   // specified in INSTANTIATE_TYPED_TEST_SUITE_P(Prefix, TestSuite,
00715   // Types).  Valid values for 'index' are [0, N - 1] where N is the
00716   // length of Types.
00717   static bool Register(const char* prefix, const CodeLocation& code_location,
00718                        const char* case_name, const char* test_names, int index,
00719                        const std::vector<std::string>& type_names =
00720                            GenerateNames<DefaultNameGenerator, Types>()) {
00721     typedef typename Types::Head Type;
00722     typedef Fixture<Type> FixtureClass;
00723     typedef typename GTEST_BIND_(TestSel, Type) TestClass;
00724
00725     // First, registers the first type-parameterized test in the type
00726     // list.
```

```
00727     MakeAndRegisterTestInfo(
00728         (std::string(prefix) + (prefix[0] == '\0' ? "" : "/") + case_name +
00729          "/" + type_names[static_cast<size_t>(index)])
00730             .c_str(),
00731         StripTrailingSpaces(GetPrefixUntilComma(test_names)).c_str(),
00732         GetTypeName<Type>().c_str(),
00733         nullptr,  // No value parameter.
00734         code_location, GetTypeId<FixtureClass>(),
00735         SuiteApiResolver<TestClass>::GetSetUpCaseOrSuite(
00736             code_location.file.c_str(), code_location.line),
00737         SuiteApiResolver<TestClass>::GetTearDownCaseOrSuite(
00738             code_location.file.c_str(), code_location.line),
00739         new TestFactoryImpl<TestClass>);
00740
00741     // Next, recurses (at compile time) with the tail of the type list.
00742     return TypeParameterizedTest<Fixture, TestSel,
00743                                  typename Types::Tail>::Register(prefix,
00744                                                                  code_location,
00745                                                                  case_name,
00746                                                                  test_names,
00747                                                                  index + 1,
00748                                                                  type_names);
00749   }
00750 };
00751
00752 // The base case for the compile time recursion.
00753 template <GTEST_TEMPLATE_ Fixture, class TestSel>
00754 class TypeParameterizedTest<Fixture, TestSel, internal::None> {
00755  public:
00756   static bool Register(const char* /*prefix*/, const CodeLocation&,
00757                        const char* /*case_name*/, const char* /*test_names*/,
00758                        int /*index*/,
00759                        const std::vector<std::string>& =
00760                            std::vector<std::string>() /*type_names*/) {
00761     return true;
00762   }
00763 };
00764
00765 GTEST_API_ void RegisterTypeParameterizedTestSuite(const char* test_suite_name,
00766                                                    CodeLocation code_location);
00767 GTEST_API_ void RegisterTypeParameterizedTestSuiteInstantiation(
00768     const char* case_name);
00769
00770 // TypeParameterizedTestSuite<Fixture, Tests, Types>::Register()
00771 // registers *all combinations* of 'Tests' and 'Types' with Google
00772 // Test.  The return value is insignificant - we just need to return
00773 // something such that we can call this function in a namespace scope.
00774 template <GTEST_TEMPLATE_ Fixture, typename Tests, typename Types>
00775 class TypeParameterizedTestSuite {
00776  public:
00777   static bool Register(const char* prefix, CodeLocation code_location,
00778                        const TypedTestSuitePState* state, const char* case_name,
00779                        const char* test_names,
00780                        const std::vector<std::string>& type_names =
00781                            GenerateNames<DefaultNameGenerator, Types>()) {
00782     RegisterTypeParameterizedTestSuiteInstantiation(case_name);
00783     std::string test_name =
00784         StripTrailingSpaces(GetPrefixUntilComma(test_names));
00785     if (!state->TestExists(test_name)) {
00786       fprintf(stderr, "Failed to get code location for test %s.%s at %s.",
00787               case_name, test_name.c_str(),
00788               FormatFileLocation(code_location.file.c_str(), code_location.line)
00789                   .c_str());
00790       fflush(stderr);
00791       posix::Abort();
00792     }
00793     const CodeLocation& test_location = state->GetCodeLocation(test_name);
00794
00795     typedef typename Tests::Head Head;
00796
00797     // First, register the first test in 'Test' for each type in 'Types'.
00798     TypeParameterizedTest<Fixture, Head, Types>::Register(
00799         prefix, test_location, case_name, test_names, 0, type_names);
00800
00801     // Next, recurses (at compile time) with the tail of the test list.
00802     return TypeParameterizedTestSuite<Fixture, typename Tests::Tail,
00803                                       Types>::Register(prefix, code_location,
00804                                                        state, case_name,
00805                                                        SkipComma(test_names),
00806                                                        type_names);
00807   }
00808 };
00809
00810 // The base case for the compile time recursion.
00811 template <GTEST_TEMPLATE_ Fixture, typename Types>
00812 class TypeParameterizedTestSuite<Fixture, internal::None, Types> {
00813  public:
```

```
00814   static bool Register(const char* /*prefix*/, const CodeLocation&,
00815                        const TypedTestSuitePState* /*state*/,
00816                        const char* /*case_name*/, const char* /*test_names*/,
00817                        const std::vector<std::string>& =
00818                            std::vector<std::string>() /*type_names*/) {
00819     return true;
00820   }
00821 };
00822
00823 // Returns the current OS stack trace as an std::string.
00824 //
00825 // The maximum number of stack frames to be included is specified by
00826 // the gtest_stack_trace_depth flag.  The skip_count parameter
00827 // specifies the number of top frames to be skipped, which doesn't
00828 // count against the number of frames to be included.
00829 //
00830 // For example, if Foo() calls Bar(), which in turn calls
00831 // GetCurrentOsStackTraceExceptTop(..., 1), Foo() will be included in
00832 // the trace but Bar() and GetCurrentOsStackTraceExceptTop() won't.
00833 GTEST_API_ std::string GetCurrentOsStackTraceExceptTop(int skip_count);
00834
00835 // Helpers for suppressing warnings on unreachable code or constant
00836 // condition.
00837
00838 // Always returns true.
00839 GTEST_API_ bool AlwaysTrue();
00840
00841 // Always returns false.
00842 inline bool AlwaysFalse() { return !AlwaysTrue(); }
00843
00844 // Helper for suppressing false warning from Clang on a const char*
00845 // variable declared in a conditional expression always being NULL in
00846 // the else branch.
00847 struct GTEST_API_ ConstCharPtr {
00848   ConstCharPtr(const char* str) : value(str) {}
00849   operator bool() const { return true; }
00850   const char* value;
00851 };
00852
00853 // Helper for declaring std::string within 'if' statement
00854 // in pre C++17 build environment.
00855 struct TrueWithString {
00856   TrueWithString() = default;
00857   explicit TrueWithString(const char* str) : value(str) {}
00858   explicit TrueWithString(const std::string& str) : value(str) {}
00859   explicit operator bool() const { return true; }
00860   std::string value;
00861 };
00862
00863 // A simple Linear Congruential Generator for generating random
00864 // numbers with a uniform distribution.  Unlike rand() and srand(), it
00865 // doesn't use global state (and therefore can't interfere with user
00866 // code).  Unlike rand_r(), it's portable.  An LCG isn't very random,
00867 // but it's good enough for our purposes.
00868 class GTEST_API_ Random {
00869  public:
00870   static const uint32_t kMaxRange = 1u << 31;
00871
00872   explicit Random(uint32_t seed) : state_(seed) {}
00873
00874   void Reseed(uint32_t seed) { state_ = seed; }
00875
00876   // Generates a random number from [0, range).  Crashes if 'range' is
00877   // 0 or greater than kMaxRange.
00878   uint32_t Generate(uint32_t range);
00879
00880  private:
00881   uint32_t state_;
00882   Random(const Random&) = delete;
00883   Random& operator=(const Random&) = delete;
00884 };
00885
00886 // Turns const U&, U&, const U, and U all into U.
00887 #define GTEST_REMOVE_REFERENCE_AND_CONST_(T) \
00888   typename std::remove_const<typename std::remove_reference<T>::type>::type
00889
00890 // HasDebugStringAndShortDebugString<T>::value is a compile-time bool constant
00891 // that's true if and only if T has methods DebugString() and ShortDebugString()
00892 // that return std::string.
00893 template <typename T>
00894 class HasDebugStringAndShortDebugString {
00895  private:
00896   template <typename C>
00897   static auto CheckDebugString(C*) -> typename std::is_same<
00898       std::string, decltype(std::declval<const C>().DebugString())>::type;
00899   template <typename>
00900   static std::false_type CheckDebugString(...);
```

```
00901
00902   template <typename C>
00903   static auto CheckShortDebugString(C*) -> typename std::is_same<
00904       std::string, decltype(std::declval<const C>().ShortDebugString())>::type;
00905   template <typename>
00906   static std::false_type CheckShortDebugString(...);
00907
00908   using HasDebugStringType = decltype(CheckDebugString<T>(nullptr));
00909   using HasShortDebugStringType = decltype(CheckShortDebugString<T>(nullptr));
00910
00911  public:
00912   static constexpr bool value =
00913       HasDebugStringType::value && HasShortDebugStringType::value;
00914 };
00915
00916 template <typename T>
00917 constexpr bool HasDebugStringAndShortDebugString<T>::value;
00918
00919 // When the compiler sees expression IsContainerTest<C>(0), if C is an
00920 // STL-style container class, the first overload of IsContainerTest
00921 // will be viable (since both C::iterator* and C::const_iterator* are
00922 // valid types and NULL can be implicitly converted to them).  It will
00923 // be picked over the second overload as 'int' is a perfect match for
00924 // the type of argument 0.  If C::iterator or C::const_iterator is not
00925 // a valid type, the first overload is not viable, and the second
00926 // overload will be picked.  Therefore, we can determine whether C is
00927 // a container class by checking the type of IsContainerTest<C>(0).
00928 // The value of the expression is insignificant.
00929 //
00930 // In C++11 mode we check the existence of a const_iterator and that an
00931 // iterator is properly implemented for the container.
00932 //
00933 // For pre-C++11 that we look for both C::iterator and C::const_iterator.
00934 // The reason is that C++ injects the name of a class as a member of the
00935 // class itself (e.g. you can refer to class iterator as either
00936 // 'iterator' or 'iterator::iterator').  If we look for C::iterator
00937 // only, for example, we would mistakenly think that a class named
00938 // iterator is an STL container.
00939 //
00940 // Also note that the simpler approach of overloading
00941 // IsContainerTest(typename C::const_iterator*) and
00942 // IsContainerTest(...) doesn't work with Visual Age C++ and Sun C++.
00943 typedef int IsContainer;
00944 template <class C,
00945           class Iterator = decltype(::std::declval<const C&>().begin()),
00946           class = decltype(::std::declval<const C&>().end()),
00947           class = decltype(++::std::declval<Iterator&>()),
00948           class = decltype(*::std::declval<Iterator>()),
00949           class = typename C::const_iterator>
00950 IsContainer IsContainerTest(int /* dummy */) {
00951   return 0;
00952 }
00953
00954 typedef char IsNotContainer;
00955 template <class C>
00956 IsNotContainer IsContainerTest(long /* dummy */) {
00957   return '\0';
00958 }
00959
00960 // Trait to detect whether a type T is a hash table.
00961 // The heuristic used is that the type contains an inner type `hasher` and does
00962 // not contain an inner type `reverse_iterator`.
00963 // If the container is iterable in reverse, then order might actually matter.
00964 template <typename T>
00965 struct IsHashTable {
00966  private:
00967   template <typename U>
00968   static char test(typename U::hasher*, typename U::reverse_iterator*);
00969   template <typename U>
00970   static int test(typename U::hasher*, ...);
00971   template <typename U>
00972   static char test(...);
00973
00974  public:
00975   static const bool value = sizeof(test<T>(nullptr, nullptr)) == sizeof(int);
00976 };
00977
00978 template <typename T>
00979 const bool IsHashTable<T>::value;
00980
00981 template <typename C,
00982           bool = sizeof(IsContainerTest<C>(0)) == sizeof(IsContainer)>
00983 struct IsRecursiveContainerImpl;
00984
00985 template <typename C>
00986 struct IsRecursiveContainerImpl<C, false> : public std::false_type {};
00987
```

```
00988 // Since the IsRecursiveContainerImpl depends on the IsContainerTest we need to
00989 // obey the same inconsistencies as the IsContainerTest, namely check if
00990 // something is a container is relying on only const_iterator in C++11 and
00991 // is relying on both const_iterator and iterator otherwise
00992 template <typename C>
00993 struct IsRecursiveContainerImpl<C, true> {
00994   using value_type = decltype(*std::declval<typename C::const_iterator>());
00995   using type =
00996       std::is_same<typename std::remove_const<
00997                        typename std::remove_reference<value_type>::type>::type,
00998                    C>;
00999 };
01000
01001 // IsRecursiveContainer<Type> is a unary compile-time predicate that
01002 // evaluates whether C is a recursive container type. A recursive container
01003 // type is a container type whose value_type is equal to the container type
01004 // itself. An example for a recursive container type is
01005 // boost::filesystem::path, whose iterator has a value_type that is equal to
01006 // boost::filesystem::path.
01007 template <typename C>
01008 struct IsRecursiveContainer : public IsRecursiveContainerImpl<C>::type {};
01009
01010 // Utilities for native arrays.
01011
01012 // ArrayEq() compares two k-dimensional native arrays using the
01013 // elements' operator==, where k can be any integer >= 0.  When k is
01014 // 0, ArrayEq() degenerates into comparing a single pair of values.
01015
01016 template <typename T, typename U>
01017 bool ArrayEq(const T* lhs, size_t size, const U* rhs);
01018
01019 // This generic version is used when k is 0.
01020 template <typename T, typename U>
01021 inline bool ArrayEq(const T& lhs, const U& rhs) {
01022   return lhs == rhs;
01023 }
01024
01025 // This overload is used when k >= 1.
01026 template <typename T, typename U, size_t N>
01027 inline bool ArrayEq(const T (&lhs)[N], const U (&rhs)[N]) {
01028   return internal::ArrayEq(lhs, N, rhs);
01029 }
01030
01031 // This helper reduces code bloat.  If we instead put its logic inside
01032 // the previous ArrayEq() function, arrays with different sizes would
01033 // lead to different copies of the template code.
01034 template <typename T, typename U>
01035 bool ArrayEq(const T* lhs, size_t size, const U* rhs) {
01036   for (size_t i = 0; i != size; i++) {
01037     if (!internal::ArrayEq(lhs[i], rhs[i])) return false;
01038   }
01039   return true;
01040 }
01041
01042 // Finds the first element in the iterator range [begin, end) that
01043 // equals elem.  Element may be a native array type itself.
01044 template <typename Iter, typename Element>
01045 Iter ArrayAwareFind(Iter begin, Iter end, const Element& elem) {
01046   for (Iter it = begin; it != end; ++it) {
01047     if (internal::ArrayEq(*it, elem)) return it;
01048   }
01049   return end;
01050 }
01051
01052 // CopyArray() copies a k-dimensional native array using the elements'
01053 // operator=, where k can be any integer >= 0.  When k is 0,
01054 // CopyArray() degenerates into copying a single value.
01055
01056 template <typename T, typename U>
01057 void CopyArray(const T* from, size_t size, U* to);
01058
01059 // This generic version is used when k is 0.
01060 template <typename T, typename U>
01061 inline void CopyArray(const T& from, U* to) {
01062   *to = from;
01063 }
01064
01065 // This overload is used when k >= 1.
01066 template <typename T, typename U, size_t N>
01067 inline void CopyArray(const T (&from)[N], U (*to)[N]) {
01068   internal::CopyArray(from, N, *to);
01069 }
01070
01071 // This helper reduces code bloat.  If we instead put its logic inside
01072 // the previous CopyArray() function, arrays with different sizes
01073 // would lead to different copies of the template code.
01074 template <typename T, typename U>
```

```
01075 void CopyArray(const T* from, size_t size, U* to) {
01076   for (size_t i = 0; i != size; i++) {
01077     internal::CopyArray(from[i], to + i);
01078   }
01079 }
01080
01081 // The relation between an NativeArray object (see below) and the
01082 // native array it represents.
01083 // We use 2 different structs to allow non-copyable types to be used, as long
01084 // as RelationToSourceReference() is passed.
01085 struct RelationToSourceReference {};
01086 struct RelationToSourceCopy {};
01087
01088 // Adapts a native array to a read-only STL-style container.  Instead
01089 // of the complete STL container concept, this adaptor only implements
01090 // members useful for Google Mock's container matchers.  New members
01091 // should be added as needed.  To simplify the implementation, we only
01092 // support Element being a raw type (i.e. having no top-level const or
01093 // reference modifier).  It's the client's responsibility to satisfy
01094 // this requirement.  Element can be an array type itself (hence
01095 // multi-dimensional arrays are supported).
01096 template <typename Element>
01097 class NativeArray {
01098  public:
01099   // STL-style container typedefs.
01100   typedef Element value_type;
01101   typedef Element* iterator;
01102   typedef const Element* const_iterator;
01103
01104   // Constructs from a native array. References the source.
01105   NativeArray(const Element* array, size_t count, RelationToSourceReference) {
01106     InitRef(array, count);
01107   }
01108
01109   // Constructs from a native array. Copies the source.
01110   NativeArray(const Element* array, size_t count, RelationToSourceCopy) {
01111     InitCopy(array, count);
01112   }
01113
01114   // Copy constructor.
01115   NativeArray(const NativeArray& rhs) {
01116     (this->*rhs.clone_)(rhs.array_, rhs.size_);
01117   }
01118
01119   ~NativeArray() {
01120     if (clone_ != &NativeArray::InitRef) delete[] array_;
01121   }
01122
01123   // STL-style container methods.
01124   size_t size() const { return size_; }
01125   const_iterator begin() const { return array_; }
01126   const_iterator end() const { return array_ + size_; }
01127   bool operator==(const NativeArray& rhs) const {
01128     return size() == rhs.size() && ArrayEq(begin(), size(), rhs.begin());
01129   }
01130
01131  private:
01132   static_assert(!std::is_const<Element>::value, "Type must not be const");
01133   static_assert(!std::is_reference<Element>::value,
01134                 "Type must not be a reference");
01135
01136   // Initializes this object with a copy of the input.
01137   void InitCopy(const Element* array, size_t a_size) {
01138     Element* const copy = new Element[a_size];
01139     CopyArray(array, a_size, copy);
01140     array_ = copy;
01141     size_ = a_size;
01142     clone_ = &NativeArray::InitCopy;
01143   }
01144
01145   // Initializes this object with a reference of the input.
01146   void InitRef(const Element* array, size_t a_size) {
01147     array_ = array;
01148     size_ = a_size;
01149     clone_ = &NativeArray::InitRef;
01150   }
01151
01152   const Element* array_;
01153   size_t size_;
01154   void (NativeArray::*clone_)(const Element*, size_t);
01155 };
01156
01157 // Backport of std::index_sequence.
01158 template <size_t... Is>
01159 struct IndexSequence {
01160   using type = IndexSequence;
01161 };
```

```
01162
01163 // Double the IndexSequence, and one if plus_one is true.
01164 template <bool plus_one, typename T, size_t sizeofT>
01165 struct DoubleSequence;
01166 template <size_t... I, size_t sizeofT>
01167 struct DoubleSequence<true, IndexSequence<I...>, sizeofT> {
01168   using type = IndexSequence<I..., (sizeofT + I)..., 2 * sizeofT>;
01169 };
01170 template <size_t... I, size_t sizeofT>
01171 struct DoubleSequence<false, IndexSequence<I...>, sizeofT> {
01172   using type = IndexSequence<I..., (sizeofT + I)...>;
01173 };
01174
01175 // Backport of std::make_index_sequence.
01176 // It uses O(ln(N)) instantiation depth.
01177 template <size_t N>
01178 struct MakeIndexSequenceImpl
01179     : DoubleSequence<N % 2 == 1, typename MakeIndexSequenceImpl<N / 2>::type,
01180                      N / 2>::type {};
01181
01182 template <>
01183 struct MakeIndexSequenceImpl<0> : IndexSequence<> {};
01184
01185 template <size_t N>
01186 using MakeIndexSequence = typename MakeIndexSequenceImpl<N>::type;
01187
01188 template <typename... T>
01189 using IndexSequenceFor = typename MakeIndexSequence<sizeof...(T)>::type;
01190
01191 template <size_t>
01192 struct Ignore {
01193   Ignore(...);  // NOLINT
01194 };
01195
01196 template <typename>
01197 struct ElemFromListImpl;
01198 template <size_t... I>
01199 struct ElemFromListImpl<IndexSequence<I...>> {
01200   // We make Ignore a template to solve a problem with MSVC.
01201   // A non-template Ignore would work fine with `decltype(Ignore(I))...`, but
01202   // MSVC doesn't understand how to deal with that pack expansion.
01203   // Use `0 * I` to have a single instantiation of Ignore.
01204   template <typename R>
01205   static R Apply(Ignore<0 * I>..., R (*)(), ...);
01206 };
01207
01208 template <size_t N, typename... T>
01209 struct ElemFromList {
01210   using type =
01211       decltype(ElemFromListImpl<typename MakeIndexSequence<N>::type>::Apply(
01212           static_cast<T (*)()>(nullptr)...));
01213 };
01214
01215 struct FlatTupleConstructTag {};
01216
01217 template <typename... T>
01218 class FlatTuple;
01219
01220 template <typename Derived, size_t I>
01221 struct FlatTupleElemBase;
01222
01223 template <typename... T, size_t I>
01224 struct FlatTupleElemBase<FlatTuple<T...>, I> {
01225   using value_type = typename ElemFromList<I, T...>::type;
01226   FlatTupleElemBase() = default;
01227   template <typename Arg>
01228   explicit FlatTupleElemBase(FlatTupleConstructTag, Arg&& t)
01229       : value(std::forward<Arg>(t)) {}
01230   value_type value;
01231 };
01232
01233 template <typename Derived, typename Idx>
01234 struct FlatTupleBase;
01235
01236 template <size_t... Idx, typename... T>
01237 struct FlatTupleBase<FlatTuple<T...>, IndexSequence<Idx...>>
01238     : FlatTupleElemBase<FlatTuple<T...>, Idx>... {
01239   using Indices = IndexSequence<Idx...>;
01240   FlatTupleBase() = default;
01241   template <typename... Args>
01242   explicit FlatTupleBase(FlatTupleConstructTag, Args&&... args)
01243       : FlatTupleElemBase<FlatTuple<T...>, Idx>(FlatTupleConstructTag{},
01244                                                 std::forward<Args>(args))... {}
01245
01246   template <size_t I>
01247   const typename ElemFromList<I, T...>::type& Get() const {
01248     return FlatTupleElemBase<FlatTuple<T...>, I>::value;
```

```
01249   }
01250
01251   template <size_t I>
01252   typename ElemFromList<I, T...>::type& Get() {
01253     return FlatTupleElemBase<FlatTuple<T...>, I>::value;
01254   }
01255
01256   template <typename F>
01257   auto Apply(F&& f) -> decltype(std::forward<F>(f)(this->Get<Idx>()...)) {
01258     return std::forward<F>(f)(Get<Idx>()...);
01259   }
01260
01261   template <typename F>
01262   auto Apply(F&& f) const -> decltype(std::forward<F>(f)(this->Get<Idx>()...)) {
01263     return std::forward<F>(f)(Get<Idx>()...);
01264   }
01265 };
01266
01267 // Analog to std::tuple but with different tradeoffs.
01268 // This class minimizes the template instantiation depth, thus allowing more
01269 // elements than std::tuple would. std::tuple has been seen to require an
01270 // instantiation depth of more than 10x the number of elements in some
01271 // implementations.
01272 // FlatTuple and ElemFromList are not recursive and have a fixed depth
01273 // regardless of T...
01274 // MakeIndexSequence, on the other hand, it is recursive but with an
01275 // instantiation depth of O(ln(N)).
01276 template <typename... T>
01277 class FlatTuple
01278     : private FlatTupleBase<FlatTuple<T...>,
01279                             typename MakeIndexSequence<sizeof...(T)>::type> {
01280   using Indices = typename FlatTupleBase<
01281       FlatTuple<T...>, typename MakeIndexSequence<sizeof...(T)>::type>::Indices;
01282
01283  public:
01284   FlatTuple() = default;
01285   template <typename... Args>
01286   explicit FlatTuple(FlatTupleConstructTag tag, Args&&... args)
01287       : FlatTuple::FlatTupleBase(tag, std::forward<Args>(args)...) {}
01288
01289   using FlatTuple::FlatTupleBase::Apply;
01290   using FlatTuple::FlatTupleBase::Get;
01291 };
01292
01293 // Utility functions to be called with static_assert to induce deprecation
01294 // warnings.
01295 GTEST_INTERNAL_DEPRECATED(
01296     "INSTANTIATE_TEST_CASE_P is deprecated, please use "
01297     "INSTANTIATE_TEST_SUITE_P")
01298 constexpr bool InstantiateTestCase_P_IsDeprecated() { return true; }
01299
01300 GTEST_INTERNAL_DEPRECATED(
01301     "TYPED_TEST_CASE_P is deprecated, please use "
01302     "TYPED_TEST_SUITE_P")
01303 constexpr bool TypedTestCase_P_IsDeprecated() { return true; }
01304
01305 GTEST_INTERNAL_DEPRECATED(
01306     "TYPED_TEST_CASE is deprecated, please use "
01307     "TYPED_TEST_SUITE")
01308 constexpr bool TypedTestCaseIsDeprecated() { return true; }
01309
01310 GTEST_INTERNAL_DEPRECATED(
01311     "REGISTER_TYPED_TEST_CASE_P is deprecated, please use "
01312     "REGISTER_TYPED_TEST_SUITE_P")
01313 constexpr bool RegisterTypedTestCase_P_IsDeprecated() { return true; }
01314
01315 GTEST_INTERNAL_DEPRECATED(
01316     "INSTANTIATE_TYPED_TEST_CASE_P is deprecated, please use "
01317     "INSTANTIATE_TYPED_TEST_SUITE_P")
01318 constexpr bool InstantiateTypedTestCase_P_IsDeprecated() { return true; }
01319
01320 }  // namespace internal
01321 }  // namespace testing
01322
01323 namespace std {
01324 // Some standard library implementations use `struct tuple_size` and some use
01325 // `class tuple_size`. Clang warns about the mismatch.
01326 // https://reviews.llvm.org/D55466
01327 #ifdef __clang__
01328 #pragma clang diagnostic push
01329 #pragma clang diagnostic ignored "-Wmismatched-tags"
01330 #endif
01331 template <typename... Ts>
01332 struct tuple_size<testing::internal::FlatTuple<Ts...>>
01333     : std::integral_constant<size_t, sizeof...(Ts)> {};
01334 #ifdef __clang__
01335 #pragma clang diagnostic pop
```

```
01336 #endif
01337 }  // namespace std
01338
01339 #define GTEST_MESSAGE_AT_(file, line, message, result_type)             \
01340   ::testing::internal::AssertHelper(result_type, file, line, message) = \
01341       ::testing::Message()
01342
01343 #define GTEST_MESSAGE_(message, result_type) \
01344   GTEST_MESSAGE_AT_(__FILE__, __LINE__, message, result_type)
01345
01346 #define GTEST_FATAL_FAILURE_(message) \
01347   return GTEST_MESSAGE_(message, ::testing::TestPartResult::kFatalFailure)
01348
01349 #define GTEST_NONFATAL_FAILURE_(message) \
01350   GTEST_MESSAGE_(message, ::testing::TestPartResult::kNonFatalFailure)
01351
01352 #define GTEST_SUCCESS_(message) \
01353   GTEST_MESSAGE_(message, ::testing::TestPartResult::kSuccess)
01354
01355 #define GTEST_SKIP_(message) \
01356   return GTEST_MESSAGE_(message, ::testing::TestPartResult::kSkip)
01357
01358 // Suppress MSVC warning 4072 (unreachable code) for the code following
01359 // statement if it returns or throws (or doesn't return or throw in some
01360 // situations).
01361 // NOTE: The "else" is important to keep this expansion to prevent a top-level
01362 // "else" from attaching to our "if".
01363 #define GTEST_SUPPRESS_UNREACHABLE_CODE_WARNING_BELOW_(statement) \
01364   if (::testing::internal::AlwaysTrue()) {                        \
01365     statement;                                                    \
01366   } else                    /* NOLINT */                        \
01367     static_assert(true, "")  // User must have a semicolon after expansion.
01368
01369 #if GTEST_HAS_EXCEPTIONS
01370
01371 namespace testing {
01372 namespace internal {
01373
01374 class NeverThrown {
01375  public:
01376   const char* what() const noexcept {
01377     return "this exception should never be thrown";
01378   }
01379 };
01380
01381 }  // namespace internal
01382 }  // namespace testing
01383
01384 #if GTEST_HAS_RTTI
01385
01386 #define GTEST_EXCEPTION_TYPE_(e) ::testing::internal::GetTypeName(typeid(e))
01387
01388 #else  // GTEST_HAS_RTTI
01389
01390 #define GTEST_EXCEPTION_TYPE_(e) \
01391   std::string { "an std::exception-derived error" }
01392
01393 #endif  // GTEST_HAS_RTTI
01394
01395 #define GTEST_TEST_THROW_CATCH_STD_EXCEPTION_(statement, expected_exception)   \
01396   catch (typename std::conditional<                                           \
01397           std::is_same<typename std::remove_cv<typename std::remove_reference<  \
01398                            expected_exception>::type>::type,                  \
01399                        std::exception>::value,                                \
01400           const ::testing::internal::NeverThrown&, const std::exception&>::type \
01401               e) {                                                           \
01402     gtest_msg.value = "Expected: " #statement                                 \
01403                       " throws an exception of type " #expected_exception     \
01404                       ".\n  Actual: it throws ";                              \
01405     gtest_msg.value += GTEST_EXCEPTION_TYPE_(e);                              \
01406     gtest_msg.value += " with description \"";                                \
01407     gtest_msg.value += e.what();                                              \
01408     gtest_msg.value += "\".";                                                 \
01409     goto GTEST_CONCAT_TOKEN_(gtest_label_testthrow_, __LINE__);               \
01410   }
01411
01412 #else  // GTEST_HAS_EXCEPTIONS
01413
01414 #define GTEST_TEST_THROW_CATCH_STD_EXCEPTION_(statement, expected_exception)
01415
01416 #endif  // GTEST_HAS_EXCEPTIONS
01417
01418 #define GTEST_TEST_THROW_(statement, expected_exception, fail)              \
01419   GTEST_AMBIGUOUS_ELSE_BLOCKER_                                             \
01420   if (::testing::internal::TrueWithString gtest_msg{}) {                    \
01421     bool gtest_caught_expected = false;                                     \
01422     try {                                                                   \
```

```
01423       GTEST_SUPPRESS_UNREACHABLE_CODE_WARNING_BELOW_(statement);           \
01424     } catch (expected_exception const&) {                                  \
01425       gtest_caught_expected = true;                                        \
01426     }                                                                      \
01427     GTEST_TEST_THROW_CATCH_STD_EXCEPTION_(statement, expected_exception)    \
01428     catch (...) {                                                          \
01429       gtest_msg.value = "Expected: " #statement                            \
01430                         " throws an exception of type " #expected_exception \
01431                         ".\n  Actual: it throws a different type.";         \
01432       goto GTEST_CONCAT_TOKEN_(gtest_label_testthrow_, __LINE__);          \
01433     }                                                                      \
01434     if (!gtest_caught_expected) {                                          \
01435       gtest_msg.value = "Expected: " #statement                            \
01436                         " throws an exception of type " #expected_exception \
01437                         ".\n  Actual: it throws nothing.";                  \
01438       goto GTEST_CONCAT_TOKEN_(gtest_label_testthrow_, __LINE__);          \
01439     }                                                                      \
01440   } else /*NOLINT*/                                                        \
01441     GTEST_CONCAT_TOKEN_(gtest_label_testthrow_, __LINE__)                  \
01442         : fail(gtest_msg.value.c_str())
01443
01444 #if GTEST_HAS_EXCEPTIONS
01445
01446 #define GTEST_TEST_NO_THROW_CATCH_STD_EXCEPTION_()                \
01447   catch (std::exception const& e) {                               \
01448     gtest_msg.value = "it throws ";                               \
01449     gtest_msg.value += GTEST_EXCEPTION_TYPE_(e);                  \
01450     gtest_msg.value += " with description \"";                    \
01451     gtest_msg.value += e.what();                                  \
01452     gtest_msg.value += "\".";                                     \
01453     goto GTEST_CONCAT_TOKEN_(gtest_label_testnothrow_, __LINE__); \
01454   }
01455
01456 #else  // GTEST_HAS_EXCEPTIONS
01457
01458 #define GTEST_TEST_NO_THROW_CATCH_STD_EXCEPTION_()
01459
01460 #endif  // GTEST_HAS_EXCEPTIONS
01461
01462 #define GTEST_TEST_NO_THROW_(statement, fail)                          \
01463   GTEST_AMBIGUOUS_ELSE_BLOCKER_                                        \
01464   if (::testing::internal::TrueWithString gtest_msg{}) {               \
01465     try {                                                              \
01466       GTEST_SUPPRESS_UNREACHABLE_CODE_WARNING_BELOW_(statement);       \
01467     }                                                                  \
01468     GTEST_TEST_NO_THROW_CATCH_STD_EXCEPTION_()                         \
01469     catch (...) {                                                      \
01470       gtest_msg.value = "it throws.";                                  \
01471       goto GTEST_CONCAT_TOKEN_(gtest_label_testnothrow_, __LINE__);    \
01472     }                                                                  \
01473   } else                                                               \
01474     GTEST_CONCAT_TOKEN_(gtest_label_testnothrow_, __LINE__)            \
01475         : fail(("Expected: " #statement " doesn't throw an exception.\n" \
01476                 "  Actual: " +                                         \
01477                 gtest_msg.value)                                       \
01478                    .c_str())
01479
01480 #define GTEST_TEST_ANY_THROW_(statement, fail)                     \
01481   GTEST_AMBIGUOUS_ELSE_BLOCKER_                                    \
01482   if (::testing::internal::AlwaysTrue()) {                         \
01483     bool gtest_caught_any = false;                                 \
01484     try {                                                          \
01485       GTEST_SUPPRESS_UNREACHABLE_CODE_WARNING_BELOW_(statement);   \
01486     } catch (...) {                                                \
01487       gtest_caught_any = true;                                     \
01488     }                                                              \
01489     if (!gtest_caught_any) {                                       \
01490       goto GTEST_CONCAT_TOKEN_(gtest_label_testanythrow_, __LINE__); \
01491     }                                                              \
01492   } else                                                           \
01493     GTEST_CONCAT_TOKEN_(gtest_label_testanythrow_, __LINE__)       \
01494         : fail("Expected: " #statement                             \
01495                " throws an exception.\n"                           \
01496                "  Actual: it doesn't.")
01497
01498 // Implements Boolean test assertions such as EXPECT_TRUE. expression can be
01499 // either a boolean expression or an AssertionResult. text is a textual
01500 // representation of expression as it was passed into the EXPECT_TRUE.
01501 #define GTEST_TEST_BOOLEAN_(expression, text, actual, expected, fail) \
01502   GTEST_AMBIGUOUS_ELSE_BLOCKER_                                       \
01503   if (const ::testing::AssertionResult gtest_ar_ =                    \
01504           ::testing::AssertionResult(expression))                     \
01505     ;                                                                 \
01506   else                                                                \
01507     fail(::testing::internal::GetBoolAssertionFailureMessage(         \
01508              gtest_ar_, text, #actual, #expected)                     \
01509              .c_str())
```

```
01510
01511 #define GTEST_TEST_NO_FATAL_FAILURE_(statement, fail)                          \
01512   GTEST_AMBIGUOUS_ELSE_BLOCKER_                                               \
01513   if (::testing::internal::AlwaysTrue()) {                                    \
01514     ::testing::internal::HasNewFatalFailureHelper gtest_fatal_failure_checker; \
01515     GTEST_SUPPRESS_UNREACHABLE_CODE_WARNING_BELOW_(statement);                \
01516     if (gtest_fatal_failure_checker.has_new_fatal_failure()) {                \
01517       goto GTEST_CONCAT_TOKEN_(gtest_label_testnofatal_, __LINE__);           \
01518     }                                                                         \
01519   } else                                                                      \
01520     GTEST_CONCAT_TOKEN_(gtest_label_testnofatal_, __LINE__)                   \
01521         : fail("Expected: " #statement                                        \
01522                " doesn't generate new fatal "                                 \
01523                "failures in the current thread.\n"                            \
01524                "  Actual: it does.")
01525
01526 // Expands to the name of the class that implements the given test.
01527 #define GTEST_TEST_CLASS_NAME_(test_suite_name, test_name) \
01528   test_suite_name##_##test_name##_Test
01529
01530 // Helper macro for defining tests.
01531 #define GTEST_TEST_(test_suite_name, test_name, parent_class, parent_id)       \
01532   static_assert(sizeof(GTEST_STRINGIFY_(test_suite_name)) > 1,                 \
01533                 "test_suite_name must not be empty");                          \
01534   static_assert(sizeof(GTEST_STRINGIFY_(test_name)) > 1,                       \
01535                 "test_name must not be empty");                                \
01536   class GTEST_TEST_CLASS_NAME_(test_suite_name, test_name)                     \
01537       : public parent_class {                                                  \
01538    public:                                                                     \
01539     GTEST_TEST_CLASS_NAME_(test_suite_name, test_name)() = default;           \
01540     ~GTEST_TEST_CLASS_NAME_(test_suite_name, test_name)() override = default; \
01541     GTEST_TEST_CLASS_NAME_(test_suite_name, test_name)                        \
01542     (const GTEST_TEST_CLASS_NAME_(test_suite_name, test_name) &) = delete;    \
01543     GTEST_TEST_CLASS_NAME_(test_suite_name, test_name) & operator=(           \
01544         const GTEST_TEST_CLASS_NAME_(test_suite_name,                         \
01545                                      test_name) &) = delete; /* NOLINT */     \
01546     GTEST_TEST_CLASS_NAME_(test_suite_name, test_name)                        \
01547     (GTEST_TEST_CLASS_NAME_(test_suite_name, test_name) &&) noexcept = delete; \
01548     GTEST_TEST_CLASS_NAME_(test_suite_name, test_name) & operator=(           \
01549         GTEST_TEST_CLASS_NAME_(test_suite_name,                               \
01550                                test_name) &&) noexcept = delete; /* NOLINT */ \
01551                                                                               \
01552    private:                                                                   \
01553     void TestBody() override;                                                 \
01554     static ::testing::TestInfo* const test_info_ GTEST_ATTRIBUTE_UNUSED_;     \
01555   };                                                                          \
01556                                                                               \
01557   ::testing::TestInfo* const GTEST_TEST_CLASS_NAME_(test_suite_name,          \
01558                                                     test_name)::test_info_ =  \
01559       ::testing::internal::MakeAndRegisterTestInfo(                           \
01560           #test_suite_name, #test_name, nullptr, nullptr,                     \
01561           ::testing::internal::CodeLocation(__FILE__, __LINE__), (parent_id), \
01562           ::testing::internal::SuiteApiResolver<                              \
01563               parent_class>::GetSetUpCaseOrSuite(__FILE__, __LINE__),         \
01564           ::testing::internal::SuiteApiResolver<                              \
01565               parent_class>::GetTearDownCaseOrSuite(__FILE__, __LINE__),      \
01566           new ::testing::internal::TestFactoryImpl<GTEST_TEST_CLASS_NAME_(    \
01567               test_suite_name, test_name)>);                                  \
01568   void GTEST_TEST_CLASS_NAME_(test_suite_name, test_name)::TestBody()
01569
01570 #endif  // GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_INTERNAL_H_
```

18.278 gtest-param-util.h

```
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```
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00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // Type and function utilities for implementing parameterized tests.
00031
00032 // IWYU pragma: private, include "gtest/gtest.h"
00033 // IWYU pragma: friend gtest/.*
00034 // IWYU pragma: friend gmock/.*
00035
00036 #ifndef GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_PARAM_UTIL_H_
00037 #define GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_PARAM_UTIL_H_
00038
00039 #include <ctype.h>
00040
00041 #include <cassert>
00042 #include <iterator>
00043 #include <memory>
00044 #include <set>
00045 #include <tuple>
00046 #include <type_traits>
00047 #include <utility>
00048 #include <vector>
00049
00050 #include "gtest/gtest-printers.h"
00051 #include "gtest/gtest-test-part.h"
00052 #include "gtest/internal/gtest-internal.h"
00053 #include "gtest/internal/gtest-port.h"
00054
00055 namespace testing {
00056     // Input to a parameterized test name generator, describing a test parameter.
00057     // Consists of the parameter value and the integer parameter index.
00058
00059     template <class ParamType>
00060     struct TestParamInfo {
00061
00062         TestParamInfo(const ParamType& a_param, size_t an_index)
00063         : param(a_param), index(an_index) {
00064         }
00065         ParamType param;
00066         size_t index;
00067     };
00068
00069     // A builtin parameterized test name generator which returns the result of
00070     // testing::PrintToString.
00071
00072     struct PrintToStringParamName {
00073
00074         template <class ParamType>
00075         std::string operator()(const TestParamInfo<ParamType>& info) const {
00076             return PrintToString(info.param);
00077         }
00078     };
00079
00080     namespace internal {
00081
00082         // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00083         // Utility Functions
00084
00085         // Outputs a message explaining invalid registration of different
00086         // fixture class for the same test suite. This may happen when
00087         // TEST_P macro is used to define two tests with the same name
00088         // but in different namespaces.
00089         GTEST_API_ void ReportInvalidTestSuiteType(const char* test_suite_name,
00090                 CodeLocation code_location);
00091
00092         template <typename>
00093         class ParamGeneratorInterface;
00094         template <typename>
00095         class ParamGenerator;
00096
00097         // Interface for iterating over elements provided by an implementation
00098         // of ParamGeneratorInterface<T>.
00099
00100         template <typename T>
00101         class ParamIteratorInterface {
00102         public:
00103
00104             virtual ~ParamIteratorInterface() {
00105             }
00106             // A pointer to the base generator instance.
00107             // Used only for the purposes of iterator comparison
00108             // to make sure that two iterators belong to the same generator.
```

```
00109             virtual const ParamGeneratorInterface<T>* BaseGenerator() const = 0;
00110             // Advances iterator to point to the next element
00111             // provided by the generator. The caller is responsible
00112             // for not calling Advance() on an iterator equal to
00113             // BaseGenerator()->End().
00114             virtual void Advance() = 0;
00115             // Clones the iterator object. Used for implementing copy semantics
00116             // of ParamIterator<T>.
00117             virtual ParamIteratorInterface* Clone() const = 0;
00118             // Dereferences the current iterator and provides (read-only) access
00119             // to the pointed value. It is the caller's responsibility not to call
00120             // Current() on an iterator equal to BaseGenerator()->End().
00121             // Used for implementing ParamGenerator<T>::operator*().
00122             virtual const T* Current() const = 0;
00123             // Determines whether the given iterator and other point to the same
00124             // element in the sequence generated by the generator.
00125             // Used for implementing ParamGenerator<T>::operator==().
00126             virtual bool Equals(const ParamIteratorInterface& other) const = 0;
00127         };
00128
00129         // Class iterating over elements provided by an implementation of
00130         // ParamGeneratorInterface<T>. It wraps ParamIteratorInterface<T>
00131         // and implements the const forward iterator concept.
00132
00133         template <typename T>
00134         class ParamIterator {
00135         public:
00136             typedef T value_type;
00137             typedef const T& reference;
00138             typedef ptrdiff_t difference_type;
00139
00140             // ParamIterator assumes ownership of the impl_ pointer.
00141
00142             ParamIterator(const ParamIterator& other) : impl_(other.impl_->Clone()) {
00143             }
00144
00145             ParamIterator& operator=(const ParamIterator& other) {
00146                 if (this != &other) impl_.reset(other.impl_->Clone());
00147                 return *this;
00148             }
00149
00150             const T& operator*() const {
00151                 return *impl_->Current();
00152             }
00153
00154             const T* operator->() const {
00155                 return impl_->Current();
00156             }
00157             // Prefix version of operator++.
00158
00159             ParamIterator& operator++() {
00160                 impl_->Advance();
00161                 return *this;
00162             }
00163             // Postfix version of operator++.
00164
00165             ParamIterator operator++(int /*unused*/) {
00166                 ParamIteratorInterface<T>* clone = impl_->Clone();
00167                 impl_->Advance();
00168                 return ParamIterator(clone);
00169             }
00170
00171             bool operator==(const ParamIterator& other) const {
00172                 return impl_.get() == other.impl_.get() || impl_->Equals(*other.impl_);
00173             }
00174
00175             bool operator!=(const ParamIterator& other) const {
00176                 return !(*this == other);
00177             }
00178
00179         private:
00180             friend class ParamGenerator<T>;
00181
00182             explicit ParamIterator(ParamIteratorInterface<T>* impl) : impl_(impl) {
00183             }
00184             std::unique_ptr<ParamIteratorInterface<T>> impl_;
00185         };
00186
00187         // ParamGeneratorInterface<T> is the binary interface to access generators
00188         // defined in other translation units.
00189
00190         template <typename T>
00191         class ParamGeneratorInterface {
00192         public:
00193             typedef T ParamType;
00194
00195             virtual ~ParamGeneratorInterface() {
```

```
00196             }
00197
00198             // Generator interface definition
00199             virtual ParamIteratorInterface<T>* Begin() const = 0;
00200             virtual ParamIteratorInterface<T>* End() const = 0;
00201         };
00202
00203         // Wraps ParamGeneratorInterface<T> and provides general generator syntax
00204         // compatible with the STL Container concept.
00205         // This class implements copy initialization semantics and the contained
00206         // ParamGeneratorInterface<T> instance is shared among all copies
00207         // of the original object. This is possible because that instance is immutable.
00208
00209         template <typename T>
00210         class ParamGenerator {
00211         public:
00212             typedef ParamIterator<T> iterator;
00213
00214             explicit ParamGenerator(ParamGeneratorInterface<T>* impl) : impl_(impl) {
00215             }
00216
00217             ParamGenerator(const ParamGenerator& other) : impl_(other.impl_) {
00218             }
00219
00220             ParamGenerator& operator=(const ParamGenerator& other) {
00221                 impl_ = other.impl_;
00222                 return *this;
00223             }
00224
00225             iterator begin() const {
00226                 return iterator(impl_->Begin());
00227             }
00228
00229             iterator end() const {
00230                 return iterator(impl_->End());
00231             }
00232
00233         private:
00234             std::shared_ptr<const ParamGeneratorInterface<T» impl_;
00235         };
00236
00237         // Generates values from a range of two comparable values. Can be used to
00238         // generate sequences of user-defined types that implement operator+() and
00239         // operator<().
00240         // This class is used in the Range() function.
00241
00242         template <typename T, typename IncrementT>
00243         class RangeGenerator : public ParamGeneratorInterface<T> {
00244         public:
00245
00246             RangeGenerator(T begin, T end, IncrementT step)
00247             : begin_(begin),
00248             end_(end),
00249             step_(step),
00250             end_index_(CalculateEndIndex(begin, end, step)) {
00251             }
00252
00253             ~RangeGenerator() override {
00254             }
00255
00256             ParamIteratorInterface<T>* Begin() const override {
00257                 return new Iterator(this, begin_, 0, step_);
00258             }
00259
00260             ParamIteratorInterface<T>* End() const override {
00261                 return new Iterator(this, end_, end_index_, step_);
00262             }
00263
00264         private:
00265
00266             class Iterator : public ParamIteratorInterface<T> {
00267             public:
00268
00269                 Iterator(const ParamGeneratorInterface<T>* base, T value, int index,
00270                         IncrementT step)
00271                 : base_(base), value_(value), index_(index), step_(step) {
00272                 }
00273
00274                 ~Iterator() override {
00275                 }
00276
00277                 const ParamGeneratorInterface<T>* BaseGenerator() const override {
00278                     return base_;
00279                 }
00280
00281                 void Advance() override {
00282                     value_ = static_cast<T> (value_ + step_);
```

```
00283                   index_++;
00284               }
00285
00286               ParamIteratorInterface<T>* Clone() const override {
00287                   return new Iterator(*this);
00288               }
00289
00290               const T* Current() const override {
00291                   return &value_;
00292               }
00293
00294               bool Equals(const ParamIteratorInterface<T>& other) const override {
00295                   // Having the same base generator guarantees that the other
00296                   // iterator is of the same type and we can downcast.
00297                   GTEST_CHECK_(BaseGenerator() == other.BaseGenerator())
00298                           << "The program attempted to compare iterators "
00299                           << "from different generators." << std::endl;
00300                   const int other_index =
00301                           CheckedDowncastToActualType<const Iterator>(&other)->index_;
00302                   return index_ == other_index;
00303               }
00304
00305           private:
00306
00307               Iterator(const Iterator& other)
00308               : ParamIteratorInterface<T>(),
00309               base_(other.base_),
00310               value_(other.value_),
00311               index_(other.index_),
00312               step_(other.step_) {
00313               }
00314
00315               // No implementation - assignment is unsupported.
00316               void operator=(const Iterator& other);
00317
00318               const ParamGeneratorInterface<T> * const base_;
00319               T value_;
00320               int index_;
00321               const IncrementT step_;
00322           }; // class RangeGenerator::Iterator
00323
00324           static int CalculateEndIndex(const T& begin, const T& end,
00325                   const IncrementT& step) {
00326               int end_index = 0;
00327               for (T i = begin; i < end; i = static_cast<T> (i + step)) end_index++;
00328               return end_index;
00329           }
00330
00331           // No implementation - assignment is unsupported.
00332           void operator=(const RangeGenerator& other);
00333
00334           const T begin_;
00335           const T end_;
00336           const IncrementT step_;
00337           // The index for the end() iterator. All the elements in the generated
00338           // sequence are indexed (0-based) to aid iterator comparison.
00339           const int end_index_;
00340       }; // class RangeGenerator
00341
00342       // Generates values from a pair of STL-style iterators. Used in the
00343       // ValuesIn() function. The elements are copied from the source range
00344       // since the source can be located on the stack, and the generator
00345       // is likely to persist beyond that stack frame.
00346
00347       template <typename T>
00348       class ValuesInIteratorRangeGenerator : public ParamGeneratorInterface<T> {
00349       public:
00350
00351           template <typename ForwardIterator>
00352           ValuesInIteratorRangeGenerator(ForwardIterator begin, ForwardIterator end)
00353           : container_(begin, end) {
00354           }
00355
00356           ~ValuesInIteratorRangeGenerator() override {
00357           }
00358
00359           ParamIteratorInterface<T>* Begin() const override {
00360               return new Iterator(this, container_.begin());
00361           }
00362
00363           ParamIteratorInterface<T>* End() const override {
00364               return new Iterator(this, container_.end());
00365           }
00366
00367       private:
00368           typedef typename ::std::vector<T> ContainerType;
00369
```

```
00370           class Iterator : public ParamIteratorInterface<T> {
00371           public:
00372
00373               Iterator(const ParamGeneratorInterface<T>* base,
00374                        typename ContainerType::const_iterator iterator)
00375               : base_(base), iterator_(iterator) {
00376               }
00377
00378               ~Iterator() override {
00379               }
00380
00381               const ParamGeneratorInterface<T>* BaseGenerator() const override {
00382                   return base_;
00383               }
00384
00385               void Advance() override {
00386                   ++iterator_;
00387                   value_.reset();
00388               }
00389
00390               ParamIteratorInterface<T>* Clone() const override {
00391                   return new Iterator(*this);
00392               }
00393               // We need to use cached value referenced by iterator_ because *iterator_
00394               // can return a temporary object (and of type other then T), so just
00395               // having "return &*iterator_;" doesn't work.
00396               // value_ is updated here and not in Advance() because Advance()
00397               // can advance iterator_ beyond the end of the range, and we cannot
00398               // detect that fact. The client code, on the other hand, is
00399               // responsible for not calling Current() on an out-of-range iterator.
00400
00401               const T* Current() const override {
00402                   if (value_.get() == nullptr) value_.reset(new T(*iterator_));
00403                   return value_.get();
00404               }
00405
00406               bool Equals(const ParamIteratorInterface<T>& other) const override {
00407                   // Having the same base generator guarantees that the other
00408                   // iterator is of the same type and we can downcast.
00409                   GTEST_CHECK_(BaseGenerator() == other.BaseGenerator())
00410                           << "The program attempted to compare iterators "
00411                           << "from different generators." << std::endl;
00412                   return iterator_ ==
00413                           CheckedDowncastToActualType<const Iterator>(&other)->iterator_;
00414               }
00415
00416           private:
00417
00418               Iterator(const Iterator& other)
00419               // The explicit constructor call suppresses a false warning
00420               // emitted by gcc when supplied with the -Wextra option.
00421               : ParamIteratorInterface<T>(),
00422               base_(other.base_),
00423               iterator_(other.iterator_) {
00424               }
00425
00426               const ParamGeneratorInterface<T> * const base_;
00427               typename ContainerType::const_iterator iterator_;
00428               // A cached value of *iterator_. We keep it here to allow access by
00429               // pointer in the wrapping iterator's operator->().
00430               // value_ needs to be mutable to be accessed in Current().
00431               // Use of std::unique_ptr helps manage cached value's lifetime,
00432               // which is bound by the lifespan of the iterator itself.
00433               mutable std::unique_ptr<const T> value_;
00434           }; // class ValuesInIteratorRangeGenerator::Iterator
00435
00436           // No implementation - assignment is unsupported.
00437           void operator=(const ValuesInIteratorRangeGenerator& other);
00438
00439           const ContainerType container_;
00440       }; // class ValuesInIteratorRangeGenerator
00441
00442       // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00443       //
00444       // Default parameterized test name generator, returns a string containing the
00445       // integer test parameter index.
00446
00447       template <class ParamType>
00448       std::string DefaultParamName(const TestParamInfo<ParamType>& info) {
00449           Message name_stream;
00450           name_stream << info.index;
00451           return name_stream.GetString();
00452       }
00453
00454       template <typename T = int>
00455       void TestNotEmpty() {
00456           static_assert(sizeof (T) == 0, "Empty arguments are not allowed.");
```

```
00457         }
00458
00459         template <typename T = int>
00460         void TestNotEmpty(const T&) {
00461         }
00462
00463         // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00464         //
00465         // Stores a parameter value and later creates tests parameterized with that
00466         // value.
00467
00468         template <class TestClass>
00469         class ParameterizedTestFactory : public TestFactoryBase {
00470         public:
00471             typedef typename TestClass::ParamType ParamType;
00472
00473             explicit ParameterizedTestFactory(ParamType parameter)
00474             : parameter_(parameter) {
00475             }
00476
00477             Test* CreateTest() override {
00478                 TestClass::SetParam(&parameter_);
00479                 return new TestClass();
00480             }
00481
00482         private:
00483             const ParamType parameter_;
00484
00485             ParameterizedTestFactory(const ParameterizedTestFactory&) = delete;
00486             ParameterizedTestFactory& operator=(const ParameterizedTestFactory&) = delete;
00487         };
00488
00489         // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00490         //
00491         // TestMetaFactoryBase is a base class for meta-factories that create
00492         // test factories for passing into MakeAndRegisterTestInfo function.
00493
00494         template <class ParamType>
00495         class TestMetaFactoryBase {
00496         public:
00497
00498             virtual ~TestMetaFactoryBase() {
00499             }
00500
00501             virtual TestFactoryBase* CreateTestFactory(ParamType parameter) = 0;
00502         };
00503
00504         // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00505         //
00506         // TestMetaFactory creates test factories for passing into
00507         // MakeAndRegisterTestInfo function. Since MakeAndRegisterTestInfo receives
00508         // ownership of test factory pointer, same factory object cannot be passed
00509         // into that method twice. But ParameterizedTestSuiteInfo is going to call
00510         // it for each Test/Parameter value combination. Thus it needs meta factory
00511         // creator class.
00512
00513         template <class TestSuite>
00514         class TestMetaFactory
00515         : public TestMetaFactoryBase<typename TestSuite::ParamType> {
00516         public:
00517             using ParamType = typename TestSuite::ParamType;
00518
00519             TestMetaFactory() {
00520             }
00521
00522             TestFactoryBase* CreateTestFactory(ParamType parameter) override {
00523                 return new ParameterizedTestFactory<TestSuite>(parameter);
00524             }
00525
00526         private:
00527             TestMetaFactory(const TestMetaFactory&) = delete;
00528             TestMetaFactory& operator=(const TestMetaFactory&) = delete;
00529         };
00530
00531         // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00532         //
00533         // ParameterizedTestSuiteInfoBase is a generic interface
00534         // to ParameterizedTestSuiteInfo classes. ParameterizedTestSuiteInfoBase
00535         // accumulates test information provided by TEST_P macro invocations
00536         // and generators provided by INSTANTIATE_TEST_SUITE_P macro invocations
00537         // and uses that information to register all resulting test instances
00538         // in RegisterTests method. The ParameterizeTestSuiteRegistry class holds
00539         // a collection of pointers to the ParameterizedTestSuiteInfo objects
00540         // and calls RegisterTests() on each of them when asked.
00541
00542         class ParameterizedTestSuiteInfoBase {
00543         public:
```

```
00544
00545           virtual ~ParameterizedTestSuiteInfoBase() {
00546           }
00547
00548           // Base part of test suite name for display purposes.
00549           virtual const std::string& GetTestSuiteName() const = 0;
00550           // Test suite id to verify identity.
00551           virtual TypeId GetTestSuiteTypeId() const = 0;
00552           // UnitTest class invokes this method to register tests in this
00553           // test suite right before running them in RUN_ALL_TESTS macro.
00554           // This method should not be called more than once on any single
00555           // instance of a ParameterizedTestSuiteInfoBase derived class.
00556           virtual void RegisterTests() = 0;
00557
00558       protected:
00559
00560           ParameterizedTestSuiteInfoBase() {
00561           }
00562
00563       private:
00564           ParameterizedTestSuiteInfoBase(const ParameterizedTestSuiteInfoBase&) =
00565           delete;
00566           ParameterizedTestSuiteInfoBase& operator=(
00567                   const ParameterizedTestSuiteInfoBase&) = delete;
00568       };
00569
00570       // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00571       //
00572       // Report a the name of a test_suit as safe to ignore
00573       // as the side effect of construction of this type.
00574
00575       struct GTEST_API_ MarkAsIgnored {
00576           explicit MarkAsIgnored(const char* test_suite);
00577       };
00578
00579       GTEST_API_ void InsertSyntheticTestCase(const std::string& name,
00580               CodeLocation location, bool has_test_p);
00581
00582       // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00583       //
00584       // ParameterizedTestSuiteInfo accumulates tests obtained from TEST_P
00585       // macro invocations for a particular test suite and generators
00586       // obtained from INSTANTIATE_TEST_SUITE_P macro invocations for that
00587       // test suite. It registers tests with all values generated by all
00588       // generators when asked.
00589
00590       template <class TestSuite>
00591       class ParameterizedTestSuiteInfo : public ParameterizedTestSuiteInfoBase {
00592       public:
00593           // ParamType and GeneratorCreationFunc are private types but are required
00594           // for declarations of public methods AddTestPattern() and
00595           // AddTestSuiteInstantiation().
00596           using ParamType = typename TestSuite::ParamType;
00597           // A function that returns an instance of appropriate generator type.
00598           typedef ParamGenerator<ParamType>(GeneratorCreationFunc) ();
00599           using ParamNameGeneratorFunc = std::string(const TestParamInfo<ParamType>&);
00600
00601           explicit ParameterizedTestSuiteInfo(const char* name,
00602                   CodeLocation code_location)
00603           : test_suite_name_(name), code_location_(code_location) {
00604           }
00605
00606           // Test suite base name for display purposes.
00607
00608           const std::string& GetTestSuiteName() const override {
00609               return test_suite_name_;
00610           }
00611           // Test suite id to verify identity.
00612
00613           TypeId GetTestSuiteTypeId() const override {
00614               return GetTypeId<TestSuite>();
00615           }
00616           // TEST_P macro uses AddTestPattern() to record information
00617           // about a single test in a LocalTestInfo structure.
00618           // test_suite_name is the base name of the test suite (without invocation
00619           // prefix). test_base_name is the name of an individual test without
00620           // parameter index. For the test SequenceA/FooTest.DoBar/1 FooTest is
00621           // test suite base name and DoBar is test base name.
00622
00623           void AddTestPattern(const char* test_suite_name, const char* test_base_name,
00624                   TestMetaFactoryBase<ParamType>* meta_factory,
00625                   CodeLocation code_location) {
00626               tests_.push_back(std::shared_ptr<TestInfo>(new TestInfo(
00627                       test_suite_name, test_base_name, meta_factory, code_location)));
00628           }
00629           // INSTANTIATE_TEST_SUITE_P macro uses AddGenerator() to record information
00630           // about a generator.
```

```
00631
00632             int AddTestSuiteInstantiation(const std::string& instantiation_name,
00633                     GeneratorCreationFunc* func,
00634                     ParamNameGeneratorFunc* name_func,
00635                     const char* file, int line) {
00636                 instantiations_.push_back(
00637                         InstantiationInfo(instantiation_name, func, name_func, file, line));
00638                 return 0; // Return value used only to run this method in namespace scope.
00639             }
00640             // UnitTest class invokes this method to register tests in this test suite
00641             // right before running tests in RUN_ALL_TESTS macro.
00642             // This method should not be called more than once on any single
00643             // instance of a ParameterizedTestSuiteInfoBase derived class.
00644             // UnitTest has a guard to prevent from calling this method more than once.
00645
00646             void RegisterTests() override {
00647                 bool generated_instantiations = false;
00648
00649                 for (typename TestInfoContainer::iterator test_it = tests_.begin();
00650                         test_it != tests_.end(); ++test_it) {
00651                     std::shared_ptr<TestInfo> test_info = *test_it;
00652                     for (typename InstantiationContainer::iterator gen_it =
00653                             instantiations_.begin();
00654                             gen_it != instantiations_.end(); ++gen_it) {
00655                         const std::string& instantiation_name = gen_it->name;
00656                         ParamGenerator<ParamType> generator((*gen_it->generator)());
00657                         ParamNameGeneratorFunc* name_func = gen_it->name_func;
00658                         const char* file = gen_it->file;
00659                         int line = gen_it->line;
00660
00661                         std::string test_suite_name;
00662                         if (!instantiation_name.empty())
00663                             test_suite_name = instantiation_name + "/";
00664                         test_suite_name += test_info->test_suite_base_name;
00665
00666                         size_t i = 0;
00667                         std::set<std::string> test_param_names;
00668                         for (typename ParamGenerator<ParamType>::iterator param_it =
00669                                 generator.begin();
00670                                 param_it != generator.end(); ++param_it, ++i) {
00671                             generated_instantiations = true;
00672
00673                             Message test_name_stream;
00674
00675                             std::string param_name =
00676                                     name_func(TestParamInfo<ParamType>(*param_it, i));
00677
00678                             GTEST_CHECK_(IsValidParamName(param_name))
00679                                     << "Parameterized test name '" << param_name
00680                                     << "' is invalid, in " << file << " line " << line << std::endl;
00681
00682                             GTEST_CHECK_(test_param_names.count(param_name) == 0)
00683                                     << "Duplicate parameterized test name '" << param_name << "', in "
00684                                     << file << " line " << line << std::endl;
00685
00686                             test_param_names.insert(param_name);
00687
00688                             if (!test_info->test_base_name.empty()) {
00689                                 test_name_stream << test_info->test_base_name << "/";
00690                             }
00691                             test_name_stream << param_name;
00692                             MakeAndRegisterTestInfo(
00693                                     test_suite_name.c_str(), test_name_stream.GetString().c_str(),
00694                                     nullptr, // No type parameter.
00695                                     PrintToString(*param_it).c_str(), test_info->code_location,
00696                                     GetTestSuiteTypeId(),
00697                                     SuiteApiResolver<TestSuite>::GetSetUpCaseOrSuite(file, line),
00698                                     SuiteApiResolver<TestSuite>::GetTearDownCaseOrSuite(file, line),
00699                                     test_info->test_meta_factory->CreateTestFactory(*param_it));
00700                         } // for param_it
00701                     } // for gen_it
00702                 } // for test_it
00703
00704                 if (!generated_instantiations) {
00705                     // There are no generaotrs, or they all generate nothing ...
00706                     InsertSyntheticTestCase(GetTestSuiteName(), code_location_,
00707                             !tests_.empty());
00708                 }
00709             } // RegisterTests
00710
00711         private:
00712             // LocalTestInfo structure keeps information about a single test registered
00713             // with TEST_P macro.
00714
00715             struct TestInfo {
00716
00717                 TestInfo(const char* a_test_suite_base_name, const char* a_test_base_name,
```

```
00718                       TestMetaFactoryBase<ParamType>* a_test_meta_factory,
00719                       CodeLocation a_code_location)
00720               : test_suite_base_name(a_test_suite_base_name),
00721               test_base_name(a_test_base_name),
00722               test_meta_factory(a_test_meta_factory),
00723               code_location(a_code_location) {
00724               }
00725
00726               const std::string test_suite_base_name;
00727               const std::string test_base_name;
00728               const std::unique_ptr<TestMetaFactoryBase<ParamType» test_meta_factory;
00729               const CodeLocation code_location;
00730           };
00731           using TestInfoContainer = ::std::vector<std::shared_ptr<TestInfo»;
00732           // Records data received from INSTANTIATE_TEST_SUITE_P macros:
00733           //  <Instantiation name, Sequence generator creation function,
00734           //     Name generator function, Source file, Source line>
00735
00736           struct InstantiationInfo {
00737
00738               InstantiationInfo(const std::string& name_in,
00739                       GeneratorCreationFunc* generator_in,
00740                       ParamNameGeneratorFunc* name_func_in, const char* file_in,
00741                       int line_in)
00742               : name(name_in),
00743               generator(generator_in),
00744               name_func(name_func_in),
00745               file(file_in),
00746               line(line_in) {
00747               }
00748
00749               std::string name;
00750               GeneratorCreationFunc* generator;
00751               ParamNameGeneratorFunc* name_func;
00752               const char* file;
00753               int line;
00754           };
00755           typedef ::std::vector<InstantiationInfo> InstantiationContainer;
00756
00757           static bool IsValidParamName(const std::string& name) {
00758               // Check for empty string
00759               if (name.empty()) return false;
00760
00761               // Check for invalid characters
00762               for (std::string::size_type index = 0; index < name.size(); ++index) {
00763                   if (!IsAlNum(name[index]) && name[index] != '_') return false;
00764               }
00765
00766               return true;
00767           }
00768
00769           const std::string test_suite_name_;
00770           CodeLocation code_location_;
00771           TestInfoContainer tests_;
00772           InstantiationContainer instantiations_;
00773
00774           ParameterizedTestSuiteInfo(const ParameterizedTestSuiteInfo&) = delete;
00775           ParameterizedTestSuiteInfo& operator=(const ParameterizedTestSuiteInfo&) =
00776                   delete;
00777       }; // class ParameterizedTestSuiteInfo
00778
00779       //  Legacy API is deprecated but still available
00780 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00781       template <class TestCase>
00782       using ParameterizedTestCaseInfo = ParameterizedTestSuiteInfo<TestCase>;
00783 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00784
00785       // INTERNAL IMPLEMENTATION - DO NOT USE IN USER CODE.
00786       //
00787       // ParameterizedTestSuiteRegistry contains a map of
00788       // ParameterizedTestSuiteInfoBase classes accessed by test suite names. TEST_P
00789       // and INSTANTIATE_TEST_SUITE_P macros use it to locate their corresponding
00790       // ParameterizedTestSuiteInfo descriptors.
00791
00792       class ParameterizedTestSuiteRegistry {
00793       public:
00794
00795           ParameterizedTestSuiteRegistry() {
00796           }
00797
00798           ~ParameterizedTestSuiteRegistry() {
00799               for (auto& test_suite_info : test_suite_infos_) {
00800                   delete test_suite_info;
00801               }
00802           }
00803
00804           // Looks up or creates and returns a structure containing information about
```

```
00805             // tests and instantiations of a particular test suite.
00806
00807             template <class TestSuite>
00808             ParameterizedTestSuiteInfo<TestSuite>* GetTestSuitePatternHolder(
00809                     const char* test_suite_name, CodeLocation code_location) {
00810                 ParameterizedTestSuiteInfo<TestSuite>* typed_test_info = nullptr;
00811                 for (auto& test_suite_info : test_suite_infos_) {
00812                     if (test_suite_info->GetTestSuiteName() == test_suite_name) {
00813                         if (test_suite_info->GetTestSuiteTypeId() != GetTypeId<TestSuite>()) {
00814                             // Complain about incorrect usage of Google Test facilities
00815                             // and terminate the program since we cannot guaranty correct
00816                             // test suite setup and tear-down in this case.
00817                             ReportInvalidTestSuiteType(test_suite_name, code_location);
00818                             posix::Abort();
00819                         } else {
00820                             // At this point we are sure that the object we found is of the same
00821                             // type we are looking for, so we downcast it to that type
00822                             // without further checks.
00823                             typed_test_info = CheckedDowncastToActualType<
00824                                     ParameterizedTestSuiteInfo < TestSuite » (test_suite_info);
00825                         }
00826                         break;
00827                     }
00828                 }
00829                 if (typed_test_info == nullptr) {
00830                     typed_test_info = new ParameterizedTestSuiteInfo<TestSuite>(
00831                             test_suite_name, code_location);
00832                     test_suite_infos_.push_back(typed_test_info);
00833                 }
00834                 return typed_test_info;
00835             }
00836
00837             void RegisterTests() {
00838                 for (auto& test_suite_info : test_suite_infos_) {
00839                     test_suite_info->RegisterTests();
00840                 }
00841             }
00842             //  Legacy API is deprecated but still available
00843 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00844
00845             template <class TestCase>
00846             ParameterizedTestCaseInfo<TestCase>* GetTestCasePatternHolder(
00847                     const char* test_case_name, CodeLocation code_location) {
00848                 return GetTestSuitePatternHolder<TestCase>(test_case_name, code_location);
00849             }
00850
00851 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00852
00853         private:
00854             using TestSuiteInfoContainer = ::std::vector<ParameterizedTestSuiteInfoBase*>;
00855
00856             TestSuiteInfoContainer test_suite_infos_;
00857
00858             ParameterizedTestSuiteRegistry(const ParameterizedTestSuiteRegistry&) =
00859             delete;
00860             ParameterizedTestSuiteRegistry& operator=(
00861                     const ParameterizedTestSuiteRegistry&) = delete;
00862         };
00863
00864         // Keep track of what type-parameterized test suite are defined and
00865         // where as well as which are intatiated. This allows susequently
00866         // identifying suits that are defined but never used.
00867
00868         class TypeParameterizedTestSuiteRegistry {
00869         public:
00870             // Add a suite definition
00871             void RegisterTestSuite(const char* test_suite_name,
00872                     CodeLocation code_location);
00873
00874             // Add an instantiation of a suit.
00875             void RegisterInstantiation(const char* test_suite_name);
00876
00877             // For each suit repored as defined but not reported as instantiation,
00878             // emit a test that reports that fact (configurably, as an error).
00879             void CheckForInstantiations();
00880
00881         private:
00882
00883             struct TypeParameterizedTestSuiteInfo {
00884
00885                 explicit TypeParameterizedTestSuiteInfo(CodeLocation c)
00886                 : code_location(c), instantiated(false) {
00887                 }
00888
00889                 CodeLocation code_location;
00890                 bool instantiated;
00891             };
```

```
00892
00893             std::map<std::string, TypeParameterizedTestSuiteInfo> suites_;
00894         };
00895
00896     } // namespace internal
00897
00898     // Forward declarations of ValuesIn(), which is implemented in
00899     // include/gtest/gtest-param-test.h.
00900     template <class Container>
00901     internal::ParamGenerator<typename Container::value_type> ValuesIn(
00902             const Container& container);
00903
00904     namespace internal {
00905         // Used in the Values() function to provide polymorphic capabilities.
00906
00907 #ifdef _MSC_VER
00908 #pragma warning(push)
00909 #pragma warning(disable : 4100)
00910 #endif
00911
00912         template <typename... Ts>
00913         class ValueArray {
00914         public:
00915
00916             explicit ValueArray(Ts... v) : v_(FlatTupleConstructTag{}, std::move(v)...) {
00917             }
00918
00919             template <typename T>
00920             operator ParamGenerator<T>() const { // NOLINT
00921                 return ValuesIn(MakeVector<T>(MakeIndexSequence<sizeof...(Ts)>()));
00922             }
00923
00924         private:
00925
00926             template <typename T, size_t... I>
00927             std::vector<T> MakeVector(IndexSequence<I...>) const {
00928                 return std::vector<T>{static_cast<T> (v_.template Get<I>())...};
00929             }
00930
00931             FlatTuple<Ts...> v_;
00932         };
00933
00934 #ifdef _MSC_VER
00935 #pragma warning(pop)
00936 #endif
00937
00938         template <typename... T>
00939         class CartesianProductGenerator
00940         : public ParamGeneratorInterface<::std::tuple<T...>>
00941         {
00942             public:
00943             typedef ::std::tuple<T...> ParamType;
00944
00945             CartesianProductGenerator(const std::tuple<ParamGenerator<T>...>& g)
00946                     : generators_(g) {
00947             }
00948
00949             ~CartesianProductGenerator() override {
00950             }
00951
00952             ParamIteratorInterface<ParamType>* Begin() const override {
00953                 return new Iterator(this, generators_, false);
00954             }
00955
00956             ParamIteratorInterface<ParamType>* End() const override {
00957                 return new Iterator(this, generators_, true);
00958             }
00959
00960             private:
00961             template <class I>
00962             class IteratorImpl;
00963             template <size_t... I>
00964             class IteratorImpl<IndexSequence<I...>>
00965             : public ParamIteratorInterface<ParamType>
00966             {
00967                 public:
00968
00969                 IteratorImpl(const ParamGeneratorInterface<ParamType>* base,
00970                         const std::tuple<ParamGenerator<T>...>& generators,
00971                         bool is_end)
00972                         : base_(base),
00973                         begin_(std::get<I>(generators).begin()...),
00974                         end_(std::get<I>(generators).end()...),
00975                         current_(is_end ? end_ : begin_) {
00976                     ComputeCurrentValue();
00977                 }
00978
```

```
00979                 ~IteratorImpl() override {
00980                 }
00981
00982                 const ParamGeneratorInterface<ParamType>* BaseGenerator() const override {
00983                     return base_;
00984                 }
00985                 // Advance should not be called on beyond-of-range iterators
00986                 // so no component iterators must be beyond end of range, either.
00987
00988                 void Advance() override {
00989                     assert(!AtEnd());
00990                     // Advance the last iterator.
00991                     ++std::get<sizeof...(T) - 1 > (current_);
00992                     // if that reaches end, propagate that up.
00993                     AdvanceIfEnd<sizeof...(T) - 1 > ();
00994                     ComputeCurrentValue();
00995                 }
00996
00997                 ParamIteratorInterface<ParamType>* Clone() const override {
00998                     return new IteratorImpl(*this);
00999                 }
01000
01001                 const ParamType * Current() const override {
01002                     return current_value_.get();
01003                 }
01004
01005                 bool Equals(const ParamIteratorInterface<ParamType>& other) const override {
01006                     // Having the same base generator guarantees that the other
01007                     // iterator is of the same type and we can downcast.
01008                     GTEST_CHECK_(BaseGenerator() == other.BaseGenerator())
01009                             << "The program attempted to compare iterators "
01010                             << "from different generators." << std::endl;
01011                     const IteratorImpl* typed_other =
01012                             CheckedDowncastToActualType<const IteratorImpl>(&other);
01013
01014                     // We must report iterators equal if they both point beyond their
01015                     // respective ranges. That can happen in a variety of fashions,
01016                     // so we have to consult AtEnd().
01017                     if (AtEnd() && typed_other->AtEnd()) return true;
01018
01019                     bool same = true;
01020                     bool dummy[] = {
01021                         (same = same && std::get<I>(current_) ==
01022                         std::get<I>(typed_other->current_))...
01023                     };
01024                     (void) dummy;
01025                     return same;
01026                 }
01027
01028                 private:
01029
01030                 template <size_t ThisI>
01031                         void AdvanceIfEnd() {
01032                     if (std::get<ThisI>(current_) != std::get<ThisI>(end_)) return;
01033
01034                     bool last = ThisI == 0;
01035                     if (last) {
01036                         // We are done. Nothing else to propagate.
01037                         return;
01038                     }
01039
01040                     constexpr size_t NextI = ThisI - (ThisI != 0);
01041                     std::get<ThisI>(current_) = std::get<ThisI>(begin_);
01042                     ++std::get<NextI>(current_);
01043                     AdvanceIfEnd<NextI>();
01044                 }
01045
01046                 void ComputeCurrentValue() {
01047                     if (!AtEnd())
01048                         current_value_ = std::make_shared<ParamType>(*std::get<I>(current_)...);
01049                 }
01050
01051                 bool AtEnd() const {
01052                     bool at_end = false;
01053                     bool dummy[] = {
01054                         (at_end = at_end || std::get<I>(current_) == std::get<I>(end_))...
01055                     };
01056                     (void) dummy;
01057                     return at_end;
01058                 }
01059
01060                 const ParamGeneratorInterface<ParamType> * const base_;
01061                 std::tuple<typename ParamGenerator<T>::iterator...> begin_;
01062                 std::tuple<typename ParamGenerator<T>::iterator...> end_;
01063                 std::tuple<typename ParamGenerator<T>::iterator...> current_;
01064                 std::shared_ptr<ParamType> current_value_;
01065             };
```

```
01066
01067             using Iterator = IteratorImpl<typename MakeIndexSequence<sizeof...(T)>::type>;
01068
01069             std::tuple<ParamGenerator<T>...> generators_;
01070         };
01071
01072         template <class... Gen>
01073         class CartesianProductHolder {
01074         public:
01075
01076             CartesianProductHolder(const Gen&... g) : generators_(g...) {
01077             }
01078
01079             template <typename... T>
01080             operator ParamGenerator<::std::tuple<T...>>() const {
01081                 return ParamGenerator<::std::tuple<T...>>(
01082                         new CartesianProductGenerator<T...>(generators_));
01083             }
01084
01085         private:
01086             std::tuple<Gen...> generators_;
01087         };
01088
01089     } // namespace internal
01090 } // namespace testing
01091
01092 #endif  // GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_PARAM_UTIL_H_
```

18.279 gtest-port-arch.h

```
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00029
00030 // The Google C++ Testing and Mocking Framework (Google Test)
00031 //
00032 // This header file defines the GTEST_OS_* macro.
00033 // It is separate from gtest-port.h so that custom/gtest-port.h can include it.
00034
00035 #ifndef GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_PORT_ARCH_H_
00036 #define GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_PORT_ARCH_H_
00037
00038 // Determines the platform on which Google Test is compiled.
00039 #ifdef __CYGWIN__
00040 #define GTEST_OS_CYGWIN 1
00041 #elif defined(__MINGW__) || defined(__MINGW32__) || defined(__MINGW64__)
00042 #define GTEST_OS_WINDOWS_MINGW 1
00043 #define GTEST_OS_WINDOWS 1
00044 #elif defined _WIN32
00045 #define GTEST_OS_WINDOWS 1
00046 #ifdef _WIN32_WCE
00047 #define GTEST_OS_WINDOWS_MOBILE 1
00048 #elif defined(WINAPI_FAMILY)
00049 #include <winapifamily.h>
00050 #if WINAPI_FAMILY_PARTITION(WINAPI_PARTITION_DESKTOP)
00051 #define GTEST_OS_WINDOWS_DESKTOP 1
00052 #elif WINAPI_FAMILY_PARTITION(WINAPI_PARTITION_PHONE_APP)
00053 #define GTEST_OS_WINDOWS_PHONE 1
00054 #elif WINAPI_FAMILY_PARTITION(WINAPI_PARTITION_APP)
00055 #define GTEST_OS_WINDOWS_RT 1
```

```
00056 #elif WINAPI_FAMILY_PARTITION(WINAPI_PARTITION_TV_TITLE)
00057 #define GTEST_OS_WINDOWS_PHONE 1
00058 #define GTEST_OS_WINDOWS_TV_TITLE 1
00059 #else
00060 // WINAPI_FAMILY defined but no known partition matched.
00061 // Default to desktop.
00062 #define GTEST_OS_WINDOWS_DESKTOP 1
00063 #endif
00064 #else
00065 #define GTEST_OS_WINDOWS_DESKTOP 1
00066 #endif  // _WIN32_WCE
00067 #elif defined __OS2__
00068 #define GTEST_OS_OS2 1
00069 #elif defined __APPLE__
00070 #define GTEST_OS_MAC 1
00071 #include <TargetConditionals.h>
00072 #if TARGET_OS_IPHONE
00073 #define GTEST_OS_IOS 1
00074 #endif
00075 #elif defined __DragonFly__
00076 #define GTEST_OS_DRAGONFLY 1
00077 #elif defined __FreeBSD__
00078 #define GTEST_OS_FREEBSD 1
00079 #elif defined __Fuchsia__
00080 #define GTEST_OS_FUCHSIA 1
00081 #elif defined(__GNU__)
00082 #define GTEST_OS_GNU_HURD 1
00083 #elif defined(__GLIBC__) && defined(__FreeBSD_kernel__)
00084 #define GTEST_OS_GNU_KFREEBSD 1
00085 #elif defined __linux__
00086 #define GTEST_OS_LINUX 1
00087 #if defined __ANDROID__
00088 #define GTEST_OS_LINUX_ANDROID 1
00089 #endif
00090 #elif defined __MVS__
00091 #define GTEST_OS_ZOS 1
00092 #elif defined(__sun) && defined(__SVR4)
00093 #define GTEST_OS_SOLARIS 1
00094 #elif defined(_AIX)
00095 #define GTEST_OS_AIX 1
00096 #elif defined(__hpux)
00097 #define GTEST_OS_HPUX 1
00098 #elif defined __native_client__
00099 #define GTEST_OS_NACL 1
00100 #elif defined __NetBSD__
00101 #define GTEST_OS_NETBSD 1
00102 #elif defined __OpenBSD__
00103 #define GTEST_OS_OPENBSD 1
00104 #elif defined __QNX__
00105 #define GTEST_OS_QNX 1
00106 #elif defined(__HAIKU__)
00107 #define GTEST_OS_HAIKU 1
00108 #elif defined ESP8266
00109 #define GTEST_OS_ESP8266 1
00110 #elif defined ESP32
00111 #define GTEST_OS_ESP32 1
00112 #elif defined(__XTENSA__)
00113 #define GTEST_OS_XTENSA 1
00114 #endif  // __CYGWIN__
00115
00116 #endif  // GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_PORT_ARCH_H_
```

18.280 gtest-string.h

```
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00029
00030 // The Google C++ Testing and Mocking Framework (Google Test)
00031 //
00032 // This header file declares the String class and functions used internally by
00033 // Google Test.  They are subject to change without notice. They should not used
00034 // by code external to Google Test.
00035 //
00036 // This header file is #included by gtest-internal.h.
00037 // It should not be #included by other files.
00038
00039 // IWYU pragma: private, include "gtest/gtest.h"
00040 // IWYU pragma: friend gtest/.*
00041 // IWYU pragma: friend gmock/.*
00042
00043 #ifndef GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_STRING_H_
00044 #define GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_STRING_H_
00045
00046 #ifdef __BORLANDC__
00047 // string.h is not guaranteed to provide strcpy on C++ Builder.
00048 #include <mem.h>
00049 #endif
00050
00051 #include <string.h>
00052
00053 #include <cstdint>
00054 #include <string>
00055
00056 #include "gtest/internal/gtest-port.h"
00057
00058 namespace testing {
00059     namespace internal {
00060
00061         // String - an abstract class holding static string utilities.
00062
00063         class GTEST_API_ String {
00064         public:
00065             // Static utility methods
00066
00067             // Clones a 0-terminated C string, allocating memory using new.  The
00068             // caller is responsible for deleting the return value using
00069             // delete[].  Returns the cloned string, or NULL if the input is
00070             // NULL.
00071             //
00072             // This is different from strdup() in string.h, which allocates
00073             // memory using malloc().
00074             static const char* CloneCString(const char* c_str);
00075
00076 #if GTEST_OS_WINDOWS_MOBILE
00077             // Windows CE does not have the 'ANSI' versions of Win32 APIs. To be
00078             // able to pass strings to Win32 APIs on CE we need to convert them
00079             // to 'Unicode', UTF-16.
00080
00081             // Creates a UTF-16 wide string from the given ANSI string, allocating
00082             // memory using new. The caller is responsible for deleting the return
00083             // value using delete[]. Returns the wide string, or NULL if the
00084             // input is NULL.
00085             //
00086             // The wide string is created using the ANSI codepage (CP_ACP) to
00087             // match the behaviour of the ANSI versions of Win32 calls and the
00088             // C runtime.
00089             static LPCWSTR AnsiToUtf16(const char* c_str);
00090
00091             // Creates an ANSI string from the given wide string, allocating
00092             // memory using new. The caller is responsible for deleting the return
00093             // value using delete[]. Returns the ANSI string, or NULL if the
00094             // input is NULL.
00095             //
00096             // The returned string is created using the ANSI codepage (CP_ACP) to
00097             // match the behaviour of the ANSI versions of Win32 calls and the
00098             // C runtime.
00099             static const char* Utf16ToAnsi(LPCWSTR utf16_str);
00100 #endif
00101
00102             // Compares two C strings.  Returns true if and only if they have the same
00103             // content.
00104             //
00105             // Unlike strcmp(), this function can handle NULL argument(s).  A
00106             // NULL C string is considered different to any non-NULL C string,
00107             // including the empty string.
00108             static bool CStringEquals(const char* lhs, const char* rhs);
```

```
00109
00110             // Converts a wide C string to a String using the UTF-8 encoding.
00111             // NULL will be converted to "(null)".  If an error occurred during
00112             // the conversion, "(failed to convert from wide string)" is
00113             // returned.
00114             static std::string ShowWideCString(const wchar_t* wide_c_str);
00115
00116             // Compares two wide C strings.  Returns true if and only if they have the
00117             // same content.
00118             //
00119             // Unlike wcscmp(), this function can handle NULL argument(s).  A
00120             // NULL C string is considered different to any non-NULL C string,
00121             // including the empty string.
00122             static bool WideCStringEquals(const wchar_t* lhs, const wchar_t* rhs);
00123
00124             // Compares two C strings, ignoring case.  Returns true if and only if
00125             // they have the same content.
00126             //
00127             // Unlike strcasecmp(), this function can handle NULL argument(s).
00128             // A NULL C string is considered different to any non-NULL C string,
00129             // including the empty string.
00130             static bool CaseInsensitiveCStringEquals(const char* lhs, const char* rhs);
00131
00132             // Compares two wide C strings, ignoring case.  Returns true if and only if
00133             // they have the same content.
00134             //
00135             // Unlike wcscasecmp(), this function can handle NULL argument(s).
00136             // A NULL C string is considered different to any non-NULL wide C string,
00137             // including the empty string.
00138             // NB: The implementations on different platforms slightly differ.
00139             // On windows, this method uses _wcsicmp which compares according to LC_CTYPE
00140             // environment variable. On GNU platform this method uses wcscasecmp
00141             // which compares according to LC_CTYPE category of the current locale.
00142             // On MacOS X, it uses towlower, which also uses LC_CTYPE category of the
00143             // current locale.
00144             static bool CaseInsensitiveWideCStringEquals(const wchar_t* lhs,
00145                     const wchar_t* rhs);
00146
00147             // Returns true if and only if the given string ends with the given suffix,
00148             // ignoring case. Any string is considered to end with an empty suffix.
00149             static bool EndsWithCaseInsensitive(const std::string& str,
00150                     const std::string& suffix);
00151
00152             // Formats an int value as "%02d".
00153             static std::string FormatIntWidth2(int value); // "%02d" for width == 2
00154
00155             // Formats an int value to given width with leading zeros.
00156             static std::string FormatIntWidthN(int value, int width);
00157
00158             // Formats an int value as "%X".
00159             static std::string FormatHexInt(int value);
00160
00161             // Formats an int value as "%X".
00162             static std::string FormatHexUInt32(uint32_t value);
00163
00164             // Formats a byte as "%02X".
00165             static std::string FormatByte(unsigned char value);
00166
00167         private:
00168             String(); // Not meant to be instantiated.
00169         }; // class String
00170
00171         // Gets the content of the stringstream's buffer as an std::string.  Each '\0'
00172         // character in the buffer is replaced with "\\0".
00173         GTEST_API_ std::string StringStreamToString(::std::stringstream* stream);
00174
00175     } // namespace internal
00176 } // namespace testing
00177
00178 #endif  // GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_STRING_H_
```

18.281 gtest-type-util.h

```
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00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
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00029
00030 // Type utilities needed for implementing typed and type-parameterized
00031 // tests.
00032
00033 // IWYU pragma: private, include "gtest/gtest.h"
00034 // IWYU pragma: friend gtest/.*
00035 // IWYU pragma: friend gmock/.*
00036
00037 #ifndef GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_TYPE_UTIL_H_
00038 #define GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_TYPE_UTIL_H_
00039
00040 #include "gtest/internal/gtest-port.h"
00041
00042 // #ifdef __GNUC__ is too general here.  It is possible to use gcc without using
00043 // libstdc++ (which is where cxxabi.h comes from).
00044 #if GTEST_HAS_CXXABI_H_
00045 #include <cxxabi.h>
00046 #elif defined(__HP_aCC)
00047 #include <acxx_demangle.h>
00048 #endif  // GTEST_HASH_CXXABI_H_
00049
00050 namespace testing {
00051     namespace internal {
00052
00053         // Canonicalizes a given name with respect to the Standard C++ Library.
00054         // This handles removing the inline namespace within `std` that is
00055         // used by various standard libraries (e.g., `std::__1`).  Names outside
00056         // of namespace std are returned unmodified.
00057
00058         inline std::string CanonicalizeForStdLibVersioning(std::string s) {
00059             static const char prefix[] = "std::__";
00060             if (s.compare(0, strlen(prefix), prefix) == 0) {
00061                 std::string::size_type end = s.find("::", strlen(prefix));
00062                 if (end != s.npos) {
00063                     // Erase everything between the initial `std` and the second `::`.
00064                     s.erase(strlen("std"), end - strlen("std"));
00065                 }
00066             }
00067             return s;
00068         }
00069
00070 #if GTEST_HAS_RTTI
00071         // GetTypeName(const std::type_info&) returns a human-readable name of type T.
00072
00073         inline std::string GetTypeName(const std::type_info& type) {
00074             const char* const name = type.name();
00075 #if GTEST_HAS_CXXABI_H_ || defined(__HP_aCC)
00076             int status = 0;
00077             // gcc's implementation of typeid(T).name() mangles the type name,
00078             // so we have to demangle it.
00079 #if GTEST_HAS_CXXABI_H_
00080             using abi::__cxa_demangle;
00081 #endif  // GTEST_HAS_CXXABI_H_
00082             char* const readable_name = __cxa_demangle(name, nullptr, nullptr, &status);
00083             const std::string name_str(status == 0 ? readable_name : name);
00084             free(readable_name);
00085             return CanonicalizeForStdLibVersioning(name_str);
00086 #else
00087             return name;
00088 #endif  // GTEST_HAS_CXXABI_H_ || __HP_aCC
00089         }
00090 #endif  // GTEST_HAS_RTTI
00091
00092         // GetTypeName<T>() returns a human-readable name of type T if and only if
00093         // RTTI is enabled, otherwise it returns a dummy type name.
00094         // NB: This function is also used in Google Mock, so don't move it inside of
00095         // the typed-test-only section below.
00096
00097         template <typename T>
00098         std::string GetTypeName() {
00099 #if GTEST_HAS_RTTI
```

```
00100             return GetTypeName(typeid (T));
00101 #else
00102             return "<type>";
00103 #endif  // GTEST_HAS_RTTI
00104         }
00105
00106         // A unique type indicating an empty node
00107
00108         struct None {
00109         };
00110
00111 #define GTEST_TEMPLATE_ \
00112   template <typename T> \
00113   class
00114
00115         // The template "selector" struct TemplateSel<Tmpl> is used to
00116         // represent Tmpl, which must be a class template with one type
00117         // parameter, as a type.  TemplateSel<Tmpl>::Bind<T>::type is defined
00118         // as the type Tmpl<T>.  This allows us to actually instantiate the
00119         // template "selected" by TemplateSel<Tmpl>.
00120         //
00121         // This trick is necessary for simulating typedef for class templates,
00122         // which C++ doesn't support directly.
00123
00124         template <GTEST_TEMPLATE_ Tmpl>
00125         struct TemplateSel {
00126
00127             template <typename T>
00128             struct Bind {
00129                 typedef Tmpl<T> type;
00130             };
00131         };
00132
00133 #define GTEST_BIND_(TmplSel, T) TmplSel::template Bind<T>::type
00134
00135         template <GTEST_TEMPLATE_ Head_, GTEST_TEMPLATE_... Tail_>
00136         struct Templates {
00137             using Head = TemplateSel<Head_>;
00138             using Tail = Templates<Tail_...>;
00139         };
00140
00141         template <GTEST_TEMPLATE_ Head_>
00142         struct Templates<Head_> {
00143             using Head = TemplateSel<Head_>;
00144             using Tail = None;
00145         };
00146
00147         // Tuple-like type lists
00148
00149         template <typename Head_, typename... Tail_>
00150         struct Types {
00151             using Head = Head_;
00152             using Tail = Types<Tail_...>;
00153         };
00154
00155         template <typename Head_>
00156         struct Types<Head_> {
00157             using Head = Head_;
00158             using Tail = None;
00159         };
00160
00161         // Helper metafunctions to tell apart a single type from types
00162         // generated by ::testing::Types
00163
00164         template <typename... Ts>
00165         struct ProxyTypeList {
00166             using type = Types<Ts...>;
00167         };
00168
00169         template <typename>
00170         struct is_proxy_type_list : std::false_type {
00171         };
00172
00173         template <typename... Ts>
00174         struct is_proxy_type_list<ProxyTypeList<Ts...>> : std::true_type
00175         {
00176         };
00177
00178         // Generator which conditionally creates type lists.
00179         // It recognizes if a requested type list should be created
00180         // and prevents creating a new type list nested within another one.
00181
00182         template <typename T>
00183         struct GenerateTypeList {
00184         private:
00185             using proxy = typename std::conditional<is_proxy_type_list<T>::value, T,
00186                     ProxyTypeList<T>>::type;
```

```
00187
00188         public:
00189             using type = typename proxy::type;
00190         };
00191
00192     } // namespace internal
00193
00194     template <typename... Ts>
00195     using Types = internal::ProxyTypeList<Ts...>;
00196
00197 } // namespace testing
00198
00199 #endif  // GOOGLETEST_INCLUDE_GTEST_INTERNAL_GTEST_TYPE_UTIL_H_
```

18.282 gtest-assertion-result.cc

```
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00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // The Google C++ Testing and Mocking Framework (Google Test)
00031 //
00032 // This file defines the AssertionResult type.
00033
00034 #include "gtest/gtest-assertion-result.h"
00035
00036 #include <string>
00037 #include <utility>
00038
00039 #include "gtest/gtest-message.h"
00040
00041 namespace testing {
00042
00043     // AssertionResult constructors.
00044     // Used in EXPECT_TRUE/FALSE(assertion_result).
00045
00046     AssertionResult::AssertionResult(const AssertionResult& other)
00047     : success_(other.success_),
00048     message_(other.message_.get() != nullptr
00049     ? new ::std::string(*other.message_)
00050     : static_cast< ::std::string*> (nullptr)) {
00051     }
00052
00053     // Swaps two AssertionResults.
00054
00055     void AssertionResult::swap(AssertionResult& other) {
00056         using std::swap;
00057         swap(success_, other.success_);
00058         swap(message_, other.message_);
00059     }
00060
00061     // Returns the assertion's negation. Used with EXPECT/ASSERT_FALSE.
00062
00063     AssertionResult AssertionResult::operator!() const {
00064         AssertionResult negation(!success_);
00065         if (message_.get() != nullptr) negation << *message_;
00066         return negation;
00067     }
00068
00069     // Makes a successful assertion result.
```

```
00070
00071     AssertionResult AssertionSuccess() {
00072         return AssertionResult(true);
00073     }
00074
00075     // Makes a failed assertion result.
00076
00077     AssertionResult AssertionFailure() {
00078         return AssertionResult(false);
00079     }
00080
00081     // Makes a failed assertion result with the given failure message.
00082     // Deprecated; use AssertionFailure() << message.
00083
00084     AssertionResult AssertionFailure(const Message& message) {
00085         return AssertionFailure() << message;
00086     }
00087
00088 } // namespace testing
```

18.283 gtest-death-test.cc

```
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00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
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00029
00030 //
00031 // This file implements death tests.
00032
00033 #include "gtest/gtest-death-test.h"
00034
00035 #include <functional>
00036 #include <utility>
00037
00038 #include "gtest/internal/custom/gtest.h"
00039 #include "gtest/internal/gtest-port.h"
00040
00041 #if GTEST_HAS_DEATH_TEST
00042
00043 #if GTEST_OS_MAC
00044 #include <crt_externs.h>
00045 #endif  // GTEST_OS_MAC
00046
00047 #include <errno.h>
00048 #include <fcntl.h>
00049 #include <limits.h>
00050
00051 #if GTEST_OS_LINUX
00052 #include <signal.h>
00053 #endif  // GTEST_OS_LINUX
00054
00055 #include <stdarg.h>
00056
00057 #if GTEST_OS_WINDOWS
00058 #include <windows.h>
00059 #else
00060 #include <sys/mman.h>
00061 #include <sys/wait.h>
00062 #endif  // GTEST_OS_WINDOWS
00063
```

```
00064 #if GTEST_OS_QNX
00065 #include <spawn.h>
00066 #endif  // GTEST_OS_QNX
00067
00068 #if GTEST_OS_FUCHSIA
00069 #include <lib/fdio/fd.h>
00070 #include <lib/fdio/io.h>
00071 #include <lib/fdio/spawn.h>
00072 #include <lib/zx/channel.h>
00073 #include <lib/zx/port.h>
00074 #include <lib/zx/process.h>
00075 #include <lib/zx/socket.h>
00076 #include <zircon/processargs.h>
00077 #include <zircon/syscalls.h>
00078 #include <zircon/syscalls/policy.h>
00079 #include <zircon/syscalls/port.h>
00080 #endif  // GTEST_OS_FUCHSIA
00081
00082 #endif  // GTEST_HAS_DEATH_TEST
00083
00084 #include "gtest/gtest-message.h"
00085 #include "gtest/internal/gtest-string.h"
00086 #include "src/gtest-internal-inl.h"
00087
00088 namespace testing {
00089
00090     // Constants.
00091
00092     // The default death test style.
00093     //
00094     // This is defined in internal/gtest-port.h as "fast", but can be overridden by
00095     // a definition in internal/custom/gtest-port.h. The recommended value, which is
00096     // used internally at Google, is "threadsafe".
00097     static const char kDefaultDeathTestStyle[] = GTEST_DEFAULT_DEATH_TEST_STYLE;
00098
00099 } // namespace testing
00100
00101 GTEST_DEFINE_string_(
00102         death_test_style,
00103         testing::internal::StringFromGTestEnv("death_test_style",
00104         testing::kDefaultDeathTestStyle),
00105         "Indicates how to run a death test in a forked child process: "
00106         "\"threadsafe\" (child process re-executes the test binary "
00107         "from the beginning, running only the specific death test) or "
00108         "\"fast\" (child process runs the death test immediately "
00109         "after forking).");
00110
00111 GTEST_DEFINE_bool_(
00112         death_test_use_fork,
00113         testing::internal::BoolFromGTestEnv("death_test_use_fork", false),
00114         "Instructs to use fork()/_exit() instead of clone() in death tests. "
00115         "Ignored and always uses fork() on POSIX systems where clone() is not "
00116         "implemented. Useful when running under valgrind or similar tools if "
00117         "those do not support clone(). Valgrind 3.3.1 will just fail if "
00118         "it sees an unsupported combination of clone() flags. "
00119         "It is not recommended to use this flag w/o valgrind though it will "
00120         "work in 99% of the cases. Once valgrind is fixed, this flag will "
00121         "most likely be removed.");
00122
00123 GTEST_DEFINE_string_(
00124         internal_run_death_test, "",
00125         "Indicates the file, line number, temporal index of "
00126         "the single death test to run, and a file descriptor to "
00127         "which a success code may be sent, all separated by "
00128         "the '|' characters.  This flag is specified if and only if the "
00129         "current process is a sub-process launched for running a thread-safe "
00130         "death test.  FOR INTERNAL USE ONLY.");
00131
00132 namespace testing {
00133
00134 #if GTEST_HAS_DEATH_TEST
00135
00136     namespace internal {
00137
00138         // Valid only for fast death tests. Indicates the code is running in the
00139         // child process of a fast style death test.
00140 #if !GTEST_OS_WINDOWS && !GTEST_OS_FUCHSIA
00141         static bool g_in_fast_death_test_child = false;
00142 #endif
00143
00144         // Returns a Boolean value indicating whether the caller is currently
00145         // executing in the context of the death test child process.  Tools such as
00146         // Valgrind heap checkers may need this to modify their behavior in death
00147         // tests.  IMPORTANT: This is an internal utility.  Using it may break the
00148         // implementation of death tests.  User code MUST NOT use it.
00149
00150         bool InDeathTestChild() {
```

```
00151 #if GTEST_OS_WINDOWS || GTEST_OS_FUCHSIA
00152
00153             // On Windows and Fuchsia, death tests are thread-safe regardless of the value
00154             // of the death_test_style flag.
00155             return !GTEST_FLAG_GET(internal_run_death_test).empty();
00156
00157 #else
00158
00159             if (GTEST_FLAG_GET(death_test_style) == "threadsafe")
00160                 return !GTEST_FLAG_GET(internal_run_death_test).empty();
00161             else
00162                 return g_in_fast_death_test_child;
00163 #endif
00164         }
00165
00166     } // namespace internal
00167
00168     // ExitedWithCode constructor.
00169
00170     ExitedWithCode::ExitedWithCode(int exit_code) : exit_code_(exit_code) {
00171     }
00172
00173     // ExitedWithCode function-call operator.
00174
00175     bool ExitedWithCode::operator()(int exit_status) const {
00176 #if GTEST_OS_WINDOWS || GTEST_OS_FUCHSIA
00177
00178         return exit_status == exit_code_;
00179
00180 #else
00181
00182         return WIFEXITED(exit_status) && WEXITSTATUS(exit_status) == exit_code_;
00183
00184 #endif  // GTEST_OS_WINDOWS || GTEST_OS_FUCHSIA
00185     }
00186
00187 #if !GTEST_OS_WINDOWS && !GTEST_OS_FUCHSIA
00188     // KilledBySignal constructor.
00189
00190     KilledBySignal::KilledBySignal(int signum) : signum_(signum) {
00191     }
00192
00193     // KilledBySignal function-call operator.
00194
00195     bool KilledBySignal::operator()(int exit_status) const {
00196 #if defined(GTEST_KILLED_BY_SIGNAL_OVERRIDE_)
00197         {
00198             bool result;
00199             if (GTEST_KILLED_BY_SIGNAL_OVERRIDE_(signum_, exit_status, &result)) {
00200                 return result;
00201             }
00202         }
00203 #endif  // defined(GTEST_KILLED_BY_SIGNAL_OVERRIDE_)
00204         return WIFSIGNALED(exit_status) && WTERMSIG(exit_status) == signum_;
00205     }
00206 #endif  // !GTEST_OS_WINDOWS && !GTEST_OS_FUCHSIA
00207
00208     namespace internal {
00209
00210         // Utilities needed for death tests.
00211
00212         // Generates a textual description of a given exit code, in the format
00213         // specified by wait(2).
00214
00215         static std::string ExitSummary(int exit_code) {
00216             Message m;
00217
00218 #if GTEST_OS_WINDOWS || GTEST_OS_FUCHSIA
00219
00220             m « "Exited with exit status " « exit_code;
00221
00222 #else
00223
00224             if (WIFEXITED(exit_code)) {
00225                 m « "Exited with exit status " « WEXITSTATUS(exit_code);
00226             } else if (WIFSIGNALED(exit_code)) {
00227                 m « "Terminated by signal " « WTERMSIG(exit_code);
00228             }
00229 #ifdef WCOREDUMP
00230             if (WCOREDUMP(exit_code)) {
00231                 m « " (core dumped)";
00232             }
00233 #endif
00234 #endif  // GTEST_OS_WINDOWS || GTEST_OS_FUCHSIA
00235
00236             return m.GetString();
00237         }
```

```
00238
00239         // Returns true if exit_status describes a process that was terminated
00240         // by a signal, or exited normally with a nonzero exit code.
00241
00242         bool ExitedUnsuccessfully(int exit_status) {
00243             return !ExitedWithCode(0)(exit_status);
00244         }
00245
00246 #if !GTEST_OS_WINDOWS && !GTEST_OS_FUCHSIA
00247         // Generates a textual failure message when a death test finds more than
00248         // one thread running, or cannot determine the number of threads, prior
00249         // to executing the given statement.  It is the responsibility of the
00250         // caller not to pass a thread_count of 1.
00251
00252         static std::string DeathTestThreadWarning(size_t thread_count) {
00253             Message msg;
00254             msg « "Death tests use fork(), which is unsafe particularly"
00255                     « " in a threaded context. For this test, " « GTEST_NAME_ « " ";
00256             if (thread_count == 0) {
00257                 msg « "couldn't detect the number of threads.";
00258             } else {
00259                 msg « "detected " « thread_count « " threads.";
00260             }
00261             msg « " See "
00262                     "https://github.com/google/googletest/blob/main/docs/"
00263                     "advanced.md#death-tests-and-threads"
00264                     « " for more explanation and suggested solutions, especially if"
00265                     « " this is the last message you see before your test times out.";
00266             return msg.GetString();
00267         }
00268 #endif  // !GTEST_OS_WINDOWS && !GTEST_OS_FUCHSIA
00269
00270         // Flag characters for reporting a death test that did not die.
00271         static const char kDeathTestLived = 'L';
00272         static const char kDeathTestReturned = 'R';
00273         static const char kDeathTestThrew = 'T';
00274         static const char kDeathTestInternalError = 'I';
00275
00276 #if GTEST_OS_FUCHSIA
00277
00278         // File descriptor used for the pipe in the child process.
00279         static const int kFuchsiaReadPipeFd = 3;
00280
00281 #endif
00282
00283         // An enumeration describing all of the possible ways that a death test can
00284         // conclude.  DIED means that the process died while executing the test
00285         // code; LIVED means that process lived beyond the end of the test code;
00286         // RETURNED means that the test statement attempted to execute a return
00287         // statement, which is not allowed; THREW means that the test statement
00288         // returned control by throwing an exception.  IN_PROGRESS means the test
00289         // has not yet concluded.
00290
00291         enum DeathTestOutcome {
00292             IN_PROGRESS, DIED, LIVED, RETURNED, THREW
00293         };
00294
00295         // Routine for aborting the program which is safe to call from an
00296         // exec-style death test child process, in which case the error
00297         // message is propagated back to the parent process.  Otherwise, the
00298         // message is simply printed to stderr.  In either case, the program
00299         // then exits with status 1.
00300         [[noreturn]] static void DeathTestAbort(const std::string& message) {
00301             // On a POSIX system, this function may be called from a threadsafe-style
00302             // death test child process, which operates on a very small stack.  Use
00303             // the heap for any additional non-minuscule memory requirements.
00304             const InternalRunDeathTestFlag * const flag =
00305                     GetUnitTestImpl()->internal_run_death_test_flag();
00306             if (flag != nullptr) {
00307                 FILE* parent = posix::FDOpen(flag->write_fd(), "w");
00308                 fputc(kDeathTestInternalError, parent);
00309                 fprintf(parent, "%s", message.c_str());
00310                 fflush(parent);
00311                 _exit(1);
00312             } else {
00313                 fprintf(stderr, "%s", message.c_str());
00314                 fflush(stderr);
00315                 posix::Abort();
00316             }
00317         }
00318
00319         // A replacement for CHECK that calls DeathTestAbort if the assertion
00320         // fails.
00321 #define GTEST_DEATH_TEST_CHECK_(expression)                                    \
00322   do {                                                                         \
00323     if (!::testing::internal::IsTrue(expression)) {                            \
00324       DeathTestAbort(::std::string("CHECK failed: File ") + __FILE__ +   \
```

```
00325                     ", line " +                                           \
00326                     ::testing::internal::StreamableToString(__LINE__) + \
00327                     ": " + #expression);                                  \
00328     }                                                                     \
00329   } while (::testing::internal::AlwaysFalse())
00330
00331         // This macro is similar to GTEST_DEATH_TEST_CHECK_, but it is meant for
00332         // evaluating any system call that fulfills two conditions: it must return
00333         // -1 on failure, and set errno to EINTR when it is interrupted and
00334         // should be tried again.  The macro expands to a loop that repeatedly
00335         // evaluates the expression as long as it evaluates to -1 and sets
00336         // errno to EINTR.  If the expression evaluates to -1 but errno is
00337         // something other than EINTR, DeathTestAbort is called.
00338 #define GTEST_DEATH_TEST_CHECK_SYSCALL_(expression)                        \
00339   do {                                                                    \
00340     int gtest_retval;                                                     \
00341     do {                                                                  \
00342       gtest_retval = (expression);                                        \
00343     } while (gtest_retval == -1 && errno == EINTR);                       \
00344     if (gtest_retval == -1) {                                             \
00345       DeathTestAbort(::std::string("CHECK failed: File ") + __FILE__ +    \
00346                     ", line " +                                           \
00347                     ::testing::internal::StreamableToString(__LINE__) + \
00348                     ": " + #expression + " != -1");                       \
00349     }                                                                     \
00350   } while (::testing::internal::AlwaysFalse())
00351
00352         // Returns the message describing the last system error in errno.
00353
00354         std::string GetLastErrnoDescription() {
00355             return errno == 0 ? "" : posix::StrError(errno);
00356         }
00357
00358         // This is called from a death test parent process to read a failure
00359         // message from the death test child process and log it with the FATAL
00360         // severity. On Windows, the message is read from a pipe handle. On other
00361         // platforms, it is read from a file descriptor.
00362
00363         static void FailFromInternalError(int fd) {
00364             Message error;
00365             char buffer[256];
00366             int num_read;
00367
00368             do {
00369                 while ((num_read = posix::Read(fd, buffer, 255)) > 0) {
00370                     buffer[num_read] = '\0';
00371                     error << buffer;
00372                 }
00373             } while (num_read == -1 && errno == EINTR);
00374
00375             if (num_read == 0) {
00376                 GTEST_LOG_(FATAL) << error.GetString();
00377             } else {
00378                 const int last_error = errno;
00379                 GTEST_LOG_(FATAL) << "Error while reading death test internal: "
00380                         << GetLastErrnoDescription() << " [" << last_error << "]";
00381             }
00382         }
00383
00384         // Death test constructor.  Increments the running death test count
00385         // for the current test.
00386
00387         DeathTest::DeathTest() {
00388             TestInfo * const info = GetUnitTestImpl()->current_test_info();
00389             if (info == nullptr) {
00390                 DeathTestAbort(
00391                         "Cannot run a death test outside of a TEST or "
00392                         "TEST_F construct");
00393             }
00394         }
00395
00396         // Creates and returns a death test by dispatching to the current
00397         // death test factory.
00398
00399         bool DeathTest::Create(const char* statement,
00400                 Matcher<const std::string&> matcher, const char* file,
00401                 int line, DeathTest** test) {
00402             return GetUnitTestImpl()->death_test_factory()->Create(
00403                     statement, std::move(matcher), file, line, test);
00404         }
00405
00406         const char* DeathTest::LastMessage() {
00407             return last_death_test_message_.c_str();
00408         }
00409
00410         void DeathTest::set_last_death_test_message(const std::string& message) {
00411             last_death_test_message_ = message;
```

```
00412         }
00413
00414         std::string DeathTest::last_death_test_message_;
00415
00416         // Provides cross platform implementation for some death functionality.
00417
00418         class DeathTestImpl : public DeathTest {
00419         protected:
00420
00421             DeathTestImpl(const char* a_statement, Matcher<const std::string&> matcher)
00422             : statement_(a_statement),
00423             matcher_(std::move(matcher)),
00424             spawned_(false),
00425             status_(-1),
00426             outcome_(IN_PROGRESS),
00427             read_fd_(-1),
00428             write_fd_(-1) {
00429             }
00430
00431             // read_fd_ is expected to be closed and cleared by a derived class.
00432
00433             ~DeathTestImpl() override {
00434                 GTEST_DEATH_TEST_CHECK_(read_fd_ == -1);
00435             }
00436
00437             void Abort(AbortReason reason) override;
00438             bool Passed(bool status_ok) override;
00439
00440             const char* statement() const {
00441                 return statement_;
00442             }
00443
00444             bool spawned() const {
00445                 return spawned_;
00446             }
00447
00448             void set_spawned(bool is_spawned) {
00449                 spawned_ = is_spawned;
00450             }
00451
00452             int status() const {
00453                 return status_;
00454             }
00455
00456             void set_status(int a_status) {
00457                 status_ = a_status;
00458             }
00459
00460             DeathTestOutcome outcome() const {
00461                 return outcome_;
00462             }
00463
00464             void set_outcome(DeathTestOutcome an_outcome) {
00465                 outcome_ = an_outcome;
00466             }
00467
00468             int read_fd() const {
00469                 return read_fd_;
00470             }
00471
00472             void set_read_fd(int fd) {
00473                 read_fd_ = fd;
00474             }
00475
00476             int write_fd() const {
00477                 return write_fd_;
00478             }
00479
00480             void set_write_fd(int fd) {
00481                 write_fd_ = fd;
00482             }
00483
00484             // Called in the parent process only. Reads the result code of the death
00485             // test child process via a pipe, interprets it to set the outcome_
00486             // member, and closes read_fd_.  Outputs diagnostics and terminates in
00487             // case of unexpected codes.
00488             void ReadAndInterpretStatusByte();
00489
00490             // Returns stderr output from the child process.
00491             virtual std::string GetErrorLogs();
00492
00493         private:
00494             // The textual content of the code this object is testing.  This class
00495             // doesn't own this string and should not attempt to delete it.
00496             const char* const statement_;
00497             // A matcher that's expected to match the stderr output by the child process.
00498             Matcher<const std::string&> matcher_;
```

```
00499             // True if the death test child process has been successfully spawned.
00500             bool spawned_;
00501             // The exit status of the child process.
00502             int status_;
00503             // How the death test concluded.
00504             DeathTestOutcome outcome_;
00505             // Descriptor to the read end of the pipe to the child process.  It is
00506             // always -1 in the child process.  The child keeps its write end of the
00507             // pipe in write_fd_.
00508             int read_fd_;
00509             // Descriptor to the child's write end of the pipe to the parent process.
00510             // It is always -1 in the parent process.  The parent keeps its end of the
00511             // pipe in read_fd_.
00512             int write_fd_;
00513         };
00514
00515         // Called in the parent process only. Reads the result code of the death
00516         // test child process via a pipe, interprets it to set the outcome_
00517         // member, and closes read_fd_.  Outputs diagnostics and terminates in
00518         // case of unexpected codes.
00519
00520         void DeathTestImpl::ReadAndInterpretStatusByte() {
00521             char flag;
00522             int bytes_read;
00523
00524             // The read() here blocks until data is available (signifying the
00525             // failure of the death test) or until the pipe is closed (signifying
00526             // its success), so it's okay to call this in the parent before
00527             // the child process has exited.
00528             do {
00529                 bytes_read = posix::Read(read_fd(), &flag, 1);
00530             } while (bytes_read == -1 && errno == EINTR);
00531
00532             if (bytes_read == 0) {
00533                 set_outcome(DIED);
00534             } else if (bytes_read == 1) {
00535                 switch (flag) {
00536                     case kDeathTestReturned:
00537                         set_outcome(RETURNED);
00538                         break;
00539                     case kDeathTestThrew:
00540                         set_outcome(THREW);
00541                         break;
00542                     case kDeathTestLived:
00543                         set_outcome(LIVED);
00544                         break;
00545                     case kDeathTestInternalError:
00546                         FailFromInternalError(read_fd()); // Does not return.
00547                         break;
00548                     default:
00549                         GTEST_LOG_(FATAL) << "Death test child process reported "
00550                                           << "unexpected status byte ("
00551                                           << static_cast<unsigned int> (flag) << ")";
00552                 }
00553             } else {
00554                 GTEST_LOG_(FATAL) << "Read from death test child process failed: "
00555                                   << GetLastErrnoDescription();
00556             }
00557             GTEST_DEATH_TEST_CHECK_SYSCALL_(posix::Close(read_fd()));
00558             set_read_fd(-1);
00559         }
00560
00561         std::string DeathTestImpl::GetErrorLogs() {
00562             return GetCapturedStderr();
00563         }
00564
00565         // Signals that the death test code which should have exited, didn't.
00566         // Should be called only in a death test child process.
00567         // Writes a status byte to the child's status file descriptor, then
00568         // calls _exit(1).
00569
00570         void DeathTestImpl::Abort(AbortReason reason) {
00571             // The parent process considers the death test to be a failure if
00572             // it finds any data in our pipe.  So, here we write a single flag byte
00573             // to the pipe, then exit.
00574             const char status_ch = reason == TEST_DID_NOT_DIE ? kDeathTestLived
00575                     : reason == TEST_THREW_EXCEPTION ? kDeathTestThrew
00576                     : kDeathTestReturned;
00577
00578             GTEST_DEATH_TEST_CHECK_SYSCALL_(posix::Write(write_fd(), &status_ch, 1));
00579             // We are leaking the descriptor here because on some platforms (i.e.,
00580             // when built as Windows DLL), destructors of global objects will still
00581             // run after calling _exit(). On such systems, write_fd_ will be
00582             // indirectly closed from the destructor of UnitTestImpl, causing double
00583             // close if it is also closed here. On debug configurations, double close
00584             // may assert. As there are no in-process buffers to flush here, we are
00585             // relying on the OS to close the descriptor after the process terminates
```

```
00586             // when the destructors are not run.
00587             _exit(1); // Exits w/o any normal exit hooks (we were supposed to crash)
00588         }
00589
00590         // Returns an indented copy of stderr output for a death test.
00591         // This makes distinguishing death test output lines from regular log lines
00592         // much easier.
00593
00594         static ::std::string FormatDeathTestOutput(const ::std::string& output) {
00595             ::std::string ret;
00596             for (size_t at = 0;;) {
00597                 const size_t line_end = output.find('\n', at);
00598                 ret += "[  DEATH   ] ";
00599                 if (line_end == ::std::string::npos) {
00600                     ret += output.substr(at);
00601                     break;
00602                 }
00603                 ret += output.substr(at, line_end + 1 - at);
00604                 at = line_end + 1;
00605             }
00606             return ret;
00607         }
00608
00609         // Assesses the success or failure of a death test, using both private
00610         // members which have previously been set, and one argument:
00611         //
00612         // Private data members:
00613         //   outcome:  An enumeration describing how the death test
00614         //             concluded: DIED, LIVED, THREW, or RETURNED.  The death test
00615         //             fails in the latter three cases.
00616         //   status:   The exit status of the child process. On *nix, it is in the
00617         //             in the format specified by wait(2). On Windows, this is the
00618         //             value supplied to the ExitProcess() API or a numeric code
00619         //             of the exception that terminated the program.
00620         //   matcher_: A matcher that's expected to match the stderr output by the child
00621         //             process.
00622         //
00623         // Argument:
00624         //   status_ok: true if exit_status is acceptable in the context of
00625         //              this particular death test, which fails if it is false
00626         //
00627         // Returns true if and only if all of the above conditions are met.  Otherwise,
00628         // the first failing condition, in the order given above, is the one that is
00629         // reported. Also sets the last death test message string.
00630
00631         bool DeathTestImpl::Passed(bool status_ok) {
00632             if (!spawned()) return false;
00633
00634             const std::string error_message = GetErrorLogs();
00635
00636             bool success = false;
00637             Message buffer;
00638
00639             buffer << "Death test: " << statement() << "\n";
00640             switch (outcome()) {
00641                 case LIVED:
00642                     buffer << "    Result: failed to die.\n"
00643                            << " Error msg:\n"
00644                            << FormatDeathTestOutput(error_message);
00645                     break;
00646                 case THREW:
00647                     buffer << "    Result: threw an exception.\n"
00648                            << " Error msg:\n"
00649                            << FormatDeathTestOutput(error_message);
00650                     break;
00651                 case RETURNED:
00652                     buffer << "    Result: illegal return in test statement.\n"
00653                            << " Error msg:\n"
00654                            << FormatDeathTestOutput(error_message);
00655                     break;
00656                 case DIED:
00657                     if (status_ok) {
00658                         if (matcher_.Matches(error_message)) {
00659                             success = true;
00660                         } else {
00661                             std::ostringstream stream;
00662                             matcher_.DescribeTo(&stream);
00663                             buffer << "    Result: died but not with expected error.\n"
00664                                    << "  Expected: " << stream.str() << "\n"
00665                                    << "Actual msg:\n"
00666                                    << FormatDeathTestOutput(error_message);
00667                         }
00668                     } else {
00669                         buffer << "    Result: died but not with expected exit code:\n"
00670                                << "            " << ExitSummary(status()) << "\n"
00671                                << "Actual msg:\n"
00672                                << FormatDeathTestOutput(error_message);
```

```
00673                     }
00674                     break;
00675                 case IN_PROGRESS:
00676                 default:
00677                     GTEST_LOG_(FATAL)
00678                             << "DeathTest::Passed somehow called before conclusion of test";
00679             }
00680
00681             DeathTest::set_last_death_test_message(buffer.GetString());
00682             return success;
00683         }
00684
00685 #if GTEST_OS_WINDOWS
00686         // WindowsDeathTest implements death tests on Windows. Due to the
00687         // specifics of starting new processes on Windows, death tests there are
00688         // always threadsafe, and Google Test considers the
00689         // --gtest_death_test_style=fast setting to be equivalent to
00690         // --gtest_death_test_style=threadsafe there.
00691         //
00692         // A few implementation notes:  Like the Linux version, the Windows
00693         // implementation uses pipes for child-to-parent communication. But due to
00694         // the specifics of pipes on Windows, some extra steps are required:
00695         //
00696         // 1. The parent creates a communication pipe and stores handles to both
00697         //    ends of it.
00698         // 2. The parent starts the child and provides it with the information
00699         //    necessary to acquire the handle to the write end of the pipe.
00700         // 3. The child acquires the write end of the pipe and signals the parent
00701         //    using a Windows event.
00702         // 4. Now the parent can release the write end of the pipe on its side. If
00703         //    this is done before step 3, the object's reference count goes down to
00704         //    0 and it is destroyed, preventing the child from acquiring it. The
00705         //    parent now has to release it, or read operations on the read end of
00706         //    the pipe will not return when the child terminates.
00707         // 5. The parent reads child's output through the pipe (outcome code and
00708         //    any possible error messages) from the pipe, and its stderr and then
00709         //    determines whether to fail the test.
00710         //
00711         // Note: to distinguish Win32 API calls from the local method and function
00712         // calls, the former are explicitly resolved in the global namespace.
00713         //
00714
00715         class WindowsDeathTest : public DeathTestImpl {
00716         public:
00717
00718             WindowsDeathTest(const char* a_statement, Matcher<const std::string&> matcher,
00719                     const char* file, int line)
00720             : DeathTestImpl(a_statement, std::move(matcher)),
00721             file_(file),
00722             line_(line) {
00723             }
00724
00725             // All of these virtual functions are inherited from DeathTest.
00726             virtual int Wait();
00727             virtual TestRole AssumeRole();
00728
00729         private:
00730             // The name of the file in which the death test is located.
00731             const char* const file_;
00732             // The line number on which the death test is located.
00733             const int line_;
00734             // Handle to the write end of the pipe to the child process.
00735             AutoHandle write_handle_;
00736             // Child process handle.
00737             AutoHandle child_handle_;
00738             // Event the child process uses to signal the parent that it has
00739             // acquired the handle to the write end of the pipe. After seeing this
00740             // event the parent can release its own handles to make sure its
00741             // ReadFile() calls return when the child terminates.
00742             AutoHandle event_handle_;
00743         };
00744
00745         // Waits for the child in a death test to exit, returning its exit
00746         // status, or 0 if no child process exists.  As a side effect, sets the
00747         // outcome data member.
00748
00749         int WindowsDeathTest::Wait() {
00750             if (!spawned()) return 0;
00751
00752             // Wait until the child either signals that it has acquired the write end
00753             // of the pipe or it dies.
00754             const HANDLE wait_handles[2] = {child_handle_.Get(), event_handle_.Get()};
00755             switch (::WaitForMultipleObjects(2, wait_handles,
00756                     FALSE, // Waits for any of the handles.
00757                     INFINITE)) {
00758                 case WAIT_OBJECT_0:
00759                 case WAIT_OBJECT_0 + 1:
```

```
00760                 break;
00761             default:
00762                 GTEST_DEATH_TEST_CHECK_(false); // Should not get here.
00763         }
00764
00765         // The child has acquired the write end of the pipe or exited.
00766         // We release the handle on our side and continue.
00767         write_handle_.Reset();
00768         event_handle_.Reset();
00769
00770         ReadAndInterpretStatusByte();
00771
00772         // Waits for the child process to exit if it haven't already. This
00773         // returns immediately if the child has already exited, regardless of
00774         // whether previous calls to WaitForMultipleObjects synchronized on this
00775         // handle or not.
00776         GTEST_DEATH_TEST_CHECK_(WAIT_OBJECT_0 ==
00777                 ::WaitForSingleObject(child_handle_.Get(), INFINITE));
00778         DWORD status_code;
00779         GTEST_DEATH_TEST_CHECK_(
00780                 ::GetExitCodeProcess(child_handle_.Get(), &status_code) != FALSE);
00781         child_handle_.Reset();
00782         set_status(static_cast<int> (status_code));
00783         return status();
00784       }
00785
00786       // The AssumeRole process for a Windows death test.  It creates a child
00787       // process with the same executable as the current process to run the
00788       // death test.  The child process is given the --gtest_filter and
00789       // --gtest_internal_run_death_test flags such that it knows to run the
00790       // current death test only.
00791
00792       DeathTest::TestRole WindowsDeathTest::AssumeRole() {
00793         const UnitTestImpl * const impl = GetUnitTestImpl();
00794         const InternalRunDeathTestFlag * const flag =
00795                 impl->internal_run_death_test_flag();
00796         const TestInfo * const info = impl->current_test_info();
00797         const int death_test_index = info->result()->death_test_count();
00798
00799         if (flag != nullptr) {
00800             // ParseInternalRunDeathTestFlag() has performed all the necessary
00801             // processing.
00802             set_write_fd(flag->write_fd());
00803             return EXECUTE_TEST;
00804         }
00805
00806         // WindowsDeathTest uses an anonymous pipe to communicate results of
00807         // a death test.
00808         SECURITY_ATTRIBUTES handles_are_inheritable = {sizeof (SECURITY_ATTRIBUTES),
00809             nullptr, TRUE};
00810         HANDLE read_handle, write_handle;
00811         GTEST_DEATH_TEST_CHECK_(::CreatePipe(&read_handle, &write_handle,
00812                 &handles_are_inheritable,
00813                 0) // Default buffer size.
00814                 != FALSE);
00815         set_read_fd(
00816                 ::_open_osfhandle(reinterpret_cast<intptr_t> (read_handle), O_RDONLY));
00817         write_handle_.Reset(write_handle);
00818         event_handle_.Reset(::CreateEvent(
00819                 &handles_are_inheritable,
00820                 TRUE, // The event will automatically reset to non-signaled state.
00821                 FALSE, // The initial state is non-signalled.
00822                 nullptr)); // The even is unnamed.
00823         GTEST_DEATH_TEST_CHECK_(event_handle_.Get() != nullptr);
00824         const std::string filter_flag = std::string("--") + GTEST_FLAG_PREFIX_ +
00825                 "filter=" + info->test_suite_name() + "." +
00826                 info->name();
00827         const std::string internal_flag =
00828                 std::string("--") + GTEST_FLAG_PREFIX_ +
00829                 "internal_run_death_test=" + file_ + "|" + StreamableToString(line_) +
00830                 "|" + StreamableToString(death_test_index) + "|" +
00831                 StreamableToString(static_cast<unsigned int> (::GetCurrentProcessId())) +
00832                 // size_t has the same width as pointers on both 32-bit and 64-bit
00833                 // Windows platforms.
00834                 // See http://msdn.microsoft.com/en-us/library/tcxf1dw6.aspx.
00835                 "|" + StreamableToString(reinterpret_cast<size_t> (write_handle)) + "|" +
00836                 StreamableToString(reinterpret_cast<size_t> (event_handle_.Get()));
00837
00838         char executable_path[_MAX_PATH + 1]; // NOLINT
00839         GTEST_DEATH_TEST_CHECK_(_MAX_PATH + 1 != ::GetModuleFileNameA(nullptr,
00840                 executable_path,
00841                 _MAX_PATH));
00842
00843         std::string command_line = std::string(::GetCommandLineA()) + " " +
00844                 filter_flag + " \"" + internal_flag + "\"";
00845
00846         DeathTest::set_last_death_test_message("");
```

```
00847
00848             CaptureStderr();
00849             // Flush the log buffers since the log streams are shared with the child.
00850             FlushInfoLog();
00851
00852             // The child process will share the standard handles with the parent.
00853             STARTUPINFOA startup_info;
00854             memset(&startup_info, 0, sizeof (STARTUPINFO));
00855             startup_info.dwFlags = STARTF_USESTDHANDLES;
00856             startup_info.hStdInput = ::GetStdHandle(STD_INPUT_HANDLE);
00857             startup_info.hStdOutput = ::GetStdHandle(STD_OUTPUT_HANDLE);
00858             startup_info.hStdError = ::GetStdHandle(STD_ERROR_HANDLE);
00859
00860             PROCESS_INFORMATION process_info;
00861             GTEST_DEATH_TEST_CHECK_(
00862                     ::CreateProcessA(
00863                     executable_path, const_cast<char*> (command_line.c_str()),
00864                     nullptr, // Returned process handle is not inheritable.
00865                     nullptr, // Returned thread handle is not inheritable.
00866                     TRUE, // Child inherits all inheritable handles (for write_handle_).
00867                     0x0, // Default creation flags.
00868                     nullptr, // Inherit the parent's environment.
00869                     UnitTest::GetInstance()->original_working_dir(), &startup_info,
00870                     &process_info) != FALSE);
00871             child_handle_.Reset(process_info.hProcess);
00872             ::CloseHandle(process_info.hThread);
00873             set_spawned(true);
00874             return OVERSEE_TEST;
00875         }
00876
00877 #elif GTEST_OS_FUCHSIA
00878
00879         class FuchsiaDeathTest : public DeathTestImpl {
00880         public:
00881
00882             FuchsiaDeathTest(const char* a_statement, Matcher<const std::string&> matcher,
00883                     const char* file, int line)
00884             : DeathTestImpl(a_statement, std::move(matcher)),
00885             file_(file),
00886             line_(line) {
00887             }
00888
00889             // All of these virtual functions are inherited from DeathTest.
00890             int Wait() override;
00891             TestRole AssumeRole() override;
00892             std::string GetErrorLogs() override;
00893
00894         private:
00895             // The name of the file in which the death test is located.
00896             const char* const file_;
00897             // The line number on which the death test is located.
00898             const int line_;
00899             // The stderr data captured by the child process.
00900             std::string captured_stderr_;
00901
00902             zx::process child_process_;
00903             zx::channel exception_channel_;
00904             zx::socket stderr_socket_;
00905         };
00906
00907         // Utility class for accumulating command-line arguments.
00908
00909         class Arguments {
00910         public:
00911
00912             Arguments() {
00913                 args_.push_back(nullptr);
00914             }
00915
00916             ~Arguments() {
00917                 for (std::vector<char*>::iterator i = args_.begin(); i != args_.end();
00918                         ++i) {
00919                     free(*i);
00920                 }
00921             }
00922
00923             void AddArgument(const char* argument) {
00924                 args_.insert(args_.end() - 1, posix::StrDup(argument));
00925             }
00926
00927             template <typename Str>
00928             void AddArguments(const ::std::vector<Str>& arguments) {
00929                 for (typename ::std::vector<Str>::const_iterator i = arguments.begin();
00930                         i != arguments.end(); ++i) {
00931                     args_.insert(args_.end() - 1, posix::StrDup(i->c_str()));
00932                 }
00933             }
```

```
00934
00935             char* const* Argv() {
00936                 return &args_[0];
00937             }
00938
00939             int size() {
00940                 return static_cast<int> (args_.size()) - 1;
00941             }
00942
00943         private:
00944             std::vector<char*> args_;
00945         };
00946
00947         // Waits for the child in a death test to exit, returning its exit
00948         // status, or 0 if no child process exists.  As a side effect, sets the
00949         // outcome data member.
00950
00951         int FuchsiaDeathTest::Wait() {
00952             const int kProcessKey = 0;
00953             const int kSocketKey = 1;
00954             const int kExceptionKey = 2;
00955
00956             if (!spawned()) return 0;
00957
00958             // Create a port to wait for socket/task/exception events.
00959             zx_status_t status_zx;
00960             zx::port port;
00961             status_zx = zx::port::create(0, &port);
00962             GTEST_DEATH_TEST_CHECK_(status_zx == ZX_OK);
00963
00964             // Register to wait for the child process to terminate.
00965             status_zx =
00966                     child_process_.wait_async(port, kProcessKey, ZX_PROCESS_TERMINATED, 0);
00967             GTEST_DEATH_TEST_CHECK_(status_zx == ZX_OK);
00968
00969             // Register to wait for the socket to be readable or closed.
00970             status_zx = stderr_socket_.wait_async(
00971                     port, kSocketKey, ZX_SOCKET_READABLE | ZX_SOCKET_PEER_CLOSED, 0);
00972             GTEST_DEATH_TEST_CHECK_(status_zx == ZX_OK);
00973
00974             // Register to wait for an exception.
00975             status_zx = exception_channel_.wait_async(port, kExceptionKey,
00976                     ZX_CHANNEL_READABLE, 0);
00977             GTEST_DEATH_TEST_CHECK_(status_zx == ZX_OK);
00978
00979             bool process_terminated = false;
00980             bool socket_closed = false;
00981             do {
00982                 zx_port_packet_t packet = {};
00983                 status_zx = port.wait(zx::time::infinite(), &packet);
00984                 GTEST_DEATH_TEST_CHECK_(status_zx == ZX_OK);
00985
00986                 if (packet.key == kExceptionKey) {
00987                     // Process encountered an exception. Kill it directly rather than
00988                     // letting other handlers process the event. We will get a kProcessKey
00989                     // event when the process actually terminates.
00990                     status_zx = child_process_.kill();
00991                     GTEST_DEATH_TEST_CHECK_(status_zx == ZX_OK);
00992                 } else if (packet.key == kProcessKey) {
00993                     // Process terminated.
00994                     GTEST_DEATH_TEST_CHECK_(ZX_PKT_IS_SIGNAL_ONE(packet.type));
00995                     GTEST_DEATH_TEST_CHECK_(packet.signal.observed & ZX_PROCESS_TERMINATED);
00996                     process_terminated = true;
00997                 } else if (packet.key == kSocketKey) {
00998                     GTEST_DEATH_TEST_CHECK_(ZX_PKT_IS_SIGNAL_ONE(packet.type));
00999                     if (packet.signal.observed & ZX_SOCKET_READABLE) {
01000                         // Read data from the socket.
01001                         constexpr size_t kBufferSize = 1024;
01002                         do {
01003                             size_t old_length = captured_stderr_.length();
01004                             size_t bytes_read = 0;
01005                             captured_stderr_.resize(old_length + kBufferSize);
01006                             status_zx =
01007                                     stderr_socket_.read(0, &captured_stderr_.front() + old_length,
01008                                     kBufferSize, &bytes_read);
01009                             captured_stderr_.resize(old_length + bytes_read);
01010                         } while (status_zx == ZX_OK);
01011                         if (status_zx == ZX_ERR_PEER_CLOSED) {
01012                             socket_closed = true;
01013                         } else {
01014                             GTEST_DEATH_TEST_CHECK_(status_zx == ZX_ERR_SHOULD_WAIT);
01015                             status_zx = stderr_socket_.wait_async(
01016                                     port, kSocketKey, ZX_SOCKET_READABLE | ZX_SOCKET_PEER_CLOSED, 0);
01017                             GTEST_DEATH_TEST_CHECK_(status_zx == ZX_OK);
01018                         }
01019                     } else {
01020                         GTEST_DEATH_TEST_CHECK_(packet.signal.observed & ZX_SOCKET_PEER_CLOSED);
```

```
01021                       socket_closed = true;
01022                   }
01023               }
01024           } while (!process_terminated && !socket_closed);
01025
01026           ReadAndInterpretStatusByte();
01027
01028           zx_info_process_t buffer;
01029           status_zx = child_process_.get_info(ZX_INFO_PROCESS, &buffer, sizeof (buffer),
01030                   nullptr, nullptr);
01031           GTEST_DEATH_TEST_CHECK_(status_zx == ZX_OK);
01032
01033           GTEST_DEATH_TEST_CHECK_(buffer.flags & ZX_INFO_PROCESS_FLAG_EXITED);
01034           set_status(static_cast<int> (buffer.return_code));
01035           return status();
01036       }
01037
01038       // The AssumeRole process for a Fuchsia death test.  It creates a child
01039       // process with the same executable as the current process to run the
01040       // death test.  The child process is given the --gtest_filter and
01041       // --gtest_internal_run_death_test flags such that it knows to run the
01042       // current death test only.
01043
01044       DeathTest::TestRole FuchsiaDeathTest::AssumeRole() {
01045           const UnitTestImpl * const impl = GetUnitTestImpl();
01046           const InternalRunDeathTestFlag * const flag =
01047                   impl->internal_run_death_test_flag();
01048           const TestInfo * const info = impl->current_test_info();
01049           const int death_test_index = info->result()->death_test_count();
01050
01051           if (flag != nullptr) {
01052               // ParseInternalRunDeathTestFlag() has performed all the necessary
01053               // processing.
01054               set_write_fd(kFuchsiaReadPipeFd);
01055               return EXECUTE_TEST;
01056           }
01057
01058           // Flush the log buffers since the log streams are shared with the child.
01059           FlushInfoLog();
01060
01061           // Build the child process command line.
01062           const std::string filter_flag = std::string("--") + GTEST_FLAG_PREFIX_ +
01063                   "filter=" + info->test_suite_name() + "." +
01064                   info->name();
01065           const std::string internal_flag = std::string("--") + GTEST_FLAG_PREFIX_ +
01066                   kInternalRunDeathTestFlag + "=" + file_ +
01067                   "|" + StreamableToString(line_) + "|" +
01068                   StreamableToString(death_test_index);
01069           Arguments args;
01070           args.AddArguments(GetInjectableArgvs());
01071           args.AddArgument(filter_flag.c_str());
01072           args.AddArgument(internal_flag.c_str());
01073
01074           // Build the pipe for communication with the child.
01075           zx_status_t status;
01076           zx_handle_t child_pipe_handle;
01077           int child_pipe_fd;
01078           status = fdio_pipe_half(&child_pipe_fd, &child_pipe_handle);
01079           GTEST_DEATH_TEST_CHECK_(status == ZX_OK);
01080           set_read_fd(child_pipe_fd);
01081
01082           // Set the pipe handle for the child.
01083           fdio_spawn_action_t spawn_actions[2] = {};
01084           fdio_spawn_action_t* add_handle_action = &spawn_actions[0];
01085           add_handle_action->action = FDIO_SPAWN_ACTION_ADD_HANDLE;
01086           add_handle_action->h.id = PA_HND(PA_FD, kFuchsiaReadPipeFd);
01087           add_handle_action->h.handle = child_pipe_handle;
01088
01089           // Create a socket pair will be used to receive the child process' stderr.
01090           zx::socket stderr_producer_socket;
01091           status = zx::socket::create(0, &stderr_producer_socket, &stderr_socket_);
01092           GTEST_DEATH_TEST_CHECK_(status >= 0);
01093           int stderr_producer_fd = -1;
01094           status =
01095                   fdio_fd_create(stderr_producer_socket.release(), &stderr_producer_fd);
01096           GTEST_DEATH_TEST_CHECK_(status >= 0);
01097
01098           // Make the stderr socket nonblocking.
01099           GTEST_DEATH_TEST_CHECK_(fcntl(stderr_producer_fd, F_SETFL, 0) == 0);
01100
01101           fdio_spawn_action_t* add_stderr_action = &spawn_actions[1];
01102           add_stderr_action->action = FDIO_SPAWN_ACTION_CLONE_FD;
01103           add_stderr_action->fd.local_fd = stderr_producer_fd;
01104           add_stderr_action->fd.target_fd = STDERR_FILENO;
01105
01106           // Create a child job.
01107           zx_handle_t child_job = ZX_HANDLE_INVALID;
```

```
01108             status = zx_job_create(zx_job_default(), 0, &child_job);
01109             GTEST_DEATH_TEST_CHECK_(status == ZX_OK);
01110             zx_policy_basic_t policy;
01111             policy.condition = ZX_POL_NEW_ANY;
01112             policy.policy = ZX_POL_ACTION_ALLOW;
01113             status = zx_job_set_policy(child_job, ZX_JOB_POL_RELATIVE, ZX_JOB_POL_BASIC,
01114                     &policy, 1);
01115             GTEST_DEATH_TEST_CHECK_(status == ZX_OK);
01116
01117             // Create an exception channel attached to the |child_job|, to allow
01118             // us to suppress the system default exception handler from firing.
01119             status = zx_task_create_exception_channel(
01120                     child_job, 0, exception_channel_.reset_and_get_address());
01121             GTEST_DEATH_TEST_CHECK_(status == ZX_OK);
01122
01123             // Spawn the child process.
01124             status = fdio_spawn_etc(child_job, FDIO_SPAWN_CLONE_ALL, args.Argv()[0],
01125                     args.Argv(), nullptr, 2, spawn_actions,
01126                     child_process_.reset_and_get_address(), nullptr);
01127             GTEST_DEATH_TEST_CHECK_(status == ZX_OK);
01128
01129             set_spawned(true);
01130             return OVERSEE_TEST;
01131         }
01132
01133         std::string FuchsiaDeathTest::GetErrorLogs() {
01134             return captured_stderr_;
01135         }
01136
01137 #else  // We are neither on Windows, nor on Fuchsia.
01138
01139         // ForkingDeathTest provides implementations for most of the abstract
01140         // methods of the DeathTest interface.  Only the AssumeRole method is
01141         // left undefined.
01142
01143         class ForkingDeathTest : public DeathTestImpl {
01144         public:
01145             ForkingDeathTest(const char* statement, Matcher<const std::string&> matcher);
01146
01147             // All of these virtual functions are inherited from DeathTest.
01148             int Wait() override;
01149
01150         protected:
01151
01152             void set_child_pid(pid_t child_pid) {
01153                 child_pid_ = child_pid;
01154             }
01155
01156         private:
01157             // PID of child process during death test; 0 in the child process itself.
01158             pid_t child_pid_;
01159         };
01160
01161         // Constructs a ForkingDeathTest.
01162
01163         ForkingDeathTest::ForkingDeathTest(const char* a_statement,
01164                 Matcher<const std::string&> matcher)
01165         : DeathTestImpl(a_statement, std::move(matcher)), child_pid_(-1) {
01166         }
01167
01168         // Waits for the child in a death test to exit, returning its exit
01169         // status, or 0 if no child process exists.  As a side effect, sets the
01170         // outcome data member.
01171
01172         int ForkingDeathTest::Wait() {
01173             if (!spawned()) return 0;
01174
01175             ReadAndInterpretStatusByte();
01176
01177             int status_value;
01178             GTEST_DEATH_TEST_CHECK_SYSCALL_(waitpid(child_pid_, &status_value, 0));
01179             set_status(status_value);
01180             return status_value;
01181         }
01182
01183         // A concrete death test class that forks, then immediately runs the test
01184         // in the child process.
01185
01186         class NoExecDeathTest : public ForkingDeathTest {
01187         public:
01188
01189             NoExecDeathTest(const char* a_statement, Matcher<const std::string&> matcher)
01190             : ForkingDeathTest(a_statement, std::move(matcher)) {
01191             }
01192             TestRole AssumeRole() override;
01193         };
01194
```

```
01195         // The AssumeRole process for a fork-and-run death test.  It implements a
01196         // straightforward fork, with a simple pipe to transmit the status byte.
01197
01198         DeathTest::TestRole NoExecDeathTest::AssumeRole() {
01199             const size_t thread_count = GetThreadCount();
01200             if (thread_count != 1) {
01201                 GTEST_LOG_(WARNING) << DeathTestThreadWarning(thread_count);
01202             }
01203
01204             int pipe_fd[2];
01205             GTEST_DEATH_TEST_CHECK_(pipe(pipe_fd) != -1);
01206
01207             DeathTest::set_last_death_test_message("");
01208             CaptureStderr();
01209             // When we fork the process below, the log file buffers are copied, but the
01210             // file descriptors are shared.  We flush all log files here so that closing
01211             // the file descriptors in the child process doesn't throw off the
01212             // synchronization between descriptors and buffers in the parent process.
01213             // This is as close to the fork as possible to avoid a race condition in case
01214             // there are multiple threads running before the death test, and another
01215             // thread writes to the log file.
01216             FlushInfoLog();
01217
01218             const pid_t child_pid = fork();
01219             GTEST_DEATH_TEST_CHECK_(child_pid != -1);
01220             set_child_pid(child_pid);
01221             if (child_pid == 0) {
01222                 GTEST_DEATH_TEST_CHECK_SYSCALL_(close(pipe_fd[0]));
01223                 set_write_fd(pipe_fd[1]);
01224                 // Redirects all logging to stderr in the child process to prevent
01225                 // concurrent writes to the log files.  We capture stderr in the parent
01226                 // process and append the child process' output to a log.
01227                 LogToStderr();
01228                 // Event forwarding to the listeners of event listener API mush be shut
01229                 // down in death test subprocesses.
01230                 GetUnitTestImpl()->listeners()->SuppressEventForwarding();
01231                 g_in_fast_death_test_child = true;
01232                 return EXECUTE_TEST;
01233             } else {
01234                 GTEST_DEATH_TEST_CHECK_SYSCALL_(close(pipe_fd[1]));
01235                 set_read_fd(pipe_fd[0]);
01236                 set_spawned(true);
01237                 return OVERSEE_TEST;
01238             }
01239         }
01240
01241         // A concrete death test class that forks and re-executes the main
01242         // program from the beginning, with command-line flags set that cause
01243         // only this specific death test to be run.
01244
01245         class ExecDeathTest : public ForkingDeathTest {
01246         public:
01247
01248             ExecDeathTest(const char* a_statement, Matcher<const std::string&> matcher,
01249                     const char* file, int line)
01250             : ForkingDeathTest(a_statement, std::move(matcher)),
01251             file_(file),
01252             line_(line) {
01253             }
01254             TestRole AssumeRole() override;
01255
01256         private:
01257
01258             static ::std::vector<std::string> GetArgvsForDeathTestChildProcess() {
01259                 ::std::vector<std::string> args = GetInjectableArgvs();
01260 #if defined(GTEST_EXTRA_DEATH_TEST_COMMAND_LINE_ARGS_)
01261                 ::std::vector<std::string> extra_args =
01262                         GTEST_EXTRA_DEATH_TEST_COMMAND_LINE_ARGS_();
01263                 args.insert(args.end(), extra_args.begin(), extra_args.end());
01264 #endif  // defined(GTEST_EXTRA_DEATH_TEST_COMMAND_LINE_ARGS_)
01265                 return args;
01266             }
01267             // The name of the file in which the death test is located.
01268             const char* const file_;
01269             // The line number on which the death test is located.
01270             const int line_;
01271         };
01272
01273         // Utility class for accumulating command-line arguments.
01274
01275         class Arguments {
01276         public:
01277
01278             Arguments() {
01279                 args_.push_back(nullptr);
01280             }
01281
```

```
01282             ~Arguments() {
01283                 for (std::vector<char*>::iterator i = args_.begin(); i != args_.end();
01284                         ++i) {
01285                     free(*i);
01286                 }
01287             }
01288
01289             void AddArgument(const char* argument) {
01290                 args_.insert(args_.end() - 1, posix::StrDup(argument));
01291             }
01292
01293             template <typename Str>
01294             void AddArguments(const ::std::vector<Str>& arguments) {
01295                 for (typename ::std::vector<Str>::const_iterator i = arguments.begin();
01296                         i != arguments.end(); ++i) {
01297                     args_.insert(args_.end() - 1, posix::StrDup(i->c_str()));
01298                 }
01299             }
01300
01301             char* const* Argv() {
01302                 return &args_[0];
01303             }
01304
01305         private:
01306             std::vector<char*> args_;
01307         };
01308
01309         // A struct that encompasses the arguments to the child process of a
01310         // threadsafe-style death test process.
01311
01312         struct ExecDeathTestArgs {
01313             char* const* argv; // Command-line arguments for the child's call to exec
01314             int close_fd; // File descriptor to close; the read end of a pipe
01315         };
01316
01317 #if GTEST_OS_QNX
01318         extern "C" char** environ;
01319 #else   // GTEST_OS_QNX
01320         // The main function for a threadsafe-style death test child process.
01321         // This function is called in a clone()-ed process and thus must avoid
01322         // any potentially unsafe operations like malloc or libc functions.
01323
01324         static int ExecDeathTestChildMain(void* child_arg) {
01325             ExecDeathTestArgs * const args = static_cast<ExecDeathTestArgs*> (child_arg);
01326             GTEST_DEATH_TEST_CHECK_SYSCALL_(close(args->close_fd));
01327
01328             // We need to execute the test program in the same environment where
01329             // it was originally invoked.  Therefore we change to the original
01330             // working directory first.
01331             const char* const original_dir =
01332                     UnitTest::GetInstance()->original_working_dir();
01333             // We can safely call chdir() as it's a direct system call.
01334             if (chdir(original_dir) != 0) {
01335                 DeathTestAbort(std::string("chdir(\"") + original_dir +
01336                         "\") failed: " + GetLastErrnoDescription());
01337                 return EXIT_FAILURE;
01338             }
01339
01340             // We can safely call execv() as it's almost a direct system call. We
01341             // cannot use execvp() as it's a libc function and thus potentially
01342             // unsafe.  Since execv() doesn't search the PATH, the user must
01343             // invoke the test program via a valid path that contains at least
01344             // one path separator.
01345             execv(args->argv[0], args->argv);
01346             DeathTestAbort(std::string("execv(") + args->argv[0] + ", ...) in " +
01347                     original_dir + " failed: " + GetLastErrnoDescription());
01348             return EXIT_FAILURE;
01349         }
01350 #endif  // GTEST_OS_QNX
01351
01352 #if GTEST_HAS_CLONE
01353         // Two utility routines that together determine the direction the stack
01354         // grows.
01355         // This could be accomplished more elegantly by a single recursive
01356         // function, but we want to guard against the unlikely possibility of
01357         // a smart compiler optimizing the recursion away.
01358         //
01359         // GTEST_NO_INLINE_ is required to prevent GCC 4.6 from inlining
01360         // StackLowerThanAddress into StackGrowsDown, which then doesn't give
01361         // correct answer.
01362         static void StackLowerThanAddress(const void* ptr,
01363                 bool* result) GTEST_NO_INLINE_;
01364         // Make sure sanitizers do not tamper with the stack here.
01365         // Ideally, we want to use `__builtin_frame_address` instead of a local variable
01366         // address with sanitizer disabled, but it does not work when the
01367         // compiler optimizes the stack frame out, which happens on PowerPC targets.
01368         // HWAddressSanitizer add a random tag to the MSB of the local variable address,
```

```
01369         // making comparison result unpredictable.
01370
01371         GTEST_ATTRIBUTE_NO_SANITIZE_ADDRESS_
01372         GTEST_ATTRIBUTE_NO_SANITIZE_HWADDRESS_
01373         static void StackLowerThanAddress(const void* ptr, bool* result) {
01374             int dummy = 0;
01375             *result = std::less<const void*>()(&dummy, ptr);
01376         }
01377
01378         // Make sure AddressSanitizer does not tamper with the stack here.
01379
01380         GTEST_ATTRIBUTE_NO_SANITIZE_ADDRESS_
01381         GTEST_ATTRIBUTE_NO_SANITIZE_HWADDRESS_
01382         static bool StackGrowsDown() {
01383             int dummy = 0;
01384             bool result;
01385             StackLowerThanAddress(&dummy, &result);
01386             return result;
01387         }
01388 #endif  // GTEST_HAS_CLONE
01389
01390         // Spawns a child process with the same executable as the current process in
01391         // a thread-safe manner and instructs it to run the death test.  The
01392         // implementation uses fork(2) + exec.  On systems where clone(2) is
01393         // available, it is used instead, being slightly more thread-safe.  On QNX,
01394         // fork supports only single-threaded environments, so this function uses
01395         // spawn(2) there instead.  The function dies with an error message if
01396         // anything goes wrong.
01397
01398         static pid_t ExecDeathTestSpawnChild(char* const* argv, int close_fd) {
01399             ExecDeathTestArgs args = {argv, close_fd};
01400             pid_t child_pid = -1;
01401
01402 #if GTEST_OS_QNX
01403             // Obtains the current directory and sets it to be closed in the child
01404             // process.
01405             const int cwd_fd = open(".", O_RDONLY);
01406             GTEST_DEATH_TEST_CHECK_(cwd_fd != -1);
01407             GTEST_DEATH_TEST_CHECK_SYSCALL_(fcntl(cwd_fd, F_SETFD, FD_CLOEXEC));
01408             // We need to execute the test program in the same environment where
01409             // it was originally invoked.  Therefore we change to the original
01410             // working directory first.
01411             const char* const original_dir =
01412                     UnitTest::GetInstance()->original_working_dir();
01413             // We can safely call chdir() as it's a direct system call.
01414             if (chdir(original_dir) != 0) {
01415                 DeathTestAbort(std::string("chdir(\"") + original_dir +
01416                         "\") failed: " + GetLastErrnoDescription());
01417                 return EXIT_FAILURE;
01418             }
01419
01420             int fd_flags;
01421             // Set close_fd to be closed after spawn.
01422             GTEST_DEATH_TEST_CHECK_SYSCALL_(fd_flags = fcntl(close_fd, F_GETFD));
01423             GTEST_DEATH_TEST_CHECK_SYSCALL_(
01424                     fcntl(close_fd, F_SETFD, fd_flags | FD_CLOEXEC));
01425             struct inheritance inherit = {0};
01426             // spawn is a system call.
01427             child_pid = spawn(args.argv[0], 0, nullptr, &inherit, args.argv, environ);
01428             // Restores the current working directory.
01429             GTEST_DEATH_TEST_CHECK_(fchdir(cwd_fd) != -1);
01430             GTEST_DEATH_TEST_CHECK_SYSCALL_(close(cwd_fd));
01431
01432 #else  // GTEST_OS_QNX
01433 #if GTEST_OS_LINUX
01434             // When a SIGPROF signal is received while fork() or clone() are executing,
01435             // the process may hang. To avoid this, we ignore SIGPROF here and re-enable
01436             // it after the call to fork()/clone() is complete.
01437             struct sigaction saved_sigprof_action;
01438             struct sigaction ignore_sigprof_action;
01439             memset(&ignore_sigprof_action, 0, sizeof (ignore_sigprof_action));
01440             sigemptyset(&ignore_sigprof_action.sa_mask);
01441             ignore_sigprof_action.sa_handler = SIG_IGN;
01442             GTEST_DEATH_TEST_CHECK_SYSCALL_(
01443                     sigaction(SIGPROF, &ignore_sigprof_action, &saved_sigprof_action));
01444 #endif  // GTEST_OS_LINUX
01445
01446 #if GTEST_HAS_CLONE
01447             const bool use_fork = GTEST_FLAG_GET(death_test_use_fork);
01448
01449             if (!use_fork) {
01450                 static const bool stack_grows_down = StackGrowsDown();
01451                 const auto stack_size = static_cast<size_t> (getpagesize() * 2);
01452                 // MMAP_ANONYMOUS is not defined on Mac, so we use MAP_ANON instead.
01453                 void* const stack = mmap(nullptr, stack_size, PROT_READ | PROT_WRITE,
01454                         MAP_ANON | MAP_PRIVATE, -1, 0);
01455                 GTEST_DEATH_TEST_CHECK_(stack != MAP_FAILED);
```

```
01456
01457                 // Maximum stack alignment in bytes:  For a downward-growing stack, this
01458                 // amount is subtracted from size of the stack space to get an address
01459                 // that is within the stack space and is aligned on all systems we care
01460                 // about.  As far as I know there is no ABI with stack alignment greater
01461                 // than 64.  We assume stack and stack_size already have alignment of
01462                 // kMaxStackAlignment.
01463                 const size_t kMaxStackAlignment = 64;
01464                 void* const stack_top =
01465                         static_cast<char*> (stack) +
01466                         (stack_grows_down ? stack_size - kMaxStackAlignment : 0);
01467                 GTEST_DEATH_TEST_CHECK_(
01468                         static_cast<size_t> (stack_size) > kMaxStackAlignment &&
01469                         reinterpret_cast<uintptr_t> (stack_top) % kMaxStackAlignment == 0);
01470
01471                 child_pid = clone(&ExecDeathTestChildMain, stack_top, SIGCHLD, &args);
01472
01473                 GTEST_DEATH_TEST_CHECK_(munmap(stack, stack_size) != -1);
01474             }
01475 #else
01476             const bool use_fork = true;
01477 #endif  // GTEST_HAS_CLONE
01478
01479             if (use_fork && (child_pid = fork()) == 0) {
01480                 ExecDeathTestChildMain(&args);
01481                 _exit(0);
01482             }
01483 #endif  // GTEST_OS_QNX
01484 #if GTEST_OS_LINUX
01485             GTEST_DEATH_TEST_CHECK_SYSCALL_(
01486                     sigaction(SIGPROF, &saved_sigprof_action, nullptr));
01487 #endif  // GTEST_OS_LINUX
01488
01489             GTEST_DEATH_TEST_CHECK_(child_pid != -1);
01490             return child_pid;
01491         }
01492
01493         // The AssumeRole process for a fork-and-exec death test.  It re-executes the
01494         // main program from the beginning, setting the --gtest_filter
01495         // and --gtest_internal_run_death_test flags to cause only the current
01496         // death test to be re-run.
01497
01498         DeathTest::TestRole ExecDeathTest::AssumeRole() {
01499             const UnitTestImpl * const impl = GetUnitTestImpl();
01500             const InternalRunDeathTestFlag * const flag =
01501                     impl->internal_run_death_test_flag();
01502             const TestInfo * const info = impl->current_test_info();
01503             const int death_test_index = info->result()->death_test_count();
01504
01505             if (flag != nullptr) {
01506                 set_write_fd(flag->write_fd());
01507                 return EXECUTE_TEST;
01508             }
01509
01510             int pipe_fd[2];
01511             GTEST_DEATH_TEST_CHECK_(pipe(pipe_fd) != -1);
01512             // Clear the close-on-exec flag on the write end of the pipe, lest
01513             // it be closed when the child process does an exec:
01514             GTEST_DEATH_TEST_CHECK_(fcntl(pipe_fd[1], F_SETFD, 0) != -1);
01515
01516             const std::string filter_flag = std::string("--") + GTEST_FLAG_PREFIX_ +
01517                     "filter=" + info->test_suite_name() + "." +
01518                     info->name();
01519             const std::string internal_flag = std::string("--") + GTEST_FLAG_PREFIX_ +
01520                     "internal_run_death_test=" + file_ + "|" +
01521                     StreamableToString(line_) + "|" +
01522                     StreamableToString(death_test_index) + "|" +
01523                     StreamableToString(pipe_fd[1]);
01524             Arguments args;
01525             args.AddArguments(GetArgvsForDeathTestChildProcess());
01526             args.AddArgument(filter_flag.c_str());
01527             args.AddArgument(internal_flag.c_str());
01528
01529             DeathTest::set_last_death_test_message("");
01530
01531             CaptureStderr();
01532             // See the comment in NoExecDeathTest::AssumeRole for why the next line
01533             // is necessary.
01534             FlushInfoLog();
01535
01536             const pid_t child_pid = ExecDeathTestSpawnChild(args.Argv(), pipe_fd[0]);
01537             GTEST_DEATH_TEST_CHECK_SYSCALL_(close(pipe_fd[1]));
01538             set_child_pid(child_pid);
01539             set_read_fd(pipe_fd[0]);
01540             set_spawned(true);
01541             return OVERSEE_TEST;
01542         }
```

```
01543
01544 #endif  // !GTEST_OS_WINDOWS
01545
01546         // Creates a concrete DeathTest-derived class that depends on the
01547         // --gtest_death_test_style flag, and sets the pointer pointed to
01548         // by the "test" argument to its address.  If the test should be
01549         // skipped, sets that pointer to NULL.  Returns true, unless the
01550         // flag is set to an invalid value.
01551
01552         bool DefaultDeathTestFactory::Create(const char* statement,
01553                 Matcher<const std::string&> matcher,
01554                 const char* file, int line,
01555                 DeathTest** test) {
01556             UnitTestImpl * const impl = GetUnitTestImpl();
01557             const InternalRunDeathTestFlag * const flag =
01558                     impl->internal_run_death_test_flag();
01559             const int death_test_index =
01560                     impl->current_test_info()->increment_death_test_count();
01561
01562             if (flag != nullptr) {
01563                 if (death_test_index > flag->index()) {
01564                     DeathTest::set_last_death_test_message(
01565                             "Death test count (" + StreamableToString(death_test_index) +
01566                             ") somehow exceeded expected maximum (" +
01567                             StreamableToString(flag->index()) + ")");
01568                     return false;
01569                 }
01570
01571                 if (!(flag->file() == file && flag->line() == line &&
01572                         flag->index() == death_test_index)) {
01573                     *test = nullptr;
01574                     return true;
01575                 }
01576             }
01577
01578 #if GTEST_OS_WINDOWS
01579
01580             if (GTEST_FLAG_GET(death_test_style) == "threadsafe" ||
01581                     GTEST_FLAG_GET(death_test_style) == "fast") {
01582                 *test = new WindowsDeathTest(statement, std::move(matcher), file, line);
01583             }
01584
01585 #elif GTEST_OS_FUCHSIA
01586
01587             if (GTEST_FLAG_GET(death_test_style) == "threadsafe" ||
01588                     GTEST_FLAG_GET(death_test_style) == "fast") {
01589                 *test = new FuchsiaDeathTest(statement, std::move(matcher), file, line);
01590             }
01591
01592 #else
01593
01594             if (GTEST_FLAG_GET(death_test_style) == "threadsafe") {
01595                 *test = new ExecDeathTest(statement, std::move(matcher), file, line);
01596             } else if (GTEST_FLAG_GET(death_test_style) == "fast") {
01597                 *test = new NoExecDeathTest(statement, std::move(matcher));
01598             }
01599 #endif  // GTEST_OS_WINDOWS
01600
01601             else { // NOLINT - this is more readable than unbalanced brackets inside #if.
01602                 DeathTest::set_last_death_test_message("Unknown death test style \"" +
01603                         GTEST_FLAG_GET(death_test_style) +
01604                         "\" encountered");
01605                 return false;
01606             }
01607
01608             return true;
01609         }
01610
01611 #if GTEST_OS_WINDOWS
01612         // Recreates the pipe and event handles from the provided parameters,
01613         // signals the event, and returns a file descriptor wrapped around the pipe
01614         // handle. This function is called in the child process only.
01615
01616         static int GetStatusFileDescriptor(unsigned int parent_process_id,
01617                 size_t write_handle_as_size_t,
01618                 size_t event_handle_as_size_t) {
01619             AutoHandle parent_process_handle(::OpenProcess(PROCESS_DUP_HANDLE,
01620                     FALSE, // Non-inheritable.
01621                     parent_process_id));
01622             if (parent_process_handle.Get() == INVALID_HANDLE_VALUE) {
01623                 DeathTestAbort("Unable to open parent process " +
01624                         StreamableToString(parent_process_id));
01625             }
01626
01627             GTEST_CHECK_(sizeof (HANDLE) <= sizeof (size_t));
01628
01629             const HANDLE write_handle = reinterpret_cast<HANDLE> (write_handle_as_size_t);
```

```
01630             HANDLE dup_write_handle;
01631
01632             // The newly initialized handle is accessible only in the parent
01633             // process. To obtain one accessible within the child, we need to use
01634             // DuplicateHandle.
01635             if (!::DuplicateHandle(parent_process_handle.Get(), write_handle,
01636                     ::GetCurrentProcess(), &dup_write_handle,
01637                     0x0, // Requested privileges ignored since
01638                     // DUPLICATE_SAME_ACCESS is used.
01639                     FALSE, // Request non-inheritable handler.
01640                     DUPLICATE_SAME_ACCESS)) {
01641                 DeathTestAbort("Unable to duplicate the pipe handle " +
01642                         StreamableToString(write_handle_as_size_t) +
01643                         " from the parent process " +
01644                         StreamableToString(parent_process_id));
01645             }
01646
01647             const HANDLE event_handle = reinterpret_cast<HANDLE> (event_handle_as_size_t);
01648             HANDLE dup_event_handle;
01649
01650             if (!::DuplicateHandle(parent_process_handle.Get(), event_handle,
01651                     ::GetCurrentProcess(), &dup_event_handle, 0x0, FALSE,
01652                     DUPLICATE_SAME_ACCESS)) {
01653                 DeathTestAbort("Unable to duplicate the event handle " +
01654                         StreamableToString(event_handle_as_size_t) +
01655                         " from the parent process " +
01656                         StreamableToString(parent_process_id));
01657             }
01658
01659             const int write_fd =
01660                     ::_open_osfhandle(reinterpret_cast<intptr_t> (dup_write_handle), O_APPEND);
01661             if (write_fd == -1) {
01662                 DeathTestAbort("Unable to convert pipe handle " +
01663                         StreamableToString(write_handle_as_size_t) +
01664                         " to a file descriptor");
01665             }
01666
01667             // Signals the parent that the write end of the pipe has been acquired
01668             // so the parent can release its own write end.
01669             ::SetEvent(dup_event_handle);
01670
01671             return write_fd;
01672         }
01673 #endif  // GTEST_OS_WINDOWS
01674
01675         // Returns a newly created InternalRunDeathTestFlag object with fields
01676         // initialized from the GTEST_FLAG(internal_run_death_test) flag if
01677         // the flag is specified; otherwise returns NULL.
01678
01679         InternalRunDeathTestFlag* ParseInternalRunDeathTestFlag() {
01680             if (GTEST_FLAG_GET(internal_run_death_test) == "") return nullptr;
01681
01682             // GTEST_HAS_DEATH_TEST implies that we have ::std::string, so we
01683             // can use it here.
01684             int line = -1;
01685             int index = -1;
01686             ::std::vector< ::std::string> fields;
01687             SplitString(GTEST_FLAG_GET(internal_run_death_test), '|', &fields);
01688             int write_fd = -1;
01689
01690 #if GTEST_OS_WINDOWS
01691
01692             unsigned int parent_process_id = 0;
01693             size_t write_handle_as_size_t = 0;
01694             size_t event_handle_as_size_t = 0;
01695
01696             if (fields.size() != 6 || !ParseNaturalNumber(fields[1], &line) ||
01697                     !ParseNaturalNumber(fields[2], &index) ||
01698                     !ParseNaturalNumber(fields[3], &parent_process_id) ||
01699                     !ParseNaturalNumber(fields[4], &write_handle_as_size_t) ||
01700                     !ParseNaturalNumber(fields[5], &event_handle_as_size_t)) {
01701                 DeathTestAbort("Bad --gtest_internal_run_death_test flag: " +
01702                         GTEST_FLAG_GET(internal_run_death_test));
01703             }
01704             write_fd = GetStatusFileDescriptor(parent_process_id, write_handle_as_size_t,
01705                     event_handle_as_size_t);
01706
01707 #elif GTEST_OS_FUCHSIA
01708
01709             if (fields.size() != 3 || !ParseNaturalNumber(fields[1], &line) ||
01710                     !ParseNaturalNumber(fields[2], &index)) {
01711                 DeathTestAbort("Bad --gtest_internal_run_death_test flag: " +
01712                         GTEST_FLAG_GET(internal_run_death_test));
01713             }
01714
01715 #else
01716
```

```
01717             if (fields.size() != 4 || !ParseNaturalNumber(fields[1], &line) ||
01718                     !ParseNaturalNumber(fields[2], &index) ||
01719                     !ParseNaturalNumber(fields[3], &write_fd)) {
01720                 DeathTestAbort("Bad --gtest_internal_run_death_test flag: " +
01721                         GTEST_FLAG_GET(internal_run_death_test));
01722             }
01723
01724 #endif  // GTEST_OS_WINDOWS
01725
01726             return new InternalRunDeathTestFlag(fields[0], line, index, write_fd);
01727         }
01728
01729     } // namespace internal
01730
01731 #endif  // GTEST_HAS_DEATH_TEST
01732
01733 } // namespace testing
```

18.284 gtest-filepath.cc

```
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00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 #include "gtest/internal/gtest-filepath.h"
00031
00032 #include <stdlib.h>
00033
00034 #include "gtest/gtest-message.h"
00035 #include "gtest/internal/gtest-port.h"
00036
00037 #if GTEST_OS_WINDOWS_MOBILE
00038 #include <windows.h>
00039 #elif GTEST_OS_WINDOWS
00040 #include <direct.h>
00041 #include <io.h>
00042 #else
00043 #include <limits.h>
00044
00045 #include <climits>  // Some Linux distributions define PATH_MAX here.
00046 #endif              // GTEST_OS_WINDOWS_MOBILE
00047
00048 #include "gtest/internal/gtest-string.h"
00049
00050 #if GTEST_OS_WINDOWS
00051 #define GTEST_PATH_MAX_ _MAX_PATH
00052 #elif defined(PATH_MAX)
00053 #define GTEST_PATH_MAX_ PATH_MAX
00054 #elif defined(_XOPEN_PATH_MAX)
00055 #define GTEST_PATH_MAX_ _XOPEN_PATH_MAX
00056 #else
00057 #define GTEST_PATH_MAX_ _POSIX_PATH_MAX
00058 #endif  // GTEST_OS_WINDOWS
00059
00060 namespace testing {
00061     namespace internal {
00062
00063 #if GTEST_OS_WINDOWS
00064         // On Windows, '\\' is the standard path separator, but many tools and the
00065         // Windows API also accept '/' as an alternate path separator. Unless otherwise
```

```
00066         // noted, a file path can contain either kind of path separators, or a mixture
00067         // of them.
00068         const char kPathSeparator = '\\';
00069         const char kAlternatePathSeparator = '/';
00070         const char kAlternatePathSeparatorString[] = "/";
00071 #if GTEST_OS_WINDOWS_MOBILE
00072         // Windows CE doesn't have a current directory. You should not use
00073         // the current directory in tests on Windows CE, but this at least
00074         // provides a reasonable fallback.
00075         const char kCurrentDirectoryString[] = "\\";
00076         // Windows CE doesn't define INVALID_FILE_ATTRIBUTES
00077         const DWORD kInvalidFileAttributes = 0xffffffff;
00078 #else
00079         const char kCurrentDirectoryString[] = ".\\";
00080 #endif  // GTEST_OS_WINDOWS_MOBILE
00081 #else
00082         const char kPathSeparator = '/';
00083         const char kCurrentDirectoryString[] = "./";
00084 #endif  // GTEST_OS_WINDOWS
00085
00086         // Returns whether the given character is a valid path separator.
00087
00088         static bool IsPathSeparator(char c) {
00089 #if GTEST_HAS_ALT_PATH_SEP_
00090             return (c == kPathSeparator) || (c == kAlternatePathSeparator);
00091 #else
00092             return c == kPathSeparator;
00093 #endif
00094         }
00095
00096         // Returns the current working directory, or "" if unsuccessful.
00097
00098         FilePath FilePath::GetCurrentDir() {
00099 #if GTEST_OS_WINDOWS_MOBILE || GTEST_OS_WINDOWS_PHONE ||         \
00100     GTEST_OS_WINDOWS_RT || GTEST_OS_ESP8266 || GTEST_OS_ESP32 || \
00101     GTEST_OS_XTENSA
00102             // These platforms do not have a current directory, so we just return
00103             // something reasonable.
00104             return FilePath(kCurrentDirectoryString);
00105 #elif GTEST_OS_WINDOWS
00106             char cwd[GTEST_PATH_MAX_ + 1] = {'\0'};
00107             return FilePath(_getcwd(cwd, sizeof (cwd)) == nullptr ? "" : cwd);
00108 #else
00109             char cwd[GTEST_PATH_MAX_ + 1] = {'\0'};
00110             char* result = getcwd(cwd, sizeof (cwd));
00111 #if GTEST_OS_NACL
00112             // getcwd will likely fail in NaCl due to the sandbox, so return something
00113             // reasonable. The user may have provided a shim implementation for getcwd,
00114             // however, so fallback only when failure is detected.
00115             return FilePath(result == nullptr ? kCurrentDirectoryString : cwd);
00116 #endif  // GTEST_OS_NACL
00117             return FilePath(result == nullptr ? "" : cwd);
00118 #endif  // GTEST_OS_WINDOWS_MOBILE
00119         }
00120
00121         // Returns a copy of the FilePath with the case-insensitive extension removed.
00122         // Example: FilePath("dir/file.exe").RemoveExtension("EXE") returns
00123         // FilePath("dir/file"). If a case-insensitive extension is not
00124         // found, returns a copy of the original FilePath.
00125
00126         FilePath FilePath::RemoveExtension(const char* extension) const {
00127             const std::string dot_extension = std::string(".") + extension;
00128             if (String::EndsWithCaseInsensitive(pathname_, dot_extension)) {
00129                 return FilePath(
00130                         pathname_.substr(0, pathname_.length() - dot_extension.length()));
00131             }
00132             return *this;
00133         }
00134
00135         // Returns a pointer to the last occurrence of a valid path separator in
00136         // the FilePath. On Windows, for example, both '/' and '\' are valid path
00137         // separators. Returns NULL if no path separator was found.
00138
00139         const char* FilePath::FindLastPathSeparator() const {
00140             const char* const last_sep = strrchr(c_str(), kPathSeparator);
00141 #if GTEST_HAS_ALT_PATH_SEP_
00142             const char* const last_alt_sep = strrchr(c_str(), kAlternatePathSeparator);
00143             // Comparing two pointers of which only one is NULL is undefined.
00144             if (last_alt_sep != nullptr &&
00145                     (last_sep == nullptr || last_alt_sep > last_sep)) {
00146                 return last_alt_sep;
00147             }
00148 #endif
00149             return last_sep;
00150         }
00151
00152         // Returns a copy of the FilePath with the directory part removed.
```

```
00153         // Example: FilePath("path/to/file").RemoveDirectoryName() returns
00154         // FilePath("file"). If there is no directory part ("just_a_file"), it returns
00155         // the FilePath unmodified. If there is no file part ("just_a_dir/") it
00156         // returns an empty FilePath ("").
00157         // On Windows platform, '\' is the path separator, otherwise it is '/'.
00158 
00159         FilePath FilePath::RemoveDirectoryName() const {
00160             const char* const last_sep = FindLastPathSeparator();
00161             return last_sep ? FilePath(last_sep + 1) : * this;
00162         }
00163 
00164         // RemoveFileName returns the directory path with the filename removed.
00165         // Example: FilePath("path/to/file").RemoveFileName() returns "path/to/".
00166         // If the FilePath is "a_file" or "/a_file", RemoveFileName returns
00167         // FilePath("./") or, on Windows, FilePath(".\\"). If the filepath does
00168         // not have a file, like "just/a/dir/", it returns the FilePath unmodified.
00169         // On Windows platform, '\' is the path separator, otherwise it is '/'.
00170 
00171         FilePath FilePath::RemoveFileName() const {
00172             const char* const last_sep = FindLastPathSeparator();
00173             std::string dir;
00174             if (last_sep) {
00175                 dir = std::string(c_str(), static_cast<size_t> (last_sep + 1 - c_str()));
00176             } else {
00177                 dir = kCurrentDirectoryString;
00178             }
00179             return FilePath(dir);
00180         }
00181 
00182         // Helper functions for naming files in a directory for xml output.
00183 
00184         // Given directory = "dir", base_name = "test", number = 0,
00185         // extension = "xml", returns "dir/test.xml". If number is greater
00186         // than zero (e.g., 12), returns "dir/test_12.xml".
00187         // On Windows platform, uses \ as the separator rather than /.
00188 
00189         FilePath FilePath::MakeFileName(const FilePath& directory,
00190                 const FilePath& base_name, int number,
00191                 const char* extension) {
00192             std::string file;
00193             if (number == 0) {
00194                 file = base_name.string() + "." + extension;
00195             } else {
00196                 file =
00197                         base_name.string() + "_" + StreamableToString(number) + "." + extension;
00198             }
00199             return ConcatPaths(directory, FilePath(file));
00200         }
00201 
00202         // Given directory = "dir", relative_path = "test.xml", returns "dir/test.xml".
00203         // On Windows, uses \ as the separator rather than /.
00204 
00205         FilePath FilePath::ConcatPaths(const FilePath& directory,
00206                 const FilePath& relative_path) {
00207             if (directory.IsEmpty()) return relative_path;
00208             const FilePath dir(directory.RemoveTrailingPathSeparator());
00209             return FilePath(dir.string() + kPathSeparator + relative_path.string());
00210         }
00211 
00212         // Returns true if pathname describes something findable in the file-system,
00213         // either a file, directory, or whatever.
00214 
00215         bool FilePath::FileOrDirectoryExists() const {
00216 #if GTEST_OS_WINDOWS_MOBILE
00217             LPCWSTR unicode = String::AnsiToUtf16(pathname_.c_str());
00218             const DWORD attributes = GetFileAttributes(unicode);
00219             delete[] unicode;
00220             return attributes != kInvalidFileAttributes;
00221 #else
00222             posix::StatStruct file_stat{};
00223             return posix::Stat(pathname_.c_str(), &file_stat) == 0;
00224 #endif  // GTEST_OS_WINDOWS_MOBILE
00225         }
00226 
00227         // Returns true if pathname describes a directory in the file-system
00228         // that exists.
00229 
00230         bool FilePath::DirectoryExists() const {
00231             bool result = false;
00232 #if GTEST_OS_WINDOWS
00233             // Don't strip off trailing separator if path is a root directory on
00234             // Windows (like "C:\\").
00235             const FilePath & path(IsRootDirectory() ? * this
00236                     : RemoveTrailingPathSeparator());
00237 #else
00238             const FilePath & path(*this);
00239 #endif
```

```
00240
00241 #if GTEST_OS_WINDOWS_MOBILE
00242             LPCWSTR unicode = String::AnsiToUtf16(path.c_str());
00243             const DWORD attributes = GetFileAttributes(unicode);
00244             delete[] unicode;
00245             if ((attributes != kInvalidFileAttributes) &&
00246                     (attributes & FILE_ATTRIBUTE_DIRECTORY)) {
00247                 result = true;
00248             }
00249 #else
00250             posix::StatStruct file_stat{};
00251             result =
00252                     posix::Stat(path.c_str(), &file_stat) == 0 && posix::IsDir(file_stat);
00253 #endif  // GTEST_OS_WINDOWS_MOBILE
00254
00255             return result;
00256         }
00257
00258         // Returns true if pathname describes a root directory. (Windows has one
00259         // root directory per disk drive.)
00260
00261         bool FilePath::IsRootDirectory() const {
00262 #if GTEST_OS_WINDOWS
00263             return pathname_.length() == 3 && IsAbsolutePath();
00264 #else
00265             return pathname_.length() == 1 && IsPathSeparator(pathname_.c_str()[0]);
00266 #endif
00267         }
00268
00269         // Returns true if pathname describes an absolute path.
00270
00271         bool FilePath::IsAbsolutePath() const {
00272             const char* const name = pathname_.c_str();
00273 #if GTEST_OS_WINDOWS
00274             return pathname_.length() >= 3 &&
00275                     ((name[0] >= 'a' && name[0] <= 'z') ||
00276                     (name[0] >= 'A' && name[0] <= 'Z')) &&
00277                     name[1] == ':' && IsPathSeparator(name[2]);
00278 #else
00279             return IsPathSeparator(name[0]);
00280 #endif
00281         }
00282
00283         // Returns a pathname for a file that does not currently exist. The pathname
00284         // will be directory/base_name.extension or
00285         // directory/base_name_<number>.extension if directory/base_name.extension
00286         // already exists. The number will be incremented until a pathname is found
00287         // that does not already exist.
00288         // Examples: 'dir/foo_test.xml' or 'dir/foo_test_1.xml'.
00289         // There could be a race condition if two or more processes are calling this
00290         // function at the same time -- they could both pick the same filename.
00291
00292         FilePath FilePath::GenerateUniqueFileName(const FilePath& directory,
00293                 const FilePath& base_name,
00294                 const char* extension) {
00295             FilePath full_pathname;
00296             int number = 0;
00297             do {
00298                 full_pathname.Set(MakeFileName(directory, base_name, number++, extension));
00299             } while (full_pathname.FileOrDirectoryExists());
00300             return full_pathname;
00301         }
00302
00303         // Returns true if FilePath ends with a path separator, which indicates that
00304         // it is intended to represent a directory. Returns false otherwise.
00305         // This does NOT check that a directory (or file) actually exists.
00306
00307         bool FilePath::IsDirectory() const {
00308             return !pathname_.empty() &&
00309                     IsPathSeparator(pathname_.c_str()[pathname_.length() - 1]);
00310         }
00311
00312         // Create directories so that path exists. Returns true if successful or if
00313         // the directories already exist; returns false if unable to create directories
00314         // for any reason.
00315
00316         bool FilePath::CreateDirectoriesRecursively() const {
00317             if (!this->IsDirectory()) {
00318                 return false;
00319             }
00320
00321             if (pathname_.length() == 0 || this->DirectoryExists()) {
00322                 return true;
00323             }
00324
00325             const FilePath parent(this->RemoveTrailingPathSeparator().RemoveFileName());
00326             return parent.CreateDirectoriesRecursively() && this->CreateFolder();
```

```
00327         }
00328
00329         // Create the directory so that path exists. Returns true if successful or
00330         // if the directory already exists; returns false if unable to create the
00331         // directory for any reason, including if the parent directory does not
00332         // exist. Not named "CreateDirectory" because that's a macro on Windows.
00333
00334         bool FilePath::CreateFolder() const {
00335 #if GTEST_OS_WINDOWS_MOBILE
00336             FilePath removed_sep(this->RemoveTrailingPathSeparator());
00337             LPCWSTR unicode = String::AnsiToUtf16(removed_sep.c_str());
00338             int result = CreateDirectory(unicode, nullptr) ? 0 : -1;
00339             delete[] unicode;
00340 #elif GTEST_OS_WINDOWS
00341             int result = _mkdir(pathname_.c_str());
00342 #elif GTEST_OS_ESP8266 || GTEST_OS_XTENSA
00343             // do nothing
00344             int result = 0;
00345 #else
00346             int result = mkdir(pathname_.c_str(), 0777);
00347 #endif  // GTEST_OS_WINDOWS_MOBILE
00348
00349             if (result == -1) {
00350                 return this->DirectoryExists(); // An error is OK if the directory exists.
00351             }
00352             return true; // No error.
00353         }
00354
00355         // If input name has a trailing separator character, remove it and return the
00356         // name, otherwise return the name string unmodified.
00357         // On Windows platform, uses \ as the separator, other platforms use /.
00358
00359         FilePath FilePath::RemoveTrailingPathSeparator() const {
00360             return IsDirectory() ? FilePath(pathname_.substr(0, pathname_.length() - 1))
00361                     : * this;
00362         }
00363
00364         // Removes any redundant separators that might be in the pathname.
00365         // For example, "bar///foo" becomes "bar/foo". Does not eliminate other
00366         // redundancies that might be in a pathname involving "." or "..".
00367
00368         void FilePath::Normalize() {
00369             auto out = pathname_.begin();
00370
00371             for (const char character : pathname_) {
00372                 if (!IsPathSeparator(character)) {
00373                     *(out++) = character;
00374                 } else if (out == pathname_.begin() || *std::prev(out) != kPathSeparator) {
00375                     *(out++) = kPathSeparator;
00376                 } else {
00377                     continue;
00378                 }
00379             }
00380
00381             pathname_.erase(out, pathname_.end());
00382         }
00383
00384     } // namespace internal
00385 } // namespace testing
```

18.285 gtest-internal-inl.h

```
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```

```
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00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // Utility functions and classes used by the Google C++ testing framework.//
00031 // This file contains purely Google Test's internal implementation.  Please
00032 // DO NOT #INCLUDE IT IN A USER PROGRAM.
00033
00034 #ifndef GOOGLETEST_SRC_GTEST_INTERNAL_INL_H_
00035 #define GOOGLETEST_SRC_GTEST_INTERNAL_INL_H_
00036
00037 #ifndef _WIN32_WCE
00038 #include <errno.h>
00039 #endif  // !_WIN32_WCE
00040 #include <stddef.h>
00041 #include <stdlib.h>  // For strtoll/_strtoul64/malloc/free.
00042 #include <string.h>  // For memmove.
00043
00044 #include <algorithm>
00045 #include <cstdint>
00046 #include <memory>
00047 #include <string>
00048 #include <vector>
00049
00050 #include "gtest/internal/gtest-port.h"
00051
00052 #if GTEST_CAN_STREAM_RESULTS_
00053 #include <arpa/inet.h>  // NOLINT
00054 #include <netdb.h>      // NOLINT
00055 #endif
00056
00057 #if GTEST_OS_WINDOWS
00058 #include <windows.h>  // NOLINT
00059 #endif                // GTEST_OS_WINDOWS
00060
00061 #include "gtest/gtest-spi.h"
00062 #include "gtest/gtest.h"
00063
00064 GTEST_DISABLE_MSC_WARNINGS_PUSH_(4251 \
00065 /* class A needs to have dll-interface to be used by clients of class B */)
00066
00067 // Declares the flags.
00068 //
00069 // We don't want the users to modify this flag in the code, but want
00070 // Google Test's own unit tests to be able to access it. Therefore we
00071 // declare it here as opposed to in gtest.h.
00072 GTEST_DECLARE_bool_(death_test_use_fork);
00073
00074 namespace testing {
00075     namespace internal {
00076
00077         // The value of GetTestTypeId() as seen from within the Google Test
00078         // library.  This is solely for testing GetTestTypeId().
00079         GTEST_API_ extern const TypeId kTestTypeIdInGoogleTest;
00080
00081         // A valid random seed must be in [1, kMaxRandomSeed].
00082         const int kMaxRandomSeed = 99999;
00083
00084         // g_help_flag is true if and only if the --help flag or an equivalent form
00085         // is specified on the command line.
00086         GTEST_API_ extern bool g_help_flag;
00087
00088         // Returns the current time in milliseconds.
00089         GTEST_API_ TimeInMillis GetTimeInMillis();
00090
00091         // Returns true if and only if Google Test should use colors in the output.
00092         GTEST_API_ bool ShouldUseColor(bool stdout_is_tty);
00093
00094         // Formats the given time in milliseconds as seconds.
00095         GTEST_API_ std::string FormatTimeInMillisAsSeconds(TimeInMillis ms);
00096
00097         // Converts the given time in milliseconds to a date string in the ISO 8601
00098         // format, without the timezone information.  N.B.: due to the use the
00099         // non-reentrant localtime() function, this function is not thread safe.  Do
00100         // not use it in any code that can be called from multiple threads.
00101         GTEST_API_ std::string FormatEpochTimeInMillisAsIso8601(TimeInMillis ms);
00102
00103         // Parses a string for an Int32 flag, in the form of "--flag=value".
00104         //
00105         // On success, stores the value of the flag in *value, and returns
00106         // true.  On failure, returns false without changing *value.
00107         GTEST_API_ bool ParseFlag(const char* str, const char* flag, int32_t* value);
00108
00109         // Returns a random seed in range [1, kMaxRandomSeed] based on the
00110         // given --gtest_random_seed flag value.
```

```
00111
00112         inline int GetRandomSeedFromFlag(int32_t random_seed_flag) {
00113             const unsigned int raw_seed =
00114                     (random_seed_flag == 0) ? static_cast<unsigned int> (GetTimeInMillis())
00115                     : static_cast<unsigned int> (random_seed_flag);
00116
00117             // Normalizes the actual seed to range [1, kMaxRandomSeed] such that
00118             // it's easy to type.
00119             const int normalized_seed =
00120                     static_cast<int> ((raw_seed - 1U) %
00121                     static_cast<unsigned int> (kMaxRandomSeed)) +
00122                     1;
00123             return normalized_seed;
00124         }
00125
00126         // Returns the first valid random seed after 'seed'.  The behavior is
00127         // undefined if 'seed' is invalid.  The seed after kMaxRandomSeed is
00128         // considered to be 1.
00129
00130         inline int GetNextRandomSeed(int seed) {
00131             GTEST_CHECK_(1 <= seed && seed <= kMaxRandomSeed)
00132                     << "Invalid random seed " << seed << " - must be in [1, "
00133                     << kMaxRandomSeed << "].";
00134             const int next_seed = seed + 1;
00135             return (next_seed > kMaxRandomSeed) ? 1 : next_seed;
00136         }
00137
00138         // This class saves the values of all Google Test flags in its c'tor, and
00139         // restores them in its d'tor.
00140
00141         class GTestFlagSaver {
00142         public:
00143             // The c'tor.
00144
00145             GTestFlagSaver() {
00146                 also_run_disabled_tests_ = GTEST_FLAG_GET(also_run_disabled_tests);
00147                 break_on_failure_ = GTEST_FLAG_GET(break_on_failure);
00148                 catch_exceptions_ = GTEST_FLAG_GET(catch_exceptions);
00149                 color_ = GTEST_FLAG_GET(color);
00150                 death_test_style_ = GTEST_FLAG_GET(death_test_style);
00151                 death_test_use_fork_ = GTEST_FLAG_GET(death_test_use_fork);
00152                 fail_fast_ = GTEST_FLAG_GET(fail_fast);
00153                 filter_ = GTEST_FLAG_GET(filter);
00154                 internal_run_death_test_ = GTEST_FLAG_GET(internal_run_death_test);
00155                 list_tests_ = GTEST_FLAG_GET(list_tests);
00156                 output_ = GTEST_FLAG_GET(output);
00157                 brief_ = GTEST_FLAG_GET(brief);
00158                 print_time_ = GTEST_FLAG_GET(print_time);
00159                 print_utf8_ = GTEST_FLAG_GET(print_utf8);
00160                 random_seed_ = GTEST_FLAG_GET(random_seed);
00161                 repeat_ = GTEST_FLAG_GET(repeat);
00162                 recreate_environments_when_repeating_ =
00163                         GTEST_FLAG_GET(recreate_environments_when_repeating);
00164                 shuffle_ = GTEST_FLAG_GET(shuffle);
00165                 stack_trace_depth_ = GTEST_FLAG_GET(stack_trace_depth);
00166                 stream_result_to_ = GTEST_FLAG_GET(stream_result_to);
00167                 throw_on_failure_ = GTEST_FLAG_GET(throw_on_failure);
00168             }
00169
00170             // The d'tor is not virtual.  DO NOT INHERIT FROM THIS CLASS.
00171
00172             ~GTestFlagSaver() {
00173                 GTEST_FLAG_SET(also_run_disabled_tests, also_run_disabled_tests_);
00174                 GTEST_FLAG_SET(break_on_failure, break_on_failure_);
00175                 GTEST_FLAG_SET(catch_exceptions, catch_exceptions_);
00176                 GTEST_FLAG_SET(color, color_);
00177                 GTEST_FLAG_SET(death_test_style, death_test_style_);
00178                 GTEST_FLAG_SET(death_test_use_fork, death_test_use_fork_);
00179                 GTEST_FLAG_SET(filter, filter_);
00180                 GTEST_FLAG_SET(fail_fast, fail_fast_);
00181                 GTEST_FLAG_SET(internal_run_death_test, internal_run_death_test_);
00182                 GTEST_FLAG_SET(list_tests, list_tests_);
00183                 GTEST_FLAG_SET(output, output_);
00184                 GTEST_FLAG_SET(brief, brief_);
00185                 GTEST_FLAG_SET(print_time, print_time_);
00186                 GTEST_FLAG_SET(print_utf8, print_utf8_);
00187                 GTEST_FLAG_SET(random_seed, random_seed_);
00188                 GTEST_FLAG_SET(repeat, repeat_);
00189                 GTEST_FLAG_SET(recreate_environments_when_repeating,
00190                         recreate_environments_when_repeating_);
00191                 GTEST_FLAG_SET(shuffle, shuffle_);
00192                 GTEST_FLAG_SET(stack_trace_depth, stack_trace_depth_);
00193                 GTEST_FLAG_SET(stream_result_to, stream_result_to_);
00194                 GTEST_FLAG_SET(throw_on_failure, throw_on_failure_);
00195             }
00196
00197         private:
```

```
00198             // Fields for saving the original values of flags.
00199             bool also_run_disabled_tests_;
00200             bool break_on_failure_;
00201             bool catch_exceptions_;
00202             std::string color_;
00203             std::string death_test_style_;
00204             bool death_test_use_fork_;
00205             bool fail_fast_;
00206             std::string filter_;
00207             std::string internal_run_death_test_;
00208             bool list_tests_;
00209             std::string output_;
00210             bool brief_;
00211             bool print_time_;
00212             bool print_utf8_;
00213             int32_t random_seed_;
00214             int32_t repeat_;
00215             bool recreate_environments_when_repeating_;
00216             bool shuffle_;
00217             int32_t stack_trace_depth_;
00218             std::string stream_result_to_;
00219             bool throw_on_failure_;
00220         } GTEST_ATTRIBUTE_UNUSED_;
00221
00222         // Converts a Unicode code point to a narrow string in UTF-8 encoding.
00223         // code_point parameter is of type UInt32 because wchar_t may not be
00224         // wide enough to contain a code point.
00225         // If the code_point is not a valid Unicode code point
00226         // (i.e. outside of Unicode range U+0 to U+10FFFF) it will be converted
00227         // to "(Invalid Unicode 0xXXXXXXXX)".
00228         GTEST_API_ std::string CodePointToUtf8(uint32_t code_point);
00229
00230         // Converts a wide string to a narrow string in UTF-8 encoding.
00231         // The wide string is assumed to have the following encoding:
00232         //   UTF-16 if sizeof(wchar_t) == 2 (on Windows, Cygwin)
00233         //   UTF-32 if sizeof(wchar_t) == 4 (on Linux)
00234         // Parameter str points to a null-terminated wide string.
00235         // Parameter num_chars may additionally limit the number
00236         // of wchar_t characters processed. -1 is used when the entire string
00237         // should be processed.
00238         // If the string contains code points that are not valid Unicode code points
00239         // (i.e. outside of Unicode range U+0 to U+10FFFF) they will be output
00240         // as '(Invalid Unicode 0xXXXXXXXX)'. If the string is in UTF16 encoding
00241         // and contains invalid UTF-16 surrogate pairs, values in those pairs
00242         // will be encoded as individual Unicode characters from Basic Normal Plane.
00243         GTEST_API_ std::string WideStringToUtf8(const wchar_t* str, int num_chars);
00244
00245         // Reads the GTEST_SHARD_STATUS_FILE environment variable, and creates the file
00246         // if the variable is present. If a file already exists at this location, this
00247         // function will write over it. If the variable is present, but the file cannot
00248         // be created, prints an error and exits.
00249         void WriteToShardStatusFileIfNeeded();
00250
00251         // Checks whether sharding is enabled by examining the relevant
00252         // environment variable values. If the variables are present,
00253         // but inconsistent (e.g., shard_index >= total_shards), prints
00254         // an error and exits. If in_subprocess_for_death_test, sharding is
00255         // disabled because it must only be applied to the original test
00256         // process. Otherwise, we could filter out death tests we intended to execute.
00257         GTEST_API_ bool ShouldShard(const char* total_shards_str,
00258                 const char* shard_index_str,
00259                 bool in_subprocess_for_death_test);
00260
00261         // Parses the environment variable var as a 32-bit integer. If it is unset,
00262         // returns default_val. If it is not a 32-bit integer, prints an error and
00263         // and aborts.
00264         GTEST_API_ int32_t Int32FromEnvOrDie(const char* env_var, int32_t default_val);
00265
00266         // Given the total number of shards, the shard index, and the test id,
00267         // returns true if and only if the test should be run on this shard. The test id
00268         // is some arbitrary but unique non-negative integer assigned to each test
00269         // method. Assumes that 0 <= shard_index < total_shards.
00270         GTEST_API_ bool ShouldRunTestOnShard(int total_shards, int shard_index,
00271                 int test_id);
00272
00273         // STL container utilities.
00274
00275         // Returns the number of elements in the given container that satisfy
00276         // the given predicate.
00277
00278         template <class Container, typename Predicate>
00279         inline int CountIf(const Container& c, Predicate predicate) {
00280             // Implemented as an explicit loop since std::count_if() in libCstd on
00281             // Solaris has a non-standard signature.
00282             int count = 0;
00283             for (auto it = c.begin(); it != c.end(); ++it) {
00284                 if (predicate(*it)) ++count;
```

```
00285             }
00286             return count;
00287         }
00288
00289         // Applies a function/functor to each element in the container.
00290
00291         template <class Container, typename Functor>
00292         void ForEach(const Container& c, Functor functor) {
00293             std::for_each(c.begin(), c.end(), functor);
00294         }
00295
00296         // Returns the i-th element of the vector, or default_value if i is not
00297         // in range [0, v.size()).
00298
00299         template <typename E>
00300         inline E GetElementOr(const std::vector<E>& v, int i, E default_value) {
00301             return (i < 0 || i >= static_cast<int> (v.size())) ? default_value
00302                     : v[static_cast<size_t> (i)];
00303         }
00304
00305         // Performs an in-place shuffle of a range of the vector's elements.
00306         // 'begin' and 'end' are element indices as an STL-style range;
00307         // i.e. [begin, end) are shuffled, where 'end' == size() means to
00308         // shuffle to the end of the vector.
00309
00310         template <typename E>
00311         void ShuffleRange(internal::Random* random, int begin, int end,
00312                 std::vector<E>* v) {
00313             const int size = static_cast<int> (v->size());
00314             GTEST_CHECK_(0 <= begin && begin <= size)
00315                     << "Invalid shuffle range start " << begin << ": must be in range [0, "
00316                     << size << "].";
00317             GTEST_CHECK_(begin <= end && end <= size)
00318                     << "Invalid shuffle range finish " << end << ": must be in range ["
00319                     << begin << ", " << size << "].";
00320
00321             // Fisher-Yates shuffle, from
00322             // http://en.wikipedia.org/wiki/Fisher-Yates_shuffle
00323             for (int range_width = end - begin; range_width >= 2; range_width--) {
00324                 const int last_in_range = begin + range_width - 1;
00325                 const int selected =
00326                         begin +
00327                         static_cast<int> (random->Generate(static_cast<uint32_t> (range_width)));
00328                 std::swap((*v)[static_cast<size_t> (selected)],
00329                         (*v)[static_cast<size_t> (last_in_range)]);
00330             }
00331         }
00332
00333         // Performs an in-place shuffle of the vector's elements.
00334
00335         template <typename E>
00336         inline void Shuffle(internal::Random* random, std::vector<E>* v) {
00337             ShuffleRange(random, 0, static_cast<int> (v->size()), v);
00338         }
00339
00340         // A function for deleting an object.  Handy for being used as a
00341         // functor.
00342
00343         template <typename T>
00344         static void Delete(T* x) {
00345             delete x;
00346         }
00347
00348         // A predicate that checks the key of a TestProperty against a known key.
00349         //
00350         // TestPropertyKeyIs is copyable.
00351
00352         class TestPropertyKeyIs {
00353         public:
00354             // Constructor.
00355             //
00356             // TestPropertyKeyIs has NO default constructor.
00357
00358             explicit TestPropertyKeyIs(const std::string& key) : key_(key) {
00359             }
00360
00361             // Returns true if and only if the test name of test property matches on key_.
00362
00363             bool operator()(const TestProperty& test_property) const {
00364                 return test_property.key() == key_;
00365             }
00366
00367         private:
00368             std::string key_;
00369         };
00370
00371         // Class UnitTestOptions.
```

```
00372         //
00373         // This class contains functions for processing options the user
00374         // specifies when running the tests.  It has only static members.
00375         //
00376         // In most cases, the user can specify an option using either an
00377         // environment variable or a command line flag.  E.g. you can set the
00378         // test filter using either GTEST_FILTER or --gtest_filter.  If both
00379         // the variable and the flag are present, the latter overrides the
00380         // former.
00381
00382         class GTEST_API_ UnitTestOptions {
00383         public:
00384             // Functions for processing the gtest_output flag.
00385
00386             // Returns the output format, or "" for normal printed output.
00387             static std::string GetOutputFormat();
00388
00389             // Returns the absolute path of the requested output file, or the
00390             // default (test_detail.xml in the original working directory) if
00391             // none was explicitly specified.
00392             static std::string GetAbsolutePathToOutputFile();
00393
00394             // Functions for processing the gtest_filter flag.
00395
00396             // Returns true if and only if the user-specified filter matches the test
00397             // suite name and the test name.
00398             static bool FilterMatchesTest(const std::string& test_suite_name,
00399                     const std::string& test_name);
00400
00401 #if GTEST_OS_WINDOWS
00402             // Function for supporting the gtest_catch_exception flag.
00403
00404             // Returns EXCEPTION_EXECUTE_HANDLER if Google Test should handle the
00405             // given SEH exception, or EXCEPTION_CONTINUE_SEARCH otherwise.
00406             // This function is useful as an __except condition.
00407             static int GTestShouldProcessSEH(DWORD exception_code);
00408 #endif  // GTEST_OS_WINDOWS
00409
00410             // Returns true if "name" matches the ':' separated list of glob-style
00411             // filters in "filter".
00412             static bool MatchesFilter(const std::string& name, const char* filter);
00413         };
00414
00415         // Returns the current application's name, removing directory path if that
00416         // is present.  Used by UnitTestOptions::GetOutputFile.
00417         GTEST_API_ FilePath GetCurrentExecutableName();
00418
00419         // The role interface for getting the OS stack trace as a string.
00420
00421         class OsStackTraceGetterInterface {
00422         public:
00423
00424             OsStackTraceGetterInterface() {
00425             }
00426
00427             virtual ~OsStackTraceGetterInterface() {
00428             }
00429
00430             // Returns the current OS stack trace as an std::string.  Parameters:
00431             //
00432             //   max_depth  - the maximum number of stack frames to be included
00433             //                in the trace.
00434             //   skip_count - the number of top frames to be skipped; doesn't count
00435             //                against max_depth.
00436             virtual std::string CurrentStackTrace(int max_depth, int skip_count) = 0;
00437
00438             // UponLeavingGTest() should be called immediately before Google Test calls
00439             // user code. It saves some information about the current stack that
00440             // CurrentStackTrace() will use to find and hide Google Test stack frames.
00441             virtual void UponLeavingGTest() = 0;
00442
00443             // This string is inserted in place of stack frames that are part of
00444             // Google Test's implementation.
00445             static const char* const kElidedFramesMarker;
00446
00447         private:
00448             OsStackTraceGetterInterface(const OsStackTraceGetterInterface&) = delete;
00449             OsStackTraceGetterInterface& operator=(const OsStackTraceGetterInterface&) =
00450                     delete;
00451         };
00452
00453         // A working implementation of the OsStackTraceGetterInterface interface.
00454
00455         class OsStackTraceGetter : public OsStackTraceGetterInterface {
00456         public:
00457
00458             OsStackTraceGetter() {
```

```
00459             }
00460
00461             std::string CurrentStackTrace(int max_depth, int skip_count) override;
00462             void UponLeavingGTest() override;
00463
00464         private:
00465 #if GTEST_HAS_ABSL
00466             Mutex mutex_; // Protects all internal state.
00467
00468             // We save the stack frame below the frame that calls user code.
00469             // We do this because the address of the frame immediately below
00470             // the user code changes between the call to UponLeavingGTest()
00471             // and any calls to the stack trace code from within the user code.
00472             void* caller_frame_ = nullptr;
00473 #endif  // GTEST_HAS_ABSL
00474
00475             OsStackTraceGetter(const OsStackTraceGetter&) = delete;
00476             OsStackTraceGetter& operator=(const OsStackTraceGetter&) = delete;
00477         };
00478
00479         // Information about a Google Test trace point.
00480
00481         struct TraceInfo {
00482             const char* file;
00483             int line;
00484             std::string message;
00485         };
00486
00487         // This is the default global test part result reporter used in UnitTestImpl.
00488         // This class should only be used by UnitTestImpl.
00489
00490         class DefaultGlobalTestPartResultReporter
00491         : public TestPartResultReporterInterface {
00492         public:
00493             explicit DefaultGlobalTestPartResultReporter(UnitTestImpl* unit_test);
00494             // Implements the TestPartResultReporterInterface. Reports the test part
00495             // result in the current test.
00496             void ReportTestPartResult(const TestPartResult& result) override;
00497
00498         private:
00499             UnitTestImpl * const unit_test_;
00500
00501             DefaultGlobalTestPartResultReporter(
00502                     const DefaultGlobalTestPartResultReporter&) = delete;
00503             DefaultGlobalTestPartResultReporter& operator=(
00504                     const DefaultGlobalTestPartResultReporter&) = delete;
00505         };
00506
00507         // This is the default per thread test part result reporter used in
00508         // UnitTestImpl. This class should only be used by UnitTestImpl.
00509
00510         class DefaultPerThreadTestPartResultReporter
00511         : public TestPartResultReporterInterface {
00512         public:
00513             explicit DefaultPerThreadTestPartResultReporter(UnitTestImpl* unit_test);
00514             // Implements the TestPartResultReporterInterface. The implementation just
00515             // delegates to the current global test part result reporter of *unit_test_.
00516             void ReportTestPartResult(const TestPartResult& result) override;
00517
00518         private:
00519             UnitTestImpl * const unit_test_;
00520
00521             DefaultPerThreadTestPartResultReporter(
00522                     const DefaultPerThreadTestPartResultReporter&) = delete;
00523             DefaultPerThreadTestPartResultReporter& operator=(
00524                     const DefaultPerThreadTestPartResultReporter&) = delete;
00525         };
00526
00527         // The private implementation of the UnitTest class.  We don't protect
00528         // the methods under a mutex, as this class is not accessible by a
00529         // user and the UnitTest class that delegates work to this class does
00530         // proper locking.
00531
00532         class GTEST_API_ UnitTestImpl {
00533         public:
00534             explicit UnitTestImpl(UnitTest* parent);
00535             virtual ~UnitTestImpl();
00536
00537             // There are two different ways to register your own TestPartResultReporter.
00538             // You can register your own reporter to listen either only for test results
00539             // from the current thread or for results from all threads.
00540             // By default, each per-thread test result reporter just passes a new
00541             // TestPartResult to the global test result reporter, which registers the
00542             // test part result for the currently running test.
00543
00544             // Returns the global test part result reporter.
00545             TestPartResultReporterInterface* GetGlobalTestPartResultReporter();
```

```
00546
00547             // Sets the global test part result reporter.
00548             void SetGlobalTestPartResultReporter(
00549                     TestPartResultReporterInterface* reporter);
00550
00551             // Returns the test part result reporter for the current thread.
00552             TestPartResultReporterInterface* GetTestPartResultReporterForCurrentThread();
00553
00554             // Sets the test part result reporter for the current thread.
00555             void SetTestPartResultReporterForCurrentThread(
00556                     TestPartResultReporterInterface* reporter);
00557
00558             // Gets the number of successful test suites.
00559             int successful_test_suite_count() const;
00560
00561             // Gets the number of failed test suites.
00562             int failed_test_suite_count() const;
00563
00564             // Gets the number of all test suites.
00565             int total_test_suite_count() const;
00566
00567             // Gets the number of all test suites that contain at least one test
00568             // that should run.
00569             int test_suite_to_run_count() const;
00570
00571             // Gets the number of successful tests.
00572             int successful_test_count() const;
00573
00574             // Gets the number of skipped tests.
00575             int skipped_test_count() const;
00576
00577             // Gets the number of failed tests.
00578             int failed_test_count() const;
00579
00580             // Gets the number of disabled tests that will be reported in the XML report.
00581             int reportable_disabled_test_count() const;
00582
00583             // Gets the number of disabled tests.
00584             int disabled_test_count() const;
00585
00586             // Gets the number of tests to be printed in the XML report.
00587             int reportable_test_count() const;
00588
00589             // Gets the number of all tests.
00590             int total_test_count() const;
00591
00592             // Gets the number of tests that should run.
00593             int test_to_run_count() const;
00594
00595             // Gets the time of the test program start, in ms from the start of the
00596             // UNIX epoch.
00597
00598             TimeInMillis start_timestamp() const {
00599                 return start_timestamp_;
00600             }
00601
00602             // Gets the elapsed time, in milliseconds.
00603
00604             TimeInMillis elapsed_time() const {
00605                 return elapsed_time_;
00606             }
00607
00608             // Returns true if and only if the unit test passed (i.e. all test suites
00609             // passed).
00610
00611             bool Passed() const {
00612                 return !Failed();
00613             }
00614
00615             // Returns true if and only if the unit test failed (i.e. some test suite
00616             // failed or something outside of all tests failed).
00617
00618             bool Failed() const {
00619                 return failed_test_suite_count() > 0 || ad_hoc_test_result()->Failed();
00620             }
00621
00622             // Gets the i-th test suite among all the test suites. i can range from 0 to
00623             // total_test_suite_count() - 1. If i is not in that range, returns NULL.
00624
00625             const TestSuite* GetTestSuite(int i) const {
00626                 const int index = GetElementOr(test_suite_indices_, i, -1);
00627                 return index < 0 ? nullptr : test_suites_[static_cast<size_t> (i)];
00628             }
00629
00630             //  Legacy API is deprecated but still available
00631 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00632
```

```
00633             const TestCase* GetTestCase(int i) const {
00634                 return GetTestSuite(i);
00635             }
00636 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00637
00638             // Gets the i-th test suite among all the test suites. i can range from 0 to
00639             // total_test_suite_count() - 1. If i is not in that range, returns NULL.
00640
00641             TestSuite* GetMutableSuiteCase(int i) {
00642                 const int index = GetElementOr(test_suite_indices_, i, -1);
00643                 return index < 0 ? nullptr : test_suites_[static_cast<size_t> (index)];
00644             }
00645
00646             // Provides access to the event listener list.
00647
00648             TestEventListeners* listeners() {
00649                 return &listeners_;
00650             }
00651
00652             // Returns the TestResult for the test that's currently running, or
00653             // the TestResult for the ad hoc test if no test is running.
00654             TestResult* current_test_result();
00655
00656             // Returns the TestResult for the ad hoc test.
00657
00658             const TestResult* ad_hoc_test_result() const {
00659                 return &ad_hoc_test_result_;
00660             }
00661
00662             // Sets the OS stack trace getter.
00663             //
00664             // Does nothing if the input and the current OS stack trace getter
00665             // are the same; otherwise, deletes the old getter and makes the
00666             // input the current getter.
00667             void set_os_stack_trace_getter(OsStackTraceGetterInterface* getter);
00668
00669             // Returns the current OS stack trace getter if it is not NULL;
00670             // otherwise, creates an OsStackTraceGetter, makes it the current
00671             // getter, and returns it.
00672             OsStackTraceGetterInterface* os_stack_trace_getter();
00673
00674             // Returns the current OS stack trace as an std::string.
00675             //
00676             // The maximum number of stack frames to be included is specified by
00677             // the gtest_stack_trace_depth flag.  The skip_count parameter
00678             // specifies the number of top frames to be skipped, which doesn't
00679             // count against the number of frames to be included.
00680             //
00681             // For example, if Foo() calls Bar(), which in turn calls
00682             // CurrentOsStackTraceExceptTop(1), Foo() will be included in the
00683             // trace but Bar() and CurrentOsStackTraceExceptTop() won't.
00684             std::string CurrentOsStackTraceExceptTop(int skip_count)
00685             GTEST_NO_INLINE_ GTEST_NO_TAIL_CALL_;
00686
00687             // Finds and returns a TestSuite with the given name.  If one doesn't
00688             // exist, creates one and returns it.
00689             //
00690             // Arguments:
00691             //
00692             //   test_suite_name: name of the test suite
00693             //   type_param:      the name of the test's type parameter, or NULL if
00694             //                    this is not a typed or a type-parameterized test.
00695             //   set_up_tc:       pointer to the function that sets up the test suite
00696             //   tear_down_tc:    pointer to the function that tears down the test suite
00697             TestSuite* GetTestSuite(const char* test_suite_name, const char* type_param,
00698                     internal::SetUpTestSuiteFunc set_up_tc,
00699                     internal::TearDownTestSuiteFunc tear_down_tc);
00700
00701             //  Legacy API is deprecated but still available
00702 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00703
00704             TestCase* GetTestCase(const char* test_case_name, const char* type_param,
00705                     internal::SetUpTestSuiteFunc set_up_tc,
00706                     internal::TearDownTestSuiteFunc tear_down_tc) {
00707                 return GetTestSuite(test_case_name, type_param, set_up_tc, tear_down_tc);
00708             }
00709 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
00710
00711             // Adds a TestInfo to the unit test.
00712             //
00713             // Arguments:
00714             //
00715             //   set_up_tc:    pointer to the function that sets up the test suite
00716             //   tear_down_tc: pointer to the function that tears down the test suite
00717             //   test_info:    the TestInfo object
00718
00719             void AddTestInfo(internal::SetUpTestSuiteFunc set_up_tc,
```

```
00720                     internal::TearDownTestSuiteFunc tear_down_tc,
00721                     TestInfo* test_info) {
00722 #if GTEST_HAS_DEATH_TEST
00723                 // In order to support thread-safe death tests, we need to
00724                 // remember the original working directory when the test program
00725                 // was first invoked.  We cannot do this in RUN_ALL_TESTS(), as
00726                 // the user may have changed the current directory before calling
00727                 // RUN_ALL_TESTS().  Therefore we capture the current directory in
00728                 // AddTestInfo(), which is called to register a TEST or TEST_F
00729                 // before main() is reached.
00730                 if (original_working_dir_.IsEmpty()) {
00731                     original_working_dir_.Set(FilePath::GetCurrentDir());
00732                     GTEST_CHECK_(!original_working_dir_.IsEmpty())
00733                             << "Failed to get the current working directory.";
00734                 }
00735 #endif  // GTEST_HAS_DEATH_TEST
00736
00737                 GetTestSuite(test_info->test_suite_name(), test_info->type_param(),
00738                         set_up_tc, tear_down_tc)
00739                         ->AddTestInfo(test_info);
00740             }
00741
00742             // Returns ParameterizedTestSuiteRegistry object used to keep track of
00743             // value-parameterized tests and instantiate and register them.
00744
00745             internal::ParameterizedTestSuiteRegistry& parameterized_test_registry() {
00746                 return parameterized_test_registry_;
00747             }
00748
00749             std::set<std::string>* ignored_parameterized_test_suites() {
00750                 return &ignored_parameterized_test_suites_;
00751             }
00752
00753             // Returns TypeParameterizedTestSuiteRegistry object used to keep track of
00754             // type-parameterized tests and instantiations of them.
00755
00756             internal::TypeParameterizedTestSuiteRegistry&
00757             type_parameterized_test_registry() {
00758                 return type_parameterized_test_registry_;
00759             }
00760
00761             // Sets the TestSuite object for the test that's currently running.
00762
00763             void set_current_test_suite(TestSuite* a_current_test_suite) {
00764                 current_test_suite_ = a_current_test_suite;
00765             }
00766
00767             // Sets the TestInfo object for the test that's currently running.  If
00768             // current_test_info is NULL, the assertion results will be stored in
00769             // ad_hoc_test_result_.
00770
00771             void set_current_test_info(TestInfo* a_current_test_info) {
00772                 current_test_info_ = a_current_test_info;
00773             }
00774
00775             // Registers all parameterized tests defined using TEST_P and
00776             // INSTANTIATE_TEST_SUITE_P, creating regular tests for each test/parameter
00777             // combination. This method can be called more then once; it has guards
00778             // protecting from registering the tests more then once.  If
00779             // value-parameterized tests are disabled, RegisterParameterizedTests is
00780             // present but does nothing.
00781             void RegisterParameterizedTests();
00782
00783             // Runs all tests in this UnitTest object, prints the result, and
00784             // returns true if all tests are successful.  If any exception is
00785             // thrown during a test, this test is considered to be failed, but
00786             // the rest of the tests will still be run.
00787             bool RunAllTests();
00788
00789             // Clears the results of all tests, except the ad hoc tests.
00790
00791             void ClearNonAdHocTestResult() {
00792                 ForEach(test_suites_, TestSuite::ClearTestSuiteResult);
00793             }
00794
00795             // Clears the results of ad-hoc test assertions.
00796
00797             void ClearAdHocTestResult() {
00798                 ad_hoc_test_result_.Clear();
00799             }
00800
00801             // Adds a TestProperty to the current TestResult object when invoked in a
00802             // context of a test or a test suite, or to the global property set. If the
00803             // result already contains a property with the same key, the value will be
00804             // updated.
00805             void RecordProperty(const TestProperty& test_property);
00806
```

```
00807             enum ReactionToSharding {
00808                 HONOR_SHARDING_PROTOCOL, IGNORE_SHARDING_PROTOCOL
00809             };
00810
00811             // Matches the full name of each test against the user-specified
00812             // filter to decide whether the test should run, then records the
00813             // result in each TestSuite and TestInfo object.
00814             // If shard_tests == HONOR_SHARDING_PROTOCOL, further filters tests
00815             // based on sharding variables in the environment.
00816             // Returns the number of tests that should run.
00817             int FilterTests(ReactionToSharding shard_tests);
00818
00819             // Prints the names of the tests matching the user-specified filter flag.
00820             void ListTestsMatchingFilter();
00821
00822             const TestSuite* current_test_suite() const {
00823                 return current_test_suite_;
00824             }
00825
00826             TestInfo* current_test_info() {
00827                 return current_test_info_;
00828             }
00829
00830             const TestInfo* current_test_info() const {
00831                 return current_test_info_;
00832             }
00833
00834             // Returns the vector of environments that need to be set-up/torn-down
00835             // before/after the tests are run.
00836
00837             std::vector<Environment*>& environments() {
00838                 return environments_;
00839             }
00840
00841             // Getters for the per-thread Google Test trace stack.
00842
00843             std::vector<TraceInfo>& gtest_trace_stack() {
00844                 return *(gtest_trace_stack_.pointer());
00845             }
00846
00847             const std::vector<TraceInfo>& gtest_trace_stack() const {
00848                 return gtest_trace_stack_.get();
00849             }
00850
00851 #if GTEST_HAS_DEATH_TEST
00852
00853             void InitDeathTestSubprocessControlInfo() {
00854                 internal_run_death_test_flag_.reset(ParseInternalRunDeathTestFlag());
00855             }
00856             // Returns a pointer to the parsed --gtest_internal_run_death_test
00857             // flag, or NULL if that flag was not specified.
00858             // This information is useful only in a death test child process.
00859             // Must not be called before a call to InitGoogleTest.
00860
00861             const InternalRunDeathTestFlag* internal_run_death_test_flag() const {
00862                 return internal_run_death_test_flag_.get();
00863             }
00864
00865             // Returns a pointer to the current death test factory.
00866
00867             internal::DeathTestFactory* death_test_factory() {
00868                 return death_test_factory_.get();
00869             }
00870
00871             void SuppressTestEventsIfInSubprocess();
00872
00873             friend class ReplaceDeathTestFactory;
00874 #endif  // GTEST_HAS_DEATH_TEST
00875
00876             // Initializes the event listener performing XML output as specified by
00877             // UnitTestOptions. Must not be called before InitGoogleTest.
00878             void ConfigureXmlOutput();
00879
00880 #if GTEST_CAN_STREAM_RESULTS_
00881             // Initializes the event listener for streaming test results to a socket.
00882             // Must not be called before InitGoogleTest.
00883             void ConfigureStreamingOutput();
00884 #endif
00885
00886             // Performs initialization dependent upon flag values obtained in
00887             // ParseGoogleTestFlagsOnly.  Is called from InitGoogleTest after the call to
00888             // ParseGoogleTestFlagsOnly.  In case a user neglects to call InitGoogleTest
00889             // this function is also called from RunAllTests.  Since this function can be
00890             // called more than once, it has to be idempotent.
00891             void PostFlagParsingInit();
00892
00893             // Gets the random seed used at the start of the current test iteration.
```

```
00894
00895           int random_seed() const {
00896               return random_seed_;
00897           }
00898
00899           // Gets the random number generator.
00900
00901           internal::Random* random() {
00902               return &random_;
00903           }
00904
00905           // Shuffles all test suites, and the tests within each test suite,
00906           // making sure that death tests are still run first.
00907           void ShuffleTests();
00908
00909           // Restores the test suites and tests to their order before the first shuffle.
00910           void UnshuffleTests();
00911
00912           // Returns the value of GTEST_FLAG(catch_exceptions) at the moment
00913           // UnitTest::Run() starts.
00914
00915           bool catch_exceptions() const {
00916               return catch_exceptions_;
00917           }
00918
00919       private:
00920           friend class ::testing::UnitTest;
00921
00922           // Used by UnitTest::Run() to capture the state of
00923           // GTEST_FLAG(catch_exceptions) at the moment it starts.
00924
00925           void set_catch_exceptions(bool value) {
00926               catch_exceptions_ = value;
00927           }
00928
00929           // The UnitTest object that owns this implementation object.
00930           UnitTest * const parent_;
00931
00932           // The working directory when the first TEST() or TEST_F() was
00933           // executed.
00934           internal::FilePath original_working_dir_;
00935
00936           // The default test part result reporters.
00937           DefaultGlobalTestPartResultReporter default_global_test_part_result_reporter_;
00938           DefaultPerThreadTestPartResultReporter
00939           default_per_thread_test_part_result_reporter_;
00940
00941           // Points to (but doesn't own) the global test part result reporter.
00942           TestPartResultReporterInterface* global_test_part_result_reporter_;
00943
00944           // Protects read and write access to global_test_part_result_reporter_.
00945           internal::Mutex global_test_part_result_reporter_mutex_;
00946
00947           // Points to (but doesn't own) the per-thread test part result reporter.
00948           internal::ThreadLocal<TestPartResultReporterInterface*>
00949           per_thread_test_part_result_reporter_;
00950
00951           // The vector of environments that need to be set-up/torn-down
00952           // before/after the tests are run.
00953           std::vector<Environment*> environments_;
00954
00955           // The vector of TestSuites in their original order.  It owns the
00956           // elements in the vector.
00957           std::vector<TestSuite*> test_suites_;
00958
00959           // Provides a level of indirection for the test suite list to allow
00960           // easy shuffling and restoring the test suite order.  The i-th
00961           // element of this vector is the index of the i-th test suite in the
00962           // shuffled order.
00963           std::vector<int> test_suite_indices_;
00964
00965           // ParameterizedTestRegistry object used to register value-parameterized
00966           // tests.
00967           internal::ParameterizedTestSuiteRegistry parameterized_test_registry_;
00968           internal::TypeParameterizedTestSuiteRegistry
00969           type_parameterized_test_registry_;
00970
00971           // The set holding the name of parameterized
00972           // test suites that may go uninstantiated.
00973           std::set<std::string> ignored_parameterized_test_suites_;
00974
00975           // Indicates whether RegisterParameterizedTests() has been called already.
00976           bool parameterized_tests_registered_;
00977
00978           // Index of the last death test suite registered.  Initially -1.
00979           int last_death_test_suite_;
00980
```

```
00981             // This points to the TestSuite for the currently running test.  It
00982             // changes as Google Test goes through one test suite after another.
00983             // When no test is running, this is set to NULL and Google Test
00984             // stores assertion results in ad_hoc_test_result_.  Initially NULL.
00985             TestSuite* current_test_suite_;
00986
00987             // This points to the TestInfo for the currently running test.  It
00988             // changes as Google Test goes through one test after another.  When
00989             // no test is running, this is set to NULL and Google Test stores
00990             // assertion results in ad_hoc_test_result_.  Initially NULL.
00991             TestInfo* current_test_info_;
00992
00993             // Normally, a user only writes assertions inside a TEST or TEST_F,
00994             // or inside a function called by a TEST or TEST_F.  Since Google
00995             // Test keeps track of which test is current running, it can
00996             // associate such an assertion with the test it belongs to.
00997             //
00998             // If an assertion is encountered when no TEST or TEST_F is running,
00999             // Google Test attributes the assertion result to an imaginary "ad hoc"
01000             // test, and records the result in ad_hoc_test_result_.
01001             TestResult ad_hoc_test_result_;
01002
01003             // The list of event listeners that can be used to track events inside
01004             // Google Test.
01005             TestEventListeners listeners_;
01006
01007             // The OS stack trace getter.  Will be deleted when the UnitTest
01008             // object is destructed.  By default, an OsStackTraceGetter is used,
01009             // but the user can set this field to use a custom getter if that is
01010             // desired.
01011             OsStackTraceGetterInterface* os_stack_trace_getter_;
01012
01013             // True if and only if PostFlagParsingInit() has been called.
01014             bool post_flag_parse_init_performed_;
01015
01016             // The random number seed used at the beginning of the test run.
01017             int random_seed_;
01018
01019             // Our random number generator.
01020             internal::Random random_;
01021
01022             // The time of the test program start, in ms from the start of the
01023             // UNIX epoch.
01024             TimeInMillis start_timestamp_;
01025
01026             // How long the test took to run, in milliseconds.
01027             TimeInMillis elapsed_time_;
01028
01029 #if GTEST_HAS_DEATH_TEST
01030             // The decomposed components of the gtest_internal_run_death_test flag,
01031             // parsed when RUN_ALL_TESTS is called.
01032             std::unique_ptr<InternalRunDeathTestFlag> internal_run_death_test_flag_;
01033             std::unique_ptr<internal::DeathTestFactory> death_test_factory_;
01034 #endif  // GTEST_HAS_DEATH_TEST
01035
01036             // A per-thread stack of traces created by the SCOPED_TRACE() macro.
01037             internal::ThreadLocal<std::vector<TraceInfo> > gtest_trace_stack_;
01038
01039             // The value of GTEST_FLAG(catch_exceptions) at the moment RunAllTests()
01040             // starts.
01041             bool catch_exceptions_;
01042
01043             UnitTestImpl(const UnitTestImpl&) = delete;
01044             UnitTestImpl& operator=(const UnitTestImpl&) = delete;
01045         }; // class UnitTestImpl
01046
01047         // Convenience function for accessing the global UnitTest
01048         // implementation object.
01049
01050         inline UnitTestImpl* GetUnitTestImpl() {
01051             return UnitTest::GetInstance()->impl();
01052         }
01053
01054 #if GTEST_USES_SIMPLE_RE
01055
01056         // Internal helper functions for implementing the simple regular
01057         // expression matcher.
01058         GTEST_API_ bool IsInSet(char ch, const char* str);
01059         GTEST_API_ bool IsAsciiDigit(char ch);
01060         GTEST_API_ bool IsAsciiPunct(char ch);
01061         GTEST_API_ bool IsRepeat(char ch);
01062         GTEST_API_ bool IsAsciiWhiteSpace(char ch);
01063         GTEST_API_ bool IsAsciiWordChar(char ch);
01064         GTEST_API_ bool IsValidEscape(char ch);
01065         GTEST_API_ bool AtomMatchesChar(bool escaped, char pattern, char ch);
01066         GTEST_API_ bool ValidateRegex(const char* regex);
01067         GTEST_API_ bool MatchRegexAtHead(const char* regex, const char* str);
```

```
01068         GTEST_API_ bool MatchRepetitionAndRegexAtHead(bool escaped, char ch,
01069                 char repeat, const char* regex,
01070                 const char* str);
01071         GTEST_API_ bool MatchRegexAnywhere(const char* regex, const char* str);
01072
01073 #endif  // GTEST_USES_SIMPLE_RE
01074
01075         // Parses the command line for Google Test flags, without initializing
01076         // other parts of Google Test.
01077         GTEST_API_ void ParseGoogleTestFlagsOnly(int* argc, char** argv);
01078         GTEST_API_ void ParseGoogleTestFlagsOnly(int* argc, wchar_t** argv);
01079
01080 #if GTEST_HAS_DEATH_TEST
01081
01082         // Returns the message describing the last system error, regardless of the
01083         // platform.
01084         GTEST_API_ std::string GetLastErrnoDescription();
01085
01086         // Attempts to parse a string into a positive integer pointed to by the
01087         // number parameter.  Returns true if that is possible.
01088         // GTEST_HAS_DEATH_TEST implies that we have ::std::string, so we can use
01089         // it here.
01090
01091         template <typename Integer>
01092         bool ParseNaturalNumber(const ::std::string& str, Integer* number) {
01093             // Fail fast if the given string does not begin with a digit;
01094             // this bypasses strtoXXX's "optional leading whitespace and plus
01095             // or minus sign" semantics, which are undesirable here.
01096             if (str.empty() || !IsDigit(str[0])) {
01097                 return false;
01098             }
01099             errno = 0;
01100
01101             char* end;
01102             // BiggestConvertible is the largest integer type that system-provided
01103             // string-to-number conversion routines can return.
01104             using BiggestConvertible = unsigned long long; // NOLINT
01105
01106             const BiggestConvertible parsed = strtoull(str.c_str(), &end, 10); // NOLINT
01107             const bool parse_success = *end == '\0' && errno == 0;
01108
01109             GTEST_CHECK_(sizeof (Integer) <= sizeof (parsed));
01110
01111             const Integer result = static_cast<Integer> (parsed);
01112             if (parse_success && static_cast<BiggestConvertible> (result) == parsed) {
01113                 *number = result;
01114                 return true;
01115             }
01116             return false;
01117         }
01118 #endif  // GTEST_HAS_DEATH_TEST
01119
01120         // TestResult contains some private methods that should be hidden from
01121         // Google Test user but are required for testing. This class allow our tests
01122         // to access them.
01123         //
01124         // This class is supplied only for the purpose of testing Google Test's own
01125         // constructs. Do not use it in user tests, either directly or indirectly.
01126
01127         class TestResultAccessor {
01128         public:
01129
01130             static void RecordProperty(TestResult* test_result,
01131                     const std::string& xml_element,
01132                     const TestProperty& property) {
01133                 test_result->RecordProperty(xml_element, property);
01134             }
01135
01136             static void ClearTestPartResults(TestResult* test_result) {
01137                 test_result->ClearTestPartResults();
01138             }
01139
01140             static const std::vector<testing::TestPartResult>& test_part_results(
01141                     const TestResult& test_result) {
01142                 return test_result.test_part_results();
01143             }
01144         };
01145
01146 #if GTEST_CAN_STREAM_RESULTS_
01147
01148         // Streams test results to the given port on the given host machine.
01149
01150         class StreamingListener : public EmptyTestEventListener {
01151         public:
01152             // Abstract base class for writing strings to a socket.
01153
01154             class AbstractSocketWriter {
```

```
01155             public:
01156
01157                 virtual ~AbstractSocketWriter() {
01158                 }
01159
01160                 // Sends a string to the socket.
01161                 virtual void Send(const std::string& message) = 0;
01162
01163                 // Closes the socket.
01164
01165                 virtual void CloseConnection() {
01166                 }
01167
01168                 // Sends a string and a newline to the socket.
01169
01170                 void SendLn(const std::string& message) {
01171                     Send(message + "\n");
01172                 }
01173             };
01174
01175             // Concrete class for actually writing strings to a socket.
01176
01177             class SocketWriter : public AbstractSocketWriter {
01178             public:
01179
01180                 SocketWriter(const std::string& host, const std::string& port)
01181                 : sockfd_(-1), host_name_(host), port_num_(port) {
01182                     MakeConnection();
01183                 }
01184
01185                 ~SocketWriter() override {
01186                     if (sockfd_ != -1) CloseConnection();
01187                 }
01188
01189                 // Sends a string to the socket.
01190
01191                 void Send(const std::string& message) override {
01192                     GTEST_CHECK_(sockfd_ != -1)
01193                             << "Send() can be called only when there is a connection.";
01194
01195                     const auto len = static_cast<size_t> (message.length());
01196                     if (write(sockfd_, message.c_str(), len) != static_cast<ssize_t> (len)) {
01197                         GTEST_LOG_(WARNING) << "stream_result_to: failed to stream to "
01198                                 << host_name_ << ":" << port_num_;
01199                     }
01200                 }
01201
01202             private:
01203                 // Creates a client socket and connects to the server.
01204                 void MakeConnection();
01205
01206                 // Closes the socket.
01207
01208                 void CloseConnection() override {
01209                     GTEST_CHECK_(sockfd_ != -1)
01210                             << "CloseConnection() can be called only when there is a connection.";
01211
01212                     close(sockfd_);
01213                     sockfd_ = -1;
01214                 }
01215
01216                 int sockfd_; // socket file descriptor
01217                 const std::string host_name_;
01218                 const std::string port_num_;
01219
01220                 SocketWriter(const SocketWriter&) = delete;
01221                 SocketWriter& operator=(const SocketWriter&) = delete;
01222             }; // class SocketWriter
01223
01224             // Escapes '=', '&', '%', and '\n' characters in str as "%xx".
01225             static std::string UrlEncode(const char* str);
01226
01227             StreamingListener(const std::string& host, const std::string& port)
01228             : socket_writer_(new SocketWriter(host, port)) {
01229                 Start();
01230             }
01231
01232             explicit StreamingListener(AbstractSocketWriter* socket_writer)
01233             : socket_writer_(socket_writer) {
01234                 Start();
01235             }
01236
01237             void OnTestProgramStart(const UnitTest& /* unit_test */) override {
01238                 SendLn("event=TestProgramStart");
01239             }
01240
01241             void OnTestProgramEnd(const UnitTest& unit_test) override {
```

```
01242                 // Note that Google Test current only report elapsed time for each
01243                 // test iteration, not for the entire test program.
01244                 SendLn("event=TestProgramEnd&passed=" + FormatBool(unit_test.Passed()));
01245
01246                 // Notify the streaming server to stop.
01247                 socket_writer_->CloseConnection();
01248             }
01249
01250             void OnTestIterationStart(const UnitTest& /* unit_test */,
01251                     int iteration) override {
01252                 SendLn("event=TestIterationStart&iteration=" +
01253                         StreamableToString(iteration));
01254             }
01255
01256             void OnTestIterationEnd(const UnitTest& unit_test,
01257                     int /* iteration */) override {
01258                 SendLn("event=TestIterationEnd&passed=" + FormatBool(unit_test.Passed()) +
01259                         "&elapsed_time=" + StreamableToString(unit_test.elapsed_time()) +
01260                         "ms");
01261             }
01262
01263             // Note that "event=TestCaseStart" is a wire format and has to remain
01264             // "case" for compatibility
01265
01266             void OnTestSuiteStart(const TestSuite& test_suite) override {
01267                 SendLn(std::string("event=TestCaseStart&name=") + test_suite.name());
01268             }
01269
01270             // Note that "event=TestCaseEnd" is a wire format and has to remain
01271             // "case" for compatibility
01272
01273             void OnTestSuiteEnd(const TestSuite& test_suite) override {
01274                 SendLn("event=TestCaseEnd&passed=" + FormatBool(test_suite.Passed()) +
01275                         "&elapsed_time=" + StreamableToString(test_suite.elapsed_time()) +
01276                         "ms");
01277             }
01278
01279             void OnTestStart(const TestInfo& test_info) override {
01280                 SendLn(std::string("event=TestStart&name=") + test_info.name());
01281             }
01282
01283             void OnTestEnd(const TestInfo& test_info) override {
01284                 SendLn("event=TestEnd&passed=" +
01285                         FormatBool((test_info.result())->Passed()) + "&elapsed_time=" +
01286                         StreamableToString((test_info.result())->elapsed_time()) + "ms");
01287             }
01288
01289             void OnTestPartResult(const TestPartResult& test_part_result) override {
01290                 const char* file_name = test_part_result.file_name();
01291                 if (file_name == nullptr) file_name = "";
01292                 SendLn("event=TestPartResult&file=" + UrlEncode(file_name) +
01293                         "&line=" + StreamableToString(test_part_result.line_number()) +
01294                         "&message=" + UrlEncode(test_part_result.message()));
01295             }
01296
01297         private:
01298             // Sends the given message and a newline to the socket.
01299
01300             void SendLn(const std::string& message) {
01301                 socket_writer_->SendLn(message);
01302             }
01303
01304             // Called at the start of streaming to notify the receiver what
01305             // protocol we are using.
01306
01307             void Start() {
01308                 SendLn("gtest_streaming_protocol_version=1.0");
01309             }
01310
01311             std::string FormatBool(bool value) {
01312                 return value ? "1" : "0";
01313             }
01314
01315             const std::unique_ptr<AbstractSocketWriter> socket_writer_;
01316
01317             StreamingListener(const StreamingListener&) = delete;
01318             StreamingListener& operator=(const StreamingListener&) = delete;
01319         }; // class StreamingListener
01320
01321 #endif  // GTEST_CAN_STREAM_RESULTS_
01322
01323     } // namespace internal
01324 } // namespace testing
01325
01326 GTEST_DISABLE_MSC_WARNINGS_POP_() //  4251
01327
01328 #endif  // GOOGLETEST_SRC_GTEST_INTERNAL_INL_H_
```

18.286 gtest-matchers.cc

```
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00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // The Google C++ Testing and Mocking Framework (Google Test)
00031 //
00032 // This file implements just enough of the matcher interface to allow
00033 // EXPECT_DEATH and friends to accept a matcher argument.
00034
00035 #include "gtest/gtest-matchers.h"
00036
00037 #include <string>
00038
00039 #include "gtest/internal/gtest-internal.h"
00040 #include "gtest/internal/gtest-port.h"
00041
00042 namespace testing {
00043
00044     // Constructs a matcher that matches a const std::string& whose value is
00045     // equal to s.
00046
00047     Matcher<const std::string&>::Matcher(const std::string& s) {
00048         *this = Eq(s);
00049     }
00050
00051     // Constructs a matcher that matches a const std::string& whose value is
00052     // equal to s.
00053
00054     Matcher<const std::string&>::Matcher(const char* s) {
00055         *this = Eq(std::string(s));
00056     }
00057
00058     // Constructs a matcher that matches a std::string whose value is equal to
00059     // s.
00060
00061     Matcher<std::string>::Matcher(const std::string& s) {
00062         *this = Eq(s);
00063     }
00064
00065     // Constructs a matcher that matches a std::string whose value is equal to
00066     // s.
00067
00068     Matcher<std::string>::Matcher(const char* s) {
00069         *this = Eq(std::string(s));
00070     }
00071
00072 #if GTEST_INTERNAL_HAS_STRING_VIEW
00073     // Constructs a matcher that matches a const StringView& whose value is
00074     // equal to s.
00075
00076     Matcher<const internal::StringView&>::Matcher(const std::string& s) {
00077         *this = Eq(s);
00078     }
00079
00080     // Constructs a matcher that matches a const StringView& whose value is
00081     // equal to s.
00082
00083     Matcher<const internal::StringView&>::Matcher(const char* s) {
00084         *this = Eq(std::string(s));
00085     }
```

```
00086
00087     // Constructs a matcher that matches a const StringView& whose value is
00088     // equal to s.
00089
00090     Matcher<const internal::StringView&>::Matcher(internal::StringView s) {
00091         *this = Eq(std::string(s));
00092     }
00093
00094     // Constructs a matcher that matches a StringView whose value is equal to
00095     // s.
00096
00097     Matcher<internal::StringView>::Matcher(const std::string& s) {
00098         *this = Eq(s);
00099     }
00100
00101     // Constructs a matcher that matches a StringView whose value is equal to
00102     // s.
00103
00104     Matcher<internal::StringView>::Matcher(const char* s) {
00105         *this = Eq(std::string(s));
00106     }
00107
00108     // Constructs a matcher that matches a StringView whose value is equal to
00109     // s.
00110
00111     Matcher<internal::StringView>::Matcher(internal::StringView s) {
00112         *this = Eq(std::string(s));
00113     }
00114 #endif  // GTEST_INTERNAL_HAS_STRING_VIEW
00115
00116 } // namespace testing
```

18.287 gtest-port.cc

```
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00029
00030 #include "gtest/internal/gtest-port.h"
00031
00032 #include <limits.h>
00033 #include <stdio.h>
00034 #include <stdlib.h>
00035 #include <string.h>
00036
00037 #include <cstdint>
00038 #include <fstream>
00039 #include <memory>
00040
00041 #if GTEST_OS_WINDOWS
00042 #include <io.h>
00043 #include <sys/stat.h>
00044 #include <windows.h>
00045
00046 #include <map>  // Used in ThreadLocal.
00047 #ifdef _MSC_VER
00048 #include <crtdbg.h>
00049 #endif  // _MSC_VER
00050 #else
00051 #include <unistd.h>
```

```
00052 #endif  // GTEST_OS_WINDOWS
00053
00054 #if GTEST_OS_MAC
00055 #include <mach/mach_init.h>
00056 #include <mach/task.h>
00057 #include <mach/vm_map.h>
00058 #endif  // GTEST_OS_MAC
00059
00060 #if GTEST_OS_DRAGONFLY || GTEST_OS_FREEBSD || GTEST_OS_GNU_KFREEBSD || \
00061     GTEST_OS_NETBSD || GTEST_OS_OPENBSD
00062 #include <sys/sysctl.h>
00063 #if GTEST_OS_DRAGONFLY || GTEST_OS_FREEBSD || GTEST_OS_GNU_KFREEBSD
00064 #include <sys/user.h>
00065 #endif
00066 #endif
00067
00068 #if GTEST_OS_QNX
00069 #include <devctl.h>
00070 #include <fcntl.h>
00071 #include <sys/procfs.h>
00072 #endif  // GTEST_OS_QNX
00073
00074 #if GTEST_OS_AIX
00075 #include <procinfo.h>
00076 #include <sys/types.h>
00077 #endif  // GTEST_OS_AIX
00078
00079 #if GTEST_OS_FUCHSIA
00080 #include <zircon/process.h>
00081 #include <zircon/syscalls.h>
00082 #endif  // GTEST_OS_FUCHSIA
00083
00084 #include "gtest/gtest-message.h"
00085 #include "gtest/gtest-spi.h"
00086 #include "gtest/internal/gtest-internal.h"
00087 #include "gtest/internal/gtest-string.h"
00088 #include "src/gtest-internal-inl.h"
00089
00090 namespace testing {
00091     namespace internal {
00092
00093 #if GTEST_OS_LINUX || GTEST_OS_GNU_HURD
00094
00095         namespace {
00096
00097             template <typename T>
00098             T ReadProcFileField(const std::string& filename, int field) {
00099                 std::string dummy;
00100                 std::ifstream file(filename.c_str());
00101                 while (field-- > 0) {
00102                     file » dummy;
00103                 }
00104                 T output = 0;
00105                 file » output;
00106                 return output;
00107             }
00108         } // namespace
00109
00110         // Returns the number of active threads, or 0 when there is an error.
00111
00112         size_t GetThreadCount() {
00113             const std::string filename =
00114                     (Message() « "/proc/" « getpid() « "/stat").GetString();
00115             return ReadProcFileField<size_t>(filename, 19);
00116         }
00117
00118 #elif GTEST_OS_MAC
00119
00120         size_t GetThreadCount() {
00121             const task_t task = mach_task_self();
00122             mach_msg_type_number_t thread_count;
00123             thread_act_array_t thread_list;
00124             const kern_return_t status = task_threads(task, &thread_list, &thread_count);
00125             if (status == KERN_SUCCESS) {
00126                 // task_threads allocates resources in thread_list and we need to free them
00127                 // to avoid leaks.
00128                 vm_deallocate(task, reinterpret_cast<vm_address_t> (thread_list),
00129                               sizeof (thread_t) * thread_count);
00130                 return static_cast<size_t> (thread_count);
00131             } else {
00132                 return 0;
00133             }
00134         }
00135
00136 #elif GTEST_OS_DRAGONFLY || GTEST_OS_FREEBSD || GTEST_OS_GNU_KFREEBSD || \
00137     GTEST_OS_NETBSD
00138
```

```
00139 #if GTEST_OS_NETBSD
00140 #undef KERN_PROC
00141 #define KERN_PROC KERN_PROC2
00142 #define kinfo_proc kinfo_proc2
00143 #endif
00144
00145 #if GTEST_OS_DRAGONFLY
00146 #define KP_NLWP(kp) (kp.kp_nthreads)
00147 #elif GTEST_OS_FREEBSD || GTEST_OS_GNU_KFREEBSD
00148 #define KP_NLWP(kp) (kp.ki_numthreads)
00149 #elif GTEST_OS_NETBSD
00150 #define KP_NLWP(kp) (kp.p_nlwps)
00151 #endif
00152
00153         // Returns the number of threads running in the process, or 0 to indicate that
00154         // we cannot detect it.
00155
00156         size_t GetThreadCount() {
00157             int mib[] = {
00158                 CTL_KERN,
00159                 KERN_PROC,
00160                 KERN_PROC_PID,
00161                 getpid(),
00162 #if GTEST_OS_NETBSD
00163                 sizeof (struct kinfo_proc),
00164                 1,
00165 #endif
00166             };
00167             u_int miblen = sizeof (mib) / sizeof (mib[0]);
00168             struct kinfo_proc info;
00169             size_t size = sizeof (info);
00170             if (sysctl(mib, miblen, &info, &size, NULL, 0)) {
00171                 return 0;
00172             }
00173             return static_cast<size_t> (KP_NLWP(info));
00174         }
00175 #elif GTEST_OS_OPENBSD
00176
00177         // Returns the number of threads running in the process, or 0 to indicate that
00178         // we cannot detect it.
00179
00180         size_t GetThreadCount() {
00181             int mib[] = {
00182                 CTL_KERN,
00183                 KERN_PROC,
00184                 KERN_PROC_PID | KERN_PROC_SHOW_THREADS,
00185                 getpid(),
00186                 sizeof (struct kinfo_proc),
00187                 0,
00188             };
00189             u_int miblen = sizeof (mib) / sizeof (mib[0]);
00190
00191             // get number of structs
00192             size_t size;
00193             if (sysctl(mib, miblen, NULL, &size, NULL, 0)) {
00194                 return 0;
00195             }
00196
00197             mib[5] = static_cast<int> (size / static_cast<size_t> (mib[4]));
00198
00199             // populate array of structs
00200             struct kinfo_proc info[mib[5]];
00201             if (sysctl(mib, miblen, &info, &size, NULL, 0)) {
00202                 return 0;
00203             }
00204
00205             // exclude empty members
00206             size_t nthreads = 0;
00207             for (size_t i = 0; i < size / static_cast<size_t> (mib[4]); i++) {
00208                 if (info[i].p_tid != -1) nthreads++;
00209             }
00210             return nthreads;
00211         }
00212
00213 #elif GTEST_OS_QNX
00214
00215         // Returns the number of threads running in the process, or 0 to indicate that
00216         // we cannot detect it.
00217
00218         size_t GetThreadCount() {
00219             const int fd = open("/proc/self/as", O_RDONLY);
00220             if (fd < 0) {
00221                 return 0;
00222             }
00223             procfs_info process_info;
00224             const int status =
00225                     devctl(fd, DCMD_PROC_INFO, &process_info, sizeof (process_info), nullptr);
```

```
00226             close(fd);
00227             if (status == EOK) {
00228                 return static_cast<size_t> (process_info.num_threads);
00229             } else {
00230                 return 0;
00231             }
00232         }
00233
00234 #elif GTEST_OS_AIX
00235
00236         size_t GetThreadCount() {
00237             struct procentry64 entry;
00238             pid_t pid = getpid();
00239             int status = getprocs64(&entry, sizeof (entry), nullptr, 0, &pid, 1);
00240             if (status == 1) {
00241                 return entry.pi_thcount;
00242             } else {
00243                 return 0;
00244             }
00245         }
00246
00247 #elif GTEST_OS_FUCHSIA
00248
00249         size_t GetThreadCount() {
00250             int dummy_buffer;
00251             size_t avail;
00252             zx_status_t status =
00253                     zx_object_get_info(zx_process_self(), ZX_INFO_PROCESS_THREADS,
00254                     &dummy_buffer, 0, nullptr, &avail);
00255             if (status == ZX_OK) {
00256                 return avail;
00257             } else {
00258                 return 0;
00259             }
00260         }
00261
00262 #else
00263
00264         size_t GetThreadCount() {
00265             // There's no portable way to detect the number of threads, so we just
00266             // return 0 to indicate that we cannot detect it.
00267             return 0;
00268         }
00269
00270 #endif  // GTEST_OS_LINUX
00271
00272 #if GTEST_IS_THREADSAFE && GTEST_OS_WINDOWS
00273
00274         AutoHandle::AutoHandle() : handle_(INVALID_HANDLE_VALUE) {
00275         }
00276
00277         AutoHandle::AutoHandle(Handle handle) : handle_(handle) {
00278         }
00279
00280         AutoHandle::~AutoHandle() {
00281             Reset();
00282         }
00283
00284         AutoHandle::Handle AutoHandle::Get() const {
00285             return handle_;
00286         }
00287
00288         void AutoHandle::Reset() {
00289             Reset(INVALID_HANDLE_VALUE);
00290         }
00291
00292         void AutoHandle::Reset(HANDLE handle) {
00293             // Resetting with the same handle we already own is invalid.
00294             if (handle_ != handle) {
00295                 if (IsCloseable()) {
00296                     ::CloseHandle(handle_);
00297                 }
00298                 handle_ = handle;
00299             } else {
00300                 GTEST_CHECK_(!IsCloseable())
00301                         << "Resetting a valid handle to itself is likely a programmer error "
00302                         "and thus not allowed.";
00303             }
00304         }
00305
00306         bool AutoHandle::IsCloseable() const {
00307             // Different Windows APIs may use either of these values to represent an
00308             // invalid handle.
00309             return handle_ != nullptr && handle_ != INVALID_HANDLE_VALUE;
00310         }
00311
00312         Mutex::Mutex()
```

```
00313         : owner_thread_id_(0),
00314         type_(kDynamic),
00315         critical_section_init_phase_(0),
00316         critical_section_(new CRITICAL_SECTION) {
00317             ::InitializeCriticalSection(critical_section_);
00318         }
00319
00320         Mutex::~Mutex() {
00321             // Static mutexes are leaked intentionally. It is not thread-safe to try
00322             // to clean them up.
00323             if (type_ == kDynamic) {
00324                 ::DeleteCriticalSection(critical_section_);
00325                 delete critical_section_;
00326                 critical_section_ = nullptr;
00327             }
00328         }
00329
00330         void Mutex::Lock() {
00331             ThreadSafeLazyInit();
00332             ::EnterCriticalSection(critical_section_);
00333             owner_thread_id_ = ::GetCurrentThreadId();
00334         }
00335
00336         void Mutex::Unlock() {
00337             ThreadSafeLazyInit();
00338             // We don't protect writing to owner_thread_id_ here, as it's the
00339             // caller's responsibility to ensure that the current thread holds the
00340             // mutex when this is called.
00341             owner_thread_id_ = 0;
00342             ::LeaveCriticalSection(critical_section_);
00343         }
00344
00345         // Does nothing if the current thread holds the mutex. Otherwise, crashes
00346         // with high probability.
00347
00348         void Mutex::AssertHeld() {
00349             ThreadSafeLazyInit();
00350             GTEST_CHECK_(owner_thread_id_ == ::GetCurrentThreadId())
00351                     << "The current thread is not holding the mutex @" << this;
00352         }
00353
00354         namespace {
00355
00356 #ifdef _MSC_VER
00357             // Use the RAII idiom to flag mem allocs that are intentionally never
00358             // deallocated. The motivation is to silence the false positive mem leaks
00359             // that are reported by the debug version of MS's CRT which can only detect
00360             // if an alloc is missing a matching deallocation.
00361             // Example:
00362             //    MemoryIsNotDeallocated memory_is_not_deallocated;
00363             //    critical_section_ = new CRITICAL_SECTION;
00364             //
00365
00366             class MemoryIsNotDeallocated {
00367             public:
00368
00369                 MemoryIsNotDeallocated() : old_crtdbg_flag_(0) {
00370                     old_crtdbg_flag_ = _CrtSetDbgFlag(_CRTDBG_REPORT_FLAG);
00371                     // Set heap allocation block type to _IGNORE_BLOCK so that MS debug CRT
00372                     // doesn't report mem leak if there's no matching deallocation.
00373                     (void) _CrtSetDbgFlag(old_crtdbg_flag_ & ~_CRTDBG_ALLOC_MEM_DF);
00374                 }
00375
00376                 ~MemoryIsNotDeallocated() {
00377                     // Restore the original _CRTDBG_ALLOC_MEM_DF flag
00378                     (void) _CrtSetDbgFlag(old_crtdbg_flag_);
00379                 }
00380
00381             private:
00382                 int old_crtdbg_flag_;
00383
00384                 MemoryIsNotDeallocated(const MemoryIsNotDeallocated&) = delete;
00385                 MemoryIsNotDeallocated& operator=(const MemoryIsNotDeallocated&) = delete;
00386             };
00387 #endif  // _MSC_VER
00388
00389         } // namespace
00390
00391         // Initializes owner_thread_id_ and critical_section_ in static mutexes.
00392
00393         void Mutex::ThreadSafeLazyInit() {
00394             // Dynamic mutexes are initialized in the constructor.
00395             if (type_ == kStatic) {
00396                 switch (
00397                         ::InterlockedCompareExchange(&critical_section_init_phase_, 1L, 0L)) {
00398                     case 0:
00399                         // If critical_section_init_phase_ was 0 before the exchange, we
```

```
00400                         // are the first to test it and need to perform the initialization.
00401                         owner_thread_id_ = 0;
00402                       {
00403                         // Use RAII to flag that following mem alloc is never deallocated.
00404 #ifdef _MSC_VER
00405                         MemoryIsNotDeallocated memory_is_not_deallocated;
00406 #endif  // _MSC_VER
00407                         critical_section_ = new CRITICAL_SECTION;
00408                       }
00409                         ::InitializeCriticalSection(critical_section_);
00410                         // Updates the critical_section_init_phase_ to 2 to signal
00411                         // initialization complete.
00412                         GTEST_CHECK_(::InterlockedCompareExchange(&critical_section_init_phase_,
00413                                 2L, 1L) == 1L);
00414                         break;
00415                       case 1:
00416                         // Somebody else is already initializing the mutex; spin until they
00417                         // are done.
00418                         while (::InterlockedCompareExchange(&critical_section_init_phase_, 2L,
00419                                 2L) != 2L) {
00420                           // Possibly yields the rest of the thread's time slice to other
00421                           // threads.
00422                           ::Sleep(0);
00423                         }
00424                         break;
00425
00426                       case 2:
00427                         break; // The mutex is already initialized and ready for use.
00428
00429                       default:
00430                         GTEST_CHECK_(false)
00431                                 << "Unexpected value of critical_section_init_phase_ "
00432                                 << "while initializing a static mutex.";
00433                     }
00434                   }
00435                 }
00436
00437                 namespace {
00438
00439                   class ThreadWithParamSupport : public ThreadWithParamBase {
00440                   public:
00441
00442                     static HANDLE CreateThread(Runnable* runnable,
00443                             Notification* thread_can_start) {
00444                       ThreadMainParam* param = new ThreadMainParam(runnable, thread_can_start);
00445                       DWORD thread_id;
00446                       HANDLE thread_handle = ::CreateThread(
00447                               nullptr, // Default security.
00448                               0, // Default stack size.
00449                               &ThreadWithParamSupport::ThreadMain,
00450                               param, // Parameter to ThreadMainStatic
00451                               0x0, // Default creation flags.
00452                               &thread_id); // Need a valid pointer for the call to work under Win98.
00453                       GTEST_CHECK_(thread_handle != nullptr)
00454                               << "CreateThread failed with error " << ::GetLastError() << ".";
00455                       if (thread_handle == nullptr) {
00456                         delete param;
00457                       }
00458                       return thread_handle;
00459                     }
00460
00461                   private:
00462
00463                     struct ThreadMainParam {
00464
00465                       ThreadMainParam(Runnable* runnable, Notification* thread_can_start)
00466                       : runnable_(runnable), thread_can_start_(thread_can_start) {
00467                       }
00468                       std::unique_ptr<Runnable> runnable_;
00469                       // Does not own.
00470                       Notification* thread_can_start_;
00471                     };
00472
00473                     static DWORD WINAPI ThreadMain(void* ptr) {
00474                       // Transfers ownership.
00475                       std::unique_ptr<ThreadMainParam> param(static_cast<ThreadMainParam*> (ptr));
00476                       if (param->thread_can_start_ != nullptr)
00477                         param->thread_can_start_->WaitForNotification();
00478                       param->runnable_->Run();
00479                       return 0;
00480                     }
00481
00482                     // Prohibit instantiation.
00483                     ThreadWithParamSupport();
00484
00485                     ThreadWithParamSupport(const ThreadWithParamSupport&) = delete;
00486                     ThreadWithParamSupport& operator=(const ThreadWithParamSupport&) = delete;
```

```
00487             };
00488
00489         } // namespace
00490
00491         ThreadWithParamBase::ThreadWithParamBase(Runnable* runnable,
00492                 Notification* thread_can_start)
00493         : thread_(
00494         ThreadWithParamSupport::CreateThread(runnable, thread_can_start)) {
00495         }
00496
00497         ThreadWithParamBase::~ThreadWithParamBase() {
00498             Join();
00499         }
00500
00501         void ThreadWithParamBase::Join() {
00502             GTEST_CHECK_(::WaitForSingleObject(thread_.Get(), INFINITE) == WAIT_OBJECT_0)
00503                     << "Failed to join the thread with error " << ::GetLastError() << ".";
00504         }
00505
00506         // Maps a thread to a set of ThreadIdToThreadLocals that have values
00507         // instantiated on that thread and notifies them when the thread exits.  A
00508         // ThreadLocal instance is expected to persist until all threads it has
00509         // values on have terminated.
00510
00511         class ThreadLocalRegistryImpl {
00512         public:
00513             // Registers thread_local_instance as having value on the current thread.
00514             // Returns a value that can be used to identify the thread from other threads.
00515
00516             static ThreadLocalValueHolderBase* GetValueOnCurrentThread(
00517                     const ThreadLocalBase* thread_local_instance) {
00518 #ifdef _MSC_VER
00519                 MemoryIsNotDeallocated memory_is_not_deallocated;
00520 #endif  // _MSC_VER
00521                 DWORD current_thread = ::GetCurrentThreadId();
00522                 MutexLock lock(&mutex_);
00523                 ThreadIdToThreadLocals * const thread_to_thread_locals =
00524                         GetThreadLocalsMapLocked();
00525                 ThreadIdToThreadLocals::iterator thread_local_pos =
00526                         thread_to_thread_locals->find(current_thread);
00527                 if (thread_local_pos == thread_to_thread_locals->end()) {
00528                     thread_local_pos =
00529                             thread_to_thread_locals
00530                             ->insert(std::make_pair(current_thread, ThreadLocalValues()))
00531                             .first;
00532                     StartWatcherThreadFor(current_thread);
00533                 }
00534                 ThreadLocalValues& thread_local_values = thread_local_pos->second;
00535                 ThreadLocalValues::iterator value_pos =
00536                         thread_local_values.find(thread_local_instance);
00537                 if (value_pos == thread_local_values.end()) {
00538                     value_pos =
00539                             thread_local_values
00540                             .insert(std::make_pair(
00541                             thread_local_instance,
00542                             std::shared_ptr<ThreadLocalValueHolderBase>(
00543                             thread_local_instance->NewValueForCurrentThread())))
00544                             .first;
00545                 }
00546                 return value_pos->second.get();
00547             }
00548
00549             static void OnThreadLocalDestroyed(
00550                     const ThreadLocalBase* thread_local_instance) {
00551                 std::vector<std::shared_ptr<ThreadLocalValueHolderBase> > value_holders;
00552                 // Clean up the ThreadLocalValues data structure while holding the lock, but
00553                 // defer the destruction of the ThreadLocalValueHolderBases.
00554                 {
00555                     MutexLock lock(&mutex_);
00556                     ThreadIdToThreadLocals * const thread_to_thread_locals =
00557                             GetThreadLocalsMapLocked();
00558                     for (ThreadIdToThreadLocals::iterator it =
00559                             thread_to_thread_locals->begin();
00560                             it != thread_to_thread_locals->end(); ++it) {
00561                         ThreadLocalValues& thread_local_values = it->second;
00562                         ThreadLocalValues::iterator value_pos =
00563                                 thread_local_values.find(thread_local_instance);
00564                         if (value_pos != thread_local_values.end()) {
00565                             value_holders.push_back(value_pos->second);
00566                             thread_local_values.erase(value_pos);
00567                             // This 'if' can only be successful at most once, so theoretically we
00568                             // could break out of the loop here, but we don't bother doing so.
00569                         }
00570                     }
00571                 }
00572                 // Outside the lock, let the destructor for 'value_holders' deallocate the
00573                 // ThreadLocalValueHolderBases.
```

```
00574             }
00575
00576             static void OnThreadExit(DWORD thread_id) {
00577                 GTEST_CHECK_(thread_id != 0) << ::GetLastError();
00578                 std::vector<std::shared_ptr<ThreadLocalValueHolderBase> > value_holders;
00579                 // Clean up the ThreadIdToThreadLocals data structure while holding the
00580                 // lock, but defer the destruction of the ThreadLocalValueHolderBases.
00581                 {
00582                     MutexLock lock(&mutex_);
00583                     ThreadIdToThreadLocals * const thread_to_thread_locals =
00584                             GetThreadLocalsMapLocked();
00585                     ThreadIdToThreadLocals::iterator thread_local_pos =
00586                             thread_to_thread_locals->find(thread_id);
00587                     if (thread_local_pos != thread_to_thread_locals->end()) {
00588                         ThreadLocalValues& thread_local_values = thread_local_pos->second;
00589                         for (ThreadLocalValues::iterator value_pos =
00590                                 thread_local_values.begin();
00591                                 value_pos != thread_local_values.end(); ++value_pos) {
00592                             value_holders.push_back(value_pos->second);
00593                         }
00594                         thread_to_thread_locals->erase(thread_local_pos);
00595                     }
00596                 }
00597                 // Outside the lock, let the destructor for 'value_holders' deallocate the
00598                 // ThreadLocalValueHolderBases.
00599             }
00600
00601         private:
00602             // In a particular thread, maps a ThreadLocal object to its value.
00603             typedef std::map<const ThreadLocalBase*,
00604             std::shared_ptr<ThreadLocalValueHolderBase> >
00605             ThreadLocalValues;
00606             // Stores all ThreadIdToThreadLocals having values in a thread, indexed by
00607             // thread's ID.
00608             typedef std::map<DWORD, ThreadLocalValues> ThreadIdToThreadLocals;
00609
00610             // Holds the thread id and thread handle that we pass from
00611             // StartWatcherThreadFor to WatcherThreadFunc.
00612             typedef std::pair<DWORD, HANDLE> ThreadIdAndHandle;
00613
00614             static void StartWatcherThreadFor(DWORD thread_id) {
00615                 // The returned handle will be kept in thread_map and closed by
00616                 // watcher_thread in WatcherThreadFunc.
00617                 HANDLE thread =
00618                         ::OpenThread(SYNCHRONIZE | THREAD_QUERY_INFORMATION, FALSE, thread_id);
00619                 GTEST_CHECK_(thread != nullptr);
00620                 // We need to pass a valid thread ID pointer into CreateThread for it
00621                 // to work correctly under Win98.
00622                 DWORD watcher_thread_id;
00623                 HANDLE watcher_thread = ::CreateThread(
00624                         nullptr, // Default security.
00625                         0, // Default stack size
00626                         &ThreadLocalRegistryImpl::WatcherThreadFunc,
00627                         reinterpret_cast<LPVOID> (new ThreadIdAndHandle(thread_id, thread)),
00628                         CREATE_SUSPENDED, &watcher_thread_id);
00629                 GTEST_CHECK_(watcher_thread != nullptr)
00630                         << "CreateThread failed with error " << ::GetLastError() << ".";
00631                 // Give the watcher thread the same priority as ours to avoid being
00632                 // blocked by it.
00633                 ::SetThreadPriority(watcher_thread,
00634                         ::GetThreadPriority(::GetCurrentThread()));
00635                 ::ResumeThread(watcher_thread);
00636                 ::CloseHandle(watcher_thread);
00637             }
00638
00639             // Monitors exit from a given thread and notifies those
00640             // ThreadIdToThreadLocals about thread termination.
00641
00642             static DWORD WINAPI WatcherThreadFunc(LPVOID param) {
00643                 const ThreadIdAndHandle* tah =
00644                         reinterpret_cast<const ThreadIdAndHandle*> (param);
00645                 GTEST_CHECK_(::WaitForSingleObject(tah->second, INFINITE) == WAIT_OBJECT_0);
00646                 OnThreadExit(tah->first);
00647                 ::CloseHandle(tah->second);
00648                 delete tah;
00649                 return 0;
00650             }
00651
00652             // Returns map of thread local instances.
00653
00654             static ThreadIdToThreadLocals* GetThreadLocalsMapLocked() {
00655                 mutex_.AssertHeld();
00656 #ifdef _MSC_VER
00657                 MemoryIsNotDeallocated memory_is_not_deallocated;
00658 #endif  // _MSC_VER
00659                 static ThreadIdToThreadLocals* map = new ThreadIdToThreadLocals();
00660                 return map;
```

```
00661             }
00662
00663             // Protects access to GetThreadLocalsMapLocked() and its return value.
00664             static Mutex mutex_;
00665             // Protects access to GetThreadMapLocked() and its return value.
00666             static Mutex thread_map_mutex_;
00667         };
00668
00669         Mutex ThreadLocalRegistryImpl::mutex_(Mutex::kStaticMutex); // NOLINT
00670         Mutex ThreadLocalRegistryImpl::thread_map_mutex_(
00671                 Mutex::kStaticMutex); // NOLINT
00672
00673         ThreadLocalValueHolderBase* ThreadLocalRegistry::GetValueOnCurrentThread(
00674                 const ThreadLocalBase* thread_local_instance) {
00675             return ThreadLocalRegistryImpl::GetValueOnCurrentThread(
00676                     thread_local_instance);
00677         }
00678
00679         void ThreadLocalRegistry::OnThreadLocalDestroyed(
00680                 const ThreadLocalBase* thread_local_instance) {
00681             ThreadLocalRegistryImpl::OnThreadLocalDestroyed(thread_local_instance);
00682         }
00683
00684 #endif  // GTEST_IS_THREADSAFE && GTEST_OS_WINDOWS
00685
00686 #if GTEST_USES_POSIX_RE
00687
00688         // Implements RE.  Currently only needed for death tests.
00689
00690         RE::~RE() {
00691             if (is_valid_) {
00692                 // regfree'ing an invalid regex might crash because the content
00693                 // of the regex is undefined. Since the regex's are essentially
00694                 // the same, one cannot be valid (or invalid) without the other
00695                 // being so too.
00696                 regfree(&partial_regex_);
00697                 regfree(&full_regex_);
00698             }
00699             free(const_cast<char*> (pattern_));
00700         }
00701
00702         // Returns true if and only if regular expression re matches the entire str.
00703
00704         bool RE::FullMatch(const char* str, const RE& re) {
00705             if (!re.is_valid_) return false;
00706
00707             regmatch_t match;
00708             return regexec(&re.full_regex_, str, 1, &match, 0) == 0;
00709         }
00710
00711         // Returns true if and only if regular expression re matches a substring of
00712         // str (including str itself).
00713
00714         bool RE::PartialMatch(const char* str, const RE& re) {
00715             if (!re.is_valid_) return false;
00716
00717             regmatch_t match;
00718             return regexec(&re.partial_regex_, str, 1, &match, 0) == 0;
00719         }
00720
00721         // Initializes an RE from its string representation.
00722
00723         void RE::Init(const char* regex) {
00724             pattern_ = posix::StrDup(regex);
00725
00726             // Reserves enough bytes to hold the regular expression used for a
00727             // full match.
00728             const size_t full_regex_len = strlen(regex) + 10;
00729             char* const full_pattern = new char[full_regex_len];
00730
00731             snprintf(full_pattern, full_regex_len, "^(%s)$", regex);
00732             is_valid_ = regcomp(&full_regex_, full_pattern, REG_EXTENDED) == 0;
00733             // We want to call regcomp(&partial_regex_, ...) even if the
00734             // previous expression returns false.  Otherwise partial_regex_ may
00735             // not be properly initialized can may cause trouble when it's
00736             // freed.
00737             //
00738             // Some implementation of POSIX regex (e.g. on at least some
00739             // versions of Cygwin) doesn't accept the empty string as a valid
00740             // regex.  We change it to an equivalent form "()" to be safe.
00741             if (is_valid_) {
00742                 const char* const partial_regex = (*regex == '\0') ? "()" : regex;
00743                 is_valid_ = regcomp(&partial_regex_, partial_regex, REG_EXTENDED) == 0;
00744             }
00745             EXPECT_TRUE(is_valid_)
00746                     << "Regular expression \"" << regex
00747                     << "\" is not a valid POSIX Extended regular expression.";
```

```
00748
00749             delete[] full_pattern;
00750         }
00751
00752 #elif GTEST_USES_SIMPLE_RE
00753
00754         // Returns true if and only if ch appears anywhere in str (excluding the
00755         // terminating '\0' character).
00756
00757         bool IsInSet(char ch, const char* str) {
00758             return ch != '\0' && strchr(str, ch) != nullptr;
00759         }
00760
00761         // Returns true if and only if ch belongs to the given classification.
00762         // Unlike similar functions in <ctype.h>, these aren't affected by the
00763         // current locale.
00764
00765         bool IsAsciiDigit(char ch) {
00766             return '0' <= ch && ch <= '9';
00767         }
00768
00769         bool IsAsciiPunct(char ch) {
00770             return IsInSet(ch, "^-!\"#$%&'()*+,./:;<=>?@[\\]_`{|}~");
00771         }
00772
00773         bool IsRepeat(char ch) {
00774             return IsInSet(ch, "?*+");
00775         }
00776
00777         bool IsAsciiWhiteSpace(char ch) {
00778             return IsInSet(ch, " \f\n\r\t\v");
00779         }
00780
00781         bool IsAsciiWordChar(char ch) {
00782             return ('a' <= ch && ch <= 'z') || ('A' <= ch && ch <= 'Z') ||
00783                     ('0' <= ch && ch <= '9') || ch == '_';
00784         }
00785
00786         // Returns true if and only if "\\c" is a supported escape sequence.
00787
00788         bool IsValidEscape(char c) {
00789             return (IsAsciiPunct(c) || IsInSet(c, "dDfnrsStvwW"));
00790         }
00791
00792         // Returns true if and only if the given atom (specified by escaped and
00793         // pattern) matches ch.  The result is undefined if the atom is invalid.
00794
00795         bool AtomMatchesChar(bool escaped, char pattern_char, char ch) {
00796             if (escaped) { // "\\p" where p is pattern_char.
00797                 switch (pattern_char) {
00798                     case 'd':
00799                         return IsAsciiDigit(ch);
00800                     case 'D':
00801                         return !IsAsciiDigit(ch);
00802                     case 'f':
00803                         return ch == '\f';
00804                     case 'n':
00805                         return ch == '\n';
00806                     case 'r':
00807                         return ch == '\r';
00808                     case 's':
00809                         return IsAsciiWhiteSpace(ch);
00810                     case 'S':
00811                         return !IsAsciiWhiteSpace(ch);
00812                     case 't':
00813                         return ch == '\t';
00814                     case 'v':
00815                         return ch == '\v';
00816                     case 'w':
00817                         return IsAsciiWordChar(ch);
00818                     case 'W':
00819                         return !IsAsciiWordChar(ch);
00820                 }
00821                 return IsAsciiPunct(pattern_char) && pattern_char == ch;
00822             }
00823
00824             return (pattern_char == '.' && ch != '\n') || pattern_char == ch;
00825         }
00826
00827         // Helper function used by ValidateRegex() to format error messages.
00828
00829         static std::string FormatRegexSyntaxError(const char* regex, int index) {
00830             return (Message() << "Syntax error at index " << index
00831                     << " in simple regular expression \"" << regex << "\": ")
00832                     .GetString();
00833         }
00834
```

```
00835         // Generates non-fatal failures and returns false if regex is invalid;
00836         // otherwise returns true.
00837
00838         bool ValidateRegex(const char* regex) {
00839             if (regex == nullptr) {
00840                 ADD_FAILURE() << "NULL is not a valid simple regular expression.";
00841                 return false;
00842             }
00843
00844             bool is_valid = true;
00845
00846             // True if and only if ?, *, or + can follow the previous atom.
00847             bool prev_repeatable = false;
00848             for (int i = 0; regex[i]; i++) {
00849                 if (regex[i] == '\\') { // An escape sequence
00850                     i++;
00851                     if (regex[i] == '\0') {
00852                         ADD_FAILURE() << FormatRegexSyntaxError(regex, i - 1)
00853                                       << "'\\' cannot appear at the end.";
00854                         return false;
00855                     }
00856
00857                     if (!IsValidEscape(regex[i])) {
00858                         ADD_FAILURE() << FormatRegexSyntaxError(regex, i - 1)
00859                                       << "invalid escape sequence \"\\" << regex[i] << "\".";
00860                         is_valid = false;
00861                     }
00862                     prev_repeatable = true;
00863                 } else { // Not an escape sequence.
00864                     const char ch = regex[i];
00865
00866                     if (ch == '^' && i > 0) {
00867                         ADD_FAILURE() << FormatRegexSyntaxError(regex, i)
00868                                       << "'^' can only appear at the beginning.";
00869                         is_valid = false;
00870                     } else if (ch == '$' && regex[i + 1] != '\0') {
00871                         ADD_FAILURE() << FormatRegexSyntaxError(regex, i)
00872                                       << "'$' can only appear at the end.";
00873                         is_valid = false;
00874                     } else if (IsInSet(ch, "()[]{}|")) {
00875                         ADD_FAILURE() << FormatRegexSyntaxError(regex, i) << "'" << ch
00876                                       << "' is unsupported.";
00877                         is_valid = false;
00878                     } else if (IsRepeat(ch) && !prev_repeatable) {
00879                         ADD_FAILURE() << FormatRegexSyntaxError(regex, i) << "'" << ch
00880                                       << "' can only follow a repeatable token.";
00881                         is_valid = false;
00882                     }
00883
00884                     prev_repeatable = !IsInSet(ch, "^$?*+");
00885                 }
00886             }
00887
00888             return is_valid;
00889         }
00890
00891         // Matches a repeated regex atom followed by a valid simple regular
00892         // expression.  The regex atom is defined as c if escaped is false,
00893         // or \c otherwise.  repeat is the repetition meta character (?, *,
00894         // or +).  The behavior is undefined if str contains too many
00895         // characters to be indexable by size_t, in which case the test will
00896         // probably time out anyway.  We are fine with this limitation as
00897         // std::string has it too.
00898
00899         bool MatchRepetitionAndRegexAtHead(bool escaped, char c, char repeat,
00900                 const char* regex, const char* str) {
00901             const size_t min_count = (repeat == '+') ? 1 : 0;
00902             const size_t max_count = (repeat == '?') ? 1 : static_cast<size_t> (-1) - 1;
00903             // We cannot call numeric_limits::max() as it conflicts with the
00904             // max() macro on Windows.
00905
00906             for (size_t i = 0; i <= max_count; ++i) {
00907                 // We know that the atom matches each of the first i characters in str.
00908                 if (i >= min_count && MatchRegexAtHead(regex, str + i)) {
00909                     // We have enough matches at the head, and the tail matches too.
00910                     // Since we only care about *whether* the pattern matches str
00911                     // (as opposed to *how* it matches), there is no need to find a
00912                     // greedy match.
00913                     return true;
00914                 }
00915                 if (str[i] == '\0' || !AtomMatchesChar(escaped, c, str[i])) return false;
00916             }
00917             return false;
00918         }
00919
00920         // Returns true if and only if regex matches a prefix of str. regex must
00921         // be a valid simple regular expression and not start with "^", or the
```

```
00922         // result is undefined.
00923
00924         bool MatchRegexAtHead(const char* regex, const char* str) {
00925             if (*regex == '\0') // An empty regex matches a prefix of anything.
00926                 return true;
00927
00928             // "$" only matches the end of a string.  Note that regex being
00929             // valid guarantees that there's nothing after "$" in it.
00930             if (*regex == '$') return *str == '\0';
00931
00932             // Is the first thing in regex an escape sequence?
00933             const bool escaped = *regex == '\\';
00934             if (escaped) ++regex;
00935             if (IsRepeat(regex[1])) {
00936                 // MatchRepetitionAndRegexAtHead() calls MatchRegexAtHead(), so
00937                 // here's an indirect recursion.  It terminates as the regex gets
00938                 // shorter in each recursion.
00939                 return MatchRepetitionAndRegexAtHead(escaped, regex[0], regex[1], regex + 2,
00940                         str);
00941             } else {
00942                 // regex isn't empty, isn't "$", and doesn't start with a
00943                 // repetition.  We match the first atom of regex with the first
00944                 // character of str and recurse.
00945                 return (*str != '\0') && AtomMatchesChar(escaped, *regex, *str) &&
00946                         MatchRegexAtHead(regex + 1, str + 1);
00947             }
00948         }
00949
00950         // Returns true if and only if regex matches any substring of str.  regex must
00951         // be a valid simple regular expression, or the result is undefined.
00952         //
00953         // The algorithm is recursive, but the recursion depth doesn't exceed
00954         // the regex length, so we won't need to worry about running out of
00955         // stack space normally.  In rare cases the time complexity can be
00956         // exponential with respect to the regex length + the string length,
00957         // but usually it's must faster (often close to linear).
00958
00959         bool MatchRegexAnywhere(const char* regex, const char* str) {
00960             if (regex == nullptr || str == nullptr) return false;
00961
00962             if (*regex == '^') return MatchRegexAtHead(regex + 1, str);
00963
00964             // A successful match can be anywhere in str.
00965             do {
00966                 if (MatchRegexAtHead(regex, str)) return true;
00967             } while (*str++ != '\0');
00968             return false;
00969         }
00970
00971         // Implements the RE class.
00972
00973         RE::~RE() {
00974             free(const_cast<char*> (pattern_));
00975             free(const_cast<char*> (full_pattern_));
00976         }
00977
00978         // Returns true if and only if regular expression re matches the entire str.
00979
00980         bool RE::FullMatch(const char* str, const RE& re) {
00981             return re.is_valid_ && MatchRegexAnywhere(re.full_pattern_, str);
00982         }
00983
00984         // Returns true if and only if regular expression re matches a substring of
00985         // str (including str itself).
00986
00987         bool RE::PartialMatch(const char* str, const RE& re) {
00988             return re.is_valid_ && MatchRegexAnywhere(re.pattern_, str);
00989         }
00990
00991         // Initializes an RE from its string representation.
00992
00993         void RE::Init(const char* regex) {
00994             pattern_ = full_pattern_ = nullptr;
00995             if (regex != nullptr) {
00996                 pattern_ = posix::StrDup(regex);
00997             }
00998
00999             is_valid_ = ValidateRegex(regex);
01000             if (!is_valid_) {
01001                 // No need to calculate the full pattern when the regex is invalid.
01002                 return;
01003             }
01004
01005             const size_t len = strlen(regex);
01006             // Reserves enough bytes to hold the regular expression used for a
01007             // full match: we need space to prepend a '^', append a '$', and
01008             // terminate the string with '\0'.
```

```
01009             char* buffer = static_cast<char*> (malloc(len + 3));
01010             full_pattern_ = buffer;
01011
01012             if (*regex != '^')
01013                 *buffer++ = '^'; // Makes sure full_pattern_ starts with '^'.
01014
01015             // We don't use snprintf or strncpy, as they trigger a warning when
01016             // compiled with VC++ 8.0.
01017             memcpy(buffer, regex, len);
01018             buffer += len;
01019
01020             if (len == 0 || regex[len - 1] != '$')
01021                 *buffer++ = '$'; // Makes sure full_pattern_ ends with '$'.
01022
01023             *buffer = '\0';
01024         }
01025
01026 #endif  // GTEST_USES_POSIX_RE
01027
01028         const char kUnknownFile[] = "unknown file";
01029
01030         // Formats a source file path and a line number as they would appear
01031         // in an error message from the compiler used to compile this code.
01032
01033         GTEST_API_::std::string FormatFileLocation(const char* file, int line) {
01034             const std::string file_name(file == nullptr ? kUnknownFile : file);
01035
01036             if (line < 0) {
01037                 return file_name + ":";
01038             }
01039 #ifdef _MSC_VER
01040             return file_name + "(" + StreamableToString(line) + "):";
01041 #else
01042             return file_name + ":" + StreamableToString(line) + ":";
01043 #endif  // _MSC_VER
01044         }
01045
01046         // Formats a file location for compiler-independent XML output.
01047         // Although this function is not platform dependent, we put it next to
01048         // FormatFileLocation in order to contrast the two functions.
01049         // Note that FormatCompilerIndependentFileLocation() does NOT append colon
01050         // to the file location it produces, unlike FormatFileLocation().
01051
01052         GTEST_API_::std::string FormatCompilerIndependentFileLocation(const char* file,
01053                 int line) {
01054             const std::string file_name(file == nullptr ? kUnknownFile : file);
01055
01056             if (line < 0)
01057                 return file_name;
01058             else
01059                 return file_name + ":" + StreamableToString(line);
01060         }
01061
01062         GTestLog::GTestLog(GTestLogSeverity severity, const char* file, int line)
01063         : severity_(severity) {
01064             const char* const marker = severity == GTEST_INFO ? "[  INFO ]"
01065                     : severity == GTEST_WARNING ? "[WARNING]"
01066                     : severity == GTEST_ERROR ? "[ ERROR ]"
01067                     : "[ FATAL ]";
01068             GetStream() << ::std::endl
01069                     << marker << " " << FormatFileLocation(file, line).c_str()
01070                     << ": ";
01071         }
01072
01073         // Flushes the buffers and, if severity is GTEST_FATAL, aborts the program.
01074
01075         GTestLog::~GTestLog() {
01076             GetStream() << ::std::endl;
01077             if (severity_ == GTEST_FATAL) {
01078                 fflush(stderr);
01079                 posix::Abort();
01080             }
01081         }
01082
01083         // Disable Microsoft deprecation warnings for POSIX functions called from
01084         // this class (creat, dup, dup2, and close)
01085
01086         GTEST_DISABLE_MSC_DEPRECATED_PUSH_()
01087
01088 #if GTEST_HAS_STREAM_REDIRECTION
01089
01090         // Object that captures an output stream (stdout/stderr).
01091         class CapturedStream {
01092         public:
01093             // The ctor redirects the stream to a temporary file.
01094
01095             explicit CapturedStream(int fd) : fd_(fd), uncaptured_fd_(dup(fd)) {
```

```
01096 #if GTEST_OS_WINDOWS
01097                 char temp_dir_path[MAX_PATH + 1] = {'\0'}; // NOLINT
01098                 char temp_file_path[MAX_PATH + 1] = {'\0'}; // NOLINT
01099
01100                 ::GetTempPathA(sizeof (temp_dir_path), temp_dir_path);
01101                 const UINT success = ::GetTempFileNameA(temp_dir_path, "gtest_redir",
01102                         0, // Generate unique file name.
01103                         temp_file_path);
01104                 GTEST_CHECK_(success != 0)
01105                         << "Unable to create a temporary file in " << temp_dir_path;
01106                 const int captured_fd = creat(temp_file_path, _S_IREAD | _S_IWRITE);
01107                 GTEST_CHECK_(captured_fd != -1)
01108                         << "Unable to open temporary file " << temp_file_path;
01109                 filename_ = temp_file_path;
01110 #else
01111                 // There's no guarantee that a test has write access to the current
01112                 // directory, so we create the temporary file in a temporary directory.
01113                 std::string name_template;
01114
01115 #if GTEST_OS_LINUX_ANDROID
01116                 // Note: Android applications are expected to call the framework's
01117                 // Context.getExternalStorageDirectory() method through JNI to get
01118                 // the location of the world-writable SD Card directory. However,
01119                 // this requires a Context handle, which cannot be retrieved
01120                 // globally from native code. Doing so also precludes running the
01121                 // code as part of a regular standalone executable, which doesn't
01122                 // run in a Dalvik process (e.g. when running it through 'adb shell').
01123                 //
01124                 // The location /data/local/tmp is directly accessible from native code.
01125                 // '/sdcard' and other variants cannot be relied on, as they are not
01126                 // guaranteed to be mounted, or may have a delay in mounting.
01127                 name_template = "/data/local/tmp/";
01128 #elif GTEST_OS_IOS
01129                 char user_temp_dir[PATH_MAX + 1];
01130
01131                 // Documented alternative to NSTemporaryDirectory() (for obtaining creating
01132                 // a temporary directory) at
01133                 //
      https://developer.apple.com/library/archive/documentation/Security/Conceptual/SecureCodingGuide/Articles/RaceConditions
01134                 //
01135                 // _CS_DARWIN_USER_TEMP_DIR (as well as _CS_DARWIN_USER_CACHE_DIR) is not
01136                 // documented in the confstr() man page at
01137                 //
      https://developer.apple.com/library/archive/documentation/System/Conceptual/ManPages_iPhoneOS/man3/confstr.3.html#//app
01138                 // but are still available, according to the WebKit patches at
01139                 // https://trac.webkit.org/changeset/262004/webkit
01140                 // https://trac.webkit.org/changeset/263705/webkit
01141                 //
01142                 // The confstr() implementation falls back to getenv("TMPDIR"). See
01143                 // https://opensource.apple.com/source/Libc/Libc-1439.100.3/gen/confstr.c.auto.html
01144                 ::confstr(_CS_DARWIN_USER_TEMP_DIR, user_temp_dir, sizeof (user_temp_dir));
01145
01146                 name_template = user_temp_dir;
01147                 if (name_template.back() != GTEST_PATH_SEP_[0])
01148                     name_template.push_back(GTEST_PATH_SEP_[0]);
01149 #else
01150                 name_template = "/tmp/";
01151 #endif
01152                 name_template.append("gtest_captured_stream.XXXXXX");
01153
01154                 // mkstemp() modifies the string bytes in place, and does not go beyond the
01155                 // string's length. This results in well-defined behavior in C++17.
01156                 //
01157                 // The const_cast is needed below C++17. The constraints on std::string
01158                 // implementations in C++11 and above make assumption behind the const_cast
01159                 // fairly safe.
01160                 const int captured_fd = ::mkstemp(const_cast<char*> (name_template.data()));
01161                 if (captured_fd == -1) {
01162                     GTEST_LOG_(WARNING)
01163                             << "Failed to create tmp file " << name_template
01164                             << " for test; does the test have access to the /tmp directory?";
01165                 }
01166                 filename_ = std::move(name_template);
01167 #endif  // GTEST_OS_WINDOWS
01168                 fflush(nullptr);
01169                 dup2(captured_fd, fd_);
01170                 close(captured_fd);
01171             }
01172
01173             ~CapturedStream() {
01174                 remove(filename_.c_str());
01175             }
01176
01177             std::string GetCapturedString() {
01178                 if (uncaptured_fd_ != -1) {
01179                     // Restores the original stream.
01180                     fflush(nullptr);
```

```
01181                     dup2(uncaptured_fd_, fd_);
01182                     close(uncaptured_fd_);
01183                     uncaptured_fd_ = -1;
01184                 }
01185
01186                 FILE * const file = posix::FOpen(filename_.c_str(), "r");
01187                 if (file == nullptr) {
01188                     GTEST_LOG_(FATAL) << "Failed to open tmp file " << filename_
01189                             << " for capturing stream.";
01190                 }
01191                 const std::string content = ReadEntireFile(file);
01192                 posix::FClose(file);
01193                 return content;
01194             }
01195
01196         private:
01197             const int fd_; // A stream to capture.
01198             int uncaptured_fd_;
01199             // Name of the temporary file holding the stderr output.
01200             ::std::string filename_;
01201
01202             CapturedStream(const CapturedStream&) = delete;
01203             CapturedStream& operator=(const CapturedStream&) = delete;
01204         };
01205
01206         GTEST_DISABLE_MSC_DEPRECATED_POP_()
01207
01208         static CapturedStream* g_captured_stderr = nullptr;
01209         static CapturedStream* g_captured_stdout = nullptr;
01210
01211         // Starts capturing an output stream (stdout/stderr).
01212
01213         static void CaptureStream(int fd, const char* stream_name,
01214                 CapturedStream** stream) {
01215             if (*stream != nullptr) {
01216                 GTEST_LOG_(FATAL) << "Only one " << stream_name
01217                         << " capturer can exist at a time.";
01218             }
01219             *stream = new CapturedStream(fd);
01220         }
01221
01222         // Stops capturing the output stream and returns the captured string.
01223
01224         static std::string GetCapturedStream(CapturedStream** captured_stream) {
01225             const std::string content = (*captured_stream)->GetCapturedString();
01226
01227             delete *captured_stream;
01228             *captured_stream = nullptr;
01229
01230             return content;
01231         }
01232
01233 #if defined(_MSC_VER) || defined(__BORLANDC__)
01234         // MSVC and C++Builder do not provide a definition of STDERR_FILENO.
01235         const int kStdOutFileno = 1;
01236         const int kStdErrFileno = 2;
01237 #else
01238         const int kStdOutFileno = STDOUT_FILENO;
01239         const int kStdErrFileno = STDERR_FILENO;
01240 #endif  // defined(_MSC_VER) || defined(__BORLANDC__)
01241
01242         // Starts capturing stdout.
01243
01244         void CaptureStdout() {
01245             CaptureStream(kStdOutFileno, "stdout", &g_captured_stdout);
01246         }
01247
01248         // Starts capturing stderr.
01249
01250         void CaptureStderr() {
01251             CaptureStream(kStdErrFileno, "stderr", &g_captured_stderr);
01252         }
01253
01254         // Stops capturing stdout and returns the captured string.
01255
01256         std::string GetCapturedStdout() {
01257             return GetCapturedStream(&g_captured_stdout);
01258         }
01259
01260         // Stops capturing stderr and returns the captured string.
01261
01262         std::string GetCapturedStderr() {
01263             return GetCapturedStream(&g_captured_stderr);
01264         }
01265
01266 #endif  // GTEST_HAS_STREAM_REDIRECTION
01267
```

```
01268         size_t GetFileSize(FILE* file) {
01269             fseek(file, 0, SEEK_END);
01270             return static_cast<size_t> (ftell(file));
01271         }
01272
01273         std::string ReadEntireFile(FILE* file) {
01274             const size_t file_size = GetFileSize(file);
01275             char* const buffer = new char[file_size];
01276
01277             size_t bytes_last_read = 0; // # of bytes read in the last fread()
01278             size_t bytes_read = 0; // # of bytes read so far
01279
01280             fseek(file, 0, SEEK_SET);
01281
01282             // Keeps reading the file until we cannot read further or the
01283             // pre-determined file size is reached.
01284             do {
01285                 bytes_last_read =
01286                         fread(buffer + bytes_read, 1, file_size - bytes_read, file);
01287                 bytes_read += bytes_last_read;
01288             } while (bytes_last_read > 0 && bytes_read < file_size);
01289
01290             const std::string content(buffer, bytes_read);
01291             delete[] buffer;
01292
01293             return content;
01294         }
01295
01296 #if GTEST_HAS_DEATH_TEST
01297         static const std::vector<std::string>* g_injected_test_argvs =
01298                 nullptr; // Owned.
01299
01300         std::vector<std::string> GetInjectableArgvs() {
01301             if (g_injected_test_argvs != nullptr) {
01302                 return *g_injected_test_argvs;
01303             }
01304             return GetArgvs();
01305         }
01306
01307         void SetInjectableArgvs(const std::vector<std::string>* new_argvs) {
01308             if (g_injected_test_argvs != new_argvs) delete g_injected_test_argvs;
01309             g_injected_test_argvs = new_argvs;
01310         }
01311
01312         void SetInjectableArgvs(const std::vector<std::string>& new_argvs) {
01313             SetInjectableArgvs(
01314                     new std::vector<std::string>(new_argvs.begin(), new_argvs.end()));
01315         }
01316
01317         void ClearInjectableArgvs() {
01318             delete g_injected_test_argvs;
01319             g_injected_test_argvs = nullptr;
01320         }
01321 #endif  // GTEST_HAS_DEATH_TEST
01322
01323 #if GTEST_OS_WINDOWS_MOBILE
01324         namespace posix {
01325
01326             void Abort() {
01327                 DebugBreak();
01328                 TerminateProcess(GetCurrentProcess(), 1);
01329             }
01330         } // namespace posix
01331 #endif  // GTEST_OS_WINDOWS_MOBILE
01332
01333         // Returns the name of the environment variable corresponding to the
01334         // given flag.  For example, FlagToEnvVar("foo") will return
01335         // "GTEST_FOO" in the open-source version.
01336
01337         static std::string FlagToEnvVar(const char* flag) {
01338             const std::string full_flag =
01339                     (Message() « GTEST_FLAG_PREFIX_ « flag).GetString();
01340
01341             Message env_var;
01342             for (size_t i = 0; i != full_flag.length(); i++) {
01343                 env_var « ToUpper(full_flag.c_str()[i]);
01344             }
01345
01346             return env_var.GetString();
01347         }
01348
01349         // Parses 'str' for a 32-bit signed integer.  If successful, writes
01350         // the result to *value and returns true; otherwise leaves *value
01351         // unchanged and returns false.
01352
01353         bool ParseInt32(const Message& src_text, const char* str, int32_t* value) {
01354             // Parses the environment variable as a decimal integer.
```

```
01355             char* end = nullptr;
01356             const long long_value = strtol(str, &end, 10); // NOLINT
01357
01358             // Has strtol() consumed all characters in the string?
01359             if (*end != '\0') {
01360                 // No - an invalid character was encountered.
01361                 Message msg;
01362                 msg << "WARNING: " << src_text
01363                         << " is expected to be a 32-bit integer, but actually"
01364                         << " has value \"" << str << "\".\n";
01365                 printf("%s", msg.GetString().c_str());
01366                 fflush(stdout);
01367                 return false;
01368             }
01369
01370             // Is the parsed value in the range of an int32_t?
01371             const auto result = static_cast<int32_t> (long_value);
01372             if (long_value == LONG_MAX || long_value == LONG_MIN ||
01373                     // The parsed value overflows as a long.  (strtol() returns
01374                     // LONG_MAX or LONG_MIN when the input overflows.)
01375                     result != long_value
01376                     // The parsed value overflows as an int32_t.
01377                     ) {
01378                 Message msg;
01379                 msg << "WARNING: " << src_text
01380                         << " is expected to be a 32-bit integer, but actually"
01381                         << " has value " << str << ", which overflows.\n";
01382                 printf("%s", msg.GetString().c_str());
01383                 fflush(stdout);
01384                 return false;
01385             }
01386
01387             *value = result;
01388             return true;
01389         }
01390
01391         // Reads and returns the Boolean environment variable corresponding to
01392         // the given flag; if it's not set, returns default_value.
01393         //
01394         // The value is considered true if and only if it's not "0".
01395
01396         bool BoolFromGTestEnv(const char* flag, bool default_value) {
01397 #if defined(GTEST_GET_BOOL_FROM_ENV_)
01398             return GTEST_GET_BOOL_FROM_ENV_(flag, default_value);
01399 #else
01400             const std::string env_var = FlagToEnvVar(flag);
01401             const char* const string_value = posix::GetEnv(env_var.c_str());
01402             return string_value == nullptr ? default_value
01403                     : strcmp(string_value, "0") != 0;
01404 #endif  // defined(GTEST_GET_BOOL_FROM_ENV_)
01405         }
01406
01407         // Reads and returns a 32-bit integer stored in the environment
01408         // variable corresponding to the given flag; if it isn't set or
01409         // doesn't represent a valid 32-bit integer, returns default_value.
01410
01411         int32_t Int32FromGTestEnv(const char* flag, int32_t default_value) {
01412 #if defined(GTEST_GET_INT32_FROM_ENV_)
01413             return GTEST_GET_INT32_FROM_ENV_(flag, default_value);
01414 #else
01415             const std::string env_var = FlagToEnvVar(flag);
01416             const char* const string_value = posix::GetEnv(env_var.c_str());
01417             if (string_value == nullptr) {
01418                 // The environment variable is not set.
01419                 return default_value;
01420             }
01421
01422             int32_t result = default_value;
01423             if (!ParseInt32(Message() << "Environment variable " << env_var, string_value,
01424                     &result)) {
01425                 printf("The default value %s is used.\n",
01426                         (Message() << default_value).GetString().c_str());
01427                 fflush(stdout);
01428                 return default_value;
01429             }
01430
01431             return result;
01432 #endif  // defined(GTEST_GET_INT32_FROM_ENV_)
01433         }
01434
01435         // As a special case for the 'output' flag, if GTEST_OUTPUT is not
01436         // set, we look for XML_OUTPUT_FILE, which is set by the Bazel build
01437         // system.  The value of XML_OUTPUT_FILE is a filename without the
01438         // "xml:" prefix of GTEST_OUTPUT.
01439         // Note that this is meant to be called at the call site so it does
01440         // not check that the flag is 'output'
01441         // In essence this checks an env variable called XML_OUTPUT_FILE
```

```
01442         // and if it is set we prepend "xml:" to its value, if it not set we return ""
01443
01444         std::string OutputFlagAlsoCheckEnvVar() {
01445             std::string default_value_for_output_flag = "";
01446             const char* xml_output_file_env = posix::GetEnv("XML_OUTPUT_FILE");
01447             if (nullptr != xml_output_file_env) {
01448                 default_value_for_output_flag = std::string("xml:") + xml_output_file_env;
01449             }
01450             return default_value_for_output_flag;
01451         }
01452
01453         // Reads and returns the string environment variable corresponding to
01454         // the given flag; if it's not set, returns default_value.
01455
01456         const char* StringFromGTestEnv(const char* flag, const char* default_value) {
01457 #if defined(GTEST_GET_STRING_FROM_ENV_)
01458             return GTEST_GET_STRING_FROM_ENV_(flag, default_value);
01459 #else
01460             const std::string env_var = FlagToEnvVar(flag);
01461             const char* const value = posix::GetEnv(env_var.c_str());
01462             return value == nullptr ? default_value : value;
01463 #endif  // defined(GTEST_GET_STRING_FROM_ENV_)
01464         }
01465
01466     } // namespace internal
01467 } // namespace testing
```

18.288 gtest-printers.cc

```
00001 // Copyright 2007, Google Inc.
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00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 // Google Test - The Google C++ Testing and Mocking Framework
00031 //
00032 // This file implements a universal value printer that can print a
00033 // value of any type T:
00034 //
00035 //   void ::testing::internal::UniversalPrinter<T>::Print(value, ostream_ptr);
00036 //
00037 // It uses the « operator when possible, and prints the bytes in the
00038 // object otherwise.  A user can override its behavior for a class
00039 // type Foo by defining either operator«(::std::ostream&, const Foo&)
00040 // or void PrintTo(const Foo&, ::std::ostream*) in the namespace that
00041 // defines Foo.
00042
00043 #include "gtest/gtest-printers.h"
00044
00045 #include <stdio.h>
00046
00047 #include <cctype>
00048 #include <cstdint>
00049 #include <cwchar>
00050 #include <ostream>  // NOLINT
00051 #include <string>
00052 #include <type_traits>
00053
00054 #include "gtest/internal/gtest-port.h"
00055 #include "src/gtest-internal-inl.h"
00056
```

```
00057 namespace testing {
00058
00059     namespace {
00060
00061         using ::std::ostream;
00062
00063         // Prints a segment of bytes in the given object.
00064
00065         GTEST_ATTRIBUTE_NO_SANITIZE_MEMORY_
00066         GTEST_ATTRIBUTE_NO_SANITIZE_ADDRESS_
00067         GTEST_ATTRIBUTE_NO_SANITIZE_HWADDRESS_
00068         GTEST_ATTRIBUTE_NO_SANITIZE_THREAD_
00069         void PrintByteSegmentInObjectTo(const unsigned char* obj_bytes, size_t start,
00070                 size_t count, ostream* os) {
00071             char text[5] = "";
00072             for (size_t i = 0; i != count; i++) {
00073                 const size_t j = start + i;
00074                 if (i != 0) {
00075                     // Organizes the bytes into groups of 2 for easy parsing by
00076                     // human.
00077                     if ((j % 2) == 0)
00078                         *os << ' ';
00079                     else
00080                         *os << '-';
00081                 }
00082                 GTEST_SNPRINTF_(text, sizeof (text), "%02X", obj_bytes[j]);
00083                 *os << text;
00084             }
00085         }
00086
00087         // Prints the bytes in the given value to the given ostream.
00088
00089         void PrintBytesInObjectToImpl(const unsigned char* obj_bytes, size_t count,
00090                 ostream* os) {
00091             // Tells the user how big the object is.
00092             *os << count << "-byte object <";
00093
00094             const size_t kThreshold = 132;
00095             const size_t kChunkSize = 64;
00096             // If the object size is bigger than kThreshold, we'll have to omit
00097             // some details by printing only the first and the last kChunkSize
00098             // bytes.
00099             if (count < kThreshold) {
00100                 PrintByteSegmentInObjectTo(obj_bytes, 0, count, os);
00101             } else {
00102                 PrintByteSegmentInObjectTo(obj_bytes, 0, kChunkSize, os);
00103                 *os << " ... ";
00104                 // Rounds up to 2-byte boundary.
00105                 const size_t resume_pos = (count - kChunkSize + 1) / 2 * 2;
00106                 PrintByteSegmentInObjectTo(obj_bytes, resume_pos, count - resume_pos, os);
00107             }
00108             *os << ">";
00109         }
00110
00111         // Helpers for widening a character to char32_t. Since the standard does not
00112         // specify if char / wchar_t is signed or unsigned, it is important to first
00113         // convert it to the unsigned type of the same width before widening it to
00114         // char32_t.
00115
00116         template <typename CharType>
00117         char32_t ToChar32(CharType in) {
00118             return static_cast<char32_t> (
00119                     static_cast<typename std::make_unsigned<CharType>::type> (in));
00120         }
00121
00122     } // namespace
00123
00124     namespace internal {
00125
00126         // Delegates to PrintBytesInObjectToImpl() to print the bytes in the
00127         // given object.  The delegation simplifies the implementation, which
00128         // uses the << operator and thus is easier done outside of the
00129         // ::testing::internal namespace, which contains a << operator that
00130         // sometimes conflicts with the one in STL.
00131
00132         void PrintBytesInObjectTo(const unsigned char* obj_bytes, size_t count,
00133                 ostream* os) {
00134             PrintBytesInObjectToImpl(obj_bytes, count, os);
00135         }
00136
00137         // Depending on the value of a char (or wchar_t), we print it in one
00138         // of three formats:
00139         //   - as is if it's a printable ASCII (e.g. 'a', '2', ' '),
00140         //   - as a hexadecimal escape sequence (e.g. '\x7F'), or
00141         //   - as a special escape sequence (e.g. '\r', '\n').
00142
00143         enum CharFormat {
```

```
00144             kAsIs, kHexEscape, kSpecialEscape
00145         };
00146
00147         // Returns true if c is a printable ASCII character.  We test the
00148         // value of c directly instead of calling isprint(), which is buggy on
00149         // Windows Mobile.
00150
00151         inline bool IsPrintableAscii(char32_t c) {
00152             return 0x20 <= c && c <= 0x7E;
00153         }
00154
00155         // Prints c (of type char, char8_t, char16_t, char32_t, or wchar_t) as a
00156         // character literal without the quotes, escaping it when necessary; returns how
00157         // c was formatted.
00158
00159         template <typename Char>
00160         static CharFormat PrintAsCharLiteralTo(Char c, ostream* os) {
00161             const char32_t u_c = ToChar32(c);
00162             switch (u_c) {
00163                 case L'\0':
00164                     *os << "\\0";
00165                     break;
00166                 case L'\'':
00167                     *os << "\\'";
00168                     break;
00169                 case L'\\':
00170                     *os << "\\\\";
00171                     break;
00172                 case L'\a':
00173                     *os << "\\a";
00174                     break;
00175                 case L'\b':
00176                     *os << "\\b";
00177                     break;
00178                 case L'\f':
00179                     *os << "\\f";
00180                     break;
00181                 case L'\n':
00182                     *os << "\\n";
00183                     break;
00184                 case L'\r':
00185                     *os << "\\r";
00186                     break;
00187                 case L'\t':
00188                     *os << "\\t";
00189                     break;
00190                 case L'\v':
00191                     *os << "\\v";
00192                     break;
00193                 default:
00194                     if (IsPrintableAscii(u_c)) {
00195                         *os << static_cast<char> (c);
00196                         return kAsIs;
00197                     } else {
00198                         ostream::fmtflags flags = os->flags();
00199                         *os << "\\x" << std::hex << std::uppercase << static_cast<int> (u_c);
00200                         os->flags(flags);
00201                         return kHexEscape;
00202                     }
00203             }
00204             return kSpecialEscape;
00205         }
00206
00207         // Prints a char32_t c as if it's part of a string literal, escaping it when
00208         // necessary; returns how c was formatted.
00209
00210         static CharFormat PrintAsStringLiteralTo(char32_t c, ostream* os) {
00211             switch (c) {
00212                 case L'\"':
00213                     *os << "'";
00214                     return kAsIs;
00215                 case L'"':
00216                     *os << "\\\"";
00217                     return kSpecialEscape;
00218                 default:
00219                     return PrintAsCharLiteralTo(c, os);
00220             }
00221         }
00222
00223         static const char* GetCharWidthPrefix(char) {
00224             return "";
00225         }
00226
00227         static const char* GetCharWidthPrefix(signed char) {
00228             return "";
00229         }
00230
```

```
00231         static const char* GetCharWidthPrefix(unsigned char) {
00232             return "";
00233         }
00234
00235 #ifdef __cpp_char8_t
00236
00237         static const char* GetCharWidthPrefix(char8_t) {
00238             return "u8";
00239         }
00240 #endif
00241
00242         static const char* GetCharWidthPrefix(char16_t) {
00243             return "u";
00244         }
00245
00246         static const char* GetCharWidthPrefix(char32_t) {
00247             return "U";
00248         }
00249
00250         static const char* GetCharWidthPrefix(wchar_t) {
00251             return "L";
00252         }
00253
00254         // Prints a char c as if it's part of a string literal, escaping it when
00255         // necessary; returns how c was formatted.
00256
00257         static CharFormat PrintAsStringLiteralTo(char c, ostream* os) {
00258             return PrintAsStringLiteralTo(ToChar32(c), os);
00259         }
00260
00261 #ifdef __cpp_char8_t
00262
00263         static CharFormat PrintAsStringLiteralTo(char8_t c, ostream* os) {
00264             return PrintAsStringLiteralTo(ToChar32(c), os);
00265         }
00266 #endif
00267
00268         static CharFormat PrintAsStringLiteralTo(char16_t c, ostream* os) {
00269             return PrintAsStringLiteralTo(ToChar32(c), os);
00270         }
00271
00272         static CharFormat PrintAsStringLiteralTo(wchar_t c, ostream* os) {
00273             return PrintAsStringLiteralTo(ToChar32(c), os);
00274         }
00275
00276         // Prints a character c (of type char, char8_t, char16_t, char32_t, or wchar_t)
00277         // and its code. '\0' is printed as "'\\0'", other unprintable characters are
00278         // also properly escaped using the standard C++ escape sequence.
00279
00280         template <typename Char>
00281         void PrintCharAndCodeTo(Char c, ostream* os) {
00282             // First, print c as a literal in the most readable form we can find.
00283             *os << GetCharWidthPrefix(c) << "'";
00284             const CharFormat format = PrintAsCharLiteralTo(c, os);
00285             *os << "'";
00286
00287             // To aid user debugging, we also print c's code in decimal, unless
00288             // it's 0 (in which case c was printed as '\\0', making the code
00289             // obvious).
00290             if (c == 0) return;
00291             *os << " (" << static_cast<int> (c);
00292
00293             // For more convenience, we print c's code again in hexadecimal,
00294             // unless c was already printed in the form '\x##' or the code is in
00295             // [1, 9].
00296             if (format == kHexEscape || (1 <= c && c <= 9)) {
00297                 // Do nothing.
00298             } else {
00299                 *os << ", 0x" << String::FormatHexInt(static_cast<int> (c));
00300             }
00301             *os << ")";
00302         }
00303
00304         void PrintTo(unsigned char c, ::std::ostream* os) {
00305             PrintCharAndCodeTo(c, os);
00306         }
00307
00308         void PrintTo(signed char c, ::std::ostream* os) {
00309             PrintCharAndCodeTo(c, os);
00310         }
00311
00312         // Prints a wchar_t as a symbol if it is printable or as its internal
00313         // code otherwise and also as its code.  L'\0' is printed as "L'\\0'".
00314
00315         void PrintTo(wchar_t wc, ostream* os) {
00316             PrintCharAndCodeTo(wc, os);
00317         }
```

```
00318
00319         // TODO(dcheng): Consider making this delegate to PrintCharAndCodeTo() as well.
00320
00321         void PrintTo(char32_t c, ::std::ostream* os) {
00322             *os << std::hex << "U+" << std::uppercase << std::setfill('0') << std::setw(4)
00323                     << static_cast<uint32_t> (c);
00324         }
00325
00326         // gcc/clang __{u,}int128_t
00327 #if defined(__SIZEOF_INT128__)
00328
00329         void PrintTo(__uint128_t v, ::std::ostream* os) {
00330             if (v == 0) {
00331                 *os << "0";
00332                 return;
00333             }
00334
00335             // Buffer large enough for ceil(log10(2^128))==39 and the null terminator
00336             char buf[40];
00337             char* p = buf + sizeof (buf);
00338
00339             // Some configurations have a __uint128_t, but no support for built in
00340             // division. Do manual long division instead.
00341
00342             uint64_t high = static_cast<uint64_t> (v >> 64);
00343             uint64_t low = static_cast<uint64_t> (v);
00344
00345             *--p = 0;
00346             while (high != 0 || low != 0) {
00347                 uint64_t high_mod = high % 10;
00348                 high = high / 10;
00349                 // This is the long division algorithm specialized for a divisor of 10 and
00350                 // only two elements.
00351                 // Notable values:
00352                 //   2^64 / 10 == 1844674407370955161
00353                 //   2^64 % 10 == 6
00354                 const uint64_t carry = 6 * high_mod + low % 10;
00355                 low = low / 10 + high_mod * 1844674407370955161 + carry / 10;
00356
00357                 char digit = static_cast<char> (carry % 10);
00358                 *--p = '0' + digit;
00359             }
00360             *os << p;
00361         }
00362
00363         void PrintTo(__int128_t v, ::std::ostream* os) {
00364             __uint128_t uv = static_cast<__uint128_t> (v);
00365             if (v < 0) {
00366                 *os << "-";
00367                 uv = -uv;
00368             }
00369             PrintTo(uv, os);
00370         }
00371 #endif  // __SIZEOF_INT128__
00372
00373         // Prints the given array of characters to the ostream.  CharType must be either
00374         // char, char8_t, char16_t, char32_t, or wchar_t.
00375         // The array starts at begin, the length is len, it may include '\0' characters
00376         // and may not be NUL-terminated.
00377
00378         template <typename CharType>
00379         GTEST_ATTRIBUTE_NO_SANITIZE_MEMORY_ GTEST_ATTRIBUTE_NO_SANITIZE_ADDRESS_
00380         GTEST_ATTRIBUTE_NO_SANITIZE_HWADDRESS_
00381         GTEST_ATTRIBUTE_NO_SANITIZE_THREAD_ static CharFormat
00382         PrintCharsAsStringTo(const CharType* begin, size_t len, ostream* os) {
00383             const char* const quote_prefix = GetCharWidthPrefix(*begin);
00384             *os << quote_prefix << "\"";
00385             bool is_previous_hex = false;
00386             CharFormat print_format = kAsIs;
00387             for (size_t index = 0; index < len; ++index) {
00388                 const CharType cur = begin[index];
00389                 if (is_previous_hex && IsXDigit(cur)) {
00390                     // Previous character is of '\x..' form and this character can be
00391                     // interpreted as another hexadecimal digit in its number. Break string to
00392                     // disambiguate.
00393                     *os << "\" " << quote_prefix << "\"";
00394                 }
00395                 is_previous_hex = PrintAsStringLiteralTo(cur, os) == kHexEscape;
00396                 // Remember if any characters required hex escaping.
00397                 if (is_previous_hex) {
00398                     print_format = kHexEscape;
00399                 }
00400             }
00401             *os << "\"";
00402             return print_format;
00403         }
00404
```

```
00405         // Prints a (const) char/wchar_t array of 'len' elements, starting at address
00406         // 'begin'.  CharType must be either char or wchar_t.
00407
00408         template <typename CharType>
00409         GTEST_ATTRIBUTE_NO_SANITIZE_MEMORY_ GTEST_ATTRIBUTE_NO_SANITIZE_ADDRESS_
00410         GTEST_ATTRIBUTE_NO_SANITIZE_HWADDRESS_
00411         GTEST_ATTRIBUTE_NO_SANITIZE_THREAD_ static void
00412         UniversalPrintCharArray(const CharType* begin, size_t len,
00413                 ostream* os) {
00414             // The code
00415             //   const char kFoo[] = "foo";
00416             // generates an array of 4, not 3, elements, with the last one being '\0'.
00417             //
00418             // Therefore when printing a char array, we don't print the last element if
00419             // it's '\0', such that the output matches the string literal as it's
00420             // written in the source code.
00421             if (len > 0 && begin[len - 1] == '\0') {
00422                 PrintCharsAsStringTo(begin, len - 1, os);
00423                 return;
00424             }
00425
00426             // If, however, the last element in the array is not '\0', e.g.
00427             //    const char kFoo[] = { 'f', 'o', 'o' };
00428             // we must print the entire array.  We also print a message to indicate
00429             // that the array is not NUL-terminated.
00430             PrintCharsAsStringTo(begin, len, os);
00431             *os << " (no terminating NUL)";
00432         }
00433
00434         // Prints a (const) char array of 'len' elements, starting at address 'begin'.
00435
00436         void UniversalPrintArray(const char* begin, size_t len, ostream* os) {
00437             UniversalPrintCharArray(begin, len, os);
00438         }
00439
00440 #ifdef __cpp_char8_t
00441         // Prints a (const) char8_t array of 'len' elements, starting at address
00442         // 'begin'.
00443
00444         void UniversalPrintArray(const char8_t* begin, size_t len, ostream* os) {
00445             UniversalPrintCharArray(begin, len, os);
00446         }
00447 #endif
00448
00449         // Prints a (const) char16_t array of 'len' elements, starting at address
00450         // 'begin'.
00451
00452         void UniversalPrintArray(const char16_t* begin, size_t len, ostream* os) {
00453             UniversalPrintCharArray(begin, len, os);
00454         }
00455
00456         // Prints a (const) char32_t array of 'len' elements, starting at address
00457         // 'begin'.
00458
00459         void UniversalPrintArray(const char32_t* begin, size_t len, ostream* os) {
00460             UniversalPrintCharArray(begin, len, os);
00461         }
00462
00463         // Prints a (const) wchar_t array of 'len' elements, starting at address
00464         // 'begin'.
00465
00466         void UniversalPrintArray(const wchar_t* begin, size_t len, ostream* os) {
00467             UniversalPrintCharArray(begin, len, os);
00468         }
00469
00470         namespace {
00471
00472             // Prints a null-terminated C-style string to the ostream.
00473
00474             template <typename Char>
00475             void PrintCStringTo(const Char* s, ostream* os) {
00476                 if (s == nullptr) {
00477                     *os << "NULL";
00478                 } else {
00479                     *os << ImplicitCast_<const void*>(s) << " pointing to ";
00480                     PrintCharsAsStringTo(s, std::char_traits<Char>::length(s), os);
00481                 }
00482             }
00483
00484         } // anonymous namespace
00485
00486         void PrintTo(const char* s, ostream* os) {
00487             PrintCStringTo(s, os);
00488         }
00489
00490 #ifdef __cpp_char8_t
00491
```

```
00492         void PrintTo(const char8_t* s, ostream* os) {
00493             PrintCStringTo(s, os);
00494         }
00495 #endif
00496
00497         void PrintTo(const char16_t* s, ostream* os) {
00498             PrintCStringTo(s, os);
00499         }
00500
00501         void PrintTo(const char32_t* s, ostream* os) {
00502             PrintCStringTo(s, os);
00503         }
00504
00505         // MSVC compiler can be configured to define whar_t as a typedef
00506         // of unsigned short. Defining an overload for const wchar_t* in that case
00507         // would cause pointers to unsigned shorts be printed as wide strings,
00508         // possibly accessing more memory than intended and causing invalid
00509         // memory accesses. MSVC defines _NATIVE_WCHAR_T_DEFINED symbol when
00510         // wchar_t is implemented as a native type.
00511 #if !defined(_MSC_VER) || defined(_NATIVE_WCHAR_T_DEFINED)
00512         // Prints the given wide C string to the ostream.
00513
00514         void PrintTo(const wchar_t* s, ostream* os) {
00515             PrintCStringTo(s, os);
00516         }
00517 #endif  // wchar_t is native
00518
00519         namespace {
00520
00521             bool ContainsUnprintableControlCodes(const char* str, size_t length) {
00522                 const unsigned char* s = reinterpret_cast<const unsigned char*> (str);
00523
00524                 for (size_t i = 0; i < length; i++) {
00525                     unsigned char ch = *s++;
00526                     if (std::iscntrl(ch)) {
00527                         switch (ch) {
00528                             case '\t':
00529                             case '\n':
00530                             case '\r':
00531                                 break;
00532                             default:
00533                                 return true;
00534                         }
00535                     }
00536                 }
00537                 return false;
00538             }
00539
00540             bool IsUTF8TrailByte(unsigned char t) {
00541                 return 0x80 <= t && t <= 0xbf;
00542             }
00543
00544             bool IsValidUTF8(const char* str, size_t length) {
00545                 const unsigned char* s = reinterpret_cast<const unsigned char*> (str);
00546
00547                 for (size_t i = 0; i < length;) {
00548                     unsigned char lead = s[i++];
00549
00550                     if (lead <= 0x7f) {
00551                         continue; // single-byte character (ASCII) 0..7F
00552                     }
00553                     if (lead < 0xc2) {
00554                         return false; // trail byte or non-shortest form
00555                     } else if (lead <= 0xdf && (i + 1) <= length && IsUTF8TrailByte(s[i])) {
00556                         ++i; // 2-byte character
00557                     } else if (0xe0 <= lead && lead <= 0xef && (i + 2) <= length &&
00558                             IsUTF8TrailByte(s[i]) && IsUTF8TrailByte(s[i + 1]) &&
00559                             // check for non-shortest form and surrogate
00560                             (lead != 0xe0 || s[i] >= 0xa0) &&
00561                             (lead != 0xed || s[i] < 0xa0)) {
00562                         i += 2; // 3-byte character
00563                     } else if (0xf0 <= lead && lead <= 0xf4 && (i + 3) <= length &&
00564                             IsUTF8TrailByte(s[i]) && IsUTF8TrailByte(s[i + 1]) &&
00565                             IsUTF8TrailByte(s[i + 2]) &&
00566                             // check for non-shortest form
00567                             (lead != 0xf0 || s[i] >= 0x90) &&
00568                             (lead != 0xf4 || s[i] < 0x90)) {
00569                         i += 3; // 4-byte character
00570                     } else {
00571                         return false;
00572                     }
00573                 }
00574                 return true;
00575             }
00576
00577             void ConditionalPrintAsText(const char* str, size_t length, ostream* os) {
00578                 if (!ContainsUnprintableControlCodes(str, length) &&
```

```
00579                         IsValidUTF8(str, length)) {
00580                       *os « "\n    As Text: \"" « str « "\"";
00581                   }
00582               }
00583
00584         } // anonymous namespace
00585
00586         void PrintStringTo(const ::std::string& s, ostream* os) {
00587             if (PrintCharsAsStringTo(s.data(), s.size(), os) == kHexEscape) {
00588                 if (GTEST_FLAG_GET(print_utf8)) {
00589                     ConditionalPrintAsText(s.data(), s.size(), os);
00590                 }
00591             }
00592         }
00593
00594 #ifdef __cpp_char8_t
00595
00596         void PrintU8StringTo(const ::std::u8string& s, ostream* os) {
00597             PrintCharsAsStringTo(s.data(), s.size(), os);
00598         }
00599 #endif
00600
00601         void PrintU16StringTo(const ::std::u16string& s, ostream* os) {
00602             PrintCharsAsStringTo(s.data(), s.size(), os);
00603         }
00604
00605         void PrintU32StringTo(const ::std::u32string& s, ostream* os) {
00606             PrintCharsAsStringTo(s.data(), s.size(), os);
00607         }
00608
00609 #if GTEST_HAS_STD_WSTRING
00610
00611         void PrintWideStringTo(const ::std::wstring& s, ostream* os) {
00612             PrintCharsAsStringTo(s.data(), s.size(), os);
00613         }
00614 #endif  // GTEST_HAS_STD_WSTRING
00615
00616     } // namespace internal
00617
00618 } // namespace testing
```

18.289 gtest-test-part.cc

```
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00029
00030 //
00031 // The Google C++ Testing and Mocking Framework (Google Test)
00032
00033 #include "gtest/gtest-test-part.h"
00034
00035 #include "gtest/internal/gtest-port.h"
00036 #include "src/gtest-internal-inl.h"
00037
00038 namespace testing {
00039
00040     using internal::GetUnitTestImpl;
00041
00042     // Gets the summary of the failure message by omitting the stack trace
```

```
00043     // in it.
00044
00045     std::string TestPartResult::ExtractSummary(const char* message) {
00046         const char* const stack_trace = strstr(message, internal::kStackTraceMarker);
00047         return stack_trace == nullptr ? message : std::string(message, stack_trace);
00048     }
00049
00050     // Prints a TestPartResult object.
00051
00052     std::ostream& operator<<(std::ostream& os, const TestPartResult& result) {
00053         return os << internal::FormatFileLocation(result.file_name(),
00054                 result.line_number())
00055                 << " "
00056                 << (result.type() == TestPartResult::kSuccess ? "Success"
00057                 : result.type() == TestPartResult::kSkip ? "Skipped"
00058                 : result.type() == TestPartResult::kFatalFailure
00059                 ? "Fatal failure"
00060                 : "Non-fatal failure")
00061                 << ":\n"
00062                 << result.message() << std::endl;
00063     }
00064
00065     // Appends a TestPartResult to the array.
00066
00067     void TestPartResultArray::Append(const TestPartResult& result) {
00068         array_.push_back(result);
00069     }
00070
00071     // Returns the TestPartResult at the given index (0-based).
00072
00073     const TestPartResult& TestPartResultArray::GetTestPartResult(int index) const {
00074         if (index < 0 || index >= size()) {
00075             printf("\nInvalid index (%d) into TestPartResultArray.\n", index);
00076             internal::posix::Abort();
00077         }
00078
00079         return array_[static_cast<size_t> (index)];
00080     }
00081
00082     // Returns the number of TestPartResult objects in the array.
00083
00084     int TestPartResultArray::size() const {
00085         return static_cast<int> (array_.size());
00086     }
00087
00088     namespace internal {
00089
00090         HasNewFatalFailureHelper::HasNewFatalFailureHelper()
00091         : has_new_fatal_failure_(false),
00092         original_reporter_(
00093         GetUnitTestImpl()->GetTestPartResultReporterForCurrentThread()) {
00094             GetUnitTestImpl()->SetTestPartResultReporterForCurrentThread(this);
00095         }
00096
00097         HasNewFatalFailureHelper::~HasNewFatalFailureHelper() {
00098             GetUnitTestImpl()->SetTestPartResultReporterForCurrentThread(
00099                     original_reporter_);
00100         }
00101
00102         void HasNewFatalFailureHelper::ReportTestPartResult(
00103                 const TestPartResult& result) {
00104             if (result.fatally_failed()) has_new_fatal_failure_ = true;
00105             original_reporter_->ReportTestPartResult(result);
00106         }
00107
00108     } // namespace internal
00109
00110 } // namespace testing
```

18.290 gtest-typed-test.cc

```
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00029
00030 #include "gtest/gtest-typed-test.h"
00031
00032 #include "gtest/gtest.h"
00033
00034 namespace testing {
00035     namespace internal {
00036
00037         // Skips to the first non-space char in str. Returns an empty string if str
00038         // contains only whitespace characters.
00039
00040         static const char* SkipSpaces(const char* str) {
00041             while (IsSpace(*str)) str++;
00042             return str;
00043         }
00044
00045         static std::vector<std::string> SplitIntoTestNames(const char* src) {
00046             std::vector<std::string> name_vec;
00047             src = SkipSpaces(src);
00048             for (; src != nullptr; src = SkipComma(src)) {
00049                 name_vec.push_back(StripTrailingSpaces(GetPrefixUntilComma(src)));
00050             }
00051             return name_vec;
00052         }
00053
00054         // Verifies that registered_tests match the test names in
00055         // registered_tests_; returns registered_tests if successful, or
00056         // aborts the program otherwise.
00057
00058         const char* TypedTestSuitePState::VerifyRegisteredTestNames(
00059                 const char* test_suite_name, const char* file, int line,
00060                 const char* registered_tests) {
00061             RegisterTypeParameterizedTestSuite(test_suite_name, CodeLocation(file, line));
00062
00063             typedef RegisteredTestsMap::const_iterator RegisteredTestIter;
00064             registered_ = true;
00065
00066             std::vector<std::string> name_vec = SplitIntoTestNames(registered_tests);
00067
00068             Message errors;
00069
00070             std::set<std::string> tests;
00071             for (std::vector<std::string>::const_iterator name_it = name_vec.begin();
00072                     name_it != name_vec.end(); ++name_it) {
00073                 const std::string& name = *name_it;
00074                 if (tests.count(name) != 0) {
00075                     errors << "Test " << name << " is listed more than once.\n";
00076                     continue;
00077                 }
00078
00079                 if (registered_tests_.count(name) != 0) {
00080                     tests.insert(name);
00081                 } else {
00082                     errors << "No test named " << name
00083                             << " can be found in this test suite.\n";
00084                 }
00085             }
00086
00087             for (RegisteredTestIter it = registered_tests_.begin();
00088                     it != registered_tests_.end(); ++it) {
00089                 if (tests.count(it->first) == 0) {
00090                     errors << "You forgot to list test " << it->first << ".\n";
00091                 }
00092             }
00093
00094             const std::string& errors_str = errors.GetString();
00095             if (errors_str != "") {
00096                 fprintf(stderr, "%s %s", FormatFileLocation(file, line).c_str(),
00097                         errors_str.c_str());
00098                 fflush(stderr);
00099                 posix::Abort();
00100             }
00101
```

```
00102             return registered_tests;
00103         }
00104
00105     } // namespace internal
00106 } // namespace testing
```

18.291 gtest.cc

```
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00003 //
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00027 // (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE
00028 // OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
00029
00030 //
00031 // The Google C++ Testing and Mocking Framework (Google Test)
00032
00033 #include "gtest/gtest.h"
00034
00035 #include <ctype.h>
00036 #include <stdarg.h>
00037 #include <stdio.h>
00038 #include <stdlib.h>
00039 #include <time.h>
00040 #include <wchar.h>
00041 #include <wctype.h>
00042
00043 #include <algorithm>
00044 #include <chrono>  // NOLINT
00045 #include <cmath>
00046 #include <cstdint>
00047 #include <initializer_list>
00048 #include <iomanip>
00049 #include <iterator>
00050 #include <limits>
00051 #include <list>
00052 #include <map>
00053 #include <ostream>  // NOLINT
00054 #include <sstream>
00055 #include <unordered_set>
00056 #include <vector>
00057
00058 #include "gtest/gtest-assertion-result.h"
00059 #include "gtest/gtest-spi.h"
00060 #include "gtest/internal/custom/gtest.h"
00061
00062 #if GTEST_OS_LINUX
00063
00064 #include <fcntl.h>   // NOLINT
00065 #include <limits.h>  // NOLINT
00066 #include <sched.h>   // NOLINT
00067 // Declares vsnprintf().  This header is not available on Windows.
00068 #include <strings.h>   // NOLINT
00069 #include <sys/mman.h>  // NOLINT
00070 #include <sys/time.h>  // NOLINT
00071 #include <unistd.h>    // NOLINT
00072
00073 #include <string>
00074
00075 #elif GTEST_OS_ZOS
00076 #include <sys/time.h>  // NOLINT
00077
```

```
00078 // On z/OS we additionally need strings.h for strcasecmp.
00079 #include <strings.h>   // NOLINT
00080
00081 #elif GTEST_OS_WINDOWS_MOBILE  // We are on Windows CE.
00082
00083 #include <windows.h>  // NOLINT
00084 #undef min
00085
00086 #elif GTEST_OS_WINDOWS  // We are on Windows proper.
00087
00088 #include <windows.h>  // NOLINT
00089 #undef min
00090
00091 #ifdef _MSC_VER
00092 #include <crtdbg.h>  // NOLINT
00093 #endif
00094
00095 #include <io.h>         // NOLINT
00096 #include <sys/stat.h>   // NOLINT
00097 #include <sys/timeb.h>  // NOLINT
00098 #include <sys/types.h>  // NOLINT
00099
00100 #if GTEST_OS_WINDOWS_MINGW
00101 #include <sys/time.h>  // NOLINT
00102 #endif                 // GTEST_OS_WINDOWS_MINGW
00103
00104 #else
00105
00106 // cpplint thinks that the header is already included, so we want to
00107 // silence it.
00108 #include <sys/time.h>  // NOLINT
00109 #include <unistd.h>    // NOLINT
00110
00111 #endif  // GTEST_OS_LINUX
00112
00113 #if GTEST_HAS_EXCEPTIONS
00114 #include <stdexcept>
00115 #endif
00116
00117 #if GTEST_CAN_STREAM_RESULTS_
00118 #include <arpa/inet.h>   // NOLINT
00119 #include <netdb.h>       // NOLINT
00120 #include <sys/socket.h>  // NOLINT
00121 #include <sys/types.h>   // NOLINT
00122 #endif
00123
00124 #include "src/gtest-internal-inl.h"
00125
00126 #if GTEST_OS_WINDOWS
00127 #define vsnprintf _vsnprintf
00128 #endif  // GTEST_OS_WINDOWS
00129
00130 #if GTEST_OS_MAC
00131 #ifndef GTEST_OS_IOS
00132 #include <crt_externs.h>
00133 #endif
00134 #endif
00135
00136 #if GTEST_HAS_ABSL
00137 #include "absl/debugging/failure_signal_handler.h"
00138 #include "absl/debugging/stacktrace.h"
00139 #include "absl/debugging/symbolize.h"
00140 #include "absl/flags/parse.h"
00141 #include "absl/flags/usage.h"
00142 #include "absl/strings/str_cat.h"
00143 #include "absl/strings/str_replace.h"
00144 #endif  // GTEST_HAS_ABSL
00145
00146 namespace testing {
00147
00148     using internal::CountIf;
00149     using internal::ForEach;
00150     using internal::GetElementOr;
00151     using internal::Shuffle;
00152
00153     // Constants.
00154
00155     // A test whose test suite name or test name matches this filter is
00156     // disabled and not run.
00157     static const char kDisableTestFilter[] = "DISABLED_*:*/DISABLED_*";
00158
00159     // A test suite whose name matches this filter is considered a death
00160     // test suite and will be run before test suites whose name doesn't
00161     // match this filter.
00162     static const char kDeathTestSuiteFilter[] = "*DeathTest:*DeathTest/*";
00163
00164     // A test filter that matches everything.
```

```
00165     static const char kUniversalFilter[] = "*";
00166
00167     // The default output format.
00168     static const char kDefaultOutputFormat[] = "xml";
00169     // The default output file.
00170     static const char kDefaultOutputFile[] = "test_detail";
00171
00172     // The environment variable name for the test shard index.
00173     static const char kTestShardIndex[] = "GTEST_SHARD_INDEX";
00174     // The environment variable name for the total number of test shards.
00175     static const char kTestTotalShards[] = "GTEST_TOTAL_SHARDS";
00176     // The environment variable name for the test shard status file.
00177     static const char kTestShardStatusFile[] = "GTEST_SHARD_STATUS_FILE";
00178
00179     namespace internal {
00180
00181         // The text used in failure messages to indicate the start of the
00182         // stack trace.
00183         const char kStackTraceMarker[] = "\nStack trace:\n";
00184
00185         // g_help_flag is true if and only if the --help flag or an equivalent form
00186         // is specified on the command line.
00187         bool g_help_flag = false;
00188
00189         // Utility function to Open File for Writing
00190
00191         static FILE* OpenFileForWriting(const std::string& output_file) {
00192             FILE* fileout = nullptr;
00193             FilePath output_file_path(output_file);
00194             FilePath output_dir(output_file_path.RemoveFileName());
00195
00196             if (output_dir.CreateDirectoriesRecursively()) {
00197                 fileout = posix::FOpen(output_file.c_str(), "w");
00198             }
00199             if (fileout == nullptr) {
00200                 GTEST_LOG_(FATAL) << "Unable to open file \"" << output_file << "\"";
00201             }
00202             return fileout;
00203         }
00204
00205     } // namespace internal
00206
00207     // Bazel passes in the argument to '--test_filter' via the TESTBRIDGE_TEST_ONLY
00208     // environment variable.
00209
00210     static const char* GetDefaultFilter() {
00211         const char* const testbridge_test_only =
00212                 internal::posix::GetEnv("TESTBRIDGE_TEST_ONLY");
00213         if (testbridge_test_only != nullptr) {
00214             return testbridge_test_only;
00215         }
00216         return kUniversalFilter;
00217     }
00218
00219     // Bazel passes in the argument to '--test_runner_fail_fast' via the
00220     // TESTBRIDGE_TEST_RUNNER_FAIL_FAST environment variable.
00221
00222     static bool GetDefaultFailFast() {
00223         const char* const testbridge_test_runner_fail_fast =
00224                 internal::posix::GetEnv("TESTBRIDGE_TEST_RUNNER_FAIL_FAST");
00225         if (testbridge_test_runner_fail_fast != nullptr) {
00226             return strcmp(testbridge_test_runner_fail_fast, "1") == 0;
00227         }
00228         return false;
00229     }
00230
00231 } // namespace testing
00232
00233 GTEST_DEFINE_bool_(
00234         fail_fast,
00235         testing::internal::BoolFromGTestEnv("fail_fast",
00236         testing::GetDefaultFailFast()),
00237         "True if and only if a test failure should stop further test execution.");
00238
00239 GTEST_DEFINE_bool_(
00240         also_run_disabled_tests,
00241         testing::internal::BoolFromGTestEnv("also_run_disabled_tests", false),
00242         "Run disabled tests too, in addition to the tests normally being run.");
00243
00244 GTEST_DEFINE_bool_(
00245         break_on_failure,
00246         testing::internal::BoolFromGTestEnv("break_on_failure", false),
00247         "True if and only if a failed assertion should be a debugger "
00248         "break-point.");
00249
00250 GTEST_DEFINE_bool_(catch_exceptions,
00251         testing::internal::BoolFromGTestEnv("catch_exceptions",
```

```
00252         true),
00253         "True if and only if " GTEST_NAME_
00254         " should catch exceptions and treat them as test failures.");
00255
00256 GTEST_DEFINE_string_(
00257         color, testing::internal::StringFromGTestEnv("color", "auto"),
00258         "Whether to use colors in the output.  Valid values: yes, no, "
00259         "and auto.  'auto' means to use colors if the output is "
00260         "being sent to a terminal and the TERM environment variable "
00261         "is set to a terminal type that supports colors.");
00262
00263 GTEST_DEFINE_string_(
00264         filter,
00265         testing::internal::StringFromGTestEnv("filter",
00266         testing::GetDefaultFilter()),
00267         "A colon-separated list of glob (not regex) patterns "
00268         "for filtering the tests to run, optionally followed by a "
00269         "'-' and a : separated list of negative patterns (tests to "
00270         "exclude).  A test is run if it matches one of the positive "
00271         "patterns and does not match any of the negative patterns.");
00272
00273 GTEST_DEFINE_bool_(
00274         install_failure_signal_handler,
00275         testing::internal::BoolFromGTestEnv("install_failure_signal_handler",
00276         false),
00277         "If true and supported on the current platform, " GTEST_NAME_
00278         " should "
00279         "install a signal handler that dumps debugging information when fatal "
00280         "signals are raised.");
00281
00282 GTEST_DEFINE_bool_(list_tests, false, "List all tests without running them.");
00283
00284 // The net priority order after flag processing is thus:
00285 //   --gtest_output command line flag
00286 //   GTEST_OUTPUT environment variable
00287 //   XML_OUTPUT_FILE environment variable
00288 //   ''
00289 GTEST_DEFINE_string_(
00290         output,
00291         testing::internal::StringFromGTestEnv(
00292         "output", testing::internal::OutputFlagAlsoCheckEnvVar().c_str()),
00293         "A format (defaults to \"xml\" but can be specified to be \"json\"), "
00294         "optionally followed by a colon and an output file name or directory. "
00295         "A directory is indicated by a trailing pathname separator. "
00296         "Examples: \"xml:filename.xml\", \"xml::directoryname/\". "
00297         "If a directory is specified, output files will be created "
00298         "within that directory, with file-names based on the test "
00299         "executable's name and, if necessary, made unique by adding "
00300         "digits.");
00301
00302 GTEST_DEFINE_bool_(
00303         brief, testing::internal::BoolFromGTestEnv("brief", false),
00304         "True if only test failures should be displayed in text output.");
00305
00306 GTEST_DEFINE_bool_(print_time,
00307         testing::internal::BoolFromGTestEnv("print_time", true),
00308         "True if and only if " GTEST_NAME_
00309         " should display elapsed time in text output.");
00310
00311 GTEST_DEFINE_bool_(print_utf8,
00312         testing::internal::BoolFromGTestEnv("print_utf8", true),
00313         "True if and only if " GTEST_NAME_
00314         " prints UTF8 characters as text.");
00315
00316 GTEST_DEFINE_int32_(
00317         random_seed, testing::internal::Int32FromGTestEnv("random_seed", 0),
00318         "Random number seed to use when shuffling test orders.  Must be in range "
00319         "[1, 99999], or 0 to use a seed based on the current time.");
00320
00321 GTEST_DEFINE_int32_(
00322         repeat, testing::internal::Int32FromGTestEnv("repeat", 1),
00323         "How many times to repeat each test.  Specify a negative number "
00324         "for repeating forever.  Useful for shaking out flaky tests.");
00325
00326 GTEST_DEFINE_bool_(
00327         recreate_environments_when_repeating,
00328         testing::internal::BoolFromGTestEnv("recreate_environments_when_repeating",
00329         false),
00330         "Controls whether global test environments are recreated for each repeat "
00331         "of the tests. If set to false the global test environments are only set "
00332         "up once, for the first iteration, and only torn down once, for the last. "
00333         "Useful for shaking out flaky tests with stable, expensive test "
00334         "environments. If --gtest_repeat is set to a negative number, meaning "
00335         "there is no last run, the environments will always be recreated to avoid "
00336         "leaks.");
00337
00338 GTEST_DEFINE_bool_(show_internal_stack_frames, false,
```

```
00339         "True if and only if " GTEST_NAME_
00340         " should include internal stack frames when "
00341         "printing test failure stack traces.");
00342
00343 GTEST_DEFINE_bool_(shuffle,
00344         testing::internal::BoolFromGTestEnv("shuffle", false),
00345         "True if and only if " GTEST_NAME_
00346         " should randomize tests' order on every run.");
00347
00348 GTEST_DEFINE_int32_(
00349         stack_trace_depth,
00350         testing::internal::Int32FromGTestEnv("stack_trace_depth",
00351         testing::kMaxStackTraceDepth),
00352         "The maximum number of stack frames to print when an "
00353         "assertion fails.  The valid range is 0 through 100, inclusive.");
00354
00355 GTEST_DEFINE_string_(
00356         stream_result_to,
00357         testing::internal::StringFromGTestEnv("stream_result_to", ""),
00358         "This flag specifies the host name and the port number on which to stream "
00359         "test results. Example: \"localhost:555\". The flag is effective only on "
00360         "Linux.");
00361
00362 GTEST_DEFINE_bool_(
00363         throw_on_failure,
00364         testing::internal::BoolFromGTestEnv("throw_on_failure", false),
00365         "When this flag is specified, a failed assertion will throw an exception "
00366         "if exceptions are enabled or exit the program with a non-zero code "
00367         "otherwise. For use with an external test framework.");
00368
00369 #if GTEST_USE_OWN_FLAGFILE_FLAG_
00370 GTEST_DEFINE_string_(
00371         flagfile, testing::internal::StringFromGTestEnv("flagfile", ""),
00372         "This flag specifies the flagfile to read command-line flags from.");
00373 #endif  // GTEST_USE_OWN_FLAGFILE_FLAG_
00374
00375 namespace testing {
00376     namespace internal {
00377
00378         const uint32_t Random::kMaxRange;
00379
00380         // Generates a random number from [0, range), using a Linear
00381         // Congruential Generator (LCG).  Crashes if 'range' is 0 or greater
00382         // than kMaxRange.
00383
00384         uint32_t Random::Generate(uint32_t range) {
00385             // These constants are the same as are used in glibc's rand(3).
00386             // Use wider types than necessary to prevent unsigned overflow diagnostics.
00387             state_ = static_cast<uint32_t> (1103515245ULL * state_ + 12345U) % kMaxRange;
00388
00389             GTEST_CHECK_(range > 0) << "Cannot generate a number in the range [0, 0).";
00390             GTEST_CHECK_(range <= kMaxRange)
00391                     << "Generation of a number in [0, " << range << ") was requested, "
00392                     << "but this can only generate numbers in [0, " << kMaxRange << ").";
00393
00394             // Converting via modulus introduces a bit of downward bias, but
00395             // it's simple, and a linear congruential generator isn't too good
00396             // to begin with.
00397             return state_ % range;
00398         }
00399
00400         // GTestIsInitialized() returns true if and only if the user has initialized
00401         // Google Test.  Useful for catching the user mistake of not initializing
00402         // Google Test before calling RUN_ALL_TESTS().
00403
00404         static bool GTestIsInitialized() {
00405             return GetArgvs().size() > 0;
00406         }
00407
00408         // Iterates over a vector of TestSuites, keeping a running sum of the
00409         // results of calling a given int-returning method on each.
00410         // Returns the sum.
00411
00412         static int SumOverTestSuiteList(const std::vector<TestSuite*>& case_list,
00413                 int (TestSuite::*method)() const) {
00414             int sum = 0;
00415             for (size_t i = 0; i < case_list.size(); i++) {
00416                 sum += (case_list[i]->*method)();
00417             }
00418             return sum;
00419         }
00420
00421         // Returns true if and only if the test suite passed.
00422
00423         static bool TestSuitePassed(const TestSuite* test_suite) {
00424             return test_suite->should_run() && test_suite->Passed();
00425         }
```

```
00426
00427         // Returns true if and only if the test suite failed.
00428
00429         static bool TestSuiteFailed(const TestSuite* test_suite) {
00430             return test_suite->should_run() && test_suite->Failed();
00431         }
00432
00433         // Returns true if and only if test_suite contains at least one test that
00434         // should run.
00435
00436         static bool ShouldRunTestSuite(const TestSuite* test_suite) {
00437             return test_suite->should_run();
00438         }
00439
00440         // AssertHelper constructor.
00441
00442         AssertHelper::AssertHelper(TestPartResult::Type type, const char* file,
00443                 int line, const char* message)
00444         : data_(new AssertHelperData(type, file, line, message)) {
00445         }
00446
00447         AssertHelper::~AssertHelper() {
00448             delete data_;
00449         }
00450
00451         // Message assignment, for assertion streaming support.
00452
00453         void AssertHelper::operator=(const Message& message) const {
00454             UnitTest::GetInstance()->AddTestPartResult(
00455                     data_->type, data_->file, data_->line,
00456                     AppendUserMessage(data_->message, message),
00457                     UnitTest::GetInstance()->impl()->CurrentOsStackTraceExceptTop(1)
00458                     // Skips the stack frame for this function itself.
00459                     ); // NOLINT
00460         }
00461
00462         namespace {
00463
00464             // When TEST_P is found without a matching INSTANTIATE_TEST_SUITE_P
00465             // to creates test cases for it, a synthetic test case is
00466             // inserted to report ether an error or a log message.
00467             //
00468             // This configuration bit will likely be removed at some point.
00469             constexpr bool kErrorOnUninstantiatedParameterizedTest = true;
00470             constexpr bool kErrorOnUninstantiatedTypeParameterizedTest = true;
00471
00472             // A test that fails at a given file/line location with a given message.
00473
00474             class FailureTest : public Test {
00475             public:
00476
00477                 explicit FailureTest(const CodeLocation& loc, std::string error_message,
00478                         bool as_error)
00479                 : loc_(loc),
00480                 error_message_(std::move(error_message)),
00481                 as_error_(as_error) {
00482                 }
00483
00484                 void TestBody() override {
00485                     if (as_error_) {
00486                         AssertHelper(TestPartResult::kNonFatalFailure, loc_.file.c_str(),
00487                                 loc_.line, "") = Message() << error_message_;
00488                     } else {
00489                         std::cout << error_message_ << std::endl;
00490                     }
00491                 }
00492
00493             private:
00494                 const CodeLocation loc_;
00495                 const std::string error_message_;
00496                 const bool as_error_;
00497             };
00498
00499         } // namespace
00500
00501         std::set<std::string>* GetIgnoredParameterizedTestSuites() {
00502             return UnitTest::GetInstance()->impl()->ignored_parameterized_test_suites();
00503         }
00504
00505         // Add a given test_suit to the list of them allow to go un-instantiated.
00506
00507         MarkAsIgnored::MarkAsIgnored(const char* test_suite) {
00508             GetIgnoredParameterizedTestSuites()->insert(test_suite);
00509         }
00510
00511         // If this parameterized test suite has no instantiations (and that
00512         // has not been marked as okay), emit a test case reporting that.
```

```
00513
00514         void InsertSyntheticTestCase(const std::string& name, CodeLocation location,
00515                 bool has_test_p) {
00516             const auto& ignored = *GetIgnoredParameterizedTestSuites();
00517             if (ignored.find(name) != ignored.end()) return;
00518
00519             const char kMissingInstantiation[] = //
00520                     " is defined via TEST_P, but never instantiated. None of the test cases "
00521                     "will run. Either no INSTANTIATE_TEST_SUITE_P is provided or the only "
00522                     "ones provided expand to nothing."
00523                     "\n\n"
00524                     "Ideally, TEST_P definitions should only ever be included as part of "
00525                     "binaries that intend to use them. (As opposed to, for example, being "
00526                     "placed in a library that may be linked in to get other utilities.)";
00527
00528             const char kMissingTestCase[] = //
00529                     " is instantiated via INSTANTIATE_TEST_SUITE_P, but no tests are "
00530                     "defined via TEST_P . No test cases will run."
00531                     "\n\n"
00532                     "Ideally, INSTANTIATE_TEST_SUITE_P should only ever be invoked from "
00533                     "code that always depend on code that provides TEST_P. Failing to do "
00534                     "so is often an indication of dead code, e.g. the last TEST_P was "
00535                     "removed but the rest got left behind.";
00536
00537             std::string message =
00538                     "Parameterized test suite " + name +
00539                     (has_test_p ? kMissingInstantiation : kMissingTestCase) +
00540                     "\n\n"
00541                     "To suppress this error for this test suite, insert the following line "
00542                     "(in a non-header) in the namespace it is defined in:"
00543                     "\n\n"
00544                     "GTEST_ALLOW_UNINSTANTIATED_PARAMETERIZED_TEST(" +
00545                     name + ");";
00546
00547             std::string full_name = "UninstantiatedParameterizedTestSuite<" + name + ">";
00548             RegisterTest(//
00549                     "GoogleTestVerification", full_name.c_str(),
00550                     nullptr, // No type parameter.
00551                     nullptr, // No value parameter.
00552                     location.file.c_str(), location.line, [message, location] {
00553                         return new FailureTest(location, message,
00554                                 kErrorOnUninstantiatedParameterizedTest);
00555                     });
00556         }
00557
00558         void RegisterTypeParameterizedTestSuite(const char* test_suite_name,
00559                 CodeLocation code_location) {
00560             GetUnitTestImpl()->type_parameterized_test_registry().RegisterTestSuite(
00561                     test_suite_name, code_location);
00562         }
00563
00564         void RegisterTypeParameterizedTestSuiteInstantiation(const char* case_name) {
00565             GetUnitTestImpl()->type_parameterized_test_registry().RegisterInstantiation(
00566                     case_name);
00567         }
00568
00569         void TypeParameterizedTestSuiteRegistry::RegisterTestSuite(
00570                 const char* test_suite_name, CodeLocation code_location) {
00571             suites_.emplace(std::string(test_suite_name),
00572                     TypeParameterizedTestSuiteInfo(code_location));
00573         }
00574
00575         void TypeParameterizedTestSuiteRegistry::RegisterInstantiation(
00576                 const char* test_suite_name) {
00577             auto it = suites_.find(std::string(test_suite_name));
00578             if (it != suites_.end()) {
00579                 it->second.instantiated = true;
00580             } else {
00581                 GTEST_LOG_(ERROR) << "Unknown type parameterized test suit '"
00582                         << test_suite_name << "'";
00583             }
00584         }
00585
00586         void TypeParameterizedTestSuiteRegistry::CheckForInstantiations() {
00587             const auto& ignored = *GetIgnoredParameterizedTestSuites();
00588             for (const auto& testcase : suites_) {
00589                 if (testcase.second.instantiated) continue;
00590                 if (ignored.find(testcase.first) != ignored.end()) continue;
00591
00592                 std::string message =
00593                         "Type parameterized test suite " + testcase.first +
00594                         " is defined via REGISTER_TYPED_TEST_SUITE_P, but never instantiated "
00595                         "via INSTANTIATE_TYPED_TEST_SUITE_P. None of the test cases will run."
00596                         "\n\n"
00597                         "Ideally, TYPED_TEST_P definitions should only ever be included as "
00598                         "part of binaries that intend to use them. (As opposed to, for "
00599                         "example, being placed in a library that may be linked in to get other "
```

```
00600                         "utilities.)"
00601                         "\n\n"
00602                         "To suppress this error for this test suite, insert the following line "
00603                         "(in a non-header) in the namespace it is defined in:"
00604                         "\n\n"
00605                         "GTEST_ALLOW_UNINSTANTIATED_PARAMETERIZED_TEST(" +
00606                         testcase.first + ");";
00607
00608                 std::string full_name =
00609                         "UninstantiatedTypeParameterizedTestSuite<" + testcase.first + ">";
00610                 RegisterTest(//
00611                         "GoogleTestVerification", full_name.c_str(),
00612                         nullptr, // No type parameter.
00613                         nullptr, // No value parameter.
00614                         testcase.second.code_location.file.c_str(),
00615                         testcase.second.code_location.line, [message, testcase] {
00616                             return new FailureTest(testcase.second.code_location, message,
00617                                     kErrorOnUninstantiatedTypeParameterizedTest);
00618                         });
00619             }
00620         }
00621
00622         // A copy of all command line arguments.  Set by InitGoogleTest().
00623         static ::std::vector<std::string> g_argvs;
00624
00625         ::std::vector<std::string> GetArgvs() {
00626 #if defined(GTEST_CUSTOM_GET_ARGVS_)
00627             // GTEST_CUSTOM_GET_ARGVS_() may return a container of std::string or
00628             // ::string. This code converts it to the appropriate type.
00629             const auto& custom = GTEST_CUSTOM_GET_ARGVS_();
00630             return ::std::vector<std::string>(custom.begin(), custom.end());
00631 #else   // defined(GTEST_CUSTOM_GET_ARGVS_)
00632             return g_argvs;
00633 #endif  // defined(GTEST_CUSTOM_GET_ARGVS_)
00634         }
00635
00636         // Returns the current application's name, removing directory path if that
00637         // is present.
00638
00639         FilePath GetCurrentExecutableName() {
00640             FilePath result;
00641
00642 #if GTEST_OS_WINDOWS || GTEST_OS_OS2
00643             result.Set(FilePath(GetArgvs()[0]).RemoveExtension("exe"));
00644 #else
00645             result.Set(FilePath(GetArgvs()[0]));
00646 #endif  // GTEST_OS_WINDOWS
00647
00648             return result.RemoveDirectoryName();
00649         }
00650
00651         // Functions for processing the gtest_output flag.
00652
00653         // Returns the output format, or "" for normal printed output.
00654
00655         std::string UnitTestOptions::GetOutputFormat() {
00656             std::string s = GTEST_FLAG_GET(output);
00657             const char* const gtest_output_flag = s.c_str();
00658             const char* const colon = strchr(gtest_output_flag, ':');
00659             return (colon == nullptr)
00660                     ? std::string(gtest_output_flag)
00661                     : std::string(gtest_output_flag,
00662                     static_cast<size_t> (colon - gtest_output_flag));
00663         }
00664
00665         // Returns the name of the requested output file, or the default if none
00666         // was explicitly specified.
00667
00668         std::string UnitTestOptions::GetAbsolutePathToOutputFile() {
00669             std::string s = GTEST_FLAG_GET(output);
00670             const char* const gtest_output_flag = s.c_str();
00671
00672             std::string format = GetOutputFormat();
00673             if (format.empty()) format = std::string(kDefaultOutputFormat);
00674
00675             const char* const colon = strchr(gtest_output_flag, ':');
00676             if (colon == nullptr)
00677                 return internal::FilePath::MakeFileName(
00678                     internal::FilePath(
00679                     UnitTest::GetInstance()->original_working_dir()),
00680                     internal::FilePath(kDefaultOutputFile), 0, format.c_str())
00681                 .string();
00682
00683             internal::FilePath output_name(colon + 1);
00684             if (!output_name.IsAbsolutePath())
00685                 output_name = internal::FilePath::ConcatPaths(
00686                     internal::FilePath(UnitTest::GetInstance()->original_working_dir()),
```

```
00687                     internal::FilePath(colon + 1));
00688
00689             if (!output_name.IsDirectory()) return output_name.string();
00690
00691             internal::FilePath result(internal::FilePath::GenerateUniqueFileName(
00692                     output_name, internal::GetCurrentExecutableName(),
00693                     GetOutputFormat().c_str()));
00694             return result.string();
00695         }
00696
00697         // Returns true if and only if the wildcard pattern matches the string. Each
00698         // pattern consists of regular characters, single-character wildcards (?), and
00699         // multi-character wildcards (*).
00700         //
00701         // This function implements a linear-time string globbing algorithm based on
00702         // https://research.swtch.com/glob.
00703
00704         static bool PatternMatchesString(const std::string& name_str,
00705                 const char* pattern, const char* pattern_end) {
00706             const char* name = name_str.c_str();
00707             const char* const name_begin = name;
00708             const char* const name_end = name + name_str.size();
00709
00710             const char* pattern_next = pattern;
00711             const char* name_next = name;
00712
00713             while (pattern < pattern_end || name < name_end) {
00714                 if (pattern < pattern_end) {
00715                     switch (*pattern) {
00716                         default: // Match an ordinary character.
00717                             if (name < name_end && *name == *pattern) {
00718                                 ++pattern;
00719                                 ++name;
00720                                 continue;
00721                             }
00722                             break;
00723                         case '?': // Match any single character.
00724                             if (name < name_end) {
00725                                 ++pattern;
00726                                 ++name;
00727                                 continue;
00728                             }
00729                             break;
00730                         case '*':
00731                             // Match zero or more characters. Start by skipping over the wildcard
00732                             // and matching zero characters from name. If that fails, restart and
00733                             // match one more character than the last attempt.
00734                             pattern_next = pattern;
00735                             name_next = name + 1;
00736                             ++pattern;
00737                             continue;
00738                     }
00739                 }
00740                 // Failed to match a character. Restart if possible.
00741                 if (name_begin < name_next && name_next <= name_end) {
00742                     pattern = pattern_next;
00743                     name = name_next;
00744                     continue;
00745                 }
00746                 return false;
00747             }
00748             return true;
00749         }
00750
00751         namespace {
00752
00753             bool IsGlobPattern(const std::string& pattern) {
00754                 return std::any_of(pattern.begin(), pattern.end(),
00755                         [](const char c) {
00756                             return c == '?' || c == '*'; });
00757             }
00758
00759             class UnitTestFilter {
00760             public:
00761                 UnitTestFilter() = default;
00762
00763                 // Constructs a filter from a string of patterns separated by `:'.
00764
00765                 explicit UnitTestFilter(const std::string& filter) {
00766                     // By design "" filter matches "" string.
00767                     std::vector<std::string> all_patterns;
00768                     SplitString(filter, ':', &all_patterns);
00769                     const auto exact_match_patterns_begin = std::partition(
00770                             all_patterns.begin(), all_patterns.end(), &IsGlobPattern);
00771
00772                     glob_patterns_.reserve(static_cast<size_t> (
00773                             std::distance(all_patterns.begin(), exact_match_patterns_begin)));
```

```
00774                   std::move(all_patterns.begin(), exact_match_patterns_begin,
00775                           std::inserter(glob_patterns_, glob_patterns_.begin()));
00776                   std::move(
00777                           exact_match_patterns_begin, all_patterns.end(),
00778                           std::inserter(exact_match_patterns_, exact_match_patterns_.begin()));
00779               }
00780
00781               // Returns true if and only if name matches at least one of the patterns in
00782               // the filter.
00783
00784               bool MatchesName(const std::string& name) const {
00785                   return exact_match_patterns_.count(name) > 0 ||
00786                           std::any_of(glob_patterns_.begin(), glob_patterns_.end(),
00787                           [&name](const std::string & pattern) {
00788                               return PatternMatchesString(
00789                                       name, pattern.c_str(),
00790                                       pattern.c_str() + pattern.size());
00791                           });
00792               }
00793
00794           private:
00795               std::vector<std::string> glob_patterns_;
00796               std::unordered_set<std::string> exact_match_patterns_;
00797           };
00798
00799           class PositiveAndNegativeUnitTestFilter {
00800           public:
00801               // Constructs a positive and a negative filter from a string. The string
00802               // contains a positive filter optionally followed by a '-' character and a
00803               // negative filter. In case only a negative filter is provided the positive
00804               // filter will be assumed "*".
00805               // A filter is a list of patterns separated by ':'.
00806
00807               explicit PositiveAndNegativeUnitTestFilter(const std::string& filter) {
00808                   std::vector<std::string> positive_and_negative_filters;
00809
00810                   // NOTE: `SplitString` always returns a non-empty container.
00811                   SplitString(filter, '-', &positive_and_negative_filters);
00812                   const auto& positive_filter = positive_and_negative_filters.front();
00813
00814                   if (positive_and_negative_filters.size() > 1) {
00815                       positive_filter_ = UnitTestFilter(
00816                               positive_filter.empty() ? kUniversalFilter : positive_filter);
00817
00818                       // TODO(b/214626361): Fail on multiple '-' characters
00819                       // For the moment to preserve old behavior we concatenate the rest of the
00820                       // string parts with `-` as separator to generate the negative filter.
00821                       auto negative_filter_string = positive_and_negative_filters[1];
00822                       for (std::size_t i = 2; i < positive_and_negative_filters.size(); i++)
00823                           negative_filter_string =
00824                               negative_filter_string + '-' + positive_and_negative_filters[i];
00825                       negative_filter_ = UnitTestFilter(negative_filter_string);
00826                   } else {
00827                       // In case we don't have a negative filter and positive filter is ""
00828                       // we do not use kUniversalFilter by design as opposed to when we have a
00829                       // negative filter.
00830                       positive_filter_ = UnitTestFilter(positive_filter);
00831                   }
00832               }
00833
00834               // Returns true if and only if test name (this is generated by appending test
00835               // suit name and test name via a '.' character) matches the positive filter
00836               // and does not match the negative filter.
00837
00838               bool MatchesTest(const std::string& test_suite_name,
00839                       const std::string& test_name) const {
00840                   return MatchesName(test_suite_name + "." + test_name);
00841               }
00842
00843               // Returns true if and only if name matches the positive filter and does not
00844               // match the negative filter.
00845
00846               bool MatchesName(const std::string& name) const {
00847                   return positive_filter_.MatchesName(name) &&
00848                           !negative_filter_.MatchesName(name);
00849               }
00850
00851           private:
00852               UnitTestFilter positive_filter_;
00853               UnitTestFilter negative_filter_;
00854           };
00855       } // namespace
00856
00857       bool UnitTestOptions::MatchesFilter(const std::string& name_str,
00858               const char* filter) {
00859           return UnitTestFilter(filter).MatchesName(name_str);
00860       }
```

```
00861
00862         // Returns true if and only if the user-specified filter matches the test
00863         // suite name and the test name.
00864
00865         bool UnitTestOptions::FilterMatchesTest(const std::string& test_suite_name,
00866                 const std::string& test_name) {
00867             // Split --gtest_filter at '-', if there is one, to separate into
00868             // positive filter and negative filter portions
00869             return PositiveAndNegativeUnitTestFilter(GTEST_FLAG_GET(filter))
00870                     .MatchesTest(test_suite_name, test_name);
00871         }
00872
00873 #if GTEST_HAS_SEH
00874         // Returns EXCEPTION_EXECUTE_HANDLER if Google Test should handle the
00875         // given SEH exception, or EXCEPTION_CONTINUE_SEARCH otherwise.
00876         // This function is useful as an __except condition.
00877
00878         int UnitTestOptions::GTestShouldProcessSEH(DWORD exception_code) {
00879             // Google Test should handle a SEH exception if:
00880             //   1. the user wants it to, AND
00881             //   2. this is not a breakpoint exception, AND
00882             //   3. this is not a C++ exception (VC++ implements them via SEH,
00883             //      apparently).
00884             //
00885             // SEH exception code for C++ exceptions.
00886             // (see http://support.microsoft.com/kb/185294 for more information).
00887             const DWORD kCxxExceptionCode = 0xe06d7363;
00888
00889             bool should_handle = true;
00890
00891             if (!GTEST_FLAG_GET(catch_exceptions))
00892                 should_handle = false;
00893             else if (exception_code == EXCEPTION_BREAKPOINT)
00894                 should_handle = false;
00895             else if (exception_code == kCxxExceptionCode)
00896                 should_handle = false;
00897
00898             return should_handle ? EXCEPTION_EXECUTE_HANDLER : EXCEPTION_CONTINUE_SEARCH;
00899         }
00900 #endif  // GTEST_HAS_SEH
00901
00902     } // namespace internal
00903
00904     // The c'tor sets this object as the test part result reporter used by
00905     // Google Test.  The 'result' parameter specifies where to report the
00906     // results. Intercepts only failures from the current thread.
00907
00908     ScopedFakeTestPartResultReporter::ScopedFakeTestPartResultReporter(
00909             TestPartResultArray* result)
00910     : intercept_mode_(INTERCEPT_ONLY_CURRENT_THREAD), result_(result) {
00911         Init();
00912     }
00913
00914     // The c'tor sets this object as the test part result reporter used by
00915     // Google Test.  The 'result' parameter specifies where to report the
00916     // results.
00917
00918     ScopedFakeTestPartResultReporter::ScopedFakeTestPartResultReporter(
00919             InterceptMode intercept_mode, TestPartResultArray* result)
00920     : intercept_mode_(intercept_mode), result_(result) {
00921         Init();
00922     }
00923
00924     void ScopedFakeTestPartResultReporter::Init() {
00925         internal::UnitTestImpl * const impl = internal::GetUnitTestImpl();
00926         if (intercept_mode_ == INTERCEPT_ALL_THREADS) {
00927             old_reporter_ = impl->GetGlobalTestPartResultReporter();
00928             impl->SetGlobalTestPartResultReporter(this);
00929         } else {
00930             old_reporter_ = impl->GetTestPartResultReporterForCurrentThread();
00931             impl->SetTestPartResultReporterForCurrentThread(this);
00932         }
00933     }
00934
00935     // The d'tor restores the test part result reporter used by Google Test
00936     // before.
00937
00938     ScopedFakeTestPartResultReporter::~ScopedFakeTestPartResultReporter() {
00939         internal::UnitTestImpl * const impl = internal::GetUnitTestImpl();
00940         if (intercept_mode_ == INTERCEPT_ALL_THREADS) {
00941             impl->SetGlobalTestPartResultReporter(old_reporter_);
00942         } else {
00943             impl->SetTestPartResultReporterForCurrentThread(old_reporter_);
00944         }
00945     }
00946
00947     // Increments the test part result count and remembers the result.
```

```
00948     // This method is from the TestPartResultReporterInterface interface.
00949
00950     void ScopedFakeTestPartResultReporter::ReportTestPartResult(
00951             const TestPartResult& result) {
00952         result_->Append(result);
00953     }
00954
00955     namespace internal {
00956
00957         // Returns the type ID of ::testing::Test.  We should always call this
00958         // instead of GetTypeId< ::testing::Test>() to get the type ID of
00959         // testing::Test.  This is to work around a suspected linker bug when
00960         // using Google Test as a framework on Mac OS X.  The bug causes
00961         // GetTypeId< ::testing::Test>() to return different values depending
00962         // on whether the call is from the Google Test framework itself or
00963         // from user test code.  GetTestTypeId() is guaranteed to always
00964         // return the same value, as it always calls GetTypeId<>() from the
00965         // gtest.cc, which is within the Google Test framework.
00966
00967         TypeId GetTestTypeId() {
00968             return GetTypeId<Test>();
00969         }
00970
00971         // The value of GetTestTypeId() as seen from within the Google Test
00972         // library.  This is solely for testing GetTestTypeId().
00973         extern const TypeId kTestTypeIdInGoogleTest = GetTestTypeId();
00974
00975         // This predicate-formatter checks that 'results' contains a test part
00976         // failure of the given type and that the failure message contains the
00977         // given substring.
00978
00979         static AssertionResult HasOneFailure(const char* /* results_expr */,
00980                 const char* /* type_expr */,
00981                 const char* /* substr_expr */,
00982                 const TestPartResultArray& results,
00983                 TestPartResult::Type type,
00984                 const std::string& substr) {
00985             const std::string expected(type == TestPartResult::kFatalFailure
00986                     ? "1 fatal failure"
00987                     : "1 non-fatal failure");
00988             Message msg;
00989             if (results.size() != 1) {
00990                 msg << "Expected: " << expected << "\n"
00991                         << "  Actual: " << results.size() << " failures";
00992                 for (int i = 0; i < results.size(); i++) {
00993                     msg << "\n" << results.GetTestPartResult(i);
00994                 }
00995                 return AssertionFailure() << msg;
00996             }
00997
00998             const TestPartResult& r = results.GetTestPartResult(0);
00999             if (r.type() != type) {
01000                 return AssertionFailure() << "Expected: " << expected << "\n"
01001                         << "  Actual:\n"
01002                         << r;
01003             }
01004
01005             if (strstr(r.message(), substr.c_str()) == nullptr) {
01006                 return AssertionFailure()
01007                         << "Expected: " << expected << " containing \"" << substr << "\"\n"
01008                         << "  Actual:\n"
01009                         << r;
01010             }
01011
01012             return AssertionSuccess();
01013         }
01014
01015         // The constructor of SingleFailureChecker remembers where to look up
01016         // test part results, what type of failure we expect, and what
01017         // substring the failure message should contain.
01018
01019         SingleFailureChecker::SingleFailureChecker(const TestPartResultArray* results,
01020                 TestPartResult::Type type,
01021                 const std::string& substr)
01022         : results_(results), type_(type), substr_(substr) {
01023         }
01024
01025         // The destructor of SingleFailureChecker verifies that the given
01026         // TestPartResultArray contains exactly one failure that has the given
01027         // type and contains the given substring.  If that's not the case, a
01028         // non-fatal failure will be generated.
01029
01030         SingleFailureChecker::~SingleFailureChecker() {
01031             EXPECT_PRED_FORMAT3(HasOneFailure, *results_, type_, substr_);
01032         }
01033
01034         DefaultGlobalTestPartResultReporter::DefaultGlobalTestPartResultReporter(
```

```
01035             UnitTestImpl* unit_test)
01036         : unit_test_(unit_test) {
01037         }
01038
01039         void DefaultGlobalTestPartResultReporter::ReportTestPartResult(
01040             const TestPartResult& result) {
01041           unit_test_->current_test_result()->AddTestPartResult(result);
01042           unit_test_->listeners()->repeater()->OnTestPartResult(result);
01043         }
01044
01045         DefaultPerThreadTestPartResultReporter::DefaultPerThreadTestPartResultReporter(
01046             UnitTestImpl* unit_test)
01047         : unit_test_(unit_test) {
01048         }
01049
01050         void DefaultPerThreadTestPartResultReporter::ReportTestPartResult(
01051             const TestPartResult& result) {
01052           unit_test_->GetGlobalTestPartResultReporter()->ReportTestPartResult(result);
01053         }
01054
01055         // Returns the global test part result reporter.
01056
01057         TestPartResultReporterInterface*
01058         UnitTestImpl::GetGlobalTestPartResultReporter() {
01059           internal::MutexLock lock(&global_test_part_result_reporter_mutex_);
01060           return global_test_part_result_reporter_;
01061         }
01062
01063         // Sets the global test part result reporter.
01064
01065         void UnitTestImpl::SetGlobalTestPartResultReporter(
01066             TestPartResultReporterInterface* reporter) {
01067           internal::MutexLock lock(&global_test_part_result_reporter_mutex_);
01068           global_test_part_result_reporter_ = reporter;
01069         }
01070
01071         // Returns the test part result reporter for the current thread.
01072
01073         TestPartResultReporterInterface*
01074         UnitTestImpl::GetTestPartResultReporterForCurrentThread() {
01075           return per_thread_test_part_result_reporter_.get();
01076         }
01077
01078         // Sets the test part result reporter for the current thread.
01079
01080         void UnitTestImpl::SetTestPartResultReporterForCurrentThread(
01081             TestPartResultReporterInterface* reporter) {
01082           per_thread_test_part_result_reporter_.set(reporter);
01083         }
01084
01085         // Gets the number of successful test suites.
01086
01087         int UnitTestImpl::successful_test_suite_count() const {
01088           return CountIf(test_suites_, TestSuitePassed);
01089         }
01090
01091         // Gets the number of failed test suites.
01092
01093         int UnitTestImpl::failed_test_suite_count() const {
01094           return CountIf(test_suites_, TestSuiteFailed);
01095         }
01096
01097         // Gets the number of all test suites.
01098
01099         int UnitTestImpl::total_test_suite_count() const {
01100           return static_cast<int> (test_suites_.size());
01101         }
01102
01103         // Gets the number of all test suites that contain at least one test
01104         // that should run.
01105
01106         int UnitTestImpl::test_suite_to_run_count() const {
01107           return CountIf(test_suites_, ShouldRunTestSuite);
01108         }
01109
01110         // Gets the number of successful tests.
01111
01112         int UnitTestImpl::successful_test_count() const {
01113           return SumOverTestSuiteList(test_suites_, &TestSuite::successful_test_count);
01114         }
01115
01116         // Gets the number of skipped tests.
01117
01118         int UnitTestImpl::skipped_test_count() const {
01119           return SumOverTestSuiteList(test_suites_, &TestSuite::skipped_test_count);
01120         }
01121
```

```
01122         // Gets the number of failed tests.
01123
01124         int UnitTestImpl::failed_test_count() const {
01125             return SumOverTestSuiteList(test_suites_, &TestSuite::failed_test_count);
01126         }
01127
01128         // Gets the number of disabled tests that will be reported in the XML report.
01129
01130         int UnitTestImpl::reportable_disabled_test_count() const {
01131             return SumOverTestSuiteList(test_suites_,
01132                     &TestSuite::reportable_disabled_test_count);
01133         }
01134
01135         // Gets the number of disabled tests.
01136
01137         int UnitTestImpl::disabled_test_count() const {
01138             return SumOverTestSuiteList(test_suites_, &TestSuite::disabled_test_count);
01139         }
01140
01141         // Gets the number of tests to be printed in the XML report.
01142
01143         int UnitTestImpl::reportable_test_count() const {
01144             return SumOverTestSuiteList(test_suites_, &TestSuite::reportable_test_count);
01145         }
01146
01147         // Gets the number of all tests.
01148
01149         int UnitTestImpl::total_test_count() const {
01150             return SumOverTestSuiteList(test_suites_, &TestSuite::total_test_count);
01151         }
01152
01153         // Gets the number of tests that should run.
01154
01155         int UnitTestImpl::test_to_run_count() const {
01156             return SumOverTestSuiteList(test_suites_, &TestSuite::test_to_run_count);
01157         }
01158
01159         // Returns the current OS stack trace as an std::string.
01160         //
01161         // The maximum number of stack frames to be included is specified by
01162         // the gtest_stack_trace_depth flag.  The skip_count parameter
01163         // specifies the number of top frames to be skipped, which doesn't
01164         // count against the number of frames to be included.
01165         //
01166         // For example, if Foo() calls Bar(), which in turn calls
01167         // CurrentOsStackTraceExceptTop(1), Foo() will be included in the
01168         // trace but Bar() and CurrentOsStackTraceExceptTop() won't.
01169
01170         std::string UnitTestImpl::CurrentOsStackTraceExceptTop(int skip_count) {
01171             return os_stack_trace_getter()->CurrentStackTrace(
01172                     static_cast<int> (GTEST_FLAG_GET(stack_trace_depth)), skip_count + 1
01173                     // Skips the user-specified number of frames plus this function
01174                     // itself.
01175                     ); // NOLINT
01176         }
01177
01178         // A helper class for measuring elapsed times.
01179
01180         class Timer {
01181         public:
01182
01183             Timer() : start_(std::chrono::steady_clock::now()) {
01184             }
01185
01186             // Return time elapsed in milliseconds since the timer was created.
01187
01188             TimeInMillis Elapsed() {
01189                 return std::chrono::duration_cast<std::chrono::milliseconds>(
01190                         std::chrono::steady_clock::now() - start_)
01191                         .count();
01192             }
01193
01194         private:
01195             std::chrono::steady_clock::time_point start_;
01196         };
01197
01198         // Returns a timestamp as milliseconds since the epoch. Note this time may jump
01199         // around subject to adjustments by the system, to measure elapsed time use
01200         // Timer instead.
01201
01202         TimeInMillis GetTimeInMillis() {
01203             return std::chrono::duration_cast<std::chrono::milliseconds>(
01204                     std::chrono::system_clock::now() -
01205                     std::chrono::system_clock::from_time_t(0))
01206                     .count();
01207         }
01208
```

```
01209         // Utilities
01210
01211         // class String.
01212
01213 #if GTEST_OS_WINDOWS_MOBILE
01214         // Creates a UTF-16 wide string from the given ANSI string, allocating
01215         // memory using new. The caller is responsible for deleting the return
01216         // value using delete[]. Returns the wide string, or NULL if the
01217         // input is NULL.
01218
01219         LPCWSTR String::AnsiToUtf16(const char* ansi) {
01220             if (!ansi) return nullptr;
01221             const int length = strlen(ansi);
01222             const int unicode_length =
01223                     MultiByteToWideChar(CP_ACP, 0, ansi, length, nullptr, 0);
01224             WCHAR* unicode = new WCHAR[unicode_length + 1];
01225             MultiByteToWideChar(CP_ACP, 0, ansi, length, unicode, unicode_length);
01226             unicode[unicode_length] = 0;
01227             return unicode;
01228         }
01229
01230         // Creates an ANSI string from the given wide string, allocating
01231         // memory using new. The caller is responsible for deleting the return
01232         // value using delete[]. Returns the ANSI string, or NULL if the
01233         // input is NULL.
01234
01235         const char* String::Utf16ToAnsi(LPCWSTR utf16_str) {
01236             if (!utf16_str) return nullptr;
01237             const int ansi_length = WideCharToMultiByte(CP_ACP, 0, utf16_str, -1, nullptr,
01238                     0, nullptr, nullptr);
01239             char* ansi = new char[ansi_length + 1];
01240             WideCharToMultiByte(CP_ACP, 0, utf16_str, -1, ansi, ansi_length, nullptr,
01241                     nullptr);
01242             ansi[ansi_length] = 0;
01243             return ansi;
01244         }
01245
01246 #endif  // GTEST_OS_WINDOWS_MOBILE
01247
01248         // Compares two C strings.  Returns true if and only if they have the same
01249         // content.
01250         //
01251         // Unlike strcmp(), this function can handle NULL argument(s).  A NULL
01252         // C string is considered different to any non-NULL C string,
01253         // including the empty string.
01254
01255         bool String::CStringEquals(const char* lhs, const char* rhs) {
01256             if (lhs == nullptr) return rhs == nullptr;
01257
01258             if (rhs == nullptr) return false;
01259
01260             return strcmp(lhs, rhs) == 0;
01261         }
01262
01263 #if GTEST_HAS_STD_WSTRING
01264
01265         // Converts an array of wide chars to a narrow string using the UTF-8
01266         // encoding, and streams the result to the given Message object.
01267
01268         static void StreamWideCharsToMessage(const wchar_t* wstr, size_t length,
01269                 Message* msg) {
01270             for (size_t i = 0; i != length;) { // NOLINT
01271                 if (wstr[i] != L'\0') {
01272                     *msg << WideStringToUtf8(wstr + i, static_cast<int> (length - i));
01273                     while (i != length && wstr[i] != L'\0') i++;
01274                 } else {
01275                     *msg << '\0';
01276                     i++;
01277                 }
01278             }
01279         }
01280
01281 #endif  // GTEST_HAS_STD_WSTRING
01282
01283         void SplitString(const ::std::string& str, char delimiter,
01284                 ::std::vector< ::std::string>* dest) {
01285             ::std::vector< ::std::string> parsed;
01286             ::std::string::size_type pos = 0;
01287             while (::testing::internal::AlwaysTrue()) {
01288                 const ::std::string::size_type colon = str.find(delimiter, pos);
01289                 if (colon == ::std::string::npos) {
01290                     parsed.push_back(str.substr(pos));
01291                     break;
01292                 } else {
01293                     parsed.push_back(str.substr(pos, colon - pos));
01294                     pos = colon + 1;
01295                 }
```

```
01296             }
01297             dest->swap(parsed);
01298         }
01299
01300     } // namespace internal
01301
01302     // Constructs an empty Message.
01303     // We allocate the stringstream separately because otherwise each use of
01304     // ASSERT/EXPECT in a procedure adds over 200 bytes to the procedure's
01305     // stack frame leading to huge stack frames in some cases; gcc does not reuse
01306     // the stack space.
01307
01308     Message::Message() : ss_(new ::std::stringstream) {
01309         // By default, we want there to be enough precision when printing
01310         // a double to a Message.
01311         *ss_ << std::setprecision(std::numeric_limits<double>::digits10 + 2);
01312     }
01313
01314     // These two overloads allow streaming a wide C string to a Message
01315     // using the UTF-8 encoding.
01316
01317     Message& Message::operator<<(const wchar_t* wide_c_str) {
01318         return *this << internal::String::ShowWideCString(wide_c_str);
01319     }
01320
01321     Message& Message::operator<<(wchar_t* wide_c_str) {
01322         return *this << internal::String::ShowWideCString(wide_c_str);
01323     }
01324
01325 #if GTEST_HAS_STD_WSTRING
01326     // Converts the given wide string to a narrow string using the UTF-8
01327     // encoding, and streams the result to this Message object.
01328
01329     Message& Message::operator<<(const ::std::wstring& wstr) {
01330         internal::StreamWideCharsToMessage(wstr.c_str(), wstr.length(), this);
01331         return *this;
01332     }
01333 #endif  // GTEST_HAS_STD_WSTRING
01334
01335     // Gets the text streamed to this object so far as an std::string.
01336     // Each '\0' character in the buffer is replaced with "\\0".
01337
01338     std::string Message::GetString() const {
01339         return internal::StringStreamToString(ss_.get());
01340     }
01341
01342     namespace internal {
01343
01344         namespace edit_distance {
01345
01346             std::vector<EditType> CalculateOptimalEdits(const std::vector<size_t>& left,
01347                     const std::vector<size_t>& right) {
01348                 std::vector<std::vector<double> > costs(
01349                         left.size() + 1, std::vector<double>(right.size() + 1));
01350                 std::vector<std::vector<EditType> > best_move(
01351                         left.size() + 1, std::vector<EditType>(right.size() + 1));
01352
01353                 // Populate for empty right.
01354                 for (size_t l_i = 0; l_i < costs.size(); ++l_i) {
01355                     costs[l_i][0] = static_cast<double> (l_i);
01356                     best_move[l_i][0] = kRemove;
01357                 }
01358                 // Populate for empty left.
01359                 for (size_t r_i = 1; r_i < costs[0].size(); ++r_i) {
01360                     costs[0][r_i] = static_cast<double> (r_i);
01361                     best_move[0][r_i] = kAdd;
01362                 }
01363
01364                 for (size_t l_i = 0; l_i < left.size(); ++l_i) {
01365                     for (size_t r_i = 0; r_i < right.size(); ++r_i) {
01366                         if (left[l_i] == right[r_i]) {
01367                             // Found a match. Consume it.
01368                             costs[l_i + 1][r_i + 1] = costs[l_i][r_i];
01369                             best_move[l_i + 1][r_i + 1] = kMatch;
01370                             continue;
01371                         }
01372
01373                         const double add = costs[l_i + 1][r_i];
01374                         const double remove = costs[l_i][r_i + 1];
01375                         const double replace = costs[l_i][r_i];
01376                         if (add < remove && add < replace) {
01377                             costs[l_i + 1][r_i + 1] = add + 1;
01378                             best_move[l_i + 1][r_i + 1] = kAdd;
01379                         } else if (remove < add && remove < replace) {
01380                             costs[l_i + 1][r_i + 1] = remove + 1;
01381                             best_move[l_i + 1][r_i + 1] = kRemove;
01382                         } else {
```

```
01383                           // We make replace a little more expensive than add/remove to lower
01384                           // their priority.
01385                           costs[l_i + 1][r_i + 1] = replace + 1.00001;
01386                           best_move[l_i + 1][r_i + 1] = kReplace;
01387                       }
01388                   }
01389               }
01390
01391               // Reconstruct the best path. We do it in reverse order.
01392               std::vector<EditType> best_path;
01393               for (size_t l_i = left.size(), r_i = right.size(); l_i > 0 || r_i > 0;) {
01394                   EditType move = best_move[l_i][r_i];
01395                   best_path.push_back(move);
01396                   l_i -= move != kAdd;
01397                   r_i -= move != kRemove;
01398               }
01399               std::reverse(best_path.begin(), best_path.end());
01400               return best_path;
01401           }
01402
01403           namespace {
01404
01405               // Helper class to convert string into ids with deduplication.
01406
01407               class InternalStrings {
01408               public:
01409
01410                   size_t GetId(const std::string& str) {
01411                       IdMap::iterator it = ids_.find(str);
01412                       if (it != ids_.end()) return it->second;
01413                       size_t id = ids_.size();
01414                       return ids_[str] = id;
01415                   }
01416
01417               private:
01418                   typedef std::map<std::string, size_t> IdMap;
01419                   IdMap ids_;
01420               };
01421
01422           } // namespace
01423
01424           std::vector<EditType> CalculateOptimalEdits(
01425                   const std::vector<std::string>& left,
01426                   const std::vector<std::string>& right) {
01427               std::vector<size_t> left_ids, right_ids;
01428               {
01429                   InternalStrings intern_table;
01430                   for (size_t i = 0; i < left.size(); ++i) {
01431                       left_ids.push_back(intern_table.GetId(left[i]));
01432                   }
01433                   for (size_t i = 0; i < right.size(); ++i) {
01434                       right_ids.push_back(intern_table.GetId(right[i]));
01435                   }
01436               }
01437               return CalculateOptimalEdits(left_ids, right_ids);
01438           }
01439
01440           namespace {
01441
01442               // Helper class that holds the state for one hunk and prints it out to the
01443               // stream.
01444               // It reorders adds/removes when possible to group all removes before all
01445               // adds. It also adds the hunk header before printint into the stream.
01446
01447               class Hunk {
01448               public:
01449
01450                   Hunk(size_t left_start, size_t right_start)
01451                   : left_start_(left_start),
01452                   right_start_(right_start),
01453                   adds_(),
01454                   removes_(),
01455                   common_() {
01456                   }
01457
01458                   void PushLine(char edit, const char* line) {
01459                       switch (edit) {
01460                           case ' ':
01461                               ++common_;
01462                               FlushEdits();
01463                               hunk_.push_back(std::make_pair(' ', line));
01464                               break;
01465                           case '-':
01466                               ++removes_;
01467                               hunk_removes_.push_back(std::make_pair('-', line));
01468                               break;
01469                           case '+':
```

```
01470                             ++adds_;
01471                             hunk_adds_.push_back(std::make_pair('+', line));
01472                             break;
01473                         }
01474                     }
01475
01476                     void PrintTo(std::ostream* os) {
01477                         PrintHeader(os);
01478                         FlushEdits();
01479                         for (std::list<std::pair<char, const char*> >::const_iterator it =
01480                                 hunk_.begin();
01481                                 it != hunk_.end(); ++it) {
01482                             *os << it->first << it->second << "\n";
01483                         }
01484                     }
01485
01486                     bool has_edits() const {
01487                         return adds_ || removes_;
01488                     }
01489
01490                 private:
01491
01492                     void FlushEdits() {
01493                         hunk_.splice(hunk_.end(), hunk_removes_);
01494                         hunk_.splice(hunk_.end(), hunk_adds_);
01495                     }
01496
01497                     // Print a unified diff header for one hunk.
01498                     // The format is
01499                     //   "@@ -<left_start>,<left_length> +<right_start>,<right_length> @@"
01500                     // where the left/right parts are omitted if unnecessary.
01501
01502                     void PrintHeader(std::ostream* ss) const {
01503                         *ss << "@@ ";
01504                         if (removes_) {
01505                             *ss << "-" << left_start_ << "," << (removes_ + common_);
01506                         }
01507                         if (removes_ && adds_) {
01508                             *ss << " ";
01509                         }
01510                         if (adds_) {
01511                             *ss << "+" << right_start_ << "," << (adds_ + common_);
01512                         }
01513                         *ss << " @@\n";
01514                     }
01515
01516                     size_t left_start_, right_start_;
01517                     size_t adds_, removes_, common_;
01518                     std::list<std::pair<char, const char*> > hunk_, hunk_adds_, hunk_removes_;
01519                 };
01520
01521             } // namespace
01522
01523             // Create a list of diff hunks in Unified diff format.
01524             // Each hunk has a header generated by PrintHeader above plus a body with
01525             // lines prefixed with ' ' for no change, '-' for deletion and '+' for
01526             // addition.
01527             // 'context' represents the desired unchanged prefix/suffix around the diff.
01528             // If two hunks are close enough that their contexts overlap, then they are
01529             // joined into one hunk.
01530
01531             std::string CreateUnifiedDiff(const std::vector<std::string>& left,
01532                     const std::vector<std::string>& right,
01533                     size_t context) {
01534                 const std::vector<EditType> edits = CalculateOptimalEdits(left, right);
01535
01536                 size_t l_i = 0, r_i = 0, edit_i = 0;
01537                 std::stringstream ss;
01538                 while (edit_i < edits.size()) {
01539                     // Find first edit.
01540                     while (edit_i < edits.size() && edits[edit_i] == kMatch) {
01541                         ++l_i;
01542                         ++r_i;
01543                         ++edit_i;
01544                     }
01545
01546                     // Find the first line to include in the hunk.
01547                     const size_t prefix_context = std::min(l_i, context);
01548                     Hunk hunk(l_i - prefix_context + 1, r_i - prefix_context + 1);
01549                     for (size_t i = prefix_context; i > 0; --i) {
01550                         hunk.PushLine(' ', left[l_i - i].c_str());
01551                     }
01552
01553                     // Iterate the edits until we found enough suffix for the hunk or the input
01554                     // is over.
01555                     size_t n_suffix = 0;
01556                     for (; edit_i < edits.size(); ++edit_i) {
```

```
01557                         if (n_suffix >= context) {
01558                             // Continue only if the next hunk is very close.
01559                             auto it = edits.begin() + static_cast<int> (edit_i);
01560                             while (it != edits.end() && *it == kMatch) ++it;
01561                             if (it == edits.end() ||
01562                                     static_cast<size_t> (it - edits.begin()) - edit_i >= context) {
01563                                 // There is no next edit or it is too far away.
01564                                 break;
01565                             }
01566                         }
01567
01568                         EditType edit = edits[edit_i];
01569                         // Reset count when a non match is found.
01570                         n_suffix = edit == kMatch ? n_suffix + 1 : 0;
01571
01572                         if (edit == kMatch || edit == kRemove || edit == kReplace) {
01573                             hunk.PushLine(edit == kMatch ? ' ' : '-', left[l_i].c_str());
01574                         }
01575                         if (edit == kAdd || edit == kReplace) {
01576                             hunk.PushLine('+', right[r_i].c_str());
01577                         }
01578
01579                         // Advance indices, depending on edit type.
01580                         l_i += edit != kAdd;
01581                         r_i += edit != kRemove;
01582                     }
01583
01584                     if (!hunk.has_edits()) {
01585                         // We are done. We don't want this hunk.
01586                         break;
01587                     }
01588
01589                     hunk.PrintTo(&ss);
01590                 }
01591                 return ss.str();
01592             }
01593
01594         } // namespace edit_distance
01595
01596         namespace {
01597
01598             // The string representation of the values received in EqFailure() are already
01599             // escaped. Split them on escaped '\n' boundaries. Leave all other escaped
01600             // characters the same.
01601
01602             std::vector<std::string> SplitEscapedString(const std::string& str) {
01603                 std::vector<std::string> lines;
01604                 size_t start = 0, end = str.size();
01605                 if (end > 2 && str[0] == '"' && str[end - 1] == '"') {
01606                     ++start;
01607                     --end;
01608                 }
01609                 bool escaped = false;
01610                 for (size_t i = start; i + 1 < end; ++i) {
01611                     if (escaped) {
01612                         escaped = false;
01613                         if (str[i] == 'n') {
01614                             lines.push_back(str.substr(start, i - start - 1));
01615                             start = i + 1;
01616                         }
01617                     } else {
01618                         escaped = str[i] == '\\';
01619                     }
01620                 }
01621                 lines.push_back(str.substr(start, end - start));
01622                 return lines;
01623             }
01624
01625         } // namespace
01626
01627         // Constructs and returns the message for an equality assertion
01628         // (e.g. ASSERT_EQ, EXPECT_STREQ, etc) failure.
01629         //
01630         // The first four parameters are the expressions used in the assertion
01631         // and their values, as strings.  For example, for ASSERT_EQ(foo, bar)
01632         // where foo is 5 and bar is 6, we have:
01633         //
01634         //   lhs_expression: "foo"
01635         //   rhs_expression: "bar"
01636         //   lhs_value:      "5"
01637         //   rhs_value:      "6"
01638         //
01639         // The ignoring_case parameter is true if and only if the assertion is a
01640         // *_STRCASEEQ*.  When it's true, the string "Ignoring case" will
01641         // be inserted into the message.
01642
01643         AssertionResult EqFailure(const char* lhs_expression,
```

```
01644                const char* rhs_expression,
01645                const std::string& lhs_value,
01646                const std::string& rhs_value, bool ignoring_case) {
01647            Message msg;
01648            msg « "Expected equality of these values:";
01649            msg « "\n  " « lhs_expression;
01650            if (lhs_value != lhs_expression) {
01651                msg « "\n    Which is: " « lhs_value;
01652            }
01653            msg « "\n  " « rhs_expression;
01654            if (rhs_value != rhs_expression) {
01655                msg « "\n    Which is: " « rhs_value;
01656            }
01657
01658            if (ignoring_case) {
01659                msg « "\nIgnoring case";
01660            }
01661
01662            if (!lhs_value.empty() && !rhs_value.empty()) {
01663                const std::vector<std::string> lhs_lines = SplitEscapedString(lhs_value);
01664                const std::vector<std::string> rhs_lines = SplitEscapedString(rhs_value);
01665                if (lhs_lines.size() > 1 || rhs_lines.size() > 1) {
01666                    msg « "\nWith diff:\n"
01667                            « edit_distance::CreateUnifiedDiff(lhs_lines, rhs_lines);
01668                }
01669            }
01670
01671            return AssertionFailure() « msg;
01672        }
01673
01674        // Constructs a failure message for Boolean assertions such as EXPECT_TRUE.
01675
01676        std::string GetBoolAssertionFailureMessage(
01677                const AssertionResult& assertion_result, const char* expression_text,
01678                const char* actual_predicate_value, const char* expected_predicate_value) {
01679            const char* actual_message = assertion_result.message();
01680            Message msg;
01681            msg « "Value of: " « expression_text
01682                    « "\n  Actual: " « actual_predicate_value;
01683            if (actual_message[0] != '\0') msg « " (" « actual_message « ")";
01684            msg « "\nExpected: " « expected_predicate_value;
01685            return msg.GetString();
01686        }
01687
01688        // Helper function for implementing ASSERT_NEAR.
01689
01690        AssertionResult DoubleNearPredFormat(const char* expr1, const char* expr2,
01691                const char* abs_error_expr, double val1,
01692                double val2, double abs_error) {
01693            const double diff = fabs(val1 - val2);
01694            if (diff <= abs_error) return AssertionSuccess();
01695
01696            // Find the value which is closest to zero.
01697            const double min_abs = std::min(fabs(val1), fabs(val2));
01698            // Find the distance to the next double from that value.
01699            const double epsilon =
01700                    nextafter(min_abs, std::numeric_limits<double>::infinity()) - min_abs;
01701            // Detect the case where abs_error is so small that EXPECT_NEAR is
01702            // effectively the same as EXPECT_EQUAL, and give an informative error
01703            // message so that the situation can be more easily understood without
01704            // requiring exotic floating-point knowledge.
01705            // Don't do an epsilon check if abs_error is zero because that implies
01706            // that an equality check was actually intended.
01707            if (!(std::isnan)(val1) && !(std::isnan)(val2) && abs_error > 0 &&
01708                    abs_error < epsilon) {
01709                return AssertionFailure()
01710                        « "The difference between " « expr1 « " and " « expr2 « " is "
01711                        « diff « ", where\n"
01712                        « expr1 « " evaluates to " « val1 « ",\n"
01713                        « expr2 « " evaluates to " « val2 « ".\nThe abs_error parameter "
01714                        « abs_error_expr « " evaluates to " « abs_error
01715                        « " which is smaller than the minimum distance between doubles for "
01716                        "numbers of this magnitude which is "
01717                        « epsilon
01718                        « ", thus making this EXPECT_NEAR check equivalent to "
01719                        "EXPECT_EQUAL. Consider using EXPECT_DOUBLE_EQ instead.";
01720            }
01721            return AssertionFailure()
01722                    « "The difference between " « expr1 « " and " « expr2 « " is "
01723                    « diff « ", which exceeds " « abs_error_expr « ", where\n"
01724                    « expr1 « " evaluates to " « val1 « ",\n"
01725                    « expr2 « " evaluates to " « val2 « ", and\n"
01726                    « abs_error_expr « " evaluates to " « abs_error « ".";
01727        }
01728
01729        // Helper template for implementing FloatLE() and DoubleLE().
01730
```

```
01731         template <typename RawType>
01732         AssertionResult FloatingPointLE(const char* expr1, const char* expr2,
01733                 RawType val1, RawType val2) {
01734             // Returns success if val1 is less than val2,
01735             if (val1 < val2) {
01736                 return AssertionSuccess();
01737             }
01738
01739             // or if val1 is almost equal to val2.
01740             const FloatingPoint<RawType> lhs(val1), rhs(val2);
01741             if (lhs.AlmostEquals(rhs)) {
01742                 return AssertionSuccess();
01743             }
01744
01745             // Note that the above two checks will both fail if either val1 or
01746             // val2 is NaN, as the IEEE floating-point standard requires that
01747             // any predicate involving a NaN must return false.
01748
01749             ::std::stringstream val1_ss;
01750             val1_ss << std::setprecision(std::numeric_limits<RawType>::digits10 + 2)
01751                     << val1;
01752
01753             ::std::stringstream val2_ss;
01754             val2_ss << std::setprecision(std::numeric_limits<RawType>::digits10 + 2)
01755                     << val2;
01756
01757             return AssertionFailure()
01758                     << "Expected: (" << expr1 << ") <= (" << expr2 << ")\n"
01759                     << "  Actual: " << StringStreamToString(&val1_ss) << " vs "
01760                     << StringStreamToString(&val2_ss);
01761         }
01762
01763     } // namespace internal
01764
01765     // Asserts that val1 is less than, or almost equal to, val2.  Fails
01766     // otherwise.  In particular, it fails if either val1 or val2 is NaN.
01767
01768     AssertionResult FloatLE(const char* expr1, const char* expr2, float val1,
01769             float val2) {
01770         return internal::FloatingPointLE<float>(expr1, expr2, val1, val2);
01771     }
01772
01773     // Asserts that val1 is less than, or almost equal to, val2.  Fails
01774     // otherwise.  In particular, it fails if either val1 or val2 is NaN.
01775
01776     AssertionResult DoubleLE(const char* expr1, const char* expr2, double val1,
01777             double val2) {
01778         return internal::FloatingPointLE<double>(expr1, expr2, val1, val2);
01779     }
01780
01781     namespace internal {
01782
01783         // The helper function for {ASSERT|EXPECT}_STREQ.
01784
01785         AssertionResult CmpHelperSTREQ(const char* lhs_expression,
01786                 const char* rhs_expression, const char* lhs,
01787                 const char* rhs) {
01788             if (String::CStringEquals(lhs, rhs)) {
01789                 return AssertionSuccess();
01790             }
01791
01792             return EqFailure(lhs_expression, rhs_expression, PrintToString(lhs),
01793                     PrintToString(rhs), false);
01794         }
01795
01796         // The helper function for {ASSERT|EXPECT}_STRCASEEQ.
01797
01798         AssertionResult CmpHelperSTRCASEEQ(const char* lhs_expression,
01799                 const char* rhs_expression, const char* lhs,
01800                 const char* rhs) {
01801             if (String::CaseInsensitiveCStringEquals(lhs, rhs)) {
01802                 return AssertionSuccess();
01803             }
01804
01805             return EqFailure(lhs_expression, rhs_expression, PrintToString(lhs),
01806                     PrintToString(rhs), true);
01807         }
01808
01809         // The helper function for {ASSERT|EXPECT}_STRNE.
01810
01811         AssertionResult CmpHelperSTRNE(const char* s1_expression,
01812                 const char* s2_expression, const char* s1,
01813                 const char* s2) {
01814             if (!String::CStringEquals(s1, s2)) {
01815                 return AssertionSuccess();
01816             } else {
01817                 return AssertionFailure()
```

```
01818                       << "Expected: (" << s1_expression << ") != (" << s2_expression
01819                       << "), actual: \"" << s1 << "\" vs \"" << s2 << "\"";
01820           }
01821       }
01822
01823       // The helper function for {ASSERT|EXPECT}_STRCASENE.
01824
01825       AssertionResult CmpHelperSTRCASENE(const char* s1_expression,
01826               const char* s2_expression, const char* s1,
01827               const char* s2) {
01828           if (!String::CaseInsensitiveCStringEquals(s1, s2)) {
01829               return AssertionSuccess();
01830           } else {
01831               return AssertionFailure()
01832                       << "Expected: (" << s1_expression << ") != (" << s2_expression
01833                       << ") (ignoring case), actual: \"" << s1 << "\" vs \"" << s2 << "\"";
01834           }
01835       }
01836
01837   } // namespace internal
01838
01839   namespace {
01840
01841       // Helper functions for implementing IsSubString() and IsNotSubstring().
01842
01843       // This group of overloaded functions return true if and only if needle
01844       // is a substring of haystack.  NULL is considered a substring of
01845       // itself only.
01846
01847       bool IsSubstringPred(const char* needle, const char* haystack) {
01848           if (needle == nullptr || haystack == nullptr) return needle == haystack;
01849
01850           return strstr(haystack, needle) != nullptr;
01851       }
01852
01853       bool IsSubstringPred(const wchar_t* needle, const wchar_t* haystack) {
01854           if (needle == nullptr || haystack == nullptr) return needle == haystack;
01855
01856           return wcsstr(haystack, needle) != nullptr;
01857       }
01858
01859       // StringType here can be either ::std::string or ::std::wstring.
01860
01861       template <typename StringType>
01862       bool IsSubstringPred(const StringType& needle, const StringType& haystack) {
01863           return haystack.find(needle) != StringType::npos;
01864       }
01865
01866       // This function implements either IsSubstring() or IsNotSubstring(),
01867       // depending on the value of the expected_to_be_substring parameter.
01868       // StringType here can be const char*, const wchar_t*, ::std::string,
01869       // or ::std::wstring.
01870
01871       template <typename StringType>
01872       AssertionResult IsSubstringImpl(bool expected_to_be_substring,
01873               const char* needle_expr,
01874               const char* haystack_expr,
01875               const StringType& needle,
01876               const StringType& haystack) {
01877           if (IsSubstringPred(needle, haystack) == expected_to_be_substring)
01878               return AssertionSuccess();
01879
01880           const bool is_wide_string = sizeof (needle[0]) > 1;
01881           const char* const begin_string_quote = is_wide_string ? "L\"" : "\"";
01882           return AssertionFailure()
01883                   << "Value of: " << needle_expr << "\n"
01884                   << "  Actual: " << begin_string_quote << needle << "\"\n"
01885                   << "Expected: " << (expected_to_be_substring ? "" : "not ")
01886                   << "a substring of " << haystack_expr << "\n"
01887                   << "Which is: " << begin_string_quote << haystack << "\"";
01888       }
01889
01890   } // namespace
01891
01892   // IsSubstring() and IsNotSubstring() check whether needle is a
01893   // substring of haystack (NULL is considered a substring of itself
01894   // only), and return an appropriate error message when they fail.
01895
01896   AssertionResult IsSubstring(const char* needle_expr, const char* haystack_expr,
01897           const char* needle, const char* haystack) {
01898       return IsSubstringImpl(true, needle_expr, haystack_expr, needle, haystack);
01899   }
01900
01901   AssertionResult IsSubstring(const char* needle_expr, const char* haystack_expr,
01902           const wchar_t* needle, const wchar_t* haystack) {
01903       return IsSubstringImpl(true, needle_expr, haystack_expr, needle, haystack);
01904   }
```

```
01905
01906     AssertionResult IsNotSubstring(const char* needle_expr,
01907             const char* haystack_expr, const char* needle,
01908             const char* haystack) {
01909         return IsSubstringImpl(false, needle_expr, haystack_expr, needle, haystack);
01910     }
01911
01912     AssertionResult IsNotSubstring(const char* needle_expr,
01913             const char* haystack_expr, const wchar_t* needle,
01914             const wchar_t* haystack) {
01915         return IsSubstringImpl(false, needle_expr, haystack_expr, needle, haystack);
01916     }
01917
01918     AssertionResult IsSubstring(const char* needle_expr, const char* haystack_expr,
01919             const ::std::string& needle,
01920             const ::std::string& haystack) {
01921         return IsSubstringImpl(true, needle_expr, haystack_expr, needle, haystack);
01922     }
01923
01924     AssertionResult IsNotSubstring(const char* needle_expr,
01925             const char* haystack_expr,
01926             const ::std::string& needle,
01927             const ::std::string& haystack) {
01928         return IsSubstringImpl(false, needle_expr, haystack_expr, needle, haystack);
01929     }
01930
01931 #if GTEST_HAS_STD_WSTRING
01932
01933     AssertionResult IsSubstring(const char* needle_expr, const char* haystack_expr,
01934             const ::std::wstring& needle,
01935             const ::std::wstring& haystack) {
01936         return IsSubstringImpl(true, needle_expr, haystack_expr, needle, haystack);
01937     }
01938
01939     AssertionResult IsNotSubstring(const char* needle_expr,
01940             const char* haystack_expr,
01941             const ::std::wstring& needle,
01942             const ::std::wstring& haystack) {
01943         return IsSubstringImpl(false, needle_expr, haystack_expr, needle, haystack);
01944     }
01945 #endif  // GTEST_HAS_STD_WSTRING
01946
01947     namespace internal {
01948
01949 #if GTEST_OS_WINDOWS
01950
01951         namespace {
01952
01953             // Helper function for IsHRESULT{SuccessFailure} predicates
01954
01955             AssertionResult HRESULTFailureHelper(const char* expr, const char* expected,
01956                     long hr) { // NOLINT
01957 #if GTEST_OS_WINDOWS_MOBILE || GTEST_OS_WINDOWS_TV_TITLE
01958
01959                 // Windows CE doesn't support FormatMessage.
01960                 const char error_text[] = "";
01961
01962 #else
01963
01964                 // Looks up the human-readable system message for the HRESULT code
01965                 // and since we're not passing any params to FormatMessage, we don't
01966                 // want inserts expanded.
01967                 const DWORD kFlags =
01968                         FORMAT_MESSAGE_FROM_SYSTEM | FORMAT_MESSAGE_IGNORE_INSERTS;
01969                 const DWORD kBufSize = 4096;
01970                 // Gets the system's human readable message string for this HRESULT.
01971                 char error_text[kBufSize] = {'\0'};
01972                 DWORD message_length = ::FormatMessageA(kFlags,
01973                         0, // no source, we're asking system
01974                         static_cast<DWORD> (hr), // the error
01975                         0, // no line width restrictions
01976                         error_text, // output buffer
01977                         kBufSize, // buf size
01978                         nullptr); // no arguments for inserts
01979                 // Trims tailing white space (FormatMessage leaves a trailing CR-LF)
01980                 for (; message_length && IsSpace(error_text[message_length - 1]);
01981                         --message_length) {
01982                     error_text[message_length - 1] = '\0';
01983                 }
01984
01985 #endif  // GTEST_OS_WINDOWS_MOBILE
01986
01987                 const std::string error_hex("0x" + String::FormatHexInt(hr));
01988                 return ::testing::AssertionFailure()
01989                         << "Expected: " << expr << " " << expected << ".\n"
01990                         << "  Actual: " << error_hex << " " << error_text << "\n";
01991             }
```

```
01992
01993         } // namespace
01994
01995         AssertionResult IsHRESULTSuccess(const char* expr, long hr) { // NOLINT
01996             if (SUCCEEDED(hr)) {
01997                 return AssertionSuccess();
01998             }
01999             return HRESULTFailureHelper(expr, "succeeds", hr);
02000         }
02001
02002         AssertionResult IsHRESULTFailure(const char* expr, long hr) { // NOLINT
02003             if (FAILED(hr)) {
02004                 return AssertionSuccess();
02005             }
02006             return HRESULTFailureHelper(expr, "fails", hr);
02007         }
02008
02009 #endif  // GTEST_OS_WINDOWS
02010
02011         // Utility functions for encoding Unicode text (wide strings) in
02012         // UTF-8.
02013
02014         // A Unicode code-point can have up to 21 bits, and is encoded in UTF-8
02015         // like this:
02016         //
02017         // Code-point length   Encoding
02018         //   0 -  7 bits       0xxxxxxx
02019         //   8 - 11 bits       110xxxxx 10xxxxxx
02020         //  12 - 16 bits       1110xxxx 10xxxxxx 10xxxxxx
02021         //  17 - 21 bits       11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
02022
02023         // The maximum code-point a one-byte UTF-8 sequence can represent.
02024         constexpr uint32_t kMaxCodePoint1 = (static_cast<uint32_t> (1) << 7) - 1;
02025
02026         // The maximum code-point a two-byte UTF-8 sequence can represent.
02027         constexpr uint32_t kMaxCodePoint2 = (static_cast<uint32_t> (1) << (5 + 6)) - 1;
02028
02029         // The maximum code-point a three-byte UTF-8 sequence can represent.
02030         constexpr uint32_t kMaxCodePoint3 =
02031                 (static_cast<uint32_t> (1) << (4 + 2 * 6)) - 1;
02032
02033         // The maximum code-point a four-byte UTF-8 sequence can represent.
02034         constexpr uint32_t kMaxCodePoint4 =
02035                 (static_cast<uint32_t> (1) << (3 + 3 * 6)) - 1;
02036
02037         // Chops off the n lowest bits from a bit pattern.  Returns the n
02038         // lowest bits.  As a side effect, the original bit pattern will be
02039         // shifted to the right by n bits.
02040
02041         inline uint32_t ChopLowBits(uint32_t* bits, int n) {
02042             const uint32_t low_bits = *bits & ((static_cast<uint32_t> (1) << n) - 1);
02043             *bits >>= n;
02044             return low_bits;
02045         }
02046
02047         // Converts a Unicode code point to a narrow string in UTF-8 encoding.
02048         // code_point parameter is of type uint32_t because wchar_t may not be
02049         // wide enough to contain a code point.
02050         // If the code_point is not a valid Unicode code point
02051         // (i.e. outside of Unicode range U+0 to U+10FFFF) it will be converted
02052         // to "(Invalid Unicode 0xXXXXXXXX)".
02053
02054         std::string CodePointToUtf8(uint32_t code_point) {
02055             if (code_point > kMaxCodePoint4) {
02056                 return "(Invalid Unicode 0x" + String::FormatHexUInt32(code_point) + ")";
02057             }
02058
02059             char str[5]; // Big enough for the largest valid code point.
02060             if (code_point <= kMaxCodePoint1) {
02061                 str[1] = '\0';
02062                 str[0] = static_cast<char> (code_point); // 0xxxxxxx
02063             } else if (code_point <= kMaxCodePoint2) {
02064                 str[2] = '\0';
02065                 str[1] = static_cast<char> (0x80 | ChopLowBits(&code_point, 6)); // 10xxxxxx
02066                 str[0] = static_cast<char> (0xC0 | code_point); // 110xxxxx
02067             } else if (code_point <= kMaxCodePoint3) {
02068                 str[3] = '\0';
02069                 str[2] = static_cast<char> (0x80 | ChopLowBits(&code_point, 6)); // 10xxxxxx
02070                 str[1] = static_cast<char> (0x80 | ChopLowBits(&code_point, 6)); // 10xxxxxx
02071                 str[0] = static_cast<char> (0xE0 | code_point); // 1110xxxx
02072             } else { // code_point <= kMaxCodePoint4
02073                 str[4] = '\0';
02074                 str[3] = static_cast<char> (0x80 | ChopLowBits(&code_point, 6)); // 10xxxxxx
02075                 str[2] = static_cast<char> (0x80 | ChopLowBits(&code_point, 6)); // 10xxxxxx
02076                 str[1] = static_cast<char> (0x80 | ChopLowBits(&code_point, 6)); // 10xxxxxx
02077                 str[0] = static_cast<char> (0xF0 | code_point); // 11110xxx
02078             }
```

```
02079             return str;
02080         }
02081
02082         // The following two functions only make sense if the system
02083         // uses UTF-16 for wide string encoding. All supported systems
02084         // with 16 bit wchar_t (Windows, Cygwin) do use UTF-16.
02085
02086         // Determines if the arguments constitute UTF-16 surrogate pair
02087         // and thus should be combined into a single Unicode code point
02088         // using CreateCodePointFromUtf16SurrogatePair.
02089
02090         inline bool IsUtf16SurrogatePair(wchar_t first, wchar_t second) {
02091             return sizeof (wchar_t) == 2 && (first & 0xFC00) == 0xD800 &&
02092                     (second & 0xFC00) == 0xDC00;
02093         }
02094
02095         // Creates a Unicode code point from UTF16 surrogate pair.
02096
02097         inline uint32_t CreateCodePointFromUtf16SurrogatePair(wchar_t first,
02098                 wchar_t second) {
02099             const auto first_u = static_cast<uint32_t> (first);
02100             const auto second_u = static_cast<uint32_t> (second);
02101             const uint32_t mask = (1 « 10) - 1;
02102             return (sizeof (wchar_t) == 2)
02103                     ? (((first_u & mask) « 10) | (second_u & mask)) + 0x10000
02104                     :
02105                     // This function should not be called when the condition is
02106                     // false, but we provide a sensible default in case it is.
02107                     first_u;
02108         }
02109
02110         // Converts a wide string to a narrow string in UTF-8 encoding.
02111         // The wide string is assumed to have the following encoding:
02112         //   UTF-16 if sizeof(wchar_t) == 2 (on Windows, Cygwin)
02113         //   UTF-32 if sizeof(wchar_t) == 4 (on Linux)
02114         // Parameter str points to a null-terminated wide string.
02115         // Parameter num_chars may additionally limit the number
02116         // of wchar_t characters processed. -1 is used when the entire string
02117         // should be processed.
02118         // If the string contains code points that are not valid Unicode code points
02119         // (i.e. outside of Unicode range U+0 to U+10FFFF) they will be output
02120         // as '(Invalid Unicode 0xXXXXXXXX)'. If the string is in UTF16 encoding
02121         // and contains invalid UTF-16 surrogate pairs, values in those pairs
02122         // will be encoded as individual Unicode characters from Basic Normal Plane.
02123
02124         std::string WideStringToUtf8(const wchar_t* str, int num_chars) {
02125             if (num_chars == -1) num_chars = static_cast<int> (wcslen(str));
02126
02127             ::std::stringstream stream;
02128             for (int i = 0; i < num_chars; ++i) {
02129                 uint32_t unicode_code_point;
02130
02131                 if (str[i] == L'\0') {
02132                     break;
02133                 } else if (i + 1 < num_chars && IsUtf16SurrogatePair(str[i], str[i + 1])) {
02134                     unicode_code_point =
02135                             CreateCodePointFromUtf16SurrogatePair(str[i], str[i + 1]);
02136                     i++;
02137                 } else {
02138                     unicode_code_point = static_cast<uint32_t> (str[i]);
02139                 }
02140
02141                 stream « CodePointToUtf8(unicode_code_point);
02142             }
02143             return StringStreamToString(&stream);
02144         }
02145
02146         // Converts a wide C string to an std::string using the UTF-8 encoding.
02147         // NULL will be converted to "(null)".
02148
02149         std::string String::ShowWideCString(const wchar_t* wide_c_str) {
02150             if (wide_c_str == nullptr) return "(null)";
02151
02152             return internal::WideStringToUtf8(wide_c_str, -1);
02153         }
02154
02155         // Compares two wide C strings.  Returns true if and only if they have the
02156         // same content.
02157         //
02158         // Unlike wcscmp(), this function can handle NULL argument(s).  A NULL
02159         // C string is considered different to any non-NULL C string,
02160         // including the empty string.
02161
02162         bool String::WideCStringEquals(const wchar_t* lhs, const wchar_t* rhs) {
02163             if (lhs == nullptr) return rhs == nullptr;
02164
02165             if (rhs == nullptr) return false;
```

```
02166
02167             return wcscmp(lhs, rhs) == 0;
02168         }
02169
02170         // Helper function for *_STREQ on wide strings.
02171
02172         AssertionResult CmpHelperSTREQ(const char* lhs_expression,
02173                 const char* rhs_expression, const wchar_t* lhs,
02174                 const wchar_t* rhs) {
02175             if (String::WideCStringEquals(lhs, rhs)) {
02176                 return AssertionSuccess();
02177             }
02178
02179             return EqFailure(lhs_expression, rhs_expression, PrintToString(lhs),
02180                     PrintToString(rhs), false);
02181         }
02182
02183         // Helper function for *_STRNE on wide strings.
02184
02185         AssertionResult CmpHelperSTRNE(const char* s1_expression,
02186                 const char* s2_expression, const wchar_t* s1,
02187                 const wchar_t* s2) {
02188             if (!String::WideCStringEquals(s1, s2)) {
02189                 return AssertionSuccess();
02190             }
02191
02192             return AssertionFailure()
02193                     << "Expected: (" << s1_expression << ") != (" << s2_expression
02194                     << "), actual: " << PrintToString(s1) << " vs " << PrintToString(s2);
02195         }
02196
02197         // Compares two C strings, ignoring case.  Returns true if and only if they have
02198         // the same content.
02199         //
02200         // Unlike strcasecmp(), this function can handle NULL argument(s).  A
02201         // NULL C string is considered different to any non-NULL C string,
02202         // including the empty string.
02203
02204         bool String::CaseInsensitiveCStringEquals(const char* lhs, const char* rhs) {
02205             if (lhs == nullptr) return rhs == nullptr;
02206             if (rhs == nullptr) return false;
02207             return posix::StrCaseCmp(lhs, rhs) == 0;
02208         }
02209
02210         // Compares two wide C strings, ignoring case.  Returns true if and only if they
02211         // have the same content.
02212         //
02213         // Unlike wcscasecmp(), this function can handle NULL argument(s).
02214         // A NULL C string is considered different to any non-NULL wide C string,
02215         // including the empty string.
02216         // NB: The implementations on different platforms slightly differ.
02217         // On windows, this method uses _wcsicmp which compares according to LC_CTYPE
02218         // environment variable. On GNU platform this method uses wcscasecmp
02219         // which compares according to LC_CTYPE category of the current locale.
02220         // On MacOS X, it uses towlower, which also uses LC_CTYPE category of the
02221         // current locale.
02222
02223         bool String::CaseInsensitiveWideCStringEquals(const wchar_t* lhs,
02224                 const wchar_t* rhs) {
02225             if (lhs == nullptr) return rhs == nullptr;
02226
02227             if (rhs == nullptr) return false;
02228
02229 #if GTEST_OS_WINDOWS
02230             return _wcsicmp(lhs, rhs) == 0;
02231 #elif GTEST_OS_LINUX && !GTEST_OS_LINUX_ANDROID
02232             return wcscasecmp(lhs, rhs) == 0;
02233 #else
02234             // Android, Mac OS X and Cygwin don't define wcscasecmp.
02235             // Other unknown OSes may not define it either.
02236             wint_t left, right;
02237             do {
02238                 left = towlower(static_cast<wint_t> (*lhs++));
02239                 right = towlower(static_cast<wint_t> (*rhs++));
02240             } while (left && left == right);
02241             return left == right;
02242 #endif  // OS selector
02243         }
02244
02245         // Returns true if and only if str ends with the given suffix, ignoring case.
02246         // Any string is considered to end with an empty suffix.
02247
02248         bool String::EndsWithCaseInsensitive(const std::string& str,
02249                 const std::string& suffix) {
02250             const size_t str_len = str.length();
02251             const size_t suffix_len = suffix.length();
02252             return (str_len >= suffix_len) &&
```

```
02253                     CaseInsensitiveCStringEquals(str.c_str() + str_len - suffix_len,
02254                     suffix.c_str());
02255         }
02256
02257         // Formats an int value as "%02d".
02258
02259         std::string String::FormatIntWidth2(int value) {
02260             return FormatIntWidthN(value, 2);
02261         }
02262
02263         // Formats an int value to given width with leading zeros.
02264
02265         std::string String::FormatIntWidthN(int value, int width) {
02266             std::stringstream ss;
02267             ss « std::setfill('0') « std::setw(width) « value;
02268             return ss.str();
02269         }
02270
02271         // Formats an int value as "%X".
02272
02273         std::string String::FormatHexUInt32(uint32_t value) {
02274             std::stringstream ss;
02275             ss « std::hex « std::uppercase « value;
02276             return ss.str();
02277         }
02278
02279         // Formats an int value as "%X".
02280
02281         std::string String::FormatHexInt(int value) {
02282             return FormatHexUInt32(static_cast<uint32_t> (value));
02283         }
02284
02285         // Formats a byte as "%02X".
02286
02287         std::string String::FormatByte(unsigned char value) {
02288             std::stringstream ss;
02289             ss « std::setfill('0') « std::setw(2) « std::hex « std::uppercase
02290                     « static_cast<unsigned int> (value);
02291             return ss.str();
02292         }
02293
02294         // Converts the buffer in a stringstream to an std::string, converting NUL
02295         // bytes to "\\0" along the way.
02296
02297         std::string StringStreamToString(::std::stringstream* ss) {
02298             const ::std::string& str = ss->str();
02299             const char* const start = str.c_str();
02300             const char* const end = start + str.length();
02301
02302             std::string result;
02303             result.reserve(static_cast<size_t> (2 * (end - start)));
02304             for (const char* ch = start; ch != end; ++ch) {
02305                 if (*ch == '\0') {
02306                     result += "\\0"; // Replaces NUL with "\\0";
02307                 } else {
02308                     result += *ch;
02309                 }
02310             }
02311
02312             return result;
02313         }
02314
02315         // Appends the user-supplied message to the Google-Test-generated message.
02316
02317         std::string AppendUserMessage(const std::string& gtest_msg,
02318                 const Message& user_msg) {
02319             // Appends the user message if it's non-empty.
02320             const std::string user_msg_string = user_msg.GetString();
02321             if (user_msg_string.empty()) {
02322                 return gtest_msg;
02323             }
02324             if (gtest_msg.empty()) {
02325                 return user_msg_string;
02326             }
02327             return gtest_msg + "\n" + user_msg_string;
02328         }
02329
02330     } // namespace internal
02331
02332     // class TestResult
02333
02334     // Creates an empty TestResult.
02335
02336     TestResult::TestResult()
02337     : death_test_count_(0), start_timestamp_(0), elapsed_time_(0) {
02338     }
02339
```

```
02340     // D'tor.
02341
02342     TestResult::~TestResult() {
02343     }
02344
02345     // Returns the i-th test part result among all the results. i can
02346     // range from 0 to total_part_count() - 1. If i is not in that range,
02347     // aborts the program.
02348
02349     const TestPartResult& TestResult::GetTestPartResult(int i) const {
02350         if (i < 0 || i >= total_part_count()) internal::posix::Abort();
02351         return test_part_results_.at(static_cast<size_t> (i));
02352     }
02353
02354     // Returns the i-th test property. i can range from 0 to
02355     // test_property_count() - 1. If i is not in that range, aborts the
02356     // program.
02357
02358     const TestProperty& TestResult::GetTestProperty(int i) const {
02359         if (i < 0 || i >= test_property_count()) internal::posix::Abort();
02360         return test_properties_.at(static_cast<size_t> (i));
02361     }
02362
02363     // Clears the test part results.
02364
02365     void TestResult::ClearTestPartResults() {
02366         test_part_results_.clear();
02367     }
02368
02369     // Adds a test part result to the list.
02370
02371     void TestResult::AddTestPartResult(const TestPartResult& test_part_result) {
02372         test_part_results_.push_back(test_part_result);
02373     }
02374
02375     // Adds a test property to the list. If a property with the same key as the
02376     // supplied property is already represented, the value of this test_property
02377     // replaces the old value for that key.
02378
02379     void TestResult::RecordProperty(const std::string& xml_element,
02380             const TestProperty& test_property) {
02381         if (!ValidateTestProperty(xml_element, test_property)) {
02382             return;
02383         }
02384         internal::MutexLock lock(&test_properties_mutex_);
02385         const std::vector<TestProperty>::iterator property_with_matching_key =
02386                 std::find_if(test_properties_.begin(), test_properties_.end(),
02387                 internal::TestPropertyKeyIs(test_property.key()));
02388         if (property_with_matching_key == test_properties_.end()) {
02389             test_properties_.push_back(test_property);
02390             return;
02391         }
02392         property_with_matching_key->SetValue(test_property.value());
02393     }
02394
02395     // The list of reserved attributes used in the <testsuites> element of XML
02396     // output.
02397     static const char* const kReservedTestSuitesAttributes[] = {
02398         "disabled", "errors", "failures", "name",
02399         "random_seed", "tests", "time", "timestamp"
02400     };
02401
02402     // The list of reserved attributes used in the <testsuite> element of XML
02403     // output.
02404     static const char* const kReservedTestSuiteAttributes[] = {
02405         "disabled", "errors", "failures", "name",
02406         "tests", "time", "timestamp", "skipped"
02407     };
02408
02409     // The list of reserved attributes used in the <testcase> element of XML output.
02410     static const char* const kReservedTestCaseAttributes[] = {
02411         "classname", "name", "status", "time",
02412         "type_param", "value_param", "file", "line"
02413     };
02414
02415     // Use a slightly different set for allowed output to ensure existing tests can
02416     // still RecordProperty("result") or "RecordProperty(timestamp")
02417     static const char* const kReservedOutputTestCaseAttributes[] = {
02418         "classname", "name", "status", "time", "type_param",
02419         "value_param", "file", "line", "result", "timestamp"
02420     };
02421
02422     template <size_t kSize>
02423     std::vector<std::string> ArrayAsVector(const char* const (&array)[kSize]) {
02424         return std::vector<std::string>(array, array + kSize);
02425     }
02426
```

```
02427     static std::vector<std::string> GetReservedAttributesForElement(
02428             const std::string& xml_element) {
02429         if (xml_element == "testsuites") {
02430             return ArrayAsVector(kReservedTestSuitesAttributes);
02431         } else if (xml_element == "testsuite") {
02432             return ArrayAsVector(kReservedTestSuiteAttributes);
02433         } else if (xml_element == "testcase") {
02434             return ArrayAsVector(kReservedTestCaseAttributes);
02435         } else {
02436             GTEST_CHECK_(false) << "Unrecognized xml_element provided: " << xml_element;
02437         }
02438         // This code is unreachable but some compilers may not realizes that.
02439         return std::vector<std::string>();
02440     }
02441
02442     // TODO(jdesprez): Merge the two getReserved attributes once skip is improved
02443
02444     static std::vector<std::string> GetReservedOutputAttributesForElement(
02445             const std::string& xml_element) {
02446         if (xml_element == "testsuites") {
02447             return ArrayAsVector(kReservedTestSuitesAttributes);
02448         } else if (xml_element == "testsuite") {
02449             return ArrayAsVector(kReservedTestSuiteAttributes);
02450         } else if (xml_element == "testcase") {
02451             return ArrayAsVector(kReservedOutputTestCaseAttributes);
02452         } else {
02453             GTEST_CHECK_(false) << "Unrecognized xml_element provided: " << xml_element;
02454         }
02455         // This code is unreachable but some compilers may not realizes that.
02456         return std::vector<std::string>();
02457     }
02458
02459     static std::string FormatWordList(const std::vector<std::string>& words) {
02460         Message word_list;
02461         for (size_t i = 0; i < words.size(); ++i) {
02462             if (i > 0 && words.size() > 2) {
02463                 word_list << ", ";
02464             }
02465             if (i == words.size() - 1) {
02466                 word_list << "and ";
02467             }
02468             word_list << "'" << words[i] << "'";
02469         }
02470         return word_list.GetString();
02471     }
02472
02473     static bool ValidateTestPropertyName(
02474             const std::string& property_name,
02475             const std::vector<std::string>& reserved_names) {
02476         if (std::find(reserved_names.begin(), reserved_names.end(), property_name) !=
02477                 reserved_names.end()) {
02478             ADD_FAILURE() << "Reserved key used in RecordProperty(): " << property_name
02479                     << " (" << FormatWordList(reserved_names)
02480                     << " are reserved by " << GTEST_NAME_ << ")";
02481             return false;
02482         }
02483         return true;
02484     }
02485
02486     // Adds a failure if the key is a reserved attribute of the element named
02487     // xml_element.  Returns true if the property is valid.
02488
02489     bool TestResult::ValidateTestProperty(const std::string& xml_element,
02490             const TestProperty& test_property) {
02491         return ValidateTestPropertyName(test_property.key(),
02492                 GetReservedAttributesForElement(xml_element));
02493     }
02494
02495     // Clears the object.
02496
02497     void TestResult::Clear() {
02498         test_part_results_.clear();
02499         test_properties_.clear();
02500         death_test_count_ = 0;
02501         elapsed_time_ = 0;
02502     }
02503
02504     // Returns true off the test part was skipped.
02505
02506     static bool TestPartSkipped(const TestPartResult& result) {
02507         return result.skipped();
02508     }
02509
02510     // Returns true if and only if the test was skipped.
02511
02512     bool TestResult::Skipped() const {
02513         return !Failed() && CountIf(test_part_results_, TestPartSkipped) > 0;
```

```
02514     }
02515
02516     // Returns true if and only if the test failed.
02517
02518     bool TestResult::Failed() const {
02519         for (int i = 0; i < total_part_count(); ++i) {
02520             if (GetTestPartResult(i).failed()) return true;
02521         }
02522         return false;
02523     }
02524
02525     // Returns true if and only if the test part fatally failed.
02526
02527     static bool TestPartFatallyFailed(const TestPartResult& result) {
02528         return result.fatally_failed();
02529     }
02530
02531     // Returns true if and only if the test fatally failed.
02532
02533     bool TestResult::HasFatalFailure() const {
02534         return CountIf(test_part_results_, TestPartFatallyFailed) > 0;
02535     }
02536
02537     // Returns true if and only if the test part non-fatally failed.
02538
02539     static bool TestPartNonfatallyFailed(const TestPartResult& result) {
02540         return result.nonfatally_failed();
02541     }
02542
02543     // Returns true if and only if the test has a non-fatal failure.
02544
02545     bool TestResult::HasNonfatalFailure() const {
02546         return CountIf(test_part_results_, TestPartNonfatallyFailed) > 0;
02547     }
02548
02549     // Gets the number of all test parts.  This is the sum of the number
02550     // of successful test parts and the number of failed test parts.
02551
02552     int TestResult::total_part_count() const {
02553         return static_cast<int> (test_part_results_.size());
02554     }
02555
02556     // Returns the number of the test properties.
02557
02558     int TestResult::test_property_count() const {
02559         return static_cast<int> (test_properties_.size());
02560     }
02561
02562     // class Test
02563
02564     // Creates a Test object.
02565
02566     // The c'tor saves the states of all flags.
02567
02568     Test::Test() : gtest_flag_saver_(new GTEST_FLAG_SAVER_) {
02569     }
02570
02571     // The d'tor restores the states of all flags.  The actual work is
02572     // done by the d'tor of the gtest_flag_saver_ field, and thus not
02573     // visible here.
02574
02575     Test::~Test() {
02576     }
02577
02578     // Sets up the test fixture.
02579     //
02580     // A sub-class may override this.
02581
02582     void Test::SetUp() {
02583     }
02584
02585     // Tears down the test fixture.
02586     //
02587     // A sub-class may override this.
02588
02589     void Test::TearDown() {
02590     }
02591
02592     // Allows user supplied key value pairs to be recorded for later output.
02593
02594     void Test::RecordProperty(const std::string& key, const std::string& value) {
02595         UnitTest::GetInstance()->RecordProperty(key, value);
02596     }
02597
02598     // Allows user supplied key value pairs to be recorded for later output.
02599
02600     void Test::RecordProperty(const std::string& key, int value) {
```

```
02601         Message value_message;
02602         value_message << value;
02603         RecordProperty(key, value_message.GetString().c_str());
02604     }
02605
02606     namespace internal {
02607
02608         void ReportFailureInUnknownLocation(TestPartResult::Type result_type,
02609                 const std::string& message) {
02610             // This function is a friend of UnitTest and as such has access to
02611             // AddTestPartResult.
02612             UnitTest::GetInstance()->AddTestPartResult(
02613                     result_type,
02614                     nullptr, // No info about the source file where the exception occurred.
02615                     -1, // We have no info on which line caused the exception.
02616                     message,
02617                     ""); // No stack trace, either.
02618         }
02619
02620     } // namespace internal
02621
02622     // Google Test requires all tests in the same test suite to use the same test
02623     // fixture class.  This function checks if the current test has the
02624     // same fixture class as the first test in the current test suite.  If
02625     // yes, it returns true; otherwise it generates a Google Test failure and
02626     // returns false.
02627
02628     bool Test::HasSameFixtureClass() {
02629         internal::UnitTestImpl * const impl = internal::GetUnitTestImpl();
02630         const TestSuite * const test_suite = impl->current_test_suite();
02631
02632         // Info about the first test in the current test suite.
02633         const TestInfo * const first_test_info = test_suite->test_info_list()[0];
02634         const internal::TypeId first_fixture_id = first_test_info->fixture_class_id_;
02635         const char* const first_test_name = first_test_info->name();
02636
02637         // Info about the current test.
02638         const TestInfo * const this_test_info = impl->current_test_info();
02639         const internal::TypeId this_fixture_id = this_test_info->fixture_class_id_;
02640         const char* const this_test_name = this_test_info->name();
02641
02642         if (this_fixture_id != first_fixture_id) {
02643             // Is the first test defined using TEST?
02644             const bool first_is_TEST = first_fixture_id == internal::GetTestTypeId();
02645             // Is this test defined using TEST?
02646             const bool this_is_TEST = this_fixture_id == internal::GetTestTypeId();
02647
02648             if (first_is_TEST || this_is_TEST) {
02649                 // Both TEST and TEST_F appear in same test suite, which is incorrect.
02650                 // Tell the user how to fix this.
02651
02652                 // Gets the name of the TEST and the name of the TEST_F.  Note
02653                 // that first_is_TEST and this_is_TEST cannot both be true, as
02654                 // the fixture IDs are different for the two tests.
02655                 const char* const TEST_name =
02656                         first_is_TEST ? first_test_name : this_test_name;
02657                 const char* const TEST_F_name =
02658                         first_is_TEST ? this_test_name : first_test_name;
02659
02660                 ADD_FAILURE()
02661                         << "All tests in the same test suite must use the same test fixture\n"
02662                         << "class, so mixing TEST_F and TEST in the same test suite is\n"
02663                         << "illegal.  In test suite " << this_test_info->test_suite_name()
02664                         << ",\n"
02665                         << "test " << TEST_F_name << " is defined using TEST_F but\n"
02666                         << "test " << TEST_name << " is defined using TEST.  You probably\n"
02667                         << "want to change the TEST to TEST_F or move it to another test\n"
02668                         << "case.";
02669             } else {
02670                 // Two fixture classes with the same name appear in two different
02671                 // namespaces, which is not allowed. Tell the user how to fix this.
02672                 ADD_FAILURE()
02673                         << "All tests in the same test suite must use the same test fixture\n"
02674                         << "class.  However, in test suite "
02675                         << this_test_info->test_suite_name() << ",\n"
02676                         << "you defined test " << first_test_name << " and test "
02677                         << this_test_name << "\n"
02678                         << "using two different test fixture classes.  This can happen if\n"
02679                         << "the two classes are from different namespaces or translation\n"
02680                         << "units and have the same name.  You should probably rename one\n"
02681                         << "of the classes to put the tests into different test suites.";
02682             }
02683             return false;
02684         }
02685
02686         return true;
02687     }
```

```
02688
02689 #if GTEST_HAS_SEH
02690
02691     // Adds an "exception thrown" fatal failure to the current test.  This
02692     // function returns its result via an output parameter pointer because VC++
02693     // prohibits creation of objects with destructors on stack in functions
02694     // using __try (see error C2712).
02695
02696     static std::string* FormatSehExceptionMessage(DWORD exception_code,
02697             const char* location) {
02698         Message message;
02699         message << "SEH exception with code 0x" << std::setbase(16) << exception_code
02700                 << std::setbase(10) << " thrown in " << location << ".";
02701
02702         return new std::string(message.GetString());
02703     }
02704
02705 #endif  // GTEST_HAS_SEH
02706
02707     namespace internal {
02708
02709 #if GTEST_HAS_EXCEPTIONS
02710
02711         // Adds an "exception thrown" fatal failure to the current test.
02712
02713         static std::string FormatCxxExceptionMessage(const char* description,
02714                 const char* location) {
02715             Message message;
02716             if (description != nullptr) {
02717                 message << "C++ exception with description \"" << description << "\"";
02718             } else {
02719                 message << "Unknown C++ exception";
02720             }
02721             message << " thrown in " << location << ".";
02722
02723             return message.GetString();
02724         }
02725
02726         static std::string PrintTestPartResultToString(
02727                 const TestPartResult& test_part_result);
02728
02729         GoogleTestFailureException::GoogleTestFailureException(
02730                 const TestPartResult& failure)
02731         : ::std::runtime_error(PrintTestPartResultToString(failure).c_str()) {
02732         }
02733
02734 #endif  // GTEST_HAS_EXCEPTIONS
02735
02736         // We put these helper functions in the internal namespace as IBM's xlC
02737         // compiler rejects the code if they were declared static.
02738
02739         // Runs the given method and handles SEH exceptions it throws, when
02740         // SEH is supported; returns the 0-value for type Result in case of an
02741         // SEH exception.  (Microsoft compilers cannot handle SEH and C++
02742         // exceptions in the same function.  Therefore, we provide a separate
02743         // wrapper function for handling SEH exceptions.)
02744
02745         template <class T, typename Result>
02746         Result HandleSehExceptionsInMethodIfSupported(T* object, Result(T::*method)(),
02747                 const char* location) {
02748 #if GTEST_HAS_SEH
02749             __try
02750             {
02751                 return (object->*method)();
02752             }
02753
02754             __except(internal::UnitTestOptions::GTestShouldProcessSEH(// NOLINT
02755                     GetExceptionCode())) {
02756                 // We create the exception message on the heap because VC++ prohibits
02757                 // creation of objects with destructors on stack in functions using __try
02758                 // (see error C2712).
02759                 std::string* exception_message =
02760                         FormatSehExceptionMessage(GetExceptionCode(), location);
02761                 internal::ReportFailureInUnknownLocation(TestPartResult::kFatalFailure,
02762                         *exception_message);
02763                 delete exception_message;
02764                 return static_cast<Result> (0);
02765             }
02766 #else
02767             (void) location;
02768             return (object->*method)();
02769 #endif  // GTEST_HAS_SEH
02770         }
02771
02772         // Runs the given method and catches and reports C++ and/or SEH-style
02773         // exceptions, if they are supported; returns the 0-value for type
02774         // Result in case of an SEH exception.
```

```
02775
02776         template <class T, typename Result>
02777         Result HandleExceptionsInMethodIfSupported(T* object, Result(T::*method)(),
02778                 const char* location) {
02779             // NOTE: The user code can affect the way in which Google Test handles
02780             // exceptions by setting GTEST_FLAG(catch_exceptions), but only before
02781             // RUN_ALL_TESTS() starts. It is technically possible to check the flag
02782             // after the exception is caught and either report or re-throw the
02783             // exception based on the flag's value:
02784             //
02785             // try {
02786             //   // Perform the test method.
02787             // } catch (...) {
02788             //   if (GTEST_FLAG_GET(catch_exceptions))
02789             //     // Report the exception as failure.
02790             //   else
02791             //     throw;  // Re-throws the original exception.
02792             // }
02793             //
02794             // However, the purpose of this flag is to allow the program to drop into
02795             // the debugger when the exception is thrown. On most platforms, once the
02796             // control enters the catch block, the exception origin information is
02797             // lost and the debugger will stop the program at the point of the
02798             // re-throw in this function -- instead of at the point of the original
02799             // throw statement in the code under test.  For this reason, we perform
02800             // the check early, sacrificing the ability to affect Google Test's
02801             // exception handling in the method where the exception is thrown.
02802             if (internal::GetUnitTestImpl()->catch_exceptions()) {
02803 #if GTEST_HAS_EXCEPTIONS
02804                 try {
02805                     return HandleSehExceptionsInMethodIfSupported(object, method, location);
02806                 } catch (const AssertionException&) { // NOLINT
02807                     // This failure was reported already.
02808                 } catch (const internal::GoogleTestFailureException&) { // NOLINT
02809                     // This exception type can only be thrown by a failed Google
02810                     // Test assertion with the intention of letting another testing
02811                     // framework catch it.  Therefore we just re-throw it.
02812                     throw;
02813                 } catch (const std::exception& e) { // NOLINT
02814                     internal::ReportFailureInUnknownLocation(
02815                             TestPartResult::kFatalFailure,
02816                             FormatCxxExceptionMessage(e.what(), location));
02817                 } catch (...) { // NOLINT
02818                     internal::ReportFailureInUnknownLocation(
02819                             TestPartResult::kFatalFailure,
02820                             FormatCxxExceptionMessage(nullptr, location));
02821                 }
02822                 return static_cast<Result> (0);
02823 #else
02824                 return HandleSehExceptionsInMethodIfSupported(object, method, location);
02825 #endif  // GTEST_HAS_EXCEPTIONS
02826             } else {
02827                 return (object->*method)();
02828             }
02829         }
02830
02831     } // namespace internal
02832
02833     // Runs the test and updates the test result.
02834
02835     void Test::Run() {
02836         if (!HasSameFixtureClass()) return;
02837
02838         internal::UnitTestImpl * const impl = internal::GetUnitTestImpl();
02839         impl->os_stack_trace_getter()->UponLeavingGTest();
02840         internal::HandleExceptionsInMethodIfSupported(this, &Test::SetUp, "SetUp()");
02841         // We will run the test only if SetUp() was successful and didn't call
02842         // GTEST_SKIP().
02843         if (!HasFatalFailure() && !IsSkipped()) {
02844             impl->os_stack_trace_getter()->UponLeavingGTest();
02845             internal::HandleExceptionsInMethodIfSupported(this, &Test::TestBody,
02846                     "the test body");
02847         }
02848
02849         // However, we want to clean up as much as possible.  Hence we will
02850         // always call TearDown(), even if SetUp() or the test body has
02851         // failed.
02852         impl->os_stack_trace_getter()->UponLeavingGTest();
02853         internal::HandleExceptionsInMethodIfSupported(this, &Test::TearDown,
02854                 "TearDown()");
02855     }
02856
02857     // Returns true if and only if the current test has a fatal failure.
02858
02859     bool Test::HasFatalFailure() {
02860         return internal::GetUnitTestImpl()->current_test_result()->HasFatalFailure();
02861     }
```

```
02862
02863     // Returns true if and only if the current test has a non-fatal failure.
02864
02865     bool Test::HasNonfatalFailure() {
02866         return internal::GetUnitTestImpl()
02867                 ->current_test_result()
02868                 ->HasNonfatalFailure();
02869     }
02870
02871     // Returns true if and only if the current test was skipped.
02872
02873     bool Test::IsSkipped() {
02874         return internal::GetUnitTestImpl()->current_test_result()->Skipped();
02875     }
02876
02877     // class TestInfo
02878
02879     // Constructs a TestInfo object. It assumes ownership of the test factory
02880     // object.
02881
02882     TestInfo::TestInfo(const std::string& a_test_suite_name,
02883             const std::string& a_name, const char* a_type_param,
02884             const char* a_value_param,
02885             internal::CodeLocation a_code_location,
02886             internal::TypeId fixture_class_id,
02887             internal::TestFactoryBase* factory)
02888     : test_suite_name_(a_test_suite_name),
02889     name_(a_name),
02890     type_param_(a_type_param ? new std::string(a_type_param) : nullptr),
02891     value_param_(a_value_param ? new std::string(a_value_param) : nullptr),
02892     location_(a_code_location),
02893     fixture_class_id_(fixture_class_id),
02894     should_run_(false),
02895     is_disabled_(false),
02896     matches_filter_(false),
02897     is_in_another_shard_(false),
02898     factory_(factory),
02899     result_() {
02900     }
02901
02902     // Destructs a TestInfo object.
02903
02904     TestInfo::~TestInfo() {
02905         delete factory_;
02906     }
02907
02908     namespace internal {
02909
02910         // Creates a new TestInfo object and registers it with Google Test;
02911         // returns the created object.
02912         //
02913         // Arguments:
02914         //
02915         //   test_suite_name:  name of the test suite
02916         //   name:             name of the test
02917         //   type_param:       the name of the test's type parameter, or NULL if
02918         //                     this is not a typed or a type-parameterized test.
02919         //   value_param:      text representation of the test's value parameter,
02920         //                     or NULL if this is not a value-parameterized test.
02921         //   code_location:    code location where the test is defined
02922         //   fixture_class_id: ID of the test fixture class
02923         //   set_up_tc:        pointer to the function that sets up the test suite
02924         //   tear_down_tc:     pointer to the function that tears down the test suite
02925         //   factory:          pointer to the factory that creates a test object.
02926         //                     The newly created TestInfo instance will assume
02927         //                     ownership of the factory object.
02928
02929         TestInfo* MakeAndRegisterTestInfo(
02930                 const char* test_suite_name, const char* name, const char* type_param,
02931                 const char* value_param, CodeLocation code_location,
02932                 TypeId fixture_class_id, SetUpTestSuiteFunc set_up_tc,
02933                 TearDownTestSuiteFunc tear_down_tc, TestFactoryBase* factory) {
02934             TestInfo * const test_info =
02935                     new TestInfo(test_suite_name, name, type_param, value_param,
02936                     code_location, fixture_class_id, factory);
02937             GetUnitTestImpl()->AddTestInfo(set_up_tc, tear_down_tc, test_info);
02938             return test_info;
02939         }
02940
02941         void ReportInvalidTestSuiteType(const char* test_suite_name,
02942                 CodeLocation code_location) {
02943             Message errors;
02944             errors
02945                     << "Attempted redefinition of test suite " << test_suite_name << ".\n"
02946                     << "All tests in the same test suite must use the same test fixture\n"
02947                     << "class.  However, in test suite " << test_suite_name << ", you tried\n"
02948                     << "to define a test using a fixture class different from the one\n"
```

```
02949                   << "used earlier. This can happen if the two fixture classes are\n"
02950                   << "from different namespaces and have the same name. You should\n"
02951                   << "probably rename one of the classes to put the tests into different\n"
02952                   << "test suites.";
02953
02954           GTEST_LOG_(ERROR) << FormatFileLocation(code_location.file.c_str(),
02955                   code_location.line)
02956                   << " " << errors.GetString();
02957       }
02958   } // namespace internal
02959
02960   namespace {
02961
02962       // A predicate that checks the test name of a TestInfo against a known
02963       // value.
02964       //
02965       // This is used for implementation of the TestSuite class only.  We put
02966       // it in the anonymous namespace to prevent polluting the outer
02967       // namespace.
02968       //
02969       // TestNameIs is copyable.
02970
02971       class TestNameIs {
02972       public:
02973           // Constructor.
02974           //
02975           // TestNameIs has NO default constructor.
02976
02977           explicit TestNameIs(const char* name) : name_(name) {
02978           }
02979
02980           // Returns true if and only if the test name of test_info matches name_.
02981
02982           bool operator()(const TestInfo* test_info) const {
02983               return test_info && test_info->name() == name_;
02984           }
02985
02986       private:
02987           std::string name_;
02988       };
02989
02990   } // namespace
02991
02992   namespace internal {
02993
02994       // This method expands all parameterized tests registered with macros TEST_P
02995       // and INSTANTIATE_TEST_SUITE_P into regular tests and registers those.
02996       // This will be done just once during the program runtime.
02997
02998       void UnitTestImpl::RegisterParameterizedTests() {
02999           if (!parameterized_tests_registered_) {
03000               parameterized_test_registry_.RegisterTests();
03001               type_parameterized_test_registry_.CheckForInstantiations();
03002               parameterized_tests_registered_ = true;
03003           }
03004       }
03005
03006   } // namespace internal
03007
03008   // Creates the test object, runs it, records its result, and then
03009   // deletes it.
03010
03011   void TestInfo::Run() {
03012       TestEventListener* repeater = UnitTest::GetInstance()->listeners().repeater();
03013       if (!should_run_) {
03014           if (is_disabled_ && matches_filter_) repeater->OnTestDisabled(*this);
03015           return;
03016       }
03017
03018       // Tells UnitTest where to store test result.
03019       internal::UnitTestImpl * const impl = internal::GetUnitTestImpl();
03020       impl->set_current_test_info(this);
03021
03022       // Notifies the unit test event listeners that a test is about to start.
03023       repeater->OnTestStart(*this);
03024       result_.set_start_timestamp(internal::GetTimeInMillis());
03025       internal::Timer timer;
03026       impl->os_stack_trace_getter()->UponLeavingGTest();
03027
03028       // Creates the test object.
03029       Test * const test = internal::HandleExceptionsInMethodIfSupported(
03030               factory_, &internal::TestFactoryBase::CreateTest,
03031               "the test fixture's constructor");
03032
03033       // Runs the test if the constructor didn't generate a fatal failure or invoke
03034       // GTEST_SKIP().
03035       // Note that the object will not be null
```

```
03036         if (!Test::HasFatalFailure() && !Test::IsSkipped()) {
03037             // This doesn't throw as all user code that can throw are wrapped into
03038             // exception handling code.
03039             test->Run();
03040         }
03041
03042         if (test != nullptr) {
03043             // Deletes the test object.
03044             impl->os_stack_trace_getter()->UponLeavingGTest();
03045             internal::HandleExceptionsInMethodIfSupported(
03046                     test, &Test::DeleteSelf_, "the test fixture's destructor");
03047         }
03048
03049         result_.set_elapsed_time(timer.Elapsed());
03050
03051         // Notifies the unit test event listener that a test has just finished.
03052         repeater->OnTestEnd(*this);
03053
03054         // Tells UnitTest to stop associating assertion results to this
03055         // test.
03056         impl->set_current_test_info(nullptr);
03057     }
03058
03059     // Skip and records a skipped test result for this object.
03060
03061     void TestInfo::Skip() {
03062         if (!should_run_) return;
03063
03064         internal::UnitTestImpl * const impl = internal::GetUnitTestImpl();
03065         impl->set_current_test_info(this);
03066
03067         TestEventListener* repeater = UnitTest::GetInstance()->listeners().repeater();
03068
03069         // Notifies the unit test event listeners that a test is about to start.
03070         repeater->OnTestStart(*this);
03071
03072         const TestPartResult test_part_result =
03073                 TestPartResult(TestPartResult::kSkip, this->file(), this->line(), "");
03074         impl->GetTestPartResultReporterForCurrentThread()->ReportTestPartResult(
03075                 test_part_result);
03076
03077         // Notifies the unit test event listener that a test has just finished.
03078         repeater->OnTestEnd(*this);
03079         impl->set_current_test_info(nullptr);
03080     }
03081
03082     // class TestSuite
03083
03084     // Gets the number of successful tests in this test suite.
03085
03086     int TestSuite::successful_test_count() const {
03087         return CountIf(test_info_list_, TestPassed);
03088     }
03089
03090     // Gets the number of successful tests in this test suite.
03091
03092     int TestSuite::skipped_test_count() const {
03093         return CountIf(test_info_list_, TestSkipped);
03094     }
03095
03096     // Gets the number of failed tests in this test suite.
03097
03098     int TestSuite::failed_test_count() const {
03099         return CountIf(test_info_list_, TestFailed);
03100     }
03101
03102     // Gets the number of disabled tests that will be reported in the XML report.
03103
03104     int TestSuite::reportable_disabled_test_count() const {
03105         return CountIf(test_info_list_, TestReportableDisabled);
03106     }
03107
03108     // Gets the number of disabled tests in this test suite.
03109
03110     int TestSuite::disabled_test_count() const {
03111         return CountIf(test_info_list_, TestDisabled);
03112     }
03113
03114     // Gets the number of tests to be printed in the XML report.
03115
03116     int TestSuite::reportable_test_count() const {
03117         return CountIf(test_info_list_, TestReportable);
03118     }
03119
03120     // Get the number of tests in this test suite that should run.
03121
03122     int TestSuite::test_to_run_count() const {
```

```
03123         return CountIf(test_info_list_, ShouldRunTest);
03124     }
03125
03126     // Gets the number of all tests.
03127
03128     int TestSuite::total_test_count() const {
03129         return static_cast<int> (test_info_list_.size());
03130     }
03131
03132     // Creates a TestSuite with the given name.
03133     //
03134     // Arguments:
03135     //
03136     //   a_name:       name of the test suite
03137     //   a_type_param: the name of the test suite's type parameter, or NULL if
03138     //                 this is not a typed or a type-parameterized test suite.
03139     //   set_up_tc:    pointer to the function that sets up the test suite
03140     //   tear_down_tc: pointer to the function that tears down the test suite
03141
03142     TestSuite::TestSuite(const char* a_name, const char* a_type_param,
03143             internal::SetUpTestSuiteFunc set_up_tc,
03144             internal::TearDownTestSuiteFunc tear_down_tc)
03145     : name_(a_name),
03146     type_param_(a_type_param ? new std::string(a_type_param) : nullptr),
03147     set_up_tc_(set_up_tc),
03148     tear_down_tc_(tear_down_tc),
03149     should_run_(false),
03150     start_timestamp_(0),
03151     elapsed_time_(0) {
03152     }
03153
03154     // Destructor of TestSuite.
03155
03156     TestSuite::~TestSuite() {
03157         // Deletes every Test in the collection.
03158         ForEach(test_info_list_, internal::Delete<TestInfo>);
03159     }
03160
03161     // Returns the i-th test among all the tests. i can range from 0 to
03162     // total_test_count() - 1. If i is not in that range, returns NULL.
03163
03164     const TestInfo* TestSuite::GetTestInfo(int i) const {
03165         const int index = GetElementOr(test_indices_, i, -1);
03166         return index < 0 ? nullptr : test_info_list_[static_cast<size_t> (index)];
03167     }
03168
03169     // Returns the i-th test among all the tests. i can range from 0 to
03170     // total_test_count() - 1. If i is not in that range, returns NULL.
03171
03172     TestInfo* TestSuite::GetMutableTestInfo(int i) {
03173         const int index = GetElementOr(test_indices_, i, -1);
03174         return index < 0 ? nullptr : test_info_list_[static_cast<size_t> (index)];
03175     }
03176
03177     // Adds a test to this test suite.  Will delete the test upon
03178     // destruction of the TestSuite object.
03179
03180     void TestSuite::AddTestInfo(TestInfo* test_info) {
03181         test_info_list_.push_back(test_info);
03182         test_indices_.push_back(static_cast<int> (test_indices_.size()));
03183     }
03184
03185     // Runs every test in this TestSuite.
03186
03187     void TestSuite::Run() {
03188         if (!should_run_) return;
03189
03190         internal::UnitTestImpl * const impl = internal::GetUnitTestImpl();
03191         impl->set_current_test_suite(this);
03192
03193         TestEventListener* repeater = UnitTest::GetInstance()->listeners().repeater();
03194
03195         // Call both legacy and the new API
03196         repeater->OnTestSuiteStart(*this);
03197         //  Legacy API is deprecated but still available
03198 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03199         repeater->OnTestCaseStart(*this);
03200 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03201
03202         impl->os_stack_trace_getter()->UponLeavingGTest();
03203         internal::HandleExceptionsInMethodIfSupported(
03204                 this, &TestSuite::RunSetUpTestSuite, "SetUpTestSuite()");
03205
03206         const bool skip_all = ad_hoc_test_result().Failed();
03207
03208         start_timestamp_ = internal::GetTimeInMillis();
03209         internal::Timer timer;
```

```
03210         for (int i = 0; i < total_test_count(); i++) {
03211             if (skip_all) {
03212                 GetMutableTestInfo(i)->Skip();
03213             } else {
03214                 GetMutableTestInfo(i)->Run();
03215             }
03216             if (GTEST_FLAG_GET(fail_fast) &&
03217                     GetMutableTestInfo(i)->result()->Failed()) {
03218                 for (int j = i + 1; j < total_test_count(); j++) {
03219                     GetMutableTestInfo(j)->Skip();
03220                 }
03221                 break;
03222             }
03223         }
03224         elapsed_time_ = timer.Elapsed();
03225
03226         impl->os_stack_trace_getter()->UponLeavingGTest();
03227         internal::HandleExceptionsInMethodIfSupported(
03228                 this, &TestSuite::RunTearDownTestSuite, "TearDownTestSuite()");
03229
03230         // Call both legacy and the new API
03231         repeater->OnTestSuiteEnd(*this);
03232         //  Legacy API is deprecated but still available
03233 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03234         repeater->OnTestCaseEnd(*this);
03235 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03236
03237         impl->set_current_test_suite(nullptr);
03238     }
03239
03240     // Skips all tests under this TestSuite.
03241
03242     void TestSuite::Skip() {
03243         if (!should_run_) return;
03244
03245         internal::UnitTestImpl * const impl = internal::GetUnitTestImpl();
03246         impl->set_current_test_suite(this);
03247
03248         TestEventListener* repeater = UnitTest::GetInstance()->listeners().repeater();
03249
03250         // Call both legacy and the new API
03251         repeater->OnTestSuiteStart(*this);
03252         //  Legacy API is deprecated but still available
03253 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03254         repeater->OnTestCaseStart(*this);
03255 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03256
03257         for (int i = 0; i < total_test_count(); i++) {
03258             GetMutableTestInfo(i)->Skip();
03259         }
03260
03261         // Call both legacy and the new API
03262         repeater->OnTestSuiteEnd(*this);
03263         // Legacy API is deprecated but still available
03264 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03265         repeater->OnTestCaseEnd(*this);
03266 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03267
03268         impl->set_current_test_suite(nullptr);
03269     }
03270
03271     // Clears the results of all tests in this test suite.
03272
03273     void TestSuite::ClearResult() {
03274         ad_hoc_test_result_.Clear();
03275         ForEach(test_info_list_, TestInfo::ClearTestResult);
03276     }
03277
03278     // Shuffles the tests in this test suite.
03279
03280     void TestSuite::ShuffleTests(internal::Random* random) {
03281         Shuffle(random, &test_indices_);
03282     }
03283
03284     // Restores the test order to before the first shuffle.
03285
03286     void TestSuite::UnshuffleTests() {
03287         for (size_t i = 0; i < test_indices_.size(); i++) {
03288             test_indices_[i] = static_cast<int> (i);
03289         }
03290     }
03291
03292     // Formats a countable noun.  Depending on its quantity, either the
03293     // singular form or the plural form is used. e.g.
03294     //
03295     // FormatCountableNoun(1, "formula", "formuli") returns "1 formula".
03296     // FormatCountableNoun(5, "book", "books") returns "5 books".
```

```
03297
03298     static std::string FormatCountableNoun(int count, const char* singular_form,
03299             const char* plural_form) {
03300         return internal::StreamableToString(count) + " " +
03301                 (count == 1 ? singular_form : plural_form);
03302     }
03303
03304     // Formats the count of tests.
03305
03306     static std::string FormatTestCount(int test_count) {
03307         return FormatCountableNoun(test_count, "test", "tests");
03308     }
03309
03310     // Formats the count of test suites.
03311
03312     static std::string FormatTestSuiteCount(int test_suite_count) {
03313         return FormatCountableNoun(test_suite_count, "test suite", "test suites");
03314     }
03315
03316     // Converts a TestPartResult::Type enum to human-friendly string
03317     // representation.  Both kNonFatalFailure and kFatalFailure are translated
03318     // to "Failure", as the user usually doesn't care about the difference
03319     // between the two when viewing the test result.
03320
03321     static const char* TestPartResultTypeToString(TestPartResult::Type type) {
03322         switch (type) {
03323             case TestPartResult::kSkip:
03324                 return "Skipped\n";
03325             case TestPartResult::kSuccess:
03326                 return "Success";
03327
03328             case TestPartResult::kNonFatalFailure:
03329             case TestPartResult::kFatalFailure:
03330 #ifdef _MSC_VER
03331                 return "error: ";
03332 #else
03333                 return "Failure\n";
03334 #endif
03335             default:
03336                 return "Unknown result type";
03337         }
03338     }
03339
03340     namespace internal {
03341         namespace {
03342
03343             enum class GTestColor {
03344                 kDefault, kRed, kGreen, kYellow
03345             };
03346         } // namespace
03347
03348         // Prints a TestPartResult to an std::string.
03349
03350         static std::string PrintTestPartResultToString(
03351                 const TestPartResult& test_part_result) {
03352             return (Message() << internal::FormatFileLocation(
03353                     test_part_result.file_name(),
03354                     test_part_result.line_number())
03355                     << " "
03356                     << TestPartResultTypeToString(test_part_result.type())
03357                     << test_part_result.message())
03358                     .GetString();
03359         }
03360
03361         // Prints a TestPartResult.
03362
03363         static void PrintTestPartResult(const TestPartResult& test_part_result) {
03364             const std::string& result = PrintTestPartResultToString(test_part_result);
03365             printf("%s\n", result.c_str());
03366             fflush(stdout);
03367             // If the test program runs in Visual Studio or a debugger, the
03368             // following statements add the test part result message to the Output
03369             // window such that the user can double-click on it to jump to the
03370             // corresponding source code location; otherwise they do nothing.
03371 #if GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_MOBILE
03372             // We don't call OutputDebugString*() on Windows Mobile, as printing
03373             // to stdout is done by OutputDebugString() there already - we don't
03374             // want the same message printed twice.
03375             ::OutputDebugStringA(result.c_str());
03376             ::OutputDebugStringA("\n");
03377 #endif
03378         }
03379
03380         // class PrettyUnitTestResultPrinter
03381 #if GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_MOBILE && !GTEST_OS_WINDOWS_PHONE && \
03382     !GTEST_OS_WINDOWS_RT && !GTEST_OS_WINDOWS_MINGW
03383
```

```
03384         // Returns the character attribute for the given color.
03385
03386         static WORD GetColorAttribute(GTestColor color) {
03387             switch (color) {
03388                 case GTestColor::kRed:
03389                     return FOREGROUND_RED;
03390                 case GTestColor::kGreen:
03391                     return FOREGROUND_GREEN;
03392                 case GTestColor::kYellow:
03393                     return FOREGROUND_RED | FOREGROUND_GREEN;
03394                 default:
03395                     return 0;
03396             }
03397         }
03398
03399         static int GetBitOffset(WORD color_mask) {
03400             if (color_mask == 0) return 0;
03401
03402             int bitOffset = 0;
03403             while ((color_mask & 1) == 0) {
03404                 color_mask >>= 1;
03405                 ++bitOffset;
03406             }
03407             return bitOffset;
03408         }
03409
03410         static WORD GetNewColor(GTestColor color, WORD old_color_attrs) {
03411             // Let's reuse the BG
03412             static const WORD background_mask = BACKGROUND_BLUE | BACKGROUND_GREEN |
03413                     BACKGROUND_RED | BACKGROUND_INTENSITY;
03414             static const WORD foreground_mask = FOREGROUND_BLUE | FOREGROUND_GREEN |
03415                     FOREGROUND_RED | FOREGROUND_INTENSITY;
03416             const WORD existing_bg = old_color_attrs & background_mask;
03417
03418             WORD new_color =
03419                     GetColorAttribute(color) | existing_bg | FOREGROUND_INTENSITY;
03420             static const int bg_bitOffset = GetBitOffset(background_mask);
03421             static const int fg_bitOffset = GetBitOffset(foreground_mask);
03422
03423             if (((new_color & background_mask) >> bg_bitOffset) ==
03424                     ((new_color & foreground_mask) >> fg_bitOffset)) {
03425                 new_color ^= FOREGROUND_INTENSITY; // invert intensity
03426             }
03427             return new_color;
03428         }
03429
03430 #else
03431
03432         // Returns the ANSI color code for the given color. GTestColor::kDefault is
03433         // an invalid input.
03434
03435         static const char* GetAnsiColorCode(GTestColor color) {
03436             switch (color) {
03437                 case GTestColor::kRed:
03438                     return "1";
03439                 case GTestColor::kGreen:
03440                     return "2";
03441                 case GTestColor::kYellow:
03442                     return "3";
03443                 default:
03444                     return nullptr;
03445             }
03446         }
03447
03448 #endif  // GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_MOBILE
03449
03450         // Returns true if and only if Google Test should use colors in the output.
03451
03452         bool ShouldUseColor(bool stdout_is_tty) {
03453             std::string c = GTEST_FLAG_GET(color);
03454             const char* const gtest_color = c.c_str();
03455
03456             if (String::CaseInsensitiveCStringEquals(gtest_color, "auto")) {
03457 #if GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_MINGW
03458                 // On Windows the TERM variable is usually not set, but the
03459                 // console there does support colors.
03460                 return stdout_is_tty;
03461 #else
03462                 // On non-Windows platforms, we rely on the TERM variable.
03463                 const char* const term = posix::GetEnv("TERM");
03464                 const bool term_supports_color =
03465                         String::CStringEquals(term, "xterm") ||
03466                         String::CStringEquals(term, "xterm-color") ||
03467                         String::CStringEquals(term, "xterm-256color") ||
03468                         String::CStringEquals(term, "screen") ||
03469                         String::CStringEquals(term, "screen-256color") ||
03470                         String::CStringEquals(term, "tmux") ||
```

```
03471                         String::CStringEquals(term, "tmux-256color") ||
03472                         String::CStringEquals(term, "rxvt-unicode") ||
03473                         String::CStringEquals(term, "rxvt-unicode-256color") ||
03474                         String::CStringEquals(term, "linux") ||
03475                         String::CStringEquals(term, "cygwin");
03476                 return stdout_is_tty && term_supports_color;
03477 #endif  // GTEST_OS_WINDOWS
03478             }
03479
03480             return String::CaseInsensitiveCStringEquals(gtest_color, "yes") ||
03481                     String::CaseInsensitiveCStringEquals(gtest_color, "true") ||
03482                     String::CaseInsensitiveCStringEquals(gtest_color, "t") ||
03483                     String::CStringEquals(gtest_color, "1");
03484             // We take "yes", "true", "t", and "1" as meaning "yes".  If the
03485             // value is neither one of these nor "auto", we treat it as "no" to
03486             // be conservative.
03487         }
03488
03489         // Helpers for printing colored strings to stdout. Note that on Windows, we
03490         // cannot simply emit special characters and have the terminal change colors.
03491         // This routine must actually emit the characters rather than return a string
03492         // that would be colored when printed, as can be done on Linux.
03493
03494         GTEST_ATTRIBUTE_PRINTF_(2, 3)
03495         static void ColoredPrintf(GTestColor color, const char* fmt, ...) {
03496             va_list args;
03497             va_start(args, fmt);
03498
03499             static const bool in_color_mode =
03500                     ShouldUseColor(posix::IsATTY(posix::FileNo(stdout)) != 0);
03501             const bool use_color = in_color_mode && (color != GTestColor::kDefault);
03502
03503             if (!use_color) {
03504                 vprintf(fmt, args);
03505                 va_end(args);
03506                 return;
03507             }
03508
03509 #if GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_MOBILE && !GTEST_OS_WINDOWS_PHONE && \
03510     !GTEST_OS_WINDOWS_RT && !GTEST_OS_WINDOWS_MINGW
03511             const HANDLE stdout_handle = GetStdHandle(STD_OUTPUT_HANDLE);
03512
03513             // Gets the current text color.
03514             CONSOLE_SCREEN_BUFFER_INFO buffer_info;
03515             GetConsoleScreenBufferInfo(stdout_handle, &buffer_info);
03516             const WORD old_color_attrs = buffer_info.wAttributes;
03517             const WORD new_color = GetNewColor(color, old_color_attrs);
03518
03519             // We need to flush the stream buffers into the console before each
03520             // SetConsoleTextAttribute call lest it affect the text that is already
03521             // printed but has not yet reached the console.
03522             fflush(stdout);
03523             SetConsoleTextAttribute(stdout_handle, new_color);
03524
03525             vprintf(fmt, args);
03526
03527             fflush(stdout);
03528             // Restores the text color.
03529             SetConsoleTextAttribute(stdout_handle, old_color_attrs);
03530 #else
03531             printf("\033[0;3%sm", GetAnsiColorCode(color));
03532             vprintf(fmt, args);
03533             printf("\033[m"); // Resets the terminal to default.
03534 #endif  // GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_MOBILE
03535             va_end(args);
03536         }
03537
03538         // Text printed in Google Test's text output and --gtest_list_tests
03539         // output to label the type parameter and value parameter for a test.
03540         static const char kTypeParamLabel[] = "TypeParam";
03541         static const char kValueParamLabel[] = "GetParam()";
03542
03543         static void PrintFullTestCommentIfPresent(const TestInfo& test_info) {
03544             const char* const type_param = test_info.type_param();
03545             const char* const value_param = test_info.value_param();
03546
03547             if (type_param != nullptr || value_param != nullptr) {
03548                 printf(", where ");
03549                 if (type_param != nullptr) {
03550                     printf("%s = %s", kTypeParamLabel, type_param);
03551                     if (value_param != nullptr) printf(" and ");
03552                 }
03553                 if (value_param != nullptr) {
03554                     printf("%s = %s", kValueParamLabel, value_param);
03555                 }
03556             }
03557         }
```

```
03558
03559         // This class implements the TestEventListener interface.
03560         //
03561         // Class PrettyUnitTestResultPrinter is copyable.
03562
03563         class PrettyUnitTestResultPrinter : public TestEventListener {
03564         public:
03565
03566             PrettyUnitTestResultPrinter() {
03567             }
03568
03569             static void PrintTestName(const char* test_suite, const char* test) {
03570                 printf("%s.%s", test_suite, test);
03571             }
03572
03573             // The following methods override what's in the TestEventListener class.
03574
03575             void OnTestProgramStart(const UnitTest& /*unit_test*/) override {
03576             }
03577             void OnTestIterationStart(const UnitTest& unit_test, int iteration) override;
03578             void OnEnvironmentsSetUpStart(const UnitTest& unit_test) override;
03579
03580             void OnEnvironmentsSetUpEnd(const UnitTest& /*unit_test*/) override {
03581             }
03582 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03583             void OnTestCaseStart(const TestCase& test_case) override;
03584 #else
03585             void OnTestSuiteStart(const TestSuite& test_suite) override;
03586 #endif  // OnTestCaseStart
03587
03588             void OnTestStart(const TestInfo& test_info) override;
03589             void OnTestDisabled(const TestInfo& test_info) override;
03590
03591             void OnTestPartResult(const TestPartResult& result) override;
03592             void OnTestEnd(const TestInfo& test_info) override;
03593 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03594             void OnTestCaseEnd(const TestCase& test_case) override;
03595 #else
03596             void OnTestSuiteEnd(const TestSuite& test_suite) override;
03597 #endif  // GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03598
03599             void OnEnvironmentsTearDownStart(const UnitTest& unit_test) override;
03600
03601             void OnEnvironmentsTearDownEnd(const UnitTest& /*unit_test*/) override {
03602             }
03603             void OnTestIterationEnd(const UnitTest& unit_test, int iteration) override;
03604
03605             void OnTestProgramEnd(const UnitTest& /*unit_test*/) override {
03606             }
03607
03608         private:
03609             static void PrintFailedTests(const UnitTest& unit_test);
03610             static void PrintFailedTestSuites(const UnitTest& unit_test);
03611             static void PrintSkippedTests(const UnitTest& unit_test);
03612         };
03613
03614         // Fired before each iteration of tests starts.
03615
03616         void PrettyUnitTestResultPrinter::OnTestIterationStart(
03617                 const UnitTest& unit_test, int iteration) {
03618             if (GTEST_FLAG_GET(repeat) != 1)
03619                 printf("\nRepeating all tests (iteration %d) . . .\n\n", iteration + 1);
03620
03621             std::string f = GTEST_FLAG_GET(filter);
03622             const char* const filter = f.c_str();
03623
03624             // Prints the filter if it's not *.  This reminds the user that some
03625             // tests may be skipped.
03626             if (!String::CStringEquals(filter, kUniversalFilter)) {
03627                 ColoredPrintf(GTestColor::kYellow, "Note: %s filter = %s\n", GTEST_NAME_,
03628                         filter);
03629             }
03630
03631             if (internal::ShouldShard(kTestTotalShards, kTestShardIndex, false)) {
03632                 const int32_t shard_index = Int32FromEnvOrDie(kTestShardIndex, -1);
03633                 ColoredPrintf(GTestColor::kYellow, "Note: This is test shard %d of %s.\n",
03634                         static_cast<int> (shard_index) + 1,
03635                         internal::posix::GetEnv(kTestTotalShards));
03636             }
03637
03638             if (GTEST_FLAG_GET(shuffle)) {
03639                 ColoredPrintf(GTestColor::kYellow,
03640                         "Note: Randomizing tests' orders with a seed of %d .\n",
03641                         unit_test.random_seed());
03642             }
03643
03644             ColoredPrintf(GTestColor::kGreen, "[==========] ");
```

```
03645             printf("Running %s from %s.\n",
03646                    FormatTestCount(unit_test.test_to_run_count()).c_str(),
03647                    FormatTestSuiteCount(unit_test.test_suite_to_run_count()).c_str());
03648             fflush(stdout);
03649         }
03650 
03651         void PrettyUnitTestResultPrinter::OnEnvironmentsSetUpStart(
03652                 const UnitTest& /*unit_test*/) {
03653             ColoredPrintf(GTestColor::kGreen, "[----------] ");
03654             printf("Global test environment set-up.\n");
03655             fflush(stdout);
03656         }
03657 
03658 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03659 
03660         void PrettyUnitTestResultPrinter::OnTestCaseStart(const TestCase& test_case) {
03661             const std::string counts =
03662                    FormatCountableNoun(test_case.test_to_run_count(), "test", "tests");
03663             ColoredPrintf(GTestColor::kGreen, "[----------] ");
03664             printf("%s from %s", counts.c_str(), test_case.name());
03665             if (test_case.type_param() == nullptr) {
03666                 printf("\n");
03667             } else {
03668                 printf(", where %s = %s\n", kTypeParamLabel, test_case.type_param());
03669             }
03670             fflush(stdout);
03671         }
03672 #else
03673 
03674         void PrettyUnitTestResultPrinter::OnTestSuiteStart(
03675                 const TestSuite& test_suite) {
03676             const std::string counts =
03677                    FormatCountableNoun(test_suite.test_to_run_count(), "test", "tests");
03678             ColoredPrintf(GTestColor::kGreen, "[----------] ");
03679             printf("%s from %s", counts.c_str(), test_suite.name());
03680             if (test_suite.type_param() == nullptr) {
03681                 printf("\n");
03682             } else {
03683                 printf(", where %s = %s\n", kTypeParamLabel, test_suite.type_param());
03684             }
03685             fflush(stdout);
03686         }
03687 #endif  // GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03688 
03689         void PrettyUnitTestResultPrinter::OnTestStart(const TestInfo& test_info) {
03690             ColoredPrintf(GTestColor::kGreen, "[ RUN      ] ");
03691             PrintTestName(test_info.test_suite_name(), test_info.name());
03692             printf("\n");
03693             fflush(stdout);
03694         }
03695 
03696         void PrettyUnitTestResultPrinter::OnTestDisabled(const TestInfo& test_info) {
03697             ColoredPrintf(GTestColor::kYellow, "[ DISABLED ] ");
03698             PrintTestName(test_info.test_suite_name(), test_info.name());
03699             printf("\n");
03700             fflush(stdout);
03701         }
03702 
03703         // Called after an assertion failure.
03704 
03705         void PrettyUnitTestResultPrinter::OnTestPartResult(
03706                 const TestPartResult& result) {
03707             switch (result.type()) {
03708                 // If the test part succeeded, we don't need to do anything.
03709                 case TestPartResult::kSuccess:
03710                     return;
03711                 default:
03712                     // Print failure message from the assertion
03713                     // (e.g. expected this and got that).
03714                     PrintTestPartResult(result);
03715                     fflush(stdout);
03716             }
03717         }
03718 
03719         void PrettyUnitTestResultPrinter::OnTestEnd(const TestInfo& test_info) {
03720             if (test_info.result()->Passed()) {
03721                 ColoredPrintf(GTestColor::kGreen, "[       OK ] ");
03722             } else if (test_info.result()->Skipped()) {
03723                 ColoredPrintf(GTestColor::kGreen, "[  SKIPPED ] ");
03724             } else {
03725                 ColoredPrintf(GTestColor::kRed, "[  FAILED  ] ");
03726             }
03727             PrintTestName(test_info.test_suite_name(), test_info.name());
03728             if (test_info.result()->Failed()) PrintFullTestCommentIfPresent(test_info);
03729 
03730             if (GTEST_FLAG_GET(print_time)) {
03731                 printf(" (%s ms)\n",
```

```
03732                         internal::StreamableToString(test_info.result()->elapsed_time())
03733                         .c_str());
03734             } else {
03735                 printf("\n");
03736             }
03737             fflush(stdout);
03738         }
03739
03740 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03741
03742         void PrettyUnitTestResultPrinter::OnTestCaseEnd(const TestCase& test_case) {
03743             if (!GTEST_FLAG_GET(print_time)) return;
03744
03745             const std::string counts =
03746                     FormatCountableNoun(test_case.test_to_run_count(), "test", "tests");
03747             ColoredPrintf(GTestColor::kGreen, "[----------] ");
03748             printf("%s from %s (%s ms total)\n\n", counts.c_str(), test_case.name(),
03749                     internal::StreamableToString(test_case.elapsed_time()).c_str());
03750             fflush(stdout);
03751         }
03752 #else
03753
03754         void PrettyUnitTestResultPrinter::OnTestSuiteEnd(const TestSuite& test_suite) {
03755             if (!GTEST_FLAG_GET(print_time)) return;
03756
03757             const std::string counts =
03758                     FormatCountableNoun(test_suite.test_to_run_count(), "test", "tests");
03759             ColoredPrintf(GTestColor::kGreen, "[----------] ");
03760             printf("%s from %s (%s ms total)\n\n", counts.c_str(), test_suite.name(),
03761                     internal::StreamableToString(test_suite.elapsed_time()).c_str());
03762             fflush(stdout);
03763         }
03764 #endif  // GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03765
03766         void PrettyUnitTestResultPrinter::OnEnvironmentsTearDownStart(
03767                 const UnitTest& /*unit_test*/) {
03768             ColoredPrintf(GTestColor::kGreen, "[----------] ");
03769             printf("Global test environment tear-down\n");
03770             fflush(stdout);
03771         }
03772
03773         // Internal helper for printing the list of failed tests.
03774
03775         void PrettyUnitTestResultPrinter::PrintFailedTests(const UnitTest& unit_test) {
03776             const int failed_test_count = unit_test.failed_test_count();
03777             ColoredPrintf(GTestColor::kRed, "[  FAILED  ] ");
03778             printf("%s, listed below:\n", FormatTestCount(failed_test_count).c_str());
03779
03780             for (int i = 0; i < unit_test.total_test_suite_count(); ++i) {
03781                 const TestSuite& test_suite = *unit_test.GetTestSuite(i);
03782                 if (!test_suite.should_run() || (test_suite.failed_test_count() == 0)) {
03783                     continue;
03784                 }
03785                 for (int j = 0; j < test_suite.total_test_count(); ++j) {
03786                     const TestInfo& test_info = *test_suite.GetTestInfo(j);
03787                     if (!test_info.should_run() || !test_info.result()->Failed()) {
03788                         continue;
03789                     }
03790                     ColoredPrintf(GTestColor::kRed, "[  FAILED  ] ");
03791                     printf("%s.%s", test_suite.name(), test_info.name());
03792                     PrintFullTestCommentIfPresent(test_info);
03793                     printf("\n");
03794                 }
03795             }
03796             printf("\n%2d FAILED %s\n", failed_test_count,
03797                     failed_test_count == 1 ? "TEST" : "TESTS");
03798         }
03799
03800         // Internal helper for printing the list of test suite failures not covered by
03801         // PrintFailedTests.
03802
03803         void PrettyUnitTestResultPrinter::PrintFailedTestSuites(
03804                 const UnitTest& unit_test) {
03805             int suite_failure_count = 0;
03806             for (int i = 0; i < unit_test.total_test_suite_count(); ++i) {
03807                 const TestSuite& test_suite = *unit_test.GetTestSuite(i);
03808                 if (!test_suite.should_run()) {
03809                     continue;
03810                 }
03811                 if (test_suite.ad_hoc_test_result().Failed()) {
03812                     ColoredPrintf(GTestColor::kRed, "[  FAILED  ] ");
03813                     printf("%s: SetUpTestSuite or TearDownTestSuite\n", test_suite.name());
03814                     ++suite_failure_count;
03815                 }
03816             }
03817             if (suite_failure_count > 0) {
03818                 printf("\n%2d FAILED TEST %s\n", suite_failure_count,
```

```
03819                       suite_failure_count == 1 ? "SUITE" : "SUITES");
03820       }
03821     }
03822
03823     // Internal helper for printing the list of skipped tests.
03824
03825     void PrettyUnitTestResultPrinter::PrintSkippedTests(const UnitTest& unit_test) {
03826       const int skipped_test_count = unit_test.skipped_test_count();
03827       if (skipped_test_count == 0) {
03828         return;
03829       }
03830
03831       for (int i = 0; i < unit_test.total_test_suite_count(); ++i) {
03832         const TestSuite& test_suite = *unit_test.GetTestSuite(i);
03833         if (!test_suite.should_run() || (test_suite.skipped_test_count() == 0)) {
03834           continue;
03835         }
03836         for (int j = 0; j < test_suite.total_test_count(); ++j) {
03837           const TestInfo& test_info = *test_suite.GetTestInfo(j);
03838           if (!test_info.should_run() || !test_info.result()->Skipped()) {
03839             continue;
03840           }
03841           ColoredPrintf(GTestColor::kGreen, "[  SKIPPED ] ");
03842           printf("%s.%s", test_suite.name(), test_info.name());
03843           printf("\n");
03844         }
03845       }
03846     }
03847
03848     void PrettyUnitTestResultPrinter::OnTestIterationEnd(const UnitTest& unit_test,
03849         int /*iteration*/) {
03850       ColoredPrintf(GTestColor::kGreen, "[==========] ");
03851       printf("%s from %s ran.",
03852           FormatTestCount(unit_test.test_to_run_count()).c_str(),
03853           FormatTestSuiteCount(unit_test.test_suite_to_run_count()).c_str());
03854       if (GTEST_FLAG_GET(print_time)) {
03855         printf(" (%s ms total)",
03856             internal::StreamableToString(unit_test.elapsed_time()).c_str());
03857       }
03858       printf("\n");
03859       ColoredPrintf(GTestColor::kGreen, "[  PASSED  ] ");
03860       printf("%s.\n", FormatTestCount(unit_test.successful_test_count()).c_str());
03861
03862       const int skipped_test_count = unit_test.skipped_test_count();
03863       if (skipped_test_count > 0) {
03864         ColoredPrintf(GTestColor::kGreen, "[  SKIPPED ] ");
03865         printf("%s, listed below:\n", FormatTestCount(skipped_test_count).c_str());
03866         PrintSkippedTests(unit_test);
03867       }
03868
03869       if (!unit_test.Passed()) {
03870         PrintFailedTests(unit_test);
03871         PrintFailedTestSuites(unit_test);
03872       }
03873
03874       int num_disabled = unit_test.reportable_disabled_test_count();
03875       if (num_disabled && !GTEST_FLAG_GET(also_run_disabled_tests)) {
03876         if (unit_test.Passed()) {
03877           printf("\n"); // Add a spacer if no FAILURE banner is displayed.
03878         }
03879         ColoredPrintf(GTestColor::kYellow, "  YOU HAVE %d DISABLED %s\n\n",
03880             num_disabled, num_disabled == 1 ? "TEST" : "TESTS");
03881       }
03882       // Ensure that Google Test output is printed before, e.g., heapchecker output.
03883       fflush(stdout);
03884     }
03885
03886     // End PrettyUnitTestResultPrinter
03887
03888     // This class implements the TestEventListener interface.
03889     //
03890     // Class BriefUnitTestResultPrinter is copyable.
03891
03892     class BriefUnitTestResultPrinter : public TestEventListener {
03893     public:
03894
03895       BriefUnitTestResultPrinter() {
03896       }
03897
03898       static void PrintTestName(const char* test_suite, const char* test) {
03899         printf("%s.%s", test_suite, test);
03900       }
03901
03902       // The following methods override what's in the TestEventListener class.
03903
03904       void OnTestProgramStart(const UnitTest& /*unit_test*/) override {
03905       }
```

```
03906
03907             void OnTestIterationStart(const UnitTest& /*unit_test*/,
03908                     int /*iteration*/) override {
03909             }
03910
03911             void OnEnvironmentsSetUpStart(const UnitTest& /*unit_test*/) override {
03912             }
03913
03914             void OnEnvironmentsSetUpEnd(const UnitTest& /*unit_test*/) override {
03915             }
03916 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03917
03918             void OnTestCaseStart(const TestCase& /*test_case*/) override {
03919             }
03920 #else
03921
03922             void OnTestSuiteStart(const TestSuite& /*test_suite*/) override {
03923             }
03924 #endif  // OnTestCaseStart
03925
03926             void OnTestStart(const TestInfo& /*test_info*/) override {
03927             }
03928
03929             void OnTestDisabled(const TestInfo& /*test_info*/) override {
03930             }
03931
03932             void OnTestPartResult(const TestPartResult& result) override;
03933             void OnTestEnd(const TestInfo& test_info) override;
03934 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03935
03936             void OnTestCaseEnd(const TestCase& /*test_case*/) override {
03937             }
03938 #else
03939
03940             void OnTestSuiteEnd(const TestSuite& /*test_suite*/) override {
03941             }
03942 #endif  // GTEST_REMOVE_LEGACY_TEST_CASEAPI_
03943
03944             void OnEnvironmentsTearDownStart(const UnitTest& /*unit_test*/) override {
03945             }
03946
03947             void OnEnvironmentsTearDownEnd(const UnitTest& /*unit_test*/) override {
03948             }
03949             void OnTestIterationEnd(const UnitTest& unit_test, int iteration) override;
03950
03951             void OnTestProgramEnd(const UnitTest& /*unit_test*/) override {
03952             }
03953         };
03954
03955         // Called after an assertion failure.
03956
03957         void BriefUnitTestResultPrinter::OnTestPartResult(
03958                 const TestPartResult& result) {
03959             switch (result.type()) {
03960                     // If the test part succeeded, we don't need to do anything.
03961                 case TestPartResult::kSuccess:
03962                     return;
03963                 default:
03964                     // Print failure message from the assertion
03965                     // (e.g. expected this and got that).
03966                     PrintTestPartResult(result);
03967                     fflush(stdout);
03968             }
03969         }
03970
03971         void BriefUnitTestResultPrinter::OnTestEnd(const TestInfo& test_info) {
03972             if (test_info.result()->Failed()) {
03973                 ColoredPrintf(GTestColor::kRed, "[  FAILED  ] ");
03974                 PrintTestName(test_info.test_suite_name(), test_info.name());
03975                 PrintFullTestCommentIfPresent(test_info);
03976
03977                 if (GTEST_FLAG_GET(print_time)) {
03978                     printf(" (%s ms)\n",
03979                             internal::StreamableToString(test_info.result()->elapsed_time())
03980                             .c_str());
03981                 } else {
03982                     printf("\n");
03983                 }
03984                 fflush(stdout);
03985             }
03986         }
03987
03988         void BriefUnitTestResultPrinter::OnTestIterationEnd(const UnitTest& unit_test,
03989                 int /*iteration*/) {
03990             ColoredPrintf(GTestColor::kGreen, "[==========] ");
03991             printf("%s from %s ran.",
03992                     FormatTestCount(unit_test.test_to_run_count()).c_str(),
```

```
03993                   FormatTestSuiteCount(unit_test.test_suite_to_run_count()).c_str());
03994           if (GTEST_FLAG_GET(print_time)) {
03995               printf(" (%s ms total)",
03996                       internal::StreamableToString(unit_test.elapsed_time()).c_str());
03997           }
03998           printf("\n");
03999           ColoredPrintf(GTestColor::kGreen, "[  PASSED  ] ");
04000           printf("%s.\n", FormatTestCount(unit_test.successful_test_count()).c_str());
04001
04002           const int skipped_test_count = unit_test.skipped_test_count();
04003           if (skipped_test_count > 0) {
04004               ColoredPrintf(GTestColor::kGreen, "[  SKIPPED ] ");
04005               printf("%s.\n", FormatTestCount(skipped_test_count).c_str());
04006           }
04007
04008           int num_disabled = unit_test.reportable_disabled_test_count();
04009           if (num_disabled && !GTEST_FLAG_GET(also_run_disabled_tests)) {
04010               if (unit_test.Passed()) {
04011                   printf("\n"); // Add a spacer if no FAILURE banner is displayed.
04012               }
04013               ColoredPrintf(GTestColor::kYellow, "  YOU HAVE %d DISABLED %s\n\n",
04014                       num_disabled, num_disabled == 1 ? "TEST" : "TESTS");
04015           }
04016           // Ensure that Google Test output is printed before, e.g., heapchecker output.
04017           fflush(stdout);
04018       }
04019
04020       // End BriefUnitTestResultPrinter
04021
04022       // class TestEventRepeater
04023       //
04024       // This class forwards events to other event listeners.
04025
04026       class TestEventRepeater : public TestEventListener {
04027       public:
04028
04029           TestEventRepeater() : forwarding_enabled_(true) {
04030           }
04031           ~TestEventRepeater() override;
04032           void Append(TestEventListener* listener);
04033           TestEventListener* Release(TestEventListener* listener);
04034
04035           // Controls whether events will be forwarded to listeners_. Set to false
04036           // in death test child processes.
04037
04038           bool forwarding_enabled() const {
04039               return forwarding_enabled_;
04040           }
04041
04042           void set_forwarding_enabled(bool enable) {
04043               forwarding_enabled_ = enable;
04044           }
04045
04046           void OnTestProgramStart(const UnitTest& unit_test) override;
04047           void OnTestIterationStart(const UnitTest& unit_test, int iteration) override;
04048           void OnEnvironmentsSetUpStart(const UnitTest& unit_test) override;
04049           void OnEnvironmentsSetUpEnd(const UnitTest& unit_test) override;
04050           //  Legacy API is deprecated but still available
04051 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
04052           void OnTestCaseStart(const TestSuite& parameter) override;
04053 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
04054           void OnTestSuiteStart(const TestSuite& parameter) override;
04055           void OnTestStart(const TestInfo& test_info) override;
04056           void OnTestDisabled(const TestInfo& test_info) override;
04057           void OnTestPartResult(const TestPartResult& result) override;
04058           void OnTestEnd(const TestInfo& test_info) override;
04059           //  Legacy API is deprecated but still available
04060 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
04061           void OnTestCaseEnd(const TestCase& parameter) override;
04062 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
04063           void OnTestSuiteEnd(const TestSuite& parameter) override;
04064           void OnEnvironmentsTearDownStart(const UnitTest& unit_test) override;
04065           void OnEnvironmentsTearDownEnd(const UnitTest& unit_test) override;
04066           void OnTestIterationEnd(const UnitTest& unit_test, int iteration) override;
04067           void OnTestProgramEnd(const UnitTest& unit_test) override;
04068
04069       private:
04070           // Controls whether events will be forwarded to listeners_. Set to false
04071           // in death test child processes.
04072           bool forwarding_enabled_;
04073           // The list of listeners that receive events.
04074           std::vector<TestEventListener*> listeners_;
04075
04076           TestEventRepeater(const TestEventRepeater&) = delete;
04077           TestEventRepeater& operator=(const TestEventRepeater&) = delete;
04078       };
04079
```

```
04080         TestEventRepeater::~TestEventRepeater() {
04081             ForEach(listeners_, Delete<TestEventListener>);
04082         }
04083
04084         void TestEventRepeater::Append(TestEventListener* listener) {
04085             listeners_.push_back(listener);
04086         }
04087
04088         TestEventListener* TestEventRepeater::Release(TestEventListener* listener) {
04089             for (size_t i = 0; i < listeners_.size(); ++i) {
04090                 if (listeners_[i] == listener) {
04091                     listeners_.erase(listeners_.begin() + static_cast<int> (i));
04092                     return listener;
04093                 }
04094             }
04095
04096             return nullptr;
04097         }
04098
04099         // Since most methods are very similar, use macros to reduce boilerplate.
04100         // This defines a member that forwards the call to all listeners.
04101 #define GTEST_REPEATER_METHOD_(Name, Type)             \
04102   void TestEventRepeater::Name(const Type& parameter) { \
04103     if (forwarding_enabled_) {                         \
04104       for (size_t i = 0; i < listeners_.size(); i++) {  \
04105         listeners_[i]->Name(parameter);                \
04106       }                                                \
04107     }                                                  \
04108   }
04109         // This defines a member that forwards the call to all listeners in reverse
04110         // order.
04111 #define GTEST_REVERSE_REPEATER_METHOD_(Name, Type)      \
04112   void TestEventRepeater::Name(const Type& parameter) { \
04113     if (forwarding_enabled_) {                         \
04114       for (size_t i = listeners_.size(); i != 0; i--) { \
04115         listeners_[i - 1]->Name(parameter);            \
04116       }                                                \
04117     }                                                  \
04118   }
04119
04120         GTEST_REPEATER_METHOD_(OnTestProgramStart, UnitTest)
04121         GTEST_REPEATER_METHOD_(OnEnvironmentsSetUpStart, UnitTest)
04122         //  Legacy API is deprecated but still available
04123 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
04124         GTEST_REPEATER_METHOD_(OnTestCaseStart, TestSuite)
04125 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
04126         GTEST_REPEATER_METHOD_(OnTestSuiteStart, TestSuite)
04127         GTEST_REPEATER_METHOD_(OnTestStart, TestInfo)
04128         GTEST_REPEATER_METHOD_(OnTestDisabled, TestInfo)
04129         GTEST_REPEATER_METHOD_(OnTestPartResult, TestPartResult)
04130         GTEST_REPEATER_METHOD_(OnEnvironmentsTearDownStart, UnitTest)
04131         GTEST_REVERSE_REPEATER_METHOD_(OnEnvironmentsSetUpEnd, UnitTest)
04132         GTEST_REVERSE_REPEATER_METHOD_(OnEnvironmentsTearDownEnd, UnitTest)
04133         GTEST_REVERSE_REPEATER_METHOD_(OnTestEnd, TestInfo)
04134         //  Legacy API is deprecated but still available
04135 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
04136         GTEST_REVERSE_REPEATER_METHOD_(OnTestCaseEnd, TestSuite)
04137 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
04138         GTEST_REVERSE_REPEATER_METHOD_(OnTestSuiteEnd, TestSuite)
04139         GTEST_REVERSE_REPEATER_METHOD_(OnTestProgramEnd, UnitTest)
04140
04141 #undef GTEST_REPEATER_METHOD_
04142 #undef GTEST_REVERSE_REPEATER_METHOD_
04143
04144         void TestEventRepeater::OnTestIterationStart(const UnitTest& unit_test,
04145                 int iteration) {
04146             if (forwarding_enabled_) {
04147                 for (size_t i = 0; i < listeners_.size(); i++) {
04148                     listeners_[i]->OnTestIterationStart(unit_test, iteration);
04149                 }
04150             }
04151         }
04152
04153         void TestEventRepeater::OnTestIterationEnd(const UnitTest& unit_test,
04154                 int iteration) {
04155             if (forwarding_enabled_) {
04156                 for (size_t i = listeners_.size(); i > 0; i--) {
04157                     listeners_[i - 1]->OnTestIterationEnd(unit_test, iteration);
04158                 }
04159             }
04160         }
04161
04162         // End TestEventRepeater
04163
04164         // This class generates an XML output file.
04165
04166         class XmlUnitTestResultPrinter : public EmptyTestEventListener {
```

```
04167         public:
04168             explicit XmlUnitTestResultPrinter(const char* output_file);
04169
04170             void OnTestIterationEnd(const UnitTest& unit_test, int iteration) override;
04171             void ListTestsMatchingFilter(const std::vector<TestSuite*>& test_suites);
04172
04173             // Prints an XML summary of all unit tests.
04174             static void PrintXmlTestsList(std::ostream* stream,
04175                     const std::vector<TestSuite*>& test_suites);
04176
04177         private:
04178             // Is c a whitespace character that is normalized to a space character
04179             // when it appears in an XML attribute value?
04180
04181             static bool IsNormalizableWhitespace(unsigned char c) {
04182                 return c == '\t' || c == '\n' || c == '\r';
04183             }
04184
04185             // May c appear in a well-formed XML document?
04186             // https://www.w3.org/TR/REC-xml/#charsets
04187
04188             static bool IsValidXmlCharacter(unsigned char c) {
04189                 return IsNormalizableWhitespace(c) || c >= 0x20;
04190             }
04191
04192             // Returns an XML-escaped copy of the input string str.  If
04193             // is_attribute is true, the text is meant to appear as an attribute
04194             // value, and normalizable whitespace is preserved by replacing it
04195             // with character references.
04196             static std::string EscapeXml(const std::string& str, bool is_attribute);
04197
04198             // Returns the given string with all characters invalid in XML removed.
04199             static std::string RemoveInvalidXmlCharacters(const std::string& str);
04200
04201             // Convenience wrapper around EscapeXml when str is an attribute value.
04202
04203             static std::string EscapeXmlAttribute(const std::string& str) {
04204                 return EscapeXml(str, true);
04205             }
04206
04207             // Convenience wrapper around EscapeXml when str is not an attribute value.
04208
04209             static std::string EscapeXmlText(const char* str) {
04210                 return EscapeXml(str, false);
04211             }
04212
04213             // Verifies that the given attribute belongs to the given element and
04214             // streams the attribute as XML.
04215             static void OutputXmlAttribute(std::ostream* stream,
04216                     const std::string& element_name,
04217                     const std::string& name,
04218                     const std::string& value);
04219
04220             // Streams an XML CDATA section, escaping invalid CDATA sequences as needed.
04221             static void OutputXmlCDataSection(::std::ostream* stream, const char* data);
04222
04223             // Streams a test suite XML stanza containing the given test result.
04224             //
04225             // Requires: result.Failed()
04226             static void OutputXmlTestSuiteForTestResult(::std::ostream* stream,
04227                     const TestResult& result);
04228
04229             // Streams an XML representation of a TestResult object.
04230             static void OutputXmlTestResult(::std::ostream* stream,
04231                     const TestResult& result);
04232
04233             // Streams an XML representation of a TestInfo object.
04234             static void OutputXmlTestInfo(::std::ostream* stream,
04235                     const char* test_suite_name,
04236                     const TestInfo& test_info);
04237
04238             // Prints an XML representation of a TestSuite object
04239             static void PrintXmlTestSuite(::std::ostream* stream,
04240                     const TestSuite& test_suite);
04241
04242             // Prints an XML summary of unit_test to output stream out.
04243             static void PrintXmlUnitTest(::std::ostream* stream,
04244                     const UnitTest& unit_test);
04245
04246             // Produces a string representing the test properties in a result as space
04247             // delimited XML attributes based on the property key="value" pairs.
04248             // When the std::string is not empty, it includes a space at the beginning,
04249             // to delimit this attribute from prior attributes.
04250             static std::string TestPropertiesAsXmlAttributes(const TestResult& result);
04251
04252             // Streams an XML representation of the test properties of a TestResult
04253             // object.
```

```
04254             static void OutputXmlTestProperties(std::ostream* stream,
04255                     const TestResult& result);
04256
04257             // The output file.
04258             const std::string output_file_;
04259
04260             XmlUnitTestResultPrinter(const XmlUnitTestResultPrinter&) = delete;
04261             XmlUnitTestResultPrinter& operator=(const XmlUnitTestResultPrinter&) = delete;
04262         };
04263
04264         // Creates a new XmlUnitTestResultPrinter.
04265
04266         XmlUnitTestResultPrinter::XmlUnitTestResultPrinter(const char* output_file)
04267         : output_file_(output_file) {
04268             if (output_file_.empty()) {
04269                 GTEST_LOG_(FATAL) << "XML output file may not be null";
04270             }
04271         }
04272
04273         // Called after the unit test ends.
04274
04275         void XmlUnitTestResultPrinter::OnTestIterationEnd(const UnitTest& unit_test,
04276                 int /*iteration*/) {
04277             FILE* xmlout = OpenFileForWriting(output_file_);
04278             std::stringstream stream;
04279             PrintXmlUnitTest(&stream, unit_test);
04280             fprintf(xmlout, "%s", StringStreamToString(&stream).c_str());
04281             fclose(xmlout);
04282         }
04283
04284         void XmlUnitTestResultPrinter::ListTestsMatchingFilter(
04285                 const std::vector<TestSuite*>& test_suites) {
04286             FILE* xmlout = OpenFileForWriting(output_file_);
04287             std::stringstream stream;
04288             PrintXmlTestsList(&stream, test_suites);
04289             fprintf(xmlout, "%s", StringStreamToString(&stream).c_str());
04290             fclose(xmlout);
04291         }
04292
04293         // Returns an XML-escaped copy of the input string str.  If is_attribute
04294         // is true, the text is meant to appear as an attribute value, and
04295         // normalizable whitespace is preserved by replacing it with character
04296         // references.
04297         //
04298         // Invalid XML characters in str, if any, are stripped from the output.
04299         // It is expected that most, if not all, of the text processed by this
04300         // module will consist of ordinary English text.
04301         // If this module is ever modified to produce version 1.1 XML output,
04302         // most invalid characters can be retained using character references.
04303
04304         std::string XmlUnitTestResultPrinter::EscapeXml(const std::string& str,
04305                 bool is_attribute) {
04306             Message m;
04307
04308             for (size_t i = 0; i < str.size(); ++i) {
04309                 const char ch = str[i];
04310                 switch (ch) {
04311                     case '<':
04312                         m << "&lt;";
04313                         break;
04314                     case '>':
04315                         m << "&gt;";
04316                         break;
04317                     case '&':
04318                         m << "&amp;";
04319                         break;
04320                     case '\'':
04321                         if (is_attribute)
04322                             m << "&apos;";
04323                         else
04324                             m << '\'';
04325                         break;
04326                     case '"':
04327                         if (is_attribute)
04328                             m << "&quot;";
04329                         else
04330                             m << '"';
04331                         break;
04332                     default:
04333                         if (IsValidXmlCharacter(static_cast<unsigned char> (ch))) {
04334                             if (is_attribute &&
04335                                     IsNormalizableWhitespace(static_cast<unsigned char> (ch)))
04336                                 m << "&#x" << String::FormatByte(static_cast<unsigned char> (ch))
04337                                 << ";";
04338                             else
04339                                 m << ch;
04340                         }
```

```
04341                         break;
04342                 }
04343             }
04344
04345             return m.GetString();
04346         }
04347
04348         // Returns the given string with all characters invalid in XML removed.
04349         // Currently invalid characters are dropped from the string. An
04350         // alternative is to replace them with certain characters such as . or ?.
04351
04352         std::string XmlUnitTestResultPrinter::RemoveInvalidXmlCharacters(
04353                 const std::string& str) {
04354             std::string output;
04355             output.reserve(str.size());
04356             for (std::string::const_iterator it = str.begin(); it != str.end(); ++it)
04357                 if (IsValidXmlCharacter(static_cast<unsigned char> (*it)))
04358                     output.push_back(*it);
04359
04360             return output;
04361         }
04362
04363         // The following routines generate an XML representation of a UnitTest
04364         // object.
04365         //
04366         // This is how Google Test concepts map to the DTD:
04367         //
04368         // <testsuites name="AllTests">        <-- corresponds to a UnitTest object
04369         //   <testsuite name="testcase-name">  <-- corresponds to a TestSuite object
04370         //     <testcase name="test-name">     <-- corresponds to a TestInfo object
04371         //       <failure message="...">...</failure>
04372         //       <failure message="...">...</failure>
04373         //       <failure message="...">...</failure>
04374         //                                     <-- individual assertion failures
04375         //     </testcase>
04376         //   </testsuite>
04377         // </testsuites>
04378
04379         // Formats the given time in milliseconds as seconds.
04380
04381         std::string FormatTimeInMillisAsSeconds(TimeInMillis ms) {
04382             ::std::stringstream ss;
04383             ss << (static_cast<double> (ms) * 1e-3);
04384             return ss.str();
04385         }
04386
04387         static bool PortableLocaltime(time_t seconds, struct tm* out) {
04388 #if defined(_MSC_VER)
04389             return localtime_s(out, &seconds) == 0;
04390 #elif defined(__MINGW32__) || defined(__MINGW64__)
04391             // MINGW <time.h> provides neither localtime_r nor localtime_s, but uses
04392             // Windows' localtime(), which has a thread-local tm buffer.
04393             struct tm* tm_ptr = localtime(&seconds); // NOLINT
04394             if (tm_ptr == nullptr) return false;
04395             *out = *tm_ptr;
04396             return true;
04397 #elif defined(__STDC_LIB_EXT1__)
04398             // Uses localtime_s when available as localtime_r is only available from
04399             // C23 standard.
04400             return localtime_s(&seconds, out) != nullptr;
04401 #else
04402             return localtime_r(&seconds, out) != nullptr;
04403 #endif
04404         }
04405
04406         // Converts the given epoch time in milliseconds to a date string in the ISO
04407         // 8601 format, without the timezone information.
04408
04409         std::string FormatEpochTimeInMillisAsIso8601(TimeInMillis ms) {
04410             struct tm time_struct;
04411             if (!PortableLocaltime(static_cast<time_t> (ms / 1000), &time_struct))
04412                 return "";
04413             // YYYY-MM-DDThh:mm:ss.sss
04414             return StreamableToString(time_struct.tm_year + 1900) + "-" +
04415                     String::FormatIntWidth2(time_struct.tm_mon + 1) + "-" +
04416                     String::FormatIntWidth2(time_struct.tm_mday) + "T" +
04417                     String::FormatIntWidth2(time_struct.tm_hour) + ":" +
04418                     String::FormatIntWidth2(time_struct.tm_min) + ":" +
04419                     String::FormatIntWidth2(time_struct.tm_sec) + "." +
04420                     String::FormatIntWidthN(static_cast<int> (ms % 1000), 3);
04421         }
04422
04423         // Streams an XML CDATA section, escaping invalid CDATA sequences as needed.
04424
04425         void XmlUnitTestResultPrinter::OutputXmlCDataSection(::std::ostream* stream,
04426                 const char* data) {
04427             const char* segment = data;
```

```
04428             *stream << "<![CDATA[";
04429             for (;;) {
04430                 const char* const next_segment = strstr(segment, "]]>");
04431                 if (next_segment != nullptr) {
04432                     stream->write(segment,
04433                             static_cast<std::streamsize> (next_segment - segment));
04434                     *stream << "]]>]]&gt;<![CDATA[";
04435                     segment = next_segment + strlen("]]>");
04436                 } else {
04437                     *stream << segment;
04438                     break;
04439                 }
04440             }
04441             *stream << "]]>";
04442         }
04443
04444         void XmlUnitTestResultPrinter::OutputXmlAttribute(
04445                 std::ostream* stream, const std::string& element_name,
04446                 const std::string& name, const std::string& value) {
04447             const std::vector<std::string>& allowed_names =
04448                     GetReservedOutputAttributesForElement(element_name);
04449
04450             GTEST_CHECK_(std::find(allowed_names.begin(), allowed_names.end(), name) !=
04451                     allowed_names.end())
04452                     << "Attribute " << name << " is not allowed for element <" << element_name
04453                     << ">.";
04454
04455             *stream << " " << name << "=\"" << EscapeXmlAttribute(value) << "\"";
04456         }
04457
04458         // Streams a test suite XML stanza containing the given test result.
04459
04460         void XmlUnitTestResultPrinter::OutputXmlTestSuiteForTestResult(
04461                 ::std::ostream* stream, const TestResult& result) {
04462             // Output the boilerplate for a minimal test suite with one test.
04463             *stream << "  <testsuite";
04464             OutputXmlAttribute(stream, "testsuite", "name", "NonTestSuiteFailure");
04465             OutputXmlAttribute(stream, "testsuite", "tests", "1");
04466             OutputXmlAttribute(stream, "testsuite", "failures", "1");
04467             OutputXmlAttribute(stream, "testsuite", "disabled", "0");
04468             OutputXmlAttribute(stream, "testsuite", "skipped", "0");
04469             OutputXmlAttribute(stream, "testsuite", "errors", "0");
04470             OutputXmlAttribute(stream, "testsuite", "time",
04471                     FormatTimeInMillisAsSeconds(result.elapsed_time()));
04472             OutputXmlAttribute(
04473                     stream, "testsuite", "timestamp",
04474                     FormatEpochTimeInMillisAsIso8601(result.start_timestamp()));
04475             *stream << ">";
04476
04477             // Output the boilerplate for a minimal test case with a single test.
04478             *stream << "    <testcase";
04479             OutputXmlAttribute(stream, "testcase", "name", "");
04480             OutputXmlAttribute(stream, "testcase", "status", "run");
04481             OutputXmlAttribute(stream, "testcase", "result", "completed");
04482             OutputXmlAttribute(stream, "testcase", "classname", "");
04483             OutputXmlAttribute(stream, "testcase", "time",
04484                     FormatTimeInMillisAsSeconds(result.elapsed_time()));
04485             OutputXmlAttribute(
04486                     stream, "testcase", "timestamp",
04487                     FormatEpochTimeInMillisAsIso8601(result.start_timestamp()));
04488
04489             // Output the actual test result.
04490             OutputXmlTestResult(stream, result);
04491
04492             // Complete the test suite.
04493             *stream << "  </testsuite>\n";
04494         }
04495
04496         // Prints an XML representation of a TestInfo object.
04497
04498         void XmlUnitTestResultPrinter::OutputXmlTestInfo(::std::ostream* stream,
04499                 const char* test_suite_name,
04500                 const TestInfo& test_info) {
04501             const TestResult& result = *test_info.result();
04502             const std::string kTestsuite = "testcase";
04503
04504             if (test_info.is_in_another_shard()) {
04505                 return;
04506             }
04507
04508             *stream << "    <testcase";
04509             OutputXmlAttribute(stream, kTestsuite, "name", test_info.name());
04510
04511             if (test_info.value_param() != nullptr) {
04512                 OutputXmlAttribute(stream, kTestsuite, "value_param",
04513                         test_info.value_param());
04514             }
```

```
04515             if (test_info.type_param() != nullptr) {
04516                 OutputXmlAttribute(stream, kTestsuite, "type_param",
04517                         test_info.type_param());
04518             }
04519
04520             OutputXmlAttribute(stream, kTestsuite, "file", test_info.file());
04521             OutputXmlAttribute(stream, kTestsuite, "line",
04522                     StreamableToString(test_info.line()));
04523             if (GTEST_FLAG_GET(list_tests)) {
04524                 *stream << " />\n";
04525                 return;
04526             }
04527
04528             OutputXmlAttribute(stream, kTestsuite, "status",
04529                     test_info.should_run() ? "run" : "notrun");
04530             OutputXmlAttribute(stream, kTestsuite, "result",
04531                     test_info.should_run()
04532                     ? (result.Skipped() ? "skipped" : "completed")
04533                     : "suppressed");
04534             OutputXmlAttribute(stream, kTestsuite, "time",
04535                     FormatTimeInMillisAsSeconds(result.elapsed_time()));
04536             OutputXmlAttribute(
04537                     stream, kTestsuite, "timestamp",
04538                     FormatEpochTimeInMillisAsIso8601(result.start_timestamp()));
04539             OutputXmlAttribute(stream, kTestsuite, "classname", test_suite_name);
04540
04541             OutputXmlTestResult(stream, result);
04542         }
04543
04544         void XmlUnitTestResultPrinter::OutputXmlTestResult(::std::ostream* stream,
04545                 const TestResult& result) {
04546             int failures = 0;
04547             int skips = 0;
04548             for (int i = 0; i < result.total_part_count(); ++i) {
04549                 const TestPartResult& part = result.GetTestPartResult(i);
04550                 if (part.failed()) {
04551                     if (++failures == 1 && skips == 0) {
04552                         *stream << ">\n";
04553                     }
04554                     const std::string location =
04555                             internal::FormatCompilerIndependentFileLocation(part.file_name(),
04556                             part.line_number());
04557                     const std::string summary = location + "\n" + part.summary();
04558                     *stream << "      <failure message=\"" << EscapeXmlAttribute(summary)
04559                             << "\" type=\"\">";
04560                     const std::string detail = location + "\n" + part.message();
04561                     OutputXmlCDataSection(stream, RemoveInvalidXmlCharacters(detail).c_str());
04562                     *stream << "</failure>\n";
04563                 } else if (part.skipped()) {
04564                     if (++skips == 1 && failures == 0) {
04565                         *stream << ">\n";
04566                     }
04567                     const std::string location =
04568                             internal::FormatCompilerIndependentFileLocation(part.file_name(),
04569                             part.line_number());
04570                     const std::string summary = location + "\n" + part.summary();
04571                     *stream << "      <skipped message=\""
04572                             << EscapeXmlAttribute(summary.c_str()) << "\">";
04573                     const std::string detail = location + "\n" + part.message();
04574                     OutputXmlCDataSection(stream, RemoveInvalidXmlCharacters(detail).c_str());
04575                     *stream << "</skipped>\n";
04576                 }
04577             }
04578
04579             if (failures == 0 && skips == 0 && result.test_property_count() == 0) {
04580                 *stream << " />\n";
04581             } else {
04582                 if (failures == 0 && skips == 0) {
04583                     *stream << ">\n";
04584                 }
04585                 OutputXmlTestProperties(stream, result);
04586                 *stream << "    </testcase>\n";
04587             }
04588         }
04589
04590         // Prints an XML representation of a TestSuite object
04591
04592         void XmlUnitTestResultPrinter::PrintXmlTestSuite(std::ostream* stream,
04593                 const TestSuite& test_suite) {
04594             const std::string kTestsuite = "testsuite";
04595             *stream << "  <" << kTestsuite;
04596             OutputXmlAttribute(stream, kTestsuite, "name", test_suite.name());
04597             OutputXmlAttribute(stream, kTestsuite, "tests",
04598                     StreamableToString(test_suite.reportable_test_count()));
04599             if (!GTEST_FLAG_GET(list_tests)) {
04600                 OutputXmlAttribute(stream, kTestsuite, "failures",
04601                         StreamableToString(test_suite.failed_test_count()));
```

```
04602                 OutputXmlAttribute(
04603                         stream, kTestsuite, "disabled",
04604                         StreamableToString(test_suite.reportable_disabled_test_count()));
04605                 OutputXmlAttribute(stream, kTestsuite, "skipped",
04606                         StreamableToString(test_suite.skipped_test_count()));
04607
04608                 OutputXmlAttribute(stream, kTestsuite, "errors", "0");
04609
04610                 OutputXmlAttribute(stream, kTestsuite, "time",
04611                         FormatTimeInMillisAsSeconds(test_suite.elapsed_time()));
04612                 OutputXmlAttribute(
04613                         stream, kTestsuite, "timestamp",
04614                         FormatEpochTimeInMillisAsIso8601(test_suite.start_timestamp()));
04615                 *stream << TestPropertiesAsXmlAttributes(test_suite.ad_hoc_test_result());
04616             }
04617             *stream << ">\n";
04618             for (int i = 0; i < test_suite.total_test_count(); ++i) {
04619                 if (test_suite.GetTestInfo(i)->is_reportable())
04620                     OutputXmlTestInfo(stream, test_suite.name(), *test_suite.GetTestInfo(i));
04621             }
04622             *stream << "  </" << kTestsuite << ">\n";
04623         }
04624
04625         // Prints an XML summary of unit_test to output stream out.
04626
04627         void XmlUnitTestResultPrinter::PrintXmlUnitTest(std::ostream* stream,
04628                 const UnitTest& unit_test) {
04629             const std::string kTestsuites = "testsuites";
04630
04631             *stream << "<?xml version=\"1.0\" encoding=\"UTF-8\"?>\n";
04632             *stream << "<" << kTestsuites;
04633
04634             OutputXmlAttribute(stream, kTestsuites, "tests",
04635                     StreamableToString(unit_test.reportable_test_count()));
04636             OutputXmlAttribute(stream, kTestsuites, "failures",
04637                     StreamableToString(unit_test.failed_test_count()));
04638             OutputXmlAttribute(
04639                     stream, kTestsuites, "disabled",
04640                     StreamableToString(unit_test.reportable_disabled_test_count()));
04641             OutputXmlAttribute(stream, kTestsuites, "errors", "0");
04642             OutputXmlAttribute(stream, kTestsuites, "time",
04643                     FormatTimeInMillisAsSeconds(unit_test.elapsed_time()));
04644             OutputXmlAttribute(
04645                     stream, kTestsuites, "timestamp",
04646                     FormatEpochTimeInMillisAsIso8601(unit_test.start_timestamp()));
04647
04648             if (GTEST_FLAG_GET(shuffle)) {
04649                 OutputXmlAttribute(stream, kTestsuites, "random_seed",
04650                         StreamableToString(unit_test.random_seed()));
04651             }
04652             *stream << TestPropertiesAsXmlAttributes(unit_test.ad_hoc_test_result());
04653
04654             OutputXmlAttribute(stream, kTestsuites, "name", "AllTests");
04655             *stream << ">\n";
04656
04657             for (int i = 0; i < unit_test.total_test_suite_count(); ++i) {
04658                 if (unit_test.GetTestSuite(i)->reportable_test_count() > 0)
04659                     PrintXmlTestSuite(stream, *unit_test.GetTestSuite(i));
04660             }
04661
04662             // If there was a test failure outside of one of the test suites (like in a
04663             // test environment) include that in the output.
04664             if (unit_test.ad_hoc_test_result().Failed()) {
04665                 OutputXmlTestSuiteForTestResult(stream, unit_test.ad_hoc_test_result());
04666             }
04667
04668             *stream << "</" << kTestsuites << ">\n";
04669         }
04670
04671         void XmlUnitTestResultPrinter::PrintXmlTestsList(
04672                 std::ostream* stream, const std::vector<TestSuite*>& test_suites) {
04673             const std::string kTestsuites = "testsuites";
04674
04675             *stream << "<?xml version=\"1.0\" encoding=\"UTF-8\"?>\n";
04676             *stream << "<" << kTestsuites;
04677
04678             int total_tests = 0;
04679             for (auto test_suite : test_suites) {
04680                 total_tests += test_suite->total_test_count();
04681             }
04682             OutputXmlAttribute(stream, kTestsuites, "tests",
04683                     StreamableToString(total_tests));
04684             OutputXmlAttribute(stream, kTestsuites, "name", "AllTests");
04685             *stream << ">\n";
04686
04687             for (auto test_suite : test_suites) {
04688                 PrintXmlTestSuite(stream, *test_suite);
```

```
04689             }
04690             *stream << "</" << kTestsuites << ">\n";
04691         }
04692
04693         // Produces a string representing the test properties in a result as space
04694         // delimited XML attributes based on the property key="value" pairs.
04695
04696         std::string XmlUnitTestResultPrinter::TestPropertiesAsXmlAttributes(
04697                 const TestResult& result) {
04698             Message attributes;
04699             for (int i = 0; i < result.test_property_count(); ++i) {
04700                 const TestProperty& property = result.GetTestProperty(i);
04701                 attributes << " " << property.key() << "="
04702                         << "\"" << EscapeXmlAttribute(property.value()) << "\"";
04703             }
04704             return attributes.GetString();
04705         }
04706
04707         void XmlUnitTestResultPrinter::OutputXmlTestProperties(
04708                 std::ostream* stream, const TestResult& result) {
04709             const std::string kProperties = "properties";
04710             const std::string kProperty = "property";
04711
04712             if (result.test_property_count() <= 0) {
04713                 return;
04714             }
04715
04716             *stream << "      <" << kProperties << ">\n";
04717             for (int i = 0; i < result.test_property_count(); ++i) {
04718                 const TestProperty& property = result.GetTestProperty(i);
04719                 *stream << "        <" << kProperty;
04720                 *stream << " name=\"" << EscapeXmlAttribute(property.key()) << "\"";
04721                 *stream << " value=\"" << EscapeXmlAttribute(property.value()) << "\"";
04722                 *stream << "/>\n";
04723             }
04724             *stream << "      </" << kProperties << ">\n";
04725         }
04726
04727         // End XmlUnitTestResultPrinter
04728
04729         // This class generates an JSON output file.
04730
04731         class JsonUnitTestResultPrinter : public EmptyTestEventListener {
04732         public:
04733             explicit JsonUnitTestResultPrinter(const char* output_file);
04734
04735             void OnTestIterationEnd(const UnitTest& unit_test, int iteration) override;
04736
04737             // Prints an JSON summary of all unit tests.
04738             static void PrintJsonTestList(::std::ostream* stream,
04739                     const std::vector<TestSuite*>& test_suites);
04740
04741         private:
04742             // Returns an JSON-escaped copy of the input string str.
04743             static std::string EscapeJson(const std::string& str);
04744
04747             static void OutputJsonKey(std::ostream* stream,
04748                     const std::string& element_name,
04749                     const std::string& name, const std::string& value,
04750                     const std::string& indent, bool comma = true);
04751             static void OutputJsonKey(std::ostream* stream,
04752                     const std::string& element_name,
04753                     const std::string& name, int value,
04754                     const std::string& indent, bool comma = true);
04755
04756             // Streams a test suite JSON stanza containing the given test result.
04757             //
04758             // Requires: result.Failed()
04759             static void OutputJsonTestSuiteForTestResult(::std::ostream* stream,
04760                     const TestResult& result);
04761
04762             // Streams a JSON representation of a TestResult object.
04763             static void OutputJsonTestResult(::std::ostream* stream,
04764                     const TestResult& result);
04765
04766             // Streams a JSON representation of a TestInfo object.
04767             static void OutputJsonTestInfo(::std::ostream* stream,
04768                     const char* test_suite_name,
04769                     const TestInfo& test_info);
04770
04771             // Prints a JSON representation of a TestSuite object
04772             static void PrintJsonTestSuite(::std::ostream* stream,
04773                     const TestSuite& test_suite);
04774
04775             // Prints a JSON summary of unit_test to output stream out.
04776             static void PrintJsonUnitTest(::std::ostream* stream,
04777                     const UnitTest& unit_test);
```

```
04778
04779           // Produces a string representing the test properties in a result as
04780           // a JSON dictionary.
04781           static std::string TestPropertiesAsJson(const TestResult& result,
04782                   const std::string& indent);
04783
04784           // The output file.
04785           const std::string output_file_;
04786
04787           JsonUnitTestResultPrinter(const JsonUnitTestResultPrinter&) = delete;
04788           JsonUnitTestResultPrinter& operator=(const JsonUnitTestResultPrinter&) =
04789                   delete;
04790       };
04791
04792       // Creates a new JsonUnitTestResultPrinter.
04793
04794       JsonUnitTestResultPrinter::JsonUnitTestResultPrinter(const char* output_file)
04795       : output_file_(output_file) {
04796           if (output_file_.empty()) {
04797               GTEST_LOG_(FATAL) << "JSON output file may not be null";
04798           }
04799       }
04800
04801       void JsonUnitTestResultPrinter::OnTestIterationEnd(const UnitTest& unit_test,
04802               int /*iteration*/) {
04803           FILE* jsonout = OpenFileForWriting(output_file_);
04804           std::stringstream stream;
04805           PrintJsonUnitTest(&stream, unit_test);
04806           fprintf(jsonout, "%s", StringStreamToString(&stream).c_str());
04807           fclose(jsonout);
04808       }
04809
04810       // Returns an JSON-escaped copy of the input string str.
04811
04812       std::string JsonUnitTestResultPrinter::EscapeJson(const std::string& str) {
04813           Message m;
04814
04815           for (size_t i = 0; i < str.size(); ++i) {
04816               const char ch = str[i];
04817               switch (ch) {
04818                   case '\\':
04819                   case '"':
04820                   case '/':
04821                       m << '\\' << ch;
04822                       break;
04823                   case '\b':
04824                       m << "\\b";
04825                       break;
04826                   case '\t':
04827                       m << "\\t";
04828                       break;
04829                   case '\n':
04830                       m << "\\n";
04831                       break;
04832                   case '\f':
04833                       m << "\\f";
04834                       break;
04835                   case '\r':
04836                       m << "\\r";
04837                       break;
04838                   default:
04839                       if (ch < ' ') {
04840                           m << "\\u00" << String::FormatByte(static_cast<unsigned char> (ch));
04841                       } else {
04842                           m << ch;
04843                       }
04844                       break;
04845               }
04846           }
04847
04848           return m.GetString();
04849       }
04850
04851       // The following routines generate an JSON representation of a UnitTest
04852       // object.
04853
04854       // Formats the given time in milliseconds as seconds.
04855
04856       static std::string FormatTimeInMillisAsDuration(TimeInMillis ms) {
04857           ::std::stringstream ss;
04858           ss << (static_cast<double> (ms) * 1e-3) << "s";
04859           return ss.str();
04860       }
04861
04862       // Converts the given epoch time in milliseconds to a date string in the
04863       // RFC3339 format, without the timezone information.
04864
```

```
04865         static std::string FormatEpochTimeInMillisAsRFC3339(TimeInMillis ms) {
04866             struct tm time_struct;
04867             if (!PortableLocaltime(static_cast<time_t> (ms / 1000), &time_struct))
04868                 return "";
04869             // YYYY-MM-DDThh:mm:ss
04870             return StreamableToString(time_struct.tm_year + 1900) + "-" +
04871                     String::FormatIntWidth2(time_struct.tm_mon + 1) + "-" +
04872                     String::FormatIntWidth2(time_struct.tm_mday) + "T" +
04873                     String::FormatIntWidth2(time_struct.tm_hour) + ":" +
04874                     String::FormatIntWidth2(time_struct.tm_min) + ":" +
04875                     String::FormatIntWidth2(time_struct.tm_sec) + "Z";
04876         }
04877
04878         static inline std::string Indent(size_t width) {
04879             return std::string(width, ' ');
04880         }
04881
04882         void JsonUnitTestResultPrinter::OutputJsonKey(std::ostream* stream,
04883                 const std::string& element_name,
04884                 const std::string& name,
04885                 const std::string& value,
04886                 const std::string& indent,
04887                 bool comma) {
04888             const std::vector<std::string>& allowed_names =
04889                     GetReservedOutputAttributesForElement(element_name);
04890
04891             GTEST_CHECK_(std::find(allowed_names.begin(), allowed_names.end(), name) !=
04892                     allowed_names.end())
04893                     << "Key \"" << name << "\" is not allowed for value \"" << element_name
04894                     << "\".";
04895
04896             *stream << indent << "\"" << name << "\": \"" << EscapeJson(value) << "\"";
04897             if (comma) *stream << ",\n";
04898         }
04899
04900         void JsonUnitTestResultPrinter::OutputJsonKey(
04901                 std::ostream* stream, const std::string& element_name,
04902                 const std::string& name, int value, const std::string& indent, bool comma) {
04903             const std::vector<std::string>& allowed_names =
04904                     GetReservedOutputAttributesForElement(element_name);
04905
04906             GTEST_CHECK_(std::find(allowed_names.begin(), allowed_names.end(), name) !=
04907                     allowed_names.end())
04908                     << "Key \"" << name << "\" is not allowed for value \"" << element_name
04909                     << "\".";
04910
04911             *stream << indent << "\"" << name << "\": " << StreamableToString(value);
04912             if (comma) *stream << ",\n";
04913         }
04914
04915         // Streams a test suite JSON stanza containing the given test result.
04916
04917         void JsonUnitTestResultPrinter::OutputJsonTestSuiteForTestResult(
04918                 ::std::ostream* stream, const TestResult& result) {
04919             // Output the boilerplate for a new test suite.
04920             *stream << Indent(4) << "{\n";
04921             OutputJsonKey(stream, "testsuite", "name", "NonTestSuiteFailure", Indent(6));
04922             OutputJsonKey(stream, "testsuite", "tests", 1, Indent(6));
04923             if (!GTEST_FLAG_GET(list_tests)) {
04924                 OutputJsonKey(stream, "testsuite", "failures", 1, Indent(6));
04925                 OutputJsonKey(stream, "testsuite", "disabled", 0, Indent(6));
04926                 OutputJsonKey(stream, "testsuite", "skipped", 0, Indent(6));
04927                 OutputJsonKey(stream, "testsuite", "errors", 0, Indent(6));
04928                 OutputJsonKey(stream, "testsuite", "time",
04929                         FormatTimeInMillisAsDuration(result.elapsed_time()),
04930                         Indent(6));
04931                 OutputJsonKey(stream, "testsuite", "timestamp",
04932                         FormatEpochTimeInMillisAsRFC3339(result.start_timestamp()),
04933                         Indent(6));
04934             }
04935             *stream << Indent(6) << "\"testsuite\": [\n";
04936
04937             // Output the boilerplate for a new test case.
04938             *stream << Indent(8) << "{\n";
04939             OutputJsonKey(stream, "testcase", "name", "", Indent(10));
04940             OutputJsonKey(stream, "testcase", "status", "RUN", Indent(10));
04941             OutputJsonKey(stream, "testcase", "result", "COMPLETED", Indent(10));
04942             OutputJsonKey(stream, "testcase", "timestamp",
04943                     FormatEpochTimeInMillisAsRFC3339(result.start_timestamp()),
04944                     Indent(10));
04945             OutputJsonKey(stream, "testcase", "time",
04946                     FormatTimeInMillisAsDuration(result.elapsed_time()),
04947                     Indent(10));
04948             OutputJsonKey(stream, "testcase", "classname", "", Indent(10), false);
04949             *stream << TestPropertiesAsJson(result, Indent(10));
04950
04951             // Output the actual test result.
```

```
04952             OutputJsonTestResult(stream, result);
04953
04954             // Finish the test suite.
04955             *stream << "\n" << Indent(6) << "]\n" << Indent(4) << "}";
04956         }
04957
04958         // Prints a JSON representation of a TestInfo object.
04959
04960         void JsonUnitTestResultPrinter::OutputJsonTestInfo(::std::ostream* stream,
04961                 const char* test_suite_name,
04962                 const TestInfo& test_info) {
04963             const TestResult& result = *test_info.result();
04964             const std::string kTestsuite = "testcase";
04965             const std::string kIndent = Indent(10);
04966
04967             *stream << Indent(8) << "{\n";
04968             OutputJsonKey(stream, kTestsuite, "name", test_info.name(), kIndent);
04969
04970             if (test_info.value_param() != nullptr) {
04971                 OutputJsonKey(stream, kTestsuite, "value_param", test_info.value_param(),
04972                         kIndent);
04973             }
04974             if (test_info.type_param() != nullptr) {
04975                 OutputJsonKey(stream, kTestsuite, "type_param", test_info.type_param(),
04976                         kIndent);
04977             }
04978
04979             OutputJsonKey(stream, kTestsuite, "file", test_info.file(), kIndent);
04980             OutputJsonKey(stream, kTestsuite, "line", test_info.line(), kIndent, false);
04981             if (GTEST_FLAG_GET(list_tests)) {
04982                 *stream << "\n" << Indent(8) << "}";
04983                 return;
04984             } else {
04985                 *stream << ",\n";
04986             }
04987
04988             OutputJsonKey(stream, kTestsuite, "status",
04989                     test_info.should_run() ? "RUN" : "NOTRUN", kIndent);
04990             OutputJsonKey(stream, kTestsuite, "result",
04991                     test_info.should_run()
04992                     ? (result.Skipped() ? "SKIPPED" : "COMPLETED")
04993                     : "SUPPRESSED",
04994                     kIndent);
04995             OutputJsonKey(stream, kTestsuite, "timestamp",
04996                     FormatEpochTimeInMillisAsRFC3339(result.start_timestamp()),
04997                     kIndent);
04998             OutputJsonKey(stream, kTestsuite, "time",
04999                     FormatTimeInMillisAsDuration(result.elapsed_time()), kIndent);
05000             OutputJsonKey(stream, kTestsuite, "classname", test_suite_name, kIndent,
05001                     false);
05002             *stream << TestPropertiesAsJson(result, kIndent);
05003
05004             OutputJsonTestResult(stream, result);
05005         }
05006
05007         void JsonUnitTestResultPrinter::OutputJsonTestResult(::std::ostream* stream,
05008                 const TestResult& result) {
05009             const std::string kIndent = Indent(10);
05010
05011             int failures = 0;
05012             for (int i = 0; i < result.total_part_count(); ++i) {
05013                 const TestPartResult& part = result.GetTestPartResult(i);
05014                 if (part.failed()) {
05015                     *stream << ",\n";
05016                     if (++failures == 1) {
05017                         *stream << kIndent << "\""
05018                                 << "failures"
05019                                 << "\": [\n";
05020                     }
05021                     const std::string location =
05022                             internal::FormatCompilerIndependentFileLocation(part.file_name(),
05023                             part.line_number());
05024                     const std::string message = EscapeJson(location + "\n" + part.message());
05025                     *stream << kIndent << "  {\n"
05026                             << kIndent << "    \"failure\": \"" << message << "\",\n"
05027                             << kIndent << "    \"type\": \"\"\n"
05028                             << kIndent << "  }";
05029                 }
05030             }
05031
05032             if (failures > 0) *stream << "\n" << kIndent << "]";
05033             *stream << "\n" << Indent(8) << "}";
05034         }
05035
05036         // Prints an JSON representation of a TestSuite object
05037
05038         void JsonUnitTestResultPrinter::PrintJsonTestSuite(
```

```
05039               std::ostream* stream, const TestSuite& test_suite) {
05040           const std::string kTestsuite = "testsuite";
05041           const std::string kIndent = Indent(6);
05042
05043           *stream << Indent(4) << "{\n";
05044           OutputJsonKey(stream, kTestsuite, "name", test_suite.name(), kIndent);
05045           OutputJsonKey(stream, kTestsuite, "tests", test_suite.reportable_test_count(),
05046                   kIndent);
05047           if (!GTEST_FLAG_GET(list_tests)) {
05048               OutputJsonKey(stream, kTestsuite, "failures",
05049                       test_suite.failed_test_count(), kIndent);
05050               OutputJsonKey(stream, kTestsuite, "disabled",
05051                       test_suite.reportable_disabled_test_count(), kIndent);
05052               OutputJsonKey(stream, kTestsuite, "errors", 0, kIndent);
05053               OutputJsonKey(
05054                       stream, kTestsuite, "timestamp",
05055                       FormatEpochTimeInMillisAsRFC3339(test_suite.start_timestamp()),
05056                       kIndent);
05057               OutputJsonKey(stream, kTestsuite, "time",
05058                       FormatTimeInMillisAsDuration(test_suite.elapsed_time()),
05059                       kIndent, false);
05060               *stream << TestPropertiesAsJson(test_suite.ad_hoc_test_result(), kIndent)
05061                       << ",\n";
05062           }
05063
05064           *stream << kIndent << "\"" << kTestsuite << "\": [\n";
05065
05066           bool comma = false;
05067           for (int i = 0; i < test_suite.total_test_count(); ++i) {
05068               if (test_suite.GetTestInfo(i)->is_reportable()) {
05069                   if (comma) {
05070                       *stream << ",\n";
05071                   } else {
05072                       comma = true;
05073                   }
05074                   OutputJsonTestInfo(stream, test_suite.name(), *test_suite.GetTestInfo(i));
05075               }
05076           }
05077           *stream << "\n" << kIndent << "]\n" << Indent(4) << "}";
05078       }
05079
05080       // Prints a JSON summary of unit_test to output stream out.
05081
05082       void JsonUnitTestResultPrinter::PrintJsonUnitTest(std::ostream* stream,
05083               const UnitTest& unit_test) {
05084           const std::string kTestsuites = "testsuites";
05085           const std::string kIndent = Indent(2);
05086           *stream << "{\n";
05087
05088           OutputJsonKey(stream, kTestsuites, "tests", unit_test.reportable_test_count(),
05089                   kIndent);
05090           OutputJsonKey(stream, kTestsuites, "failures", unit_test.failed_test_count(),
05091                   kIndent);
05092           OutputJsonKey(stream, kTestsuites, "disabled",
05093                   unit_test.reportable_disabled_test_count(), kIndent);
05094           OutputJsonKey(stream, kTestsuites, "errors", 0, kIndent);
05095           if (GTEST_FLAG_GET(shuffle)) {
05096               OutputJsonKey(stream, kTestsuites, "random_seed", unit_test.random_seed(),
05097                       kIndent);
05098           }
05099           OutputJsonKey(stream, kTestsuites, "timestamp",
05100                   FormatEpochTimeInMillisAsRFC3339(unit_test.start_timestamp()),
05101                   kIndent);
05102           OutputJsonKey(stream, kTestsuites, "time",
05103                   FormatTimeInMillisAsDuration(unit_test.elapsed_time()), kIndent,
05104                   false);
05105
05106           *stream << TestPropertiesAsJson(unit_test.ad_hoc_test_result(), kIndent)
05107                   << ",\n";
05108
05109           OutputJsonKey(stream, kTestsuites, "name", "AllTests", kIndent);
05110           *stream << kIndent << "\"" << kTestsuites << "\": [\n";
05111
05112           bool comma = false;
05113           for (int i = 0; i < unit_test.total_test_suite_count(); ++i) {
05114               if (unit_test.GetTestSuite(i)->reportable_test_count() > 0) {
05115                   if (comma) {
05116                       *stream << ",\n";
05117                   } else {
05118                       comma = true;
05119                   }
05120                   PrintJsonTestSuite(stream, *unit_test.GetTestSuite(i));
05121               }
05122           }
05123
05124           // If there was a test failure outside of one of the test suites (like in a
05125           // test environment) include that in the output.
```

```
05126             if (unit_test.ad_hoc_test_result().Failed()) {
05127                 OutputJsonTestSuiteForTestResult(stream, unit_test.ad_hoc_test_result());
05128             }
05129
05130             *stream << "\n"
05131                     << kIndent << "]\n"
05132                     << "}\n";
05133         }
05134
05135         void JsonUnitTestResultPrinter::PrintJsonTestList(
05136                 std::ostream* stream, const std::vector<TestSuite*>& test_suites) {
05137             const std::string kTestsuites = "testsuites";
05138             const std::string kIndent = Indent(2);
05139             *stream << "{\n";
05140             int total_tests = 0;
05141             for (auto test_suite : test_suites) {
05142                 total_tests += test_suite->total_test_count();
05143             }
05144             OutputJsonKey(stream, kTestsuites, "tests", total_tests, kIndent);
05145
05146             OutputJsonKey(stream, kTestsuites, "name", "AllTests", kIndent);
05147             *stream << kIndent << "\"" << kTestsuites << "\": [\n";
05148
05149             for (size_t i = 0; i < test_suites.size(); ++i) {
05150                 if (i != 0) {
05151                     *stream << ",\n";
05152                 }
05153                 PrintJsonTestSuite(stream, *test_suites[i]);
05154             }
05155
05156             *stream << "\n"
05157                     << kIndent << "]\n"
05158                     << "}\n";
05159         }
05160         // Produces a string representing the test properties in a result as
05161         // a JSON dictionary.
05162
05163         std::string JsonUnitTestResultPrinter::TestPropertiesAsJson(
05164                 const TestResult& result, const std::string& indent) {
05165             Message attributes;
05166             for (int i = 0; i < result.test_property_count(); ++i) {
05167                 const TestProperty& property = result.GetTestProperty(i);
05168                 attributes << ",\n"
05169                         << indent << "\"" << property.key() << "\": "
05170                         << "\"" << EscapeJson(property.value()) << "\"";
05171             }
05172             return attributes.GetString();
05173         }
05174
05175         // End JsonUnitTestResultPrinter
05176
05177 #if GTEST_CAN_STREAM_RESULTS_
05178
05179         // Checks if str contains '=', '&', '%' or '\n' characters. If yes,
05180         // replaces them by "%xx" where xx is their hexadecimal value. For
05181         // example, replaces "=" with "%3D".  This algorithm is O(strlen(str))
05182         // in both time and space -- important as the input str may contain an
05183         // arbitrarily long test failure message and stack trace.
05184
05185         std::string StreamingListener::UrlEncode(const char* str) {
05186             std::string result;
05187             result.reserve(strlen(str) + 1);
05188             for (char ch = *str; ch != '\0'; ch = *++str) {
05189                 switch (ch) {
05190                     case '%':
05191                     case '=':
05192                     case '&':
05193                     case '\n':
05194                         result.append("%" + String::FormatByte(static_cast<unsigned char> (ch)));
05195                         break;
05196                     default:
05197                         result.push_back(ch);
05198                         break;
05199                 }
05200             }
05201             return result;
05202         }
05203
05204         void StreamingListener::SocketWriter::MakeConnection() {
05205             GTEST_CHECK_(sockfd_ == -1)
05206                     << "MakeConnection() can't be called when there is already a connection.";
05207
05208             addrinfo hints;
05209             memset(&hints, 0, sizeof (hints));
05210             hints.ai_family = AF_UNSPEC; // To allow both IPv4 and IPv6 addresses.
05211             hints.ai_socktype = SOCK_STREAM;
05212             addrinfo* servinfo = nullptr;
```

```
05213
05214             // Use the getaddrinfo() to get a linked list of IP addresses for
05215             // the given host name.
05216             const int error_num =
05217                     getaddrinfo(host_name_.c_str(), port_num_.c_str(), &hints, &servinfo);
05218             if (error_num != 0) {
05219                 GTEST_LOG_(WARNING) << "stream_result_to: getaddrinfo() failed: "
05220                         << gai_strerror(error_num);
05221             }
05222
05223             // Loop through all the results and connect to the first we can.
05224             for (addrinfo* cur_addr = servinfo; sockfd_ == -1 && cur_addr != nullptr;
05225                     cur_addr = cur_addr->ai_next) {
05226                 sockfd_ = socket(cur_addr->ai_family, cur_addr->ai_socktype,
05227                         cur_addr->ai_protocol);
05228                 if (sockfd_ != -1) {
05229                     // Connect the client socket to the server socket.
05230                     if (connect(sockfd_, cur_addr->ai_addr, cur_addr->ai_addrlen) == -1) {
05231                         close(sockfd_);
05232                         sockfd_ = -1;
05233                     }
05234                 }
05235             }
05236
05237             freeaddrinfo(servinfo); // all done with this structure
05238
05239             if (sockfd_ == -1) {
05240                 GTEST_LOG_(WARNING) << "stream_result_to: failed to connect to "
05241                         << host_name_ << ":" << port_num_;
05242             }
05243         }
05244
05245         // End of class Streaming Listener
05246 #endif  // GTEST_CAN_STREAM_RESULTS__
05247
05248         // class OsStackTraceGetter
05249
05250         const char* const OsStackTraceGetterInterface::kElidedFramesMarker =
05251                 "... " GTEST_NAME_ " internal frames ...";
05252
05253         std::string OsStackTraceGetter::CurrentStackTrace(int max_depth, int skip_count)
05254         GTEST_LOCK_EXCLUDED_(mutex_) {
05255 #if GTEST_HAS_ABSL
05256             std::string result;
05257
05258             if (max_depth <= 0) {
05259                 return result;
05260             }
05261
05262             max_depth = std::min(max_depth, kMaxStackTraceDepth);
05263
05264             std::vector<void*> raw_stack(max_depth);
05265             // Skips the frames requested by the caller, plus this function.
05266             const int raw_stack_size =
05267                     absl::GetStackTrace(&raw_stack[0], max_depth, skip_count + 1);
05268
05269             void* caller_frame = nullptr;
05270             {
05271                 MutexLock lock(&mutex_);
05272                 caller_frame = caller_frame_;
05273             }
05274
05275             for (int i = 0; i < raw_stack_size; ++i) {
05276                 if (raw_stack[i] == caller_frame &&
05277                         !GTEST_FLAG_GET(show_internal_stack_frames)) {
05278                     // Add a marker to the trace and stop adding frames.
05279                     absl::StrAppend(&result, kElidedFramesMarker, "\n");
05280                     break;
05281                 }
05282
05283                 char tmp[1024];
05284                 const char* symbol = "(unknown)";
05285                 if (absl::Symbolize(raw_stack[i], tmp, sizeof (tmp))) {
05286                     symbol = tmp;
05287                 }
05288
05289                 char line[1024];
05290                 snprintf(line, sizeof (line), "  %p: %s\n", raw_stack[i], symbol);
05291                 result += line;
05292             }
05293
05294             return result;
05295
05296 #else   // !GTEST_HAS_ABSL
05297             static_cast<void> (max_depth);
05298             static_cast<void> (skip_count);
05299             return "";
```

```
05300 #endif  // GTEST_HAS_ABSL
05301         }
05302
05303         void OsStackTraceGetter::UponLeavingGTest() GTEST_LOCK_EXCLUDED_(mutex_) {
05304 #if GTEST_HAS_ABSL
05305             void* caller_frame = nullptr;
05306             if (absl::GetStackTrace(&caller_frame, 1, 3) <= 0) {
05307                 caller_frame = nullptr;
05308             }
05309
05310             MutexLock lock(&mutex_);
05311             caller_frame_ = caller_frame;
05312 #endif  // GTEST_HAS_ABSL
05313         }
05314
05315         // A helper class that creates the premature-exit file in its
05316         // constructor and deletes the file in its destructor.
05317
05318         class ScopedPrematureExitFile {
05319         public:
05320
05321             explicit ScopedPrematureExitFile(const char* premature_exit_filepath)
05322             : premature_exit_filepath_(
05323             premature_exit_filepath ? premature_exit_filepath : "") {
05324                 // If a path to the premature-exit file is specified...
05325                 if (!premature_exit_filepath_.empty()) {
05326                     // create the file with a single "0" character in it.  I/O
05327                     // errors are ignored as there's nothing better we can do and we
05328                     // don't want to fail the test because of this.
05329                     FILE* pfile = posix::FOpen(premature_exit_filepath_.c_str(), "w");
05330                     fwrite("0", 1, 1, pfile);
05331                     fclose(pfile);
05332                 }
05333             }
05334
05335             ~ScopedPrematureExitFile() {
05336 #if !defined GTEST_OS_ESP8266
05337                 if (!premature_exit_filepath_.empty()) {
05338                     int retval = remove(premature_exit_filepath_.c_str());
05339                     if (retval) {
05340                         GTEST_LOG_(ERROR) << "Failed to remove premature exit filepath \""
05341                                           << premature_exit_filepath_ << "\" with error "
05342                                           << retval;
05343                     }
05344                 }
05345 #endif
05346             }
05347
05348         private:
05349             const std::string premature_exit_filepath_;
05350
05351             ScopedPrematureExitFile(const ScopedPrematureExitFile&) = delete;
05352             ScopedPrematureExitFile& operator=(const ScopedPrematureExitFile&) = delete;
05353         };
05354
05355     } // namespace internal
05356
05357     // class TestEventListeners
05358
05359     TestEventListeners::TestEventListeners()
05360     : repeater_(new internal::TestEventRepeater()),
05361     default_result_printer_(nullptr),
05362     default_xml_generator_(nullptr) {
05363     }
05364
05365     TestEventListeners::~TestEventListeners() {
05366         delete repeater_;
05367     }
05368
05369     // Returns the standard listener responsible for the default console
05370     // output.  Can be removed from the listeners list to shut down default
05371     // console output.  Note that removing this object from the listener list
05372     // with Release transfers its ownership to the user.
05373
05374     void TestEventListeners::Append(TestEventListener* listener) {
05375         repeater_->Append(listener);
05376     }
05377
05378     // Removes the given event listener from the list and returns it.  It then
05379     // becomes the caller's responsibility to delete the listener. Returns
05380     // NULL if the listener is not found in the list.
05381
05382     TestEventListener* TestEventListeners::Release(TestEventListener* listener) {
05383         if (listener == default_result_printer_)
05384             default_result_printer_ = nullptr;
05385         else if (listener == default_xml_generator_)
05386             default_xml_generator_ = nullptr;
```

```
05387         return repeater_->Release(listener);
05388     }
05389
05390     // Returns repeater that broadcasts the TestEventListener events to all
05391     // subscribers.
05392
05393     TestEventListener* TestEventListeners::repeater() {
05394         return repeater_;
05395     }
05396
05397     // Sets the default_result_printer attribute to the provided listener.
05398     // The listener is also added to the listener list and previous
05399     // default_result_printer is removed from it and deleted. The listener can
05400     // also be NULL in which case it will not be added to the list. Does
05401     // nothing if the previous and the current listener objects are the same.
05402
05403     void TestEventListeners::SetDefaultResultPrinter(TestEventListener* listener) {
05404         if (default_result_printer_ != listener) {
05405             // It is an error to pass this method a listener that is already in the
05406             // list.
05407             delete Release(default_result_printer_);
05408             default_result_printer_ = listener;
05409             if (listener != nullptr) Append(listener);
05410         }
05411     }
05412
05413     // Sets the default_xml_generator attribute to the provided listener.  The
05414     // listener is also added to the listener list and previous
05415     // default_xml_generator is removed from it and deleted. The listener can
05416     // also be NULL in which case it will not be added to the list. Does
05417     // nothing if the previous and the current listener objects are the same.
05418
05419     void TestEventListeners::SetDefaultXmlGenerator(TestEventListener* listener) {
05420         if (default_xml_generator_ != listener) {
05421             // It is an error to pass this method a listener that is already in the
05422             // list.
05423             delete Release(default_xml_generator_);
05424             default_xml_generator_ = listener;
05425             if (listener != nullptr) Append(listener);
05426         }
05427     }
05428
05429     // Controls whether events will be forwarded by the repeater to the
05430     // listeners in the list.
05431
05432     bool TestEventListeners::EventForwardingEnabled() const {
05433         return repeater_->forwarding_enabled();
05434     }
05435
05436     void TestEventListeners::SuppressEventForwarding() {
05437         repeater_->set_forwarding_enabled(false);
05438     }
05439
05440     // class UnitTest
05441
05442     // Gets the singleton UnitTest object.  The first time this method is
05443     // called, a UnitTest object is constructed and returned.  Consecutive
05444     // calls will return the same object.
05445     //
05446     // We don't protect this under mutex_ as a user is not supposed to
05447     // call this before main() starts, from which point on the return
05448     // value will never change.
05449
05450     UnitTest* UnitTest::GetInstance() {
05451         // CodeGear C++Builder insists on a public destructor for the
05452         // default implementation.  Use this implementation to keep good OO
05453         // design with private destructor.
05454
05455 #if defined(__BORLANDC__)
05456         static UnitTest * const instance = new UnitTest;
05457         return instance;
05458 #else
05459         static UnitTest instance;
05460         return &instance;
05461 #endif  // defined(__BORLANDC__)
05462     }
05463
05464     // Gets the number of successful test suites.
05465
05466     int UnitTest::successful_test_suite_count() const {
05467         return impl()->successful_test_suite_count();
05468     }
05469
05470     // Gets the number of failed test suites.
05471
05472     int UnitTest::failed_test_suite_count() const {
05473         return impl()->failed_test_suite_count();
```

```
05474     }
05475
05476     // Gets the number of all test suites.
05477
05478     int UnitTest::total_test_suite_count() const {
05479         return impl()->total_test_suite_count();
05480     }
05481
05482     // Gets the number of all test suites that contain at least one test
05483     // that should run.
05484
05485     int UnitTest::test_suite_to_run_count() const {
05486         return impl()->test_suite_to_run_count();
05487     }
05488
05489     //  Legacy API is deprecated but still available
05490 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
05491
05492     int UnitTest::successful_test_case_count() const {
05493         return impl()->successful_test_suite_count();
05494     }
05495
05496     int UnitTest::failed_test_case_count() const {
05497         return impl()->failed_test_suite_count();
05498     }
05499
05500     int UnitTest::total_test_case_count() const {
05501         return impl()->total_test_suite_count();
05502     }
05503
05504     int UnitTest::test_case_to_run_count() const {
05505         return impl()->test_suite_to_run_count();
05506     }
05507 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
05508
05509     // Gets the number of successful tests.
05510
05511     int UnitTest::successful_test_count() const {
05512         return impl()->successful_test_count();
05513     }
05514
05515     // Gets the number of skipped tests.
05516
05517     int UnitTest::skipped_test_count() const {
05518         return impl()->skipped_test_count();
05519     }
05520
05521     // Gets the number of failed tests.
05522
05523     int UnitTest::failed_test_count() const {
05524         return impl()->failed_test_count();
05525     }
05526
05527     // Gets the number of disabled tests that will be reported in the XML report.
05528
05529     int UnitTest::reportable_disabled_test_count() const {
05530         return impl()->reportable_disabled_test_count();
05531     }
05532
05533     // Gets the number of disabled tests.
05534
05535     int UnitTest::disabled_test_count() const {
05536         return impl()->disabled_test_count();
05537     }
05538
05539     // Gets the number of tests to be printed in the XML report.
05540
05541     int UnitTest::reportable_test_count() const {
05542         return impl()->reportable_test_count();
05543     }
05544
05545     // Gets the number of all tests.
05546
05547     int UnitTest::total_test_count() const {
05548         return impl()->total_test_count();
05549     }
05550
05551     // Gets the number of tests that should run.
05552
05553     int UnitTest::test_to_run_count() const {
05554         return impl()->test_to_run_count();
05555     }
05556
05557     // Gets the time of the test program start, in ms from the start of the
05558     // UNIX epoch.
05559
05560     internal::TimeInMillis UnitTest::start_timestamp() const {
```

```
05561         return impl()->start_timestamp();
05562     }
05563
05564     // Gets the elapsed time, in milliseconds.
05565
05566     internal::TimeInMillis UnitTest::elapsed_time() const {
05567         return impl()->elapsed_time();
05568     }
05569
05570     // Returns true if and only if the unit test passed (i.e. all test suites
05571     // passed).
05572
05573     bool UnitTest::Passed() const {
05574         return impl()->Passed();
05575     }
05576
05577     // Returns true if and only if the unit test failed (i.e. some test suite
05578     // failed or something outside of all tests failed).
05579
05580     bool UnitTest::Failed() const {
05581         return impl()->Failed();
05582     }
05583
05584     // Gets the i-th test suite among all the test suites. i can range from 0 to
05585     // total_test_suite_count() - 1. If i is not in that range, returns NULL.
05586
05587     const TestSuite* UnitTest::GetTestSuite(int i) const {
05588         return impl()->GetTestSuite(i);
05589     }
05590
05591     //  Legacy API is deprecated but still available
05592 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
05593
05594     const TestCase* UnitTest::GetTestCase(int i) const {
05595         return impl()->GetTestCase(i);
05596     }
05597 #endif  //  GTEST_REMOVE_LEGACY_TEST_CASEAPI_
05598
05599     // Returns the TestResult containing information on test failures and
05600     // properties logged outside of individual test suites.
05601
05602     const TestResult& UnitTest::ad_hoc_test_result() const {
05603         return *impl()->ad_hoc_test_result();
05604     }
05605
05606     // Gets the i-th test suite among all the test suites. i can range from 0 to
05607     // total_test_suite_count() - 1. If i is not in that range, returns NULL.
05608
05609     TestSuite* UnitTest::GetMutableTestSuite(int i) {
05610         return impl()->GetMutableSuiteCase(i);
05611     }
05612
05613     // Returns the list of event listeners that can be used to track events
05614     // inside Google Test.
05615
05616     TestEventListeners& UnitTest::listeners() {
05617         return *impl()->listeners();
05618     }
05619
05620     // Registers and returns a global test environment.  When a test
05621     // program is run, all global test environments will be set-up in the
05622     // order they were registered.  After all tests in the program have
05623     // finished, all global test environments will be torn-down in the
05624     // *reverse* order they were registered.
05625     //
05626     // The UnitTest object takes ownership of the given environment.
05627     //
05628     // We don't protect this under mutex_, as we only support calling it
05629     // from the main thread.
05630
05631     Environment* UnitTest::AddEnvironment(Environment* env) {
05632         if (env == nullptr) {
05633             return nullptr;
05634         }
05635
05636         impl_->environments().push_back(env);
05637         return env;
05638     }
05639
05640     // Adds a TestPartResult to the current TestResult object.  All Google Test
05641     // assertion macros (e.g. ASSERT_TRUE, EXPECT_EQ, etc) eventually call
05642     // this to report their results.  The user code should use the
05643     // assertion macros instead of calling this directly.
05644
05645     void UnitTest::AddTestPartResult(TestPartResult::Type result_type,
05646             const char* file_name, int line_number,
05647             const std::string& message,
```

```
05648             const std::string& os_stack_trace)
05649     GTEST_LOCK_EXCLUDED_(mutex_) {
05650         Message msg;
05651         msg << message;
05652
05653         internal::MutexLock lock(&mutex_);
05654         if (impl_->gtest_trace_stack().size() > 0) {
05655             msg << "\n" << GTEST_NAME_ << " trace:";
05656
05657             for (size_t i = impl_->gtest_trace_stack().size(); i > 0; --i) {
05658                 const internal::TraceInfo& trace = impl_->gtest_trace_stack()[i - 1];
05659                 msg << "\n"
05660                         << internal::FormatFileLocation(trace.file, trace.line) << " "
05661                         << trace.message;
05662             }
05663         }
05664
05665         if (os_stack_trace.c_str() != nullptr && !os_stack_trace.empty()) {
05666             msg << internal::kStackTraceMarker << os_stack_trace;
05667         }
05668
05669         const TestPartResult result = TestPartResult(
05670                 result_type, file_name, line_number, msg.GetString().c_str());
05671         impl_->GetTestPartResultReporterForCurrentThread()->ReportTestPartResult(
05672                 result);
05673
05674         if (result_type != TestPartResult::kSuccess &&
05675                 result_type != TestPartResult::kSkip) {
05676             // gtest_break_on_failure takes precedence over
05677             // gtest_throw_on_failure.  This allows a user to set the latter
05678             // in the code (perhaps in order to use Google Test assertions
05679             // with another testing framework) and specify the former on the
05680             // command line for debugging.
05681             if (GTEST_FLAG_GET(break_on_failure)) {
05682 #if GTEST_OS_WINDOWS && !GTEST_OS_WINDOWS_PHONE && !GTEST_OS_WINDOWS_RT
05683                 // Using DebugBreak on Windows allows gtest to still break into a debugger
05684                 // when a failure happens and both the --gtest_break_on_failure and
05685                 // the --gtest_catch_exceptions flags are specified.
05686                 DebugBreak();
05687 #elif (!defined(__native_client__)) &&            \
05688     ((defined(__clang__) || defined(__GNUC__)) && \
05689      (defined(__x86_64__) || defined(__i386__)))
05690                 // with clang/gcc we can achieve the same effect on x86 by invoking int3
05691                 asm("int3");
05692 #else
05693                 // Dereference nullptr through a volatile pointer to prevent the compiler
05694                 // from removing. We use this rather than abort() or __builtin_trap() for
05695                 // portability: some debuggers don't correctly trap abort().
05696                 *static_cast<volatile int*> (nullptr) = 1;
05697 #endif  // GTEST_OS_WINDOWS
05698             } else if (GTEST_FLAG_GET(throw_on_failure)) {
05699 #if GTEST_HAS_EXCEPTIONS
05700                 throw internal::GoogleTestFailureException(result);
05701 #else
05702                 // We cannot call abort() as it generates a pop-up in debug mode
05703                 // that cannot be suppressed in VC 7.1 or below.
05704                 exit(1);
05705 #endif
05706             }
05707         }
05708     }
05709
05710     // Adds a TestProperty to the current TestResult object when invoked from
05711     // inside a test, to current TestSuite's ad_hoc_test_result_ when invoked
05712     // from SetUpTestSuite or TearDownTestSuite, or to the global property set
05713     // when invoked elsewhere.  If the result already contains a property with
05714     // the same key, the value will be updated.
05715
05716     void UnitTest::RecordProperty(const std::string& key,
05717             const std::string& value) {
05718         impl_->RecordProperty(TestProperty(key, value));
05719     }
05720
05721     // Runs all tests in this UnitTest object and prints the result.
05722     // Returns 0 if successful, or 1 otherwise.
05723     //
05724     // We don't protect this under mutex_, as we only support calling it
05725     // from the main thread.
05726
05727     int UnitTest::Run() {
05728         const bool in_death_test_child_process =
05729                 GTEST_FLAG_GET(internal_run_death_test).length() > 0;
05730
05731         // Google Test implements this protocol for catching that a test
05732         // program exits before returning control to Google Test:
05733         //
05734         //   1. Upon start, Google Test creates a file whose absolute path
```

```
05735         //      is specified by the environment variable
05736         //      TEST_PREMATURE_EXIT_FILE.
05737         //   2. When Google Test has finished its work, it deletes the file.
05738         //
05739         // This allows a test runner to set TEST_PREMATURE_EXIT_FILE before
05740         // running a Google-Test-based test program and check the existence
05741         // of the file at the end of the test execution to see if it has
05742         // exited prematurely.
05743
05744         // If we are in the child process of a death test, don't
05745         // create/delete the premature exit file, as doing so is unnecessary
05746         // and will confuse the parent process.  Otherwise, create/delete
05747         // the file upon entering/leaving this function.  If the program
05748         // somehow exits before this function has a chance to return, the
05749         // premature-exit file will be left undeleted, causing a test runner
05750         // that understands the premature-exit-file protocol to report the
05751         // test as having failed.
05752         const internal::ScopedPrematureExitFile premature_exit_file(
05753                 in_death_test_child_process
05754                 ? nullptr
05755                 : internal::posix::GetEnv("TEST_PREMATURE_EXIT_FILE"));
05756
05757         // Captures the value of GTEST_FLAG(catch_exceptions).  This value will be
05758         // used for the duration of the program.
05759         impl()->set_catch_exceptions(GTEST_FLAG_GET(catch_exceptions));
05760
05761 #if GTEST_OS_WINDOWS
05762         // Either the user wants Google Test to catch exceptions thrown by the
05763         // tests or this is executing in the context of death test child
05764         // process. In either case the user does not want to see pop-up dialogs
05765         // about crashes - they are expected.
05766         if (impl()->catch_exceptions() || in_death_test_child_process) {
05767 #if !GTEST_OS_WINDOWS_MOBILE && !GTEST_OS_WINDOWS_PHONE && !GTEST_OS_WINDOWS_RT
05768             // SetErrorMode doesn't exist on CE.
05769             SetErrorMode(SEM_FAILCRITICALERRORS | SEM_NOALIGNMENTFAULTEXCEPT |
05770                          SEM_NOGPFAULTERRORBOX | SEM_NOOPENFILEERRORBOX);
05771 #endif  // !GTEST_OS_WINDOWS_MOBILE
05772
05773 #if (defined(_MSC_VER) || GTEST_OS_WINDOWS_MINGW) && !GTEST_OS_WINDOWS_MOBILE
05774             // Death test children can be terminated with _abort().  On Windows,
05775             // _abort() can show a dialog with a warning message.  This forces the
05776             // abort message to go to stderr instead.
05777             _set_error_mode(_OUT_TO_STDERR);
05778 #endif
05779
05780 #if defined(_MSC_VER) && !GTEST_OS_WINDOWS_MOBILE
05781             // In the debug version, Visual Studio pops up a separate dialog
05782             // offering a choice to debug the aborted program. We need to suppress
05783             // this dialog or it will pop up for every EXPECT/ASSERT_DEATH statement
05784             // executed. Google Test will notify the user of any unexpected
05785             // failure via stderr.
05786             if (!GTEST_FLAG_GET(break_on_failure))
05787                 _set_abort_behavior(
05788                     0x0, // Clear the following flags:
05789                     _WRITE_ABORT_MSG | _CALL_REPORTFAULT); // pop-up window, core dump.
05790
05791             // In debug mode, the Windows CRT can crash with an assertion over invalid
05792             // input (e.g. passing an invalid file descriptor).  The default handling
05793             // for these assertions is to pop up a dialog and wait for user input.
05794             // Instead ask the CRT to dump such assertions to stderr non-interactively.
05795             if (!IsDebuggerPresent()) {
05796                 (void) _CrtSetReportMode(_CRT_ASSERT,
05797                         _CRTDBG_MODE_FILE | _CRTDBG_MODE_DEBUG);
05798                 (void) _CrtSetReportFile(_CRT_ASSERT, _CRTDBG_FILE_STDERR);
05799             }
05800 #endif
05801         }
05802 #endif  // GTEST_OS_WINDOWS
05803
05804         return internal::HandleExceptionsInMethodIfSupported(
05805                 impl(), &internal::UnitTestImpl::RunAllTests,
05806                 "auxiliary test code (environments or event listeners)")
05807                 ? 0
05808                 : 1;
05809     }
05810
05811     // Returns the working directory when the first TEST() or TEST_F() was
05812     // executed.
05813
05814     const char* UnitTest::original_working_dir() const {
05815         return impl_->original_working_dir_.c_str();
05816     }
05817
05818     // Returns the TestSuite object for the test that's currently running,
05819     // or NULL if no test is running.
05820
05821     const TestSuite* UnitTest::current_test_suite() const
```

```
05822     GTEST_LOCK_EXCLUDED_(mutex_) {
05823         internal::MutexLock lock(&mutex_);
05824         return impl_->current_test_suite();
05825     }
05826
05827     // Legacy API is still available but deprecated
05828 #ifndef GTEST_REMOVE_LEGACY_TEST_CASEAPI_
05829
05830     const TestCase* UnitTest::current_test_case() const
05831     GTEST_LOCK_EXCLUDED_(mutex_) {
05832         internal::MutexLock lock(&mutex_);
05833         return impl_->current_test_suite();
05834     }
05835 #endif
05836
05837     // Returns the TestInfo object for the test that's currently running,
05838     // or NULL if no test is running.
05839
05840     const TestInfo* UnitTest::current_test_info() const
05841     GTEST_LOCK_EXCLUDED_(mutex_) {
05842         internal::MutexLock lock(&mutex_);
05843         return impl_->current_test_info();
05844     }
05845
05846     // Returns the random seed used at the start of the current test run.
05847
05848     int UnitTest::random_seed() const {
05849         return impl_->random_seed();
05850     }
05851
05852     // Returns ParameterizedTestSuiteRegistry object used to keep track of
05853     // value-parameterized tests and instantiate and register them.
05854
05855     internal::ParameterizedTestSuiteRegistry&
05856     UnitTest::parameterized_test_registry() GTEST_LOCK_EXCLUDED_(mutex_) {
05857         return impl_->parameterized_test_registry();
05858     }
05859
05860     // Creates an empty UnitTest.
05861
05862     UnitTest::UnitTest() {
05863         impl_ = new internal::UnitTestImpl(this);
05864     }
05865
05866     // Destructor of UnitTest.
05867
05868     UnitTest::~UnitTest() {
05869         delete impl_;
05870     }
05871
05872     // Pushes a trace defined by SCOPED_TRACE() on to the per-thread
05873     // Google Test trace stack.
05874
05875     void UnitTest::PushGTestTrace(const internal::TraceInfo& trace)
05876     GTEST_LOCK_EXCLUDED_(mutex_) {
05877         internal::MutexLock lock(&mutex_);
05878         impl_->gtest_trace_stack().push_back(trace);
05879     }
05880
05881     // Pops a trace from the per-thread Google Test trace stack.
05882
05883     void UnitTest::PopGTestTrace() GTEST_LOCK_EXCLUDED_(mutex_) {
05884         internal::MutexLock lock(&mutex_);
05885         impl_->gtest_trace_stack().pop_back();
05886     }
05887
05888     namespace internal {
05889
05890         UnitTestImpl::UnitTestImpl(UnitTest* parent)
05891         : parent_(parent),
05892         GTEST_DISABLE_MSC_WARNINGS_PUSH_(4355 /* using this in initializer */)
05893         default_global_test_part_result_reporter_(this),
05894         default_per_thread_test_part_result_reporter_(this),
05895         GTEST_DISABLE_MSC_WARNINGS_POP_() global_test_part_result_reporter_(
05896         &default_global_test_part_result_reporter_),
05897         per_thread_test_part_result_reporter_(
05898         &default_per_thread_test_part_result_reporter_),
05899         parameterized_test_registry_(),
05900         parameterized_tests_registered_(false),
05901         last_death_test_suite_(-1),
05902         current_test_suite_(nullptr),
05903         current_test_info_(nullptr),
05904         ad_hoc_test_result_(),
05905         os_stack_trace_getter_(nullptr),
05906         post_flag_parse_init_performed_(false),
05907         random_seed_(0), // Will be overridden by the flag before first use.
05908         random_(0), // Will be reseeded before first use.
```

```
05909         start_timestamp_(0),
05910         elapsed_time_(0),
05911 #if GTEST_HAS_DEATH_TEST
05912         death_test_factory_(new DefaultDeathTestFactory),
05913 #endif
05914         // Will be overridden by the flag before first use.
05915         catch_exceptions_(false) {
05916             listeners()->SetDefaultResultPrinter(new PrettyUnitTestResultPrinter);
05917         }
05918
05919         UnitTestImpl::~UnitTestImpl() {
05920             // Deletes every TestSuite.
05921             ForEach(test_suites_, internal::Delete<TestSuite>);
05922
05923             // Deletes every Environment.
05924             ForEach(environments_, internal::Delete<Environment>);
05925
05926             delete os_stack_trace_getter_;
05927         }
05928
05929         // Adds a TestProperty to the current TestResult object when invoked in a
05930         // context of a test, to current test suite's ad_hoc_test_result when invoke
05931         // from SetUpTestSuite/TearDownTestSuite, or to the global property set
05932         // otherwise.  If the result already contains a property with the same key,
05933         // the value will be updated.
05934
05935         void UnitTestImpl::RecordProperty(const TestProperty& test_property) {
05936             std::string xml_element;
05937             TestResult* test_result; // TestResult appropriate for property recording.
05938
05939             if (current_test_info_ != nullptr) {
05940                 xml_element = "testcase";
05941                 test_result = &(current_test_info_->result_);
05942             } else if (current_test_suite_ != nullptr) {
05943                 xml_element = "testsuite";
05944                 test_result = &(current_test_suite_->ad_hoc_test_result_);
05945             } else {
05946                 xml_element = "testsuites";
05947                 test_result = &ad_hoc_test_result_;
05948             }
05949             test_result->RecordProperty(xml_element, test_property);
05950         }
05951
05952 #if GTEST_HAS_DEATH_TEST
05953         // Disables event forwarding if the control is currently in a death test
05954         // subprocess. Must not be called before InitGoogleTest.
05955
05956         void UnitTestImpl::SuppressTestEventsIfInSubprocess() {
05957             if (internal_run_death_test_flag_.get() != nullptr)
05958                 listeners()->SuppressEventForwarding();
05959         }
05960 #endif  // GTEST_HAS_DEATH_TEST
05961
05962         // Initializes event listeners performing XML output as specified by
05963         // UnitTestOptions. Must not be called before InitGoogleTest.
05964
05965         void UnitTestImpl::ConfigureXmlOutput() {
05966             const std::string& output_format = UnitTestOptions::GetOutputFormat();
05967             if (output_format == "xml") {
05968                 listeners()->SetDefaultXmlGenerator(new XmlUnitTestResultPrinter(
05969                         UnitTestOptions::GetAbsolutePathToOutputFile().c_str()));
05970             } else if (output_format == "json") {
05971                 listeners()->SetDefaultXmlGenerator(new JsonUnitTestResultPrinter(
05972                         UnitTestOptions::GetAbsolutePathToOutputFile().c_str()));
05973             } else if (output_format != "") {
05974                 GTEST_LOG_(WARNING) << "WARNING: unrecognized output format \""
05975                         << output_format << "\" ignored.";
05976             }
05977         }
05978
05979 #if GTEST_CAN_STREAM_RESULTS_
05980         // Initializes event listeners for streaming test results in string form.
05981         // Must not be called before InitGoogleTest.
05982
05983         void UnitTestImpl::ConfigureStreamingOutput() {
05984             const std::string& target = GTEST_FLAG_GET(stream_result_to);
05985             if (!target.empty()) {
05986                 const size_t pos = target.find(':');
05987                 if (pos != std::string::npos) {
05988                     listeners()->Append(
05989                             new StreamingListener(target.substr(0, pos), target.substr(pos + 1)));
05990                 } else {
05991                     GTEST_LOG_(WARNING) << "unrecognized streaming target \"" << target
05992                             << "\" ignored.";
05993                 }
05994             }
05995         }
```

```
05996 #endif  // GTEST_CAN_STREAM_RESULTS_
05997
05998         // Performs initialization dependent upon flag values obtained in
05999         // ParseGoogleTestFlagsOnly.  Is called from InitGoogleTest after the call to
06000         // ParseGoogleTestFlagsOnly.  In case a user neglects to call InitGoogleTest
06001         // this function is also called from RunAllTests.  Since this function can be
06002         // called more than once, it has to be idempotent.
06003
06004         void UnitTestImpl::PostFlagParsingInit() {
06005             // Ensures that this function does not execute more than once.
06006             if (!post_flag_parse_init_performed_) {
06007                 post_flag_parse_init_performed_ = true;
06008
06009 #if defined(GTEST_CUSTOM_TEST_EVENT_LISTENER_)
06010                 // Register to send notifications about key process state changes.
06011                 listeners()->Append(new GTEST_CUSTOM_TEST_EVENT_LISTENER_());
06012 #endif  // defined(GTEST_CUSTOM_TEST_EVENT_LISTENER_)
06013
06014 #if GTEST_HAS_DEATH_TEST
06015                 InitDeathTestSubprocessControlInfo();
06016                 SuppressTestEventsIfInSubprocess();
06017 #endif  // GTEST_HAS_DEATH_TEST
06018
06019                 // Registers parameterized tests. This makes parameterized tests
06020                 // available to the UnitTest reflection API without running
06021                 // RUN_ALL_TESTS.
06022                 RegisterParameterizedTests();
06023
06024                 // Configures listeners for XML output. This makes it possible for users
06025                 // to shut down the default XML output before invoking RUN_ALL_TESTS.
06026                 ConfigureXmlOutput();
06027
06028                 if (GTEST_FLAG_GET(brief)) {
06029                     listeners()->SetDefaultResultPrinter(new BriefUnitTestResultPrinter);
06030                 }
06031
06032 #if GTEST_CAN_STREAM_RESULTS_
06033                 // Configures listeners for streaming test results to the specified server.
06034                 ConfigureStreamingOutput();
06035 #endif  // GTEST_CAN_STREAM_RESULTS_
06036
06037 #if GTEST_HAS_ABSL
06038                 if (GTEST_FLAG_GET(install_failure_signal_handler)) {
06039                     absl::FailureSignalHandlerOptions options;
06040                     absl::InstallFailureSignalHandler(options);
06041                 }
06042 #endif  // GTEST_HAS_ABSL
06043             }
06044         }
06045
06046         // A predicate that checks the name of a TestSuite against a known
06047         // value.
06048         //
06049         // This is used for implementation of the UnitTest class only.  We put
06050         // it in the anonymous namespace to prevent polluting the outer
06051         // namespace.
06052         //
06053         // TestSuiteNameIs is copyable.
06054
06055         class TestSuiteNameIs {
06056         public:
06057             // Constructor.
06058
06059             explicit TestSuiteNameIs(const std::string& name) : name_(name) {
06060             }
06061
06062             // Returns true if and only if the name of test_suite matches name_.
06063
06064             bool operator()(const TestSuite* test_suite) const {
06065                 return test_suite != nullptr &&
06066                         strcmp(test_suite->name(), name_.c_str()) == 0;
06067             }
06068
06069         private:
06070             std::string name_;
06071         };
06072
06073         // Finds and returns a TestSuite with the given name.  If one doesn't
06074         // exist, creates one and returns it.  It's the CALLER'S
06075         // RESPONSIBILITY to ensure that this function is only called WHEN THE
06076         // TESTS ARE NOT SHUFFLED.
06077         //
06078         // Arguments:
06079         //
06080         //   test_suite_name: name of the test suite
06081         //   type_param:      the name of the test suite's type parameter, or NULL if
06082         //                    this is not a typed or a type-parameterized test suite.
```

```
06083         //   set_up_tc:       pointer to the function that sets up the test suite
06084         //   tear_down_tc:    pointer to the function that tears down the test suite
06085
06086         TestSuite* UnitTestImpl::GetTestSuite(
06087                 const char* test_suite_name, const char* type_param,
06088                 internal::SetUpTestSuiteFunc set_up_tc,
06089                 internal::TearDownTestSuiteFunc tear_down_tc) {
06090             // Can we find a TestSuite with the given name?
06091             const auto test_suite =
06092                     std::find_if(test_suites_.rbegin(), test_suites_.rend(),
06093                     TestSuiteNameIs(test_suite_name));
06094
06095             if (test_suite != test_suites_.rend()) return *test_suite;
06096
06097             // No.  Let's create one.
06098             auto* const new_test_suite =
06099                     new TestSuite(test_suite_name, type_param, set_up_tc, tear_down_tc);
06100
06101             const UnitTestFilter death_test_suite_filter(kDeathTestSuiteFilter);
06102             // Is this a death test suite?
06103             if (death_test_suite_filter.MatchesName(test_suite_name)) {
06104                 // Yes.  Inserts the test suite after the last death test suite
06105                 // defined so far.  This only works when the test suites haven't
06106                 // been shuffled.  Otherwise we may end up running a death test
06107                 // after a non-death test.
06108                 ++last_death_test_suite_;
06109                 test_suites_.insert(test_suites_.begin() + last_death_test_suite_,
06110                         new_test_suite);
06111             } else {
06112                 // No.  Appends to the end of the list.
06113                 test_suites_.push_back(new_test_suite);
06114             }
06115
06116             test_suite_indices_.push_back(static_cast<int> (test_suite_indices_.size()));
06117             return new_test_suite;
06118         }
06119
06120         // Helpers for setting up / tearing down the given environment.  They
06121         // are for use in the ForEach() function.
06122
06123         static void SetUpEnvironment(Environment* env) {
06124             env->SetUp();
06125         }
06126
06127         static void TearDownEnvironment(Environment* env) {
06128             env->TearDown();
06129         }
06130
06131         // Runs all tests in this UnitTest object, prints the result, and
06132         // returns true if all tests are successful.  If any exception is
06133         // thrown during a test, the test is considered to be failed, but the
06134         // rest of the tests will still be run.
06135         //
06136         // When parameterized tests are enabled, it expands and registers
06137         // parameterized tests first in RegisterParameterizedTests().
06138         // All other functions called from RunAllTests() may safely assume that
06139         // parameterized tests are ready to be counted and run.
06140
06141         bool UnitTestImpl::RunAllTests() {
06142             // True if and only if Google Test is initialized before RUN_ALL_TESTS() is
06143             // called.
06144             const bool gtest_is_initialized_before_run_all_tests = GTestIsInitialized();
06145
06146             // Do not run any test if the --help flag was specified.
06147             if (g_help_flag) return true;
06148
06149             // Repeats the call to the post-flag parsing initialization in case the
06150             // user didn't call InitGoogleTest.
06151             PostFlagParsingInit();
06152
06153             // Even if sharding is not on, test runners may want to use the
06154             // GTEST_SHARD_STATUS_FILE to query whether the test supports the sharding
06155             // protocol.
06156             internal::WriteToShardStatusFileIfNeeded();
06157
06158             // True if and only if we are in a subprocess for running a thread-safe-style
06159             // death test.
06160             bool in_subprocess_for_death_test = false;
06161
06162 #if GTEST_HAS_DEATH_TEST
06163             in_subprocess_for_death_test =
06164                     (internal_run_death_test_flag_.get() != nullptr);
06165 #if defined(GTEST_EXTRA_DEATH_TEST_CHILD_SETUP_)
06166             if (in_subprocess_for_death_test) {
06167                 GTEST_EXTRA_DEATH_TEST_CHILD_SETUP_();
06168             }
06169 #endif  // defined(GTEST_EXTRA_DEATH_TEST_CHILD_SETUP_)
```

```
06170 #endif  // GTEST_HAS_DEATH_TEST
06171
06172             const bool should_shard = ShouldShard(kTestTotalShards, kTestShardIndex,
06173                     in_subprocess_for_death_test);
06174
06175             // Compares the full test names with the filter to decide which
06176             // tests to run.
06177             const bool has_tests_to_run =
06178                     FilterTests(should_shard ? HONOR_SHARDING_PROTOCOL
06179                     : IGNORE_SHARDING_PROTOCOL) > 0;
06180
06181             // Lists the tests and exits if the --gtest_list_tests flag was specified.
06182             if (GTEST_FLAG_GET(list_tests)) {
06183                 // This must be called *after* FilterTests() has been called.
06184                 ListTestsMatchingFilter();
06185                 return true;
06186             }
06187
06188             random_seed_ = GetRandomSeedFromFlag(GTEST_FLAG_GET(random_seed));
06189
06190             // True if and only if at least one test has failed.
06191             bool failed = false;
06192
06193             TestEventListener* repeater = listeners()->repeater();
06194
06195             start_timestamp_ = GetTimeInMillis();
06196             repeater->OnTestProgramStart(*parent_);
06197
06198             // How many times to repeat the tests?  We don't want to repeat them
06199             // when we are inside the subprocess of a death test.
06200             const int repeat = in_subprocess_for_death_test ? 1 : GTEST_FLAG_GET(repeat);
06201
06202             // Repeats forever if the repeat count is negative.
06203             const bool gtest_repeat_forever = repeat < 0;
06204
06205             // Should test environments be set up and torn down for each repeat, or only
06206             // set up on the first and torn down on the last iteration? If there is no
06207             // "last" iteration because the tests will repeat forever, always recreate the
06208             // environments to avoid leaks in case one of the environments is using
06209             // resources that are external to this process. Without this check there would
06210             // be no way to clean up those external resources automatically.
06211             const bool recreate_environments_when_repeating =
06212                     GTEST_FLAG_GET(recreate_environments_when_repeating) ||
06213                     gtest_repeat_forever;
06214
06215             for (int i = 0; gtest_repeat_forever || i != repeat; i++) {
06216                 // We want to preserve failures generated by ad-hoc test
06217                 // assertions executed before RUN_ALL_TESTS().
06218                 ClearNonAdHocTestResult();
06219
06220                 Timer timer;
06221
06222                 // Shuffles test suites and tests if requested.
06223                 if (has_tests_to_run && GTEST_FLAG_GET(shuffle)) {
06224                     random()->Reseed(static_cast<uint32_t> (random_seed_));
06225                     // This should be done before calling OnTestIterationStart(),
06226                     // such that a test event listener can see the actual test order
06227                     // in the event.
06228                     ShuffleTests();
06229                 }
06230
06231                 // Tells the unit test event listeners that the tests are about to start.
06232                 repeater->OnTestIterationStart(*parent_, i);
06233
06234                 // Runs each test suite if there is at least one test to run.
06235                 if (has_tests_to_run) {
06236                     // Sets up all environments beforehand. If test environments aren't
06237                     // recreated for each iteration, only do so on the first iteration.
06238                     if (i == 0 || recreate_environments_when_repeating) {
06239                         repeater->OnEnvironmentsSetUpStart(*parent_);
06240                         ForEach(environments_, SetUpEnvironment);
06241                         repeater->OnEnvironmentsSetUpEnd(*parent_);
06242                     }
06243
06244                     // Runs the tests only if there was no fatal failure or skip triggered
06245                     // during global set-up.
06246                     if (Test::IsSkipped()) {
06247                         // Emit diagnostics when global set-up calls skip, as it will not be
06248                         // emitted by default.
06249                         TestResult& test_result =
06250                                 *internal::GetUnitTestImpl()->current_test_result();
06251                         for (int j = 0; j < test_result.total_part_count(); ++j) {
06252                             const TestPartResult& test_part_result =
06253                                     test_result.GetTestPartResult(j);
06254                             if (test_part_result.type() == TestPartResult::kSkip) {
06255                                 const std::string& result = test_part_result.message();
06256                                 printf("%s\n", result.c_str());
```

```
06257                           }
06258                       }
06259                       fflush(stdout);
06260                   } else if (!Test::HasFatalFailure()) {
06261                       for (int test_index = 0; test_index < total_test_suite_count();
06262                               test_index++) {
06263                           GetMutableSuiteCase(test_index)->Run();
06264                           if (GTEST_FLAG_GET(fail_fast) &&
06265                                   GetMutableSuiteCase(test_index)->Failed()) {
06266                               for (int j = test_index + 1; j < total_test_suite_count(); j++) {
06267                                   GetMutableSuiteCase(j)->Skip();
06268                               }
06269                               break;
06270                           }
06271                       }
06272                   } else if (Test::HasFatalFailure()) {
06273                       // If there was a fatal failure during the global setup then we know we
06274                       // aren't going to run any tests. Explicitly mark all of the tests as
06275                       // skipped to make this obvious in the output.
06276                       for (int test_index = 0; test_index < total_test_suite_count();
06277                               test_index++) {
06278                           GetMutableSuiteCase(test_index)->Skip();
06279                       }
06280                   }
06281
06282                   // Tears down all environments in reverse order afterwards. If test
06283                   // environments aren't recreated for each iteration, only do so on the
06284                   // last iteration.
06285                   if (i == repeat - 1 || recreate_environments_when_repeating) {
06286                       repeater->OnEnvironmentsTearDownStart(*parent_);
06287                       std::for_each(environments_.rbegin(), environments_.rend(),
06288                               TearDownEnvironment);
06289                       repeater->OnEnvironmentsTearDownEnd(*parent_);
06290                   }
06291               }
06292
06293               elapsed_time_ = timer.Elapsed();
06294
06295               // Tells the unit test event listener that the tests have just finished.
06296               repeater->OnTestIterationEnd(*parent_, i);
06297
06298               // Gets the result and clears it.
06299               if (!Passed()) {
06300                   failed = true;
06301               }
06302
06303               // Restores the original test order after the iteration.  This
06304               // allows the user to quickly repro a failure that happens in the
06305               // N-th iteration without repeating the first (N - 1) iterations.
06306               // This is not enclosed in "if (GTEST_FLAG(shuffle)) { ... }", in
06307               // case the user somehow changes the value of the flag somewhere
06308               // (it's always safe to unshuffle the tests).
06309               UnshuffleTests();
06310
06311               if (GTEST_FLAG_GET(shuffle)) {
06312                   // Picks a new random seed for each iteration.
06313                   random_seed_ = GetNextRandomSeed(random_seed_);
06314               }
06315           }
06316
06317           repeater->OnTestProgramEnd(*parent_);
06318
06319           if (!gtest_is_initialized_before_run_all_tests) {
06320               ColoredPrintf(
06321                       GTestColor::kRed,
06322                       "\nIMPORTANT NOTICE - DO NOT IGNORE:\n"
06323                       "This test program did NOT call " GTEST_INIT_GOOGLE_TEST_NAME_
06324                       "() before calling RUN_ALL_TESTS(). This is INVALID. Soon " GTEST_NAME_
06325                       " will start to enforce the valid usage. "
06326                       "Please fix it ASAP, or IT WILL START TO FAIL.\n"); // NOLINT
06327 #if GTEST_FOR_GOOGLE_
06328               ColoredPrintf(GTestColor::kRed,
06329                       "For more details, see http://wiki/Main/ValidGUnitMain.\n");
06330 #endif  // GTEST_FOR_GOOGLE_
06331           }
06332
06333           return !failed;
06334       }
06335
06336       // Reads the GTEST_SHARD_STATUS_FILE environment variable, and creates the file
06337       // if the variable is present. If a file already exists at this location, this
06338       // function will write over it. If the variable is present, but the file cannot
06339       // be created, prints an error and exits.
06340
06341       void WriteToShardStatusFileIfNeeded() {
06342           const char* const test_shard_file = posix::GetEnv(kTestShardStatusFile);
06343           if (test_shard_file != nullptr) {
```

```
06344               FILE * const file = posix::FOpen(test_shard_file, "w");
06345               if (file == nullptr) {
06346                   ColoredPrintf(GTestColor::kRed,
06347                           "Could not write to the test shard status file \"%s\" "
06348                           "specified by the %s environment variable.\n",
06349                           test_shard_file, kTestShardStatusFile);
06350                   fflush(stdout);
06351                   exit(EXIT_FAILURE);
06352               }
06353               fclose(file);
06354           }
06355       }
06356
06357       // Checks whether sharding is enabled by examining the relevant
06358       // environment variable values. If the variables are present,
06359       // but inconsistent (i.e., shard_index >= total_shards), prints
06360       // an error and exits. If in_subprocess_for_death_test, sharding is
06361       // disabled because it must only be applied to the original test
06362       // process. Otherwise, we could filter out death tests we intended to execute.
06363
06364       bool ShouldShard(const char* total_shards_env, const char* shard_index_env,
06365               bool in_subprocess_for_death_test) {
06366           if (in_subprocess_for_death_test) {
06367               return false;
06368           }
06369
06370           const int32_t total_shards = Int32FromEnvOrDie(total_shards_env, -1);
06371           const int32_t shard_index = Int32FromEnvOrDie(shard_index_env, -1);
06372
06373           if (total_shards == -1 && shard_index == -1) {
06374               return false;
06375           } else if (total_shards == -1 && shard_index != -1) {
06376               const Message msg = Message() << "Invalid environment variables: you have "
06377                       << kTestShardIndex << " = " << shard_index
06378                       << ", but have left " << kTestTotalShards
06379                       << " unset.\n";
06380               ColoredPrintf(GTestColor::kRed, "%s", msg.GetString().c_str());
06381               fflush(stdout);
06382               exit(EXIT_FAILURE);
06383           } else if (total_shards != -1 && shard_index == -1) {
06384               const Message msg = Message()
06385                       << "Invalid environment variables: you have "
06386                       << kTestTotalShards << " = " << total_shards
06387                       << ", but have left " << kTestShardIndex << " unset.\n";
06388               ColoredPrintf(GTestColor::kRed, "%s", msg.GetString().c_str());
06389               fflush(stdout);
06390               exit(EXIT_FAILURE);
06391           } else if (shard_index < 0 || shard_index >= total_shards) {
06392               const Message msg =
06393                       Message() << "Invalid environment variables: we require 0 <= "
06394                       << kTestShardIndex << " < " << kTestTotalShards
06395                       << ", but you have " << kTestShardIndex << "=" << shard_index
06396                       << ", " << kTestTotalShards << "=" << total_shards << ".\n";
06397               ColoredPrintf(GTestColor::kRed, "%s", msg.GetString().c_str());
06398               fflush(stdout);
06399               exit(EXIT_FAILURE);
06400           }
06401
06402           return total_shards > 1;
06403       }
06404
06405       // Parses the environment variable var as an Int32. If it is unset,
06406       // returns default_val. If it is not an Int32, prints an error
06407       // and aborts.
06408
06409       int32_t Int32FromEnvOrDie(const char* var, int32_t default_val) {
06410           const char* str_val = posix::GetEnv(var);
06411           if (str_val == nullptr) {
06412               return default_val;
06413           }
06414
06415           int32_t result;
06416           if (!ParseInt32(Message() << "The value of environment variable " << var,
06417                   str_val, &result)) {
06418               exit(EXIT_FAILURE);
06419           }
06420           return result;
06421       }
06422
06423       // Given the total number of shards, the shard index, and the test id,
06424       // returns true if and only if the test should be run on this shard. The test id
06425       // is some arbitrary but unique non-negative integer assigned to each test
06426       // method. Assumes that 0 <= shard_index < total_shards.
06427
06428       bool ShouldRunTestOnShard(int total_shards, int shard_index, int test_id) {
06429           return (test_id % total_shards) == shard_index;
06430       }
```

```
06431
06432         // Compares the name of each test with the user-specified filter to
06433         // decide whether the test should be run, then records the result in
06434         // each TestSuite and TestInfo object.
06435         // If shard_tests == true, further filters tests based on sharding
06436         // variables in the environment - see
06437         // https://github.com/google/googletest/blob/main/docs/advanced.md
06438         // . Returns the number of tests that should run.
06439
06440         int UnitTestImpl::FilterTests(ReactionToSharding shard_tests) {
06441             const int32_t total_shards = shard_tests == HONOR_SHARDING_PROTOCOL
06442                     ? Int32FromEnvOrDie(kTestTotalShards, -1)
06443                     : -1;
06444             const int32_t shard_index = shard_tests == HONOR_SHARDING_PROTOCOL
06445                     ? Int32FromEnvOrDie(kTestShardIndex, -1)
06446                     : -1;
06447
06448             const PositiveAndNegativeUnitTestFilter gtest_flag_filter(
06449                     GTEST_FLAG_GET(filter));
06450             const UnitTestFilter disable_test_filter(kDisableTestFilter);
06451             // num_runnable_tests are the number of tests that will
06452             // run across all shards (i.e., match filter and are not disabled).
06453             // num_selected_tests are the number of tests to be run on
06454             // this shard.
06455             int num_runnable_tests = 0;
06456             int num_selected_tests = 0;
06457             for (auto* test_suite : test_suites_) {
06458                 const std::string& test_suite_name = test_suite->name();
06459                 test_suite->set_should_run(false);
06460
06461                 for (size_t j = 0; j < test_suite->test_info_list().size(); j++) {
06462                     TestInfo * const test_info = test_suite->test_info_list()[j];
06463                     const std::string test_name(test_info->name());
06464                     // A test is disabled if test suite name or test name matches
06465                     // kDisableTestFilter.
06466                     const bool is_disabled =
06467                             disable_test_filter.MatchesName(test_suite_name) ||
06468                             disable_test_filter.MatchesName(test_name);
06469                     test_info->is_disabled_ = is_disabled;
06470
06471                     const bool matches_filter =
06472                             gtest_flag_filter.MatchesTest(test_suite_name, test_name);
06473                     test_info->matches_filter_ = matches_filter;
06474
06475                     const bool is_runnable =
06476                             (GTEST_FLAG_GET(also_run_disabled_tests) || !is_disabled) &&
06477                             matches_filter;
06478
06479                     const bool is_in_another_shard =
06480                             shard_tests != IGNORE_SHARDING_PROTOCOL &&
06481                             !ShouldRunTestOnShard(total_shards, shard_index, num_runnable_tests);
06482                     test_info->is_in_another_shard_ = is_in_another_shard;
06483                     const bool is_selected = is_runnable && !is_in_another_shard;
06484
06485                     num_runnable_tests += is_runnable;
06486                     num_selected_tests += is_selected;
06487
06488                     test_info->should_run_ = is_selected;
06489                     test_suite->set_should_run(test_suite->should_run() || is_selected);
06490                 }
06491             }
06492             return num_selected_tests;
06493         }
06494
06495         // Prints the given C-string on a single line by replacing all '\n'
06496         // characters with string "\\n".  If the output takes more than
06497         // max_length characters, only prints the first max_length characters
06498         // and "...".
06499
06500         static void PrintOnOneLine(const char* str, int max_length) {
06501             if (str != nullptr) {
06502                 for (int i = 0; *str != '\0'; ++str) {
06503                     if (i >= max_length) {
06504                         printf("...");
06505                         break;
06506                     }
06507                     if (*str == '\n') {
06508                         printf("\\n");
06509                         i += 2;
06510                     } else {
06511                         printf("%c", *str);
06512                         ++i;
06513                     }
06514                 }
06515             }
06516         }
06517
```

```
06518         // Prints the names of the tests matching the user-specified filter flag.
06519
06520         void UnitTestImpl::ListTestsMatchingFilter() {
06521             // Print at most this many characters for each type/value parameter.
06522             const int kMaxParamLength = 250;
06523
06524             for (auto* test_suite : test_suites_) {
06525                 bool printed_test_suite_name = false;
06526
06527                 for (size_t j = 0; j < test_suite->test_info_list().size(); j++) {
06528                     const TestInfo * const test_info = test_suite->test_info_list()[j];
06529                     if (test_info->matches_filter_) {
06530                         if (!printed_test_suite_name) {
06531                             printed_test_suite_name = true;
06532                             printf("%s.", test_suite->name());
06533                             if (test_suite->type_param() != nullptr) {
06534                                 printf("  # %s = ", kTypeParamLabel);
06535                                 // We print the type parameter on a single line to make
06536                                 // the output easy to parse by a program.
06537                                 PrintOnOneLine(test_suite->type_param(), kMaxParamLength);
06538                             }
06539                             printf("\n");
06540                         }
06541                         printf("  %s", test_info->name());
06542                         if (test_info->value_param() != nullptr) {
06543                             printf("  # %s = ", kValueParamLabel);
06544                             // We print the value parameter on a single line to make the
06545                             // output easy to parse by a program.
06546                             PrintOnOneLine(test_info->value_param(), kMaxParamLength);
06547                         }
06548                         printf("\n");
06549                     }
06550                 }
06551             }
06552             fflush(stdout);
06553             const std::string& output_format = UnitTestOptions::GetOutputFormat();
06554             if (output_format == "xml" || output_format == "json") {
06555                 FILE* fileout = OpenFileForWriting(
06556                         UnitTestOptions::GetAbsolutePathToOutputFile().c_str());
06557                 std::stringstream stream;
06558                 if (output_format == "xml") {
06559                     XmlUnitTestResultPrinter(
06560                             UnitTestOptions::GetAbsolutePathToOutputFile().c_str())
06561                             .PrintXmlTestsList(&stream, test_suites_);
06562                 } else if (output_format == "json") {
06563                     JsonUnitTestResultPrinter(
06564                             UnitTestOptions::GetAbsolutePathToOutputFile().c_str())
06565                             .PrintJsonTestList(&stream, test_suites_);
06566                 }
06567                 fprintf(fileout, "%s", StringStreamToString(&stream).c_str());
06568                 fclose(fileout);
06569             }
06570         }
06571
06572         // Sets the OS stack trace getter.
06573         //
06574         // Does nothing if the input and the current OS stack trace getter are
06575         // the same; otherwise, deletes the old getter and makes the input the
06576         // current getter.
06577
06578         void UnitTestImpl::set_os_stack_trace_getter(
06579                 OsStackTraceGetterInterface* getter) {
06580             if (os_stack_trace_getter_ != getter) {
06581                 delete os_stack_trace_getter_;
06582                 os_stack_trace_getter_ = getter;
06583             }
06584         }
06585
06586         // Returns the current OS stack trace getter if it is not NULL;
06587         // otherwise, creates an OsStackTraceGetter, makes it the current
06588         // getter, and returns it.
06589
06590         OsStackTraceGetterInterface* UnitTestImpl::os_stack_trace_getter() {
06591             if (os_stack_trace_getter_ == nullptr) {
06592 #ifdef GTEST_OS_STACK_TRACE_GETTER_
06593                 os_stack_trace_getter_ = new GTEST_OS_STACK_TRACE_GETTER_;
06594 #else
06595                 os_stack_trace_getter_ = new OsStackTraceGetter;
06596 #endif  // GTEST_OS_STACK_TRACE_GETTER_
06597             }
06598
06599             return os_stack_trace_getter_;
06600         }
06601
06602         // Returns the most specific TestResult currently running.
06603
06604         TestResult* UnitTestImpl::current_test_result() {
```

```
06605             if (current_test_info_ != nullptr) {
06606                 return &current_test_info_->result_;
06607             }
06608             if (current_test_suite_ != nullptr) {
06609                 return &current_test_suite_->ad_hoc_test_result_;
06610             }
06611             return &ad_hoc_test_result_;
06612         }
06613
06614         // Shuffles all test suites, and the tests within each test suite,
06615         // making sure that death tests are still run first.
06616
06617         void UnitTestImpl::ShuffleTests() {
06618             // Shuffles the death test suites.
06619             ShuffleRange(random(), 0, last_death_test_suite_ + 1, &test_suite_indices_);
06620
06621             // Shuffles the non-death test suites.
06622             ShuffleRange(random(), last_death_test_suite_ + 1,
06623                     static_cast<int> (test_suites_.size()), &test_suite_indices_);
06624
06625             // Shuffles the tests inside each test suite.
06626             for (auto& test_suite : test_suites_) {
06627                 test_suite->ShuffleTests(random());
06628             }
06629         }
06630
06631         // Restores the test suites and tests to their order before the first shuffle.
06632
06633         void UnitTestImpl::UnshuffleTests() {
06634             for (size_t i = 0; i < test_suites_.size(); i++) {
06635                 // Unshuffles the tests in each test suite.
06636                 test_suites_[i]->UnshuffleTests();
06637                 // Resets the index of each test suite.
06638                 test_suite_indices_[i] = static_cast<int> (i);
06639             }
06640         }
06641
06642         // Returns the current OS stack trace as an std::string.
06643         //
06644         // The maximum number of stack frames to be included is specified by
06645         // the gtest_stack_trace_depth flag.  The skip_count parameter
06646         // specifies the number of top frames to be skipped, which doesn't
06647         // count against the number of frames to be included.
06648         //
06649         // For example, if Foo() calls Bar(), which in turn calls
06650         // GetCurrentOsStackTraceExceptTop(..., 1), Foo() will be included in
06651         // the trace but Bar() and GetCurrentOsStackTraceExceptTop() won't.
06652
06653         GTEST_NO_INLINE_ GTEST_NO_TAIL_CALL_ std::string
06654         GetCurrentOsStackTraceExceptTop(int skip_count) {
06655             // We pass skip_count + 1 to skip this wrapper function in addition
06656             // to what the user really wants to skip.
06657             return GetUnitTestImpl()->CurrentOsStackTraceExceptTop(skip_count + 1);
06658         }
06659
06660         // Used by the GTEST_SUPPRESS_UNREACHABLE_CODE_WARNING_BELOW_ macro to
06661         // suppress unreachable code warnings.
06662         namespace {
06663
06664             class ClassUniqueToAlwaysTrue {
06665             };
06666         } // namespace
06667
06668         bool IsTrue(bool condition) {
06669             return condition;
06670         }
06671
06672         bool AlwaysTrue() {
06673 #if GTEST_HAS_EXCEPTIONS
06674             // This condition is always false so AlwaysTrue() never actually throws,
06675             // but it makes the compiler think that it may throw.
06676             if (IsTrue(false)) throw ClassUniqueToAlwaysTrue();
06677 #endif  // GTEST_HAS_EXCEPTIONS
06678             return true;
06679         }
06680
06681         // If *pstr starts with the given prefix, modifies *pstr to be right
06682         // past the prefix and returns true; otherwise leaves *pstr unchanged
06683         // and returns false.  None of pstr, *pstr, and prefix can be NULL.
06684
06685         bool SkipPrefix(const char* prefix, const char** pstr) {
06686             const size_t prefix_len = strlen(prefix);
06687             if (strncmp(*pstr, prefix, prefix_len) == 0) {
06688                 *pstr += prefix_len;
06689                 return true;
06690             }
06691             return false;
```

```
06692         }
06693
06694         // Parses a string as a command line flag.  The string should have
06695         // the format "--flag=value".  When def_optional is true, the "=value"
06696         // part can be omitted.
06697         //
06698         // Returns the value of the flag, or NULL if the parsing failed.
06699
06700         static const char* ParseFlagValue(const char* str, const char* flag_name,
06701                 bool def_optional) {
06702             // str and flag must not be NULL.
06703             if (str == nullptr || flag_name == nullptr) return nullptr;
06704
06705             // The flag must start with "--" followed by GTEST_FLAG_PREFIX_.
06706             const std::string flag_str =
06707                     std::string("--") + GTEST_FLAG_PREFIX_ + flag_name;
06708             const size_t flag_len = flag_str.length();
06709             if (strncmp(str, flag_str.c_str(), flag_len) != 0) return nullptr;
06710
06711             // Skips the flag name.
06712             const char* flag_end = str + flag_len;
06713
06714             // When def_optional is true, it's OK to not have a "=value" part.
06715             if (def_optional && (flag_end[0] == '\0')) {
06716                 return flag_end;
06717             }
06718
06719             // If def_optional is true and there are more characters after the
06720             // flag name, or if def_optional is false, there must be a '=' after
06721             // the flag name.
06722             if (flag_end[0] != '=') return nullptr;
06723
06724             // Returns the string after "=".
06725             return flag_end + 1;
06726         }
06727
06728         // Parses a string for a bool flag, in the form of either
06729         // "--flag=value" or "--flag".
06730         //
06731         // In the former case, the value is taken as true as long as it does
06732         // not start with '0', 'f', or 'F'.
06733         //
06734         // In the latter case, the value is taken as true.
06735         //
06736         // On success, stores the value of the flag in *value, and returns
06737         // true.  On failure, returns false without changing *value.
06738
06739         static bool ParseFlag(const char* str, const char* flag_name, bool* value) {
06740             // Gets the value of the flag as a string.
06741             const char* const value_str = ParseFlagValue(str, flag_name, true);
06742
06743             // Aborts if the parsing failed.
06744             if (value_str == nullptr) return false;
06745
06746             // Converts the string value to a bool.
06747             *value = !(*value_str == '0' || *value_str == 'f' || *value_str == 'F');
06748             return true;
06749         }
06750
06751         // Parses a string for an int32_t flag, in the form of "--flag=value".
06752         //
06753         // On success, stores the value of the flag in *value, and returns
06754         // true.  On failure, returns false without changing *value.
06755
06756         bool ParseFlag(const char* str, const char* flag_name, int32_t* value) {
06757             // Gets the value of the flag as a string.
06758             const char* const value_str = ParseFlagValue(str, flag_name, false);
06759
06760             // Aborts if the parsing failed.
06761             if (value_str == nullptr) return false;
06762
06763             // Sets *value to the value of the flag.
06764             return ParseInt32(Message() << "The value of flag --" << flag_name, value_str,
06765                     value);
06766         }
06767
06768         // Parses a string for a string flag, in the form of "--flag=value".
06769         //
06770         // On success, stores the value of the flag in *value, and returns
06771         // true.  On failure, returns false without changing *value.
06772
06773         template <typename String>
06774         static bool ParseFlag(const char* str, const char* flag_name, String* value) {
06775             // Gets the value of the flag as a string.
06776             const char* const value_str = ParseFlagValue(str, flag_name, false);
06777
06778             // Aborts if the parsing failed.
```

```
06779             if (value_str == nullptr) return false;
06780
06781             // Sets *value to the value of the flag.
06782             *value = value_str;
06783             return true;
06784         }
06785
06786         // Determines whether a string has a prefix that Google Test uses for its
06787         // flags, i.e., starts with GTEST_FLAG_PREFIX_ or GTEST_FLAG_PREFIX_DASH_.
06788         // If Google Test detects that a command line flag has its prefix but is not
06789         // recognized, it will print its help message. Flags starting with
06790         // GTEST_INTERNAL_PREFIX_ followed by "internal_" are considered Google Test
06791         // internal flags and do not trigger the help message.
06792
06793         static bool HasGoogleTestFlagPrefix(const char* str) {
06794             return (SkipPrefix("--", &str) || SkipPrefix("-", &str) ||
06795                     SkipPrefix("/", &str)) &&
06796                     !SkipPrefix(GTEST_FLAG_PREFIX_ "internal_", &str) &&
06797                     (SkipPrefix(GTEST_FLAG_PREFIX_, &str) ||
06798                     SkipPrefix(GTEST_FLAG_PREFIX_DASH_, &str));
06799         }
06800
06801         // Prints a string containing code-encoded text.  The following escape
06802         // sequences can be used in the string to control the text color:
06803         //
06804         //   @@    prints a single '@' character.
06805         //   @R    changes the color to red.
06806         //   @G    changes the color to green.
06807         //   @Y    changes the color to yellow.
06808         //   @D    changes to the default terminal text color.
06809         //
06810
06811         static void PrintColorEncoded(const char* str) {
06812             GTestColor color = GTestColor::kDefault; // The current color.
06813
06814             // Conceptually, we split the string into segments divided by escape
06815             // sequences.  Then we print one segment at a time.  At the end of
06816             // each iteration, the str pointer advances to the beginning of the
06817             // next segment.
06818             for (;;) {
06819                 const char* p = strchr(str, '@');
06820                 if (p == nullptr) {
06821                     ColoredPrintf(color, "%s", str);
06822                     return;
06823                 }
06824
06825                 ColoredPrintf(color, "%s", std::string(str, p).c_str());
06826
06827                 const char ch = p[1];
06828                 str = p + 2;
06829                 if (ch == '@') {
06830                     ColoredPrintf(color, "@");
06831                 } else if (ch == 'D') {
06832                     color = GTestColor::kDefault;
06833                 } else if (ch == 'R') {
06834                     color = GTestColor::kRed;
06835                 } else if (ch == 'G') {
06836                     color = GTestColor::kGreen;
06837                 } else if (ch == 'Y') {
06838                     color = GTestColor::kYellow;
06839                 } else {
06840                     --str;
06841                 }
06842             }
06843         }
06844
06845         static const char kColorEncodedHelpMessage[] =
06846                 "This program contains tests written using " GTEST_NAME_
06847                 ". You can use the\n"
06848                 "following command line flags to control its behavior:\n"
06849                 "\n"
06850                 "Test Selection:\n"
06851                 "  @G--" GTEST_FLAG_PREFIX_
06852                 "list_tests@D\n"
06853                 "      List the names of all tests instead of running them. The name of\n"
06854                 "      TEST(Foo, Bar) is \"Foo.Bar\".\n"
06855                 "  @G--" GTEST_FLAG_PREFIX_
06856                 "filter=@YPOSITIVE_PATTERNS"
06857                 "[@G-@YNEGATIVE_PATTERNS]@D\n"
06858                 "      Run only the tests whose name matches one of the positive patterns "
06859                 "but\n"
06860                 "      none of the negative patterns. '?' matches any single character; "
06861                 "'*'\n"
06862                 "      matches any substring; ':' separates two patterns.\n"
06863                 "  @G--" GTEST_FLAG_PREFIX_
06864                 "also_run_disabled_tests@D\n"
06865                 "      Run all disabled tests too.\n"
```

```
06866                 "\n"
06867                 "Test Execution:\n"
06868                 "  @G--" GTEST_FLAG_PREFIX_
06869                 "repeat=@Y[COUNT]@D\n"
06870                 "      Run the tests repeatedly; use a negative count to repeat forever.\n"
06871                 "  @G--" GTEST_FLAG_PREFIX_
06872                 "shuffle@D\n"
06873                 "      Randomize tests' orders on every iteration.\n"
06874                 "  @G--" GTEST_FLAG_PREFIX_
06875                 "random_seed=@Y[NUMBER]@D\n"
06876                 "      Random number seed to use for shuffling test orders (between 1 and\n"
06877                 "      99999, or 0 to use a seed based on the current time).\n"
06878                 "  @G--" GTEST_FLAG_PREFIX_
06879                 "recreate_environments_when_repeating@D\n"
06880                 "      Sets up and tears down the global test environment on each repeat\n"
06881                 "      of the test.\n"
06882                 "\n"
06883                 "Test Output:\n"
06884                 "  @G--" GTEST_FLAG_PREFIX_
06885                 "color=@Y(@Gyes@Y|@Gno@Y|@Gauto@Y)@D\n"
06886                 "      Enable/disable colored output. The default is @Gauto@D.\n"
06887                 "  @G--" GTEST_FLAG_PREFIX_
06888                 "brief=1@D\n"
06889                 "      Only print test failures.\n"
06890                 "  @G--" GTEST_FLAG_PREFIX_
06891                 "print_time=0@D\n"
06892                 "      Don't print the elapsed time of each test.\n"
06893                 "  @G--" GTEST_FLAG_PREFIX_
06894                 "output=@Y(@Gjson@Y|@Gxml@Y)[@G:@YDIRECTORY_PATH@G" GTEST_PATH_SEP_
06895                 "@Y|@G:@YFILE_PATH]@D\n"
06896                 "      Generate a JSON or XML report in the given directory or with the "
06897                 "given\n"
06898                 "      file name. @YFILE_PATH@D defaults to @Gtest_detail.xml@D.\n"
06899 #if GTEST_CAN_STREAM_RESULTS_
06900                 "  @G--" GTEST_FLAG_PREFIX_
06901                 "stream_result_to=@YHOST@G:@YPORT@D\n"
06902                 "      Stream test results to the given server.\n"
06903 #endif  // GTEST_CAN_STREAM_RESULTS_
06904                 "\n"
06905                 "Assertion Behavior:\n"
06906 #if GTEST_HAS_DEATH_TEST && !GTEST_OS_WINDOWS
06907                 "  @G--" GTEST_FLAG_PREFIX_
06908                 "death_test_style=@Y(@Gfast@Y|@Gthreadsafe@Y)@D\n"
06909                 "      Set the default death test style.\n"
06910 #endif  // GTEST_HAS_DEATH_TEST && !GTEST_OS_WINDOWS
06911                 "  @G--" GTEST_FLAG_PREFIX_
06912                 "break_on_failure@D\n"
06913                 "      Turn assertion failures into debugger break-points.\n"
06914                 "  @G--" GTEST_FLAG_PREFIX_
06915                 "throw_on_failure@D\n"
06916                 "      Turn assertion failures into C++ exceptions for use by an external\n"
06917                 "      test framework.\n"
06918                 "  @G--" GTEST_FLAG_PREFIX_
06919                 "catch_exceptions=0@D\n"
06920                 "      Do not report exceptions as test failures. Instead, allow them\n"
06921                 "      to crash the program or throw a pop-up (on Windows).\n"
06922                 "\n"
06923                 "Except for @G--" GTEST_FLAG_PREFIX_
06924                 "list_tests@D, you can alternatively set "
06925                 "the corresponding\n"
06926                 "environment variable of a flag (all letters in upper-case). For example, "
06927                 "to\n"
06928                 "disable colored text output, you can either specify "
06929                 "@G--" GTEST_FLAG_PREFIX_
06930                 "color=no@D or set\n"
06931                 "the @G" GTEST_FLAG_PREFIX_UPPER_
06932                 "COLOR@D environment variable to @Gno@D.\n"
06933                 "\n"
06934                 "For more information, please read the " GTEST_NAME_
06935                 " documentation at\n"
06936                 "@G" GTEST_PROJECT_URL_ "@D. If you find a bug in " GTEST_NAME_
06937                 "\n"
06938                 "(not one in your own code or tests), please report it to\n"
06939                 "@G<" GTEST_DEV_EMAIL_ ">@D.\n";
06940
06941         static bool ParseGoogleTestFlag(const char* const arg) {
06942 #define GTEST_INTERNAL_PARSE_FLAG(flag_name)  \
06943   do {                                        \
06944     auto value = GTEST_FLAG_GET(flag_name);   \
06945     if (ParseFlag(arg, #flag_name, &value)) { \
06946       GTEST_FLAG_SET(flag_name, value);       \
06947       return true;                            \
06948     }                                         \
06949   } while (false)
06950
06951             GTEST_INTERNAL_PARSE_FLAG(also_run_disabled_tests);
06952             GTEST_INTERNAL_PARSE_FLAG(break_on_failure);
```

```
06953             GTEST_INTERNAL_PARSE_FLAG(catch_exceptions);
06954             GTEST_INTERNAL_PARSE_FLAG(color);
06955             GTEST_INTERNAL_PARSE_FLAG(death_test_style);
06956             GTEST_INTERNAL_PARSE_FLAG(death_test_use_fork);
06957             GTEST_INTERNAL_PARSE_FLAG(fail_fast);
06958             GTEST_INTERNAL_PARSE_FLAG(filter);
06959             GTEST_INTERNAL_PARSE_FLAG(internal_run_death_test);
06960             GTEST_INTERNAL_PARSE_FLAG(list_tests);
06961             GTEST_INTERNAL_PARSE_FLAG(output);
06962             GTEST_INTERNAL_PARSE_FLAG(brief);
06963             GTEST_INTERNAL_PARSE_FLAG(print_time);
06964             GTEST_INTERNAL_PARSE_FLAG(print_utf8);
06965             GTEST_INTERNAL_PARSE_FLAG(random_seed);
06966             GTEST_INTERNAL_PARSE_FLAG(repeat);
06967             GTEST_INTERNAL_PARSE_FLAG(recreate_environments_when_repeating);
06968             GTEST_INTERNAL_PARSE_FLAG(shuffle);
06969             GTEST_INTERNAL_PARSE_FLAG(stack_trace_depth);
06970             GTEST_INTERNAL_PARSE_FLAG(stream_result_to);
06971             GTEST_INTERNAL_PARSE_FLAG(throw_on_failure);
06972             return false;
06973         }
06974
06975 #if GTEST_USE_OWN_FLAGFILE_FLAG_
06976
06977         static void LoadFlagsFromFile(const std::string& path) {
06978             FILE* flagfile = posix::FOpen(path.c_str(), "r");
06979             if (!flagfile) {
06980                 GTEST_LOG_(FATAL) << "Unable to open file \"" << GTEST_FLAG_GET(flagfile)
06981                         << "\"";
06982             }
06983             std::string contents(ReadEntireFile(flagfile));
06984             posix::FClose(flagfile);
06985             std::vector<std::string> lines;
06986             SplitString(contents, '\n', &lines);
06987             for (size_t i = 0; i < lines.size(); ++i) {
06988                 if (lines[i].empty()) continue;
06989                 if (!ParseGoogleTestFlag(lines[i].c_str())) g_help_flag = true;
06990             }
06991         }
06992 #endif  // GTEST_USE_OWN_FLAGFILE_FLAG_
06993
06994         // Parses the command line for Google Test flags, without initializing
06995         // other parts of Google Test.  The type parameter CharType can be
06996         // instantiated to either char or wchar_t.
06997
06998         template <typename CharType>
06999         void ParseGoogleTestFlagsOnlyImpl(int* argc, CharType** argv) {
07000             std::string flagfile_value;
07001             for (int i = 1; i < *argc; i++) {
07002                 const std::string arg_string = StreamableToString(argv[i]);
07003                 const char* const arg = arg_string.c_str();
07004
07005                 using internal::ParseFlag;
07006
07007                 bool remove_flag = false;
07008                 if (ParseGoogleTestFlag(arg)) {
07009                     remove_flag = true;
07010 #if GTEST_USE_OWN_FLAGFILE_FLAG_
07011                 } else if (ParseFlag(arg, "flagfile", &flagfile_value)) {
07012                     GTEST_FLAG_SET(flagfile, flagfile_value);
07013                     LoadFlagsFromFile(flagfile_value);
07014                     remove_flag = true;
07015 #endif  // GTEST_USE_OWN_FLAGFILE_FLAG_
07016                 } else if (arg_string == "--help" || HasGoogleTestFlagPrefix(arg)) {
07017                     // Both help flag and unrecognized Google Test flags (excluding
07018                     // internal ones) trigger help display.
07019                     g_help_flag = true;
07020                 }
07021
07022                 if (remove_flag) {
07023                     // Shift the remainder of the argv list left by one.  Note
07024                     // that argv has (*argc + 1) elements, the last one always being
07025                     // NULL.  The following loop moves the trailing NULL element as
07026                     // well.
07027                     for (int j = i; j != *argc; j++) {
07028                         argv[j] = argv[j + 1];
07029                     }
07030
07031                     // Decrements the argument count.
07032                     (*argc)--;
07033
07034                     // We also need to decrement the iterator as we just removed
07035                     // an element.
07036                     i--;
07037                 }
07038             }
07039
```

```
07040             if (g_help_flag) {
07041                 // We print the help here instead of in RUN_ALL_TESTS(), as the
07042                 // latter may not be called at all if the user is using Google
07043                 // Test with another testing framework.
07044                 PrintColorEncoded(kColorEncodedHelpMessage);
07045             }
07046         }
07047 
07048         // Parses the command line for Google Test flags, without initializing
07049         // other parts of Google Test.
07050 
07051         void ParseGoogleTestFlagsOnly(int* argc, char** argv) {
07052 #if GTEST_HAS_ABSL
07053             if (*argc > 0) {
07054                 // absl::ParseCommandLine() requires *argc > 0.
07055                 auto positional_args = absl::flags_internal::ParseCommandLineImpl(
07056                         *argc, argv, absl::flags_internal::ArgvListAction::kRemoveParsedArgs,
07057                         absl::flags_internal::UsageFlagsAction::kHandleUsage,
07058                         absl::flags_internal::OnUndefinedFlag::kReportUndefined);
07059                 // Any command-line positional arguments not part of any command-line flag
07060                 // (or arguments to a flag) are copied back out to argv, with the program
07061                 // invocation name at position 0, and argc is resized. This includes
07062                 // positional arguments after the flag-terminating delimiter '--'.
07063                 // See https://abseil.io/docs/cpp/guides/flags.
07064                 std::copy(positional_args.begin(), positional_args.end(), argv);
07065                 if (static_cast<int> (positional_args.size()) < *argc) {
07066                     argv[positional_args.size()] = nullptr;
07067                     *argc = static_cast<int> (positional_args.size());
07068                 }
07069             }
07070 #else
07071             ParseGoogleTestFlagsOnlyImpl(argc, argv);
07072 #endif
07073 
07074             // Fix the value of *_NSGetArgc() on macOS, but if and only if
07075             // *_NSGetArgv() == argv
07076             // Only applicable to char** version of argv
07077 #if GTEST_OS_MAC
07078 #ifndef GTEST_OS_IOS
07079             if (*_NSGetArgv() == argv) {
07080                 *_NSGetArgc() = *argc;
07081             }
07082 #endif
07083 #endif
07084         }
07085 
07086         void ParseGoogleTestFlagsOnly(int* argc, wchar_t** argv) {
07087             ParseGoogleTestFlagsOnlyImpl(argc, argv);
07088         }
07089 
07090         // The internal implementation of InitGoogleTest().
07091         //
07092         // The type parameter CharType can be instantiated to either char or
07093         // wchar_t.
07094 
07095         template <typename CharType>
07096         void InitGoogleTestImpl(int* argc, CharType** argv) {
07097             // We don't want to run the initialization code twice.
07098             if (GTestIsInitialized()) return;
07099 
07100             if (*argc <= 0) return;
07101 
07102             g_argvs.clear();
07103             for (int i = 0; i != *argc; i++) {
07104                 g_argvs.push_back(StreamableToString(argv[i]));
07105             }
07106 
07107 #if GTEST_HAS_ABSL
07108             absl::InitializeSymbolizer(g_argvs[0].c_str());
07109 
07110             // When using the Abseil Flags library, set the program usage message to the
07111             // help message, but remove the color-encoding from the message first.
07112             absl::SetProgramUsageMessage(absl::StrReplaceAll(
07113                     kColorEncodedHelpMessage,{
07114                 {"@D", ""},
07115                 {"@R", ""},
07116                 {"@G", ""},
07117                 {"@Y", ""},
07118                 {"@@", "@"}
07119             }));
07120 #endif  // GTEST_HAS_ABSL
07121 
07122             ParseGoogleTestFlagsOnly(argc, argv);
07123             GetUnitTestImpl()->PostFlagParsingInit();
07124         }
07125 
07126     } // namespace internal
```

```
07127
07128     // Initializes Google Test.  This must be called before calling
07129     // RUN_ALL_TESTS().  In particular, it parses a command line for the
07130     // flags that Google Test recognizes.  Whenever a Google Test flag is
07131     // seen, it is removed from argv, and *argc is decremented.
07132     //
07133     // No value is returned.  Instead, the Google Test flag variables are
07134     // updated.
07135     //
07136     // Calling the function for the second time has no user-visible effect.
07137
07138     void InitGoogleTest(int* argc, char** argv) {
07139 #if defined(GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_)
07140         GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_(argc, argv);
07141 #else   // defined(GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_)
07142         internal::InitGoogleTestImpl(argc, argv);
07143 #endif  // defined(GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_)
07144     }
07145
07146     // This overloaded version can be used in Windows programs compiled in
07147     // UNICODE mode.
07148
07149     void InitGoogleTest(int* argc, wchar_t** argv) {
07150 #if defined(GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_)
07151         GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_(argc, argv);
07152 #else   // defined(GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_)
07153         internal::InitGoogleTestImpl(argc, argv);
07154 #endif  // defined(GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_)
07155     }
07156
07157     // This overloaded version can be used on Arduino/embedded platforms where
07158     // there is no argc/argv.
07159
07160     void InitGoogleTest() {
07161         // Since Arduino doesn't have a command line, fake out the argc/argv arguments
07162         int argc = 1;
07163         const auto arg0 = "dummy";
07164         char* argv0 = const_cast<char*> (arg0);
07165         char** argv = &argv0;
07166
07167 #if defined(GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_)
07168         GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_(&argc, argv);
07169 #else   // defined(GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_)
07170         internal::InitGoogleTestImpl(&argc, argv);
07171 #endif  // defined(GTEST_CUSTOM_INIT_GOOGLE_TEST_FUNCTION_)
07172     }
07173
07174 #if !defined(GTEST_CUSTOM_TEMPDIR_FUNCTION_)
07175     // Return value of first environment variable that is set and contains
07176     // a non-empty string. If there are none, return the "fallback" string.
07177     // Since we like the temporary directory to have a directory separator suffix,
07178     // add it if not provided in the environment variable value.
07179
07180     static std::string GetTempDirFromEnv(
07181             std::initializer_list<const char*> environment_variables,
07182             const char* fallback, char separator) {
07183         for (const char* variable_name : environment_variables) {
07184             const char* value = internal::posix::GetEnv(variable_name);
07185             if (value != nullptr && value[0] != '\0') {
07186                 if (value[strlen(value) - 1] != separator) {
07187                     return std::string(value).append(1, separator);
07188                 }
07189                 return value;
07190             }
07191         }
07192         return fallback;
07193     }
07194 #endif
07195
07196     std::string TempDir() {
07197 #if defined(GTEST_CUSTOM_TEMPDIR_FUNCTION_)
07198         return GTEST_CUSTOM_TEMPDIR_FUNCTION_();
07199 #elif GTEST_OS_WINDOWS || GTEST_OS_WINDOWS_MOBILE
07200         return GetTempDirFromEnv({"TEST_TMPDIR", "TEMP"}, "\\temp\\", '\\');
07201 #elif GTEST_OS_LINUX_ANDROID
07202         return GetTempDirFromEnv({"TEST_TMPDIR", "TMPDIR"}, "/data/local/tmp/", '/');
07203 #else
07204         return GetTempDirFromEnv({"TEST_TMPDIR", "TMPDIR"}, "/tmp/", '/');
07205 #endif
07206     }
07207
07208     // Class ScopedTrace
07209
07210     // Pushes the given source file location and message onto a per-thread
07211     // trace stack maintained by Google Test.
07212
07213     void ScopedTrace::PushTrace(const char* file, int line, std::string message) {
```

```
07214         internal::TraceInfo trace;
07215         trace.file = file;
07216         trace.line = line;
07217         trace.message.swap(message);
07218
07219         UnitTest::GetInstance()->PushGTestTrace(trace);
07220     }
07221
07222     // Pops the info pushed by the c'tor.
07223
07224     ScopedTrace::~ScopedTrace() GTEST_LOCK_EXCLUDED_(&UnitTest::mutex_) {
07225         UnitTest::GetInstance()->PopGTestTrace();
07226     }
07227
07228 } // namespace testing
```

18.292 ConnectionManager.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ConnectionManager.h
00009  * Author: rlcancian
00010  *
00011  * Created on 1 de Julho de 2019, 18:39
00012  */
00013
00014 #ifndef CONNECTIONMANAGER_H
00015 #define CONNECTIONMANAGER_H
00016
00017 #include <utility>
00018 #include <map>
00019 #include "../util/List.h"
00020
00021 //namespace GenesysKernel {
00022
00023 class ModelComponent;
00024
00025 //typedef unsigned int ConnectionPort;
00029 struct ConnectionChannel { /*< actually a port is only a uint, but it could be more complex, including
      type, presence or absence of data, etc */
00030     unsigned int portNumber = 0;
00031     std::string portDescription = "";
00032 };
00033
00034
00040 struct Connection {
00041     ModelComponent* component;
00042     ConnectionChannel channel;
00043 };
00044
00051 class ConnectionManager {
00052 public:
00054     ConnectionManager();
00055     virtual ~ConnectionManager();
00056 public:
00061     unsigned int size();
00066     Connection* getFrontConnection();
00072     Connection* getConnectionAtPort(unsigned int rank);
00081     void insert(ModelComponent* component, unsigned int inputPortNumber = 0);
00087     void insert(Connection* connection);
00094     void insertAtPort(unsigned int port, Connection* connection);
00100     void remove(Connection* connection);
00106     void removeAtPort(unsigned int port);
00111     std::map<unsigned int, Connection*>* connections() const;
00112     //void setCurrentOutputConnections(unsigned int _currentOutputConnections);
00117     unsigned int getCurrentOutputConnectionsSize() const;
00122     void setMaxOutputConnections(unsigned int _maxOutputConnections);
00127     unsigned int getMaxOutputConnections() const;
00132     void setMinOutputConnections(unsigned int _minOutputConnections);
00137     unsigned int getMinOutputConnections() const;
00138     //void setCurrentInputConnections(unsigned int _currentInputConnections);
00143     unsigned int getCurrentInputConnectionsSize() const;
00148     void setMaxInputConnections(unsigned int _maxInputConnections);
00153     unsigned int getMaxInputConnections() const;
00158     void setMinInputConnections(unsigned int _minInputConnections);
00163     unsigned int getMinInputConnections() const;
00164 private:
00165     std::map<unsigned int, Connection*>* _nextConnections = new std::map<unsigned int, Connection*>();
00166     unsigned int _minInputConnections = 0;
00167     unsigned int _maxInputConnections = 0;
```

```
00168     unsigned int _currentInputConnections = 0;
00169     unsigned int _minOutputConnections = 0;
00170     unsigned int _maxOutputConnections = 0;
00171 };
00172 //namespace\\}
00173 #endif /* CONNECTIONMANAGER_H */
```

18.293 DefineGetterSetter.h

```
00001 #ifndef DEFINEGETTERSETTER_H
00002 #define DEFINEGETTERSETTER_H
00003
00004 #endif
```

18.294 Attribute.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Attribute.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 25 de Setembro de 2018, 16:37
00012  */
00013
00014 #ifndef ATTRIBUTE_H
00015 #define ATTRIBUTE_H
00016
00017 #include <string>
00018 #include <list>
00019 #include <map>
00020 #include <utility>
00021 #include <vector>
00022 #include "../model/ModelDataDefinition.h"
00023 #include "../model/ModelDataManager.h"
00024 #include "../PluginInformation.h"
00025 #include "SparseValueStore.h"
00026
00027 //namespace GenesysKernel {
00028
00049 class Attribute : public ModelDataDefinition {
00050 public:
00052     Attribute(Model* model, std::string name = "", std::string dataDefinitionTypename =
      Util::TypeOf<Attribute>());
00054     virtual ~Attribute() override;
00055 public:
00056     virtual std::string show() override;
00057 public: // public static methods
00058     static PluginInformation* GetPluginInformation();
00059     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00060     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00061 public:
00063     virtual std::string getInitialValuesText() const;
00065     virtual void setInitialValuesText(std::string valuesText);
00067     double getInitialValue(std::string index = "");
00069     void setInitialValue(double value, std::string index = "");
00071     void setInitialValues(const std::vector<std::pair<std::string,double» values);
00073     void copyInitialValuesFrom(const Attribute& source);
00075     void insertDimentionSize(unsigned int size);
00077     std::list<unsigned int>* getDimensionSizes() const;
00079     std::map<std::string, double>* getInitialValues() const;
00081     SparseValueStore* getInitialValueStore();
00083     bool isInitialValuesTextValid() const;
00084 protected:
00085     virtual bool _loadInstance(PersistenceRecord *fields) override;
00086     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00087 protected:
00088     virtual bool _check(std::string& errorMessage) override;
00089 private:
00090     void _syncInitialValuesTextFromStore();
00091     SparseValueStore* _initialValues = new SparseValueStore();
00092     std::string _initialValuesText = "0";
00093     bool _initialValuesTextValid = true;
00094     std::string _initialValuesTextErrorMessage;
00095 };
00096 //namespace\\}
00097 #endif /* ATTRIBUTE_H */
```

18.295 Counter.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   CounterDefaultImpl1.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 29 de Maio de 2019, 11:24
00012  */
00013
00014 #ifndef COUNTERDEFAULTIMPL1_H
00015 #define COUNTERDEFAULTIMPL1_H
00016
00017 #include "../model/ModelDataDefinition.h"
00018 #include "../model/ModelDataManager.h"
00019 #include "../Plugin.h"
00020 //namespace GenesysKernel {
00021
00028 class Counter : public ModelDataDefinition {
00029 public:
00031     Counter(Model* model, std::string name = "", ModelDataDefinition* parent = nullptr);
00032     virtual ~Counter() override;
00033 public:
00034     virtual std::string show() override;
00035 public: // public static methods
00036     static PluginInformation* GetPluginInformation();
00037     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00038     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00039 public:
00044     void clear();
00050     void incCountValue(/*int*/double value = 1.0);
00055     double /*unsigned long*/ getCountValue() const;
00060     ModelDataDefinition* getParent() const;
00061 protected:
00062     virtual bool _loadInstance(PersistenceRecord *fields) override;
00063     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00064 protected:
00065     virtual bool _check(std::string& errorMessage) override;
00066     virtual void _initBetweenReplications() override;
00067 private:
00068     ModelDataDefinition* _parent;
00069     double /*unsigned long*/ _count = 0;
00070     SimulationResponse* _countValueResponse = nullptr;
00071 };
00072 //namespace\}
00073 #endif /* COUNTERDEFAULTIMPL1_H */
```

18.296 Entity.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Entity.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Junho de 2018, 16:30
00012  */
00013
00014 #ifndef ENTITY_H
00015 #define ENTITY_H
00016
00017 #include <string>
00018 #include <map>
00019
00020 #include "../../util/Util.h"
00021 #include "../../util/List.h"
00022 #include "../model/ModelDataDefinition.h"
00023 #include "EntityType.h"
00024 #include "SparseValueStore.h"
00025 //namespace GenesysKernel {
00026
00047 class Entity : public ModelDataDefinition {
00048 private: // no one can create or destry entities directlly. This can be done one throught friend class
      Model
00050     Entity(Model* model, std::string name = "", bool insertIntoModel = true);
```

```
00051     virtual ~Entity();
00052     // friend Entity* Model::createEntity(std::string name, bool insertIntoModel); // It would be
      better, but Model is not known at this point of compilaton
00053     friend class Model;
00054 public:
00055     virtual std::string show() override;
00056
00057 public: // g & s
00063     void setEntityTypeName(std::string entityTypeName); //*!< indirect access to EntityType
00068     std::string getEntityTypeName() const;
00074     void setEntityType(EntityType* entityType); //*!< direct access to EntityType
00079     EntityType* getEntityType() const;
00080 public:
00087     double getAttributeValue(std::string attributeName, std::string index="");
00094     double getAttributeValue(Util::identification attributeID, std::string index="");
00103     void setAttributeValue(std::string attributeName, double value, std::string index="", bool
      createIfNotFound=false);
00111     void setAttributeValue(Util::identification attributeID, double value, std::string index="");
00113     void copyAttributeValues(std::string destinationAttributeName, std::string sourceAttributeName);
00118     Util::identification entityNumber() const;
00119 protected:
00120     virtual bool _loadInstance(PersistenceRecord *fields) override;
00121     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00122     virtual bool _check(std::string& errorMessage) override;
00123 private:
00125     SparseValueStore* _ensureAttributeStore(unsigned int rank);
00126
00127     Util::identification _entityNumber;
00128     EntityType* _entityType = nullptr;
00129     List<SparseValueStore*>* _attributeValues = new List<SparseValueStore*>();
00130 };
00131 //namespace\\}
00132 #endif /* ENTITY_H */
```

18.297 EntityType.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   EntityType.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 14 de Agosto de 2018, 19:24
00012  */
00013
00014 #ifndef ENTITYTYPE_H
00015 #define ENTITYTYPE_H
00016
00017 #include <string>
00018 #include "../model/ModelDataDefinition.h"
00019 #include "StatisticsCollector.h"
00020 #include "../model/ModelDataManager.h"
00021 #include "../Plugin.h"
00022
00023 //#include "Model.h"
00024 //namespace GenesysKernel {
00025
00046 class EntityType : public ModelDataDefinition {
00047 public:
00048     EntityType(Model* model, std::string name = "");
00049     virtual ~EntityType();
00050 public:
00051     virtual std::string show() override;
00052 public: //static
00053     static PluginInformation* GetPluginInformation();
00054     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00055     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00056 public: //get & set
00057     // @ToDo: (importante): Change categories Waiting, Others, Transfer, etc to customizable ones
00058     void setInitialWaitingCost(double _initialWaitingCost);
00059     double initialWaitingCost() const;
00060     void setInitialOtherCost(double _initialOtherCost);
00061     double initialOtherCost() const;
00062     void setInitialNVACost(double _initialNVACost);
00063     double initialNVACost() const;
00064     void setInitialVACost(double _initialVACost);
00065     double initialVACost() const;
00066     void setInitialPicture(std::string _initialPicture);
00067     std::string initialPicture() const;
00068 public: //get
00074     StatisticsCollector* addGetStatisticsCollector(std::string name);
```

```
00075
00076 protected: // must be overriden
00077     virtual bool _loadInstance(PersistenceRecord *fields) override;
00078     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00079 protected: // could be ovveriden
00080     //virtual bool _check(std::string& errorMessage) override;
00081     virtual void _initBetweenReplications() override;
00082     virtual void _createInternalStatisticReporters() override;
00083 private:
00084     void _initCostsAndStatistics();
00085 private:
00086
00087     const struct DEFAULT_VALUES {
00088         const std::string initialPicture = "report";
00089         const double initialCost = 0.0;
00090     } DEFAULT;
00091     std::string _initialPicture = DEFAULT.initialPicture;
00092     double _initialVACost = DEFAULT.initialCost;
00093     double _initialNVACost = DEFAULT.initialCost;
00094     double _initialOtherCost = DEFAULT.initialCost;
00095     double _initialWaitingCost = DEFAULT.initialCost;
00096 private: //1:n
00097     List<StatisticsCollector*>* _statisticsCollectors = new List<StatisticsCollector*>();
00098 };
00099 //namespace\\}
00100 #endif /* ENTITYTYPE_H */
```

18.298 SparseValueStore.h

```
00001 /*
00002  * File:   SparseValueStore.h
00003  * Author: rafael.luiz.cancian
00004  *
00005  * Created on 16 de Abril de 2026
00006  */
00007
00008 #ifndef SPARSEVALUESTORE_H
00009 #define SPARSEVALUESTORE_H
00010
00011 #include "../Persistence.h"
00012 #include "../../util/Util.h"
00013
00014 #include <list>
00015 #include <map>
00016 #include <algorithm>
00017 #include <cctype>
00018 #include <sstream>
00019 #include <string>
00020 #include <utility>
00021 #include <vector>
00022
00032 class SparseValueStore {
00033 public:
00035     static std::string makeIndexKey(const std::vector<unsigned int>& indexes);
00036
00038     static std::string makeIndexKeyFromDoubles(const std::vector<double>& indexes);
00039
00041     static std::string appendIndexKey(const std::string& currentKey, unsigned int index);
00042
00044     static std::string appendIndexKeyFromDouble(const std::string& currentKey, double index);
00045
00047     static std::string normalizeIndexKey(const std::string& index);
00048
00050     double value(const std::string& index = "") const;
00051
00053     void setValue(double value, const std::string& index = "");
00054
00056     void setValues(const std::map<std::string, double>& values);
00057
00059     void copyFrom(const SparseValueStore& other);
00060
00062     void insertDimensionSize(unsigned int size);
00063
00065     std::list<unsigned int>* dimensionSizes();
00066
00068     const std::list<unsigned int>* dimensionSizes() const;
00069
00071     std::map<std::string, double>* values();
00072
00074     const std::map<std::string, double>* values() const;
00075
00077     void clear();
00078
00080     void loadDimensions(PersistenceRecord* fields);
00081
```

```
00083     void saveDimensions(PersistenceRecord* fields, bool saveDefaultValues) const;
00084
00086     void loadValues(PersistenceRecord* fields,
00087                     const std::string& countField,
00088                     const std::string& positionPrefix,
00089                     const std::string& valuePrefix);
00090
00092     void saveValues(PersistenceRecord* fields,
00093                     const std::string& countField,
00094                     const std::string& positionPrefix,
00095                     const std::string& valuePrefix,
00096                     bool saveDefaultValues) const;
00097
00099     std::string showValues() const;
00100
00101 private:
00102     static std::string trimCopy(const std::string& text) {
00103         std::size_t first = 0;
00104         while (first < text.size() && std::isspace(static_cast<unsigned char>(text[first]))) {
00105             ++first;
00106         }
00107         std::size_t last = text.size();
00108         while (last > first && std::isspace(static_cast<unsigned char>(text[last - 1]))) {
00109             --last;
00110         }
00111         return text.substr(first, last - first);
00112     }
00113
00114     // Returns true when every comma-separated token in index consists of digits only
00115     // (or when index is empty, representing the scalar position).
00116     static bool isNumericKey(const std::string& index) {
00117         if (index.empty()) {
00118             return true;
00119         }
00120         for (char c : index) {
00121             if (!std::isdigit(static_cast<unsigned char>(c)) && c != ',' && c != ' ' && c != '\t') {
00122                 return false;
00123             }
00124         }
00125         return true;
00126     }
00127
00128     static std::vector<unsigned int> parseIndex(const std::string& index) {
00129         std::vector<unsigned int> parsed;
00130         if (index.empty()) {
00131             return parsed;
00132         }
00133         std::stringstream input(index);
00134         std::string token;
00135         while (std::getline(input, token, ',')) {
00136             token = trimCopy(token);
00137             if (token.empty()) {
00138                 continue;
00139             }
00140             parsed.push_back(static_cast<unsigned int>(std::stoul(token)));
00141         }
00142         return parsed;
00143     }
00144
00145     static std::size_t productOfDimensions(const std::list<unsigned int>& dimensions) {
00146         std::size_t product = 1u;
00147         for (unsigned int dimension : dimensions) {
00148             product *= static_cast<std::size_t>(dimension);
00149         }
00150         return product;
00151     }
00152
00153     static std::vector<std::size_t> stridesFor(const std::list<unsigned int>& dimensions) {
00154         std::vector<std::size_t> strides;
00155         strides.reserve(dimensions.size());
00156         std::size_t stride = 1u;
00157         for (auto it = dimensions.rbegin(); it != dimensions.rend(); ++it) {
00158             strides.push_back(stride);
00159             stride *= static_cast<std::size_t>(*it);
00160         }
00161         std::reverse(strides.begin(), strides.end());
00162         return strides;
00163     }
00164
00165     std::size_t linearIndex(const std::vector<unsigned int>& indexes) const {
00166         if (_dimensionSizes.empty()) {
00167             return indexes.empty() ? 0u : static_cast<std::size_t>(-1);
00168         }
00169         if (indexes.size() != _dimensionSizes.size()) {
00170             return static_cast<std::size_t>(-1);
00171         }
00172         const std::vector<std::size_t> strides = stridesFor(_dimensionSizes);
```

```
00173         std::size_t linear = 0u;
00174         auto dimIt = _dimensionSizes.begin();
00175         for (std::size_t i = 0; i < indexes.size(); ++i, ++dimIt) {
00176             if (indexes[i] >= *dimIt) {
00177                 return static_cast<std::size_t>(-1);
00178             }
00179             linear += static_cast<std::size_t>(indexes[i]) * strides[i];
00180         }
00181         return linear;
00182     }
00183
00184     void rebuildDenseFromCache() {
00185         const std::size_t size = _dimensionSizes.empty() ? 1u : productOfDimensions(_dimensionSizes);
00186         _denseValues.assign(size, 0.0);
00187         for (const auto& entry : _values) {
00188             if (entry.first.empty()) {
00189                 continue;
00190             }
00191             // Skip non-numeric string keys -- they live only in the sparse map.
00192             if (!isNumericKey(entry.first)) {
00193                 continue;
00194             }
00195             const std::vector<unsigned int> indexes = parseIndex(entry.first);
00196             const std::size_t linear = linearIndex(indexes);
00197             if (linear != static_cast<std::size_t>(-1) && linear < _denseValues.size()) {
00198                 _denseValues[linear] = entry.second;
00199             }
00200         }
00201         _denseActive = true;
00202     }
00203
00204     void syncCacheFromDense() {
00205         // Preserve non-numeric string keys across the rebuild.
00206         std::map<std::string, double> stringKeyed;
00207         for (const auto& entry : _values) {
00208             if (!entry.first.empty() && !isNumericKey(entry.first)) {
00209                 stringKeyed.insert(entry);
00210             }
00211         }
00212         _values.clear();
00213         if (_hasScalarValue) {
00214             _values[""] = _scalarValue;
00215         }
00216         if (!_denseActive) {
00217             _values.insert(stringKeyed.begin(), stringKeyed.end());
00218             return;
00219         }
00220         if (_dimensionSizes.empty()) {
00221             if (!_denseValues.empty() && !_hasScalarValue) {
00222                 _values[""] = _denseValues.front();
00223             }
00224             _values.insert(stringKeyed.begin(), stringKeyed.end());
00225             return;
00226         }
00227         std::vector<unsigned int> indexes;
00228         indexes.reserve(_dimensionSizes.size());
00229         const std::vector<std::size_t> strides = stridesFor(_dimensionSizes);
00230         for (std::size_t linear = 0; linear < _denseValues.size(); ++linear) {
00231             if (_denseValues[linear] == 0.0) {
00232                 continue;
00233             }
00234             std::size_t remainder = linear;
00235             indexes.clear();
00236             for (std::size_t i = 0; i < strides.size(); ++i) {
00237                 const unsigned int index = static_cast<unsigned int>(remainder / strides[i]);
00238                 indexes.push_back(index);
00239                 remainder %= strides[i];
00240             }
00241             _values[makeIndexKey(indexes)] = _denseValues[linear];
00242         }
00243         _values.insert(stringKeyed.begin(), stringKeyed.end());
00244     }
00245
00246     void ensureDenseForIndex(const std::vector<unsigned int>& indexes) {
00247         if (indexes.empty() && _dimensionSizes.empty()) {
00248             if (!_denseActive) {
00249                 _denseActive = true;
00250                 _denseValues.assign(1u, 0.0);
00251             }
00252             return;
00253         }
00254
00255         if (_dimensionSizes.empty()) {
00256             _dimensionSizes.clear();
00257             for (unsigned int index : indexes) {
00258                 _dimensionSizes.push_back(index + 1u);
00259             }
```

```
00260             rebuildDenseFromCache();
00261             return;
00262         }
00263 
00264         if (indexes.size() != _dimensionSizes.size()) {
00265             return;
00266         }
00267 
00268         bool needsResize = false;
00269         auto dimIt = _dimensionSizes.begin();
00270         for (std::size_t i = 0; i < indexes.size(); ++i, ++dimIt) {
00271             if (indexes[i] >= *dimIt) {
00272                 *dimIt = indexes[i] + 1u;
00273                 needsResize = true;
00274             }
00275         }
00276         if (needsResize || !_denseActive) {
00277             rebuildDenseFromCache();
00278         }
00279     }
00280 
00281     bool _denseActive = false;
00282     bool _hasScalarValue = false;
00283     double _scalarValue = 0.0;
00284     std::vector<double> _denseValues;
00285     std::list<unsigned int> _dimensionSizes;
00286     std::map<std::string, double> _values;
00287 };
00288 
00289 inline std::string SparseValueStore::makeIndexKey(const std::vector<unsigned int>& indexes) {
00290     std::ostringstream key;
00291     bool first = true;
00292     for (const unsigned int index : indexes) {
00293         if (!first) {
00294             key « ",";
00295         }
00296         first = false;
00297         key « index;
00298     }
00299     return key.str();
00300 }
00301 
00302 inline std::string SparseValueStore::makeIndexKeyFromDoubles(const std::vector<double>& indexes) {
00303     std::vector<unsigned int> unsignedIndexes;
00304     unsignedIndexes.reserve(indexes.size());
00305     for (const double index : indexes) {
00306         unsignedIndexes.push_back(static_cast<unsigned int>(index));
00307     }
00308     return makeIndexKey(unsignedIndexes);
00309 }
00310 
00311 inline std::string SparseValueStore::appendIndexKey(const std::string& currentKey, unsigned int index)
    {
00312     const std::string nextPart = std::to_string(index);
00313     if (currentKey.empty()) {
00314         return nextPart;
00315     }
00316     return currentKey + "," + nextPart;
00317 }
00318 
00319 inline std::string SparseValueStore::appendIndexKeyFromDouble(const std::string& currentKey, double
    index) {
00320     return appendIndexKey(currentKey, static_cast<unsigned int>(index));
00321 }
00322 
00323 inline std::string SparseValueStore::normalizeIndexKey(const std::string& index) {
00324     return index;
00325 }
00326 
00327 inline double SparseValueStore::value(const std::string& index) const {
00328     if (index.empty() && _hasScalarValue) {
00329         return _scalarValue;
00330     }
00331     // Non-numeric string keys (e.g. "slot", "pi") bypass dense storage.
00332     if (!index.empty() && !isNumericKey(index)) {
00333         const auto it = _values.find(index);
00334         return (it != _values.end()) ? it->second : 0.0;
00335     }
00336     if (_denseActive) {
00337         const std::vector<unsigned int> indexes = parseIndex(index);
00338         const std::size_t linear = linearIndex(indexes);
00339         if (linear == static_cast<std::size_t>(-1) || linear >= _denseValues.size()) {
00340             // Dimension count mismatch or out of bounds -- fall back to sparse map.
00341             const auto it = _values.find(normalizeIndexKey(index));
00342             return (it != _values.end()) ? it->second : 0.0;
00343         }
00344         return _denseValues[linear];
```

```
00345     }
00346     const auto it = _values.find(normalizeIndexKey(index));
00347     if (it == _values.end()) {
00348         return 0.0;
00349     }
00350     return it->second;
00351 }
00352
00353 inline void SparseValueStore::setValue(double value, const std::string& index) {
00354     if (index.empty()) {
00355         _scalarValue = value;
00356         _hasScalarValue = true;
00357         _values[""] = value;
00358         if (_denseActive && _dimensionSizes.empty()) {
00359             if (_denseValues.empty()) {
00360                 _denseValues.assign(1u, 0.0);
00361             }
00362             _denseValues.front() = value;
00363         }
00364         return;
00365     }
00366     // Non-numeric string keys (e.g. "slot", "pi") are stored directly in the
00367     // sparse map without touching the dense storage or dimension metadata.
00368     if (!isNumericKey(index)) {
00369         _values[index] = value;
00370         return;
00371     }
00372     const std::vector<unsigned int> indexes = parseIndex(index);
00373     ensureDenseForIndex(indexes);
00374     if (_denseActive) {
00375         const std::size_t linear = linearIndex(indexes);
00376         if (linear != static_cast<std::size_t>(-1)) {
00377             if (linear >= _denseValues.size()) {
00378                 _denseValues.resize(linear + 1u, 0.0);
00379             }
00380             _denseValues[linear] = value;
00381             syncCacheFromDense();
00382             return;
00383         }
00384     }
00385     _values[normalizeIndexKey(index)] = value;
00386 }
00387
00388 inline void SparseValueStore::setValues(const std::map<std::string, double>& values) {
00389     _values = values;
00390     const auto scalarIt = _values.find("");
00391     _hasScalarValue = scalarIt != _values.end();
00392     if (_hasScalarValue) {
00393         _scalarValue = scalarIt->second;
00394     }
00395     if (_dimensionSizes.empty()) {
00396         if (_values.size() == 1u && _hasScalarValue) {
00397             _denseActive = true;
00398             _denseValues.assign(1u, _scalarValue);
00399         } else {
00400             _denseActive = false;
00401             _denseValues.clear();
00402         }
00403         return;
00404     }
00405     rebuildDenseFromCache();
00406 }
00407
00408 inline void SparseValueStore::copyFrom(const SparseValueStore& other) {
00409     *this = other;
00410 }
00411
00412 inline void SparseValueStore::insertDimensionSize(unsigned int size) {
00413     _dimensionSizes.insert(_dimensionSizes.end(), size);
00414     rebuildDenseFromCache();
00415 }
00416
00417 inline std::list<unsigned int>* SparseValueStore::dimensionSizes() {
00418     return &_dimensionSizes;
00419 }
00420
00421 inline const std::list<unsigned int>* SparseValueStore::dimensionSizes() const {
00422     return &_dimensionSizes;
00423 }
00424
00425 inline std::map<std::string, double>* SparseValueStore::values() {
00426     return &_values;
00427 }
00428
00429 inline const std::map<std::string, double>* SparseValueStore::values() const {
00430     return &_values;
00431 }
```

```
00432
00433 inline void SparseValueStore::clear() {
00434     _dimensionSizes.clear();
00435     _values.clear();
00436     _hasScalarValue = false;
00437     _scalarValue = 0.0;
00438     _denseValues.clear();
00439     _denseActive = false;
00440 }
00441
00442 inline void SparseValueStore::loadDimensions(PersistenceRecord* fields) {
00443     _dimensionSizes.clear();
00444     if (fields == nullptr) {
00445         return;
00446     }
00447     const unsigned int count = fields->loadField("dimensions", 0u);
00448     for (unsigned int i = 0; i < count; i++) {
00449         insertDimensionSize(fields->loadField("dimension" + Util::StrIndex(i), 0u));
00450     }
00451     if (!_dimensionSizes.empty()) {
00452         rebuildDenseFromCache();
00453     }
00454 }
00455
00456 inline void SparseValueStore::saveDimensions(PersistenceRecord* fields, bool saveDefaultValues) const
    {
00457     if (fields == nullptr) {
00458         return;
00459     }
00460     unsigned int i = 0;
00461     fields->saveField("dimensions", static_cast<unsigned int>(_dimensionSizes.size()), 0u,
    saveDefaultValues);
00462     for (const unsigned int dimension : _dimensionSizes) {
00463         fields->saveField("dimension" + Util::StrIndex(i), dimension, 1u, saveDefaultValues);
00464         i++;
00465     }
00466 }
00467
00468 inline void SparseValueStore::loadValues(PersistenceRecord* fields,
00469                                          const std::string& countField,
00470                                          const std::string& positionPrefix,
00471                                          const std::string& valuePrefix) {
00472     _values.clear();
00473     if (fields == nullptr) {
00474         return;
00475     }
00476     const unsigned int count = fields->loadField(countField, 0u);
00477     for (unsigned int i = 0; i < count; i++) {
00478         const std::string position = fields->loadField(positionPrefix + Util::StrIndex(i), "");
00479         const double loadedValue = fields->loadField(valuePrefix + Util::StrIndex(i), 0.0);
00480         setValue(loadedValue, position);
00481     }
00482 }
00483
00484 inline void SparseValueStore::saveValues(PersistenceRecord* fields,
00485                                          const std::string& countField,
00486                                          const std::string& positionPrefix,
00487                                          const std::string& valuePrefix,
00488                                          bool saveDefaultValues) const {
00489     if (fields == nullptr) {
00490         return;
00491     }
00492     unsigned int i = 0;
00493     fields->saveField(countField, static_cast<unsigned int>(_values.size()), 0u, saveDefaultValues);
00494     for (const auto& value : _values) {
00495         fields->saveField(positionPrefix + Util::StrIndex(i), value.first, "0", saveDefaultValues);
00496         fields->saveField(valuePrefix + Util::StrIndex(i), value.second, 0.0, saveDefaultValues);
00497         i++;
00498     }
00499 }
00500
00501 inline std::string SparseValueStore::showValues() const {
00502     if (_denseActive && _values.empty() && !_denseValues.empty() && _dimensionSizes.empty()) {
00503         return "{" + std::to_string(_denseValues.front()) + "}";
00504     }
00505     std::string text = "{";
00506     for (const auto& value : _values) {
00507         text += value.first + "=" + Util::StrTruncIfInt(std::to_string(value.second)) + ", ";
00508     }
00509     if (!_values.empty()) {
00510         text = text.substr(0, text.length() - 2);
00511     }
00512     text += "}";
00513     return text;
00514 }
00515
00516 #endif /* SPARSEVALUESTORE_H */
```

18.299 StatisticsCollector.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   StatisticsCollector.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 30 de Agosto de 2018, 17:24
00012  */
00013
00014 #ifndef STATISTICSCOLLECTOR_H
00015 #define STATISTICSCOLLECTOR_H
00016
00017 #include "../model/ModelDataDefinition.h"
00018 #include "../../statistics/Statistics_if.h"
00019 #include "../model/ModelDataManager.h"
00020 #include "../Plugin.h"
00021
00022 #include <vector>
00023
00024 //namespace GenesysKernel {
00025
00029 class StatisticsCollector : public ModelDataDefinition {//, public Statistics_if {
00030 public:
00031     StatisticsCollector(Model* model, std::string name = "", ModelDataDefinition* parent = nullptr,
      bool insertIntoModel = true);
00032     virtual ~StatisticsCollector() override;
00033 public:
00034     virtual std::string show() override;
00035 public:
00040     ModelDataDefinition* getParent() const;
00045     Statistics_if* getStatistics() const;
00046
00047 public:
00048     static PluginInformation* GetPluginInformation();
00049     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00050     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00051
00052 protected:
00053     virtual bool _loadInstance(PersistenceRecord *fields) override;
00054     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00055 protected:
00056     virtual bool _check(std::string& errorMessage) override;
00057
00058 private:
00059     void _initStaticsAndCollector();
00060     void _addOwnedResponse(SimulationResponse* response);
00061 private:
00062     ModelDataDefinition* _parent;
00063     Statistics_if* _statistics = nullptr;
00064     std::vector<SimulationResponse*> _ownedResponses;
00065 };
00066 //namespace\}
00067 #endif /* STATISTICSCOLLECTOR_H */
```

18.300 Event.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Event.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Junho de 2018, 19:41
00012  */
00013
00014 #ifndef EVENT_H
00015 #define EVENT_H
00016
00017 #include <string>
00018 #include <functional>
00019
00020 #include "model/ModelDataDefinition.h"
00021 #include "essentialPlugins/Entity.h"
00022 #include "model/ModelComponent.h"
```

```
00023
00024 //namespace GenesysKernel {
00025
00030 class Event {//: public ModelDataDefinition {
00031 public:
00032     Event(double time, Entity* entity, ModelComponent* component, unsigned int
      componentinputPortNumber = 0);
00033     Event(double time, Entity* entity, Connection* connection);
00034     virtual ~Event() = default;
00035 public:
00040     double getTime() const;
00045     ModelComponent* getComponent() const;
00050     Entity* getEntity() const;
00055     unsigned int getComponentinputPortNumber() const;
00060     void setEntity(Entity* _entity);
00061 public: // virtual
00062     virtual std::string show();
00063 protected:
00064     double _time;
00065 private:
00066     Entity* _entity;
00067     ModelComponent* _component;
00068     unsigned int _componentinputPortNumber;
00069 };
00070
00071 //****************************************************** 2022/07/05
00072
00073 typedef std::function<void(void*) > InternalEventHandler;
00074
00075 class InternalEvent: public Event { // @ToDo: (importante): BAD SW ENGINEERING Fix it
00076 public:
00077     InternalEvent(double time, std::string description): Event(time, nullptr, nullptr, 0){
00078         _description = description;
00079         _time = time;
00080     }
00081     virtual ~InternalEvent() = default;
00082
00083     template<typename Class>
00084     void setEventHandler(Class* object, void (Class::*function)(void*), void* parameter) {
00085         _object = object;
00086         _handler= std::bind(function, object, std::placeholders::_1);
00087         _parameter = parameter;
00088     }
00089
00090     virtual std::string show() override {
00091         std::string dataDefinitionName = static_cast<ModelDataDefinition*>(_object)->getName();
00092         std::string message = "time=" + std::to_string(_time) + //Util::StrTimeUnit(???)+
00093                 ",description=\"" + _description+"\""+
00094                 ",block=\"" + dataDefinitionName+"\"";
00095         return message;
00096     }
00097
00098     const InternalEventHandler& eventHandler() const {
00099         return _handler;
00100     }
00101
00102     const std::string& description() const {
00103         return _description;
00104     }
00105
00106     void dispatchEvent() const {
00107         _handler(_parameter);
00108     }
00109
00110     void* object() const {
00111         return _object;
00112     }
00113
00114     void* parameter() const {
00115         return _parameter;
00116     }
00117
00118
00119 private:
00120     InternalEventHandler _handler;
00121     std::string _description;
00122     void* _parameter;
00123     void* _object;
00124 };
00125
00126 //namespace\\}
00127 #endif /* EVENT_H */
```

18.301 ExperimentManager.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ExperimentManager.h
00009  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00010  *
00011  * Created on 4 de maio de 2021, 11:43
00012  */
00013
00014 #ifndef EXPERIMENTMANAGER_H
00015 #define EXPERIMENTMANAGER_H
00016
00017 //#include "Simulator.h"
00018 #include "Persistence.h"
00019 //#include "SimulationControl.h"
00020 //#include "SimulationResponse.h"
00021 #include "SimulationExperiment.h"
00022 #include "../util/List.h"
00023
00024 class Simulator;
00025
00041 class ExperimentManager {// : PersistentObject_base {
00042 public:
00043
00044     /*  class ModelSimulationInsOuts {
00045         public:
00046             ModelSimulationInsOuts() = default;
00047             struct ControlResponseInfos {
00048                 std::string type;
00049                 std::string parent;
00050                 std::string name;
00051                 double defaultValue;
00052             };
00053
00054             List<ControlResponseInfos*>* getResponsesInfos() const {
00055                 return _responsesInfos;
00056             }
00057
00058             std::string getModelfilename() const {
00059                 return _experimentfilename;
00060             }
00061
00062             List<ControlResponseInfos*>* getControlInfos() const {
00063                 return _controlInfos;
00064             }
00065         private:
00066             std::string _modelfilename;
00067             List<ControlResponseInfos*>* _responsesInfos = new List<ControlResponseInfos*>();
00068             List<ControlResponseInfos*>* _controlInfos = new List<ControlResponseInfos*>();
00069         }; */
00070
00071 public:
00072     ExperimentManager(Simulator* simulator);
00073     // Release owned experiment instances and container storage during simulator shutdown.
00074     virtual ~ExperimentManager();
00075
00076 public:
00077     SimulationExperiment* newSimulationExperiment();
00078     void insert(SimulationExperiment* experiment);
00079     void remove(SimulationExperiment* experiment);
00080     void setCurrent(SimulationExperiment* experiment);
00081     // @ToDo: (importante): Implement persistence once SimulationExperiment serialization is defined.
00082     bool saveSimulationExperiment(std::string filename);
00083     // @ToDo: (importante): Implement loading once SimulationExperiment deserialization and scenario
      execution are defined.
00084     bool loadSimulationExperiment(std::string filename);
00085     unsigned int size();
00086 public:
00087     SimulationExperiment* front();
00088     SimulationExperiment* current();
00089     SimulationExperiment* next();
00090     //SimulationExperiment* end();
00091 public:
00092     List<SimulationExperiment*>* getExperiments() const;
00093 private:
00094     Simulator* _simulator;
00095     SimulationExperiment* _currentSimulationExperiment;
00096     List<SimulationExperiment*>* _experiments = new List<SimulationExperiment*>();
00097
00098 };
00099
```

```
00100 #endif /* EXPERIMENTMANAGER_H */
```

18.302 ExperimentManagerDefaultImpl1.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ProcessAnalyserDefaultImpl1.h
00009  * Author: rlcancian
00010  *
00011  * Created on 20 de Maio de 2019, 20:45
00012  */
00013
00014 #ifndef PROCESSANALYSERDEFAULTIMPL1_H
00015 #define PROCESSANALYSERDEFAULTIMPL1_H
00016
00017 #include "ExperimetManager_if.h"
00018 #include "SimulationScenario.h"
00019
00020 class ExperimentManagerDefaultImpl1 : public ExperimentManager_if {
00021 public:
00022     ExperimentManagerDefaultImpl1();
00023     virtual ~ExperimentManagerDefaultImpl1() = default;
00024 public:
00025     virtual List<SimulationScenario*>* getScenarios() const override;
00026     //virtual List<SimulationControl*>* getControls() const;
00027     //virtual List<SimulationResponse*>* getResponses() const;
00028     //virtual List<SimulationControl*>* extractControlsFromModel(std::string modelFilename) const;
00029     //virtual List<SimulationResponse*>* extractResponsesFromModel(std::string modelFilename) const;
00030     virtual void startSimulationOfScenario(SimulationScenario* scenario) override;
00031     virtual void startExperiment() override;
00032     virtual void stopExperiment() override;
00033     virtual void addTraceSimulationHandler(traceSimulationProcessListener
      traceSimulationProcessListener) override;
00034 private:
00035     //List<SimulationControl*>* _controls = new List<SimulationControl*>();
00036 };
00037
00038 #endif /* PROCESSANALYSERDEFAULTIMPL1_H */
```

18.303 ExperimetManager_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ExperimentManager_if (old ProcessAnalyser_if.h)
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 10 de Outubro de 2018, 14:26
00012  */
00013
00014 #ifndef EXPERIMENTMANAGER_IF_H
00015 #define EXPERIMENTMANAGER_IF_H
00016
00017 #include "../util/List.h"
00018 #include "SimulationScenario.h"
00019 #include "PropertyGenesys.h"
00020 #include "TraceManager.h"
00021
00028 class ExperimentManager_if {
00029 public:
00030     virtual ~ExperimentManager_if() = default;
00032     virtual List<SimulationScenario*>* getScenarios() const = 0;
00033     //virtual List<SimulationControl*>* getControls() const = 0;
00034     //virtual List<SimulationResponse*>* getResponses() const = 0;
00035     //virtual List<SimulationControl*>* extractControlsFromModel(std::string modelFilename) const = 0;
00036     //virtual List<SimulationResponse*>* extractResponsesFromModel(std::string modelFilename) const =
      0;
00038     virtual void startSimulationOfScenario(SimulationScenario* scenario) = 0;
00040     virtual void startExperiment() = 0;
00042     virtual void stopExperiment() = 0;
00044     virtual void addTraceSimulationHandler(traceSimulationProcessListener
      traceSimulationProcessListener) = 0;
00045 };
```

```
00046
00047 #endif /* EXPERIMENTMANAGER_IF_H */
```

18.304 GenesysPropertyIntrospection.h

```
00001 #pragma once
00002
00003 #include <string>
00004 #include <vector>
00005
00006 #include "SimulationControlAndResponse.h"
00007 #include "../util/Util.h"
00008
00009 enum class GenesysPropertyKind {
00010     Unknown,
00011     Boolean,
00012     Integer,
00013     UnsignedInteger,
00014     UnsignedShort,
00015     Double,
00016     String,
00017     TimeUnit,
00018     Enum,
00019     Object,
00020     List
00021 };
00022
00023 struct GenesysPropertyDescriptor {
00024     SimulationControl* control = nullptr;
00025
00026     std::string ownerClassName;
00027     std::string ownerElementName;
00028     std::string displayName;
00029     std::string technicalTypeName;
00030
00031     GenesysPropertyKind kind = GenesysPropertyKind::Unknown;
00032
00033     bool readOnly = true;
00034     bool isList = false;
00035     bool isClass = false;
00036     bool isEnum = false;
00037     bool isInlineObject = false;
00038     bool isModelDataDefinitionReference = false;
00039     bool supportsInlineExpansion = false;
00040     bool supportsListEditor = false;
00041     bool supportsExistingObjectSelection = false;
00042     bool supportsObjectCreation = false;
00043     bool supportsNewListElementCreation = false;
00044     SimulationControlEditorHint editorHint = SimulationControlEditorHint::Default;
00045
00046     std::string currentValue;
00047     std::vector<std::string> choices;
00048     std::string currentReferenceType;
00049     std::vector<std::string> creatableReferenceTypes;
00050     // For polymorphic list controls, this is the currently selected concrete element type.
00051     //
00052     // Example: a Set physically stores ModelDataDefinition* but may currently be a Set of Resource.
00053     // Empty means the list has not been typed yet.
00054     std::string currentListElementType;
00055     // For polymorphic list controls, these are the concrete element types that the kernel allows
00056     // the GUI to select/create. Empty means the list should use its legacy single-type behavior.
00057     std::vector<std::string> creatableListElementTypes;
00058 };
00059
00060 class GenesysPropertyIntrospection {
00061 public:
00062     static GenesysPropertyDescriptor describe(SimulationControl* control);
00063     static std::vector<GenesysPropertyDescriptor> describe(List<SimulationControl*>* controls);
00064
00065     static GenesysPropertyKind deduceKind(const SimulationControl* control);
00066
00067     static bool setValue(
00068         SimulationControl* control,
00069         const std::string& value,
00070         bool remove = false,
00071         std::string* errorMessage = nullptr
00072         );
00073
00074     static std::vector<SimulationControl*> children(
00075         SimulationControl* control,
00076         int index = 0
00077         );
00078 };
```

18.305 LicenceManager.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   LicenceManager.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 29 de Maio de 2019, 11:45
00012  */
00013
00014 #ifndef LICENCEMANAGER_H
00015 #define LICENCEMANAGER_H
00016
00017 #include <string>
00018
00019 //namespace GenesysKernel {
00020 class Simulator;
00021
00025 class LicenceManager {
00026 public:
00027     LicenceManager(Simulator* simulator);
00028     virtual ~LicenceManager() = default;
00029     const std::string showLicence() const;
00030     const std::string showLimits() const;
00031     const std::string showActivationCode() const;
00032     bool lookforActivationCode();
00033     bool insertActivationCode();
00034     void removeActivationCode();
00035     unsigned int getModelComponentsLimit();
00036     unsigned int getModelDatasLimit();
00037     unsigned int getEntityLimit();
00038     unsigned int getHostsLimit();
00039     unsigned int getThreadsLimit();
00040 private:
00041     void setDefaultLicenceAndLimits();
00042 private:
00043     Simulator* _simulator;
00044     std::string _licence;
00045     std::string _activationCode;
00046     unsigned int _components, _datadefinitions, _entities, _hosts, _threads;
00047 };
00048 //namespace\\}
00049 #endif /* LICENCEMANAGER_H */
00050
```

18.306 Model.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   SimulationModel.h
00009  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00010  *
00011  * Created on 21 de Junho de 2018, 15:01
00012  */
00013
00014 #ifndef SIMULATIONMODEL_H
00015 #define SIMULATIONMODEL_H
00016
00017 #include <list>
00018 #include <string>
00019
00020 #include "../../util/List.h"
00021 #include "ModelComponent.h"
00022 #include "../Event.h"
00023 #include "ModelChecker_if.h"
00024 #include "../Parser_if.h"
00025 #include "../persistence/Persistence_if.h"
00026 #include "ModelDataManager.h"
00027 #include "ModelComponentManager.h"
00028 #include "../TraceManager.h"
00029 #include "../OnEventManager.h"
00030 #include "ModelInfo.h"
00031 #include "ModelSimulation.h"
00032 #include "../SimulationControlAndResponse.h"
```

```
00033
00034 //namespace GenesysKernel {
00035 class Simulator;
00036
00043 class Model {
00044 public:
00050     Model(Simulator* simulator, unsigned int level = 0);
00054     virtual ~Model();
00055 public: // model control
00056     //void showReports();
00062     bool save(std::string filename);
00068     bool load(std::string filename);
00076     bool check();
00080     void clear();
00085     std::string showLanguage();
00089     void show();
00097     bool insert(ModelDataDefinition* elemOrComp);
00104     void remove(ModelDataDefinition* elemOrComp);
00110     std::list<ModelDataDefinition*> collectDataDefinitionsRemovedWith(const
      std::list<ModelDataDefinition*>& roots) const;
00117     Entity* createEntity(std::string name, bool insertIntoModel = true);
00122     void removeEntity(Entity* entity); //, bool collectStatistics);
00131     void sendEntityToComponent(Entity* entity, Connection* connection, double timeDelay = 0.0);
00141     void sendEntityToComponent(Entity* entity, ModelComponent* component, double timeDelay = 0.0,
      unsigned int componentinputPortNumber = 0);
00149     double parseExpression(const std::string expression);
00160     double parseExpression(const std::string expression, bool& success, std::string& errorMessage);
00170     bool checkExpression(const std::string expression, const std::string expressionName, std::string&
      errorMessage);
00176     void checkReferencesToDataDefinitions(std::string expression, std::map<std::string,
      std::list<std::string>*>* referencedDataDefinitions);
00177
00178 public: // only gets
00183     Util::identification getId() const;
00188     bool hasChanged() const;
00193     void setHasChanged(bool hasChanged);
00194     // 1:1
00202     OnEventManager* getOnEventManager() const;
00210     ModelDataManager* getDataManager() const;
00215     ModelComponentManager* getComponentManager() const;
00220     ModelInfo* getInfos() const;
00225     Simulator* getParentSimulator() const;
00231     ModelSimulation* getSimulation() const;
00232     // 1:n
00241     List<Event*>* getFutureEvents() const;
00249     List<SimulationControl*>* getControls() const;
00257     List<SimulationResponse*>* getResponses() const;
00258
00259 public: // gets and sets
00264     void setTracer(TraceManager* _traceManager);
00269     TraceManager* getTracer() const;
00274     Persistence_if* getPersistence() const;
00281     unsigned int getLevel() const;
00282
00283 private:
00284     void _showConnections() const;
00285     void _showComponents() const;
00286     void _showElements() const;
00287     void _showSimulationControls() const;
00288     void _showSimulationResponses() const;
00289     void _destroyFutureEvents();
00290     void _destroyTransientEntities();
00291     void _destroyComponents();
00292     void _destroyModelDataDefinitions();
00293     void _clearOrphanedDataDefinitions();
00294     void _createInternalDataDefinitions();
00295 private:
00296     bool _hasChanged = false;
00297 private: // read only public access (gets)
00298     Util::identification _id;
00299     unsigned int _level = 0;
00300     Simulator* _parentSimulator;
00301     // 1:1 (associted classes)
00302     TraceManager* _traceManager;
00303     OnEventManager* _eventManager;
00304     ModelDataManager* _modeldataManager;
00305     ModelComponentManager* _componentManager;
00306     ModelInfo* _modelInfo;
00307     ModelSimulation* _simulation;
00308     Persistence_if* _modelPersistence;
00309
00310     // 1:n
00311     //List<ModelComponent*>* _components;
00312     List<Event*>* _futureEvents;
00313     // for process analyser
00314     List<SimulationResponse*>* _responses;
00315     List<SimulationControl*>* _controls;
```

```
00316
00317 private: // no public access (no gets / sets)
00318     ModelChecker_if* _modelChecker;
00319     Parser_if* _parser;
00320 };
00321 //namespace\\}
00322 #endif /* SIMULATIONMODEL_H */
```

18.307 ModelChecker_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ModelChecker_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 23 de Agosto de 2018, 15:48
00012  */
00013
00014 #ifndef MODELCHECKER_IF_H
00015 #define MODELCHECKER_IF_H
00016
00017 #include <typeinfo>
00018 #include <string>
00019
00020 //#include "Model.h"
00021 //class Model;
00022
00026 class ModelChecker_if {
00027 public:
00028     virtual ~ModelChecker_if() = default;
00033     virtual bool checkAll() = 0;
00038     virtual bool checkConnected() = 0;
00043     virtual bool checkSymbols() = 0;
00048     virtual bool checkActivationCode() = 0;
00053     virtual bool checkLimits() = 0;
00054 };
00055
00056 #endif /* MODELCHECKER_IF_H */
```

18.308 ModelCheckerDefaultImpl1.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ModelCheckerDefaultImpl1.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 8 de Agosto de 2018, 18:44
00012  */
00013
00014 #ifndef MODELCHECKERDEFAULTIMPL1_H
00015 #define MODELCHECKERDEFAULTIMPL1_H
00016
00017 #include "ModelChecker_if.h"
00018 #include "Model.h"
00019 #include "../PluginManager.h"
00020
00021 //namespace GenesysKernel {
00022
00023 class ModelCheckerDefaultImpl1 : public ModelChecker_if {
00024 public:
00025     ModelCheckerDefaultImpl1(Model* model);
00026     virtual ~ModelCheckerDefaultImpl1() = default;
00027 public:
00028     virtual bool checkAll() override;
00029     virtual bool checkConnected() override;
00030     virtual bool checkSymbols() override;
00031     virtual bool checkActivationCode() override;
00032     virtual bool checkLimits() override;
00033 private:
00034     void _recursiveConnectedTo(PluginManager* pluginManager, ModelComponent* comp,
      List<ModelComponent*>* visited, List<ModelComponent*>* unconnected, bool* drenoFound);
00035     void _showResult(bool result, std::string checking);
```

```
00036 private:
00037     Model* _model;
00038 };
00039 //namespace\\}
00040 #endif /* MODELCHECKERDEFAULTIMPL1_H */
00041
```

18.309 ModelComponent.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Element.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Junho de 2018, 15:56
00012  */
00013
00014 #ifndef MODELCOMPONENT_H
00015 #define MODELCOMPONENT_H
00016
00017 #include <string>
00018 #include <list>
00019
00020 #include "../Plugin.h"
00021 #include "../../util/List.h"
00022 #include "../essentialPlugins/Entity.h"
00023 #include "ModelDataDefinition.h"
00024 #include "../ConnectionManager.h"
00025 //namespace GenesysKernel {
00026
00027 class Model;
00028 class Event;
00029
00034 class ModelComponent : public ModelDataDefinition {
00035 public:
00042     ModelComponent(Model* model, std::string componentTypename, std::string name = "");
00046     virtual ~ModelComponent();
00047
00048 public:
00059     ConnectionManager* getConnectionManager() const;
00065     void connectTo(ModelComponent* component, unsigned int inputPortNumber = 0);
00070     bool hasBreakpointAt();
00075     void setDescription(std::string _description);
00080     std::string getDescription() const;
00081
00086     void remove(ModelComponent * comp);
00087     // ...
00088
00089 public:
00094     virtual std::string show() override;
00095
00096 public:
00103     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00104     // new static methods for all ModelComponents
00110     static bool Check(ModelComponent* component);
00115     static void CreateInternalData(ModelComponent* component);
00121     static void SaveInstance(PersistenceRecord *fields, ModelComponent* component);
00129     static void DispatchEvent(Event* event);
00130
00131 protected:
00132     virtual bool _loadInstance(PersistenceRecord *fields) override;
00133     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00134     // new virtual methods for all ModelComponents
00142     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) = 0;
00143
00144 protected:
00146     // virtual bool _check(std::string& errorMessage);
00148     // virtual ParserChangesInformation* _getParserChangesInformation();
00150     // virtual void _initBetweenReplications();
00152     // virtual void _createInternalAndAttachedData(); /* legacy combined lifecycle hook */
00154     // virtual void _addSimulationControl(SimulationControl* property);
00155
00156 protected: // new protected attributes for all ModelComponents
00157     ConnectionManager* _connections = new ConnectionManager();
00158
00159 private:
00160     // ...
00161
00162 private:
00163
```

```
00164     // Default values for the attributes. Used on initing, loading and saving
00165     const struct DEFAULT_VALUES {
00166         const unsigned int nextSize = 1;              // No need for attribute. Taken from
      _connections
00167         const unsigned int nextinputPortNumber = 0;
00168         const std::string description = "";
00169     } DEFAULT;
00170     std::string _description = DEFAULT.description;
00171 
00172 private:
00173     //...
00174 
00175 private:
00176     // ...
00177 
00178 };
00179 //namespace\\}
00180 #endif /* MODELCOMPONENT_H */
```

18.310 ModelComponentManager.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
00007 /*
00008  * File:   ModelComponentManager.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 28 de Maio de 2019, 10:41
00012  */
00013 
00014 #ifndef COMPONENTMANAGER_H
00015 #define COMPONENTMANAGER_H
00016 
00017 #include "ModelComponent.h"
00018 #include "../SourceModelComponent.h"
00019 
00020 //namespace GenesysKernel {
00021 //class Model;
00022 
00029 class ModelComponentManager {
00030 public:
00032     ModelComponentManager(Model* model);
00033     virtual ~ModelComponentManager();
00034 public:
00040     bool insert(ModelComponent* comp);
00045     void remove(ModelComponent* comp);
00051     ModelComponent* find(std::string name);
00057     ModelComponent* find(Util::identification id);
00062     void clear();
00063 public:
00068     unsigned int getNumberOfComponents();
00073     std::list<ModelComponent*>::iterator begin();
00078     std::list<ModelComponent*>::iterator end();
00083     ModelComponent* front();
00088     ModelComponent* next();
00093     bool hasChanged() const;
00098     void setHasChanged(bool _hasChanged);
00099 public:
00104     std::list<SourceModelComponent*>* getSourceComponents();
00109     std::list<ModelComponent*>* getTransferInComponents();
00114     std::list<ModelComponent*>* getAllComponents() const;
00115 private:
00116     List<ModelComponent*>* _components;
00117     Model* _parentModel;
00118     bool _hasChanged = false;
00119     std::list<ModelComponent*>::iterator _componentIterator;
00120 };
00121 //namespace\\}
00122 #endif /* COMPONENTMANAGER_H */
```

18.311 ModelDataDefinition.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
```

```
00007 /*
00008  * File:   ModelDataDefinition.h
00009  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00010  *
00011  * Created on 21 de Junho de 2018, 19:40
00012  */
00013
00014 #ifndef MODELELEMENT_H
00015 #define MODELELEMENT_H
00016
00017 #include <string>
00018 #include <list>
00019 #include <vector>
00020 #include <map>
00021
00022 #include "../../util/List.h"
00023 #include "../ParserChangesInformation.h"
00024 #include "../Persistence.h"
00025 //#include "PropertyGenesys.h"
00026 #include "ModelDataDefinitionAssociations.h"
00027 #include "../SimulationControlAndResponse.h"
00028 #include "../TraceManager.h"
00029
00030 //namespace GenesysKernel {
00031 class Model;
00032
00033
00038 class ModelDataDefinition : PersistentObject_base {
00039 public:
00047     ModelDataDefinition(Model* model, std::string datadefinitionTypename, std::string name = "",
00048                         bool insertIntoModel = true);
00049     //ModelDataDefinition(Model* model, std::string datadefinitionTypename, std::string name = "",
      bool insertIntoModel = true);
00050     //ModelDataDefinition(const ModelDataDefinition &orig);
00054     virtual ~ModelDataDefinition();
00055
00056 public: // get & set
00061     Util::identification getId() const;
00066     void setName(const std::string& name);
00071     const std::string& getName() const;
00076     void setLabel(const std::string& label);
00081     const std::string& getLabel() const;
00086     const std::string& getClassname() const;
00091     Model* getParentModel() const;
00096     bool isReportStatistics() const;
00101     void setReportStatistics(bool reportStatistics);
00102
00103 public:
00109     ModelDataDefinition* getInternalData(const std::string& name) const;
00114     std::map<std::string, ModelDataDefinition*>* getInternalData() const;
00119     std::map<std::string, ModelDataDefinition*>* getAttachedData() const;
00120     //ModelDataDefinition* getInternalData(std::string key) const;
00125     bool hasChanged() const;
00130     void setHasChanged(bool hasChanged);
00139     unsigned int getLevel() const;
00147     void setModelLevel(unsigned int _modelLevel);
00154     List<SimulationControl*>* getSimulationControls() const;
00159     TraceManager::Level getTraceLevelSpecific() const;
00164     void setTraceLevelSpecific(TraceManager::Level level);
00170     void defineTraceLevelSpecific(TraceManager::Level traceLevelspecific, bool
      traceLevelSpecificEnabled = true);
00175     bool isTraceLevelSpecificEnabled() const;
00180     void setTraceLevelSpecificEnabled(bool traceLevelSpecificEnabled);
00181
00182 public: // public static methods
00194     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields, bool
      insertIntoModel);
00195     // @ToDo: (pequena alteração): return ModelComponent* ?
00205     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00213     static void SaveInstance(PersistenceRecord* fields, ModelDataDefinition* modeldatum);
00222     static bool Check(ModelDataDefinition* modeldatum, std::string& errorMessage);
00227     static void InitBetweenReplications(ModelDataDefinition* modeldatum);
00228     //static PluginInformation* GetPluginInformation();
00229
00230 public:
00238     static void CreateInternalData(ModelDataDefinition* modeldatum);
00247     static bool CreateRelatedDataElements(ModelDataDefinition* modeldatum, std::string& errorMessage);
00248     /* This class methood is responsible for invoking the protected method _initBetweenReplication(),
      which clears all statistics, attributes, counters and other stuff before starting a new repliction */
00249
00250 public: // public virtual methods
00258     virtual std::string show();
00261 protected:
00262     bool _getSaveDefaultsOption();
00263
00264 protected:
00265     virtual bool _loadInstance(PersistenceRecord* fields) override;
```

```
00266     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00267
00268 protected:
00269     virtual bool _check(std::string& errorMessage);
00271     virtual ParserChangesInformation* _getParserChangesInformation();
00272     virtual void _initBetweenReplications();
00273     /*< A ModelDataDefinition or ModelComponent that includes (internal) ou refers to (attach) other
      ModelDataDefinition must register them inside this method. */
00274     virtual void _addSimulationControl(SimulationControl* control);
00275     //virtual void _addSimulationResponse(SimulationControl* response);
00276
00277 /*@Todo: Fix this mess about diffent kind of internal and attached data related to report statistics,
00278  * mandatory and optional editable internal or attached dataelements that are properties,
00279  * and, I don't know, maybe other kind of internal and attached nedded stuff?
00280  */
00281 protected:
00283     virtual void _createInternalAndAttachedData(); // final; //@Todo: will be final
00284 protected:
00285     virtual void _createInternalStatisticReporters();
00286     virtual void _createAttachedAttributes();
00287     virtual void _createNonEditableDataDefinitions();
00288     virtual void _createEditableDataDefinitions();
00289 protected:
00290     void _templateCreateInternalStatisticReporters();
00291     void _templateCreateAttachedAttributes();
00292     void _templateCreateNonEditableDataDefinitions();
00293     void _templateCreateEditableDataDefinitions();
00294 protected:
00295     void _internaStatisticReportersClear();
00296     void _internalDataClear();
00297     void _internalDataInsert(const std::string& key, ModelDataDefinition* child);
00298     void _internalDataRemove(const std::string& key);
00299     void _attachedDataClear();
00300     void _attachedDataInsert(const std::string& key, ModelDataDefinition* data);
00301     void _attachedDataRemove(const std::string& key);
00302     void _attachedAttributesInsert(const std::vector<std::string>& neededNames);
00303     void _statisticReporterInsert(const std::string& key, ModelDataDefinition* data);
00304     void _statisticReporterRemove(const std::string& key);
00305     void _statisticReportersClear();
00306     void _mandatoryAttachedAttributeInsert(const std::string& key, ModelDataDefinition* data);
00307     void _mandatoryAttachedAttributeRemove(const std::string& key);
00308     void _mandatoryAttachedAttributesClear();
00309     void _mandatoryEditableDataDefinitionInsert(const std::string& key, ModelDataDefinition* data);
00310     void _mandatoryEditableDataDefinitionRemove(const std::string& key);
00311     void _mandatoryEditableDataDefinitionsClear();
00312     void _optionalEditableDataDefinitionInsert(const std::string& key, ModelDataDefinition* data);
00313     void _optionalEditableDataDefinitionRemove(const std::string& key);
00314     void _optionalEditableDataDefinitionsClear();
00315     void _mandatoryNonEditableDataDefinitionInsert(const std::string& key, ModelDataDefinition* data);
00316     void _mandatoryNonEditableDataDefinitionRemove(const std::string& key);
00317     void _mandatoryNonEditableDataDefinitionsClear();
00318     void _optionalNonEditableDataDefinitionInsert(const std::string& key, ModelDataDefinition* data);
00319     void _optionalNonEditableDataDefinitionRemove(const std::string& key);
00320     void _optionalNonEditableDataDefinitionsClear();
00321 protected:
00323     void _checkCreateAttachedReferencedDataDefinition(const std::string& expression);
00324     //(std::map<std::string, std::list<std::string>*>* referencedDataDefinitions);
00325
00326
00327 private:
00328     // name is now private. So changes in name must be throught setName, wich gives oportunity to
      rename internelElements, SimulationControls and SimulationResponses
00329     std::string _name;
00330     std::string _label;
00331
00332 private:
00333     friend class ModelDataDefinitionAssociations;
00334     ModelDataDefinitionAssociations _dataAssociations;
00335
00336 protected:
00337     Util::identification _id;
00338     std::string _typename;
00339     bool _reportStatistics;
00340     bool _hasChanged;
00341     unsigned int _modelLevel = 0; // 0=root; otherwise ID of the ModelDataDefinition defining this
      editing context.
00342     Model* _parentModel;
00343
00344 private:
00345     bool _checkSpecificTraceLevel(TraceManager::Level level);
00346
00347 protected:
00348     void trace(const std::string& text, TraceManager::Level level = TraceManager::Level::L8_detailed);
00349     void traceError(const std::string& text, const std::exception& e);
00350     void traceError(const std::string& text, TraceManager::Level level =
      TraceManager::Level::L1_errorFatal);
00351     void traceReport(const std::string& text, TraceManager::Level level =
```

```
    TraceManager::Level::L2_results);
00352     void traceSimulation(void* thisobject, double time, Entity* entity, ModelComponent* component,
    const std::string& text,
00353                          TraceManager::Level level = TraceManager::Level::L8_detailed);
00362     void traceSimulation(void* thisobject, const std::string& text,
00363                          TraceManager::Level level = TraceManager::Level::L8_detailed);
00364     void traceSimulation(void* thisobject, TraceManager::Level level, const std::string& text);
00365
00366 protected:
00367     //List<SimulationControl*>* _simulationResponses = new List<SimulationControl*>();
00368     List<SimulationControl*>* _simulationControls = new List<SimulationControl*>();
00369     TraceManager::Level _traceLevelSpecific;
00370     bool _traceLevelSpecificEnabled = false;
00371     //PropertyListG* _propertiesG = new PropertyListG();
00372 };
00373
00374 //namespace\\}
00375
00376 #endif /* MODELELEMENT_H */
```

18.312 ModelDataDefinitionAssociations.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ModelDataDefinitionAssociations.h
00009  * Author: Codex
00010  *
00011  * Created on 6 de Maio de 2026
00012  */
00013
00014 #ifndef MODELDATADEFINITIONASSOCIATIONS_H
00015 #define MODELDATADEFINITIONASSOCIATIONS_H
00016
00017 #include <map>
00018 #include <string>
00019 #include <vector>
00020
00021 class Attribute;
00022 class Model;
00023 class ModelDataDefinition;
00024
00032 class ModelDataDefinitionAssociations {
00033 public:
00034     explicit ModelDataDefinitionAssociations(ModelDataDefinition* owner);
00035     ~ModelDataDefinitionAssociations();
00036
00037 public:
00038     std::map<std::string, ModelDataDefinition*>* getInternalData() const;
00039     std::map<std::string, ModelDataDefinition*>* getAttachedData() const;
00040     ModelDataDefinition* getInternalData(const std::string& key) const;
00041
00042     void clearAll();
00043     void clearInternalData();
00044     void clearAttachedData();
00045
00046     void internalDataInsert(const std::string& key, ModelDataDefinition* data);
00047     void internalDataRemove(const std::string& key);
00048     void attachedDataInsert(const std::string& key, ModelDataDefinition* data);
00049     void attachedDataRemove(const std::string& key);
00050     void attachedAttributesInsert(const std::vector<std::string>& neededNames);
00051     void checkCreateAttachedReferencedDataDefinition(const std::string& expression);
00052
00053     void createInternalAndAttachedData();
00054     void createInternalStatisticReporters();
00055     void createAttachedAttributes();
00056     void createNonEditableDataDefinitions();
00057     void createEditableDataDefinitions();
00058
00059     void insertStatisticReporter(const std::string& key, ModelDataDefinition* data);
00060     void removeStatisticReporter(const std::string& key);
00061     void clearStatisticReporters();
00062
00063     void insertMandatoryAttachedAttribute(const std::string& key, ModelDataDefinition* data);
00064     void removeMandatoryAttachedAttribute(const std::string& key);
00065     void clearMandatoryAttachedAttributes();
00066
00067     void insertMandatoryEditableDataDefinition(const std::string& key, ModelDataDefinition* data);
00068     void removeMandatoryEditableDataDefinition(const std::string& key);
00069     void clearMandatoryEditableDataDefinitions();
00070
```

```
00071     void insertOptionalEditableDataDefinition(const std::string& key, ModelDataDefinition* data);
00072     void removeOptionalEditableDataDefinition(const std::string& key);
00073     void clearOptionalEditableDataDefinitions();
00074
00075     void insertMandatoryNonEditableDataDefinition(const std::string& key, ModelDataDefinition* data);
00076     void removeMandatoryNonEditableDataDefinition(const std::string& key);
00077     void clearMandatoryNonEditableDataDefinitions();
00078
00079     void insertOptionalNonEditableDataDefinition(const std::string& key, ModelDataDefinition* data);
00080     void removeOptionalNonEditableDataDefinition(const std::string& key);
00081     void clearOptionalNonEditableDataDefinitions();
00082
00083 private:
00084     void _deleteOwnedInternalData(ModelDataDefinition* data);
00085     void _rebuildLegacyViews();
00086
00087 private:
00088     ModelDataDefinition* _owner = nullptr;
00089     std::map<std::string, ModelDataDefinition*> _internalData;
00090     std::map<std::string, ModelDataDefinition*> _attachedData;
00091     std::map<std::string, ModelDataDefinition*> _statisticReporters;
00092     std::map<std::string, ModelDataDefinition*> _mandatoryAttachedAttributes;
00093     std::map<std::string, ModelDataDefinition*> _mandatoryEditableDataDefinitions;
00094     std::map<std::string, ModelDataDefinition*> _optionalEditableDataDefinitions;
00095     std::map<std::string, ModelDataDefinition*> _mandatoryNonEditableDataDefinitions;
00096     std::map<std::string, ModelDataDefinition*> _optionalNonEditableDataDefinitions;
00097 };
00098
00099 #endif /* MODELDATADEFINITIONASSOCIATIONS_H */
```

18.313 ModelDataManager.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ElementManager.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 7 de Novembro de 2018, 12:48
00012  */
00013
00014 #ifndef ELEMENTMANAGER_H
00015 #define ELEMENTMANAGER_H
00016
00017 #include <list>
00018 #include <map>
00019 #include "../../util/List.h"
00020 #include "ModelDataDefinition.h"
00021
00022 //namespace GenesysKernel {
00023 class Model;
00024
00031 class ModelDataManager {
00032 public:
00034     ModelDataManager(Model* model);
00035     virtual ~ModelDataManager();
00036 public:
00042     bool insert(ModelDataDefinition* anElement);
00047     void remove(ModelDataDefinition* anElement);
00054     bool insert(std::string datadefinitionTypename, ModelDataDefinition* anElement);
00060     void remove(std::string datadefinitionTypename, ModelDataDefinition* anElement);
00069     bool check(std::string datadefinitionTypename, ModelDataDefinition* anElement, std::string
      expressionName, std::string& errorMessage);
00079     bool check(std::string datadefinitionTypename, std::string elementName, std::string
      expressionName, bool mandatory, std::string& errorMessage);
00083     void clear();
00084 public:
00091     ModelDataDefinition* getDataDefinition(std::string datadefinitionTypename, Util::identification
      id);
00098     ModelDataDefinition* getDataDefinition(std::string datadefinitionTypename, std::string name);
00104     unsigned int getNumberOfDataDefinitions(std::string datadefinitionTypename);
00109     unsigned int getNumberOfDataDefinitions();
00116     int getRankOf(std::string datadefinitionTypename, std::string name);
00121     std::list<std::string> getDataDefinitionClassnames() const;
00122
00123     //private:
00124 public:
00125     // @ToDo: (importante): Should be removed and replaced by GetElement(elementType, rank)
00126     List<ModelDataDefinition*>* getDataDefinitionList(std::string datadefinitionTypename) const;
00127 public:
00129     void show();
```

```
00131     Model* getParentModel() const;
00133     bool hasChanged() const;
00135     void setHasChanged(bool _hasChanged);
00136 private:
00137     std::map<std::string, List<ModelDataDefinition*>*>* _datadefinitions;
00138     Model* _parentModel;
00139     bool _hasChanged = false;
00140 };
00141 //namespace\\}
00142 #endif /* ELEMENTMANAGER_H */
```

18.314 ModelInfo.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ModelInfo.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 7 de Novembro de 2018, 18:13
00012  */
00013
00014 #ifndef MODELINFO_H
00015 #define MODELINFO_H
00016
00017 #include <string>
00018 #include "../../util/Util.h"
00019 #include "../Persistence.h"
00020
00021 //namespace GenesysKernel {
00022
00026 class ModelInfo {
00027 public:
00028     ModelInfo();
00029     virtual ~ModelInfo() = default;
00030 public:
00031     std::string show();
00032 public: // gets and sets
00037     void setName(std::string _name);
00042     std::string getName() const;
00047     void setAnalystName(std::string _analystName);
00052     std::string getAnalystName() const;
00057     void setDescription(std::string _description);
00062     std::string getDescription() const;
00067     void setProjectTitle(std::string _projectTitle);
00072     std::string getProjectTitle() const;
00077     void setVersion(std::string _version);
00082     std::string getVersion() const;
00083 public:
00088     void loadInstance(PersistenceRecord *fields);
00093     void saveInstance(PersistenceRecord *fields);
00098     bool hasChanged() const;
00100     void setHasChanged(bool hasChanged);
00101 private: // with public access (get & set)
00102     // model general information
00103     std::string _name;
00104     std::string _analystName = "";
00105     std::string _description = "";
00106     std::string _projectTitle = "";
00107     std::string _version = "1.0";
00108     bool _hasChanged = false;
00109 };
00110 //namespace\\}
00111 #endif /* MODELINFO_H */
```

18.315 ModelManager.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ModelManager.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 31 de Maio de 2019, 08:37
```

```
00012  */
00013
00014 #ifndef MODELMANAGER_H
00015 #define MODELMANAGER_H
00016
00017 #include "Model.h"
00018 #include "../TraceManager.h"
00019 #include <vector>
00020
00021 //namespace GenesysKernel {
00022
00026 class ModelManager {
00027 public:
00032     ModelManager(Simulator* simulator);
00036     virtual ~ModelManager();
00037 public:
00042     Model* newModel();
00047     void insert(Model* model);
00052     void remove(Model* model);
00058     bool setCurrent(Model* model);
00064     bool saveModel(std::string filename);
00070     Model* loadModel(std::string filename);
00076     Model* createFromLanguage(std::string modelSpecification);
00081     unsigned int size();
00082 public:
00087     std::vector<Model*> models() const;
00093     bool hasModel(Model* model) const;
00099     int indexOf(Model* model) const;
00105     Model* modelAt(unsigned int index) const;
00110     int currentIndex() const;
00115     Model* front();
00120     Model* last();
00125     Model* current();
00130     Model* next();
00135     Model* previous();
00140     bool canGoNext() const;
00145     bool canGoPrevious() const;
00146     //Model* end();
00147 private:
00148     List<Model*>* _models = new List<Model*>();
00149     Model* _currentModel = nullptr;
00150 private:
00151     Simulator* _simulator;
00152 };
00153 //namespace\\}
00154 #endif /* MODELMANAGER_H */
```

18.316 ModelSimulation.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ModelSimulation.h
00009  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00010  *
00011  * Created on 7 de Novembro de 2018, 18:04
00012  */
00013
00014 #ifndef MODELSIMULATION_H
00015 #define MODELSIMULATION_H
00016
00017 #include <chrono>
00018 #include <memory>
00019 #include "../SimulationControlAndResponse.h"
00020 #include "../Event.h"
00021 #include "../essentialPlugins/Entity.h"
00022 #include "ModelInfo.h"
00023 #include "../SimulationReporter_if.h"
00024 #include "../OnEventManager.h"
00025 //#include "Counter.h"
00026 //namespace GenesysKernel {
00027
00028 //#include "Model.h" // for friend functions
00029 class Model;
00030
00047 class ModelSimulation { // 202104 to be subjected to SimulationScenario
00048 public:
00049     ModelSimulation(Model* model);
00050     virtual ~ModelSimulation();
00051 public:
00052     std::string show();
```

```
00053 // @ToDo: (importante): Keep ModelSimulation as the stable
00054 // execution core while the higher-level experiment abstractions mature.
00055 // Any migration of responsibilities to SimulationExperiment/SimulationScenario
00056 // should preserve the already functional replication-based workflow here.
00057 public: // simulation control
00058     // @ToDo: (importante): Revisit how this class should interact with the unfinished higher-level
      experiment layer.
00059     void start();
00060     void pause();
00061     void step();
00062     void stop();
00063 public: // old modelInfos
00064     void setNumberOfReplications(unsigned int _numberOfReplications);
00065     unsigned int getNumberOfReplications() const;
00066     //void setReplicationLength(double _replicationLength);
00067     void setReplicationLength(double _replicationLength, Util::TimeUnit _replicationLengthTimeUnit =
      Util::TimeUnit::unknown);
00068     double getReplicationLength() const;
00069     void setReplicationLengthTimeUnit(Util::TimeUnit _replicationLengthTimeUnit);
00070     Util::TimeUnit getReplicationLengthTimeUnit() const;
00071     void setReplicationReportBaseTimeUnit(Util::TimeUnit _replicationReportBaseTimeUnit);
00072     Util::TimeUnit getReplicationBaseTimeUnit() const;
00073     //void setWarmUpPeriod(double warmUpPeriod);
00074     void setWarmUpPeriod(double warmUpPeriod, Util::TimeUnit warmUpPeriodTimeUnit =
      Util::TimeUnit::unknown);
00075     double getWarmUpPeriod() const;
00076     void setWarmUpPeriodTimeUnit(Util::TimeUnit _warmUpPeriodTimeUnit);
00077     Util::TimeUnit getWarmUpPeriodTimeUnit() const;
00078     void setTerminatingCondition(std::string _terminatingCondition);
00079     std::string getTerminatingCondition() const;
00080 public: // gets and sets
00081     void setPauseOnEvent(bool _pauseOnEvent);
00082     bool isPauseOnEvent() const;
00083     void setStepByStep(bool _stepByStep);
00084     bool isStepByStep() const;
00085     void setInitializeStatistics(bool _initializeStatistics);
00086     bool isInitializeStatistics() const;
00087     void setInitializeSystem(bool _initializeSystem);
00088     bool isInitializeSystem() const;
00089     void setPauseOnReplication(bool _pauseBetweenReplications);
00090     bool isPauseOnReplication() const;
00091 public: // gets and sets ModelSubParts
00092     void setReporter(SimulationReporter_if* _simulationReporter);
00093     SimulationReporter_if* getReporter() const;
00094 public: // only gets
00095     double getSimulatedTime() const;
00096     bool isRunning() const;
00097     bool isPaused() const;
00098     unsigned int getCurrentReplicationNumber() const;
00099     void setShowReportsAfterReplication(bool showReportsAfterReplication);
00100     bool isShowReportsAfterReplication() const;
00101     void setShowReportsAfterSimulation(bool showReportsAfterSimulation);
00102     bool isShowReportsAfterSimulation() const;
00103     List<double>* getBreakpointsOnTime() const;
00104     List<Entity*>* getBreakpointsOnEntity() const;
00105     List<ModelComponent*>* getBreakpointsOnComponent() const;
00106     const List<ModelDataDefinition*>* getSimulationStatisticsAggregates() const;
00107 public:
00108     void loadInstance(PersistenceRecord *fields);
00109     void saveInstance(PersistenceRecord *fields, bool saveDefaults);
00110     Event* getCurrentEvent() const;
00111     void setShowSimulationResposesInReport(bool _showSimulationResposesInReport);
00112     bool isShowSimulationResposesInReport() const;
00113     void setShowSimulationControlsInReport(bool _showSimulationControlsInReport);
00114     bool isShowSimulationControlsInReport() const;
00115     /*
00116      * PRIVATE
00117      */
00118 private: // simulation control
00119     void _initSimulation();
00120     void _initReplication();
00121     void _clearStatistics();
00122     void _checkWarmUpTime(Event* nextEvent);
00123     void _stepSimulation();
00124     void _replicationEnded();
00125     void _simulationEnded();
00126 private:
00127     void _dispatchEvent(Event* event);
00128     bool _checkBreakpointAt(Event* event);
00129     bool _isReplicationEndCondition();
00130     void _actualizeSimulationStatistics();
00131     void _showSimulationHeader();
00132     void _traceReplicationEnded();
00133 private:
00134     std::unique_ptr<SimulationEvent> _createSimulationEvent(void* thiscustomObject = nullptr);
00135     //friend void Model::removeEntity(Entity* entity);
00136     //friend void Model::sendEntityToComponent(Entity* entity, ModelComponent* component, double
```

```
    timeDelay, unsigned int componentinputPortNumber);
00137     friend class Model;
00138 private:
00139     double _simulatedTime = 0.0;
00140     double _justTriggeredBreakpointsOnTime = 0.0;
00141     // @ToDo: (pequena alteração): list of double double _breakOnTimes;
00142     // @ToDo: (pequena alteração): list of modules _breakOnModules;
00143     bool _stepByStep = false;
00144     bool _pauseOnReplication = false;
00145     bool _pauseOnEvent = false;
00146     bool _initializeStatisticsBetweenReplications = true;
00147     bool _initializeSystem = true;
00148     bool _isRunning = false;
00149     bool _isPaused = false;
00150     bool _pauseRequested = false;
00151     bool _stopRequested = false;
00152     bool _simulationIsInitiated = false;
00153     bool _replicationIsInitiaded = false;
00154     bool _hasChanged = false;
00155 private:
00156
00157     const struct DEFAULT_VALUES {
00158         const unsigned int numberOfReplications = 1;
00159         const double replicationLength = 60.0; // by default, 60s = 1min
00160         const Util::TimeUnit replicationLengthTimeUnit = Util::TimeUnit::second;
00161         const Util::TimeUnit replicationBeseTimeUnit = Util::TimeUnit::second;
00162         const Util::TimeUnit replicationReportBaseTimeUnit = Util::TimeUnit::second;
00163         const double warmUpPeriod = 0.0;
00164         const Util::TimeUnit warmUpPeriodTimeUnit = Util::TimeUnit::second;
00165         const std::string terminatingCondition = "";
00166         const bool initializeStatisticsBetweenReplications = true;
00167         const bool initializeSystem = true;
00168         const bool showReportsAfterSimulation = true;
00169         const bool showReportsAfterReplication = true;
00170         const bool showSimulationControlsInReport = false;
00171         const bool showSimulationResposesInReport = false;
00172     } DEFAULT;
00173     unsigned int _numberOfReplications = DEFAULT.numberOfReplications;
00174     double _replicationLength = DEFAULT.replicationLength;
00175     Util::TimeUnit _replicationLengthTimeUnit = DEFAULT.replicationLengthTimeUnit;
00176     Util::TimeUnit _replicationBaseTimeUnit = DEFAULT.replicationReportBaseTimeUnit;
00177     double _warmUpPeriod = DEFAULT.warmUpPeriod;
00178     Util::TimeUnit _warmUpPeriodTimeUnit = DEFAULT.warmUpPeriodTimeUnit;
00179     std::string _terminatingCondition = DEFAULT.terminatingCondition;
00180     bool _showReportsAfterSimulation = DEFAULT.showReportsAfterSimulation;
00181     bool _showReportsAfterReplication = DEFAULT.showReportsAfterReplication;
00182     bool _showSimulationControlsInReport = DEFAULT.showSimulationControlsInReport;
00183     bool _showSimulationResposesInReport = DEFAULT.showSimulationResposesInReport;
00184     //
00185     double _replicationTimeScaleFactorToBase; // a scale that converts ReplicationLenghtTimeUnit to
    ReplicationBaseTimeUnit. Future events are in "times" of unit ReplicationBaseTimeUnit
00186     std::chrono::system_clock::time_point _startRealSimulationTimeSimulation;
00187     std::chrono::system_clock::time_point _startRealSimulationTimeReplication;
00188 private:
00189     Event* _currentEvent = nullptr;
00190     unsigned int _currentReplicationNumber;
00191 private:
00192     const std::string _cte_stCountSimulNamePrefix = ""; //Simul.";
00193 private:
00194     Model* _model;
00195     ModelInfo* _info;
00196     SimulationReporter_if* _simulationReporter;
00197     bool _ownsSimulationReporter = true;
00198     List<SimulationControl*>* _ownedControls = new List<SimulationControl*>();
00199     // @ToDo: (importante): Change List below to a MAP, associating every CstatOuCounter in the
    replication to the equivalent in the simulation
00200     List<ModelDataDefinition*>* _cstatsAndCountersSimulation = new List<ModelDataDefinition*>();
00201     std::map<ModelDataDefinition*, ModelDataDefinition*>* _cstatsAndCountersMapSimulation = new
    std::map<ModelDataDefinition*, ModelDataDefinition*>();
00202     List<double>* _breakpointsOnTime = new List<double>();
00203     List<ModelComponent*>* _breakpointsOnComponent = new List<ModelComponent*>();
00204     List<Entity*>* _breakpointsOnEntity = new List<Entity*>();
00205     ModelComponent* _justTriggeredBreakpointsOnComponent = nullptr;
00206     Entity* _justTriggeredBreakpointsOnEntity = nullptr;
00207 };
00208 //namespace\\}
00209 #endif /* MODELSIMULATION_H */
```

18.317 OnEventManager.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
```

```
00006
00007 /*
00008  * File:   OnEventManager.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 7 de Novembro de 2018, 12:28
00012  */
00013
00014 #ifndef ONEVENTMANAGER_H
00015 #define ONEVENTMANAGER_H
00016
00017 #include "../util/List.h"
00018 #include "Event.h"
00019
00020 //namespace GenesysKernel {
00021 /*
00022  * used to get and set values no matter the class (for process analyser)
00023  * should be a wait to invoke a getter or setter no matter the class (a pointer to a member function
      without specifying the class
00024  */
00025 //typedef double (*memberFunctionGetDoubleVarHandler)(); //template<> ... typedef double
      (T::*getDoubleVarHandler)() or something like that
00026 //typedef void (*memberFunctionSetDoubleVarHandler)(double);
00027
00028
00031 class ModelEvent {
00032 public:
00033     ModelEvent(Model* model){
00034         _model = model;
00035     };
00036 public:
00037     const Model* getModel() {
00038         return _model;
00039     }
00040 private:
00041     Model* _model;
00042 };
00043 typedef void (*modelEventHandler)(ModelEvent*);
00044 // for handlers that are class members (methods)
00045 typedef std::function<void(ModelEvent*) > modelEventHandlerMethod;
00046
00047
00050 class SimulationEvent {
00051 public:
00052
00053     void setSimulatedTime(double simulatedTime) {
00054         this->simulatedTime = simulatedTime;
00055     }
00056
00057     double getSimulatedTime() const {
00058         return simulatedTime;
00059     }
00060
00061     void setStopRequested(bool stopRequested) {
00062         this->stopRequested = stopRequested;
00063     }
00064
00065     bool isStopRequested() const {
00066         return stopRequested;
00067     }
00068
00069     void setPauseRequested(bool pauseRequested) {
00070         this->pauseRequested = pauseRequested;
00071     }
00072
00073     bool isPauseRequested() const {
00074         return pauseRequested;
00075     }
00076
00077     void setPaused(bool Paused) {
00078         _isPaused = Paused;
00079     }
00080
00081     bool isPaused() const {
00082         return _isPaused;
00083     }
00084
00085     void setRunning(bool Running) {
00086         _isRunning = Running;
00087     }
00088
00089     bool isRunning() const {
00090         return _isRunning;
00091     }
00092
00093     void setCustomObject(void* customObject) {
00094         this->customObject = customObject;
```

```
00095     }
00096
00097     void* getCustomObject() const {
00098         return customObject;
00099     }
00100
00101     void setCurrentReplicationNumber(unsigned int currentReplicationNumber) {
00102         this->currentReplicationNumber = currentReplicationNumber;
00103     }
00104
00105     unsigned int getCurrentReplicationNumber() const {
00106         return currentReplicationNumber;
00107     }
00108
00109     void setCurrentEvent(Event* currentEvent) {
00110         this->currentEvent = currentEvent;
00111     }
00112
00113     Event* getCurrentEvent() const {
00114         return currentEvent;
00115     }
00116     //void setCurrentinputPortNumber(unsigned int currentinputPortNumber) {
00117     //  this->currentinputPortNumber = currentinputPortNumber;
00118     //}
00119     //unsigned int getCurrentinputPortNumber() const {
00120     //  return currentinputPortNumber;
00121     //}
00122     //void setCurrentComponent(ModelComponent* currentComponent) {
00123     //  this->currentComponent = currentComponent;
00124     //}
00125     //ModelComponent* getCurrentComponent() const {
00126     //  return currentComponent;
00127     //}
00128     //void setCurrentEntity(Entity* currentEntity) {
00129     //  this->currentEntity = currentEntity;
00130     //}
00131     //Entity* getCurrentEntity() const {
00132     //  return currentEntity;
00133     //}
00134
00135     void setEntityCreated(Entity* entityCreated) {
00136         this->entityCreated = entityCreated;
00137     }
00138
00139     Entity* getEntityCreated() const {
00140         return entityCreated;
00141     }
00142
00143     void setDestinationComponent(ModelComponent* destinationComponent) {
00144         this->destinationComponent = destinationComponent;
00145     }
00146
00147     ModelComponent* getDestinationComponent() const {
00148         return destinationComponent;
00149     }
00150
00151     void setEntityMoveTimeDelay(double entityMoveTimeDelay) {
00152         this->entityMoveTimeDelay = entityMoveTimeDelay;
00153     }
00154
00155     double getEntityMoveTimeDelay() const {
00156         return entityMoveTimeDelay;
00157     }
00158 private:
00159
00160     SimulationEvent() {
00161     }
00162     friend class ModelSimulation;
00163 private:
00164     Entity* entityCreated = nullptr;
00165     Event* currentEvent = nullptr;
00166     //Entity* currentEntity = nullptr;
00167     //unsigned int currentinputPortNumber = 0;
00168     //ModelComponent* currentComponent = nullptr;
00169     ModelComponent* destinationComponent = nullptr;
00170     void* customObject = nullptr;
00171     unsigned int currentReplicationNumber = 0;
00172     double simulatedTime = 0.0;
00173     double entityMoveTimeDelay = 0.0;
00174     bool _isRunning = false;
00175     bool _isPaused = false;
00176     bool pauseRequested = false;
00177     bool stopRequested = false;
00178 };
00179 typedef void (*simulationEventHandler)(SimulationEvent*);
00180 // for handlers that are class members (methods)
00181 typedef std::function<void(SimulationEvent*) > simulationEventHandlerMethod;
```

```
00182
00187 class OnEventManager {
00188 public:
00189     OnEventManager();
00190     virtual ~OnEventManager();
00191 public: // event listeners (handlers)
00192     void addOnModelCheckSucessHandler(modelEventHandler EventHandler);
00193     void addOnModelLoadHandler(modelEventHandler EventHandler);
00194     void addOnModelSaveHandler(modelEventHandler EventHandler);
00195     void addOnReplicationStartHandler(simulationEventHandler EventHandler);
00196     void addOnReplicationStepHandler(simulationEventHandler EventHandler);
00197     void addOnReplicationEndHandler(simulationEventHandler EventHandler);
00198     void addOnProcessEventHandler(simulationEventHandler EventHandler);
00199     void addOnAfterProcessEventHandler(simulationEventHandler EventHandler);
00200     void addOnEntityCreateHandler(simulationEventHandler EventHandler);
00201     void addOnEntityMoveHandler(simulationEventHandler EventHandler);
00202     void addOnEntityRemoveHandler(simulationEventHandler EventHandler);
00203     void addOnSimulationStartHandler(simulationEventHandler EventHandler);
00204     void addOnSimulationPausedHandler(simulationEventHandler EventHandler);
00205     void addOnSimulationResumeHandler(simulationEventHandler EventHandler);
00206     void addOnSimulationEndHandler(simulationEventHandler EventHandler);
00207     void addOnBreakpointHandler(simulationEventHandler EventHandler);
00208 public: // event listeners (method handlers)
00209     // for handlers that are class members (methods)
00210     template<typename Class> void addOnModelCheckSuccessHandler(Class * object, void
    (Class::*function)(ModelEvent*));
00211     template<typename Class> void addOnModelLoadHandler(Class * object, void
    (Class::*function)(ModelEvent*));
00212     template<typename Class> void addOnModelSaveHandler(Class * object, void
    (Class::*function)(ModelEvent*));
00213     template<typename Class> void addOnReplicationStartHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00214     template<typename Class> void addOnReplicationStepHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00215     template<typename Class> void addOnReplicationEndHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00216     template<typename Class> void addOnProcessEventHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00217     template<typename Class> void addOnAfterProcessEventHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00218     template<typename Class> void addOnEntityCreateHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00219     template<typename Class> void addOnEntityMoveHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00220     template<typename Class> void addOnEntityRemoveHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00221     template<typename Class> void addOnSimulationStartHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00222     template<typename Class> void addOnSimulationPausedHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00223     template<typename Class> void addOnSimulationResumeHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00224     template<typename Class> void addOnSimulationEndHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00225     template<typename Class> void addOnBreakpointHandler(Class * object, void
    (Class::*function)(SimulationEvent*));
00226
00227 public:
00228     void NotifyModelCheckSuccessHandlers(ModelEvent* se);
00229     void NotifyModelLoadHandlers(ModelEvent* se);
00230     void NotifyModelSaveHandlers(ModelEvent* se);
00231     void NotifyReplicationStartHandlers(SimulationEvent* se);
00232     void NotifyReplicationStepHandlers(SimulationEvent* se);
00233     void NotifyReplicationEndHandlers(SimulationEvent* se);
00234     void NotifyProcessEventHandlers(SimulationEvent* se);
00235     void NotifyAfterProcessEventHandlers(SimulationEvent* se);
00236     void NotifyEntityCreateHandlers(SimulationEvent* se);
00237     void NotifyEntityMoveHandlers(SimulationEvent* se);
00238     void NotifyEntityRemoveHandlers(SimulationEvent* se);
00239     void NotifySimulationStartHandlers(SimulationEvent* se);
00240     void NotifySimulationPausedHandlers(SimulationEvent* se);
00241     void NotifySimulationResumeHandlers(SimulationEvent* se);
00242     void NotifySimulationEndHandlers(SimulationEvent* se);
00243     void NotifyBreakpointHandlers(SimulationEvent* se);
00244
00245 private:
00246     void _NotifyHandlers(List<modelEventHandler>* list, ModelEvent* se);
00247     void _NotifyHandlers(List<simulationEventHandler>* list, SimulationEvent* se);
00248     void _NotifyHandlerMethods(List<modelEventHandlerMethod>* list, ModelEvent* se);
00249     void _NotifyHandlerMethods(List<simulationEventHandlerMethod>* list, SimulationEvent* se);
00250     void _addOnHandler(List<modelEventHandler>* list, modelEventHandler EventHandler);
00251     void _addOnHandler(List<simulationEventHandler>* list, simulationEventHandler EventHandler);
00252 private: // events listener
00253     List<modelEventHandler>* _onModelCheckSuccessHandlers = new List<modelEventHandler>();
00254     List<modelEventHandler>* _onModelLoadHandlers = new List<modelEventHandler>();
00255     List<modelEventHandler>* _onModelSaveHandlers = new List<modelEventHandler>();
00256     List<simulationEventHandler>* _onReplicationStartHandlers = new List<simulationEventHandler>();
```

```
00257     List<simulationEventHandler>* _onReplicationStepHandlers = new List<simulationEventHandler>();
00258     List<simulationEventHandler>* _onReplicationEndHandlers = new List<simulationEventHandler>();
00259     List<simulationEventHandler>* _onProcessEventHandlers = new List<simulationEventHandler>();
00260     List<simulationEventHandler>* _onAfterProcessEventHandlers = new List<simulationEventHandler>();
00261     List<simulationEventHandler>* _onEntityCreateHandlers = new List<simulationEventHandler>();
00262     List<simulationEventHandler>* _onEntityMoveHandlers = new List<simulationEventHandler>();
00263     List<simulationEventHandler>* _onEntityRemoveHandlers = new List<simulationEventHandler>();
00264     List<simulationEventHandler>* _onSimulationStartHandlers = new List<simulationEventHandler>();
00265     List<simulationEventHandler>* _onSimulationPausedHandlers = new List<simulationEventHandler>();
00266     List<simulationEventHandler>* _onSimulationResumeHandlers = new List<simulationEventHandler>();
00267     List<simulationEventHandler>* _onSimulationEndHandlers = new List<simulationEventHandler>();
00268     List<simulationEventHandler>* _onBreakpointHandlers = new List<simulationEventHandler>();
00269 private: // events listener for handlers that are class members (methods)
00270     List<modelEventHandlerMethod>* _onModelCheckSuccessHandlerMethods = new
      List<modelEventHandlerMethod>();
00271     List<modelEventHandlerMethod>* _onModelLoadHandlerMethods = new List<modelEventHandlerMethod>();
00272     List<modelEventHandlerMethod>* _onModelSaveHandlerMethods = new List<modelEventHandlerMethod>();
00273     List<simulationEventHandlerMethod>* _onReplicationStartHandlerMethods = new
      List<simulationEventHandlerMethod>();
00274     List<simulationEventHandlerMethod>* _onReplicationStepHandlerMethods = new
      List<simulationEventHandlerMethod>();
00275     List<simulationEventHandlerMethod>* _onReplicationEndHandlerMethods = new
      List<simulationEventHandlerMethod>();
00276     List<simulationEventHandlerMethod>* _onProcessEventHandlerMethods = new
      List<simulationEventHandlerMethod>();
00277     List<simulationEventHandlerMethod>* _onAfterProcessEventHandlerMethods = new
      List<simulationEventHandlerMethod>();
00278     List<simulationEventHandlerMethod>* _onEntityCreateHandlerMethods = new
      List<simulationEventHandlerMethod>();
00279     List<simulationEventHandlerMethod>* _onEntityMoveHandlerMethods = new
      List<simulationEventHandlerMethod>();
00280     List<simulationEventHandlerMethod>* _onEntityRemoveHandlerMethods = new
      List<simulationEventHandlerMethod>();
00281     List<simulationEventHandlerMethod>* _onSimulationStartHandlerMethods = new
      List<simulationEventHandlerMethod>();
00282     List<simulationEventHandlerMethod>* _onSimulationPausedHandlerMethods = new
      List<simulationEventHandlerMethod>();
00283     List<simulationEventHandlerMethod>* _onSimulationResumeHandlerMethods = new
      List<simulationEventHandlerMethod>();
00284     List<simulationEventHandlerMethod>* _onSimulationEndHandlerMethods = new
      List<simulationEventHandlerMethod>();
00285     List<simulationEventHandlerMethod>* _onBreakpointHandlerMethods = new
      List<simulationEventHandlerMethod>();
00286 };
00287
00288 //
00289 // implementation for template methods
00290 //
00291 //  // @ToDo: (importante): (!!) Complicated: if handlerMethod already insert, should not insert it
      again. Problem to solve <...> for function
00292 //if (_onProcessEventHandlerMethods->find(handlerMethod) ==
      _onProcessEventHandlerMethods->list()->end())
00293 // trying unique to solve the issue
00294 //this->_onProcessEventHandlerMethods->list()->unique(); // does not work
00295 //  @ToDo: (importante): probabily to override == operator for type simulationEventHandlerMethod
00296 // ...
00297
00298 template<typename Class> void OnEventManager::addOnModelCheckSuccessHandler(Class * object, void
      (Class::*function)(ModelEvent*)) {
00299     modelEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00300     this->_onModelCheckSuccessHandlerMethods->insert(handlerMethod);
00301 }
00302
00303 template<typename Class> void OnEventManager::addOnModelLoadHandler(Class * object, void
      (Class::*function)(ModelEvent*)) {
00304     modelEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00305     this->_onModelLoadHandlerMethods->insert(handlerMethod);
00306 }
00307
00308 template<typename Class> void OnEventManager::addOnModelSaveHandler(Class * object, void
      (Class::*function)(ModelEvent*)) {
00309     modelEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00310     this->_onModelSaveHandlerMethods->insert(handlerMethod);
00311 }
00312 template<typename Class> void OnEventManager::addOnReplicationStartHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00313     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00314     this->_onReplicationStartHandlerMethods->insert(handlerMethod);
00315 }
00316
00317 template<typename Class> void OnEventManager::addOnReplicationStepHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00318     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00319     this->_onReplicationStepHandlerMethods->insert(handlerMethod);
00320 }
00321
00322 template<typename Class> void OnEventManager::addOnReplicationEndHandler(Class * object, void
```

```
      (Class::*function)(SimulationEvent*)) {
00323     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00324     this->_onReplicationEndHandlerMethods->insert(handlerMethod);
00325 }
00326
00327 template<typename Class> void OnEventManager::addOnProcessEventHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00328     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00329     this->_onProcessEventHandlerMethods->insert(handlerMethod);
00330 }
00331
00332 template<typename Class> void OnEventManager::addOnAfterProcessEventHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00333     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00334     this->_onAfterProcessEventHandlerMethods->insert(handlerMethod);
00335 }
00336
00337 template<typename Class> void OnEventManager::addOnEntityCreateHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00338     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00339     this->_onEntityCreateHandlerMethods->insert(handlerMethod);
00340 }
00341
00342 template<typename Class> void OnEventManager::addOnEntityMoveHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00343     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00344     this->_onEntityMoveHandlerMethods->insert(handlerMethod);
00345 }
00346
00347 template<typename Class> void OnEventManager::addOnEntityRemoveHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00348     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00349     this->_onEntityRemoveHandlerMethods->insert(handlerMethod);
00350 }
00351
00352 template<typename Class> void OnEventManager::addOnSimulationStartHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00353     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00354     this->_onSimulationStartHandlerMethods->insert(handlerMethod);
00355 }
00356
00357 template<typename Class> void OnEventManager::addOnSimulationPausedHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00358     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00359     this->_onSimulationPausedHandlerMethods->insert(handlerMethod);
00360 }
00361
00362 template<typename Class> void OnEventManager::addOnSimulationResumeHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00363     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00364     this->_onSimulationResumeHandlerMethods->insert(handlerMethod);
00365 }
00366
00367 template<typename Class> void OnEventManager::addOnSimulationEndHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00368     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00369     this->_onSimulationEndHandlerMethods->insert(handlerMethod);
00370 }
00371
00372 template<typename Class> void OnEventManager::addOnBreakpointHandler(Class * object, void
      (Class::*function)(SimulationEvent*)) {
00373     simulationEventHandlerMethod handlerMethod = std::bind(function, object, std::placeholders::_1);
00374     this->_onBreakpointHandlerMethods->insert(handlerMethod);
00375 }
00376
00377
00378 /*
00379 Private Sub ModelLogic_DocumentOpen()
00380 Private Sub ModelLogic_DocumentSave()
00381 Private Sub ModelLogic_OnClearStatistics()
00382 Private Sub ModelLogic_OnFileClose(ByVal filesElementNumber As Long, ByVal fileName As String)
00383 Private Sub ModelLogic_OnFileRead(ByVal filesElementNumber As Long, ByVal fileName As String, ByVal
      numberOfItems As Long, readBuffer() As Double)
00384 Private Sub ModelLogic_OnFileReadAsVariant(ByVal filesElementNumber As Long, ByVal fileName As String,
      ByVal numberOfItems As Long, readBuffer() As Variant)
00385 Private Sub ModelLogic_OnFileWrite(ByVal filesElementNumber As Long, ByVal fileName As String, ByVal
      numberOfItems As Long, writeBuffer() As Double)
00386 Private Sub ModelLogic_OnFileWriteAsVariant(ByVal filesElementNumber As Long, ByVal fileName As
      String, ByVal numberOfItems As Long, writeBuffer() As Variant)
00387 Private Sub ModelLogic_OnKeystroke(ByVal keyCode As Long)
00388 Private Function ModelLogic_RealTimeGetMaxTimeAdvance() As Long
00389 Private Function ModelLogic_RealTimeInitialize(ByVal processName As String, ByVal remoteProcessName As
      String) As Long
00390 Private Sub ModelLogic_RealTimeInitializeMaxTimeAdvance()
00391 Private Function ModelLogic_RealTimeReceive(message As String) As Long
00392 Private Function ModelLogic_RealTimeSend(ByVal message As String) As Long
00393 Private Function ModelLogic_RealTimeTerminate() As Long
```

```
00394 Private Sub ModelLogic_RunBegin()
00395 Private Sub ModelLogic_RunBeginReplication()
00396 Private Sub ModelLogic_RunBeginSimulation()
00397 Private Sub ModelLogic_RunBreak()
00398 Private Sub ModelLogic_RunEnd()
00399 Private Sub ModelLogic_RunEndReplication()
00400 Private Sub ModelLogic_RunEndSimulation()
00401 Private Sub ModelLogic_RunFastForward()
00402 Private Sub ModelLogic_RunPause()
00403 Private Sub ModelLogic_RunRestart()
00404 Private Sub ModelLogic_RunResume()
00405 Private Sub ModelLogic_RunStep()
00406 Private Sub ModelLogic_SimanError(ByVal routineName As String, msgType As Long, msgNum As Long)
00407 Private Sub ModelLogic_UserContinuousEquations()
00408 Private Function ModelLogic_UserFunction(ByVal entityID As Long, ByVal functionID As Long) As Double
00409 Private Function ModelLogic_UserRule(ByVal entityID As Long, ByVal ruleID As Long) As Double
00410 Private Sub VBA_Block_1_Fire()
00411  */
00412
00413 //namespace\\}
00414 #endif /* ONEVENTMANAGER_H */
```

18.318 Parser_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Parser_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 23 de Agosto de 2018, 15:57
00012  */
00013
00014 #ifndef PARSER_IF_H
00015 #define PARSER_IF_H
00016
00017 #include <string>
00018 #include "../statistics/Sampler_if.h"
00019
00020 class genesyspp_driver;
00021
00029 class Parser_if {
00030 public:
00031     virtual ~Parser_if() = default;
00038     virtual double parse(const std::string expression) = 0; // may throw exception
00047     virtual double parse(const std::string expression, bool& success, std::string& errorMessage) = 0;
    // does not throw exceptions
00053     virtual std::string getErrorMessage() = 0; // to get error message in the case of thrown exception
00064     virtual void setSampler(Sampler_if* sampler) = 0;
00071     virtual Sampler_if* getSampler() const = 0;
00072     // @ToDo: (pequena alteração): Implement a method to get the TOKENS parsed. Example: If parsing
    "nq(queue1) < var1" return a list containing something like "fNQ tLPAR eQUEUE tRPAR tLESS eVARIABLE"
00078     virtual genesyspp_driver getParser() const = 0;
00079 };
00080
00081 #endif /* PARSER_IF_H */
```

18.319 ParserChangesInformation.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ParserChangesInformation.h
00009  * Author: rlcancian
00010  *
00011  * Created on 11 de Setembro de 2019, 20:42
00012  */
00013
00014 #ifndef PARSERCHANGESINFORMATION_H
00015 #define PARSERCHANGESINFORMATION_H
00016
00017 #include <string>
00018 #include <list>
00019
```

```
00020 class ParserChangesInformation {
00021 public:
00022     ParserChangesInformation();
00023     virtual ~ParserChangesInformation() = default;
00024
00025 public: // gets and sets
00026     // @ToDo: (importante): Review the consistency of this class public API names.
00027     // There are mixed naming patterns such as getincludes() and getfunctionProdutions().
00028     // Any rename may break existing callers and should be planned carefully.
00029     std::string getincludes() const;
00030     void setIncludes(const std::string &newIncludes);
00031
00032     std::string gettokens() const;
00033     void setTokens(const std::string &newTokens);
00034
00035     std::string gettypeObjs() const;
00036     void setTypeObjs(const std::string &newTypeObjs);
00037
00038     std::string getexpressions() const;
00039     void setExpressions(const std::string &newExpressions);
00040
00041     std::string getexpressionProductions() const;
00042     void setExpressionProductions(const std::string &newExpressionProductions);
00043
00044     std::string getassignments() const;
00045     void setAssignments(const std::string &newAssignments);
00046
00047     // @ToDo: (importante): Confirm whether "Produtions" should actually be "Productions".
00048     // Do not rename it directly without mapping impacts on existing callers.
00049     std::string getfunctionProdutions() const;
00050     void setFunctionProdutions(const std::string &newFunctionProdutions);
00051
00052 private:
00053     std::string _includes = "";
00054     std::string _tokens = "";
00055     std::string _typeObjs = "";
00056     std::string _expressions = "";
00057     std::string _expressionProductions = "";
00058     std::string _assignments = "";
00059     std::string _functionProdutions = "";
00060
00061 };
00062
00063 #endif /* PARSERCHANGESINFORMATION_H */
```

18.320 ParserDefaultImpl1.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ParserDefaultImpl1.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 2 de Agosto de 2018, 01:25
00012  */
00013
00014 #ifndef PARSERDEFAULTIMPL1_H
00015 #define PARSERDEFAULTIMPL1_H
00016
00017 #include "Parser_if.h"
00018 //namespace GenesysKernel {
00019 class Model;
00020
00021 class ParserDefaultImpl1 : public Parser_if {
00022 public:
00023     ParserDefaultImpl1(Model* model);
00024     virtual ~ParserDefaultImpl1() = default;
00025 public:
00026     virtual double parse(const std::string expression) override; // may throw exception
00027     virtual double parse(const std::string expression, bool& success, std::string& errorMessage)
      override;
00028     virtual std::string getErrorMessage() override;
00029 private:
00030     Model* _model;
00031 };
00032 //namespace\\}
00033 #endif /* PARSERDEFAULTIMPL1_H */
00034
```

18.321 ParserDefaultImpl2.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ParserDefaultImpl2.h
00009  * Author: Joao Ortigara (20180801-20181207) rafael.luiz.cancian (20181208-...(
00010  *
00011  *
00012  */
00013
00014 #ifndef PARSERDEFAULTIMPL2_H
00015 #define PARSERDEFAULTIMPL2_H
00016
00017 #include <string>
00018 #include "Parser_if.h"
00019 #include "model/Model.h"
00020 #include "../../parser/Genesys++-driver.h"
00021
00022
00023 //namespace GenesysKernel {
00024
00025 class ParserDefaultImpl2 : public Parser_if {
00026 public:
00034     ParserDefaultImpl2(Model* model, Sampler_if* sampler, bool throws = false);
00042     virtual ~ParserDefaultImpl2();
00043 public:
00044     virtual double parse(const std::string expression) override; // may throw exception
00045     virtual double parse(const std::string expression, bool& success, std::string& errorMessage)
      override;
00046     virtual std::string getErrorMessage() override;
00054     virtual void setSampler(Sampler_if* _sampler) override;
00055     virtual Sampler_if* getSampler() const override;
00056     virtual genesyspp_driver getParser() const override;
00057 private:
00058     void _setSamplerInternal(Sampler_if* sampler, bool ownsSampler);
00059 private:
00060     Model* _model;
00061     genesyspp_driver _wrapper;
00062     bool _ownsSampler = false;
00063 };
00064 //namespace\\}
00065 #endif /* PARSERDEFAULTIMPL2_H */
```

18.322 ParserManager.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ParserManager.h
00009  * Author: rlcancian
00010  *
00011  * Created on 11 de Setembro de 2019, 20:34
00012  */
00013
00014 #ifndef PARSERMANAGER_H
00015 #define PARSERMANAGER_H
00016
00017 #include "ParserChangesInformation.h"
00018
00019 //namespace GenesysKernel {
00020
00021 class ParserManager {
00022 public:
00023
00027     struct NewParser {
00028         std::string bisonFilename;
00029         std::string flexFilename;
00030         std::string compiledParserFilename;
00031     };
00032
00036     struct GenerateNewParserResult {
00037         bool result = false;
00038         std::string bisonMessages;
00039         std::string lexMessages;
00040         std::string compilationMessages;
```

```
00041         NewParser newParser;
00042     };
00043 public:
00044     ParserManager();
00045     virtual ~ParserManager() = default;
00046 public:
00052     // @ToDo: (importante): Implement parser generation workflow and define the expected
      filesystem/toolchain contract.
00053     ParserManager::GenerateNewParserResult generateNewParser(ParserChangesInformation* changes);
00059     // @ToDo: (importante): Implement parser loading/activation and define ownership/lifetime rules
      for connected parsers.
00060     bool connectNewParser(ParserManager::NewParser newParser);
00061 private:
00062 };
00063 //namespace\\}
00064 #endif /* PARSERMANAGER_H */
```

18.323 Persistence.h

```
00001 #ifndef PERSISTENCE_H
00002 #define PERSISTENCE_H
00003
00004 #include <string>
00005 #include <utility>
00006 #include <unordered_map>
00007 #include<memory>
00008
00009 #include "../util/Util.h"
00010 #include "persistence/Persistence_if.h"
00011
00012
00022 class PersistenceRecord {
00023 public:
00024     using Payload = std::string;
00026     struct Entry {
00027         enum Kind { text, numeric };
00028         std::string first;
00029         Payload second;
00030         Kind kind{Kind::text};
00031     };
00032     using Iterator = std::unordered_map<std::string, Entry>::iterator;
00033
00034     // this is the actual abstract interface
00035
00036     virtual ~PersistenceRecord();
00037
00038     virtual PersistenceRecord* newInstance();
00039
00040     virtual std::string loadField(const std::string& key, const std::string& defaultValue = "");
00041     virtual double loadField(const std::string& key, double defaultValue);
00042     virtual unsigned int loadField(const std::string& key, unsigned int defaultValue);
00043     virtual int loadField(const std::string& key, int defaultValue);
00044     virtual Util::TimeUnit loadField(const std::string& key, Util::TimeUnit defaultValue);
00045
00046     virtual void saveField(const std::string& key, const std::string& value, const std::string&
      defaultValue, const bool saveIfDefault = false);
00047     virtual void saveField(const std::string& key, const std::string& value);
00048     virtual void saveField(const std::string& key, double value, const double defaultValue, const bool
      saveIfDefault = false);
00049     virtual void saveField(const std::string& key, unsigned int value, const unsigned int
      defaultValue, const bool saveIfDefault = false);
00050     virtual void saveField(const std::string& key, unsigned int value);
00051     virtual void saveField(const std::string& key, int value, const int defaultValue, const bool
      saveIfDefault = false);
00052     virtual void saveField(const std::string& key, Util::TimeUnit value, const Util::TimeUnit
      defaultValue, const bool saveIfDefault = false);
00053
00054     // these are only needed because existing code does not abstract away the STL container API
00055     // still, subclasses implementing this interface may use them to insert post-processed entries
00056
00057     explicit PersistenceRecord(Persistence_if& config);
00058
00060     std::size_t size() const;
00061
00063     void insert(Entry entry);
00064
00066     void insert(Iterator it, Iterator end);
00067
00069     void erase(const std::string& key);
00070
00072     void clear();
00073
00075     Iterator find(const std::string& key);
00076
00078     Iterator begin();
```

```
00079
00081     Iterator end();
00082
00083 private:
00084     std::unordered_map<std::string, Entry> _fields;
00085     Persistence_if& _config;
00086 };
00087
00088
00094 class PersistentObject_base {
00095 public:
00096     virtual ~PersistentObject_base() {};
00097
00098 protected:
00106     virtual bool _loadInstance(PersistenceRecord *fields) = 0;
00107
00114     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues = false) = 0;
00115 };
00116
00117 #endif /* PERSISTENCE_H */
```

18.324 CppSerializer.h

```
00001 #ifndef CPPSERIALIZER_H
00002 #define CPPSERIALIZER_H
00003
00004 #include <unordered_map>
00005 #include<memory>
00006
00007 #include "Serializer.h"
00008 #include "../model/Model.h"
00009
00010 class CppSerializer : public Serializer {
00011 public:
00012     explicit CppSerializer(Model *model);
00013
00014 public: // Serializer interface
00015     PersistenceRecord* newPersistenceRecord() override;
00016     bool dump(std::ostream& output) override;
00017     bool load(std::istream& input) override;
00018     bool get(const std::string& name, PersistenceRecord *entry) override;
00019     bool put(const std::string name, const std::string type, const Util::identification id,
      PersistenceRecord *fields) override;
00020     int for_each(std::function<int(const std::string&) > delegate) override;
00021
00022 private:
00023     Model *_model{};
00024     std::unordered_map<std::string, std::unique_ptr<PersistenceRecord» _metaobjects
00025     {
00026     };
00027     std::unordered_map<std::string, std::unique_ptr<PersistenceRecord» _components
00028     {
00029     };
00030 };
00031
00032 #endif // CPPSERIALIZER_H
```

18.325 GenSerializer.h

```
00001 #ifndef GENSERIALIZER_H
00002 #define GENSERIALIZER_H
00003
00004 #include <unordered_map>
00005 #include<memory>
00006
00007 #include "Serializer.h"
00008 #include "../model/Model.h"
00009
00010 class GenSerializer : public Serializer {
00011 public:
00012     explicit GenSerializer(Model *model);
00013
00014 public: // Serializer interface
00015     PersistenceRecord* newPersistenceRecord() override;
00016     bool dump(std::ostream& output) override;
00017     bool load(std::istream& input) override;
00018     bool get(const std::string& name, PersistenceRecord *entry) override;
00019     bool put(const std::string name, const std::string type, const Util::identification id,
      PersistenceRecord *fields) override;
00020     int for_each(std::function<int(const std::string&) > delegate) override;
00021
00022 private:
```

```
00023     static std::string linearize(PersistenceRecord *fields);
00024     friend class PersistenceDefaultImpl2;
00025
00026 private:
00027     Model *_model{};
00028     std::unordered_map<std::string, std::unique_ptr<PersistenceRecord» _components
00029     {
00030     };
00031 };
00032
00033 #endif // GENSERIALIZER_H
```

18.326 JsonSerializer.h

```
00001 #ifndef JSONSERIALIZER_H
00002 #define JSONSERIALIZER_H
00003
00004 #include <unordered_map>
00005 #include<memory>
00006
00007 #include "Serializer.h"
00008 #include "../model/Model.h"
00009
00010 class JsonSerializer : public Serializer {
00011 public:
00012     explicit JsonSerializer(Model *model);
00013
00014 public: // Serializer interface
00015     PersistenceRecord* newPersistenceRecord() override;
00016     bool dump(std::ostream& output) override;
00017     bool load(std::istream& input) override;
00018     bool get(const std::string& name, PersistenceRecord *entry) override;
00019     bool put(const std::string name, const std::string type, const Util::identification id,
      PersistenceRecord *fields) override;
00020     int for_each(std::function<int(const std::string&) > delegate) override;
00021
00022 private:
00023     Model *_model{};
00024     std::unordered_map<std::string, std::unique_ptr<PersistenceRecord» _metaobjects
00025     {
00026     };
00027     std::unordered_map<std::string, std::unique_ptr<PersistenceRecord» _components
00028     {
00029     };
00030 };
00031
00032 #endif // JSONSERIALIZER_H
```

18.327 Persistence_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Persistence_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 24 de Agosto de 2018, 19:22
00012  */
00013
00014 #ifndef MODELPERSISTENCE_IF_H
00015 #define MODELPERSISTENCE_IF_H
00016
00017 #include <string>
00018 #include <map>
00019
00020
00021 // forward decl
00022 class PersistenceRecord;
00023
00031 class Persistence_if {
00032 public:
00033     virtual ~Persistence_if() = default;
00034
00035     enum class Options : int {
00036         SAVEDEFAULTS = 1, HIDEIDKEY = 2, HIDETYPEKEY = 4, HIDENAMEKEY = 8, SORTALPHLY = 16
00037     };
00038
00039 public:
```

```
00041     virtual bool save(std::string filename) = 0;
00043     virtual bool load(std::string filename) = 0;
00044
00045 public:
00047     virtual bool hasChanged() = 0;
00049     virtual void setHasChanged(bool hasChanged) = 0;
00051     virtual bool getOption(Persistence_if::Options option) = 0;
00053     virtual void setOption(Persistence_if::Options option, bool value) = 0;
00054 public:
00056     virtual std::string getFormatedField(PersistenceRecord *fields) = 0;
00057 };
00058
00059 #endif /* MODELPERSISTENCE_IF_H */
```

18.328 PersistenceDefaultImpl2.h

```
00001 #ifndef MODELPERSISTENCEDEFAULTIMPL_H
00002 #define MODELPERSISTENCEDEFAULTIMPL_H
00003
00004 #include "Persistence_if.h"
00005 #include "../model/Model.h"
00006
00007 class PersistenceDefaultImpl2 : public Persistence_if {
00008 public:
00009     PersistenceDefaultImpl2(Model* model);
00010
00011 public: // Persistence_if interface
00012     bool save(std::string filename) override;
00013     bool load(std::string filename) override;
00014     bool hasChanged() override;
00015     void setHasChanged(bool hasChanged) override;
00016     bool getOption(Persistence_if::Options option) override;
00017     void setOption(Persistence_if::Options option, bool value) override;
00018     std::string getFormatedField(PersistenceRecord *fields) override;
00019
00020 private:
00021     Model* _model;
00022     unsigned short _options{0};
00023     bool _dirty{false};
00024 };
00025
00026 #endif /* MODELPERSISTENCEDEFAULTIMPL_H */
```

18.329 PersistencePartialLoadImpl1.h

```
00001 #ifndef MODELPERSISTENCEPARTIALLOADIMPL1_H
00002 #define MODELPERSISTENCEPARTIALLOADIMPL1_H
00003
00004 #include "PersistenceDefaultImpl2.h"
00005 #include "../model/Model.h"
00006
00007 class PersistencePartialLoadImpl1 : public PersistenceDefaultImpl2 {
00008 public:
00009     explicit PersistencePartialLoadImpl1(Model* model);
00010
00011 public:
00012     bool load(std::string filename) override;
00013
00014 private:
00015     Model* _model;
00016 };
00017
00018 #endif /* MODELPERSISTENCEPARTIALLOADIMPL1_H */
```

18.330 Serializer.h

```
00001 #ifndef MODELSERIALIZER_H
00002 #define MODELSERIALIZER_H
00003
00004 #include <iostream>
00005 #include <functional>
00006
00007 #include "../Persistence.h"
00008 #include "../../util/Util.h"
00009
00023 class Serializer {
00024 public:
00025     virtual ~Serializer();
00026
```

```
00033     virtual PersistenceRecord* newPersistenceRecord() = 0;
00034
00042     virtual bool dump(std::ostream& output) = 0;
00043
00049     virtual bool load(std::istream& input) = 0;
00050
00059     virtual bool get(const std::string& name, PersistenceRecord *entry) = 0;
00060
00071     virtual bool put(const std::string name, const std::string type, const Util::identification id,
      PersistenceRecord *fields) = 0;
00072
00080     virtual int for_each(std::function<int(const std::string&) > delegate) = 0;
00081 };
00082
00083 #endif // MODELSERIALIZER_H
```

18.331 XmlSerializer.h

```
00001 #ifndef XMLSERIALIZER_H
00002 #define XMLSERIALIZER_H
00003
00004 #include <unordered_map>
00005 #include<memory>
00006
00007 #include "Serializer.h"
00008 #include "../model/Model.h"
00009
00010 class XmlSerializer : public Serializer {
00011 public:
00012     explicit XmlSerializer(Model *model);
00013
00014 public: // Serializer interface
00015     PersistenceRecord* newPersistenceRecord() override;
00016     bool dump(std::ostream& output) override;
00017     bool load(std::istream& input) override;
00018     bool get(const std::string& name, PersistenceRecord *entry) override;
00019     bool put(const std::string name, const std::string type, const Util::identification id,
      PersistenceRecord *fields) override;
00020     int for_each(std::function<int(const std::string&) > delegate) override;
00021
00022 private:
00023     Model *_model{};
00024     std::unordered_map<std::string, std::unique_ptr<PersistenceRecord» _metaobjects
00025     {
00026     };
00027     std::unordered_map<std::string, std::unique_ptr<PersistenceRecord» _components
00028     {
00029     };
00030 };
00031
00032 #endif // XMLSERIALIZER_H
```

18.332 Plugin.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Plugin.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Junho de 2018, 12:58
00012  */
00013
00014 #ifndef PLUGIN_H
00015 #define PLUGIN_H
00016
00017 #include <string>
00018 #include <functional>
00019 #include <list>
00020
00021 #include "../util/Util.h"
00022 #include "Persistence.h"
00023 #include "PluginInformation.h"
00024
00025 //namespace GenesysKernel {
00026
00031 class Plugin {
00032 public:
```

```
00037     Plugin(std::string filename_so_dll);
00042     Plugin(StaticGetPluginInformation getInformation); // @ToDo: (importante): temporary. Just while
    compiled together
00046     virtual ~Plugin();
00047 public:
00052     std::string show();
00053 public:
00058     bool isIsValidPlugin() const;
00063     PluginInformation* getPluginInfo() const;
00072     ModelDataDefinition* loadNew(Model* model, PersistenceRecord *fields);
00079     bool loadAndInsertNew(Model* model, PersistenceRecord *fields);
00086     ModelDataDefinition* newInstance(Model* model, std::string name = "");
00087
00088 private:
00089     ModelComponent* _loadNewComponent(Model* model, PersistenceRecord *fields);
00090     ModelDataDefinition* _loadNewElement(Model* model, PersistenceRecord *fields);
00091
00092 private: // read only
00093     // Default invalid state avoids reading indeterminate validity after construction failures.
00094     bool _isValidPlugin = false;
00095     // Store plugin metadata pointer with a safe default for invalid plugin construction paths.
00096     PluginInformation* _pluginInfo = nullptr;
00097 private:
00098     StaticGetPluginInformation _StatMethodGetInformation;
00099 };
00100 //namespace\\}
00101 #endif /* PLUGIN_H */
```

18.333 PluginConnector_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   PluginConnector_if.h
00009  * Author: rlcancian
00010  *
00011  * Created on 9 de Setembro de 2019, 19:17
00012  */
00013
00014 #ifndef PLUGINCONNECTOR_IF_H
00015 #define PLUGINCONNECTOR_IF_H
00016
00017 #include<string>
00018 #include "Plugin.h"
00019 #include "../util/List.h"
00020
00021 //namespace GenesysKernel {
00022
00029 class PluginConnector_if {
00030 public:
00031     virtual ~PluginConnector_if() = default;
00038     virtual Plugin* check(const std::string dynamicLibraryFilename) = 0;
00045     virtual Plugin* connect(const std::string dynamicLibraryFilename) = 0;
00052     virtual List<std::string>* find() = 0;
00053
00060     virtual bool disconnect(const std::string dynamicLibraryFilename) = 0;
00067     virtual bool disconnect(Plugin* plugin) = 0;
00068 };
00069 //namespace\\}
00070 #endif /* PLUGINCONNECTOR_IF_H */
```

18.334 PluginInformation.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   PluginInformation.h
00009  * Author: rlcancian
00010  *
00011  * Created on 9 de Setembro de 2019, 20:02
00012  */
00013
00014 #ifndef PLUGININFORMATION_H
00015 #define PLUGININFORMATION_H
```

```
00016
00017 #include <map>
00018 #include <list>
00019 #include <string>
00020
00021 //namespace GenesysKernel {
00022 class ModelDataDefinition;
00023 class ModelComponent;
00024 class Model;
00025 class ModelDataManager;
00026 class PersistenceRecord;
00027
00028 typedef ModelComponent* (*StaticLoaderComponentInstance)(Model*, PersistenceRecord*);
00029 typedef ModelDataDefinition* (*StaticLoaderDataDefinitionInstance)(Model*, PersistenceRecord*);
00030 typedef ModelDataDefinition* (*StaticConstructorDataDefinitionInstance)(Model*, std::string);
00031 class PluginInformation;
00032 typedef PluginInformation* (*StaticGetPluginInformation)();
00033
00034 class SystemDependency {
00035 public:
00036     enum class OS {
00037         Any,
00038         Linux,
00039         Windows,
00040         MacOS,
00041         Unknown
00042     };
00043
00044     SystemDependency() = default;
00045     SystemDependency(OS os, std::string name, std::string installCommand = "", std::string
      checkCommand = "");
00046
00047     OS getOS() const;
00048     std::string getName() const;
00049     std::string getInstallCommand() const;
00050     std::string getCheckCommand() const;
00051     std::string show() const;
00052
00053     static std::string osToString(OS os);
00054
00055 private:
00056     OS _os = OS::Any;
00057     std::string _name = "";
00058     std::string _installCommand = "";
00059     std::string _checkCommand = "";
00060 };
00061
00062 class PluginInformation {
00063 public:
00064     PluginInformation(std::string pluginTypename, StaticLoaderComponentInstance componentloader,
      StaticConstructorDataDefinitionInstance elementConstructor);
00065     PluginInformation(std::string pluginTypename, StaticLoaderDataDefinitionInstance elementloader,
      StaticConstructorDataDefinitionInstance elementConstructor);
00066     // Release owned metadata containers allocated alongside plugin-information objects.
00067     virtual ~PluginInformation();
00068 public:
00069     // gets
00070     StaticLoaderDataDefinitionInstance getDataDefinitionLoader() const;
00071     StaticLoaderComponentInstance GetComponentLoader() const;
00072     StaticConstructorDataDefinitionInstance getDataDefinitionConstructor() const;
00073     bool isGenerateReport() const;
00074     bool isComponent() const;
00075     bool isSendTransfer() const;
00076     bool isReceiveTransfer() const;
00077     bool isSink() const;
00078     bool isSource() const;
00079     std::string getObservation() const;
00080     std::string getVersion() const;
00081     std::string getDate() const;
00082     std::string getAuthor() const;
00083     std::string getPluginTypename() const;
00084     // sets
00085     void insertDynamicLibFileDependence(std::string filename);
00086     // @ToDo: (importante): Clarify ownership for the incoming list pointer.
00087     // This setter currently replaces the internal raw pointer without documenting transfer semantics.
00088     void setDynamicLibFilenameDependencies(std::list<std::string>* dynamicLibFilenameDependencies);
00089     std::list<std::string>* getDynamicLibFilenameDependencies() const;
00090     void insertSystemDependency(const SystemDependency& dependency);
00091     const std::list<SystemDependency>* getSystemDependencies() const;
00092     bool hasSystemDependencies() const;
00093     void setGenerateReport(bool generateReport);
00094     void setSendTransfer(bool sendTransfer);
00095     void setReceiveTransfer(bool receiveTransfer);
00096     void setSink(bool Sink);
00097     void setSource(bool Source);
00098     void setObservation(std::string observation);
00099     void setVersion(std::string version);
```

```
00100     void setDate(std::string date);
00101     void setAuthor(std::string author);
00102     void setMaximumOutputs(unsigned short _maximumOutputs);
00103     unsigned short getMaximumOutputs() const;
00104     void setMinimumOutputs(unsigned short _minimumOutputs);
00105     unsigned short getMinimumOutputs() const;
00106     void setMaximumInputs(unsigned short _maximumInputs);
00107     unsigned short getMaximumInputs() const;
00108     void setMinimumInputs(unsigned short _minimumInputs);
00109     unsigned short getMinimumInputs() const;
00110     void setDescriptionHelp(std::string _descriptionHelp);
00111     std::string getDescriptionHelp() const;
00112     // @ToDo: (importante): Clarify ownership for the incoming fields map.
00113     // This setter currently replaces the internal raw pointer without documenting lifetime
      expectations.
00114     void setFields(std::map<std::string, std::string>* _fiewlds);
00115     std::map<std::string, std::string>* getFields() const;
00116     void setLanguageTemplate(std::string _languageTemplate);
00117     std::string getLanguageTemplate() const;
00118     void setCategory(std::string _category);
00119     std::string getCategory() const;
00120     static std::string categoryFolderName(const std::string& category);
00121 public:
00122 private:
00123     std::string _author = "prof. Dr. Ing. Rafael Luiz Cancian";
00124     std::string _date = "01/08/2018";
00125     std::string _version = "0.9.1";
00126     std::string _category = "DiscreteProcessing";
00127     std::string _observation = "First implementation not fully completed nor tested. Use with
      caution.";
00128     std::string _descriptionHelp = "";
00129     std::string _languageTemplate = "";
00130     bool _isSource = false;
00131     bool _isSink = false;
00132     bool _receiveTransfer = false;
00133     bool _sendTransfer = false;
00134     bool _generateReport = false;
00135     std::list<std::string>* _dynamicLibFilenameDependencies = new std::list<std::string>();
00136     std::list<SystemDependency>* _systemDependencies = new std::list<SystemDependency>();
00137     std::map<std::string, std::string>* _fields = new std::map<std::string, std::string>();
00138     // set from constructor
00139     std::string _pluginTypename;
00140     bool _isComponent = false;
00141     unsigned short _minimumInputs = 1;
00142     unsigned short _maximumInputs = 1;
00143     unsigned short _minimumOutputs = 1;
00144     unsigned short _maximumOutputs = 1;
00145     StaticLoaderComponentInstance _componentloader;
00146     StaticLoaderDataDefinitionInstance _elementloader;
00147     StaticConstructorDataDefinitionInstance _elementConstructor;
00148 };
00149 //namespace\\}
00150
00151 #endif /* PLUGININFORMATION_H */
```

18.335 PluginManager.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   PluginManager.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 30 de Maio de 2019, 17:49
00012  */
00013
00014 #ifndef PLUGINMANAGER_H
00015 #define PLUGINMANAGER_H
00016
00017 #include <functional>
00018 #include <string>
00019 #include <vector>
00020
00021 #include "model/Model.h"
00022 #include "../util/List.h"
00023 #include "../util/Util.h"
00024 #include "Plugin.h"
00025 #include "PluginConnector_if.h"
00026 #include "SystemDependencyResolver.h"
00027
00028 class Simulator;
```

```
00029 class Model;
00030
00039 struct PluginInsertionOptions {
00041     std::function<bool(const SystemDependencyCheckResult&)> confirmSystemDependencyInstallation;
00042 };
00043
00051 class PluginLoadIssue {
00052 public:
00054     enum class Reason {
00055         InvalidPlugin,
00056         MissingSystemDependency,
00057         DynamicDependencyFailure,
00058         ConnectionFailure,
00059         InsertionFailure,
00060         Exception
00061     };
00062
00064     PluginLoadIssue();// fix //  = default;
00066     PluginLoadIssue(std::string filename,
00067                     std::string pluginTypename,
00068                     Reason reason,
00069                     std::string message,
00070                     SystemDependencyCheckResult systemDependencyResult = {}) noexcept;
00071
00073     const std::string& getFilename() const;
00075     const std::string& getPluginTypename() const;
00077     Reason getReason() const;
00079     const std::string& getMessage() const;
00081     const SystemDependencyCheckResult& getSystemDependencyResult() const;
00083     bool hasSystemDependencyResult() const;
00085     static std::string reasonToString(Reason reason);
00087     std::string diagnosticText() const;
00088
00089 private:
00090     std::string _filename;
00091     std::string _pluginTypename;
00092     Reason _reason;
00093     std::string _message;
00094     SystemDependencyCheckResult _systemDependencyResult;
00095 };
00096
00097 //namespace GenesysKernel {
00098
00106 class PluginManager {
00107 public:
00109     PluginManager(Simulator* simulator);
00116     PluginManager(Simulator* simulator, PluginConnector_if* pluginConnector, SystemCommandExecutor_if*
      systemCommandExecutor);
00118     virtual ~PluginManager();
00119
00120 public:
00122     std::string show();
00123
00124 public:
00126     List<Plugin*>* completePluginsFieldsAndTemplates();
00127
00128 public:
00130     bool check(const std::string& dynamicLibraryFilename);
00132     SystemDependencyCheckResult checkSystemDependencies(const std::string& dynamicLibraryFilename);
00134     List<std::string>* discoverPluginFilenames() const;
00136     List<PluginLoadIssue>* getPluginLoadIssues() const;
00138     void clearPluginLoadIssues();
00140     void clearPluginLoadIssue(const std::string& dynamicLibraryFilename);
00142     Plugin* insert(const std::string& dynamicLibraryFilename);
00144     Plugin* insert(const std::string& dynamicLibraryFilename, const PluginInsertionOptions& options);
00146     bool remove(const std::string& dynamicLibraryFilename);
00148     bool remove(Plugin* plugin);
00150     Plugin* find(const std::string& pluginTypeName);
00156     std::vector<std::string> getDataDefinitionPluginTypenames() const;
00158     std::string sourceIncludePathFor(const std::string& pluginTypeName);
00160     List<Plugin*>* autoInsertPlugins(const std::string& pluginsListFilename,
00161                                      const bool lookForPluginsIfFilenameNotFound = true);
00163     List<Plugin*>* autoInsertPlugins(const std::string& pluginsListFilename,
00164                                      const bool lookForPluginsIfFilenameNotFound,
00165                                      const PluginInsertionOptions& options);
00167     List<Plugin*>* autoInsertPlugins();
00169     List<Plugin*>* autoInsertPlugins(const PluginInsertionOptions& options);
00170
00171 public:
00173     Plugin* front();
00175     Plugin* next();
00177     Plugin* last();
00179     unsigned int size();
00181     Plugin* getAtRank(unsigned int rank);
00182
00183 public:
00185     ModelDataDefinition* newInstance(const std::string& pluginTypename, Model* model, std::string name
```

```
= "");
00186
00187     template <typename T>
00188     T* newInstance(Model* model, std::string name = "") {
00189         name = Util::StrReplace(name, " ", "_");
00190         Plugin* plugin;
00191         std::string pluginTypename = Util::TypeOf<T>();
00192         // @ToDo: (pequena alteração): Use Find method??
00193         for (unsigned short i = 0; i < _plugins->size(); i++) {
00194             plugin = _plugins->getAtRank(i);
00195             if (plugin->getPluginInfo()->getPluginTypename() == pluginTypename) {
00196                 T* instance;
00197                 StaticConstructorDataDefinitionInstance constructor = plugin->getPluginInfo()->
00198
getDataDefinitionConstructor();
00199                 instance = static_cast<T*>(constructor(model, name));
00200                 if (model != nullptr) {
00201                     ModelDataDefinition::CreateInternalData(instance);
00202                 }
00203                 return instance;
00204             }
00205         }
00206         // Keep this header free of Simulator implementation details to avoid include cycles.
00207         return nullptr;
00208     }
00209
00210 private:
00212     bool _insert(Plugin* plugin, const PluginInsertionOptions& options, const std::string&
dynamicLibraryFilename);
00214     bool _preflightAndMaybeInstallSystemDependencies(PluginInformation* plugInfo,
00215                                                      const PluginInsertionOptions& options,
00216                                                      SystemDependencyCheckResult* blockingResult =
nullptr,
00217                                                      std::string* failureMessage = nullptr);
00219     void _recordLoadIssue(const PluginLoadIssue& issue);
00221     void _removeLoadIssue(const std::string& dynamicLibraryFilename, const std::string&
pluginTypename);
00223     void _insertDefaultKernelElements();
00225     List<Plugin*>* _autoFindPlugins(const PluginInsertionOptions& options);
00226
00227 private:
00228     List<Plugin*>* _plugins;
00229     List<PluginLoadIssue>* _pluginLoadIssues;
00230     Simulator* _simulator;
00231     PluginConnector_if* _pluginConnector;
00232     SystemCommandExecutor_if* _systemCommandExecutor;
00233 };
00234
00235 //namespace\\}
00236 #endif /* PLUGINMANAGER_H */
```

18.336 PropertyGenesys.h

```
00001 #ifndef PROPERTY_H
00002 #define PROPERTY_H
00003
00004 #include "SimulationControlAndResponse.h"
00005 #include "model/ModelComponent.h"
00006
00007 class PropertyEditorGenesys {
00008 public:
00009     PropertyEditorGenesys() = default;
00010     virtual ~PropertyEditorGenesys() = default;
00011
00012 public:
00013     std::list<ModelComponent*> getElements() {
00014         return _elements;
00015     }
00016
00017 public:
00018     void changeSimulationControl(SimulationControl* control, const std::string& value, bool remove =
false) {
00019         if (control != nullptr) {
00020             control->setValue(value, remove);
00021         }
00022     }
00023
00024     SimulationControl* findSimulationControl(const std::string& id, const std::string& attribute) {
00025         for (auto element : _elements) {
00026             if (std::to_string(element->getId()) == id) {
00027                 for (auto control : *element->getSimulationControls()->list()) {
00028                     if (control->getName() == attribute) {
00029                         return control;
00030                     }
00031                 }
```

```
00032             }
00033         }
00034         return nullptr;
00035     }
00036
00037     // Legacy compatibility wrapper. Prefer changeSimulationControl.
00038     //void changeProperty(SimulationControl* control, const std::string& value, bool remove = false) {
00039     //    changeSimulationControl(control, value, remove);
00040     //}
00041
00042     // Legacy compatibility wrapper. Prefer findSimulationControl.
00043     //SimulationControl* findProperty(const std::string& id, const std::string& attribute) {
00044     //    return findSimulationControl(id, attribute);
00045     //}
00046
00047     void addElement(ModelComponent* component) {
00048         _elements.push_back(component);
00049     }
00050
00051 private:
00052     std::list<ModelComponent*> _elements;
00053 };
00054
00055 #endif
```

18.337 ScenarioExperiment_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   ScenarioExperiment_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 10 de Outubro de 2018, 14:52
00012  */
00013
00014 #ifndef SCENARIOEXPERIMENT_IF_H
00015 #define SCENARIOEXPERIMENT_IF_H
00016
00017 class ScenarioExperiment_if {
00018 public:
00019     virtual ~ScenarioExperiment_if() = default;
00020 };
00021
00022 #endif /* SCENARIOEXPERIMENT_IF_H */
```

18.338 SimulationControlAndResponse.h

```
00001 #pragma once
00002
00003 #include <string>
00004 #include <sstream>
00005 #include <functional>
00006 #include <list>
00007 #include <vector>
00008 #include <stdexcept>
00009 #include <type_traits>
00010 #include "../util/Util.h"
00011 #include "../util/List.h"
00012
00013 //namespace GenesysKernel {
00014
00015 class ModelDataDefinition;
00016
00017 class SourceCodeString {
00018 public:
00019     SourceCodeString() = default;
00020     SourceCodeString(const std::string& value) : _value(value) {}
00021     SourceCodeString(std::string&& value) : _value(std::move(value)) {}
00022
00023     const std::string& str() const { return _value; }
00024     operator std::string() const { return _value; }
00025
00026 private:
00027     std::string _value;
00028 };
00029
00030 inline std::ostream& operator<<(std::ostream& os, const SourceCodeString& value) {
```

```
00031     os « value.str();
00032     return os;
00033 }
00034
00035
00047 class PropertyGenesysBase {
00048 public:
00049     PropertyGenesysBase(std::string className, std::string elementName, std::string propertyName,
      std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false) {
00050         _className=className;
00051         _elementName=elementName;
00052         _propertyName=propertyName;
00053         _whatsThis=whatsThis;
00054         _isList=isList;
00055         _isClass=isClass;
00056         _isEnum=isEnum;
00057     }
00058     virtual ~PropertyGenesysBase() = default;
00059 public:
00060     std::string whatsThis() const { return _whatsThis; }
00061     std::string getClassname() const { return _className; }
00062     std::string getElementName() const { return _elementName; }
00063     std::string getName() const { return _propertyName; }
00064     std::string propertyType() const { return _propertyType; }
00065     bool getIsList() const { return _isList; }
00066     bool getIsClass() const { return _isClass; }
00067     bool getIsEnum() const { return _isEnum; }
00068     void setWhatsThis(const std::string text);
00069     void setName(std::string name) { _propertyName = name; }
00070 protected:
00071     std::string _elementName;
00072     std::string _className;
00073     std::string _propertyName;
00074     std::string _propertyType;
00075     std::string _whatsThis;
00076     bool _isList;
00077     bool _isClass;
00078     bool _isEnum;
00079 };
00080
00081 class PropertyGenesys: public PropertyGenesysBase {
00082 public:
00083     PropertyGenesys(std::string className, std::string elementName, std::string propertyName,
      std::string whatsThis="", std::string propertyId="", std::string toolTip="", std::string statusTip="")
00084         : PropertyGenesysBase(className, elementName, propertyName, whatsThis){
00085         _propertyId=propertyId;
00086         _toolTip=toolTip;
00087         _statusTip=statusTip;
00088     }
00089     virtual ~PropertyGenesys() = default;
00090 public:
00091     std::list<PropertyGenesys *> subProperties() const { return _subProperties; };
00092     std::string toolTip() const { return _toolTip; }
00093     std::string statusTip() const { return _statusTip; }
00094     std::string propertyName() const { return _propertyName; }
00095     std::string propertyId() const { return _propertyId; }
00096     bool isEnabled() const { return _isEnabled; }
00097     bool isModified() const { return _isModified; }
00098     bool isSubProperty()const { return _isSubProperty; }
00099     bool hasValue() const { return _hasValue; }
00100     std::string valueText() const { return "?"; } // @ToDo: (importante): implement value text
00101     void setToolTip(const std::string text);
00102     void setStatusTip(const std::string text);
00103     void setWhatsThis(const std::string text);
00104     void setPropertyName(const std::string text);
00105     void setPropertyId(const std::string text);
00106     void setEnabled(bool enable);
00107     void setModified(bool modified);
00108 public:
00109     virtual bool compare(PropertyGenesys* otherProperty)const { return true; } // @ToDo: (importante):
      implement comparison
00110 public:
00111     void addSubProperty(PropertyGenesys *property) {}
00112     void insertSubProperty(PropertyGenesys *property, PropertyGenesys *afterProperty) {}
00113     void removeSubProperty(PropertyGenesys *property) {}
00114 protected:
00115     void propertyChanged() {}
00116 protected:
00117     //friend class QtAbstractPropertyManager;
00118     std::string _className;
00119     std::string _propertyName;
00120     std::string _whatsThis;
00121     std::string _propertyId;
00122     std::string _toolTip;
00123     std::string _statusTip;
00124     std::list<PropertyGenesys*> _subProperties;
00125     bool _isEnabled;
```

```
00126     bool _isModified;
00127     bool _hasValue;
00128     bool _isSubProperty;
00129 };
00130
00137 class SimulationResponse: public PropertyGenesysBase {
00138 public:
00139     SimulationResponse(std::string className, std::string elementName, std::string propertyName,
      std::string whatsThis="", bool isList=false, bool isClass=false, bool isEnum=false)
00140         : PropertyGenesysBase(className, elementName, propertyName, whatsThis, isList, isClass,
      isEnum) {
00141     }
00142     virtual ~SimulationResponse() = default;
00143 public:
00144     std::string show() const {
00145         std::string msg = "classname="+_className+ ", elementName="+_elementName+", name=\"" +
      _propertyName + "\"";
00146         msg += ", value="+getValue();
00147         return msg;
00148     }
00149     virtual std::string getValue() const = 0;
00150     virtual List<std::string>* getStrValues() { return nullptr; };
00151 };
00152
00153 enum class SimulationControlEditorHint {
00154     Default,
00155     CodeEditor,
00156     MultiLineText
00157 };
00158
00165 class SimulationControl: public SimulationResponse {
00166 public:
00167     SimulationControl(std::string className, std::string elementName, std::string propertyName,
      std::string whatsThis="", bool isList=false,  bool isClass=false, bool isEnum=false)
00168         : SimulationResponse(className, elementName, propertyName, whatsThis, isList, isClass,
      isEnum){
00169     }
00170     std::string show() const {
00171         std::string msg = "classname="+_className+ ", elementName="+_elementName+",
      name=\""+_propertyName+"\"";
00172         if (_readonly) {
00173             msg += " (readonly)";
00174         }
00175         //msg += ", type="+_propertyType;
00176         msg += ", value="+getValue();
00177         return msg;
00178     }
00179     bool isReadOnly() const { return _readonly; }
00180 public:
00181     virtual void setValue(std::string value, bool remove=false) = 0;
00188     virtual List<SimulationControl*>* getChildSimulationControls(int index=0) { return nullptr; };
00189     virtual bool hasObjectInstance() const { return true; }
00190     virtual bool ensureObjectInstance() { return hasObjectInstance(); }
00191     virtual bool isModelDataDefinitionReference() const { return false; }
00192     virtual ModelDataDefinition* getReferencedModelDataDefinition() const { return nullptr; }
00193     // This method validates a proposed string value before GUI editors commit it.
00194     virtual bool validateProposedValue(const std::string& value, std::string& errorMessage) const {
00195         (void)value;
00196         errorMessage.clear();
00197         return true;
00198     }
00199     // This method exposes the preferred editor affordance for GUI-side property editing.
00200     virtual SimulationControlEditorHint preferredEditorHint() const { return _preferredEditorHint; }
00201     void setPreferredEditorHint(SimulationControlEditorHint hint) { _preferredEditorHint = hint; }
00202     // This method exposes the ModelDataDefinition represented by a class-list row, when any.
00203     //
00204     // It lets generic tooling classify list members themselves as editable references without
00205     // knowing the concrete owner class. Non ModelDataDefinition lists keep the default nullptr.
00206     virtual ModelDataDefinition* getListElementModelDataDefinition(int index) const {
00207         (void)index;
00208         return nullptr;
00209     }
00210     // This method exposes whether the property should be rendered as an inline expandable object
      tree.
00211     virtual bool supportsInlineExpansion() const { return getIsClass() && !getIsList(); }
00212     // This method exposes whether the property should be edited by the dedicated list editor.
00213     virtual bool supportsListEditor() const { return getIsList(); }
00214     // This method exposes whether the property supports choosing an existing object instance.
00215     virtual bool supportsExistingObjectSelection() const { return false; }
00216     // This method exposes whether the property supports creating a new object instance.
00217     virtual bool supportsObjectCreation() const { return false; }
00218     // This method exposes whether the property supports typed creation of a new list element.
00219     virtual bool supportsNewListElementCreation() const { return false; }
00220     // This method exposes which concrete element types can be created for a polymorphic list.
00221     //
00222     // A nullptr return means the list has no typed-creation contract and should use the legacy
00223     // single "create new element" action. Callers own the returned List when it is not nullptr.
```

```
00224     virtual List<std::string>* getCreatableListElementTypes() { return nullptr; }
00225     // This method exposes the concrete type currently accepted by a polymorphic list.
00226     //
00227     // Empty string means no concrete type has been selected yet. Implementations may lock the
00228     // current type once the list has members, while still allowing it to be configured while empty.
00229     virtual std::string getCurrentListElementType() const { return ""; }
00230     // This method configures the concrete type currently accepted by a polymorphic list.
00231     //
00232     // The default implementation rejects the operation. Implementations that support it should
00233     // validate the requested type against getCreatableListElementTypes() and their own invariants.
00234     virtual bool setCurrentListElementType(const std::string& typeName) {
00235         (void)typeName;
00236         return false;
00237     }
00238     // This method exposes whether the property is an inline object (and not a ModelDataDefinition
      reference).
00239     virtual bool isInlineObjectProperty() const {
00240         return supportsInlineExpansion() && !isModelDataDefinitionReference();
00241     }
00242     // This method provides an explicit object-creation operation for class-like properties.
00243     virtual bool createObjectInstance(const std::string& value = "") {
00244         (void)value;
00245         return false;
00246     }
00247     virtual List<std::string>* getCreatableReferenceTypes() { return nullptr; }
00248     virtual std::string getCurrentReferenceType() const { return ""; }
00249     virtual bool setCurrentReferenceType(const std::string& typeName) {
00250         (void)typeName;
00251         return false;
00252     }
00253     virtual bool createObjectInstanceOfType(const std::string& typeName, const std::string& value =
      "") {
00254         if (typeName.empty()) {
00255             return createObjectInstance(value);
00256         }
00257         (void)value;
00258         return false;
00259     }
00260     // This method provides an explicit typed list-element creation operation for list-like
      properties.
00261     virtual bool createNewListElement(const std::string& value = "") {
00262         (void)value;
00263         return false;
00264     }
00265     // This method creates a list element of a requested concrete type.
00266     //
00267     // The default keeps old controls working: when no concrete type is provided, it delegates to
00268     // createNewListElement(); otherwise the operation is unsupported.
00269     virtual bool createNewListElementOfType(const std::string& typeName, const std::string& value =
      "") {
00270         if (typeName.empty()) {
00271             return createNewListElement(value);
00272         }
00273         (void)value;
00274         return false;
00275     }
00276     virtual List<SimulationControl*>* getEditableChildSimulationControls(int index=0) {
00277         if (supportsInlineExpansion() && !hasObjectInstance()) {
00278             if (!supportsObjectCreation()) {
00279                 return nullptr;
00280             }
00281             if (!ensureObjectInstance()) {
00282                 return nullptr;
00283             }
00284         }
00285         return getChildSimulationControls(index);
00286     }
00287 protected:
00288     void _ensureWritable(const char* operation) const {
00289         if (_readonly) {
00290             throw std::logic_error(std::string("Cannot ") + operation + " readonly SimulationControl
      \"" + _propertyName + "\"");
00291         }
00292     }
00293     bool _readonly = false;
00294     SimulationControlEditorHint _preferredEditorHint = SimulationControlEditorHint::Default;
00295 };
00296
00297 // ------------------------------------------------------------
00298
00299 template<typename Class, typename T>
00300 std::function<T()> DefineSimulationGetter(Class* object, T(Class::*function)() const) {
00301     return std::bind(function, object);
00302 }
00303
00304 template<typename Class, typename T>
00305 std::function<T()> DefineSimulationGetter(Class* object, T(Class::*function)()) {
```

```
00306     return std::bind(function, object);
00307 }
00308
00309 template<typename Class, typename T>
00310 std::function<void(T)> DefineSimulationSetter(Class* object, void (Class::*function)(T)) {
00311     return std::bind(function, object, std::placeholders::_1);
00312 }
00313
00314 typedef std::function<std::string()> GetterString;
00315 typedef std::function<void(std::string)> SetterString;
00316 typedef std::function<bool(const std::string&, std::string&)> ValidatorString;
00317 class SimulationControlString: public SimulationControl {
00318 public:
00319 //  SimulationControlString(GetterString getter, std::string className, std::string elementName,
      std::string propertyName, std::string whatsThis="") : SimulationControl(className, elementName,
      propertyName, whatsThis) {
00320 //      SimulationControlString(getter, nullptr, className, propertyName, whatsThis);
00321 //  }
00322     SimulationControlString(GetterString getter, SetterString setter, std::string className,
      std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool
      isClass=false, bool isEnum=false, ValidatorString validator=nullptr) : SimulationControl(className,
      elementName, propertyName, whatsThis, isList, isClass, isEnum) {
00323         _getter= getter;
00324         _setter = setter;
00325         _validator = validator;
00326         _readonly = setter == nullptr;
00327         _propertyType = Util::TypeOf<std::string>();
00328     }
00329 public:
00330     virtual std::string getValue() const override { return _getter(); }
00331     virtual bool validateProposedValue(const std::string& value, std::string& errorMessage) const
      override {
00332         if (!_validator) {
00333             errorMessage.clear();
00334             return true;
00335         }
00336         return _validator(value, errorMessage);
00337     }
00338     virtual void setValue(std::string value, bool remove=false) override {
00339         _ensureWritable("set value of");
00340         if (!_setter) {
00341             throw std::logic_error("SimulationControlString setter is not defined");
00342         }
00343         _setter(value);
00344     };
00345 private:
00346     GetterString _getter;
00347     SetterString _setter;
00348     ValidatorString _validator;
00349 };
00350
00351 typedef std::function<SourceCodeString()> GetterSourceCodeString;
00352 typedef std::function<void(SourceCodeString)> SetterSourceCodeString;
00353 class SimulationControlSourceCodeString: public SimulationControl {
00354 public:
00355     SimulationControlSourceCodeString(GetterSourceCodeString getter, SetterSourceCodeString setter,
      std::string className, std::string elementName, std::string propertyName, std::string whatsThis="",
      bool isList=false, bool isClass=false, bool isEnum=false, ValidatorString validator=nullptr) :
      SimulationControl(className, elementName, propertyName, whatsThis, isList, isClass, isEnum) {
00356         _getter = getter;
00357         _setter = setter;
00358         _validator = validator;
00359         _readonly = setter == nullptr;
00360         _propertyType = Util::TypeOf<SourceCodeString>();
00361         _preferredEditorHint = SimulationControlEditorHint::CodeEditor;
00362     }
00363 public:
00364     virtual std::string getValue() const override { return _getter().str(); }
00365     virtual bool validateProposedValue(const std::string& value, std::string& errorMessage) const
      override {
00366         if (!_validator) {
00367             errorMessage.clear();
00368             return true;
00369         }
00370         return _validator(value, errorMessage);
00371     }
00372     virtual void setValue(std::string value, bool remove=false) override {
00373         (void)remove;
00374         _ensureWritable("set value of");
00375         if (!_setter) {
00376             throw std::logic_error("SimulationControlSourceCodeString setter is not defined");
00377         }
00378         _setter(SourceCodeString(std::move(value)));
00379     };
00380 private:
00381     GetterSourceCodeString _getter;
00382     SetterSourceCodeString _setter;
```

```
00383     ValidatorString _validator;
00384 };
00385
00386
00387 // ----------------------------------------------------------
00388
00389 typedef std::function<double()> GetterDouble;
00390 typedef std::function<void(double)> SetterDouble;
00391
00392 class SimulationResponseDouble: public SimulationResponse {
00393 public:
00394     SimulationResponseDouble(GetterDouble getter, std::string className, std::string elementName,
      std::string propertyName, std::string whatsThis="", bool isList=false, bool isClass=false, bool
      isEnum=false)
00395         : SimulationResponse(className, elementName, propertyName, whatsThis, isList, isClass, isEnum)
      {
00396         _getter = getter;
00397         _propertyType = Util::TypeOf<double>();
00398     }
00399 public:
00400     virtual std::string getValue() const override { return std::to_string(_getter()); }
00401 private:
00402     GetterDouble _getter;
00403 };
00404
00405 class SimulationControlDouble: public SimulationControl {
00406 public:
00407 //  SimulationControlDouble(GetterDouble getter, std::string className, std::string elementName,
      std::string propertyName, std::string whatsThis="") : SimulationControl(className, elementName,
      propertyName, whatsThis) {
00408 //      SimulationControlDouble(getter, nullptr, className, propertyName, whatsThis);
00409 //  }
00410     SimulationControlDouble(GetterDouble getter, SetterDouble setter, std::string className,
      std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool
      isClass=false, bool isEnum=false) : SimulationControl(className, elementName, propertyName, whatsThis,
      isList, isClass, isEnum){
00411         _getter= getter;
00412         _setter = setter;
00413         _readonly = setter == nullptr;
00414         _propertyType = Util::TypeOf<double>();
00415     }
00416 public:
00417     virtual std::string getValue() const override { return std::to_string(_getter()); }
00418     virtual void setValue(std::string value, bool remove=false) override {
00419         _ensureWritable("set value of");
00420         if (!_setter) {
00421             throw std::logic_error("SimulationControlDouble setter is not defined");
00422         }
00423         _setter(std::stod(value));
00424     };
00425 private:
00426     GetterDouble _getter;
00427     SetterDouble _setter;
00428 };
00429
00430 // ----------------------------------------------------------
00431
00432 typedef std::function<bool()> GetterBool;
00433 typedef std::function<void(bool)> SetterBool;
00434 class SimulationControlBool: public SimulationControl {
00435 public:
00436 //  SimulationControlBool(GetterBool getter, std::string className, std::string elementName,
      std::string propertyName, std::string whatsThis="") : SimulationControl(className, elementName,
      propertyName, whatsThis) {
00437 //      SimulationControlBool(getter, nullptr, className, propertyName, whatsThis);
00438 //  }
00439     SimulationControlBool(GetterBool getter, SetterBool setter, std::string className, std::string
      elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool
      isClass=false, bool isEnum=false) : SimulationControl(className, elementName, propertyName, whatsThis,
      isList, isClass, isEnum){
00440         _getter= getter;
00441         _setter = setter;
00442         _readonly = setter == nullptr;
00443         _propertyType = Util::TypeOf<bool>();
00444     }
00445 public:
00446     virtual std::string getValue() const override { return std::to_string(_getter()); }
00447     virtual void setValue(std::string value, bool remove=false) override {
00448         _ensureWritable("set value of");
00449         if (!_setter) {
00450             throw std::logic_error("SimulationControlBool setter is not defined");
00451         }
00452         std::istringstream auxStr(value);
00453         auxStr » std::boolalpha;
00454         bool boolVal = false;
00455         if (!(auxStr » boolVal)) {
00456             boolVal = std::stoi(value) != 0;
```

```
00457         }
00458         _setter(boolVal);
00459     };
00460 private:
00461     GetterBool _getter;
00462     SetterBool _setter;
00463 };
00464 
00465 
00466 // ------------------------------------------------------------
00467 
00468 typedef std::function<unsigned int()> GetterUInt;
00469 typedef std::function<void(unsigned int)> SetterUInt;
00470 class SimulationControlUInt: public SimulationControl {
00471 public:
00472 //  SimulationControlUInt(GetterUInt getter, std::string className, std::string elementName,
      std::string propertyName, std::string whatsThis="") : SimulationControl(className, elementName,
      propertyName, whatsThis) {
00473 //      SimulationControlUInt(getter, nullptr, className, propertyName, whatsThis);
00474 //  }
00475     SimulationControlUInt(GetterUInt getter, SetterUInt setter, std::string className, std::string
      elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool
      isClass=false, bool isEnum=false) : SimulationControl(className, elementName, propertyName, whatsThis,
      isList, isClass, isEnum){
00476         _getter= getter;
00477         _setter = setter;
00478         _readonly = setter == nullptr;
00479         _propertyType = Util::TypeOf<unsigned int>();
00480     }
00481 public:
00482     virtual std::string getValue() const override { return std::to_string(_getter()); }
00483     virtual void setValue(std::string value, bool remove=false) override {
00484         _ensureWritable("set value of");
00485         if (!_setter) {
00486             throw std::logic_error("SimulationControlUInt setter is not defined");
00487         }
00488         _setter(std::stoul(value));
00489     };
00490 private:
00491     GetterUInt _getter;
00492     SetterUInt _setter;
00493 };
00494 
00495 // ------------------------------------------------------------
00496 
00497 typedef std::function<unsigned short()> GetterUShort;
00498 typedef std::function<void(unsigned short)> SetterUShort;
00499 class SimulationControlUShort: public SimulationControl {
00500 public:
00501 //  SimulationControlUShort(GetterUShort getter, std::string className, std::string elementName,
      std::string propertyName, std::string whatsThis="") : SimulationControl(className, elementName,
      propertyName, whatsThis) {
00502 //      SimulationControlUShort(getter, nullptr, className, propertyName, whatsThis);
00503 //  }
00504         SimulationControlUShort(GetterUShort getter, SetterUShort setter, std::string className,
      std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool
      isClass=false, bool isEnum=false) : SimulationControl(className, elementName, propertyName, whatsThis,
      isList, isClass, isEnum){
00505             _getter= getter;
00506             _setter = setter;
00507             _readonly = setter == nullptr;
00508             _propertyType = Util::TypeOf<unsigned short>();
00509         }
00510 public:
00511     virtual std::string getValue() const override { return std::to_string(_getter()); }
00512     virtual void setValue(std::string value, bool remove=false) override {
00513         _ensureWritable("set value of");
00514         if (!_setter) {
00515             throw std::logic_error("SimulationControlUShort setter is not defined");
00516         }
00517         _setter(std::stoul(value));
00518     };
00519 private:
00520     GetterUShort _getter;
00521     SetterUShort _setter;
00522 };
00523 
00524 
00525 // ------------------------------------------------------------
00526 
00527 typedef std::function<int()> GetterInt;
00528 typedef std::function<void(int)> SetterInt;
00529 class SimulationControlInt: public SimulationControl {
00530 public:
00531 //  SimulationControlInt(GetterInt getter, std::string className, std::string elementName, std::string
      propertyName, std::string whatsThis="") : SimulationControl(className, elementName, propertyName,
      whatsThis) {
```

```
00532 //      SimulationControlInt(getter, nullptr, className, propertyName, whatsThis);
00533 //  }
00534         SimulationControlInt(GetterInt getter, SetterInt setter, std::string className, std::string
      elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool
      isClass=false, bool isEnum=false) : SimulationControl(className, elementName, propertyName, whatsThis,
      isList, isClass, isEnum){
00535             _getter= getter;
00536             _setter = setter;
00537             _readonly = setter == nullptr;
00538             _propertyType = Util::TypeOf<int>();
00539         }
00540 public:
00541     virtual std::string getValue() const override { return std::to_string(_getter()); }
00542     virtual void setValue(std::string value, bool remove=false) override {
00543         _ensureWritable("set value of");
00544         if (!_setter) {
00545             throw std::logic_error("SimulationControlInt setter is not defined");
00546         }
00547         _setter(std::stoi(value));
00548     };
00549 private:
00550     GetterInt _getter;
00551     SetterInt _setter;
00552 };
00553
00554 // ------------------------------------------------------------
00555
00556 typedef std::function<Util::TimeUnit()> GetterTimeUnit;
00557 typedef std::function<void(Util::TimeUnit)> SetterTimeUnit;
00558 class SimulationControlTimeUnit: public SimulationControl {
00559 public:
00560 //  SimulationControlTimeUnit(GetterTimeUnit getter, std::string className, std::string elementName,
      std::string propertyName, std::string whatsThis="") : SimulationControl(className, elementName,
      propertyName, whatsThis) {
00561 //      SimulationControlTimeUnit(getter, nullptr, className, propertyName, whatsThis);
00562 //  }
00563     SimulationControlTimeUnit(GetterTimeUnit getter, SetterTimeUnit setter, std::string className,
      std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool
      isClass=false, bool isEnum=false) : SimulationControl(className, elementName, propertyName, whatsThis,
      isList, isClass, isEnum){
00564         _getter= getter;
00565         _setter = setter;
00566         _readonly = setter == nullptr;
00567         _propertyType = Util::TypeOf<Util::TimeUnit>();
00568     }
00569 public:
00570     virtual std::string getValue() const override {
00571         int intVal = static_cast<int>(_getter());
00572         return std::to_string(intVal);
00573     }
00574     virtual void setValue(std::string value, bool remove=false) override {
00575         _ensureWritable("set value of");
00576         if (!_setter) {
00577             throw std::logic_error("SimulationControlTimeUnit setter is not defined");
00578         }
00579         int intVal = std::stoul(value);
00580         _setter(static_cast<Util::TimeUnit>(intVal));
00581     };
00582 private:
00583     GetterTimeUnit _getter;
00584     SetterTimeUnit _setter;
00585 };
00586
00587
00588 template <typename T>
00589 using GetterGeneric = std::function<T()>;
00590 template <typename T>
00591 using SetterGeneric = std::function<void(T)>;
00592 template <typename T>
00593 using AdderGeneric = std::function<void(T)>;
00594 template <typename T>
00595 using RemoverGeneric = std::function<void(T)>;
00596
00597 template <typename T>
00598 class SimulationControlGeneric: public SimulationControl {
00599 public:
00600     SimulationControlGeneric(GetterGeneric<T> getter, SetterGeneric<T> setter, std::string className,
      std::string elementName, std::string propertyName, std::string whatsThis="", bool isList=false, bool
      isClass=false, bool isEnum=false) : SimulationControl(className, elementName, propertyName, whatsThis,
      isList, isClass, isEnum){
00601         _getter= getter;
00602         _setter = setter;
00603         _readonly = setter == nullptr;
00604         _propertyType = Util::TypeOf<T>();
00605     }
00606 public:
00607     virtual std::string getValue() const override {
```

```
00608         T tVal = static_cast<T>(_getter());
00609
00610         std::string strVal;
00611         std::ostringstream auxStr;
00612         auxStr « tVal;
00613         strVal = auxStr.str();
00614
00615         return strVal;
00616     }
00617
00618     virtual void setValue(std::string value, bool remove=false) override {
00619         _ensureWritable("set value of");
00620         if (!_setter) {
00621             throw std::logic_error("SimulationControlGeneric setter is not defined");
00622         }
00623         if constexpr (std::is_same_v<T, std::string>) {
00624             _setter(value);
00625         } else {
00626             std::istringstream auxStr(value);
00627             T tVal{};
00628             auxStr » tVal;
00629             _setter(static_cast<T>(tVal));
00630         }
00631     };
00632
00633 private:
00634     GetterGeneric<T> _getter;
00635     SetterGeneric<T> _setter;
00636 };
00637
00638 // @ToDo: (pequena alteração): remove typename E
00639 template <typename T, typename E>
00640 class SimulationControlGenericEnum: public SimulationControl {
00641 public:
00642     SimulationControlGenericEnum(GetterGeneric<T> getter, SetterGeneric<T> setter, std::string
      className, std::string elementName, std::string propertyName, std::string whatsThis="", bool
      isList=false, bool isClass=false, bool isEnum=true) : SimulationControl(className, elementName,
      propertyName, whatsThis, isList, isClass, isEnum){
00643         static_assert(std::is_enum<T>::value, "SimulationControlGenericEnum requires T to be an
      enum");
00644         _getter= getter;
00645         _setter = setter;
00646         _readonly = setter == nullptr;
00647         _propertyType = Util::TypeOf<T>();
00648         _isEnum=true;
00649     }
00650 public:
00651     virtual std::string getValue() const override {
00652         int intVal = static_cast<int>(_getter());
00653         int max_i = static_cast<int>(T::num_elements);
00654         if (intVal < 0 || intVal >= max_i) {
00655             return "";
00656         }
00657         return E::convertEnumToStr(static_cast<T>(intVal));
00658     }
00659
00660         virtual void setValue(std::string value, bool remove=false) override {
00661         _ensureWritable("set value of");
00662         if (!_setter) {
00663             throw std::logic_error("SimulationControlGenericEnum setter is not defined");
00664         }
00665         int intVal = std::stoul(value);
00666         _setter(static_cast<T>(intVal));
00667     };
00668
00669     virtual List<std::string>* getStrValues() override {
00670         List<std::string>* strOptions = new List<std::string>();
00671         int max_i = static_cast<int>(T::num_elements);;
00672         for (int i=0; i<max_i; i++) {
00673             std::string value = E::convertEnumToStr((T)i);
00674             strOptions->insert(value);
00675         }
00676         return strOptions;
00677     }
00678
00679 private:
00680     GetterGeneric<T> _getter;
00681     SetterGeneric<T> _setter;
00682 };
00683
00684 // @ToDo: (pequena alteração): remove typename C
00685 template <typename T, typename M, typename C>
00686 class SimulationControlGenericClass: public SimulationControl {
00687 public:
00688     using Creator = std::function<T(M, const std::string&)>;
00689     SimulationControlGenericClass(M model, GetterGeneric<T> getter, SetterGeneric<T> setter,
      std::string className, std::string elementName, std::string propertyName, std::string whatsThis="",
```

```
    bool isList=false, bool isClass=true, bool isEnum=false, Creator creator=nullptr) :
    SimulationControl(className, elementName, propertyName, whatsThis, isList, isClass, isEnum){
00690         static_assert(std::is_pointer<T>::value, "SimulationControlGenericClass requires pointer type
    T");
00691         _model = model;
00692         _getter= getter;
00693         _setter = setter;
00694         _creator = creator;
00695         _readonly = setter == nullptr;
00696         _propertyType = Util::TypeOf<C>();
00697     }
00698 public:
00699     virtual std::string getValue() const override {
00700         T tVal = static_cast<T>(_getter());
00701         std::string strVal;
00702 
00703         if (tVal != nullptr) {
00704             strVal = tVal->getName();
00705         } else {
00706             strVal = "";
00707         };
00708 
00709         return strVal;
00710     }
00711 
00712     virtual void setValue(std::string value, bool remove=false) override {
00713         _ensureWritable("set value of");
00714         if (!_setter) {
00715             throw std::logic_error("SimulationControlGenericClass setter is not defined");
00716         }
00717         if (remove) {
00718             _setter(nullptr);
00719             return;
00720         }
00721         bool exists = false;
00722         // value.pop_back();
00723         T newVal;
00724 
00725         for (auto modeldata : *_model->getDataManager()->getDataDefinitionList(_propertyType)->list())
    {
00726             if (modeldata->getName() == value) {
00727                 exists = true;
00728                 newVal = static_cast<T>(modeldata);
00729                 break;
00730             };
00731         };
00732 
00733         if (!exists) {
00734             if (_creator != nullptr) {
00735                 newVal = _creator(_model, value);
00736             } else {
00737                 newVal = new C(_model, value);
00738             }
00739             if (newVal == nullptr) {
00740                 throw std::logic_error("SimulationControlGenericClass creator returned null");
00741             }
00742             _model->getDataManager()->insert(newVal);
00743         }
00744 
00745         _setter(newVal);
00746     };
00747 
00748     virtual bool isModelDataDefinitionReference() const override { return true; }
00749     virtual ModelDataDefinition* getReferencedModelDataDefinition() const override {
00750         return static_cast<T>(_getter());
00751     }
00752     // This method marks class references as inline-expandable in the universal contract.
00753     virtual bool supportsInlineExpansion() const override { return true; }
00754     // This method marks class references as selectable from existing ModelDataDefinition instances.
00755     virtual bool supportsExistingObjectSelection() const override { return true; }
00756     // This method marks class references as creatable when writable.
00757     virtual bool supportsObjectCreation() const override { return !_readonly; }
00758     // This method marks class references as non-inline object payloads.
00759     virtual bool isInlineObjectProperty() const override { return false; }
00760     // Property Editor advanced editing experiment, step 1:
00761     // ModelDataDefinition references must report whether the referenced object
00762     // actually exists. Previously the base implementation always returned true,
00763     // so the GUI could not distinguish "empty reference" from "loaded object".
00764     // Revert this override to restore the old always-available reference behavior.
00765     virtual bool hasObjectInstance() const override {
00766         return static_cast<T>(_getter()) != nullptr;
00767     }
00768     // Property Editor advanced editing experiment, step 1:
00769     // Do not auto-create ModelDataDefinition references while expanding the tree.
00770     // Unlike inline helper objects, Queue/Resource/Set references need an explicit
00771     // name or an existing target chosen by the user before creation is safe.
00772     // Revert this override if implicit expansion should create references again.
```

```
00773     virtual bool ensureObjectInstance() override {
00774         return hasObjectInstance();
00775     }
00776     // This method performs explicit object creation for class references.
00777     virtual bool createObjectInstance(const std::string& value = "") override {
00778         _ensureWritable("create instance for");
00779         std::string name = value;
00780         if (name.empty()) {
00781             name = getValue();
00782         }
00783         if (name.empty()) {
00784             return false;
00785         }
00786         setValue(name, false);
00787         return hasObjectInstance();
00788     }
00789 
00790     virtual List<std::string>* getStrValues() override {
00791         List<std::string>* strOptions = new List<std::string>();
00792         for (auto modeldata : *_model->getDataManager()->getDataDefinitionList(_propertyType)->list())
      {
00793             if (modeldata != nullptr) {
00794                 strOptions->insert(modeldata->getName());
00795             }
00796         }
00797         return strOptions;
00798     }
00799 
00800     virtual List<SimulationControl*>* getChildSimulationControls(int index=0) override {
00801         T tVal = static_cast<T>(_getter());
00802 
00803         if (tVal) {
00804             return tVal->getSimulationControls();
00805         } else {
00806             return nullptr;
00807         }
00808     }
00809 
00810 private:
00811     M _model;
00812     GetterGeneric<T> _getter;
00813     SetterGeneric<T> _setter;
00814     Creator _creator;
00815 };
00816 
00817 // @ToDo: (pequena alteração): remove typename C
00818 template <typename T, typename M, typename C>
00819 class SimulationControlGenericClassNotDC: public SimulationControl {
00820 public:
00821     using Creator = std::function<T(M)>;
00822     using NamedCreator = std::function<T(M, const std::string&)>;
00823     SimulationControlGenericClassNotDC(M model, GetterGeneric<T> getter, SetterGeneric<T> setter,
      std::string className, std::string elementName, std::string propertyName, std::string whatsThis="",
      bool isList=false, bool isClass=true, bool isEnum=false, Creator creator=nullptr, NamedCreator
      namedCreator=nullptr) : SimulationControl(className, elementName, propertyName, whatsThis, isList,
      isClass, isEnum){
00824         static_assert(std::is_pointer<T>::value, "SimulationControlGenericClassNotDC requires pointer
      type T");
00825         _model = model;
00826         _getter= getter;
00827         _setter = setter;
00828         _creator = creator;
00829         _namedCreator = namedCreator;
00830         _readonly = setter == nullptr;
00831         _propertyType = Util::TypeOf<C>();
00832     }
00833 public:
00834     virtual std::string getValue() const override {
00835         T tVal = static_cast<T>(_getter());
00836         std::string strVal;
00837 
00838         if (tVal != nullptr) {
00839             strVal = tVal->getName();
00840         } else {
00841             strVal = "";
00842         };
00843 
00844         return strVal;
00845     }
00846 
00847     virtual void setValue(std::string value, bool remove=false) override {
00848         _ensureWritable("set value of");
00849         if (!_setter) {
00850             throw std::logic_error("SimulationControlGenericClassNotDC setter is not defined");
00851         }
00852         T newVal = _createNewInstance(value);
00853 
```

```
00854         _setter(newVal);
00855     };
00856
00857     virtual bool hasObjectInstance() const override {
00858         return static_cast<T>(_getter()) != nullptr;
00859     }
00860     // This method marks inline class payloads as expandable in the universal contract.
00861     virtual bool supportsInlineExpansion() const override { return true; }
00862     // This method marks inline class payloads as creatable when writable.
00863     virtual bool supportsObjectCreation() const override { return !_readonly; }
00864     // This method marks inline class payloads as non-reference object payloads.
00865     virtual bool isInlineObjectProperty() const override { return true; }
00866
00867     virtual bool ensureObjectInstance() override {
00868         _ensureWritable("ensure instance of");
00869         if (!_setter) {
00870             throw std::logic_error("SimulationControlGenericClassNotDC setter is not defined");
00871         }
00872
00873         T current = static_cast<T>(_getter());
00874         if (current != nullptr) {
00875             return true;
00876         }
00877
00878         T newVal;
00879         newVal = _createNewInstance("");
00880         _setter(newVal);
00881         return static_cast<T>(_getter()) != nullptr;
00882     }
00883     // This method performs explicit object creation for non-DataManager classes.
00884     virtual bool createObjectInstance(const std::string& value = "") override {
00885         if (value.empty()) {
00886             return ensureObjectInstance();
00887         }
00888         setValue(value, false);
00889         return hasObjectInstance();
00890     }
00891
00892     virtual List<SimulationControl*>* getChildSimulationControls(int index=0) override {
00893         T tVal = static_cast<T>(_getter());
00894
00895         if (tVal) {
00896             return tVal->getSimulationControls();
00897         } else {
00898             return nullptr;
00899         }
00900     }
00901
00902 private:
00903     T _createNewInstance(const std::string& value) const {
00904         if (_namedCreator != nullptr) {
00905             return _namedCreator(_model, value);
00906         }
00907         if (_creator != nullptr) {
00908             return _creator(_model);
00909         }
00910         return new C(_model, value);
00911     }
00912
00913 private:
00914     M _model;
00915     GetterGeneric<T> _getter;
00916     SetterGeneric<T> _setter;
00917     Creator _creator;
00918     NamedCreator _namedCreator;
00919 };
00920
00921 template <typename T, typename M, typename C>
00922 class SimulationControlGenericList: public SimulationControl {
00923 public:
00924     SimulationControlGenericList(M model, GetterGeneric<List<T>*> getter, AdderGeneric<T> adder,
      RemoverGeneric<T> remover, std::string className, std::string elementName, std::string propertyName,
      std::string whatsThis="", bool isList=true, bool isClass=false, bool isEnum=false) :
      SimulationControl(className, elementName, propertyName, whatsThis, isList, isClass, isEnum){
00925         _model = model;
00926         _getter= getter;
00927         _adder = adder;
00928         _remover = remover;
00929         _readonly = adder == nullptr;
00930         _propertyType = Util::TypeOf<C>();
00931     }
00932 public:
00933     virtual std::string getValue() const override {
00934         List<T>* tVal = static_cast<List<T>*>(_getter());
00935         std::string strVal = std::to_string(tVal->size()) + " Items";
00936
00937         return strVal;
```

```
00938     }
00939
00940     virtual void setValue(std::string value, bool remove=false) override {
00941         _ensureWritable(remove ? "remove value from" : "add value to");
00942         T newVal{};
00943         if constexpr (std::is_same_v<T, std::string>) {
00944             newVal = value;
00945         } else {
00946             std::istringstream auxStr(value);
00947             auxStr » newVal;
00948             newVal = static_cast<T>(newVal);
00949         }
00950
00951             if (remove) {
00952                 if (!_remover) {
00953                     throw std::logic_error("SimulationControlGenericList remover is not defined");
00954                 }
00955                 _remover(newVal);
00956             } else {
00957                 if (!_adder) {
00958                     throw std::logic_error("SimulationControlGenericList adder is not defined");
00959                 }
00960                 bool exists = false;
00961                 List<T>* tVal = static_cast<List<T>*>(_getter());
00962                 for (auto element : *tVal->list()) {
00963                     std::ostringstream auxElement;
00964                     auxElement « element;
00965                     if (value == auxElement.str()) {
00966                         exists = true;
00967                         break;
00968                     }
00969                 }
00970
00971             if (!exists) {
00972                 _adder(newVal);
00973             }
00974         }
00975     };
00976
00977     virtual List<std::string>* getStrValues() override {
00978         List<T>* tVal = static_cast<List<T>*>(_getter());
00979         List<std::string>* strOptions = new List<std::string>();
00980
00981         for (auto element : *tVal->list()) {
00982             std::string strVal;
00983             std::ostringstream auxStr;
00984             auxStr « element;
00985             strVal = auxStr.str();
00986
00987             strOptions->insert(strVal);
00988         }
00989         return strOptions;
00990     }
00991     // This method marks scalar lists as editable through the list editor in the universal contract.
00992     virtual bool supportsListEditor() const override { return true; }
00993
00994 private:
00995     M _model;
00996     GetterGeneric<List<T>*> _getter;
00997     AdderGeneric<T> _adder;
00998     RemoverGeneric<T> _remover;
00999 };
01000
01001 template <typename T, typename M, typename C>
01002 class SimulationControlGenericListPointer: public SimulationControl {
01003 public:
01004     using Creator = std::function<T(M, const std::string&)>;
01005     using TypedCreator = std::function<T(M)>;
01006     using ElementTypeNamesGetter = std::function<std::vector<std::string>()>;
01007     using CurrentElementTypeGetter = std::function<std::string()>;
01008     using CurrentElementTypeSetter = std::function<bool(const std::string&)>;
01009     using TypedNamedCreator = std::function<T(M, const std::string&, const std::string&)>;
01010     using ListElementDataDefinitionGetter = std::function<ModelDataDefinition*(int)>;
01011     SimulationControlGenericListPointer(M model, GetterGeneric<List<T>*> getter, AdderGeneric<T>
      adder, RemoverGeneric<T> remover, std::string className, std::string elementName, std::string
      propertyName, std::string whatsThis="", bool isList=true, bool isClass=true, bool isEnum=false,
      Creator creator=nullptr, TypedCreator typedCreator=nullptr, ElementTypeNamesGetter
      elementTypeNamesGetter=nullptr, CurrentElementTypeGetter currentElementTypeGetter=nullptr,
      CurrentElementTypeSetter currentElementTypeSetter=nullptr, TypedNamedCreator
      typedNamedCreator=nullptr, ListElementDataDefinitionGetter listElementDataDefinitionGetter=nullptr) :
      SimulationControl(className, elementName, propertyName, whatsThis, isList, isClass, isEnum){
01012         static_assert(std::is_pointer<T>::value, "SimulationControlGenericListPointer requires pointer
      type T");
01013         _model = model;
01014         _getter= getter;
01015         _adder = adder;
01016         _remover = remover;
```

```
01017         _creator = creator;
01018         _typedCreator = typedCreator;
01019         _elementTypeNamesGetter = elementTypeNamesGetter;
01020         _currentElementTypeGetter = currentElementTypeGetter;
01021         _currentElementTypeSetter = currentElementTypeSetter;
01022         _typedNamedCreator = typedNamedCreator;
01023         _listElementDataDefinitionGetter = listElementDataDefinitionGetter;
01024         _readonly = adder == nullptr;
01025         _propertyType = Util::TypeOf<C>();
01026     }
01027 public:
01028     virtual std::string getValue() const override {
01029         List<T>* tVal = static_cast<List<T>*>(_getter());
01030         std::string strVal = std::to_string(tVal->size()) + " Items";
01031 
01032         return strVal;
01033     }
01034 
01035         virtual void setValue(std::string value, bool remove=false) override {
01036         _ensureWritable(remove ? "remove value from" : "add value to");
01037         List<T>* tVal = static_cast<List<T>*>(_getter());
01038         T existingVal = nullptr;
01039         for (auto element : *tVal->list()) {
01040             if (element != nullptr && element->getName() == value) {
01041                 existingVal = element;
01042                 break;
01043             }
01044         }
01045 
01046             if (remove) {
01047                 if (!_remover) {
01048                     throw std::logic_error("SimulationControlGenericListPointer remover is not
      defined");
01049                 }
01050                 if (existingVal != nullptr) {
01051                     _remover(existingVal);
01052                 }
01053             } else {
01054                 if (!_adder) {
01055                     throw std::logic_error("SimulationControlGenericListPointer adder is not
      defined");
01056                 }
01057                 if (existingVal == nullptr) {
01058                     if (!createNewListElement(value)) {
01059                         throw std::logic_error("SimulationControlGenericListPointer could not create a
      new list element");
01060                     }
01061             }
01062         }
01063     };
01064     // This method marks pointer lists as editable through the list editor in the universal contract.
01065     virtual bool supportsListEditor() const override { return true; }
01066     // This method marks pointer lists as supporting explicit typed element creation when writable.
01067     virtual bool supportsNewListElementCreation() const override { return !_readonly; }
01068     // This method exposes concrete types for polymorphic pointer lists such as Set members.
01069     //
01070     // When no provider is configured, the list keeps the legacy single-type creation behavior.
01071     virtual List<std::string>* getCreatableListElementTypes() override {
01072         if (_elementTypeNamesGetter == nullptr) {
01073             return nullptr;
01074         }
01075         List<std::string>* types = new List<std::string>();
01076         for (const std::string& typeName : _elementTypeNamesGetter()) {
01077             if (!typeName.empty()) {
01078                 types->insert(typeName);
01079             }
01080         }
01081         return types;
01082     }
01083     // This method reports the selected concrete type for typed polymorphic lists.
01084     virtual std::string getCurrentListElementType() const override {
01085         if (_currentElementTypeGetter != nullptr) {
01086             return _currentElementTypeGetter();
01087         }
01088         return "";
01089     }
01090     // This method lets the GUI configure the selected concrete type before creating members.
01091     virtual bool setCurrentListElementType(const std::string& typeName) override {
01092         if (_currentElementTypeSetter == nullptr) {
01093             return false;
01094         }
01095         return _currentElementTypeSetter(typeName);
01096     }
01097     // This method creates and inserts a typed list element using the configured creator or the
      default constructor path.
01098     virtual bool createNewListElement(const std::string& value = "") override {
01099         if (_typedNamedCreator != nullptr) {
```

```
01100             return createNewListElementOfType(getCurrentListElementType(), value);
01101         }
01102         _ensureWritable("create list element for");
01103         if (!_adder) {
01104             throw std::logic_error("SimulationControlGenericListPointer adder is not defined");
01105         }
01106         T newVal = _createElement(value);
01107         if (newVal == nullptr) {
01108             return false;
01109         }
01110         _adder(newVal);
01111         return true;
01112     }
01113     // This method creates and inserts a member of a caller-selected concrete type.
01114     //
01115     // It is intentionally part of the control contract instead of GUI code so each kernel owner can
01116     // validate type compatibility, naming and model-registration rules in one place.
01117     virtual bool createNewListElementOfType(const std::string& typeName, const std::string& value =
      "") override {
01118         _ensureWritable("create typed list element for");
01119         if (!_adder) {
01120             throw std::logic_error("SimulationControlGenericListPointer adder is not defined");
01121         }
01122         if (_typedNamedCreator == nullptr) {
01123             return SimulationControl::createNewListElementOfType(typeName, value);
01124         }
01125         std::string effectiveTypeName = typeName;
01126         if (effectiveTypeName.empty()) {
01127             effectiveTypeName = getCurrentListElementType();
01128         }
01129         if (effectiveTypeName.empty()) {
01130             return false;
01131         }
01132         if (_currentElementTypeSetter != nullptr && !_currentElementTypeSetter(effectiveTypeName)) {
01133             return false;
01134         }
01135         T newVal = _typedNamedCreator(_model, effectiveTypeName, value);
01136         if (newVal == nullptr) {
01137             return false;
01138         }
01139         _adder(newVal);
01140         return true;
01141     }
01142
01143     virtual List<SimulationControl*>* getChildSimulationControls(int index=0) override {
01144             List<T>* tVal = static_cast<List<T>*>(_getter());
01145             T selectedElement = nullptr;
01146
01147         int current_index = 0;
01148         for (auto element : *tVal->list()) {
01149             if (current_index == index) {
01150                 selectedElement = element;
01151             }
01152             current_index++;
01153         }
01154
01155         if (selectedElement) {
01156             if constexpr (requires(T element, M model) { element->ensureSimulationControls(model); })
      {
01157                 selectedElement->ensureSimulationControls(_model);
01158             }
01159             return selectedElement->getSimulationControls();
01160         } else {
01161             return nullptr;
01162         }
01163     }
01164
01165     virtual ModelDataDefinition* getListElementModelDataDefinition(int index) const override {
01166         if (_listElementDataDefinitionGetter == nullptr) {
01167             return nullptr;
01168         }
01169         return _listElementDataDefinitionGetter(index);
01170     }
01171
01172     virtual List<std::string>* getStrValues() override {
01173         List<T>* tVal = static_cast<List<T>*>(_getter());
01174         List<std::string>* strOptions = new List<std::string>();
01175
01176         for (auto element : *tVal->list()) {
01177             std::string strVal;
01178             if (element != nullptr) {
01179                 strVal = element->getName();
01180             } else {
01181                 strVal = "";
01182             };
01183
01184             strOptions->insert(strVal);
```

```
01185         }
01186         return strOptions;
01187     }
01188
01189 private:
01190     T _createElement(const std::string& value) const {
01191         if (_typedCreator != nullptr) {
01192             return _typedCreator(_model);
01193         }
01194         if (_creator != nullptr) {
01195             return _creator(_model, value);
01196         }
01197         return new C(_model, value);
01198     }
01199
01200 private:
01201     M _model;
01202     GetterGeneric<List<T>*> _getter;
01203     AdderGeneric<T> _adder;
01204     RemoverGeneric<T> _remover;
01205     Creator _creator;
01206     TypedCreator _typedCreator;
01207     ElementTypeNamesGetter _elementTypeNamesGetter;
01208     CurrentElementTypeGetter _currentElementTypeGetter;
01209     CurrentElementTypeSetter _currentElementTypeSetter;
01210     TypedNamedCreator _typedNamedCreator;
01211     ListElementDataDefinitionGetter _listElementDataDefinitionGetter;
01212 };
01213
01214 //namespace\\}
```

18.339 SimulationExperiment.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   SimulationExperiment.h
00009  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00010  *
00011  * Created on 4 de maio de 2021, 15:10
00012  */
00013
00014 #ifndef SIMULATIONEXPERIMENT_H
00015 #define SIMULATIONEXPERIMENT_H
00016
00030 class SimulationExperiment {
00031 public:
00032     SimulationExperiment();
00033     virtual ~SimulationExperiment() = default;
00034 private:
00035
00036 };
00037
00038 #endif /* SIMULATIONEXPERIMENT_H */
00039
```

18.340 SimulationReporter_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   SimulationReporter_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 8 de Agosto de 2018, 10:56
00012  */
00013
00014 #ifndef SIMULATIONREPORTER_IF_H
00015 #define SIMULATIONREPORTER_IF_H
00016
00017 #include "../util/List.h"
00018 //#include "StatisticsCollector.h"
00019
00026 class SimulationReporter_if {
```

```
00027 public:
00028     virtual ~SimulationReporter_if() = default;
00033     virtual void showReplicationStatistics() = 0;
00038     virtual void showSimulationStatistics() = 0;
00043     virtual void showSimulationResponses() = 0;
00048     virtual void showSimulationControls() = 0;
00049     //virtual void setFormat(std::string format)=0;
00050 };
00051
00052 #endif /* SIMULATIONREPORTER_IF_H */
```

18.341 SimulationReporterDefaultImpl1.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   SimulationReporterDefaultImpl1.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 8 de Agosto de 2018, 10:59
00012  */
00013
00014 #ifndef SIMULATIONREPORTERDEFAULTIMPL1_H
00015 #define SIMULATIONREPORTERDEFAULTIMPL1_H
00016
00017 #include "SimulationReporter_if.h"
00018 #include "model/ModelSimulation.h"
00019 #include "model/Model.h"
00020
00021 //namespace GenesysKernel {
00022
00026 class SimulationReporterDefaultImpl1 : public SimulationReporter_if {
00027 public:
00028     SimulationReporterDefaultImpl1(ModelSimulation* simulation, Model* model,
      List<ModelDataDefinition*>* statsCountersSimulation);
00029     virtual ~SimulationReporterDefaultImpl1() = default;
00030 public:
00031     virtual void showReplicationStatistics() override;
00032     virtual void showSimulationStatistics() override;
00033     virtual void showSimulationResponses() override;
00034     virtual void showSimulationControls() override;
00035 private:
00036     ModelSimulation* _simulation;
00037     Model* _model;
00038 private:
00039     List<ModelDataDefinition*>* _statsCountersSimulation;
00040 private:
00041     const unsigned short _w = 9;
00042     const unsigned short _nameW = 36;
00043 };
00044 //namespace\\}
00045 #endif /* SIMULATIONREPORTERDEFAULTIMPL1_H */
00046
```

18.342 SimulationScenario.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   SimulationScenario.h
00009  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00010  *
00011  * Created on 10 de Outubro de 2018, 18:21
00012  */
00013
00014 #ifndef SIMULATIONSCENARIO_H
00015 #define SIMULATIONSCENARIO_H
00016
00017 #include <string>
00018 #include <list>
00019
00020 class Simulator;
00021
00038 class SimulationScenario {
```

```
00039 public:
00040     SimulationScenario();
00041     virtual ~SimulationScenario();
00042 public: // results
00043     bool startSimulation(Simulator *sim, std::string& errorMessage);
00044     std::list<std::pair<std::string, double>*>* getResponseValues() const;
00045     std::list<std::pair<std::string, double>*>* getControlValues() const;
00046     double getResponseValue(const std::string& responseName);
00047 public: // gets and sets
00048     void setModelFilename(std::string _modelFilename);
00049     std::string getModelFilename() const;
00050     void setScenarioName(std::string _name);
00051     std::string getScenarioName() const;
00052     void setScenarioDescription(std::string _scenarioDescription);
00053     std::string getScenarioDescription() const;
00054     std::list<std::string>* getSelectedControls() const; // access to the list to insert or remove
      controls
00055     double getControlValue(const std::string& controlName);
00056     std::list<std::string>* getSelectedResponses() const; // access to the list to insert or remove
      responses
00057     // @ToDo: (importante): Clarify ownership and replacement semantics for selected controls.
00058     // This setter currently allocates a new list and replaces the stored pointer.
00059     void setSelectedControls(std::list<std::string>* selectedControls);
00060     void setControl(std::string name, double value) const;
00061 private:
00062     std::string _scenarioName;
00063     std::string _scenarioDescription;
00064     std::string _modelFilename;
00065     std::list<std::string>* _selectedControls = new std::list<std::string>();
00066     std::list<std::string>* _selectedResponses = new std::list<std::string>();
00067     std::list<std::pair<std::string, double>*>* _controlValues = new std::list<std::pair<std::string,
      double>*>();
00068     // @ToDo: (importante): Define initialization and ownership rules for response values.
00069     std::list<std::pair<std::string, double>*>* _responseValues = new std::list<std::pair<std::string,
      double>*>();
00070 };
00071
00072 #endif /* SIMULATIONSCENARIO_H */
```

18.343 Simulator.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Genesys.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Junho de 2018, 12:48
00012  */
00013
00014 #ifndef GENESYS_H
00015 #define GENESYS_H
00016
00017 #include <string>
00018 #include <iostream>
00019 #include "model/Model.h"
00020 #include "Plugin.h"
00021 #include "../util/List.h"
00022 #include "LicenceManager.h"
00023 #include "PluginManager.h"
00024 #include "model/ModelManager.h"
00025 //#include "ToolManager.h"
00026 #include "ParserManager.h"
00027 #include "ExperimentManager.h"
00028
00029 //namespace GenesysKernel {
00030
00031 class PluginManager;
00032
00033 /* https://www.doxygen.nl/manual/commands.html
00034  * Bried description \class \implements
00035  *
00036  * Detailed description
00037  * \param \return
00038  */
00039
00044 class Simulator {
00045     typedef void (*eventHandler)();
00046 public:
00047     Simulator();
00048     virtual ~Simulator();
```

```
00049 public: // only get
00054     std::string getVersion() const;
00059     unsigned int getVersionNumber() const;
00064     std::string getName() const;
00069     LicenceManager* getLicenceManager() const;
00074     PluginManager* getPluginManager() const;
00079     ModelManager* getModelManager() const;
00084     TraceManager* getTraceManager() const;
00089     ParserManager* getParserManager() const;
00094     ExperimentManager* getExperimentManager() const;
00095
00096 private:
00097     List<Plugin*>* _completePluginsFieldsAndTemplate();
00098     friend class PluginManager; // @ToDo: (pequena alteração): should be only member function
      PluginManager::completePluginsFieldsAndTemplate()
00099
00100 private: // attributes 1:1 objects
00101     LicenceManager* _licenceManager;
00102     PluginManager* _pluginManager;
00103     ModelManager* _modelManager;
00104     TraceManager* _traceManager;
00105     ParserManager* _parserManager;
00106     ExperimentManager* _experimentManager;
00107
00108 private: // attributes 1:1 native
00109     const std::string _name = "GenESyS - GENeric and Expansible SYstem Simulator";
00110     const std::string _versionName = "anaiera";
00111     const unsigned int _versionNumber = 260705;
00112 };
00113 //namespace\\}
00114
00115 // passing a "C" class factory function which instantiates the class.
00116 typedef /*GenesysKernel::*/Simulator* GenesysSimulator;
00117 #endif /* GENESYS_H */
```

18.344 SimulatorFacade.h

```
00001 /*
00002  * SimulatorFacade.h
00003  *
00004  * High-level non-owning facade over Simulator and its main composed services.
00005  *
00006  * This header delegates calls to Simulator, its managers, the current Model and
00007  * the current Model sub-services without forcing callers to navigate manually
00008  * through getPluginManager(), getModelManager(), getTraceManager(), etc.
00009  *
00010  * All methods tolerate a null _simulator or missing current model:
00011  *   - pointer returns  -> nullptr
00012  *   - bool returns     -> false
00013  *   - numeric returns  -> 0
00014  *   - string returns   -> ""
00015  *   - void methods     -> no-op
00016  *
00017  * Method prefixes resolve name conflicts between delegates:
00018  *   (none)            Simulator direct methods
00019  *   licence*          LicenceManager
00020  *   plugin*           PluginManager
00021  *   model*            ModelManager / Model
00022  *   info*             ModelInfo
00023  *   sim*              ModelSimulation
00024  *   data*             ModelDataManager
00025  *   component*        ModelComponentManager
00026  *   event*            OnEventManager
00027  *   trace*, setTrace* TraceManager
00028  *   parser*           ParserManager
00029  *   simulationExp*    ExperimentManager
00030  */
00031
00032 #ifndef SIMULATORFACADE_H
00033 #define SIMULATORFACADE_H
00034
00035 #include "Simulator.h"
00036 #include "LicenceManager.h"
00037 #include "PluginManager.h"
00038 #include "TraceManager.h"
00039 #include "ParserManager.h"
00040 #include "ParserChangesInformation.h"
00041 #include "ExperimentManager.h"
00042 #include "SimulationExperiment.h"
00043 #include "OnEventManager.h"
00044 #include "model/ModelManager.h"
00045 #include "model/Model.h"
00046 #include "model/ModelInfo.h"
00047 #include "model/ModelSimulation.h"
00048 #include "model/ModelDataManager.h"
```

```
00049 #include "model/ModelComponentManager.h"
00050 #include "model/ModelComponent.h"
00051 #include "model/ModelDataDefinition.h"
00052
00060 class SimulatorFacade {
00061 public:
00062     explicit SimulatorFacade(Simulator* simulator) : _simulator(simulator) {}
00063     virtual ~SimulatorFacade() = default;
00064
00065     // -------------------------------------------------------------------------
00066     // Simulator -- direct methods
00067     // -------------------------------------------------------------------------
00068
00069     std::string getVersion() const {
00070         return _simulator != nullptr ? _simulator->getVersion() : "";
00071     }
00072
00073     unsigned int getVersionNumber() const {
00074         return _simulator != nullptr ? _simulator->getVersionNumber() : 0u;
00075     }
00076
00077     std::string getName() const {
00078         return _simulator != nullptr ? _simulator->getName() : "";
00079     }
00080
00081     // -------------------------------------------------------------------------
00082     // LicenceManager
00083     // -------------------------------------------------------------------------
00084
00085     const std::string licenceShow() const {
00086         LicenceManager* m = licenceManager_();
00087         return m != nullptr ? m->showLicence() : "";
00088     }
00089
00090     const std::string licenceShowLimits() const {
00091         LicenceManager* m = licenceManager_();
00092         return m != nullptr ? m->showLimits() : "";
00093     }
00094
00095     const std::string licenceShowActivationCode() const {
00096         LicenceManager* m = licenceManager_();
00097         return m != nullptr ? m->showActivationCode() : "";
00098     }
00099
00100     bool licenceLookforActivationCode() {
00101         LicenceManager* m = licenceManager_();
00102         return m != nullptr ? m->lookforActivationCode() : false;
00103     }
00104
00105     bool licenceInsertActivationCode() {
00106         LicenceManager* m = licenceManager_();
00107         return m != nullptr ? m->insertActivationCode() : false;
00108     }
00109
00110     void licenceRemoveActivationCode() {
00111         LicenceManager* m = licenceManager_();
00112         if (m != nullptr) m->removeActivationCode();
00113     }
00114
00115     unsigned int licenceGetModelComponentsLimit() {
00116         LicenceManager* m = licenceManager_();
00117         return m != nullptr ? m->getModelComponentsLimit() : 0u;
00118     }
00119
00120     unsigned int licenceGetModelDatasLimit() {
00121         LicenceManager* m = licenceManager_();
00122         return m != nullptr ? m->getModelDatasLimit() : 0u;
00123     }
00124
00125     unsigned int licenceGetEntityLimit() {
00126         LicenceManager* m = licenceManager_();
00127         return m != nullptr ? m->getEntityLimit() : 0u;
00128     }
00129
00130     unsigned int licenceGetHostsLimit() {
00131         LicenceManager* m = licenceManager_();
00132         return m != nullptr ? m->getHostsLimit() : 0u;
00133     }
00134
00135     unsigned int licenceGetThreadsLimit() {
00136         LicenceManager* m = licenceManager_();
00137         return m != nullptr ? m->getThreadsLimit() : 0u;
00138     }
00139
00140     // -------------------------------------------------------------------------
00141     // PluginManager
00142     // -------------------------------------------------------------------------
```

```
00143
00144     std::string showPlugins() {
00145         PluginManager* m = pluginManager_();
00146         return m != nullptr ? m->show() : "";
00147     }
00148
00149     List<Plugin*>* completePluginsFieldsAndTemplates() {
00150         PluginManager* m = pluginManager_();
00151         return m != nullptr ? m->completePluginsFieldsAndTemplates() : nullptr;
00152     }
00153
00154     bool pluginCheck(const std::string& dynamicLibraryFilename) {
00155         PluginManager* m = pluginManager_();
00156         return m != nullptr ? m->check(dynamicLibraryFilename) : false;
00157     }
00158
00159     SystemDependencyCheckResult pluginCheckSystemDependencies(const std::string&
      dynamicLibraryFilename) {
00160         PluginManager* m = pluginManager_();
00161         return m != nullptr ? m->checkSystemDependencies(dynamicLibraryFilename) :
      SystemDependencyCheckResult{};
00162     }
00163
00164     List<std::string>* discoverPluginFilenames() const {
00165         PluginManager* m = pluginManager_();
00166         return m != nullptr ? m->discoverPluginFilenames() : nullptr;
00167     }
00168
00169     List<PluginLoadIssue>* getPluginLoadIssues() const {
00170         PluginManager* m = pluginManager_();
00171         return m != nullptr ? m->getPluginLoadIssues() : nullptr;
00172     }
00173
00174     void clearPluginLoadIssues() {
00175         PluginManager* m = pluginManager_();
00176         if (m != nullptr) m->clearPluginLoadIssues();
00177     }
00178
00179     void clearPluginLoadIssue(const std::string& dynamicLibraryFilename) {
00180         PluginManager* m = pluginManager_();
00181         if (m != nullptr) m->clearPluginLoadIssue(dynamicLibraryFilename);
00182     }
00183
00184     Plugin* insertPlugin(const std::string& dynamicLibraryFilename) {
00185         PluginManager* m = pluginManager_();
00186         return m != nullptr ? m->insert(dynamicLibraryFilename) : nullptr;
00187     }
00188
00189     Plugin* insertPlugin(const std::string& dynamicLibraryFilename, const PluginInsertionOptions&
      options) {
00190         PluginManager* m = pluginManager_();
00191         return m != nullptr ? m->insert(dynamicLibraryFilename, options) : nullptr;
00192     }
00193
00194     bool removePlugin(const std::string& dynamicLibraryFilename) {
00195         PluginManager* m = pluginManager_();
00196         return m != nullptr ? m->remove(dynamicLibraryFilename) : false;
00197     }
00198
00199     bool removePlugin(Plugin* plugin) {
00200         PluginManager* m = pluginManager_();
00201         return m != nullptr ? m->remove(plugin) : false;
00202     }
00203
00204     Plugin* findPlugin(const std::string& pluginTypeName) {
00205         PluginManager* m = pluginManager_();
00206         return m != nullptr ? m->find(pluginTypeName) : nullptr;
00207     }
00208
00209     std::vector<std::string> getDataDefinitionPluginTypenames() const {
00210         PluginManager* m = pluginManager_();
00211         return m != nullptr ? m->getDataDefinitionPluginTypenames() : std::vector<std::string>{};
00212     }
00213
00214     std::string pluginSourceIncludePathFor(const std::string& pluginTypeName) {
00215         PluginManager* m = pluginManager_();
00216         return m != nullptr ? m->sourceIncludePathFor(pluginTypeName) : "";
00217     }
00218
00219     List<Plugin*>* autoInsertPlugins(const std::string& pluginsListFilename, const bool
      lookForPluginsIfFilenameNotFound = true) {
00220         PluginManager* m = pluginManager_();
00221         return m != nullptr ? m->autoInsertPlugins(pluginsListFilename,
      lookForPluginsIfFilenameNotFound) : nullptr;
00222     }
00223
00224     List<Plugin*>* autoInsertPlugins(const std::string& pluginsListFilename, const bool
```

```
    lookForPluginsIfFilenameNotFound, const PluginInsertionOptions& options) {
00225         PluginManager* m = pluginManager_();
00226         return m != nullptr ? m->autoInsertPlugins(pluginsListFilename,
    lookForPluginsIfFilenameNotFound, options) : nullptr;
00227     }
00228
00229     List<Plugin*>* autoInsertPlugins() {
00230         PluginManager* m = pluginManager_();
00231         return m != nullptr ? m->autoInsertPlugins() : nullptr;
00232     }
00233
00234     List<Plugin*>* autoInsertPlugins(const PluginInsertionOptions& options) {
00235         PluginManager* m = pluginManager_();
00236         return m != nullptr ? m->autoInsertPlugins(options) : nullptr;
00237     }
00238
00239     Plugin* firstPlugin() {
00240         PluginManager* m = pluginManager_();
00241         return m != nullptr ? m->front() : nullptr;
00242     }
00243
00244     Plugin* nextPlugin() {
00245         PluginManager* m = pluginManager_();
00246         return m != nullptr ? m->next() : nullptr;
00247     }
00248
00249     Plugin* lastPlugin() {
00250         PluginManager* m = pluginManager_();
00251         return m != nullptr ? m->last() : nullptr;
00252     }
00253
00254     unsigned int pluginCount() {
00255         PluginManager* m = pluginManager_();
00256         return m != nullptr ? m->size() : 0u;
00257     }
00258
00259     Plugin* getPluginAtRank(unsigned int rank) {
00260         PluginManager* m = pluginManager_();
00261         return m != nullptr ? m->getAtRank(rank) : nullptr;
00262     }
00263
00264     ModelDataDefinition* newInstance(const std::string& pluginTypename, Model* model, std::string name
    = "") {
00265         PluginManager* m = pluginManager_();
00266         return m != nullptr ? m->newInstance(pluginTypename, model, name) : nullptr;
00267     }
00268
00269     template <typename T>
00270     T* newInstance(Model* model, std::string name = "") {
00271         PluginManager* m = pluginManager_();
00272         return m != nullptr ? m->newInstance<T>(model, name) : nullptr;
00273     }
00274
00275     // ------------------------------------------------------------------------
00276     // ModelManager
00277     // ------------------------------------------------------------------------
00278
00279     Model* newModel() {
00280         ModelManager* m = modelManager_();
00281         return m != nullptr ? m->newModel() : nullptr;
00282     }
00283
00284     void insertModel(Model* model) {
00285         ModelManager* m = modelManager_();
00286         if (m != nullptr) m->insert(model);
00287     }
00288
00289     void removeModel(Model* model) {
00290         ModelManager* m = modelManager_();
00291         if (m != nullptr) m->remove(model);
00292     }
00293
00294     bool setCurrentModel(Model* model) {
00295         ModelManager* m = modelManager_();
00296         return m != nullptr ? m->setCurrent(model) : false;
00297     }
00298
00299     bool saveModel(std::string filename) {
00300         ModelManager* m = modelManager_();
00301         return m != nullptr ? m->saveModel(filename) : false;
00302     }
00303
00304     Model* loadModel(std::string filename) {
00305         ModelManager* m = modelManager_();
00306         return m != nullptr ? m->loadModel(filename) : nullptr;
00307     }
00308
```

```
00309     Model* createModelFromLanguage(std::string modelSpecification) {
00310         ModelManager* m = modelManager_();
00311         return m != nullptr ? m->createFromLanguage(modelSpecification) : nullptr;
00312     }
00313
00314     unsigned int modelCount() {
00315         ModelManager* m = modelManager_();
00316         return m != nullptr ? m->size() : 0u;
00317     }
00318
00319     std::vector<Model*> models() const {
00320         ModelManager* m = modelManager_();
00321         return m != nullptr ? m->models() : std::vector<Model*>{};
00322     }
00323
00324     bool hasModel(Model* model) const {
00325         ModelManager* m = modelManager_();
00326         return m != nullptr ? m->hasModel(model) : false;
00327     }
00328
00329     int indexOfModel(Model* model) const {
00330         ModelManager* m = modelManager_();
00331         return m != nullptr ? m->indexOf(model) : -1;
00332     }
00333
00334     Model* modelAt(unsigned int index) const {
00335         ModelManager* m = modelManager_();
00336         return m != nullptr ? m->modelAt(index) : nullptr;
00337     }
00338
00339     int currentModelIndex() const {
00340         ModelManager* m = modelManager_();
00341         return m != nullptr ? m->currentIndex() : -1;
00342     }
00343
00344     Model* firstModel() {
00345         ModelManager* m = modelManager_();
00346         return m != nullptr ? m->front() : nullptr;
00347     }
00348
00349     Model* lastModel() {
00350         ModelManager* m = modelManager_();
00351         return m != nullptr ? m->last() : nullptr;
00352     }
00353
00354     Model* currentModel() {
00355         return currentModel_();
00356     }
00357
00358     Model* nextModel() {
00359         ModelManager* m = modelManager_();
00360         return m != nullptr ? m->next() : nullptr;
00361     }
00362
00363     Model* previousModel() {
00364         ModelManager* m = modelManager_();
00365         return m != nullptr ? m->previous() : nullptr;
00366     }
00367
00368     bool canGoNextModel() const {
00369         ModelManager* m = modelManager_();
00370         return m != nullptr ? m->canGoNext() : false;
00371     }
00372
00373     bool canGoPreviousModel() const {
00374         ModelManager* m = modelManager_();
00375         return m != nullptr ? m->canGoPrevious() : false;
00376     }
00377
00378     // ------------------------------------------------------------------------
00379     // TraceManager
00380     // ------------------------------------------------------------------------
00381
00382     static std::string traceConvertEnumToStr(TraceManager::Level level) {
00383         return TraceManager::convertEnumToStr(level);
00384     }
00385
00386     void addTraceHandler(traceListener traceListener) {
00387         TraceManager* m = traceManager_();
00388         if (m != nullptr) m->addTraceHandler(traceListener);
00389     }
00390
00391     void addTraceReportHandler(traceListener traceReportListener) {
00392         TraceManager* m = traceManager_();
00393         if (m != nullptr) m->addTraceReportHandler(traceReportListener);
00394     }
00395
```

```
00396     void addTraceSimulationHandler(traceSimulationListener traceSimulationListener) {
00397         TraceManager* m = traceManager_();
00398         if (m != nullptr) m->addTraceSimulationHandler(traceSimulationListener);
00399     }
00400
00401     void addTraceErrorHandler(traceErrorListener traceErrorListener) {
00402         TraceManager* m = traceManager_();
00403         if (m != nullptr) m->addTraceErrorHandler(traceErrorListener);
00404     }
00405
00406     template<typename Class>
00407     void addTraceHandler(Class* object, void (Class::*function)(TraceEvent)) {
00408         TraceManager* m = traceManager_();
00409         if (m != nullptr) m->addTraceHandler(object, function);
00410     }
00411
00412     template<typename Class>
00413     void addTraceErrorHandler(Class* object, void (Class::*function)(TraceErrorEvent)) {
00414         TraceManager* m = traceManager_();
00415         if (m != nullptr) m->addTraceErrorHandler(object, function);
00416     }
00417
00418     template<typename Class>
00419     void addTraceReportHandler(Class* object, void (Class::*function)(TraceEvent)) {
00420         TraceManager* m = traceManager_();
00421         if (m != nullptr) m->addTraceReportHandler(object, function);
00422     }
00423
00424     template<typename Class>
00425     void addTraceSimulationHandler(Class* object, void (Class::*function)(TraceSimulationEvent)) {
00426         TraceManager* m = traceManager_();
00427         if (m != nullptr) m->addTraceSimulationHandler(object, function);
00428     }
00429
00430     void addTraceSimulationExceptionRuleModelData(void* thisobject) {
00431         TraceManager* m = traceManager_();
00432         if (m != nullptr) m->addTraceSimulationExceptionRuleModelData(thisobject);
00433     }
00434
00435     void beginTraceShutdown() {
00436         TraceManager* m = traceManager_();
00437         if (m != nullptr) m->beginShutdown();
00438     }
00439
00440     void trace(const std::string& text, TraceManager::Level level = TraceManager::Level::L8_detailed)
      {
00441         TraceManager* m = traceManager_();
00442         if (m != nullptr) m->trace(text, level);
00443     }
00444
00445     void traceError(const std::string& text, const std::exception& e) {
00446         TraceManager* m = traceManager_();
00447         if (m != nullptr) m->traceError(text, e);
00448     }
00449
00450     void traceError(const std::string& text, TraceManager::Level level =
      TraceManager::Level::L1_errorFatal) {
00451         TraceManager* m = traceManager_();
00452         if (m != nullptr) m->traceError(text, level);
00453     }
00454
00455     void traceReport(const std::string& text, TraceManager::Level level =
      TraceManager::Level::L2_results) {
00456         TraceManager* m = traceManager_();
00457         if (m != nullptr) m->traceReport(text, level);
00458     }
00459
00460     void traceSimulation(void* thisobject, double time, Entity* entity, ModelComponent* component,
      const std::string& text, TraceManager::Level level = TraceManager::Level::L8_detailed, bool showAnyway
      = false) {
00461         TraceManager* m = traceManager_();
00462         if (m != nullptr) m->traceSimulation(thisobject, time, entity, component, text, level,
      showAnyway);
00463     }
00464
00465     void traceSimulation(void* thisobject, const std::string& text, TraceManager::Level level =
      TraceManager::Level::L8_detailed, bool showAnyway = false) {
00466         TraceManager* m = traceManager_();
00467         if (m != nullptr) m->traceSimulation(thisobject, text, level, showAnyway);
00468     }
00469
00470     List<std::string>* traceGetErrorMessages() const {
00471         TraceManager* m = traceManager_();
00472         return m != nullptr ? m->errorMessages() : nullptr;
00473     }
00474
00475     void setTraceLevel(TraceManager::Level traceLevel) {
```

```
00476         TraceManager* m = traceManager_();
00477         if (m != nullptr) m->setTraceLevel(traceLevel);
00478     }
00479
00480     TraceManager::Level getTraceLevel() const {
00481         TraceManager* m = traceManager_();
00482         return m != nullptr ? m->getTraceLevel() : TraceManager::Level::L0_noTraces;
00483     }
00484
00485     void setTraceSimulationRuleAllAllowed(bool traceSimulationRuleAllAllowed) {
00486         TraceManager* m = traceManager_();
00487         if (m != nullptr) m->setTraceSimulationRuleAllAllowed(traceSimulationRuleAllAllowed);
00488     }
00489
00490     bool isTraceSimulationRuleAllAllowed() const {
00491         TraceManager* m = traceManager_();
00492         return m != nullptr ? m->isTraceSimulationRuleAllAllowed() : false;
00493     }
00494
00495     // ----------------------------------------------------------------------
00496     // ParserManager
00497     // ----------------------------------------------------------------------
00498
00499     ParserManager::GenerateNewParserResult parserGenerateNewParser(ParserChangesInformation* changes)
      {
00500         ParserManager* m = parserManager_();
00501         return m != nullptr ? m->generateNewParser(changes) :
      ParserManager::GenerateNewParserResult{};
00502     }
00503
00504     bool parserConnectNewParser(ParserManager::NewParser newParser) {
00505         ParserManager* m = parserManager_();
00506         return m != nullptr ? m->connectNewParser(newParser) : false;
00507     }
00508
00509     // ----------------------------------------------------------------------
00510     // ExperimentManager
00511     // ----------------------------------------------------------------------
00512
00513     SimulationExperiment* newSimulationExperiment() {
00514         ExperimentManager* m = experimentManager_();
00515         return m != nullptr ? m->newSimulationExperiment() : nullptr;
00516     }
00517
00518     void insertSimulationExperiment(SimulationExperiment* experiment) {
00519         ExperimentManager* m = experimentManager_();
00520         if (m != nullptr) m->insert(experiment);
00521     }
00522
00523     void removeSimulationExperiment(SimulationExperiment* experiment) {
00524         ExperimentManager* m = experimentManager_();
00525         if (m != nullptr) m->remove(experiment);
00526     }
00527
00528     void setCurrentSimulationExperiment(SimulationExperiment* experiment) {
00529         ExperimentManager* m = experimentManager_();
00530         if (m != nullptr) m->setCurrent(experiment);
00531     }
00532
00533     bool saveSimulationExperiment(std::string filename) {
00534         ExperimentManager* m = experimentManager_();
00535         return m != nullptr ? m->saveSimulationExperiment(filename) : false;
00536     }
00537
00538     bool loadSimulationExperiment(std::string filename) {
00539         ExperimentManager* m = experimentManager_();
00540         return m != nullptr ? m->loadSimulationExperiment(filename) : false;
00541     }
00542
00543     unsigned int simulationExperimentCount() {
00544         ExperimentManager* m = experimentManager_();
00545         return m != nullptr ? m->size() : 0u;
00546     }
00547
00548     SimulationExperiment* firstSimulationExperiment() {
00549         ExperimentManager* m = experimentManager_();
00550         return m != nullptr ? m->front() : nullptr;
00551     }
00552
00553     SimulationExperiment* currentSimulationExperiment() {
00554         ExperimentManager* m = experimentManager_();
00555         return m != nullptr ? m->current() : nullptr;
00556     }
00557
00558     SimulationExperiment* nextSimulationExperiment() {
00559         ExperimentManager* m = experimentManager_();
00560         return m != nullptr ? m->next() : nullptr;
```

```
00561     }
00562
00563     List<SimulationExperiment*>* getSimulationExperiments() const {
00564         ExperimentManager* m = experimentManager_();
00565         return m != nullptr ? m->getExperiments() : nullptr;
00566     }
00567
00568     // -------------------------------------------------------------------------
00569     // Model -- operations on current model
00570     // -------------------------------------------------------------------------
00571
00572     bool modelSave(std::string filename) {
00573         Model* model = currentModel_();
00574         return model != nullptr ? model->save(filename) : false;
00575     }
00576
00577     bool modelLoad(std::string filename) {
00578         Model* model = currentModel_();
00579         return model != nullptr ? model->load(filename) : false;
00580     }
00581
00582     bool modelCheck() {
00583         Model* model = currentModel_();
00584         return model != nullptr ? model->check() : false;
00585     }
00586
00587     void modelClear() {
00588         Model* model = currentModel_();
00589         if (model != nullptr) model->clear();
00590     }
00591
00592     std::string modelShowLanguage() {
00593         Model* model = currentModel_();
00594         return model != nullptr ? model->showLanguage() : "";
00595     }
00596
00597     void modelShow() {
00598         Model* model = currentModel_();
00599         if (model != nullptr) model->show();
00600     }
00601
00602     bool modelInsert(ModelDataDefinition* elemOrComp) {
00603         Model* model = currentModel_();
00604         return model != nullptr ? model->insert(elemOrComp) : false;
00605     }
00606
00607     void modelRemove(ModelDataDefinition* elemOrComp) {
00608         Model* model = currentModel_();
00609         if (model != nullptr) model->remove(elemOrComp);
00610     }
00611
00612     std::list<ModelDataDefinition*> modelCollectDataDefinitionsRemovedWith(const
      std::list<ModelDataDefinition*>& roots) const {
00613         Model* model = currentModel_();
00614         return model != nullptr ? model->collectDataDefinitionsRemovedWith(roots) :
      std::list<ModelDataDefinition*>{};
00615     }
00616
00617     Entity* modelCreateEntity(std::string name, bool insertIntoModel = true) {
00618         Model* model = currentModel_();
00619         return model != nullptr ? model->createEntity(name, insertIntoModel) : nullptr;
00620     }
00621
00622     void modelRemoveEntity(Entity* entity) {
00623         Model* model = currentModel_();
00624         if (model != nullptr) model->removeEntity(entity);
00625     }
00626
00627     void modelSendEntityToComponent(Entity* entity, Connection* connection, double timeDelay = 0.0) {
00628         Model* model = currentModel_();
00629         if (model != nullptr) model->sendEntityToComponent(entity, connection, timeDelay);
00630     }
00631
00632     void modelSendEntityToComponent(Entity* entity, ModelComponent* component, double timeDelay = 0.0,
      unsigned int componentinputPortNumber = 0) {
00633         Model* model = currentModel_();
00634         if (model != nullptr) model->sendEntityToComponent(entity, component, timeDelay,
      componentinputPortNumber);
00635     }
00636
00637     double modelParseExpression(const std::string expression) {
00638         Model* model = currentModel_();
00639         return model != nullptr ? model->parseExpression(expression) : 0.0;
00640     }
00641
00642     double modelParseExpression(const std::string expression, bool& success, std::string&
      errorMessage) {
```

```
00643         Model* model = currentModel_();
00644         if (model != nullptr) return model->parseExpression(expression, success, errorMessage);
00645         success = false; errorMessage = ""; return 0.0;
00646     }
00647
00648     bool modelCheckExpression(const std::string expression, const std::string expressionName,
      std::string& errorMessage) {
00649         Model* model = currentModel_();
00650         return model != nullptr ? model->checkExpression(expression, expressionName, errorMessage) :
      false;
00651     }
00652
00653     void modelCheckReferencesToDataDefinitions(std::string expression, std::map<std::string,
      std::list<std::string>*>* referencedDataDefinitions) {
00654         Model* model = currentModel_();
00655         if (model != nullptr) model->checkReferencesToDataDefinitions(expression,
      referencedDataDefinitions);
00656     }
00657
00658     Util::identification modelGetId() const {
00659         Model* model = currentModel_();
00660         return model != nullptr ? model->getId() : static_cast<Util::identification>(0);
00661     }
00662
00663     bool modelHasChanged() const {
00664         Model* model = currentModel_();
00665         return model != nullptr ? model->hasChanged() : false;
00666     }
00667
00668     void modelSetHasChanged(bool hasChanged) {
00669         Model* model = currentModel_();
00670         if (model != nullptr) model->setHasChanged(hasChanged);
00671     }
00672
00673     List<Event*>* modelGetFutureEvents() const {
00674         Model* model = currentModel_();
00675         return model != nullptr ? model->getFutureEvents() : nullptr;
00676     }
00677
00678     List<SimulationControl*>* modelGetControls() const {
00679         Model* model = currentModel_();
00680         return model != nullptr ? model->getControls() : nullptr;
00681     }
00682
00683     List<SimulationResponse*>* modelGetResponses() const {
00684         Model* model = currentModel_();
00685         return model != nullptr ? model->getResponses() : nullptr;
00686     }
00687
00688     Persistence_if* modelGetPersistence() const {
00689         Model* model = currentModel_();
00690         return model != nullptr ? model->getPersistence() : nullptr;
00691     }
00692
00693     unsigned int modelGetLevel() const {
00694         Model* model = currentModel_();
00695         return model != nullptr ? model->getLevel() : 0u;
00696     }
00697
00698     // ----------------------------------------------------------------------------
00699     // ModelInfo -- metadata of current model
00700     // ----------------------------------------------------------------------------
00701
00702     std::string infoShow() {
00703         ModelInfo* info = modelInfo_();
00704         return info != nullptr ? info->show() : "";
00705     }
00706
00707     void infoSetName(std::string name) {
00708         ModelInfo* info = modelInfo_();
00709         if (info != nullptr) info->setName(name);
00710     }
00711
00712     std::string infoGetName() const {
00713         ModelInfo* info = modelInfo_();
00714         return info != nullptr ? info->getName() : "";
00715     }
00716
00717     void infoSetAnalystName(std::string analystName) {
00718         ModelInfo* info = modelInfo_();
00719         if (info != nullptr) info->setAnalystName(analystName);
00720     }
00721
00722     std::string infoGetAnalystName() const {
00723         ModelInfo* info = modelInfo_();
00724         return info != nullptr ? info->getAnalystName() : "";
00725     }
```

```
00726
00727     void infoSetDescription(std::string description) {
00728         ModelInfo* info = modelInfo_();
00729         if (info != nullptr) info->setDescription(description);
00730     }
00731
00732     std::string infoGetDescription() const {
00733         ModelInfo* info = modelInfo_();
00734         return info != nullptr ? info->getDescription() : "";
00735     }
00736
00737     void infoSetProjectTitle(std::string projectTitle) {
00738         ModelInfo* info = modelInfo_();
00739         if (info != nullptr) info->setProjectTitle(projectTitle);
00740     }
00741
00742     std::string infoGetProjectTitle() const {
00743         ModelInfo* info = modelInfo_();
00744         return info != nullptr ? info->getProjectTitle() : "";
00745     }
00746
00747     void infoSetVersion(std::string version) {
00748         ModelInfo* info = modelInfo_();
00749         if (info != nullptr) info->setVersion(version);
00750     }
00751
00752     std::string infoGetVersion() const {
00753         ModelInfo* info = modelInfo_();
00754         return info != nullptr ? info->getVersion() : "";
00755     }
00756
00757     void infoLoadInstance(PersistenceRecord* fields) {
00758         ModelInfo* info = modelInfo_();
00759         if (info != nullptr) info->loadInstance(fields);
00760     }
00761
00762     void infoSaveInstance(PersistenceRecord* fields) {
00763         ModelInfo* info = modelInfo_();
00764         if (info != nullptr) info->saveInstance(fields);
00765     }
00766
00767     bool infoHasChanged() const {
00768         ModelInfo* info = modelInfo_();
00769         return info != nullptr ? info->hasChanged() : false;
00770     }
00771
00772     void infoSetHasChanged(bool hasChanged) {
00773         ModelInfo* info = modelInfo_();
00774         if (info != nullptr) info->setHasChanged(hasChanged);
00775     }
00776
00777     // ------------------------------------------------------------------------
00778     // ModelSimulation -- simulation control on current model
00779     // ------------------------------------------------------------------------
00780
00781     std::string simShow() {
00782         ModelSimulation* s = modelSimulation_();
00783         return s != nullptr ? s->show() : "";
00784     }
00785
00786     void simStart() {
00787         ModelSimulation* s = modelSimulation_();
00788         if (s != nullptr) s->start();
00789     }
00790
00791     void simPause() {
00792         ModelSimulation* s = modelSimulation_();
00793         if (s != nullptr) s->pause();
00794     }
00795
00796     void simStep() {
00797         ModelSimulation* s = modelSimulation_();
00798         if (s != nullptr) s->step();
00799     }
00800
00801     void simStop() {
00802         ModelSimulation* s = modelSimulation_();
00803         if (s != nullptr) s->stop();
00804     }
00805
00806     void simSetNumberOfReplications(unsigned int numberOfReplications) {
00807         ModelSimulation* s = modelSimulation_();
00808         if (s != nullptr) s->setNumberOfReplications(numberOfReplications);
00809     }
00810
00811     unsigned int simGetNumberOfReplications() const {
00812         ModelSimulation* s = modelSimulation_();
```

```
00813         return s != nullptr ? s->getNumberOfReplications() : 0u;
00814     }
00815
00816     void simSetReplicationLength(double replicationLength, Util::TimeUnit replicationLengthTimeUnit =
      Util::TimeUnit::unknown) {
00817         ModelSimulation* s = modelSimulation_();
00818         if (s != nullptr) s->setReplicationLength(replicationLength, replicationLengthTimeUnit);
00819     }
00820
00821     double simGetReplicationLength() const {
00822         ModelSimulation* s = modelSimulation_();
00823         return s != nullptr ? s->getReplicationLength() : 0.0;
00824     }
00825
00826     void simSetReplicationLengthTimeUnit(Util::TimeUnit replicationLengthTimeUnit) {
00827         ModelSimulation* s = modelSimulation_();
00828         if (s != nullptr) s->setReplicationLengthTimeUnit(replicationLengthTimeUnit);
00829     }
00830
00831     Util::TimeUnit simGetReplicationLengthTimeUnit() const {
00832         ModelSimulation* s = modelSimulation_();
00833         return s != nullptr ? s->getReplicationLengthTimeUnit() : Util::TimeUnit::unknown;
00834     }
00835
00836     void simSetReplicationReportBaseTimeUnit(Util::TimeUnit replicationReportBaseTimeUnit) {
00837         ModelSimulation* s = modelSimulation_();
00838         if (s != nullptr) s->setReplicationReportBaseTimeUnit(replicationReportBaseTimeUnit);
00839     }
00840
00841     Util::TimeUnit simGetReplicationBaseTimeUnit() const {
00842         ModelSimulation* s = modelSimulation_();
00843         return s != nullptr ? s->getReplicationBaseTimeUnit() : Util::TimeUnit::unknown;
00844     }
00845
00846     void simSetWarmUpPeriod(double warmUpPeriod, Util::TimeUnit warmUpPeriodTimeUnit =
      Util::TimeUnit::unknown) {
00847         ModelSimulation* s = modelSimulation_();
00848         if (s != nullptr) s->setWarmUpPeriod(warmUpPeriod, warmUpPeriodTimeUnit);
00849     }
00850
00851     double simGetWarmUpPeriod() const {
00852         ModelSimulation* s = modelSimulation_();
00853         return s != nullptr ? s->getWarmUpPeriod() : 0.0;
00854     }
00855
00856     void simSetWarmUpPeriodTimeUnit(Util::TimeUnit warmUpPeriodTimeUnit) {
00857         ModelSimulation* s = modelSimulation_();
00858         if (s != nullptr) s->setWarmUpPeriodTimeUnit(warmUpPeriodTimeUnit);
00859     }
00860
00861     Util::TimeUnit simGetWarmUpPeriodTimeUnit() const {
00862         ModelSimulation* s = modelSimulation_();
00863         return s != nullptr ? s->getWarmUpPeriodTimeUnit() : Util::TimeUnit::unknown;
00864     }
00865
00866     void simSetTerminatingCondition(std::string terminatingCondition) {
00867         ModelSimulation* s = modelSimulation_();
00868         if (s != nullptr) s->setTerminatingCondition(terminatingCondition);
00869     }
00870
00871     std::string simGetTerminatingCondition() const {
00872         ModelSimulation* s = modelSimulation_();
00873         return s != nullptr ? s->getTerminatingCondition() : "";
00874     }
00875
00876     void simSetPauseOnEvent(bool pauseOnEvent) {
00877         ModelSimulation* s = modelSimulation_();
00878         if (s != nullptr) s->setPauseOnEvent(pauseOnEvent);
00879     }
00880
00881     bool simIsPauseOnEvent() const {
00882         ModelSimulation* s = modelSimulation_();
00883         return s != nullptr ? s->isPauseOnEvent() : false;
00884     }
00885
00886     void simSetStepByStep(bool stepByStep) {
00887         ModelSimulation* s = modelSimulation_();
00888         if (s != nullptr) s->setStepByStep(stepByStep);
00889     }
00890
00891     bool simIsStepByStep() const {
00892         ModelSimulation* s = modelSimulation_();
00893         return s != nullptr ? s->isStepByStep() : false;
00894     }
00895
00896     void simSetInitializeStatistics(bool initializeStatistics) {
00897         ModelSimulation* s = modelSimulation_();
```

```
00898         if (s != nullptr) s->setInitializeStatistics(initializeStatistics);
00899     }
00900
00901     bool simIsInitializeStatistics() const {
00902         ModelSimulation* s = modelSimulation_();
00903         return s != nullptr ? s->isInitializeStatistics() : false;
00904     }
00905
00906     void simSetInitializeSystem(bool initializeSystem) {
00907         ModelSimulation* s = modelSimulation_();
00908         if (s != nullptr) s->setInitializeSystem(initializeSystem);
00909     }
00910
00911     bool simIsInitializeSystem() const {
00912         ModelSimulation* s = modelSimulation_();
00913         return s != nullptr ? s->isInitializeSystem() : false;
00914     }
00915
00916     void simSetPauseOnReplication(bool pauseBetweenReplications) {
00917         ModelSimulation* s = modelSimulation_();
00918         if (s != nullptr) s->setPauseOnReplication(pauseBetweenReplications);
00919     }
00920
00921     bool simIsPauseOnReplication() const {
00922         ModelSimulation* s = modelSimulation_();
00923         return s != nullptr ? s->isPauseOnReplication() : false;
00924     }
00925
00926     void simSetReporter(SimulationReporter_if* simulationReporter) {
00927         ModelSimulation* s = modelSimulation_();
00928         if (s != nullptr) s->setReporter(simulationReporter);
00929     }
00930
00931     SimulationReporter_if* simGetReporter() const {
00932         ModelSimulation* s = modelSimulation_();
00933         return s != nullptr ? s->getReporter() : nullptr;
00934     }
00935
00936     double simGetSimulatedTime() const {
00937         ModelSimulation* s = modelSimulation_();
00938         return s != nullptr ? s->getSimulatedTime() : 0.0;
00939     }
00940
00941     bool simIsRunning() const {
00942         ModelSimulation* s = modelSimulation_();
00943         return s != nullptr ? s->isRunning() : false;
00944     }
00945
00946     bool simIsPaused() const {
00947         ModelSimulation* s = modelSimulation_();
00948         return s != nullptr ? s->isPaused() : false;
00949     }
00950
00951     unsigned int simGetCurrentReplicationNumber() const {
00952         ModelSimulation* s = modelSimulation_();
00953         return s != nullptr ? s->getCurrentReplicationNumber() : 0u;
00954     }
00955
00956     void simSetShowReportsAfterReplication(bool showReportsAfterReplication) {
00957         ModelSimulation* s = modelSimulation_();
00958         if (s != nullptr) s->setShowReportsAfterReplication(showReportsAfterReplication);
00959     }
00960
00961     bool simIsShowReportsAfterReplication() const {
00962         ModelSimulation* s = modelSimulation_();
00963         return s != nullptr ? s->isShowReportsAfterReplication() : false;
00964     }
00965
00966     void simSetShowReportsAfterSimulation(bool showReportsAfterSimulation) {
00967         ModelSimulation* s = modelSimulation_();
00968         if (s != nullptr) s->setShowReportsAfterSimulation(showReportsAfterSimulation);
00969     }
00970
00971     bool simIsShowReportsAfterSimulation() const {
00972         ModelSimulation* s = modelSimulation_();
00973         return s != nullptr ? s->isShowReportsAfterSimulation() : false;
00974     }
00975
00976     List<double>* simGetBreakpointsOnTime() const {
00977         ModelSimulation* s = modelSimulation_();
00978         return s != nullptr ? s->getBreakpointsOnTime() : nullptr;
00979     }
00980
00981     List<Entity*>* simGetBreakpointsOnEntity() const {
00982         ModelSimulation* s = modelSimulation_();
00983         return s != nullptr ? s->getBreakpointsOnEntity() : nullptr;
00984     }
```

```
00985
00986     List<ModelComponent*>* simGetBreakpointsOnComponent() const {
00987         ModelSimulation* s = modelSimulation_();
00988         return s != nullptr ? s->getBreakpointsOnComponent() : nullptr;
00989     }
00990
00991     const List<ModelDataDefinition*>* simGetSimulationStatisticsAggregates() const {
00992         ModelSimulation* s = modelSimulation_();
00993         return s != nullptr ? s->getSimulationStatisticsAggregates() : nullptr;
00994     }
00995
00996     void simLoadInstance(PersistenceRecord* fields) {
00997         ModelSimulation* s = modelSimulation_();
00998         if (s != nullptr) s->loadInstance(fields);
00999     }
01000
01001     void simSaveInstance(PersistenceRecord* fields, bool saveDefaults) {
01002         ModelSimulation* s = modelSimulation_();
01003         if (s != nullptr) s->saveInstance(fields, saveDefaults);
01004     }
01005
01006     Event* simGetCurrentEvent() const {
01007         ModelSimulation* s = modelSimulation_();
01008         return s != nullptr ? s->getCurrentEvent() : nullptr;
01009     }
01010
01011     void simSetShowSimulationResposesInReport(bool showSimulationResposesInReport) {
01012         ModelSimulation* s = modelSimulation_();
01013         if (s != nullptr) s->setShowSimulationResposesInReport(showSimulationResposesInReport);
01014     }
01015
01016     bool simIsShowSimulationResposesInReport() const {
01017         ModelSimulation* s = modelSimulation_();
01018         return s != nullptr ? s->isShowSimulationResposesInReport() : false;
01019     }
01020
01021     void simSetShowSimulationControlsInReport(bool showSimulationControlsInReport) {
01022         ModelSimulation* s = modelSimulation_();
01023         if (s != nullptr) s->setShowSimulationControlsInReport(showSimulationControlsInReport);
01024     }
01025
01026     bool simIsShowSimulationControlsInReport() const {
01027         ModelSimulation* s = modelSimulation_();
01028         return s != nullptr ? s->isShowSimulationControlsInReport() : false;
01029     }
01030
01031     // ------------------------------------------------------------------------
01032     // ModelDataManager -- data definitions in current model
01033     // ------------------------------------------------------------------------
01034
01035     bool dataInsert(ModelDataDefinition* anElement) {
01036         ModelDataManager* m = modelDataManager_();
01037         return m != nullptr ? m->insert(anElement) : false;
01038     }
01039
01040     void dataRemove(ModelDataDefinition* anElement) {
01041         ModelDataManager* m = modelDataManager_();
01042         if (m != nullptr) m->remove(anElement);
01043     }
01044
01045     bool dataInsert(std::string datadefinitionTypename, ModelDataDefinition* anElement) {
01046         ModelDataManager* m = modelDataManager_();
01047         return m != nullptr ? m->insert(datadefinitionTypename, anElement) : false;
01048     }
01049
01050     void dataRemove(std::string datadefinitionTypename, ModelDataDefinition* anElement) {
01051         ModelDataManager* m = modelDataManager_();
01052         if (m != nullptr) m->remove(datadefinitionTypename, anElement);
01053     }
01054
01055     bool dataCheck(std::string datadefinitionTypename, ModelDataDefinition* anElement, std::string
      expressionName, std::string& errorMessage) {
01056         ModelDataManager* m = modelDataManager_();
01057         return m != nullptr ? m->check(datadefinitionTypename, anElement, expressionName,
      errorMessage) : false;
01058     }
01059
01060     bool dataCheck(std::string datadefinitionTypename, std::string elementName, std::string
      expressionName, bool mandatory, std::string& errorMessage) {
01061         ModelDataManager* m = modelDataManager_();
01062         return m != nullptr ? m->check(datadefinitionTypename, elementName, expressionName, mandatory,
      errorMessage) : false;
01063     }
01064
01065     void dataClear() {
01066         ModelDataManager* m = modelDataManager_();
01067         if (m != nullptr) m->clear();
```

```
01068     }
01069
01070     ModelDataDefinition* dataGetDataDefinition(std::string datadefinitionTypename,
      Util::identification id) {
01071         ModelDataManager* m = modelDataManager_();
01072         return m != nullptr ? m->getDataDefinition(datadefinitionTypename, id) : nullptr;
01073     }
01074
01075     ModelDataDefinition* dataGetDataDefinition(std::string datadefinitionTypename, std::string name) {
01076         ModelDataManager* m = modelDataManager_();
01077         return m != nullptr ? m->getDataDefinition(datadefinitionTypename, name) : nullptr;
01078     }
01079
01080     unsigned int dataGetNumberOfDataDefinitions(std::string datadefinitionTypename) {
01081         ModelDataManager* m = modelDataManager_();
01082         return m != nullptr ? m->getNumberOfDataDefinitions(datadefinitionTypename) : 0u;
01083     }
01084
01085     unsigned int dataGetNumberOfDataDefinitions() {
01086         ModelDataManager* m = modelDataManager_();
01087         return m != nullptr ? m->getNumberOfDataDefinitions() : 0u;
01088     }
01089
01090     int dataGetRankOf(std::string datadefinitionTypename, std::string name) {
01091         ModelDataManager* m = modelDataManager_();
01092         return m != nullptr ? m->getRankOf(datadefinitionTypename, name) : -1;
01093     }
01094
01095     std::list<std::string> dataGetDataDefinitionClassnames() const {
01096         ModelDataManager* m = modelDataManager_();
01097         return m != nullptr ? m->getDataDefinitionClassnames() : std::list<std::string>{};
01098     }
01099
01100     List<ModelDataDefinition*>* dataGetDataDefinitionList(std::string datadefinitionTypename) const {
01101         ModelDataManager* m = modelDataManager_();
01102         return m != nullptr ? m->getDataDefinitionList(datadefinitionTypename) : nullptr;
01103     }
01104
01105     void dataShow() {
01106         ModelDataManager* m = modelDataManager_();
01107         if (m != nullptr) m->show();
01108     }
01109
01110     bool dataHasChanged() const {
01111         ModelDataManager* m = modelDataManager_();
01112         return m != nullptr ? m->hasChanged() : false;
01113     }
01114
01115     void dataSetHasChanged(bool hasChanged) {
01116         ModelDataManager* m = modelDataManager_();
01117         if (m != nullptr) m->setHasChanged(hasChanged);
01118     }
01119
01120     // ------------------------------------------------------------------------
01121     // ModelComponentManager -- components in current model
01122     // ------------------------------------------------------------------------
01123
01124     bool componentInsert(ModelComponent* comp) {
01125         ModelComponentManager* m = modelComponentManager_();
01126         return m != nullptr ? m->insert(comp) : false;
01127     }
01128
01129     void componentRemove(ModelComponent* comp) {
01130         ModelComponentManager* m = modelComponentManager_();
01131         if (m != nullptr) m->remove(comp);
01132     }
01133
01134     ModelComponent* componentFind(std::string name) {
01135         ModelComponentManager* m = modelComponentManager_();
01136         return m != nullptr ? m->find(name) : nullptr;
01137     }
01138
01139     ModelComponent* componentFind(Util::identification id) {
01140         ModelComponentManager* m = modelComponentManager_();
01141         return m != nullptr ? m->find(id) : nullptr;
01142     }
01143
01144     void componentClear() {
01145         ModelComponentManager* m = modelComponentManager_();
01146         if (m != nullptr) m->clear();
01147     }
01148
01149     unsigned int componentGetNumberOfComponents() {
01150         ModelComponentManager* m = modelComponentManager_();
01151         return m != nullptr ? m->getNumberOfComponents() : 0u;
01152     }
01153
```

```
01154     std::list<ModelComponent*>::iterator componentBegin() {
01155         ModelComponentManager* m = modelComponentManager_();
01156         static std::list<ModelComponent*> _empty;
01157         return m != nullptr ? m->begin() : _empty.begin();
01158     }
01159
01160     std::list<ModelComponent*>::iterator componentEnd() {
01161         ModelComponentManager* m = modelComponentManager_();
01162         static std::list<ModelComponent*> _empty;
01163         return m != nullptr ? m->end() : _empty.end();
01164     }
01165
01166     ModelComponent* componentFront() {
01167         ModelComponentManager* m = modelComponentManager_();
01168         return m != nullptr ? m->front() : nullptr;
01169     }
01170
01171     ModelComponent* componentNext() {
01172         ModelComponentManager* m = modelComponentManager_();
01173         return m != nullptr ? m->next() : nullptr;
01174     }
01175
01176     bool componentHasChanged() const {
01177         ModelComponentManager* m = modelComponentManager_();
01178         return m != nullptr ? m->hasChanged() : false;
01179     }
01180
01181     void componentSetHasChanged(bool hasChanged) {
01182         ModelComponentManager* m = modelComponentManager_();
01183         if (m != nullptr) m->setHasChanged(hasChanged);
01184     }
01185
01186     std::list<SourceModelComponent*>* componentGetSourceComponents() {
01187         ModelComponentManager* m = modelComponentManager_();
01188         return m != nullptr ? m->getSourceComponents() : nullptr;
01189     }
01190
01191     std::list<ModelComponent*>* componentGetTransferInComponents() {
01192         ModelComponentManager* m = modelComponentManager_();
01193         return m != nullptr ? m->getTransferInComponents() : nullptr;
01194     }
01195
01196     std::list<ModelComponent*>* componentGetAllComponents() const {
01197         ModelComponentManager* m = modelComponentManager_();
01198         return m != nullptr ? m->getAllComponents() : nullptr;
01199     }
01200
01201     // ------------------------------------------------------------------------
01202     // OnEventManager -- event handlers on current model
01203     // ------------------------------------------------------------------------
01204
01205     void eventAddOnModelCheckSucessHandler(modelEventHandler EventHandler) {
01206         OnEventManager* m = onEventManager_();
01207         if (m != nullptr) m->addOnModelCheckSucessHandler(EventHandler);
01208     }
01209
01210     void eventAddOnModelLoadHandler(modelEventHandler EventHandler) {
01211         OnEventManager* m = onEventManager_();
01212         if (m != nullptr) m->addOnModelLoadHandler(EventHandler);
01213     }
01214
01215     void eventAddOnModelSaveHandler(modelEventHandler EventHandler) {
01216         OnEventManager* m = onEventManager_();
01217         if (m != nullptr) m->addOnModelSaveHandler(EventHandler);
01218     }
01219
01220     void eventAddOnReplicationStartHandler(simulationEventHandler EventHandler) {
01221         OnEventManager* m = onEventManager_();
01222         if (m != nullptr) m->addOnReplicationStartHandler(EventHandler);
01223     }
01224
01225     void eventAddOnReplicationStepHandler(simulationEventHandler EventHandler) {
01226         OnEventManager* m = onEventManager_();
01227         if (m != nullptr) m->addOnReplicationStepHandler(EventHandler);
01228     }
01229
01230     void eventAddOnReplicationEndHandler(simulationEventHandler EventHandler) {
01231         OnEventManager* m = onEventManager_();
01232         if (m != nullptr) m->addOnReplicationEndHandler(EventHandler);
01233     }
01234
01235     void eventAddOnProcessEventHandler(simulationEventHandler EventHandler) {
01236         OnEventManager* m = onEventManager_();
01237         if (m != nullptr) m->addOnProcessEventHandler(EventHandler);
01238     }
01239
01240     void eventAddOnAfterProcessEventHandler(simulationEventHandler EventHandler) {
```

```
01241         OnEventManager* m = onEventManager_();
01242         if (m != nullptr) m->addOnAfterProcessEventHandler(EventHandler);
01243     }
01244
01245     void eventAddOnEntityCreateHandler(simulationEventHandler EventHandler) {
01246         OnEventManager* m = onEventManager_();
01247         if (m != nullptr) m->addOnEntityCreateHandler(EventHandler);
01248     }
01249
01250     void eventAddOnEntityMoveHandler(simulationEventHandler EventHandler) {
01251         OnEventManager* m = onEventManager_();
01252         if (m != nullptr) m->addOnEntityMoveHandler(EventHandler);
01253     }
01254
01255     void eventAddOnEntityRemoveHandler(simulationEventHandler EventHandler) {
01256         OnEventManager* m = onEventManager_();
01257         if (m != nullptr) m->addOnEntityRemoveHandler(EventHandler);
01258     }
01259
01260     void eventAddOnSimulationStartHandler(simulationEventHandler EventHandler) {
01261         OnEventManager* m = onEventManager_();
01262         if (m != nullptr) m->addOnSimulationStartHandler(EventHandler);
01263     }
01264
01265     void eventAddOnSimulationPausedHandler(simulationEventHandler EventHandler) {
01266         OnEventManager* m = onEventManager_();
01267         if (m != nullptr) m->addOnSimulationPausedHandler(EventHandler);
01268     }
01269
01270     void eventAddOnSimulationResumeHandler(simulationEventHandler EventHandler) {
01271         OnEventManager* m = onEventManager_();
01272         if (m != nullptr) m->addOnSimulationResumeHandler(EventHandler);
01273     }
01274
01275     void eventAddOnSimulationEndHandler(simulationEventHandler EventHandler) {
01276         OnEventManager* m = onEventManager_();
01277         if (m != nullptr) m->addOnSimulationEndHandler(EventHandler);
01278     }
01279
01280     void eventAddOnBreakpointHandler(simulationEventHandler EventHandler) {
01281         OnEventManager* m = onEventManager_();
01282         if (m != nullptr) m->addOnBreakpointHandler(EventHandler);
01283     }
01284
01285     template<typename Class>
01286     void eventAddOnModelCheckSuccessHandler(Class* object, void (Class::*function)(ModelEvent*)) {
01287         OnEventManager* m = onEventManager_();
01288         if (m != nullptr) m->addOnModelCheckSuccessHandler(object, function);
01289     }
01290
01291     template<typename Class>
01292     void eventAddOnModelLoadHandler(Class* object, void (Class::*function)(ModelEvent*)) {
01293         OnEventManager* m = onEventManager_();
01294         if (m != nullptr) m->addOnModelLoadHandler(object, function);
01295     }
01296
01297     template<typename Class>
01298     void eventAddOnModelSaveHandler(Class* object, void (Class::*function)(ModelEvent*)) {
01299         OnEventManager* m = onEventManager_();
01300         if (m != nullptr) m->addOnModelSaveHandler(object, function);
01301     }
01302
01303     template<typename Class>
01304     void eventAddOnReplicationStartHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01305         OnEventManager* m = onEventManager_();
01306         if (m != nullptr) m->addOnReplicationStartHandler(object, function);
01307     }
01308
01309     template<typename Class>
01310     void eventAddOnReplicationStepHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01311         OnEventManager* m = onEventManager_();
01312         if (m != nullptr) m->addOnReplicationStepHandler(object, function);
01313     }
01314
01315     template<typename Class>
01316     void eventAddOnReplicationEndHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01317         OnEventManager* m = onEventManager_();
01318         if (m != nullptr) m->addOnReplicationEndHandler(object, function);
01319     }
01320
01321     template<typename Class>
01322     void eventAddOnProcessEventHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01323         OnEventManager* m = onEventManager_();
01324         if (m != nullptr) m->addOnProcessEventHandler(object, function);
01325     }
01326
01327     template<typename Class>
```

```
01328     void eventAddOnAfterProcessEventHandler(Class* object, void (Class::*function)(SimulationEvent*))
      {
01329         OnEventManager* m = onEventManager_();
01330         if (m != nullptr) m->addOnAfterProcessEventHandler(object, function);
01331     }
01332
01333     template<typename Class>
01334     void eventAddOnEntityCreateHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01335         OnEventManager* m = onEventManager_();
01336         if (m != nullptr) m->addOnEntityCreateHandler(object, function);
01337     }
01338
01339     template<typename Class>
01340     void eventAddOnEntityMoveHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01341         OnEventManager* m = onEventManager_();
01342         if (m != nullptr) m->addOnEntityMoveHandler(object, function);
01343     }
01344
01345     template<typename Class>
01346     void eventAddOnEntityRemoveHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01347         OnEventManager* m = onEventManager_();
01348         if (m != nullptr) m->addOnEntityRemoveHandler(object, function);
01349     }
01350
01351     template<typename Class>
01352     void eventAddOnSimulationStartHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01353         OnEventManager* m = onEventManager_();
01354         if (m != nullptr) m->addOnSimulationStartHandler(object, function);
01355     }
01356
01357     template<typename Class>
01358     void eventAddOnSimulationPausedHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01359         OnEventManager* m = onEventManager_();
01360         if (m != nullptr) m->addOnSimulationPausedHandler(object, function);
01361     }
01362
01363     template<typename Class>
01364     void eventAddOnSimulationResumeHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01365         OnEventManager* m = onEventManager_();
01366         if (m != nullptr) m->addOnSimulationResumeHandler(object, function);
01367     }
01368
01369     template<typename Class>
01370     void eventAddOnSimulationEndHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01371         OnEventManager* m = onEventManager_();
01372         if (m != nullptr) m->addOnSimulationEndHandler(object, function);
01373     }
01374
01375     template<typename Class>
01376     void eventAddOnBreakpointHandler(Class* object, void (Class::*function)(SimulationEvent*)) {
01377         OnEventManager* m = onEventManager_();
01378         if (m != nullptr) m->addOnBreakpointHandler(object, function);
01379     }
01380
01381     void eventNotifyModelCheckSuccessHandlers(ModelEvent* se) {
01382         OnEventManager* m = onEventManager_();
01383         if (m != nullptr) m->NotifyModelCheckSuccessHandlers(se);
01384     }
01385
01386     void eventNotifyModelLoadHandlers(ModelEvent* se) {
01387         OnEventManager* m = onEventManager_();
01388         if (m != nullptr) m->NotifyModelLoadHandlers(se);
01389     }
01390
01391     void eventNotifyModelSaveHandlers(ModelEvent* se) {
01392         OnEventManager* m = onEventManager_();
01393         if (m != nullptr) m->NotifyModelSaveHandlers(se);
01394     }
01395
01396     void eventNotifyReplicationStartHandlers(SimulationEvent* se) {
01397         OnEventManager* m = onEventManager_();
01398         if (m != nullptr) m->NotifyReplicationStartHandlers(se);
01399     }
01400
01401     void eventNotifyReplicationStepHandlers(SimulationEvent* se) {
01402         OnEventManager* m = onEventManager_();
01403         if (m != nullptr) m->NotifyReplicationStepHandlers(se);
01404     }
01405
01406     void eventNotifyReplicationEndHandlers(SimulationEvent* se) {
01407         OnEventManager* m = onEventManager_();
01408         if (m != nullptr) m->NotifyReplicationEndHandlers(se);
01409     }
01410
01411     void eventNotifyProcessEventHandlers(SimulationEvent* se) {
01412         OnEventManager* m = onEventManager_();
01413         if (m != nullptr) m->NotifyProcessEventHandlers(se);
```

```
01414     }
01415
01416     void eventNotifyAfterProcessEventHandlers(SimulationEvent* se) {
01417         OnEventManager* m = onEventManager_();
01418         if (m != nullptr) m->NotifyAfterProcessEventHandlers(se);
01419     }
01420
01421     void eventNotifyEntityCreateHandlers(SimulationEvent* se) {
01422         OnEventManager* m = onEventManager_();
01423         if (m != nullptr) m->NotifyEntityCreateHandlers(se);
01424     }
01425
01426     void eventNotifyEntityMoveHandlers(SimulationEvent* se) {
01427         OnEventManager* m = onEventManager_();
01428         if (m != nullptr) m->NotifyEntityMoveHandlers(se);
01429     }
01430
01431     void eventNotifyEntityRemoveHandlers(SimulationEvent* se) {
01432         OnEventManager* m = onEventManager_();
01433         if (m != nullptr) m->NotifyEntityRemoveHandlers(se);
01434     }
01435
01436     void eventNotifySimulationStartHandlers(SimulationEvent* se) {
01437         OnEventManager* m = onEventManager_();
01438         if (m != nullptr) m->NotifySimulationStartHandlers(se);
01439     }
01440
01441     void eventNotifySimulationPausedHandlers(SimulationEvent* se) {
01442         OnEventManager* m = onEventManager_();
01443         if (m != nullptr) m->NotifySimulationPausedHandlers(se);
01444     }
01445
01446     void eventNotifySimulationResumeHandlers(SimulationEvent* se) {
01447         OnEventManager* m = onEventManager_();
01448         if (m != nullptr) m->NotifySimulationResumeHandlers(se);
01449     }
01450
01451     void eventNotifySimulationEndHandlers(SimulationEvent* se) {
01452         OnEventManager* m = onEventManager_();
01453         if (m != nullptr) m->NotifySimulationEndHandlers(se);
01454     }
01455
01456     void eventNotifyBreakpointHandlers(SimulationEvent* se) {
01457         OnEventManager* m = onEventManager_();
01458         if (m != nullptr) m->NotifyBreakpointHandlers(se);
01459     }
01460
01461 private:
01462     Simulator* _simulator;
01463
01464     LicenceManager* licenceManager_() const {
01465         return _simulator != nullptr ? _simulator->getLicenceManager() : nullptr;
01466     }
01467
01468     PluginManager* pluginManager_() const {
01469         return _simulator != nullptr ? _simulator->getPluginManager() : nullptr;
01470     }
01471
01472     ModelManager* modelManager_() const {
01473         return _simulator != nullptr ? _simulator->getModelManager() : nullptr;
01474     }
01475
01476     TraceManager* traceManager_() const {
01477         return _simulator != nullptr ? _simulator->getTraceManager() : nullptr;
01478     }
01479
01480     ParserManager* parserManager_() const {
01481         return _simulator != nullptr ? _simulator->getParserManager() : nullptr;
01482     }
01483
01484     ExperimentManager* experimentManager_() const {
01485         return _simulator != nullptr ? _simulator->getExperimentManager() : nullptr;
01486     }
01487
01488     Model* currentModel_() const {
01489         ModelManager* m = modelManager_();
01490         return m != nullptr ? m->current() : nullptr;
01491     }
01492
01493     ModelInfo* modelInfo_() const {
01494         Model* model = currentModel_();
01495         return model != nullptr ? model->getInfos() : nullptr;
01496     }
01497
01498     ModelSimulation* modelSimulation_() const {
01499         Model* model = currentModel_();
01500         return model != nullptr ? model->getSimulation() : nullptr;
```

```
01501     }
01502
01503     ModelDataManager* modelDataManager_() const {
01504         Model* model = currentModel_();
01505         return model != nullptr ? model->getDataManager() : nullptr;
01506     }
01507
01508     ModelComponentManager* modelComponentManager_() const {
01509         Model* model = currentModel_();
01510         return model != nullptr ? model->getComponentManager() : nullptr;
01511     }
01512
01513     OnEventManager* onEventManager_() const {
01514         Model* model = currentModel_();
01515         return model != nullptr ? model->getOnEventManager() : nullptr;
01516     }
01517 };
01518
01519 #endif /* SIMULATORFACADE_H */
```

18.345 SinkModelComponent.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   SinkModelComponent.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 14 de Agosto de 2018, 14:29
00012  */
00013
00014 #ifndef SINKMODELCOMPONENT_H
00015 #define SINKMODELCOMPONENT_H
00016
00017 #include "model/ModelComponent.h"
00018
00019 //namespace GenesysKernel {
00020
00025 class SinkModelComponent : public ModelComponent {
00026 public:
00027     SinkModelComponent(Model* model, std::string componentTypename, std::string name = "");
00028     virtual ~SinkModelComponent() = default;
00029 public:
00030 protected:
00031     virtual bool _loadInstance(PersistenceRecord *fields) override;
00032     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00033 protected:
00034     //virtual void _initBetweenReplications();
00035     virtual bool _check(std::string& errorMessage) override;
00036 private:
00037 };
00038 //namespace\\}
00039 #endif /* SINKMODELCOMPONENT_H */
00040
```

18.346 SourceModelComponent.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   SourceModelCOmponent.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Junho de 2018, 19:50
00012  */
00013
00014 #ifndef SOURCEMODELCOMPONENT_H
00015 #define SOURCEMODELCOMPONENT_H
00016
00017 #include "model/ModelComponent.h"
00018 #include <string>
00019 #include <limits>
00020 //namespace GenesysKernel {
00021
```

```
00035 class SourceModelComponent : public ModelComponent {
00036 public:
00037     SourceModelComponent(Model* model, std::string componentTypename, std::string name = "");
00038     virtual ~SourceModelComponent() = default;
00039 public: // get & set
00044     void setFirstCreation(double _firstCreation);
00049     double getFirstCreation() const;
00054     void setEntityType(EntityType* _entityType);
00059     void setEntityTypeName(std::string entityTypeName);
00064     EntityType* getEntityType() const;
00069     void setTimeUnit(Util::TimeUnit _timeUnit);
00074     Util::TimeUnit getTimeUnit() const;
00080     void setTimeBetweenCreationsExpression(std::string _timeBetweenCreations, Util::TimeUnit
      _timeUnit=Util::TimeUnit::unknown);
00085     std::string getTimeBetweenCreationsExpression() const;
00090     void setMaxCreations(unsigned long _maxCreations);
00095     void setMaxCreations(std::string _maxCreationsExpression);
00100     std::string getMaxCreations() const;
00105     unsigned int getEntitiesCreated() const;
00110     void setEntitiesCreated(unsigned int _entitiesCreated);
00115     void setEntitiesPerCreation(unsigned int _entitiesPerCreation);
00120     unsigned int getEntitiesPerCreation() const;
00121 public:
00122     virtual std::string show() override;
00123
00124 protected:
00125     virtual bool _loadInstance(PersistenceRecord *fields) override;
00126     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00127 protected:
00128     virtual void _initBetweenReplications() override;
00129     virtual bool _check(std::string& errorMessage) override;
00130 protected:
00131     //virtual void _createInternalStatisticReporters();
00132     virtual void _createEditableDataDefinitions();
00133     virtual void _createAttachedAttributes();
00134 protected: // get & set
00135     EntityType* _entityType = nullptr;
00136
00137     const struct DEFAULT_VALUES {
00138         const double firstCreation = 0.0;
00139         const unsigned int entitiesPerCreation = 1;
00140         const std::string maxCreationsExpression = std::to_string(std::numeric_limits<unsigned
      int>::max());
00141         const std::string timeBetweenCreationsExpression = "EXPO(1.0)";
00142         const Util::TimeUnit timeBetweenCreationsTimeUnit = Util::TimeUnit::second;
00143         //
00144         const std::string entityTypename = "entitytype";
00145     } DEFAULT;
00146     double _firstCreation = DEFAULT.firstCreation;
00147     unsigned int _entitiesPerCreation = DEFAULT.entitiesPerCreation;
00148     std::string _maxCreationsExpression = DEFAULT.maxCreationsExpression;
00149     std::string _timeBetweenCreationsExpression = DEFAULT.timeBetweenCreationsExpression;
00150     Util::TimeUnit _timeBetweenCreationsTimeUnit = DEFAULT.timeBetweenCreationsTimeUnit;
00151     unsigned int _entitiesCreatedSoFar = 0;
00152 };
00153 //namespace\\}
00154 #endif /* SOURCEMODELCOMPONENT_H */
```

18.347 SystemDependencyResolver.h

```
00001 /*
00002  * File:   SystemDependencyResolver.h
00003  * Author: rafael.luiz.cancian
00004  *
00005  * Created on 16 de Abril de 2026
00006  */
00007
00008 #ifndef SYSTEMDEPENDENCYRESOLVER_H
00009 #define SYSTEMDEPENDENCYRESOLVER_H
00010
00011 #include "PluginInformation.h"
00012
00013 #include <list>
00014 #include <string>
00015
00019 struct SystemCommandResult {
00021     bool started = false;
00023     int exitCode = -1;
00025     std::string output = "";
00027     std::string errorMessage = "";
00028
00030     bool succeeded() const;
00031 };
00032
00036 class SystemCommandExecutor_if {
```

```
00037 public:
00039     virtual ~SystemCommandExecutor_if() = default;
00041     virtual SystemCommandResult run(const std::string& command) = 0;
00042 };
00043
00047 class SystemShellCommandExecutor : public SystemCommandExecutor_if {
00048 public:
00050     SystemCommandResult run(const std::string& command) override;
00051 };
00052
00056 class SystemDependencyCheckEntry {
00057 public:
00059     enum class Status {
00060         Satisfied,
00061         Missing,
00062         NotVerifiable,
00063         IgnoredDifferentOS
00064     };
00065
00067     SystemDependencyCheckEntry(SystemDependency dependency, Status status, SystemCommandResult
      checkResult = {}, std::string message = "");
00068
00070     const SystemDependency& dependency() const;
00072     Status status() const;
00074     const SystemCommandResult& checkResult() const;
00076     const std::string& message() const;
00078     bool blocksInsertion() const;
00080     bool canAttemptInstall() const;
00082     std::string diagnosticText() const;
00084     static std::string statusToString(Status status);
00085
00086 private:
00087     SystemDependency _dependency;
00088     Status _status = Status::NotVerifiable;
00089     SystemCommandResult _checkResult;
00090     std::string _message = "";
00091 };
00092
00096 class SystemDependencyCheckResult {
00097 public:
00099     void add(SystemDependencyCheckEntry entry);
00101     const std::list<SystemDependencyCheckEntry>& entries() const;
00103     bool canInsertPlugin() const;
00105     bool hasBlockingEntries() const;
00107     bool canAttemptInstallForAllMissing() const;
00109     std::string summary() const;
00111     std::string diagnosticText(bool includeSatisfied = false) const;
00112
00113 private:
00114     std::list<SystemDependencyCheckEntry> _entries;
00115 };
00116
00120 class SystemDependencyInstallEntry {
00121 public:
00123     SystemDependencyInstallEntry(SystemDependency dependency, SystemCommandResult installResult);
00125     const SystemDependency& dependency() const;
00127     const SystemCommandResult& installResult() const;
00129     std::string diagnosticText() const;
00130
00131 private:
00132     SystemDependency _dependency;
00133     SystemCommandResult _installResult;
00134 };
00135
00139 class SystemDependencyInstallResult {
00140 public:
00142     void add(SystemDependencyInstallEntry entry);
00144     const std::list<SystemDependencyInstallEntry>& entries() const;
00146     bool succeeded() const;
00148     std::string summary() const;
00150     std::string diagnosticText() const;
00151
00152 private:
00153     std::list<SystemDependencyInstallEntry> _entries;
00154 };
00155
00163 class SystemDependencyResolver {
00164 public:
00166     static SystemDependency::OS currentOS();
00168     static bool appliesToCurrentOS(const SystemDependency& dependency);
00170     static SystemDependencyCheckResult evaluate(const std::list<SystemDependency>* dependencies,
00171                                                 SystemCommandExecutor_if& executor);
00173     static SystemDependencyInstallResult installMissingDependencies(const SystemDependencyCheckResult&
      checkResult,
00174                                                                     SystemCommandExecutor_if&
      executor);
00175 };
```

```
00176
00177 #endif /* SYSTEMDEPENDENCYRESOLVER_H */
```

18.348 TraceManager.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   TraceManager.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 7 de Novembro de 2018, 11:59
00012  */
00013
00014 #ifndef TRACEMANAGER_H
00015 #define TRACEMANAGER_H
00016
00017 #include "../util/List.h"
00018 #include "../util/Util.h"
00019 #include <functional>
00020 //namespace GenesysKernel {
00021 class Model;
00022 class Entity;
00023 class ModelComponent;
00024 class Simulator;
00025
00026 class TraceEvent;
00027 class TraceErrorEvent;
00028 class TraceSimulationEvent;
00029 class TraceSimulationProcess;
00030
00031 // for handlers that are simple functions
00032 typedef void (*traceListener)(TraceEvent);
00033 typedef void (*traceErrorListener)(TraceErrorEvent);
00034 typedef void (*traceSimulationListener)(TraceSimulationEvent);
00035 typedef void (*traceSimulationProcessListener)(TraceSimulationProcess);
00036 // for handlers that are class members (methods)
00037 typedef std::function<void(TraceEvent)> traceListenerMethod;
00038 typedef std::function<void(TraceErrorEvent)> traceErrorListenerMethod;
00039 typedef std::function<void(TraceSimulationEvent)> traceSimulationListenerMethod;
00040 typedef std::function<void(TraceSimulationProcess)> traceSimulationProcessListenerMethod;
00041
00046 class TraceManager {
00047 public:
00048
00049     enum class Level : int {
00050         L0_noTraces = 0,
00051         L1_errorFatal = 1,
00052         L2_results = 2,
00053         L3_errorRecover = 3, // or progress
00054         L4_warning = 4,
00055         L5_event = 5,
00056         L6_arrival = 6,
00057         L7_internal = 7,
00058         L8_detailed = 8,
00059         L9_mostDetailed = 9,
00060         num_elements = 10
00061     };
00062 public:
00063     static std::string convertEnumToStr(Level level);
00064     TraceManager(Simulator* simulator); //(Model* model);
00065     virtual ~TraceManager();
00066 public: // add trace handlers
00067     // for handlers that are simple functions
00072     void addTraceHandler(traceListener traceListener);
00077     void addTraceReportHandler(traceListener traceReportListener);
00082     void addTraceSimulationHandler(traceSimulationListener traceSimulationListener);
00087     void addTraceErrorHandler(traceErrorListener traceErrorListener);
00088     // for handlers that are class members (methods)
00093     template<typename Class> void addTraceHandler(Class* object, void (Class::*function)(TraceEvent));
00098     template<typename Class> void addTraceErrorHandler(Class* object, void
      (Class::*function)(TraceErrorEvent));
00103     template<typename Class> void addTraceReportHandler(Class* object, void
      (Class::*function)(TraceEvent));
00108     template<typename Class> void addTraceSimulationHandler(Class* object, void
      (Class::*function)(TraceSimulationEvent));
00109
00110 public: // add ModelDataDefinition (or subclasses) allowed (or restrcted) to traceSimulation
00111     void addTraceSimulationExceptionRuleModelData(void* thisobject);
00112 public: // traces (invoke trace handlers)
00113     void trace(TraceManager::Level level, const std::string& text);
```

```
00114     void traceSimulation(void* thisobject, TraceManager::Level level, double time, Entity* entity,
      ModelComponent* component, const std::string& text);
00115     void traceSimulation(void* thisobject, TraceManager::Level level, const std::string& text);
00116 public: // traces
00117     // Disable callbacks and clear handler lists before simulator teardown destroys GUI objects.
00118     void beginShutdown();
00119     void trace(const std::string& text, TraceManager::Level level = TraceManager::Level::L8_detailed);
00120     void traceError(const std::string& text, const std::exception& e);
00121     void traceError(const std::string& text, TraceManager::Level level =
      TraceManager::Level::L1_errorFatal);
00122     void traceReport(const std::string& text, TraceManager::Level level =
      TraceManager::Level::L2_results);
00123     void traceSimulation(void* thisobject, double time, Entity* entity, ModelComponent* component,
      const std::string& text, TraceManager::Level level = TraceManager::Level::L8_detailed, bool showAnyway
      = false);
00124     void traceSimulation(void* thisobject, const std::string& text, TraceManager::Level level =
      TraceManager::Level::L8_detailed, bool showAnyway = false);
00125
00126 public:
00131     List<std::string>* errorMessages() const;
00136     void setTraceLevel(TraceManager::Level _traceLevel);
00141     TraceManager::Level getTraceLevel() const;
00146     Simulator* getParentSimulator() const;
00151     void setTraceSimulationRuleAllAllowed(bool _traceSimulationRuleAllAllowed);
00156     bool isTraceSimulationRuleAllAllowed() const;
00157
00158 private:
00159     //void _addHandler(List<traceListener>* list, )
00160     bool _traceConditionPassed(TraceManager::Level level);
00161     bool _traceSimulationConditionPassed(TraceManager::Level level, void* thisobject, bool showAnyway
      = false);
00162 private: // trace listener
00163     // for handlers that are simple functions
00164     List<traceListener>* _traceHandlers;
00165     List<traceErrorListener>* _traceErrorHandlers;
00166     List<traceListener>* _traceReportHandlers;
00167     List<traceSimulationListener>* _traceSimulationHandlers;
00168     // for handlers that are class members (methods)
00169     List<traceListenerMethod>* _traceHandlersMethod;
00170     List<traceErrorListenerMethod>* _traceErrorHandlersMethod;
00171     List<traceListenerMethod>* _traceReportHandlersMethod;
00172     List<traceSimulationListenerMethod>* _traceSimulationHandlersMethod;
00173 private:
00174     List<void*>* _traceSimulationExceptionRule;
00175 private:
00176     Simulator* _simulator;
00177 private:
00178     TraceManager::Level _traceLevel; // = TraceManager::Level::L9_mostDetailed;
00179     bool _shuttingDown;
00180     double _lastTimeTraceSimulation;
00181     Util::identification _lastEntityTraceSimulation;
00182     Util::identification _lastModuleTraceSimulation;
00183     bool _traceSimulationRuleAllAllowed;
00184     List<std::string>* _errorMessages;
00185
00186 };
00187
00188 // implementation for template methods
00189
00190 template<typename Class> void TraceManager::addTraceHandler(Class* object, void
      (Class::*function)(TraceEvent)) {
00191     this->_traceHandlersMethod->insert(std::bind(function, object, std::placeholders::_1));
00192 }
00193
00194 template<typename Class> void TraceManager::addTraceErrorHandler(Class* object, void
      (Class::*function)(TraceErrorEvent)) {
00195     this->_traceErrorHandlersMethod->insert(std::bind(function, object, std::placeholders::_1));
00196 }
00197
00198 template<typename Class> void TraceManager::addTraceReportHandler(Class* object, void
      (Class::*function)(TraceEvent)) {
00199     this->_traceReportHandlersMethod->insert(std::bind(function, object, std::placeholders::_1));
00200 }
00201
00202 template<typename Class> void TraceManager::addTraceSimulationHandler(Class* object, void
      (Class::*function)(TraceSimulationEvent)) {
00203     this->_traceSimulationHandlersMethod->insert(std::bind(function, object, std::placeholders::_1));
00204 }
00205
00206 class TraceEvent {
00207 public:
00208
00209     TraceEvent(TraceManager::Level level, const std::string& text) {
00210         _tracelevel = level;
00211         _text = text;
00212     }
00213
```

```
00214     TraceEvent(const std::string& text, TraceManager::Level level = TraceManager::Level::L8_detailed)
      {
00215         _tracelevel = level;
00216         _text = text;
00217     }
00218 
00219     TraceManager::Level getTracelevel() const {
00220         return _tracelevel;
00221     }
00222 
00223     const std::string& getText() const {
00224         return _text;
00225     }
00226 private:
00227     TraceManager::Level _tracelevel;
00228     std::string _text;
00229 };
00230 
00231 class TraceErrorEvent : public TraceEvent {
00232 public:
00233 
00234     TraceErrorEvent(const std::string& text, const std::exception& e) : TraceEvent(text,
      TraceManager::Level::L3_errorRecover), _exceptionMessage(e.what()) {
00235     }
00236 
00237     TraceErrorEvent(const std::string& text, TraceManager::Level level) : TraceEvent(text, level),
      _exceptionMessage("") {
00238     }
00239 
00240     const std::string& getExceptionMessage() const {
00241         return _exceptionMessage;
00242     }
00243 private:
00244     std::string _exceptionMessage;
00245 };
00246 
00247 class TraceSimulationEvent : public TraceEvent {
00248 public:
00249 
00250     TraceSimulationEvent(TraceManager::Level level, double time, Entity* entity, ModelComponent*
      component, const std::string& text) : TraceEvent(text, level) {
00251         _time = time;
00252         _entity = entity;
00253         _component = component;
00254     }
00255 
00256     ModelComponent* getComponent() const {
00257         return _component;
00258     }
00259 
00260     Entity* getEntity() const {
00261         return _entity;
00262     }
00263 
00264     double getTime() const {
00265         return _time;
00266     }
00267 
00268 private:
00269     double _time;
00270     Entity* _entity;
00271     ModelComponent* _component;
00272 };
00273 
00278 class TraceSimulationProcess : public TraceEvent {
00279 public:
00280 
00281     TraceSimulationProcess(const std::string& text, TraceManager::Level level =
      TraceManager::Level::L8_detailed) : TraceEvent(text, level) {
00282     }
00283 };
00284 
00285 
00286 //namespace\}
00287 #endif /* TRACEMANAGER_H */
```

18.349 Collector_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
00007 /*
```

```
00008  * File:   Collector_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 14 de Agosto de 2018, 14:16
00012  */
00013
00014 #ifndef COLLECTOR_IF_H
00015 #define COLLECTOR_IF_H
00016
00017 #include <string>
00018 #include <functional>
00019
00022 typedef std::function<void(double, double) > CollectorAddValueHandler;
00023
00024 template<typename Class>
00025 CollectorAddValueHandler setCollectorAddValueHandler(void (Class::*function)(double, double), Class *
      object) {
00026     return std::bind(function, object, std::placeholders::_1, std::placeholders::_2);
00027 }
00028
00029 typedef std::function<void() > CollectorClearHandler;
00030
00031 template<typename Class>
00032 CollectorClearHandler setCollectorClearHandler(void (Class::*function)(), Class * object) {
00033     return std::bind(function, object);
00034 }
00035
00042 class Collector_if {
00043 public:
00044     virtual ~Collector_if() = default;
00049     virtual void clear() = 0;
00056     virtual void addValue(double value, double weight=1) = 0;
00062     virtual double getLastValue() = 0;
00068     virtual unsigned long numElements() = 0;
00069 public:
00075     virtual void setAddValueHandler(CollectorAddValueHandler addValueHandler) = 0;
00081     virtual void setClearHandler(CollectorClearHandler clearHandler) = 0;
00082 };
00083
00084 #endif /* COLLECTOR_IF_H */
```

18.350 CollectorDatafile_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   CollectorDatafile_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 30 de Agosto de 2018, 16:45
00012  */
00013
00014 #ifndef COLLECTORDATAFILE_IF_H
00015 #define COLLECTORDATAFILE_IF_H
00016
00017 #include "Collector_if.h"
00018
00025 class CollectorDatafile_if : public Collector_if {
00026 public:
00027     virtual ~CollectorDatafile_if() = default;
00033     virtual double getValue(unsigned int rank) = 0;
00037     virtual void seekFirstValue() = 0;
00042     virtual double getNextValue() = 0;
00048     virtual std::string getDataFilename() = 0;
00054     virtual void setDataFilename(std::string filename) = 0;
00055 };
00056
00057 #endif /* COLLECTORDATAFILE_IF_H */
```

18.351 CollectorDatafileDefaultImpl1.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
```

```
00008  * File:   CollectorDatafileDefaultImpl1.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 1 de Agosto de 2018, 20:58
00012  */
00013
00014 #ifndef COLLECTORDATAFILEDEFAULTIMPL1_H
00015 #define COLLECTORDATAFILEDEFAULTIMPL1_H
00016
00017 #include <limits>
00018 #include <string>
00019
00020 #include "CollectorDatafile_if.h"
00021
00028 class CollectorDatafileDefaultImpl1 : public CollectorDatafile_if {
00029 public:
00031     CollectorDatafileDefaultImpl1();
00032     virtual ~CollectorDatafileDefaultImpl1() = default;
00033 public: // inherited from Collector_if
00035     virtual void clear() override;
00037     virtual void addValue(double value, double weight=1) override;
00039     virtual double getLastValue() override;
00041     virtual unsigned long numElements() override;
00042 public:
00044     virtual double getValue(unsigned int num) override;
00046     virtual double getNextValue() override;
00048     virtual void seekFirstValue() override;
00050     virtual std::string getDataFilename() override;
00052     virtual void setDataFilename(std::string filename) override;
00053 public:
00055     virtual void setAddValueHandler(CollectorAddValueHandler addValueHandler) override;
00057     virtual void setClearHandler(CollectorClearHandler clearHandler) override;
00058 private:
00059     // Cache file metadata and sequential read position to avoid stale state.
00060     std::string _filename;
00061     double _lastValue = std::numeric_limits<double>::quiet_NaN();
00062     unsigned int _numElements = 0;
00063     unsigned int _nextReadIndex = 0;
00064 };
00065
00066 #endif /* COLLECTORDATAFILEDEFAULTIMPL1_H */
```

18.352 CollectorDefaultImpl1.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   CollectorDefaultImpl1.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 1 de Agosto de 2018, 20:43
00012  */
00013
00014 #ifndef COLLECTORDEFAULTIMPL1_H
00015 #define COLLECTORDEFAULTIMPL1_H
00016
00017 #include "Collector_if.h"
00018 //namespace GenesysKernel {
00019
00026 class CollectorDefaultImpl1 : public Collector_if {
00027 public:
00029     CollectorDefaultImpl1();
00030     virtual ~CollectorDefaultImpl1() = default;
00031 public:
00033     virtual void clear() override;
00035     virtual void addValue(double value, double weight=1) override;
00037     virtual double getLastValue() override;
00039     virtual unsigned long numElements() override;
00040 public:
00042     virtual void setAddValueHandler(CollectorAddValueHandler addValueHandler) override;
00044     virtual void setClearHandler(CollectorClearHandler clearHandler) override;
00045 private:
00046     double _lastValue;
00047     unsigned long _numElements = 0;
00048     CollectorAddValueHandler _addValueHandler = nullptr;
00049     CollectorClearHandler _clearHandler = nullptr;
00050 };
00051 //namespace\\}
00052 #endif /* COLLECTORDEFAULTIMPL1_H */
```

18.353 Sampler_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Sampler_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 14 de Agosto de 2018, 13:20
00012  */
00013
00014 #ifndef Sampler_IF_H
00015 #define Sampler_IF_H
00016
00024 class Sampler_if {
00025 public:
00026     virtual ~Sampler_if() = default;
00027
00034     struct RNG_Parameters {
00035         virtual ~RNG_Parameters() = default;
00036     };
00037 public: // RNG
00042     virtual double random() = 0;
00043 public: // continuous probability distributions
00052     virtual double sampleBeta(double alpha, double beta, double infLimit, double supLimit) = 0;
00054     virtual double sampleBeta(double alpha, double beta) = 0;
00056     virtual double sampleErlang(double mean, int M, double offset = 0.0) = 0;
00058     virtual double sampleExponential(double mean, double offset = 0.0) = 0;
00059     //virtual double sampleGamma(double mean, double alpha, double offset=0.0) = 0;
00061     virtual double sampleGamma(double alpha, double beta, double offset = 0.0) = 0;
00063     virtual double sampleGumbell(double mode, double scale) = 0;
00065     virtual double sampleLogNormal(double mean, double stddev, double offset = 0.0) = 0;
00067     virtual double sampleNormal(double mean, double stddev) = 0;
00069     virtual double sampleTriangular(double min, double mode, double max) = 0;
00071     virtual double sampleUniform(double min, double max) = 0;
00073     virtual double sampleWeibull(double alpha, double scale) = 0;
00074 public: // discrete probability distributions
00075     // @ToDo: (pequena alteração): Poisson, si vous plait!!!!
00077     virtual double sampleBinomial(int trials, double p) = 0;
00079     virtual double sampleBernoulli(double p) = 0;
00081     virtual double sampleDiscrete(double prob, double value, ...) = 0;
00083     virtual double sampleDiscrete(double *prob, double *value, int size) = 0;
00085     virtual double sampleGeometric(double p) = 0;
00086 public:
00088     virtual void setRNGparameters(RNG_Parameters* param) = 0;
00090     virtual RNG_Parameters* getRNGparameters() const = 0;
00091 };
00092
00093 #endif /* Sampler_IF_H */
```

18.354 SamplerDefaultImpl1.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   SamplerDefaultImpl1.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 2 de Agosto de 2018, 01:10
00012  */
00013
00014 #ifndef SAMPLERDEFAULTIMPL1_H
00015 #define SAMPLERDEFAULTIMPL1_H
00016
00017 #include "Sampler_if.h"
00018 #include <stdint.h>
00019
00020 //namespace GenesysKernel {
00021
00028 class SamplerDefaultImpl1 : public Sampler_if {
00029 public:
00030
00032     struct DefaultImpl1RNG_Parameters : public RNG_Parameters {
00034         uint32_t seed = 16021974;
00036         uint32_t a = 279470273u; // multiplier
00038         uint32_t m = 0xfffffffb; // module
```

```
00040         uint32_t c = 0; // increment
00041         ~DefaultImpl1RNG_Parameters() = default;
00042     };
00043 public:
00045     SamplerDefaultImpl1();
00047     virtual ~SamplerDefaultImpl1();
00048 public: // RNG
00050     virtual double random() override;
00051 public: // continuous probability distributions
00053     virtual double sampleBeta(double alpha, double beta, double infLimit, double supLimit) override;
00055     virtual double sampleBeta(double alpha, double beta) override;
00057     virtual double sampleErlang(double mean, int M, double offset = 0.0) override;
00059     virtual double sampleExponential(double mean, double offset = 0.0) override;
00060     //virtual double sampleGamma(double mean, double alpha, double offset=0.0);
00062     virtual double sampleGamma(double alpha, double beta, double offset = 0.0) override;
00064     virtual double sampleGumbell(double mode, double scale) override;
00066     virtual double sampleLogNormal(double mean, double stddev, double offset = 0.0) override;
00068     virtual double sampleNormal(double mean, double stddev) override;
00070     virtual double sampleTriangular(double min, double mode, double max) override;
00072     virtual double sampleUniform(double min, double max) override;
00074     virtual double sampleWeibull(double alpha, double scale) override;
00075 public: // discrete probability distributions
00076     // @ToDo: (pequena alteração): Poisson, si vous plait!!!!
00078     virtual double sampleBinomial(int trials, double p) override;
00080     virtual double sampleBernoulli(double p) override;
00082     virtual double sampleDiscrete(double prob, double value, ...) override;
00084     virtual double sampleDiscrete(double *prob, double *value, int size) override;
00086     virtual double sampleGeometric(double p) override;
00087 public:
00089     void reset();
00090 public:
00091     virtual void setRNGparameters(RNG_Parameters* param) override;
00092     virtual RNG_Parameters* getRNGparameters() const override;
00093 private:
00094     RNG_Parameters* _param = new DefaultImpl1RNG_Parameters();
00095     bool _ownsParam = true;
00096     uint32_t _xi;
00097     bool _normalflag;
00098     double _lastnormal;
00099 };
00100
00101
00102 /*
00103
00104  (*                          mrand.ras
00105
00106               Random generator for Borland Delphi 3.0
00107
00108 Copyrigth(c) 1998 Private enterprise - firm SOFTLAND, Ukraine
00109
00110
00111 1.  What is it?
00112
00113     TMultiRand is a component intended for generating random variates of
00114     different distributions:
00115
00116                       Uniform
00117                       Triangular
00118                       Exponential
00119                       Erlang
00120                       Normal
00121                       Gamma
00122                       Beta
00123                       Weibull
00124                       Lognormal
00125
00126 2. Licensing.
00127
00128   a) This software is under Copyrigth(c) 1998 SOFTLAND,Ukraine
00129   b) This software is for free usage and distribution.
00130   c) The  conditions of the distribution are
00131         - remaining the source code, comments and manual without changes
00132         - supplying the manual
00133   e) The user assumes all risks associated with this software.
00134      Softland is not responsible for any loss or damage incured
00135      during the use of this software.
00136
00137
00138 3. Manual
00139
00140   The manual for this component is in mrand.pdf
00141
00142 4. Where is it used?
00143
00144 This component is used in the discrete-event simulation system Delsi 1.0,
00145 which is implemented as a set of components for Borland Delphi.
00146
```

```
00147 Find more info at  http://www.softland.rovno.ua/delsi.htm
00148
00149 5. What is Softland ?
00150
00151 Softland is Ukrainian software firm.
00152
00153 Our skills:
00154
00155             - Delhpi, C++
00156             - SQL
00157             - Computer simulation of queuing systems
00158             - Web-design
00159
00160 We are looking for the remote development job in areas of
00161 general client/server applications and computer simulation.
00162 The rate of our service is extremely unexpensive.
00163
00164 Find more info at http://www.softland.rovno.ua
00165
00166 *)
00167
00168 unit Mrand;
00169
00170 interface
00171
00172 uses WinTypes, WinProcs, Classes, Dialogs;
00173
00174 const
00175
00176   MODULUS = 2147483647;
00177
00178 type
00179
00180 TMRand = class(TComponent)
00181 private
00182   FSeed: comp;
00183   FSeedInit: comp;
00184   FMultiplier: comp;
00185   NormalFlag: boolean;
00186   NormalResult: real;
00187   function Uni01: real;
00188   procedure SetSeed(Value: comp);
00189   function GetSeed: comp;
00190   procedure SetMultiplier(Value: comp);
00191   function GetMultiplier: comp;
00192   function GammaJonk(Alpha: real): real;
00193 public
00194   constructor Create(Owner: TComponent); override;
00195   procedure Reset;
00196   function Uniform(LowerBound: real; UpperBound: real): real;
00197   function Triangular(Min: real; Mode: real; Max: real): real;
00198   function Exponential(Mean: real): real;
00199   function Erlang(Mean: real; M: integer): real;
00200   function Normal(Mean: real; Deviation: real): real;
00201   function Gamma(Mean: real; Alpha: real): real;
00202   function Beta(ShapeAlpha: real; ShapeBeta: real;
00203                 LowerBound: real; UpperBound: real): real;
00204   function Weibull(Alpha: real; Scale: real): real;
00205   function Lognormal(Mean: real; Deviation: real): real;
00206 published
00207   property Seed: comp read GetSeed write SetSeed;
00208   property Multiplier: comp read GetMultiplier write SetMultiplier;
00209 end;
00210
00211
00212 procedure Register;
00213
00214 implementation
00215
00216
00217 constructor TMRand.Create(Owner: TComponent);
00218 begin
00219    inherited Create(Owner);
00220    FSeed:=1000000000;
00221    FMultiplier:=950706376;
00222    NormalFlag:=True;
00223 end;
00224
00225 procedure TMRand.Reset;
00226 begin
00227    NormalFlag:=True;
00228    FSeed:=FSeedInit;
00229 end;
00230
00231
00232 procedure TMRand.SetSeed(Value: comp);
00233 begin
```

```
00234    if (Value<1) or (Value>2147483646) then
00235    begin
00236       MessageDlg(Name+'Seed must be in the range 2...2147483646',mtError,[mbOK],0);
00237       FSeed:=1000000000;
00238    end
00239    else
00240       FSeed:=Value;
00241    FSeedInit:=FSeed;
00242 end;
00243
00244 function TMRand.GetSeed: comp;
00245 begin
00246    if (FSeed<1) or (FSeed>2147483646) then
00247    begin
00248       FSeed:=1000000000;
00249       FSeedInit:=FSeed;
00250    end;
00251    Result:=FSeed;
00252 end;
00253
00254
00255 procedure TMRand.SetMultiplier(Value: comp);
00256 begin
00257    if (Value<2) or (Value>2147483646) then
00258    begin
00259       MessageDlg(Name+'Multuplier must be in the range 2...2147483646',mtError,[mbOK],0);
00260       FMultiplier:=950706376;
00261    end
00262    else
00263       FMultiplier:=Value;
00264 end;
00265
00266 function TMRand.GetMultiplier: comp;
00267 begin
00268    if (FMultiplier<2) or (FMultiplier>2147483646) then
00269       FMultiplier:=950706376;
00270    Result:=FMultiplier;
00271 end;
00272
00273 function TMRand.Uni01: real;
00274 begin
00275    FSeed:=FSeed*FMultiplier;
00276    FSeed:=FSeed-Int(FSeed/MODULUS)*MODULUS;
00277    Result:=FSeed/MODULUS;
00278 end;
00279
00280
00281 {  Uniform distribution with the range LowerBound....UpperBound  }
00282
00283 function TMRand.Uniform(LowerBound: real; UpperBound: real): real;
00284 begin
00285    if (LowerBound<0) or (UpperBound<0) or (UpperBound<LowerBound) then
00286    begin
00287       MessageDlg('Incorrect parameters of '+Name+'.Uniform',mtError,[mbOK],0);
00288       Result:=0.0;
00289       Exit;
00290    end;
00291    Result:=LowerBound+(UpperBound-LowerBound)*Uni01
00292 end;
00293
00294
00295 {  Exponential distribution with mean "Mean"  }
00296
00297 function TMRand.Exponential(Mean: real): real;
00298 begin
00299    if Mean>=0.0 then
00300       Result:=Mean*(-ln(Uni01))
00301    else
00302       begin
00303          MessageDlg('Incorrect parameter of '+Name+'.Exponential',mtError,[mbOK],0);
00304          Result:=0;
00305       end;
00306 end;
00307
00308
00309 {  M-Erlang distribution with mean "Mean"  }
00310
00311 function TMRand.Erlang(Mean: real; M: integer): real;
00312 var
00313    i: integer;
00314    P: real;
00315 begin
00316    if (Mean<0.0) or (M<=0) then
00317    begin
00318       MessageDlg('Incorrect parameters of '+Name+'.Erlang',mtError,[mbOK],0);
00319       Result:=0.0;
00320       Exit;
```

```
00321    end;
00322
00323    P:=1;
00324    for i:=1 to M do
00325      P:=P*Uni01;
00326    Result:=(Mean/M)*(-ln(P));
00327 end;
00328
00329
00330 {  Normal distribution with mean "Mean" and deviation "Deviation"   }
00331
00332 function TMRand.Normal(Mean: real; Deviation: real): real;
00333 var U1,U2: real;
00334     W,Y: real;
00335 begin
00336    if (Mean<=0.0) or (Deviation<=0.0) then
00337    begin
00338       MessageDlg('Incorrect parameter Deviation of '+Name+'.Normal',mtError,[mbOK],0);
00339       Result:=0.0;
00340       Exit;
00341    end;
00342
00343    if NormalFlag then
00344    begin
00345       repeat
00346         U1:=2*Uni01-1.0;
00347         U2:=2*Uni01-1.0;
00348         W:=U1*U1+U2*U2;
00349       until not (W>=1.0);
00350       Y:= sqrt((-2* ln (W)) / W);
00351       Result:=Mean+U1*Y*Deviation;
00352       NormalResult:=Mean+U2*Y*Deviation;
00353       NormalFlag:=False;
00354    end
00355    else
00356    begin
00357       Result:=NormalResult;
00358       NormalFlag:=True;
00359    end;
00360 end;
00361
00362
00363 { Gamma distribution for Alpha<1 and beta=1 }
00364
00365 function TMRand.GammaJonk(Alpha: real ): real;
00366 var R,R1,R2,X,Y:real;
00367 begin
00368    repeat
00369      repeat
00370        R1:=Uni01;
00371        R2:=Uni01;
00372      until ((R1>1e-30) and (R2>1e-30));
00373
00374      if (ln(R2)/alpha<-1e3) then X:=0 else X:= exp(ln(R2)/alpha);
00375      if (ln(R1)/(1-alpha)<-1e3) then Y:=0 else Y:=exp(ln(R1)/(1-alpha));
00376
00377    until (X+Y<=1);
00378
00379    repeat
00380      R:=Uni01;
00381    until (R>1e-20);
00382
00383    Result:=-ln(R)*X/(Y+X);
00384 end;
00385
00386
00387 {  Gamma distribution with the mean "Mean" and shape-parameter "Alpha" }
00388
00389 function TMRand.Gamma(Mean: real; Alpha: real): real;
00390    var
00391    i: integer;
00392    P: real;
00393    IntAlpha: integer;
00394    OstAlpha: real;
00395 begin
00396
00397    if (Mean<=0.0) or (Alpha<=0.0) then
00398    begin
00399       MessageDlg('Incorrect parameters of '+Name+'.Gamma',mtError,[mbOK],0);
00400       Result:=0.0;
00401       Exit;
00402    end;
00403
00404    if Alpha<1.0 then
00405       Result:=(Mean/Alpha)*GammaJonk(Alpha)
00406    else
00407       if Alpha=1.0 then
```

```
00408          Result:=Mean*(-ln(Uni01))
00409       else
00410          begin
00411             IntAlpha:=Round(Int(Alpha));
00412             OstAlpha:=Alpha-IntAlpha;
00413             repeat
00414                P:=1;
00415                for i:=1 to IntAlpha do
00416                   P:=P*Uni01;
00417             until P>0;
00418             if OstAlpha>0 then
00419                Result:=(Mean/Alpha)*((-ln(P))+GammaJonk(OstAlpha))
00420             else
00421                Result:=(Mean/Alpha)*(-ln(P));
00422
00423          end;
00424 end;
00425
00426
00427 { Beta distribution with alpha-parameter "ShapeAlpha,
00428                          beta-perameter "ShapeBeta,
00429                          lower bound "LowerBound and
00430                          upper bound "UpperBound
00431 }
00432
00433 function TMRand.Beta(ShapeAlpha: real; ShapeBeta: real;
00434                          LowerBound: real; UpperBound: real): real;
00435 var
00436    X,Y1,Y2: real;
00437 begin
00438
00439    if (ShapeAlpha<=0.0) or (ShapeBeta<=0.0) or (LowerBound>UpperBound) or
00440       (LowerBound<0) or (UpperBound<0)
00441    then
00442    begin
00443       MessageDlg('Incorrect parameters of '+Name+'.Beta',mtError,[mbOK],0);
00444       Result:=0.0;
00445       Exit;
00446    end;
00447
00448    repeat
00449      Y1:=Gamma(ShapeAlpha,ShapeAlpha);
00450      Y2:=Gamma(ShapeBeta,ShapeBeta);
00451      X:=Y1/(Y1+Y2);
00452    until (X>=0) and (X<=1.0);
00453    Result:= LowerBound + (UpperBound - LowerBound ) * X;
00454 end;
00455
00456
00457 {  Weibull distribution with shape-parameter "Alpha" and scale "Scale" }
00458
00459 function TMRand.Weibull(Alpha: real; Scale: real): real;
00460 begin
00461    if (Alpha<=0.0) or (Scale<=0.0) then
00462    begin
00463       MessageDlg('Incorrect parameters of '+Name+'.Weibull',mtError,[mbOK],0);
00464       Result:=0.0;
00465       Exit;
00466    end;
00467    Result:= exp(ln(Scale*(-ln(Uni01)))/Alpha);
00468 end;
00469
00470
00471 {  Lognormal distribution with mean "Mean" and deviation "Deviation"   }
00472
00473 function TMRand.Lognormal(Mean: real; Deviation: real): real;
00474 var
00475    MeanNorm,DispNorm: real;
00476 begin
00477
00478    if (Mean<=0.0) or (Deviation<=0.0) then
00479    begin
00480       MessageDlg('Incorrect parameters of '+Name+'.Lognormal',mtError,[mbOK],0);
00481       Result:=0.0;
00482       Exit;
00483    end;
00484
00485    DispNorm:=ln((Deviation*Deviation)/(Mean*Mean)+1.0);
00486    MeanNorm:=ln(Mean)-0.5*DispNorm;
00487    Result:=Exp(Normal(MeanNorm,SQRT(DispNorm)));
00488 end;
00489
00490
00491 { Triangular distribution with minimal value "Min", maximal - "Max" and mode "Mode"}
00492
00493 function TMRand.Triangular(Min: real; Mode: real; Max: real): real;
00494 var
```

```
00495     Part1,Part2,Full,R: real;
00496 begin
00497
00498    if (Min>Mode) or (Max<Mode) or (Min>Max) or
00499       (Min<0) or (Max<0) or (Mode<0) then
00500    begin
00501       MessageDlg('Incorrect parameters of '+Name+'.Triangular',mtError,[mbOK],0);
00502       Result:=0.0;
00503       Exit;
00504    end;
00505
00506    Part1:=Mode-Min;
00507    Part2:=Max-Mode;
00508    Full:=Max-Min;
00509    R:=Uni01;
00510    if R<=Part1/Full then
00511       Result:=Min+sqrt(Part1*Full*R)
00512    else
00513       Result:=Max-sqrt(Part2*Full*(1.0-R));
00514 end;
00515
00516
00517 procedure Register;
00518 begin
00519    RegisterComponents('MRand', [TMRand]);
00520 end;
00521
00522
00523 end.
00524
00525  */
00526
00527
00528
00529 //namespace\\}
00530 #endif /* SAMPLERDEFAULTIMPL1_H */
```

18.355 SorttFile.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   SorttFile.h
00008  * Author: rlcancian
00009  *
00010  * Created on 20 de novembro de 2019, 22:27
00011  */
00012
00013 #ifndef SORTTFILE_H
00014 #define SORTTFILE_H
00015
00016 #include <string>
00017
00024 class SortFile {
00025 public:
00027     SortFile();
00028     ~SortFile() = default;
00029 public:
00034     bool sort();
00039     void setDataFilename(std::string filename);
00040 private:
00042     void run(int start, int end);
00044     void swap(int position1, int position2);
00046     int partition(int start, int end);
00048     void addValue(double value, int position);
00050     double getValue(unsigned long position);
00051 private:
00052     std::string _filename;
00053     double _numElements = 0;
00054 };
00055
00056 #endif /* SORTTFILE_H */
```

18.356 Statistics_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
```

```
00005  */
00006
00007 /*
00008  * File:   Statistics_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 14 de Agosto de 2018, 13:47
00012  */
00013
00014 #ifndef STATISTICS_IF_H
00015 #define STATISTICS_IF_H
00016
00017 #include <string>
00018 #include "Collector_if.h"
00019
00026 class Statistics_if {
00027 public:
00028     virtual ~Statistics_if() = default;
00030     virtual Collector_if* getCollector() const = 0;
00032     virtual void setCollector(Collector_if* collector) = 0;
00033 public:
00039     virtual unsigned int numElements() = 0;
00045     virtual double min() = 0;
00051     virtual double max() = 0;
00057     virtual double average() = 0;
00063     virtual double variance() = 0;
00069     virtual double stddeviation() = 0;
00075     virtual double variationCoef() = 0;
00081     virtual double halfWidthConfidenceInterval() = 0;
00087     virtual double confidenceLevel() = 0;
00094     virtual unsigned int newSampleSize(double halfWidth) = 0;
00100     virtual void setConfidenceLevel(double confidencelevel) = 0;
00101 };
00102
00103 #endif /* STATISTICS_IF_H */
```

18.357 StatisticsDataFile_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   StatisticsDataFile.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 22 de Novembro de 2018, 01:16
00012  */
00013
00014 #ifndef STATISTICSDATAFILE_IF_H
00015 #define STATISTICSDATAFILE_IF_H
00016
00017 #include "CollectorDatafile_if.h"
00018 #include "Statistics_if.h"
00019
00026 class StatisticsDatafile_if : public Statistics_if {
00027     //public:
00028     //    virtual CollectorDatafile_if* getCollector() = 0;
00029     //    virtual void setCollector(Collector_if* collector) = 0;
00030 public:
00031     virtual ~StatisticsDatafile_if() = default;
00037     virtual double mode() = 0;
00043     virtual double mediane() = 0;
00050     virtual double quartil(unsigned short num) = 0;
00057     virtual double decil(unsigned short num) = 0;
00064     virtual double centil(unsigned short num) = 0;
00070     virtual void setHistogramNumClasses(unsigned short num) = 0;
00076     virtual unsigned short histogramNumClasses() = 0;
00083     virtual double histogramClassLowerLimit(unsigned short classNum) = 0;
00090     virtual unsigned int histogramClassFrequency(unsigned short classNum) = 0;
00091
00092 };
00093
00094 #endif /* STATISTICSDATAFILE_IF_H */
```

18.358 StatisticsDataFileDefaultImpl.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
```

```
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   StatisticsDataFileDefaultImpl.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 22 de Novembro de 2018, 01:24
00012  */
00013
00014 #ifndef STATISTICSDATAFILEDEFAULTIMPL_H
00015 #define STATISTICSDATAFILEDEFAULTIMPL_H
00016
00017 #include "StatisticsDataFile_if.h"
00018 #include "SorttFile.h"
00019 #include <limits.h>
00020 #include <map>
00021 #include <string>
00022
00023 typedef double valueType;
00024
00031 class StatisticsDatafileDefaultImpl1 : public StatisticsDatafile_if {
00032 public:
00034     StatisticsDatafileDefaultImpl1();
00036     virtual ~StatisticsDatafileDefaultImpl1();
00037 public:
00039     virtual Collector_if* getCollector()  const override;
00041     virtual void setCollector(Collector_if* collector) override;
00042 public:
00044     virtual unsigned int numElements() override;
00046     virtual double min() override;
00048     virtual double max() override;
00050     virtual double average() override;
00052     virtual double variance() override;
00054     virtual double stddeviation() override;
00056     virtual double variationCoef() override;
00058     virtual double halfWidthConfidenceInterval() override;
00060     virtual double confidenceLevel() override;
00062     virtual unsigned int newSampleSize(double halfWidth) override;
00064     virtual void setConfidenceLevel(double confidencelevel) override;
00065 public:
00067     virtual double mode() override;
00069     virtual double mediane() override;
00071     virtual double quartil(unsigned short num) override;
00073     virtual double decil(unsigned short num) override;
00075     virtual double centil(unsigned short num) override;
00077     virtual void setHistogramNumClasses(unsigned short num) override;
00079     virtual unsigned short histogramNumClasses() override;
00081     virtual double histogramClassLowerLimit(unsigned short classNum) override;
00083     virtual unsigned int histogramClassFrequency(unsigned short classNum) override;
00084 private:
00086     void _invalidateCachedResults();
00088     void _resetSortedFileState();
00090     void _sortFile();
00092     bool _hasNewValue();
00094     double _getNormalProbability(double confidenceLevel);
00095 private:
00096     CollectorDatafile_if* _collector;
00097     bool _ownsCollector = true;
00098     CollectorDatafile_if* _collectorSorted;
00099     bool _ownsCollectorSorted = true;
00100     //CollectorDatafile_if* _collector;
00101     SortFile * sort = new SortFile();
00102     bool _ownsSort = true;
00103     double _confidenceLevel = 0.95;
00104     unsigned long _numElements = 0;
00105     std::map<double, double> _z;
00106     bool _fileSorted = false;
00107     bool _fileSortedCreated = false;
00108     std::string _sortedSourceFilename;
00109     //unsigned long _numElements = 0;
00110     bool _numElementsCalculated = false;
00111     double _max = INT_MIN;
00112     bool _maxCalculated = false;
00113     double _min = INT_MAX;
00114     bool _minCalculated = false;
00115     bool _modeCalculated = false;
00116     double _mode = 0;
00117     bool _medianeCalculated = false;
00118     double _mediane = 0;
00119     double _average = 0;
00120     bool _averageCalculated = false;
00121     double _variance = 0;
00122     bool _varianceCalculated = false;
00123     double _sumSquareElements = 0;
00124     double _sumElements = 0;
00125     double _stddeviation = 0;
```

```
00126     bool _stddeviationCalculated = false;
00127     double _variationCoef = 0;
00128     bool _variationCoefCalculated = false;
00129     bool _halfWidthConfidenceIntervalCalculated = false;
00130     double _lastConfidenceLevel = 0;
00131     double _halfWidthConfidenceInterval = 0;
00132     bool _newSampleSizeCalculated = false;
00133     double _lastNewSampleSizeConfidenceLevel = 0;
00134     double _lastNewSampleSizeHalfWidth = 0;
00135     double _newSampleSize = 0;
00136     bool _quartilCalculated = false;
00137     unsigned short _lastQuartilNum = 0;
00138     double _quartil = 0;
00139     bool _decilCalculated = false;
00140     unsigned short _lastDecilNum = 0;
00141     double _decil = 0;
00142     bool _centilCalculated = false;
00143     unsigned short _lastCentilNum = 0;
00144     double _centil = 0;
00145     bool _histogramNumClassesCalculated = false;
00146     unsigned short _histogramNumClasses = 1;
00147     bool _histogramClassLowerLimitCalculated = false;
00148     unsigned short _LastClassNumHistogramClassLowerLimit = 0;
00149     double _histogramClassLowerLimit = 0;
00150     bool _histogramClassFrequencyCalculated = false;
00151     unsigned short _lastClassNumHistogramClassFrequency = 0;
00152     unsigned int _histogramClassFrequency = 0;
00153     bool _proportionCalculed = false;
00154     double _proportion = 0;
00155 };
00156
00157 #endif /* STATISTICSDATAFILEDEFAULTIMPL_H */
```

18.359 StatisticsDefaultImpl1.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   StatisticsDefaultImpl1.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 1 de Agosto de 2018, 21:03
00012  */
00013
00014 #ifndef STATISTICSDEFAULTIMPL1_H
00015 #define STATISTICSDEFAULTIMPL1_H
00016
00017 #include "Statistics_if.h"
00018 #include "Collector_if.h"
00019 #include <map>
00020 //namespace GenesysKernel {
00021
00028 class StatisticsDefaultImpl1 : public Statistics_if {
00029 public:
00031     StatisticsDefaultImpl1();
00033     StatisticsDefaultImpl1(Collector_if* collector);
00034     virtual ~StatisticsDefaultImpl1() override;
00035 public:
00037     virtual Collector_if* getCollector() const override;
00039     virtual void setCollector(Collector_if* collector) override;
00040 public:
00042     virtual unsigned int numElements() override;
00044     virtual double min() override;
00046     virtual double max() override;
00048     virtual double average() override;
00050     virtual double variance() override;
00052     virtual double stddeviation() override;
00054     virtual double variationCoef() override;
00056     virtual double halfWidthConfidenceInterval() override;
00058     virtual double confidenceLevel() override;
00060     virtual unsigned int newSampleSize(double halfWidth) override;
00062     virtual void setConfidenceLevel(double confidencelevel) override;
00063 private:
00065     void collectorAddHandler(double newValue, double newWeight);
00067     void collectorClearHandler();
00069     void _bindCollectorHandlers(Collector_if* collector);
00071     void _resetStateForCurrentCollector();
00073     void initStatistics();
00074 private:
00075     Collector_if* _collector;
00076     bool _ownsCollector = false;
```

```
00077     bool _onlineStateValid = true;
00078     unsigned long _elems;
00079     double _sumData;
00080     double _sumDataSquare;
00081     double _sumWeight;
00082     double _sumWeightSquare;
00083     double _min;
00084     double _max;
00085     double _average;
00086     double _variance;
00087     double _unweightedvariance;
00088     double _unbiasedVariance;
00089     double _stddeviation;
00090     double _variationCoef;
00091     double _confidenceLevel = 0.95;
00092     double _criticalTn_1 = 1.96;
00093     double _halfWidth;
00094     const std::map<double, double> _supportedConfidenceLevels{
00095         {0.50, 0.0},
00096         {0.80, 1.282},
00097         {0.90, 1.645},
00098         {0.95, 1.96},
00099         {0.975, 2.06},
00100         {0.98, 2.326},
00101         {0.99, 2.576},
00102         {0.995, 2.807}
00103     };
00104 };
00105 //namespace\\}
00106 #endif /* STATISTICSDEFAULTIMPL1_H */
```

18.360 TraitsKernel.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Traits.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 14 de Agosto de 2018, 19:36
00012  */
00013
00014
00015 #ifndef TRAITSKERNEL_H
00016 #define TRAITSKERNEL_H
00017
00018
00019 #include "simulator/persistence/PersistenceDefaultImpl2.h"
00020 #include "simulator/persistence/PersistencePartialLoadImpl1.h"
00021 #include "simulator/SimulationReporterDefaultImpl1.h"
00022 #include "simulator/essentialPlugins/Counter.h"
00023 #include "simulator/model/ModelCheckerDefaultImpl1.h"
00024 #include "simulator/ParserDefaultImpl2.h"
00025 #include "simulator/SimulationReporter_if.h"
00026
00027 #include "statistics/CollectorDatafileDefaultImpl1.h"
00028 #include "statistics/CollectorDefaultImpl1.h"
00029 #include "statistics/StatisticsDefaultImpl1.h"
00030 #include "statistics/StatisticsDataFileDefaultImpl.h"
00031 #include "statistics/SamplerDefaultImpl1.h"
00032
00033 #include "../plugins/PluginConnectorStaticImpl1.h"
00034 #include "../plugins/PluginConnectorDummyImpl1.h"
00035
00036
00037 //namespace GenesysKernel {
00038
00039 template <typename T>
00040 struct TraitsKernel {
00041 };
00042
00043 /*
00044  *  Simulation and Simulation Parts
00045  */
00046
00047 template <>
00048 struct TraitsKernel<Simulator> {
00049     static const TraceManager::Level traceLevel = TraceManager::Level::L9_mostDetailed;
00050 };
00051
00052 template <>
```

```
00053 struct TraitsKernel<SimulationReporter_if> {
00054     typedef SimulationReporterDefaultImpl1 Implementation;
00055     typedef Counter CounterImplementation;
00056     static const TraceManager::Level traceLevel = TraceManager::Level::L2_results;
00057 };
00058
00059 template <>
00060 struct TraitsKernel<PluginConnector_if> {
00061     //typedef PluginConnectorStaticImpl1 Implementation;
00062     typedef PluginConnectorDummyImpl1 Implementation;
00063     //typedef PluginConnectorDynamicLibraryLoader Implementation;
00064     static const TraceManager::Level traceLevel = TraceManager::Level::L4_warning;
00065 };
00066
00067 template <>
00068 struct TraitsKernel<Parser_if> {
00069     typedef ParserDefaultImpl2 Implementation;
00070 };
00071
00072 /*
00073  *  Model and Model Parts
00074  */
00075
00076 template <>
00077 struct TraitsKernel<Model> {
00078     typedef StatisticsDefaultImpl1 StatisticsCollector_StatisticsImplementation;
00079     typedef CollectorDefaultImpl1 StatisticsCollector_CollectorImplementation;
00080     static const TraceManager::Level traceLevel = TraceManager::Level::L5_event;
00081 };
00082
00083 template <>
00084 struct TraitsKernel<ModelComponent> {
00085     static constexpr bool reportStatistics = true;
00086     static const TraceManager::Level traceLevel = TraceManager::Level::L2_results;
00087 };
00088
00089 template <>
00090 struct TraitsKernel<ModelDataDefinition> {
00091     static constexpr bool reportStatistics = true;
00092     static const TraceManager::Level traceLevel = TraceManager::Level::L2_results;
00093 };
00094
00095 template <>
00096 struct TraitsKernel<ModelChecker_if> {
00097     typedef ModelCheckerDefaultImpl1 Implementation;
00098     static const TraceManager::Level traceLevel = TraceManager::Level::L2_results;
00099 };
00100
00101 template <>
00102 struct TraitsKernel<Persistence_if> {
00103     typedef PersistencePartialLoadImpl1 Implementation;
00104     static const TraceManager::Level traceLevel = TraceManager::Level::L2_results;
00105 };
00106
00107 /*
00108  *  Statistics
00109  */
00110 template <>
00111 struct TraitsKernel<Statistics_if> {
00112     typedef double DataType;
00113     // @ToDo: (importante): not used yet. Change all classes that collect statistics to this type (so
      classes that deal with erros and bit limits can be assigned to it
00114     static constexpr double SignificanceLevel = 0.05;
00115 };
00116
00117 template <>
00118 struct TraitsKernel<StatisticsDatafile_if> {
00119     typedef StatisticsDatafileDefaultImpl1 Implementation;
00120     typedef CollectorDatafileDefaultImpl1 CollectorImplementation;
00121     static constexpr double SignificanceLevel = 0.05;
00122 };
00123
00124 template <>
00125 struct TraitsKernel<Sampler_if> {
00126     typedef SamplerDefaultImpl1 Implementation;
00127     typedef SamplerDefaultImpl1::DefaultImpl1RNG_Parameters Parameters;
00128 };
00129
00130 template <>
00131 struct TraitsKernel<Collector_if> {
00132     typedef CollectorDatafileDefaultImpl1 Implementation;
00133     typedef double DataType;
00134     // @ToDo: (importante): not used yet. Change all classes that collect statistics to this type (so
      classes that deal with erros and bit limits can be assigned to it
00135 };
00136
00137 //namespace\\}
```

```
00138
00139 #endif /* TRAITSKERNEL_H */
```

18.361 Exact.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Exact.h
00009  * Author: rlcancian
00010  *
00011  * Created on 12 de agosto de 2021, 18:55
00012  */
00013
00014 #ifndef Exact_H
00015 #define Exact_H
00016
00017 #include <iostream>
00018 #include <iomanip>
00019
00020 class Exact {
00021     //friend ostream &operator«(ostream&, const Exact &);
00022     //friend istream &operator»(istream&, Exact &);
00023 public:
00024
00025     Exact() {
00026     };
00027
00028     Exact(int num, int den = 1) {
00029         _num = num;
00030         _den = den;
00031     };
00032
00033     const Exact &operator+(const Exact &right) {
00034         Exact e;
00035         e._num = this->_num * right._den + right._num * this->_den;
00036         e._den = this->_den * right._den;
00037         return &e;
00038     }
00039
00040     const Exact &operator+=(const Exact &right) {
00041         this->_num = this->_num * right._den + right._num * this->_den;
00042         this->_den *= right._den;
00043     }
00044
00045     const Exact &operator-=(const Exact &right) {
00046         this->_num = this->_num * right._den - right._num * this->_den;
00047         this->_den *= right._den;
00048     }
00049
00050     const Exact &operator*=(const Exact &right) {
00051         this->_num *= right._num;
00052         this->_den *= right._den;
00053     }
00054
00055     const Exact &operator/=(const Exact &right) {
00056         this->_num *= right._num;
00057         this->_den *= right._den;
00058     }
00059
00060     const Exact &operator=(const Exact &right) {
00061         this->_num = right._num;
00062         this->_den = right._den;
00063     }
00064
00065     const Exact &operator=(const int &right) {
00066         this->_num = right;
00067         this->_den = 1;
00068     }
00069
00070     bool operator==(const Exact &right) {
00071         // verify proportionality is missing
00072         return (this->_num == right._num) && (this->_den == right._den);
00073     }
00074
00075     bool operator==(const double &right) {
00076         // verify proportionality is missing
00077         return (this->_num / this->_den) == right;
00078     }
00079
00080     bool operator<(const Exact &right) {
```

```
00081         // verify proportionality is missing
00082         return (this->_num / this->_den) < (right._num / right._den);
00083     }
00084 
00085     bool operator<(const double &right) {
00086         // verify proportionality is missing
00087         return (this->_num / this->_den) < right;
00088     }
00089 
00090     bool operator<=(const Exact &right) {
00091         // verify proportionality is missing
00092         return (this->_num / this->_den) <= (right._num / right._den);
00093     }
00094 
00095     bool operator<=(const double &right) {
00096         // verify proportionality is missing
00097         return (this->_num / this->_den) <= right;
00098     }
00099 
00100     bool operator>(const Exact &right) {
00101         // verify proportionality is missing
00102         return (this->_num / this->_den) > (right._num / right._den);
00103     }
00104 
00105     bool operator>(const double &right) {
00106         // verify proportionality is missing
00107         return (this->_num / this->_den) > right;
00108     }
00109 
00110     bool operator>=(const Exact &right) {
00111         // verify proportionality is missing
00112         return (this->_num / this->_den) >= (right._num / right._den);
00113     }
00114 
00115     bool operator>=(const double &right) {
00116         // verify proportionality is missing
00117         return (this->_num / this->_den) >= right;
00118     }
00119 
00120     void Simplify() {
00121     }
00122 
00123     double toFloat() {
00124         return _num / _den;
00125     };
00126 protected:
00127 private:
00128     double _num, _den;
00129 };
00130 
00131 
00132 /*
00133 ostream &operator«(ostream &output, const Exact &num) {
00134     output « num._num « "/" « num._den;
00135     return output;
00136 }
00137 
00138 /*
00139 istream &operator»(istream &input, Exact &num) {
00140     input » num._num;
00141     input » num._dem;
00142     return input;
00143 }
00144  */
00145 
00146 
00147 #endif /* Exact_H */
00148 
```

18.362 List.h

```
00001 #ifndef GENLIST_H
00002 #define GENLIST_H
00003 /*
00004  * To change this license header, choose License Headers in Project Properties.
00005  * To change this template file, choose Tools | Templates
00006  * and open the template in the editor.
00007  */
00008 
00009 /*
00010  * File:   List.h
00011  * Author: rafael.luiz.cancian
00012  *
00013  * Created on 21 de Junho de 2018, 12:55
00014  */
```

```
00015
00016
00017
00018 #include <string>
00019 #include <list>
00020 #include <map>
00021 #include <iterator>
00022 #include <functional>
00023 #include <algorithm>
00024
00025
00032 template <typename T>
00033 class List {
00034 public:
00035     using CompFunct = std::function<bool(const T, const T) >;
00036 public:
00038     List();
00040     List(const List<T>& origin);
00041     List<T>& operator=(const List<T>& origin);
00042     virtual ~List();
00043 public: // direct access to list
00045     unsigned int size();
00047     bool empty();
00049     void clear();
00051     void pop_front();
00052     template<class Compare>
00054     void sort(Compare comp);
00056     std::list<T>* list() const;
00057 public: // new methods
00059     T create();
00060     template<typename U>
00062     T create(U arg);
00064     std::string show();
00066     typename std::list<T>::iterator find(T element);
00067     //int rankOf(T modeldatum); //!< returns the position (1st position=0) of the modeldatum if found,
      or negative value if not found
00068 public: // improved (easier) methods
00070     void insert(T element);
00072     void remove(T element);
00074     void setAtRank(unsigned int rank, T element);
00076     T getAtRank(unsigned int rank);
00078     T next();
00080     T front();
00082     T last();
00084     T previous();
00086     T current(); // get current modeldatum on the list (the last used)
00088     void setSortFunc(CompFunct _sortFunc);
00089 protected:
00090     //std::map<Util::identitifcation, T>* _map;
00091     std::list<T>* _list;
00092     CompFunct _sortFunc{[](const T, const T) {
00093             return false;
00094         }};
00095     typename std::list<T>::iterator _it;
00096 };
00097
00098 template <typename T>
00099 List<T>::List() {
00100     //_map = new std::map<Util::identitifcation, T>();
00101     _list = new std::list<T>();
00102     _it = _list->begin();
00103 }
00104
00105 template <typename T>
00106 List<T>::List(const List<T>& origin) {
00107     _list = new std::list<T>(*origin._list);
00108     _sortFunc = origin._sortFunc;
00109     _it = _list->begin();
00110 }
00111
00112 template <typename T>
00113 List<T>& List<T>::operator=(const List<T>& origin) {
00114     if (this != &origin) {
00115         *_list = *origin._list;
00116         _sortFunc = origin._sortFunc;
00117         _it = _list->begin();
00118     }
00119     return *this;
00120 }
00121
00122 template <typename T>
00123 List<T>::~List() {
00124     delete _list;
00125 }
00126
00127 template <typename T>
00128 std::list<T>* List<T>::list() const {
```

```
00129     return _list;
00130 }
00131
00132 template <typename T>
00133 unsigned int List<T>::size() {
00134     return _list->size();
00135 }
00136
00137 //template <typename T>
00138 //List<T>::List(const List& orig) {
00139 //}
00140
00141 //template <typename T>
00142 //List<T>::~List() {
00143 //}
00144
00145 template <typename T>
00146 std::string List<T>::show() {
00147     int i = 0;
00148     std::string text = "{";
00149     for (typename std::list<T>::iterator it = _list->begin(); it != _list->end(); it++, i++) {
00150         text += "[" + std::to_string(i) + "]=(" + (*it)->show() + "),";
00151     }
00152     text += "}";
00153     return text;
00154 }
00155
00156 template <typename T>
00157 void List<T>::insert(T element) {
00158     const bool wasEmpty = _list->empty();
00159     _list->insert(std::upper_bound(_list->begin(), _list->end(), element, _sortFunc), element);
00160     if (wasEmpty) {
00161         _it = _list->begin();
00162     }
00163 }
00164
00165 template <typename T>
00166 bool List<T>::empty() {
00167     return _list->empty();
00168 }
00169
00170 template <typename T>
00171 void List<T>::pop_front() {
00172     if (_list->empty()) {
00173         return;
00174     }
00175     const bool cursorAtFront = _it == _list->begin();
00176     _list->pop_front();
00177     if (cursorAtFront) {
00178         _it = _list->begin(); // if it points to the removed modeldatum, then changes to begin
00179     }
00180 }
00181
00182 template <typename T>
00183 void List<T>::remove(T element) {
00184     // Reset the internal cursor after erasure to avoid dereferencing a stale iterator during
    recursive removals.
00185     _list->remove(element);
00186     _it = _list->begin();
00187 }
00188
00189 template <typename T>
00190 T List<T>::create() {
00191     return new T();
00192 }
00193
00194 template <typename T>
00195 void List<T>::clear() {
00196     _list->clear();
00197     _it = _list->begin();
00198 }
00199
00200 template <typename T>
00201 T List<T>::getAtRank(unsigned int rank) {
00202     unsigned int thisRank = 0;
00203     for (typename std::list<T>::iterator it = _list->begin(); it != _list->end(); it++) {
00204         if (rank == thisRank) {
00205             return (*it);
00206         } else {
00207             thisRank++;
00208         }
00209     }
00210     return T{};
00211 }
00212
00213 template <typename T>
00214 void List<T>::setAtRank(unsigned int rank, T element) {
```

```
00215     if (rank == _list->size()) {
00216         _list->insert(_list->end(), element);
00217     } else {
00218         unsigned int thisRank = 0;
00219         for (typename std::list<T>::iterator it = _list->begin(); it != _list->end(); it++) {
00220             if (rank == thisRank) {
00221                 *it = element;
00222                 return;
00223             } else {
00224                 thisRank++;
00225             }
00226         }
00227     }
00228 }
00229
00230 template <typename T>
00231 T List<T>::next() {
00232     if (_list->empty() || _it == _list->end()) {
00233         return T{};
00234     }
00235     _it++;
00236     if (_it != _list->end())
00237         return (*_it);
00238     else
00239         return T{};
00240
00241 }
00242
00243 template <typename T>
00244 typename std::list<T>::iterator List<T>::find(const T element) {
00245     for (typename std::list<T>::iterator it = _list->begin(); it != _list->end(); it++) {
00246         if ((*it) == element) {
00247             return it;
00248         }
00249     }
00250     return _list->end();
00251 }
00252
00253 /*
00254 template <typename T>
00255 int List<T>::rankOf(T modeldatum) {
00256     int rank = 0;
00257     for (typename std::list<T>::iterator it = _list->begin(); it != _list->end(); it++) {
00258     if ((*it) == modeldatum) {
00259         return rank;
00260     } else
00261         rank++;
00262     }
00263     return -1; // not found -> negative rank
00264 }
00265  */
00266
00267 template <typename T>
00268 T List<T>::front() {
00269     if (_list->empty()) {
00270         return T{};
00271     }
00272     _it = _list->begin();
00273     //if (_it != _list->end())
00274     return (*_it);
00275     //else
00276     //return dynamic_cast<T>(nullptr);
00277 }
00278
00279 template <typename T>
00280 T List<T>::last() {
00281     if (_list->empty()) {
00282         return T{};
00283     }
00284     _it = _list->end();
00285     _it--;
00286     return (*_it);
00287     //else return nullptr;
00288 }
00289
00290 template <typename T>
00291 T List<T>::previous() {
00292     if (_list->empty() || _it == _list->begin()) {
00293         return T{};
00294     }
00295     _it--;
00296     return (*_it);
00297 }
00298
00299 template <typename T>
00300 T List<T>::current() {
00301     if (_list->empty() || _it == _list->end()) {
```

```
00302         return T{};
00303     }
00304     return (*_it);
00305 }
00306
00307 template <typename T>
00308 void List<T>::setSortFunc(CompFunct _sortFunc) {
00309     this->_sortFunc = _sortFunc;
00310 }
00311
00312 template <typename T>
00313 template<typename U>
00314 T List<T>::create(U arg) {
00315     return T(arg);
00316 }
00317
00318 template <typename T>
00319 template<class Compare>
00320 void List<T>::sort(Compare comp) {
00321     _list->sort(comp);
00322 }
00323
00324 #endif
```

18.363 ListObservable.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   TManager.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Junho de 2018, 12:55
00012  */
00013
00014 #ifndef LISTOBSERVABLE_H
00015 #define LISTOBSERVABLE_H
00016
00017 #include <string>
00018 #include <list>
00019 #include <map>
00020 #include <iterator>
00021 #include <functional>
00022 #include <algorithm>
00023 #include "Util.h"
00024 #include "../simulator/ModelElement.h"
00025
00026 #include "List.h"
00027
00034 template <typename T>
00035 class ListObservable : public List {
00036 public:
00037     using CompFunct = std::function<bool(const T, const T) >;
00038 public:
00040     ListObservable();
00041     virtual ~ListObservable() = default;
00042 public: // direct access to list
00044     void clear();
00046     void pop_front();
00047 public: // improved (easier) methods
00049     void insert(T element);
00051     void remove(T element);
00053     void setAtRank(unsigned int rank, T element);
00054 };
00055
00056 template <typename T>
00057 ListObservable<T>::ListObservable() {
00058     //_map = new std::map<Util::identitifcation, T>();
00059     _list = new std::list<T>();
00060     _it = _list->begin();
00061 }
00062
00063 template <typename T>
00064 std::list<T>* ListObservable<T>::list() const {
00065     return _list;
00066 }
00067
00068 template <typename T>
00069 unsigned int ListObservable<T>::size() {
00070     return _list->size();
00071 }
```

```
00072
00073 //template <typename T>
00074 //ListObservable<T>::ListObservable(const ListObservable& orig) {
00075 //}
00076
00077 //template <typename T>
00078 //ListObservable<T>::~ListObservable() {
00079 //}
00080
00081 template <typename T>
00082 std::string ListObservable<T>::show() {
00083     int i = 0;
00084     std::string text = "{";
00085     for (typename std::list<T>::iterator it = _list->begin(); it != _list->end(); it++, i++) {
00086         text += "[" + std::to_string(i) + "]=(" + (*it)->show() + "),";
00087     }
00088     text += "}";
00089     return text;
00090 }
00091
00092 template <typename T>
00093 void ListObservable<T>::insert(T element) {
00094     _list->insert(std::upper_bound(_list->begin(), _list->end(), element, _sortFunc), element);
00095 }
00096
00097 template <typename T>
00098 bool ListObservable<T>::empty() {
00099     return _list->empty();
00100 }
00101
00102 template <typename T>
00103 void ListObservable<T>::pop_front() {
00104     typename std::list<T>::iterator itTemp = _list->begin();
00105     _list->pop_front();
00106     if (_it == itTemp) { /* @ToDo: (importante): check this */
00107         _it = _list->begin(); // if it points to the removed element, then changes to begin
00108     }
00109 }
00110
00111 template <typename T>
00112 void ListObservable<T>::remove(T element) {
00113     _list->remove(element);
00114     if ((*_it) == element) { /* @ToDo: (importante): check this */
00115         _it = _list->begin(); // if it points to the removed element, then changes to begin
00116     }
00117 }
00118
00119 template <typename T>
00120 T ListObservable<T>::create() {
00121     return new T();
00122 }
00123
00124 template <typename T>
00125 void ListObservable<T>::clear() {
00126     _list->clear();
00127 }
00128
00129 template <typename T>
00130 T ListObservable<T>::getAtRank(unsigned int rank) {
00131     unsigned int thisRank = 0;
00132     for (typename std::list<T>::iterator it = _list->begin(); it != _list->end(); it++) {
00133         if (rank == thisRank) {
00134             return (*it);
00135         } else {
00136             thisRank++;
00137         }
00138     }
00139     return 0; /* @ToDo: (importante): Invalid return depends on T. If T is pointer, nullptr works
      fine. If T is double, it does not. I just let (*it), buut it is not nice */
00140 }
00141
00142 template <typename T>
00143 void ListObservable<T>::setAtRank(unsigned int rank, T element) {
00144     if (rank == _list->size()) {
00145         _list->insert(_list->end(), element);
00146     } else {
00147         unsigned int thisRank = 0;
00148         for (typename std::list<T>::iterator it = _list->begin(); it != _list->end(); it++) {
00149             if (rank == thisRank) {
00150                 *it = element;
00151                 return;
00152             } else {
00153                 thisRank++;
00154             }
00155         }
00156     }
00157 }
```

```
00158
00159 template <typename T>
00160 T ListObservable<T>::next() {
00161     _it++;
00162     if (_it != _list->end())
00163         return (*_it);
00164     else
00165         return nullptr;
00166
00167 }
00168
00169 template <typename T>
00170 typename std::list<T>::iterator ListObservable<T>::find(T element) {
00171     for (typename std::list<T>::iterator it = _list->begin(); it != _list->end(); it++) {
00172         if ((*it) == element) {
00173             return it;
00174         }
00175     }
00176     return _list->end(); /* @ToDo: (importante): check nullptr or invalid iterator when not found */
00177     //return nullptr;
00178 }
00179
00180 /*
00181 template <typename T>
00182 int ListObservable<T>::rankOf(T element) {
00183     int rank = 0;
00184     for (typename std::list<T>::iterator it = _list->begin(); it != _list->end(); it++) {
00185     if ((*it) == element) {
00186         return rank;
00187     } else
00188         rank++;
00189     }
00190     return -1; // not found -> negative rank
00191 }
00192  */
00193
00194 template <typename T>
00195 T ListObservable<T>::front() {
00196     _it = _list->begin();
00197     //if (_it != _list->end())
00198     return (*_it);
00199     //else
00200     //return dynamic_cast<T>(nullptr);
00201 }
00202
00203 template <typename T>
00204 T ListObservable<T>::last() {
00205     _it = _list->end();
00206     _it--;
00207     //if (_it != _list->end()) // @ToDo: (importante): CHECK!!!
00208     return (*_it);
00209     //else return nullptr;
00210 }
00211
00212 template <typename T>
00213 T ListObservable<T>::previous() {
00214     _it--; // @ToDo: (importante): CHECK!!!
00215     return (*_it);
00216 }
00217
00218 template <typename T>
00219 T ListObservable<T>::current() {
00220     /* @ToDo: (importante): To implement (i thing it's just to check). Must actualize _it on other
      methods when other elements are accessed */
00221     return (*_it);
00222 }
00223
00224 template <typename T>
00225 void ListObservable<T>::setSortFunc(CompFunct _sortFunc) {
00226     this->_sortFunc = _sortFunc;
00227 }
00228
00229 template <typename T>
00230 template<typename U>
00231 T ListObservable<T>::create(U arg) {
00232     return T(arg);
00233 }
00234
00235 template <typename T>
00236 template<class Compare>
00237 void ListObservable<T>::sort(Compare comp) {
00238     _list->sort(comp);
00239 }
00240
00241 #endif /* LISTOBSERVABLE_H */
```

18.364 Util.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Util.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Junho de 2018, 13:02
00012  */
00013
00014
00015 //
00016 // VINCULADOR
00017 // SAIDA
00018 // ${CND_DISTDIR}/${CND_CONF}/${CND_PLATFORM}/genesysterminalapplication
00019 //
00020
00021
00022 #ifndef UTIL_H
00023 #define UTIL_H
00024
00025 #include <map>
00026 #include <typeinfo>
00027 #include <string>
00028 #include <list>
00029 #include <cctype>
00030 #include <algorithm>
00031 #include <cctype>
00032 #include <locale>
00033 #include <stdio.h>
00034 // dir
00035 #include <limits.h>
00036 #include <unistd.h>
00037 #include <vector>
00038 #include <dirent.h>
00039 #include <errno.h>
00040 #include <iostream>
00041 #include <sys/stat.h>
00042 #include <cstring>
00043
00044 #include <iostream>
00045 #include <string>
00046 //namespace GenesysKernel {
00047
00048 class Util {
00049 public:
00050     typedef unsigned long identification;
00051     typedef unsigned int rank;
00052
00053     // @ToDo: (pequena alteração): Should be insde ModelSimulation, where time goes on
00054
00058     enum class TimeUnit : int {
00059         unknown = 0,
00060         picosecond = 1,
00061         nanosecond = 2,
00062         microsecond = 3,
00063         milisecond = 4,
00064         second = 5,
00065         minute = 6,
00066         hour = 7,
00067         day = 8,
00068         week = 9,
00069         num_elements = 10
00070     };
00071
00077     static std::string StrTimeUnitShort(Util::TimeUnit timeUnit);
00083     static std::string StrTimeUnitLong(Util::TimeUnit timeUnit);
00089     static std::string convertEnumToStr(Util::TimeUnit timeUnit);
00090
00091     // @ToDo: (pequena alteração): check: here? Shouldn't it be on SimulationReport interface?
00092     enum class AllocationType : int {
00093         ValueAdded = 0, NonValueAdded = 1, Transfer = 2, Wait = 3, Others = 4, num_elements = 5
00094     };
00095
00101     static std::string StrAllocation(Util::AllocationType allocation);
00107     static std::string convertEnumToStr(Util::AllocationType allocation);
00108
00109     enum class TimeFormat : unsigned int {
00110         twelve = 12,
00111         twentyFour = 24,
00112     };
00113
```

```
00114 private:
00115     static unsigned int _S_indentation;
00116     static Util::identification _S_lastId;
00117     static std::map<std::string, Util::identification> _S_lastIdOfType;
00118     static std::map<std::string, std::string> _S_TypeOf;
00119
00120 public: // indentation and string
00125     static void SetIndent(const unsigned short indent);
00129     static void IncIndent();
00133     static void DecIndent();
00140     static void SepKeyVal(std::string str, std::string& key, std::string& value);
00145     static std::string Indent();
00152     static std::string SetW(std::string text, unsigned short width);
00158     static std::string StrTruncIfInt(double value);
00164     static std::string StrTruncIfInt(std::string strValue);
00170     static std::string Trim(std::string str);
00178     static std::string StrReplace(std::string text, std::string searchFor, std::string replaceBy);
00184     static std::string StrReplaceSpecialChars(std::string text);
00190     static std::string StrIndex(int index);
00191     //static char* Str2CharPtr(std::string str);
00196     static void Trimwithin(std::string& str);
00197
00198 public: // show strucutres
00204     static std::string Map2str(std::map<std::string, std::string>* mapss);
00210     static std::string Map2str(std::map<std::string, double>* mapss);
00216     static std::string List2str(std::list<unsigned int>* list);
00217
00218 public:
00219     // identitification // @ToDo: (importante): CHECK ALL, since some should be private and available
    to FRIEND classes in the kernel
00224     static Util::identification GenerateNewId();
00230     static Util::identification GenerateNewIdOfType(std::string objtype);
00236     static Util::identification GetLastIdOfType(std::string objtype);
00241     static void ResetIdOfType(std::string objtype);
00245     static void ResetAllIds();
00246
00247 public: // simulation support
00254     static double TimeUnitConvert(Util::TimeUnit timeUnit1, Util::TimeUnit timeUnit2);
00255
00256 public: // files
00261     static char DirSeparator();
00267     static std::string FilenameFromFullFilename(const std::string& s);
00272     static void FileDelete(const std::string& filename);
00278     static std::string PathFromFullFilename(const std::string& s);
00283     static std::string RunningPath();
00291     static std::vector<std::string> ListFiles(std::string dir, std::string fileFilter = "",
00292                                               mode_t attribFilter = S_IFREG & S_IFDIR);
00298     static bool FileExists(const std::string& name);
00299
00300     //public: // operating system specifics
00301     class CommandResult {
00302     public:
00303         CommandResult() {
00304         }
00305
00306         bool success = false;
00307         std::string commandStdOutput = "";
00308         std::string commandErrOutput = "";
00309         std::string workingDirectory = "./";
00310         std::string destinationPath = "./";
00311     };
00312
00318     static Util::CommandResult ExecuteCommand(std::string command);
00319
00320 public: // template implementations
00321
00327     template <typename T>
00328     static std::string TypeOf() {
00329         std::string name = typeid(T).name();
00330         std::map<std::string, std::string>::iterator it = _S_TypeOf.find(name);
00331         if (it != _S_TypeOf.end()) {
00332             return (*it).second;
00333         }
00334         else {
00335             std::string newname(name);
00336             while (std::isdigit(newname[0])) {
00337                 newname.erase(0, 1);
00338             }
00339             _S_TypeOf.insert(std::pair<std::string, std::string>(name, newname));
00340             return newname;
00341         }
00342     }
00343
00349     template <class T>
00350     static Util::identification GenerateNewIdOfType() {
00351         std::string objtype = Util::TypeOf<T>();
00352         std::map<std::string, Util::identification>::iterator it =
```

```
    Util::_S_lastIdOfType.find(objtype);
00353         if (it == Util::_S_lastIdOfType.end()) {
00354             // a new one. create the pair
00355             Util::_S_lastIdOfType.insert(std::pair<std::string, Util::identification>(objtype, 0));
00356             it = Util::_S_lastIdOfType.find(objtype);
00357         }
00358         Util::identification next = (*it).second;
00359         next++;
00360         (*it).second = next;
00361         return (*it).second;
00362     }
00363
00364 private:
00365     Util();
00366     virtual ~Util() = default;
00367 };
00368
00369 //namespace\\}
00370 #endif /* UTIL_H */
```

18.365 Genesys++-driver.h

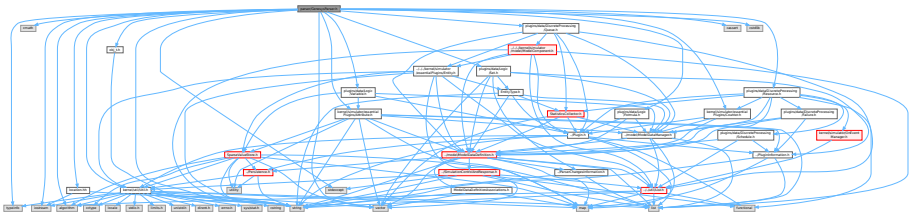
```
00001 #ifndef GENESYSPP_DRIVER_H
00002 #define GENESYSPP_DRIVER_H
00003
00004 #include <string>
00005 #include <map>
00006 #include "GenesysParser.h"
00007 #include "../kernel/simulator/model/Model.h"
00008 #include "kernel/simulator/SimulationControlAndResponse.h"
00009 #include "kernel/util/Util.h"
00010 #include "kernel/statistics/Sampler_if.h"
00011 // Tell Flex the lexer's prototype ...
00012
00013
00014 #define YY_DECL \
00015   yy::genesyspp_parser::symbol_type yylex (genesyspp_driver& driver)
00016
00017 /* # define YY_DECL \
00018 //  yy::genesyspp_parser::symbol_type yylex (semantic_type* yylval, location_type* yylloc,
    genesyspp_driver& driver)
00019  */
00020 // ... and declare it for the parser's sake.
00021 YY_DECL;
00022 // Conducting the whole scanning and parsing of Calc++.
00023
00024 //namespace GenesysKernel {
00025
00026 class genesyspp_driver {
00027 public:
00028     genesyspp_driver();
00029     genesyspp_driver(/*GenesysKernel::*/Model* model, Sampler_if* sampler, bool throws = false);
00030     virtual ~genesyspp_driver();
00031     genesyspp_driver(const genesyspp_driver& other);
00032     genesyspp_driver& operator=(const genesyspp_driver& other);
00033     genesyspp_driver(genesyspp_driver&& other) noexcept;
00034     genesyspp_driver& operator=(genesyspp_driver&& other) noexcept;
00035 public:
00036     // Handling the scanner.
00037     void scan_begin_file();
00038     void scan_end_file();
00039     void scan_begin_str();
00040     void scan_end_str();
00041
00042     // Run the parser on file F.
00043     // Return 0 on success.
00044     int parse_file(const std::string& f);
00045     int parse_str(const std::string& str);
00046
00047     // Error handling.
00048     void error(const yy::location& l, const std::string& m);
00049     void error(const std::string& m);
00050
00051     double getResult();
00052     void setResult(double value);
00053
00054     bool getThrowsException();
00055     void setThrowsException(bool throws);
00056
00057     void setErrorMessage(std::string message);
00058     std::string getErrorMessage();
00059
00060     /*GenesysKernel::*/Model* getModel();
00061     //Sampler_if* getProbs();
00062
```

```
00063     std::string getFile();
00064     void setFile(std::string f);
00065
00066     std::string getStrToParse();
00067     void setStrToParse(std::string str);
00068
00069 public: // Sampler
00070     void setSampler(Sampler_if* _sampler);
00071     Sampler_if* getSampler() const;
00072
00073 public: // trying to get infos about ModelDataElements refered in expressions (so they are not
      considered orphans
00074     void setRegisterReferedDataElements(bool value);
00075     bool getRegisterReferedDataElements();
00076     std::map<std::string, std::list<std::string>*>* getReferedDataElements();
00077     void clearReferedDataElements();
00078     void addRefered(std::pair<std::string,std::string> referedElement);
00079
00080 public: // SimulationResponse / SimulationControl helpers for parser semantics
00081     SimulationResponse* findSimulationResponse(const std::string& name) const;
00082     SimulationControl* findSimulationControl(const std::string& name) const;
00083     double getSimulationResponseValueAsDouble(const std::string& name) const;
00084     double getSimulationControlValueAsDouble(const std::string& name) const;
00085     double stringToDoubleOrWarn(const std::string& sourceType, const std::string& symbolName, const
      std::string& valueText) const;
00086     void traceWarning(const std::string& message) const;
00087
00088 private:
00089     /*GenesysKernel::*/Model* _model = nullptr;
00090     Sampler_if* _sampler = nullptr;
00091     std::map<std::string, std::list<std::string>*>* _referedDataElements = new std::map<std::string,
      std::list<std::string>*>(); // maps each dataelement class referenced to a list of referenced names
00092     bool _isRegisterReferedDataElements = false;
00093 private:
00094     double result = 0;
00095     std::string file;
00096     std::string str_to_parse;
00097     bool throwsException = true;
00098     std::string errorMessage;
00099 };
00100 //}
00101 #endif // ! GENESYSPP_DRIVER_H
```

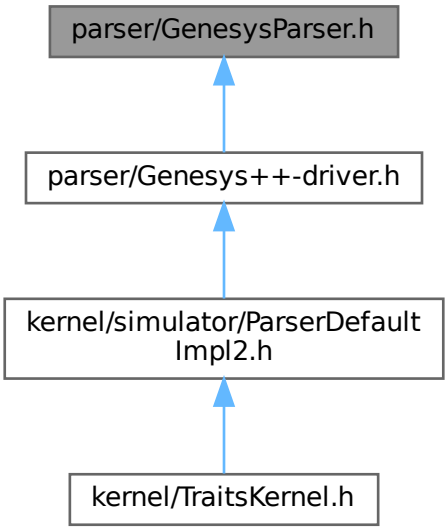
18.366 parser/GenesysParser.h File Reference

```
#include <string>
#include <cmath>
#include <algorithm>
#include "obj_t.h"
#include "kernel/util/Util.h"
#include "kernel/simulator/essentialPlugins/Attribute.h"
#include "kernel/simulator/essentialPlugins/Counter.h"
#include "plugins/data/Logic/Variable.h"
#include "plugins/data/DiscreteProcessing/Queue.h"
#include "plugins/data/Logic/Formula.h"
#include "plugins/data/DiscreteProcessing/Resource.h"
#include "plugins/data/Logic/Set.h"
#include <cassert>
#include <cstdlib>
#include <iostream>
#include <stdexcept>
#include <vector>
#include "location.hh"
#include <typeinfo>
```

Include dependency graph for GenesysParser.h:



This graph shows which files directly or indirectly include this file:



Classes

- class yy::genesyspp_parser
  A Bison parser.
- class yy::genesyspp_parser::value_type
- struct yy::genesyspp_parser::syntax_error
  Syntax errors thrown from user actions.
- struct yy::genesyspp_parser::token
  Token kinds.
- struct yy::genesyspp_parser::symbol_kind
  Symbol kinds.
- struct yy::genesyspp_parser::basic_symbol< Base >
- struct yy::genesyspp_parser::by_kind
  Type access provider for token (enum) based symbols.
- struct yy::genesyspp_parser::symbol_type
  "External" symbols: returned by the scanner.
- class yy::genesyspp_parser::context
- class yy::genesyspp_parser::stack< T, S >::slice
  Present a slice of the top of a stack.

Macros

- #define **YY_CPLUSPLUS** 199711L
- #define **YY_MOVE**
- #define **YY_MOVE_OR_COPY** copy
- #define **YY_MOVE_REF**(Type) Type&
- #define **YY_RVREF**(Type) const Type&
- #define **YY_COPY**(Type) const Type&
- #define **YY_NOEXCEPT**
- #define **YY_NOTHROW** throw ()
- #define **YY_CONSTEXPR**
- #define **YY_ASSERT** assert
- #define **YY_ATTRIBUTE_PURE**
- #define **YY_ATTRIBUTE_UNUSED**
- #define **YY_USE**(E) ((void) (E))
- #define **YY_INITIAL_VALUE**(Value) Value
- #define **YY_IGNORE_MAYBE_UNINITIALIZED_BEGIN**
- #define **YY_IGNORE_MAYBE_UNINITIALIZED_END**
- #define **YY_IGNORE_USELESS_CAST_BEGIN**
- #define **YY_IGNORE_USELESS_CAST_END**
- #define **YY_CAST**(Type, Val) ((Type) (Val))
- #define **YY_REINTERPRET_CAST**(Type, Val) ((Type) (Val))
- #define **YYDEBUG** 1

18.366.1 Detailed Description

Define the yy::parser class.

18.367 GenesysParser.h

Go to the documentation of this file.

```
00001 // A Bison parser, made by GNU Bison 3.8.2.
00002
00003 // Skeleton interface for Bison LALR(1) parsers in C++
00004
00005 // Copyright (C) 2002-2015, 2018-2021 Free Software Foundation, Inc.
00006
00007 // This program is free software: you can redistribute it and/or modify
00008 // it under the terms of the GNU General Public License as published by
00009 // the Free Software Foundation, either version 3 of the License, or
00010 // (at your option) any later version.
00011
00012 // This program is distributed in the hope that it will be useful,
00013 // but WITHOUT ANY WARRANTY; without even the implied warranty of
00014 // MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
00015 // GNU General Public License for more details.
00016
00017 // You should have received a copy of the GNU General Public License
00018 // along with this program.  If not, see <https://www.gnu.org/licenses/>.
00019
00020 // As a special exception, you may create a larger work that contains
00021 // part or all of the Bison parser skeleton and distribute that work
00022 // under terms of your choice, so long as that work isn't itself a
00023 // parser generator using the skeleton or a modified version thereof
00024 // as a parser skeleton.  Alternatively, if you modify or redistribute
00025 // the parser skeleton itself, you may (at your option) remove this
00026 // special exception, which will cause the skeleton and the resulting
00027 // Bison output files to be licensed under the GNU General Public
00028 // License without this special exception.
00029
00030 // This special exception was added by the Free Software Foundation in
00031 // version 2.2 of Bison.
00032
00033
00039 // C++ LALR(1) parser skeleton written by Akim Demaille.
00040
00041 // DO NOT RELY ON FEATURES THAT ARE NOT DOCUMENTED in the manual,
00042 // especially those whose name start with YY_ or yy_.  They are
00043 // private implementation details that can be changed or removed.
```

```
00044
00045 #ifndef YY_YY_GENESYSPARSER_H_INCLUDED
00046 # define YY_YY_GENESYSPARSER_H_INCLUDED
00047 // "%code requires" blocks.
00048 #line 9 "bisonparser.yy"
00049
00050     #include <string>
00051     #include <cmath>
00052     #include <algorithm>
00053     #include "obj_t.h"
00054     #include "kernel/util/Util.h"
00055     #include "kernel/simulator/essentialPlugins/Attribute.h"
00056     #include "kernel/simulator/essentialPlugins/Counter.h"
00057
00058     /****begin_Includes_plugins****/
00059
00061         #include "plugins/data/Logic/Variable.h"
00065         #include "plugins/data/DiscreteProcessing/Queue.h"
00069         #include "plugins/data/Logic/Formula.h"
00073         #include "plugins/data/DiscreteProcessing/Resource.h"
00077         #include "plugins/data/Logic/Set.h"
00080     /****end_Includes_plugins****/
00081
00082 #ifdef YYDEBUG
00083   yydebug = 1;
00084 #endif
00085
00086     class genesyspp_driver;
00087
00088 #line 89 "../GenesysParser.h"
00089
00090 # include <cassert>
00091 # include <cstdlib> // std::abort
00092 # include <iostream>
00093 # include <stdexcept>
00094 # include <string>
00095 # include <vector>
00096
00097 #if defined __cplusplus
00098 # define YY_CPLUSPLUS __cplusplus
00099 #else
00100 # define YY_CPLUSPLUS 199711L
00101 #endif
00102
00103 // Support move semantics when possible.
00104 #if 201103L <= YY_CPLUSPLUS
00105 # define YY_MOVE           std::move
00106 # define YY_MOVE_OR_COPY   move
00107 # define YY_MOVE_REF(Type) Type&&
00108 # define YY_RVREF(Type)    Type&&
00109 # define YY_COPY(Type)     Type
00110 #else
00111 # define YY_MOVE
00112 # define YY_MOVE_OR_COPY   copy
00113 # define YY_MOVE_REF(Type) Type&
00114 # define YY_RVREF(Type)    const Type&
00115 # define YY_COPY(Type)     const Type&
00116 #endif
00117
00118 // Support noexcept when possible.
00119 #if 201103L <= YY_CPLUSPLUS
00120 # define YY_NOEXCEPT noexcept
00121 # define YY_NOTHROW
00122 #else
00123 # define YY_NOEXCEPT
00124 # define YY_NOTHROW throw ()
00125 #endif
00126
00127 // Support constexpr when possible.
00128 #if 201703 <= YY_CPLUSPLUS
00129 # define YY_CONSTEXPR constexpr
00130 #else
00131 # define YY_CONSTEXPR
00132 #endif
00133 # include "location.hh"
00134 #include <typeinfo>
00135 #ifndef YY_ASSERT
00136 # include <cassert>
00137 # define YY_ASSERT assert
00138 #endif
00139
00140
00141 #ifndef YY_ATTRIBUTE_PURE
00142 # if defined __GNUC__ && 2 < __GNUC__ + (96 <= __GNUC_MINOR__)
00143 #  define YY_ATTRIBUTE_PURE __attribute__ ((__pure__))
00144 # else
00145 #  define YY_ATTRIBUTE_PURE
```

```
00146 # endif
00147 #endif
00148
00149 #ifndef YY_ATTRIBUTE_UNUSED
00150 # if defined __GNUC__ && 2 < __GNUC__ + (7 <= __GNUC_MINOR__)
00151 #  define YY_ATTRIBUTE_UNUSED __attribute__ ((__unused__))
00152 # else
00153 #  define YY_ATTRIBUTE_UNUSED
00154 # endif
00155 #endif
00156
00157 /* Suppress unused-variable warnings by "using" E.  */
00158 #if ! defined lint || defined __GNUC__
00159 # define YY_USE(E) ((void) (E))
00160 #else
00161 # define YY_USE(E) /* empty */
00162 #endif
00163
00164 /* Suppress an incorrect diagnostic about yylval being uninitialized.  */
00165 #if defined __GNUC__ && ! defined __ICC && 406 <= __GNUC__ * 100 + __GNUC_MINOR__
00166 # if __GNUC__ * 100 + __GNUC_MINOR__ < 407
00167 #  define YY_IGNORE_MAYBE_UNINITIALIZED_BEGIN                           \
00168     _Pragma ("GCC diagnostic push")                                     \
00169     _Pragma ("GCC diagnostic ignored \"-Wuninitialized\"")
00170 # else
00171 #  define YY_IGNORE_MAYBE_UNINITIALIZED_BEGIN                           \
00172     _Pragma ("GCC diagnostic push")                                     \
00173     _Pragma ("GCC diagnostic ignored \"-Wuninitialized\"")              \
00174     _Pragma ("GCC diagnostic ignored \"-Wmaybe-uninitialized\"")
00175 # endif
00176 # define YY_IGNORE_MAYBE_UNINITIALIZED_END      \
00177     _Pragma ("GCC diagnostic pop")
00178 #else
00179 # define YY_INITIAL_VALUE(Value) Value
00180 #endif
00181 #ifndef YY_IGNORE_MAYBE_UNINITIALIZED_BEGIN
00182 # define YY_IGNORE_MAYBE_UNINITIALIZED_BEGIN
00183 # define YY_IGNORE_MAYBE_UNINITIALIZED_END
00184 #endif
00185 #ifndef YY_INITIAL_VALUE
00186 # define YY_INITIAL_VALUE(Value) /* Nothing. */
00187 #endif
00188
00189 #if defined __cplusplus && defined __GNUC__ && ! defined __ICC && 6 <= __GNUC__
00190 # define YY_IGNORE_USELESS_CAST_BEGIN                          \
00191     _Pragma ("GCC diagnostic push")                            \
00192     _Pragma ("GCC diagnostic ignored \"-Wuseless-cast\"")
00193 # define YY_IGNORE_USELESS_CAST_END            \
00194     _Pragma ("GCC diagnostic pop")
00195 #endif
00196 #ifndef YY_IGNORE_USELESS_CAST_BEGIN
00197 # define YY_IGNORE_USELESS_CAST_BEGIN
00198 # define YY_IGNORE_USELESS_CAST_END
00199 #endif
00200
00201 # ifndef YY_CAST
00202 #  ifdef __cplusplus
00203 #   define YY_CAST(Type, Val) static_cast<Type> (Val)
00204 #   define YY_REINTERPRET_CAST(Type, Val) reinterpret_cast<Type> (Val)
00205 #  else
00206 #   define YY_CAST(Type, Val) ((Type) (Val))
00207 #   define YY_REINTERPRET_CAST(Type, Val) ((Type) (Val))
00208 #  endif
00209 # endif
00210 # ifndef YY_NULLPTR
00211 #  if defined __cplusplus
00212 #   if 201103L <= __cplusplus
00213 #    define YY_NULLPTR nullptr
00214 #   else
00215 #    define YY_NULLPTR 0
00216 #   endif
00217 #  else
00218 #   define YY_NULLPTR ((void*)0)
00219 #  endif
00220 # endif
00221
00222 /* Debug traces.  */
00223 #ifndef YYDEBUG
00224 # define YYDEBUG 1
00225 #endif
00226
00227 namespace yy {
00228 #line 229 "../GenesysParser.h"
00229
00230
00231
00232
```

```
00234   class genesyspp_parser
00235   {
00236   public:
00237 #ifdef YYSTYPE
00238 # ifdef __GNUC__
00239 #  pragma GCC message "bison: do not #define YYSTYPE in C++, use %define api.value.type"
00240 # endif
00241     typedef YYSTYPE value_type;
00242 #else
00248   class value_type
00249   {
00250   public:
00252     typedef value_type self_type;
00253
00255     value_type () YY_NOEXCEPT
00256       : yyraw_ ()
00257       , yytypeid_ (YY_NULLPTR)
00258     {}
00259
00261     template <typename T>
00262     value_type (YY_RVREF (T) t)
00263       : yytypeid_ (&typeid (T))
00264     {
00265       YY_ASSERT (sizeof (T) <= size);
00266       new (yyas_<T> ()) T (YY_MOVE (t));
00267     }
00268
00269 #if 201103L <= YY_CPLUSPLUS
00271     value_type (const self_type&) = delete;
00273     self_type& operator= (const self_type&) = delete;
00274 #endif
00275
00277     ~value_type () YY_NOEXCEPT
00278     {
00279       YY_ASSERT (!yytypeid_);
00280     }
00281
00282 # if 201103L <= YY_CPLUSPLUS
00284     template <typename T, typename... U>
00285     T&
00286     emplace (U&&... u)
00287     {
00288       YY_ASSERT (!yytypeid_);
00289       YY_ASSERT (sizeof (T) <= size);
00290       yytypeid_ = & typeid (T);
00291       return *new (yyas_<T> ()) T (std::forward <U>(u)...);
00292     }
00293 # else
00295     template <typename T>
00296     T&
00297     emplace ()
00298     {
00299       YY_ASSERT (!yytypeid_);
00300       YY_ASSERT (sizeof (T) <= size);
00301       yytypeid_ = & typeid (T);
00302       return *new (yyas_<T> ()) T ();
00303     }
00304
00306     template <typename T>
00307     T&
00308     emplace (const T& t)
00309     {
00310       YY_ASSERT (!yytypeid_);
00311       YY_ASSERT (sizeof (T) <= size);
00312       yytypeid_ = & typeid (T);
00313       return *new (yyas_<T> ()) T (t);
00314     }
00315 # endif
00316
00319     template <typename T>
00320     T&
00321     build ()
00322     {
00323       return emplace<T> ();
00324     }
00325
00328     template <typename T>
00329     T&
00330     build (const T& t)
00331     {
00332       return emplace<T> (t);
00333     }
00334
00336     template <typename T>
00337     T&
00338     as () YY_NOEXCEPT
00339     {
```

```
00340       YY_ASSERT (yytypeid_);
00341       YY_ASSERT (*yytypeid_ == typeid (T));
00342       YY_ASSERT (sizeof (T) <= size);
00343       return *yyas_<T> ();
00344     }
00345
00347     template <typename T>
00348     const T&
00349     as () const YY_NOEXCEPT
00350     {
00351       YY_ASSERT (yytypeid_);
00352       YY_ASSERT (*yytypeid_ == typeid (T));
00353       YY_ASSERT (sizeof (T) <= size);
00354       return *yyas_<T> ();
00355     }
00356
00365     template <typename T>
00366     void
00367     swap (self_type& that) YY_NOEXCEPT
00368     {
00369       YY_ASSERT (yytypeid_);
00370       YY_ASSERT (*yytypeid_ == *that.yytypeid_);
00371       std::swap (as<T> (), that.as<T> ());
00372     }
00373
00377     template <typename T>
00378     void
00379     move (self_type& that)
00380     {
00381 # if 201103L <= YY_CPLUSPLUS
00382       emplace<T> (std::move (that.as<T> ()));
00383 # else
00384       emplace<T> ();
00385       swap<T> (that);
00386 # endif
00387       that.destroy<T> ();
00388     }
00389
00390 # if 201103L <= YY_CPLUSPLUS
00392     template <typename T>
00393     void
00394     move (self_type&& that)
00395     {
00396       emplace<T> (std::move (that.as<T> ()));
00397       that.destroy<T> ();
00398     }
00399 #endif
00400
00402     template <typename T>
00403     void
00404     copy (const self_type& that)
00405     {
00406       emplace<T> (that.as<T> ());
00407     }
00408
00410     template <typename T>
00411     void
00412     destroy ()
00413     {
00414       as<T> ().~T ();
00415       yytypeid_ = YY_NULLPTR;
00416     }
00417
00418   private:
00419 #if YY_CPLUSPLUS < 201103L
00421     value_type (const self_type&);
00423     self_type& operator= (const self_type&);
00424 #endif
00425
00427     template <typename T>
00428     T*
00429     yyas_ () YY_NOEXCEPT
00430     {
00431       void *yyp = yyraw_;
00432       return static_cast<T*> (yyp);
00433      }
00434
00436     template <typename T>
00437     const T*
00438     yyas_ () const YY_NOEXCEPT
00439     {
00440       const void *yyp = yyraw_;
00441       return static_cast<const T*> (yyp);
00442      }
00443
00445     union union_type
00446     {
```

```
00447       // NUMD
00448       // NUMH
00449       // CTEZERO
00450       // oLE
00451       // oGE
00452       // oEQ
00453       // oNE
00454       // oAND
00455       // oOR
00456       // oNAND
00457       // oXOR
00458       // oNOT
00459       // fSIN
00460       // fCOS
00461       // fROUND
00462       // fMOD
00463       // fTRUNC
00464       // fFRAC
00465       // fEXP
00466       // fSQRT
00467       // fLOG
00468       // fLN
00469       // mathMIN
00470       // mathMAX
00471       // fVAL
00472       // fEVAL
00473       // fLENG
00474       // fRND1
00475       // fEXPO
00476       // fNORM
00477       // fUNIF
00478       // fWEIB
00479       // fLOGN
00480       // fGAMM
00481       // fERLA
00482       // fTRIA
00483       // fBETA
00484       // fDISC
00485       // fTNOW
00486       // fTFIN
00487       // fMAXREP
00488       // fNUMREP
00489       // fIDENT
00490       // simulEntitiesWIP
00491       // cIF
00492       // cELSE
00493       // cFOR
00494       // cTO
00495       // cDO
00496       // ATRIB
00497       // CSTAT
00498       // COUNTER
00499       // SIMRESP
00500       // SIMCTRL
00501       // fTAVG
00502       // fCOUNT
00503       // ILLEGAL
00504       // RESOURCE
00505       // fNR
00506       // fMR
00507       // fIRF
00508       // fRESSEIZES
00509       // fSTATE
00510       // fSETSUM
00511       // fRESUTIL
00512       // QUEUE
00513       // fNQ
00514       // fFIRSTINQ
00515       // fLASTINQ
00516       // fSAQUE
00517       // fAQUE
00518       // fENTATRANK
00519       // SET
00520       // fNUMSET
00521       // VARI
00522       // FORM
00523       // fNUMGR
00524       // fATRGR
00525       // input
00526       // expression
00527       // logicalOr
00528       // logicalXor
00529       // logicalAnd
00530       // logicalNot
00531       // relational
00532       // additive
00533       // multiplicative
```

```
00534       // power
00535       // unary
00536       // primary
00537       // number
00538       // command
00539       // commandIF
00540       // commandFOR
00541       // function
00542       // kernelFunction
00543       // elementFunction
00544       // trigonFunction
00545       // mathFunction
00546       // probFunction
00547       // userFunction
00548       // listaparm
00549       // illegal
00550       // attribute
00551       // simulationResponse
00552       // simulationControl
00553       // variable
00554       // formula
00555       // assigment
00556       // pluginFunction
00557       char dummy1[sizeof (obj_t)];
00558
00559       // indexList
00560       char dummy2[sizeof (std::string)];
00561     };
00562
00564     enum { size = sizeof (union_type) };
00565
00567     union
00568     {
00570       long double yyalign_me_;
00572       char yyraw_[size];
00573     };
00574
00576     const std::type_info *yytypeid_;
00577   };
00578
00579 #endif
00581     typedef value_type semantic_type;
00582
00584     typedef location location_type;
00585
00587     struct syntax_error : std::runtime_error
00588     {
00589       syntax_error (const location_type& l, const std::string& m)
00590         : std::runtime_error (m)
00591         , location (l)
00592       {}
00593
00594       syntax_error (const syntax_error& s)
00595         : std::runtime_error (s.what ())
00596         , location (s.location)
00597       {}
00598
00599       ~syntax_error () YY_NOEXCEPT YY_NOTHROW;
00600
00601       location_type location;
00602     };
00603
00605     struct token
00606     {
00607       enum token_kind_type
00608       {
00609         YYEMPTY = -2,
00610     END = 0,                        // "end of file"
00611     YYerror = 256,                  // error
00612     YYUNDEF = 257,                  // "invalid token"
00613     NUMD = 258,                     // NUMD
00614     NUMH = 259,                     // NUMH
00615     CTEZERO = 260,                  // CTEZERO
00616     oLE = 261,                      // oLE
00617     oGE = 262,                      // oGE
00618     oEQ = 263,                      // oEQ
00619     oNE = 264,                      // oNE
00620     oAND = 265,                     // oAND
00621     oOR = 266,                      // oOR
00622     oNAND = 267,                    // oNAND
00623     oXOR = 268,                     // oXOR
00624     oNOT = 269,                     // oNOT
00625     fSIN = 270,                     // fSIN
00626     fCOS = 271,                     // fCOS
00627     fROUND = 272,                   // fROUND
00628     fMOD = 273,                     // fMOD
00629     fTRUNC = 274,                   // fTRUNC
```

```
00630     fFRAC = 275,                  // fFRAC
00631     fEXP = 276,                   // fEXP
00632     fSQRT = 277,                  // fSQRT
00633     fLOG = 278,                   // fLOG
00634     fLN = 279,                    // fLN
00635     mathMIN = 280,                // mathMIN
00636     mathMAX = 281,                // mathMAX
00637     fVAL = 282,                   // fVAL
00638     fEVAL = 283,                  // fEVAL
00639     fLENG = 284,                  // fLENG
00640     fRND1 = 285,                  // fRND1
00641     fEXPO = 286,                  // fEXPO
00642     fNORM = 287,                  // fNORM
00643     fUNIF = 288,                  // fUNIF
00644     fWEIB = 289,                  // fWEIB
00645     fLOGN = 290,                  // fLOGN
00646     fGAMM = 291,                  // fGAMM
00647     fERLA = 292,                  // fERLA
00648     fTRIA = 293,                  // fTRIA
00649     fBETA = 294,                  // fBETA
00650     fDISC = 295,                  // fDISC
00651     fTNOW = 296,                  // fTNOW
00652     fTFIN = 297,                  // fTFIN
00653     fMAXREP = 298,                // fMAXREP
00654     fNUMREP = 299,                // fNUMREP
00655     fIDENT = 300,                 // fIDENT
00656     simulEntitiesWIP = 301,       // simulEntitiesWIP
00657     cIF = 302,                    // cIF
00658     cELSE = 303,                  // cELSE
00659     cFOR = 304,                   // cFOR
00660     cTO = 305,                    // cTO
00661     cDO = 306,                    // cDO
00662     ATRIB = 307,                  // ATRIB
00663     CSTAT = 308,                  // CSTAT
00664     COUNTER = 309,                // COUNTER
00665     SIMRESP = 310,                // SIMRESP
00666     SIMCTRL = 311,                // SIMCTRL
00667     fTAVG = 312,                  // fTAVG
00668     fCOUNT = 313,                 // fCOUNT
00669     ILLEGAL = 314,                // ILLEGAL
00670     RESOURCE = 315,               // RESOURCE
00671     fNR = 316,                    // fNR
00672     fMR = 317,                    // fMR
00673     fIRF = 318,                   // fIRF
00674     fRESSEIZES = 319,             // fRESSEIZES
00675     fSTATE = 320,                 // fSTATE
00676     fSETSUM = 321,                // fSETSUM
00677     fRESUTIL = 322,               // fRESUTIL
00678     QUEUE = 323,                  // QUEUE
00679     fNQ = 324,                    // fNQ
00680     fFIRSTINQ = 325,              // fFIRSTINQ
00681     fLASTINQ = 326,               // fLASTINQ
00682     fSAQUE = 327,                 // fSAQUE
00683     fAQUE = 328,                  // fAQUE
00684     fENTATRANK = 329,             // fENTATRANK
00685     SET = 330,                    // SET
00686     fNUMSET = 331,                // fNUMSET
00687     VARI = 332,                   // VARI
00688     FORM = 333,                   // FORM
00689     fNUMGR = 334,                 // fNUMGR
00690     fATRGR = 335,                 // fATRGR
00691     LPAREN = 336,                 // "("
00692     RPAREN = 337,                 // ")"
00693     LBRACKET = 338,               // "["
00694     RBRACKET = 339,               // "]"
00695     PLUS = 340,                   // "+"
00696     MINUS = 341,                  // "-"
00697     STAR = 342,                   // "*"
00698     POWER = 343,                  // "^"
00699     SLASH = 344,                  // "/"
00700     LESS = 345,                   // "<"
00701     GREATER = 346,                // ">"
00702     ASSIGN = 347,                 // "="
00703     COMMA = 348                   // ","
00704       };
00706       typedef token_kind_type yytokentype;
00707     };
00708
00710     typedef token::token_kind_type token_kind_type;
00711
00713     typedef token_kind_type token_type;
00714
00716     struct symbol_kind
00717     {
00718       enum symbol_kind_type
00719       {
00720         YYNTOKENS = 95,
```

```
00721         S_YYEMPTY = -2,
00722         S_YYEOF = 0,                            // "end of file"
00723         S_YYerror = 1,                          // error
00724         S_YYUNDEF = 2,                          // "invalid token"
00725         S_NUMD = 3,                             // NUMD
00726         S_NUMH = 4,                             // NUMH
00727         S_CTEZERO = 5,                          // CTEZERO
00728         S_oLE = 6,                              // oLE
00729         S_oGE = 7,                              // oGE
00730         S_oEQ = 8,                              // oEQ
00731         S_oNE = 9,                              // oNE
00732         S_oAND = 10,                            // oAND
00733         S_oOR = 11,                             // oOR
00734         S_oNAND = 12,                           // oNAND
00735         S_oXOR = 13,                            // oXOR
00736         S_oNOT = 14,                            // oNOT
00737         S_fSIN = 15,                            // fSIN
00738         S_fCOS = 16,                            // fCOS
00739         S_fROUND = 17,                          // fROUND
00740         S_fMOD = 18,                            // fMOD
00741         S_fTRUNC = 19,                          // fTRUNC
00742         S_fFRAC = 20,                           // fFRAC
00743         S_fEXP = 21,                            // fEXP
00744         S_fSQRT = 22,                           // fSQRT
00745         S_fLOG = 23,                            // fLOG
00746         S_fLN = 24,                             // fLN
00747         S_mathMIN = 25,                         // mathMIN
00748         S_mathMAX = 26,                         // mathMAX
00749         S_fVAL = 27,                            // fVAL
00750         S_fEVAL = 28,                           // fEVAL
00751         S_fLENG = 29,                           // fLENG
00752         S_fRND1 = 30,                           // fRND1
00753         S_fEXPO = 31,                           // fEXPO
00754         S_fNORM = 32,                           // fNORM
00755         S_fUNIF = 33,                           // fUNIF
00756         S_fWEIB = 34,                           // fWEIB
00757         S_fLOGN = 35,                           // fLOGN
00758         S_fGAMM = 36,                           // fGAMM
00759         S_fERLA = 37,                           // fERLA
00760         S_fTRIA = 38,                           // fTRIA
00761         S_fBETA = 39,                           // fBETA
00762         S_fDISC = 40,                           // fDISC
00763         S_fTNOW = 41,                           // fTNOW
00764         S_fTFIN = 42,                           // fTFIN
00765         S_fMAXREP = 43,                         // fMAXREP
00766         S_fNUMREP = 44,                         // fNUMREP
00767         S_fIDENT = 45,                          // fIDENT
00768         S_simulEntitiesWIP = 46,                // simulEntitiesWIP
00769         S_cIF = 47,                             // cIF
00770         S_cELSE = 48,                           // cELSE
00771         S_cFOR = 49,                            // cFOR
00772         S_cTO = 50,                             // cTO
00773         S_cDO = 51,                             // cDO
00774         S_ATRIB = 52,                           // ATRIB
00775         S_CSTAT = 53,                           // CSTAT
00776         S_COUNTER = 54,                         // COUNTER
00777         S_SIMRESP = 55,                         // SIMRESP
00778         S_SIMCTRL = 56,                         // SIMCTRL
00779         S_fTAVG = 57,                           // fTAVG
00780         S_fCOUNT = 58,                          // fCOUNT
00781         S_ILLEGAL = 59,                         // ILLEGAL
00782         S_RESOURCE = 60,                        // RESOURCE
00783         S_fNR = 61,                             // fNR
00784         S_fMR = 62,                             // fMR
00785         S_fIRF = 63,                            // fIRF
00786         S_fRESSEIZES = 64,                      // fRESSEIZES
00787         S_fSTATE = 65,                          // fSTATE
00788         S_fSETSUM = 66,                         // fSETSUM
00789         S_fRESUTIL = 67,                        // fRESUTIL
00790         S_QUEUE = 68,                           // QUEUE
00791         S_fNQ = 69,                             // fNQ
00792         S_fFIRSTINQ = 70,                       // fFIRSTINQ
00793         S_fLASTINQ = 71,                        // fLASTINQ
00794         S_fSAQUE = 72,                          // fSAQUE
00795         S_fAQUE = 73,                           // fAQUE
00796         S_fENTATRANK = 74,                      // fENTATRANK
00797         S_SET = 75,                             // SET
00798         S_fNUMSET = 76,                         // fNUMSET
00799         S_VARI = 77,                            // VARI
00800         S_FORM = 78,                            // FORM
00801         S_fNUMGR = 79,                          // fNUMGR
00802         S_fATRGR = 80,                          // fATRGR
00803         S_LPAREN = 81,                          // "("
00804         S_RPAREN = 82,                          // ")"
00805         S_LBRACKET = 83,                        // "["
00806         S_RBRACKET = 84,                        // "]"
00807         S_PLUS = 85,                            // "+"
```

```
00808         S_MINUS = 86,                           // "-"
00809         S_STAR = 87,                            // "*"
00810         S_POWER = 88,                           // "^"
00811         S_SLASH = 89,                           // "/"
00812         S_LESS = 90,                            // "<"
00813         S_GREATER = 91,                         // ">"
00814         S_ASSIGN = 92,                          // "="
00815         S_COMMA = 93,                           // ","
00816         S_94_USER_ = 94,                        // "USER"
00817         S_YYACCEPT = 95,                        // $accept
00818         S_input = 96,                           // input
00819         S_expression = 97,                      // expression
00820         S_logicalOr = 98,                       // logicalOr
00821         S_logicalXor = 99,                      // logicalXor
00822         S_logicalAnd = 100,                     // logicalAnd
00823         S_logicalNot = 101,                     // logicalNot
00824         S_relational = 102,                     // relational
00825         S_additive = 103,                       // additive
00826         S_multiplicative = 104,                 // multiplicative
00827         S_power = 105,                          // power
00828         S_unary = 106,                          // unary
00829         S_primary = 107,                        // primary
00830         S_number = 108,                         // number
00831         S_command = 109,                        // command
00832         S_commandIF = 110,                      // commandIF
00833         S_commandFOR = 111,                     // commandFOR
00834         S_function = 112,                       // function
00835         S_kernelFunction = 113,                 // kernelFunction
00836         S_elementFunction = 114,                // elementFunction
00837         S_trigonFunction = 115,                 // trigonFunction
00838         S_mathFunction = 116,                   // mathFunction
00839         S_probFunction = 117,                   // probFunction
00840         S_userFunction = 118,                   // userFunction
00841         S_listaparm = 119,                      // listaparm
00842         S_indexList = 120,                      // indexList
00843         S_illegal = 121,                        // illegal
00844         S_attribute = 122,                      // attribute
00845         S_simulationResponse = 123,             // simulationResponse
00846         S_simulationControl = 124,              // simulationControl
00847         S_variable = 125,                       // variable
00848         S_formula = 126,                        // formula
00849         S_assigment = 127,                      // assigment
00850         S_pluginFunction = 128                  // pluginFunction
00851       };
00852     };
00853
00855     typedef symbol_kind::symbol_kind_type symbol_kind_type;
00856
00858     static const symbol_kind_type YYNTOKENS = symbol_kind::YYNTOKENS;
00859
00866     template <typename Base>
00867     struct basic_symbol : Base
00868     {
00870       typedef Base super_type;
00871
00873       basic_symbol () YY_NOEXCEPT
00874         : value ()
00875         , location ()
00876       {}
00877
00878 #if 201103L <= YY_CPLUSPLUS
00880       basic_symbol (basic_symbol&& that)
00881         : Base (std::move (that))
00882         , value ()
00883         , location (std::move (that.location))
00884       {
00885         switch (this->kind ())
00886     {
00887       case symbol_kind::S_NUMD: // NUMD
00888       case symbol_kind::S_NUMH: // NUMH
00889       case symbol_kind::S_CTEZERO: // CTEZERO
00890       case symbol_kind::S_oLE: // oLE
00891       case symbol_kind::S_oGE: // oGE
00892       case symbol_kind::S_oEQ: // oEQ
00893       case symbol_kind::S_oNE: // oNE
00894       case symbol_kind::S_oAND: // oAND
00895       case symbol_kind::S_oOR: // oOR
00896       case symbol_kind::S_oNAND: // oNAND
00897       case symbol_kind::S_oXOR: // oXOR
00898       case symbol_kind::S_oNOT: // oNOT
00899       case symbol_kind::S_fSIN: // fSIN
00900       case symbol_kind::S_fCOS: // fCOS
00901       case symbol_kind::S_fROUND: // fROUND
00902       case symbol_kind::S_fMOD: // fMOD
00903       case symbol_kind::S_fTRUNC: // fTRUNC
00904       case symbol_kind::S_fFRAC: // fFRAC
00905       case symbol_kind::S_fEXP: // fEXP
```

```
00906       case symbol_kind::S_fSQRT: // fSQRT
00907       case symbol_kind::S_fLOG: // fLOG
00908       case symbol_kind::S_fLN: // fLN
00909       case symbol_kind::S_mathMIN: // mathMIN
00910       case symbol_kind::S_mathMAX: // mathMAX
00911       case symbol_kind::S_fVAL: // fVAL
00912       case symbol_kind::S_fEVAL: // fEVAL
00913       case symbol_kind::S_fLENG: // fLENG
00914       case symbol_kind::S_fRND1: // fRND1
00915       case symbol_kind::S_fEXPO: // fEXPO
00916       case symbol_kind::S_fNORM: // fNORM
00917       case symbol_kind::S_fUNIF: // fUNIF
00918       case symbol_kind::S_fWEIB: // fWEIB
00919       case symbol_kind::S_fLOGN: // fLOGN
00920       case symbol_kind::S_fGAMM: // fGAMM
00921       case symbol_kind::S_fERLA: // fERLA
00922       case symbol_kind::S_fTRIA: // fTRIA
00923       case symbol_kind::S_fBETA: // fBETA
00924       case symbol_kind::S_fDISC: // fDISC
00925       case symbol_kind::S_fTNOW: // fTNOW
00926       case symbol_kind::S_fTFIN: // fTFIN
00927       case symbol_kind::S_fMAXREP: // fMAXREP
00928       case symbol_kind::S_fNUMREP: // fNUMREP
00929       case symbol_kind::S_fIDENT: // fIDENT
00930       case symbol_kind::S_simulEntitiesWIP: // simulEntitiesWIP
00931       case symbol_kind::S_cIF: // cIF
00932       case symbol_kind::S_cELSE: // cELSE
00933       case symbol_kind::S_cFOR: // cFOR
00934       case symbol_kind::S_cTO: // cTO
00935       case symbol_kind::S_cDO: // cDO
00936       case symbol_kind::S_ATRIB: // ATRIB
00937       case symbol_kind::S_CSTAT: // CSTAT
00938       case symbol_kind::S_COUNTER: // COUNTER
00939       case symbol_kind::S_SIMRESP: // SIMRESP
00940       case symbol_kind::S_SIMCTRL: // SIMCTRL
00941       case symbol_kind::S_fTAVG: // fTAVG
00942       case symbol_kind::S_fCOUNT: // fCOUNT
00943       case symbol_kind::S_ILLEGAL: // ILLEGAL
00944       case symbol_kind::S_RESOURCE: // RESOURCE
00945       case symbol_kind::S_fNR: // fNR
00946       case symbol_kind::S_fMR: // fMR
00947       case symbol_kind::S_fIRF: // fIRF
00948       case symbol_kind::S_fRESSEIZES: // fRESSEIZES
00949       case symbol_kind::S_fSTATE: // fSTATE
00950       case symbol_kind::S_fSETSUM: // fSETSUM
00951       case symbol_kind::S_fRESUTIL: // fRESUTIL
00952       case symbol_kind::S_QUEUE: // QUEUE
00953       case symbol_kind::S_fNQ: // fNQ
00954       case symbol_kind::S_fFIRSTINQ: // fFIRSTINQ
00955       case symbol_kind::S_fLASTINQ: // fLASTINQ
00956       case symbol_kind::S_fSAQUE: // fSAQUE
00957       case symbol_kind::S_fAQUE: // fAQUE
00958       case symbol_kind::S_fENTATRANK: // fENTATRANK
00959       case symbol_kind::S_SET: // SET
00960       case symbol_kind::S_fNUMSET: // fNUMSET
00961       case symbol_kind::S_VARI: // VARI
00962       case symbol_kind::S_FORM: // FORM
00963       case symbol_kind::S_fNUMGR: // fNUMGR
00964       case symbol_kind::S_fATRGR: // fATRGR
00965       case symbol_kind::S_input: // input
00966       case symbol_kind::S_expression: // expression
00967       case symbol_kind::S_logicalOr: // logicalOr
00968       case symbol_kind::S_logicalXor: // logicalXor
00969       case symbol_kind::S_logicalAnd: // logicalAnd
00970       case symbol_kind::S_logicalNot: // logicalNot
00971       case symbol_kind::S_relational: // relational
00972       case symbol_kind::S_additive: // additive
00973       case symbol_kind::S_multiplicative: // multiplicative
00974       case symbol_kind::S_power: // power
00975       case symbol_kind::S_unary: // unary
00976       case symbol_kind::S_primary: // primary
00977       case symbol_kind::S_number: // number
00978       case symbol_kind::S_command: // command
00979       case symbol_kind::S_commandIF: // commandIF
00980       case symbol_kind::S_commandFOR: // commandFOR
00981       case symbol_kind::S_function: // function
00982       case symbol_kind::S_kernelFunction: // kernelFunction
00983       case symbol_kind::S_elementFunction: // elementFunction
00984       case symbol_kind::S_trigonFunction: // trigonFunction
00985       case symbol_kind::S_mathFunction: // mathFunction
00986       case symbol_kind::S_probFunction: // probFunction
00987       case symbol_kind::S_userFunction: // userFunction
00988       case symbol_kind::S_listaparm: // listaparm
00989       case symbol_kind::S_illegal: // illegal
00990       case symbol_kind::S_attribute: // attribute
00991       case symbol_kind::S_simulationResponse: // simulationResponse
00992       case symbol_kind::S_simulationControl: // simulationControl
```

```
00993       case symbol_kind::S_variable: // variable
00994       case symbol_kind::S_formula: // formula
00995       case symbol_kind::S_assigment: // assigment
00996       case symbol_kind::S_pluginFunction: // pluginFunction
00997         value.move< obj_t > (std::move (that.value));
00998         break;
00999
01000       case symbol_kind::S_indexList: // indexList
01001         value.move< std::string > (std::move (that.value));
01002         break;
01003
01004       default:
01005         break;
01006     }
01007
01008       }
01009 #endif
01010
01012       basic_symbol (const basic_symbol& that);
01013
01015 #if 201103L <= YY_CPLUSPLUS
01016       basic_symbol (typename Base::kind_type t, location_type&& l)
01017         : Base (t)
01018         , location (std::move (l))
01019       {}
01020 #else
01021       basic_symbol (typename Base::kind_type t, const location_type& l)
01022         : Base (t)
01023         , location (l)
01024       {}
01025 #endif
01026
01027 #if 201103L <= YY_CPLUSPLUS
01028       basic_symbol (typename Base::kind_type t, obj_t&& v, location_type&& l)
01029         : Base (t)
01030         , value (std::move (v))
01031         , location (std::move (l))
01032       {}
01033 #else
01034       basic_symbol (typename Base::kind_type t, const obj_t& v, const location_type& l)
01035         : Base (t)
01036         , value (v)
01037         , location (l)
01038       {}
01039 #endif
01040
01041 #if 201103L <= YY_CPLUSPLUS
01042       basic_symbol (typename Base::kind_type t, std::string&& v, location_type&& l)
01043         : Base (t)
01044         , value (std::move (v))
01045         , location (std::move (l))
01046       {}
01047 #else
01048       basic_symbol (typename Base::kind_type t, const std::string& v, const location_type& l)
01049         : Base (t)
01050         , value (v)
01051         , location (l)
01052       {}
01053 #endif
01054
01056       ~basic_symbol ()
01057       {
01058         clear ();
01059       }
01060
01061
01062
01064       void clear () YY_NOEXCEPT
01065       {
01066         // User destructor.
01067         symbol_kind_type yykind = this->kind ();
01068         basic_symbol<Base>& yysym = *this;
01069         (void) yysym;
01070         switch (yykind)
01071         {
01072        default:
01073           break;
01074         }
01075
01076         // Value type destructor.
01077 switch (yykind)
01078     {
01079       case symbol_kind::S_NUMD: // NUMD
01080       case symbol_kind::S_NUMH: // NUMH
01081       case symbol_kind::S_CTEZERO: // CTEZERO
01082       case symbol_kind::S_oLE: // oLE
01083       case symbol_kind::S_oGE: // oGE
```

```
01084       case symbol_kind::S_oEQ: // oEQ
01085       case symbol_kind::S_oNE: // oNE
01086       case symbol_kind::S_oAND: // oAND
01087       case symbol_kind::S_oOR: // oOR
01088       case symbol_kind::S_oNAND: // oNAND
01089       case symbol_kind::S_oXOR: // oXOR
01090       case symbol_kind::S_oNOT: // oNOT
01091       case symbol_kind::S_fSIN: // fSIN
01092       case symbol_kind::S_fCOS: // fCOS
01093       case symbol_kind::S_fROUND: // fROUND
01094       case symbol_kind::S_fMOD: // fMOD
01095       case symbol_kind::S_fTRUNC: // fTRUNC
01096       case symbol_kind::S_fFRAC: // fFRAC
01097       case symbol_kind::S_fEXP: // fEXP
01098       case symbol_kind::S_fSQRT: // fSQRT
01099       case symbol_kind::S_fLOG: // fLOG
01100       case symbol_kind::S_fLN: // fLN
01101       case symbol_kind::S_mathMIN: // mathMIN
01102       case symbol_kind::S_mathMAX: // mathMAX
01103       case symbol_kind::S_fVAL: // fVAL
01104       case symbol_kind::S_fEVAL: // fEVAL
01105       case symbol_kind::S_fLENG: // fLENG
01106       case symbol_kind::S_fRND1: // fRND1
01107       case symbol_kind::S_fEXPO: // fEXPO
01108       case symbol_kind::S_fNORM: // fNORM
01109       case symbol_kind::S_fUNIF: // fUNIF
01110       case symbol_kind::S_fWEIB: // fWEIB
01111       case symbol_kind::S_fLOGN: // fLOGN
01112       case symbol_kind::S_fGAMM: // fGAMM
01113       case symbol_kind::S_fERLA: // fERLA
01114       case symbol_kind::S_fTRIA: // fTRIA
01115       case symbol_kind::S_fBETA: // fBETA
01116       case symbol_kind::S_fDISC: // fDISC
01117       case symbol_kind::S_fTNOW: // fTNOW
01118       case symbol_kind::S_fTFIN: // fTFIN
01119       case symbol_kind::S_fMAXREP: // fMAXREP
01120       case symbol_kind::S_fNUMREP: // fNUMREP
01121       case symbol_kind::S_fIDENT: // fIDENT
01122       case symbol_kind::S_simulEntitiesWIP: // simulEntitiesWIP
01123       case symbol_kind::S_cIF: // cIF
01124       case symbol_kind::S_cELSE: // cELSE
01125       case symbol_kind::S_cFOR: // cFOR
01126       case symbol_kind::S_cTO: // cTO
01127       case symbol_kind::S_cDO: // cDO
01128       case symbol_kind::S_ATRIB: // ATRIB
01129       case symbol_kind::S_CSTAT: // CSTAT
01130       case symbol_kind::S_COUNTER: // COUNTER
01131       case symbol_kind::S_SIMRESP: // SIMRESP
01132       case symbol_kind::S_SIMCTRL: // SIMCTRL
01133       case symbol_kind::S_fTAVG: // fTAVG
01134       case symbol_kind::S_fCOUNT: // fCOUNT
01135       case symbol_kind::S_ILLEGAL: // ILLEGAL
01136       case symbol_kind::S_RESOURCE: // RESOURCE
01137       case symbol_kind::S_fNR: // fNR
01138       case symbol_kind::S_fMR: // fMR
01139       case symbol_kind::S_fIRF: // fIRF
01140       case symbol_kind::S_fRESSEIZES: // fRESSEIZES
01141       case symbol_kind::S_fSTATE: // fSTATE
01142       case symbol_kind::S_fSETSUM: // fSETSUM
01143       case symbol_kind::S_fRESUTIL: // fRESUTIL
01144       case symbol_kind::S_QUEUE: // QUEUE
01145       case symbol_kind::S_fNQ: // fNQ
01146       case symbol_kind::S_fFIRSTINQ: // fFIRSTINQ
01147       case symbol_kind::S_fLASTINQ: // fLASTINQ
01148       case symbol_kind::S_fSAQUE: // fSAQUE
01149       case symbol_kind::S_fAQUE: // fAQUE
01150       case symbol_kind::S_fENTATRANK: // fENTATRANK
01151       case symbol_kind::S_SET: // SET
01152       case symbol_kind::S_fNUMSET: // fNUMSET
01153       case symbol_kind::S_VARI: // VARI
01154       case symbol_kind::S_FORM: // FORM
01155       case symbol_kind::S_fNUMGR: // fNUMGR
01156       case symbol_kind::S_fATRGR: // fATRGR
01157       case symbol_kind::S_input: // input
01158       case symbol_kind::S_expression: // expression
01159       case symbol_kind::S_logicalOr: // logicalOr
01160       case symbol_kind::S_logicalXor: // logicalXor
01161       case symbol_kind::S_logicalAnd: // logicalAnd
01162       case symbol_kind::S_logicalNot: // logicalNot
01163       case symbol_kind::S_relational: // relational
01164       case symbol_kind::S_additive: // additive
01165       case symbol_kind::S_multiplicative: // multiplicative
01166       case symbol_kind::S_power: // power
01167       case symbol_kind::S_unary: // unary
01168       case symbol_kind::S_primary: // primary
01169       case symbol_kind::S_number: // number
01170       case symbol_kind::S_command: // command
```

```
01171       case symbol_kind::S_commandIF: // commandIF
01172       case symbol_kind::S_commandFOR: // commandFOR
01173       case symbol_kind::S_function: // function
01174       case symbol_kind::S_kernelFunction: // kernelFunction
01175       case symbol_kind::S_elementFunction: // elementFunction
01176       case symbol_kind::S_trigonFunction: // trigonFunction
01177       case symbol_kind::S_mathFunction: // mathFunction
01178       case symbol_kind::S_probFunction: // probFunction
01179       case symbol_kind::S_userFunction: // userFunction
01180       case symbol_kind::S_listaparm: // listaparm
01181       case symbol_kind::S_illegal: // illegal
01182       case symbol_kind::S_attribute: // attribute
01183       case symbol_kind::S_simulationResponse: // simulationResponse
01184       case symbol_kind::S_simulationControl: // simulationControl
01185       case symbol_kind::S_variable: // variable
01186       case symbol_kind::S_formula: // formula
01187       case symbol_kind::S_assigment: // assigment
01188       case symbol_kind::S_pluginFunction: // pluginFunction
01189         value.template destroy< obj_t > ();
01190         break;
01191
01192       case symbol_kind::S_indexList: // indexList
01193         value.template destroy< std::string > ();
01194         break;
01195
01196       default:
01197         break;
01198     }
01199
01200         Base::clear ();
01201       }
01202
01204       std::string name () const YY_NOEXCEPT
01205       {
01206         return genesyspp_parser::symbol_name (this->kind ());
01207       }
01208
01210       symbol_kind_type type_get () const YY_NOEXCEPT;
01211
01213       bool empty () const YY_NOEXCEPT;
01214
01216       void move (basic_symbol& s);
01217
01219       value_type value;
01220
01222       location_type location;
01223
01224     private:
01225 #if YY_CPLUSPLUS < 201103L
01227       basic_symbol& operator= (const basic_symbol& that);
01228 #endif
01229     };
01230
01232     struct by_kind
01233     {
01235       typedef token_kind_type kind_type;
01236
01238       by_kind () YY_NOEXCEPT;
01239
01240 #if 201103L <= YY_CPLUSPLUS
01242       by_kind (by_kind&& that) YY_NOEXCEPT;
01243 #endif
01244
01246       by_kind (const by_kind& that) YY_NOEXCEPT;
01247
01249       by_kind (kind_type t) YY_NOEXCEPT;
01250
01251
01252
01254       void clear () YY_NOEXCEPT;
01255
01257       void move (by_kind& that);
01258
01261       symbol_kind_type kind () const YY_NOEXCEPT;
01262
01264       symbol_kind_type type_get () const YY_NOEXCEPT;
01265
01268       symbol_kind_type kind_;
01269     };
01270
01272     typedef by_kind by_type;
01273
01275     struct symbol_type : basic_symbol<by_kind>
01276     {
01278       typedef basic_symbol<by_kind> super_type;
01279
01281       symbol_type () YY_NOEXCEPT {}
```

```
01282
01284 #if 201103L <= YY_CPLUSPLUS
01285       symbol_type (int tok, location_type l)
01286         : super_type (token_kind_type (tok), std::move (l))
01287 #else
01288       symbol_type (int tok, const location_type& l)
01289         : super_type (token_kind_type (tok), l)
01290 #endif
01291       {
01292 #if !defined _MSC_VER || defined __clang__
01293         YY_ASSERT (tok == token::END
01294                    || (token::YYerror <= tok && tok <= token::YYUNDEF)
01295                    || (token::LPAREN <= tok && tok <= 349));
01296 #endif
01297       }
01298 #if 201103L <= YY_CPLUSPLUS
01299       symbol_type (int tok, obj_t v, location_type l)
01300         : super_type (token_kind_type (tok), std::move (v), std::move (l))
01301 #else
01302       symbol_type (int tok, const obj_t& v, const location_type& l)
01303         : super_type (token_kind_type (tok), v, l)
01304 #endif
01305       {
01306 #if !defined _MSC_VER || defined __clang__
01307         YY_ASSERT ((token::NUMD <= tok && tok <= token::fATRGR));
01308 #endif
01309       }
01310     };
01311
01313     genesyspp_parser (genesyspp_driver& driver_yyarg);
01314     virtual ~genesyspp_parser ();
01315
01316 #if 201103L <= YY_CPLUSPLUS
01318     genesyspp_parser (const genesyspp_parser&) = delete;
01320     genesyspp_parser& operator= (const genesyspp_parser&) = delete;
01321 #endif
01322
01325     int operator() ();
01326
01329     virtual int parse ();
01330
01331 #if YYDEBUG
01333     std::ostream& debug_stream () const YY_ATTRIBUTE_PURE;
01335     void set_debug_stream (std::ostream &);
01336
01338     typedef int debug_level_type;
01340     debug_level_type debug_level () const YY_ATTRIBUTE_PURE;
01342     void set_debug_level (debug_level_type l);
01343 #endif
01344
01348     virtual void error (const location_type& loc, const std::string& msg);
01349
01351     void error (const syntax_error& err);
01352
01355     static std::string symbol_name (symbol_kind_type yysymbol);
01356
01357     // Implementation of make_symbol for each token kind.
01358 #if 201103L <= YY_CPLUSPLUS
01359       static
01360       symbol_type
01361       make_END (location_type l)
01362       {
01363         return symbol_type (token::END, std::move (l));
01364       }
01365 #else
01366       static
01367       symbol_type
01368       make_END (const location_type& l)
01369       {
01370         return symbol_type (token::END, l);
01371       }
01372 #endif
01373 #if 201103L <= YY_CPLUSPLUS
01374       static
01375       symbol_type
01376       make_YYerror (location_type l)
01377       {
01378         return symbol_type (token::YYerror, std::move (l));
01379       }
01380 #else
01381       static
01382       symbol_type
01383       make_YYerror (const location_type& l)
01384       {
01385         return symbol_type (token::YYerror, l);
01386       }
01387 #endif
```

```
01388 #if 201103L <= YY_CPLUSPLUS
01389       static
01390       symbol_type
01391       make_YYUNDEF (location_type l)
01392       {
01393         return symbol_type (token::YYUNDEF, std::move (l));
01394       }
01395 #else
01396       static
01397       symbol_type
01398       make_YYUNDEF (const location_type& l)
01399       {
01400         return symbol_type (token::YYUNDEF, l);
01401       }
01402 #endif
01403 #if 201103L <= YY_CPLUSPLUS
01404       static
01405       symbol_type
01406       make_NUMD (obj_t v, location_type l)
01407       {
01408         return symbol_type (token::NUMD, std::move (v), std::move (l));
01409       }
01410 #else
01411       static
01412       symbol_type
01413       make_NUMD (const obj_t& v, const location_type& l)
01414       {
01415         return symbol_type (token::NUMD, v, l);
01416       }
01417 #endif
01418 #if 201103L <= YY_CPLUSPLUS
01419       static
01420       symbol_type
01421       make_NUMH (obj_t v, location_type l)
01422       {
01423         return symbol_type (token::NUMH, std::move (v), std::move (l));
01424       }
01425 #else
01426       static
01427       symbol_type
01428       make_NUMH (const obj_t& v, const location_type& l)
01429       {
01430         return symbol_type (token::NUMH, v, l);
01431       }
01432 #endif
01433 #if 201103L <= YY_CPLUSPLUS
01434       static
01435       symbol_type
01436       make_CTEZERO (obj_t v, location_type l)
01437       {
01438         return symbol_type (token::CTEZERO, std::move (v), std::move (l));
01439       }
01440 #else
01441       static
01442       symbol_type
01443       make_CTEZERO (const obj_t& v, const location_type& l)
01444       {
01445         return symbol_type (token::CTEZERO, v, l);
01446       }
01447 #endif
01448 #if 201103L <= YY_CPLUSPLUS
01449       static
01450       symbol_type
01451       make_oLE (obj_t v, location_type l)
01452       {
01453         return symbol_type (token::oLE, std::move (v), std::move (l));
01454       }
01455 #else
01456       static
01457       symbol_type
01458       make_oLE (const obj_t& v, const location_type& l)
01459       {
01460         return symbol_type (token::oLE, v, l);
01461       }
01462 #endif
01463 #if 201103L <= YY_CPLUSPLUS
01464       static
01465       symbol_type
01466       make_oGE (obj_t v, location_type l)
01467       {
01468         return symbol_type (token::oGE, std::move (v), std::move (l));
01469       }
01470 #else
01471       static
01472       symbol_type
01473       make_oGE (const obj_t& v, const location_type& l)
01474       {
```

```
01475         return symbol_type (token::oGE, v, l);
01476       }
01477 #endif
01478 #if 201103L <= YY_CPLUSPLUS
01479       static
01480       symbol_type
01481       make_oEQ (obj_t v, location_type l)
01482       {
01483         return symbol_type (token::oEQ, std::move (v), std::move (l));
01484       }
01485 #else
01486       static
01487       symbol_type
01488       make_oEQ (const obj_t& v, const location_type& l)
01489       {
01490         return symbol_type (token::oEQ, v, l);
01491       }
01492 #endif
01493 #if 201103L <= YY_CPLUSPLUS
01494       static
01495       symbol_type
01496       make_oNE (obj_t v, location_type l)
01497       {
01498         return symbol_type (token::oNE, std::move (v), std::move (l));
01499       }
01500 #else
01501       static
01502       symbol_type
01503       make_oNE (const obj_t& v, const location_type& l)
01504       {
01505         return symbol_type (token::oNE, v, l);
01506       }
01507 #endif
01508 #if 201103L <= YY_CPLUSPLUS
01509       static
01510       symbol_type
01511       make_oAND (obj_t v, location_type l)
01512       {
01513         return symbol_type (token::oAND, std::move (v), std::move (l));
01514       }
01515 #else
01516       static
01517       symbol_type
01518       make_oAND (const obj_t& v, const location_type& l)
01519       {
01520         return symbol_type (token::oAND, v, l);
01521       }
01522 #endif
01523 #if 201103L <= YY_CPLUSPLUS
01524       static
01525       symbol_type
01526       make_oOR (obj_t v, location_type l)
01527       {
01528         return symbol_type (token::oOR, std::move (v), std::move (l));
01529       }
01530 #else
01531       static
01532       symbol_type
01533       make_oOR (const obj_t& v, const location_type& l)
01534       {
01535         return symbol_type (token::oOR, v, l);
01536       }
01537 #endif
01538 #if 201103L <= YY_CPLUSPLUS
01539       static
01540       symbol_type
01541       make_oNAND (obj_t v, location_type l)
01542       {
01543         return symbol_type (token::oNAND, std::move (v), std::move (l));
01544       }
01545 #else
01546       static
01547       symbol_type
01548       make_oNAND (const obj_t& v, const location_type& l)
01549       {
01550         return symbol_type (token::oNAND, v, l);
01551       }
01552 #endif
01553 #if 201103L <= YY_CPLUSPLUS
01554       static
01555       symbol_type
01556       make_oXOR (obj_t v, location_type l)
01557       {
01558         return symbol_type (token::oXOR, std::move (v), std::move (l));
01559       }
01560 #else
01561       static
```

```
01562       symbol_type
01563       make_oXOR (const obj_t& v, const location_type& l)
01564       {
01565         return symbol_type (token::oXOR, v, l);
01566       }
01567 #endif
01568 #if 201103L <= YY_CPLUSPLUS
01569       static
01570       symbol_type
01571       make_oNOT (obj_t v, location_type l)
01572       {
01573         return symbol_type (token::oNOT, std::move (v), std::move (l));
01574       }
01575 #else
01576       static
01577       symbol_type
01578       make_oNOT (const obj_t& v, const location_type& l)
01579       {
01580         return symbol_type (token::oNOT, v, l);
01581       }
01582 #endif
01583 #if 201103L <= YY_CPLUSPLUS
01584       static
01585       symbol_type
01586       make_fSIN (obj_t v, location_type l)
01587       {
01588         return symbol_type (token::fSIN, std::move (v), std::move (l));
01589       }
01590 #else
01591       static
01592       symbol_type
01593       make_fSIN (const obj_t& v, const location_type& l)
01594       {
01595         return symbol_type (token::fSIN, v, l);
01596       }
01597 #endif
01598 #if 201103L <= YY_CPLUSPLUS
01599       static
01600       symbol_type
01601       make_fCOS (obj_t v, location_type l)
01602       {
01603         return symbol_type (token::fCOS, std::move (v), std::move (l));
01604       }
01605 #else
01606       static
01607       symbol_type
01608       make_fCOS (const obj_t& v, const location_type& l)
01609       {
01610         return symbol_type (token::fCOS, v, l);
01611       }
01612 #endif
01613 #if 201103L <= YY_CPLUSPLUS
01614       static
01615       symbol_type
01616       make_fROUND (obj_t v, location_type l)
01617       {
01618         return symbol_type (token::fROUND, std::move (v), std::move (l));
01619       }
01620 #else
01621       static
01622       symbol_type
01623       make_fROUND (const obj_t& v, const location_type& l)
01624       {
01625         return symbol_type (token::fROUND, v, l);
01626       }
01627 #endif
01628 #if 201103L <= YY_CPLUSPLUS
01629       static
01630       symbol_type
01631       make_fMOD (obj_t v, location_type l)
01632       {
01633         return symbol_type (token::fMOD, std::move (v), std::move (l));
01634       }
01635 #else
01636       static
01637       symbol_type
01638       make_fMOD (const obj_t& v, const location_type& l)
01639       {
01640         return symbol_type (token::fMOD, v, l);
01641       }
01642 #endif
01643 #if 201103L <= YY_CPLUSPLUS
01644       static
01645       symbol_type
01646       make_fTRUNC (obj_t v, location_type l)
01647       {
01648         return symbol_type (token::fTRUNC, std::move (v), std::move (l));
```

```
01649       }
01650 #else
01651       static
01652       symbol_type
01653       make_fTRUNC (const obj_t& v, const location_type& l)
01654       {
01655         return symbol_type (token::fTRUNC, v, l);
01656       }
01657 #endif
01658 #if 201103L <= YY_CPLUSPLUS
01659       static
01660       symbol_type
01661       make_fFRAC (obj_t v, location_type l)
01662       {
01663         return symbol_type (token::fFRAC, std::move (v), std::move (l));
01664       }
01665 #else
01666       static
01667       symbol_type
01668       make_fFRAC (const obj_t& v, const location_type& l)
01669       {
01670         return symbol_type (token::fFRAC, v, l);
01671       }
01672 #endif
01673 #if 201103L <= YY_CPLUSPLUS
01674       static
01675       symbol_type
01676       make_fEXP (obj_t v, location_type l)
01677       {
01678         return symbol_type (token::fEXP, std::move (v), std::move (l));
01679       }
01680 #else
01681       static
01682       symbol_type
01683       make_fEXP (const obj_t& v, const location_type& l)
01684       {
01685         return symbol_type (token::fEXP, v, l);
01686       }
01687 #endif
01688 #if 201103L <= YY_CPLUSPLUS
01689       static
01690       symbol_type
01691       make_fSQRT (obj_t v, location_type l)
01692       {
01693         return symbol_type (token::fSQRT, std::move (v), std::move (l));
01694       }
01695 #else
01696       static
01697       symbol_type
01698       make_fSQRT (const obj_t& v, const location_type& l)
01699       {
01700         return symbol_type (token::fSQRT, v, l);
01701       }
01702 #endif
01703 #if 201103L <= YY_CPLUSPLUS
01704       static
01705       symbol_type
01706       make_fLOG (obj_t v, location_type l)
01707       {
01708         return symbol_type (token::fLOG, std::move (v), std::move (l));
01709       }
01710 #else
01711       static
01712       symbol_type
01713       make_fLOG (const obj_t& v, const location_type& l)
01714       {
01715         return symbol_type (token::fLOG, v, l);
01716       }
01717 #endif
01718 #if 201103L <= YY_CPLUSPLUS
01719       static
01720       symbol_type
01721       make_fLN (obj_t v, location_type l)
01722       {
01723         return symbol_type (token::fLN, std::move (v), std::move (l));
01724       }
01725 #else
01726       static
01727       symbol_type
01728       make_fLN (const obj_t& v, const location_type& l)
01729       {
01730         return symbol_type (token::fLN, v, l);
01731       }
01732 #endif
01733 #if 201103L <= YY_CPLUSPLUS
01734       static
01735       symbol_type
```

```
01736       make_mathMIN (obj_t v, location_type l)
01737       {
01738         return symbol_type (token::mathMIN, std::move (v), std::move (l));
01739       }
01740 #else
01741       static
01742       symbol_type
01743       make_mathMIN (const obj_t& v, const location_type& l)
01744       {
01745         return symbol_type (token::mathMIN, v, l);
01746       }
01747 #endif
01748 #if 201103L <= YY_CPLUSPLUS
01749       static
01750       symbol_type
01751       make_mathMAX (obj_t v, location_type l)
01752       {
01753         return symbol_type (token::mathMAX, std::move (v), std::move (l));
01754       }
01755 #else
01756       static
01757       symbol_type
01758       make_mathMAX (const obj_t& v, const location_type& l)
01759       {
01760         return symbol_type (token::mathMAX, v, l);
01761       }
01762 #endif
01763 #if 201103L <= YY_CPLUSPLUS
01764       static
01765       symbol_type
01766       make_fVAL (obj_t v, location_type l)
01767       {
01768         return symbol_type (token::fVAL, std::move (v), std::move (l));
01769       }
01770 #else
01771       static
01772       symbol_type
01773       make_fVAL (const obj_t& v, const location_type& l)
01774       {
01775         return symbol_type (token::fVAL, v, l);
01776       }
01777 #endif
01778 #if 201103L <= YY_CPLUSPLUS
01779       static
01780       symbol_type
01781       make_fEVAL (obj_t v, location_type l)
01782       {
01783         return symbol_type (token::fEVAL, std::move (v), std::move (l));
01784       }
01785 #else
01786       static
01787       symbol_type
01788       make_fEVAL (const obj_t& v, const location_type& l)
01789       {
01790         return symbol_type (token::fEVAL, v, l);
01791       }
01792 #endif
01793 #if 201103L <= YY_CPLUSPLUS
01794       static
01795       symbol_type
01796       make_fLENG (obj_t v, location_type l)
01797       {
01798         return symbol_type (token::fLENG, std::move (v), std::move (l));
01799       }
01800 #else
01801       static
01802       symbol_type
01803       make_fLENG (const obj_t& v, const location_type& l)
01804       {
01805         return symbol_type (token::fLENG, v, l);
01806       }
01807 #endif
01808 #if 201103L <= YY_CPLUSPLUS
01809       static
01810       symbol_type
01811       make_fRND1 (obj_t v, location_type l)
01812       {
01813         return symbol_type (token::fRND1, std::move (v), std::move (l));
01814       }
01815 #else
01816       static
01817       symbol_type
01818       make_fRND1 (const obj_t& v, const location_type& l)
01819       {
01820         return symbol_type (token::fRND1, v, l);
01821       }
01822 #endif
```

```
01823 #if 201103L <= YY_CPLUSPLUS
01824       static
01825       symbol_type
01826       make_fEXPO (obj_t v, location_type l)
01827       {
01828         return symbol_type (token::fEXPO, std::move (v), std::move (l));
01829       }
01830 #else
01831       static
01832       symbol_type
01833       make_fEXPO (const obj_t& v, const location_type& l)
01834       {
01835         return symbol_type (token::fEXPO, v, l);
01836       }
01837 #endif
01838 #if 201103L <= YY_CPLUSPLUS
01839       static
01840       symbol_type
01841       make_fNORM (obj_t v, location_type l)
01842       {
01843         return symbol_type (token::fNORM, std::move (v), std::move (l));
01844       }
01845 #else
01846       static
01847       symbol_type
01848       make_fNORM (const obj_t& v, const location_type& l)
01849       {
01850         return symbol_type (token::fNORM, v, l);
01851       }
01852 #endif
01853 #if 201103L <= YY_CPLUSPLUS
01854       static
01855       symbol_type
01856       make_fUNIF (obj_t v, location_type l)
01857       {
01858         return symbol_type (token::fUNIF, std::move (v), std::move (l));
01859       }
01860 #else
01861       static
01862       symbol_type
01863       make_fUNIF (const obj_t& v, const location_type& l)
01864       {
01865         return symbol_type (token::fUNIF, v, l);
01866       }
01867 #endif
01868 #if 201103L <= YY_CPLUSPLUS
01869       static
01870       symbol_type
01871       make_fWEIB (obj_t v, location_type l)
01872       {
01873         return symbol_type (token::fWEIB, std::move (v), std::move (l));
01874       }
01875 #else
01876       static
01877       symbol_type
01878       make_fWEIB (const obj_t& v, const location_type& l)
01879       {
01880         return symbol_type (token::fWEIB, v, l);
01881       }
01882 #endif
01883 #if 201103L <= YY_CPLUSPLUS
01884       static
01885       symbol_type
01886       make_fLOGN (obj_t v, location_type l)
01887       {
01888         return symbol_type (token::fLOGN, std::move (v), std::move (l));
01889       }
01890 #else
01891       static
01892       symbol_type
01893       make_fLOGN (const obj_t& v, const location_type& l)
01894       {
01895         return symbol_type (token::fLOGN, v, l);
01896       }
01897 #endif
01898 #if 201103L <= YY_CPLUSPLUS
01899       static
01900       symbol_type
01901       make_fGAMM (obj_t v, location_type l)
01902       {
01903         return symbol_type (token::fGAMM, std::move (v), std::move (l));
01904       }
01905 #else
01906       static
01907       symbol_type
01908       make_fGAMM (const obj_t& v, const location_type& l)
01909       {
```

```
01910         return symbol_type (token::fGAMM, v, l);
01911       }
01912 #endif
01913 #if 201103L <= YY_CPLUSPLUS
01914       static
01915       symbol_type
01916       make_fERLA (obj_t v, location_type l)
01917       {
01918         return symbol_type (token::fERLA, std::move (v), std::move (l));
01919       }
01920 #else
01921       static
01922       symbol_type
01923       make_fERLA (const obj_t& v, const location_type& l)
01924       {
01925         return symbol_type (token::fERLA, v, l);
01926       }
01927 #endif
01928 #if 201103L <= YY_CPLUSPLUS
01929       static
01930       symbol_type
01931       make_fTRIA (obj_t v, location_type l)
01932       {
01933         return symbol_type (token::fTRIA, std::move (v), std::move (l));
01934       }
01935 #else
01936       static
01937       symbol_type
01938       make_fTRIA (const obj_t& v, const location_type& l)
01939       {
01940         return symbol_type (token::fTRIA, v, l);
01941       }
01942 #endif
01943 #if 201103L <= YY_CPLUSPLUS
01944       static
01945       symbol_type
01946       make_fBETA (obj_t v, location_type l)
01947       {
01948         return symbol_type (token::fBETA, std::move (v), std::move (l));
01949       }
01950 #else
01951       static
01952       symbol_type
01953       make_fBETA (const obj_t& v, const location_type& l)
01954       {
01955         return symbol_type (token::fBETA, v, l);
01956       }
01957 #endif
01958 #if 201103L <= YY_CPLUSPLUS
01959       static
01960       symbol_type
01961       make_fDISC (obj_t v, location_type l)
01962       {
01963         return symbol_type (token::fDISC, std::move (v), std::move (l));
01964       }
01965 #else
01966       static
01967       symbol_type
01968       make_fDISC (const obj_t& v, const location_type& l)
01969       {
01970         return symbol_type (token::fDISC, v, l);
01971       }
01972 #endif
01973 #if 201103L <= YY_CPLUSPLUS
01974       static
01975       symbol_type
01976       make_fTNOW (obj_t v, location_type l)
01977       {
01978         return symbol_type (token::fTNOW, std::move (v), std::move (l));
01979       }
01980 #else
01981       static
01982       symbol_type
01983       make_fTNOW (const obj_t& v, const location_type& l)
01984       {
01985         return symbol_type (token::fTNOW, v, l);
01986       }
01987 #endif
01988 #if 201103L <= YY_CPLUSPLUS
01989       static
01990       symbol_type
01991       make_fTFIN (obj_t v, location_type l)
01992       {
01993         return symbol_type (token::fTFIN, std::move (v), std::move (l));
01994       }
01995 #else
01996       static
```

```
01997       symbol_type
01998       make_fTFIN (const obj_t& v, const location_type& l)
01999       {
02000         return symbol_type (token::fTFIN, v, l);
02001       }
02002 #endif
02003 #if 201103L <= YY_CPLUSPLUS
02004       static
02005       symbol_type
02006       make_fMAXREP (obj_t v, location_type l)
02007       {
02008         return symbol_type (token::fMAXREP, std::move (v), std::move (l));
02009       }
02010 #else
02011       static
02012       symbol_type
02013       make_fMAXREP (const obj_t& v, const location_type& l)
02014       {
02015         return symbol_type (token::fMAXREP, v, l);
02016       }
02017 #endif
02018 #if 201103L <= YY_CPLUSPLUS
02019       static
02020       symbol_type
02021       make_fNUMREP (obj_t v, location_type l)
02022       {
02023         return symbol_type (token::fNUMREP, std::move (v), std::move (l));
02024       }
02025 #else
02026       static
02027       symbol_type
02028       make_fNUMREP (const obj_t& v, const location_type& l)
02029       {
02030         return symbol_type (token::fNUMREP, v, l);
02031       }
02032 #endif
02033 #if 201103L <= YY_CPLUSPLUS
02034       static
02035       symbol_type
02036       make_fIDENT (obj_t v, location_type l)
02037       {
02038         return symbol_type (token::fIDENT, std::move (v), std::move (l));
02039       }
02040 #else
02041       static
02042       symbol_type
02043       make_fIDENT (const obj_t& v, const location_type& l)
02044       {
02045         return symbol_type (token::fIDENT, v, l);
02046       }
02047 #endif
02048 #if 201103L <= YY_CPLUSPLUS
02049       static
02050       symbol_type
02051       make_simulEntitiesWIP (obj_t v, location_type l)
02052       {
02053         return symbol_type (token::simulEntitiesWIP, std::move (v), std::move (l));
02054       }
02055 #else
02056       static
02057       symbol_type
02058       make_simulEntitiesWIP (const obj_t& v, const location_type& l)
02059       {
02060         return symbol_type (token::simulEntitiesWIP, v, l);
02061       }
02062 #endif
02063 #if 201103L <= YY_CPLUSPLUS
02064       static
02065       symbol_type
02066       make_cIF (obj_t v, location_type l)
02067       {
02068         return symbol_type (token::cIF, std::move (v), std::move (l));
02069       }
02070 #else
02071       static
02072       symbol_type
02073       make_cIF (const obj_t& v, const location_type& l)
02074       {
02075         return symbol_type (token::cIF, v, l);
02076       }
02077 #endif
02078 #if 201103L <= YY_CPLUSPLUS
02079       static
02080       symbol_type
02081       make_cELSE (obj_t v, location_type l)
02082       {
02083         return symbol_type (token::cELSE, std::move (v), std::move (l));
```

```
02084       }
02085 #else
02086       static
02087       symbol_type
02088       make_cELSE (const obj_t& v, const location_type& l)
02089       {
02090         return symbol_type (token::cELSE, v, l);
02091       }
02092 #endif
02093 #if 201103L <= YY_CPLUSPLUS
02094       static
02095       symbol_type
02096       make_cFOR (obj_t v, location_type l)
02097       {
02098         return symbol_type (token::cFOR, std::move (v), std::move (l));
02099       }
02100 #else
02101       static
02102       symbol_type
02103       make_cFOR (const obj_t& v, const location_type& l)
02104       {
02105         return symbol_type (token::cFOR, v, l);
02106       }
02107 #endif
02108 #if 201103L <= YY_CPLUSPLUS
02109       static
02110       symbol_type
02111       make_cTO (obj_t v, location_type l)
02112       {
02113         return symbol_type (token::cTO, std::move (v), std::move (l));
02114       }
02115 #else
02116       static
02117       symbol_type
02118       make_cTO (const obj_t& v, const location_type& l)
02119       {
02120         return symbol_type (token::cTO, v, l);
02121       }
02122 #endif
02123 #if 201103L <= YY_CPLUSPLUS
02124       static
02125       symbol_type
02126       make_cDO (obj_t v, location_type l)
02127       {
02128         return symbol_type (token::cDO, std::move (v), std::move (l));
02129       }
02130 #else
02131       static
02132       symbol_type
02133       make_cDO (const obj_t& v, const location_type& l)
02134       {
02135         return symbol_type (token::cDO, v, l);
02136       }
02137 #endif
02138 #if 201103L <= YY_CPLUSPLUS
02139       static
02140       symbol_type
02141       make_ATRIB (obj_t v, location_type l)
02142       {
02143         return symbol_type (token::ATRIB, std::move (v), std::move (l));
02144       }
02145 #else
02146       static
02147       symbol_type
02148       make_ATRIB (const obj_t& v, const location_type& l)
02149       {
02150         return symbol_type (token::ATRIB, v, l);
02151       }
02152 #endif
02153 #if 201103L <= YY_CPLUSPLUS
02154       static
02155       symbol_type
02156       make_CSTAT (obj_t v, location_type l)
02157       {
02158         return symbol_type (token::CSTAT, std::move (v), std::move (l));
02159       }
02160 #else
02161       static
02162       symbol_type
02163       make_CSTAT (const obj_t& v, const location_type& l)
02164       {
02165         return symbol_type (token::CSTAT, v, l);
02166       }
02167 #endif
02168 #if 201103L <= YY_CPLUSPLUS
02169       static
02170       symbol_type
```

```
02171       make_COUNTER (obj_t v, location_type l)
02172       {
02173         return symbol_type (token::COUNTER, std::move (v), std::move (l));
02174       }
02175 #else
02176       static
02177       symbol_type
02178       make_COUNTER (const obj_t& v, const location_type& l)
02179       {
02180         return symbol_type (token::COUNTER, v, l);
02181       }
02182 #endif
02183 #if 201103L <= YY_CPLUSPLUS
02184       static
02185       symbol_type
02186       make_SIMRESP (obj_t v, location_type l)
02187       {
02188         return symbol_type (token::SIMRESP, std::move (v), std::move (l));
02189       }
02190 #else
02191       static
02192       symbol_type
02193       make_SIMRESP (const obj_t& v, const location_type& l)
02194       {
02195         return symbol_type (token::SIMRESP, v, l);
02196       }
02197 #endif
02198 #if 201103L <= YY_CPLUSPLUS
02199       static
02200       symbol_type
02201       make_SIMCTRL (obj_t v, location_type l)
02202       {
02203         return symbol_type (token::SIMCTRL, std::move (v), std::move (l));
02204       }
02205 #else
02206       static
02207       symbol_type
02208       make_SIMCTRL (const obj_t& v, const location_type& l)
02209       {
02210         return symbol_type (token::SIMCTRL, v, l);
02211       }
02212 #endif
02213 #if 201103L <= YY_CPLUSPLUS
02214       static
02215       symbol_type
02216       make_fTAVG (obj_t v, location_type l)
02217       {
02218         return symbol_type (token::fTAVG, std::move (v), std::move (l));
02219       }
02220 #else
02221       static
02222       symbol_type
02223       make_fTAVG (const obj_t& v, const location_type& l)
02224       {
02225         return symbol_type (token::fTAVG, v, l);
02226       }
02227 #endif
02228 #if 201103L <= YY_CPLUSPLUS
02229       static
02230       symbol_type
02231       make_fCOUNT (obj_t v, location_type l)
02232       {
02233         return symbol_type (token::fCOUNT, std::move (v), std::move (l));
02234       }
02235 #else
02236       static
02237       symbol_type
02238       make_fCOUNT (const obj_t& v, const location_type& l)
02239       {
02240         return symbol_type (token::fCOUNT, v, l);
02241       }
02242 #endif
02243 #if 201103L <= YY_CPLUSPLUS
02244       static
02245       symbol_type
02246       make_ILLEGAL (obj_t v, location_type l)
02247       {
02248         return symbol_type (token::ILLEGAL, std::move (v), std::move (l));
02249       }
02250 #else
02251       static
02252       symbol_type
02253       make_ILLEGAL (const obj_t& v, const location_type& l)
02254       {
02255         return symbol_type (token::ILLEGAL, v, l);
02256       }
02257 #endif
```

```
02258 #if 201103L <= YY_CPLUSPLUS
02259       static
02260       symbol_type
02261       make_RESOURCE (obj_t v, location_type l)
02262       {
02263         return symbol_type (token::RESOURCE, std::move (v), std::move (l));
02264       }
02265 #else
02266       static
02267       symbol_type
02268       make_RESOURCE (const obj_t& v, const location_type& l)
02269       {
02270         return symbol_type (token::RESOURCE, v, l);
02271       }
02272 #endif
02273 #if 201103L <= YY_CPLUSPLUS
02274       static
02275       symbol_type
02276       make_fNR (obj_t v, location_type l)
02277       {
02278         return symbol_type (token::fNR, std::move (v), std::move (l));
02279       }
02280 #else
02281       static
02282       symbol_type
02283       make_fNR (const obj_t& v, const location_type& l)
02284       {
02285         return symbol_type (token::fNR, v, l);
02286       }
02287 #endif
02288 #if 201103L <= YY_CPLUSPLUS
02289       static
02290       symbol_type
02291       make_fMR (obj_t v, location_type l)
02292       {
02293         return symbol_type (token::fMR, std::move (v), std::move (l));
02294       }
02295 #else
02296       static
02297       symbol_type
02298       make_fMR (const obj_t& v, const location_type& l)
02299       {
02300         return symbol_type (token::fMR, v, l);
02301       }
02302 #endif
02303 #if 201103L <= YY_CPLUSPLUS
02304       static
02305       symbol_type
02306       make_fIRF (obj_t v, location_type l)
02307       {
02308         return symbol_type (token::fIRF, std::move (v), std::move (l));
02309       }
02310 #else
02311       static
02312       symbol_type
02313       make_fIRF (const obj_t& v, const location_type& l)
02314       {
02315         return symbol_type (token::fIRF, v, l);
02316       }
02317 #endif
02318 #if 201103L <= YY_CPLUSPLUS
02319       static
02320       symbol_type
02321       make_fRESSEIZES (obj_t v, location_type l)
02322       {
02323         return symbol_type (token::fRESSEIZES, std::move (v), std::move (l));
02324       }
02325 #else
02326       static
02327       symbol_type
02328       make_fRESSEIZES (const obj_t& v, const location_type& l)
02329       {
02330         return symbol_type (token::fRESSEIZES, v, l);
02331       }
02332 #endif
02333 #if 201103L <= YY_CPLUSPLUS
02334       static
02335       symbol_type
02336       make_fSTATE (obj_t v, location_type l)
02337       {
02338         return symbol_type (token::fSTATE, std::move (v), std::move (l));
02339       }
02340 #else
02341       static
02342       symbol_type
02343       make_fSTATE (const obj_t& v, const location_type& l)
02344       {
```

```
02345         return symbol_type (token::fSTATE, v, l);
02346       }
02347 #endif
02348 #if 201103L <= YY_CPLUSPLUS
02349       static
02350       symbol_type
02351       make_fSETSUM (obj_t v, location_type l)
02352       {
02353         return symbol_type (token::fSETSUM, std::move (v), std::move (l));
02354       }
02355 #else
02356       static
02357       symbol_type
02358       make_fSETSUM (const obj_t& v, const location_type& l)
02359       {
02360         return symbol_type (token::fSETSUM, v, l);
02361       }
02362 #endif
02363 #if 201103L <= YY_CPLUSPLUS
02364       static
02365       symbol_type
02366       make_fRESUTIL (obj_t v, location_type l)
02367       {
02368         return symbol_type (token::fRESUTIL, std::move (v), std::move (l));
02369       }
02370 #else
02371       static
02372       symbol_type
02373       make_fRESUTIL (const obj_t& v, const location_type& l)
02374       {
02375         return symbol_type (token::fRESUTIL, v, l);
02376       }
02377 #endif
02378 #if 201103L <= YY_CPLUSPLUS
02379       static
02380       symbol_type
02381       make_QUEUE (obj_t v, location_type l)
02382       {
02383         return symbol_type (token::QUEUE, std::move (v), std::move (l));
02384       }
02385 #else
02386       static
02387       symbol_type
02388       make_QUEUE (const obj_t& v, const location_type& l)
02389       {
02390         return symbol_type (token::QUEUE, v, l);
02391       }
02392 #endif
02393 #if 201103L <= YY_CPLUSPLUS
02394       static
02395       symbol_type
02396       make_fNQ (obj_t v, location_type l)
02397       {
02398         return symbol_type (token::fNQ, std::move (v), std::move (l));
02399       }
02400 #else
02401       static
02402       symbol_type
02403       make_fNQ (const obj_t& v, const location_type& l)
02404       {
02405         return symbol_type (token::fNQ, v, l);
02406       }
02407 #endif
02408 #if 201103L <= YY_CPLUSPLUS
02409       static
02410       symbol_type
02411       make_fFIRSTINQ (obj_t v, location_type l)
02412       {
02413         return symbol_type (token::fFIRSTINQ, std::move (v), std::move (l));
02414       }
02415 #else
02416       static
02417       symbol_type
02418       make_fFIRSTINQ (const obj_t& v, const location_type& l)
02419       {
02420         return symbol_type (token::fFIRSTINQ, v, l);
02421       }
02422 #endif
02423 #if 201103L <= YY_CPLUSPLUS
02424       static
02425       symbol_type
02426       make_fLASTINQ (obj_t v, location_type l)
02427       {
02428         return symbol_type (token::fLASTINQ, std::move (v), std::move (l));
02429       }
02430 #else
02431       static
```

```
02432       symbol_type
02433       make_fLASTINQ (const obj_t& v, const location_type& l)
02434       {
02435         return symbol_type (token::fLASTINQ, v, l);
02436       }
02437 #endif
02438 #if 201103L <= YY_CPLUSPLUS
02439       static
02440       symbol_type
02441       make_fSAQUE (obj_t v, location_type l)
02442       {
02443         return symbol_type (token::fSAQUE, std::move (v), std::move (l));
02444       }
02445 #else
02446       static
02447       symbol_type
02448       make_fSAQUE (const obj_t& v, const location_type& l)
02449       {
02450         return symbol_type (token::fSAQUE, v, l);
02451       }
02452 #endif
02453 #if 201103L <= YY_CPLUSPLUS
02454       static
02455       symbol_type
02456       make_fAQUE (obj_t v, location_type l)
02457       {
02458         return symbol_type (token::fAQUE, std::move (v), std::move (l));
02459       }
02460 #else
02461       static
02462       symbol_type
02463       make_fAQUE (const obj_t& v, const location_type& l)
02464       {
02465         return symbol_type (token::fAQUE, v, l);
02466       }
02467 #endif
02468 #if 201103L <= YY_CPLUSPLUS
02469       static
02470       symbol_type
02471       make_fENTATRANK (obj_t v, location_type l)
02472       {
02473         return symbol_type (token::fENTATRANK, std::move (v), std::move (l));
02474       }
02475 #else
02476       static
02477       symbol_type
02478       make_fENTATRANK (const obj_t& v, const location_type& l)
02479       {
02480         return symbol_type (token::fENTATRANK, v, l);
02481       }
02482 #endif
02483 #if 201103L <= YY_CPLUSPLUS
02484       static
02485       symbol_type
02486       make_SET (obj_t v, location_type l)
02487       {
02488         return symbol_type (token::SET, std::move (v), std::move (l));
02489       }
02490 #else
02491       static
02492       symbol_type
02493       make_SET (const obj_t& v, const location_type& l)
02494       {
02495         return symbol_type (token::SET, v, l);
02496       }
02497 #endif
02498 #if 201103L <= YY_CPLUSPLUS
02499       static
02500       symbol_type
02501       make_fNUMSET (obj_t v, location_type l)
02502       {
02503         return symbol_type (token::fNUMSET, std::move (v), std::move (l));
02504       }
02505 #else
02506       static
02507       symbol_type
02508       make_fNUMSET (const obj_t& v, const location_type& l)
02509       {
02510         return symbol_type (token::fNUMSET, v, l);
02511       }
02512 #endif
02513 #if 201103L <= YY_CPLUSPLUS
02514       static
02515       symbol_type
02516       make_VARI (obj_t v, location_type l)
02517       {
02518         return symbol_type (token::VARI, std::move (v), std::move (l));
```

```
02519       }
02520 #else
02521       static
02522       symbol_type
02523       make_VARI (const obj_t& v, const location_type& l)
02524       {
02525         return symbol_type (token::VARI, v, l);
02526       }
02527 #endif
02528 #if 201103L <= YY_CPLUSPLUS
02529       static
02530       symbol_type
02531       make_FORM (obj_t v, location_type l)
02532       {
02533         return symbol_type (token::FORM, std::move (v), std::move (l));
02534       }
02535 #else
02536       static
02537       symbol_type
02538       make_FORM (const obj_t& v, const location_type& l)
02539       {
02540         return symbol_type (token::FORM, v, l);
02541       }
02542 #endif
02543 #if 201103L <= YY_CPLUSPLUS
02544       static
02545       symbol_type
02546       make_fNUMGR (obj_t v, location_type l)
02547       {
02548         return symbol_type (token::fNUMGR, std::move (v), std::move (l));
02549       }
02550 #else
02551       static
02552       symbol_type
02553       make_fNUMGR (const obj_t& v, const location_type& l)
02554       {
02555         return symbol_type (token::fNUMGR, v, l);
02556       }
02557 #endif
02558 #if 201103L <= YY_CPLUSPLUS
02559       static
02560       symbol_type
02561       make_fATRGR (obj_t v, location_type l)
02562       {
02563         return symbol_type (token::fATRGR, std::move (v), std::move (l));
02564       }
02565 #else
02566       static
02567       symbol_type
02568       make_fATRGR (const obj_t& v, const location_type& l)
02569       {
02570         return symbol_type (token::fATRGR, v, l);
02571       }
02572 #endif
02573 #if 201103L <= YY_CPLUSPLUS
02574       static
02575       symbol_type
02576       make_LPAREN (location_type l)
02577       {
02578         return symbol_type (token::LPAREN, std::move (l));
02579       }
02580 #else
02581       static
02582       symbol_type
02583       make_LPAREN (const location_type& l)
02584       {
02585         return symbol_type (token::LPAREN, l);
02586       }
02587 #endif
02588 #if 201103L <= YY_CPLUSPLUS
02589       static
02590       symbol_type
02591       make_RPAREN (location_type l)
02592       {
02593         return symbol_type (token::RPAREN, std::move (l));
02594       }
02595 #else
02596       static
02597       symbol_type
02598       make_RPAREN (const location_type& l)
02599       {
02600         return symbol_type (token::RPAREN, l);
02601       }
02602 #endif
02603 #if 201103L <= YY_CPLUSPLUS
02604       static
02605       symbol_type
```

```
02606       make_LBRACKET (location_type l)
02607       {
02608         return symbol_type (token::LBRACKET, std::move (l));
02609       }
02610 #else
02611       static
02612       symbol_type
02613       make_LBRACKET (const location_type& l)
02614       {
02615         return symbol_type (token::LBRACKET, l);
02616       }
02617 #endif
02618 #if 201103L <= YY_CPLUSPLUS
02619       static
02620       symbol_type
02621       make_RBRACKET (location_type l)
02622       {
02623         return symbol_type (token::RBRACKET, std::move (l));
02624       }
02625 #else
02626       static
02627       symbol_type
02628       make_RBRACKET (const location_type& l)
02629       {
02630         return symbol_type (token::RBRACKET, l);
02631       }
02632 #endif
02633 #if 201103L <= YY_CPLUSPLUS
02634       static
02635       symbol_type
02636       make_PLUS (location_type l)
02637       {
02638         return symbol_type (token::PLUS, std::move (l));
02639       }
02640 #else
02641       static
02642       symbol_type
02643       make_PLUS (const location_type& l)
02644       {
02645         return symbol_type (token::PLUS, l);
02646       }
02647 #endif
02648 #if 201103L <= YY_CPLUSPLUS
02649       static
02650       symbol_type
02651       make_MINUS (location_type l)
02652       {
02653         return symbol_type (token::MINUS, std::move (l));
02654       }
02655 #else
02656       static
02657       symbol_type
02658       make_MINUS (const location_type& l)
02659       {
02660         return symbol_type (token::MINUS, l);
02661       }
02662 #endif
02663 #if 201103L <= YY_CPLUSPLUS
02664       static
02665       symbol_type
02666       make_STAR (location_type l)
02667       {
02668         return symbol_type (token::STAR, std::move (l));
02669       }
02670 #else
02671       static
02672       symbol_type
02673       make_STAR (const location_type& l)
02674       {
02675         return symbol_type (token::STAR, l);
02676       }
02677 #endif
02678 #if 201103L <= YY_CPLUSPLUS
02679       static
02680       symbol_type
02681       make_POWER (location_type l)
02682       {
02683         return symbol_type (token::POWER, std::move (l));
02684       }
02685 #else
02686       static
02687       symbol_type
02688       make_POWER (const location_type& l)
02689       {
02690         return symbol_type (token::POWER, l);
02691       }
02692 #endif
```

```
02693 #if 201103L <= YY_CPLUSPLUS
02694       static
02695       symbol_type
02696       make_SLASH (location_type l)
02697       {
02698         return symbol_type (token::SLASH, std::move (l));
02699       }
02700 #else
02701       static
02702       symbol_type
02703       make_SLASH (const location_type& l)
02704       {
02705         return symbol_type (token::SLASH, l);
02706       }
02707 #endif
02708 #if 201103L <= YY_CPLUSPLUS
02709       static
02710       symbol_type
02711       make_LESS (location_type l)
02712       {
02713         return symbol_type (token::LESS, std::move (l));
02714       }
02715 #else
02716       static
02717       symbol_type
02718       make_LESS (const location_type& l)
02719       {
02720         return symbol_type (token::LESS, l);
02721       }
02722 #endif
02723 #if 201103L <= YY_CPLUSPLUS
02724       static
02725       symbol_type
02726       make_GREATER (location_type l)
02727       {
02728         return symbol_type (token::GREATER, std::move (l));
02729       }
02730 #else
02731       static
02732       symbol_type
02733       make_GREATER (const location_type& l)
02734       {
02735         return symbol_type (token::GREATER, l);
02736       }
02737 #endif
02738 #if 201103L <= YY_CPLUSPLUS
02739       static
02740       symbol_type
02741       make_ASSIGN (location_type l)
02742       {
02743         return symbol_type (token::ASSIGN, std::move (l));
02744       }
02745 #else
02746       static
02747       symbol_type
02748       make_ASSIGN (const location_type& l)
02749       {
02750         return symbol_type (token::ASSIGN, l);
02751       }
02752 #endif
02753 #if 201103L <= YY_CPLUSPLUS
02754       static
02755       symbol_type
02756       make_COMMA (location_type l)
02757       {
02758         return symbol_type (token::COMMA, std::move (l));
02759       }
02760 #else
02761       static
02762       symbol_type
02763       make_COMMA (const location_type& l)
02764       {
02765         return symbol_type (token::COMMA, l);
02766       }
02767 #endif
02768
02769
02770     class context
02771     {
02772     public:
02773       context (const genesyspp_parser& yyparser, const symbol_type& yyla);
02774       const symbol_type& lookahead () const YY_NOEXCEPT { return yyla_; }
02775       symbol_kind_type token () const YY_NOEXCEPT { return yyla_.kind (); }
02776       const location_type& location () const YY_NOEXCEPT { return yyla_.location; }
02777
02781       int expected_tokens (symbol_kind_type yyarg[], int yyargn) const;
02782
```

```
02783     private:
02784       const genesyspp_parser& yyparser_;
02785       const symbol_type& yyla_;
02786     };
02787
02788   private:
02789 #if YY_CPLUSPLUS < 201103L
02791     genesyspp_parser (const genesyspp_parser&);
02793     genesyspp_parser& operator= (const genesyspp_parser&);
02794 #endif
02795
02796
02798     typedef short state_type;
02799
02801     int yy_syntax_error_arguments_ (const context& yyctx,
02802                                     symbol_kind_type yyarg[], int yyargn) const;
02803
02806     virtual std::string yysyntax_error_ (const context& yyctx) const;
02810     static state_type yy_lr_goto_state_ (state_type yystate, int yysym);
02811
02814     static bool yy_pact_value_is_default_ (int yyvalue) YY_NOEXCEPT;
02815
02818     static bool yy_table_value_is_error_ (int yyvalue) YY_NOEXCEPT;
02819
02820     static const short yypact_ninf_;
02821     static const signed char yytable_ninf_;
02822
02826     static symbol_kind_type yytranslate_ (int t) YY_NOEXCEPT;
02827
02829     static std::string yytnamerr_ (const char *yystr);
02830
02832     static const char* const yytname_[];
02833
02834
02835     // Tables.
02836     // YYPACT[STATE-NUM] -- Index in YYTABLE of the portion describing
02837     // STATE-NUM.
02838     static const short yypact_[];
02839
02840     // YYDEFACT[STATE-NUM] -- Default reduction number in state STATE-NUM.
02841     // Performed when YYTABLE does not specify something else to do.  Zero
02842     // means the default is an error.
02843     static const signed char yydefact_[];
02844
02845     // YYPGOTO[NTERM-NUM].
02846     static const short yypgoto_[];
02847
02848     // YYDEFGOTO[NTERM-NUM].
02849     static const unsigned char yydefgoto_[];
02850
02851     // YYTABLE[YYPACT[STATE-NUM]] -- What to do in state STATE-NUM.  If
02852     // positive, shift that token.  If negative, reduce the rule whose
02853     // number is the opposite.  If YYTABLE_NINF, syntax error.
02854     static const short yytable_[];
02855
02856     static const short yycheck_[];
02857
02858     // YYSTOS[STATE-NUM] -- The symbol kind of the accessing symbol of
02859     // state STATE-NUM.
02860     static const unsigned char yystos_[];
02861
02862     // YYR1[RULE-NUM] -- Symbol kind of the left-hand side of rule RULE-NUM.
02863     static const unsigned char yyr1_[];
02864
02865     // YYR2[RULE-NUM] -- Number of symbols on the right-hand side of rule RULE-NUM.
02866     static const signed char yyr2_[];
02867
02868
02869 #if YYDEBUG
02870     // YYRLINE[YYN] -- Source line where rule number YYN was defined.
02871     static const short yyrline_[];
02873     virtual void yy_reduce_print_ (int r) const;
02875     virtual void yy_stack_print_ () const;
02876
02878     int yydebug_;
02880     std::ostream* yycdebug_;
02881
02885     template <typename Base>
02886     void yy_print_ (std::ostream& yyo, const basic_symbol<Base>& yysym) const;
02887 #endif
02888
02893     template <typename Base>
02894     void yy_destroy_ (const char* yymsg, basic_symbol<Base>& yysym) const;
02895
02896   private:
02898     struct by_state
02899     {
```

```
02901       by_state () YY_NOEXCEPT;
02902
02904       typedef state_type kind_type;
02905
02907       by_state (kind_type s) YY_NOEXCEPT;
02908
02910       by_state (const by_state& that) YY_NOEXCEPT;
02911
02913       void clear () YY_NOEXCEPT;
02914
02916       void move (by_state& that);
02917
02920       symbol_kind_type kind () const YY_NOEXCEPT;
02921
02924       enum { empty_state = 0 };
02925
02928       state_type state;
02929     };
02930
02932     struct stack_symbol_type : basic_symbol<by_state>
02933     {
02935       typedef basic_symbol<by_state> super_type;
02937       stack_symbol_type ();
02939       stack_symbol_type (YY_RVREF (stack_symbol_type) that);
02941       stack_symbol_type (state_type s, YY_MOVE_REF (symbol_type) sym);
02942 #if YY_CPLUSPLUS < 201103L
02945       stack_symbol_type& operator= (stack_symbol_type& that);
02946
02949       stack_symbol_type& operator= (const stack_symbol_type& that);
02950 #endif
02951     };
02952
02954     template <typename T, typename S = std::vector<T> >
02955     class stack
02956     {
02957     public:
02958       // Hide our reversed order.
02959       typedef typename S::iterator iterator;
02960       typedef typename S::const_iterator const_iterator;
02961       typedef typename S::size_type size_type;
02962       typedef typename std::ptrdiff_t index_type;
02963
02964       stack (size_type n = 200) YY_NOEXCEPT
02965         : seq_ (n)
02966       {}
02967
02968 #if 201103L <= YY_CPLUSPLUS
02970       stack (const stack&) = delete;
02972       stack& operator= (const stack&) = delete;
02973 #endif
02974
02978       const T&
02979       operator[] (index_type i) const
02980       {
02981         return seq_[size_type (size () - 1 - i)];
02982       }
02983
02987       T&
02988       operator[] (index_type i)
02989       {
02990         return seq_[size_type (size () - 1 - i)];
02991       }
02992
02996       void
02997       push (YY_MOVE_REF (T) t)
02998       {
02999         seq_.push_back (T ());
03000         operator[] (0).move (t);
03001       }
03002
03004       void
03005       pop (std::ptrdiff_t n = 1) YY_NOEXCEPT
03006       {
03007         for (; 0 < n; --n)
03008           seq_.pop_back ();
03009       }
03010
03012       void
03013       clear () YY_NOEXCEPT
03014       {
03015         seq_.clear ();
03016       }
03017
03019       index_type
03020       size () const YY_NOEXCEPT
03021       {
03022         return index_type (seq_.size ());
```

```
03023       }
03024
03026       const_iterator
03027       begin () const YY_NOEXCEPT
03028       {
03029         return seq_.begin ();
03030       }
03031
03033       const_iterator
03034       end () const YY_NOEXCEPT
03035       {
03036         return seq_.end ();
03037       }
03038
03040       class slice
03041       {
03042       public:
03043         slice (const stack& stack, index_type range) YY_NOEXCEPT
03044           : stack_ (stack)
03045           , range_ (range)
03046         {}
03047
03048         const T&
03049         operator[] (index_type i) const
03050         {
03051           return stack_[range_ - i];
03052         }
03053
03054       private:
03055         const stack& stack_;
03056         index_type range_;
03057       };
03058
03059     private:
03060 #if YY_CPLUSPLUS < 201103L
03062       stack (const stack&);
03064       stack& operator= (const stack&);
03065 #endif
03067       S seq_;
03068     };
03069
03070
03072     typedef stack<stack_symbol_type> stack_type;
03073
03075     stack_type yystack_;
03076
03082     void yypush_ (const char* m, YY_MOVE_REF (stack_symbol_type) sym);
03083
03090     void yypush_ (const char* m, state_type s, YY_MOVE_REF (symbol_type) sym);
03091
03093     void yypop_ (int n = 1) YY_NOEXCEPT;
03094
03096     enum
03097     {
03098       yylast_ = 493,
03099       yynnts_ = 34,
03100       yyfinal_ = 142
03101     };
03102
03103
03104     // User arguments.
03105     genesyspp_driver& driver;
03106
03107   };
03108
03109   inline
03110   genesyspp_parser::symbol_kind_type
03111   genesyspp_parser::yytranslate_ (int t) YY_NOEXCEPT
03112   {
03113     // YYTRANSLATE[TOKEN-NUM] -- Symbol number corresponding to
03114     // TOKEN-NUM as returned by yylex.
03115     static
03116     const signed char
03117     translate_table[] =
03118     {
03119        0,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03120        2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03121        2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03122        2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03123        2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03124        2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03125        2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03126        2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03127        2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03128        2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03129        2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03130        2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
```

```
03131       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03132       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03133       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03134       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03135       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03136       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03137       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03138       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03139       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03140       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03141       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03142       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03143       2,     2,     2,     2,     2,     2,     2,     2,     2,     2,
03144       2,     2,     2,     2,     2,     2,     1,     2,     3,     4,
03145       5,     6,     7,     8,     9,    10,    11,    12,    13,    14,
03146      15,    16,    17,    18,    19,    20,    21,    22,    23,    24,
03147      25,    26,    27,    28,    29,    30,    31,    32,    33,    34,
03148      35,    36,    37,    38,    39,    40,    41,    42,    43,    44,
03149      45,    46,    47,    48,    49,    50,    51,    52,    53,    54,
03150      55,    56,    57,    58,    59,    60,    61,    62,    63,    64,
03151      65,    66,    67,    68,    69,    70,    71,    72,    73,    74,
03152      75,    76,    77,    78,    79,    80,    81,    82,    83,    84,
03153      85,    86,    87,    88,    89,    90,    91,    92,    93,    94
03154    };
03155    // Last valid token kind.
03156    const int code_max = 349;
03157
03158    if (t <= 0)
03159      return symbol_kind::S_YYEOF;
03160    else if (t <= code_max)
03161      return static_cast <symbol_kind_type> (translate_table[t]);
03162    else
03163      return symbol_kind::S_YYUNDEF;
03164  }
03165
03166  // basic_symbol.
03167  template <typename Base>
03168  genesyspp_parser::basic_symbol<Base>::basic_symbol (const basic_symbol& that)
03169    : Base (that)
03170    , value ()
03171    , location (that.location)
03172  {
03173    switch (this->kind ())
03174    {
03175      case symbol_kind::S_NUMD: // NUMD
03176      case symbol_kind::S_NUMH: // NUMH
03177      case symbol_kind::S_CTEZERO: // CTEZERO
03178      case symbol_kind::S_oLE: // oLE
03179      case symbol_kind::S_oGE: // oGE
03180      case symbol_kind::S_oEQ: // oEQ
03181      case symbol_kind::S_oNE: // oNE
03182      case symbol_kind::S_oAND: // oAND
03183      case symbol_kind::S_oOR: // oOR
03184      case symbol_kind::S_oNAND: // oNAND
03185      case symbol_kind::S_oXOR: // oXOR
03186      case symbol_kind::S_oNOT: // oNOT
03187      case symbol_kind::S_fSIN: // fSIN
03188      case symbol_kind::S_fCOS: // fCOS
03189      case symbol_kind::S_fROUND: // fROUND
03190      case symbol_kind::S_fMOD: // fMOD
03191      case symbol_kind::S_fTRUNC: // fTRUNC
03192      case symbol_kind::S_fFRAC: // fFRAC
03193      case symbol_kind::S_fEXP: // fEXP
03194      case symbol_kind::S_fSQRT: // fSQRT
03195      case symbol_kind::S_fLOG: // fLOG
03196      case symbol_kind::S_fLN: // fLN
03197      case symbol_kind::S_mathMIN: // mathMIN
03198      case symbol_kind::S_mathMAX: // mathMAX
03199      case symbol_kind::S_fVAL: // fVAL
03200      case symbol_kind::S_fEVAL: // fEVAL
03201      case symbol_kind::S_fLENG: // fLENG
03202      case symbol_kind::S_fRND1: // fRND1
03203      case symbol_kind::S_fEXPO: // fEXPO
03204      case symbol_kind::S_fNORM: // fNORM
03205      case symbol_kind::S_fUNIF: // fUNIF
03206      case symbol_kind::S_fWEIB: // fWEIB
03207      case symbol_kind::S_fLOGN: // fLOGN
03208      case symbol_kind::S_fGAMM: // fGAMM
03209      case symbol_kind::S_fERLA: // fERLA
03210      case symbol_kind::S_fTRIA: // fTRIA
03211      case symbol_kind::S_fBETA: // fBETA
03212      case symbol_kind::S_fDISC: // fDISC
03213      case symbol_kind::S_fTNOW: // fTNOW
03214      case symbol_kind::S_fTFIN: // fTFIN
03215      case symbol_kind::S_fMAXREP: // fMAXREP
03216      case symbol_kind::S_fNUMREP: // fNUMREP
03217      case symbol_kind::S_fIDENT: // fIDENT
```

```
03218       case symbol_kind::S_simulEntitiesWIP: // simulEntitiesWIP
03219       case symbol_kind::S_cIF: // cIF
03220       case symbol_kind::S_cELSE: // cELSE
03221       case symbol_kind::S_cFOR: // cFOR
03222       case symbol_kind::S_cTO: // cTO
03223       case symbol_kind::S_cDO: // cDO
03224       case symbol_kind::S_ATRIB: // ATRIB
03225       case symbol_kind::S_CSTAT: // CSTAT
03226       case symbol_kind::S_COUNTER: // COUNTER
03227       case symbol_kind::S_SIMRESP: // SIMRESP
03228       case symbol_kind::S_SIMCTRL: // SIMCTRL
03229       case symbol_kind::S_fTAVG: // fTAVG
03230       case symbol_kind::S_fCOUNT: // fCOUNT
03231       case symbol_kind::S_ILLEGAL: // ILLEGAL
03232       case symbol_kind::S_RESOURCE: // RESOURCE
03233       case symbol_kind::S_fNR: // fNR
03234       case symbol_kind::S_fMR: // fMR
03235       case symbol_kind::S_fIRF: // fIRF
03236       case symbol_kind::S_fRESSEIZES: // fRESSEIZES
03237       case symbol_kind::S_fSTATE: // fSTATE
03238       case symbol_kind::S_fSETSUM: // fSETSUM
03239       case symbol_kind::S_fRESUTIL: // fRESUTIL
03240       case symbol_kind::S_QUEUE: // QUEUE
03241       case symbol_kind::S_fNQ: // fNQ
03242       case symbol_kind::S_fFIRSTINQ: // fFIRSTINQ
03243       case symbol_kind::S_fLASTINQ: // fLASTINQ
03244       case symbol_kind::S_fSAQUE: // fSAQUE
03245       case symbol_kind::S_fAQUE: // fAQUE
03246       case symbol_kind::S_fENTATRANK: // fENTATRANK
03247       case symbol_kind::S_SET: // SET
03248       case symbol_kind::S_fNUMSET: // fNUMSET
03249       case symbol_kind::S_VARI: // VARI
03250       case symbol_kind::S_FORM: // FORM
03251       case symbol_kind::S_fNUMGR: // fNUMGR
03252       case symbol_kind::S_fATRGR: // fATRGR
03253       case symbol_kind::S_input: // input
03254       case symbol_kind::S_expression: // expression
03255       case symbol_kind::S_logicalOr: // logicalOr
03256       case symbol_kind::S_logicalXor: // logicalXor
03257       case symbol_kind::S_logicalAnd: // logicalAnd
03258       case symbol_kind::S_logicalNot: // logicalNot
03259       case symbol_kind::S_relational: // relational
03260       case symbol_kind::S_additive: // additive
03261       case symbol_kind::S_multiplicative: // multiplicative
03262       case symbol_kind::S_power: // power
03263       case symbol_kind::S_unary: // unary
03264       case symbol_kind::S_primary: // primary
03265       case symbol_kind::S_number: // number
03266       case symbol_kind::S_command: // command
03267       case symbol_kind::S_commandIF: // commandIF
03268       case symbol_kind::S_commandFOR: // commandFOR
03269       case symbol_kind::S_function: // function
03270       case symbol_kind::S_kernelFunction: // kernelFunction
03271       case symbol_kind::S_elementFunction: // elementFunction
03272       case symbol_kind::S_trigonFunction: // trigonFunction
03273       case symbol_kind::S_mathFunction: // mathFunction
03274       case symbol_kind::S_probFunction: // probFunction
03275       case symbol_kind::S_userFunction: // userFunction
03276       case symbol_kind::S_listaparm: // listaparm
03277       case symbol_kind::S_illegal: // illegal
03278       case symbol_kind::S_attribute: // attribute
03279       case symbol_kind::S_simulationResponse: // simulationResponse
03280       case symbol_kind::S_simulationControl: // simulationControl
03281       case symbol_kind::S_variable: // variable
03282       case symbol_kind::S_formula: // formula
03283       case symbol_kind::S_assigment: // assigment
03284       case symbol_kind::S_pluginFunction: // pluginFunction
03285         value.copy< obj_t > (YY_MOVE (that.value));
03286         break;
03287
03288       case symbol_kind::S_indexList: // indexList
03289         value.copy< std::string > (YY_MOVE (that.value));
03290         break;
03291
03292       default:
03293         break;
03294     }
03295
03296   }
03297
03298
03299
03300
03301   template <typename Base>
03302   genesyspp_parser::symbol_kind_type
03303   genesyspp_parser::basic_symbol<Base>::type_get () const YY_NOEXCEPT
03304   {
```

```
03305     return this->kind ();
03306   }
03307
03308
03309   template <typename Base>
03310   bool
03311   genesyspp_parser::basic_symbol<Base>::empty () const YY_NOEXCEPT
03312   {
03313     return this->kind () == symbol_kind::S_YYEMPTY;
03314   }
03315
03316   template <typename Base>
03317   void
03318   genesyspp_parser::basic_symbol<Base>::move (basic_symbol& s)
03319   {
03320     super_type::move (s);
03321     switch (this->kind ())
03322     {
03323       case symbol_kind::S_NUMD: // NUMD
03324       case symbol_kind::S_NUMH: // NUMH
03325       case symbol_kind::S_CTEZERO: // CTEZERO
03326       case symbol_kind::S_oLE: // oLE
03327       case symbol_kind::S_oGE: // oGE
03328       case symbol_kind::S_oEQ: // oEQ
03329       case symbol_kind::S_oNE: // oNE
03330       case symbol_kind::S_oAND: // oAND
03331       case symbol_kind::S_oOR: // oOR
03332       case symbol_kind::S_oNAND: // oNAND
03333       case symbol_kind::S_oXOR: // oXOR
03334       case symbol_kind::S_oNOT: // oNOT
03335       case symbol_kind::S_fSIN: // fSIN
03336       case symbol_kind::S_fCOS: // fCOS
03337       case symbol_kind::S_fROUND: // fROUND
03338       case symbol_kind::S_fMOD: // fMOD
03339       case symbol_kind::S_fTRUNC: // fTRUNC
03340       case symbol_kind::S_fFRAC: // fFRAC
03341       case symbol_kind::S_fEXP: // fEXP
03342       case symbol_kind::S_fSQRT: // fSQRT
03343       case symbol_kind::S_fLOG: // fLOG
03344       case symbol_kind::S_fLN: // fLN
03345       case symbol_kind::S_mathMIN: // mathMIN
03346       case symbol_kind::S_mathMAX: // mathMAX
03347       case symbol_kind::S_fVAL: // fVAL
03348       case symbol_kind::S_fEVAL: // fEVAL
03349       case symbol_kind::S_fLENG: // fLENG
03350       case symbol_kind::S_fRND1: // fRND1
03351       case symbol_kind::S_fEXPO: // fEXPO
03352       case symbol_kind::S_fNORM: // fNORM
03353       case symbol_kind::S_fUNIF: // fUNIF
03354       case symbol_kind::S_fWEIB: // fWEIB
03355       case symbol_kind::S_fLOGN: // fLOGN
03356       case symbol_kind::S_fGAMM: // fGAMM
03357       case symbol_kind::S_fERLA: // fERLA
03358       case symbol_kind::S_fTRIA: // fTRIA
03359       case symbol_kind::S_fBETA: // fBETA
03360       case symbol_kind::S_fDISC: // fDISC
03361       case symbol_kind::S_fTNOW: // fTNOW
03362       case symbol_kind::S_fTFIN: // fTFIN
03363       case symbol_kind::S_fMAXREP: // fMAXREP
03364       case symbol_kind::S_fNUMREP: // fNUMREP
03365       case symbol_kind::S_fIDENT: // fIDENT
03366       case symbol_kind::S_simulEntitiesWIP: // simulEntitiesWIP
03367       case symbol_kind::S_cIF: // cIF
03368       case symbol_kind::S_cELSE: // cELSE
03369       case symbol_kind::S_cFOR: // cFOR
03370       case symbol_kind::S_cTO: // cTO
03371       case symbol_kind::S_cDO: // cDO
03372       case symbol_kind::S_ATRIB: // ATRIB
03373       case symbol_kind::S_CSTAT: // CSTAT
03374       case symbol_kind::S_COUNTER: // COUNTER
03375       case symbol_kind::S_SIMRESP: // SIMRESP
03376       case symbol_kind::S_SIMCTRL: // SIMCTRL
03377       case symbol_kind::S_fTAVG: // fTAVG
03378       case symbol_kind::S_fCOUNT: // fCOUNT
03379       case symbol_kind::S_ILLEGAL: // ILLEGAL
03380       case symbol_kind::S_RESOURCE: // RESOURCE
03381       case symbol_kind::S_fNR: // fNR
03382       case symbol_kind::S_fMR: // fMR
03383       case symbol_kind::S_fIRF: // fIRF
03384       case symbol_kind::S_fRESSEIZES: // fRESSEIZES
03385       case symbol_kind::S_fSTATE: // fSTATE
03386       case symbol_kind::S_fSETSUM: // fSETSUM
03387       case symbol_kind::S_fRESUTIL: // fRESUTIL
03388       case symbol_kind::S_QUEUE: // QUEUE
03389       case symbol_kind::S_fNQ: // fNQ
03390       case symbol_kind::S_fFIRSTINQ: // fFIRSTINQ
03391       case symbol_kind::S_fLASTINQ: // fLASTINQ
```

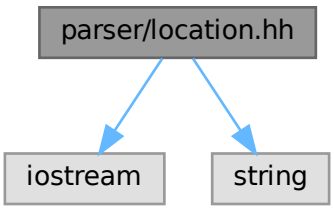
```
03392       case symbol_kind::S_fSAQUE: // fSAQUE
03393       case symbol_kind::S_fAQUE: // fAQUE
03394       case symbol_kind::S_fENTATRANK: // fENTATRANK
03395       case symbol_kind::S_SET: // SET
03396       case symbol_kind::S_fNUMSET: // fNUMSET
03397       case symbol_kind::S_VARI: // VARI
03398       case symbol_kind::S_FORM: // FORM
03399       case symbol_kind::S_fNUMGR: // fNUMGR
03400       case symbol_kind::S_fATRGR: // fATRGR
03401       case symbol_kind::S_input: // input
03402       case symbol_kind::S_expression: // expression
03403       case symbol_kind::S_logicalOr: // logicalOr
03404       case symbol_kind::S_logicalXor: // logicalXor
03405       case symbol_kind::S_logicalAnd: // logicalAnd
03406       case symbol_kind::S_logicalNot: // logicalNot
03407       case symbol_kind::S_relational: // relational
03408       case symbol_kind::S_additive: // additive
03409       case symbol_kind::S_multiplicative: // multiplicative
03410       case symbol_kind::S_power: // power
03411       case symbol_kind::S_unary: // unary
03412       case symbol_kind::S_primary: // primary
03413       case symbol_kind::S_number: // number
03414       case symbol_kind::S_command: // command
03415       case symbol_kind::S_commandIF: // commandIF
03416       case symbol_kind::S_commandFOR: // commandFOR
03417       case symbol_kind::S_function: // function
03418       case symbol_kind::S_kernelFunction: // kernelFunction
03419       case symbol_kind::S_elementFunction: // elementFunction
03420       case symbol_kind::S_trigonFunction: // trigonFunction
03421       case symbol_kind::S_mathFunction: // mathFunction
03422       case symbol_kind::S_probFunction: // probFunction
03423       case symbol_kind::S_userFunction: // userFunction
03424       case symbol_kind::S_listaparm: // listaparm
03425       case symbol_kind::S_illegal: // illegal
03426       case symbol_kind::S_attribute: // attribute
03427       case symbol_kind::S_simulationResponse: // simulationResponse
03428       case symbol_kind::S_simulationControl: // simulationControl
03429       case symbol_kind::S_variable: // variable
03430       case symbol_kind::S_formula: // formula
03431       case symbol_kind::S_assigment: // assigment
03432       case symbol_kind::S_pluginFunction: // pluginFunction
03433         value.move< obj_t > (YY_MOVE (s.value));
03434         break;
03435
03436       case symbol_kind::S_indexList: // indexList
03437         value.move< std::string > (YY_MOVE (s.value));
03438         break;
03439
03440       default:
03441         break;
03442     }
03443
03444     location = YY_MOVE (s.location);
03445   }
03446
03447   // by_kind.
03448   inline
03449   genesyspp_parser::by_kind::by_kind () YY_NOEXCEPT
03450     : kind_ (symbol_kind::S_YYEMPTY)
03451   {}
03452
03453 #if 201103L <= YY_CPLUSPLUS
03454   inline
03455   genesyspp_parser::by_kind::by_kind (by_kind&& that) YY_NOEXCEPT
03456     : kind_ (that.kind_)
03457   {
03458     that.clear ();
03459   }
03460 #endif
03461
03462   inline
03463   genesyspp_parser::by_kind::by_kind (const by_kind& that) YY_NOEXCEPT
03464     : kind_ (that.kind_)
03465   {}
03466
03467   inline
03468   genesyspp_parser::by_kind::by_kind (token_kind_type t) YY_NOEXCEPT
03469     : kind_ (yytranslate_ (t))
03470   {}
03471
03472
03473
03474   inline
03475   void
03476   genesyspp_parser::by_kind::clear () YY_NOEXCEPT
03477   {
03478     kind_ = symbol_kind::S_YYEMPTY;
```

```
03479   }
03480
03481   inline
03482   void
03483   genesyspp_parser::by_kind::move (by_kind& that)
03484   {
03485     kind_ = that.kind_;
03486     that.clear ();
03487   }
03488
03489   inline
03490   genesyspp_parser::symbol_kind_type
03491   genesyspp_parser::by_kind::kind () const YY_NOEXCEPT
03492   {
03493     return kind_;
03494   }
03495
03496
03497   inline
03498   genesyspp_parser::symbol_kind_type
03499   genesyspp_parser::by_kind::type_get () const YY_NOEXCEPT
03500   {
03501     return this->kind ();
03502   }
03503
03504
03505 } // yy
03506 #line 3507 "../GenesysParser.h"
03507
03508
03509
03510
03511 #endif // !YY_YY_GENESYSPARSER_H_INCLUDED
```

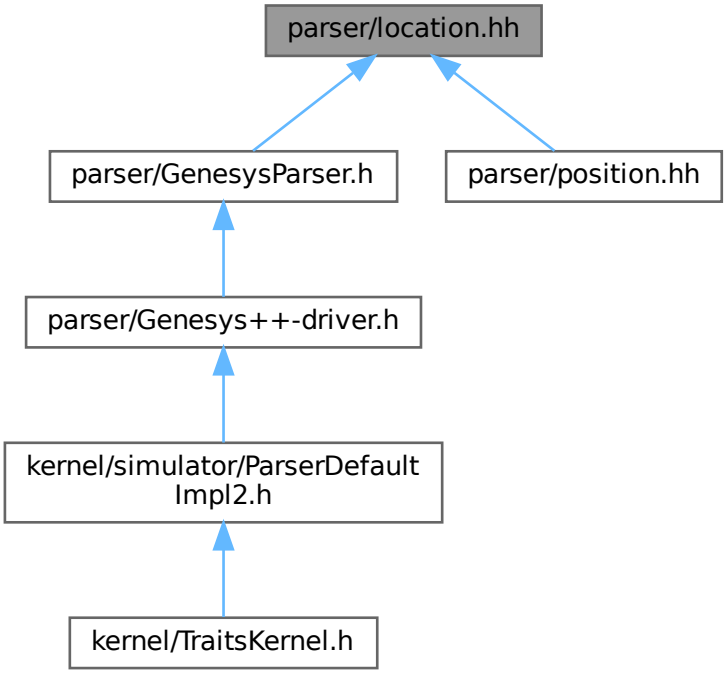
18.368 parser/location.hh File Reference

```
#include <iostream>
#include <string>
```

Include dependency graph for location.hh:



This graph shows which files directly or indirectly include this file:



Classes

- class yy::position

  A point in a source file.
- class yy::location

  Two points in a source file.

Macros

- #define **YY_NULLPTR** ((void∗)0)

Functions

- position & **yy::operator+=** (position &res, position::counter_type width)

  Add width columns, in place.
- position **yy::operator+** (position res, position::counter_type width)

  Add width columns.
- position & **yy::operator-=** (position &res, position::counter_type width)

  Subtract width columns, in place.
- position **yy::operator-** (position res, position::counter_type width)

  Subtract width columns.
- template<typename YYChar >
  std::basic_ostream< YYChar > & yy::operator<< (std::basic_ostream< YYChar > &ostr, const position &pos)

  Intercept output stream redirection.

- location & **yy::operator+=** (location &res, const location &end)

  Join two locations, in place.
- location **yy::operator+** (location res, const location &end)

  Join two locations.
- location & **yy::operator+=** (location &res, location::counter_type width)

  Add width columns to the end position, in place.
- location **yy::operator+** (location res, location::counter_type width)

  Add width columns to the end position.
- location & **yy::operator-=** (location &res, location::counter_type width)

  Subtract width columns to the end position, in place.
- location **yy::operator-** (location res, location::counter_type width)

  Subtract width columns to the end position.
- template<typename YYChar >
  std::basic_ostream< YYChar > & yy::operator<< (std::basic_ostream< YYChar > &ostr, const location &loc)

  Intercept output stream redirection.

18.368.1 Detailed Description

Define the yy::location class.

18.368.2 Function Documentation

18.368.2.1 operator<<() [1/2]

```
template<typename YYChar >
std::basic_ostream< YYChar > & yy::operator<< (
             std::basic_ostream< YYChar > & ostr,
             const location & loc )
```

Intercept output stream redirection.

Parameters

ostr	the destination output stream
loc	a reference to the location to redirect

Avoid duplicate information.

18.368.2.2 operator<<() [2/2]

```
template<typename YYChar >
std::basic_ostream< YYChar > & yy::operator<< (
             std::basic_ostream< YYChar > & ostr,
             const position & pos )
```

Intercept output stream redirection.

Parameters

ostr	the destination output stream
pos	a reference to the position to redirect

18.369 location.hh

Go to the documentation of this file.

```
00001 // A Bison parser, made by GNU Bison 3.8.2.
00002
00003 // Locations for Bison parsers in C++
00004
00005 // Copyright (C) 2002-2015, 2018-2021 Free Software Foundation, Inc.
00006
00007 // This program is free software: you can redistribute it and/or modify
00008 // it under the terms of the GNU General Public License as published by
00009 // the Free Software Foundation, either version 3 of the License, or
00010 // (at your option) any later version.
00011
00012 // This program is distributed in the hope that it will be useful,
00013 // but WITHOUT ANY WARRANTY; without even the implied warranty of
00014 // MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
00015 // GNU General Public License for more details.
00016
00017 // You should have received a copy of the GNU General Public License
00018 // along with this program.  If not, see <https://www.gnu.org/licenses/>.
00019
00020 // As a special exception, you may create a larger work that contains
00021 // part or all of the Bison parser skeleton and distribute that work
00022 // under terms of your choice, so long as that work isn't itself a
00023 // parser generator using the skeleton or a modified version thereof
00024 // as a parser skeleton.  Alternatively, if you modify or redistribute
00025 // the parser skeleton itself, you may (at your option) remove this
00026 // special exception, which will cause the skeleton and the resulting
00027 // Bison output files to be licensed under the GNU General Public
00028 // License without this special exception.
00029
00030 // This special exception was added by the Free Software Foundation in
00031 // version 2.2 of Bison.
00032
00038 #ifndef YY_YY_LOCATION_HH_INCLUDED
00039 # define YY_YY_LOCATION_HH_INCLUDED
00040
00041 # include <iostream>
00042 # include <string>
00043
00044 # ifndef YY_NULLPTR
00045 #  if defined __cplusplus
00046 #   if 201103L <= __cplusplus
00047 #    define YY_NULLPTR nullptr
00048 #   else
00049 #    define YY_NULLPTR 0
00050 #   endif
00051 #  else
00052 #   define YY_NULLPTR ((void*)0)
00053 #  endif
00054 # endif
00055
00056 namespace yy {
00057 #line 58 "../location.hh"
00058
00060   class position
00061   {
00062   public:
00064     typedef const std::string filename_type;
00066     typedef int counter_type;
00067
00069     explicit position (filename_type* f = YY_NULLPTR,
00070                        counter_type l = 1,
00071                        counter_type c = 1)
00072       : filename (f)
00073       , line (l)
00074       , column (c)
00075     {}
00076
00077
00079     void initialize (filename_type* fn = YY_NULLPTR,
00080                      counter_type l = 1,
00081                      counter_type c = 1)
00082     {
00083       filename = fn;
00084       line = l;
00085       column = c;
00086     }
00087
00091     void lines (counter_type count = 1)
00092     {
00093       if (count)
00094         {
00095           column = 1;
00096           line = add_ (line, count, 1);
00097         }
00098     }
00099
00101     void columns (counter_type count = 1)
```

```
00102     {
00103       column = add_ (column, count, 1);
00104     }
00108     filename_type* filename;
00110     counter_type line;
00112     counter_type column;
00113
00114   private:
00116     static counter_type add_ (counter_type lhs, counter_type rhs, counter_type min)
00117     {
00118       return lhs + rhs < min ? min : lhs + rhs;
00119     }
00120   };
00121
00123   inline position&
00124   operator+= (position& res, position::counter_type width)
00125   {
00126     res.columns (width);
00127     return res;
00128   }
00129
00131   inline position
00132   operator+ (position res, position::counter_type width)
00133   {
00134     return res += width;
00135   }
00136
00138   inline position&
00139   operator-= (position& res, position::counter_type width)
00140   {
00141     return res += -width;
00142   }
00143
00145   inline position
00146   operator- (position res, position::counter_type width)
00147   {
00148     return res -= width;
00149   }
00150
00155   template <typename YYChar>
00156   std::basic_ostream<YYChar>&
00157   operator« (std::basic_ostream<YYChar>& ostr, const position& pos)
00158   {
00159     if (pos.filename)
00160       ostr « *pos.filename « ':';
00161     return ostr « pos.line « '.' « pos.column;
00162   }
00163
00165   class location
00166   {
00167   public:
00169     typedef position::filename_type filename_type;
00171     typedef position::counter_type counter_type;
00172
00174     location (const position& b, const position& e)
00175       : begin (b)
00176       , end (e)
00177     {}
00178
00180     explicit location (const position& p = position ())
00181       : begin (p)
00182       , end (p)
00183     {}
00184
00186     explicit location (filename_type* f,
00187                        counter_type l = 1,
00188                        counter_type c = 1)
00189       : begin (f, l, c)
00190       , end (f, l, c)
00191     {}
00192
00193
00195     void initialize (filename_type* f = YY_NULLPTR,
00196                      counter_type l = 1,
00197                      counter_type c = 1)
00198     {
00199       begin.initialize (f, l, c);
00200       end = begin;
00201     }
00202
00205   public:
00207     void step ()
00208     {
00209       begin = end;
00210     }
00211
00213     void columns (counter_type count = 1)
```

```
00214     {
00215       end += count;
00216     }
00217
00219     void lines (counter_type count = 1)
00220     {
00221       end.lines (count);
00222     }
00226   public:
00228     position begin;
00230     position end;
00231   };
00232
00234   inline location&
00235   operator+= (location& res, const location& end)
00236   {
00237     res.end = end.end;
00238     return res;
00239   }
00240
00242   inline location
00243   operator+ (location res, const location& end)
00244   {
00245     return res += end;
00246   }
00247
00249   inline location&
00250   operator+= (location& res, location::counter_type width)
00251   {
00252     res.columns (width);
00253     return res;
00254   }
00255
00257   inline location
00258   operator+ (location res, location::counter_type width)
00259   {
00260     return res += width;
00261   }
00262
00264   inline location&
00265   operator-= (location& res, location::counter_type width)
00266   {
00267     return res += -width;
00268   }
00269
00271   inline location
00272   operator- (location res, location::counter_type width)
00273   {
00274     return res -= width;
00275   }
00276
00283   template <typename YYChar>
00284   std::basic_ostream<YYChar>&
00285   operator« (std::basic_ostream<YYChar>& ostr, const location& loc)
00286   {
00287     location::counter_type end_col
00288       = 0 < loc.end.column ? loc.end.column - 1 : 0;
00289     ostr « loc.begin;
00290     if (loc.end.filename
00291         && (!loc.begin.filename
00292             || *loc.begin.filename != *loc.end.filename))
00293       ostr « '-' « loc.end.filename « ':' « loc.end.line « '.' « end_col;
00294     else if (loc.begin.line < loc.end.line)
00295       ostr « '-' « loc.end.line « '.' « end_col;
00296     else if (loc.begin.column < end_col)
00297       ostr « '-' « end_col;
00298     return ostr;
00299   }
00300
00301 } // yy
00302 #line 303 "../location.hh"
00303
00304 #endif // !YY_YY_LOCATION_HH_INCLUDED
```

18.370 obj_t.h

```
00001 #ifndef OBJ_T_H
00002 #define OBJ_T_H
00003 #include <string>
00004
00005 class obj_t {
00006 public:
00007     obj_t();
00008     virtual ~obj_t();
00009     obj_t(double v, std::string t);
```

```
00010     obj_t(double v, std::string t, unsigned long uid);
00011     double valor;
00012     std::string tipo;
00013     unsigned long id;
00014     bool isReference;
00015     bool isAttributeReference;
00016 };
00017
00018 #endif
```

18.371 position.hh

```
00001 // A Bison parser, made by GNU Bison 3.8.2.
00002
00003 // Starting with Bison 3.2, this file is useless: the structure it
00004 // used to define is now defined in "location.hh".
00005 //
00006 // To get rid of this file:
00007 // 1. add '%require "3.2"' (or newer) to your grammar file
00008 // 2. remove references to this file from your build system
00009 // 3. if you used to include it, include "location.hh" instead.
00010
00011 #include "location.hh"
```

18.372 stack.hh

```
00001 // A Bison parser, made by GNU Bison 3.8.2.
00002
00003 // Starting with Bison 3.2, this file is useless: the structure it
00004 // used to define is now defined with the parser itself.
00005 //
00006 // To get rid of this file:
00007 // 1. add '%require "3.2"' (or newer) to your grammar file
00008 // 2. remove references to this file from your build system.
```

18.373 AIAssistant.h

```
00001 #ifndef AIASSISTANTCOMPONENT_H
00002 #define AIASSISTANTCOMPONENT_H
00003
00004 #include "kernel/simulator/model/ModelComponent.h"
00005 #include "plugins/data/AI/AISupport.h"
00006
00007 #include <map>
00008
00009 class AIAssistant : public ModelComponent {
00010 public:
00011     AIAssistant(Model* model, std::string name = "");
00012     virtual ~AIAssistant() override = default;
00013
00014     static PluginInformation* GetPluginInformation();
00015     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00016     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00017
00018     void setAISupport(AISupport* support);
00019     AISupport* getAISupport() const;
00020     void setAISupportName(std::string value);
00021     std::string getAISupportName() const;
00022     void setPromptTemplate(std::string value);
00023     std::string getPromptTemplate() const;
00024     void setConversationScope(std::string value);
00025     std::string getConversationScope() const;
00026     void setConversationKeyExpression(std::string value);
00027     std::string getConversationKeyExpression() const;
00028     void setActionMappings(std::string value);
00029     std::string getActionMappings() const;
00030     void setDelayExpression(std::string value);
00031     std::string getDelayExpression() const;
00032     void setDefaultOutputPort(unsigned int value);
00033     unsigned int getDefaultOutputPort() const;
00034     void setErrorOutputPort(unsigned int value);
00035     unsigned int getErrorOutputPort() const;
00036     void setAllowDirectOutputPort(bool value);
00037     bool isAllowDirectOutputPort() const;
00038     void setMinimumConfidence(double value);
00039     double getMinimumConfidence() const;
00040     void setSaveConfidenceAttribute(std::string value);
00041     std::string getSaveConfidenceAttribute() const;
00042     void setSaveSuccessAttribute(std::string value);
```

```
00043     std::string getSaveSuccessAttribute() const;
00044     std::string getLastPrompt() const;
00045     std::string getLastResponse() const;
00046     std::string getLastError() const;
00047     bool renderPrompt(std::string& result, std::string& error);
00048 
00049     virtual std::string show() override;
00050 
00051 protected:
00052     virtual bool _loadInstance(PersistenceRecord* fields) override;
00053     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00054     virtual bool _check(std::string& errorMessage) override;
00055     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00056 
00057 private:
00058     struct ParsedResponse {
00059         bool valid = false;
00060         std::string action;
00061         int outputPort = -1;
00062         double confidence = 1.0;
00063         std::string explanation;
00064         std::map<std::string, double> values;
00065         std::string error;
00066     };
00067 
00068     AISupport* _resolveSupport();
00069     bool _renderPrompt(std::string& result, std::string& error);
00070     std::string _conversationKey(Entity* entity, std::string& error);
00071     ParsedResponse _parseResponse(const std::string& content) const;
00072     std::map<std::string, unsigned int> _parseActionMappings() const;
00073     unsigned int _selectPort(const ParsedResponse& response, bool& success) const;
00074     void _forward(Entity* entity, unsigned int port, double delay);
00075     void _fail(Entity* entity, const std::string& message);
00076 
00077     const struct DEFAULT_VALUES {
00078         std::string aiSupportName = "";
00079         std::string promptTemplate = "Decide the route for this entity. Return only JSON with action,
      confidence, explanation, and numeric values fields.";
00080         std::string conversationScope = "PerEntity";
00081         std::string conversationKeyExpression = "Entity.Id";
00082         std::string actionMappings = "route:0";
00083         std::string delayExpression = "0";
00084         unsigned int defaultOutputPort = 0;
00085         unsigned int errorOutputPort = 0;
00086         bool allowDirectOutputPort = false;
00087         double minimumConfidence = 0.0;
00088         std::string saveConfidenceAttribute = "";
00089         std::string saveSuccessAttribute = "";
00090     } DEFAULT;
00091 
00092     AISupport* _aiSupport = nullptr;
00093     std::string _aiSupportName = DEFAULT.aiSupportName;
00094     std::string _promptTemplate = DEFAULT.promptTemplate;
00095     std::string _conversationScope = DEFAULT.conversationScope;
00096     std::string _conversationKeyExpression = DEFAULT.conversationKeyExpression;
00097     std::string _actionMappings = DEFAULT.actionMappings;
00098     std::string _delayExpression = DEFAULT.delayExpression;
00099     unsigned int _defaultOutputPort = DEFAULT.defaultOutputPort;
00100     unsigned int _errorOutputPort = DEFAULT.errorOutputPort;
00101     bool _allowDirectOutputPort = DEFAULT.allowDirectOutputPort;
00102     double _minimumConfidence = DEFAULT.minimumConfidence;
00103     std::string _saveConfidenceAttribute = DEFAULT.saveConfidenceAttribute;
00104     std::string _saveSuccessAttribute = DEFAULT.saveSuccessAttribute;
00105     std::string _lastPrompt;
00106     std::string _lastResponse;
00107     std::string _lastError;
00108 };
00109 
00110 #endif
```

18.374 MarkovChain.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
00007 /*
00008  * File:   MarkovChain.h
00009  * Author: rlcancian
00010  *
00011  * Created on 24 de Outubro de 2019, 18:26
00012  */
00013 
```

```
00014 #ifndef MARKOVCHAIN_H
00015 #define MARKOVCHAIN_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "kernel/statistics/Sampler_if.h"
00019
00020 class MarkovChain : public ModelComponent {
00021 public: // constructors
00022     MarkovChain(Model* model, std::string name = "");
00023     virtual ~MarkovChain() override;
00024 public: // virtual
00025     virtual std::string show() override;
00026 public: // static
00027     static PluginInformation* GetPluginInformation();
00028     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00029     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00030 public: // get and set
00031     void setTransitionProbabilityMatrix(ModelDataDefinition* transitionMatrix);
00032     ModelDataDefinition* getTransitionMatrix() const;
00033     void setInitialDistribution(ModelDataDefinition* initialDistribution);
00034     ModelDataDefinition* getInitialDistribution() const;
00035     void setCurrentState(ModelDataDefinition* currentState);
00036     ModelDataDefinition* getCurrentState() const;
00037 protected: // virtual
00038     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00039     virtual bool _loadInstance(PersistenceRecord *fields) override;
00040     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00041 protected: // virtual
00042     virtual void _initBetweenReplications() override;
00043     virtual bool _check(std::string& errorMessage) override;
00044 //  virtual void _createInternalAndAttachedData() override;
00045 protected:
00046     // virtual void _createInternalStatisticReporters() override;
00047     // virtual void _createNonEditableDataDefinitions() override;
00048     virtual void _createEditableDataDefinitions() override;
00049     // virtual void _createAttachedAttributes() override;
00050
00051 private: // methods
00052     unsigned int _drawNextState(Entity* entity, unsigned int currentState);
00053     double _readStateValue(Entity* entity) const;
00054     void _writeStateValue(Entity* entity, double value);
00055     double _readTransitionProbability(Entity* entity, unsigned int fromState, unsigned int toState)
    const;
00056     unsigned int _stateCount() const;
00057 private: // attributes 1:1
00058     ModelDataDefinition* _transitionProbMatrix = nullptr;
00059     ModelDataDefinition* _initialDistribution = nullptr;
00060     ModelDataDefinition* _currentState = nullptr;
00061 private: // attributes 1:n
00062     Sampler_if* _sampler = nullptr;
00063 };
00064
00065 #endif /* MARKOVCHAIN_H */
```

18.375 BacteriaColony.h

```
00001 /*
00002  * File:   BacteriaColony.h
00003  * Author: rlcancian
00004  *
00005  * Created on 17 de Abril de 2022
00006  */
00007
00008 #ifndef BACTERIACOLONY_H
00009 #define BACTERIACOLONY_H
00010
00011 #include "../../../kernel/simulator/model/ModelComponent.h"
00012 #include "kernel/simulator/Plugin.h"
00013 #include "kernel/util/Util.h"
00014 #include "plugins/data/BiochemicalSimulation/BioNetwork.h"
00015 #include "plugins/data/BiochemicalSimulation/GroProgram.h"
00016 #include "plugins/data/BiochemicalSimulation/GroProgramRuntime.h"
00017 #include "plugins/data/BiochemicalSimulation/BacteriaSignalGrid.h"
00018
00019 #include <cstddef>
00020 #include <map>
00021 #include <string>
00022 #include <vector>
00023
00032 class BacteriaColony : public ModelComponent {
00033 public:
00034     struct BarrierSegment {
00035         double x1 = 0.0;
00036         double y1 = 0.0;
00037         double x2 = 0.0;
```

```
00038         double y2 = 0.0;
00039     };
00040
00041     struct BacteriumState {
00042         unsigned int id = 0;
00043         unsigned int parentId = 0;
00044         unsigned int generation = 0;
00045         std::string programName = "";
00046         std::vector<double> programArguments;
00047         unsigned int divisionCount = 0;
00048         double birthTime = 0.0;
00049         double lastUpdateTime = 0.0;
00050         double lastDivisionTime = 0.0;
00051         double positionX = 0.0;
00052         double positionY = 0.0;
00053         double directionRadians = 0.0;
00054         double volume = 1.0;
00055         double size = 1.0;
00056         double gfp = 0.0;
00057         double rfp = 0.0;
00058         double yfp = 0.0;
00059         double cfp = 0.0;
00060         double speed = 0.1;
00061         unsigned int tickCount = 0;
00062         unsigned int gridX = 0;
00063         unsigned int gridY = 0;
00064         bool hasExplicitGridPosition = false;
00065         bool positionInitialized = false;
00066         bool justDivided = false;
00067         bool daughter = false;
00068         bool alive = true;
00069         std::map<std::string, double> runtimeVariables;
00070     };
00071
00072 public:
00073     BacteriaColony(Model* model, std::string name = "");
00074     virtual ~BacteriaColony() override = default;
00075
00076 public:
00077     virtual std::string show() override;
00078
00079 public: // static plugin interface
00080     static PluginInformation* GetPluginInformation();
00081     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00082     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00083
00084 public:
00086     void setGroProgram(GroProgram* groProgram);
00088     GroProgram* getGroProgram() const;
00090     void setSimulationStep(double simulationStep);
00092     double getSimulationStep() const;
00094     void setNumSteps(unsigned int numSteps);
00096     unsigned int getNumSteps() const;
00098     void setColonyTimeUnit(Util::TimeUnit colonyTimeUnit);
00100     Util::TimeUnit getColonyTimeUnit() const;
00102     double getColonyTime() const;
00104     void setInitialPopulation(unsigned int initialPopulation);
00106     unsigned int getInitialPopulation() const;
00108     unsigned int getPopulationSize() const;
00110     void setAutomaticDivisionEnabled(bool enabled);
00112     bool getAutomaticDivisionEnabled() const;
00114     void setDivisionThresholdVolume(double threshold);
00116     double getDivisionThresholdVolume() const;
00118     void setBioNetwork(BioNetwork* bioNetwork);
00120     BioNetwork* getBioNetwork() const;
00122     void setSignalGrid(BacteriaSignalGrid* signalGrid);
00124     BacteriaSignalGrid* getSignalGrid() const;
00126     std::size_t getInternalBacteriaCount() const;
00128     const BacteriumState& getBacteriumState(std::size_t index) const;
00130     double getBacteriumAge(std::size_t index) const;
00132     double getBacteriumPositionX(std::size_t index) const;
00134     double getBacteriumPositionY(std::size_t index) const;
00136     double getBacteriumDirectionRadians(std::size_t index) const;
00138     double getBacteriumVolume(std::size_t index) const;
00140     double getBacteriumSize(std::size_t index) const;
00142     double getBacteriumGfp(std::size_t index) const;
00144     double getBacteriumRfp(std::size_t index) const;
00146     double getBacteriumRuntimeVariableValue(std::size_t index, const std::string& variableName) const;
00148     bool hasBacteriumRuntimeVariable(std::size_t index, const std::string& variableName) const;
00150     double getRuntimeVariableValue(const std::string& variableName) const;
00152     bool hasRuntimeVariable(const std::string& variableName) const;
00154     double getSignalValueAt(unsigned int x, unsigned int y) const;
00156     unsigned int getSignalChannelCount() const;
00158     double getAdditionalSignalValueAt(unsigned int channel, unsigned int x, unsigned int y) const;
00160     std::vector<std::vector<double» getSignalMatrix() const;
00162     std::string getSignalMatrixDump(std::size_t maxRows = 0, std::size_t maxColumns = 0) const;
00164     double getBacteriumLocalSignal(std::size_t index) const;
```

```
00166     void setChemostatMode(bool enabled);
00168     bool getChemostatMode() const;
00170     void addBarrier(double x1, double y1, double x2, double y2);
00172     const std::vector<BarrierSegment>& getBarriers() const;
00174     const std::vector<double>& getMappedCellValues() const;
00176     const std::string& getMappedCellExpression() const;
00178     void setGridWidth(unsigned int gridWidth);
00180     unsigned int getGridWidth() const;
00182     void setGridHeight(unsigned int gridHeight);
00184     unsigned int getGridHeight() const;
00186     GroProgramRuntime::ExecutionResult executeGroProgram();
00187
00188 protected:
00189     virtual bool _loadInstance(PersistenceRecord* fields) override;
00190     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00191     virtual bool _check(std::string& errorMessage) override;
00192     virtual void _initBetweenReplications() override;
00193     // virtual void _createInternalAndAttachedData() override;
00194     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00195
00196
00197 protected:
00198     // virtual void _createInternalStatisticReporters() override;
00199     // virtual void _createNonEditableDataDefinitions() override;
00200     // virtual void _createEditableDataDefinitions() override;
00201     virtual void _createAttachedAttributes() override;
00202
00203 private:
00204     const struct DEFAULT_VALUES {
00205         const std::string groProgramName = "";
00206         const std::string bioNetworkName = "";
00207         const std::string signalGridName = "";
00208         const double simulationStep = 0.01;
00209         const unsigned int numSteps = 100;
00210         const Util::TimeUnit colonyTimeUnit = Util::TimeUnit::second;
00211         const unsigned int initialPopulation = 1;
00212         const unsigned int gridWidth = 1;
00213         const unsigned int gridHeight = 1;
00214         const bool automaticDivisionEnabled = false;
00215         const double divisionThresholdVolume = 2.0;
00216     } DEFAULT;
00217
00218     GroProgram* _groProgram = nullptr;
00219     BioNetwork* _bioNetwork = nullptr;
00220     BacteriaSignalGrid* _signalGrid = nullptr;
00221     double _simulationStep = DEFAULT.simulationStep;
00222     unsigned int _currentStep = 0;
00223     unsigned int _numSteps = DEFAULT.numSteps;
00224     Util::TimeUnit _colonyTimeUnit = DEFAULT.colonyTimeUnit;
00225     unsigned int _initialPopulation = DEFAULT.initialPopulation;
00226     unsigned int _populationSize = DEFAULT.initialPopulation;
00227     unsigned int _gridWidth = DEFAULT.gridWidth;
00228     unsigned int _gridHeight = DEFAULT.gridHeight;
00229     bool _automaticDivisionEnabled = DEFAULT.automaticDivisionEnabled;
00230     double _divisionThresholdVolume = DEFAULT.divisionThresholdVolume;
00231     unsigned int _nextBacteriumId = 1;
00232     unsigned int _colonyTickCount = 0;
00233     bool _chemostatMode = false;
00234     std::vector<BacteriumState> _bacteria;
00235     std::vector<unsigned int> _bacteriaDividedThisStep;
00236     std::map<std::string, double> _runtimeVariables;
00237     std::vector<BarrierSegment> _barriers;
00238     std::vector<double> _mappedCellValues;
00239     std::string _mappedCellExpression;
00240     std::vector<double> _signalField;
00241     struct AdditionalSignalChannel {
00242         double diffusionRate = 0.0;
00243         double decayRate = 0.0;
00244         bool coefficientsDeclared = false;
00245         std::string declarationScope;
00246         std::vector<double> field;
00247     };
00248     // Channel 0/1 (no "signal()" handle, or the first declaration) always
00249     // means the legacy single field above, so every pre-multi-channel
00250     // program/fixture is unaffected. Channel N >= 2 is the (N-2)th entry
00251     // here: a genuinely independent field with its own diffusion/decay
00252     // rate, created lazily the first time a program declares a second (or
00253     // later) "signal(...)" channel. Purely runtime state, not persisted:
00254     // dynamically-declared channels are reconstructed by re-running the
00255     // Gro program's own declarations, matching the original Gro model
00256     // where signals are created by the running program, not by the
00257     // modeler.
00258     std::vector<AdditionalSignalChannel> _additionalSignalChannels;
00259     // Coefficients declared by the first "signal(kdiff,kdeg)" statement
00260     // (handle 1). Honored by _applySignalFieldStep() only when no
00261     // BacteriaSignalGrid is attached; when one is attached, its persisted
00262     // diffusionRate/decayRate remain authoritative for the legacy field
```

```
00263     // and these are stored but not used. Default 0.0/0.0 preserves the
00264     // prior "no diffusion/decay" behavior when no such declaration ever
00265     // runs. Pure runtime state, not persisted, same lifecycle as
00266     // _additionalSignalChannels.
00267     double _legacyChannelDiffusionRate = 0.0;
00268     double _legacyChannelDecayRate = 0.0;
00269     bool _legacyChannelCoefficientsDeclared = false;
00270     std::string _legacyChannelDeclarationScope;
00271     // One-shot latch: emits at most one diagnostic per replication when a
00272     // declared first-channel coefficient differs from an attached
00273     // BacteriaSignalGrid's persisted value, instead of once per step.
00274     bool _legacyChannelCoefficientMismatchWarned = false;
00275     struct GroSeedDefinition {
00276         double positionX = 0.0;
00277         double positionY = 0.0;
00278         std::string programName = "";
00279         std::vector<double> programArguments;
00280     };
00281     std::vector<GroSeedDefinition> _groSeedDefinitions;
00282     std::string _groSeedSignature = "";
00283     bool _groSeedsApplied = false;
00284
00285 private:
00286     // SignalGrid remains the authoritative source for spatial configuration
00287     // when attached; colony timing is always read from ModelSimulation.
00288     bool _usesSignalGridDimensions() const;
00289     void _synchronizeGridDimensionsFromSignalGrid();
00290     bool _resetRuntimeSignalField(std::string& errorMessage);
00291     bool _collectBioNetworkSpecies(std::vector<BioSpecies*>& species, std::string& errorMessage)
      const;
00292     bool _collectBioNetworkSpeciesByIdentifier(std::map<std::string, BioSpecies*>&
      speciesByIdentifier,
00293                                                std::string& errorMessage) const;
00294     void _appendBioNetworkContextVariables(GroProgramRuntimeState& runtimeState, std::string&
      errorMessage) const;
00295     bool _applyBioNetworkAssignments(const std::map<std::string, double>& assignedVariables,
      std::string& errorMessage);
00296     void _removeBioNetworkAssignmentVariables(std::map<std::string, double>& variables) const;
00297     bool _refreshRuntimeConfigurationFromGroProgram(std::string& errorMessage);
00298     void _ensureGroSeededBacteriaInitialized();
00299     void _rebuildInternalBacteriaFromGroSeeds();
00300     std::size_t _signalIndex(unsigned int x, unsigned int y) const;
00301     double _signalValueAt(unsigned int x, unsigned int y) const;
00302     void _setSignalValueAt(unsigned int x, unsigned int y, double value);
00303     void _addSignalAt(unsigned int x, unsigned int y, double value);
00304     std::vector<std::vector<double>> _buildSignalMatrix() const;
00305     std::string _buildSignalFieldDump(std::size_t maxRows = 0, std::size_t maxColumns = 0) const;
00306     void _captureSignalFieldSnapshot(GroProgramRuntime::ExecutionResult& result,
00307                                      std::size_t maxRows = 0,
00308                                      std::size_t maxColumns = 0) const;
00309     double _computeNeighborSignalSum(unsigned int x, unsigned int y) const;
00310     unsigned int _computeLocalBacteriaCount(unsigned int x, unsigned int y) const;
00311     bool _ensureAdditionalSignalChannel(unsigned int channel, double diffusionRate, double decayRate,
00312                                         const std::string& declarationScope,
00313                                         std::string& errorMessage);
00314     double _additionalSignalValueAt(unsigned int channel, unsigned int x, unsigned int y) const;
00315     void _setAdditionalSignalValueAt(unsigned int channel, unsigned int x, unsigned int y, double
      value);
00316     void _addAdditionalSignalAt(unsigned int channel, unsigned int x, unsigned int y, double value);
00317     void _applyAdditionalSignalChannelsStep();
00318     void _resizeAdditionalSignalChannelsToGrid();
00319     // Handle 0/1 always resolves to the legacy single field; handle N >= 2
00320     // only resolves if that channel has already been declared (i.e. an
00321     // EnsureSignalChannel mutation for it was already applied in this same
00322     // mutation batch/program order). Rejects negative/non-integer/
00323     // not-yet-declared handles instead of silently aliasing channel 0.
00324     bool _tryResolveSignalChannelHandle(double handleValue, unsigned int& channel, std::string&
      errorMessage) const;
00325     void _relaxLegacySignalField(double diffusionRate, double decayRate);
00326     void _applySignalFieldStep();
00327     void _applyBacteriumSignalMutations(const BacteriumState& bacterium,
00328                                         const std::vector<GroProgramRuntime::SignalMutation>&
      mutations);
00329     void _rebuildInternalBacteria(unsigned int populationSize);
00330     void _resizeInternalBacteria(unsigned int populationSize);
00331     bool _applyRuntimePopulationMutations(const std::vector<GroProgramRuntime::PopulationMutation>&
      mutations,
00332                                           unsigned int finalPopulationSize,
00333                                           std::string& errorMessage);
00334     bool _divideBacterium(unsigned int bacteriumId);
00335     void _applyAutomaticDivisions(const std::vector<unsigned int>& eligibleBacteriumIds);
00336     bool _applyColonyMutations(const std::vector<GroProgramRuntime::ColonyMutation>& mutations,
00337                                GroProgramRuntime::ExecutionResult& result,
00338                                bool allowStructureMutations,
00339                                std::string& errorMessage);
00340     bool _executeSeededNamedGroPrograms(const GroProgramIr& ir, GroProgramRuntime::ExecutionResult&
      result);
```

```
00341     bool _executeBacteriumScopedGroProgram(const GroProgramIr& ir, GroProgramRuntime::ExecutionResult&
      result);
00342     bool _containsBacteriumScopedOnlyUnsupportedCommand(const std::vector<GroProgramIr::Command>&
      commands,
00343                                                         std::string& unsupportedCommand) const;
00344     std::size_t _findBacteriumIndexById(unsigned int bacteriumId) const;
00345     GroProgramRuntimeState _createBacteriumRuntimeState(const BacteriumState& bacterium,
00346                                                         std::size_t bacteriumIndex) const;
00347     void _applyBacteriumScopedPopulationMutations(unsigned int bacteriumId,
00348                                                   const
      std::vector<GroProgramRuntime::PopulationMutation>& mutations,
00349                                                   GroProgramRuntime::ExecutionResult& result);
00350     void _appendBacterium(unsigned int parentId = 0, unsigned int generation = 0,
00351                           const std::string& programName = "");
00352     void _appendBacterium(const GroSeedDefinition& seedDefinition);
00353     void _removeBacteria(unsigned int amount);
00354     bool _removeBacteriumById(unsigned int bacteriumId);
00355     void _refreshBacteriaUpdateTime();
00356     void _applyBacteriumGrowth(BacteriumState& bacterium) const;
00357     void _rebuildBacteriaGridPositions();
00358     void _assignBacteriumGridPosition(BacteriumState& bacterium, std::size_t index) const;
00359     void _initializeBacteriumPhenotype(BacteriumState& bacterium, std::size_t index) const;
00360     void _syncBacteriumSpatialState(BacteriumState& bacterium, const GroProgramRuntimeState&
      runtimeState) const;
00361     void _updateBacteriumSpatialMotion(BacteriumState& bacterium) const;
00362     void _separateBacteriaBodies();
00363     void _setBacteriumPosition(BacteriumState& bacterium, double x, double y) const;
00364     double _clampPositionX(double value) const;
00365     double _clampPositionY(double value) const;
00366     void _bindProgramArguments(GroProgramRuntimeState& runtimeState,
00367                                const std::vector<std::string>& parameterNames,
00368                                const std::vector<double>& arguments) const;
00369 };
00370
00371 #endif /* BACTERIACOLONY_H */
```

18.376 BioRunnerCommand.h

```
00001 /*
00002  * File:   BioRunnerCommand.h
00003  * Author: rlcancian
00004  *
00005  * Component that executes BioSimulatorRunner commands when entities arrive.
00006  */
00007
00008 #ifndef BIORUNNERCOMMAND_H
00009 #define BIORUNNERCOMMAND_H
00010
00011 #include <string>
00012
00013 #include "../../../kernel/simulator/model/ModelComponent.h"
00014 #include "kernel/simulator/Plugin.h"
00015 #include "plugins/data/BiochemicalSimulation/BioSimulatorRunner.h"
00016
00024 class BioRunnerCommand : public ModelComponent {
00025 public:
00026     BioRunnerCommand(Model* model, std::string name = "");
00027     virtual ~BioRunnerCommand() override = default;
00028
00029 public:
00030     static PluginInformation* GetPluginInformation();
00031     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00032     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00033
00034 public:
00035     void setRunner(BioSimulatorRunner* runner);
00036     BioSimulatorRunner* getRunner() const;
00037     void setCommand(std::string command);
00038     std::string getCommand() const;
00039     void setLastSucceeded(bool lastSucceeded);
00040     bool getLastSucceeded() const;
00041     void setLastStatus(std::string lastStatus);
00042     std::string getLastStatus() const;
00043     void setLastMessage(std::string lastMessage);
00044     std::string getLastMessage() const;
00045
00046     virtual std::string show() override;
00047
00048 protected:
00049     virtual bool _loadInstance(PersistenceRecord* fields) override;
00050     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00051     virtual bool _check(std::string& errorMessage) override;
00052     // virtual void _createInternalAndAttachedData() override;
00053     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00054
```

```
00055
00056 protected:
00057     // virtual void _createInternalStatisticReporters() override;
00058     // virtual void _createNonEditableDataDefinitions() override;
00059     virtual void _createEditableDataDefinitions() override;
00060     // virtual void _createAttachedAttributes() override;
00061
00062 private:
00063     void _forwardEntity(Entity* entity);
00064
00065 private:
00066     const struct DEFAULT_VALUES {
00067         std::string runnerName = "";
00068         std::string command = "validateModel()";
00069         bool lastSucceeded = false;
00070         std::string lastStatus = "";
00071         std::string lastMessage = "";
00072     } DEFAULT;
00073
00074     BioSimulatorRunner* _runner = nullptr;
00075     std::string _command = DEFAULT.command;
00076     bool _lastSucceeded = DEFAULT.lastSucceeded;
00077     std::string _lastStatus = DEFAULT.lastStatus;
00078     std::string _lastMessage = DEFAULT.lastMessage;
00079 };
00080
00081 #endif /* BIORUNNERCOMMAND_H */
```

18.377 BioSimulate.h

```
00001 /*
00002  * File:   BioSimulate.h
00003  * Author: rlcancian
00004  *
00005  * Component that executes BioNetwork simulation when entities arrive.
00006  */
00007
00008 #ifndef BIOSIMULATE_H
00009 #define BIOSIMULATE_H
00010
00011 #include <string>
00012
00013 #include "../../../kernel/simulator/model/ModelComponent.h"
00014 #include "kernel/simulator/Plugin.h"
00015 #include "plugins/data/BiochemicalSimulation/BioNetwork.h"
00016
00024 class BioSimulate : public ModelComponent {
00025 public:
00026     BioSimulate(Model* model, std::string name = "");
00027     virtual ~BioSimulate() override = default;
00028
00029 public:
00030     static PluginInformation* GetPluginInformation();
00031     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00032     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00033
00034 public:
00035     void setBioNetwork(BioNetwork* bioNetwork);
00036     BioNetwork* getBioNetwork() const;
00037     void setUseNetworkTimeWindow(bool useNetworkTimeWindow);
00038     bool getUseNetworkTimeWindow() const;
00039     void setStartTime(double startTime);
00040     double getStartTime() const;
00041     void setStopTime(double stopTime);
00042     double getStopTime() const;
00043     void setStepSize(double stepSize);
00044     double getStepSize() const;
00045     void setLastSucceeded(bool lastSucceeded);
00046     bool getLastSucceeded() const;
00047     void setLastMessage(std::string lastMessage);
00048     std::string getLastMessage() const;
00049
00050     virtual std::string show() override;
00051
00052 protected:
00053     virtual bool _loadInstance(PersistenceRecord* fields) override;
00054     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00055     virtual bool _check(std::string& errorMessage) override;
00056     // virtual void _createInternalAndAttachedData() override;
00057     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00058
00059
00060 protected:
00061     // virtual void _createInternalStatisticReporters() override;
00062     // virtual void _createNonEditableDataDefinitions() override;
```

```
00063     virtual void _createEditableDataDefinitions() override;
00064     // virtual void _createAttachedAttributes() override;
00065
00066 private:
00067     void _forwardEntity(Entity* entity);
00068
00069 private:
00070     const struct DEFAULT_VALUES {
00071         std::string bioNetworkName = "";
00072         bool useNetworkTimeWindow = true;
00073         double startTime = 0.0;
00074         double stopTime = 1.0;
00075         double stepSize = 0.1;
00076         bool lastSucceeded = false;
00077         std::string lastMessage = "";
00078     } DEFAULT;
00079
00080     BioNetwork* _bioNetwork = nullptr;
00081     bool _useNetworkTimeWindow = DEFAULT.useNetworkTimeWindow;
00082     double _startTime = DEFAULT.startTime;
00083     double _stopTime = DEFAULT.stopTime;
00084     double _stepSize = DEFAULT.stepSize;
00085     bool _lastSucceeded = DEFAULT.lastSucceeded;
00086     std::string _lastMessage = DEFAULT.lastMessage;
00087 };
00088
00089 #endif /* BIOSIMULATE_H */
```

18.378 BioSteadyState.h

```
00001 /*
00002  * File:   BioSteadyState.h
00003  * Author: rlcancian
00004  *
00005  * Component that evaluates BioNetwork steady-state status.
00006  */
00007
00008 #ifndef BIOSTEADYSTATE_H
00009 #define BIOSTEADYSTATE_H
00010
00011 #include <string>
00012
00013 #include "../../../kernel/simulator/model/ModelComponent.h"
00014 #include "kernel/simulator/Plugin.h"
00015 #include "plugins/data/BiochemicalSimulation/BioNetwork.h"
00016
00023 class BioSteadyState : public ModelComponent {
00024 public:
00025     BioSteadyState(Model* model, std::string name = "");
00026     virtual ~BioSteadyState() override = default;
00027
00028 public:
00029     static PluginInformation* GetPluginInformation();
00030     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00031     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00032
00033 public:
00034     void setBioNetwork(BioNetwork* bioNetwork);
00035     BioNetwork* getBioNetwork() const;
00036     void setTolerance(double tolerance);
00037     double getTolerance() const;
00038     void setRunSimulationBeforeCheck(bool runSimulationBeforeCheck);
00039     bool getRunSimulationBeforeCheck() const;
00040     void setLastSteady(bool lastSteady);
00041     bool getLastSteady() const;
00042     void setLastMaxAbsoluteDerivative(double lastMaxAbsoluteDerivative);
00043     double getLastMaxAbsoluteDerivative() const;
00044     void setLastMessage(std::string lastMessage);
00045     std::string getLastMessage() const;
00046
00047     virtual std::string show() override;
00048
00049 protected:
00050     virtual bool _loadInstance(PersistenceRecord* fields) override;
00051     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00052     virtual bool _check(std::string& errorMessage) override;
00053     // virtual void _createInternalAndAttachedData() override;
00054     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00055
00056
00057 protected:
00058     // virtual void _createInternalStatisticReporters() override;
00059     // virtual void _createNonEditableDataDefinitions() override;
00060     // virtual void _createEditableDataDefinitions() override;
00061     virtual void _createAttachedAttributes() override;
```

```
00062
00063 private:
00064     void _forwardEntity(Entity* entity);
00065
00066 private:
00067     const struct DEFAULT_VALUES {
00068         std::string bioNetworkName = "";
00069         double tolerance = 1e-6;
00070         bool runSimulationBeforeCheck = false;
00071         bool lastSteady = false;
00072         double lastMaxAbsoluteDerivative = 0.0;
00073         std::string lastMessage = "";
00074     } DEFAULT;
00075
00076     BioNetwork* _bioNetwork = nullptr;
00077     double _tolerance = DEFAULT.tolerance;
00078     bool _runSimulationBeforeCheck = DEFAULT.runSimulationBeforeCheck;
00079     bool _lastSteady = DEFAULT.lastSteady;
00080     double _lastMaxAbsoluteDerivative = DEFAULT.lastMaxAbsoluteDerivative;
00081     std::string _lastMessage = DEFAULT.lastMessage;
00082 };
00083
00084 #endif /* BIOSTEADYSTATE_H */
```

18.379 GeneticCircuitSimulate.h

```
00001 #ifndef GENETICCIRCUITSIMULATE_H
00002 #define GENETICCIRCUITSIMULATE_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelComponent.h"
00007 #include "kernel/simulator/Plugin.h"
00008 #include "plugins/data/BiochemicalSimulation/GeneticCircuit.h"
00009
00010 class GeneticCircuitPart;
00011
00012 struct GeneticCircuitSimulationSummary {
00013     bool succeeded = false;
00014     unsigned int sampleCount = 0;
00015     double totalExpression = 0.0;
00016     std::string message = "";
00017 };
00018
00022 class GeneticCircuitSimulate : public ModelComponent {
00023 public:
00024     GeneticCircuitSimulate(Model* model, std::string name = "");
00025     virtual ~GeneticCircuitSimulate() override = default;
00026
00027 public:
00028     static PluginInformation* GetPluginInformation();
00029     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00030     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00031
00032 public:
00033     void setGeneticCircuit(GeneticCircuit* geneticCircuit);
00034     GeneticCircuit* getGeneticCircuit() const;
00035     void setStartTime(double startTime);
00036     double getStartTime() const;
00037     void setStopTime(double stopTime);
00038     double getStopTime() const;
00039     void setStepSize(double stepSize);
00040     double getStepSize() const;
00041     void setApplyRegulation(bool applyRegulation);
00042     bool getApplyRegulation() const;
00043     void setLastSucceeded(bool lastSucceeded);
00044     bool getLastSucceeded() const;
00045     void setLastSampleCount(unsigned int lastSampleCount);
00046     unsigned int getLastSampleCount() const;
00047     void setLastTotalExpression(double lastTotalExpression);
00048     double getLastTotalExpression() const;
00049     void setLastMessage(std::string lastMessage);
00050     std::string getLastMessage() const;
00051     static bool simulateCircuit(Model* model, GeneticCircuit* geneticCircuit, double startTime, double
      stopTime,
00052                                 double stepSize, bool applyRegulation,
      GeneticCircuitSimulationSummary* summary,
00053                                 std::string& errorMessage);
00054     bool simulate(std::string& errorMessage);
00055
00056     virtual std::string show() override;
00057
00058 protected:
00059     virtual bool _loadInstance(PersistenceRecord* fields) override;
00060     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
```

```
00061     virtual bool _check(std::string& errorMessage) override;
00062     // virtual void _createInternalAndAttachedData() override;
00063     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00064
00065 protected:
00066     // virtual void _createInternalStatisticReporters() override;
00067     // virtual void _createNonEditableDataDefinitions() override;
00068     virtual void _createEditableDataDefinitions() override;
00069     // virtual void _createAttachedAttributes() override;
00070
00071 private:
00072     double _computeRegulationMultiplier(const GeneticCircuitPart* part) const;
00073     void _executeStep(double timeStep);
00074     void _forwardEntity(Entity* entity);
00075
00076 private:
00077     const struct DEFAULT_VALUES {
00078         std::string geneticCircuitName = "";
00079         double startTime = 0.0;
00080         double stopTime = 10.0;
00081         double stepSize = 1.0;
00082         bool applyRegulation = true;
00083         bool lastSucceeded = false;
00084         unsigned int lastSampleCount = 0;
00085         double lastTotalExpression = 0.0;
00086         std::string lastMessage = "";
00087     } DEFAULT;
00088
00089     GeneticCircuit* _geneticCircuit = nullptr;
00090     double _startTime = DEFAULT.startTime;
00091     double _stopTime = DEFAULT.stopTime;
00092     double _stepSize = DEFAULT.stepSize;
00093     bool _applyRegulation = DEFAULT.applyRegulation;
00094     bool _lastSucceeded = DEFAULT.lastSucceeded;
00095     unsigned int _lastSampleCount = DEFAULT.lastSampleCount;
00096     double _lastTotalExpression = DEFAULT.lastTotalExpression;
00097     std::string _lastMessage = DEFAULT.lastMessage;
00098 };
00099
00100 #endif /* GENETICCIRCUITSIMULATE_H */
```

18.380 GeneticExpressionStep.h

```
00001 #ifndef GENETICEXPRESSIONSTEP_H
00002 #define GENETICEXPRESSIONSTEP_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelComponent.h"
00007 #include "kernel/simulator/Plugin.h"
00008 #include "plugins/data/BiochemicalSimulation/GeneticCircuit.h"
00009
00010 class GeneticCircuitPart;
00011
00015 class GeneticExpressionStep : public ModelComponent {
00016 public:
00017     GeneticExpressionStep(Model* model, std::string name = "");
00018     virtual ~GeneticExpressionStep() override = default;
00019
00020 public:
00021     static PluginInformation* GetPluginInformation();
00022     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00023     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00024
00025 public:
00026     void setGeneticCircuit(GeneticCircuit* geneticCircuit);
00027     GeneticCircuit* getGeneticCircuit() const;
00028     void setTimeStep(double timeStep);
00029     double getTimeStep() const;
00030     void setApplyRegulation(bool applyRegulation);
00031     bool getApplyRegulation() const;
00032     void setLastSucceeded(bool lastSucceeded);
00033     bool getLastSucceeded() const;
00034     void setLastTotalExpression(double lastTotalExpression);
00035     double getLastTotalExpression() const;
00036     void setLastMessage(std::string lastMessage);
00037     std::string getLastMessage() const;
00038
00039     virtual std::string show() override;
00040
00041 protected:
00042     virtual bool _loadInstance(PersistenceRecord* fields) override;
00043     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00044     virtual bool _check(std::string& errorMessage) override;
00045     // virtual void _createInternalAndAttachedData() override;
```

```
00046     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00047 protected:
00048     // virtual void _createInternalStatisticReporters() override;
00049     // virtual void _createNonEditableDataDefinitions() override;
00050     virtual void _createEditableDataDefinitions() override;
00051     // virtual void _createAttachedAttributes() override;
00052
00053 private:
00054     double _computeRegulationMultiplier(const GeneticCircuitPart* part) const;
00055     void _forwardEntity(Entity* entity);
00056
00057 private:
00058     const struct DEFAULT_VALUES {
00059         std::string geneticCircuitName = "";
00060         double timeStep = 1.0;
00061         bool applyRegulation = true;
00062         bool lastSucceeded = false;
00063         double lastTotalExpression = 0.0;
00064         std::string lastMessage = "";
00065     } DEFAULT;
00066
00067     GeneticCircuit* _geneticCircuit = nullptr;
00068     double _timeStep = DEFAULT.timeStep;
00069     bool _applyRegulation = DEFAULT.applyRegulation;
00070     bool _lastSucceeded = DEFAULT.lastSucceeded;
00071     double _lastTotalExpression = DEFAULT.lastTotalExpression;
00072     std::string _lastMessage = DEFAULT.lastMessage;
00073 };
00074
00075 #endif /* GENETICEXPRESSIONSTEP_H */
```

18.381 MetabolicFluxBalance.h

```
00001 #ifndef METABOLICFLUXBALANCE_H
00002 #define METABOLICFLUXBALANCE_H
00003
00004 #include <map>
00005 #include <string>
00006
00007 #include "../../../kernel/simulator/model/ModelComponent.h"
00008 #include "kernel/simulator/Plugin.h"
00009 #include "plugins/data/BiochemicalSimulation/MetabolicNetwork.h"
00010
00038 class MetabolicFluxBalance : public ModelComponent {
00039 public:
00040     MetabolicFluxBalance(Model* model, std::string name = "");
00041     virtual ~MetabolicFluxBalance() override = default;
00042
00043 public:
00044     static PluginInformation* GetPluginInformation();
00045     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00046     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00047
00048 public:
00049     void setMetabolicNetwork(MetabolicNetwork* metabolicNetwork);
00050     MetabolicNetwork* getMetabolicNetwork() const;
00051     void setObjectiveReactionName(std::string objectiveReactionName);
00052     std::string getObjectiveReactionName() const;
00053     void setLastSucceeded(bool lastSucceeded);
00054     bool getLastSucceeded() const;
00055     void setLastObjectiveValue(double lastObjectiveValue);
00056     double getLastObjectiveValue() const;
00057     void setLastMessage(std::string lastMessage);
00058     std::string getLastMessage() const;
00059     const std::map<std::string, double>& getLastFluxes() const;
00060
00061     virtual std::string show() override;
00062
00063 protected:
00064     virtual bool _loadInstance(PersistenceRecord* fields) override;
00065     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00066     virtual bool _check(std::string& errorMessage) override;
00067     // virtual void _createInternalAndAttachedData() override;
00068     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00069
00070 protected:
00071     // virtual void _createInternalStatisticReporters() override;
00072     // virtual void _createNonEditableDataDefinitions() override;
00073     virtual void _createEditableDataDefinitions() override;
00074     // virtual void _createAttachedAttributes() override;
00075
00076 private:
00077     void _forwardEntity(Entity* entity);
00078
00079 private:
```

```
00080     const struct DEFAULT_VALUES {
00081         std::string metabolicNetworkName = "";
00082         std::string objectiveReactionName = "";
00083         bool lastSucceeded = false;
00084         double lastObjectiveValue = 0.0;
00085         std::string lastMessage = "";
00086     } DEFAULT;
00087
00088     MetabolicNetwork* _metabolicNetwork = nullptr;
00089     std::string _objectiveReactionName = DEFAULT.objectiveReactionName;
00090     bool _lastSucceeded = DEFAULT.lastSucceeded;
00091     double _lastObjectiveValue = DEFAULT.lastObjectiveValue;
00092     std::string _lastMessage = DEFAULT.lastMessage;
00093     std::map<std::string, double> _lastFluxes;
00094 };
00095
00096 #endif /* METABOLICFLUXBALANCE_H */
```

18.382 ContinuousSystemComponent.h

```
00001 /*
00002  * File:   ContinuousSystemComponent.h
00003  * Author: GenESyS
00004  *
00005  * ModelComponent that drives continuous-time simulation by delegating
00006  * numerical integration to an ODESolver data definition.
00007  */
00008
00009 #ifndef CONTINUOUSSYSTEMCOMPONENT_H
00010 #define CONTINUOUSSYSTEMCOMPONENT_H
00011
00012 #include <string>
00013
00014 #include "kernel/simulator/model/ModelComponent.h"
00015 #include "plugins/data/Continuous/ODESolver.h"
00016
00026 class ContinuousSystemComponent : public ModelComponent {
00027 public: // constructors
00028     ContinuousSystemComponent(Model* model, std::string name = "");
00029     virtual ~ContinuousSystemComponent() override = default;
00030
00031 public: // virtual
00032     virtual std::string show() override;
00033
00034 public: // static -- required by the plugin framework
00035     static PluginInformation*  GetPluginInformation();
00036     static ModelComponent*     LoadInstance(Model* model, PersistenceRecord* fields);
00037     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00038     static void                SaveInstance(PersistenceRecord* fields, ModelComponent* component);
00039     static bool                Check(ModelComponent* component);
00040     static void                CreateInternalData(ModelComponent* component);
00041     static void                DispatchEvent(Event* event);
00042
00043 public: // getters & setters
00044     void        setOdeSolver(ODESolver* solver);
00045     ODESolver*  getOdeSolver() const;
00046     void        setOdeSolverName(std::string name);
00047     std::string getOdeSolverName() const;
00048
00049 protected: // virtual -- must be overridden
00050     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00051     virtual bool _loadInstance(PersistenceRecord* fields) override;
00052     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00053     virtual bool _check(std::string& errorMessage) override;
00054     virtual void _createEditableDataDefinitions() override;
00055
00056 private:
00057     const struct DEFAULT_VALUES {
00058         const std::string odeSolverName = "";
00059     } DEFAULT;
00060
00061     ODESolver*  _odeSolver     = nullptr;
00062     std::string _odeSolverName = DEFAULT.odeSolverName;
00063 };
00064
00065 #endif /* CONTINUOUSSYSTEMCOMPONENT_H */
```

18.383 DiffEquations.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
```

```
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   DiffEquations.h
00009  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00010  *
00011  * Created on
00012  */
00013
00014 #pragma once
00015
00016 #include "../../../kernel/simulator/model/ModelComponent.h"
00017 #include "../../data/ExternalIntegration/CppCompiler.h"
00018
00019 //extern "C" typedef void (*initBetweenReplications_t)(Model* model);
00020 extern "C" typedef void (*diffEquationsOnDispatchEvent_t)(double* stateVars, int n, double t0, double
      tTarget, double step);
00021
00025 class DiffEquations : public ModelComponent {
00026 public:
00027     DiffEquations(Model* model, std::string name = "");
00028     virtual ~DiffEquations() override;
00029
00030 public:
00031     List<std::string> *getEquations() const;
00032     List<std::string> *getStateVariableNames() const;
00033     double getprecision() const;
00034     void setPrecision(double newPrecision);
00035     double getmaxSteps() const;
00036     void setMaxSteps(double newMaxSteps);
00037     double getfinalTime() const;
00038     void setFinalTime(double newFinalTime);
00039     List<double> *getvaluesOnFinalTime() const;
00040
00041 public:
00042     virtual std::string show() override;
00043
00044 public:
00045     static PluginInformation* GetPluginInformation();
00046     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00047     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00048
00049 protected:
00050     virtual bool _loadInstance(PersistenceRecord *fields) override;
00051     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00052     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00053
00054
00055 protected:
00057     virtual bool _check(std::string& errorMessage) override;
00059     // virtual ParserChangesInformation* _getParserChangesInformation();
00061     //virtual void _initBetweenReplications();
00063     // virtual void _addSimulationControl(SimulationControl* property);
00064 protected:
00065     // virtual void _createInternalStatisticReporters() override;
00066     // virtual void _createNonEditableDataDefinitions() override;
00067     virtual void _createEditableDataDefinitions() override;
00068     // virtual void _createAttachedAttributes() override;
00069
00070 protected:
00071
00072 private:
00073     List<std::string> *_equations = new List<std::string>();
00074     List<std::string> *_stateVariableNames = new List<std::string>();
00075     List<double> *_valuesOnFinalTime = new List<double>();
00076     // pointers to functions on a shared libraty
00077     diffEquationsOnDispatchEvent_t dispatchEvent_SharedLibHandler;
00078     //initBetweenReplications_t initBetweenReplications_SharedLibHandler;
00079     // ...
00080
00081
00082 private:
00083
00084     // Default values for the attributes. Used on initing, loading and saving
00085     const struct DEFAULT_VALUES {
00086         const double precision = 1e-6;
00087         const double maxSteps = 1e3;
00088         const double finalTime = 1.0;
00089     } DEFAULT;
00090     double _precision = DEFAULT.precision;
00091     double _maxSteps = DEFAULT.maxSteps;
00092     double _finalTime = DEFAULT.finalTime;
00093
00094 private:
00095     CppCompiler *_cppCompiler = nullptr;
00096
```

```
00097 private:
00098 };
```

18.384 DiffusionSimulate.h

```
00001 /*
00002  * File:   DiffusionSimulate.h
00003  * Author: GenESyS DCS (Tema 8.2)
00004  *
00005  * Component that advances a DiffusionField (continuous N-D diffusion) when
00006  * entities arrive, mirroring the BioSimulate -> BioNetwork pattern.
00007  */
00008
00009 #ifndef DIFFUSIONSIMULATE_H
00010 #define DIFFUSIONSIMULATE_H
00011
00012 #include <string>
00013
00014 #include "../../../kernel/simulator/model/ModelComponent.h"
00015 #include "kernel/simulator/Plugin.h"
00016 #include "plugins/data/Continuous/DiffusionField.h"
00017
00029 class DiffusionSimulate : public ModelComponent {
00030 public:
00031     DiffusionSimulate(Model* model, std::string name = "");
00032     virtual ~DiffusionSimulate() override = default;
00033
00034 public:
00035     static PluginInformation* GetPluginInformation();
00036     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00037     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00038
00039 public:
00040     void setDiffusionField(DiffusionField* diffusionField);
00041     DiffusionField* getDiffusionField() const;
00042     void setUseFieldTimeWindow(bool useFieldTimeWindow);
00043     bool getUseFieldTimeWindow() const;
00044     void setStartTime(double startTime);
00045     double getStartTime() const;
00046     void setStopTime(double stopTime);
00047     double getStopTime() const;
00048     void setStepSize(double stepSize);
00049     double getStepSize() const;
00050     void setLastSucceeded(bool lastSucceeded);
00051     bool getLastSucceeded() const;
00052     void setLastMessage(std::string lastMessage);
00053     std::string getLastMessage() const;
00054
00055     virtual std::string show() override;
00056
00057 protected:
00058     virtual bool _loadInstance(PersistenceRecord* fields) override;
00059     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00060     virtual bool _check(std::string& errorMessage) override;
00061     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00062
00063 protected:
00064     virtual void _createEditableDataDefinitions() override;
00065
00066 private:
00067     void _forwardEntity(Entity* entity);
00068
00069 private:
00070     const struct DEFAULT_VALUES {
00071         std::string diffusionFieldName = "";
00072         bool useFieldTimeWindow = true;
00073         double startTime = 0.0;
00074         double stopTime = 1.0;
00075         double stepSize = 0.1;
00076         bool lastSucceeded = false;
00077         std::string lastMessage = "";
00078     } DEFAULT;
00079
00080     DiffusionField* _diffusionField = nullptr;
00081     bool _useFieldTimeWindow = DEFAULT.useFieldTimeWindow;
00082     double _startTime = DEFAULT.startTime;
00083     double _stopTime = DEFAULT.stopTime;
00084     double _stepSize = DEFAULT.stepSize;
00085     bool _lastSucceeded = DEFAULT.lastSucceeded;
00086     std::string _lastMessage = DEFAULT.lastMessage;
00087 };
00088
00089 #endif /* DIFFUSIONSIMULATE_H */
```

18.385 LSODE.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   LSODE.h
00009  * Author: rlcancian
00010  *
00011  * Created on 22 de Outubro de 2019, 22:28
00012  */
00013
00014 #ifndef LSODE_H
00015 #define LSODE_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../data/Logic/Formula.h"
00019 #include "../../data/Logic/Variable.h"
00020
00024 class LSODE : public ModelComponent {
00025 public: // constructors
00026     LSODE(Model* model, std::string name = "");
00027     virtual ~LSODE() override;
00028 public: // virtual
00029     virtual std::string show() override;
00030 public: // static
00031     static PluginInformation* GetPluginInformation();
00032     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00033     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00034 public: // g&s
00035     void setTimeVariable(Variable* _timeVariable);
00036     Variable* getTimeVariable() const;
00037     void setStep(double _step);
00038     double getStep() const;
00039     void setVariable(Variable* _variables);
00040     Variable* getVariable() const;
00041     List<std::string>* getDiffEquations() const;
00042     void addDiffEquation(std::string newDiffEquation);
00043     void removeDiffEquation(std::string diffEquation);
00044     void setFilename(std::string filename);
00045     std::string getFileName() const;
00046 protected: // virtual
00047     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00048     virtual bool _loadInstance(PersistenceRecord *fields) override;
00049     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00050 protected: // virtual
00051     //virtual void _initBetweenReplications();
00052     // virtual void _createInternalStatisticReporters() override;
00053     // virtual void _createNonEditableDataDefinitions() override;
00054     // virtual void _createEditableDataDefinitions() override;
00055     // virtual void _createAttachedAttributes() override;
00056     virtual bool _check(std::string& errorMessage) override;
00057     //virtual void _createInternalAndAttachedData();
00058
00059 protected:
00060
00061 private: // methods
00062     bool _doStep();
00063 private: // attributes 1:1
00064
00065     const struct DEFAULT_VALUES {
00066         const std::string filename = "";
00067         const double step = 0.0;
00068     } DEFAULT;
00069     List<std::string>* _diffEquations = new List<std::string>();
00070     Variable* _variable = nullptr;
00071     Variable* _timeVariable = nullptr;
00072     std::string _filename = DEFAULT.filename;
00073     double _step = DEFAULT.step;
00074 private: // attributes 1:n
00075 };
00076
00077 #endif /* LSODE_H */
```

18.386 OLD_ODEelement.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
```

```
00006
00007 /*
00008  * File:   ODE.h
00009  * Author: rlcancian
00010  *
00011  * Created on 13 de Junho de 2019, 19:12
00012  */
00013
00014 #ifndef OLD_ODEELEMENT_H
00015 #define OLD_ODEELEMENT_H
00016
00017 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00018 #include "kernel/simulator/model/ModelDataManager.h"
00019 #include "kernel/simulator/Plugin.h"
00020
00021 class ODEfunction {
00022 public:
00023     ODEfunction(std::string expression, double initialPoint, double initialValue) {
00024         this->expression = expression;
00025         this->initialPoint = initialPoint;
00026         this->initialValue = initialValue;
00027     }
00028     std::string expression;
00029     double initialPoint;
00030     double initialValue;
00031 };
00032
00033 class OLD_ODEelement : public ModelDataDefinition {
00034 public:
00035     OLD_ODEelement(Model* model, std::string name = "");
00036     virtual ~OLD_ODEelement() = default;
00037 public:
00038     virtual std::string show() override;
00039 public:
00040     static PluginInformation* GetPluginInformation();
00041     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00042     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00043 public:
00044     double solve();
00045     void setStepH(double _h);
00046     double getStepH() const;
00047     void setEndTime(double _endTime);
00048     double getEndTime() const;
00049     List<ODEfunction*>* getODEfunctions() const;
00050     void addODEfunction(ODEfunction* newFunction);
00051     void removeODEfunction(ODEfunction* function);
00052 protected: // must be overriden
00053     virtual bool _loadInstance(PersistenceRecord *fields) override;
00054     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00055     virtual bool _check(std::string& errorMessage) override;
00056
00057 protected:
00058     // virtual void _createInternalStatisticReporters() override;
00059     // virtual void _createNonEditableDataDefinitions() override;
00060     // virtual void _createEditableDataDefinitions() override;
00061     // virtual void _createAttachedAttributes() override;
00062
00063 private:
00064
00065 private:
00066     List<ODEfunction*>* _ODEfunctions = new List<ODEfunction*>();
00067     double _stepH = 0.1;
00068     double _endTime;
00069 };
00070
00071 #endif /* OLD_ODEELEMENT_H */
```

18.387 Decide.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Decide.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 9 de Agosto de 2018, 20:39
00012  */
00013
00014 #ifndef DECIDE_H
00015 #define DECIDE_H
00016
```

```
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../../kernel/simulator/essentialPlugins/Counter.h"
00019
00038 class Decide : public ModelComponent {
00039 public:
00040     Decide(Model* model, std::string name = "");
00041     virtual ~Decide() override;
00042 public:
00043     List<std::string>* getConditions() const;
00044     void addConditions(std::string newCondition);
00045     void removeConditions(std::string condition);
00046 public:
00047     virtual std::string show() override;
00048 public:
00049     static PluginInformation* GetPluginInformation();
00050     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00051     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00052 protected:
00053     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00054     virtual bool _loadInstance(PersistenceRecord *fields) override;
00055     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00056 protected:
00057     //virtual void _initBetweenReplications();
00058     virtual bool _check(std::string& errorMessage) override;
00059 protected:
00060     virtual void _createInternalStatisticReporters() override;
00061     // virtual void _createNonEditableDataDefinitions() override;
00062     // virtual void _createEditableDataDefinitions() override;
00063     // virtual void _createAttachedAttributes() override;
00064
00065 protected:
00066
00067 private:
00068     List<std::string>* _conditions = new List<std::string>();
00069     List<Counter*>* _numberOuts = nullptr;
00070 private:
00071
00072 };
00073
00074 #endif /* DECIDE_H */
```

18.388 DropOff.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   DropOff.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:15
00012  */
00013
00014 #ifndef DROPOFF_H
00015 #define DROPOFF_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018
00030 class DropOff : public ModelComponent {
00031 public: // constructors
00032     DropOff(Model* model, std::string name = "");
00033     virtual ~DropOff() = default;
00034 public: // virtual
00035     virtual std::string show() override;
00036 public:
00037     void setQuantityExpression(std::string quantityExpression);
00038     std::string getQuantityExpression() const;
00039     void setStartingRankExpression(std::string startingRankExpression);
00040     std::string getStartingRankExpression() const;
00041 public: // static
00042     static PluginInformation* GetPluginInformation();
00043     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00044     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00045 protected: // virtual
00046     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00047     virtual bool _loadInstance(PersistenceRecord *fields) override;
00048 protected: // virtual
00049     //virtual void _initBetweenReplications();
00050     // virtual void _createInternalStatisticReporters() override;
00051     // virtual void _createNonEditableDataDefinitions() override;
00052     // virtual void _createEditableDataDefinitions() override;
00053     // virtual void _createAttachedAttributes() override;
```

```
00054     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00055     virtual bool _check(std::string& errorMessage) override;
00056
00057 protected:
00058
00059 private: // methods
00060 private: // attributes 1:1
00061     const struct DEFAULT_VALUES {
00062         const std::string quantityExpression = "1";
00063         const std::string startingRankExpression = "1";
00064     } DEFAULT;
00065     std::string _quantityExpression = DEFAULT.quantityExpression;
00066     std::string _startingRankExpression = DEFAULT.startingRankExpression;
00067 private: // attributes 1:n
00068 };
00069
00070
00071 #endif /* DROPOFF_H */
```

18.389 PickUp.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   PickUp.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:15
00012  */
00013
00014 #ifndef PICKUP_H
00015 #define PICKUP_H
00016
00017 #include "plugins/components/Decisions/Remove.h"
00018
00028 class PickUp : public Remove {
00029 public: // constructors
00030     PickUp(Model* model, std::string name = "");
00031     virtual ~PickUp() = default;
00032 public: // static
00033     static PluginInformation* GetPluginInformation();
00034     //static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00035     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00036 };
00037
00038
00039 #endif /* PICKUP_H */
00040
```

18.390 Remove.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Remove.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:20
00012  */
00013
00014 #ifndef REMOVE_H
00015 #define REMOVE_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018
00033 class Remove : public ModelComponent {
00034 public:
00035
00036     enum class RemoveFromType : int {
00037         QUEUE = 0, ENTITYGROUP = 1, num_elements = 2
00038     };
00039 public:
00040     static std::string convertEnumToStr(RemoveFromType type);
00041 public: // constructors
```

```
00042     Remove(Model* model, std::string name = "");
00043     virtual ~Remove() = default;
00044 public: // virtual
00045     virtual std::string show() override;
00046 public:
00047     void setRemoveStartRank(std::string _removeFromRank);
00048     std::string getRemoveStartRank() const;
00049     void setRemoveEndRank(std::string _removeEndRank);
00050     std::string getRemoveEndRank() const;
00051     void setRemoveFromType(Remove::RemoveFromType _removeFromType);
00052     Remove::RemoveFromType getRemoveFromType() const;
00053     void setRemoveFrom(ModelDataDefinition* _removeFrom);
00054     ModelDataDefinition* getRemoveFrom() const;
00055
00056 public: // static
00057     static PluginInformation* GetPluginInformation();
00058     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00059     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00060 protected: // must be overriden
00061     virtual bool _loadInstance(PersistenceRecord *fields) override;
00062     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00063     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00064 protected: // could be overriden by derived classes
00065     virtual bool _check(std::string& errorMessage) override;
00067     //virtual ParserChangesInformation* _getParserChangesInformation();
00068     //virtual void _initBetweenReplications();
00069 protected:
00070     // virtual void _createInternalStatisticReporters() override;
00071     // virtual void _createNonEditableDataDefinitions() override;
00072     virtual void _createEditableDataDefinitions() override;
00073     // virtual void _createAttachedAttributes() override;
00075     // virtual void _createInternalAndAttachedData() override;
00076     //virtual void _addSimulationControl(SimulationControl* property);
00077
00078 protected:
00079
00080 private: // methods
00081 private: // attributes 1:1
00082
00083     const struct DEFAULT_VALUES {
00084         const Remove::RemoveFromType removeFromType = Remove::RemoveFromType::QUEUE;
00085         const std::string removeStartRank = "";
00086         const std::string removeEndRank = "";
00087     } DEFAULT;
00088     ModelDataDefinition* _removeFrom = nullptr;
00089     Remove::RemoveFromType _removeFromType = DEFAULT.removeFromType;
00090     std::string _removeStartRank = DEFAULT.removeStartRank;
00091     std::string _removeEndRank = DEFAULT.removeStartRank;
00092     //std::string _removeFromRank = DEFAULT.removeFromRank;
00093
00094 private: // attributes 1:n
00095 };
00096
00097
00098 #endif /* REMOVE_H */
```

18.391 Search.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Search.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:20
00012  */
00013
00014 #ifndef SEARCH_H
00015 #define SEARCH_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018
00036 class Search : public ModelComponent {
00037 public:
00038     enum class SearchInType : int {
00039         QUEUE = 0, ENTITYGROUP = 1, num_elements = 2
00040     };
00041 public:
00042     static std::string convertEnumToStr(SearchInType type);
00043 public: // constructors
00044     Search(Model* model, std::string name = "");
```

```
00045     virtual ~Search() = default;
00046 public: // virtual
00047     virtual std::string show() override;
00048 public:
00049     void setSaveFounRankAttribute(std::string _saveFounRankAttribute);
00050     std::string getSaveFounRankAttribute() const;
00051     void setSearchCondition(std::string _searchCondition);
00052     std::string getSearchCondition() const;
00053     void setEndRank(std::string _endRank);
00054     std::string getEndRank() const;
00055     void setStartRank(std::string _startRank);
00056     std::string getStartRank() const;
00057     void setSearchInName(std::string searchInName);
00058     std::string getSearchInName() const;
00059     void setSearchIn(ModelDataDefinition* _searchIn);
00060     ModelDataDefinition* getSearchIn() const;
00061     void setSearchInType(Search::SearchInType _searchInType);
00062     Search::SearchInType getSearchInType() const;
00063 public: // static
00064     static PluginInformation* GetPluginInformation();
00065     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00066     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00067
00068 protected: // must be overriden
00069     virtual bool _loadInstance(PersistenceRecord *fields) override;
00070     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00071     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00072 protected: // could be overriden by derived classes
00073     virtual bool _check(std::string& errorMessage) override;
00075     //virtual ParserChangesInformation* _getParserChangesInformation();
00076     //virtual void _initBetweenReplications();
00077 protected:
00078     // virtual void _createInternalStatisticReporters() override;
00079     // virtual void _createNonEditableDataDefinitions() override;
00080      virtual void _createEditableDataDefinitions() override;
00081     // virtual void _createAttachedAttributes() override;
00083     // virtual void _createInternalAndAttachedData() override;
00084     //virtual void _addSimulationControl(SimulationControl* property);
00085
00086 protected:
00087
00088 private: // methods
00089 private: // attributes 1:1
00090
00091     const struct DEFAULT_VALUES {
00092         const SearchInType searchInType = SearchInType::QUEUE;
00093         const std::string startRank = "";
00094         const std::string endRank = "";
00095         const std::string searchCondition = "";
00096         const std::string saveFounRankAttribute = "";
00097     } DEFAULT;
00098     ModelDataDefinition* _searchIn = nullptr;
00099     Search::SearchInType _searchInType = DEFAULT.searchInType;
00100     std::string _startRank = DEFAULT.startRank;
00101     std::string _endRank = DEFAULT.endRank;
00102     std::string _searchCondition = DEFAULT.searchCondition;
00103     std::string _saveFounRankAttribute = DEFAULT.saveFounRankAttribute;
00104
00105 private: // attributes 1:n
00106 };
00107
00108
00109 #endif /* SEARCH_H */
```

18.392 QueueableItem.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   QueueableItem.h
00009  * Author: rlcancian
00010  *
00011  * Created on 23 de abril de 2021, 15:09
00012  */
00013
00014 #ifndef QUEUEABLEITEM_H
00015 #define QUEUEABLEITEM_H
00016
00017 #include "plugins/data/DiscreteProcessing/Queue.h"
00018 #include "../../../data/Logic/Set.h"
00019 #include "../../../../kernel/simulator/model/ModelDataManager.h"
```

```
00020
00021 // @TODO should inhere from a common base to SeizeableItem
00022
00023 class QueueableItem {
00024 public:
00025
00026     enum class QueueableType : int {
00027         QUEUE = 0, SET = 1, num_elements = 2 //, HOLD = 2, num_elements = 3
00028     };
00029 public:
00030     static std::string convertEnumToStr(QueueableType type);
00031 public:
00032     QueueableItem(ModelDataDefinition* queueOrSet, QueueableItem::QueueableType queueableType =
      QueueableItem::QueueableType::QUEUE, std::string index = "0");
00033     QueueableItem(Model* model, std::string queueName);
00034 public:
00035     bool loadInstance(PersistenceRecord *fields);
00036     void saveInstance(PersistenceRecord *fields, bool saveDefaultValues);
00037 public:
00038     std::string show();
00039     void setIndex(std::string index);
00040     std::string getIndex() const;
00041     std::string getQueueableName() const;
00042     std::string getName() const;
00043     void setQueue(Queue* resource);
00044     Queue* getQueue() const;
00045     void setSet(Set* set);
00046     Set* getSet() const;
00047     //
00048     void setQueueableType(QueueableType queueableType);
00049     QueueableType getQueueableType() const;
00050     ModelDataDefinition* getQueueable() const;
00051     void setElementManager(ModelDataManager* _modeldataManager);
00052     //void setComponentManager(ModelComponentManager* _componentManager);
00053     List<SimulationControl*>* getSimulationControls() const;
00054     void _addSimulationControl(SimulationControl* control);
00055
00056     std::string getTypeDC() {return _typeDC;};
00057 private:
00058     void _ensureSimulationControls(Model* model);
00059
00060 private:
00061
00062     const struct DEFAULT_VALUES {
00063         const std::string quantityExpression = "1";
00064         const QueueableItem::QueueableType queueableType = QueueableItem::QueueableType::QUEUE;
00065         const std::string saveAttribute = "";
00066         const std::string index = "0";
00067     } DEFAULT;
00068
00069     QueueableType _queueableType = DEFAULT.queueableType;
00070     ModelDataDefinition* _queueOrSet = nullptr;
00071     std::string _queueableName;
00072     std::string _index = DEFAULT.index;
00073     std::string _typeDC = Util::TypeOf<Queue>();
00074 private:
00075     //ModelComponentManager* _componentManager;
00076     ModelDataManager* _modeldataManager = nullptr;
00077     List<SimulationControl*>* _simulationControls = new List<SimulationControl*>();
00078 };
00079
00080
00081 #endif /* QUEUEABLEITEM_H */
```

18.393 SeizableItem.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   SeizableItem.h
00009  * Author: rlcancian
00010  *
00011  * Created on 10 de abril de 2021, 08:45
00012  */
00013
00014 #ifndef SEIZABLEITEM_H
00015 #define SEIZABLEITEM_H
00016
00017 #include "plugins/data/DiscreteProcessing/Resource.h"
00018 #include "../../../data/Logic/Set.h"
00019 #include "../../../../kernel/simulator/model/ModelDataManager.h"
```

```
00020
00021 // @TODO should inhere from a common base to QueueableItem
00022
00023 class SeizableItem {
00024 public:
00025     enum class SelectionRule : int {
00026         CYCLICAL = 0, RANDOM = 1, SPECIFICMEMBER = 2, LARGESTREMAININGCAPACITY = 3, SMALLESTNUMBERBUSY
      = 4,
00027         PREFEREDORDER = 5, num_elements = 6
00028     };
00029
00030     enum class SeizableType : int {
00031         RESOURCE = 0, SET = 1, num_elements = 2
00032     };
00033
00034     static std::string convertEnumToStr(SelectionRule rule);
00035     static std::string convertEnumToStr(SeizableType type);
00036
00037 public:
00038     SeizableItem(ModelDataDefinition* resourceOrSet, std::string quantityExpression = "1",
00039                  SeizableItem::SelectionRule selectionRule =
      SeizableItem::SelectionRule::LARGESTREMAININGCAPACITY,
00040                  std::string saveAttribute = "", std::string index = "0");
00041     SeizableItem(Model* model, std::string resourceName, std::string quantityExpression = "1",
00042                  SeizableItem::SelectionRule selectionRule =
      SeizableItem::SelectionRule::LARGESTREMAININGCAPACITY,
00043                  std::string saveAttribute = "", std::string index = "0");
00044     SeizableItem(SeizableItem* original);
00045
00046 public:
00047     bool loadInstance(PersistenceRecord* fields, unsigned int parentIndex);
00048     void saveInstance(PersistenceRecord* fields, unsigned int parentIndex, bool saveDefault);
00049     bool loadInstance(PersistenceRecord* fields);
00050     void saveInstance(PersistenceRecord* fields, bool saveDefaults);
00051
00052 public:
00053     std::string show();
00054     void setIndex(std::string index);
00055     std::string getIndex() const;
00056     void setSaveAttribute(std::string saveAttribute);
00057     std::string getSaveAttribute() const;
00058     void setSelectionRule(SelectionRule selectionRule);
00059     SelectionRule getSelectionRule() const;
00060     void setQuantityExpression(std::string quantityExpression);
00061     std::string getQuantityExpression() const;
00062     std::string getResourceName() const;
00063     std::string getName() const;
00064     void setResource(Resource* resource);
00065     Resource* getResource() const;
00066     void setSet(Set* set);
00067     Set* getSet() const;
00068     //void setSeizableType(SeizableType resourceType);
00069     SeizableType getSeizableType() const;
00070     void setLastMemberSeized(unsigned int lastMemberSeized);
00071     unsigned int getLastMemberSeized() const;
00072     ModelDataDefinition* getSeizable() const;
00073     void setElementManager(ModelDataManager* modeldataManager);
00074     void setLastPreferedOrder(unsigned int lastPreferedOrder);
00075     unsigned int getLastPreferedOrder() const;
00076     //void setComponentManager(ModelComponentManager* _componentManager);
00077     List<SimulationControl*>* getSimulationControls() const;
00078     void _addSimulationControl(SimulationControl* control);
00079
00080     std::string getTypeDC() { return _typeDC; };
00081
00082 private:
00083     void _ensureSimulationControls(Model* model);
00084
00085 private:
00086     const struct DEFAULT_VALUES {
00087         const std::string quantityExpression = "1";
00088         const SeizableItem::SeizableType seizableType = SeizableItem::SeizableType::RESOURCE;
00089         const SeizableItem::SelectionRule selectionRule =
      SeizableItem::SelectionRule::LARGESTREMAININGCAPACITY;
00090         const std::string saveAttribute = "";
00091         const std::string index = "0";
00092     } DEFAULT;
00093
00094     SeizableType _seizableType = DEFAULT.seizableType;
00095     SelectionRule _selectionRule = DEFAULT.selectionRule;
00096     std::string _saveAttribute = DEFAULT.saveAttribute;
00097     std::string _index = DEFAULT.index;
00098     ModelDataDefinition* _resourceOrSet = nullptr;
00099     std::string _seizableName;
00100     std::string _quantityExpression;
00101     unsigned int _lastMemberSeized = 0;
00102     unsigned int _lastPreferedOrder = 0;
```

```
00103     std::string _typeDC = Util::TypeOf<Resource>();
00104
00105 private:
00106     ModelDataManager* _modeldataManager;
00107     List<SimulationControl*>* _simulationControls = new List<SimulationControl*>();
00108 };
00109
00110 #endif /* SEIZABLEITEM_H */
```

18.394 Buffer.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Buffer.h
00009  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00010  *
00011  * Created on
00012  */
00013
00014 #ifndef BUFFER_H
00015 #define BUFFER_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../data/Synchronization/SignalData.h"
00019
00023 class Buffer : public ModelComponent {
00024 public:
00025     enum class AdvanceOn : int {
00026         NewArrivals = 0, Signal = 1, num_elements = 2
00027     };
00028     enum class ArrivalOnFullBufferRule : int {
00029         Dispose = 0, SendToBulkPort = 1, ReplaceLastPosition = 2, num_elements = 3
00030     };
00031     static std::string convertEnumToStr(AdvanceOn advance);
00032     static std::string convertEnumToStr(ArrivalOnFullBufferRule arrival);
00033 public: // constructors
00034     Buffer(Model* model, std::string name = "");
00035     virtual ~Buffer() override;
00036 public: // virtual
00037     virtual std::string show() override;
00038 public: // static
00039     static PluginInformation* GetPluginInformation();
00040     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00041     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00042 public:
00043     Buffer::ArrivalOnFullBufferRule getarrivalOnFullBufferRule() const;
00044     void setArrivalOnFullBufferRule(Buffer::ArrivalOnFullBufferRule newArrivalOnFullBufferRule);
00045     Buffer::AdvanceOn getadvanceOn() const;
00046     void setAdvanceOn(Buffer::AdvanceOn newAdvanceOn);
00047     unsigned int getcapacity() const;
00048     void setCapacity(unsigned int newCapacity);
00049     SignalData *getsignal() const;
00050     void setSignal(SignalData *newSignal);
00051
00052 protected: // must be overriden
00053     virtual bool _loadInstance(PersistenceRecord *fields) override;
00054     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00055     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00056 protected: // could be overriden by derived classes
00057     virtual bool _check(std::string& errorMessage) override;
00059     virtual ParserChangesInformation* _getParserChangesInformation() override;
00060     virtual void _initBetweenReplications() override;
00062     // virtual void _createInternalAndAttachedData() override;
00063     virtual void _addSimulationControl(SimulationControl* property) override;
00064
00065 protected:
00066     // virtual void _createInternalStatisticReporters() override;
00067     // virtual void _createNonEditableDataDefinitions() override;
00068     virtual void _createEditableDataDefinitions() override;
00069     // virtual void _createAttachedAttributes() override;
00070
00071 private: // methods
00072     unsigned int _handlerForSignalDataEvent(SignalData* signalData);
00073     Entity* _advance(Entity* enteringEntity);
00074 private: // attributes 1:1
00075
00076     const struct DEFAULT_VALUES {
00077         const AdvanceOn advanceOn = AdvanceOn::NewArrivals;
00078         const ArrivalOnFullBufferRule arrivalOnFullBufferRule = ArrivalOnFullBufferRule::Dispose;
00079         const unsigned int capacity = 1;
```

```
00080     } DEFAULT;
00081     ArrivalOnFullBufferRule _arrivalOnFullBufferRule = DEFAULT.arrivalOnFullBufferRule;
00082     AdvanceOn _advanceOn = DEFAULT.advanceOn;
00083     unsigned int _capacity = DEFAULT.capacity;
00084 private:
00085     std::vector<Entity*>* _buffer = new std::vector<Entity*>;
00086 private: // attached
00087     SignalData* _attachedSignal = nullptr;
00088     SignalData* _signalWithRegisteredHandler = nullptr;
00089 };
00090
00091 #endif /* BUFFER_H */
```

18.395 Clone.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Clone.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 30 de Novembro de 2021, 18:50
00012  */
00013
00014 #ifndef CLONE_H
00015 #define CLONE_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../../kernel/simulator/essentialPlugins/Counter.h"
00019
00038 class Clone : public ModelComponent {
00039 public: // constructors
00040     Clone(Model* model, std::string name = "");
00041     virtual ~Clone() = default;
00042 public: // virtual
00043     virtual std::string show() override;
00044 public: //
00045     void setNumClonesExpression(std::string numClones);
00046     std::string getNumClonesExpression() const;
00047 public: // static
00048     static PluginInformation* GetPluginInformation();
00049     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00050     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00051 protected: // must be overriden
00052     virtual bool _loadInstance(PersistenceRecord *fields) override;
00053     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00054     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00055 protected: // could be overriden .
00056     virtual bool _check(std::string& errorMessage) override;
00057     virtual void _initBetweenReplications() override;
00058     // virtual void _createInternalAndAttachedData() override;
00059     //virtual ParserChangesInformation* _getParserChangesInformation();
00060
00061 protected:
00062     virtual void _createInternalStatisticReporters() override;
00063     // virtual void _createNonEditableDataDefinitions() override;
00064     // virtual void _createEditableDataDefinitions() override;
00065     // virtual void _createAttachedAttributes() override;
00066
00067 private: // methods
00068 private: // attributes 1:1
00069
00070     const struct DEFAULT_VALUES {
00071         const std::string numClonesExpression = "1";
00072     } DEFAULT;
00073     std::string _numClonesExpression = DEFAULT.numClonesExpression;
00074     Counter* _counter = nullptr;
00075 private: // attributes 1:n
00076 };
00077
00078 #endif /* CLONE_H */
```

18.396 Delay.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
```

```
00005  */
00006
00007 /*
00008  * File:   Delay.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Junho de 2018, 19:49
00012  */
00013
00014 #ifndef DELAY_H
00015 #define DELAY_H
00016
00017 #include <string>
00018 #include <vector>
00019 #include "../../../kernel/simulator/model/ModelComponent.h"
00020 #include "kernel/simulator/Plugin.h"
00021
00043 class Delay : public ModelComponent {
00044 public:
00045     Delay(Model* model, std::string name = "");
00046     virtual ~Delay() = default;
00047 public:
00048     //void setDelayExpression(std::string _delayExpression);
00049     void setDelayExpression(std::string _delayExpression, Util::TimeUnit
      _delayTimeUnit=Util::TimeUnit::unknown);
00050     std::string delayExpression() const;
00051     void setDelay(double delay);
00052     double delay() const;
00053     void setDelayTimeUnit(Util::TimeUnit _delayTimeUnit);
00054     Util::TimeUnit delayTimeUnit() const;
00055     void setAllocation(Util::AllocationType allocation);
00056     Util::AllocationType getAllocation() const;
00057 public:
00058     virtual std::string show() override;
00059 public:
00060     static PluginInformation* GetPluginInformation();
00061     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00062     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00063 //public: // properties
00064     //using property_owner_t = Delay;
00065     //decl_property(DelayExpression, decl_get(std::string){return delayExpression();} void
      decl_set(std::string val){setDelayExpression(val);});
00066
00067 protected:
00068     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00069     virtual bool _loadInstance(PersistenceRecord *fields) override;
00070     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00071 protected:
00072     //virtual void _initBetweenReplications();
00073     virtual bool _check(std::string& errorMessage) override;
00074 protected:
00075     virtual void _createInternalStatisticReporters() override;
00076     // virtual void _createNonEditableDataDefinitions() override;
00077     // virtual void _createEditableDataDefinitions() override;
00078     virtual void _createAttachedAttributes() override;
00079 public:
00080     const struct DEFAULT_VALUES {
00081         const std::string delayExpression = "1.0";
00082         const Util::TimeUnit delayTimeUnit = Util::TimeUnit::second;
00083         const Util::AllocationType allocation = Util::AllocationType::Wait;
00084
00085     } DEFAULT;
00086
00087 protected:
00088
00089 private:
00090     std::string _delayExpression = DEFAULT.delayExpression;
00091     Util::TimeUnit _delayTimeUnit = DEFAULT.delayTimeUnit;
00092     Util::AllocationType _allocation = DEFAULT.allocation;
00093 private:
00094     std::vector<std::string> _allAllocationAttachedAttributeNames() const;
00095     std::string _allocationAttachedAttributeName(Util::AllocationType allocation) const;
00096     void _reconcileAllocationAttachedAttributes();
00097     void _initCStats();
00098 private: // inner internal elements
00099     friend class DelayProbe;
00100     StatisticsCollector* _cstatWaitTime = nullptr;
00101 };
00102 //enable_this_owner(Delay, DelayExpression);
00103
00104 #endif /* DELAY_H */
```

18.397 Process.h

```
00001 /*
```

```
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Process.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 22 de Maio de 2019, 18:41
00012  */
00013
00014 #ifndef PROCESS_H
00015 #define PROCESS_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "plugins/components/DiscreteProcessing/Seize.h"
00019 #include "plugins/components/DiscreteProcessing/Delay.h"
00020 #include "plugins/components/DiscreteProcessing/Release.h"
00021
00047 class Process : public ModelComponent {
00048 public: // constructors
00049     Process(Model* model, std::string name = "");
00050     virtual ~Process() = default;
00051 public: // virtual
00052     virtual std::string show() override;
00053 public: // static
00054     static PluginInformation* GetPluginInformation();
00055     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00056     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00057 public:
00058     void setPriority(unsigned short _priority);
00059     unsigned short getPriority() const;
00060     void setPriorityExpression(std::string priority);
00061     std::string getPriorityExpression() const;
00062     void setAllocationType(Util::AllocationType _allocationType);
00063     Util::AllocationType getAllocationType() const;
00064     List<SeizableItem*>* getSeizeRequests() const;
00065     void addSeizeRequest(SeizableItem* newRequest);
00066     void removeSeizeRequest(SeizableItem* request);
00067     void setQueueableItem(QueueableItem* _queueableItem);
00068     QueueableItem* getQueueableItem() const;
00069     void setDelayExpression(std::string _delayExpression);
00070     void setDelayExpression(std::string _delayExpression, Util::TimeUnit _delayTimeUnit);
00071     std::string delayExpression() const;
00072     void setDelayTimeUnit(Util::TimeUnit _delayTimeUnit);
00073     Util::TimeUnit delayTimeUnit() const;
00074 protected: // virtual
00075     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00076     // virtual void _initBetweenReplications();
00077     virtual bool _loadInstance(PersistenceRecord *fields) override;
00078     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00079 protected: // virtual
00080     virtual bool _check(std::string& errorMessage) override;
00081 protected: // virtual
00082     // virtual void _createInternalStatisticReporters() override;
00083     virtual void _createNonEditableDataDefinitions() override;
00084     virtual void _createEditableDataDefinitions() override;
00085     // virtual void _createAttachedAttributes() override;
00086     // virtual void _createInternalAndAttachedData() override;
00087
00088 protected:
00089
00090 private: // methods
00091     void _adjustConnections();
00092     void _ensureInternalComponents();
00093     void _reconcileInternalComponents();
00094     bool _hasConsistentInternalChain(std::string& errorMessage) const;
00095 private: // attributes 1:1
00096     Seize* _seize = nullptr;
00097     Delay* _delay = nullptr;
00098     Release* _release = nullptr;
00099     bool _flagConstructing;
00100     //ModelComponent* _nextComponent = nullptr;
00101     //unsigned int _nextInput;
00102 private: // attributes 1:n
00103 };
00104
00105 #endif /* PROCESS_H */
```

18.398 Release.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
```

```
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Release.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Agosto de 2018, 16:17
00012  */
00013
00014 #ifndef RELEASE_H
00015 #define RELEASE_H
00016
00017 #include <string>
00018
00019 #include "../../../kernel/simulator/model/ModelComponent.h"
00020 #include "kernel/simulator/Plugin.h"
00021 #include "auxiliar/SeizableItem.h"
00022 #include "plugins/data/DiscreteProcessing/Resource.h"
00023
00040 class Release : public ModelComponent {
00041 public:
00042     Release(Model* model, std::string name = "");
00043     virtual ~Release() override;
00044 public:
00045     virtual std::string show() override;
00046 public: //static
00047     static PluginInformation* GetPluginInformation();
00048     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00049     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00050 public: // get & set
00051     void setPriority(unsigned short _priority);
00052     unsigned short priority() const;
00053 public: // gets
00054     List<SeizableItem*>* getReleaseRequests() const;
00055     void addReleaseRequests(SeizableItem* newRequest);
00056     void removeReleaseRequests(SeizableItem* request);
00057
00058 protected:
00059     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00060     virtual bool _loadInstance(PersistenceRecord *fields) override;
00061     virtual void _initBetweenReplications() override;
00062     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00063     virtual bool _check(std::string& errorMessage) override;
00064     // virtual void _createInternalAndAttachedData() override;
00065
00066 protected:
00067     // virtual void _createInternalStatisticReporters() override;
00068     // virtual void _createNonEditableDataDefinitions() override;
00069     virtual void _createEditableDataDefinitions() override;
00070     // virtual void _createAttachedAttributes() override;
00071
00072 private:
00073     Resource* _getResourceFromSeizableItem(SeizableItem* seizable, Entity* entity);
00074 private:
00075
00076     const struct DEFAULT_VALUES {
00077         const unsigned short priority = 0;
00078         const unsigned int releaseRequestSize = 1;
00079     } DEFAULT;
00080     unsigned short _priority = DEFAULT.priority;
00081     List<SeizableItem*>* _releaseRequests = new List<SeizableItem*>();
00082 };
00083
00084 #endif /* RELEASE_H */
```

18.399 Seize.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Seize.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Agosto de 2018, 16:17
00012  */
00013
00014 #ifndef SEIZE_H
00015 #define SEIZE_H
00016
00017 #include <string>
```

```
00018 #include "../../../kernel/simulator/model/ModelComponent.h"
00019 #include "../../../kernel/simulator/model/Model.h"
00020 #include "plugins/data/DiscreteProcessing/Resource.h"
00021 #include "plugins/data/DiscreteProcessing/Queue.h"
00022 #include "kernel/simulator/Plugin.h"
00023 #include "auxiliar/SeizableItem.h"
00024 #include "auxiliar/QueueableItem.h"
00025
00026 class WaitingResource : public Waiting {
00027 public:
00028
00029     WaitingResource(Entity* entity, double timeStartedWaiting, unsigned int quantity, ModelComponent*
      thisComponent, unsigned int thisComponentOutputPort = 0) : Waiting(entity, timeStartedWaiting,
      thisComponent, thisComponentOutputPort) {
00030         _quantity = quantity;
00031     }
00032
00033     WaitingResource(const WaitingResource& orig) : Waiting(orig) {
00034     }
00035
00036     virtual ~WaitingResource() = default;
00037 public:
00038
00039     virtual std::string show() override {
00040         return Waiting::show() +
00041                 ",quantity=" + std::to_string(this->_quantity);
00042     }
00043 public:
00044
00045     unsigned int getQuantity() const {
00046         return _quantity;
00047     }
00048
00049 protected:
00050 private:
00051     unsigned int _quantity;
00052 };
00053
00076 class Seize : public ModelComponent {
00077 public:
00078     Seize(Model* model, std::string name = "");
00079     virtual ~Seize() override;
00080 public:
00081     virtual std::string show() override;
00082 public:
00083     static PluginInformation* GetPluginInformation();
00084     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00085     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00086 public: // get & set
00087     void setPriority(unsigned short _priority);
00088     unsigned short getPriority() const;
00089     void setAllocationType(Util::AllocationType _allocationType);
00090     Util::AllocationType getAllocationType() const;
00091     // indirect access to Queue* and Resource*
00092     //void setResourceName(std::string _resourceName) throw ();
00093     //std::string getResourceName() const;
00094     //void setQueueName(std::string queueName) throw ();
00095     //std::string getQueueName() const;
00096     void setQueue(Queue* queue);
00097     //Queue* getQueue() const;
00098     List<SeizableItem*>* getSeizeRequests() const;
00099     void addRequest(SeizableItem* newRequest);
00100     void removeRequest(SeizableItem* request);
00101     void setQueueableItem(QueueableItem* _queueableItem);
00102     QueueableItem* getQueueableItem() const;
00103     void setPriorityExpression(std::string _priorityExpression);
00104     std::string getPriorityExpression() const;
00105 protected:
00106     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00107     virtual bool _loadInstance(PersistenceRecord *fields) override;
00108     virtual void _initBetweenReplications() override;
00109     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00110     virtual bool _check(std::string& errorMessage) override;
00111 protected:
00112     // virtual void _createInternalStatisticReporters() override;
00113     // virtual void _createNonEditableDataDefinitions() override;
00114     virtual void _createEditableDataDefinitions() override;
00115     // virtual void _createAttachedAttributes() override;
00116     // virtual void _createInternalAndAttachedData() override;
00117 private:
00118     void _handlerForResourceEvent(Resource* resource);
00119     Resource* _getResourceFromSeizableItem(SeizableItem* seizable, Entity* entity, unsigned
      int*indexPtr);
00120     Queue* _getQueue() const;
00121 public:
00122
00123     const struct DEFAULT_VALUES {
```

```
00124         const Util::AllocationType allocationType = Util::AllocationType::Others;
00125         const unsigned short priority = 0;
00126         const std::string priorityExpression = "";
00127         const unsigned int seizeRequestSize = 1;
00128     } DEFAULT;
00129 private:
00130     Util::AllocationType _allocationType = DEFAULT.allocationType; // uint ? enum?
00131     unsigned short _priority = DEFAULT.priority;
00132     std::string _priorityExpression = DEFAULT.priorityExpression;
00133     QueueableItem* _queueableItem = nullptr; // usually has a queue, but not always (it could be a
      hold or a set)
00134     List<SeizableItem*>* _seizeRequests = new List<SeizableItem*>();
00135 };
00136
00137 #endif /* SEIZE_H */
```

18.400 SPICECircuit.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   SPICECircuit.h
00009  * Author: bnmfw & joaomai
00010  *
00011  * Created on 10 de outubro de 2023, 23:00
00012  */
00013
00014 #pragma once
00015
00016 #include "../../../kernel/simulator/model/ModelComponent.h"
00017 #include "../../data/ExternalIntegration/SPICERunner.h"
00018 #include "plugins/components/ElectronicsSimulation/SPICENode.h"
00019
00023 class SPICECircuit : public ModelComponent {
00024 public:
00025     SPICECircuit(Model* model, std::string name = "");
00026     virtual ~SPICECircuit() = default;
00027
00028 public:
00029     void setRunner(SPICERunner* runner);
00030     SPICERunner* getRunner();
00031
00032 public:
00033     virtual std::string show() override;
00034
00035 public:
00036     static PluginInformation* GetPluginInformation();
00037     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00038     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00039     // Added
00040     static std::vector<std::string> split(std::string word, char split_char);
00041
00042 protected:
00043     void BuildCircuit(std::string description, unsigned int id, std::vector<std::string> the_params =
      {""}, std::string the_model = "", std::string the_model_file = "");
00044     void UpdateConnections();
00045     virtual void Build() { id = counter++; build = [this]() {this->Build();};}
00046     virtual bool _loadInstance(PersistenceRecord *fields) override;
00047     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00048     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00049     // virtual void _createInternalAndAttachedData() override;
00050
00051
00052 protected:
00054     // virtual bool _check(std::string& errorMessage);
00056     // virtual ParserChangesInformation* _getParserChangesInformation();
00058     // virtual void _initBetweenReplications();
00059     // virtual void _createInternalStatisticReporters() override;
00060     // virtual void _createEditableDataDefinitions() override;
00061     virtual void _createNonEditableDataDefinitions() override;
00062     // virtual void _createAttachedAttributes() override;
00064     // virtual void _addSimulationControl(SimulationControl* property);
00065 protected:
00066
00067 private:
00068     // Spice subcircuit declaration, if one exists
00069     std::string spice_template;
00070     // Name of component
00071     std::string spice_name;
00072     // Params of the component, such as resistance or voltage
00073     std::vector<std::string> params;
```

```
00074     // Pin map of the component
00075     std::vector<std::string> pins;
00076     // Instace of the component in spice
00077     std::string *spice_instance;
00078     // Model of the component, if one exists (the model name should also be the name of the .cir file)
00079     std::string model;
00080     std::string model_file;
00081     // Spice compiler
00082     SPICERunner *compiler = nullptr;
00083 
00084     static unsigned int counter;
00085     bool plain_circuit;
00086 protected:
00087     unsigned int id;
00088     std::function<void()> build;
00089 
00090 private:
00091 
00093     const struct DEFAULT_VALUES {
00094         const std::string someString = "Test";
00095         const unsigned int someUint = 1;
00096     } DEFAULT;
00097     std::string _someString = DEFAULT.someString;
00098     unsigned int _someUint = DEFAULT.someUint;
00099 
00100 private:
00101 
00102 private:
00103     // ...
00104 };
00105 
00106 class Resistor : public SPICECircuit {
00107 public:
00108     Resistor(Model* model, std::string name = ""): SPICECircuit(model, name) {
00109         id = counter++;
00110         build = [this]() {this->Build();};
00111     }
00112     void Build() override;
00113     void setResistance(float resistance);
00114 
00115     static PluginInformation* GetPluginInformation();
00116     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00117     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00118 
00119 protected:
00120     virtual bool _loadInstance(PersistenceRecord *fields) override;
00121     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00122 
00123 private:
00124     const struct DEFAULT_VALUES {
00125         const std::string someString = "Test";
00126         const unsigned int someUint = 1;
00127         const float someResistance = 1000;
00128     } DEFAULT;
00129 
00130     static unsigned int counter;
00131     float resistance = DEFAULT.someResistance;
00132 };
00133 
00134 class Vsource : public SPICECircuit {
00135 public:
00136     Vsource(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build =
      [this]() {this->Build();};}
00137     void Build() override;
00138     void setVoltage(float voltage);
00139 
00140     static PluginInformation* GetPluginInformation();
00141     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00142     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00143 
00144 protected:
00145     virtual bool _loadInstance(PersistenceRecord *fields) override;
00146     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00147 
00148 private:
00149     const struct DEFAULT_VALUES {
00150         const std::string someString = "Test";
00151         const unsigned int someUint = 1;
00152         const float someVoltage = 5;
00153     } DEFAULT;
00154 
00155     float voltage = DEFAULT.someVoltage;
00156     static unsigned int counter;
00157 };
00158 
00159 class Vpulse : public SPICECircuit {
00160 public:
```

```
00161     Vpulse(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build =
      [this]() {this->Build();};}
00162     void Build() override;
00163     void setVoltage(float voltage);
00164     void setFreq(float freq);
00165     void setSlope(float slope);
00166
00167     static PluginInformation* GetPluginInformation();
00168     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00169     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00170
00171 protected:
00172     virtual bool _loadInstance(PersistenceRecord *fields) override;
00173     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00174
00175 private:
00176     const struct DEFAULT_VALUES {
00177         const std::string someString = "Test";
00178         const unsigned int someUint = 1;
00179         const float someVoltage = 5;
00180         const float someFreq = 60;
00181         const float someSlope = 1e-11;
00182     } DEFAULT;
00183
00184     static unsigned int counter;
00185     float voltage = DEFAULT.someVoltage;
00186     float freq = DEFAULT.someFreq;
00187     float slope = DEFAULT.someSlope;
00188 };
00189
00190 class Vsine : public SPICECircuit {
00191 public:
00192     Vsine(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build =
      [this]() {this->Build();};}
00193     void Build() override;
00194     void setVoltage(float voltage);
00195     void setFreq(float freq);
00196
00197     static PluginInformation* GetPluginInformation();
00198     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00199     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00200
00201 protected:
00202     virtual bool _loadInstance(PersistenceRecord *fields) override;
00203     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00204
00205 private:
00206     const struct DEFAULT_VALUES {
00207         const std::string someString = "Test";
00208         const unsigned int someUint = 1;
00209         const float someVoltage = 5;
00210         const float someFreq = 60;
00211     } DEFAULT;
00212
00213     static unsigned int counter;
00214     float voltage = DEFAULT.someVoltage;
00215     float freq = DEFAULT.someFreq;
00216 };
00217
00218 class Capacitor : public SPICECircuit {
00219 public:
00220     Capacitor(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build
      = [this]() {this->Build();};}
00221     void setCapacitance(float capacitance);
00222     void Build() override;
00223
00224     static PluginInformation* GetPluginInformation();
00225     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00226     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00227
00228 protected:
00229     virtual bool _loadInstance(PersistenceRecord *fields) override;
00230     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00231
00232 private:
00233     const struct DEFAULT_VALUES {
00234         const std::string someString = "Test";
00235         const unsigned int someUint = 1;
00236         const float someCapacitance = 0.01;
00237     } DEFAULT;
00238
00239     static unsigned int counter;
00240     float capacitance = DEFAULT.someCapacitance;
00241 };
00242
00243 class Diode : public SPICECircuit {
00244 public:
```

```
00245     Diode(Model* model, std::string name = ""): SPICECircuit(model, name) {
00246         id = counter++; build = [this]() {this->Build();};
00247     }
00248     void Build() override;
00249     void setElectricalModel(std::string electricalModel);
00250 
00251     static PluginInformation* GetPluginInformation();
00252     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00253     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00254 
00255 protected:
00256     virtual bool _loadInstance(PersistenceRecord *fields) override;
00257     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00258 
00259 private:
00260     const struct DEFAULT_VALUES {
00261         const std::string someString = "Test";
00262         const unsigned int someUint = 1;
00263         const std::string someElectricalModel = "diode";
00264     } DEFAULT;
00265 
00266     static unsigned int counter;
00267     std::string electricalModel = DEFAULT.someElectricalModel;
00268 };
00269 
00270 class PMOS: public SPICECircuit {
00271 public:
00272     PMOS(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build =
      [this]() {this->Build();};}
00273     void Build() override;
00274     void setWidth(float width);
00275     void setLength(float length);
00276     void setElectricalModel(std::string electricalModel);
00277 
00278     static PluginInformation* GetPluginInformation();
00279     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00280     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00281 
00282 protected:
00283     virtual bool _loadInstance(PersistenceRecord *fields) override;
00284     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00285 
00286 private:
00287     const struct DEFAULT_VALUES {
00288         const std::string someString = "Test";
00289         const unsigned int someUint = 1;
00290         const float someWidth = 140e-9;
00291         const float someLength = 32e-9;
00292         const std::string someElectricalModel = "32nmCMOS";
00293     } DEFAULT;
00294 
00295     static unsigned int counter;
00296     float width = DEFAULT.someWidth;
00297     float length = DEFAULT.someLength;
00298     std::string electricalModel = DEFAULT.someElectricalModel;
00299 };
00300 
00301 class NMOS: public SPICECircuit {
00302 public:
00303     NMOS(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build =
      [this]() {this->Build();};}
00304     void Build() override;
00305     void setWidth(float width);
00306     void setLength(float length);
00307     void setElectricalModel(std::string electricalModel);
00308 
00309     static PluginInformation* GetPluginInformation();
00310     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00311     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00312 
00313 protected:
00314     virtual bool _loadInstance(PersistenceRecord *fields) override;
00315     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00316 
00317 private:
00318     const struct DEFAULT_VALUES {
00319         const std::string someString = "Test";
00320         const unsigned int someUint = 1;
00321         const float someWidth = 70e-9;
00322         const float someLength = 32e-9;
00323         const std::string someElectricalModel = "32nmCMOS";
00324     } DEFAULT;
00325 
00326     static unsigned int counter;
00327     float width = DEFAULT.someWidth;
00328     float length = DEFAULT.someLength;
00329     std::string electricalModel = DEFAULT.someElectricalModel;
```

```
00330 };
00331
00332 class NOT : public SPICECircuit {
00333 public:
00334     NOT(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build =
      [this]() {this->Build();};}
00335     void Build() override;
00336     void setPMOSWidth(float PMOSWidth);
00337     void setPMOSLength(float length);
00338     void setNMOSWidth(float NMOSWidth);
00339     void setNMOSLength(float NMOSLength);
00340     void setElectricalModel(std::string electricalModel);
00341
00342     static PluginInformation* GetPluginInformation();
00343     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00344     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00345
00346 protected:
00347     virtual bool _loadInstance(PersistenceRecord *fields) override;
00348     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00349
00350 private:
00351     const struct DEFAULT_VALUES {
00352         const std::string someString = "Test";
00353         const unsigned int someUint = 1;
00354         const float somePMOSWidth = 140e-9;
00355         const float somePMOSLength = 32e-9;
00356         const float someNMOSWidth = 70e-9;
00357         const float someNMOSLength = 32e-9;
00358         const std::string someElectricalModel = "32nmCMOS";
00359     } DEFAULT;
00360
00361     static unsigned int counter;
00362     float PMOSWidth = DEFAULT.somePMOSWidth;
00363     float PMOSLength = DEFAULT.somePMOSLength;
00364     float NMOSWidth = DEFAULT.someNMOSWidth;
00365     float NMOSLength = DEFAULT.someNMOSLength;
00366     std::string electricalModel = DEFAULT.someElectricalModel;
00367 };
00368
00369 class NAND : public SPICECircuit {
00370 public:
00371     NAND(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build =
      [this]() {this->Build();};}
00372     void Build() override;
00373     void setPMOSWidth(float PMOSWidth);
00374     void setPMOSLength(float length);
00375     void setNMOSWidth(float NMOSWidth);
00376     void setNMOSLength(float NMOSLength);
00377     void setElectricalModel(std::string electricalModel);
00378
00379     static PluginInformation* GetPluginInformation();
00380     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00381     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00382
00383 protected:
00384     virtual bool _loadInstance(PersistenceRecord *fields) override;
00385     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00386
00387 private:
00388     const struct DEFAULT_VALUES {
00389         const std::string someString = "Test";
00390         const unsigned int someUint = 1;
00391         const float somePMOSWidth = 140e-9;
00392         const float somePMOSLength = 32e-9;
00393         const float someNMOSWidth = 70e-9;
00394         const float someNMOSLength = 32e-9;
00395         const std::string someElectricalModel = "32nmCMOS";
00396     } DEFAULT;
00397
00398     static unsigned int counter;
00399     float PMOSWidth = DEFAULT.somePMOSWidth;
00400     float PMOSLength = DEFAULT.somePMOSLength;
00401     float NMOSWidth = DEFAULT.someNMOSWidth;
00402     float NMOSLength = DEFAULT.someNMOSLength;
00403     std::string electricalModel = DEFAULT.someElectricalModel;
00404 };
00405
00406 class AND : public SPICECircuit {
00407 public:
00408     AND(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build =
      [this]() {this->Build();};}
00409     void Build() override;
00410     void setPMOSWidth(float PMOSWidth);
00411     void setPMOSLength(float length);
00412     void setNMOSWidth(float NMOSWidth);
00413     void setNMOSLength(float NMOSLength);
```

```
00414     void setElectricalModel(std::string electricalModel);
00415
00416     static PluginInformation* GetPluginInformation();
00417     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00418     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00419
00420 protected:
00421     virtual bool _loadInstance(PersistenceRecord *fields) override;
00422     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00423
00424 private:
00425     const struct DEFAULT_VALUES {
00426         const std::string someString = "Test";
00427         const unsigned int someUint = 1;
00428         const float somePMOSWidth = 140e-9;
00429         const float somePMOSLength = 32e-9;
00430         const float someNMOSWidth = 70e-9;
00431         const float someNMOSLength = 32e-9;
00432         const std::string someElectricalModel = "32nmCMOS";
00433     } DEFAULT;
00434
00435     static unsigned int counter;
00436     float PMOSWidth = DEFAULT.somePMOSWidth;
00437     float PMOSLength = DEFAULT.somePMOSLength;
00438     float NMOSWidth = DEFAULT.someNMOSWidth;
00439     float NMOSLength = DEFAULT.someNMOSLength;
00440     std::string electricalModel = DEFAULT.someElectricalModel;
00441 };
00442
00443 class NOR : public SPICECircuit {
00444 public:
00445     NOR(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build =
      [this]() {this->Build();};}
00446     void Build() override;
00447     void setPMOSWidth(float PMOSWidth);
00448     void setPMOSLength(float length);
00449     void setNMOSWidth(float NMOSWidth);
00450     void setNMOSLength(float NMOSLength);
00451     void setElectricalModel(std::string electricalModel);
00452
00453     static PluginInformation* GetPluginInformation();
00454     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00455     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00456
00457 protected:
00458     virtual bool _loadInstance(PersistenceRecord *fields) override;
00459     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00460
00461 private:
00462     const struct DEFAULT_VALUES {
00463         const std::string someString = "Test";
00464         const unsigned int someUint = 1;
00465         const float somePMOSWidth = 140e-9;
00466         const float somePMOSLength = 32e-9;
00467         const float someNMOSWidth = 70e-9;
00468         const float someNMOSLength = 32e-9;
00469         const std::string someElectricalModel = "32nmCMOS";
00470     } DEFAULT;
00471
00472     static unsigned int counter;
00473     float PMOSWidth = DEFAULT.somePMOSWidth;
00474     float PMOSLength = DEFAULT.somePMOSLength;
00475     float NMOSWidth = DEFAULT.someNMOSWidth;
00476     float NMOSLength = DEFAULT.someNMOSLength;
00477     std::string electricalModel = DEFAULT.someElectricalModel;
00478 };
00479
00480 class OR : public SPICECircuit {
00481 public:
00482     OR(Model* model, std::string name = ""): SPICECircuit(model, name)  { id = counter++; build =
      [this]() {this->Build();};}
00483     void Build() override;
00484     void setPMOSWidth(float PMOSWidth);
00485     void setPMOSLength(float length);
00486     void setNMOSWidth(float NMOSWidth);
00487     void setNMOSLength(float NMOSLength);
00488     void setElectricalModel(std::string electricalModel);
00489
00490     static PluginInformation* GetPluginInformation();
00491     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00492     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00493
00494 protected:
00495     virtual bool _loadInstance(PersistenceRecord *fields) override;
00496     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00497
00498 private:
```

```
00499     const struct DEFAULT_VALUES {
00500         const std::string someString = "Test";
00501         const unsigned int someUint = 1;
00502         const float somePMOSWidth = 140e-9;
00503         const float somePMOSLength = 32e-9;
00504         const float someNMOSWidth = 70e-9;
00505         const float someNMOSLength = 32e-9;
00506         const std::string someElectricalModel = "32nmCMOS";
00507     } DEFAULT;
00508
00509     static unsigned int counter;
00510     float PMOSWidth = DEFAULT.somePMOSWidth;
00511     float PMOSLength = DEFAULT.somePMOSLength;
00512     float NMOSWidth = DEFAULT.someNMOSWidth;
00513     float NMOSLength = DEFAULT.someNMOSLength;
00514     std::string electricalModel = DEFAULT.someElectricalModel;
00515 };
00516
00517 class XOR : public SPICECircuit {
00518 public:
00519     XOR(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build =
      [this]() {this->Build();};}
00520     void Build() override;
00521     void setPMOSWidth(float PMOSWidth);
00522     void setPMOSLength(float length);
00523     void setNMOSWidth(float NMOSWidth);
00524     void setNMOSLength(float NMOSLength);
00525     void setElectricalModel(std::string electricalModel);
00526
00527     static PluginInformation* GetPluginInformation();
00528     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00529     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00530
00531 protected:
00532     virtual bool _loadInstance(PersistenceRecord *fields) override;
00533     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00534
00535 private:
00536     const struct DEFAULT_VALUES {
00537         const std::string someString = "Test";
00538         const unsigned int someUint = 1;
00539         const float somePMOSWidth = 140e-9;
00540         const float somePMOSLength = 32e-9;
00541         const float someNMOSWidth = 70e-9;
00542         const float someNMOSLength = 32e-9;
00543         const std::string someElectricalModel = "32nmCMOS";
00544     } DEFAULT;
00545
00546     static unsigned int counter;
00547     float PMOSWidth = DEFAULT.somePMOSWidth;
00548     float PMOSLength = DEFAULT.somePMOSLength;
00549     float NMOSWidth = DEFAULT.someNMOSWidth;
00550     float NMOSLength = DEFAULT.someNMOSLength;
00551     std::string electricalModel = DEFAULT.someElectricalModel;
00552 };
00553
00554 class XNOR : public SPICECircuit {
00555 public:
00556     XNOR(Model* model, std::string name = ""): SPICECircuit(model, name) { id = counter++; build =
      [this]() {this->Build();};}
00557     void Build() override;
00558     void setPMOSWidth(float PMOSWidth);
00559     void setPMOSLength(float length);
00560     void setNMOSWidth(float NMOSWidth);
00561     void setNMOSLength(float NMOSLength);
00562     void setElectricalModel(std::string electricalModel);
00563
00564     static PluginInformation* GetPluginInformation();
00565     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00566     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00567
00568 protected:
00569     virtual bool _loadInstance(PersistenceRecord *fields) override;
00570     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00571
00572 private:
00573     const struct DEFAULT_VALUES {
00574         const std::string someString = "Test";
00575         const unsigned int someUint = 1;
00576         const float somePMOSWidth = 140e-9;
00577         const float somePMOSLength = 32e-9;
00578         const float someNMOSWidth = 70e-9;
00579         const float someNMOSLength = 32e-9;
00580         const std::string someElectricalModel = "32nmCMOS";
00581     } DEFAULT;
00582
00583     static unsigned int counter;
```

```
00584     float PMOSWidth = DEFAULT.somePMOSWidth;
00585     float PMOSLength = DEFAULT.somePMOSLength;
00586     float NMOSWidth = DEFAULT.someNMOSWidth;
00587     float NMOSLength = DEFAULT.someNMOSLength;
00588     std::string electricalModel = DEFAULT.someElectricalModel;
00589 };
```

18.401 SPICENode.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   SPICENode.h
00009  * Author: bnmfw & joaomai
00010  *
00011  * Created on 10 de outubro de 2023, 23:00
00012  */
00013
00014 #pragma once
00015
00016 #include "../../../kernel/simulator/model/ModelComponent.h"
00017 #include "../../data/ExternalIntegration/SPICERunner.h"
00018 #include "plugins/components/ElectronicsSimulation/SPICENode.h"
00019
00023 class SPICENode : public ModelComponent {
00024 public:
00025     SPICENode(Model* model, std::string name = "");
00026     virtual ~SPICENode() = default;
00027
00028 public:
00029     std::string getNodeName();
00030
00031 public:
00032     virtual std::string show() override;
00033
00034 public:
00035     static PluginInformation* GetPluginInformation();
00036     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00037     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00038     operator std::string() const;
00039
00040 protected:
00041     virtual bool _loadInstance(PersistenceRecord *fields) override;
00042     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00043     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00044
00045
00046 protected:
00048     // virtual bool _check(std::string& errorMessage);
00050     // virtual ParserChangesInformation* _getParserChangesInformation();
00052     // virtual void _initBetweenReplications();
00053     // virtual void _createInternalStatisticReporters() override;
00054     // virtual void _createNonEditableDataDefinitions() override;
00055     // virtual void _createEditableDataDefinitions() override;
00056     // virtual void _createAttachedAttributes() override;
00058     // virtual void _createInternalAndAttachedData(); /*< A ModelDataDefinition or ModelComponent that
      includes (internal) ou refers to (attach) other ModelDataDefinition must register them inside this
      method. */
00060     // virtual void _addSimulationControl(SimulationControl* property);
00061
00062
00063 protected:
00064
00065 private:
00066     std::string node_name;
00067     static int std_node_count;
00068     static std::set<std::string> used_names;
00069
00070
00071 private:
00072
00074     const struct DEFAULT_VALUES {
00075         const std::string someString = "Test";
00076         const unsigned int someUint = 1;
00077     } DEFAULT;
00078     std::string _someString = DEFAULT.someString;
00079     unsigned int _someUint = DEFAULT.someUint;
00080
00081 private:
00082
00083 private:
```

```
00084     // ...
00085 };
```

18.402 CppForG.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   CppForG.h
00008  * Author: rlcancian
00009  *
00010  * Created on 11 de janeiro de 2022, 22:37
00011  */
00012
00013 #pragma once
00014
00015 #include "../../../kernel/simulator/model/ModelComponent.h"
00016 #include "kernel/simulator/Simulator.h"
00017 #include "../../data/ExternalIntegration/CppCompiler.h"
00018
00019 extern "C" typedef void (*initBetweenReplications_t)(Model* model);
00020 extern "C" typedef void (*onDispatchEvent_t)(Simulator* simulator, Model* model, Entity* entity);
00021
00025 class CppForG : public ModelComponent {
00026 public: // constructors
00027     CppForG(Model* model, std::string name = "");
00028     virtual ~CppForG() = default;
00029 public: // virtual
00030     virtual std::string show() override;
00031 public: // static
00032     static PluginInformation* GetPluginInformation();
00033     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00034     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00035 public:
00036     void setCppCompiler(CppCompiler* _cppCompiler);
00037     CppCompiler* getCppCompiler() const;
00038     void setInitBetweenReplicationCode(std::string _initBetweenReplicationCode);
00039     std::string getInitBetweenReplicationCode() const;
00040     void setOnDispatchEventCode(std::string _onDispatchEventCode);
00041     std::string getOnDispatchEventCode() const;
00042     void setIncludesCode(std::string _includesCode);
00043     std::string getIncludesCode() const;
00044 protected: // must be overriden
00045     virtual bool _loadInstance(PersistenceRecord *fields) override;
00046     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00047     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00048 protected: // could be overriden by derived classes
00049     virtual bool _check(std::string& errorMessage) override;
00051     //virtual ParserChangesInformation* _getParserChangesInformation();
00052     virtual void _initBetweenReplications() override;
00054     // virtual void _createInternalAndAttachedData() override;
00055     //virtual void _addSimulationControl(SimulationControl* property);
00056 protected:
00057     // virtual void _createInternalStatisticReporters() override;
00058     virtual void _createNonEditableDataDefinitions() override;
00059     // virtual void _createAttachedAttributes() override;
00060
00061 private: // methods
00062
00063 private: // attributes 1:1
00064
00065     const struct DEFAULT_VALUES {
00066         std::string onDispatchEventCode = "";
00067         std::string initBetweenReplicationCode = "";
00068         std::string includesCode = "";
00069     } DEFAULT;
00070     std::string _onDispatchEventCode = DEFAULT.onDispatchEventCode;
00071     std::string _initBetweenReplicationCode = DEFAULT.initBetweenReplicationCode;
00072     std::string _includesCode = DEFAULT.includesCode;
00073     //
00074     std::string _sourceFilename;
00075     std::string _outputFilename;
00076     // pointers to functions on a shared libraty
00077     onDispatchEvent_t dispatchEvent_SharedLibHandler;
00078     initBetweenReplications_t initBetweenReplications_SharedLibHandler;
00079 private: // internal (TODO: should be Attached??)
00080     CppCompiler* _cppCompiler = nullptr;
00081 private: // attributes 1:n
00082 };
```

18.403 PythonForG.h

```
00001 #ifndef PYTHONFORG_H
00002 #define PYTHONFORG_H
00003
00004 #include "../../../kernel/simulator/model/ModelComponent.h"
00005 #include "../../../kernel/simulator/SimulationControlAndResponse.h"
00006 #include "plugins/data/ExternalIntegration/PythonRuntime.h"
00007
00008 class PythonForG : public ModelComponent {
00009 public:
00010     PythonForG(Model* model, std::string name = "");
00011     virtual ~PythonForG() override = default;
00012
00013 public:
00014     static PluginInformation* GetPluginInformation();
00015     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00016     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00017
00018 public:
00019     void setInitBetweenReplicationCode(std::string initBetweenReplicationCode);
00020     std::string getInitBetweenReplicationCode() const;
00021     void setInitBetweenReplicationCodeProperty(SourceCodeString initBetweenReplicationCode);
00022     SourceCodeString getInitBetweenReplicationCodeProperty() const;
00023     void setOnDispatchEventCode(std::string onDispatchEventCode);
00024     std::string getOnDispatchEventCode() const;
00025     void setOnDispatchEventCodeProperty(SourceCodeString onDispatchEventCode);
00026     SourceCodeString getOnDispatchEventCodeProperty() const;
00027     void setForwardEntityOnError(bool forwardEntityOnError);
00028     bool isForwardEntityOnError() const;
00029     PythonRuntime* getPythonRuntime() const;
00030
00031     virtual std::string show() override;
00032
00033 protected:
00034     virtual bool _loadInstance(PersistenceRecord* fields) override;
00035     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00036     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00037     virtual bool _check(std::string& errorMessage) override;
00038     virtual void _initBetweenReplications() override;
00039     virtual void _createNonEditableDataDefinitions() override;
00040
00041 private:
00042     bool _validateInitCode(const std::string& candidate, std::string& errorMessage) const;
00043     bool _validateDispatchCode(const std::string& candidate, std::string& errorMessage) const;
00044
00045 private:
00046     const struct DEFAULT_VALUES {
00047         std::string initBetweenReplicationCode = "";
00048         std::string onDispatchEventCode = "";
00049         bool forwardEntityOnError = true;
00050     } DEFAULT;
00051
00052     std::string _initBetweenReplicationCode = DEFAULT.initBetweenReplicationCode;
00053     std::string _onDispatchEventCode = DEFAULT.onDispatchEventCode;
00054     bool _forwardEntityOnError = DEFAULT.forwardEntityOnError;
00055     PythonRuntime* _pythonRuntime = nullptr;
00056 };
00057
00058 #endif /* PYTHONFORG_H */
```

18.404 RSimulator.h

```
00001 /*
00002  * File:   RSimulator.h
00003  * Author: GenESyS
00004  *
00005  * Component that executes R commands when entities arrive.
00006  */
00007
00008 #ifndef RSIMULATOR_H
00009 #define RSIMULATOR_H
00010
00011 #include <list>
00012 #include <string>
00013
00014 #include "../../../kernel/simulator/model/ModelComponent.h"
00015 #include "kernel/simulator/Plugin.h"
00016 #include "plugins/data/ExternalIntegration/RSimulatorRunner.h"
00017
00025 class RSimulator : public ModelComponent {
00026 public:
00028     RSimulator(Model* model, std::string name = "");
00030     virtual ~RSimulator() = default;
```

```
00031
00032 public:
00034     void setRExecutable(std::string rExecutable);
00036     std::string getRExecutable() const;
00038     void setWorkingDirectory(std::string workingDirectory);
00040     std::string getWorkingDirectory() const;
00042     void setPreludeScript(std::string preludeScript);
00044     std::string getPreludeScript() const;
00046     void insertCommand(std::string command);
00048     void setCommands(std::list<std::string> commands);
00050     List<std::string>* getCommands() const;
00052     void setCommandList(std::string commandList);
00054     std::string getCommandList() const;
00056     RSimulatorRunner* getRunner();
00058     virtual std::string show() override;
00059
00060 public:
00062     static PluginInformation* GetPluginInformation();
00064     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00066     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00067
00068 protected:
00070     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00072     virtual bool _loadInstance(PersistenceRecord *fields) override;
00074     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00076     virtual bool _check(std::string& errorMessage) override;
00078     // virtual void _createInternalAndAttachedData() override;
00079
00080
00081 protected:
00082     // virtual void _createInternalStatisticReporters() override;
00083     // virtual void _createNonEditableDataDefinitions() override;
00084     virtual void _createEditableDataDefinitions() override;
00085     // virtual void _createAttachedAttributes() override;
00086
00087 private:
00089     void _ensureRunner();
00091     void _syncRunnerConfiguration();
00093     bool _isNonEmptyCommand(const std::string& command) const;
00094
00095 private:
00096     const struct DEFAULT_VALUES {
00097         std::string rExecutable = "Rscript";
00098         std::string workingDirectory = "";
00099         std::string preludeScript = "";
00100     } DEFAULT;
00101
00102     std::string _rExecutable = DEFAULT.rExecutable;
00103     std::string _workingDirectory = DEFAULT.workingDirectory;
00104     std::string _preludeScript = DEFAULT.preludeScript;
00105     List<std::string>* _commands = new List<std::string>();
00106     RSimulatorRunner* _runner = nullptr;
00107 };
00108
00109 #endif /* RSIMULATOR_H */
```

18.405 Batch.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Batch.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:14
00012  */
00013
00014 #ifndef BATCH_H
00015 #define BATCH_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "plugins/data/DiscreteProcessing/Queue.h"
00019 #include "../../data/Grouping/EntityGroup.h"
00034 class Batch : public ModelComponent {
00035 public:
00036
00037     enum class BatchType : int {
00038         Temporary = 0, Permanent = 1, num_elements = 2
00039     };
00040
00041     enum class Rule : int {
```

```
00042         Any = 0, ByAttribute = 1, num_elements = 2
00043     };
00044
00045     enum class GroupedAttribs : int {
00046         FirstEntity = 0, LastEntity = 1, SumAttributes = 2, num_elements = 3
00047     };
00048 public:
00049     static std::string convertEnumToStr(BatchType type);
00050     static std::string convertEnumToStr(Rule rule);
00051     static std::string convertEnumToStr(GroupedAttribs attribs);
00052 public: // constructors
00053     Batch(Model* model, std::string name = "");
00054     virtual ~Batch() = default;
00055 public: // virtual
00056     virtual std::string show() override;
00057 public: // static
00058     static PluginInformation* GetPluginInformation();
00059     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00060     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00061 public:
00062     void setGroupedEntityType(EntityType* groupedEntityType);
00063     void setGroupedEntityTypeName(std::string groupedEntityTypeName);
00064     EntityType* getGroupedEntityType() const;
00065     void setBatchType(Batch::BatchType batchType);
00066     Batch::BatchType getBatchType() const;
00067     void setAttributeName(std::string attributeName);
00068     std::string getAttributeName() const;
00069     void setBatchSize(std::string batchSize);
00070     std::string getBatchSize() const;
00071     void setRule(Batch::Rule _rule);
00072     Batch::Rule getRule() const;
00073     void setGroupedAttributes(Batch::GroupedAttribs _groupedAttributes);
00074     Batch::GroupedAttribs getGroupedAttributes() const;
00075 protected: // virtual should
00076     //virtual void _initBetweenReplications();
00077     // virtual void _createInternalStatisticReporters() override;
00078     virtual void _createNonEditableDataDefinitions() override;
00079     virtual void _createEditableDataDefinitions() override;
00080     virtual void _createAttachedAttributes() override;
00081     // virtual void _createInternalAndAttachedData() override;
00082     virtual bool _check(std::string& errorMessage) override;
00083 protected: // virtual must
00084     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00085     virtual bool _loadInstance(PersistenceRecord *fields) override;
00086     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00087
00088 protected:
00089
00090 private: // methods
00091 private: // attributes 1:1
00092
00093     const struct DEFAULT_VALUES {
00094         const Batch::BatchType batchType = Batch::BatchType::Temporary;
00095         const Batch::Rule rule = Batch::Rule::Any;
00096         const Batch::GroupedAttribs groupedAttributes = Batch::GroupedAttribs::FirstEntity;
00097         const std::string batchSize = "2";
00098         const std::string attributeName = "";
00099     } DEFAULT;
00100     Batch::BatchType _batchType = DEFAULT.batchType;
00101     Batch::Rule _rule = DEFAULT.rule;
00102     Batch::GroupedAttribs _groupedAttributes = DEFAULT.groupedAttributes;
00103     std::string _batchSize = DEFAULT.batchSize;
00104     std::string _attributeName = DEFAULT.attributeName;
00105 private: // attributes 1:1
00106     EntityType* _groupedEntityType = nullptr;
00107     EntityGroup* _entityGroup = nullptr;
00108     Queue* _queue = nullptr;
00109 private: // attributes 1:n
00110     // count number of batches?
00111 };
00112
00113
00114 #endif /* BATCH_H */
```

18.406 Separate.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Separate.h
00009  * Author: rlcancian
```

```
00010  *
00011  * Created on 03 de Junho de 2019, 15:14
00012  */
00013
00014 #ifndef SEPARATE_H
00015 #define SEPARATE_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018
00041 class Separate : public ModelComponent {
00042 public: // constructors
00043     Separate(Model* model, std::string name = "");
00044     virtual ~Separate() = default;
00045 public: // virtual
00046     virtual std::string show() override;
00047 public: // static
00048     static PluginInformation* GetPluginInformation();
00049     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00050     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00051 protected: // virtual
00052     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00053     virtual bool _loadInstance(PersistenceRecord *fields) override;
00054     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00055 protected: // virtual
00056     //virtual void _initBetweenReplications();
00057     // virtual void _createInternalStatisticReporters() override;
00058     // virtual void _createNonEditableDataDefinitions() override;
00059     // virtual void _createEditableDataDefinitions() override;
00060     // virtual void _createAttachedAttributes() override;
00061     virtual bool _check(std::string& errorMessage) override;
00062
00063 protected:
00064
00065 private: // methods
00066 private: // attributes 1:1
00067 private: // attributes 1:n
00068 };
00069
00070
00071 #endif /* SEPARATE_H */
```

18.407 Record.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Record.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 9 de Agosto de 2018, 13:52
00012  */
00013
00014 #ifndef RECORD_H
00015 #define RECORD_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include <string>
00019
00043 class Record : public ModelComponent {
00044 public:
00045     Record(Model* model, std::string name = "");
00046     virtual ~Record();
00047
00048 public:
00049     void setFilename(std::string filename);
00050     std::string getFileName() const;
00051     void setExpression(const std::string expression);
00052     std::string getExpression() const;
00056     void setExpressionName(std::string expressionName);
00057     std::string getExpressionName() const;
00061     void setDatasetName(std::string datasetName);
00062     std::string getDatasetName() const;
00066     void setRandomVariableName(std::string randomVariableName);
00067     std::string getRandomVariableName() const;
00071     void setVariableType(std::string variableType);
00072     std::string getVariableType() const;
00076     void setDatasetDescription(std::string description);
00077     std::string getDatasetDescription() const;
00078     StatisticsCollector* getCstatExpression() const;
00079     bool getTimeDependent() const;
00080     void setTimeDependent(bool timeDependent);
```

```
00081
00082 public:
00083     virtual std::string show() override;
00084
00085 public:
00086     static PluginInformation* GetPluginInformation();
00087     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00088     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00089
00090 protected:
00091     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00092     virtual bool _loadInstance(PersistenceRecord *fields) override;
00093     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00094
00095 protected:
00096     virtual void _initBetweenReplications() override;
00097     virtual bool _check(std::string& errorMessage) override;
00098     // virtual void _createInternalAndAttachedData() override;
00099
00100
00101 protected:
00102     virtual void _createInternalStatisticReporters() override;
00103     // virtual void _createNonEditableDataDefinitions() override;
00104     // virtual void _createEditableDataDefinitions() override;
00105     // virtual void _createAttachedAttributes() override;
00106
00107 private:
00108     const struct DEFAULT_VALUES {
00109         const bool timeDependent = false;
00110         const std::string expression = "";
00111         const std::string expressionName = "";
00112         const std::string datasetName = "";
00113         const std::string randomVariableName = "";
00114         const std::string variableType = "Continuous numeric";
00115         const std::string description = "";
00116         const std::string filename = "";
00117     } DEFAULT;
00118     bool _timeDependent = DEFAULT.timeDependent;
00119     std::string _expression = DEFAULT.expression;
00120     std::string _expressionName = DEFAULT.expressionName;
00121     std::string _datasetName = DEFAULT.datasetName;
00122     std::string _randomVariableName = DEFAULT.randomVariableName;
00123     std::string _variableType = DEFAULT.variableType;
00124     std::string _description = DEFAULT.description;
00125     std::string _filename = DEFAULT.filename;
00126 private:
00127     const std::string _separator = " ";
00128     StatisticsCollector* _cstatExpression = nullptr;
00129     /* @TODO: Create an internal class to agregate ExpressionStatisticsColelctor, and change Record to
      got a list of it, so Record can record a set of expressions into a set of files */
00130 };
00131
00132 #endif /* RECORD_H */
```

18.408 Write.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Write.h
00009  * Author: rlcancian
00010  *
00011  * Created on 11 de Setembro de 2019, 13:06
00012  */
00013
00014 #ifndef WRITE_H
00015 #define WRITE_H
00016
00017 #include <list>
00018 #include <fstream>
00019 #include "../../../kernel/simulator/model/ModelComponent.h"
00020
00035 class Write : public ModelComponent {
00036 public:
00037     enum class WriteToType : int {
00038         SCREEN = 0, FILE = 1, num_elements = 2
00039     };
00040 public:
00041     static std::string convertEnumToStr(WriteToType type);
00042 public: // constructors
00043     Write(Model* model, std::string name = "");
```

```
00044     virtual ~Write() override;
00045
00046 public:
00047     void insertText(std::list<std::string> texts);
00048     void setFilename(std::string _filename);
00049     std::string filename() const;
00050     void setWriteToType(WriteToType _writeToType);
00051     Write::WriteToType writeToType() const;
00052
00053 public: // virtual
00054     virtual std::string show() override;
00055
00056 public: // static
00057     static PluginInformation* GetPluginInformation();
00058     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00059     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00060
00061 protected: // virtual
00062     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00063     virtual bool _loadInstance(PersistenceRecord *fields) override;
00064     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00065
00066 protected: // virtual
00067     virtual bool _check(std::string& errorMessage) override;
00068     virtual void _initBetweenReplications() override;
00069
00070
00071 protected:
00072     // virtual void _createInternalStatisticReporters() override;
00073     // virtual void _createNonEditableDataDefinitions() override;
00074     // virtual void _createEditableDataDefinitions() override;
00075     // virtual void _createAttachedAttributes() override;
00076
00077 private: // methods
00078 private: // attributes 1:1
00079     const struct DEFAULT_VALUES {
00080         const WriteToType writeToType = Write::WriteToType::SCREEN;
00081         const std::string filename = "";
00082     } DEFAULT;
00083     WriteToType _writeToType = DEFAULT.writeToType;
00084     std::string _filename = DEFAULT.filename;
00085     std::ofstream _savefile;
00086
00087 private: // attributes 1:n
00088     List<std::string>* _writeElements = new List<std::string>();
00089 };
00090
00091
00092 #endif /* WRITE_H */
```

18.409 Assign.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Assign.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 31 de Agosto de 2018, 10:10
00012  */
00013
00014 #ifndef ASSIGN_H
00015 #define ASSIGN_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../../kernel/simulator/model/Model.h"
00019 #include "kernel/simulator/Plugin.h"
00020 #include "../../../kernel/simulator/essentialPlugins/Attribute.h"
00021 #include "../../data/Logic/AssignmentItem.h"
00022 #include "../../data/Logic/Variable.h"
00023
00046 class Assign : public ModelComponent {
00047 public:
00048     Assign(Model* model, std::string name = "");
00049     virtual ~Assign() override;
00050 public:
00051     virtual std::string show() override;
00052 public:
00053     static PluginInformation* GetPluginInformation();
00054     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00055     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
```

```
00056 public:
00057     List<Assignment*>* getAssignments() const;
00058     void addAssignment(Assignment* newAssignment);
00059     void removeAssignment(Assignment* assignment);
00060 protected: // must be overriden
00061     virtual bool _loadInstance(PersistenceRecord *fields) override;
00062     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00063     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00064 protected: // could be overriden by derived classes
00065     virtual bool _check(std::string& errorMessage) override;
00067     //virtual ParserChangesInformation* _getParserChangesInformation();
00068     //virtual void _initBetweenReplications();
00069 protected:
00070     // virtual void _createInternalStatisticReporters() override;
00071     // virtual void _createNonEditableDataDefinitions() override;
00072     virtual void _createEditableDataDefinitions() override;
00073     //virtual void _createAttachedAttributes() override;
00075     // virtual void _createInternalAndAttachedData() override;
00076     //virtual void _addSimulationControl(SimulationControl* property);
00077
00078 protected:
00079
00080 private:
00081     static std::string _destinationBaseName(const std::string& destination);
00082     std::string _destinationIndex(const std::string& destination);
00083     void _prepareAssignment(Assignment* assignment);
00084 private:
00085
00086     const struct DEFAULT_VALUES {
00087         const unsigned int assignmentsSize = 1;
00088     } DEFAULT;
00089     List<Assignment*>* _assignments = new List<Assignment*>();
00090 };
00091
00092 #endif /* ASSIGN_H */
```

18.410 Create.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Create.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Junho de 2018, 20:12
00012  */
00013
00014 #ifndef CREATE_H
00015 #define CREATE_H
00016
00017 #include <string>
00018 #include <limits>
00019 #include "kernel/simulator/SourceModelComponent.h"
00020 #include "../../../kernel/simulator/essentialPlugins/EntityType.h"
00021 #include "../../../kernel/simulator/essentialPlugins/Counter.h"
00022 #include "kernel/simulator/Plugin.h"
00023
00024 #include "../../data/Logic/Formula.h"
00025 #include "plugins/data/DiscreteProcessing/Schedule.h"
00026
00050 class Create : public SourceModelComponent {
00051 public:
00052     Create(Model* model, std::string name = "");
00053     virtual ~Create() = default;
00054 public: // virtual
00055     virtual std::string show() override;
00056 public:
00057     void setTimeBetweenCreationsFormula(Formula* _timeBetweenCreationsFormula);
00058     Formula* getTimeBetweenCreationsFormula() const;
00059     void setTimeBetweenCreationsSchedule(Schedule* _timeBetweenCreationsSchedule);
00060     Schedule* getTimeBetweenCreationsSchedule() const;
00061 public: // static
00062     static PluginInformation* GetPluginInformation();
00063     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00064     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00065
00066     double testePropertyCreateDouble() const;
00067     void setTestePropertyCreateDouble(double newTestePropertyCreateDouble);
00068
00069 protected:
00070     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
```

```
00071     virtual bool _loadInstance(PersistenceRecord *fields) override;
00072     virtual void _initBetweenReplications() override;
00073     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00074     virtual bool _check(std::string& errorMessage) override;
00075     virtual void _createInternalStatisticReporters() override;
00076     virtual void _createEditableDataDefinitions() override;
00077 private:
00078     double _lastArrival = -1.0;
00079 private: // internal elements
00080     Counter* _numberOut = nullptr; // internal modeldatum
00081 private: // attached elements
00082     Schedule* _timeBetweenCreationsSchedule = nullptr;
00083     Formula* _timeBetweenCreationsFormula = nullptr;
00084 };
00085
00086 #endif /* CREATE_H */
```

18.411 Dispose.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Dispose.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Junho de 2018, 20:13
00012  */
00013
00014 #ifndef DISPOSE_H
00015 #define DISPOSE_H
00016
00017 #include "kernel/simulator/SinkModelComponent.h"
00018 #include "../../../kernel/simulator/essentialPlugins/Counter.h"
00019 #include "kernel/simulator/Plugin.h"
00020
00037 class Dispose : public SinkModelComponent {
00038 public:
00039     Dispose(Model* model, std::string name = "");
00040     virtual ~Dispose() = default;
00041 public:
00042     virtual std::string show() override;
00043 public:
00044     static PluginInformation* GetPluginInformation();
00045     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00046     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00047 protected:
00048     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00049     virtual bool _loadInstance(PersistenceRecord *fields) override;
00050     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00051 protected:
00052     virtual void _initBetweenReplications() override;
00053     virtual bool _check(std::string& errorMessage) override;
00054 protected:
00055     virtual void _createInternalStatisticReporters() override;
00056     // virtual void _createNonEditableDataDefinitions() override;
00057     // virtual void _createEditableDataDefinitions() override;
00058     virtual void _createAttachedAttributes() override;
00059 protected:
00060 private: // internal elements
00061     Counter* _numberOut = nullptr;
00062 };
00063
00064 #endif /* DISPOSE_H */
```

18.412 Access.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Access.h
00009  * Author: rlcancian
00010  *
00011  * Created on 11 de Setembro de 2019, 13:14
00012  */
```

```
00013
00014 #ifndef ACCESS_H
00015 #define ACCESS_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../data/MaterialHandling/Conveyor.h"
00019
00054 class Access : public ModelComponent {
00055 public: // constructors
00056     Access(Model* model, std::string name = "");
00057     virtual ~Access() = default;
00058 public: // virtual
00059     virtual std::string show() override;
00060 public: // static
00061     static PluginInformation* GetPluginInformation();
00062     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00063     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00064 public:
00065     void setConveyor(Conveyor* conveyor);
00066     Conveyor* getConveyor() const;
00067     void setQuantityExpression(std::string quantityExpression);
00068     std::string getQuantityExpression() const;
00069 protected: // virtual
00070     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00071     virtual bool _loadInstance(PersistenceRecord *fields) override;
00072     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00073 protected: // virtual
00074     //virtual void _initBetweenReplications();
00075     // virtual void _createInternalStatisticReporters() override;
00076     // virtual void _createNonEditableDataDefinitions() override;
00077     // virtual void _createEditableDataDefinitions() override;
00078     // virtual void _createAttachedAttributes() override;
00079     virtual bool _check(std::string& errorMessage) override;
00080
00081 protected:
00082
00083 private: // methods
00084 private: // attributes 1:1
00085     Conveyor* _conveyor = nullptr;
00086     std::string _quantityExpression = "1";
00087 private: // attributes 1:n
00088 };
00089
00090
00091 #endif /* ACCESS_H */
```

18.413 Enter.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Enter.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:14
00012  */
00013
00014 #ifndef ENTER_H
00015 #define ENTER_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../data/MaterialHandling/Station.h"
00019 #include "../../../kernel/simulator/essentialPlugins/Counter.h"
00020
00099 class Enter : public ModelComponent {
00100 public:
00101     Enter(Model* model, std::string name = "");
00102     virtual ~Enter() = default;
00103 public:
00104     virtual std::string show() override;
00105 public:
00106     static PluginInformation* GetPluginInformation();
00107     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00108     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00109 public:
00110     void setStation(Station* _station);
00111     void setStationName(std::string stationName);
00112     Station* getStation() const;
00113 protected:
00114     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00115     virtual bool _loadInstance(PersistenceRecord *fields) override;
```

```
00116     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00117 protected:
00118     //virtual void _initBetweenReplications();
00119     virtual bool _check(std::string& errorMessage) override;
00120 
00121      virtual void _createInternalStatisticReporters() override;
00122     // virtual void _createNonEditableDataDefinitions() override;
00123      virtual void _createEditableDataDefinitions() override;
00124     // virtual void _createAttachedAttributes() override;
00125     // virtual void _createInternalAndAttachedData() override;
00126 
00127 protected:
00128 
00129 private: // association
00130     Station* _station = nullptr;
00131 private: // internal elements
00132     Counter* _numberIn = nullptr;
00133 };
00134 
00135 #endif /* ENTER_H */
```

18.414 Exit.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
00007 /*
00008  * File:   Exit.h
00009  * Author: rlcancian
00010  *
00011  * Created on 11 de Setembro de 2019, 13:15
00012  */
00013 
00014 #ifndef EXIT_H
00015 #define EXIT_H
00016 
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../data/MaterialHandling/Conveyor.h"
00019 
00038 class Exit : public ModelComponent {
00039 public: // constructors
00040     Exit(Model* model, std::string name = "");
00041     virtual ~Exit() = default;
00042 public: // virtual
00043     virtual std::string show() override;
00044 public: // static
00045     static PluginInformation* GetPluginInformation();
00046     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00047     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00048 public:
00049     void setConveyor(Conveyor* conveyor);
00050     Conveyor* getConveyor() const;
00051     void setQuantityExpression(std::string quantityExpression);
00052     std::string getQuantityExpression() const;
00053 protected: // virtual
00054     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00055     virtual bool _loadInstance(PersistenceRecord *fields) override;
00056     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00057 protected: // virtual
00058     //virtual void _initBetweenReplications();
00059     // virtual void _createInternalStatisticReporters() override;
00060     // virtual void _createNonEditableDataDefinitions() override;
00061     // virtual void _createEditableDataDefinitions() override;
00062     // virtual void _createAttachedAttributes() override;
00063     virtual bool _check(std::string& errorMessage) override;
00064 
00065 protected:
00066 
00067 private: // methods
00068 private: // attributes 1:1
00069     Conveyor* _conveyor = nullptr;
00070     std::string _quantityExpression = "1";
00071 private: // attributes 1:n
00072 };
00073 
00074 
00075 #endif /* EXIT_H */
```

18.415 Leave.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Leave.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:14
00012  */
00013
00014 #ifndef LEAVE_H
00015 #define LEAVE_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../data/MaterialHandling/Station.h"
00019 #include "../../../kernel/simulator/essentialPlugins/Counter.h"
00020
00094 class Leave : public ModelComponent {
00095 public:
00096     Leave(Model* model, std::string name = "");
00097     virtual ~Leave() = default;
00098 public:
00099     virtual std::string show() override;
00100 public:
00101     static PluginInformation* GetPluginInformation();
00102     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00103     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00104 public:
00105     void setStation(Station* _station);
00106     void setStationName(std::string stationName);
00107     Station* getStation() const;
00108 protected:
00109     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00110     virtual bool _loadInstance(PersistenceRecord *fields) override;
00111     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00112 protected:
00113     //virtual void _initBetweenReplications();
00114 protected:
00115      virtual void _createInternalStatisticReporters() override;
00116     // virtual void _createNonEditableDataDefinitions() override;
00117      virtual void _createEditableDataDefinitions() override;
00118     // virtual void _createAttachedAttributes() override;
00119     virtual bool _check(std::string& errorMessage) override;
00120     // virtual void _createInternalAndAttachedData() override;
00121
00122 protected:
00123
00124 private: // association
00125     Station* _station = nullptr;
00126 private: // internal elements
00127     Counter* _numberIn = nullptr;
00128 };
00129
00130 #endif /* LEAVE_H */
```

18.416 Move.h

```
00001 #ifndef MOVE_H
00002 #define MOVE_H
00003
00004 #include "../../../kernel/simulator/model/ModelComponent.h"
00005 #include "../../data/MaterialHandling/Station.h"
00006 #include "../../data/MaterialHandling/Transporter.h"
00007 #include "../../../kernel/simulator/essentialPlugins/Counter.h"
00008
00017 class Move : public ModelComponent {
00018 public:
00019     Move(Model* model, std::string name = "");
00020     virtual ~Move() override = default;
00021 public:
00022     virtual std::string show() override;
00023 public: // static
00024     static PluginInformation* GetPluginInformation();
00025     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00026     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00027 public:
00028     void setTransporter(Transporter* transporter);
00029     Transporter* getTransporter() const;
00030     void setStation(Station* station);
```

```
00031     Station* getStation() const;
00032     void setStationExpression(std::string stationExpression);
00033     std::string getStationExpression() const;
00034 protected:
00035     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00036     virtual bool _loadInstance(PersistenceRecord* fields) override;
00037     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00038     virtual bool _check(std::string& errorMessage) override;
00039     virtual void _createInternalStatisticReporters() override;
00040     virtual void _createEditableDataDefinitions() override;
00041     virtual void _createAttachedAttributes() override;
00042 private:
00043     Station* _resolveDestination() const;
00044     Station* _resolveSource(Entity* entity) const;
00045 private:
00046     const struct DEFAULT_VALUES {
00047         const std::string stationExpression = "";
00048     } DEFAULT;
00049     Transporter* _transporter = nullptr;
00050     Station* _station = nullptr;
00051     std::string _stationExpression = DEFAULT.stationExpression;
00052     Counter* _numberIn = nullptr;
00053 };
00054
00055 #endif /* MOVE_H */
```

18.417 PickableStationItem.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   PickableStationItem.h
00008  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00009  *
00010  * Created on 1 de dezembro de 2022, 14:00
00011  */
00012
00013 #ifndef PICKABLESTATIONITEM_H
00014 #define PICKABLESTATIONITEM_H
00015
00016 #include "../../../kernel/simulator/model/Model.h"
00017 #include "plugins/data/DiscreteProcessing/Queue.h"
00018 #include "plugins/data/DiscreteProcessing/Resource.h"
00019 #include "../../data/MaterialHandling/Station.h"
00020
00021 class PickableStationItem {
00022 public:
00023     PickableStationItem(Model* model, std::string stationName, std::string queueName = "", std::string
      resourceName = "");
00024     PickableStationItem(Model* model, std::string stationName, std::string expression);
00025     PickableStationItem(Station* station, Queue* queue);
00026     PickableStationItem(Station* station, Resource* resource);
00027     PickableStationItem(Station* station, std::string expression);
00028     PickableStationItem(Station* station);
00029     virtual ~PickableStationItem() = default;
00030 public:
00031     std::string getExpression() const;
00032     Queue* getQueue() const;
00033     Resource* getResource() const;
00034     Station* getStation() const;
00035     void setExpression(std::string _expression);
00036     void setQueue(Queue* _queue);
00037     void setResource(Resource* _resource);
00038     void setStation(Station* _station);
00039
00040     std::string getName();
00041     void _addSimulationControl(SimulationControl* control);
00042     List<SimulationControl*>* getSimulationControls() const;
00043
00044     std::string getTypeDC() {return _typeDC;};
00045 private:
00046 private:
00047     Station* _station = nullptr;
00048     Resource* _resource = nullptr;
00049     Queue* _queue = nullptr;
00050     std::string _expression = "";
00051     std::string _stationName = "";
00052
00053     std::string _typeDC = Util::TypeOf<Station>();
00054     List<SimulationControl*>* _simulationControls = new List<SimulationControl*>();
00055 };
00056
```

```
00057 #endif /* PICKABLESTATIONITEM_H */
```

18.418 PickStation.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   PickStation.h
00009  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00010  *
00011  * Created on 11 de Setembro de 2019, 13:08
00012  */
00013
00014 #ifndef PICKSTATION_H
00015 #define PICKSTATION_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00019 #include "kernel/util/List.h"
00020 #include "plugins/data/DiscreteProcessing/Queue.h"
00021 #include "plugins/data/DiscreteProcessing/Resource.h"
00022
00023 #include "PickableStationItem.h"
00024
00068 class PickStation : public ModelComponent {
00069 public:
00070
00071     enum class TestCondition : int {
00072         MINIMUM = 0, MAXIMUM = 1, num_elements = 2
00073     };
00074
00075     //enum class PickCondition : int {
00076     //  NUMBER_BUSY_RESOURCE = 1, NUMBER_IN_QUEUE = 2, EXPRESSION = 3
00077     //};
00078 public:
00079     static std::string convertEnumToStr(TestCondition condition);
00080 public: // constructors
00081     PickStation(Model* model, std::string name = "");
00082     virtual ~PickStation() override;
00083 public: // virtual
00084     virtual std::string show() override;
00085 public:
00086     //void setPickCondition(PickStation::PickCondition _pickCondition);
00087     //PickStation::PickCondition getPickCondition() const;
00088     void setTestCondition(PickStation::TestCondition _testCondition);
00089     PickStation::TestCondition getTestCondition() const;
00090     void setSaveAttribute(std::string _saveAttribute);
00091     std::string getSaveAttribute() const;
00092     List<PickableStationItem*>* getPickableStationItens() const;
00093     void addPickableStationItem(PickableStationItem* newItem);
00094     void removePickableStationItem(PickableStationItem* item);
00095     void setPickConditionExpression(bool _pickConditionExpression);
00096     bool isPickConditionExpression() const;
00097     void setPickConditionNumberInQueue(bool _pickConditionNumberInQueue);
00098     bool isPickConditionNumberInQueue() const;
00099     void setPickConditionNumberBusyResource(bool _pickConditionNumberBusyResource);
00100     bool isPickConditionNumberBusyResource() const;
00101 public: // static
00102     static PluginInformation* GetPluginInformation();
00103     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00104     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00105
00106
00107 protected: // must be overriden
00108     virtual bool _loadInstance(PersistenceRecord *fields) override;
00109     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00110     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00111 protected: // could be overriden by derived classes
00112     virtual bool _check(std::string& errorMessage) override;
00114     virtual ParserChangesInformation* _getParserChangesInformation() override;
00115     virtual void _initBetweenReplications() override;
00117     // virtual void _createInternalAndAttachedData() override;
00118     virtual void _addSimulationControl(SimulationControl* property) override;
00119
00120
00121 protected:
00122     // virtual void _createInternalStatisticReporters() override;
00123     // virtual void _createNonEditableDataDefinitions() override;
00124     virtual void _createEditableDataDefinitions() override;
00125     // virtual void _createAttachedAttributes() override;
00126
```

```
00127 private: // methods
00128 private: // attributes 1:1
00129
00130     const struct DEFAULT_VALUES {
00131         const std::string saveAttribute = "";
00132         const PickStation::TestCondition testCondition = TestCondition::MINIMUM;
00133         bool pickConditionNumberBusyResource = false;
00134         bool pickConditionNumberInQueue = false;
00135         bool pickConditionExpression = true;
00136         //const PickStation::PickCondition pickCondition = PickCondition::NUMBER_BUSY_RESOURCE;
00137     } DEFAULT;
00138     std::string _saveAttribute = DEFAULT.saveAttribute;
00139     PickStation::TestCondition _testCondition = DEFAULT.testCondition;
00140     //PickStation::PickCondition _pickCondition = DEFAULT.pickCondition;
00141     bool _pickConditionNumberBusyResource = DEFAULT.pickConditionNumberBusyResource;
00142     bool _pickConditionNumberInQueue = DEFAULT.pickConditionNumberInQueue;
00143     bool _pickConditionExpression = DEFAULT.pickConditionExpression;
00144
00145     //DummyElement* _internalDataDefinition = nullptr;
00146 private: // attributes 1:n
00147     List<PickableStationItem*>* _pickableStationItens = new List<PickableStationItem*>();
00148 };
00149
00150
00151 #endif /* PICKSTATION_H */
```

18.419 Route.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Route.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:15
00012  */
00013
00014 #ifndef ROUTE_H
00015 #define ROUTE_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../../kernel/simulator/essentialPlugins/Counter.h"
00019 #include "../../data/MaterialHandling/Station.h"
00020 #include "../../data/MaterialHandling/Sequence.h"
00021 #include "../../data/Logic/Label.h"
00022
00055 class Route : public ModelComponent {
00056 public:
00057
00058     enum class DestinationType : int {
00059         Station = 0, Sequence = 1, Label = 2, num_elements = 3
00060     };
00061 public:
00062     static std::string convertEnumToStr(DestinationType type);
00063 public:
00064     Route(Model* model, std::string name = "");
00065     virtual ~Route() = default;
00066 public:
00067     virtual std::string show() override;
00068 public:
00069     static PluginInformation* GetPluginInformation();
00070     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00071     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00072 public:
00073     void setStationName(std::string stationName);
00074     void setStation(Station* _station);
00075     Station* getStation() const;
00076     void setStationExpression(std::string _stationExpression);
00077     std::string getStationExpression() const;
00078     void setRouteTimeExpression(std::string _routeTimeExpression);
00079     void setRouteTimeExpression(std::string _routeTimeExpression, Util::TimeUnit _routeTimeTimeUnit);
00080     std::string getRouteTimeExpression() const;
00081     void setRouteTimeTimeUnit(Util::TimeUnit _routeTimeTimeUnit);
00082     Util::TimeUnit getRouteTimeTimeUnit() const;
00083     void setRouteDestinationType(DestinationType _routeDestinationType);
00084     Route::DestinationType getRouteDestinationType() const;
00085     void setLabel(Label* _label);
00086     Label* getLabel() const;
00087 protected:
00088     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00089     virtual bool _loadInstance(PersistenceRecord *fields) override;
```

```
00090     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00091 protected:
00092     //virtual void _initBetweenReplications();
00093     virtual bool _check(std::string& errorMessage) override;
00094 
00095 protected:
00096      virtual void _createInternalStatisticReporters() override;
00097     // virtual void _createNonEditableDataDefinitions() override;
00098      virtual void _createEditableDataDefinitions() override;
00099      virtual void _createAttachedAttributes() override;
00100     // virtual void _createInternalAndAttachedData() override;
00101 
00102 protected:
00103 
00104 private:
00105 
00106     const struct DEFAULT_VALUES {
00107         const std::string routeTimeExpression = "0.0";
00108         const std::string stationExpression = "";
00109         const Util::TimeUnit routeTimeTimeUnit = Util::TimeUnit::second;
00110         const Route::DestinationType routeDestinationType = DestinationType::Station;
00111     } DEFAULT;
00112     std::string _routeTimeExpression = DEFAULT.routeTimeExpression;
00113     std::string _stationExpression = DEFAULT.stationExpression;
00114     Util::TimeUnit _routeTimeTimeUnit = DEFAULT.routeTimeTimeUnit;
00115     Route::DestinationType _routeDestinationType = DEFAULT.routeDestinationType;
00116 private: // association
00117     Station* _station = nullptr;
00118     Label* _label = nullptr;
00119 private: // internal elements
00120     Counter* _numberIn = nullptr;
00121 };
00122 
00123 #endif /* ROUTE_H */
```

18.420 Start.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
00007 /*
00008  * File:   Start.h
00009  * Author: rlcancian
00010  *
00011  * Created on 11 de Setembro de 2019, 13:15
00012  */
00013 
00014 #ifndef START_H
00015 #define START_H
00016 
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../data/MaterialHandling/Conveyor.h"
00019 
00040 class Start : public ModelComponent {
00041 public: // constructors
00042     Start(Model* model, std::string name = "");
00043     virtual ~Start() = default;
00044 public: // virtual
00045     virtual std::string show() override;
00046 public: // static
00047     static PluginInformation* GetPluginInformation();
00048     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00049     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00050 public:
00051     void setConveyor(Conveyor* conveyor);
00052     Conveyor* getConveyor() const;
00053     void setVelocityExpression(std::string velocityExpression);
00054     std::string getVelocityExpression() const;
00055 protected: // virtual
00056     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00057     virtual bool _loadInstance(PersistenceRecord *fields) override;
00058     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00059 protected: // virtual
00060     //virtual void _initBetweenReplications();
00061     // virtual void _createInternalStatisticReporters() override;
00062     // virtual void _createNonEditableDataDefinitions() override;
00063     // virtual void _createEditableDataDefinitions() override;
00064     // virtual void _createAttachedAttributes() override;
00065     virtual bool _check(std::string& errorMessage) override;
00066 
00067 protected:
00068 
```

```
00069 private: // methods
00070 private: // attributes 1:1
00071     Conveyor* _conveyor = nullptr;
00072     std::string _velocityExpression = "";
00073 private: // attributes 1:n
00074 };
00075
00076
00077 #endif /* START_H */
```

18.421 Stop.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Stop.h
00009  * Author: rlcancian
00010  *
00011  * Created on 11 de Setembro de 2019, 13:15
00012  */
00013
00014 #ifndef STOP_H
00015 #define STOP_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../data/MaterialHandling/Conveyor.h"
00019
00037 class Stop : public ModelComponent {
00038 public: // constructors
00039     Stop(Model* model, std::string name = "");
00040     virtual ~Stop() = default;
00041 public: // virtual
00042     virtual std::string show() override;
00043 public: // static
00044     static PluginInformation* GetPluginInformation();
00045     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00046     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00047 public:
00048     void setConveyor(Conveyor* conveyor);
00049     Conveyor* getConveyor() const;
00050 protected: // virtual
00051     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00052     virtual bool _loadInstance(PersistenceRecord *fields) override;
00053     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00054 protected: // virtual
00055     //virtual void _initBetweenReplications();
00056     // virtual void _createInternalStatisticReporters() override;
00057     // virtual void _createNonEditableDataDefinitions() override;
00058     // virtual void _createEditableDataDefinitions() override;
00059     // virtual void _createAttachedAttributes() override;
00060     virtual bool _check(std::string& errorMessage) override;
00061
00062 protected:
00063
00064 private: // methods
00065 private: // attributes 1:1
00066     Conveyor* _conveyor = nullptr;
00067 private: // attributes 1:n
00068 };
00069
00070
00071 #endif /* STOP_H */
```

18.422 Store.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Store.h
00009  * Author: rlcancian
00010  *
00011  * Created on 11 de Setembro de 2019, 13:07
00012  */
00013
```

```
00014 #ifndef STORE_H
00015 #define STORE_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../data/MaterialHandling/Storage.h"
00019
00032 class Store : public ModelComponent {
00033 public: // constructors
00034     Store(Model* model, std::string name = "");
00035     virtual ~Store() = default;
00036 public: // virtual
00037     virtual std::string show() override;
00038 public:
00039     void setStorage(Storage* storage);
00040     Storage* getStorage() const;
00041     void setQuantityExpression(std::string quantityExpression);
00042     std::string getQuantityExpression() const;
00043 public: // static
00044     static PluginInformation* GetPluginInformation();
00045     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00046     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00047 protected: // virtual
00048     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00049     virtual bool _loadInstance(PersistenceRecord *fields) override;
00050     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00051 protected: // virtual
00052     //virtual void _initBetweenReplications();
00053     // virtual void _createInternalStatisticReporters() override;
00054     // virtual void _createNonEditableDataDefinitions() override;
00055     virtual void _createEditableDataDefinitions() override;
00056     // virtual void _createAttachedAttributes() override;
00057     virtual bool _check(std::string& errorMessage) override;
00058
00059 protected:
00060
00061 private: // methods
00062 private: // attributes 1:1
00063     const struct DEFAULT_VALUES {
00064         const std::string quantityExpression = "1";
00065     } DEFAULT;
00066     Storage* _storage = nullptr;
00067     std::string _quantityExpression = DEFAULT.quantityExpression;
00068 private: // attributes 1:n
00069 };
00070
00071
00072 #endif /* STORE_H */
```

18.423 Unstore.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Unstore.h
00009  * Author: rlcancian
00010  *
00011  * Created on 11 de Setembro de 2019, 13:08
00012  */
00013
00014 #ifndef UNSTORE_H
00015 #define UNSTORE_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../data/MaterialHandling/Storage.h"
00019
00031 class Unstore : public ModelComponent {
00032 public: // constructors
00033     Unstore(Model* model, std::string name = "");
00034     virtual ~Unstore() = default;
00035 public: // virtual
00036     virtual std::string show() override;
00037 public:
00038     void setStorage(Storage* storage);
00039     Storage* getStorage() const;
00040     void setQuantityExpression(std::string quantityExpression);
00041     std::string getQuantityExpression() const;
00042 public: // static
00043     static PluginInformation* GetPluginInformation();
00044     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00045     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00046 protected: // virtual
```

```
00047     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00048     virtual bool _loadInstance(PersistenceRecord *fields) override;
00049     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00050 protected: // virtual
00051     //virtual void _initBetweenReplications();
00052     // virtual void _createInternalStatisticReporters() override;
00053     // virtual void _createNonEditableDataDefinitions() override;
00054     virtual void _createEditableDataDefinitions() override;
00055     // virtual void _createAttachedAttributes() override;
00056     virtual bool _check(std::string& errorMessage) override;
00057
00058 protected:
00059
00060 private: // methods
00061 private: // attributes 1:1
00062     const struct DEFAULT_VALUES {
00063         const std::string quantityExpression = "1";
00064     } DEFAULT;
00065     Storage* _storage = nullptr;
00066     std::string _quantityExpression = DEFAULT.quantityExpression;
00067 private: // attributes 1:n
00068 };
00069
00070
00071 #endif /* UNSTORE_H */
```

18.424 Boundary_Closed.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Boundary_Closed.h
00008  * Author: cancian
00009  *
00010  * Created on 10 de abril de 2023, 16:22
00011  */
00012 #pragma once
00013
00014 #include "plugins/components/ModalModel/CellularAutomata/BoundaryCondition.h"
00015 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00016 #include "plugins/components/ModalModel/CellularAutomata/Lattice.h"
00017 #include "plugins/components/ModalModel/CellularAutomata/State.h"
00018
00019 class Boundary_Closed : public BoundaryCondition {
00020 public:
00021
00022     Boundary_Closed(Lattice* lattice = nullptr, Neighborhood* neighborhood = nullptr)
00023     : BoundaryCondition(lattice, neighborhood) {
00024         fixedCell = new Cell();
00025     }
00026
00027     Boundary_Closed(const Boundary_Closed& orig)
00028     : BoundaryCondition(orig) {
00029     }
00030     virtual ~Boundary_Closed() = default;
00031 public:
00032     virtual Cell* getNeighborCell(std::vector<int> cellNDimPosition, std::vector<int>
    neighborNDimPosition) override {
00033         long neighborCellNumber = lattice->cellNDimPosition2Number(neighborNDimPosition);
00034         if (neighborCellNumber < 0) { // out of bounds
00035             int pos, maxPos;
00036             for (unsigned short dim = 0; dim < neighborNDimPosition.size(); dim++) {
00037                 pos = neighborNDimPosition.at(dim);
00038                 maxPos = (int) (lattice->getDimension(dim));
00039                 if (pos < 0 || pos >= maxPos) { // out of bound
00040                     // Keep periodic wrapping valid for any radius, including large negative offsets.
00041                     pos = ((pos % maxPos) + maxPos) % maxPos;
00042                     // position corrected
00043                     neighborNDimPosition.at(dim) = pos;
00044                 }
00045             }
00046             //std::cout << " bond fixed to " <<Util::showNDimPosition(&neighborNDimPosition);
00047             neighborCellNumber = lattice->cellNDimPosition2Number(neighborNDimPosition);
00048         }
00049         return lattice->getCell(neighborCellNumber);
00050     }
00051 public:
00052
00053     void setFixedState(const State& fixedState) {
00054         fixedCell->setCurrentState(fixedState);
00055     }
00056
```

```
00057     const State& getFixedState() const {
00058         return fixedCell->getCurrentState();
00059     }
00060 private:
00061     Cell* fixedCell;
00062 };
```

18.425 Boundary_Fixed.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Boundary_Fixed.h
00008  * Author: cancian
00009  *
00010  * Created on 6 de abril de 2023, 13:48
00011  */
00012 #pragma once
00013
00014 #include "plugins/components/ModalModel/CellularAutomata/BoundaryCondition.h"
00015 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00016 #include "plugins/components/ModalModel/CellularAutomata/Lattice.h"
00017 #include "plugins/components/ModalModel/CellularAutomata/State.h"
00018
00019 class Boundary_Fixed : public BoundaryCondition {
00020 public:
00021
00022     Boundary_Fixed(Lattice* lattice = nullptr, Neighborhood* neighborhood = nullptr)
00023     : BoundaryCondition(lattice, neighborhood) {
00024         fixedCell = new Cell();
00025         fixedCell->setCellNumber(-99);
00026     }
00027
00028     Boundary_Fixed(const Boundary_Fixed& orig)
00029     : BoundaryCondition(orig) {
00030     }
00031     virtual ~Boundary_Fixed() = default;
00032 public:
00033
00034     virtual Cell* getNeighborCell(std::vector<int> cellNDimPosition, std::vector<int>
      neighborNDimPosition) override {
00035         long neighborCellNumber = lattice->cellNDimPosition2Number(neighborNDimPosition);
00036         if (neighborCellNumber < 0) { // out of bounds
00037             //std::cout « " bond fixed to fixedCell";
00038             return fixedCell;
00039         }
00040         return lattice->getCell(neighborCellNumber);
00041     }
00042 public:
00043
00044     void setFixedState(const State& fixedState) {
00045         fixedCell->setCurrentState(fixedState);
00046     }
00047
00048     const State& getFixedState() const {
00049         return fixedCell->getCurrentState();
00050     }
00051 private:
00052     Cell* fixedCell;
00053 };
```

18.426 BoundaryCondition.h

```
00001 #pragma once
00002
00003 #include "plugins/components/ModalModel/CellularAutomata/Lattice.h"
00004 //#include "plugins/components/ModalModel/CellularAutomata/Neighborhood.h"
00005 #include <string>
00006 #include <vector>
00007
00008
00009 class Neighborhood;
00010
00011 class BoundaryCondition {
00012 public:
00013     BoundaryCondition(Lattice* lattice = nullptr, Neighborhood* neighborhood=nullptr){
00014         this->lattice = lattice;
00015         this->neighborhood = neighborhood;
00016     }
```

```
00017     BoundaryCondition(const BoundaryCondition& orig){}
00018     virtual ~BoundaryCondition()=default;
00019 public:
00020     virtual std::string show() {
00021         return "todo Boundary.show()";
00022     }
00023     virtual Cell* getNeighborCell(std::vector<int> cellPosition, std::vector<int>
      neighborCellPosition) = 0;
00024 public:
00025     void setNeighborhood(Neighborhood* neighborhood){
00026         this->neighborhood = neighborhood;
00027     }
00028     Neighborhood* getNeighborhood() const{
00029         return neighborhood;
00030     }
00031     void setLattice(Lattice* lattice){
00032         this->lattice = lattice;
00033     }
00034     Lattice* getLattice() const{
00035         return lattice;
00036     }
00037 protected:
00038     Lattice* lattice;
00039     Neighborhood* neighborhood;
00040 private:
00041 };
```

18.427 Cell.h

```
00001 #pragma once
00002
00003
00004
00005 #include "plugins/components/ModalModel/CellularAutomata/State.h"
00006 #include "plugins/components/ModalModel/CellularAutomata/StateSet.h"
00007 #include <memory>
00008 #include <vector>
00009 #include <string>
00010
00011 class State;
00012
00013 class Cell {
00014 public:  // constructors
00015     Cell(StateSet* stateSet = nullptr, long cellNum=0, std::vector<int> position={});
00016     Cell(long cellNum, std::vector<int> position);
00017     Cell(const Cell& orig);
00018     Cell& operator=(const Cell& orig);
00019     virtual ~Cell()=default;
00020 public: // methods
00021     virtual std::string show();
00022     bool updateState();
00023     bool isUpdatePending();
00024     virtual void draw();
00025 public: // getters
00026     void setCellNumber(long cellNum);
00027     long getCellNumber() const;
00028     void setPosition(std::vector<int> position);
00029     std::vector<int> getPosition() const;
00030     void setStateSet(StateSet* stateSet);
00031     StateSet* getStateSet() const;
00032     const State& getPreviousState() const;
00033     bool setCurrentState(const State& currentState);
00034     const State& getCurrentState() const;
00035     bool setNextState(const State& nextState);
00036     const State& getNextState() const;
00037     void setNeighbors(std::vector<Cell*> neighbors);
00038     std::vector<Cell*> getNeighbors() const;
00039 protected:
00040     bool acceptsState(const State& state) const;
00041     std::unique_ptr<State> previousState;
00042     std::unique_ptr<State> currentState;
00043     std::unique_ptr<State> nextState;
00044     StateSet* stateSet = nullptr;
00045     bool updatePending;
00046     std::vector<int> position;
00047     long cellNum;
00048     std::vector<Cell*> neighbors;
00049 protected: // for other CA types
00050 private:
00051 };
```

18.428 CellularAutomata.h

```
00001 #pragma once
00002
00003 #include "plugins/components/ModalModel/CellularAutomata/Lattice.h"
00004 #include "plugins/components/ModalModel/CellularAutomata/StateSet.h"
00005 #include "plugins/components/ModalModel/CellularAutomata/Neighborhood.h"
00006 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00007 #include <string>
00008 #include <functional>
00009 #include <list>
00010 #include <thread>
00011
00012 class CellularAutomata;
00013
00014 class CellularAutomataEvent { // event for observers
00015 public:
00016     CellularAutomataEvent(CellularAutomata* ca) {
00017         this->ca = ca;
00018     }
00019     CellularAutomata* ca;
00020 };
00021
00022 typedef std::function<void(CellularAutomataEvent) > CellularAutomataEventObserver;
00023
00024 class CellularAutomata {
00025 public:
00026     CellularAutomata(std::string name = "", Lattice* lattice = nullptr, StateSet* stateSet = nullptr,
      Neighborhood* neighborhood = nullptr, LocalRule* localRule = nullptr);
00027     CellularAutomata(const CellularAutomata& orig);
00028     virtual ~CellularAutomata() = default;
00029 public:
00030     //virtual std::string show();
00031     virtual bool init();
00032     virtual void step();
00033     virtual void start();
00034     virtual void pause();
00035     template<typename Class> void addObserver(Class * object, void
      (Class::*function)(CellularAutomataEvent));
00036 public:
00037     std::string getName() const;
00038     void setName(const std::string name);
00039     void setLocalRule(LocalRule* localRule);
00040     LocalRule* getLocalRule() const;
00041     void setNeighborhood(Neighborhood* neighborhood);
00042     Neighborhood* getNeighborhood() const;
00043     void setStateSet(StateSet* stateSet);
00044     StateSet* getStateSet() const;
00045     void setLattice(Lattice* lattice);
00046     Lattice* getLattice() const;
00047     void setSimulationLength(unsigned int simulationLength);
00048     unsigned int getSimulationLength() const;
00049     unsigned int getSimulatedTime() const;
00050     bool isRunning() const;
00051 protected:
00052     void notifyObservers(CellularAutomataEvent event);
00053     virtual void applyLocalRule() = 0;
00054 protected:
00055     Lattice* lattice;
00056     StateSet* stateSet;
00057     Neighborhood* neighborhood;
00058     LocalRule* localRule;
00059     std::string name;
00060     unsigned int simulatedTime;
00061     unsigned int simulationLength;
00062     bool running;
00063 private:
00064     std::list<CellularAutomataEventObserver>* observers = new
      std::list<CellularAutomataEventObserver>();
00065 };
00066
00067 template<typename Class> void CellularAutomata::addObserver(Class * object, void
      (Class::*function)(CellularAutomataEvent)) {
00068     CellularAutomataEventObserver observer = std::bind(function, object, std::placeholders::_1);
00069     observers->emplace_back(observer);
00070 }
```

18.429 CellularAutomata_1DTimed.h

```
00001 #pragma once
00002
00003 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00004 #include "plugins/components/ModalModel/CellularAutomata/CellularAutomataBase.h"
00005 #include "plugins/components/ModalModel/CellularAutomata/Lattice.h"
```

```
00006 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00007 #include "plugins/components/ModalModel/CellularAutomata/State.h"
00008
00010 class CellularAutomata_1DTimed: public CellularAutomataBase {
00011 public:
00012     CellularAutomata_1DTimed(Lattice* lattice = nullptr, StateSet* stateSet = nullptr, Neighborhood*
      neighborhood = nullptr, LocalRule* localRule = nullptr)
00013     :CellularAutomataBase(lattice, stateSet, neighborhood, localRule) { }
00014     CellularAutomata_1DTimed(const CellularAutomata_1DTimed& orig)
00015     : CellularAutomataBase(orig){ }
00016     virtual ~CellularAutomata_1DTimed() = default;
00017 protected:
00018     virtual void applyLocalRule() override {
00019         Cell* cell;
00020         int nextRow;
00021         int maxRows = std::min<int>(lattice->getDimension(1)-2, lattice->getDimension(1)-2); //
      simulatedTime);
00022         for (int row=maxRows; row>=0; row--) {
00023             nextRow = row + 1;
00024             // move cell states to the next row
00025             for (int col=0; col<lattice->getDimension(0); col++) { // move to row bellow
00026                 cell = lattice->getCell({col,row});
00027                 lattice->setCellState({col,nextRow}, const_cast<State*>(&cell->getCurrentState()));
00028             }
00029             if (row==0) {
00030                 // apply the same local rule for all unidimensional cells at the same time
00031                 for (int col=0; col<lattice->getDimension(0); col++) {
00032                     cell = lattice->getCell({col,row});
00033                     localRule->applyRule(cell);
00034                 }
00035             }
00036         }
00037         // alfter changing the nextState of all unidimensional cells, update them (currentState =
      nextState)
00038         for (int col=0; col<lattice->getDimension(0); col++) {
00039             cell = lattice->getCell({col,0});
00040             cell->updateState();
00041         }
00042     }
00043 protected:
00044 };
```

18.430 CellularAutomata_Classic.h

```
00001 #pragma once
00002
00003 #include "plugins/components/ModalModel/CellularAutomata/CellularAutomataBase.h"
00004 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00005
00007 class CellularAutomata_Classic: public CellularAutomataBase {
00008 public:
00009     CellularAutomata_Classic(Lattice* lattice = nullptr, StateSet* stateSet = nullptr, Neighborhood*
      neighborhood = nullptr, LocalRule* localRule = nullptr)
00010     :CellularAutomataBase(lattice, stateSet, neighborhood, localRule) { }
00011     CellularAutomata_Classic(const CellularAutomata_Classic& orig): CellularAutomataBase(orig){ }
00012     virtual ~CellularAutomata_Classic() = default;
00013 protected:
00014     virtual void applyLocalRule() override {
00015         Cell* cell;
00016         // apply the same local rule to all cells at the same time
00017         for (long cellNumber = 0; cellNumber < lattice->getCellsSize(); cellNumber++) {
00018             cell = lattice->getCell(cellNumber);
00019             localRule->applyRule(cell);
00020         }
00021         // alfter changing the nextState of all cells, update them (currentState = nextState)
00022         for (long cellNumber = 0; cellNumber < lattice->getCellsSize(); cellNumber++) {
00023             cell = lattice->getCell(cellNumber);
00024             cell->updateState();
00025         }
00026     }
00027 };
```

18.431 CellularAutomataAllHeaders.h

```
00001 #pragma once
00002
00003 #include "plugins/components/ModalModel/CellularAutomata/BoundaryCondition.h"
00004 #include "plugins/components/ModalModel/CellularAutomata/Boundary_Closed.h"
00005 #include "plugins/components/ModalModel/CellularAutomata/Boundary_Fixed.h"
00006 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00007 #include "plugins/components/ModalModel/CellularAutomata/CellularAutomata.h"
00008 #include "plugins/components/ModalModel/CellularAutomata/CellularAutomataBase.h"
```

```
00009 #include "plugins/components/ModalModel/CellularAutomata/CellularAutomata_1DTimed.h"
00010 #include "plugins/components/ModalModel/CellularAutomata/CellularAutomata_Classic.h"
00011 #include "plugins/components/ModalModel/CellularAutomata/Lattice.h"
00012 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00013 #include "plugins/components/ModalModel/CellularAutomata/LocalRule_Elementary.h"
00014 #include "plugins/components/ModalModel/CellularAutomata/LocalRule_FlorestalFire.h"
00015 #include "plugins/components/ModalModel/CellularAutomata/LocalRule_GameOfLife.h"
00016 #include "plugins/components/ModalModel/CellularAutomata/LocalRule_Growty.h"
00017 #include "plugins/components/ModalModel/CellularAutomata/LocalRule_HppLatticeGas.h"
00018 #include "plugins/components/ModalModel/CellularAutomata/LocalRule_SandRock_Student.h"
00019 #include "plugins/components/ModalModel/CellularAutomata/LocalRule_ShowNeighbordhood.h"
00020 #include "plugins/components/ModalModel/CellularAutomata/Neighborhood.h"
00021 #include "plugins/components/ModalModel/CellularAutomata/Neighborhood_Center.h"
00022 #include "plugins/components/ModalModel/CellularAutomata/Neighborhood_Moore.h"
00023 #include "plugins/components/ModalModel/CellularAutomata/Neighborhood_VonNeumann.h"
00024 #include "plugins/components/ModalModel/CellularAutomata/State.h"
00025 #include "plugins/components/ModalModel/CellularAutomata/StateSet.h"
00026 #include "plugins/components/ModalModel/CellularAutomata/StateSet_Enumerable.h"
00027 #include "plugins/components/ModalModel/CellularAutomata/State_Numerical.h"
```

18.432 CellularAutomataBase.h

```
00001 #pragma once
00002
00003
00004 #include <string>
00005 #include <functional>
00006 #include <list>
00007 #include <thread>
00008
00009 class Lattice;
00010 class StateSet;
00011 class Neighborhood;
00012 class LocalRule;
00013
00014
00015 class CellularAutomataBase {
00016 public:
00017     CellularAutomataBase(Lattice* lattice = nullptr, StateSet* stateSet = nullptr, Neighborhood*
      neighborhood = nullptr, LocalRule* localRule = nullptr);
00018     CellularAutomataBase(const CellularAutomataBase& orig);
00019     virtual ~CellularAutomataBase() = default;
00020 public:
00021     //virtual std::string show();
00022     virtual bool init();
00023     virtual void step();
00024 public:
00025     void setLocalRule(LocalRule* localRule);
00026     LocalRule* getLocalRule() const;
00027     void setNeighborhood(Neighborhood* neighborhood);
00028     Neighborhood* getNeighborhood() const;
00029     void setStateSet(StateSet* stateSet);
00030     StateSet* getStateSet() const;
00031     void setLattice(Lattice* lattice);
00032     Lattice* getLattice() const;
00033 protected:
00034     virtual void applyLocalRule() = 0;
00035 protected:
00036     Lattice* lattice;
00037     StateSet* stateSet;
00038     Neighborhood* neighborhood;
00039     LocalRule* localRule;
00040 };
```

18.433 Lattice.h

```
00001 #pragma once
00002
00003 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00004 #include <vector>
00005 #include <string>
00006
00007 class CellularAutomataBase;
00008
00009 enum class LatticeType : int {
00010     RETICULAR = 1, TRIANGULAR = 2, HEXAGONAL = 3, NETWORK = 4, USERDEFINED = 5
00011 };
00012
00014 class Lattice {
00015
00016 public:
```

```
00017     Lattice(CellularAutomataBase* parentCellularAutomata, Cell *progenitorCell = nullptr,
      std::vector<unsigned short> dimensions={}, LatticeType latticeType = LatticeType::RETICULAR);
00018     Lattice(const Lattice& orig);
00019     virtual ~Lattice() = default;
00020 public:
00021     virtual std::string show();
00022     bool init();
00023     virtual long cellNDimPosition2Number(const std::vector<int> position);
00024     virtual std::vector<int> cellNumber2NDimPosition(const long cellNumber);
00025     Cell* getCell(const long cellNumber);
00026     Cell* getCell(const std::vector<int> position);
00027     void setCell(const std::vector<int> position, Cell* cell);
00028     bool setCellState(std::vector<int> position, State* state = nullptr, Cell* cell = nullptr);
00029     bool setCellState(long cellNumber, State* state = nullptr, Cell* cell = nullptr);
00030 public:
00031     void setProgenitorCell(Cell *progenitorCell);
00032     Cell* getProgenitorCell() const;
00033     bool setNumCells(unsigned short dimension, unsigned int numCells, bool createDimensions = true);
00034     unsigned short getNumDimensions();
00035     unsigned int getNumCell(unsigned short dimension);
00036     LatticeType getLatticeType() const;
00037     unsigned long getCellsSize() const;
00038     unsigned long getTotalCells() const;
00039     unsigned long calculateTotalCells();
00040     bool isInit() const;
00041     void setDimensions(std::vector<unsigned short> dimensions);
00042     std::vector<unsigned short> getDimensions() const;
00043     unsigned short getDimension(unsigned short dimensionNumber) const;
00044     std::vector<Cell*> getCells() const;
00045 protected:
00046     CellularAutomataBase* parentCellularAutomata;
00047     std::vector<Cell*> cells;
00048     std::vector<unsigned short> dimensions;
00049     Cell *progenitorCell;
00050     LatticeType latticeType;
00051     bool hasBeenInit;
00052     unsigned long totalCells;
00053 };
```

18.434 LocalRule.h

```
00001 #pragma once
00002 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00003 #include "plugins/components/ModalModel/CellularAutomata/CellularAutomataBase.h"
00004
00005
00006 class LocalRule {
00007 public:
00008     LocalRule(CellularAutomataBase* parentCellularAutomata,  StateSet* stateSet=nullptr){
00009         this->parentCellularAutomata = parentCellularAutomata;
00010         parentCellularAutomata->setLocalRule(this);
00011         if (stateSet != nullptr)
00012             this->stateSet = stateSet;
00013         else
00014             this->stateSet = parentCellularAutomata->getStateSet();
00015     }
00016     LocalRule(const LocalRule& orig){}
00017     virtual ~LocalRule()=default;
00018 public:
00019     virtual void applyRule(Cell* cell) = 0;
00020 public:
00021     void setStateSet(StateSet* stateSet){
00022         this->stateSet = stateSet;
00023     }
00024 protected:
00025     CellularAutomataBase* parentCellularAutomata;
00026     StateSet* stateSet;
00027 };
```

18.435 LocalRule_Elementary.h

```
00001 #pragma once
00002
00003 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00004 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00005 #include "plugins/components/ModalModel/CellularAutomata/StateSet_Enumerable.h"
00006 #include <cmath>
00007
00008 class LocalRule_Elementary: public LocalRule {
00009 public:
00010     LocalRule_Elementary(CellularAutomataBase* parentCellularAutomata, uint8_t ruleNumber=30,
      StateSet* stateSet=nullptr)
```

```
00011     :LocalRule(parentCellularAutomata, stateSet) {
00012         this->ruleNumber = ruleNumber;
00013     }
00014     LocalRule_Elementary(const LocalRule_Elementary& orig): LocalRule(orig) { }
00015     virtual ~LocalRule_Elementary()=default;
00016 public:
00017     virtual void applyRule(Cell* cell) override {
00018         long number = 0;
00019         int bit, power = 2;
00020         for (Cell* neigh : cell->getNeighbors()) {
00021             number += neigh->getCurrentState().getValue() * std::pow(2, power--);
00022             if (power==1)
00023                 number += std::pow(2, power--)*cell->getCurrentState().getValue();
00024         }
00025         bit = (ruleNumber & (uint8_t)std::pow(2,number)) » number;
00026         cell->setNextState(State(bit));
00027     }
00028 public:
00029     uint8_t getRuleNumber() const { return ruleNumber; }
00030     void setRuleNumber(const uint8_t ruleNumber) {this->ruleNumber = ruleNumber;}
00031 private:
00032     uint8_t ruleNumber;
00033 };
```

18.436 LocalRule_FlorestalFire.h

```
00001 #pragma once
00002
00003 #include <random>
00004 #include <vector>
00005
00006 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00007 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00008 #include "plugins/components/ModalModel/CellularAutomata/StateSet_Enumerable.h"
00009
00010 class LocalRule_FlorestalFire : public LocalRule {
00011 public:
00012
00013     LocalRule_FlorestalFire(CellularAutomataBase* parentCellularAutomata)
00014     : LocalRule(parentCellularAutomata) {
00015     }
00016
00017     LocalRule_FlorestalFire(const LocalRule_FlorestalFire& orig)
00018     : LocalRule(orig) {
00019     }
00020     virtual ~LocalRule_FlorestalFire() = default;
00021 public:
00022
00023     virtual void applyRule(Cell* cell) override {
00024         if (cell == nullptr || stateSet == nullptr) {
00025             return;
00026         }
00027
00028         auto* enumerableStateSet = dynamic_cast<StateSet_Enumerable*>(stateSet);
00029         if (enumerableStateSet == nullptr || enumerableStateSet->getStatesSize() < 4) {
00030             cell->setNextState(cell->getCurrentState());
00031             return;
00032         }
00033
00034         std::random_device randomDevice;
00035         std::default_random_engine engine(randomDevice());
00036         std::uniform_real_distribution<double> uniform(0.0, 1.0);
00037         std::vector<unsigned int> sumNeighStates(enumerableStateSet->getStatesSize(), 0);
00038
00039         for (Cell* neigh : cell->getNeighbors()) {
00040             const long stateValue = neigh->getCurrentState().getValue();
00041             if (stateValue >= 0 && static_cast<size_t>(stateValue) < sumNeighStates.size()) {
00042                 sumNeighStates.at(static_cast<size_t>(stateValue))++;
00043             }
00044         }
00045
00046         const long currentState = cell->getCurrentState().getValue();
00047         cell->setNextState(cell->getCurrentState());
00048
00049         switch (currentState) {
00050             case 0: // soil
00051                 if (sumNeighStates.size() > 1 && uniform(engine) < pSoil2Grass + sumNeighStates.at(1)
      * pSoil2Grass) {
00052                     cell->setNextState(*enumerableStateSet->getState(1));
00053                 }
00054                 break;
00055             case 1: // grass
00056                 if (sumNeighStates.size() > 2 && uniform(engine) < pGrass2Tree + sumNeighStates.at(2)
      * pGrass2Tree) {
00057                     cell->setNextState(*enumerableStateSet->getState(2));
```

```
00058                 } else if (sumNeighStates.size() > 3 &&
00059                             uniform(engine) < pGrass2Fire + (sumNeighStates.at(3) > 0 ?
    pGrassPropagateFire : 0.0)) {
00060                     cell->setNextState(*enumerableStateSet->getState(3));
00061                 }
00062                 break;
00063             case 2: // tree
00064                 if (sumNeighStates.size() > 3 &&
00065                     uniform(engine) < pTree2Fire + (sumNeighStates.at(3) > 0 ? pTreePropagateFire :
    0.0)) {
00066                     cell->setNextState(*enumerableStateSet->getState(3));
00067                 } else if (sumNeighStates.size() > 0 && uniform(engine) < pTree2Soil +
    sumNeighStates.at(0) * pTree2Soil) {
00068                     cell->setNextState(*enumerableStateSet->getState(0));
00069                 }
00070                 break;
00071             case 3: // fire
00072                 cell->setNextState(*enumerableStateSet->getState(0));
00073                 break;
00074             default:
00075                 break;
00076         }
00077     }
00078 protected:
00079 private:
00080     double pSoil2Grass = 0.01;
00081     double pGrass2Tree = 0.001;
00082     double pGrass2Fire = 0.00001;
00083     double pTree2Fire =  0.00001;
00084     double pGrassPropagateFire = 0.05;
00085     double pTreePropagateFire = 0.5;
00086     double pTree2Soil = 0.0005;
00087 };
```

18.437 LocalRule_GameOfLife.h

```
00001 #pragma once
00002
00003 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00004 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00005 #include "plugins/components/ModalModel/CellularAutomata/State.h"
00006
00007 class LocalRule_GameOfLife: public LocalRule {
00008 public:
00009     LocalRule_GameOfLife(CellularAutomataBase* parentCellularAutomata)
00010     :LocalRule(parentCellularAutomata) { }
00011     LocalRule_GameOfLife(const LocalRule_GameOfLife& orig): LocalRule(orig) { }
00012     virtual ~LocalRule_GameOfLife()=default;
00013 public:
00014     virtual void applyRule(Cell* cell) override {
00015         unsigned int living = 0;
00016         for (Cell* neigh : cell->getNeighbors()) {
00017             living += neigh->getCurrentState().getValue();
00018         }
00019         if (cell->getCurrentState().getValue() == 1) {
00020             if (living < 2 || living > 3)
00021                 cell->setNextState(State(0));
00022             else
00023                 cell->setNextState(cell->getCurrentState());
00024         } else if (living == 3)
00025             cell->setNextState(State(1));
00026         else
00027             cell->setNextState(cell->getCurrentState());
00028     }
00029 };
```

18.438 LocalRule_Growty.h

```
00001 #pragma once
00002
00003 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00004 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00005
00006 class LocalRule_Growty : public LocalRule {
00007 public:
00008     LocalRule_Growty(CellularAutomataBase* parentCellularAutomata)
00009     : LocalRule(parentCellularAutomata) {  }
00010     LocalRule_Growty(const LocalRule_Growty& orig): LocalRule(orig) {}
00011     virtual ~LocalRule_Growty() = default;
00012 public:
00013     virtual void applyRule(Cell* cell) override {
00014         unsigned int sum = 0;
```

```
00015         for (Cell* neigh : cell->getNeighbors()) {
00016             sum += neigh->getCurrentState().getValue();
00017         }
00018         if (sum <= 3 || sum == 5)
00019             cell->setNextState(State(0));
00020         else
00021             cell->setNextState(State(1));
00022     }
00023 };
```

18.439 LocalRule_HppLatticeGas.h

```
00001 #pragma once
00002
00003 #include <iostream>
00004 #include <vector>
00005
00006 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00007 #include "plugins/components/ModalModel/CellularAutomata/Lattice.h"
00008 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00009 #include "plugins/components/ModalModel/CellularAutomata/State.h"
00010 #include "plugins/components/ModalModel/CellularAutomata/StateSet.h"
00011
00012 class LocalRule_HppLatticeGas : public LocalRule {
00013 public:
00014     // Simplified HPP lattice gas: particles stream on the four von Neumann
00015     // directions and head-on two-particle collisions rotate by 90 degrees.
00016     // Conceptual reference: local PDF "2018_Book_Adamatzky A. Cellular Automata...Systems Science 2ed
      2018.pdf",
00017     // physical systems / lattice-gas cellular automata discussion.
00018     explicit LocalRule_HppLatticeGas(CellularAutomataBase* parentCellularAutomata)
00019         : LocalRule(parentCellularAutomata) {
00020     }
00021
00022     void applyRule(Cell* cell) override {
00023         if (cell == nullptr || parentCellularAutomata == nullptr ||
      parentCellularAutomata->getLattice() == nullptr) {
00024             return;
00025         }
00026         if (stateSet != nullptr && dynamic_cast<HppLatticeGasStateSet*>(stateSet) == nullptr) {
00027             std::cerr « "LocalRule_HppLatticeGas requires HppLatticeGasStateSet." « std::endl;
00028             cell->setNextState(cell->getCurrentState());
00029             return;
00030         }
00031
00032         Lattice* lattice = parentCellularAutomata->getLattice();
00033         const std::vector<int> position = cell->getPosition();
00034         if (position.size() < 2 || lattice->getNumDimensions() < 2) {
00035             cell->setNextState(cell->getCurrentState());
00036             return;
00037         }
00038         // sem tempo para acessar por Von Neuman
00039         Cell* northCell = lattice->getCell(wrappedPosition({position[0], position[1] - 1}, lattice));
00040         Cell* eastCell = lattice->getCell(wrappedPosition({position[0] + 1, position[1]}, lattice));
00041         Cell* southCell = lattice->getCell(wrappedPosition({position[0], position[1] + 1}, lattice));
00042         Cell* westCell = lattice->getCell(wrappedPosition({position[0] - 1, position[1]}, lattice));
00043
00044         HppLatticeGasState incoming;
00045         incoming.set(HppLatticeGasState::North, hasDirection(southCell, HppLatticeGasState::North));
00046         incoming.set(HppLatticeGasState::East, hasDirection(westCell, HppLatticeGasState::East));
00047         incoming.set(HppLatticeGasState::South, hasDirection(northCell, HppLatticeGasState::South));
00048         incoming.set(HppLatticeGasState::West, hasDirection(eastCell, HppLatticeGasState::West));
00049
00050         const bool verticalCollision = incoming.has(HppLatticeGasState::North) &&
00051                                        incoming.has(HppLatticeGasState::South) &&
00052                                        !incoming.has(HppLatticeGasState::East) &&
00053                                        !incoming.has(HppLatticeGasState::West);
00054         const bool horizontalCollision = incoming.has(HppLatticeGasState::East) &&
00055                                          incoming.has(HppLatticeGasState::West) &&
00056                                          !incoming.has(HppLatticeGasState::North) &&
00057                                          !incoming.has(HppLatticeGasState::South);
00058
00059         if (verticalCollision) {
00060             cell->setNextState(HppLatticeGasState({{false, true, false, true}}));
00061         } else if (horizontalCollision) {
00062             cell->setNextState(HppLatticeGasState({{true, false, true, false}}));
00063         } else {
00064             cell->setNextState(incoming);
00065         }
00066     }
00067
00068 private:
00069     static std::vector<int> wrappedPosition(std::vector<int> position, Lattice* lattice) {
00070         for (unsigned short dimension = 0; dimension < position.size(); ++dimension) {
00071             const int dimensionSize = static_cast<int>(lattice->getDimension(dimension));
```

```
00072             position[dimension] = ((position[dimension] % dimensionSize) + dimensionSize) %
      dimensionSize;
00073         }
00074         return position;
00075     }
00076
00077     static bool hasDirection(Cell* cell, HppLatticeGasState::Direction direction) {
00078         if (cell == nullptr) {
00079             return false;
00080         }
00081         const auto* hpp = cell->getCurrentState().as<HppLatticeGasState>();
00082         if (hpp != nullptr) {
00083             return hpp->has(direction);
00084         }
00085         return (cell->getCurrentState().getValue() & (1L << direction)) != 0;
00086     }
00087 };
```

18.440 LocalRule_SandRock.h

```
00001 // Aluno: Matheus Antonio de Souza - 21203363
00002 #pragma once
00003
00004 #include <vector>
00005
00006 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00007 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00008 #include "plugins/components/ModalModel/CellularAutomata/StateSet_Enumerable.h"
00009
00010 class LocalRule_SandRock_Student : public LocalRule {
00011 public:
00012     LocalRule_SandRock_Student(CellularAutomataBase* parentCellularAutomata)
00013         : LocalRule(parentCellularAutomata) {
00014     }
00015
00016     LocalRule_SandRock_Student(const LocalRule_SandRock_Student& orig)
00017         : LocalRule(orig) {
00018     }
00019     virtual ~LocalRule_SandRock_Student() = default;
00020     static constexpr long EMPTY = 0;
00021     static constexpr long SAND = 1;
00022     static constexpr long ROCK = 2;
00023 public:
00024
00025     virtual void applyRule(Cell* cell) override {
00026         if (cell == nullptr || stateSet == nullptr) {
00027             return;
00028         }
00029
00030         auto* enumerableStateSet = dynamic_cast<StateSet_Enumerable*>(stateSet);
00031         if (enumerableStateSet == nullptr || enumerableStateSet->getStatesSize() < 3) {
00032             cell->setNextState(cell->getCurrentState());
00033             return;
00034         }
00035
00036         auto pos = cell->getPosition();
00037         // Cada célula possui uma preferência entre cair para esquerda ou para direita
00038         // Neste caso esta sendo utilizado um padrão xadrez que é visualmente agradável
00039         bool preferLeft = ((pos[0] + pos[1]) & 1) != 0;
00040
00041         const auto& nbrs = cell->getNeighbors();
00042         // 0   5   10  14  19
00043         // 1   6   11  15  20
00044         // 2   7       16  21
00045         // 3   8   12  17  22
00046         // 4   9   13  18  23
00047
00048         // nw2 nnw n2  nne ne2
00049         // wnw nw   n  ne  ene
00050         // w2  w       e   e2
00051         // wsw sw   s  se  ese
00052         // sw2 ssw s2  sse se2
00053
00054         long nw = nbrs[6]->getCurrentState().getValue();
00055         long w = nbrs[7]->getCurrentState().getValue();
00056         long sw = nbrs[8]->getCurrentState().getValue();
00057
00058         long n = nbrs[11]->getCurrentState().getValue();
00059         long s = nbrs[12]->getCurrentState().getValue();
00060
00061         long ne = nbrs[15]->getCurrentState().getValue();
00062         long e = nbrs[16]->getCurrentState().getValue();
00063         long se = nbrs[17]->getCurrentState().getValue();
00064
00065         const long wnw = nbrs[1]->getCurrentState().getValue();
```

```
00066         const long w2  = nbrs[2]->getCurrentState().getValue();
00067         const long wsw = nbrs[3]->getCurrentState().getValue();
00068 
00069         const long ene = nbrs[20]->getCurrentState().getValue();
00070         const long e2  = nbrs[21]->getCurrentState().getValue();
00071         const long ese = nbrs[22]->getCurrentState().getValue();
00072 
00073         const long currentState = cell->getCurrentState().getValue();
00074 
00075         // A estratétia funciona da seguinte maneira para uma célula areia:
00076         // 1) Se a célula de baixo (s) estiver livre, a areia cai verticalmente (vira vazio)
00077         // 2) Se (s) está bloqueado, e preferLeft
00078         //     - tenta ir para (sw).
00079         //     - se falhar, tenta (se).
00080         // 3) Analogamente, em caso de !preferLeft
00081         //     - tenta ir para (se).
00082         //     - se falhar, tenta (sw).
00083         // 4) Não há movimentos validos, permanece onde está
00084         //
00085         // Do ponto de vista de uma célula vazia, ela segue a mesma lógica observando os grãos de
      areia de cima
00086         // que tentariam escorregar para sua posição, e se o grão cederia ou não a preferencia.
00087         // Isto garante a conservação de matéria. Pelo menos nos casos que testei.
00088         // Um comportamento que não consegui reproduzir é o de preencher completamente um recipiente
      dependendo de
00089         // seu formato.
00090         switch (currentState) {
00091         case SAND: {
00092             // 1) Queda vertical tem prioridade máxima.
00093             if (s == EMPTY) {
00094                 cell->setNextState(*enumerableStateSet->getState(EMPTY));
00095                 return;
00096             }
00097 
00098             // 2) Tentativa de escorregamento diagonal.
00099             bool canSW = (sw == EMPTY) && (w != SAND && w != ROCK);
00100             bool canSE = (se == EMPTY) && (e != SAND && e != ROCK);
00101 
00102             if (preferLeft) {
00103                 if (canSW) {
00104                     cell->setNextState(*enumerableStateSet->getState(EMPTY));
00105                     return;
00106                 }
00107                 // Fallback para SE: só escorrega para direita (se) se não ha uma areia a direita(e2)
      com
00108                 // preferencia para cair em SE. Por ser preferLeft ser definido em xadrez, sei que ela
00109                 // também é preferLeft e cederá passagem para ela.
00110                 if (canSE && !(e2 == SAND && ese != EMPTY)) {
00111                     cell->setNextState(*enumerableStateSet->getState(EMPTY));
00112                     return;
00113                 }
00114             } else {
00115                 if (canSE) {
00116                     cell->setNextState(*enumerableStateSet->getState(EMPTY));
00117                     return;
00118                 }
00119                 // Fallback para SW: análogo, cedo se w2 quer meu sw.
00120                 if (canSW && !(w2 == SAND && wsw != EMPTY)) {
00121                     cell->setNextState(*enumerableStateSet->getState(EMPTY));
00122                     return;
00123                 }
00124             }
00125 
00126             // Não consigo me mover: permaneço SAND.
00127             cell->setNextState(*enumerableStateSet->getState(SAND));
00128             break;
00129         }
00130         case EMPTY: {
00131             // 1) não cai verticalmente.
00132             if (n == SAND) {
00133                 cell->setNextState(*enumerableStateSet->getState(SAND));
00134                 return;
00135             }
00136 
00137             // 2) se há rock em cima não tem como areia escorregar pra aqui
00138             if (n == ROCK) {
00139                 cell->setNextState(*enumerableStateSet->getState(EMPTY));
00140                 return;
00141             }
00142 
00143             // 3) escorregamento diagonal.
00144             // (Sabemos n != SAND, então diagonais não competem com queda vertical)
00145             bool nw_fills = false;
00146             if (nw == SAND && w != EMPTY) {
00147                 if (preferLeft) {
00148                     // por xadrez, nw prefere seu sw(w2)
00149                     const bool canSW_of_nw = (w2 == EMPTY) && (wnw == EMPTY);
```

```
00150                     //
00151                     const bool fallback_safe = !(ne == SAND && e != EMPTY);
00152
00153                     if (!canSW_of_nw && fallback_safe) {
00154                         nw_fills = true;
00155                     }
00156                 } else {
00157                     // nw prefere seu se, direção preferida.
00158                     nw_fills = true;
00159                 }
00160             }
00161
00162             // simétrico
00163             bool ne_fills = false;
00164             if (ne == SAND && e != EMPTY) {
00165                 if (preferLeft) {
00166                     ne_fills = true;
00167                 } else {
00168                     const bool canSE_of_ne = (e2 == EMPTY) && (ene == EMPTY);
00169                     const bool fallback_safe = !(nw == SAND && w != EMPTY);
00170                     if (!canSE_of_ne && fallback_safe) {
00171                         ne_fills = true;
00172                     }
00173                 }
00174             }
00175
00176             if (nw_fills || ne_fills) {
00177                 cell->setNextState(*enumerableStateSet->getState(SAND));
00178             } else {
00179                 cell->setNextState(*enumerableStateSet->getState(EMPTY));
00180             }
00181             break;
00182         }
00183         case ROCK: {
00184             // objeto imóvel, sempre é pedra
00185             cell->setNextState(*enumerableStateSet->getState(ROCK));
00186             break;
00187         }
00188         default:
00189             cell->setNextState(cell->getCurrentState());
00190             break;
00191         }
00192     }
00193 protected:
00194 private:
00195 };
```

18.441 LocalRule_SandRock_Student.h

```
00001 // Aluno: Matheus Antonio de Souza - 21203363
00002 #pragma once
00003
00004 #include <vector>
00005
00006 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00007 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00008 #include "plugins/components/ModalModel/CellularAutomata/StateSet_Enumerable.h"
00009
00010 class LocalRule_SandRock_Student : public LocalRule {
00011 public:
00012     LocalRule_SandRock_Student(CellularAutomataBase* parentCellularAutomata)
00013         : LocalRule(parentCellularAutomata) {
00014     }
00015
00016     LocalRule_SandRock_Student(const LocalRule_SandRock_Student& orig)
00017         : LocalRule(orig) {
00018     }
00019     virtual ~LocalRule_SandRock_Student() = default;
00020     static constexpr long EMPTY = 0;
00021     static constexpr long SAND = 1;
00022     static constexpr long ROCK = 2;
00023 public:
00024
00025     virtual void applyRule(Cell* cell) override {
00026         if (cell == nullptr || stateSet == nullptr) {
00027             return;
00028         }
00029
00030         auto* enumerableStateSet = dynamic_cast<StateSet_Enumerable*>(stateSet);
00031         if (enumerableStateSet == nullptr || enumerableStateSet->getStatesSize() < 3) {
00032             cell->setNextState(cell->getCurrentState());
00033             return;
00034         }
00035
00036         auto pos = cell->getPosition();
```

```
00037         // Cada célula possui uma preferência entre cair para esquerda ou para direita
00038         // Neste caso esta sendo utilizado um padrão xadrez que é visualmente agradável
00039         bool preferLeft = ((pos[0] + pos[1]) & 1) != 0;
00040 
00041         const auto& nbrs = cell->getNeighbors();
00042         // 0   5   10  14  19
00043         // 1   6   11  15  20
00044         // 2   7       16  21
00045         // 3   8   12  17  22
00046         // 4   9   13  18  23
00047 
00048         // nw2 nnw n2  nne ne2
00049         // wnw nw   n  ne  ene
00050         // w2  w       e   e2
00051         // wsw sw   s  se  ese
00052         // sw2 ssw s2  sse se2
00053 
00054         long nw = nbrs[6]->getCurrentState().getValue();
00055         long w = nbrs[7]->getCurrentState().getValue();
00056         long sw = nbrs[8]->getCurrentState().getValue();
00057 
00058         long n = nbrs[11]->getCurrentState().getValue();
00059         long s = nbrs[12]->getCurrentState().getValue();
00060 
00061         long ne = nbrs[15]->getCurrentState().getValue();
00062         long e = nbrs[16]->getCurrentState().getValue();
00063         long se = nbrs[17]->getCurrentState().getValue();
00064 
00065         const long wnw = nbrs[1]->getCurrentState().getValue();
00066         const long w2  = nbrs[2]->getCurrentState().getValue();
00067         const long wsw = nbrs[3]->getCurrentState().getValue();
00068 
00069         const long ene = nbrs[20]->getCurrentState().getValue();
00070         const long e2  = nbrs[21]->getCurrentState().getValue();
00071         const long ese = nbrs[22]->getCurrentState().getValue();
00072 
00073         const long currentState = cell->getCurrentState().getValue();
00074 
00075         // A estratétia funciona da seguinte maneira para uma célula areia:
00076         // 1) Se a célula de baixo (s) estiver livre, a areia cai verticalmente (vira vazio)
00077         // 2) Se (s) está bloqueado, e preferLeft
00078         //     - tenta ir para (sw).
00079         //     - se falhar, tenta (se).
00080         // 3) Analogamente, em caso de !preferLeft
00081         //     - tenta ir para (se).
00082         //     - se falhar, tenta (sw).
00083         // 4) Não há movimentos validos, permanece onde está
00084         //
00085         // Do ponto de vista de uma célula vazia, ela segue a mesma lógica observando os grãos de
      areia de cima
00086         // que tentariam escorregar para sua posição, e se o grão cederia ou não a preferencia.
00087         // Isto garante a conservação de matéria. Pelo menos nos casos que testei.
00088         // Um comportamento que não consegui reproduzir é o de preencher completamente um recipiente
      dependendo de
00089         // seu formato.
00090         switch (currentState) {
00091         case SAND: {
00092             // 1) Queda vertical tem prioridade máxima.
00093             if (s == EMPTY) {
00094                 cell->setNextState(*enumerableStateSet->getState(EMPTY));
00095                 return;
00096             }
00097 
00098             // 2) Tentativa de escorregamento diagonal.
00099             bool canSW = (sw == EMPTY) && (w != SAND && w != ROCK);
00100             bool canSE = (se == EMPTY) && (e != SAND && e != ROCK);
00101 
00102             if (preferLeft) {
00103                 if (canSW) {
00104                     cell->setNextState(*enumerableStateSet->getState(EMPTY));
00105                     return;
00106                 }
00107                 // Fallback para SE: só escorrega para direita (se) se não ha uma areia a direita(e2)
      com
00108                 // preferencia para cair em SE. Por ser preferLeft ser definido em xadrez, sei que ela
00109                 // também é preferLeft e cederá passagem para ela.
00110                 if (canSE && !(e2 == SAND && ese != EMPTY)) {
00111                     cell->setNextState(*enumerableStateSet->getState(EMPTY));
00112                     return;
00113                 }
00114             } else {
00115                 if (canSE) {
00116                     cell->setNextState(*enumerableStateSet->getState(EMPTY));
00117                     return;
00118                 }
00119                 // Fallback para SW: análogo, cedo se w2 quer meu sw.
00120                 if (canSW && !(w2 == SAND && wsw != EMPTY)) {
```

```
00121                     cell->setNextState(*enumerableStateSet->getState(EMPTY));
00122                     return;
00123                 }
00124             }
00125
00126             // Não consigo me mover: permaneço SAND.
00127             cell->setNextState(*enumerableStateSet->getState(SAND));
00128             break;
00129         }
00130         case EMPTY: {
00131             // 1) não cai verticalmente.
00132             if (n == SAND) {
00133                 cell->setNextState(*enumerableStateSet->getState(SAND));
00134                 return;
00135             }
00136
00137             // 2) se há rock em cima não tem como areia escorregar pra aqui
00138             if (n == ROCK) {
00139                 cell->setNextState(*enumerableStateSet->getState(EMPTY));
00140                 return;
00141             }
00142
00143             // 3) escorregamento diagonal.
00144             // (Sabemos n != SAND, então diagonais não competem com queda vertical)
00145             bool nw_fills = false;
00146             if (nw == SAND && w != EMPTY) {
00147                 if (preferLeft) {
00148                     // por xadrez, nw prefere seu sw(w2)
00149                     const bool canSW_of_nw = (w2 == EMPTY) && (wnw == EMPTY);
00150                     //
00151                     const bool fallback_safe = !(ne == SAND && e != EMPTY);
00152
00153                     if (!canSW_of_nw && fallback_safe) {
00154                         nw_fills = true;
00155                     }
00156                 } else {
00157                     // nw prefere seu se, direção preferida.
00158                     nw_fills = true;
00159                 }
00160             }
00161
00162             // simétrico
00163             bool ne_fills = false;
00164             if (ne == SAND && e != EMPTY) {
00165                 if (preferLeft) {
00166                     ne_fills = true;
00167                 } else {
00168                     const bool canSE_of_ne = (e2 == EMPTY) && (ene == EMPTY);
00169                     const bool fallback_safe = !(nw == SAND && w != EMPTY);
00170                     if (!canSE_of_ne && fallback_safe) {
00171                         ne_fills = true;
00172                     }
00173                 }
00174             }
00175
00176             if (nw_fills || ne_fills) {
00177                 cell->setNextState(*enumerableStateSet->getState(SAND));
00178             } else {
00179                 cell->setNextState(*enumerableStateSet->getState(EMPTY));
00180             }
00181             break;
00182         }
00183         case ROCK: {
00184             // objeto imóvel, sempre é pedra
00185             cell->setNextState(*enumerableStateSet->getState(ROCK));
00186             break;
00187         }
00188         default:
00189             cell->setNextState(cell->getCurrentState());
00190             break;
00191         }
00192     }
00193 protected:
00194 private:
00195 };
```

18.442 LocalRule_ShowNeighbordhood.h

```
00001 #pragma once
00002
00003 #include "plugins/components/ModalModel/CellularAutomata/LocalRule.h"
00004 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00005 #include "plugins/components/ModalModel/CellularAutomata/StateSet_Enumerable.h"
00006
00007 class LocalRule_ShowNeighbordhood : public LocalRule {
```

```
00008 public:
00009     LocalRule_ShowNeighbordhood(CellularAutomataBase* parentCellularAutomata)
00010     : LocalRule(parentCellularAutomata) {  }
00011     LocalRule_ShowNeighbordhood(const LocalRule_ShowNeighbordhood& orig): LocalRule(orig) {}
00012     virtual ~LocalRule_ShowNeighbordhood() = default;
00013 public:
00014     virtual void applyRule(Cell* cell) {
00015         auto* enumerableStateSet = dynamic_cast<StateSet_Enumerable*>(stateSet);
00016         if (cell == nullptr || enumerableStateSet == nullptr || enumerableStateSet->getStatesSize() <
    3)
00017             return;
00018         if (cell->getCurrentState()==*enumerableStateSet->getState(1))
00019             cell->setNextState(cell->getCurrentState());
00020         else {
00021             for (Cell* n: cell->getNeighbors()) {
00022                 if (n != nullptr && n->getCurrentState()==*enumerableStateSet->getState(1)) {
00023                     cell->setNextState(*enumerableStateSet->getState(2));
00024                     return;
00025                 }
00026             }
00027             cell->setNextState(*enumerableStateSet->getState(0));
00028         }
00029     }
00030 };
```

18.443 Neighborhood.h

```
00001 #pragma once
00002
00003 #include <string>
00004 #include <utility>
00005 #include <vector>
00006
00007
00008 class Cell;
00009 #include "plugins/components/ModalModel/CellularAutomata/BoundaryCondition.h"
00010 #include "plugins/components/ModalModel/CellularAutomata/CellularAutomataBase.h"
00011
00013 class Neighborhood {
00014 public:
00015     Neighborhood(CellularAutomataBase* parentCellularAutomata, unsigned short radius=1,
    BoundaryCondition* boundary= nullptr, bool includeCellItself=false){
00016         this->parentCellularAutomata = parentCellularAutomata;
00017         parentCellularAutomata->setNeighborhood(this);
00018         this->radius = radius;
00019         this->boundary = boundary;
00020         this->includeCellItself = includeCellItself;
00021     }
00022     Neighborhood(const Neighborhood& orig){}
00023     virtual ~Neighborhood()=default;
00024 public:
00025     virtual std::string show(){
00026         return "-";
00027     }
00028     virtual std::vector<Cell*> getNeighbors(Cell* cell) = 0;
00029 public:
00030     void setRadius(unsigned short radius) {
00031         this->radius = radius;
00032     }
00033     unsigned short getRadius() const{
00034         return radius;
00035     }
00036     void setBoundary(BoundaryCondition* boundary) {
00037         this->boundary = boundary;
00038         boundary->setNeighborhood(this);
00039     }
00040     BoundaryCondition* getBoundary() const{
00041         return boundary;
00042     }
00043     void setIncludeCellItself(bool includeCellItself){
00044         this->includeCellItself = includeCellItself;
00045     }
00046     bool getIncludeCellItself() const{
00047         return includeCellItself;
00048     }
00049 public:
00050 protected:
00051     Cell* getNeighborCell(std::vector<int> cellPosition, std::vector<std::pair<unsigned short, int>>
    dimensionChanges){
00052         std::vector<int> neighborPosition;
00053         neighborPosition = cellPosition;
00054         for (std::pair<unsigned short, int> change : dimensionChanges) {
00055             neighborPosition.at(change.first) += change.second;
00056         }
```

```
00057         //std::cout « "Neig pos from "«Util::showNDimPosition(cellPosition)« " is "
      «Util::showNDimPosition(&neighborPosition);
00058         Cell* neighCell = boundary->getNeighborCell(cellPosition, neighborPosition);
00059         //std::cout « std::endl;
00060         return neighCell;
00061     }
00062 protected:
00063     std::string name;
00064     CellularAutomataBase* parentCellularAutomata;
00065     BoundaryCondition* boundary;
00066     unsigned short radius;
00067     bool includeCellItself;
00068 private:
00069 };
```

18.444 Neighborhood_Center.h

```
00001 #pragma once
00002
00003 #include <string>
00004 #include <vector>
00005
00006 #include "plugins/components/ModalModel/CellularAutomata/Neighborhood.h"
00007 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00008 #include "plugins/components/ModalModel/CellularAutomata/CellularAutomataBase.h"
00009
00010 class Neighborhood_Center: public Neighborhood {
00011 public:
00012     Neighborhood_Center(CellularAutomataBase* parentCellularAutomata, unsigned short radius=1,
      BoundaryCondition* boundary= nullptr, bool includeCellItself=false)
00013     :Neighborhood(parentCellularAutomata, radius, boundary, includeCellItself) {
00014         this->name = "Center";
00015     }
00016
00017     Neighborhood_Center(const Neighborhood_Center& orig)
00018     : Neighborhood(orig) {    }
00019     virtual ~Neighborhood_Center()=default;
00020 public:
00021     virtual std::vector<Cell*> getNeighbors(Cell* cell) override {
00022         std::vector<Cell*> neighbors;
00023         //std::string m = "Center::getNeighbors(CellPos=" +
      Util::showNDimPosition(cell->getPosition())+")=";
00024         //unsigned short numDimensions = parentCellularAutomata->getLattice()->getNumDimensions();
00025         //std::vector<int>* cellPosition = cell->getPosition();
00026         for (int r = 1; r <= radius; r++) {
00027             int d = 0;
00028             //for (int d = 0; r < numDimensions; d++) {
00029                 neighbors.emplace_back(getNeighborCell(cell->getPosition(), {{d, -r}}));
00030                 neighbors.emplace_back(getNeighborCell(cell->getPosition(), {{d, r}}));
00031             //}
00032         }
00033         if (includeCellItself)
00034             neighbors.emplace_back(cell);
00035         return neighbors;
00036     }
00037 protected:
00038 private:
00039 };
```

18.445 Neighborhood_Moore.h

```
00001 /*
00002  * File:   Neighborhood_Moore.h
00003  * Author: cancian
00004  *
00005  * Created on 6 de abril de 2023, 14:27
00006  */
00007
00008 #pragma once
00009
00010 #include "plugins/components/ModalModel/CellularAutomata/Lattice.h"
00011 #include "plugins/components/ModalModel/CellularAutomata/Neighborhood.h"
00012 #include "plugins/components/ModalModel/CellularAutomata/Cell.h"
00013
00014 #include <vector>
00015
00016 class Neighborhood_Moore : public Neighborhood {
00017 public:
00018     Neighborhood_Moore(CellularAutomataBase* parentCellularAutomata, unsigned short radius = 1,
      BoundaryCondition* boundary = nullptr)
00019         :Neighborhood(parentCellularAutomata, radius, boundary) { this->name = "Moore"; }
00020     Neighborhood_Moore(const Neighborhood_Moore& orig):Neighborhood(orig) {    }
```

```
00021     virtual ~Neighborhood_Moore() = default;
00022 public:
00023     virtual std::string show() override {
00024         return "todo";
00025     }
00026     virtual std::vector<Cell*> getNeighbors(Cell* cell)override {
00027         std::vector<Cell*> neighbors;
00028         unsigned short numDimensions = parentCellularAutomata->getLattice()->getNumDimensions();
00029         std::vector<int> cellPosition = cell->getPosition();
00030 
00031         std::vector<int> offset(numDimensions, 0);
00032         collectNeighbors(cellPosition, offset, 0, neighbors);
00033 
00034         if (includeCellItself)
00035             neighbors.emplace_back(cell);
00036         return neighbors;
00037     }
00038 private:
00039     void collectNeighbors(const std::vector<int>& cellPosition, std::vector<int>& offset, unsigned
      short dimension, std::vector<Cell*>& neighbors) {
00040         if (dimension == offset.size()) {
00041             bool isZeroOffset = true;
00042             std::vector<std::pair<unsigned short, int» dimensionChanges;
00043             for (unsigned short dim = 0; dim < offset.size(); ++dim) {
00044                 if (offset.at(dim) != 0) {
00045                     isZeroOffset = false;
00046                     dimensionChanges.emplace_back(dim, offset.at(dim));
00047                 }
00048             }
00049             if (!isZeroOffset) {
00050                 // Offsets are visited from dimension 0 to n-1.
00051                 neighbors.emplace_back(getNeighborCell(cellPosition, dimensionChanges));
00052             }
00053             return;
00054         }
00055 
00056         for (int delta = -static_cast<int>(radius); delta <= static_cast<int>(radius); ++delta) {
00057             offset.at(dimension) = delta;
00058             collectNeighbors(cellPosition, offset, dimension + 1, neighbors);
00059         }
00060     }
00061 };
```

18.446 Neighborhood_VonNeumann.h

```
00001 /*
00002  * File:   Neighborhood_VonNeumann.h
00003  * Author: cancian
00004  *
00005  * Created on 6 de abril de 2023, 14:27
00006  */
00007 
00008 #pragma once
00009 
00010 #include <string>
00011 #include <vector>
00012 #include <cstdlib>
00013 
00014 #include "plugins/components/ModalModel/CellularAutomata/Neighborhood.h"
00015 #include "plugins/components/ModalModel/CellularAutomata/Lattice.h"
00016 
00017 class Neighborhood_VonNeumann : public Neighborhood {
00018 public:
00019 
00020     Neighborhood_VonNeumann(CellularAutomataBase* parentCellularAutomata, unsigned short radius = 1,
      BoundaryCondition* boundary = nullptr)
00021     :Neighborhood(parentCellularAutomata, radius, boundary) {
00022          this->name = "Von Neumann";
00023     }
00024 
00025     Neighborhood_VonNeumann(const Neighborhood_VonNeumann& orig)
00026     :Neighborhood(orig) {
00027     }
00028     virtual ~Neighborhood_VonNeumann() = default;
00029 public:
00030 
00031     virtual std::string show() override {
00032         return "-";
00033     }
00034 
00035     virtual std::vector<Cell*> getNeighbors(Cell* cell)override {
00036         std::vector<Cell*> neighbors;
00037         unsigned short numDimensions = parentCellularAutomata->getLattice()->getNumDimensions();
00038         std::vector<int> cellPosition = cell->getPosition();
00039 
```

```
00040         std::vector<int> offset(numDimensions, 0);
00041         collectNeighbors(cellPosition, offset, 0, neighbors);
00042 
00043         if (includeCellItself)
00044             neighbors.emplace_back(cell);
00045         return neighbors;
00046     }
00047 private:
00048     void collectNeighbors(const std::vector<int>& cellPosition, std::vector<int>& offset, unsigned
      short dimension, std::vector<Cell*>& neighbors) {
00049         if (dimension == offset.size()) {
00050             bool isZeroOffset = true;
00051             unsigned int manhattanDistance = 0;
00052             std::vector<std::pair<unsigned short, int» dimensionChanges;
00053             for (unsigned short dim = 0; dim < offset.size(); ++dim) {
00054                 int delta = offset.at(dim);
00055                 manhattanDistance += static_cast<unsigned int>(std::abs(delta));
00056                 if (delta != 0) {
00057                     isZeroOffset = false;
00058                     dimensionChanges.emplace_back(dim, delta);
00059                 }
00060             }
00061             if (!isZeroOffset && manhattanDistance <= radius) {
00062                 // Offsets are visited from dimension 0 to n-1.
00063                 neighbors.emplace_back(getNeighborCell(cellPosition, dimensionChanges));
00064             }
00065             return;
00066         }
00067 
00068         for (int delta = -static_cast<int>(radius); delta <= static_cast<int>(radius); ++delta) {
00069             offset.at(dimension) = delta;
00070             collectNeighbors(cellPosition, offset, dimension + 1, neighbors);
00071         }
00072     }
00073 };
```

18.447 State.h

```
00001 #pragma once
00002 
00003 #include <array>
00004 #include <memory>
00005 #include <sstream>
00006 #include <string>
00007 #include <typeinfo>
00008 #include <utility>
00009 #include <vector>
00010 
00011 /*
00012 template<class T>
00013 class State {
00014 public:
00015     State<T>() {}
00016     State<T>(T value) {_value=value;}
00017     State<T>(const State& orig){}
00018     virtual ~State<T>() = default;
00019 public:
00020     T getValue(){return _value;}
00021     void setValue(T value) {_value=value;}
00022 protected:
00023     T _value;
00024 private:
00025 };
00026 */
00027 
00028 class State {
00029 public:
00030     State() : _value(0) {}
00031     State(long value) : _value(value) {}
00032     State(const State& orig) : _value(orig._value) {}
00033     virtual ~State() = default;
00034 public:
00035     virtual std::unique_ptr<State> clone() const { return std::make_unique<State>(*this); }
00036     virtual bool equals(const State& other) const { return typeName() == other.typeName() &&
      getValue() == other.getValue(); }
00037     virtual std::string toString() const { return std::to_string(_value); }
00038     virtual std::string show() const { return toString(); }
00039     virtual std::string typeName() const { return "State"; }
00040     virtual long getValue() const { return _value; }
00041     virtual void setValue(long value) { _value=value; }
00042 
00043     template <typename T>
00044     const T* as() const noexcept { return dynamic_cast<const T*>(this); }
00045 
00046     template <typename T>
```

```
00047     T* as() noexcept { return dynamic_cast<T*>(this); }
00048 protected:
00049     long _value;
00050 private:
00051 };
00052
00053 inline bool operator==(const State& left, const State& right) {
00054     return left.equals(right);
00055 }
00056
00057 inline bool operator!=(const State& left, const State& right) {
00058     return !(left == right);
00059 }
00060
00061 class RealState : public State {
00062 public:
00063     explicit RealState(double value = 0.0)
00064         : State(static_cast<long>(value)), _realValue(value) {
00065     }
00066
00067     std::unique_ptr<State> clone() const override { return std::make_unique<RealState>(*this); }
00068     bool equals(const State& other) const override {
00069         const auto* real = other.as<RealState>();
00070         return real != nullptr && _realValue == real->_realValue;
00071     }
00072     std::string toString() const override {
00073         std::ostringstream stream;
00074         stream « _realValue;
00075         return stream.str();
00076     }
00077     std::string typeName() const override { return "RealState"; }
00078     long getValue() const override { return static_cast<long>(_realValue); }
00079     void setValue(long value) override { _realValue = static_cast<double>(value); _value = value; }
00080     double realValue() const { return _realValue; }
00081     void setRealValue(double value) { _realValue = value; _value = static_cast<long>(value); }
00082 private:
00083     double _realValue = 0.0;
00084 };
00085
00086 class CompositeState : public State {
00087 public:
00088     using Field = std::pair<std::string, std::unique_ptr<State»;
00089
00090     CompositeState() = default;
00091     CompositeState(const CompositeState& orig) : State(orig) {
00092         for (const auto& field : orig._fields) {
00093             _fields.emplace_back(field.first, field.second != nullptr ? field.second->clone() :
      nullptr);
00094         }
00095     }
00096     CompositeState& operator=(const CompositeState& orig) {
00097         if (this == &orig) {
00098             return *this;
00099         }
00100         _value = orig._value;
00101         _fields.clear();
00102         for (const auto& field : orig._fields) {
00103             _fields.emplace_back(field.first, field.second != nullptr ? field.second->clone() :
      nullptr);
00104         }
00105         return *this;
00106     }
00107
00108     std::unique_ptr<State> clone() const override { return std::make_unique<CompositeState>(*this); }
00109     bool equals(const State& other) const override {
00110         const auto* composite = other.as<CompositeState>();
00111         if (composite == nullptr || _fields.size() != composite->_fields.size()) {
00112             return false;
00113         }
00114         for (size_t index = 0; index < _fields.size(); ++index) {
00115             const auto& left = _fields.at(index);
00116             const auto& right = composite->_fields.at(index);
00117             if (left.first != right.first || left.second == nullptr || right.second == nullptr ||
      *left.second != *right.second) {
00118                 return false;
00119             }
00120         }
00121         return true;
00122     }
00123     std::string toString() const override {
00124         std::string text = "{";
00125         for (size_t index = 0; index < _fields.size(); ++index) {
00126             if (index > 0) {
00127                 text += ",";
00128             }
00129             text += _fields.at(index).first + ":" + (_fields.at(index).second != nullptr ?
      _fields.at(index).second->toString() : "null");
```

```
00130         }
00131         return text + "}";
00132     }
00133     std::string typeName() const override { return "CompositeState"; }
00134     bool addField(std::string name, const State& state) {
00135         _fields.emplace_back(std::move(name), state.clone());
00136         return true;
00137     }
00138     const std::vector<Field>& fields() const { return _fields; }
00139 private:
00140     std::vector<Field> _fields;
00141 };
00142
00143 class HppLatticeGasState : public State {
00144 public:
00145     enum Direction : size_t { North = 0, East = 1, South = 2, West = 3 };
00146
00147     HppLatticeGasState() = default;
00148     explicit HppLatticeGasState(std::array<bool, 4> occupations)
00149         : State(maskFromOccupations(occupations)), _occupations(occupations) {
00150     }
00151
00152     std::unique_ptr<State> clone() const override { return
      std::make_unique<HppLatticeGasState>(*this); }
00153     bool equals(const State& other) const override {
00154         const auto* hpp = other.as<HppLatticeGasState>();
00155         return hpp != nullptr && _occupations == hpp->_occupations;
00156     }
00157     std::string toString() const override {
00158         return std::string("{N:") + (_occupations[North] ? "1" : "0") +
00159                ",E:" + (_occupations[East] ? "1" : "0") +
00160                ",S:" + (_occupations[South] ? "1" : "0") +
00161                ",W:" + (_occupations[West] ? "1" : "0") + "}";
00162     }
00163     std::string typeName() const override { return "HppLatticeGasState"; }
00164     long getValue() const override { return maskFromOccupations(_occupations); }
00165     void setValue(long value) override {
00166         _occupations = {{
00167             (value & (1L << North)) != 0,
00168             (value & (1L << East)) != 0,
00169             (value & (1L << South)) != 0,
00170             (value & (1L << West)) != 0,
00171         }};
00172         _value = getValue();
00173     }
00174     bool has(Direction direction) const { return _occupations[direction]; }
00175     void set(Direction direction, bool occupied) { _occupations[direction] = occupied; _value =
      getValue(); }
00176     const std::array<bool, 4>& occupations() const { return _occupations; }
00177     static long maskFromOccupations(const std::array<bool, 4>& occupations) {
00178         long mask = 0;
00179         for (size_t index = 0; index < occupations.size(); ++index) {
00180             if (occupations[index]) {
00181                 mask |= (1L << index);
00182             }
00183         }
00184         return mask;
00185     }
00186 private:
00187     std::array<bool, 4> _occupations{{false, false, false, false}};
00188 };
```

18.448 State_Numerical.h

```
00001 #pragma once
00002
00003 #include "plugins/components/ModalModel/CellularAutomata/State.h"
00004
00005 /*
00006 class State_Numerical: public State {
00007 public:
00008     State_Numerical(long value)
00009             : State(ColorCA(0,0,0), ' ', value, ""){
00010                 this->longValue = value;
00011             }
00012     State_Numerical(const State_Numerical& orig)
00013     :State(orig) {
00014
00015     }
00016     virtual ~State_Numerical() = default;
00017 public:
00018     virtual ColorCA getColor() const override {
00019         int llongValue = 0;
00020         if (longValue)
00021             llongValue = 1<<9;
```

```
00022         uint8_t south = longValue&((1«3)|(1«5)|(1«7))?0:255;
00023         uint8_t r = longValue&((1«0)|(1«4)|(1«5))?0:south; // west : south
00024         uint8_t g = longValue&((1«1)|(1«6)|(1«7))?0:south; // east : south
00025         uint8_t b = longValue&((1«2)|(1«4)|(1«6))?0:south; // north : south
00026         //std::cout « longValue « ": "« (int)r « ","«(int)g«","«(int)b«std::endl;
00027         return ColorCA(r,g,b);
00028     }
00029
00030 protected:
00031 private:
00032 };
00033 */
```

18.449 StateSet.h

```
00001 #pragma once
00002
00003 #include <limits>
00004 #include <memory>
00005 #include <optional>
00006 #include <sstream>
00007 #include <string>
00008 #include <vector>
00009 #include "plugins/components/ModalModel/CellularAutomata/State.h"
00010
00011 class CellularAutomataBase;
00012
00013 class StateSet {
00014 public:
00015     StateSet(CellularAutomataBase* parentCellularAutomata);
00016     StateSet(const StateSet& orig);
00017     virtual ~StateSet()=default;
00018 public:
00019     virtual bool contains(const State& state) const;
00020     virtual std::unique_ptr<State> createDefaultState() const;
00021     virtual std::unique_ptr<State> parseState(const std::string& text) const;
00022     virtual bool isFinite() const;
00023     virtual bool isDiscrete() const;
00024     virtual bool isContinuous() const;
00025     virtual bool isComposite() const;
00026     virtual std::vector<std::unique_ptr<State» enumerateStates() const;
00027     virtual std::string name() const;
00028 protected:
00029     CellularAutomataBase* parentCellularAutomata;
00030 private:
00031 };
00032
00033 class IntegerStateSet : public StateSet {
00034 public:
00035     IntegerStateSet(CellularAutomataBase* parentCellularAutomata,
00036                     std::optional<long> minimum = std::nullopt,
00037                     std::optional<long> maximum = std::nullopt)
00038         : StateSet(parentCellularAutomata), _minimum(minimum), _maximum(maximum) {
00039     }
00040
00041     bool contains(const State& state) const override {
00042         if (state.as<RealState>() != nullptr || state.as<CompositeState>() != nullptr ||
      state.as<HppLatticeGasState>() != nullptr) {
00043             return false;
00044         }
00045         const long value = state.getValue();
00046         return (!_minimum.has_value() || value >= *_minimum) && (!_maximum.has_value() || value <=
      *_maximum);
00047     }
00048     std::unique_ptr<State> createDefaultState() const override {
00049         return std::make_unique<State>(_minimum.value_or(0));
00050     }
00051     std::unique_ptr<State> parseState(const std::string& text) const override {
00052         try {
00053             auto state = std::make_unique<State>(std::stol(text));
00054             return contains(*state) ? std::move(state) : nullptr;
00055         } catch (...) {
00056             return nullptr;
00057         }
00058     }
00059     bool isFinite() const override { return _minimum.has_value() && _maximum.has_value(); }
00060     bool isDiscrete() const override { return true; }
00061     std::vector<std::unique_ptr<State» enumerateStates() const override {
00062         std::vector<std::unique_ptr<State» states;
00063         if (!isFinite()) {
00064             return states;
00065         }
00066         for (long value = *_minimum; value <= *_maximum; ++value) {
00067             states.emplace_back(std::make_unique<State>(value));
00068             if (value == std::numeric_limits<long>::max()) {
```

```
00069                 break;
00070             }
00071         }
00072         return states;
00073     }
00074     std::string name() const override { return "IntegerStateSet"; }
00075 private:
00076     std::optional<long> _minimum;
00077     std::optional<long> _maximum;
00078 };
00079
00080 class NaturalStateSet : public IntegerStateSet {
00081 public:
00082     explicit NaturalStateSet(CellularAutomataBase* parentCellularAutomata, std::optional<long> maximum
      = std::nullopt)
00083         : IntegerStateSet(parentCellularAutomata, 0, maximum) {
00084     }
00085     std::string name() const override { return "NaturalStateSet"; }
00086 };
00087
00088 class RealIntervalStateSet : public StateSet {
00089 public:
00090     RealIntervalStateSet(CellularAutomataBase* parentCellularAutomata,
00091                          std::optional<double> minimum = std::nullopt,
00092                          std::optional<double> maximum = std::nullopt)
00093         : StateSet(parentCellularAutomata), _minimum(minimum), _maximum(maximum) {
00094     }
00095
00096     bool contains(const State& state) const override {
00097         const auto* real = state.as<RealState>();
00098         const double value = real != nullptr ? real->realValue() :
      static_cast<double>(state.getValue());
00099         return (!_minimum.has_value() || value >= *_minimum) && (!_maximum.has_value() || value <=
      *_maximum);
00100     }
00101     std::unique_ptr<State> createDefaultState() const override {
00102         return std::make_unique<RealState>(_minimum.value_or(0.0));
00103     }
00104     std::unique_ptr<State> parseState(const std::string& text) const override {
00105         try {
00106             auto state = std::make_unique<RealState>(std::stod(text));
00107             return contains(*state) ? std::move(state) : nullptr;
00108         } catch (...) {
00109             return nullptr;
00110         }
00111     }
00112     bool isContinuous() const override { return true; }
00113     std::string name() const override { return "RealIntervalStateSet"; }
00114 private:
00115     std::optional<double> _minimum;
00116     std::optional<double> _maximum;
00117 };
00118
00119 class CompositeStateSet : public StateSet {
00120 public:
00121     explicit CompositeStateSet(CellularAutomataBase* parentCellularAutomata)
00122         : StateSet(parentCellularAutomata) {
00123     }
00124
00125     bool contains(const State& state) const override { return state.as<CompositeState>() != nullptr; }
00126     std::unique_ptr<State> createDefaultState() const override { return
      std::make_unique<CompositeState>(); }
00127     bool isComposite() const override { return true; }
00128     std::string name() const override { return "CompositeStateSet"; }
00129 };
00130
00131 class HppLatticeGasStateSet : public StateSet {
00132 public:
00133     explicit HppLatticeGasStateSet(CellularAutomataBase* parentCellularAutomata)
00134         : StateSet(parentCellularAutomata) {
00135     }
00136
00137     bool contains(const State& state) const override { return state.as<HppLatticeGasState>() !=
      nullptr; }
00138     std::unique_ptr<State> createDefaultState() const override { return
      std::make_unique<HppLatticeGasState>(); }
00139     std::unique_ptr<State> parseState(const std::string& text) const override {
00140         try {
00141             auto state = std::make_unique<HppLatticeGasState>();
00142             state->setValue(std::stol(text));
00143             return state;
00144         } catch (...) {
00145             return nullptr;
00146         }
00147     }
00148     bool isFinite() const override { return true; }
00149     bool isDiscrete() const override { return true; }
```

```
00150     bool isComposite() const override { return true; }
00151     std::vector<std::unique_ptr<State» enumerateStates() const override {
00152         std::vector<std::unique_ptr<State» states;
00153         for (long mask = 0; mask < 16; ++mask) {
00154             auto state = std::make_unique<HppLatticeGasState>();
00155             state->setValue(mask);
00156             states.emplace_back(std::move(state));
00157         }
00158         return states;
00159     }
00160     std::string name() const override { return "HppLatticeGasStateSet"; }
00161 };
```

18.450 StateSet_Enumerable.h

```
00001 #pragma once
00002
00003 #include <memory>
00004 #include <string>
00005 #include <vector>
00006 #include "plugins/components/ModalModel/CellularAutomata/State.h"
00007 #include "plugins/components/ModalModel/CellularAutomata/StateSet.h"
00008
00009 class CellularAutomataBase;
00010
00011 class StateSet_Enumerable: public StateSet{
00012 public:
00013     StateSet_Enumerable(CellularAutomataBase* parentCellularAutomata, std::vector<State*> states =
      {});
00014     StateSet_Enumerable(const StateSet_Enumerable& orig);
00015     virtual ~StateSet_Enumerable()=default;
00016 public:
00017     virtual std::string show();
00018     bool contains(const State& state) const override;
00019     std::unique_ptr<State> createDefaultState() const override;
00020     std::unique_ptr<State> parseState(const std::string& text) const override;
00021     bool isFinite() const override;
00022     bool isDiscrete() const override;
00023     std::vector<std::unique_ptr<State» enumerateStates() const override;
00024     std::string name() const override;
00025     unsigned int size();
00026     void addState(State* state);
00027     void addState(const State& state);
00028     unsigned int getStatesSize();
00029     State* getState(unsigned int rank);
00030     const State* getState(unsigned int rank) const;
00031     State* getState(std::string name);
00032     void setStates(std::vector<State*> states);
00033     std::vector<State*> getStates() const;
00034 protected:
00035     CellularAutomataBase* parentCellularAutomata;
00036     std::vector<std::unique_ptr<State» ownedStates;
00037     std::vector<State*> states;
00038 private:
00039 };
```

18.451 CellularAutomataComp.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Dummy.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 22 de Maio de 2019, 18:41
00012  */
00013
00014 #pragma once
00015
00016 #include "../../../kernel/simulator/model/ModelComponent.h"
00017 #include "plugins/components/ModalModel/CellularAutomata/CellularAutomataBase.h"
00018 #include "plugins/components/ModalModel/CellularAutomata/Lattice.h"
00019 #include "plugins/components/ModalModel/CellularAutomata/Neighborhood.h"
00020 #include "plugins/components/ModalModel/CellularAutomata/StateSet.h"
00021
00022 class BoundaryCondition;
00023
00027 class CellularAutomataComp : public ModelComponent {
```

```
00028 public:
00029     enum class CellularAutomataType : int {
00030         CLASSIC = 1, TIMED_1D = 2, ASYNCHRONOUS = 3, NONUNIFORMRULE = 4, NONUNIFORMNEIGHBOOR = 5,
      USERDEFINED = 6
00031     };
00032
00033     enum class LatticeType : int {
00034         RETICULAR = 1, TRIANGULAR = 2, HEXAGONAL = 3, NETWORK = 4, USERDEFINED = 5
00035     };
00036
00037     enum class NeighboorhoodType : int {
00038         CENTERED = 1, BACKWARD = 2, FORWARD = 3, VONNEUMANN = 4, MOORE = 5, USERDEFINED = 6
00039     };
00040
00041     enum class BoundaryType : int {
00042         FIXED = 1, CLOSED = 2, REFLEXIVE = 3, ADIABATIC = 4, USERDEFINED = 5
00043     };
00044
00045     enum class StateSetType : int {
00046         ENUMERATED = 1, INTEGERBASED = 2, BITBASED = 3, DOUBLEBASED = 4, USERDEFINED = 5
00047     };
00048
00049     enum class LocalRuleType : int {
00050         ELEMENTAR_CA = 1, GAME_OF_LIFE = 2, BIASED_COMPETITION = 3, HPP = 4, SAND_ROCK = 5,
      USERDEFINED = 99
00051     };
00052
00053     enum class AutomataType : int {
00054         Temporary = 0, Permanent = 1
00055     };
00056
00057 public:
00058     CellularAutomataComp(Model* model, std::string name = "");
00059     virtual ~CellularAutomataComp() = default;
00060
00061 public:
00062     CellularAutomataComp::CellularAutomataType getCellularAutomataType() const;
00063     void setCellularAutomataType(CellularAutomataComp::CellularAutomataType newCellularAutomataType);
00064
00065     CellularAutomataComp::LatticeType getLatticeType() const;
00066     void setLatticeType(CellularAutomataComp::LatticeType newLatticeStructure);
00067
00068     CellularAutomataComp::NeighboorhoodType getNeighboorhoodType() const;
00069     void setNeighboorhoodType(CellularAutomataComp::NeighboorhoodType newNeighboorhood);
00070
00071     CellularAutomataComp::BoundaryType geBoundaryType() const;
00072     void setBoundaryType(CellularAutomataComp::BoundaryType newBoundary);
00073
00074     CellularAutomataComp::StateSetType getStateSetType() const;
00075     void setStateSetType(CellularAutomataComp::StateSetType newStateSetType);
00076
00077     //CellularAutomataBase *getcellularAutomata() const;
00078     Lattice *getlattice() const;
00079     Neighborhood *getNeighboorhood() const;
00080     //BoundaryCondition *getBoundary() const;
00081     StateSet *getStateSet() const;
00082
00083 public:
00084     virtual std::string show() override;
00085
00086 public:
00087     static PluginInformation* GetPluginInformation();
00088     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00089     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00090
00091     CellularAutomataComp::LocalRuleType getlocalRuleType() const;
00092     void setLocalRuleType(CellularAutomataComp::LocalRuleType newLocalRuleType);
00093
00094     LocalRule *getlocalRule() const;
00095
00096 protected:
00097     virtual bool _loadInstance(PersistenceRecord *fields) override;
00098     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00099     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00100
00101
00102 protected:
00104     virtual bool _check(std::string* errorMessage);
00106     // virtual ParserChangesInformation* _getParserChangesInformation();
00108     virtual void _initBetweenReplications() override;
00110     // virtual void _createInternalAndAttachedData(); /*< A ModelDataDefinition or ModelComponent that
      includes (internal) ou refers to (attach) other ModelDataDefinition must register them inside this
      method. */
00112     // virtual void _addSimulationControl(SimulationControl* property);
00113
00114
00115 protected:
```

```
00116     // virtual void _createInternalStatisticReporters() override;
00117     // virtual void _createNonEditableDataDefinitions() override;
00118     // virtual void _createEditableDataDefinitions() override;
00119     // virtual void _createAttachedAttributes() override;
00120
00121 private:
00122     // ...
00123
00124 private:
00125
00126     // Default values for the attributes. Used on initing, loading and saving
00127     const struct DEFAULT_VALUES {
00128         const CellularAutomataType cellularAutomataType = CellularAutomataType::CLASSIC;
00129         const LatticeType latticeType = LatticeType::RETICULAR;
00130         const NeighboorhoodType neighboorhoodType = NeighboorhoodType::VONNEUMANN;
00131         const BoundaryType boundaryType = BoundaryType::FIXED;
00132         const StateSetType stateSetType = StateSetType::ENUMERATED;
00133         const LocalRuleType localRuleType = LocalRuleType::GAME_OF_LIFE;
00134     } DEFAULT;
00135     CellularAutomataComp::CellularAutomataType _cellularAutomataType = DEFAULT.cellularAutomataType;
00136     CellularAutomataComp::LatticeType _latticeType = DEFAULT.latticeType;
00137     CellularAutomataComp::NeighboorhoodType _neighboorhoodType = DEFAULT.neighboorhoodType;
00138     CellularAutomataComp::BoundaryType _boundaryType = DEFAULT.boundaryType;
00139     CellularAutomataComp::StateSetType _stateSetType = DEFAULT.stateSetType;
00140     CellularAutomataComp::LocalRuleType _localRuleType = DEFAULT.localRuleType;
00141
00142 private:
00143     CellularAutomataBase* _cellularAutomata = nullptr;
00144     Lattice* _lattice = nullptr;
00145     Neighborhood* _neighboorhood = nullptr;
00146     BoundaryCondition* _boundary = nullptr;
00147     StateSet* _stateSet = nullptr;
00148     LocalRule* _localRule = nullptr;
00149
00150 private:
00151
00152 private:
00153     // ...
00154 };
```

18.452 DefaultNode.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:DefaultNode.h
00009  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00010  *
00011  * Created on 01 de Julho de 2025, 14:26
00012  */
00013
00014 #pragma once
00015
00016 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00017 #include <functional>
00018 #include <string>
00019 //#include "plugins/data/Template/DummyElement.h"
00020
00021 class DefaultNode;
00022 class Model;
00023 class Entity;
00024 class PluginInformation;
00025
00026 class DefaultNodeTransition {
00027 public:
00028     enum class TransitionKind : int { DETERMINISTIC = 0, PROBABILISTIC = 1 };
00029
00030 public:
00031     DefaultNodeTransition(DefaultNode* source, DefaultNode* destination, std::string name = "");
00032     virtual ~DefaultNodeTransition() = default;
00033
00034 public:
00035     void setSource(DefaultNode* source);
00036     DefaultNode* getSource() const;
00037     void setDestination(DefaultNode* destination);
00038     DefaultNode* getDestination() const;
00039     void setName(std::string name);
00040     std::string getName() const;
00041     void setGuardExpression(std::string guardExpression);
00042     std::string getGuardExpression() const;
00043     void setOutputExpression(std::string outputExpression);
```

```
00044     std::string getOutputExpression() const;
00045     void setInputEvent(std::string inputEvent);
00046     std::string getInputEvent() const;
00047     void setPriority(unsigned int priority);
00048     unsigned int getPriority() const;
00049     void setProbability(double probability);
00050     double getProbability() const;
00051     void setTransitionKind(TransitionKind transitionKind);
00052     TransitionKind getTransitionKind() const;
00053 
00054 public:
00055     virtual bool canFire(Model* model, Entity* entity) const;
00056     virtual void execute(Model* model, Entity* entity) const;
00057 
00058 
00059 protected:
00060 
00061 private:
00062     DefaultNode* _source = nullptr;
00063     DefaultNode* _destination = nullptr;
00064     std::string _name = "";
00065     std::string _guardExpression = "";
00066     std::string _outputExpression = "";
00067     std::string _inputEvent = "";
00068     unsigned int _priority = 0;
00069     double _probability = 1.0;
00070     TransitionKind _transitionKind = TransitionKind::DETERMINISTIC;
00071 };
00072 
00076 class DefaultNode : public ModelDataDefinition {
00077 public:
00078     DefaultNode(Model* model, std::string componentTypename, std::string name);
00079     DefaultNode(Model* model, std::string name = "");
00080     virtual ~DefaultNode() override;
00081 
00082 public:
00083     void addTransition(DefaultNodeTransition* transition);
00084     void removeTransition(DefaultNodeTransition* transition);
00085     List<DefaultNodeTransition*>* getTransitions() const;
00086     void setInitialNode(bool initialNode);
00087     bool isInitialNode() const;
00088     void setFinalNode(bool finalNode);
00089     bool isFinalNode() const;
00090 
00091 public:
00092     virtual std::string show() override;
00093 
00094 public:
00095     static PluginInformation* GetPluginInformation();
00096     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00097     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00098 
00099 protected:
00100     virtual bool _loadInstance(PersistenceRecord *fields) override;
00101     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00102 protected:
00104     // virtual bool _check(std::string& errorMessage);
00106     // virtual ParserChangesInformation* _getParserChangesInformation();
00108     // virtual void _initBetweenReplications();
00109     // virtual void _createInternalStatisticReporters() override;
00110     // virtual void _createNonEditableDataDefinitions() override;
00111     // virtual void _createEditableDataDefinitions() override;
00112     // virtual void _createAttachedAttributes() override;
00114     // virtual void _createInternalAndAttachedData(); /*< A ModelDataDefinition or ModelComponent that
      includes (internal) ou refers to (attach) other ModelDataDefinition must register them inside this
      method. */
00116     // virtual void _addSimulationControl(SimulationControl* property);
00117 
00118 private:
00119     // ...
00120 
00121 private:
00122 
00124     const struct DEFAULT_VALUES {
00125         const bool initialNode = false;
00126         const bool finalNode = false;
00127     } DEFAULT;
00128     bool _initialNode = DEFAULT.initialNode;
00129     bool _finalNode = DEFAULT.finalNode;
00130     List<DefaultNodeTransition*>* _transitions = new List<DefaultNodeTransition*>();
00131 
00132 private:
00133     //DummyElement* _internalDataDefinition = nullptr;
00134 
00135 private:
00136     // ...
00137 };
```

18.453 DefaultTransitionExtensions.h

```
00001 #pragma once
00002
00003 #include "DefaultNode.h"
00004 #include <string>
00005
00006 class EFSMTransition : public DefaultNodeTransition {
00007 public:
00008     EFSMTransition(DefaultNode* source, DefaultNode* destination, std::string name = "");
00009     virtual ~EFSMTransition() = default;
00010
00011 public:
00012     void setTriggerEvent(std::string triggerEvent);
00013     std::string getTriggerEvent() const;
00014     void setProbabilityExpression(std::string probabilityExpression);
00015     std::string getProbabilityExpression() const;
00016
00017 public:
00018     virtual bool canFire(Model* model, Entity* entity) const override;
00019     virtual void execute(Model* model, Entity* entity) const override;
00020
00021 private:
00022     std::string _triggerEvent = "";
00023     std::string _probabilityExpression = "";
00024 };
00025
```

18.454 FSMState.h

```
00001 #pragma once
00002
00003 #include "DefaultNode.h"
00004
00005 class FSMState : public DefaultNode {
00006 public:
00007     FSMState(Model* model, std::string name = "");
00008     virtual ~FSMState() = default;
00009
00010 public:
00011     void setEntryActionExpression(std::string expression);
00012     std::string getEntryActionExpression() const;
00013     void setExitActionExpression(std::string expression);
00014     std::string getExitActionExpression() const;
00015
00016 public: // static
00017     static PluginInformation* GetPluginInformation();
00018     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00019     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00020
00021 protected:
00022     virtual bool _loadInstance(PersistenceRecord *fields) override;
00023     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00024
00025 private:
00026     const struct DEFAULT_VALUES {
00027         const std::string entryActionExpression = "";
00028         const std::string exitActionExpression = "";
00029     } DEFAULT;
00030     std::string _entryActionExpression = DEFAULT.entryActionExpression;
00031     std::string _exitActionExpression = DEFAULT.exitActionExpression;
00032 };
```

18.455 ModalModelDefault.h

```
00001 /*
00002  * File:    ModalModelDefault.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 01 de Julho de 2025, 14:26
00006  */
00007
00008 #pragma once
00009
00010 #include "../../../kernel/simulator/model/ModelComponent.h"
00011 #include <string>
00012 #include <vector>
00013
00014 class DefaultNetwork;
00015 class NetworkActivationResult;
00016
00024 class ModalModelDefault : public ModelComponent {
```

```
00025 public:
00026     ModalModelDefault(Model* model, std::string name = "");
00027     virtual ~ModalModelDefault() override = default;
00028 
00029 public:
00030     virtual void setMaxTransitionsPerDispatch(unsigned int maxTransitionsPerDispatch);
00031     virtual unsigned int getMaxTransitionsPerDispatch() const;
00032     std::string getTimeDelayExpressionPerDispatch();
00033     void setTimeDelayExpressionPerDispatch(const std::string time_delay_expression_per_dispatch);
00034     Util::TimeUnit getTimeDelayPerDispatchTimeUnit();
00035     void setTimeDelayPerDispatchTimeUnit(const Util::TimeUnit time_delay_per_dispatch_time_unit);
00036     DefaultNetwork* getNetwork() const;
00037     void setNetwork(DefaultNetwork* network);
00038     std::string getNetworkName() const;
00039     void setNetworkName(const std::string& networkName);
00040     void setInputBinding(unsigned int port, const std::string& expression);
00041     std::string getInputBinding(unsigned int port) const;
00042     void setOutputBinding(unsigned int port, const std::string& attributeName);
00043     std::string getOutputBinding(unsigned int port) const;
00044 
00045 public:
00046     virtual void addOutputExpressionReference(ModelDataDefinition* expressionReference);
00047 
00048 public:
00049     virtual std::string show() override;
00050 
00051 public:
00052     static PluginInformation* GetPluginInformation();
00053     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00054     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00055 
00056 protected:
00057     virtual bool _loadInstance(PersistenceRecord *fields) override;
00058     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00059     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00060     virtual bool _check(std::string& errorMessage) override;
00061     virtual void _initBetweenReplications() override;
00062     virtual void _createAttachedAttributes() override;
00063 
00064 protected:
00065     const struct DEFAULT_VALUES {
00066         const std::string timeDelayExpressionPerDispatch = "0";
00067         const Util::TimeUnit timeDelayPerDispatchTimeUnit = Util::TimeUnit::second;
00068         const unsigned int maxTransitionsPerDispatch = 1;
00069     } DEFAULT;
00070     unsigned int _maxTransitionsPerDispatch = DEFAULT.maxTransitionsPerDispatch;
00071     std::string _timeDelayExpressionPerDispatch = DEFAULT.timeDelayExpressionPerDispatch;
00072     Util::TimeUnit _timeDelayPerDispatchTimeUnit = DEFAULT.timeDelayPerDispatchTimeUnit;
00073 
00074 private:
00075     DefaultNetwork* _resolveNetworkReference();
00076     void _syncBindingsToNetwork();
00077     bool _dispatchNetworkActivation(Entity* entity, unsigned int inputPortNumber);
00078     Entity* _cloneEntity(Entity* entity);
00079     void _writeOutputBinding(Entity* entity, unsigned int outputPort, double value);
00080     void _routeNetworkOutputs(Entity* entity, const NetworkActivationResult& result);
00081 
00082 private:
00083     DefaultNetwork* _network = nullptr;
00084     std::string _networkName = "";
00085     std::vector<std::string> _inputBindings;
00086     std::vector<std::string> _outputBindings;
00087 };
```

18.456 PetriPlace.h

```
00001 #pragma once
00002 
00003 #include "DefaultNode.h"
00004 #include <map>
00005 
00006 class PetriPlace : public DefaultNode {
00007 public:
00008     PetriPlace(Model* model, std::string name = "");
00009     virtual ~PetriPlace() override;
00010 
00011 public:
00012     unsigned int getTokens(std::string color = "default") const;
00013     void setTokens(unsigned int quantity, std::string color = "default");
00014     void addTokens(unsigned int quantity, std::string color = "default");
00015     bool removeTokens(unsigned int quantity, std::string color = "default");
00016     void clearTokens();
00017     std::map<std::string, unsigned int>* getAllTokens();
00018 
00019 public: // static
```

```
00020     static PluginInformation* GetPluginInformation();
00021     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00022     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00023 
00024 protected:
00025     virtual bool _loadInstance(PersistenceRecord *fields) override;
00026     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00027 
00028 private:
00029     std::map<std::string, unsigned int>* _tokensByColor = new std::map<std::string, unsigned int>();
00030 };
```

18.457 Submodel.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
00007 /*
00008  * File:   Submodel.h
00009  * Author: rlcancian
00010  *
00011  * Created on 11 de Setembro de 2019, 13:04
00012  */
00013 
00014 #ifndef SUBMODEL_H
00015 #define SUBMODEL_H
00016 
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 
00022 class Submodel : public ModelComponent {
00023 public: // constructors
00024     Submodel(Model* model, std::string name = "");
00025     virtual ~Submodel() = default;
00026 public: // virtual
00027     virtual std::string show() override;
00028 public: // static
00029     static PluginInformation* GetPluginInformation();
00030     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00031     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00032 protected: // virtual
00033     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00034     virtual bool _loadInstance(PersistenceRecord *fields) override;
00035     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00036 protected: // virtual
00037     //virtual void _initBetweenReplications();
00038     // virtual void _createInternalStatisticReporters() override;
00039     // virtual void _createNonEditableDataDefinitions() override;
00040     // virtual void _createEditableDataDefinitions() override;
00041     // virtual void _createAttachedAttributes() override;
00042     virtual bool _check(std::string& errorMessage) override;
00043 
00044 protected:
00045 
00046 private: // methods
00047 private: // attributes 1:1
00048 private: // attributes 1:n
00049 };
00050 
00051 
00052 #endif /* SUBMODEL_H */
```

18.458 Match.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
00007 /*
00008  * File:   Match.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:14
00012  */
00013 
00014 #ifndef MATCH_H
00015 #define MATCH_H
00016 
```

```
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "plugins/data/DiscreteProcessing/Queue.h"
00019
00031 class Match : public ModelComponent {
00032 public:
00033
00034     enum class Rule : int {
00035         Any = 0, ByAttribute = 1, num_elements = 2
00036     };
00037 public:
00038     static std::string convertEnumToStr(Rule rule);
00039 public: // constructors
00040     Match(Model* model, std::string name = "");
00041     virtual ~Match() override;
00042 public: // virtual
00043     virtual std::string show() override;
00044 public: // static
00045     static PluginInformation* GetPluginInformation();
00046     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00047     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00048 public: // virtual
00049     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00050     virtual bool _loadInstance(PersistenceRecord *fields) override;
00051     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00052 public:
00053     void setRule(Match::Rule _rule);
00054     Match::Rule getRule() const;
00055     void setAttributeName(std::string _attributeName);
00056     std::string getAttributeName() const;
00057     void setMatchSize(std::string _matchSize);
00058     std::string getMatchSize() const;
00059     void setNumberOfQueues(unsigned int _numberOfQueues);
00060     unsigned int getNumberOfQueues() const;
00061 protected: // virtual
00062     //virtual void _initBetweenReplications();
00063     // virtual void _createInternalStatisticReporters() override;
00064     // virtual void _createNonEditableDataDefinitions() override;
00065     virtual void _createEditableDataDefinitions() override;
00066     // virtual void _createAttachedAttributes() override;
00067     virtual bool _check(std::string& errorMessage) override;
00068     // virtual void _createInternalAndAttachedData() override;
00069
00070 protected:
00071
00072 private: // methods
00073 private: // attributes 1:1
00074
00075     const struct DEFAULT_VALUES {
00076         const Match::Rule rule = Match::Rule::Any;
00077         const unsigned int numberOfQueues = 2;
00078         const std::string matchSize = "1";
00079         const std::string attributeName = "";
00080     } DEFAULT;
00081     Match::Rule _rule = DEFAULT.rule;
00082     unsigned int _numberOfQueues = DEFAULT.numberOfQueues;
00083     std::string _matchSize = DEFAULT.matchSize;
00084     std::string _attributeName = DEFAULT.attributeName;
00085 private: // attributes 1:1
00086     std::map<Queue*, std::map<double, unsigned int>*>* _entitiesByAttrib = new std::map<Queue*,
      std::map<double, unsigned int>*>();
00087 private: // attributes 1:n
00088     List<Queue*>* _queues = new List<Queue*>();
00089 };
00090
00091
00092 #endif /* MATCH_H */
```

18.459 Signal.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Signal.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:20
00012  */
00013
00014 #ifndef SIGNAL_H
00015 #define SIGNAL_H
00016
```

```
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "../../data/Synchronization/SignalData.h"
00019
00029 class Signal : public ModelComponent {
00030 public: // constructors
00031     Signal(Model* model, std::string name = "");
00032     virtual ~Signal() = default;
00033 public: // virtual
00034     virtual std::string show() override;
00035 public:
00036     void setSignalData(SignalData* signal);
00037     const std::string&limitExpression() const;
00038     void setLimitExpression(const std::string&newLimitExpression);
00039 public: // static
00040     static PluginInformation* GetPluginInformation();
00041     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00042     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00043
00044 protected: // must be overriden
00045     virtual bool _loadInstance(PersistenceRecord *fields) override;
00046     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00047     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00048 protected: // could be overriden .
00049     virtual bool _check(std::string& errorMessage) override;
00050     virtual void _initBetweenReplications() override;
00051     // virtual void _createInternalAndAttachedData() override;
00052     //virtual ParserChangesInformation* _getParserChangesInformation();
00053
00054 protected:
00055     // virtual void _createInternalStatisticReporters() override;
00056     // virtual void _createNonEditableDataDefinitions() override;
00057     virtual void _createEditableDataDefinitions() override;
00058     // virtual void _createAttachedAttributes() override;
00059
00060 private: // methods
00061 private: // attributes 1:1
00062     const struct DEFAULT_VALUES {
00063         const std::string limitExpression = "1";
00064     } DEFAULT;
00065     std::string _limitExpression = DEFAULT.limitExpression;
00066     unsigned int _signalsTriggered = 0;
00067 private: // attributes 1:n
00068 private: // attached
00069     SignalData* _signalData = nullptr;
00070
00071 };
00072
00073
00074 #endif /* SIGNAL_H */
```

18.460 Wait.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Wait.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:20
00012  */
00013
00014 #ifndef WAIT_H
00015 #define WAIT_H
00016
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "plugins/data/DiscreteProcessing/Queue.h"
00019 #include "../../data/Synchronization/SignalData.h"
00020 #include "kernel/simulator/OnEventManager.h"
00021
00038 class Wait : public ModelComponent {
00039 public:
00040
00041     enum class WaitType : int {
00042         WaitForSignal = 0, InfiniteHold = 1, ScanForCondition = 2, num_elements = 3
00043     };
00044 public:
00045     static std::string convertEnumToStr(WaitType type);
00046 public: // constructors
00047     Wait(Model* model, std::string name = "");
00048     virtual ~Wait() = default;
00049 public: // virtual
```

```
00050     virtual std::string show() override;
00051 public: //
00052     void setSignalData(SignalData* signal);
00053     void setWaitType(WaitType _watType);
00054     Wait::WaitType getWaitType() const;
00055     void setCondition(std::string _condition);
00056     std::string getCondition() const;
00057     Queue* getQueue() const;
00058     void setQueue(Queue* queue);
00059 public: // static
00060     static PluginInformation* GetPluginInformation();
00061     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00062     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00063     std::string getlimitExpression() const;
00064     void setLimitExpression(const std::string &newLimitExpression);
00065
00066 protected: // must be overriden
00067     virtual bool _loadInstance(PersistenceRecord *fields) override;
00068     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00069     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00070 protected: // could be overriden .
00071     virtual bool _check(std::string& errorMessage) override;
00072     virtual void _initBetweenReplications() override;
00073     // virtual void _createInternalAndAttachedData() override;
00074     //virtual ParserChangesInformation* _getParserChangesInformation();
00075
00076 protected:
00077     //virtual void _createInternalStatisticReporters() override;
00078     virtual void _createNonEditableDataDefinitions() override;
00079     virtual void _createEditableDataDefinitions() override;
00080     //virtual void _createAttachedAttributes() override;
00081
00082 private: // methods
00083     unsigned int _handlerForSignalDataEvent(SignalData* signalData);
00084     void _handlerForAfterProcessEventEvent(SimulationEvent* event);
00085 private: // attributes 1:1
00086
00087     const struct DEFAULT_VALUES {
00088         const WaitType waitType = Wait::WaitType::WaitForSignal;
00089         const std::string condition = "";
00090         const std::string limitExpression = "0";
00091     } DEFAULT;
00092     WaitType _waitType = DEFAULT.waitType;
00093     std::string _condition = DEFAULT.condition;
00094     std::string limitExpression = DEFAULT.limitExpression;
00095 private: // internal
00096     Queue *_queue = nullptr; // @TODO: It should be a QueueableItem, (Queue or Set)
00097 private: // attached
00098     SignalData* _signalData = nullptr;
00099     bool _isScanConditionHandlerRegistered = false; // local guard to avoid duplicate registration in
      repeated checks
00100 private: // attributes 1:n
00101 };
00102
00103
00104 #endif /* WAIT_H */
```

18.461 DummyComponent.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Dummy.h
00009  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00010  *
00011  * Created on 22 de Maio de 2019, 18:41
00012  */
00013
00014 #pragma once
00015
00016 #include "../../../kernel/simulator/model/ModelComponent.h"
00017 #include "plugins/data/Template/DummyElement.h"
00018
00022 class DummyComponent : public ModelComponent {
00023 public:
00024     DummyComponent(Model* model, std::string name = "");
00025     virtual ~DummyComponent() = default;
00026
00027 public:
00028     // ...
00029
```

```
00030 public:
00031     virtual std::string show() override;
00032 
00033 public:
00034     static PluginInformation* GetPluginInformation();
00035     static ModelComponent* LoadInstance(Model* model, PersistenceRecord *fields);
00036     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00037 
00038 protected:
00039     virtual bool _loadInstance(PersistenceRecord *fields) override;
00040     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00041     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00042 
00043 
00044 protected:
00046     // virtual bool _check(std::string& errorMessage);
00048     // virtual ParserChangesInformation* _getParserChangesInformation();
00050     // virtual void _initBetweenReplications();
00051     // virtual void _createInternalStatisticReporters() override;
00052     // virtual void _createNonEditableDataDefinitions() override;
00053     // virtual void _createEditableDataDefinitions() override;
00054     // virtual void _createAttachedAttributes() override;
00056     // virtual void _createInternalAndAttachedData(); /*< A ModelDataDefinition or ModelComponent that
      includes (internal) ou refers to (attach) other ModelDataDefinition must register them inside this
      method. */
00058     // virtual void _addSimulationControl(SimulationControl* property);
00059 
00060 
00061 protected:
00062 
00063 private:
00064     // ...
00065 
00066 private:
00067 
00069     const struct DEFAULT_VALUES {
00070         const std::string someString = "Test";
00071         const unsigned int someUint = 1;
00072     } DEFAULT;
00073     std::string _someString = DEFAULT.someString;
00074     unsigned int _someUint = DEFAULT.someUint;
00075 
00076 private:
00077     DummyElement* _internalDataDefinition = nullptr;
00078 
00079 private:
00080     // ...
00081 };
```

18.462 BioStateProjectionComponent.h

```
00001 #ifndef BIONETWORKPROJECTIONCOMPONENT_H
00002 #define BIONETWORKPROJECTIONCOMPONENT_H
00003 
00004 #include <string>
00005 #include <vector>
00006 
00007 #include "../../../kernel/simulator/model/ModelComponent.h"
00008 #include "kernel/simulator/Plugin.h"
00009 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00010 
00011 class BioSpecies;
00012 
00022 class BioStateProjectionComponent : public ModelComponent {
00023 public:
00024     enum class ProjectionKind {
00025         MoleculeCount,
00026         MetaboliteAmount
00027     };
00028 
00029     struct Projection {
00030         std::string sourceSpeciesName;
00031         std::string targetKey;
00032         ProjectionKind kind = ProjectionKind::MoleculeCount;
00033         double scale = 1.0;
00034         double offset = 0.0;
00035     };
00036 
00037 public:
00038     BioStateProjectionComponent(Model* model, std::string name = "");
00039     virtual ~BioStateProjectionComponent() override = default;
00040 
00041 public:
00042     static PluginInformation* GetPluginInformation();
00043     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
```

```
00044     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00045
00046 public:
00047     void setWholeCellState(WholeCellState* state);
00048     WholeCellState* getWholeCellState() const;
00049     void clearProjections();
00050     void addMoleculeProjection(const std::string& sourceSpeciesName, const std::string& targetKey =
      "",
00051                                double scale = 1.0, double offset = 0.0);
00052     void addMetaboliteProjection(const std::string& sourceSpeciesName, const std::string& targetKey =
      "",
00053                                  double scale = 1.0, double offset = 0.0);
00054     unsigned int getProjectionCount() const;
00055     void setLastSucceeded(bool lastSucceeded);
00056     bool getLastSucceeded() const;
00057     void setLastMessage(std::string lastMessage);
00058     std::string getLastMessage() const;
00059
00060     virtual std::string show() override;
00061
00062 protected:
00063     virtual bool _loadInstance(PersistenceRecord* fields) override;
00064     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00065     virtual bool _check(std::string& errorMessage) override;
00066     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00067     virtual void _createEditableDataDefinitions() override;
00068
00069 private:
00070     void _forwardEntity(Entity* entity);
00071     void _syncEditableDataDefinitions();
00072
00073 private:
00074     const struct DEFAULT_VALUES {
00075         std::string wholeCellStateName = "";
00076         bool lastSucceeded = false;
00077         std::string lastMessage = "";
00078     } DEFAULT;
00079
00080     WholeCellState* _wholeCellState = nullptr;
00081     std::vector<Projection> _projections;
00082     bool _lastSucceeded = DEFAULT.lastSucceeded;
00083     std::string _lastMessage = DEFAULT.lastMessage;
00084 };
00085
00086 #endif /* BIONETWORKPROJECTIONCOMPONENT_H */
```

18.463 CellCycleCheckpointComponent.h

```
00001 #ifndef CELLCYCLECHECKPOINTCOMPONENT_H
00002 #define CELLCYCLECHECKPOINTCOMPONENT_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelComponent.h"
00007 #include "kernel/simulator/Plugin.h"
00008 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00009
00027 class CellCycleCheckpointComponent : public ModelComponent {
00028 public:
00029     CellCycleCheckpointComponent(Model* model, std::string name = "");
00030     virtual ~CellCycleCheckpointComponent() override = default;
00031
00032 public:
00033     static PluginInformation* GetPluginInformation();
00034     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00035     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00036
00037 public:
00038     void setWholeCellState(WholeCellState* state);
00039     WholeCellState* getWholeCellState() const;
00040     void setDeltaT(double deltaT);
00041     double getDeltaT() const;
00042     void setAdvanceWholeCellClock(bool advance);
00043     bool getAdvanceWholeCellClock() const;
00044     void setEnergyMetaboliteKey(std::string key);
00045     std::string getEnergyMetaboliteKey() const;
00046     void setCompartmentEnergyRegion(std::string region);
00047     std::string getCompartmentEnergyRegion() const;
00048     void setCompartmentEnergyMetaboliteKey(std::string key);
00049     std::string getCompartmentEnergyMetaboliteKey() const;
00050     void setStarvationAtpThreshold(double threshold);
00051     double getStarvationAtpThreshold() const;
00052     void setCompartmentStarvationThreshold(double threshold);
00053     double getCompartmentStarvationThreshold() const;
00054     void setLethalStarvationSteps(unsigned int steps);
```

```
00055     unsigned int getLethalStarvationSteps() const;
00056     void setDivisionMassThreshold(double threshold);
00057     double getDivisionMassThreshold() const;
00058     void setFtsZRingKey(std::string key);
00059     std::string getFtsZRingKey() const;
00060     void setFtsZThreshold(double threshold);
00061     double getFtsZThreshold() const;
00062     void setCriticalPathwayActivityKey(std::string key);
00063     std::string getCriticalPathwayActivityKey() const;
00064     void setCriticalPathwayActivityThreshold(double threshold);
00065     double getCriticalPathwayActivityThreshold() const;
00066     unsigned int getStarvationStreak() const;
00067
00068     virtual std::string show() override;
00069
00070 protected:
00071     virtual bool _loadInstance(PersistenceRecord* fields) override;
00072     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00073     virtual bool _check(std::string& errorMessage) override;
00074     virtual void _initBetweenReplications() override;
00075     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00076
00077 protected:
00078     virtual void _createEditableDataDefinitions() override;
00079
00080 private:
00081     bool _isDivisionReady() const;
00082     void _updateLifecyclePhase();
00083     void _forwardEntity(Entity* entity);
00084
00085 private:
00086     const struct DEFAULT_VALUES {
00087         std::string wholeCellStateName = "";
00088         double deltaT = 60.0;
00089         bool advanceWholeCellClock = true;
00090         std::string energyMetaboliteKey = "ATP";
00091         std::string compartmentEnergyRegion = "";
00092         std::string compartmentEnergyMetaboliteKey = "";
00093         double starvationAtpThreshold = 0.25;
00094         double compartmentStarvationThreshold = 0.0;
00095         unsigned int lethalStarvationSteps = 0u;
00096         double divisionMassThreshold = 0.0;
00097         std::string ftsZRingKey = "FtsZ_ring_completion";
00098         double ftsZThreshold = 0.0;
00099         std::string criticalPathwayActivityKey = "";
00100         double criticalPathwayActivityThreshold = 0.0;
00101     } DEFAULT;
00102
00103     WholeCellState* _wholeCellState = nullptr;
00104     double _deltaT = DEFAULT.deltaT;
00105     bool _advanceWholeCellClock = DEFAULT.advanceWholeCellClock;
00106     std::string _energyMetaboliteKey = DEFAULT.energyMetaboliteKey;
00107     std::string _compartmentEnergyRegion = DEFAULT.compartmentEnergyRegion;
00108     std::string _compartmentEnergyMetaboliteKey = DEFAULT.compartmentEnergyMetaboliteKey;
00109     double _starvationAtpThreshold = DEFAULT.starvationAtpThreshold;
00110     double _compartmentStarvationThreshold = DEFAULT.compartmentStarvationThreshold;
00111     unsigned int _lethalStarvationSteps = DEFAULT.lethalStarvationSteps;
00112     double _divisionMassThreshold = DEFAULT.divisionMassThreshold;
00113     std::string _ftsZRingKey = DEFAULT.ftsZRingKey;
00114     double _ftsZThreshold = DEFAULT.ftsZThreshold;
00115     std::string _criticalPathwayActivityKey = DEFAULT.criticalPathwayActivityKey;
00116     double _criticalPathwayActivityThreshold = DEFAULT.criticalPathwayActivityThreshold;
00117     unsigned int _starvationStreak = 0u;
00118 };
00119
00120 #endif /* CELLCYCLECHECKPOINTCOMPONENT_H */
```

18.464 CellDivisionEvent.h

```
00001 #ifndef CELLDIVISIONEVENT_H
00002 #define CELLDIVISIONEVENT_H
00003
00004 #include <random>
00005 #include <string>
00006
00007 #include "../../../kernel/simulator/model/ModelComponent.h"
00008 #include "kernel/simulator/Plugin.h"
00009 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00010
00039 class CellDivisionEvent : public ModelComponent {
00040 public:
00041     CellDivisionEvent(Model* model, std::string name = "");
00042     virtual ~CellDivisionEvent() override = default;
00043
00044 public:
```

```
00045     static PluginInformation* GetPluginInformation();
00046     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00047     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00048 
00049 public:
00050     void setWholeCellState(WholeCellState* state);
00051     WholeCellState* getWholeCellState() const;
00052     void setDivisionMassThreshold(double threshold);
00053     double getDivisionMassThreshold() const;
00054     void setFtsZThreshold(double threshold);
00055     double getFtsZThreshold() const;
00056     void setFtsZRingKey(std::string key);
00057     std::string getFtsZRingKey() const;
00058     void setRandomSeed(unsigned int seed);
00059     unsigned int getRandomSeed() const;
00060     bool getLastDivided() const;
00061     int getDivisionCount() const;
00062 
00063     virtual std::string show() override;
00064 
00065 protected:
00066     virtual bool _loadInstance(PersistenceRecord* fields) override;
00067     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00068     virtual bool _check(std::string& errorMessage) override;
00069     virtual void _initBetweenReplications() override;
00070     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00071 
00072 protected:
00073     virtual void _createEditableDataDefinitions() override;
00074 
00075 private:
00076     bool _checkDivisionCondition() const;
00077     void _applyDivision();
00078     void _forwardEntity(Entity* entity, unsigned int port);
00079 
00080 private:
00081     const struct DEFAULT_VALUES {
00082         std::string wholeCellStateName  = "";
00083         double divisionMassThreshold    = 7.86e-15; // 2 × 3.93e-15 g (M. genitalium,
      parameters.json)
00084         double ftsZThreshold            = 0.0;      // 0 = mass-only trigger; >0 requires FtsZ ring
00085         std::string ftsZRingKey         = "FtsZ_ring_completion";
00086         unsigned int randomSeed         = 44u;
00087     } DEFAULT;
00088 
00089     WholeCellState* _wholeCellState    = nullptr;
00090     double _divisionMassThreshold      = DEFAULT.divisionMassThreshold;
00091     double _ftsZThreshold              = DEFAULT.ftsZThreshold;
00092     std::string _ftsZRingKey           = DEFAULT.ftsZRingKey;
00093     unsigned int _randomSeed           = DEFAULT.randomSeed;
00094     bool _lastDivided                  = false;
00095     int _divisionCount                 = 0;
00096     std::mt19937 _rng{DEFAULT.randomSeed};
00097 };
00098 
00099 #endif /* CELLDIVISIONEVENT_H */
```

18.465 CellFateDecisionComponent.h

```
00001 #ifndef CELLFATEDECISIONCOMPONENT_H
00002 #define CELLFATEDECISIONCOMPONENT_H
00003 
00004 #include <string>
00005 
00006 #include "../../../kernel/simulator/model/ModelComponent.h"
00007 #include "kernel/simulator/Plugin.h"
00008 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00009 
00024 class CellFateDecisionComponent : public ModelComponent {
00025 public:
00026     CellFateDecisionComponent(Model* model, std::string name = "");
00027     virtual ~CellFateDecisionComponent() override = default;
00028 
00029 public:
00030     static PluginInformation* GetPluginInformation();
00031     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00032     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00033 
00034 public:
00035     void setWholeCellState(WholeCellState* state);
00036     WholeCellState* getWholeCellState() const;
00037     void setDivisionReadyPhase(std::string phase);
00038     std::string getDivisionReadyPhase() const;
00039     void setStarvedPhase(std::string phase);
00040     std::string getStarvedPhase() const;
```

```
00041     void setArrestedPhase(std::string phase);
00042     std::string getArrestedPhase() const;
00043     void setDeadPhase(std::string phase);
00044     std::string getDeadPhase() const;
00045     unsigned int getLastRoutedPort() const;
00046 
00047     virtual std::string show() override;
00048 
00049 protected:
00050     virtual bool _loadInstance(PersistenceRecord* fields) override;
00051     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00052     virtual bool _check(std::string& errorMessage) override;
00053     virtual void _initBetweenReplications() override;
00054     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00055 
00056 protected:
00057     virtual void _createEditableDataDefinitions() override;
00058 
00059 private:
00060     unsigned int _selectOutputPort() const;
00061     void _forwardEntity(Entity* entity, unsigned int port);
00062 
00063 private:
00064     const struct DEFAULT_VALUES {
00065         std::string wholeCellStateName = "";
00066         std::string divisionReadyPhase = "division_ready";
00067         std::string starvedPhase = "starved";
00068         std::string arrestedPhase = "arrested";
00069         std::string deadPhase = "dead";
00070         unsigned int lastRoutedPort = 0u;
00071     } DEFAULT;
00072 
00073     WholeCellState* _wholeCellState = nullptr;
00074     std::string _divisionReadyPhase = DEFAULT.divisionReadyPhase;
00075     std::string _starvedPhase = DEFAULT.starvedPhase;
00076     std::string _arrestedPhase = DEFAULT.arrestedPhase;
00077     std::string _deadPhase = DEFAULT.deadPhase;
00078     unsigned int _lastRoutedPort = DEFAULT.lastRoutedPort;
00079 };
00080 
00081 #endif /* CELLFATEDECISIONCOMPONENT_H */
```

18.466 CellGrowthComponent.h

```
00001 #ifndef CELLGROWTHCOMPONENT_H
00002 #define CELLGROWTHCOMPONENT_H
00003 
00004 #include <string>
00005 
00006 #include "../../../kernel/simulator/model/ModelComponent.h"
00007 #include "kernel/simulator/Plugin.h"
00008 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00009 
00022 class CellGrowthComponent : public ModelComponent {
00023 public:
00024     CellGrowthComponent(Model* model, std::string name = "");
00025     virtual ~CellGrowthComponent() override = default;
00026 
00027 public:
00028     static PluginInformation* GetPluginInformation();
00029     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00030     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00031 
00032 public:
00033     void setWholeCellState(WholeCellState* state);
00034     WholeCellState* getWholeCellState() const;
00035     void setGrowthRate(double rate);
00036     double getGrowthRate() const;
00037     void setEnergyMetaboliteKey(std::string key);
00038     std::string getEnergyMetaboliteKey() const;
00039     void setEnergyHalfSaturation(double value);
00040     double getEnergyHalfSaturation() const;
00041     void setDensity(double density);
00042     double getDensity() const;
00043     void setDeltaT(double dt);
00044     double getDeltaT() const;
00045 
00046     virtual std::string show() override;
00047 
00048 protected:
00049     virtual bool _loadInstance(PersistenceRecord* fields) override;
00050     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00051     virtual bool _check(std::string& errorMessage) override;
00052     virtual void _initBetweenReplications() override;
00053     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
```

```
00054
00055 protected:
00056     virtual void _createEditableDataDefinitions() override;
00057
00058 private:
00059     void _forwardEntity(Entity* entity);
00060
00061 private:
00062     const struct DEFAULT_VALUES {
00063         std::string wholeCellStateName = "";
00064         double growthRate = 2.1393e-05;  // /s -- M. genitalium (parameters.json)
00065         std::string energyMetaboliteKey = "";
00066         double energyHalfSaturation = 0.0; // 0 disables energy gating
00067         double density    = 1100.0;      // kg/m^3 -- states.Geometry.density
00068         double deltaT     = 60.0;        // s -- must match clock step
00069     } DEFAULT;
00070
00071     WholeCellState* _wholeCellState = nullptr;
00072     double _growthRate = DEFAULT.growthRate;
00073     std::string _energyMetaboliteKey = DEFAULT.energyMetaboliteKey;
00074     double _energyHalfSaturation = DEFAULT.energyHalfSaturation;
00075     double _density    = DEFAULT.density;
00076     double _deltaT     = DEFAULT.deltaT;
00077 };
00078
00079 #endif /* CELLGROWTHCOMPONENT_H */
```

18.467 CompartmentExchangeComponent.h

```
00001 #ifndef COMPARTMENTEXCHANGECOMPONENT_H
00002 #define COMPARTMENTEXCHANGECOMPONENT_H
00003
00004 #include <string>
00005 #include <vector>
00006
00007 #include "../../../kernel/simulator/model/ModelComponent.h"
00008 #include "kernel/simulator/Plugin.h"
00009 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00010
00024 class CompartmentExchangeComponent : public ModelComponent {
00025 public:
00026     struct ExchangeRule {
00027         std::string label;
00028         std::string sourceRegion;
00029         std::string sourceMetaboliteKey;
00030         std::string targetRegion;
00031         std::string targetMetaboliteKey;
00032         double exchangeFraction = 0.0;
00033         double baselineAmount = 0.0;
00034         std::string driverPathwayKey;
00035         double driverScale = 1.0;
00036         double maxTransferAmount = 0.0;
00037         bool conserveMass = true;
00038     };
00039
00040 public:
00041     CompartmentExchangeComponent(Model* model, std::string name = "");
00042     virtual ~CompartmentExchangeComponent() override = default;
00043
00044 public:
00045     static PluginInformation* GetPluginInformation();
00046     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00047     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00048
00049 public:
00050     void setWholeCellState(WholeCellState* state);
00051     WholeCellState* getWholeCellState() const;
00052     void setSourceRegion(std::string region);
00053     std::string getSourceRegion() const;
00054     void setSourceMetaboliteKey(std::string key);
00055     std::string getSourceMetaboliteKey() const;
00056     void setTargetRegion(std::string region);
00057     std::string getTargetRegion() const;
00058     void setTargetMetaboliteKey(std::string key);
00059     std::string getTargetMetaboliteKey() const;
00060     void setExchangeFraction(double fraction);
00061     double getExchangeFraction() const;
00062     void setBaselineAmount(double amount);
00063     double getBaselineAmount() const;
00064     void setDriverPathwayKey(std::string key);
00065     std::string getDriverPathwayKey() const;
00066     void setDriverScale(double scale);
00067     double getDriverScale() const;
00068     void setMaxTransferAmount(double amount);
00069     double getMaxTransferAmount() const;
```

```
00070     void setConserveMass(bool conserve);
00071     bool getConserveMass() const;
00072     void clearExchangeRules();
00073     void addExchangeRule(const std::string& label,
00074                          const std::string& sourceRegion,
00075                          const std::string& sourceMetaboliteKey,
00076                          const std::string& targetRegion,
00077                          const std::string& targetMetaboliteKey,
00078                          double exchangeFraction,
00079                          double baselineAmount = 0.0,
00080                          const std::string& driverPathwayKey = "",
00081                          double driverScale = 1.0,
00082                          double maxTransferAmount = 0.0,
00083                          bool conserveMass = true);
00084     unsigned int getExchangeRuleCount() const;
00085     const std::vector<ExchangeRule>& getExchangeRules() const;
00086     double getLastTransferAmount() const;
00087
00088     virtual std::string show() override;
00089
00090 protected:
00091     virtual bool _loadInstance(PersistenceRecord* fields) override;
00092     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00093     virtual bool _check(std::string& errorMessage) override;
00094     virtual void _initBetweenReplications() override;
00095     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00096     virtual void _createEditableDataDefinitions() override;
00097
00098 private:
00099     ExchangeRule _buildLegacyRule() const;
00100     std::vector<ExchangeRule> _getEffectiveRules() const;
00101     bool _checkRule(const ExchangeRule& rule, std::string& errorMessage, unsigned int index) const;
00102     double _computeTransferAmount(const ExchangeRule& rule) const;
00103     void _applyRule(const ExchangeRule& rule);
00104     double _getPoolAmount(const std::string& region, const std::string& key) const;
00105     void _setPoolAmount(const std::string& region, const std::string& key, double amount);
00106     void _forwardEntity(Entity* entity);
00107
00108 private:
00109     const struct DEFAULT_VALUES {
00110         std::string wholeCellStateName = "";
00111         std::string sourceRegion = "";
00112         std::string sourceMetaboliteKey = "";
00113         std::string targetRegion = "";
00114         std::string targetMetaboliteKey = "";
00115         double exchangeFraction = 0.0;
00116         double baselineAmount = 0.0;
00117         std::string driverPathwayKey = "";
00118         double driverScale = 1.0;
00119         double maxTransferAmount = 0.0;
00120         bool conserveMass = true;
00121         double lastTransferAmount = 0.0;
00122     } DEFAULT;
00123
00124     WholeCellState* _wholeCellState = nullptr;
00125     std::string _sourceRegion = DEFAULT.sourceRegion;
00126     std::string _sourceMetaboliteKey = DEFAULT.sourceMetaboliteKey;
00127     std::string _targetRegion = DEFAULT.targetRegion;
00128     std::string _targetMetaboliteKey = DEFAULT.targetMetaboliteKey;
00129     double _exchangeFraction = DEFAULT.exchangeFraction;
00130     double _baselineAmount = DEFAULT.baselineAmount;
00131     std::string _driverPathwayKey = DEFAULT.driverPathwayKey;
00132     double _driverScale = DEFAULT.driverScale;
00133     double _maxTransferAmount = DEFAULT.maxTransferAmount;
00134     bool _conserveMass = DEFAULT.conserveMass;
00135     std::vector<ExchangeRule> _exchangeRules;
00136     double _lastTransferAmount = DEFAULT.lastTransferAmount;
00137 };
00138
00139 #endif /* COMPARTMENTEXCHANGECOMPONENT_H */
```

18.468 EukaryoticCellCycleComponent.h

```
00001 #ifndef EUKARYOTICCELLCYCLECOMPONENT_H
00002 #define EUKARYOTICCELLCYCLECOMPONENT_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelComponent.h"
00007 #include "kernel/simulator/Plugin.h"
00008 #include "plugins/data/WholeCellModeling/BioCompartment.h"
00009 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00010
00039 class EukaryoticCellCycleComponent : public ModelComponent {
00040 public:
```

```
00041     EukaryoticCellCycleComponent(Model* model, std::string name = "");
00042     virtual ~EukaryoticCellCycleComponent() override = default;
00043
00044 public:
00045     static PluginInformation* GetPluginInformation();
00046     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00047     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00048
00049 public:
00050     void setWholeCellState(WholeCellState* state);
00051     WholeCellState* getWholeCellState() const;
00052     void setDeltaT(double deltaT);
00053     double getDeltaT() const;
00054     void setAdvanceWholeCellClock(bool advance);
00055     bool getAdvanceWholeCellClock() const;
00056
00057     void setCytosolCompartmentName(std::string name);
00058     std::string getCytosolCompartmentName() const;
00059     void setNucleusCompartmentName(std::string name);
00060     std::string getNucleusCompartmentName() const;
00061     void setMitochondrionCompartmentName(std::string name);
00062     std::string getMitochondrionCompartmentName() const;
00063     void setBudCompartmentName(std::string name);
00064     std::string getBudCompartmentName() const;
00065
00066     void setEnergyMetaboliteKey(std::string key);
00067     std::string getEnergyMetaboliteKey() const;
00068     void setCytosolicEnergyMetaboliteKey(std::string key);
00069     std::string getCytosolicEnergyMetaboliteKey() const;
00070     void setMitochondrialEnergyMetaboliteKey(std::string key);
00071     std::string getMitochondrialEnergyMetaboliteKey() const;
00072     void setBudEnergyMetaboliteKey(std::string key);
00073     std::string getBudEnergyMetaboliteKey() const;
00074
00075     void setGrowthFluxKey(std::string key);
00076     std::string getGrowthFluxKey() const;
00077     void setRespirationFluxKey(std::string key);
00078     std::string getRespirationFluxKey() const;
00079
00080     void setBudProgressKey(std::string key);
00081     std::string getBudProgressKey() const;
00082     void setDnaReplicationProgressKey(std::string key);
00083     std::string getDnaReplicationProgressKey() const;
00084     void setSpindleAssemblyProgressKey(std::string key);
00085     std::string getSpindleAssemblyProgressKey() const;
00086     void setMitoticExitProgressKey(std::string key);
00087     std::string getMitoticExitProgressKey() const;
00088
00089     void setGlobalAtpThreshold(double threshold);
00090     double getGlobalAtpThreshold() const;
00091     void setCytosolicAtpThreshold(double threshold);
00092     double getCytosolicAtpThreshold() const;
00093     void setMitochondrialAtpThreshold(double threshold);
00094     double getMitochondrialAtpThreshold() const;
00095     void setBudAtpThreshold(double threshold);
00096     double getBudAtpThreshold() const;
00097     void setGrowthFluxThreshold(double threshold);
00098     double getGrowthFluxThreshold() const;
00099     void setRespirationFluxThreshold(double threshold);
00100     double getRespirationFluxThreshold() const;
00101
00102     void setBudProgressRate(double rate);
00103     double getBudProgressRate() const;
00104     void setDnaReplicationRate(double rate);
00105     double getDnaReplicationRate() const;
00106     void setSpindleAssemblyRate(double rate);
00107     double getSpindleAssemblyRate() const;
00108     void setMitoticExitRate(double rate);
00109     double getMitoticExitRate() const;
00110
00111     void setBudProgressThreshold(double threshold);
00112     double getBudProgressThreshold() const;
00113     void setDnaReplicationThreshold(double threshold);
00114     double getDnaReplicationThreshold() const;
00115     void setSpindleAssemblyThreshold(double threshold);
00116     double getSpindleAssemblyThreshold() const;
00117     void setMitoticExitThreshold(double threshold);
00118     double getMitoticExitThreshold() const;
00119
00120     virtual std::string show() override;
00121
00122 protected:
00123     virtual bool _loadInstance(PersistenceRecord* fields) override;
00124     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00125     virtual bool _check(std::string& errorMessage) override;
00126     virtual void _initBetweenReplications() override;
00127     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
```

```
00128     virtual void _createEditableDataDefinitions() override;
00129
00130 private:
00131     double _normalizedSignal(double value, double threshold) const;
00132     double _averageSignal(double a, double b) const;
00133     double _advanceProgress(const std::string& key, double increment) const;
00134     void _updateLifecyclePhase();
00135     void _forwardEntity(Entity* entity);
00136     bool _validateReferencedCompartment(const std::string& name, std::string* errorMessage) const;
00137     BioCompartment* _findCompartment(const std::string& name) const;
00138
00139 private:
00140     const struct DEFAULT_VALUES {
00141         std::string wholeCellStateName = "";
00142         double deltaT = 60.0;
00143         bool advanceWholeCellClock = true;
00144         std::string cytosolCompartmentName = "cytosol";
00145         std::string nucleusCompartmentName = "nucleus";
00146         std::string mitochondrionCompartmentName = "mitochondria";
00147         std::string budCompartmentName = "bud";
00148         std::string energyMetaboliteKey = "ATP";
00149         std::string cytosolicEnergyMetaboliteKey = "ATP_c";
00150         std::string mitochondrialEnergyMetaboliteKey = "ATP_m";
00151         std::string budEnergyMetaboliteKey = "ATP_bud";
00152         std::string growthFluxKey = "biomass_flux";
00153         std::string respirationFluxKey = "respiration_flux";
00154         std::string budProgressKey = "bud_growth_progress";
00155         std::string dnaReplicationProgressKey = "dna_replication_progress";
00156         std::string spindleAssemblyProgressKey = "spindle_assembly_progress";
00157         std::string mitoticExitProgressKey = "mitotic_exit_progress";
00158         double globalAtpThreshold = 0.50;
00159         double cytosolicAtpThreshold = 0.40;
00160         double mitochondrialAtpThreshold = 0.35;
00161         double budAtpThreshold = 0.20;
00162         double growthFluxThreshold = 4.0;
00163         double respirationFluxThreshold = 1.0;
00164         double budProgressRate = 0.30;
00165         double dnaReplicationRate = 0.24;
00166         double spindleAssemblyRate = 0.22;
00167         double mitoticExitRate = 0.20;
00168         double budProgressThreshold = 1.0;
00169         double dnaReplicationThreshold = 1.0;
00170         double spindleAssemblyThreshold = 1.0;
00171         double mitoticExitThreshold = 1.0;
00172     } DEFAULT;
00173
00174     WholeCellState* _wholeCellState = nullptr;
00175     double _deltaT = DEFAULT.deltaT;
00176     bool _advanceWholeCellClock = DEFAULT.advanceWholeCellClock;
00177     std::string _cytosolCompartmentName = DEFAULT.cytosolCompartmentName;
00178     std::string _nucleusCompartmentName = DEFAULT.nucleusCompartmentName;
00179     std::string _mitochondrionCompartmentName = DEFAULT.mitochondrionCompartmentName;
00180     std::string _budCompartmentName = DEFAULT.budCompartmentName;
00181     std::string _energyMetaboliteKey = DEFAULT.energyMetaboliteKey;
00182     std::string _cytosolicEnergyMetaboliteKey = DEFAULT.cytosolicEnergyMetaboliteKey;
00183     std::string _mitochondrialEnergyMetaboliteKey = DEFAULT.mitochondrialEnergyMetaboliteKey;
00184     std::string _budEnergyMetaboliteKey = DEFAULT.budEnergyMetaboliteKey;
00185     std::string _growthFluxKey = DEFAULT.growthFluxKey;
00186     std::string _respirationFluxKey = DEFAULT.respirationFluxKey;
00187     std::string _budProgressKey = DEFAULT.budProgressKey;
00188     std::string _dnaReplicationProgressKey = DEFAULT.dnaReplicationProgressKey;
00189     std::string _spindleAssemblyProgressKey = DEFAULT.spindleAssemblyProgressKey;
00190     std::string _mitoticExitProgressKey = DEFAULT.mitoticExitProgressKey;
00191     double _globalAtpThreshold = DEFAULT.globalAtpThreshold;
00192     double _cytosolicAtpThreshold = DEFAULT.cytosolicAtpThreshold;
00193     double _mitochondrialAtpThreshold = DEFAULT.mitochondrialAtpThreshold;
00194     double _budAtpThreshold = DEFAULT.budAtpThreshold;
00195     double _growthFluxThreshold = DEFAULT.growthFluxThreshold;
00196     double _respirationFluxThreshold = DEFAULT.respirationFluxThreshold;
00197     double _budProgressRate = DEFAULT.budProgressRate;
00198     double _dnaReplicationRate = DEFAULT.dnaReplicationRate;
00199     double _spindleAssemblyRate = DEFAULT.spindleAssemblyRate;
00200     double _mitoticExitRate = DEFAULT.mitoticExitRate;
00201     double _budProgressThreshold = DEFAULT.budProgressThreshold;
00202     double _dnaReplicationThreshold = DEFAULT.dnaReplicationThreshold;
00203     double _spindleAssemblyThreshold = DEFAULT.spindleAssemblyThreshold;
00204     double _mitoticExitThreshold = DEFAULT.mitoticExitThreshold;
00205 };
00206
00207 #endif /* EUKARYOTICCELLCYCLECOMPONENT_H */
```

18.469 FtsZPolymerizationComponent.h

```
00001 #ifndef FTSZPOLYMERIZATIONCOMPONENT_H
00002 #define FTSZPOLYMERIZATIONCOMPONENT_H
```

```
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelComponent.h"
00007 #include "kernel/simulator/Plugin.h"
00008 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00009
00035 class FtsZPolymerizationComponent : public ModelComponent {
00036 public:
00037     FtsZPolymerizationComponent(Model* model, std::string name = "");
00038     virtual ~FtsZPolymerizationComponent() override = default;
00039
00040 public:
00041     static PluginInformation* GetPluginInformation();
00042     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00043     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00044
00045 public:
00046     void setWholeCellState(WholeCellState* state);
00047     WholeCellState* getWholeCellState() const;
00048     void setActivationFwd(double rate);
00049     double getActivationFwd() const;
00050     void setActivationRev(double rate);
00051     double getActivationRev() const;
00052     void setVolumeNorm(double norm);
00053     double getVolumeNorm() const;
00054     void setDeltaT(double dt);
00055     double getDeltaT() const;
00056     void setFtsZRingKey(std::string key);
00057     std::string getFtsZRingKey() const;
00058     void setFtsZMonomerKey(std::string key);
00059     std::string getFtsZMonomerKey() const;
00060
00061     virtual std::string show() override;
00062
00063 protected:
00064     virtual bool _loadInstance(PersistenceRecord* fields) override;
00065     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00066     virtual bool _check(std::string& errorMessage) override;
00067     virtual void _initBetweenReplications() override;
00068     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00069
00070 protected:
00071     virtual void _createEditableDataDefinitions() override;
00072
00073 private:
00074     void _forwardEntity(Entity* entity);
00075
00076 private:
00077     const struct DEFAULT_VALUES {
00078         std::string wholeCellStateName = "";
00079         double activationFwd  = 1.1;    // /s -- processes.FtsZPolymerization.activationFwd
00080         double activationRev  = 0.01;   // /s -- derived from Karr 2012 ratio
00081         double volumeNorm     = 3.94e-5; // dimensionless -- calibrated for 35% at ~12000 s
00082         double deltaT         = 60.0;   // s -- must match clock step
00083         std::string ftsZRingKey   = "FtsZ_ring_completion";
00084         std::string ftsZMonomerKey = "FtsZ_monomer";
00085     } DEFAULT;
00086
00087     WholeCellState* _wholeCellState  = nullptr;
00088     double _activationFwd   = DEFAULT.activationFwd;
00089     double _activationRev   = DEFAULT.activationRev;
00090     double _volumeNorm      = DEFAULT.volumeNorm;
00091     double _deltaT          = DEFAULT.deltaT;
00092     std::string _ftsZRingKey    = DEFAULT.ftsZRingKey;
00093     std::string _ftsZMonomerKey = DEFAULT.ftsZMonomerKey;
00094     int _nRef = 0;  // reference monomer count captured on first step
00095 };
00096
00097 #endif /* FTSZPOLYMERIZATIONCOMPONENT_H */
```

18.470 MetabolicStateProjectionComponent.h

```
00001 #ifndef METABOLICSTATEPROJECTIONCOMPONENT_H
00002 #define METABOLICSTATEPROJECTIONCOMPONENT_H
00003
00004 #include <string>
00005 #include <vector>
00006
00007 #include "../../../kernel/simulator/model/ModelComponent.h"
00008 #include "kernel/simulator/Plugin.h"
00009 #include "plugins/components/BiochemicalSimulation/MetabolicFluxBalance.h"
00010 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00011
00027 class MetabolicStateProjectionComponent : public ModelComponent {
```

```
00028 public:
00029     enum class ProjectionKind {
00030         MetaboliteAmount,
00031         CompartmentMetaboliteAmount,
00032         PathwayActivity
00033     };
00034
00035     enum class ProjectionUpdateMode {
00036         Overwrite,
00037         Accumulate,
00038         Turnover
00039     };
00040
00041     struct Projection {
00042         std::string reactionName;
00043         std::string targetKey;
00044         ProjectionKind kind = ProjectionKind::MetaboliteAmount;
00045         ProjectionUpdateMode updateMode = ProjectionUpdateMode::Overwrite;
00046         double scale = 1.0;
00047         double offset = 0.0;
00048         double turnoverFraction = 0.0;
00049         std::string compartmentName;
00050     };
00051
00052 public:
00053     MetabolicStateProjectionComponent(Model* model, std::string name = "");
00054     virtual ~MetabolicStateProjectionComponent() override = default;
00055
00056 public:
00057     static PluginInformation* GetPluginInformation();
00058     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00059     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00060
00061 public:
00062     void setWholeCellState(WholeCellState* state);
00063     WholeCellState* getWholeCellState() const;
00064     void setFluxBalanceComponent(MetabolicFluxBalance* component);
00065     MetabolicFluxBalance* getFluxBalanceComponent() const;
00066     void setFluxBalanceComponentName(std::string componentName);
00067     std::string getFluxBalanceComponentName() const;
00068
00069     void clearObjectiveProjection();
00070     void setObjectiveAsMetabolite(const std::string& targetKey, double scale = 1.0, double offset =
      0.0,
00071                                   ProjectionUpdateMode updateMode = ProjectionUpdateMode::Overwrite,
00072                                   double turnoverFraction = 0.0);
00073     void setObjectiveAsCompartmentMetabolite(const std::string& compartmentName, const std::string&
      targetKey,
00074                                              double scale = 1.0, double offset = 0.0,
00075                                              ProjectionUpdateMode updateMode =
      ProjectionUpdateMode::Overwrite,
00076                                              double turnoverFraction = 0.0);
00077     void setObjectiveAsPathwayActivity(const std::string& targetKey, double scale = 1.0, double offset
      = 0.0,
00078                                        ProjectionUpdateMode updateMode =
      ProjectionUpdateMode::Overwrite,
00079                                        double turnoverFraction = 0.0);
00080     bool hasObjectiveProjection() const;
00081
00082     void clearFluxProjections();
00083     void addMetaboliteProjection(const std::string& reactionName, const std::string& targetKey,
00084                                  double scale = 1.0, double offset = 0.0,
00085                                  ProjectionUpdateMode updateMode = ProjectionUpdateMode::Overwrite,
00086                                  double turnoverFraction = 0.0);
00087     void addCompartmentMetaboliteProjection(const std::string& reactionName, const std::string&
      compartmentName,
00088                                             const std::string& targetKey, double scale = 1.0, double
      offset = 0.0,
00089                                             ProjectionUpdateMode updateMode =
      ProjectionUpdateMode::Overwrite,
00090                                             double turnoverFraction = 0.0);
00091     void addPathwayProjection(const std::string& reactionName, const std::string& targetKey,
00092                               double scale = 1.0, double offset = 0.0,
00093                               ProjectionUpdateMode updateMode = ProjectionUpdateMode::Overwrite,
00094                               double turnoverFraction = 0.0);
00095     unsigned int getProjectionCount() const;
00096
00097     void setLastSucceeded(bool lastSucceeded);
00098     bool getLastSucceeded() const;
00099     void setLastMessage(std::string lastMessage);
00100     std::string getLastMessage() const;
00101
00102     virtual std::string show() override;
00103
00104 protected:
00105     virtual bool _loadInstance(PersistenceRecord* fields) override;
00106     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
```

```
00107     virtual bool _check(std::string& errorMessage) override;
00108     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00109     virtual void _createEditableDataDefinitions() override;
00110
00111 private:
00112     void _forwardEntity(Entity* entity);
00113     void _syncEditableDataDefinitions();
00114     void _resolveFluxBalanceComponent();
00115     void _configureObjectiveProjection(ProjectionKind kind, const std::string& targetKey, const
      std::string& compartmentName,
00116                                        double scale, double offset, ProjectionUpdateMode updateMode,
00117                                        double turnoverFraction);
00118     void _appendProjection(const Projection& projection);
00119     void _applyProjection(const Projection& projection, double value);
00120     std::string _resolveCompartmentName(const Projection& projection) const;
00121     double _resolveCurrentValue(const Projection& projection) const;
00122
00123 private:
00124     const struct DEFAULT_VALUES {
00125         std::string wholeCellStateName = "";
00126         std::string fluxBalanceComponentName = "";
00127         bool objectiveProjectionEnabled = false;
00128         std::string objectiveTargetKey = "";
00129         std::string objectiveKind = "MetaboliteAmount";
00130         std::string objectiveUpdateMode = "Overwrite";
00131         double objectiveScale = 1.0;
00132         double objectiveOffset = 0.0;
00133         double objectiveTurnoverFraction = 0.0;
00134         std::string objectiveCompartmentName = "";
00135         bool lastSucceeded = false;
00136         std::string lastMessage = "";
00137     } DEFAULT;
00138
00139     WholeCellState* _wholeCellState = nullptr;
00140     MetabolicFluxBalance* _fluxBalanceComponent = nullptr;
00141     std::string _fluxBalanceComponentName = DEFAULT.fluxBalanceComponentName;
00142     bool _objectiveProjectionEnabled = DEFAULT.objectiveProjectionEnabled;
00143     Projection _objectiveProjection;
00144     std::vector<Projection> _projections;
00145     bool _lastSucceeded = DEFAULT.lastSucceeded;
00146     std::string _lastMessage = DEFAULT.lastMessage;
00147 };
00148
00149 #endif /* METABOLICSTATEPROJECTIONCOMPONENT_H */
```

18.471 MetabolicSubmodelComponent.h

```
00001 #ifndef METABOLICSUBMODELCOMPONENT_H
00002 #define METABOLICSUBMODELCOMPONENT_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelComponent.h"
00007 #include "kernel/simulator/Plugin.h"
00008 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00009
00029 class MetabolicSubmodelComponent : public ModelComponent {
00030 public:
00031     MetabolicSubmodelComponent(Model* model, std::string name = "");
00032     virtual ~MetabolicSubmodelComponent() override = default;
00033
00034 public:
00035     static PluginInformation* GetPluginInformation();
00036     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00037     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00038
00039 public:
00040     void setWholeCellState(WholeCellState* state);
00041     WholeCellState* getWholeCellState() const;
00042     void setAtpYieldRate(double rate);
00043     double getAtpYieldRate() const;
00044     void setConsumptionFraction(double frac);
00045     double getConsumptionFraction() const;
00046     void setAtpPoolMax(double maxPool);
00047     double getAtpPoolMax() const;
00048     void setDeltaT(double dt);
00049     double getDeltaT() const;
00050
00051     virtual std::string show() override;
00052
00053 protected:
00054     virtual bool _loadInstance(PersistenceRecord* fields) override;
00055     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00056     virtual bool _check(std::string& errorMessage) override;
00057     virtual void _initBetweenReplications() override;
```

```
00058     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00059
00060 protected:
00061     virtual void _createEditableDataDefinitions() override;
00062
00063 private:
00064     void _forwardEntity(Entity* entity);
00065
00066 private:
00067     const struct DEFAULT_VALUES {
00068         std::string wholeCellStateName = "";
00069         // mmol ATP per fL per second -- calibrated for M. genitalium basal metabolism
00070         double atpYieldRate       = 1.0e-3;
00071         double consumptionFraction = 0.1;   // fraction of ATP pool consumed per step
00072         double atpPoolMax         = 10.0;   // mmol -- approximate saturation
00073         double deltaT             = 60.0;   // s
00074     } DEFAULT;
00075
00076     WholeCellState* _wholeCellState  = nullptr;
00077     double _atpYieldRate       = DEFAULT.atpYieldRate;
00078     double _consumptionFraction = DEFAULT.consumptionFraction;
00079     double _atpPoolMax         = DEFAULT.atpPoolMax;
00080     double _deltaT             = DEFAULT.deltaT;
00081 };
00082
00083 #endif /* METABOLICSUBMODELCOMPONENT_H */
```

18.472 PathwayStressResponseComponent.h

```
00001 #ifndef PATHWAYSTRESSRESPONSECOMPONENT_H
00002 #define PATHWAYSTRESSRESPONSECOMPONENT_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelComponent.h"
00007 #include "kernel/simulator/Plugin.h"
00008 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00009
00026 class PathwayStressResponseComponent : public ModelComponent {
00027 public:
00028     PathwayStressResponseComponent(Model* model, std::string name = "");
00029     virtual ~PathwayStressResponseComponent() override = default;
00030
00031 public:
00032     static PluginInformation* GetPluginInformation();
00033     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00034     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00035
00036 public:
00037     void setWholeCellState(WholeCellState* state);
00038     WholeCellState* getWholeCellState() const;
00039     void setMonitoredPathwayKey(std::string key);
00040     std::string getMonitoredPathwayKey() const;
00041     void setStressThreshold(double threshold);
00042     double getStressThreshold() const;
00043     void setArrestAfterSteps(unsigned int steps);
00044     unsigned int getArrestAfterSteps() const;
00045     void setDeathAfterSteps(unsigned int steps);
00046     unsigned int getDeathAfterSteps() const;
00047     void setArrestPhase(std::string phase);
00048     std::string getArrestPhase() const;
00049     void setDeadPhase(std::string phase);
00050     std::string getDeadPhase() const;
00051     void setRecoveryPhase(std::string phase);
00052     std::string getRecoveryPhase() const;
00053     void setResetStreakOnRecovery(bool reset);
00054     bool getResetStreakOnRecovery() const;
00055     unsigned int getStressStreak() const;
00056     double getLastObservedActivity() const;
00057
00058     virtual std::string show() override;
00059
00060 protected:
00061     virtual bool _loadInstance(PersistenceRecord* fields) override;
00062     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00063     virtual bool _check(std::string& errorMessage) override;
00064     virtual void _initBetweenReplications() override;
00065     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00066     virtual void _createEditableDataDefinitions() override;
00067
00068 private:
00069     void _forwardEntity(Entity* entity);
00070
00071 private:
00072     const struct DEFAULT_VALUES {
```

```
00073         std::string wholeCellStateName = "";
00074         std::string monitoredPathwayKey = "biomass_objective";
00075         double stressThreshold = 0.0;
00076         unsigned int arrestAfterSteps = 0u;
00077         unsigned int deathAfterSteps = 0u;
00078         std::string arrestPhase = "arrested";
00079         std::string deadPhase = "dead";
00080         std::string recoveryPhase = "growth";
00081         bool resetStreakOnRecovery = true;
00082         unsigned int stressStreak = 0u;
00083         double lastObservedActivity = 0.0;
00084     } DEFAULT;
00085
00086     WholeCellState* _wholeCellState = nullptr;
00087     std::string _monitoredPathwayKey = DEFAULT.monitoredPathwayKey;
00088     double _stressThreshold = DEFAULT.stressThreshold;
00089     unsigned int _arrestAfterSteps = DEFAULT.arrestAfterSteps;
00090     unsigned int _deathAfterSteps = DEFAULT.deathAfterSteps;
00091     std::string _arrestPhase = DEFAULT.arrestPhase;
00092     std::string _deadPhase = DEFAULT.deadPhase;
00093     std::string _recoveryPhase = DEFAULT.recoveryPhase;
00094     bool _resetStreakOnRecovery = DEFAULT.resetStreakOnRecovery;
00095     unsigned int _stressStreak = DEFAULT.stressStreak;
00096     double _lastObservedActivity = DEFAULT.lastObservedActivity;
00097 };
00098
00099 #endif /* PATHWAYSTRESSRESPONSECOMPONENT_H */
```

18.473 ResourceAllocationComponent.h

```
00001 #ifndef RESOURCEALLOCATIONCOMPONENT_H
00002 #define RESOURCEALLOCATIONCOMPONENT_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelComponent.h"
00007 #include "kernel/simulator/Plugin.h"
00008 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00009
00031 class ResourceAllocationComponent : public ModelComponent {
00032 public:
00033     ResourceAllocationComponent(Model* model, std::string name = "");
00034     virtual ~ResourceAllocationComponent() override = default;
00035
00036 public:
00037     static PluginInformation* GetPluginInformation();
00038     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00039     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00040
00041 public:
00042     void setWholeCellState(WholeCellState* state);
00043     WholeCellState* getWholeCellState() const;
00044     void setRnapCountKey(std::string key);
00045     std::string getRnapCountKey() const;
00046     void setRibosomeCountKey(std::string key);
00047     std::string getRibosomeCountKey() const;
00048     void setMRNASpeciesPrefix(std::string prefix);
00049     std::string getMRNASpeciesPrefix() const;
00050     void setProteinSpeciesPrefix(std::string prefix);
00051     std::string getProteinSpeciesPrefix() const;
00052
00053     virtual std::string show() override;
00054
00055 protected:
00056     virtual bool _loadInstance(PersistenceRecord* fields) override;
00057     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00058     virtual bool _check(std::string& errorMessage) override;
00059     virtual void _initBetweenReplications() override;
00060     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00061
00062 protected:
00063     virtual void _createEditableDataDefinitions() override;
00064
00065 private:
00066     void _forwardEntity(Entity* entity);
00067
00068 private:
00069     const struct DEFAULT_VALUES {
00070         std::string wholeCellStateName   = "";
00071         std::string rnapCountKey         = "RNAP_free";
00072         std::string ribosomeCountKey     = "ribosome_free";
00073         std::string mRNASpeciesPrefix    = "mRNA_";
00074         std::string proteinSpeciesPrefix = "protein_";
00075     } DEFAULT;
00076
```

```
00077     WholeCellState* _wholeCellState  = nullptr;
00078     std::string _rnapCountKey        = DEFAULT.rnapCountKey;
00079     std::string _ribosomeCountKey    = DEFAULT.ribosomeCountKey;
00080     std::string _mRNASpeciesPrefix   = DEFAULT.mRNASpeciesPrefix;
00081     std::string _proteinSpeciesPrefix = DEFAULT.proteinSpeciesPrefix;
00082 };
00083
00084 #endif /* RESOURCEALLOCATIONCOMPONENT_H */
```

18.474 StochasticReactionComponent.h

```
00001 #ifndef STOCHASTICREACTIONCOMPONENT_H
00002 #define STOCHASTICREACTIONCOMPONENT_H
00003
00004 #include <map>
00005 #include <random>
00006 #include <string>
00007 #include <vector>
00008
00009 #include "../../../kernel/simulator/model/ModelComponent.h"
00010 #include "kernel/simulator/Plugin.h"
00011 #include "plugins/data/WholeCellModeling/StochasticReactionRule.h"
00012 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00013
00032 class StochasticReactionComponent : public ModelComponent {
00033 public:
00034     StochasticReactionComponent(Model* model, std::string name = "");
00035     virtual ~StochasticReactionComponent() override = default;
00036
00037 public:
00038     static PluginInformation* GetPluginInformation();
00039     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00040     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00041
00042 public:
00043     void setWholeCellState(WholeCellState* state);
00044     WholeCellState* getWholeCellState() const;
00045     void setTimeWindow(double timeWindow);
00046     double getTimeWindow() const;
00047     void setAdvanceWholeCellClock(bool advance);
00048     bool getAdvanceWholeCellClock() const;
00049     void setRandomSeed(unsigned int seed);
00050     unsigned int getRandomSeed() const;
00051     int getLastReactionCount() const;
00052     double getLastSimulatedTime() const;
00053
00054     virtual std::string show() override;
00055
00056 protected:
00057     virtual bool _loadInstance(PersistenceRecord* fields) override;
00058     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00059     virtual bool _check(std::string& errorMessage) override;
00060     virtual void _initBetweenReplications() override;
00061     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00062
00063 protected:
00064     virtual void _createEditableDataDefinitions() override;
00065
00066 private:
00067     struct GillespieState {
00068         std::map<std::string, int> counts;
00069         double simulatedTime = 0.0;
00070         int reactionCount = 0;
00071     };
00072
00073     GillespieState _runGillespie(
00074         const std::map<std::string, int>& initialCounts,
00075         const std::vector<StochasticReactionRule*>& rules,
00076         const std::map<std::string, int>& resourceBudget,
00077         double timeWindow);
00078
00079     void _collectCountsFromState(std::map<std::string, int>& counts) const;
00080     void _applyCountsToState(const std::map<std::string, int>& counts);
00081     std::vector<StochasticReactionRule*> _collectRules() const;
00082
00083     void _forwardEntity(Entity* entity);
00084
00085 private:
00086         const struct DEFAULT_VALUES {
00087             std::string wholeCellStateName = "";
00088             double timeWindow = 1.0;
00089             bool advanceWholeCellClock = true;
00090             unsigned int randomSeed = 42u;
00091         } DEFAULT;
00092
```

```
00093     WholeCellState* _wholeCellState = nullptr;
00094     double _timeWindow = DEFAULT.timeWindow;
00095     bool _advanceWholeCellClock = DEFAULT.advanceWholeCellClock;
00096     unsigned int _randomSeed = DEFAULT.randomSeed;
00097     int _lastReactionCount = 0;
00098     double _lastSimulatedTime = 0.0;
00099
00100     std::mt19937 _rng{DEFAULT.randomSeed};
00101 };
00102
00103 #endif /* STOCHASTICREACTIONCOMPONENT_H */
```

18.475 StochasticTranscription.h

```
00001 #ifndef STOCHASTICTRANSCRIPTION_H
00002 #define STOCHASTICTRANSCRIPTION_H
00003
00004 #include <random>
00005 #include <string>
00006
00007 #include "../../../kernel/simulator/model/ModelComponent.h"
00008 #include "kernel/simulator/Plugin.h"
00009 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00010
00031 class StochasticTranscription : public ModelComponent {
00032 public:
00033     StochasticTranscription(Model* model, std::string name = "");
00034     virtual ~StochasticTranscription() override = default;
00035
00036 public:
00037     static PluginInformation* GetPluginInformation();
00038     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00039     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00040
00041 public:
00042     void setWholeCellState(WholeCellState* state);
00043     WholeCellState* getWholeCellState() const;
00044     void setElongationRate(double rate);
00045     double getElongationRate() const;
00046     void setMeanGeneLength(double length);
00047     double getMeanGeneLength() const;
00048     void setBindingProbability(double prob);
00049     double getBindingProbability() const;
00050     void setTimeWindow(double window);
00051     double getTimeWindow() const;
00052     void setMRNASpeciesPrefix(std::string prefix);
00053     std::string getMRNASpeciesPrefix() const;
00054     void setRnapCountKey(std::string key);
00055     std::string getRnapCountKey() const;
00056     void setRandomSeed(unsigned int seed);
00057     unsigned int getRandomSeed() const;
00058     int getLastSynthesizedCount() const;
00059
00060     virtual std::string show() override;
00061
00062 protected:
00063     virtual bool _loadInstance(PersistenceRecord* fields) override;
00064     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00065     virtual bool _check(std::string& errorMessage) override;
00066     virtual void _initBetweenReplications() override;
00067     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00068
00069 protected:
00070     virtual void _createEditableDataDefinitions() override;
00071
00072 private:
00073     void _forwardEntity(Entity* entity);
00074
00075 private:
00076     const struct DEFAULT_VALUES {
00077         std::string wholeCellStateName  = "";
00078         double elongationRate           = 50.0;   // nt/s -- M. genitalium RNA pol (parameters.json)
00079         double meanGeneLength           = 900.0;  // nt   -- approximate M. genitalium average
00080         double bindingProbability       = 1.0;    // dimensionless; scale to calibrate output
00081         double timeWindow               = 1.0;    // s
00082         std::string mRNASpeciesPrefix   = "mRNA_";
00083         std::string rnapCountKey        = "RNAP_free";
00084         unsigned int randomSeed         = 42u;
00085     } DEFAULT;
00086
00087     WholeCellState* _wholeCellState   = nullptr;
00088     double _elongationRate            = DEFAULT.elongationRate;
00089     double _meanGeneLength            = DEFAULT.meanGeneLength;
00090     double _bindingProbability        = DEFAULT.bindingProbability;
00091     double _timeWindow                = DEFAULT.timeWindow;
```

```
00092     std::string _mRNASpeciesPrefix    = DEFAULT.mRNASpeciesPrefix;
00093     std::string _rnapCountKey         = DEFAULT.rnapCountKey;
00094     unsigned int _randomSeed          = DEFAULT.randomSeed;
00095     int _lastSynthesizedCount         = 0;
00096     std::mt19937 _rng{DEFAULT.randomSeed};
00097 };
00098
00099 #endif /* STOCHASTICTRANSCRIPTION_H */
```

18.476 StochasticTranslation.h

```
00001 #ifndef STOCHASTICTRANSLATION_H
00002 #define STOCHASTICTRANSLATION_H
00003
00004 #include <random>
00005 #include <string>
00006
00007 #include "../../../kernel/simulator/model/ModelComponent.h"
00008 #include "kernel/simulator/Plugin.h"
00009 #include "plugins/data/WholeCellModeling/WholeCellState.h"
00010
00029 class StochasticTranslation : public ModelComponent {
00030 public:
00031     StochasticTranslation(Model* model, std::string name = "");
00032     virtual ~StochasticTranslation() override = default;
00033
00034 public:
00035     static PluginInformation* GetPluginInformation();
00036     static ModelComponent* LoadInstance(Model* model, PersistenceRecord* fields);
00037     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00038
00039 public:
00040     void setWholeCellState(WholeCellState* state);
00041     WholeCellState* getWholeCellState() const;
00042     void setElongationRate(double rate);
00043     double getElongationRate() const;
00044     void setMeanProteinLength(double length);
00045     double getMeanProteinLength() const;
00046     void setTimeWindow(double window);
00047     double getTimeWindow() const;
00048     void setMRNASpeciesPrefix(std::string prefix);
00049     std::string getMRNASpeciesPrefix() const;
00050     void setProteinSpeciesPrefix(std::string prefix);
00051     std::string getProteinSpeciesPrefix() const;
00052     void setRibosomeCountKey(std::string key);
00053     std::string getRibosomeCountKey() const;
00054     void setRandomSeed(unsigned int seed);
00055     unsigned int getRandomSeed() const;
00056     int getLastSynthesizedCount() const;
00057
00058     virtual std::string show() override;
00059
00060 protected:
00061     virtual bool _loadInstance(PersistenceRecord* fields) override;
00062     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00063     virtual bool _check(std::string& errorMessage) override;
00064     virtual void _initBetweenReplications() override;
00065     virtual void _onDispatchEvent(Entity* entity, unsigned int inputPortNumber) override;
00066
00067 protected:
00068     virtual void _createEditableDataDefinitions() override;
00069
00070 private:
00071     void _forwardEntity(Entity* entity);
00072
00073 private:
00074     const struct DEFAULT_VALUES {
00075         std::string wholeCellStateName  = "";
00076         double elongationRate           = 16.0;    // AA/s -- M. genitalium ribosome (parameters.json)
00077         double meanProteinLength        = 300.0;   // AA   -- approximate M. genitalium average
00078         double timeWindow               = 1.0;     // s
00079         std::string mRNASpeciesPrefix   = "mRNA_";
00080         std::string proteinSpeciesPrefix= "prot_";
00081         std::string ribosomeCountKey    = "ribosome_free";
00082         unsigned int randomSeed         = 43u;
00083     } DEFAULT;
00084
00085     WholeCellState* _wholeCellState     = nullptr;
00086     double _elongationRate              = DEFAULT.elongationRate;
00087     double _meanProteinLength           = DEFAULT.meanProteinLength;
00088     double _timeWindow                  = DEFAULT.timeWindow;
00089     std::string _mRNASpeciesPrefix      = DEFAULT.mRNASpeciesPrefix;
00090     std::string _proteinSpeciesPrefix   = DEFAULT.proteinSpeciesPrefix;
00091     std::string _ribosomeCountKey       = DEFAULT.ribosomeCountKey;
00092     unsigned int _randomSeed            = DEFAULT.randomSeed;
```

```
00093     int _lastSynthesizedCount           = 0;
00094     std::mt19937 _rng{DEFAULT.randomSeed};
00095 };
00096
00097 #endif /* STOCHASTICTRANSLATION_H */
```

18.477 AISupport.h

```
00001 #ifndef AISUPPORT_H
00002 #define AISUPPORT_H
00003
00004 #include "kernel/simulator/model/ModelDataDefinition.h"
00005 #include "kernel/simulator/PluginInformation.h"
00006 #include "tools/AIAssistant/AIConversationService.h"
00007
00008 class AISupport : public ModelDataDefinition {
00009 public:
00010     AISupport(Model* model, std::string name = "");
00011     virtual ~AISupport() override = default;
00012
00013     static PluginInformation* GetPluginInformation();
00014     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00015     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00016
00017     AIConversationResponse converse(const std::string& conversationKey, const std::string& prompt);
00018     void clearConversation(const std::string& conversationKey);
00019     void clearAllConversations();
00020
00021     void setProvider(std::string value);
00022     std::string getProvider() const;
00023     void setModel(std::string value);
00024     std::string getModel() const;
00025     void setBaseUrl(std::string value);
00026     std::string getBaseUrl() const;
00027     void setApiKeyEnvironmentVariable(std::string value);
00028     std::string getApiKeyEnvironmentVariable() const;
00029     void setSystemInstructions(std::string value);
00030     std::string getSystemInstructions() const;
00031     void setTemperature(double value);
00032     double getTemperature() const;
00033     void setMaxOutputTokens(unsigned int value);
00034     unsigned int getMaxOutputTokens() const;
00035     void setTimeoutSeconds(unsigned int value);
00036     unsigned int getTimeoutSeconds() const;
00037     void setHighReasoningMode(bool value);
00038     bool isHighReasoningMode() const;
00039     void setConversationEnabled(bool value);
00040     bool isConversationEnabled() const;
00041     void setMaxHistoryMessages(unsigned int value);
00042     unsigned int getMaxHistoryMessages() const;
00043     void setClearHistoryBetweenReplications(bool value);
00044     bool isClearHistoryBetweenReplications() const;
00045     unsigned long getCallCount() const;
00046     unsigned long getFailureCount() const;
00047
00048     virtual std::string show() override;
00049
00050 protected:
00051     virtual bool _loadInstance(PersistenceRecord* fields) override;
00052     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00053     virtual bool _check(std::string& errorMessage) override;
00054     virtual void _initBetweenReplications() override;
00055
00056 private:
00057     void _configurationChanged();
00058     AIConversationConfiguration _makeConfiguration() const;
00059
00060     const struct DEFAULT_VALUES {
00061         std::string provider = "Local";
00062         std::string model = "llama3.2";
00063         std::string baseUrl = "";
00064         std::string apiKeyEnvironmentVariable = "";
00065         std::string systemInstructions = "You are a decision service inside a GenESyS simulation.
      Return only valid JSON.";
00066         double temperature = 0.2;
00067         unsigned int maxOutputTokens = 512;
00068         unsigned int timeoutSeconds = 120;
00069         bool highReasoningMode = false;
00070         bool conversationEnabled = true;
00071         unsigned int maxHistoryMessages = 20;
00072         bool clearHistoryBetweenReplications = true;
00073     } DEFAULT;
00074
00075     std::string _provider = DEFAULT.provider;
00076     std::string _model = DEFAULT.model;
```

```
00077     std::string _baseUrl = DEFAULT.baseUrl;
00078     std::string _apiKeyEnvironmentVariable = DEFAULT.apiKeyEnvironmentVariable;
00079     std::string _systemInstructions = DEFAULT.systemInstructions;
00080     double _temperature = DEFAULT.temperature;
00081     unsigned int _maxOutputTokens = DEFAULT.maxOutputTokens;
00082     unsigned int _timeoutSeconds = DEFAULT.timeoutSeconds;
00083     bool _highReasoningMode = DEFAULT.highReasoningMode;
00084     bool _conversationEnabled = DEFAULT.conversationEnabled;
00085     unsigned int _maxHistoryMessages = DEFAULT.maxHistoryMessages;
00086     bool _clearHistoryBetweenReplications = DEFAULT.clearHistoryBetweenReplications;
00087     unsigned long _callCount = 0;
00088     unsigned long _failureCount = 0;
00089     AIConversationService _service;
00090 };
00091
00092 #endif
```

18.478 BacteriaSignalGrid.h

```
00001 /*
00002  * File:   BacteriaSignalGrid.h
00003  * Author: rlcancian
00004  *
00005  * Created on 28 de Abril de 2022
00006  */
00007
00008 #ifndef BACTERIASIGNALGRID_H
00009 #define BACTERIASIGNALGRID_H
00010
00011 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00012 #include "kernel/simulator/Plugin.h"
00013
00014 #include <string>
00015 #include <vector>
00016
00024 class BacteriaSignalGrid : public ModelDataDefinition {
00025 public:
00026     BacteriaSignalGrid(Model* model, std::string name = "");
00027     virtual ~BacteriaSignalGrid() override = default;
00028
00029 public:
00030     virtual std::string show() override;
00031
00032 public: // static plugin interface
00033     static PluginInformation* GetPluginInformation();
00034     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00035     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00036
00037 public:
00038     void setWidth(unsigned int width);
00039     unsigned int getWidth() const;
00040     void setHeight(unsigned int height);
00041     unsigned int getHeight() const;
00042     void setInitialSignal(double initialSignal);
00043     double getInitialSignal() const;
00044     void setDiffusionRate(double diffusionRate);
00045     double getDiffusionRate() const;
00046     void setDecayRate(double decayRate);
00047     double getDecayRate() const;
00048     void setInitialValues(std::string initialValues);
00049     std::string getInitialValues() const;
00050     unsigned int getCellCount() const;
00051     bool buildInitialField(std::vector<double>& values, std::string& errorMessage) const;
00052
00053 protected:
00054     virtual bool _loadInstance(PersistenceRecord* fields) override;
00055     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00056     virtual bool _check(std::string& errorMessage) override;
00057
00058 protected:
00059     // virtual void _createInternalStatisticReporters() override;
00060     // virtual void _createNonEditableDataDefinitions() override;
00061     // virtual void _createEditableDataDefinitions() override;
00062     // virtual void _createAttachedAttributes() override;
00063
00064 private:
00065     const struct DEFAULT_VALUES {
00066         const unsigned int width = 32;
00067         const unsigned int height = 32;
00068         const double initialSignal = 0.0;
00069         const double diffusionRate = 0.18;
00070         const double decayRate = 0.02;
00071         const std::string initialValues = "";
00072     } DEFAULT;
00073
```

```
00074     unsigned int _width = DEFAULT.width;
00075     unsigned int _height = DEFAULT.height;
00076     double _initialSignal = DEFAULT.initialSignal;
00077     double _diffusionRate = DEFAULT.diffusionRate;
00078     double _decayRate = DEFAULT.decayRate;
00079     std::string _initialValues = DEFAULT.initialValues;
00080 };
00081
00082 #endif /* BACTERIASIGNALGRID_H */
```

18.479 BioNetwork.h

```
00001 #ifndef BIONETWORK_H
00002 #define BIONETWORK_H
00003
00004 #include <string>
00005 #include <vector>
00006
00007 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00008 #include "kernel/simulator/PluginInformation.h"
00009 #include "tools/Biochemical/BioSimulationAnalysis.h"
00010 #include "tools/Biochemical/BioSimulationResult.h"
00011
00012 class BioReaction;
00013 class BioSpecies;
00014 class MassActionOdeSystem;
00015
00025 class BioNetwork : public ModelDataDefinition {
00026 public:
00027     BioNetwork(Model* model, std::string name = "");
00028     virtual ~BioNetwork() override = default;
00029
00030 public:
00031     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00032     static PluginInformation* GetPluginInformation();
00033     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00034
00035 public:
00036     void setStartTime(double startTime);
00037     double getStartTime() const;
00038     void setStopTime(double stopTime);
00039     double getStopTime() const;
00040     void setStepSize(double stepSize);
00041     double getStepSize() const;
00042     void setCurrentTime(double currentTime);
00043     double getCurrentTime() const;
00044     void setAutoSchedule(bool autoSchedule);
00045     bool getAutoSchedule() const;
00046     void setLastStatus(std::string lastStatus);
00047     std::string getLastStatus() const;
00048     void setLastErrorMessage(std::string lastErrorMessage);
00049     std::string getLastErrorMessage() const;
00050     void setLastResponsePayload(std::string lastResponsePayload);
00051     std::string getLastResponsePayload() const;
00052     void addSpecies(std::string speciesName);
00053     void addReaction(std::string reactionName);
00054     void clearSpecies();
00055     void clearReactions();
00056     const std::vector<std::string>& getSpeciesNames() const;
00057     const std::vector<std::string>& getReactionNames() const;
00058     const BioSimulationResult& getLastSimulationResult() const;
00059     bool getSpeciesTimeCourseDataset(const std::string& speciesName, SimulationResultsDataset*
    dataset,
00060                                      std::string* errorMessage = nullptr) const;
00061     bool buildOdeSystemForAnalysis(MassActionOdeSystem* system, std::string& errorMessage) const;
00062     bool getStoichiometryMatrix(BioStoichiometryMatrix* matrix, std::string* errorMessage = nullptr)
    const;
00063     bool getReactionRateTimeCourse(BioReactionRateTimeCourse* timeCourse, std::string* errorMessage =
    nullptr) const;
00064     bool checkLastSampleSteadyState(double tolerance, BioSteadyStateCheck* check, std::string*
    errorMessage = nullptr) const;
00065     bool scanLocalParameterSensitivity(double relativeStep, double absoluteStep, BioSensitivityScan*
    scan,
00066                                        std::string* errorMessage = nullptr) const;
00067
00068     bool simulate(std::string& errorMessage);
00069     bool simulate(double startTime, double stopTime, double stepSize, std::string& errorMessage);
00070     bool advanceOneStep(std::string& errorMessage);
00071
00072     virtual std::string show() override;
00073
00074 protected:
00075     virtual bool _loadInstance(PersistenceRecord *fields) override;
00076     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00077     virtual bool _check(std::string& errorMessage) override;
```

```
00078     virtual void _initBetweenReplications() override;
00079
00080
00081 protected:
00082     // virtual void _createInternalStatisticReporters() override;
00083     // virtual void _createNonEditableDataDefinitions() override;
00091     virtual void _createEditableDataDefinitions() override;
00092     // virtual void _createAttachedAttributes() override;
00093
00094 private:
00095     void handleInternalEvent(void* parameter);
00096     void scheduleNextInternalEvent();
00097     bool collectSpecies(std::vector<BioSpecies*>& species, std::string& errorMessage) const;
00098     bool collectReactions(std::vector<BioReaction*>& reactions, std::string& errorMessage) const;
00099     bool buildSystem(const std::vector<BioSpecies*>& species, const std::vector<BioReaction*>&
      reactions, MassActionOdeSystem* system, std::string& errorMessage) const;
00100     void updatePayload(const std::vector<BioSpecies*>& species);
00101     void configureSimulationResult(const std::vector<BioSpecies*>& species);
00102     void appendSimulationResultSample(const std::vector<BioSpecies*>& species);
00103
00104 private:
00105     const struct DEFAULT_VALUES {
00106         double startTime = 0.0;
00107         double stopTime = 1.0;
00108         double stepSize = 0.1;
00109         double currentTime = 0.0;
00110         bool autoSchedule = false;
00111         std::string lastStatus = "Idle";
00112         std::string lastErrorMessage = "";
00113         std::string lastResponsePayload = "";
00114     } DEFAULT;
00115
00116     double _startTime = DEFAULT.startTime;
00117     double _stopTime = DEFAULT.stopTime;
00118     double _stepSize = DEFAULT.stepSize;
00119     double _currentTime = DEFAULT.currentTime;
00120     bool _autoSchedule = DEFAULT.autoSchedule;
00121     std::string _lastStatus = DEFAULT.lastStatus;
00122     std::string _lastErrorMessage = DEFAULT.lastErrorMessage;
00123     std::string _lastResponsePayload = DEFAULT.lastResponsePayload;
00124     std::vector<std::string> _speciesNames;
00125     std::vector<std::string> _reactionNames;
00126     BioSimulationResult _lastSimulationResult;
00127 };
00128
00129 #endif /* BIONETWORK_H */
```

18.480 BioParameter.h

```
00001 #ifndef BIOPARAMETER_H
00002 #define BIOPARAMETER_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00007 #include "kernel/simulator/PluginInformation.h"
00008
00016 class BioParameter : public ModelDataDefinition {
00017 public:
00018     BioParameter(Model* model, std::string name = "");
00019     virtual ~BioParameter() override = default;
00020
00021 public:
00022     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00023     static PluginInformation* GetPluginInformation();
00024     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00025
00026 public:
00027     void setValue(double value);
00028     double getValue() const;
00029     void setUnit(std::string unit);
00030     std::string getUnit() const;
00031
00032     virtual std::string show() override;
00033
00034 protected:
00035     virtual bool _loadInstance(PersistenceRecord *fields) override;
00036     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00037     virtual bool _check(std::string& errorMessage) override;
00038
00039
00040 protected:
00041     // virtual void _createInternalStatisticReporters() override;
00042     // virtual void _createNonEditableDataDefinitions() override;
00043     // virtual void _createEditableDataDefinitions() override;
```

```
00044     // virtual void _createAttachedAttributes() override;
00045
00046 private:
00047     const struct DEFAULT_VALUES {
00048         double value = 0.0;
00049         std::string unit = "";
00050     } DEFAULT;
00051
00052     double _value = DEFAULT.value;
00053     std::string _unit = DEFAULT.unit;
00054 };
00055
00056 #endif /* BIOPARAMETER_H */
```

18.481 BioReaction.h

```
00001 #ifndef BIOREACTION_H
00002 #define BIOREACTION_H
00003
00004 #include <string>
00005 #include <vector>
00006
00007 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00008 #include "kernel/simulator/PluginInformation.h"
00009
00023 class BioReaction : public ModelDataDefinition {
00024 public:
00025     struct StoichiometricTerm {
00026         std::string speciesName;
00027         double stoichiometry = 1.0;
00028     };
00029
00030 public:
00036     BioReaction(Model* model, std::string name = "");
00040     virtual ~BioReaction() override = default;
00041
00042 public:
00049     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00054     static PluginInformation* GetPluginInformation();
00061     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00062
00063 public:
00069     void addReactant(std::string speciesName, double stoichiometry = 1.0);
00075     void addProduct(std::string speciesName, double stoichiometry = 1.0);
00080     void addModifier(std::string speciesName);
00084     void clearReactants();
00088     void clearProducts();
00092     void clearModifiers();
00097     const std::vector<StoichiometricTerm>& getReactants() const;
00102     const std::vector<StoichiometricTerm>& getProducts() const;
00107     const std::vector<std::string>& getModifiers() const;
00112     void setRateConstant(double rateConstant);
00117     double getRateConstant() const;
00122     void setRateConstantParameterName(std::string rateConstantParameterName);
00127     std::string getRateConstantParameterName() const;
00132     double resolveRateConstant() const;
00137     void setReverseRateConstant(double reverseRateConstant);
00142     double getReverseRateConstant() const;
00147     void setReverseRateConstantParameterName(std::string reverseRateConstantParameterName);
00152     std::string getReverseRateConstantParameterName() const;
00157     double resolveReverseRateConstant() const;
00162     void setKineticLawExpression(std::string kineticLawExpression);
00167     std::string getKineticLawExpression() const;
00172     void setReverseKineticLawExpression(std::string reverseKineticLawExpression);
00177     std::string getReverseKineticLawExpression() const;
00182     void setReversible(bool reversible);
00187     bool isReversible() const;
00188
00193     virtual std::string show() override;
00194
00195 protected:
00196     virtual bool _loadInstance(PersistenceRecord *fields) override;
00197     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00198     virtual bool _check(std::string& errorMessage) override;
00199     // virtual void _createInternalAndAttachedData() override;
00200
00201
00202 protected:
00203     // virtual void _createInternalStatisticReporters() override;
00204     // virtual void _createNonEditableDataDefinitions() override;
00205     // virtual void _createEditableDataDefinitions() override;
00206     virtual void _createAttachedAttributes() override;
00207
00208 private:
```

```
00209     bool checkTerms(const std::vector<StoichiometricTerm>& terms, const std::string& side,
      std::string& errorMessage) const;
00210     bool checkModifiers(std::string& errorMessage) const;
00211     bool validateKineticLawExpression(std::string& errorMessage) const;
00212     bool validateKineticLawExpression(const std::string& expression, const std::string& label,
      std::string& errorMessage) const;
00213     bool hasParticipantSpecies(const std::string& speciesName) const;
00214     bool resolveKineticLawSymbol(const std::string& symbolName, double& value) const;
00215     void syncParticipantAttachedSpecies();
00216
00217 private:
00218     const struct DEFAULT_VALUES {
00219         double rateConstant = 0.0;
00220         std::string rateConstantParameterName = "";
00221         double reverseRateConstant = 0.0;
00222         std::string reverseRateConstantParameterName = "";
00223         std::string kineticLawExpression = "";
00224         std::string reverseKineticLawExpression = "";
00225         bool reversible = false;
00226     } DEFAULT;
00227
00228     std::vector<StoichiometricTerm> _reactants;
00229     std::vector<StoichiometricTerm> _products;
00230     std::vector<std::string> _modifiers;
00231     double _rateConstant = DEFAULT.rateConstant;
00232     std::string _rateConstantParameterName = DEFAULT.rateConstantParameterName;
00233     double _reverseRateConstant = DEFAULT.reverseRateConstant;
00234     std::string _reverseRateConstantParameterName = DEFAULT.reverseRateConstantParameterName;
00235     std::string _kineticLawExpression = DEFAULT.kineticLawExpression;
00236     std::string _reverseKineticLawExpression = DEFAULT.reverseKineticLawExpression;
00237     bool _reversible = DEFAULT.reversible;
00238 };
00239
00240 #endif /* BIOREACTION_H */
```

18.482 BioSBMLBridge.h

```
00001 #ifndef BIOSBMLBRIDGE_H
00002 #define BIOSBMLBRIDGE_H
00003
00004 #include <string>
00005 #include <vector>
00006
00007 class Model;
00008
00009 struct BioSBMLImportResult {
00010     std::string networkName;
00011     unsigned int speciesImported = 0;
00012     unsigned int parametersImported = 0;
00013     unsigned int reactionsImported = 0;
00014     std::vector<std::string> warnings;
00015 };
00016
00017 struct BioSBMLExportResult {
00018     std::string networkName;
00019     unsigned int speciesExported = 0;
00020     unsigned int parametersExported = 0;
00021     unsigned int reactionsExported = 0;
00022     std::vector<std::string> warnings;
00023 };
00024
00025 class BioSBMLBridge {
00026 public:
00027     static bool importFromString(Model* model,
00028                                  const std::string& sbmlText,
00029                                  const std::string& preferredNetworkName,
00030                                  BioSBMLImportResult* result,
00031                                  std::string& errorMessage);
00032
00033     static bool exportToString(Model* model,
00034                                const std::string& requestedNetworkName,
00035                                std::string* sbmlText,
00036                                BioSBMLExportResult* result,
00037                                std::string& errorMessage);
00038 };
00039
00040 #endif /* BIOSBMLBRIDGE_H */
```

18.483 BioSimulatorRunner.h

```
00001 /*
00002  * File:   BioSimulatorRunner.h
```

```
00003  * Author: rlcancian
00004  *
00005  * Structural biochemical simulator runner data definition.
00006  */
00007
00008 #ifndef BIOSIMULATORRUNNER_H
00009 #define BIOSIMULATORRUNNER_H
00010
00011 #include <string>
00012
00013 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00014 #include "kernel/simulator/PluginInformation.h"
00015
00016 class BioSimulatorRunner : public ModelDataDefinition {
00017 public:
00018     BioSimulatorRunner(Model* model, std::string name = "");
00019     virtual ~BioSimulatorRunner() = default;
00020
00021 public:
00022     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00023     static PluginInformation* GetPluginInformation();
00024     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00025
00026 public:
00027     void setBackend(std::string backend);
00028     std::string getBackend() const;
00029     void setModelSourceType(std::string modelSourceType);
00030     std::string getModelSourceType() const;
00031     void setModelSource(std::string modelSource);
00032     std::string getModelSource() const;
00033     void setCommand(std::string command);
00034     std::string getCommand() const;
00035     void setLastStatus(std::string lastStatus);
00036     std::string getLastStatus() const;
00037     void setLastErrorMessage(std::string lastErrorMessage);
00038     std::string getLastErrorMessage() const;
00039     void setLastResponsePayload(std::string lastResponsePayload);
00040     std::string getLastResponsePayload() const;
00041     void setLastResponseFilename(std::string lastResponseFilename);
00042     std::string getLastResponseFilename() const;
00043     void setTargetBioNetworkName(std::string targetBioNetworkName);
00044     std::string getTargetBioNetworkName() const;
00045     void setWorkingDirectory(std::string workingDirectory);
00046     std::string getWorkingDirectory() const;
00047     void setWorkingInputFilename(std::string workingInputFilename);
00048     std::string getWorkingInputFilename() const;
00049     void setWorkingOutputFilename(std::string workingOutputFilename);
00050     std::string getWorkingOutputFilename() const;
00051     void setEndpointOrLibrary(std::string endpointOrLibrary);
00052     std::string getEndpointOrLibrary() const;
00053     void setTimeoutSeconds(unsigned int timeoutSeconds);
00054     unsigned int getTimeoutSeconds() const;
00055     void setAutoValidateModel(bool autoValidateModel);
00056     bool getAutoValidateModel() const;
00057
00058     bool executeCommand(std::string& errorMessage);
00059
00060     virtual std::string show() override;
00061
00062 protected:
00063     virtual bool _loadInstance(PersistenceRecord *fields) override;
00064     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00065     virtual bool _check(std::string& errorMessage) override;
00066     // virtual void _createInternalAndAttachedData() override;
00067
00068
00069 protected:
00070     // virtual void _createInternalStatisticReporters() override;
00071     // virtual void _createNonEditableDataDefinitions() override;
00072     // virtual void _createEditableDataDefinitions() override;
00073     // virtual void _createAttachedAttributes() override;
00074
00075 private:
00076     const struct DEFAULT_VALUES {
00077         std::string backend = "RoadRunnerEmbedded";
00078         std::string modelSourceType = "SBMLString";
00079         std::string modelSource = "";
00080         std::string command = "";
00081         std::string lastStatus = "Idle";
00082         std::string lastErrorMessage = "";
00083         std::string lastResponsePayload = "";
00084         std::string lastResponseFilename = "";
00085         std::string targetBioNetworkName = "";
00086         std::string workingDirectory = "";
00087         std::string workingInputFilename = "biosim_input.xml";
00088         std::string workingOutputFilename = "biosim_output.json";
00089         std::string endpointOrLibrary = "";
```

```
00090         unsigned int timeoutSeconds = 30;
00091         bool autoValidateModel = true;
00092     } DEFAULT;
00093 
00094     std::string _backend = DEFAULT.backend;
00095     std::string _modelSourceType = DEFAULT.modelSourceType;
00096     std::string _modelSource = DEFAULT.modelSource;
00097     std::string _command = DEFAULT.command;
00098     std::string _lastStatus = DEFAULT.lastStatus;
00099     std::string _lastErrorMessage = DEFAULT.lastErrorMessage;
00100     std::string _lastResponsePayload = DEFAULT.lastResponsePayload;
00101     std::string _lastResponseFilename = DEFAULT.lastResponseFilename;
00102     std::string _targetBioNetworkName = DEFAULT.targetBioNetworkName;
00103     std::string _workingDirectory = DEFAULT.workingDirectory;
00104     std::string _workingInputFilename = DEFAULT.workingInputFilename;
00105     std::string _workingOutputFilename = DEFAULT.workingOutputFilename;
00106     std::string _endpointOrLibrary = DEFAULT.endpointOrLibrary;
00107     unsigned int _timeoutSeconds = DEFAULT.timeoutSeconds;
00108     bool _autoValidateModel = DEFAULT.autoValidateModel;
00109 };
00110 
00111 #endif /* BIOSIMULATORRUNNER_H */
```

18.484 BioSpecies.h

```
00001 #ifndef BIOSPECIES_H
00002 #define BIOSPECIES_H
00003 
00004 #include <string>
00005 
00006 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00007 #include "kernel/simulator/PluginInformation.h"
00008 
00017 class BioSpecies : public ModelDataDefinition {
00018 public:
00019     BioSpecies(Model* model, std::string name = "");
00020     virtual ~BioSpecies() override = default;
00021 
00022 public:
00023     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00024     static PluginInformation* GetPluginInformation();
00025     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00026 
00027 public:
00028     void setInitialAmount(double initialAmount);
00029     double getInitialAmount() const;
00030     void setAmount(double amount);
00031     double getAmount() const;
00032     void setConstant(bool constant);
00033     bool isConstant() const;
00034     void setBoundaryCondition(bool boundaryCondition);
00035     bool isBoundaryCondition() const;
00036     void setUnit(std::string unit);
00037     std::string getUnit() const;
00038 
00039     virtual std::string show() override;
00040 
00041 protected:
00042     virtual bool _loadInstance(PersistenceRecord *fields) override;
00043     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00044     virtual bool _check(std::string& errorMessage) override;
00045     virtual void _initBetweenReplications() override;
00046 
00047 
00048 protected:
00049     // virtual void _createInternalStatisticReporters() override;
00050     // virtual void _createNonEditableDataDefinitions() override;
00051     // virtual void _createEditableDataDefinitions() override;
00052     // virtual void _createAttachedAttributes() override;
00053 
00054 private:
00055     const struct DEFAULT_VALUES {
00056         double initialAmount = 0.0;
00057         double amount = 0.0;
00058         bool constant = false;
00059         bool boundaryCondition = false;
00060         std::string unit = "";
00061     } DEFAULT;
00062 
00063     double _initialAmount = DEFAULT.initialAmount;
00064     double _amount = DEFAULT.amount;
00065     bool _constant = DEFAULT.constant;
00066     bool _boundaryCondition = DEFAULT.boundaryCondition;
00067     std::string _unit = DEFAULT.unit;
00068 };
```

```
00069
00070 #endif /* BIOSPECIES_H */
```

18.485 GeneticCircuit.h

```
00001 #ifndef GENETICCIRCUIT_H
00002 #define GENETICCIRCUIT_H
00003
00004 #include <string>
00005 #include <vector>
00006
00007 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00008 #include "kernel/simulator/PluginInformation.h"
00009
00017 class GeneticCircuit : public ModelDataDefinition {
00018 public:
00019     GeneticCircuit(Model* model, std::string name = "");
00020     virtual ~GeneticCircuit() override = default;
00021
00022 public:
00023     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00024     static PluginInformation* GetPluginInformation();
00025     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00026
00027 public:
00028     void addPart(std::string partName);
00029     void addRegulation(std::string regulationName);
00030     void clearParts();
00031     void clearRegulations();
00032     const std::vector<std::string>& getPartNames() const;
00033     const std::vector<std::string>& getRegulationNames() const;
00034     void setHostOrganism(std::string hostOrganism);
00035     std::string getHostOrganism() const;
00036     void setCompartment(std::string compartment);
00037     std::string getCompartment() const;
00038     void setEnabled(bool enabled);
00039     bool isEnabled() const;
00040
00041     virtual std::string show() override;
00042
00043 protected:
00044     virtual bool _loadInstance(PersistenceRecord* fields) override;
00045     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00046     virtual bool _check(std::string& errorMessage) override;
00047
00048 protected:
00049     // virtual void _createInternalStatisticReporters() override;
00050     // virtual void _createNonEditableDataDefinitions() override;
00051     // virtual void _createEditableDataDefinitions() override;
00052     // virtual void _createAttachedAttributes() override;
00053
00054 private:
00055     bool _checkPartNames(std::string& errorMessage) const;
00056     bool _checkRegulationNames(std::string& errorMessage) const;
00057
00058 private:
00059     const struct DEFAULT_VALUES {
00060         std::string hostOrganism = "";
00061         std::string compartment = "";
00062         bool enabled = true;
00063     } DEFAULT;
00064
00065     std::vector<std::string> _partNames;
00066     std::vector<std::string> _regulationNames;
00067     std::string _hostOrganism = DEFAULT.hostOrganism;
00068     std::string _compartment = DEFAULT.compartment;
00069     bool _enabled = DEFAULT.enabled;
00070 };
00071
00072 #endif /* GENETICCIRCUIT_H */
```

18.486 GeneticCircuitPart.h

```
00001 #ifndef GENETICCIRCUITPART_H
00002 #define GENETICCIRCUITPART_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00007 #include "kernel/simulator/PluginInformation.h"
00008
00016 class GeneticCircuitPart : public ModelDataDefinition {
```

```
00017 public:
00018     GeneticCircuitPart(Model* model, std::string name = "");
00019     virtual ~GeneticCircuitPart() override = default;
00020 
00021 public:
00022     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00023     static PluginInformation* GetPluginInformation();
00024     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00025 
00026 public:
00027     void setPartType(std::string partType);
00028     std::string getPartType() const;
00029     void setSequence(std::string sequence);
00030     std::string getSequence() const;
00031     void setProductSpeciesName(std::string productSpeciesName);
00032     std::string getProductSpeciesName() const;
00033     void setCopyNumber(double copyNumber);
00034     double getCopyNumber() const;
00035     void setBasalExpressionRate(double basalExpressionRate);
00036     double getBasalExpressionRate() const;
00037     void setDegradationRate(double degradationRate);
00038     double getDegradationRate() const;
00039     void setEnabled(bool enabled);
00040     bool isEnabled() const;
00041 
00042     virtual std::string show() override;
00043 
00044 protected:
00045     virtual bool _loadInstance(PersistenceRecord* fields) override;
00046     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00047     virtual bool _check(std::string& errorMessage) override;
00048 
00049 protected:
00050     // virtual void _createInternalStatisticReporters() override;
00051     // virtual void _createNonEditableDataDefinitions() override;
00052     // virtual void _createEditableDataDefinitions() override;
00053     // virtual void _createAttachedAttributes() override;
00054 
00055 private:
00056     const struct DEFAULT_VALUES {
00057         std::string partType = "Promoter";
00058         std::string sequence = "";
00059         std::string productSpeciesName = "";
00060         double copyNumber = 1.0;
00061         double basalExpressionRate = 0.0;
00062         double degradationRate = 0.0;
00063         bool enabled = true;
00064     } DEFAULT;
00065 
00066     std::string _partType = DEFAULT.partType;
00067     std::string _sequence = DEFAULT.sequence;
00068     std::string _productSpeciesName = DEFAULT.productSpeciesName;
00069     double _copyNumber = DEFAULT.copyNumber;
00070     double _basalExpressionRate = DEFAULT.basalExpressionRate;
00071     double _degradationRate = DEFAULT.degradationRate;
00072     bool _enabled = DEFAULT.enabled;
00073 };
00074 
00075 #endif /* GENETICCIRCUITPART_H */
```

18.487 GeneticRegulation.h

```
00001 #ifndef GENETICREGULATION_H
00002 #define GENETICREGULATION_H
00003 
00004 #include <string>
00005 
00006 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00007 #include "kernel/simulator/PluginInformation.h"
00008 
00012 class GeneticRegulation : public ModelDataDefinition {
00013 public:
00014     GeneticRegulation(Model* model, std::string name = "");
00015     virtual ~GeneticRegulation() override = default;
00016 
00017 public:
00018     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00019     static PluginInformation* GetPluginInformation();
00020     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00021 
00022 public:
00023     void setRegulatorSpeciesName(std::string regulatorSpeciesName);
00024     std::string getRegulatorSpeciesName() const;
00025     void setTargetPartName(std::string targetPartName);
00026     std::string getTargetPartName() const;
```

```
00027     void setRegulationType(std::string regulationType);
00028     std::string getRegulationType() const;
00029     void setHillCoefficient(double hillCoefficient);
00030     double getHillCoefficient() const;
00031     void setDissociationConstant(double dissociationConstant);
00032     double getDissociationConstant() const;
00033     void setMaxFoldChange(double maxFoldChange);
00034     double getMaxFoldChange() const;
00035     void setLeakiness(double leakiness);
00036     double getLeakiness() const;
00037     void setEnabled(bool enabled);
00038     bool isEnabled() const;
00039 
00040     virtual std::string show() override;
00041 
00042 protected:
00043     virtual bool _loadInstance(PersistenceRecord* fields) override;
00044     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00045     virtual bool _check(std::string& errorMessage) override;
00046 
00047 protected:
00048     // virtual void _createInternalStatisticReporters() override;
00049     // virtual void _createNonEditableDataDefinitions() override;
00050     // virtual void _createEditableDataDefinitions() override;
00051     // virtual void _createAttachedAttributes() override;
00052 
00053 private:
00054     const struct DEFAULT_VALUES {
00055         std::string regulatorSpeciesName = "";
00056         std::string targetPartName = "";
00057         std::string regulationType = "Activation";
00058         double hillCoefficient = 1.0;
00059         double dissociationConstant = 1.0;
00060         double maxFoldChange = 1.0;
00061         double leakiness = 0.0;
00062         bool enabled = true;
00063     } DEFAULT;
00064 
00065     std::string _regulatorSpeciesName = DEFAULT.regulatorSpeciesName;
00066     std::string _targetPartName = DEFAULT.targetPartName;
00067     std::string _regulationType = DEFAULT.regulationType;
00068     double _hillCoefficient = DEFAULT.hillCoefficient;
00069     double _dissociationConstant = DEFAULT.dissociationConstant;
00070     double _maxFoldChange = DEFAULT.maxFoldChange;
00071     double _leakiness = DEFAULT.leakiness;
00072     bool _enabled = DEFAULT.enabled;
00073 };
00074 
00075 #endif /* GENETICREGULATION_H */
```

18.488 GroProgram.h

```
00001 /*
00002  * File:   GroProgram.h
00003  * Author: rlcancian
00004  *
00005  * Created on 17 de Abril de 2022
00006  */
00007 
00008 #ifndef GROPROGRAM_H
00009 #define GROPROGRAM_H
00010 
00011 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00012 #include "kernel/simulator/Plugin.h"
00013 #include <string>
00014 
00023 class GroProgram : public ModelDataDefinition {
00024 public:
00025     GroProgram(Model* model, std::string name = "");
00026     virtual ~GroProgram() override = default;
00027 
00028 public:
00029     virtual std::string show() override;
00030 
00031 public: // static plugin interface
00032     static PluginInformation* GetPluginInformation();
00033     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00034     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00035 
00036 public:
00038     void setSourceCode(std::string sourceCode);
00040     void setSourceCodeProperty(SourceCodeString sourceCode);
00042     std::string getSourceCode() const;
00044     SourceCodeString getSourceCodeProperty() const;
00046     bool createDefaultGroProgram(const std::string& filename = "");
```

```
00048     bool validateSyntax(std::string& errorMessage) const;
00049
00050 protected:
00051     virtual bool _loadInstance(PersistenceRecord* fields) override;
00052     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00053     virtual bool _check(std::string& errorMessage) override;
00054
00055
00056 protected:
00057     // virtual void _createInternalStatisticReporters() override;
00058     // virtual void _createNonEditableDataDefinitions() override;
00059     // virtual void _createEditableDataDefinitions() override;
00060     // virtual void _createAttachedAttributes() override;
00061
00062 private:
00063     bool _validateSourceCodeSyntax(const std::string& sourceCode, std::string& errorMessage) const;
00064
00065     private:
00066     const struct DEFAULT_VALUES {
00067         const std::string sourceCode = R"(
00068 include gro
00069 set ( "dt", 0.1 );
00070 set ( "population_max", 2000 );
00071 program p() := {
00072   skip();
00073 };
00074 program q() := {
00075   set ( "ecoli_growth_rate", 0.1 );
00076   set ( "ecoli_division_size_mean", 2.0 );
00077   set ( "ecoli_division_size_variance", 0.2 );
00078 };
00079 program r() := {
00080   selected : { message ( 1, tostring ( volume ) ) }
00081 };
00082 program s() := {
00083   set ( "ecoli_division_size_mean", 1000 ); // essentially turns off gro's division machinery
00084   rate(1) & volume > 3.14 : {
00085     divide()
00086   }
00087 };
00088 ecoli ( [ x := 0, y := 0 ], program q() ); // Try p(), q(), r() or s() here. They are all a bit
      different.
00089
00090 //----------------
00091 set ( "dt", 0.1 );
00092 set ( "ecoli_growth_rate", 0.1 );
00093
00094 s0 := signal ( 1, 0.2 ); // create two new signals
00095 s1 := signal ( 1, 0.2 );
00096
00097 // These functions provide various growth rates in different states. a and b
00098 // are the intensities of signaling molecules.
00099 fun on a b .  0.1;
00100 fun off a b . 0.0;
00101 fun and a b . if a > 0.1 & b > 0.1 then 0.1 else 0.0 end;
00102
00103 //
00104 // The state program. Parameters are
00105 //
00106 //   this: the state index
00107 //   m_next, d_next: states to go to after division if mother or daughter cell
00108 //   t_next: state to go to after timer is up
00109 //   tf: timer length (use -1 for infinity)
00110 //   gr: growth rate function
00111 //   sigs: rates to emit signals
00112 //
00113 program state ( this, m_next, d_next, t_next, tf, gr, sigs ) := {
00114
00115   needs q, t;
00116   active := false;
00117   needs event;
00118
00119   true : { active := ( q = this ); }
00120
00121   !event & active & just_divided & !daughter & q != m_next : { q := m_next, t := 0, event := true }
00122   !event & active & just_divided & daughter  & q != d_next:  { q := d_next, t := 0, event := true }
00123   !event & active & tf > 0 & t > tf & q != t_next :          { q := t_next, t := 0, event := true }
00124
00125   active : {
00126     set ( "ecoli_growth_rate", gr ( get_signal ( 0 ) ) ( get_signal ( 1 ) ) ),
00127     emit_signal ( 0, sigs[0] ),
00128     emit_signal ( 1, sigs[1] )
00129   }
00130
00131 };
00132
00133 //
```

```
00134 // State machine program: coordinates states.
00135 //
00136 program sm() := {
00137
00138   q := 0;
00139   t := 0;
00140   event := false;
00141
00142   true : { t := t + dt, event := false }
00143
00144 };
00145
00146 //
00147 // The morphogenesis state machine example.
00148 //
00149 program p() := sm() + state ( 0, 1, 2, 0, -1, on, { 0, 0 } )  sharing q, t, event
00150                     + state ( 1, 1, 3, 5, 60, on, { 50, 0 } ) sharing q, t, event
00151                     + state ( 2, 2, 3, 6, 60, on, { 0, 50 } ) sharing q, t, event
00152                     + state ( 3, 3, 3, 3, -1, and, { 0, 0 } ) sharing q, t, event
00153                     + state ( 5, 5, 5, 7, 40, on, { 0, 0 } )  sharing q, t, event
00154                     + state ( 6, 6, 6, 8, 40, on, { 0, 0 } )  sharing q, t, event
00155                     + state ( 7, 7, 7, 7,  0, off, { 0, 0 } ) sharing q, t, event
00156                     + state ( 8, 8, 8, 8,  0, off, { 0, 0 } ) sharing q, t, event;
00157
00158 // A helper function.
00159 fun test x y . if x = y then 1 else 0 end;
00160
00161 //
00162 // Makes cells in different states report different colors.
00163 //
00164 program report() := {
00165
00166   gfp := 0; rfp := 0; yfp := 0; cfp := 0;
00167   needs q;
00168
00169   true : {
00170     rfp := 100 * test ( q ) 3;
00171     yfp := 100 * ( test ( q ) 1 + test ( q ) 2 ),
00172     cfp := 100 * ( test ( q ) 5 + test ( q ) 7 );
00173     gfp := 100 * ( test ( q ) 6 + test ( q ) 8 );
00174   }
00175
00176   selected : { message ( 2, tostring ( { id, q } ) ) }
00177
00178 };
00179
00180 //
00181 // Hides the state variable in all() -- see below -- so that it is not cut in half when the
00182 // cell divides.
00183 //
00184 program hide() := {
00185
00186   needs gfp;
00187   needs rfp;
00188   needs yfp;
00189   needs cfp;
00190
00191   skip();
00192
00193 };
00194
00195 //
00196 // All the programs stuck together
00197 //
00198 program all() := p() + report() sharing q, gfp, rfp, yfp, cfp + hide() sharing gfp, rfp, yfp, cfp;
00199
00200 //
00201 // The main program makes the simulation loop
00202 //
00203 program main() := {
00204
00205   t := 1;
00206
00207   true : { t := t + dt }
00208
00209   t > 120 : {
00210     reset(),
00211     ecoli ( [], program all() ),
00212     t := 0
00213   }
00214
00215 };
00216
00217 // associate the program with a cell
00218 ecoli ( [], program all() );
00219
00220
```

```
00221
00222 )";
00223
00224     } DEFAULT;
00225
00226     std::string _sourceCode = DEFAULT.sourceCode;
00227 };
00228
00229 #endif /* GROPROGRAM_H */
00230
00231
00232 /*
00233
00234
00235
00236 //
00237 // gro is protected by the UW OPEN SOURCE LICENSE, which is summarized here.
00238 // Please see the file LICENSE.txt for the complete license.
00239 //
00240 // THE SOFTWARE (AS DEFINED BELOW) AND HARDWARE DESIGNS (AS DEFINED BELOW) IS PROVIDED
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00247 // CONSIDERED A CONTRACT, THE UNIVERSITY OF WASHINGTON ("UW") GRANTS YOU THE RIGHTS
00248 // CONTAINED HERE IN CONSIDERATION OF YOUR ACCEPTANCE OF SUCH TERMS AND CONDITIONS.
00249 //
00250 // TERMS AND CONDITIONS FOR USE, REPRODUCTION, AND DISTRIBUTION
00251 //
00252
00253 include gro
00254
00255 set ( "dt", 0.1 );
00256 set ( "ecoli_growth_rate", 0.1 );
00257
00258 s0 := signal ( 1, 0.2 ); // create two new signals
00259 s1 := signal ( 1, 0.2 );
00260
00261 // These functions provide various growth rates in different states. a and b
00262 // are the intensities of signaling molecules.
00263 fun on a b .  0.1;
00264 fun off a b . 0.0;
00265 fun and a b . if a > 0.1 & b > 0.1 then 0.1 else 0.0 end;
00266
00267 //
00268 // The state program. Parameters are
00269 //
00270 //   this: the state index
00271 //   m_next, d_next: states to go to after division if mother or daughter cell
00272 //   t_next: state to go to after timer is up
00273 //   tf: timer length (use -1 for infinity)
00274 //   gr: growth rate function
00275 //   sigs: rates to emit signals
00276 //
00277 program state ( this, m_next, d_next, t_next, tf, gr, sigs ) := {
00278
00279   needs q, t;
00280   active := false;
00281   needs event;
00282
00283   true : { active := ( q = this ); }
00284
00285   !event & active & just_divided & !daughter & q != m_next : { q := m_next, t := 0, event := true }
00286   !event & active & just_divided & daughter  & q != d_next:  { q := d_next, t := 0, event := true }
00287   !event & active & tf > 0 & t > tf & q != t_next :          { q := t_next, t := 0, event := true }
00288
00289   active : {
00290     set ( "ecoli_growth_rate", gr ( get_signal ( 0 ) ) ( get_signal ( 1 ) ) ),
00291     emit_signal ( 0, sigs[0] ),
00292     emit_signal ( 1, sigs[1] )
00293   }
00294
00295 };
00296
00297 //
00298 // State machine program: coordinates states.
00299 //
00300 program sm() := {
00301
00302   q := 0;
00303   t := 0;
00304   event := false;
00305
00306   true : { t := t + dt, event := false }
00307
```

```
00308 };
00309
00310 //
00311 // The morphogenesis state machine example.
00312 //
00313 program p() := sm() + state ( 0, 1, 2, 0, -1, on, { 0, 0 } )  sharing q, t, event
00314                     + state ( 1, 1, 3, 5, 60, on, { 50, 0 } ) sharing q, t, event
00315                     + state ( 2, 2, 3, 6, 60, on, { 0, 50 } ) sharing q, t, event
00316                     + state ( 3, 3, 3, 3, -1, and, { 0, 0 } ) sharing q, t, event
00317                     + state ( 5, 5, 5, 7, 40, on, { 0, 0 } )  sharing q, t, event
00318                     + state ( 6, 6, 6, 8, 40, on, { 0, 0 } )  sharing q, t, event
00319                     + state ( 7, 7, 7, 7,  0, off, { 0, 0 } ) sharing q, t, event
00320                     + state ( 8, 8, 8, 8,  0, off, { 0, 0 } ) sharing q, t, event;
00321
00322 // A helper function.
00323 fun test x y . if x = y then 1 else 0 end;
00324
00325 //
00326 // Makes cells in different states report different colors.
00327 //
00328 program report() := {
00329
00330   gfp := 0; rfp := 0; yfp := 0; cfp := 0;
00331   needs q;
00332
00333   true : {
00334     rfp := 100 * test ( q ) 3;
00335     yfp := 100 * ( test ( q ) 1 + test ( q ) 2 ),
00336     cfp := 100 * ( test ( q ) 5 + test ( q ) 7 );
00337     gfp := 100 * ( test ( q ) 6 + test ( q ) 8 );
00338   }
00339
00340   selected : { message ( 2, tostring ( { id, q } ) ) }
00341
00342 };
00343
00344 //
00345 // Hides the state variable in all() -- see below -- so that it is not cut in half when the
00346 // cell divides.
00347 //
00348 program hide() := {
00349
00350   needs gfp;
00351   needs rfp;
00352   needs yfp;
00353   needs cfp;
00354
00355   skip();
00356
00357 };
00358
00359 //
00360 // All the programs stuck together
00361 //
00362 program all() := p() + report() sharing q, gfp, rfp, yfp, cfp + hide() sharing gfp, rfp, yfp, cfp;
00363
00364 //
00365 // The main program makes the simulation loop
00366 //
00367 program main() := {
00368
00369   t := 1;
00370
00371   true : { t := t + dt }
00372
00373   t > 120 : {
00374     reset(),
00375     ecoli ( [], program all() ),
00376     t := 0
00377   }
00378
00379 };
00380
00381 // associate the program with a cell
00382 ecoli ( [], program all() );
00383 */
00384
00385 /*
00386 include gro
00387 set ("dt",0.1);
00388 s := signal (1,1.05);
00389 k0 := 2.5;  // base rate of oscillation
00390 kb := 5;    // cell-cell feedback strength
00391 tr := 10;   // refractory period length
00392 se := 650;  // signal emit magnitude
00393 GO := 0;
00394 WAIT := 1;
```

```
00395 program oscillator(g0) := {
00396 gfp := 0.5*volume*g0;
00397 p := [mode := GO, t := 0, x := g0];
00398 true : {gfp := 0.5*volume*(g0 + p.x/150)}
00399 // advance the oscillation phase
00400 p.mode = GO & rate (k0) : {p.x := p.x + 0.01*(150-p.x)}
00401 p.mode = GO & rate(kb*get_signal(s)) : {p.x := p.x + 0.01*(150-p.x)}
00402 // phase reset
00403 p.x > 100 : {
00404 p.x := 0,
00405 p.mode := WAIT,
00406 emit_signal(s, se)
00407 }
00408 // refractory period timer
00409 p.mode = WAIT : {p.t := p.t+dt}
00410 p.mode = WAIT & p.t > tr : {
00411 p.mode := GO,
00412 p.t := 0
00413 }
00414 };
00415 ecoli ( [x:=0, y:=0], program oscillator(1));
00416 program main() := {
00417 //n := 0;
00418 //true : {snapshot("movie/oscillator/" <> tostring(n) <> ".tif"), n:=n+1}
00419 skip();
00420 };
00421 */
```

18.489 GroProgramAst.h

```
00001 /*
00002  * File:   GroProgramAst.h
00003  * Author: rlcancian
00004  *
00005  * Created on 17 de Abril de 2022
00006  */
00007
00008 #ifndef GROPROGRAMAST_H
00009 #define GROPROGRAMAST_H
00010
00011 #include <string>
00012 #include <vector>
00013
00021 struct GroProgramAst {
00022     enum class SourceForm {
00023         RawStatements,
00024         ProgramBlock
00025     };
00026
00027     struct Statement {
00028         std::string sourceText = "";
00029     };
00030
00031     struct NamedProgram {
00032         std::string name = "";
00033         std::vector<std::string> parameters;
00034         std::string bodySource = "";
00035         std::vector<Statement> statements;
00036     };
00037
00038     SourceForm sourceForm = SourceForm::RawStatements;
00039     std::string programName = "";
00040     std::vector<std::string> programParameters;
00041     std::string bodySource = "";
00042     std::vector<Statement> statements;
00043     std::vector<NamedProgram> namedPrograms;
00044
00045     bool isProgramBlock() const {
00046         return sourceForm == SourceForm::ProgramBlock;
00047     }
00048
00049     bool hasNamedPrograms() const {
00050         return !namedPrograms.empty();
00051     }
00052 };
00053
00054 #endif /* GROPROGRAMAST_H */
```

18.490 GroProgramCompiler.h

```
00001 /*
00002  * File:   GroProgramCompiler.h
```

```
00003  * Author: rlcancian
00004  *
00005  * Created on 17 de Abril de 2022
00006  */
00007
00008 #ifndef GROPROGRAMCOMPILER_H
00009 #define GROPROGRAMCOMPILER_H
00010
00011 #include "plugins/data/BiochemicalSimulation/GroProgramAst.h"
00012 #include "plugins/data/BiochemicalSimulation/GroProgramIr.h"
00013
00017 class GroProgramCompiler {
00018 public:
00020     GroProgramIr compile(const GroProgramAst& ast) const;
00021 };
00022
00023 #endif /* GROPROGRAMCOMPILER_H */
```

18.491 GroProgramIr.h

```
00001 /*
00002  * File:   GroProgramIr.h
00003  * Author: rlcancian
00004  *
00005  * Created on 17 de Abril de 2022
00006  */
00007
00008 #ifndef GROPROGRAMIR_H
00009 #define GROPROGRAMIR_H
00010
00011 #include "plugins/data/BiochemicalSimulation/GroProgramAst.h"
00012
00013 #include <map>
00014 #include <string>
00015 #include <vector>
00016
00024 struct GroProgramIr {
00025     struct Command {
00026         enum class Kind {
00027             RawStatement,
00028             FunctionCall,
00029             Assignment,
00030             IfStatement
00031         };
00032
00033         Kind kind = Kind::RawStatement;
00034         std::string sourceText = "";
00035         std::string functionName = "";
00036         std::vector<std::string> arguments;
00037         std::string assignmentTarget = "";
00038         std::string expressionText = "";
00039         bool assignmentOnlyIfUnset = false;
00040         std::vector<Command> thenCommands;
00041         std::vector<Command> elseCommands;
00042
00043         bool isFunctionCall() const {
00044             return kind == Kind::FunctionCall;
00045         }
00046
00047         bool isAssignment() const {
00048             return kind == Kind::Assignment;
00049         }
00050
00051         bool isIfStatement() const {
00052             return kind == Kind::IfStatement;
00053         }
00054     };
00055
00056     struct NamedProgramDefinition {
00057         std::vector<std::string> parameters;
00058         std::string sourceText = "";
00059         std::vector<Command> commands;
00060     };
00061
00062     GroProgramAst::SourceForm sourceForm = GroProgramAst::SourceForm::RawStatements;
00063     std::string programName = "";
00064     std::vector<Command> commands;
00065     std::map<std::string, NamedProgramDefinition> namedPrograms;
00066
00067     bool isProgramBlock() const {
00068         return sourceForm == GroProgramAst::SourceForm::ProgramBlock;
00069     }
00070
00071     bool hasNamedProgram(const std::string& name) const {
00072         return namedPrograms.find(name) != namedPrograms.end();
```

```
00073     }
00074 };
00075
00076 #endif /* GROPROGRAMIR_H */
```

18.492 GroProgramParser.h

```
00001 /*
00002  * File:   GroProgramParser.h
00003  * Author: rlcancian
00004  *
00005  * Created on 17 de Abril de 2022
00006  */
00007
00008 #ifndef GROPROGRAMPARSER_H
00009 #define GROPROGRAMPARSER_H
00010
00011 #include "plugins/data/BiochemicalSimulation/GroProgramAst.h"
00012
00013 #include <string>
00014
00022 class GroProgramParser {
00023 public:
00024     struct Result {
00025         bool accepted = false;
00026         std::string errorMessage = "";
00027         GroProgramAst ast;
00028     };
00029
00030 public:
00032     Result parse(const std::string& sourceCode) const;
00033 };
00034
00035 #endif /* GROPROGRAMPARSER_H */
```

18.493 GroProgramRuntime.h

```
00001 /*
00002  * File:   GroProgramRuntime.h
00003  * Author: rlcancian
00004  *
00005  * Created on 17 de Abril de 2022
00006  */
00007
00008 #ifndef GROPROGRAMRUNTIME_H
00009 #define GROPROGRAMRUNTIME_H
00010
00011 #include "plugins/data/BiochemicalSimulation/GroProgramIr.h"
00012
00013 #include <functional>
00014 #include <map>
00015 #include <string>
00016 #include <vector>
00017
00021 struct GroProgramRuntimeState {
00022     double colonyTime = 0.0;
00023     double simulationStep = 1.0;
00024     unsigned int populationSize = 1;
00025     unsigned int tickCount = 0;
00026     std::map<std::string, double> contextVariables;
00027     std::map<std::string, double> variables;
00028     // Optional host-provided RNG bridge used to preserve command-order
00029     // semantics for stochastic commands such as tumble(). Standalone runtime
00030     // callers retain the deterministic fallback when no host sampler exists.
00031     std::function<bool(double&, std::string&)> sampleUnitInterval;
00032     // Counts "VAR := signal(kdiff, kdeg);" assignments executed so far in
00033     // this pass, in source order. The Nth assignment yields handle N within
00034     // one program. BacteriaColony separately rejects that ordinal being
00035     // declared by a different program scope in the same replication.
00036     unsigned int signalDeclarationOrdinal = 0;
00037     // Highest signal channel handle already established in an EARLIER pass
00038     // (e.g. a global "signal(...)" declaration executed by the colony-wide
00039     // prelude before this bacterium-scoped pass began). A named program
00040     // that only *consumes* a global handle (get_signal/emit_signal/
00041     // absorb_signal) without itself containing a "signal(...)" declaration
00042     // has signalDeclarationOrdinal == 0 in its own fresh pass, so handle
00043     // validation must also accept anything already known from history.
00044     // The caller (BacteriaColony) is responsible for setting this from its
00045     // own persistent channel bookkeeping; it defaults to 0 (no prior
00046     // history) for callers that run self-contained IR with no such colony.
00047     unsigned int knownSignalChannelCount = 0;
00048     // Stable source scope for signal() declarations. All passes of one named
```

```
00049     // program share this identity; global prelude declarations use "<global>".
00050     std::string signalDeclarationScope;
00051 };
00052
00061 class GroProgramRuntime {
00062 public:
00063     enum class PopulationMutationType {
00064         Grow,
00065         Divide,
00066         Die,
00067         SetPopulation
00068     };
00069
00070     enum class SignalMutationType {
00071         Emit,
00072         Consume,
00073         Set
00074     };
00075
00076     enum class MotionMutationType {
00077         Run,
00078         Tumble
00079     };
00080
00081     struct PopulationMutation {
00082         PopulationMutationType type = PopulationMutationType::Grow;
00083         unsigned int value = 0;
00084         unsigned int previousPopulationSize = 0;
00085         unsigned int resultingPopulationSize = 0;
00086     };
00087
00088     struct SignalMutation {
00089         SignalMutationType type = SignalMutationType::Emit;
00090         double value = 0.0;
00091         // Channel 0 is the colony's single default/legacy field (backward
00092         // compatible with pre-multi-channel behavior). Channel N >= 1 is the
00093         // Nth distinct "signal(...)" declaration encountered in program
00094         // order, matching GroProgramCompiler/GroProgramRuntime's ordinal
00095         // channel-handle assignment (see executeCommands' Assignment case).
00096         unsigned int channel = 0;
00097     };
00098
00099     struct MotionMutation {
00100         MotionMutationType type = MotionMutationType::Run;
00101         double value = 0.0;
00102         double previousDirection = 0.0;
00103         double resultingDirection = 0.0;
00104         double previousSpeed = 0.0;
00105         double resultingSpeed = 0.0;
00106     };
00107
00108         struct ColonyMutation {
00109             enum class Type {
00110                 Reset,
00111                 SpawnSeed,
00112             SetChemostatMode,
00113             AddBarrier,
00114             MapToCells,
00115             SetSignalAt,
00116             SetSignalRect,
00117             SetSignalGridWidth,
00118             SetSignalGridHeight,
00119             GetSignalMatrix,
00120             DumpSignalField,
00121             EnsureSignalChannel
00122         };
00123
00124             Type type = Type::Reset;
00125             std::vector<std::string> arguments;
00126             std::vector<double> numericArguments;
00127             std::string expressionText;
00128             std::string signalDeclarationScope;
00129             std::size_t previewRows = 0;
00130             std::size_t previewColumns = 0;
00131         };
00132
00133     struct ExecutionResult {
00134         bool succeeded = true;
00135         std::string errorMessage = "";
00136         unsigned int executedCommands = 0;
00137         std::vector<PopulationMutation> populationMutations;
00138         std::vector<SignalMutation> signalMutations;
00139         std::vector<MotionMutation> motionMutations;
00140         std::vector<ColonyMutation> colonyMutations;
00141         std::vector<double> mappedCellValues;
00142         std::string mappedCellExpression;
00143         std::vector<double> signalMatrixValues;
```

```
00144         unsigned int signalMatrixWidth = 0;
00145         unsigned int signalMatrixHeight = 0;
00146         std::string signalFieldDump;
00147         std::vector<std::string> messages;
00148         std::map<std::string, double> assignedVariables;
00149         std::vector<std::string> unsupportedCommands;
00150         std::vector<std::string> skippedRawStatements;
00151     };
00152 
00153 public:
00155     ExecutionResult execute(const GroProgramIr& ir, GroProgramRuntimeState& state) const;
00157     static bool validateSignalDeclarationPlacement(const GroProgramIr& ir, std::string& errorMessage);
00159     static bool evaluateExpression(const std::string& expressionText,
00160                                    const GroProgramRuntimeState& state,
00161                                    double& value,
00162                                    std::string& errorMessage);
00163 };
00164 
00165 #endif /* GROPROGRAMRUNTIME_H */
```

18.494 MetabolicNetwork.h

```
00001 #ifndef METABOLICNETWORK_H
00002 #define METABOLICNETWORK_H
00003 
00004 #include <string>
00005 #include <vector>
00006 
00007 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00008 #include "kernel/simulator/PluginInformation.h"
00009 
00016 class MetabolicNetwork : public ModelDataDefinition {
00017 public:
00018     MetabolicNetwork(Model* model, std::string name = "");
00019     virtual ~MetabolicNetwork() override = default;
00020 
00021 public:
00022     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00023     static PluginInformation* GetPluginInformation();
00024     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00025 
00026 public:
00027     void addReaction(std::string reactionName);
00028     void addExchangeSpecies(std::string speciesName);
00029     void clearReactions();
00030     void clearExchangeSpecies();
00031     const std::vector<std::string>& getReactionNames() const;
00032     const std::vector<std::string>& getExchangeSpeciesNames() const;
00033     void setObjectiveReactionName(std::string objectiveReactionName);
00034     std::string getObjectiveReactionName() const;
00035     void setObjectiveSense(std::string objectiveSense);
00036     std::string getObjectiveSense() const;
00037     void setCompartment(std::string compartment);
00038     std::string getCompartment() const;
00039     void setEnabled(bool enabled);
00040     bool isEnabled() const;
00041 
00042     virtual std::string show() override;
00043 
00044 protected:
00045     virtual bool _loadInstance(PersistenceRecord* fields) override;
00046     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00047     virtual bool _check(std::string& errorMessage) override;
00048     virtual void _createEditableDataDefinitions() override;
00049 
00050 protected:
00051     // virtual void _createInternalStatisticReporters() override;
00052     // virtual void _createNonEditableDataDefinitions() override;
00053     // virtual void _createAttachedAttributes() override;
00054 
00055 private:
00056     bool _checkReactionNames(std::string& errorMessage) const;
00057     bool _checkExchangeSpeciesNames(std::string& errorMessage) const;
00058 
00059 private:
00060     const struct DEFAULT_VALUES {
00061         std::string objectiveReactionName = "";
00062         std::string objectiveSense = "Maximize";
00063         std::string compartment = "";
00064         bool enabled = true;
00065     } DEFAULT;
00066 
00067     std::vector<std::string> _reactionNames;
00068     std::vector<std::string> _exchangeSpeciesNames;
00069     std::string _objectiveReactionName = DEFAULT.objectiveReactionName;
```

```
00070     std::string _objectiveSense = DEFAULT.objectiveSense;
00071     std::string _compartment = DEFAULT.compartment;
00072     bool _enabled = DEFAULT.enabled;
00073 };
00074
00075 #endif /* METABOLICNETWORK_H */
```

18.495 MetabolicReaction.h

```
00001 #ifndef METABOLICREACTION_H
00002 #define METABOLICREACTION_H
00003
00004 #include <string>
00005 #include <vector>
00006
00007 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00008 #include "kernel/simulator/PluginInformation.h"
00009
00013 class MetabolicReaction : public ModelDataDefinition {
00014 public:
00015     struct StoichiometricTerm {
00016         std::string speciesName;
00017         double stoichiometry = 1.0;
00018     };
00019
00020 public:
00021     MetabolicReaction(Model* model, std::string name = "");
00022     virtual ~MetabolicReaction() override = default;
00023
00024 public:
00025     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00026     static PluginInformation* GetPluginInformation();
00027     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00028
00029 public:
00030     void addReactant(std::string speciesName, double stoichiometry = 1.0);
00031     void addProduct(std::string speciesName, double stoichiometry = 1.0);
00032     void clearReactants();
00033     void clearProducts();
00034     const std::vector<StoichiometricTerm>& getReactants() const;
00035     const std::vector<StoichiometricTerm>& getProducts() const;
00036     void setLowerBound(double lowerBound);
00037     double getLowerBound() const;
00038     void setUpperBound(double upperBound);
00039     double getUpperBound() const;
00040     void setObjectiveCoefficient(double objectiveCoefficient);
00041     double getObjectiveCoefficient() const;
00042     void setReversible(bool reversible);
00043     bool isReversible() const;
00044     void setGeneRule(std::string geneRule);
00045     std::string getGeneRule() const;
00046
00047     virtual std::string show() override;
00048
00049 protected:
00050     virtual bool _loadInstance(PersistenceRecord* fields) override;
00051     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00052     virtual bool _check(std::string& errorMessage) override;
00053     virtual void _createEditableDataDefinitions() override;
00054
00055 protected:
00056     // virtual void _createInternalStatisticReporters() override;
00057     // virtual void _createNonEditableDataDefinitions() override;
00058     // virtual void _createAttachedAttributes() override;
00059
00060 private:
00061     bool _checkTerms(const std::vector<StoichiometricTerm>& terms, const std::string& side,
      std::string& errorMessage) const;
00062
00063 private:
00064     const struct DEFAULT_VALUES {
00065         double lowerBound = 0.0;
00066         double upperBound = 1000.0;
00067         double objectiveCoefficient = 0.0;
00068         bool reversible = false;
00069         std::string geneRule = "";
00070     } DEFAULT;
00071
00072     std::vector<StoichiometricTerm> _reactants;
00073     std::vector<StoichiometricTerm> _products;
00074     double _lowerBound = DEFAULT.lowerBound;
00075     double _upperBound = DEFAULT.upperBound;
00076     double _objectiveCoefficient = DEFAULT.objectiveCoefficient;
00077     bool _reversible = DEFAULT.reversible;
00078     std::string _geneRule = DEFAULT.geneRule;
```

```
00079 };
00080
00081 #endif /* METABOLICREACTION_H */
```

18.496 DiffusionField.h

```
00001 #ifndef DIFFUSIONFIELD_H
00002 #define DIFFUSIONFIELD_H
00003
00004 #include <string>
00005 #include <vector>
00006
00007 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00008 #include "kernel/simulator/PluginInformation.h"
00009
00010 class DiffusionMethodOfLinesSystem;
00011
00012 // Plugin that solves the diffusion (heat) equation du/dt = D*Laplacian(u) in N
00013 // dimensions.
00014 //
00015 // It works in two steps. First the Method of Lines turns the PDE into a system
00016 // of ODEs (central differences in space - see DiffusionMethodOfLinesSystem).
00017 // Then it integrates that system in time with a solver from OdeSolverFactory, so
00018 // the user can pick RK4 or Dormand-Prince 5(4), exactly like in BioNetwork.
00019 //
00020 // N is set by the Dimensions control; the grid has PointsPerDimension points and
00021 // length DomainLength on each axis. The boundary is Dirichlet (fixed value) or
00022 // Neumann (no flux, conserves mass), and the starting field is one of a few
00023 // named shapes (sine mode, Gaussian bump, constant).
00024 //
00025 // The class follows the same layout as BioNetwork (controls, save/load, _check,
00026 // simulate / advanceOneStep) to stay consistent with the rest of the code. The
00027 // grid doesn't change during a run, so each step just rebuilds the small system
00028 // object and reuses the stored field. The solver is never hard-coded - it comes
00029 // from the OdeSolver name through the factory.
00030 //
00031 // Equation representation (design choice): the diffusion PDE is encoded as a
00032 // dedicated C++ class (DiffusionMethodOfLinesSystem), not as a parser expression
00033 // nor as dynamically compiled C++. This is a deliberate scope decision - a
00034 // specific, well-tested class for diffusion - leaving a general user-defined
00035 // equation mechanism (parser or CppCompiler) as a future extension.
00036 //
00037 // Time semantics: with AutoSchedule enabled, the field steps once per kernel
00038 // InternalEvent (see scheduleNextInternalEvent), so the continuous solve advances
00039 // in sync with the discrete-event clock; otherwise simulate() runs the whole time
00040 // window synchronously when driven by the DiffusionSimulate component.
00041 class DiffusionField : public ModelDataDefinition {
00042 public:
00043     DiffusionField(Model* model, std::string name = "");
00044     virtual ~DiffusionField() override = default;
00045
00046 public:
00047     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00048     static PluginInformation* GetPluginInformation();
00049     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00050
00051 public:
00052     // --- configuration ---
00053     void setDimensions(unsigned int dimensions);
00054     unsigned int getDimensions() const;
00055     void setPointsPerDimension(unsigned int pointsPerDimension);
00056     unsigned int getPointsPerDimension() const;
00057     void setDomainLength(double domainLength);
00058     double getDomainLength() const;
00059     void setDiffusionCoefficient(double diffusionCoefficient);
00060     double getDiffusionCoefficient() const;
00061     void setBoundaryCondition(std::string boundaryCondition);
00062     std::string getBoundaryCondition() const;
00063     void setInitialCondition(std::string initialCondition);
00064     std::string getInitialCondition() const;
00065     void setInitialParameter(double initialParameter);
00066     double getInitialParameter() const;
00067     void setStartTime(double startTime);
00068     double getStartTime() const;
00069     void setStopTime(double stopTime);
00070     double getStopTime() const;
00071     void setStepSize(double stepSize);
00072     double getStepSize() const;
00073     void setCurrentTime(double currentTime);
00074     double getCurrentTime() const;
00075     // When true, the field schedules itself on the kernel's event queue and
00076     // advances one step per InternalEvent, staying in sync with simulated time.
00077     void setAutoSchedule(bool autoSchedule);
00078     bool getAutoSchedule() const;
00079     void setOdeSolver(std::string odeSolver);
```

```
00080     std::string getOdeSolver() const;
00081     void setLastStatus(std::string lastStatus);
00082     std::string getLastStatus() const;
00083     void setLastErrorMessage(std::string lastErrorMessage);
00084     std::string getLastErrorMessage() const;
00085 
00086     // --- diagnostics (read-only outputs) ---
00087     double getTotalMass() const;
00088     double getMaxValue() const;
00089     double getL2Norm() const;
00090     const std::vector<double>& getField() const;
00092     double getFieldValue(const std::vector<unsigned int>& multiIndex) const;
00093 
00094     // --- execution ---
00095     bool simulate(std::string& errorMessage);
00096     bool simulate(double startTime, double stopTime, double stepSize, std::string& errorMessage);
00097     bool advanceOneStep(std::string& errorMessage);
00098 
00099     // --- display ---
00100     std::string showFieldAsAscii() const;
00101 
00102     virtual std::string show() override;
00103 
00104 protected:
00105     virtual bool _loadInstance(PersistenceRecord* fields) override;
00106     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00107     virtual bool _check(std::string& errorMessage) override;
00108     virtual void _initBetweenReplications() override;
00109 
00110 private:
00111     bool buildSystem(DiffusionMethodOfLinesSystem* system, std::string& errorMessage) const;
00112     void buildInitialField(const DiffusionMethodOfLinesSystem& system, std::vector<double>& field)
      const;
00113     void refreshDiagnostics(const DiffusionMethodOfLinesSystem& system);
00114     // Event-driven advance (mirrors BioNetwork): each InternalEvent advances one
00115     // step and reschedules the next, coupling the continuous solve to the kernel clock.
00116     void handleInternalEvent(void* parameter);
00117     void scheduleNextInternalEvent();
00118 
00119 private:
00120     const struct DEFAULT_VALUES {
00121         unsigned int dimensions = 2;
00122         unsigned int pointsPerDimension = 21;
00123         double domainLength = 1.0;
00124         double diffusionCoefficient = 0.1;
00125         std::string boundaryCondition = "Dirichlet";
00126         std::string initialCondition = "SineMode";
00127         double initialParameter = 1.0;
00128         double startTime = 0.0;
00129         double stopTime = 0.1;
00130         double stepSize = 0.01;
00131         double currentTime = 0.0;
00132         bool autoSchedule = false;
00133         std::string odeSolver = "RungeKutta4";
00134         std::string lastStatus = "Idle";
00135         std::string lastErrorMessage = "";
00136     } DEFAULT;
00137 
00138     unsigned int _dimensions = DEFAULT.dimensions;
00139     unsigned int _pointsPerDimension = DEFAULT.pointsPerDimension;
00140     double _domainLength = DEFAULT.domainLength;
00141     double _diffusionCoefficient = DEFAULT.diffusionCoefficient;
00142     std::string _boundaryCondition = DEFAULT.boundaryCondition;
00143     std::string _initialCondition = DEFAULT.initialCondition;
00144     double _initialParameter = DEFAULT.initialParameter;
00145     double _startTime = DEFAULT.startTime;
00146     double _stopTime = DEFAULT.stopTime;
00147     double _stepSize = DEFAULT.stepSize;
00148     double _currentTime = DEFAULT.currentTime;
00149     bool _autoSchedule = DEFAULT.autoSchedule;
00150     std::string _odeSolver = DEFAULT.odeSolver;
00151     std::string _lastStatus = DEFAULT.lastStatus;
00152     std::string _lastErrorMessage = DEFAULT.lastErrorMessage;
00153 
00154     std::vector<double> _field;       // current flattened grid state
00155     double _lastTotalMass = 0.0;
00156     double _lastMaxValue = 0.0;
00157     double _lastL2Norm = 0.0;
00158 };
00159 
00160 #endif /* DIFFUSIONFIELD_H */
```

18.497 ODESolver.h

```
00001 /*
```

```
00002  * File:   ODESolver.h
00003  * Author: GenESyS
00004  *
00005  * Created on 13 de maio de 2026
00006  */
00007
00008 #ifndef ODESOLVER_H
00009 #define ODESOLVER_H
00010
00011 #include <string>
00012 #include <vector>
00013
00014 #include "kernel/simulator/model/ModelDataDefinition.h"
00015 #include "kernel/simulator/PluginInformation.h"
00016 #include "tools/Continuous/OdeSolver_if.h"
00017 #include "tools/Continuous/OdeSystem_if.h"
00018 #include "tools/Continuous/RungeKutta4OdeSolver.h"
00019
00020 class ODESolver : public ModelDataDefinition {
00021 public:
00022     ODESolver(Model* model, std::string name = "");
00023     virtual ~ODESolver() = default;
00024 public: // static
00025     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00026     static PluginInformation* GetPluginInformation();
00027     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00028     static void SaveInstance(PersistenceRecord *fields, ModelDataDefinition* dataDef);
00029     static bool Check(ModelDataDefinition* dataDef, std::string& errorMessage);
00030     static void InitBetweenReplications(ModelDataDefinition* dataDef);
00031     static void CreateInternalData(ModelDataDefinition* dataDef);
00032 public:
00033     virtual std::string show() override;
00034 public: // gets & sets
00035     void setTimeVariableName(std::string timeVariableName);
00036     std::string getTimeVariableName() const;
00037     void setStateVariableNames(std::vector<std::string> stateVariableNames);
00038     std::vector<std::string> getStateVariableNames() const;
00039     void setEquationExpressions(std::vector<std::string> equationExpressions);
00040     std::vector<std::string> getEquationExpressions() const;
00041     void setStep(double step);
00042     double getStep() const;
00043     void setPrecision(double precision);
00044     double getPrecision() const;
00045     void setMaxSteps(int maxSteps);
00046     int getMaxSteps() const;
00047     void setCurrentTime(double currentTime);
00048     double getCurrentTime() const;
00049     void setStateValues(std::vector<double> stateValues);
00050     std::vector<double> getStateValues() const;
00051     void setInitialStateValues(std::vector<double> initialStateValues);
00052     std::vector<double> getInitialStateValues() const;
00053     void setAutoAdvance(bool autoAdvance);
00054     bool isAutoAdvance() const;
00055     void setOutputFile(std::string outputFile);
00056     std::string getOutputFile() const;
00057 public: // new methods
00058     void integrate(double targetTime);
00059     void resetState();
00060     double getStateValue(unsigned int index) const;
00061     void setStateValue(unsigned int index, double value);
00062     void synchronizeStateFromVariables();
00063     void writeStateToVariables() const;
00064
00065 protected: // must be overriden
00066     virtual bool _loadInstance(PersistenceRecord *fields) override;
00067     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00068 protected: // could be overriden
00069     virtual bool _check(std::string& errorMessage) override;
00070     virtual void _initBetweenReplications() override;
00071     virtual void _createEditableDataDefinitions() override;
00072
00073 private:
00074     const struct DEFAULT_VALUES {
00075         const std::string timeVariableName = "t";
00076         const double step = 0.01;
00077         const double precision = 1e-6;
00078         const int maxSteps = 100000;
00079         const double currentTime = 0.0;
00080         const bool autoAdvance = true;
00081         const std::string outputFile = "";
00082     } DEFAULT;
00083     std::string _timeVariableName = DEFAULT.timeVariableName;
00084     std::vector<std::string> _stateVariableNames;
00085     std::vector<std::string> _equationExpressions;
00086     double _step = DEFAULT.step;
00087     double _precision = DEFAULT.precision;
00088     int _maxSteps = DEFAULT.maxSteps;
```

```
00089     double _currentTime = DEFAULT.currentTime;
00090     bool _autoAdvance = DEFAULT.autoAdvance;
00091     std::string _outputFile = DEFAULT.outputFile;
00092     std::vector<double> _stateValues;
00093     std::vector<double> _initialStateValues;
00094     RungeKutta4OdeSolver _solver;
00095 
00096     std::vector<double> _loadStateVector(PersistenceRecord* fields, const std::string& prefix) const;
00097     void _saveStateVector(PersistenceRecord* fields,
00098                           const std::string& prefix,
00099                           const std::vector<double>& values,
00100                           bool saveDefaultValues) const;
00101     double _readVariableValue(const std::string& variableName) const;
00102     // Self-scheduling that couples the continuous integration to the discrete event calendar:
00103     // each internal event advances the ODE by one step and schedules the next, so the continuous
00104     // state evolves with the simulation clock instead of only when a discrete entity arrives.
00105     void _handleInternalEvent(void* parameter);
00106     void _scheduleNextInternalEvent();
00107     // Optional CSV dump of the trajectory: when _outputFile is set, one row per integration
00108     // step is written (t and each state variable), so any model can export its trajectory.
00109     void _writeOutputHeader() const;
00110     void _appendOutputRow() const;
00111 };
00112 
00113 #endif /* ODESOLVER_H */
```

18.498 Failure.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
00007 /*
00008  * File:   Failure.h
00009  * Author: rlcancian
00010  *
00011  * Created on 20 de Failureembro de 2019, 20:07
00012  */
00013 
00014 #ifndef FAILURE_H
00015 #define FAILURE_H
00016 
00017 
00018 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00019 #include "../../../kernel/simulator/model/ModelDataManager.h"
00020 #include "kernel/simulator/PluginInformation.h"
00021 
00022 class Resource;
00023 
00044 class Failure : public ModelDataDefinition {
00045 public:
00046 
00047     enum class FailureType : int {
00048         COUNT = 0, TIME = 1, num_elements = 2
00049     };
00050 
00051     enum class FailureRule : int {
00052         IGNORE = 0, PREEMPT = 1, WAIT = 2, num_elements = 3
00053     };
00054 public:
00055     static std::string convertEnumToStr(FailureType type);
00056     static std::string convertEnumToStr(FailureRule rule);
00057 public:
00058     Failure(Model* model, std::string name = "");
00059     virtual ~Failure() override;
00060 public: // static
00061     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00062     static PluginInformation* GetPluginInformation();
00063     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00064 public:
00065     virtual std::string show() override;
00066     void checkIsGoingToFailByCount(Resource* resource);
00067 public: // gets & sets
00068     void setFailureType(FailureType _failureType);
00069     Failure::FailureType getFailureType() const;
00070     void setCountExpression(std::string countExpression);
00071     std::string getCountExpression() const;
00072     void setDownTimeTimeUnit(Util::TimeUnit downTimeTimeUnit);
00073     Util::TimeUnit getDownTimeTimeUnit() const;
00074     void setDownTimeExpression(std::string downTimeExpression);
00075     std::string getDownTimeExpression() const;
00076     void setUpTimeTimeUnit(Util::TimeUnit upTimeTimeUnit);
00077     Util::TimeUnit getUpTimeTimeUnit() const;
```

```
00078     void setUpTimeExpression(std::string upTimeExpression);
00079     std::string getUpTimeExpression() const;
00080     void setFailureRule(FailureRule _failureRule);
00081     FailureRule getFailureRule() const;
00082 
00083     List<Resource*>*falingResources() const;
00084     void addResource(Resource* newResource);
00085     void removeResource(Resource* resource);
00086 
00087 protected: // must be overriden
00088     virtual bool _loadInstance(PersistenceRecord *fields) override;
00089     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00090 protected: // could be overriden .
00091     virtual bool _check(std::string& errorMessage) override;
00092     virtual void _initBetweenReplications() override;
00093     //virtual ParserChangesInformation* _getParserChangesInformation();
00094 protected:
00095     // virtual void _createInternalStatisticReporters() override;
00096     // virtual void _createNonEditableDataDefinitions() override;
00097     // virtual void _createEditableDataDefinitions() override;
00098     // virtual void _createAttachedAttributes() override;
00099 
00100 private: // simulation internal event handlers
00101     void _onFailureActiveEventHandler(void* resourcePtr);
00102     void _onFailureFailEventHandler(void* resourcePtr);
00103 private:
00104     void _scheduleActivation(Resource* resource);
00105 private:
00106 
00107     const struct DEFAULT_VALUES {
00108         const FailureType failureType = FailureType::TIME;
00109         const FailureRule failureRule = FailureRule::IGNORE;
00110         const std::string countExpression = "";
00111         const std::string upTimeExpression = "";
00112         const Util::TimeUnit upTimeTimeUnit = Util::TimeUnit::second;
00113         const std::string downTimeExpression = "";
00114         const Util::TimeUnit downTimeTimeUnit = Util::TimeUnit::second;
00115     } DEFAULT;
00116     FailureType _failureType = DEFAULT.failureType;
00117     FailureRule _failureRule = DEFAULT.failureRule;
00118     std::string _countExpression = DEFAULT.countExpression;
00119     std::string _upTimeExpression = DEFAULT.upTimeExpression;
00120     Util::TimeUnit _upTimeTimeUnit = DEFAULT.upTimeTimeUnit;
00121     std::string _downTimeExpression = DEFAULT.downTimeExpression;
00122     Util::TimeUnit _downTimeTimeUnit = DEFAULT.downTimeTimeUnit;
00123 private:
00124     std::map<Resource*, unsigned int>* _releaseCounts = new std::map<Resource*, unsigned int>();
00125     List<Resource*>* _falingResources = new List<Resource*>();
00126 };
00127 
00128 #endif /* FAILURE_H */
```

18.499 Queue.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
00007 /*
00008  * File:   Queue.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Agosto de 2018, 17:12
00012  */
00013 
00014 #ifndef QUEUE_H
00015 #define QUEUE_H
00016 
00017 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00018 #include "kernel/util/List.h"
00019 #include "../../../kernel/simulator/essentialPlugins/Entity.h"
00020 #include "../../../kernel/simulator/model/ModelDataManager.h"
00021 #include "../../../kernel/simulator/essentialPlugins/StatisticsCollector.h"
00022 #include "kernel/simulator/Plugin.h"
00023 #include "../../../kernel/simulator/model/ModelComponent.h"
00024 
00025 class Waiting {
00026 public:
00027     Waiting(Entity* entity, double timeStartedWaiting, ModelComponent* thisComponent, unsigned int
    thisComponentOutputPort = 0) {
00028         _entity = entity;
00029         _thisComponent = thisComponent;
00030         _timeStartedWaiting = timeStartedWaiting;
```

```
00031         _thisComponentOutputPort = thisComponentOutputPort;
00032     }
00033
00034     virtual ~Waiting() = default;
00035 public:
00036     virtual std::string show() {
00037         return //ModelDataDefinition::show()+
00038         ",entity=" + std::to_string(_entity->getId()) +
00039                 ",component=\"" + _thisComponent->getName() + "\"" +
00040                 ",inputPort=\"" + std::to_string(_thisComponentOutputPort) + "\"" +
00041                 ",timeStartedWaiting=" + std::to_string(_timeStartedWaiting);
00042     }
00043 public:
00044     double getTimeStartedWaiting() const {
00045         return _timeStartedWaiting;
00046     }
00047     ModelComponent* geComponent() const {
00048         return _thisComponent;
00049     }
00050     Entity* getEntity() const {
00051         return _entity;
00052     }
00053     unsigned int geComponentOutputPort() const {
00054         return _thisComponentOutputPort;
00055     }
00056     unsigned long long getArrivalOrder() const {
00057         return _arrivalOrder;
00058     }
00059     void setArrivalOrder(unsigned long long arrivalOrder) {
00060         _arrivalOrder = arrivalOrder;
00061     }
00062
00063 protected:
00064 private:
00065     Entity* _entity;
00066     ModelComponent* _thisComponent;
00067     double _timeStartedWaiting;
00068     unsigned int _thisComponentOutputPort;
00069     unsigned long long _arrivalOrder = 0;
00070 };
00071
00094 class Queue : public ModelDataDefinition {
00095 public:
00096
00097     enum class OrderRule : int {
00098         FIFO = 0, LIFO = 1, HIGHESTVALUE = 2, SMALLESTVALUE = 3, num_elements = 4
00099     };
00100 public:
00101     static std::string convertEnumToStr(OrderRule rule);
00102 public:
00103     Queue(Model* model, std::string name = "");
00104     virtual ~Queue();
00105 public:
00106     virtual std::string show() override;
00107 public: // static
00108     static PluginInformation* GetPluginInformation();
00109     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00110     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00111 public:
00112     void insertElement(Waiting* modeldatum);
00113     void removeElement(Waiting* modeldatum);
00114     unsigned int size();
00115     Waiting* first();
00116     Waiting* getAtRank(unsigned int rank);
00117     void setAttributeName(std::string _attributeName);
00118     std::string getAttributeName() const;
00119     void setOrderRule(OrderRule _orderRule);
00120     Queue::OrderRule getOrderRule() const;
00121     void setOrderRuleInt(int orderRule);
00122     int getOrderRuleInt() const;
00123 public: // to implement SIMAN functions
00124     double sumAttributesFromWaiting(Util::identification attributeID); // use to implement SIMAN SAQUE
      function
00125     double getAttributeFromWaitingRank(unsigned int rank, Util::identification attributeID);
00126     //public:
00127     //  void initBetweenReplications();
00128 protected: // must be overriden
00129     virtual bool _loadInstance(PersistenceRecord *fields) override;
00130     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00131 protected: // could be overriden
00132     virtual bool _check(std::string& errorMessage) override;
00133     virtual void _initBetweenReplications() override;
00134     virtual ParserChangesInformation* _getParserChangesInformation() override;
00135 protected:
00136     virtual void _createInternalStatisticReporters() override;
00137     // virtual void _createNonEditableDataDefinitions() override;
00138     // virtual void _createEditableDataDefinitions() override;
```

```
00139     // virtual void _createAttachedAttributes() override;
00140
00141 private:
00142     void _initCStats();
00143     void _configureListComparator();
00144 private:
00145     List<Waiting*>* _list = new List<Waiting*>();
00146     double _lastTimeNumberInQueueChanged = 0.0;
00147     unsigned long long _nextArrivalOrder = 0;
00148 private: //1::1
00149
00150     const struct DEFAULT_VALUES {
00151         const OrderRule orderRule = OrderRule::FIFO;
00152         const std::string attributeName = "";
00153     } DEFAULT;
00154     OrderRule _orderRule = DEFAULT.orderRule;
00155     std::string _attributeName = DEFAULT.attributeName;
00156 private: // inner internal elements
00157     StatisticsCollector* _cstatNumberInQueue = nullptr;
00158     StatisticsCollector* _cstatTimeInQueue = nullptr;
00159 };
00160
00161 #endif /* QUEUE_H */
```

18.500 Resource.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Resource.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 21 de Agosto de 2018, 16:52
00012  */
00013
00014 #ifndef RESOURCE_H
00015 #define RESOURCE_H
00016
00017 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00018 #include "../../../kernel/simulator/essentialPlugins/StatisticsCollector.h"
00019 #include "../../../kernel/simulator/model/ModelDataManager.h"
00020 #include "../../../kernel/simulator/essentialPlugins/Counter.h"
00021 #include "kernel/simulator/Plugin.h"
00022 #include "kernel/simulator/OnEventManager.h"
00023
00024 #include "plugins/data/DiscreteProcessing/Failure.h"
00025 #include "plugins/data/DiscreteProcessing/Schedule.h"
00026
00027 #include <functional>
00028
00029
00030
00031 class SeizableItem;
00032
00060 class Resource : public ModelDataDefinition {
00061 public:
00062     typedef std::function<void(Resource*) > ResourceEventHandler;
00063     typedef std::pair<std::pair<ResourceEventHandler, ModelComponent*>, unsigned int>
      SortedResourceEventHandler;
00064
00065     template<typename Class>
00066     static ResourceEventHandler SetResourceEventHandler(void (Class::*function)(Resource*), Class *
      object) {
00067         return std::bind(function, object, std::placeholders::_1);
00068     }
00069
00070     enum class ResourceState : int {
00071         IDLE = 0, BUSY = 1, FAILED = 2, INACTIVE = 3, OTHER = 4, num_elements = 5
00072     };
00073 public:
00074     static std::string convertEnumToStr(ResourceState state);
00075 public:
00076     //Resource(Model* model);
00077     Resource(Model* model, std::string name = "");
00078     virtual ~Resource() override;
00079 public:
00080     virtual std::string show() override;
00081 public: // static
00082     static PluginInformation* GetPluginInformation();
00083     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00084     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
```

```
00085 public:
00086     bool seize(unsigned int quantity, double priority = 0);
00087     void release(unsigned int quantity);
00088     void insertFailure(Failure* failure);
00089     void removeFailure(Failure* failure);
00090     double getInstantCapacityUtilization() const;
00091     double getCapacityUtilization() const;
00092     double getSeizedUtilization() const;
00093     double getLastTimeSeized() const; // used only by "Release" component
00094     void addReleaseResourceEventHandler(ResourceEventHandler eventHandler, ModelComponent* component,
      unsigned int priority);
00095 public: // g&s
00096     void setResourceState(ResourceState _resourceState);
00097     Resource::ResourceState getResourceState() const;
00098     void setCapacity(unsigned int capacity);
00099     unsigned int getCapacity() const;
00100     void setCostBusyTimeUnit(double _costBusyTimeUnit);
00101     double getCostBusyTimeUnit() const;
00102     void setCostIdleTimeUnit(double _costIdleTimeUnit);
00103     double getCostIdleTimeUnit() const;
00104     void setCostPerUse(double _costPerUse);
00105     double getCostPerUse() const;
00106     void setCapacitySchedule(Schedule* _capacitySchedule);
00107     Schedule* getCapacitySchedule() const;
00108     unsigned int getNumberBusy() const;
00109
00110 protected: // protected must override
00111     virtual bool _loadInstance(PersistenceRecord *fields) override;
00112     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00113 protected: // protected could override
00114     virtual bool _check(std::string& errorMessage) override;
00115     // virtual void _createInternalAndAttachedData() override;
00116     virtual void _initBetweenReplications() override;
00117
00118
00119 protected:
00120     // virtual void _createInternalStatisticReporters() override;
00121     // virtual void _createNonEditableDataDefinitions() override;
00122     // virtual void _createEditableDataDefinitions() override;
00123     virtual void _createAttachedAttributes() override;
00124
00125 private: //methods
00126     void _notifyReleaseEventHandlers();
00127     void _onReplicationEnd(SimulationEvent* se);
00128     void _fail();
00129     void _active();
00130     void _checkFailByCount();
00131     void _initStatisticsAndAccounting();
00132     friend class Failure;
00133     friend class ResourceTestProbe;
00134
00135 private:
00136
00137     const struct DEFAULT_VALUES {
00138         const unsigned int capacity = 1;
00139         const double cost = 1.0;
00140         const ResourceState resourceState = ResourceState::IDLE;
00141     } DEFAULT;
00142     unsigned int _capacity = DEFAULT.capacity;
00143     double _costBusyTimeUnit = DEFAULT.cost;
00144     double _costIdleTimeUnit = DEFAULT.cost;
00145     double _costPerUse = DEFAULT.cost;
00146     ResourceState _resourceState = DEFAULT.resourceState;
00147 private: // only gets
00148     unsigned int _numberBusy = 0;
00149     double _lastTimeSeized = 0.0; // @TODO: It won't work for resources with capacity>1, when not all
      capacity is seized and them some more are seized. Seized time of first units will be lost. I don't
      have a solution so far
00150     double _lastTimeReleased = 0.0;
00151     double _lastTimeFailed = 0.0;
00152     double _lastTimeCapacityEvaluated = 0.0;
00153     double _lastTimeAnythingNumberBusy = 0.0;
00154     double _lastTimeIdle = 0.0;
00155     double _lastTimeBusy = 0.0;
00156     double _sumNumberBusyOverTime = 0.0;
00157     double _sumCapacityOverTime = 0.0;
00158     bool _isActive = true;
00159     bool _replicationEndHandlerRegistered = false;
00160 private: // not gets nor sets
00161     unsigned int _originalCapacity; // used for failing purposes, when _capacity changes to 0
00162 private: //1::n
00163     List<SortedResourceEventHandler*>* _resourceEventHandlers = new
      List<SortedResourceEventHandler*>();
00164     List<Failure*>* _failures = new List<Failure*>();
00165 private: // attached elements
00166     Schedule* _capacitySchedule = nullptr;
00167 private: // internal elements
```

```
00168     StatisticsCollector* _cstatTimeSeized = nullptr;
00169     StatisticsCollector* _cstatTimeFailed = nullptr;
00170     StatisticsCollector* _cstatProportionSeized = nullptr;
00171     StatisticsCollector* _cstatCapacityUtilization = nullptr;
00172     Counter* _counterTotalTimeSeized = nullptr;
00173     Counter* _counterTotalTimeFailed = nullptr;
00174     Counter* _counterNumSeizes = nullptr;
00175     Counter* _counterNumReleases = nullptr;
00176     Counter* _counterTotalCostPerUse = nullptr;
00177     Counter* _counterTotalCostBusy = nullptr;
00178     Counter* _counterTotalCostIdle = nullptr;
00179 };
00180
00181 #endif /* RESOURCE_H */
```

18.501 Schedule.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   Schedule.h
00008  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00009  *
00010  * Created on 14 de dezembro de 2022, 04:24
00011  */
00012
00013 #ifndef SCHEDULE_H
00014 #define SCHEDULE_H
00015
00016 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00017 #include "kernel/simulator/PluginInformation.h"
00018 #include "kernel/util/List.h"
00019
00032 class SchedulableItem {
00033 public:
00034
00035     enum class Rule : int {
00036         IGNORE = 1, PREEMPT = 2, WAIT = 3
00037     };
00038
00039 public:
00040
00041     SchedulableItem(std::string expression, double duration, SchedulableItem::Rule rule =
      SchedulableItem::Rule::IGNORE) {
00042         this->expression = expression;
00043         this->duration = duration;
00044         this->rule = rule;
00045     }
00046 public:
00047
00048     void setDuration(double duration) {
00049         this->duration = duration;
00050     }
00051
00052     double getDuration() const {
00053         return duration;
00054     }
00055
00056     void setExpression(std::string expression) {
00057         this->expression = expression;
00058     }
00059
00060     std::string getExpression() const {
00061         return expression;
00062     }
00063
00064     void setRule(SchedulableItem::Rule rule) {
00065         this->rule = rule;
00066     }
00067
00068     SchedulableItem::Rule getRule() const {
00069         return rule;
00070     }
00071
00072 protected:
00073 private:
00074     std::string expression;
00075     double duration;
00076     SchedulableItem::Rule rule;
00077 };
00078
00079 //class SchedulableResourceItem : public SchedulableItem {
```

```
00080 //};
00081
00082 class Schedule : public ModelDataDefinition {
00083 public:
00084     Schedule(Model* model, std::string name = "");
00085     virtual ~Schedule() override;
00086 public: // static
00087     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00088     static PluginInformation* GetPluginInformation();
00089     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00090 public: // virtual
00091     virtual std::string show() override;
00092 public:
00093     std::string getExpression();
00094     List<SchedulableItem*>* getSchedulableItems() const;
00095     void setRepeatAfterLast(bool _repeatAfterLast);
00096     bool isRepeatAfterLast() const;
00097 protected: // must be overriden
00098     virtual bool _loadInstance(PersistenceRecord *fields) override;
00099     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00100 protected: // could be overriden by derived classes
00101     virtual bool _check(std::string& errorMessage) override;
00103     virtual ParserChangesInformation* _getParserChangesInformation() override;
00104     virtual void _initBetweenReplications() override;
00105     virtual void _addSimulationControl(SimulationControl* property) override;
00106 protected:
00107     // virtual void _createInternalStatisticReporters() override;
00108     // virtual void _createNonEditableDataDefinitions() override;
00109     // virtual void _createEditableDataDefinitions() override;
00110     // virtual void _createAttachedAttributes() override;
00111
00112 private:
00113
00114     const struct DEFAULT_VALUES {
00115         //const std::string someString = "Test";
00116         //const unsigned int someUint = 1;
00117         bool repeatAfterLast = true;
00118     } DEFAULT;
00119     bool _repeatAfterLast = DEFAULT.repeatAfterLast;
00120     List<SchedulableItem*>* _schedulableItems = new List<SchedulableItem*>();
00121     //std::string _someString = DEFAULT.someString;
00122     //unsigned int _someUint = DEFAULT.someUint;
00123 };
00124
00125 #endif /* SCHEDULE_H */
```

18.502 CppCompiler.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   DynamicLinkedCode.h
00008  * Author: rlcancian
00009  *
00010  * Created on 11 de janeiro de 2022, 22:24
00011  */
00012
00013 #ifndef CPPCOMPILER_H
00014 #define CPPCOMPILER_H
00015
00016 #ifndef OPEN_MAX
00017 #define OPEN_MAX 256
00018 #endif
00019
00020 #include <string>
00021
00022 #include <sys/types.h>
00023 #include <limits.h>
00024 #include <sys/stat.h>
00025 //#include <stdio.h>
00026 #include <unistd.h>
00027 //#include <errno.h>
00028 #include <paths.h>
00029
00030 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00031 #include "kernel/simulator/PluginInformation.h"
00032
00033
00034 class CompilerThread {
00035 public:
00036     void operator()();
00037 };
```

```
00038
00039 class CppCompiler : public ModelDataDefinition {
00040 public:
00041
00042     class CompilationResult {
00043     public:
00044         CompilationResult() { }
00045         bool success = false;
00046         std::string compilationStdOutput = "";
00047         std::string compilationErrOutput = "";
00048         std::string destinationPath = "";
00049         unsigned int lineNumber = 0;
00050     };
00051
00052 public:
00053     CppCompiler(Model* model, std::string name = "");
00054     virtual ~CppCompiler() = default;
00055 public: // static
00056     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00057     static PluginInformation* GetPluginInformation();
00058     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00059 public:
00060     virtual std::string show() override;
00061 public: // gets & sets
00062     void setSourceFilename(std::string sourceFilename);
00063     std::string getSourceFilename() const;
00064     bool IsLibraryLoaded() const;
00065     void setFlagsExecutable(std::string _flagsExecutable);
00066     std::string getFlagsExecutable() const;
00067     void setLibraryLoaded(bool libraryLoaded);
00068     void setFlagsStaticLibrary(std::string _flagsStaticLibrary);
00069     std::string getFlagsStaticLibrary() const;
00070     void setFlagsDynamicLibrary(std::string _flagsDynamicLibrary);
00071     std::string getFlagsDynamicLibrary() const;
00072     void setFlagsGeneral(std::string _flagsGeneral);
00073     std::string getFlagsGeneral() const;
00074     void setCompilerCommand(std::string _compilerCommand);
00075     std::string getCompilerCommand() const;
00076     void setOutputDir(std::string _outputDir);
00077     std::string getOutputDir() const;
00078     void setTempDir(std::string _tempDir);
00079     std::string getTempDir() const;
00080     void setOutputFilename(std::string _outputFilename);
00081     std::string getOutputFilename() const;
00082     void setObjectFiles(std::string _objectFiles);
00083     std::string getObjectFiles() const;
00084 public: // new methods
00085     CompilationResult compileToExecutable();
00086     CompilationResult compileToDynamicLibrary();
00087     CompilationResult compileToStaticLibrary();
00088     bool loadLibrary(std::string& errorMessage);
00089     bool unloadLibrary();
00090     void* getDynamicLibraryHandler() const;
00091
00092 protected: // must be overriden
00093     virtual bool _loadInstance(PersistenceRecord *fields) override;
00094     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00095 protected: // could be overriden
00096     virtual bool _check(std::string& errorMessage) override;
00097     // virtual ParserChangesInformation* _getParserChangesInformation();
00098     virtual void _initBetweenReplications() override;
00099     // virtual void _createInternalAndAttachedData() override;
00100 protected:
00101     CompilationResult _invokeCompiler(std::string command);  //system(command.c_str());
00102     std::string _read(std::string filename);
00103
00104 protected:
00105     // virtual void _createInternalStatisticReporters() override;
00106     // virtual void _createNonEditableDataDefinitions() override;
00107     // virtual void _createEditableDataDefinitions() override;
00108     // virtual void _createAttachedAttributes() override;
00109
00110 private:
00111
00112     const struct DEFAULT_VALUES {
00113         const std::string sourceFilename = "";
00114         const std::string tempDir = ".temp/";
00115         const std::string outputDir = ".temp/";
00116         const std::string outputFilename = "";
00117         const std::string compiler = "g++";
00118         const std::string flagsGeneral = "-v -w -std=c++14";
00119         const std::string flagsDynamicLibrary = "-shared -fPIC";
00120         const std::string flagsStaticLibrary = "-fPIC";
00121         const std::string flagsExecutable = "";
00122         const std::string objectFiles = "";
00123     } DEFAULT;
00124     std::string _sourceFilename = DEFAULT.sourceFilename;
```

```
00125     std::string _outputFilename = DEFAULT.outputFilename;
00126     std::string _tempDir = DEFAULT.tempDir;
00127     std::string _outputDir = DEFAULT.outputDir;
00128     std::string _compilerCommand = DEFAULT.compiler;
00129     std::string _flagsGeneral = DEFAULT.flagsGeneral;
00130     std::string _flagsDynamicLibrary = DEFAULT.flagsDynamicLibrary;
00131     std::string _flagsStaticLibrary = DEFAULT.flagsStaticLibrary;
00132     std::string _flagsExecutable = DEFAULT.flagsExecutable;
00133     std::string _objectFiles = DEFAULT.objectFiles;
00134     //
00135     bool _libraryLoaded = false;
00136     bool _compiledToDynamicLibrary = false;
00137     void* _dynamicLibraryHandle = nullptr;
00138 };
00139
00140 #endif /* CPPCOMPILER_H */
```

18.503 OctaveRunner.h

```
00001 /*
00002  * File:   OctaveRunner.h
00003  * Author: rlcancian
00004  *
00005  * Octave integration data definition.
00006  */
00007
00008 #ifndef OCTAVERUNNER_H
00009 #define OCTAVERUNNER_H
00010
00011 #include <string>
00012
00013 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00014 #include "kernel/simulator/PluginInformation.h"
00015
00023 class OctaveRunner : public ModelDataDefinition {
00024 public:
00026     OctaveRunner(Model* model, std::string name = "");
00028     virtual ~OctaveRunner() = default;
00029
00030 public:
00032     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00034     static PluginInformation* GetPluginInformation();
00036     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00037
00038 public:
00040     void setOctaveExecutable(std::string octaveExecutable);
00042     std::string getOctaveExecutable() const;
00044     void setWorkingDirectory(std::string workingDirectory);
00046     std::string getWorkingDirectory() const;
00048     void setPreludeScript(std::string preludeScript);
00050     std::string getPreludeScript() const;
00052     void setCommand(std::string command);
00054     std::string getCommand() const;
00056     void setLastStatus(std::string lastStatus);
00058     std::string getLastStatus() const;
00060     void setLastExitCode(int lastExitCode);
00062     int getLastExitCode() const;
00064     void setLastStdout(std::string lastStdout);
00066     std::string getLastStdout() const;
00068     void setLastStderr(std::string lastStderr);
00070     std::string getLastStderr() const;
00072     void setLastResponsePayload(std::string lastResponsePayload);
00074     std::string getLastResponsePayload() const;
00076     void setLastScriptFilename(std::string lastScriptFilename);
00078     std::string getLastScriptFilename() const;
00080     void setLastResponseFilename(std::string lastResponseFilename);
00082     std::string getLastResponseFilename() const;
00083
00085     bool executeCommand(std::string& errorMessage);
00086
00088     virtual std::string show() override;
00089
00090 protected:
00092     virtual bool _loadInstance(PersistenceRecord *fields) override;
00094     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00096     virtual bool _check(std::string& errorMessage) override;
00098     // virtual void _createInternalAndAttachedData() override;
00099
00100
00101 protected:
00102     // virtual void _createInternalStatisticReporters() override;
00103     // virtual void _createNonEditableDataDefinitions() override;
00104     // virtual void _createEditableDataDefinitions() override;
00105     // virtual void _createAttachedAttributes() override;
00106
```

```
00107 private:
00108     const struct DEFAULT_VALUES {
00109         std::string octaveExecutable = "Ocavescript";
00110         std::string workingDirectory = "";
00111         std::string preludeScript = "";
00112         std::string command = "";
00113         std::string lastStatus = "Idle";
00114         int lastExitCode = -1;
00115         std::string lastStdout = "";
00116         std::string lastStderr = "";
00117         std::string lastResponsePayload = "";
00118         std::string lastScriptFilename = "";
00119         std::string lastResponseFilename = "";
00120     } DEFAULT;
00121
00122     std::string _octaveExecutable = DEFAULT.octaveExecutable;
00123     std::string _workingDirectory = DEFAULT.workingDirectory;
00124     std::string _preludeScript = DEFAULT.preludeScript;
00125     std::string _command = DEFAULT.command;
00126     std::string _lastStatus = DEFAULT.lastStatus;
00127     int _lastExitCode = DEFAULT.lastExitCode;
00128     std::string _lastStdout = DEFAULT.lastStdout;
00129     std::string _lastStderr = DEFAULT.lastStderr;
00130     std::string _lastResponsePayload = DEFAULT.lastResponsePayload;
00131     std::string _lastScriptFilename = DEFAULT.lastScriptFilename;
00132     std::string _lastResponseFilename = DEFAULT.lastResponseFilename;
00133 };
00134
00135 #endif /* OCTAVERUNNER_H */
```

18.504 PythonRuntime.h

```
00001 #ifndef PYTHONRUNTIME_H
00002 #define PYTHONRUNTIME_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00007 #include "kernel/simulator/PluginInformation.h"
00008
00009 class Entity;
00010 class ModelComponent;
00011
00012 class PythonRuntime : public ModelDataDefinition {
00013 public:
00014     PythonRuntime(Model* model, std::string name = "");
00015     virtual ~PythonRuntime() override;
00016
00017 public:
00018     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00019     static PluginInformation* GetPluginInformation();
00020     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00021
00022 public:
00023     void setPythonExecutable(std::string pythonExecutable);
00024     std::string getPythonExecutable() const;
00025     void setFacadeObjectName(std::string facadeObjectName);
00026     std::string getFacadeObjectName() const;
00027     void setCaptureOutput(bool captureOutput);
00028     bool isCaptureOutput() const;
00029     void setLastStatus(std::string lastStatus);
00030     std::string getLastStatus() const;
00031     void setLastErrorMessage(std::string lastErrorMessage);
00032     std::string getLastErrorMessage() const;
00033     void setLastStdout(std::string lastStdout);
00034     std::string getLastStdout() const;
00035     void setLastStderr(std::string lastStderr);
00036     std::string getLastStderr() const;
00037
00038     bool validateHooks(const std::string& initCode,
00039                        const std::string& onDispatchCode,
00040                        std::string& errorMessage) const;
00041     bool executeInitHook(ModelComponent* component,
00042                          const std::string& initCode,
00043                          const std::string& onDispatchCode,
00044                          std::string& errorMessage);
00045     bool executeOnDispatchHook(ModelComponent* component,
00046                                Entity* entity,
00047                                const std::string& initCode,
00048                                const std::string& onDispatchCode,
00049                                std::string& errorMessage);
00050     void resetExecutionState();
00051
00052     virtual std::string show() override;
00053
```

```
00054 protected:
00055     virtual bool _loadInstance(PersistenceRecord* fields) override;
00056     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00057     virtual bool _check(std::string& errorMessage) override;
00058 
00059 private:
00060     bool _validateIdentifier(std::string& errorMessage) const;
00061     void _invalidatePreparedHooks();
00062     bool _prepareHooks(ModelComponent* component,
00063                        const std::string& initCode,
00064                        const std::string& onDispatchCode,
00065                        std::string& errorMessage);
00066 
00067 private:
00068     const struct DEFAULT_VALUES {
00069         std::string pythonExecutable = "python3";
00070         std::string facadeObjectName = "simulator";
00071         bool captureOutput = true;
00072         std::string lastStatus = "Idle";
00073         std::string lastErrorMessage = "";
00074         std::string lastStdout = "";
00075         std::string lastStderr = "";
00076     } DEFAULT;
00077 
00078     std::string _pythonExecutable = DEFAULT.pythonExecutable;
00079     std::string _facadeObjectName = DEFAULT.facadeObjectName;
00080     bool _captureOutput = DEFAULT.captureOutput;
00081     std::string _lastStatus = DEFAULT.lastStatus;
00082     std::string _lastErrorMessage = DEFAULT.lastErrorMessage;
00083     std::string _lastStdout = DEFAULT.lastStdout;
00084     std::string _lastStderr = DEFAULT.lastStderr;
00085     std::string _preparedSignature = "";
00086 
00087 #if GENESYS_HAS_PYTHON_INTEGRATION
00088     void* _preparedGlobals = nullptr;
00089     void* _preparedInitFunction = nullptr;
00090     void* _preparedDispatchFunction = nullptr;
00091     void* _preparedFacadeObject = nullptr;
00092 #endif
00093 };
00094 
00095 #endif /* PYTHONRUNTIME_H */
```

18.505 RSimulatorRunner.h

```
00001 /*
00002  * File:   RSimulatorRunner.h
00003  * Author: GenESyS
00004  *
00005  * R integration data definition.
00006  */
00007 
00008 #ifndef RSIMULATORRUNNER_H
00009 #define RSIMULATORRUNNER_H
00010 
00011 #include <string>
00012 
00013 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00014 #include "kernel/simulator/PluginInformation.h"
00015 
00023 class RSimulatorRunner : public ModelDataDefinition {
00024 public:
00026     RSimulatorRunner(Model* model, std::string name = "");
00028     virtual ~RSimulatorRunner() = default;
00029 
00030 public:
00032     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00034     static PluginInformation* GetPluginInformation();
00036     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00037 
00038 public:
00040     void setRExecutable(std::string rExecutable);
00042     std::string getRExecutable() const;
00044     void setWorkingDirectory(std::string workingDirectory);
00046     std::string getWorkingDirectory() const;
00048     void setPreludeScript(std::string preludeScript);
00050     std::string getPreludeScript() const;
00052     void setCommand(std::string command);
00054     std::string getCommand() const;
00056     void setLastStatus(std::string lastStatus);
00058     std::string getLastStatus() const;
00060     void setLastExitCode(int lastExitCode);
00062     int getLastExitCode() const;
00064     void setLastStdout(std::string lastStdout);
00066     std::string getLastStdout() const;
```

```
00068     void setLastStderr(std::string lastStderr);
00070     std::string getLastStderr() const;
00072     void setLastResponsePayload(std::string lastResponsePayload);
00074     std::string getLastResponsePayload() const;
00076     void setLastScriptFilename(std::string lastScriptFilename);
00078     std::string getLastScriptFilename() const;
00080     void setLastResponseFilename(std::string lastResponseFilename);
00082     std::string getLastResponseFilename() const;
00083
00085     bool executeCommand(std::string& errorMessage);
00086
00088     virtual std::string show() override;
00089
00090 protected:
00092     virtual bool _loadInstance(PersistenceRecord *fields) override;
00094     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00096     virtual bool _check(std::string& errorMessage) override;
00098     // virtual void _createInternalAndAttachedData() override;
00099
00100
00101 protected:
00102     // virtual void _createInternalStatisticReporters() override;
00103     // virtual void _createNonEditableDataDefinitions() override;
00104     // virtual void _createEditableDataDefinitions() override;
00105     // virtual void _createAttachedAttributes() override;
00106
00107 private:
00108     const struct DEFAULT_VALUES {
00109         std::string rExecutable = "Rscript";
00110         std::string workingDirectory = "";
00111         std::string preludeScript = "";
00112         std::string command = "";
00113         std::string lastStatus = "Idle";
00114         int lastExitCode = -1;
00115         std::string lastStdout = "";
00116         std::string lastStderr = "";
00117         std::string lastResponsePayload = "";
00118         std::string lastScriptFilename = "";
00119         std::string lastResponseFilename = "";
00120     } DEFAULT;
00121
00122     std::string _rExecutable = DEFAULT.rExecutable;
00123     std::string _workingDirectory = DEFAULT.workingDirectory;
00124     std::string _preludeScript = DEFAULT.preludeScript;
00125     std::string _command = DEFAULT.command;
00126     std::string _lastStatus = DEFAULT.lastStatus;
00127     int _lastExitCode = DEFAULT.lastExitCode;
00128     std::string _lastStdout = DEFAULT.lastStdout;
00129     std::string _lastStderr = DEFAULT.lastStderr;
00130     std::string _lastResponsePayload = DEFAULT.lastResponsePayload;
00131     std::string _lastScriptFilename = DEFAULT.lastScriptFilename;
00132     std::string _lastResponseFilename = DEFAULT.lastResponseFilename;
00133 };
00134
00135 #endif /* RSIMULATORRUNNER_H */
```

18.506 SPICERunner.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   SPICERunner.h
00008  * Author: bnmfw & joaomai
00009  *
00010  * Created on 10 de outubro de 2023, 23:00
00011  */
00012
00013 #ifndef SPICERUNNER_H
00014 #define SPICERUNNER_H
00015
00016 #include <iostream>
00017 #include <fstream>
00018 #include <sstream>
00019 #include <string>
00020 #include <vector>
00021 #include <map>
00022 #include <set>
00023 #include <algorithm>
00024 #include <functional>
00025
00026 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00027 #include "kernel/simulator/PluginInformation.h"
```

```
00028
00029 const std::vector<std::pair<std::string, float>>
      spice_unit_map{{"T",1e12},{"G",1e9},{"Meg",1e6},{"K",1e3},{"",1e0},{"m",1e-3},{"u",1e-6},{"n",1e-9},{"p",1e-12},{"f",1e-
00030 inline std::string uc(double value){
00031     float precision = 0.9999;
00032     for (auto element: spice_unit_map){
00033         if (value/element.second < precision) continue;
00034         return std::to_string(value/element.second) + element.first + " ";
00035     }
00036     return std::to_string(value/1e-18) + "a ";
00037 };
00038
00039 class SPICERunner : public ModelDataDefinition {
00040 public:
00041     SPICERunner(Model* model, std::string name = "");
00042     virtual ~SPICERunner();
00043
00044 public:
00045     std::string CompileSpiceFile();
00046     void SendComponent(std::string *instance, std::string subcircuit = "", std::string model = "");
00047     void SendCallback(std::function<void()> callback);
00048     void ConfigSim(double duration, double step);
00049     void Run();
00050     void PlotV(std::string net);
00051     template<typename... Args> void PlotV(std::string net, Args... args);
00052     void PlotVRelative(std::string comparison_base, std::string net);
00053     template<typename... Args> void PlotVRelative(std::string comparison_base, std::string net,
      Args... args);
00054     void PlotI(std::string net);
00055     template<typename... Args> void PlotI(std::string net, Args... args);
00056     void PlotIRelative(std::string comparison_base, std::string net);
00057     template<typename... Args> void PlotIRelative(std::string comparison_base, std::string net,
      Args... args);
00058     double* MeasurePeak(std::string label, std::string peak, std::string quantity, std::string node,
      float start, float finish);
00059     double* MeasureTrigTarg(std::string label, std::string trig, float trig_value, std::string
      trig_inclin, std::string targ, float targ_value, std::string targ_inclin);
00060     void ParseOutput();
00061     void setRunnerCommand(std::string runnerCommand);
00062     std::string getRunnerCommand() const;
00063     void setModelsPath(std::string modelsPath);
00064     std::string getModelsPath() const;
00065     void setWorkingInputFilename(std::string workingInputFilename);
00066     std::string getWorkingInputFilename() const;
00067     void setWorkingOutputFilename(std::string workingOutputFilename);
00068     std::string getWorkingOutputFilename() const;
00069     void setWorkingDirectory(std::string workingDirectory);
00070     std::string getWorkingDirectory() const;
00071     std::string getLastRunCommand() const;
00072
00073 public:
00074     virtual std::string show() override;
00075
00076 public:
00077     static PluginInformation* GetPluginInformation();
00078     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00079     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00080     // Added
00081     static std::vector<std::string> split(std::string word, char split_char);
00082
00083 protected:
00084     virtual bool _loadInstance(PersistenceRecord *fields) override;
00085     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00086
00087 protected:
00089     virtual bool _check(std::string& errorMessage) override;
00091     // virtual ParserChangesInformation* _getParserChangesInformation();
00093     // virtual void _initBetweenReplications();
00094     // virtual void _createInternalStatisticReporters() override;
00095     // virtual void _createNonEditableDataDefinitions() override;
00096     // virtual void _createEditableDataDefinitions() override;
00097     // virtual void _createAttachedAttributes() override;
00099     // virtual void _createInternalAndAttachedData(); /*< A ModelDataDefinition or ModelComponent that
      includes (internal) ou refers to (attach) other ModelDataDefinition must register them inside this
      method. */
00101     // virtual void _addSimulationControl(SimulationControl* property);
00102
00103
00104 protected:
00105
00106 private:
00107     // ...
00108
00109 private:
00110     const struct DEFAULT_VALUES {
00111         const std::string runnerCommand = "ngspice";
00112         const std::string modelsPath = "./";
```

```
00113         const std::string workingInputFilename = "input.cir";
00114         const std::string workingOutputFilename = "output";
00115         const std::string workingDirectory = "";
00116     } DEFAULT;
00117     std::string _runnerCommand = DEFAULT.runnerCommand;
00118     std::string _modelsPath = DEFAULT.modelsPath;
00119     std::string _workingInputFilename = DEFAULT.workingInputFilename;
00120     std::string _workingOutputFilename = DEFAULT.workingOutputFilename;
00121     std::string _workingDirectory = DEFAULT.workingDirectory;
00122     std::string _lastRunCommand = "";
00123
00124     std::set<std::string> subcircuits;
00125     std::set<std::string> models;
00126
00127     std::vector<std::function<void()» subscribers;
00128     std::vector<std::string*> instances;
00129
00130     // Plot
00131     std::vector<std::string> vplots;
00132     std::vector<std::string> iplots;
00133
00134     // Measure
00135     std::vector<std::string> measures;
00136     std::map<std::string, double*> promises;
00137
00138     std::string config;
00139
00140 private:
00141     // DummyElement* _internalDataDefinition = nullptr;
00142
00143 private:
00144     // ...
00145 };
00146
00147 #endif /* SPICERUNNER_H */
```

18.507 EntityGroup.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Group.h
00009  * Author: rlcancian
00010  *
00011  * Created on 12 de Junho de 2019, 19:00
00012  */
00013
00014 #ifndef ENTITYGROUP_H
00015 #define ENTITYGROUP_H
00016
00017 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00018 #include "../../../kernel/simulator/essentialPlugins/Entity.h"
00019 #include "kernel/util/List.h"
00020 #include <list>
00021 #include <map>
00022
00023 /*
00024  The EntityGroup represents a map of entities grouped togheter somehow.
00025  * The entities in this group are kept into a internal queue.
00026  * Usually one entity representing all grouped entities is generated by ModelComponents and the
      EntityGroup it represented is accessed through an attribute.
00027  * The Key of this map is the ID of the representative entity
00028  */
00029 class EntityGroup : public ModelDataDefinition {
00030 public:
00031     EntityGroup(Model* model, std::string name = "");
00032     virtual ~EntityGroup();
00033 public:
00034     virtual std::string show() override;
00035 public:
00036     static PluginInformation* GetPluginInformation();
00037     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00038     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00039 public:
00040     void insertElement(unsigned int idKey, Entity* modeldatum);
00041     void removeElement(unsigned int idKey, Entity* modeldatum);
00042     List<Entity*>* getGroup(unsigned int idKey);
00043 protected:
00044     virtual bool _loadInstance(PersistenceRecord *fields) override;
00045     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00046     virtual bool _check(std::string& errorMessage) override;
```

```
00047     virtual void _initBetweenReplications() override;
00048     // virtual void _createInternalAndAttachedData() override;
00049
00050 protected:
00051     virtual void _createInternalStatisticReporters() override;
00052     // virtual void _createNonEditableDataDefinitions() override;
00053     // virtual void _createEditableDataDefinitions() override;
00054     virtual void _createAttachedAttributes() override;
00055
00056 private:
00057     void _clearGroups();
00058     void _initCStats();
00059 private: //1::n
00060     std::map<unsigned int, List<Entity*>*>* _groupMap = new std::map<unsigned int, List<Entity*>*>();
00061     std::list<List<Entity*>*> _detachedEmptyGroups;
00062 private: // child inner modeldatum
00063     StatisticsCollector* _cstatNumberInGroup = nullptr;
00064 };
00065
00066 #endif /* ENTITYGROUP_H */
```

18.508 File.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   File.h
00009  * Author: rlcancian
00010  *
00011  * Created on 20 de Fileembro de 2019, 20:07
00012  */
00013
00014 #ifndef FILE_H
00015 #define FILE_H
00016
00017
00018 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00019 #include "../../../kernel/simulator/model/ModelDataManager.h"
00020 //#include "ParserChangesInformation.h"
00021 #include "kernel/simulator/PluginInformation.h"
00022
00041 class File : public ModelDataDefinition {
00042 public:
00043     enum class AccessMode : unsigned int {
00044         Read = 0,
00045         Write = 1,
00046         Append = 2,
00047         ReadWrite = 3
00048     };
00049
00050     File(Model* model, std::string name = "");
00051     virtual ~File() = default;
00052 public: // static
00053     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00054     static PluginInformation* GetPluginInformation();
00055     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00056 public:
00057     void setSystemFilename(std::string systemFilename);
00058     std::string getSystemFilename() const;
00059     std::string getFilenameOnly() const;
00060     std::string getPathOnly() const;
00061
00062     void setAccessMode(AccessMode accessMode);
00063     AccessMode getAccessMode() const;
00064     void setAccessModeAsString(std::string accessMode);
00065     std::string getAccessModeAsString() const;
00066
00067     virtual std::string show() override;
00068
00069 protected: // must be overriden
00070     virtual bool _loadInstance(PersistenceRecord *fields) override;
00071     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00072 protected: // could be overriden
00073     virtual bool _check(std::string& errorMessage) override;
00074     virtual ParserChangesInformation* _getParserChangesInformation() override;
00075
00076
00077 protected:
00078     // virtual void _createInternalStatisticReporters() override;
00079     // virtual void _createNonEditableDataDefinitions() override;
00080     // virtual void _createEditableDataDefinitions() override;
```

```
00081     // virtual void _createAttachedAttributes() override;
00082
00083 private:
00084     static std::string _accessModeToString(AccessMode accessMode);
00085     static bool _stringToAccessMode(const std::string& accessMode, AccessMode* parsedAccessMode);
00086     static std::string _normalizePathSeparators(const std::string& filename);
00087
00088 private:
00089     const struct DEFAULT_VALUES {
00090         std::string systemFilename = "";
00091         AccessMode accessMode = AccessMode::Read;
00092     } DEFAULT;
00093     std::string _systemFilename = DEFAULT.systemFilename;
00094     AccessMode _accessMode = DEFAULT.accessMode;
00095     bool _accessModeWasInvalid = false;
00096 };
00097
00098 #endif /* FILE_H */
```

18.509 AssignmentItem.h

```
00001 /*
00002  * The MIT License
00003  *
00004  * Copyright 2022 rlcancian.
00005  *
00006  * Permission is hereby granted, free of charge, to any person obtaining a copy
00007  * of this software and associated documentation files (the "Software"), to deal
00008  * in the Software without restriction, including without limitation the rights
00009  * to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
00010  * copies of the Software, and to permit persons to whom the Software is
00011  * furnished to do so, subject to the following conditions:
00012  *
00013  * The above copyright notice and this permission notice shall be included in
00014  * all copies or substantial portions of the Software.
00015  *
00016  * THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR
00017  * IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
00018  * FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
00019  * AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
00020  * LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
00021  * OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN
00022  * THE SOFTWARE.
00023  */
00024
00025 /*
00026  * File:   AssignmentItem.h
00027  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00028  *
00029  * Created on 2 de fevereiro de 2022, 14:12
00030  */
00031
00032 #ifndef ASSIGNMENTITEM_H
00033 #define ASSIGNMENTITEM_H
00034
00035 #include <functional>
00036 #include "../../../kernel/simulator/model/Model.h"
00037
00050 class Assignment {
00051 public:
00052     Assignment(Model* model, std::string destination, std::string expression="1", bool
      isAttributeNotVariable = true);
00053     Assignment(std::string destination, std::string expression, bool isAttributeNotVariable = true);
00054     virtual ~Assignment();
00055     void setDestination(std::string _destination);
00056     std::string getDestination() const;
00057     void setExpression(std::string _expression);
00058     std::string getExpression() const;
00059     void setAttributeNotVariable(bool isAttributeNotVariable);
00060     bool isAttributeNotVariable() const;
00061     void setChangeCallback(std::function<void(Assignment&)> callback);
00062
00063     // TODO: don't have name
00064     std::string getName() const;
00065
00066     List<SimulationControl*>* getSimulationControls() const;
00067     void ensureSimulationControls(Model* model);
00068     void _addSimulationControl(SimulationControl* control);
00069
00070     std::string getTypeDC() {return _typeDC;};
00071 public:
00072     bool loadInstance(PersistenceRecord *fields, unsigned int parentIndex);
00073     void saveInstance(PersistenceRecord *fields, unsigned int parentIndex, bool saveDefault);
00074 private:
00075     void _updateTypeDC();
```

```
00076     std::string _destination = "";
00077     std::string _expression = "";
00078     bool _isAttributeNotVariable = true;
00079     std::string _typeDC;
00080     std::function<void(Assignment&)> _changeCallback = nullptr;
00081 
00082     List<SimulationControl*>* _simulationControls = new List<SimulationControl*>();
00083 };
00084 
00085 
00086 #endif /* ASSIGNMENTITEM_H */
```

18.510 Formula.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
00007 /*
00008  * File:   Formula.h
00009  * Author: rlcancian
00010  *
00011  * Created on 20 de Junho de 2019, 00:56
00012  */
00013 
00014 #ifndef FORMULA_H
00015 #define FORMULA_H
00016 
00017 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00018 #include "../../../kernel/simulator/model/ModelDataManager.h"
00019 #include "kernel/simulator/Plugin.h"
00020 //#include "kernel/simulator/Parser_if.h"
00021 
00039 class Formula : public ModelDataDefinition {
00040 public:
00046     Formula(Model* model, std::string name = "");
00050     virtual ~Formula() override;
00051 public: // virtual
00056     virtual std::string show() override;
00057 public:
00062     unsigned int size();
00068     void setExpression(std::string formulaExpression, std::string index = "");
00069     //void setExpression(std::string formulaExpression);
00075     std::string getExpression(std::string index = "");
00076     //std::string getExpression();
00077     //double getValue();
00083     double getValue(std::string index);
00084 public: // statics
00089     static PluginInformation* GetPluginInformation();
00096     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00103     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00104 protected: // must be overriden by derived classes
00105     virtual bool _loadInstance(PersistenceRecord *fields) override;
00106     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00107 protected: // could be overriden by derived classes
00108     virtual bool _check(std::string& errorMessage) override;
00109     //virtual void _addSimulationControl(SimulationControl* property);
00111     //virtual ParserChangesInformation* _getParserChangesInformation();
00112     //virtual void _initBetweenReplications();
00113     // virtual void _createInternalStatisticReporters() override;
00114     // virtual void _createNonEditableDataDefinitions() override;
00115     // virtual void _createEditableDataDefinitions() override;
00116     virtual void _createAttachedAttributes() override;
00117 
00118 private:
00119     std::map<std::string, std::string>* _formulaExpressions = new std::map<std::string,
      std::string>(); // map<index,formula>
00120 private:
00122 };
00123 
00124 #endif /* FORMULA_H */
```

18.511 Label.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005 
00006 /*
```

```
00007  * File:   Label.h
00008  * Author: rlcancian
00009  *
00010  * Created on 15 de janeiro de 2022, 10:13
00011  */
00012
00013 #ifndef LABEL_H
00014 #define LABEL_H
00015 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00016 #include "../../../kernel/simulator/essentialPlugins/Entity.h"
00017
00018 class Label : public ModelDataDefinition {
00019 public:
00020     Label(Model* model, std::string name = "");
00021     virtual ~Label() = default;
00022 public: // static
00023     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00024     static PluginInformation* GetPluginInformation();
00025     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00026 public:
00027     virtual std::string show() override;
00028     void setLabel(std::string _label);
00029     std::string getLabel() const;
00030     void setEnterIntoLabelComponent(ModelComponent* enteringLabelComponent);
00031     ModelComponent* getEnterIntoLabelComponent() const;
00032     void sendEntityToLabelComponent(Entity* entity, double timeDelay);
00033 protected: // must be overriden
00034     virtual bool _loadInstance(PersistenceRecord *fields) override;
00035     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00036 protected: // could be overriden
00037     virtual bool _check(std::string& errorMessage) override;
00038     //virtual ParserChangesInformation* _getParserChangesInformation();
00039     //virtual void _initBetweenReplications();
00040     // virtual void _createInternalStatisticReporters() override;
00041     // virtual void _createNonEditableDataDefinitions() override;
00042     // virtual void _createEditableDataDefinitions() override;
00043     // virtual void _createAttachedAttributes() override;
00044     //virtual void _createInternalAndAttachedData();
00045
00046 protected:
00047
00048 private:
00049     std::string _label;
00050     ModelComponent* _enteringLabelComponent = nullptr;
00051 };
00052
00053 #endif /* LABEL_H */
```

18.512 Set.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Set.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:11
00012  */
00013
00014 #ifndef SET_H
00015 #define SET_H
00016
00017 #include <vector>
00018
00019 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00020 #include "../../../kernel/simulator/model/ModelDataManager.h"
00021 #include "kernel/simulator/ParserChangesInformation.h"
00022 #include "kernel/simulator/PluginInformation.h"
00023 #include "../../../kernel/simulator/essentialPlugins/EntityType.h"
00024
00047 class Set : public ModelDataDefinition {
00048 public:
00049     Set(Model* model, std::string name = "", std::vector<std::string> allowedElementTypes = {});
00050     virtual ~Set() override;
00051 public: // static
00052     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00053     static PluginInformation* GetPluginInformation();
00054     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00055 public:
00056     virtual std::string show() override;
00057 public:
```

```
00058     //void setSetOfType(std::string _setOfType);
00059     std::string getSetOfType() const;
00065     bool setSetOfType(const std::string& setOfType);
00067     bool canChangeSetOfType() const;
00074     void setAllowedElementTypes(const std::vector<std::string>& allowedElementTypes);
00076     void addAllowedElementType(const std::string& allowedElementType);
00078     std::vector<std::string> getAllowedElementTypes() const;
00079     List<ModelDataDefinition*>* getElementSet() const;
00080
00081     void addElementSet(ModelDataDefinition* newElement);
00082     void removeElementSet(ModelDataDefinition* element);
00089     ModelDataDefinition* createElementSetOfType(const std::string& typeName, const std::string& name =
      "");
00090
00091 protected: // must be overriden
00092     virtual bool _loadInstance(PersistenceRecord *fields) override;
00093     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00094 protected: // could be overriden
00095     virtual bool _check(std::string& errorMessage) override;
00096     virtual ParserChangesInformation* _getParserChangesInformation() override;
00097 protected:
00098     // virtual void _createInternalStatisticReporters() override;
00099     // virtual void _createNonEditableDataDefinitions() override;
00100     // virtual void _createEditableDataDefinitions() override;
00101     virtual void _createAttachedAttributes() override;
00102
00103 private:
00104     //ElementManager* _elems;
00105
00106     const struct DEFAULT_VALUES {
00107         const unsigned int membersSize = 0;
00108         const std::string setOfType = "";
00109     } DEFAULT;
00110     List<ModelDataDefinition*>* _elementSet = new List<ModelDataDefinition*>();
00111     std::string _setOfType = DEFAULT.setOfType;
00112     std::vector<std::string> _allowedElementTypes;
00113 };
00114
00115 #endif /* SET_H */
```

18.513 Variable.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Variable.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 4 de Setembro de 2018, 18:28
00012  */
00013
00014 #ifndef VARIABLE_H
00015 #define VARIABLE_H
00016
00017 #include "../../../kernel/simulator/essentialPlugins/Attribute.h"
00018 #include "../../../kernel/simulator/model/ModelDataManager.h"
00019 #include "kernel/simulator/Plugin.h"
00020 #include "../../../kernel/simulator/essentialPlugins/SparseValueStore.h"
00021
00041 class Variable : public Attribute {
00042 public:
00043     Variable(Model* model, std::string name = "");
00044     virtual ~Variable() override;
00045 public:
00046     virtual std::string show() override;
00047 public: //static
00048     static PluginInformation* GetPluginInformation();
00049     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00050     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00051 public:
00053     double getValue(std::string index="");
00055     void setValue(double value, std::string index="");
00057     void copyValuesFrom(const Variable& source);
00059     std::string getInitialValuesText() const override;
00061     void setInitialValuesText(std::string valuesText) override;
00063     void insertDimentionSize(unsigned int size);
00065     std::map<std::string, double> *getValues() const;
00067     SparseValueStore* getValueStore();
00068     ModelDataDefinition* get_scope();
00069     void set_scope(ModelDataDefinition* const scope);
00070
```

```
00071 protected:
00072     virtual bool _loadInstance(PersistenceRecord *fields) override;
00073     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00074     virtual bool _check(std::string& errorMessage) override;
00075     virtual void _initBetweenReplications() override;
00076
00077
00078 protected:
00079     // virtual void _createInternalStatisticReporters() override;
00080     // virtual void _createNonEditableDataDefinitions() override;
00081     // virtual void _createEditableDataDefinitions() override;
00082     // virtual void _createAttachedAttributes() override;
00083
00084 private:
00085     //const struct DEFAULT_VALUES { } DEFAULT;
00086     SparseValueStore* _values = new SparseValueStore();
00087     ModelDataDefinition* scope = nullptr; // not used so far
00088 };
00089
00090 #endif /* VARIABLE_H */
```

18.514 Conveyor.h

```
00001 #ifndef CONVEYOR_H
00002 #define CONVEYOR_H
00003
00004 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00005 #include "kernel/simulator/PluginInformation.h"
00006 #include "plugins/data/MaterialHandling/Segment.h"
00007
00016 class Conveyor : public ModelDataDefinition {
00017 public:
00018     Conveyor(Model* model, std::string name = "");
00019     virtual ~Conveyor() override = default;
00020 public:
00021     virtual std::string show() override;
00022 public: // static
00023     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00024     static PluginInformation* GetPluginInformation();
00025     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00026 public:
00027     void setSegment(Segment* segment);
00028     Segment* getSegment() const;
00029     void setVelocity(double velocity);
00030     double getVelocity() const;
00031     void setCapacity(unsigned int capacity);
00032     unsigned int getCapacity() const;
00033     void setInitiallyActive(bool initiallyActive);
00034     bool isInitiallyActive() const;
00035     void setActive(bool active);
00036     bool isActive() const;
00037     unsigned int getCurrentAllocation() const;
00038     bool access(unsigned int quantity = 1);
00039     bool exit(unsigned int quantity = 1);
00040     double getDistanceBetween(const std::string& fromStationName, const std::string& toStationName)
      const;
00041 protected:
00042     virtual bool _loadInstance(PersistenceRecord* fields) override;
00043     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00044     virtual bool _check(std::string& errorMessage) override;
00045     virtual void _initBetweenReplications() override;
00046     virtual void _createEditableDataDefinitions() override;
00047 private:
00048     const struct DEFAULT_VALUES {
00049         const double velocity = 1.0;
00050         const unsigned int capacity = 1;
00051         const bool initiallyActive = true;
00052     } DEFAULT;
00053     Segment* _segment = nullptr;
00054     double _velocity = DEFAULT.velocity;
00055     unsigned int _capacity = DEFAULT.capacity;
00056     bool _initiallyActive = DEFAULT.initiallyActive;
00057     bool _active = DEFAULT.initiallyActive;
00058     unsigned int _currentAllocation = 0;
00059 };
00060
00061 #endif /* CONVEYOR_H */
```

18.515 Distance.h

```
00001 /*
00002  * File:   Distance.h
```

```
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 30 de Agosto de 2026
00006  */
00007
00008 #ifndef DISTANCE_H
00009 #define DISTANCE_H
00010
00011 #include <string>
00012 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00013 #include "kernel/simulator/PluginInformation.h"
00014 #include "kernel/util/List.h"
00015
00021 class DistanceEntry {
00022 public:
00023     DistanceEntry(std::string fromStationName, std::string toStationName, double length, bool
      bidirectional = true) {
00024         this->fromStationName = fromStationName;
00025         this->toStationName = toStationName;
00026         this->length = length;
00027         this->bidirectional = bidirectional;
00028     }
00029 public:
00030     std::string fromStationName;
00031     std::string toStationName;
00032     double length;
00033     bool bidirectional;
00034 };
00035
00051 class Distance : public ModelDataDefinition {
00052 public:
00053     Distance(Model* model, std::string name = "");
00054     virtual ~Distance() override;
00055 public: // static
00056     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00057     static PluginInformation* GetPluginInformation();
00058     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00059 public: // virtual
00060     virtual std::string show() override;
00061 public:
00062     List<DistanceEntry*>* getEntries() const;
00063     void insertEntry(DistanceEntry* entry);
00065     double getDistanceBetween(const std::string& fromStationName, const std::string& toStationName)
      const;
00066 protected: // must be overriden
00067     virtual bool _loadInstance(PersistenceRecord *fields) override;
00068     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00069 protected: // could be overriden by derived classes
00070     virtual bool _check(std::string& errorMessage) override;
00071 private:
00072     List<DistanceEntry*>* _entries = new List<DistanceEntry*>();
00073 };
00074
00075 #endif /* DISTANCE_H */
```

18.516 Segment.h

```
00001 /*
00002  * File:   Segment.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 30 de Agosto de 2026
00006  */
00007
00008 #ifndef SEGMENT_H
00009 #define SEGMENT_H
00010
00011 #include <string>
00012 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00013 #include "kernel/simulator/PluginInformation.h"
00014 #include "kernel/util/List.h"
00015
00020 class SegmentStep {
00021 public:
00022     SegmentStep(std::string stationName, double lengthToNext) {
00023         this->stationName = stationName;
00024         this->lengthToNext = lengthToNext;
00025     }
00026 public:
00027     std::string stationName;
00028     double lengthToNext;
00029 };
00030
00047 class Segment : public ModelDataDefinition {
00048 public:
```

```
00049     Segment(Model* model, std::string name = "");
00050     virtual ~Segment() override;
00051 public: // static
00052     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00053     static PluginInformation* GetPluginInformation();
00054     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00055 public: // virtual
00056     virtual std::string show() override;
00057 public:
00058     List<SegmentStep*>* getSteps() const;
00059     void insertStep(SegmentStep* step);
00061     double getDistanceBetween(const std::string& fromStationName, const std::string& toStationName)
      const;
00062 protected: // must be overriden
00063     virtual bool _loadInstance(PersistenceRecord *fields) override;
00064     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00065 protected: // could be overriden by derived classes
00066     virtual bool _check(std::string& errorMessage) override;
00067 private:
00068     List<SegmentStep*>* _steps = new List<SegmentStep*>();
00069 };
00070
00071 #endif /* SEGMENT_H */
```

18.517 Sequence.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Sequence.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:12
00012  */
00013
00014 #ifndef SEQUENCE_H
00015 #define SEQUENCE_H
00016
00017 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00018 #include "../../../kernel/simulator/model/ModelDataManager.h"
00019 #include "kernel/simulator/PluginInformation.h"
00020 #include "Station.h"
00021 #include "../Logic/Label.h"
00022 #include "../Logic/AssignmentItem.h"
00023
00024 class SequenceStep : public PersistentObject_base {
00025 public:
00026     SequenceStep(Station* station, std::list<Assignment*>* assignments = nullptr);
00027     SequenceStep(Label* label, std::list<Assignment*>* assignments = nullptr);
00028     SequenceStep(Model* model, std::string stationOrLabelName, bool isStation = true,
      std::list<Assignment*>* assignments = nullptr);
00029     virtual ~SequenceStep() override;
00030 public: // virtual
00031
00032     virtual bool _loadInstance(PersistenceRecord *fields, unsigned int parentIndex);
00033     virtual void _saveInstance(PersistenceRecord *fields, unsigned int parentIndex, bool
      saveDefaultValues);
00034     virtual bool _loadInstance(PersistenceRecord *fields) override;
00035     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00036
00037 public:
00038
00039     std::list<Assignment*>* getAssignments() const;
00040     void setStation(Station* _station);
00041     Station* getStation() const;
00042     void setElementManager(ModelDataManager* _modeldataManager);
00043     void setLabel(Label* _label);
00044     Label* getLabel() const;
00045
00046 protected:
00047 private:
00048
00049     const struct DEFAULT_VALUES {
00050         const unsigned int assignmentsSize = 0;
00051     } DEFAULT;
00052     Station* _station = nullptr;
00053     Label* _label = nullptr;
00054     std::list<Assignment*>* _assignments = new std::list<Assignment*>();
00055 private:
00056     ModelDataManager* _modeldataManager = nullptr;
00057 };
```

```
00058
00079 class Sequence : public ModelDataDefinition {
00080 public:
00081
00082
00083 public:
00084     Sequence(Model* model, std::string name = "");
00085     virtual ~Sequence() override;
00086 public:
00087     virtual std::string show() override;
00088 public: // static
00089     static PluginInformation* GetPluginInformation();
00090     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00091     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00092 public:
00093     List<SequenceStep*>* getSteps() const;
00094 protected:
00095     virtual bool _loadInstance(PersistenceRecord *fields) override;
00096     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00097     virtual bool _check(std::string& errorMessage) override;
00098 protected:
00099     // virtual void _createInternalStatisticReporters() override;
00100     // virtual void _createNonEditableDataDefinitions() override;
00101     // virtual void _createEditableDataDefinitions() override;
00102     virtual void _createAttachedAttributes() override;
00103 private:
00104     List<SequenceStep*>* _steps = new List<SequenceStep*>();
00105 };
00106
00107 #endif /* SEQUENCE_H */
```

18.518 Station.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Station.h
00009  * Author: rlcancian
00010  *
00011  * Created on 03 de Junho de 2019, 15:12
00012  */
00013
00014 #ifndef STATION_H
00015 #define STATION_H
00016
00017 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00018 #include "../../../kernel/simulator/model/ModelDataManager.h"
00019 #include "kernel/simulator/Plugin.h"
00020 #include "../../../kernel/simulator/essentialPlugins/Entity.h"
00021
00043 class Station : public ModelDataDefinition {
00044 public:
00045     Station(Model* model, std::string name = "");
00046     virtual ~Station();
00047 public:
00048     virtual std::string show() override;
00049 public: // static
00050     static PluginInformation* GetPluginInformation();
00051     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00052     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00053 public:
00054     void initBetweenReplications();
00055     void enter(Entity* entity);
00056     void leave(Entity* entity);
00057     void setEnterIntoStationComponent(ModelComponent* _enterIntoStationComponent);
00058     ModelComponent* getEnterIntoStationComponent() const;
00059 protected:
00060     virtual bool _loadInstance(PersistenceRecord *fields) override;
00061     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00062     virtual bool _check(std::string& errorMessage) override;
00063     // virtual void _createInternalAndAttachedData() override;
00064     virtual void _initBetweenReplications() override;
00065
00066 protected:
00067     virtual void _createInternalStatisticReporters() override;
00068     // virtual void _createNonEditableDataDefinitions() override;
00069     // virtual void _createEditableDataDefinitions() override;
00070     virtual void _createAttachedAttributes() override;
00071
00072 private:
00073     void _initCStats();
```

```
00074     unsigned int _numberInStation = 0;
00075     ModelComponent* _enterIntoStationComponent = nullptr;
00076 private: // inner elements
00077     StatisticsCollector* _cstatNumberInStation = nullptr;
00078     StatisticsCollector* _cstatTimeInStation = nullptr;
00079     friend class StationTestProbe;
00080 };
00081
00082 #endif /* STATION_H */
```

18.519 Storage.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Storage.h
00009  * Author: rlcancian
00010  *
00011  * Created on 20 de Storageembro de 2019, 20:06
00012  */
00013
00014 #ifndef STORAGE_H
00015 #define STORAGE_H
00016
00017
00018 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00019 #include "../../../kernel/simulator/model/ModelDataManager.h"
00020 //#include "kernel/simulator/ParserChangesInformation.h"
00021 #include "kernel/simulator/PluginInformation.h"
00022
00038 class Storage : public ModelDataDefinition {
00039 public:
00040     Storage(Model* model, std::string name = "");
00041     virtual ~Storage() = default;
00042 public: // static
00043     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00044     static PluginInformation* GetPluginInformation();
00045     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00046 public:
00047     virtual std::string show() override;
00048     void setTotalArea(double _totalArea);
00049     double getTotalArea() const;
00050     void setCapacity(unsigned int _capacity);
00051     unsigned int getCapacity() const;
00052     void setUnitsPerArea(double _unitsPerArea);
00053     double getUnitsPerArea() const;
00054     unsigned int getCurrentOccupation() const;
00055     bool store(unsigned int quantity = 1);
00056     bool unstore(unsigned int quantity = 1);
00057
00058 protected: // must be overriden
00059     virtual bool _loadInstance(PersistenceRecord *fields) override;
00060     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00061 protected: // could be overriden
00062     virtual bool _check(std::string& errorMessage) override;
00063     virtual ParserChangesInformation* _getParserChangesInformation() override;
00064     virtual void _initBetweenReplications() override;
00065
00066 protected:
00067     // virtual void _createInternalStatisticReporters() override;
00068     // virtual void _createNonEditableDataDefinitions() override;
00069     // virtual void _createEditableDataDefinitions() override;
00070     // virtual void _createAttachedAttributes() override;
00071
00072 private:
00073
00074     const struct DEFAULT_VALUES {
00075         const double totalArea = 1;
00076         const unsigned int capacity = 10;
00077         const double unitsPerArea = 1;
00078     } DEFAULT;
00079     double _totalArea = DEFAULT.totalArea;
00080     unsigned int _capacity = DEFAULT.capacity;
00081     double _unitsPerArea = DEFAULT.unitsPerArea;
00082     unsigned int _currentOccupation = 0;
00083
00084 private:
00085     //@TODO: Add statisticCollector for ProportionOfStorageUsage
00086 };
00087 #endif /* STORAGE_H */
```

18.520 Transporter.h

```
00001 #ifndef TRANSPORTER_H
00002 #define TRANSPORTER_H
00003
00004 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00005 #include "kernel/simulator/Event.h"
00006 #include "kernel/simulator/PluginInformation.h"
00007 #include "plugins/data/MaterialHandling/Distance.h"
00008 #include "plugins/data/MaterialHandling/Station.h"
00009
00010 class Transporter : public ModelDataDefinition {
00011 public:
00012     struct TravelCompletion {
00013         Station* destination = nullptr;
00014     };
00015 public:
00016     Transporter(Model* model, std::string name = "");
00017     virtual ~Transporter() override = default;
00018 public:
00019     virtual std::string show() override;
00020 public: // static
00021     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00022     static PluginInformation* GetPluginInformation();
00023     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00024 public:
00025     void setDistance(Distance* distance);
00026     Distance* getDistance() const;
00027     void setInitialStation(Station* station);
00028     Station* getInitialStation() const;
00029     Station* getCurrentStation() const;
00030     void setSpeed(double speed);
00031     double getSpeed() const;
00032     void setInitiallyActive(bool initiallyActive);
00033     bool isInitiallyActive() const;
00034     void setActive(bool active);
00035     bool isActive() const;
00036     bool isBusy() const;
00037     bool reserve();
00038     void releaseAt(Station* station);
00039     double getTravelTime(const Station* fromStation, const Station* toStation) const;
00040     void handleTravelCompletion(void* parameter);
00041 protected:
00042     virtual bool _loadInstance(PersistenceRecord* fields) override;
00043     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00044     virtual bool _check(std::string& errorMessage) override;
00045     virtual void _initBetweenReplications() override;
00046     virtual void _createEditableDataDefinitions() override;
00047 private:
00048     const struct DEFAULT_VALUES {
00049         const double speed = 1.0;
00050         const bool initiallyActive = true;
00051     } DEFAULT;
00052     Distance* _distance = nullptr;
00053     Station* _initialStation = nullptr;
00054     Station* _currentStation = nullptr;
00055     double _speed = DEFAULT.speed;
00056     bool _initiallyActive = DEFAULT.initiallyActive;
00057     bool _active = DEFAULT.initiallyActive;
00058     bool _busy = false;
00059 };
00060
00061 #endif /* TRANSPORTER_H */
```

18.521 ColoredPetriNetNetwork.h

```
00001 /*
00002  * File:   ColoredPetriNetNetwork.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 31 de Agosto de 2026
00006  */
00007
00008 #ifndef COLOREDPETRINETNETWORK_H
00009 #define COLOREDPETRINETNETWORK_H
00010
00011 #include "DefaultNetwork.h"
00012 #include "plugins/components/ModalModel/PetriPlace.h"
00013 #include "plugins/data/ModalModel/CPNArc.h"
00014 #include "plugins/data/ModalModel/CPNTransition.h"
00015
00016 #include <map>
00017 #include <string>
00018 #include <vector>
```

```
00019
00029 class ColoredPetriNetNetwork : public DefaultNetwork {
00030 public:
00031     enum class FiringMode : unsigned int {
00032         SingleDeterministic = 0
00033     };
00034
00035 public:
00036     ColoredPetriNetNetwork(Model* model, std::string name = "");
00037     virtual ~ColoredPetriNetNetwork() override;
00038
00039 public: // static
00040     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00041     static PluginInformation* GetPluginInformation();
00042     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00043
00044 public:
00045     void addPlace(PetriPlace* place);
00046     void removePlace(PetriPlace* place);
00047     List<PetriPlace*>* getPlaces() const;
00048     void addTransition(CPNTransition* transition);
00049     void removeTransition(CPNTransition* transition);
00050     List<CPNTransition*>* getTransitions() const;
00051     bool addArc(CPNArc* arc);
00052     void removeArc(CPNArc* arc);
00053     List<CPNArc*>* getArcs() const;
00054     std::vector<CPNArc*> getInputArcs(CPNTransition* transition) const;
00055     std::vector<CPNArc*> getOutputArcs(CPNTransition* transition) const;
00056     bool isEnabled(CPNTransition* transition) const;
00057     CPNTransition* firstEnabledTransition() const;
00058     bool fire(CPNTransition* transition);
00059     void setInitialTokens(PetriPlace* place, std::string color, unsigned int quantity);
00060     unsigned int getInitialTokens(PetriPlace* place, std::string color = "default") const;
00061     void captureCurrentMarkingAsInitial();
00062     void restoreInitialMarking();
00063     FiringMode getFiringMode() const;
00064     void setFiringMode(FiringMode firingMode);
00065     virtual std::string show() override;
00066
00067 protected:
00068     virtual bool _loadInstance(PersistenceRecord* fields) override;
00069     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00070     virtual bool _check(std::string& errorMessage) override;
00071     virtual void _initBetweenReplications() override;
00072     virtual NetworkActivationResult _activate(const NetworkActivationFrame& frame) override;
00073
00074 private:
00075     bool _hasPlace(PetriPlace* place) const;
00076     bool _hasTransition(CPNTransition* transition) const;
00077     bool _hasArc(CPNArc* arc) const;
00078     bool _guardAllows(CPNTransition* transition) const;
00079     void _removeIncidentArcs(PetriPlace* place);
00080     void _removeIncidentArcs(CPNTransition* transition);
00081
00082 private:
00083     List<PetriPlace*>* _places = new List<PetriPlace*>();
00084     List<CPNTransition*>* _transitions = new List<CPNTransition*>();
00085     List<CPNArc*>* _arcs = new List<CPNArc*>();
00086     std::map<PetriPlace*, std::map<std::string, unsigned int>> _initialMarking;
00087     FiringMode _firingMode = FiringMode::SingleDeterministic;
00088 };
00089
00090 #endif /* COLOREDPETRINETNETWORK_H */
```

18.522 CPNArc.h

```
00001 /*
00002  * File:   CPNArc.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 31 de Agosto de 2026
00006  */
00007
00008 #ifndef CPNARC_H
00009 #define CPNARC_H
00010
00011 #include "kernel/simulator/model/ModelDataDefinition.h"
00012 #include "kernel/simulator/PluginInformation.h"
00013
00014 #include <map>
00015 #include <string>
00016
00017 class CPNTransition;
00018 class PetriPlace;
00019
```

```
00027 class CPNArc : public ModelDataDefinition {
00028 public:
00029     enum class Direction : unsigned int {
00030         PlaceToTransition = 0,
00031         TransitionToPlace = 1
00032     };
00033
00034 public:
00035     CPNArc(Model* model, std::string name = "");
00036     CPNArc(Model* model, PetriPlace* place, CPNTransition* transition, Direction direction,
      std::string name = "");
00037     virtual ~CPNArc() override;
00038
00039 public:
00040     static PluginInformation* GetPluginInformation();
00041     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00042     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00043
00044 public:
00045     void setPlace(PetriPlace* place);
00046     PetriPlace* getPlace() const;
00047     void setTransition(CPNTransition* transition);
00048     CPNTransition* getTransition() const;
00049     void setDirection(Direction direction);
00050     Direction getDirection() const;
00051     bool isInputArc() const;
00052     bool isOutputArc() const;
00053     void setInscription(std::string color, unsigned int quantity);
00054     unsigned int getInscription(std::string color = "default") const;
00055     const std::map<std::string, unsigned int>& getInscriptions() const;
00056     void clearInscriptions();
00057     std::string getPlaceName() const;
00058     std::string getTransitionName() const;
00059     virtual std::string show() override;
00060
00061 protected:
00062     virtual bool _loadInstance(PersistenceRecord* fields) override;
00063     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00064     virtual bool _check(std::string& errorMessage) override;
00065
00066 private:
00067     void _syncEndpointNames();
00068
00069 private:
00070     PetriPlace* _place = nullptr;
00071     CPNTransition* _transition = nullptr;
00072     std::string _placeName = "";
00073     std::string _transitionName = "";
00074     Direction _direction = Direction::PlaceToTransition;
00075     std::map<std::string, unsigned int>* _inscriptions = new std::map<std::string, unsigned int>();
00076 };
00077
00078 #endif /* CPNARC_H */
```

18.523 CPNTransition.h

```
00001 /*
00002  * File:   CPNTransition.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 31 de Agosto de 2026
00006  */
00007
00008 #ifndef CPNTRANSITION_H
00009 #define CPNTRANSITION_H
00010
00011 #include "plugins/components/ModalModel/DefaultNode.h"
00012
00019 class CPNTransition : public DefaultNode {
00020 public:
00021     CPNTransition(Model* model, std::string name = "");
00022     virtual ~CPNTransition() override = default;
00023
00024 public:
00025     void setGuardExpression(std::string guardExpression);
00026     std::string getGuardExpression() const;
00027     void setPriority(unsigned int priority);
00028     unsigned int getPriority() const;
00029     virtual std::string show() override;
00030
00031 public: // static
00032     static PluginInformation* GetPluginInformation();
00033     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00034     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00035
```

```
00036 protected:
00037     virtual bool _loadInstance(PersistenceRecord* fields) override;
00038     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00039     virtual bool _check(std::string& errorMessage) override;
00040 
00041 private:
00042     std::string _guardExpression = "";
00043     unsigned int _priority = 0;
00044 };
00045 
00046 #endif /* CPNTRANSITION_H */
```

18.524 DefaultNetwork.h

```
00001 /*
00002  * File:   DefaultNetwork.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 30 de Agosto de 2026
00006  */
00007 
00008 #ifndef DEFAULTNETWORK_H
00009 #define DEFAULTNETWORK_H
00010 
00011 #include <string>
00012 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00013 #include "kernel/simulator/PluginInformation.h"
00014 #include "kernel/util/List.h"
00015 #include "NetworkActivation.h"
00016 
00017 class Counter;
00018 
00045 class DefaultNetwork : public ModelDataDefinition {
00046 public:
00047     DefaultNetwork(Model* model, std::string name = "", std::string dataDefinitionTypename =
      Util::TypeOf<DefaultNetwork>());
00048     virtual ~DefaultNetwork() override;
00049 public: // static
00050     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00051     static PluginInformation* GetPluginInformation();
00052     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00053 public: // virtual
00054     virtual std::string show() override;
00055 public: // port schema
00057     unsigned int addInputPort(const std::string& portName);
00059     unsigned int addOutputPort(const std::string& portName);
00060     unsigned int getNumInputPorts() const;
00061     unsigned int getNumOutputPorts() const;
00062     std::string getInputPortName(unsigned int index) const;
00063     std::string getOutputPortName(unsigned int index) const;
00065     int getInputPortIndex(const std::string& portName) const;
00067     int getOutputPortIndex(const std::string& portName) const;
00068 public: // activation
00070     double getActivationCount() const;
00079     NetworkActivationResult activate(const NetworkActivationFrame& frame);
00080 protected: // must be overriden
00081     virtual bool _loadInstance(PersistenceRecord *fields) override;
00082     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00083 protected: // could be overriden by derived classes
00084     virtual bool _check(std::string& errorMessage) override;
00085     virtual void _initBetweenReplications() override;
00086     virtual void _createInternalStatisticReporters() override;
00095     virtual NetworkActivationResult _activate(const NetworkActivationFrame& frame);
00096 private:
00097     List<std::string>* _inputPortNames = new List<std::string>();
00098     List<std::string>* _outputPortNames = new List<std::string>();
00099     Counter* _activationCounter = nullptr;
00100 };
00101 
00102 #endif /* DEFAULTNETWORK_H */
```

18.525 DirectedAcyclicGraphNetwork.h

```
00001 /*
00002  * File:   DirectedAcyclicGraphNetwork.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 30 de Agosto de 2026
00006  */
00007 
00008 #ifndef DIRECTEDACYCLICGRAPHNETWORK_H
00009 #define DIRECTEDACYCLICGRAPHNETWORK_H
```

```
00010
00011 #include "DirectedGraphNetwork.h"
00012
00019 class DirectedAcyclicGraphNetwork : public DirectedGraphNetwork {
00020 public:
00021     DirectedAcyclicGraphNetwork(Model* model, std::string name = "");
00022     virtual ~DirectedAcyclicGraphNetwork() override = default;
00023
00024 public:
00025     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00026     static PluginInformation* GetPluginInformation();
00027     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00028
00029 public:
00030     virtual bool addEdge(GraphEdge* edge) override;
00031
00032 protected:
00033     virtual bool _check(std::string& errorMessage) override;
00034 };
00035
00036 #endif /* DIRECTEDACYCLICGRAPHNETWORK_H */
```

18.526 DirectedGraphNetwork.h

```
00001 /*
00002  * File:   DirectedGraphNetwork.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 30 de Agosto de 2026
00006  */
00007
00008 #ifndef DIRECTEDGRAPHNETWORK_H
00009 #define DIRECTEDGRAPHNETWORK_H
00010
00011 #include "GraphNetwork.h"
00012
00020 class DirectedGraphNetwork : public GraphNetwork {
00021 public:
00022     DirectedGraphNetwork(Model* model, std::string name = "", std::string dataDefinitionTypename =
      Util::TypeOf<DirectedGraphNetwork>());
00023     virtual ~DirectedGraphNetwork() override = default;
00024
00025 public:
00026     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00027     static PluginInformation* GetPluginInformation();
00028     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00029
00030 public:
00031     virtual bool isDirected() const override;
00032     virtual std::vector<GraphNode*> getPredecessors(GraphNode* node) const override;
00033     virtual std::vector<GraphNode*> getSuccessors(GraphNode* node) const override;
00034     virtual std::vector<GraphEdge*> getIncomingEdges(GraphNode* node) const override;
00035     virtual std::vector<GraphEdge*> getOutgoingEdges(GraphNode* node) const override;
00036     virtual unsigned int inDegree(GraphNode* node) const override;
00037     virtual unsigned int outDegree(GraphNode* node) const override;
00038     std::vector<std::vector<GraphNode*>> stronglyConnectedComponents() const;
00039     virtual bool hasCycle() const override;
00040     virtual GraphTopologicalOrderResult topologicalOrder() const override;
00041
00042 protected:
00043     virtual bool _edgeConnects(GraphEdge* edge, GraphNode* source, GraphNode* destination) const
      override;
00044     virtual GraphNode* _oppositeNode(GraphEdge* edge, GraphNode* node) const override;
00045 };
00046
00047 #endif /* DIRECTEDGRAPHNETWORK_H */
```

18.527 EFSMNetwork.h

```
00001 /*
00002  * File:   EFSMNetwork.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 30 de Agosto de 2026
00006  */
00007
00008 #ifndef EFSMNETWORK_H
00009 #define EFSMNETWORK_H
00010
00011 #include "DefaultNetwork.h"
00012 #include "plugins/components/ModalModel/DefaultTransitionExtensions.h"
00013 #include "plugins/components/ModalModel/FSMState.h"
```

```
00014
00015 #include <memory>
00016 #include <string>
00017 #include <unordered_map>
00018
00019 class Sampler_if;
00020
00039 class EFSMNetwork : public DefaultNetwork {
00040 public:
00041     enum class ConflictPolicy : unsigned int {
00042         MODEL_ERROR = 0,
00043         NONDETERMINISTIC_CHOICE = 1,
00044         DETERMINISTIC_PRIORITY = 2
00045     };
00046
00047     EFSMNetwork(Model* model, std::string name = "");
00048     virtual ~EFSMNetwork() override;
00049
00050 public: // static
00051     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00052     static PluginInformation* GetPluginInformation();
00053     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00054
00055 public:
00056     void addState(FSMState* state);
00057     void removeState(FSMState* state);
00058     List<FSMState*>* getStates() const;
00059     void addTransition(EFSMTransition* transition);
00060     void removeTransition(EFSMTransition* transition);
00066     std::unique_ptr<EFSMTransition> detachTransition(EFSMTransition* transition, unsigned int* ordinal
      = nullptr);
00070     bool restoreTransition(EFSMTransition* transition, unsigned int ordinal);
00071     List<EFSMTransition*>* getTransitions() const;
00072     void setInitialState(FSMState* state);
00073     FSMState* getInitialState() const;
00074     FSMState* getCurrentState() const;
00075     void setCurrentState(FSMState* state);
00076     void setConflictPolicy(ConflictPolicy policy);
00077     ConflictPolicy getConflictPolicy() const;
00078     static std::string convertConflictPolicyToString(ConflictPolicy policy);
00079     static ConflictPolicy convertStringToConflictPolicy(const std::string& policy);
00080
00081 public:
00082     virtual std::string show() override;
00083
00084 protected:
00085     virtual bool _loadInstance(PersistenceRecord* fields) override;
00086     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00087     virtual bool _check(std::string& errorMessage) override;
00088     virtual void _initBetweenReplications() override;
00089     virtual NetworkActivationResult _activate(const NetworkActivationFrame& frame) override;
00090
00091 private:
00092     FSMState* _resolveInitialState();
00093     void _destroyOwnedTransitions();
00094
00095 private:
00096     List<FSMState*>* _states = new List<FSMState*>();
00097     List<EFSMTransition*>* _transitions = new List<EFSMTransition*>();
00098     FSMState* _initialState = nullptr;
00099     FSMState* _currentState = nullptr;
00100     ConflictPolicy _conflictPolicy = ConflictPolicy::DETERMINISTIC_PRIORITY;
00101     Sampler_if* _sampler = nullptr;
00102 };
00103
00104 #endif /* EFSMNETWORK_H */
```

18.528 GraphEdge.h

```
00001 /*
00002  * File:   GraphEdge.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 30 de Agosto de 2026
00006  */
00007
00008 #ifndef GRAPHEDGE_H
00009 #define GRAPHEDGE_H
00010
00011 #include "kernel/simulator/model/ModelDataDefinition.h"
00012 #include "kernel/simulator/PluginInformation.h"
00013
00014 #include <string>
00015
00016 class GraphNode;
```

```
00017
00026 class GraphEdge : public ModelDataDefinition {
00027 public:
00028     GraphEdge(Model* model, std::string name = "");
00029     GraphEdge(Model* model, GraphNode* source, GraphNode* destination, std::string name = "");
00030     virtual ~GraphEdge() override = default;
00031
00032 public:
00033     static PluginInformation* GetPluginInformation();
00034     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00035     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00036
00037 public:
00038     void setSource(GraphNode* source);
00039     GraphNode* getSource() const;
00040     void setDestination(GraphNode* destination);
00041     GraphNode* getDestination() const;
00042     void setDirected(bool directed);
00043     bool isDirected() const;
00044     void setWeight(double weight);
00045     void clearWeight();
00046     bool hasWeight() const;
00047     double getWeight(double defaultWeight = 1.0) const;
00048     std::string getSourceNodeName() const;
00049     std::string getDestinationNodeName() const;
00050     virtual std::string show() override;
00051
00052 protected:
00053     virtual bool _loadInstance(PersistenceRecord* fields) override;
00054     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00055     virtual bool _check(std::string& errorMessage) override;
00056
00057 private:
00058     void _syncEndpointNames();
00059
00060 private:
00061     const struct DEFAULT_VALUES {
00062         const bool directed = false;
00063         const bool hasWeight = false;
00064         const double weight = 1.0;
00065     } DEFAULT;
00066
00067     GraphNode* _source = nullptr;
00068     GraphNode* _destination = nullptr;
00069     std::string _sourceNodeName = "";
00070     std::string _destinationNodeName = "";
00071     bool _directed = DEFAULT.directed;
00072     bool _hasWeight = DEFAULT.hasWeight;
00073     double _weight = DEFAULT.weight;
00074 };
00075
00076 #endif /* GRAPHEDGE_H */
```

18.529 GraphNetwork.h

```
00001 /*
00002  * File:   GraphNetwork.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 30 de Agosto de 2026
00006  */
00007
00008 #ifndef GRAPHNETWORK_H
00009 #define GRAPHNETWORK_H
00010
00011 #include "DefaultNetwork.h"
00012
00013 #include <map>
00014 #include <string>
00015 #include <vector>
00016
00017 class GraphEdge;
00018 class GraphNode;
00019
00020 struct GraphSearchTreeEntry {
00021     GraphNode* node = nullptr;
00022     GraphNode* predecessor = nullptr;
00023     GraphEdge* edge = nullptr;
00024     unsigned int distance = 0;
00025 };
00026
00033 struct GraphTraversalResult {
00034     std::vector<GraphNode*> visitOrder;
00035     std::vector<GraphSearchTreeEntry> tree;
00036
```

```
00037     bool visited(GraphNode* node) const;
00038     GraphNode* predecessorOf(GraphNode* node) const;
00039     GraphEdge* edgeTo(GraphNode* node) const;
00040     unsigned int distanceTo(GraphNode* node) const;
00041 };
00042
00049 struct GraphPathResult {
00050     bool reachable = false;
00051     double distance = 0.0;
00052     std::vector<GraphNode*> nodes;
00053     std::vector<GraphEdge*> edges;
00054     std::string errorMessage = "";
00055 };
00056
00057 struct GraphTopologicalOrderResult {
00058     bool acyclic = false;
00059     std::vector<GraphNode*> order;
00060     std::string errorMessage = "";
00061 };
00062
00078 class GraphNetwork : public DefaultNetwork {
00079 public:
00080     GraphNetwork(Model* model, std::string name = "", std::string dataDefinitionTypename =
      Util::TypeOf<GraphNetwork>());
00081     virtual ~GraphNetwork() override = default;
00082
00083 public:
00084     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00085     static PluginInformation* GetPluginInformation();
00086     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00087
00088 public:
00089     virtual bool isDirected() const;
00090     virtual bool addNode(GraphNode* node);
00091     virtual bool removeNode(GraphNode* node);
00092     bool containsNode(GraphNode* node) const;
00093     GraphNode* findNodeByName(const std::string& name) const;
00094     std::vector<GraphNode*> getNodes() const;
00095
00096     virtual bool addEdge(GraphEdge* edge);
00097     virtual bool removeEdge(GraphEdge* edge);
00098     bool containsEdge(GraphEdge* edge) const;
00099     GraphEdge* findEdgeByName(const std::string& name) const;
00100     std::vector<GraphEdge*> getEdges() const;
00101     bool hasEdge(GraphNode* source, GraphNode* destination) const;
00102     std::vector<GraphEdge*> getEdgesBetween(GraphNode* source, GraphNode* destination) const;
00103
00104     std::vector<GraphNode*> getNeighbors(GraphNode* node) const;
00105     std::vector<GraphEdge*> getIncidentEdges(GraphNode* node) const;
00106     unsigned int degree(GraphNode* node) const;
00107     virtual std::vector<GraphNode*> getPredecessors(GraphNode* node) const;
00108     virtual std::vector<GraphNode*> getSuccessors(GraphNode* node) const;
00109     virtual std::vector<GraphEdge*> getIncomingEdges(GraphNode* node) const;
00110     virtual std::vector<GraphEdge*> getOutgoingEdges(GraphNode* node) const;
00111     virtual unsigned int inDegree(GraphNode* node) const;
00112     virtual unsigned int outDegree(GraphNode* node) const;
00113
00114     GraphTraversalResult breadthFirstSearch(GraphNode* source) const;
00115     GraphTraversalResult depthFirstSearch(GraphNode* source) const;
00116     bool hasPath(GraphNode* source, GraphNode* destination) const;
00117     std::vector<GraphNode*> reachableNodes(GraphNode* source) const;
00118     GraphPathResult shortestPathUnweighted(GraphNode* source, GraphNode* destination) const;
00119     GraphPathResult shortestPathDijkstra(GraphNode* source, GraphNode* destination) const;
00120     std::vector<std::vector<GraphNode*>> connectedComponents() const;
00121     virtual bool hasCycle() const;
00122     virtual GraphTopologicalOrderResult topologicalOrder() const;
00123     virtual std::string show() override;
00124
00125 protected:
00126     virtual bool _loadInstance(PersistenceRecord* fields) override;
00127     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00128     virtual bool _check(std::string& errorMessage) override;
00129     virtual void _initBetweenReplications() override;
00130
00131 protected:
00132     virtual bool _edgeConnects(GraphEdge* edge, GraphNode* source, GraphNode* destination) const;
00133     virtual GraphNode* _oppositeNode(GraphEdge* edge, GraphNode* node) const;
00134
00135 private:
00136     void _removeIncidentEdges(GraphNode* node);
00137     bool _hasNegativeWeight(std::string& errorMessage) const;
00138     GraphPathResult _reconstructPath(GraphNode* source,
00139                                      GraphNode* destination,
00140                                      const std::map<GraphNode*, GraphNode*>& predecessors,
00141                                      const std::map<GraphNode*, GraphEdge*>& predecessorEdges,
00142                                      double distance) const;
00143
```

```
00144 private:
00145     std::vector<GraphNode*> _nodes;
00146     std::vector<GraphEdge*> _edges;
00147 };
00148
00149 #endif /* GRAPHNETWORK_H */
```

18.530 GraphNode.h

```
00001 /*
00002  * File:   GraphNode.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 30 de Agosto de 2026
00006  */
00007
00008 #ifndef GRAPHNODE_H
00009 #define GRAPHNODE_H
00010
00011 #include "plugins/components/ModalModel/DefaultNode.h"
00012
00020 class GraphNode : public DefaultNode {
00021 public:
00022     GraphNode(Model* model, std::string name = "");
00023     virtual ~GraphNode() override = default;
00024
00025 public:
00026     static PluginInformation* GetPluginInformation();
00027     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00028     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00029 };
00030
00031 #endif /* GRAPHNODE_H */
```

18.531 MarkovChainNetwork.h

```
00001 /*
00002  * File:   MarkovChainNetwork.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 31 de Agosto de 2026
00006  */
00007
00008 #ifndef MARKOVCHAINNETWORK_H
00009 #define MARKOVCHAINNETWORK_H
00010
00011 #include "DefaultNetwork.h"
00012 #include "plugins/data/ModalModel/MarkovState.h"
00013
00014 #include <string>
00015 #include <vector>
00016
00017 class Sampler_if;
00018
00031 class MarkovChainNetwork : public DefaultNetwork {
00032 public:
00033     class MarkovTransition {
00034     public:
00035         MarkovTransition(MarkovState* source = nullptr, MarkovState* destination = nullptr, double
      probability = 1.0, std::string name = "");
00036         virtual ~MarkovTransition() = default;
00037
00038     public:
00039         void setSource(MarkovState* source);
00040         MarkovState* getSource() const;
00041         void setDestination(MarkovState* destination);
00042         MarkovState* getDestination() const;
00043         void setProbability(double probability);
00044         double getProbability() const;
00045         void setName(std::string name);
00046         std::string getName() const;
00047
00048     private:
00049         MarkovState* _source = nullptr;
00050         MarkovState* _destination = nullptr;
00051         double _probability = 1.0;
00052         std::string _name = "";
00053     };
00054
00055 public:
00056     MarkovChainNetwork(Model* model, std::string name = "");
00057     virtual ~MarkovChainNetwork() override;
```

```
00058
00059 public: // static
00060     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00061     static PluginInformation* GetPluginInformation();
00062     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00063
00064 public:
00065     void addState(MarkovState* state);
00066     void removeState(MarkovState* state);
00067     List<MarkovState*>* getStates() const;
00068     void addTransition(MarkovTransition* transition);
00069     void removeTransition(MarkovTransition* transition);
00070     List<MarkovTransition*>* getTransitions() const;
00071     std::vector<MarkovTransition*> getOutgoingTransitions(MarkovState* state) const;
00072     void setInitialState(MarkovState* state);
00073     MarkovState* getInitialState() const;
00074     void setCurrentState(MarkovState* state);
00075     MarkovState* getCurrentState() const;
00076     unsigned int getStateIndex(MarkovState* state) const;
00077     MarkovState* getStateAt(unsigned int index) const;
00078     double getProbabilityTolerance() const;
00079     void setProbabilityTolerance(double tolerance);
00080     void resetSampler();
00081
00082 public:
00083     virtual std::string show() override;
00084
00085 protected:
00086     virtual bool _loadInstance(PersistenceRecord* fields) override;
00087     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00088     virtual bool _check(std::string& errorMessage) override;
00089     virtual void _initBetweenReplications() override;
00090     virtual NetworkActivationResult _activate(const NetworkActivationFrame& frame) override;
00091
00092 private:
00093     MarkovState* _resolveInitialState();
00094     void _destroyOwnedTransitions();
00095     unsigned int _sampleNextStateIndex(const std::vector<MarkovTransition*>& outgoing);
00096     bool _hasState(MarkovState* state) const;
00097
00098 private:
00099     List<MarkovState*>* _states = new List<MarkovState*>();
00100     List<MarkovTransition*>* _transitions = new List<MarkovTransition*>();
00101     MarkovState* _initialState = nullptr;
00102     MarkovState* _currentState = nullptr;
00103     Sampler_if* _sampler = nullptr;
00104     double _probabilityTolerance = 1e-9;
00105 };
00106
00107 #endif /* MARKOVCHAINNETWORK_H */
```

18.532 MarkovState.h

```
00001 /*
00002  * File:   MarkovState.h
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 31 de Agosto de 2026
00006  */
00007
00008 #ifndef MARKOVSTATE_H
00009 #define MARKOVSTATE_H
00010
00011 #include "plugins/components/ModalModel/DefaultNode.h"
00012
00019 class MarkovState : public DefaultNode {
00020 public:
00021     MarkovState(Model* model, std::string name = "");
00022     virtual ~MarkovState() override = default;
00023
00024 public:
00025     static PluginInformation* GetPluginInformation();
00026     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00027     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00028 };
00029
00030 #endif /* MARKOVSTATE_H */
```

18.533 NetworkActivation.h

```
00001 /*
00002  * File:   NetworkActivation.h
```

```
00003  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00004  *
00005  * Created on 30 de Agosto de 2026
00006  */
00007
00008 #ifndef NETWORKACTIVATION_H
00009 #define NETWORKACTIVATION_H
00010
00011 #include <vector>
00012
00028 struct NetworkPortValue {
00029     bool present = false;
00030     double value = 0.0;
00031 };
00032
00047 class NetworkActivationFrame {
00048 public:
00049     explicit NetworkActivationFrame(unsigned int numInputs = 0);
00050 public:
00052     void reset(unsigned int numInputs);
00053     unsigned int size() const;
00055     void setPresent(unsigned int index, double value);
00057     void setAbsent(unsigned int index);
00058     bool isPresent(unsigned int index) const;
00060     double getValue(unsigned int index) const;
00061 private:
00062     std::vector<NetworkPortValue> _inputs;
00063 };
00064
00074 class NetworkActivationResult {
00075 public:
00076     explicit NetworkActivationResult(unsigned int numOutputs = 0);
00077 public:
00079     void reset(unsigned int numOutputs);
00080     unsigned int size() const;
00081     void setPresent(unsigned int index, double value);
00082     void setAbsent(unsigned int index);
00083     bool isPresent(unsigned int index) const;
00084     double getValue(unsigned int index) const;
00086     unsigned int countPresent() const;
00087 private:
00088     std::vector<NetworkPortValue> _outputs;
00089 };
00090
00091 #endif /* NETWORKACTIVATION_H */
```

18.534 SignalData.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   SignalData.h
00008  * Author: rlcancian
00009  *
00010  * Created on 25 de maio de 2022, 14:37
00011  */
00012
00013 #ifndef SIGNALDATA_H
00014 #define SIGNALDATA_H
00015
00016 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00017 #include "../../../kernel/simulator/model/ModelComponent.h"
00018 #include "kernel/util/List.h"
00019
00020 class SignalData : public ModelDataDefinition {
00021 public:
00022     typedef std::function<unsigned int(SignalData*) > SignalDataEventHandler; //< Returns the total
      number of freed entities.
00023     typedef std::pair<SignalDataEventHandler, ModelComponent*> PairSignalDataEventHandler;
00024     template<typename Class>
00025     static SignalDataEventHandler SetSignalDataEventHandler(unsigned int
      (Class::*function)(SignalData*), Class * object) {
00026         return std::bind(function, object, std::placeholders::_1);
00027     }
00028 public:
00029     SignalData(Model* model, std::string name = "");
00030     virtual ~SignalData() override;
00031 public: // static
00032     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00033     static PluginInformation* GetPluginInformation();
00034     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00035 public: //virtual
```

```
00036     virtual std::string show() override;
00037 public:
00038     unsigned int generateSignal(double signalValue, unsigned int limit);
00039     void addSignalDataEventHandler(SignalDataEventHandler eventHandler, ModelComponent* component);
00040     void removeSignalDataEventHandler(ModelComponent* component);
00041     bool hasSignalDataEventHandler(ModelComponent* component) const;
00042     unsigned int remainsToLimit() const;
00043     void decreaseRemainLimit();
00044 
00045 protected: // must be overriden
00046     virtual bool _loadInstance(PersistenceRecord *fields) override;
00047     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00048 protected: // could be overriden .
00049     virtual bool _check(std::string& errorMessage) override;
00050     virtual void _initBetweenReplications() override;
00051     //virtual void _createInternalAndAttachedData();
00052     //virtual ParserChangesInformation* _getParserChangesInformation();
00053 
00054 protected:
00055     // virtual void _createInternalStatisticReporters() override;
00056     // virtual void _createNonEditableDataDefinitions() override;
00057     // virtual void _createEditableDataDefinitions() override;
00058     // virtual void _createAttachedAttributes() override;
00059 
00060 private: // methods
00061     unsigned int  _notifySignalDataEventHandlers();
00062 private: //1::1
00063     unsigned int _remainsToLimit = 0;
00064 private: //1::n
00065     List<PairSignalDataEventHandler*>* _signalDataEventHandlers = new
      List<PairSignalDataEventHandler*>();
00066 };
00067 
00068 #endif /* SIGNALDATA_H */
```

18.535 DummyElement.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005 
00006 /*
00007  * File:   DummyElement.h
00008  * Author: Prof. Rafael Luiz Cancian, Dr. Eng.
00009  *
00010  * Created on 11 de janeiro de 2022, 22:26
00011  */
00012 
00013 #ifndef DUMMYELEMENT_H
00014 #define DUMMYELEMENT_H
00015 
00016 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00017 #include "kernel/simulator/PluginInformation.h"
00018 
00019 class DummyElement : public ModelDataDefinition {
00020 public:
00021     DummyElement(Model* model, std::string name = "");
00022     virtual ~DummyElement() = default;
00023 
00024 public:
00025     void setSomeString(const std::string& someString);
00026     std::string getSomeString() const;
00027     void setSomeUint(unsigned int someUint);
00028     unsigned int getSomeUint() const;
00029 
00030 public:
00031     virtual std::string show() override;
00032 
00033 public:
00034     static PluginInformation* GetPluginInformation();
00035     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord *fields);
00036     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00037 
00038 protected:
00039     virtual bool _loadInstance(PersistenceRecord *fields) override;
00040     virtual void _saveInstance(PersistenceRecord *fields, bool saveDefaultValues) override;
00041     virtual bool _check(std::string& errorMessage) override;
00042 
00043 protected:
00045     // virtual bool _check(std::string& errorMessage);
00047     // virtual ParserChangesInformation* _getParserChangesInformation();
00049     // virtual void _initBetweenReplications();
00050     // virtual void _createInternalStatisticReporters() override;
00051     // virtual void _createNonEditableDataDefinitions() override;
```

```
00052     // virtual void _createEditableDataDefinitions() override;
00053     // virtual void _createAttachedAttributes() override;
00055     // virtual void _createInternalAndAttachedData(); /*< A ModelDataDefinition or ModelComponent that
      includes (internal) ou refers to (attach) other ModelDataDefinition must register them inside this
      method. */
00057     // virtual void _addSimulationControl(SimulationControl* property);
00058
00059
00060 protected:
00061
00062 private:
00063     // ...
00064
00065 private:
00066     const struct DEFAULT_VALUES {
00067         const std::string someString = "Test";
00068         const unsigned int someUint = 1;
00069     } DEFAULT;
00070     std::string _someString = DEFAULT.someString;
00071     unsigned int _someUint = DEFAULT.someUint;
00072
00073 private:
00074     // DummyElement* _internalDataDefinition = nullptr;
00075
00076 private:
00077     // ...
00078 };
00079
00080 #endif /* DUMMYELEMENT_H */
```

18.536 BioCompartment.h

```
00001 #ifndef BIOCOMPARTMENT_H
00002 #define BIOCOMPARTMENT_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00007 #include "kernel/simulator/PluginInformation.h"
00008
00025 class BioCompartment : public ModelDataDefinition {
00026 public:
00027     BioCompartment(Model* model, std::string name = "");
00028     virtual ~BioCompartment() override = default;
00029
00030 public:
00031     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00032     static PluginInformation* GetPluginInformation();
00033     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00034
00035 public:
00036     void setCompartmentType(std::string type);
00037     std::string getCompartmentType() const;
00038     void setParentCompartmentName(std::string parentName);
00039     std::string getParentCompartmentName() const;
00040     void setMembraneBounded(bool membraneBounded);
00041     bool isMembraneBounded() const;
00042     void setVolumeFraction(double volumeFraction);
00043     double getVolumeFraction() const;
00044     void setCopyNumber(unsigned int copyNumber);
00045     unsigned int getCopyNumber() const;
00046     void setRole(std::string role);
00047     std::string getRole() const;
00048
00049     virtual std::string show() override;
00050
00051 protected:
00052     virtual bool _loadInstance(PersistenceRecord* fields) override;
00053     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00054     virtual bool _check(std::string& errorMessage) override;
00055     virtual void _createEditableDataDefinitions() override;
00056
00057 private:
00058     const struct DEFAULT_VALUES {
00059         std::string compartmentType = "cytosol";
00060         std::string parentCompartmentName = "";
00061         bool membraneBounded = false;
00062         double volumeFraction = 1.0;
00063         unsigned int copyNumber = 1u;
00064         std::string role = "generic";
00065     } DEFAULT;
00066
00067     std::string _compartmentType = DEFAULT.compartmentType;
00068     std::string _parentCompartmentName = DEFAULT.parentCompartmentName;
00069     bool _membraneBounded = DEFAULT.membraneBounded;
```

```
00070     double _volumeFraction = DEFAULT.volumeFraction;
00071     unsigned int _copyNumber = DEFAULT.copyNumber;
00072     std::string _role = DEFAULT.role;
00073 };
00074
00075 #endif /* BIOCOMPARTMENT_H */
```

18.537 MolecularSpecies.h

```
00001 #ifndef MOLECULARSPECIES_H
00002 #define MOLECULARSPECIES_H
00003
00004 #include <string>
00005
00006 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00007 #include "kernel/simulator/PluginInformation.h"
00008
00016 class MolecularSpecies : public ModelDataDefinition {
00017 public:
00018     MolecularSpecies(Model* model, std::string name = "");
00019     virtual ~MolecularSpecies() override = default;
00020
00021 public:
00022     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00023     static PluginInformation* GetPluginInformation();
00024     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00025
00026 public:
00027     void setInitialCount(int initialCount);
00028     int getInitialCount() const;
00029     void setCount(int count);
00030     int getCount() const;
00031     void setCompartment(std::string compartment);
00032     std::string getCompartment() const;
00033     void setUnit(std::string unit);
00034     std::string getUnit() const;
00035
00036     virtual std::string show() override;
00037
00038 protected:
00039     virtual bool _loadInstance(PersistenceRecord* fields) override;
00040     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00041     virtual bool _check(std::string& errorMessage) override;
00042     virtual void _initBetweenReplications() override;
00043
00044 private:
00045     const struct DEFAULT_VALUES {
00046         int initialCount = 0;
00047         int count = 0;
00048         std::string compartment = "cytoplasm";
00049         std::string unit = "molecules";
00050     } DEFAULT;
00051
00052     int _initialCount = DEFAULT.initialCount;
00053     int _count = DEFAULT.count;
00054     std::string _compartment = DEFAULT.compartment;
00055     std::string _unit = DEFAULT.unit;
00056 };
00057
00058 #endif /* MOLECULARSPECIES_H */
```

18.538 StochasticReactionRule.h

```
00001 #ifndef STOCHASTICREACTIONRULE_H
00002 #define STOCHASTICREACTIONRULE_H
00003
00004 #include <string>
00005 #include <vector>
00006 #include <map>
00007
00008 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00009 #include "kernel/simulator/PluginInformation.h"
00010
00022 class StochasticReactionRule : public ModelDataDefinition {
00023 public:
00024     struct StoichiometricTerm {
00025         std::string speciesName;
00026         int stoichiometry = 1;
00027     };
00028
00029 public:
00030     StochasticReactionRule(Model* model, std::string name = "");
```

```
00031     virtual ~StochasticReactionRule() override = default;
00032
00033 public:
00034     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00035     static PluginInformation* GetPluginInformation();
00036     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00037
00038 public:
00039     void addReactant(std::string speciesName, int stoichiometry = 1);
00040     void addProduct(std::string speciesName, int stoichiometry = 1);
00041     void clearReactants();
00042     void clearProducts();
00043     const std::vector<StoichiometricTerm>& getReactants() const;
00044     const std::vector<StoichiometricTerm>& getProducts() const;
00045
00046     void setRateConstant(double rateConstant);
00047     double getRateConstant() const;
00048     void setCompartment(std::string compartment);
00049     std::string getCompartment() const;
00050
00051     double computePropensity(const std::map<std::string, int>& counts) const;
00052
00053     virtual std::string show() override;
00054
00055 protected:
00056     virtual bool _loadInstance(PersistenceRecord* fields) override;
00057     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00058     virtual bool _check(std::string& errorMessage) override;
00059
00060 private:
00061     static double _combinatorial(int n, int k);
00062
00063 private:
00064     const struct DEFAULT_VALUES {
00065         double rateConstant = 0.0;
00066         std::string compartment = "cytoplasm";
00067     } DEFAULT;
00068
00069     std::vector<StoichiometricTerm> _reactants;
00070     std::vector<StoichiometricTerm> _products;
00071     double _rateConstant = DEFAULT.rateConstant;
00072     std::string _compartment = DEFAULT.compartment;
00073 };
00074
00075 #endif /* STOCHASTICREACTIONRULE_H */
```

18.539 WholeCellState.h

```
00001 #ifndef WHOLECELLSTATE_H
00002 #define WHOLECELLSTATE_H
00003
00004 #include <map>
00005 #include <string>
00006 #include <vector>
00007
00008 #include "../../../kernel/simulator/model/ModelDataDefinition.h"
00009 #include "kernel/simulator/PluginInformation.h"
00010
00025 class WholeCellState : public ModelDataDefinition {
00026 public:
00027     WholeCellState(Model* model, std::string name = "");
00028     virtual ~WholeCellState() override = default;
00029
00030 public:
00031     static ModelDataDefinition* LoadInstance(Model* model, PersistenceRecord* fields);
00032     static PluginInformation* GetPluginInformation();
00033     static ModelDataDefinition* NewInstance(Model* model, std::string name = "");
00034
00035 public:
00036     // Molecule count access (integer, for SSA)
00037     void setMoleculeCount(const std::string& speciesName, int count);
00038     int getMoleculeCount(const std::string& speciesName) const;
00039     bool hasMoleculeCount(const std::string& speciesName) const;
00040     const std::map<std::string, int>& getMoleculeCounts() const;
00041
00042     // Metabolite pool access (continuous, for FBA output)
00043     void setMetaboliteAmount(const std::string& metaboliteName, double amount);
00044     double getMetaboliteAmount(const std::string& metaboliteName) const;
00045     const std::map<std::string, double>& getMetabolitePool() const;
00046
00047     // Compartment-specific metabolite pool access (continuous, for compartment-aware metabolism)
00048     void setCompartmentMetaboliteAmount(const std::string& compartmentName, const std::string&
      metaboliteName, double amount);
00049     double getCompartmentMetaboliteAmount(const std::string& compartmentName, const std::string&
      metaboliteName) const;
```

```
00050     bool hasCompartmentMetaboliteAmount(const std::string& compartmentName, const std::string&
      metaboliteName) const;
00051     const std::map<std::string, double>& getCompartmentMetabolitePool() const;
00052
00053     // Pathway activity access (continuous, for FBA objective/flux summaries)
00054     void setPathwayActivity(const std::string& pathwayName, double activity);
00055     double getPathwayActivity(const std::string& pathwayName) const;
00056     bool hasPathwayActivity(const std::string& pathwayName) const;
00057     const std::map<std::string, double>& getPathwayActivities() const;
00058
00059     // Resource budget access (integer, set by fair allocation before SSA)
00060     void setResourceBudget(const std::string& speciesName, int budget);
00061     int getResourceBudget(const std::string& speciesName) const;
00062     void clearResourceBudget();
00063
00064     // Cell geometry
00065     void setCellVolume(double volume);
00066     double getCellVolume() const;
00067     void setCellMass(double mass);
00068     double getCellMass() const;
00069
00070     // Simulation progress
00071     void setCurrentTime(double time);
00072     double getCurrentTime() const;
00073     void setStepCount(int stepCount);
00074     int getStepCount() const;
00075     void incrementStep();
00076
00077     // Lifecycle metadata
00078     void setLifecyclePhase(const std::string& phase);
00079     std::string getLifecyclePhase() const;
00080     void setGenerationCount(int generationCount);
00081     int getGenerationCount() const;
00082     void setViable(bool viable);
00083     bool isViable() const;
00084     void setLastDivisionTime(double time);
00085     double getLastDivisionTime() const;
00086
00087     // JSON loading from CovertLab/WholeCell data files (MIT)
00088     bool loadFixedConstants(const std::string& jsonPath);
00089     bool loadParameters(const std::string& jsonPath);
00090
00091     // JSON file paths (persisted so the model file remembers them)
00092     void setFixedConstantsPath(std::string path);
00093     std::string getFixedConstantsPath() const;
00094     void setParametersPath(std::string path);
00095     std::string getParametersPath() const;
00096
00097     virtual std::string show() override;
00098
00099 protected:
00100     virtual bool _loadInstance(PersistenceRecord* fields) override;
00101     virtual void _saveInstance(PersistenceRecord* fields, bool saveDefaultValues) override;
00102     virtual bool _check(std::string& errorMessage) override;
00103     virtual void _initBetweenReplications() override;
00104
00105 private:
00106         const struct DEFAULT_VALUES {
00107             double cellVolume = 1.0e-15;  // 1 fL, typical M. genitalium volume
00108             double cellMass = 1.0e-13;   // ~100 fg, typical M. genitalium mass
00109             double currentTime = 0.0;
00110             int stepCount = 0;
00111             std::string lifecyclePhase = "newborn";
00112             int generationCount = 0;
00113             bool viable = true;
00114             double lastDivisionTime = -1.0;
00115             std::string fixedConstantsPath = "";
00116             std::string parametersPath = "";
00117         } DEFAULT;
00118
00119     std::map<std::string, int>    _moleculeCounts;
00120     std::map<std::string, int>    _initialMoleculeCounts;
00121     std::map<std::string, double> _metabolitePool;
00122     std::map<std::string, double> _initialMetabolitePool;
00123     std::map<std::string, double> _compartmentMetabolitePool;
00124     std::map<std::string, double> _initialCompartmentMetabolitePool;
00125     std::map<std::string, double> _pathwayActivities;
00126     std::map<std::string, double> _initialPathwayActivities;
00127     std::map<std::string, int>    _resourceBudget;
00128
00129         double _cellVolume        = DEFAULT.cellVolume;
00130         double _cellMass          = DEFAULT.cellMass;
00131         double _initialCellVolume = DEFAULT.cellVolume;
00132         double _initialCellMass   = DEFAULT.cellMass;
00133         double _currentTime = DEFAULT.currentTime;
00134         int    _stepCount   = DEFAULT.stepCount;
00135         std::string _lifecyclePhase = DEFAULT.lifecyclePhase;
```

```
00136         std::string _initialLifecyclePhase = DEFAULT.lifecyclePhase;
00137         int _generationCount = DEFAULT.generationCount;
00138         int _initialGenerationCount = DEFAULT.generationCount;
00139         bool _viable = DEFAULT.viable;
00140         bool _initialViable = DEFAULT.viable;
00141         double _lastDivisionTime = DEFAULT.lastDivisionTime;
00142         double _initialLastDivisionTime = DEFAULT.lastDivisionTime;
00143         std::string _fixedConstantsPath = DEFAULT.fixedConstantsPath;
00144         std::string _parametersPath     = DEFAULT.parametersPath;
00145 };
00146
00147 #endif /* WHOLECELLSTATE_H */
```

18.540 PluginConnectorDummyImpl1.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   PluginConnectorDummyImpl1.h
00009  * Author: rlcancian
00010  *
00011  * Created on 9 de Setembro de 2019, 19:24
00012  */
00013
00014 #ifndef PLUGINCONNECTORDUMMYIMPL1_H
00015 #define PLUGINCONNECTORDUMMYIMPL1_H
00016
00017 #include "kernel/simulator/PluginConnector_if.h"
00018 #include "kernel/util/List.h"
00019 //namespace GenesysKernel {
00020
00027 class PluginConnectorDummyImpl1 : public PluginConnector_if {
00028 public:
00029     PluginConnectorDummyImpl1();
00030     virtual ~PluginConnectorDummyImpl1() = default;
00031 public:
00037     Plugin* check(const std::string dynamicLibraryFilename) override;
00043     Plugin* connect(const std::string dynamicLibraryFilename) override;
00048     List<std::string>* find() override;
00054     bool disconnect(const std::string dynamicLibraryFilename) override;
00060     bool disconnect(Plugin* plugin) override;
00061 private:
00062     void _connect(List<Plugin*>* plugins, Plugin* plugin);
00063 };
00064 //namespace\\}
00065 #endif /* PLUGINCONNECTORDUMMYIMPL1_H */
```

18.541 PluginConnectorStaticImpl1.h

```
00001 #ifndef PLUGINCONNECTORSTATICIMPL1_H
00002 #define PLUGINCONNECTORSTATICIMPL1_H
00003
00004 #include "kernel/simulator/PluginConnector_if.h"
00005
00012 class PluginConnectorStaticImpl1 : public PluginConnector_if {
00013 public:
00014     PluginConnectorStaticImpl1() = default;
00015     virtual ~PluginConnectorStaticImpl1() = default;
00016
00022     Plugin* check(const std::string dynamicLibraryFilename) override;
00028     Plugin* connect(const std::string dynamicLibraryFilename) override;
00033     List<std::string>* find() override;
00039     bool disconnect(const std::string dynamicLibraryFilename) override;
00045     bool disconnect(Plugin* plugin) override;
00046 };
00047
00048 #endif
```

18.542 AIAssistant_if.h

```
00001 /*
00002  * File:   AIAssistant_if.h
00003  * Author: Genesys Team
00004  *
00005  * Created on 28 de Maio de 2026
```

```
00006  */
00007
00008 #ifndef AIASSISTANT_IF_H
00009 #define AIASSISTANT_IF_H
00010
00011 #include "AIAuditLog.h"
00012
00013 #include <string>
00014 #include <vector>
00015
00016 // Non-owning handles to kernel objects.
00017 // Implementation files that call facade methods must include the full headers.
00018 class SimulatorFacade;
00019 class AIProviderClient_if;
00020
00021 // ---------------------------------------------------------------------------
00022 // Enumerations
00023 // ---------------------------------------------------------------------------
00024
00028 enum class AIProvider {
00029     Undefined,
00030     OpenAI,
00031     Anthropic,
00032     Local,
00033     CustomHttp
00034 };
00035
00039 enum class ExecutionState {
00040     NotConfigured,
00041     Ready,
00042     Running,
00043     Finished,
00044     Failed,
00045     Cancelled,
00046     NotImplemented
00047 };
00048
00057 enum class ReasoningEffort {
00058     Low,
00059     Medium,
00060     High
00061 };
00062
00063 // ---------------------------------------------------------------------------
00064 // Configuration structures
00065 // ---------------------------------------------------------------------------
00066
00084 struct AIAssistantConfiguration {
00085     AIProvider   provider                = AIProvider::Undefined;
00086     std::string  providerName;
00087     std::string  modelName;
00088     std::string  baseUrl;
00089     std::string  apiKeyEnvironmentVariable;
00090     std::string  organizationId;
00091     std::string  projectId;
00092     ReasoningEffort reasoningEffort        = ReasoningEffort::Medium;
00093     double       temperature               = 0.2;
00094     unsigned int maxOutputTokens           = 4096;
00095     unsigned int timeoutSeconds            = 120;
00096     bool         allowNetworkAccess        = false;
00097     bool         allowModelMutation        = false;
00098     bool         allowSimulationExecution  = false;
00099     bool         allowFileSystemWrites     = false;
00100
00110     bool         sandboxEnabled            = false;
00111
00126     bool         dryRun                    = false;
00127 };
00128
00129 // ---------------------------------------------------------------------------
00130 // Request / response structures
00131 // ---------------------------------------------------------------------------
00132
00142 struct AIAssistantRequest {
00143     std::string  userPrompt;
00144     std::string  systemContext;
00145     std::string  modelDescription;
00146     std::string  experimentDescription;
00156     std::string  modelPlan;
00157     bool         buildModel            = false;
00158     bool         configureSimulation   = false;
00159     bool         runSimulation         = false;
00160     bool         collectResults        = false;
00161 };
00162
00166 struct AIAssistantResponse {
00167     bool          success              = false;
```

```
00168     ExecutionState state               = ExecutionState::NotImplemented;
00169     std::string   message;
00170     std::string   generatedPlan;
00171     std::string   generatedModelLanguage;
00172     std::string   diagnostics;
00173 };
00174
00175 // ----------------------------------------------------------------------
00176 // Knowledge base structures
00177 // ----------------------------------------------------------------------
00178
00190 struct GenesysKnowledgeItem {
00191     std::string name;
00192     std::string category;
00193     std::string typeName;
00194     std::string description;
00195     std::string usage;
00196     std::string constraints;
00197 };
00198
00202 struct GenesysKnowledgeBaseSummary {
00203     unsigned int numberOfPlugins         = 0;
00204     unsigned int numberOfComponents      = 0;
00205     unsigned int numberOfDataDefinitions = 0;
00206     std::string  summaryMarkdown;
00207 };
00208
00209 // ----------------------------------------------------------------------
00210 // Interface
00211 // ----------------------------------------------------------------------
00212
00237 class AIAssistant_if {
00238 public:
00239     virtual ~AIAssistant_if() = default;
00240
00241     // ----------------------------------------------------------------------
00242     // Configuration
00243     // ----------------------------------------------------------------------
00244
00252     virtual void setConfiguration(const AIAssistantConfiguration& configuration) = 0;
00253
00260     virtual AIAssistantConfiguration getConfiguration() const = 0;
00261
00263     virtual void setProvider(AIProvider provider) = 0;
00265     virtual AIProvider getProvider() const = 0;
00266
00268     virtual void setProviderName(const std::string& providerName) = 0;
00270     virtual std::string getProviderName() const = 0;
00271
00273     virtual void setModelName(const std::string& modelName) = 0;
00275     virtual std::string getModelName() const = 0;
00276
00284     virtual void setBaseUrl(const std::string& baseUrl) = 0;
00286     virtual std::string getBaseUrl() const = 0;
00287
00289     virtual void setReasoningEffort(ReasoningEffort effort) = 0;
00291     virtual ReasoningEffort getReasoningEffort() const = 0;
00292
00300     virtual void setTemperature(double temperature) = 0;
00302     virtual double getTemperature() const = 0;
00303
00305     virtual void setMaxOutputTokens(unsigned int maxOutputTokens) = 0;
00307     virtual unsigned int getMaxOutputTokens() const = 0;
00308
00310     virtual void setTimeoutSeconds(unsigned int timeoutSeconds) = 0;
00312     virtual unsigned int getTimeoutSeconds() const = 0;
00313
00314     // ----------------------------------------------------------------------
00315     // Security and secrets
00316     // ----------------------------------------------------------------------
00317
00330     virtual void setApiKey(const std::string& apiKey) = 0;
00331
00342     virtual void clearApiKey() = 0;
00343
00345     virtual bool hasApiKey() const = 0;
00346
00355     virtual std::string getMaskedApiKey() const = 0;
00356
00364     virtual void setApiKeyEnvironmentVariable(const std::string& variableName) = 0;
00366     virtual std::string getApiKeyEnvironmentVariable() const = 0;
00367
00378     virtual bool loadApiKeyFromEnvironment() = 0;
00379
00380     // ----------------------------------------------------------------------
00381     // Integration with GenESyS
00382     // ----------------------------------------------------------------------
```

```
00383
00392     virtual void setSimulatorFacade(SimulatorFacade* simulatorFacade) = 0;
00394     virtual SimulatorFacade* getSimulatorFacade() const = 0;
00396     virtual bool hasSimulatorFacade() const = 0;
00397
00398     // ------------------------------------------------------------------
00399     // Provider client
00400     // ------------------------------------------------------------------
00401
00414     virtual void setProviderClient(AIProviderClient_if* client) = 0;
00415
00419     virtual AIProviderClient_if* getProviderClient() const = 0;
00420
00422     virtual bool hasProviderClient() const = 0;
00423
00424     // ------------------------------------------------------------------
00425     // Knowledge base
00426     // ------------------------------------------------------------------
00427
00429     virtual void clearKnowledgeBase() = 0;
00430
00432     virtual void addKnowledgeItem(const GenesysKnowledgeItem& item) = 0;
00433
00435     virtual std::vector<GenesysKnowledgeItem> getKnowledgeItems() const = 0;
00436
00438     virtual GenesysKnowledgeBaseSummary getKnowledgeBaseSummary() const = 0;
00439
00447     virtual std::string exportKnowledgeBaseAsMarkdown() const = 0;
00448
00466     virtual bool refreshKnowledgeBaseFromSimulator(std::string* message = nullptr) = 0;
00467
00468     // ------------------------------------------------------------------
00469     // Workflow
00470     // ------------------------------------------------------------------
00471
00490     virtual bool checkReady(std::string* message = nullptr) const = 0;
00491
00498     virtual AIAssistantResponse analyzePrompt(const AIAssistantRequest& request) = 0;
00499
00506     virtual AIAssistantResponse createModelPlan(const AIAssistantRequest& request) = 0;
00507
00523     virtual AIAssistantResponse buildModel(const AIAssistantRequest& request) = 0;
00524
00539     virtual AIAssistantResponse configureSimulation(const AIAssistantRequest& request) = 0;
00540
00549     virtual AIAssistantResponse runSimulation(const AIAssistantRequest& request) = 0;
00550
00558     virtual AIAssistantResponse collectResults(const AIAssistantRequest& request) = 0;
00559
00568     virtual AIAssistantResponse execute(const AIAssistantRequest& request) = 0;
00569
00570     // ------------------------------------------------------------------
00571     // Diagnostics
00572     // ------------------------------------------------------------------
00573
00575     virtual ExecutionState getState() const = 0;
00576
00578     virtual std::string getLastError() const = 0;
00579
00586     virtual std::string getLastDiagnostics() const = 0;
00587
00589     virtual void clearDiagnostics() = 0;
00590
00591     // ------------------------------------------------------------------
00592     // Audit log
00593     // ------------------------------------------------------------------
00594
00602     virtual std::vector<AuditEntry> getAuditEntries(unsigned int maxEntries = 200) const = 0;
00603
00605     virtual std::string getAuditLogPath() const = 0;
00606
00612     virtual unsigned int exportAuditLog(const std::string& csvPath) const = 0;
00613 };
00614
00615 #endif /* AIASSISTANT_IF_H */
```

18.543 AIAssistantDefaultImpl.h

```
00001 /*
00002  * File:   AIAssistantDefaultImpl.h
00003  * Author: Genesys Team
00004  *
00005  * Created on 28 de Maio de 2026
00006  */
00007
```

```
00008 #ifndef AIASSISTANTDEFAULTIMPL_H
00009 #define AIASSISTANTDEFAULTIMPL_H
00010
00011 #include "AIAssistant_if.h"
00012 #include "AIAuditLog.h"
00013 #include "AIProviderClient_if.h"
00014
00015 #include <memory>
00016 #include <string>
00017 #include <vector>
00018
00045 class AIAssistantDefaultImpl : public AIAssistant_if {
00046 public:
00047     AIAssistantDefaultImpl();
00048     virtual ~AIAssistantDefaultImpl();
00049
00050     // Not copyable: would propagate the in-memory API key and the non-owning
00051     // SimulatorFacade pointer without a clear ownership contract.
00052     AIAssistantDefaultImpl(const AIAssistantDefaultImpl&) = delete;
00053     AIAssistantDefaultImpl& operator=(const AIAssistantDefaultImpl&) = delete;
00054
00055 public:
00056     // ------------------------------------------------------------------------
00057     // Configuration
00058     // ------------------------------------------------------------------------
00059     virtual void setConfiguration(const AIAssistantConfiguration& configuration) override;
00060     virtual AIAssistantConfiguration getConfiguration() const override;
00061
00062     virtual void setProvider(AIProvider provider) override;
00063     virtual AIProvider getProvider() const override;
00064
00065     virtual void setProviderName(const std::string& providerName) override;
00066     virtual std::string getProviderName() const override;
00067
00068     virtual void setModelName(const std::string& modelName) override;
00069     virtual std::string getModelName() const override;
00070
00071     virtual void setBaseUrl(const std::string& baseUrl) override;
00072     virtual std::string getBaseUrl() const override;
00073
00074     virtual void setReasoningEffort(ReasoningEffort effort) override;
00075     virtual ReasoningEffort getReasoningEffort() const override;
00076
00077     virtual void setTemperature(double temperature) override;
00078     virtual double getTemperature() const override;
00079
00080     virtual void setMaxOutputTokens(unsigned int maxOutputTokens) override;
00081     virtual unsigned int getMaxOutputTokens() const override;
00082
00083     virtual void setTimeoutSeconds(unsigned int timeoutSeconds) override;
00084     virtual unsigned int getTimeoutSeconds() const override;
00085
00086     // ------------------------------------------------------------------------
00087     // Security and secrets
00088     // ------------------------------------------------------------------------
00089     virtual void setApiKey(const std::string& apiKey) override;
00090     virtual void clearApiKey() override;
00091     virtual bool hasApiKey() const override;
00092     virtual std::string getMaskedApiKey() const override;
00093
00094     virtual void setApiKeyEnvironmentVariable(const std::string& variableName) override;
00095     virtual std::string getApiKeyEnvironmentVariable() const override;
00096     virtual bool loadApiKeyFromEnvironment() override;
00097
00098     // ------------------------------------------------------------------------
00099     // Integration with GenESyS
00100     // ------------------------------------------------------------------------
00101     virtual void setSimulatorFacade(SimulatorFacade* simulatorFacade) override;
00102     virtual SimulatorFacade* getSimulatorFacade() const override;
00103     virtual bool hasSimulatorFacade() const override;
00104
00105     // ------------------------------------------------------------------------
00106     // Provider client
00107     // ------------------------------------------------------------------------
00108     virtual void setProviderClient(AIProviderClient_if* client) override;
00109     virtual AIProviderClient_if* getProviderClient() const override;
00110     virtual bool hasProviderClient() const override;
00111
00130     bool createAndAttachProviderClient();
00131
00132     // ------------------------------------------------------------------------
00133     // Knowledge base
00134     // ------------------------------------------------------------------------
00135     virtual void clearKnowledgeBase() override;
00136     virtual void addKnowledgeItem(const GenesysKnowledgeItem& item) override;
00137     virtual std::vector<GenesysKnowledgeItem> getKnowledgeItems() const override;
00138     virtual GenesysKnowledgeBaseSummary getKnowledgeBaseSummary() const override;
```

```
00139     virtual std::string exportKnowledgeBaseAsMarkdown() const override;
00140     virtual bool refreshKnowledgeBaseFromSimulator(std::string* message = nullptr) override;
00141
00142     // ------------------------------------------------------------------------
00143     // Workflow
00144     // ------------------------------------------------------------------------
00145     virtual bool checkReady(std::string* message = nullptr) const override;
00146
00147     virtual AIAssistantResponse analyzePrompt(const AIAssistantRequest& request) override;
00148     virtual AIAssistantResponse createModelPlan(const AIAssistantRequest& request) override;
00149     virtual AIAssistantResponse buildModel(const AIAssistantRequest& request) override;
00150     virtual AIAssistantResponse configureSimulation(const AIAssistantRequest& request) override;
00151     virtual AIAssistantResponse runSimulation(const AIAssistantRequest& request) override;
00152     virtual AIAssistantResponse collectResults(const AIAssistantRequest& request) override;
00153     virtual AIAssistantResponse execute(const AIAssistantRequest& request) override;
00154
00155     // ------------------------------------------------------------------------
00156     // Diagnostics
00157     // ------------------------------------------------------------------------
00158     virtual ExecutionState getState() const override;
00159     virtual std::string getLastError() const override;
00160     virtual std::string getLastDiagnostics() const override;
00161     virtual void clearDiagnostics() override;
00162
00163     // ------------------------------------------------------------------------
00164     // Audit log
00165     // ------------------------------------------------------------------------
00166     virtual std::vector<AuditEntry> getAuditEntries(unsigned int maxEntries) const override;
00167     virtual std::string getAuditLogPath() const override;
00168     virtual unsigned int exportAuditLog(const std::string& csvPath) const override;
00169
00170 private:
00171     // ------------------------------------------------------------------------
00172     // Private helpers
00173     // ------------------------------------------------------------------------
00174
00178     bool _providerRequiresApiKey() const;
00179
00183     bool _providerRequiresBaseUrl() const;
00184
00191     AIAssistantResponse _notImplementedResponse(const std::string& operationName,
00192                                                 unsigned int plannedStage) const;
00193
00199     void _updateState();
00200
00209     void _appendAudit(const std::string& operation,
00210                       const AIAssistantResponse& resp,
00211                       long long durationMs,
00212                       const std::string& promptPreview);
00213
00214 private:
00215     // ------------------------------------------------------------------------
00216     // Private data members
00217     // ------------------------------------------------------------------------
00218
00219     AIAssistantConfiguration _config;
00220
00230     std::string _apiKey;
00231
00233     SimulatorFacade* _simulatorFacade = nullptr;
00234
00243     AIProviderClient_if* _providerClient    = nullptr;
00244     bool                 _ownsProviderClient = false;
00245
00247     std::vector<GenesysKnowledgeItem> _knowledgeItems;
00248
00250     ExecutionState _state = ExecutionState::NotConfigured;
00251
00253     std::string _lastError;
00254
00256     std::string _lastDiagnostics;
00257
00259     AIAuditLog _auditLog;
00260 };
00261
00262 #endif /* AIASSISTANTDEFAULTIMPL_H */
```

18.544 AIAuditLog.h

```
00001 /*
00002  * File:   AIAuditLog.h
00003  * Author: Genesys Team
00004  *
00005  * Created on 28 de Maio de 2026
00006  */
```

```
00007
00008 #ifndef AIAUDITLOG_H
00009 #define AIAUDITLOG_H
00010
00011 #include <chrono>
00012 #include <string>
00013 #include <vector>
00014
00022 struct AuditEntry {
00023     std::chrono::system_clock::time_point timestamp;
00024     std::string  operation;
00025     std::string  provider;
00026     std::string  modelName;
00027     std::string  promptPreview;
00028     bool         success   = false;
00029     bool         dryRun    = false;
00030     long long    durationMs = 0;
00031     std::string  errorSummary;
00032 };
00033
00042 class AIAuditLog {
00043 public:
00044     explicit AIAuditLog(const std::string& logFilePath = "");
00045     ~AIAuditLog() = default;
00046
00047     AIAuditLog(const AIAuditLog&)            = delete;
00048     AIAuditLog& operator=(const AIAuditLog&) = delete;
00049
00051     void append(const AuditEntry& entry);
00052
00057     std::vector<AuditEntry> loadRecentEntries(unsigned int maxEntries = 200) const;
00058
00060     std::string logFilePath() const { return _logFilePath; }
00061
00066     unsigned int exportCsv(const std::string& csvPath) const;
00067
00069     static std::string defaultLogFilePath();
00070
00071 private:
00072     static std::string _encodeEntry(const AuditEntry& entry);
00073     static bool        _decodeEntry(const std::string& line, AuditEntry& out);
00074     void               _ensureDirectoryExists() const;
00075
00076     std::string _logFilePath;
00077 };
00078
00079 #endif /* AIAUDITLOG_H */
```

18.545 AIConversationService.h

```
00001 #ifndef AICONVERSATIONSERVICE_H
00002 #define AICONVERSATIONSERVICE_H
00003
00004 #include "AIProviderClient_if.h"
00005
00006 #include <map>
00007 #include <memory>
00008 #include <string>
00009 #include <vector>
00010
00011 enum class AIConversationProvider { OpenAI, Anthropic, Local };
00012
00013 struct AIConversationConfiguration {
00014     AIConversationProvider provider = AIConversationProvider::Local;
00015     std::string model = "llama3.2";
00016     std::string baseUrl;
00017     std::string apiKeyEnvironmentVariable;
00018     std::string systemInstructions;
00019     double temperature = 0.2;
00020     unsigned int maxOutputTokens = 512;
00021     unsigned int timeoutSeconds = 120;
00022     bool highReasoningMode = false;
00023     bool conversationEnabled = true;
00024     unsigned int maxHistoryMessages = 20;
00025 };
00026
00027 struct AIConversationResponse {
00028     bool success = false;
00029     std::string content;
00030     std::string errorMessage;
00031     int httpStatusCode = 0;
00032     unsigned int inputTokens = 0;
00033     unsigned int outputTokens = 0;
00034 };
00035
```

```
00036 class AIConversationService {
00037 public:
00038     AIConversationService() = default;
00039     ~AIConversationService() = default;
00040 
00041     void setConfiguration(const AIConversationConfiguration& configuration);
00042     const AIConversationConfiguration& getConfiguration() const;
00043     bool initialize(std::string& errorMessage);
00044     void setProviderClient(std::unique_ptr<AIProviderClient_if> client);
00045     AIConversationResponse send(const std::string& conversationKey, const std::string& prompt);
00046     void clearConversation(const std::string& conversationKey);
00047     void clearAllConversations();
00048     bool isInitialized() const;
00049 
00050 private:
00051     std::vector<ChatMessage> _buildMessages(const std::string& key, const std::string& prompt) const;
00052     void _appendHistory(const std::string& key, const ChatMessage& message);
00053 
00054     AIConversationConfiguration _configuration;
00055     std::unique_ptr<AIProviderClient_if> _client;
00056     std::map<std::string, std::vector<ChatMessage» _histories;
00057 };
00058 
00059 #endif
```

18.546 AIProviderClient_if.h

```
00001 /*
00002  * File:   AIProviderClient_if.h
00003  * Author: Genesys Team
00004  *
00005  * Created on 28 de Maio de 2026
00006  */
00007 
00008 #ifndef AIPROVIDERCLIENT_IF_H
00009 #define AIPROVIDERCLIENT_IF_H
00010 
00011 #include <string>
00012 #include <vector>
00013 
00014 // ---------------------------------------------------------------------------
00015 // LLM chat structures
00016 // ---------------------------------------------------------------------------
00017 
00027 struct ChatMessage {
00028     std::string role;
00029     std::string content;
00030 };
00031 
00035 struct ProviderRequest {
00036     std::string              model;
00037     std::vector<ChatMessage> messages;
00038     double                   temperature       = 0.2;
00039     unsigned int             maxTokens         = 4096;
00040     bool                     highReasoningMode = false;
00041 };
00042 
00046 struct ProviderResponse {
00047     bool         success        = false;
00048     std::string  content;
00049     std::string  errorMessage;
00050     int          httpStatusCode = 0;
00051     unsigned int inputTokens    = 0;
00052     unsigned int outputTokens   = 0;
00053 };
00054 
00055 // ---------------------------------------------------------------------------
00056 // Interface
00057 // ---------------------------------------------------------------------------
00058 
00077 class AIProviderClient_if {
00078 public:
00079     virtual ~AIProviderClient_if() = default;
00080 
00081     // -----------------------------------------------------------------------
00082     // Provider identification
00083     // -----------------------------------------------------------------------
00084 
00086     virtual std::string getProviderName() const = 0;
00087 
00095     virtual bool isAvailable() const = 0;
00096 
00097     // -----------------------------------------------------------------------
00098     // Configuration
00099     // -----------------------------------------------------------------------
```

```
00100
00102     virtual void setBaseUrl(const std::string& url) = 0;
00104     virtual std::string getBaseUrl() const = 0;
00105
00107     virtual void setTimeoutSeconds(unsigned int seconds) = 0;
00109     virtual unsigned int getTimeoutSeconds() const = 0;
00110
00111     // ------------------------------------------------------------------------
00112     // Secrets -- same security contract as AIAssistant_if
00113     // ------------------------------------------------------------------------
00114
00122     virtual void setApiKey(const std::string& apiKey) = 0;
00123
00127     virtual void clearApiKey() = 0;
00128
00130     virtual bool hasApiKey() const = 0;
00131
00132     // ------------------------------------------------------------------------
00133     // Request
00134     // ------------------------------------------------------------------------
00135
00144     virtual ProviderResponse send(const ProviderRequest& request) = 0;
00145 };
00146
00147 #endif /* AIPROVIDERCLIENT_IF_H */
```

18.547 AISecretStore.h

```
00001 /*
00002  * File:   AISecretStore.h
00003  * Author: Genesys Team
00004  *
00005  * Created on 28 de Maio de 2026
00006  */
00007
00008 #ifndef AISECRETSTORE_H
00009 #define AISECRETSTORE_H
00010
00011 #include <optional>
00012 #include <string>
00013
00038 class AISecretStore {
00039 public:
00040     AISecretStore() = delete;
00041
00049     static bool isAvailable();
00050
00056     static bool save(const std::string& service,
00057                      const std::string& account,
00058                      const std::string& secret);
00059
00064     static std::optional<std::string> load(const std::string& service,
00065                                            const std::string& account);
00066
00071     static bool remove(const std::string& service,
00072                        const std::string& account);
00073
00074 private:
00075     static std::string _shellEscape(const std::string& s);
00076     static std::string _pipeRead(const std::string& cmd);
00077 };
00078
00079 #endif /* AISECRETSTORE_H */
```

18.548 AnthropicProviderClientImpl.h

```
00001 /*
00002  * File:   AnthropicProviderClientImpl.h
00003  * Author: Genesys Team
00004  *
00005  * Created on 28 de Maio de 2026
00006  */
00007
00008 #ifndef ANTHROPICPROVIDERCLIENTIMPL_H
00009 #define ANTHROPICPROVIDERCLIENTIMPL_H
00010
00011 #include "HttpProviderClientBase.h"
00012
00035 class AnthropicProviderClientImpl : public HttpProviderClientBase {
00036 public:
00040     AnthropicProviderClientImpl();
00041     virtual ~AnthropicProviderClientImpl() = default;
```

```
00042
00043     // ---------------------------------------------------------------------
00044     // Provider identification
00045     // ---------------------------------------------------------------------
00046
00047     std::string getProviderName() const override;
00048
00050     bool isAvailable() const override;
00051
00052     // ---------------------------------------------------------------------
00053     // Anthropic-specific configuration
00054     // ---------------------------------------------------------------------
00055
00064     void setApiVersion(const std::string& version);
00066     std::string getApiVersion() const;
00067
00075     void setThinkingBudgetTokens(unsigned int budget);
00077     unsigned int getThinkingBudgetTokens() const;
00078
00079 protected:
00080     std::string _buildRequestBody(const ProviderRequest& request) const override;
00081     ProviderResponse _parseResponseBody(const std::string& rawBody,
00082                                         int httpStatus) const override;
00083     std::string _authorizationHeader() const override;
00084     std::vector<std::string> _extraRequestHeaders() const override;
00085
00086 private:
00087     std::string  _apiVersion           = "2023-06-01";
00088     unsigned int _thinkingBudgetTokens = 10000;
00089 };
00090
00091 #endif /* ANTHROPICPROVIDERCLIENTIMPL_H */
```

18.549 HttpProviderClientBase.h

```
00001 /*
00002  * File:   HttpProviderClientBase.h
00003  * Author: Genesys Team
00004  *
00005  * Created on 28 de Maio de 2026
00006  */
00007
00008 #ifndef HTTPPROVIDERCLIENTBASE_H
00009 #define HTTPPROVIDERCLIENTBASE_H
00010
00011 #include "AIProviderClient_if.h"
00012
00013 #include <string>
00014 #include <vector>
00015
00049 class HttpProviderClientBase : public AIProviderClient_if {
00050 public:
00051     HttpProviderClientBase();
00052     virtual ~HttpProviderClientBase();
00053
00054     // Not copyable: API key in member string, non-owning semantics.
00055     HttpProviderClientBase(const HttpProviderClientBase&) = delete;
00056     HttpProviderClientBase& operator=(const HttpProviderClientBase&) = delete;
00057
00058     // ---------------------------------------------------------------------
00059     // AIProviderClient_if -- common implementations
00060     // ---------------------------------------------------------------------
00061
00062     bool isAvailable() const override;
00063
00064     void setBaseUrl(const std::string& url) override;
00065     std::string getBaseUrl() const override;
00066
00067     void setTimeoutSeconds(unsigned int seconds) override;
00068     unsigned int getTimeoutSeconds() const override;
00069
00070     void setApiKey(const std::string& apiKey) override;
00071     void clearApiKey() override;
00072     bool hasApiKey() const override;
00073
00087     ProviderResponse send(const ProviderRequest& request) override;
00088
00089 protected:
00090     // ---------------------------------------------------------------------
00091     // Subclass hooks (pure virtual)
00092     // ---------------------------------------------------------------------
00093
00095     virtual std::string _buildRequestBody(const ProviderRequest& request) const = 0;
00096
00104     virtual ProviderResponse _parseResponseBody(const std::string& rawBody,
```

```
00105                                               int httpStatus) const = 0;
00106
00118     virtual std::string _authorizationHeader() const = 0;
00119
00130     virtual std::vector<std::string> _extraRequestHeaders() const;
00131
00132 public:
00133     // ------------------------------------------------------------------------
00134     // JSON utilities -- public static; usable by any caller that needs
00135     // lightweight JSON field extraction without a full parser dependency.
00136     // ------------------------------------------------------------------------
00137
00139     static std::string _escapeJsonString(const std::string& s);
00140
00145     static std::string _extractJsonString(const std::string& json,
00146                                           const std::string& key);
00147
00152     static int _extractJsonInt(const std::string& json, const std::string& key);
00153
00158     static double _extractJsonDouble(const std::string& json, const std::string& key);
00159
00161     static std::string _buildMessagesArray(const std::vector<ChatMessage>& messages);
00162
00167     static std::string _extractFirstArrayElement(const std::string& json,
00168                                                  const std::string& arrayKey);
00169
00170 protected:
00171     std::string  _baseUrl;
00172     unsigned int _timeoutSeconds = 30;
00173
00183     std::string _apiKey;
00184 };
00185
00186 #endif /* HTTPPROVIDERCLIENTBASE_H */
```

18.550 LocalProviderClientImpl.h

```
00001 /*
00002  * File:   LocalProviderClientImpl.h
00003  * Author: Genesys Team
00004  *
00005  * Created on 28 de Maio de 2026
00006  */
00007
00008 #ifndef LOCALPROVIDERCLIENTIMPL_H
00009 #define LOCALPROVIDERCLIENTIMPL_H
00010
00011 #include "HttpProviderClientBase.h"
00012
00034 class LocalProviderClientImpl : public HttpProviderClientBase {
00035 public:
00039     LocalProviderClientImpl();
00040     virtual ~LocalProviderClientImpl() = default;
00041
00042     // ------------------------------------------------------------------------
00043     // Provider identification
00044     // ------------------------------------------------------------------------
00045
00046     std::string getProviderName() const override;
00047
00054     bool isAvailable() const override;
00055
00056 protected:
00057     std::string _buildRequestBody(const ProviderRequest& request) const override;
00058     ProviderResponse _parseResponseBody(const std::string& rawBody,
00059                                         int httpStatus) const override;
00060
00064     std::string _authorizationHeader() const override;
00065 };
00066
00067 #endif /* LOCALPROVIDERCLIENTIMPL_H */
```

18.551 OpenAIProviderClientImpl.h

```
00001 /*
00002  * File:   OpenAIProviderClientImpl.h
00003  * Author: Genesys Team
00004  *
00005  * Created on 28 de Maio de 2026
00006  */
00007
00008 #ifndef OPENAIPROVIDERIMPL_H
```

```
00009 #define OPENAIPROVIDERIMPL_H
00010
00011 #include "HttpProviderClientBase.h"
00012
00031 class OpenAIProviderClientImpl : public HttpProviderClientBase {
00032 public:
00039     OpenAIProviderClientImpl();
00040     virtual ~OpenAIProviderClientImpl() = default;
00041
00042     // ------------------------------------------------------------------------
00043     // Provider identification
00044     // ------------------------------------------------------------------------
00045
00046     std::string getProviderName() const override;
00047
00051     bool isAvailable() const override;
00052
00053     // ------------------------------------------------------------------------
00054     // Optional per-request headers
00055     // ------------------------------------------------------------------------
00056
00058     void setOrganizationId(const std::string& orgId);
00060     std::string getOrganizationId() const;
00061
00063     void setProjectId(const std::string& projectId);
00065     std::string getProjectId() const;
00066
00067 protected:
00068     std::string _buildRequestBody(const ProviderRequest& request) const override;
00069     ProviderResponse _parseResponseBody(const std::string& rawBody,
00070                                         int httpStatus) const override;
00071     std::string _authorizationHeader() const override;
00072     std::vector<std::string> _extraRequestHeaders() const override;
00073
00074 private:
00075     std::string _organizationId;
00076     std::string _projectId;
00077 };
00078
00079 #endif /* OPENAIPROVIDERIMPL_H */
```

18.552 BioKineticLawExpression.h

```
00001 #ifndef BIOKINETICLAWEXPRESSION_H
00002 #define BIOKINETICLAWEXPRESSION_H
00003
00004 #include <algorithm>
00005 #include <cctype>
00006 #include <cmath>
00007 #include <functional>
00008 #include <string>
00009 #include <vector>
00010
00020 class BioKineticLawExpression {
00021 public:
00022     using SymbolResolver = std::function<bool(const std::string& name, double& value)>;
00023
00024     bool evaluate(const std::string& expression, SymbolResolver resolver, double& value, std::string&
      errorMessage) const {
00025         Parser parser(expression, resolver, errorMessage);
00026         value = parser.parseExpression();
00027         parser.skipWhitespace();
00028         if (!errorMessage.empty()) {
00029             return false;
00030         }
00031         if (!parser.atEnd()) {
00032             errorMessage = "Unexpected token at position " + std::to_string(parser.position()) + ".";
00033             return false;
00034         }
00035         if (!std::isfinite(value)) {
00036             errorMessage = "Expression evaluated to a non-finite value.";
00037             return false;
00038         }
00039         return true;
00040     }
00041
00042 private:
00043     class Parser {
00044     public:
00045         Parser(const std::string& expression, SymbolResolver resolver, std::string& errorMessage)
00046                 : _expression(expression), _resolver(resolver), _errorMessage(errorMessage) {
00047         }
00048
00049         double parseExpression() {
00050             double value = parseTerm();
```

```
00051             while (_errorMessage.empty()) {
00052                 skipWhitespace();
00053                 if (consume('+')) {
00054                     value += parseTerm();
00055                 } else if (consume('-')) {
00056                     value -= parseTerm();
00057                 } else {
00058                     break;
00059                 }
00060             }
00061             return value;
00062         }
00063 
00064         void skipWhitespace() {
00065             while (_position < _expression.size() && std::isspace(static_cast<unsigned
      char>(_expression[_position])) != 0) {
00066                 ++_position;
00067             }
00068         }
00069 
00070         bool atEnd() const {
00071             return _position >= _expression.size();
00072         }
00073 
00074         unsigned int position() const {
00075             return static_cast<unsigned int>(_position);
00076         }
00077 
00078     private:
00079         double parseTerm() {
00080             double value = parsePower();
00081             while (_errorMessage.empty()) {
00082                 skipWhitespace();
00083                 if (consume('*')) {
00084                     value *= parsePower();
00085                 } else if (consume('/')) {
00086                     const double denominator = parsePower();
00087                     if (denominator == 0.0) {
00088                         _errorMessage = "Division by zero.";
00089                         return 0.0;
00090                     }
00091                     value /= denominator;
00092                 } else {
00093                     break;
00094                 }
00095             }
00096             return value;
00097         }
00098 
00099         double parsePower() {
00100             double value = parseUnary();
00101             skipWhitespace();
00102             if (consume('^')) {
00103                 value = std::pow(value, parsePower());
00104             }
00105             return value;
00106         }
00107 
00108         double parseUnary() {
00109             skipWhitespace();
00110             if (consume('+')) {
00111                 return parseUnary();
00112             }
00113             if (consume('-')) {
00114                 return -parseUnary();
00115             }
00116             return parsePrimary();
00117         }
00118 
00119         double parsePrimary() {
00120             skipWhitespace();
00121             if (_position >= _expression.size()) {
00122                 _errorMessage = "Unexpected end of expression.";
00123                 return 0.0;
00124             }
00125             if (consume('(')) {
00126                 double value = parseExpression();
00127                 skipWhitespace();
00128                 if (!consume(')')) {
00129                     _errorMessage = "Missing closing parenthesis.";
00130                     return 0.0;
00131                 }
00132                 return value;
00133             }
00134             if (std::isdigit(static_cast<unsigned char>(_expression[_position])) != 0 ||
      _expression[_position] == '.') {
00135                 return parseNumber();
```

```
00136             }
00137             if (isIdentifierStart(_expression[_position])) {
00138                 const std::string identifier = parseIdentifier();
00139                 skipWhitespace();
00140                 if (consume('(')) {
00141                     return parseFunction(identifier);
00142                 }
00143                 double value = 0.0;
00144                 if (_resolver == nullptr || !_resolver(identifier, value)) {
00145                     _errorMessage = "Unknown kinetic-law symbol \"" + identifier + "\".";
00146                     return 0.0;
00147                 }
00148                 return value;
00149             }
00150             _errorMessage = "Unexpected token at position " + std::to_string(_position) + ".";
00151             return 0.0;
00152         }
00153
00154         double parseNumber() {
00155             const size_t start = _position;
00156             while (_position < _expression.size() &&
00157                    (std::isdigit(static_cast<unsigned char>(_expression[_position])) != 0 ||
    _expression[_position] == '.')) {
00158                 ++_position;
00159             }
00160             if (_position < _expression.size() && (_expression[_position] == 'e' ||
    _expression[_position] == 'E')) {
00161                 ++_position;
00162                 if (_position < _expression.size() && (_expression[_position] == '+' ||
    _expression[_position] == '-')) {
00163                     ++_position;
00164                 }
00165                 while (_position < _expression.size() && std::isdigit(static_cast<unsigned
    char>(_expression[_position])) != 0) {
00166                     ++_position;
00167                 }
00168             }
00169             try {
00170                 return std::stod(_expression.substr(start, _position - start));
00171             } catch (...) {
00172                 _errorMessage = "Invalid numeric literal at position " + std::to_string(start) + ".";
00173                 return 0.0;
00174             }
00175         }
00176
00177         std::string parseIdentifier() {
00178             const size_t start = _position;
00179             ++_position;
00180             while (_position < _expression.size() && isIdentifierChar(_expression[_position])) {
00181                 ++_position;
00182             }
00183             return _expression.substr(start, _position - start);
00184         }
00185
00186         double parseFunction(const std::string& name) {
00187             std::vector<double> args;
00188             skipWhitespace();
00189             if (!consume(')')) {
00190                 while (_errorMessage.empty()) {
00191                     args.push_back(parseExpression());
00192                     skipWhitespace();
00193                     if (consume(')')) {
00194                         break;
00195                     }
00196                     if (!consume(',')) {
00197                         _errorMessage = "Missing comma or closing parenthesis in function call.";
00198                         return 0.0;
00199                     }
00200                 }
00201             }
00202             return applyFunction(name, args);
00203         }
00204
00205         double applyFunction(const std::string& name, const std::vector<double>& args) {
00206             if (name == "abs" && args.size() == 1) {
00207                 return std::fabs(args[0]);
00208             }
00209             if (name == "exp" && args.size() == 1) {
00210                 return std::exp(args[0]);
00211             }
00212             if (name == "log" && args.size() == 1) {
00213                 if (args[0] <= 0.0) {
00214                     _errorMessage = "log() requires a positive argument.";
00215                     return 0.0;
00216                 }
00217                 return std::log(args[0]);
00218             }
```

```
00219             if (name == "sqrt" && args.size() == 1) {
00220                 if (args[0] < 0.0) {
00221                     _errorMessage = "sqrt() requires a non-negative argument.";
00222                     return 0.0;
00223                 }
00224                 return std::sqrt(args[0]);
00225             }
00226             if (name == "min" && args.size() == 2) {
00227                 return std::min(args[0], args[1]);
00228             }
00229             if (name == "max" && args.size() == 2) {
00230                 return std::max(args[0], args[1]);
00231             }
00232             if (name == "pow" && args.size() == 2) {
00233                 return std::pow(args[0], args[1]);
00234             }
00235             _errorMessage = "Unsupported function \"" + name + "\" or invalid argument count.";
00236             return 0.0;
00237         }
00238
00239         bool consume(char expected) {
00240             skipWhitespace();
00241             if (_position < _expression.size() && _expression[_position] == expected) {
00242                 ++_position;
00243                 return true;
00244             }
00245             return false;
00246         }
00247
00248         bool isIdentifierStart(char ch) const {
00249             return std::isalpha(static_cast<unsigned char>(ch)) != 0 || ch == '_';
00250         }
00251
00252         bool isIdentifierChar(char ch) const {
00253             return std::isalnum(static_cast<unsigned char>(ch)) != 0 || ch == '_' || ch == '.';
00254         }
00255
00256     private:
00257         const std::string& _expression;
00258         SymbolResolver _resolver;
00259         std::string& _errorMessage;
00260         size_t _position = 0;
00261     };
00262 };
00263
00264 #endif /* BIOKINETICLAWEXPRESSION_H */
```

18.553 BioSimulationAnalysis.h

```
00001 #ifndef BIOSIMULATIONANALYSIS_H
00002 #define BIOSIMULATIONANALYSIS_H
00003
00004 #include <algorithm>
00005 #include <cmath>
00006 #include <string>
00007 #include <vector>
00008
00009 #include "tools/Biochemical/BioSimulationResult.h"
00010 #include "tools/Biochemical/MassActionOdeSystem.h"
00011
00012 struct BioStoichiometryMatrix {
00013     std::vector<std::string> speciesNames;
00014     std::vector<std::string> reactionNames;
00015     std::vector<std::vector<double» coefficients;
00016
00017     double coefficient(unsigned int speciesIndex, unsigned int reactionIndex) const {
00018         if (speciesIndex >= coefficients.size() || reactionIndex >= reactionNames.size()) {
00019             return 0.0;
00020         }
00021         return coefficients[speciesIndex][reactionIndex];
00022     }
00023 };
00024
00025 enum class BioReactionRateKind {
00026     Forward,
00027     Reverse,
00028     Net
00029 };
00030
00031 struct BioReactionRateSample {
00032     double time = 0.0;
00033     std::vector<double> forwardRates;
00034     std::vector<double> reverseRates;
00035     std::vector<double> netRates;
00036 };
```

```
00037
00038 struct BioReactionRateTimeCourse {
00039     std::vector<std::string> reactionNames;
00040     std::vector<BioReactionRateSample> samples;
00041
00042     bool hasReaction(const std::string& reactionName) const {
00043         return std::find(reactionNames.begin(), reactionNames.end(), reactionName) !=
      reactionNames.end();
00044     }
00045
00046     bool toDataset(const std::string& reactionName, BioReactionRateKind kind,
      SimulationResultsDataset* dataset,
00047                    std::string* errorMessage = nullptr) const {
00048         if (dataset == nullptr) {
00049             if (errorMessage != nullptr) {
00050                 *errorMessage = "Invalid dataset output parameter.";
00051             }
00052             return false;
00053         }
00054
00055         const auto it = std::find(reactionNames.begin(), reactionNames.end(), reactionName);
00056         if (it == reactionNames.end()) {
00057             if (errorMessage != nullptr) {
00058                 *errorMessage = "BioReactionRateTimeCourse does not contain reaction \"" +
      reactionName + "\".";
00059             }
00060             return false;
00061         }
00062         const unsigned int reactionIndex = static_cast<unsigned int>(it - reactionNames.begin());
00063         const std::string suffix = rateKindName(kind);
00064
00065         SimulationResultsDataset converted;
00066         converted.datasetName = reactionName + "." + suffix;
00067         converted.randomVariableName = reactionName + "." + suffix;
00068         converted.description = "Bio simulation reaction-rate time-course dataset.";
00069         converted.variableType = "Continuous numeric";
00070         converted.source = "BioReactionRateTimeCourse";
00071         converted.sourceDescription = "BioReactionRateTimeCourse";
00072         converted.expression = reactionName + "." + suffix;
00073         converted.expressionName = reactionName + "." + suffix;
00074         converted.formatKind = SimulationResultsDatasetFormat::GuiTabular;
00075         converted.recordFile = false;
00076         converted.timeDependent = true;
00077         for (unsigned int i = 0; i < samples.size(); ++i) {
00078             const std::vector<double>* values = valuesForKind(samples[i], kind);
00079             if (reactionIndex >= values->size()) {
00080                 if (errorMessage != nullptr) {
00081                     *errorMessage = "BioReactionRateTimeCourse sample is missing reaction \"" +
      reactionName + "\".";
00082                 }
00083                 return false;
00084             }
00085             SimulationResultsObservation observation;
00086             observation.replication = 1;
00087             observation.time = samples[i].time;
00088             observation.hasTime = true;
00089             observation.value = (*values)[reactionIndex];
00090             observation.sourceLine = i + 1;
00091             converted.observations.push_back(observation);
00092         }
00093
00094         *dataset = converted;
00095         return true;
00096     }
00097
00098 private:
00099     static std::string rateKindName(BioReactionRateKind kind) {
00100         switch (kind) {
00101             case BioReactionRateKind::Forward:
00102                 return "forwardRate";
00103             case BioReactionRateKind::Reverse:
00104                 return "reverseRate";
00105             case BioReactionRateKind::Net:
00106                 return "netRate";
00107         }
00108         return "netRate";
00109     }
00110
00111     static const std::vector<double>* valuesForKind(const BioReactionRateSample& sample,
      BioReactionRateKind kind) {
00112         switch (kind) {
00113             case BioReactionRateKind::Forward:
00114                 return &sample.forwardRates;
00115             case BioReactionRateKind::Reverse:
00116                 return &sample.reverseRates;
00117             case BioReactionRateKind::Net:
00118                 return &sample.netRates;
```

```
00119         }
00120         return &sample.netRates;
00121     }
00122 };
00123
00124 struct BioSpeciesDerivative {
00125     std::string speciesName;
00126     double derivative = 0.0;
00127 };
00128
00129 struct BioSteadyStateCheck {
00130     bool steady = false;
00131     double maxAbsoluteDerivative = 0.0;
00132     double tolerance = 0.0;
00133     std::vector<BioSpeciesDerivative> derivatives;
00134 };
00135
00136 struct BioParameterSensitivityEntry {
00137     std::string parameterName;
00138     double baseValue = 0.0;
00139     double delta = 0.0;
00140     double maxAbsoluteSensitivity = 0.0;
00141     std::vector<BioSpeciesDerivative> derivativeSensitivities;
00142 };
00143
00144 struct BioSensitivityScan {
00145     double time = 0.0;
00146     std::vector<std::string> speciesNames;
00147     std::vector<BioParameterSensitivityEntry> entries;
00148 };
00149
00150 class BioSimulationAnalysis {
00151 public:
00152     static BioStoichiometryMatrix buildStoichiometryMatrix(const MassActionOdeSystem& system) {
00153         BioStoichiometryMatrix matrix;
00154         matrix.speciesNames.reserve(system.species().size());
00155         matrix.reactionNames.reserve(system.reactions().size());
00156         matrix.coefficients.assign(system.species().size(),
      std::vector<double>(system.reactions().size(), 0.0));
00157
00158         for (const MassActionOdeSystem::Species& species : system.species()) {
00159             matrix.speciesNames.push_back(species.name);
00160         }
00161         for (unsigned int reactionIndex = 0; reactionIndex < system.reactions().size();
      ++reactionIndex) {
00162             const MassActionOdeSystem::Reaction& reaction = system.reactions()[reactionIndex];
00163             matrix.reactionNames.push_back(reaction.name);
00164             for (const MassActionOdeSystem::StoichiometricTerm& reactant : reaction.reactants) {
00165                 if (reactant.speciesIndex < matrix.coefficients.size()) {
00166                     matrix.coefficients[reactant.speciesIndex][reactionIndex] -=
      reactant.stoichiometry;
00167                 }
00168             }
00169             for (const MassActionOdeSystem::StoichiometricTerm& product : reaction.products) {
00170                 if (product.speciesIndex < matrix.coefficients.size()) {
00171                     matrix.coefficients[product.speciesIndex][reactionIndex] += product.stoichiometry;
00172                 }
00173             }
00174         }
00175         return matrix;
00176     }
00177
00178     static bool buildReactionRateTimeCourse(const MassActionOdeSystem& system, const
      BioSimulationResult& result,
00179                                             BioReactionRateTimeCourse* timeCourse,
00180                                             std::string* errorMessage = nullptr) {
00181         if (timeCourse == nullptr) {
00182             if (errorMessage != nullptr) {
00183                 *errorMessage = "Invalid reaction rate time-course output parameter.";
00184             }
00185             return false;
00186         }
00187
00188         BioReactionRateTimeCourse converted;
00189         converted.reactionNames.reserve(system.reactions().size());
00190         for (const MassActionOdeSystem::Reaction& reaction : system.reactions()) {
00191             converted.reactionNames.push_back(reaction.name);
00192         }
00193
00194         for (const BioSimulationSample& sample : result.getSamples()) {
00195             std::vector<double> y;
00196             if (!stateFromSample(system, sample, &y, errorMessage)) {
00197                 return false;
00198             }
00199             BioReactionRateSample rateSample;
00200             rateSample.time = sample.time;
00201             rateSample.forwardRates.reserve(system.reactions().size());
```

```
00202             rateSample.reverseRates.reserve(system.reactions().size());
00203             rateSample.netRates.reserve(system.reactions().size());
00204             for (unsigned int reactionIndex = 0; reactionIndex < system.reactions().size();
      ++reactionIndex) {
00205                 const double forwardRate = system.evaluateForwardRate(reactionIndex, y.data());
00206                 const double reverseRate = system.evaluateReverseRate(reactionIndex, y.data());
00207                 rateSample.forwardRates.push_back(forwardRate);
00208                 rateSample.reverseRates.push_back(reverseRate);
00209                 rateSample.netRates.push_back(forwardRate - reverseRate);
00210             }
00211             converted.samples.push_back(rateSample);
00212         }
00213
00214         *timeCourse = converted;
00215         return true;
00216     }
00217
00218     static bool checkSteadyState(const MassActionOdeSystem& system, const BioSimulationSample& sample,
      double tolerance,
00219                                  BioSteadyStateCheck* check, std::string* errorMessage = nullptr) {
00220         if (check == nullptr) {
00221             if (errorMessage != nullptr) {
00222                 *errorMessage = "Invalid steady-state output parameter.";
00223             }
00224             return false;
00225         }
00226         if (tolerance < 0.0) {
00227             if (errorMessage != nullptr) {
00228                 *errorMessage = "Steady-state tolerance must be non-negative.";
00229             }
00230             return false;
00231         }
00232
00233         std::vector<double> y;
00234         if (!stateFromSample(system, sample, &y, errorMessage)) {
00235             return false;
00236         }
00237         std::vector<double> dydt(system.dimension(), 0.0);
00238         system.evaluate(sample.time, y.data(), dydt.data());
00239
00240         BioSteadyStateCheck converted;
00241         converted.tolerance = tolerance;
00242         converted.derivatives.reserve(system.species().size());
00243         for (unsigned int i = 0; i < system.species().size(); ++i) {
00244             const double absoluteDerivative = std::abs(dydt[i]);
00245             converted.maxAbsoluteDerivative = std::max(converted.maxAbsoluteDerivative,
      absoluteDerivative);
00246             converted.derivatives.push_back({system.species()[i].name, dydt[i]});
00247         }
00248         converted.steady = converted.maxAbsoluteDerivative <= tolerance;
00249
00250         *check = converted;
00251         return true;
00252     }
00253
00254     static bool scanLocalParameterSensitivity(const MassActionOdeSystem& system, const
      BioSimulationSample& sample,
00255                                               double relativeStep, double absoluteStep,
      BioSensitivityScan* scan,
00256                                               std::string* errorMessage = nullptr) {
00257         if (scan == nullptr) {
00258             if (errorMessage != nullptr) {
00259                 *errorMessage = "Invalid sensitivity scan output parameter.";
00260             }
00261             return false;
00262         }
00263         if (relativeStep < 0.0 || absoluteStep <= 0.0) {
00264             if (errorMessage != nullptr) {
00265                 *errorMessage = "Sensitivity scan steps must satisfy relativeStep >= 0 and
      absoluteStep > 0.";
00266             }
00267             return false;
00268         }
00269
00270         std::vector<double> y;
00271         if (!stateFromSample(system, sample, &y, errorMessage)) {
00272             return false;
00273         }
00274         std::vector<double> baseDerivatives;
00275         evaluateDerivatives(system, sample.time, y, &baseDerivatives);
00276
00277         BioSensitivityScan converted;
00278         converted.time = sample.time;
00279         converted.speciesNames.reserve(system.species().size());
00280         for (const MassActionOdeSystem::Species& species : system.species()) {
00281             converted.speciesNames.push_back(species.name);
00282         }
```

```
00283
00284         const std::vector<ParameterTarget> targets = parameterTargets(system);
00285         for (const ParameterTarget& target : targets) {
00286             std::vector<MassActionOdeSystem::Reaction> reactions = system.reactions();
00287             double baseValue = 0.0;
00288             if (!applyPerturbation(target, relativeStep, absoluteStep, &reactions, &baseValue)) {
00289                 continue;
00290             }
00291
00292             MassActionOdeSystem perturbed(system.species(), reactions);
00293             std::vector<double> perturbedDerivatives;
00294             evaluateDerivatives(perturbed, sample.time, y, &perturbedDerivatives);
00295
00296             const double delta = deltaForValue(baseValue, relativeStep, absoluteStep);
00297             BioParameterSensitivityEntry entry;
00298             entry.parameterName = target.name;
00299             entry.baseValue = baseValue;
00300             entry.delta = delta;
00301             entry.derivativeSensitivities.reserve(system.species().size());
00302             for (unsigned int i = 0; i < system.species().size(); ++i) {
00303                 const double sensitivity = (perturbedDerivatives[i] - baseDerivatives[i]) / delta;
00304                 entry.maxAbsoluteSensitivity = std::max(entry.maxAbsoluteSensitivity,
    std::abs(sensitivity));
00305                 entry.derivativeSensitivities.push_back({system.species()[i].name, sensitivity});
00306             }
00307             converted.entries.push_back(entry);
00308         }
00309
00310         *scan = converted;
00311         return true;
00312     }
00313
00314 private:
00315     enum class ParameterTargetKind {
00316         ForwardRateConstant,
00317         ReverseRateConstant,
00318         NamedParameter
00319     };
00320
00321     struct ParameterTarget {
00322         ParameterTargetKind kind = ParameterTargetKind::NamedParameter;
00323         unsigned int reactionIndex = 0;
00324         std::string name;
00325     };
00326
00327     static void evaluateDerivatives(const MassActionOdeSystem& system, double time, const
    std::vector<double>& y,
00328                                     std::vector<double>* derivatives) {
00329         derivatives->assign(system.dimension(), 0.0);
00330         system.evaluate(time, y.data(), derivatives->data());
00331     }
00332
00333     static std::vector<ParameterTarget> parameterTargets(const MassActionOdeSystem& system) {
00334         std::vector<ParameterTarget> targets;
00335         std::vector<std::string> namedParameters;
00336         for (unsigned int reactionIndex = 0; reactionIndex < system.reactions().size();
    ++reactionIndex) {
00337             const MassActionOdeSystem::Reaction& reaction = system.reactions()[reactionIndex];
00338             if (reaction.kineticLawExpression.empty()) {
00339                 targets.push_back({ParameterTargetKind::ForwardRateConstant, reactionIndex,
    reaction.name + ".rateConstant"});
00340             }
00341             if (reaction.reversible && reaction.reverseKineticLawExpression.empty()) {
00342                 targets.push_back({ParameterTargetKind::ReverseRateConstant, reactionIndex,
    reaction.name + ".reverseRateConstant"});
00343             }
00344             for (const auto& parameter : reaction.parameters) {
00345                 if (std::find(namedParameters.begin(), namedParameters.end(), parameter.first) ==
    namedParameters.end()) {
00346                     namedParameters.push_back(parameter.first);
00347                     targets.push_back({ParameterTargetKind::NamedParameter, reactionIndex,
    parameter.first});
00348                 }
00349             }
00350         }
00351         return targets;
00352     }
00353
00354     static double deltaForValue(double value, double relativeStep, double absoluteStep) {
00355         return std::max(std::abs(value) * relativeStep, absoluteStep);
00356     }
00357
00358     static bool applyPerturbation(const ParameterTarget& target, double relativeStep, double
    absoluteStep,
00359                                   std::vector<MassActionOdeSystem::Reaction>* reactions, double*
    baseValue) {
00360         if (reactions == nullptr || baseValue == nullptr || target.reactionIndex >= reactions->size())
```

```
    {
00361             return false;
00362         }
00363         if (target.kind == ParameterTargetKind::ForwardRateConstant) {
00364             *baseValue = (*reactions)[target.reactionIndex].rateConstant;
00365             (*reactions)[target.reactionIndex].rateConstant += deltaForValue(*baseValue, relativeStep,
    absoluteStep);
00366             return true;
00367         }
00368         if (target.kind == ParameterTargetKind::ReverseRateConstant) {
00369             *baseValue = (*reactions)[target.reactionIndex].reverseRateConstant;
00370             (*reactions)[target.reactionIndex].reverseRateConstant += deltaForValue(*baseValue,
    relativeStep, absoluteStep);
00371             return true;
00372         }
00373
00374         bool found = false;
00375         for (MassActionOdeSystem::Reaction& reaction : *reactions) {
00376             for (auto& parameter : reaction.parameters) {
00377                 if (parameter.first == target.name) {
00378                     if (!found) {
00379                         *baseValue = parameter.second;
00380                     }
00381                     parameter.second += deltaForValue(parameter.second, relativeStep, absoluteStep);
00382                     found = true;
00383                 }
00384             }
00385         }
00386         return found;
00387     }
00388
00389     static bool stateFromSample(const MassActionOdeSystem& system, const BioSimulationSample& sample,
00390                                 std::vector<double>* y, std::string* errorMessage) {
00391         if (y == nullptr) {
00392             if (errorMessage != nullptr) {
00393                 *errorMessage = "Invalid state output parameter.";
00394             }
00395             return false;
00396         }
00397         y->assign(system.species().size(), 0.0);
00398         for (unsigned int speciesIndex = 0; speciesIndex < system.species().size(); ++speciesIndex) {
00399             const std::string& speciesName = system.species()[speciesIndex].name;
00400             bool found = false;
00401             for (const BioSimulationSpeciesAmount& amount : sample.species) {
00402                 if (amount.speciesName == speciesName) {
00403                     (*y)[speciesIndex] = amount.amount;
00404                     found = true;
00405                     break;
00406                 }
00407             }
00408             if (!found) {
00409                 if (errorMessage != nullptr) {
00410                     *errorMessage = "BioSimulationSample is missing species \"" + speciesName + "\".";
00411                 }
00412                 return false;
00413             }
00414         }
00415         return true;
00416     }
00417 };
00418
00419 #endif /* BIOSIMULATIONANALYSIS_H */
```

18.554 BioSimulationResult.h

```
00001 #ifndef BIOSIMULATIONRESULT_H
00002 #define BIOSIMULATIONRESULT_H
00003
00004 #include <algorithm>
00005 #include <string>
00006 #include <vector>
00007
00008 #include "tools/Statistics/SimulationResultsDataset.h"
00009
00010 struct BioSimulationSpeciesAmount {
00011     std::string speciesName;
00012     double amount = 0.0;
00013 };
00014
00015 struct BioSimulationSample {
00016     double time = 0.0;
00017     std::vector<BioSimulationSpeciesAmount> species;
00018 };
00019
00020 class BioSimulationResult {
```

```
00021 public:
00022     void clear() {
00023         _networkName.clear();
00024         _startTime = 0.0;
00025         _stopTime = 0.0;
00026         _stepSize = 0.0;
00027         _speciesNames.clear();
00028         _samples.clear();
00029     }
00030
00031     void configure(std::string networkName, double startTime, double stopTime, double stepSize,
00032                    const std::vector<std::string>& speciesNames) {
00033         _networkName = networkName;
00034         _startTime = startTime;
00035         _stopTime = stopTime;
00036         _stepSize = stepSize;
00037         _speciesNames = speciesNames;
00038         _samples.clear();
00039     }
00040
00041     void appendSample(double time, const std::vector<BioSimulationSpeciesAmount>& speciesAmounts) {
00042         BioSimulationSample sample;
00043         sample.time = time;
00044         sample.species = speciesAmounts;
00045         _samples.push_back(sample);
00046     }
00047
00048     bool empty() const {
00049         return _samples.empty();
00050     }
00051
00052     unsigned int sampleCount() const {
00053         return static_cast<unsigned int>(_samples.size());
00054     }
00055
00056     const std::string& getNetworkName() const {
00057         return _networkName;
00058     }
00059
00060     double getStartTime() const {
00061         return _startTime;
00062     }
00063
00064     double getStopTime() const {
00065         return _stopTime;
00066     }
00067
00068     double getStepSize() const {
00069         return _stepSize;
00070     }
00071
00072     const std::vector<std::string>& getSpeciesNames() const {
00073         return _speciesNames;
00074     }
00075
00076     const std::vector<BioSimulationSample>& getSamples() const {
00077         return _samples;
00078     }
00079
00080     bool hasSpecies(const std::string& speciesName) const {
00081         return std::find(_speciesNames.begin(), _speciesNames.end(), speciesName) !=
      _speciesNames.end();
00082     }
00083
00084     bool toDataset(const std::string& speciesName, SimulationResultsDataset* dataset,
00085                    std::string* errorMessage = nullptr) const {
00086         if (dataset == nullptr) {
00087             if (errorMessage != nullptr) {
00088                 *errorMessage = "Invalid dataset output parameter.";
00089             }
00090             return false;
00091         }
00092         if (!hasSpecies(speciesName)) {
00093             if (errorMessage != nullptr) {
00094                 *errorMessage = "BioSimulationResult does not contain species \"" + speciesName +
      "\".";
00095             }
00096             return false;
00097         }
00098
00099         SimulationResultsDataset converted;
00100         converted.datasetName = _networkName + "." + speciesName;
00101         converted.randomVariableName = speciesName;
00102         converted.description = "Bio simulation species time-course dataset.";
00103         converted.variableType = "Continuous numeric";
00104         converted.source = _networkName;
00105         converted.sourceDescription = _networkName;
```

```
00106         converted.expression = speciesName;
00107         converted.expressionName = speciesName;
00108         converted.formatKind = SimulationResultsDatasetFormat::GuiTabular;
00109         converted.recordFile = false;
00110         converted.timeDependent = true;
00111         for (unsigned int i = 0; i < _samples.size(); ++i) {
00112             bool found = false;
00113             const double value = valueForSpecies(_samples[i], speciesName, &found);
00114             if (!found) {
00115                 if (errorMessage != nullptr) {
00116                     *errorMessage = "BioSimulationResult sample is missing species \"" + speciesName +
      "\".";
00117                 }
00118                 return false;
00119             }
00120             SimulationResultsObservation observation;
00121             observation.replication = 1;
00122             observation.time = _samples[i].time;
00123             observation.hasTime = true;
00124             observation.value = value;
00125             observation.sourceLine = i + 1;
00126             converted.observations.push_back(observation);
00127         }
00128 
00129         *dataset = converted;
00130         return true;
00131     }
00132 
00133     std::vector<SimulationResultsDataset> toDatasets() const {
00134         std::vector<SimulationResultsDataset> datasets;
00135         for (const std::string& speciesName : _speciesNames) {
00136             SimulationResultsDataset dataset;
00137             if (toDataset(speciesName, &dataset)) {
00138                 datasets.push_back(dataset);
00139             }
00140         }
00141         return datasets;
00142     }
00143 
00144 private:
00145     double valueForSpecies(const BioSimulationSample& sample, const std::string& speciesName, bool*
      found) const {
00146         for (const BioSimulationSpeciesAmount& amount : sample.species) {
00147             if (amount.speciesName == speciesName) {
00148                 if (found != nullptr) {
00149                     *found = true;
00150                 }
00151                 return amount.amount;
00152             }
00153         }
00154         if (found != nullptr) {
00155             *found = false;
00156         }
00157         return 0.0;
00158     }
00159 
00160 private:
00161     std::string _networkName = "";
00162     double _startTime = 0.0;
00163     double _stopTime = 0.0;
00164     double _stepSize = 0.0;
00165     std::vector<std::string> _speciesNames;
00166     std::vector<BioSimulationSample> _samples;
00167 };
00168 
00169 #endif /* BIOSIMULATIONRESULT_H */
```

18.555 GlpkFluxBalanceSolver.h

```
00001 #ifndef GLPKFLUXBALANCESOLVER_H
00002 #define GLPKFLUXBALANCESOLVER_H
00003 
00004 #include "tools/Biochemical/MetabolicFluxBalanceSolver.h"
00005 
00006 #ifdef GENESYS_HAVE_GLPK
00007 
00008 #include <glpk.h>
00009 #include <cstring>
00010 #include <limits>
00011 
00039 class GlpkFluxBalanceSolver {
00040 public:
00041     static MetabolicFluxBalanceSolver::Solution solve(const MetabolicFluxBalanceSolver::Problem&
      problem) {
00042         MetabolicFluxBalanceSolver::Solution solution;
```

```
00043         const std::size_t nCols = problem.objective.size();
00044         const std::size_t nRows = problem.stoichiometry.size();
00045 
00046         if (nCols == 0u) {
00047             solution.errorMessage = "Empty objective vector.";
00048             return solution;
00049         }
00050         if (problem.lowerBounds.size() != nCols || problem.upperBounds.size() != nCols) {
00051             solution.errorMessage = "Invalid flux bound vector dimensions.";
00052             return solution;
00053         }
00054 
00055         glp_prob* lp = glp_create_prob();
00056         glp_set_prob_name(lp, "fba");
00057         glp_set_obj_dir(lp, problem.maximize ? GLP_MAX : GLP_MIN);
00058 
00059         // Add constraint rows (one per metabolite: S_i * v = 0)
00060         if (nRows > 0u) {
00061             glp_add_rows(lp, static_cast<int>(nRows));
00062             for (std::size_t i = 0; i < nRows; ++i) {
00063                 glp_set_row_bnds(lp, static_cast<int>(i) + 1, GLP_FX, 0.0, 0.0);
00064             }
00065         }
00066 
00067         // Add flux variables (columns) with bounds and objective coefficients
00068         glp_add_cols(lp, static_cast<int>(nCols));
00069         for (std::size_t j = 0; j < nCols; ++j) {
00070             const int col = static_cast<int>(j) + 1;
00071             const double lb = problem.lowerBounds[j];
00072             const double ub = problem.upperBounds[j];
00073             const double kInf = std::numeric_limits<double>::infinity();
00074             if (lb == ub) {
00075                 glp_set_col_bnds(lp, col, GLP_FX, lb, ub);
00076             } else if (lb <= -kInf && ub >= kInf) {
00077                 glp_set_col_bnds(lp, col, GLP_FR, 0.0, 0.0);
00078             } else if (lb <= -kInf) {
00079                 glp_set_col_bnds(lp, col, GLP_UP, 0.0, ub);
00080             } else if (ub >= kInf) {
00081                 glp_set_col_bnds(lp, col, GLP_LO, lb, 0.0);
00082             } else {
00083                 glp_set_col_bnds(lp, col, GLP_DB, lb, ub);
00084             }
00085             glp_set_obj_coef(lp, col, problem.objective[j]);
00086         }
00087 
00088         // Load stoichiometry matrix as sparse triplets (GLPK uses 1-based indices)
00089         // Count non-zeros first
00090         std::size_t nnz = 0u;
00091         for (const auto& row : problem.stoichiometry) {
00092             for (double val : row) {
00093                 if (val != 0.0) ++nnz;
00094             }
00095         }
00096 
00097         if (nnz > 0u && nRows > 0u) {
00098             std::vector<int> ia(nnz + 1u);
00099             std::vector<int> ja(nnz + 1u);
00100             std::vector<double> ar(nnz + 1u);
00101             std::size_t k = 1u;
00102             for (std::size_t i = 0; i < nRows; ++i) {
00103                 for (std::size_t j = 0; j < nCols && j < problem.stoichiometry[i].size(); ++j) {
00104                     const double val = problem.stoichiometry[i][j];
00105                     if (val != 0.0) {
00106                         ia[k] = static_cast<int>(i) + 1;
00107                         ja[k] = static_cast<int>(j) + 1;
00108                         ar[k] = val;
00109                         ++k;
00110                     }
00111                 }
00112             }
00113             glp_load_matrix(lp, static_cast<int>(nnz), ia.data(), ja.data(), ar.data());
00114         }
00115 
00116         // Solve with simplex, messages suppressed
00117         glp_smcp parm;
00118         glp_init_smcp(&parm);
00119         parm.msg_lev = GLP_MSG_OFF;
00120         const int status = glp_simplex(lp, &parm);
00121 
00122         if (status == 0 && glp_get_status(lp) == GLP_OPT) {
00123             solution.feasible = true;
00124             solution.objectiveValue = glp_get_obj_val(lp);
00125             solution.fluxes.resize(nCols);
00126             for (std::size_t j = 0; j < nCols; ++j) {
00127                 solution.fluxes[j] = glp_get_col_prim(lp, static_cast<int>(j) + 1);
00128             }
00129         } else {
```

```
00130             const int glpStatus = glp_get_status(lp);
00131             if (glpStatus == GLP_INFEAS || glpStatus == GLP_NOFEAS) {
00132                 solution.errorMessage = "GLPK: problem is infeasible.";
00133             } else if (glpStatus == GLP_UNBND) {
00134                 solution.errorMessage = "GLPK: problem is unbounded.";
00135             } else {
00136                 solution.errorMessage = "GLPK: simplex failed (status=" + std::to_string(status) + ",
      glpStatus=" + std::to_string(glpStatus) + ").";
00137             }
00138         }
00139
00140         glp_delete_prob(lp);
00141         return solution;
00142     }
00143 };
00144
00145 #endif /* GENESYS_HAVE_GLPK */
00146
00147 #endif /* GLPKFLUXBALANCESOLVER_H */
```

18.556 MassActionOdeSystem.h

```
00001 #ifndef MASSACTIONODESYSTEM_H
00002 #define MASSACTIONODESYSTEM_H
00003
00004 #include <algorithm>
00005 #include <cmath>
00006 #include <string>
00007 #include <utility>
00008 #include <vector>
00009
00010 #include "BioKineticLawExpression.h"
00011 #include "tools/Continuous/OdeSystem_if.h"
00012
00016 class MassActionOdeSystem : public OdeSystem_if {
00017 public:
00018     struct Species {
00019         std::string name;
00020         double initialAmount = 0.0;
00021         bool boundaryCondition = false;
00022         bool constant = false;
00023     };
00024
00025     struct StoichiometricTerm {
00026         unsigned int speciesIndex = 0;
00027         double stoichiometry = 1.0;
00028     };
00029
00030     struct Reaction {
00031         std::string name;
00032         double rateConstant = 0.0;
00033         std::vector<StoichiometricTerm> reactants;
00034         std::vector<StoichiometricTerm> products;
00035         std::string kineticLawExpression;
00036         std::vector<std::pair<std::string, double» parameters;
00037         std::string reverseKineticLawExpression;
00038         double reverseRateConstant = 0.0;
00039         bool reversible = false;
00040     };
00041
00042     MassActionOdeSystem() = default;
00043
00044     MassActionOdeSystem(std::vector<Species> species, std::vector<Reaction> reactions)
00045             : _species(std::move(species)), _reactions(std::move(reactions)) {
00046     }
00047
00048     unsigned int dimension() const override {
00049         return static_cast<unsigned int>(_species.size());
00050     }
00051
00052     void evaluate(double, const double* y, double* dydt) const override {
00053         const unsigned int n = dimension();
00054         if (y == nullptr || dydt == nullptr) {
00055             return;
00056         }
00057
00058         for (unsigned int i = 0; i < n; ++i) {
00059             dydt[i] = 0.0;
00060         }
00061
00062         for (const Reaction& reaction : _reactions) {
00063             const double forwardRate = evaluateRate(reaction, reaction.kineticLawExpression,
      reaction.rateConstant, reaction.reactants, y);
00064             const double reverseRate = reaction.reversible ? evaluateRate(reaction,
      reaction.reverseKineticLawExpression, reaction.reverseRateConstant, reaction.products, y) : 0.0;
```

```
00065
00066             for (const StoichiometricTerm& reactant : reaction.reactants) {
00067                 applyDerivative(reactant.speciesIndex, reactant.stoichiometry * (reverseRate -
      forwardRate), dydt);
00068             }
00069             for (const StoichiometricTerm& product : reaction.products) {
00070                 applyDerivative(product.speciesIndex, product.stoichiometry * (forwardRate -
      reverseRate), dydt);
00071             }
00072         }
00073     }
00074
00075     const std::vector<Species>& species() const {
00076         return _species;
00077     }
00078
00079     const std::vector<Reaction>& reactions() const {
00080         return _reactions;
00081     }
00082
00083     double evaluateForwardRate(unsigned int reactionIndex, const double* y) const {
00084         if (reactionIndex >= _reactions.size() || y == nullptr) {
00085             return 0.0;
00086         }
00087         const Reaction& reaction = _reactions[reactionIndex];
00088         return evaluateRate(reaction, reaction.kineticLawExpression, reaction.rateConstant,
      reaction.reactants, y);
00089     }
00090
00091     double evaluateReverseRate(unsigned int reactionIndex, const double* y) const {
00092         if (reactionIndex >= _reactions.size() || y == nullptr) {
00093             return 0.0;
00094         }
00095         const Reaction& reaction = _reactions[reactionIndex];
00096         if (!reaction.reversible) {
00097             return 0.0;
00098         }
00099         return evaluateRate(reaction, reaction.reverseKineticLawExpression,
      reaction.reverseRateConstant, reaction.products, y);
00100     }
00101
00102     double evaluateNetRate(unsigned int reactionIndex, const double* y) const {
00103         return evaluateForwardRate(reactionIndex, y) - evaluateReverseRate(reactionIndex, y);
00104     }
00105
00106 private:
00107     double evaluateRate(const Reaction& reaction, const std::string& kineticLawExpression, double
      rateConstant, const std::vector<StoichiometricTerm>& massActionTerms, const double* y) const {
00108         if (kineticLawExpression.empty()) {
00109             return evaluateMassActionRate(rateConstant, massActionTerms, y);
00110         }
00111
00112         BioKineticLawExpression expression;
00113         double rate = 0.0;
00114         std::string errorMessage;
00115         const bool ok = expression.evaluate(kineticLawExpression,
00116                 [this, &reaction, y](const std::string& name, double& value) {
00117                     for (unsigned int i = 0; i < _species.size(); ++i) {
00118                         if (_species[i].name == name) {
00119                             value = std::max(0.0, y[i]);
00120                             return true;
00121                         }
00122                     }
00123                     for (const auto& parameter : reaction.parameters) {
00124                         if (parameter.first == name) {
00125                             value = parameter.second;
00126                             return true;
00127                         }
00128                     }
00129                     return false;
00130                 },
00131                 rate, errorMessage);
00132         return ok ? std::max(0.0, rate) : 0.0;
00133     }
00134
00135     double evaluateMassActionRate(double rateConstant, const std::vector<StoichiometricTerm>& terms,
      const double* y) const {
00136         double rate = rateConstant;
00137         for (const StoichiometricTerm& term : terms) {
00138             if (term.speciesIndex >= _species.size()) {
00139                 return 0.0;
00140             }
00141             const double amount = std::max(0.0, y[term.speciesIndex]);
00142             rate *= std::pow(amount, term.stoichiometry);
00143         }
00144         return std::max(0.0, rate);
00145     }
```

```
00146
00147     void applyDerivative(unsigned int speciesIndex, double value, double* dydt) const {
00148         if (speciesIndex >= _species.size() || _species[speciesIndex].boundaryCondition ||
      _species[speciesIndex].constant) {
00149             return;
00150         }
00151         dydt[speciesIndex] += value;
00152     }
00153
00154 private:
00155     std::vector<Species> _species;
00156     std::vector<Reaction> _reactions;
00157 };
00158
00159 #endif /* MASSACTIONODESYSTEM_H */
```

18.557 MetabolicFluxBalanceSolver.h

```
00001 #ifndef METABOLICFLUXBALANCESOLVER_H
00002 #define METABOLICFLUXBALANCESOLVER_H
00003
00004 #include <algorithm>
00005 #include <cmath>
00006 #include <limits>
00007 #include <string>
00008 #include <vector>
00009
00023 class MetabolicFluxBalanceSolver {
00024 public:
00025     struct Problem {
00026         std::vector<std::vector<double» stoichiometry;
00027         std::vector<double> lowerBounds;
00028         std::vector<double> upperBounds;
00029         std::vector<double> objective;
00030         bool maximize = true;
00031     };
00032
00033     struct Solution {
00034         bool feasible = false;
00035         double objectiveValue = 0.0;
00036         std::vector<double> fluxes;
00037         std::string errorMessage;
00038     };
00039
00040 public:
00041     static Solution solve(const Problem& problem) {
00042         Solution solution;
00043         const std::size_t variableCount = problem.objective.size();
00044         if (variableCount == 0u) {
00045             solution.errorMessage = "Empty objective vector.";
00046             return solution;
00047         }
00048         if (problem.lowerBounds.size() != variableCount || problem.upperBounds.size() !=
      variableCount) {
00049             solution.errorMessage = "Invalid flux bound vector dimensions.";
00050             return solution;
00051         }
00052         for (const std::vector<double>& row : problem.stoichiometry) {
00053             if (row.size() != variableCount) {
00054                 solution.errorMessage = "Invalid stoichiometry matrix dimensions.";
00055                 return solution;
00056             }
00057         }
00058         for (std::size_t i = 0; i < variableCount; ++i) {
00059             if (problem.upperBounds[i] < problem.lowerBounds[i]) {
00060                 solution.errorMessage = "Upper bound smaller than lower bound.";
00061                 return solution;
00062             }
00063         }
00064
00065         std::vector<std::vector<double» constraints = problem.stoichiometry;
00066         _removeRedundantRows(&constraints);
00067         const std::size_t rank = constraints.size();
00068         if (rank > variableCount) {
00069             solution.errorMessage = "Stoichiometry rank exceeds variable count.";
00070             return solution;
00071         }
00072
00073         std::vector<std::size_t> basis;
00074         for (std::size_t i = 0; i < rank; ++i) {
00075             basis.push_back(i);
00076         }
00077
00078         bool found = false;
00079         double bestObjective = problem.maximize ? -std::numeric_limits<double>::infinity()
```

```
00080                                                 : std::numeric_limits<double>::infinity();
00081         std::vector<double> bestFluxes(variableCount, 0.0);
00082
00083         do {
00084             if (!_isIndependentBasis(constraints, basis)) {
00085                 continue;
00086             }
00087
00088             std::vector<std::size_t> nonBasis;
00089             for (std::size_t i = 0; i < variableCount; ++i) {
00090                 if (std::find(basis.begin(), basis.end(), i) == basis.end()) {
00091                     nonBasis.push_back(i);
00092                 }
00093             }
00094
00095             const std::size_t combinations = std::size_t{1} << nonBasis.size();
00096             for (std::size_t mask = 0; mask < combinations; ++mask) {
00097                 std::vector<double> fluxes(variableCount, 0.0);
00098                 for (std::size_t k = 0; k < nonBasis.size(); ++k) {
00099                     const std::size_t variableIndex = nonBasis[k];
00100                     const bool chooseUpper = ((mask >> k) & 1u) != 0u;
00101                     fluxes[variableIndex] = chooseUpper ? problem.upperBounds[variableIndex] :
      problem.lowerBounds[variableIndex];
00102                 }
00103
00104                 std::vector<std::vector<double>> basisMatrix(rank, std::vector<double>(rank, 0.0));
00105                 std::vector<double> rhs(rank, 0.0);
00106                 for (std::size_t row = 0; row < rank; ++row) {
00107                     for (std::size_t col = 0; col < rank; ++col) {
00108                         basisMatrix[row][col] = constraints[row][basis[col]];
00109                     }
00110                     for (std::size_t variableIndex : nonBasis) {
00111                         rhs[row] -= constraints[row][variableIndex] * fluxes[variableIndex];
00112                     }
00113                 }
00114
00115                 std::vector<double> basisValues;
00116                 if (!_solveLinearSystem(basisMatrix, rhs, &basisValues)) {
00117                     continue;
00118                 }
00119                 for (std::size_t col = 0; col < rank; ++col) {
00120                     fluxes[basis[col]] = basisValues[col];
00121                 }
00122                 if (!_withinBounds(fluxes, problem.lowerBounds, problem.upperBounds)) {
00123                     continue;
00124                 }
00125                 if (!_satisfiesConstraints(constraints, fluxes)) {
00126                     continue;
00127                 }
00128
00129                 const double objectiveValue = _dot(problem.objective, fluxes);
00130                 if (!found ||
00131                     (problem.maximize && objectiveValue > bestObjective + 1e-9) ||
00132                     (!problem.maximize && objectiveValue < bestObjective - 1e-9)) {
00133                     found = true;
00134                     bestObjective = objectiveValue;
00135                     bestFluxes = fluxes;
00136                 }
00137             }
00138         } while (_nextBasis(variableCount, rank, &basis));
00139
00140         if (!found) {
00141             solution.errorMessage = "No feasible bounded flux solution found.";
00142             return solution;
00143         }
00144
00145         solution.feasible = true;
00146         solution.objectiveValue = bestObjective;
00147         solution.fluxes = std::move(bestFluxes);
00148         return solution;
00149     }
00150
00151 private:
00152     static void _removeRedundantRows(std::vector<std::vector<double>>* matrix) {
00153         if (matrix == nullptr) {
00154             return;
00155         }
00156         std::vector<std::vector<double>> filtered;
00157         for (const std::vector<double>& row : *matrix) {
00158             bool allZero = true;
00159             for (double value : row) {
00160                 if (std::fabs(value) > 1e-9) {
00161                     allZero = false;
00162                     break;
00163                 }
00164             }
00165             if (!allZero) {
```

```
00166                 filtered.push_back(row);
00167             }
00168         }
00169 
00170         std::vector<std::vector<double» independent;
00171         for (const std::vector<double>& row : filtered) {
00172             std::vector<std::vector<double» candidate = independent;
00173             candidate.push_back(row);
00174             if (_matrixRank(candidate) > independent.size()) {
00175                 independent.push_back(row);
00176             }
00177         }
00178         *matrix = independent;
00179     }
00180 
00181     static bool _isIndependentBasis(const std::vector<std::vector<double»& constraints,
00182                                     const std::vector<std::size_t>& basis) {
00183         const std::size_t rank = constraints.size();
00184         if (basis.size() != rank) {
00185             return false;
00186         }
00187         if (rank == 0u) {
00188             return true;
00189         }
00190         std::vector<std::vector<double» basisMatrix(rank, std::vector<double>(rank, 0.0));
00191         for (std::size_t row = 0; row < rank; ++row) {
00192             for (std::size_t col = 0; col < rank; ++col) {
00193                 basisMatrix[row][col] = constraints[row][basis[col]];
00194             }
00195         }
00196         return _matrixRank(basisMatrix) == rank;
00197     }
00198 
00199     static bool _solveLinearSystem(std::vector<std::vector<double» matrix,
00200                                    std::vector<double> rhs,
00201                                    std::vector<double>* solution) {
00202         if (solution == nullptr) {
00203             return false;
00204         }
00205         const std::size_t n = matrix.size();
00206         if (rhs.size() != n) {
00207             return false;
00208         }
00209         for (const std::vector<double>& row : matrix) {
00210             if (row.size() != n) {
00211                 return false;
00212             }
00213         }
00214 
00215         for (std::size_t pivot = 0; pivot < n; ++pivot) {
00216             std::size_t pivotRow = pivot;
00217             double bestAbs = std::fabs(matrix[pivot][pivot]);
00218             for (std::size_t row = pivot + 1; row < n; ++row) {
00219                 const double candidate = std::fabs(matrix[row][pivot]);
00220                 if (candidate > bestAbs) {
00221                     bestAbs = candidate;
00222                     pivotRow = row;
00223                 }
00224             }
00225             if (bestAbs <= 1e-9) {
00226                 return false;
00227             }
00228             if (pivotRow != pivot) {
00229                 std::swap(matrix[pivotRow], matrix[pivot]);
00230                 std::swap(rhs[pivotRow], rhs[pivot]);
00231             }
00232 
00233             const double pivotValue = matrix[pivot][pivot];
00234             for (std::size_t col = pivot; col < n; ++col) {
00235                 matrix[pivot][col] /= pivotValue;
00236             }
00237             rhs[pivot] /= pivotValue;
00238 
00239             for (std::size_t row = 0; row < n; ++row) {
00240                 if (row == pivot) {
00241                     continue;
00242                 }
00243                 const double factor = matrix[row][pivot];
00244                 if (std::fabs(factor) <= 1e-12) {
00245                     continue;
00246                 }
00247                 for (std::size_t col = pivot; col < n; ++col) {
00248                     matrix[row][col] -= factor * matrix[pivot][col];
00249                 }
00250                 rhs[row] -= factor * rhs[pivot];
00251             }
00252         }
```

```
00253
00254         *solution = rhs;
00255         return true;
00256     }
00257
00258     static std::size_t _matrixRank(std::vector<std::vector<double» matrix) {
00259         if (matrix.empty()) {
00260             return 0u;
00261         }
00262         const std::size_t rows = matrix.size();
00263         const std::size_t cols = matrix.front().size();
00264         std::size_t rank = 0u;
00265         std::size_t pivotRow = 0u;
00266         for (std::size_t col = 0; col < cols && pivotRow < rows; ++col) {
00267             std::size_t bestRow = pivotRow;
00268             double bestAbs = std::fabs(matrix[pivotRow][col]);
00269             for (std::size_t row = pivotRow + 1; row < rows; ++row) {
00270                 const double candidate = std::fabs(matrix[row][col]);
00271                 if (candidate > bestAbs) {
00272                     bestAbs = candidate;
00273                     bestRow = row;
00274                 }
00275             }
00276             if (bestAbs <= 1e-9) {
00277                 continue;
00278             }
00279             if (bestRow != pivotRow) {
00280                 std::swap(matrix[bestRow], matrix[pivotRow]);
00281             }
00282             const double pivotValue = matrix[pivotRow][col];
00283             for (std::size_t activeCol = col; activeCol < cols; ++activeCol) {
00284                 matrix[pivotRow][activeCol] /= pivotValue;
00285             }
00286             for (std::size_t row = pivotRow + 1; row < rows; ++row) {
00287                 const double factor = matrix[row][col];
00288                 if (std::fabs(factor) <= 1e-12) {
00289                     continue;
00290                 }
00291                 for (std::size_t activeCol = col; activeCol < cols; ++activeCol) {
00292                     matrix[row][activeCol] -= factor * matrix[pivotRow][activeCol];
00293                 }
00294             }
00295             ++pivotRow;
00296             ++rank;
00297         }
00298         return rank;
00299     }
00300
00301     static bool _withinBounds(const std::vector<double>& fluxes,
00302                               const std::vector<double>& lowerBounds,
00303                               const std::vector<double>& upperBounds) {
00304         for (std::size_t i = 0; i < fluxes.size(); ++i) {
00305             if (fluxes[i] < lowerBounds[i] - 1e-7 || fluxes[i] > upperBounds[i] + 1e-7) {
00306                 return false;
00307             }
00308         }
00309         return true;
00310     }
00311
00312     static bool _satisfiesConstraints(const std::vector<std::vector<double»& constraints,
00313                                       const std::vector<double>& fluxes) {
00314         for (const std::vector<double>& row : constraints) {
00315             double total = 0.0;
00316             for (std::size_t i = 0; i < row.size(); ++i) {
00317                 total += row[i] * fluxes[i];
00318             }
00319             if (std::fabs(total) > 1e-6) {
00320                 return false;
00321             }
00322         }
00323         return true;
00324     }
00325
00326     static double _dot(const std::vector<double>& a, const std::vector<double>& b) {
00327         double value = 0.0;
00328         for (std::size_t i = 0; i < a.size(); ++i) {
00329             value += a[i] * b[i];
00330         }
00331         return value;
00332     }
00333
00334     static bool _nextBasis(std::size_t variableCount,
00335                            std::size_t basisSize,
00336                            std::vector<std::size_t>* basis) {
00337         if (basis == nullptr) {
00338             return false;
00339         }
```

```
00340         if (basisSize == 0u) {
00341             return false;
00342         }
00343         for (std::size_t i = basisSize; i-- > 0;) {
00344             const std::size_t maxValue = variableCount - basisSize + i;
00345             if ((*basis)[i] < maxValue) {
00346                 ++(*basis)[i];
00347                 for (std::size_t j = i + 1u; j < basisSize; ++j) {
00348                     (*basis)[j] = (*basis)[j - 1u] + 1u;
00349                 }
00350                 return true;
00351             }
00352         }
00353         return false;
00354     }
00355 };
00356
00357 #endif /* METABOLICFLUXBALANCESOLVER_H */
```

18.558 WholeCellParameterReader.h

```
00001 #ifndef WHOLECELLPARAMETERREADER_H
00002 #define WHOLECELLPARAMETERREADER_H
00003
00004 #include <cctype>
00005 #include <fstream>
00006 #include <map>
00007 #include <sstream>
00008 #include <string>
00009
00030 class WholeCellParameterReader {
00031 public:
00032     static std::map<std::string, double> read(const std::string& filePath) {
00033         std::map<std::string, double> result;
00034         std::ifstream file(filePath);
00035         if (!file.is_open()) return result;
00036         std::ostringstream oss;
00037         oss << file.rdbuf();
00038         const std::string json = oss.str();
00039         _parseTopLevel(json, result);
00040         return result;
00041     }
00042
00043 private:
00044     static void _skipWhitespace(const std::string& s, std::size_t& pos) {
00045         while (pos < s.size() && std::isspace(static_cast<unsigned char>(s[pos]))) ++pos;
00046     }
00047
00048     static bool _parseString(const std::string& s, std::size_t& pos, std::string& out) {
00049         _skipWhitespace(s, pos);
00050         if (pos >= s.size() || s[pos] != '"') return false;
00051         ++pos;
00052         out.clear();
00053         while (pos < s.size() && s[pos] != '"') {
00054             if (s[pos] == '\\') {
00055                 ++pos;
00056                 if (pos < s.size()) { out += s[pos]; ++pos; }
00057             } else {
00058                 out += s[pos++];
00059             }
00060         }
00061         if (pos < s.size()) ++pos;
00062         return true;
00063     }
00064
00065     static bool _parseDouble(const std::string& s, std::size_t& pos, double& out) {
00066         _skipWhitespace(s, pos);
00067         const std::size_t start = pos;
00068         if (pos < s.size() && (s[pos] == '-' || s[pos] == '+')) ++pos;
00069         bool hasDigit = false;
00070         while (pos < s.size() && std::isdigit(static_cast<unsigned char>(s[pos]))) { ++pos; hasDigit =
      true; }
00071         if (pos < s.size() && s[pos] == '.') {
00072             ++pos;
00073             while (pos < s.size() && std::isdigit(static_cast<unsigned char>(s[pos]))) { ++pos;
      hasDigit = true; }
00074         }
00075         if (pos < s.size() && (s[pos] == 'e' || s[pos] == 'E')) {
00076             ++pos;
00077             if (pos < s.size() && (s[pos] == '+' || s[pos] == '-')) ++pos;
00078             while (pos < s.size() && std::isdigit(static_cast<unsigned char>(s[pos]))) ++pos;
00079         }
00080         if (!hasDigit) return false;
00081         try { out = std::stod(s.substr(start, pos - start)); } catch (...) { return false; }
00082         return true;
```

```
00083     }
00084
00085     static void _skipValue(const std::string& s, std::size_t& pos) {
00086         _skipWhitespace(s, pos);
00087         if (pos >= s.size()) return;
00088         if (s[pos] == '"') {
00089             std::string dummy; _parseString(s, pos, dummy);
00090         } else if (s[pos] == '{') {
00091             int depth = 1; ++pos;
00092             while (pos < s.size() && depth > 0) {
00093                 if (s[pos] == '{') ++depth;
00094                 else if (s[pos] == '}') --depth;
00095                 else if (s[pos] == '"') { std::string d; _parseString(s, pos, d); continue; }
00096                 ++pos;
00097             }
00098         } else if (s[pos] == '[') {
00099             int depth = 1; ++pos;
00100             while (pos < s.size() && depth > 0) {
00101                 if (s[pos] == '[') ++depth;
00102                 else if (s[pos] == ']') --depth;
00103                 else if (s[pos] == '"') { std::string d; _parseString(s, pos, d); continue; }
00104                 ++pos;
00105             }
00106         } else {
00107             while (pos < s.size() && s[pos] != ',' && s[pos] != '}' && s[pos] != ']') ++pos;
00108         }
00109     }
00110
00111     // Parse {"key": value, ...} and call handler for each key-value pair
00112     static void _parseObject(const std::string& s, std::size_t& pos,
00113                              const std::string& prefix,
00114                              std::map<std::string, double>& out) {
00115         _skipWhitespace(s, pos);
00116         if (pos >= s.size() || s[pos] != '{') return;
00117         ++pos;
00118         while (pos < s.size()) {
00119             _skipWhitespace(s, pos);
00120             if (pos >= s.size() || s[pos] == '}') { if (pos < s.size()) ++pos; break; }
00121             if (s[pos] == ',') { ++pos; continue; }
00122             std::string key;
00123             if (!_parseString(s, pos, key)) { ++pos; continue; }
00124             _skipWhitespace(s, pos);
00125             if (pos < s.size() && s[pos] == ':') ++pos;
00126             _skipWhitespace(s, pos);
00127             const std::string fullKey = prefix.empty() ? key : prefix + "." + key;
00128             if (pos < s.size() && s[pos] == '{') {
00129                 // nested object -- recurse one level
00130                 _parseObject(s, pos, fullKey, out);
00131             } else if (pos < s.size() && s[pos] != '[' && s[pos] != '"') {
00132                 // scalar value -- try as double
00133                 double val = 0.0;
00134                 const std::size_t before = pos;
00135                 if (_parseDouble(s, pos, val)) {
00136                     out[fullKey] = val;
00137                 } else {
00138                     pos = before;
00139                     _skipValue(s, pos);
00140                 }
00141             } else {
00142                 // array or string -- skip
00143                 _skipValue(s, pos);
00144             }
00145         }
00146     }
00147
00148     static void _parseTopLevel(const std::string& json, std::map<std::string, double>& out) {
00149         std::size_t pos = 0;
00150         _skipWhitespace(json, pos);
00151         if (pos >= json.size() || json[pos] != '{') return;
00152         ++pos;
00153         while (pos < json.size()) {
00154             _skipWhitespace(json, pos);
00155             if (json[pos] == '}') break;
00156             if (json[pos] == ',') { ++pos; continue; }
00157             std::string section;
00158             if (!_parseString(json, pos, section)) { ++pos; continue; }
00159             _skipWhitespace(json, pos);
00160             if (pos < json.size() && json[pos] == ':') ++pos;
00161             _parseObject(json, pos, section, out);
00162         }
00163     }
00164 };
00165
00166 #endif /* WHOLECELLPARAMETERREADER_H */
```

18.559 DiffusionMethodOfLinesSystem.h

```
00001 #ifndef DIFFUSIONMETHODOFLINESSYSTEM_H
00002 #define DIFFUSIONMETHODOFLINESSYSTEM_H
00003
00004 #include <cmath>
00005 #include <cstddef>
00006 #include <limits>
00007 #include <vector>
00008
00009 #include "OdeSystem_if.h"
00010
00011 class DiffusionMethodOfLinesSystem : public OdeSystem_if {
00012 public:
00013     static constexpr double kPi = 3.14159265358979323846;
00014
00015     enum class Boundary {
00016         Dirichlet,
00017         Neumann
00018     };
00019
00020     DiffusionMethodOfLinesSystem() = default;
00021
00022     DiffusionMethodOfLinesSystem(std::vector<unsigned int> shape,
00023                                  std::vector<double> spacing,
00024                                  double diffusionCoefficient,
00025                                  Boundary boundary)
00026         : _shape(std::move(shape)),
00027           _spacing(std::move(spacing)),
00028           _D(diffusionCoefficient),
00029           _boundary(boundary) {
00030         rebuild();
00031     }
00032
00033     unsigned int dimension() const override { return _totalNodes; }
00034
00035     void evaluate(double, const double* y, double* dydt) const override {
00036         if (!_valid || y == nullptr || dydt == nullptr) {
00037             return;
00038         }
00039         const unsigned int N = static_cast<unsigned int>(_shape.size());
00040         for (unsigned int node = 0; node < _totalNodes; ++node) {
00041             if (_boundary == Boundary::Dirichlet && isBoundaryNode(node)) {
00042                 dydt[node] = 0.0;
00043                 continue;
00044             }
00045             double laplacian = 0.0;
00046             for (unsigned int d = 0; d < N; ++d) {
00047                 const unsigned int m = coordinate(node, d);
00048                 const unsigned int last = _shape[d] - 1u;
00049                 const std::size_t s = _stride[d];
00050                 double secondDifference;
00051                 if (m == 0u) {
00052                     secondDifference = 2.0 * (y[node + s] - y[node]);
00053                 } else if (m == last) {
00054                     secondDifference = 2.0 * (y[node - s] - y[node]);
00055                 } else {
00056                     secondDifference = y[node + s] - 2.0 * y[node] + y[node - s];
00057                 }
00058                 laplacian += secondDifference * _invH2[d];
00059             }
00060             dydt[node] = _D * laplacian;
00061         }
00062     }
00063
00064     bool isValid() const { return _valid; }
00065     unsigned int totalNodes() const { return _totalNodes; }
00066     const std::vector<unsigned int>& shape() const { return _shape; }
00067     const std::vector<double>& spacing() const { return _spacing; }
00068     double diffusionCoefficient() const { return _D; }
00069     Boundary boundary() const { return _boundary; }
00070
00071     unsigned int coordinate(unsigned int node, unsigned int d) const {
00072         return static_cast<unsigned int>((node / _stride[d]) % _shape[d]);
00073     }
00074
00075     unsigned int nodeIndex(const std::vector<unsigned int>& multiIndex) const {
00076         std::size_t idx = 0;
00077         for (std::size_t d = 0; d < _shape.size(); ++d) {
00078             idx += static_cast<std::size_t>(multiIndex[d]) * _stride[d];
00079         }
00080         return static_cast<unsigned int>(idx);
00081     }
00082
00083     bool isBoundaryNode(unsigned int node) const {
00084         for (unsigned int d = 0; d < _shape.size(); ++d) {
00085             const unsigned int m = coordinate(node, d);
```

```
00086             if (m == 0u || m == _shape[d] - 1u) {
00087                 return true;
00088             }
00089         }
00090         return false;
00091     }
00092
00093     void fillSineModes(const std::vector<unsigned int>& modes, double amplitude, double* y) const {
00094         const unsigned int N = static_cast<unsigned int>(_shape.size());
00095         for (unsigned int node = 0; node < _totalNodes; ++node) {
00096             double value = amplitude;
00097             for (unsigned int d = 0; d < N; ++d) {
00098                 const unsigned int m = coordinate(node, d);
00099                 const double L = (_shape[d] - 1u) * _spacing[d];
00100                 const double x = m * _spacing[d];
00101                 value *= std::sin(modes[d] * kPi * x / L);
00102             }
00103             y[node] = value;
00104         }
00105     }
00106
00107     void fillGaussian(double amplitude, double sigma, double* y) const {
00108         const unsigned int N = static_cast<unsigned int>(_shape.size());
00109         const double twoSigma2 = 2.0 * sigma * sigma;
00110         for (unsigned int node = 0; node < _totalNodes; ++node) {
00111             double r2 = 0.0;
00112             for (unsigned int d = 0; d < N; ++d) {
00113                 const unsigned int m = coordinate(node, d);
00114                 const double L = (_shape[d] - 1u) * _spacing[d];
00115                 const double x = m * _spacing[d];
00116                 const double dx = x - 0.5 * L;
00117                 r2 += dx * dx;
00118             }
00119             y[node] = amplitude * std::exp(-r2 / twoSigma2);
00120         }
00121     }
00122
00123     double sineModeDecayRate(const std::vector<unsigned int>& modes) const {
00124         double lambda = 0.0;
00125         for (std::size_t d = 0; d < _shape.size(); ++d) {
00126             const double s = std::sin(modes[d] * kPi / (2.0 * (_shape[d] - 1u)));
00127             lambda += (4.0 / (_spacing[d] * _spacing[d])) * s * s;
00128         }
00129         return _D * lambda;
00130     }
00131
00132     double totalMass(const double* y) const {
00133         double mass = 0.0;
00134         double cellVolume = 1.0;
00135         for (double h : _spacing) {
00136             cellVolume *= h;
00137         }
00138         const unsigned int N = static_cast<unsigned int>(_shape.size());
00139         for (unsigned int node = 0; node < _totalNodes; ++node) {
00140             double weight = 1.0;
00141             for (unsigned int d = 0; d < N; ++d) {
00142                 const unsigned int m = coordinate(node, d);
00143                 if (m == 0u || m == _shape[d] - 1u) {
00144                     weight *= 0.5;
00145                 }
00146             }
00147             mass += weight * y[node];
00148         }
00149         return mass * cellVolume;
00150     }
00151
00152     double l2Norm(const double* y) const {
00153         double sum = 0.0;
00154         for (unsigned int i = 0; i < _totalNodes; ++i) {
00155             sum += y[i] * y[i];
00156         }
00157         return std::sqrt(sum);
00158     }
00159
00160     double maxValue(const double* y) const {
00161         double mx = -std::numeric_limits<double>::infinity();
00162         for (unsigned int i = 0; i < _totalNodes; ++i) {
00163             if (y[i] > mx) {
00164                 mx = y[i];
00165             }
00166         }
00167         return mx;
00168     }
00169
00170 private:
00171     void rebuild() {
00172         _valid = false;
```

```
00173         _totalNodes = 0;
00174         _stride.clear();
00175         _invH2.clear();
00176         const std::size_t N = _shape.size();
00177         if (N == 0 || _spacing.size() != N || _D < 0.0) {
00178             return;
00179         }
00180         for (std::size_t d = 0; d < N; ++d) {
00181             if (_shape[d] < 3u || _spacing[d] <= 0.0) {
00182                 return;
00183             }
00184         }
00185 
00186         _stride.assign(N, 1);
00187         for (std::size_t d = N - 1; d-- > 0;) {
00188             _stride[d] = _stride[d + 1] * _shape[d + 1];
00189         }
00190         std::size_t total = 1;
00191         for (unsigned int n : _shape) {
00192             if (total > std::numeric_limits<std::size_t>::max() / n) {
00193                 return;
00194             }
00195             total *= n;
00196         }
00197         if (total > std::numeric_limits<unsigned int>::max()) {
00198             return;
00199         }
00200         _totalNodes = static_cast<unsigned int>(total);
00201 
00202         _invH2.assign(N, 0.0);
00203         for (std::size_t d = 0; d < N; ++d) {
00204             _invH2[d] = 1.0 / (_spacing[d] * _spacing[d]);
00205         }
00206         _valid = true;
00207     }
00208 
00209 private:
00210     std::vector<unsigned int> _shape;
00211     std::vector<double> _spacing;
00212     double _D = 0.0;
00213     Boundary _boundary = Boundary::Dirichlet;
00214     bool _valid = false;
00215     unsigned int _totalNodes = 0;
00216     std::vector<std::size_t> _stride;
00217     std::vector<double> _invH2;
00218 };
00219 
00220 #endif /* DIFFUSIONMETHODOFLINESSYSTEM_H */
```

18.560 DormandPrince54OdeSolver.h

```
00001 #ifndef DORMANDPRINCE54ODESOLVER_H
00002 #define DORMANDPRINCE54ODESOLVER_H
00003 
00004 #include <algorithm>
00005 #include <cmath>
00006 #include <limits>
00007 #include <vector>
00008 
00009 #include "OdeSolver_if.h"
00010 
00011 class DormandPrince54OdeSolver : public OdeSolver_if {
00012 public:
00013     explicit DormandPrince54OdeSolver(double relativeTolerance = 1e-6,
00014                                       double absoluteTolerance = 1e-9,
00015                                       double safetyFactor = 0.9,
00016                                       double minScale = 0.2,
00017                                       double maxScale = 10.0,
00018                                       unsigned int maxSubsteps = 100000u)
00019         : _rtol(relativeTolerance),
00020           _atol(absoluteTolerance),
00021           _safety(safetyFactor),
00022           _minScale(minScale),
00023           _maxScale(maxScale),
00024           _maxSubsteps(maxSubsteps) {}
00025 
00026     double getRelativeTolerance() const { return _rtol; }
00027     double getAbsoluteTolerance() const { return _atol; }
00028 
00029     bool advance(const OdeSystem_if& system, double t0, double dt,
00030                  const double* y0, double* y1) const override {
00031         const unsigned int n = system.dimension();
00032         if (n == 0 || y0 == nullptr || y1 == nullptr || dt < 0.0) {
00033             return false;
00034         }
```

```
00035         for (unsigned int i = 0; i < n; ++i) {
00036             y1[i] = y0[i];
00037         }
00038         if (dt == 0.0) {
00039             return true;
00040         }
00041 
00042         std::vector<double> y(y0, y0 + n), yNext(n, 0.0), yErr(n, 0.0), tmp(n, 0.0);
00043         std::vector<double> k1(n, 0.0), k2(n, 0.0), k3(n, 0.0), k4(n, 0.0);
00044         std::vector<double> k5(n, 0.0), k6(n, 0.0), k7(n, 0.0);
00045         double t = t0;
00046         const double tEnd = t0 + dt;
00047         double h = dt;
00048         system.evaluate(t, y.data(), k1.data());
00049         bool firstSameAsLast = false;
00050 
00051         unsigned int steps = 0;
00052         while (t < tEnd) {
00053             if (++steps > _maxSubsteps) {
00054                 return false;
00055             }
00056             const double remaining = tEnd - t;
00057             if (remaining <= stepFloor(t)) {
00058                 break;
00059             }
00060             if (h > remaining) {
00061                 h = remaining;
00062             }
00063             if (firstSameAsLast) {
00064                 k1 = k7;
00065             }
00066 
00067             for (unsigned int i = 0; i < n; ++i) tmp[i] = y[i] + h * A21 * k1[i];
00068             system.evaluate(t + C2 * h, tmp.data(), k2.data());
00069             for (unsigned int i = 0; i < n; ++i) tmp[i] = y[i] + h * (A31 * k1[i] + A32 * k2[i]);
00070             system.evaluate(t + C3 * h, tmp.data(), k3.data());
00071             for (unsigned int i = 0; i < n; ++i) tmp[i] = y[i] + h * (A41 * k1[i] + A42 * k2[i] + A43
      * k3[i]);
00072             system.evaluate(t + C4 * h, tmp.data(), k4.data());
00073             for (unsigned int i = 0; i < n; ++i) tmp[i] = y[i] + h * (A51 * k1[i] + A52 * k2[i] + A53
      * k3[i] + A54 * k4[i]);
00074             system.evaluate(t + C5 * h, tmp.data(), k5.data());
00075             for (unsigned int i = 0; i < n; ++i) tmp[i] = y[i] + h * (A61 * k1[i] + A62 * k2[i] + A63
      * k3[i] + A64 * k4[i] + A65 * k5[i]);
00076             system.evaluate(t + C6 * h, tmp.data(), k6.data());
00077             for (unsigned int i = 0; i < n; ++i) yNext[i] = y[i] + h * (B5_1 * k1[i] + B5_3 * k3[i] +
      B5_4 * k4[i] + B5_5 * k5[i] + B5_6 * k6[i]);
00078             system.evaluate(t + h, yNext.data(), k7.data());
00079 
00080             for (unsigned int i = 0; i < n; ++i) {
00081                 yErr[i] = h * (E1 * k1[i] + E3 * k3[i] + E4 * k4[i] + E5 * k5[i] + E6 * k6[i] + E7 *
      k7[i]);
00082             }
00083 
00084             double errNorm = 0.0;
00085             for (unsigned int i = 0; i < n; ++i) {
00086                 if (!std::isfinite(yNext[i]) || !std::isfinite(yErr[i])) {
00087                     return false;
00088                 }
00089                 const double sc = _atol + _rtol * std::max(std::fabs(y[i]), std::fabs(yNext[i]));
00090                 const double ratio = (sc > 0.0) ? (yErr[i] / sc) : 0.0;
00091                 errNorm += ratio * ratio;
00092             }
00093             errNorm = std::sqrt(errNorm / static_cast<double>(n));
00094 
00095             if (errNorm <= 1.0) {
00096                 t += h;
00097                 y = yNext;
00098                 firstSameAsLast = true;
00099                 h *= scaleFactor(errNorm);
00100             } else {
00101                 h *= scaleFactor(errNorm);
00102                 firstSameAsLast = false;
00103                 if (h < stepFloor(t)) {
00104                     return false;
00105                 }
00106             }
00107         }
00108 
00109         for (unsigned int i = 0; i < n; ++i) {
00110             y1[i] = y[i];
00111         }
00112         return true;
00113     }
00114 
00115 private:
00116     double scaleFactor(double errNorm) const {
```

```
00117         if (errNorm == 0.0) {
00118             return _maxScale;
00119         }
00120         return std::min(_maxScale, std::max(_minScale, _safety * std::pow(errNorm, -0.2)));
00121     }
00122
00123     static double stepFloor(double t) {
00124         return 16.0 * std::numeric_limits<double>::epsilon() * (std::fabs(t) + 1.0);
00125     }
00126
00127 private:
00128     double _rtol;
00129     double _atol;
00130     double _safety;
00131     double _minScale;
00132     double _maxScale;
00133     unsigned int _maxSubsteps;
00134
00135     static constexpr double C2 = 1.0 / 5.0;
00136     static constexpr double C3 = 3.0 / 10.0;
00137     static constexpr double C4 = 4.0 / 5.0;
00138     static constexpr double C5 = 8.0 / 9.0;
00139     static constexpr double C6 = 1.0;
00140     static constexpr double A21 = 1.0 / 5.0;
00141     static constexpr double A31 = 3.0 / 40.0;
00142     static constexpr double A32 = 9.0 / 40.0;
00143     static constexpr double A41 = 44.0 / 45.0;
00144     static constexpr double A42 = -56.0 / 15.0;
00145     static constexpr double A43 = 32.0 / 9.0;
00146     static constexpr double A51 = 19372.0 / 6561.0;
00147     static constexpr double A52 = -25360.0 / 2187.0;
00148     static constexpr double A53 = 64448.0 / 6561.0;
00149     static constexpr double A54 = -212.0 / 729.0;
00150     static constexpr double A61 = 9017.0 / 3168.0;
00151     static constexpr double A62 = -355.0 / 33.0;
00152     static constexpr double A63 = 46732.0 / 5247.0;
00153     static constexpr double A64 = 49.0 / 176.0;
00154     static constexpr double A65 = -5103.0 / 18656.0;
00155     static constexpr double B5_1 = 35.0 / 384.0;
00156     static constexpr double B5_3 = 500.0 / 1113.0;
00157     static constexpr double B5_4 = 125.0 / 192.0;
00158     static constexpr double B5_5 = -2187.0 / 6784.0;
00159     static constexpr double B5_6 = 11.0 / 84.0;
00160     static constexpr double E1 = 35.0 / 384.0 - 5179.0 / 57600.0;
00161     static constexpr double E3 = 500.0 / 1113.0 - 7571.0 / 16695.0;
00162     static constexpr double E4 = 125.0 / 192.0 - 393.0 / 640.0;
00163     static constexpr double E5 = -2187.0 / 6784.0 - (-92097.0 / 339200.0);
00164     static constexpr double E6 = 11.0 / 84.0 - 187.0 / 2100.0;
00165     static constexpr double E7 = -1.0 / 40.0;
00166 };
00167
00168 #endif /* DORMANDPRINCE54ODESOLVER_H */
```

18.561 OdeSolver_if.h

```
00001 #ifndef ODESOLVER_IF_H
00002 #define ODESOLVER_IF_H
00003
00004 #include "OdeSystem_if.h"
00005
00013 class OdeSolver_if {
00014 public:
00015     virtual ~OdeSolver_if() = default;
00016     virtual bool advance(const OdeSystem_if& system, double t0, double dt, const double* y0, double*
      y1) const = 0;
00017 };
00018
00019 #endif /* ODESOLVER_IF_H */
```

18.562 OdeSolverFactory.h

```
00001 #ifndef ODESOLVERFACTORY_H
00002 #define ODESOLVERFACTORY_H
00003
00004 #include <functional>
00005 #include <map>
00006 #include <memory>
00007 #include <string>
00008 #include <vector>
00009
00010 #include "DormandPrince54OdeSolver.h"
00011 #include "OdeSolver_if.h"
```

```
00012 #include "RungeKutta4OdeSolver.h"
00013
00014 enum class OdeSolverType {
00015     RungeKutta4,
00016     DormandPrince54
00017 };
00018
00019 class OdeSolverFactory {
00020 public:
00021     using Creator = std::function<std::unique_ptr<OdeSolver_if>()>;
00022
00023     static const std::string& keyRungeKutta4() {
00024         static const std::string k = "RungeKutta4";
00025         return k;
00026     }
00027
00028     static const std::string& keyDormandPrince54() {
00029         static const std::string k = "DormandPrince54";
00030         return k;
00031     }
00032
00033     static const std::string& defaultKey() {
00034         return keyRungeKutta4();
00035     }
00036
00037     static std::unique_ptr<OdeSolver_if> create(OdeSolverType type) {
00038         switch (type) {
00039             case OdeSolverType::DormandPrince54:
00040                 return std::make_unique<DormandPrince54OdeSolver>();
00041             case OdeSolverType::RungeKutta4:
00042             default:
00043                 return std::make_unique<RungeKutta4OdeSolver>();
00044         }
00045     }
00046
00047     static std::unique_ptr<OdeSolver_if> create(const std::string& key) {
00048         auto& reg = registry();
00049         auto it = reg.find(key);
00050         if (it != reg.end()) {
00051             return it->second();
00052         }
00053         return std::make_unique<RungeKutta4OdeSolver>();
00054     }
00055
00056     static bool isRegistered(const std::string& key) {
00057         return registry().find(key) != registry().end();
00058     }
00059
00060     static std::vector<std::string> availableKeys() {
00061         std::vector<std::string> keys;
00062         keys.reserve(registry().size());
00063         for (const auto& entry : registry()) {
00064             keys.push_back(entry.first);
00065         }
00066         return keys;
00067     }
00068
00069     static bool registerCreator(const std::string& key, Creator creator) {
00070         return registry().emplace(key, std::move(creator)).second;
00071     }
00072
00073     static std::string toKey(OdeSolverType type) {
00074         switch (type) {
00075             case OdeSolverType::DormandPrince54:
00076                 return keyDormandPrince54();
00077             case OdeSolverType::RungeKutta4:
00078             default:
00079                 return keyRungeKutta4();
00080         }
00081     }
00082
00083     static bool fromKey(const std::string& key, OdeSolverType& out) {
00084         if (key == keyDormandPrince54()) {
00085             out = OdeSolverType::DormandPrince54;
00086             return true;
00087         }
00088         if (key == keyRungeKutta4()) {
00089             out = OdeSolverType::RungeKutta4;
00090             return true;
00091         }
00092         return false;
00093     }
00094
00095 private:
00096     static std::map<std::string, Creator>& registry() {
00097         static std::map<std::string, Creator> reg = [] {
00098             std::map<std::string, Creator> initial;
```

```
00099             initial.emplace(keyRungeKutta4(),
00100                             [] { return std::make_unique<RungeKutta4OdeSolver>(); });
00101             initial.emplace(keyDormandPrince54(),
00102                             [] { return std::make_unique<DormandPrince54OdeSolver>(); });
00103             return initial;
00104         }();
00105         return reg;
00106     }
00107 };
00108
00109 #endif /* ODESOLVERFACTORY_H */
```

18.563 OdeSystem_if.h

```
00001 #ifndef ODESYSTEM_IF_H
00002 #define ODESYSTEM_IF_H
00003
00007 class OdeSystem_if {
00008 public:
00009     virtual ~OdeSystem_if() = default;
00010     virtual unsigned int dimension() const = 0;
00011     virtual void evaluate(double t, const double* y, double* dydt) const = 0;
00012 };
00013
00014 #endif /* ODESYSTEM_IF_H */
```

18.564 Quadrature_if.h

```
00001 #ifndef QUADRATURE_IF_H
00002 #define QUADRATURE_IF_H
00003
00010 class Quadrature_if {
00011 public:
00012     virtual ~Quadrature_if() = default;
00013     virtual double integrate(double min, double max, double (*f)(double), double precision) const = 0;
00014 };
00015
00016 #endif /* QUADRATURE_IF_H */
```

18.565 RootFinder_if.h

```
00001 #ifndef ROOTFINDER_IF_H
00002 #define ROOTFINDER_IF_H
00003
00007 class RootFinder_if {
00008 public:
00009     virtual ~RootFinder_if() = default;
00010     virtual double findRoot(double min, double max, double (*f)(double), double tolerance) const = 0;
00011 };
00012
00013 #endif /* ROOTFINDER_IF_H */
```

18.566 RungeKutta4OdeSolver.h

```
00001 #ifndef RUNGEKUTTA4ODESOLVER_H
00002 #define RUNGEKUTTA4ODESOLVER_H
00003
00004 #include <vector>
00005
00006 #include "OdeSolver_if.h"
00007
00011 class RungeKutta4OdeSolver : public OdeSolver_if {
00012 public:
00013     // Advances the ODE system by one step dt using classical RK4.
00014     // y0: state at t0 (input), y1: state at t0+dt (output). Returns false if inputs are invalid.
00015     bool advance(const OdeSystem_if& system, double t0, double dt, const double* y0, double* y1) const
      override {
00016         const unsigned int n = system.dimension();
00017         if (n == 0 || y0 == nullptr || y1 == nullptr || dt < 0.0) {
00018             return false;
00019         }
00020
00021         std::vector<double> k1(n, 0.0);
00022         std::vector<double> k2(n, 0.0);
00023         std::vector<double> k3(n, 0.0);
00024         std::vector<double> k4(n, 0.0);
00025         std::vector<double> temp(n, 0.0);
```

```
00026
00027         // k1: slope at the beginning of the interval
00028         system.evaluate(t0, y0, k1.data());
00029         for (unsigned int i = 0; i < n; ++i) {
00030             temp[i] = y0[i] + 0.5 * dt * k1[i];
00031         }
00032
00033         // k2: slope at the midpoint, using k1 to estimate the state there
00034         system.evaluate(t0 + 0.5 * dt, temp.data(), k2.data());
00035         for (unsigned int i = 0; i < n; ++i) {
00036             temp[i] = y0[i] + 0.5 * dt * k2[i];
00037         }
00038
00039         // k3: slope at the midpoint again, using the improved k2 estimate
00040         system.evaluate(t0 + 0.5 * dt, temp.data(), k3.data());
00041         for (unsigned int i = 0; i < n; ++i) {
00042             temp[i] = y0[i] + dt * k3[i];
00043         }
00044
00045         // k4: slope at the end of the interval
00046         system.evaluate(t0 + dt, temp.data(), k4.data());
00047
00048         // weighted average: k1 and k4 count once, k2 and k3 count twice
00049         for (unsigned int i = 0; i < n; ++i) {
00050             y1[i] = y0[i] + (dt / 6.0) * (k1[i] + 2.0 * k2[i] + 2.0 * k3[i] + k4[i]);
00051         }
00052         return true;
00053     }
00054 };
00055
00056 #endif /* RUNGEKUTTA4ODESOLVER_H */
```

18.567 Solver_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Integrator_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 14 de Agosto de 2018, 13:54
00012  */
00013
00014 #ifndef SOLVER_IF_H
00015 #define SOLVER_IF_H
00016
00038 class Solver_if {
00039 public:
00040     virtual ~Solver_if() = default;
00041     virtual void setPrecision(double e) = 0;
00042     virtual double getPrecision() = 0;
00043     virtual void setMaxSteps(double steps) = 0;
00044     virtual double getMaxSteps() = 0;
00045     virtual double integrate(double min, double max, double (*f)(double, double), double p2) = 0;
00046     virtual double integrate(double min, double max, double (*f)(double, double, double), double p2,
      double p3) = 0;
00047     virtual double integrate(double min, double max, double (*f)(double, double, double, double),
      double p2, double p3, double p4) = 0;
00048     virtual double integrate(double min, double max, double (*f)(double, double, double, double,
      double), double p2, double p3, double p4, double p5) = 0;
00049     virtual double derivate(double initPoint, double initValue, double (*f)(double, double), double
      p2) = 0;
00050     virtual double derivate(double initPoint, double initValue, double (*f)(double, double, double),
      double p2, double p3) = 0;
00051     virtual double derivate(double initPoint, double initValue, double (*f)(double, double, double,
      double), double p2, double p3, double p4) = 0;
00052     virtual double derivate(double initPoint, double initValue, double (*f)(double, double, double,
      double, double), double p2, double p3, double p4, double p5) = 0;
00053 };
00054
00055 #endif /* SOLVER_IF_H */
```

18.568 SolverDefaultImpl1.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
```

```
00005
00006 /*
00007  * File:   IntegratorDefaultImpl1.h
00008  * Author: rlcancian
00009  *
00010  * Created on 23 de novembro de 2021, 18:36
00011  */
00012
00013 #ifndef SOLVERDEFAULTIMPL1_IF_H
00014 #define SOLVERDEFAULTIMPL1_IF_H
00015
00016 #include "Solver_if.h"
00017
00034 class SolverDefaultImpl1 : public Solver_if {
00035 public:
00036     SolverDefaultImpl1(double precision = 1e-6, unsigned int steps = 1e3);
00037     virtual ~SolverDefaultImpl1() = default;
00038 public:
00039     virtual void setPrecision(double e) override;
00040     virtual double getPrecision() override;
00041     virtual void setMaxSteps(double steps) override;
00042     virtual double getMaxSteps() override;
00043     virtual double integrate(double min, double max, double (*f)(double, double), double p2) override;
00044     virtual double integrate(double min, double max, double (*f)(double, double, double), double p2,
      double p3) override;
00045     virtual double integrate(double min, double max, double (*f)(double, double, double, double),
      double p2, double p3, double p4) override;
00046     virtual double integrate(double min, double max, double (*f)(double, double, double, double,
      double), double p2, double p3, double p4, double p5) override;
00047     virtual double derivate(double initPoint, double initValue, double (*f)(double, double), double
      p2) override;
00048     virtual double derivate(double initPoint, double initValue, double (*f)(double, double, double),
      double p2, double p3) override;
00049     virtual double derivate(double initPoint, double initValue, double (*f)(double, double, double,
      double), double p2, double p3, double p4) override;
00050     virtual double derivate(double initPoint, double initValue, double (*f)(double, double, double,
      double, double), double p2, double p3, double p4, double p5) override;
00051 private:
00052     double _precision;
00053     unsigned int _numSteps;
00054     double _stepSize;
00055 };
00056
00057 #endif /* SOLVERDEFAULTIMPL1_IF_H */
```

18.569 FactorialDesign.h

```
00001 #ifndef FACTORIAL_DESIGN_H
00002 #define FACTORIAL_DESIGN_H
00003
00004 #include <vector>
00005 #include <string>
00006 #include <tuple>
00007
00008 // Class representing a Fractional Factorial Design of Experiments
00009 class FactorialDesign {
00010 public:
00011     // Design parameters
00012     int k; // Number of factors investigated
00013     int p; // Size of the fraction of the full factorial
00014     int r; // Number of replications
00015     int NUM_FACTORS; // Derived number of factors
00016     int NUM_ROWS;  // Number of rows in the sign table
00017
00018     // Storage of experimental results
00019     std::vector<std::vector<int» resultsMatrix;
00020
00021     // Constructor and main methods
00022     FactorialDesign(int k, int p, int r);
00023     bool checkFeasibility();
00024     void generateIndexCombinations();
00025     void createColumnLabels();
00026     void generateInputs();
00027     void calculateResultsMean();
00028     void createSignTable();
00029     void calculateStatistics();
00030     double linearRegression(const std::vector<int>& xs);
00031     void showSignTable();
00032     void showImpacts();
00033     void showLinearRegressionModel();
00034     void showResiduals();
00035
00036     // Getters
00037     const std::vector<std::vector<int»& getIndexCombinations() const {
00038         return indexCombinations;
```

```
00039     }
00040
00041     const std::vector<std::string>& getColumnLabels() const {
00042         return columnLabels;
00043     }
00044
00045     const std::vector<std::vector<int>& getInputs() const {
00046         return inputs;
00047     }
00048
00049     const std::vector<double>& getResultsMean() const {
00050         return resultsMean;
00051     }
00052
00053     const std::vector<std::tuple<int, std::vector<int>, std::vector<int>, double>& getSignTable()
      const {
00054         return signTable;
00055     }
00056
00057     const std::vector<double>& getEffectSums() const {
00058         return effectSums;
00059     }
00060
00061     const std::vector<double>& getEffectAverages() const {
00062         return effectAverages;
00063     }
00064
00065     const std::vector<int>& getSquareSums() const {
00066         return squareSums;
00067     }
00068
00069     double getSquareSumError() const {
00070         return squareSumError;
00071     }
00072
00073     double getSquareSumTotal() const {
00074         return squareSumTotal;
00075     }
00076
00077     const std::vector<double>& getFactorVariations() const {
00078         return factorVariations;
00079     }
00080
00081     double getErrorVariation() const {
00082         return errorVariation;
00083     }
00084
00085     // Setters
00086     void setResultsMatrix(const std::vector<std::vector<int>& matrix) {
00087         resultsMatrix = matrix;
00088     }
00089
00090 private:
00091     // Internal data structures
00092     std::vector<std::vector<int> indexCombinations;
00093     std::vector<std::string> factorLabels;
00094     std::vector<std::string> columnLabels;
00095     std::vector<std::vector<int> inputs;
00096     std::vector<double> resultsMean;
00097     std::vector<std::tuple<int, std::vector<int>, std::vector<int>, double> signTable;
00098     std::vector<double> effectSums;
00099     std::vector<double> effectAverages;
00100     std::vector<int> squareSums;
00101     double squareSumTotal;
00102     std::vector<double> factorVariations;
00103     double squareSumError;
00104     double errorVariation;
00105
00106     // Helper method
00107     std::vector<std::vector<int> generateCombinations(const std::vector<int>& elements, int length);
00108 };
00109
00110 #endif
```

18.570 Optimizer_if.h

```
00001 /*
00002  * File:   Optimizer_if.h
00003  * Author: Genesys Team
00004  *
00005  * Created on 12 de Abril de 2026
00006  */
00007
00008 #ifndef OPTIMIZER_IF_H
00009 #define OPTIMIZER_IF_H
```

```
00010
00011 #include <cstddef>
00012 #include <string>
00013 #include "kernel/util/List.h"
00014
00015 class Model;
00016 class SimulationControl;
00017 class SimulationResponse;
00018
00037 class Optimizer_if {
00038 public:
00039     virtual ~Optimizer_if() = default;
00040
00044     enum class ObjectiveSense {
00045         MINIMIZE,
00046         MAXIMIZE
00047     };
00048
00052     enum class ExecutionState {
00053         NOT_READY,
00054         READY,
00055         RUNNING,
00056         PAUSED,
00057         STOPPED,
00058         FINISHED,
00059         ERROR
00060     };
00061
00069     struct ObjectiveDefinition {
00070         std::string name;
00071         ObjectiveSense sense;
00072         std::string expression;
00073         double weight = 1.0;
00074         bool enabled = true;
00075     };
00076
00084     struct ConstraintDefinition {
00085         std::string name;
00086         std::string expression;
00087         bool enabled = true;
00088     };
00089
00093     struct OptimizationSettings {
00094         unsigned int maxIterations = 100;
00095         unsigned int maxSimulations = 0;
00096         unsigned int replicationsPerSolution = 1;
00097         unsigned int bestSolutionsToKeep = 10;
00098         unsigned int randomSeed = 0;
00099         double improvementTolerance = 0.0;
00100         double timeLimitSeconds = 0.0;
00101     };
00102
00110     struct SolutionSummary {
00111         unsigned int iteration = 0;
00112         unsigned int simulationNumber = 0;
00113         bool feasible = false;
00114         double aggregatedObjectiveValue = 0.0;
00115         std::string description;
00116     };
00117
00121     virtual void setModel(Model* model) = 0;
00122
00129     virtual bool setModelFilename(const std::string& filename) = 0;
00130
00134     virtual Model* getModel() const = 0;
00135
00139     virtual std::string getModelFilename() const = 0;
00140
00144     virtual List<SimulationControl*>* getAvailableControls() const = 0;
00145
00149     virtual List<SimulationResponse*>* getAvailableResponses() const = 0;
00150
00154     virtual void setSelectedControls(List<SimulationControl*>* controls) = 0;
00155
00159     virtual void setSelectedResponses(List<SimulationResponse*>* responses) = 0;
00160
00164     virtual List<SimulationControl*>* getSelectedControls() const = 0;
00165
00169     virtual List<SimulationResponse*>* getSelectedResponses() const = 0;
00170
00174     virtual void addObjective(const ObjectiveDefinition& objective) = 0;
00175
00179     virtual void clearObjectives() = 0;
00180
00184     virtual List<ObjectiveDefinition>* getObjectives() const = 0;
00185
00189     virtual void addConstraint(const ConstraintDefinition& constraint) = 0;
```

```
00190
00194     virtual void clearConstraints() = 0;
00195
00199     virtual List<ConstraintDefinition>* getConstraints() const = 0;
00200
00204     virtual void setSettings(const OptimizationSettings& settings) = 0;
00205
00209     virtual OptimizationSettings getSettings() const = 0;
00210
00217     virtual bool checkReady(std::string* message = nullptr) const = 0;
00218
00223     virtual bool start() = 0;
00224
00229     virtual bool step() = 0;
00230
00234     virtual bool pause() = 0;
00235
00239     virtual bool resume() = 0;
00240
00244     virtual bool stop() = 0;
00245
00249     virtual ExecutionState getExecutionState() const = 0;
00250
00254     virtual unsigned int getCurrentIteration() const = 0;
00255
00259     virtual unsigned int getTotalSimulations() const = 0;
00260
00264     virtual List<SolutionSummary>* getBestSolutions() const = 0;
00265
00269     virtual const SolutionSummary* getCurrentBestSolution() const = 0;
00270
00274     virtual void resetResults() = 0;
00275 };
00276
00277 #endif /* OPTIMIZER_IF_H */
```

18.571 OptimizerDefaultImpl1.h

```
00001 /*
00002  * File:   OptimizerDefaultImpl1.h
00003  * Author: Genesys Team
00004  *
00005  * Created on 12 de Abril de 2026
00006  */
00007
00008 #ifndef OPTIMIZERDEFAULTIMPL1_H
00009 #define OPTIMIZERDEFAULTIMPL1_H
00010
00011 #include "Optimizer_if.h"
00012 #include <string>
00013
00030 class OptimizerDefaultImpl1 : public Optimizer_if {
00031 public:
00032     OptimizerDefaultImpl1();
00033     virtual ~OptimizerDefaultImpl1();
00034
00035     OptimizerDefaultImpl1(const OptimizerDefaultImpl1&) = delete;
00036     OptimizerDefaultImpl1& operator=(const OptimizerDefaultImpl1&) = delete;
00037     OptimizerDefaultImpl1(OptimizerDefaultImpl1&&) = delete;
00038     OptimizerDefaultImpl1& operator=(OptimizerDefaultImpl1&&) = delete;
00039
00040 public:
00041     virtual void setModel(Model* model) override;
00042     virtual bool setModelFilename(const std::string& filename) override;
00043     virtual Model* getModel() const override;
00044     virtual std::string getModelFilename() const override;
00045
00046     virtual List<SimulationControl*>* getAvailableControls() const override;
00047     virtual List<SimulationResponse*>* getAvailableResponses() const override;
00048     virtual void setSelectedControls(List<SimulationControl*>* controls) override;
00049     virtual void setSelectedResponses(List<SimulationResponse*>* responses) override;
00050     virtual List<SimulationControl*>* getSelectedControls() const override;
00051     virtual List<SimulationResponse*>* getSelectedResponses() const override;
00052
00053     virtual void addObjective(const ObjectiveDefinition& objective) override;
00054     virtual void clearObjectives() override;
00055     virtual List<ObjectiveDefinition>* getObjectives() const override;
00056
00057     virtual void addConstraint(const ConstraintDefinition& constraint) override;
00058     virtual void clearConstraints() override;
00059     virtual List<ConstraintDefinition>* getConstraints() const override;
00060
00061     virtual void setSettings(const OptimizationSettings& settings) override;
00062     virtual OptimizationSettings getSettings() const override;
00063     virtual bool checkReady(std::string* message = nullptr) const override;
```

```
00064
00065     virtual bool start() override;
00066     virtual bool step() override;
00067     virtual bool pause() override;
00068     virtual bool resume() override;
00069     virtual bool stop() override;
00070
00071     virtual ExecutionState getExecutionState() const override;
00072     virtual unsigned int getCurrentIteration() const override;
00073     virtual unsigned int getTotalSimulations() const override;
00074     virtual List<SolutionSummary>* getBestSolutions() const override;
00075     virtual const SolutionSummary* getCurrentBestSolution() const override;
00076     virtual void resetResults() override;
00077
00078 private:
00079     void _clearRuntimeState();
00080     void _refreshAvailableElementsFromModel();
00081     void _copyControlList(List<SimulationControl*>* source, List<SimulationControl*>* target);
00082     void _copyResponseList(List<SimulationResponse*>* source, List<SimulationResponse*>* target);
00083     bool _hasEnabledObjective() const;
00084     void _updateReadyState();
00085
00086 private:
00087     Model* _model = nullptr;
00088     std::string _modelFilename;
00089     List<SimulationControl*>* _availableControls = nullptr;
00090     List<SimulationResponse*>* _availableResponses = nullptr;
00091     List<SimulationControl*>* _selectedControls = nullptr;
00092     List<SimulationResponse*>* _selectedResponses = nullptr;
00093     List<ObjectiveDefinition>* _objectives = nullptr;
00094     List<ConstraintDefinition>* _constraints = nullptr;
00095     List<SolutionSummary>* _bestSolutions = nullptr;
00096     OptimizationSettings _settings;
00097     ExecutionState _executionState = ExecutionState::NOT_READY;
00098     unsigned int _currentIteration = 0;
00099     unsigned int _totalSimulations = 0;
00100     std::string _lastErrorMessage;
00101     const SolutionSummary* _currentBestSolution = nullptr;
00102 };
00103
00104 #endif /* OPTIMIZERDEFAULTIMPL1_H */
```

18.572 ContinuousDistribution_if.h

```
00001 #ifndef CONTINUOUSDISTRIBUTION_IF_H
00002 #define CONTINUOUSDISTRIBUTION_IF_H
00003
00004 #include "Distribution_if.h"
00005
00014 class ContinuousDistribution_if : public Distribution_if {
00015 public:
00016     virtual ~ContinuousDistribution_if() = default;
00017     virtual double pdf(double x) const = 0;
00018     virtual double cdf(double x) const = 0;
00019     virtual double quantile(double probability) const = 0;
00020 };
00021
00022 #endif /* CONTINUOUSDISTRIBUTION_IF_H */
```

18.573 DataAnalyser_if.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/file.h to edit this template
00004  */
00005
00006 /*
00007  * File:   DataAnalyser.h
00008  * Author: rlcancian
00009  *
00010  * Created on 17 de maio de 2022, 11:06
00011  */
00012
00013 #ifndef DATAANALYSERIF_H
00014 #define DATAANALYSERIF_H
00015
00016 #include <string>
00017 #include "Fitter_if.h"
00018 #include "HypothesisTester_if.h"
00019 #include "kernel/statistics/Sampler_if.h"
00020 #include "kernel/simulator/ExperimetManager_if.h"
00021
```

```
00042 class DataAnalyser_if {
00043 public:
00044     virtual ~DataAnalyser_if() = default;
00045     virtual bool loadDataSet(std::string datafilename) = 0;
00046     virtual bool saveDataSet(std::string datasetname) = 0;
00047     virtual void newDataSet(std::string datasetname, std::string datafilename) = 0;
00048     virtual Fitter_if* fitter() = 0;
00049     virtual Sampler_if* sampler() = 0;
00050     virtual ExperimentManager_if* experimenter() = 0;
00051     virtual HypothesisTester_if* tester() = 0;
00052 };
00053
00054
00055 #endif /* DATAANALYSERIF_H */
```

18.574 DataSet_if.h

```
00001 #ifndef DATASET_IF_H
00002 #define DATASET_IF_H
00003
00004 #include <string>
00005
00016 class DataSet_if {
00017 public:
00018     virtual ~DataSet_if() = default;
00019     virtual unsigned int size() const = 0;
00020     virtual double value(unsigned int index) const = 0;
00021     virtual bool isLoaded() const = 0;
00022     virtual std::string sourceDescription() const = 0;
00023 };
00024
00025 #endif /* DATASET_IF_H */
```

18.575 DiscreteDistribution_if.h

```
00001 #ifndef DISCRETEDISTRIBUTION_IF_H
00002 #define DISCRETEDISTRIBUTION_IF_H
00003
00004 #include "Distribution_if.h"
00005
00014 class DiscreteDistribution_if : public Distribution_if {
00015 public:
00016     virtual ~DiscreteDistribution_if() = default;
00017     virtual double pmf(unsigned int k) const = 0;
00018     virtual double cdf(unsigned int k) const = 0;
00019     virtual unsigned int quantile(double probability) const = 0;
00020 };
00021
00022 #endif /* DISCRETEDISTRIBUTION_IF_H */
```

18.576 Distribution_if.h

```
00001 #ifndef DISTRIBUTION_IF_H
00002 #define DISTRIBUTION_IF_H
00003
00004 #include <string>
00005
00009 class Distribution_if {
00010 public:
00011     virtual ~Distribution_if() = default;
00012     virtual std::string name() const = 0;
00013 };
00014
00015 #endif /* DISTRIBUTION_IF_H */
```

18.577 Fitter_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   Fitter_if.h
00009  * Author: cancian
00010  *
```

```
00011  * Created on 14 de Agosto de 2018, 14:05
00012  */
00013
00014 #ifndef FITTER_IF_H
00015 #define FITTER_IF_H
00016
00017 #include <string>
00018
00051 class Fitter_if {
00052 public:
00053     virtual ~Fitter_if() = default;
00054     virtual bool isNormalDistributed(double confidencelevel) = 0;
00055     virtual void fitUniform(double *sqrerror, double *min, double *max) = 0;
00056     virtual void fitTriangular(double *sqrerror, double *min, double *mo, double *max) = 0;
00057     virtual void fitNormal(double *sqrerror, double *avg, double *stddev) = 0;
00058     virtual void fitExpo(double *sqrerror, double *avg1) = 0;
00059     virtual void fitErlang(double *sqrerror, double *avg, double *m) = 0;
00060     virtual void fitBeta(double *sqrerror, double *alpha, double *beta, double *infLimit, double
      *supLimit) = 0;
00061     virtual void fitWeibull(double *sqrerror, double *alpha, double *scale) = 0;
00062     virtual void fitAll(double *sqrerror, std::string *name) = 0;
00063 public:
00067     virtual void setDataFilename(std::string dataFilename) = 0;
00071     virtual std::string getDataFilename() = 0;
00072 };
00073
00074 #endif /* FITTER_IF_H */
```

18.578 FitterDefaultImpl.h

```
00001 #ifndef FITTERDEFAULTIMPL_H
00002 #define FITTERDEFAULTIMPL_H
00003
00004 #include "Fitter_if.h"
00005 #include "ProbabilityDistributionBase.h"
00006 #include "tools/Continuous/SolverDefaultImpl1.h"
00007
00008 #include <algorithm>
00009 #include <cmath>
00010 #include <fstream>
00011 #include <functional>
00012 #include <limits>
00013 #include <numeric>
00014 #include <string>
00015 #include <vector>
00016
00036 class FitterDefaultImpl : public Fitter_if {
00037 public:
00038     FitterDefaultImpl() = default;
00039     virtual ~FitterDefaultImpl() = default;
00040
00041 public:
00042     virtual bool isNormalDistributed(double confidencelevel) override {
00043         double sqrerror = std::numeric_limits<double>::infinity();
00044         double avg = _nan();
00045         double stddev = _nan();
00046         fitNormal(&sqrerror, &avg, &stddev);
00047         if (!std::isfinite(sqrerror)) {
00048             return false;
00049         }
00050         if (_count < 8) {
00051             return false;
00052         }
00053         double cl = confidencelevel;
00054         if (cl > 1.0) {
00055             cl /= 100.0;
00056         }
00057         cl = std::max(0.0, std::min(1.0, cl));
00058         const double normalizedSse = sqrerror / static_cast<double>(_count);
00059         const double threshold = 0.008 + (1.0 - cl) * 0.02;
00060         return normalizedSse <= threshold;
00061     }
00062
00063     virtual void fitUniform(double *sqrerror, double *min, double *max) override {
00064         if (!_ensureDataLoaded() || _count < 2 || !(_sampleMax > _sampleMin)) {
00065             _setFailure3(sqrerror, min, max);
00066             return;
00067         }
00068         const double fittedMin = _sampleMin;
00069         const double fittedMax = _sampleMax;
00070
00071         if (min != nullptr) {
00072             *min = fittedMin;
00073         }
00074         if (max != nullptr) {
```

```
00075             *max = fittedMax;
00076         }
00077         _setSse(sqrerror, [fittedMin, fittedMax](double x) {
00078             if (x <= fittedMin) {
00079                 return 0.0;
00080             }
00081             if (x >= fittedMax) {
00082                 return 1.0;
00083             }
00084             return (x - fittedMin) / (fittedMax - fittedMin);
00085         });
00086     }
00087
00088     virtual void fitTriangular(double *sqrerror, double *min, double *mo, double *max) override {
00089         if (!_ensureDataLoaded() || _count < 2 || !(_sampleMax > _sampleMin)) {
00090             _setFailure4(sqrerror, min, mo, max);
00091             return;
00092         }
00093
00094         const double fittedMin = _sampleMin;
00095         const double fittedMax = _sampleMax;
00096         const double range = fittedMax - fittedMin;
00097         double mode = 3.0 * _sampleMean - fittedMin - fittedMax;
00098         mode = std::max(fittedMin, std::min(fittedMax, mode));
00099
00100         const double eps = range * 1e-9;
00101         if (mode <= fittedMin) {
00102             mode = fittedMin + eps;
00103         }
00104         if (mode >= fittedMax) {
00105             mode = fittedMax - eps;
00106         }
00107         if (!(mode > fittedMin) || !(mode < fittedMax)) {
00108             _setFailure4(sqrerror, min, mo, max);
00109             return;
00110         }
00111
00112         if (min != nullptr) {
00113             *min = fittedMin;
00114         }
00115         if (mo != nullptr) {
00116             *mo = mode;
00117         }
00118         if (max != nullptr) {
00119             *max = fittedMax;
00120         }
00121
00122         _setSse(sqrerror, [fittedMin, mode, fittedMax](double x) {
00123             if (x <= fittedMin) {
00124                 return 0.0;
00125             }
00126             if (x >= fittedMax) {
00127                 return 1.0;
00128             }
00129             if (x <= mode) {
00130                 const double num = (x - fittedMin) * (x - fittedMin);
00131                 const double den = (fittedMax - fittedMin) * (mode - fittedMin);
00132                 return num / den;
00133             }
00134             const double num = (fittedMax - x) * (fittedMax - x);
00135             const double den = (fittedMax - fittedMin) * (fittedMax - mode);
00136             return 1.0 - num / den;
00137         });
00138     }
00139
00140     virtual void fitNormal(double *sqrerror, double *avg, double *stddev) override {
00141         if (!_ensureDataLoaded() || _count < 2 || !(_sampleStddev > 0.0) ||
      !std::isfinite(_sampleStddev)) {
00142             _setFailure3(sqrerror, avg, stddev);
00143             return;
00144         }
00145         if (avg != nullptr) {
00146             *avg = _sampleMean;
00147         }
00148         if (stddev != nullptr) {
00149             *stddev = _sampleStddev;
00150         }
00151         _setSse(sqrerror, [this](double x) {
00152             const double z = (x - _sampleMean) / (_sampleStddev * std::sqrt(2.0));
00153             return 0.5 * std::erfc(-z);
00154         });
00155     }
00156
00157     virtual void fitExpo(double *sqrerror, double *avg1) override {
00158         if (!_ensureDataLoaded() || !(_sampleMean > 0.0) || _hasNegativeData) {
00159             _setFailure2(sqrerror, avg1);
00160             return;
```

```
00161         }
00162         if (avg1 != nullptr) {
00163             *avg1 = _sampleMean;
00164         }
00165         _setSse(sqrerror, [this](double x) {
00166             if (x < 0.0) {
00167                 return 0.0;
00168             }
00169             return 1.0 - std::exp(-x / _sampleMean);
00170         });
00171     }
00172 
00173     virtual void fitErlang(double *sqrerror, double *avg, double *m) override {
00174         if (!_ensureDataLoaded() || !(_sampleMean > 0.0) || !(_sampleVariance > 0.0) ||
      _hasNegativeData) {
00175             _setFailure3(sqrerror, avg, m);
00176             return;
00177         }
00178         long int mInt = static_cast<long int>(std::llround((_sampleMean * _sampleMean) /
      _sampleVariance));
00179         if (mInt < 1) {
00180             mInt = 1;
00181         }
00182         const double scale = _sampleMean / static_cast<double>(mInt);
00183         if (!(scale > 0.0) || !std::isfinite(scale)) {
00184             _setFailure3(sqrerror, avg, m);
00185             return;
00186         }
00187         if (avg != nullptr) {
00188             *avg = _sampleMean;
00189         }
00190         if (m != nullptr) {
00191             *m = static_cast<double>(mInt);
00192         }
00193         _setSse(sqrerror, [mInt, scale](double x) {
00194             if (x < 0.0) {
00195                 return 0.0;
00196             }
00197             const double z = x / scale;
00198             double sum = 1.0;
00199             double term = 1.0;
00200             for (long int k = 1; k < mInt; ++k) {
00201                 term *= z / static_cast<double>(k);
00202                 sum += term;
00203             }
00204             return 1.0 - std::exp(-z) * sum;
00205         });
00206     }
00207 
00208     virtual void fitBeta(double *sqrerror, double *alpha, double *beta, double *infLimit, double
      *supLimit) override {
00209         double fittedAlpha = _nan();
00210         double fittedBeta = _nan();
00211         double fittedInf = _nan();
00212         double fittedSup = _nan();
00213         if (!_estimateScaledBetaMoments(&fittedAlpha, &fittedBeta, &fittedInf, &fittedSup)) {
00214             _setFailure5(sqrerror, alpha, beta, infLimit, supLimit);
00215             return;
00216         }
00217 
00218         if (alpha != nullptr) {
00219             *alpha = fittedAlpha;
00220         }
00221         if (beta != nullptr) {
00222             *beta = fittedBeta;
00223         }
00224         if (infLimit != nullptr) {
00225             *infLimit = fittedInf;
00226         }
00227         if (supLimit != nullptr) {
00228             *supLimit = fittedSup;
00229         }
00230 
00231         _setSse(sqrerror, [this, fittedAlpha, fittedBeta, fittedInf, fittedSup](double x) {
00232             return _betaCdfScaled(x, fittedAlpha, fittedBeta, fittedInf, fittedSup);
00233         });
00234     }
00235 
00236     virtual void fitWeibull(double *sqrerror, double *alpha, double *scale) override {
00237         if (!_ensureDataLoaded() || _count < 2 || _hasNegativeData || !(_sampleMean > 0.0) ||
      !(_sampleVariance > 0.0)) {
00238             _setFailure3(sqrerror, alpha, scale);
00239             return;
00240         }
00241 
00242         const double cv = _sampleStddev / _sampleMean;
00243         double fittedShape = _nan();
```

```
00244         if (!_estimateWeibullShapeFromCv(cv, &fittedShape)) {
00245             _setFailure3(sqrerror, alpha, scale);
00246             return;
00247         }
00248 
00249         const double gammaTerm = std::tgamma(1.0 + 1.0 / fittedShape);
00250         const double fittedScale = _sampleMean / gammaTerm;
00251         if (!_isFinitePositive(fittedShape) || !_isFinitePositive(fittedScale)) {
00252             _setFailure3(sqrerror, alpha, scale);
00253             return;
00254         }
00255 
00256         if (alpha != nullptr) {
00257             *alpha = fittedShape;
00258         }
00259         if (scale != nullptr) {
00260             *scale = fittedScale;
00261         }
00262 
00263         _setSse(sqrerror, [this, fittedShape, fittedScale](double x) {
00264             return _weibullCdf(x, fittedShape, fittedScale);
00265         });
00266     }
00267 
00268     virtual void fitAll(double *sqrerror, std::string *name) override {
00269         if (!_ensureDataLoaded()) {
00270             if (sqrerror != nullptr) {
00271                 *sqrerror = std::numeric_limits<double>::infinity();
00272             }
00273             if (name != nullptr) {
00274                 *name = "invalid-dataset";
00275             }
00276             return;
00277         }
00278 
00279         double bestError = std::numeric_limits<double>::infinity();
00280         std::string bestName = "no-valid-fit";
00281 
00282         auto consider = [&bestError, &bestName](double error, const std::string& candidate) {
00283             if (std::isfinite(error) && error < bestError) {
00284                 bestError = error;
00285                 bestName = candidate;
00286             }
00287         };
00288 
00289         double error = std::numeric_limits<double>::infinity();
00290         double p1, p2, p3, p4;
00291 
00292         fitUniform(&error, &p1, &p2);
00293         consider(error, "uniform");
00294 
00295         fitTriangular(&error, &p1, &p2, &p3);
00296         consider(error, "triangular");
00297 
00298         fitNormal(&error, &p1, &p2);
00299         consider(error, "normal");
00300 
00301         fitExpo(&error, &p1);
00302         consider(error, "exponential");
00303 
00304         fitErlang(&error, &p1, &p2);
00305         consider(error, "erlang");
00306 
00307         fitBeta(&error, &p1, &p2, &p3, &p4);
00308         consider(error, "beta");
00309 
00310         fitWeibull(&error, &p1, &p2);
00311         consider(error, "weibull");
00312 
00313         if (sqrerror != nullptr) {
00314             *sqrerror = bestError;
00315         }
00316         if (name != nullptr) {
00317             *name = bestName;
00318         }
00319     }
00320 
00321     virtual void setDataFilename(std::string dataFilename) override {
00322         if (_dataFilename != dataFilename) {
00323             _dataFilename = dataFilename;
00324             _invalidateCache();
00325         }
00326     }
00327 
00328     virtual std::string getDataFilename() override {
00329         return _dataFilename;
00330     }
```

```
00331
00332 private:
00333     void _invalidateCache() {
00334         _cacheLoaded = false;
00335         _cacheUsable = false;
00336         _data.clear();
00337         _sortedData.clear();
00338         _count = 0;
00339         _sampleMin = _nan();
00340         _sampleMax = _nan();
00341         _sampleMean = _nan();
00342         _sampleVariance = _nan();
00343         _sampleStddev = _nan();
00344         _hasNegativeData = false;
00345     }
00346
00347     bool _ensureDataLoaded() {
00348         if (!_cacheLoaded) {
00349             _cacheUsable = _loadDataFromBinaryFile();
00350             _cacheLoaded = true;
00351         }
00352         return _cacheUsable;
00353     }
00354
00355     bool _loadDataFromBinaryFile() {
00356         _invalidateCache();
00357         if (_dataFilename.empty()) {
00358             return false;
00359         }
00360
00361         std::ifstream file(_dataFilename.c_str(), std::ios::in | std::ios::binary | std::ios::ate);
00362         if (!file.good()) {
00363             return false;
00364         }
00365
00366         const std::streamsize bytes = file.tellg();
00367         if (bytes <= 0 || (bytes % static_cast<std::streamsize>(sizeof(double))) != 0) {
00368             return false;
00369         }
00370
00371         file.seekg(0, std::ios::beg);
00372         const std::size_t n = static_cast<std::size_t>(bytes /
      static_cast<std::streamsize>(sizeof(double)));
00373         std::vector<double> values(n, 0.0);
00374         file.read(reinterpret_cast<char*>(values.data()), bytes);
00375         if (!file || file.gcount() != bytes) {
00376             return false;
00377         }
00378
00379         for (double value : values) {
00380             if (!std::isfinite(value)) {
00381                 return false;
00382             }
00383         }
00384
00385         _data = values;
00386         _sortedData = _data;
00387         std::sort(_sortedData.begin(), _sortedData.end());
00388         _count = _data.size();
00389         if (_count == 0) {
00390             return false;
00391         }
00392
00393         _sampleMin = _sortedData.front();
00394         _sampleMax = _sortedData.back();
00395         const double sum = std::accumulate(_data.begin(), _data.end(), 0.0);
00396         _sampleMean = sum / static_cast<double>(_count);
00397         _hasNegativeData = (_sampleMin < 0.0);
00398
00399         if (_count >= 2) {
00400             double sq = 0.0;
00401             for (double x : _data) {
00402                 const double d = x - _sampleMean;
00403                 sq += d * d;
00404             }
00405             _sampleVariance = sq / static_cast<double>(_count - 1U);
00406             _sampleStddev = std::sqrt(_sampleVariance);
00407         } else {
00408             _sampleVariance = _nan();
00409             _sampleStddev = _nan();
00410         }
00411         return true;
00412     }
00413
00414     void _setSse(double* sqrerror, const std::function<double(double)>& cdf) {
00415         if (sqrerror == nullptr) {
00416             return;
```

```
00417         }
00418         if (_sortedData.empty()) {
00419             *sqrerror = std::numeric_limits<double>::infinity();
00420             return;
00421         }
00422         double sse = 0.0;
00423         const std::size_t n = _sortedData.size();
00424         for (std::size_t i = 0; i < n; ++i) {
00425             const double pi = (static_cast<double>(i) + 0.5) / static_cast<double>(n);
00426             double model = cdf(_sortedData[i]);
00427             if (!std::isfinite(model)) {
00428                 *sqrerror = std::numeric_limits<double>::infinity();
00429                 return;
00430             }
00431             model = std::max(0.0, std::min(1.0, model));
00432             const double d = model - pi;
00433             sse += d * d;
00434         }
00435         *sqrerror = sse;
00436     }
00437
00438     static double _clampUnitInterval(double value) {
00439         return std::max(0.0, std::min(1.0, value));
00440     }
00441
00442     static bool _isFinitePositive(double value) {
00443         return std::isfinite(value) && value > 0.0;
00444     }
00445
00446     static double _betaPdfSafe(double x, double alpha, double beta) {
00447         if (!(_isFinitePositive(alpha) && _isFinitePositive(beta))) {
00448             return std::numeric_limits<double>::quiet_NaN();
00449         }
00450         const double eps = 1e-12;
00451         const double xc = std::max(eps, std::min(1.0 - eps, x));
00452         const double pdf = ProbabilityDistributionBase::beta(xc, alpha, beta);
00453         if (!std::isfinite(pdf) || pdf < 0.0) {
00454             return std::numeric_limits<double>::quiet_NaN();
00455         }
00456         return pdf;
00457     }
00458
00459     double _betaCdfScaled(double x, double alpha, double beta, double infLimit, double supLimit) const
    {
00460         if (!(_isFinitePositive(alpha) && _isFinitePositive(beta))) {
00461             return _nan();
00462         }
00463         if (!std::isfinite(infLimit) || !std::isfinite(supLimit) || !(supLimit > infLimit)) {
00464             return _nan();
00465         }
00466         if (x <= infLimit) {
00467             return 0.0;
00468         }
00469         if (x >= supLimit) {
00470             return 1.0;
00471         }
00472         const double y = (x - infLimit) / (supLimit - infLimit);
00473         if (y <= 0.0) {
00474             return 0.0;
00475         }
00476         if (y >= 1.0) {
00477             return 1.0;
00478         }
00479         SolverDefaultImpl1 solver(1e-5, 2000);
00480         const double cdf = solver.integrate(0.0, y, &_betaPdfSafe, alpha, beta);
00481         if (!std::isfinite(cdf)) {
00482             return _nan();
00483         }
00484         return _clampUnitInterval(cdf);
00485     }
00486
00487     double _weibullCdf(double x, double alpha, double scale) const {
00488         if (!(_isFinitePositive(alpha) && _isFinitePositive(scale))) {
00489             return _nan();
00490         }
00491         if (x < 0.0) {
00492             return 0.0;
00493         }
00494         const double ratio = x / scale;
00495         const double p = std::pow(ratio, alpha);
00496         if (!std::isfinite(p)) {
00497             return _nan();
00498         }
00499         return _clampUnitInterval(1.0 - std::exp(-p));
00500     }
00501
00502     bool _estimateScaledBetaMoments(double* alpha, double* beta, double* infLimit, double* supLimit) {
```

```
00503         if (!_ensureDataLoaded() || _count < 2 || !(_sampleMax > _sampleMin) || !(_sampleVariance > 0.0)) {
00504             return false;
00505         }
00506 
00507         const double a = _sampleMin;
00508         const double b = _sampleMax;
00509         const double range = b - a;
00510         if (!(range > 0.0) || !std::isfinite(range)) {
00511             return false;
00512         }
00513 
00514         const double eps = 1e-12;
00515         double sum = 0.0;
00516         for (double x : _data) {
00517             const double y = (x - a) / range;
00518             const double yc = std::max(eps, std::min(1.0 - eps, y));
00519             sum += yc;
00520         }
00521         const double n = static_cast<double>(_data.size());
00522         const double m = sum / n;
00523         if (!(m > 0.0 && m < 1.0) || !std::isfinite(m)) {
00524             return false;
00525         }
00526 
00527         double varSum = 0.0;
00528         for (double x : _data) {
00529             const double y = (x - a) / range;
00530             const double yc = std::max(eps, std::min(1.0 - eps, y));
00531             const double d = yc - m;
00532             varSum += d * d;
00533         }
00534         const double v = varSum / static_cast<double>(_data.size() - 1U);
00535         if (!(v > 0.0) || !std::isfinite(v)) {
00536             return false;
00537         }
00538 
00539         const double common = (m * (1.0 - m) / v) - 1.0;
00540         const double aShape = m * common;
00541         const double bShape = (1.0 - m) * common;
00542         if (!(common > 0.0) || !_isFinitePositive(aShape) || !_isFinitePositive(bShape)) {
00543             return false;
00544         }
00545 
00546         if (alpha != nullptr) {
00547             *alpha = aShape;
00548         }
00549         if (beta != nullptr) {
00550             *beta = bShape;
00551         }
00552         if (infLimit != nullptr) {
00553             *infLimit = a;
00554         }
00555         if (supLimit != nullptr) {
00556             *supLimit = b;
00557         }
00558         return true;
00559     }
00560 
00561     bool _estimateWeibullShapeFromCv(double cv, double* shape) const {
00562         if (!(cv > 0.0) || !std::isfinite(cv) || shape == nullptr) {
00563             return false;
00564         }
00565         const double cv2 = cv * cv;
00566         auto g = [cv2](double k) {
00567             const double g1 = std::tgamma(1.0 + 1.0 / k);
00568             const double g2 = std::tgamma(1.0 + 2.0 / k);
00569             if (!(std::isfinite(g1) && std::isfinite(g2) && g1 > 0.0)) {
00570                 return std::numeric_limits<double>::quiet_NaN();
00571             }
00572             return (g2 / (g1 * g1)) - 1.0 - cv2;
00573         };
00574 
00575         double lo = 0.1;
00576         double hi = 100.0;
00577         double flo = g(lo);
00578         double fhi = g(hi);
00579         if (!(std::isfinite(flo) && std::isfinite(fhi))) {
00580             return false;
00581         }
00582         if (flo * fhi > 0.0) {
00583             return false;
00584         }
00585 
00586         for (int i = 0; i < 100; ++i) {
00587             const double mid = 0.5 * (lo + hi);
00588             const double fmid = g(mid);
```

```
00589             if (!std::isfinite(fmid)) {
00590                 return false;
00591             }
00592             if (std::abs(fmid) < 1e-10 || (hi - lo) < 1e-8) {
00593                 *shape = mid;
00594                 return true;
00595             }
00596             if (flo * fmid > 0.0) {
00597                 lo = mid;
00598                 flo = fmid;
00599             } else {
00600                 hi = mid;
00601                 fhi = fmid;
00602             }
00603         }
00604 
00605         *shape = 0.5 * (lo + hi);
00606         return _isFinitePositive(*shape);
00607     }
00608 
00609     static double _nan() {
00610         return std::numeric_limits<double>::quiet_NaN();
00611     }
00612 
00613     void _setFailure2(double* v1, double* v2) {
00614         if (v1 != nullptr) {
00615             *v1 = std::numeric_limits<double>::infinity();
00616         }
00617         if (v2 != nullptr) {
00618             *v2 = _nan();
00619         }
00620     }
00621 
00622     void _setFailure3(double* v1, double* v2, double* v3) {
00623         _setFailure2(v1, v2);
00624         if (v3 != nullptr) {
00625             *v3 = _nan();
00626         }
00627     }
00628 
00629     void _setFailure4(double* v1, double* v2, double* v3, double* v4) {
00630         _setFailure3(v1, v2, v3);
00631         if (v4 != nullptr) {
00632             *v4 = _nan();
00633         }
00634     }
00635 
00636     void _setFailure5(double* v1, double* v2, double* v3, double* v4, double* v5) {
00637         _setFailure4(v1, v2, v3, v4);
00638         if (v5 != nullptr) {
00639             *v5 = _nan();
00640         }
00641     }
00642 
00643 private:
00644     std::string _dataFilename = "";
00645 
00646     bool _cacheLoaded = false;
00647     bool _cacheUsable = false;
00648     std::vector<double> _data;
00649     std::vector<double> _sortedData;
00650 
00651     std::size_t _count = 0;
00652     double _sampleMin = _nan();
00653     double _sampleMax = _nan();
00654     double _sampleMean = _nan();
00655     double _sampleVariance = _nan();
00656     double _sampleStddev = _nan();
00657     bool _hasNegativeData = false;
00658 };
00659 
00660 #endif /* FITTERDEFAULTIMPL_H */
```

18.579 FitterDummyImpl.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006 
00007 /*
00008  * File:   FitterDummyImpl.h
00009  * Author: cancian
00010  *
```

```
00011  * Created on 23 de Agosto de 2018, 15:36
00012  */
00013
00014 #ifndef FITTERDUMMYIMPL_H
00015 #define FITTERDUMMYIMPL_H
00016
00017 #include "Fitter_if.h"
00018
00033 class FitterDummyImpl : public Fitter_if {
00034 public:
00035     FitterDummyImpl();
00036     FitterDummyImpl(const FitterDummyImpl& orig);
00037     ~FitterDummyImpl();
00038 public:
00039     bool isNormalDistributed(double confidencelevel);
00040     void fitUniform(double *sqrerror, double *min, double *max);
00041     void fitTriangular(double *sqrerror, double *min, double *mo, double *max);
00042     void fitNormal(double *sqrerror, double *avg, double *stddev);
00043     void fitExpo(double *sqrerror, double *avg1);
00044     void fitErlang(double *sqrerror, double *avg, double *m);
00045     void fitBeta(double *sqrerror, double *alpha, double *beta, double *infLimit, double *supLimit);
00046     void fitWeibull(double *sqrerror, double *alpha, double *scale);
00047     void fitAll(double *sqrerror, std::string *name);
00048 public:
00049     void setDataFilename(std::string dataFilename);
00050     std::string getDataFilename();
00051 private:
00052     std::string _dataFilename = "";
00053 };
00054
00055 #endif /* FITTERDUMMYIMPL_H */
```

18.580 HypothesisTester_if.h

```
00001 /*
00002  * To change this license header, choose License Headers in Project Properties.
00003  * To change this template file, choose Tools | Templates
00004  * and open the template in the editor.
00005  */
00006
00007 /*
00008  * File:   HypothesisTester_if.h
00009  * Author: rafael.luiz.cancian
00010  *
00011  * Created on 23 de Agosto de 2018, 19:04
00012  */
00013
00014 #ifndef HYPOTHESISTESTER_IF_H
00015 #define HYPOTHESISTESTER_IF_H
00016
00017 #include <string>
00018 #include "kernel/statistics/StatisticsDataFile_if.h"
00019
00020 typedef bool (*checkProportionFunction)(double value);
00021
00036 class HypothesisTester_if {
00037 public:
00038     virtual ~HypothesisTester_if() = default;
00039
00043     class ConfidenceInterval {
00044     public:
00045
00046         ConfidenceInterval(double inferiorLimit, double superiorLimit, double e0) {
00047             if (inferiorLimit <= superiorLimit) {
00048                 _infLim = inferiorLimit;
00049                 _supLim = superiorLimit;
00050                 _e0 = e0;
00051             } else {
00052                 _infLim = superiorLimit;
00053                 _supLim = inferiorLimit;
00054             }
00055         }
00056
00057         double inferiorLimit() {
00058             return _infLim;
00059         }
00060
00061         double superiorLimit() {
00062             return _supLim;
00063         }
00064
00065         double halfWidth() {
00066             return _e0;
00067         }
00068
```

```
00069     private:
00070         double _infLim, _supLim, _e0;
00071     };
00072
00077     enum H1Comparition {
00078         DIFFERENT = 1,
00079         LESS_THAN = 2,
00080         GREATER_THAN = 3
00081     };
00082
00089     class TestResult {
00090     public:
00091
00092         TestResult(double pvalue, bool rejectH0, double acceptanceInferiorLimit, double
      acceptanceSuperiorLimit, double testStat) {
00093             _pvalue = pvalue;
00094             _rejectH0 = rejectH0;
00095             _acceptanceInferiorLimit = acceptanceInferiorLimit;
00096             _acceptanceSuperiorLimit = acceptanceSuperiorLimit;
00097             _testStat = testStat;
00098         }
00099
00100         inline bool rejectH0() const {
00101             return _rejectH0;
00102         }
00103
00104         inline bool acceptH0() const {
00105             return !_rejectH0;
00106         }
00107
00108         inline double pValue() const {
00109             return _pvalue;
00110         }
00111
00112         inline double testStat() const {
00113             return _testStat;
00114         }
00115
00116         inline double acceptanceInferiorLimit() const {
00117             return _acceptanceInferiorLimit;
00118         }
00119
00120         inline double acceptanceSuperiorLimit() const {
00121             return _acceptanceSuperiorLimit;
00122         }
00123
00124     private:
00125         double _pvalue, _acceptanceInferiorLimit, _acceptanceSuperiorLimit, _testStat;
00126         bool _rejectH0;
00127     };
00128
00129
00130 public:
00131     // One-population confidence intervals.
00132     virtual HypothesisTester_if::ConfidenceInterval averageConfidenceInterval(double avg, double
      stddev, unsigned int n, double confidenceLevel) = 0;
00133     virtual HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval(double prop, unsigned
      int n, double confidenceLevel) = 0;
00134     virtual HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval(double prop, unsigned
      int n, int N, double confidenceLevel) = 0;
00135     virtual HypothesisTester_if::ConfidenceInterval varianceConfidenceInterval(double var, unsigned
      int n, double confidenceLevel) = 0;
00136     // Two-population confidence intervals.
00137     virtual HypothesisTester_if::ConfidenceInterval averageDifferenceConfidenceInterval(double avg1,
      double stddev1, unsigned int n1, double avg2, double stddev2, unsigned int n2, double confidenceLevel)
      = 0;
00138     virtual HypothesisTester_if::ConfidenceInterval proportionDifferenceConfidenceInterval(double
      avg1, double stddev1, unsigned int n1, double avg2, double stddev2, unsigned int n2, double
      confidenceLevel) = 0;
00139     virtual HypothesisTester_if::ConfidenceInterval varianceRatioConfidenceInterval(double var1,
      unsigned int n1, double var2, unsigned int n2, double confidenceLevel) = 0;
00140     // File-based confidence intervals.
00141     virtual HypothesisTester_if::ConfidenceInterval averageConfidenceInterval(std::string
      sampleDataFilename, double confidenceLevel) = 0;
00142     virtual HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval(std::string
      sampleDataFilename, checkProportionFunction function, double confidenceLevel) = 0;
00143     virtual HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval(std::string
      sampleDataFilename, checkProportionFunction function, double N, double confidenceLevel) = 0;
00144     virtual HypothesisTester_if::ConfidenceInterval varianceConfidenceInterval(std::string
      sampleDataFilename, double confidenceLevel) = 0;
00145     // determine sample size based on confidence interval
00146     virtual unsigned int estimateSampleSize(double avg, double stddev, double desiredE0, double
      confidenceLevel) = 0;
00147
00148     // One-population tests.
00149     virtual HypothesisTester_if::TestResult testAverage(double avg, double stddev, unsigned int n,
      double avgSample, double confidenceLevel, HypothesisTester_if::H1Comparition comp) = 0;
```

```
00150     virtual HypothesisTester_if::TestResult testProportion(double prop, unsigned int n, double
      proptest, double confidenceLevel, HypothesisTester_if::H1Comparition comp) = 0;
00151     virtual HypothesisTester_if::TestResult testVariance(double var, unsigned int n, double vartest,
      double confidenceLevel, HypothesisTester_if::H1Comparition comp) = 0;
00152     // Two-population tests.
00153     virtual HypothesisTester_if::TestResult testAverage(double avg1, double stddev1, unsigned int n1,
      double avg2, double stddev2, unsigned int n2, double confidenceLevel,
      HypothesisTester_if::H1Comparition comp) = 0;
00154     virtual HypothesisTester_if::TestResult testProportion(double prop1, unsigned int n1, double
      prop2, unsigned int n2, double confidenceLevel, HypothesisTester_if::H1Comparition comp) = 0;
00155     virtual HypothesisTester_if::TestResult testVariance(double var1, unsigned int n1, double var2,
      unsigned int n2, double confidenceLevel, HypothesisTester_if::H1Comparition comp) = 0;
00156     // One-population file-based tests.
00157     virtual HypothesisTester_if::TestResult testAverage(std::string sampleDataFilename, double
      avgSample, double confidenceLevel, HypothesisTester_if::H1Comparition comp) = 0;
00158     virtual HypothesisTester_if::TestResult testProportion(std::string sampleDataFilename,
      checkProportionFunction function, double proptest, double confidenceLevel,
      HypothesisTester_if::H1Comparition comp) = 0;
00159     virtual HypothesisTester_if::TestResult testVariance(std::string sampleDataFilename, double
      vartest, double confidenceLevel, HypothesisTester_if::H1Comparition comp) = 0;
00160     // Two-population file-based tests.
00161     virtual HypothesisTester_if::TestResult testAverage(std::string firstSampleDataFilename,
      std::string secondSampleDataFilename, double confidenceLevel, HypothesisTester_if::H1Comparition comp)
      = 0;
00162     virtual HypothesisTester_if::TestResult testProportion(std::string firstSampleDataFilename,
      std::string secondSampleDataFilename, checkProportionFunction function, double confidenceLevel,
      HypothesisTester_if::H1Comparition comp) = 0;
00163     virtual HypothesisTester_if::TestResult testVariance(std::string firstSampleDataFilename,
      std::string secondSampleDataFilename, double confidenceLevel, HypothesisTester_if::H1Comparition comp)
      = 0;
00164 };
00165
00166 #endif /* HYPOTHESISTESTER_IF_H */
```

18.581 HypothesisTesterDefaultImpl1.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   HypothesisTesterDefaultImpl1.h
00008  * Author: rlcancian
00009  *
00010  * Created on 24 de novembro de 2021, 02:52
00011  */
00012
00013 #ifndef HYPOTHESISTESTERDEFAULTIMPL1_H
00014 #define HYPOTHESISTESTERDEFAULTIMPL1_H
00015
00016 #include "HypothesisTester_if.h"
00017
00033 class HypothesisTesterDefaultImpl1 : public HypothesisTester_if {
00034 public:
00035     HypothesisTesterDefaultImpl1();
00036     ~HypothesisTesterDefaultImpl1() = default;
00037 public:
00038     // confidence intervals of parameters
00039     HypothesisTester_if::ConfidenceInterval averageConfidenceInterval(double avg, double stddev,
      unsigned int n, double confidenceLevel) override;
00040     HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval(double prop, unsigned int n,
      double confidenceLevel) override;
00041     HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval(double prop, unsigned int n,
      int N, double confidenceLevel) override;
00042     HypothesisTester_if::ConfidenceInterval varianceConfidenceInterval(double var, unsigned int n,
      double confidenceLevel) override;
00043     // confidence intervals of differences of parameters
00044     HypothesisTester_if::ConfidenceInterval averageDifferenceConfidenceInterval(double avg1, double
      stddev1, unsigned int n1, double avg2, double stddev2, unsigned int n2, double confidenceLevel)
      override;
00045     HypothesisTester_if::ConfidenceInterval proportionDifferenceConfidenceInterval(double avg1, double
      stddev1, unsigned int n1, double avg2, double stddev2, unsigned int n2, double confidenceLevel)
      override;
00046     HypothesisTester_if::ConfidenceInterval varianceRatioConfidenceInterval(double var1, unsigned int
      n1, double var2, unsigned int n2, double confidenceLevel) override;
00047     // confidence intervals based on datafile
00048     HypothesisTester_if::ConfidenceInterval averageConfidenceInterval(std::string sampleDataFilename,
      double confidenceLevel) override;
00049     HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval(std::string
      sampleDataFilename, checkProportionFunction function, double confidenceLevel) override;
00050     HypothesisTester_if::ConfidenceInterval proportionConfidenceInterval(std::string
      sampleDataFilename, checkProportionFunction function, double N, double confidenceLevel) override;
00051     HypothesisTester_if::ConfidenceInterval varianceConfidenceInterval(std::string sampleDataFilename,
      double confidenceLevel) override;
```

```
00052     // determine sample size based on confidence interval
00053     unsigned int estimateSampleSize(double avg, double stddev, double desiredE0, double
      confidenceLevel) override;
00054     // one population
00055     HypothesisTester_if::TestResult testAverage(double avg, double stddev, unsigned int n, double
      avgSample, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override;
00056     HypothesisTester_if::TestResult testProportion(double prop, unsigned int n, double proptest,
      double confidenceLevel, HypothesisTester_if::H1Comparition comp) override;
00057     HypothesisTester_if::TestResult testVariance(double var, unsigned int n, double vartest, double
      confidenceLevel, HypothesisTester_if::H1Comparition comp) override;
00058     // two populations
00059     HypothesisTester_if::TestResult testAverage(double avg1, double stddev1, unsigned int n1, double
      avg2, double stddev2, unsigned int n2, double confidenceLevel, HypothesisTester_if::H1Comparition
      comp) override;
00060     HypothesisTester_if::TestResult testProportion(double prop1, unsigned int n1, double prop2,
      unsigned int n2, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override;
00061     HypothesisTester_if::TestResult testVariance(double var1, unsigned int n1, double var2, unsigned
      int n2, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override;
00062     // one population based on datafile
00063     HypothesisTester_if::TestResult testAverage(std::string sampleDataFilename, double avgSample,
      double confidenceLevel, HypothesisTester_if::H1Comparition comp) override;
00064     HypothesisTester_if::TestResult testProportion(std::string sampleDataFilename,
      checkProportionFunction function, double proptest, double confidenceLevel,
      HypothesisTester_if::H1Comparition comp) override;
00065     HypothesisTester_if::TestResult testVariance(std::string sampleDataFilename, double vartest,
      double confidenceLevel, HypothesisTester_if::H1Comparition comp) override;
00066     // two populations based on datafile
00067     HypothesisTester_if::TestResult testAverage(std::string firstSampleDataFilename, std::string
      secondSampleDataFilename, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override;
00068     HypothesisTester_if::TestResult testProportion(std::string firstSampleDataFilename, std::string
      secondSampleDataFilename, checkProportionFunction function, double confidenceLevel,
      HypothesisTester_if::H1Comparition comp) override;
00069     HypothesisTester_if::TestResult testVariance(std::string firstSampleDataFilename, std::string
      secondSampleDataFilename, double confidenceLevel, HypothesisTester_if::H1Comparition comp) override;
00070 };
00071
00072 #endif /* HYPOTHESISTESTERDEFAULTIMPL1_H */
```

18.582 ProbabilityDistribution.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   ProbabilityDistribution.h
00008  * Author: rlcancian
00009  *
00010  * Created on 22 de novembro de 2021, 17:24
00011  */
00012
00013 #ifndef PROBABILITYDISTRIBUTION_H
00014 #define PROBABILITYDISTRIBUTION_H
00015
00016 #include "ProbabilityDistributionBase.h"
00017
00033 class ProbabilityDistribution : public ProbabilityDistributionBase {
00034 public:
00035     static double inverseChi2(double cumulativeProbability, double degreeFreedom);
00036     static double inverseFFisherSnedecor(double cumulativeProbability, double d1, double d2);
00037     static double inverseNormal(double cumulativeProbability, double mean, double stddev);
00038     static double inverseTStudent(double cumulativeProbability, double mean, double stddev, double
      degreeFreedom);
00039 private:
00040     static double _findInverseChi2(double a, double fa, double b, double fb, unsigned int recursions,
      double cumulativeProbability, double degreeFreedom);
00041     static double _findInverseFFisherSnedecor(double a, double fa, double b, double fb, unsigned int
      recursions, double cumulativeProbability, double d1, double d2);
00042     static double _findInverseNormal(double a, double fa, double b, double fb, unsigned int
      recursions, double cumulativeProbability, double mean, double stddev);
00043     static double _findInverseTStudent(double a, double fa, double b, double fb, unsigned int
      recursions, double cumulativeProbability, double mean, double stddev, double degreeFreedom);
00044 private:
00049     static std::map<std::string, double>* memory; // = std::map<std::string, double>();
00053     static Solver_if* integrator;
00054 };
00055 #endif /* PROBABILITYDISTRIBUTION_H */
```

18.583 ProbabilityDistributionBase.h

```
00001 /*
```

```
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/class.h to edit this template
00004  */
00005
00006 /*
00007  * File:   BaseProbabilityDistribution.h
00008  * Author: rlcancian
00009  *
00010  * Created on 4 de agosto de 2022, 11:03
00011  */
00012
00013 #ifndef PROBABILITYDISTRIBUTIONBASE_H
00014 #define PROBABILITYDISTRIBUTIONBASE_H
00015
00016 #include "tools/Continuous/Solver_if.h"
00017 #include <string>
00018 #include <map>
00019
00031 class ProbabilityDistributionBase {
00032 public:
00034     static double beta(double x, double alpha, double beta);
00036     static double chi2(double x, double degreeFreedom);
00038     static double erlang(double x, double shape, double scale); // int M
00040     static double exponential(double x, double mean);
00042     static double fisherSnedecor(double x, double d1, double d2);
00044     static double gamma(double x, double shape, double scale);
00046     static double logNormal(double x, double mean, double stddev);
00048     static double normal(double x, double mean, double stddev);
00050     static double poisson(double x, double mean);
00052     static double triangular(double x, double min, double mode, double max);
00054     static double tStudent(double x, double mean, double stddev, double degreeFreedom);
00056     static double uniform(double x, double min, double max);
00058     static double weibull(double x, double shape, double scale);
00059 protected:
00061     static double _gammaFunction(double x);
00063     static double _betaFunction(double x, double y);
00065     static double _factorial(double x);
00066 };
00067
00068 #endif /* PROBABILITYDISTRIBUTIONBASE_H */
```

18.584 SimulationResultsDataset.h

```
00001 #ifndef SIMULATIONRESULTSDATASET_H
00002 #define SIMULATIONRESULTSDATASET_H
00003
00004 #include <string>
00005 #include <vector>
00006
00007 enum class SimulationResultsDatasetFormat {
00008     Unknown,
00009     RawNumeric,
00010     RecordLegacy,
00011     RecordEnriched,
00012     GuiTabular
00013 };
00014
00015 struct SimulationResultsObservation {
00016     unsigned int replication = 1;
00017     double time = 0.0;
00018     bool hasTime = false;
00019     double value = 0.0;
00020     unsigned int sourceLine = 0;
00021 };
00022
00023 struct SimulationResultsDataset {
00024     std::string datasetName;
00025     std::string randomVariableName;
00026     std::string description;
00027     std::string variableType = "Continuous numeric";
00028     std::string source;
00029     std::string sourceDescription;
00030     std::string expression;
00031     std::string expressionName;
00032     std::string formatVersion;
00033     SimulationResultsDatasetFormat formatKind = SimulationResultsDatasetFormat::Unknown;
00034     bool recordFile = false;
00035     bool timeDependent = false;
00036     std::vector<SimulationResultsObservation> observations;
00037     std::vector<std::string> previewLines;
00038
00042     std::string formatKindName() const;
00043
00047     bool isRecordFormat() const;
00048
```

```
00052     std::vector<double> values() const;
00053
00057     std::vector<unsigned int> replications() const;
00058
00062     std::vector<double> valuesForReplication(unsigned int replication) const;
00063 };
00064
00065 class SimulationResultsDatasetParser {
00066 public:
00074     static bool loadFromTextFile(const std::string& fileName, SimulationResultsDataset* dataset,
00075                                  std::string* errorMessage = nullptr);
00076 };
00077
00078 #endif /* SIMULATIONRESULTSDATASET_H */
```

18.585 TraitsTools.h

```
00001 /*
00002  * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
00003  * Click nbfs://nbhost/SystemFileSystem/Templates/cppFiles/file.h to edit this template
00004  */
00005
00006 /*
00007  * File:   TraitsTools.h
00008  * Author: rlcancian
00009  *
00010  * Created on 3 de maio de 2022, 18:57
00011  */
00012
00013 #ifndef TRAITSTOOLS_H
00014 #define TRAITSTOOLS_H
00015
00016 // TOOLS
00017 #include "AIAssistant/AIAssistant_if.h"
00018 #include "AIAssistant/AIAssistantDefaultImpl.h"
00019 #include "Continuous/SolverDefaultImpl1.h"
00020 #include "Statistics/HypothesisTesterDefaultImpl1.h"
00021 #include "Statistics/HypothesisTester_if.h"
00022 #include "Statistics/FitterDefaultImpl.h"
00023 #include "Statistics/Fitter_if.h"
00024
00039 template <typename T>
00040 struct TraitsTools {
00041 };
00042
00046 template <> struct TraitsTools<Solver_if> {
00047     typedef SolverDefaultImpl1 Implementation;
00048     static constexpr double Precision = 1e-5;
00049     static constexpr unsigned int MaxSteps = 1e2;
00050 };
00051
00055 template <> struct TraitsTools<HypothesisTester_if> {
00056     typedef HypothesisTesterDefaultImpl1 Implementation;
00057     static constexpr unsigned int ConfidenceLevel = 95;
00058 };
00059
00063 template <> struct TraitsTools<Fitter_if> {
00064     typedef FitterDefaultImpl Implementation;
00065 };
00066
00070 template <> struct TraitsTools<AIAssistant_if> {
00071     typedef AIAssistantDefaultImpl Implementation;
00072 };
00073
00074 #endif /* TRAITSTOOLS_H */
```

Index

_activate
ColoredPetriNetNetwork, 423
DefaultNetwork, 528
EFSMNetwork, 643
MarkovChainNetwork, 970
_addSimulationControl
Buffer, 338
ModelDataDefinition, 1052
PickStation, 1246
Schedule, 2125
_apiKey
HttpProviderClientBase, 882
_authorizationHeader
AnthropicProviderClientImpl, 201
HttpProviderClientBase, 880
LocalProviderClientImpl, 931
OpenAIProviderClientImpl, 1170
_betaFunction
ProbabilityDistributionBase, 1280
_buildRequestBody
AnthropicProviderClientImpl, 202
HttpProviderClientBase, 880
LocalProviderClientImpl, 931
OpenAIProviderClientImpl, 1170
_check
Access, 122
AIAssistant, 130
AISupport, 172
Assign, 211
Attribute, 217
BacteriaColony, 226
BacteriaSignalGrid, 233
Batch, 247
BioCompartment, 253
BioNetwork, 259
BioParameter, 268
BioReaction, 276
BioRunnerCommand, 290
BioSimulate, 299
BioSimulatorRunner, 307
BioSpecies, 312
BioStateProjectionComponent, 319
BioSteadyState, 326
Buffer, 338
CellCycleCheckpointComponent, 359
CellDivisionEvent, 365
CellFateDecisionComponent, 372
CellGrowthComponent, 378
CellularAutomataComp, 391
Clone, 405
ColoredPetriNetNetwork, 423
CompartmentExchangeComponent, 431
ContinuousSystemComponent, 453
Conveyor, 458
Counter, 465
CPNArc, 475
CPNTransition, 481
CppCompiler, 486
CppForG, 492
Create, 503
Decide, 519
DefaultNetwork, 528
Delay, 542
DiffEquations, 567
DiffusionField, 572
DiffusionSimulate, 580
DirectedAcyclicGraphNetwork, 594
Dispose, 609
Distance, 614
DropOff, 624
DummyElement, 635
EFSMNetwork, 643
Enter, 661
Entity, 666
EntityGroup, 673
EukaryoticCellCycleComponent, 686
Exit, 701
Failure, 715
File, 726
Formula, 744
FtsZPolymerizationComponent, 757
GeneticCircuit, 779
GeneticCircuitPart, 786
GeneticCircuitSimulate, 792
GeneticExpressionStep, 798
GeneticRegulation, 803
GraphEdge, 813
GraphNetwork, 842
GroProgram, 855
Label, 911
Leave, 921
LSODE, 953
MarkovChain, 964
MarkovChainNetwork, 970
Match, 984
MetabolicFluxBalance, 990
MetabolicNetwork, 996
MetabolicReaction, 1001

MetabolicStateProjectionComponent, 1008
MetabolicSubmodelComponent, 1014
ModalModelDefault, 1022
ModelDataDefinition, 1052
MolecularSpecies, 1100
Move, 1106
OctaveRunner, 1154
ODESolver, 1159
OLD_ODEelement, 1165
PathwayStressResponseComponent, 1224
PickStation, 1246
Process, 1288
PythonForG, 1304
PythonRuntime, 1310
Queue, 2026
Record, 2039
Release, 2059
Remove, 2066
Resource, 2078
ResourceAllocationComponent, 2084
Route, 2092
RSimulator, 2099
RSimulatorRunner, 2105
Schedule, 2125
Search, 2132
Segment, 2138
Seize, 2145
Separate, 2152
Sequence, 2157
Set, 2169
Signal, 2175
SignalData, 2181
SinkModelComponent, 2269
SourceModelComponent, 2430
SPICERunner, 2451
Start, 2457
Station, 2469
StatisticsCollector, 2482
StochasticReactionComponent, 2500
StochasticReactionRule, 2505
StochasticTranscription, 2511
StochasticTranslation, 2518
Stop, 2526
Storage, 2531
Store, 2537
Submodel, 2544
Transporter, 2597
Unstore, 2618
Variable, 2643
Wait, 2668
WholeCellState, 2701
Write, 2717

_connections
ModelComponent, 1044

_createAttachedAttributes
BacteriaColony, 226
Batch, 247
BioReaction, 276
BioSteadyState, 326
Delay, 542
Dispose, 609
EntityGroup, 673
Formula, 744
ModalModelDefault, 1022
ModelDataDefinition, 1052
Move, 1106
Resource, 2078
Route, 2092
Sequence, 2157
Set, 2169
SourceModelComponent, 2430
Station, 2469

_createEditableDataDefinitions
Assign, 211
Batch, 247
BioCompartment, 253
BioNetwork, 259
BioRunnerCommand, 290
BioSimulate, 299
BioStateProjectionComponent, 319
Buffer, 338
CellCycleCheckpointComponent, 359
CellDivisionEvent, 365
CellFateDecisionComponent, 372
CellGrowthComponent, 378
CompartmentExchangeComponent, 431
ContinuousSystemComponent, 453
Conveyor, 458
Create, 503
DiffEquations, 567
DiffusionSimulate, 580
Enter, 661
EukaryoticCellCycleComponent, 686
FtsZPolymerizationComponent, 757
GeneticCircuitSimulate, 792
GeneticExpressionStep, 798
Leave, 921
MarkovChain, 964
Match, 984
MetabolicFluxBalance, 990
MetabolicNetwork, 996
MetabolicReaction, 1001
MetabolicStateProjectionComponent, 1008
MetabolicSubmodelComponent, 1014
ModelDataDefinition, 1052
Move, 1106
ODESolver, 1159
PathwayStressResponseComponent, 1224
PickStation, 1246
Process, 1288
Release, 2059
Remove, 2066
ResourceAllocationComponent, 2084
Route, 2093
RSimulator, 2099
Search, 2132

Seize, 2145
Signal, 2175
SourceModelComponent, 2430
StochasticReactionComponent, 2500
StochasticTranscription, 2511
StochasticTranslation, 2518
Store, 2537
Transporter, 2597
Unstore, 2618
Wait, 2668
_createInternalAndAttachedData
ModelDataDefinition, 1052
_createInternalStatisticReporters
Clone, 405
Create, 503
Decide, 519
DefaultNetwork, 528
Delay, 542
Dispose, 609
Enter, 661
EntityGroup, 673
EntityType, 678
Leave, 921
Move, 1106
Queue, 2026
Record, 2039
Route, 2093
Station, 2469
_createNonEditableDataDefinitions
Batch, 248
CppForG, 492
ModelDataDefinition, 1052
Process, 1288
PythonForG, 1304
SPICECircuit, 2440
Wait, 2668
_edgeConnects
DirectedGraphNetwork, 601
_extraRequestHeaders
AnthropicProviderClientImpl, 202
HttpProviderClientBase, 880
OpenAIProviderClientImpl, 1170
_factorial
ProbabilityDistributionBase, 1280
_gammaFunction
ProbabilityDistributionBase, 1280
_getParserChangesInformation
Buffer, 339
File, 726
ModelDataDefinition, 1052
PickStation, 1246
Queue, 2026
Schedule, 2126
Set, 2169
Storage, 2531
_getSaveDefaultsOption
ModelDataDefinition, 1052
_initBetweenReplications
AISupport, 172
BacteriaColony, 226
BioNetwork, 259
BioSpecies, 312
Buffer, 339
CellCycleCheckpointComponent, 359
CellDivisionEvent, 366
CellFateDecisionComponent, 372
CellGrowthComponent, 378
CellularAutomataComp, 391
Clone, 405
ColoredPetriNetNetwork, 423
CompartmentExchangeComponent, 431
Conveyor, 458
Counter, 465
CppCompiler, 486
CppForG, 492
Create, 503
DefaultNetwork, 528
DiffusionField, 572
Dispose, 609
EFSMNetwork, 643
EntityGroup, 673
EntityType, 678
EukaryoticCellCycleComponent, 687
Failure, 715
FtsZPolymerizationComponent, 757
GraphNetwork, 842
MarkovChain, 964
MarkovChainNetwork, 970
MetabolicSubmodelComponent, 1014
ModalModelDefault, 1022
ModelDataDefinition, 1053
MolecularSpecies, 1100
ODESolver, 1159
PathwayStressResponseComponent, 1224
PickStation, 1246
PythonForG, 1305
Queue, 2026
Record, 2039
Release, 2060
Resource, 2078
ResourceAllocationComponent, 2084
Schedule, 2126
Seize, 2146
Signal, 2176
SignalData, 2181
SourceModelComponent, 2430
Station, 2469
StochasticReactionComponent, 2500
StochasticTranscription, 2512
StochasticTranslation, 2518
Storage, 2531
Transporter, 2597
Variable, 2643
Wait, 2668
WholeCellState, 2701
Write, 2717

_loadInstance
Access, 122
AIAssistant, 130
AISupport, 172
AND, 179
Assign, 211
Attribute, 217
BacteriaColony, 226
BacteriaSignalGrid, 233
Batch, 248
BioCompartment, 254
BioNetwork, 259
BioParameter, 268
BioReaction, 276
BioRunnerCommand, 290
BioSimulate, 299
BioSimulatorRunner, 307
BioSpecies, 312
BioStateProjectionComponent, 319
BioSteadyState, 326
Buffer, 339
Capacitor, 349
CellCycleCheckpointComponent, 359
CellDivisionEvent, 366
CellFateDecisionComponent, 372
CellGrowthComponent, 378
CellularAutomataComp, 391
Clone, 405
ColoredPetriNetNetwork, 423
CompartmentExchangeComponent, 431
ContinuousSystemComponent, 453
Conveyor, 459
Counter, 465
CPNArc, 475
CPNTransition, 481
CppCompiler, 486
CppForG, 492
Create, 503
Decide, 519
DefaultNetwork, 529
DefaultNode, 534
Delay, 542
DiffEquations, 567
DiffusionField, 572
DiffusionSimulate, 580
Diode, 587
Dispose, 609
Distance, 614
DropOff, 624
DummyComponent, 630
DummyElement, 635
EFSMNetwork, 643
Enter, 661
Entity, 666
EntityGroup, 673
EntityType, 678
EukaryoticCellCycleComponent, 687
Exit, 701
Failure, 715
File, 726
Formula, 744
FSMState, 752
FtsZPolymerizationComponent, 758
GeneticCircuit, 779
GeneticCircuitPart, 786
GeneticCircuitSimulate, 792
GeneticExpressionStep, 798
GeneticRegulation, 803
GraphEdge, 813
GraphNetwork, 842
GroProgram, 855
Label, 911
Leave, 921
LSODE, 953
MarkovChain, 964
MarkovChainNetwork, 970
Match, 984
MetabolicFluxBalance, 990
MetabolicNetwork, 996
MetabolicReaction, 1001
MetabolicStateProjectionComponent, 1008
MetabolicSubmodelComponent, 1014
ModalModelDefault, 1022
ModelComponent, 1040
ModelDataDefinition, 1053
MolecularSpecies, 1100
Move, 1106
NAND, 1114
NMOS, 1131
NOR, 1138
NOT, 1144
OctaveRunner, 1154
ODESolver, 1159
OLD_ODEelement, 1165
OR, 1194
PathwayStressResponseComponent, 1224
PersistentObject_base, 1234
PetriPlace, 1240
PickStation, 1246
PMOS, 1275
Process, 1288
PythonForG, 1305
PythonRuntime, 1310
Queue, 2026
Record, 2039
Release, 2060
Remove, 2066
Resistor, 2072
Resource, 2078
ResourceAllocationComponent, 2084
Route, 2093
RSimulator, 2100
RSimulatorRunner, 2105
Schedule, 2126
Search, 2133
Segment, 2138

Seize, 2146
Separate, 2152
Sequence, 2157
SequenceStep, 2159
Set, 2169
Signal, 2176
SignalData, 2181
SinkModelComponent, 2269
SourceModelComponent, 2430
SPICECircuit, 2440
SPICENode, 2446
SPICERunner, 2451
Start, 2457
Station, 2469
StatisticsCollector, 2482
StochasticReactionComponent, 2500
StochasticReactionRule, 2505
StochasticTranscription, 2512
StochasticTranslation, 2518
Stop, 2526
Storage, 2531
Store, 2538
Submodel, 2544
Transporter, 2598
Unstore, 2619
Variable, 2643
Vpulse, 2650
Vsine, 2656
Vsource, 2662
Wait, 2669
WholeCellState, 2701
Write, 2717
XNOR, 2727
XOR, 2733

_onDispatchEvent
Access, 122
AIAssistant, 130
Assign, 211
BacteriaColony, 226
Batch, 248
BioRunnerCommand, 290
BioSimulate, 299
BioStateProjectionComponent, 320
BioSteadyState, 326
Buffer, 339
CellCycleCheckpointComponent, 359
CellDivisionEvent, 366
CellFateDecisionComponent, 372
CellGrowthComponent, 378
CellularAutomataComp, 392
Clone, 405
CompartmentExchangeComponent, 431
ContinuousSystemComponent, 453
CppForG, 492
Create, 504
Decide, 519
Delay, 542
DiffEquations, 567
DiffusionSimulate, 580
Dispose, 609
DropOff, 625
DummyComponent, 630
Enter, 661
EukaryoticCellCycleComponent, 687
Exit, 701
FtsZPolymerizationComponent, 758
GeneticCircuitSimulate, 792
GeneticExpressionStep, 798
Leave, 921
LSODE, 953
MarkovChain, 964
Match, 984
MetabolicFluxBalance, 991
MetabolicStateProjectionComponent, 1008
MetabolicSubmodelComponent, 1014
ModalModelDefault, 1022
ModelComponent, 1040
Move, 1106
PathwayStressResponseComponent, 1224
PickStation, 1247
Process, 1288
PythonForG, 1305
Record, 2039
RecordingContinuousSystemComponent, 2046
RecordingLSODE, 2052
Release, 2060
Remove, 2066
ResourceAllocationComponent, 2085
Route, 2093
RSimulator, 2100
Search, 2133
Seize, 2146
Separate, 2152
Signal, 2176
SPICECircuit, 2440
SPICENode, 2446
Start, 2457
StochasticReactionComponent, 2500
StochasticTranscription, 2512
StochasticTranslation, 2518
Stop, 2526
Store, 2538
Submodel, 2544
Unstore, 2619
Wait, 2669
Write, 2717

_oppositeNode
DirectedGraphNetwork, 601

_parseResponseBody
AnthropicProviderClientImpl, 202
HttpProviderClientBase, 880
LocalProviderClientImpl, 931
OpenAIProviderClientImpl, 1171

_saveInstance
Access, 123
AIAssistant, 130

AISupport, 172
AND, 179
Assign, 211
Attribute, 217
BacteriaColony, 227
BacteriaSignalGrid, 233
Batch, 248
BioCompartment, 254
BioNetwork, 260
BioParameter, 269
BioReaction, 277
BioRunnerCommand, 291
BioSimulate, 299
BioSimulatorRunner, 307
BioSpecies, 312
BioStateProjectionComponent, 320
BioSteadyState, 326
Buffer, 339
Capacitor, 349
CellCycleCheckpointComponent, 359
CellDivisionEvent, 366
CellFateDecisionComponent, 372
CellGrowthComponent, 379
CellularAutomataComp, 392
Clone, 406
ColoredPetriNetNetwork, 423
CompartmentExchangeComponent, 432
ContinuousSystemComponent, 453
Conveyor, 459
Counter, 465
CPNArc, 475
CPNTransition, 481
CppCompiler, 486
CppForG, 493
Create, 504
Decide, 520
DefaultNetwork, 529
DefaultNode, 534
Delay, 542
DiffEquations, 567
DiffusionField, 573
DiffusionSimulate, 581
Diode, 587
Dispose, 609
Distance, 614
DropOff, 625
DummyComponent, 631
DummyElement, 635
EFSMNetwork, 644
Enter, 662
Entity, 667
EntityGroup, 673
EntityType, 679
EukaryoticCellCycleComponent, 687
Exit, 702
Failure, 715
File, 726
Formula, 745
FSMState, 752
FtsZPolymerizationComponent, 758
GeneticCircuit, 779
GeneticCircuitPart, 786
GeneticCircuitSimulate, 792
GeneticExpressionStep, 798
GeneticRegulation, 803
GraphEdge, 813
GraphNetwork, 842
GroProgram, 855
Label, 911
Leave, 922
LSODE, 953
MarkovChain, 964
MarkovChainNetwork, 970
Match, 984
MetabolicFluxBalance, 991
MetabolicNetwork, 996
MetabolicReaction, 1001
MetabolicStateProjectionComponent, 1008
MetabolicSubmodelComponent, 1015
ModalModelDefault, 1022
ModelComponent, 1041
ModelDataDefinition, 1053
MolecularSpecies, 1100
Move, 1106
NAND, 1114
NMOS, 1131
NOR, 1138
NOT, 1144
OctaveRunner, 1154
ODESolver, 1159
OLD_ODEelement, 1166
OR, 1194
PathwayStressResponseComponent, 1225
PersistentObject_base, 1234
PetriPlace, 1240
PickStation, 1247
PMOS, 1275
Process, 1288
PythonForG, 1305
PythonRuntime, 1310
Queue, 2027
Record, 2039
Release, 2060
Remove, 2066
Resistor, 2072
Resource, 2078
ResourceAllocationComponent, 2085
Route, 2093
RSimulator, 2100
RSimulatorRunner, 2106
Schedule, 2126
Search, 2133
Segment, 2138
Seize, 2146
Separate, 2152
Sequence, 2157

SequenceStep, 2159
Set, 2169
Signal, 2176
SignalData, 2181
SinkModelComponent, 2269
SourceModelComponent, 2430
SPICECircuit, 2440
SPICENode, 2446
SPICERunner, 2451
Start, 2458
Station, 2469
StatisticsCollector, 2482
StochasticReactionComponent, 2500
StochasticReactionRule, 2505
StochasticTranscription, 2512
StochasticTranslation, 2518
Stop, 2526
Storage, 2532
Store, 2538
Submodel, 2545
Transporter, 2598
Unstore, 2619
Variable, 2643
Vpulse, 2650
Vsine, 2656
Vsource, 2662
Wait, 2669
WholeCellState, 2701
Write, 2718
XNOR, 2727
XOR, 2733
_sortFunc
List< T >, 925
~GraphicalAutomaticPositioningStrategy
GraphicalAutomaticPositioningStrategy, 815
~ParserDefaultImpl2
ParserDefaultImpl2, 1214
~QtAbstractPropertyBrowser
QtAbstractPropertyBrowser, 1814
~QtAbstractPropertyManager
QtAbstractPropertyManager, 1820
~QtBoolPropertyManager
QtBoolPropertyManager, 1827
~QtButtonPropertyBrowser
QtButtonPropertyBrowser, 1832
~QtCharEditorFactory
QtCharEditorFactory, 1836
~QtCharPropertyManager
QtCharPropertyManager, 1839
~QtCheckBoxFactory
QtCheckBoxFactory, 1842
~QtColorEditorFactory
QtColorEditorFactory, 1846
~QtColorPropertyManager
QtColorPropertyManager, 1850
~QtCursorEditorFactory
QtCursorEditorFactory, 1855
~QtCursorPropertyManager
QtCursorPropertyManager, 1857
~QtDateEditFactory
QtDateEditFactory, 1862
~QtDatePropertyManager
QtDatePropertyManager, 1865
~QtDateTimeEditFactory
QtDateTimeEditFactory, 1871
~QtDateTimePropertyManager
QtDateTimePropertyManager, 1874
~QtDoublePropertyManager
QtDoublePropertyManager, 1878
~QtDoubleSpinBoxFactory
QtDoubleSpinBoxFactory, 1884
~QtEnumEditorFactory
QtEnumEditorFactory, 1888
~QtEnumPropertyManager
QtEnumPropertyManager, 1891
~QtFlagPropertyManager
QtFlagPropertyManager, 1896
~QtFontEditorFactory
QtFontEditorFactory, 1900
~QtFontPropertyManager
QtFontPropertyManager, 1904
~QtGroupBoxPropertyBrowser
QtGroupBoxPropertyBrowser, 1909
~QtGroupPropertyManager
QtGroupPropertyManager, 1911
~QtIntPropertyManager
QtIntPropertyManager, 1914
~QtKeySequenceEditorFactory
QtKeySequenceEditorFactory, 1920
~QtKeySequencePropertyManager
QtKeySequencePropertyManager, 1923
~QtLineEditFactory
QtLineEditFactory, 1926
~QtLocalePropertyManager
QtLocalePropertyManager, 1929
~QtPointFPropertyManager
QtPointFPropertyManager, 1935
~QtPointPropertyManager
QtPointPropertyManager, 1940
~QtProperty
QtProperty, 1944
~QtRectFPropertyManager
QtRectFPropertyManager, 1953
~QtRectPropertyManager
QtRectPropertyManager, 1959
~QtScrollBarFactory
QtScrollBarFactory, 1964
~QtSizeFPropertyManager
QtSizeFPropertyManager, 1967
~QtSizePolicyPropertyManager
QtSizePolicyPropertyManager, 1973
~QtSizePropertyManager
QtSizePropertyManager, 1977
~QtSliderFactory
QtSliderFactory, 1983
~QtSpinBoxFactory

QtSpinBoxFactory, 1986
~QtStringPropertyManager
QtStringPropertyManager, 1990
~QtTimeEditFactory
QtTimeEditFactory, 1995
~QtTimePropertyManager
QtTimePropertyManager, 1998
~QtTreePropertyBrowser
QtTreePropertyBrowser, 2003
~QtVariantEditorFactory
QtVariantEditorFactory, 2008
~QtVariantProperty
QtVariantProperty, 2011
~QtVariantPropertyManager
QtVariantPropertyManager, 2014

accept
DialogSimulationConfigure, 558
accepted
QCPSelectionRect, 1749
Access, 117
_check, 122
_loadInstance, 122
_onDispatchEvent, 122
_saveInstance, 123
show, 123
activate
DefaultNetwork, 529
actualizeModelSimLanguage
ModelLanguageSynchronizer, 1085
add
QCPDataContainer< DataType >, 1477
addAxes
QCPAxisRect, 1375
addAxis
QCPAxisRect, 1375
addConstraint
Optimizer_if, 1177
OptimizerDefaultImpl1, 1184
addCppCodeLine
CppModelExporter, 494
addData
QCPBars, 1417
QCPCurve, 1469, 1470
QCPErrorBars, 1496, 1497
QCPFinancial, 1509
QCPGraph, 1522
QCPStatisticalBox, 1757
addDataRange
QCPDataSelection, 1487
addEdge
DirectedAcyclicGraphNetwork, 594
addElement
QCPLayoutGrid, 1624
QCPLayoutInset, 1635
addGetStatisticsCollector
EntityType, 679
addGraph
QCustomPlot, 1782
addItem
QCPLegend, 1645
addKnowledgeItem
AIAssistant_if, 134
AIAssistantDefaultImpl, 147
addLayer
QCustomPlot, 1783
addModifier
BioReaction, 277
addObjective
Optimizer_if, 1177
OptimizerDefaultImpl1, 1184
addProduct
BioReaction, 277
addProperty
QtAbstractPropertyBrowser, 1814
QtAbstractPropertyManager, 1821
QtVariantPropertyManager, 2015
addPropertyManager
QtAbstractEditorFactory< PropertyManager >, 1809
addRadialAxis
QCPPolarAxisAngular, 1677
addReactant
BioReaction, 277
addSubProperty
QtProperty, 1944
addTick
QCPAxisTickerText, 1405
addTicks
QCPAxisTickerText, 1405, 1406
addToLegend
QCPAbstractPlottable, 1332
addTraceErrorHandler
TraceManager, 2581
addTraceHandler
TraceManager, 2581
addTraceReportHandler
TraceManager, 2581, 2582
addTraceSimulationHandler
ExperimentManager_if, 703
ExperimentManagerDefaultImpl1, 705
TraceManager, 2582
AddUndoCommand, 124
addValue
Collector_if, 408
CollectorDatafileDefaultImpl1, 414
CollectorDefaultImpl1, 416
adjustDotName
GraphvizModelExporter, 850
adjusted
QCPDataRange, 1483
advance
DormandPrince54OdeSolver, 618
RungeKutta4OdeSolver, 2107
aeAll
QCP, 113
aeAxes

QCP, 112
aeFills
QCP, 112
aeGrid
QCP, 112
aeItems
QCP, 112
aeLegend
QCP, 112
aeLegendItems
QCP, 112
aeNone
QCP, 113
aeOther
QCP, 113
aePlottables
QCP, 112
aeScatters
QCP, 112
aeSubGrid
QCP, 112
aeZeroLine
QCP, 113
afterLayout
QCustomPlot, 1783
afterReplot
QCustomPlot, 1783
AIAssistant, 125
_check, 130
_loadInstance, 130
_onDispatchEvent, 130
_saveInstance, 130
show, 131
AIAssistant_if, 131
addKnowledgeItem, 134
analyzePrompt, 134
buildModel, 134
checkReady, 134
clearApiKey, 135
clearDiagnostics, 135
clearKnowledgeBase, 135
collectResults, 135
configureSimulation, 135
createModelPlan, 136
execute, 136
exportAuditLog, 136
exportKnowledgeBaseAsMarkdown, 136
getApiKeyEnvironmentVariable, 136
getAuditEntries, 136
getAuditLogPath, 137
getBaseUrl, 137
getConfiguration, 137
getKnowledgeBaseSummary, 137
getKnowledgeItems, 137
getLastDiagnostics, 137
getLastError, 137
getMaskedApiKey, 137
getMaxOutputTokens, 137
getModelName, 138
getProvider, 138
getProviderClient, 138
getProviderName, 138
getReasoningEffort, 138
getSimulatorFacade, 138
getState, 138
getTemperature, 138
getTimeoutSeconds, 138
hasApiKey, 138
hasProviderClient, 139
hasSimulatorFacade, 139
loadApiKeyFromEnvironment, 139
refreshKnowledgeBaseFromSimulator, 139
runSimulation, 139
setApiKey, 140
setApiKeyEnvironmentVariable, 140
setBaseUrl, 140
setConfiguration, 140
setMaxOutputTokens, 140
setModelName, 140
setProvider, 140
setProviderClient, 141
setProviderName, 141
setReasoningEffort, 141
setSimulatorFacade, 141
setTemperature, 141
setTimeoutSeconds, 141
AIAssistantConfiguration, 142
dryRun, 143
sandboxEnabled, 143
AIAssistantDefaultImpl, 143
addKnowledgeItem, 147
analyzePrompt, 147
buildModel, 147
checkReady, 147
clearApiKey, 148
clearDiagnostics, 148
clearKnowledgeBase, 148
collectResults, 148
configureSimulation, 148
createAndAttachProviderClient, 148
createModelPlan, 149
execute, 149
exportAuditLog, 149
exportKnowledgeBaseAsMarkdown, 149
getApiKeyEnvironmentVariable, 149
getAuditEntries, 150
getAuditLogPath, 150
getBaseUrl, 150
getConfiguration, 150
getKnowledgeBaseSummary, 150
getKnowledgeItems, 150
getLastDiagnostics, 150
getLastError, 150
getMaskedApiKey, 150
getMaxOutputTokens, 151
getModelName, 151

getProvider, 151
getProviderClient, 151
getProviderName, 151
getReasoningEffort, 151
getSimulatorFacade, 151
getState, 151
getTemperature, 151
getTimeoutSeconds, 151
hasApiKey, 152
hasProviderClient, 152
hasSimulatorFacade, 152
loadApiKeyFromEnvironment, 152
refreshKnowledgeBaseFromSimulator, 152
runSimulation, 152
setApiKey, 153
setApiKeyEnvironmentVariable, 153
setBaseUrl, 153
setConfiguration, 153
setMaxOutputTokens, 153
setModelName, 153
setProvider, 154
setProviderClient, 154
setProviderName, 154
setReasoningEffort, 154
setSimulatorFacade, 154
setTemperature, 154
setTimeoutSeconds, 154
AIAssistantRequest, 155
modelPlan, 155
AIAssistantResponse, 156
AIAssistantWindow, 156
AIAuditLog, 157
exportCsv, 158
loadRecentEntries, 158
AIConversationConfiguration, 158
AIConversationResponse, 158
AIConversationService, 159
AIProviderClient_if, 159
clearApiKey, 160
getBaseUrl, 160
getProviderName, 160
getTimeoutSeconds, 160
hasApiKey, 160
isAvailable, 160
send, 160
setApiKey, 161
setBaseUrl, 161
setTimeoutSeconds, 161
Airport_Extended1, 161
main, 163
AirportSecurityExample, 163
main, 164
AirportSecurityExampleExtended, 165
main, 166
AISecretStore, 166
isAvailable, 167
load, 167
remove, 167
save, 167
AISupport, 168
_check, 172
_initBetweenReplications, 172
_loadInstance, 172
_saveInstance, 172
show, 172
alpha
QCPColorMapData, 1444
analyzePrompt
AIAssistant_if, 134
AIAssistantDefaultImpl, 147
anchor
QCPAbstractItem, 1315
AnchorMode
QCPLabelPainterPrivate, 1593
anchorPixelPosition
QCPItemBracket, 1537
QCPItemEllipse, 1548
QCPItemPixmap, 1558
QCPItemRect, 1570
QCPItemText, 1580
AnchorReferenceType
QCPLabelPainterPrivate, 1593
anchors
QCPAbstractItem, 1315
AnchorSide
QCPLabelPainterPrivate, 1593
AND, 173
_loadInstance, 179
_saveInstance, 179
Build, 179
AnElectronicAssemblyAndTestSystem, 179
main, 181
angle
QCPVector2D, 1774
AngleReference
QCPPolarAxisRadial, 1697
AnimationAttribute, 181
AnimationCounter, 182
AnimationEntity, 183
AnimationEvent, 185
AnimationExpression, 186
AnimationPlaceholder, 187
AnimationPlot, 188
AnimationPlot::DataSeries, 513
AnimationQueue, 190
AnimationQueueDisplay, 190
AnimationResource, 192
AnimationStation, 193
AnimationStatistics, 194
AnimationTimer, 195
AnimationTransition, 196
AnimationVariable, 198
AnthropicProviderClientImpl, 199
_authorizationHeader, 201
_buildRequestBody, 202
_extraRequestHeaders, 202

_parseResponseBody, 202
getProviderName, 202
isAvailable, 202
setApiVersion, 202
setThinkingBudgetTokens, 202
AntialiasedElement
QCP, 112
Análise técnica do parser (Bison/Flex), 23
ApiRouter, 203
ApiRouter, 203
handle, 203
append
QCPBarsGroup, 1426
applications/BaseGenesysTerminalApplication.h, 2735
applications/GenesysApplication_if.h, 2736
applications/gui/ai_assistant/AIAssistantWindow.h, 2736
applications/gui/genesys/actions/AddUndoCommand.h, 2737
applications/gui/genesys/actions/DeleteUndoCommand.h, 2738
applications/gui/genesys/actions/EFSMInitialStateUndoCommand.h, 2738
applications/gui/genesys/actions/EFSMStateMembershipUndoCommand.h, 2739
applications/gui/genesys/actions/EFSMStatePropertiesUndoCommand.h, 2739
applications/gui/genesys/actions/EFSMTransitionPropertiesUndoCommand.h, 2740
applications/gui/genesys/actions/EFSMTransitionUndoCommand.h, 2740
applications/gui/genesys/actions/GroupUndoCommand.h, 2741
applications/gui/genesys/actions/ModalModelBindingsUndoCommand.h, 2741
applications/gui/genesys/actions/MoveUndoCommand.h, 2742
applications/gui/genesys/actions/PasteUndoCommand.h, 2742
applications/gui/genesys/actions/UngroupUndoCommand.h, 2742
applications/gui/genesys/animations/AnimationCounter.h, 2743
applications/gui/genesys/animations/AnimationPlaceholder.h, 2743
applications/gui/genesys/animations/AnimationQueue.h, 2745
applications/gui/genesys/animations/AnimationTimer.h, 2746
applications/gui/genesys/animations/AnimationTransition.h, 2746
applications/gui/genesys/animations/AnimationVariable.h, 2747
applications/gui/genesys/BaseMVController.h, 2748
applications/gui/genesys/cellular_automata/CellularAutomataDimensionBuilder.h, 2749
applications/gui/genesys/cellular_automata/CellularAutomataInitialStateWidget.h, 2751
applications/gui/genesys/cellular_automata/CellularAutomataStateColorM... 2751
applications/gui/genesys/cellular_automata/CellularAutomataViewerContro... 2752
applications/gui/genesys/cellular_automata/CellularAutomataViewerWindo... 2753
applications/gui/genesys/codeeditor/CodeEditor.h, 2753
applications/gui/genesys/codeeditor/LineNumberArea.h, 2754
applications/gui/genesys/controllers/DialogUtilityController.h, 2754
applications/gui/genesys/controllers/EditCommandController.h, 2755
applications/gui/genesys/controllers/GraphicalContextMenuController.h, 2756
applications/gui/genesys/controllers/ModelInspectorController.h, 2757
applications/gui/genesys/controllers/ModelLifecycleController.h, 2757
applications/gui/genesys/controllers/PluginCatalogController.h, 2758
applications/gui/genesys/controllers/PropertyEditorController.h, 2759
applications/gui/genesys/controllers/SceneToolController.h, 2760
applications/gui/genesys/controllers/SimulationCommandController.h, 2761
applications/gui/genesys/controllers/SimulationController.h, 2762
applications/gui/genesys/controllers/SimulationEventController.h, 2762
applications/gui/genesys/controllers/TraceConsoleController.h, 2763
applications/gui/genesys/dialogs/dialogBreakpoint.h, 2764
applications/gui/genesys/dialogs/DialogFind.h, 2764
applications/gui/genesys/dialogs/Dialogmodelinformation.h, 2765
applications/gui/genesys/dialogs/dialogpluginmanager.h, 2766
applications/gui/genesys/dialogs/DialogReportIssue.h, 2767
applications/gui/genesys/dialogs/DialogSelectCounter.h, 2767
applications/gui/genesys/dialogs/DialogSelectVariable.h, 2768
applications/gui/genesys/dialogs/dialogsimulationconfigure.h, 2768
applications/gui/genesys/dialogs/dialogsystempreferences.h, 2769
applications/gui/genesys/dialogs/DialogTimerConfigure.h, 2769
applications/gui/genesys/extensions/BacteriaColonyViewerCoordinates.h, 2770
applications/gui/genesys/extensions/GuiExtensionConfigurationDialog.h, 2770
applications/gui/genesys/extensions/GuiExtensionContracts.h, 2771

applications/gui/genesys/extensions/GuiExtensionManager.h, 2772
applications/gui/genesys/extensions/GuiExtensionPluginCatalog.h, 2773
applications/gui/genesys/graphicals/EFSMTransitionGraphicalItem.h, 2773
applications/gui/genesys/graphicals/GraphicalAnimateExpression.h, 2773
applications/gui/genesys/graphicals/GraphicalAssociation.h, 2774
applications/gui/genesys/graphicals/GraphicalComponentPainter.h, 2774
applications/gui/genesys/graphicals/GraphicalConnection.h, 2774
applications/gui/genesys/graphicals/GraphicalConnectionStyle.h, 2775
applications/gui/genesys/graphicals/GraphicalDiagramConnection.h, 2776
applications/gui/genesys/graphicals/GraphicalImageAnimation.h, 2776
applications/gui/genesys/graphicals/GraphicalModelComponent.h, 2777
applications/gui/genesys/graphicals/GraphicalModelDataDefinition.h, 2779
applications/gui/genesys/graphicals/GraphicalModelItemRenderStrategy.h, 2780
applications/gui/genesys/graphicals/ModelGraphicsScene.h, 2781
applications/gui/genesys/graphicals/ModelGraphicsView.h, 2786
applications/gui/genesys/GuiCrashDiagnostics.h, 2788
applications/gui/genesys/GuiScopeTrace.h, 2788
applications/gui/genesys/guithememanager.h, 2788
applications/gui/genesys/mainwindow.cpp, 2789
applications/gui/genesys/mainwindow.h, 2790
applications/gui/genesys/mainwindow_controller.cpp, 2797
applications/gui/genesys/mainwindow_modelrepresentation.cpp, 2798
applications/gui/genesys/mainwindow_scene.cpp, 2799
applications/gui/genesys/mainwindow_simulator.cpp, 2800
applications/gui/genesys/MovedGuiCompatibilityIncludes.h, 2800
applications/gui/genesys/propertyeditor/ComboBoxEnum.h, 2801
applications/gui/genesys/propertyeditor/DataComponentEditor.h, 2801
applications/gui/genesys/propertyeditor/DataComponentProperty.h, 2802
applications/gui/genesys/propertyeditor/ObjectPropertyBrowser.h, 2803
applications/gui/genesys/propertyeditor/PropertyEditor.h, 2806
applications/gui/genesys/propertyeditor/qtpropertybrowser/qteditorfactory.h, 2806
applications/gui/genesys/propertyeditor/qtpropertybrowser/qtenumpropertymanager.h, 2807
applications/gui/genesys/propertyeditor/qtpropertybrowser/qtgroupboxpropertybrowser.h, 2812
applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowser.h, 2813
applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertybrowserutils_p.h, 2816
applications/gui/genesys/propertyeditor/qtpropertybrowser/qtpropertymanager.h, 2820
applications/gui/genesys/propertyeditor/qtpropertybrowser/qttreepropertybrowser.h, 2830
applications/gui/genesys/propertyeditor/qtpropertybrowser/qtvariantproperty.h, 2831
applications/gui/genesys/qcustomplot.h, 2833
applications/gui/genesys/services/CppModelExporter.h, 2932
applications/gui/genesys/services/EFSMNetworkDiagramCodec.h, 2932
applications/gui/genesys/services/EFSMNetworkDiagramLayout.h, 2933
applications/gui/genesys/services/EFSMNetworkDiagramProjection.h, 2933
applications/gui/genesys/services/GraphicalAutomaticPositioningStrategy.h, 2933
applications/gui/genesys/services/GraphicalDataDefinitionLayout.h, 2934
applications/gui/genesys/services/GraphicalModelBuilder.h, 2934
applications/gui/genesys/services/GraphicalModelSerializer.h, 2935
applications/gui/genesys/services/GraphvizModelExporter.h, 2936
applications/gui/genesys/services/IssueReportRelayService.h, 2937
applications/gui/genesys/services/ModelLanguageSynchronizer.h, 2937
applications/gui/genesys/services/ToolLauncher.h, 2938
applications/gui/genesys/systempreferences.h, 2938
applications/gui/genesys/tools/dataanalyzer/DataAnalyzerWindow.h, 2939
applications/gui/genesys/tools/expressionbuilder/ExpressionBuilder.h, 2941
applications/gui/genesys/tools/optimizer/OptimizerWindow.h, 2942
applications/gui/genesys/TraitsGUI.h, 2944
applications/gui/genesys/UtilGUI.h, 2945
applications/gui/httpworker/WebWorkerDialog.h, 2946
applications/launcher/ArchiveExtractor.h, 2946
applications/launcher/DispatchController.h, 2947
applications/launcher/LauncherConfig.h, 2947
applications/launcher/LauncherController.h, 2947
applications/launcher/LauncherLogger.h, 2948
applications/launcher/NetworkTransport.h, 2948
applications/launcher/PlatformInfo.h, 2949
applications/launcher/ProcessLauncher.h, 2949
applications/launcher/RuntimePaths.h, 2950
applications/launcher/RuntimeSelector.h, 2950
applications/launcher/UpdateChecker.h, 2951
applications/launcher/UpdateDialog.h, 2952

applications/launcher/UpdateDownloader.h, 2953
applications/launcher/UpdateManifest.h, 2953
applications/launcher/UpdateVerifier.h, 2954
applications/launcher/UserRuntimeManager.h, 2955
applications/launcher/Version.h, 2955
applications/modelSpecific/arenaExamples/Airport_Extended.h, 2956
applications/modelSpecific/arenaExamples/AirportSecurityAlternate.h, 2956
applications/modelSpecific/arenaExamples/AirportSecurityAlternateExtended.h, 2956
applications/modelSpecific/arenaExamples/Assembly_Line.h, 2956
applications/modelSpecific/arenaExamples/Banking_Transactions.h, 2957
applications/modelSpecific/arenaExamples/Example_BasicOperations.h, 2957
applications/modelSpecific/arenaExamples/Example_PortModel.h, 2957
applications/modelSpecific/arenaExamples/Example_PublicTransport.h, 2958
applications/modelSpecific/arenaSmarts/Smart_AddingResources.h, 2958
applications/modelSpecific/arenaSmarts/Smart_AlternatingEntityCreation.h, 2958
applications/modelSpecific/arenaSmarts/Smart_ArrivalsElementStopsEntitiesSpecifyingAttributes.h, 2959
applications/modelSpecific/arenaSmarts/Smart_ArrivalsEnablingTypeVsAttributes.h, 2959
applications/modelSpecific/arenaSmarts/Smart_AssignExample.h, 2960
applications/modelSpecific/arenaSmarts/Smart_AutomaticStatisticsCollection.h, 2960
applications/modelSpecific/arenaSmarts/Smart_BasicModelLogic.h, 2960
applications/modelSpecific/arenaSmarts/Smart_BatchAndSeparate.h, 2961
applications/modelSpecific/arenaSmarts/Smart_ContinuousFlowEntities.h, 2961
applications/modelSpecific/arenaSmarts/Smart_Create.h, 2961
applications/modelSpecific/arenaSmarts/Smart_DecideNWayByChance.h, 2962
applications/modelSpecific/arenaSmarts/Smart_DefiningAttributeAssignments.h, 2962
applications/modelSpecific/arenaSmarts/Smart_DefiningCapacities.h, 2962
applications/modelSpecific/arenaSmarts/Smart_DefiningResourceCapacities.h, 2963
applications/modelSpecific/arenaSmarts/Smart_DelayBasedOnResources.h, 2963
applications/modelSpecific/arenaSmarts/Smart_EntitiesProcessedByPriority.h, 2964
applications/modelSpecific/arenaSmarts/Smart_EvaluatingConditionsBeforeSpecifiedAmounts.h, 2964
applications/modelSpecific/arenaSmarts/Smart_ExpressionApp.h, 2964
applications/modelSpecific/arenaSmarts/Smart_InventoryAppHoldingCosts.h, 2965
applications/modelSpecific/arenaSmarts/Smart_MaxArrivalsField.h, 2965
applications/modelSpecific/arenaSmarts/Smart_ModelRunUntil1000Parts.h, 2965
applications/modelSpecific/arenaSmarts/Smart_ModuleDisplayVariables.h, 2966
applications/modelSpecific/arenaSmarts/Smart_OverlappingResources.h, 2966
applications/modelSpecific/arenaSmarts/Smart_ParallelProcessingOfEntities.h, 2966
applications/modelSpecific/arenaSmarts/Smart_PlacingEntitiesInQueueSet.h, 2967
applications/modelSpecific/arenaSmarts/Smart_PriorityExample.h, 2967
applications/modelSpecific/arenaSmarts/Smart_ProcessArena.h, 2968
applications/modelSpecific/arenaSmarts/Smart_Record_Arena.h, 2968
applications/modelSpecific/arenaSmarts/Smart_RemovingAndReorderingEntities.h, 2968
applications/modelSpecific/arenaSmarts/Smart_ResourceCosting.h, 2969
applications/modelSpecific/arenaSmarts/Smart_ResourceScheduleCosting.h, 2969
applications/modelSpecific/arenaSmarts/Smart_ResourceSets.h, 2969
applications/modelSpecific/arenaSmarts/Smart_SeizingMultipleSimultaneously.h, 2970
applications/modelSpecific/arenaSmarts/Smart_SelectingRouteBasedOnPriority.h, 2970
applications/modelSpecific/arenaSmarts/Smart_SelectingShorterQueue.h, 2970
applications/modelSpecific/arenaSmarts/Smart_SynchronizingParallelEntities.h, 2971
applications/modelSpecific/arenaSmarts/Smart_ValueAdded.h, 2971
applications/modelSpecific/arenaSmarts/Smart_WaitForSignal.h, 2971
applications/modelSpecific/book/Book_Cap02_Example01.h, 2972
applications/modelSpecific/smarts/AI/StartAIAssistant.h, 2972
applications/modelSpecific/smarts/AnalyticalModeling/Smart_MarkovChain.h, 2972
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_Bacteria.h, 2973
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioKinetics.h, 2973
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioNetwork.h, 2973
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioReversible.h, 2974
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_BioReversible2.h, 2974
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_GroColonies.h, 2974
applications/modelSpecific/smarts/BiochemicalSimulation/Smart_GroColonies2.h,

2974
applications/modelSpecific/smarts/Continuous/Smart_LSODE.h,
2975
applications/modelSpecific/smarts/Continuous/Smart_ODEApp.h,
2975
applications/modelSpecific/smarts/DiscreteProcessing/Smart_Buffer.h,
2975
applications/modelSpecific/smarts/DiscreteProcessing/Smart_Clerk.h,
2976
applications/modelSpecific/smarts/DiscreteProcessing/Smart_Delay.h,
2976
applications/modelSpecific/smarts/DiscreteProcessing/Smart_Failure.h,
2977
applications/modelSpecific/smarts/DiscreteProcessing/Smart_HoldScanObserve.h,
2977
applications/modelSpecific/smarts/DiscreteProcessing/Smart_Process.h,
2977
applications/modelSpecific/smarts/DiscreteProcessing/Smart_ProcessSet.h,
2977
applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayRelease.h,
2978
applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayReleaseMulti.h,
2978
applications/modelSpecific/smarts/DiscreteProcessing/Smart_SeizeDelayReleaseNoDataDefs.h,
2979
applications/modelSpecific/smarts/ExternalIntegration/Smart_CppFormula.h,
2979
applications/modelSpecific/smarts/ExternalIntegration/Smart_Python.h,
2979
applications/modelSpecific/smarts/ExternalIntegration/Smart_RSimulator.h,
2979
applications/modelSpecific/smarts/Grouping/Smart_BatchSeparate.h,
2980
applications/modelSpecific/smarts/InputOutput/Smart_Recorder.h,
2980
applications/modelSpecific/smarts/Logic/Smart_AssignWriteStatistic.h,
2981
applications/modelSpecific/smarts/Logic/Smart_Attribute_Variable.h,
2981
applications/modelSpecific/smarts/Logic/Smart_ModelInfo.h,
2981
applications/modelSpecific/smarts/Logic/Smart_Parser.h,
2982
applications/modelSpecific/smarts/Logic/Smart_ParserModelFunctions.h,
2982
applications/modelSpecific/smarts/Logic/Smart_Plugin.h,
2982
applications/modelSpecific/smarts/MaterialHandling/Smart_RouteStation.h,
2983
applications/modelSpecific/smarts/MaterialHandling/Smart_Sequence.h,
2983
applications/modelSpecific/smarts/ModalModel/Smart_CellularAutomata.h,
2984
applications/modelSpecific/smarts/ModalModel/Smart_CellularAutomata3.h,
2984
applications/modelSpecific/smarts/ModalModel/Smart_DefaultModalModel.h,
2984
applications/modelSpecific/smarts/ModalModel/Smart_Dummy.h,
2985
applications/modelSpecific/smarts/ModalModel/Smart_FSMState.h,
2985
applications/modelSpecific/smarts/ModalModel/Smart_Modalmodel.h,
2985
applications/modelSpecific/smarts/ModalModel/Smart_ModalModelDefaul
2985
applications/modelSpecific/smarts/ModalModel/Smart_ModalModelFSM.h
2986
applications/modelSpecific/smarts/ModalModel/Smart_ModalModelPetriN
2986
applications/modelSpecific/smarts/ModalModel/Smart_PetriPlace.h,
2986
applications/modelSpecific/smarts/Synchronization/Smart_OnEvent.h,
2986
applications/modelSpecific/smarts/Synchronization/Smart_SimulationCont
2987
applications/modelSpecific/smarts/Synchronization/Smart_WaitScanCond
2987
applications/modelSpecific/smarts/Synchronization/Smart_WaitSignal.h,
2987
applications/modelSpecific/teaching/AnElectronicAssemblyAndTestSystem
2988
applications/modelSpecific/teaching/BufferFIFO.h, 2988
applications/modelSpecific/teaching/ContinuousModel.h,
2989
applications/modelSpecific/teaching/FullSimulationOfComplexModel.h,
2989
applications/modelSpecific/teaching/Half_Adder.h, 2990
applications/modelSpecific/teaching/Loja/Loja01.h,
2990
applications/modelSpecific/teaching/OperatingSystem02.h,
2990
applications/modelSpecific/teaching/OperatingSystem03.h,
2991
applications/modelSpecific/teaching/Rectifier.h, 2991
applications/modelSpecific/wcm/WcmBioNetworkBridge.h,
2991
applications/modelSpecific/wcm/WcmCompartmentTransportWholeCellMv
2992
applications/modelSpecific/wcm/WcmFluxBalanceWholeCellMvp.h,
2992
applications/modelSpecific/wcm/WcmGeneExpression.h,
2992
applications/modelSpecific/wcm/WcmLifecycleStress.h,
2992
applications/modelSpecific/wcm/WcmMgenitaliumKarr2012.h,
2992
applications/modelSpecific/wcm/WcmPathwayStressWholeCellMvp.h,
2993
applications/modelSpecific/wcm/WcmSimpleSsa.h,
2993
applications/modelSpecific/wcm/WcmWholeCellMvp.h,
2993
applications/modelSpecific/wcm/WcmYeastDidacticWholeCellMvp.h,
2993
applications/modelSpecific/wcm/WcmYeastEukaryoticWholeCellMvp.h,
2993

applications/shell/GenesysShell.h, 2994
applications/shell/GenesysShell_if.h, 2995
applications/worker/api/ApiRouter.h, 2996
applications/worker/auth/TokenService.h, 2996
applications/worker/BaseGenesysWebApplication.h, 2997
applications/worker/http/HttpRequest.h, 2997
applications/worker/http/HttpResponse.h, 2997
applications/worker/http/SimpleHttpServer.h, 2997
applications/worker/service/SimulatorSessionService.h, 2998
applications/worker/service/WebWorkerRuntime.h, 3000
applications/worker/session/SessionContext.h, 3001
applications/worker/session/SessionManager.h, 3002
applications/worker/worker/WorkerJob.h, 3002
applications/worker/worker/WorkerJobManager.h, 3002
applyApplicationTheme
    GuiThemeManager, 869
applyBrush
    QCPSelectionDecorator, 1740
applyDefaultAntialiasingHint
    QCPAbstractItem, 1315
    QCPAbstractLegendItem, 1323
    QCPAbstractPlottable, 1332
    QCPAxis, 1353
    QCPAxisRect, 1375
    QCPColorScale, 1452
    QCPGrid, 1530
    QCPLayoutElement, 1614
    QCPLegend, 1645
    QCPPolarAxisAngular, 1677
    QCPPolarAxisRadial, 1698
    QCPPolarGraph, 1715
    QCPPolarGrid, 1721
    QCPSelectionRect, 1750
    QCPTextElement, 1768
applyLocalRule
    CellularAutomata_1DTimed, 382
    CellularAutomata_Classic, 384
    CellularAutomataBase, 385
applyModelGraphicsTheme
    GuiThemeManager, 869
applyPen
    QCPSelectionDecorator, 1741
applyRule
    LocalRule, 933
    LocalRule_CopyCurrentState, 935
    LocalRule_Elementary, 936
    LocalRule_FlorestalFire, 937
    LocalRule_GameOfLife, 939
    LocalRule_Growty, 940
    LocalRule_HppLatticeGas, 941
    LocalRule_SandRock_Student, 943
    LocalRule_ShowNeighbordhood, 944
applyTo
    QCPScatterStyle, 1736
arAbsolute
    QCPPolarAxisRadial, 1698
arAngularAxis
    QCPPolarAxisRadial, 1698
Architecture notes for <tt>source/tools</tt>, 31
arcPosition
    GraphicalDataDefinitionLayout, 821
Assembly_Line, 204
    main, 205
Assign, 205
    _check, 211
    _createEditableDataDefinitions, 211
    _loadInstance, 211
    _onDispatchEvent, 211
    _saveInstance, 211
    show, 211
Assignment, 212
at
    QCPDataContainer< DataType >, 1478
atBottom
    QCPAxis, 1352
atLeft
    QCPAxis, 1352
atRight
    QCPAxis, 1352
atTop
    QCPAxis, 1352
Attribute, 212
    _check, 217
    _loadInstance, 217
    _saveInstance, 217
    getInitialValuesText, 218
    setInitialValuesText, 218
    show, 218
attributeChanged
    QtVariantPropertyManager, 2015
attributes
    QtVariantPropertyManager, 2015
attributeType
    QtVariantPropertyManager, 2015
attributeValue
    QtVariantProperty, 2011
    QtVariantPropertyManager, 2015
AuditEntry, 218
average
    Statistics_if, 2476
    StatisticsDatafileDefaultImpl1, 2489
    StatisticsDefaultImpl1, 2493
averageConfidenceInterval
    HypothesisTesterDefaultImpl1, 887, 888
averageDifferenceConfidenceInterval
    HypothesisTesterDefaultImpl1, 888
axes
    QCPAxisRect, 1375
axis
    QCPAxisRect, 1375
    QCPColorScale, 1452
axisClick
    QCustomPlot, 1783

axisCount
QCPAxisRect, 1376
axisDoubleClick
QCustomPlot, 1784
axisRect
QCustomPlot, 1784
axisRectAt
QCustomPlot, 1784
axisRectCount
QCustomPlot, 1784
axisRects
QCustomPlot, 1785
AxisType
QCPAxis, 1352

backgroundColor
QtTreePropertyBrowser, 2003
BacteriaColony, 219
_check, 226
_createAttachedAttributes, 226
_initBetweenReplications, 226
_loadInstance, 226
_onDispatchEvent, 226
_saveInstance, 227
show, 227
BacteriaColony::BacteriumState, 233
BacteriaColony::BarrierSegment, 236
BacteriaColonyViewerGuiExtensionPlugin, 227
displayName, 228
extensionId, 228
registerContributions, 228
requiredModelPlugins, 228
version, 228
BacteriaSignalGrid, 229
_check, 233
_loadInstance, 233
_saveInstance, 233
show, 233
Banking_Transactions, 234
main, 236
barAbove
QCPBars, 1417
barBelow
QCPBars, 1417
bars
QCPBarsGroup, 1426
BaseGenesysTerminalApplication, 236
main, 237
setDefaultEventHandlers, 238
BaseGenesysWebApplication, 238
main, 239
BaseMVController, 239
Batch, 242
_check, 247
_createAttachedAttributes, 247
_createEditableDataDefinitions, 247
_createNonEditableDataDefinitions, 248
_loadInstance, 248
_onDispatchEvent, 248
_saveInstance, 248
show, 248
beforeReplot
QCustomPlot, 1785
Begin
testing::internal::CartesianProductGenerator< T >, 351
testing::internal::RangeGenerator< T, IncrementT >, 2029
testing::internal::ValuesInIteratorRangeGenerator< T >, 2637
begin
ModelComponentManager, 1045
QCPDataContainer< DataType >, 1478
QCPPainter, 1659
beginDownload
genesys::launcher::UpdateDialog, 2621
beta
ProbabilityDistributionBase, 1280
BioCompartment, 249
_check, 253
_createEditableDataDefinitions, 253
_loadInstance, 254
_saveInstance, 254
show, 254
BioKineticLawExpression, 254
BioNetwork, 255
_check, 259
_createEditableDataDefinitions, 259
_initBetweenReplications, 259
_loadInstance, 259
_saveInstance, 260
show, 260
BioNetworkAwareGuiExtensionPlugin, 260
displayName, 261
extensionId, 261
registerContributions, 261
requiredModelPlugins, 261
version, 261
BioNetworkEditorGuiExtensionPlugin, 262
displayName, 262
extensionId, 262
registerContributions, 262
requiredModelPlugins, 263
version, 263
BioNetworkExecutionGuiExtensionPlugin, 263
displayName, 264
extensionId, 264
registerContributions, 264
requiredModelPlugins, 264
version, 264
BioParameter, 264
_check, 268
_loadInstance, 268
_saveInstance, 269
show, 269
BioParameterManagerGuiExtensionPlugin, 269
displayName, 270

extensionId, 270
registerContributions, 270
requiredModelPlugins, 270
version, 270
BioParameterSensitivityEntry, 271
BioReaction, 271
_check, 276
_createAttachedAttributes, 276
_loadInstance, 276
_saveInstance, 277
addModifier, 277
addProduct, 277
addReactant, 277
BioReaction, 276
getKineticLawExpression, 277
getModifiers, 278
GetPluginInformation, 278
getProducts, 278
getRateConstant, 278
getRateConstantParameterName, 278
getReactants, 278
getReverseKineticLawExpression, 278
getReverseRateConstant, 279
getReverseRateConstantParameterName, 279
isReversible, 279
LoadInstance, 279
NewInstance, 279
resolveRateConstant, 280
resolveReverseRateConstant, 280
setKineticLawExpression, 280
setRateConstant, 280
setRateConstantParameterName, 280
setReverseKineticLawExpression, 281
setReverseRateConstant, 281
setReverseRateConstantParameterName, 281
setReversible, 281
show, 281
BioReaction::StoichiometricTerm, 2520
BioReactionKineticsGuiExtensionPlugin, 282
displayName, 282
extensionId, 282
registerContributions, 282
requiredModelPlugins, 283
version, 283
BioReactionRateSample, 283
BioReactionRateTimeCourse, 283
BioReactionStoichiometryGuiExtensionPlugin, 284
displayName, 284
extensionId, 284
registerContributions, 284
requiredModelPlugins, 285
version, 285
BioRunnerCommand, 285
_check, 290
_createEditableDataDefinitions, 290
_loadInstance, 290
_onDispatchEvent, 290
_saveInstance, 291
show, 291
BioSBMLBridge, 291
BioSBMLExportResult, 291
BioSBMLImportResult, 292
BioSBMLInteroperabilityGuiExtensionPlugin, 292
displayName, 293
extensionId, 293
registerContributions, 293
requiredModelPlugins, 293
version, 293
BioSensitivityScan, 293
BioSimulate, 293
_check, 299
_createEditableDataDefinitions, 299
_loadInstance, 299
_onDispatchEvent, 299
_saveInstance, 299
show, 299
BioSimulationAnalysis, 300
BioSimulationResult, 300
BioSimulationResultsGuiExtensionPlugin, 301
displayName, 301
extensionId, 301
registerContributions, 301
requiredModelPlugins, 302
version, 302
BioSimulationSample, 302
BioSimulationSpeciesAmount, 302
BioSimulatorRunner, 303
_check, 307
_loadInstance, 307
_saveInstance, 307
show, 307
BioSpecies, 308
_check, 312
_initBetweenReplications, 312
_loadInstance, 312
_saveInstance, 312
show, 312
BioSpeciesDerivative, 312
BioSpeciesManagerGuiExtensionPlugin, 313
displayName, 313
extensionId, 313
registerContributions, 313
requiredModelPlugins, 314
version, 314
BioStateProjectionComponent, 314
_check, 319
_createEditableDataDefinitions, 319
_loadInstance, 319
_onDispatchEvent, 320
_saveInstance, 320
show, 320
BioStateProjectionComponent::Projection, 1291
BioSteadyState, 320
_check, 326
_createAttachedAttributes, 326
_loadInstance, 326

_onDispatchEvent, 326
_saveInstance, 326
show, 326
BioSteadyStateCheck, 327
BioStoichiometryMatrix, 327
Book_Cap02_Example01, 327
main, 329
bottom
QCPAxisRect, 1376
QCPPolarAxisAngular, 1677
bottomLeft
QCPAxisRect, 1376
QCPPolarAxisAngular, 1678
bottomRight
QCPAxisRect, 1376
QCPPolarAxisAngular, 1678
Boundary_Closed, 329
getNeighborCell, 330
Boundary_Fixed, 330
getNeighborCell, 331
BoundaryCondition, 332
getNeighborCell, 333
bounded
QCPDataRange, 1483
QCPRange, 1728
boundingRect
GraphicalConnection, 819
BracketStyle
QCPItemBracket, 1537
QCPSelectionDecoratorBracket, 1744
browser
QtBrowserItem, 1829
bsCalligraphic
QCPItemBracket, 1537
bsCurly
QCPItemBracket, 1537
bsEllipse
QCPSelectionDecoratorBracket, 1744
bsHalfEllipse
QCPSelectionDecoratorBracket, 1744
bsPlus
QCPSelectionDecoratorBracket, 1745
bsRound
QCPItemBracket, 1537
bsSquare
QCPItemBracket, 1537
bsSquareBracket
QCPSelectionDecoratorBracket, 1744
bsUserStyle
QCPSelectionDecoratorBracket, 1745
Buffer, 333
_addSimulationControl, 338
_check, 338
_createEditableDataDefinitions, 338
_getParserChangesInformation, 339
_initBetweenReplications, 339
_loadInstance, 339
_onDispatchEvent, 339
_saveInstance, 339
show, 339
BufferFIFO, 340
main, 341
Build
AND, 179
Capacitor, 350
Diode, 587
NAND, 1114
NMOS, 1131
NOR, 1138
NOT, 1144
OR, 1194
PMOS, 1275
Resistor, 2073
Vpulse, 2651
Vsine, 2657
Vsource, 2663
XNOR, 2727
XOR, 2733
build
yy::genesyspp_parser::value_type, 2636
buildModel
AIAssistant_if, 134
AIAssistantDefaultImpl, 147

calculateAutoMargin
QCPAxisRect, 1376
QCPColorScaleAxisRectPrivate, 1461
calculatedBackgroundColor
QtTreePropertyBrowser, 2003
calculateResultsMean
FactorialDesign, 708
calculateStatistics
FactorialDesign, 708
cancel
QCPSelectionRect, 1750
canceled
QCPSelectionRect, 1750
canFire
EFSMTransition, 649
canGoNext
ModelManager, 1088
canGoPrevious
ModelManager, 1088
Capacitor, 344
_loadInstance, 349
_saveInstance, 349
Build, 350
Cell, 352
CellCycleCheckpointComponent, 353
_check, 359
_createEditableDataDefinitions, 359
_initBetweenReplications, 359
_loadInstance, 359
_onDispatchEvent, 359
_saveInstance, 359
show, 359
CellDivisionEvent, 360

_check, 365
_createEditableDataDefinitions, 365
_initBetweenReplications, 366
_loadInstance, 366
_onDispatchEvent, 366
_saveInstance, 366
show, 366
CellFateDecisionComponent, 367
_check, 372
_createEditableDataDefinitions, 372
_initBetweenReplications, 372
_loadInstance, 372
_onDispatchEvent, 372
_saveInstance, 372
show, 373
CellGrowthComponent, 373
_check, 378
_createEditableDataDefinitions, 378
_initBetweenReplications, 378
_loadInstance, 378
_onDispatchEvent, 378
_saveInstance, 379
show, 379
cellToCoord
QCPColorMapData, 1444
Cellular Automata Viewer, 3
CellularAutomata, 380
CellularAutomata_1DTimed, 381
applyLocalRule, 382
CellularAutomata_Classic, 382
applyLocalRule, 384
CellularAutomataBase, 384
applyLocalRule, 385
CellularAutomataComp, 386
_check, 391
_initBetweenReplications, 391
_loadInstance, 391
_onDispatchEvent, 392
_saveInstance, 392
show, 392
CellularAutomataDemoBuilder, 392
CellularAutomataDemoModel, 393
CellularAutomataDemoSettings, 393
CellularAutomataEvent, 394
CellularAutomataGuiExtensionPlugin, 395
displayName, 395
extensionId, 395
registerContributions, 395
requiredModelPlugins, 396
version, 396
CellularAutomataLatticeWidget, 396
CellularAutomataStateColorMap, 397
CellularAutomataViewerController, 397
CellularAutomataViewerWindow, 399
center
QCPAxisRect, 1376
QCPPolarAxisAngular, 1678
QCPRange, 1729
centil
StatisticsDatafile_if, 2485
StatisticsDatafileDefaultImpl1, 2489
changed
QCPSelectionRect, 1750
ChartStyle
QCPFinancial, 1508
ChatMessage, 399
Check
ModelComponent, 1041
ModelDataDefinition, 1054
check
Model, 1025
ModelDataManager, 1063
PluginConnector_if, 1259
PluginConnectorDummyImpl1, 1261
PluginConnectorStaticImpl1, 1264
checkActivationCode
ModelChecker_if, 1032
ModelCheckerDefaultImpl1, 1034
checkAll
ModelChecker_if, 1032
ModelCheckerDefaultImpl1, 1034
checkConnected
ModelChecker_if, 1032
ModelCheckerDefaultImpl1, 1034
checkExpression
Model, 1025
checkFeasibility
FactorialDesign, 709
checkLimits
ModelChecker_if, 1033
ModelCheckerDefaultImpl1, 1035
checkReady
AIAssistant_if, 134
AIAssistantDefaultImpl, 147
Optimizer_if, 1178
OptimizerDefaultImpl1, 1184
checkReferencesToDataDefinitions
Model, 1025
checkSymbols
ModelChecker_if, 1033
ModelCheckerDefaultImpl1, 1035
chi2
ProbabilityDistributionBase, 1280
children
QCPLayer, 1596
QtBrowserItem, 1829
ciHSV
QCPColorGradient, 1429
ciRGB
QCPColorGradient, 1429
clear
Collector_if, 409
CollectorDatafileDefaultImpl1, 414
CollectorDefaultImpl1, 416
Counter, 465
ModelComponentManager, 1045

QCPAbstractPaintBuffer, 1326
QCPAxisTickerText, 1406
QCPBarsGroup, 1426
QCPColorMapData, 1444
QCPDataContainer< DataType >, 1478
QCPDataSelection, 1487
QCPLayout, 1608
QCPMarginGroup, 1654
QCPPaintBufferPixmap, 1656
QtAbstractPropertyBrowser, 1814
QtAbstractPropertyManager, 1821
clearAlpha
QCPColorMapData, 1445
clearApiKey
AIAssistant_if, 135
AIAssistantDefaultImpl, 148
AIProviderClient_if, 160
HttpProviderClientBase, 881
clearColorStops
QCPColorGradient, 1430
clearConstraints
Optimizer_if, 1178
OptimizerDefaultImpl1, 1184
clearDiagnostics
AIAssistant_if, 135
AIAssistantDefaultImpl, 148
clearGraphs
QCustomPlot, 1785
clearItems
QCPLegend, 1645
QCustomPlot, 1785
clearKnowledgeBase
AIAssistant_if, 135
AIAssistantDefaultImpl, 148
clearObjectives
Optimizer_if, 1178
OptimizerDefaultImpl1, 1184
clearPlottables
QCustomPlot, 1785
clicked
QCPTextElement, 1768
clipRect
QCPAbstractItem, 1315
QCPAbstractLegendItem, 1323
QCPAbstractPlottable, 1333
QCPPolarGraph, 1715
Clone, 400
_check, 405
_createInternalStatisticReporters, 405
_initBetweenReplications, 405
_loadInstance, 405
_onDispatchEvent, 405
_saveInstance, 406
show, 406
clone
CompositeState, 435
HppLatticeGasState, 875
RealState, 2033
CodeEditor, 406
collapsed
QtButtonPropertyBrowser, 1832
QtTreePropertyBrowser, 2003
collectDataDefinitionsRemovedWith
Model, 1026
Collector_if, 407
addValue, 408
clear, 409
getLastValue, 409
numElements, 409
setAddValueHandler, 409
setClearHandler, 409
CollectorDatafile_if, 410
getDataFilename, 411
getNextValue, 411
getValue, 411
seekFirstValue, 411
setDataFilename, 412
CollectorDatafileDefaultImpl1, 412
addValue, 414
clear, 414
getDataFilename, 414
getLastValue, 414
getNextValue, 414
getValue, 414
numElements, 414
seekFirstValue, 414
setAddValueHandler, 414
setClearHandler, 414
setDataFilename, 415
CollectorDefaultImpl1, 415
addValue, 416
clear, 416
getLastValue, 416
numElements, 416
setAddValueHandler, 416
setClearHandler, 416
collectResults
AIAssistant_if, 135
AIAssistantDefaultImpl, 148
ColoredPetriNetNetwork, 417
_activate, 423
_check, 423
_initBetweenReplications, 423
_loadInstance, 423
_saveInstance, 423
show, 423
ColorInterpolation
QCPColorGradient, 1429
colorize
QCPColorGradient, 1430, 1431
colorMaps
QCPColorScale, 1452
columnCount
QCPLayoutGrid, 1624
ComboBoxEnum, 424
CompartmentExchangeComponent, 425

_check, 431
_createEditableDataDefinitions, 431
_initBetweenReplications, 431
_loadInstance, 431
_onDispatchEvent, 431
_saveInstance, 432
show, 432
CompartmentExchangeComponent::ExchangeRule, 695
CompilerThread, 433
ComponentItem, 433
CompositeState, 434
clone, 435
equals, 435
toString, 435
typeName, 435
CompositeStateSet, 435
contains, 436
createDefaultState, 436
isComposite, 436
name, 436
confidenceLevel
Statistics_if, 2476
StatisticsDatafileDefaultImpl1, 2489
StatisticsDefaultImpl1, 2493
configureSimulation
AIAssistant_if, 135
AIAssistantDefaultImpl, 148
SimulatorSessionService, 2260
confirmUpdate
genesys::launcher::UpdateDialog, 2621
connect
PluginConnector_if, 1259
PluginConnectorDummyImpl1, 1261
PluginConnectorStaticImpl1, 1264
Connection, 437
ConnectionChannel, 438
ConnectionManager, 438
connections, 439
getConnectionAtPort, 439
getCurrentInputConnectionsSize, 439
getCurrentOutputConnectionsSize, 440
getFrontConnection, 440
getMaxInputConnections, 440
getMaxOutputConnections, 440
getMinInputConnections, 440
getMinOutputConnections, 440
insert, 440, 441
insertAtPort, 441
remove, 441
removeAtPort, 441
setMaxInputConnections, 441
setMaxOutputConnections, 442
setMinInputConnections, 442
setMinOutputConnections, 442
size, 442
connections
ConnectionManager, 439
connectNewParser
ParserManager, 1217
connectPropertyManager
QtAbstractEditorFactory< PropertyManager >, 1809
connectTo
ModelComponent, 1041
constBegin
QCPDataContainer< DataType >, 1478
constEnd
QCPDataContainer< DataType >, 1478
constraint
QtRectFPropertyManager, 1953
QtRectPropertyManager, 1959
constraintChanged
QtRectFPropertyManager, 1954
QtRectPropertyManager, 1959
contains
CompositeStateSet, 436
HppLatticeGasStateSet, 877
IntegerStateSet, 894
QCPBarsGroup, 1426
QCPDataRange, 1484
QCPDataSelection, 1487
QCPRange, 1729
RealIntervalStateSet, 2031
StateSet_Enumerable, 2464
ContinuousDistribution_if, 444
ContinuousModel, 446
Simulate, 447
ContinuousSystemComponent, 447
_check, 453
_createEditableDataDefinitions, 453
_loadInstance, 453
_onDispatchEvent, 453
_saveInstance, 453
show, 453
convertEnumToStr
Util, 2626, 2627
Conveyor, 454
_check, 458
_createEditableDataDefinitions, 458
_initBetweenReplications, 458
_loadInstance, 459
_saveInstance, 459
show, 459
coordsToPixels
QCPAbstractPlottable, 1333
QCPPolarGraph, 1715
coordToCell
QCPColorMapData, 1445
coordToPixel
QCPAxis, 1353
QCPPolarAxisAngular, 1678
QCPPolarAxisRadial, 1698
copyFrom
QCPSelectionDecorator, 1741
CoreGuiExtensionPlugin, 459

displayName, 460
extensionId, 460
registerContributions, 460
requiredModelPlugins, 460
version, 460
Counter, 461
_check, 465
_initBetweenReplications, 465
_loadInstance, 465
_saveInstance, 465
clear, 465
getCountValue, 465
getParent, 465
incCountValue, 465
show, 466
CountingSampler, 466
getRNGparameters, 468
random, 468
sampleBernoulli, 468
sampleBeta, 468
sampleBinomial, 469
sampleDiscrete, 469
sampleErlang, 469
sampleExponential, 469
sampleGamma, 469
sampleGeometric, 469
sampleGumbell, 470
sampleLogNormal, 470
sampleNormal, 470
sampleTriangular, 470
sampleUniform, 470
sampleWeibull, 470
setRNGparameters, 470
CPNArc, 471
_check, 475
_loadInstance, 475
_saveInstance, 475
show, 475
CPNTransition, 476
_check, 481
_loadInstance, 481
_saveInstance, 481
show, 481
CppCompiler, 482
_check, 486
_initBetweenReplications, 486
_loadInstance, 486
_saveInstance, 486
show, 487
CppCompiler::CompilationResult, 432
CppForG, 487
_check, 492
_createNonEditableDataDefinitions, 492
_initBetweenReplications, 492
_loadInstance, 492
_onDispatchEvent, 492
_saveInstance, 493
show, 493
CppModelExporter, 493
addCppCodeLine, 494
CppSerializer, 494
dump, 495
for_each, 495
get, 496
load, 496
newPersistenceRecord, 496
put, 496
Create, 497
_check, 503
_createEditableDataDefinitions, 503
_createInternalStatisticReporters, 503
_initBetweenReplications, 503
_loadInstance, 503
_onDispatchEvent, 504
_saveInstance, 504
show, 504
createAndAttachProviderClient
AIAssistantDefaultImpl, 148
createColumnLabels
FactorialDesign, 709
createDefaultState
CompositeStateSet, 436
HppLatticeGasStateSet, 877
IntegerStateSet, 894
RealIntervalStateSet, 2031
StateSet_Enumerable, 2464
createEditor
QtAbstractEditorFactory< PropertyManager >, 1809
QtAbstractEditorFactoryBase, 1812
QtAbstractPropertyBrowser, 1814
createElementSetOfType
Set, 2170
createEntity
Model, 1026
createFromLanguage
ModelManager, 1088
CreateInternalData
ModelComponent, 1042
ModelDataDefinition, 1054
createModelImage
GraphvizModelExporter, 850
createModelPlan
AIAssistant_if, 136
AIAssistantDefaultImpl, 149
createNewListElement
SimulationControlGenericListPointer< T, M, C >, 2215
createNewListElementOfType
SimulationControlGenericListPointer< T, M, C >, 2215
createObjectInstance
SimulationControlGenericClass< T, M, C >, 2200
SimulationControlGenericClassNotDC< T, M, C >, 2205
createProperty

QtAbstractPropertyManager, 1821
QtVariantPropertyManager, 2016
createQueuedJob
WorkerJobManager, 2707
CreateRelatedDataElements
ModelDataDefinition, 1054
createSession
SessionManager, 2164
SimulatorSessionService, 2260
createSignTable
FactorialDesign, 709
CreateTest
testing::internal::ParameterizedTestFactory< TestClass >, 1197
testing::internal::TestFactoryImpl< TestClass >, 2564
CreateTestFactory
testing::internal::TestMetaFactory< TestSuite >, 2566
createTickVector
QCPAxisTicker, 1387
QCPAxisTickerDateTime, 1391
QCPAxisTickerLog, 1399
QCPAxisTickerText, 1406
createWorkerJob
SimulatorSessionService, 2260
csCandlestick
QCPFinancial, 1508
csOhlc
QCPFinancial, 1508
current
ModelManager, 1088
currentIndex
ModelManager, 1088
currentItem
QtAbstractPropertyBrowser, 1815
currentItemChanged
QtAbstractPropertyBrowser, 1815
currentLayer
QCustomPlot, 1786
currentSimulation
SimulationController, 2221
CurrentStackTrace
testing::internal::OsStackTraceGetter, 1195
Customization Points, 21

data
QCPBars, 1418
QCPColorMap, 1438
QCPCurve, 1470
QCPErrorBars, 1497
QCPFinancial, 1510
QCPGraph, 1522
QCPStatisticalBox, 1757
DataAnalyser_if, 508
DataAnalyzerWindow, 509
DataAnalyzerWindow::DoePreviewPlot, 616
DataAnalyzerWindow::HistogramPreview, 873
DataComponentEditor, 510
DataComponentProperty, 511
dataCount
QCPAbstractPlottable1D< DataType >, 1344
QCPErrorBars, 1497
QCPPlottableInterface1D, 1663
DataDefinitionItem, 512
dataDefinitionPosition
GraphicalAutomaticPositioningStrategy, 815
DataDefinitionRelationItem, 512
dataMainKey
QCPAbstractPlottable1D< DataType >, 1344
QCPErrorBars, 1497
QCPPlottableInterface1D, 1663
dataMainValue
QCPAbstractPlottable1D< DataType >, 1344
QCPErrorBars, 1497
QCPPlottableInterface1D, 1663
dataPixelPosition
QCPAbstractPlottable1D< DataType >, 1344
QCPBars, 1418
QCPErrorBars, 1497
QCPPlottableInterface1D, 1663
dataPointCount
QCPDataSelection, 1487
dataRange
QCPDataContainer< DataType >, 1478
QCPDataSelection, 1487
dataRangeChanged
QCPColorMap, 1438
QCPColorScale, 1453
dataRangeCount
QCPDataSelection, 1488
dataRanges
QCPDataSelection, 1488
dataScaleTypeChanged
QCPColorMap, 1438
QCPColorScale, 1453
DataSet_if, 514
dataSortKey
QCPAbstractPlottable1D< DataType >, 1344
QCPErrorBars, 1498
QCPPlottableInterface1D, 1663
dataValueRange
QCPAbstractPlottable1D< DataType >, 1345
QCPErrorBars, 1498
QCPPlottableInterface1D, 1663
dateTimeToKey
QCPAxisTickerDateTime, 1391
Decide, 514
_check, 519
_createInternalStatisticReporters, 519
_loadInstance, 519
_onDispatchEvent, 519
_saveInstance, 520
show, 520
decil
StatisticsDatafile_if, 2485
StatisticsDatafileDefaultImpl1, 2489

decimals
QtDoublePropertyManager, 1879
QtPointFPropertyManager, 1935
QtRectFPropertyManager, 1954
QtSizeFPropertyManager, 1967
decimalsChanged
QtDoublePropertyManager, 1879
QtPointFPropertyManager, 1936
QtRectFPropertyManager, 1954
QtSizeFPropertyManager, 1968
DefaultNetwork, 523
_activate, 528
_check, 528
_createInternalStatisticReporters, 528
_initBetweenReplications, 528
_loadInstance, 529
_saveInstance, 529
activate, 529
show, 529
DefaultNode, 529
_loadInstance, 534
_saveInstance, 534
DefaultNode, 534
show, 534
DefaultNodeTransition, 535
defineTraceLevelSpecific
ModelDataDefinition, 1055
Delay, 536
_check, 542
_createAttachedAttributes, 542
_createInternalStatisticReporters, 542
_loadInstance, 542
_onDispatchEvent, 542
_saveInstance, 542
show, 543
Delay::DEFAULT_VALUES, 520
DelayProbe, 543
DeleteUndoCommand, 548
derivate
SolverDefaultImpl1, 2421
descriptionToolTip
QtProperty, 1944
deselectAll
QCustomPlot, 1786
deselectEvent
QCPAbstractItem, 1316
QCPAbstractLegendItem, 1323
QCPAbstractPlottable, 1333
QCPAxis, 1353
QCPLegend, 1645
QCPPolarAxisRadial, 1699
QCPPolarGraph, 1715
QCPTextElement, 1769
detachTransition
EFSMNetwork, 644
dialogBreakpoint, 549
dialogBreakpoint::MVCResult, 1108
DialogFind, 550
DialogModelInformation, 551
DialogPluginManager, 552
DialogPluginManager, 553
DialogReportIssue, 553
DialogSelectCounter, 555
DialogSelectVariable, 556
DialogSimulationConfigure, 557
accept, 558
DialogSimulationConfigure, 557
reject, 558
DialogSystemPreferences, 558
DialogTimerConfigure, 559
DialogUtilityController, 559
onActionAboutReportIssueTriggered, 561
onActionEditReplaceTriggered, 561
onActionToolsParserGrammarCheckerTriggered, 561
DiffEquations, 561
_check, 567
_createEditableDataDefinitions, 567
_loadInstance, 567
_onDispatchEvent, 567
_saveInstance, 567
getEquations, 567
show, 567
DiffusionField, 568
_check, 572
_initBetweenReplications, 572
_loadInstance, 572
_saveInstance, 573
show, 573
DiffusionMethodOfLinesSystem, 573
dimension, 574
evaluate, 574
DiffusionSimulate, 575
_check, 580
_createEditableDataDefinitions, 580
_loadInstance, 580
_onDispatchEvent, 580
_saveInstance, 581
show, 581
dimension
DiffusionMethodOfLinesSystem, 574
HarmonicOscillator, 872
MassActionOdeSystem, 978
Diode, 582
_loadInstance, 587
_saveInstance, 587
Build, 587
DirectedAcyclicGraphNetwork, 588
_check, 594
addEdge, 594
DirectedGraphNetwork, 594
_edgeConnects, 601
_oppositeNode, 601
getIncomingEdges, 601
getOutgoingEdges, 601
getPredecessors, 601

getSuccessors, 601
hasCycle, 601
inDegree, 601
isDirected, 601
outDegree, 602
topologicalOrder, 602
DirSeparator
Util, 2627
disconnect
PluginConnector_if, 1259, 1260
PluginConnectorDummyImpl1, 1262
PluginConnectorStaticImpl1, 1264
disconnectPropertyManager
QtAbstractEditorFactory< PropertyManager >, 1809
DiscreteDistribution_if, 602
DispatchEvent
ModelComponent, 1042
displayName
BacteriaColonyViewerGuiExtensionPlugin, 228
BioNetworkAwareGuiExtensionPlugin, 261
BioNetworkEditorGuiExtensionPlugin, 262
BioNetworkExecutionGuiExtensionPlugin, 264
BioParameterManagerGuiExtensionPlugin, 270
BioReactionKineticsGuiExtensionPlugin, 282
BioReactionStoichiometryGuiExtensionPlugin, 284
BioSBMLInteroperabilityGuiExtensionPlugin, 293
BioSimulationResultsGuiExtensionPlugin, 301
BioSpeciesManagerGuiExtensionPlugin, 313
CellularAutomataGuiExtensionPlugin, 395
CoreGuiExtensionPlugin, 460
GeneticCircuitGuiExtensionPlugin, 781
GuiExtensionConfigurationGuiExtensionPlugin, 863
ModelPluginInspectorGuiExtensionPlugin, 1093
Dispose, 603
_check, 609
_createAttachedAttributes, 609
_createInternalStatisticReporters, 609
_initBetweenReplications, 609
_loadInstance, 609
_onDispatchEvent, 609
_saveInstance, 609
show, 610
Distance, 610
_check, 614
_loadInstance, 614
_saveInstance, 614
show, 615
DistanceEntry, 615
distanceSquaredToLine
QCPVector2D, 1774
distanceToStraightLine
QCPVector2D, 1774
Distribution_if, 615
donePainting
QCPAbstractPaintBuffer, 1326
DormandPrince54OdeSolver, 617
advance, 618
dot
QCPVector2D, 1774
doubleClicked
QCPTextElement, 1769
download
genesys::launcher::QtNetworkTransport, 1933
draw
QCPAbstractItem, 1316
QCPAbstractLegendItem, 1323
QCPAbstractPaintBuffer, 1326
QCPAbstractPlottable, 1333
QCPAxis, 1353
QCPAxisRect, 1376
QCPBars, 1418
QCPColorMap, 1438
QCPColorScaleAxisRectPrivate, 1461
QCPCurve, 1470
QCPErrorBars, 1498
QCPFinancial, 1510
QCPGraph, 1522
QCPGrid, 1530
QCPItemBracket, 1538
QCPItemCurve, 1543
QCPItemEllipse, 1548
QCPItemLine, 1553
QCPItemPixmap, 1558
QCPItemRect, 1570
QCPItemStraightLine, 1575
QCPItemText, 1580
QCPItemTracer, 1588
QCPLayoutElement, 1614
QCPLegend, 1645
QCPPaintBufferPixmap, 1656
QCPPlottableLegendItem, 1669
QCPPolarAxisAngular, 1678
QCPPolarAxisRadial, 1699
QCPPolarGraph, 1715
QCPPolarGrid, 1721
QCPPolarLegendItem, 1727
QCPSelectionRect, 1750
QCPStatisticalBox, 1757
QCPTextElement, 1769
drawBracket
QCPSelectionDecoratorBracket, 1745
drawDecoration
QCPSelectionDecorator, 1741
QCPSelectionDecoratorBracket, 1745
DrawingItem, 619
drawLegendIcon
QCPBars, 1418
QCPColorMap, 1438
QCPCurve, 1470
QCPErrorBars, 1498
QCPFinancial, 1510
QCPGraph, 1522
QCPStatisticalBox, 1758
drawLine

QCPPainter, 1659
drawPolyline
QCPAbstractPlottable1D< DataType >, 1345
drawShape
QCPScatterStyle, 1737
drawStatisticalBox
QCPStatisticalBox, 1758
DropOff, 619
_check, 624
_loadInstance, 624
_onDispatchEvent, 625
_saveInstance, 625
show, 625
dryRun
AIAssistantConfiguration, 143
DummyComponent, 625
_loadInstance, 630
_onDispatchEvent, 630
_saveInstance, 631
DummyComponent, 630
show, 631
DummyElement, 631
_check, 635
_loadInstance, 635
_saveInstance, 635
show, 635
dump
CppSerializer, 495
GenSerializer, 805
JsonSerializer, 906
Serializer, 2160
XmlSerializer, 2719

EditCommandController, 636
Editing an EFSM Network in the graphical interface, 1
editItem
QtTreePropertyBrowser, 2004
EditorFactoryPrivate< Editor >, 637
EFSMInitialStateUndoCommand, 637
EFSMNetwork, 638
_activate, 643
_check, 643
_initBetweenReplications, 643
_loadInstance, 643
_saveInstance, 644
detachTransition, 644
restoreTransition, 644
show, 644
EFSMNetworkDiagramCodec, 105
EFSMNetworkDiagramLayout, 644
efsmNetworkDiagramLayout
GraphicalModelSerializer, 835
EFSMNetworkDiagramLayout::StatePosition, 2461
EFSMNetworkDiagramLayout::TransitionLayout, 2593
EFSMStateMembershipUndoCommand, 645
EFSMStatePropertiesUndoCommand, 646
EFSMStatePropertiesUndoCommand::Properties, 1292
EFSMTransition, 647
canFire, 649
execute, 649
EFSMTransitionGraphicsItem, 649
EFSMTransitionPropertiesUndoCommand, 650
EFSMTransitionPropertiesUndoCommand::Properties, 1292
EFSMTransitionUndoCommand, 651
element
QCPLayoutGrid, 1625
elementAt
QCPLayout, 1608
QCPLayoutGrid, 1625
QCPLayoutInset, 1635
elementCount
QCPLayout, 1609
QCPLayoutGrid, 1625
QCPLayoutInset, 1635
elements
QCPAxisRect, 1376
QCPLayout, 1609
QCPLayoutElement, 1614
QCPLayoutGrid, 1625
QCPMarginGroup, 1654
QCPPolarAxisAngular, 1678
End
testing::internal::CartesianProductGenerator< T >, 351
testing::internal::RangeGenerator< T, IncrementT >, 2029
testing::internal::ValuesInIteratorRangeGenerator< T >, 2638
end
ModelComponentManager, 1045
QCPDataContainer< DataType >, 1478
EndingStyle
QCPLineEnding, 1651
enforceType
QCPDataSelection, 1488
ensureObjectInstance
SimulationControlGenericClass< T, M, C >, 2200
SimulationControlGenericClassNotDC< T, M, C >, 2205
ensureReady
SimulationController, 2221
Enter, 655
_check, 661
_createEditableDataDefinitions, 661
_createInternalStatisticReporters, 661
_loadInstance, 661
_onDispatchEvent, 661
_saveInstance, 662
show, 662
Entity, 662
_check, 666
_loadInstance, 666
_saveInstance, 667
entityNumber, 667
getAttributeValue, 667
getEntityType, 668

getEntityTypeName, 668
setAttributeValue, 668
setEntityType, 668
setEntityTypeName, 669
show, 669
EntityGroup, 669
_check, 673
_createAttachedAttributes, 673
_createInternalStatisticReporters, 673
_initBetweenReplications, 673
_loadInstance, 673
_saveInstance, 673
show, 674
entityNumber
Entity, 667
EntityType, 674
_createInternalStatisticReporters, 678
_initBetweenReplications, 678
_loadInstance, 678
_saveInstance, 679
addGetStatisticsCollector, 679
show, 679
enumerateStates
HppLatticeGasStateSet, 877
IntegerStateSet, 894
StateSet_Enumerable, 2464
enumIcons
QtEnumPropertyManager, 1891
enumIconsChanged
QtEnumPropertyManager, 1891
enumNames
QtEnumPropertyManager, 1892
enumNamesChanged
QtEnumPropertyManager, 1892
enumTypeId
QtVariantPropertyManager, 2016
epAllowCosmetic
QCP, 113
epNoCosmetic
QCP, 113
equals
CompositeState, 435
HppLatticeGasState, 875
RealState, 2033
erlang
ProbabilityDistributionBase, 1280
error
yy::genesyspp_parser, 770
errorMessages
TraceManager, 2582
ErrorType
QCPErrorBars, 1496
esBar
QCPLineEnding, 1651
esDiamond
QCPLineEnding, 1651
esDisc
QCPLineEnding, 1651
esFlatArrow
QCPLineEnding, 1651
esHalfBar
QCPLineEnding, 1651
esLineArrow
QCPLineEnding, 1651
esNone
QCPLineEnding, 1651
esSkewedBar
QCPLineEnding, 1651
esSpikeArrow
QCPLineEnding, 1651
esSquare
QCPLineEnding, 1651
estimateSampleSize
HypothesisTesterDefaultImpl1, 888
etKeyError
QCPErrorBars, 1496
etValueError
QCPErrorBars, 1496
EukaryoticCellCycleComponent, 680
_check, 686
_createEditableDataDefinitions, 686
_initBetweenReplications, 687
_loadInstance, 687
_onDispatchEvent, 687
_saveInstance, 687
show, 687
evaluate
DiffusionMethodOfLinesSystem, 574
HarmonicOscillator, 872
MassActionOdeSystem, 978
Event, 688
getComponent, 688
getComponentinputPortNumber, 689
getEntity, 689
getTime, 689
setEntity, 689
Exact, 689
Example_BasicOrderShipping, 690
main, 692
Example_PortModel, 692
main, 693
Example_PublicTransport, 694
main, 695
execute
AIAssistant_if, 136
AIAssistantDefaultImpl, 149
EFSMTransition, 649
ExecuteCommand
Util, 2627
Exit, 696
_check, 701
_loadInstance, 701
_onDispatchEvent, 701
_saveInstance, 702
show, 702
expand

QCPRange, 1729
expanded
QCPDataRange, 1484
QCPRange, 1729
QtButtonPropertyBrowser, 1832
QtTreePropertyBrowser, 2004
expandTo
QCPLayoutGrid, 1625
expected_tokens
yy::genesyspp_parser::context, 444
ExperimentManager, 702
ExperimentManager_if, 703
addTraceSimulationHandler, 703
getScenarios, 703
startExperiment, 704
startSimulationOfScenario, 704
stopExperiment, 704
ExperimentManagerDefaultImpl1, 704
addTraceSimulationHandler, 705
getScenarios, 705
startExperiment, 705
startSimulationOfScenario, 705
stopExperiment, 706
exponential
ProbabilityDistributionBase, 1280
exportAuditLog
AIAssistant_if, 136
AIAssistantDefaultImpl, 149
exportCsv
AIAuditLog, 158
exportKnowledgeBaseAsMarkdown
AIAssistant_if, 136
AIAssistantDefaultImpl, 149
ExportPen
QCP, 113
ExpressionBuilder, 706
ExpressionCategory, 707
ExpressionParameter, 707
ExpressionTemplate, 707
extensionId
BacteriaColonyViewerGuiExtensionPlugin, 228
BioNetworkAwareGuiExtensionPlugin, 261
BioNetworkEditorGuiExtensionPlugin, 262
BioNetworkExecutionGuiExtensionPlugin, 264
BioParameterManagerGuiExtensionPlugin, 270
BioReactionKineticsGuiExtensionPlugin, 282
BioReactionStoichiometryGuiExtensionPlugin, 284
BioSBMLInteroperabilityGuiExtensionPlugin, 293
BioSimulationResultsGuiExtensionPlugin, 301
BioSpeciesManagerGuiExtensionPlugin, 313
CellularAutomataGuiExtensionPlugin, 395
CoreGuiExtensionPlugin, 460
GeneticCircuitGuiExtensionPlugin, 781
GuiExtensionConfigurationGuiExtensionPlugin, 863
ModelPluginInspectorGuiExtensionPlugin, 1093
extract
genesys::launcher::TarZstdArchiveExtractor, 2560
FactorialDesign, 707
calculateResultsMean, 708
calculateStatistics, 708
checkFeasibility, 709
createColumnLabels, 709
createSignTable, 709
generateIndexCombinations, 709
generateInputs, 709
linearRegression, 709
showImpacts, 710
showLinearRegressionModel, 710
showResiduals, 710
showSignTable, 710
Failure, 710
_check, 715
_initBetweenReplications, 715
_loadInstance, 715
_saveInstance, 715
show, 716
FakeModelPersistence, 716
getFormatedField, 717
getOption, 717
hasChanged, 717
load, 717
save, 717
setHasChanged, 718
setOption, 718
FakeModelPersistenceB, 718
getFormatedField, 719
getOption, 719
hasChanged, 719
load, 719
save, 719
setHasChanged, 720
setOption, 720
FakeSystemCommandExecutor, 720
run, 721
fields
Json, 904
File, 721
_check, 726
_getParserChangesInformation, 726
_loadInstance, 726
_saveInstance, 726
show, 726
FileDelete
Util, 2627
FileExists
Util, 2627
FilenameFromFullFilename
Util, 2628
fill
QCPColorMapData, 1445
fillAlpha
QCPColorMapData, 1445
FillOrder
QCPLayoutGrid, 1624
find

ModelComponentManager, 1045, 1046
PluginConnector_if, 1260
PluginConnectorDummyImpl1, 1262
PluginConnectorStaticImpl1, 1265
findBegin
QCPAbstractPlottable1D< DataType >, 1345
QCPDataContainer< DataType >, 1478
QCPErrorBars, 1498
QCPPlottableInterface1D, 1664
findEnd
QCPAbstractPlottable1D< DataType >, 1345
QCPDataContainer< DataType >, 1479
QCPErrorBars, 1498
QCPPlottableInterface1D, 1664
FindLineEdit, 727
finishDownload
genesys::launcher::UpdateDialog, 2622
fisherSnedecor
ProbabilityDistributionBase, 1281
fitAll
FitterDefaultImpl, 731
FitterDummyImpl, 734
fitBeta
FitterDefaultImpl, 731
FitterDummyImpl, 734
fitErlang
FitterDefaultImpl, 731
FitterDummyImpl, 734
fitExpo
FitterDefaultImpl, 731
FitterDummyImpl, 734
fitNormal
FitterDefaultImpl, 731
FitterDummyImpl, 734
Fitter_if, 728
getDataFilename, 729
setDataFilename, 729
FitterDefaultImpl, 730
fitAll, 731
fitBeta, 731
fitErlang, 731
fitExpo, 731
fitNormal, 731
fitTriangular, 732
fitUniform, 732
fitWeibull, 732
getDataFilename, 732
isNormalDistributed, 732
setDataFilename, 732
FitterDummyImpl, 732
fitAll, 734
fitBeta, 734
fitErlang, 734
fitExpo, 734
fitNormal, 734
fitTriangular, 734
fitUniform, 734
fitWeibull, 735
getDataFilename, 735
isNormalDistributed, 735
setDataFilename, 735
fitTriangular
FitterDefaultImpl, 732
FitterDummyImpl, 734
fitUniform
FitterDefaultImpl, 732
FitterDummyImpl, 734
fitWeibull
FitterDefaultImpl, 732
FitterDummyImpl, 735
flagNames
QtFlagPropertyManager, 1896
flagNamesChanged
QtFlagPropertyManager, 1896
flagTypeId
QtVariantPropertyManager, 2016
foColumnsFirst
QCPLayoutGrid, 1624
for_each
CppSerializer, 495
GenSerializer, 806
JsonSerializer, 906
Serializer, 2160
XmlSerializer, 2719
formatKindName
SimulationResultsDataset, 2246
Formula, 740
_check, 744
_createAttachedAttributes, 744
_loadInstance, 744
_saveInstance, 745
Formula, 744
getExpression, 745
GetPluginInformation, 745
getValue, 745
LoadInstance, 746
NewInstance, 746
setExpression, 746
show, 746
size, 747
foRowsFirst
QCPLayoutGrid, 1624
FractionStyle
QCPAxisTickerPi, 1402
fromSortKey
QCPBarsData, 1422
QCPCurveData, 1474
QCPFinancialData, 1515
QCPGraphData, 1526
QCPStatisticalBoxData, 1762
front
ModelComponentManager, 1046
ModelManager, 1089
fsAsciiFractions
QCPAxisTickerPi, 1402
fsFloatingPoint

QCPAxisTickerPi, 1402
FSMState, 747
_loadInstance, 752
_saveInstance, 752
fsUnicodeFractions
QCPAxisTickerPi, 1402
FtsZPolymerizationComponent, 752
_check, 757
_createEditableDataDefinitions, 757
_initBetweenReplications, 757
_loadInstance, 758
_onDispatchEvent, 758
_saveInstance, 758
show, 758
FullSimulationOfComplexModel, 759
main, 760

gamma
ProbabilityDistributionBase, 1281
generate
QCPAxisTicker, 1387
generate_kernel_simulator_test_stubs, 105
generate_kernel_simulator_test_stubs.MethodEntry, 1015
generateGraphicalModelFromModel
GraphicalModelBuilder, 824
generateIndexCombinations
FactorialDesign, 709
generateInputs
FactorialDesign, 709
GenerateNewId
Util, 2628
GenerateNewIdOfType
Util, 2628
generateNewParser
ParserManager, 1217
GenESyS Launcher and Per-User Runtime Dispatcher, 5
Genesys MCP Connector, 15
Genesys Worker Application, 17
genesys::launcher::DispatchController, 603
genesys::launcher::DownloadResult, 619
genesys::launcher::ExtractionResult, 707
genesys::launcher::GpgvSignatureVerifier, 807
verify, 808
genesys::launcher::IArchiveExtractor, 892
genesys::launcher::INetworkTransport, 893
genesys::launcher::IProcessLauncher, 897
genesys::launcher::ISignatureVerifier, 899
genesys::launcher::IUpdateUi, 904
genesys::launcher::LauncherConfig, 914
genesys::launcher::LauncherController, 914
genesys::launcher::LauncherLogger, 915
genesys::launcher::LauncherUpdateResult, 915
genesys::launcher::ManifestParseResult, 958
genesys::launcher::NetworkRequestOptions, 1125
genesys::launcher::NetworkResponse, 1125
genesys::launcher::PlatformInfo, 1253
genesys::launcher::ProcessLauncher, 1289
launchReplacingProcess, 1290
genesys::launcher::QtNetworkTransport, 1933
download, 1933
genesys::launcher::RuntimeInstallResult, 2107
genesys::launcher::RuntimePaths, 2107
genesys::launcher::RuntimeSelection, 2108
genesys::launcher::RuntimeSelector, 2108
genesys::launcher::TarZstdArchiveExtractor, 2560
extract, 2560
genesys::launcher::UpdateChecker, 2620
genesys::launcher::UpdateCheckResult, 2620
genesys::launcher::UpdateDialog, 2621
beginDownload, 2621
confirmUpdate, 2621
finishDownload, 2622
showError, 2622
showInfo, 2622
updateDownloadProgress, 2622
genesys::launcher::UpdateDownloader, 2622
genesys::launcher::UpdateManifest, 2623
genesys::launcher::UpdateManifestParser, 2623
genesys::launcher::UpdatePlatform, 2624
genesys::launcher::UpdateVerifier, 2624
genesys::launcher::UserRuntimeManager, 2624
genesys::launcher::VerificationResult, 2644
genesys::launcher::Version, 2644
genesys_auth
genesys_mcp.server, 107
genesys_configure_simulation
genesys_mcp.server, 107
genesys_create_job
genesys_mcp.server, 108
genesys_get_model
genesys_mcp.server, 108
genesys_health
genesys_mcp.server, 108
genesys_import_model_language
genesys_mcp.server, 108
genesys_info
genesys_mcp.server, 108
genesys_job_result
genesys_mcp.server, 109
genesys_job_status
genesys_mcp.server, 109
genesys_load_model
genesys_mcp.server, 109
genesys_mcp, 106
genesys_mcp.__main__, 106
genesys_mcp.client, 106
genesys_mcp.client.GenesysClient, 763
genesys_mcp.client.GenesysClientError, 764
genesys_mcp.server, 106
genesys_auth, 107
genesys_configure_simulation, 107
genesys_create_job, 108
genesys_get_model, 108
genesys_health, 108
genesys_import_model_language, 108

genesys_info, 108
genesys_job_result, 109
genesys_job_status, 109
genesys_load_model, 109
genesys_new_model, 109
genesys_run_job, 109
genesys_run_simulation, 110
genesys_save_model, 110
genesys_simulation_status, 110
genesys_step_simulation, 110
genesys_worker_capabilities, 110
genesys_worker_info, 111
mcp, 111
genesys_new_model
genesys_mcp.server, 109
genesys_run_job
genesys_mcp.server, 109
genesys_run_simulation
genesys_mcp.server, 110
genesys_save_model
genesys_mcp.server, 110
genesys_simulation_status
genesys_mcp.server, 110
genesys_step_simulation
genesys_mcp.server, 110
genesys_worker_capabilities
genesys_mcp.server, 110
genesys_worker_info
genesys_mcp.server, 111
GenesysApplication_if, 762
main, 763
GenesysKnowledgeBaseSummary, 765
GenesysKnowledgeItem, 765
genesyspp_driver, 766
GenesysPropertyDescriptor, 771
GenesysPropertyIntrospection, 772
GenesysShell, 772
main, 774
GenesysShell_if, 774
GeneticCircuit, 775
_check, 779
_loadInstance, 779
_saveInstance, 779
show, 780
GeneticCircuitGuiExtensionPlugin, 780
displayName, 781
extensionId, 781
registerContributions, 781
requiredModelPlugins, 781
version, 781
GeneticCircuitPart, 782
_check, 786
_loadInstance, 786
_saveInstance, 786
show, 786
GeneticCircuitSimulate, 786
_check, 792
_createEditableDataDefinitions, 792
_loadInstance, 792
_onDispatchEvent, 792
_saveInstance, 792
show, 792
GeneticCircuitSimulationSummary, 793
GeneticExpressionStep, 793
_check, 798
_createEditableDataDefinitions, 798
_loadInstance, 798
_onDispatchEvent, 798
_saveInstance, 798
show, 799
GeneticRegulation, 799
_check, 803
_loadInstance, 803
_saveInstance, 803
show, 804
GenSerializer, 804
dump, 805
for_each, 806
get, 806
load, 806
newPersistenceRecord, 806
put, 806
get
CppSerializer, 496
GenSerializer, 806
JsonSerializer, 906
Serializer, 2161
XmlSerializer, 2720
getAllComponents
ModelComponentManager, 1046
getAnalystName
ModelInfo, 1080
getApiKeyEnvironmentVariable
AIAssistant_if, 136
AIAssistantDefaultImpl, 149
getAttachedData
ModelDataDefinition, 1055
getAttributeValue
Entity, 667
getAuditEntries
AIAssistant_if, 136
AIAssistantDefaultImpl, 150
getAuditLogPath
AIAssistant_if, 137
AIAssistantDefaultImpl, 150
getAvailableControls
Optimizer_if, 1178
OptimizerDefaultImpl1, 1184
getAvailableResponses
Optimizer_if, 1178
OptimizerDefaultImpl1, 1184
getBaseUrl
AIAssistant_if, 137
AIAssistantDefaultImpl, 150
AIProviderClient_if, 160
HttpProviderClientBase, 881

getBestSolutions
Optimizer_if, 1178
OptimizerDefaultImpl1, 1184
getChildSimulationControls
SimulationControl, 2188
SimulationControlGenericClass< T, M, C >, 2200
SimulationControlGenericClassNotDC< T, M, C >, 2205
SimulationControlGenericListPointer< T, M, C >, 2215
getClassname
ModelDataDefinition, 1055
getCollector
Statistics_if, 2476
StatisticsDatafileDefaultImpl1, 2489
StatisticsDefaultImpl1, 2493
getComponent
Event, 688
getComponentinputPortNumber
Event, 689
getComponentManager
Model, 1026
getConfiguration
AIAssistant_if, 137
AIAssistantDefaultImpl, 150
getConnectionAtPort
ConnectionManager, 439
getConnectionManager
ModelComponent, 1042
getConstraints
Optimizer_if, 1178
OptimizerDefaultImpl1, 1185
getControls
Model, 1026
getCountValue
Counter, 465
getCreatableListElementTypes
SimulationControlGenericListPointer< T, M, C >, 2216
getCurrentBestSolution
Optimizer_if, 1178
OptimizerDefaultImpl1, 1185
getCurrentInputConnectionsSize
ConnectionManager, 439
getCurrentIteration
Optimizer_if, 1179
OptimizerDefaultImpl1, 1185
getCurrentListElementType
SimulationControlGenericListPointer< T, M, C >, 2216
getCurrentOutputConnectionsSize
ConnectionManager, 440
getDataDefinition
ModelDataManager, 1063, 1064
getDataDefinitionClassnames
ModelDataManager, 1064
getDataDefinitionPluginTypenames
PluginManager, 1269
getDataFilename
CollectorDatafile_if, 411
CollectorDatafileDefaultImpl1, 414
Fitter_if, 729
FitterDefaultImpl, 732
FitterDummyImpl, 735
getDataManager
Model, 1026
getDataSegments
QCPAbstractPlottable1D< DataType >, 1345
getDescription
ModelComponent, 1042
ModelInfo, 1080
getEntitiesCreated
SourceModelComponent, 2431
getEntitiesPerCreation
SourceModelComponent, 2431
getEntity
Event, 689
getEntityType
Entity, 668
SourceModelComponent, 2431
getEntityTypeName
Entity, 668
getEquations
DiffEquations, 567
getErrorMessage
Parser_if, 1203
ParserDefaultImpl1, 1212
ParserDefaultImpl2, 1214
getExecutionState
Optimizer_if, 1179
OptimizerDefaultImpl1, 1185
getExperimentManager
Simulator, 2250
getExpression
Formula, 745
getFinalScatterStyle
QCPSelectionDecorator, 1741
getFirstCreation
SourceModelComponent, 2431
getFormatedField
FakeModelPersistence, 717
FakeModelPersistenceB, 719
Persistence_if, 1226
PersistenceDefaultImpl2, 1229
getFrontConnection
ConnectionManager, 440
getFutureEvents
Model, 1027
getId
Model, 1027
ModelDataDefinition, 1055
getIncomingEdges
DirectedGraphNetwork, 601
getInfos
Model, 1027
getInitialValuesText

Attribute, 218
Variable, 2644
getInternalData
ModelDataDefinition, 1055
getJob
WorkerJobManager, 2707
getKeyRange
QCPAbstractPlottable, 1333
QCPBars, 1418
QCPColorMap, 1438
QCPCurve, 1470
QCPErrorBars, 1499
QCPFinancial, 1510
QCPGraph, 1522
QCPStatisticalBox, 1758
getKineticLawExpression
BioReaction, 277
getKnowledgeBaseSummary
AIAssistant_if, 137
AIAssistantDefaultImpl, 150
getKnowledgeItems
AIAssistant_if, 137
AIAssistantDefaultImpl, 150
getLabel
ModelDataDefinition, 1056
getLastDiagnostics
AIAssistant_if, 137
AIAssistantDefaultImpl, 150
getLastError
AIAssistant_if, 137
AIAssistantDefaultImpl, 150
GetLastIdOfType
Util, 2629
getLastValue
Collector_if, 409
CollectorDatafileDefaultImpl1, 414
CollectorDefaultImpl1, 416
getLevel
Model, 1027
ModelDataDefinition, 1056
getLicenceManager
Simulator, 2250
getListElementModelDataDefinition
SimulationControlGenericListPointer< T, M, C >, 2216
getMaskedApiKey
AIAssistant_if, 137
AIAssistantDefaultImpl, 150
getMaxCreations
SourceModelComponent, 2431
getMaxInputConnections
ConnectionManager, 440
getMaxOutputConnections
ConnectionManager, 440
getMaxOutputTokens
AIAssistant_if, 137
AIAssistantDefaultImpl, 151
getMaxSteps
SolverDefaultImpl1, 2422
getMinInputConnections
ConnectionManager, 440
getMinOutputConnections
ConnectionManager, 440
getModel
Optimizer_if, 1179
OptimizerDefaultImpl1, 1185
getModelFilename
Optimizer_if, 1179
OptimizerDefaultImpl1, 1185
getModelManager
Simulator, 2250
getModelName
AIAssistant_if, 138
AIAssistantDefaultImpl, 151
getModifiers
BioReaction, 278
getName
ModelDataDefinition, 1056
ModelInfo, 1081
Simulator, 2250
getNeighborCell
Boundary_Closed, 330
Boundary_Fixed, 331
BoundaryCondition, 333
getNeighbors
Neighborhood, 1118
Neighborhood_Center, 1120
Neighborhood_Moore, 1121
Neighborhood_VonNeumann, 1123
getNextValue
CollectorDatafile_if, 411
CollectorDatafileDefaultImpl1, 414
getNumberOfComponents
ModelComponentManager, 1046
getNumberOfDataDefinitions
ModelDataManager, 1064
getObjectives
Optimizer_if, 1179
OptimizerDefaultImpl1, 1185
getOnEventManager
Model, 1027
getOption
FakeModelPersistence, 717
FakeModelPersistenceB, 719
Persistence_if, 1226
PersistenceDefaultImpl2, 1229
getOutgoingEdges
DirectedGraphNetwork, 601
getParent
Counter, 465
StatisticsCollector, 2483
getParentModel
ModelDataDefinition, 1056
getParentSimulator
Model, 1027
TraceManager, 2582

getParser
Parser_if, 1203
ParserDefaultImpl2, 1214
getParserManager
Simulator, 2250
getPartAt
QCPAxis, 1354
QCPPolarAxisAngular, 1678
QCPPolarAxisRadial, 1699
getPersistence
Model, 1028
getPluginInfo
Plugin, 1255
GetPluginInformation
BioReaction, 278
Formula, 745
getPluginManager
Simulator, 2250
getPrecision
SolverDefaultImpl1, 2422
getPredecessors
DirectedGraphNetwork, 601
getProducts
BioReaction, 278
getProjectTitle
ModelInfo, 1081
getProvider
AIAssistant_if, 138
AIAssistantDefaultImpl, 151
getProviderClient
AIAssistant_if, 138
AIAssistantDefaultImpl, 151
getProviderName
AIAssistant_if, 138
AIAssistantDefaultImpl, 151
AIProviderClient_if, 160
AnthropicProviderClientImpl, 202
LocalProviderClientImpl, 931
OpenAIProviderClientImpl, 1171
getRankOf
ModelDataManager, 1064
getRateConstant
BioReaction, 278
getRateConstantParameterName
BioReaction, 278
getReactants
BioReaction, 278
getReasoningEffort
AIAssistant_if, 138
AIAssistantDefaultImpl, 151
getReferencedModelDataDefinition
SimulationControlGenericClass< T, M, C >, 2201
getResponses
Model, 1028
getResponseValues
SimulationScenario, 2248
getReverseKineticLawExpression
BioReaction, 278
getReverseRateConstant
BioReaction, 279
getReverseRateConstantParameterName
BioReaction, 279
getRNGparameters
CountingSampler, 468
Sampler_if, 2110
SamplerDefaultImpl1, 2115
getSampler
Parser_if, 1203
ParserDefaultImpl2, 1214
getScenarios
ExperimentManager_if, 703
ExperimentManagerDefaultImpl1, 705
getSelectedControls
Optimizer_if, 1179
OptimizerDefaultImpl1, 1185
getSelectedResponses
Optimizer_if, 1179
OptimizerDefaultImpl1, 1185
getSessionByToken
SessionManager, 2164
getSettings
Optimizer_if, 1179
OptimizerDefaultImpl1, 1186
getSimulation
Model, 1028
getSimulationControls
ModelDataDefinition, 1056
getSimulationStatus
SimulatorSessionService, 2260
getSimulatorFacade
AIAssistant_if, 138
AIAssistantDefaultImpl, 151
getSourceComponents
ModelComponentManager, 1046
getState
AIAssistant_if, 138
AIAssistantDefaultImpl, 151
getStatistics
StatisticsCollector, 2483
getStrValues
SimulationControlGenericClass< T, M, C >, 2201
SimulationControlGenericEnum< T, E >, 2209
SimulationControlGenericList< T, M, C >, 2212
SimulationControlGenericListPointer< T, M, C >, 2216
getSubTickCount
QCPAxisTicker, 1388
QCPAxisTickerDateTime, 1392
QCPAxisTickerLog, 1399
QCPAxisTickerPi, 1402
QCPAxisTickerText, 1406
QCPAxisTickerTime, 1410
getSuccessors
DirectedGraphNetwork, 601
getTemperature
AIAssistant_if, 138

AIAssistantDefaultImpl, 151
GetTestSuiteName
testing::internal::ParameterizedTestSuiteInfo< TestSuite >, 1199
GetTestSuiteTypeId
testing::internal::ParameterizedTestSuiteInfo< TestSuite >, 1199
getTickLabel
QCPAxisTicker, 1388
QCPAxisTickerDateTime, 1392
QCPAxisTickerPi, 1402
QCPAxisTickerText, 1406
QCPAxisTickerTime, 1410
getTickStep
QCPAxisTicker, 1388
QCPAxisTickerDateTime, 1392
QCPAxisTickerFixed, 1397
QCPAxisTickerPi, 1402
QCPAxisTickerText, 1406
QCPAxisTickerTime, 1410
getTime
Event, 689
getTimeBetweenCreationsExpression
SourceModelComponent, 2431
getTimeoutSeconds
AIAssistant_if, 138
AIAssistantDefaultImpl, 151
AIProviderClient_if, 160
HttpProviderClientBase, 881
getTimeUnit
SourceModelComponent, 2431
getTotalSimulations
Optimizer_if, 1179
OptimizerDefaultImpl1, 1186
getTraceLevel
TraceManager, 2583
getTraceLevelSpecific
ModelDataDefinition, 1056
getTraceManager
Simulator, 2250
getTracer
Model, 1028
getTransferInComponents
ModelComponentManager, 1047
getValue
CollectorDatafile_if, 411
CollectorDatafileDefaultImpl1, 414
Formula, 745
HppLatticeGasState, 875
RealState, 2033
SimulationControlBool, 2191
SimulationControlDouble, 2194
SimulationControlGeneric< T >, 2197
SimulationControlGenericClass< T, M, C >, 2201
SimulationControlGenericClassNotDC< T, M, C >, 2205
SimulationControlGenericEnum< T, E >, 2209
SimulationControlGenericList< T, M, C >, 2212
SimulationControlGenericListPointer< T, M, C >, 2216
SimulationControlInt, 2220
SimulationControlSourceCodeString, 2225
SimulationControlString, 2228
SimulationControlTimeUnit, 2231
SimulationControlUInt, 2234
SimulationControlUShort, 2237
SimulationResponseDouble, 2246
TestSimulationControl, 2571
TestSimulationResponse, 2573
getValueRange
QCPAbstractPlottable, 1334
QCPBars, 1418
QCPColorMap, 1439
QCPCurve, 1471
QCPErrorBars, 1499
QCPFinancial, 1510
QCPGraph, 1523
QCPStatisticalBox, 1758
getVersion
ModelInfo, 1081
Simulator, 2251
getVersionNumber
Simulator, 2251
getVisibleDataBounds
QCPGraph, 1523
getWorkerCapabilities
SimulatorSessionService, 2261
getWorkerInfo
SimulatorSessionService, 2261
getWorkerJob
SimulatorSessionService, 2261
getWorkerJobResult
SimulatorSessionService, 2261
gpCandy
QCPColorGradient, 1429
gpCold
QCPColorGradient, 1429
gpGeography
QCPColorGradient, 1429
gpGrayscale
QCPColorGradient, 1429
gpHot
QCPColorGradient, 1429
gpHues
QCPColorGradient, 1430
gpIon
QCPColorGradient, 1429
gpJet
QCPColorGradient, 1430
gpNight
QCPColorGradient, 1429
gpPolar
QCPColorGradient, 1430
gpSpectrum
QCPColorGradient, 1430
gpThermal

QCPColorGradient, 1429
gradientChanged
QCPColorMap, 1439
QCPColorScale, 1453
GradientPreset
QCPColorGradient, 1429
graph
QCustomPlot, 1786
graphCount
QCustomPlot, 1786
GraphEdge, 808
_check, 813
_loadInstance, 813
_saveInstance, 813
show, 813
GraphicalAnimateExpression, 814
GraphicalAssociation, 814
GraphicalAutomaticPositioningStrategy, 814
~GraphicalAutomaticPositioningStrategy, 815
dataDefinitionPosition, 815
GraphicalAutomaticPositioningStrategyFactory, 815
GraphicalComponentPort, 816
GraphicalComponentPort, 817
GraphicalConnection, 817
boundingRect, 819
GraphicalConnection, 819
paint, 819
updateDimensionsAndPosition, 820
GraphicalConnectionStyle, 820
GraphicalContextMenuController, 820
GraphicalDataDefinitionLayout, 821
arcPosition, 821
GraphicalDiagramConnection, 822
GraphicalImageAnimation, 823
GraphicalModelBuilder, 823
generateGraphicalModelFromModel, 824
GraphicalModelBuilder, 824
recursivalyGenerateGraphicalModelFromModel, 824
setModelLevelFilter, 825
synchronizeGraphicalDataDefinitionsLayer, 825
GraphicalModelComponent, 825
renderContext, 830
sceneEvent, 830
GraphicalModelDataDefinition, 830
GraphicalModelEvent, 832
GraphicalModelItemRenderContext, 833
GraphicalModelItemRenderer, 834
GraphicalModelItemRenderStrategy, 834
GraphicalModelSerializer, 834
efsmNetworkDiagramLayout, 835
GraphicalModelSerializer, 835
loadGraphicalModel, 835
saveGraphicalModel, 836
saveTextModel, 836
setEFSMNetworkDiagramLayout, 836
GraphNetwork, 836
_check, 842
_initBetweenReplications, 842
_loadInstance, 842
_saveInstance, 842
show, 843
GraphNode, 843
GraphPathResult, 848
graphs
QCPAxis, 1354
QCPAxisRect, 1377
GraphSearchTreeEntry, 848
GraphTopologicalOrderResult, 849
GraphTraversalResult, 849
GraphvizModelExporter, 849
adjustDotName, 850
createModelImage, 850
grid
QCPAxis, 1354
GridType
QCPPolarGrid, 1721
GroProgram, 851
_check, 855
_loadInstance, 855
_saveInstance, 855
show, 856
GroProgramAst, 856
GroProgramAst::NamedProgram, 1108
GroProgramAst::Statement, 2461
GroProgramCompiler, 857
GroProgramIr, 857
GroProgramIr::Command, 424
GroProgramIr::NamedProgramDefinition, 1108
GroProgramParser, 858
GroProgramParser::Result, 2086
GroProgramRuntime, 858
validateSignalDeclarationPlacement, 859
GroProgramRuntime::ColonyMutation, 417
GroProgramRuntime::ExecutionResult, 696
GroProgramRuntime::MotionMutation, 1100
GroProgramRuntime::PopulationMutation, 1276
GroProgramRuntime::SignalMutation, 2181
GroProgramRuntimeState, 859
GroupItem, 859
groupTypeId
QtVariantPropertyManager, 2016
GroupUndoCommand, 860
gtest/gtest/gtest-assertion-result.h, 3003
gtest/gtest/gtest-death-test.h, 3006
gtest/gtest/gtest-matchers.h, 3010
gtest/gtest/gtest-message.h, 3023
gtest/gtest/gtest-param-test.h, 3026
gtest/gtest/gtest-printers.h, 3032
gtest/gtest/gtest-spi.h, 3045
gtest/gtest/gtest-test-part.h, 3048
gtest/gtest/gtest-typed-test.h, 3051
gtest/gtest/gtest.h, 3055
gtest/gtest/gtest_pred_impl.h, 3085
gtest/gtest/gtest_prod.h, 3088
gtest/gtest/internal/custom/gtest-port.h, 3089

gtest/gtest/internal/custom/gtest-printers.h, 3045
gtest/gtest/internal/custom/gtest.h, 3084
gtest/gtest/internal/gtest-death-test-internal.h, 3121
gtest/gtest/internal/gtest-filepath.h, 3125
gtest/gtest/internal/gtest-internal.h, 3127
gtest/gtest/internal/gtest-param-util.h, 3145
gtest/gtest/internal/gtest-port-arch.h, 3158
gtest/gtest/internal/gtest-port.h, 3089
gtest/gtest/internal/gtest-string.h, 3159
gtest/gtest/internal/gtest-type-util.h, 3161
gtest/src/gtest-assertion-result.cc, 3164
gtest/src/gtest-death-test.cc, 3165
gtest/src/gtest-filepath.cc, 3185
gtest/src/gtest-internal-inl.h, 3189
gtest/src/gtest-matchers.cc, 3205
gtest/src/gtest-port.cc, 3206
gtest/src/gtest-printers.cc, 3223
gtest/src/gtest-test-part.cc, 3230
gtest/src/gtest-typed-test.cc, 3231
gtest/src/gtest.cc, 3233
GuiActionContribution, 861
GuiCrashDiagnostics, 111
GuiDockContribution, 861
GuiExtensionConfigurationDialog, 862
GuiExtensionConfigurationGuiExtensionPlugin, 863
  displayName, 863
  extensionId, 863
  registerContributions, 863
  requiredModelPlugins, 864
  version, 864
GuiExtensionManager, 864
GuiExtensionPlugin, 865
GuiExtensionPluginAutoRegistrar, 866
GuiExtensionPluginCatalog, 866
GuiExtensionPresentationMetadata, 866
GuiExtensionRegistry, 866
GuiExtensionRuntimeContext, 867
GuiScopeTrace, 867
  GuiScopeTrace, 867
GuiThemeManager, 869
  applyApplicationTheme, 869
  applyModelGraphicsTheme, 869
GuiWindowContribution, 870

Half_Adder, 870
  main, 871
halfWidthConfidenceInterval
  Statistics_if, 2476
  StatisticsDatafileDefaultImpl1, 2490
  StatisticsDefaultImpl1, 2493
handle
  ApiRouter, 203
HarmonicOscillator, 872
  dimension, 872
  evaluate, 872
hasAnchor
  QCPAbstractItem, 1316
hasApiKey
  AIAssistant_if, 138
  AIAssistantDefaultImpl, 152
  AIProviderClient_if, 160
  HttpProviderClientBase, 881
hasBreakpointAt
  ModelComponent, 1042
hasChanged
  FakeModelPersistence, 717
  FakeModelPersistenceB, 719
  Model, 1028
  ModelComponentManager, 1047
  ModelDataDefinition, 1057
  ModelInfo, 1081
  Persistence_if, 1227
  PersistenceDefaultImpl2, 1229
hasCurrentModelSimulation
  SimulationController, 2221
hasCycle
  DirectedGraphNetwork, 601
hasElement
  QCPLayoutGrid, 1626
hasInvalidatedPaintBuffers
  QCustomPlot, 1786
hasItem
  QCPLegend, 1645
  QCustomPlot, 1786
hasItemWithPlottable
  QCPLegend, 1645
hasModel
  ModelManager, 1089
hasObjectInstance
  SimulationControlGenericClass< T, M, C >, 2201
  SimulationControlGenericClassNotDC< T, M, C >, 2205
hasPlottable
  QCustomPlot, 1787
hasProviderClient
  AIAssistant_if, 139
  AIAssistantDefaultImpl, 152
hasSimulatorFacade
  AIAssistant_if, 139
  AIAssistantDefaultImpl, 152
hasValue
  QtAbstractPropertyManager, 1821
  QtGroupPropertyManager, 1911
  QtProperty, 1944
  QtVariantPropertyManager, 2016
header
  HttpRequest, 883
height
  QCPAxisRect, 1377
  QCPPolarAxisAngular, 1679
histogramClassFrequency
  StatisticsDatafile_if, 2485
  StatisticsDatafileDefaultImpl1, 2490
histogramClassLowerLimit
  StatisticsDatafile_if, 2486
  StatisticsDatafileDefaultImpl1, 2490
histogramNumClasses

StatisticsDatafile_if, 2486
StatisticsDatafileDefaultImpl1, 2490
HppLatticeGasState, 874
clone, 875
equals, 875
getValue, 875
setValue, 875
toString, 875
typeName, 875
HppLatticeGasStateSet, 876
contains, 877
createDefaultState, 877
enumerateStates, 877
isComposite, 877
isDiscrete, 877
isFinite, 877
name, 877
parseState, 877
HttpProviderClientBase, 878
_apiKey, 882
_authorizationHeader, 880
_buildRequestBody, 880
_extraRequestHeaders, 880
_parseResponseBody, 880
clearApiKey, 881
getBaseUrl, 881
getTimeoutSeconds, 881
hasApiKey, 881
isAvailable, 881
send, 881
setApiKey, 882
setBaseUrl, 882
setTimeoutSeconds, 882
HttpRequest, 882
header, 883
HttpResponse, 883
HypothesisTester_if, 884
HypothesisTester_if::ConfidenceInterval, 437
HypothesisTester_if::TestResult, 2567
HypothesisTesterDefaultImpl1, 885
averageConfidenceInterval, 887, 888
averageDifferenceConfidenceInterval, 888
estimateSampleSize, 888
proportionConfidenceInterval, 888, 889
proportionDifferenceConfidenceInterval, 889
testAverage, 889
testProportion, 890
testVariance, 890, 891
varianceConfidenceInterval, 891
varianceRatioConfidenceInterval, 891

iconMapTypeId
QtVariantPropertyManager, 2016
importModelFromLanguage
SimulatorSessionService, 2262
iMultiSelect
QCP, 113
incCountValue
Counter, 465
inDegree
DirectedGraphNetwork, 601
Indent
Util, 2629
index
QCPLayer, 1596
indexOf
ModelManager, 1089
indexToRowCol
QCPLayoutGrid, 1626
InitBetweenReplications
ModelDataDefinition, 1057
initializeProperty
QtAbstractPropertyManager, 1821
QtBoolPropertyManager, 1827
QtCharPropertyManager, 1839
QtColorPropertyManager, 1850
QtCursorPropertyManager, 1858
QtDatePropertyManager, 1865
QtDateTimePropertyManager, 1874
QtDoublePropertyManager, 1879
QtEnumPropertyManager, 1892
QtFlagPropertyManager, 1896
QtFontPropertyManager, 1904
QtGroupPropertyManager, 1911
QtIntPropertyManager, 1915
QtKeySequencePropertyManager, 1923
QtLocalePropertyManager, 1929
QtPointFPropertyManager, 1936
QtPointPropertyManager, 1940
QtRectFPropertyManager, 1954
QtRectPropertyManager, 1959
QtSizeFPropertyManager, 1968
QtSizePolicyPropertyManager, 1973
QtSizePropertyManager, 1977
QtStringPropertyManager, 1990
QtTimePropertyManager, 1998
QtVariantPropertyManager, 2017
iNone
QCP, 113
insert
ConnectionManager, 440, 441
Model, 1028
ModelComponentManager, 1047
ModelDataManager, 1066
ModelManager, 1089
PersistenceRecord, 1232
QCPBarsGroup, 1426
insertAtPort
ConnectionManager, 441
insertColumn
QCPLayoutGrid, 1626
insertProperty
QtAbstractPropertyBrowser, 1815
insertRow
QCPLayoutGrid, 1626
insertSubProperty
QtProperty, 1944

insetAlignment
QCPLayoutInset, 1636
insetLayout
QCPAxisRect, 1377
QCPPolarAxisAngular, 1679
InsetPlacement
QCPLayoutInset, 1634
insetPlacement
QCPLayoutInset, 1636
insetRect
QCPLayoutInset, 1636
IntegerStateSet, 893
contains, 894
createDefaultState, 894
enumerateStates, 894
isDiscrete, 895
isFinite, 895
name, 895
parseState, 895
integrate
SolverDefaultImpl1, 2422
Interaction
QCP, 113
interface1D
QCPAbstractPlottable, 1334
QCPAbstractPlottable1D< DataType >, 1346
QCPErrorBars, 1499
InternalEvent, 895
show, 896
intersection
QCPDataRange, 1484
QCPDataSelection, 1488
intersects
QCPDataRange, 1484
inverse
QCPDataSelection, 1488
inverseChi2
ProbabilityDistribution, 1278
inverseFFisherSnedecor
ProbabilityDistribution, 1278
inverseNormal
ProbabilityDistribution, 1278
inverseTStudent
ProbabilityDistribution, 1279
inverted
QCPColorGradient, 1431
ipBorderAligned
QCPLayoutInset, 1635
ipFree
QCPLayoutInset, 1635
iRangeDrag
QCP, 113
iRangeZoom
QCP, 113
isActive
QCPSelectionRect, 1750
isAvailable
AIProviderClient_if, 160
AISecretStore, 167
AnthropicProviderClientImpl, 202
HttpProviderClientBase, 881
LocalProviderClientImpl, 931
OpenAIProviderClientImpl, 1171
isComposite
CompositeStateSet, 436
HppLatticeGasStateSet, 877
isContinuous
RealIntervalStateSet, 2031
isDirected
DirectedGraphNetwork, 601
isDiscrete
HppLatticeGasStateSet, 877
IntegerStateSet, 895
StateSet_Enumerable, 2464
iSelectAxes
QCP, 113
iSelectItems
QCP, 113
iSelectLegend
QCP, 113
iSelectOther
QCP, 113
iSelectPlottables
QCP, 113
iSelectPlottablesBeyondAxisRect
QCP, 113
isEmpty
QCPBarsGroup, 1426
QCPColorMapData, 1445
QCPDataContainer< DataType >, 1479
QCPDataRange, 1484
QCPDataSelection, 1489
QCPMarginGroup, 1654
isEnabled
QtProperty, 1945
isExpanded
QtButtonPropertyBrowser, 1832
QtTreePropertyBrowser, 2004
isFinite
HppLatticeGasStateSet, 877
IntegerStateSet, 895
StateSet_Enumerable, 2464
isInlineObjectProperty
SimulationControlGenericClass< T, M, C >, 2201
SimulationControlGenericClassNotDC< T, M, C >, 2205
isIsValidPlugin
Plugin, 1255
isItemVisible
QtTreePropertyBrowser, 2004
isModelDataDefinitionReference
SimulationControlGenericClass< T, M, C >, 2201
isModified
QtProperty, 1945
isNone
QCPScatterStyle, 1737

isNormalDistributed
FitterDefaultImpl, 732
FitterDummyImpl, 735
isNull
QCPVector2D, 1774
isPenDefined
QCPScatterStyle, 1737
isPropertyTypeSupported
QtVariantPropertyManager, 2017
isRecordFormat
SimulationResultsDataset, 2246
isReportStatistics
ModelDataDefinition, 1057
isReversible
BioReaction, 279
isRunning
ModelSimulation, 1095
IssueReportDraft, 902
IssueReportRelayService, 902
IssueReportRuntimeContext, 903
IssueReportSubmissionResult, 903
isTraceLevelSpecificEnabled
ModelDataDefinition, 1057
isTraceSimulationRuleAllAllowed
TraceManager, 2583
isValid
QCPDataRange, 1484
item
QCPLegend, 1646
QCustomPlot, 1787
itemAt
QCustomPlot, 1787
itemChanged
QtAbstractPropertyBrowser, 1815
QtButtonPropertyBrowser, 1832
QtGroupBoxPropertyBrowser, 1909
QtTreePropertyBrowser, 2004
itemClick
QCustomPlot, 1788
itemCount
QCPLegend, 1646
QCustomPlot, 1788
itemDoubleClick
QCustomPlot, 1788
itemInserted
QtAbstractPropertyBrowser, 1816
QtButtonPropertyBrowser, 1833
QtGroupBoxPropertyBrowser, 1909
QtTreePropertyBrowser, 2004
itemRemoved
QtAbstractPropertyBrowser, 1816
QtButtonPropertyBrowser, 1833
QtGroupBoxPropertyBrowser, 1909
QtTreePropertyBrowser, 2005
items
QCPAxis, 1354
QCPAxisRect, 1377
QtAbstractPropertyBrowser, 1816
itemWithPlottable
QCPLegend, 1646

Json, 904
fields, 904
sequence, 904
JsonSerializer, 905
dump, 906
for_each, 906
get, 906
load, 906
newPersistenceRecord, 907
put, 907

kElidedFramesMarker
testing::internal::OsStackTraceGetterInterface, 1196
Kernel Simulator Unit Tests - Build & Run Guide, 29
kernel/simulator/ConnectionManager.h, 3316
kernel/simulator/DefineGetterSetter.h, 3317
kernel/simulator/essentialPlugins/Attribute.h, 3317
kernel/simulator/essentialPlugins/Counter.h, 3318
kernel/simulator/essentialPlugins/Entity.h, 3318
kernel/simulator/essentialPlugins/EntityType.h, 3319
kernel/simulator/essentialPlugins/SparseValueStore.h, 3320
kernel/simulator/essentialPlugins/StatisticsCollector.h, 3326
kernel/simulator/Event.h, 3326
kernel/simulator/ExperimentManager.h, 3328
kernel/simulator/ExperimentManagerDefaultImpl1.h, 3329
kernel/simulator/ExperimetManager_if.h, 3329
kernel/simulator/GenesysPropertyIntrospection.h, 3330
kernel/simulator/LicenceManager.h, 3331
kernel/simulator/model/Model.h, 3331
kernel/simulator/model/ModelChecker_if.h, 3333
kernel/simulator/model/ModelCheckerDefaultImpl1.h, 3333
kernel/simulator/model/ModelComponent.h, 3334
kernel/simulator/model/ModelComponentManager.h, 3335
kernel/simulator/model/ModelDataDefinition.h, 3335
kernel/simulator/model/ModelDataDefinitionAssociations.h, 3338
kernel/simulator/model/ModelDataManager.h, 3339
kernel/simulator/model/ModelInfo.h, 3340
kernel/simulator/model/ModelManager.h, 3340
kernel/simulator/model/ModelSimulation.h, 3341
kernel/simulator/OnEventManager.h, 3343
kernel/simulator/Parser_if.h, 3349
kernel/simulator/ParserChangesInformation.h, 3349
kernel/simulator/ParserDefaultImpl1.h, 3350
kernel/simulator/ParserDefaultImpl2.h, 3351
kernel/simulator/ParserManager.h, 3351
kernel/simulator/Persistence.h, 3352
kernel/simulator/persistence/CppSerializer.h, 3353
kernel/simulator/persistence/GenSerializer.h, 3353
kernel/simulator/persistence/JsonSerializer.h, 3354

kernel/simulator/persistence/Persistence_if.h, 3354
kernel/simulator/persistence/PersistenceDefaultImpl2.h, 3355
kernel/simulator/persistence/PersistencePartialLoadImpl1.h, 3355
kernel/simulator/persistence/Serializer.h, 3355
kernel/simulator/persistence/XmlSerializer.h, 3356
kernel/simulator/Plugin.h, 3356
kernel/simulator/PluginConnector_if.h, 3357
kernel/simulator/PluginInformation.h, 3357
kernel/simulator/PluginManager.h, 3359
kernel/simulator/PropertyGenesys.h, 3361
kernel/simulator/ScenarioExperiment_if.h, 3362
kernel/simulator/SimulationControlAndResponse.h, 3362
kernel/simulator/SimulationExperiment.h, 3377
kernel/simulator/SimulationReporter_if.h, 3377
kernel/simulator/SimulationReporterDefaultImpl1.h, 3378
kernel/simulator/SimulationScenario.h, 3378
kernel/simulator/Simulator.h, 3379
kernel/simulator/SimulatorFacade.h, 3380
kernel/simulator/SinkModelComponent.h, 3398
kernel/simulator/SourceModelComponent.h, 3398
kernel/simulator/SystemDependencyResolver.h, 3399
kernel/simulator/TraceManager.h, 3401
kernel/statistics/Collector_if.h, 3403
kernel/statistics/CollectorDatafile_if.h, 3404
kernel/statistics/CollectorDatafileDefaultImpl1.h, 3404
kernel/statistics/CollectorDefaultImpl1.h, 3405
kernel/statistics/Sampler_if.h, 3406
kernel/statistics/SamplerDefaultImpl1.h, 3406
kernel/statistics/SorttFile.h, 3412
kernel/statistics/Statistics_if.h, 3412
kernel/statistics/StatisticsDataFile_if.h, 3413
kernel/statistics/StatisticsDataFileDefaultImpl.h, 3413
kernel/statistics/StatisticsDefaultImpl1.h, 3415
kernel/TraitsKernel.h, 3416
kernel/util/Exact.h, 3418
kernel/util/List.h, 3419
kernel/util/ListObservable.h, 3423
kernel/util/Util.h, 3426
KernelAccessorProbe, 907
keyRange
    QCPDataContainer< DataType >, 1479
keyToDateTime
    QCPAxisTickerDateTime, 1392
kFractionBitCount
    testing::internal::FloatingPoint< RawType >, 739
kFractionBitMask
    testing::internal::FloatingPoint< RawType >, 739
kind
    yy::genesyspp_parser::by_kind, 343
kind_
    yy::genesyspp_parser::by_kind, 343

Label, 908
    _check, 911
    _loadInstance, 911
    _saveInstance, 911
    show, 912
LabelMode
    QCPPolarAxisAngular, 1677
LabelSide
    QCPAxis, 1352
last
    ModelManager, 1089
Lattice, 913
launchReplacingProcess
    genesys::launcher::ProcessLauncher, 1290
layer
    QCustomPlot, 1788
layerChanged
    QCPLayerable, 1601
layerCount
    QCustomPlot, 1789
LayerInsertMode
    QCustomPlot, 1782
LayerMode
    QCPLayer, 1596
layout
    QCPLayoutElement, 1615
layoutChanged
    QCPAxisRect, 1377
layoutElementAt
    QCustomPlot, 1789
Leave, 915
    _check, 921
    _createEditableDataDefinitions, 921
    _createInternalStatisticReporters, 921
    _loadInstance, 921
    _onDispatchEvent, 921
    _saveInstance, 922
    show, 922
left
    QCPAxisRect, 1377
    QCPPolarAxisAngular, 1679
legend
    QCustomPlot, 1806
legendClick
    QCustomPlot, 1789
legendDoubleClick
    QCustomPlot, 1789
length
    QCPDataRange, 1485
    QCPVector2D, 1774
lengthSquared
    QCPVector2D, 1775
LicenceManager, 922
limAbove
    QCustomPlot, 1782
limBelow
    QCustomPlot, 1782
limitIteratorsToDataRange
    QCPDataContainer< DataType >, 1479
linearRegression
    FactorialDesign, 709

LineNumberArea, 923
LineStyle
QCPCurve, 1469
QCPGraph, 1521
QCPPolarGraph, 1714
List< T >, 923
_sortFunc, 925
List2str
Util, 2629
ListFiles
Util, 2629
ListObservable< T >, 925
lmBuffered
QCPLayer, 1596
lmLogical
QCPLayer, 1596
load
AISecretStore, 167
CppSerializer, 496
FakeModelPersistence, 717
FakeModelPersistenceB, 719
GenSerializer, 806
JsonSerializer, 906
Model, 1029
Persistence_if, 1227
PersistenceDefaultImpl2, 1229
PersistencePartialLoadImpl1, 1231
Serializer, 2161
SystemPreferences, 2558
XmlSerializer, 2720
loadAndInsertNew
Plugin, 1255
loadApiKeyFromEnvironment
AIAssistant_if, 139
AIAssistantDefaultImpl, 152
loadFromTextFile
SimulationResultsDatasetParser, 2247
loadGraphicalModel
GraphicalModelSerializer, 835
LoadInstance
BioReaction, 279
Formula, 746
ModelComponent, 1043
ModelDataDefinition, 1057
loadInstance
ModelInfo, 1081
loadModel
ModelManager, 1090
SimulatorSessionService, 2262
loadNew
Plugin, 1255
loadPreset
QCPColorGradient, 1431
loadRecentEntries
AIAuditLog, 158
LocalProviderClientImpl, 928
_authorizationHeader, 931
_buildRequestBody, 931
_parseResponseBody, 931
getProviderName, 931
isAvailable, 931
LocalRule, 932
applyRule, 933
LocalRule_CopyCurrentState, 934
applyRule, 935
LocalRule_Elementary, 935
applyRule, 936
LocalRule_FlorestalFire, 936
applyRule, 937
LocalRule_GameOfLife, 938
applyRule, 939
LocalRule_Growty, 939
applyRule, 940
LocalRule_HppLatticeGas, 940
applyRule, 941
LocalRule_SandRock_Student, 942
applyRule, 943
LocalRule_ShowNeighbordhood, 943
applyRule, 944
location.hh
operator<<, 3471
logNormal
ProbabilityDistributionBase, 1281
Loja01, 946
main, 947
lsImpulse
QCPGraph, 1521
lsInside
QCPAxis, 1352
lsLine
QCPCurve, 1469
QCPGraph, 1521
QCPPolarGraph, 1714
lsNone
QCPCurve, 1469
QCPGraph, 1521
QCPPolarGraph, 1714
LSODE, 947
_check, 953
_loadInstance, 953
_onDispatchEvent, 953
_saveInstance, 953
show, 953
LSODEInitialState, 954
LSODESample, 954
lsOutside
QCPAxis, 1352
lsStepCenter
QCPGraph, 1521
lsStepLeft
QCPGraph, 1521
lsStepRight
QCPGraph, 1521

main
Airport_Extended1, 163
AirportSecurityExample, 164

AirportSecurityExampleExtended, 166
AnElectronicAssemblyAndTestSystem, 181
Assembly_Line, 205
Banking_Transactions, 236
BaseGenesysTerminalApplication, 237
BaseGenesysWebApplication, 239
Book_Cap02_Example01, 329
BufferFIFO, 341
Example_BasicOrderShipping, 692
Example_PortModel, 693
Example_PublicTransport, 695
FullSimulationOfComplexModel, 760
GenesysApplication_if, 763
GenesysShell, 774
Half_Adder, 871
Loja01, 947
OperatingSystem02, 1173
OperatingSystem03, 1175
Rectifier, 2054
Smart_AddingResource, 2272
Smart_AlternatingEntityCreation, 2274
Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime, 2276
Smart_ArrivalsEntityTypeVsAttribute, 2278
Smart_AssignExample, 2279
Smart_AssignWriteSeizes, 2281
Smart_Attribute_Variable, 2283
Smart_AutomaticStatisticsCollection, 2285
Smart_BacteriaColony_GRO, 2286
Smart_BasicModeling, 2288
Smart_BatchAndSeparate, 2290
Smart_BatchSeparate, 2291
Smart_BioKineticLawRegulation, 2293
Smart_BioNetworkAnalysis, 2295
Smart_BioReversibleKineticLaw, 2296
Smart_BioReversibleMassAction, 2298
Smart_Buffer, 2300
Smart_CellularAutomata, 2301
Smart_CellularAutomata1, 2303
Smart_Clone, 2305
Smart_ContinuousFlowEntities, 2306
Smart_CppForG, 2308
Smart_Create, 2310
Smart_DecideNWayByChance, 2311
Smart_DefaultModalModel, 2313
Smart_DefiningAttributesAsStrings, 2315
Smart_DefiningControlLogic, 2316
Smart_DefiningResourceCapacity, 2318
Smart_Delay, 2320
Smart_DelayBasedOnReplication, 2322
Smart_Dummy, 2324
Smart_EntitiesProcessedByPriority, 2326
Smart_EvaluatingConditionsBeforeEnteringQueue, 2328
Smart_Expression, 2329
Smart_Failures, 2331
Smart_FSMState, 2333
Smart_GroColonyGrowth, 2334
Smart_GroColonyLifecycle, 2336
Smart_HoldSearchRemove, 2338
Smart_InventoryAndHoldingCosts, 2340
Smart_LSODE, 2341
Smart_MarkovChain, 2343
Smart_MaxArrivalsField, 2345
Smart_ModalModel, 2346
Smart_ModalModelDefaultNetwork, 2348
Smart_ModalModelFSM, 2350
Smart_ModalModelPetriNet, 2351
Smart_ModelInfoModelSimulation, 2353
Smart_ModelRunUntil1000Parts, 2355
Smart_ModuleDisplayVariables, 2356
Smart_ODE, 2358
Smart_OnEvent, 2360
Smart_OverlappingResources, 2363
Smart_ParallelProcessingOfEntities, 2365
Smart_Parser, 2366
Smart_ParserModelFunctions, 2368
Smart_PetriPlace, 2370
Smart_PlacingEntitiesInQueueSets, 2371
Smart_Plugin, 2373
Smart_PriorityExample, 2375
Smart_Process, 2376
Smart_ProcessArena, 2378
Smart_ProcessSet, 2380
Smart_PythonForG, 2381
Smart_R_Simulator, 2383
Smart_Record, 2385
Smart_Record_Arena, 2387
Smart_RemovingAndReorderingEntitiesInAQueue, 2389
Smart_ResourceCosting, 2390
Smart_ResourceScheduleCosting, 2392
Smart_ResourceSets, 2394
Smart_RouteStation, 2395
Smart_SeizeDelayRelease, 2397
Smart_SeizeDelayReleaseMany, 2399
Smart_SeizeDelayReleaseNoDataDefs, 2401
Smart_SeizingMultipleSimultaneosly, 2402
Smart_SelectingRouteBasedOnProbability, 2404
Smart_SelectingShorterQueue, 2406
Smart_Sequence, 2407
Smart_SimulationControlResponse, 2409
Smart_SynchronizingParallelEntities, 2411
Smart_ValueAdded, 2412
Smart_WaitForSignal, 2414
Smart_WaitScanCondition, 2416
Smart_WaitSignal, 2417
StartAIAssistant, 2460
WcmBioNetworkBridge, 2673
WcmCompartmentTransportWholeCellMvp, 2675
WcmFluxBalanceWholeCellMvp, 2677
WcmGeneExpression, 2679
WcmLifecycleStress, 2681
WcmMgenitaliumKarr2012, 2683
WcmPathwayStressWholeCellMvp, 2685
WcmSimpleSsa, 2687

WcmWholeCellMvp, 2689
WcmYeastDidacticWholeCellMvp, 2691
WcmYeastEukaryoticWholeCellMvp, 2693
mainKey
QCPBarsData, 1422
QCPCurveData, 1474
QCPFinancialData, 1515
QCPGraphData, 1526
QCPStatisticalBoxData, 1762
mainValue
QCPBarsData, 1422
QCPCurveData, 1474
QCPFinancialData, 1515
QCPGraphData, 1527
QCPStatisticalBoxData, 1763
MainWindow, 954
makeNonCosmetic
QCPPainter, 1660
managerDestroyed
QtAbstractEditorFactory< PropertyManager >, 1810
Map2str
Util, 2630
MarginSide
QCP, 114
marginSideToAxisType
QCPAxis, 1354
MarkovChain, 959
_check, 964
_createEditableDataDefinitions, 964
_initBetweenReplications, 964
_loadInstance, 964
_onDispatchEvent, 964
_saveInstance, 964
show, 965
MarkovChainNetwork, 965
_activate, 970
_check, 970
_initBetweenReplications, 970
_loadInstance, 970
_saveInstance, 970
show, 972
MarkovChainNetwork::MarkovTransition, 977
MarkovState, 972
MassActionOdeSystem, 977
dimension, 978
evaluate, 978
MassActionOdeSystem::Reaction, 2030
MassActionOdeSystem::Species, 2435
MassActionOdeSystem::StoichiometricTerm, 2520
Match, 978
_check, 984
_createEditableDataDefinitions, 984
_loadInstance, 984
_onDispatchEvent, 984
_saveInstance, 984
show, 984
max
Statistics_if, 2476
StatisticsDatafileDefaultImpl1, 2490
StatisticsDefaultImpl1, 2493
maximum
QtDatePropertyManager, 1865
QtDoublePropertyManager, 1879
QtIntPropertyManager, 1915
QtSizeFPropertyManager, 1968
QtSizePropertyManager, 1977
maximumOuterSizeHint
QCPLayoutElement, 1615
QCPLayoutGrid, 1627
QCPTextElement, 1769
maxRange
QCPRange, 1732
mcp
genesys_mcp.server, 111
mediane
StatisticsDatafile_if, 2486
StatisticsDatafileDefaultImpl1, 2490
MetabolicFluxBalance, 985
_check, 990
_createEditableDataDefinitions, 990
_loadInstance, 990
_onDispatchEvent, 991
_saveInstance, 991
show, 991
MetabolicFluxBalanceSolver, 991
MetabolicFluxBalanceSolver::Problem, 1282
MetabolicFluxBalanceSolver::Solution, 2418
MetabolicNetwork, 992
_check, 996
_createEditableDataDefinitions, 996
_loadInstance, 996
_saveInstance, 996
show, 996
MetabolicReaction, 997
_check, 1001
_createEditableDataDefinitions, 1001
_loadInstance, 1001
_saveInstance, 1001
show, 1001
MetabolicReaction::StoichiometricTerm, 2520
MetabolicStateProjectionComponent, 1002
_check, 1008
_createEditableDataDefinitions, 1008
_loadInstance, 1008
_onDispatchEvent, 1008
_saveInstance, 1008
show, 1009
MetabolicStateProjectionComponent::Projection, 1291
MetabolicSubmodelComponent, 1009
_check, 1014
_createEditableDataDefinitions, 1014
_initBetweenReplications, 1014
_loadInstance, 1014
_onDispatchEvent, 1014
_saveInstance, 1015

show, 1015
min
Statistics_if, 2476
StatisticsDatafileDefaultImpl1, 2490
StatisticsDefaultImpl1, 2494
minimum
QtDatePropertyManager, 1866
QtDoublePropertyManager, 1879
QtIntPropertyManager, 1915
QtSizeFPropertyManager, 1968
QtSizePropertyManager, 1978
minimumOuterSizeHint
QCPLayoutElement, 1615
QCPLayoutGrid, 1627
QCPPlottableLegendItem, 1669
QCPPolarLegendItem, 1727
QCPTextElement, 1769
minRange
QCPRange, 1732
ModalModelBindingsUndoCommand, 1016
ModalModelBindingsUndoCommand::Bindings, 249
ModalModelDefault, 1016
_check, 1022
_createAttachedAttributes, 1022
_initBetweenReplications, 1022
_loadInstance, 1022
_onDispatchEvent, 1022
_saveInstance, 1022
show, 1023
ModalModelDefault::DEFAULT_VALUES, 520
mode
StatisticsDatafile_if, 2486
StatisticsDatafileDefaultImpl1, 2490
Model, 1023
check, 1025
checkExpression, 1025
checkReferencesToDataDefinitions, 1025
collectDataDefinitionsRemovedWith, 1026
createEntity, 1026
getComponentManager, 1026
getControls, 1026
getDataManager, 1026
getFutureEvents, 1027
getId, 1027
getInfos, 1027
getLevel, 1027
getOnEventManager, 1027
getParentSimulator, 1027
getPersistence, 1028
getResponses, 1028
getSimulation, 1028
getTracer, 1028
hasChanged, 1028
insert, 1028
load, 1029
Model, 1025
parseExpression, 1029
remove, 1030
removeEntity, 1030
save, 1030
sendEntityToComponent, 1030
setHasChanged, 1031
setTracer, 1031
showLanguage, 1031
modelAt
ModelManager, 1090
ModelChecker_if, 1031
checkActivationCode, 1032
checkAll, 1032
checkConnected, 1032
checkLimits, 1033
checkSymbols, 1033
ModelCheckerDefaultImpl1, 1033
checkActivationCode, 1034
checkAll, 1034
checkConnected, 1034
checkLimits, 1035
checkSymbols, 1035
ModelComponent, 1035
_connections, 1044
_loadInstance, 1040
_onDispatchEvent, 1040
_saveInstance, 1041
Check, 1041
connectTo, 1041
CreateInternalData, 1042
DispatchEvent, 1042
getConnectionManager, 1042
getDescription, 1042
hasBreakpointAt, 1042
LoadInstance, 1043
ModelComponent, 1040
remove, 1043
SaveInstance, 1043
setDescription, 1043
show, 1043
ModelComponentManager, 1044
begin, 1045
clear, 1045
end, 1045
find, 1045, 1046
front, 1046
getAllComponents, 1046
getNumberOfComponents, 1046
getSourceComponents, 1046
getTransferInComponents, 1047
hasChanged, 1047
insert, 1047
next, 1047
remove, 1047
setHasChanged, 1048
ModelDataDefinition, 1048
_addSimulationControl, 1052
_check, 1052
_createAttachedAttributes, 1052
_createEditableDataDefinitions, 1052

_createInternalAndAttachedData, 1052
_createNonEditableDataDefinitions, 1052
_getParserChangesInformation, 1052
_getSaveDefaultsOption, 1052
_initBetweenReplications, 1053
_loadInstance, 1053
_saveInstance, 1053
Check, 1054
CreateInternalData, 1054
CreateRelatedDataElements, 1054
defineTraceLevelSpecific, 1055
getAttachedData, 1055
getClassname, 1055
getId, 1055
getInternalData, 1055
getLabel, 1056
getLevel, 1056
getName, 1056
getParentModel, 1056
getSimulationControls, 1056
getTraceLevelSpecific, 1056
hasChanged, 1057
InitBetweenReplications, 1057
isReportStatistics, 1057
isTraceLevelSpecificEnabled, 1057
LoadInstance, 1057
ModelDataDefinition, 1051
NewInstance, 1058
SaveInstance, 1058
setHasChanged, 1058
setLabel, 1058
setModelLevel, 1059
setName, 1059
setReportStatistics, 1059
setTraceLevelSpecific, 1059
setTraceLevelSpecificEnabled, 1059
show, 1059
traceSimulation, 1060
ModelDataDefinitionAssociations, 1060
ModelDataManager, 1061
check, 1063
getDataDefinition, 1063, 1064
getDataDefinitionClassnames, 1064
getNumberOfDataDefinitions, 1064
getRankOf, 1064
insert, 1066
remove, 1066
ModelEvent, 1067
ModelGraphicsScene, 1067
notifyGraphicalModelChange, 1076
setModelLevelFilter, 1076
ModelGraphicsScene::GRID, 851
ModelGraphicsView, 1076
ModelGraphicsView, 1078
notifySceneGraphicalModelEventHandler, 1079
ModelInfo, 1080
getAnalystName, 1080
getDescription, 1080
getName, 1081
getProjectTitle, 1081
getVersion, 1081
hasChanged, 1081
loadInstance, 1081
saveInstance, 1081
setAnalystName, 1082
setDescription, 1082
setName, 1082
setProjectTitle, 1082
setVersion, 1082
ModelInspectorController, 1083
ModelInspectorController::DataDefinitionSelection, 513
ModelLanguageSynchronizer, 1084
actualizeModelSimLanguage, 1085
ModelLanguageSynchronizer, 1085
setSimulationModelBasedOnText, 1085
ModelLifecycleController, 1085
ModelLifecycleController::Callbacks, 343
ModelManager, 1086
canGoNext, 1088
canGoPrevious, 1088
createFromLanguage, 1088
current, 1088
currentIndex, 1088
front, 1089
hasModel, 1089
indexOf, 1089
insert, 1089
last, 1089
loadModel, 1090
modelAt, 1090
ModelManager, 1088
models, 1090
newModel, 1090
next, 1090
previous, 1090
remove, 1091
saveModel, 1091
setCurrent, 1091
size, 1091
modelPlan
AIAssistantRequest, 155
ModelPluginInspectorGuiExtensionPlugin, 1092
displayName, 1093
extensionId, 1093
registerContributions, 1093
requiredModelPlugins, 1093
version, 1093
models
ModelManager, 1090
ModelSimulation, 1094
isRunning, 1095
start, 1095
MolecularSpecies, 1095
_check, 1100
_initBetweenReplications, 1100
_loadInstance, 1100

_saveInstance, 1100
show, 1100
mouseDoubleClick
QCustomPlot, 1789
mouseDoubleClickEvent
QCPLayerable, 1601
QCPTextElement, 1769
mouseMove
QCustomPlot, 1790
mouseMoveEvent
QCPAxis, 1354
QCPAxisRect, 1377
QCPColorScale, 1453
QCPColorScaleAxisRectPrivate, 1461
QCPLayerable, 1601
QCPPolarAxisAngular, 1679
QCPPolarAxisRadial, 1699
mousePress
QCustomPlot, 1790
mousePressEvent
QCPAxis, 1355
QCPAxisRect, 1378
QCPColorScale, 1453
QCPColorScaleAxisRectPrivate, 1462
QCPLayerable, 1601
QCPPolarAxisAngular, 1679
QCPPolarAxisRadial, 1699
QCPTextElement, 1769
mouseRelease
QCustomPlot, 1790
mouseReleaseEvent
QCPAxis, 1355
QCPAxisRect, 1378
QCPColorScale, 1454
QCPColorScaleAxisRectPrivate, 1462
QCPLayerable, 1602
QCPPolarAxisAngular, 1680
QCPPolarAxisRadial, 1700
QCPTextElement, 1770
mouseWheel
QCustomPlot, 1790
Move, 1101
_check, 1106
_createAttachedAttributes, 1106
_createEditableDataDefinitions, 1106
_createInternalStatisticReporters, 1106
_loadInstance, 1106
_onDispatchEvent, 1106
_saveInstance, 1106
show, 1106
move
yy::genesyspp_parser::value_type, 2636
moveAbove
QCPBars, 1419
moveBelow
QCPBars, 1419
moveLayer
QCustomPlot, 1790
moveRange
QCPAxis, 1355
QCPPolarAxisAngular, 1680
QCPPolarAxisRadial, 1700
MoveUndoCommand, 1107
msAll
QCP, 114
msBottom
QCP, 114
msLeft
QCP, 114
msNone
QCP, 114
msRight
QCP, 114
msTop
QCP, 114

name
CompositeStateSet, 436
HppLatticeGasStateSet, 877
IntegerStateSet, 895
NaturalStateSet, 1116
RealIntervalStateSet, 2031
StateSet_Enumerable, 2464
NAND, 1109
_loadInstance, 1114
_saveInstance, 1114
Build, 1114
NanHandling
QCPColorGradient, 1430
NaturalStateSet, 1115
name, 1116
Neighborhood, 1117
getNeighbors, 1118
Neighborhood_Center, 1118
getNeighbors, 1120
Neighborhood_Moore, 1120
getNeighbors, 1121
show, 1121
Neighborhood_VonNeumann, 1122
getNeighbors, 1123
show, 1123
NetworkActivationFrame, 1123
NetworkActivationResult, 1124
NetworkPortValue, 1124
NewInstance
BioReaction, 279
Formula, 746
ModelDataDefinition, 1058
newInstance
Plugin, 1255
newModel
ModelManager, 1090
newPersistenceRecord
CppSerializer, 496
GenSerializer, 806
JsonSerializer, 907
Serializer, 2161

XmlSerializer, 2720
newSampleSize
Statistics_if, 2476
StatisticsDatafileDefaultImpl1, 2490
StatisticsDefaultImpl1, 2494
next
ModelComponentManager, 1047
ModelManager, 1090
nhHighestColor
QCPColorGradient, 1430
nhLowestColor
QCPColorGradient, 1430
nhNanColor
QCPColorGradient, 1430
nhNone
QCPColorGradient, 1430
nhTransparent
QCPColorGradient, 1430
NMOS, 1126
_loadInstance, 1131
_saveInstance, 1131
Build, 1131
NOR, 1132
_loadInstance, 1138
_saveInstance, 1138
Build, 1138
normal
ProbabilityDistributionBase, 1281
normalize
QCPRange, 1730
QCPVector2D, 1775
normalized
QCPVector2D, 1775
NOT, 1139
_loadInstance, 1144
_saveInstance, 1144
Build, 1144
notifyGraphicalModelChange
ModelGraphicsScene, 1076
notifySceneGraphicalModelEventHandler
ModelGraphicsView, 1079
numElements
Collector_if, 409
CollectorDatafileDefaultImpl1, 414
CollectorDefaultImpl1, 416
Statistics_if, 2477
StatisticsDatafileDefaultImpl1, 2491
StatisticsDefaultImpl1, 2494

obj_t, 1145
ObjectPropertyBrowser, 1146
OctaveRunner, 1149
_check, 1154
_loadInstance, 1154
_saveInstance, 1154
show, 1154
ODEfunction, 1154
ODESolver, 1155
_check, 1159
_createEditableDataDefinitions, 1159
_initBetweenReplications, 1159
_loadInstance, 1159
_saveInstance, 1159
show, 1160
OdeSolver_if, 1160
OdeSolverFactory, 1161
OdeSystem_if, 1161
OLD_ODEelement, 1162
_check, 1165
_loadInstance, 1165
_saveInstance, 1166
show, 1166
onActionAboutReportIssueTriggered
DialogUtilityController, 561
onActionDrawLineTriggered
SceneToolController, 2120
onActionEditReplaceTriggered
DialogUtilityController, 561
onActionShowGridTriggered
SceneToolController, 2120
onActionToolsParserGrammarCheckerTriggered
DialogUtilityController, 561
onActionZoomActualSizeTriggered
SceneToolController, 2121
onActionZoomAllTriggered
SceneToolController, 2121
onBreakpointHandler
Smart_OnEvent, 2360
onEntityCreateHandler
Smart_OnEvent, 2360
onEntityMoveHandler
Smart_OnEvent, 2360
onEntityRemoveHandler
Smart_OnEvent, 2360
OnEnvironmentsSetUpEnd
testing::EmptyTestEventListener, 654
OnEnvironmentsSetUpStart
testing::EmptyTestEventListener, 654
OnEnvironmentsTearDownEnd
testing::EmptyTestEventListener, 654
OnEnvironmentsTearDownStart
testing::EmptyTestEventListener, 654
OnEventManager, 1166
onHorizontalSliderZoomGraphicalValueChanged
SceneToolController, 2121
onProcessEventHandler
Smart_OnEvent, 2360
onReplicationEndHandler
Smart_OnEvent, 2360
onReplicationStartHandler
Smart_OnEvent, 2360
onReplicationStepHandler
Smart_OnEvent, 2360
onSimulationEndHandler
Smart_OnEvent, 2360
onSimulationPausedHandler
Smart_OnEvent, 2361

onSimulationResumeHandler
Smart_OnEvent, 2361
onSimulationStartHandler
Smart_OnEvent, 2361
OnTestCaseEnd
testing::EmptyTestEventListener, 654
OnTestCaseStart
testing::EmptyTestEventListener, 654
OnTestDisabled
testing::EmptyTestEventListener, 654
OnTestEnd
testing::EmptyTestEventListener, 654
OnTestIterationEnd
testing::EmptyTestEventListener, 654
OnTestIterationStart
testing::EmptyTestEventListener, 654
OnTestPartResult
testing::EmptyTestEventListener, 655
OnTestProgramEnd
testing::EmptyTestEventListener, 655
OnTestProgramStart
testing::EmptyTestEventListener, 655
OnTestStart
testing::EmptyTestEventListener, 655
OnTestSuiteEnd
testing::EmptyTestEventListener, 655
OnTestSuiteStart
testing::EmptyTestEventListener, 655
OpenAIProviderClientImpl, 1168
_authorizationHeader, 1170
_buildRequestBody, 1170
_extraRequestHeaders, 1170
_parseResponseBody, 1171
getProviderName, 1171
isAvailable, 1171
OpenAIProviderClientImpl, 1170
OperatingSystem02, 1172
main, 1173
OperatingSystem03, 1173
main, 1175
operator<<
location.hh, 3471
QCPDataRange, 1485
QCPDataSelection, 1491
QCPRange, 1732
QCPVector2D, 1776
operator()
yy::genesyspp_parser, 770
operator+
QCPDataSelection, 1490
QCPRange, 1731
operator+=
QCPDataSelection, 1489
QCPRange, 1730
QCPVector2D, 1775
operator-
QCPDataSelection, 1490
QCPRange, 1731
operator-=
QCPDataSelection, 1489
QCPRange, 1730
QCPVector2D, 1775
operator/
QCPRange, 1732
operator/=
QCPRange, 1730
QCPVector2D, 1775
operator=
QCPColorMapData, 1445
operator==
QCPDataSelection, 1489
operator∗
QCPRange, 1731
operator∗=
QCPRange, 1730
QCPVector2D, 1775
opposite
QCPAxis, 1355
Optimizer_if, 1175
addConstraint, 1177
addObjective, 1177
checkReady, 1178
clearConstraints, 1178
clearObjectives, 1178
getAvailableControls, 1178
getAvailableResponses, 1178
getBestSolutions, 1178
getConstraints, 1178
getCurrentBestSolution, 1178
getCurrentIteration, 1179
getExecutionState, 1179
getModel, 1179
getModelFilename, 1179
getObjectives, 1179
getSelectedControls, 1179
getSelectedResponses, 1179
getSettings, 1179
getTotalSimulations, 1179
pause, 1179
resetResults, 1180
resume, 1180
setModel, 1180
setModelFilename, 1180
setSelectedControls, 1180
setSelectedResponses, 1180
setSettings, 1180
start, 1180
step, 1181
stop, 1181
Optimizer_if::ConstraintDefinition, 443
Optimizer_if::ObjectiveDefinition, 1145
Optimizer_if::OptimizationSettings, 1175
Optimizer_if::SolutionSummary, 2418
OptimizerDefaultImpl1, 1181
addConstraint, 1184
addObjective, 1184

checkReady, 1184
clearConstraints, 1184
clearObjectives, 1184
getAvailableControls, 1184
getAvailableResponses, 1184
getBestSolutions, 1184
getConstraints, 1185
getCurrentBestSolution, 1185
getCurrentIteration, 1185
getExecutionState, 1185
getModel, 1185
getModelFilename, 1185
getObjectives, 1185
getSelectedControls, 1185
getSelectedResponses, 1185
getSettings, 1186
getTotalSimulations, 1186
pause, 1186
resetResults, 1186
resume, 1186
setModel, 1186
setModelFilename, 1186
setSelectedControls, 1186
setSelectedResponses, 1187
setSettings, 1187
start, 1187
step, 1187
stop, 1187
OptimizerWindow, 1188
OptimizerWindow::ProgressPlot, 1290
OR, 1189
_loadInstance, 1194
_saveInstance, 1194
Build, 1194
orientation
QCPAxis, 1355, 1356
outDegree
DirectedGraphNetwork, 602
outerRect
QCPLayoutElement, 1615

paint
GraphicalConnection, 819
PainterMode
QCPPainter, 1659
parent
QtBrowserItem, 1829
parentAnchor
QCPItemPosition, 1563
parentLayerable
QCPLayerable, 1602
parentPlotInitialized
QCPLayoutElement, 1615
QCPLegend, 1646
parse
Parser_if, 1203
ParserDefaultImpl1, 1212
ParserDefaultImpl2, 1214, 1215
yy::genesyspp_parser, 770
parseExpression
Model, 1029
parser/Genesys++-driver.h, 3428
parser/GenesysParser.h, 3429, 3431
parser/location.hh, 3469, 3471
parser/obj_t.h, 3474
parser/position.hh, 3475
parser/stack.hh, 3475
Parser_if, 1202
getErrorMessage, 1203
getParser, 1203
getSampler, 1203
parse, 1203
setSampler, 1204
ParserAttributeEventInjector, 1204
ParserAttributeInternalEventOwner, 1205
ParserAttributeProcessObserver, 1210
ParserChangesInformation, 1210
ParserDefaultImpl1, 1211
getErrorMessage, 1212
parse, 1212
ParserDefaultImpl2, 1213
∼ParserDefaultImpl2, 1214
getErrorMessage, 1214
getParser, 1214
getSampler, 1214
parse, 1214, 1215
ParserDefaultImpl2, 1214
setSampler, 1215
ParserExpressionsTest, 1216
SetUp, 1217
ParserManager, 1217
connectNewParser, 1217
generateNewParser, 1217
ParserManager::GenerateNewParserResult, 760
ParserManager::NewParser, 1125
parseState
HppLatticeGasStateSet, 877
IntegerStateSet, 895
RealIntervalStateSet, 2032
StateSet_Enumerable, 2464
PasteUndoCommand, 1218
PathFromFullFilename
Util, 2630
PathwayStressResponseComponent, 1219
_check, 1224
_createEditableDataDefinitions, 1224
_initBetweenReplications, 1224
_loadInstance, 1224
_onDispatchEvent, 1224
_saveInstance, 1225
show, 1225
pause
Optimizer_if, 1179
OptimizerDefaultImpl1, 1186
perpendicular
QCPVector2D, 1775
Persistence_if, 1225

getFormatedField, 1226
getOption, 1226
hasChanged, 1227
load, 1227
save, 1227
setHasChanged, 1227
setOption, 1227
PersistenceDefaultImpl2, 1228
getFormatedField, 1229
getOption, 1229
hasChanged, 1229
load, 1229
save, 1229
setHasChanged, 1229
setOption, 1229
PersistencePartialLoadImpl1, 1230
load, 1231
PersistenceRecord, 1231
insert, 1232
PersistenceRecord::Entry, 679
PersistentObject_base, 1233
_loadInstance, 1234
_saveInstance, 1234
PetriPlace, 1235
_loadInstance, 1240
_saveInstance, 1240
phCacheLabels
QCP, 114
phFastPolylines
QCP, 114
phImmediateRefresh
QCP, 114
phNone
QCP, 114
PickableStationItem, 1240
PickStation, 1241
_addSimulationControl, 1246
_check, 1246
_createEditableDataDefinitions, 1246
_getParserChangesInformation, 1246
_initBetweenReplications, 1246
_loadInstance, 1246
_onDispatchEvent, 1247
_saveInstance, 1247
show, 1247
PickUp, 1247
pixelOrientation
QCPAxis, 1356
pixelPosition
QCPItemAnchor, 1533
QCPItemPosition, 1563
pixelsToCoords
QCPAbstractPlottable, 1334
pixelToCoord
QCPAxis, 1356
QCPPolarAxisAngular, 1680
QCPPolarAxisRadial, 1700
plotLayout
QCustomPlot, 1790
plottable
QCustomPlot, 1791
plottableAt
QCustomPlot, 1791
plottableClick
QCustomPlot, 1791
plottableCount
QCustomPlot, 1792
plottableDoubleClick
QCustomPlot, 1792
plottables
QCPAxis, 1356
QCPAxisRect, 1378
PlottingHint
QCP, 114
Plugin, 1253
getPluginInfo, 1255
isIsValidPlugin, 1255
loadAndInsertNew, 1255
loadNew, 1255
newInstance, 1255
Plugin, 1254
show, 1257
PluginCatalogController, 1257
PluginConnector_if, 1258
check, 1259
connect, 1259
disconnect, 1259, 1260
find, 1260
PluginConnectorDummyImpl1, 1260
check, 1261
connect, 1261
disconnect, 1262
find, 1262
PluginConnectorStaticImpl1, 1263
check, 1264
connect, 1264
disconnect, 1264
find, 1265
PluginInformation, 1265
PluginInsertionOptions, 1266
PluginLoadIssue, 1266
PluginManager, 1267
getDataDefinitionPluginTypenames, 1269
PluginManager, 1269
plugins/components/AI/AIAssistant.h, 3475
plugins/components/AnalyticalModeling/MarkovChain.h, 3476
plugins/components/BiochemicalSimulation/BacteriaColony.h, 3477
plugins/components/BiochemicalSimulation/BioRunnerCommand.h, 3481
plugins/components/BiochemicalSimulation/BioSimulate.h, 3482
plugins/components/BiochemicalSimulation/BioSteadyState.h, 3483

plugins/components/BiochemicalSimulation/GeneticCircuitSimulate.h, 3484
plugins/components/BiochemicalSimulation/GeneticExpressionStep.h, 3485
plugins/components/BiochemicalSimulation/MetabolicFluxBalance.h, 3486
plugins/components/Continuous/ContinuousSystemComponents.h, 3487
plugins/components/Continuous/DiffEquations.h, 3487
plugins/components/Continuous/DiffusionSimulate.h, 3489
plugins/components/Continuous/LSODE.h, 3490
plugins/components/Continuous/OLD_ODEelement.h, 3490
plugins/components/Decisions/Decide.h, 3491
plugins/components/Decisions/DropOff.h, 3492
plugins/components/Decisions/PickUp.h, 3493
plugins/components/Decisions/Remove.h, 3493
plugins/components/Decisions/Search.h, 3494
plugins/components/DiscreteProcessing/auxiliar/QueueableItem.h, 3495
plugins/components/DiscreteProcessing/auxiliar/SeizableItem.h, 3496
plugins/components/DiscreteProcessing/Buffer.h, 3498
plugins/components/DiscreteProcessing/Clone.h, 3499
plugins/components/DiscreteProcessing/Delay.h, 3499
plugins/components/DiscreteProcessing/Process.h, 3500
plugins/components/DiscreteProcessing/Release.h, 3501
plugins/components/DiscreteProcessing/Seize.h, 3502
plugins/components/ElectronicsSimulation/SPICECircuit.h, 3504
plugins/components/ElectronicsSimulation/SPICENode.h, 3511
plugins/components/ExternalIntegration/CppForG.h, 3512
plugins/components/ExternalIntegration/PythonForG.h, 3513
plugins/components/ExternalIntegration/RSimulator.h, 3513
plugins/components/Grouping/Batch.h, 3514
plugins/components/Grouping/Separate.h, 3515
plugins/components/InputOutput/Record.h, 3516
plugins/components/InputOutput/Write.h, 3517
plugins/components/Logic/Assign.h, 3518
plugins/components/Logic/Create.h, 3519
plugins/components/Logic/Dispose.h, 3520
plugins/components/MaterialHandling/Access.h, 3520
plugins/components/MaterialHandling/Enter.h, 3521
plugins/components/MaterialHandling/Exit.h, 3522
plugins/components/MaterialHandling/Leave.h, 3523
plugins/components/MaterialHandling/Move.h, 3523
plugins/components/MaterialHandling/PickableStationItem.h, 3524
plugins/components/MaterialHandling/PickStation.h, 3525
plugins/components/MaterialHandling/Route.h, 3526
plugins/components/MaterialHandling/Start.h, 3527
plugins/components/MaterialHandling/Stop.h, 3528
plugins/components/MaterialHandling/Store.h, 3528
plugins/components/MaterialHandling/Unstore.h, 3529
plugins/components/ModalModel/CellularAutomata/Boundary_Closed.h, 3530
plugins/components/ModalModel/CellularAutomata/Boundary_Fixed.h, 3531
plugins/components/ModalModel/CellularAutomata/BoundaryCondition.h, 3531
plugins/components/ModalModel/CellularAutomata/Cell.h, 3532
plugins/components/ModalModel/CellularAutomata/CellularAutomata.h, 3533
plugins/components/ModalModel/CellularAutomata/CellularAutomata_1DT
3533
plugins/components/ModalModel/CellularAutomata/CellularAutomata_Clas
3534
plugins/components/ModalModel/CellularAutomata/CellularAutomataAllHe
3534
plugins/components/ModalModel/CellularAutomata/CellularAutomataBase
3535
plugins/components/ModalModel/CellularAutomata/Lattice.h, 3535
plugins/components/ModalModel/CellularAutomata/LocalRule.h, 3536
plugins/components/ModalModel/CellularAutomata/LocalRule_Elementary
3536
plugins/components/ModalModel/CellularAutomata/LocalRule_FlorestalFi
3537
plugins/components/ModalModel/CellularAutomata/LocalRule_GameOfLif
3538
plugins/components/ModalModel/CellularAutomata/LocalRule_Growty.h, 3538
plugins/components/ModalModel/CellularAutomata/LocalRule_HppLattice
3539
plugins/components/ModalModel/CellularAutomata/LocalRule_SandRock.
3540
plugins/components/ModalModel/CellularAutomata/LocalRule_SandRock_
3542
plugins/components/ModalModel/CellularAutomata/LocalRule_ShowNeigh
3544
plugins/components/ModalModel/CellularAutomata/Neighborhood.h, 3545
plugins/components/ModalModel/CellularAutomata/Neighborhood_Center
3546
plugins/components/ModalModel/CellularAutomata/Neighborhood_Moore.
3546
plugins/components/ModalModel/CellularAutomata/Neighborhood_VonNe
3547
plugins/components/ModalModel/CellularAutomata/State.h, 3548
plugins/components/ModalModel/CellularAutomata/State_Numerical.h, 3550
plugins/components/ModalModel/CellularAutomata/StateSet.h, 3551
plugins/components/ModalModel/CellularAutomata/StateSet_Enumerable
3553

plugins/components/ModalModel/CellularAutomataComp.h, 3553
plugins/components/ModalModel/DefaultNode.h, 3555
plugins/components/ModalModel/DefaultTransitionExtensions.h, 3557
plugins/components/ModalModel/FSMState.h, 3557
plugins/components/ModalModel/ModalModelDefault.h, 3557
plugins/components/ModalModel/PetriPlace.h, 3558
plugins/components/ModalModel/Submodel.h, 3559
plugins/components/Synchronization/Match.h, 3559
plugins/components/Synchronization/Signal.h, 3560
plugins/components/Synchronization/Wait.h, 3561
plugins/components/Template/DummyComponent.h, 3562
plugins/components/WholeCellModeling/BioStateProjectionComponent.h, 3563
plugins/components/WholeCellModeling/CellCycleCheckpointComponent.h, 3564
plugins/components/WholeCellModeling/CellDivisionEvent.h, 3565
plugins/components/WholeCellModeling/CellFateDecisionComponent.h, 3566
plugins/components/WholeCellModeling/CellGrowthComponent.h, 3567
plugins/components/WholeCellModeling/CompartmentExchangeComponent.h, 3568
plugins/components/WholeCellModeling/EukaryoticCellCycleComponent.h, 3569
plugins/components/WholeCellModeling/FtsZPolymerizationComponent.h, 3571
plugins/components/WholeCellModeling/MetabolicStateProjectionComponent.h, 3572
plugins/components/WholeCellModeling/MetabolicSubmodelComponent.h, 3574
plugins/components/WholeCellModeling/PathwayStressResponseComponent.h, 3575
plugins/components/WholeCellModeling/ResourceAllocationComponent.h, 3576
plugins/components/WholeCellModeling/StochasticReactionComponent.h, 3577
plugins/components/WholeCellModeling/StochasticTranscriptionComponent.h, 3578
plugins/components/WholeCellModeling/StochasticTranslationComponent.h, 3579
plugins/data/AI/AISupport.h, 3580
plugins/data/BiochemicalSimulation/BacteriaSignalGrid.h, 3581
plugins/data/BiochemicalSimulation/BioNetwork.h, 3582
plugins/data/BiochemicalSimulation/BioParameter.h, 3583
plugins/data/BiochemicalSimulation/BioReaction.h, 3584
plugins/data/BiochemicalSimulation/BioSBMLBridge.h, 3585
plugins/data/BiochemicalSimulation/BioSimulatorRunner.h, 3585
plugins/data/BiochemicalSimulation/BioSpecies.h, 3587
plugins/data/BiochemicalSimulation/GeneticCircuit.h, 3588
plugins/data/BiochemicalSimulation/GeneticCircuitPart.h, 3588
plugins/data/BiochemicalSimulation/GeneticRegulation.h, 3589
plugins/data/BiochemicalSimulation/GroProgram.h, 3590
plugins/data/BiochemicalSimulation/GroProgramAst.h, 3595
plugins/data/BiochemicalSimulation/GroProgramCompiler.h, 3595
plugins/data/BiochemicalSimulation/GroProgramIr.h, 3596
plugins/data/BiochemicalSimulation/GroProgramParser.h, 3597
plugins/data/BiochemicalSimulation/GroProgramRuntime.h, 3597
plugins/data/BiochemicalSimulation/MetabolicNetwork.h, 3599
plugins/data/BiochemicalSimulation/MetabolicReaction.h, 3600
plugins/data/Continuous/DiffusionField.h, 3601
plugins/data/Continuous/ODESolver.h, 3602
plugins/data/DiscreteProcessing/Failure.h, 3604
plugins/data/DiscreteProcessing/Queue.h, 3605
plugins/data/DiscreteProcessing/Resource.h, 3607
plugins/data/DiscreteProcessing/Schedule.h, 3609
plugins/data/ExternalIntegration/CppCompiler.h, 3610
plugins/data/ExternalIntegration/OctaveRunner.h, 3612
plugins/data/ExternalIntegration/PythonRuntime.h, 3613
plugins/data/ExternalIntegration/RSimulatorRunner.h, 3614
plugins/data/ExternalIntegration/SPICERunner.h, 3615
plugins/data/Grouping/EntityGroup.h, 3617
plugins/data/Input/Output/File.h, 3618
plugins/data/Logic/AssignmentItem.h, 3619
plugins/data/Logic/Formula.h, 3620
plugins/data/Logic/Label.h, 3620
plugins/data/Logic/Set.h, 3621
plugins/data/Logic/Variable.h, 3622
plugins/data/MaterialHandling/Conveyor.h, 3623
plugins/data/MaterialHandling/Distance.h, 3623
plugins/data/MaterialHandling/Segment.h, 3624
plugins/data/MaterialHandling/Sequence.h, 3625
plugins/data/MaterialHandling/Station.h, 3626
plugins/data/MaterialHandling/Storage.h, 3627
plugins/data/MaterialHandling/Transporter.h, 3628
plugins/data/ModalModel/ColoredPetriNetNetwork.h, 3628
plugins/data/ModalModel/CPNArc.h, 3629
plugins/data/ModalModel/CPNTransition.h, 3630
plugins/data/ModalModel/DefaultNetwork.h, 3631
plugins/data/ModalModel/DirectedAcyclicGraphNetwork.h, 3631
plugins/data/ModalModel/DirectedGraphNetwork.h, 3632
plugins/data/ModalModel/EFSMNetwork.h, 3632

plugins/data/ModalModel/GraphEdge.h, 3633
plugins/data/ModalModel/GraphNetwork.h, 3634
plugins/data/ModalModel/GraphNode.h, 3636
plugins/data/ModalModel/MarkovChainNetwork.h, 3636
plugins/data/ModalModel/MarkovState.h, 3637
plugins/data/ModalModel/NetworkActivation.h, 3637
plugins/data/Synchronization/SignalData.h, 3638
plugins/data/Template/DummyElement.h, 3639
plugins/data/WholeCellModeling/BioCompartment.h, 3640
plugins/data/WholeCellModeling/MolecularSpecies.h, 3641
plugins/data/WholeCellModeling/StochasticReactionRule.h, 3641
plugins/data/WholeCellModeling/WholeCellState.h, 3642
plugins/PluginConnectorDummyImpl1.h, 3644
plugins/PluginConnectorStaticImpl1.h, 3644
pmDefault
QCPPainter, 1659
pmNoCaching
QCPPainter, 1659
pmNonCosmetic
QCPPainter, 1659
PMOS, 1270
_loadInstance, 1275
_saveInstance, 1275
Build, 1275
pmVectorized
QCPPainter, 1659
poisson
ProbabilityDistributionBase, 1281
position
QCPAbstractItem, 1316
positions
QCPAbstractItem, 1316
PositionType
QCPItemPosition, 1562
previous
ModelManager, 1090
ProbabilityDistribution, 1277
inverseChi2, 1278
inverseFFisherSnedecor, 1278
inverseNormal, 1278
inverseTStudent, 1279
ProbabilityDistributionBase, 1279
_betaFunction, 1280
_factorial, 1280
_gammaFunction, 1280
beta, 1280
chi2, 1280
erlang, 1280
exponential, 1280
fisherSnedecor, 1281
gamma, 1281
logNormal, 1281
normal, 1281
poisson, 1281
triangular, 1281
tStudent, 1281
uniform, 1282
weibull, 1282
Process, 1282
_check, 1288
_createEditableDataDefinitions, 1288
_createNonEditableDataDefinitions, 1288
_loadInstance, 1288
_onDispatchEvent, 1288
_saveInstance, 1288
show, 1289
properties
QtAbstractPropertyBrowser, 1816
QtAbstractPropertyManager, 1822
propertiesWithoutValueMarked
QtTreePropertyBrowser, 2005
property
QtBrowserItem, 1829
propertyChanged
QtAbstractPropertyManager, 1822
propertyDestroyed
QtAbstractPropertyManager, 1822
PropertyEditor, 1292
setModelBlock, 1293
PropertyEditorController, 1293
PropertyEditorGenesys, 1294
PropertyGenesys, 1294
PropertyGenesysBase, 1296
propertyInserted
QtAbstractPropertyManager, 1822
propertyManager
QtAbstractEditorFactory< PropertyManager >, 1810
QtProperty, 1945
propertyManagers
QtAbstractEditorFactory< PropertyManager >, 1810
propertyName
QtProperty, 1945
propertyRemoved
QtAbstractPropertyManager, 1822
propertyType
QtVariantProperty, 2012
QtVariantPropertyManager, 2017
proportionConfidenceInterval
HypothesisTesterDefaultImpl1, 888, 889
proportionDifferenceConfidenceInterval
HypothesisTesterDefaultImpl1, 889
ProviderRequest, 1298
ProviderResponse, 1298
ptAbsolute
QCPItemPosition, 1562
ptAxisRectRatio
QCPItemPosition, 1562
ptPlotCoords
QCPItemPosition, 1562
ptViewportRatio

QCPItemPosition, 1562
pushRecentModelFile
SystemPreferences, 2558
put
CppSerializer, 496
GenSerializer, 806
JsonSerializer, 907
Serializer, 2161
XmlSerializer, 2720
PythonForG, 1300
_check, 1304
_createNonEditableDataDefinitions, 1304
_initBetweenReplications, 1305
_loadInstance, 1305
_onDispatchEvent, 1305
_saveInstance, 1305
show, 1305
PythonRuntime, 1306
_check, 1310
_loadInstance, 1310
_saveInstance, 1310
show, 1310

QCP, 111
aeAll, 113
aeAxes, 112
aeFills, 112
aeGrid, 112
aeItems, 112
aeLegend, 112
aeLegendItems, 112
aeNone, 113
aeOther, 113
aePlottables, 112
aeScatters, 112
aeSubGrid, 112
aeZeroLine, 113
AntialiasedElement, 112
epAllowCosmetic, 113
epNoCosmetic, 113
ExportPen, 113
iMultiSelect, 113
iNone, 113
Interaction, 113
iRangeDrag, 113
iRangeZoom, 113
iSelectAxes, 113
iSelectItems, 113
iSelectLegend, 113
iSelectOther, 113
iSelectPlottables, 113
iSelectPlottablesBeyondAxisRect, 113
MarginSide, 114
msAll, 114
msBottom, 114
msLeft, 114
msNone, 114
msRight, 114
msTop, 114
phCacheLabels, 114
phFastPolylines, 114
phImmediateRefresh, 114
phNone, 114
PlottingHint, 114
ResolutionUnit, 114
ruDotsPerCentimeter, 114
ruDotsPerInch, 114
ruDotsPerMeter, 114
sdBoth, 116
sdNegative, 116
sdPositive, 116
SelectionRectMode, 114
SelectionType, 115
SignDomain, 115
srmCustom, 115
srmNone, 115
srmSelect, 115
srmZoom, 115
stDataRange, 115
stMultipleDataRanges, 115
stNone, 115
stSingleData, 115
stWhole, 115
QCPAbstractItem, 1311
anchor, 1315
anchors, 1315
applyDefaultAntialiasingHint, 1315
clipRect, 1315
deselectEvent, 1316
draw, 1316
hasAnchor, 1316
position, 1316
positions, 1316
QCPAbstractItem, 1315
selectEvent, 1316
selectionCategory, 1316
selectionChanged, 1317
selectTest, 1317
setClipAxisRect, 1317
setClipToAxisRect, 1317
setSelectable, 1318
setSelected, 1318
QCPAbstractLegendItem, 1318
applyDefaultAntialiasingHint, 1323
clipRect, 1323
deselectEvent, 1323
draw, 1323
QCPAbstractLegendItem, 1322
selectEvent, 1323
selectionCategory, 1323
selectionChanged, 1323
selectTest, 1323
setFont, 1323
setSelectable, 1324
setSelected, 1324
setSelectedFont, 1324
setSelectedTextColor, 1324

setTextColor, 1324
QCPAbstractPaintBuffer, 1325
clear, 1326
donePainting, 1326
draw, 1326
QCPAbstractPaintBuffer, 1326
reallocateBuffer, 1326
setDevicePixelRatio, 1326
setInvalidated, 1327
setSize, 1327
startPainting, 1327
QCPAbstractPlottable, 1327
addToLegend, 1332
applyDefaultAntialiasingHint, 1332
clipRect, 1333
coordsToPixels, 1333
deselectEvent, 1333
draw, 1333
getKeyRange, 1333
getValueRange, 1334
interface1D, 1334
pixelsToCoords, 1334
QCPAbstractPlottable, 1332
removeFromLegend, 1335
rescaleAxes, 1335
rescaleKeyAxis, 1335
rescaleValueAxis, 1335
selectableChanged, 1336
selected, 1336
selectEvent, 1336
selection, 1336
selectionCategory, 1336
selectionChanged, 1336
selectionDecorator, 1337
selectTest, 1337
setAntialiasedFill, 1337
setAntialiasedScatters, 1338
setBrush, 1338
setKeyAxis, 1338
setName, 1338
setPen, 1338
setSelectable, 1338
setSelection, 1339
setSelectionDecorator, 1339
setValueAxis, 1339
QCPAbstractPlottable1D
QCPAbstractPlottable1D< DataType >, 1344
QCPAbstractPlottable1D< DataType >, 1340
dataCount, 1344
dataMainKey, 1344
dataMainValue, 1344
dataPixelPosition, 1344
dataSortKey, 1344
dataValueRange, 1345
drawPolyline, 1345
findBegin, 1345
findEnd, 1345
getDataSegments, 1345
interface1D, 1346
QCPAbstractPlottable1D, 1344
selectTest, 1346
selectTestRect, 1346
sortKeyIsMainKey, 1346
QCPAxis, 1347
applyDefaultAntialiasingHint, 1353
atBottom, 1352
atLeft, 1352
atRight, 1352
atTop, 1352
AxisType, 1352
coordToPixel, 1353
deselectEvent, 1353
draw, 1353
getPartAt, 1354
graphs, 1354
grid, 1354
items, 1354
LabelSide, 1352
lsInside, 1352
lsOutside, 1352
marginSideToAxisType, 1354
mouseMoveEvent, 1354
mousePressEvent, 1355
mouseReleaseEvent, 1355
moveRange, 1355
opposite, 1355
orientation, 1355, 1356
pixelOrientation, 1356
pixelToCoord, 1356
plottables, 1356
QCPAxis, 1353
rangeChanged, 1356
rescale, 1357
scaleRange, 1357
ScaleType, 1352
scaleTypeChanged, 1357
selectableChanged, 1357
SelectablePart, 1353
selectEvent, 1357
selectionCategory, 1358
selectionChanged, 1358
selectTest, 1358
setBasePen, 1358
setLabel, 1359
setLabelColor, 1359
setLabelFont, 1359
setLabelPadding, 1359
setLowerEnding, 1359
setNumberFormat, 1359
setNumberPrecision, 1360
setOffset, 1360
setPadding, 1360
setRange, 1360, 1361
setRangeLower, 1361
setRangeReversed, 1361
setRangeUpper, 1361

setScaleRatio, 1362
setScaleType, 1362
setSelectableParts, 1362
setSelectedBasePen, 1362
setSelectedLabelColor, 1362
setSelectedLabelFont, 1362
setSelectedParts, 1363
setSelectedSubTickPen, 1363
setSelectedTickLabelColor, 1363
setSelectedTickLabelFont, 1363
setSelectedTickPen, 1363
setSubTickLength, 1364
setSubTickLengthIn, 1364
setSubTickLengthOut, 1364
setSubTickPen, 1364
setSubTicks, 1364
setTicker, 1365
setTickLabelColor, 1365
setTickLabelFont, 1365
setTickLabelPadding, 1365
setTickLabelRotation, 1365
setTickLabels, 1365
setTickLabelSide, 1366
setTickLength, 1366
setTickLengthIn, 1366
setTickLengthOut, 1366
setTickPen, 1366
setTicks, 1366
setUpperEnding, 1367
spAxis, 1353
spAxisLabel, 1353
spNone, 1353
spTickLabels, 1353
stLinear, 1353
stLogarithmic, 1353
ticker, 1367
wheelEvent, 1367
QCPAxisPainterPrivate, 1368
QCPAxisPainterPrivate, 1369
QCPAxisPainterPrivate::CachedLabel, 343
QCPAxisPainterPrivate::TickLabelData, 2575
QCPAxisRect, 1370
addAxes, 1375
addAxis, 1375
applyDefaultAntialiasingHint, 1375
axes, 1375
axis, 1375
axisCount, 1376
bottom, 1376
bottomLeft, 1376
bottomRight, 1376
calculateAutoMargin, 1376
center, 1376
draw, 1376
elements, 1376
graphs, 1377
height, 1377
insetLayout, 1377
items, 1377
layoutChanged, 1377
left, 1377
mouseMoveEvent, 1377
mousePressEvent, 1378
mouseReleaseEvent, 1378
plottables, 1378
QCPAxisRect, 1374
rangeDragAxes, 1379
rangeDragAxis, 1379
rangeZoomAxes, 1379
rangeZoomAxis, 1379
rangeZoomFactor, 1379
removeAxis, 1379
right, 1380
setBackground, 1380
setBackgroundScaled, 1381
setBackgroundScaledMode, 1381
setRangeDrag, 1381
setRangeDragAxes, 1381, 1382
setRangeZoom, 1382
setRangeZoomAxes, 1382, 1383
setRangeZoomFactor, 1383
setupFullAxesBox, 1383
size, 1383
top, 1384
topLeft, 1384
topRight, 1384
update, 1384
wheelEvent, 1384
width, 1384
zoom, 1384, 1385
QCPAxisTicker, 1385
createTickVector, 1387
generate, 1387
getSubTickCount, 1388
getTickLabel, 1388
getTickStep, 1388
QCPAxisTicker, 1387
setTickCount, 1388
setTickOrigin, 1388
setTickStepStrategy, 1388
TickStepStrategy, 1387
tssMeetTickCount, 1387
tssReadability, 1387
QCPAxisTickerDateTime, 1389
createTickVector, 1391
dateTimeToKey, 1391
getSubTickCount, 1392
getTickLabel, 1392
getTickStep, 1392
keyToDateTime, 1392
QCPAxisTickerDateTime, 1391
setDateTimeFormat, 1392
setDateTimeSpec, 1393
setTickOrigin, 1393, 1394
setTimeZone, 1394
QCPAxisTickerFixed, 1394

getTickStep, 1397
QCPAxisTickerFixed, 1396
ScaleStrategy, 1396
setScaleStrategy, 1397
setTickStep, 1397
ssMultiples, 1396
ssNone, 1396
ssPowers, 1396
QCPAxisTickerLog, 1397
createTickVector, 1399
getSubTickCount, 1399
QCPAxisTickerLog, 1399
setLogBase, 1399
setSubTickCount, 1399
QCPAxisTickerPi, 1400
FractionStyle, 1402
fsAsciiFractions, 1402
fsFloatingPoint, 1402
fsUnicodeFractions, 1402
getSubTickCount, 1402
getTickLabel, 1402
getTickStep, 1402
QCPAxisTickerPi, 1402
setFractionStyle, 1403
setPeriodicity, 1403
setPiSymbol, 1403
setPiValue, 1403
QCPAxisTickerText, 1403
addTick, 1405
addTicks, 1405, 1406
clear, 1406
createTickVector, 1406
getSubTickCount, 1406
getTickLabel, 1406
getTickStep, 1406
QCPAxisTickerText, 1405
setSubTickCount, 1407
setTicks, 1407
ticks, 1407
QCPAxisTickerTime, 1407
getSubTickCount, 1410
getTickLabel, 1410
getTickStep, 1410
QCPAxisTickerTime, 1410
setFieldWidth, 1410
setTimeFormat, 1411
TimeUnit, 1410
tuDays, 1410
tuHours, 1410
tuMilliseconds, 1410
tuMinutes, 1410
tuSeconds, 1410
QCPBars, 1411
addData, 1417
barAbove, 1417
barBelow, 1417
data, 1418
dataPixelPosition, 1418
draw, 1418
drawLegendIcon, 1418
getKeyRange, 1418
getValueRange, 1418
moveAbove, 1419
moveBelow, 1419
QCPBars, 1417
selectTest, 1419
selectTestRect, 1420
setBarsGroup, 1420
setBaseValue, 1420
setData, 1420
setStackingGap, 1421
setWidth, 1421
setWidthType, 1421
WidthType, 1416
wtAbsolute, 1417
wtAxisRectRatio, 1417
wtPlotCoords, 1417
QCPBarsData, 1421
fromSortKey, 1422
mainKey, 1422
mainValue, 1422
QCPBarsData, 1422
sortKey, 1422
sortKeyIsMainKey, 1423
valueRange, 1423
QCPBarsGroup, 1423
append, 1426
bars, 1426
clear, 1426
contains, 1426
insert, 1426
isEmpty, 1426
QCPBarsGroup, 1425
remove, 1427
setSpacing, 1427
setSpacingType, 1427
size, 1427
SpacingType, 1425
stAbsolute, 1425
stAxisRectRatio, 1425
stPlotCoords, 1425
QCPColorGradient, 1427
ciHSV, 1429
ciRGB, 1429
clearColorStops, 1430
ColorInterpolation, 1429
colorize, 1430, 1431
gpCandy, 1429
gpCold, 1429
gpGeography, 1429
gpGrayscale, 1429
gpHot, 1429
gpHues, 1430
gpIon, 1429
gpJet, 1430
gpNight, 1429

gpPolar, 1430
gpSpectrum, 1430
gpThermal, 1429
GradientPreset, 1429
inverted, 1431
loadPreset, 1431
NanHandling, 1430
nhHighestColor, 1430
nhLowestColor, 1430
nhNanColor, 1430
nhNone, 1430
nhTransparent, 1430
QCPColorGradient, 1430
setColorInterpolation, 1431
setColorStopAt, 1432
setColorStops, 1432
setLevelCount, 1432
setNanColor, 1432
setNanHandling, 1432
setPeriodic, 1432
QCPColorMap, 1433
data, 1438
dataRangeChanged, 1438
dataScaleTypeChanged, 1438
draw, 1438
drawLegendIcon, 1438
getKeyRange, 1438
getValueRange, 1439
gradientChanged, 1439
QCPColorMap, 1438
rescaleDataRange, 1439
selectTest, 1440
setColorScale, 1440
setData, 1440
setDataRange, 1441
setDataScaleType, 1441
setGradient, 1441
setInterpolate, 1441
setTightBoundary, 1441
updateLegendIcon, 1441
QCPColorMapData, 1442
alpha, 1444
cellToCoord, 1444
clear, 1444
clearAlpha, 1445
coordToCell, 1445
fill, 1445
fillAlpha, 1445
isEmpty, 1445
operator=, 1445
QCPColorMapData, 1444
recalculateDataBounds, 1445
setAlpha, 1446
setCell, 1446
setData, 1446
setKeyRange, 1447
setKeySize, 1447
setRange, 1447
setSize, 1447
setValueRange, 1447
setValueSize, 1448
QCPColorScale, 1448
applyDefaultAntialiasingHint, 1452
axis, 1452
colorMaps, 1452
dataRangeChanged, 1453
dataScaleTypeChanged, 1453
gradientChanged, 1453
mouseMoveEvent, 1453
mousePressEvent, 1453
mouseReleaseEvent, 1454
QCPColorScale, 1452
rescaleDataRange, 1454
setBarWidth, 1454
setDataRange, 1454
setDataScaleType, 1454
setGradient, 1455
setLabel, 1455
setRangeDrag, 1455
setRangeZoom, 1455
setType, 1455
update, 1455
wheelEvent, 1456
QCPColorScaleAxisRectPrivate, 1457
calculateAutoMargin, 1461
draw, 1461
mouseMoveEvent, 1461
mousePressEvent, 1462
mouseReleaseEvent, 1462
QCPColorScaleAxisRectPrivate, 1461
update, 1462
wheelEvent, 1463
QCPCurve, 1463
addData, 1469, 1470
data, 1470
draw, 1470
drawLegendIcon, 1470
getKeyRange, 1470
getValueRange, 1471
LineStyle, 1469
lsLine, 1469
lsNone, 1469
QCPCurve, 1469
selectTest, 1471
setData, 1471, 1472
setLineStyle, 1472
setScatterSkip, 1472
setScatterStyle, 1472
QCPCurveData, 1473
fromSortKey, 1474
mainKey, 1474
mainValue, 1474
QCPCurveData, 1474
sortKey, 1474
sortKeyIsMainKey, 1474
valueRange, 1474

QCPDataContainer
QCPDataContainer< DataType >, 1477
QCPDataContainer< DataType >, 1475
add, 1477
at, 1478
begin, 1478
clear, 1478
constBegin, 1478
constEnd, 1478
dataRange, 1478
end, 1478
findBegin, 1478
findEnd, 1479
isEmpty, 1479
keyRange, 1479
limitIteratorsToDataRange, 1479
QCPDataContainer, 1477
qcpLessThanSortKey, 1482
remove, 1480
removeAfter, 1480
removeBefore, 1480
set, 1480, 1481
setAutoSqueeze, 1481
size, 1481
sort, 1481
squeeze, 1481
valueRange, 1482
QCPDataRange, 1482
adjusted, 1483
bounded, 1483
contains, 1484
expanded, 1484
intersection, 1484
intersects, 1484
isEmpty, 1484
isValid, 1484
length, 1485
operator<<, 1485
QCPDataRange, 1483
setBegin, 1485
setEnd, 1485
size, 1485
QCPDataSelection, 1486
addDataRange, 1487
clear, 1487
contains, 1487
dataPointCount, 1487
dataRange, 1487
dataRangeCount, 1488
dataRanges, 1488
enforceType, 1488
intersection, 1488
inverse, 1488
isEmpty, 1489
operator<<, 1491
operator+, 1490
operator+=, 1489
operator-, 1490
operator-=, 1489
operator==, 1489
QCPDataSelection, 1487
simplify, 1489
span, 1489
QCPErrorBars, 1491
addData, 1496, 1497
data, 1497
dataCount, 1497
dataMainKey, 1497
dataMainValue, 1497
dataPixelPosition, 1497
dataSortKey, 1498
dataValueRange, 1498
draw, 1498
drawLegendIcon, 1498
ErrorType, 1496
etKeyError, 1496
etValueError, 1496
findBegin, 1498
findEnd, 1498
getKeyRange, 1499
getValueRange, 1499
interface1D, 1499
QCPErrorBars, 1496
selectTest, 1500
selectTestRect, 1500
setData, 1500, 1501
setDataPlottable, 1501
setErrorType, 1501
setSymbolGap, 1501
setWhiskerWidth, 1501
sortKeyIsMainKey, 1501
QCPErrorBarsData, 1502
QCPErrorBarsData, 1502
QCPFinancial, 1503
addData, 1509
ChartStyle, 1508
csCandlestick, 1508
csOhlc, 1508
data, 1510
draw, 1510
drawLegendIcon, 1510
getKeyRange, 1510
getValueRange, 1510
QCPFinancial, 1509
selectTest, 1511
selectTestRect, 1511
setBrushNegative, 1511
setBrushPositive, 1511
setChartStyle, 1512
setData, 1512
setPenNegative, 1512
setPenPositive, 1513
setTwoColored, 1513
setWidth, 1513
setWidthType, 1513
timeSeriesToOhlc, 1513

WidthType, 1509
wtAbsolute, 1509
wtAxisRectRatio, 1509
wtPlotCoords, 1509
QCPFinancialData, 1514
fromSortKey, 1515
mainKey, 1515
mainValue, 1515
QCPFinancialData, 1514
sortKey, 1515
sortKeyIsMainKey, 1515
valueRange, 1515
QCPGraph, 1516
addData, 1522
data, 1522
draw, 1522
drawLegendIcon, 1522
getKeyRange, 1522
getValueRange, 1523
getVisibleDataBounds, 1523
LineStyle, 1521
lsImpulse, 1521
lsLine, 1521
lsNone, 1521
lsStepCenter, 1521
lsStepLeft, 1521
lsStepRight, 1521
QCPGraph, 1521
selectTest, 1523
setAdaptiveSampling, 1524
setChannelFillGraph, 1524
setData, 1524
setLineStyle, 1525
setScatterSkip, 1525
setScatterStyle, 1525
QCPGraphData, 1525
fromSortKey, 1526
mainKey, 1526
mainValue, 1527
QCPGraphData, 1526
sortKey, 1527
sortKeyIsMainKey, 1527
valueRange, 1527
QCPGrid, 1527
applyDefaultAntialiasingHint, 1530
draw, 1530
QCPGrid, 1530
setAntialiasedSubGrid, 1530
setAntialiasedZeroLine, 1530
setPen, 1531
setSubGridPen, 1531
setSubGridVisible, 1531
setZeroLinePen, 1531
QCPItemAnchor, 1531
pixelPosition, 1533
QCPItemAnchor, 1533
toQCPItemPosition, 1533
QCPItemBracket, 1533
anchorPixelPosition, 1537
BracketStyle, 1537
bsCalligraphic, 1537
bsCurly, 1537
bsRound, 1537
bsSquare, 1537
draw, 1538
QCPItemBracket, 1537
selectTest, 1538
setLength, 1538
setPen, 1538
setSelectedPen, 1539
setStyle, 1539
QCPItemCurve, 1539
draw, 1543
QCPItemCurve, 1542
selectTest, 1543
setHead, 1543
setPen, 1543
setSelectedPen, 1544
setTail, 1544
QCPItemEllipse, 1544
anchorPixelPosition, 1548
draw, 1548
QCPItemEllipse, 1548
selectTest, 1548
setBrush, 1548
setPen, 1549
setSelectedBrush, 1549
setSelectedPen, 1549
QCPItemLine, 1549
draw, 1553
QCPItemLine, 1553
selectTest, 1553
setHead, 1553
setPen, 1554
setSelectedPen, 1554
setTail, 1554
QCPItemPixmap, 1554
anchorPixelPosition, 1558
draw, 1558
QCPItemPixmap, 1558
selectTest, 1558
setPen, 1559
setPixmap, 1559
setScaled, 1559
setSelectedPen, 1559
QCPItemPosition, 1560
parentAnchor, 1563
pixelPosition, 1563
PositionType, 1562
ptAbsolute, 1562
ptAxisRectRatio, 1562
ptPlotCoords, 1562
ptViewportRatio, 1562
QCPItemPosition, 1563
setAxes, 1563
setAxisRect, 1563

setCoords, 1563
setParentAnchor, 1564
setParentAnchorX, 1564
setParentAnchorY, 1564
setPixelPosition, 1564
setType, 1565
setTypeX, 1565
setTypeY, 1565
toQCPItemPosition, 1565
type, 1566
QCPItemRect, 1566
anchorPixelPosition, 1570
draw, 1570
QCPItemRect, 1570
selectTest, 1570
setBrush, 1570
setPen, 1571
setSelectedBrush, 1571
setSelectedPen, 1571
QCPItemStraightLine, 1571
draw, 1575
QCPItemStraightLine, 1575
selectTest, 1575
setPen, 1575
setSelectedPen, 1575
QCPItemText, 1576
anchorPixelPosition, 1580
draw, 1580
QCPItemText, 1580
selectTest, 1580
setBrush, 1581
setColor, 1581
setFont, 1581
setPadding, 1581
setPen, 1582
setPositionAlignment, 1582
setRotation, 1582
setSelectedBrush, 1582
setSelectedColor, 1582
setSelectedFont, 1582
setSelectedPen, 1583
setText, 1583
setTextAlignment, 1583
QCPItemTracer, 1583
draw, 1588
QCPItemTracer, 1587
selectTest, 1588
setBrush, 1588
setGraph, 1588
setGraphKey, 1589
setInterpolating, 1589
setPen, 1589
setSelectedBrush, 1589
setSelectedPen, 1589
setSize, 1590
setStyle, 1590
TracerStyle, 1587
tsCircle, 1587
tsCrosshair, 1587
tsNone, 1587
tsPlus, 1587
tsSquare, 1587
updatePosition, 1590
QCPLabelPainterPrivate, 1591
AnchorMode, 1593
AnchorReferenceType, 1593
AnchorSide, 1593
QCPLabelPainterPrivate, 1593
QCPLabelPainterPrivate::CachedLabel, 343
QCPLabelPainterPrivate::LabelData, 912
QCPLayer, 1593
children, 1596
index, 1596
LayerMode, 1596
lmBuffered, 1596
lmLogical, 1596
QCPLayer, 1596
replot, 1596
setMode, 1596
setVisible, 1597
QCPLayerable, 1597
layerChanged, 1601
mouseDoubleClickEvent, 1601
mouseMoveEvent, 1601
mousePressEvent, 1601
mouseReleaseEvent, 1602
parentLayerable, 1602
QCPLayerable, 1600
realVisibility, 1602
selectTest, 1602
setAntialiased, 1603
setLayer, 1603
setVisible, 1603
wheelEvent, 1604
QCPLayout, 1604
clear, 1608
elementAt, 1608
elementCount, 1609
elements, 1609
QCPLayout, 1608
remove, 1609
removeAt, 1609
simplify, 1609
sizeConstraintsChanged, 1610
take, 1610
takeAt, 1610
update, 1610
QCPLayoutElement, 1611
applyDefaultAntialiasingHint, 1614
draw, 1614
elements, 1614
layout, 1615
maximumOuterSizeHint, 1615
minimumOuterSizeHint, 1615
outerRect, 1615
parentPlotInitialized, 1615

QCPLayoutElement, 1614
rect, 1615
scrInnerRect, 1614
scrOuterRect, 1614
selectTest, 1616
setAutoMargins, 1616
setMarginGroup, 1616
setMargins, 1616
setMaximumSize, 1617
setMinimumMargins, 1617
setMinimumSize, 1617
setOuterRect, 1618
setSizeConstraintRect, 1618
SizeConstraintRect, 1614
update, 1618
UpdatePhase, 1614
upLayout, 1614
upMargins, 1614
upPreparation, 1614
QCPLayoutGrid, 1618
addElement, 1624
columnCount, 1624
element, 1625
elementAt, 1625
elementCount, 1625
elements, 1625
expandTo, 1625
FillOrder, 1624
foColumnsFirst, 1624
foRowsFirst, 1624
hasElement, 1626
indexToRowCol, 1626
insertColumn, 1626
insertRow, 1626
maximumOuterSizeHint, 1627
minimumOuterSizeHint, 1627
QCPLayoutGrid, 1624
rowColToIndex, 1627
rowCount, 1627
setColumnSpacing, 1627
setColumnStretchFactor, 1628
setColumnStretchFactors, 1628
setFillOrder, 1628
setRowSpacing, 1628
setRowStretchFactor, 1629
setRowStretchFactors, 1629
setWrap, 1629
simplify, 1629
take, 1629
takeAt, 1630
updateLayout, 1630
QCPLayoutInset, 1630
addElement, 1635
elementAt, 1635
elementCount, 1635
insetAlignment, 1636
InsetPlacement, 1634
insetPlacement, 1636
insetRect, 1636
ipBorderAligned, 1635
ipFree, 1635
QCPLayoutInset, 1635
selectTest, 1636
setInsetAlignment, 1636
setInsetPlacement, 1636
setInsetRect, 1637
simplify, 1637
take, 1637
takeAt, 1637
updateLayout, 1637
QCPLegend, 1638
addItem, 1645
applyDefaultAntialiasingHint, 1645
clearItems, 1645
deselectEvent, 1645
draw, 1645
hasItem, 1645
hasItemWithPlottable, 1645
item, 1646
itemCount, 1646
itemWithPlottable, 1646
parentPlotInitialized, 1646
QCPLegend, 1645
removeItem, 1646
SelectablePart, 1644
selectedItems, 1647
selectEvent, 1647
selectionCategory, 1647
selectionChanged, 1647
selectTest, 1647
setBorderPen, 1647
setBrush, 1648
setFont, 1648
setIconBorderPen, 1648
setIconSize, 1648
setIconTextPadding, 1648
setSelectableParts, 1648
setSelectedBorderPen, 1649
setSelectedBrush, 1649
setSelectedFont, 1649
setSelectedIconBorderPen, 1649
setSelectedParts, 1649
setSelectedTextColor, 1650
setTextColor, 1650
spItems, 1644
spLegendBox, 1644
spNone, 1644
qcpLessThanSortKey
QCPDataContainer< DataType >, 1482
QCPLineEnding, 1650
EndingStyle, 1651
esBar, 1651
esDiamond, 1651
esDisc, 1651
esFlatArrow, 1651
esHalfBar, 1651

esLineArrow, 1651
esNone, 1651
esSkewedBar, 1651
esSpikeArrow, 1651
esSquare, 1651
QCPLineEnding, 1651
realLength, 1652
setInverted, 1652
setLength, 1652
setStyle, 1652
setWidth, 1652
QCPMarginGroup, 1653
clear, 1654
elements, 1654
isEmpty, 1654
QCPMarginGroup, 1654
QCPPaintBufferPixmap, 1655
clear, 1656
draw, 1656
QCPPaintBufferPixmap, 1656
reallocateBuffer, 1656
startPainting, 1657
QCPPainter, 1657
begin, 1659
drawLine, 1659
makeNonCosmetic, 1660
PainterMode, 1659
pmDefault, 1659
pmNoCaching, 1659
pmNonCosmetic, 1659
pmVectorized, 1659
QCPPainter, 1659
restore, 1660
save, 1660
setAntialiasing, 1660
setMode, 1660
setModes, 1660
setPen, 1660, 1661
QCPPlottableInterface1D, 1661
dataCount, 1663
dataMainKey, 1663
dataMainValue, 1663
dataPixelPosition, 1663
dataSortKey, 1663
dataValueRange, 1663
findBegin, 1664
findEnd, 1664
selectTestRect, 1664
sortKeyIsMainKey, 1665
QCPPlottableLegendItem, 1665
draw, 1669
minimumOuterSizeHint, 1669
QCPPlottableLegendItem, 1669
QCPPolarAxisAngular, 1670
addRadialAxis, 1677
applyDefaultAntialiasingHint, 1677
bottom, 1677
bottomLeft, 1678
bottomRight, 1678
center, 1678
coordToPixel, 1678
draw, 1678
elements, 1678
getPartAt, 1678
height, 1679
insetLayout, 1679
LabelMode, 1677
left, 1679
mouseMoveEvent, 1679
mousePressEvent, 1679
mouseReleaseEvent, 1680
moveRange, 1680
pixelToCoord, 1680
QCPPolarAxisAngular, 1677
radialAxes, 1680
radialAxis, 1680
radialAxisCount, 1680
removeRadialAxis, 1681
rescale, 1681
right, 1681
scaleRange, 1681
SelectablePart, 1677
selectionCategory, 1681
selectTest, 1682
setBackground, 1682
setBackgroundScaled, 1683
setBackgroundScaledMode, 1683
setBasePen, 1683
setLabel, 1683
setLabelColor, 1683
setLabelFont, 1683
setLabelPadding, 1684
setNumberFormat, 1684
setNumberPrecision, 1684
setRange, 1684, 1685
setRangeLower, 1685
setRangeReversed, 1685
setRangeUpper, 1685
setSelectableParts, 1686
setSelectedBasePen, 1686
setSelectedLabelColor, 1686
setSelectedLabelFont, 1686
setSelectedParts, 1686
setSelectedSubTickPen, 1686
setSelectedTickLabelColor, 1687
setSelectedTickLabelFont, 1687
setSelectedTickPen, 1687
setSubTickLength, 1687
setSubTickLengthIn, 1687
setSubTickLengthOut, 1687
setSubTickPen, 1688
setSubTicks, 1688
setTicker, 1688
setTickLabelColor, 1688
setTickLabelFont, 1688
setTickLabelPadding, 1689

setTickLabelRotation, 1689
setTickLabels, 1689
setTickLength, 1689
setTickLengthIn, 1689
setTickLengthOut, 1689
setTickPen, 1690
setTicks, 1690
size, 1690
spAxis, 1677
spAxisLabel, 1677
spNone, 1677
spTickLabels, 1677
top, 1690
topLeft, 1690
topRight, 1690
update, 1690
wheelEvent, 1691
width, 1691
QCPPolarAxisRadial, 1691
AngleReference, 1697
applyDefaultAntialiasingHint, 1698
arAbsolute, 1698
arAngularAxis, 1698
coordToPixel, 1698
deselectEvent, 1699
draw, 1699
getPartAt, 1699
mouseMoveEvent, 1699
mousePressEvent, 1699
mouseReleaseEvent, 1700
moveRange, 1700
pixelToCoord, 1700
QCPPolarAxisRadial, 1698
rangeChanged, 1700
rescale, 1701
scaleRange, 1701
ScaleType, 1698
scaleTypeChanged, 1701
selectableChanged, 1701
SelectablePart, 1698
selectEvent, 1701
selectionCategory, 1702
selectionChanged, 1702
selectTest, 1702
setBasePen, 1702
setLabel, 1703
setLabelColor, 1703
setLabelFont, 1703
setLabelPadding, 1703
setNumberFormat, 1703
setNumberPrecision, 1704
setRange, 1704
setRangeLower, 1704
setRangeReversed, 1705
setRangeUpper, 1705
setScaleType, 1705
setSelectableParts, 1705
setSelectedBasePen, 1705
setSelectedLabelColor, 1705
setSelectedLabelFont, 1706
setSelectedParts, 1706
setSelectedSubTickPen, 1706
setSelectedTickLabelColor, 1706
setSelectedTickLabelFont, 1706
setSelectedTickPen, 1707
setSubTickLength, 1707
setSubTickLengthIn, 1707
setSubTickLengthOut, 1707
setSubTickPen, 1707
setSubTicks, 1707
setTicker, 1708
setTickLabelColor, 1708
setTickLabelFont, 1708
setTickLabelPadding, 1708
setTickLabelRotation, 1708
setTickLabels, 1708
setTickLength, 1709
setTickLengthIn, 1709
setTickLengthOut, 1709
setTickPen, 1709
setTicks, 1709
spAxis, 1698
spAxisLabel, 1698
spNone, 1698
spTickLabels, 1698
stLinear, 1698
stLogarithmic, 1698
ticker, 1709
wheelEvent, 1710
QCPPolarGraph, 1710
applyDefaultAntialiasingHint, 1715
clipRect, 1715
coordsToPixels, 1715
deselectEvent, 1715
draw, 1715
LineStyle, 1714
lsLine, 1714
lsNone, 1714
QCPPolarGraph, 1714
selectEvent, 1715
selectionCategory, 1715
selectTest, 1715
setAntialiasedFill, 1716
setAntialiasedScatters, 1716
setBrush, 1716
setData, 1716, 1717
setKeyAxis, 1717
setLineStyle, 1717
setName, 1717
setPen, 1717
setScatterStyle, 1718
setSelectable, 1718
setSelection, 1718
setValueAxis, 1718
QCPPolarGrid, 1719
applyDefaultAntialiasingHint, 1721

draw, 1721
GridType, 1721
QCPPolarGrid, 1721
setAngularPen, 1722
setAngularSubGridPen, 1722
setAntialiasedSubGrid, 1722
setAntialiasedZeroLine, 1722
QCPPolarLegendItem, 1722
draw, 1727
minimumOuterSizeHint, 1727
QCPRange, 1727
bounded, 1728
center, 1729
contains, 1729
expand, 1729
expanded, 1729
maxRange, 1732
minRange, 1732
normalize, 1730
operator<<, 1732
operator+, 1731
operator+=, 1730
operator-, 1731
operator-=, 1730
operator/, 1732
operator/=, 1730
operator*, 1731
operator*=, 1730
QCPRange, 1728
sanitizedForLinScale, 1730
sanitizedForLogScale, 1730
size, 1730
validRange, 1731
QCPScatterStyle, 1733
applyTo, 1736
drawShape, 1737
isNone, 1737
isPenDefined, 1737
QCPScatterStyle, 1735, 1736
ScatterProperty, 1734
ScatterShape, 1734
setBrush, 1737
setCustomPath, 1738
setFromOther, 1738
setPen, 1738
setPixmap, 1738
setShape, 1738
setSize, 1738
spAll, 1734
spBrush, 1734
spNone, 1734
spPen, 1734
spShape, 1734
spSize, 1734
ssCircle, 1735
ssCross, 1735
ssCrossCircle, 1735
ssCrossSquare, 1735
ssCustom, 1735
ssDiamond, 1735
ssDisc, 1735
ssDot, 1735
ssNone, 1735
ssPeace, 1735
ssPixmap, 1735
ssPlus, 1735
ssPlusCircle, 1735
ssPlusSquare, 1735
ssSquare, 1735
ssStar, 1735
ssTriangle, 1735
ssTriangleInverted, 1735
undefinePen, 1738
QCPSelectionDecorator, 1739
applyBrush, 1740
applyPen, 1741
copyFrom, 1741
drawDecoration, 1741
getFinalScatterStyle, 1741
QCPSelectionDecorator, 1740
setBrush, 1741
setPen, 1741
setScatterStyle, 1741
setUsedScatterProperties, 1742
QCPSelectionDecoratorBracket, 1742
BracketStyle, 1744
bsEllipse, 1744
bsHalfEllipse, 1744
bsPlus, 1745
bsSquareBracket, 1744
bsUserStyle, 1745
drawBracket, 1745
drawDecoration, 1745
QCPSelectionDecoratorBracket, 1745
setBracketBrush, 1745
setBracketHeight, 1745
setBracketPen, 1745
setBracketStyle, 1745
setBracketWidth, 1746
setTangentAverage, 1746
setTangentToData, 1746
QCPSelectionRect, 1746
accepted, 1749
applyDefaultAntialiasingHint, 1750
cancel, 1750
canceled, 1750
changed, 1750
draw, 1750
isActive, 1750
QCPSelectionRect, 1749
range, 1750
setBrush, 1750
setPen, 1751
started, 1751
QCPStatisticalBox, 1751
addData, 1757

data, 1757
draw, 1757
drawLegendIcon, 1758
drawStatisticalBox, 1758
getKeyRange, 1758
getValueRange, 1758
QCPStatisticalBox, 1757
selectTest, 1759
selectTestRect, 1759
setData, 1759
setMedianPen, 1760
setOutlierStyle, 1760
setWhiskerAntialiased, 1760
setWhiskerBarPen, 1760
setWhiskerPen, 1760
setWhiskerWidth, 1761
setWidth, 1761
QCPStatisticalBoxData, 1761
fromSortKey, 1762
mainKey, 1762
mainValue, 1763
QCPStatisticalBoxData, 1762
sortKey, 1763
sortKeyIsMainKey, 1763
valueRange, 1763
QCPTextElement, 1763
applyDefaultAntialiasingHint, 1768
clicked, 1768
deselectEvent, 1769
doubleClicked, 1769
draw, 1769
maximumOuterSizeHint, 1769
minimumOuterSizeHint, 1769
mouseDoubleClickEvent, 1769
mousePressEvent, 1769
mouseReleaseEvent, 1770
QCPTextElement, 1767, 1768
selectEvent, 1770
selectionChanged, 1770
selectTest, 1770
setFont, 1770
setSelectable, 1770
setSelected, 1771
setSelectedFont, 1771
setSelectedTextColor, 1771
setText, 1771
setTextColor, 1771
setTextFlags, 1771
QCPVector2D, 1772
angle, 1774
distanceSquaredToLine, 1774
distanceToStraightLine, 1774
dot, 1774
isNull, 1774
length, 1774
lengthSquared, 1775
normalize, 1775
normalized, 1775
operator<<, 1776
operator+=, 1775
operator-=, 1775
operator/=, 1775
operator*=, 1775
perpendicular, 1775
QCPVector2D, 1773
setX, 1776
setY, 1776
toPoint, 1776
toPointF, 1776
QCustomPlot, 1777
addGraph, 1782
addLayer, 1783
afterLayout, 1783
afterReplot, 1783
axisClick, 1783
axisDoubleClick, 1784
axisRect, 1784
axisRectAt, 1784
axisRectCount, 1784
axisRects, 1785
beforeReplot, 1785
clearGraphs, 1785
clearItems, 1785
clearPlottables, 1785
currentLayer, 1786
deselectAll, 1786
graph, 1786
graphCount, 1786
hasInvalidatedPaintBuffers, 1786
hasItem, 1786
hasPlottable, 1787
item, 1787
itemAt, 1787
itemClick, 1788
itemCount, 1788
itemDoubleClick, 1788
layer, 1788
layerCount, 1789
LayerInsertMode, 1782
layoutElementAt, 1789
legend, 1806
legendClick, 1789
legendDoubleClick, 1789
limAbove, 1782
limBelow, 1782
mouseDoubleClick, 1789
mouseMove, 1790
mousePress, 1790
mouseRelease, 1790
mouseWheel, 1790
moveLayer, 1790
plotLayout, 1790
plottable, 1791
plottableAt, 1791
plottableClick, 1791
plottableCount, 1792

plottableDoubleClick, 1792
QCustomPlot, 1782
RefreshPriority, 1782
removeGraph, 1792
removeItem, 1792, 1793
removeLayer, 1793
removePlottable, 1793
replot, 1793
replotTime, 1794
rescaleAxes, 1794
rpImmediateRefresh, 1782
rpQueuedRefresh, 1782
rpQueuedReplot, 1782
rpRefreshHint, 1782
saveBmp, 1794
saveJpg, 1795
savePdf, 1796
savePng, 1796
saveRastered, 1797
selectedAxes, 1798
selectedGraphs, 1798
selectedItems, 1798
selectedLegends, 1798
selectedPlottables, 1798
selectionChangedByUser, 1798
selectionRect, 1799
setAntialiasedElement, 1799
setAntialiasedElements, 1799
setAutoAddPlottableToLegend, 1799
setBackground, 1799, 1800
setBackgroundScaled, 1800
setBackgroundScaledMode, 1800
setBufferDevicePixelRatio, 1801
setCurrentLayer, 1801
setInteraction, 1801
setInteractions, 1801
setMultiSelectModifier, 1802
setNoAntialiasingOnDrag, 1802
setNotAntialiasedElement, 1803
setNotAntialiasedElements, 1803
setOpenGl, 1803
setPlottingHint, 1804
setPlottingHints, 1804
setSelectionRect, 1804
setSelectionRectMode, 1804
setSelectionTolerance, 1805
setViewport, 1805
toPainter, 1805
toPixmap, 1805
xAxis, 1806
xAxis2, 1806
yAxis, 1806
yAxis2, 1807
QtAbstractEditorFactory
QtAbstractEditorFactory< PropertyManager >, 1808
QtAbstractEditorFactory< PropertyManager >, 1807
addPropertyManager, 1809
connectPropertyManager, 1809
createEditor, 1809
disconnectPropertyManager, 1809
managerDestroyed, 1810
propertyManager, 1810
propertyManagers, 1810
QtAbstractEditorFactory, 1808
removePropertyManager, 1810
QtAbstractEditorFactoryBase, 1811
createEditor, 1812
QtAbstractEditorFactoryBase, 1812
QtAbstractPropertyBrowser, 1813
~QtAbstractPropertyBrowser, 1814
addProperty, 1814
clear, 1814
createEditor, 1814
currentItem, 1815
currentItemChanged, 1815
insertProperty, 1815
itemChanged, 1815
itemInserted, 1816
itemRemoved, 1816
items, 1816
properties, 1816
QtAbstractPropertyBrowser, 1814
removeProperty, 1816
setCurrentItem, 1817
setFactoryForManager, 1817
topLevelItem, 1817
topLevelItems, 1817
unsetFactoryForManager, 1817
QtAbstractPropertyBrowserPrivate, 1818
QtAbstractPropertyManager, 1819
~QtAbstractPropertyManager, 1820
addProperty, 1821
clear, 1821
createProperty, 1821
hasValue, 1821
initializeProperty, 1821
properties, 1822
propertyChanged, 1822
propertyDestroyed, 1822
propertyInserted, 1822
propertyRemoved, 1822
QtAbstractPropertyManager, 1820
uninitializeProperty, 1823
valueIcon, 1823
valueText, 1823
QtAbstractPropertyManagerPrivate, 1824
QtBoolEdit, 1824
QtBoolPropertyManager, 1825
~QtBoolPropertyManager, 1827
initializeProperty, 1827
QtBoolPropertyManager, 1827
setValue, 1827
uninitializeProperty, 1827
value, 1827
valueChanged, 1827

valueIcon, 1828
valueText, 1828
QtBoolPropertyManagerPrivate, 1828
QtBoolPropertyManagerPrivate::Data, 505
QtBrowserItem, 1828
browser, 1829
children, 1829
parent, 1829
property, 1829
QtBrowserItemPrivate, 1829
QtButtonPropertyBrowser, 1830
~QtButtonPropertyBrowser, 1832
collapsed, 1832
expanded, 1832
isExpanded, 1832
itemChanged, 1832
itemInserted, 1833
itemRemoved, 1833
QtButtonPropertyBrowser, 1832
setExpanded, 1833
QtButtonPropertyBrowserPrivate, 1833
QtButtonPropertyBrowserPrivate::WidgetItem, 2702
QtCharEdit, 1834
QtCharEditorFactory, 1835
~QtCharEditorFactory, 1836
QtCharEditorFactory, 1836
QtCharEditorFactoryPrivate, 1837
QtCharPropertyManager, 1838
~QtCharPropertyManager, 1839
initializeProperty, 1839
QtCharPropertyManager, 1839
setValue, 1839
uninitializeProperty, 1840
value, 1840
valueChanged, 1840
valueText, 1840
QtCharPropertyManagerPrivate, 1840
QtCheckBoxFactory, 1841
~QtCheckBoxFactory, 1842
QtCheckBoxFactory, 1842
QtCheckBoxFactoryPrivate, 1843
QtColorEditorFactory, 1844
~QtColorEditorFactory, 1846
QtColorEditorFactory, 1846
QtColorEditorFactoryPrivate, 1846
QtColorEditWidget, 1848
QtColorPropertyManager, 1849
~QtColorPropertyManager, 1850
initializeProperty, 1850
QtColorPropertyManager, 1850
setValue, 1850
subIntPropertyManager, 1850
uninitializeProperty, 1851
value, 1851
valueChanged, 1851
valueIcon, 1851
valueText, 1851
QtColorPropertyManagerPrivate, 1852
QtCursorDatabase, 1853
QtCursorEditorFactory, 1853
~QtCursorEditorFactory, 1855
QtCursorEditorFactory, 1855
QtCursorEditorFactoryPrivate, 1855
QtCursorPropertyManager, 1856
~QtCursorPropertyManager, 1857
initializeProperty, 1858
QtCursorPropertyManager, 1857
setValue, 1858
uninitializeProperty, 1858
value, 1858
valueChanged, 1858
valueIcon, 1858
valueText, 1858
QtCursorPropertyManagerPrivate, 1859
QtDateEdit, 1859
QtDateEditFactory, 1860
~QtDateEditFactory, 1862
QtDateEditFactory, 1862
QtDateEditFactoryPrivate, 1862
QtDatePropertyManager, 1864
~QtDatePropertyManager, 1865
initializeProperty, 1865
maximum, 1865
minimum, 1866
QtDatePropertyManager, 1865
rangeChanged, 1866
setMaximum, 1866
setMinimum, 1866
setRange, 1866
setValue, 1867
uninitializeProperty, 1867
value, 1867
valueChanged, 1867
valueText, 1867
QtDatePropertyManagerPrivate, 1868
QtDatePropertyManagerPrivate::Data, 505
QtDateTimeEdit, 1868
QtDateTimeEditFactory, 1870
~QtDateTimeEditFactory, 1871
QtDateTimeEditFactory, 1871
QtDateTimeEditFactoryPrivate, 1872
QtDateTimePropertyManager, 1873
~QtDateTimePropertyManager, 1874
initializeProperty, 1874
QtDateTimePropertyManager, 1874
setValue, 1874
uninitializeProperty, 1875
value, 1875
valueChanged, 1875
valueText, 1875
QtDateTimePropertyManagerPrivate, 1875
QtDateTimePropertyManagerPrivate::Data, 505
QtDoubleEdit, 1876
QtDoublePropertyManager, 1877
~QtDoublePropertyManager, 1878
decimals, 1879

decimalsChanged, 1879
initializeProperty, 1879
maximum, 1879
minimum, 1879
QtDoublePropertyManager, 1878
rangeChanged, 1879
setDecimals, 1879
setMaximum, 1880
setMinimum, 1880
setRange, 1880
setSingleStep, 1880
setValue, 1881
singleStep, 1881
singleStepChanged, 1881
uninitializeProperty, 1881
value, 1881
valueChanged, 1881
valueText, 1882
QtDoublePropertyManagerPrivate, 1882
QtDoublePropertyManagerPrivate::Data, 505
QtDoubleSpinBoxFactory, 1883
~QtDoubleSpinBoxFactory, 1884
QtDoubleSpinBoxFactory, 1884
QtDoubleSpinBoxFactoryPrivate, 1885
QtEnumEditorFactory, 1886
~QtEnumEditorFactory, 1888
QtEnumEditorFactory, 1888
QtEnumEditorFactoryPrivate, 1888
QtEnumPropertyManager, 1890
~QtEnumPropertyManager, 1891
enumIcons, 1891
enumIconsChanged, 1891
enumNames, 1892
enumNamesChanged, 1892
initializeProperty, 1892
QtEnumPropertyManager, 1891
setEnumIcons, 1892
setEnumNames, 1892
setValue, 1893
uninitializeProperty, 1893
value, 1893
valueChanged, 1893
valueIcon, 1893
valueText, 1893
QtEnumPropertyManagerPrivate, 1894
QtEnumPropertyManagerPrivate::Data, 506
QtEnumPropertyType, 1894
QtFlagPropertyManager, 1894
~QtFlagPropertyManager, 1896
flagNames, 1896
flagNamesChanged, 1896
initializeProperty, 1896
QtFlagPropertyManager, 1896
setFlagNames, 1896
setValue, 1896
subBoolPropertyManager, 1897
uninitializeProperty, 1897
value, 1897
valueChanged, 1897
valueText, 1897
QtFlagPropertyManagerPrivate, 1898
QtFlagPropertyManagerPrivate::Data, 506
QtFlagPropertyType, 1898
QtFontEditorFactory, 1899
~QtFontEditorFactory, 1900
QtFontEditorFactory, 1900
QtFontEditorFactoryPrivate, 1901
QtFontEditWidget, 1902
QtFontPropertyManager, 1903
~QtFontPropertyManager, 1904
initializeProperty, 1904
QtFontPropertyManager, 1904
setValue, 1904
subBoolPropertyManager, 1905
subEnumPropertyManager, 1905
subIntPropertyManager, 1905
uninitializeProperty, 1905
value, 1905
valueChanged, 1905
valueIcon, 1906
valueText, 1906
QtFontPropertyManagerPrivate, 1906
QtGroupBoxPropertyBrowser, 1907
~QtGroupBoxPropertyBrowser, 1909
itemChanged, 1909
itemInserted, 1909
itemRemoved, 1909
QtGroupBoxPropertyBrowser, 1909
QtGroupBoxPropertyBrowserPrivate, 1909
QtGroupBoxPropertyBrowserPrivate::WidgetItem, 2703
QtGroupPropertyManager, 1910
~QtGroupPropertyManager, 1911
hasValue, 1911
initializeProperty, 1911
QtGroupPropertyManager, 1911
uninitializeProperty, 1911
QtGroupPropertyType, 1912
QtIntEdit, 1912
QtIntPropertyManager, 1913
~QtIntPropertyManager, 1914
initializeProperty, 1915
maximum, 1915
minimum, 1915
QtIntPropertyManager, 1914
rangeChanged, 1915
setMaximum, 1915
setMinimum, 1915
setRange, 1916
setSingleStep, 1916
setValue, 1916
singleStep, 1916
singleStepChanged, 1916
uninitializeProperty, 1917
value, 1917
valueChanged, 1917
valueText, 1917

QtIntPropertyManagerPrivate, 1917
QtIntPropertyManagerPrivate::Data, 506
QtKeySequenceEditorFactory, 1918
~QtKeySequenceEditorFactory, 1920
QtKeySequenceEditorFactory, 1920
QtKeySequenceEditorFactoryPrivate, 1920
QtKeySequencePropertyManager, 1922
~QtKeySequencePropertyManager, 1923
initializeProperty, 1923
QtKeySequencePropertyManager, 1923
setValue, 1923
uninitializeProperty, 1923
value, 1924
valueChanged, 1924
valueText, 1924
QtKeySequencePropertyManagerPrivate, 1924
QtLineEditFactory, 1925
~QtLineEditFactory, 1926
QtLineEditFactory, 1926
QtLineEditFactoryPrivate, 1927
QtLocalePropertyManager, 1928
~QtLocalePropertyManager, 1929
initializeProperty, 1929
QtLocalePropertyManager, 1929
setValue, 1929
subEnumPropertyManager, 1930
uninitializeProperty, 1930
value, 1930
valueChanged, 1930
valueText, 1930
QtLocalePropertyManagerPrivate, 1931
QtMetaEnumProvider, 1931
QtMetaEnumWrapper, 1932
QtPointFPropertyManager, 1934
~QtPointFPropertyManager, 1935
decimals, 1935
decimalsChanged, 1936
initializeProperty, 1936
QtPointFPropertyManager, 1935
setDecimals, 1936
setValue, 1936
subDoublePropertyManager, 1936
uninitializeProperty, 1936
value, 1937
valueChanged, 1937
valueText, 1937
QtPointFPropertyManagerPrivate, 1938
QtPointFPropertyManagerPrivate::Data, 507
QtPointPropertyManager, 1939
~QtPointPropertyManager, 1940
initializeProperty, 1940
QtPointPropertyManager, 1940
setValue, 1940
subIntPropertyManager, 1940
uninitializeProperty, 1941
value, 1941
valueChanged, 1941
valueText, 1941
QtPointPropertyManagerPrivate, 1942
QtProperty, 1943
~QtProperty, 1944
addSubProperty, 1944
descriptionToolTip, 1944
hasValue, 1944
insertSubProperty, 1944
isEnabled, 1945
isModified, 1945
propertyManager, 1945
propertyName, 1945
QtProperty, 1944
removeSubProperty, 1945
setDescriptionToolTip, 1945
setEnabled, 1946
setModified, 1946
setPropertyName, 1946
setStatusTip, 1946
setValueToolTip, 1946
setWhatsThis, 1947
statusTip, 1947
subProperties, 1947
valueIcon, 1947
valueText, 1947
valueToolTip, 1947
whatsThis, 1948
QtPropertyBrowserUtils, 1948
QtPropertyEditorDelegate, 1949
QtPropertyEditorView, 1950
QtPropertyPrivate, 1951
QtRectFPropertyManager, 1952
~QtRectFPropertyManager, 1953
constraint, 1953
constraintChanged, 1954
decimals, 1954
decimalsChanged, 1954
initializeProperty, 1954
QtRectFPropertyManager, 1953
setConstraint, 1954
setDecimals, 1954
setValue, 1955
subDoublePropertyManager, 1955
uninitializeProperty, 1955
value, 1955
valueChanged, 1955
valueText, 1956
QtRectFPropertyManagerPrivate, 1956
QtRectFPropertyManagerPrivate::Data, 507
QtRectPropertyManager, 1957
~QtRectPropertyManager, 1959
constraint, 1959
constraintChanged, 1959
initializeProperty, 1959
QtRectPropertyManager, 1959
setConstraint, 1959
setValue, 1960
subIntPropertyManager, 1960
uninitializeProperty, 1960

value, 1960
valueChanged, 1960
valueText, 1960
QtRectPropertyManagerPrivate, 1961
QtRectPropertyManagerPrivate::Data, 507
QtScrollBarFactory, 1962
~QtScrollBarFactory, 1964
QtScrollBarFactory, 1964
QtScrollBarFactoryPrivate, 1964
QtSizeFPropertyManager, 1966
~QtSizeFPropertyManager, 1967
decimals, 1967
decimalsChanged, 1968
initializeProperty, 1968
maximum, 1968
minimum, 1968
QtSizeFPropertyManager, 1967
rangeChanged, 1968
setDecimals, 1968
setMaximum, 1969
setMinimum, 1969
setRange, 1969
setValue, 1969
subDoublePropertyManager, 1970
uninitializeProperty, 1970
value, 1970
valueChanged, 1970
valueText, 1970
QtSizeFPropertyManagerPrivate, 1971
QtSizeFPropertyManagerPrivate::Data, 507
QtSizePolicyPropertyManager, 1972
~QtSizePolicyPropertyManager, 1973
initializeProperty, 1973
QtSizePolicyPropertyManager, 1973
setValue, 1973
subEnumPropertyManager, 1973
subIntPropertyManager, 1974
uninitializeProperty, 1974
value, 1974
valueChanged, 1974
valueText, 1974
QtSizePolicyPropertyManagerPrivate, 1975
QtSizePropertyManager, 1976
~QtSizePropertyManager, 1977
initializeProperty, 1977
maximum, 1977
minimum, 1978
QtSizePropertyManager, 1977
rangeChanged, 1978
setMaximum, 1978
setMinimum, 1978
setRange, 1978
setValue, 1979
subIntPropertyManager, 1979
uninitializeProperty, 1979
value, 1979
valueChanged, 1979
valueText, 1980
QtSizePropertyManagerPrivate, 1980
QtSizePropertyManagerPrivate::Data, 507
QtSliderFactory, 1981
~QtSliderFactory, 1983
QtSliderFactory, 1983
QtSliderFactoryPrivate, 1983
QtSpinBoxFactory, 1985
~QtSpinBoxFactory, 1986
QtSpinBoxFactory, 1986
QtSpinBoxFactoryPrivate, 1987
QtStringEdit, 1988
QtStringPropertyManager, 1989
~QtStringPropertyManager, 1990
initializeProperty, 1990
QtStringPropertyManager, 1990
regExp, 1990
regExpChanged, 1991
setRegExp, 1991
setValue, 1991
uninitializeProperty, 1991
value, 1991
valueChanged, 1991
valueText, 1992
QtStringPropertyManagerPrivate, 1992
QtStringPropertyManagerPrivate::Data, 508
QtTimeEdit, 1992
QtTimeEditFactory, 1994
~QtTimeEditFactory, 1995
QtTimeEditFactory, 1995
QtTimeEditFactoryPrivate, 1996
QtTimePropertyManager, 1997
~QtTimePropertyManager, 1998
initializeProperty, 1998
QtTimePropertyManager, 1998
setValue, 1998
uninitializeProperty, 1999
value, 1999
valueChanged, 1999
valueText, 1999
QtTimePropertyManagerPrivate, 1999
QtTimePropertyManagerPrivate::Data, 508
QtTreePropertyBrowser, 2000
~QtTreePropertyBrowser, 2003
backgroundColor, 2003
calculatedBackgroundColor, 2003
collapsed, 2003
editItem, 2004
expanded, 2004
isExpanded, 2004
isItemVisible, 2004
itemChanged, 2004
itemInserted, 2004
itemRemoved, 2005
propertiesWithoutValueMarked, 2005
QtTreePropertyBrowser, 2003
ResizeMode, 2003
setBackgroundColor, 2005
setExpanded, 2005

setItemVisible, 2005
QtTreePropertyBrowserPrivate, 2006
QtVariantEditorFactory, 2007
~QtVariantEditorFactory, 2008
QtVariantEditorFactory, 2008
QtVariantEditorFactoryPrivate, 2009
QtVariantProperty, 2010
~QtVariantProperty, 2011
attributeValue, 2011
propertyType, 2012
QtVariantProperty, 2011
setAttribute, 2012
setValue, 2012
value, 2012
valueType, 2012
QtVariantPropertyManager, 2013
~QtVariantPropertyManager, 2014
addProperty, 2015
attributeChanged, 2015
attributes, 2015
attributeType, 2015
attributeValue, 2015
createProperty, 2016
enumTypeId, 2016
flagTypeId, 2016
groupTypeId, 2016
hasValue, 2016
iconMapTypeId, 2016
initializeProperty, 2017
isPropertyTypeSupported, 2017
propertyType, 2017
QtVariantPropertyManager, 2014
setAttribute, 2017
setValue, 2017
uninitializeProperty, 2018
value, 2018
valueChanged, 2018
valueIcon, 2018
valueText, 2018
valueType, 2019
variantProperty, 2019
QtVariantPropertyManagerPrivate, 2019
QtVariantPropertyPrivate, 2021
Quadrature_if, 2021
quartil
StatisticsDatafile_if, 2486
StatisticsDatafileDefaultImpl1, 2491
Queue, 2022
_check, 2026
_createInternalStatisticReporters, 2026
_getParserChangesInformation, 2026
_initBetweenReplications, 2026
_loadInstance, 2026
_saveInstance, 2027
show, 2027
QueueableItem, 2027

radialAxes
QCPPolarAxisAngular, 1680
radialAxis
QCPPolarAxisAngular, 1680
radialAxisCount
QCPPolarAxisAngular, 1680
random
CountingSampler, 468
Sampler_if, 2110
SamplerDefaultImpl1, 2115
range
QCPSelectionRect, 1750
rangeChanged
QCPAxis, 1356
QCPPolarAxisRadial, 1700
QtDatePropertyManager, 1866
QtDoublePropertyManager, 1879
QtIntPropertyManager, 1915
QtSizeFPropertyManager, 1968
QtSizePropertyManager, 1978
rangeDragAxes
QCPAxisRect, 1379
rangeDragAxis
QCPAxisRect, 1379
rangeZoomAxes
QCPAxisRect, 1379
rangeZoomAxis
QCPAxisRect, 1379
rangeZoomFactor
QCPAxisRect, 1379
RealIntervalStateSet, 2030
contains, 2031
createDefaultState, 2031
isContinuous, 2031
name, 2031
parseState, 2032
realLength
QCPLineEnding, 1652
reallocateBuffer
QCPAbstractPaintBuffer, 1326
QCPPaintBufferPixmap, 1656
RealState, 2032
clone, 2033
equals, 2033
getValue, 2033
setValue, 2033
toString, 2033
typeName, 2033
realVisibility
QCPLayerable, 1602
recalculateDataBounds
QCPColorMapData, 1445
Record, 2034
_check, 2039
_createInternalStatisticReporters, 2039
_initBetweenReplications, 2039
_loadInstance, 2039
_onDispatchEvent, 2039
_saveInstance, 2039
setDatasetDescription, 2040

setDatasetName, 2040
setExpressionName, 2040
setRandomVariableName, 2040
setVariableType, 2040
show, 2040
RecordingContinuousSystemComponent, 2041
_onDispatchEvent, 2046
RecordingLSODE, 2047
_onDispatchEvent, 2052
rect
QCPLayoutElement, 1615
Rectifier, 2053
main, 2054
recursivalyGenerateGraphicalModelFromModel
GraphicalModelBuilder, 824
refreshKnowledgeBaseFromSimulator
AIAssistant_if, 139
AIAssistantDefaultImpl, 152
RefreshPriority
QCustomPlot, 1782
regExp
QtStringPropertyManager, 1990
regExpChanged
QtStringPropertyManager, 1991
registerContributions
BacteriaColonyViewerGuiExtensionPlugin, 228
BioNetworkAwareGuiExtensionPlugin, 261
BioNetworkEditorGuiExtensionPlugin, 262
BioNetworkExecutionGuiExtensionPlugin, 264
BioParameterManagerGuiExtensionPlugin, 270
BioReactionKineticsGuiExtensionPlugin, 282
BioReactionStoichiometryGuiExtensionPlugin, 284
BioSBMLInteroperabilityGuiExtensionPlugin, 293
BioSimulationResultsGuiExtensionPlugin, 301
BioSpeciesManagerGuiExtensionPlugin, 313
CellularAutomataGuiExtensionPlugin, 395
CoreGuiExtensionPlugin, 460
GeneticCircuitGuiExtensionPlugin, 781
GuiExtensionConfigurationGuiExtensionPlugin, 863
ModelPluginInspectorGuiExtensionPlugin, 1093
RegisterTests
testing::internal::ParameterizedTestSuiteInfo< TestSuite >, 1199
reject
DialogSimulationConfigure, 558
Release, 2054
_check, 2059
_createEditableDataDefinitions, 2059
_initBetweenReplications, 2060
_loadInstance, 2060
_onDispatchEvent, 2060
_saveInstance, 2060
show, 2060
Remove, 2061
_check, 2066
_createEditableDataDefinitions, 2066
_loadInstance, 2066
_onDispatchEvent, 2066
_saveInstance, 2066
show, 2067
remove
AISecretStore, 167
ConnectionManager, 441
Model, 1030
ModelComponent, 1043
ModelComponentManager, 1047
ModelDataManager, 1066
ModelManager, 1091
QCPBarsGroup, 1427
QCPDataContainer< DataType >, 1480
QCPLayout, 1609
removeAfter
QCPDataContainer< DataType >, 1480
removeAt
QCPLayout, 1609
removeAtPort
ConnectionManager, 441
removeAxis
QCPAxisRect, 1379
removeBefore
QCPDataContainer< DataType >, 1480
removeEntity
Model, 1030
removeFromLegend
QCPAbstractPlottable, 1335
removeGraph
QCustomPlot, 1792
removeItem
QCPLegend, 1646
QCustomPlot, 1792, 1793
removeLayer
QCustomPlot, 1793
removePlottable
QCustomPlot, 1793
removeProperty
QtAbstractPropertyBrowser, 1816
removePropertyManager
QtAbstractEditorFactory< PropertyManager >, 1810
removeRadialAxis
QCPPolarAxisAngular, 1681
removeSubProperty
QtProperty, 1945
renderContext
GraphicalModelComponent, 830
replications
SimulationResultsDataset, 2246
replot
QCPLayer, 1596
QCustomPlot, 1793
replotTime
QCustomPlot, 1794
requiredModelPlugins
BacteriaColonyViewerGuiExtensionPlugin, 228
BioNetworkAwareGuiExtensionPlugin, 261

BioNetworkEditorGuiExtensionPlugin, 263
BioNetworkExecutionGuiExtensionPlugin, 264
BioParameterManagerGuiExtensionPlugin, 270
BioReactionKineticsGuiExtensionPlugin, 283
BioReactionStoichiometryGuiExtensionPlugin, 285
BioSBMLInteroperabilityGuiExtensionPlugin, 293
BioSimulationResultsGuiExtensionPlugin, 302
BioSpeciesManagerGuiExtensionPlugin, 314
CellularAutomataGuiExtensionPlugin, 396
CoreGuiExtensionPlugin, 460
GeneticCircuitGuiExtensionPlugin, 781
GuiExtensionConfigurationGuiExtensionPlugin, 864
ModelPluginInspectorGuiExtensionPlugin, 1093
rescale
QCPAxis, 1357
QCPPolarAxisAngular, 1681
QCPPolarAxisRadial, 1701
rescaleAxes
QCPAbstractPlottable, 1335
QCustomPlot, 1794
rescaleDataRange
QCPColorMap, 1439
QCPColorScale, 1454
rescaleKeyAxis
QCPAbstractPlottable, 1335
rescaleValueAxis
QCPAbstractPlottable, 1335
ResetIdOfType
Util, 2630
resetResults
Optimizer_if, 1180
OptimizerDefaultImpl1, 1186
Resistor, 2067
_loadInstance, 2072
_saveInstance, 2072
Build, 2073
ResizeMode
QtTreePropertyBrowser, 2003
ResolutionUnit
QCP, 114
resolveRateConstant
BioReaction, 280
resolveReverseRateConstant
BioReaction, 280
Resource, 2073
_check, 2078
_createAttachedAttributes, 2078
_initBetweenReplications, 2078
_loadInstance, 2078
_saveInstance, 2078
Resource, 2078
show, 2079
ResourceAllocationComponent, 2079
_check, 2084
_createEditableDataDefinitions, 2084
_initBetweenReplications, 2084
_loadInstance, 2084
_onDispatchEvent, 2085
_saveInstance, 2085
show, 2085
ResourceTestProbe, 2085
restart
WebWorkerRuntime, 2695
restore
QCPPainter, 1660
restoreTransition
EFSMNetwork, 644
resume
Optimizer_if, 1180
OptimizerDefaultImpl1, 1186
right
QCPAxisRect, 1380
QCPPolarAxisAngular, 1681
RootFinder_if, 2087
Route, 2087
_check, 2092
_createAttachedAttributes, 2092
_createEditableDataDefinitions, 2093
_createInternalStatisticReporters, 2093
_loadInstance, 2093
_onDispatchEvent, 2093
_saveInstance, 2093
show, 2093
rowColToIndex
QCPLayoutGrid, 1627
rowCount
QCPLayoutGrid, 1627
rpImmediateRefresh
QCustomPlot, 1782
rpQueuedRefresh
QCustomPlot, 1782
rpQueuedReplot
QCustomPlot, 1782
rpRefreshHint
QCustomPlot, 1782
RSimulator, 2094
_check, 2099
_createEditableDataDefinitions, 2099
_loadInstance, 2100
_onDispatchEvent, 2100
_saveInstance, 2100
show, 2100
RSimulatorRunner, 2100
_check, 2105
_loadInstance, 2105
_saveInstance, 2106
show, 2106
ruDotsPerCentimeter
QCP, 114
ruDotsPerInch
QCP, 114
ruDotsPerMeter
QCP, 114
run
FakeSystemCommandExecutor, 721

SystemCommandExecutor_if, 2552
SystemShellCommandExecutor, 2559
RungeKutta4OdeSolver, 2106
advance, 2107
RunningPath
Util, 2631
runSimulation
AIAssistant_if, 139
AIAssistantDefaultImpl, 152
SimulatorSessionService, 2262
runWorkerJob
SimulatorSessionService, 2262

Sample, 2109
sampleBernoulli
CountingSampler, 468
Sampler_if, 2110
SamplerDefaultImpl1, 2115
sampleBeta
CountingSampler, 468
Sampler_if, 2110, 2111
SamplerDefaultImpl1, 2115
sampleBinomial
CountingSampler, 469
Sampler_if, 2111
SamplerDefaultImpl1, 2116
sampleDiscrete
CountingSampler, 469
Sampler_if, 2111
SamplerDefaultImpl1, 2116
sampleErlang
CountingSampler, 469
Sampler_if, 2111
SamplerDefaultImpl1, 2116
sampleExponential
CountingSampler, 469
Sampler_if, 2112
SamplerDefaultImpl1, 2116
sampleGamma
CountingSampler, 469
Sampler_if, 2112
SamplerDefaultImpl1, 2116
sampleGeometric
CountingSampler, 469
Sampler_if, 2112
SamplerDefaultImpl1, 2116
sampleGumbell
CountingSampler, 470
Sampler_if, 2112
SamplerDefaultImpl1, 2117
sampleLogNormal
CountingSampler, 470
Sampler_if, 2112
SamplerDefaultImpl1, 2117
sampleNormal
CountingSampler, 470
Sampler_if, 2112
SamplerDefaultImpl1, 2117
Sampler_if, 2109
getRNGparameters, 2110
random, 2110
sampleBernoulli, 2110
sampleBeta, 2110, 2111
sampleBinomial, 2111
sampleDiscrete, 2111
sampleErlang, 2111
sampleExponential, 2112
sampleGamma, 2112
sampleGeometric, 2112
sampleGumbell, 2112
sampleLogNormal, 2112
sampleNormal, 2112
sampleTriangular, 2112
sampleUniform, 2113
sampleWeibull, 2113
setRNGparameters, 2113
Sampler_if::RNG_Parameters, 2086
SamplerDefaultImpl1, 2113
getRNGparameters, 2115
random, 2115
sampleBernoulli, 2115
sampleBeta, 2115
sampleBinomial, 2116
sampleDiscrete, 2116
sampleErlang, 2116
sampleExponential, 2116
sampleGamma, 2116
sampleGeometric, 2116
sampleGumbell, 2117
sampleLogNormal, 2117
sampleNormal, 2117
sampleTriangular, 2117
sampleUniform, 2117
sampleWeibull, 2117
setRNGparameters, 2117
SamplerDefaultImpl1::DefaultImpl1RNG_Parameters, 522
sampleTriangular
CountingSampler, 470
Sampler_if, 2112
SamplerDefaultImpl1, 2117
sampleUniform
CountingSampler, 470
Sampler_if, 2113
SamplerDefaultImpl1, 2117
sampleWeibull
CountingSampler, 470
Sampler_if, 2113
SamplerDefaultImpl1, 2117
sandboxEnabled
AIAssistantConfiguration, 143
sanitizedForLinScale
QCPRange, 1730
sanitizedForLogScale
QCPRange, 1730
save
AISecretStore, 167

FakeModelPersistence, 717
FakeModelPersistenceB, 719
Model, 1030
Persistence_if, 1227
PersistenceDefaultImpl2, 1229
QCPPainter, 1660
SystemPreferences, 2558
saveBmp
QCustomPlot, 1794
saveCurrentModel
SimulatorSessionService, 2263
saveGraphicalModel
GraphicalModelSerializer, 836
SaveInstance
ModelComponent, 1043
ModelDataDefinition, 1058
saveInstance
ModelInfo, 1081
saveJpg
QCustomPlot, 1795
saveModel
ModelManager, 1091
savePdf
QCustomPlot, 1796
savePng
QCustomPlot, 1796
saveRastered
QCustomPlot, 1797
saveTextModel
GraphicalModelSerializer, 836
scaleRange
QCPAxis, 1357
QCPPolarAxisAngular, 1681
QCPPolarAxisRadial, 1701
ScaleStrategy
QCPAxisTickerFixed, 1396
ScaleType
QCPAxis, 1352
QCPPolarAxisRadial, 1698
scaleTypeChanged
QCPAxis, 1357
QCPPolarAxisRadial, 1701
ScatterProperty
QCPScatterStyle, 1734
ScatterShape
QCPScatterStyle, 1734
ScenarioExperiment_if, 2118
sceneEvent
GraphicalModelComponent, 830
SceneToolController, 2118
onActionDrawLineTriggered, 2120
onActionShowGridTriggered, 2120
onActionZoomActualSizeTriggered, 2121
onActionZoomAllTriggered, 2121
onHorizontalSliderZoomGraphicalValueChanged, 2121
SceneToolController, 2120
SchedulableItem, 2121
Schedule, 2122
_addSimulationControl, 2125
_check, 2125
_getParserChangesInformation, 2126
_initBetweenReplications, 2126
_loadInstance, 2126
_saveInstance, 2126
show, 2126
scrInnerRect
QCPLayoutElement, 1614
scrOuterRect
QCPLayoutElement, 1614
sdBoth
QCP, 116
sdNegative
QCP, 116
sdPositive
QCP, 116
Search, 2127
_check, 2132
_createEditableDataDefinitions, 2132
_loadInstance, 2133
_onDispatchEvent, 2133
_saveInstance, 2133
show, 2133
seekFirstValue
CollectorDatafile_if, 411
CollectorDatafileDefaultImpl1, 414
Segment, 2134
_check, 2138
_loadInstance, 2138
_saveInstance, 2138
show, 2138
SegmentStep, 2138
SeizableItem, 2139
Seize, 2140
_check, 2145
_createEditableDataDefinitions, 2145
_initBetweenReplications, 2146
_loadInstance, 2146
_onDispatchEvent, 2146
_saveInstance, 2146
show, 2146
Seize::DEFAULT_VALUES, 521
selectableChanged
QCPAbstractPlottable, 1336
QCPAxis, 1357
QCPPolarAxisRadial, 1701
SelectablePart
QCPAxis, 1353
QCPLegend, 1644
QCPPolarAxisAngular, 1677
QCPPolarAxisRadial, 1698
selected
QCPAbstractPlottable, 1336
selectedAxes
QCustomPlot, 1798
selectedGraphs

QCustomPlot, 1798
selectedItems
QCPLegend, 1647
QCustomPlot, 1798
selectedLegends
QCustomPlot, 1798
selectedPlottables
QCustomPlot, 1798
selectEvent
QCPAbstractItem, 1316
QCPAbstractLegendItem, 1323
QCPAbstractPlottable, 1336
QCPAxis, 1357
QCPLegend, 1647
QCPPolarAxisRadial, 1701
QCPPolarGraph, 1715
QCPTextElement, 1770
selection
QCPAbstractPlottable, 1336
selectionCategory
QCPAbstractItem, 1316
QCPAbstractLegendItem, 1323
QCPAbstractPlottable, 1336
QCPAxis, 1358
QCPLegend, 1647
QCPPolarAxisAngular, 1681
QCPPolarAxisRadial, 1702
QCPPolarGraph, 1715
selectionChanged
QCPAbstractItem, 1317
QCPAbstractLegendItem, 1323
QCPAbstractPlottable, 1336
QCPAxis, 1358
QCPLegend, 1647
QCPPolarAxisRadial, 1702
QCPTextElement, 1770
selectionChangedByUser
QCustomPlot, 1798
selectionDecorator
QCPAbstractPlottable, 1337
selectionRect
QCustomPlot, 1799
SelectionRectMode
QCP, 114
SelectionType
QCP, 115
selectTest
QCPAbstractItem, 1317
QCPAbstractLegendItem, 1323
QCPAbstractPlottable, 1337
QCPAbstractPlottable1D< DataType >, 1346
QCPAxis, 1358
QCPBars, 1419
QCPColorMap, 1440
QCPCurve, 1471
QCPErrorBars, 1500
QCPFinancial, 1511
QCPGraph, 1523
QCPItemBracket, 1538
QCPItemCurve, 1543
QCPItemEllipse, 1548
QCPItemLine, 1553
QCPItemPixmap, 1558
QCPItemRect, 1570
QCPItemStraightLine, 1575
QCPItemText, 1580
QCPItemTracer, 1588
QCPLayerable, 1602
QCPLayoutElement, 1616
QCPLayoutInset, 1636
QCPLegend, 1647
QCPPolarAxisAngular, 1682
QCPPolarAxisRadial, 1702
QCPPolarGraph, 1715
QCPStatisticalBox, 1759
QCPTextElement, 1770
selectTestRect
QCPAbstractPlottable1D< DataType >, 1346
QCPBars, 1420
QCPErrorBars, 1500
QCPFinancial, 1511
QCPPlottableInterface1D, 1664
QCPStatisticalBox, 1759
send
AIProviderClient_if, 160
HttpProviderClientBase, 881
sendEntityToComponent
Model, 1030
Separate, 2147
_check, 2152
_loadInstance, 2152
_onDispatchEvent, 2152
_saveInstance, 2152
show, 2152
SepKeyVal
Util, 2631
Sequence, 2153
_check, 2157
_createAttachedAttributes, 2157
_loadInstance, 2157
_saveInstance, 2157
show, 2157
sequence
Json, 904
SequenceStep, 2158
_loadInstance, 2159
_saveInstance, 2159
Serializer, 2159
dump, 2160
for_each, 2160
get, 2161
load, 2161
newPersistenceRecord, 2161
put, 2161
serve
SimpleHttpServer, 2182

SessionContext, 2163
SessionManager, 2163
createSession, 2164
getSessionByToken, 2164
SessionManager, 2164
Set, 2165
_check, 2169
_createAttachedAttributes, 2169
_getParserChangesInformation, 2169
_loadInstance, 2169
_saveInstance, 2169
createElementSetOfType, 2170
setAllowedElementTypes, 2170
setSetOfType, 2170
show, 2170
set
QCPDataContainer< DataType >, 1480, 1481
setAdaptiveSampling
QCPGraph, 1524
setAddValueHandler
Collector_if, 409
CollectorDatafileDefaultImpl1, 414
CollectorDefaultImpl1, 416
setAllowedElementTypes
Set, 2170
setAlpha
QCPColorMapData, 1446
setAnalystName
ModelInfo, 1082
setAngularPen
QCPPolarGrid, 1722
setAngularSubGridPen
QCPPolarGrid, 1722
setAntialiased
QCPLayerable, 1603
setAntialiasedElement
QCustomPlot, 1799
setAntialiasedElements
QCustomPlot, 1799
setAntialiasedFill
QCPAbstractPlottable, 1337
QCPPolarGraph, 1716
setAntialiasedScatters
QCPAbstractPlottable, 1338
QCPPolarGraph, 1716
setAntialiasedSubGrid
QCPGrid, 1530
QCPPolarGrid, 1722
setAntialiasedZeroLine
QCPGrid, 1530
QCPPolarGrid, 1722
setAntialiasing
QCPPainter, 1660
setApiKey
AIAssistant_if, 140
AIAssistantDefaultImpl, 153
AIProviderClient_if, 161
HttpProviderClientBase, 882
setApiKeyEnvironmentVariable
AIAssistant_if, 140
AIAssistantDefaultImpl, 153
setApiVersion
AnthropicProviderClientImpl, 202
setAttribute
QtVariantProperty, 2012
QtVariantPropertyManager, 2017
setAttributeValue
Entity, 668
setAutoAddPlottableToLegend
QCustomPlot, 1799
setAutoMargins
QCPLayoutElement, 1616
setAutoSqueeze
QCPDataContainer< DataType >, 1481
setAxes
QCPItemPosition, 1563
setAxisRect
QCPItemPosition, 1563
setBackground
QCPAxisRect, 1380
QCPPolarAxisAngular, 1682
QCustomPlot, 1799, 1800
setBackgroundColor
QtTreePropertyBrowser, 2005
setBackgroundScaled
QCPAxisRect, 1381
QCPPolarAxisAngular, 1683
QCustomPlot, 1800
setBackgroundScaledMode
QCPAxisRect, 1381
QCPPolarAxisAngular, 1683
QCustomPlot, 1800
setBarsGroup
QCPBars, 1420
setBarWidth
QCPColorScale, 1454
setBasePen
QCPAxis, 1358
QCPPolarAxisAngular, 1683
QCPPolarAxisRadial, 1702
setBaseUrl
AIAssistant_if, 140
AIAssistantDefaultImpl, 153
AIProviderClient_if, 161
HttpProviderClientBase, 882
setBaseValue
QCPBars, 1420
setBegin
QCPDataRange, 1485
setBorderPen
QCPLegend, 1647
setBracketBrush
QCPSelectionDecoratorBracket, 1745
setBracketHeight
QCPSelectionDecoratorBracket, 1745
setBracketPen

QCPSelectionDecoratorBracket, 1745
setBracketStyle
QCPSelectionDecoratorBracket, 1745
setBracketWidth
QCPSelectionDecoratorBracket, 1746
setBrush
QCPAbstractPlottable, 1338
QCPItemEllipse, 1548
QCPItemRect, 1570
QCPItemText, 1581
QCPItemTracer, 1588
QCPLegend, 1648
QCPPolarGraph, 1716
QCPScatterStyle, 1737
QCPSelectionDecorator, 1741
QCPSelectionRect, 1750
setBrushNegative
QCPFinancial, 1511
setBrushPositive
QCPFinancial, 1511
setBufferDevicePixelRatio
QCustomPlot, 1801
setCell
QCPColorMapData, 1446
setChannelFillGraph
QCPGraph, 1524
setChartStyle
QCPFinancial, 1512
setClearHandler
Collector_if, 409
CollectorDatafileDefaultImpl1, 414
CollectorDefaultImpl1, 416
setClipAxisRect
QCPAbstractItem, 1317
setClipToAxisRect
QCPAbstractItem, 1317
setCollector
Statistics_if, 2477
StatisticsDatafileDefaultImpl1, 2491
StatisticsDefaultImpl1, 2494
setColor
QCPItemText, 1581
setColorInterpolation
QCPColorGradient, 1431
setColorScale
QCPColorMap, 1440
setColorStopAt
QCPColorGradient, 1432
setColorStops
QCPColorGradient, 1432
setColumnSpacing
QCPLayoutGrid, 1627
setColumnStretchFactor
QCPLayoutGrid, 1628
setColumnStretchFactors
QCPLayoutGrid, 1628
setConfidenceLevel
Statistics_if, 2477
StatisticsDatafileDefaultImpl1, 2491
StatisticsDefaultImpl1, 2494
setConfiguration
AIAssistant_if, 140
AIAssistantDefaultImpl, 153
setConstraint
QtRectFPropertyManager, 1954
QtRectPropertyManager, 1959
setCoords
QCPItemPosition, 1563
setCurrent
ModelManager, 1091
setCurrentItem
QtAbstractPropertyBrowser, 1817
setCurrentLayer
QCustomPlot, 1801
setCurrentListElementType
SimulationControlGenericListPointer< T, M, C >, 2216
setCustomPath
QCPScatterStyle, 1738
setData
QCPBars, 1420
QCPColorMap, 1440
QCPColorMapData, 1446
QCPCurve, 1471, 1472
QCPErrorBars, 1500, 1501
QCPFinancial, 1512
QCPGraph, 1524
QCPPolarGraph, 1716, 1717
QCPStatisticalBox, 1759
setDataFilename
CollectorDatafile_if, 412
CollectorDatafileDefaultImpl1, 415
Fitter_if, 729
FitterDefaultImpl, 732
FitterDummyImpl, 735
SortFile, 2423
setDataPlottable
QCPErrorBars, 1501
setDataRange
QCPColorMap, 1441
QCPColorScale, 1454
setDataScaleType
QCPColorMap, 1441
QCPColorScale, 1454
setDatasetDescription
Record, 2040
setDatasetName
Record, 2040
setDateTimeFormat
QCPAxisTickerDateTime, 1392
setDateTimeSpec
QCPAxisTickerDateTime, 1393
setDecimals
QtDoublePropertyManager, 1879
QtPointFPropertyManager, 1936
QtRectFPropertyManager, 1954

QtSizeFPropertyManager, 1968
setDefaultEventHandlers
BaseGenesysTerminalApplication, 238
setDescription
ModelComponent, 1043
ModelInfo, 1082
setDescriptionToolTip
QtProperty, 1945
setDevicePixelRatio
QCPAbstractPaintBuffer, 1326
setEFSMNetworkDiagramLayout
GraphicalModelSerializer, 836
setEnabled
QtProperty, 1946
setEnd
QCPDataRange, 1485
setEntitiesCreated
SourceModelComponent, 2432
setEntitiesPerCreation
SourceModelComponent, 2432
setEntity
Event, 689
setEntityType
Entity, 668
SourceModelComponent, 2432
setEntityTypeName
Entity, 669
SourceModelComponent, 2432
setEnumIcons
QtEnumPropertyManager, 1892
setEnumNames
QtEnumPropertyManager, 1892
setErrorType
QCPErrorBars, 1501
setExpanded
QtButtonPropertyBrowser, 1833
QtTreePropertyBrowser, 2005
setExpression
Formula, 746
setExpressionName
Record, 2040
setFactoryForManager
QtAbstractPropertyBrowser, 1817
setFieldWidth
QCPAxisTickerTime, 1410
setFillOrder
QCPLayoutGrid, 1628
setFirstCreation
SourceModelComponent, 2432
setFlagNames
QtFlagPropertyManager, 1896
setFont
QCPAbstractLegendItem, 1323
QCPItemText, 1581
QCPLegend, 1648
QCPTextElement, 1770
setFractionStyle
QCPAxisTickerPi, 1403
setFromOther
QCPScatterStyle, 1738
setGradient
QCPColorMap, 1441
QCPColorScale, 1455
setGraph
QCPItemTracer, 1588
setGraphKey
QCPItemTracer, 1589
setHasChanged
FakeModelPersistence, 718
FakeModelPersistenceB, 720
Model, 1031
ModelComponentManager, 1048
ModelDataDefinition, 1058
Persistence_if, 1227
PersistenceDefaultImpl2, 1229
setHead
QCPItemCurve, 1543
QCPItemLine, 1553
setHistogramNumClasses
StatisticsDatafile_if, 2487
StatisticsDatafileDefaultImpl1, 2491
setIconBorderPen
QCPLegend, 1648
setIconSize
QCPLegend, 1648
setIconTextPadding
QCPLegend, 1648
SetIndent
Util, 2631
setInitialValuesText
Attribute, 218
Variable, 2644
setInsetAlignment
QCPLayoutInset, 1636
setInsetPlacement
QCPLayoutInset, 1636
setInsetRect
QCPLayoutInset, 1637
setInteraction
QCustomPlot, 1801
setInteractions
QCustomPlot, 1801
setInterpolate
QCPColorMap, 1441
setInterpolating
QCPItemTracer, 1589
setInvalidated
QCPAbstractPaintBuffer, 1327
setInverted
QCPLineEnding, 1652
setItemVisible
QtTreePropertyBrowser, 2005
setKeyAxis
QCPAbstractPlottable, 1338
QCPPolarGraph, 1717
setKeyRange

QCPColorMapData, 1447
setKeySize
QCPColorMapData, 1447
setKineticLawExpression
BioReaction, 280
setLabel
ModelDataDefinition, 1058
QCPAxis, 1359
QCPColorScale, 1455
QCPPolarAxisAngular, 1683
QCPPolarAxisRadial, 1703
setLabelColor
QCPAxis, 1359
QCPPolarAxisAngular, 1683
QCPPolarAxisRadial, 1703
setLabelFont
QCPAxis, 1359
QCPPolarAxisAngular, 1683
QCPPolarAxisRadial, 1703
setLabelPadding
QCPAxis, 1359
QCPPolarAxisAngular, 1684
QCPPolarAxisRadial, 1703
setLayer
QCPLayerable, 1603
setLength
QCPItemBracket, 1538
QCPLineEnding, 1652
setLevelCount
QCPColorGradient, 1432
setLineStyle
QCPCurve, 1472
QCPGraph, 1525
QCPPolarGraph, 1717
setLogBase
QCPAxisTickerLog, 1399
setLowerEnding
QCPAxis, 1359
setMarginGroup
QCPLayoutElement, 1616
setMargins
QCPLayoutElement, 1616
setMaxCreations
SourceModelComponent, 2433
setMaximum
QtDatePropertyManager, 1866
QtDoublePropertyManager, 1880
QtIntPropertyManager, 1915
QtSizeFPropertyManager, 1969
QtSizePropertyManager, 1978
setMaximumSize
QCPLayoutElement, 1617
setMaxInputConnections
ConnectionManager, 441
setMaxOutputConnections
ConnectionManager, 442
setMaxOutputTokens
AIAssistant_if, 140
AIAssistantDefaultImpl, 153
setMaxSteps
SolverDefaultImpl1, 2422
setMedianPen
QCPStatisticalBox, 1760
setMessage
WorkerJobManager, 2708
setMinimum
QtDatePropertyManager, 1866
QtDoublePropertyManager, 1880
QtIntPropertyManager, 1915
QtSizeFPropertyManager, 1969
QtSizePropertyManager, 1978
setMinimumMargins
QCPLayoutElement, 1617
setMinimumSize
QCPLayoutElement, 1617
setMinInputConnections
ConnectionManager, 442
setMinOutputConnections
ConnectionManager, 442
setMode
QCPLayer, 1596
QCPPainter, 1660
setModel
Optimizer_if, 1180
OptimizerDefaultImpl1, 1186
setModelBlock
PropertyEditor, 1293
setModelFilename
Optimizer_if, 1180
OptimizerDefaultImpl1, 1186
setModelLevel
ModelDataDefinition, 1059
setModelLevelFilter
GraphicalModelBuilder, 825
ModelGraphicsScene, 1076
setModelName
AIAssistant_if, 140
AIAssistantDefaultImpl, 153
setModes
QCPPainter, 1660
setModified
QtProperty, 1946
setMultiSelectModifier
QCustomPlot, 1802
setName
ModelDataDefinition, 1059
ModelInfo, 1082
QCPAbstractPlottable, 1338
QCPPolarGraph, 1717
setNanColor
QCPColorGradient, 1432
setNanHandling
QCPColorGradient, 1432
setNoAntialiasingOnDrag
QCustomPlot, 1802
setNotAntialiasedElement

QCustomPlot, 1803
setNotAntialiasedElements
QCustomPlot, 1803
setNumberFormat
QCPAxis, 1359
QCPPolarAxisAngular, 1684
QCPPolarAxisRadial, 1703
setNumberPrecision
QCPAxis, 1360
QCPPolarAxisAngular, 1684
QCPPolarAxisRadial, 1704
setOffset
QCPAxis, 1360
setOpenGl
QCustomPlot, 1803
setOption
FakeModelPersistence, 718
FakeModelPersistenceB, 720
Persistence_if, 1227
PersistenceDefaultImpl2, 1229
setOuterRect
QCPLayoutElement, 1618
setOutlierStyle
QCPStatisticalBox, 1760
setPadding
QCPAxis, 1360
QCPItemText, 1581
setParentAnchor
QCPItemPosition, 1564
setParentAnchorX
QCPItemPosition, 1564
setParentAnchorY
QCPItemPosition, 1564
setPen
QCPAbstractPlottable, 1338
QCPGrid, 1531
QCPItemBracket, 1538
QCPItemCurve, 1543
QCPItemEllipse, 1549
QCPItemLine, 1554
QCPItemPixmap, 1559
QCPItemRect, 1571
QCPItemStraightLine, 1575
QCPItemText, 1582
QCPItemTracer, 1589
QCPPainter, 1660, 1661
QCPPolarGraph, 1717
QCPScatterStyle, 1738
QCPSelectionDecorator, 1741
QCPSelectionRect, 1751
setPenNegative
QCPFinancial, 1512
setPenPositive
QCPFinancial, 1513
setPeriodic
QCPColorGradient, 1432
setPeriodicity
QCPAxisTickerPi, 1403
setPiSymbol
QCPAxisTickerPi, 1403
setPiValue
QCPAxisTickerPi, 1403
setPixelPosition
QCPItemPosition, 1564
setPixmap
QCPItemPixmap, 1559
QCPScatterStyle, 1738
setPlottingHint
QCustomPlot, 1804
setPlottingHints
QCustomPlot, 1804
setPositionAlignment
QCPItemText, 1582
setPrecision
SolverDefaultImpl1, 2423
setProjectTitle
ModelInfo, 1082
setPropertyName
QtProperty, 1946
setProvider
AIAssistant_if, 140
AIAssistantDefaultImpl, 154
setProviderClient
AIAssistant_if, 141
AIAssistantDefaultImpl, 154
setProviderName
AIAssistant_if, 141
AIAssistantDefaultImpl, 154
setRandomVariableName
Record, 2040
setRange
QCPAxis, 1360, 1361
QCPColorMapData, 1447
QCPPolarAxisAngular, 1684, 1685
QCPPolarAxisRadial, 1704
QtDatePropertyManager, 1866
QtDoublePropertyManager, 1880
QtIntPropertyManager, 1916
QtSizeFPropertyManager, 1969
QtSizePropertyManager, 1978
setRangeDrag
QCPAxisRect, 1381
QCPColorScale, 1455
setRangeDragAxes
QCPAxisRect, 1381, 1382
setRangeLower
QCPAxis, 1361
QCPPolarAxisAngular, 1685
QCPPolarAxisRadial, 1704
setRangeReversed
QCPAxis, 1361
QCPPolarAxisAngular, 1685
QCPPolarAxisRadial, 1705
setRangeUpper
QCPAxis, 1361
QCPPolarAxisAngular, 1685

QCPPolarAxisRadial, 1705
setRangeZoom
QCPAxisRect, 1382
QCPColorScale, 1455
setRangeZoomAxes
QCPAxisRect, 1382, 1383
setRangeZoomFactor
QCPAxisRect, 1383
setRateConstant
BioReaction, 280
setRateConstantParameterName
BioReaction, 280
setReasoningEffort
AIAssistant_if, 141
AIAssistantDefaultImpl, 154
setRegExp
QtStringPropertyManager, 1991
setReportStatistics
ModelDataDefinition, 1059
setReverseKineticLawExpression
BioReaction, 281
setReverseRateConstant
BioReaction, 281
setReverseRateConstantParameterName
BioReaction, 281
setReversible
BioReaction, 281
setRNGparameters
CountingSampler, 470
Sampler_if, 2113
SamplerDefaultImpl1, 2117
setRotation
QCPItemText, 1582
setRowSpacing
QCPLayoutGrid, 1628
setRowStretchFactor
QCPLayoutGrid, 1629
setRowStretchFactors
QCPLayoutGrid, 1629
setSampler
Parser_if, 1204
ParserDefaultImpl2, 1215
setScaled
QCPItemPixmap, 1559
setScaleRatio
QCPAxis, 1362
setScaleStrategy
QCPAxisTickerFixed, 1397
setScaleType
QCPAxis, 1362
QCPPolarAxisRadial, 1705
setScatterSkip
QCPCurve, 1472
QCPGraph, 1525
setScatterStyle
QCPCurve, 1472
QCPGraph, 1525
QCPPolarGraph, 1718
QCPSelectionDecorator, 1741
setSelectable
QCPAbstractItem, 1318
QCPAbstractLegendItem, 1324
QCPAbstractPlottable, 1338
QCPPolarGraph, 1718
QCPTextElement, 1770
setSelectableParts
QCPAxis, 1362
QCPLegend, 1648
QCPPolarAxisAngular, 1686
QCPPolarAxisRadial, 1705
setSelected
QCPAbstractItem, 1318
QCPAbstractLegendItem, 1324
QCPTextElement, 1771
setSelectedBasePen
QCPAxis, 1362
QCPPolarAxisAngular, 1686
QCPPolarAxisRadial, 1705
setSelectedBorderPen
QCPLegend, 1649
setSelectedBrush
QCPItemEllipse, 1549
QCPItemRect, 1571
QCPItemText, 1582
QCPItemTracer, 1589
QCPLegend, 1649
setSelectedColor
QCPItemText, 1582
setSelectedControls
Optimizer_if, 1180
OptimizerDefaultImpl1, 1186
setSelectedFont
QCPAbstractLegendItem, 1324
QCPItemText, 1582
QCPLegend, 1649
QCPTextElement, 1771
setSelectedIconBorderPen
QCPLegend, 1649
setSelectedLabelColor
QCPAxis, 1362
QCPPolarAxisAngular, 1686
QCPPolarAxisRadial, 1705
setSelectedLabelFont
QCPAxis, 1362
QCPPolarAxisAngular, 1686
QCPPolarAxisRadial, 1706
setSelectedParts
QCPAxis, 1363
QCPLegend, 1649
QCPPolarAxisAngular, 1686
QCPPolarAxisRadial, 1706
setSelectedPen
QCPItemBracket, 1539
QCPItemCurve, 1544
QCPItemEllipse, 1549
QCPItemLine, 1554

QCPItemPixmap, 1559
QCPItemRect, 1571
QCPItemStraightLine, 1575
QCPItemText, 1583
QCPItemTracer, 1589
setSelectedResponses
Optimizer_if, 1180
OptimizerDefaultImpl1, 1187
setSelectedSubTickPen
QCPAxis, 1363
QCPPolarAxisAngular, 1686
QCPPolarAxisRadial, 1706
setSelectedTextColor
QCPAbstractLegendItem, 1324
QCPLegend, 1650
QCPTextElement, 1771
setSelectedTickLabelColor
QCPAxis, 1363
QCPPolarAxisAngular, 1687
QCPPolarAxisRadial, 1706
setSelectedTickLabelFont
QCPAxis, 1363
QCPPolarAxisAngular, 1687
QCPPolarAxisRadial, 1706
setSelectedTickPen
QCPAxis, 1363
QCPPolarAxisAngular, 1687
QCPPolarAxisRadial, 1707
setSelection
QCPAbstractPlottable, 1339
QCPPolarGraph, 1718
setSelectionDecorator
QCPAbstractPlottable, 1339
setSelectionRect
QCustomPlot, 1804
setSelectionRectMode
QCustomPlot, 1804
setSelectionTolerance
QCustomPlot, 1805
setSetOfType
Set, 2170
setSettings
Optimizer_if, 1180
OptimizerDefaultImpl1, 1187
setShape
QCPScatterStyle, 1738
setSimulationModelBasedOnText
ModelLanguageSynchronizer, 1085
setSimulatorFacade
AIAssistant_if, 141
AIAssistantDefaultImpl, 154
setSingleStep
QtDoublePropertyManager, 1880
QtIntPropertyManager, 1916
setSize
QCPAbstractPaintBuffer, 1327
QCPColorMapData, 1447
QCPItemTracer, 1590
QCPScatterStyle, 1738
setSizeConstraintRect
QCPLayoutElement, 1618
setSnapshotFilename
WorkerJobManager, 2708
setSpacing
QCPBarsGroup, 1427
setSpacingType
QCPBarsGroup, 1427
setStackingGap
QCPBars, 1421
setState
WorkerJobManager, 2708
setStatusTip
QtProperty, 1946
setStyle
QCPItemBracket, 1539
QCPItemTracer, 1590
QCPLineEnding, 1652
setSubGridPen
QCPGrid, 1531
setSubGridVisible
QCPGrid, 1531
setSubTickCount
QCPAxisTickerLog, 1399
QCPAxisTickerText, 1407
setSubTickLength
QCPAxis, 1364
QCPPolarAxisAngular, 1687
QCPPolarAxisRadial, 1707
setSubTickLengthIn
QCPAxis, 1364
QCPPolarAxisAngular, 1687
QCPPolarAxisRadial, 1707
setSubTickLengthOut
QCPAxis, 1364
QCPPolarAxisAngular, 1687
QCPPolarAxisRadial, 1707
setSubTickPen
QCPAxis, 1364
QCPPolarAxisAngular, 1688
QCPPolarAxisRadial, 1707
setSubTicks
QCPAxis, 1364
QCPPolarAxisAngular, 1688
QCPPolarAxisRadial, 1707
setSymbolGap
QCPErrorBars, 1501
setTail
QCPItemCurve, 1544
QCPItemLine, 1554
setTangentAverage
QCPSelectionDecoratorBracket, 1746
setTangentToData
QCPSelectionDecoratorBracket, 1746
setTemperature
AIAssistant_if, 141
AIAssistantDefaultImpl, 154

setTerminalResult
WorkerJobManager, 2708
setText
QCPItemText, 1583
QCPTextElement, 1771
setTextAlignment
QCPItemText, 1583
setTextColor
QCPAbstractLegendItem, 1324
QCPLegend, 1650
QCPTextElement, 1771
setTextFlags
QCPTextElement, 1771
setThinkingBudgetTokens
AnthropicProviderClientImpl, 202
setTickCount
QCPAxisTicker, 1388
setTicker
QCPAxis, 1365
QCPPolarAxisAngular, 1688
QCPPolarAxisRadial, 1708
setTickLabelColor
QCPAxis, 1365
QCPPolarAxisAngular, 1688
QCPPolarAxisRadial, 1708
setTickLabelFont
QCPAxis, 1365
QCPPolarAxisAngular, 1688
QCPPolarAxisRadial, 1708
setTickLabelPadding
QCPAxis, 1365
QCPPolarAxisAngular, 1689
QCPPolarAxisRadial, 1708
setTickLabelRotation
QCPAxis, 1365
QCPPolarAxisAngular, 1689
QCPPolarAxisRadial, 1708
setTickLabels
QCPAxis, 1365
QCPPolarAxisAngular, 1689
QCPPolarAxisRadial, 1708
setTickLabelSide
QCPAxis, 1366
setTickLength
QCPAxis, 1366
QCPPolarAxisAngular, 1689
QCPPolarAxisRadial, 1709
setTickLengthIn
QCPAxis, 1366
QCPPolarAxisAngular, 1689
QCPPolarAxisRadial, 1709
setTickLengthOut
QCPAxis, 1366
QCPPolarAxisAngular, 1689
QCPPolarAxisRadial, 1709
setTickOrigin
QCPAxisTicker, 1388
QCPAxisTickerDateTime, 1393, 1394
setTickPen
QCPAxis, 1366
QCPPolarAxisAngular, 1690
QCPPolarAxisRadial, 1709
setTicks
QCPAxis, 1366
QCPAxisTickerText, 1407
QCPPolarAxisAngular, 1690
QCPPolarAxisRadial, 1709
setTickStep
QCPAxisTickerFixed, 1397
setTickStepStrategy
QCPAxisTicker, 1388
setTightBoundary
QCPColorMap, 1441
setTimeBetweenCreationsExpression
SourceModelComponent, 2433
setTimeFormat
QCPAxisTickerTime, 1411
setTimeoutSeconds
AIAssistant_if, 141
AIAssistantDefaultImpl, 154
AIProviderClient_if, 161
HttpProviderClientBase, 882
setTimeUnit
SourceModelComponent, 2433
setTimeZone
QCPAxisTickerDateTime, 1394
setTraceLevel
TraceManager, 2583
setTraceLevelSpecific
ModelDataDefinition, 1059
setTraceLevelSpecificEnabled
ModelDataDefinition, 1059
setTracer
Model, 1031
setTraceSimulationRuleAllAllowed
TraceManager, 2583
setTwoColored
QCPFinancial, 1513
setType
QCPColorScale, 1455
QCPItemPosition, 1565
setTypeX
QCPItemPosition, 1565
setTypeY
QCPItemPosition, 1565
SetUp
ParserExpressionsTest, 1217
setupFullAxesBox
QCPAxisRect, 1383
setUpperEnding
QCPAxis, 1367
setUsedScatterProperties
QCPSelectionDecorator, 1742
setValue
HppLatticeGasState, 875
QtBoolPropertyManager, 1827

QtCharPropertyManager, 1839
QtColorPropertyManager, 1850
QtCursorPropertyManager, 1858
QtDatePropertyManager, 1867
QtDateTimePropertyManager, 1874
QtDoublePropertyManager, 1881
QtEnumPropertyManager, 1893
QtFlagPropertyManager, 1896
QtFontPropertyManager, 1904
QtIntPropertyManager, 1916
QtKeySequencePropertyManager, 1923
QtLocalePropertyManager, 1929
QtPointFPropertyManager, 1936
QtPointPropertyManager, 1940
QtRectFPropertyManager, 1955
QtRectPropertyManager, 1960
QtSizeFPropertyManager, 1969
QtSizePolicyPropertyManager, 1973
QtSizePropertyManager, 1979
QtStringPropertyManager, 1991
QtTimePropertyManager, 1998
QtVariantProperty, 2012
QtVariantPropertyManager, 2017
RealState, 2033
SimulationControlBool, 2191
SimulationControlDouble, 2194
SimulationControlGeneric< T >, 2197
SimulationControlGenericClass< T, M, C >, 2201
SimulationControlGenericClassNotDC< T, M, C >, 2205
SimulationControlGenericEnum< T, E >, 2209
SimulationControlGenericList< T, M, C >, 2212
SimulationControlGenericListPointer< T, M, C >, 2216
SimulationControlInt, 2220
SimulationControlSourceCodeString, 2225
SimulationControlString, 2228
SimulationControlTimeUnit, 2231
SimulationControlUInt, 2234
SimulationControlUShort, 2237
TestSimulationControl, 2571
setValueAxis
QCPAbstractPlottable, 1339
QCPPolarGraph, 1718
setValueRange
QCPColorMapData, 1447
setValueSize
QCPColorMapData, 1448
setValueToolTip
QtProperty, 1946
setVariableType
Record, 2040
setVersion
ModelInfo, 1082
setViewport
QCustomPlot, 1805
setVisible
QCPLayer, 1597
QCPLayerable, 1603
SetW
Util, 2631
setWhatsThis
QtProperty, 1947
setWhiskerAntialiased
QCPStatisticalBox, 1760
setWhiskerBarPen
QCPStatisticalBox, 1760
setWhiskerPen
QCPStatisticalBox, 1760
setWhiskerWidth
QCPErrorBars, 1501
QCPStatisticalBox, 1761
setWidth
QCPBars, 1421
QCPFinancial, 1513
QCPLineEnding, 1652
QCPStatisticalBox, 1761
setWidthType
QCPBars, 1421
QCPFinancial, 1513
setWrap
QCPLayoutGrid, 1629
setX
QCPVector2D, 1776
setY
QCPVector2D, 1776
setZeroLinePen
QCPGrid, 1531
show
Access, 123
AIAssistant, 131
AISupport, 172
Assign, 211
Attribute, 218
BacteriaColony, 227
BacteriaSignalGrid, 233
Batch, 248
BioCompartment, 254
BioNetwork, 260
BioParameter, 269
BioReaction, 281
BioRunnerCommand, 291
BioSimulate, 299
BioSimulatorRunner, 307
BioSpecies, 312
BioStateProjectionComponent, 320
BioSteadyState, 326
Buffer, 339
CellCycleCheckpointComponent, 359
CellDivisionEvent, 366
CellFateDecisionComponent, 373
CellGrowthComponent, 379
CellularAutomataComp, 392
Clone, 406
ColoredPetriNetNetwork, 423
CompartmentExchangeComponent, 432

ContinuousSystemComponent, 453
Conveyor, 459
Counter, 466
CPNArc, 475
CPNTransition, 481
CppCompiler, 487
CppForG, 493
Create, 504
Decide, 520
DefaultNetwork, 529
DefaultNode, 534
Delay, 543
DiffEquations, 567
DiffusionField, 573
DiffusionSimulate, 581
Dispose, 610
Distance, 615
DropOff, 625
DummyComponent, 631
DummyElement, 635
EFSMNetwork, 644
Enter, 662
Entity, 669
EntityGroup, 674
EntityType, 679
EukaryoticCellCycleComponent, 687
Exit, 702
Failure, 716
File, 726
Formula, 746
FtsZPolymerizationComponent, 758
GeneticCircuit, 780
GeneticCircuitPart, 786
GeneticCircuitSimulate, 792
GeneticExpressionStep, 799
GeneticRegulation, 804
GraphEdge, 813
GraphNetwork, 843
GroProgram, 856
InternalEvent, 896
Label, 912
Leave, 922
LSODE, 953
MarkovChain, 965
MarkovChainNetwork, 972
Match, 984
MetabolicFluxBalance, 991
MetabolicNetwork, 996
MetabolicReaction, 1001
MetabolicStateProjectionComponent, 1009
MetabolicSubmodelComponent, 1015
ModalModelDefault, 1023
ModelComponent, 1043
ModelDataDefinition, 1059
MolecularSpecies, 1100
Move, 1106
Neighborhood_Moore, 1121
Neighborhood_VonNeumann, 1123
OctaveRunner, 1154
ODESolver, 1160
OLD_ODEelement, 1166
PathwayStressResponseComponent, 1225
PickStation, 1247
Plugin, 1257
Process, 1289
PythonForG, 1305
PythonRuntime, 1310
Queue, 2027
Record, 2040
Release, 2060
Remove, 2067
Resource, 2079
ResourceAllocationComponent, 2085
Route, 2093
RSimulator, 2100
RSimulatorRunner, 2106
Schedule, 2126
Search, 2133
Segment, 2138
Seize, 2146
Separate, 2152
Sequence, 2157
Set, 2170
Signal, 2176
SignalData, 2181
SourceModelComponent, 2433
SPICECircuit, 2441
SPICENode, 2446
SPICERunner, 2452
Start, 2458
Station, 2469
StatisticsCollector, 2483
StochasticReactionComponent, 2501
StochasticReactionRule, 2506
StochasticTranscription, 2512
StochasticTranslation, 2520
Stop, 2527
Storage, 2532
Store, 2538
Submodel, 2545
Transporter, 2598
Unstore, 2619
Variable, 2644
Wait, 2669
WaitingResource, 2671
WholeCellState, 2702
Write, 2718
showError
genesys::launcher::UpdateDialog, 2622
showImpacts
FactorialDesign, 710
showInfo
genesys::launcher::UpdateDialog, 2622
showLanguage
Model, 1031
showLinearRegressionModel

FactorialDesign, 710
showReplicationStatistics
SimulationReporter_if, 2240
SimulationReporterDefaultImpl1, 2241
showResiduals
FactorialDesign, 710
showSignTable
FactorialDesign, 710
showSimulationControls
SimulationReporter_if, 2240
SimulationReporterDefaultImpl1, 2242
showSimulationResponses
SimulationReporter_if, 2240
SimulationReporterDefaultImpl1, 2242
showSimulationStatistics
SimulationReporter_if, 2240
SimulationReporterDefaultImpl1, 2242
Signal, 2171
_check, 2175
_createEditableDataDefinitions, 2175
_initBetweenReplications, 2176
_loadInstance, 2176
_onDispatchEvent, 2176
_saveInstance, 2176
show, 2176
SignalData, 2177
_check, 2181
_initBetweenReplications, 2181
_loadInstance, 2181
_saveInstance, 2181
show, 2181
SignDomain
QCP, 115
SimpleHttpServer, 2182
serve, 2182
SimpleHttpServer, 2182
simplify
QCPDataSelection, 1489
QCPLayout, 1609
QCPLayoutGrid, 1629
QCPLayoutInset, 1637
Simulate
ContinuousModel, 447
SimulationCommandController, 2183
SimulationControl, 2185
getChildSimulationControls, 2188
SimulationControlBool, 2189
getValue, 2191
setValue, 2191
SimulationControlDouble, 2192
getValue, 2194
setValue, 2194
SimulationControlGeneric< T >, 2195
getValue, 2197
setValue, 2197
SimulationControlGenericClass< T, M, C >, 2198
createObjectInstance, 2200
ensureObjectInstance, 2200
getChildSimulationControls, 2200
getReferencedModelDataDefinition, 2201
getStrValues, 2201
getValue, 2201
hasObjectInstance, 2201
isInlineObjectProperty, 2201
isModelDataDefinitionReference, 2201
setValue, 2201
supportsExistingObjectSelection, 2201
supportsInlineExpansion, 2202
supportsObjectCreation, 2202
SimulationControlGenericClassNotDC< T, M, C >, 2202
createObjectInstance, 2205
ensureObjectInstance, 2205
getChildSimulationControls, 2205
getValue, 2205
hasObjectInstance, 2205
isInlineObjectProperty, 2205
setValue, 2205
supportsInlineExpansion, 2206
supportsObjectCreation, 2206
SimulationControlGenericEnum< T, E >, 2206
getStrValues, 2209
getValue, 2209
setValue, 2209
SimulationControlGenericList< T, M, C >, 2209
getStrValues, 2212
getValue, 2212
setValue, 2212
supportsListEditor, 2212
SimulationControlGenericListPointer< T, M, C >, 2213
createNewListElement, 2215
createNewListElementOfType, 2215
getChildSimulationControls, 2215
getCreatableListElementTypes, 2216
getCurrentListElementType, 2216
getListElementModelDataDefinition, 2216
getStrValues, 2216
getValue, 2216
setCurrentListElementType, 2216
setValue, 2216
supportsListEditor, 2217
supportsNewListElementCreation, 2217
SimulationControlInt, 2217
getValue, 2220
setValue, 2220
SimulationController, 2220
currentSimulation, 2221
ensureReady, 2221
hasCurrentModelSimulation, 2221
SimulationController, 2221
SimulationControlSourceCodeString, 2222
getValue, 2225
setValue, 2225
validateProposedValue, 2225
SimulationControlString, 2225
getValue, 2228

setValue, 2228
validateProposedValue, 2228
SimulationControlTimeUnit, 2228
getValue, 2231
setValue, 2231
SimulationControlUInt, 2231
getValue, 2234
setValue, 2234
SimulationControlUShort, 2234
getValue, 2237
setValue, 2237
SimulationEvent, 2237
SimulationEventController, 2238
SimulationEventController::Callbacks, 344
SimulationExperiment, 2239
SimulationReporter_if, 2239
showReplicationStatistics, 2240
showSimulationControls, 2240
showSimulationResponses, 2240
showSimulationStatistics, 2240
SimulationReporterDefaultImpl1, 2240
showReplicationStatistics, 2241
showSimulationControls, 2242
showSimulationResponses, 2242
showSimulationStatistics, 2242
SimulationResponse, 2242
SimulationResponseDouble, 2244
getValue, 2246
SimulationResultsDataset, 2246
formatKindName, 2246
isRecordFormat, 2246
replications, 2246
values, 2246
valuesForReplication, 2247
SimulationResultsDatasetParser, 2247
loadFromTextFile, 2247
SimulationResultsObservation, 2247
SimulationScenario, 2247
getResponseValues, 2248
Simulator, 2249
getExperimentManager, 2250
getLicenceManager, 2250
getModelManager, 2250
getName, 2250
getParserManager, 2250
getPluginManager, 2250
getTraceManager, 2250
getVersion, 2251
getVersionNumber, 2251
Simulator, 2250
SimulatorFacade, 2251
SimulatorSessionService, 2257
configureSimulation, 2260
createSession, 2260
createWorkerJob, 2260
getSimulationStatus, 2260
getWorkerCapabilities, 2261
getWorkerInfo, 2261
getWorkerJob, 2261
getWorkerJobResult, 2261
importModelFromLanguage, 2262
loadModel, 2262
runSimulation, 2262
runWorkerJob, 2262
saveCurrentModel, 2263
SimulatorSessionService, 2259
stepSimulation, 2263
tryCreateModel, 2263
tryGetCurrentModelInfo, 2264
tryGetSimulatorInfo, 2264
SimulatorSessionService::CreateSessionResult, 504
SimulatorSessionService::ModelImportResult, 1079
SimulatorSessionService::ModelInfoResult, 1083
SimulatorSessionService::ModelPersistenceResult, 1092
SimulatorSessionService::SimulationActionResult, 2183
SimulatorSessionService::SimulationConfigInput, 2184
SimulatorSessionService::SimulationConfigResult, 2185
SimulatorSessionService::SimulationStatusResult, 2248
SimulatorSessionService::SimulatorInfoResult, 2257
SimulatorSessionService::WorkerCapabilitiesResult, 2704
SimulatorSessionService::WorkerInfoResult, 2704
SimulatorSessionService::WorkerJobCreationResult, 2706
SimulatorSessionService::WorkerJobInfoResult, 2706
SimulatorSessionService::WorkerJobQueryResult, 2709
SimulatorSessionService::WorkerJobResultInfo, 2710
SimulatorSessionService::WorkerJobResultQueryResult, 2710
SimulatorSessionService::WorkerJobRunResult, 2711
singleStep
QtDoublePropertyManager, 1881
QtIntPropertyManager, 1916
singleStepChanged
QtDoublePropertyManager, 1881
QtIntPropertyManager, 1916
SinkModelComponent, 2264
_check, 2269
_loadInstance, 2269
_saveInstance, 2269
size
ConnectionManager, 442
Formula, 747
ModelManager, 1091
QCPAxisRect, 1383
QCPBarsGroup, 1427
QCPDataContainer< DataType >, 1481
QCPDataRange, 1485
QCPPolarAxisAngular, 1690
QCPRange, 1730
SizeConstraintRect
QCPLayoutElement, 1614

sizeConstraintsChanged
QCPLayout, 1610
Smart_AddingResource, 2271
main, 2272
Smart_AlternatingEntityCreation, 2272
main, 2274
Smart_ArrivalsElementStopsEntitiesArrivingAfterASetTime, 2274
main, 2276
Smart_ArrivalsEntityTypeVsAttribute, 2276
main, 2278
Smart_AssignExample, 2278
main, 2279
Smart_AssignWriteSeizes, 2280
main, 2281
Smart_Attribute_Variable, 2281
main, 2283
Smart_AutomaticStatisticsCollection, 2283
main, 2285
Smart_BacteriaColony_GRO, 2285
main, 2286
Smart_BasicModeling, 2287
main, 2288
Smart_BatchAndSeparate, 2288
main, 2290
Smart_BatchSeparate, 2290
main, 2291
Smart_BioKineticLawRegulation, 2292
main, 2293
Smart_BioNetworkAnalysis, 2293
main, 2295
Smart_BioReversibleKineticLaw, 2295
main, 2296
Smart_BioReversibleMassAction, 2297
main, 2298
Smart_Buffer, 2298
main, 2300
Smart_CellularAutomata, 2300
main, 2301
Smart_CellularAutomata1, 2302
main, 2303
Smart_Clone, 2303
main, 2305
Smart_ContinuousFlowEntities, 2305
main, 2306
Smart_CppForG, 2307
main, 2308
Smart_Create, 2308
main, 2310
Smart_DecideNWayByChance, 2310
main, 2311
Smart_DefaultModalModel, 2312
main, 2313
Smart_DefiningAttributesAsStrings, 2313
main, 2315
Smart_DefiningControlLogic, 2315
main, 2316
Smart_DefiningResourceCapacity, 2317
main, 2318
Smart_Delay, 2318
main, 2320
Smart_DelayBasedOnReplication, 2320
main, 2322
TestEventHandler, 2322
Smart_Dummy, 2322
main, 2324
Smart_EntitiesProcessedByPriority, 2324
main, 2326
Smart_EvaluatingConditionsBeforeEnteringQueue, 2326
main, 2328
Smart_Expression, 2328
main, 2329
Smart_Failures, 2330
main, 2331
Smart_FSMState, 2331
main, 2333
Smart_GroColonyGrowth, 2333
main, 2334
Smart_GroColonyLifecycle, 2335
main, 2336
Smart_HoldSearchRemove, 2336
main, 2338
Smart_InventoryAndHoldingCosts, 2338
main, 2340
Smart_LSODE, 2340
main, 2341
Smart_MarkovChain, 2342
main, 2343
Smart_MaxArrivalsField, 2343
main, 2345
Smart_ModalModel, 2345
main, 2346
Smart_ModalModelDefaultNetwork, 2347
main, 2348
Smart_ModalModelFSM, 2348
main, 2350
Smart_ModalModelPetriNet, 2350
main, 2351
Smart_ModelInfoModelSimulation, 2352
main, 2353
Smart_ModelRunUntil1000Parts, 2353
main, 2355
Smart_ModuleDisplayVariables, 2355
main, 2356
Smart_ODE, 2357
main, 2358
Smart_OnEvent, 2358
main, 2360
onBreakpointHandler, 2360
onEntityCreateHandler, 2360
onEntityMoveHandler, 2360
onEntityRemoveHandler, 2360
onProcessEventHandler, 2360
onReplicationEndHandler, 2360
onReplicationStartHandler, 2360

onReplicationStepHandler, 2360
onSimulationEndHandler, 2360
onSimulationPausedHandler, 2361
onSimulationResumeHandler, 2361
onSimulationStartHandler, 2361
Smart_OverlappingResources, 2361
main, 2363
Smart_ParallelProcessingOfEntities, 2363
main, 2365
Smart_Parser, 2365
main, 2366
Smart_ParserModelFunctions, 2367
main, 2368
Smart_PetriPlace, 2368
main, 2370
Smart_PlacingEntitiesInQueueSets, 2370
main, 2371
Smart_Plugin, 2372
main, 2373
Smart_PriorityExample, 2373
main, 2375
Smart_Process, 2375
main, 2376
Smart_ProcessArena, 2377
main, 2378
Smart_ProcessSet, 2378
main, 2380
Smart_PythonForG, 2380
main, 2381
Smart_R_Simulator, 2381
main, 2383
Smart_Record, 2384
main, 2385
Smart_Record_Arena, 2385
main, 2387
Smart_RemovingAndReorderingEntitiesInAQueue, 2387
main, 2389
Smart_ResourceCosting, 2389
main, 2390
Smart_ResourceScheduleCosting, 2391
main, 2392
Smart_ResourceSets, 2392
main, 2394
Smart_RouteStation, 2394
main, 2395
Smart_SeizeDelayRelease, 2396
main, 2397
Smart_SeizeDelayReleaseMany, 2397
main, 2399
Smart_SeizeDelayReleaseNoDataDefs, 2399
main, 2401
Smart_SeizingMultipleSimultaneosly, 2401
main, 2402
Smart_SelectingRouteBasedOnProbability, 2403
main, 2404
Smart_SelectingShorterQueue, 2404
main, 2406
Smart_Sequence, 2406
main, 2407
Smart_SimulationControlResponse, 2408
main, 2409
Smart_SynchronizingParallelEntities, 2409
main, 2411
Smart_ValueAdded, 2411
main, 2412
Smart_WaitForSignal, 2413
main, 2414
Smart_WaitScanCondition, 2414
main, 2416
Smart_WaitSignal, 2416
main, 2417
Solver_if, 2419
SolverDefaultImpl1, 2420
derivate, 2421
getMaxSteps, 2422
getPrecision, 2422
integrate, 2422
setMaxSteps, 2422
setPrecision, 2423
sort
QCPDataContainer< DataType >, 1481
SortFile, 2423
SortFile, 2423
setDataFilename, 2423
sort, 2423
sortKey
QCPBarsData, 1422
QCPCurveData, 1474
QCPFinancialData, 1515
QCPGraphData, 1527
QCPStatisticalBoxData, 1763
sortKeyIsMainKey
QCPAbstractPlottable1D< DataType >, 1346
QCPBarsData, 1423
QCPCurveData, 1474
QCPErrorBars, 1501
QCPFinancialData, 1515
QCPGraphData, 1527
QCPPlottableInterface1D, 1665
QCPStatisticalBoxData, 1763
SourceCodeString, 2424
SourceModelComponent, 2424
_check, 2430
_createAttachedAttributes, 2430
_createEditableDataDefinitions, 2430
_initBetweenReplications, 2430
_loadInstance, 2430
_saveInstance, 2430
getEntitiesCreated, 2431
getEntitiesPerCreation, 2431
getEntityType, 2431
getFirstCreation, 2431
getMaxCreations, 2431
getTimeBetweenCreationsExpression, 2431
getTimeUnit, 2431

setEntitiesCreated, 2432
setEntitiesPerCreation, 2432
setEntityType, 2432
setEntityTypeName, 2432
setFirstCreation, 2432
setMaxCreations, 2433
setTimeBetweenCreationsExpression, 2433
setTimeUnit, 2433
show, 2433
SourceModelComponent::DEFAULT_VALUES, 521
SpacingType
QCPBarsGroup, 1425
spAll
QCPScatterStyle, 1734
span
QCPDataSelection, 1489
SparseValueStore, 2434
spAxis
QCPAxis, 1353
QCPPolarAxisAngular, 1677
QCPPolarAxisRadial, 1698
spAxisLabel
QCPAxis, 1353
QCPPolarAxisAngular, 1677
QCPPolarAxisRadial, 1698
spBrush
QCPScatterStyle, 1734
SPICECircuit, 2435
_createNonEditableDataDefinitions, 2440
_loadInstance, 2440
_onDispatchEvent, 2440
_saveInstance, 2440
show, 2441
SPICENode, 2441
_loadInstance, 2446
_onDispatchEvent, 2446
_saveInstance, 2446
show, 2446
SPICERunner, 2447
_check, 2451
_loadInstance, 2451
_saveInstance, 2451
show, 2452
spItems
QCPLegend, 1644
spLegendBox
QCPLegend, 1644
spNone
QCPAxis, 1353
QCPLegend, 1644
QCPPolarAxisAngular, 1677
QCPPolarAxisRadial, 1698
QCPScatterStyle, 1734
spPen
QCPScatterStyle, 1734
spShape
QCPScatterStyle, 1734
spSize
QCPScatterStyle, 1734
spTickLabels
QCPAxis, 1353
QCPPolarAxisAngular, 1677
QCPPolarAxisRadial, 1698
squeeze
QCPDataContainer< DataType >, 1481
srmCustom
QCP, 115
srmNone
QCP, 115
srmSelect
QCP, 115
srmZoom
QCP, 115
ssCircle
QCPScatterStyle, 1735
ssCross
QCPScatterStyle, 1735
ssCrossCircle
QCPScatterStyle, 1735
ssCrossSquare
QCPScatterStyle, 1735
ssCustom
QCPScatterStyle, 1735
ssDiamond
QCPScatterStyle, 1735
ssDisc
QCPScatterStyle, 1735
ssDot
QCPScatterStyle, 1735
ssMultiples
QCPAxisTickerFixed, 1396
ssNone
QCPAxisTickerFixed, 1396
QCPScatterStyle, 1735
ssPeace
QCPScatterStyle, 1735
ssPixmap
QCPScatterStyle, 1735
ssPlus
QCPScatterStyle, 1735
ssPlusCircle
QCPScatterStyle, 1735
ssPlusSquare
QCPScatterStyle, 1735
ssPowers
QCPAxisTickerFixed, 1396
ssSquare
QCPScatterStyle, 1735
ssStar
QCPScatterStyle, 1735
ssTriangle
QCPScatterStyle, 1735
ssTriangleInverted
QCPScatterStyle, 1735
stAbsolute
QCPBarsGroup, 1425

Start, 2452
_check, 2457
_loadInstance, 2457
_onDispatchEvent, 2457
_saveInstance, 2458
show, 2458
start
ModelSimulation, 1095
Optimizer_if, 1180
OptimizerDefaultImpl1, 1187
WebWorkerRuntime, 2695
StartAIAssistant, 2458
main, 2460
started
QCPSelectionRect, 1751
startExperiment
ExperimentManager_if, 704
ExperimentManagerDefaultImpl1, 705
startPainting
QCPAbstractPaintBuffer, 1327
QCPPaintBufferPixmap, 1657
startSimulationOfScenario
ExperimentManager_if, 704
ExperimentManagerDefaultImpl1, 705
State, 2460
StateSet, 2461
StateSet_Enumerable, 2463
contains, 2464
createDefaultState, 2464
enumerateStates, 2464
isDiscrete, 2464
isFinite, 2464
name, 2464
parseState, 2464
Station, 2465
_check, 2469
_createAttachedAttributes, 2469
_createInternalStatisticReporters, 2469
_initBetweenReplications, 2469
_loadInstance, 2469
_saveInstance, 2469
show, 2469
StationTestProbe, 2470
Statistics_if, 2474
average, 2476
confidenceLevel, 2476
getCollector, 2476
halfWidthConfidenceInterval, 2476
max, 2476
min, 2476
newSampleSize, 2476
numElements, 2477
setCollector, 2477
setConfidenceLevel, 2477
stddeviation, 2477
variance, 2477
variationCoef, 2478
StatisticsCollector, 2478
_check, 2482
_loadInstance, 2482
_saveInstance, 2482
getParent, 2483
getStatistics, 2483
show, 2483
StatisticsDatafile_if, 2483
centil, 2485
decil, 2485
histogramClassFrequency, 2485
histogramClassLowerLimit, 2486
histogramNumClasses, 2486
mediane, 2486
mode, 2486
quartil, 2486
setHistogramNumClasses, 2487
StatisticsDatafileDefaultImpl1, 2487
average, 2489
centil, 2489
confidenceLevel, 2489
decil, 2489
getCollector, 2489
halfWidthConfidenceInterval, 2490
histogramClassFrequency, 2490
histogramClassLowerLimit, 2490
histogramNumClasses, 2490
max, 2490
mediane, 2490
min, 2490
mode, 2490
newSampleSize, 2490
numElements, 2491
quartil, 2491
setCollector, 2491
setConfidenceLevel, 2491
setHistogramNumClasses, 2491
stddeviation, 2491
variance, 2491
variationCoef, 2491
StatisticsDefaultImpl1, 2492
average, 2493
confidenceLevel, 2493
getCollector, 2493
halfWidthConfidenceInterval, 2493
max, 2493
min, 2494
newSampleSize, 2494
numElements, 2494
setCollector, 2494
setConfidenceLevel, 2494
StatisticsDefaultImpl1, 2493
stddeviation, 2494
variance, 2494
variationCoef, 2494
statusTip
QtProperty, 1947
stAxisRectRatio
QCPBarsGroup, 1425

std::tuple_size< testing::internal::FlatTuple< Ts... > >, 2600
stDataRange
QCP, 115
stddeviation
Statistics_if, 2477
StatisticsDatafileDefaultImpl1, 2491
StatisticsDefaultImpl1, 2494
step
Optimizer_if, 1181
OptimizerDefaultImpl1, 1187
stepSimulation
SimulatorSessionService, 2263
stLinear
QCPAxis, 1353
QCPPolarAxisRadial, 1698
stLogarithmic
QCPAxis, 1353
QCPPolarAxisRadial, 1698
stMultipleDataRanges
QCP, 115
stNone
QCP, 115
StochasticReactionComponent, 2495
_check, 2500
_createEditableDataDefinitions, 2500
_initBetweenReplications, 2500
_loadInstance, 2500
_onDispatchEvent, 2500
_saveInstance, 2500
show, 2501
StochasticReactionRule, 2501
_check, 2505
_loadInstance, 2505
_saveInstance, 2505
show, 2506
StochasticReactionRule::StoichiometricTerm, 2521
StochasticTranscription, 2506
_check, 2511
_createEditableDataDefinitions, 2511
_initBetweenReplications, 2512
_loadInstance, 2512
_onDispatchEvent, 2512
_saveInstance, 2512
show, 2512
StochasticTranslation, 2513
_check, 2518
_createEditableDataDefinitions, 2518
_initBetweenReplications, 2518
_loadInstance, 2518
_onDispatchEvent, 2518
_saveInstance, 2518
show, 2520
Stop, 2521
_check, 2526
_loadInstance, 2526
_onDispatchEvent, 2526
_saveInstance, 2526
show, 2527
stop
Optimizer_if, 1181
OptimizerDefaultImpl1, 1187
stopExperiment
ExperimentManager_if, 704
ExperimentManagerDefaultImpl1, 706
Storage, 2527
_check, 2531
_getParserChangesInformation, 2531
_initBetweenReplications, 2531
_loadInstance, 2531
_saveInstance, 2532
show, 2532
Store, 2532
_check, 2537
_createEditableDataDefinitions, 2537
_loadInstance, 2538
_onDispatchEvent, 2538
_saveInstance, 2538
show, 2538
stPlotCoords
QCPBarsGroup, 1425
StrAllocation
Util, 2631
StrIndex
Util, 2632
StrReplace
Util, 2632
StrReplaceSpecialChars
Util, 2632
StrTimeUnitLong
Util, 2632
StrTimeUnitShort
Util, 2633
StrTruncIfInt
Util, 2633
stSingleData
QCP, 115
stWhole
QCP, 115
subBoolPropertyManager
QtFlagPropertyManager, 1897
QtFontPropertyManager, 1905
subDoublePropertyManager
QtPointFPropertyManager, 1936
QtRectFPropertyManager, 1955
QtSizeFPropertyManager, 1970
subEnumPropertyManager
QtFontPropertyManager, 1905
QtLocalePropertyManager, 1930
QtSizePolicyPropertyManager, 1973
subIntPropertyManager
QtColorPropertyManager, 1850
QtFontPropertyManager, 1905
QtPointPropertyManager, 1940
QtRectPropertyManager, 1960
QtSizePolicyPropertyManager, 1974

QtSizePropertyManager, 1979
Submodel, 2539
_check, 2544
_loadInstance, 2544
_onDispatchEvent, 2544
_saveInstance, 2545
show, 2545
subProperties
QtProperty, 1947
supportsExistingObjectSelection
SimulationControlGenericClass< T, M, C >, 2201
supportsInlineExpansion
SimulationControlGenericClass< T, M, C >, 2202
SimulationControlGenericClassNotDC< T, M, C >, 2206
supportsListEditor
SimulationControlGenericList< T, M, C >, 2212
SimulationControlGenericListPointer< T, M, C >, 2217
supportsNewListElementCreation
SimulationControlGenericListPointer< T, M, C >, 2217
supportsObjectCreation
SimulationControlGenericClass< T, M, C >, 2202
SimulationControlGenericClassNotDC< T, M, C >, 2206
swap
yy::genesyspp_parser::value_type, 2636
symbol_kind_type
yy::genesyspp_parser::symbol_kind, 2547
symbol_name
yy::genesyspp_parser, 770
synchronizeGraphicalDataDefinitionsLayer
GraphicalModelBuilder, 825
SystemCommandExecutor_if, 2551
run, 2552
SystemCommandResult, 2552
SystemDependency, 2553
SystemDependencyCheckEntry, 2553
SystemDependencyCheckResult, 2554
SystemDependencyInstallEntry, 2554
SystemDependencyInstallResult, 2555
SystemDependencyResolver, 2555
SystemPreferences, 2556
load, 2558
pushRecentModelFile, 2558
save, 2558
SystemShellCommandExecutor, 2558
run, 2559

take
QCPLayout, 1610
QCPLayoutGrid, 1629
QCPLayoutInset, 1637
takeAt
QCPLayout, 1610
QCPLayoutGrid, 1630
QCPLayoutInset, 1637
testAverage
HypothesisTesterDefaultImpl1, 889
TestEventHandler
Smart_DelayBasedOnReplication, 2322
testing::EmptyTestEventListener, 653
OnEnvironmentsSetUpEnd, 654
OnEnvironmentsSetUpStart, 654
OnEnvironmentsTearDownEnd, 654
OnEnvironmentsTearDownStart, 654
OnTestCaseEnd, 654
OnTestCaseStart, 654
OnTestDisabled, 654
OnTestEnd, 654
OnTestIterationEnd, 654
OnTestIterationStart, 654
OnTestPartResult, 655
OnTestProgramEnd, 655
OnTestProgramStart, 655
OnTestStart, 655
OnTestSuiteEnd, 655
OnTestSuiteStart, 655
testing::Environment, 680
testing::internal::AssertHelper, 205
testing::internal::CartesianProductGenerator< T >, 350
Begin, 351
End, 351
testing::internal::CartesianProductHolder< Gen >, 351
testing::internal::CodeLocation, 407
testing::internal::ConstCharPtr, 443
testing::internal::ConstRef< T >, 443
testing::internal::ConstRef< T & >, 443
testing::internal::ContainerPrinter, 444
testing::internal::ConvertibleToIntegerPrinter, 454
testing::internal::ConvertibleToStringViewPrinter, 454
testing::internal::DefaultGlobalTestPartResultReporter, 521
testing::internal::DefaultNameGenerator, 523
testing::internal::DefaultPerThreadTestPartResultReporter, 536
testing::internal::DoubleSequence< false, IndexSequence< I... >, sizeofT >, 618
testing::internal::DoubleSequence< plus_one, T, sizeofT >, 618
testing::internal::DoubleSequence< true, IndexSequence< I... >, sizeofT >, 619
testing::internal::ElemFromList< N, T >, 652
testing::internal::ElemFromListImpl< IndexSequence< I... > >, 652
testing::internal::ElemFromListImpl< typename >, 652
testing::internal::FallbackPrinter, 721
testing::internal::FindFirstPrinter< T, decltype(Printer::PrintValue(std::declv
const T & >(), nullptr)), Printer, Printers... >, 727
testing::internal::FindFirstPrinter< T, E, Printer, Printers >, 727
testing::internal::FlatTuple< T >, 735
testing::internal::FlatTupleBase< Derived, Idx >, 736
testing::internal::FlatTupleBase< FlatTuple< T... >, IndexSequence< Idx... > >, 736

testing::internal::FlatTupleConstructTag, 737
testing::internal::FlatTupleElemBase< Derived, I >, 737
testing::internal::FlatTupleElemBase< FlatTuple< T... >, I >, 738
testing::internal::FloatingPoint< RawType >, 738
kFractionBitCount, 739
kFractionBitMask, 739
testing::internal::FormatForComparison< ToPrint, OtherOperand >, 739
testing::internal::FormatForComparison< ToPrint[N], OtherOperand >, 739
testing::internal::FunctionPointerPrinter, 760
testing::internal::GenerateTypeList< T >, 761
testing::internal::GTestFlagSaver, 860
testing::internal::GTestLog, 860
testing::internal::GTestMutexLock, 861
testing::internal::HasDebugStringAndShortDebugString< T >, 873
value, 873
testing::internal::Ignore< size_t >, 892
testing::internal::IgnoredValue, 892
testing::internal::IndexSequence< Is >, 892
testing::internal::internal_stream_operator_without_lexical_name_lookup::LoadTemplateSel, 947
testing::internal::internal_stream_operator_without_lexical_name_lookup::TestFactoryBase, 2539
testing::internal::is_proxy_type_list< ProxyTypeList< Ts... > >, 898
testing::internal::is_proxy_type_list< typename >, 897
testing::internal::IsHashTable< T >, 899
testing::internal::IsRecursiveContainer< C >, 900
testing::internal::IsRecursiveContainerImpl< C, bool >, 900
testing::internal::IsRecursiveContainerImpl< C, false >, 901
testing::internal::IsRecursiveContainerImpl< C, true >, 901
testing::internal::MakeIndexSequenceImpl< 0 >, 957
testing::internal::MakeIndexSequenceImpl< N >, 956
testing::internal::MarkAsIgnored, 959
testing::internal::Mutex, 1108
testing::internal::NameGeneratorSelector< Provided >, 1108
testing::internal::NativeArray< Element >, 1115
testing::internal::None, 1132
testing::internal::OsStackTraceGetter, 1195
CurrentStackTrace, 1195
UponLeavingGTest, 1195
testing::internal::OsStackTraceGetterInterface, 1196
kElidedFramesMarker, 1196
testing::internal::ParameterizedTestFactory< TestClass >, 1197
CreateTest, 1197
testing::internal::ParameterizedTestSuiteInfo< TestSuite >, 1198
GetTestSuiteName, 1199
GetTestSuiteTypeId, 1199
RegisterTests, 1199
testing::internal::ParameterizedTestSuiteInfoBase, 1199
testing::internal::ParameterizedTestSuiteRegistry, 1200
testing::internal::ParamGenerator< T >, 1200
testing::internal::ParamGeneratorInterface< T >, 1201
testing::internal::ParamIterator< T >, 1201
testing::internal::ParamIteratorInterface< T >, 1202
testing::internal::PointerPrinter, 1276
testing::internal::ProtobufPrinter, 1298
testing::internal::ProxyTypeList< Ts >, 1299
testing::internal::Random, 2028
testing::internal::RangeGenerator< T, IncrementT >, 2028
Begin, 2029
End, 2029
testing::internal::RawBytesPrinter, 2029
testing::internal::RE, 2030
testing::internal::RelationToSourceCopy, 2054
testing::internal::RelationToSourceReference, 2054
testing::internal::String, 2539
testing::internal::SuiteApiResolver< T >, 2546
testing::internal::Templates< Head_ >, 2561
testing::internal::Templates< Head_, Tail_ >, 2561
testing::internal::TemplateSel< Tmpl >, 2561
testing::internal::TemplateSel< Tmpl >::Bind< T >, 249
testing::internal::TestFactoryBase, 2563
testing::internal::TestFactoryImpl< TestClass >, 2564
CreateTest, 2564
testing::internal::TestMetaFactory< TestSuite >, 2565
CreateTestFactory, 2566
testing::internal::TestMetaFactoryBase< ParamType >, 2566
testing::internal::TestPropertyKeyIs, 2567
testing::internal::TestResultAccessor, 2568
testing::internal::ThreadLocal< T >, 2575
testing::internal::TraceInfo, 2579
testing::internal::TrueWithString, 2599
testing::internal::TypeIdHelper< T >, 2600
testing::internal::TypeParameterizedTest< Fixture, TestSel, internal::None >, 2601
testing::internal::TypeParameterizedTest< Fixture, TestSel, Types >, 2601
testing::internal::TypeParameterizedTestSuite< Fixture, internal::None, Types >, 2601
testing::internal::TypeParameterizedTestSuite< Fixture, Tests, Types >, 2601
testing::internal::TypeParameterizedTestSuiteRegistry, 2601
testing::internal::Types< Head_ >, 2602
testing::internal::Types< Head_, Tail_ >, 2602
testing::internal::TypeWithSize< 4 >, 2602
testing::internal::TypeWithSize< 8 >, 2602
testing::internal::TypeWithSize< size >, 2602
testing::internal::UnitTestImpl, 2604
testing::internal::UnitTestOptions, 2606
testing::internal::UniversalPrinter< const T >, 2607
testing::internal::UniversalPrinter< T >, 2606
testing::internal::UniversalPrinter< T & >, 2608
testing::internal::UniversalPrinter< T[N]>, 2608

testing::internal::UniversalTersePrinter< char * >, 2608
testing::internal::UniversalTersePrinter< char16_t * >, 2609
testing::internal::UniversalTersePrinter< char32_t * >, 2610
testing::internal::UniversalTersePrinter< const char * >, 2611
testing::internal::UniversalTersePrinter< const char16_t * >, 2612
testing::internal::UniversalTersePrinter< const char32_t * >, 2612
testing::internal::UniversalTersePrinter< T >, 2608
testing::internal::UniversalTersePrinter< T & >, 2613
testing::internal::UniversalTersePrinter< T[N] >, 2613
testing::internal::UniversalTersePrinter< wchar_t * >, 2613
testing::internal::ValueArray< Ts >, 2636
testing::internal::ValuesInIteratorRangeGenerator< T >, 2637
Begin, 2637
End, 2638
testing::Message, 985
testing::PrintToStringParamName, 1277
testing::ScopedTrace, 2127
testing::Test, 2561
testing::TestEventListener, 2562
testing::TestEventListeners, 2563
testing::TestInfo, 2565
testing::TestParamInfo< ParamType >, 2566
testing::TestProperty, 2567
testing::TestResult, 2568
testing::TestSuite, 2573
testing::TestWithParam< T >, 2574
testing::UnitTest, 2603
testing::WithParamInterface< T >, 2703
testProportion
HypothesisTesterDefaultImpl1, 890
TestSimulationControl, 2569
getValue, 2571
setValue, 2571
TestSimulationResponse, 2572
getValue, 2573
testVariance
HypothesisTesterDefaultImpl1, 890, 891
ticker
QCPAxis, 1367
QCPPolarAxisRadial, 1709
ticks
QCPAxisTickerText, 1407
TickStepStrategy
QCPAxisTicker, 1387
timeSeriesToOhlc
QCPFinancial, 1513
TimeUnit
QCPAxisTickerTime, 1410
TimeUnitConvert
Util, 2633
Todo List, 37
Token, 2576
TokenService, 2577
tools package, 35
tools/AIAssistant/AIAssistant_if.h, 3644
tools/AIAssistant/AIAssistantDefaultImpl.h, 3647
tools/AIAssistant/AIAuditLog.h, 3649
tools/AIAssistant/AIConversationService.h, 3650
tools/AIAssistant/AIProviderClient_if.h, 3651
tools/AIAssistant/AISecretStore.h, 3652
tools/AIAssistant/AnthropicProviderClientImpl.h, 3652
tools/AIAssistant/HttpProviderClientBase.h, 3653
tools/AIAssistant/LocalProviderClientImpl.h, 3654
tools/AIAssistant/OpenAIProviderClientImpl.h, 3654
tools/Biochemical/BioKineticLawExpression.h, 3655
tools/Biochemical/BioSimulationAnalysis.h, 3658
tools/Biochemical/BioSimulationResult.h, 3663
tools/Biochemical/GlpkFluxBalanceSolver.h, 3665
tools/Biochemical/MassActionOdeSystem.h, 3667
tools/Biochemical/MetabolicFluxBalanceSolver.h, 3669
tools/Biochemical/WholeCellParameterReader.h, 3673
tools/Continuous/DiffusionMethodOfLinesSystem.h, 3675
tools/Continuous/DormandPrince54OdeSolver.h, 3677
tools/Continuous/OdeSolver_if.h, 3679
tools/Continuous/OdeSolverFactory.h, 3679
tools/Continuous/OdeSystem_if.h, 3681
tools/Continuous/Quadrature_if.h, 3681
tools/Continuous/RootFinder_if.h, 3681
tools/Continuous/RungeKutta4OdeSolver.h, 3681
tools/Continuous/Solver_if.h, 3682
tools/Continuous/SolverDefaultImpl1.h, 3682
tools/FactorialDesign/FactorialDesign.h, 3683
tools/Optimization/Optimizer_if.h, 3684
tools/Optimization/OptimizerDefaultImpl1.h, 3686
tools/Statistics/ContinuousDistribution_if.h, 3687
tools/Statistics/DataAnalyser_if.h, 3687
tools/Statistics/DataSet_if.h, 3688
tools/Statistics/DiscreteDistribution_if.h, 3688
tools/Statistics/Distribution_if.h, 3688
tools/Statistics/Fitter_if.h, 3688
tools/Statistics/FitterDefaultImpl.h, 3689
tools/Statistics/FitterDummyImpl.h, 3696
tools/Statistics/HypothesisTester_if.h, 3697
tools/Statistics/HypothesisTesterDefaultImpl1.h, 3699
tools/Statistics/ProbabilityDistribution.h, 3700
tools/Statistics/ProbabilityDistributionBase.h, 3700
tools/Statistics/SimulationResultsDataset.h, 3701
tools/TraitsTools.h, 3702
top
QCPAxisRect, 1384
QCPPolarAxisAngular, 1690
toPainter
QCustomPlot, 1805
toPixmap
QCustomPlot, 1805
topLeft
QCPAxisRect, 1384
QCPPolarAxisAngular, 1690

topLevelItem
QtAbstractPropertyBrowser, 1817
topLevelItems
QtAbstractPropertyBrowser, 1817
toPoint
QCPVector2D, 1776
toPointF
QCPVector2D, 1776
topologicalOrder
DirectedGraphNetwork, 602
topRight
QCPAxisRect, 1384
QCPPolarAxisAngular, 1690
toQCPItemPosition
QCPItemAnchor, 1533
QCPItemPosition, 1565
toString
CompositeState, 435
HppLatticeGasState, 875
RealState, 2033
TraceConsoleController, 2577
TraceErrorEvent, 2578
TraceEvent, 2579
TraceManager, 2579
addTraceErrorHandler, 2581
addTraceHandler, 2581
addTraceReportHandler, 2581, 2582
addTraceSimulationHandler, 2582
errorMessages, 2582
getParentSimulator, 2582
getTraceLevel, 2583
isTraceSimulationRuleAllAllowed, 2583
setTraceLevel, 2583
setTraceSimulationRuleAllAllowed, 2583
TracerStyle
QCPItemTracer, 1587
traceSimulation
ModelDataDefinition, 1060
TraceSimulationEvent, 2584
TraceSimulationProcess, 2584
TraitsGUI< GAssociation >, 2586
TraitsGUI< GComponentPort >, 2586
TraitsGUI< GConnection >, 2586
TraitsGUI< GMainWindow >, 2587
TraitsGUI< GModelComponent >, 2587
TraitsGUI< GModelDataDefinition >, 2587
TraitsGUI< GModelGraphicPic >, 2588
TraitsGUI< GScene >, 2588
TraitsGUI< GView >, 2588
TraitsGUI< T >, 2585
TraitsKernel< Collector_if >, 2589
TraitsKernel< Model >, 2589
TraitsKernel< ModelChecker_if >, 2589
TraitsKernel< ModelComponent >, 2589
TraitsKernel< ModelDataDefinition >, 2589
TraitsKernel< Parser_if >, 2590
TraitsKernel< Persistence_if >, 2590
TraitsKernel< PluginConnector_if >, 2590
TraitsKernel< Sampler_if >, 2590
TraitsKernel< SimulationReporter_if >, 2590
TraitsKernel< Simulator >, 2591
TraitsKernel< Statistics_if >, 2591
TraitsKernel< StatisticsDatafile_if >, 2591
TraitsKernel< T >, 2588
TraitsTools< AIAssistant_if >, 2592
TraitsTools< Fitter_if >, 2592
TraitsTools< HypothesisTester_if >, 2592
TraitsTools< Solver_if >, 2592
TraitsTools< T >, 2591
Transporter, 2593
_check, 2597
_createEditableDataDefinitions, 2597
_initBetweenReplications, 2597
_loadInstance, 2598
_saveInstance, 2598
show, 2598
Transporter::TravelCompletion, 2599
triangular
ProbabilityDistributionBase, 1281
Trim
Util, 2634
Trimwithin
Util, 2634
tryCreateModel
SimulatorSessionService, 2263
tryGetCurrentModelInfo
SimulatorSessionService, 2264
tryGetSimulatorInfo
SimulatorSessionService, 2264
tsCircle
QCPItemTracer, 1587
tsCrosshair
QCPItemTracer, 1587
tsNone
QCPItemTracer, 1587
tsPlus
QCPItemTracer, 1587
tssMeetTickCount
QCPAxisTicker, 1387
tsSquare
QCPItemTracer, 1587
tssReadability
QCPAxisTicker, 1387
tStudent
ProbabilityDistributionBase, 1281
tuDays
QCPAxisTickerTime, 1410
tuHours
QCPAxisTickerTime, 1410
tuMilliseconds
QCPAxisTickerTime, 1410
tuMinutes
QCPAxisTickerTime, 1410
tuSeconds
QCPAxisTickerTime, 1410
type

QCPItemPosition, 1566
typeName
CompositeState, 435
HppLatticeGasState, 875
RealState, 2033
TypeOf
Util, 2634

undefinePen
QCPScatterStyle, 1738
UngroupUndoCommand, 2603
uniform
ProbabilityDistributionBase, 1282
uninitializeProperty
QtAbstractPropertyManager, 1823
QtBoolPropertyManager, 1827
QtCharPropertyManager, 1840
QtColorPropertyManager, 1851
QtCursorPropertyManager, 1858
QtDatePropertyManager, 1867
QtDateTimePropertyManager, 1875
QtDoublePropertyManager, 1881
QtEnumPropertyManager, 1893
QtFlagPropertyManager, 1897
QtFontPropertyManager, 1905
QtGroupPropertyManager, 1911
QtIntPropertyManager, 1917
QtKeySequencePropertyManager, 1923
QtLocalePropertyManager, 1930
QtPointFPropertyManager, 1936
QtPointPropertyManager, 1941
QtRectFPropertyManager, 1955
QtRectPropertyManager, 1960
QtSizeFPropertyManager, 1970
QtSizePolicyPropertyManager, 1974
QtSizePropertyManager, 1979
QtStringPropertyManager, 1991
QtTimePropertyManager, 1999
QtVariantPropertyManager, 2018
unsetFactoryForManager
QtAbstractPropertyBrowser, 1817
Unstore, 2613
_check, 2618
_createEditableDataDefinitions, 2618
_loadInstance, 2619
_onDispatchEvent, 2619
_saveInstance, 2619
show, 2619
update
QCPAxisRect, 1384
QCPColorScale, 1455
QCPColorScaleAxisRectPrivate, 1462
QCPLayout, 1610
QCPLayoutElement, 1618
QCPPolarAxisAngular, 1690
updateDimensionsAndPosition
GraphicalConnection, 820
updateDownloadProgress
genesys::launcher::UpdateDialog, 2622
updateLayout
QCPLayoutGrid, 1630
QCPLayoutInset, 1637
updateLegendIcon
QCPColorMap, 1441
UpdatePhase
QCPLayoutElement, 1614
updatePosition
QCPItemTracer, 1590
upLayout
QCPLayoutElement, 1614
upMargins
QCPLayoutElement, 1614
UponLeavingGTest
testing::internal::OsStackTraceGetter, 1195
upPreparation
QCPLayoutElement, 1614
Util, 2624
convertEnumToStr, 2626, 2627
DirSeparator, 2627
ExecuteCommand, 2627
FileDelete, 2627
FileExists, 2627
FilenameFromFullFilename, 2628
GenerateNewId, 2628
GenerateNewIdOfType, 2628
GetLastIdOfType, 2629
Indent, 2629
List2str, 2629
ListFiles, 2629
Map2str, 2630
PathFromFullFilename, 2630
ResetIdOfType, 2630
RunningPath, 2631
SepKeyVal, 2631
SetIndent, 2631
SetW, 2631
StrAllocation, 2631
StrIndex, 2632
StrReplace, 2632
StrReplaceSpecialChars, 2632
StrTimeUnitLong, 2632
StrTimeUnitShort, 2633
StrTrunclfInt, 2633
TimeUnitConvert, 2633
Trim, 2634
Trimwithin, 2634
TypeOf, 2634
Util::CommandResult, 425

validateProposedValue
SimulationControlSourceCodeString, 2225
SimulationControlString, 2228
validateSignalDeclarationPlacement
GroProgramRuntime, 859
validRange
QCPRange, 1731
value
QtBoolPropertyManager, 1827

QtCharPropertyManager, 1840
QtColorPropertyManager, 1851
QtCursorPropertyManager, 1858
QtDatePropertyManager, 1867
QtDateTimePropertyManager, 1875
QtDoublePropertyManager, 1881
QtEnumPropertyManager, 1893
QtFlagPropertyManager, 1897
QtFontPropertyManager, 1905
QtIntPropertyManager, 1917
QtKeySequencePropertyManager, 1924
QtLocalePropertyManager, 1930
QtPointFPropertyManager, 1937
QtPointPropertyManager, 1941
QtRectFPropertyManager, 1955
QtRectPropertyManager, 1960
QtSizeFPropertyManager, 1970
QtSizePolicyPropertyManager, 1974
QtSizePropertyManager, 1979
QtStringPropertyManager, 1991
QtTimePropertyManager, 1999
QtVariantProperty, 2012
QtVariantPropertyManager, 2018
testing::internal::HasDebugStringAndShortDebugString< T >, 873
valueChanged
QtBoolPropertyManager, 1827
QtCharPropertyManager, 1840
QtColorPropertyManager, 1851
QtCursorPropertyManager, 1858
QtDatePropertyManager, 1867
QtDateTimePropertyManager, 1875
QtDoublePropertyManager, 1881
QtEnumPropertyManager, 1893
QtFlagPropertyManager, 1897
QtFontPropertyManager, 1905
QtIntPropertyManager, 1917
QtKeySequencePropertyManager, 1924
QtLocalePropertyManager, 1930
QtPointFPropertyManager, 1937
QtPointPropertyManager, 1941
QtRectFPropertyManager, 1955
QtRectPropertyManager, 1960
QtSizeFPropertyManager, 1970
QtSizePolicyPropertyManager, 1974
QtSizePropertyManager, 1979
QtStringPropertyManager, 1991
QtTimePropertyManager, 1999
QtVariantPropertyManager, 2018
valueIcon
QtAbstractPropertyManager, 1823
QtBoolPropertyManager, 1828
QtColorPropertyManager, 1851
QtCursorPropertyManager, 1858
QtEnumPropertyManager, 1893
QtFontPropertyManager, 1906
QtProperty, 1947
QtVariantPropertyManager, 2018
valueRange
QCPBarsData, 1423
QCPCurveData, 1474
QCPDataContainer< DataType >, 1482
QCPFinancialData, 1515
QCPGraphData, 1527
QCPStatisticalBoxData, 1763
values
SimulationResultsDataset, 2246
valuesForReplication
SimulationResultsDataset, 2247
valueText
QtAbstractPropertyManager, 1823
QtBoolPropertyManager, 1828
QtCharPropertyManager, 1840
QtColorPropertyManager, 1851
QtCursorPropertyManager, 1858
QtDatePropertyManager, 1867
QtDateTimePropertyManager, 1875
QtDoublePropertyManager, 1882
QtEnumPropertyManager, 1893
QtFlagPropertyManager, 1897
QtFontPropertyManager, 1906
QtIntPropertyManager, 1917
QtKeySequencePropertyManager, 1924
QtLocalePropertyManager, 1930
QtPointFPropertyManager, 1937
QtPointPropertyManager, 1941
QtProperty, 1947
QtRectFPropertyManager, 1956
QtRectPropertyManager, 1960
QtSizeFPropertyManager, 1970
QtSizePolicyPropertyManager, 1974
QtSizePropertyManager, 1980
QtStringPropertyManager, 1992
QtTimePropertyManager, 1999
QtVariantPropertyManager, 2018
valueToolTip
QtProperty, 1947
valueType
QtVariantProperty, 2012
QtVariantPropertyManager, 2019
Variable, 2638
_check, 2643
_initBetweenReplications, 2643
_loadInstance, 2643
_saveInstance, 2643
getInitialValuesText, 2644
setInitialValuesText, 2644
show, 2644
variance
Statistics_if, 2477
StatisticsDatafileDefaultImpl1, 2491
StatisticsDefaultImpl1, 2494
varianceConfidenceInterval
HypothesisTesterDefaultImpl1, 891
varianceRatioConfidenceInterval
HypothesisTesterDefaultImpl1, 891

variantProperty
QtVariantPropertyManager, 2019
variationCoef
Statistics_if, 2478
StatisticsDatafileDefaultImpl1, 2491
StatisticsDefaultImpl1, 2494
verify
genesys::launcher::GpgvSignatureVerifier, 808
version
BacteriaColonyViewerGuiExtensionPlugin, 228
BioNetworkAwareGuiExtensionPlugin, 261
BioNetworkEditorGuiExtensionPlugin, 263
BioNetworkExecutionGuiExtensionPlugin, 264
BioParameterManagerGuiExtensionPlugin, 270
BioReactionKineticsGuiExtensionPlugin, 283
BioReactionStoichiometryGuiExtensionPlugin, 285
BioSBMLInteroperabilityGuiExtensionPlugin, 293
BioSimulationResultsGuiExtensionPlugin, 302
BioSpeciesManagerGuiExtensionPlugin, 314
CellularAutomataGuiExtensionPlugin, 396
CoreGuiExtensionPlugin, 460
GeneticCircuitGuiExtensionPlugin, 781
GuiExtensionConfigurationGuiExtensionPlugin, 864
ModelPluginInspectorGuiExtensionPlugin, 1093
Vpulse, 2645
_loadInstance, 2650
_saveInstance, 2650
Build, 2651
Vsine, 2651
_loadInstance, 2656
_saveInstance, 2656
Build, 2657
Vsource, 2657
_loadInstance, 2662
_saveInstance, 2662
Build, 2663

Wait, 2663
_check, 2668
_createEditableDataDefinitions, 2668
_createNonEditableDataDefinitions, 2668
_initBetweenReplications, 2668
_loadInstance, 2669
_onDispatchEvent, 2669
_saveInstance, 2669
show, 2669
Waiting, 2670
WaitingResource, 2670
show, 2671
WcmBioNetworkBridge, 2671
main, 2673
WcmCompartmentTransportWholeCellMvp, 2673
main, 2675
WcmFluxBalanceWholeCellMvp, 2675
main, 2677
WcmGeneExpression, 2677
main, 2679
WcmLifecycleStress, 2679
main, 2681
WcmMgenitaliumKarr2012, 2681
main, 2683
WcmPathwayStressWholeCellMvp, 2684
main, 2685
WcmSimpleSsa, 2685
main, 2687
WcmWholeCellMvp, 2687
main, 2689
WcmYeastDidacticWholeCellMvp, 2689
main, 2691
WcmYeastEukaryoticWholeCellMvp, 2691
main, 2693
WebWorkerDialog, 2693
WebWorkerRuntime, 2694
restart, 2695
start, 2695
WebWorkerRuntime::Snapshot, 2417
weibull
ProbabilityDistributionBase, 1282
whatsThis
QtProperty, 1948
wheelEvent
QCPAxis, 1367
QCPAxisRect, 1384
QCPColorScale, 1456
QCPColorScaleAxisRectPrivate, 1463
QCPLayerable, 1604
QCPPolarAxisAngular, 1691
QCPPolarAxisRadial, 1710
WholeCellParameterReader, 2696
WholeCellState, 2696
_check, 2701
_initBetweenReplications, 2701
_loadInstance, 2701
_saveInstance, 2701
show, 2702
width
QCPAxisRect, 1384
QCPPolarAxisAngular, 1691
WidthType
QCPBars, 1416
QCPFinancial, 1509
WorkerJob, 2705
WorkerJobManager, 2707
createQueuedJob, 2707
getJob, 2707
setMessage, 2708
setSnapshotFilename, 2708
setState, 2708
setTerminalResult, 2708
WorkerJobTerminalResult, 2711
Write, 2712
_check, 2717
_initBetweenReplications, 2717
_loadInstance, 2717
_onDispatchEvent, 2717
_saveInstance, 2718

show, 2718
wtAbsolute
QCPBars, 1417
QCPFinancial, 1509
wtAxisRectRatio
QCPBars, 1417
QCPFinancial, 1509
wtPlotCoords
QCPBars, 1417
QCPFinancial, 1509

xAxis
QCustomPlot, 1806
xAxis2
QCustomPlot, 1806
XmlSerializer, 2718
dump, 2719
for_each, 2719
get, 2720
load, 2720
newPersistenceRecord, 2720
put, 2720
XNOR, 2721
_loadInstance, 2727
_saveInstance, 2727
Build, 2727
XOR, 2728
_loadInstance, 2733
_saveInstance, 2733
Build, 2733

yAxis
QCustomPlot, 1806
yAxis2
QCustomPlot, 1807
yy::genesyspp_parser, 767
error, 770
operator(), 770
parse, 770
symbol_name, 770
yy::genesyspp_parser::basic_symbol< Base >, 240
yy::genesyspp_parser::by_kind, 341
kind, 343
kind_, 343
yy::genesyspp_parser::context, 444
expected_tokens, 444
yy::genesyspp_parser::stack< T, S >::slice, 2270
yy::genesyspp_parser::symbol_kind, 2546
symbol_kind_type, 2547
YYNTOKENS, 2547
yy::genesyspp_parser::symbol_type, 2547
yy::genesyspp_parser::syntax_error, 2550
yy::genesyspp_parser::token, 2576
yy::genesyspp_parser::value_type, 2635
build, 2636
move, 2636
swap, 2636
yy::location, 944
yy::position, 1276
yy_bs_column
yy_buffer_state, 2734
yy_bs_lineno
yy_buffer_state, 2734
yy_buffer_state, 2734
yy_bs_column, 2734
yy_bs_lineno, 2734
yy_trans_info, 2734
YYNTOKENS
yy::genesyspp_parser::symbol_kind, 2547

zoom
QCPAxisRect, 1384, 1385